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Archive chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, manuscript frames 1–1527. Catalogue snapshot: 5 October 2026.

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Gelfand Seminar

Diplomatic translation. Source frames 1–1527.

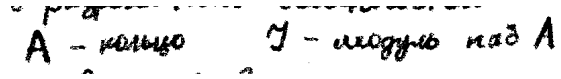
Chapter 1

GELFAND-P000001

Monday, 21/IX. Cartesian and Boolean products of groups.

“Hironaka: on resolution of singularities.” Gelfand

A is a ring, $\mathcal{J}$ is a module over A.

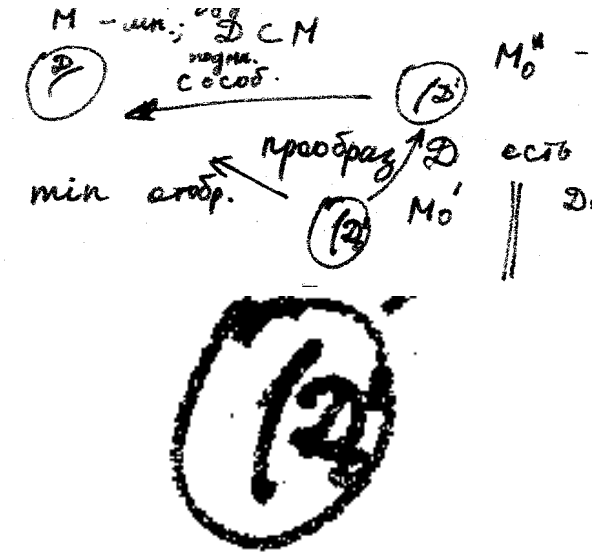


Original source details: GELFAND-P000001

Uncertain reading. The letter for the module and ideal is rendered as $\mathcal{J}$; $\mathcal{I}$ or J are possible.

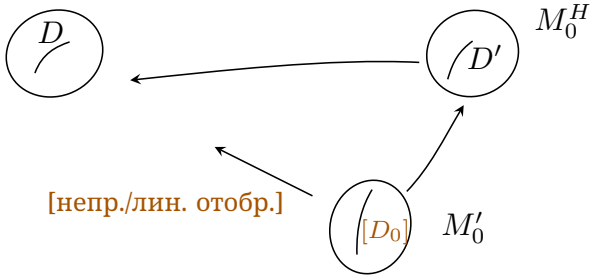
Blow-up — ? M is a manifold; $D \subset M$. [Beneath D:] a subset with singularities.

M_0^H is the blow-up. Blowing-up. The inverse image of D is D', a manifold of dimension $n-1$. D_0 is $(n-1)$-dimensional.



Original source details: GELFAND-P000001

Diagram and typed description. Three enclosed regions: D at left, D' and M_0^H at upper right, and a region labelled M_0' below. The upper and lower oblique arrows point left; a curved arrow rises from the lower region toward the upper right one. The lower-left arrow’s caption reads непр. / лин. отобр. [continuous/linear map]; the lower interior sign reads D_0 / D, with overlapping strokes. Some curve endings remain indistinct.



Typographic recreation; square brackets mark uncertain readings.

Original source details: GELFAND-P000001

Uncertain reading. The upper index on M_0 resembles H; the lower region’s label resembles M_0'.

The construction is Grothendieck’s. A is a ring, $\mathcal{J}$ is a module. Example. A is a ring of functions; $\mathcal{J}$ is the ideal of functions taking the value 0 on D.

One obtains a sheaf of rings and a sheaf of modules. A is a ring, $\mathcal{J}$ is an ideal. Construct X, O_X a sheaf of rings, $\mathcal{J}_X$ a sheaf of ideals, so that

$$(A, \mathcal{J}) \longleftarrow (X, O_X, \mathcal{J}_X).$$

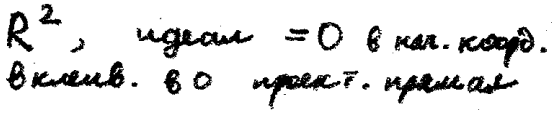
[Beneath the arrow:] linear map.

$(A, \mathcal{J})$ is a sheaf generated by 1 ideal.

$\mathcal{J}_X$ is invertible, i.e. an ideal of the ring: $\exists f \in \mathcal{J}_X$ such that $O_X f = \mathcal{J}_X$. [Above the formula, an arrow to $\exists$:] generator.

(The ideal is principal if $\mathcal{J}$ is an ideal.) Then the zero variety is 1 equation.

A is a ring, $\mathcal{J}$ is an ideal. Typical example: R^2, the ideal $= 0$ at the origin.



Original source details: GELFAND-P000001

Uncertain reading. The next line: “Gluing a projective line into 0” [source: Вклеив. в 0 проект. прямой]; the beginning may read “Оклеив.”.

A, $\mathcal{J}$ are modules.

$$\mathcal{J} \otimes_A \mathcal{J}, \qquad \mathcal{J} \otimes_A \mathcal{J} \otimes_A \mathcal{J}, \dots$$

[Above the groups of tensor products:] symmetric; symmetric.

Consider formal sums

$$\sum_{k=0}^{N} a_k z^k.$$

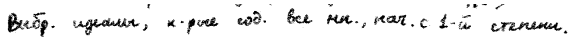
Add and multiply. $x \otimes y = y \otimes x$.

$$a_0 \in A, \qquad a_1 \in \mathcal{J}, \qquad a_2 \in \mathcal{J} \otimes_A \mathcal{J} \quad —$$

GELFAND-P000002

The ring of these sums is $S_A(\mathcal{J})$. Consider homogeneous ideals [inserted above the line:] (graded). (Together with a polynomial it contains all its homogeneous components.)

Consider all homogeneous prime ideals in $S_A(\mathcal{J})$. Выбр. [source abbreviation] ideals which contain all polynomials, starting with degree 1.



Original source details: GELFAND-P000002

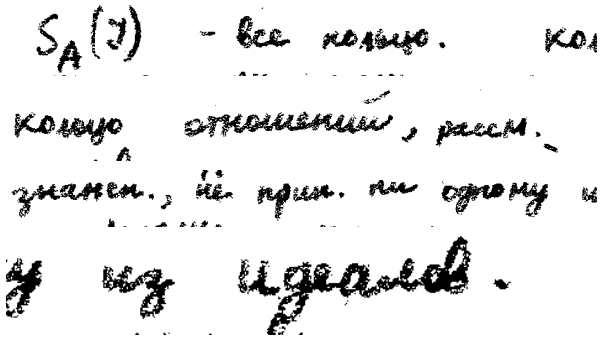
Uncertain reading. The end of the line after “contain all” is better read as “polynomials, starting with degree 1”; the middle strokes of the abbreviation “нач.” remain indistinct.

The collection of ideals is X. (The Zariski topology.) O_X, $\mathcal{J}_X$. If U is an open set, then O_U, $\mathcal{J}_U$.

Original source details: GELFAND-P000002

Uncertain reading. After “then”, the first letter has the same form as O_X above, the second as the ideal letter: O_U, $\mathcal{J}_U$; the latter may be $\mathcal{I}_U$.

$S_A(\mathcal{J})$ is the whole ring. The ring of fractions, numerators, denominators, not belonging to any of the ideals. Localize the ring at this set. Fractions of degree 1: $\mathcal{J}_U$.

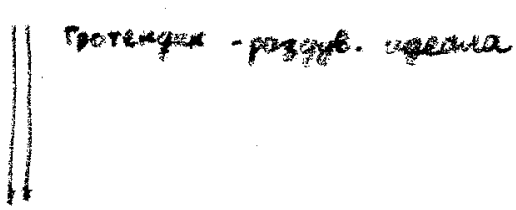


Original source details: GELFAND-P000002

Uncertain reading. In the line about fractions, the abbreviations “numerator–denominator” and whether “not belonging” applies only to the denominator are unclear. No words are supplied from the meaning.

$$\Downarrow$$

[At right:] Grothendieck — blow up ideally.



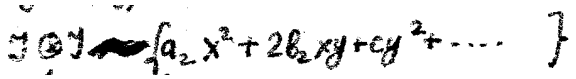
Original source details: GELFAND-P000002

Uncertain reading. The right-hand note reads “Grothendieck — blow up ideally”; the last word may be “идеально” [ideally] or “идеалы” [ideals].

A is the ring of polynomials in 2 variables. $\mathcal{J}$ is the ideal of polynomials with $P(0,0)=0$.

$$A=\{P(x,y)\},\qquad \mathcal{J}=P(x,y)=a_1x+b_1y+\cdots.$$

$$\mathcal{J}\otimes_A\mathcal{J}\ \longleftrightarrow\ \{a_2x^2+2b_2xy+cy^2+\cdots\}.$$



Original source details: GELFAND-P000002

Uncertain reading. The coefficients in the quadratic polynomial read a_2, $2b_2$, c; the index on c and the shape of the thick two-headed arrow are uncertain.

Because of the interchange, one obtains the same as the square of the ideal. Then — the ideal of polynomials equal to 0 at this point. What will it blow up into? ?

$$\sum P_k(x,y)z^k,\qquad P_k(x,y)=ax^k+\cdots.$$

[At right:] Graded Σ of tensor products.

$$A=\{P_0(x,y)\}\longrightarrow S_A(\mathcal{J}),\qquad P_0(x,y)\sim P_0(x,y).$$

The set of prime ideals obtained from the ideal $P(0,0)=0$.

$$S_A\sim\sum P_kz^k,\qquad P_k\in\mathcal{J}^k,\qquad K\sim\sum P_kz^k.$$

It suffices to find the max. ideals of the quotient ring — (to find the ideals containing the given one).

$$S_A/K\ \longleftrightarrow\ a_0+(a_1x-b_1y)z+\cdots.$$

Original source details: GELFAND-P000002

Uncertain reading. In the last formula a minus is visible before b_1y, although the next scan has a plus. The visible minus is retained; the direction of the thick arrow is also uncertain.

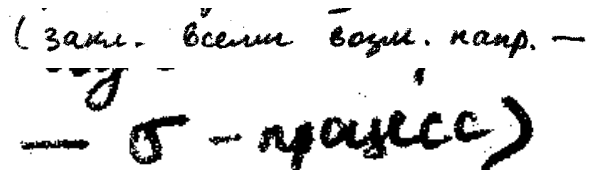
GELFAND-P000003

All homogeneous max. ideals in it will be the points of the extension. There is no constant term (then also 1).

$$(a_1x+b_1y)z.$$

Only 1 linear combination of x and y. (Not 2, since then all.)

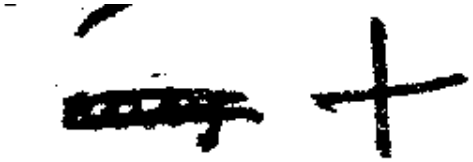
The ideals determine directions in the plane (замк. [source abbreviation] with all possible directions — the σ-process).



Original source details: GELFAND-P000003

Uncertain reading. The abbreviations in parentheses read “замк. всеми возм. напр.”; neither the first word nor the meaning of the sign σ is asserted.

Problem. In the space [cancelled:]



Original source details: GELFAND-P000003

Uncertain reading. The cancelled word before the drawing is illegible.

Original diagram

Typeset recreation

What results from gluing it in?

$$z = 0, \qquad xy = 0, \qquad \mathcal{J} = (z, xy).$$

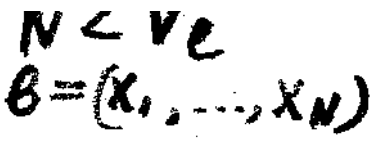
[A line is drawn beneath the system.]

Dobrushin. [Cancelled before “Find”:]

Original source details: GELFAND-P000003

Uncertain reading. The cancelled beginning of the line cannot be read confidently.

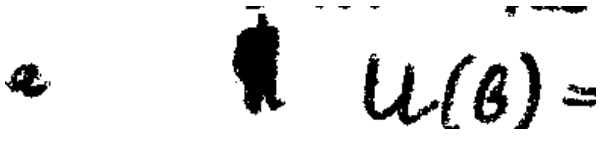
Find an example of a gas in which phase transitions occur. A cube of the integer lattice. ℓ is the length. $V_\ell = \ell^\nu$. N points, $N < V_\ell$. They can occupy positions on the lattice $\theta = (x_1, \ldots, x_N)$.



Original source details: GELFAND-P000003

Uncertain reading. The configuration sign is rendered as θ; the handwritten sign also resembles b or Cyrillic “в”.

The Ising model. [Before the formula, a densely cancelled sign:]

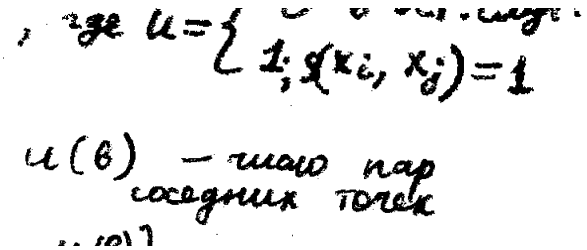


Original source details: GELFAND-P000003

Uncertain reading. The separate cancelled sign before $U(\theta)$ is illegible.

$$U(\theta) = -\frac{1}{2}\sum_{i\neq j} u(x_i, x_j), \qquad u = \begin{cases} 0 & \text{in the other cases,} \\ 1; & \rho(x_i, x_j) = 1. \end{cases}$$

$u(\theta)$ is the number of pairs of neighboring points.



Uncertain reading. In the condition for the second branch a sign resembling ρ is visible; the semicolon after 1 is retained. In the caption below, the first letter resembles lowercase u rather than U.

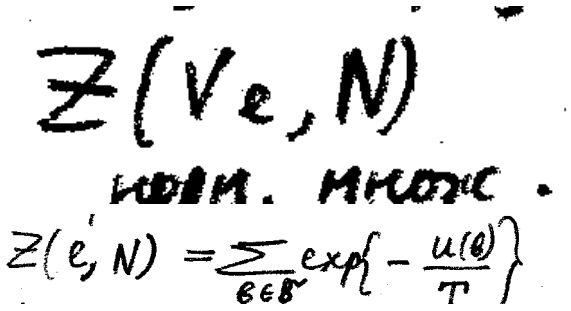
Original source details: GELFAND-P000003

The basic axiom is the Gibbs distribution.

$$\mathsf{P}(B) = \frac{\sum\limits_{\theta\in B} \exp\{-U(\theta)/T\}}{Z(V_\ell, N)}, \qquad T > 0 \text{ — temperature.}$$

[Beneath the denominator:] normalizing factor.

$$Z(\ell, N) = \sum_{\theta\in\mathcal{B}} \exp\{-U(\theta)/T\}.$$



Uncertain reading. The denominator in the first formula reads $Z(V_\ell, N)$, the next line $Z(\ell, N)$; the set sign beneath the sum resembles $\mathcal{B}$.

Original source details: GELFAND-P000003

The molecules attract one another. ([Two filled signs with arrows:] neighbors $>$ their probability.)

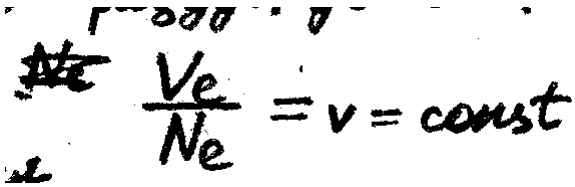
Uncertain reading. In parentheses after "The molecules attract", there are filled signs and arrows; the reading "neighbors $>$ their probability" is uncertain.

Original source details: GELFAND-P000003

$B \subset \mathcal{B}$ is a collection of configurations.

$$p(\theta) = \frac{\exp\{-U(\theta)/T\}}{Z(\ell, N)}.$$

Take the limit $\ell \to \infty$; the cube expands to ∞. N_ℓ depends on ℓ. [A cancelled sign before the fraction:]



Uncertain reading. The cancelled sign before V_ℓ/N_ℓ is illegible.

Original source details: GELFAND-P000003

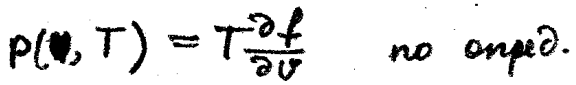
$$\frac{V_\ell}{N_\ell} = v = \text{const.}$$

The lattice grows. One can show that

$$\frac{1}{N_\ell} \log Z(\ell, N_\ell) \longrightarrow f(v, T).$$

GELFAND-P000004

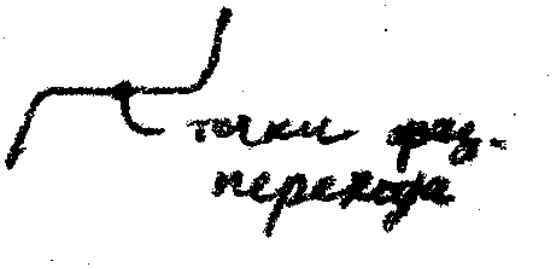
$$p(v,T) = T\frac{\partial f}{\partial v} \qquad \text{by definition.}$$



Original source details: GELFAND-P000004

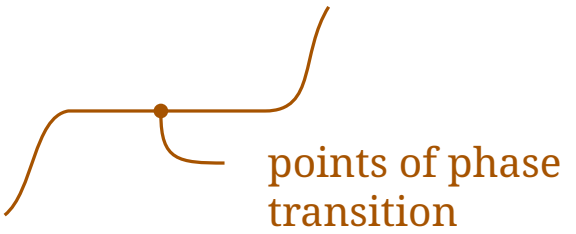
Uncertain reading. The heavily written first argument of p resembles v; no minus is visible before the derivative.

Phase transition. As for f: it is a monotone, convex function of v. Therefore $p > 0$, and p is a monotone function of v. Phase transition: $p(v, T)$ has a flat segment.



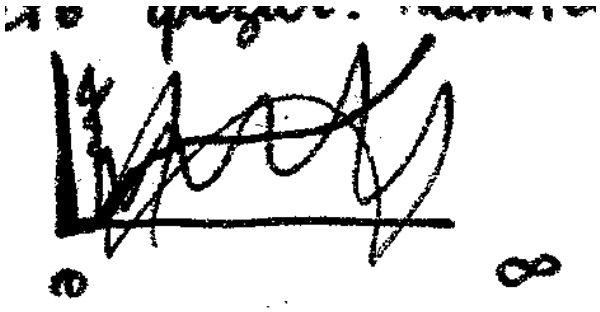
Original source details: GELFAND-P000004

Diagram and typed description. In the upper right-hand field is a bent, S-like curve with an almost horizontal middle segment and an arrow toward it; the caption reads “points of phase transition” [source abbreviation: точки фаз. перехода]. Precise axes and points in the original drawing are not identified confidently.



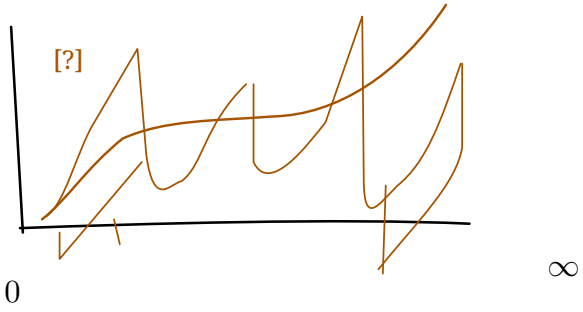
Typeset sketch; limitations are recorded in the note.

There is a physical hypothesis: 1 transition segment.

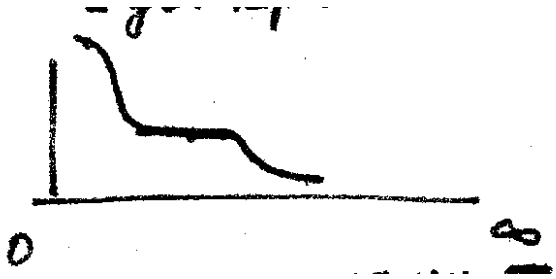


Original source details: GELFAND-P000004

Diagram and typed description. Intersecting wavy and oblique lines, horizontal and vertical axes, 0 and ∞. The strokes overlap; their exact separation into curves and the vertical label are unresolved.

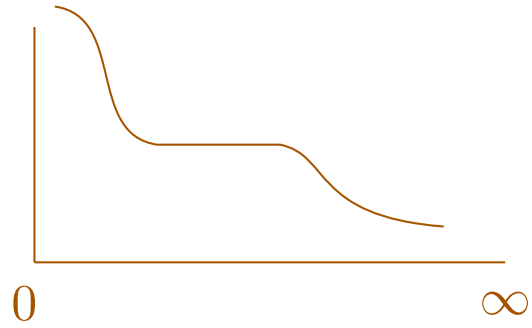


Visible strokes; [?] marks the unreadable vertical label.



Original source details: GELFAND-P000004

Diagram and typed description. Neighboring drawing at right: a clean descending curve with a flat segment. This is a separate graph, not a reconstruction of the cancelled drawing at left.

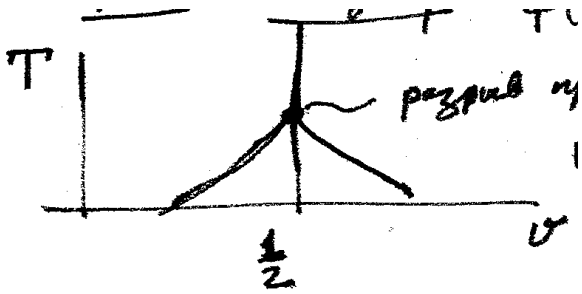


Typeset recreation.

A flat segment occurs for sufficiently small T ($< T_{\text{crit}}$).

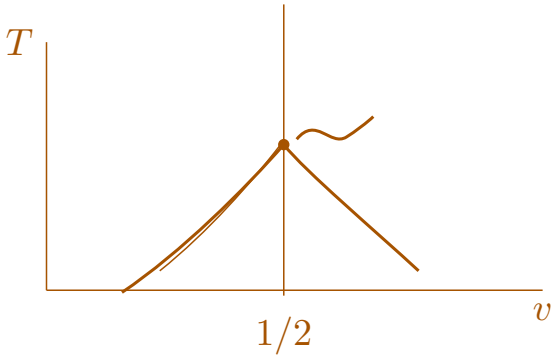
It admits an explicit solution for $n = 1$. Z and $f(v, T)$ are calculated. There are no phase transitions, which is considered physically natural.

$n = 2$. Onsager. $f(v, T)$ is calculated for particular v and T. A discontinuity of the derivative corresponds to phase transitions.



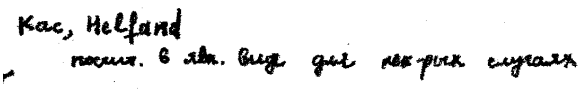
Original source details: GELFAND-P000004

Diagram and typed description. A graph with the T axis upward, the v axis to the right, and the label $1/2$ beneath a vertical through the vertex. The curve rises from the left to the vertex and descends to the right; the vertical line passes through the vertex, with a short wavy segment to its right. The precise geometry and the meaning of the wavy branch have not been established.



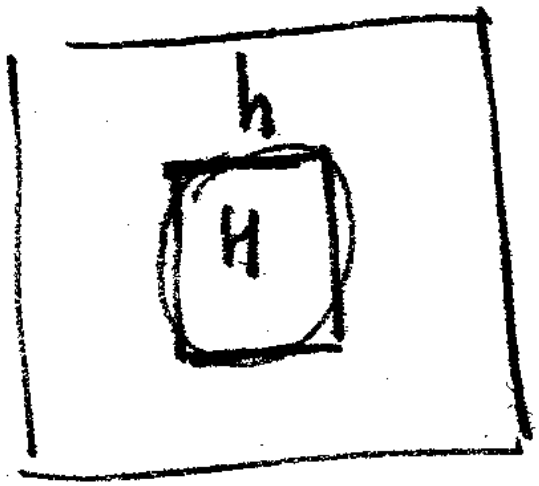
Typeset sketch; limitations are recorded in the note.

Kac, Helfand. Exact, in explicit form, for some cases [source endings: точн. в яв. виду для нек-рых случаях].



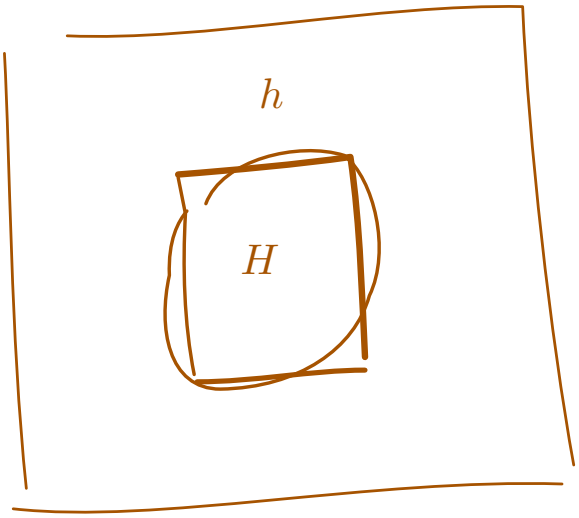
Original source details: GELFAND-P000004

Uncertain reading. The line reads “точн. в яв. виду для нек-рых случаях”; the endings may instead be “виде” and “случаев”.



Original source details: GELFAND-P000004

Diagram and typed description. Within the large frame, a small rectangular contour overlaps a rounded one; H lies inside and h above. The rounded contour protrudes to the right and left. Individual lines merge at the heavy intersections.



Visible contours and strokes; uncertain connections have not been completed.

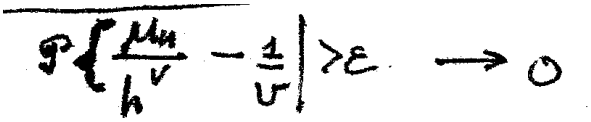
$$\frac{\mu_H}{h^\nu}.$$

μ_H is the number of particles in H — a subcube. ν is the dimension.

$$\frac{\mu_H}{h^\nu} = \frac{1}{v} \qquad \text{as in the whole space.}$$

Law of large numbers.

$$\mathsf{P}\left\{\left|\frac{\mu_H}{h^\nu} - \frac{1}{v}\right| > \varepsilon\right\} \longrightarrow 0.$$



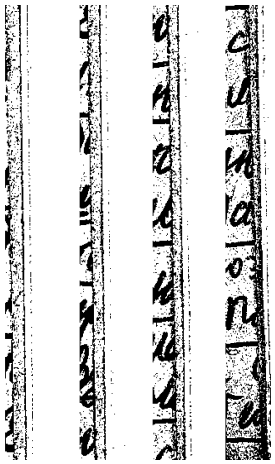
Original source details: GELFAND-P000004

Uncertain reading. In the probability formula the right vertical bar before $> \varepsilon$ is visible; a left bar before the fraction has not been read confidently.

$$\ell \to \infty, \qquad H \to \infty, \qquad d < \frac{h}{\ell} < 1 - d, \qquad d > 0.$$

Uncertain reading. At the lower right edge a small sign resembles “2” or a broken-off letter; it has not been adopted as a page number.

Original source details: GELFAND-P000004

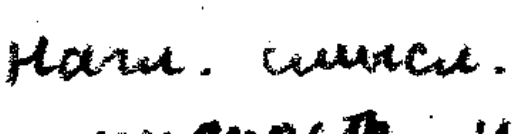


Undeciphered passage. Along the extreme left edge is a narrow strip of neighboring handwritten text, cropped to isolated endings and strokes; no connected text can be read.

Original source details: GELFAND-P000004

GELFAND-P000005

If this does not hold, a phase transition takes place. Нагл. meaning. 2 phases coexist; the particles are distributed unevenly — — liquid and gas.

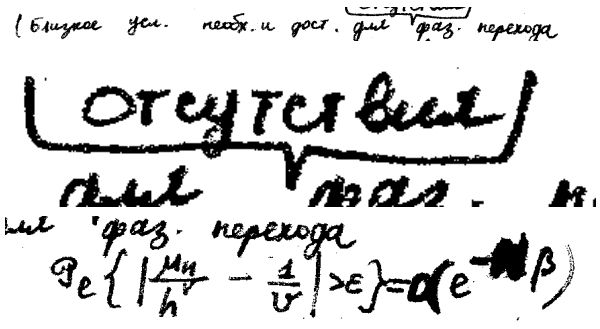


Uncertain reading. Нагл. [candidate abbreviation; the middle strokes are not fully distinguishable]

Original source details: GELFAND-P000005

[Above the line:] (law of large numbers). [Inserted above “phase transition”, with a bracket:] absence. (A closely related condition is necessary and sufficient for a phase transition

$$\mathsf{P}_\ell\left\{\left|\frac{\mu_H}{h^\nu}-\frac{1}{v}\right|>\varepsilon\right\}=O(e^{-N\beta}).$$



Uncertain reading. The exponent of e contains a heavily written sign before β; the candidate is $-N\beta$, but N is obscured by overlapping strokes. The parenthesis opened before “A closely related” is not clearly closed in the scan.

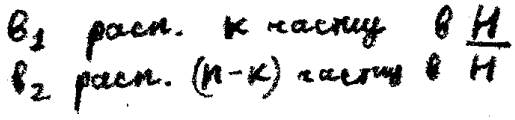
Original source details: GELFAND-P000005

[At right:] N is the number of particles.

Calculate the probability that

$$\mathsf{P}\{\mu_H=k\}=Z^{-1}\sum_{\theta\in\mathcal{B}_k}\exp\{-U(\theta)/T\}.$$

Over configurations in which H contains k particles. θ is a configuration of N particles. $\theta\in\mathcal{B}_k$. Then $\theta=(\theta_1,\theta_2)$. θ_1 is a configuration of k particles in H. θ_2 is a configuration of $(N-k)$ particles in $\overline{H}$.

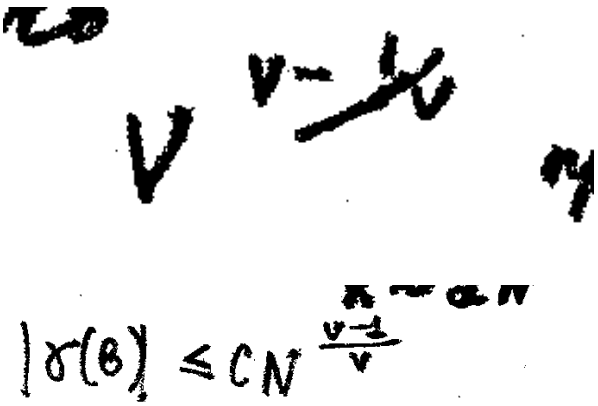


Uncertain reading. The upper bar on H in the line about θ_2 is not completely distinct; the candidate is $\overline{H}$.

Original source details: GELFAND-P000005

$$U(\theta) = U(\theta_1) + U(\theta_2) + \gamma(\theta).$$

$U(\theta_1)$ is the interaction inside. $U(\theta_2)$: neighbors outside. $\gamma(\theta)$ is the interaction across the surface. $\gamma(\theta)$ can be neglected asymptotically. (Across the boundary: of the order of the number of boundary points) $V^{\nu-1/\nu}$ as $N \to \infty$. $k \sim dN$.

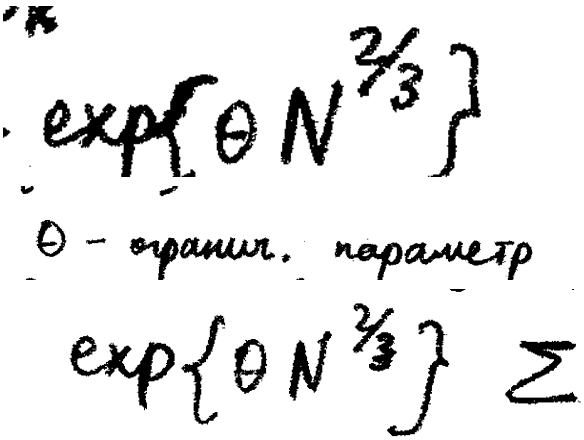


Uncertain reading. The exponent on V looks like $\nu - 1/\nu$, without a confidently distinguishable common numerator. A possible grouping is $(\nu - 1)/\nu$.

Original source details: GELFAND-P000005

$$|\gamma(\theta)| \le CN^{\frac{\nu-1}{\nu}}.$$

$$\sum \exp\{-U(\theta)/T\} = \sum_{\theta \in \mathcal{B}_k} \exp\{-U(\theta_1)/T\} \exp\{-U(\theta_2)/T\}$$
$$\times \exp\{\theta N^{2/3}\} =$$



Uncertain reading. The letter before $N^{2/3}$ resembles θ or Θ; below, the same letter is glossed as “bounded parameter”. It may differ from the configuration sign.

Original source details: GELFAND-P000005

θ is a bounded parameter.

$$= \exp\{\theta N^{2/3}\} \sum_{\theta_1} \exp\{-U(\theta_1)/T\} \sum_{\theta_2} \exp\{-U(\theta_2)/T\}.$$

[Beneath the first sum:] configurations.

GELFAND-P000006

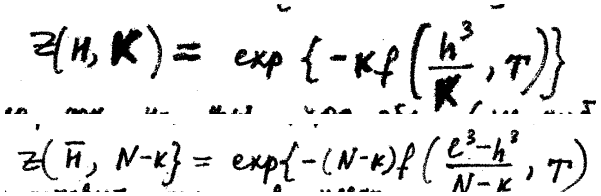
Factor 1: Z, calculated for the subcube. Factor 2 — ditto — Z for the outer cube.

$$\mathsf{P}\{\mu_H = k\} = \frac{Z(H,k)Z(\overline{H}, N-k)}{Z(V,N)} \exp\{\theta N^{2/3}\}.$$

[Asterisk.] Apply to them the asymptotic formula

$$Z(V,N) \approx \exp\{-Nf(v,T)\},$$

$$Z(H,k) = \exp\{-kf(h^3/k,T)\}.$$



Uncertain reading. The numerator of the argument of f in this line visibly has exponent 3 on h, although ν is used later.

Original source details: GELFAND-P000006

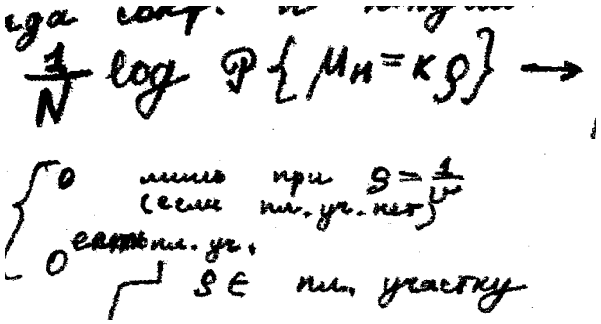
It can also be proved for well-behaved regions (not a cube). The limiting formula does not depend on the region's shape.

$$Z(\overline{H},N-k) = \exp\{-(N-k)f((\ell^3-h^3)/(N-k),T)\}.$$

Substitute there under the assumption that f has a linear segment. If f has a linear segment (the derivative has a horizontal one), then terms cancel and we obtain

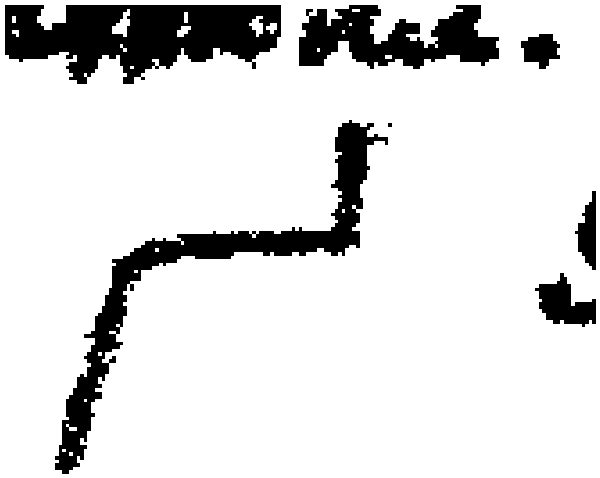
$$\frac{1}{N}\log \mathsf{P}\{\mu_H = k\rho\} \longrightarrow \begin{cases} 0 & \text{only at } \rho = 1/v \text{ (if there is no linear segment)}, \\ 0 & \rho \in \text{ the linear segment.} \end{cases}$$

[Cancelled by the second branch:]



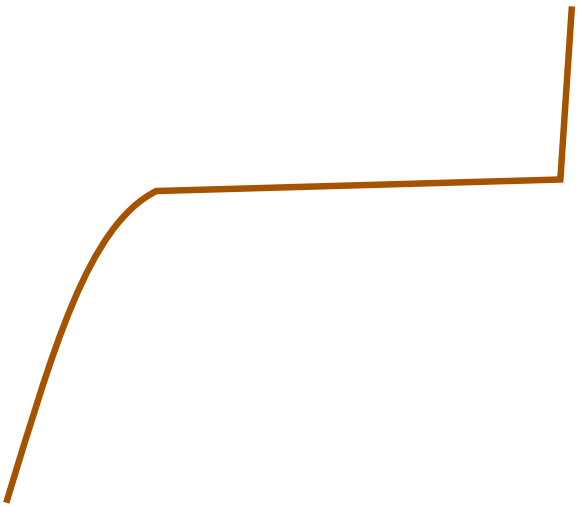
Uncertain reading. The event's argument looks like $\mu_H = k\rho$, but the final sign may be θ or δ. The density letter at right resembles ρ. The cancelled text above the second branch has not been read.

Original source details: GELFAND-P000006



Original source details: GELFAND-P000006

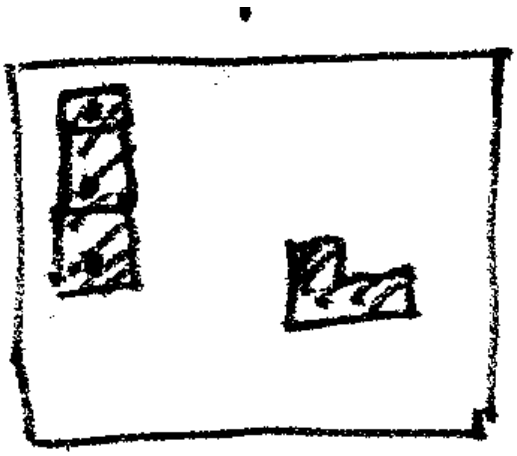
Diagram and typed description. At right of the second branch of the limit is a small stroke: from the lower left, a line rises almost vertically, then runs almost horizontally to the right, and rises again at the right end. Separate axes, arrows and labels have not been established; the sketch's mathematical meaning requires the original image.



Visible stroke; no axes or values supplied.

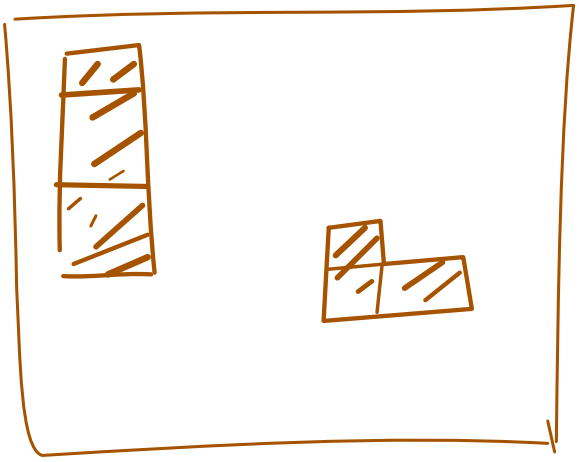
$\mathsf{P} \to 0$ exponentially. ($1/v$ follows from the convexity of f.)

Arrangement of particles. Its boundary; around each particle a cube of side 1. $\Gamma'(\theta)$ is the boundary of the occupied region. $\Gamma(\theta)$ is the length [inserted below:] area.



Original source details: GELFAND-P000006

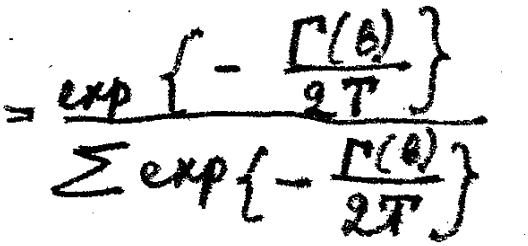
Diagram and typed description. At left is a vertical outline with two strong transverse strokes, diagonal hatching and separate short marks. At right, three cells form an L: two below and one above the left cell. In the left column, divisions merge with the hatching; the exact cell count is uncertain.



Visible contours and strokes; uncertain connections have not been completed.

Replace the Gibbs probability.

$$\mathsf{P}(\theta) = \frac{\exp\{-U(\theta)/T\}}{\sum \exp\{-U(\theta)/T\}} = \frac{\exp\{-\Gamma'(\theta)/(2T)\}}{\sum \exp\{-\Gamma'(\theta)/(2T)\}}.$$

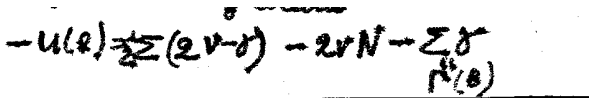


Original source details: GELFAND-P000006

Uncertain reading. The prime on Γ in the two exponents after the equality is not completely clear; the visible letter resembles Γ' or Γ.

[At right:] $2\nu - \delta$. Unoccupied squares $\Gamma'(\theta)$. [Inserted:] at neighbors.

$$-U(\theta) \geq \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta.$$



Original source details: GELFAND-P000006

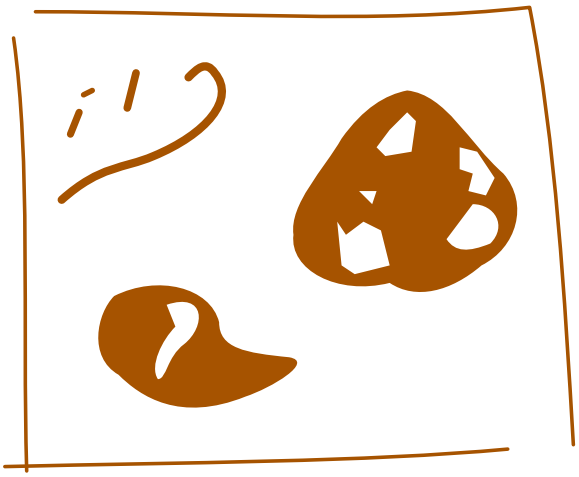
Uncertain reading. The sign after $-U(\theta)$ is written over another sign: candidates are $\geq$, $\leq$, and $=$. The sign before the final sum and the sum's index may belong to another term.

GELFAND-P000007

($U(\theta)$ is minus the number of neighbors.) A smaller boundary makes an arrangement more favorable. Force them into one corner, but this is unfavorable (there are few such arrangements in all). A finite number of clumps. (Inside the gas: liquid.)

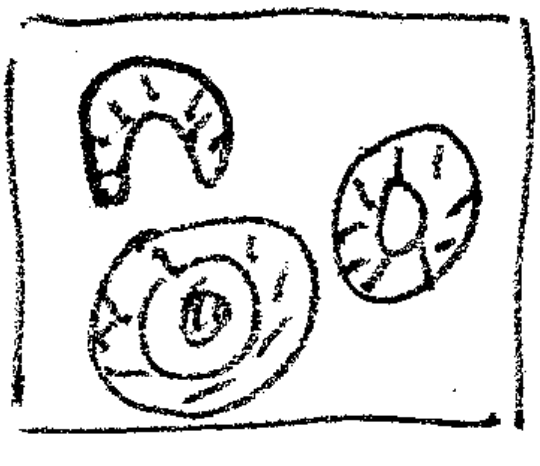
Original source details: GELFAND-P000007

Diagram and typed description. The small frame contains two dark regions with light interior gaps. At upper left are a bent heavy stroke and separate short oblique marks. These marks have not been identified as letters or arrows.



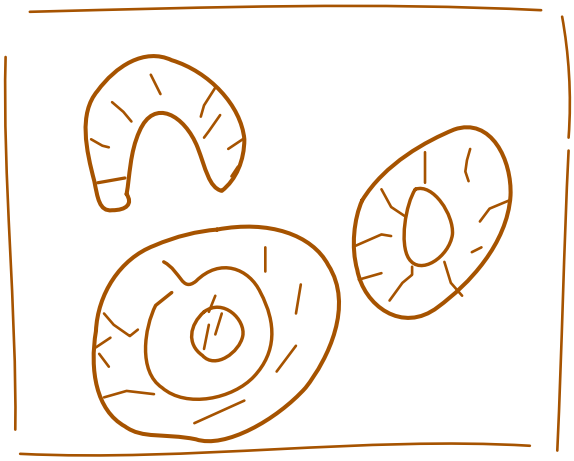
Visible contours and strokes; uncertain connections have not been completed.

And this contradicts the law of large numbers.



Original source details: GELFAND-P000007

Diagram and typed description. Inside the large frame, the upper-left horseshoe-shaped region has an open lower gap and a small outlined part at its left tip. At right is a slanted ring. The large lower-left region contains an open or broken inner contour and a small central contour. Individual hatch marks are visible; some connections are uncertain.

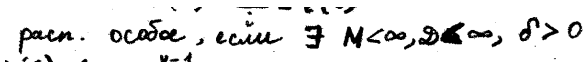


Visible contours and strokes; uncertain connections have not been completed.

The whole boundary splits into connected cycles. Each arrangement has its own number. Number them

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots, \qquad \Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

An arrangement is special if $\exists N < \infty$, $D < \infty$, $\delta > 0$.

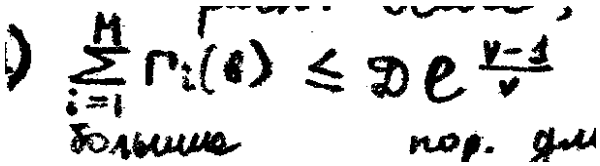


Original source details: GELFAND-P000007

Uncertain reading. In this line the first finite parameter resembles N, whereas the later summation limit resembles M.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\frac{\nu-1}{\nu}}.$$

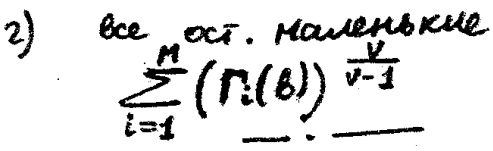
[Beneath the sum:] Large. Of the order of the boundary length. (Almost the whole cube.)



Original source details: GELFAND-P000007

Uncertain reading. The exponent on ℓ in the first condition is written as a fraction, read as $(\nu-1)/\nu$; in GELFAND-P000010 the analogous line has $\nu - 1$.

$$2) \quad \text{all the others small} \qquad \sum_{i=1}^{M} \left(\Gamma_i(\theta)\right)^{\frac{\nu}{\nu-1}}.$$



Original source details: GELFAND-P000007

Uncertain reading. The visible limits of the second sum look like 1 and M, although the caption says “all the others small”. No condition is visible after the sum.

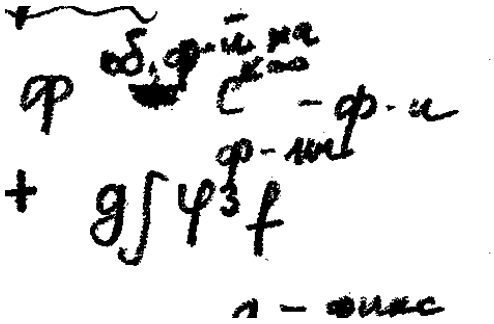
[Dividing line.]

24/IX. Thursday. 16 h, room 13-11. Prof. Segal (USA). Theory of functions in infinite-dimensional space.

$$+\Box\varphi = m^2\varphi + g\varphi^3.$$

$$F_f(\varphi) = \int (-\Box + m^2) f\varphi + g \int \varphi^3 f.$$

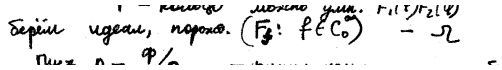
[Above φ^3:] φ — мн. [source abbreviation]. φ [arrow] C_0^∞ — functions.



Original source details: GELFAND-P000007

Uncertain reading. Above the note at right are the visible words “б, ф-й на” [fragment: functions on] and an arrow toward C_0^∞.

g is fixed. $F_f \in \Phi$ is a nonlinear functional of φ. Φ is a ring; one can multiply $F_1(\varphi)F_2(\varphi)$. Take the ideal generated by $(F_f : f \in C_0^\infty)$ — Ω.



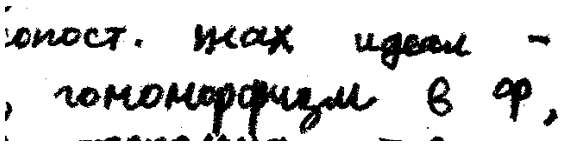
Original source details: GELFAND-P000007

Uncertain reading. The index on F in the ideal’s definition resembles f; the source sign after the parenthesis is Ω or a calligraphic J.

Let $p = \Phi/\Omega$ be the function ring on the manifold of solutions of the given partial differential equation. To obtain a solution $(1) \stackrel{\text{def}}{=}$ a point of the space of maximal ideals of the ring p. An ordinary solution φ_0 [a check mark above the word] is assigned a maximal ideal: consider the map

$$F \longrightarrow F(\varphi_0),$$

a homomorphism into Φ, F_f is zero on the ideal Ω. This is a homomorphism of the quotient ring as well, i.e. a maximal ideal.



Original source details: GELFAND-P000007

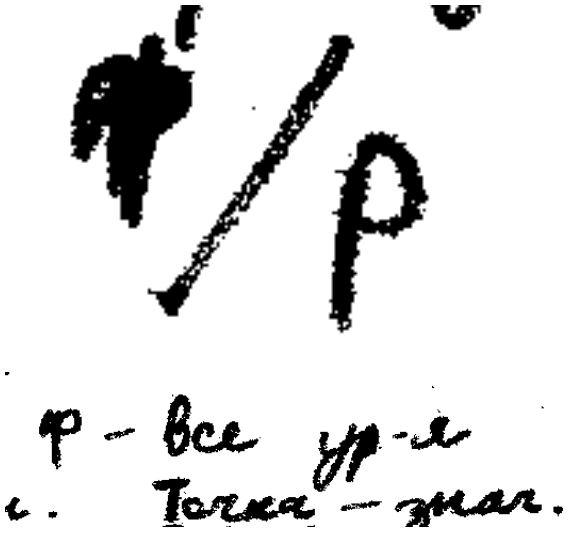
Uncertain reading. After “a homomorphism into”, the visible letter resembles Φ.

GELFAND-P000008

A maximal ideal is a generalized solution of a nonlinear equation. The algebraic case.

$$f_1(x_1, \ldots, x_n) = 0, \qquad f_k(x_1, \ldots, x_n) = 0.$$

Solutions in terms of a ring: in the ring of all polynomials. The ideal generated by these functions is p, and the quotient ring is Φ/p.

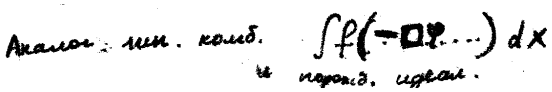


Original source details: GELFAND-P000008

Uncertain reading. In the fraction, the numerator resembles Φ, P, or a heavily written A; the candidate Φ is retained.

Φ — all equations. This is the ring of algebraic functions. A point, the value at the point, is precisely a maximal ideal. In the algebraic case, conversely as well. Analogy: linear oscillations.

$$\int f(-\Box\varphi\ldots)\,dx.$$

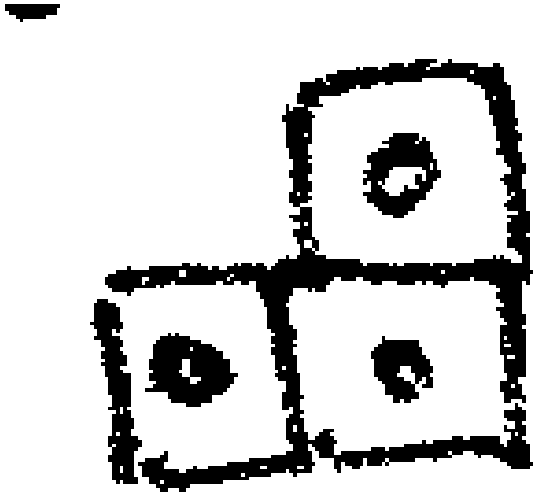


Original source details: GELFAND-P000008

Uncertain reading. The part of the integral after $-\Box$ looks like $\varphi\ldots$, but the letter is partly inked over.

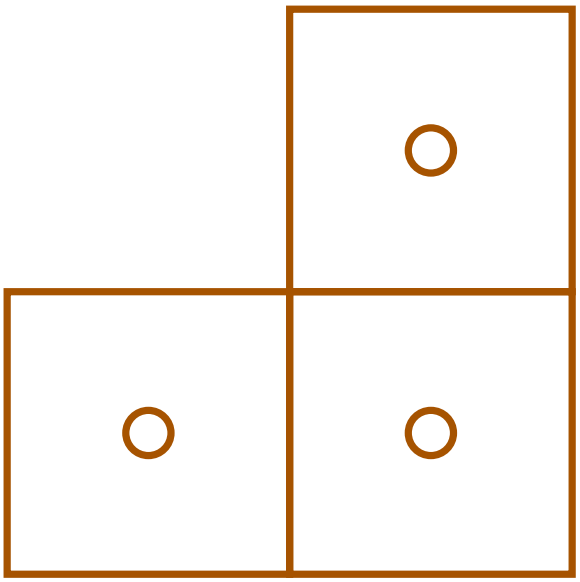
And the corresponding ideal.

Dobrushin. $\nu = 2, 3$. $V_\ell = \ell^\nu$, $N < V_\ell$, $\theta = (x_1, \ldots, x_N)$.



Original source details: GELFAND-P000008

Diagram and typed description. Original diagram and typographic recreation.



$\Gamma(\theta)$ is the boundary length. $U(\theta)$ is the number of neighbors (pairs of neighbors). The Gibbs distribution

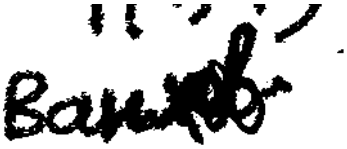
$$p(\theta) = \frac{\exp\{U(\theta)/T\}}{\sum\limits_{\theta\in\mathcal{B}(V_\ell,N)} \exp\{U(\theta)/T\}} = \frac{\exp\{-\Gamma(\theta)/(2T)\}}{\sum\limits_{\theta} \exp\{-\Gamma(\theta)/(2T)\}}.$$

[Beneath the first sum:] all arrangements.

$$(\text{since } 2U(\theta) = 2\nu N - \Gamma(\theta)).$$

Phase transition. Let the lattice grow, $\ell \to \infty$, and let $N_\ell \to \infty$. The limit $V_\ell / N_\ell \to v$ is the specific volume. It turns out that

$$\frac{1}{N} \log \sum_\theta \exp\{U(\theta)/T\} \longrightarrow f(v, T).$$



Uncertain reading. Beneath the limit the name reads “Ван Хов” or “Ванхов” [Van Hove]; the middle and ending are heavily overwritten.

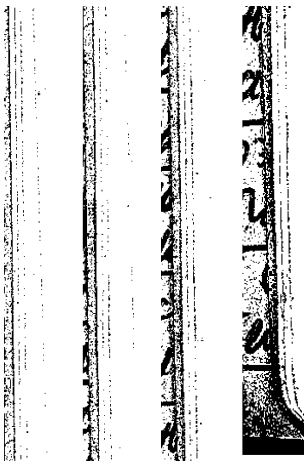
Original source details: GELFAND-P000008

(v, T) is called a point of phase transition if

$$\frac{\partial f(v,T)}{\partial v} = 0 \quad \text{in some neighborhood at } T = \text{const.}$$

Uncertain reading. Between T and const, the sign resembles $=$ or $\neq$; the candidate $=$ is retained, but the reading is not asserted.

Original source details: GELFAND-P000008



Undeciphered passage. At the extreme left edge is a narrow cropped column of handwritten endings from neighboring text; continuous words cannot be recovered.

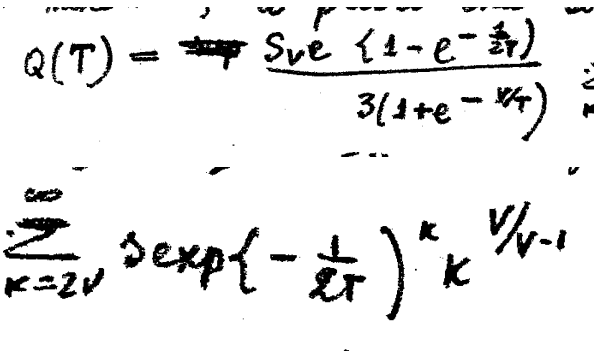
Original source details: GELFAND-P000008

GELFAND-P000009

(Since the function is linear; a priori it is convex.)

$$Q(T) = \frac{S_\nu e(1 - e^{-1/(2T)})}{3(1 + e^{-\nu/T})} \sum_{k=2\nu}^{\infty} 3 \exp\{-1/(2T)\}^k k^{\frac{\nu}{\nu-1}}.$$

[After the first equality, a cancelled fragment:]

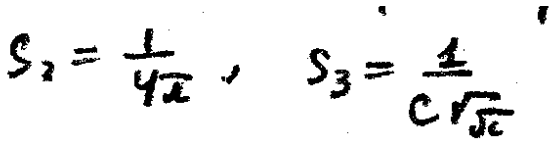


Uncertain reading. The cancelled sign after $=$ is illegible. In the numerator, the factor after S_ν resembles e; the upper summation limit resembles ∞ or a double stroke. The exponent k may apply to the factor 3 as well, but in the scan it stands by the closing brace.

Original source details: GELFAND-P000009

S_ν is the volume of a ν-dimensional sphere with surface 1.

$$S_2 = \frac{1}{4\pi}, \qquad S_3 = \frac{1}{C\sqrt{\pi}}.$$



Uncertain reading. The first factor in the denominator of S_3 resembles C, c, or 6; the written C is only a candidate.

Original source details: GELFAND-P000009

If

$$Q(T) < \frac{1}{v} < 1 - Q(T),$$

then a phase transition takes place.

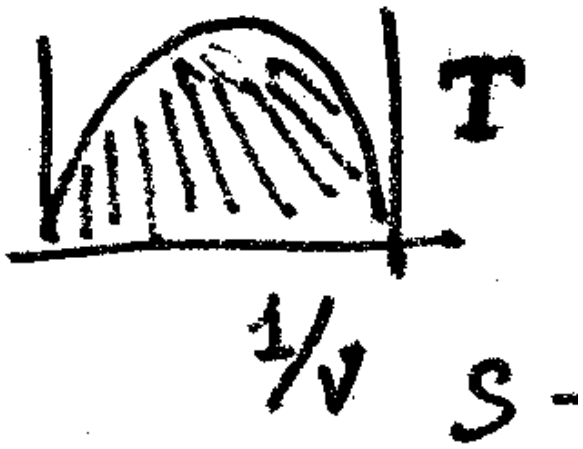
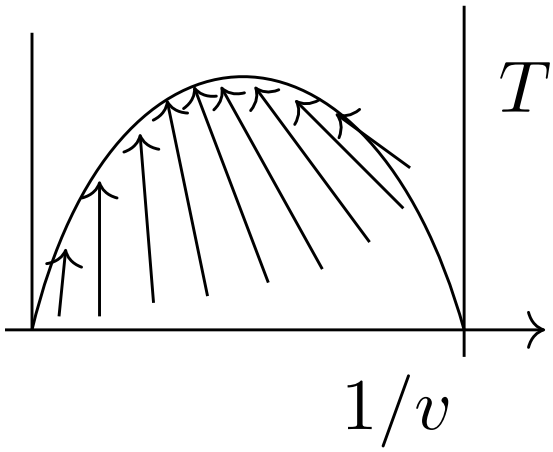


Diagram and typed description. Original diagram and typographic recreation.



Original source details: GELFAND-P000009

[Dividing line.]

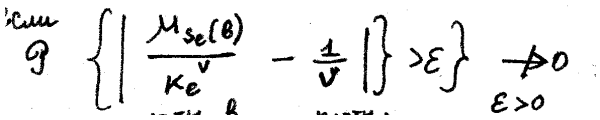
S is a subcube. $S_\ell \subset V_\ell$. K_ℓ is the side of S_ℓ.

$$0 < d < \frac{K_\ell}{\ell} < 1 - d.$$

Introduce a random variable. $\mu(S_\ell)$ is the number of points of the arrangement θ that fall in S_ℓ. $\mu_{S_\ell}(\theta)$. If

$$\mathsf{P}\left\{\left|\frac{\mu_{S_\ell}(\theta)}{K_\ell^\nu} - \frac{1}{v}\right|\right\} > \varepsilon\} \not\longrightarrow 0, \qquad \varepsilon > 0,$$

then there is a phase transition.



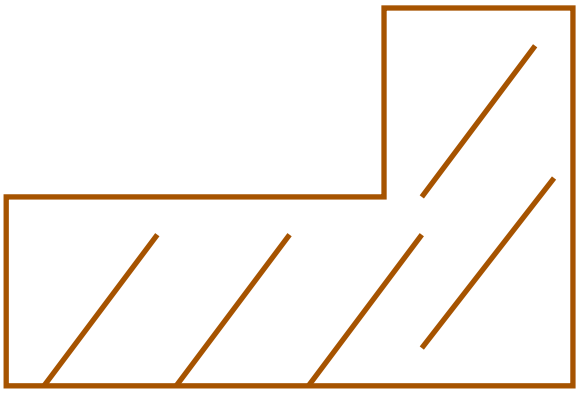
Original source details: GELFAND-P000009

Uncertain reading. After the right absolute-value bar, the scan shows a brace before $> \varepsilon$ and another after ε.

[Beneath the first fraction:] density in the subcube. [Beneath $1/v$:] density in the whole cube. And a closely related condition is necessary. (Physical meaning: coexistence of 2 phases.)

Original source details: GELFAND-P000009

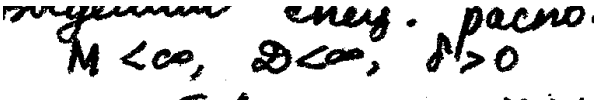
Diagram and typed description. Original diagram and typographic recreation.



The boundary splits into components — — but a connection through corners ($\nu = 2$) or an edge ($\nu = 3$) is not counted.

$$\Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

$\Gamma_i(\theta)$ is the length or area of a boundary component. Definition. Single out a special arrangement. It depends on 3 numbers, $N < \infty$, $D < \infty$, $\delta > 0$.



Original source details: GELFAND-P000009

Uncertain reading. The first letter resembles N, but admits M; the enlarged fragment does not resolve the distinction.

Number the cycles in decreasing order of length

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots .$$

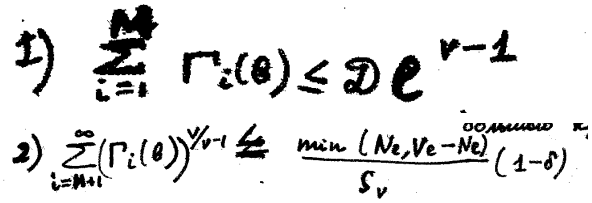
GELFAND-P000010

Conditions.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\nu-1}.$$

(Of the order of the surface of the large cube.)

$$2) \quad \sum_{i=M+1}^{\infty} \big(\Gamma_i(\theta)\big)^{\frac{\nu}{\nu-1}} \leq \frac{\min(N_\ell, V_\ell - N_\ell)}{S_\nu}(1-\delta).$$

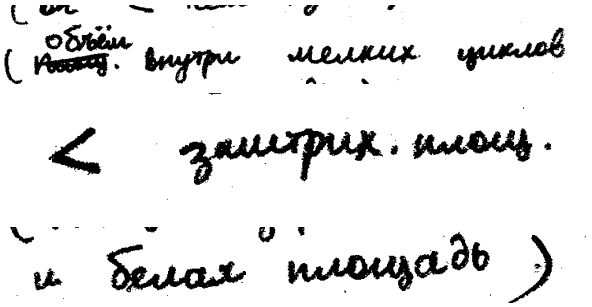


Original source details: GELFAND-P000010

Uncertain reading. The inequality sign before the fraction is heavily overwritten: candidates are $\leq$ and $<$. The parameter in the upper limit of the first sum resembles M or N.

$$S_\nu \left(\Gamma_i(\theta)\right)^{\frac{\nu}{\nu-1}}$$

Area of a circle with length Γ_i. A ball has maximal volume; this is an upper bound on the volume of the region inside the cycles. (It is $<$ what is needed to contain all occupied points.) ([A word is cancelled; inserted above:] volume. Inside small cycles $<$ the shaded area and the white area.)

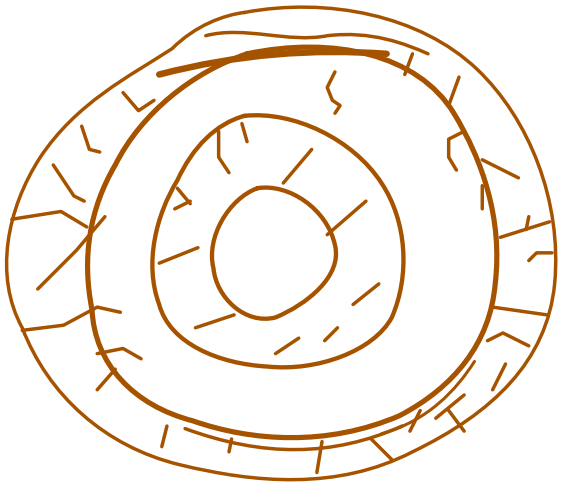


Original source details: GELFAND-P000010

Uncertain reading. The word beneath the inserted “volume” is heavily cancelled and illegible. The two lines write “Area”.

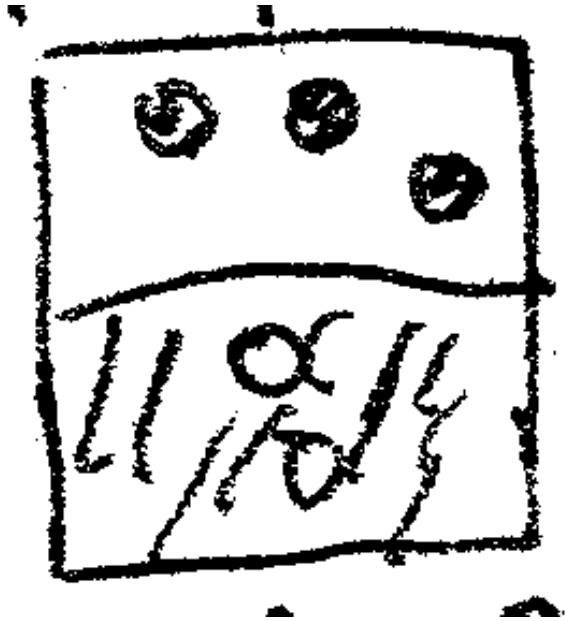
Original source details: GELFAND-P000010

Diagram and typed description. Four nested irregular rounded contours. The outer ring has uneven thick and thin strokes, retraced arcs above and at lower right, and separate ends bending inward. The inner ring has radial and oblique strokes. Lines merge in heavy areas; no additional closed boundaries or arrows have been established.



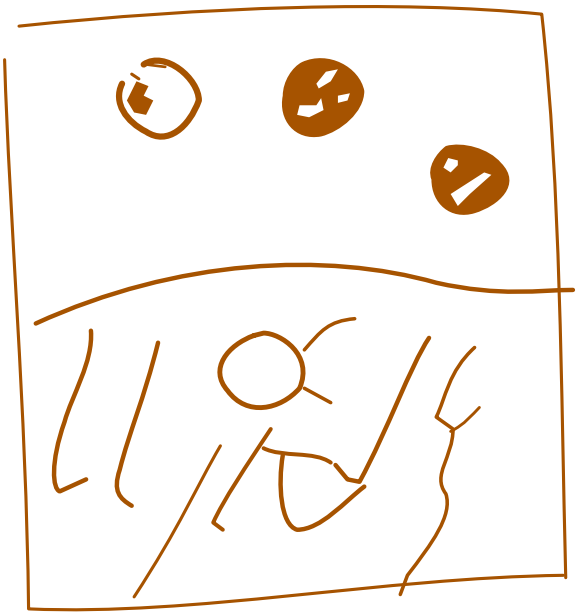
Visible irregular contours and strokes; no directions assigned.

Neither all the points nor all the empty sites can be forced into it. An example of an arrangement where the conditions do not hold.



Original source details: GELFAND-P000010

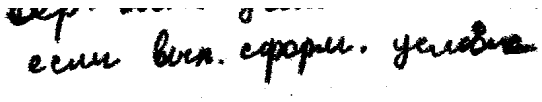
Diagram and typed description. The three upper dark regions have distinct light centres. Below the curved divider are a rounded contour with two right branches, an incomplete lower curve, several oblique and bent strokes, and a short branch on the right. Some connections at the lower intersections and gaps are uncertain.



Visible contours and strokes; uncertain connections have not been completed.

This is a phase transition. Everything reduces to 2 lemmas.

$1)$ $\exists M, D, \delta$ such that the probability that the conditions hold $\longrightarrow 1$, if the stated condition holds. [Beneath the parameters:] independent of everything.

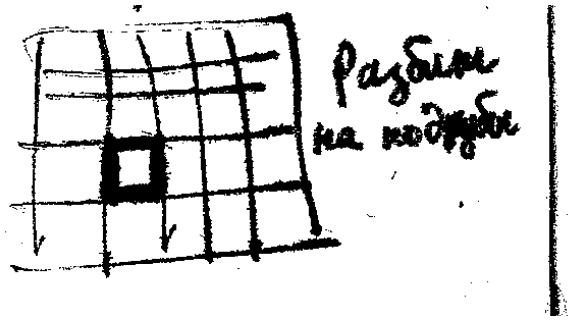


Original source details: GELFAND-P000010

Uncertain reading. In the line after “if ... holds”, the word looks like “сформ.” [stated] or “строгое” [strict]; nothing is supplied from the meaning.

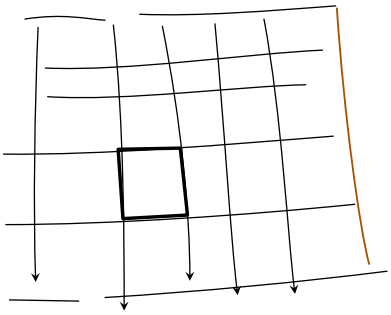
$2)$ Lemma 2. If almost all arrangements are special, then there is a phase transition. Idea of the proof of Lemma 2. G is the volume of the region enclosed by the small cycles.

$$G \leq N_\ell(1-\delta), \qquad \frac{G}{V_\ell} \leq \frac{1}{v}(1-\delta).$$



Original source details: GELFAND-P000010

Diagram and typed description. A grid of five columns and five rows; the second cell from the left in the fourth row from the top is marked. The left edge and four internal vertical lines have downward tips. Two upper horizontals stop short of the right edge; its lower endpoint is thickened and its exact shape is unclear. Caption: “Divided into subcubes”.

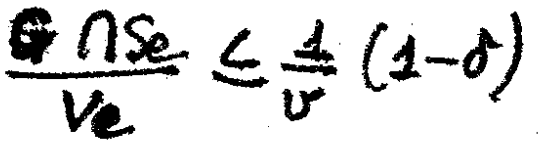


Divided into subcubes

Five downward tips; the uncertain right endpoint is a separate stroke.

There exists a subcube S_ℓ:

$$\frac{G \cap S_\ell}{V_\ell} \leq \frac{1}{v}(1-\delta).$$

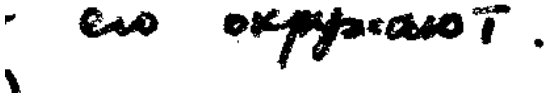


Uncertain reading. In the numerator, the heavy sign before $\cap S_\ell$ resembles G; the denominator clearly resembles V_ℓ.

Original source details: GELFAND-P000010

GELFAND-P000011

Let all large cycles surround co / ∞.



Uncertain reading. surround co / ∞. [The two-loop sign before the verb is uncertain.]

Original source details: GELFAND-P000011

2 cases.

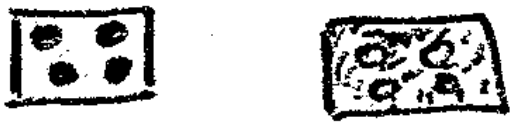


Diagram and typed description. Two small rectangular frames, each with four dark rounded marks. The right frame has light centres, an irregular chain of strokes above, and detached short and curved lines between the rounded marks and near the edges. Their exact connections and possible arrowheads are indistinct.

Visible interior strokes; connections and directions remain uncertain.

Original source details: GELFAND-P000011

Lemma. [Непрс.] a boundary of k [cancelled: точк.; inserted above/below: links] $\leq 3^k$.

Uncertain reading. Неprс. [uncertain source abbreviation]

Original source details: GELFAND-P000011

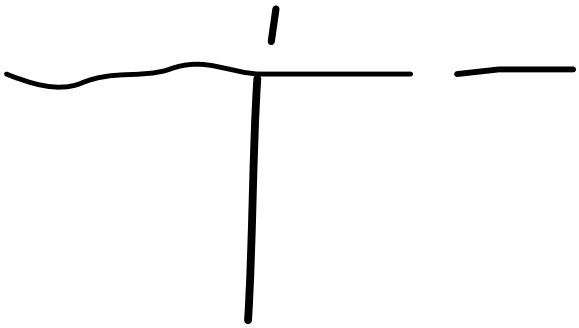
Uncertain reading. точк. [points; uncertain cancelled reading]

Original source details: GELFAND-P000011

Gain 3, loss.

Diagram and typed description. Beneath the words about gain and loss is a wavy, almost horizontal line, a gap before a separate right segment, and a long vertical segment extending downward. A detached short stroke lies above their junction; it has not been identified as a letter or arrow.

Original source details: GELFAND-P000011



The detached upper stroke is retained without assigning a letter or arrow.

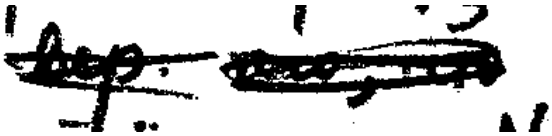
$$\exp\left\{-\frac{1}{2T}\right\}, \qquad \left(3\exp\left\{-\frac{1}{2T}\right\}\right)^k.$$

Problem. (unsolved) Let N be large. Turn with probability $\frac{1}{3}$, but returning is forbidden [Cancelled: not].

Uncertain reading. *Cancelled.* не [not; uncertain candidate for a short word]

Original source details: GELFAND-P000011

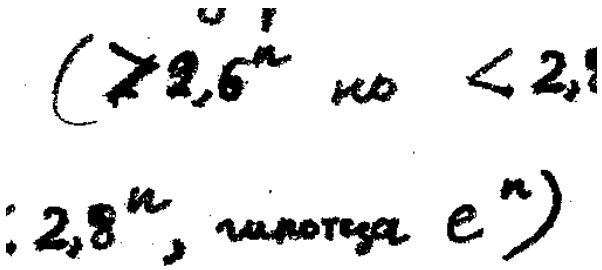
What is [Cancelled: prob. ...] the expected distance he will travel in N steps?



Uncertain reading. *Cancelled.* вер. ... [prob.; candidate for the first abbreviation; the following cancelled writing is unreadable]

Original source details: GELFAND-P000011

(If returning is allowed, then $\sim \sqrt{n}$.) If returning is allowed, then
($\geq 2{,}6^n$ but $< 2{,}8^n$, conjecture e^n).



Uncertain reading.
($\geq 2{,}6^n$ but $< 2{,}8^n$, conjecture e^n); the first sign may also read $>$; additional strokes are uncertain

Original source details: GELFAND-P000011

[Horizontal divider.]

28/IX Segal. Theory of functions on the set of solutions of a relativistic differential equation. [In the left margin: They are all written this way.]

$$u' = Au + K(u) \qquad (A) \qquad u(t), \quad -\infty < t < +\infty,$$

$$A = -A^*, \qquad u \in L^2, \qquad u(t) \in H.$$

K is a given nonlinear operator. In integral form

$$u(t) = e^{A(t-t_0)}u(t_0) + \int_{t_0}^{t} e^{A(t-s)}K(u(s))\,ds \qquad (B).$$

When $K \in C^1$, i.e. $\partial_u K = K_u$ exists in the Fréchet sense and is continuous, every solution of (B) is a solution of A (if $u(t)$ is in the domain of A).

[Horizontal divider.]

Suppose (B) has a global solution for every initial datum at t_0 (u_0). (And $K \in C^1$.) In quantum mechanics

$$\Box\varphi = f(\varphi)$$

[Beneath □: d'Alembertian.]

$$u(t) = [\varphi(\cdot, t), \dot{\varphi}(\cdot, t)], \qquad \varphi = \varphi(\vec{x}, t).$$

K is a nonlinear operator, Немыц. / Намыц.

Uncertain reading. Немыц. / Намыц. [Nemyts. / Namyts.; the final source abbreviation is uncertain]

Original source details: GELFAND-P000011

$$\|[\varphi,\omega]\| \qquad \Box\varphi = m^2\varphi + f(\varphi)$$

[Beneath the formula: A shorter form.]

$$B = \sqrt{m^2 I - \Delta}, \qquad \|[\varphi,\omega]\| = \|B\varphi\|^2_{L^2} + \|\omega\|^2_{L^2}.$$

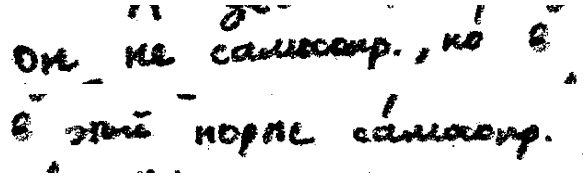
Uncertain reading. No square is visible above the norm on the left of the last formula; the right side contains squares.

Original source details: GELFAND-P000012

A is given by a matrix of order 2,

$$\begin{pmatrix} 0 & I \\ -B^2 & 0 \end{pmatrix}.$$

It is not self-adjoint, but in this norm it is self-adjoint.



Uncertain reading. Both places read “самосопр.” [self-adjoint]; $A = -A^*$ is written above.

Original source details: GELFAND-P000012

Any equation of relativistic physics can be written this way.

M is the collection of all solutions.

$$M = [u(\cdot);\ (**) \text{ satisfies}]$$

[At right/beneath the formula: of t; various B.]

Uncertain reading. различные B. [various B.; uncertain reading of the underlined note]

Original source details: GELFAND-P000012

Now let $K \in C^\infty$. M is a Banach manifold (every point has a neighbourhood homeomorphic to a Banach [Cancelled: illegible] space; the transition is a C^∞ function in the Fréchet sense).

Uncertain reading. *Cancelled.* unreadable word

Original source details: GELFAND-P000012

$$S_t : M \longrightarrow H, \qquad u(\cdot) \longmapsto u(t).$$

[At right: operator. set of sol. $\longrightarrow H$]

Uncertain reading. мн. реш. $\longrightarrow H$ / им. реш. $\longrightarrow H$ [set of sol. / им. sol.; the first source abbreviation is uncertain; the final arrow and H are visible]

Original source details: GELFAND-P000012

Theorem. There exists a unique C^∞ manifold structure for which the map S_t is a C^∞ isomorphism. $u(t, u_0)$ depends on u_0 in a C^∞ manner.

A true manifold. There is a tangent space $T_{u(\cdot)}$. The tangent map

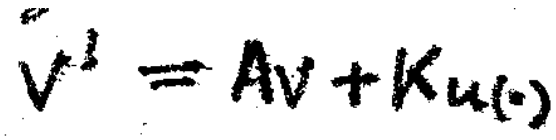
$$dS_t : T_{u(\cdot)} \longrightarrow T_{u(t)}.$$

Of course, at every point it is isomorphic to the space itself.

$$\lambda \in T_{u(\cdot)}, \qquad dS_t(\lambda) = v(t) \in H.$$

Thus a function corresponds to each tangent vector. [Underlined and cancelled: Lemma.] Denote by $v(t)$ (not the value of v at the point t); variational equation, an equation of first order:

$$v' = Av + K_{u(\cdot)}.$$



Uncertain reading. No separate factor v is visible after $K_{u(\cdot)}$ on this line.

Original source details: GELFAND-P000012

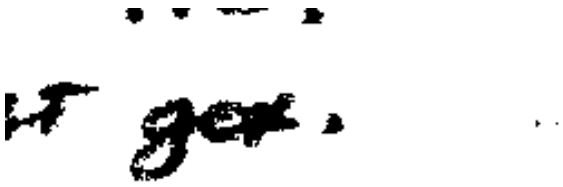
Solutions of this equation arise from tangent vectors λ, and conversely.

1) Canonical differential form on M.

2) Poisson brackets (analogy with the finite-dimensional case).

GELFAND-P000013

3) Vector fields in M. In their terms there will be зд.



Uncertain reading. зд. [the final source abbreviation is uncertain]

Original source details: GELFAND-P000013

4) Canonical commutation relations.

5) A representation and an explicit definition of the operators will be given.

5/X. Segal. I. Renormalized field equation. 1) Derivation. 2) Existence and uniqueness.

II. Definition of the scattering matrix S. A. Classical S operator. B. Extension of generalized probability distributions to the nonlinear case.

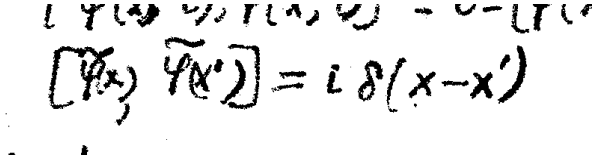
III. Discussion of interpretation in terms of particle theory.

[Horizontal divider.]

Consider the equation

$$\Box\widetilde{\varphi} = m^2\widetilde{\varphi} + \widetilde{\varphi}^3,$$

$$[\widetilde{\varphi}(\vec{x}, t), \widetilde{\varphi}(\vec{x}', t)] = 0 = [\dot{\widetilde{\varphi}}(\vec{x}, t), \dot{\widetilde{\varphi}}(\vec{x}', t)].$$



Uncertain reading. Candidates for the source line: $[\widetilde{\varphi}(x), \widetilde{\varphi}(x')] = i\delta'(x - x')$ / $[\widetilde{\varphi}(x), \dot{\widetilde{\varphi}}(x')] = i\delta(x - x')$. The dot above the second letter, the additional stroke after δ, and the exact placement of the arguments are not fully distinguishable.

Original source details: GELFAND-P000013

Loran Schwartz. There are 2 types of measures: Radon measures and simply measures [inserted above the last word: abstract]. For now, restrict to positive measures. An abstract measure on a set X, F is a σ-algebra. μ is a measure. μ is σ-additive on F.

[Horizontal line.]

Once a measure is given, one can consider the μ-measurable sets, F_μ on X, and integrate measurable functions.

$$f : X \longrightarrow F, \qquad \int_X f(x)\, d\mu(x).$$

[At right: values in a Banach space; F is a Banach space.]

[Horizontal line.]

Radon measure. X is an arbitrary locally compact space. $C(X)$ consists of continuous functions with compact supports, with the inductive-limit topology, and let $C'(X)$ be the dual space. Its elements are measures on X.

GELFAND-P000014

μ is positive if $\mu(\varphi) \geq 0$ for $\varphi \geq 0$. A measure is bounded if

$$|\mu(\varphi)| \leq C \max |\varphi|.$$

The 2nd definition is a bounded positive Radon measure. The same Lebesgue extension is performed with it. The integral also exists (with values in a Banach space).

The σ-field constructed from a Radon measure depends on μ, but there exists one independent of μ (substantially): the field of Borel sets (min containing closed and open sets).

Uncertain reading. min containing closed and open sets [the beginning of the parenthetical note is uncertain]

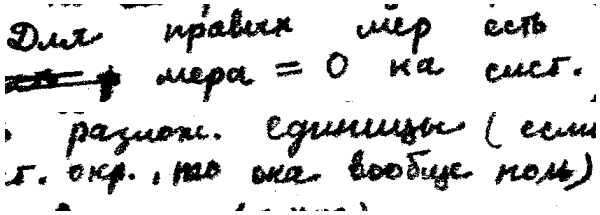
Original source details: GELFAND-P000014

[Horizontal divider.]

Why do various mathematicians prefer Radon measures? Radon measures exist only on locally compact spaces (a restriction in probability theory). But [Cancelled: two illegible words] Radon measures are more convenient.

Uncertain reading. *Cancelled.* two short unreadable words

Original source details: GELFAND-P000014



Uncertain reading. For “right” measures there is разлож. / различ. единицы [decomp. / различ. of unity] (if [cancelled beginning] a measure $= 0$ on a system of neighbourhoods, then it is zero everywhere). [The first description of the measures and the abbreviation before “unity” are uncertain.]

Original source details: GELFAND-P000014

There are 5–15 theorems true for Radon measures (only).

$$X \xrightarrow{f} Y$$

[Beneath the arrow: topological spaces.] [Cancelled.]

Uncertain reading. a short word before “Select”

Original source details: GELFAND-P000014

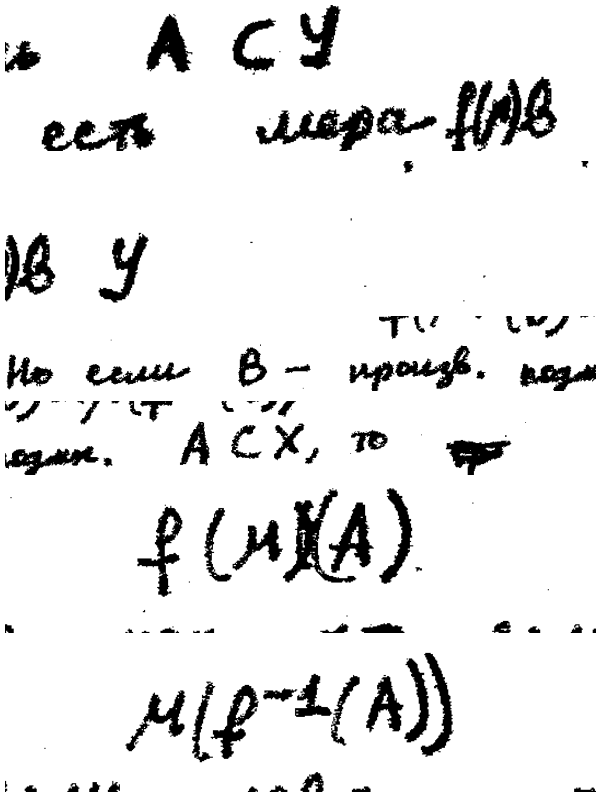
Select the Borel field on X and on Y, and an abstract measure μ on X. Let $A \subset Y$. Then there is a measure $f(\mu)$ on Y:

$$f(\mu)(B) = \mu(f^{-1}(B)).$$

But if B is an arbitrary subset $A \subset X$, then [one sign cancelled]

$$f(\mu)(A) \qquad \mu(f^{-1}(A)).$$

Uncertain reading. The source alternates between A and B, and $A \subset X$ is read before the second pair of formulas.



Original source details: GELFAND-P000014

One can show that if the left side makes sense, the right side does too and they are equal; the converse is false. [Cancelled.]

Uncertain reading. an unreadable beginning of a line

Original source details: GELFAND-P000014

$$A \text{ — } f(\mu)\text{-measurable} \implies f^{-1}(A) \ \mu\text{-measurable}$$

If μ is a Radon measure, there is equivalence.

Example (contradictory)

$$X \subseteq Y = [0,1]$$

such that X is a Lebesgue-nonmeasurable set, $Y \setminus X$ has inner measure 0 (or the outer measure of $X = 1$). A subset of X is Borel $\Longleftrightarrow$ it is an intersection with a Borel set of Y. μ is an abstract measure.

GELFAND-P000015

Definition. $\mu(D)$. $D \subset X$ is Borel.

$$D = X \cap E, \qquad E \text{ — Borel in } Y,$$

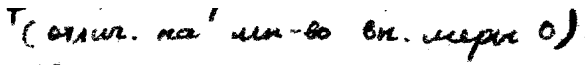
$$\mu(D) \underset{\text{def}}{=} dx(E)$$

[Beneath $dx(E)$: Lebesgue measure.]

Original source details: GELFAND-P000015

Uncertain reading. In $dx(E)$ both letters are on the baseline; the case of the second letter is not fully distinguishable: $dx(E)$ / $dX(E)$.

The measure is the same (differs on a set of вн. measure 0).



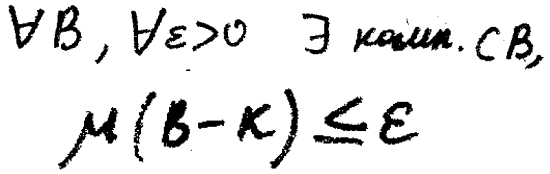
Original source details: GELFAND-P000015

Uncertain reading. отлич. на мн-во вн. меры 0 [differs on a set of вн. measure 0; uncertain parenthetical source note]

We have defined an abstract measure on X. Its image [The following line repeats: its image] is dx. Take some (non-Borel) subset of $Y \setminus X$. It is nonmeasurable, but its inverse image is empty and hence measurable.

The aim of the talk is to show that local compactness is unnecessary. Let μ be a measure on F_B; then μ is a Radon measure if it is inner compact regular.

$$\mu \text{ Radon} \Longleftrightarrow \forall B,\ \forall \varepsilon > 0\ \exists \text{ compact } \subset B, \qquad \mu(b-k) \leq \varepsilon.$$



Original source details: GELFAND-P000015

Uncertain reading. Lowercase $b - k$ is read in the difference, although $\forall B$ precedes it. No separate set letter is visible between “compact” and $\subset B$.

[Horizontal divider.]

X is a topological space. Find a measure with all the properties of locally compact Radon measures. Suppose it is defined (set up) on Borel sets. A definition of a Radon measure valid beyond locally compact spaces. Let X be completely regular (such spaces can be embedded in compact spaces of maximal ideals).

$$X, \qquad \widehat{X} \text{ — Čech completion of } X$$

(maximal ideals of the algebra of continuous functions).

$$X \xrightarrow{i} \widehat{X}.$$

Complete regularity means that this is a homeomorphic map, rather than merely an injective map. Another definition of a Radon measure on X: a Radon measure on $\widehat{X}$ for which

$$\mu(\widehat{X} \setminus X) = 0.$$

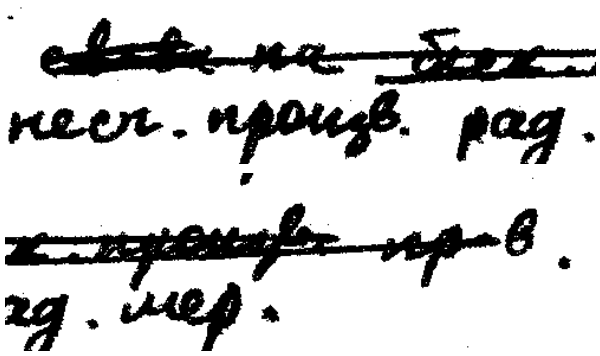
One can show equivalence. In the locally compact case.

$$\mu \text{ Rad} \iff \forall B \text{ compact } \forall \varepsilon > 0 \; \exists \text{ compact} \quad \mu(B - k) < \varepsilon.$$

Original source details: GELFAND-P000015

Uncertain reading. The second sign in the difference resembles lowercase k; the first reads as B, but its case is uncertain ($B - k$ / $b - k$). No separate set letter or inclusion condition is visible after “compact”.

All the properties of Radon measures hold. The theorem’s proof is valid in $\widehat{X}$, and the rest is almost trivial. There is a single exception: [Cancelled: properties ... Borel ...] spaces.



Original source details: GELFAND-P000015

Uncertain reading. *Cancelled.* св-ва ... бор. ... [properties ... Borel ...; visible fragmentary candidates beneath the horizontal stroke; the complete cancelled phrase is unreadable]

— one cannot define the support of arbitrary Radon measures.

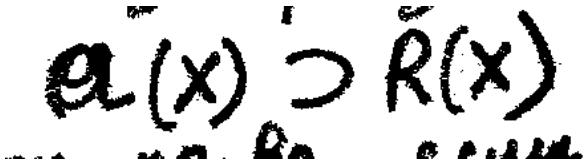
GELFAND-P000016

$X \subset K_0$ into a compact space (any one). Then an analogous definition of Radon measures exists. We obtain the same law.

[Horizontal divider.]

Compare with the definition of a nuclear space (spaces in which the 2 definitions of the tensor product coincide).

$$a(X) \supset R(X).$$



Original source details: GELFAND-P000016

Uncertain reading. The first symbol in the inclusion resembles a lowercase a with a large loop; A or a calligraphic form is not wholly excluded.

A Radon space if every abstract measure on it is a Radon measure.

3 properties of Radon spaces. 1) If X is Radon and $Y \subset X$, then Y is Radon $\iff$ it is universally measurable in X (measurable with respect to all Radon measures—for example, Borel, Suslin and analytic).

An important criterion. 2) If X is Radon, then it is universally measurable, i.e. measurable in every containing space and for every measure.

Original source details: GELFAND-P000016

Uncertain reading. The source abbreviation after “measurable ∀” reads объемл. / объем.; the ending and a possible superscript stroke are not fully distinguishable. It has not been expanded to a complete source word.

Probability theory—probability is always defined.

3) An example of a broad class of Radon spaces. Distributions, continuous functions, etc.? Let E be a T. L. C.—a separable Fréchet space or their countable inductive limit. F is of the same kind. Then $\mathrm{Hom}(E, F)$ is of the same kind (equipped with any topology between pointwise and compact convergence). In particular, the dual space, the space itself ($R \longrightarrow E$). All spaces of generalized functions, of C^∞ functions. And the operator space $\mathcal{L}(S, S)$ is also a Radon space.

Original source details: GELFAND-P000016

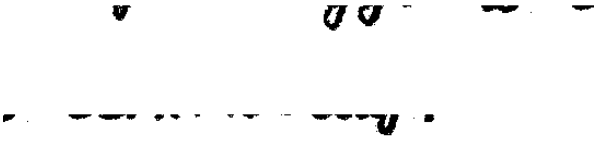
Uncertain reading. Above or beside the closing parenthesis is a short note “ф-и” [functions]. A separate stroke after $\mathcal{L}(S, S)$ cannot be securely distinguished from the note and parenthesis; the candidate $\mathcal{L}(S, S)'$ remains uncertain.

Application. Consider a stochastic process given as a probability on the space of all functions with simple (pointwise) convergence (a measure in this space). Take the subspace of continuous functions. It is measurable in R^R.

GELFAND-P000017

Hence the probability that a function is continuous always exists. Let it equal 1. Then there is a probability distribution on this subspace, and it can be defined simply as a Radon measure on C. It is concentrated on a countable number of compact sets (equicontinuous). If the probability that a function is continuous is 1, then with probability 1 all these functions have a common modulus of continuity. This is true for every stochastic process (there exists a common mod. of continuity with probability 1).

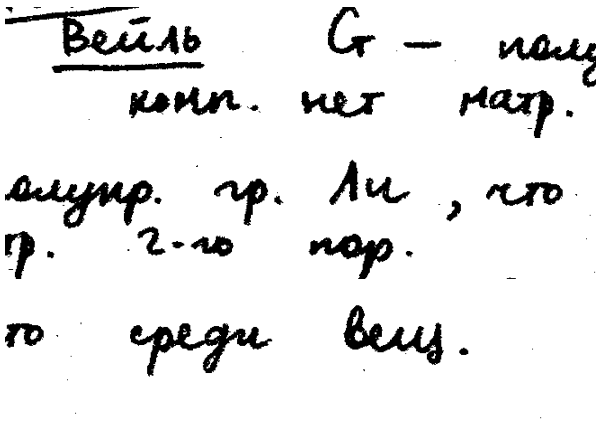
[A large unwritten area; isolated small horizontal or curved strokes are visible along its boundary in the middle of the frame.]



Original source details: GELFAND-P000017

Uncertain reading. Only lower parts of isolated letter-like strokes survive along the horizontal boundary of the large unwritten area; no complete word is readable.

$12/\mathrm{X}$. Gindikin. Weil. G is a semisimple Lie group such that among the real comp. there are no matrices of order 2.



Uncertain reading. The line reads “вещ. комп.” [real comp.] and “матр. 2-го пор.” [matr. of 2nd order].

Original source details: GELFAND-P000017

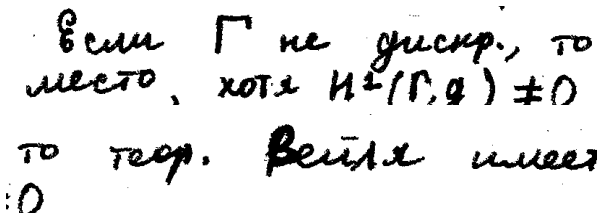
Γ is a discrete subgroup; G/Γ is compact. Consider deformations of Γ: $\gamma(t)$, with a discrete group $\Gamma^{(t)}$ for every t. A trivial method uses inner automorphisms:

$$g(t)\gamma g(t)^{-1}.$$

The quotient group is the same. Under these conditions, all deformations are trivial. It suffices to prove that

$$H^1(\Gamma, \mathfrak{g}) = 0.$$

$\mathfrak{g}$ is the Lie algebra, carrying a representation of Γ. If Γ is not discrete, Weil’s theorem holds, although $H^1(\Gamma, \mathfrak{g}) \neq 0$.



Uncertain reading. The condition reads “не дискр.” [not discrete].

Original source details: GELFAND-P000017

There is a representation $\rho(\gamma)$ in V^d. Then the dimension is

$$p = (2g-2)d + i + i'.$$

GELFAND-P000018

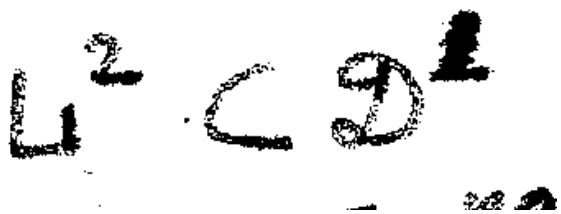
Laurent Schwartz.

E — TVS LCH QC.

[A note leads from H in the abbreviation: Hausdorff.] A locally convex topological vector space, quasi-complete (every closed bounded set is complete). $\mathcal{H}$ is a Hilbert subspace of E.

$$\mathcal{H} \subset E.$$

[Beneath the inclusion: the map is continuous; at right: in $\mathcal{H}$ there is a Hilbert metric.] The Hilbert topology does not coincide with, but is stronger than, the topology in E. Example.



Uncertain reading. The inclusion reads $L^2 \subset \mathcal{D}^1$ / $L^2 \subset \mathcal{D}'$; the heavy upper-right mark above $\mathcal{D}$ resembles both 1 and a prime.

Original source details: GELFAND-P000018

Another example is the space of constants:

$$(\alpha \mid \beta) = \alpha\overline{\beta}.$$

$\mathcal{H} \in \mathrm{Hilb}(E)$ — all Hilbert subspaces.

[Repeated in the left margin: $\mathrm{Hilb}(E)$, with an arrow to the following heading.] An abstract conical structure. Addition and multiplication by positive numbers must be defined, and an order introduced.

$$\mathcal{H} \in \mathrm{Hilb}(E), \quad \lambda \geq 0, \quad \lambda\mathcal{H}, \quad \mathcal{H}_1 + \mathcal{H}_2, \qquad \mathcal{H}_1 \leq \mathcal{H}_2.$$

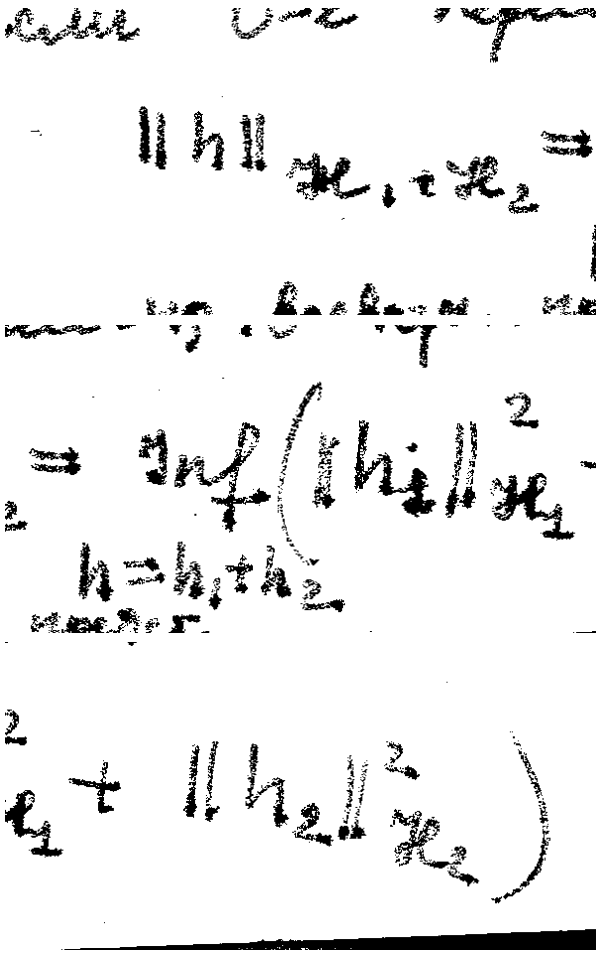
For $\lambda = 0$, $\lambda\mathcal{H}$ is the zero subspace. For $\lambda \neq 0$, $\lambda\mathcal{H}$ is the same space with scalar product

$$(h \mid k)_{\lambda\mathcal{H}} = \frac{1}{\lambda}(h \mid k)_{\mathcal{H}}.$$

$\mathcal{H}_1 + \mathcal{H}_2$. If the intersection is zero, it is a direct sum.

$$\|h\|_{\mathcal{H}_1,\mathcal{H}_2} = \inf\left(\|h_1\|^2_{\mathcal{H}_1} + \|h_2\|^2_{\mathcal{H}_2}\right), \qquad h = h_1 + h_2.$$

[Beneath the formula: over all possible representations.]



Uncertain reading. No square is visible on the left of the norm formula, whereas both norms on the right are squared. The subscript has a comma between the two Hilbert-space symbols.

Original source details: GELFAND-P000018

GELFAND-P000019

$$\mathcal{H}_1 \oplus \mathcal{H}_2.$$

A direct sum as abstract Hilbert spaces.

$$(h_1, h_2) \longmapsto h_1 + h_2.$$

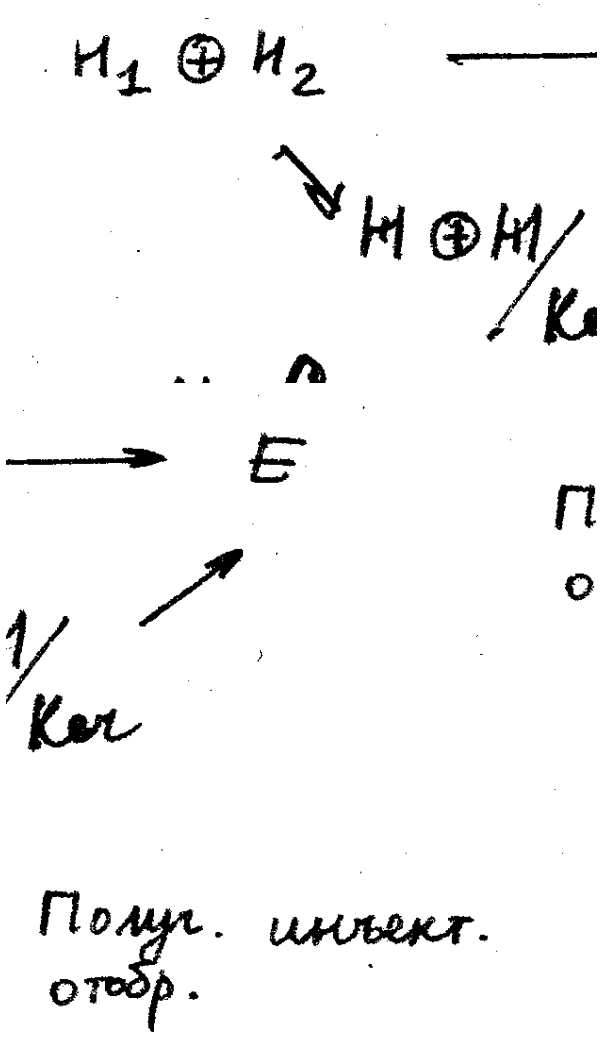
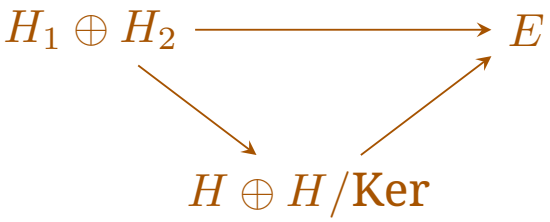


Diagram and typed description. A triangular diagram: at upper left $H_1 \oplus H_2$, at upper right E, and below the quotient $H \oplus H/\mathrm{Ker}$. The horizontal arrow runs from left to right into E; a diagonal arrow runs down from the direct sum to the quotient; the other diagonal arrow runs up from the quotient to E. The indices on the letters at the lower vertex are uncertain. At right: “Obtain an injective map.”



The lower Hs retain no reconstructed indices; the stroke above Ker is shown as a slash.

Original source details: GELFAND-P000019

The infimum is precisely the definition of the norm in a quotient space. But a quotient of a Hilbert space is itself a Hilbert space. Thus this is a Hilbert norm. Suppose

$$H_1 \subset H_2 \subset E,$$

then we say that $H_1 \le H_2$. [Below the first inclusion, an arrow i: an operator of norm ≤ 1.]

$$\|h\|_{H_1} \ge \|h\|_{H_2} > 0.$$

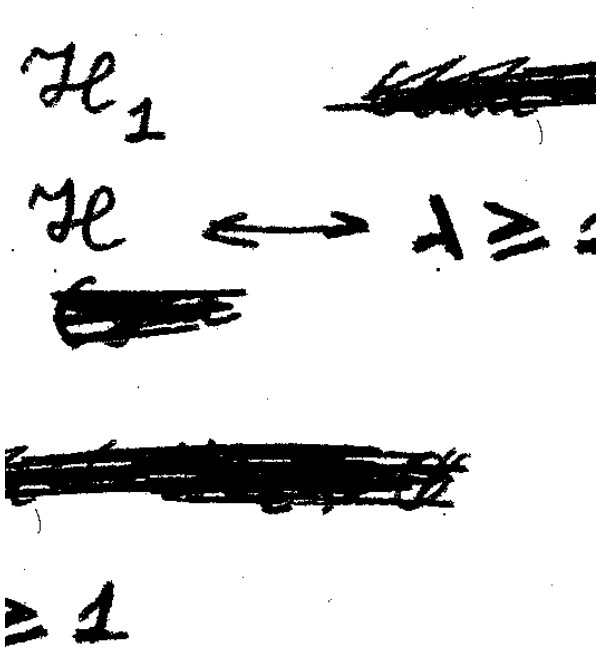
All the properties are easy to check:

$$(\lambda\mu)H = \lambda(\mu H), \qquad (\lambda+\mu)H = \lambda H + \mu H.$$

And all the others.

$$\mathcal{H}_1 + \mathcal{H}_2 \ge \mathcal{H}_1, \qquad \lambda\mathcal{H} \ge \mathcal{H} \iff \lambda \ge 1.$$

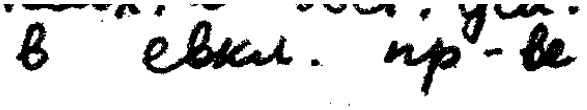
[At right of the first inequality and beneath the second line are heavily cancelled entries.]



Uncertain reading. the original letters and signs beneath the dense hatching are unreadable

Original source details: GELFAND-P000019

These are the axioms of a convex cone. There are necessary and sufficient conditions for realizing an abstract conical structure in евкл. / свкл. space.



Uncertain reading. евкл. / свкл. space [the initial letter of the source abbreviation is uncertain]

Original source details: GELFAND-P000019

GELFAND-P000020

[The first word of the heading is cancelled: Strict; inserted above: strictly convex.] abstract cone:

$$h \geq h_1 \iff \exists h_2 \text{ unique (!), such that } h = h_1 + h_2.$$

It will be proved that the cone of Hilbert subspaces can be embedded in a linear space; in particular, this condition holds. One must prove that if $H_1 \leq H$, there is a unique H_2 such that $H = H_1 + H_2$. Instead, a cone in a linear space will be constructed to which this one is isomorphic.

$$E \text{ — space of kernels}, \qquad H \in \mathcal{L}(E^*, E).$$

$\mathcal{L}$ is the space of linear operators from the antidual space (antilinear [Cancelled: illegible] functionals) to E, with the weak topology in E.

Uncertain reading. *Cancelled.* an unreadable short word

Original source details: GELFAND-P000020

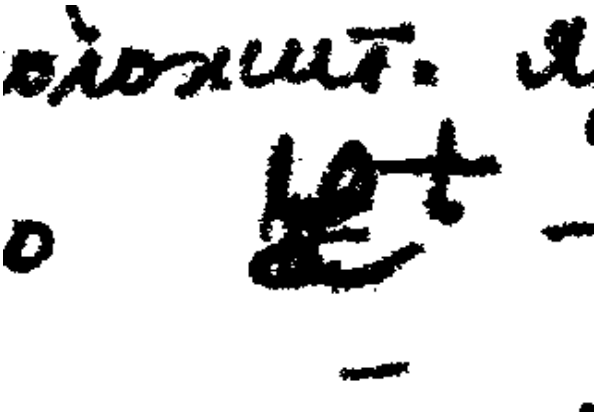
By definition,

$$H \geq 0 \underset{\text{def}}{\iff} \forall e^* \in E^*, \qquad (He^* \mid e^*)_{E,E^*} \geq 0.$$

$$e^* \in E^*, \quad f \in E, \qquad (e^* \mid f)_{E^*,E} = \overline{(f \mid e^*)_{E,E^*}}.$$

$$\mathcal{L}^+(E^*, E) \subset \mathcal{L}(E^*, E).$$

[Beneath the formula: the cone of positive kernels.] Example. E is a Hilbert space; $\mathcal{L}^+$ denotes positive Hermitian operators.



Uncertain reading. $\mathcal{L}^+$ [the corrected or overwritten symbol is uncertain]

Original source details: GELFAND-P000020

This is a strictly convex cone. The cone Hilb is isomorphic to it. Let $\mathcal{H} \in \mathrm{Hilb}(E)$. The corresponding kernel must be constructed.

$$e^* \in E^*; \qquad h \longmapsto (h \mid e^*)_{E,E^*}.$$

[Beneath the right side: e^* is antilinear.]

GELFAND-P000021

This is a continuous linear functional on H itself as well, since the embedding is continuous; but then $\exists He^* \in \mathcal{H}$ by means of which this scalar product is expressed.

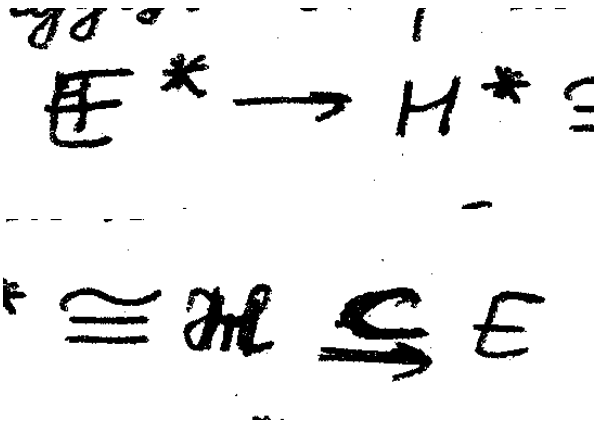
$$(h \mid e^*)_{E,E^*} = (h \mid He^*)_{\mathcal{H}}.$$

We have obtained a map $E^* \longrightarrow \mathcal{H} \longrightarrow E$. It is easy to check that this is a weakly continuous map from E^* to E.

$$\mathcal{H} \overset{c}{\longrightarrow} E \qquad \text{injectively embedded.}$$

It induces a map

$$E^* \longrightarrow H^* \cong \mathcal{H} \overset{\subset}{\longrightarrow} E.$$



Manuscript notation. The inclusion sign is written above the arrow.

Original source details: GELFAND-P000021

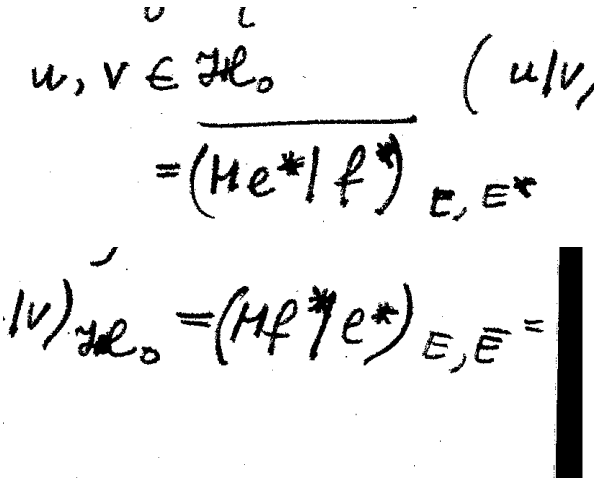
We have a map $E^* \longrightarrow E$, as required. Let $h = Hf^*$; then we obtain

$$(Hf^* \mid e^*)_{E,E^*} = (Hf^* \mid He^*)_{\mathcal{H}},$$
$$(He^* \mid e^*) \geq 0.$$

$$\mathrm{Hilb}(E) \longrightarrow \mathcal{L}^*(E^*, E).$$

One must prove that this is an isomorphism. It will be proved to be one-to-one and onto. Algebraically simple. Topological difficulties. If H is the kernel corresponding to $\mathcal{H}$, then $\forall e^* \in E^*$

$$He^* \in \mathcal{H}, \qquad \mathcal{H}_0 = \{He^* : e^* \in E^*\},$$
$$u, v \in \mathcal{H}_0 \qquad (u \mid v)_{\mathcal{H}_0} = (Hf^* \mid e^*)_{E,E^*} = \overline{(He^* \mid f^*)}_{E,E^*}.$$



Uncertain reading. The relation of u, v to e^*, f^* is not written here; a bar is visible above the last scalar product.

Original source details: GELFAND-P000021

GELFAND-P000022

Mackey's theorem: a set is bounded if it is weakly bounded. Prove that the unit ball is weakly bounded. Examples.

$$E = \mathcal{D}', \qquad E^* = \mathcal{D}, \qquad \mathcal{H} = L^2.$$

$\mathcal{L}(\mathcal{D}; \mathcal{D}')$ is the space of kernels, operators from $\mathcal{D}$ to $\mathcal{D}'$. $\mathcal{H} = L^2$ defines a natural map $\mathcal{D} \longrightarrow \mathcal{D}'$. If $\mathcal{H}$ is arbitrary, we obtain an arbitrary positive operator, but by the kernel theorem it is given by a generalized function of 2 variables.

$$H_{x,\xi} \in \mathcal{D}'(R^n \times R^n), \qquad \iint H_{x,\xi}\varphi(x)\overline{\varphi}(\xi)\,dx\,d\xi \geq 0.$$

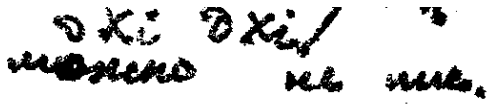
There is a correspondence between Hilbert subspaces of $\mathcal{D}'$ and such kernels. $\mathcal{H}^1$ is the space of functions, [Cancelled: illegible] L^2 with 1st derivatives.

Uncertain reading. *Cancelled.* an unreadable word before L^2

Original source details: GELFAND-P000022

$$(f \mid g)_{\mathcal{H}^1} = \int \left(\lambda f\bar{g} + \sum \frac{\partial f}{\partial x_i}\frac{\partial \bar{g}}{\partial x_i}\right) dx.$$

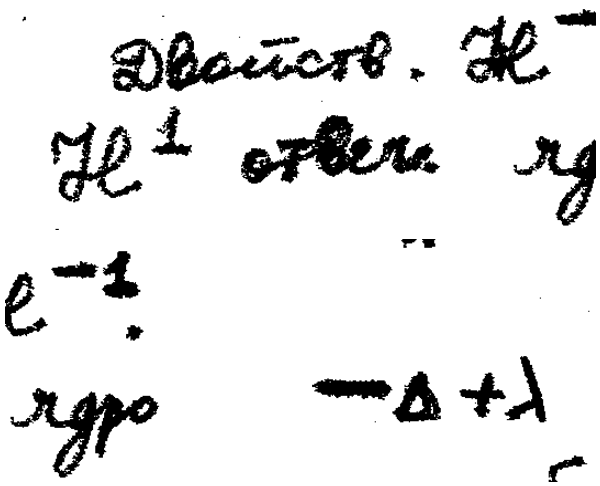
$\lambda > 0$ may be left [unwritten?]



Uncertain reading. The final abbreviation resembles «пис.» (written).

Original source details: GELFAND-P000022

The dual is $\mathcal{H}^{-1}$. To $\mathcal{H}^1$ corresponds the kernel $-\Delta + \lambda$.



Manuscript notation. The manuscript gives $-\Delta + \lambda$ as the kernel for $\mathcal{H}^1$.

Original source details: GELFAND-P000022

If we consider functions on a domain $\Omega \subset R^n$, these too are examples:

$$\mathcal{H}^{-1}, \quad \mathcal{H}^1, \quad \mathcal{H}_0^1 \quad \text{closure of } \mathcal{D}_0.$$

One can obtain existence theorems for solutions as generalized functions.

GELFAND-P000023

Uncertain reading. Several oblique strokes are cut off at the top edge; their meaning is unclear.

Original source details: GELFAND-P000023

Kushnirenko's result on the entropy of dynamical systems. Arnold. [Undeciphered cancellation] Entropy is the rate at which initial data are forgotten.

Uncertain reading. Heavy crossing-out obscures the middle of the entry. Its reading is unresolved.

Original source details: GELFAND-P000023

$$0 \qquad h \qquad +\infty.$$

Beneath 0: with slow mixing. Beneath $+\infty$: strong mixing. Entropy for abstract dynamical systems (a measure-preserving transformation T). M^n is a smooth manifold. M^n is compact. T is smooth, the measure of $M = 1$ and is given by a Riemannian metric. Then entropy is always finite and is estimated by local properties of the transformation.

$$h(T) \leq n \log \lambda.$$

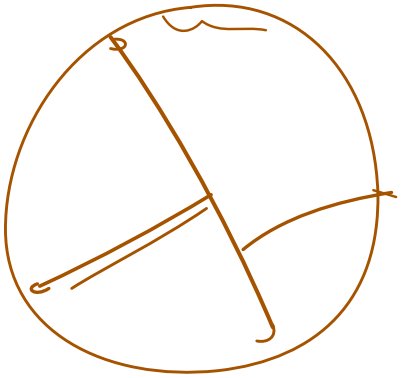
n is the dimension of the space. λ is the expansion coefficient of $(n-1)$-dimensional area. The bound is attained.

$$\lambda = \sup \frac{dS(Tx)}{dS(x)}.$$

Proof from the isoperimetric inequality. h is estimated by area. Entropy of a partition: $h(\xi) = -\sum p_i \log p_i$.

Original source details: GELFAND-P000023

Diagram and typed description. At right is a circle with a diagonal divider from upper left to lower right, a doubled branch towards the lower left and an arc to the right. A narrow light slit separates the two left strokes, with a thickened left endpoint; a separate thick mark projects beyond the circle at the right edge. These thickened marks do not establish arrow directions.

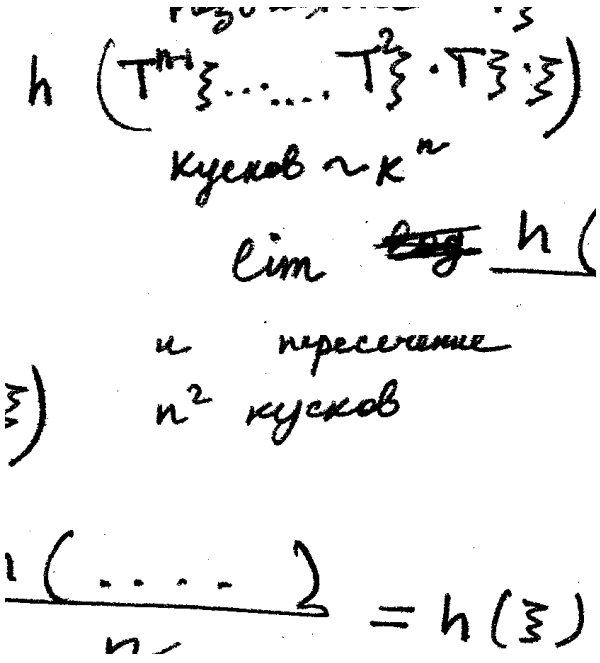


The doubled lower-left trace and terminal hooks are drawn without assigning arrow directions; their exact reading remains unresolved.

The partition $T\xi$ and the intersection.

$$h(T^{n-1}\xi \ldots\ldots T^2\xi \cdot T\xi \cdot \xi) \qquad n^2 \text{ pieces},$$
$$\text{Pieces } \sim k^n,$$
$$\lim \frac{h(\ldots)}{n} = h(\xi).$$

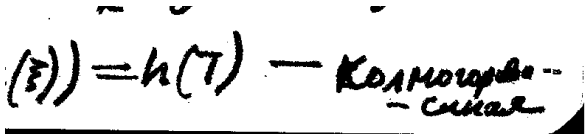
[Before the fraction, log is cancelled.]



Uncertain reading. Dots stand between $T^2\xi$, $T\xi$, ξ; alongside is written n^2 pieces. The limit's result resembles $h(\xi)$.

Original source details: GELFAND-P000023

For equal ones, $\sum \frac{1}{k} \log k = \log k$. Entropy of a dynamical system: $\sup(h(\xi)) = h(T)$—Kolmogoro-skaya [the source's split form].

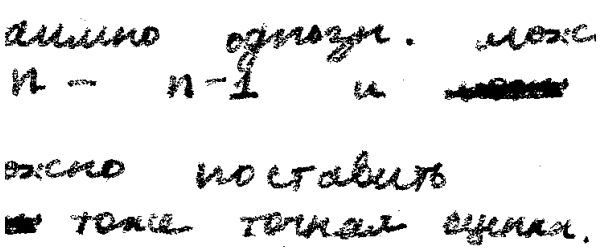


Uncertain reading. The final word is divided across lines as Колмогоро- / -ская; letters may be lost in the abbreviation.

Original source details: GELFAND-P000023

GELFAND-P000024

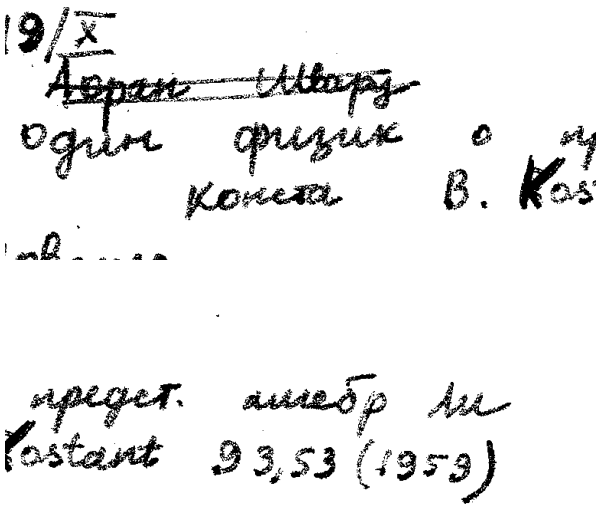
The entropy of a partition is estimated by the area of its boundary. And the boundary areas simply add! For a one-to-one transformation one can put $n - 1$ instead of n, and [Cancelled: illegible] there is again a sharp bound.



Uncertain reading. *Cancelled.* an unreadable word

Original source details: GELFAND-P000024

19/X. [Cancelled heading: Laurent Schwartz.] A physicist on representations of Lie algebras. [Kone...a?] B. Kostant 93, 53 (1959).

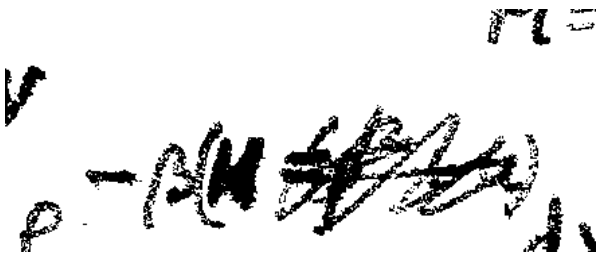


Uncertain reading. Before the Latin-script name, Коне is visible, apparently followed later by a. The intervening strokes are undeciphered.

Original source details: GELFAND-P000024

Povzner. Definition of free energy.

$$H = p^2 + u, \qquad \frac{-\ln \frac{1}{N!} \iiint_V^N e^{-\beta(H + \text{[cancelled]})}\, dx}{N}.$$



Uncertain reading. After H inside the exponent are several cancelled signs; $+$ and overlapping strokes are visible.

Original source details: GELFAND-P000024

N is the number of integrals.

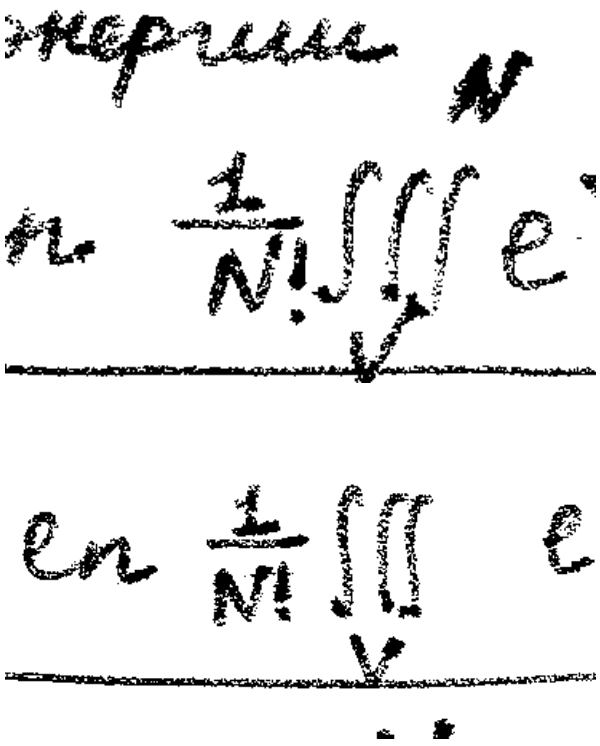
Manuscript notation. The manuscript writes «число интегралов» (number of integrals).

Original source details: GELFAND-P000024

$$\frac{V}{N} = \text{const}, \qquad \beta = \frac{1}{T},$$
$$\lim_{N\to\infty} \quad \text{and} \quad V/N = v = \text{const}.$$

free energy

$$\lim_{N\to\infty} \frac{-\ln \frac{1}{N!} \iiint_V e^{-\beta H}\, dx}{N}.$$



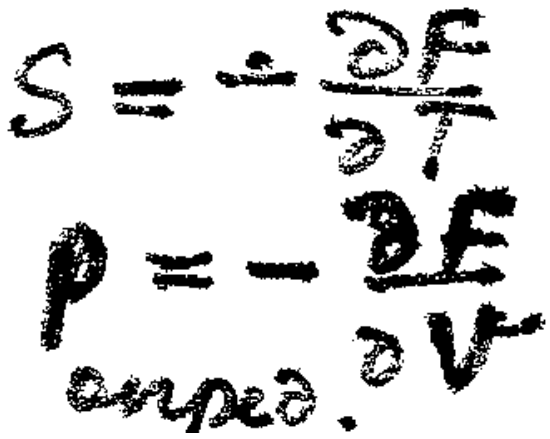
Uncertain reading. The first triple integral has N above and V below; only V is distinguishable in the repeated formula.

Original source details: GELFAND-P000024

Entropy from the thermodynamic formulas:

$$S = -\frac{\partial F}{\partial T},$$

$$\text{pressure} \quad p = -\frac{\partial F}{\partial V}.$$

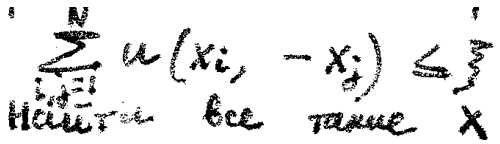


Uncertain reading. A dot is written above the minus in the formula for S. Its purpose is uncertain.

Original source details: GELFAND-P000024

Entropy can be defined as follows:

$$\frac{1}{N} \sum_{i,j=1}^{N} u(x_i, -x_j) \leq \xi.$$



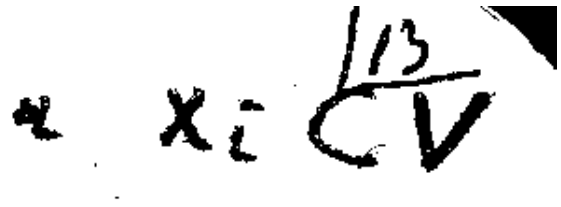
Manuscript notation. A comma follows x_i and a minus precedes x_j. The letter on the right is ξ.

Original source details: GELFAND-P000024

Find all such x.

GELFAND-P000025

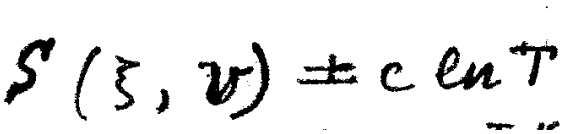
Take the volume of this region in phase space. P. In the upper-right margin: and $x_i \subset V$.



Manuscript notation. The upper margin has "and" and a curved inclusion sign without a middle bar.

Original source details: GELFAND-P000025

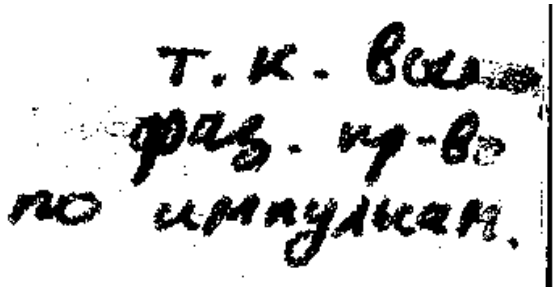
$$\lim_{N\to\infty} \frac{\ln \frac{1}{N!} P(\xi, V/N)}{N} = S(\xi, v) \pm c \ln T.$$



Manuscript notation. The manuscript writes $\pm$ before $c \ln T$.

Original source details: GELFAND-P000025

ξ is energy. In the right margin: since весь / всё [all; the source ending is uncertain] phase space in the momenta.

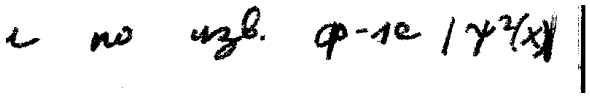


Uncertain reading. Весь / всё фаз. пр-во по импульсам [all phase space in the momenta; source grammar retained].

Original source details: GELFAND-P000025

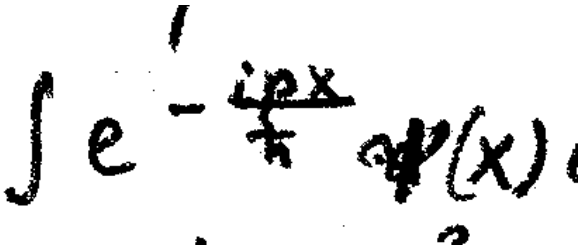
S is a convex function of 2 variables. On Wigner's work (1932) on quantum mechanics. Let $\Psi(x)$ give the probabilities by the familiar formula $|\Psi^2(x)|$ and

$$\widetilde{\Psi}(p) = \int e^{-ipx/\bar{h}} \Psi(x)\, dx.$$



Manuscript notation. The first line places the exponent inside the absolute value signs, $|\Psi^2(x)|$.

Original source details: GELFAND-P000025



Manuscript notation. The h in the exponent denominator has a cross stroke.

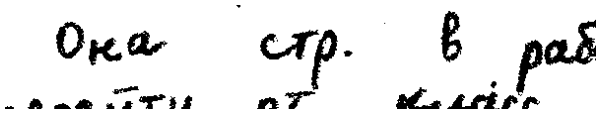
Original source details: GELFAND-P000025

If

$$\int P(x,p)\,dp = |\Psi(x)|^2,$$

$$\text{and} \quad \int P(x,p)\,dx = |\widetilde{\Psi}(p)|^2,$$

then what is P? It is [стр.; uncertain source abbreviation] in Wigner's work, and from this one can pass from classical mechanics to statistical mechanics.



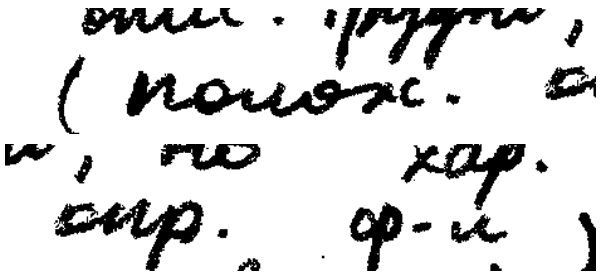
Uncertain reading. The manuscript writes «Она стр.». The abbreviation is unresolved.

Original source details: GELFAND-P000025

Kirillov on Thoma's work. Elmar Thoma. Math. Zeitschr. 84 (1964). If G is a discrete noncommutative group, its representations are difficult to describe, but the characters can be found (positive-definite functions).

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

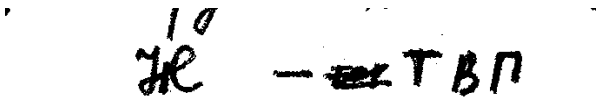
The characters are the same (the same formulas as over the real, finite and p-adic numbers).



Manuscript notation. The manuscript uses the abbreviation «полож. опр. ф-и».

Original source details: GELFAND-P000025

Laurent Schwartz. $s \in G$—a Lie group. dx is Haar measure (two-sided). $\mathcal{H}$ is a [Cancelled: illegible] topological vector space.

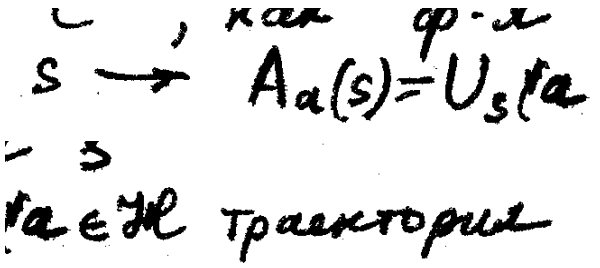


Uncertain reading. *Cancelled.* an unreadable word

Original source details: GELFAND-P000025

U_s is a representation in it. $a \in \mathcal{H}$, which is infinitely differentiable with respect to the group; its orbit is C^∞ as a function of s.

$$s \longrightarrow A_a(s) = U_s(a \in \mathcal{H} \qquad \text{trajectory.}$$



Uncertain reading. Between U_s and a is an opening delimiter stroke: (/ {. No closing delimiter is visible on this line.

Original source details: GELFAND-P000025

$A_a : G \longrightarrow \mathcal{H}$, which is C^∞. Instead of the representation U_s, one can consider U_μ and U_φ.

$$U_\mu = \int U_s \, d\mu(s).$$

Trajectory

$$U_s A_a(t) = A_a(st).$$

If there is such a curve, then it is an orbit, i.e.

$$\mathcal{H} \supset \mathcal{H}^m \supset \mathcal{H}^\infty.$$

By $\mathcal{H}^m$: the trajectory is m times differentiable. By $\mathcal{H}^\infty$: the trajectory is C^∞. One can prove that $\mathcal{H}^{+\infty}$ is dense in $\mathcal{H}$ (by averaging). Define

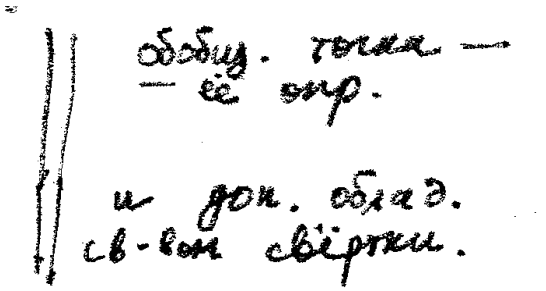
$$\mathcal{H}^{-\infty} \supset \mathcal{H} \supset \mathcal{H}^m \supset \mathcal{H}^{+\infty}.$$

Point generalized functions. We shall consider $U_s A_a(t) = A_a(st)$ as a distribution with respect to Haar measure.

$$U_\varphi A_a(\psi) = A_a(\varphi * \psi).$$

Generalized points of the representation space.

$$A : \mathcal{D}'(G; \mathcal{H}),$$
$$\forall \varphi \in \mathcal{D}(G), \quad A(\varphi) \in \mathcal{H},$$
$$U_\varphi A(\psi) = A(\varphi * \psi).$$

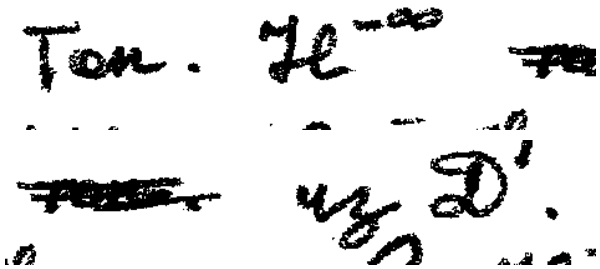


Original source details: GELFAND-P000026

Diagram and typed description.

Generalized point—
—its definition.
and до[л. / п.] [uncertain abbrev.] possess
the convolution property.
The last letter of the abbreviation resembles л or п.

In particular, ordinary points possess these properties. Topology of $\mathcal{H}^{-\infty}$ [Cancelled: illegible] from $\mathcal{D}'$. $\mathcal{H}^{-\infty}$ will be complete.

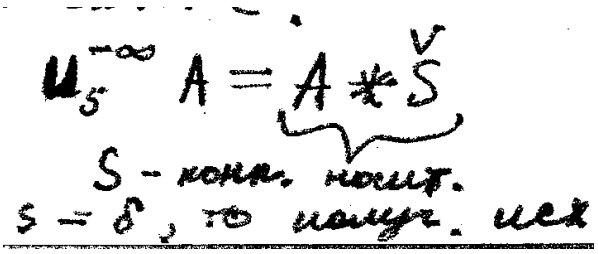


Original source details: GELFAND-P000026

Uncertain reading. *Cancelled.* an unreadable word

The group also acts in $\mathcal{H}^{-\infty}$. There the group acts C^∞.

$$U_S^{-\infty} A = \underbrace{A * \check{S}}_{S \text{ — finite supp.}}$$



Uncertain reading. The convolution and its caption both use S. The index resembles S, with a short detached stroke to its right.

Original source details: GELFAND-P000026

If $S = \delta$, then the original representation is obtained.

GELFAND-P000027

In particular, the Lie algebra also acts (δ').

. . . . . .

Problem. $\Box X = A$, an equation with constant coefficients. This equation is solvable (Ehrenpreis). Let A be Lorentz invariant. Then there exists a Lorentz invariant X.

$$\mathcal{D}X = A.$$

The same for an arbitrary group and differential operator—a problem!

Uncertain reading. In the repeated formula, the letter before X is rounded: $\mathcal{D}$ / D; in the first formula there is a rectangle □.

Original source details: GELFAND-P000027

Solution for the Lorentz group.

$$\Box X = A.$$

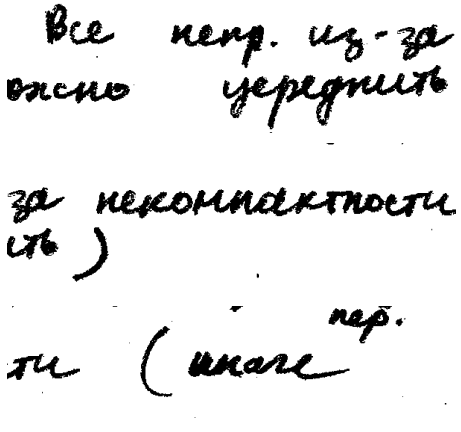
One must [Cancelled: illegible] study Lorentz invariant functions.

Uncertain reading. *Cancelled.* a short unreadable word

Original source details: GELFAND-P000027

Consider $R^n \setminus 0$. This is $T(x_1^2 + \cdots + x_n^2 - x_0^2)$ of one variable.

All непр. [unexpanded source abbreviation] because of noncompactness (otherwise one can average).

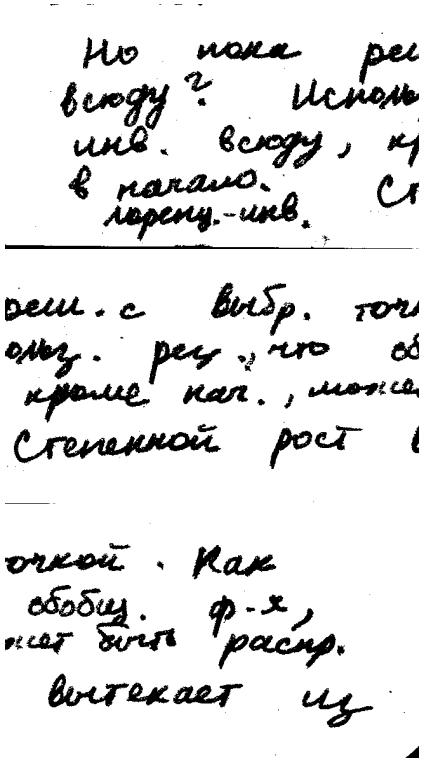


Uncertain reading. The manuscript abbreviates the word as непр. Its expansion is unresolved.

Original source details: GELFAND-P000027

For X, seek an invariant distribution of one variable. The wave operator turns into an operator of one variable, but with variable coefficients. But for one variable it is solvable. But if there is a group with 2 invariants, then instead of a singularity at 1 point one obtains lines of singularities, etc.

But so far the solution has a chosen point. How everywhere? Use the result that a generalized function, invariant everywhere except at the origin, can be extended to the origin. Power growth follows from Lorentz invariance.

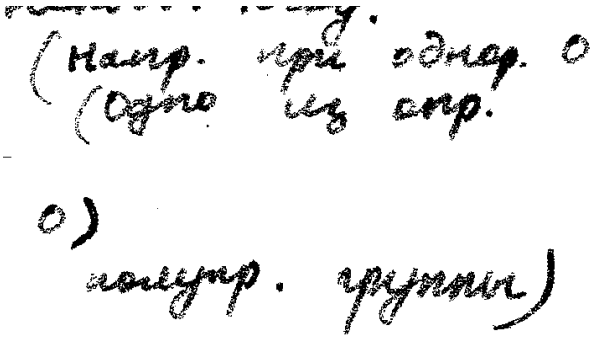


Manuscript notation. The manuscript abbreviates “solution,” “chosen,” “result,” “origin,” and “extended.”

Original source details: GELFAND-P000027

GELFAND-P000028

The Lorentz group is semisimple and the space is finite dimensional. A finite dimensional representation of a semisimple Lie group has a fixed point (e.g. at однор. 0 [uncertain source abbreviation]) (one of the definitions of a semisimple group).



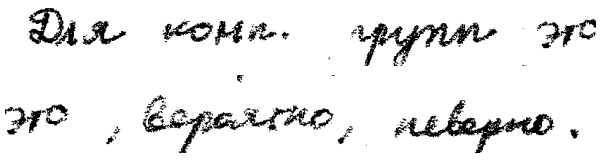
Manuscript notation. The manuscript writes «при однор. 0», with an abbreviated adjective.

Original source details: GELFAND-P000028

Another problem. G is a Lie group. A Lie operator X is an infinitesimal translation $Xf(g)$. Consider a polynomial in Lie operators. How to find elementary solutions? (fund.) $PE = \delta$.

$$P(X_1, \ldots, X_n) \qquad \text{a polynomial in Lie operators.}$$

For compact groups this is probably false.

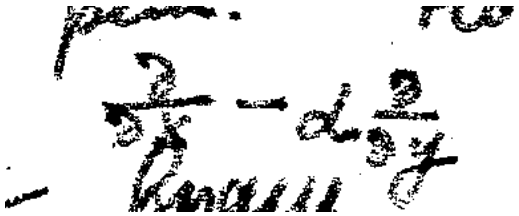


Original source details: GELFAND-P000028

Manuscript notation. The manuscript writes «комп.» for compact.

On the torus there is no elementary solution. There is no solution for the torus. But the torus has a fundamental group. And for the rotation group

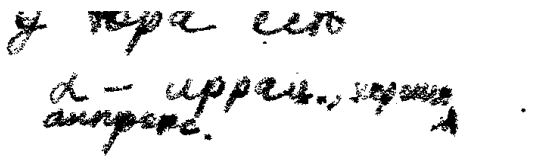
$$\frac{\partial}{\partial x} - \alpha \frac{\partial}{\partial y}.$$



Original source details: GELFAND-P000028

Manuscript notation. The manuscript writes α without a separate superscript.

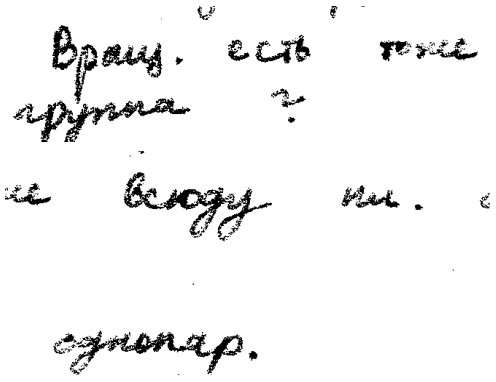
In the right margin: α—irrational, good. approx.



Original source details: GELFAND-P000028

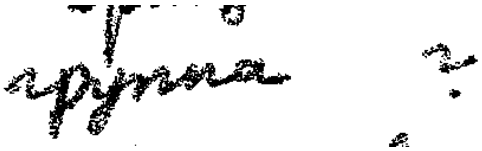
Uncertain reading. The last abbreviations in the margin are хорош. / хорошо [good/well]; аппрокс. [approx.]. Below the first is a small detached dark A/1-like stroke whose role is not clear.

For rotations, is there also everywhere an [ин. / им., source abbreviation] one-parameter group?



Original source details: GELFAND-P000028

Uncertain reading. Before «однопар.» (one-parameter), the shorter abbreviation resembles ин. or им.



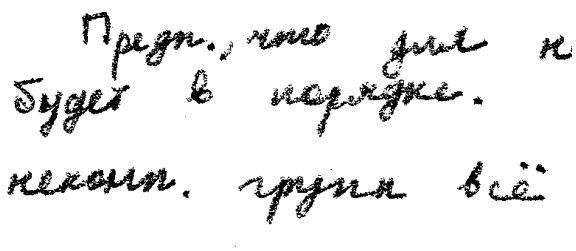
Original source details: GELFAND-P000028

Manuscript notation. The word «группа?» (group?) continues on the next line.

Nontrivial even for operators of the 1st order.

$$E(P\varphi) = \varphi(0).$$

Suppose that for a noncompact group everything will be in order. For the circle...



Manuscript notation. The manuscript writes «некомп.» for noncompact.

Original source details: GELFAND-P000028

GELFAND-P000029

2/XI. Kirillov. Elmar Thoma. 3 works, representations of countable noncommutative groups [“discrete” written below “noncommutative”]. G. There is an explicit form for certain groups of $+$ definite functions constant on conjugacy classes of elements—extreme points of the set of such functions.

$$f(g) \quad + \text{ definite, if } \quad \|f(g_i g_j^{-1})\| \gg 0 \quad + \text{ definite.}$$
$$g_1, \dots, g_n \quad \text{any collection of elements.}$$

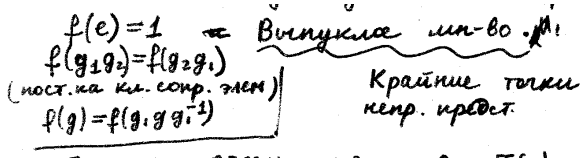


Diagram and typed description.

$f(e) = 1 \longleftarrow$ Convex set M

$$f(g_1 g_2) = f(g_2 g_1)$$

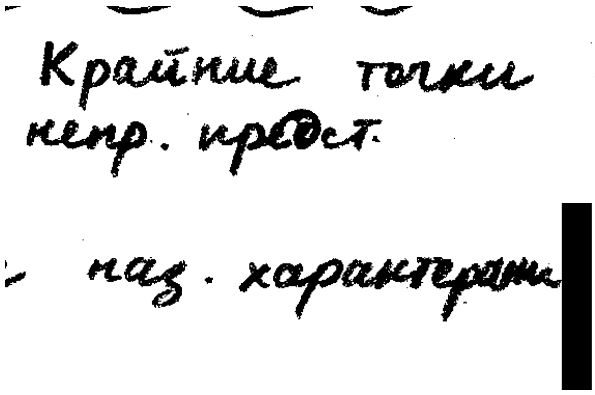
(constant on conjugacy classes of elements)

$$f(g) = f(g_1 g g_1^{-1})$$

The short stroke to the right of M is unresolved.

Original source details: GELFAND-P000029

The extreme points are called characters of irreducible representations.



Manuscript notation. The manuscript abbreviates “irreducible” as непр. and uses the fuller неприв. below.

Original source details: GELFAND-P000029

Example: $g \longrightarrow T(g)$, $\frac{1}{n} \operatorname{Sp} T(g)$ is an example of such a function. If the representation is irreducible, such functions form extreme points—easy to check. (The representation is uniquely determined by this function.) There are groups for which describing the representations is impossible (for set theoretic reasons—there are many of them), but the characters exist and are given by formulas.

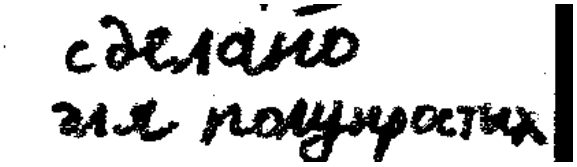
$$\begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} \quad \text{group of affine transformations of the line,}$$
$$x \longrightarrow \alpha x + \beta, \qquad \alpha, \beta \quad \text{rational.}$$

If α and $\beta \in$ a finite field (mod p), how to find all traces (irreducible representations). Representations—complex numbers. Not done if the representation is given

with $\alpha, \beta \in (\mathrm{mod}\, p)$.

$$\begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & \delta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \alpha\gamma & \alpha\delta + \beta \\ 0 & 1 \end{pmatrix}.$$

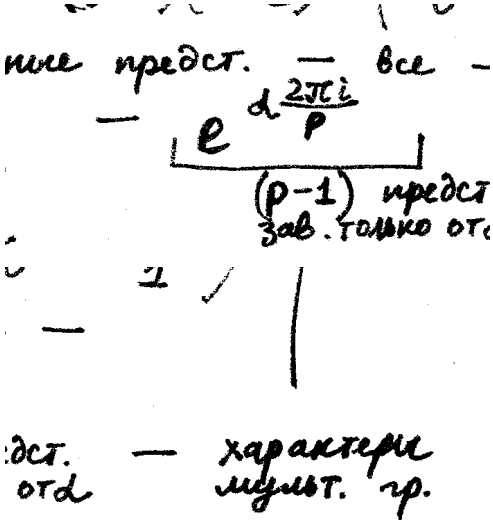
In the right margin: done for semisimple ones.



Original source details: GELFAND-P000029

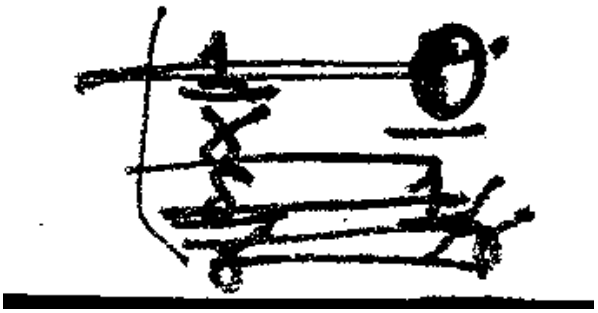
Manuscript notation. The margin reads «сделано для полупростых» (done for semisimple ones).

One dimensional representations—all—$e^{\alpha \frac{2\pi i}{p}}$. $(p-1)$ representations depending only on α—characters of the multiplicative group.



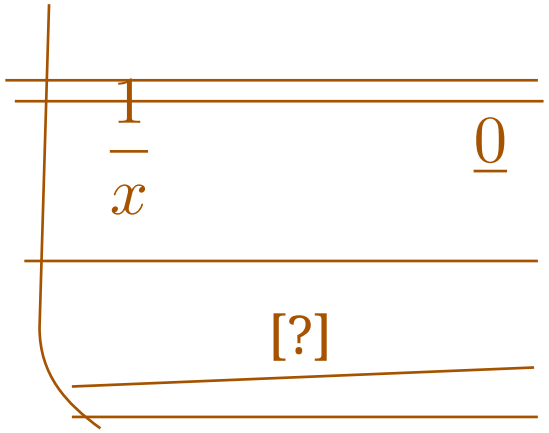
Original source details: GELFAND-P000029

Uncertain reading. In the exponent one can see $\alpha\, 2\pi i/p$.



Original source details: GELFAND-P000029

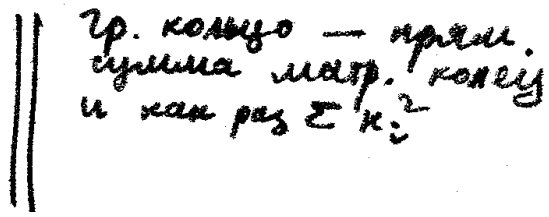
Diagram and typed description.



Two fractional expressions, 0 and horizontal strokes are crossed out; bracket-like strokes run down the left and below. The first fraction reads $1/x$. The horizontal arrow and the second fraction cannot be read securely through the cancellations.

$$\sum k_i^2 = p(p-1),$$

$$\sum{}' k_i^2 = p(p-1) - \underbrace{(p-1)}_{\text{one dim.}} = (p-1)^2.$$



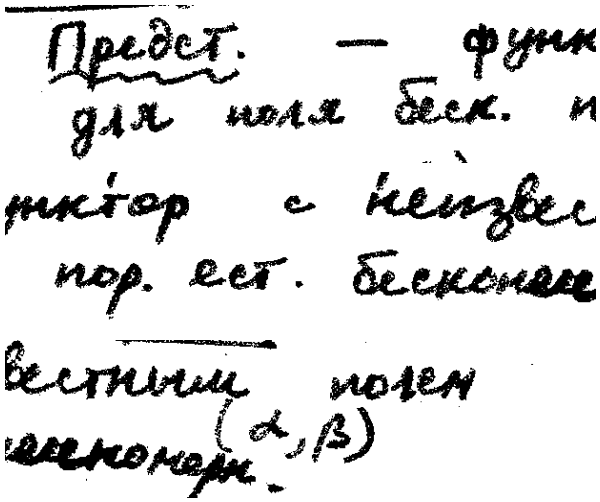
Original source details: GELFAND-P000030

Diagram and typed description.

> Group ring—direct sum of matrix rings and precisely $\sum k_i^2$.

Two vertical rules separate the note in the right margin.

Conjecture: another representation of order $p-1$. Representation—a functor with an unknown field. For a field of infinite ord. there is an infinite dimensional (α, β).

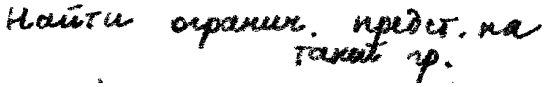


Original source details: GELFAND-P000030

Uncertain reading. Беск. пор. ест. бесконечномерн. [infinite ord., there is an infinite dimensional]; the connection of this abbreviated line to “functor with an unknown field” is not explicit.

$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}$ find the restr. of the repr. to such a group,

$$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & \beta' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \beta+\beta' \\ 0 & 1 \end{pmatrix}.$$



Original source details: GELFAND-P000030

Uncertain reading. The manuscript abbreviates the expression as «огранич. предст.». The expansion of огранич. is uncertain.

Its one dimensional representations

$$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \longrightarrow e^{\frac{2\pi i k}{p}\beta}.$$

$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}$ normal subgroup.

Transformations with [a short word cancelled]

$$\begin{pmatrix}\alpha & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & \beta\\ 0 & 1\end{pmatrix}\begin{pmatrix}\alpha^{-1} & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}1 & \alpha\beta\\ 0 & 1\end{pmatrix}.$$

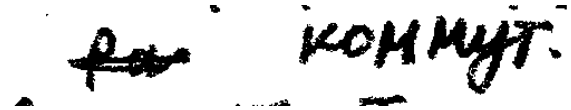
Original source details: GELFAND-P000030

Uncertain reading. Between “with” and the matrix is a small obliterated sign or word.

β is not needed (it commutes).

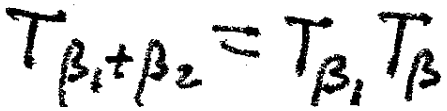
$$\begin{pmatrix}1 & \beta\\ 0 & 1\end{pmatrix}\longrightarrow T_\beta,\qquad T_{\beta_1+\beta_2}=T_{\beta_1}T_\beta.$$

[Cancelled: illegible] commute.



Original source details: GELFAND-P000030

Uncertain reading. *Cancelled.* a short word before коммут. [commute]

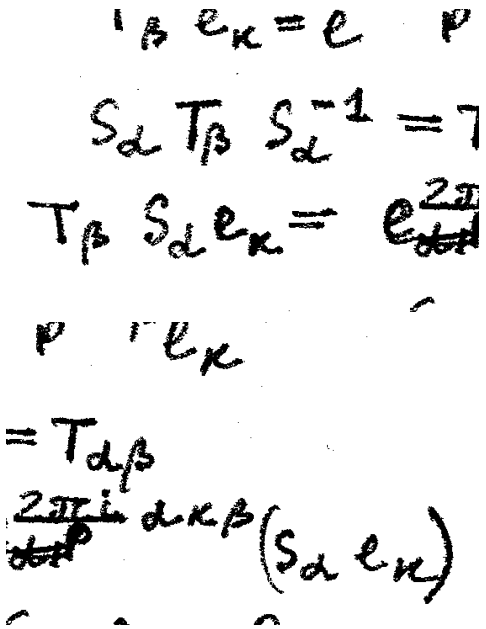


Original source details: GELFAND-P000030

Manuscript notation. In the last T_β, no numeral follows β.

Decompose the space into eigenvectors with respect to T_β.

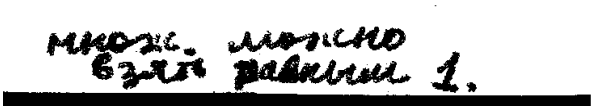
$$T_\beta e_k=e^{\frac{2\pi ik}{p}\beta}e_k,$$
$$S_\alpha T_\beta S_\alpha^{-1}=T_{\alpha\beta},$$
$$T_\beta S_\alpha e_k=e^{\frac{2\pi i}{p}\alpha k\beta}(S_\alpha e_k),$$
$$S_\alpha e_k=e_{\alpha k}.$$



Original source details: GELFAND-P000030

Uncertain reading. The source writes $S_\alpha T_\beta S_\alpha^{-1}=T_{\alpha\beta}$ and the factor $\alpha k\beta$ in the exponent beside $T_\beta S_\alpha e_k$. Cancelled signs lie below the denominator of the latter exponent.

At the lower right: multipliers can be taken equal to 1.



Original source details: GELFAND-P000030

Manuscript notation. The manuscript writes «множ. можно взять равными 1» (the multipliers can be taken equal to 1).

GELFAND-P000031

There will remain $(p-1)$ vector. It is written how it acts. A more beautiful realization.

$$x \longrightarrow \alpha x + \beta.$$

[At right: aff. line over a finite field.] $f(x)$ — functions on it.

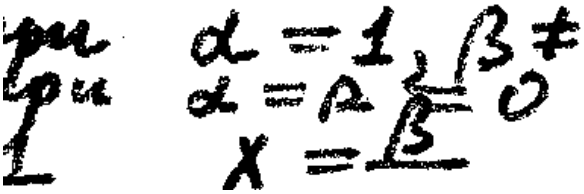
$$T_g f(x) = f(\alpha x + \beta).$$

Const is invariant, so this repr. is reduc. We shall consider functions for which

$$\sum_x f(x) = 0 \qquad (\int = 0).$$

This is a $(p-1)$-dimensional repr.

$$\mathrm{Sp}(T_g) = \begin{cases} 1 & \text{when } \alpha \neq 1, \\ 0 & \text{when } \alpha = 1,\ \beta \neq 0, \\ p & \text{when } \alpha = 1,\ \beta = 0. \end{cases}$$



Original source details: GELFAND-P000031

Uncertain reading. In the third condition, the looped sign after $\alpha =$ resembles 1, α or p; the coloured 1 is a provisional candidate.

[Repeated separately at left beneath the brace: $\alpha \neq 1$. At right: reducible.]

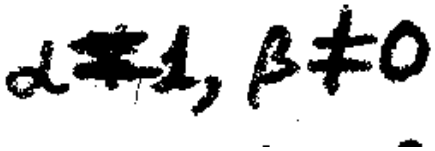
$$x = \alpha x + \beta, \qquad x = \frac{\beta}{1-\alpha}.$$

Must subtract the trace [след.; provisional].

Original source details: GELFAND-P000031

Uncertain reading. The abbreviation after “subtract” is provisionally read as след. [trace].

$$\mathrm{Sp}\, T_g = \begin{cases} 0 & \alpha \neq 1, \\ -1 & \alpha = 1,\ \beta \neq 0, \\ p-1 & \alpha = 1,\ \beta = 0. \end{cases}$$



Original source details: GELFAND-P000031

Uncertain reading. Heavy overwritten or cancelled strokes stand before 1 in the second condition. The earlier sign $\neq$ / $=$ cannot be established securely.

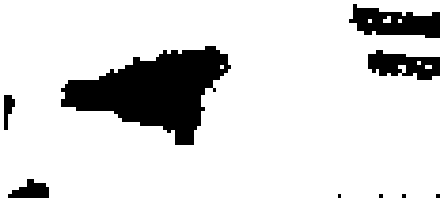
How is this related to dyn. systems?

Let α and β be real.

$$\beta \longrightarrow T_\beta, \qquad T_{\beta_1+\beta_2} = T_{\beta_1} T_{\beta_2},$$

$$S_\alpha, \qquad S_{\alpha_1\alpha_2} = S_{\alpha_1} S_{\alpha_2}, \qquad S_\alpha^{-1} T_\beta S_\alpha \blacktriangleleft = T_{\alpha\beta}.$$

Uncertain reading. Between the last S_α and $=$ is a small separate filled wedge pointing left. Its visible shape is retained; its meaning has not been established.



Original source details: GELFAND-P000031

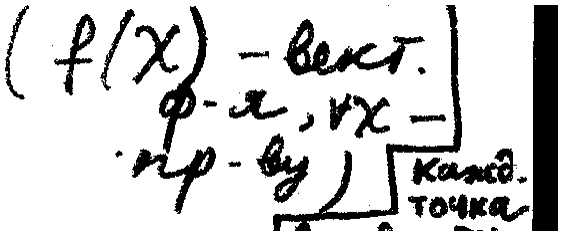
[Horizontal line.] How can they be found? They can be brought to diagonal form (the repr. is unit.).

$$T_\beta f(\chi) = \chi(\beta) f(\chi).$$

[Cancelled at right of the formula: $\chi(\beta)$.] Bring to an operator of multiplication by a function, where

$$\chi(\beta_1 + \beta_2) = \chi(\beta_1)\chi(\beta_2).$$

[In the right brace:] $f(\chi)$ — a vector function, $\forall\chi$ — to a space [пр-ву; abbreviated in the source].



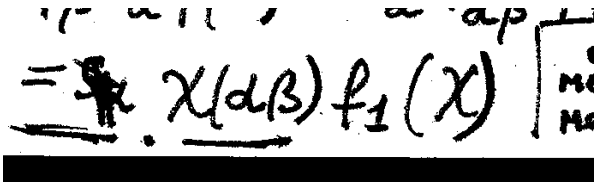
Uncertain reading. The sign before χ resembles $\forall$; пр-ву is an abbreviation for “to a space”. The source does not write out the full word.

Original source details: GELFAND-P000031

[In the left margin, rotated: dyn. syst. on the addit. group — $\longrightarrow$ the mult. group acts on it.]

$$S_\alpha f(\chi) = f_1(\chi),$$

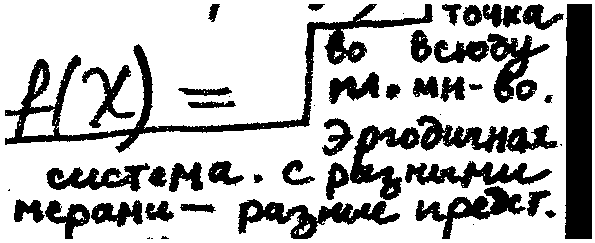
$$T_\beta f_1(\chi) = T_\beta S_\alpha f(\chi) = S_\alpha T_{\alpha\beta} f(\chi) = \chi(\alpha\beta) f_1(\chi).$$



Uncertain reading. Before $\chi(\alpha\beta)$ in the last equality, a symbol resembling S / S_α is heavily cancelled. Beneath the cancellation is $\longleftarrow$; beneath $\chi(\alpha\beta)$ is $\longrightarrow$. The cancelled sign remains unreadable.

Original source details: GELFAND-P000031

[In the lower-right margin, within a stepped outline:] Each point — into an everywhere dense [пл.; uncertain abbreviated reading] set.



Uncertain reading. The source abbreviation before мн-во [set] resembles пл. [dense]; its middle strokes remain uncertain.

Original source details: GELFAND-P000031

Ergodic system. With different measures — different repr.

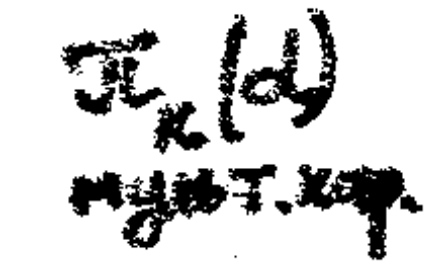
Haar measure — [непр. пр.]

Uncertain reading. The expansions of the source abbreviations непр. пр. are uncertain.

Original source details: GELFAND-P000031

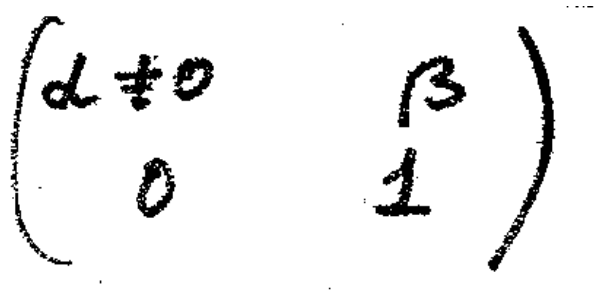
GELFAND-P000032

$\overline{\chi}_K(\alpha)$ mult. char.	$\begin{pmatrix} \alpha \neq 0 & \beta \\ 0 & 1 \end{pmatrix}$	0	0
1	$\begin{pmatrix} 1 & \beta \neq 0 \\ 0 & 1 \end{pmatrix}$	$-\dfrac{1}{p-1}$	0
1	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	1	1



Uncertain reading. The barred notation before the first matrix is provisionally read as $\overline{\chi}_K(\alpha)$. The first sign may be π, and the index may be Latin K or Cyrillic к. Beneath it is мульт. хар. [mult. char.].

Original source details: GELFAND-P000032



Uncertain reading. The upper-left entry reads $\alpha \neq 0$: the final sign is a closed loop. A separate 1 precedes each of the two lower matrices.

Original source details: GELFAND-P000032

[To the right of the first upper 0: Char. for finite groups.] If one passes to the lim. as $p \longrightarrow \infty$. [From the line about the limit, $\nearrow$ points to the lower-right 1.]

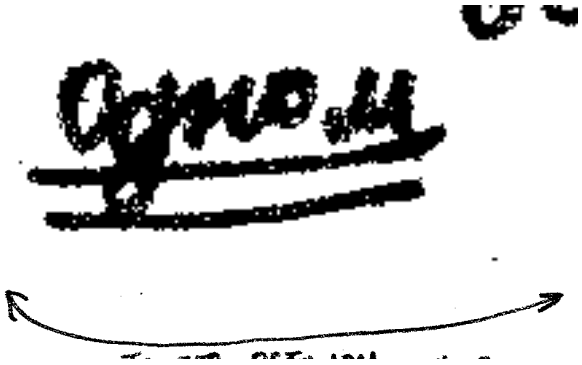


Diagram and typed description. The left marginal word has two underlines and resembles Одном / Одному [one / to one, literal candidate readings]. Below it is a long low curve with outward arrowheads at both ends; the right end is beneath the limit. The label remains unresolved.

Одном / Одному

Original source details: GELFAND-P000032

What remains for rational numbers.

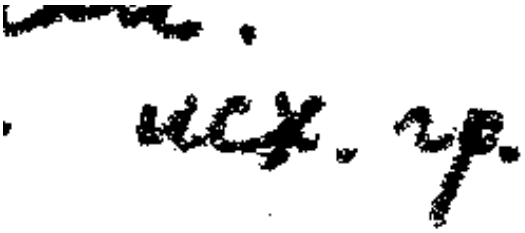
Lemma. G, $G_0 \subset G$ [beneath G_0: norm. div.]. f — pos. def., $f(e) = 1$, const. on conj. M — the set of them. Consider $f|_{G_0}$.

Also consider functions satisfying the same [a short entry is cancelled] conditions on G_0.

Uncertain reading. The short entry before the source abbreviation усл. is crossed out with heavy horizontal strokes; its letters cannot be established.

Original source details: GELFAND-P000032

Consider H_{G_0} — their extreme points. They coincide with the extreme points of the original / countab. [исх. / сч.; uncertain source abbreviation] gr.



Uncertain reading. The joined final source abbreviation resembles исх. гр. [original gr.]; сч. гр. [countab. gr.] remains an alternative. The full source words are not written.

Original source details: GELFAND-P000032

(For countable groups.) One must count how many conj. classes there are and then seek the representing. From + def. [Cancelled; provisional reading: something [чего-то; uncertain reading].] it follows that there are no other repr.

Uncertain reading. The crossed-out entry after Из + опр. [From + def.] resembles чего-то [something]; the heavy overlapping strokes do not establish a complete secure word.

Original source details: GELFAND-P000032

[In the left margin: 2nd work.]

Infinite symmetric group: permutations of the natural sequence (only a finite number of elements move — for countability). Conj. classes of elements are specified by cycles. — [A separate horizontal stroke at the left; a short entry is cancelled.]

Uncertain reading. The short sign or word before “cycle lengths” remains unreadable beneath two heavy horizontal strokes.

Original source details: GELFAND-P000032

Cycle lengths determine the conj. class of elements.

$$(n_2, n_3, \ldots),$$

only a finite number $\neq 0$.

GELFAND-P000033

Every character

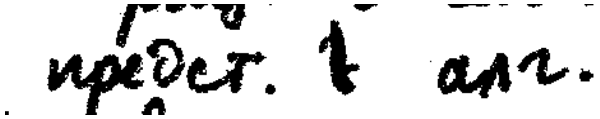
$$\chi(n_2, n_3, \ldots) = \prod a_k^{n_k}.$$

a_k — a sequence of real numbers. For a finite group it is more complicated. [Above “finite”: (arbitr.); between “group” and “more complicated,” inserted above: “char.” with a brace pointer.] (Frobenius, Young, Young diagr. — classification of tensors.) G_n — a subgr. isom. to the original — the first n stay in place. And the char. is the same (depends only on cycle lengths). From this follows multiplicativity of the char. in cycle length, and further. 3rd work.

There are countab. noncompact gr. Practically all are bad. Char. Type 1: there is a norm. div. G_0 such that G/G_0 is finite. [Inserted beneath norm. div. G_0:] isom. G. (Necessary and sufficient condition.)

Thoma $\left.\begin{matrix}\text{Math. Zeits.}\\ \text{Math. Ann.}\end{matrix}\right\}$ 1964.

These are his first 3 works. Not done: characters for the gr. of matrices of order 2 over the field of rational numbers.
What is a repr. [mark] of an alg. group with an unknown field?

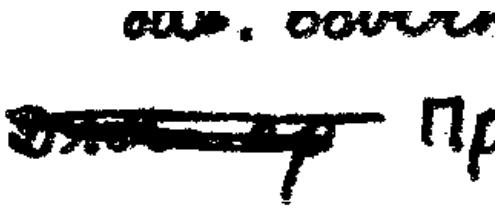


Uncertain reading. A separate upright mark stands between the source abbreviations предст. and алг. Its meaning is uncertain.

Original source details: GELFAND-P000033

$\begin{pmatrix}\cdot & \cdot\\ 0 & 1\end{pmatrix}$ Unknown field.

[At right: So that for char. of fields it gives the usual expr.] A way of writing char. of $\forall$ alg. gr. [A short entry is cancelled.]



Uncertain reading. After алг. гр. [alg. gr.] a short entry is heavily cancelled; its beginning resembles Это... [This...]. The remaining strokes do not establish the text.

Original source details: GELFAND-P000033

Repr. of types 1 and 2 are recovered from char. G is specified by a scheme of semigr. alg., assigned to $\forall$ field. Char. also.

[Cancelled; provisional reading: Function / Func. [Функция / Функц.; uncertain reading]]

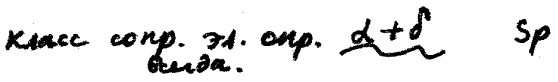
Uncertain reading. The cancelled source word resembles Функция / Функц. [Function / Func.]; the ending is obscured by horizontal strokes.

Original source details: GELFAND-P000033

Character — a functor of the field K. [At right: $\mathcal{F}(\cdot)$.] (A formula corresponds to the field K.) If there is an equality for several fields, then also always. (Similarly, e.g. only for finite ones.)

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{with } \det = 1.$$

The conj. class of elements is determined by $\alpha + \delta$. [Beneath the source words сопр. эл.: вида. / велич. [of the form / quant.; provisional]. At right: Sp.]



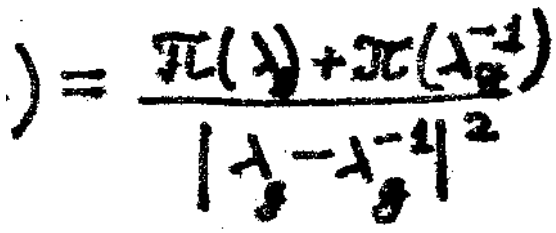
Uncertain reading. The word written beneath сопр. эл. is provisionally read as вида. or велич.

Original source details: GELFAND-P000033

GELFAND-P000034

What do we know? Char. of the gr. of real matr. with $\det = 1$. Complex —

$$\sigma(g) = \frac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda_g - \lambda_g^{-1}|^2}.$$



Uncertain reading. The lower stroke by the first λ in the numerator merges into the right parenthesis. The subscript g is provisional.

Original source details: GELFAND-P000034

[At right: Gelfand, Naimark, 1947.] $\pi(\lambda_g)$ — a mult. character of the field of complex numbers. Real.

$$\sigma(g) = \begin{cases} \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda - \lambda^{-1}|} & \text{if } \lambda \text{ is real.,} \\ 0 & \text{if } \lambda \text{ is complex.} \end{cases}$$

$$\sigma'(g) = \begin{cases} \dfrac{\pi(\lambda_{\min})}{|\lambda - \lambda^{-1}|} & \text{if } \lambda \text{ is real.,} \\ \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda - \lambda^{-1}|} - \dfrac{\pi(\lambda_g) - \pi(\lambda_g^{-1})}{\lambda_g - \lambda_g^{-1}} & \end{cases}$$

[No condition is written at right in the second row of the last brace. In the left margin, rotated:] Vanguten / Vantsuten, 1947.

Uncertain reading. The marginal surname resembles Вангутен / Ванцутен [Vanguten / Vantsuten]. Its joined middle strokes remain uncertain; the written year is 1947.

Original source details: GELFAND-P000034

[Cancelled; provisional reading: repr. [предст.; uncertain reading]]

Uncertain reading. The crossed-out source abbreviation resembles предст. [repr.]; a horizontal cancellation and two long strokes below it obscure parts of the letters.

Original source details: GELFAND-P000034

p-adic numbers.

$$\sigma(g) = \begin{cases} \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda_g - \lambda_g^{-1}|} & \text{if } \lambda_g \in K \text{ is real.}, \\ 0 & \lambda_g \overline{\in} K. \end{cases}$$

[Nonmembership is written in the source with a bar over $\in$. At right, beside a long vertical stroke:] Gelfand, Graev, 1963.

Finite field.

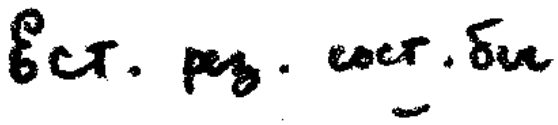
$$\sigma(g) = \begin{cases} \pi(\lambda_g) + \pi(\lambda_g^{-1}) & \lambda \in K, \\ 0 & \lambda \overline{\in} K, \end{cases}$$

[At right: Frobenius, 1895.]

$$\sigma'(g) = \begin{cases} 0 & \lambda \in K, \\ -\pi(\lambda_g) - \pi(\lambda_g^{-1}) & \lambda \overline{\in} K. \end{cases}$$

GELFAND-P000035

Nat. res. [рез.; provisional] [сост.; abbreviated] was that one must pass to the group of adeles.



Uncertain reading. The first and third source abbreviations read Ест. and сост. The middle is provisionally read as рез. [res.]; the middle strokes are joined.

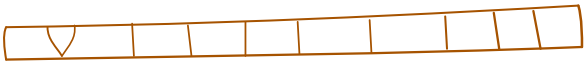
Original source details: GELFAND-P000035

Gr. of adeles — the direct $\sum$ of the gr. of all p-adic numbers. In it there are rational numbers — the diagonal.

Vishik.

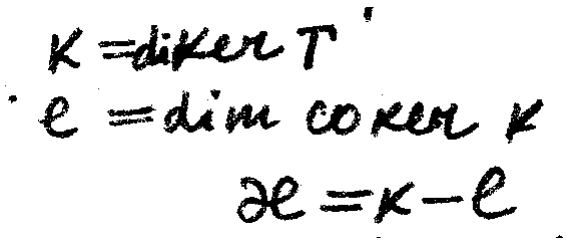
Diagram and typed description. Beneath the surname is a narrow strip with irregular long edges and transverse strokes. Two strokes near the left meet at the lower edge; the following dividers have unequal spacing. No labels are written inside.

Original source details: GELFAND-P000035



What one should work on in diff. eqns. For a lin. oper. with finite kernel and cokernel an index is introduced.

$$k = \dim \ker T, \qquad \ell = \dim \operatorname{coker} K, \qquad æ = k - \ell.$$



Original source details: GELFAND-P000035

Uncertain reading. The letter after coker resembles K. The looped index sign is represented by the provisional candidate æ; its exact letter has not been established.

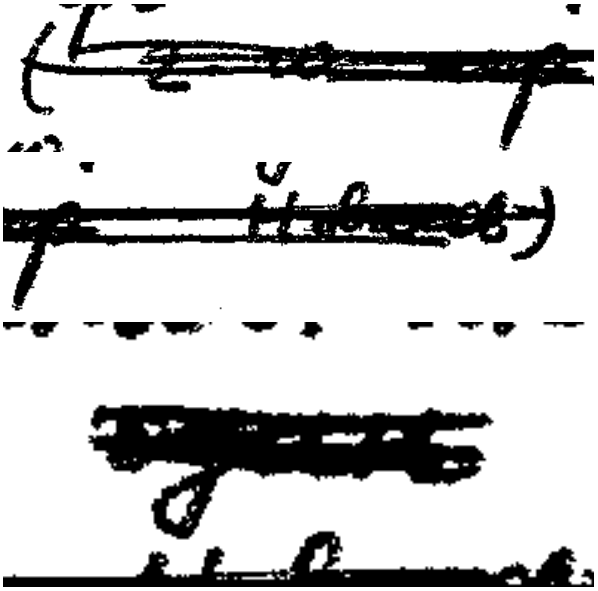
1) If replaced by $\|T - T_1\| < \varepsilon$, then æ does not change. 2) Adding a compl. cont. [вп. непр.; abbreviated reading] oper. does not change the index.

Original source details: GELFAND-P000035

Uncertain reading. The source writes Доб. вп. непр. опер. [Adding a compl. cont. oper.]. The joined abbreviations are shown in their written form; the full words are not written out.

æ allows classification of elliptic eqns. on manifolds. (Atiyah and Singer) Homotopy classification. Vishik considers that the index is the only homotopy inv.

A syst. with const. coeffs. in a ball has [a short entry is cancelled] as one more homot. inv. [Beneath this is a long cancelled parenthetical entry.]



Original source details: GELFAND-P000035

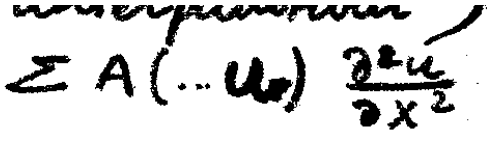
Uncertain reading. A short entry after имеет [has] is cancelled; its descending loop remains visible. Beneath it a long parenthetical entry is crossed out with two horizontal lines; its ending resembles инв. [inv.]. The text remains unreadable.

Can any one be transformed into any other (a large number of eqns. — stable). The question has an analogous meaning for a different class of nonlinear eqns. What is æ? What are the kernel and cokernel? $\sim$ a finite kernel. Quasilin. eqn.

$$Tu = v.$$

[At right: u — a manifold. It is required to be finite dimensional.] Principal part of a nonlinear eqn. Cokernel $\sim$ [a short wavy line]. [At right: In each point, linearize the eqn., and everywhere it is the same.] Problem: study the index of a nonlinear elliptic eqn. (This is an integral question.) Quasilin.

$$\sum A(\dots u_0)\frac{\partial^2 u}{\partial x^2}.$$



Original source details: GELFAND-P000035

Uncertain reading. The lower sign on u in the argument of A resembles 0 / ω; the joined dark strokes do not establish which. The dots and absence of sum indices are retained.

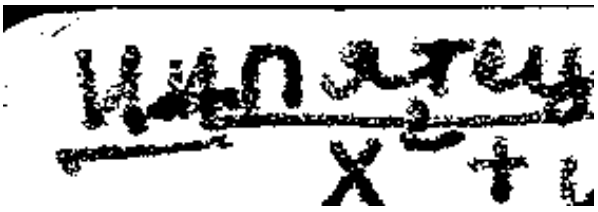
16th. Linnik — “How to use funct. anal. in probability theory”.

Original source details: GELFAND-P000035

Uncertain reading. The small marks after 16 are provisionally read as the Russian ordinal suffix ro. They could also be digits.

GELFAND-P000036

I. I. / I. M. Pyatetsky-Shapiro.



Original source details: GELFAND-P000036

Uncertain reading. I. I. / I. M. Pyatetsky-Shapiro. The second initial is merged; its exact reading is not established.

(Ilya Iosifovich). Linnik’s work.

$$x^2 + y^2 + z^2 = m.$$

[At left:] ϕ.

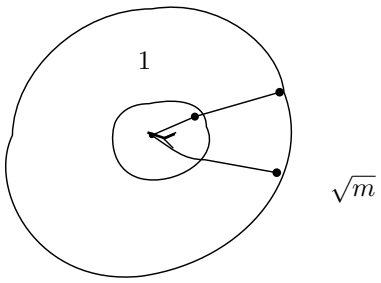
Original source details: GELFAND-P000036

Uncertain reading. A separate looped sign with a right vertical stroke in the left margin resembles ϕ. No meaning is written beside it.

For which m are there integral solutions? It is known that solutions do not always exist. (When $m = 8a + 7$, there are no solutions.) [Beneath $8a + 7$: mod 8.] How the solutions are distributed on the surface of the 3-dimensional sphere. Rad. $m^{1/2}$.

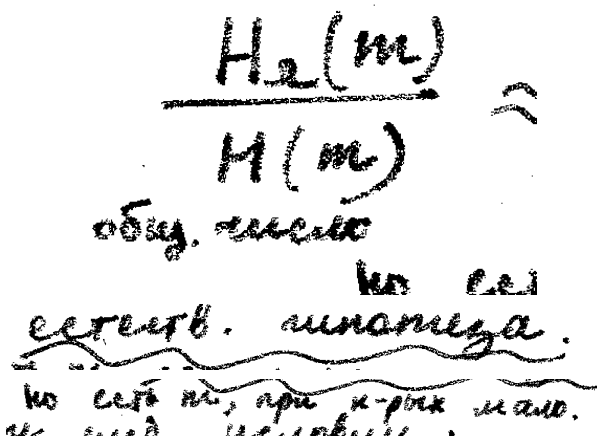
Original source details: GELFAND-P000036

Diagram and typed description. Two irregular rounded contours, labels 1 and $\sqrt{m}$, diverging lines, three heavy points and a thick inner junction with a small leftward arrow. The junction of the inner strokes is not wholly clear.



Select a region on the unit sphere and see how many points project into it.

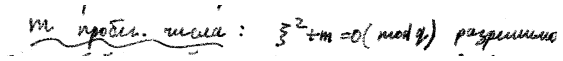
$$\frac{H_2(m)}{H(m)} \approx \text{mes}\,\Omega.$$



Original source details: GELFAND-P000036

Uncertain reading. The numerator subscript resembles 2; Ω remains an alternative. Beneath $H(m)$: total number. At right: nat. hypothesis; below: but there are m for which few. The last abbreviated entry is not wholly clear.

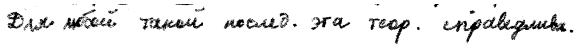
This holds under the following condition: if q is a prime m ranges over / пробл. [пробег. / пробл.; uncertain abbreviation] numbers: $\xi^2 + m = 0 \pmod q$ is solvable.



Original source details: GELFAND-P000036

Uncertain reading. Underlined: m пробег. / пробл. числа [m ranges over / пробл. numbers; uncertain abbreviated readings]. The middle and final strokes are joined; the second candidate is not expanded.

For any / small [любой / небол.; uncertain readings] such seq. this theorem holds.



Original source details: GELFAND-P000036

Uncertain reading. The joined word after For resembles любой [any]; it is followed by такой послед. [such seq.]. небол. [small; abbreviated reading] remains an alternative for the dark initial strokes.

Using q, construct a dyn. system on the sphere, whose behaviour obeys prob. laws: Fix q, consider [a short entry is cancelled] an integral point on the sphere and all orthog. matrices with entries of the form a/q.

Original source details: GELFAND-P000036

Uncertain reading. Before integral point, a short entry is cancelled with several dense strokes; the text has not been read.

(a is an integer.) There are finitely many of them, but many. They carry an integral point to a non-integral one. But among them there are finitely many carrying an integral point to an integral one. Construct such a flow (each point its own 48 orthog. matrices). e is a characteristic fn.

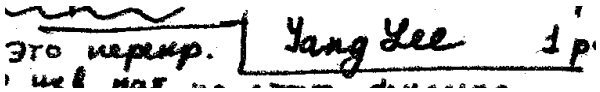
$$e(Tx) + e(T^2x).$$

[Rightward arrow: to the area.] This is a Markov chain.

GELFAND-P000037

Povzner. Naum Yakovlevich. Plan of stat. physics. Int. works.

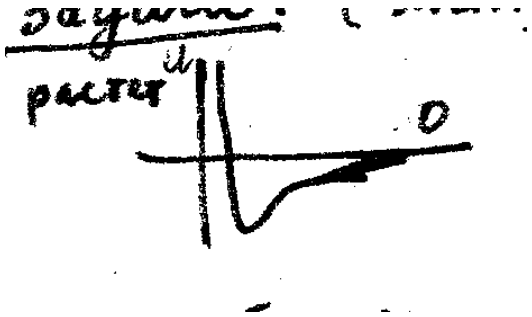
Ruelle	2–3 works
Yang Lee	1 work



Original source details: GELFAND-P000037

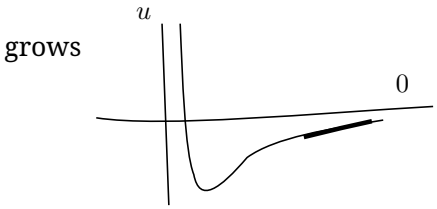
Diagram and typed description. The line steps down beside the list of works; its right segment runs beneath “Yang Lee”.

This is a revolution. Everything known mathematically about stat. physics. Kac — models. Problem. (Elem.)



Original source details: GELFAND-P000037

Diagram and typed description. A vertical with u, a horizontal with 0 at right; the curve falls steeply below the horizontal and then rises towards it at right. The right part is thickened. At left: grows. The precise junction of the thick strokes is not wholly clear.



$$\min \sum_{i,j=1}^{N} u(x_i - x_j).$$

Original source details: GELFAND-P000037

Diagram and typed description. The line beneath the sum steps upward toward the note at right.

x ranges in 2-dimensional space.

Set of x. There is a C such that the min. config. has everything in a sphere of rad. C. (Easy to show N^2, but in the case of a sphere N^3.) [Separate thick lines beneath N^2 and case.]

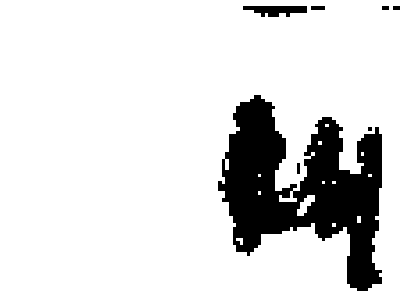
16/XI A lin. operator in Hilbert space has an inv. subspace. Solved!!! Bull. Amer. Math. Soc. Lui de Brange and ...

Original source details: GELFAND-P000037

Uncertain reading. The entry reads Lui de Brange and ...; the surname ending is looped, with a possible final s. Year: 1964.

1964. Connected with Livshits’s theory.

[Below at left:] ϕ / ψ.

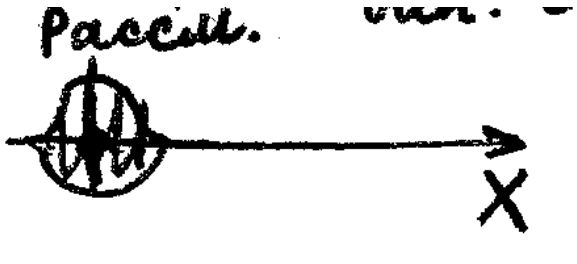


Uncertain reading. A separate sign in the left margin has two upper strokes and a right lower loop; graphic candidates are ϕ / ψ. There is no caption.

Original source details: GELFAND-P000037

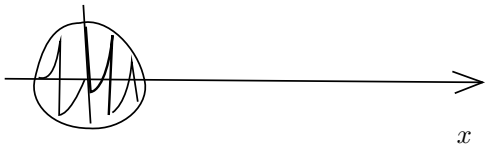
det by Schur. One must take the substitution group, and instead of all — repr. det of a + definite matrix is a + definite matrix.

Arnold. Consider a dyn. system and the corresponding quantum system.



Original source details: GELFAND-P000037

Diagram and typed description. A nearly round contour around the intersection of a vertical and a horizontal rightward arrow labelled x. Inside are irregular vertical and wavy strokes; some intersections merge.

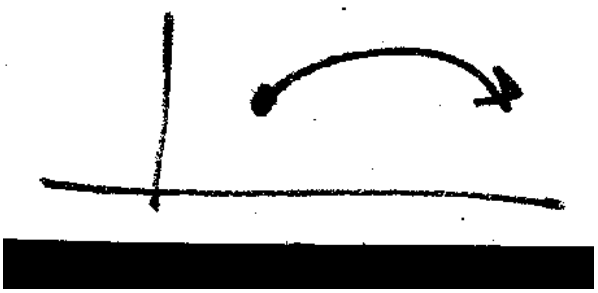


$$\psi(x)e^{\frac{i}{h}S(x)}.$$

Quantum particle.

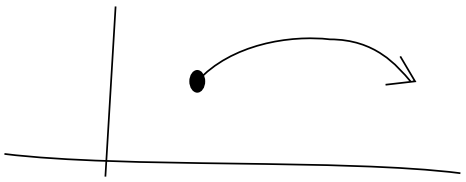
$$\psi \in \mathcal{D}(\mathbf{R}^n), \qquad S(x) = \int p\,dx.$$

h is Planck's const. Seek the solution in the same form. A semiclass. asymptotic is obtained.

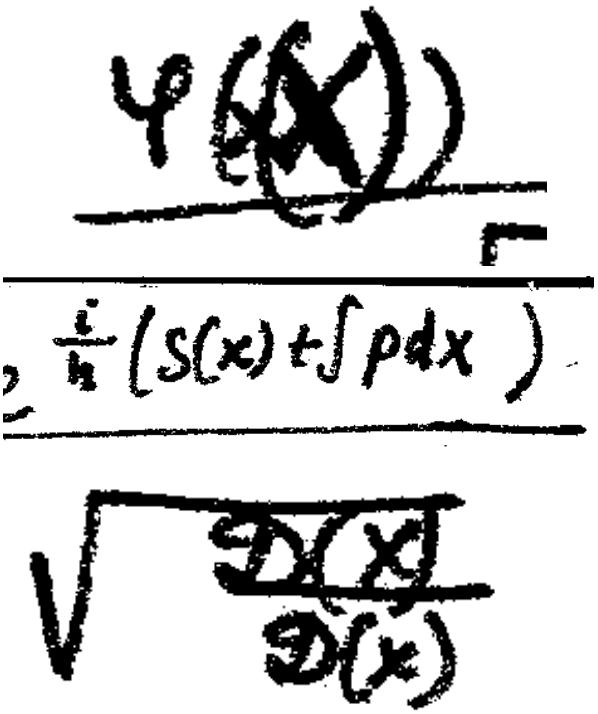


Original source details: GELFAND-P000037

Diagram and typed description. Unlabelled axes. A heavy point lies at right above the horizontal; an arch goes up and right from it, then down to a right arrowhead. No continuation beyond the frame is shown.



$$\frac{\psi(\xi(x))e^{\frac{i}{h}(S(x)+\int p\,dx)}}{\sqrt{\frac{D(X)}{D(x)}}}.$$



Uncertain reading. The argument of ψ is overwritten: $\xi(x)$ / x. The sign after $S(x)$ resembles $+$ / t. Under the radical the upper argument letter resembles X / x; no separate index is visible.

Original source details: GELFAND-P000038

[At right: norm. factor.] Holds only as $h \longrightarrow 0$, but when $D \neq 0$. (Through t, map. D is the det of the new pts. in terms of the old ones.) The solution of the question reduces to top. inv. of a manifold. (At a turning point.) Index. [At right: During mov. one obtains a cohomology class.]

Linnik. Ideals of rings of fns. from the point of view of appl. to statistics. A. Cartan — on ideals of rings of holom. fns. (47 and 50). Analyt. sheaves appear (first with Oka). Appl. of the apparatus of analyt. sheaves. What is statistics?

$$(X, A, \mu) \xrightarrow{T} (Y, B).$$

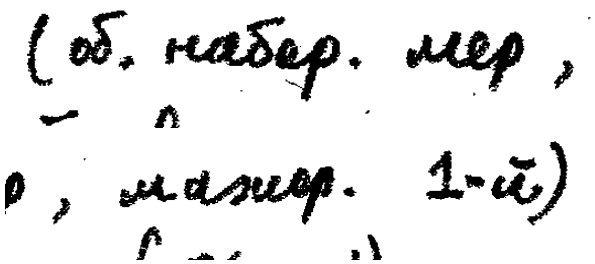
[Beneath the first parentheses: space with measure. Beneath the second: space with a σ-system.] T is a measurable mapping (observation fn.). X is what we observe. The measure on Y is directly induced [a short entry is cancelled] from X.

Uncertain reading. Between directly induced and from X, a short entry is heavily crossed out; the exact text is not established.

Original source details: GELFAND-P000038

X is Euclidean space. A — Borel sets.

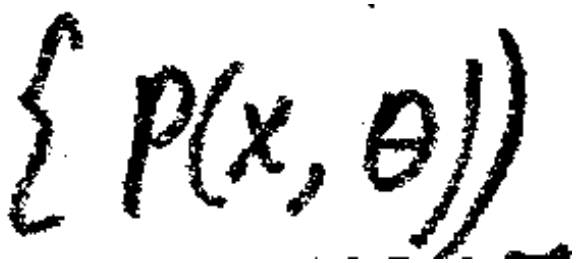
μ — measure. (об. [abbrev.] set of measures, major. [мажор.; provisional] 1st).



Uncertain reading. In parentheses: об. набор. мер, мажор. / массы 1-й / 1-ой [об.; set of measures; major. / mass; 1st, uncertain abbreviated readings]. The joined strokes before 1-й are closer to мажор.; the exact abbreviation and ending remain uncertain.

Original source details: GELFAND-P000038

There is a family of measures $\{P(x,\theta)\}$.



Manuscript notation. After $P(x,\theta)$ a second round parenthesis is written.

Original source details: GELFAND-P000038

[Beneath $P(x,\theta)$: given by a density.] We want to obtain info. about the parameter by observing X.

$$x_1, \ldots, x_n \in N(a,1).$$

[Beneath $N(a,1)$: Gaussian family. At right: 1 is precision, a is the parameter.]

$$x_i = a + \Delta_i, \qquad \overline{X} \text{ arithm. mean}$$

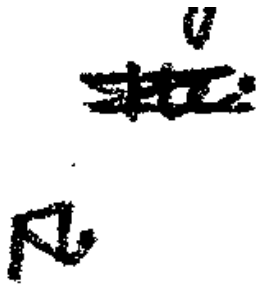
Δ_i — normal.

GELFAND-P000039

Stat. — finding estimates. Testing hypotheses $a = a_0$. $\theta \in \Omega$. (Ω — a comp. region in $\mathbf{R}^n$.) Null hypothesis $H_0 = \theta \in \omega$ (subspace). Testing with a fixed sample with randomization. A selected test is a fn. of the observations

$$0 \leq \varphi(x_1, \ldots, x_n) \leq 1.$$

[Beneath the formula: prob. meaning.] Observe with a random fn. and randomize with probabilities φ, $1 - \varphi$. [Cancellation at left and a separate sign below.]

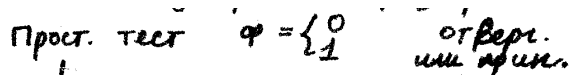


Uncertain reading. A short entry is heavily cancelled. Beneath it is a separate sign resembling π / Π; no meaning is written.

Original source details: GELFAND-P000039

Simple test

$$\varphi = \begin{cases} 0 & \text{reject} \\ 1 & \text{or accept} \end{cases}$$



Original source details: GELFAND-P000039

Manuscript notation. Two short entries stand at right of 0 and 1. There are no visible lines connecting the entries to the values.

[Downward arrow to the example:]

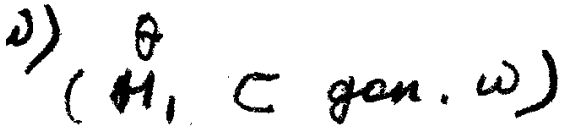
$$|\overline{x} - a_0| > \varepsilon \qquad \text{reject or accept}$$

Test quality is assessed as follows.

$$E(\varphi \mid H_0).$$

Expectation under hypothesis H_0 (param. $\in \omega$). [At right: meaningful only if 1 point.] H_1 — alternative.

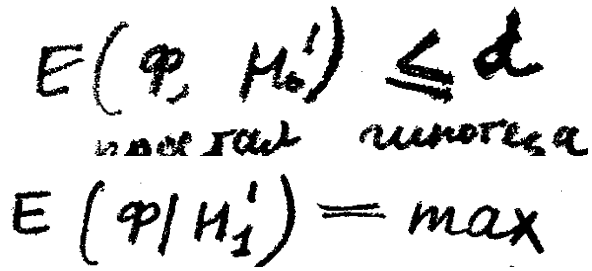
$(\overset{\theta}{H_1} \subset$ compl. $\omega)$.



Original source details: GELFAND-P000039

Uncertain reading. A separate θ stands over H_1. The following C-like sign may be $\subset$; it is followed by compl. ω.

$$E(\varphi, H_a') \leq \alpha, \qquad E(\varphi \mid H_1') = \max.$$

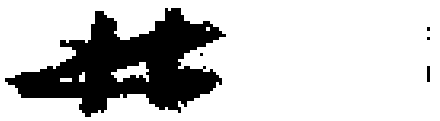


Original source details: GELFAND-P000039

Uncertain reading. At left the lower sign at H resembles a / 0, and the upper stroke resembles / / 1. There is a comma between φ and H. At right the upper stroke at H_1 is separately visible.

[Beneath the left expectation: simple hypothesis (fixed point). Beneath the right: (desirable).] Reject the true one not too often, but also reject [a short entry is cancelled] the false one often.

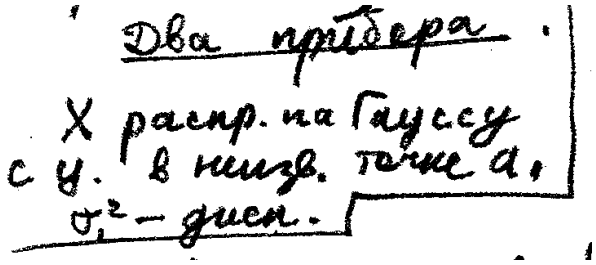
Uncertain reading. Before false, a tiny sign is heavily crossed out; the text beneath the cancellation has not been read.



Original source details: GELFAND-P000039

Similar tests. (At each point of the null hypothesis the prob. is α.)

Two examples.



Original source details: GELFAND-P000039

Diagram and typed description. A stepped lower line and a tall right edge surround the entry about the Gaussian sample.

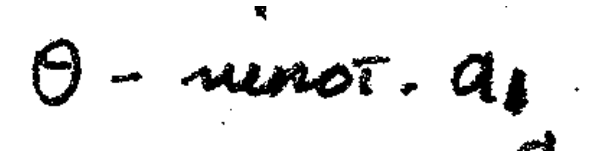
X is Gaussian distributed with centre at an unknown point a_1.
σ_1^2 is the variance.

Testing by means of inv. tests.

Original source details: GELFAND-P000039

Uncertain reading. In the continuation: null, below it a short cancellation and гип. / ил. The joined strokes after the cancellation do not give an exact reading.

θ — mult. [мног.; provisional] a_i.



Original source details: GELFAND-P000039

Uncertain reading. θ — мног. / мнот. a_i. The final stroke of the abbreviation resembles г / т; the reading is not established.

Question. $a_1 = a_2$?

$$N(a_1, \sigma_1^2), \qquad x_1, x_2, \ldots, x_n,$$
$$N(a_2, \sigma_2^2), \qquad y_1, \ldots, y_n.$$

[At right: σ_i^2 are unknown param.]

$$H_0: \quad a_1 = a_2,$$

$$E(\varphi \mid H_0') = \alpha, \qquad E(\varphi \mid H_1) = \max.$$

φ — ?

GELFAND-P000040

φ is inv. or similar if for every θ the expectation of φ is the same. (For every θ there is a prob. distrib.)

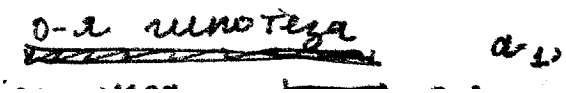
$$E(\varphi \mid H_0) = \alpha, \qquad E(\varphi \mid H_1') = \max.$$

[At right: [mark]].

Uncertain reading. There is a separate dark mark to the right of the expectations. Its reading is uncertain.

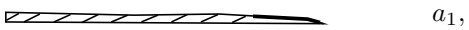
Original source details: GELFAND-P000040

H_1' is one point in the complement. Bayesian measure is a measure on parameter space. (Given beforehand — a priori distribution.) [At left a rightward arrow to the lines:] The measure depends on a 4-tuple of points $(a_1, a_2, \sigma_1, \sigma_2)$. Null hypothesis.



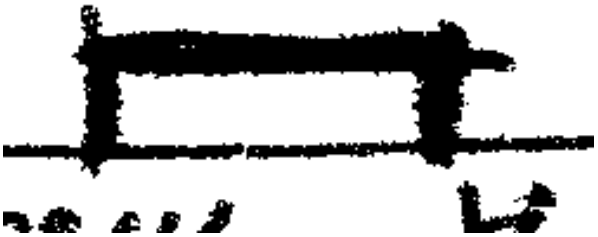
Original source details: GELFAND-P000040

Diagram and typed description. Beneath null hypothesis is an irregular narrow strip with diagonal strokes; a_1 and a comma stand separately at right. The right strokes merge with the upper edge.



$a_1,$

Bayes measure.

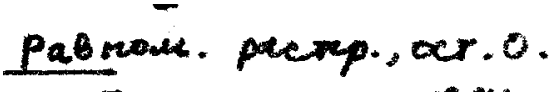


Original source details: GELFAND-P000040

Diagram and typed description. Beside Bayes measure is an empty rectangle: its thick top extends rightwards, and the lower line continues beyond both vertical sides. There is no label inside.

[At right: Unif. distrib., from [от.; provisional] 0.]

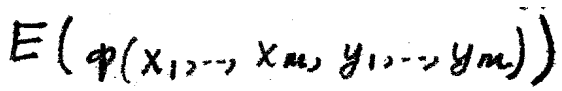
Uncertain reading. The short entry before 0 is provisionally read as от. [from].



Original source details: GELFAND-P000040

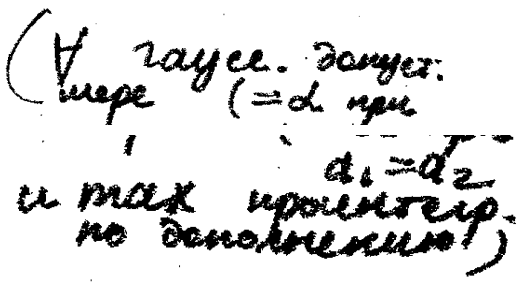
One must be able to solve for every Bayesian distrib.-measure. Result: A prob. distrib. in $X \times Y$ with a given Gaussian measure (4-param.). Find $\varphi(x, y)$ such that

$$E\big(\varphi(x_1,\dots,x_n,y_1,\dots,y_m)\big).$$



Original source details: GELFAND-P000040

Uncertain reading. The final subscript on x resembles n or m and has two joined strokes. The subscript m on y is retained.



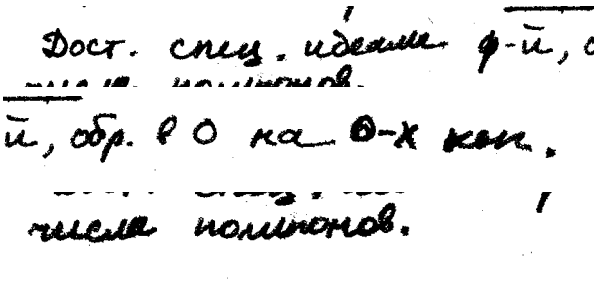
Original source details: GELFAND-P000040

Uncertain reading. At right: for every admiss. Gaussian measure ($=\alpha$ when $a_1=a_2$ and max integr. / param. [проинтегр. / параметр.; uncertain abbreviated readings] on the complement). The word after max is graphically closer to проинтегр.; neither candidate is asserted as certain.

Sufficient statistic — contains all info. about the parameters.

Problem not done. θ_1, θ_2 are complex variables. $f(\theta_1,\theta_2)$ in a ball, ideals are not known (All prim. ideals corresponding to the ideal equal to 0.) Glisson [Глиссон] — ball and centre holom.

Suff. spec. ideals of fns., van. at 0 on a θ-th fin. number of polynomials.



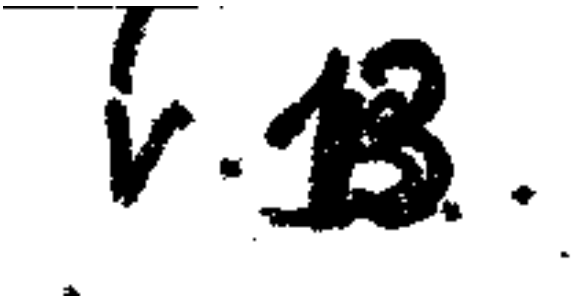
Original source details: GELFAND-P000040

Uncertain reading. The first sign after the source word на is provisionally read as θ; it may be 0.

GELFAND-P000041

Gleason — Gorin.

J. Math. Mech. v. 13. N 1 (1964).



Original source details: GELFAND-P000041

Uncertain reading. The number 13 is written over earlier strokes. The underlying numeral is uncertain.

Consider a ball $z=(z_1,z_2)$,

$$|z_1|^2+|z_2|^2\le 1$$

and fns., an. inside and cont. on the bdry., with $f(0,0)=0$. Do there exist g_1,g_2 of the same type such that

$$f=z_1g_1+z_2g_2$$

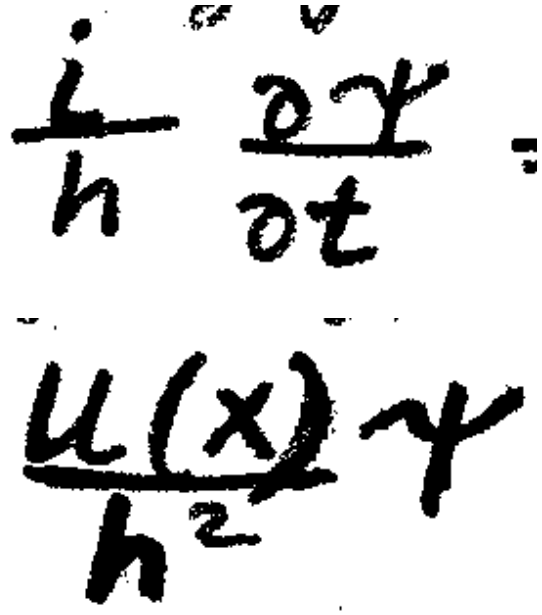
(max ideal; will z_1,z_2 be a basis? This can be done inside, but it is unknown whether it can be done on the boundary? For one var. it is trivial. For a bicylinder it is done (since the

Taylor series conv. on the bicylinder and one can take ...). Without bdry. — then the ball can be replaced by any domain of holomorphy.

23/XI. Arnold. "On Maslov's work".

Consider, in partic., the Cauchy prob. for the Schrödinger eqn. Config. space.

$$\frac{i}{h}\frac{\partial\Psi}{\partial t} = \Delta\Psi - \frac{U(x)}{h^2}\Psi$$



Uncertain reading. The factor before the derivative is written i/h; h^2 stands beneath $U(x)$ at right.

Original source details: GELFAND-P000041

$U(x)$ — potent. energy.

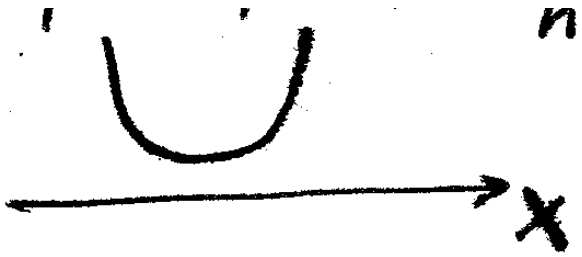
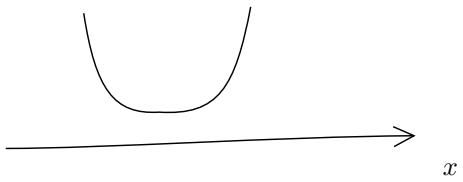


Diagram and typed description. An open-topped U-shaped curve lies above an almost horizontal line with a rightward tip labelled x; its bottom does not touch the line. No vertical axis is drawn.

Original source details: GELFAND-P000041

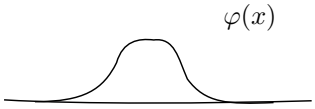


$$\Psi|_{t=0} = \varphi(x)e^{\frac{i}{h}S(x)}$$

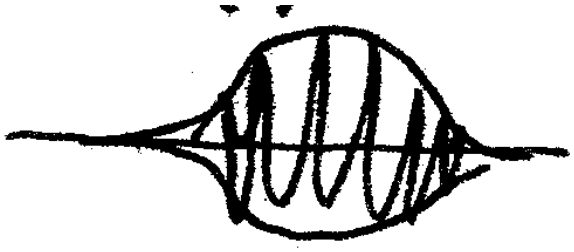
[Beneath $\varphi(x)$: fin.]

Diagram and typed description. A small rounded rise with steep sides stands over a horizontal stroke; $\varphi(x)$ is written above at right. Separate arrowheads cannot be distinguished.

Original source details: GELFAND-P000041

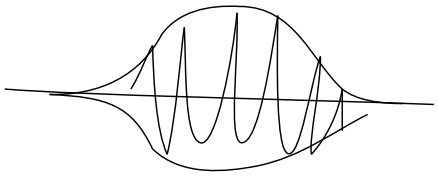


$S(x)$ can be $S(x) = px$.



Original source details: GELFAND-P000041

Diagram and typed description. Wavy and almost vertical strokes lie between rounded upper and lower contours, crossed by an almost horizontal stroke. Some individual connections at the ends and thickened parts cannot be distinguished securely.



Oscill. (rapid), [пропод.] φ.

Uncertain reading. After “Oscill. (rapid)” the manuscript writes пропод. Its expansion is uncertain. It is followed by φ.

Original source details: GELFAND-P000041

GELFAND-P000042

Seek the asympt. as $h \to 0$ in the form

$$\Psi = e^{\frac{i}{h}a(x,t)}.$$

a — a complex fn. Substitute and obtain a seq. of asympt. formulas for the solution of the prob. The first 2 terms have the form

$$\Psi(x,t) = \frac{\varphi(X(x,t)}{\sqrt{\frac{\partial x(X,t)}{\partial X}}} e^{\frac{i}{h}(S(X(x,t))+\int p\,dx)+O(h)}$$

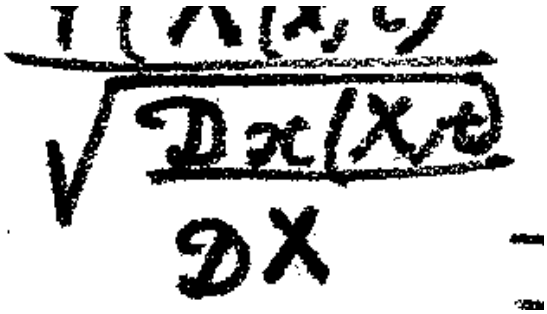
Manuscript notation. There is one closing parenthesis after t in the numerator.

Original source details: GELFAND-P000042

Uncertain reading. The entry $+O(h)$ is on the raised line after the phase parenthesis; there is no separate lower line for it.

Original source details: GELFAND-P000042

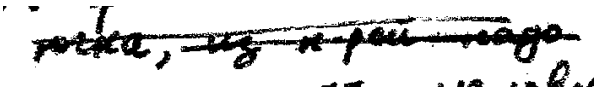
Uncertain reading. The first rounded sign in the numerator beneath the radical resembles D / ∂; the denominator shows ∂X. The upper argument is written (X,t).



Original source details: GELFAND-P000042

[From the denominator: — from unit.; see Titchmarsh and Landau.] φ — given init. amplitude. X at $t = 0 = x$.

$\underset{(x,t)}{X}$ — [cancelled: a point from which one must].



Original source details: GELFAND-P000042

Uncertain reading. Beneath the cancellation: точка, из к-рой надо / наше (a point from which one must / ours); the final letters are crossed by the stroke.

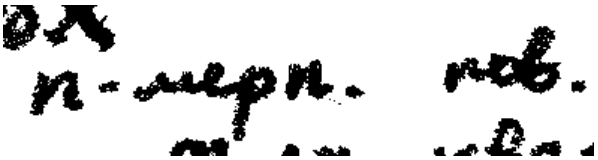
Classical init. condition.

$$p = \dot{x}, \qquad \dot{x} = \frac{\partial S}{\partial x}$$

Original source details: GELFAND-P000042

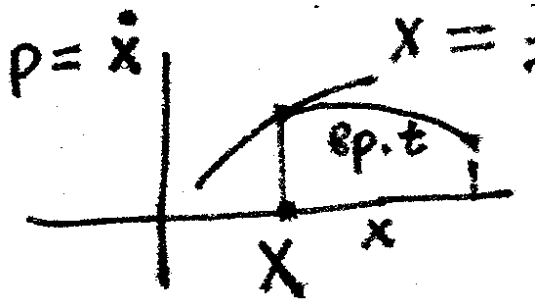
Uncertain reading. A dot is visible over each x / X. The lower sign after ∂S resembles X / x; its case is not secure. No separate subscript is visible on the first sign.

S is given at the init. instant. n-dim. пв. / пб.



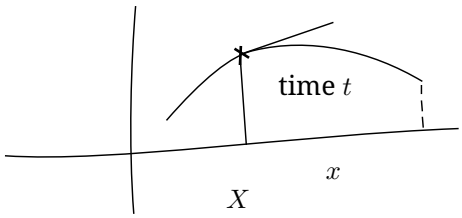
Original source details: GELFAND-P000042

Uncertain reading. After n-dim. is a short joined abbreviation resembling пв. / пб.; it is not expanded.



Original source details: GELFAND-P000042

Diagram and typed description. An almost horizontal line crosses a vertical one; an arch lies above them, with two vertical strokes from it. X is below the left stroke; x stands between the strokes. Inside: time t. An inclined stroke rises rightwards from the left junction, with a thickened start; its arrow direction is not established.



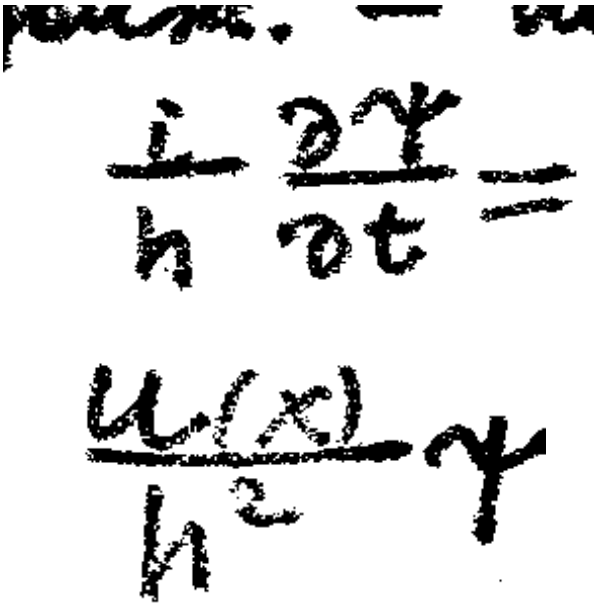
The formula is quasiclassical, valid when the Jacob. $\neq 0$. Quasicl. — when this formula holds.

$$\ddot{x} = -\frac{\partial U}{\partial x}$$

classical mechanics (x — vector). In quantum mech., $\forall t$ there is a wave fn. Ψ. $|\Psi|^2$ — prob.

Eqn. of motion — Schröd.

$$\frac{i}{h}\frac{\partial \Psi}{\partial t} = \Delta\Psi - \frac{U(x)}{h^2}\Psi$$



Uncertain reading. The repeated equation has i/h at left and $U(x)/h^2$ at right; the two fractions are separately visible.

Original source details: GELFAND-P000042

(if there is no i, then it is the heat eqn.; if initially at a point, it later spreads out). Connected with the asympt. of eigenvalues. $\Psi|_{t=0}$ is given. $\Psi(x)$

Diagram and typed description. A long almost horizontal stroke with a thickened right end follows $\Psi(x)$. There is no caption over it or secure arrowhead.

Original source details: GELFAND-P000042

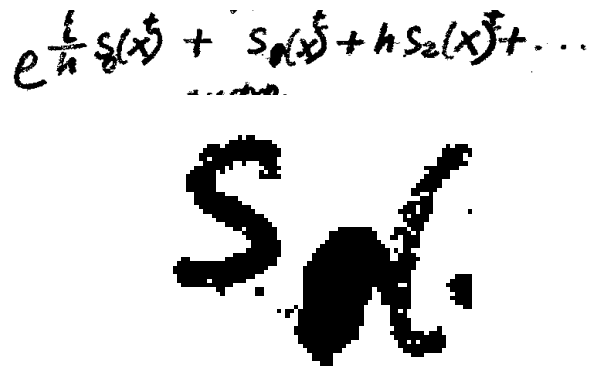
GELFAND-P000043

$S(x) = px$, then the de Broglie formula.

$$\frac{\partial S}{\partial x} = p$$

Usual method of finding the asympt. — seek the soln. in the form

$$e^{\frac{i}{h}S_0(x^t)+S_1(x^t)+hS_2(x^t)+\cdots}$$



Uncertain reading. A raised sign with a crossbar resembling t stands over x in all three arguments; readings x^t / $\dot{x}$. The subscript of the second S is blotted: S_1 / S_0.

Original source details: GELFAND-P000043

With respect to time [inserted above: diff.] $\frac{1}{h^2}$ ...The leading term must be sought as $e^{\frac{i}{h}S_0(x)}$. If S is real, then oscill.; if complex, then there is decay. Obtain a 1st-order eqn., and the 1st fn. is real, and obtain oscill. Must take the amplit., hence 2 terms. Now must choose init. data for this asympt.

Ω / δ is not allowed, since there are many extraneous frequencies (strongly localized).

Uncertain reading. Ω or δ: a loop with a horizontal lower stroke and a long left tail.

Original source details: GELFAND-P000043

$$\varphi(x,t)e^{\frac{i}{h}S(x,t)}$$

if substituted there, then for S a 1st-order eqn.

$$\frac{\partial S}{\partial t} = H\left(x, \frac{\partial S}{\partial x}\right), \qquad H = U(x) + \frac{\dot{x}^2}{2}$$

to which an ordinary eqn. corresponds. φ — density of the dyn. syst. (how they move according to class. mech.). Must take x and find X for it, so that our eqn. comes from X to x after t. The points carry φ with them.

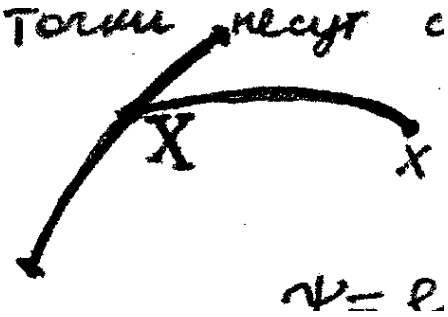
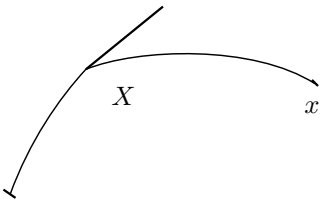


Diagram and typed description. A bending curve has X below the left junction and x at its right end. A thick inclined stroke passes through the left junction. Its arrowhead or direction is not established.



Original source details: GELFAND-P000043

$\dot{x}$ is given from x (the dyn. syst. is given). For small t, single-valued.

$$\Psi = e^{iH}$$

unit. $\mathcal{D}$ — the Jacob. is added for unitarity. e — the phase factor rotates with varying amplitude. Maslov's work, when t is arbitrary rather than small.

$t = 0$, Jacob. [an entry is cancelled] $= 1$ (identity transf.).

Uncertain reading. A heavily cancelled expression after Jacob. and before $= 1$; the individual signs are not established.

Original source details: GELFAND-P000043

The wave fn. of the oscillator. Pendulum.

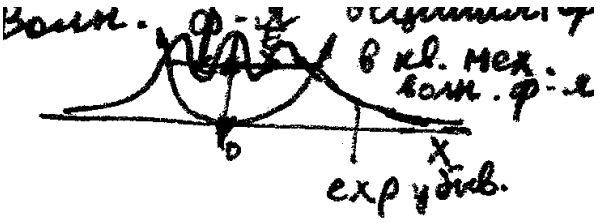
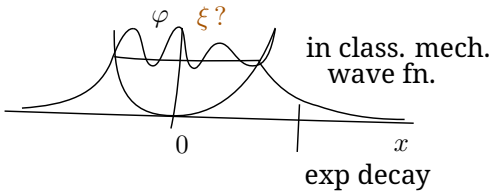


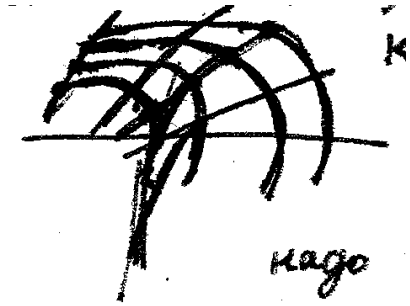
Diagram and typed description. A parabolic arch, oscillating contour, almost horizontal stroke and central stroke intersect. Visible labels: φ, 0, x; a small upper sign resembles ξ, but its lower part joins the curve. Captions: in class. mech. wave fn.; exp decay. Exact connections of dense strokes and the number of oscillations are not established.



Original source details: GELFAND-P000043

GELFAND-P000044

If t is large, it may happen that the curve of init. data (n-dim. пв.) moves.



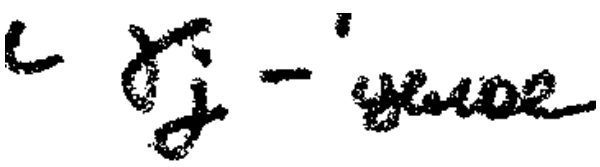
Original source details: GELFAND-P000044

Diagram and typed description. Upper-left drawing: several bending arches cross inclined strokes, an almost horizontal line and a dense almost vertical bundle. Small tips appear at the upper-left end and inside the bundle; some individual connections overlap.

If there is a vertical tangent, then $\mathcal{D} = 0$, but afterwards there are 2 points and the formula is retained [(!) / (?)], but must add terms over the points γ_j / n_j — an integer.

Uncertain reading. After retained, a small parenthesized sign with a separate lower dot: (!) / (?). Its upper part is blotted.

Original source details: GELFAND-P000044



Uncertain reading. γ_j or n_j: the upper sign has a loop and two raised strokes; j lies below. The following entries resemble n_j.

Original source details: GELFAND-P000044

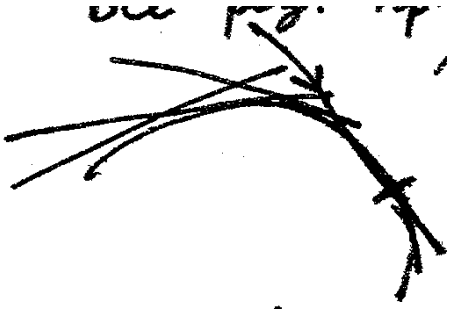
$$\int p\,dx) + i\,\frac{\pi}{2} n_j$$

[A sign between i and $\pi/2$ is cancelled.]

Original source details: GELFAND-P000044

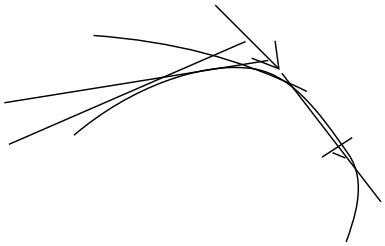
Uncertain reading. A sign between i and $\pi/2$ is heavily cancelled; the readings $+$ / $-$ are not securely established.

Maslov's result — the n_j have been found. $n_j = \pm 1$ from general considerations. Go around the branch point in the complex plane; this can determine $\oplus$ or $\ominus$. All results also apply in optics.



Original source details: GELFAND-P000044

Diagram and typed description. Several inclined rays cross a bending envelope. A thick arrow points downwards to the right at the upper tangency. Short crossing strokes lie on the lower-right branch; their connection to the envelope is uncertain.



The rays envelop a curve. We shall follow the phase along a ray. For small λ — a phase jump of $\pi/2$ at the point of tangency. Answer n_j — the Morse index.

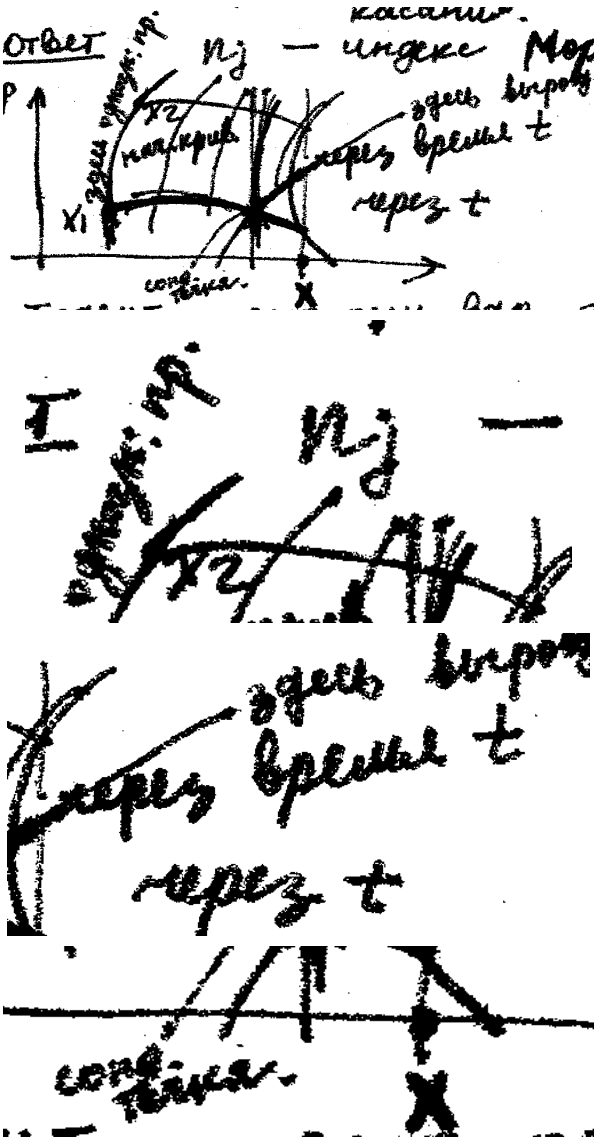
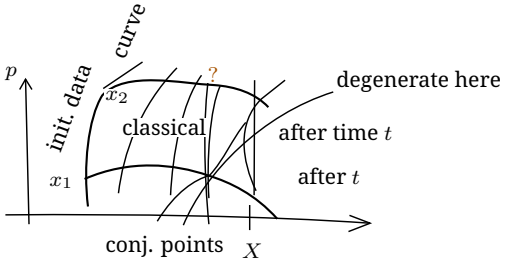


Diagram and typed description. The p axis points upwards and the horizontal axis rightwards. A family of bending and almost vertical strokes; x_1, x_2, X. Captions: curve of init. data; classical; degenerate here; after time t; after t; conj. points. A small dense sign is above the central vertical stroke; its reading and some individual intersections are not established.



Original source details: GELFAND-P000044

A trajectory is a soln. of a variational problem by the principle of least action.

$$\delta u = 0$$

$$\delta^2 u$$

a quadratic form. It has an index [a word is cancelled] (number of neg. squares).

Uncertain reading. A short word left of (number of neg. squares) is cancelled by dense strokes; its reading is not established.

Original source details: GELFAND-P000044

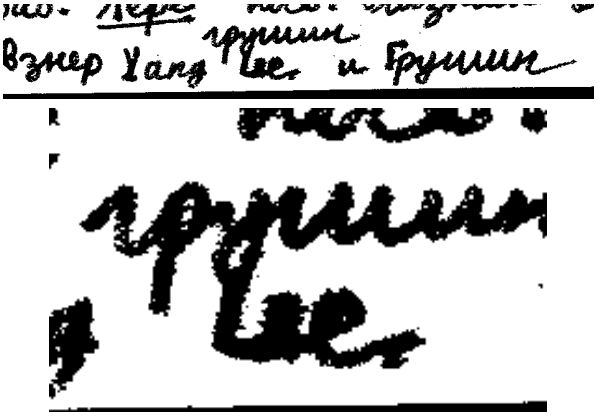
Or [a word is cancelled] the number of conjugate points.

Uncertain reading. A word above number of conjugate points is covered by several horizontal strokes; its reading is not established.

Original source details: GELFAND-P000044

The 1st part of Leray's work is devoted to related questions.

Next time Povzner Yang Lee and Brushlin [Брушлин].



Uncertain reading. The line reads "Yang Lee". Above "Lee" is a short Russian addition resembling группам [to groups]. The final surname is provisionally read as Брушлин [Brushlin].

Original source details: GELFAND-P000044

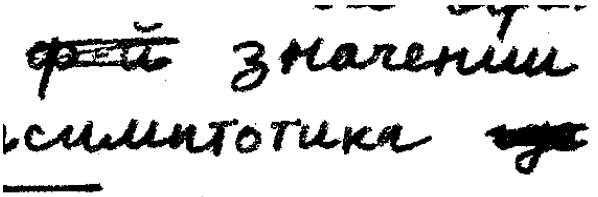
GELFAND-P000045

Berezin.

$$H = \frac{p^2}{2} + V(x)$$

(e.g. V grows). Let the surfaces $p^2 + V(x) =$ const all be bounded.

Then the asympt. of eigenvalues [an abbreviation is cancelled]:



Uncertain reading. The cancelled abbreviation before values resembles ф-й / ф-и (fns.); horizontal strokes cross the upper letters.

Original source details: GELFAND-P000045

the volume of the region where $H \leq n\pi h$ — the asymptotic [a word is cancelled] of the eigenvalues.

Uncertain reading. A short word between asymptotic and eigenvalues is covered by dense ink; its reading is not established.

Original source details: GELFAND-P000045

———— · ————

Gelfand I.

$$-y'' + p(x)y = \lambda y$$

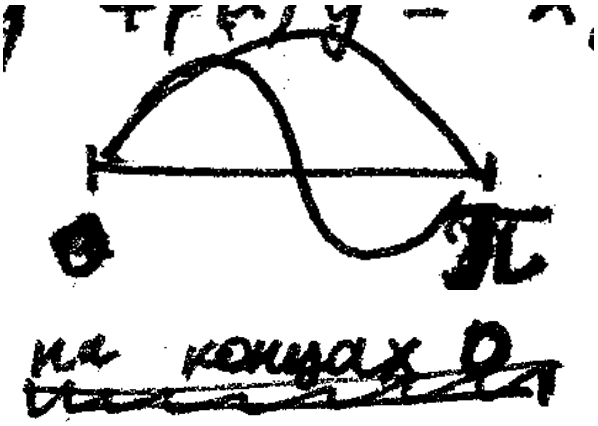
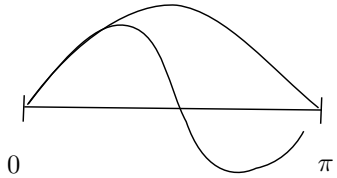


Diagram and typed description. A horizontal segment with short end strokes and labels 0, π. One arch lies above the segment; another curve begins above it, crosses it, and falls below. A dense junction lies near the right end; the nearby underline under zero at the endpoints is crossed out.



Original source details: GELFAND-P000045

Zero at the endpoints [the underline is crossed out].

$p = 0$, eigenfns., $\lambda = -n^2$.

Uncertain reading. In $\lambda = -n^2$, a separate horizontal stroke is visible between the equality sign and n; the minus is retained.

Original source details: GELFAND-P000045

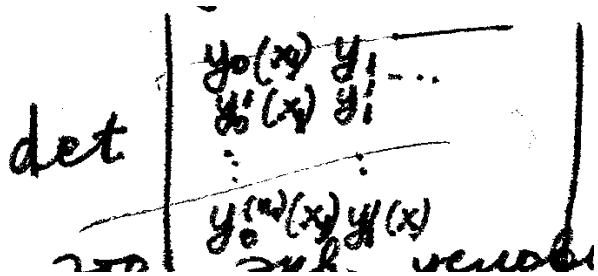
Claim. The 1st eigenfn. has no zeros (Morse index 1). The 2nd — ditto — has one zero. This holds for $\forall$ eqn. The zeros interlace. A more precise theorem. If $y_0, \dots, y_n$ are eigenfns., then

$$\sum_{i=1}^{n} a_i y_i(x)$$

will have $\leq n$ zeros!! (resembles polynomials).

[A cancelled determinant:]

$$\det \begin{vmatrix} y_0(x) & y_1 & \cdots \\ y_0'(x) & y_1' & \\ \vdots & \vdots & \\ y_0^{(n)}(x) & y'(x) & \end{vmatrix}$$

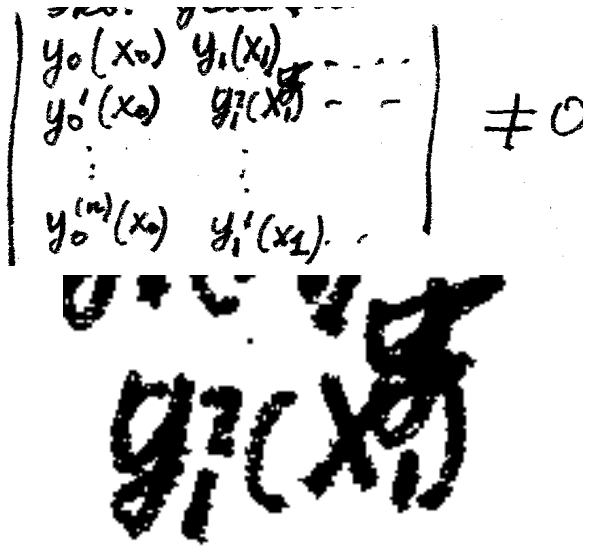


Uncertain reading. The lower second entry of the cancelled determinant resembles $y'(x)$ / $y_1^{(n)}(x)$; its subscript and derivative order are not established.

Original source details: GELFAND-P000045

This is equiv. to the condition

$$\begin{vmatrix} y_0(x_0) & y_1(x_1) & \cdots \\ y_0'(x_0) & y_1'(x_1) & \cdots \\ \vdots & \vdots & \\ y_0^{(n)}(x_0) & y_1'(x_1) & \end{vmatrix} \neq 0$$

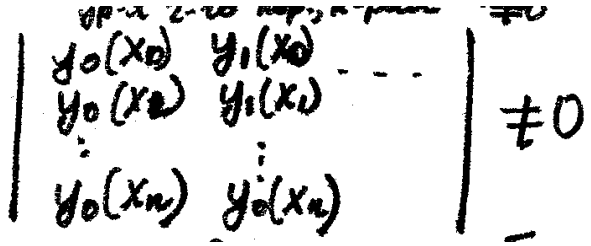


Uncertain reading. The upper sign of $y_1\prime(x_1)$ in the second entry of the second row is overwritten. A single raised stroke is visible in the lower second entry: $y_1\prime(x_1)$; a possible order (n) is not established.

Original source details: GELFAND-P000045

Or [Above the following determinant:] this is the 1st eigenfn. of the 2nd-order eqn., which is $\neq 0$.

$$\begin{vmatrix} y_0(x_0) & y_1(x_0) & \cdots \\ y_0(x_1) & y_1(x_1) & \cdots \\ \vdots & \vdots & \\ y_0(x_n) & y_0(x_n) & \end{vmatrix} \neq 0$$



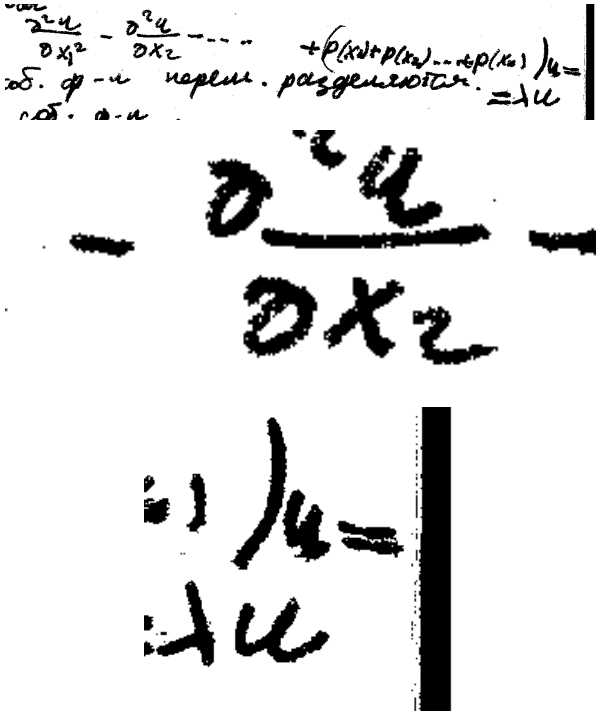
Uncertain reading. In the second entry of the bottom row the subscript is rounded, like y_0 in the first column. The dot above it aligns with the vertical ellipsis; a separate derivative stroke is uncertain.

Original source details: GELFAND-P000045

Hence a linear comb. has $\leq n$ zeros if $x_0 < x_1 < \cdots < x_n$.

Petrov. So that

$$\frac{\partial^2 u}{\partial x_1^2} - \frac{\partial^2 u}{\partial x_2} - \dots + (p(x_1) + p(x_2) \dots + p(x_n))\, u = \\ = \lambda u$$



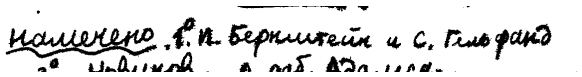
Uncertain reading. The first denominator is x_1^2; the second shows x_2 without a square. At the right edge: $u =$; below, another equality sign before λu. Both signs are retained.

Original source details: GELFAND-P000045

Eigenfns., the variables separate. Eigenfns. $u_1(x_1)u_2(x_2) \dots u_n(x_n)$. The 1st eigenfn. $\neq 0$.

GELFAND-P000046

Planned. [Number and initials unclear] Bernstein and S. Gilfand.

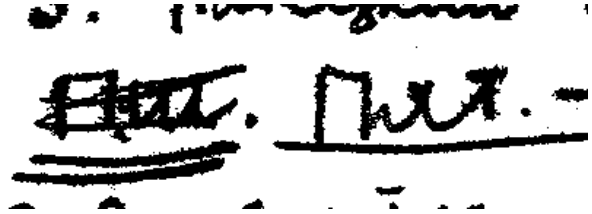


Uncertain reading. The first row begins with a long looped sign, then И / Н before Бернштейн. Readings: 1° I. Bernstein and S. Gilfand / F. N. Bernstein and S. Gilfand / 1° I. Bernstein and S. Gelfand. Full names are not established.

Original source details: GELFAND-P000046

[Separately at the right:] Dyson. 2°. Novikov, on the work of Adams. 3°. Dynkin. 4°. Vinberg. 5°. Pyatetskii-Shapiro.

[Cancelled.]

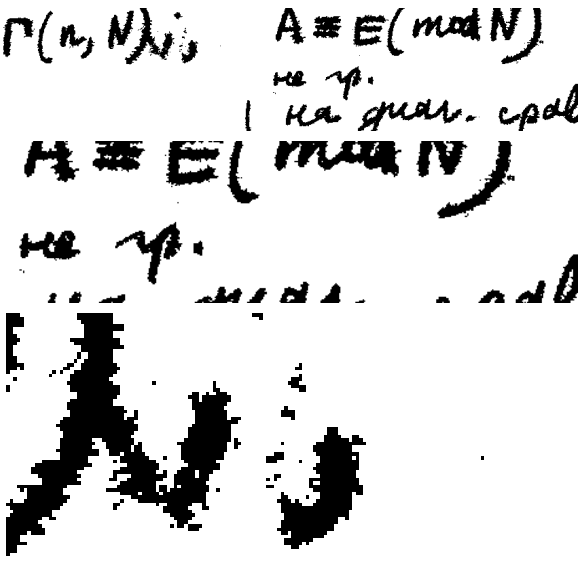


Uncertain reading. The short cancelled and doubly underlined heading before Pyat.-Shapiro resembles Пят. (Pyat.); dense strokes cover the earlier writing.

Original source details: GELFAND-P000046

Pyat.-Shapiro. Problem of Klein and Serre. Consider the gr. of integer matrices of order n, $\Gamma(n)$, with det $= 1$. [Cancelled: How] describe all subgrs. Γ' of finite index? Ex.

$$\Gamma(n, N), \qquad A \equiv E \pmod{N}$$



Uncertain reading. A small inscription below $\Gamma(n, N)$ resembles λij / i, j; below $A \equiv E$: probably не гр. (not gr.), possibly на гр. (on gr.). The letters and the attachment of the small inscription to the formula are not established.

Original source details: GELFAND-P000046

(on the diag., congruent to 1; off the diag., divisible by N).

GELFAND-P000047

Are these all the subgroups? Does Γ' always contain $\Gamma(n, N)$ (a congruence subgr.). For $n = 2$ this is false (Klein). $n \geq 3$, this is true (Serre). Connection with Lie groups. Consider the matrices

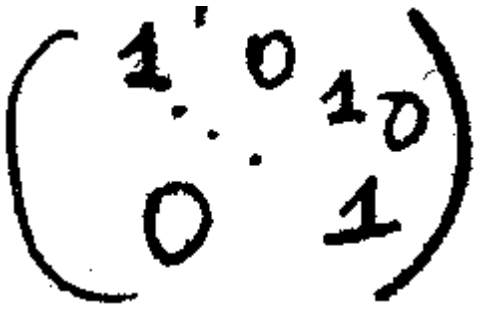


Diagram and typed description. Matrix schematic.

$$\begin{pmatrix} 1 & & 0 & & \\ & \ddots & & 1 & 0 \\ & & \ddots & & \\ 0 & & & 1 & \end{pmatrix}$$

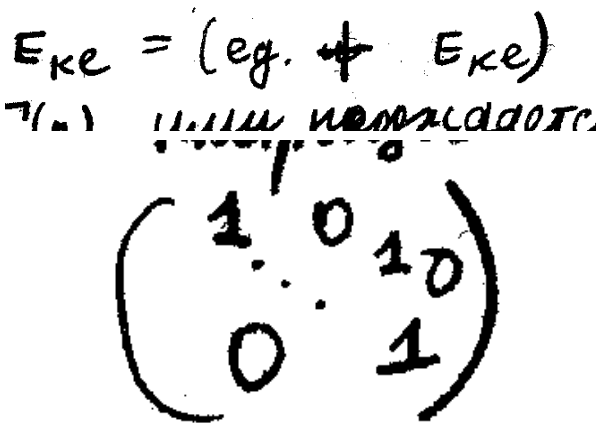
Original source details: GELFAND-P000047

[Cancelled symbol.]

Uncertain reading. A cancelled and overwritten sign before the elementary-matrix notation resembles E; the underlying strokes are not established.

Original source details: GELFAND-P000047

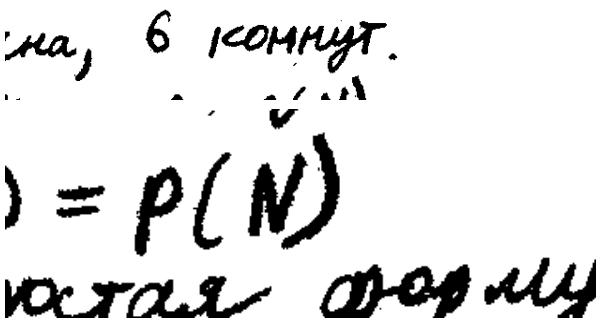
$E_{ke} = (\mathrm{id.} + E_{ke})$.



Uncertain reading. The same E-shaped sign occurs on both sides of the equality; its second subscript resembles e / ℓ. Readings $E_{ke} = (\mathrm{id.} + E_{ke})$ / $e_{k\ell} = (\mathrm{id.} + E_{k\ell})$. The schematic matrix retains the visible ones, zeros and dots; unwritten entries have not been supplied.

Original source details: GELFAND-P000047

The gr. $\Gamma(n)$ is generated by them. Take these matrices and their conjugates and raise them to the power N: E_{ke}^N, Γ^N. For $n = 2$ it has infinite index; $n \geq 3$, finite index. Γ/Γ^N. $\boxed{n = 2}$ $N = 2, 3, 4, 5$ is finite, 6 commut. / коннут. [коммут. / коннут.; uncertain abbreviation]

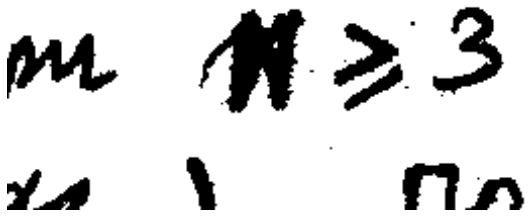


Uncertain reading. After 6, the abbreviation коммут. / коннут. (commut. / коннут.; uncertain reading). Its initial letter has separate vertical and angled strokes. Below: $p = p(N)$; the letter resembles p. The abbreviation is not expanded.

Original source details: GELFAND-P000047

Γ/Γ^N = fund. gr. of a surface, $p = p(N)$, a simple formula. Serre, Lazard, Bass (1964). Consider $a \equiv b\ (N)$, then for $n \geq 3$ / $N \geq 3$ there is a transf. $a \to b$ in Γ/Γ^N (elem. transf.).

Uncertain reading. After then for, the sign before ≥ 3 is heavily overwritten: $n \geq 3$ / $N \geq 3$. Vertical strokes resembling capital N are visible.



Original source details: GELFAND-P000047

Then cohomol. technique.

Gelfand. Tomorrow at the conf. on elem. particles, there will be a talk about

[Cancelled.]

Uncertain reading. A group designation before boxed $SU(3)$ is densely cancelled. Readings $SU(3)$ / $SU(2)$; the digit beneath the ink is not established.

Original source details: GELFAND-P000047

$\boxed{SU(3)}$. Known particles are combined into groups, transforming into one another (multiplets). Another repr. of $SU(3)$. The idea $SU(2) \underset{\text{almost}}{\approx} SO(3)$. [Above the line: fns., lin. comb.] Combine by means of $\boxed{SU(2) \times SU(3)}$. The hypothesis is known also for $SU(6)$. Remark — actually the Lorentz group (with par. translations) (the Poincaré group). [Cancelled: $SU(2)$] $\longrightarrow$ Poincaré gr. Work of Todorov. Kadyshevsky, Muradyan, Tavkhelidze. Instead of $SU(6)$ — a 108-param. group.

GELFAND-P000048

Salam proposed a 144-param. group. Fulton, Wess. A 2-step gr. with a comm. normal divisor.
$SU(3) \otimes SU(2)$ in some sense.

Uncertain reading. The abbreviation before смысле (sense) resembles как. or нек.

Original source details: GELFAND-P000048

Lorentz transf. — complex matrices.

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \det = 1$$

(spinor repr. of the Lorentz gr.). Poincaré gr.

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

Vector — Hermitian forms of order 2.

$$H = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$$

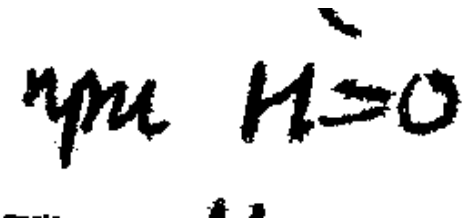
A vector in 4-dim. space; a, c real, b complex.

$$H \to AHA^*$$

Lorentz, det is preserved. Poincaré gr.

$$\begin{pmatrix} A & H \\ 0 & A^{*-1} \end{pmatrix}$$

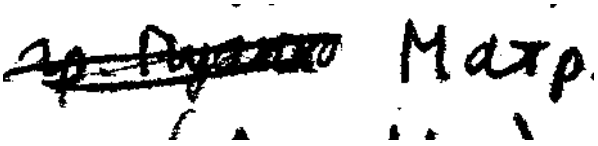
This is a group for $H \geq 0$ / $H = 0$ rot. (H adds). $H = 1$ — par. transl.



Uncertain reading. The condition on H has an upper angled stroke and a lower horizontal stroke before 0; readings $H \geq 0$ / $H = 0$.

Original source details: GELFAND-P000048

[Cancelled.]



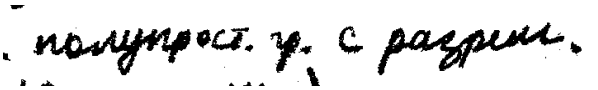
Uncertain reading. Before Matrix of order 4, a cancelled phrase begins gr.; the next word resembles Вращ. / Лоренца (rot. / Lorentz).

Original source details: GELFAND-P000048

Matrix of order 4.

$$\underline{\begin{pmatrix} A & H \\ 0 & A^{*-1} \end{pmatrix} \otimes SU(3)}$$

Obtain a semisimple gr. with solv. (108-param.).



Uncertain reading. The abbreviation after with resembles разреш. (solv.); the ending is joined.

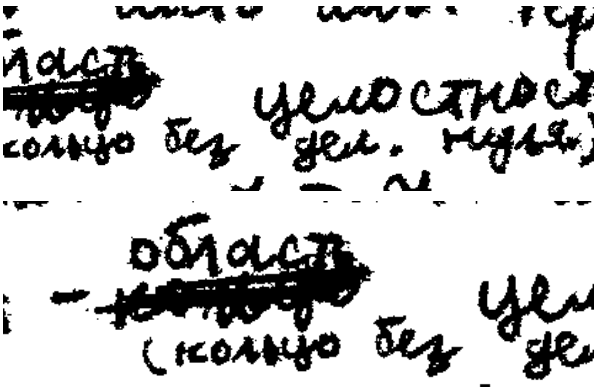
Original source details: GELFAND-P000048

One can describe what is needed to solve the theor. of elem. particles).

15/II

Kazhdan. Palam.'s result in purely alg. terms.

Let A be an integral domain [domain inserted above a cancelled word] (ring without zero divisors), I a prim. ideal.

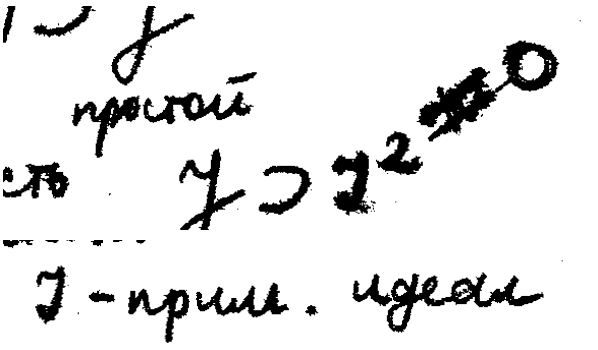


Uncertain reading. Область [domain] is inserted above a cancelled word; the candidate is кольцо [ring]. The dense strokes do not establish the whole word. The following целостности [integral, in integral domain] (ring without zero divisors) is legible.

Original source details: GELFAND-P000048

$$I \supset J$$

[Beneath J:] prime. And let $J \supset I^2$ [cancelled: $\neq$] 0.



Uncertain reading. I is introduced as a prim. ideal; prime is written beneath J in $I \supset J$. Next: $J \supset I^2$, with a cancelled sign resembling $\neq$ before 0. The letters and cancellation are retained.

Original source details: GELFAND-P000048

Every such ideal is given by diff. opers. of order 1.

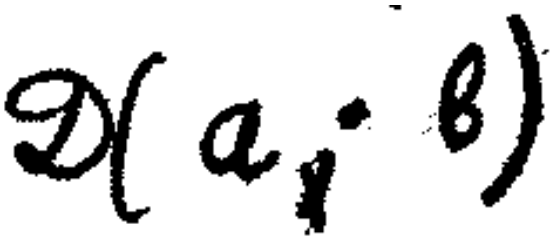
GELFAND-P000049

A diff. oper. is a A / K / k-lin. oper.

Original source details: GELFAND-P000049

Uncertain reading. The heavily overwritten sign before -lin. resembles A with a projecting upper stroke. K and k remain possible.

$$D(a \cdot b) = aD(b) + D(a)b$$

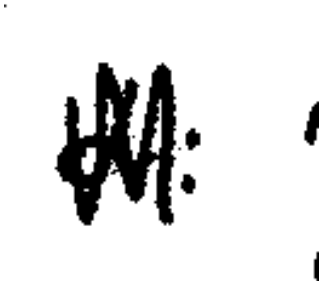


Original source details: GELFAND-P000049

Uncertain reading. Below the multiplication dot in the argument of D is a separate comma-like mark.

An ideal is the zero of some collection of diff. opers. Let A/J^2 (of order 1 because of J^2). An operator with values in the ring A/J instead of in the original ring. Diff. everywhere, and the ans. on a subset. $aD(b)$ — multiplication by a. On J the oper. is A-linear [below: diff.]. Conversely, every A-lin. oper. on J

[Cancelled sign before J]:



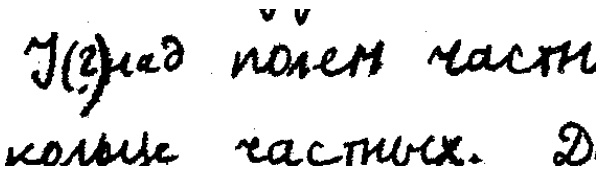
Original source details: GELFAND-P000049

Uncertain reading. To the left of J, three dense, nearly vertical strokes with a loop are cancelled, followed by a colon. An individual underlying letter is not established.

$$J \xrightarrow{f} A/J$$

Then it can be multiplied by an elem. b, again lin. [Above: bf.] $\exists b$ such that bf extends to a diff. of the whole ring. Everything follows from this.

We shall consider $J(2)$ / $J(\xi)$ / $J(\ell)$ over the field of fractions.

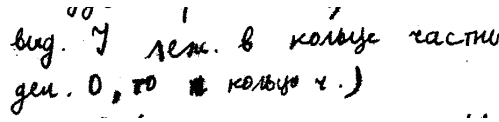


Original source details: GELFAND-P000049

Uncertain reading. The parenthesized sign after J resembles 2 / ξ / ℓ, followed by over the field of fractions. Readings $J(2)$ / $J(\xi)$ / $J(\ell)$.

Functionals exist, appar. J lies in the ring of fractions.

Multiply by b (since there are no zero divisors, then [a short sign is cancelled] ring of fr.).



Original source details: GELFAND-P000049

Uncertain reading. A short densely cancelled sign between then and ring of fr.; its reading is not established.

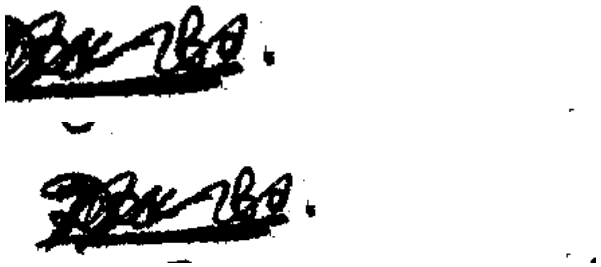
If A/J is a field, then $\forall$ functional extends to a diff. (Need to take the localization of A/J). There is a map of the field

$$A/J \longrightarrow A$$

[Beneath the arrow: hom.] such that the reverse hom. is the identity.

$$A/J \longrightarrow A \longrightarrow A/J$$

[Cancelled.]



Original source details: GELFAND-P000049

Uncertain reading. The cancelled heading before For now, arb. resembles Док-во. (Proof.); its reading remains uncertain beneath the strike.

For now, arb. $A/J \subset A$, $J \subset A$. Any elem. decomposes into a Σ (Σ as a lin. space).

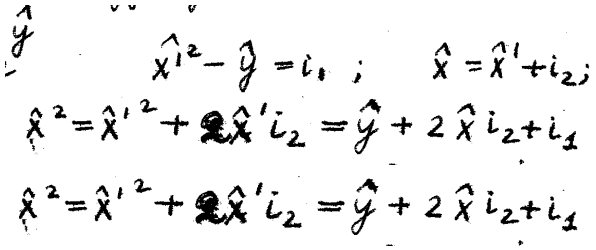
GELFAND-P000050

On J it is already given. A lin. map $J \to A/J$ extends.

On the whole [of it], 0.

A transcend. elem. anywhere.

$$\begin{aligned} y \to \hat{y}, \qquad & x^2 = y \\ \hat{x}'^2 - \hat{y} = i_1; \qquad & \hat{x} = \hat{x}' + i_2; \\ \hat{x}^2 = \hat{x}'^2 + 2\hat{x}' i_2 = \hat{y} + 2\hat{x} i_2 + i_1 & \end{aligned}$$



Original source details: GELFAND-P000050

Uncertain reading. The expansion has no i_2^2. The multiplier is $\hat{x}\prime$ in the first equality and $\hat{x}$ without a prime in the second. Both readings are retained.

Hence i_2 is found (since A/J is a field). Instead of I, J was written, and conversely.

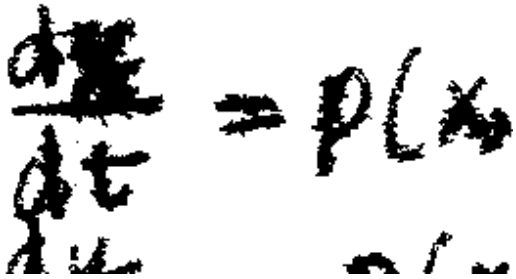
General form of a diff. oper. in the polynomial ring.

$$\sum P_i(x_1, \ldots, x_n) \frac{\partial}{\partial x_i}$$

P_i are polynomials. Alg. theory of diff. eqns. over an arb. field. Dieudonné, Grothendieck: in alg. problems, passed to an arb. field.

Ordinary eqn.

$$\frac{dx}{dt} = P(x,y)$$
$$\frac{dy}{dt} = Q(x,y)$$



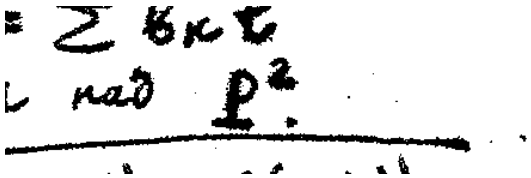
Uncertain reading. The letter in the numerator of the first equation is heavily overwritten; the visible candidate is dx, and the earlier letter is not established.

Original source details: GELFAND-P000050

t is replaced by a local ring at $\forall$ point (t a local parameter).

$$x = \sum a_k t^k$$
$$y = \sum b_k t^k$$

Diff. eqns. over P^2.



Uncertain reading. An underlined P-like letter with a raised 2: P^2 / p^2. No double stroke is visible.

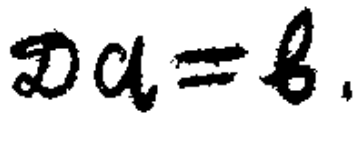
Original source details: GELFAND-P000050

$$D(x(t)) = f(x(t))$$

D is a diff. operator in the ring of elems. x. Ringed space. Points (x,y) and fibres. A ring at $\forall$ point. The fibre over each point is a power series in t (a power series in t at $\forall$ point).

GELFAND-P000051

$$Da = b.$$



Uncertain reading. In the left entry, a / α; no separate definition on this frame.

Original source details: GELFAND-P000051

Init. cond. — value in a max. ideal. Then localize. D carries over to any localization. The soln. must be an ideal. (?) The whole ring need not be given. It is known locally. x and y need not be specified.

Palamodov.

$\mathcal{P}$ — ring of polyn. $P(z_1, \dots, z_n)$

[Before P: a sign is cancelled.]

Uncertain reading. a short densely cancelled symbol

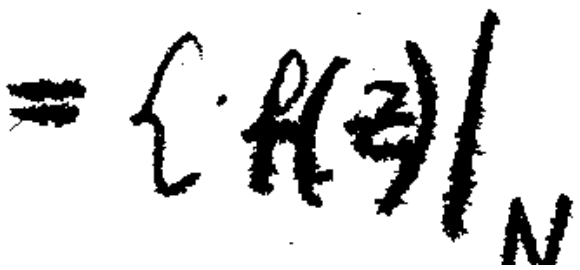
Original source details: GELFAND-P000051

$\mathcal{H}$ — space of all fns. in a domain of holom. (e.g. a ball)

$$\mathcal{J} \subset \mathcal{P}$$

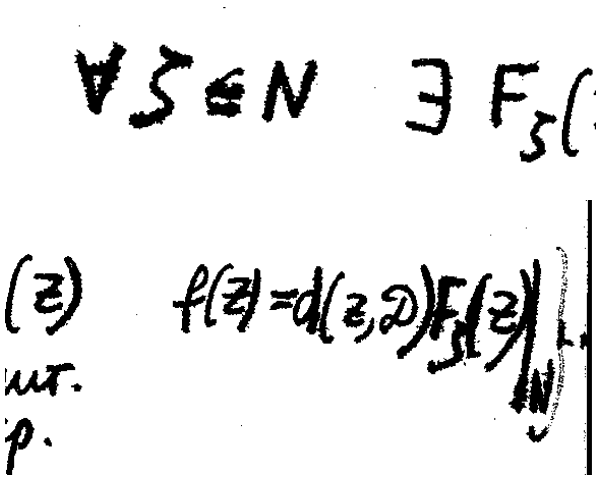
A prim. ideal with basis $P_1, \dots, P_t$ Consider $\mathcal{J}\mathcal{H}$ Quotient space $\mathcal{H}/\mathcal{J}\mathcal{H}$ Construct $\mathcal{H}\{\mathcal{J}\}$, isomorphic to it by means of the operator $d(z, D)$

$$\mathcal{H}\{\mathcal{J}\} = \{h(z)\big|_N \quad \forall \zeta \in N \quad \exists F_\zeta(z) \quad f(z) = d(z, D)F_\zeta(z)\big|_N\}$$



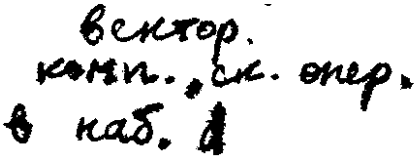
Uncertain reading. The first function symbol inside the brace resembles h with a stroke across its lower part; the separate entry on the right reads $f(z)$.

Original source details: GELFAND-P000051



Uncertain reading. The point sign after $\forall$ and the subscript of F resemble ζ / ξ; the operator letter in $d(z, D)$ resembles d / δ. The right formula reaches the frame edge; the lower N and closing brace are visible. The mark below the right-hand F matches the subscript on the first F; both are typeset as ζ.

Original source details: GELFAND-P000051



Uncertain reading. Below the first $h(z)$: vector. comp. / конх. / конк.; see / an. / lin. oper. in coll. d / λ [комп. / конх. / конк., см. / ан. / лин. опер. в наб.; uncertain abbreviations].

Original source details: GELFAND-P000051

[Below $F_\zeta(z)$: analytic in a nbd.] Local theorem. There is a theorem with growth estimates. Res.

$$\mathcal{H}\{\mathcal{J}\} \cong \mathcal{H}/\mathcal{J}\mathcal{H}$$

The isom. is natural

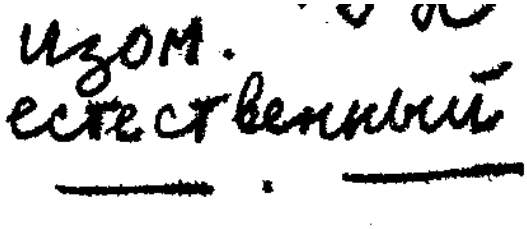


Diagram and typed description. Below the annotation are two nearly horizontal strokes with a separate dot between them.

Original source details: GELFAND-P000051

GELFAND-P000052

Dynin. X — comp. manif. Over it, vect. bundles E_0 E_1

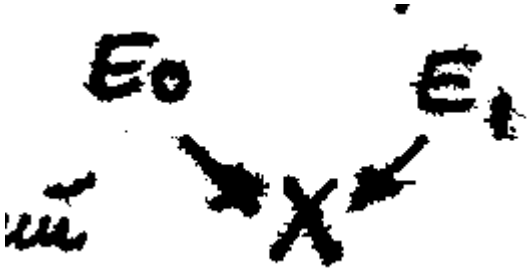
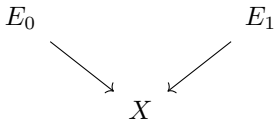


Diagram and typed description. Over X, two bundles E_0, E_1, with arrows to X.



Original source details: GELFAND-P000052

Sheaf of germs of C^∞-fns. of the 1st and 2nd bundles:

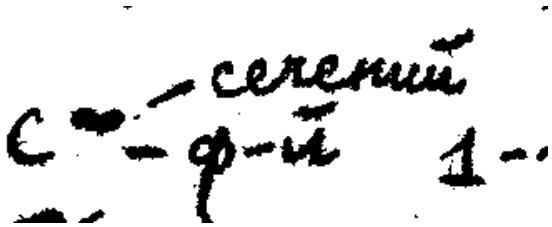
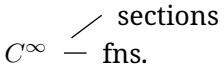


Diagram and typed description. Above fns., the word sections is inserted, with a short inclined stroke before it.



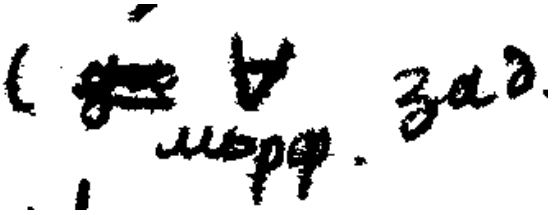
Original source details: GELFAND-P000052

$$\mathcal{E}(E_0) \qquad \mathcal{E}(E_1)$$

and morphisms

$$\mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

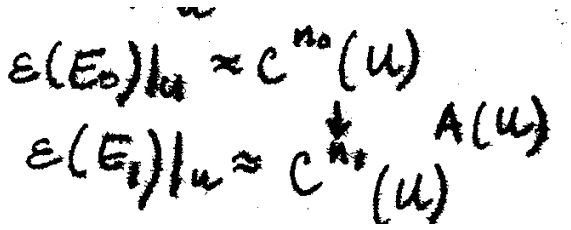
([An entry is cancelled; below it, morph.] $\forall$ given by diff. operators)



Uncertain reading. a short densely overwritten entry

Original source details: GELFAND-P000052

Locally, $\mathcal{E}(E_0)|_U$ — smooth vector fns.



Original source details: GELFAND-P000052

Diagram and typed description. Two local entries; in the first, the lower restriction sign is densely overwritten: U / a. Downward arrow from $C^{n_0}(U)$ to $C^{n_1}(U)$, with $A(U)$ at right.

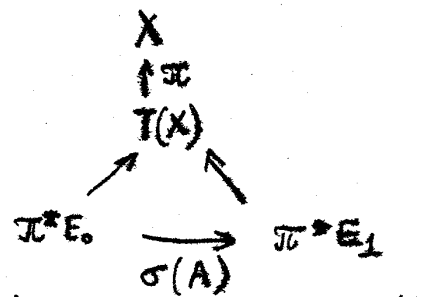
$$\begin{array}{lcl} \mathcal{E}(E_0)|_U \approx & & C^{n_0}(U) \\ & & \Big\downarrow {\scriptstyle A(U)} \\ \mathcal{E}(E_1)|_U \approx & & C^{n_1}(U) \end{array}$$

diff. oper. — a matrix. Symbol — princ. part of the matrix. Locally $\sigma(A)(x,\xi)$ — princ. part of the char. matrix.

Original source details: GELFAND-P000052

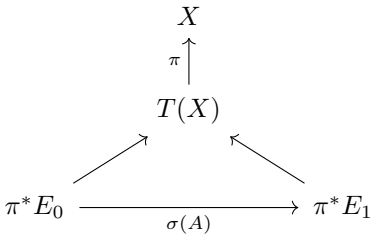
Uncertain reading. The second argument of $\sigma(A)$ is densely overwritten; forms ξ / ζ / D are visible.

(Select terms of highest degree Formalize. Tang. bundle — vect. F

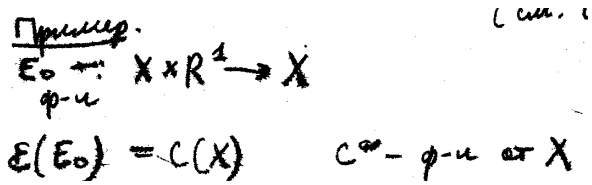


Original source details: GELFAND-P000052

Diagram and typed description. Diagram with X, $T(X)$, π, π^*E_0, π^*E_1, $\sigma(A)$.



A ellipt. if $\sigma(A)$ is a monomorphism (see Schwartz) Example.



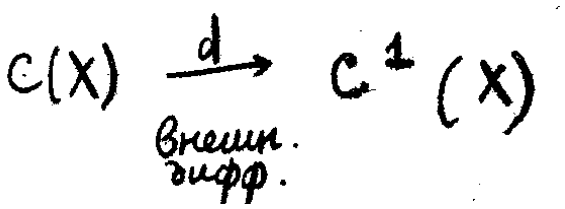
Original source details: GELFAND-P000052

Diagram and typed description. At E_0, a leftward arrow; below E_0, fns.

$$E_0 \longleftarrow X \times \mathbf{R}^1 \longrightarrow X$$
fns.

$$\mathcal{E}(E_0) = C(X) \qquad C^\infty \text{— fns. on } X$$

GELFAND-P000053



Original source details: GELFAND-P000053

Diagram and typed description. Below d, ext. diff.

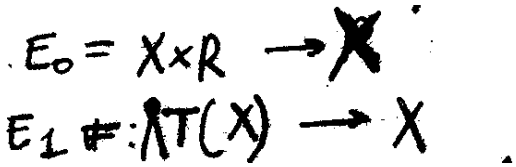
$$C(X) \xrightarrow[\text{ext. diff.}]{d} C^1(X)$$

Loc. $f(x) \longrightarrow$ grad $f(x)$ symbol — mult. by the vector $\xi \in T(X)$ This is an example of an overdet. elliptic system.

$$f(x) \xrightarrow{d} \sum \frac{\partial f}{\partial x_i}\, dx_i$$

$$f(x) \longrightarrow \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{pmatrix} f(x)$$

$$E_0 = X \times \mathbf{R} \longrightarrow X \qquad E_1 : \Lambda T(X) \longrightarrow X$$



Original source details: GELFAND-P000053

Uncertain reading. At the end of the first line, a heavily overwritten X. After E_1, a dense cancellation before the colon; the sign before $T(X)$ resembles Λ / $\wedge$. No clear star at T is visible.

[At right:] diff. forms — sections of such a bundle. Result.

$$\mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

Take

$$0 \longrightarrow \ker A \longrightarrow \mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

If A is ellipt., then

$$\dim H^i(X, \ker A) < \infty$$

special cases. 1°. Let A be ellipt. in Petrovskii's sense.

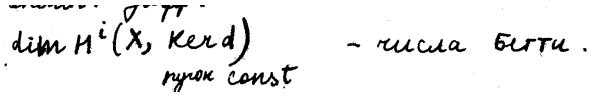
$$\mathcal{E}(\mathbf{R}^n) \xrightarrow{A} \mathcal{E}(\mathbf{R}^n)$$

Then the space of glob. solutions is finite-dimensional

$$H^0 \quad \text{— finite:} \qquad H^1 \quad \text{— defect. (fin. defect)}$$

$$H^1(X, \ker A)$$

2°. Ext. diff. oper.



Original source details: GELFAND-P000053

Diagram and typed description. Below $\ker d$, const. sheaf.

$\dim H^i(X, \underset{\text{const. sheaf}}{\ker d})$ — Betti numbers.

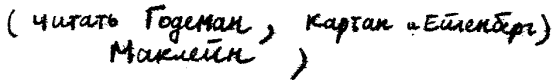
3°. Compl. an. Cartan–Serre–Kodaira theorem

GELFAND-P000054

Original source details: GELFAND-P000054

Diagram and typed description. Several letters in the heading are densely overwritten.

Proof.



Original source details: GELFAND-P000054

Diagram and typed description. Mac Lane inserted below within the parenthesis; two closing parentheses at right.

(read Godement, Cartan and Eilenberg)

Mac Lane)

$\mathcal{P} \longrightarrow$ a sheaf. How to compute homology.

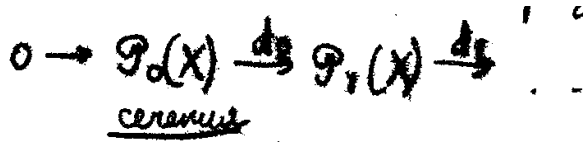
$$0 \longrightarrow \mathcal{P} \longrightarrow \mathcal{P}_0 \longrightarrow \mathcal{P}_1 \longrightarrow \cdots .$$

a fine resolution.

$$0 \longrightarrow \underset{\text{sections}}{\mathcal{P}_0(X)} \xrightarrow{d_0} \mathcal{P}_1(X) \xrightarrow{d_1} \cdots$$

Original source details: GELFAND-P000054

Uncertain reading. The arrow subscripts are heavily inked. The visible candidates are 0 and 1.



Original source details: GELFAND-P000054

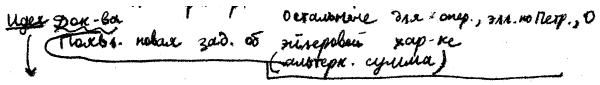
Diagram and typed description. Below $\mathcal{P}_0(X)$, sections underlined.

$\underset{\text{sections}}{\mathcal{P}_0(X)}$

$$H^1(X, \mathcal{P}) = \ker d_1 / \operatorname{Im} d_0$$

$$0 \longrightarrow \ker A \longrightarrow \mathcal{E}(\mathbf{R}^n) \xrightarrow{A} \mathcal{E}(\mathbf{R}^n)$$

this is a fine resolution. H^1 — quotient. The others, for an oper. ellipt. in Petrov.'s sense, 0.



Original source details: GELFAND-P000054

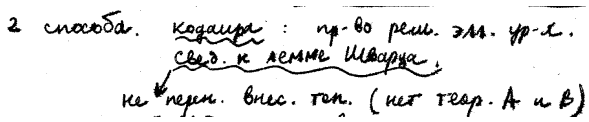
Diagram and typed description. At left, Idea underlined, beside Of proof with a wavy underline; downward arrow and a long oval/stepped line.

Idea Of proof

A new prob. arises on the Euler char.

(altern. sum)

2 methods.

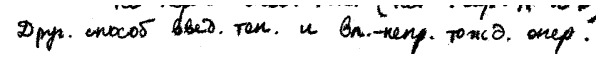


Original source details: GELFAND-P000054

Diagram and typed description. Kodaira and the line Reduc. to the Schwarz lemma have wavy lines; arrow to the following line.

Kodaira: space of solns. of an ellipt. eqn.
Reduc. to the Schwarz lemma,

does not carry over to higher hom. (no theorems A and B)



Original source details: GELFAND-P000054

Uncertain reading. Another method: introd. top. / thin. and вп.-непр. ident. oper. [введ. / ввод. топ. / тон.; the initial abbreviations are not established.]

Resolution from Ban. spaces. Banach sheaf. Sheaf $\mathcal{P}$: $U \longrightarrow \mathcal{P}(U)$ (presheaf) of Fréchet type $\mathcal{P}(U)$ — F-spaces, and the restr. maps are cont. A series of new sheaves

$$C^{p,r}(\mathcal{P})$$

Original source details: GELFAND-P000054

Diagram and typed description. The arrow from U has a wavy beginning and an arrowhead at right.

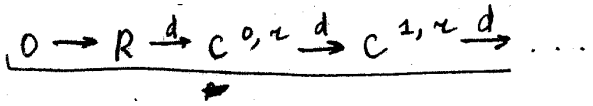
$U \rightsquigarrow C_U^{p,r}(\mathcal{P}(U))$

Vect.-valued diff. forms of degree p, smoothness r The sheaf extends, but with no add. assumpt., it does not extend.

GELFAND-P000055

For $\mathcal{P} = \mathcal{E}(E)$ Sheaf — a Fréchet space. de Rham resolution.

$$0 \longrightarrow \mathbf{R} \xrightarrow{d} C^{0,r} \xrightarrow{d} C^{1,r} \xrightarrow{d} \cdots .$$



Original source details: GELFAND-P000055

Diagram and typed description. Below the sequence, a long bracket.

[Below it, a sign is cancelled.]

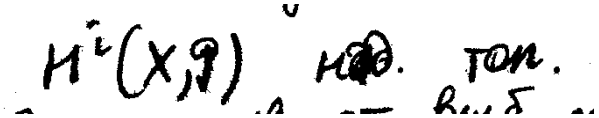
Original source details: GELFAND-P000055

Uncertain reading. a short densely cancelled sign

Coeffs. — maps of U into diff. forms. coeffs. — smooth maps $U \longrightarrow \mathcal{P}(U)$ r — smoothness exponent. Passing to the sheaf adds no new sections (but not in the gen. case, under add. assump.) Of the type of the de Rham resolution

$$0 \longrightarrow \mathcal{P} \longrightarrow \widehat{C}^{p,r}(\mathcal{P}) \xrightarrow{d} \widehat{C}^{1,r}(\mathcal{P}) \longrightarrow \cdots .$$

adjusted because of the order of smoothness Banach spaces result. $H^i(X, \mathcal{P})$ new [нов.; provisional] topol.



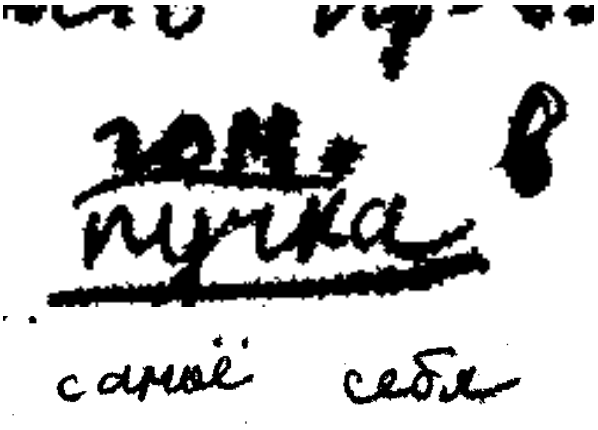
Original source details: GELFAND-P000055

Uncertain reading. The heavily overwritten abbreviation before топ. [topol.] resembles нов. [new]; над. is another possibility.

It does not depend on the choice of resolutions.

$$H^i(X, \mathcal{P})^{(r)} \longrightarrow H^i(X, \mathcal{P})^{(r-1)}$$

take the same Fréchet spaces (isomorphism) It is compl. cont. by the embedding theorem Gelfand. Problem: X — comp. with a sheaf of lin. top. spaces on it. Ex. X — a point. Just a space. Diff. oper. — endom. into itself.

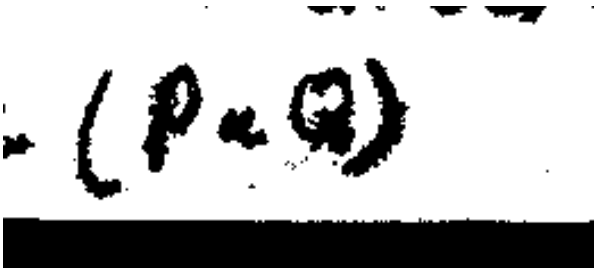


Original source details: GELFAND-P000055

Uncertain reading. The abbreviation before “into itself” is heavily overwritten and resembles endom.; below it “of the sheaf” is inserted and underlined.

endom.
of the sheaf

[At right:] (fibres preserved.) When it is of finite index. X — 1 point. $E+$ compl. cont. Riesz theorem. Then one can consider operators ellipt. of order ∞. Dynin. Hypoellipt. of the operator is needed. in the resolution, always the same sheaf ker A Merely cont. coeffs., will it be of finite index? Next time. Adams (Novikov), Berezin (P and Q)



Original source details: GELFAND-P000055

Uncertain reading. The last letter in the parenthesis resembles Q with a lower right stroke; O remains possible.

GELFAND-P000056

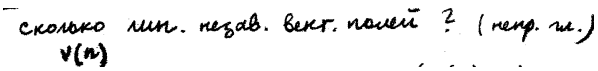
Original source details: GELFAND-P000056

Diagram and typed description. Date in a frame open at the top; below it, Adams; Novikov underlined twice.

22/II. Novikov S.P.

Adams.

S^n how many lin. indep. vect. fields? (cont. f.)



Original source details: GELFAND-P000056

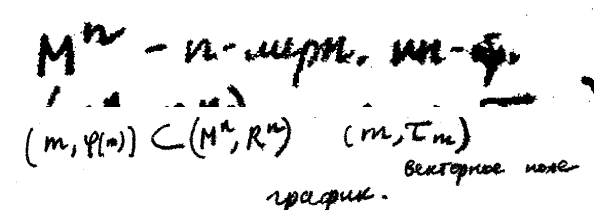
Diagram and typed description. Below how many, $\nu(n)$.

how many
$\nu(n)$

Triv. facts. Exist on odd-dim. sph. ($\nu(n) \geq 1$) on even-dim. sph., none ($\nu(n) = 0$)

$$\sum \text{ind. of sing.} = \text{Euler char.}$$

M^n — n-dim. м-о. / м-ф.



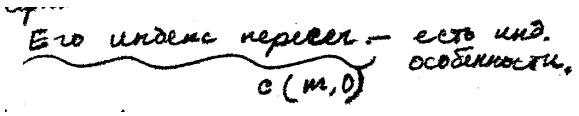
Original source details: GELFAND-P000056

Uncertain reading. м-о. / м-ф. [the last sign of the abbreviation is dense].

$(m, \varphi(m)) \subset (M^n, \mathbf{R}^n)$
graph.

(m, T_m)
vector field

$(m, 0)$ — canon. Its intersection index is the index of the singularity.

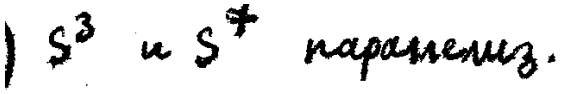


Original source details: GELFAND-P000056

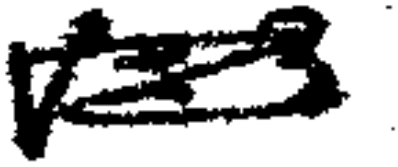
Uncertain reading. Below the line, a wavy bracket to $c(m, 0)$ / $\varepsilon(m, 0)$.

$c(m, 0)$

(Here only 1 and -1, but degree n). $E = 2$, so there are no fields. 2) S^3 and S^7 paralleliz.



Uncertain reading. Before the following equalities, a short cancelled entry; the upper sign of the second sphere resembles 7 / an asterisk.



Original source details: GELFAND-P000056

$\nu(3) = 3$, $\nu(7) = 7$ 3) $\nu(2n-1) \geq 1$ (S^{2n-1})

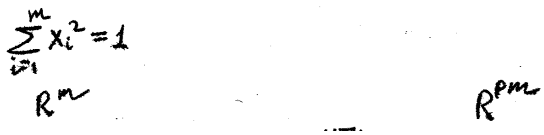
4) If on S^{2m-1} there are k indep. fields ($\nu(2m-1) \geq k$), then on S^{2pn-1}, $\nu_{2pn-1} \geq k$

Original source details: GELFAND-P000056

Uncertain reading. After p the one-humped sign resembles n more than the m on the preceding line. Readings $2pn-1$ / $2pm-1$.

In short

$$S^{m-1}, \quad \nu_m \geq k \quad \Longrightarrow \quad S^{pm-1}, \quad \nu_{pm-1} \geq k$$



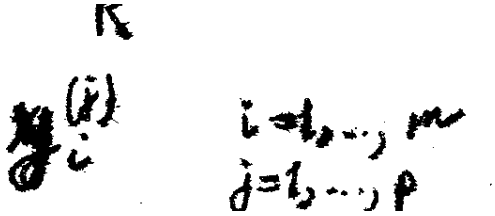
Original source details: GELFAND-P000056

Diagram and typed description. Below the sum, $\mathbf{R}^m$; at right, $\mathbf{R}^{pm}$.

$\sum_{i=1}^{m} x_i^2 = 1$ $\quad$ $\mathbf{R}^{pm}$

$\mathbf{R}^m$

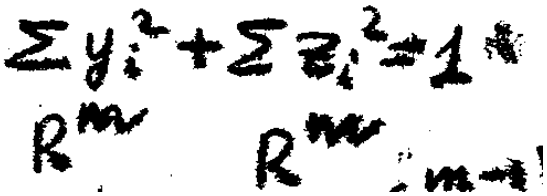
Coords. p groups of m each



Original source details: GELFAND-P000056

Uncertain reading. $x_i^{(j)}$ / $y_i^{(j)}$ [first symbol overwritten], $i = 1, \ldots, m$, $j = 1, \ldots, p$.

let $p = 2$.



Original source details: GELFAND-P000056

Diagram and typed description. Below each sum, $\mathbf{R}^m$. Asterisk after 1; no footnote on the frame.

$\sum_{\mathbf{R}^m} y_i^2 + \sum_{\mathbf{R}^m} z_i^2 = 1^*$

$$v(x) \text{ in } \mathbf{R}^m, \qquad v(x) \perp x \qquad (\text{field on } S^{m-1})$$

Uncertain reading. The initial letter in $v(x)$ resembles Latin v / ν; no separate quantifier sign is established.

Original source details: GELFAND-P000056

GELFAND-P000057

Put

$$\widetilde{v}(y,z) = \left(\|y\| v\left(\frac{y}{\|y\|}\right),\ \|z\| v\left(\frac{z}{\|z\|}\right) \right)$$

$\|y\|$ to cope with $y = 0$, all in the 2nd comp., of mod $= 1$, and $\perp$ (y, z) Indep. fields are obtained from indep. fields. Thus Hopf fields are obt. (On the right one can write other fields) can stabilize the problem: incr. dim. Was known to the classical authors (hard to construct.

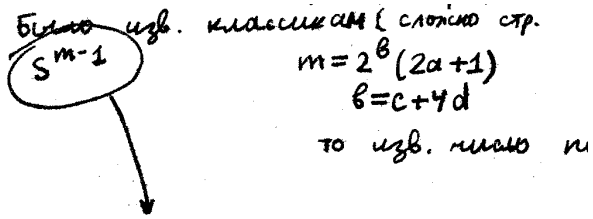
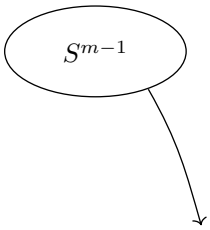


Diagram and typed description. At left, S^{m-1} in an oval. The upper stroke overlaps Was. Arrow down.



Original source details: GELFAND-P000057

$$m = 2^b(2a+1) \qquad b = c + 4d \qquad 0 \le e \le 3,$$

Uncertain reading. In the bounds $0 \le e \le 3$, the middle letter resembles e / c.

Original source details: GELFAND-P000057

then the known number of fields

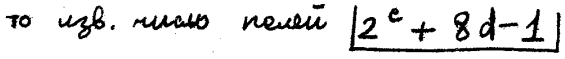
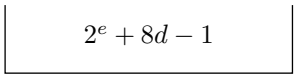


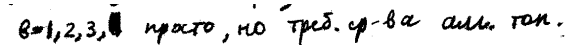
Diagram and typed description. Frame open at the top; the exponent resembles e / c.



Original source details: GELFAND-P000057

$$\nu_{m-1} \ge 2^c + 8d - 1$$

In cases 3, 7 gives max. $b = 0$ (even — triv.). $b = 1, 2, 3$, [a sign is overwritten], simple, but req. methods of alg. top.



Uncertain reading. densely overwritten sign / downward arrow

Original source details: GELFAND-P000057

$b \le 12$ Toda, but m of order 3000 Adams proved this in all dimensions.

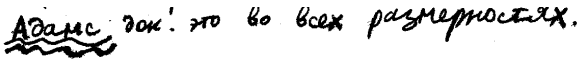


Diagram and typed description. Two wavy underlines.

Adams

Original source details: GELFAND-P000057

$*$ The reduct. part belongs to Atiyah and Toda. This leads to a problem on homot. gr. of spheres. finally

$$\nu_{m-1} = 2^c + 8d - 1$$

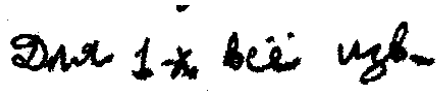
The odd factor plays no role.

Here one assertion. $m \uparrow$ by a factor of 6, then the number of fields incr. by 8

Original source details: GELFAND-P000057

Manuscript notation. The factor is written as 6.

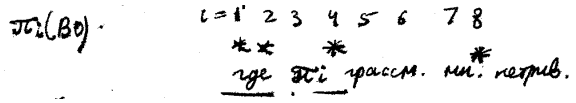
$(d+1)$ this is seen from the constr. in Bott periodicity. For the first ones all is known.



Original source details: GELFAND-P000057

Uncertain reading. After 1 are a short stroke and an x-shaped sign. The proposed reading is 1-x [the first ones].

2 parall. sph., then Bott periodicity



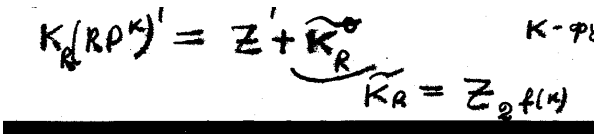
Original source details: GELFAND-P000057

Diagram and typed description. Asterisks below 1, 2, 4, 8.

$\pi_i(BO)$.

$i=$	1	2	3	4	5	6	7	8
	*	*		*				*

where π_i of Grassm. mfld. is nontriv. $\mathbf{R}P^k$ — project. space Consid.



Original source details: GELFAND-P000057

Uncertain reading. In the lower entry $f(k)$ is raised relative to 2. Readings $2^{f(k)}$ / $2f(k)$. The small upper sign on $\widetilde{K}$ resembles 0. From the second summand a bracket to the lower entry.

$K_{\mathbf{R}}(\mathbf{R}P^k) = \mathbf{Z} + \widetilde{K}^0_{\mathbf{R}}$

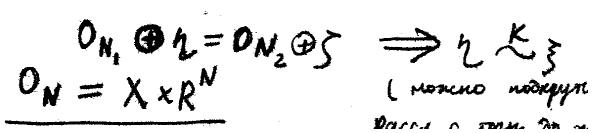
$\widetilde{K}_{\mathbf{R}} = \mathbf{Z}_{2^{f(k)}}$

K-functor, see “Матем.” [Math.] Atiyah and Hirzebruch

GELFAND-P000058

Elem. of K — real vect. bundle, but with an equiv. relation. Add. — direct Σ. Mult. — tensor prod. of 2 vect. bund. 1 elem. in K, if

$$O_{N_1} \oplus \eta = O_{N_2} \oplus \zeta \quad \Longrightarrow \quad \eta \overset{K}{\sim} \xi$$



Original source details: GELFAND-P000058

Uncertain reading. After the second sum the sign resembles ζ; after the equivalence it resembles ξ / ζ. The initial oval permits O / 0.

$$O_N = X \times \mathbf{R}^N$$

[At right:] (can twist it) Bund. up to equiv. — element of the K-funct. Coll. — K-functor. Can add, mult.! Triv. bund. (Σ of tang. and norm. — triv., there is an inverse elem.) To each space assigns a ring) Over the sphere K-functor

$$K_{\mathbf{R}}(S^i) = \pi_i(BO)$$

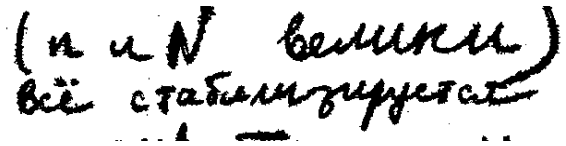
(how many homot. maps to Grassm. mfld.)

Original source details: GELFAND-P000058

Diagram and typed description. Above large, the insertion Grassm. mfld.

K-funct. of sph. — bund. of $\overset{\text{Grassm. mfld.}}{\text{large}}$ dim.

BO — Grassm. mfld. — n-dim. lin. in N-dim. sp. (n and N large)



Original source details: GELFAND-P000058

Uncertain reading. Below, everything stabilizes; the long tail of the parenthesis overlaps the end of the word.

everything stabilizes

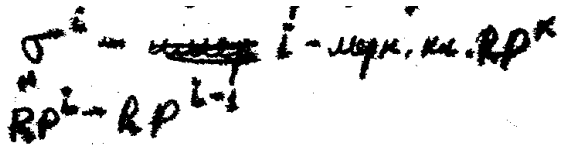
Vect. bund. with given base ⟵ evident π_i.

Original source details: GELFAND-P000058

Manuscript notation. The abbreviation before π_i reads очев. [evident].

To the mfld. corresp. tang. spher. maps. To the normal sheaf — norm. spher. map. Next an artificial device. $\mathbf{R}P^k$ subdiv. into cells (1 in each dim.) All vect. bund. over $\mathbf{R}P^k$ are constr. thus.

σ^i — i-dim. cell of $\mathbf{R}P^k$

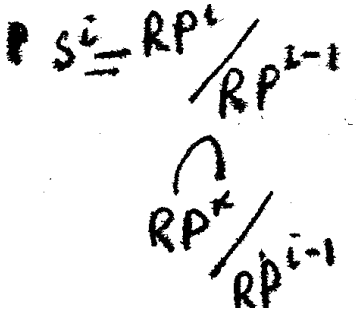


Original source details: GELFAND-P000058

Uncertain reading. After σ^i a short entry densely cancelled; below it, $\mathbf{R}P^i \longrightarrow \mathbf{R}P^{i-1}$. Above the first $\mathbf{R}P^i$ a small unidentified sign.

$\mathbf{R}P^i \longrightarrow \mathbf{R}P^{i-1}$

consid.



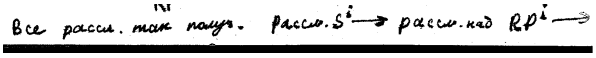
Original source details: GELFAND-P000058

Uncertain reading. Before S^i a dense short sign, its reading not established. Below, $\mathbf{R}P^k/\mathbf{R}P^{i-1}$ with a curved arrow.

$S^i = \mathbf{R}P^i/\mathbf{R}P^{i-1}$

$\mathbf{R}P^k/\mathbf{R}P^{i-1}$

All bund. are obtained thus. Bund. $S^i \longrightarrow$ bund. over $\mathbf{R}P^i \longrightarrow$



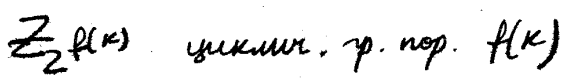
Original source details: GELFAND-P000058

Undeciphered passage. The arrow in the bottom line reaches the right edge; the next object is not visible in this frame.

GELFAND-P000059

$\longrightarrow$ bund. on $\mathbf{R}P^k$. $f(k)$ — number of asterisks to the left of k **Z** — invar. — dimension Crit. points appear on cycles

$$\mathbf{Z}_{2f(k)} \qquad \text{cyclic gr. of order } f(k)$$



Original source details: GELFAND-P000059

Uncertain reading. The index on **Z** permits $2^{f(k)}$ / $2f(k)$; the order caption visibly has $f(k)$, without a separate 2.

[At left an entry is cancelled.]

Original source details: GELFAND-P000059

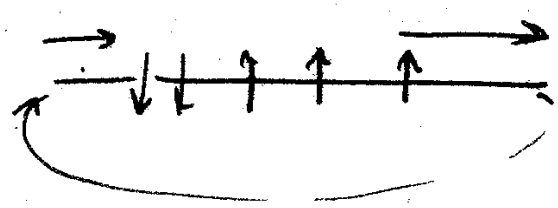
Uncertain reading. short dense cancellation

$$\mathbf{R}P^k \in \mathbf{R}P^{k+1}$$

Original source details: GELFAND-P000059

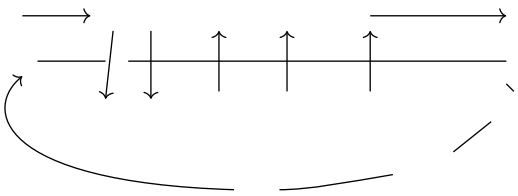
Manuscript sign. The sign has the middle stroke of $\in$. A separate $\subset$ appears lower on the page.

there is a norm. sheaf — Möbius strip (this is a generator) of the group it must be added to itself $2^{f(k)}$ times

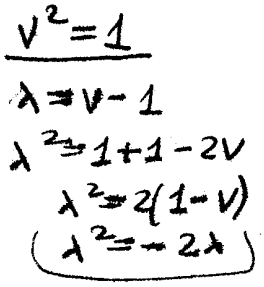


Original source details: GELFAND-P000059

Diagram and typed description. At left an almost horizontal line; five transverse arrows: two on the left down, the next three up. Two rightward arrows above the line, short on the left, long on the right. Below, an open oval arc with a gap at the lower right; its left end rises to the line and has an arrow, the right end is faint. No letter labels.



$$\underline{\nu^2 = 1} \qquad \lambda = \nu - 1$$
$$\lambda^2 = 1 + 1 - 2\nu$$
$$\lambda^2 = 2(1 - \nu)$$

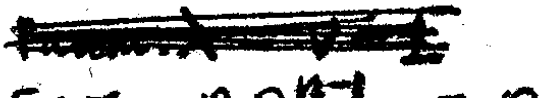


Original source details: GELFAND-P000059

Uncertain reading. The narrow sign in ν resembles ν / v / V; some equality signs are overwritten. The last line is in a rounded frame open at the top.

$$\lambda^2 = -2\lambda$$

mult. structure. [A line is cancelled.]



Original source details: GELFAND-P000059

Uncertain reading. Long, densely crossed-out line on the left below the formulas

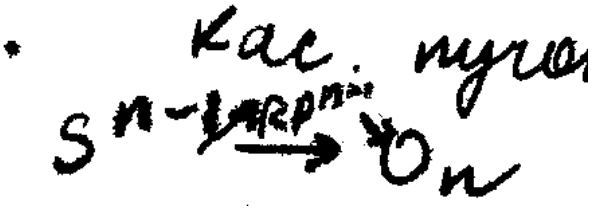
There is

Original source details: GELFAND-P000059

Diagram and typed description. Above the right side are two horizontal strokes of different lengths with a dot between them.

$$\mathbf{R}P^{n-1} \subset O_n$$

S^n is assigned to the hyperpl. of the equator $\longrightarrow$ into the orthog. group. Tang. sheaf — map.

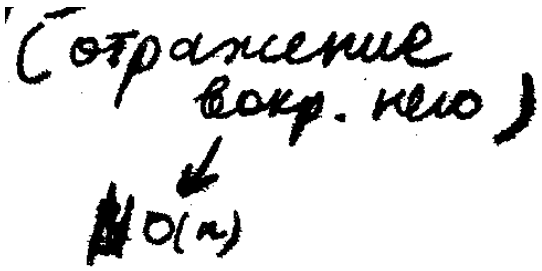


Original source details: GELFAND-P000059

Uncertain reading. Above the arrow, a tight insertion $\mathbf{R}P^{n-1}$ / $\mathbf{R}P^n$ with an additional slanted arrow pointing to O_n; some signs are dense.

$$\begin{array}{ccc} & & \mathbf{R}P^{n-1} \\ & \nearrow & \\ S^{n-1} & \longrightarrow & O_n \end{array}$$

At symmetric points, the same. it is 2-sheeted (Gluing on the equator) rotation about it



Original source details: GELFAND-P000059

Uncertain reading. From the parenthesis about reflection an arrow down to $O(n)$; before $O(n)$ a black overwritten stroke, reading not established.

(reflection about it)

$$\downarrow$$

$$O(n)$$

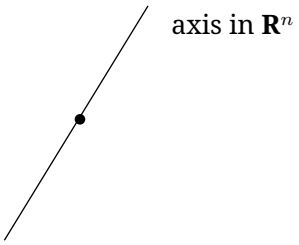
axes of proj. space.

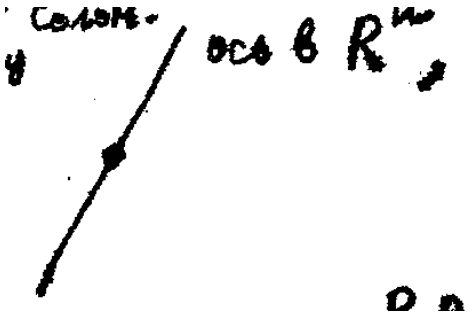
Uncertain reading. Albert Solom. / Soln.

Original source details: GELFAND-P000059

Schwarz.

Diagram and typed description. Slanted line with a point and the label axis in $\mathbf{R}^n$.

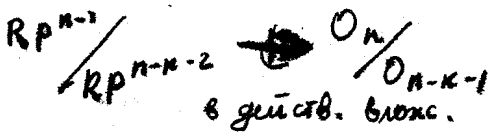




Original source details: GELFAND-P000059

$$\mathbf{R}P^{n-1}/\mathbf{R}P^{n-k-2} \longrightarrow O_n/O_{n-k-1}$$

Uncertain reading. In the lower left exponent, $n-k-2$ / $n-n-2$; the arrow between the quotients is densely overwritten.



Original source details: GELFAND-P000059

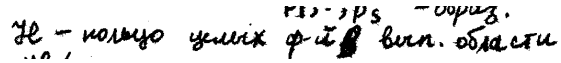
in fact an embedding.

GELFAND-P000060

1/III

V.P. Palamodov. $\mathcal{P}$ — ring of polyn. in $\mathbf{C}^n$, $\mathcal{J} \subset \mathcal{P}$ — ideal, primary, $P_1, \dots, P_s$ — gener. $\mathcal{H}$ — ring of entire funct. [a sign is cancelled] of a conv. region

Uncertain reading. after funct. a short sign is densely crossed out

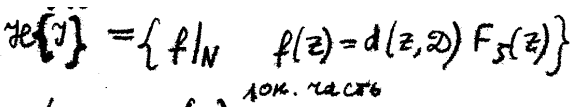


Original source details: GELFAND-P000060

$\mathcal{H}/\mathcal{J}\mathcal{H}$ is described thus

$$\mathcal{H}\{\mathcal{J}\} = \{f|_N \quad f(z) = d(z,\mathcal{D})F_\zeta(z)\}$$

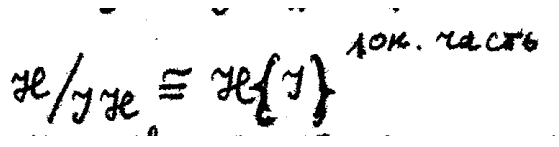
Uncertain reading. The subscript on F resembles ζ / s.



Original source details: GELFAND-P000060

$$\mathcal{H}/\mathcal{J}\mathcal{H} \cong \mathcal{H}\{\mathcal{J}\}$$ local part

Position of the note. The note “local part” is written above and to the right of the second formula.

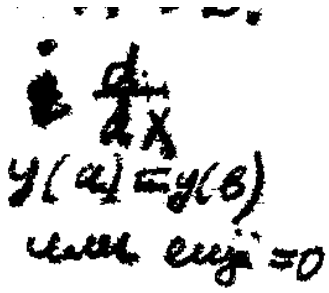


Original source details: GELFAND-P000060

Kazhdan says that this is obt. algebraically.

• Gelfand. Nelson's example. 2 self-adj. operators A and B. what changes a self-adj. oper? (not symm. closed) [In the right margin:]

$$i\frac{d}{dx}, \qquad y(a) = y(b)$$

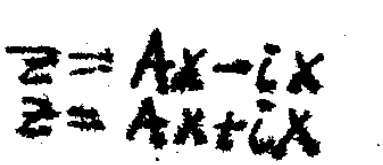


Uncertain reading. Below is written "or also $= 0$"; no argument precedes the equality sign.

Original source details: GELFAND-P000060

Essentially self-adj. A (symm.), if there are no vect. orthog. to all vect. of the form

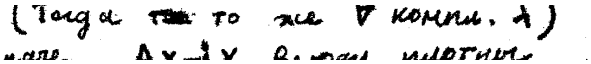
$$\overline{z} = Ax - ix, \qquad z = Ax + ix$$



Uncertain reading. The first z has a long upper bar. The upper stroke of the second may be part of the letter or a separate mark.

Original source details: GELFAND-P000060

(Then [an entry is cancelled] the same $\forall$ complex λ)

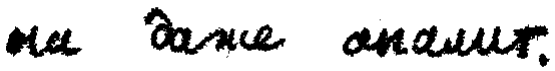


Uncertain reading. short dense cancellation

Original source details: GELFAND-P000060

otherwise $Ax - ix$ are everywhere dense $A - iE$ has inv. (in general?: if closed)

$$\left(i\frac{d}{dx} \quad \text{[on / not] even analyt.} \quad \begin{matrix} f(a) = f(b) & \text{essentially self-adj.} \\ f(a) = f(b) = 0 & \text{only symm.} \end{matrix} \right)$$

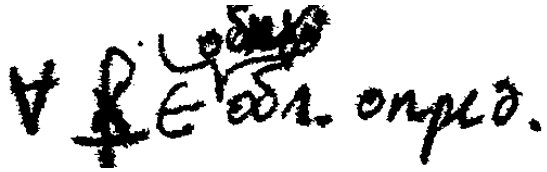


Uncertain reading. Before the conditions is written "[on / not] even analyt."; the last letter of the first word is joined.

Original source details: GELFAND-P000060

A and B essentially self-adj., $\alpha A + \beta B$ essentially self-adj., α, β — real.

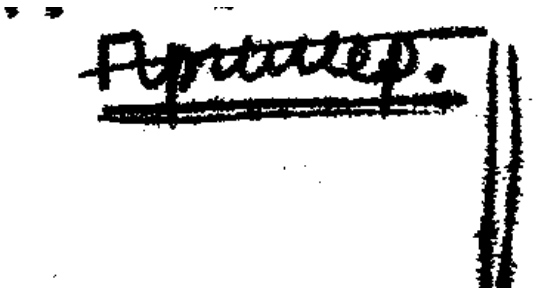
$$\forall f \in \text{domain of def.} \qquad ABf = BAf.$$



Uncertain reading. Above the domain of definition a short entry is densely overwritten; below it a curved stroke.

Original source details: GELFAND-P000060

Nevertheless A and B do not commute.



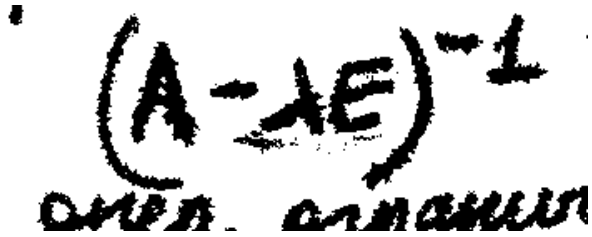
Original source details: GELFAND-P000060

Diagram and typed description. At left Example. is crossed out and doubly underlined; to its right a double vertical stroke.

~~Example.~~ ||

A and B comm., if reasonabl. funct. of them commute.

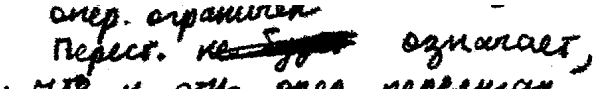
$$(A - \lambda E)^{-1}$$
oper. bounded



Original source details: GELFAND-P000060

Uncertain reading. Below λE lower strokes; whether a separate underline or parts of letters is not established.

Comm. [Cancelled: not...] also with B means that these oper. too comm.



Original source details: GELFAND-P000060

Uncertain reading. The cancelled entry begins with a mark resembling не [not]; the remaining letters are obscured.

e^{iA} — unit. e^{iB} do not commute

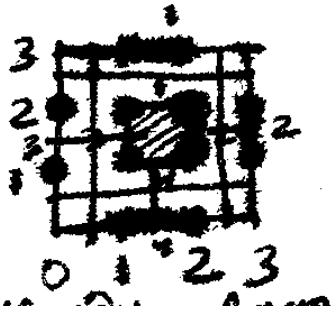
[An entry is cancelled; beside it (∀ bounded oper.).]

Original source details: GELFAND-P000060

Uncertain reading. Short densely crossed-out entry

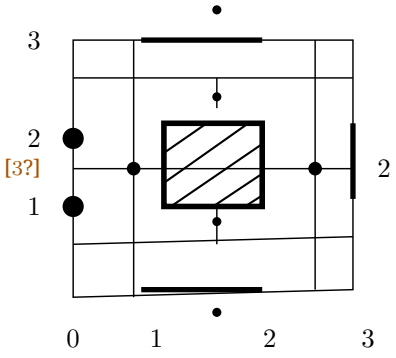
GELFAND-P000061

Nelson's example.



Original source details: GELFAND-P000061

Diagram and typed description. Square grid, heavy points and central hatching; labels $0, 1, 2, 3$ below, $1, 2, 3$ at left and 2 at right. Between 1 and 2 at left, a sign resembling 3 with a heavily inked lower stroke; the central overlapping strokes and exact gluing are not established.



From the region between the squares, remove the heavy points The segments are glued each point is glued to the nearest boundary point along the horiz. or vert. The corners are glued together. Obtain a surface of genus 4 with one removed point

Uncertain reading. The numeral has an angular top and a broad foot. Prefer 4; 1 is also possible.

Original source details: GELFAND-P000061

$$X = \frac{d}{dx}, \qquad y = \frac{d}{dy}$$

Uncertain reading. The second sign before the equality has the descender of y; the later entry is iY.

Original source details: GELFAND-P000061

iX, iY — formally self-adj. They commute But the corresp. oper. do not commute. Remove measure-0 closed traject. hitting the vertices. e^X and e^Y — unit. oper. which do not commute.

$$e^{\frac{d}{dx}} f(x) = f(x+1)$$

(Taylor formula) Prove first for analytic funct., then for the whole Hilb. space by a limiting passage (the oper. is unitary) And it has to be moved.

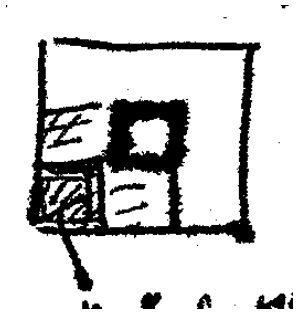
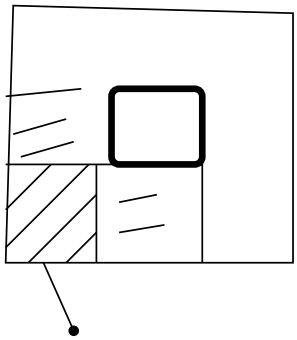


Diagram and typed description. Square with a hole; hatching to the left of the hole and in the lower left region, two short strokes to its right. A slanted line runs from the lower left region down towards the caption. The boundaries of the overlapping strokes are not established.

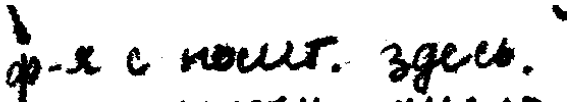
Original source details: GELFAND-P000061



e^Y — region by 1 upwards, e^X will turn round

$$e^X$$

funct. with compact. [компт.] here.

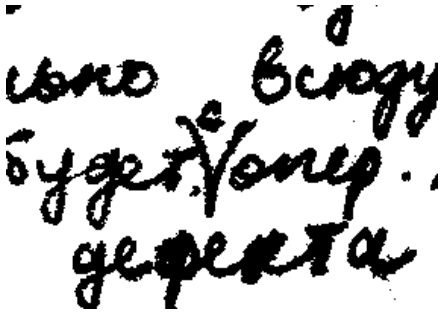


Uncertain reading. compact. / comp. [компт. / комп.]

Original source details: GELFAND-P000061

Exclude a finite number of traject. e^{tX} — shift along the traject. of oper. X locally everywhere they commute

what will happen to [Inserted above: with] the Laplace oper.?



Manuscript insertion. The letter c [with] is above a caret.

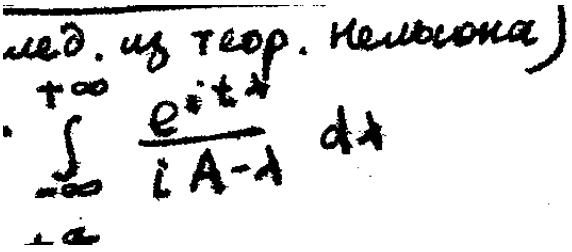
Original source details: GELFAND-P000061

deficiency index

$$\Delta f = if$$
$$\underline{\Delta = X^2 + Y^2}$$

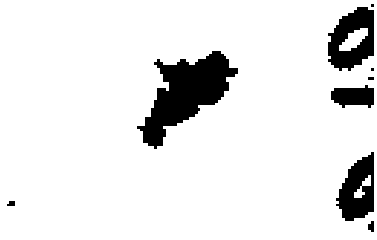
(this follows from Nelson's theorem) It is not self-adj. (has a C^∞ solution, bound. cond., and at the point only $\int$ square.) The shift oper. themselves are def. on L^2, and not on periodic funct. 3 def. e^{itA}:

$$1^\circ = \int_{-\infty}^{+\infty} \frac{e^{it\lambda}}{iA - \lambda}\, d\lambda$$
$$2^\circ = \int_{-\infty}^{+\infty} e^{it\lambda}\, dE_\lambda$$
$$3^\circ : \quad \frac{df}{dt} = iAf$$



Uncertain reading. In the first entry the denominator is $iA - \lambda$; no separate factor before the integral is visible.

Original source details: GELFAND-P000061



Uncertain reading. Before the fraction in the third line, a short dark stroke; the exact reading is not established.

Original source details: GELFAND-P000061

$\underset{\text{sol.}}{U}_{t,0} f_0 = f$ — unit. oper.

GELFAND-P000062

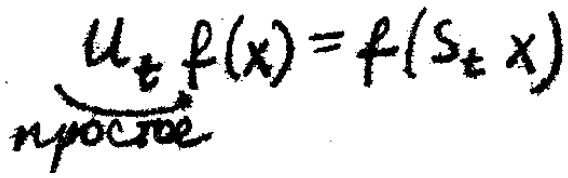
Sinai. Posit. in dyn. systems. [Полож.] For an ergod. dyn. syst. [Crossed out: a short densely crossed-out entry; inserted above: with discr. spec.] the spectrum is always simple

Uncertain reading. a short densely crossed-out entry

Original source details: GELFAND-P000062

M, μ, $\{S_t\}$ — one-param. gr. in M In $\mathcal{L}^2_\mu$ a one-param. gr. of unit. oper.

$$U_t \quad f(x) = f(S_t x)$$



Manuscript sign. Below U_t is a curve with a pointed right end; “simple” belongs to the following line.

Original source details: GELFAND-P000062

In the ergod. case every eigenvalue is simple Lebesgue spectrum: $\forall f$

$$(U_t f, f) = \int e^{it\lambda} p(\lambda)\, d\lambda$$

($p(\lambda)$ — integr.) (In general always $\int e^{it\lambda}\, d\mu$, here μ abs. cont.) Countably-mult. Leb. spec.: $\exists$ orthog. $f_1, f_2, \ldots$

$$(U_t f_i, f_j) = 0 \qquad (i \neq j)$$

and generate everything. In all known ex. the spectrum is countably-mult. Ex. Transf. of the torus.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det = 1, \quad \text{integer}$$

$$\begin{aligned} x &\longrightarrow ax + by \\ y &\longrightarrow cx + dy \end{aligned}$$

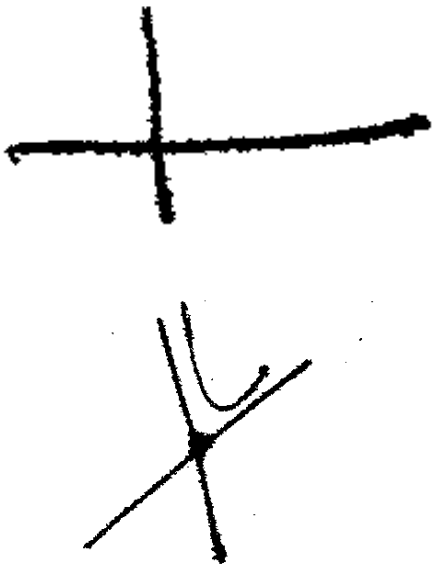
Need to find autocorr. [авток.] Expand in a Fourier series and act.

$$\begin{aligned} f(x,y) &\longrightarrow f(ax+by, cx+dy) \\ e^{2\pi i(mx+ny)} &\longrightarrow e^{2\pi i((am+cn)x+(bm+dn)y)} \end{aligned}$$

A matrix acts on the integer latt. Need to take trajectories

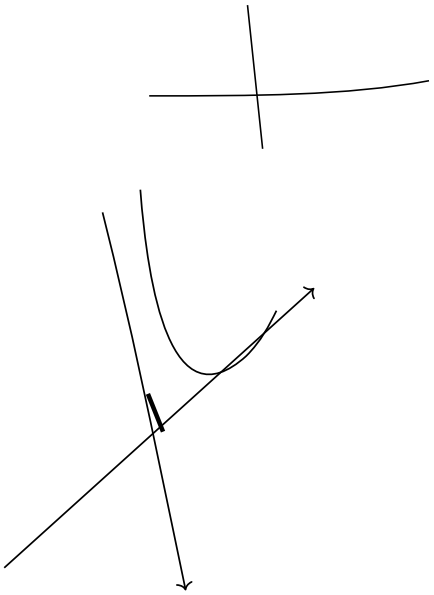
$$(m,n) \longrightarrow S(m,n) = (m', n')$$

The traject. is a hyperbola.



Original source details: GELFAND-P000062

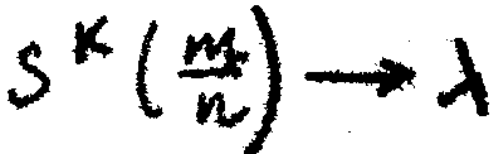
Diagram and typed description. Above, two crossing strokes without explicit arrows. Below, an oblique line and two curved strokes; thickening at the crossing. The upper right end of the oblique line and the bottom end of the long curved stroke resemble arrows; the membership of the overlapping strokes is not established.



Real eigenvalues — contraction along one axis and expansion along the other (det $= 1$)

$$S^k\left(\frac{m}{n}\right) \longrightarrow \lambda$$

(λ — eigenvalue)



Original source details: GELFAND-P000062

Uncertain reading. A horizontal stroke separates m from n. The arrow points to λ; no additional factor is visible.

The waves become ever denser and direct. Waves $\Longrightarrow$ towards an eigendir. The space decomposes into a space of funct. with frequencies on hyperbolas. On the 1st hyperbola. A sequence of orthog. vect., each passes to the next

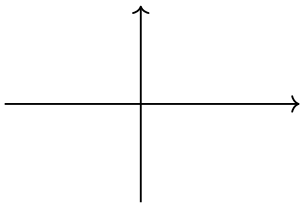
GELFAND-P000063

Original source details: GELFAND-P000063

Diagram and typed description. Above, a row of nine separate points.

(shift by 1 at the integer pts.) cont. Lebesgue spectrum. With the Bernoulli scheme — charact. of the group. Theorem. Measure $d\sigma(\lambda)$

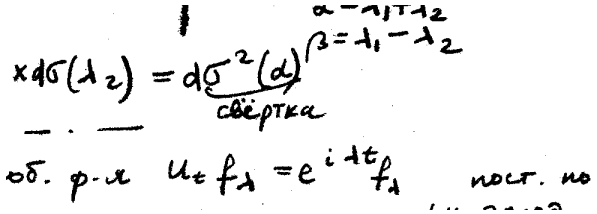
Diagram and typed description. Two crossing strokes; arrows to the right and upwards. No labels.



Original source details: GELFAND-P000063

$\sigma \times \sigma$

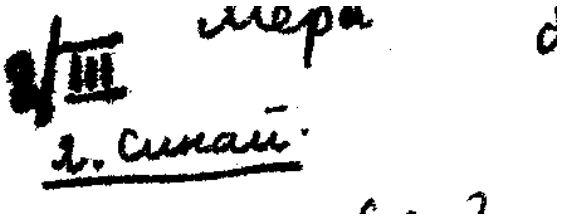
$$\alpha = \lambda_1 + \lambda_2$$
$$\beta = \lambda_1 - \lambda_2$$
$$\text{measure} \quad d\sigma(\lambda_1) \times d\sigma(\lambda_2) = \underbrace{d\sigma^2(\alpha)}_{\text{convolution}}$$



Uncertain reading. After the product of measures, $d\sigma^2(\alpha)$ is written, with convolution below; no separate factor in β is visible.

Original source details: GELFAND-P000063

8/III Ya. Sinai.



Uncertain reading. Before /III, a densely overwritten sign: 8 / 4 / 1; below, Ya. Sinai / 2. Sinai.

Original source details: GELFAND-P000063

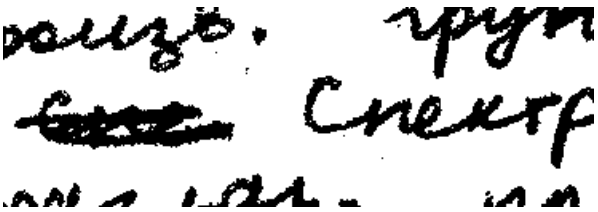
Diagram and typed description. A short horizontal stroke, a separate point, a longer almost horizontal stroke.

Original source details: GELFAND-P000063

$M, \mu, \{S_t\}$. An eigenfunct.

$$U_t f_\lambda = e^{i\lambda t} f_\lambda$$

const. in mod (for an ergod. system) 15/III Todorov SU 22/III-65 In a year there will be a math. congress in Moscow. (March — April) [what is a “simple root”? connection with matrices] Gelf. Probl. 1. Do alg. geom. over noncomm. rings!! 2. Alg. theory of diff. eqs. Povzner. Small parameter. Berezin. On funct. of many variables. Need to study not all funct., but specific ones — occurring in repr. theory and in stat. physics. Povzner a “wish” to Berezin. Kirillov. Ordinary dyn. syst. — time R or Z. Can have an arbitrary group preserving measure. Classific., invar.? [Before Spectrum, crossed out: a short entry under dense strokes.]



Uncertain reading. a short entry under dense strokes

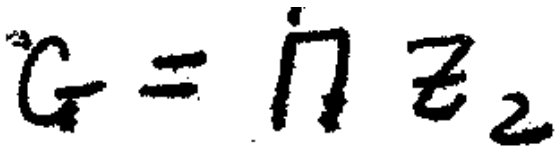
Original source details: GELFAND-P000063

Spectrum — the set of irred. repr. into which the space of funct. on M decomposes

$$\begin{matrix} M & G \\ \underset{\text{funct.}}{H} & \underset{\text{unit. oper.}}{U_g : f(x) \longrightarrow f(g^{-1}x)} \end{matrix}$$

H decomposes into irred. compon., which are called the spectrum. G comm. — repr. [прдпр.] are one-dimensional. G acts ergodically (no inv. funct. except const) For a compact gr. All transitive and reduce to describing subgroups. Ex. 1) Gordin and Whitehead.

$$G = \prod Z_2$$



Manuscript notation. The product sign is written. The next line specifies finitely many nonzero terms.

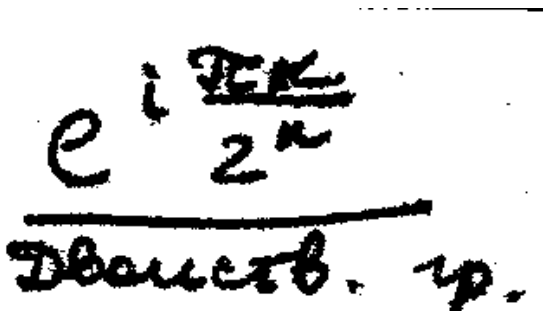
Original source details: GELFAND-P000063

Noncomp. group Sequences of 0 and 1, a finite number not 0. $\sum$ usual. For it, the problem of describing repr. of virtual subgroups is solved

GELFAND-P000064

2) Dyadic-rat. numbers on the circle.

$$e^{i\frac{\pi k}{2^n}}$$

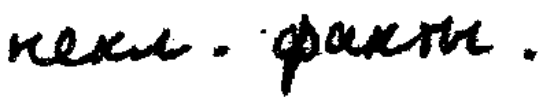


Uncertain reading. The sign before k resembles π; a separate 2 is indistinct. Readings πk / $2\pi k$.

Original source details: GELFAND-P000064

Countable. Dual gr. — 2-adic numbers.

On it, already nonclass. [некл.] facts.



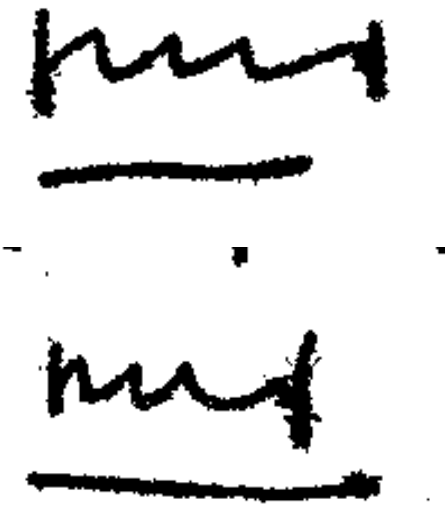
Uncertain reading. Before “facts” the abbreviation resembles некл. [nonclass.]; the final strokes are joined.

Original source details: GELFAND-P000064

Point spectrum and cont. sp. Possibly, [crossed out: ϕ / illegible] prod. of funct., decomposed by sections.

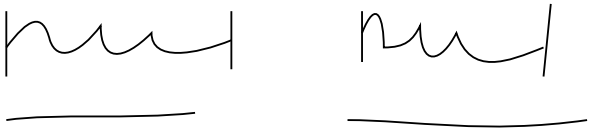
Original source details: GELFAND-P000064

Uncertain reading. ϕ / illegible



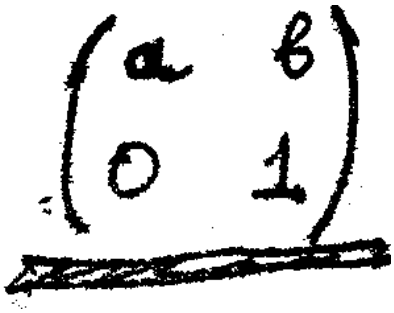
Original source details: GELFAND-P000064

Diagram and typed description. Two short sawtooth-like sections over horizontal underlines; the terminal strokes and some bends are uncertain.



Cont. sp. gives a funct. decomposed over a discrete set. Oseledets: also held for an ordinary system. 2-dim. time.

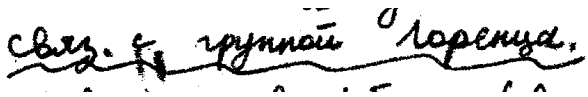
$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$



Original source details: GELFAND-P000064

Diagram and typed description. Below the matrix, a long double underline with oblique strokes.

Sinai's flows (equipped). Irred. repr. are few: one-dim. $|a|^{i\lambda}(\operatorname{sign} a)^{\varepsilon}$! And one infinite-dim. Connection with the Lorentz group.

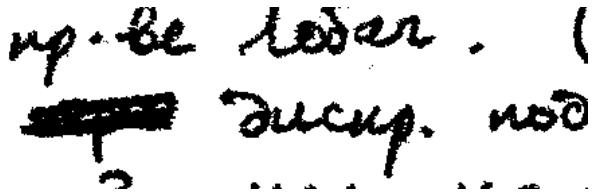


Original source details: GELFAND-P000064

Diagram and typed description. Wavy underline; below with [c], a short descending stroke with a bend.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Acts on Lobach. space (upper half-plane). About quot. by [crossed out: illegible] a discrete subgr. Will be compact, of negative curv.



Uncertain reading. illegible

Original source details: GELFAND-P000064

Tang. bundle — a 3-dim. compact manif., on which acts an ergod. Lorentz gr. Discrete spectrum because of compactness. Homologies of dyn. systems. Operator

$$U_g = \alpha(x, g) f(xg).$$

A homomorphism then. $\alpha(x, g)$ — a cocycle. $|\alpha| = 1$.

$$\alpha \in Z^1(G, \underset{\text{gr. of functions}}{F}).$$

α equiv. if a перев. [abbrev. in source] transf. exists.

Manuscript notation. The abbreviation перев. is written before преобр. [transf.].

Original source details: GELFAND-P000064

This happens if

$$\frac{\alpha}{\alpha_1} \in B^1(G, F)$$

General form of a boundary

$$\alpha_0(x, g) = \frac{\beta(xg)}{\beta(x)}$$

— homolog. to zero. Z^1/B^1 — homologies. Computed only if the gr. is transitive (then this is the set of repr. of the stabilizer subgr.).

Homol. 0, not all const, but $e^{2\pi i\lambda}$.

Manuscript notation. The λ follows immediately after i.

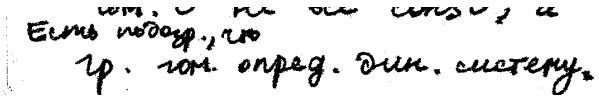
Original source details: GELFAND-P000064

Diagram and typed description. At the right, a caption with a long curved underline.

λ from the spectrum.

Original source details: GELFAND-P000064

There is a suspic. that the homol. gr. determines the dyn. system.



Manuscript notation. The last sentence begins "There is a suspic. that".

Original source details: GELFAND-P000064

GELFAND-P000065

The collection of homol. is called a repr. of a virtual subgroup (although there is no subgr.). 2-dim. homol. of the spectrum of a dyn. system. f_λ — eigenfunct. Choose 1 fixed eigenfunct.

$$f_\lambda f_\mu = C_{\lambda,\mu} f_{\lambda\mu}$$

$C_{\lambda,\mu}$ — 2-dim. cocycles on the spectrum of a dyn. system.

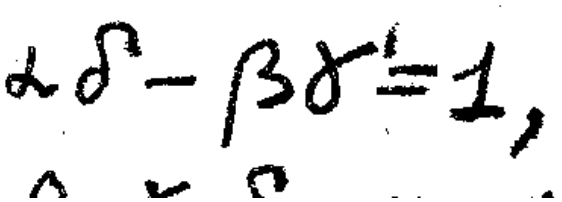
$$C_{\lambda,\mu} C_{\lambda\mu,\nu} = C_{\lambda,\mu\nu} C_{\mu,\nu}$$

Equiv. to 1

$$C_{\lambda,\mu} = \frac{a_\lambda a_\mu}{a_{\lambda\mu}}$$

In the classical case these homol. are trivial ("the syst. is determined by the spectrum"). It is not known even whether a dyn. syst. is determined by its spectrum for finite groups (Gelfand thinks not). Rotation group of the icosahedron.

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma' = 1$$



Manuscript notation. A separate raised stroke follows γ in the determinant; the matrix entry γ has no stroke.

Original source details: GELFAND-P000065

$\alpha, \beta, \gamma, \delta$ mod 5. (At a vertex, 5 triangles.) Klein. Lectures on the Icosahedron.

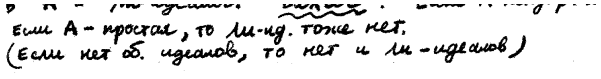
$$T_g f(x) = f(xg)$$

Need to recover X. There are shift oper., a group. But there are also oper. of mult. by a funct. (but if there is a ring, then the repr. in the ring determines the dyn. system). 29/III Viktor Nikolaevich. Connection between Lie algebras and assoc. algebras.

$$\begin{array}{ccc} L & \longrightarrow & U \\ \text{Lie alg.} & & \text{univ. envelop. alg.} \end{array}$$

Harish-Chandra — if L is simple, what can be said about U? Jacobson's problem: what is needed for U to be primitive (a faithful irred. module over it exists). Example. e_1, e_2, $[e_1, e_2] = e_1$ — U primitive. $[e_1, e_2] = 0$ — U non-primitive. (This is a polynomial ring, and a commut. primitive alg. is a field.) $A^{(-)}$ — a Lie alg. obtained from an assoc. alg. A $(ab - ba)$. Which properties of A carry over to $A^{(-)}$? In A — Lie ideals. Baxter. If A is Jacobson semisimple.

If A is simple, then there are no Lie ideals either. (If there are no ordinary ideals, then there are no Lie ideals either.)

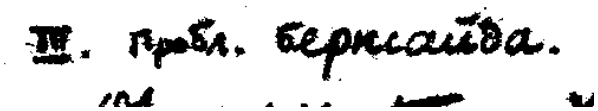


Manuscript notation. The last two lines are written below the name Baxter.

Original source details: GELFAND-P000065

GELFAND-P000066

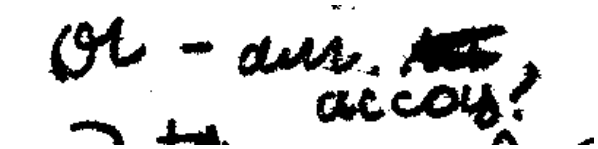
III. Burnside's probl. Kostrikin's result.



Uncertain reading. Below the top stroke, three strokes are the preferred reading, III; those at right meet.

Original source details: GELFAND-P000066

$\mathcal{A}$ — an alg. [Crossed out: illegible; inserted below: assoc.?]



Uncertain reading. illegible

Original source details: GELFAND-P000066

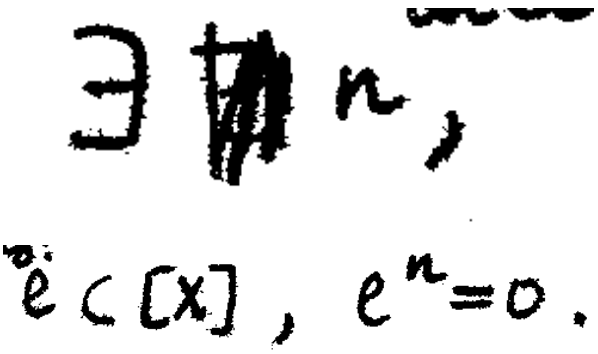
X — a finite set, [crossed out: illegible]

Uncertain reading. illegible

Original source details: GELFAND-P000066

generating $\mathcal{A}$, $[X]$ ($\supset X$.)
generated Lie alg.

$\exists$ [crossed out: $\forall$] n, $\quad e \subset [X], \quad e^n = 0.$



Uncertain reading. $\exists\, n$, an overwritten sign before n; relation $e \subset [X]$, without a middle stroke.

Original source details: GELFAND-P000066

Then $\mathcal{A}$ nilpot. (i.e. $\forall$ elem. is nilpotent).

Diagram and typed description. Two horizontal strokes of different lengths with a separate dot between them.

Original source details: GELFAND-P000066

Gelfand. Dixmier — unsolved problem: 2 rings $P(x, \frac{d}{dx})$ and $P(x, \frac{d}{dx}, y, \frac{d}{dy})$ (polynomials). Are these rings isomorphic? S. Gelfand. Dyson. 1. Orthog. ensemble.

$U^T = U$ symm.
$UU^* = 1$ and unit.

Inv. for the unit. gr.

$$U \longrightarrow VUV^T$$

(a homogeneous space of the unit. group). Density of the distribution of eigenval. $P(\theta_1, \ldots, \theta_N)d\theta_1 \ldots d\theta_N$ — measure of matrices, eigenval. (their args.) in $(\theta_1, \theta_1 + d\theta_1); \ldots$. Stabilizer subgr.: $E \longrightarrow VV^T = E$ (real orthog. matrices). An inv. measure is introduced. Turns out

$$p_1(\theta_1, \ldots, \theta_N) = C_1 \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|$$

2. All unit. matrices. (Probab. of eigenvalues!) Group — unit. [Inner autom.] $U \longrightarrow VU$ [then a crossed-out sign: illegible, resembling V^*.]

Original source details: GELFAND-P000066

Uncertain reading. illegible, resembling V^*

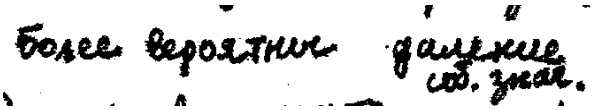
$$p_2(\theta_1, \ldots, \theta_n) = C_2 \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^2$$

Original source details: GELFAND-P000066

Uncertain reading. In the last argument of p_2, the index is n / N.

(Inv. measure of the unit. gr. $\int$ over conjugacy classes.)

Eigenval. far apart are more probable.



Original source details: GELFAND-P000066

Manuscript notation. The adjective is далёкие [far apart].

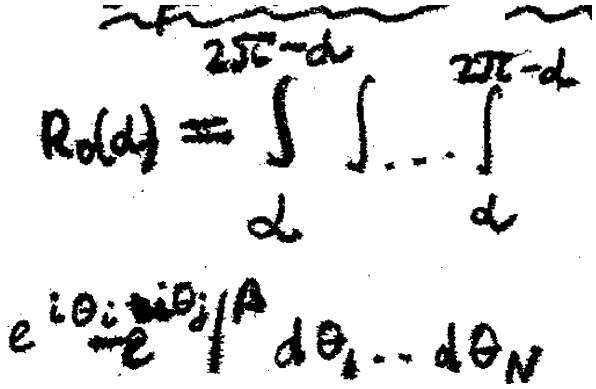
[Above θ: a circled letter: Δ / illegible; a triangular sign inside an oval.]

Original source details: GELFAND-P000066

Uncertain reading. Δ / illegible; a triangular sign inside an oval.

Consider all matr., $-\alpha < \theta < \alpha$ contains no eigenvalues. The probability of this is calculated.

$$R_\alpha(\alpha) = \int_\alpha^{2\pi-\alpha} \int \ldots \int_\alpha^{2\pi-\alpha} |e^{i\theta_i} - e^{i\theta_j}|^\beta \, d\theta_1 \ldots d\theta_N$$

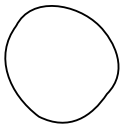


Uncertain reading. In this line, no definite $\prod$ is visible before the absolute value; the index of R resembles α, argument α / d. A dense mark precedes $i\theta_j$.

Original source details: GELFAND-P000066

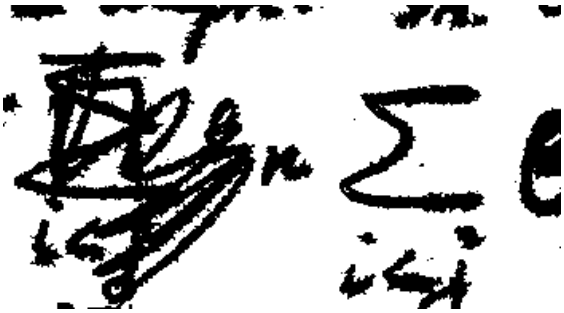
$\beta = 1, 2$. For Dyson, $\forall\beta$. Physical assumptions are made. Phys. analogy.

Diagram and typed description. A circle without labels.



Original source details: GELFAND-P000066

Conducting wire in the form of a circle, on it N charges, repelling by 2-dim. electrostat. Then the energy (potent.) = W. [Crossed out before the sum: $\prod_{i<j}$ / illegible.]



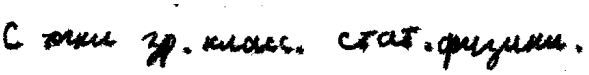
Uncertain reading. $\prod_{i<j}$ / illegible

Original source details: GELFAND-P000066

$$W = \sum_{i<j} \ln |e^{i\theta_i} - e^{i\theta_j}|$$

(Free charges.)

From the point of view of class. stat. physics.



Manuscript notation. The phrase preceding the final integral.

Original source details: GELFAND-P000066

$$\int_0^{2\pi} \int_0^{2\pi} \dots \int_0^{2\pi} e^{[-\beta W]}\, d\theta_1 \dots d\theta_N$$

GELFAND-P000067

$\beta = \frac{1}{T}$. $T = 1$ and $1/2$ correspond to the orthog. and unit. ensembles. The segment $(-\alpha, \alpha)$ is forbidden.

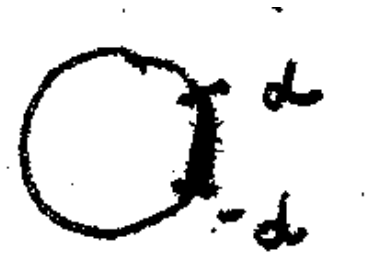
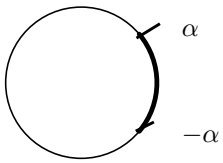


Diagram and typed description. A circle with a short dark right-hand section between the labels α and $-\alpha$. The thick terminal strokes do not establish arrows.



Original source details: GELFAND-P000067

Free energy.

$$R_\beta(\alpha) = e^{\beta[F(\alpha)-F(0))}$$

Manuscript brackets. A square bracket precedes F and a round parenthesis closes the exponent.

Original source details: GELFAND-P000067

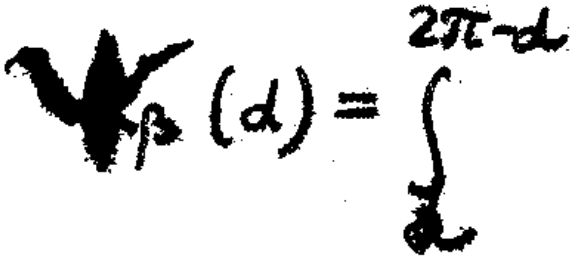
F — free energy. (This equality can be taken as its def.)

Diagram and typed description. Two horizontal strokes of different lengths with a separate dot between them.

Original source details: GELFAND-P000067

Consider matrices with no eigenval. in the interval $-\alpha < \theta < \alpha$. Their probability

$$\Psi_\beta(\alpha) = \int_\alpha^{2\pi-\alpha} \dots \int_\alpha^{2\pi-\alpha} \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^\beta \, d\theta_1 \dots d\theta_N$$



Uncertain reading. The initial letter Ψ in the integral is overwritten; the upper reading is Ψ, the lower one is not established.

Original source details: GELFAND-P000067

$$R_\beta(\alpha) = \frac{\Psi_\beta(\alpha)}{\Psi_\beta(0)}$$

(Need to calculate.) Phys. assumption. Let N be large, introduce a density of eigenval. (instead of eigenval.). $\sigma(\theta)d\theta$. $\sigma(\theta)$ — δ-funct. at the points $\theta_1, \dots, \theta_n$. $\sigma(\theta)d\theta \longrightarrow$ number of eigenval. in the interval $d\theta$.

$$R_2(\alpha) = e^{(N^2 \ln \cos \frac{\alpha}{2})}$$

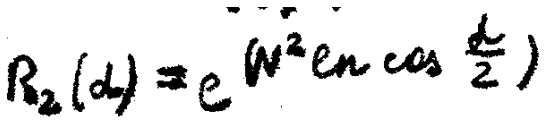
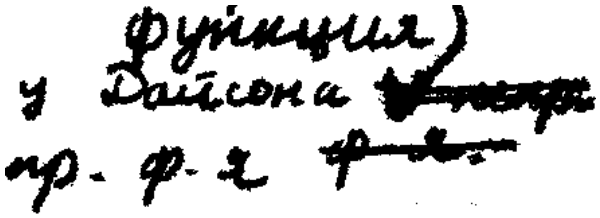


Diagram and typed description. The exponent is enclosed in round parentheses.

Original source details: GELFAND-P000067

Assumptions. 1) The distribution of eigenvalues is given by $\overline{\sigma_\alpha(\theta)}$. Under Dyson's assumpt.,

between $(-\alpha, \alpha)$ they are 0. (For $\forall$ matr., its own function.) Generalization — for Dyson [crossed out: an illegible word] any cont. funct. [crossed out: a short entry] ($N \to \infty$).



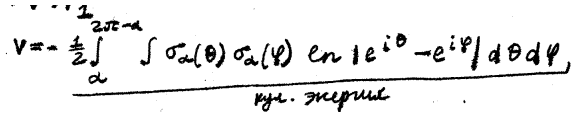
Uncertain reading. an illegible word and a short fragment

Original source details: GELFAND-P000067

2) $F = V + F_1$. def:

$$V = -\frac{1}{2}\int_{\alpha}^{2\pi-\alpha}\int \sigma_\alpha(\theta)\sigma_\alpha(\varphi)\ln|e^{i\theta} - e^{i\varphi}|\,d\theta\,d\varphi$$

Coul. energy



Manuscript annotation. Below the integral a long line turns up at the right and is labelled "Coul. energy".

Original source details: GELFAND-P000067

def:

$$F_1 = \int_{\alpha}^{2\pi-\alpha} \sigma_\alpha(\theta) f_\beta(\sigma_\alpha(\theta))\,d\theta$$

f_β — a certain funct., written out explicitly.

GELFAND-P000068

The assumpt. consists in

$$R_\beta(\alpha) = e^{\beta(F(\alpha)-F(0))}$$

where σ is a funct. giving the minimum of the functional F (with $\int_{\alpha}^{2\pi-\alpha}\sigma(\theta)d\theta = N$).

$$f_\beta = U_\beta(\sigma) - \beta^{-1}S_\beta(\sigma)$$

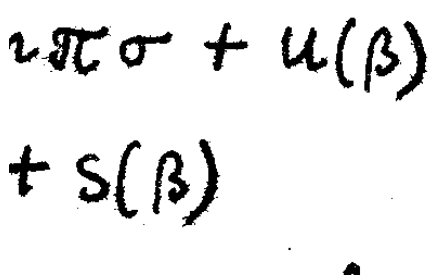
where [in U_β, the first term is crossed out: illegible]

Uncertain reading. illegible

Original source details: GELFAND-P000068

$$U_\beta(\sigma) = -\frac{1}{2}\ln \pi\sigma + u(\beta)$$
$$S_\beta(\sigma) = \ln \frac{N}{2\pi\sigma} + S(\beta)$$

Uncertain reading. In the terms independent of σ, $u(\beta)$ / $U(\beta)$ and $S(\beta)$ / $s(\beta)$.

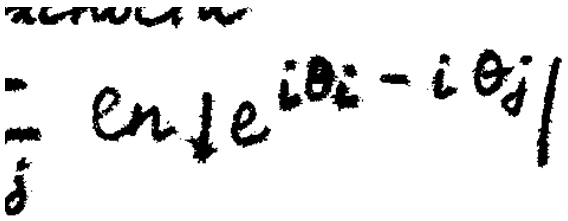


Original source details: GELFAND-P000068

U — potent. energy, S — entropy in the system. N charged points on a circle.

$$W = \sum_{i<j} \ln|e^{i\theta_i - i\theta_j}|$$

Manuscript notation. One e is written with exponent $i\theta_i - i\theta_j$.

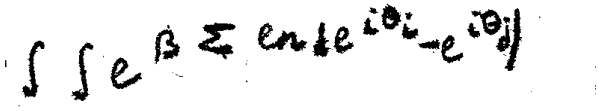


Original source details: GELFAND-P000068

$\beta = \frac{1}{T}$. T — temperature. The distribution of these points will be the same as the distribution of eigenval.

$$\int\int = \int\int e^{\beta \sum \ln|e^{i\theta_i} - e^{i\theta_j}|} \prod d\theta_j$$

Diagram and typed description. The exponent of the last integral reads $\beta \sum \ln|e^{i\theta_i} - e^{i\theta_j}|$; no minus is visible before β.

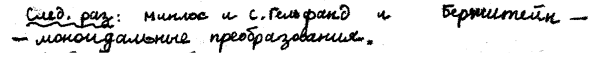


Original source details: GELFAND-P000068

The canonical $\int$ of statistical physics in the plane.

Next time: Minlos and S. Gelfand and Bernstein —
— monoidal transformations.

Manuscript notation. A dash ends the first line and another begins the second.



Original source details: GELFAND-P000068

GELFAND-P000069

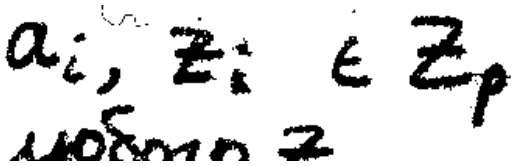
[Above the heading, only the lower strokes of a line are visible beneath a horizontal line.]

Uncertain reading. illegible

Original source details: GELFAND-P000069

5/IV. I. Bernstein. Grothendieck top. and monoidal transformations. Dioph. eq. $F(x) = 0$. F — polyn. with integer coeff., $x_1, \ldots, x_n$. Usually reduce mod p. Everything in Z_p.

$a_i, z_i \in Z_p$.



Manuscript notation. The small upper stroke belongs to the dot of the subscript i.

Original source details: GELFAND-P000069

If the sol. is known $\forall Z_p$, then there is inform. for any Z. Weil. Take F over the alg. closure of Z_p. Polyn. — surf. How many points over Z_p? $\mathcal{L}$ — closure of Z^p.

Manuscript notation. Here p is raised relative to Z. The nearby lines use Z_p.

Original source details: GELFAND-P000069

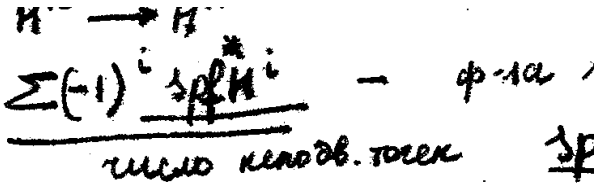
$x \longrightarrow x^p$ — homomorphism (since the field has char. p) (automorphism of the field $\mathcal{L}$). The fixed elem. of the base field: $x^p = x$ in the base field and only these (since $\leq p$ solutions).

$$(x_1, \ldots, x_n) \longrightarrow (x_1^p, \ldots, x_n^p)$$

How many fixed points? These will be in Z_p. The coeff. are fixed under the automorphism. The number of fixed points is found using the Lefschetz formula. $f : X \longrightarrow X$. X — top. space. If f is in gen. position, then one can find $H^n \longrightarrow H^n$.

$$\underbrace{\sum (-1)^i \operatorname{Sp} f^* H^i}_{\text{number of fixed points}}$$

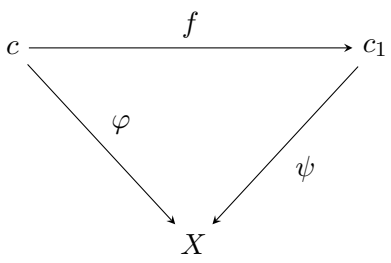
— Lefschetz formula!



Uncertain reading. The exponent over f is overwritten; the upper layer resembles *, the lower one n.

Original source details: GELFAND-P000069

Sp $f^* H^i$ (H^i — cohomol.). Closed sets — $\bigcup$ of a finite number of alg. submanifolds of smaller dimensions. (Zariski top.) It is non-Hausd. — generally bad. Grothendieck generalizes and improves. Definition. Grothendieck top. C — category, X — distinguished object (one). New categ. C_X — objects — $(c, \varphi : c \longrightarrow X)$, $c \in C$. Mapping $(c_1, \psi : c_1 \longrightarrow X)$.
pair



def commutative. [To the left of the diagonal arrow, a stroke is crossed out: illegible.]

Original source details: GELFAND-P000069

Uncertain reading. illegible

The elements of C_X are called open sets [Crossed out: unreadable sign; inserted above: (over)] X.

Original source details: GELFAND-P000069

Uncertain reading. Below "(over)", immediately before X, a sign is heavily cancelled.

A topology is introduced in X. This is no special case of ordinary top. — the reverse.

GELFAND-P000070

U — elem. of C_X — open sets. Fix a fam. of mappings
object

$$\{U_\alpha \underset{\alpha\in I}{\longrightarrow} U\}$$

There are terribly many such families. Some of these families are called coverings of U, if 1°. $\{U \xrightarrow{\text{id}} U\}$ [Crossed out and inserted above: is; the crossed-out word is illegible] a covering.

Original source details: GELFAND-P000070

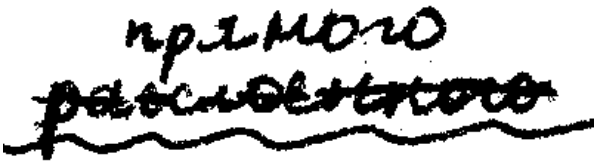
Uncertain reading. is; the crossed-out word is illegible

Example. X — space (top.). C — open sets, mapping — inclusion (if no incl., then none). C_X is almost C (incl. in X). Covering U_α — completely covers U. 2°. $\{U_\alpha \to U\}$ — cov. $\forall\alpha$ there is a covering $\{U_{\alpha\beta} \underset{\alpha\in I}{\to} U_\alpha\}$. Then $\{U_{\alpha\beta} \to U\}$ — also a covering. 3°. [Before "def." a symbol is crossed out: illegible.]

Uncertain reading. illegible

Original source details: GELFAND-P000070

Def. [crossed out: fibred; above: direct] product.

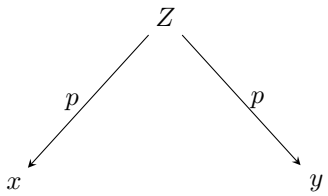


Uncertain reading. fibred

Original source details: GELFAND-P000070

[On the right:] $X \times Y$. A — categ., def. of product: $x, y \in A$. Product — object Z and

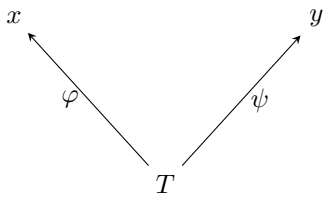
Uncertain reading. Both projections carry the same p-like sign; p / ρ, indices not established.

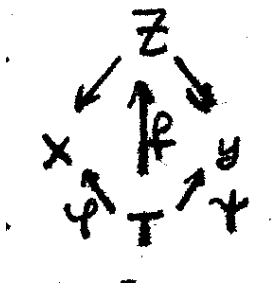


Original source details: GELFAND-P000070

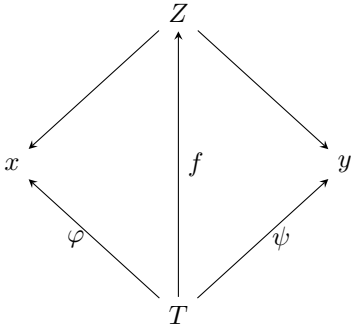
If $T \in A$, and φ, ψ are given

Diagram and typed description. The two arrows from T point to x, y; in the next diagram f points from T to Z.





then there exists a unique f



Original source details: GELFAND-P000070

(If it exists, then it is called a product.) [Crossed out: illegible.]

Uncertain reading. illegible

Original source details: GELFAND-P000070

The product is defined uniquely up to isomorphism $\overset{\longrightarrow}{\underset{1}{\longleftarrow}}$.

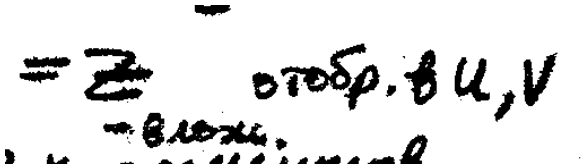
Diagram and typed description. Two opposing arrows, upper rightwards, lower leftwards; 1 below them.

Original source details: GELFAND-P000070

Example. X — top. space, elem. — open sets.

$$U \times V = U \cap V = Z$$

Mappings to U, V — incl.



Position of the note. "— incl." is below and to the right of Z, beneath "mappings to U, V".

Original source details: GELFAND-P000070

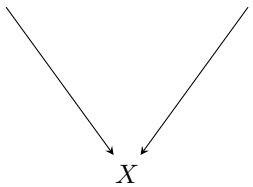
Ax. 3°. In categ. C_X there exists the product of any 2 elements. (Here this is the intersect. of $\forall$ 2 sets.)
Untangle in C.

Manuscript notation. The line reads Распутать в C [untangle in C].

Original source details: GELFAND-P000070

Look at the categ. of sets with all mappings (product rel. to X

Diagram and typed description. Two arrows with unlabelled upper objects meet at X.



Original source details: GELFAND-P000070

pairs (x, y) map to 1 element. (If X is a point, then the direct product.)

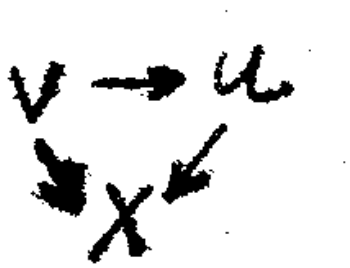
GELFAND-P000071

Ax. 4. (intersection is defined)

Given $f: V \to U$ (open sets) and a covering $\{U_\alpha \to U\}$.

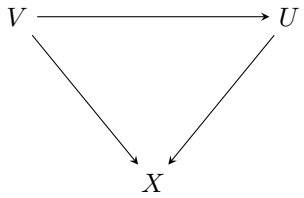
$$\{V \times_U U_\alpha \to V\} \quad \text{will also be a covering.}$$

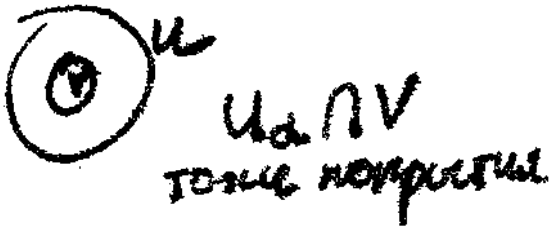
Product in categ. C_U.



Original source details: GELFAND-P000071

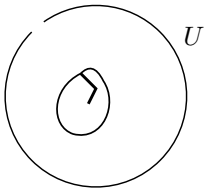
Diagram and typed description. An arrow $V \to U$; two oblique arrows from V and U meet at X. The arrows have no labels.





Original source details: GELFAND-P000071

Diagram and typed description. A circle with an external letter U; inside is a small oval with heavy strokes. No letter or numeral can be established.



$U_\alpha \cap V$

also coverings

Uncertain reading. At the extreme right edge, a separate vertical stroke resembling 1; a complete leaf number is not readable.

Original source details: GELFAND-P000071

Presheaf. $U \to P(U)$ (ab. group)

$$U \to V, \qquad P(U) \leftarrow P(V).$$

And ax. $1 \to 1$ and functor.

Not every presheaf is a sheaf. [Crossed out: This is a functor.]

Manuscript notation. The crossed-out phrase.

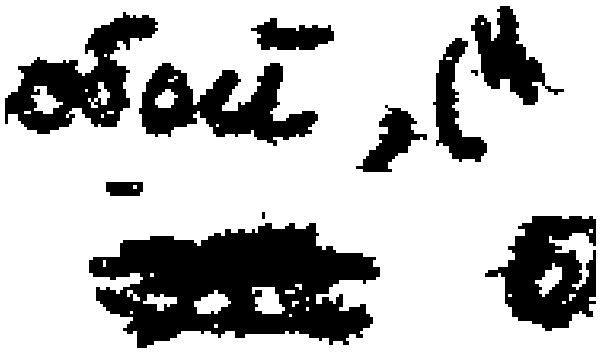
Original source details: GELFAND-P000071

Let there be a sect. $P(U)$ and a covering $\{U_\alpha \to U\}$. Then a mapping is def. $P(U) \to P(U_\alpha)$, hence $P(U) \to \prod P(U_\alpha)$.

Sheaf ax. — 1°. This is an embedding.

$$0 \to P(U) \to \prod P(U_\alpha).$$

2°. If $\xi_\alpha \in P(U_\alpha)$, $\xi_\beta \in P(U_\beta)$ are compat. with each other, [?],



Uncertain reading. after the comma, short upper strokes; before “images”, a crossed-out sign is illegible

Original source details: GELFAND-P000071

$$P(U_\alpha) \longrightarrow P(U_\alpha \times_U U_\beta) \longleftarrow P(U_\beta)$$

the images coincide, [At left: U_α-cov. U.]

Manuscript notation. The note in the left margin.

Original source details: GELFAND-P000071

then there exists [Inserted above: sect.] $\xi \in P(U)$ such that they are obtained from it.

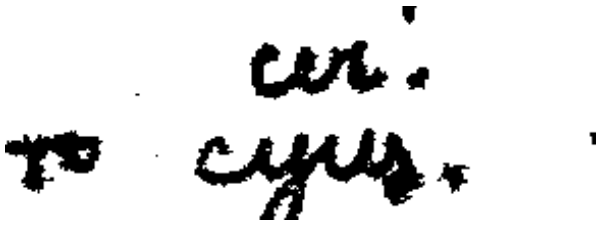
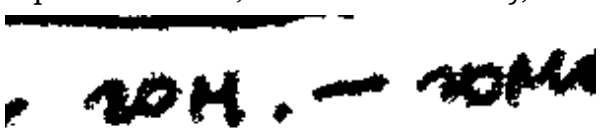


Diagram and typed description. Above “exists”, “sect.” is inserted.

Original source details: GELFAND-P000071

Top. over a point — unusual sheaves.

A point — a field, this is Galois theory, and hom.[?] — homol. of Galois groups.

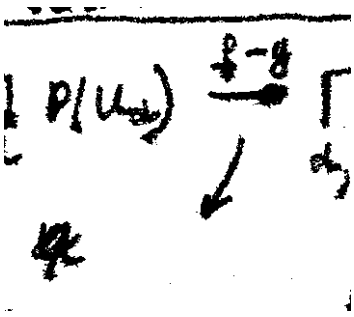


Uncertain reading. The first abbreviation reads гом. or гон.; the second reads гомол. [homol.].

Original source details: GELFAND-P000071

Sheaf ax. in the form of an exact seq.

$$0 \to P(U) \to \prod_{\alpha} P(U_\alpha) \overset{f-g}{\underset{\swarrow}{\rightrightarrows}} \prod_{\alpha,\beta} P(U_\alpha \times_U U_\beta).$$

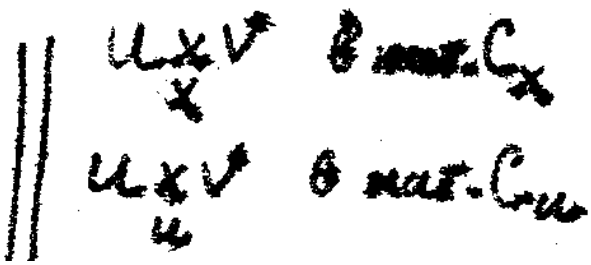


Uncertain reading. Below, at the end of the arrow pointing down and left, a densely crossed-out sign; the exact reading is not established.

Original source details: GELFAND-P000071

[In the right margin:]

$\| \; U \times_X V$ in [crossed out: illegible] C_X;
$\| \; U \times_U V$ in categ. C_U.



Uncertain reading. In the upper marginal note, between “in” and C_X, a densely overwritten fragment.

Original source details: GELFAND-P000071

2 mappings.

$$U_\alpha \times_U U_\beta \to U_\alpha,$$
$$U_\alpha \times U_\beta \to U_\beta.$$

Whence

$$P(U_\alpha) \to P(U_\alpha \times_U U_\beta),$$
$$P(U_\beta) \to P(U_\alpha \times_U U_\beta).$$

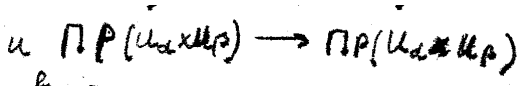
Therefore def. [?],

Uncertain reading. before $P(U_\alpha)$, densely crossed-out signs, probably a product sign

Original source details: GELFAND-P000071

$$\prod P(U_\alpha) \to \prod P(U_\alpha \times U_\beta),$$

and [?]:

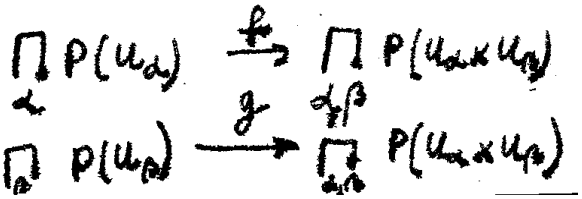


Uncertain reading. in the next line, both on the left and on the right, $\prod P(U_\alpha \times U_\beta)$ is visible

Original source details: GELFAND-P000071

$$\prod P(U_\alpha \times U_\beta) \to \prod P(U_\alpha \times U_\beta).$$

$$\prod_\alpha P(U_\alpha) \xrightarrow{f} \prod_{\alpha,\beta} P(U_\alpha \times U_\beta),$$
$$\prod_\beta P(U_\beta) \xrightarrow{g} \prod_{\alpha,\beta} P(U_\alpha \times U_\beta).$$



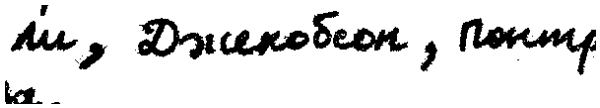
Manuscript notation. The product at right has the index α, β in both lines.

Original source details: GELFAND-P000071

GELFAND-P000072

Gindikin. Eigenval. on a symm. space.

Need to know roots. Sem. Sophus Lie, Jacobson, Pontryagin.

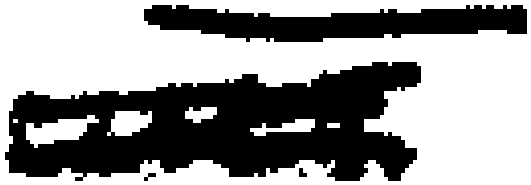


Manuscript notation. Jacobson is written Джекобсон.

Original source details: GELFAND-P000072

G. Topic: study of Dyson, $\forall$ symm. spaces.

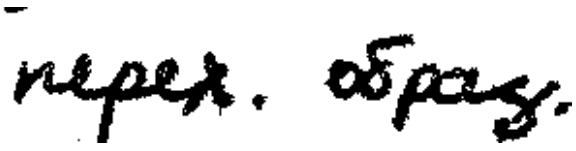
$12/\overline{\text{IV}}$. Stepin. Kirillov's probl. Construct an example [crossed out: an illegible word]



Uncertain reading. an illegible word

Original source details: GELFAND-P000072

of a dyn. syst. with cont. spectrum, dyadic-rat. numbers on the circle G. Every elem. — of finite order. (not with discr.) $g \to T_g$. Suff. to specify [перех.] images of $1/2^k$.



Manuscript notation. The manuscript reads перех. образ., both abbreviated.

Original source details: GELFAND-P000072

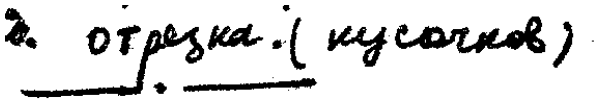
$$T^2_{1/2^k} = T_{1/2^{k-1}}.$$

$T_0 = E$, T_1 — [?], refl.

Original source details: GELFAND-P000072

Uncertain reading. a crossed-out word resembling “rotation” / “refl.”

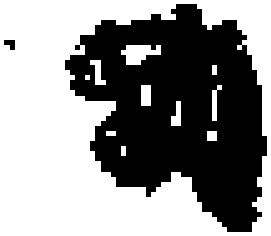
Next, rearr. of the segment. (pieces) preserving orientation.



Original source details: GELFAND-P000072

Manuscript notation. The underlined word отрезка is singular; a full stop and “(pieces)” follow.

[On the right:] $1/2^k \to T_k$.

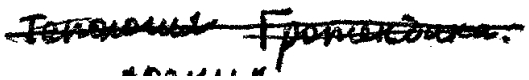


Original source details: GELFAND-P000072

Uncertain reading. At the left before the heading, a densely overwritten sign; the exact reading is not established.

What is it desirable for participants in the seminar to know?

1. Representations. Lie groups and Lie algebras. Symmetric spaces. Simple Lie groups.
2. [Crossed out: [?] Grothendieck’s[?].]



Original source details: GELFAND-P000072

Uncertain reading. The first word is unresolved; the second resembles Гротендика [Grothendieck’s].

   Sheaves. Homol. algebra (fundamentals). Vector bundles.
3. Elements of algebraic geometry.
4. Quantum mechanics (nonrelativ.). Cauchy $\int$ of several variables according to Gindikin will be incl.
5. [At the left, densely crossed out: an illegible sign.]

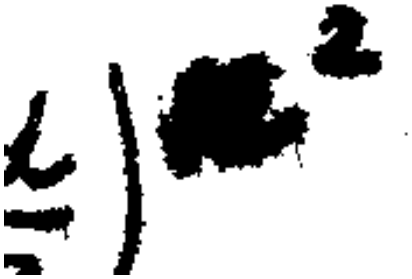
Original source details: GELFAND-P000072

Uncertain reading. an illegible sign

   Nuclear spaces and measures (Wiener measure).

S. Gelfand. $R_2(\alpha)$ calculated exactly.

$$R_2(\alpha) = \left(\cos\frac{\alpha}{2}\right)^{[?]^2}.$$



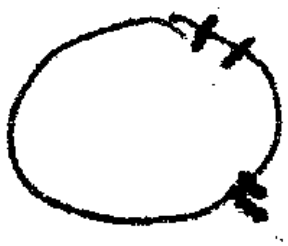
Original source details: GELFAND-P000072

Uncertain reading. The letter under the superscript 2 is overwritten. A possible reading of the exponent is n^2.

(This is asserted, but not proved.) This probl. is replaced by another for which this is the exact answer.

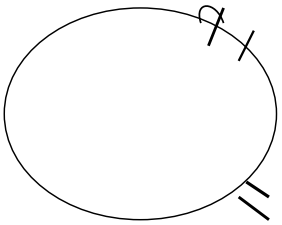
Dyson's 3rd paper. $S(\theta)$.

$$(\theta_1, \theta_1 + d\theta_1), \qquad (\theta_2, \theta_2 + d\theta_2).$$



Original source details: GELFAND-P000072

Diagram and typed description. A circle; at the upper right, two short transverse strokes and a short curved stroke; at the lower right, a heavy mark pointing outward. The marks have no labels.



Prob. that there is an eigenval. in the 1st interval, one in the 2nd, and none between. This is

$$F(\theta_1, \theta_2) d\theta_1 d\theta_2.$$

It depends on the difference.

$$F(\theta_1; \theta_2) = S(\theta_1 - \theta_2).$$

Original source details: GELFAND-P000072

Uncertain reading. An initial stroke like d and a final low 2 remain visible through the cancellation; the middle is illegible.

This ~~$S(\theta)$~~ —

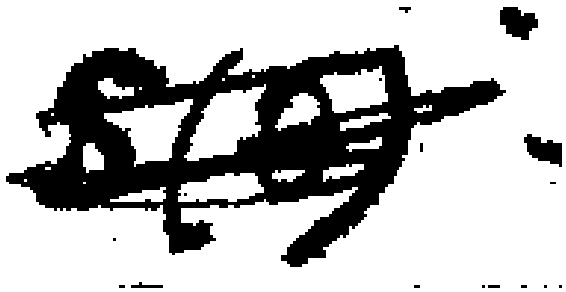


Diagram and typed description. At the bottom, $S(\theta)$ is crossed out.

Original source details: GELFAND-P000072

[On the right:] easy to prove.

$$S(\theta) = \frac{d^2 R}{d\theta^2}.$$

GELFAND-P000073

$R(2\alpha)$ — prob. that there are no eigenval. in $(-\alpha, \alpha)$.

Uncertain reading. The first R has a lower stroke. A separate 1, as in the second line, cannot be established.

Original source details: GELFAND-P000073

$$R_1(2\alpha) = \det \left| \delta_{pq} - \frac{1}{\pi} \int_{-\alpha}^{\alpha} \cos p\theta \cos q\theta \, d\theta \right|,$$
$$p, q = \frac{1}{2}, \ldots, m - \frac{1}{2}$$
$$2m \geq n$$

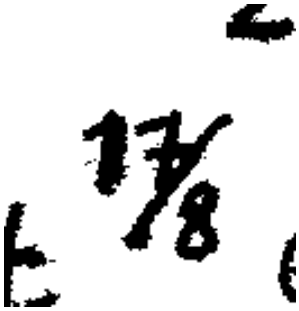
Uncertain reading. The relation sign: $2m \geq n$ / $2m = n$; its upper stroke slopes and the sign is heavily written.

Original source details: GELFAND-P000073

Asymptotics

$$R_1(t) \sim At^{17/8} \exp\left[-\frac{1}{4}t^2 - \frac{1}{2}t\right]$$

$$\underline{\underline{t = \alpha n}}.$$



Manuscript notation. The exponent on t reads $17/8$. Below the formula $t = \alpha n$ is double underlined.

Original source details: GELFAND-P000073

And other results (analogous).

Minlos. Elems. of the matrix (Hermit.) — random numbers with Gauss. distr.

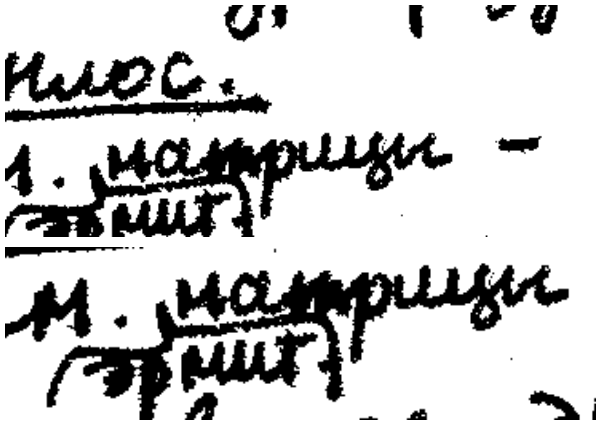


Diagram and typed description. Below the word “matrix”, “(Hermit.)” is inserted.

Original source details: GELFAND-P000073

Brownian motion.

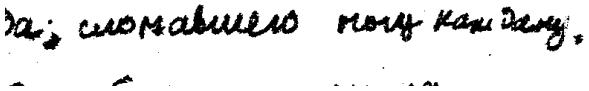
Diagram and typed description. Two horizontal strokes with a point between them.

Original source details: GELFAND-P000073

——— • ———

What is the angle of k-dim. subspaces in an n-dim. space?

Kazhdan. Res. of Gromov from Leningrad; who broke his leg while running.



Manuscript notation. The final words are сломавшего ногу на бегу.

Original source details: GELFAND-P000073

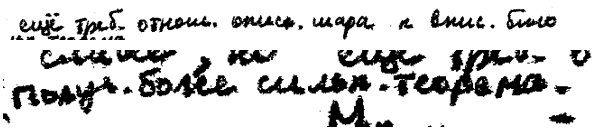
Dvoretzky’s res. $\forall \varepsilon, k\ \exists$ a space of a suff. large number of dims., such that $\forall$ [crossed out: an illegible word beneath several heavy strokes]

Uncertain reading. an illegible word beneath several heavy strokes

Original source details: GELFAND-P000073

convex centrally symm. body ∃ a section almost a ball (k-dimensional). Dvoretzky did it head-on (70 pp.). (In a space of a large number of dims., there is almost a circle in a section.)

Gromov is weaker, but also req. the ratio of the circumscr. ball to the inscr. to be



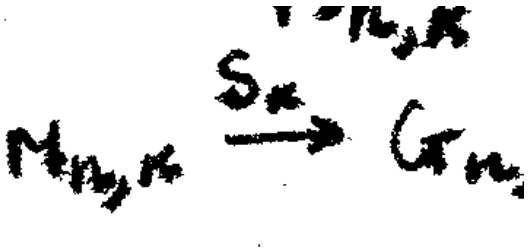
Manuscript notation. The line ends at было [to be]; “a stronger theorem is obt.” is inserted below.

Original source details: GELFAND-P000073

[Inserted below:] a stronger theorem is obt.

Def. $G_{n,k}$. $M_{n,k}$ — a k-dim. plane and a point of its unit sphere.

$$M_{n,k} \xrightarrow{S_k} G_{n,k}.$$

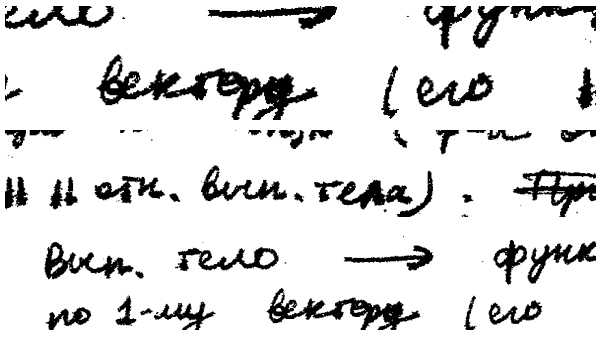


Uncertain reading. Above the arrow, S_k; the small lower index is heavily written.

Original source details: GELFAND-P000073

A bundle of spheres over $G_{n,k}$ (Grassm.).

Convex body $\to$ a function on $M_{n,k}$ (func. determ. by the 1st vector (its $\|\ \|$ rel. to the convex body).



Manuscript notation. Two pairs of vertical strokes and the abbreviation отн. [rel. to] are visible.

Original source details: GELFAND-P000073

[On the right, crossed out: a short word beginning with “П”; the remainder is illegible.]

Uncertain reading. a short word beginning with “П”; the remainder is illegible

Original source details: GELFAND-P000073

Let us expand it in sph. funcs. If n is large, then ∃ a fibre such that the first ℓ funcs. vanish (except 1). ∀ repr. of the unit. group ∃n such that ∀ funcs. on $M_{n,k}$ ∃ a fibre such that the proj. $= 0$. Conv. — compactness of the cl. of funcs.,

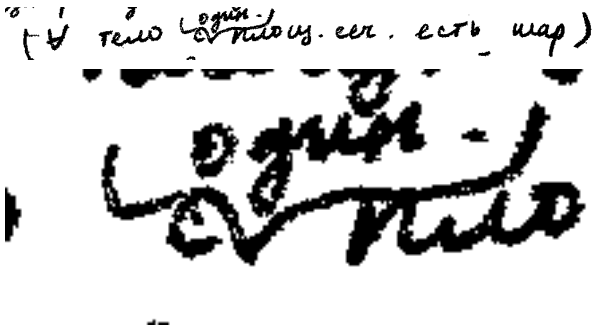
Uncertain reading. a densely overwritten sign; the exact reading is not established.

Original source details: GELFAND-P000073

Fourier coeffs. — many 0.

GELFAND-P000074

Kirillov. A convex body (centrally symm.) in R^3 (R^n). Draw plane sections through the centre. The areas of all its sects. are known. The body is recovered from the areas of the sections! ($\forall$ body with (equal) sect. areas is a ball)



Manuscript notation. один. [equal] is inserted above с [with].

Original source details: GELFAND-P000074

Consider S^2 and a func. on it. Can it be recovered if the $\int$ over great circles is known (even func.) This probl. is solved triv. using expansion in sph. funcs.

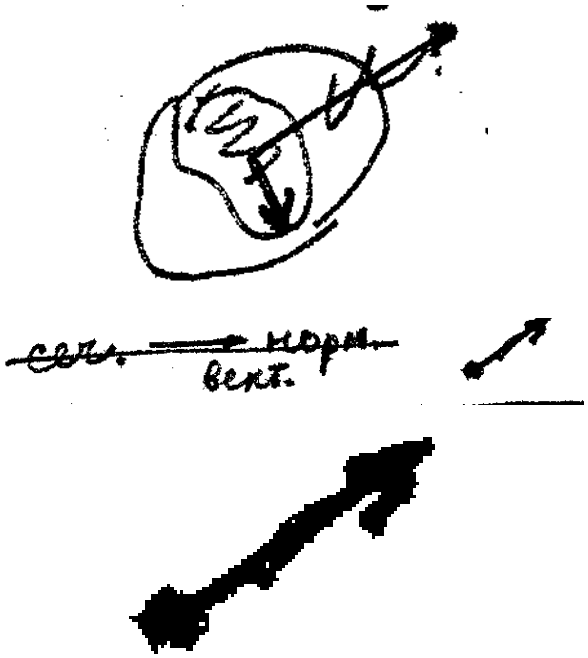
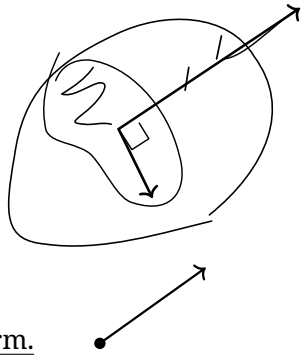
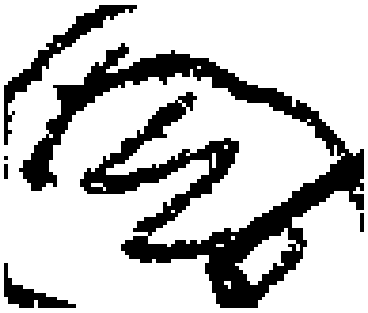


Diagram and typed description. An outer oval and an irregular inner contour; an oblique vector pointing upward to the right, a short downward arrow, an angle mark and short transverse strokes. Beside them, “sect.” and “norm.” are crossed out; “vect.” is inserted below the latter. Farther right, a separate oblique arrow has a heavy mark at its lower end.



sect. $\longrightarrow$ norm.
vect.

Original source details: GELFAND-P000074



Original source details: GELFAND-P000074

Uncertain reading. Inside the inner contour, a winding sign; its exact reading as a letter is not established.

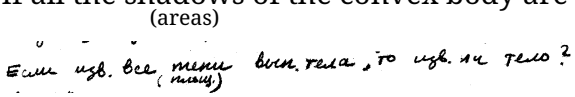
On the sphere, a vect. of the same direction. A func. on the sphere $\to$ [crossed out: an illegible entry beneath several heavy strokes]

Original source details: GELFAND-P000074

Uncertain reading. an illegible entry beneath several heavy strokes

$r^2/2$ (so that the $\int$ equals the area). A convex body $K \longrightarrow$ a func. on the sphere. We obtain a probl. of $\int$ geometry.

If all the shadows of the convex body are known, is the body known?



Original source details: GELFAND-P000074

Manuscript notation. "(areas)" is below "shadows".

Reduces.

19/IV. Faddeev. Ludwig (the younger). What is the characteristic determ. of the Sturm–Liouville oper.? On the half-line. (For a matrix — zeros — eigenval., Fredholm det.) A — a matrix, $w(\lambda) = \det(A - \lambda E)$.

Original source details: GELFAND-P000074

Diagram and typed description. Notation for the characteristic determinant: $w(\lambda)$.

Then

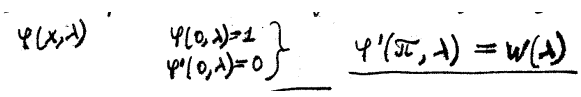
$$\frac{d}{d\lambda} \ln w(\lambda) = -\operatorname{Tr}(A - \lambda E)^{-1}.$$

(Check using eigenval.) Determ. $w(\lambda)$ up to mult const. The same in $\int$ eqns.

$$Ly = -y'' + q(x)y, \qquad y'(0) = y'(\pi) = 0.$$

The resolvent — an $\int$ oper. with a sm. kernel and the trace exists. The determ. — an entire func. of λ and its zeros — eigenval. of the St.–Liouv. probl.

$$\psi(x, \lambda), \qquad \left.\begin{matrix} \psi(0, \lambda) = 1 \\ \psi'(0, \lambda) = 0 \end{matrix}\right\} \quad \underline{\psi'(\pi, \lambda) = w(\lambda).}$$



Original source details: GELFAND-P000074

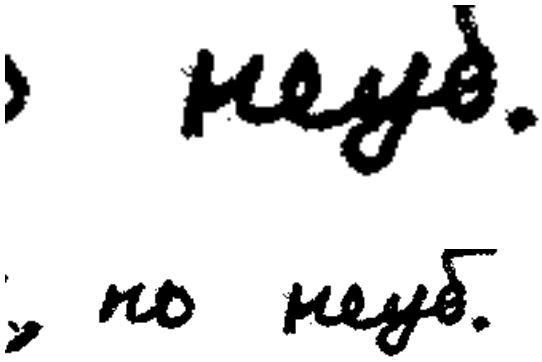
Manuscript notation. The same letter ψ occurs four times; the lower initial condition and the value at π have primes.

Original source details: GELFAND-P000074

Diagram and typed description. Two short horizontal strokes with a point between them.

———— • ————

$y(0) = 0$, $q(x) \to 0$, $x \to \infty$ suff. rapidly. The res. has no trace; the resolvent does not decay, but the non-dec.



Manuscript notation. The line after the comma reads но неуб. [but the non-dec.]; the next line begins “part”.

Original source details: GELFAND-P000074

the part has a triv. form. Regularize it.

$$r(x, x, \lambda) = \frac{1}{2i\sqrt{\lambda}} - O(1).$$

Manuscript notation. A minus precedes $O(1)$. The double rule at right separates the marginal text.

Original source details: GELFAND-P000074

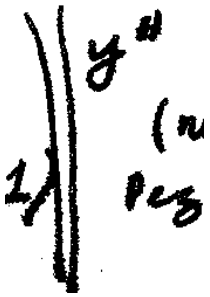
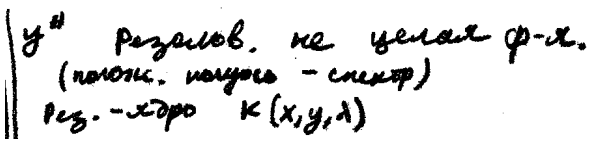


Diagram and typed description. In the right margin, a double almost vertical stroke and y''.

Original source details: GELFAND-P000074

y''

The resolvent is not an entire func. (pos. half-axis — spectrum) Res.-kernel $K(x, y, \lambda)$



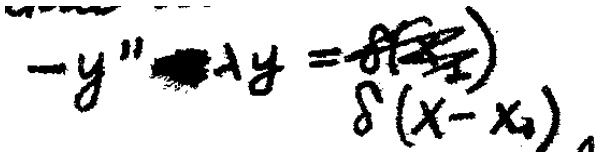
Manuscript notation. At the right of the double rule the parenthesis reads “pos. half-axis — spectrum”.

Original source details: GELFAND-P000074

GELFAND-P000075

What singularities?

$$-y'' \text{ [?] } \lambda y = \text{[crossed out: } \delta([?])\text{]}.$$
$$\delta(x-x_0)$$



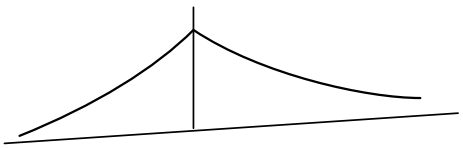
Uncertain reading. The sign before λy is overwritten. A minus is possible. Below the crossed-out right-hand side is $\delta(x - x_0)$.

Original source details: GELFAND-P000075

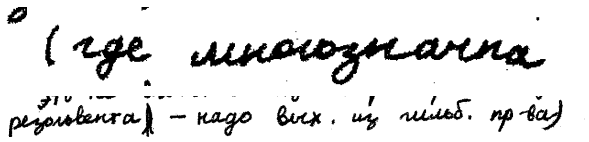
It will be the resolvent. There are 2 lin. indep. sols. So as not to grow. The 1st deriv. must have a jump.

Diagram and typed description. The horizontal axis rises slightly to the right. Two branches meet at a peak above it; the left rises, the right decreases. A vertical segment goes from the axis through the peak and a little above it. There are no labels.

Original source details: GELFAND-P000075



$\lambda > 0$ does not work (sin and cos), $e^{\pm\sqrt{-\lambda}\,x}$. When crossing the axis, $\sqrt{\ }$ changes sign.
This does not fit into the general theor. of operators. (where the resolvent is multivalued) — one must go outside the Hilbert space)



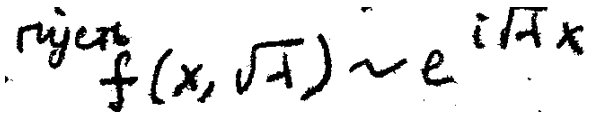
Manuscript notation. The manuscript has closing parentheses after “resolvent” and “space”.

Original source details: GELFAND-P000075

2-dim. case: ln (many-sheeted).

For the cont. spectrum

$$\overset{\text{let}}{f}(x, \sqrt{\lambda}) \sim e^{i\sqrt{\lambda}x}, \qquad x \to \infty.$$



Manuscript notation. пусть [let] sits above the beginning of the formula.

Original source details: GELFAND-P000075

Then $W(\lambda) = f(0, \sqrt{\lambda})$ (in the phys. lit., the Jost func.).

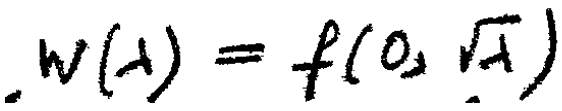
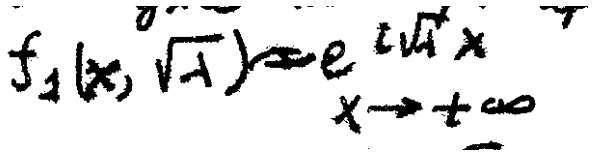


Diagram and typed description. Notation in the line with the Jost function: $W(\lambda)$.

Original source details: GELFAND-P000075

On the whole axis, it is no longer expressed through 1 solution.

$$f_1(x,\sqrt{\lambda}) \simeq e^{i\sqrt{\lambda}x}, \quad x \to +\infty,$$
$$f_2(x,\sqrt{\lambda}) \sim e^{-i\sqrt{\lambda}x}, \quad x \to -\infty.$$



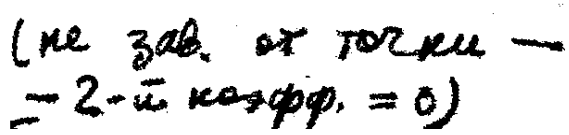
Uncertain reading. The first relation has a wave and a lower stroke, resembling $\simeq$. The second has a single wave.

Original source details: GELFAND-P000075

Their Wronskian — is the determ.

$$f_1' f_2 - f_1 f_2' = W(\lambda).$$

(does not depend on the point —
— 2nd coeff. $= 0$)



Manuscript notation. In the note at right, a dash ends the first line and another begins the second.

Original source details: GELFAND-P000075

How, knowing the determ., to write an identity for eigenval.

$$\frac{1}{2\pi i} \oint h(\lambda) \frac{W'(\lambda)}{W(\lambda)}\, d\lambda =$$

det — an entire func. of λ, discr. 0 on the $+$ axis. For good $h(\lambda)$

$$= \sum h(\lambda_i) \quad \text{over zeros } \lambda_i \text{ of } W(\lambda).$$

$h(\lambda) = \lambda^s$.

Uncertain reading. Below $h(\lambda) = \lambda^s$, a short expression is crossed out: the first sign is 1 / λ, followed by poorly distinguishable strokes.

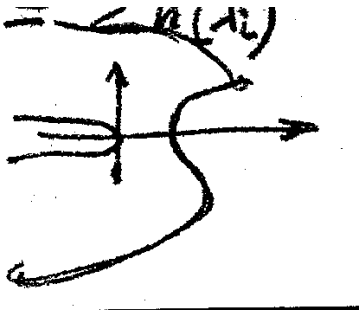
Original source details: GELFAND-P000075

[Before the next line, densely crossed out: a short entry; its exact reading is not established.]

Uncertain reading. a short entry; its exact reading is not established

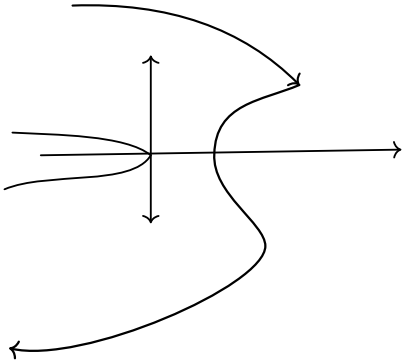
Original source details: GELFAND-P000075

Deform the contour, we obtain:



Original source details: GELFAND-P000075

Diagram and typed description. Axes with arrows upward, downward and to the right; on the left, two short lines merge at the origin. The upper curve goes downward to the right to an arrow; from its end, an S-shaped line bends to the right of the origin and goes downward to the left to another arrow. There are no labels.

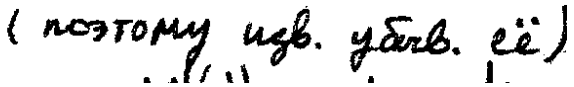


GELFAND-P000076

$$\int_{-\infty}^{0} |\lambda|^s \frac{W'(\lambda)}{W(\lambda)}\, d\lambda = \sum_i \lambda_i^s.$$

(Re s, so that the $\int$ converges.) And its analyt. cont. automatically regularizes it. ($\int$ — the jump.)

W has known asymptotics (therefore its decay is known).



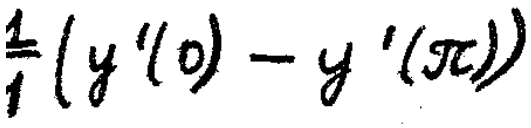
Original source details: GELFAND-P000076

Manuscript notation. The parenthesis reads поэтому изв. убыв. её.

$$\frac{W'(\lambda)}{W(\lambda)} = d_1 - \frac{d_2}{\sqrt{\lambda}} - \frac{d_3}{\lambda} - \dots$$

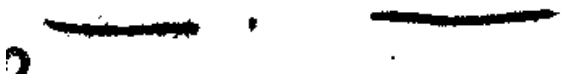
s is taken negative and then it is proved that the left and right sides — are analyt. funcs. of s.

$$\sum_n (\lambda_n - n^2 - c_0) = \frac{1}{4}\big(y'(0) - y'(\pi)\big).$$



Original source details: GELFAND-P000076

Manuscript notation. On the right the letter y has a prime at both endpoints.



Original source details: GELFAND-P000076

Diagram and typed description. Two short separating strokes with a point between them.

How to calculate the trace?

$$\sum \lambda_n = \int K(x,x)\, dx.$$

Diverges. (Sum of diag. elems.) Diag. elems.

$$-\delta''(x_1 - x_2) + q(x)\delta(x_1 - x_2).$$

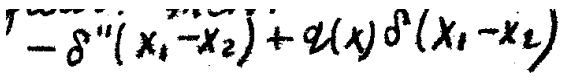
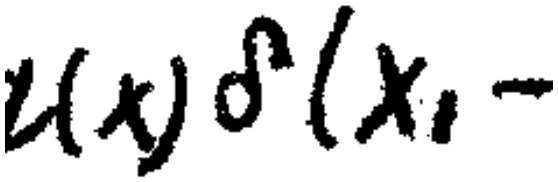


Diagram and typed description. In the diagonal element, the first δ has two primes; the second δ has none.



Original source details: GELFAND-P000076

$x_1 = x_2$: nothing is calculated. $-y''$ has eigenval. n^2!! Subtract

$$\sum(\lambda_n - n^2) = \operatorname{Sp} q(x)\delta(x_1 - x_2) = \delta(0)\int_0^\pi q(x)\,dx.$$

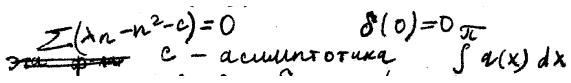
$-c_0$, so that $\int q(x)$ [crossed out: a sign resembling d and densely overwritten signs] $= 0$.

Uncertain reading. a sign resembling d and densely overwritten signs

Original source details: GELFAND-P000076

$$\sum(\lambda_n - n^2 - c) = 0, \qquad \delta(0) = 0.$$

c — asymptotics. [Separately on the right:] $\int^\pi q(x)\,dx$.



Manuscript notation. The separate integral at right has only an upper limit π. The preceding line reads $\delta(0) = 0$.

Original source details: GELFAND-P000076

[On the left, crossed out: “this is a func.” [это ф-я] / a short illegible entry.]

Uncertain reading. “this is a func.” [это ф-я] / a short illegible entry

Original source details: GELFAND-P000076

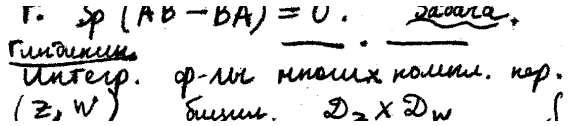
True if there is a δ-func. everywhere (periodic conds.). At the end, incomplete δ-funcs.; there one obtains $1/4$. G. $\operatorname{Sp}(AB - BA) = 0$. Problem.

Diagram and typed description. Before Gindikin’s name, two short separating strokes with a point between them.

Original source details: GELFAND-P000076

Gindikin.

Integral formulas of several complex vars. (z, w): a bicyl. $D_z \times D_w$ $\int$ formula



Manuscript notation. The heading uses the plural ф-лы. The next line uses the singular ф-ла beside the integral sign.

Original source details: GELFAND-P000076

$$f(z,w) = \frac{1}{(2\pi i)^2} \iint_{\Omega} \frac{f(\zeta,\eta)\, d\zeta\, d\eta}{(\zeta - z)(\eta - w)}.$$

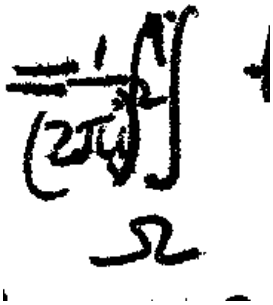
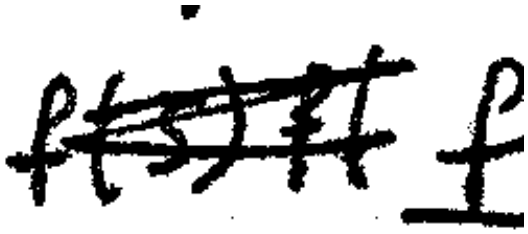


Diagram and typed description. A double integral with lower notation Ω and factor $1/(2\pi i)^2$.

Original source details: GELFAND-P000076

[In the numerator before $f(\zeta,\eta)$, crossed out: f with a parenthesis; the arguments and following signs are densely overwritten.]

Uncertain reading. f with a parenthesis; the arguments and following signs are densely overwritten



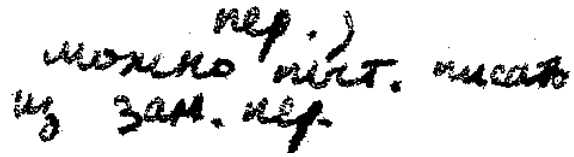
Original source details: GELFAND-P000076

The next interesting example — Weil domains. (generaliz. bicyl.) There is a set of funcs. $\chi_i(z,w)$, $i = 1,\dots,\ell$ — holom. And D_i are specified in the complex plane (1-dim.). A Weil domain: $(z,w)\colon \chi_i(z,w) \in D_i$. 2 funcs., z and w, $z \in D_z$, $w \in D_w$ — a bicylinder!

GELFAND-P000077

Let there be two of them, χ_1, χ_2 — a skew bicylinder. This is simply a bicyl. in other coords.! (χ_1, χ_2 — new vars.).

[At right:] the int. can be written from a change of vars.

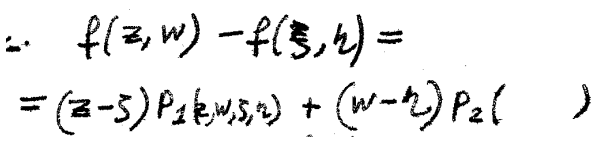


Manuscript notation. The note in the right margin.

Original source details: GELFAND-P000077

Formula: Weil.

$$f(z,w) - f(\zeta,\eta) =$$
$$= (z-\zeta)P_1(z,w,\zeta,\eta) + (w-\eta)P_2(\,)$$



Manuscript notation. Both lines have an equals sign. The parentheses after P_2 are empty.

Original source details: GELFAND-P000077

Lem. There is a decomposition.

But this is nontriv., [crossed out: a short entry beneath dense horizontal strokes; its reading is undetermined]

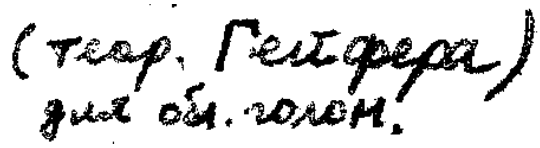
Uncertain reading. a short entry beneath dense horizontal strokes; its reading is undetermined

Original source details: GELFAND-P000077

— possible in any domain of holomorphy.

$$\forall D \quad \begin{matrix}(z,w) \in D \\ (\zeta,\eta) \in D\end{matrix}$$

[At right:] (Hefer's theor.) for a domain of holom. (Cartan's theor. B)



Manuscript notation. The right-hand note names Hefer (Гейфера).

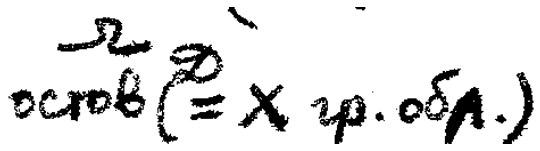
Original source details: GELFAND-P000077

Apply to χ_i:

$$\chi_i(z,w) - \chi_i(\zeta,\eta) = (z-\zeta)P_1^i(\,) + (w-\eta)P_2^i(\,).$$

$$f(z,w) = \frac{1}{(2\pi i)^2}\int_\Omega \frac{(P_1^1P_2^2 - P_1^2P_2^1)f(\zeta,\eta)\,d\zeta\,d\eta}{(\chi_1(\zeta,\eta)-\chi_1(z,w))(\chi_2(\zeta,\eta)-\chi_2(z,w))}.$$

skeleton ($\overset{\mathcal{D}}{=}$ × bdy. of dom.)



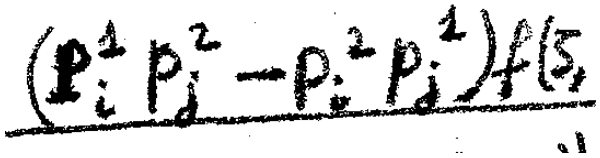
Uncertain reading. The mark above the beginning of the parenthesis resembles $\mathcal{D}$; its attachment is uncertain.

Original source details: GELFAND-P000077

If there are ℓ funcs., then the formula has the form

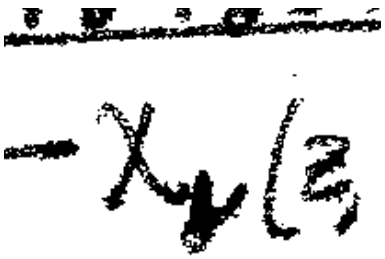
$\chi_1,\dots,\chi_\ell$

$$f(z,w) = \frac{1}{(2\pi i)^2} \sum_{i,j} \int_{\Omega_{ij}} \frac{(P_i^1 P_j^2 - P_i^2 P_j^1) f(\zeta,\eta)\, d\zeta\, d\eta}{(\chi_i(\zeta,\eta) - \chi_{i?}(z,w))(\chi_j(\zeta,\eta) - \chi_j(z,w))}.$$



Original source details: GELFAND-P000077

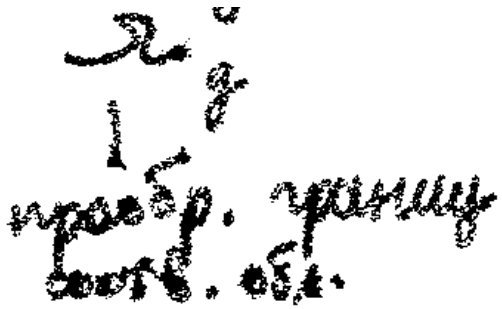
Diagram and typed description. In the numerator: $P_i^1 P_j^2 - P_i^2 P_j^1$; $1, 2$ above, i, j below.



Original source details: GELFAND-P000077

Uncertain reading. The lower index of the right-hand χ in the first denominator factor: i / j / a densely overwritten illegible sign.

Ω_{ij}
$\downarrow$
preimage of the boundary
of the corresp. dom.



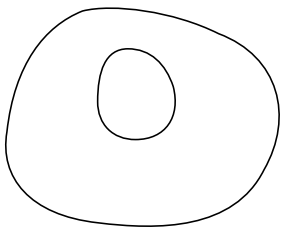
Original source details: GELFAND-P000077

Diagram and typed description. A downward arrow from Ω_{ij} to “preimage of the boundary of the corresp. dom.”.

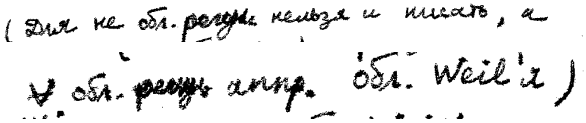
Domain $\chi_i(z,w) \in D_i$.

Original source details: GELFAND-P000077

Diagram and typed description. Two closed rounded contours, the small one inside the large one; without labels or arrows.



Decomp. into a $\sum$ of funcs. analyt. in larger doms.
(For non-dom. [?] one cannot even write, whereas $\forall$ dom. [?] is approx. by Weil doms.)



Original source details: GELFAND-P000077

Uncertain reading. The two overwritten abbreviations following “dom.” remain undeciphered.

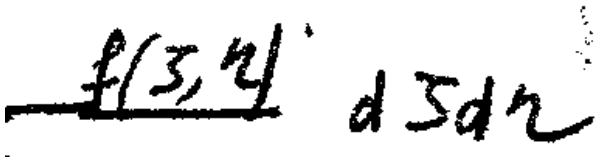
Gindikin: A contin. analogue of a Weil dom. Dom. $\chi(z,w,\alpha) \in D_\alpha$.

$$\chi(\alpha,z,w) - \chi(\alpha,\zeta,\eta) = (z-\zeta)P_1(\alpha,z,w,\zeta,\eta) + (w-\eta)P_2(\alpha,\ldots).$$

Smooth in α.

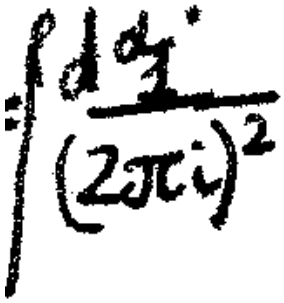
GELFAND-P000078

$$f(z,w) = \int \frac{d\alpha}{(2\pi i)^2} \int \frac{P_1(\alpha\ldots)P_2'(\ ) - P_2P_1' \quad f(\zeta,\eta)}{(\chi(\alpha,\zeta.\ - \chi(\alpha,z,\ldots)}\, d\zeta\, d\eta$$



Manuscript notation. The fraction bar extends under $f(\zeta,\eta)$; the differentials stand to its right.

Original source details: GELFAND-P000078



Uncertain reading. A separate dot appears at upper right of $d\alpha$; its meaning is uncertain.

Original source details: GELFAND-P000078

[Above the numerator:] Wronskian det. (Derivs. in α

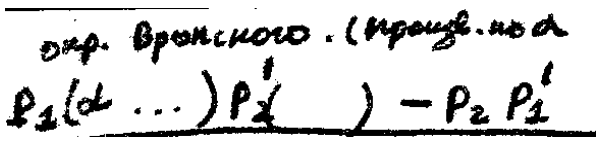


Diagram and typed description. Above the numerator,"Wronskian det. (Derivs. in α"; primes at P_2 and the last P_1.

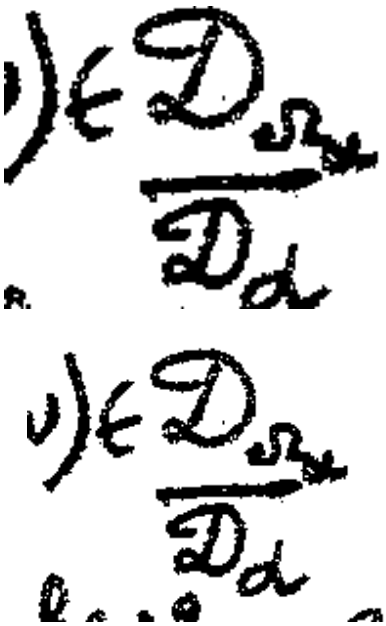
Original source details: GELFAND-P000078

Uncertain reading. One χ difference is visible. A small dot follows the first ζ, with no visible closing parenthesis before the minus. One parenthesis closes after the last ellipsis.

Original source details: GELFAND-P000078

Some new conds. on the dom. (general position, etc.).

$$\chi(\alpha, z, w) \in D_{[?]}$$
$$\overline{\qquad\qquad}$$
$$D_\alpha$$



Uncertain reading. The subscript on the upper D remains uncertain. The lower D_α is clear.

Original source details: GELFAND-P000078

∫ already over the entire boundary.

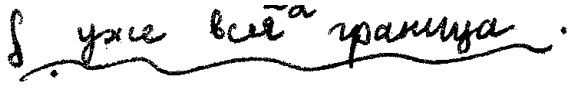
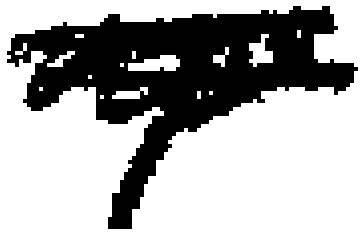


Diagram and typed description. A wavy underline beneath "∫ already over the entire boundary.".

Original source details: GELFAND-P000078

[At right:] In a partic. case [crossed out: a short entry beneath two dense horizontal strokes, with a lower slanting stroke; its reading is undetermined]

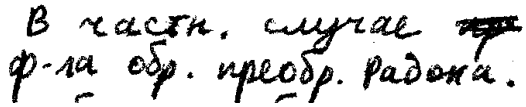
Uncertain reading. a short entry beneath two dense horizontal strokes, with a lower slanting stroke; its reading is undetermined



Original source details: GELFAND-P000078

the formula for the inverse Radon transform.

Manuscript notation. The phrase on the Radon formula occupies two lines in the right margin.

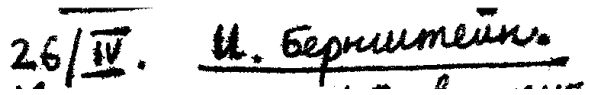


Original source details: GELFAND-P000078

χ_α — linear, then tube doms.

GELFAND-P000079

[The upper part of the sheet is without text.] $26/\overline{\overline{\text{IV}}}$. I. Bernstein.



Original source details: GELFAND-P000079

Diagram and typed description. The date $26/\text{IV}$ with two strokes above IV and one below; the name is underlined separately.

Cohom. with coeffs. in a sheaf in Grothendieck top. All coverings are fixed. In categ. C, $\forall$ 3 objects X, Y, Z, $\exists T = Z \times_X Y$.

Categ. of sets.

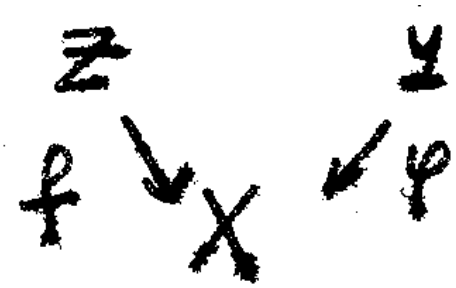
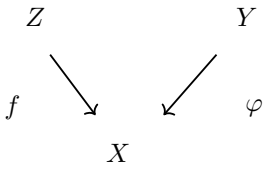
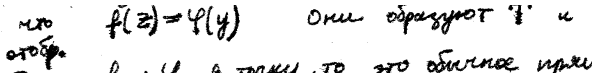


Diagram and typed description. Z and Y above X; two slanting arrows toward X, f on the left, φ on the right.



Original source details: GELFAND-P000079

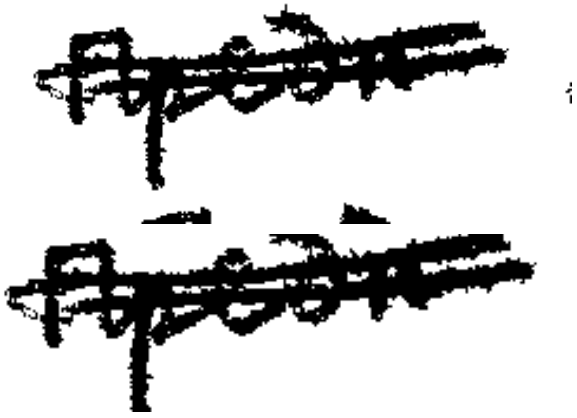
T: take $Z \times Y$ and in it those pairs (z, y) such that $f(z) = \varphi(y)$. They form T and there are [Below the line at left: maps.]



Manuscript notation. "Maps" sits at left, between the line about the pairs and the line beginning "If".

Original source details: GELFAND-P000079

If f and φ are to a point, this is the ordinary direct prod. If f and φ are embeddings in X, then $Z \times_X Y = Z \cap Y$ in X. [Crossed out: "Pres..." [Предп...] / a short entry beneath dense horizontal strokes; the ending is undetermined.]



Uncertain reading. "Pres..." [Предп...] / a short entry beneath dense horizontal strokes; the ending is undetermined

Original source details: GELFAND-P000079

Open sets — a category. A presheaf S — a contravar. functor from this categ. to the categ. of ab. groups.

$$U \to X$$
$$\Gamma(U)$$

A substantive example — the topology of a point (in fact of a field).

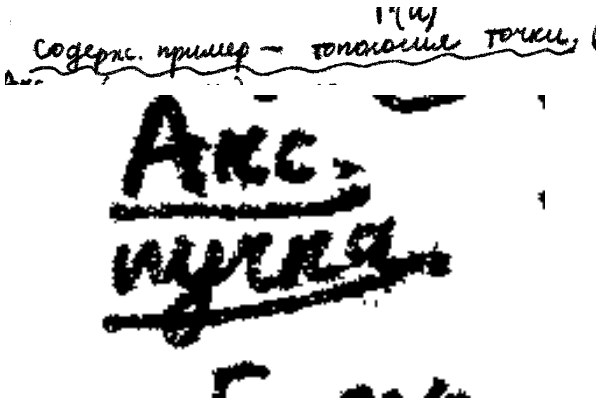


Diagram and typed description. A wavy underline beneath “A substantive example — the topology of a point”; “Ax.” and “of a sheaf.” are underlined separately.

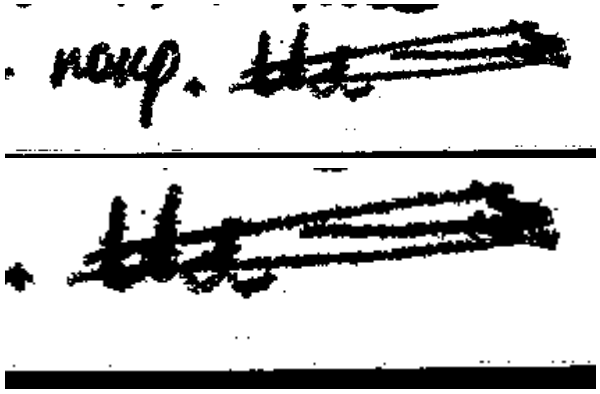
Original source details: GELFAND-P000079

Ax.
of a sheaf. $\{U_\alpha \to U\}$ a covering. There is $\Gamma(U) \to \Gamma(U_\alpha)$; hence there is $\Gamma(U) \to \prod_\alpha \Gamma(U_\alpha)$.

$$0 \to \Gamma(U) \to \prod_\alpha \Gamma(U_\alpha).$$

This is an embedding.

U_β — an elem. of the same covering. [Crossed out: U with overwritten lower signs: α / β / illegible; long horizontal strokes converge on the right.]



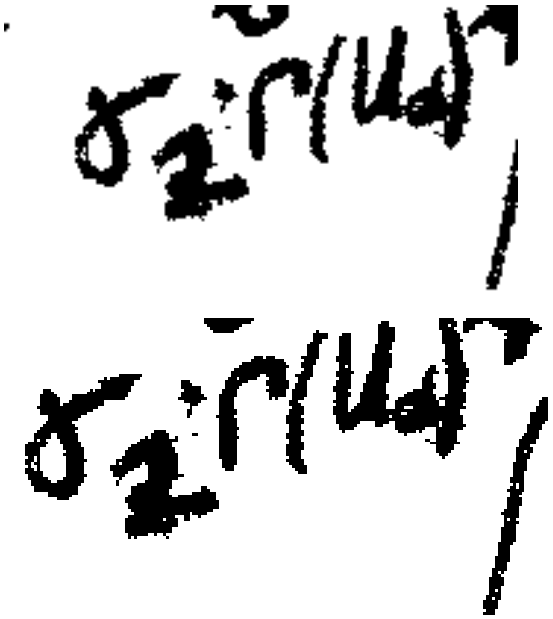
Uncertain reading. U with overwritten lower signs: α / β / illegible; long horizontal strokes converge on the right

Original source details: GELFAND-P000079

$U_\alpha \times_U U_\beta \to U_\alpha$.

GELFAND-P000080

There is $\Gamma(U_\alpha) \to \Gamma(U_\alpha \times_U U_\beta)$ and into $\prod_\beta \Gamma(U_\alpha \times_U U_\beta)$. [Below the top line, to the left of the product:] $\delta_2 : \Gamma(U_\alpha)$.

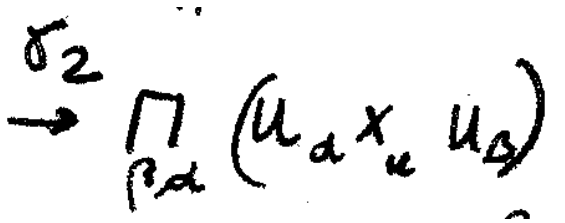


Manuscript notation. The 2 at δ is clear. A dot-like mark appears before Γ.

Original source details: GELFAND-P000080

$$0 \to \Gamma(U) \to \prod_{\alpha} \Gamma(U_\alpha) \xrightarrow{\delta_1 - \delta_2} \prod_{\alpha,\beta} \Gamma(U_\alpha \times_U U_\beta).$$

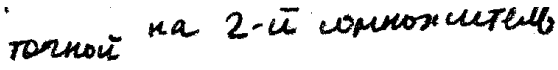
$$\prod_{\beta} \Gamma(U_\beta) \xrightarrow{\delta_2} \prod_{\beta\alpha} (U_\alpha \times_U U_\beta).$$



Uncertain reading. On the right, a parenthesis immediately after $\prod_{\beta\alpha}$, without a visible Γ.

Original source details: GELFAND-P000080

This seq. must be exact at the 2nd factor



Manuscript notation. The manuscript reads на 2-й сомножитель [at the 2nd factor].

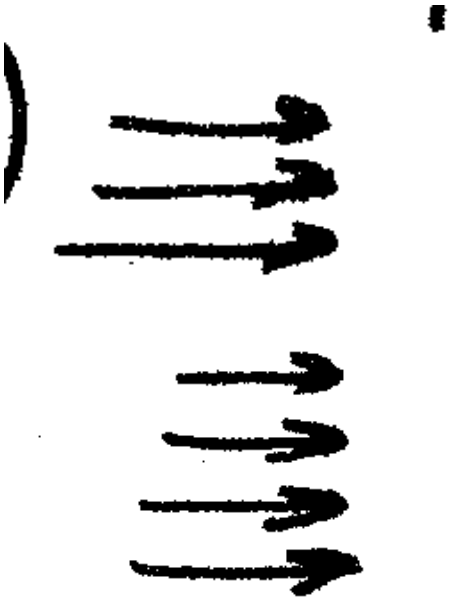
Original source details: GELFAND-P000080

(If the sect. over the pieces is 0, it is obtained from the 0th sect. over all of U.) Those agreeing on intersections can be glued.

Continuation of the seq.

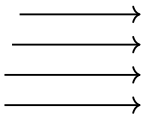
$$0 \to \Gamma(U) \to \prod_{\alpha} \Gamma(U_\alpha) \underset{\delta_2}{\overset{\delta_1}{\rightrightarrows}} \prod_{\alpha,\beta} \Gamma(U_\alpha \times U_\beta) \underset{\longrightarrow}{\overset{\longrightarrow}{\longrightarrow}} \prod_{\alpha,\beta,\gamma} \Gamma(U_\alpha \times U_\beta \times U_\gamma).$$

[On the right:] 3 maps by pairs.



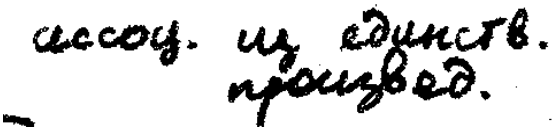
Original source details: GELFAND-P000080

Diagram and typed description. Three parallel rightward arrows between the products and, below, four separate parallel rightward arrows.



$U_\alpha \times U_\beta \times U_\gamma \to U_\beta \times U_\gamma$.

Assoc. from uniqueness
of the prod.



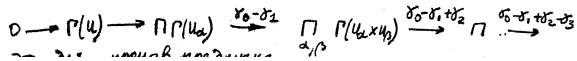
Original source details: GELFAND-P000080

Manuscript notation. "Of the prod." is below "from uniqueness".

$$\prod_{\beta,\gamma} \Gamma(U_\beta \times U_\gamma) \to \prod_{\alpha} \Gamma(U_\alpha \times U_\beta \times U_\gamma).$$

Next 4 maps.

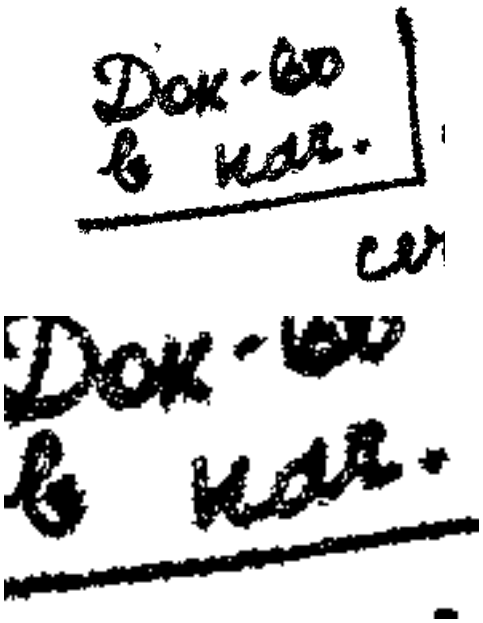
$$0 \to \Gamma(U) \to \prod \Gamma(U_\alpha) \overset{\delta_0-\delta_1}{\longrightarrow} \prod_{\alpha,\beta} \Gamma(U_\alpha \times U_\beta) \overset{\delta_0-\delta_1+\delta_2}{\longrightarrow} \prod \cdots \overset{\delta_0-\delta_1+\delta_2-\delta_3}{\longrightarrow} .$$



Original source details: GELFAND-P000080

Diagram and typed description. Upper arrow labels: $\delta_0 - \delta_1$, $\delta_0 - \delta_1 + \delta_2$, $\delta_0 - \delta_1 + \delta_2 - \delta_3$; after the last product, an ellipsis.

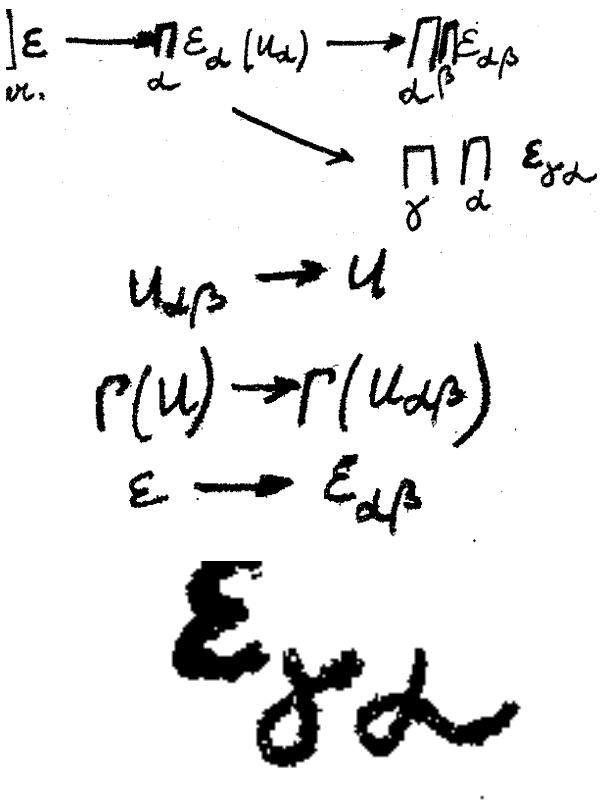
This is for an arbitrary presheaf. The compos. of 2 adjacent maps $= 0$. (Easy to check.) [On the right:] ker / Im.



Original source details: GELFAND-P000080

Uncertain reading. On the left, “Proof in cat.” / “Proof in id.”; the last letters are undetermined, with an angular stroke to the right and below.

Proof
in cat.



Original source details: GELFAND-P000080

Uncertain reading. The lower term has indices γ, α and $\varepsilon_{\gamma\alpha}$. A detached dot appears above the upper $\varepsilon_{\alpha\beta}$, with uncertain meaning.

$$\varepsilon \longrightarrow \prod_{\alpha} \varepsilon_{\alpha}(U_{\alpha}) \longrightarrow \prod_{\alpha}\prod_{\beta} \varepsilon_{\alpha\beta}$$

sect.

$$\prod_{\gamma}\prod_{\alpha} \varepsilon_{\gamma\alpha}$$

[In the right margin:]

$$U_{\alpha\beta} \to U,$$
$$\Gamma(U) \to \Gamma(U_{\alpha\beta}),$$
$$\varepsilon \to \varepsilon_{\alpha\beta}.$$

Def. the same. ker / Im.

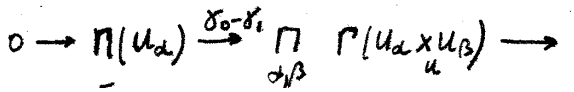
Čech cohom. covering.

Original source details: GELFAND-P000080

Manuscript notation. The last line reads Чеховские когом. покрытие.

GELFAND-P000081

$$0 \longrightarrow \prod(U_\alpha) \overset{\delta_0-\delta_1}{\longrightarrow} \prod_{\alpha\beta} \Gamma(U_\alpha \times_U U_\beta) \longrightarrow$$



Original source details: GELFAND-P000081

Uncertain reading. Before (U_α), $\prod$ is visible; a separate Γ is not established.

In this seq. sect.

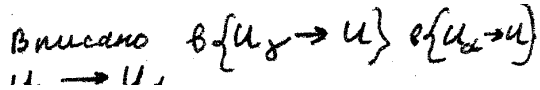
Original source details: GELFAND-P000081

Uncertain reading. The last abbreviation: “сеч.” / “cer.” (“sect.” / “seg.”); upper joined strokes.

$$H_0(S) = \Gamma(U)$$

Čech cohom. of the whole space.

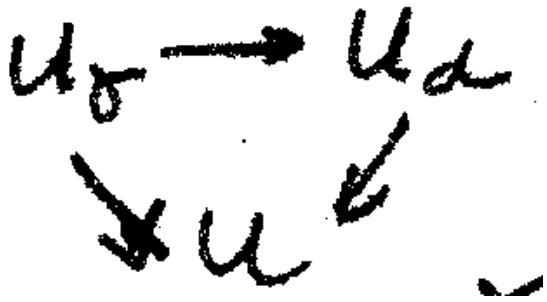
Embedded in $\{U_\gamma \to U\}$ in $\{U_\alpha \to U\}$,



Original source details: GELFAND-P000081

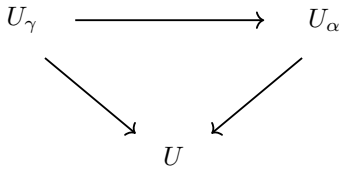
Manuscript notation. The preposition в (“in”) is written before both opening braces.

if $\forall\gamma$ a map is given.



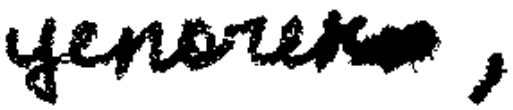
Original source details: GELFAND-P000081

Diagram and typed description. Above, $U_\gamma \to U_\alpha$; both slanting arrows point to the lower U.



Obtain a map of the chain / of chains (“цепочки” / “цепочек”), commut. with $\underset{\text{(functor!)}}{\delta}$ and therefore a map of cohomology.

Uncertain reading. of the chain / of chains (“цепочки” / “цепочек”)

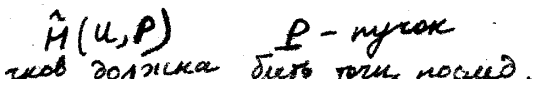


Original source details: GELFAND-P000081

Original source details: GELFAND-P000081

Diagram and typed description. Below δ, "(functor!)" is inserted.

This map does not depend on the manner of embedding. (Easy to understand.) Ind. limit $\hat{H}(U, P)$. [On the right:] $\underline{P}$ — a sheaf.

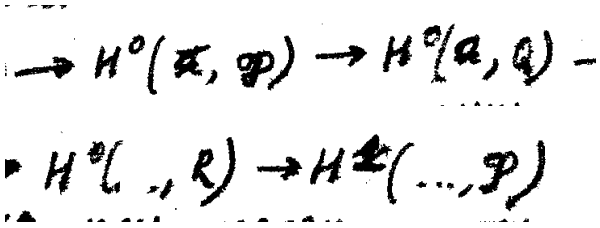


Original source details: GELFAND-P000081

Diagram and typed description. The separately written P on the right is underlined; the letter in $\hat{H}(U, P)$ has no underline.

For an exact seq. of sheaves there must be an exact seq. of cohom.

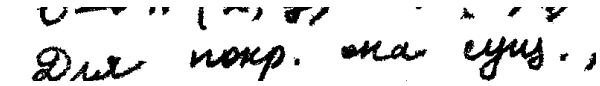
$$0 \to H^0(x, \Phi) \to H^0(a, Q) \to H^0(\ldots, R) \to H^1(\ldots, \Phi)$$



Original source details: GELFAND-P000081

Uncertain reading. The first argument x / X; the next resembles a; the sheaf letter Φ / P. The upper digit of the last H: 1, with dense overlaid strokes.

For a covering it exists, but on passing to

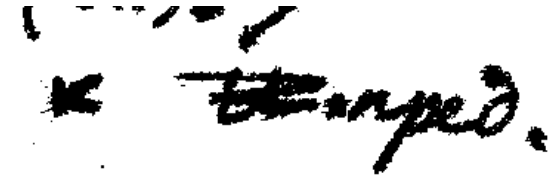


Original source details: GELFAND-P000081

Manuscript notation. The pronoun она ("it") is singular.

[crossed out before "limit": illegible]

the limit, exactness is not preserved.



Original source details: GELFAND-P000081

Uncertain reading. illegible

1. Paracomp. spaces in top. it is done. New def. (ax.) now only for sheaves. Exact seq. of presheaves

$$0 \to P \to Q \to R \to 0$$

[Before "If", a mark is crossed out: illegible.]

Original source details: GELFAND-P000081

Uncertain reading. illegible

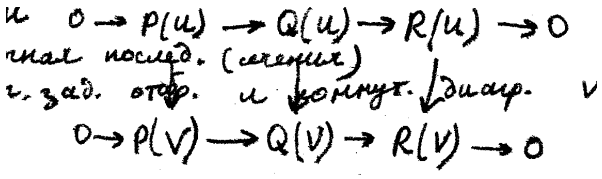
If $\forall U$

$$0 \to P(U) \to Q(U) \to R(U) \to 0$$

exact seq. (sections).

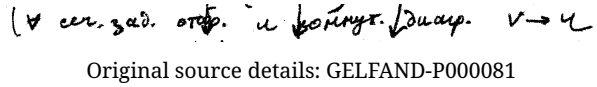
($\forall$ sect. given maps and commut. diagr. $V \to U$

$$\begin{array}{ccccccccc} & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & P(V) & \to & Q(V) & \to & R(V) & \to & 0 \end{array}$$



Original source details: GELFAND-P000081

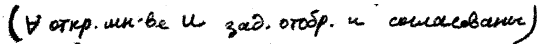
Uncertain reading. Two rows with $P(U), Q(U), R(U)$ and $P(V), Q(V), R(V)$; three downward arrows cross the intervening handwritten annotation. The abbreviation before “given”: “сеч.” / “сек.” (“sect.”).



Original source details: GELFAND-P000081

Manuscript notation. Between зад. (“given”) and и коммут. (“and commut.”) the manuscript has отобр. (“maps”). Three arrows cross the line.

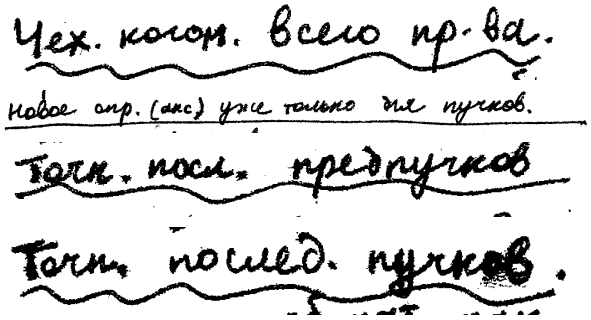
($\forall$ open set U given maps and gluing.)



Original source details: GELFAND-P000081

Undeciphered passage. After “gluing.”, one closing parenthesis is visible; the parenthesis of the preceding annotation is not closed separately.

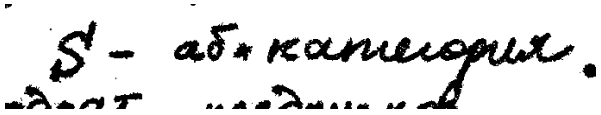
Exact seq. of sheaves.



Original source details: GELFAND-P000081

Diagram and typed description. Wavy underlines at “Čech cohom. of the whole space.”, “Exact seq. of presheaves” and “Exact seq. of sheaves.”; a straight line at “New def. (ax.) now only for sheaves.”.

S — an ab. category.



Original source details: GELFAND-P000081

Diagram and typed description. Before “ab. category”, the letter S with upper and lower terminal strokes.

Sheaves — an ab. cat. as a subcat. of presheaves. Ker and Coker are defined. Ax.: isom. of the image and coimage.

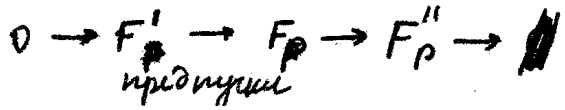
GELFAND-P000082

$$0 \to F' \to F \to F'' \to 0$$

Exact seq. (as sheaves).

$$0 \to F'_p \to F_p \to F''_p \to \text{[crossed out: } 0\text{]}$$

[Below F'_p: presheaves.]



Uncertain reading. Lower indices p / ρ; the last 0 is heavily crossed out.

Original source details: GELFAND-P000082

Not necess. 0(!) — the categ. is wider!. In another category.

$$0 \to iF' \to iF \to iF'' \longrightarrow$$

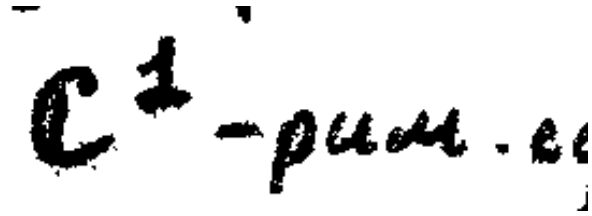
The def. of cohomology is based on this being [crossed out: illegible]

a nonexact seq.

Uncertain reading. illegible

Original source details: GELFAND-P000082

(In the difference between sheaves and presheaves — the whole theor. of cohom.) C^1 — the Riem. sph. and 2 points.



Uncertain reading. C^1 / P^1; the initial sign is C-like.

Original source details: GELFAND-P000082

Exact seq. of kogf. — exact seq. of sheaves.

Uncertain reading. “когф.” / “коэф.” / “когр.” (kogf. / koef., coeff. / kogr.); the middle letter has dense strokes.

Original source details: GELFAND-P000082

Sheaf of germs [an arrow is crossed out: arrow / illegible]

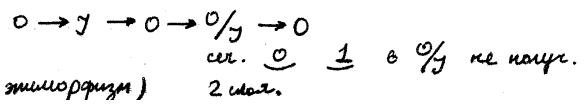
of hol. fns. $= 0$ at the 2 points. $\mathcal{J}$.

Uncertain reading. arrow / illegible

Original source details: GELFAND-P000082

$$0 \to \mathcal{J} \to O \to O/\mathcal{J} \to 0$$

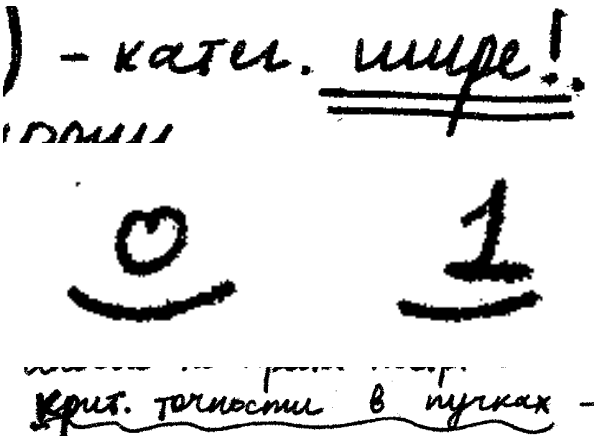
Sect. $\underset{\smile}{0}$ $\underset{\smile}{1}$ in $O/\mathcal{J}$ are not obtained.



Original source details: GELFAND-P000082

Uncertain reading. Before $0, 1$, the abbreviation “сек.” / “сеч.” (“sect.”); both numbers are underlined with arcs.

2 stalks. (Sect. is not an epimorphism.) Locally this map is onto. Sometimes the 0th sheaf is constr. from a presh. Criterion of exactness in sheaves — if locally exact.



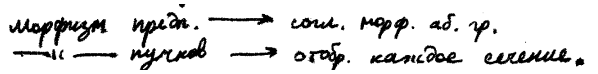
Original source details: GELFAND-P000082

Diagram and typed description. At “wider!”, a double straight line; below 0 and 1, arcs; at “Criterion of exactness in sheaves”, a wavy line.

Morphism ...

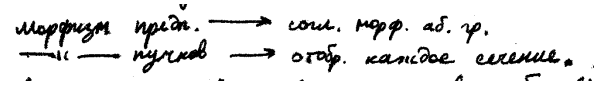
$$\text{exact seq. of sheaves} \iff \text{exact seq. of “stalks”}.$$

Morphism of presh. $\longrightarrow$ compat. morph. of ab. gr. $__$"$__$ of sheaves $\longrightarrow$ maps each section.



Original source details: GELFAND-P000082

Manuscript notation. The first line reads совм. морф. аб. гр. (“compat. morph. of ab. gr.”), the second отобр. каждое сечение (“maps each section”).

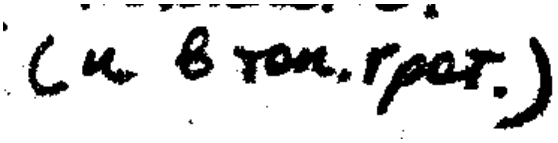


Original source details: GELFAND-P000082

Diagram and typed description. At the start of the second line, a repetition sign: two short vertical strokes between horizontal strokes.

The addit. cat. of presheaves and sheaves are abelian. If there is an exact seq. of sheaves, then the seq. is exact locally.

(also in Groth. top.)

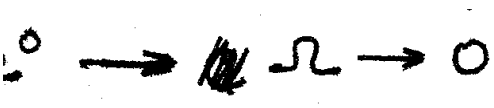


Original source details: GELFAND-P000082

Manuscript notation. The parentheses contain и в топ. грот. (“also in Groth. top.”).

(Here stalks cannot be def. in principle.) Presh. exact on fine coverings. Presh. compared on all sets, and sheaves on fine ones.

$$0 \to \underset{\text{const.}}{C} \to \underset{\text{fns.}}{\Omega^0} \to [\text{a sign is crossed out}]\Omega \to 0$$



Original source details: GELFAND-P000082

Uncertain reading. The sign before Ω is heavily crossed out; illegible.

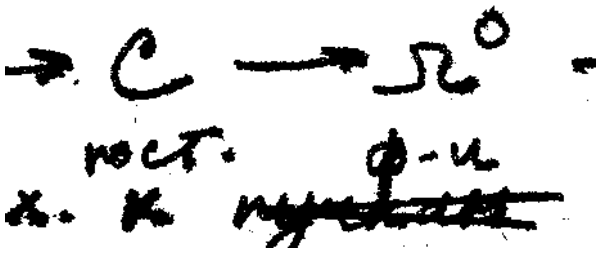
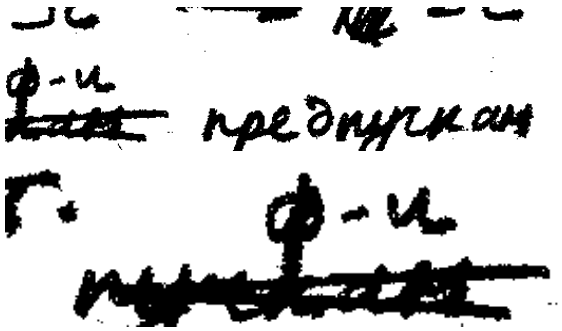


Diagram and typed description. Below C, “const.”; below Ω^0, “fns.”.

Original source details: GELFAND-P000082

Pass to [crossed out: sheaves / illegible] presheaves.

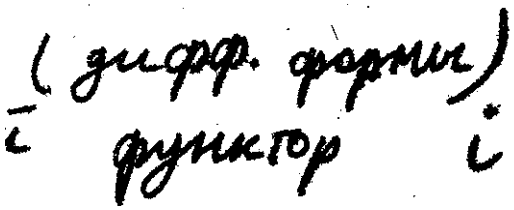


Uncertain reading. sheaves / illegible

Original source details: GELFAND-P000082

$$0 \to C \to f$$

(diff. forms)



Manuscript notation. At right, “(diff. forms)”. The dot below the end of the parenthesis is above i on the lower line.

Original source details: GELFAND-P000082

Uncertain reading. f / $\mathcal{F}$.

Original source details: GELFAND-P000082

[A short fragment is crossed out: illegible.]

Uncertain reading. illegible

Original source details: GELFAND-P000082

Cohom. $\longrightarrow$ derived functor i.

GELFAND-P000083

$$0 \to F' \to F \to F'' \to 0$$

Req. that there be an exact seq. [On the right:] H — presheaves. 1.

$$0 \to iF' \to iF \to iF'' \to H^1(F') \to H^1(F) \\ \to H^1(F'') \to H^2(F') \longrightarrow \dots$$

(Ax. of derived functors.)

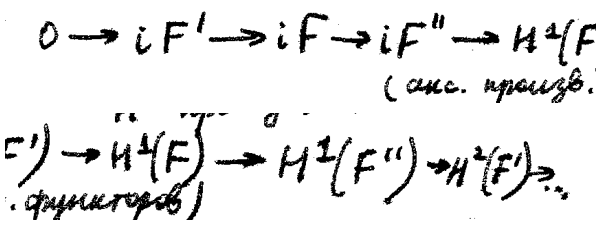


Diagram and typed description. After iF'', the terms $H^1(F')$, $H^1(F)$, $H^1(F'')$, $H^2(F')$; then an arrow and an ellipsis.

Original source details: GELFAND-P000083

$$i : S \longrightarrow \underline{P}$$

Diagram and typed description. In $i : S \longrightarrow P$, a straight line below the final P.

Original source details: GELFAND-P000083

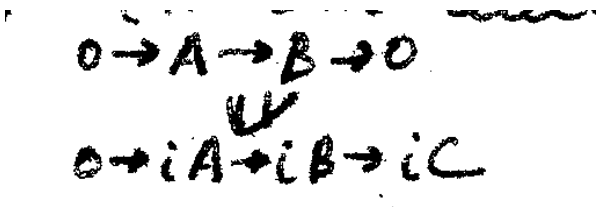
[Crossed out: illegible]
a functor (left exact).

Uncertain reading. illegible

Original source details: GELFAND-P000083

[On the right:]

$$0 \to A \to B \to 0$$
$$\Downarrow$$
$$0 \to iA \to iB \to iC$$

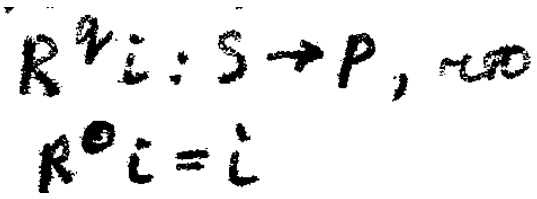


Uncertain reading. In the upper row, 0 is visible after B; in the lower row the last term is iC, without a corresponding C above.

Original source details: GELFAND-P000083

Derived functor

$$R^q i : S \to P, \quad \text{but}$$
$$R^0 i = i$$



Original source details: GELFAND-P000083

Manuscript notation. The superscript q has a descending stroke like the q in $q > 0$ below. The next line reads $R^0 i = i$.

1:

$$0 \to R^0(A) \longrightarrow \dots$$

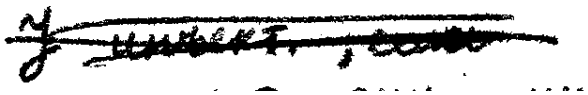
Exact seq. 2. For inject. objects S

$$R^q(S) = 0, \qquad q > 0.$$

Original source details: GELFAND-P000083

Manuscript notation. The vanishing condition shows the same descending stroke on q.

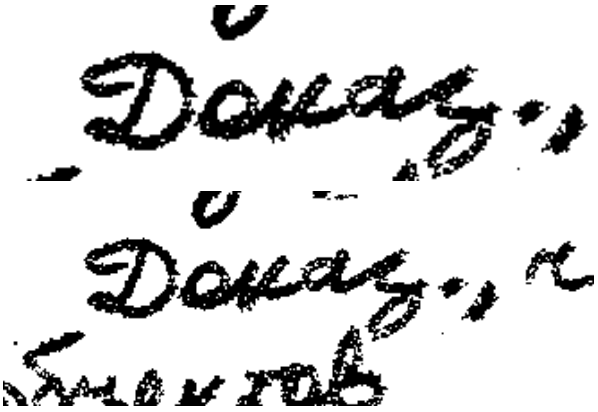
[A row is crossed out: “For inject.” / illegible.]



Original source details: GELFAND-P000083

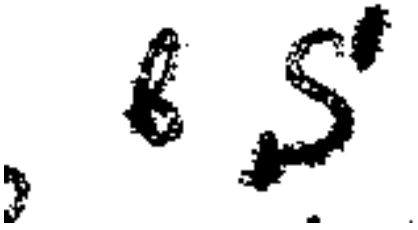
Uncertain reading. “For inject.” / illegible

Prov., that they exist, if in S' there are enough inject. objects (an embedding into an inj. obj. exists for any).



Original source details: GELFAND-P000083

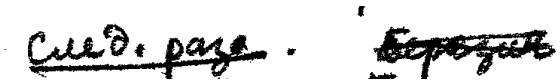
Diagram and typed description. At the start of the row, the abbreviation “Доказ.” (“Prov.”); a comma after the full stop.



Original source details: GELFAND-P000083

Diagram and typed description. In the words “if in S' there are enough”, a separate upper stroke after S.

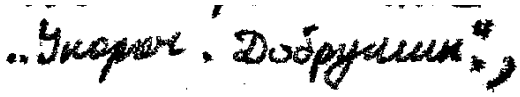
See Grothendieck, “Lectures on hom. algebra”. Val.: on the whole space, of the derived functor, is called cohom. of the whole space. Next time. ~~Berezin~~



Original source details: GELFAND-P000083

Diagram and typed description. The heading След. раза. (“Next time”) is underlined. The following “Berezin” is crossed out.

“Another form. of Plancherel for diff. oper.” Berezin and Sinai “Ukoren. Dobrushin” / “Igor. Dobrushin” (“Укорен. Добрушин” / “Игорь. Добрушин”), possibly, Halmos.

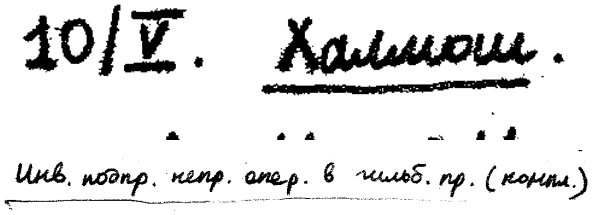


Original source details: GELFAND-P000083

Uncertain reading. “Ukoren. Dobrushin” / “Igor. Dobrushin” (“Укорен. Добрушин” / “Игорь. Добрушин”)

GELFAND-P000084

$10/\overline{\underline{V}}$. Halmos. Inv. subsp. of a cont. oper. in Hilb. sp. (compl.)



Original source details: GELFAND-P000084

Diagram and typed description. The V in the date has an upper and a lower bar; “Halmos.” and the heading are underlined.

$$A : H \to H$$

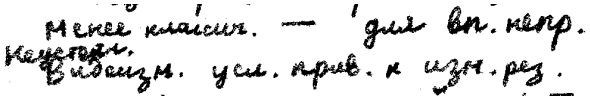
Inv. sp. — closed and inv. M. $\exists M$ — ? ($M \neq 0, H$) The gen. probl. is not solved, [crossed out: illegible]

Original source details: GELFAND-P000084

Uncertain reading. illegible

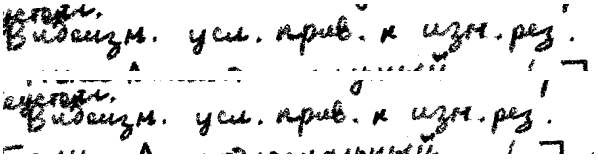
(known if A is Herm. or norm.).

Less class. — for completely cont. (von Neumann, ...). [In the margin: counterex.] Modif. the cond. leads to known res.



Original source details: GELFAND-P000084

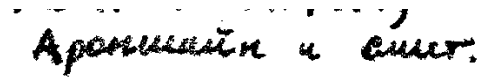
Manuscript notation. The two lines contain Менее классич. (“Less class.”) and Видоизм. усл. прив. к изв. рез. (“Modif. the cond. leads to known res.”).



Original source details: GELFAND-P000084

Manuscript notation. The first word in the line is Видоизм. (“Modif.”).

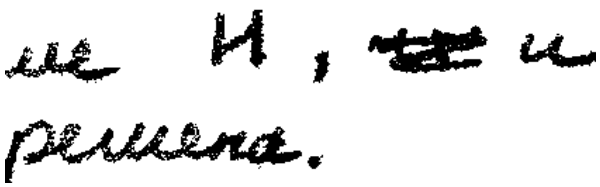
Aronszajn and Smith.



Original source details: GELFAND-P000084

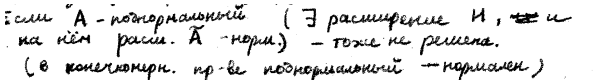
Manuscript notation. The two names are Ароншайн и Смит (“Aronszajn and Smith”).

If A is subnormal (∃ an extension of H, crossed out: illegible; and on it an ext. $\tilde{A}$ is norm.), — also not solved.



Original source details: GELFAND-P000084

Uncertain reading. illegible



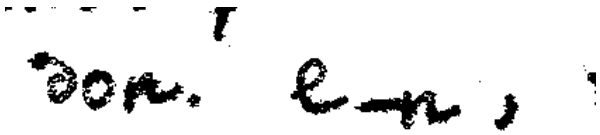
Original source details: GELFAND-P000084

Diagram and typed description. The definition is enclosed in round parentheses; a wavy stroke above A in “ext. $\tilde{A}$”.

(In a finite-dim. sp., subnormal $\longrightarrow$ normal.) In infinite dim. — the shift

$$e_1 \to e_2,\ e_2 \to e_3, \dots$$

If add / prove (“доп.” / “док.”) e_{-n}, then it will be normal.



Original source details: GELFAND-P000084

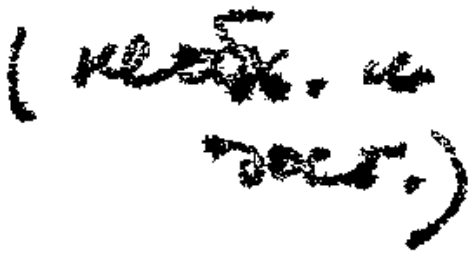
Uncertain reading. add / prove (“доп.” / “док.”)

Internal conditions of subnorm.

$$A^*A \ [?]\ AA^* \geq 0$$

Original source details: GELFAND-P000084

Uncertain reading. The horizontal mark thickens at its right end: $\longrightarrow$ / $-$.

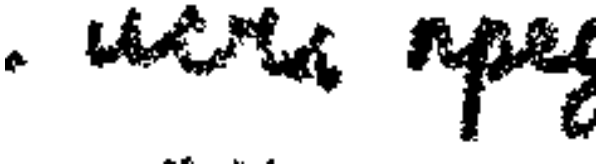


Original source details: GELFAND-P000084

Uncertain reading. (necess. and suff.); the words have overwritten strokes.

(The adj. of a subnorm. need not be subnorm.) See Halmos, Bram. Generalizations. A^2 completely cont. — positive ans. (∃ an inv. subsp.) A. Robinson, Bernstein in terms of pred. calc.

Manuscript notation. The abbreviations are исч. предик. (“pred. calc.”).



Original source details: GELFAND-P000084

Proof of the schem. method.

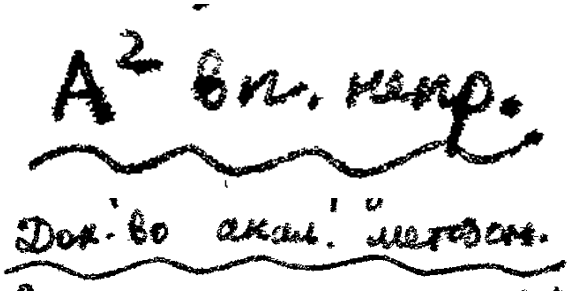
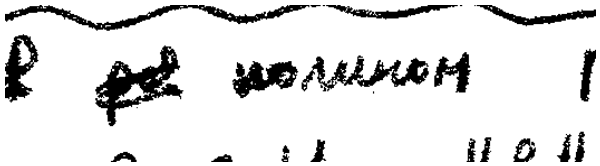


Diagram and typed description. Wavy lines below “A^2 completely cont.” and “Proof of the schem. method.”

Original source details: GELFAND-P000084

p [A sign is crossed out: $\forall$ / illegible]

a polynomial $p \neq 0$, $p(A)$ completely cont. Then $\exists$ an inv. subsp.



Uncertain reading. $\forall$ / illegible

Original source details: GELFAND-P000084

$e \in H$, $\|e\| = 1$.

Rasst. $\{e, Ae, A^2e, \ldots\} = H$

Uncertain reading. The abbreviation before the brace: “Расст.” / “Рассм.” (“Rasst.” / “Rassm.”); the final joined strokes are uncertain.

Original source details: GELFAND-P000084

(Otherwise there is a subsp.) $\therefore$ therefore.

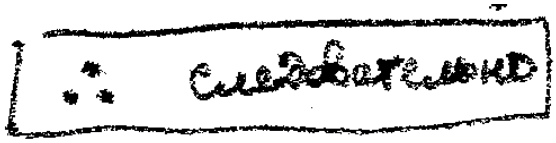
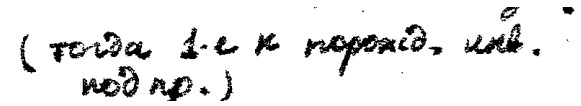


Diagram and typed description. Three dots in a triangle and the word “therefore” are enclosed in a box.

Original source details: GELFAND-P000084

$e, Ae, \ldots$ may be considered lin. indep. Otherwise there is a finite-dim. subsp. (then the 1st to the generated / the 1st to each (“1-е к порожд.” / “1-е к каждой”) inv. subsp.)

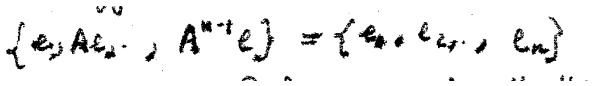


Original source details: GELFAND-P000084

Uncertain reading. the 1st to the generated / the 1st to each (“1-е к порожд.” / “1-е к каждой”)

Orthogonalize

$$\{e, Ae, \dots, A^{n-1}e\} = \{e_1, e_2, \dots, e_n\}$$



Original source details: GELFAND-P000084

Diagram and typed description. The last exponent is $n-1$; on the right, the indices 1, 2, n.

$e_1, e_2, \dots$ — $\perp$, $\| \| = 1$. The matrix has the form

$$a_{mn} = (Ae_n, e_m)$$

$$\begin{pmatrix} * & * & * \\ * & * & * \\ 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}$$

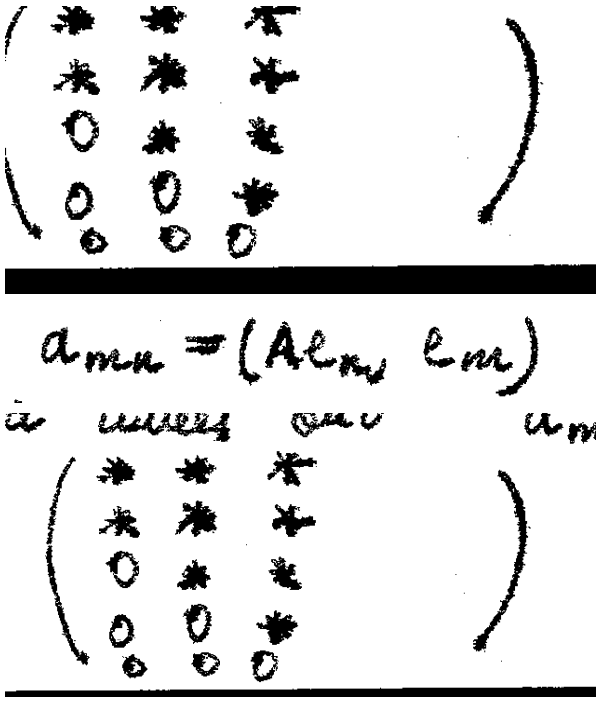


Diagram and typed description. The three lower zeros continue below the ends of the round parentheses; above, $a_{mn} = (Ae_n, e_m)$.

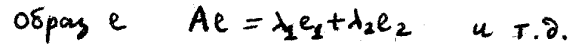
Original source details: GELFAND-P000084

GELFAND-P000085

If 1 beyond the diag., then 0. Image of e:

$$Ae = \lambda_1 e_1 + \lambda_2 e_2$$

and so on.



Original source details: GELFAND-P000085

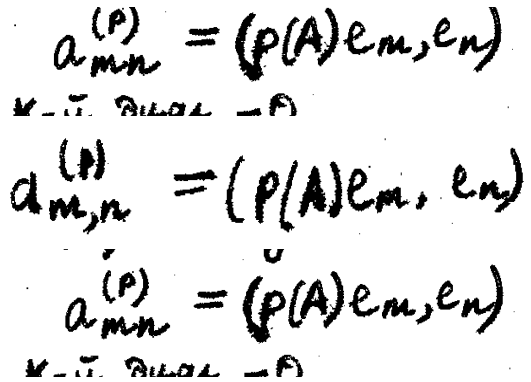
Diagram and typed description. In “Image of e” and Ae, no lower index; on the right, $\lambda_1 e_1 + \lambda_2 e_2$.

A^k has the matrix $a^{(k)}_{mn}$ (add k diagonals below). (Farther than k units from the main diag., 0.) On the kth diag.

$$a^{(k)}_{n+k,n} = \prod_{j=1}^{k} a_{n+j,n+j-1}$$

Matrix $p(A)$, deg k.

$$a^{(p)}_{mn} = (p(A)e_m, e_n)$$



Uncertain reading. For $p(A)$, the notation $(p(A)e_m, e_n)$; on P000084 it was (Ae_n, e_m).

Original source details: GELFAND-P000085

Below the kth diag. — 0. On the kth diag. — the same as for the kth power of A (up to a factor). e_n converges weakly to 0, therefore $p(A)e_n$ converges strongly to 0.

Diagram and typed description. First "converges weakly to 0", then "converges strongly to 0".

Original source details: GELFAND-P000085

$$\prod_{j=1}^{k} d_{n+j,n+j-1} \longrightarrow 0, \qquad n \to \infty.$$

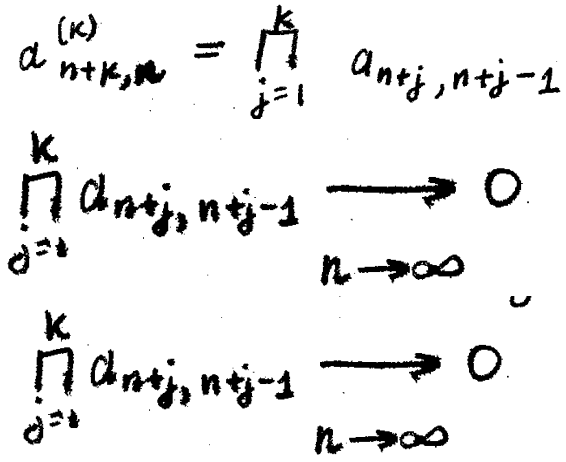


Diagram and typed description. In the first product, the letter a; in the product with an arrow to 0, the letter d. In both, the limits $j = 1$, k and the indices $n + j$, $n + j - 1$.

Original source details: GELFAND-P000085

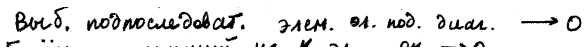
Product of numbers below the diagonal. Change both indices.

$$d^{(p)}_{m,n} = (p(A)e_m, e_n)$$

Uncertain reading. The upper sign on d: (p) / (1); a short joined stroke inside the parentheses.

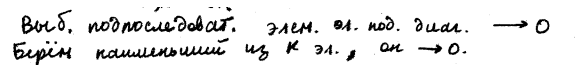
Original source details: GELFAND-P000085

Choose a subseq. of elem. elem. below the diag. $\to 0$.



Original source details: GELFAND-P000085

Manuscript notation. After "subseq." the manuscript has both элем. and эл. ("elem. elem.").

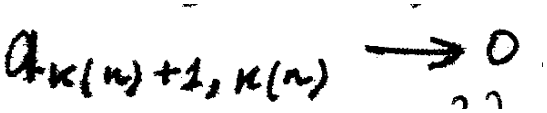


Original source details: GELFAND-P000085

Manuscript notation. The two abbreviations элем. and эл. occur consecutively.

Take the smallest of the k elem., it $\to 0$.

$$a_{k(n)+1,k(n)} \longrightarrow 0$$



Original source details: GELFAND-P000085

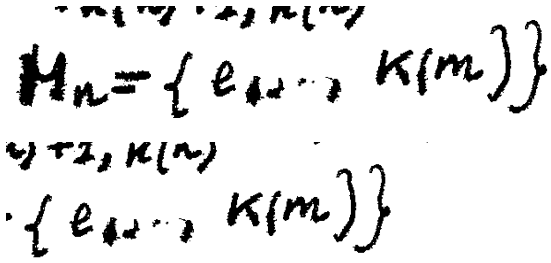
Diagram and typed description. In the row with an arrow to 0: $a_{k(n)+1,k(n)}$.

Consider. [A symbol is crossed out: illegible.]

Uncertain reading. illegible

Original source details: GELFAND-P000085

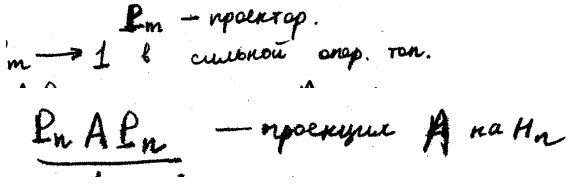
$$H_n = \{e_1, \ldots, k(m)\}$$



Original source details: GELFAND-P000085

Manuscript notation. The manuscript shows H_n and $k(m)$, with no visible e before $k(m)$.

P_m — a projector. $P_m \to 1$ in the strong oper. top. $\underline{P_n A P_n}$ — the projection of A onto H_n.



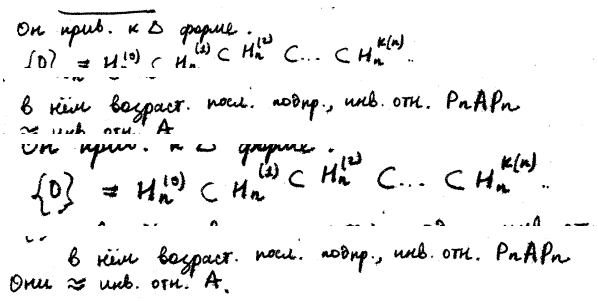
Original source details: GELFAND-P000085

Uncertain reading. First the index m, then n in $P_n A P_n$ and H_n.

It is reduced to Δ form.

$$\{0\} = H_n^{(0)} \subset H_n^{(1)} \subset H_n^{(2)} \subset \ldots \subset H_n^{k(n)}.$$

In it, an increasing seq. of subsp., inv. with respect to $P_n A P_n$. They are $\approx$ inv. with respect to A.



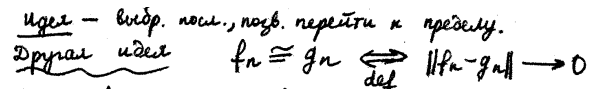
Original source details: GELFAND-P000085

Diagram and typed description. The sign Δ before “form”; in the chain, the upper (0), (1), (2) and $k(n)$; in the last row, $\approx$.

GELFAND-P000086

Idea — choose a seq., allowing passage to the limit. Another idea

$$f_n \cong g_n \quad \underset{\text{def}}{\iff} \quad \|f_n - g_n\| \to 0$$



Original source details: GELFAND-P000086

Diagram and typed description. A straight underline below “Idea”, a wavy underline below “Another idea”; def below the two-way arrow.

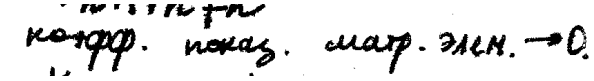
If f_n is bounded $\Rightarrow$

$$AP_nf_n \cong P_nAP_nf_n.$$

Original source details: GELFAND-P000086

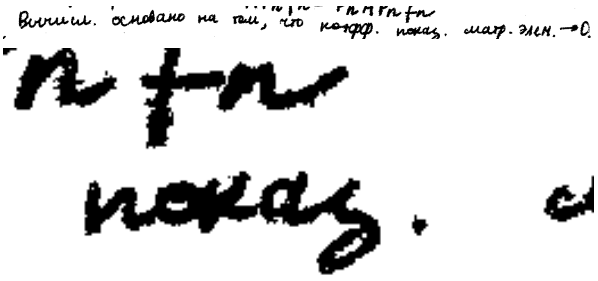
Diagram and typed description. On the left AP_nf_n, on the right $P_nAP_nf_n$.

The calc. is based on the fact that the coeff. of the shown matrix elem. $\to 0$.



Original source details: GELFAND-P000086

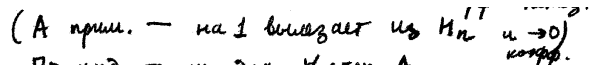
Manuscript notation. Between “coeff.” and “matrix elem.” the manuscript has показ. (“shown”).



Original source details: GELFAND-P000086

Manuscript notation. The abbreviation is показ. (“shown”).

(A appl. — goes 1 outside H_n and $\to 0$.) [Added below the end: coeff.]

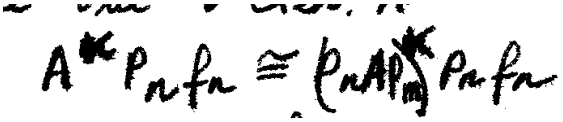


Original source details: GELFAND-P000086

Diagram and typed description. In parentheses “A appl. — goes 1 outside H_n and $\to 0$”; below the end, the addition “coeff.”.

By ind., the same for $\forall$ powers of A:

$$A^kP_nf_n \cong (P_nAP_n)^kP_nf_n.$$

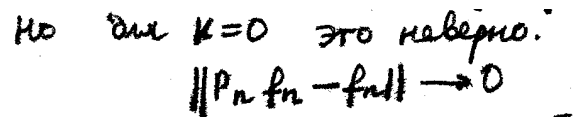


Original source details: GELFAND-P000086

Uncertain reading. Inside the parentheses the third P_n is overwritten; the exponent k has dense strokes.

But for $k = 0$ this is false.

$$\|P_n f_n - f_n\| \to 0.$$

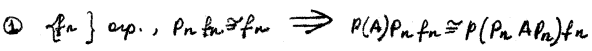


Original source details: GELFAND-P000086

Diagram and typed description. In the row "But for $k = 0$ this is false", a visible 0; below, $\|P_n f_n - f_n\| \to 0$.

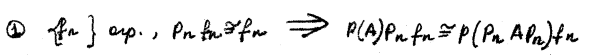
But if this req. is imposed on f_n, then this will be true for all powers (and for polyn.).
① $\{f_n\}$ bounded, $P_n f_n$ [?] f_n
⇒

$$p(A)P_n f_n \cong p(P_n A P_n) f_n.$$



Original source details: GELFAND-P000086

Manuscript notation. The 1 is circled. After "bounded" the manuscript shows $P_n f_n$, an overwritten sign, and f_n.

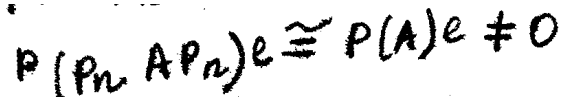


Original source details: GELFAND-P000086

Uncertain reading. $P_n f_n \cong f_n$ / $P_n f_n \neq f_n$

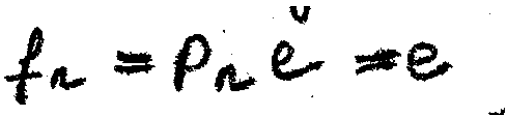
Spec. particular case

$$f_n = P_n e = e$$
$$p(P_n A P_n)e \cong p(A)e \neq 0.$$



Original source details: GELFAND-P000086

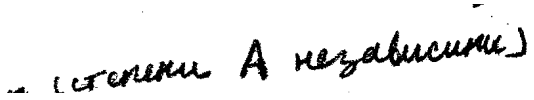
Diagram and typed description. An e follows both $p(P_n A P_n)$ and $p(A)$; at the end $\neq 0$.



Original source details: GELFAND-P000086

Manuscript notation. The second sign in $f_n = P_n e = e$ consists of two strokes.

[On the right: powers of A are independent.]

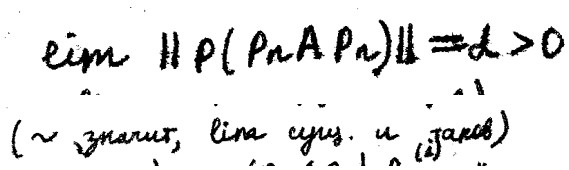


Original source details: GELFAND-P000086

Uncertain reading. Overwritten strokes near "powers of A"; the lower layer is illegible.

Hence

$$\lim \|p(P_n A P_n)\| = d > 0.$$



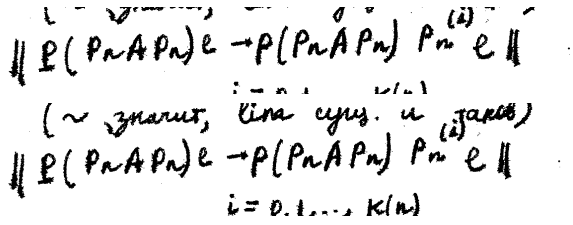
Original source details: GELFAND-P000086

Uncertain reading. After the closing parenthesis inside the norm, no confidently visible e; in the following parentheses, a single wavy sign $\sim$.

($\sim$ means, the lim exists and is such.)

$$\|p(P_n A P_n)e \longrightarrow p(P_n A P_n)P_n^{(i)} e\|$$

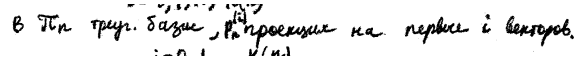
$i = 0, 1, \ldots, k(n)$.



Original source details: GELFAND-P000086

Uncertain reading. Between the terms inside the norm, a visible arrow rather than an ordinary minus sign.

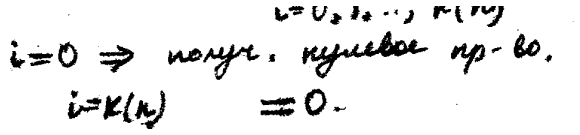
In Π_n, triang. basis, $P_n^{(i)}$ — projection onto the first i vectors.



Original source details: GELFAND-P000086

Uncertain reading. Π_n / H_n.

$i = 0, 1, \ldots, k(n)$. $i = 0 \Rightarrow$ obtain the zero subsp. $i = k(n) = 0$.



Original source details: GELFAND-P000086

Diagram and typed description. The first row $i = 0 \Rightarrow$ “obtain the zero subsp.”; the next $i = k(n) = 0$.

$$\exists n: \quad \|p(P_nAP_n)e\| > \frac{3}{4}d.$$

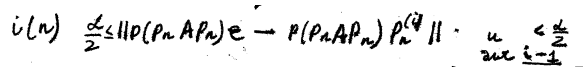
Original source details: GELFAND-P000086

Diagram and typed description. Before the norm $\exists n$:; inside it a final e; after the norm $> \frac{3}{4}d$.

$i(n)$:

$$\frac{d}{2} \le \|p(P_nAP_n)e \longrightarrow p(P_nAP_n)P_n^{(i)}\|$$

and $< \frac{d}{2}$ for $\underline{i-1}$.

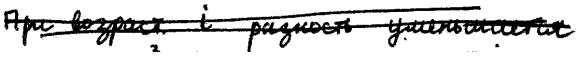


Original source details: GELFAND-P000086

Uncertain reading. The parameter d / α; in this last norm after $P_n^{(i)}$, no confidently visible e.

GELFAND-P000087

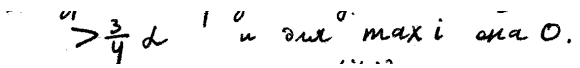
[Crossed out: As i incr., the diff. decreases.]



Original source details: GELFAND-P000087

Uncertain reading. Abbreviations under two lines: “возрастан.” / “возрастан”, “разност.” / “разность” (“incr.”, “diff.” / “difference”).

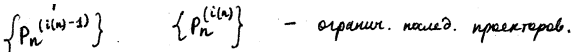
$> \frac{3}{4}d$ and for $\max i$ it is 0.



Original source details: GELFAND-P000087

Manuscript notation. The letter after the fraction is d.

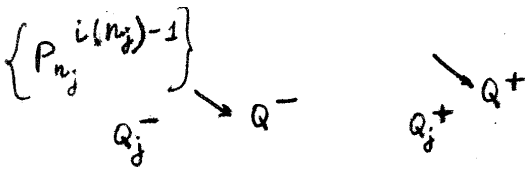
$$\{P_n^{(i(n)-1)}\}, \qquad \{P_n^{(i(n)}\}$$



Original source details: GELFAND-P000087

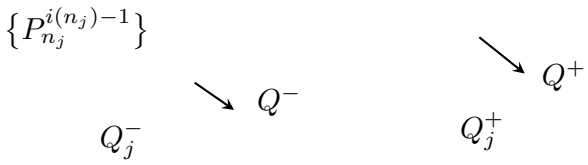
Uncertain reading. In the first exponent $(i(n)-1)$; in the second a visible opening parenthesis before $i(n)$, without a separate second closing parenthesis.

— bounded seq. of projectors. Choose a weakly conv. subseq.

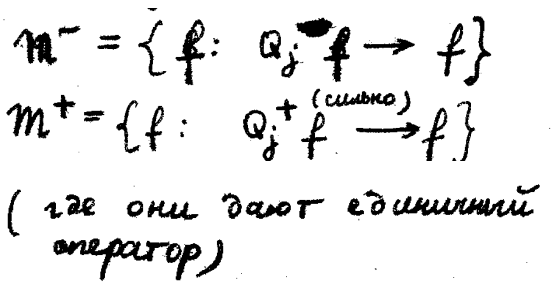


Original source details: GELFAND-P000087

Diagram and typed description. On the left $\{P_{n_j}^{i(n_j)-1}\}$, an inclined arrow to Q^-; below the left part Q_j^-. On the right an inclined arrow to Q^+, below it Q_j^+. The upper starting point of the right arrow has no separate label.



$$m^- = \{f : Q_j^- f \to f\}$$
$$m^+ = \{f : Q_j^+ f \overset{\text{(strongly)}}{\longrightarrow} f\}.$$



Original source details: GELFAND-P000087

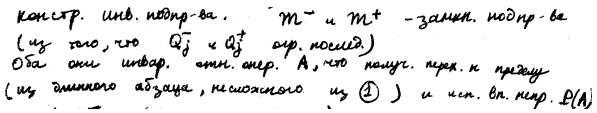
Uncertain reading. The minus above the first Q_j is overwritten; m / M in the left-hand notation.

[On the right: where they give the identity operator.] (On M^- them Q^- is 1.)

Original source details: GELFAND-P000087

Uncertain reading. M^- is inserted before "them"; the original phrase is not amended.

Constr. of inv. subsp. m^- and m^+ — closed subsp. (from the fact that Q_j^- and Q_j^+ are bounded seq.). Both are inv. with respect to the oper. A, which is obtained by passage to the limit (from the long paragraph, uncomplicated from ①) and using comp. $p(A)$.



Original source details: GELFAND-P000087

Diagram and typed description. Here lower-case m^-, m^+; the reference has a circled 1; at the end $p(A)$.

$$e \notin M^- \qquad \text{(so that } M^- \neq H\text{)}.$$

Original source details: GELFAND-P000087

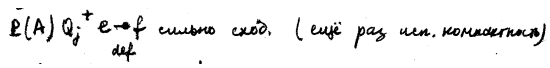
Diagram and typed description. Capital M^- in both places; before it $e \notin$, in parentheses $M^- \neq H$.

$M^+ \neq \{0\}$ — this is already less simple.

$\{Q_j^+ e\}$ converges weakly.

$$p(A)Q_j^+ e \underset{\text{def}}{\longrightarrow} f$$

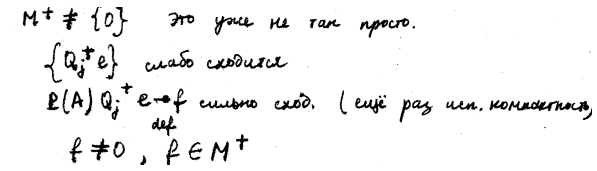
Strongly conv. (compactness used once more).



Original source details: GELFAND-P000087

Diagram and typed description. Below f the label def; before it an arrow, on the left $p(A)Q_j^+ e$.

$$f \neq 0, \qquad f \in M^+.$$



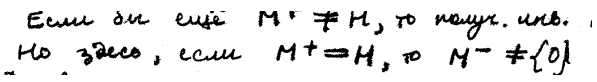
Original source details: GELFAND-P000087

Diagram and typed description. The sequence has $Q_j^+ e$ with a $+$; below, $f \neq 0$, $f \in M^+$.

If also $M^+ \neq H$, then we obtain an inv. subsp.
But here, if $M^+ = H$, then $M^- \neq \{0\}$.

Original source details: GELFAND-P000087

Manuscript notation. This line has $M^+ = H$ and $M^- \neq \{0\}$.



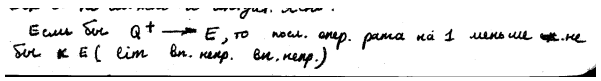
Original source details: GELFAND-P000087

Manuscript notation. After “then” there is a capital M with a minus.

The proof is not difficult and is intuitively clear. If $Q^+ \to E$, then the seq. of oper. of rank 1 less [A short fragment is crossed out: illegible.]
not cl. to E (lim comp. comp.).

Uncertain reading. illegible

Original source details: GELFAND-P000087

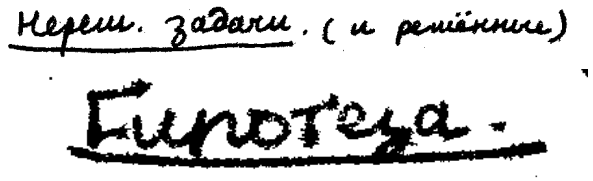


Original source details: GELFAND-P000087

Manuscript notation. The end reads не бл. к E (“not cl. to E”).

GELFAND-P000088

Unsolved problems. (and solved ones)



Original source details: GELFAND-P000088

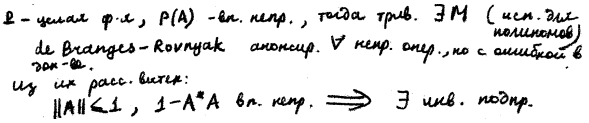
Diagram and typed description. “Unsolved problems.” is underlined; the parentheses are outside the underline. Below, “Conjecture.” is separately underlined.

p — entire fn., $p(A)$ — comp., then triv. $\exists M$ (use for polynomials). de Branges — Rovnyak announced $\forall$ cont. oper., but with an error in the proof. From their reasoning follows:

$\|A\|$ [?] 1, $\quad 1 - A^*A$ comp. $\quad \Rightarrow \quad \exists$ inv. subsp.

Manuscript notation. The lower stroke of the sign before 1 merges with the slanted branch.

Original source details: GELFAND-P000088



Uncertain reading. The sign before 1 is $<$ / $\leq$, with a thick lower branch. Then $1 - A^*A$ and $\Rightarrow \exists$.

Original source details: GELFAND-P000088

Res. of Sakhnovich, Livshits and brodsky. $A + iB$, A — self-adj., B in some sense small (nuclear, etc.).

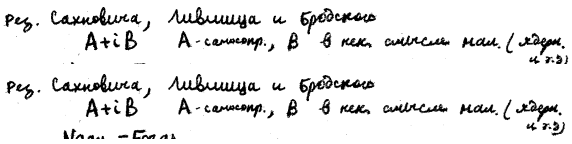


Diagram and typed description. At the start $A + iB$; "nuclear, etc." continues on the second line.

Original source details: GELFAND-P000088

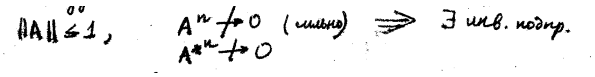
Nagy — Foias

Manuscript notation. The names are written in Latin letters, "Nagy — Foias".

Original source details: GELFAND-P000088

$$\|A\| \leq 1, \qquad A^n \not\longrightarrow 0 \quad \text{(strongly)},$$
$$A^{*n} \not\longrightarrow 0$$

$\Rightarrow \exists$ inv. subsp.



Uncertain reading. Both arrows are crossed by an inclined stroke.

Original source details: GELFAND-P000088

Triv. structure $\{0, 1\}$.

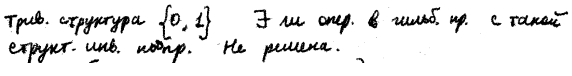


Diagram and typed description. Two elements $\{0, 1\}$; then the question about an operator in Hilb. sp. and "Not solved."

Original source details: GELFAND-P000088

$\exists$ an oper. in Hilb. sp. with such a struct. of inv. subsp.? Not solved. Another probl. Struct. $\{0, 1, 2, \ldots, \infty\}$. Does an oper. with such a struct. exist? Affirm. answer Donoghue (1957).

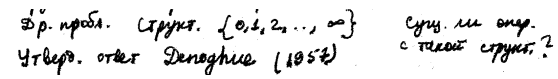


Diagram and typed description. Written $\{0, 1, 2, \ldots, \infty\}$; the date in parentheses (1957).

Original source details: GELFAND-P000088

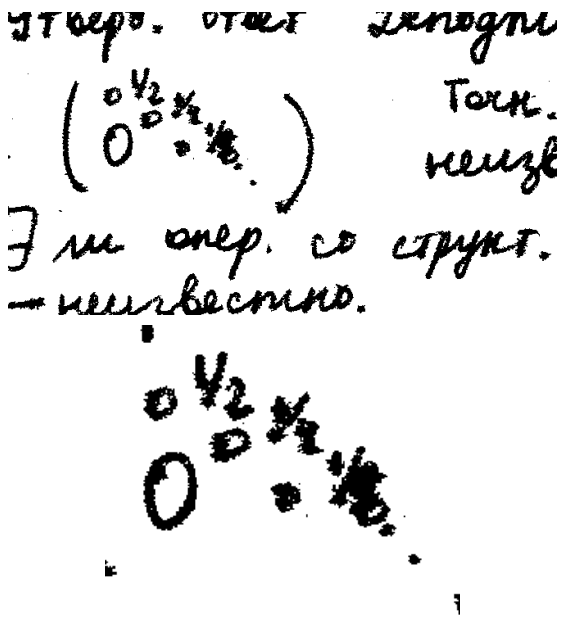


Diagram and typed description. Within round parentheses two small 0s, $1/2$, $1/4$; the third densely overwritten coefficient $1/8$ / $1/6$; a large 0 below and separate dots down to the right.

$$\begin{pmatrix} 0 & \frac{1}{2} & & \\ & 0 & \frac{1}{4} & \\ 0 & & \cdot & \frac{1}{8}\,[?] \\ & & & \cdot \\ & & & \cdot \end{pmatrix}$$

Original source details: GELFAND-P000088

Exact cond. on the numbers above the diag. are unknown (so that precisely these occur). $\exists$ an oper. with the struct. of all integers $\to$ unknown.

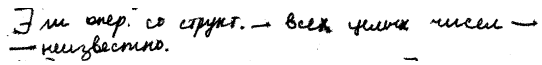


Diagram and typed description. “all integers”; then an arrow and “unknown”.

Original source details: GELFAND-P000088

$$[0,1] \longmapsto$$

— such a structure $\exists$ operator.

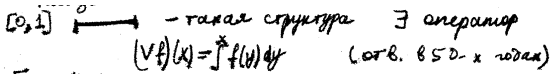


Diagram and typed description. On the left the label $[0,1]$, then a segment with two end strokes. Below $(Vf)(x) = \int_0^x f(y)\,dy$.

Original source details: GELFAND-P000088

$$(Vf)(x) = \int_0^x f(y)\,dy$$

(answ. in the 1950s). Conjecture. A measure on the line μ.

$$\overline{\qquad\qquad\qquad\qquad\qquad\qquad}$$
$$-\infty \qquad\qquad +\infty$$

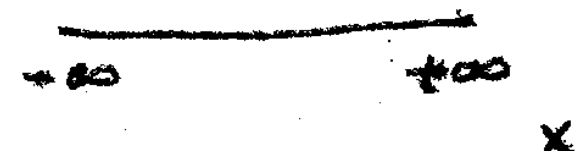
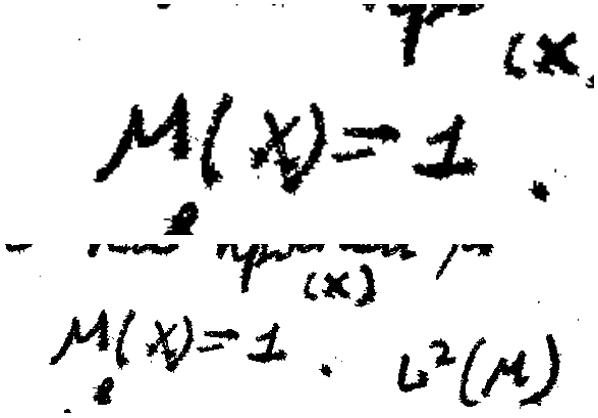


Diagram and typed description. Below the line $-\infty$ on the left, $+\infty$ on the right.

Original source details: GELFAND-P000088

$$\mu(x) = 1, \qquad L^2(\mu).$$

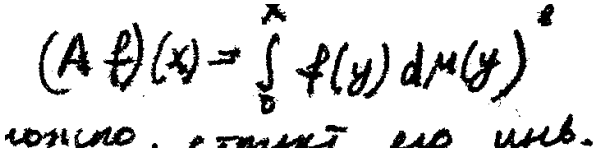
[Above the first argument: (x).]



Uncertain reading. Visible $\mu(x) = 1$ and another (x) above; μ / M. Below the first letter a separate short mark: 2 / illegible sign.

Original source details: GELFAND-P000088

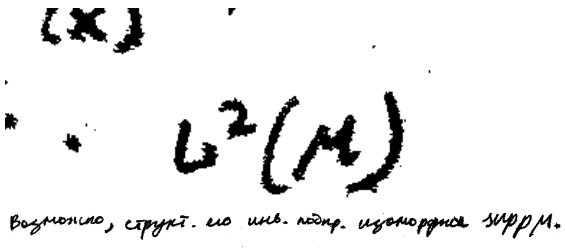
$$(Af)(x) = \int_0^x f(y)\, d\mu(y).$$



Uncertain reading. The lower limit is visible as 0.

Original source details: GELFAND-P000088

Possibly, the struct. of inv. subsp. is isomorphic to supp μ.

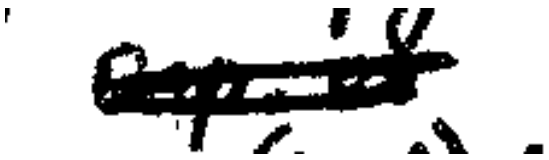


Uncertain reading. In $L^2(\mu)$ and after supp the letter μ / M.

Original source details: GELFAND-P000088

GELFAND-P000089

Possible counterexample. X — sp. with measure μ. T — transf., preserving measure (in Russian a dyn. system). φ — bounded fn. [Crossed out before the formula: Oper. / illegible.]



Uncertain reading. Oper. / illegible

Original source details: GELFAND-P000089

[Above the formula: in L^2.]

$$(Af)(x) = \varphi(x) f(Tx).$$

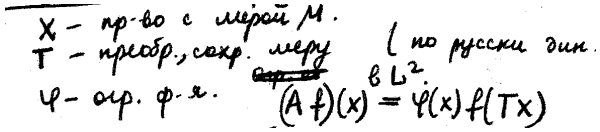
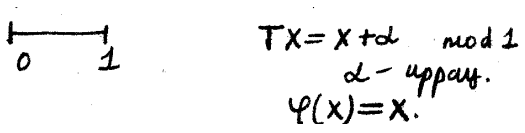


Diagram and typed description. Above X, T, φ; above the formula "in L^2"; on the right $f(Tx)$.

Original source details: GELFAND-P000089

$\exists$ examples of transf. with simple cont. spectrum.

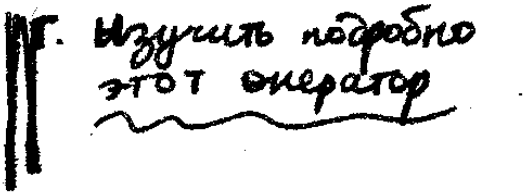


Original source details: GELFAND-P000089

Diagram and typed description. A segment with two end strokes, labels 0, 1 below. On the right $Tx = x + \alpha \pmod 1$, "α — irr.", then $\varphi(x) = x$.

$$Tx = x + \alpha \pmod 1, \qquad \alpha \text{ — irr.}$$
$$\varphi(x) = x.$$

Study in detail
this operator



Original source details: GELFAND-P000089

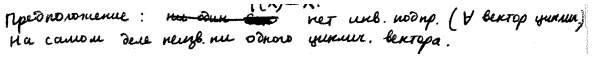
Diagram and typed description. At right, two vertical strokes with a mark like Γ. at the top. Below "this operator" a wavy underline.

Conjecture: [Crossed out: on one... / on a giv... ("на один..." / "на дан...") / illegible.] no inv. subsp. (∀ vector cyclic).

Original source details: GELFAND-P000089

Uncertain reading. on one... / on a giv... ("на один..." / "на дан...") / illegible

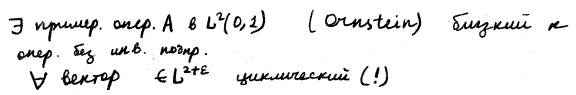
In fact, not a single cyclic vector is known.



Original source details: GELFAND-P000089

Diagram and typed description. In parentheses ∀ vector cyclic; below "not a single cyclic vector is known".

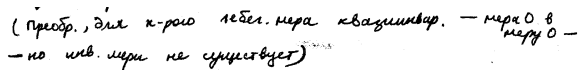
∃ an example of an oper. A in $L^2(0,1)$ (Ornstein), close to an oper. without inv. subsp. ∀ vector $\in L^{2+\varepsilon}$ cyclic (!).



Original source details: GELFAND-P000089

Diagram and typed description. In the first line $L^2(0,1)$ and the name Ornstein; below ∀, $\in L^{2+\varepsilon}$ and (!).

(A transf. for which Lebesgue measure is quasi-inv. — measure 0 to measure 0 — but an inv. measure does not exist.)

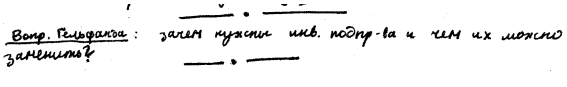


Original source details: GELFAND-P000089

Diagram and typed description. "measure 0 to measure 0"; after the line break "but an inv. measure does not exist".

— • —

Gelfand's question: why are inv. subsp. needed, and what can replace them?

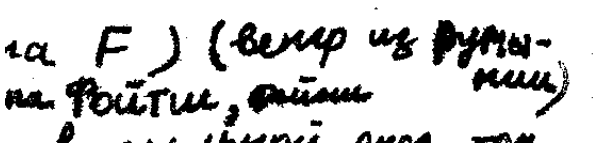


Original source details: GELFAND-P000089

Diagram and typed description. "Gelfand's question" is underlined; before and after the question, separators with a dot between two lines.

— • —

Another professor, speaking English (beginning with F) (a Hungarian from Romania), something like Foytsh / Foyshi (“Фойтш” / “Фойши”)



Original source details: GELFAND-P000089

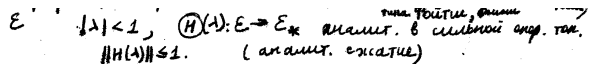
Uncertain reading. Foytsh / Foyshi (“Фойтш” / “Фойши”)

[A repeated entry is densely overwritten: oyshi / Foyshi (“ойши” / “Фойши”) / illegible.]

Original source details: GELFAND-P000089

Uncertain reading. oyshi / Foyshi (“ойши” / “Фойши”) / illegible

$$E, \qquad |\lambda| < 1, \qquad \textcircled{H}(\lambda) : E \longrightarrow E_*,$$
$$\|H(\lambda)\| \leq 1.$$



Original source details: GELFAND-P000089

Diagram and typed description. Separately E; $|\lambda| < 1$; the first letter H is circled, the right E has a lower star; below $\|H(\lambda)\| \leq 1$.

[The letter H in the first entry is circled.] Analyt. in the strong oper. top. (analyt. contraction).

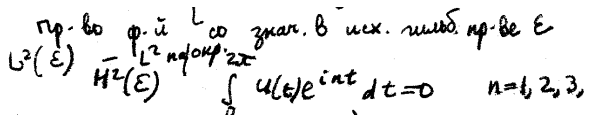
$$\{E, E_*, \textcircled{H}\}$$

[The last letter in the triple is circled.]

Original source details: GELFAND-P000089

Uncertain reading. The circled sign in the triple: H / Θ / illegible; above, the circled H has two clear upright strokes.

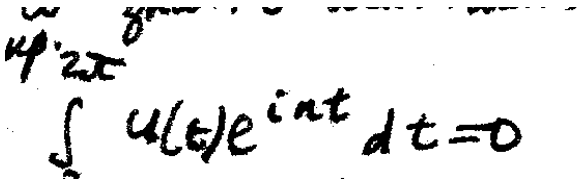
Sp. of fns. with values in the initial Hilb. sp. E. $L^2(E)$ — L^2 on the circ. 2π.



Original source details: GELFAND-P000089

Diagram and typed description. In the upper line Hilb. sp. E; then $L^2(E)$, 2π and $H^2(E)$; the exponent has $+int$.

$$H^2(E) \qquad \int_0 u(t)e^{int}\,dt = 0, \qquad n = 1, 2, 3, \ldots$$

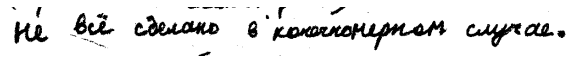


Original source details: GELFAND-P000089

Uncertain reading. The integral clearly has a lower 0; no separate upper limit is visible.

(boundary values of analyt. fns.).

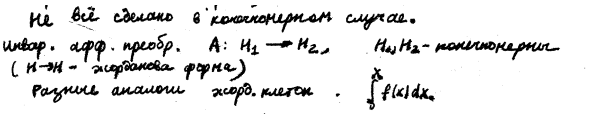
Not everything has been done in the finite-dimensional case.



Original source details: GELFAND-P000089

Manuscript notation. The line reads в конечномерном случае (“in the finite-dimensional case”).

Inv. aff. transf. $A : H_1 \longrightarrow H_2$,



Original source details: GELFAND-P000089

Manuscript notation. The abbreviation is аффф. (“aff.”). Then $A : H_1 \to H_2$, below $H \to H$ and “Jordan form”.

H_1, H_2 — finite-dimensional. ($H \to H$ — Jordan form.) Various analogues of Jordan blocks.

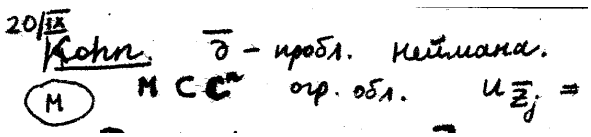
$$\int_0^x f(x)\,dx.$$

Original source details: GELFAND-P000089

Diagram and typed description. Upper limit x, lower 0; the integrand and the differential also have x.

GELFAND-P000090

$20/\overline{\text{IX}}$. Kohn. $\bar\partial$ — Neumann probl.



Original source details: GELFAND-P000090

Diagram and typed description. Date $20/\text{IX}$: strokes above and below IX; Kohn underlined; on the left M in an oval.

M

$M \subset C^n$ bounded dom.

Uncertain reading. The exponent above C is dense: n / m.

Original source details: GELFAND-P000090

$$u_{\bar{z}_j} = \frac{1}{2}(u_{x_j} + iu_{y_j}) = \alpha_j.$$

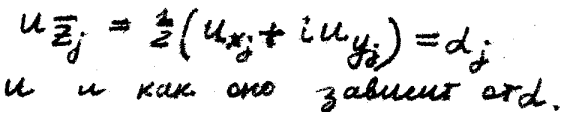


Diagram and typed description. In the first definition a bar above z_j; on the right $\frac{1}{2}$, the sign $+$ and i, then α_j.

Original source details: GELFAND-P000090

Given α_j, when does $\exists u$ and how does it depend on α. Connection with class. theorems.

$$\bar{\partial} u = \alpha,$$
$$\sum u_{z_j}\, d\bar{z}_j = \sum \alpha_j\, d\bar{z}_j$$
under cond. $\alpha_{j\bar{z}_k} = \alpha_{k\bar{z}_j} \qquad (\bar{\partial}\alpha = 0).$

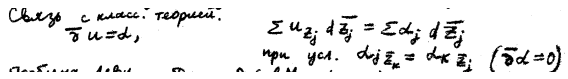
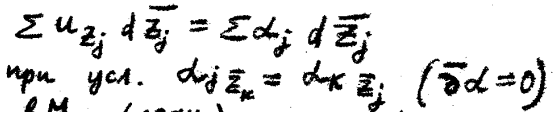


Diagram and typed description. Bars above z_j in the differentials; condition $\alpha_{j\bar{z}_k} = \alpha_{k\bar{z}_j}$, on the right $(\bar{\partial}\alpha = 0)$.

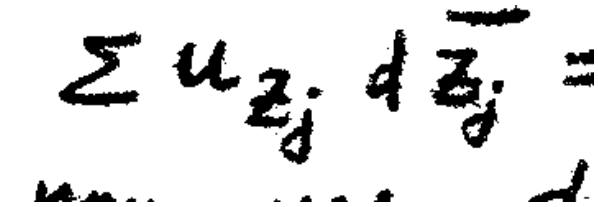
Original source details: GELFAND-P000090

Manuscript notation. In the sum no bar is visible on z_j in u_{z_j}. The bar in $d\bar{z}_j$ is visible.



Original source details: GELFAND-P000090

Manuscript notation. The left-hand sum shows u_{z_j} and $d\bar{z}_j$.



Original source details: GELFAND-P000090

Levi problem.

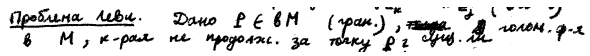
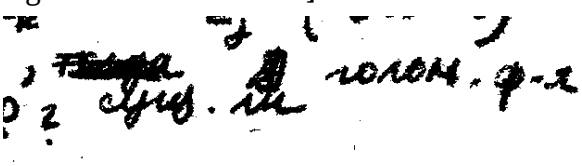


Diagram and typed description. The heading has a straight underline; in the question p, M, the sign $\exists$ and abbreviations.

Original source details: GELFAND-P000090

Given $p \in \partial M$ (bound.), [Crossed out: which (“какая”) / illegible.]
$\exists$ a holom. fn. in M, which does not continue beyond the point p? [Below, by the existential sign: Does there exist.]



Manuscript notation. Below the existential sign is Сущ. ли (“Does there exist”).

Original source details: GELFAND-P000090

Original source details: GELFAND-P000090

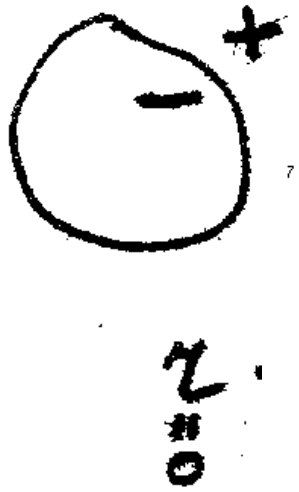
Uncertain reading. which (“какая”) / illegible

The class. Levi theor. solves this probl. locally:

Original source details: GELFAND-P000090

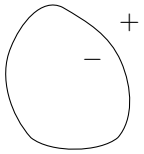
Diagram and typed description. A wavy line under the whole sentence from “The class.” to “locally:”.

$\begin{matrix} r \\ x \\ 0 \end{matrix}$ — equ. of the boundary.

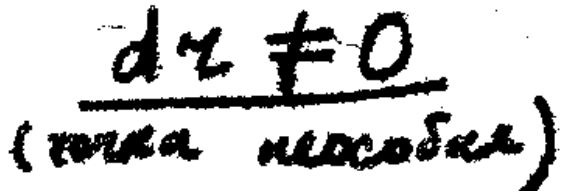


Original source details: GELFAND-P000090

Diagram and typed description. An irregular closed curve, $-$ inside and $+$ outside above right. Below r a small crossed sign resembling x, then 0; their exact reading and role are not established.



$$\underline{dr \neq 0}$$

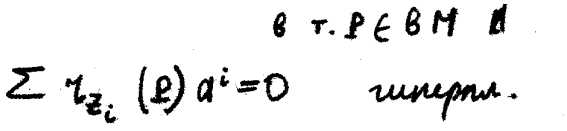


Original source details: GELFAND-P000090

Diagram and typed description. The entry $dr \neq 0$ has a straight underline; below “(point nonsingular)”.

(point nonsingular). At pt. $p \in \partial M$

$$\sum r_{z_i}(p) a^i = 0 \qquad \text{hyperpl.}$$

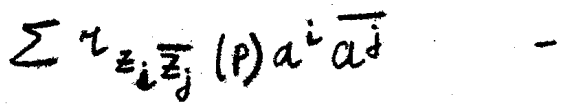


Original source details: GELFAND-P000090

Diagram and typed description. Above the formula “At pt. $p \in \partial M$”; in the formula a^i and $= 0$; on the right “hyperpl.”.

$$\sum r_{z_i \bar{z}_j}(p) a^i \bar{a}^j$$

— Levi form.



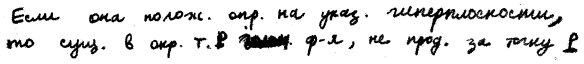
Original source details: GELFAND-P000090

Uncertain reading. The boundary function's letter resembles r / τ, the boundary sign ∂ / b.

If it is pos. def. on the indicated hyperplane,
then there exists in a neighb. of pt. p [Crossed out: illegible.]
a fn., not cont. beyond the point p.

Original source details: GELFAND-P000090

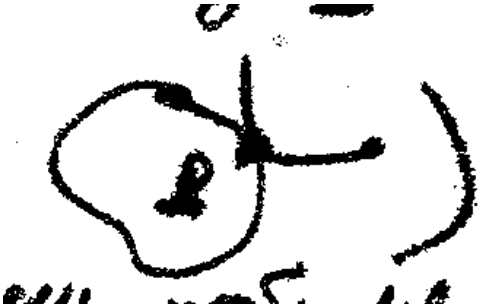
Uncertain reading. illegible



Original source details: GELFAND-P000090

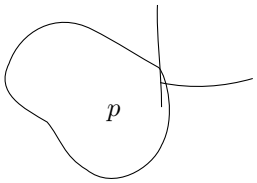
Diagram and typed description. Abbreviations "pos. def.", "on the indicated hyperplane", then "exists in a neighb. of pt. p".

(∃ a polynomial W, such that the zeros are arranged thus:



Original source details: GELFAND-P000090

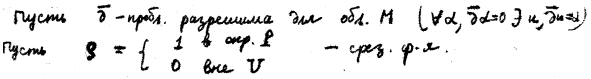
Diagram and typed description. An irregular closed curve; at the right edge a nearly vertical stroke and a stroke running out to the right. Label p inside, left and below their junction. The exact position of the dense overlapping strokes is not established.



) A good answer to that probl. ($\bar{\partial}$) leads to a glob. sol. of the Levi probl. Let the $\bar{\partial}$-probl. be solvable for the dom. M ($\forall \alpha$, $\bar{\partial}\alpha = 0$, $\exists u$, $\bar{\partial} u = \alpha$). Let

$$g = \begin{cases} 1 & \text{in a neighb. of } p, \\ 0 & \text{outside } U \end{cases}$$

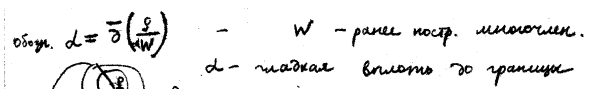
— cut-off fn.



Original source details: GELFAND-P000090

Diagram and typed description. The entry g has a brace, 1 "in a neighb. of p", 0 "outside U"; on the right "cut-off fn.".

Denote $\alpha = \bar{\partial}\left(\frac{g}{W}\right)$. W — the previously constr. polynomial. α — smooth up to the boundary.

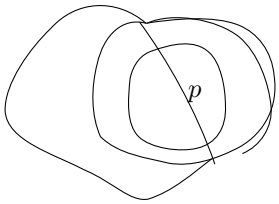


Original source details: GELFAND-P000090

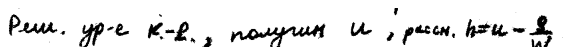
Diagram and typed description. On the left $\alpha = \bar{\partial}(g/W)$; on the right W — the previously constr. polynomial; below "smooth up to the boundary".

Original source details: GELFAND-P000090

Diagram and typed description. Three irregular nested contours and an inclined stroke; a gap at the lower right of the outer contour. p inside the small contour. No labels U or boundary are visible at the contours; dense intersections are not fully distinguishable.



Solve the equ. someh., obtain u, consider $h = u - \frac{g}{W}$.



Original source details: GELFAND-P000090

Uncertain reading. After “equ.” the abbreviation “к-л.” (someh.); the second letter is dense, the exact reading is not fully established. Then u, $h = u - \frac{g}{W}$.

h holom. (since $\bar{\partial} = 0$) [A stroke crossed out: arrow (“стрелка”) / illegible.]

Uncertain reading. arrow (“стрелка”) / illegible

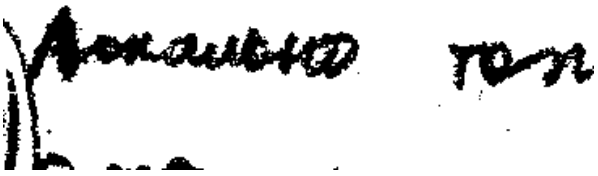
Original source details: GELFAND-P000090

u does not continue.

Original source details: GELFAND-P000090

Uncertain reading. Before “does not continue” the letter u is visible.

Analogo / Analogously (“Аналого” / “Аналогично”) is also possible, but there it continues beyond the point.

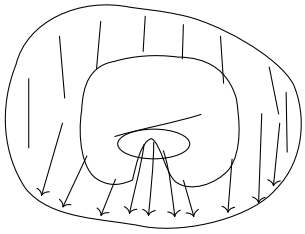


Original source details: GELFAND-P000090

Uncertain reading. Analogo / Analogously (“Аналого” / “Аналогично”)

Original source details: GELFAND-P000090

Diagram and typed description. An outer contour; an inner contour with a lower notch; hatching between them. At the notch a small oval and an oblique stroke above it; several long strokes with downward arrows. Dense intersections and some stroke ends are not fully distinguishable.

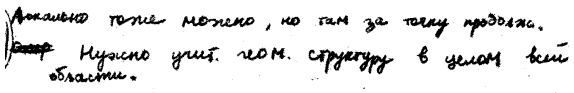


[Crossed out to the right of the drawing: illegible.]

Original source details: GELFAND-P000090

Uncertain reading. illegible

Must take into acc. the geom. structure of the entire domain as a whole.

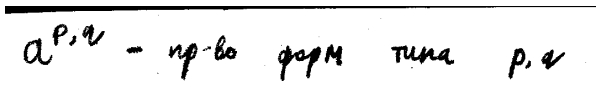


Original source details: GELFAND-P000090

Diagram and typed description. The last sentence breaks to “domain.”; above “but there it continues beyond the point”.

GELFAND-P000091

$a^{p,q}$ — space of forms of type p, q.

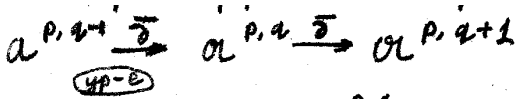


Original source details: GELFAND-P000091

Diagram and typed description. $a^{p,q}$; space of forms of type p, q.

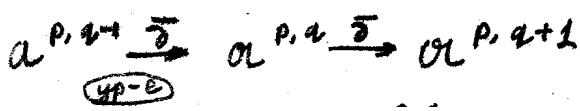
$$a^{p,q-1} \xrightarrow{\bar\partial} \alpha^{p,q} \xrightarrow{\bar\partial} \alpha^{p,q+1}.$$

eqn



Original source details: GELFAND-P000091

Diagram and typed description. Two arrows with $\bar\partial$; under the first, eqn is enclosed in an oval.



Original source details: GELFAND-P000091

Manuscript notation. The three letterforms in a single line. The first letter resembles a. The middle and last have an upper loop like the following α.

General theorems. $\alpha^{p,q}$ — Hilb. scalar product. $\varphi, \psi \in D_{\bar\partial^*}$.

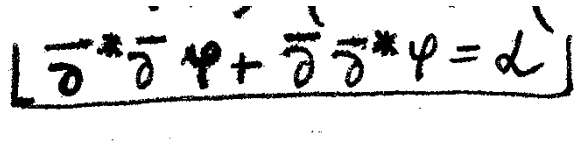
$$Q(\varphi,\psi) = (\bar\partial\varphi, \bar\partial\psi) + (\bar\partial^*\varphi, \bar\partial^*\psi).$$

[At the right:] ($\bar\partial$ on smooth forms of 1st degree without boundary conditions).

Original source details: GELFAND-P000091

Diagram and typed description. $D_{\bar\partial^*}$, two terms in $Q(\varphi,\psi)$; at the right, parentheses about smooth forms of 1st degree without boundary conditions.

$$\boxed{\bar\partial^*\bar\partial\varphi + \bar\partial\bar\partial^*\varphi = \alpha}$$



Original source details: GELFAND-P000091

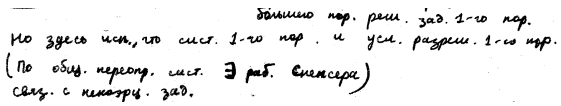
Diagram and typed description. Equation with two terms and right side α; frame at the left, bottom and right, with no top side.

With the aid of a pos. def. prob. [Crossed out: illegible] of higher order solve a prob. of 1st order.

Original source details: GELFAND-P000091

Uncertain reading. After prob., a former short expression is densely crossed out; an initial 1-/4-like stroke, the rest illegible.

But here it is used that the syst. is of 1st order and the solvability cond. is of 1st order. (On gen. overdet. syst. ∃ work of Spencer.) Connection with the Neumann prob.

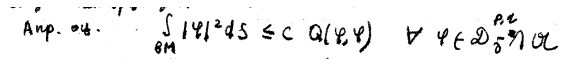


Original source details: GELFAND-P000091

Diagram and typed description. Syst. of 1st order, solvability cond. of 1st order; in parentheses ∃ work of Spencer; then connection with the Neumann prob.

A priori est.

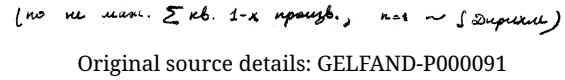
$$\int_{\partial M} |\varphi|^2\, dS \le CQ(\varphi,\varphi) \qquad \forall\varphi \in D_{\bar\partial^*} \cap \alpha^{p,q}.$$



Original source details: GELFAND-P000091

Diagram and typed description. The integral has ∂M, integrand $|\varphi|^2 dS$; $\le CQ(\varphi,\varphi)$ and $\forall\varphi \in D_{\bar\partial^*} \cap \alpha^{p,q}$.

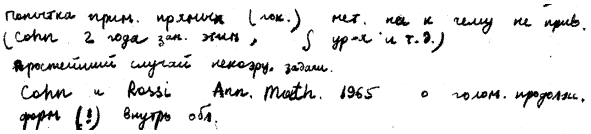
(but not min. $\sum$ of squares of 1st derivs., $n = 1 \sim \int$ Dirichlet).



Original source details: GELFAND-P000091

Diagram and typed description. Before $\sum$, min.; then squares of 1st derivs., $n = 1 \sim \int$ Dirichlet.

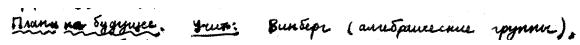
Attempt to apply projection (loc.) no, led nowhere. (Cohn spent 2 years on this, ∫ eqns and so on.) Simplest case of the Neumann problem. Cohn and Rossi Ann. Math. 1965 on holom. continuation of forms (!) into the domain.



Original source details: GELFAND-P000091

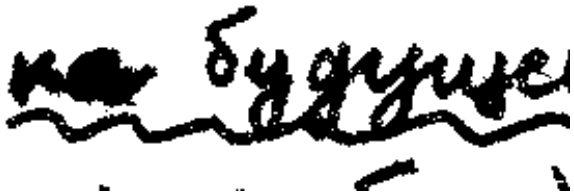
Diagram and typed description. Attempt to apply projection (loc.); Cohn in two places; 2 years, ∫ eqns; Rossi, Ann. Math. 1965; (!) after forms.

Plans for the future.



Original source details: GELFAND-P000091

Diagram and typed description. Wavy line beneath the heading Plans for the future.



Original source details: GELFAND-P000091

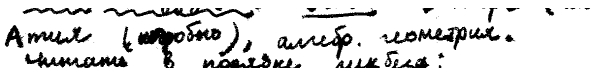
Uncertain reading. The word for in the heading is densely overwritten; the earlier strokes are illegible.

Study: Vinberg (algebraic groups),

Original source details: GELFAND-P000091

Diagram and typed description. Study: has a straight underline; Vinberg (algebraic groups).

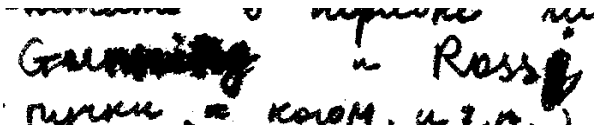
Atiy[?] (...detail[?]; the beginning in parentheses is densely overwritten), alg. geometry.



Original source details: GELFAND-P000091

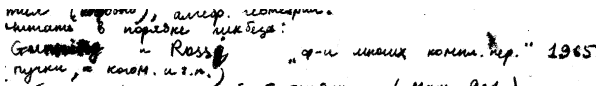
Uncertain reading. Atiy[?] (...detail[?]; the beginning in parentheses is densely overwritten), alg. geometry.

read for basic instruction: [Crossed out: Latin *G*... / Cyrillic Г...; the rest of the name is illegible.] and Rossi “functions of several complex variables” 1965 (sheaves, cohom. and so on).



Original source details: GELFAND-P000091

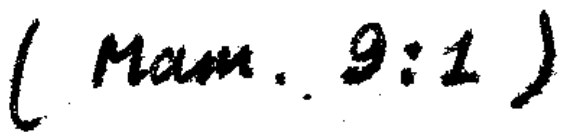
Uncertain reading. An initial Latin G is readable under the heavy strokes. Gunning is a possible reading.



Original source details: GELFAND-P000091

Diagram and typed description. After the cancelled name, and Rossi; functions of several complex variables, 1965; in parentheses, sheaves, cohom. and so on.

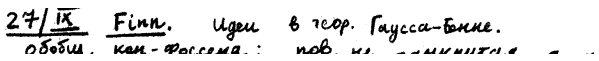
Review by Dieudonné of the work of Grothendieck (Mat. 9:1).



Original source details: GELFAND-P000091

Manuscript notation. The parenthesis reads Мат. 9:1 (“Mat. 9:1”).

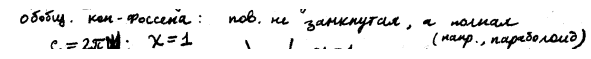
27/IX Finn. Ideas in the Gauss–Bonnet theor.



Original source details: GELFAND-P000091

Diagram and typed description. Separate straight lines under the date and Finn; short bars above and below IX.

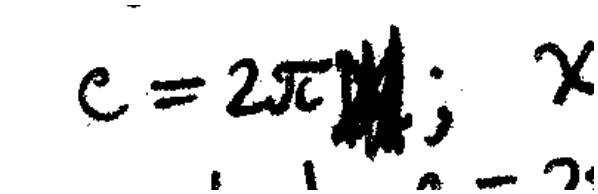
Generalization of Cohn–Vossen: the surf. is not closed, but complete (e.g., a paraboloid).



Original source details: GELFAND-P000091

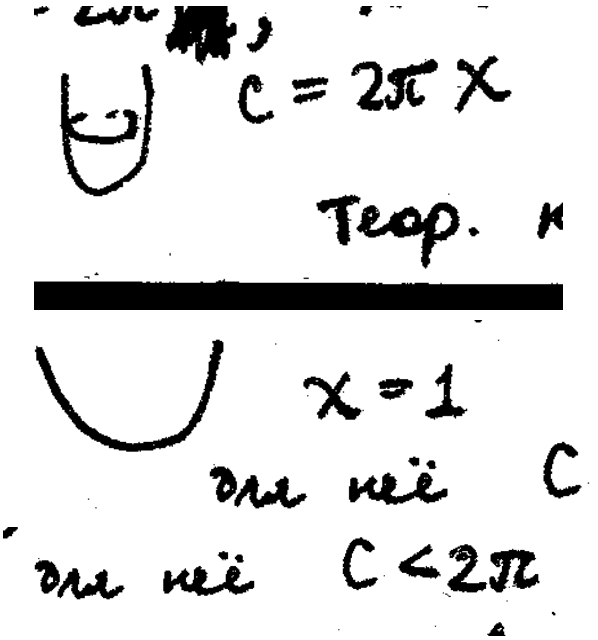
Diagram and typed description. The surf. is not closed, but complete; in parentheses, e.g., a paraboloid.

$C = 2\pi$ [Crossed-out sign: illegible]; $\chi = 1$.



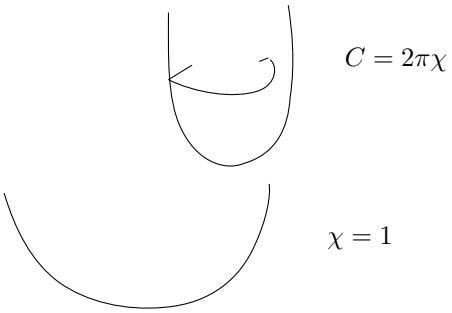
Original source details: GELFAND-P000091

Uncertain reading. After the first $C = 2\pi$ a sign is densely crossed out; its exact reading is not established.



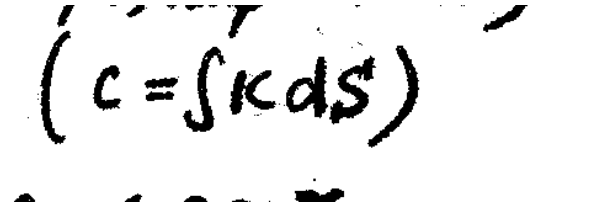
Original source details: GELFAND-P000091

Diagram and typed description. Two lines opening upward; the small one at the left has a short curved transverse stroke with thickened ends. Its exact role and direction are not established. To its right $C = 2\pi\chi$; by the large one $\chi = 1$, below, for it $C < 2\pi$.



for it $C < 2\pi$

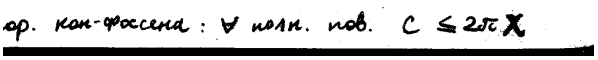
($C = \int K\, dS$).



Original source details: GELFAND-P000091

Diagram and typed description. $C = \int K dS$ is enclosed in round parentheses.

Cohn–Vossen theor.: $\forall$ complete surf. $C \leq 2\pi\chi$.

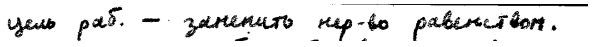


Original source details: GELFAND-P000091

Diagram and typed description. The last line has $\forall$ complete surf. and $C \leq 2\pi\chi$.

GELFAND-P000092

Aim of the work — to replace the inequality by an equality.



Original source details: GELFAND-P000092

Diagram and typed description. Aim of the work — to replace the inequality by an equality.

Next time a review by Volevich

on var., anal. bundles.

Original source details: GELFAND-P000092

Manuscript notation. The manuscript reads Волевича (“Volevich”).

$Kohn$’om[?]

Original source details: GELFAND-P000092

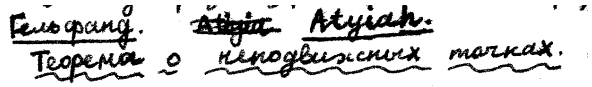
Uncertain reading. After the heavy opening stroke, ohn is legible, followed by the apostrophe and the Russian ending ом.

Gelfand. [Crossed out: Atiy...; ending illegible.] Atiyah.

Original source details: GELFAND-P000092

Uncertain reading. [Crossed out: Atiy...; ending illegible.] Atiyah.

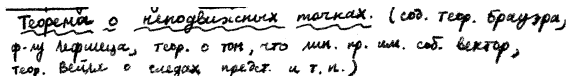
Fixed-point theorem.



Original source details: GELFAND-P000092

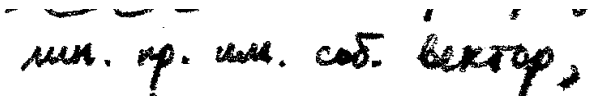
Diagram and typed description. Straight lines under Gelfand. and Atiyah.; wavy line under Fixed-point theorem.

(contains Brouwer’s theor., the Lefschetz formula, the theor. that a lin. transf. has an eigenvector, Weyl’s theor. on traces of reps. and so on.)



Original source details: GELFAND-P000092

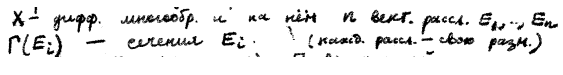
Diagram and typed description. At the beginning of the parentheses, contains; then Brouwer’s theor., the Lefschetz formula, eigenvector, Weyl’s theor. on traces of reps.



Original source details: GELFAND-P000092

Manuscript notation. The manuscript abbreviates “lin. transf.” as лин. пр.

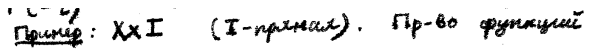
X — diff. manifold and on it n vector bundles $E_1, \ldots, E_n$. $\Gamma(E_i)$ — sections of E_i. (each bundle — its own dim.)



Original source details: GELFAND-P000092

Diagram and typed description. X, n, $E_1, \ldots, E_n$ and $\Gamma(E_i)$; in parentheses, each bundle — its own dim.; the last ending of the Russian word for its own is tight.

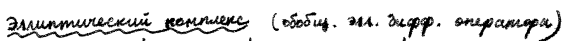
Example: $X \times I$ (I — a straight line). Space of functions.



Original source details: GELFAND-P000092

Diagram and typed description. Straight underline under Example:; $X \times I$, I — a straight line, space of functions.

Elliptic complex (generalizes ell. diff. operators).

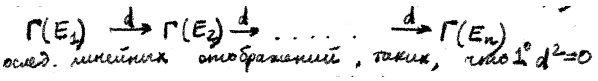


Original source details: GELFAND-P000092

Diagram and typed description. Wavy line under Elliptic complex; then generalizes ell. diff. operators.

$$\Gamma(E_1) \xrightarrow{d} \Gamma(E_2) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(E_n).$$

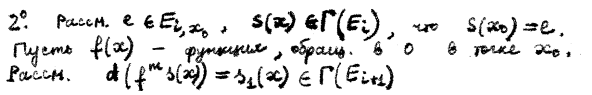
a seq. of linear maps such that 1° $d^2 = 0$.



Original source details: GELFAND-P000092

Diagram and typed description. Three arrows with d; $\Gamma(E_1)$, $\Gamma(E_2)$, ellipsis, $\Gamma(E_n)$; 1°, $d^2 = 0$.

2°. Consider $e \in E_{i,x_0}$, $s(x) \in \Gamma(E_i)$, such that $s(x_0) = e$. Let $f(x)$ be a function taking the value 0 at the point x_0. Consider $d(f^m s(x)) = s_1(x) \in \Gamma(E_{i+1})$.

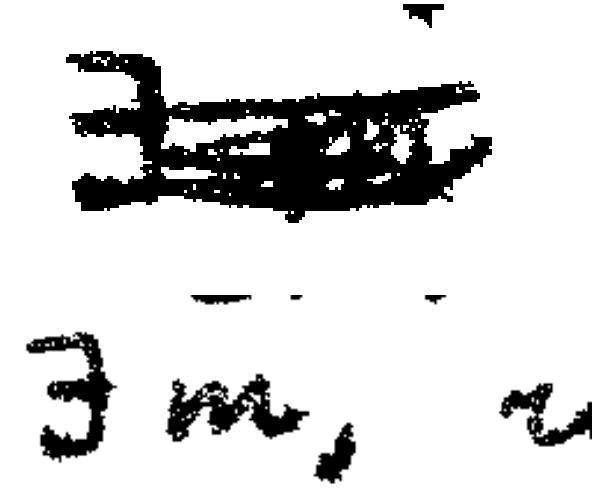


Original source details: GELFAND-P000092

Diagram and typed description. $e \in E_{i,x_0}$; $s(x_0) = e$; function $f(x)$ vanishes at x_0; $d(f^m s(x)) = s_1(x) \in \Gamma(E_{i+1})$.

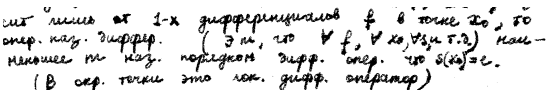
Then, if [Crossed out: $\exists$ with an illegible continuation.] $\exists m$,

such that $\forall x_0$ ($f(x_0) = 0$) $s_1(x_0)$ depends only on the 1st differentials of f at the point x_0, the operator is called differential. ($\exists m$ such that $\forall f$, $\forall x_0$, $\forall s$ and so on.) The least m is called the order of the diff. operator such that $s(x_0) = e$. (In a neighbourhood of the point this is a local diff. operator.)



Original source details: GELFAND-P000092

Uncertain reading. [Crossed out: $\exists$ with an illegible continuation.] $\exists m$,



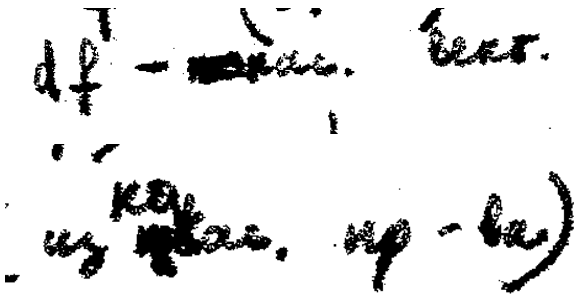
Original source details: GELFAND-P000092

Diagram and typed description. Three $\forall$ signs in parentheses; the least m, then such that $s(x_0) = e$; in the final parentheses, in a neighbourhood of the point.

$$df = (\eta, dx).$$

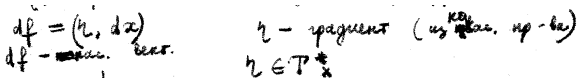
df — [Short word crossed out: illegible] vector; η — gradient (from [tang. inserted above densely overwritten strokes] space).

$\eta \in T_x^*$.



Original source details: GELFAND-P000092

Uncertain reading. Before “vector,” the ending ac. is visible after a dense stroke. The abbreviation кас. (“tang.”) is written above the correction before “space.”



Original source details: GELFAND-P000092

Diagram and typed description. $df = (\eta, dx)$; alongside, η — gradient; below, $\eta \in T_x^*$.

$$\sigma(d, \eta)e$$

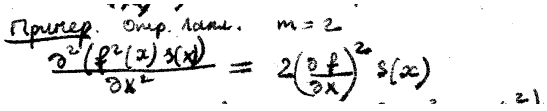
— a lin. function of e, called the symbol of the diff. operator.

Original source details: GELFAND-P000092

Uncertain reading. First argument: d added above a densely cancelled sign; the former sign is illegible.

Example. Laplace operator. $m = 2$.

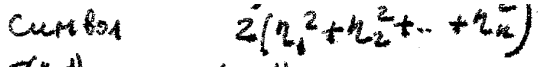
$$\frac{\partial^2 (f^2(x)s(x))}{\partial x^2} = 2\left(\frac{\partial f}{\partial x}\right)^2 s(x).$$



Original source details: GELFAND-P000092

Diagram and typed description. Straight underline under Example.; $m = 2$; $f^2(x)s(x)$ in the second derivative; on the right $2(\partial f/\partial x)^2 s(x)$.

Symbol $2(\eta_1^2 + \eta_2^2 + \cdots + \eta_n^2)$.

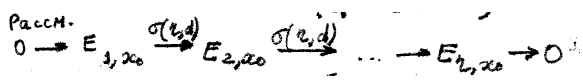


Original source details: GELFAND-P000092

Diagram and typed description. Factor 2; $\eta_1^2 + \eta_2^2 + \cdots + \eta_n^2$.

Consider.

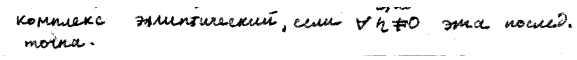
$$0 \longrightarrow E_{1,x_0} \overset{\sigma(\eta,d)}{\longrightarrow} E_{2,x_0} \overset{\sigma(\eta,d)}{\longrightarrow} \cdots \longrightarrow E_{n,x_0} \longrightarrow 0.$$



Original source details: GELFAND-P000092

Diagram and typed description. Zero endpoints; E_{1,x_0}, E_{2,x_0}, ellipsis, E_{n,x_0}; $\sigma(\eta, d)$ over the first two arrows.

The complex is elliptic if $\forall \eta \neq 0$ this seq. is exact.

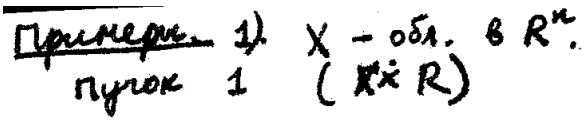


Original source details: GELFAND-P000092

Diagram and typed description. The last line has $\forall \eta \neq 0$ and this seq. is exact.

GELFAND-P000093

Examples. 1) X — a domain in R^n. sheaf 1 ($X \times R$).



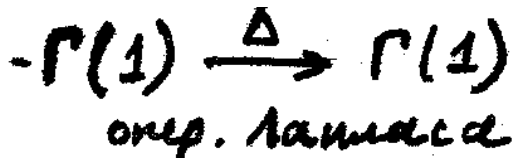
Original source details: GELFAND-P000093

Diagram and typed description. Straight underline under Examples.; 1), X, R^n, sheaf 1, $X \times R$.

Original source details: GELFAND-P000093

$$-\Gamma(1) \xrightarrow{\Delta} \Gamma(1).$$

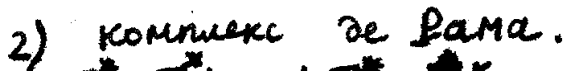
Laplace operator.



Original source details: GELFAND-P000093

Uncertain reading. A short horizontal stroke before the first $\Gamma(1)$: minus / dash.

2) de Rham complex.

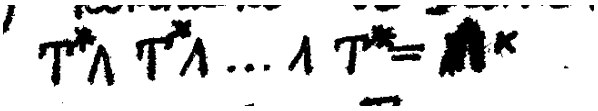


Original source details: GELFAND-P000093

Diagram and typed description. 2) de Rham complex.

Original source details: GELFAND-P000093

$$T^* \wedge T^* \wedge \cdots \wedge T^* = \Lambda^k\,[?].$$



Original source details: GELFAND-P000093

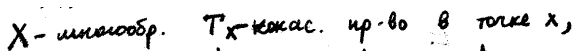
Uncertain reading. The Λ is heavily overwritten. The exponent resembles k or κ.

T — [Beginning of the word densely overwritten; former letters illegible] tang. space.

Original source details: GELFAND-P000093

Uncertain reading. [Beginning of the word densely overwritten; former letters illegible] tang. space.

X — manifold. T_x — tang. space at the point x,



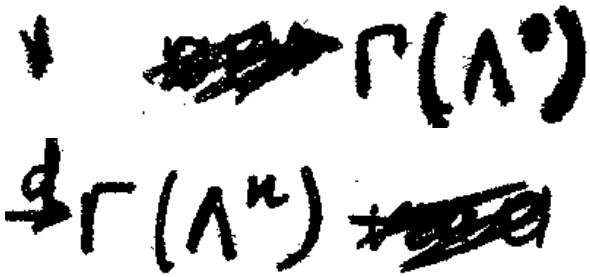
Original source details: GELFAND-P000093

Diagram and typed description. X — manifold; T_x — tang. space at the point x.

Original source details: GELFAND-P000093

Uncertain reading. [At the right: compact entry crossed out; T-/Phi-like strokes, exact reading undetermined.]

$$\Gamma(\Lambda^0) \xrightarrow{d} \Gamma(\Lambda^1) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(\Lambda^n).$$



Original source details: GELFAND-P000093

Uncertain reading. [Signs densely cancelled before the first $\Gamma(\Lambda^0)$, a separate short stroke at the left; after $\Gamma(\Lambda^n)$, an arrow and a 0-like sign are crossed out.]

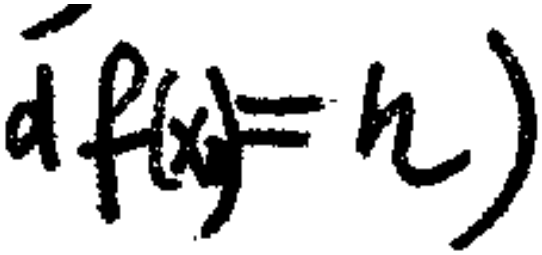
[At the right:] sections — diff. forms.

Original source details: GELFAND-P000093

Diagram and typed description. At the right, sections — diff. forms.; wavy line under diff. forms.

Symbol at the 1st step — $(\eta_1, \ldots, \eta_n)$.

$$d(s(x)f(x)) = s(x_0)\eta = e\eta \qquad (df(x) = \eta).$$

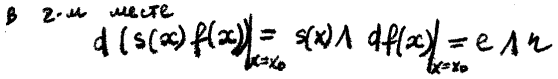


Original source details: GELFAND-P000093

Manuscript notation. The parenthesis reads $df(x) = \eta$.

In the 2nd place

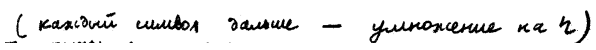
$$d(s(x)f(x))\big|_{x=x_0} = s(x) \wedge df(x)\big|_{x=x_0} = e \wedge \eta.$$



Original source details: GELFAND-P000093

Diagram and typed description. Two restrictions $x = x_0$; $s(x) \wedge df(x)$ and $e \wedge \eta$.

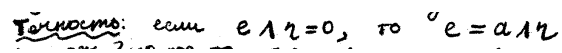
(each subsequent symbol — multiplication by η).



Original source details: GELFAND-P000093

Diagram and typed description. Each subsequent symbol — multiplication by η, in parentheses.

Exactness: if $e \wedge \eta = 0$, then $e = a \wedge \eta$.

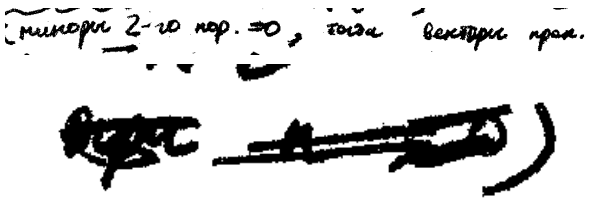


Original source details: GELFAND-P000093

Diagram and typed description. Wavy line under Exactness:; $e \wedge \eta = 0$, $e = a \wedge \eta$.

(minors of the 2nd ord. $= 0$, then the vectors are prop. [Final signs crossed out; dense entry, arrow and 0-like ending; exact reading undetermined.]

)



Original source details: GELFAND-P000093

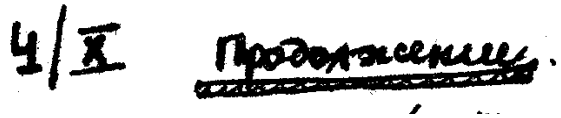
Uncertain reading. The words read “the vectors are prop.” Under the following cancellation, an arrow and a zero-like sign are visible.

3) $\vec{\partial}$ and so on.

Original source details: GELFAND-P000093

Manuscript notation. The arrow above ∂ points to the right.

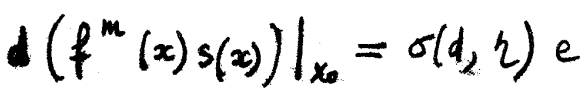
$4/\overline{\underline{X}}$ Continuation. $\Gamma(E)$ — seq.



Original source details: GELFAND-P000093

Diagram and typed description. Date $4/\overline{\underline{X}}$; Continuation. underlined with two straight lines.

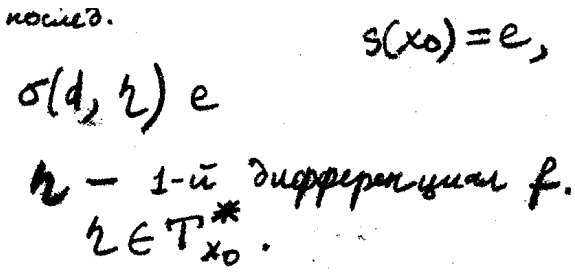
$$d(f^m(x)s(x))\big|_{x_0} = \sigma(d,\eta)e, \qquad s(x_0) = e.$$



Original source details: GELFAND-P000093

Diagram and typed description. $d(f^m(x)s(x))\big|_{x_0} = \sigma(d,\eta)e$; at the right $s(x_0) = e$.

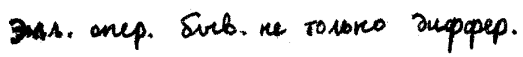
η — 1st differential of f. $\eta \in T^*_{x_0}$.



Original source details: GELFAND-P000093

Diagram and typed description. At the right $s(x_0) = e$; η — 1st differential of f; $\eta \in T^*_{x_0}$.

Ell. operators can be other than differential.



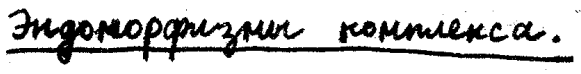
Original source details: GELFAND-P000093

Diagram and typed description. Ell. operators can be other than differential.

Original source details: GELFAND-P000093

Uncertain reading. [Short word crossed out before the heading; dense horizontal strokes and a descending stroke; former letters illegible.]

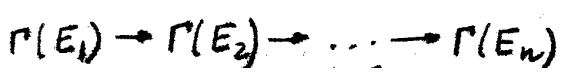
Endomorphism of a complex.



Original source details: GELFAND-P000093

Diagram and typed description. Straight underline under Endomorphism of a complex.

$$\Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n).$$



Original source details: GELFAND-P000093

Diagram and typed description. $\Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n)$.

It induces [Short sign / word densely cancelled before map; illegible.]
a map:

$$T_i^* : H^i(E) \longrightarrow H^i(E).$$

Uncertain reading. [Short sign / word densely cancelled before map; illegible.]

Original source details: GELFAND-P000093

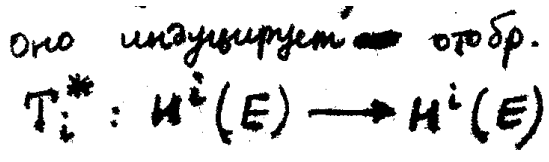
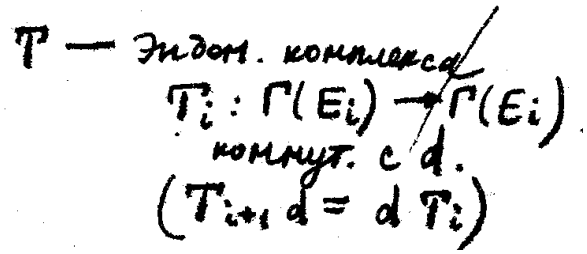


Diagram and typed description. It induces ... a map:; $T_i^* : H^i(E) \longrightarrow H^i(E)$.

Original source details: GELFAND-P000093

[At the right:] T — endom. of the complex, $T_i : \Gamma(E_i) \to \Gamma(E_i)$, commutes with d. $(T_{i+1}d = dT_i)$. ↙



Uncertain reading. At the right a slanting arrow crosses the end of the word complex and passes near the arrow between $\Gamma(E_i)$ and $\Gamma(E_i)$; its exact connection is unclear. Formulas: $T_i : \Gamma(E_i) \to \Gamma(E_i)$, $(T_{i+1}d = dT_i)$.

Original source details: GELFAND-P000093

For an ell. complex they are all finite-dimensional. (In the case of an ell. operator it has the dim. of the index.)

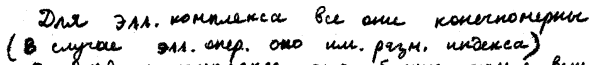


Diagram and typed description. They are all finite-dimensional; in parentheses, In the case of an ell. operator it has the dim. of the index.

Original source details: GELFAND-P000093

In the de Rham complex these are the usual cohom. with real coeffs.

Diagram and typed description. In the de Rham complex these are the usual cohom. with real coeffs.

Original source details: GELFAND-P000093

$$L(T, E) = \sum_{i=1}^{n} (-1)^i \operatorname{tr}(T_i^*).$$

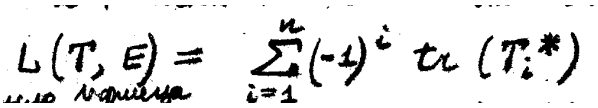


Diagram and typed description. $L(T, E)$; sum from $i = 1$ to n; $(-1)^i \operatorname{tr}(T_i^*)$.

Original source details: GELFAND-P000093

[Under L:] Lefschetz number of the homomorphism [At the right:] $\Downarrow$ for the de Rham complex this is the usual Lefschetz number. [At the bottom:] Lefschetz-Atiyah-Bott.

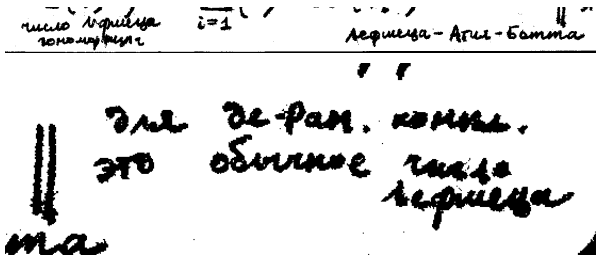


Diagram and typed description. Under L: Lefschetz number, homological; at the right a downward arrow with two strokes and for the de Rham complex this is the usual Lefschetz number; at the bottom Lefschetz-Atiyah-Bott.

Original source details: GELFAND-P000093

GELFAND-P000094

Let $f : X \longrightarrow X$.

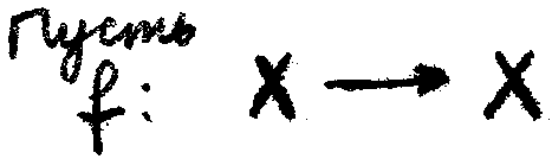
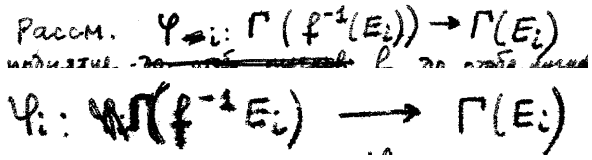


Diagram and typed description. Let; $f : X \longrightarrow X$.

Original source details: GELFAND-P000094

Consider.

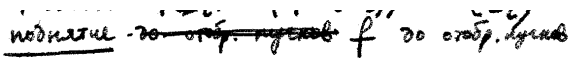
$$\psi_i : \Gamma(f^{-1}(E_i)) \longrightarrow \Gamma(E_i).$$



Uncertain reading. At the first ψ_i, a dense additional stroke precedes the clear i; former index / sign illegible. The first entry has $\Gamma(f^{-1}(E_i))$, the repeated entry $\Gamma(f^{-1}E_i)$.

Original source details: GELFAND-P000094

lifting — ~~to a map of sheaves~~ f to a map of sheaves.



Uncertain reading. "To a map of sheaves" is crossed out. In the second ψ_i formula, a short sign before Γ is densely crossed out.

Original source details: GELFAND-P000094

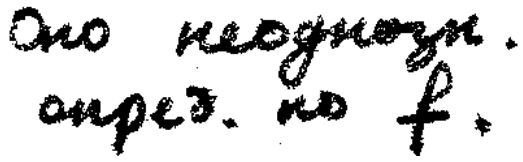
$$\psi_i : \Gamma(f^{-1}E_i) \longrightarrow \Gamma(E_i),$$

$$E_{i,f^{-1}x} \xrightarrow{\psi_{i,x}} E_{i,x}.$$

Manuscript notation. The left fibre index reads $f^{-1}x$.

Original source details: GELFAND-P000094

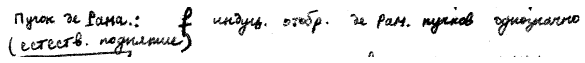
It is not uniquely determ. by f.



Original source details: GELFAND-P000094

Diagram and typed description. It is not uniquely determ. by f.

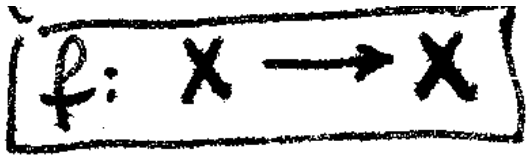
de Rham sheaf: f uniquely induces a map of de Rham sheaves. (natural lifting).



Original source details: GELFAND-P000094

Diagram and typed description. f uniquely induces a map of de Rham sheaves; natural lifting in parentheses.

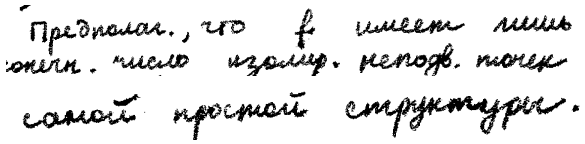
$$\boxed{f : X \longrightarrow X}.$$



Original source details: GELFAND-P000094

Diagram and typed description. Rectangular frame around $f : X \longrightarrow X$; the natural-lifting parenthesis is above its upper border.

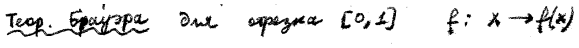
Supp. that f has only a finite number of isol. fixed points of the simplest structure.



Original source details: GELFAND-P000094

Diagram and typed description. Supp. that f has only a finite number of isol. fixed points of the simplest structure.

Brouwer theorem for the interval $[0, 1]$. $f : x \to f(x)$.

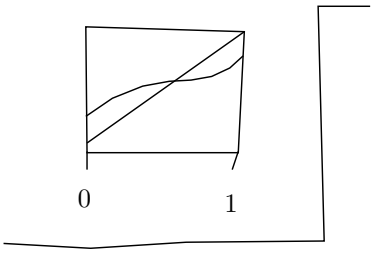


Original source details: GELFAND-P000094

Diagram and typed description. Wavy line under Brouwer theorem; $[0, 1]$; $f : x \to f(x)$.

Original source details: GELFAND-P000094

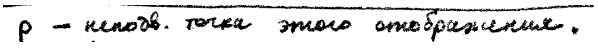
Diagram and typed description. Sketch for Brouwer's theorem. The angular line continues to the right above the three lines about the fixed point.



ρ — fixed point of this map.

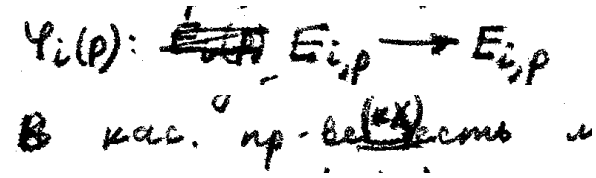
At it we have $\psi_i(\rho)$: [An E-like fibre entry crossed out; indices illegible.] $E_{i,\rho} \to E_{i,\rho}$.

In the tang. space (to[?] X) there is a linear transformation.



Original source details: GELFAND-P000094

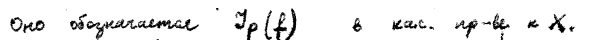
Diagram and typed description. ρ — fixed point of this map.



Original source details: GELFAND-P000094

Uncertain reading. Before the new $E_{i,\rho}$ an E-like entry with unread indices is cancelled. In the underlined parentheses after "space," X is legible and the first letter resembles к.

It is denoted $J_\rho(f)$ in the tang. space to X.



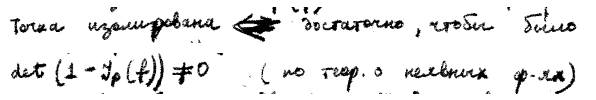
Original source details: GELFAND-P000094

Diagram and typed description. $J_\rho(f)$; in the tang. space to X.

The point is isolated $\Leftarrow$ it is sufficient that

$$\det(1 - J_\rho(f)) \neq 0.$$

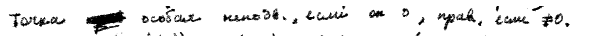
(by the implicit-function theorem).



Original source details: GELFAND-P000094

Diagram and typed description. Heavy leftward arrow $\Leftarrow$; $\det(1 - J_\rho(f)) \neq 0$; by the implicit-function theorem.

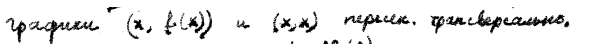
The point [short crossed-out word illegible] singular fixed, if it is 0, regular, if $\neq 0$.



Original source details: GELFAND-P000094

Uncertain reading. After point, a short word is densely cancelled; former letters cannot be recovered. Then singular fixed, if it is 0, regular, if $\neq 0$.

The graphs $(x, f(x))$ and (x, x) intersect transversally.



Original source details: GELFAND-P000094

Diagram and typed description. $(x, f(x))$ and (x, x); intersect transversally.

Form the number

$$\frac{\operatorname{tr}\psi_i(\rho)}{|\det(1-J_\rho(f))|}.$$

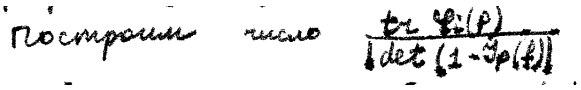


Diagram and typed description. $\operatorname{tr}\psi_i(\rho)$ above $|\det(1-J_\rho(f))|$.

Original source details: GELFAND-P000094

At the point ρ there are 2 lin. transf. $\psi_i(\rho)$ — the fibre into itself, and a transf. of the tang. space.

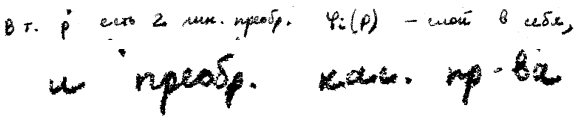


Diagram and typed description. At the point ρ — 2 lin. transf.; $\psi_i(\rho)$ — the fibre into itself; transf. of the tang. space.

Original source details: GELFAND-P000094

$$\sum(-1)^i\frac{\operatorname{tr}\psi_i(\rho)}{|\det(1-J_\rho(f))|}=\sigma(\rho).$$

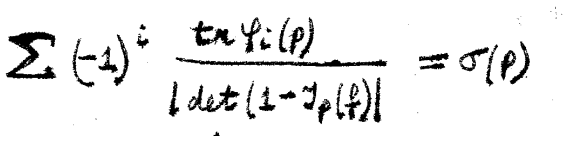


Diagram and typed description. $\sum(-1)^i$ with no visible limits; $\operatorname{tr}\psi_i(\rho)$; $|\det(1-J_\rho(f))|$; $\sigma(\rho)$.

Original source details: GELFAND-P000094

This number does not depend on the complex, but depends only on f and the induced maps in the fibres (but the map of fibres is not arbitrary, it must commute with the complex).

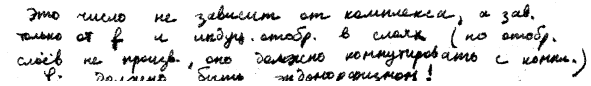


Diagram and typed description. Does not depend on the complex; f and induced maps in the fibres; in parentheses, but the map of fibres is not arbitrary, it must commute with the complex.

Original source details: GELFAND-P000094

ψ_i must be an endomorphism!

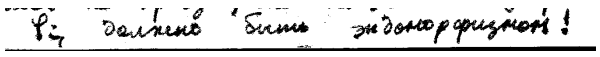


Diagram and typed description. ψ_i must be an endomorphism!

Original source details: GELFAND-P000094

GELFAND-P000095

Theorem of Atiyah–Bott.

Diagram and typed description. Only Theorem is underlined; Atiyah–Bott; number 64.

Original source details: GELFAND-P000095

Let there be an ellipt. complex d, $f : X \to X$ and its lifting ψ, which is an endomorphism of this complex. Then the formula holds

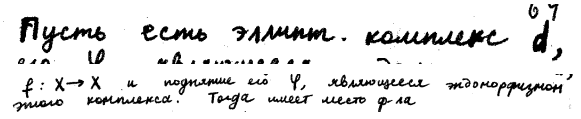


Diagram and typed description. Ellipt. complex d; $f : X \to X$; its lifting ψ, which is an endomorphism of this complex.

Original source details: GELFAND-P000095

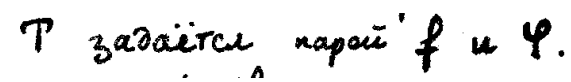
$$L(T, E) = \sum_{\rho} \sigma(\rho)$$

over all fixed points.

Original source details: GELFAND-P000095

Diagram and typed description. $L(T, E) = \sum_{\rho} \sigma(\rho)$; over all fixed points.

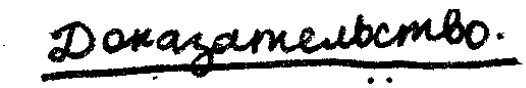
T is given by the pair f and ψ.



Original source details: GELFAND-P000095

Diagram and typed description. T is given by the pair f and ψ.

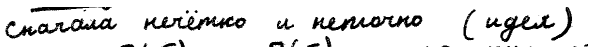
Proof.



Original source details: GELFAND-P000095

Diagram and typed description. Proof is underlined.

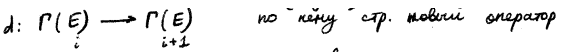
First, without precision and incompletely (ideas)



Original source details: GELFAND-P000095

Manuscript notation. The manuscript writes нечётко, with two dots over ё.

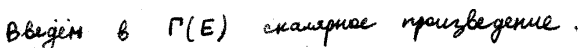
$d : \Gamma(E)_i \longrightarrow \Gamma(E)_{i+1}$, construct a new operator from it.



Original source details: GELFAND-P000095

Diagram and typed description. $d : \Gamma(E)_i \longrightarrow \Gamma(E)_{i+1}$; indices i, $i+1$ are outside the parentheses; construct a new operator from it.

Introduce a scalar product on $\Gamma(E)$.

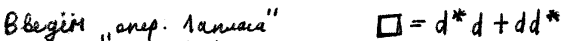


Original source details: GELFAND-P000095

Diagram and typed description. Introduce a scalar product on $\Gamma(E)$.

Introduce the “Laplacian oper.”

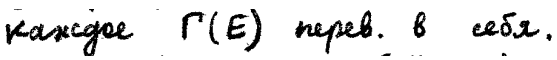
$$\Box = d^*d + dd^*.$$



Original source details: GELFAND-P000095

Diagram and typed description. Laplacian oper.; $\Box = d^*d + dd^*$.

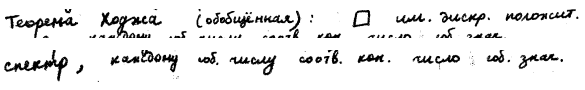
Maps each $\Gamma(E)$ into itself.



Original source details: GELFAND-P000095

Diagram and typed description. Maps each $\Gamma(E)$ into itself.

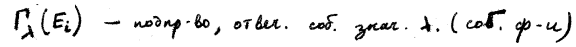
Hodge theorem (generalized): $\Box$ has a discr. positive spectrum, to each eig. number corresponds a finite number of eig. values.



Original source details: GELFAND-P000095

Diagram and typed description. Discr. positive spectrum; to each eig. number corresponds a finite number of eig. values.

$\Gamma_\lambda(E_i)$ — subspace corresp. to the eig. value λ. (eig. functions)



Original source details: GELFAND-P000095

Diagram and typed description. $\Gamma_\lambda(E_i)$; subspace corresp. to the eig. value λ; eig. functions.

$$d : \Gamma_\lambda(E_i) \longrightarrow \Gamma_\lambda(E_{i+1}).$$

Diagram and typed description. $d : \Gamma_\lambda(E_i) \longrightarrow \Gamma_\lambda(E_{i+1})$.

Original source details: GELFAND-P000095

Uncertain reading. The oval contains $\square d = d\square$ followed by a faint short mark. Its identity is unclear.

$\square d = d\square$ [?]

Original source details: GELFAND-P000095

$$d : \Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n).$$

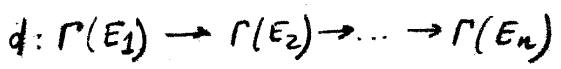


Diagram and typed description. $d : \Gamma(E_1) \to \Gamma(E_2) \to \cdots \to \Gamma(E_n)$.

Original source details: GELFAND-P000095

Now construct

$$d : \Gamma_\lambda(E_1) \longrightarrow \Gamma_\lambda(E_2) \longrightarrow \cdots \longrightarrow \Gamma_\lambda(E_n).$$

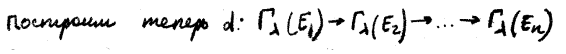


Diagram and typed description. Now construct; $d : \Gamma_\lambda(E_1) \to \Gamma_\lambda(E_2) \to \cdots \to \Gamma_\lambda(E_n)$.

Original source details: GELFAND-P000095

Γ_λ are already finite-dimensional ($d^2 = 0$), homologies arise here too.

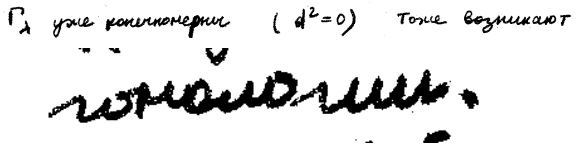


Diagram and typed description. Γ_λ are already finite-dimensional; $(d^2 = 0)$; homologies arise here too.

Original source details: GELFAND-P000095

$$H^i\{\Gamma_\lambda(E)\} = \begin{cases} H^i(E), & \lambda = 0, \\ 0, & \lambda \neq 0. \end{cases}$$

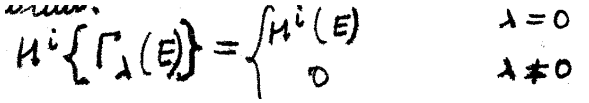
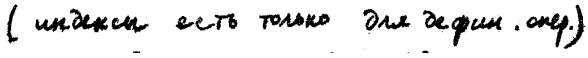


Diagram and typed description. $H^i\{\Gamma_\lambda(E)\}$; $H^i(E)$ at $\lambda = 0$, 0 at $\lambda \neq 0$.

Original source details: GELFAND-P000095

(indices exist only for defin. oper.)



Manuscript notation. The manuscript abbreviation is дефин. (“defin.”), without expansion.

Original source details: GELFAND-P000095

For the de Rham compl. — Hodge theorem — every [crossed-out expression illegible]

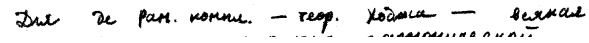


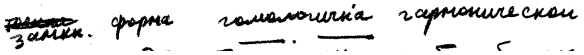
Diagram and typed description. For the de Rham compl. — Hodge theorem — every.

Original source details: GELFAND-P000095

[Crossed-out expression illegible] closed form homologous to a harmonic one.

Uncertain reading. Two densely crossed-out expressions: after every and before closed; former letters undetermined.

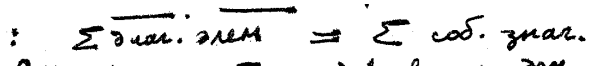
Original source details: GELFAND-P000095



Original source details: GELFAND-P000095

Diagram and typed description. Closed form homologous to a harmonic one; two separate horizontal strokes under homologous.

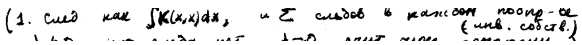
Calculate the trace: $\sum$ diag. elem. $= \sum$ eig. values.



Original source details: GELFAND-P000095

Manuscript notation. The equality sign has two strokes.

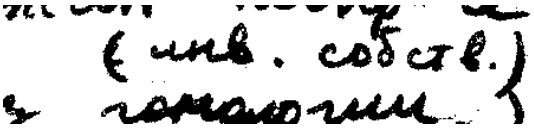
(1. Trace as $\int K(x,x)\,dx$, and $\sum$ of traces in each subspace



Original source details: GELFAND-P000095

Diagram and typed description. (1.; trace as $\int K(x,x)\,dx$; and $\sum$ of traces in each subspace.

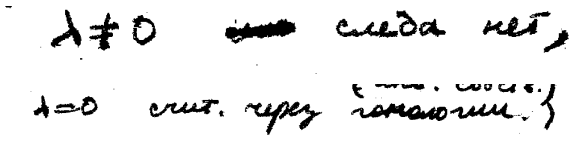
(inv. eig.)



Original source details: GELFAND-P000095

Diagram and typed description. Inv. eig. in parentheses.

$\lambda \neq 0$ [short sign densely crossed out, illegible] no trace, $\lambda = 0$ calc. through homologies. }

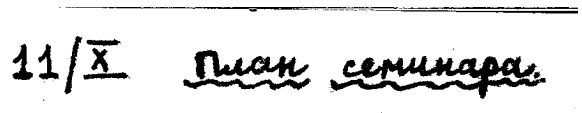


Original source details: GELFAND-P000095

Uncertain reading. A short sign is densely crossed out after $\lambda \neq 0$. A closing brace is visible at the right.

GELFAND-P000096

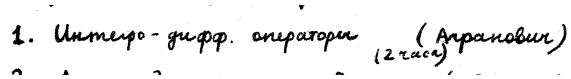
$11/\overline{\underline{X}}$ Seminar plan.



Original source details: GELFAND-P000096

Diagram and typed description. $11/\overline{\underline{X}}$; upper and lower bars at X; Seminar and plan have wavy underlining.

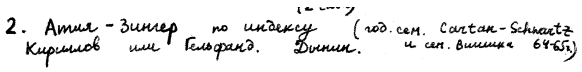
1. Integro-diff. operators (Agranovich). (2 hours)



Original source details: GELFAND-P000096

Diagram and typed description. Integro-diff. operators; Agranovich; beneath the parenthesis, (2 hours).

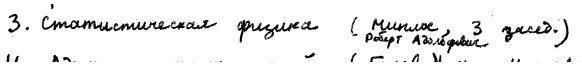
2. Atiyah–Singer on the index. Kirillov or Gelfand. Dynin. (yearly sem. Cartan-Schwartz and Vishik's sem. 64–65.)



Original source details: GELFAND-P000096

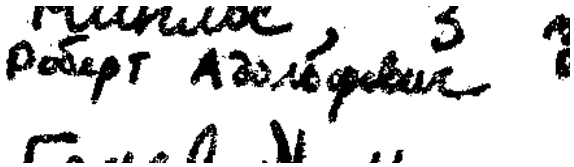
Diagram and typed description. Atiyah–Singer on the index; Kirillov or Gelfand; Dynin; yearly sem. Cartan-Schwartz and Vishik's sem. 64–65.

3. Statistical physics (Minlos , 3 meetings).
Robert Adol'fovich



Original source details: GELFAND-P000096

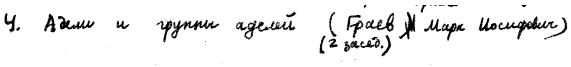
Diagram and typed description. Statistical physics; Minlos; 3 meetings; first name and patronymic below the surname.



Original source details: GELFAND-P000096

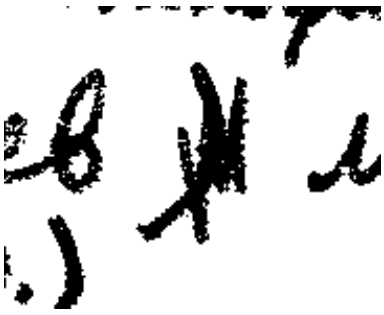
Manuscript notation. The line under Minlos reads Роберт Адольфович ("Robert Adol'fovich").

4. Adeles and adele groups (Graev [crossed-out M-like sign, illegible] Mark Iosifovich). (2 meetings)



Original source details: GELFAND-P000096

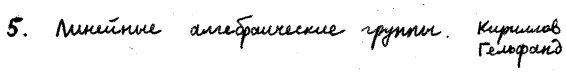
Diagram and typed description. Adeles and adele groups; Graev Mark Iosifovich; (2 meetings).



Original source details: GELFAND-P000096

Uncertain reading. Between Graev and Mark, an M-like sign is densely cancelled; former reading undetermined.

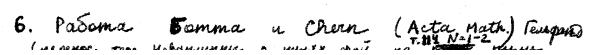
5. Linear algebraic groups. Kirillov Gelfand



Original source details: GELFAND-P000096

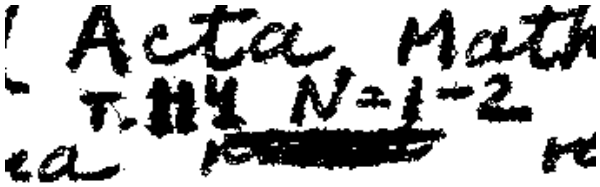
Diagram and typed description. Linear algebraic groups; Kirillov / Gelfand in a column at right.

6. Work of Bott and Chern (Acta Math.) Gelfand Vol. 114 $N = 1$–2



Original source details: GELFAND-P000096

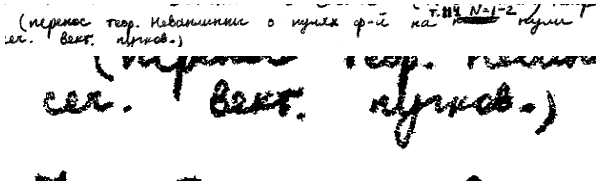
Diagram and typed description. Work of Bott and Chern; Acta Math.; Gelfand; volume notation below the journal name.



Original source details: GELFAND-P000096

Manuscript notation. Below the journal name: “vol. 114 $N = 1$–2.”

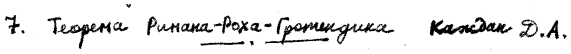
(transfer of Nevanlinna’s theorem on zeros of funct. to [densely crossed-out fragment illegible] curves sect. of vect. bundles.)



Original source details: GELFAND-P000096

Uncertain reading. The parenthesis begins with “transfer.” A fragment is densely crossed out before “curves.”

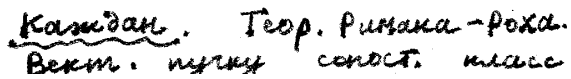
7. Riemann–Roch–Grothendieck theorem. Kazhdan D. A.



Original source details: GELFAND-P000096

Uncertain reading. Item 7 reads Каждан Д. А. (“Kazhdan D. A.”). The extent of the two underlines is approximate.

Kazhdan. Riemann–Roch theorem.



Original source details: GELFAND-P000096

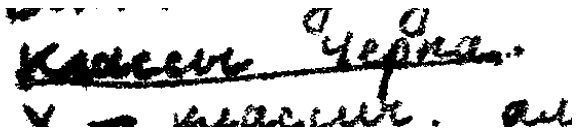
Manuscript notation. The wavy underline is under Каждан (“Kazhdan”).

To a vect. bundle is assoc. a cohom. class — Chern class.

Original source details: GELFAND-P000096

Manuscript notation. The manuscript abbreviation сопост. means “is associated.”

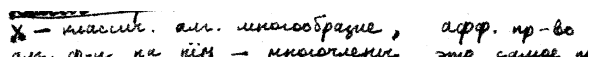
Chern classes.



Original source details: GELFAND-P000096

Diagram and typed description. Chern classes is underlined; the start of classes is written densely.

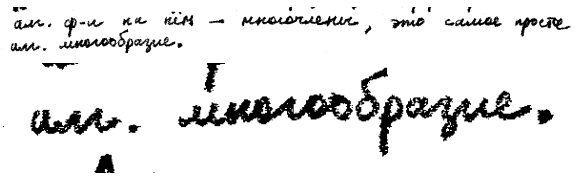
X — classical alg. variety, aff. space



Original source details: GELFAND-P000096

Diagram and typed description. X — classical alg. variety, aff. space.

alg. funct. on it — polynomials, this is the simplest alg. variety.



Manuscript notation. The line ends with “this is the simplest.”

Original source details: GELFAND-P000096

Affine alg. varieties. Let a collection of polynomials be given

$$p_1(x_1, \ldots, x_n), \ldots, p_k(x_1, \ldots, x_n)$$

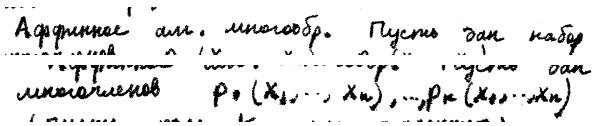


Diagram and typed description. Affine alg. varieties; collection of polynomials $p_1(x_1, \ldots, x_n), \ldots, p_k(x_1, \ldots, x_n)$.

Original source details: GELFAND-P000096

(let the field K be alg. closed)

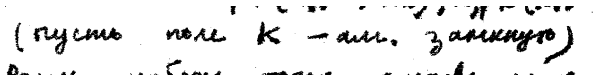


Diagram and typed description. Let the field K be alg. closed; uppercase K.

Original source details: GELFAND-P000096

Consider collections of points — satisfy an eqn., aff. alg. var.-

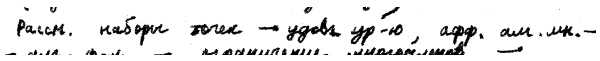


Diagram and typed description. Consider collections of points; satisfy an eqn.; aff. alg. var.- with a final dash.

Original source details: GELFAND-P000096

— alg. funct. — restrictions of polynomials —

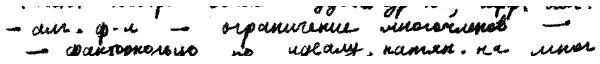


Diagram and typed description. — Alg. funct. — restrictions of polynomials —.

Original source details: GELFAND-P000096

— quotient ring by the ideal generated by polyn.

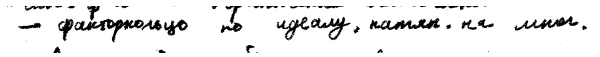


Diagram and typed description. Quotient ring by the ideal generated by polyn.

Original source details: GELFAND-P000096

Alg. prevariety — space loc. isomorphic to an aff. variety.

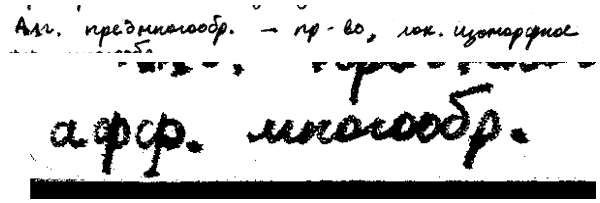


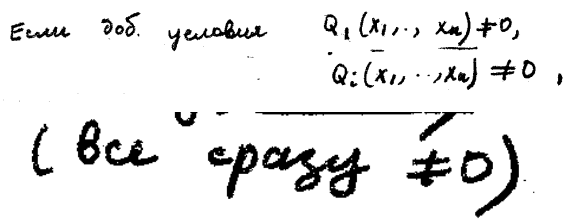
Diagram and typed description. Alg. prevariety; space loc. isomorphic to an aff. variety.

Original source details: GELFAND-P000096

GELFAND-P000097

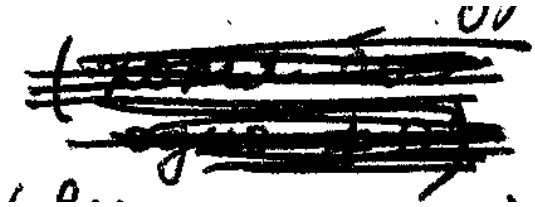
If add. conditions

$$Q_1(x_1, \ldots, x_n) \neq 0,$$
$$Q_i(x_1, \ldots, x_n) \neq 0,$$



Original source details: GELFAND-P000097

Uncertain reading. Q_1, Q_i; x_n) underlined in the first line. Between the lines below Q_1, a dot ·, and a short dash – farther right; their role is undetermined. At right, (all at once $\neq 0$).



Original source details: GELFAND-P000097

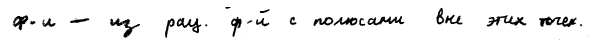
Uncertain reading. Two lines at right are densely crossed out; an initial parenthesis, loops, horizontal and slanting strokes remain visible; no reliable reading established.

(all at once $\neq 0$) then also an alg. var.

Original source details: GELFAND-P000097

Diagram and typed description. Then also an alg. var.

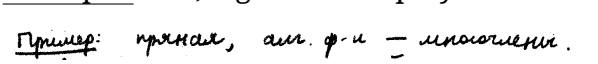
funct. — from rat. funct. with poles outside these points.



Original source details: GELFAND-P000097

Diagram and typed description. Funct. — from rat. funct. with poles outside these points.

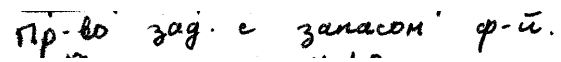
Example: line, alg. funct. — polynomials.



Original source details: GELFAND-P000097

Diagram and typed description. Example is underlined; line, alg. funct. — polynomials.

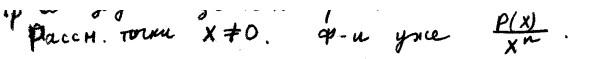
Space given with a stock of funct.



Original source details: GELFAND-P000097

Diagram and typed description. Space given with a stock of funct.

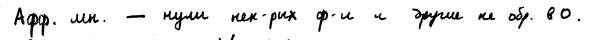
Consider points $x \neq 0$. Funct. now $\frac{P(x)}{x^n}$.



Original source details: GELFAND-P000097

Manuscript notation. The denominator is x^n, with one hump in the exponent.

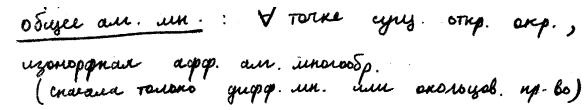
Aff. var. — zeros of some funct. and others do not vanish.



Original source details: GELFAND-P000097

Diagram and typed description. Aff. var. — zeros of some funct. and others do not vanish.

General alg. var.: ∀ point there exists an open nbhd., isomorphic to an aff. alg. variety. (at first only a diff. var. or ringed space)



Original source details: GELFAND-P000097

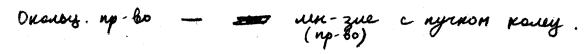
Diagram and typed description. General alg. var. underlined; ∀ point, open nbhd., isomorphic to an aff. alg. variety; below, parentheses concerning a diff. var. or ringed space.

Ringed space — [short crossed-out fragment illegible]

Original source details: GELFAND-P000097

Uncertain reading. Between the dash and var., a short sign / word is densely cancelled; reading undetermined.

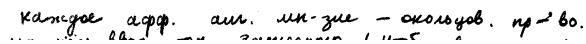
var. with a sheaf of rings.
(space)



Original source details: GELFAND-P000097

Diagram and typed description. Var. with a sheaf of rings; (space) inserted below var.

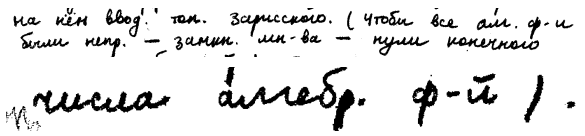
Every aff. alg. var. — ringed space.



Original source details: GELFAND-P000097

Diagram and typed description. Every aff. alg. var. — ringed space.

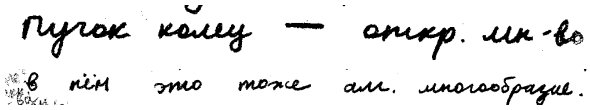
On it one introduces the Zariski top. (So that all alg. funct. are cont. — closed sets — zeros of a finite number of alg. funct.).



Original source details: GELFAND-P000097

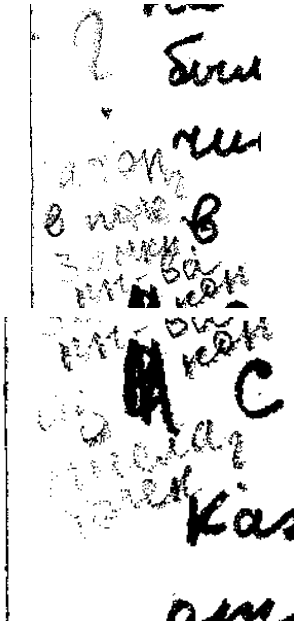
Manuscript notation. A slanted closing parenthesis and a period follow “alg. funct.”

Sheaf of rings — an open set in it this is also an alg. variety.



Original source details: GELFAND-P000097

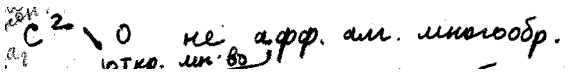
Diagram and typed description. Sheaf of rings — an open set in it this is also an alg. variety.



Original source details: GELFAND-P000097

Uncertain reading. In the left margin, faint, dotted and loop-like traces; before the following formula, a dense A-like sign. No reliable complete words read; origin of the traces undetermined.

$C^2 \setminus 0$ is not an aff. alg. variety.

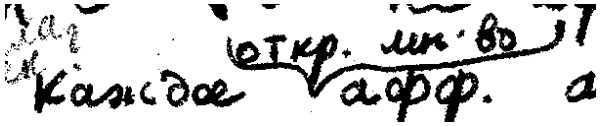


Original source details: GELFAND-P000097

Diagram and typed description. $C^2 \setminus 0$ is not an aff. alg. variety; written C, without an additional vertical bar.

open set

Every aff. alg. variety cont... [ending crossed out]



Original source details: GELFAND-P000097

Diagram and typed description. Above aff., open set is inserted with a lower brace and a downward point; insertion position retained.

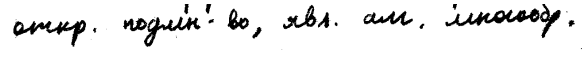
Original source details: GELFAND-P000097

Manuscript notation. The Russian word for “variety” has the genitive ending -ия.

Original source details: GELFAND-P000097

Uncertain reading. Cont...: the ending is densely crossed out; zhit-like letter tops remain above the strokes[?]. Possible whole word contains[?]; cancelled letters not securely determined.

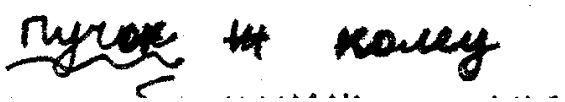
open subset, being an alg. variety.



Original source details: GELFAND-P000097

Diagram and typed description. Open subset, being an alg. variety.

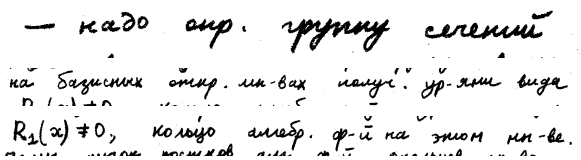
Sheaf [n-/i-like sign, illegible] of rings — must def. the group of sections



Original source details: GELFAND-P000097

Uncertain reading. Sheaf has a wavy underline. Before of rings, a separate dense n-/i-like sign; whether cancellation or insertion is undetermined.

on basic open sets obtained by eqns. of the form $R_1(x) \neq 0$, the ring of alg. funct. on this set.



Original source details: GELFAND-P000097

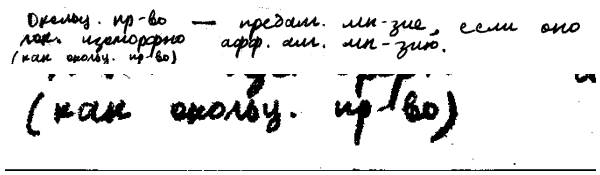
Manuscript notation. The subscript of R_1 is a one with a slanted top and a long foot.

Obtain a sheaf of germs of alg. funct. ringed space.

Original source details: GELFAND-P000097

Diagram and typed description. Obtain a sheaf of germs of alg. funct. ringed space; no separate comma visible after funct.

Ringed space — prealg. var., if it is loc. isomorphic to an aff. alg. var. (as a ringed space)



Original source details: GELFAND-P000097

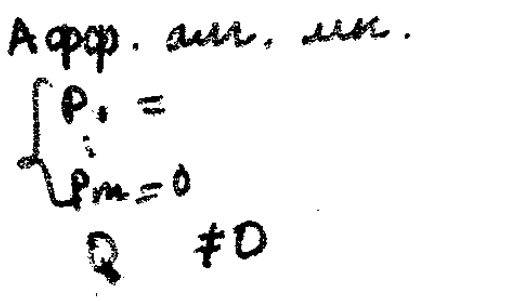
Manuscript notation. After the dash: предалг. мн-зие (“prealg. var.”).

GELFAND-P000098

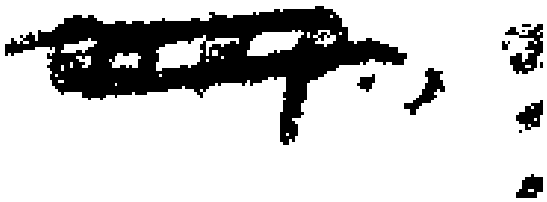
Aff. alg. var.

$$\begin{cases} p_1 = \\ \vdots \\ p_m = 0 \end{cases}$$
$$Q \neq 0$$

[Crossed out: open[?], remainder illegible] closed set.



Uncertain reading. Under the left brace, $p_1 =$ with no visible right-hand side, dots, $p_m = 0$; below, $Q \neq 0$. At right, dense cancellation before closed set; otkr.-like remnants (open)[?], full reading undetermined.



Original source details: GELFAND-P000098

$R_i(x_1, \ldots, x_n) = 0$. R_i — polynomials.

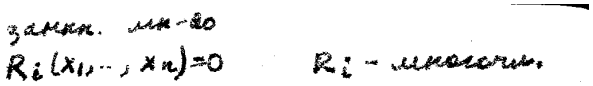


Diagram and typed description. Closed set; $R_i(x_1, \ldots, x_n) = 0$, R_i — polynomials.

Original source details: GELFAND-P000098

Basis of open sets — one condition $U_S : S \neq 0$. (such is cont. in every open — — leave 1 condition).

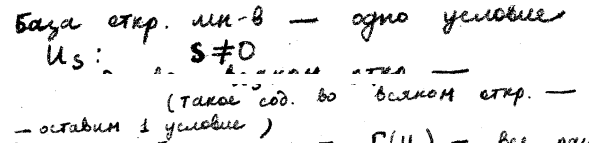


Diagram and typed description. $U_S : S \neq 0$; in parentheses two separate dashes: after open and before leave 1 condition.

Original source details: GELFAND-P000098

The group of sections is given thus —

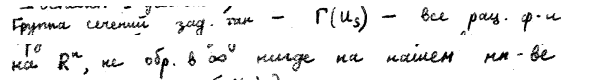


Diagram and typed description. The group of sections is given thus —; then $\Gamma(U_S)$ and rat. funct. on R^n. Written R, without an additional vertical bar.

Original source details: GELFAND-P000098

$\Gamma(U_S)$ — all rat. funct. on R^n which do not turn into ∞ anywhere on our set.

$$\Gamma(U_S) = [?] \left\{ \frac{P(x)}{Q(x)} \right\}$$

Uncertain reading. Between $\Gamma(U_S)$ and the brace, two horizontal bars with dense overwriting between them: $=$ with alteration / rewritten sign; $\neq$ remains an unestablished alternative. No definite diagonal stroke read.

Original source details: GELFAND-P000098

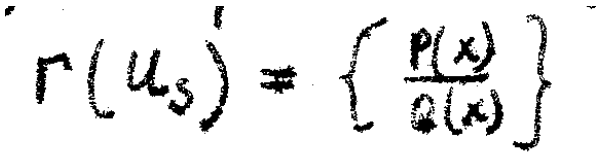


Diagram and typed description. $\Gamma(U_S)$, overwritten sign, $\left\{ \frac{P(x)}{Q(x)} \right\}$; no separate period visible after the right brace.

Original source details: GELFAND-P000098

Identify, if on U_S one value. This is a ring.

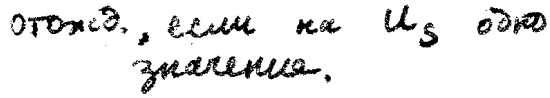
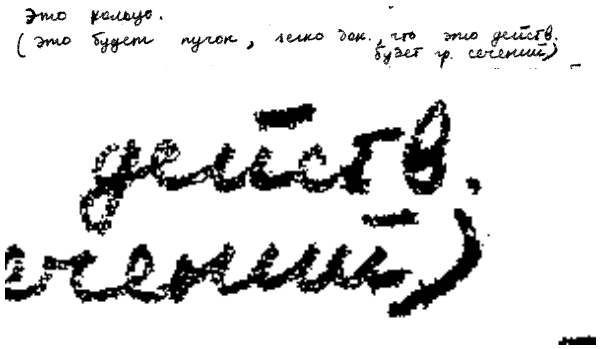


Diagram and typed description. Identify, if on U_S one value.

Original source details: GELFAND-P000098

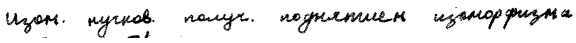
(This will be a sheaf, easy to prove that this will really be the gr. of sections.)



Uncertain reading. This is a ring. In parentheses: This will be a sheaf, easy to prove that this will really be the gr. of sections. A short horizontal stroke lies below the final letter of the translated really; its role is undetermined. The upper breve belongs to the source letter y-short.

Original source details: GELFAND-P000098

Isom. of sheaves. obtained by lifting an isomorphism



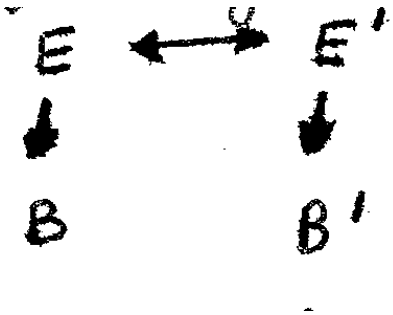
Original source details: GELFAND-P000098

Diagram and typed description. Isom. of sheaves. obtained by lifting an isomorphism; period after sheaves.

of bases. homeom. nat.

Manuscript notation. The line ends with баз. (“of bases”). The lower line reads “homeom. nat.”

Original source details: GELFAND-P000098

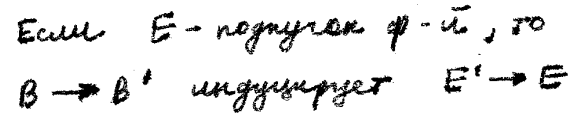


Original source details: GELFAND-P000098

Uncertain reading. Top row $E \leftrightarrow E'$, two heavy downward arrows to B, B'. The upper ends and arrowheads are dense; their exact outlines are undetermined. No horizontal arrow visible between B and B'.

$$\begin{array}{ccc} E & \longleftrightarrow & E' \\ \downarrow & & \downarrow \\ B & & B' \end{array}$$

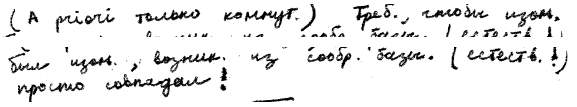
If E is a subsheaf of funct., then $B \longrightarrow B'$ induces $E' \longrightarrow E$.



Original source details: GELFAND-P000098

Diagram and typed description. In the line, $B \longrightarrow B'$ induces $E' \longrightarrow E$; the prime strokes are raised at the second letters.

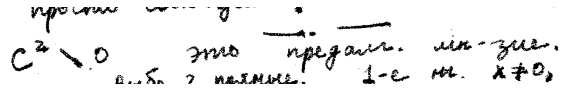
(A priori only commut.) Require that the isom. be an isom. arising from considerations of the base. (natural!) simply coincidence!



Original source details: GELFAND-P000098

Diagram and typed description. (A priori only commut.) Require that the isom. be an isom. arising from considerations of the base. (natural!) simply coincidence!

$C^2 \setminus 0$ is a prealg. var.



Original source details: GELFAND-P000098

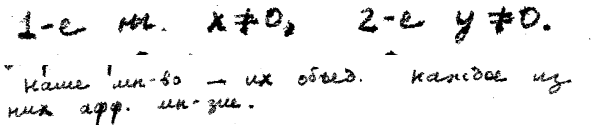
Uncertain reading. $C^2 \setminus 0$ is a prealg. var. Above prealg., two separated horizontal strokes, the right higher; a dot beside the lower. Their role and association with the words are undetermined. Written C, without an additional vertical bar.

Choose 2 straight lines.

Original source details: GELFAND-P000098

Manuscript notation. The line reads “Choose 2 straight lines.”

1st on $x \neq 0$, 2nd $y \neq 0$. Our set — their union. Each of them an aff. var.

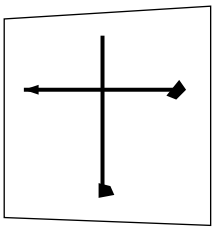


Original source details: GELFAND-P000098

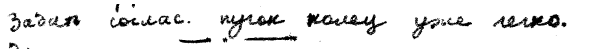
Diagram and typed description. 1st on $x \neq 0$, 2nd $y \neq 0$. Our set — their union; each of them an aff. var.

Original source details: GELFAND-P000098

Uncertain reading. Within a rectangular frame, heavy vertical and horizontal lines intersect. The left end is pointed, the right and lower ends thickened; the role of the thickenings and exact form of possible arrowheads are undetermined. No letter labels or separate circle visible at the intersection.



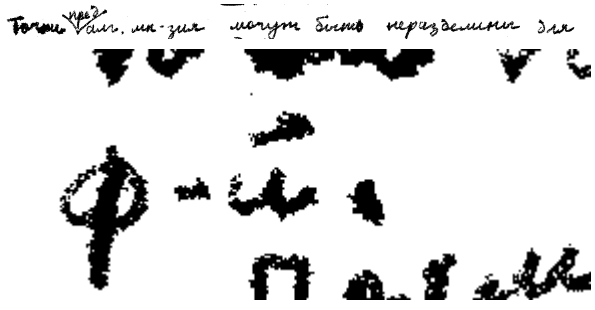
Giving a compatible sheaf of rings is now easy.



Original source details: GELFAND-P000098

Uncertain reading. Below the line about the sheaf are two separate strokes and a dot between them. Their association with the words is unclear.

Points of $\overset{\text{pre}}{\vee}$ alg. var. may be inseparable for funct.

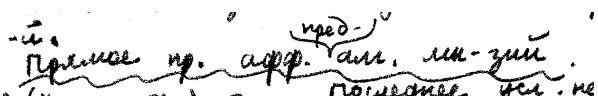


Original source details: GELFAND-P000098

Diagram and typed description. Above the gap before alg., pre is inserted with a V-shaped insertion mark. Points is written densely; next-line ending funct.

$\overbrace{\text{pre-}}$

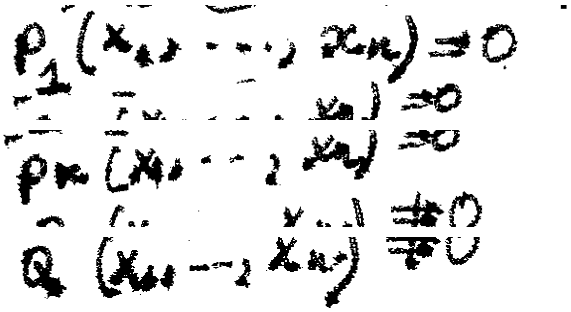
Example of pr. aff. alg. var.



Original source details: GELFAND-P000098

Diagram and typed description. The heading Example of pr. aff. alg. var. has a wavy underline. Above alg., pre- is inserted with a lower curly brace. The abbreviation pr. is not expanded.

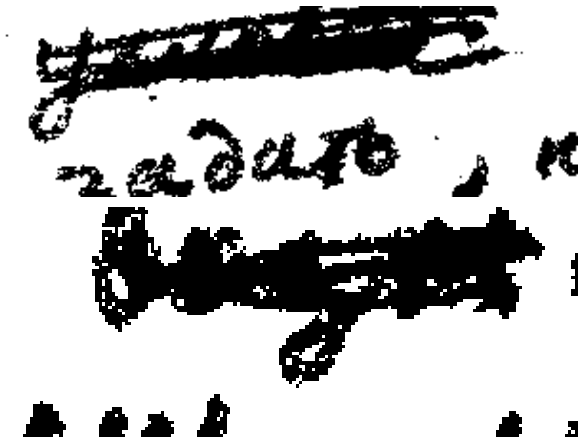
$$p_1(x_1, \ldots, x_n[?]) = 0,$$
$$, \quad - \qquad - [?]$$
$$p_k(x_1, \ldots, x_n[?]) = 0,$$
$$Q(x_1, \ldots, x_n[?]) \neq 0.$$



Uncertain reading. In the three lines, the final indices x_n[?] / x_m[?] are dense; in the second line there is an additional upper stroke. The subscript 1 on the first p has a dense horizontal base. Between the first and second lines, three separate short marks: the left comma-/dot-like, and two dashes; their role is undetermined.

Original source details: GELFAND-P000098

The last cond. is not obligatory, any [crossed-out word illegible] aff. alg. var. can be given without using these [crossed-out word illegible] conditions (∃ var., isomorphic to it).



Uncertain reading. After any, a word densely cancelled with long horizontal strokes and a loop at left; below, after these, a second densely crossed-out word. No secure letter-by-letter reading of either established.

Original source details: GELFAND-P000098

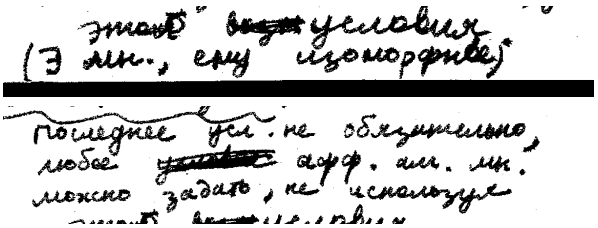


Diagram and typed description. The last cond. is not obligatory; then without using these ... conditions; below, (∃ var., isomorphic to it). The source ending usloviyakh is retained.

Original source details: GELFAND-P000098

GELFAND-P000099

Take (x, y) with the condition

$$\begin{cases} P_i(x) = 0, \\ Q(x)y - 1 = 0 \end{cases}$$

$$(x_1, \ldots, x_n, y)$$

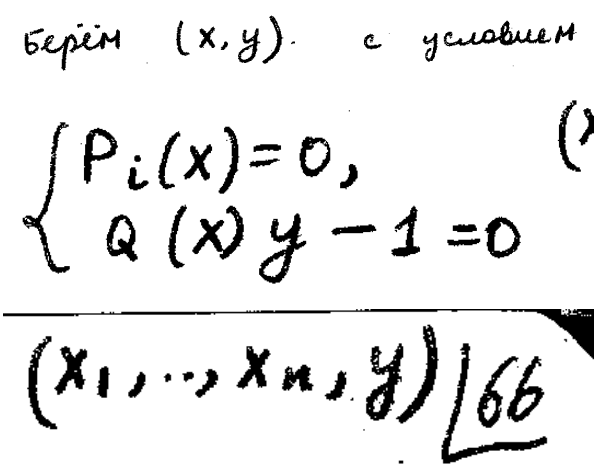


Diagram and typed description. System with a left brace: $P_i(x) = 0$, $Q(x)y - 1 = 0$. At the upper right, $(x_1, \ldots, x_n, y)$; the number 66 with an angular stroke to its left and below.

Original source details: GELFAND-P000099

They are isomorphic.

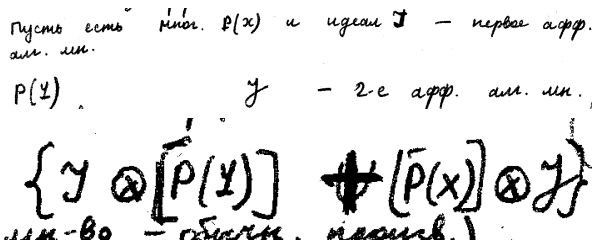
Diagram and typed description. They are isomorphic.

Original source details: GELFAND-P000099

Let there be polyn. $P(x)$ and an ideal I[?] — the first aff. alg. var. $P(y)$ J — the 2nd aff. alg. var.

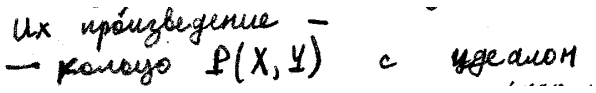
Their product — — the ring $P(x, y)$ with the ideal

$$\{I[?] \otimes [P(y)] + [?]\,[P(x)] \otimes J\}$$



Original source details: GELFAND-P000099

Uncertain reading. The manuscript has мног. ("polyn.") before $P(x)$. The first ideal sign has a top bar and a lower hook in the prose and a cursive hook in the formula. The reading I or J remains uncertain. The plus between the terms is heavily overwritten.



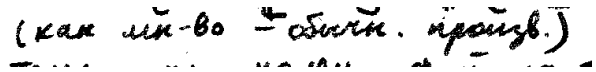
Original source details: GELFAND-P000099

Diagram and typed description. Their product — and the following line — the ring $P(x, y)$ with the ideal: two separate dashes.

Original source details: GELFAND-P000099

Manuscript notation. At the right is a circled question mark with a separate lower dot.

(as a set — the usual prod.)



Original source details: GELFAND-P000099

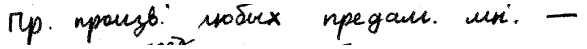
Diagram and typed description. (as a set — the usual prod.).

The ring of funct. on it — the tensor pr. of the rings of funct. on those var.

Original source details: GELFAND-P000099

Diagram and typed description. The ring of funct. on it — the tensor pr. of the rings of funct. on those var.

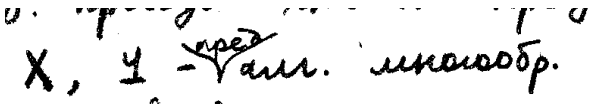
Pr. prod. of any prealg. var. — [crossed-out text illegible]



Original source details: GELFAND-P000099

Diagram and typed description. Pr. prod. of any prealg. var. —. The abbreviation Pr. is retained.

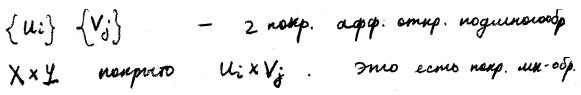
X, Y — $\overset{\text{pre}}{\vee}$ alg. varieties.



Original source details: GELFAND-P000099

Diagram and typed description. Above the gap before alg., pre is inserted, with a V-shaped insertion mark below; in the line, X, Y.

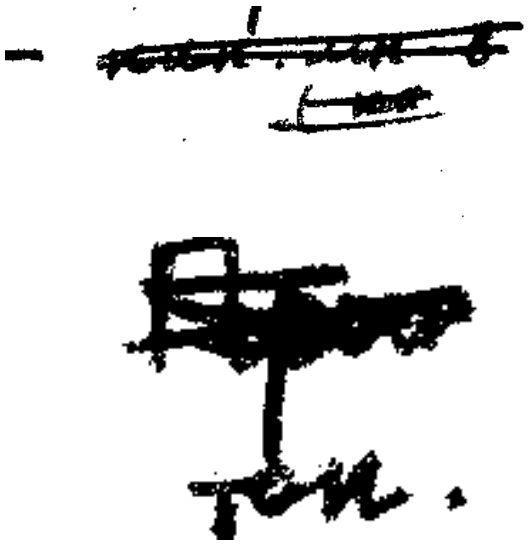
$\{U_i\}$ $\{V_j\}$ — 2 covers by aff. open subvarieties. $X \times Y$ is covered by $U_i \times V_j$. This is a cover of the var.



Original source details: GELFAND-P000099

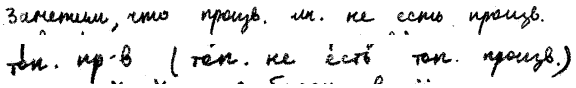
Manuscript notation. The line ends “This is a cover of the var.”

[Densely crossed-out initial text illegible] Note that the prod. of var. is not the prod. of top. spaces (the top. is not the top. of the prod.).



Original source details: GELFAND-P000099

Uncertain reading. After the dash in the product line, dense cancellation; below, a separate angular stroke and a black mark. Before Note, initial text is densely crossed out. Letter-by-letter readings of the cancellations undetermined.

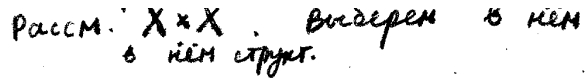


Original source details: GELFAND-P000099

Diagram and typed description. Note that the prod. of var. is not the prod. of top. spaces (the top. is not the top. of the prod.).

Consider $X \times X$. Choose in it

in it a struct.



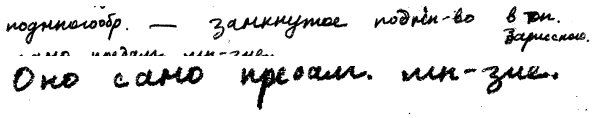
Original source details: GELFAND-P000099

Diagram and typed description. In the line: Consider $X \times X$. Choose in it; below the line, separately, in it a struct.

Prealg.[?] subvar. — a closed subset in the top.[?] of Zariski. It is itself a prealg. var.

Original source details: GELFAND-P000099

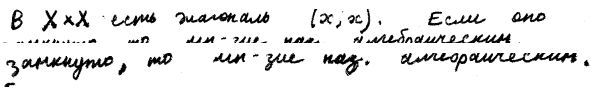
Uncertain reading. At the beginning, Prealg.[?]; the middle letters are densely overwritten; earlier strokes unread.



Original source details: GELFAND-P000099

Uncertain reading. Subvar. — a closed subset in the top.[?] of Zariski. The source abbreviation tp.[?] / top.[?] is cramped. Next line: It is itself a prealg. var.

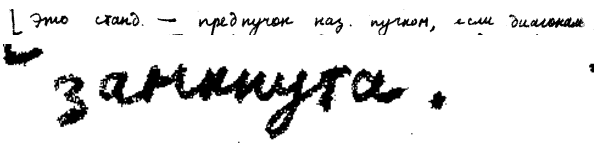
In $X \times X$ there is the diagonal (x, x). If it is closed, the var. is called algebraic.



Original source details: GELFAND-P000099

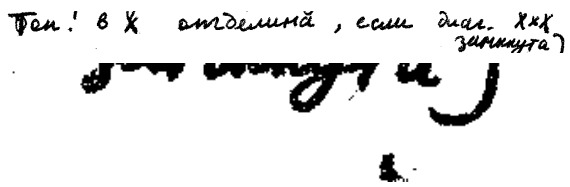
Diagram and typed description. In $X \times X$ there is the diagonal (x, x). If it is closed, the var. is called algebraic.

[This is stand. — a presheaf is called a sheaf if the diagonal is closed.
The top. in X is separated if the diag. $X \times X$ is closed.)



Original source details: GELFAND-P000099

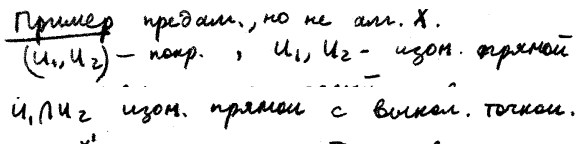
Uncertain reading. A presheaf is called a sheaf if the diagonal is closed. The first word is written predpuchok (presheaf); its cramped joined strokes are retained in the adjacent original.



Original source details: GELFAND-P000099

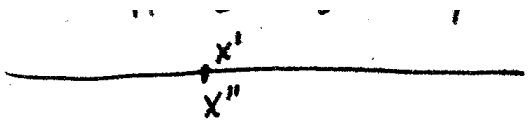
Uncertain reading. The top. in X is separated if the diag. $X \times X$ is closed.); the aside begins with a square bracket and ends with a round parenthesis. Association of the unequal brackets undetermined.

Example of a prealg. but non-alg. X. (U_1, U_2) — a cover, U_1, U_2 — isom. to a line. $U_1 \cap U_2$ isom. to a line with a point removed.



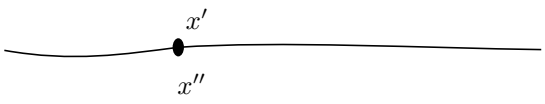
Original source details: GELFAND-P000099

Diagram and typed description. Example has a straight underline; (U_1, U_2), U_1, U_2, $U_1 \cap U_2$ and a removed point.

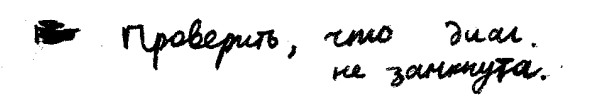


Original source details: GELFAND-P000099

Uncertain reading. One nearly horizontal line with a thickened point; above it x', below it x''. Exact outline of the thickening undetermined.



Check that the diag. is not closed.

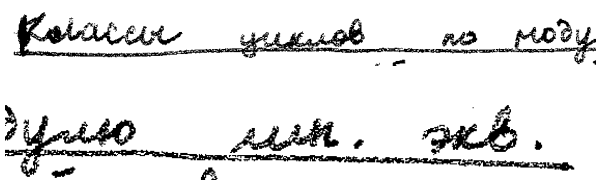


Original source details: GELFAND-P000099

Uncertain reading. Before Check, a separate dense black sign with two rightward projections; its role and exact outline undetermined. Check that the diag. is not closed.

GELFAND-P000100

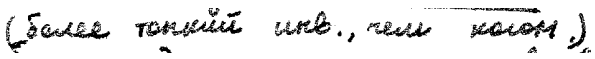
Classes of cycles modulo lin. equiv.



Original source details: GELFAND-P000100

Diagram and typed description. The entire heading Classes of cycles modulo lin. equiv. has a straight underline.

(a finer inv. than cohom.)

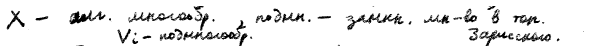


Original source details: GELFAND-P000100

Manuscript notation. Below the heading: “(a finer inv. than cohom.).”

Original source details: GELFAND-P000100

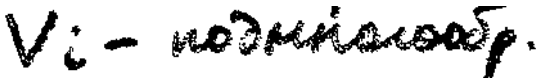
X — an alg. variety, subvar. — a closed set in the top. of Zariski.



Original source details: GELFAND-P000100

Diagram and typed description. X — an alg. variety; subvar. — a closed set in the top. of Zariski.

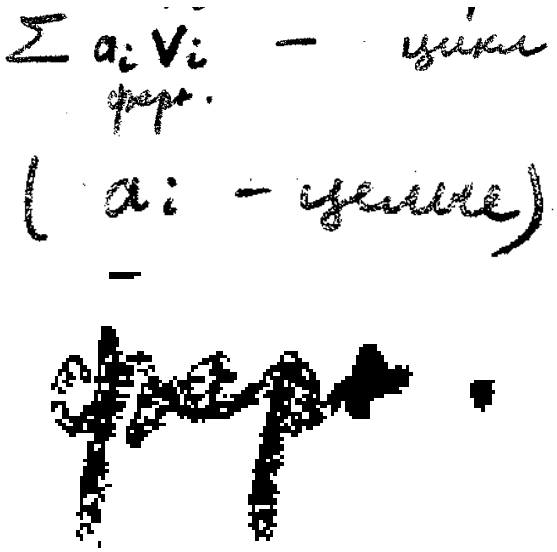
V_i — subvarieties.



Original source details: GELFAND-P000100

Diagram and typed description. V_i — subvarieties; lower index i.

$$\sum_{\text{for.[?]}} a_i V_i \text{ — cycles} \qquad (a_i \text{ — integers})$$

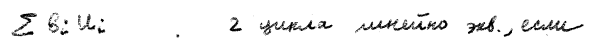


Original source details: GELFAND-P000100

Uncertain reading. Below the summand, the short source annotation for.[?] / form.[?]; after the distinct f-, o- and r-shaped letters, a cramped upper exit stroke. Formula $\sum a_i V_i$ and, at right, (a_i — integers).

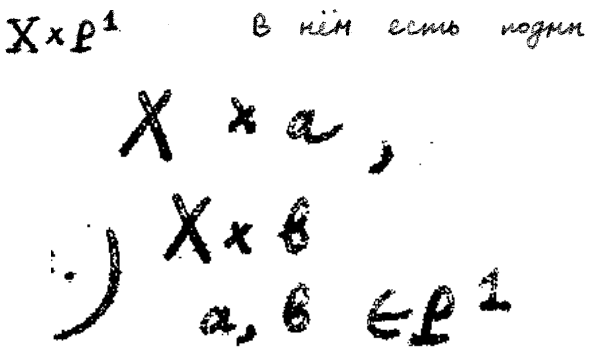
$\sum b_i U_i$. 2 cycles are linearly equiv. if

$X \times P^1$ in it there are subvar. $X \times a$, $X \times b$ $a, b \in P^1$



Original source details: GELFAND-P000100

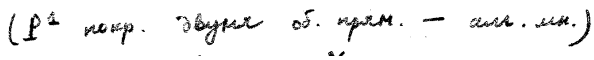
Diagram and typed description. $\sum b_i U_i$. 2 cycles are linearly equiv. if; a period after U_i.



Original source details: GELFAND-P000100

Diagram and typed description. $X \times P^1$; the words in it there are subvar.; at right $X \times a$, below it $X \times b$ and $a, b \in P^1$.

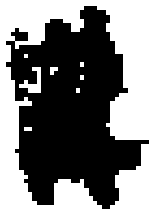
(P^1 cov. by two ord. lines — an alg. var.)



Original source details: GELFAND-P000100

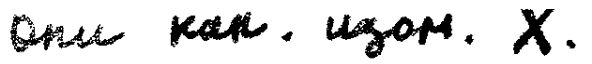
Diagram and typed description. (P^1 cov. by two ord. lines — an alg. var.); the source abbreviation ob. and both parentheses.

They are can. isom. to X.



Original source details: GELFAND-P000100

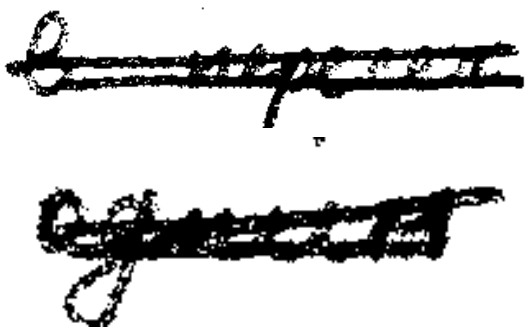
Uncertain reading. At left, a black vertical star-/letter-like mark with projections. Its approximate form is reproduced; letter reading and purpose undetermined.



Original source details: GELFAND-P000100

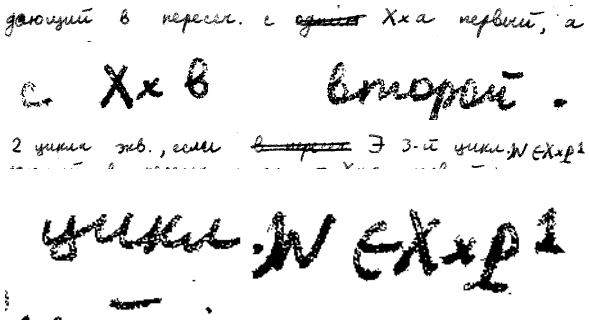
Diagram and typed description. They are can. isom. to X.

2 cycles are equiv. if [crossed out: illegible] ∃ a 3rd cycle, N[?] $\in X \times P^1$, giving in intersect. with [crossed out: illegible] $X \times a$ the first, and with $X \times b$ the second.



Original source details: GELFAND-P000100

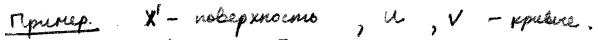
Uncertain reading. After if, an expression with an initial v-like stroke is cancelled; after with, another word with a lower loop is cancelled. Full readings of the cancellations undetermined.



Original source details: GELFAND-P000100

Uncertain reading. A comma follows “cycle.” The next sign resembles N, but the letter is uncertain.

Example. X — a surface, U, V — curves.



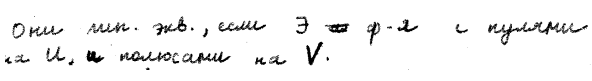
Original source details: GELFAND-P000100

Diagram and typed description. Example. is underlined including the period; X — a surface, U, V — curves.

They are lin. equiv. if ∃ [crossed-out sign illegible] a funct. with zeros on U and poles on V.

Original source details: GELFAND-P000100

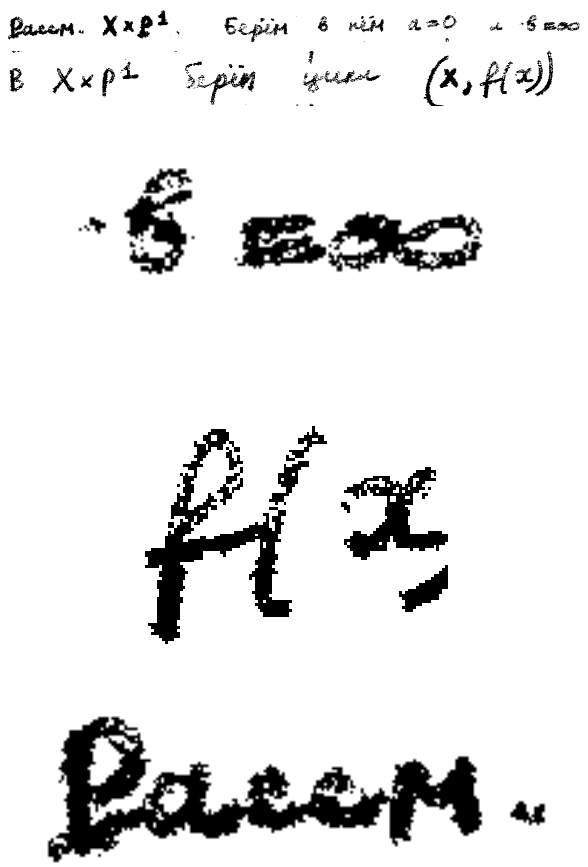
Uncertain reading. Between the existence sign and funct., a small dense sign: a closed horizontal loop with a transverse stroke; letter reading undetermined.



Original source details: GELFAND-P000100

Diagram and typed description. They are lin. equiv. if ∃ ... a funct. with zeros on U and poles on V.

Consider $X \times P^1$. Take in it $a = 0$ and $b = \infty$. In $X \times P^1$ take the cycle $(x, f(x))$



Manuscript notation. After b, two dense horizontal strokes form the equality sign, followed by ∞.

Original source details: GELFAND-P000100

Def. of Chern classes (alg. varieties)

Uncertain reading. The current reading is Chern. The strokes beneath the overwriting are indistinct.

Original source details: GELFAND-P000100

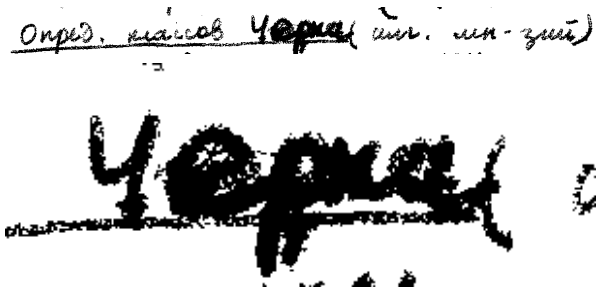


Diagram and typed description. A straight line runs under Def. of classes and the name Chern; at right, (alg. varieties).

Original source details: GELFAND-P000100

[Crossed-out word illegible] They are giv. by means of axioms.

Original source details: GELFAND-P000100

Uncertain reading. A separate word is densely crossed out before “They.”

Original source details: GELFAND-P000100

Manuscript notation. The line reads “They are giv. by means of axioms,” with пом. in the source.

[Crossed-out initial sign illegible] Functor from the category [crossed-out circled sign illegible]

Original source details: GELFAND-P000100

Uncertain reading. At left before Functor, an earlier sign with a tall left stroke is densely cancelled; after category, another sign is cancelled and enclosed by a loop. Letter readings undetermined.

Original source details: GELFAND-P000100

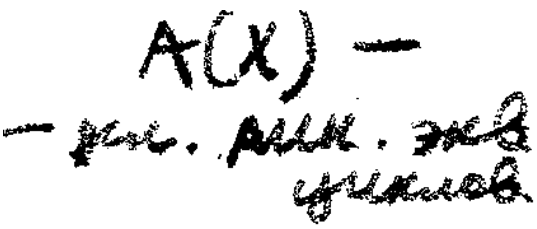
Diagram and typed description. Functor from the category; separate cancellations at right and left.

Algebr. sheaf, algebr. K-functor.

Original source details: GELFAND-P000100

Diagram and typed description. Algebr. sheaf, algebr. K-functor.

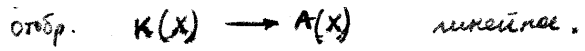
$A(X)$ —
— gr. of lin. equiv.
cycles



Original source details: GELFAND-P000100

Diagram and typed description. At right $A(X)$ —; on the next line another dash before gr. of lin. equiv., with cycles below.

The map $K(X) \longrightarrow A(X)$ is linear.



Original source details: GELFAND-P000100

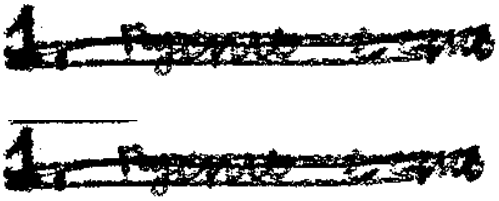
Diagram and typed description. The map $K(X) \longrightarrow A(X)$ is linear.; an arrow directed to the right.

Axioms.

Original source details: GELFAND-P000100

Diagram and typed description. Axioms. has a straight underline.

1. [Crossed-out continuation illegible]



Original source details: GELFAND-P000100

Uncertain reading. Final entry: 1. and a densely crossed-out line with two long transverse strokes. Below it, down to the lower edge, there are no further legible axioms.

GELFAND-P000101

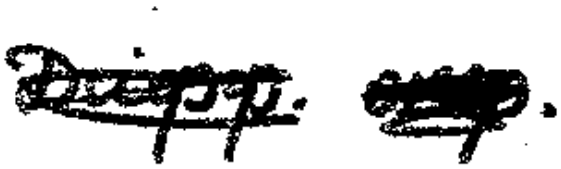
$18/\overline{X}$. I.M. Gelfand.

X

$E_1, \dots, E_n$

$$\Gamma(E_1) \xrightarrow{d} \Gamma(E_2) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(E_n)$$

[Crossed out at right: Diff.[?] oper.[?]] Ell. complex.



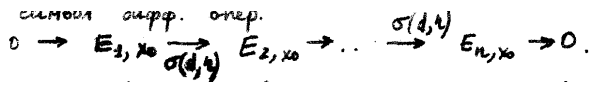
Original source details: GELFAND-P000101

Uncertain reading. Two densely crossed-out entries: Diff.[?] / Differ.[?] and oper.[?]. The distinct remnants do not establish all cancelled letters.

$$d\big(f^m(x)s(x)\big)\big|_{x_0} = \sigma(d,\eta)s(x_0), \qquad df\big|_{x_0} = \eta$$

σ — symbol of the diff. oper.

$$0 \longrightarrow E_{1,x_0} \underset{\sigma(d,\eta)}{\longrightarrow} E_{2,x_0} \longrightarrow \cdots \overset{\sigma(d,\eta)}{\longrightarrow} E_{n,x_0} \longrightarrow 0.$$



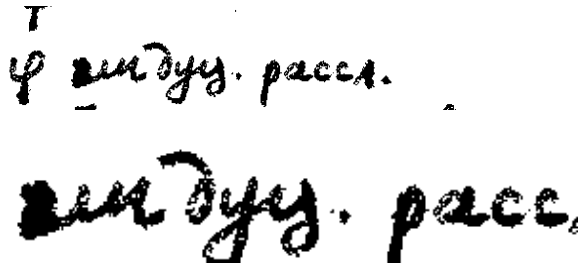
Original source details: GELFAND-P000101

Diagram and typed description. At the first arrow between fibres, $\sigma(d,\eta)$ is below; at the last, above; zeros at both ends.

Map. $T : \Gamma(E) \longrightarrow \Gamma(E)$. $\Gamma(E_i) \longrightarrow \Gamma(E_i)$

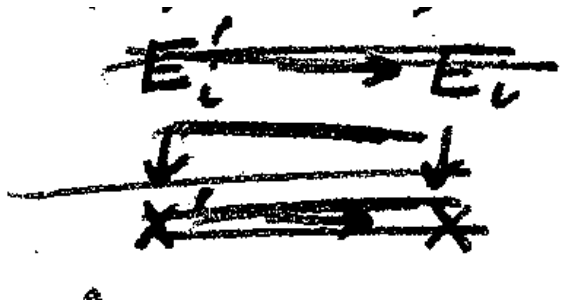
$f : X \longrightarrow X$

φ induc. bundle.



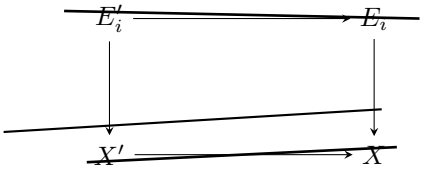
Original source details: GELFAND-P000101

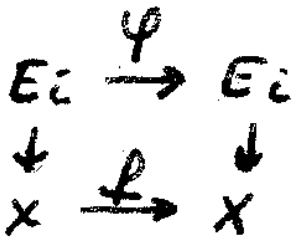
Manuscript notation. After φ the manuscript reads индуц. (“induc.”). The first stroke is dense.



Original source details: GELFAND-P000101

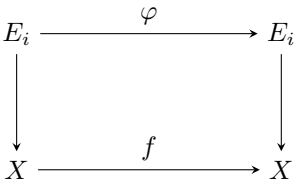
Uncertain reading. At the left of the cancelled square, E_i' and X' are legible. The heavy transverse strokes are drawn approximately.



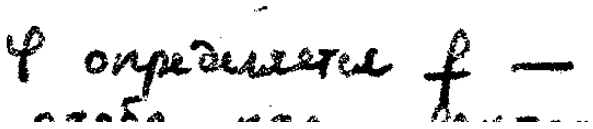


Original source details: GELFAND-P000101

Diagram and typed description. Upper arrow $E_i \xrightarrow{\varphi} E_i$, two downward arrows; lower arrow $X \xrightarrow{f} X$.



(In the simplest cases φ is determined by f — — forms f determ. the map of tang. vectors)



Original source details: GELFAND-P000101

Manuscript notation. The line reads “φ is determined by f.”

Endom., if it commut. with d.

T is an oper. in a C^∞-space. The theorem reduces to the study of the trace (it is calc. in 2 ways)

Theorem 1. The complex has finite [crossed-out word illegible] homologies.

Original source details: GELFAND-P000101

Uncertain reading. Between finite and homologies, a word is densely crossed out; an upper arch and a lower stroke remain. Full reading undetermined.

Let ω^n be a form of degree n on X, not taking the value 0. Use it to introduce a Hermitian structure. (Let it exist once and for all. Noninvariant, but simpler.)

$$\int \langle s, \alpha(s') \rangle = \int_X \overline{(s, s')}\, \omega.$$

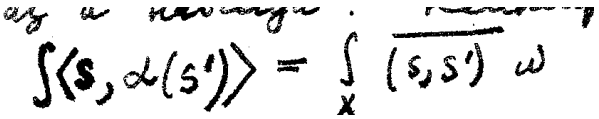
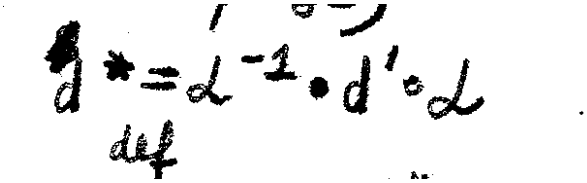


Diagram and typed description. At left, angle brackets containing $s, \alpha(s')$; at right, integral over X, a bar over (s, s'), then ω.

Original source details: GELFAND-P000101

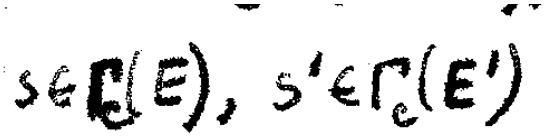
$$d^* \underset{\text{def}}{=} \alpha^{-1} \cdot d' \circ \alpha.$$



Uncertain reading. The top of the initial d is dense, with a star beside it. Below the equality, def; at right $\alpha^{-1} \cdot d' \circ \alpha$. No separate sign is securely readable inside the dense part.

Original source details: GELFAND-P000101

$s \in \Gamma(E)$, $s' \in \Gamma_c(E')$.



Uncertain reading. The second Γ has a legible c. At the first Γ, the overwritten lower stroke is unresolved.

Original source details: GELFAND-P000101

E' — the dual sheaf.

$$\Box \underset{\text{def}}{=} d^* d + d d^*$$

[A separate black mark below and to the left of the definition.]

Uncertain reading. A separate dense blot with small dots and slanting strokes; letter reading and purpose undetermined.

Original source details: GELFAND-P000101

(d' — simply the adjoint operator)

$\square$ — an ellipt. operator, taking each $\Gamma(E_i)$ into itself.

$\Gamma_\lambda(E)$ — sections satisfying the eqn. $\square s = \lambda s$. Since $\square$ is elliptic, for $\forall \lambda$ a fin. number of distinct lin. indep. eig. functs. [dense separate sign illegible].

Original source details: GELFAND-P000101

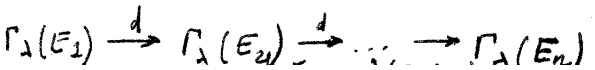
Uncertain reading. After eig. functs., a separate dense closed black entry with inner vertical strokes; an H-like reading is not established.

$$\square d = d\square$$

$$\Gamma_\lambda(E_1) \xrightarrow{d} \Gamma_\lambda(E_2) \xrightarrow{d} \cdots \longrightarrow \Gamma_\lambda(E_n)$$

Original source details: GELFAND-P000101

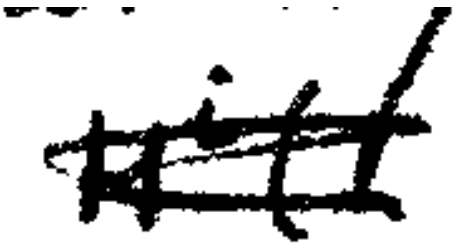
Diagram and typed description. The second closing parenthesis has a low outgoing stroke touching the next line; no separate comma established.



Original source details: GELFAND-P000101

Manuscript notation. The first two arrows are labelled d. The last arrow is unlabelled.

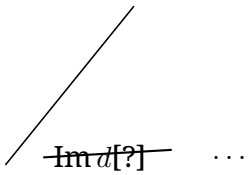
Introduce the homology groups [below, crossed-out formula illegible].

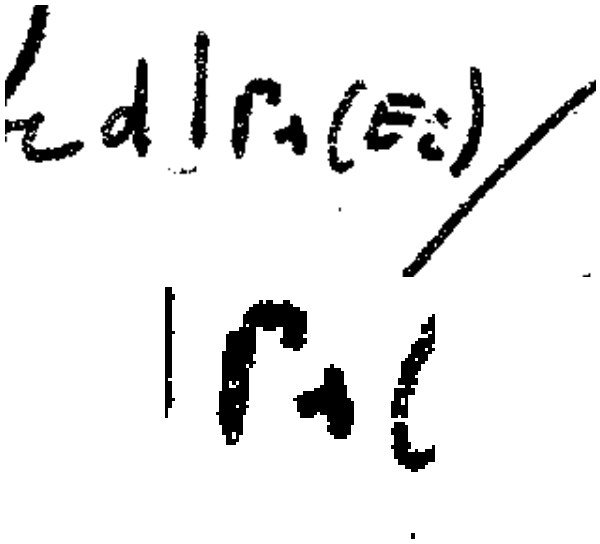


Original source details: GELFAND-P000101

Uncertain reading. Below homology groups, two long transverse strokes cross a short formula; upright bases and an upper slanting stroke remain. Full reading undetermined.

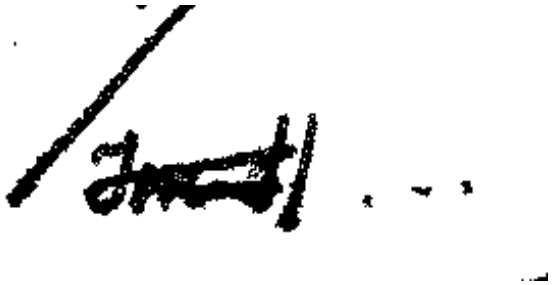
$$H^i(\Gamma_\lambda(E)) = \operatorname{Ker} d\big|_{\Gamma_{\lambda[?]}(E_i)}$$





Uncertain reading. After the vertical bar, Γ with a dense lower sign λ[?], then (E_i); the initial Ker has a capital K.

Original source details: GELFAND-P000101



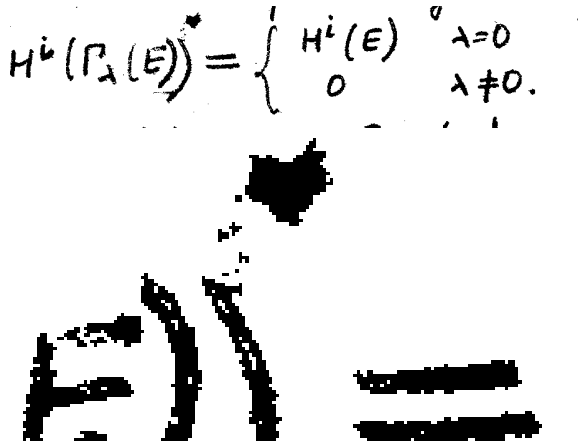
Uncertain reading. At right a slanting quotient stroke; below, densely crossed-out Im d[?], then three separate dots. The cancelled middle strokes remain unread.

Original source details: GELFAND-P000101

GELFAND-P000102

Without proof this work asserts Hodge's theorem

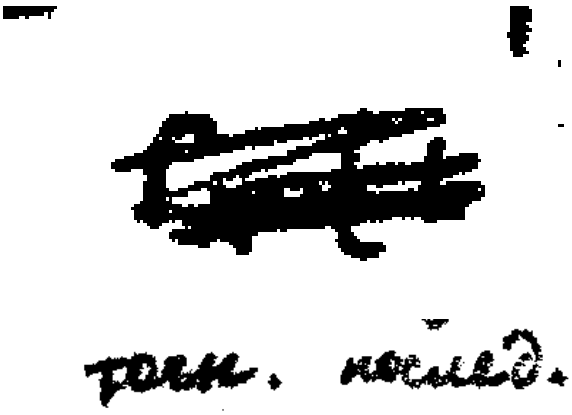
$$H^i(\Gamma_\lambda(E))^{*[?]} = \begin{cases} H^i(E) & \lambda = 0 \\ 0 & \lambda \neq 0. \end{cases}$$



Original source details: GELFAND-P000102

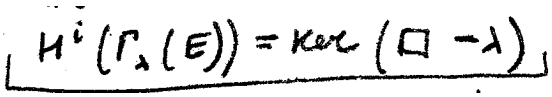
Uncertain reading. Above the right closing parentheses, a small dense star-like mark; its exact attachment to the parentheses and notation is undetermined.

Dynin. For $\lambda \neq 0$ — here 0. (d for $\lambda \neq 0$ maps eig. functs. [crossed-out word illegible] exact seq.



Original source details: GELFAND-P000102

Uncertain reading. A short entry is densely crossed out before "exact seq."



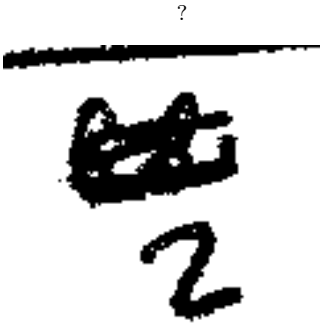
Original source details: GELFAND-P000102

Diagram and typed description. Three sides of a frame below the formula and at both edges, open at the top. At right Ker$(\Box - \lambda)$, with an initial capital K.

$$H^i(\Gamma_\lambda(E)) = \mathrm{Ker}(\Box - \lambda)$$

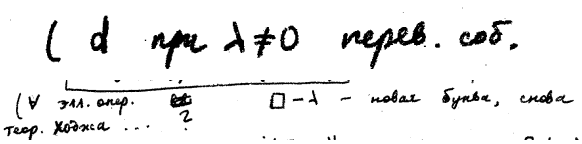
this is Hodge's theor. $\forall \lambda$

(∀ ell. oper. [crossed out: illegible] $\square - \lambda$ — a new letter, again Hodge's theor....



Original source details: GELFAND-P000102

Uncertain reading. After ell. oper., a short abbreviation is densely cancelled; a question mark lies immediately below it. Full reading undetermined.



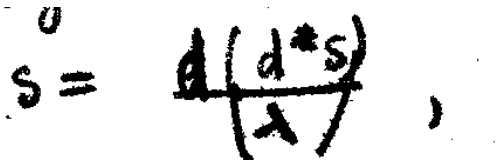
Original source details: GELFAND-P000102

Diagram and typed description. Visible opening round parentheses before d and before ∀ ell. oper.; the last line has three separate dots after Hodge. No closing parentheses read in this fragment.

Bernstein. Same. $H^i(\Gamma_\lambda(E)) = 0$, $\lambda \neq 0$. Let $s \in \Gamma_\lambda(E_i)$ and let $ds = 0$. [?] $\square s = dd^*s$. Then $\lambda s = d(d^*s)$ and $s = d\left(\frac{d^*s}{\lambda}\right)$, d^*s is an eig. funct. (d^* commut. with $\square$).

Original source details: GELFAND-P000102

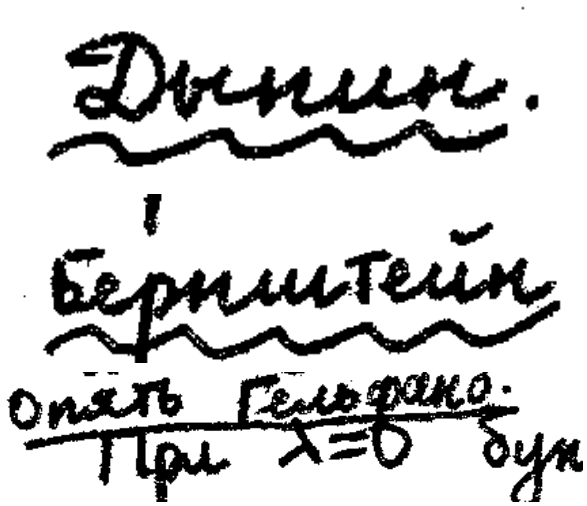
Uncertain reading. Before $\square$, a tall dense stroke with a closed upper loop and a lower slanting exit: d[?] / opening parenthesis[?]. Reading undetermined.



Original source details: GELFAND-P000102

Diagram and typed description. The entry $s = d\left(\frac{d^*s}{\lambda}\right)$: a long horizontal bar passes near the base of the outer d and across the parentheses; a separate comma at right.

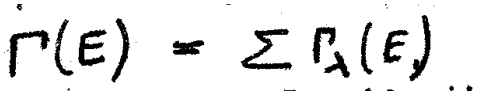
$d^*s \in C^\infty$ follows from the theor. on ell. eqns. Gelfand again. For $\lambda = 0$ transfer Hodge's proof literally.



Original source details: GELFAND-P000102

Diagram and typed description. Wavy lines below Dynin. and Bernstein.; a straight line below Gelfand again. In the line before the last heading, theor. and ell. are separately underlined.

$$\Gamma(E) = \sum \Gamma_\lambda(E) \qquad \text{(discr. spectrum)}$$



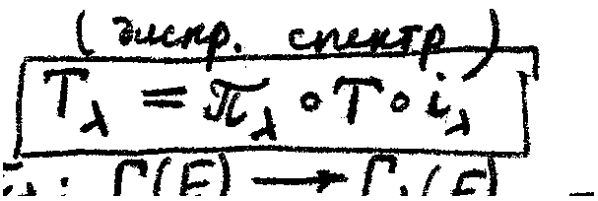
Original source details: GELFAND-P000102

Diagram and typed description. $\Gamma(E) = \sum \Gamma_\lambda(E)$; no separate index visible below the summation sign.

Besides the oper. T, introduce

$$\boxed{T_\lambda = \pi_\lambda \circ T \circ i_\lambda},$$

where $i_\lambda : \Gamma_\lambda(E) \longrightarrow \Gamma(E)$, $\pi_\lambda : \Gamma(E) \longrightarrow \Gamma_\lambda(E)$ — orthog. projection.

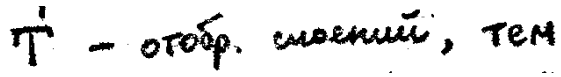


Original source details: GELFAND-P000102

Diagram and typed description. Rectangular frame around $T_\lambda = \pi_\lambda \circ T \circ i_\lambda$; a comma follows the frame. Below, the line with π_λ and orthog. projection; above the frame (discr. spectrum).

(oper. T in this cell) Lefschetz number for T.

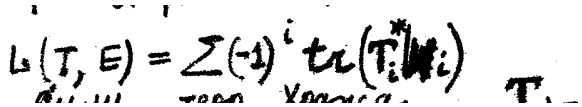
T — a map of foliations, thus a map of sections $\Gamma(E) \to \Gamma(E)$



Original source details: GELFAND-P000102

Manuscript notation. After the first “map of,” the manuscript reads слоений (“foliations”).

$$L(T, E) = \sum (-1)^i \operatorname{tr}(T_i^* | H_i[?]).$$

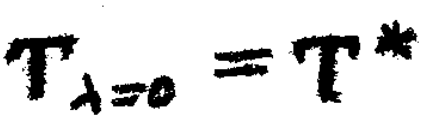


Original source details: GELFAND-P000102

Uncertain reading. After the vertical bar, an H-like letter is densely overwritten; a separate i lies at the lower right, with a dot above it. The main letter and index placement remain uncertain: H_i[?] / H^i[?].

By Hodge's theor. $T_{\lambda=0} = T^*$.

$T_0 = T^{*[?]}$



Original source details: GELFAND-P000102

Uncertain reading. At right, $T_0 = T^*$. The superscript of the second T is heavily overwritten and its individual strokes cannot be counted.

$$\sum(-1)^i \operatorname{tr}(T_0|E_i) = L(T,E)$$
$$\sum(-1)^i \operatorname{tr}(T_\lambda|E_i) = 0 \qquad \lambda \neq 0$$

[At right, two densely crossed-out entries illegible.]

Uncertain reading. Two small dense cancellations in the right margin: the upper one has a lower loop, the lower a long rising slanting stroke. Letter readings undetermined.

Original source details: GELFAND-P000102

(the trace in hom. is the same as the trace in the spaces themselves)
(By Hodge's theor. $H^i(E) = \Gamma_0(E_i)$)

$$\Box s = 0 \Longrightarrow \underline{ds = 0}.$$

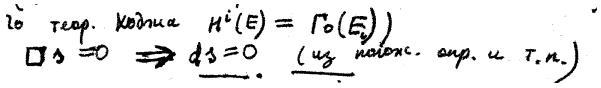
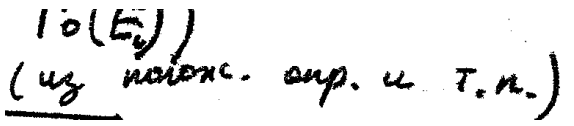


Diagram and typed description. In $\Box s = 0 \Longrightarrow ds = 0$, a thick overwritten arrow; below $ds = 0$, a horizontal line, with a separate dot at right.

Original source details: GELFAND-P000102

(from posit. def. and so on)



Manuscript notation. The parenthesis reads "from posit. def. and so on." A separate line lies below its opening.

Original source details: GELFAND-P000102

Must give meaning to $\sum(-1)^i \operatorname{tr}(T|E_i)$. Regulariz. it.
In 2 ways.

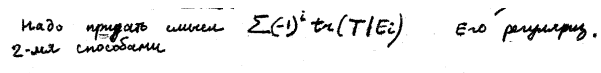
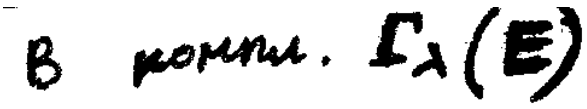


Diagram and typed description. First line: Must give meaning, the sum, and Regulariz. it at right. Below at left: In 2 ways.

Original source details: GELFAND-P000102

GELFAND-P000103

In the compl. $\Gamma_\lambda(E)$



Manuscript notation. The opening reads В компл. (“In the compl.”).

Original source details: GELFAND-P000103

$$d : \Gamma_\lambda(E_1) \longrightarrow \Gamma_\lambda(E_2) \longrightarrow \cdots$$

(finite-dimensionality) T — its map into itself. Then

$$\sum (-1)^i \operatorname{tr} T|_{E_i} = \sum (-1)^i \operatorname{tr} T|_{H_i}$$

(and $H^i = 0$)

(invar. of the Euler characteristic)

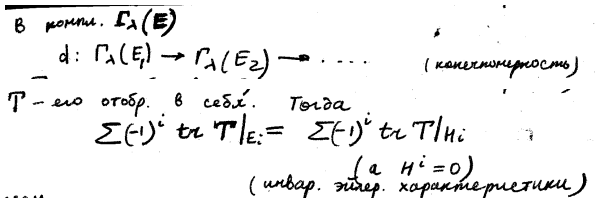


Diagram and typed description. The first chain has two arrows, then an ellipsis; finite-dimensionality in parentheses. In the trace equality, E_i and H_i have subscripts after the restriction bars; the next parenthesis has $H^i = 0$ with superscript i.

Original source details: GELFAND-P000103

Consider.

$$\sum_\lambda (1+\lambda)^{-z} \operatorname{tr}(T_\lambda | E_i) = \operatorname{tr}^z(E_i)$$

Theorem. The series conv. for $\operatorname{Re} z > \frac{1}{2} \dim X$ (on the asympt. of eig. functions) (Dunford and Schwartz, vol.2)

This is an anal. funct. of z in the half-plane.

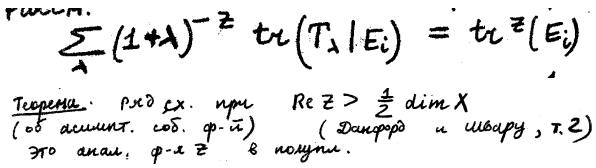


Diagram and typed description. The exponent of $(1+\lambda)$ is $-z$, with a separate short minus stroke. At right, $\operatorname{tr}^z(E_i)$. Beneath, Theorem. is underlined; $\operatorname{Re} z > \frac{1}{2} \dim X$ and two parentheses.

Original source details: GELFAND-P000103

$$L(T, E) = \sum (-1)^i \operatorname{tr}^z E_i \qquad \forall z$$

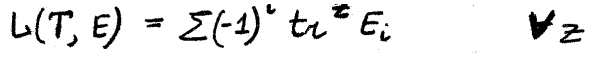


Diagram and typed description. At left, $L(T, E)$; at right the trace has superscript z, then E_i without parentheses. Farther right, a separate $\forall z$ with a dense first sign.

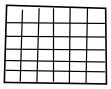
Original source details: GELFAND-P000103

(everything except convergence is proved [densely overwritten sign illegible]

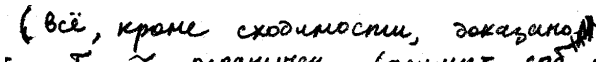
—

Original source details: GELFAND-P000103

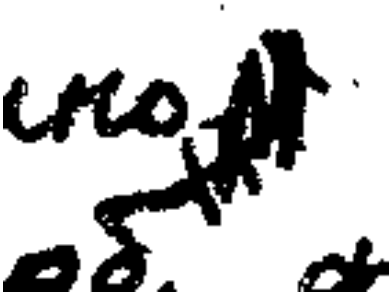
Diagram and typed description. After the dash, a small rectangular grid of horizontal and vertical strokes. The typographic recreation shows the visible lines; dense intersections and borders are not fully distinguishable. No letter label or definite role of the grid is read.



)



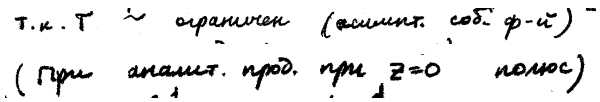
Uncertain reading. After proved, a dense vertical entry with a lower oblique looplike exit. Former letters remain inseparable even under enlargement; exact reading undetermined.



Original source details: GELFAND-P000103

Since $T\sim$ is bounded (asympt. of eig. functions)

(on anal. cont., a pole at $z=0$)



Original source details: GELFAND-P000103

Diagram and typed description. After T, a separate wavy $\sim$ sign, then bounded; asympt. of eig. functions in parentheses. Beneath, a separate parenthesis: on anal. cont., a pole at $z=0$.

Example. S^1

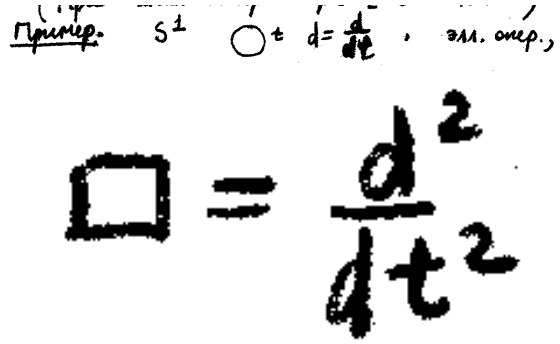


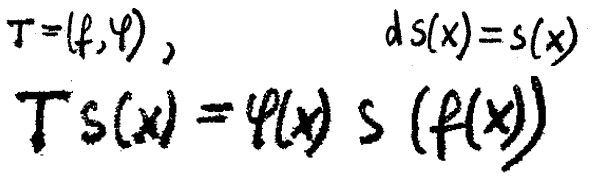
Diagram and typed description. A circle with t at right. The initial d in the first derivative's denominator is densely overwritten, followed by t; former strokes are inseparable. The entry for $\square$ is d^2/dt^2 with a positive sign.

Original source details: GELFAND-P000103

$d=\frac{d}{dt}$, ell. oper., $\square=\frac{d^2}{dt^2}$.

$T = (f, \varphi)$,

$$dS(x) = s(x)$$
$$Ts(x) = \varphi(x)s(f(x))$$



Original source details: GELFAND-P000103

Diagram and typed description. At left, $T = (f, \varphi)$ and a comma. At upper right, $dS(x) = s(x)$: the first S is uppercase, the second s lowercase. Beneath, $Ts(x) = \varphi(x)s(f(x))$.

Original source details: GELFAND-P000103

Diagram and typed description. At left beneath $T = (f, \varphi)$, a separate wavy line; below it, an almost horizontal stroke with a thickened right end. The typographic recreation preserves the general course of the lines; direction of the thickening and purpose of the sketch are undetermined. There are no axis or function labels.

Definition [?] of a pseudodiff. oper.:

Original source details: GELFAND-P000103

Uncertain reading. Between the heading and pseudodiff., a dense overwritten sign with a right vertical and several transverse strokes: P[?] / illegible. No secure letter reading established.

X — manifold. $\Gamma(E)$, $\Gamma(F)$ — 2 bundles on X

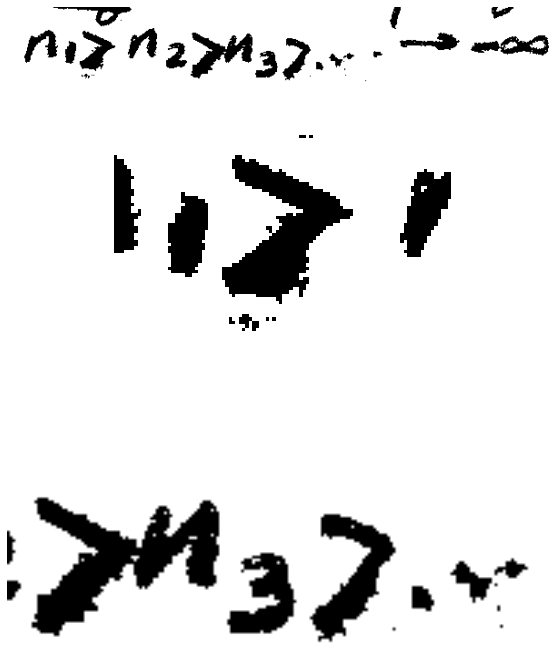
$$P : \Gamma_c(E) \longrightarrow \Gamma(F)$$

$\Gamma_c(E)$ — sections with compact support. [Crossed out: Called[?]]

Original source details: GELFAND-P000103

Uncertain reading. Before Lin. oper. is called, a short abbreviation with a long lower oblique cancellation stroke. Source Naz.[?] (Called[?]); former letters are densely joined and not securely read. A separate period follows.

A lin. oper. is called pseudodiff. if $\exists$ a seq. of numbers $n_1 > n_2 > n_3 > \cdots \to -\infty$



Uncertain reading. Three right-pointing signs. The first and third have dense lower ends and separate tiny traces; the middle one is overwritten and joins the following n. Separate lower equality strokes are not established: $>$[?] / $\geq$[?].

Original source details: GELFAND-P000103

and satisfies the following conditions

$$e^{-i\lambda g}\underline{P}(e^{i\lambda g}s)$$

$g \in \Gamma(1)$ (C^∞ funct.), 1 — trivial sheaf, $s(x) \in \Gamma_c(E)$, $dg \neq 0$ on supp s, if there is an asympt. expansion.

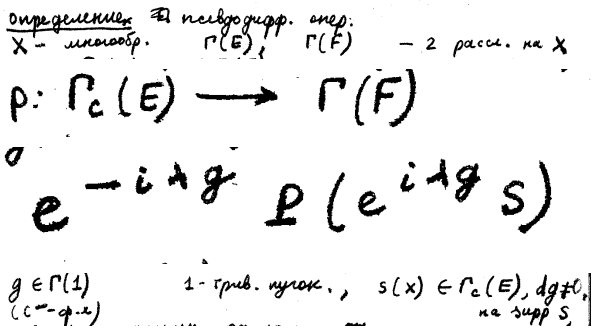


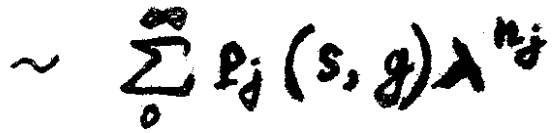
Diagram and typed description. The heading Definition has a straight underline. In the exponential expression, P has a short underline; exponents $-i\lambda g$ and $i\lambda g$. In parentheses, C^∞ funct. At right, $dg \neq 0$; beneath, on supp s and a comma.

Original source details: GELFAND-P000103

Uncertain reading. To the right of “expansion” is a heavy mark with a rightward projection. Its reading and purpose are unclear.

Original source details: GELFAND-P000103

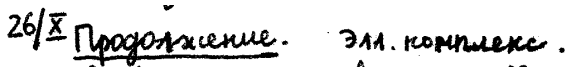
$$\sim \sum_{0}^{\infty} P_j(s,g)\lambda^{n_j}$$



Original source details: GELFAND-P000104

Diagram and typed description. Continuation of the formula: a separate $\sim$, lower limit 0 without $j =$, upper ∞, then $P_j(s,g)\lambda^{n_j}$.

$26/\overline{X}$ Continuation. Ell. complex.



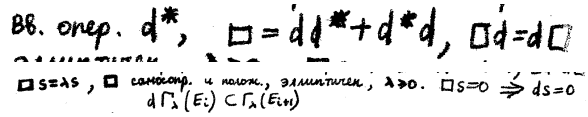
Original source details: GELFAND-P000104

Diagram and typed description. X in the date has separate over- and underlines. Continuation. has its own underline; then Ell. complex.

In E_i — a Hermitian metric, and in X — integration. Introduce oper. d^*, $\square = dd^* + d^*d$, $\square d = d\square$

$\square s = \lambda s$, $\square$ self-adj. and positive, elliptic, $\lambda > 0$. $\square s = 0 \Longrightarrow ds = 0$

$$d\Gamma_\lambda(E_i) \subset \Gamma_\lambda(E_{i+1})$$

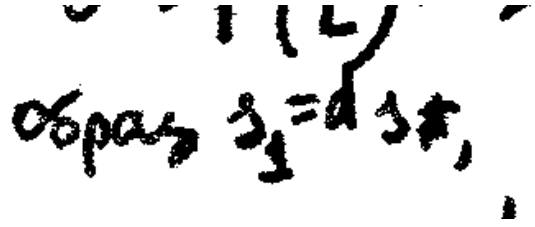


Original source details: GELFAND-P000104

Uncertain reading. At right in $\square d = d\square$, the last square's right edge is broken and faint. Its other sides are distinguishable; the exact right boundary of the sign is damaged in the held scan. Beneath, separate entries $\lambda > 0$ and $\square s = 0 \Longrightarrow ds = 0$.

$$0 \xrightarrow{d} \Gamma(E) \xrightarrow{d} \Gamma(F) \xrightarrow{d} 0$$

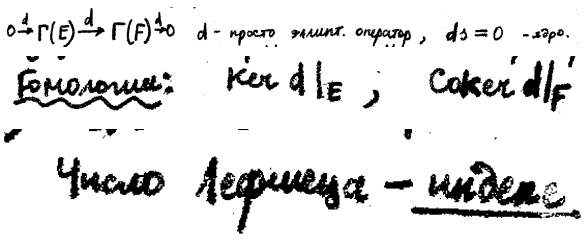
d — simply an elliptic operator, $ds = 0$ — the kernel. Image $s_1 = ds*$,



Original source details: GELFAND-P000104

Uncertain reading. The first s has a subscript one. The star after the second s lies near the baseline; its role as an index is unclear.

Homology: Ker $d|_E$, Coker $d|_F$. The Lefschetz number — the index

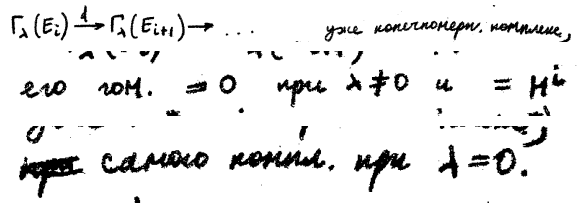


Original source details: GELFAND-P000104

Diagram and typed description. All three arrows of the first chain are labelled d. Homology: has a wavy underline; Ker and Coker begin with capitals. Index has a separate straight underline.

$$\Gamma_\lambda(E_i) \xrightarrow{d} \Gamma_\lambda(E_{i+1}) \longrightarrow \cdots$$

already a finite-dimensional complex, its hom. $= 0$ for $\lambda \neq 0$ and $= H^i$ [crossed out: for] of the complex itself for $\lambda = 0$.



Manuscript notation. After H^i, "for" is crossed out, then the line reads "of the complex itself for $\lambda = 0$."

Original source details: GELFAND-P000104

T_i^* H^i

$$\sum(-1)^i \operatorname{tr}(T_i^*) = L(T, E)$$

Compute through fixed points.

T — a map of X, commutes with d.

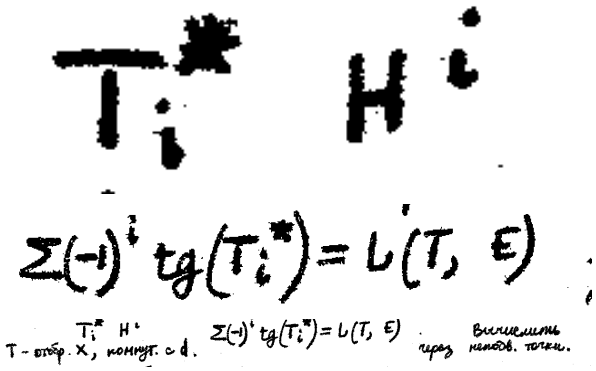


Diagram and typed description. At left, separate T_i^* and H^i, without an arrow between them. At right, an equality with $L(T, E)$. Beneath, T — a map of X, commutes with d.; at right, Compute through fixed points.

Original source details: GELFAND-P000104

T is a map of $\Gamma_\lambda(E_i)$ Introd. operators π_λ and i_λ. inclusion of Γ_λ into all of Γ, and projection onto [crossed out: illegible entry]

$\Gamma_\lambda(E_i)$.

Uncertain reading. After onto, a short densely cancelled entry with an upper loop and transverse strokes. Former letters remain inseparable; followed by a separate active $\Gamma_\lambda(E_i)$.

Original source details: GELFAND-P000104

$$T_\lambda = \pi_\lambda \cdot T \cdot i_\lambda$$

(select the blocks corresp. to Γ_λ)

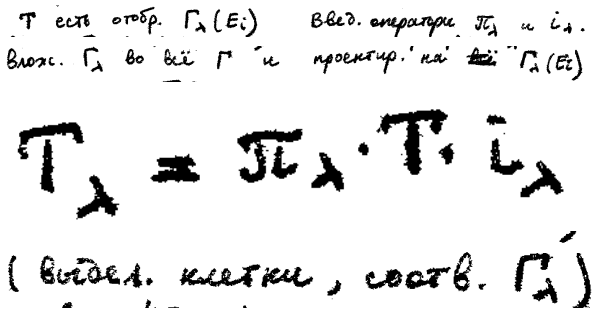


Diagram and typed description. Entries for inclusion of Γ_λ into all of Γ and projection onto $\Gamma_\lambda(E_i)$. The formula has two dots, one on each side of T. Beneath, a parenthesis about selecting blocks.

Original source details: GELFAND-P000104

Commut. unnecessary, suff. [dense overwritten sign]

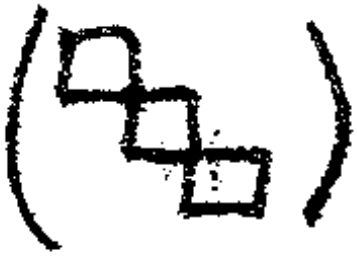
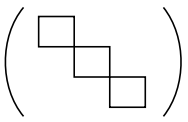


Diagram and typed description. Three empty squares descend from upper left to lower right within parentheses; neighbouring corners touch. The typographic recreation preserves the three outlines and parentheses; exact sizes and dense joints remain undetermined.



Original source details: GELFAND-P000104

in the blocks.

Uncertain reading. Between suff. and the sketch, a dense vertical entry with two dark clusters and a lower oblique exit. No letter reading established; the sign is retained separately from the blocks.

Original source details: GELFAND-P000104

Then $\boxed{T^* = T_0}$

$$0 \longrightarrow \Gamma_0(E_1) \xrightarrow{d} \Gamma_0(E_2) \xrightarrow{d} \cdots$$

all $d = 0$ since $\Box s = 0$.

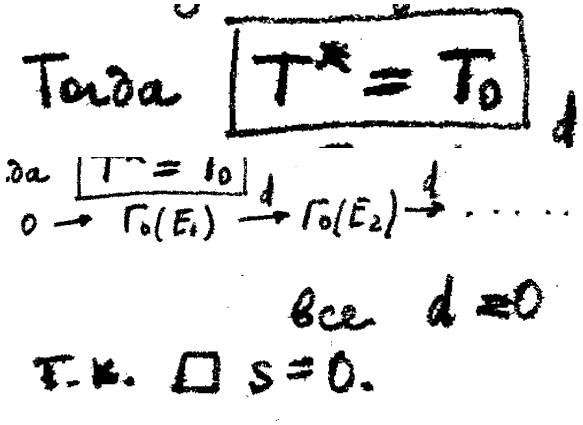


Diagram and typed description. The equality $T^* = T_0$ has a full rectangular frame. The first arrow from 0 is unlabelled, the next two have d; then an ellipsis. At right, all $d = 0$; beneath, since $\Box s = 0$.

Original source details: GELFAND-P000104

All homology in Γ_0

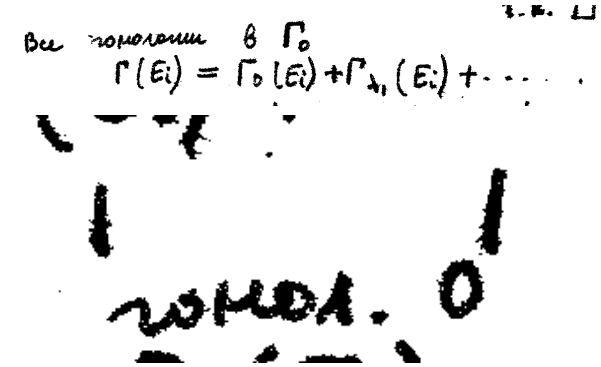


Diagram and typed description. A short downward arrow beneath $\Gamma_{\lambda_1}(E_i)$; a separate vertical stroke with a dense lower end beneath the ellipsis. Beneath, homol. 0. No arrow beneath $\Gamma_0(E_i)$. A separate arrowhead on the right stroke is not established.

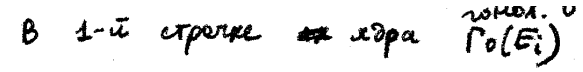
$$\Gamma(E_i) = \Gamma_0(E_i) + \quad \Gamma_{\lambda_1}(E_i) \quad + \quad \cdots$$

$$\downarrow \qquad\qquad |$$

homol. 0

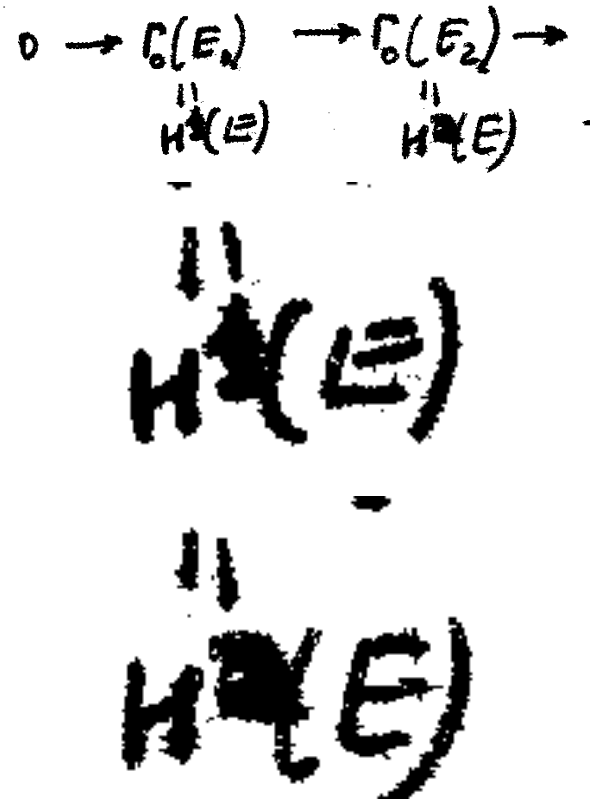
Original source details: GELFAND-P000104

In the 1st row [crossed out: illegible word] kernel $\Gamma_0(E_i)$



Uncertain reading. Before kernel, a short horizontal cancelled entry with dense joined strokes. No secure letter reading.

Original source details: GELFAND-P000104



Original source details: GELFAND-P000104

Uncertain reading. Beneath two terms of the chain, vertical double strokes and entries $H^{[?]}(E)$. Both superscripts are densely overwritten: $H^{*}(E)$ / $H^{i}(E)$ / illegible degree. Former strokes remain inseparable even under enlargement. The chain's last arrow has no visible continuation or ellipsis.

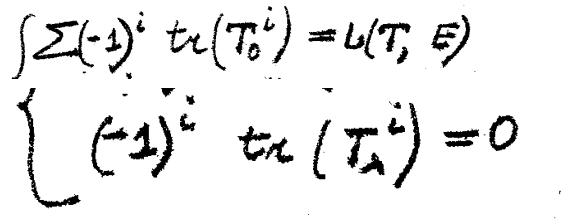
$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma_0(E_1) & \longrightarrow & \Gamma_0(E_2) & \longrightarrow \\ & & \| & & \| & \\ & & H^{[?]}(E) & & H^{[?]}(E) & \end{array}$$

Remark. π_λ commutes with d.

Original source details: GELFAND-P000104

Diagram and typed description. Remark. has a straight underline; then π_λ commutes with d.

$$\begin{cases} \sum (-1)^i \operatorname{tr}(T_0^i) = L(T, E) \\ (-1)^i \operatorname{tr}(T_\lambda^i) = 0 \end{cases}$$



Original source details: GELFAND-P000104

Uncertain reading. A shared left brace. Only the first row starts with $\sum$; no separate sum is visible before $(-1)^i$ in the second row. T has superscript i in both rows; lower signs 0 and λ.

GELFAND-P000105

If there is a seq. of spaces

$$0 \longrightarrow A_1 \xrightarrow{d} A_2 \xrightarrow{d} \cdots \xrightarrow{d} A_n \longrightarrow 0, \qquad d^2 = 0.$$

and maps T_i, comm. with d. Then

$$\sum_{\text{in chains}} (-1)^i \operatorname{tr} T_i = \sum_{\text{in homol.}} (-1)^i \operatorname{tr}(T_i^*) \qquad \text{(Euler's formula)}$$

(Euler's formula is obtained by taking $T = E$)

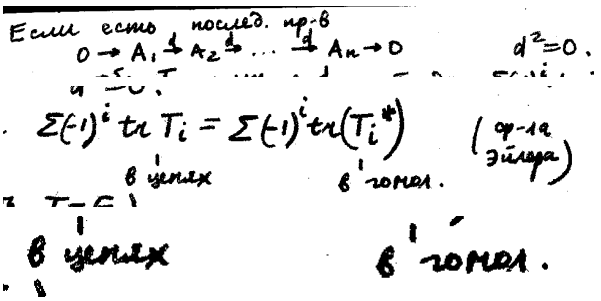


Diagram and typed description. The chain's end arrows are unlabelled, its internal arrows have d. Beneath the left sum, a separate stroke and in chains; beneath the right, a stroke and in homol. At right, a parenthesis Euler's formula.

Original source details: GELFAND-P000105

Proof: represent.

$$A_i = \underset{Z \text{ 3 summands.}}{Z} + C_Z, \qquad Z = B + \text{compl.}$$

and then consider T there.

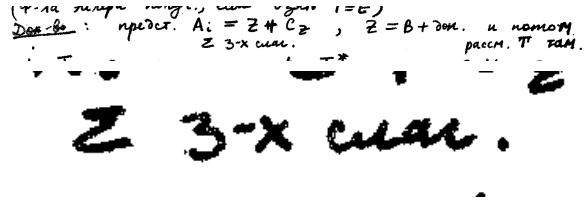
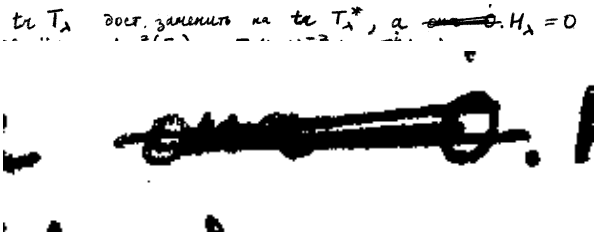


Diagram and typed description. Proof has a straight underline; the colon is to the right of the line. Beneath the first Z, a separate entry Z 3 summands. The second entry is $Z = B +$ compl.; then and then consider T there.

Original source details: GELFAND-P000105

It suffices to replace $\operatorname{tr} T_\lambda$ by $\operatorname{tr} T_\lambda^*$, and [crossed out: illegible entry] $H_\lambda = 0$.



Uncertain reading. Before $H_\lambda = 0$, a long densely cancelled entry with loops at both ends and a horizontal stroke. Former letters remain inseparable.

Original source details: GELFAND-P000105

Introduce

$$\operatorname{tr}^z(E_i) = \sum_\lambda (1+\lambda)^{-z} \operatorname{tr}(T_\lambda^i | E_i)$$

for $\operatorname{Re} z > \frac{1}{2} \dim X$,

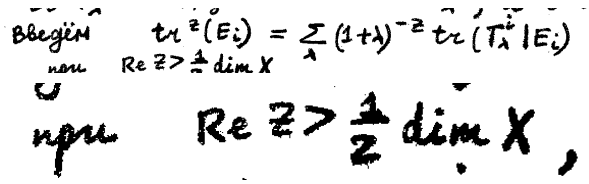
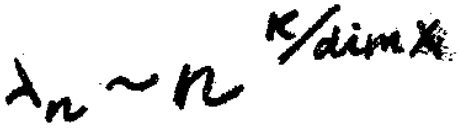


Diagram and typed description. The sum has subscript λ, the factor exponent $-z$, and T_λ a superscript i with a separate dot. The next line is for $\operatorname{Re} z > \frac{1}{2} \dim X$, followed by a comma.

Original source details: GELFAND-P000105

[At right:]

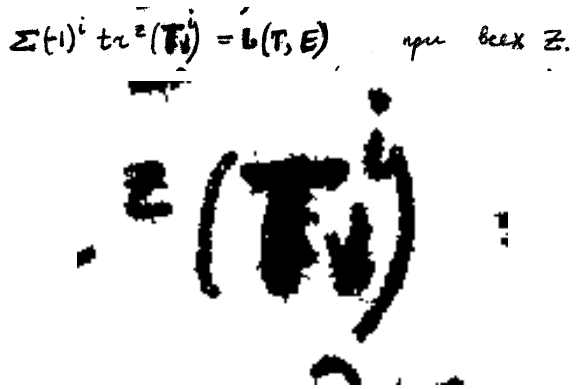
$$\lambda_n \sim n^{K/\dim X}$$



Uncertain reading. The exponent's numerator is a K-like sign with a straight left stroke and two right branches: K / κ.

Original source details: GELFAND-P000105

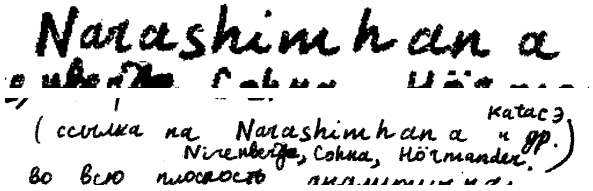
$$\sum (-1)^i \operatorname{tr}^z(T_i^i) = L(T, E) \qquad \text{for all } z.$$



Uncertain reading. Above T, a separate dot over a short stroke gives i. The dense lower strokes permit i or λ.

Original source details: GELFAND-P000105

Continuation theorem (reference to Narashimhan a and others.)



Manuscript notation. The manuscript writes "Narashimhan a," with a separate final a. Below it, the ending of Nirenberg is overwritten.

Original source details: GELFAND-P000105

[Above and others.: katase.[?]]

Uncertain reading. A separate small entry above and others.: source катасэ. / катасэ. (katase.[?]) / illegible. The last sign resembles э; some forms also permit a Latin c.

Original source details: GELFAND-P000105

[Beneath: Nirenberg, Cohna[?], Hörmander.]

Uncertain reading. The middle Latin name between commas: Cohna / Cohn a. Between h and final a, dense joined strokes; exact separation of letters is undetermined.

Original source details: GELFAND-P000105

$\mathrm{tr}^z T^i$ extends to the whole plane, is analytic throughout the plane (!) and at $z = 0$

$$\mathrm{tr}^0(T^i) = \mathrm{tr}^{[?]}(T^i)$$

where tr^b — the trace, calc. in the usual way through generalized functions.

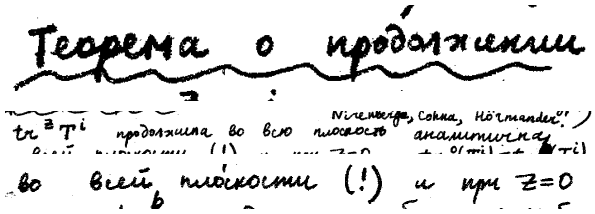
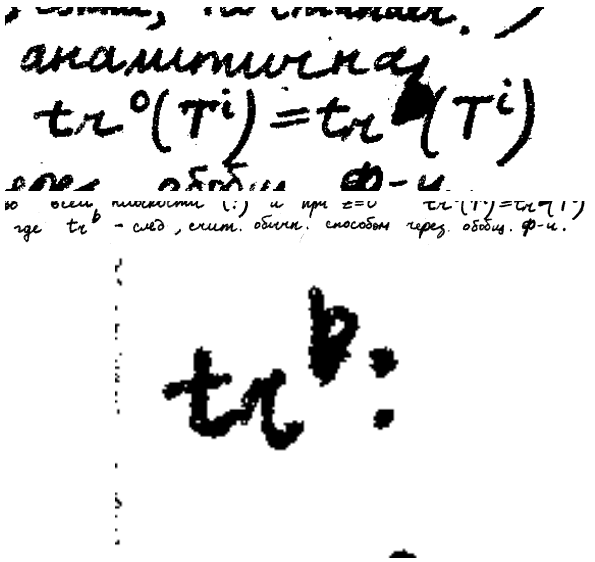


Diagram and typed description. The heading Continuation theorem has a wavy underline. The text twice mentions the whole plane, with a separate (!) and at $z = 0$. T has superscript i.

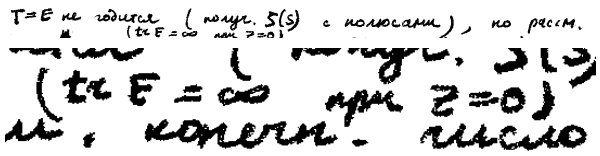
Original source details: GELFAND-P000105



Uncertain reading. The second tr's superscript is densely overwritten and joins surrounding strokes: b[?] / illegible. Beneath and later at left, tr^b is clearly written.

Original source details: GELFAND-P000105

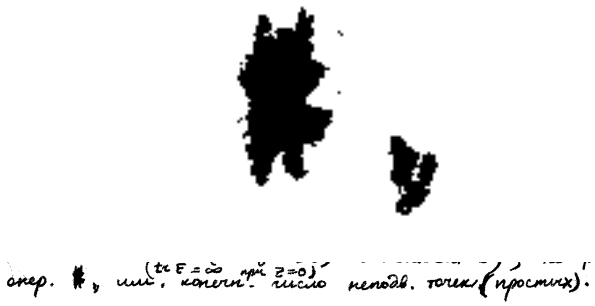
$T = E$ does not work (giving $\zeta(s)$[?] with poles), but consider [Insertion beneath $T = E$:] ($\mathrm{tr}\, E = \infty$ at $z = 0$).



Uncertain reading. Before (s), a large zigzag sign with a rounded lower exit: ζ[?] / Σ[?]. Beneath the line, a separate parenthesis $\mathrm{tr}\, E = \infty$ at $z = 0$.

Original source details: GELFAND-P000105

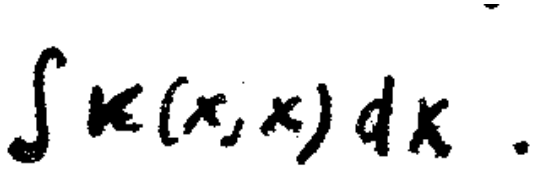
an oper. [illegible sign], having a finite no. of fixed points (simple).



Uncertain reading. After oper., a dense vertical overwritten entry with several transverse projections and lower legs; no secure letter reading. Then a comma, having a finite no. of fixed points (simple).

Original source details: GELFAND-P000105

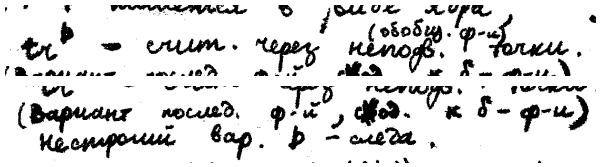
tr^b: T is given as a kernel, $\int \kappa(x,x)\,dx$.



Uncertain reading. The kernel letter has a K-/к-like form: κ / K. A short dense sign in the integral, a taller left stroke in the two-argument formula. The handwritten height does not establish alphabet or case.

Original source details: GELFAND-P000105

tr^b — calc. through fixed points. [Above fixed: (generalized functions).] (A version using a seq. of functions, conv.[?] to a δ-function) A nonrigorous variant of the b-trace.



Uncertain reading. In the parenthesis, the word before to a δ-function is densely overwritten: source сход.[?] (conv.[?]) / illegible beginning. Former strokes remain inseparable. Above the preceding fixed, a separate insertion (generalized functions).

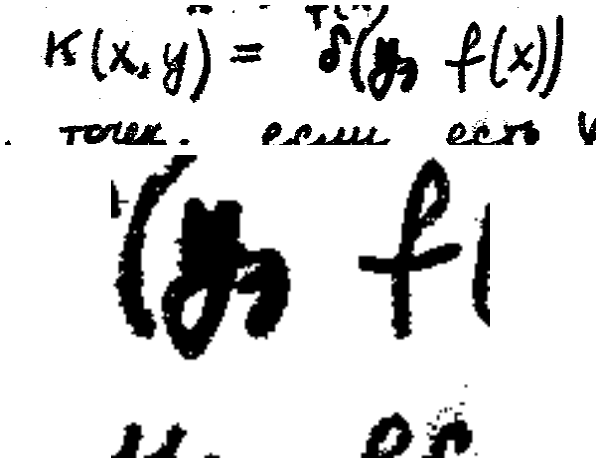
Original source details: GELFAND-P000105

[At right:] $\delta(x,a)$ — δ at the point a.

$$s(x) \longrightarrow \varphi s(f(x))$$

(transf. of traces) $x \longrightarrow f(x)$

$$\kappa(x,y) = \delta(y\,[?]\,f(x))$$



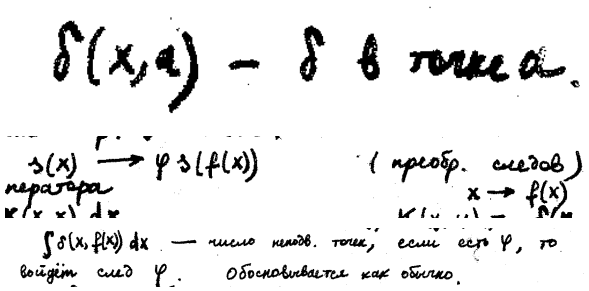
Uncertain reading. To the right of y a descending stroke in the dense ink resembles a comma. Compare $\delta(x,a)$ at right and $\delta(x,f(x))$ below. The overwritten sign remains uncertain.

Original source details: GELFAND-P000105

The operator's trace $\int \kappa(x,x)\,dx$.

$$\int \delta(x, f(x))\,dx$$

— the number of fixed points; if there is φ, the trace of φ enters. Justified as usual.

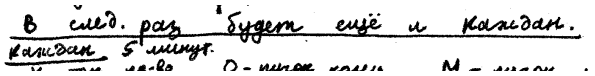


Original source details: GELFAND-P000105

Diagram and typed description. At right, $\delta(x,a)$ with a comma and an entry about the point a. The lower integral $\delta(x, f(x))$ also has a comma. At left, the map with φ; at right, a parenthesis «преобр. следов» (transf. of traces) and a separate $x \longrightarrow f(x)$.

Next time Kazhdan will also speak.

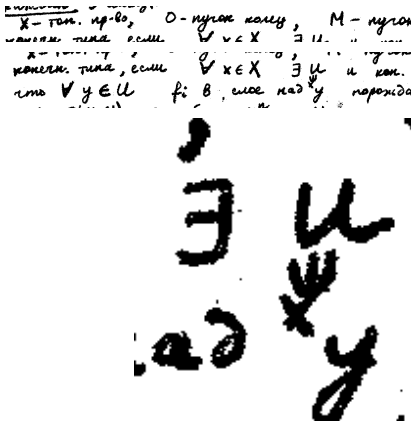
Kazhdan 5 minutes.



Original source details: GELFAND-P000105

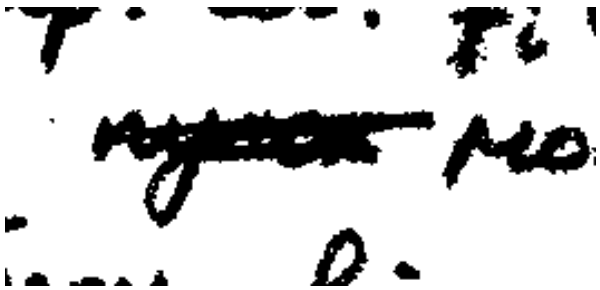
Diagram and typed description. The entire first phrase has a straight underline. Beneath, Kazhdan is separately underlined, followed by 5 minutes.

X — top. space, O — sheaf of rings, M — sheaf of modules. It is of finite type if $\forall x \in X\ \exists U \underset{x}{\Downarrow}$ and a fin. set of sections $f_i \in \Gamma(U, M)$, such that $\forall y \in U$, the f_i in the fibre over y generate [crossed out: illegible word] the module M_y.



Original source details: GELFAND-P000105

Diagram and typed description. Beneath U after $\exists$, a double downward arrow to x. This x is separate from y in the next line over y. The text first has $\forall x \in X$, then $\forall y \in U$.



Original source details: GELFAND-P000105

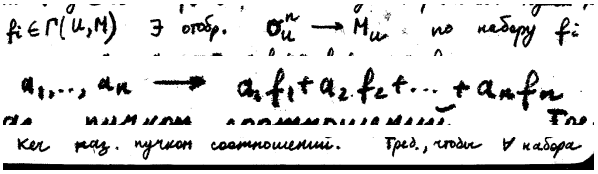
Uncertain reading. Between generate and module, a long densely cancelled word with a lower loop at left and a horizontal stroke to the right. Former letters remain inseparable.

$$f_i \in \Gamma(U, M) \qquad \exists \text{ map } O_U^n \longrightarrow M_U \qquad \text{from the set } f_i$$
$$a_1, \ldots, a_n \longrightarrow a_1 f_1 + a_2 f_2 + \cdots + a_n f_n.$$

Ker is called the sheaf of relations. Req., that $\forall$ set

GELFAND-P000106

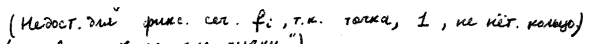
of sections the sheaf of rel. should be of finite type.



Original source details: GELFAND-P000105

Diagram and typed description. O has superscript n and subscript U; M has subscript U. The coefficient arrow points right; at right a linear combination through $a_n f_n$ and a period. On the last line the abbreviation source Треб. (Req.) has a separate following comma; the line ends $\forall$ set.

(Insuff. for fixed sections f_i, since a point, 1, a non-Noeth. ring)

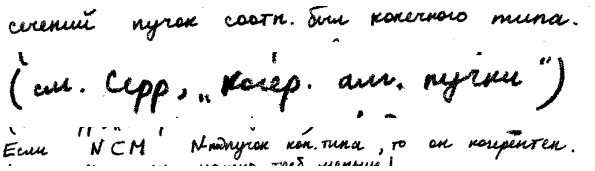


Original source details: GELFAND-P000106

Uncertain reading. Between a point, and a non-Noeth. ring, a separate 1 with a following comma. The sign has no label; its role is undetermined.

(see Serre, “Coher. alg. sheaves”)

If $N \subset M$, N — a subsheaf of fin. type, then it is coherent. (For Noeth. rings one can require less)



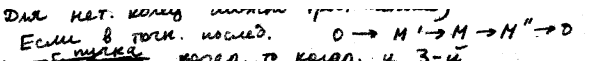
Original source details: GELFAND-P000106

Diagram and typed description. The page begins with the continuation of sections the sheaf of rel. should be of finite type. The reference has a comma after Serre, then quotation marks. The next assertion has a separate $N \subset M$, then N — a subsheaf of fin. type.

If in an exact seq.

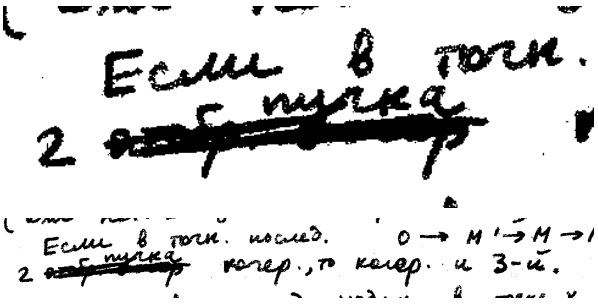
$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

2 [densely crossed out: illegible word; written above: sheaves] are coher., the 3rd is coher. too.



Original source details: GELFAND-P000106

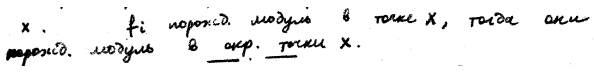
Diagram and typed description. The chain has M with one prime, M without primes, and M with two primes; all four arrows are unlabelled.



Original source details: GELFAND-P000106

Uncertain reading. Above the cancelled word, “sheaves” is legible. The former word remains obscured by the heavy strokes.

x. The f_i gen. the module at the point x; then they gen. the module in a neighb. of the point x.



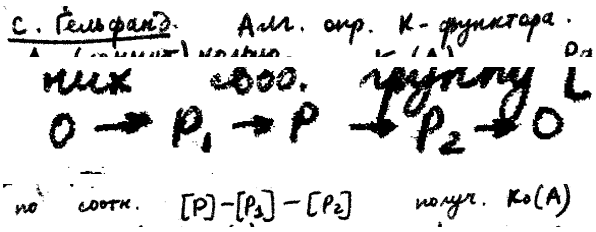
Original source details: GELFAND-P000106

Diagram and typed description. An isolated x followed by a period at the beginning. In the lower line, two separate straight strokes at neighb. and point; none beneath the last x.

S. Gelfand. Alg. def. of the K-functor. A — a (commut.) ring. $K_0(A)$. Consider left proj. modules over A and form the free group $[P]$ on them, abelian. Quotient by exact seq.

$$0 \longrightarrow P_1 \longrightarrow P \longrightarrow P_2 \longrightarrow 0$$

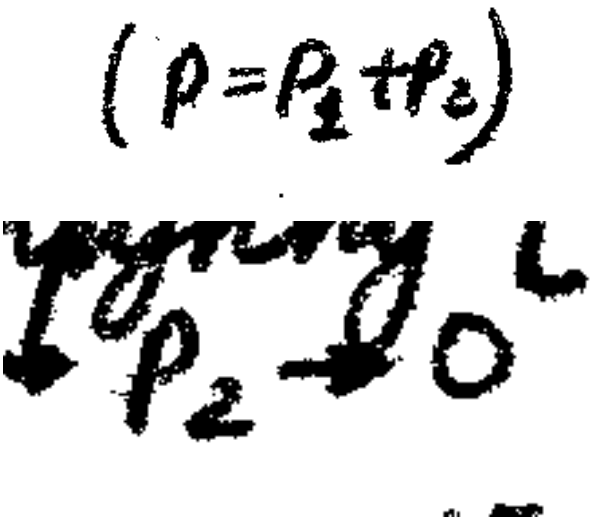
by the rel. $[P] - [P_1] - [P_2]$, obtaining $K_0(A)$



Original source details: GELFAND-P000106

Diagram and typed description. S. Gelfand. is separately underlined. The chain has P_1, P, P_2; the next line has $[P] - [P_1] - [P_2]$, with no $= 0$ written beside it, then obtaining $K_0(A)$.

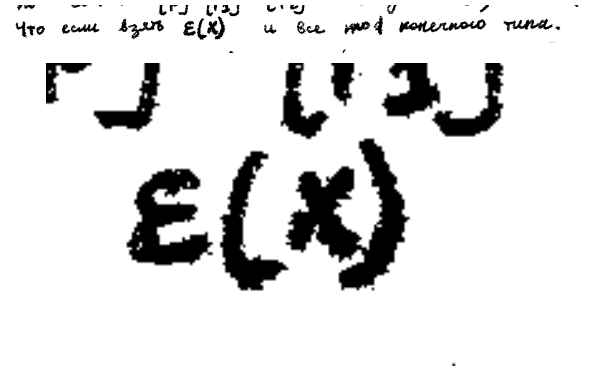
$(P = P_1 + P_2)$.



Original source details: GELFAND-P000106

Manuscript notation. The last index reads 2. Its lower stroke curves more than in the two neighboring instances of P_2.

What if one takes $E(X)$ and all modules of finite type.



Original source details: GELFAND-P000106

Uncertain reading. Before (X), a rounded E-like letter with three exits to the right: E / calligraphic E / an ε-like form. Alphabet and exact case are not established by this sign.

P — in topology, a loc. triv. bundle (sheaf of rings of cont. functions)

A proj. mod over A is loc. free in the localization at prime ideals (all if A is commut.). A proj. mod over a loc. ring is free.

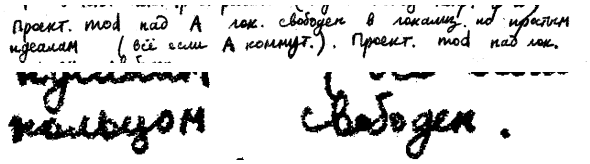


Diagram and typed description. After loc., “ring is free” is written on the next line. The parenthesis says all if A is commut.

Original source details: GELFAND-P000106

$1/\overline{XI}$ Smirnov. Positive-def. functions on a group. Describe the extreme points. (cont. representations) for groups G obtained from a nilp. alg. by introducing mult. by the Campbell — Hausdorff formula.

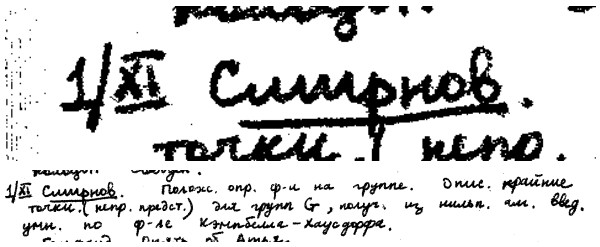


Diagram and typed description. XI in the date has separate over- and underlines. Smirnov. is separately underlined. Then Positive-def. functions on a group and the entry about the Campbell — Hausdorff formula.

Original source details: GELFAND-P000106

Gelfand. Again on Atiyah. $S(x) \in \Gamma(E_i)$, $\varphi(x)s(f^{-1}(x))$. This is $\int K(x,y)s(y)\,dy$. For fixed y, $\kappa(x,y) \in \mathrm{Hom}(E_y, E_x)$,

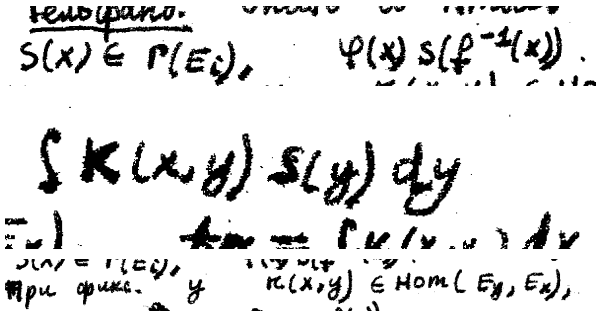
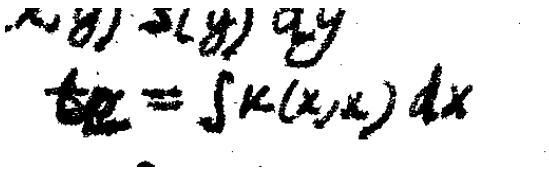


Diagram and typed description. An uppercase S(x) at the beginning; lowercase s after $\varphi(x)$ and in the integral. The first f has an explicit superscript -1. Hom has lower signs y and x. In this integral the kernel is a tall K.

Original source details: GELFAND-P000106

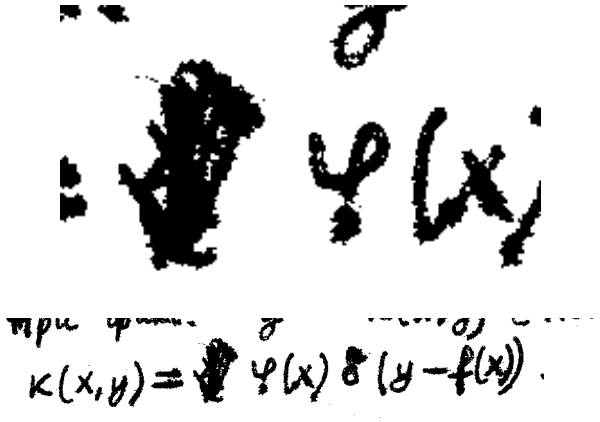
[Densely crossed out: tr[?]] $= \int \kappa(x,x)\,dx$.



Uncertain reading. Before the equality, a dense overwritten entry: tr[?] / illegible. The upper transverse stroke and lower loops merge; former lines remain inseparable.

Original source details: GELFAND-P000106

$$\kappa(x,y) = \text{[crossed out: ?]}\,\varphi(x)\delta(y-f(x)).$$



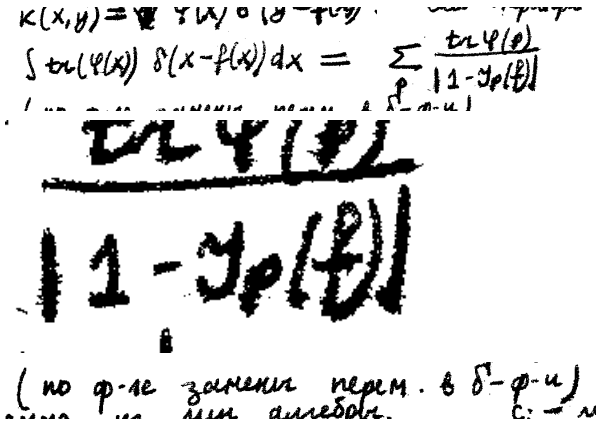
Uncertain reading. Between the equality and $\varphi(x)$, a dense cancelled vertical sign with upper and lower loops. Exact reading undetermined. A minus stroke is visible in the following δ argument: $y - f(x)$.

Original source details: GELFAND-P000106

The operator's trace $\int \operatorname{tr}(\kappa(x,x))\,dx$.

$$\int \operatorname{tr}(\varphi(x))\delta(x-f(x))\,dx = \sum_{\rho} \frac{\operatorname{tr}\varphi(\rho)}{|1-J_\rho(f)|}.$$

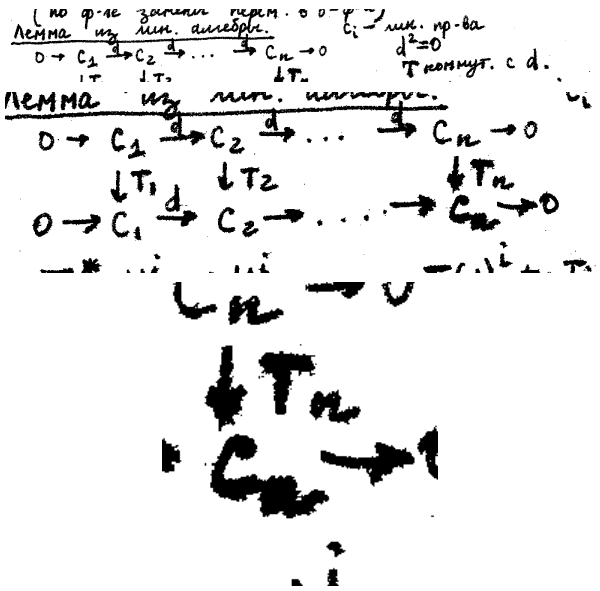
(by the change-of-variables formula for the δ-function)



Uncertain reading. The sign before subscript ρ is curved, with a short upper exit and a hooked lower part: $J_\rho(f)$[?] / $D_\rho(f)$ / $Д_\rho(f)$. Two vertical outer bars. No separate det entry is visible in the denominator.

Original source details: GELFAND-P000106

Lemma from lin. algebra. C_i — lin. spaces. $d^2 = 0$. T commutes with d.

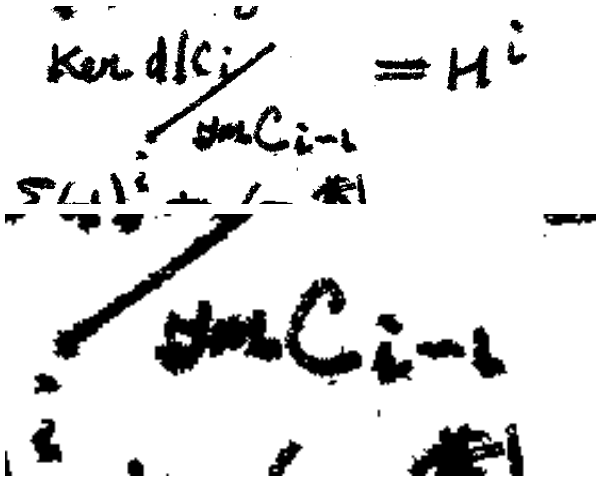


Original source details: GELFAND-P000106

Diagram and typed description. Two horizontal chains, with 0 at their ends. The upper row has three internal d labels; the lower row has d only between C_1 and C_2. Three downward vertical marks labelled T_1, T_2, T_n. The right mark has a dense diamond-like lower end; beneath it the lower C_n is densely overwritten. The typographic recreation preserves direction and labels; dense contours remain inseparable.

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & C_1 & \xrightarrow{d} & C_2 & \xrightarrow{d} & \cdots & \xrightarrow{d} & C_n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle T_1} & & \downarrow{\scriptstyle T_2} & & & & \downarrow{\scriptstyle T_n} & & \\ 0 & \longrightarrow & C_1 & \xrightarrow{d} & C_2 & \longrightarrow & \cdots & \longrightarrow & C_n & \longrightarrow & 0 \end{array}$$

$$\frac{\operatorname{Ker} d|C_i}{\operatorname{Im} C_{i-1}} = H^i.$$

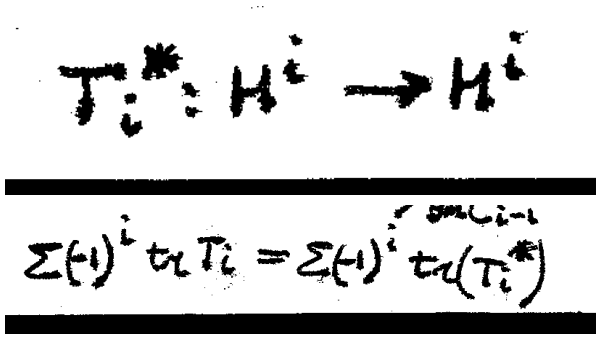


Original source details: GELFAND-P000106

Diagram and typed description. The manuscript fraction bar is oblique. Above it, Ker $d|C_i$; beneath, a dense Im C_{i-1}, with no separate d or restriction bar visible after Im. At right, $= H^i$.

$T_i^* : H^i \longrightarrow H^i$.

$$\sum (-1)^i \operatorname{tr} T_i = \sum (-1)^i \operatorname{tr}(T_i^*).$$

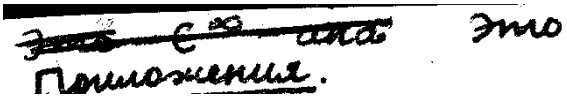


Original source details: GELFAND-P000106

Diagram and typed description. At left, $T_i^* : H^i \longrightarrow H^i$. In the last equality both sums have superscript i on (-1); tr T_i at left, tr(T_i^*) at right.

GELFAND-P000107

[Crossed out at the top: This[?] C^∞ that[?]; the rest illegible.]



Original source details: GELFAND-P000107

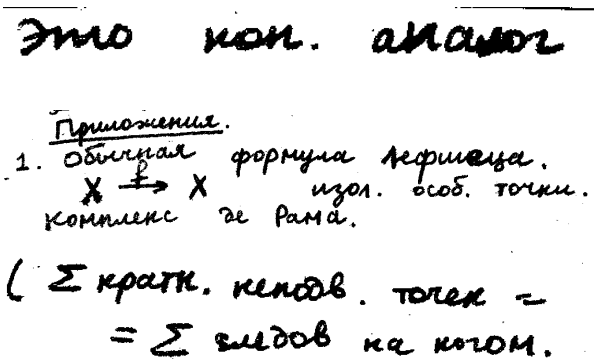
Uncertain reading. A long stroke crosses the top line; dense writing beneath it on the left, C^∞ in the middle, that[?] on the right. The word boundaries and exact reading are not established.

This is a finite analogue.

Applications.

1. Usual Lefschetz formula. $X \xrightarrow{f} X$ isol. sing. points. De Rham complex. [On the right:]

$$\left(\sum \text{mult. of fixed points} = \right.$$
$$= \sum \text{traces on cohom.}$$

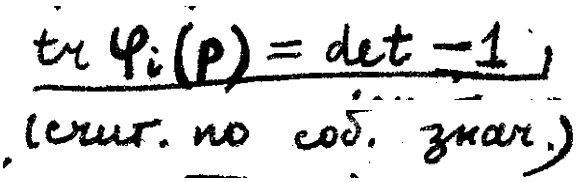


Original source details: GELFAND-P000107

Diagram and typed description. This is a finite analogue. is separate from the crossed line. Applications. is underlined. On the right an opening parenthesis precedes the sum; an equals sign ends the upper line and another begins the lower one. No closing parenthesis is visible.

$$\underline{\operatorname{tr} \varphi_i(\rho) = \det - 1};$$

(computed by eigenvalues)



Original source details: GELFAND-P000107

Manuscript notation. After det the manuscript writes -1. The long underline curves upward at the right.

[On the right:]

$$\sum \pm 1 \quad \text{over fixed points.}$$

$$\sum (\pm 1)\lambda_{i_1} \cdots \lambda_{i_k} \qquad J_\rho = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

[A diagonal ellipsis lies between λ_1 and λ_n in the matrix.]

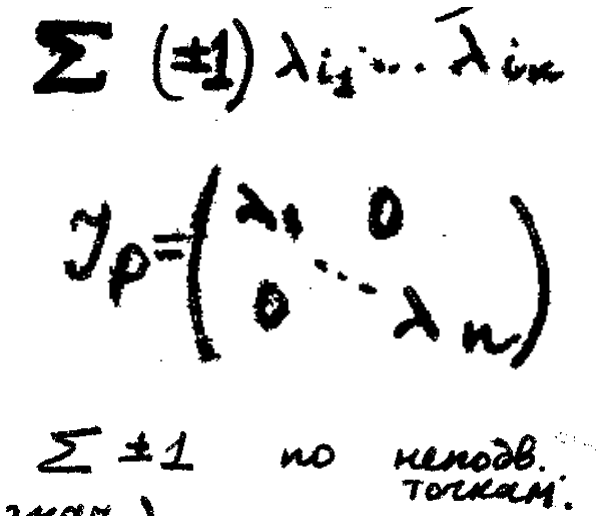


Diagram and typed description. The sum has a separate (± 1), then $\lambda_{i_1} \cdots \lambda_{i_k}$. On the right a J-like letter with subscript ρ; inside parentheses λ_1 at upper left, λ_n at lower right, zeros at the other corners and a diagonal ellipsis between them.

Original source details: GELFAND-P000107

2. Complex case. $f : X \longrightarrow X$ — holom.

$$\sum f_{i_1 \cdots i_k}\, dz_{i_1} \wedge \cdots \wedge dz_{i_k}$$

$f_{i_1 \cdots i_k}$ holom. and holom. diff. $H^{i,0}($

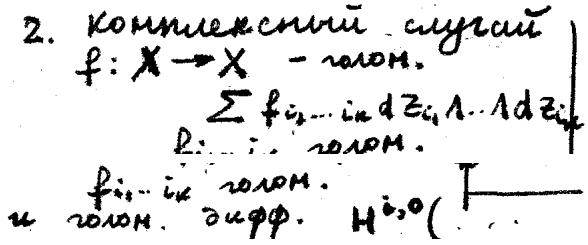
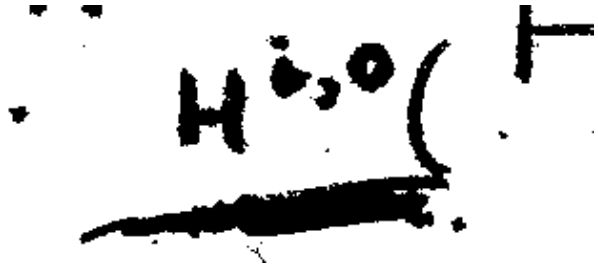


Diagram and typed description. To the right of the lines of forms a long vertical stroke downward and a horizontal stroke to the right from it; the corner has no arrows or labels. The parenthesis at $H^{i,0}($ remains open.

Original source details: GELFAND-P000107

[Densely crossed out below $H^{i,0}$(: illegible.]



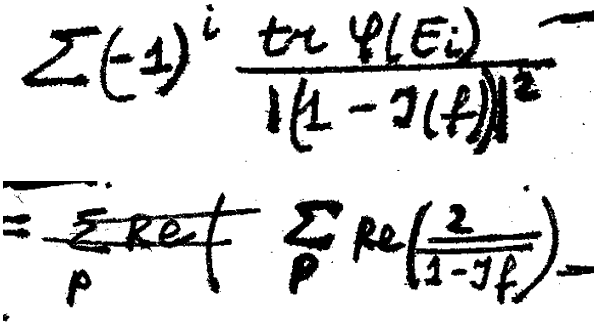
Original source details: GELFAND-P000107

Uncertain reading. Below the open parenthesis dense horizontal strokes slope upward to the right; individual letters are not distinguished.

$$\sum (-1)^i \frac{\operatorname{tr} \varphi(E_i)}{|(1 - J(f))|^2} =$$

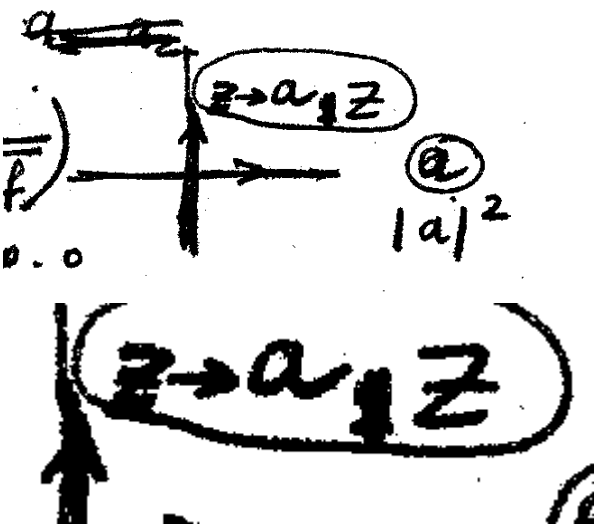
[Crossed out: $\sum_\rho \operatorname{Re}($; followed by:]

$$\sum_\rho \operatorname{Re}\left(\frac{2}{1 - J_f}\right)$$



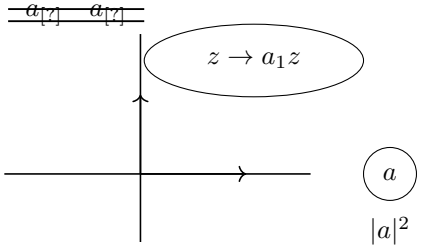
Original source details: GELFAND-P000107

Uncertain reading. The vertical strokes at the denominator are thickened: absolute value / double bars. Inside, a J-like letter / handwritten Cyrillic D; the numerator of the active sum on the right is 2, with two close horizontal lines below it.

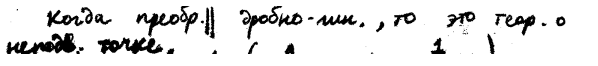


Original source details: GELFAND-P000107

Uncertain reading. Two intersecting lines, arrows upward and to the right; the horizontal line continues beyond the arrowhead. In an oval $z \to a_1 z$, separately a circled a, below it $|a|^2$. Above the vertical line crossed a-like signs with illegible lower strokes; the exact reading of this writing is not established.



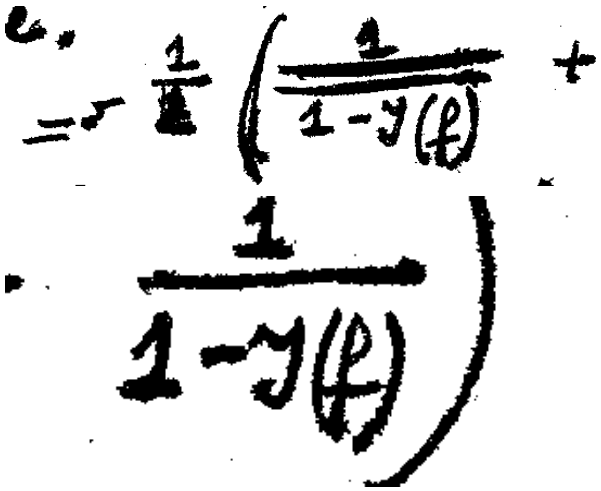
When the transf. $\|$ is fractional-linear, this is the fixed-point theorem.



Original source details: GELFAND-P000107

Diagram and typed description. Two vertical lines stand between transf. and fractional-linear. The final words fixed point. are on the next line.

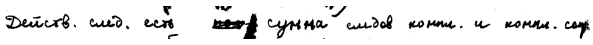
$$= \frac{1}{[?]} \left(\frac{1}{1 - J(f)} + \frac{1}{1 - J(f)} \right)$$



Uncertain reading. Below the first separate 1 a dense sign: 2[?] / 1[?] / illegible. The first inner fraction has two long horizontal lines, the second one; the first extra line may indicate conjugation of the fraction or denominator.

Original source details: GELFAND-P000107

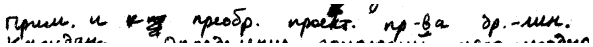
The real trace, is [crossed out: illegible] the sum of the complex and complex-conjugate traces.



Uncertain reading. Between is and the sum a densely crossed horizontal record with a vertical stroke at the right; individual letters are not distinguished.

Original source details: GELFAND-P000107

Appl. also to [heavily overwritten: illegible] fractional-linear transf. of projective space.



Uncertain reading. After also to two dense superposed signs. Under the abbreviation for space two separate short lines; the preceding overwritten writing is illegible.

Original source details: GELFAND-P000107

Kazhdan. Definition of homology of anything.

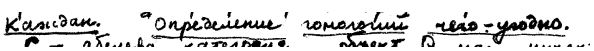


Diagram and typed description. The surname and three parts of the heading have separate underlines. The Russian words for of anything are joined by a hyphen.

Original source details: GELFAND-P000107

C — an abelian category. an object Q is called injective if the functor $F(A) = \mathrm{Hom}(A, Q)$ is exact. ($A' \subset A \; \forall f$ [crossed out: an illegible word] extends)

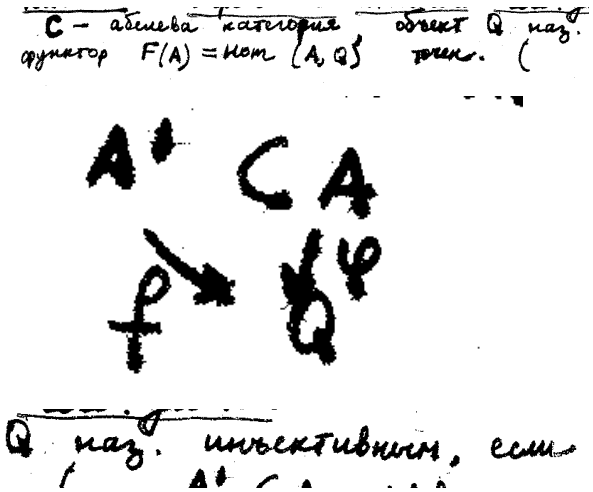
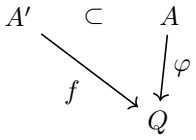
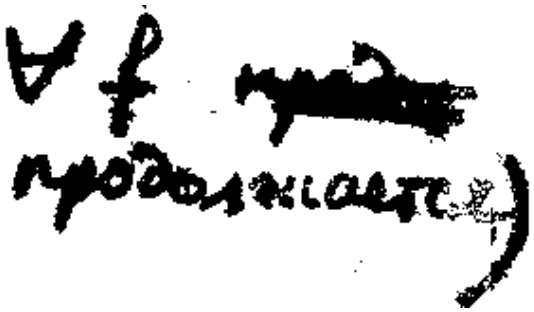


Diagram and typed description. Within the parentheses $A' \subset A$; from A' an arrow downward to the right labelled f, from A downward to Q an arrow with φ at the right. There is no superscript at $F(A)$.



Original source details: GELFAND-P000107



Uncertain reading. After $\forall f$ above extends a densely crossed word; its letters are not separated.

Original source details: GELFAND-P000107

The categ. has enough injective objects if $\forall A \; \exists$ inj. $Q \parallel 0 \longrightarrow A \longrightarrow Q$. [cat. of sheaves]

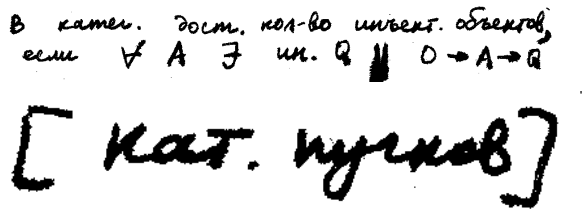
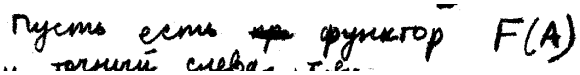


Diagram and typed description. Before $0 \to A \to Q$ two vertical lines, with the lower part thickened. On the right a separate record in square brackets cat. of sheaves.

Original source details: GELFAND-P000107

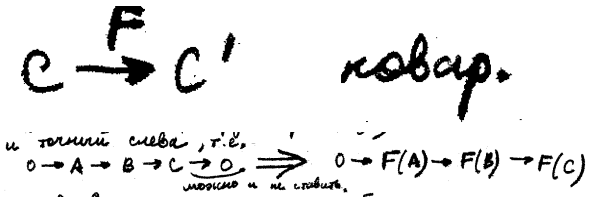
Let there be [crossed out: illegible] a functor $F(A)\ C \xrightarrow{F} C'$ covar. and left exact, i.e.

$$0 \longrightarrow A \longrightarrow B \longrightarrow \underbrace{C \longrightarrow 0}_{\text{may also be left out.}} \Longrightarrow 0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C)$$



Original source details: GELFAND-P000107

Uncertain reading. Between there be and a functor a small densely crossed record, with a downward projecting stroke; the exact reading is not established.



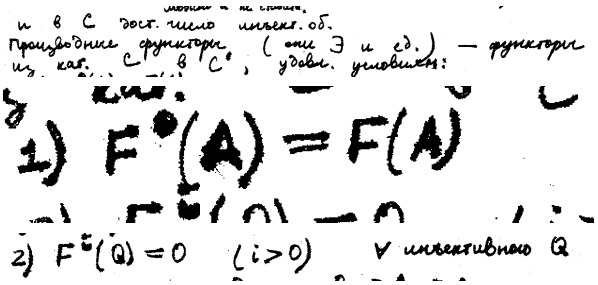
Original source details: GELFAND-P000107

Diagram and typed description. F above the arrow at $C \to C'$. Beneath $C \to 0$ in the original sequence an arc with may also be left out. In the sequence after the large arrow the last visible term is $F(C)$, followed by no dot or zero.

and C has enough inj. obj. Derived functors (they $\exists$ and are unique) — functors from cat. C to C', satisfying the conditions:

1) $F^0(A) = F(A)$

2) $F^i(Q) = 0\ (i > 0)\ \forall$ injective Q

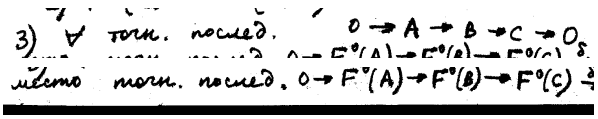


Original source details: GELFAND-P000107

Manuscript notation. In the first condition F has a rounded superscript 0. In the second, a separate dot sits above the short stroke, as in the i at right.

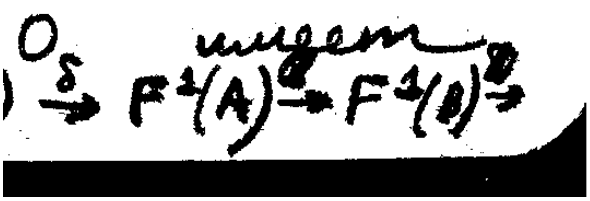
3) $\forall$ exact seq. $0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$ there is an exact seq.

$$0 \longrightarrow F^0(A) \longrightarrow F^0(B) \longrightarrow F^0(C) \xrightarrow{\delta} F^1(A) \longrightarrow F^1(B) \xrightarrow{\delta}$$



Original source details: GELFAND-P000107

Diagram and typed description. At the bottom the sequence begins 0, then three terms with superscript 0, then an arrow with δ, two terms with superscript 1 and another arrow with δ.

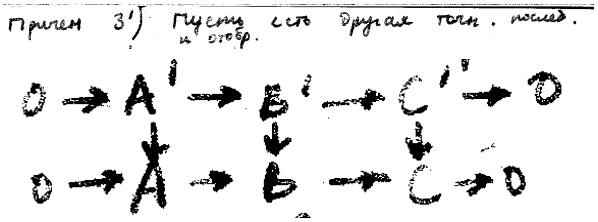


Original source details: GELFAND-P000107

Undeciphered passage. The last arrow is at the right edge; the following term is not visible in this frame. The word «имеет» above the arrows joins «место» at the start of the next line (“there is”).

GELFAND-P000108

Moreover 3') Let there be another exact seq. and a map.

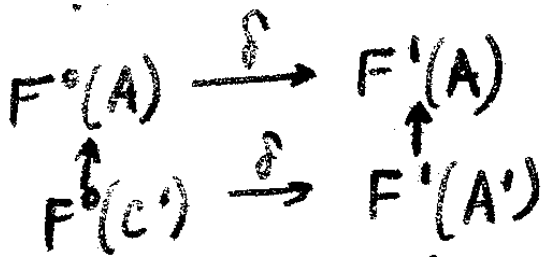


Original source details: GELFAND-P000108

Diagram and typed description. Two rows $0 \to A' \to B' \to C' \to 0$ and $0 \to A \to B \to C \to 0$; three vertical arrows downward from the primed letters. The arrows have no labels.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

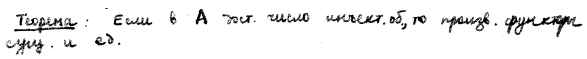
$$\begin{array}{ccc} F^0(A) & \xrightarrow{\delta} & F^1(A) \\ \uparrow & & \uparrow \\ F^0(C') & \xrightarrow{\delta} & F^1(A') \end{array}$$



Original source details: GELFAND-P000108

Manuscript notation. The upper-left entry is $F^0(A)$ and the lower-left entry is $F^0(C\prime)$. Both vertical arrows point upward.

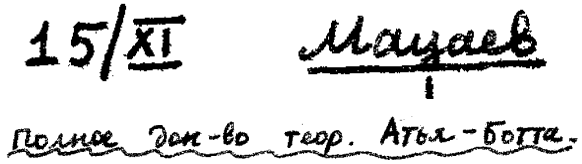
Theorem: If A has enough inj. obj., the derived functors exist and are unique.



Original source details: GELFAND-P000108

Diagram and typed description. A separate large A in the hypothesis. Theorem is underlined, with the colon to the right of the line. Exist and are unique continues the writing on the next line.

15/XI Matsaev Complete proof of the Atiyah-Bott theorem.

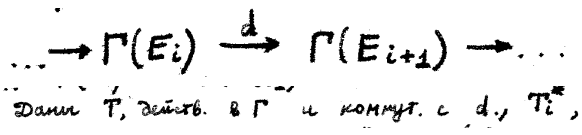


Original source details: GELFAND-P000108

Diagram and typed description. XI has lines above and below. A straight underline below the surname; a wavy underline below Complete proof of the Atiyah-Bott theorem.

$$\cdots \longrightarrow \Gamma(E_i) \xrightarrow{d} \Gamma(E_{i+1}) \longrightarrow \cdots$$

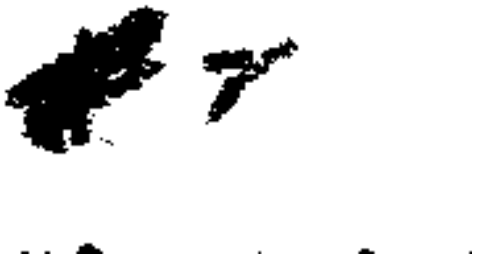
Given T, acting on Γ and commuting with d, T_i^*,



Original source details: GELFAND-P000108

Diagram and typed description. d above the middle arrow of the complex, subscript $i+1$ on the right. On the next line T_i^*, followed by a comma.

[Densely crossed out: an illegible sign; to the right a T-like sign[?].]



Uncertain reading. Dense crossed writing slopes upward to the right; farther right a separate small sign with a horizontal top and slanting stem: T[?] / illegible.

Original source details: GELFAND-P000108

$$L(T, E) = \sum (-1)^i \operatorname{tr} T_i^*$$

[On the right:] (Here we do not need z — does not depend)

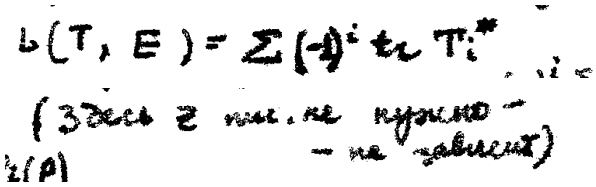
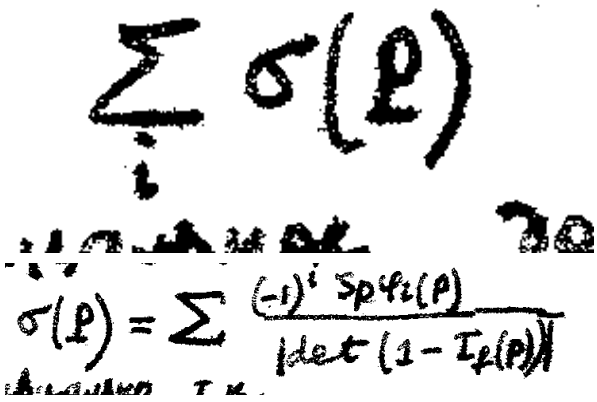


Diagram and typed description. T in the sum has a superscript star and subscript i. In the right parenthesis z, then we do not need, below — does not depend.

Original source details: GELFAND-P000108

$$\sum_i \sigma(\rho) \qquad \sigma(\rho) = \sum \frac{(-1)^i \operatorname{Sp} \varphi_i(\rho)}{|\det(1 - I_f(\rho))|}$$



Uncertain reading. The first sum has subscript i with a separate dot. The denominator has a sign with a broad base, I or J, followed by subscript f and (ρ).

Original source details: GELFAND-P000108

The formal proof is trivial, since

$$L = \sum_i (-1)^i \operatorname{tr} T_i$$

(elem. theorem of lin. algebra) $\operatorname{tr} T_i$ formally give this sum. T_i is given by the kernel $\varphi\delta(y - f(x))$. The trace is precisely $\sum$ over fixed points.

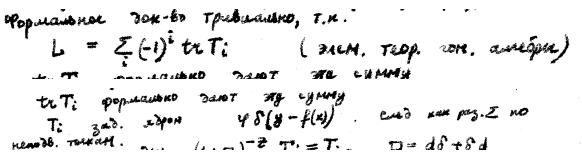


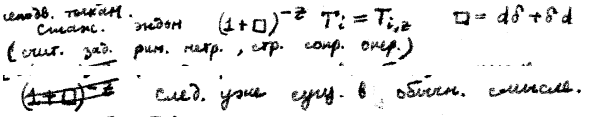
Diagram and typed description. In the formal sum T has subscript i without a star. In the kernel a horizontal minus follows y, then $f(x)$. Fixed points. is carried to the next line.

Original source details: GELFAND-P000108

Smooth by the kernel

$$(1+\Box)^{-z}T_i = T_{i,z}, \qquad \Box = d\delta + \delta d$$

(assume a Riem. metric is given, Herm. adjoint oper.) [Crossed out: $(1+\Box)^{-z}$.] The trace already exists in the usual sense.



Original source details: GELFAND-P000108

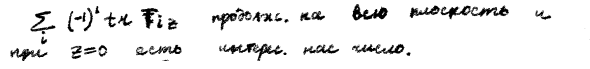
Diagram and typed description. Exponent $-z$; a comma in the right-hand $T_{i,z}$. A separate formula with a square, d and δ. Below, a crossed copy of $(1+\Box)^{-z}$.

$$\operatorname{tr} T_{iz} = \sum_{\lambda} \pi_\lambda T i_\lambda (1+\lambda)^{-z}$$

Original source details: GELFAND-P000108

Uncertain reading. The indices of the left-hand T are densely overwritten: iz / i,z. On the right π_λ, T, i_λ and $(1+\lambda)^{-z}$; no separate tr is visible after the sum.

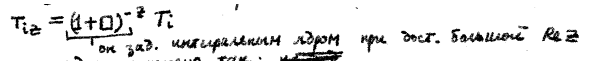
$\sum_i (-1)^i \operatorname{tr} T_{iz}$ continues to the whole plane and at $z=0$ is an integral as a number.



Original source details: GELFAND-P000108

Diagram and typed description. Subscript i in the sum; after the whole plane and, the continuation at $z=0$ is an integral as a number.

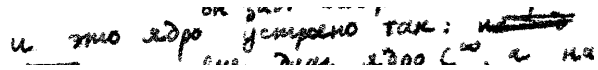
$$T_{iz} = \underbrace{(1+\Box)^{-z}}_{\text{it is given by an integral kernel for sufficiently large Re } z} T_i$$



Original source details: GELFAND-P000108

Diagram and typed description. Beneath $(1+\Box)^{-z}$ a straight bracket with a short downward stroke; the explanation below it ends in Re z.

and this kernel is arranged as follows: [crossed out: an illegible word]

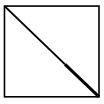


Original source details: GELFAND-P000108

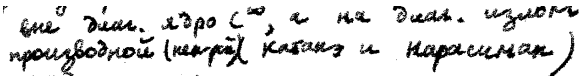
Uncertain reading. After arranged as follows: two dense, almost horizontal crossing strokes; the original letters are not separated.

Original source details: GELFAND-P000108

Uncertain reading. An unlabelled square; the diagonal goes from upper left to lower right, thickening at the latter corner. Thickening / arrowhead[?]; it has no separate label.



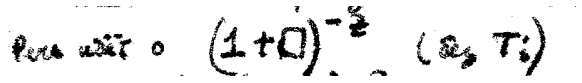
off the diag. the kernel is C^∞, and on the diag. breaks in the derivative (contr.[?] to Katz and Narasimhan)



Original source details: GELFAND-P000108

Uncertain reading. In parentheses the abbreviation contr.[?] with a dense ending, then to Katz and Narasimhan. Before the parenthesis derivative.

The discussion concerns $(1+\Box)^{-z}$ (without T_i).

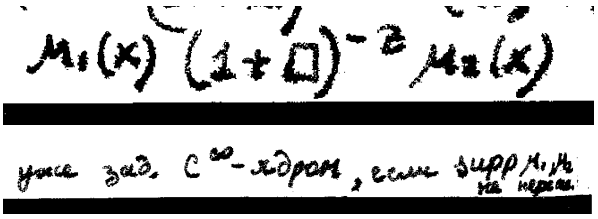


Original source details: GELFAND-P000108

Diagram and typed description. The final parenthesis reads without T_i. A separate $(1+\Box)^{-z}$ precedes it.

$$\mu_1(x)(1+\Box)^{-z}\mu_2(x)$$

is already given by a C^∞ kernel if the supports of μ_1, μ_2 do not intersect.

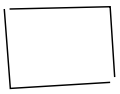


Uncertain reading. After “if” comes supp with two letters indexed 1 and 2. Underneath is «не перес.» (“do not intersect”). The letters read as μ, possibly M.

Original source details: GELFAND-P000108

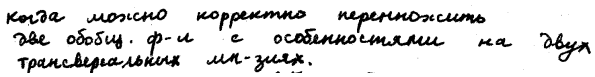
GELFAND-P000109

Diagram and typed description. An empty quadrilateral in the upper-left margin; no diagonal or labels. The side ends do not join at lower right; a small gap also at upper left.



Original source details: GELFAND-P000109

When can one correctly multiply two generalized fn.s with singularities on two transversal manifolds.



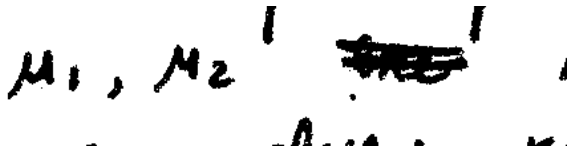
Original source details: GELFAND-P000109

Diagram and typed description. The writing starts When and ends manifolds. There is no question mark at the end.

μ_1, μ_2 [crossed out: illegible]

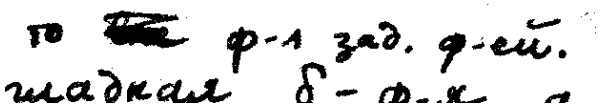
$\cap$ supp. off the curve, then [crossed out: illegible]

the fn. is given by a fn.



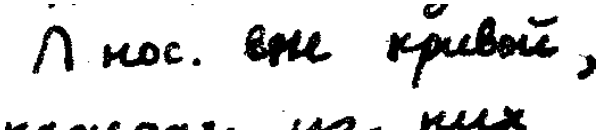
Original source details: GELFAND-P000109

Uncertain reading. Between μ_1, μ_2 and $\cap$ a short densely crossed record. The first μ has an angular top and a long lower stroke.



Original source details: GELFAND-P000109

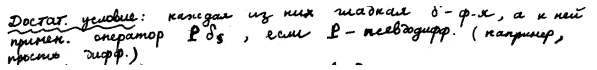
Uncertain reading. After then and before fn., dense crossed writing; the original signs are not separated.



Original source details: GELFAND-P000109

Manuscript notation. The line reads “off the curve.” The word «кривой» has a descending p and a breve over the final й.

Suff. condition: each of them is a smooth δ-fn., and to it is applied an operator $P\delta_S$, if P is pseudodiff. (for example, simply diff.)

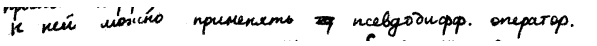


Original source details: GELFAND-P000109

Diagram and typed description. A wavy underline below Suff. condition. After operator, a separate $P\delta_S$; then if P is pseudodiff. The parenthesis ends simply diff.)

To it one can apply [densely crossed out: an illegible sign]

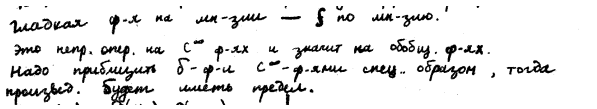
a pseudodiff. operator.



Original source details: GELFAND-P000109

Uncertain reading. Between apply and pseudodiff., a short dense sign with a lower slanting stroke.

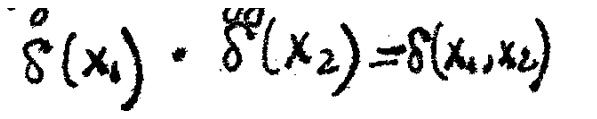
A smooth fn. on a manifold — $\int$ over the manifold. This is a cont. oper. on C^∞ fn.s and therefore on generalized fn.s. One must approximate the δ-fn.s by C^∞ fn.s in a special way, then the product will have a limit.



Original source details: GELFAND-P000109

Diagram and typed description. After the dash a separate integral sign without limits. Then two sentences on the cont. oper. and approximation of δ-fn.s; ending in limit.

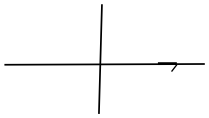
$$\delta(x_1) \cdot \delta(x_2) = \delta(x_1, x_2)$$



Original source details: GELFAND-P000109

Diagram and typed description. A dot between the first two factors; on the right $\delta(x_1, x_2)$ without a separate final period.

Uncertain reading. Intersecting vertical and horizontal lines without labels. To the right of the intersection a short thickening and slanted stroke: arrowhead[?] / unevenness of the line. No separately marked point.



Original source details: GELFAND-P000109

(supp. in 4-dim. space transvers. intersect. of diag.)

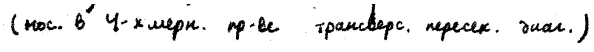


Diagram and typed description. The parenthesis below the formula: supp. in 4-dim. space transvers. intersect. of diag.

Original source details: GELFAND-P000109

Kazhdan. X_P — an open set in the Zariski top. $P \neq 0$.

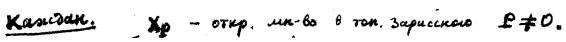
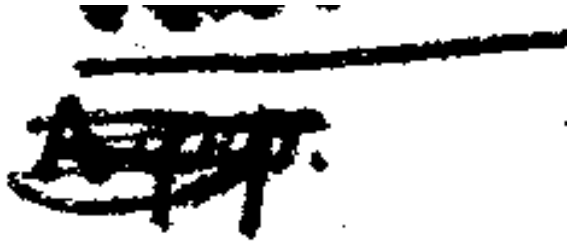


Diagram and typed description. The surname Kazhdan. is underlined. Then X_P, on the right $P \neq 0$.

Original source details: GELFAND-P000109

[On the left, crossed out: an illegible word.]



Uncertain reading. Below the underlined surname and to the left of This is a basis, several dense crossing strokes.

Original source details: GELFAND-P000109

This is a basis of open sets on an aff. alg. variety. Sect. of the sheaf on X_P: $R(x)$: $\frac{Q(x)}{P^n}$,

[A short sign crossed out: =[?] / illegible.]

they are identified if the diff. becomes 0 on the variety.

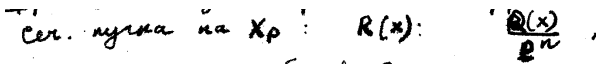


Diagram and typed description. Colons after X_P and after $R(x)$. Numerator $Q(x)$, denominator P^n; a comma on the right.

Original source details: GELFAND-P000109

Uncertain reading. Before they, a short dark sign with two almost horizontal strokes; original sign =[?].

Original source details: GELFAND-P000109

$\Gamma(X_P, O_X)$. Aff. alg. variety — a finite number of eq.s. X_P is again an aff. space.

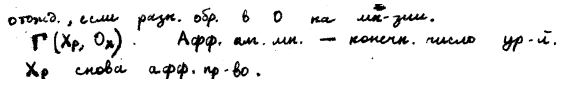
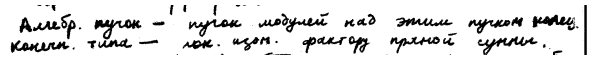


Diagram and typed description. A period after $\Gamma(X_P, O_X)$. The final line reads precisely again an aff. space.

Original source details: GELFAND-P000109

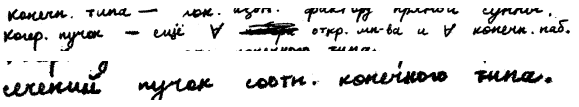
An algebraic sheaf — a sheaf of modules over this sheaf of rings. Of finite type — loc. isom. to a quotient of a direct sum.



Original source details: GELFAND-P000109

Diagram and typed description. Two lines: the definition of an algebraic sheaf and Of finite type — loc. isom. to a quotient of a direct sum.

A coher. sheaf — further ∀ [crossed out: illegible] open set and ∀ finite coll. of sections, the sheaf of rel.s is of finite type.



Original source details: GELFAND-P000109

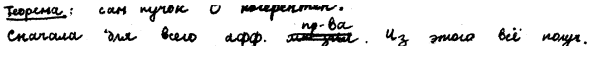
Uncertain reading. After the first ∀ a densely crossed word. Then open set and ∀ finite coll.; continuation on the next line of sections, the sheaf of rel.s is of finite type.

Theorem: the sheaf O itself is coherent.

Original source details: GELFAND-P000109

Diagram and typed description. Theorem is underlined, with the colon to the right. After the sheaf a separate O, then itself is coherent.

First for the whole aff. [crossed out: variety[?] / illegible.] [Inserted above: space.] From this everything follows autom.



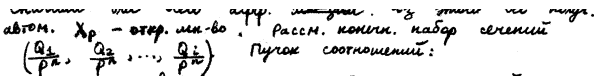
Original source details: GELFAND-P000109

Uncertain reading. Beneath the insertion space, a crossed record variety[?]. After it a period; From this everything follows continues on the same line, autom. below.

X_P — an open set. Consider a finite collection of sections

$$\left(\frac{Q_1}{P^n}, \frac{Q_2}{P^n}, \ldots, \frac{Q_i}{P^n}\right)$$

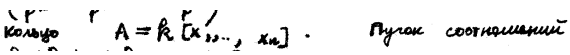
Sheaf of relations:



Original source details: GELFAND-P000109

Diagram and typed description. The tuple has subscripts $1, 2, \ldots, i$, denominators P^n; no separate period after the parenthesis. On the right Sheaf of relations:

Ring $A = k[x_1, \ldots, x_n]$. [On the right again: Sheaf of relations]



Original source details: GELFAND-P000109

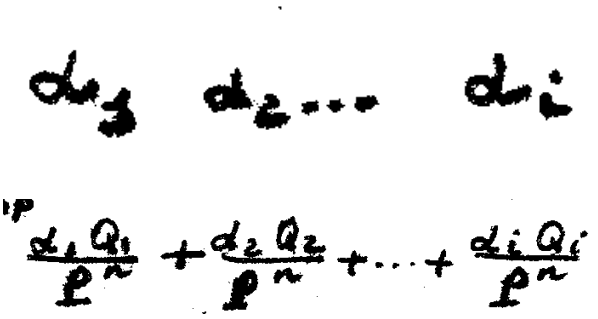
Diagram and typed description. Ring $A = k[x_1, \ldots, x_n]$ with a final period. On the right Sheaf of relations is repeated without a final sign.

$$O_X + O_X + \cdots + O_X \underset{\text{over } X_P}{\longrightarrow} O_X$$

Original source details: GELFAND-P000109

Diagram and typed description. Everywhere O_X; plus signs between the initial summands. Beneath the rightward arrow over X_P.

$$d_3\, d_2 \cdots d_i \longrightarrow \frac{d_1 Q_1}{P^n} + \frac{d_2 Q_2}{P^n} + \cdots + \frac{d_i Q_i}{P^n}$$



Manuscript notation. The first subscript on the left is 3, with a rounded lower stroke. On the right the first numerator has d_1.

Original source details: GELFAND-P000109

Kernel — the fibre of the sheaf of rel.s over a point

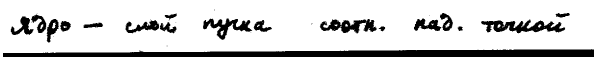


Diagram and typed description. The final line reads fibre, then of the sheaf of rel.s over a point. No separate final sign after the last word.

Original source details: GELFAND-P000109

GELFAND-P000110

Consider the following map:

$$A + A + A + \cdots + A \longrightarrow A$$

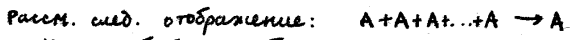
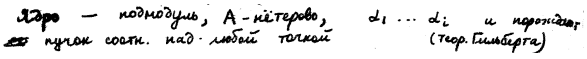


Diagram and typed description. A colon after map; no labels on the rightward arrow. No separate period after the final A.

Original source details: GELFAND-P000110

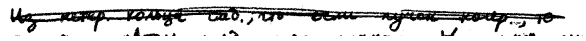
One can dispense with P, the rel.s are the same, Kernel — a submodule, A is Noetherian, $d_1 \cdots d_i$ and generate [crossed out: an illegible sign] the sheaf of rel.s over any point (Hilbert's theorem)



Uncertain reading. On the left before the sheaf of rel.s, a short crossed sign. On the preceding line d_1, ellipsis, d_i and generate; the right end of the last word reaches the image edge.

Original source details: GELFAND-P000110

[The whole line crossed out: From coher. of the ring it follows that if[?] a sheaf is coher., then; individual portions illegible.]



Uncertain reading. The full line is cancelled. After "that" there may be "if." The words at right read "a sheaf is coher., then."

Original source details: GELFAND-P000110

From this autom. follows coher. of the sheaf $\forall$ aff. variety. $Y \subset X$, Y is closed, O_X is on X, J_Y — the sheaf of ideals ($= 0$ on Y).

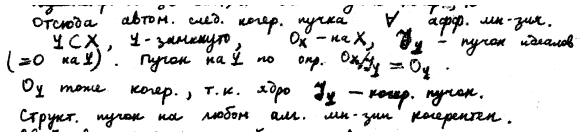
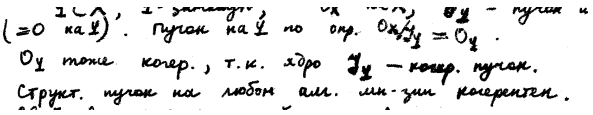


Diagram and typed description. A separate $\forall$ after coher. of the sheaf. On the next line $Y \subset X$, then Y is closed.

Original source details: GELFAND-P000110

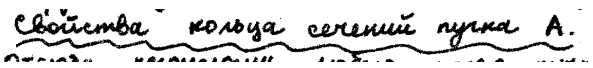
The sheaf on Y by def. $O_X/J_Y = O_Y$. O_Y is also coher., since the kernel J_Y is a coher. sheaf. The struct. sheaf on any alg. variety is coherent.



Original source details: GELFAND-P000110

Manuscript notation. The ideal has a hooked J, as in the line about the ideal below. The quotient $O_X/J_Y = O_Y$ uses a slash.

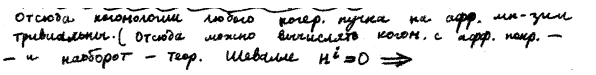
Properties of the ring of sections of the sheaf A.



Original source details: GELFAND-P000110

Diagram and typed description. The whole line Properties of the ring of sections of the sheaf A. has a wavy underline.

From this the cohomology of any coher. sheaf on an aff. variety is trivial. (From this one can compute cohom. using an aff. cover — — and conversely — Chevalley's theorem $H^i = 0 \Longrightarrow$

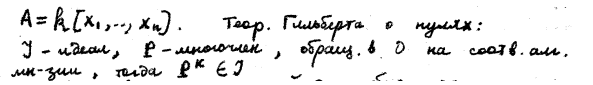


Original source details: GELFAND-P000110

Undeciphered passage. After $H^i = 0$ a double arrow pointing right; there is no continuation to its right. The parenthesis opened before the second From this is not closed.

$$A = k[x_1, \ldots, x_n].$$

Hilbert's theorem on zeros: J is an ideal, P is a polynomial vanishing on the corresp. alg. variety, then $P^k \in J$



Original source details: GELFAND-P000110

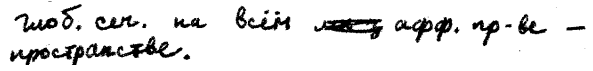
Diagram and typed description. A final period after the ring $A = k[x_1, \ldots, x_n]$. A colon after Hilbert's theorem on zeros. P has exponent k; no separate period after J.

Corollary. A is the ring of sections for any aff. variety.

Original source details: GELFAND-P000110

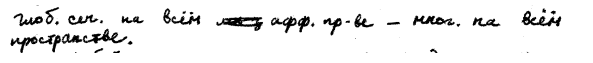
Diagram and typed description. A wavy underline beneath Corollary. Then A is the ring of sections for any aff. variety.

Glob. sect. on the whole [crossed out: illegible] aff. space — polyn. on the whole space.



Original source details: GELFAND-P000110

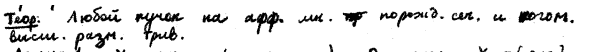
Uncertain reading. Between whole and aff., a short crossed record; its lower-right end forms a loop.



Original source details: GELFAND-P000110

Manuscript notation. After the dash the manuscript has the abbreviation «мног.» ("polyn."), as in the lemma below.

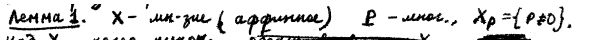
Theor. Any sheaf on an aff. variety [dense sign: illegible] is generated by sect. and its cohom. in higher dim. is triv.



Original source details: GELFAND-P000110

Uncertain reading. Between variety and is generated, a separate dense sign with a lower stroke. No word coher. is visible before sheaf in this line.

Lemma 1. X — a variety (affine), P — a polyn., $X_P = \{P \neq 0\}$.

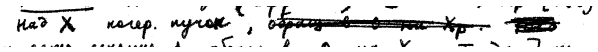


Original source details: GELFAND-P000110

Diagram and typed description. Lemma 1. has a straight underline. Affine is in parentheses. On the right $X_P = \{P \neq 0\}$.

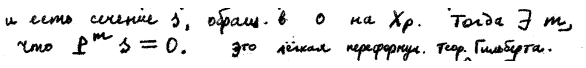
A coher. sheaf over X, [crossed out: vanishing on X_P[?].] [Separately crossed out on the right: illegible.]

and there is a section s vanishing on X_P. Then $\exists m$ such that $P^m s = 0$. This is an easy reformulation of Hilbert's theorem.



Original source details: GELFAND-P000110

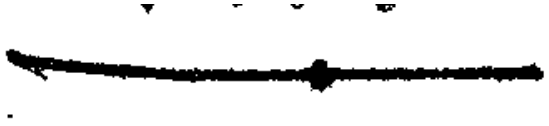
Uncertain reading. After sheaf, long crossing strokes; portions of vanishing, on X_P[?] are distinguished. Farther right a separate densely crossed record.



Original source details: GELFAND-P000110

Diagram and typed description. The uncanceled line and there is a section s vanishing on X_P. Then $\exists m$ such that $P^m s = 0$.

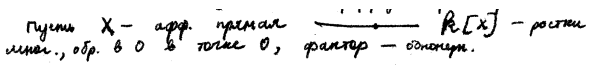
Let X be the aff. line



Original source details: GELFAND-P000110

Uncertain reading. A horizontal line with a marked point; no label at the point. At the left end a short downward slant: left arrowhead[?] / bend in the line. No arrowhead is visible on the right.

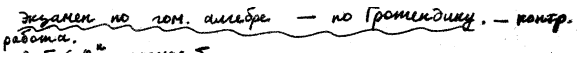
$k[x]$ — germs of polyn.s vanishing at the point 0, quotient — one-dim.



Original source details: GELFAND-P000110

Diagram and typed description. To the right of the drawing $k[x]$ — germs; on the next line of polyn.s vanishing at the point 0, quotient — one-dim.

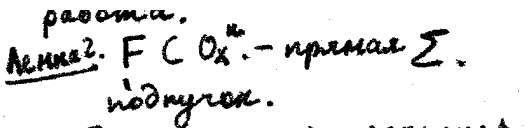
Examination in hom. algebra — following Grothendieck. — class test.



Original source details: GELFAND-P000110

Diagram and typed description. The wavy underline runs from Examination to Grothendieck. After it class, and test. is carried below without an underline.

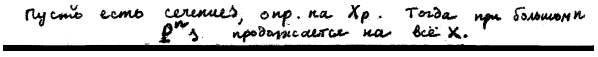
Lemma 2. $F \subset O_X^{n[?]}$ — direct $\sum$. [Below, with an insertion sign: subsheaf.]



Original source details: GELFAND-P000110

Uncertain reading. After direct, a separate $\sum$ / Σ with a period to the right. The superscript of O_X is n[?] / m-like; the small upper strokes merge. Subsheaf is inserted beneath F.

Let there be a section s defined on X_P. Then for large n, $P^n s$ extends to the whole X.



Original source details: GELFAND-P000110

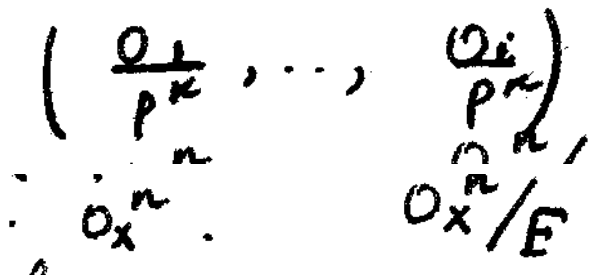
Diagram and typed description. The section s is defined on X_P. On the last line $P^n s$ extends to the whole X. A final period is visible.

GELFAND-P000111

(also Hilbert's theorem).

$$\left(\frac{O_1[?]}{p^k}, \ldots, \frac{O_i[?]}{p^k}\right)$$

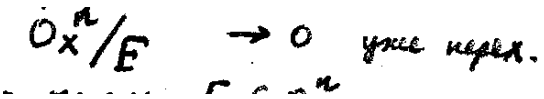
mult. by p^k, but this may not be a sect. of O_X^n.



Original source details: GELFAND-P000111

Uncertain reading. Rounded O-like signs in the numerators, with lower 1 and i; their lower portions join the fraction bars. Readings O / Q are not fully distinguished. Beneath, O_X^n and O_X^n/E.

$O_X^n/E \longrightarrow 0$ already perekh.[?]



Original source details: GELFAND-P000111

Uncertain reading. After "already," the word begins пе, has a descending p and ends in an x-like shape. The reading перех. (perekh.) remains tentative. The denominator has E with a lower stroke. The separate line below has $F \subset O_X^n$.

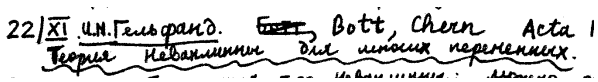
Now it already suffices for the proof that the cohom. of $F \subset O_X^n$ are zeros. X — an aff. variety, $(X_{p_1}, X_{p_2}, \ldots, X_{p_i})$ a cover. The cohomology of X with coefficients in F in this cover are zeros.

$22/\overline{XI}$. I.M. Gelfand. [Crossed out: an illegible entry.] Bott, Chern Acta Math., 1965, 114 $N = 1$–2.

Original source details: GELFAND-P000111

Uncertain reading. Before Bott, a densely canceled entry with a Cyrillic Б-like beginning; its former letters are not established. A separate comma follows the cancellation.

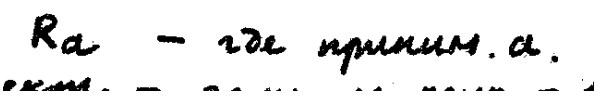
Nevanlinna theory for several variables.



Original source details: GELFAND-P000111

Diagram and typed description. XI has separate over- and underlines; the surname with initials has a straight underline. The heading Nevanlinna theory for several variables. has a long wavy underline.

Picard's theorem, the more precise theorem of Nevanlinna: one can estimate the number of zeros in a circle of radius R (may be less). R_a — where it takes a.



Original source details: GELFAND-P000111

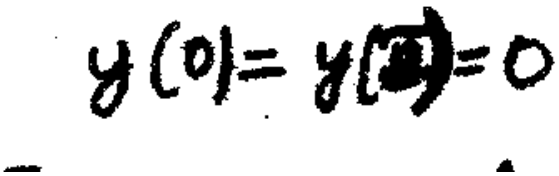
Diagram and typed description. Before the lower a, a large R-like letter with a slanting right leg. At right, a dash and where it takes a.

Most values are taken equ. often. Defect — if not taken, then 1; if less often, then a numer. char. The $\sum$ of defects does not exceed two. $\frac{a(z)}{b(z)}$ omits at most 2 values. $(a(z), b(z))$ a merom. fn. — a curve in C^2, a value — a line (only the ratio matters). This curve intersects all lines except, perhaps, two. Generalization. In 3-dim. space an entire curve $(a(z), b(z), c(z))$ intersects all planes passing through 0, except, perhaps, three. (In n-dim. a finite number.)
What is this for?

Original source details: GELFAND-P000111

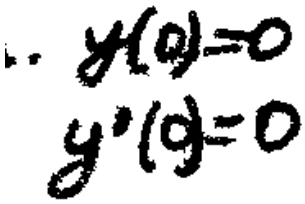
Manuscript notation. The wavy-underlined question reads «Для чего это?» (“What is this for?”).

$-y'' + p(x)y = \lambda y$ — an eqn. on the int. $[0, \pi]$ with conditions [At right:] $y(0) = 0$, $y'(0) = 0$. [Crossed out: an illegible short sign.] λ [ne sushch.; abbreviation unexpanded], $y(0) = y(\pi) = 0 \Rightarrow$ a countable number of eig. values.



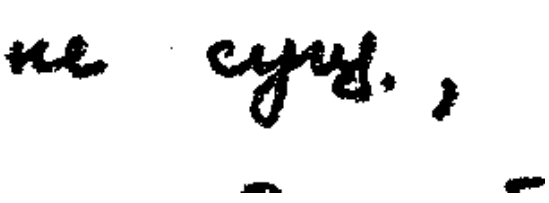
Original source details: GELFAND-P000111

Uncertain reading. The right endpoint of the interval and the argument in the second entry $y(0) = y(\pi) = 0$ are densely overwritten. The current sign has π-like strokes; the former entry is not established.



Original source details: GELFAND-P000111

Diagram and typed description. Two separate lines in the upper right field: $y(0) = 0$ and $y'(0) = 0$. Both final signs are oval, with a white centre.



Original source details: GELFAND-P000111

Uncertain reading. Written source не сущ. (ne sushch.), with a period and following comma. The expansion of this abbreviation is not established on this line.

Uncertain reading. Immediately before λ, a short dense cancellation with a loop on the left and horizontal strokes. The former sign is unread.

Original source details: GELFAND-P000111

Possibly, for most boundary conditions there are eigenvalues $y(0) = 0$, $ay'(\pi) + by(\pi) = 0$. Consider the solution

$$\begin{cases} y(0,\lambda) = 0 \\ y'(0,\lambda) = 1. \end{cases}$$

It is analytic in λ.

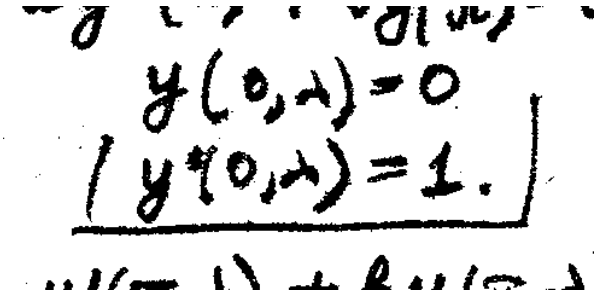
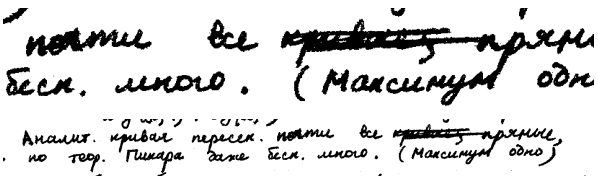


Diagram and typed description. Two lines $y(0,\lambda) = 0$, $y'(0,\lambda) = 1$.. Below them a horizontal line turns upward vertically at right. To the left of the second line a separate short curved stroke, unconnected to the lower line.

Original source details: GELFAND-P000111

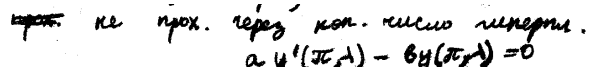
Eig. values — when $ay'(\pi,\lambda) + by(\pi,\lambda) = 0$ has a nonzero solution. An analytic curve intersects almost all [crossed out: an illegible word], lines, and by Picard's theorem even infinitely many. (At most one)



Uncertain reading. Before "all" the manuscript reads "almost." A long word after "all" is crossed out, and "lines" is written again to its right. The next line ends "(At most one)."

Original source details: GELFAND-P000111

How from general considerations do we obtain completeness (a curve in a function space) — this has not been done. One must prove a theorem in a Banach space — an analytic curve in a Banach space (possibly, false) [Crossed out: an illegible entry.] does not pass through a finite number of hyperplanes.



Uncertain reading. Before does not pass, a separate dense cancellation with a long downward stroke. The former letters are not separated.

Original source details: GELFAND-P000111

$$ay'(\pi,\lambda) - by(\pi,\lambda) = 0.$$

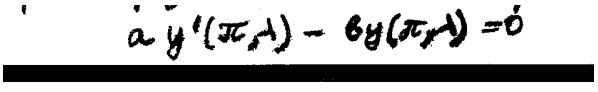
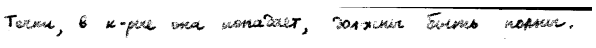


Diagram and typed description. In the last formula a separate horizontal minus stroke between ay' and by. The derivative stroke at y' is visible; both arguments are (π,λ).

Original source details: GELFAND-P000111

GELFAND-P000112

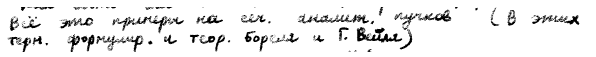
The points into which it falls must be poln.[?]



Original source details: GELFAND-P000112

Uncertain reading. The line ends poln.[?] / polny; the middle strokes are densely joined, with a separate dot at the right. The ending is not established.

What is to be done for several complex variables? All these are examples of sect. of analytic sheaves (in these terms both Borel's theorem and G. Weyl's theorem are formulated).

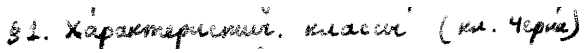


Original source details: GELFAND-P000112

Manuscript notation. Before "analytic," the abbreviation «сеч.» ("sect.") ends with a period.

There is no satisfactory theory analogous to Nevanlinna's theory. Distribution of the zeros of n functions of n variables — ?

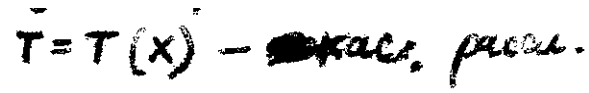
§ 1. Characteristic classes (Chern classes)



Original source details: GELFAND-P000112

Diagram and typed description. At the left are § and a separate 1 with a dot. No underline is visible below the heading. At the right, (Chern classes).

X — manifold, E — vector bundle, $T = T(X)$ — [Crossed out: unreadable entry.] tangent bundle.

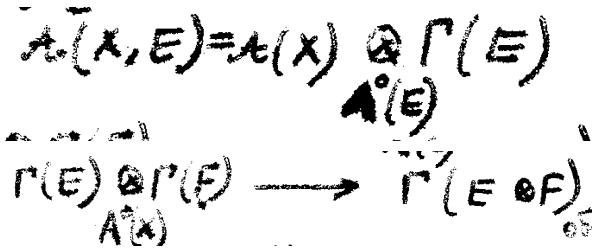


Original source details: GELFAND-P000112

Uncertain reading. Before the abbreviation for tangent a short prefix is densely crossed out; its individual earlier letters are not established.

$A^i(X)$ — sheaf of differential forms of order i. $\sum A^i(X) = A(X)$ — a ring. d — differential in $A(X)$, $A(X, E)$ — sheaf of differential forms on X with values in E.

$$A(X, E) = A(X) \otimes_{A^0(E)} \Gamma(E).$$



Original source details: GELFAND-P000112

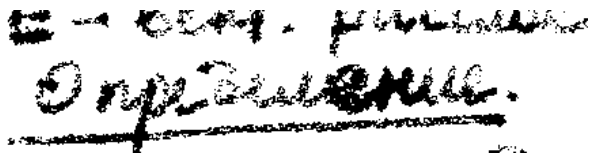
Manuscript notation. The first tensor sign has $A^0(E)$ below it, with a small rounded 0. The second has $A^0(X)$.

Notation.

$$\Gamma(E) \otimes_{A^0(X)} \Gamma(F) \longrightarrow \Gamma(E \otimes F)$$

— this multiplication is denoted simply by a dot. E — vector bundle over X. Definition. A connection on E is a differential operator D on $\Gamma(E)$ satisfying the requirement:

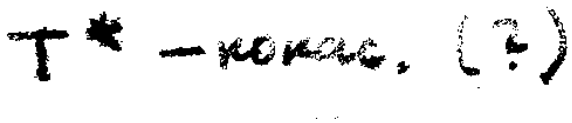
$$D : \Gamma(E) \longrightarrow \Gamma(T^* \otimes E), \qquad D(fs) = df \cdot s + f \cdot Ds, \quad s \in \Gamma(E),\ f \in A^0(X).$$



Original source details: GELFAND-P000112

Uncertain reading. In the underlined Definition. the middle letters are densely overwritten; the earlier strokes have not been read separately.

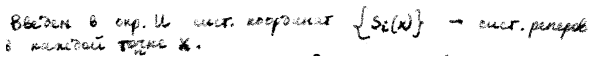
T^* — cotang. (?)



Original source details: GELFAND-P000112

Manuscript notation. The abbreviation «кокас.» (“cotang.”) is followed by a question mark in parentheses.

Introduce local coordinates $\{s_i(x)\}$ in a neighbourhood U — a syst. of frames at each point x.



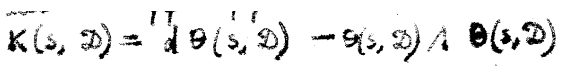
Original source details: GELFAND-P000112

Diagram and typed description. Before the brace the words local coordinates are written; after the brace, «сист. реперов» (“a syst. of frames”). The handwritten word koordinat has a long upper arc at d and a separate final stroke.

$$Ds_i = \sum_{j=1}^{n} \theta_{ij} s_j.$$

θ_{ij} — connection matrix $\theta(s, D)$. Its values — differential forms.

$$K(s, D) = d\theta(s, D) - \theta(s, D) \wedge \theta(s, D).$$

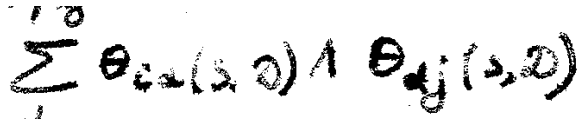


Original source details: GELFAND-P000112

Diagram and typed description. After the first $\theta(s, D)$ is a separate horizontal minus stroke; between the following matrices is $\wedge$.

Not 0, since this is a product of matrices

$$\sum_{l} \theta_{il}(s, D) \wedge \theta_{lj}(s, D).$$

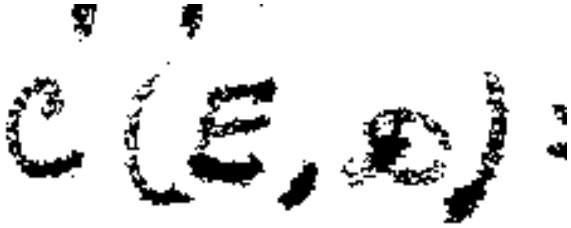


Original source details: GELFAND-P000112

Diagram and typed description. Below the sum is a separate l; the first θ has lower i, l, the second lower l, j. Both factors have arguments (s, D).

This exterior 2-form is called the curvature. 2-forms commute with one another, so a det of forms makes sense. A characteristic class is called

$$C(E, D[?]) = \text{Det}\left(1 + \frac{2\pi}{i} K(s, D)\right).$$



Uncertain reading. In $C(E, D[?])$ the second argument is pale and round, with a dense left stroke; D / a C-like letter.

Original source details: GELFAND-P000112

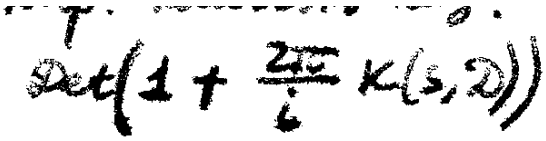


Diagram and typed description. In the fraction 2π is above the bar and i below; before the fraction is a plus, and at the right $K(s, D)$.

Original source details: GELFAND-P000112

Entries are 2-forms. [At the right:] Forms of different orders — their collection is a char. class.

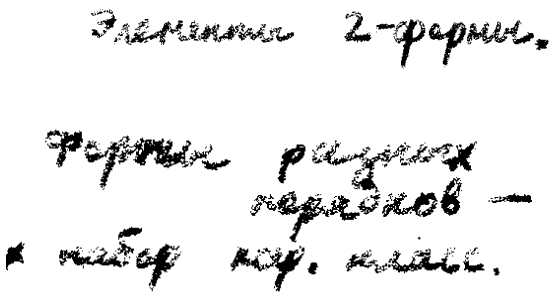


Diagram and typed description. The bottom line reads «Элементы 2-формы» (“Entries are 2-forms”). A separate entry in the right margin: Forms of different orders — their collection is a char. class.

Original source details: GELFAND-P000112

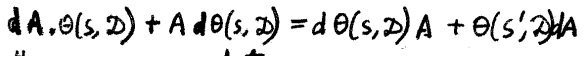
Prove that C does not depend on the choice of basis and is a closed form; in fact it does not depend on the choice of connection either. Independence of the choice of basis.

$$s'_i = \sum A_{ij} s_j, \qquad s' = As;$$
$$Ds' = dA \cdot s + A\theta(s, D)s,$$
$$Ds' = \theta(s', D)s',$$
$$dA + A\theta(s, D) = \theta(s', D)A.$$

Substitute this into the definition of K.

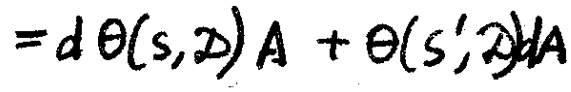
$$\underline{AK(s, D) = K(s', D)A}$$

$$dA \cdot \theta(s, D) + Ad\theta(s, D) = d\theta(s, D)A + \theta(s', D)dA.$$



Original source details: GELFAND-P000113

Uncertain reading. In the first $d\theta$ on the right the argument is s without a separate prime. In the next θ the argument is s' with a prime.



Original source details: GELFAND-P000113

Diagram and typed description. On the right between $d\theta(s, D)A$ and $\theta(s', D)dA$ is a separate plus with intersecting horizontal and vertical strokes.

"$A^{-1}KA$" the same det.

Original source details: GELFAND-P000113

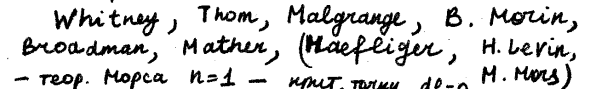
Manuscript notation. The two short strokes after the last A form the raised closing quotation mark. Two low strokes before the first A form the opening mark.

V. I. Arnold. Progress in the theory of singularities of differentiable mappings.
$f : M^m \longrightarrow N^n$ — 2 manifolds.

Original source details: GELFAND-P000113

Uncertain reading. In the target N^n the capital letter is densely overwritten; the individual earlier strokes are unread.

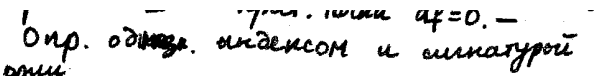
$R^m \longrightarrow R^n$, $0 \longrightarrow 0$. Whitney, Thom, Malgrange, B. Morin, Broadman, Mather, (Haefliger, H. Levin, M. Mors).



Original source details: GELFAND-P000113

Diagram and typed description. The Latin list has Broadman, H. Levin and M. Mors; before Morin is a B-like capital.

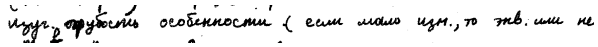
The simplest case — Morse's theorem $n = 1$ — critical points $df = 0$. ← isolated for general position. Determined by [Crossed out: an od...-like entry, continuation unreadable.] index and signature. (Morse lemma) — a model for the theory.



Original source details: GELFAND-P000113

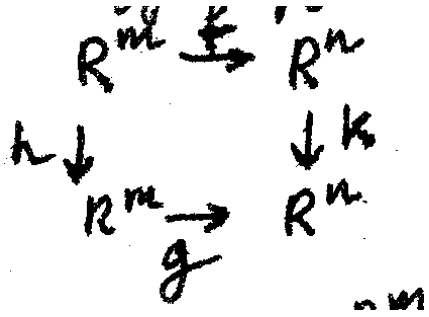
Uncertain reading. After the abbreviation for determined a short token with an od...-like beginning is cancelled. The middle and ending are densely covered by strokes.

Study the roughness of singularities (if changed slightly, equivalent or not).



Original source details: GELFAND-P000113

Uncertain reading. The word reads «грубость» ("roughness"). Its initial strokes are heavy and the earlier strokes remain inseparable.



Original source details: GELFAND-P000113

Diagram and typed description. The upper and lower rows are $R^m \to R^n$, with f above and g below. Vertical arrows point down: h on the left, k on the right.

$$\begin{array}{ccc} R^m & \xrightarrow{\ f\ } & R^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ R^m & \xrightarrow[\ g\ ]{} & R^n \end{array}$$

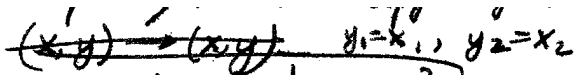
f, g are equivalent if obtained by a change of variables.

$d^2 f$ without zeros — reducible to a quadratic form. For a comp. source manifold they form an everywhere dense open set.

Original source details: GELFAND-P000113

Manuscript notation. After "for" the manuscript abbreviates compact, and after "everywhere" it abbreviates dense. The final words are "open set."

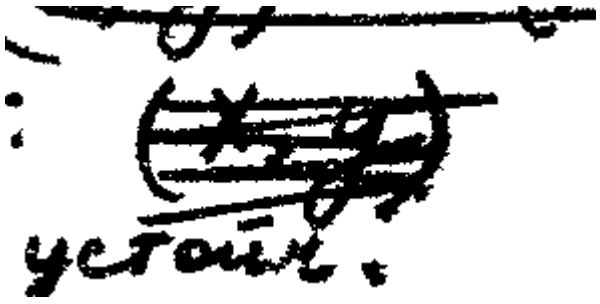
$R^m \longrightarrow R^1$ is solved. Whitney's first result (55) $R^2 \longrightarrow R^2$. An example of singularities. Singular points — rank of the differential $= 1$ and they are of 2 classes (folds and cusps). They are structurally stable (0 can be avoided). Ordinary point ~~$(x, y) \longrightarrow (x, y)$~~ $y_1 = x_1$, $y_2 = x_2$.



Original source details: GELFAND-P000113

Uncertain reading. The entry $(x, y) \longrightarrow (x, y)$ is crossed out by two long horizontal lines. At the right, separately $y_1 = x_1$, $y_2 = x_2$.

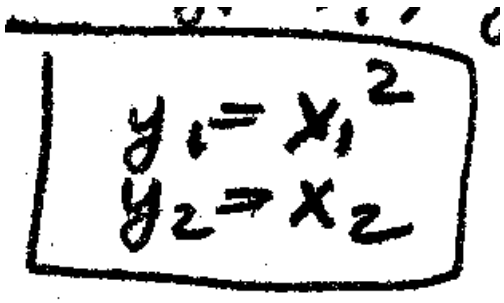
Fold point: [Crossed out: (x, y)[?], other strokes unread.] structurally stable.



Original source details: GELFAND-P000113

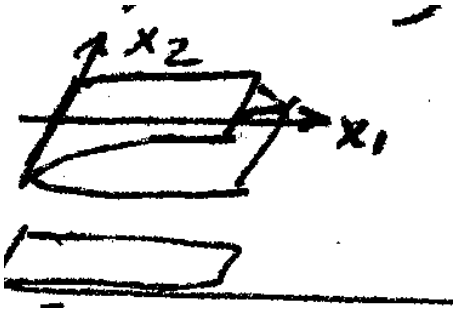
Uncertain reading. Below the heading an (x, y)-like entry is densely crossed out; below the lower slanting strokes the individual letters are not separated.

$$\boxed{\begin{aligned} y_1 &= x_1^2, \\ y_2 &\to x_2. \end{aligned}}$$



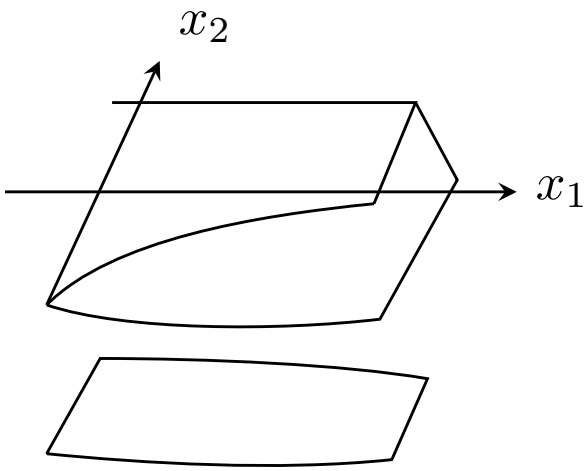
Original source details: GELFAND-P000113

Diagram and typed description. Two lines in one frame. The first is $y_1 = x_1^2$. In the second, between y_2 and x_2 is an arrow-like sign, a horizontal stroke with a point at the right.



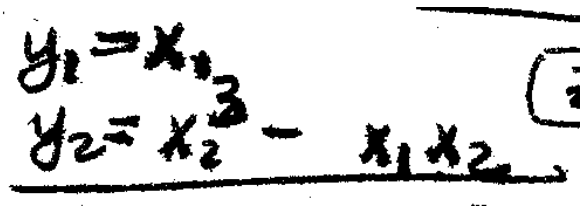
Original source details: GELFAND-P000113

Diagram and typed description. Two separate oblique strips. The upper one has a curved front edge and an internal line; the x_1 axis points right, x_2 obliquely upward. The lower strip has no labels.



Cusp point:

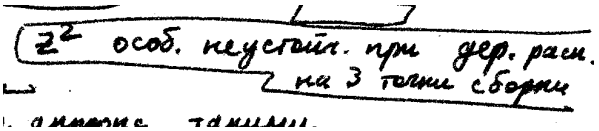
$$\begin{aligned} y_1 &\to x_1, \\ y_2 &= x_2^3 - x_1 x_2. \end{aligned}$$



Original source details: GELFAND-P000113

Manuscript notation. The first line has a right-pointing mark like the one in the second line of the fold formula. The second line reads $y_2 = x_2^3 - x_1 x_2$.

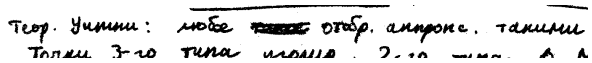
z^2 singularity unstable; under deformation splits into 3 cusp points.



Original source details: GELFAND-P000113

Diagram and typed description. At the right an entry is framed, open at the right edge, with a step in the lower outline. The text begins with z^2, and the next line has into 3 cusp points.

Whitney's theorem: any [Crossed out: unreadable entry.] mapping is approximated by such ones.



Original source details: GELFAND-P000113

Uncertain reading. Between any and the abbreviation for mapping is a short dense cancellation. The earlier letters are not established.

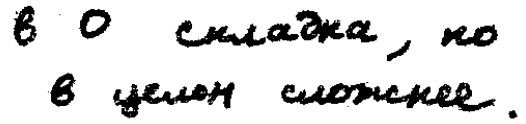
Points of the 3rd type are isolated, of the 2nd type in M^m a nonsingular 1-dimensional manifold; in N it has singularities.

GELFAND-P000114

Cusp. Jacobian

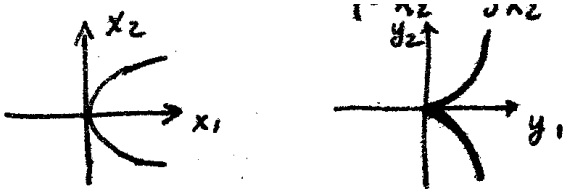
$$\begin{vmatrix} 1 & 0 \\ -x_2 & 3x_2^2 - x_1 \end{vmatrix}.$$

$3x_2^2 - x_1 = 0$ a parabola. At 0 a fold, but more complicated as a whole.



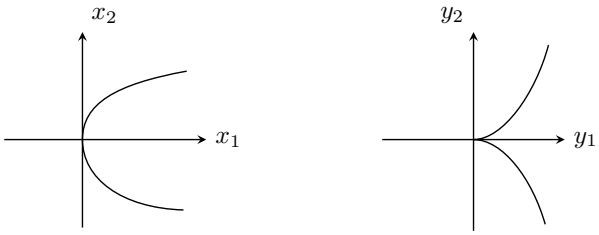
Original source details: GELFAND-P000114

Diagram and typed description. At the right are separate lines At 0 a fold, but / more complicated as a whole.



Original source details: GELFAND-P000114

Diagram and typed description. At the left a parabola with vertex at the origin, x_1 right,x_2 up. At the right two branches with a common point,directed right; y_1 right,y_2 up.



Where does the parabola go? [Crossed out: $y_2 = [?]y_1^{3/2}$.]

Original source details: GELFAND-P000114

Uncertain reading. In the cancelled entry y_2,an equals sign and $y_1^{3/2}$ are visible. The coefficient and intervening strokes are densely covered.

$y_2 = -Cy_1^{3/2}$.

$$\boxed{y_2^2 = -y_1^3}$$

$y_1 > 0$.

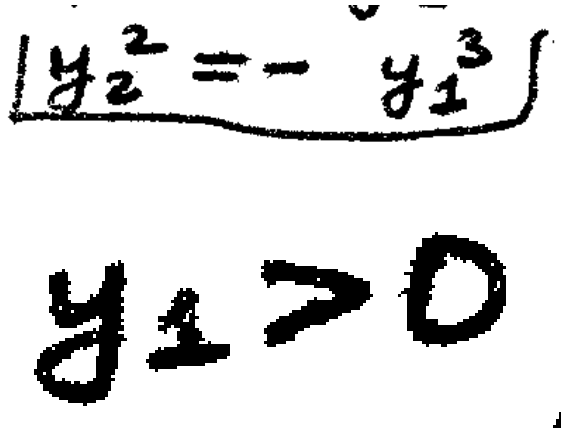


Diagram and typed description. The corner frame has $y_2^2 = -y_1^3$,with a separate minus. Below and to the left is a separate $y_1 > 0$.

Original source details: GELFAND-P000114

The mapping is not a fold on the whole parabola. Away from the point 0 a fold. Consider lines parallel to the x_2 axis. $x_1 = \text{const} \Rightarrow y_1 = \text{const}$. $y_2 = x_2^3 - Cx_2$.

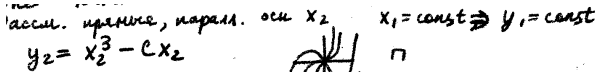
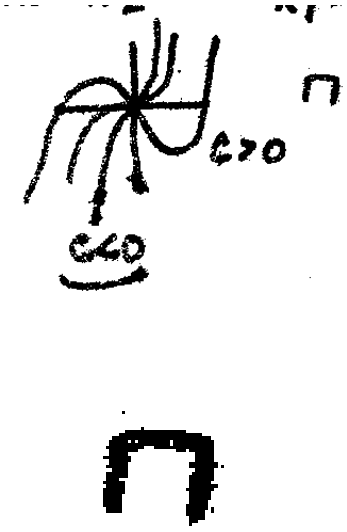
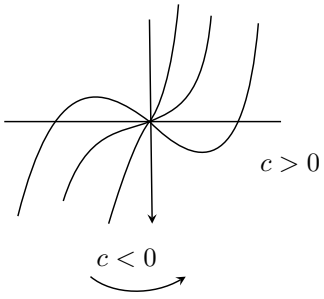


Diagram and typed description. In the line $x_1 = \text{const}$ an arrow points to $y_1 = \text{const}$. Below is $y_2 = x_2^3 - Cx_2$,with a large C-like letter before x_2.

Original source details: GELFAND-P000114



Uncertain reading. Several curves pass through the common intersection of the axes. At the right,next to the curve with turns,is $c > 0$; below,$c < 0$ and a curved arrow. At the right is a separate Cyrillic-P-/ П-like mark,meaning unestablished.



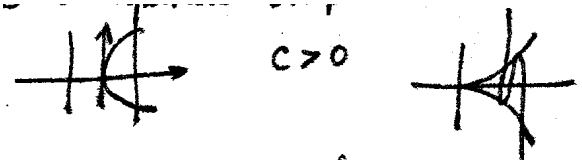
П[?]

Original source details: GELFAND-P000114

$c < 0$ the lines are mapped homeom. $c > 0$.

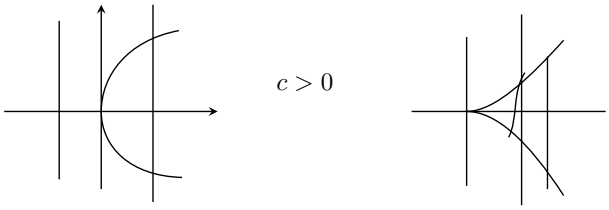
Manuscript notation. After “mapped” the manuscript has «гомеом.» (“homeom.”), followed by a separate $c > 0$ at right.

Original source details: GELFAND-P000114

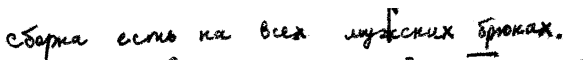


Original source details: GELFAND-P000114

Diagram and typed description. At the left a parabola,axes and two separate verticals. At the right a pointed two-branch curve with several vertical strokes and a short curved internal line. Between the sketches is $c > 0$.



There is a cusp in every pair of men's trousers.



Original source details: GELFAND-P000114

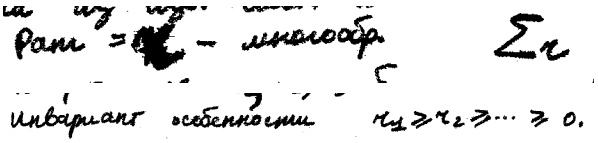
Manuscript notation. The manuscript reads «на всех мужских брюках» ("in every pair of men's trousers"). A vertical line of the drawing crosses the middle of «мужских».

Problem. [Crossed out: z^2[?] and unreadable strokes.] $z \longrightarrow z^2 + \varepsilon\overline{z}$ — general position. $\varepsilon = 0$ singularity at 0; for $\varepsilon \neq 0$ it moves, there will be folds and cusps. And only this is stable.

Original source details: GELFAND-P000114

Uncertain reading. After Problem. a z-like sign with an upper 2 is cancelled; the right part of the entry is covered by long strokes.

Scientific definition of a cusp. One of the known systems of invariants. Rank $= r$[?] — manifold Σ_r.



Original source details: GELFAND-P000114

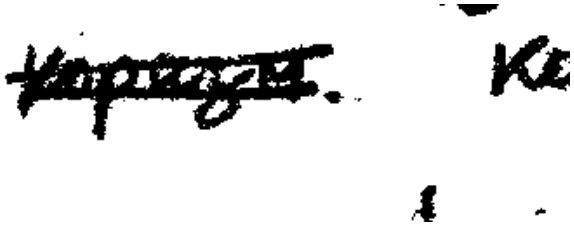
Uncertain reading. After the equals sign the letter is densely overwritten; currently r[?],with earlier round strokes unread separately. Below,similar signs occur at Σ_r and in the sequence r_1, r_2.

M^m is partitioned into manifolds. ~~Codim.~~

Codim. sing. $=$ product of coranks $(m-r)\cdot(n-r)$

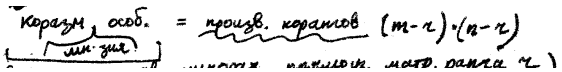
manifold

(This is a question about independent minors of a rectangular matrix of rank r.)



Original source details: GELFAND-P000114

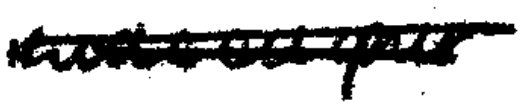
Uncertain reading. The first Codim. is crossed out by long horizontal lines. Below,a separate repetition Codim. sing.



Original source details: GELFAND-P000114

Diagram and typed description. Below sing. is a brace pointing up with the label manifold; below that,a long corner frame. Product of coranks has a wavy underline; at the right,$(m-r)\cdot(n-r)$.

Σ_r — smooth manifolds in R^m; consider our mapping on the manifolds themselves (parabola). Rank on these manifolds. Σ_{r_1,r_2}, $r_2 < r_1$ — submanifold — rank levels. Σ_{r_1,r_2,r_3}. To each [Crossed out: unreadable word.] singularity corresponds a set of numbers r_1, r_2, r_3.



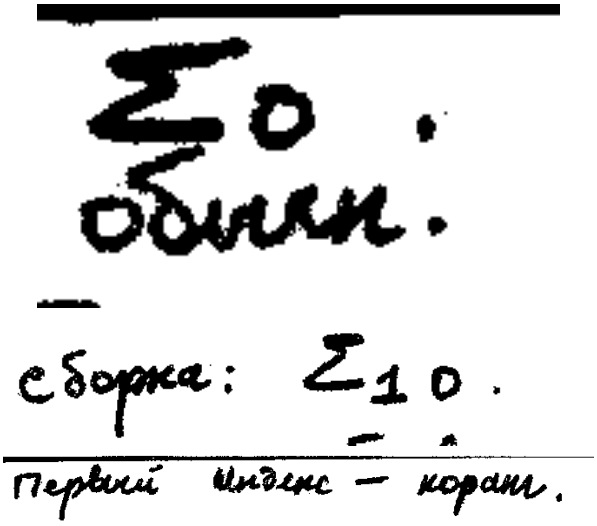
Original source details: GELFAND-P000114

Uncertain reading. After to each a long word,resembling osobennosti,is cancelled. Upper and lower horizontal strokes cover most letters.

Invariant of a singularity $r_1 \geq r_2 \geq \cdots \geq 0$.

GELFAND-P000115

Whitney sing.: $\underset{\text{ordinary}}{\Sigma_0}$. The first index is the corank. Cusp: Σ_{10}. It is proved that this is a manifold.



Original source details: GELFAND-P000115

Diagram and typed description. For Whitney,Σ_0,with ordinary below it; for the cusp,Σ_{10}. At the right,The first index is the corank.

[Crossed out: D...]

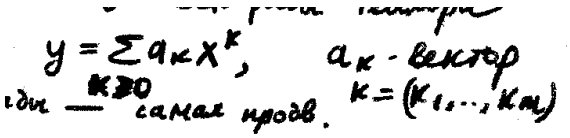
For the partition in the space of Taylor-series segments. (Brodman) — the old naive approach.

Original source details: GELFAND-P000115

Uncertain reading. Before for the partition,a short cancelled entry; a Cyrillic-D-like beginning is visible,the other letters covered by strokes.

It is known that Σ is not a complete classification. (There is an example $R^9 \longrightarrow R^9$.) General theory. Consider the space of Taylor series. (3 classes — C^∞, analytic, formal power series.) [At the right:] $R^m \longrightarrow R^n$; C^∞ has Taylor series.

$$y = \sum_{\substack{[?]\\ k \geq 0}} a_k x^k, \qquad a_k \text{ — a vector}, \quad k = (k_1, \ldots, k_m).$$



Uncertain reading. Below the sum,$k \geq 0$[?] / $k > 0$; the small strokes of the sign are joined. At the right,a_k is a vector,$k = (k_1, \ldots, k_m)$.

Original source details: GELFAND-P000115

Local theories. C^∞ — the most difficult, analytic — the most interesting, formal series — the simplest. Mappings are equivalent if there is a change. A partition into pieces arises — a stratification. Study of this partition.

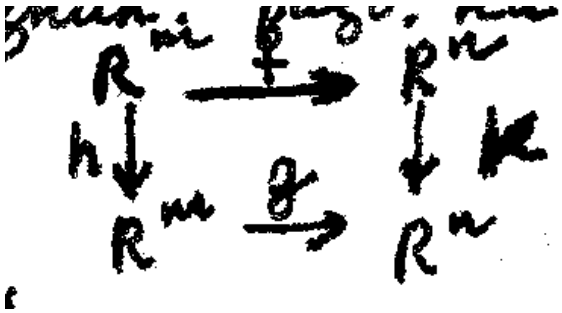


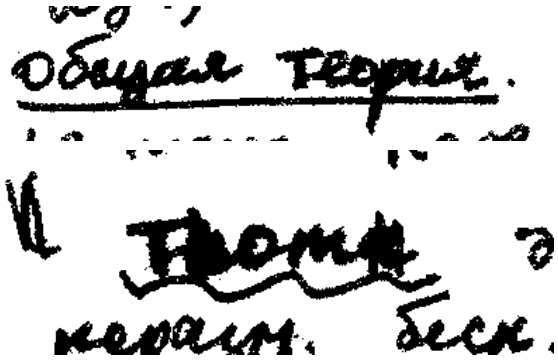
Diagram and typed description. Two rows $R^m \to R^n$: above,f; below,g; vertical arrows down,h on the left,k on the right.

$$\begin{array}{ccc} R^m & \xrightarrow{\ f\ } & R^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ R^m & \xrightarrow{\ g\ } & R^n \end{array}$$

Original source details: GELFAND-P000115

h, k — formal diffeomorphisms (1st terms with Jacobian $\neq 0$).

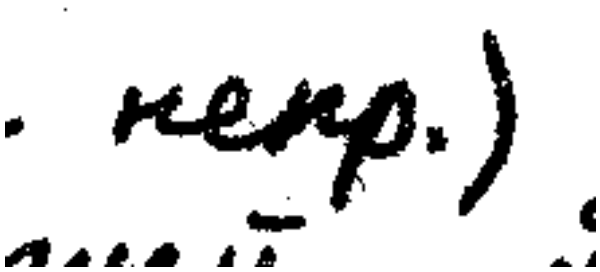
There is an example (> 9) with infinitely many invariants. ↓ Thom proved that in the space of power series a part of infinite codimension is singled out, that the rest are stable in the topological sense.



Uncertain reading. General theory. has a straight underline,with middle letters overwritten. Thom has a wavy underline; at the left,a short arrow down. Earlier strokes of the name cannot be read separately.

Original source details: GELFAND-P000115

(Thom's invariants in R^9 are invariants of smooth,but not cont.)



Manuscript notation. The final abbreviation is «непр.» ("cont."), with a descending p and a final period.

Original source details: GELFAND-P000115

(Thom does not want to publish the proof,he is working on biology. The proof was recounted,for a whole year,but nobody understood it: Malgrange, Cartan, Arnold and others.) $\bigcup X_\gamma + X_\infty$, X_γ singularities stable in the topological sense.

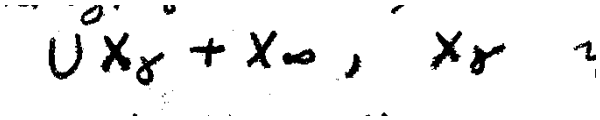


Diagram and typed description. Before X_γ is a large union sign without a lower index; between X_γ and X_∞ is a separate plus.

Original source details: GELFAND-P000115

For some dimensions Thom's analytic difficulties do not arise. (Singularities differentially stable.) Apparently the problem is solved where there is [Crossed out: unreadable word.] a classification. (In which dimensions.)

Uncertain reading. Between there is and classification,a long cancelled entry covered by several horizontal strokes.

Original source details: GELFAND-P000115

There are Σ that are differentially stable $\Sigma_{1,\dots,1,0}$. Not all Σ are differentially stable. Another result of Mother. (Milnor's student.)

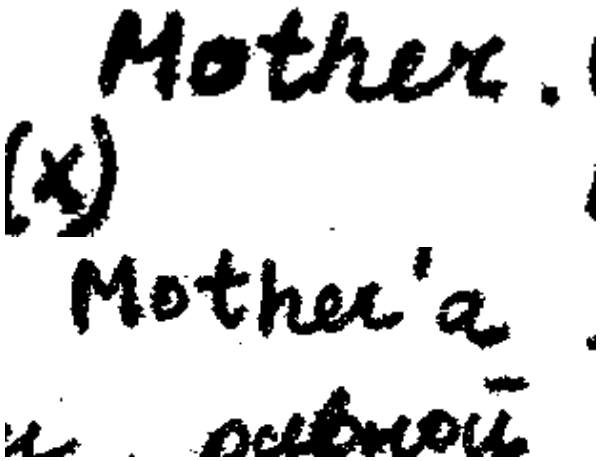
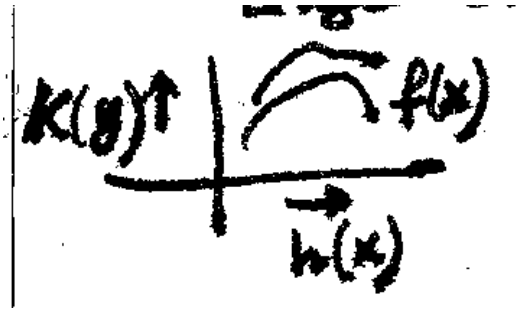


Diagram and typed description. The name has a round o: Mother. In the lower repetition,Mother'a.

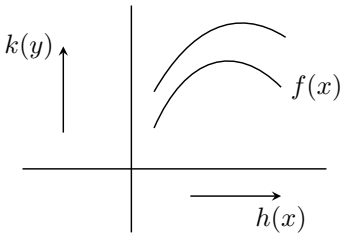
Original source details: GELFAND-P000115

$\delta y = \varphi(x)$. $h(x)$ — variation in X — a vector field.



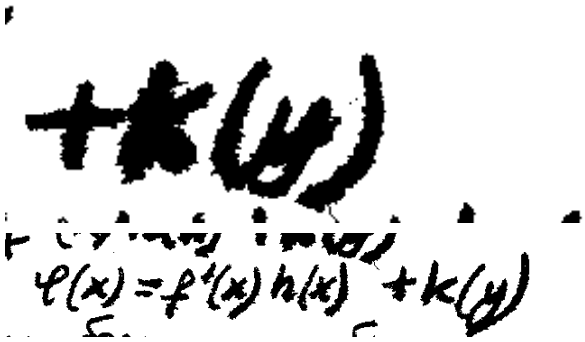
Original source details: GELFAND-P000115

Diagram and typed description. Crossing axes and two nearby arcs above the horizontal,with $f(x)$ at the lower arc. A separate horizontal arrow right below the axis is labelled $h(x)$,and a vertical arrow up at the left,$k(y)$.



Change of variables — the function changes by $f'(x)h(x) + k(y)$. In the 1st approximation one must solve the equation

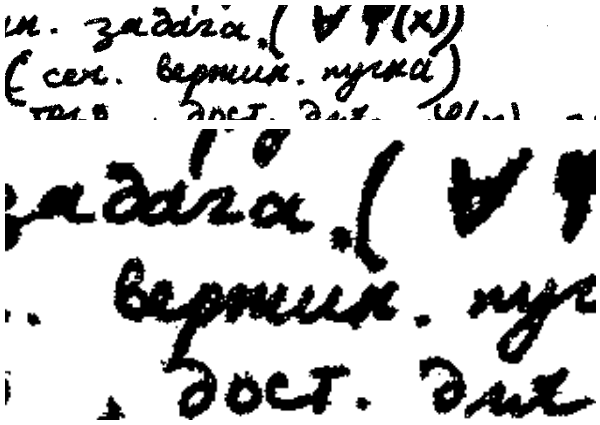
$$\varphi(x) = f'(x)h(x) + k(y).$$



Original source details: GELFAND-P000115

Uncertain reading. In the first formula,$k(y)$ is densely overwritten; below,in a separate equation,is a clean $k(y)$. The earlier strokes of the first letter are unread.

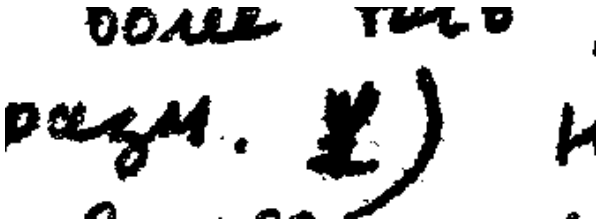
The corresponding theorem says that for stability it is necessary and sufficient that the corresponding linear problem be solvable. ($\forall\varphi(x)$) (sect. of the vershin.[?] sheaf)



Original source details: GELFAND-P000115

Uncertain reading. The first abbreviation is «сеч.» (“sect.”), and the last word is «пучка» (“of the sheaf”). The middle abbreviation tentatively reads «вершин.» (vershin.).

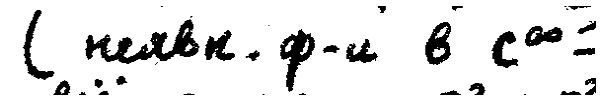
(This is Mother's 1st theorem. Moreover,it suffices for $\varphi(x)$ given by a polynomial of degree equal to dim. ℓ[?].)



Original source details: GELFAND-P000115

Uncertain reading. After dim.,a densely overwritten sign,ℓ- / N-like; earlier strokes are covered.

But the proof is incorrect. Modern technique (implicit functions in C^∞ spaces).



Original source details: GELFAND-P000115

Manuscript notation. The parenthesis abbreviates «неявн. ф-и» ("implicit functions").

All dimensions for which everything is good have been found. $R^2 \longrightarrow R^2, R^3$ (Whitney); in R^4 unknown, in R^5 already known (there are no singularities at all any more). Immersion (immersion). For $m = n$ up to 8 all mappings are stable (approx.).

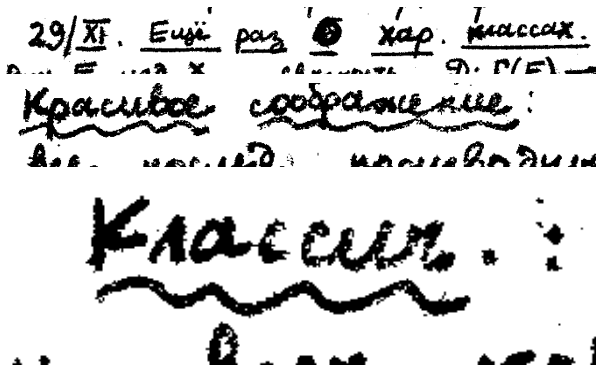
GELFAND-P000116

Thom — a continuous invariant (stable). It is not topological — it is an invariant of smooth mappings. Under continuous deformation one can deform them. 29/XI. Once again about characteristic classes. [Crossed out: unreadable word.]

(I. M. Gelfand.)

Original source details: GELFAND-P000116

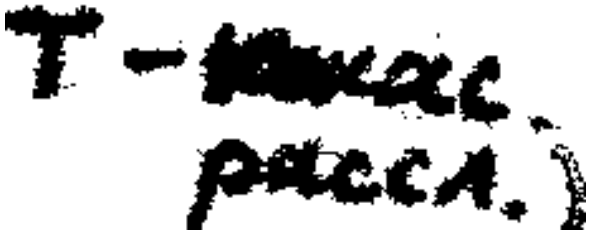
Uncertain reading. After classes.,a long densely cancelled entry; the letters are covered by horizontal strokes.



Original source details: GELFAND-P000116

Manuscript notation. The round «о» ("about") is separately underlined. Two headings below have wavy underlines.

Consider E over X,a connection $D : \Gamma(E) \to \Gamma(T^* \otimes E)$. f (function) $\in A^0$, $s \in \Gamma(E) \Rightarrow$ $D(fs) = f \cdot Ds + df \cdot s$. [At the right:] (Here T is the tang.[?] bundle.)

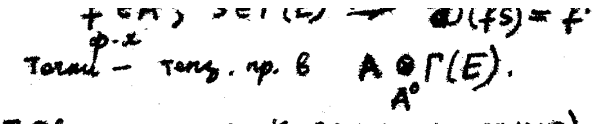


Uncertain reading. Before and at the beginning of kas.,a dense correction; the earlier strokes cannot be read separately.

Original source details: GELFAND-P000116

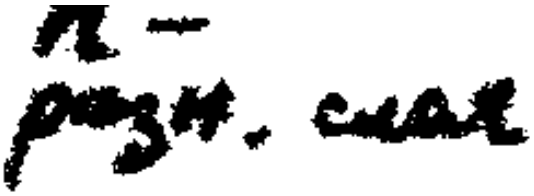
[In the margin:] ② More precisely — tensor product in $A \otimes_{A^0} \Gamma(E)$.

Diagram and typed description. Below f is the label function,joined to the letter by a short stroke. In the tensor product,A^0 is below $\otimes$. In the right margin,2 is circled.



Original source details: GELFAND-P000116

Cartan's moving frame (locally): s_i $(i = 1, \ldots, n)$, n — dim. of the fibre.
Linearly independent at $\forall$ point.



Manuscript notation. At right, «разм. слоя» ("dim. of the fibre"), with a descending p in the abbreviation.

Original source details: GELFAND-P000116

$D(s_i) = \sum \theta_{ij} s_j$,coefficients from A^1 (forms). $\|\theta_{ij}\|$ is called the connection matrix (in Cartan — ω).

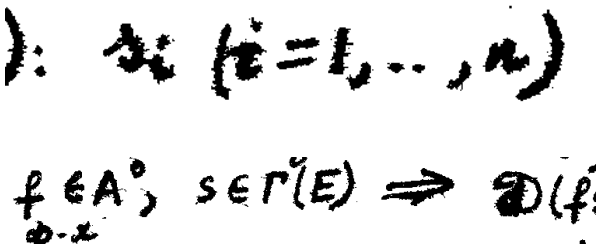


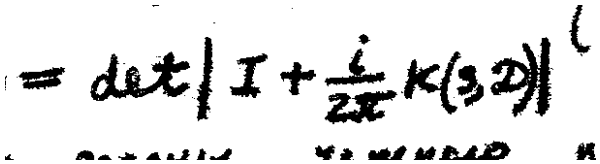
Diagram and typed description. For sections an s-like letter recurs: $s \in \Gamma(E)$,s_i,s_j; below,$s = (s_1, \ldots, s_n)$. The initial D in $D(fs)$ is overwritten.

Original source details: GELFAND-P000116

$$\|K_{ij}\| = K(s,D) = d\theta(s,D) - \theta(s,D) \wedge \theta(s,D), \qquad s = (s_1, \dots, s_n).$$

(This is a matrix.) (Matrix product,but the product is exterior.) The form

$$C(E,D) = \det \left| I + \frac{i}{2\pi} K(s,D) \right|.$$



Original source details: GELFAND-P000116

Diagram and typed description. In the determinant,before $K(s,D)$,a fraction with i in the numerator and 2π in the denominator.

The entries are forms of even order,they commute and one can take det. A $\sum$ of forms of different orders. These are the characteristic classes associated with the given bundle. This does not depend on the choice of connection. [Idea of the proof according to Gelfand: closedness has to be proved,θ[?] a closed form no longer depends on the connection,$\int$ can be expressed through $\int$ over the boundary. A variation of the connection inside has no effect.]

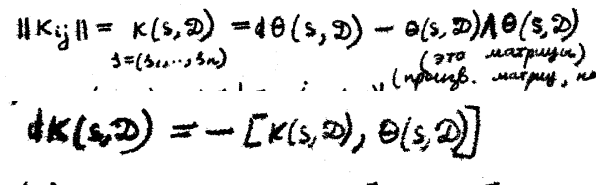
Original source details: GELFAND-P000116

Uncertain reading. Before “closed,” a dense round sign has a heavy lower stroke. Its reading remains undetermined.

It does not depend on the choice of frame. Bianchi identity:

$$dK(s,D) = -[K(s,D), \theta(s,D)].$$

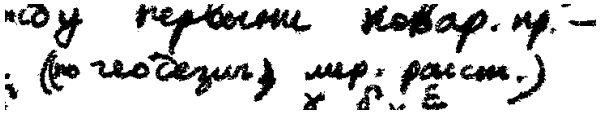
(Derivative of the curvature tensor.)



Original source details: GELFAND-P000116

Diagram and typed description. The curvature formula has a minus before $\theta \wedge \theta$. The Bianchi identity has a minus before the commutator; the order inside the brackets is $K(s,D), \theta(s,D)$.

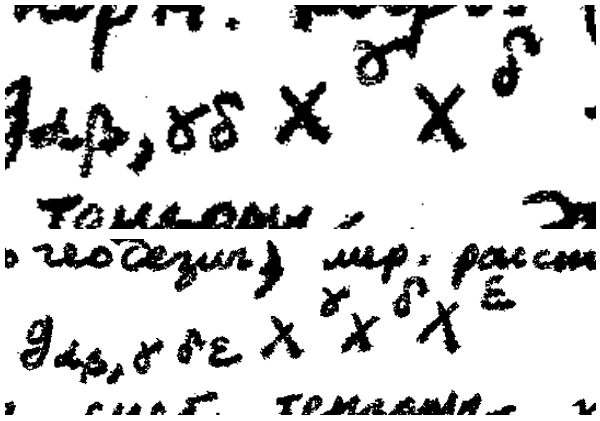
Beautiful observation. $g_{\alpha\beta}(x)$ — a Riemannian metric,we shall take all successive derivatives of $g_{\alpha\beta}(x)$ and form tensors from them. How does one find all tensors that can be formed from $g_{\alpha\beta}(x)$? Classification: they are functions of the curvature tensor and all covariant derivatives of the curvature tensor (of any order) (algebraic expressions in them). But what algebraic relations? (Between first covariant derivatives — the Bianchi identity.) Introduce normal coordinates (along geodesics, meas.[?] dist.).



Original source details: GELFAND-P000116

Uncertain reading. After “norm. coord.” the parentheses contain «по геодезич.» (“along geodesics”) and «мер.[?] расст.» (“meas.[?] dist.”). The first abbreviation after “geodesics” remains tentative.

$$g_{\alpha\beta}(x) = g_{\alpha\beta}(0) + g_{\alpha\beta,\gamma\delta}x^{\gamma}x^{\delta} + g_{\alpha\beta,\gamma\delta\varepsilon}x^{\gamma}x^{\delta}x^{\varepsilon}.$$



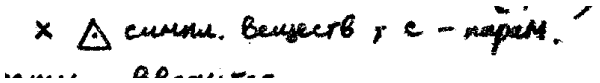
Original source details: GELFAND-P000116

Manuscript notation. After the comma the second term has $\gamma\delta$ and the third $\gamma\delta\varepsilon$. The same signs occur above the corresponding x factors.

In a normal system they are all tensors. This system of tensors,and here the algebraic relations are simpler.

GELFAND-P000117

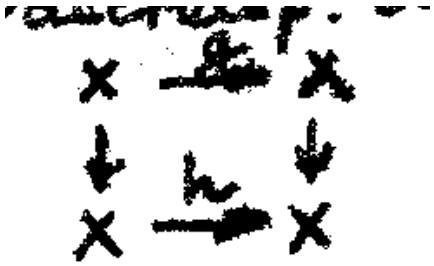
V. I. Arnold. Stable flow. $\dot{x} = f(x, C)$; g^t — a group of diffeomorphisms. There are chemical and biological models (Thom). x △ sums of substances,c — a parameter.



Original source details: GELFAND-P000117

Diagram and typed description. Between x and sums of substances,a triangle with a dot inside. The right parameter label uses lowercase c,the equation uppercase C.

Consider one (stable). A number of kogrubost is introduced.



Original source details: GELFAND-P000117

Uncertain reading. Four X's; above the upper horizontal arrow,an overwritten g^t[?],above the lower,an h-like letter. Horizontals right,verticals down,with no labels at the verticals.

$$\begin{array}{ccc} X & \xrightarrow{\;g^t[?]\;} & X \\ \downarrow & & \downarrow \\ X & \xrightarrow{\;h[?]\;} & X \end{array}$$

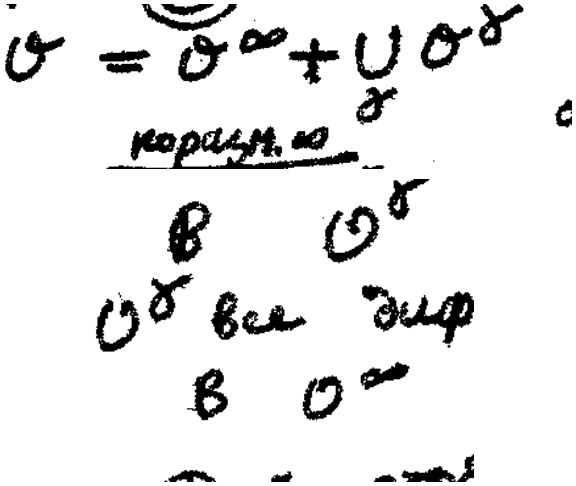
Stable if for $\forall$ nearby h there is a perturbation of X. One cannot consider only stable ones (when parameters change,one may pass through nonstable parameters). (x, C) — a point of kogrubost 1,if for nearby parameter values stable,etc. As a mapping $X \times C \longrightarrow X \times C$,stable. (How many terms of the Taylor series must be taken.) A centre — an infinite degree of nonstability.

Original source details: GELFAND-P000117

Diagram and typed description. Next to the entry about a centre,three nested closed round lines. They have no labels.

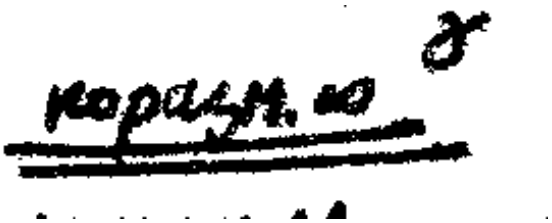
Thom's theorem on germs of Taylor series $\vartheta = \vartheta^{\infty} + \bigcup_{\gamma} \vartheta^{\gamma}$.

[Below the formula:] codim. ∞



Original source details: GELFAND-P000117

Uncertain reading. The recurring letter is ϑ[?] / θ / O-like,with a large upper loop; the superscripts are ∞ and gamma-like.



Original source details: GELFAND-P000117

Diagram and typed description. Below the formula,a separate annotation codim. ∞,underlined twice.

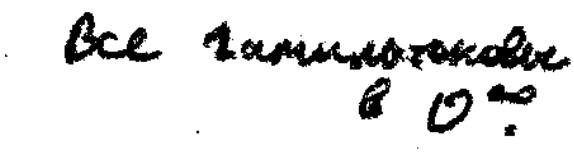
In ϑ^γ,an open subset of finite negrub.[?] In ϑ^γ singularities are described by finitely many parameters,inside [Crossed out: unreadable short token.]

ϑ^γ,all are diffeomorphic. (Tr-ie for differential equations.)

Uncertain reading. After inside,a short dense cancellation; separately at its left,another short horizontal stroke.

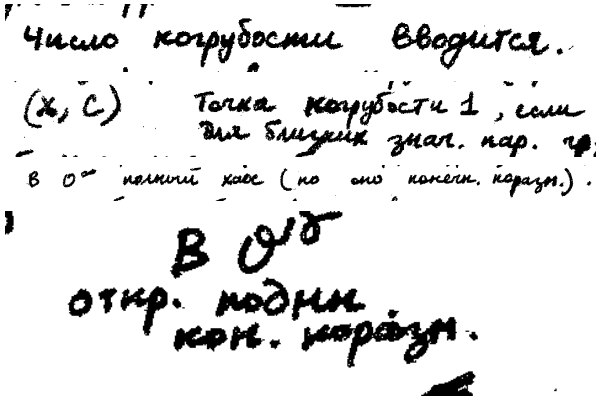
Original source details: GELFAND-P000117

In ϑ^∞,complete chaos (but they are of finite codim.[?]). All Hamiltonian ones are in ϑ^∞.



Manuscript notation. At the far right, “All Hamiltonian ones.”

Original source details: GELFAND-P000117



Uncertain reading. The first lines twice have «когрубости» (kogrubosti). The abbreviations in the right-hand note and lower parentheses remain uncertain.

Original source details: GELFAND-P000117

For self-mappings,the invariants are unknown. (F. Bendixson’s theorem: $\frac{dy}{dx} = \frac{P}{Q}$ on CP^2,then by finitely many σ-processes one reaches Bendixson singular points $x^p \frac{dy}{dx} = \lambda y + P(x,y)$. Already an invariant relative to σ-processes. A σ-process — substitute $\frac{y[?]}{x}$.

For 3 variables unknown. On a birationally equivalent manifold,simple singular points. Acta Math. 1901.)

Uncertain reading. At the beginning of the parentheses,a separate F-like Latin letter with a dot. The beginning of the next word is overwritten; the current reading is theorem.

Original source details: GELFAND-P000117

Original source details: GELFAND-P000117

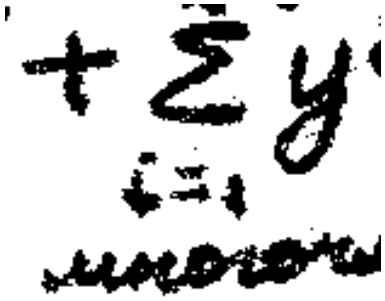
Uncertain reading. The numerator is y[?] / h-like,densely overwritten; the denominator x. Beside it,a short pointer bent down.

On the Weierstrass preparation theorem. (Malgrange.)

$$P(y) = y^n + \sum_{i=1}^{n-1} y^i a_i(x,z).$$

Original source details: GELFAND-P000117

Uncertain reading. In the first argument of a_i,a dense cancellation,currently x[?]. Below,a separate clean $a_i(x,z)$.



Original source details: GELFAND-P000117

Diagram and typed description. The first sum's lower limit is explicitly $i = 1$,the upper $n - 1$.

Distinguished polynomial $a_i(x,z)$,$a_i(0,0) = 0$.

Original source details: GELFAND-P000117

Uncertain reading. The second 0 in $a_i(0,0) = 0$ is densely overwritten; earlier strokes are covered.

Spät Ruckert.

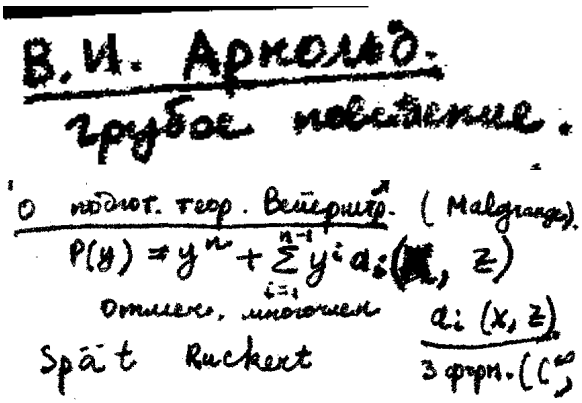


Diagram and typed description. The name V. I. Arnold. is underlined,the next line Stable flow. separate. The heading On the Weierstrass preparation theorem. is underlined; below,$a_i(x,z)$ is also underlined. The Latin entry is Spät Ruckert.

Original source details: GELFAND-P000117

$P(y,x,z)$; y — one variable, $x=(x_1,\dots,x_m)$,$z=(z_1,\dots,z_n)$ — spare ones. 3 forms (C^∞,analytic,formal series).

$$f(x,y,z)=P(y,x,z)\qquad Q(y,x,z)\qquad +\sum_0^{n-1} y^i b_i(x,z).$$

of the same class

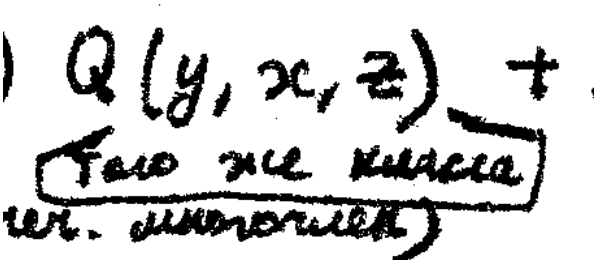
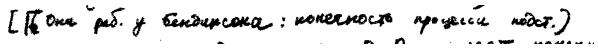


Diagram and typed description. Below $Q(y,x,z)$,a frame with the label of the same class,joined to the right of the entry by a short line. The remainder sum's lower limit is 0.

Original source details: GELFAND-P000117

(Division is possible.) (One can do this on relative sets.) [G.[?] This works for Bendixson: finiteness of the substitution process.)



Uncertain reading. After "process" is «подст.» ("substitution"). The first sign after the bracket remains uncertain.

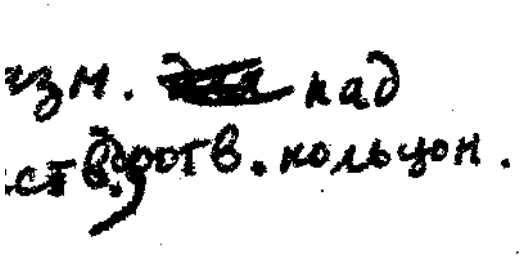
Original source details: GELFAND-P000117

The quotient by the ideal generated by P has finite dimension [Crossed out: unreadable token.] over

[Below at the right:] the corresponding ring.

Uncertain reading. Before over,a short entry is densely cancelled; the hidden letters are unread.

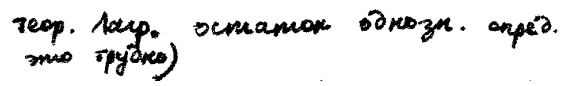
Original source details: GELFAND-P000117



Original source details: GELFAND-P000117

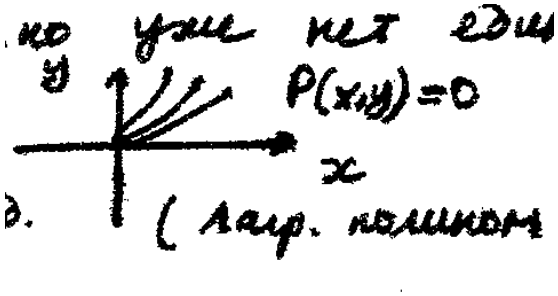
Diagram and typed description. Over stands at the end of the upper line. The corresponding ring. is below at the right,at the end of the next line.

(Malgrange: for C^∞ this is also true,but there is no longer uniqueness.) 1) Let P have n real roots near zero — by Lagrange's interpol. theorem the remainder is uniquely determ. (the Lagr. polynomial will be C^∞, but this is difficult)



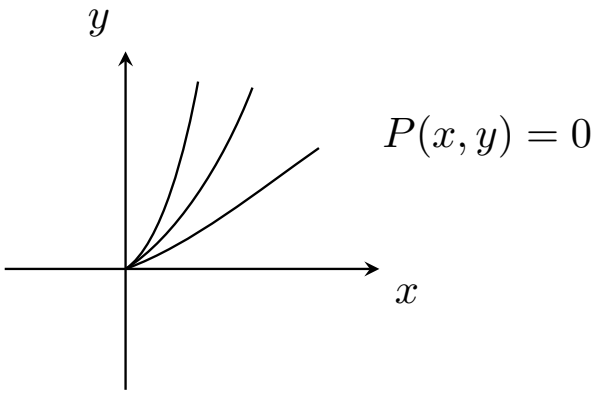
Original source details: GELFAND-P000117

Manuscript notation. The line has "Lagrange's theorem, the remainder is uniquely determ."

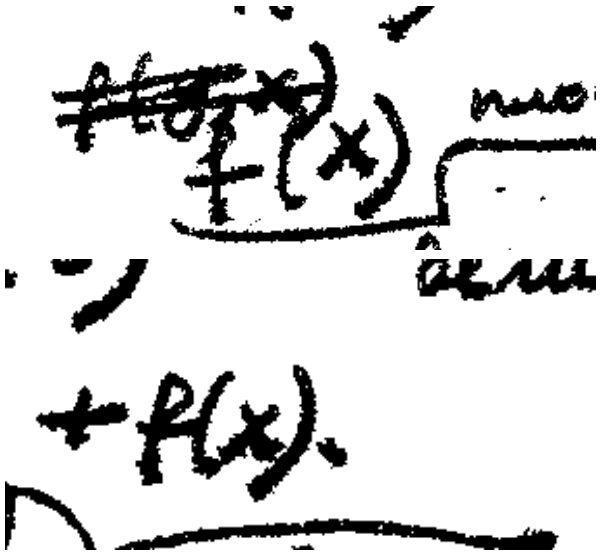


Original source details: GELFAND-P000117

Diagram and typed description. Axes:x right,y up. Three separate branches leave the origin into the first quadrant. At the right,the label $P(x,y)=0$.



2) [Crossed out: $f(y,x)$-like entry.]
$f(x)$ flat at 0 (all derivatives 0).



Original source details: GELFAND-P000117

Uncertain reading. In the cancelled entry,the arguments look like y,x. Below it,a clean $f(x)$. The first division formula ends with a separate $+f(x)$.

Divide by $y^2 + x^2$. Trivial answer.

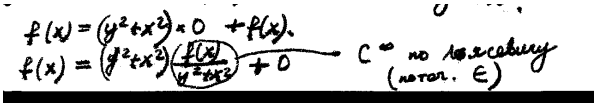
$$f(x) = (y^2 + x^2) \cdot 0 + f(x).$$

$$f(x) = (y^2 + x^2) \left(\frac{f(x)}{y^2 + x^2} \right) + 0$$

C^∞ by Łojasiewicz (potom. E[?]).

Original source details: GELFAND-P000117

Uncertain reading. In the last parentheses,potom. and a separate E- / epsilon-like sign. The abbreviation's expansion is unestablished.

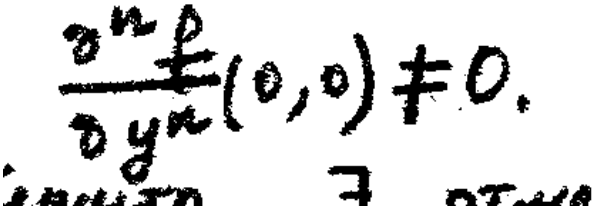


Original source details: GELFAND-P000117

Diagram and typed description. The fraction $f(x)/(y^2 + x^2)$ is circled. A curved line with a round endpoint runs right from it toward the entry C^∞ by Łojasiewicz. At the right of the equality,a separate $+0$.

GELFAND-P000118

Another formulation. $f(y, x)$,$f(0, 0) = 0$. Let $\frac{\partial^n f}{\partial y^n}(0, 0) \neq 0$. This is possible if f is not flat.

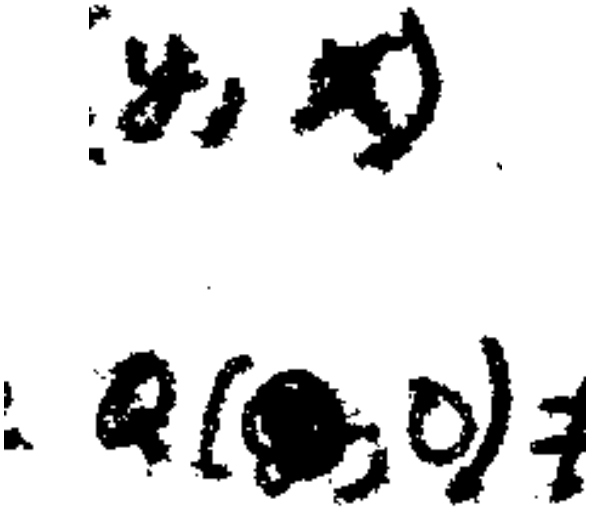


Original source details: GELFAND-P000118

Diagram and typed description. In the condition,n is the derivative order in y,followed by $(0, 0) \neq 0$.

Weierstrass's theorem:∃ a distinguished polynomial P,

$$f(y,x) = P(y,x)Q(y,x), \qquad Q(0,0) \neq 0.$$



Uncertain reading. In $P(y,x)$ the second argument is densely overwritten,the current letter x[?]. In $Q(0,0)$ the first 0 is overwritten; earlier strokes are covered.

Original source details: GELFAND-P000118

V. I. Arnold.

Diffeomorphisms of the annulus,preserving area,homotopic to the identity.

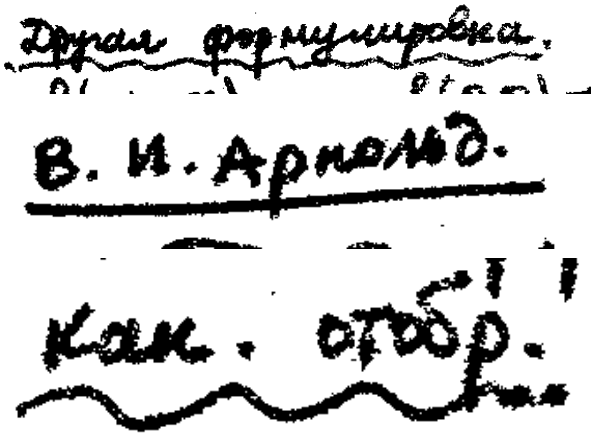
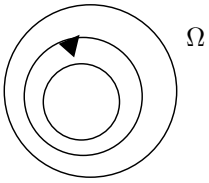


Diagram and typed description. Another formulation. has a wavy underline; Arnold's name a straight one. Canonical mappings. below also has a wavy underline.

Original source details: GELFAND-P000118

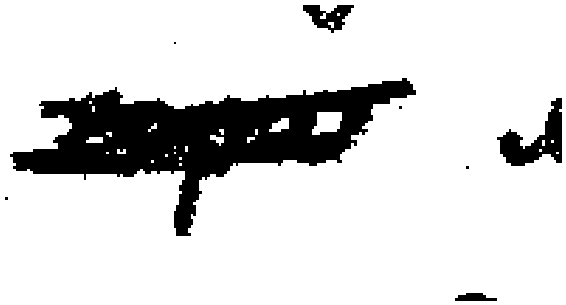
Diagram and typed description. Three nested round lines; at the top of the middle one,a dense triangular stroke. At the right,Ω.



Original source details: GELFAND-P000118

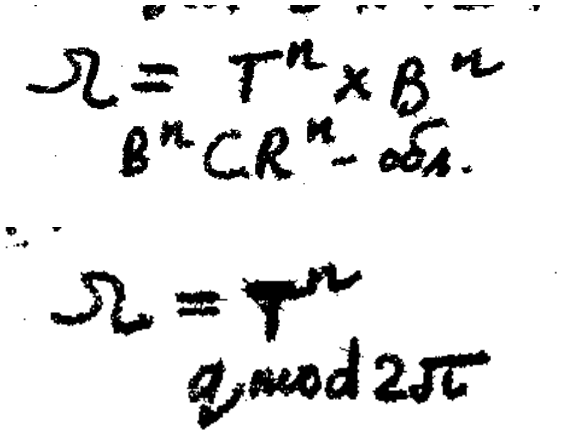
$\forall\gamma$ — a closed curve,going around 0, $\underset{\text{intersection points}}{A\gamma \cap \gamma} \geq 2$ (otherwise one does not obtain area preservation). Has significance for the 3-body problem. For n bodies [Crossed out: unreadable word.]

multidim. problem — tori, annulus $\Omega = T^n \times B^n$,$B^n \subset R^n$ — a domain. Diffeomorphism,homotopic to the identity,$A : \Omega \longrightarrow \Omega$. $\Omega = \underset{q,\,\mathrm{mod}\,2\pi}{T^n}$. In R^n,coordinates $p = (p_1, \ldots, p_n)$.



Original source details: GELFAND-P000118

Uncertain reading. After “for n bodies”, a dense cancellation before “multidim. problem”; the hidden letters are unread.

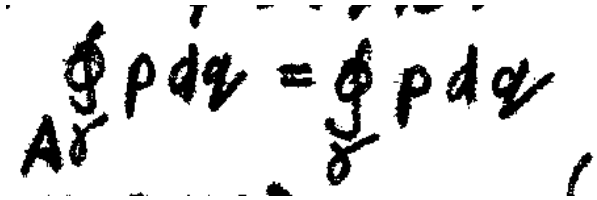


Original source details: GELFAND-P000118

Diagram and typed description. The upper entry is $\Omega = T^n \times B^n$,below a separate $\Omega = T^n$; under T^n,the label $q,\,\mathrm{mod}\,2\pi$.

Consider the form $p\,dq = p_1\,dq_1 + \cdots + p_n\,dq_n$. Canonical mappings. preserve the form:

$$\oint_{A\gamma} p\,dq = \oint_{\gamma} p\,dq.$$



Original source details: GELFAND-P000118

Diagram and typed description. Both integrals have closed contours; below the left sign,$A\gamma$,below the right,γ.

(Only these are considered.) (This is globally canonical.) For $n = 1$ this is the enclosed area. Transfer this to the n-dimensional case. In place of γ, T^n — but there it bounds nothing.
Theorem. Consider the torus $T : p = 0$,then $AT \cap T \geq 2^n$ (at least 2^n points).

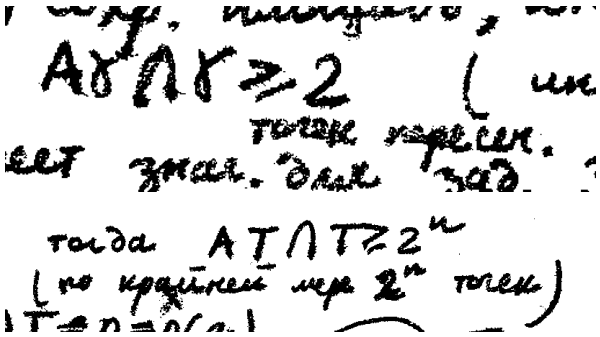
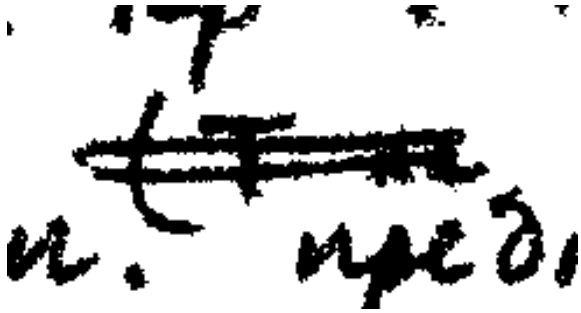


Diagram and typed description. The first entry is $A\gamma \cap \gamma \geq 2$,with intersection points below it. In the theorem,$AT \cap T \geq 2^n$; below,in words,at least 2^n points.

Original source details: GELFAND-P000118

Proof: under the additional [Crossed out: unreadable entry.]
assumption $AT \Rightarrow p = p(q)$.



Uncertain reading. Between under the additional and assumption,a short densely cancelled entry,with an opening-parenthesis-like stroke at the left.

Original source details: GELFAND-P000118

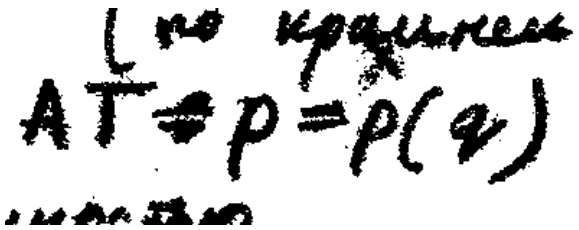
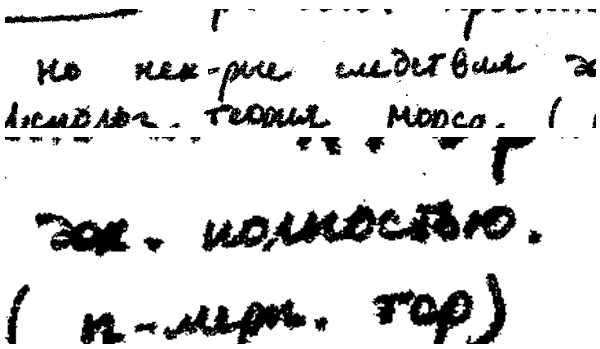


Diagram and typed description. After AT,a thick arrow-like sign right,then $p = p(q)$.

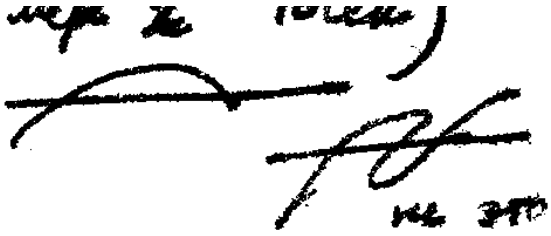
Original source details: GELFAND-P000118

But some consequences are proved completely.



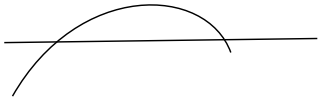
Manuscript notation. The line has the abbreviation «док.» (“proved”).

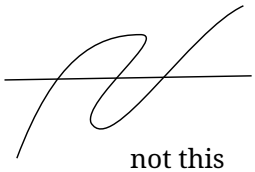
Original source details: GELFAND-P000118



Original source details: GELFAND-P000118

Diagram and typed description. Above,a straight line and a single-valued arched curve with two intersections. Below,a different straight line and a looped self-crossing curve; below it,not this.



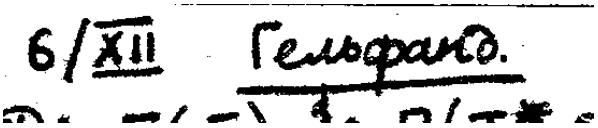


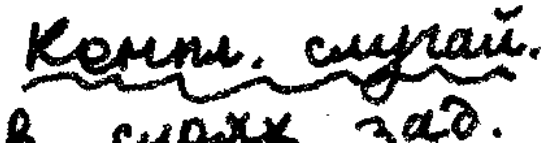
not this

Morse theory is used (n-dimensional torus)

GELFAND-P000119

6/XII Gelfand.





Original source details: GELFAND-P000119

Diagram and typed description. Straight lines above and below XII; Gelfand's name has a straight underline. Complex case. below has a wavy underline.

On X, a vector bundle E. T^* — cotangent. $D : \Gamma(E) \xrightarrow{\delta} \Gamma(T^* \otimes E)$.

Original source details: GELFAND-P000119

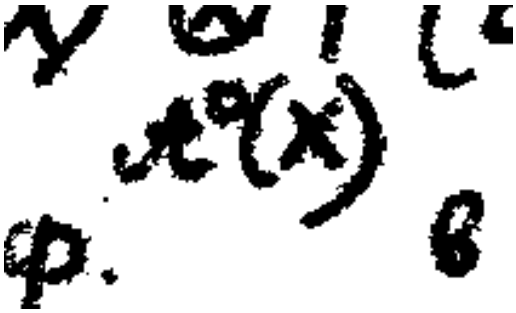
Uncertain reading. Above the arrow, a small δ- / s-like sign; its exact reading is unestablished.

$D(fs) = df \cdot s + f \cdot Ds$.

Original source details: GELFAND-P000119

Uncertain reading. The D in the last Ds is densely overwritten; the earlier strokes are unread.

$A(X) = \sum A^p(X)$ — forms. $A(X, E) = A(X) \otimes_{A^0(X)} \Gamma(E)$ — forms with coefficients in the bundle.

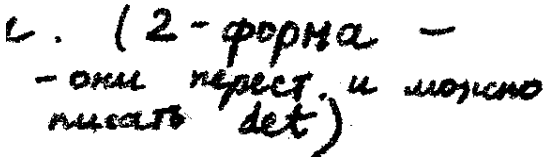


Original source details: GELFAND-P000119

Diagram and typed description. Below the tensor-product sign, $A^0(X)$.

s_i — a set of sections, a frame at every point in the fibre over x. $D(s_i) = \sum \theta_{ij} s_j$; θ — connection matrix. Curvature $K = d\theta - \theta \wedge \theta$ and characteristic classes (2-form — they commute and one can write det)

$$C(E, D) = \det\left(1 + \frac{i}{2\pi} K\right).$$



Original source details: GELFAND-P000119

Diagram and typed description. In parentheses, singular 2-form, then they commute and one can write det.

This does not depend on the choice of connection. Complex case. An analytic vector bundle is given. It is Hermitian if a Hermitian metric is given in the fibres.
$A(X) = \sum A^{p,q}(X)$; $A^{p,q}(X)$:

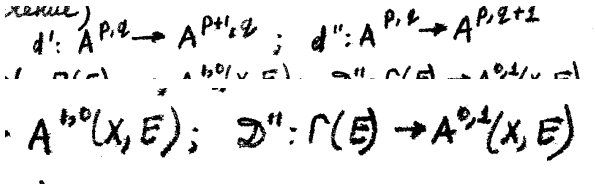
$$df_1 \wedge \cdots \wedge df_p \wedge d\overline{f_{p+1}} \wedge \cdots \wedge d\overline{f_{p+q}}.$$

(f_i — an analytic section.)

Original source details: GELFAND-P000119

Manuscript notation. The last index reads $p + q$. The bar extends over f and its index.

$$
\begin{aligned}
d &= d' + d'', \\
d' &: A^{p,q} \longrightarrow A^{p+1,q}, \\
d'' &: A^{p,q} \longrightarrow A^{p,q+1}, \\
D &= D' + D'', \\
D' &: \Gamma(E) \longrightarrow A^{1,0}(X, E), \\
D'' &: \Gamma(E) \longrightarrow A^{0,1}(X, E), \\
\Gamma(T^* \otimes E) &= A^1(X, E).
\end{aligned}
$$



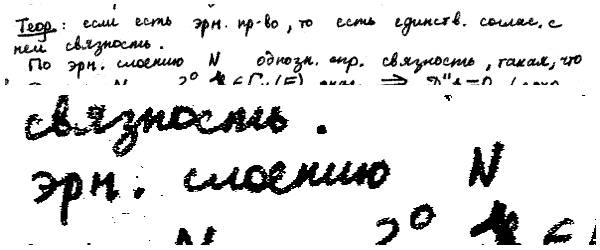
Original source details: GELFAND-P000119

Diagram and typed description. For d', at the right $p+1, q$; for d'', $p, q+1$. For D', type $1, 0$; for D'', $0, 1$.

The connection decomposes into a sum of 2 connections. s_i given, consider $N_{ij} = \langle s_i, s_j \rangle_x$ — functions of X. (Hermitian matrix.) Theorem: if there is a Hermitian space, then there is a unique connection compatible with it. From the Hermitian bundle N, a connection is uniquely determined such that

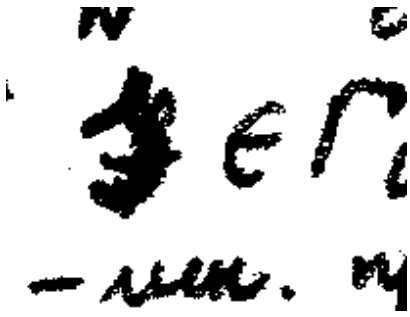
1° D preserves N; 2° $s \in \Gamma_U(E)$ analytic $\Rightarrow D''s = 0$.

(Preserves the holomorphic structure.) (Parallel transport — a linear transformation.)



Original source details: GELFAND-P000119

Diagram and typed description. At the start, Theorem is underlined. Below, From the Hermitian bundle N; the word is bundle.



Original source details: GELFAND-P000119

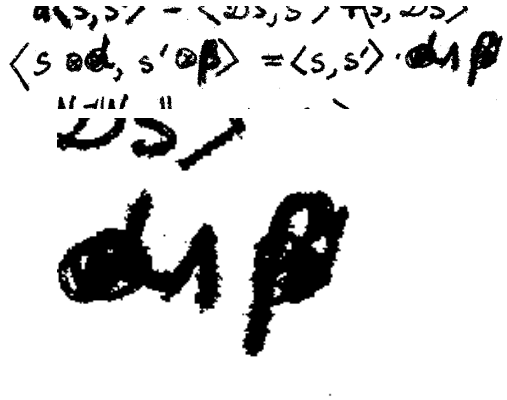
Uncertain reading. The s in item 2 is densely overwritten; the earlier sign is unread. Below Γ, subscript U.

$$d\langle s, s'\rangle = \langle Ds, s'\rangle + \langle s, Ds'\rangle.$$

(Differentiation formula.)

$$\langle s \otimes \alpha, s' \otimes \beta\rangle = \langle s, s'\rangle \cdot \alpha \wedge \beta.$$

(The scalar product of forms — a form.)



Manuscript notation. The manuscript has $\alpha \wedge \beta$ with no separate bar over β.

Original source details: GELFAND-P000119

$$N = \|N_{ik}\| = \langle s_i, s_k\rangle, \qquad dN = \theta N + N\overline{\theta}^t.$$

θ of type $1, 0$ — equivalent to condition 2. $d'N = \theta N$.

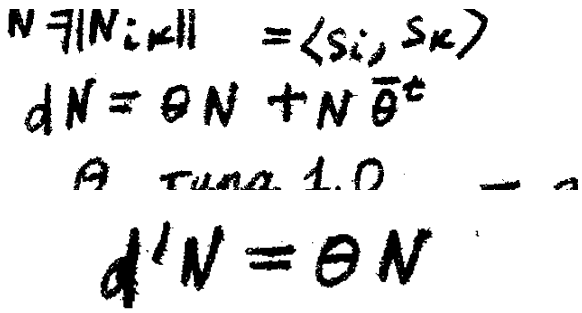


Diagram and typed description. In the matrix, indices ik; in the second term $N\overline{\theta}^t$, a bar over θ and a superscript t at its right. At the bottom, $d'N = \theta N$.

Original source details: GELFAND-P000119

GELFAND-P000120

$\theta = d'N \cdot N^{-1}$. (The connection constructed from the Hermitian metric must have this form.) And conversely, from N one easily obtains a connection.

$$d\langle s_i, s_j\rangle = \langle s_i, Ds_j\rangle + \langle Ds_i, s_j\rangle.$$

Relation between N, θ and K.

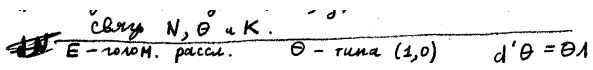


Diagram and typed description. Below Relation between N, θ and K, a long straight line.

Original source details: GELFAND-P000120

[Crossed out: a short unreadable entry.]

Uncertain reading. At the left below the heading, dense cancelled strokes; their exact letters are unestablished.

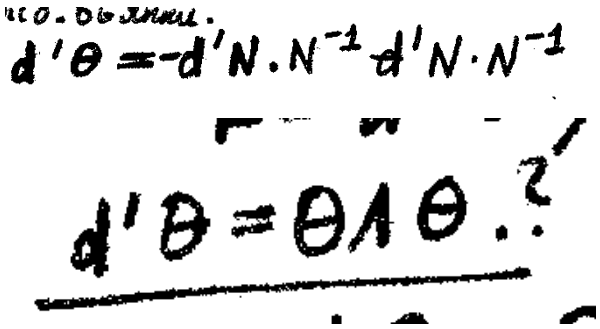
Original source details: GELFAND-P000120

E — a holomorphic bundle. θ — of type $(1,0)$.

$$K = d''\theta, \qquad d''K = 0,$$
$$d'K = -[K,\theta] \quad \text{Bianchi identity.}$$

$d'\theta = \theta \wedge \theta$. $\underline{d'\theta = \theta \wedge \theta\,?}$ $\theta = d'N \cdot N^{-1}$, differentiate.

$$d'\theta = -d'N \cdot N^{-1} d'N \cdot N^{-1}.$$



Manuscript notation. The differentiated product has a leading minus. The equality at the right has no minus. At the left its repetition with a question mark is underlined.

Original source details: GELFAND-P000120

$$K = d\theta - \theta \wedge \theta$$
$$= d'\theta + d''\theta - \theta \wedge \theta = d''\theta.$$

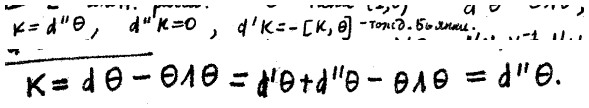


Diagram and typed description. In the identity $d'K = -[K,\theta]$, a minus before the bracket. Below, the equality chain ends with $d''\theta$.

Original source details: GELFAND-P000120

Let E be one-dimensional over an analytic manifold, a Hermitian metric at every point. $K = d''\theta$, $\theta = +d'N \cdot N^{-1}$. $N(s)$ — a function on X — positive $N(x)$. $K = d''d' \log N(x)$. The characteristic class — the same up to a factor, a form of type $(1,1)$.

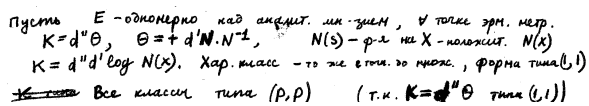
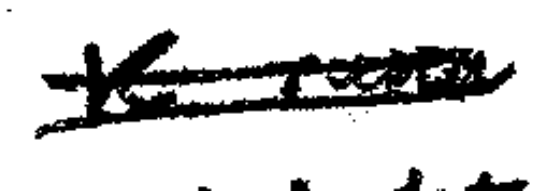


Diagram and typed description. An explicit plus before $d'N$. On one line, $N(s)$, then $N(x)$. Below, $K = d''d' \log N(x)$.

Original source details: GELFAND-P000120

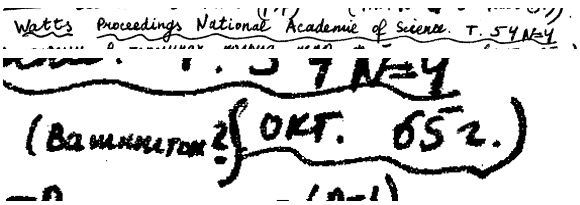
[Crossed out: a K-like beginning, the continuation unreadable.]



Original source details: GELFAND-P000120

Uncertain reading. Before All classes, a short K-like fragment is cancelled; the hidden strokes are unread.

All classes of type (p, p) (since $K = d''\theta$ is of type $(1, 1)$). Watts Proceedings National Academie of Science. vol. 54, $N = 4$ homology in terms of the ring of continuous functions. (Washington? Oct. 65)



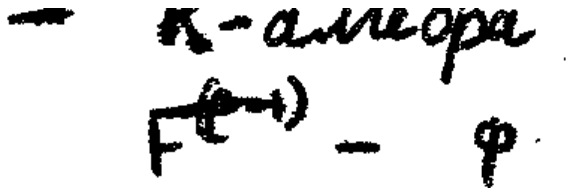
Original source details: GELFAND-P000120

Diagram and typed description. Watts and Proceedings National Academie of Science. vol. 54, $N = 4$ have wavy underlines. In parentheses, Washington?; Oct. 65 also has a wavy underline.

Λ — a K-algebra (K — a given ring). (In the application, $k = R$.)

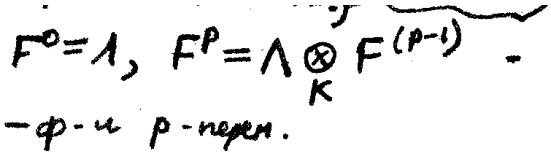
$$F^0 = \Lambda, \qquad F^p = \Lambda \otimes_K F^{(p-1)}.$$

functions of p variables. $F^{(p-1)}$ — φ.



Original source details: GELFAND-P000120

Uncertain reading. In the left annotation, exponent $p-1$ / $p+1$, with a dense intervening stroke.



Original source details: GELFAND-P000120

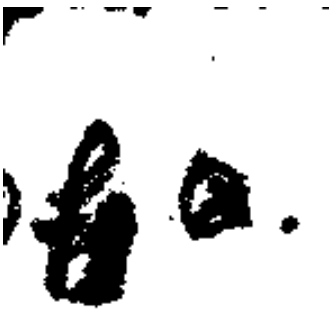
Diagram and typed description. At the right, $F^0 = \Lambda$, then $F^p = \Lambda \otimes_K F^{(p-1)}$; below them, functions of p variables.

$$\mu_F(f_0 \otimes \cdots \otimes f_p) = f_0 f_1 \cdots f_p.$$

(Restriction map to the diagonal.)

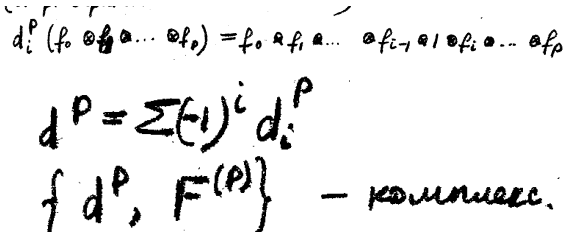
$$d_i^p(f_0 \otimes f_1 \otimes \cdots \otimes f_p) = f_0 \otimes f_1 \otimes \cdots \otimes f_{i-1} \otimes 1 \otimes f_i \otimes \cdots \otimes f_p.$$

$d^p = \sum(-1)^i d_i^p$. $\{d^p, F^{(p)}\}$ — a complex.



Original source details: GELFAND-P000120

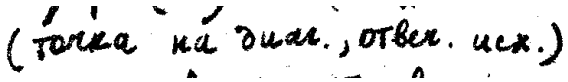
Uncertain reading. The f_1 in the left parentheses is densely overwritten; the earlier index is unread.



Original source details: GELFAND-P000120

Diagram and typed description. Between f_{i-1} and f_i, an inserted 1. The sum has no written limits; below, $\{d^p, F^{(p)}\}$.

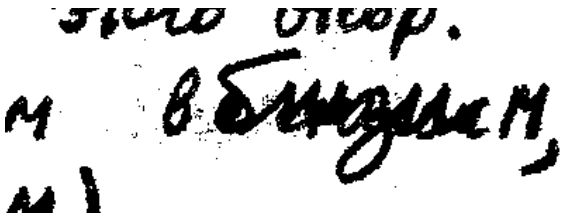
Let M be a maximal ideal. $\mu_F^{-1}(M) \Rightarrow$ a maximal ideal in $F^{(p)}$ — arises from this map. (Point on the diag., corresp. to the original.)



Original source details: GELFAND-P000120

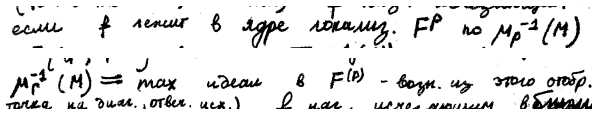
Manuscript notation. In parentheses, the final words are «отвеч. исх.» (“corresp. to the original”).

f is called vanishing near M, if f lies in the kernel of the localization of F^p at $\mu_p^{-1}(M)$.



Original source details: GELFAND-P000120

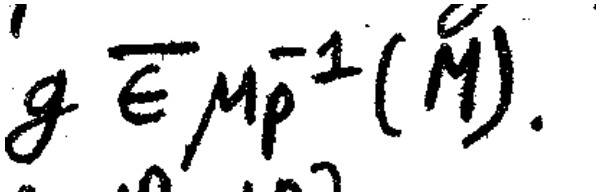
Uncertain reading. Near and the final M are densely corrected; M is visible, the earlier letters are unread.



Original source details: GELFAND-P000120

Diagram and typed description. In the first entry, $\mu_F^{-1}(M)$; in the localization line, $\mu_p^{-1}(M)$. Both p and (p) occur on F.

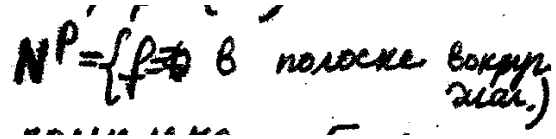
$\exists g$ such that $gf = 0$, $g \overline{\in} \mu_p^{-1}(M)$.



Original source details: GELFAND-P000120

Manuscript notation. A separate bar over $\in$ in $g \overline{\in} \mu_p^{-1}(M)$.

$N^p = \{f = 0$[?] in a strip around the diagonal $\}$.

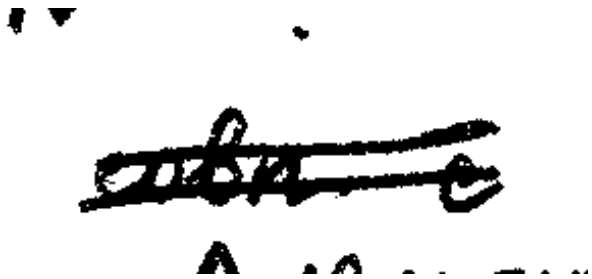


Original source details: GELFAND-P000120

Uncertain reading. After f, a dense mark resembling $= 0$. The horizontal strokes overlap the sign, and the exact reading remains uncertain.

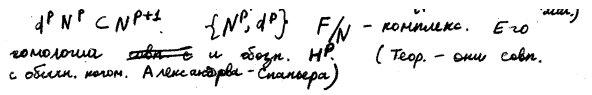
$d^p N^p \subset N^{p+1}$. $\{N^p, d^p\}$, F/N — a complex. Its homology [Crossed out: several unreadable letters.] is denoted H^p.

(Theorem — they coincide with the usual Alexander–Spanier cohomology.)



Original source details: GELFAND-P000120

Uncertain reading. After homology, several letters are cancelled by two lines.



Original source details: GELFAND-P000120

Diagram and typed description. In the line, Its homology; in the final parentheses, Alexander–Spanier cohomology. Both $\{N^p, d^p\}$ and F/N are written.

GELFAND-P000121

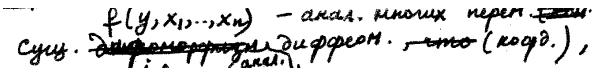
On the work of N. Levinson Duke, 1961 p. 345–353

$f(y, x_1, \ldots, x_n)$ — analytic in many variables. [Crossed out: a short unreadable entry.] There exists [Crossed out: a word resembling Diffeomorphism.] a diffeom.

[Inserted: anal.] [Crossed out: that[?].] (of coord.), such that

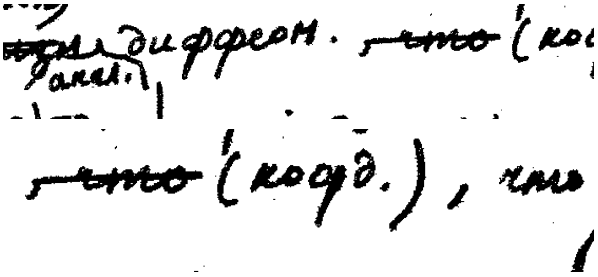
(local property)

Given $\left|\begin{array}{l} \dfrac{\partial^i f}{\partial y^i}(0,0) = 0 \\ \\ \dfrac{\partial^n f}{\partial y^n}(0,0) \neq 0 \end{array}\right.$ $\left|\begin{array}{l} i = 0, \ldots, n-1 \\ \\ d^n(x)y^n + \sum_0^{n-1} d^i(x)y^i \\ \\ d^n(0) \neq 0. \qquad d^i(0) = 0. \end{array}\right\|$



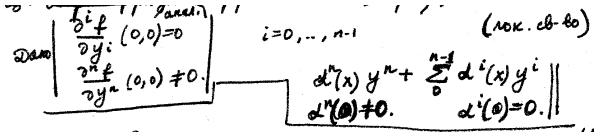
Original source details: GELFAND-P000121

Uncertain reading. After variables, a short dense cancellation. After There exists, a long cancelled word with a Diffeomorphism-like beginning; covered letters have not been supplied.



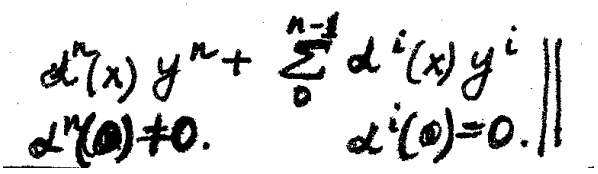
Original source details: GELFAND-P000121

Uncertain reading. A separate anal. stands below the main line near the cancelled word. Before (of coord.), a cancelled entry resembling that; after the parentheses, a separate clean that.



Original source details: GELFAND-P000121

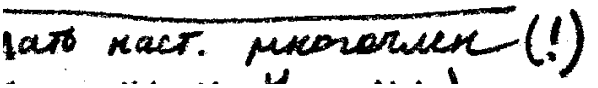
Diagram and typed description. Given at the left; the two conditions have a separate left vertical and a right vertical. A lower stepped line runs to the polynomial at the right; beyond it, two vertical lines. Above it, (local property).



Original source details: GELFAND-P000121

Diagram and typed description. The first coefficient has superscript n; the sum has lower limit 0, upper $n-1$; then $d^i(x)y^i$. Beneath, separate $d^n(0) \neq 0$ and $d^i(0) = 0$.

Also, if there are 2 variables, one can make an actual polynomial (!)



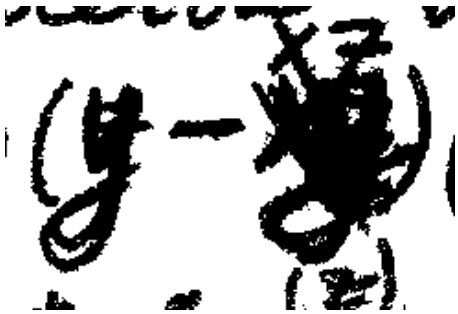
Original source details: GELFAND-P000121

Diagram and typed description. Before polynomial, the abbreviation nast., with a separate full stop. After polynomial, a separate (!).

(if $n = 3$, this is not always possible — Whitney's example)

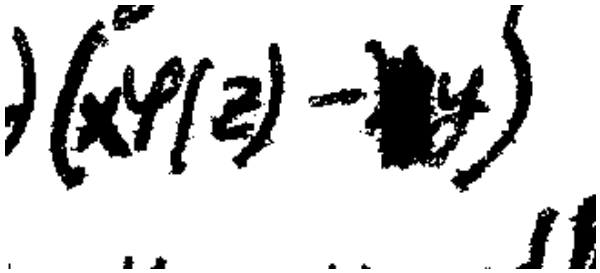
$$f(x, y, z) = xy(y - x)(y - xz[?])(x\varphi(z) - y[?]), \qquad \varphi(0) = 0,$$

$\varphi(z)$ — any transcendental function. Zeros — (4 surfaces)



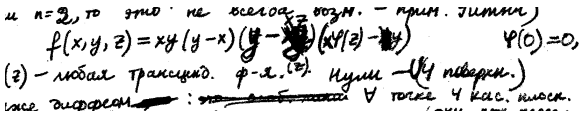
Original source details: GELFAND-P000121

Uncertain reading. In the fourth factor, after the minus, a densely corrected entry; xz is inserted above it. The covered lower strokes have not been read.



Original source details: GELFAND-P000121

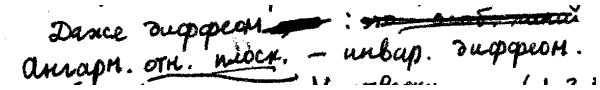
Uncertain reading. In the last factor, after the minus, a dense correction before a y-like letter; the preceding strokes have not been established separately.



Original source details: GELFAND-P000121

Diagram and typed description. In the example line, a separate $\varphi(0) = 0$. Below, after Zeros, the entry (4 surfaces); further down, at $\forall$ point 4 tangent planes.

Even a diffeom. [The end of the abbreviation is crossed out]: [Crossed out: an unreadable phrase.]



Original source details: GELFAND-P000121

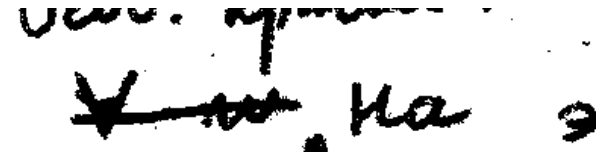
Uncertain reading. After Even, a diffeom.-like entry is visible, with its ending densely covered; after the colon, a long cancellation with several lines. In the next line, the words ratio of planes are underlined.

At $\forall$ point 4 tangent planes. (They pass through one line.) Anharmonic ratio of planes — an invariant of diffeomorphisms. Singular curve.

Original source details: GELFAND-P000121

Manuscript notation. At the left, "singular curve."

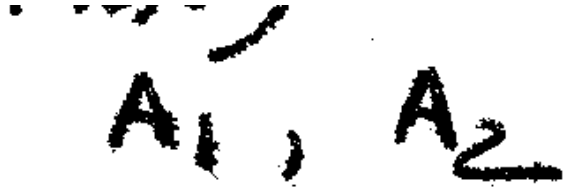
$\forall$ quadruple $(1, 2, 3, 4; \quad 1, 2, 3, 5)$ [Crossed out: $\forall$ and a short unreadable entry.]



Original source details: GELFAND-P000121

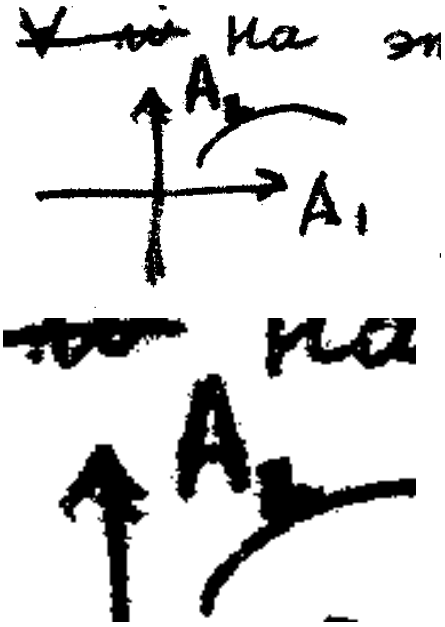
Uncertain reading. Before On this, a separately cancelled $\forall$ and a following short sign or word. These signs are crossed by one long line.

On this singular line, this is a function of z. $A_1,\ A_2$.



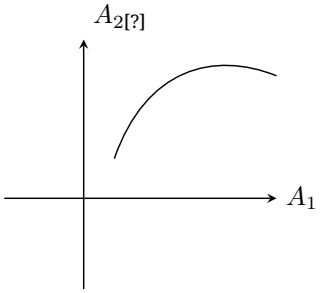
Original source details: GELFAND-P000121

Diagram and typed description. At the right, separate A_1, A_2; the lower horizontal stroke of 2 is extended.

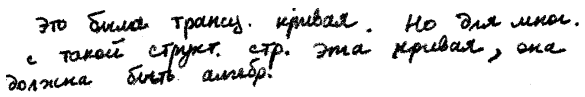


Original source details: GELFAND-P000121

Uncertain reading. The horizontal axis points right with A_1; the vertical points up. Beside its A, the lower right stroke merges with the beginning of the arc; subscript 2[?]. The arc in the first quadrant initially rises, then descends gently.



This was a transcendental curve. But for a polynomial with such a structure, str.[?] this curve, it must be algebraic.



Original source details: GELFAND-P000121

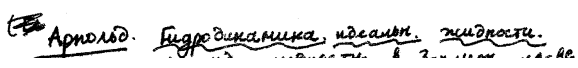
Uncertain reading. After structure, a short str.-like entry; its abbreviation has not been expanded. The final line reads must be algebraic.

[Crossed out: a short unreadable sign.]

Original source details: GELFAND-P000121

Uncertain reading. At the left before Arnold., a separate densely cancelled sign with a rounded left stroke.

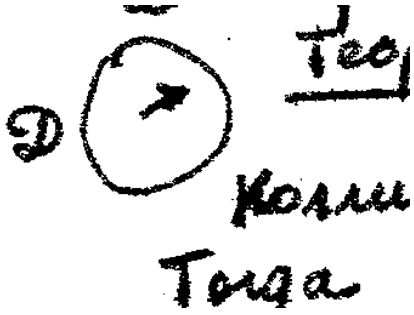
Arnold. Hydrodynamics, ideal fluid.



Original source details: GELFAND-P000121

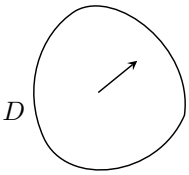
Diagram and typed description. Arnold. has a straight underline. Hydrodynamics, ideal and fluid. have separate wavy underlines.

Stationary flows of ideal fluid in 3-dimensional space. D — a bounded domain, and ideal fluid flows there (stationary flow).



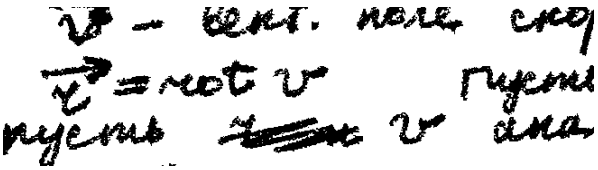
Original source details: GELFAND-P000121

Diagram and typed description. A closed rounded contour; D at the left. Inside, a short arrow points right and up.



Theorem. $\vec{v}$ — the velocity vector field (tangent on the boundary).

$$\vec{\tau}[?] = \operatorname{rot} v.$$



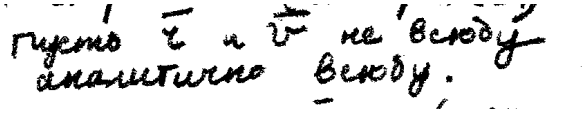
Original source details: GELFAND-P000121

Uncertain reading. The letter at the left has a rightward arrow above it. The letter is τ- / r-like; its exact reading is unestablished. After rot, a separate v with no visible overhead arrow.

Let $\vec{\tau}$[?] and $\vec{v}$ be not everywhere collinear; let [Crossed out: an unreadable word.] v be analytic everywhere.

Original source details: GELFAND-P000121

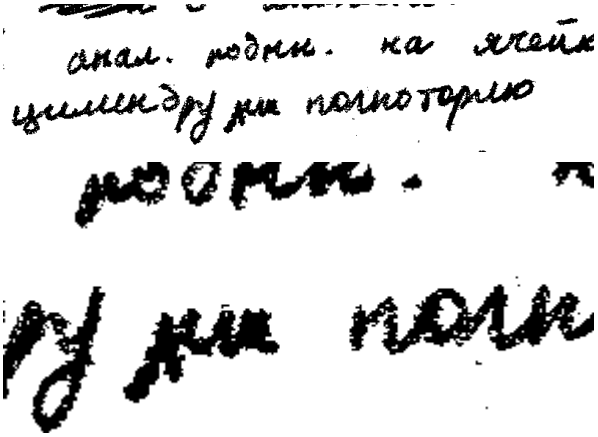
Uncertain reading. Before the last v, a separate dense cancellation; a short initial k-like stroke is visible, the other letters covered.



Original source details: GELFAND-P000121

Uncertain reading. In the condition not everywhere collinear, the first disputed letter has a short horizontal stroke overhead with no clearly separate head; v has a rightward arrow.

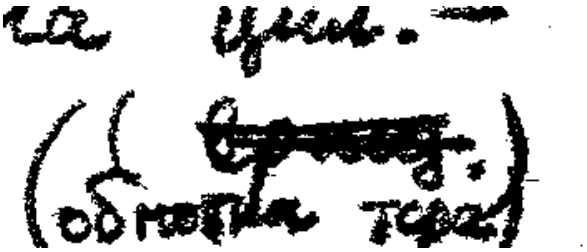
Then an analytic subset divides D into cells (a finite number) homeom. to a cylinder or a solid torus, and in them the flow is arranged in a special way. On the cylinder — rotations.



Original source details: GELFAND-P000121

Manuscript notation. After “number)” the text reads “homeom. to a cylinder or a solid torus.”

On the torus, $\dot{\varphi}_1 = \lambda_1, \quad \dot{\varphi}_2 = \lambda_2$ ([Crossed out: a word beginning vrashch...] [Below:] winding of the torus.)

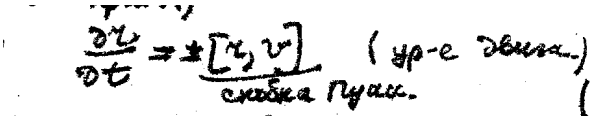


Original source details: GELFAND-P000121

Uncertain reading. In the parentheses, the upper entry is densely crossed out; vrashch...-like initial strokes are visible. The lower separate line is winding of the torus.

If everywhere collinear, there are examples when this is not so. Ideal fluid

$$\frac{\partial \tau[?]}{\partial t} = \pm \underbrace{[\overline{\tau}, v]}_{\text{Poisson bracket}} \qquad \text{(eq. of mot.)}$$



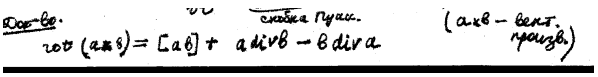
Original source details: GELFAND-P000121

Diagram and typed description. The numerator has the same τ-like letter as the first entry in the bracket. The exact letter remains uncertain. Before the bracket, a separate $\pm$. A short bar above its first letter. Below the bracket, a line with a short upturn at the right and the label Poisson bracket; at the right, (eq. of mot.).

Proof:

$$\operatorname{rot}(a * b) = [a\ b] + a \operatorname{div} b - b \operatorname{div} a$$

($a * b$ — vector product.)



Original source details: GELFAND-P000121

Diagram and typed description. Between a and b, a star-shaped sign. At the right, $[a\ b]$ with no written comma, then $+a \operatorname{div} b - b \operatorname{div} a$. In parentheses, vector product.

GELFAND-P000122

78

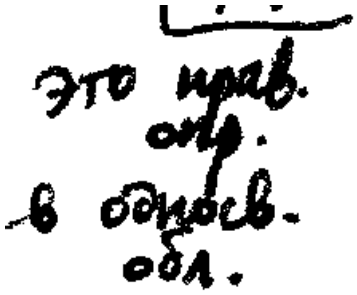
Original source details: GELFAND-P000122

Diagram and typed description. At the upper right, 78; an angular line at its left and below it, with no upper or right side.

v is called the velocity field of ideal fluid if

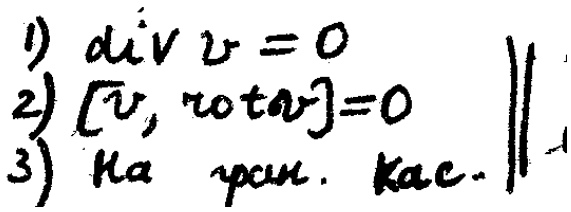
1) $\operatorname{div} v = 0$
2) $[v, \operatorname{rot} v] = 0$
3) Tangent on the boundary.

this is the correct definition in a simply connected domain.



Original source details: GELFAND-P000122

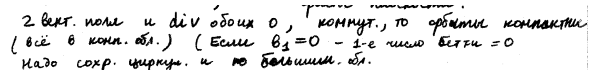
Manuscript notation. The margin reads «это прав. опр. в односв. обл.» (“this is the correct definition in a simply connected domain”).



Original source details: GELFAND-P000122

Diagram and typed description. The conditions are numbered $1), 2), 3)$. The second has $[v, \operatorname{rot} v] = 0$; the third, Tangent on the boundary. At the right, two separate parallel vertical lines.

At $\forall$ point, 2 vector fields, R^2 — the orbits of the plane. 2 vector fields and the div of both 0, commuting, then the orbits are compact (everything in a compact domain) (If $b_1 = 0$ — the 1st Betti number $= 0$ One must preserve circulation also over large domains.



Original source details: GELFAND-P000122

Diagram and typed description. In the condition, separate $b_1 = 0$ and the 1st Betti number $= 0$. An opening parenthesis before If; no closing parenthesis is visible after the last 0. The next line is One must preserve circulation also over large domains.

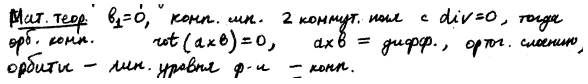
Mathematical theorem. $b_1 = 0$, compact manifold, 2 commuting fields with div $= 0$, then the orbits are compact.

$$\operatorname{rot}(a * b) = 0, \qquad a \times b = \text{diff.}, \quad \text{orthog. to the foliation},$$

orbits — level lines of a function — compact.

Original source details: GELFAND-P000122

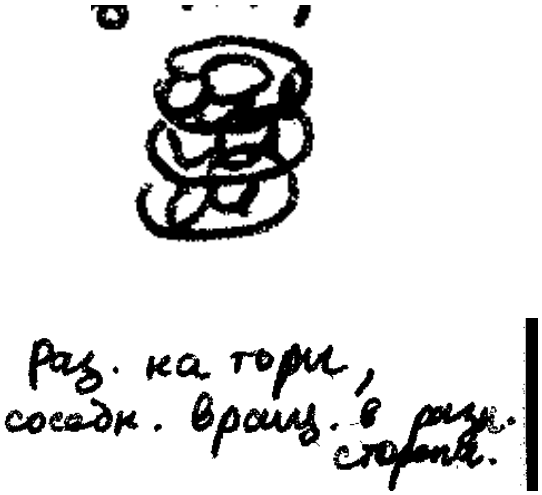
Uncertain reading. After $a \times b =$ the text has “diff.”, then “orthog. to the foliation”. The first abbreviation is unexpanded. The first product uses a star-shaped sign.



Original source details: GELFAND-P000122

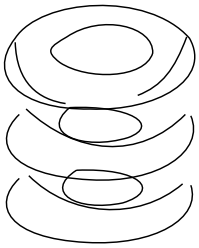
Diagram and typed description. Mathematical theorem. is underlined. After $b_1 = 0$, compact manifold; below, the rot of the product equals 0. At the end, a separate line: orbits — level lines of a function — compact.

If there is no boundary, the compact orbits are tori; if noncompact, then cylinders. The action of the line — winding. (Experiments are visible in photographs.) There are cylinders too. 2 rotating cylinders, fluid there.



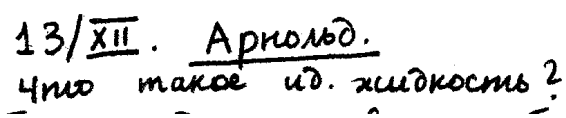
Original source details: GELFAND-P000122

Diagram and typed description. Three overlapping ring contours with inner loops and short side strokes. No separate labels or arrows are visible on the rings. At the right, Division into tori, neighbouring ones rotate in different directions.



Division into tori, neighbouring ones rotate in different directions.

$13/\overline{\text{XII}}$. Arnold. What is ideal fluid?



Original source details: GELFAND-P000122

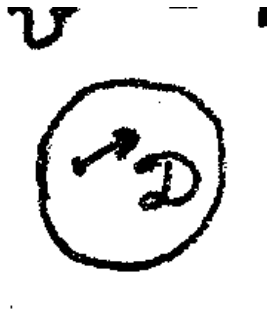
Diagram and typed description. Separate straight lines above and below XII; Arnold. is underlined separately. Below, the question What is ideal fluid?

Fluid flow in a bounded domain, [Crossed out: a short unreadable sign.] Interesting to study instability.

Uncertain reading. Before Interesting, a separate densely crossed-out short sign; its letter or meaning is unestablished.

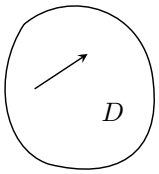
Original source details: GELFAND-P000122

v — velocity field, $\operatorname{div} v = 0$ (incompressible). (Makes sense on a Riemannian manifold.)



Original source details: GELFAND-P000122

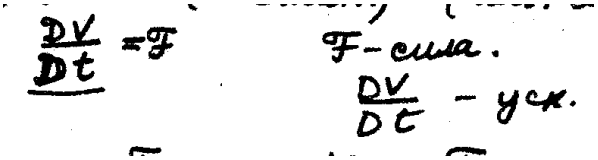
Diagram and typed description. A closed oval contour. Inside, a right-and-up arrow; D is also inside, to the right and below the arrow.



Newton's equation

$$\frac{DV}{Dt} = \mathcal{F}.$$

$\mathcal{F}$ — force. $\dfrac{DV}{Dt}$ — acceleration of a particle (not the partial derivative).



Original source details: GELFAND-P000122

Diagram and typed description. In Newton's equation, numerator DV, denominator Dt; a separate straight line beneath the whole fraction. At the right, a calligraphic F-like letter, and its repetition beside force.

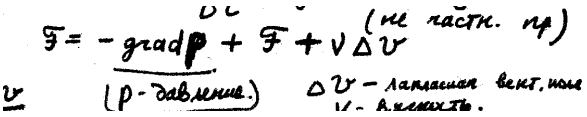
$v(x, y, z, t)$. If there are no external forces

$$\mathcal{F} = -\underline{\operatorname{grad} p[?]} + \mathcal{F} + \nu\Delta v$$

(They are potential.)

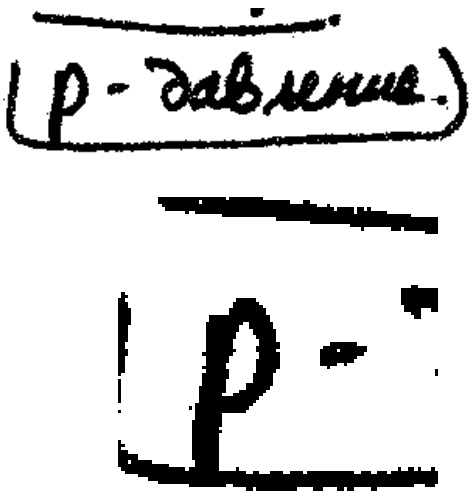
$$\boxed{p[?] \text{ — pressure.}}$$

Δv — Laplacian of the vector field. ν — viscosity.



Original source details: GELFAND-P000122

Diagram and typed description. In the equality, after the underlined grad with a p-like letter, separate $+\mathcal{F}$ and $+\nu\Delta v$. Below, Δv — Laplacian of the vector field and ν — viscosity.

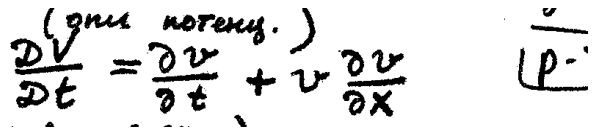


Original source details: GELFAND-P000122

Uncertain reading. In the pressure label, a p-/ρ-like letter with a long lower stroke. Around the label, left, lower and right lines; no upper side is drawn. The letter's shape differs from the disputed sign in the gradient of the system below.

$$\frac{DV}{Dt} = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}$$

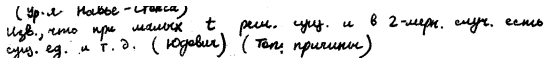
(Navier–Stokes equation)



Original source details: GELFAND-P000122

Diagram and typed description. In the left fraction, numerator DV with a capital V-like letter; the right fractions have small v. The last denominator is ∂x.

It is known that for small t a solution exists, and in the 2-dimensional case there is existence, uniqueness, etc. (Yudovich) (Top. reasons)



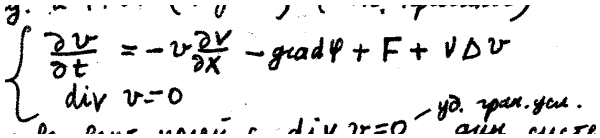
Original source details: GELFAND-P000122

Diagram and typed description. After existence, uniqueness, etc., two separate parentheses: (Yudovich) and (Top. reasons). The second parenthesis reads «Топ. причины» (“Top. reasons”).

$$\begin{cases} \dfrac{\partial v}{\partial t} = -v\dfrac{\partial v}{\partial x} - \operatorname{grad}\varphi[?] + F + \nu\Delta v \\ \operatorname{div} v = 0 \end{cases}$$

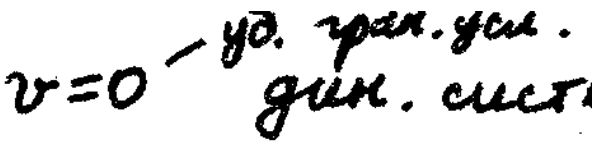
ud. bound. cond.

In the space of vector fields with $\operatorname{div} v = 0$ — a dynamical system.



Original source details: GELFAND-P000122

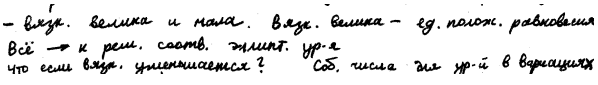
Uncertain reading. After grad, the letter is φ- / p-like, with a long stroke through the rounded part. A minus precedes it. Then a separate plain F, followed by $+\nu\Delta v$. A common brace at the left.



Original source details: GELFAND-P000122

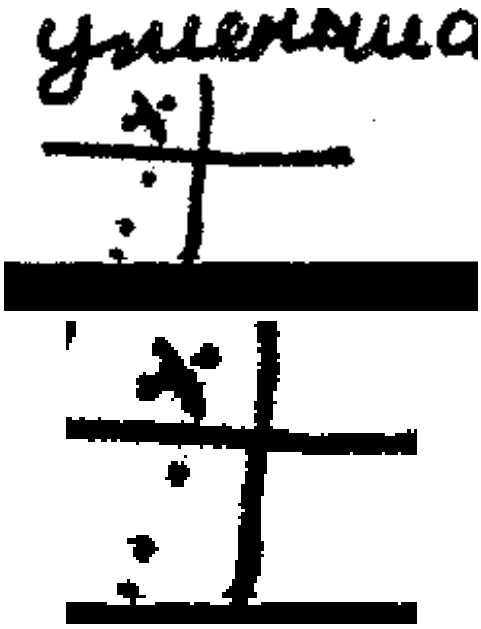
Diagram and typed description. A short oblique pointer leads from $\operatorname{div} v = 0$ to a note above the line: ud. bound. cond. The first abbreviation is ud.; it has not been expanded.

Limiting cases — — viscosity large and small. Large viscosity — a unique equilibrium position. Everything $\to$ the solution of the corresponding elliptic equation. What if viscosity decreases? Eigenvalues for the variational equations (stable solution) λ moves right.



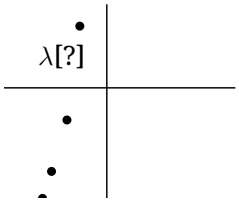
Original source details: GELFAND-P000122

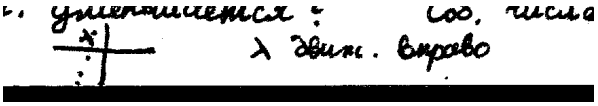
Diagram and typed description. In the question, decreases is underlined; below, (stable solution), then a separate λ moves right.



Original source details: GELFAND-P000122

Uncertain reading. Two intersecting lines with no clearly separate arrowheads. Several points at the left; above the horizontal, a dense λ-like sign and a separate point. The sign's reading is unestablished.



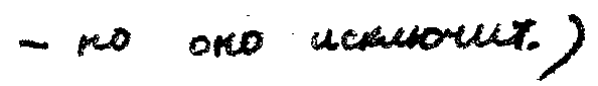


Original source details: GELFAND-P000122

Undeciphered passage. The lower edge of the scan passes through the vertical stroke and the lowest point of the diagram. Their continuation beyond the edge is not visible.

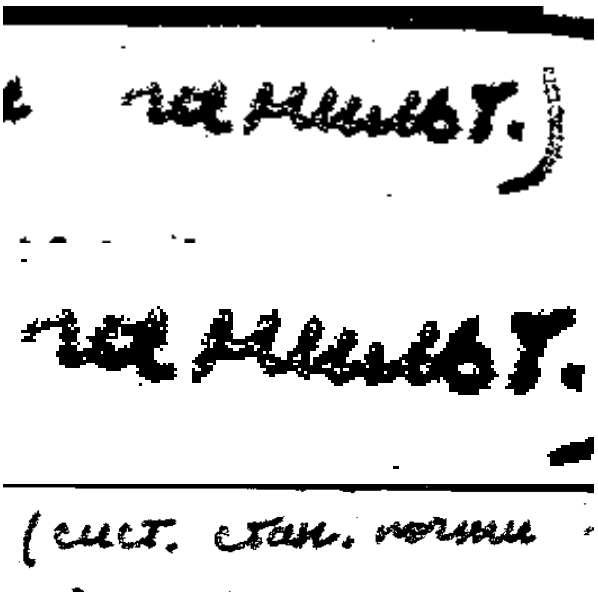
GELFAND-P000123

For $\nu = 0$, we shall be symm. about both axes (the syst. becomes almost Hamil.) (Poiseuille flow and others) — but it is exceptional.)



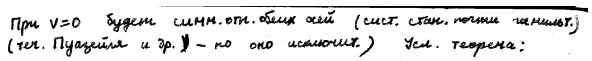
Original source details: GELFAND-P000123

Manuscript notation. After "but it is" the manuscript reads «исключит.» ("exceptional").



Original source details: GELFAND-P000123

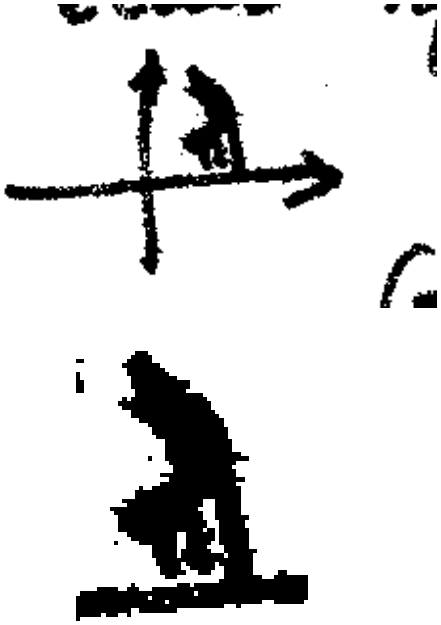
Manuscript notation. The first parenthesis reads «сист. стан. почти гамильт.» ("the syst. becomes almost Hamil.").



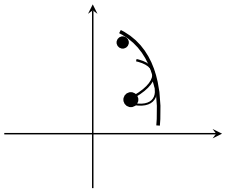
Original source details: GELFAND-P000123

Diagram and typed description. In the first line, $\nu = 0$, then we shall be symm. about both axes. In the second line a separate closing parenthesis follows “exceptional.”

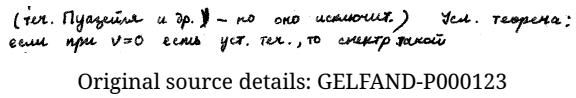
Conditional theorem: if at $\nu = 0$ there is stable flow, then the spectrum is like this



Uncertain reading. Axes with arrows up and right. To the right above the intersection, a dense curved trace with short adjoining strokes. Its reading as a letter or separate points is unestablished.



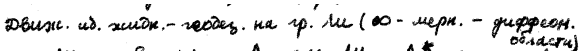
Original source details: GELFAND-P000123



Original source details: GELFAND-P000123

Diagram and typed description. Conditional theorem: above a separate line, if at $\nu = 0$ there is stable flow, then the spectrum is like this. The drawing has no readable axis labels.

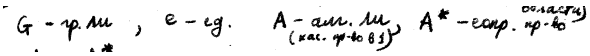
Motion of ideal fluid — geodesics on a Lie gr. (∞-dimensional — diffeomorphisms of the domain)



Original source details: GELFAND-P000123

Diagram and typed description. In the heading line, geodesics on a Lie gr.; at the right in parentheses, ∞-dimensional — diffeomorphisms of the domain. The word domain continues below.

G — Lie gr., e — identity. A — Lie alg. (tangent space at 1), A^* — dual space.



Original source details: GELFAND-P000123

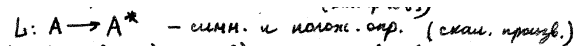
Diagram and typed description. Beside A, parentheses: tangent space at 1; the numeral 1 is written. Beside A^*, separately, dual space.

$L : A \to A^*$ — symm. and positive definite (scalar product)

$$\langle a, b\rangle = \langle La, b) = (a, Lb).$$

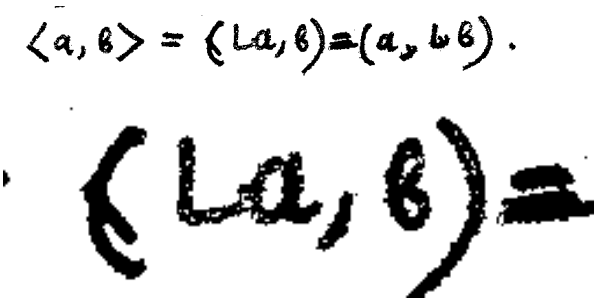
Defines a left-inv. metric on the gr.

$$\langle a, b\rangle_g = \langle \widetilde{\Lambda}_{g^{-1}} a, \widetilde{\Lambda}_{g^{-1}} b\rangle. \qquad L_g : TG_g \to T^*G_g$$



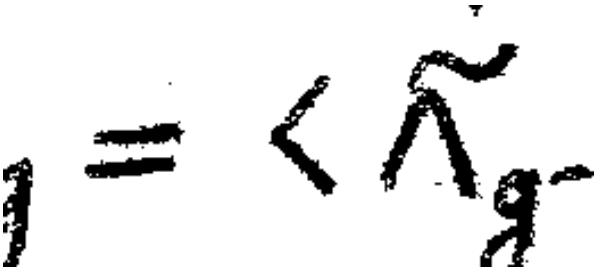
Original source details: GELFAND-P000123

Diagram and typed description. A single rightward arrow from A to A^*; after it, symm. and positive definite, then scalar product in parentheses.



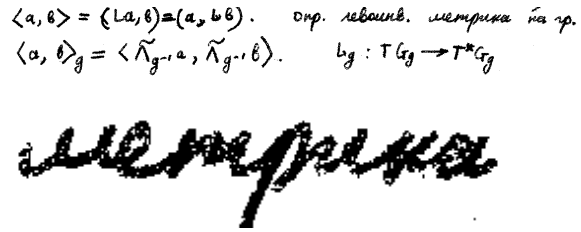
Original source details: GELFAND-P000123

Uncertain reading. The first opening bracket after the equality has an angular upper part and two lower strokes, one rounded. Its bracket type is unestablished. After b, a round closing bracket; the last pair is (a, Lb).



Original source details: GELFAND-P000123

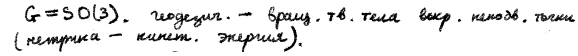
Uncertain reading. Wavy strokes above both Λ-like signs; the choice between capital Λ and small λ is not finally established. Subscripts g^{-1}.



Original source details: GELFAND-P000123

Diagram and typed description. In the line before the formula, the Russian word metrika ends in a. The left pair has subscript g; at the right, separately, $L_g : TG_g \to T^*G_g$.

$G = SO(3)$. geodesics — rotation of a rigid body about a fixed point (metric — kinetic energy).

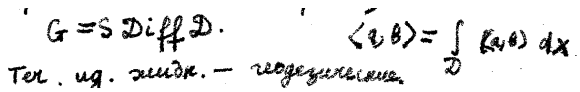


Original source details: GELFAND-P000123

Diagram and typed description. After $SO(3)$, a separate full stop; then geodesics — rotation of a rigid body about a fixed point. On the next line, parentheses: metric — kinetic energy.

$$G = \mathrm{SDiff}\, D. \qquad \langle a, b\rangle = \int_D (a, b)\, dx.$$

Ideal fluid flows — geodesics.



Original source details: GELFAND-P000123

Diagram and typed description. The integral has lower subscript D, and the integrand pair (a, b) is in round brackets. A separate next line: Ideal fluid flows — geodesics.

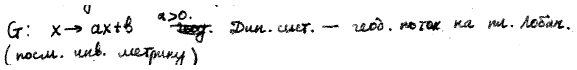
$$G: \quad x \to ax + b \qquad a > 0.$$

[Crossed out: a short unreadable sign or word beneath $a > 0$.]

Dynamical syst. — geodesic flow on the Lobachevsky plane. (Look at the inv. metric.)

Original source details: GELFAND-P000123

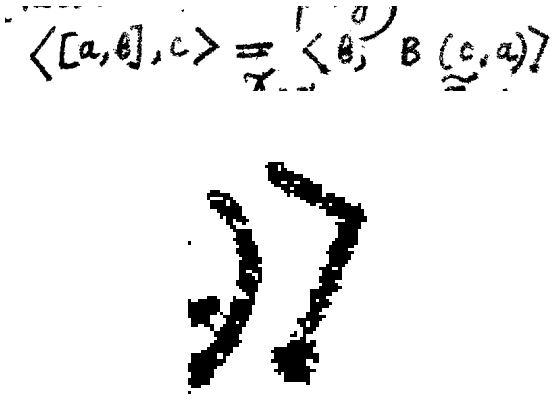
Uncertain reading. Below the active $a > 0$, a densely crossed-out entry with a lower loop; its letters and sign are unestablished.



Original source details: GELFAND-P000123

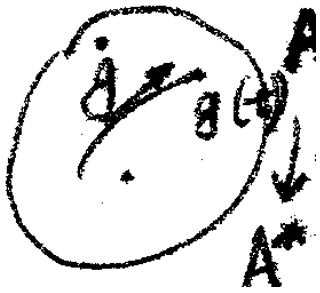
Diagram and typed description. Arrow $x \to ax + b$; above the crossed-out entry, $a > 0$. At the right, Dynamical syst. — geodesic flow on the Lobachevsky plane; below, separate parentheses: Look at the inv. metric.

$$\langle [a, b], c\rangle = \langle b, B(c, a)\rangle$$



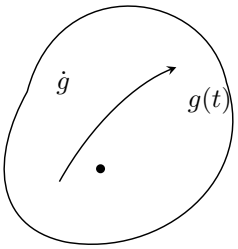
Original source details: GELFAND-P000123

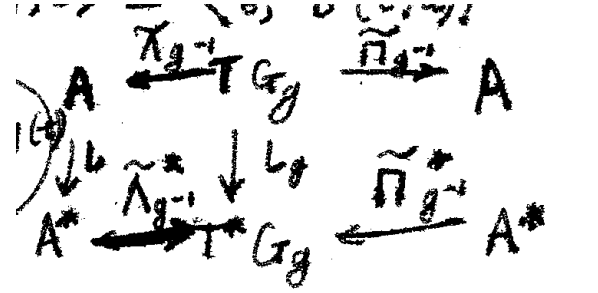
Diagram and typed description. In the first pair, $[a, b]$ and c; in the second, b and $B(c, a)$. After the round bracket, the final angle bracket, with a short thickening at the bottom.

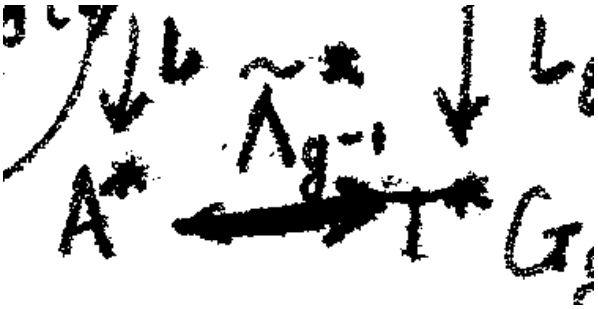


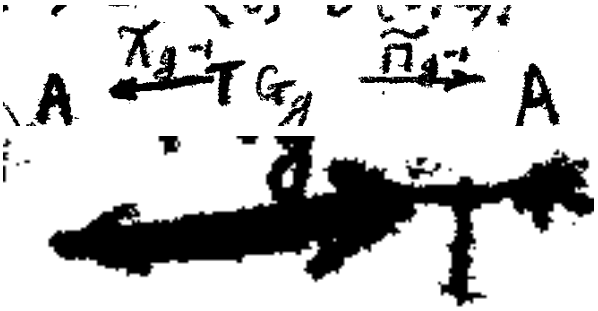
Original source details: GELFAND-P000123

Diagram and typed description. At the left, a closed oval contour; inside, a curve with an arrow, $\dot{g}$, $g(t)$ and a separate lower point. The label $g(t)$ touches the contour's right edge.



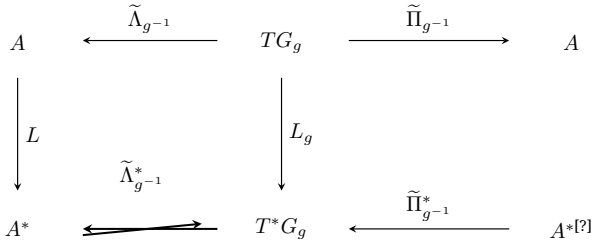






Original source details: GELFAND-P000123

Uncertain reading. Upper row $A \leftarrow TG_g \rightarrow A$, with wavy Λ- and Π-like signs above the arrows, subscripts g^{-1}. Separate downward arrows L and L_g. In the lower row, A^* at the left, T^*G_g in the middle, and at the right A with a dense star or overlaid stars: A^* / A^{**}. The lower right arrow points left. The lower left line has two overlaid tips in opposite directions; its active direction is unestablished. Wavy signs with stars and subscripts g^{-1} above the lower arrows.



1966 yr. 7/II Kushnirenko: “On invariants of dynamical systems”

I. M. Gelfand. on the envelop. alg. of a Lie algebra

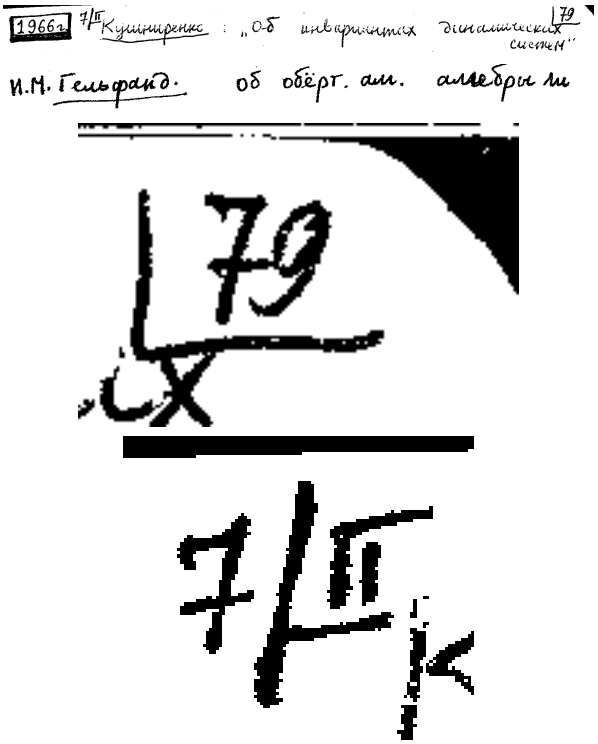


Diagram and typed description. The year is in a rectangular frame with a thickened lower stroke. At 79, left and lower frame sides. Lines above and below II in the date; Kushnirenko and Gelfand. are underlined separately from the date and initials. The Russian title ends with sistem.

Original source details: GELFAND-P000124

Every Noeth. ring (without zero divisors) — an Ore ring

$$\begin{pmatrix} \forall a, b \neq 0 & \exists x, y \neq 0, \\ & xa = yb \end{pmatrix}$$

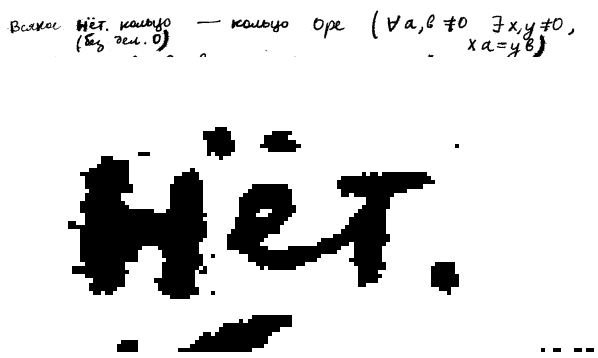


Diagram and typed description. Two dots over e in the Russian abbreviation nyot. At the right, parentheses with $\forall a, b \neq 0$, $\exists x, y \neq 0$; below, $xa = yb$.

Original source details: GELFAND-P000124

An Ore ring embeds in a skew field (these are nat. classes) (elements of the form ab^{-1})

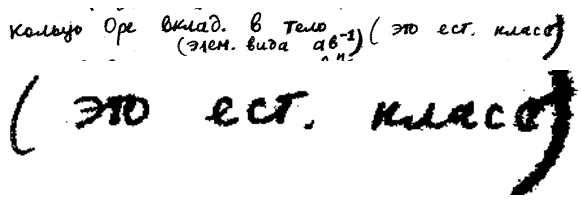


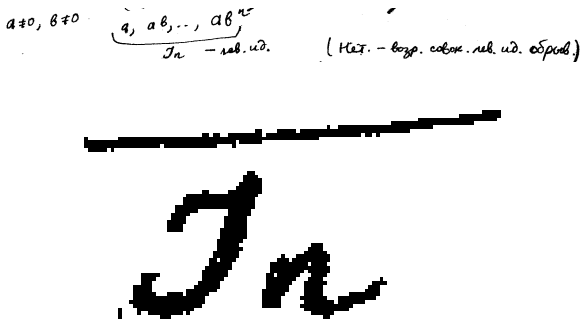
Diagram and typed description. In the right parentheses, these are nat. classes; «ест.» is an abbreviation. Below, separate parentheses: elements of the form ab^{-1}, with negative one on b.

Original source details: GELFAND-P000124

$a \neq 0, b \neq 0.$

$$\underbrace{a, ab, \ldots, ab^n}_{\mathcal{I}_n[?] \text{ — left ideal}}$$

(Noeth. — an increasing collection of left ideals stops.)



Uncertain reading. A long lower brace beneath the sequence. Below it, a capital I-/J-like letter with a lower loop and subscript n, then left ideal; its exact letter is unestablished. At the right, separate parentheses beginning Noeth.

Original source details: GELFAND-P000124

$$ab^{n+1} = x_0 a + x_1 ab + \cdots + x_n ab^n, \qquad x_0 a = (\quad) b$$

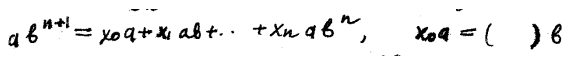
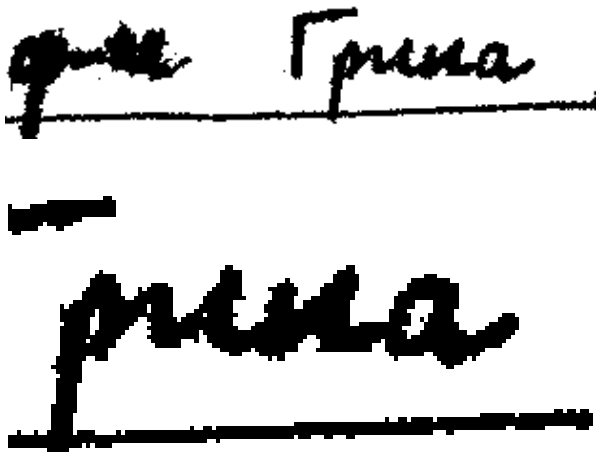


Diagram and typed description. In the first equality, powers $n+1$ and n, subscripts $0, 1, n$. In the second, after $x_0 a =$, round brackets with an empty interval, then b.

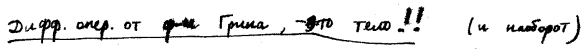
Original source details: GELFAND-P000124

Diff. oper. of [Crossed out: an f-n-like abbreviation.] Grin[?], this is a skew field!! (and conversely)



Uncertain reading. After of, a short densely cancelled entry with an f-like beginning. The following name reads Grin / Grim; its penultimate letter is unestablished.

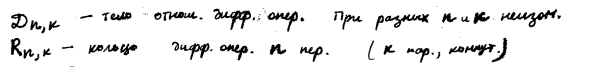
Original source details: GELFAND-P000124



Original source details: GELFAND-P000124

Diagram and typed description. The line from Diff. oper. of through skew field!! has a common underline. A comma after the disputed name; at the right, separate parentheses: and conversely. Two exclamation marks.

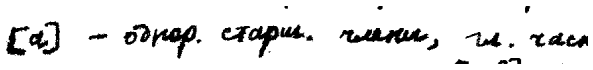
$D_{n,k}$ — skew field of ratios of diff. oper. For different n and k, nonisom. $R_{n,k}$ — ring of diff. oper. in n variables (k param., commuting).



Original source details: GELFAND-P000124

Diagram and typed description. Two different capitals, D and R, with the same subscripts n, k. Beside R, separately, n variables and parentheses: k param., commuting.

a, $[a]$ — homog. leading term, principal part.



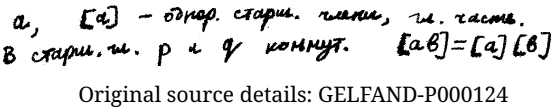
Original source details: GELFAND-P000124

Manuscript notation. Before “leading term” the manuscript has «однор.» (“homog.”).

In the leading part, p and q commute. $[ab] = [a][b]$

$$b = a_1 a_2^{-1} \qquad \text{Consider} \quad \frac{[a_1]}{[a_2]}$$

it depends only on b, and not on the manner of representation.



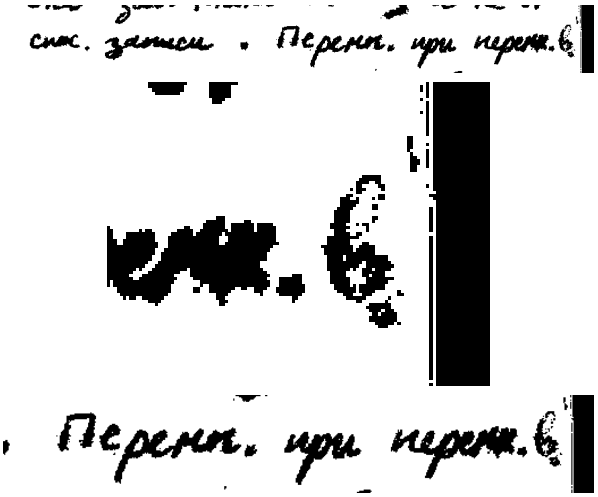
Original source details: GELFAND-P000124

Diagram and typed description. A separate comma at a, $[a]$. Square brackets around the principal parts; below, p and q, then $[ab] = [a][b]$.

Original source details: GELFAND-P000124

Diagram and typed description. In $b = a_1 a_2^{-1}$, negative one on a_2. At the right, a fraction with $[a_1]$ and $[a_2]$ in square brackets; below, it depends only on b, and not on the manner of representation.

Multipl. under multipl. of b[?]



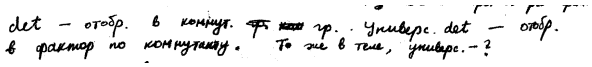
Original source details: GELFAND-P000124

Uncertain reading. Two instances of «перемн.» (“multipl.”) surround «при». The final v-/b-like letter near the edge remains uncertain.

det — a map to a commut. [Crossed out: two short signs or words.] gr. Universal det — a map to the quotient by the commutator subgroup. The same in a skew field, universal — ?

Original source details: GELFAND-P000124

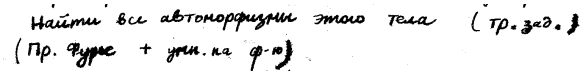
Uncertain reading. Between commut. and gr., two separate dense cancellations. The first has an f-/f-ya-like lower stroke; the second, covered kom-like strokes. The active gr. is at the right of both.



Original source details: GELFAND-P000124

Diagram and typed description. At the start, Latin det; then Universal det — a map to the quotient by the commutator subgroup. The final phrase, The same in a skew field, universal — ?, ends with a question mark.

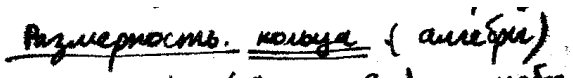
Find all automorphisms of this skew field (difficult problem). (Fourier tr. + multiplication by a function)



Original source details: GELFAND-P000124

Diagram and typed description. In the first line, all automorphisms of this skew field and parentheses: difficult problem. In the second, Fourier tr., a separate plus, then multiplication by a function.

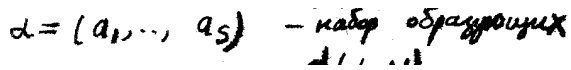
Dimension. of a ring (algebra)



Original source details: GELFAND-P000124

Diagram and typed description. Dimension. is underlined once, of a ring with two separate lower lines. Parentheses: algebra, after ring.

$\alpha = (a_1, \ldots, a_s)$ — a set of generators

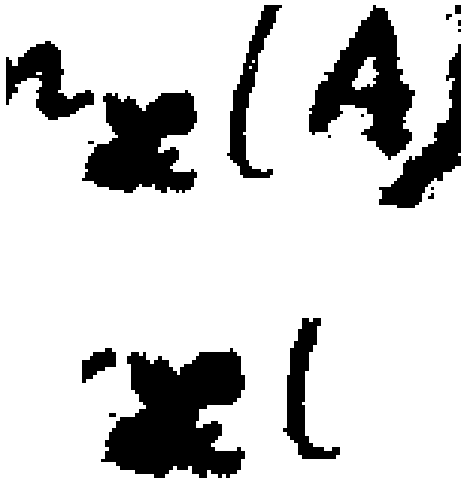


Original source details: GELFAND-P000124

Diagram and typed description. At the left, α; a tuple in round brackets, with extreme subscripts 1 and s. After the dash, a set of generators.

$$\sup_{\alpha} \overline{\lim}_{N\to\infty} \frac{d(\alpha, N)}{\ln N} = \mathrm{Dim}_{\chi[?]}(A)$$

$d(\alpha, N)$ — the number of linearly independent monomials of degree $\leq N$.



Uncertain reading. Below Dim, a dense χ- / z- / α-like sign; its exact reading is unestablished. At the right, round brackets with A.

Original source details: GELFAND-P000124

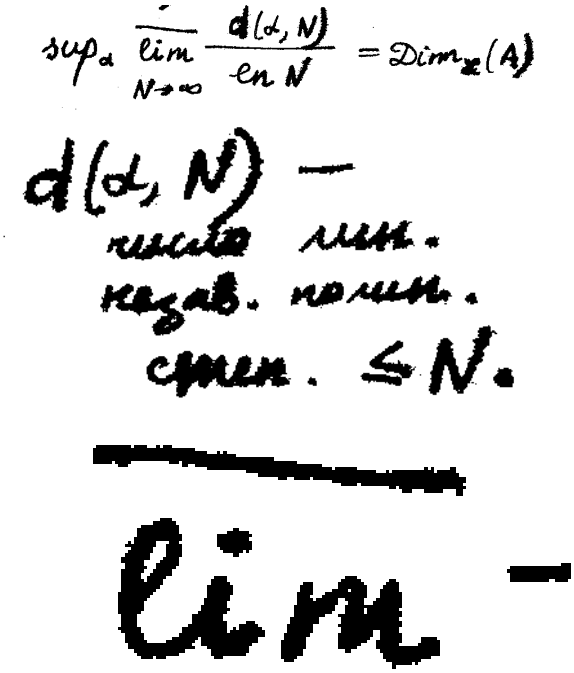


Diagram and typed description. A lower α at sup; a separate horizontal line above lim, with $N \to \infty$ below. Numerator $d(\alpha, N)$, denominator $\ln N$. At the right, separately, $d(\alpha, N)$ — the number of linearly independent monomials of degree $\leq N$.

Original source details: GELFAND-P000124

$$\dim R_{n,k} = 2n + k.$$

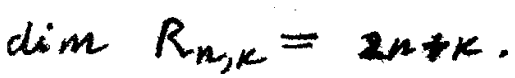
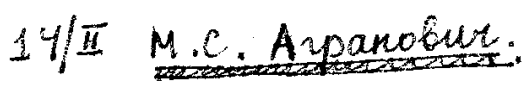


Diagram and typed description. In the final line, dim with small d, $R_{n,k}$ and $2n + k$, then a full stop.

Original source details: GELFAND-P000124

GELFAND-P000125

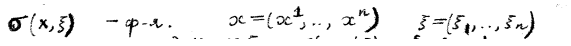
14/II M. S. Agranovich.



Original source details: GELFAND-P000125

Diagram and typed description. Lines above and below II in the date. Beneath the name M. S. Agranovich., two lines with oblique hatching between them.

$\sigma(x,\xi)$ — a fn. $x=(x^1,\dots,x^n)$, $\xi=(\xi_1,\dots,\xi_n)$.

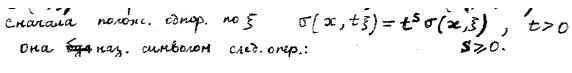


Original source details: GELFAND-P000125

Diagram and typed description. The first sign is σ; the coordinates of x have superscripts $1, n$, and those of ξ have subscripts $1, n$.

Initially pos. homog. in ξ

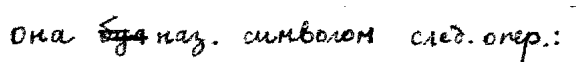
$$\sigma(x,t\xi)=t^s\sigma(x,\xi),\qquad t>0,\quad s\geq 0.$$



Original source details: GELFAND-P000125

Diagram and typed description. The exponent of t is s. After $t>0$, a separate $s\geq 0$ followed by a period.

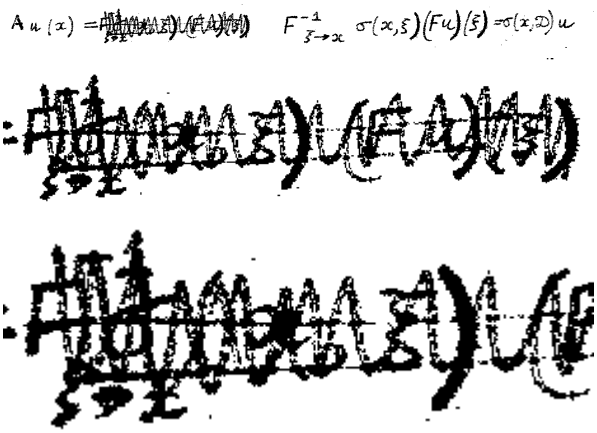
it [Crossed out: bud[?].] is called the symbol of the following oper.:



Uncertain reading. After it and before is called, a short crossed-out bud-like word; the exact reading of the covered strokes is undetermined.

Original source details: GELFAND-P000125

$$Au(x)=\text{[Crossed out: } F^{-1}_{\xi\to x}\sigma(x[?],\xi)(Fu)(\xi)]$$
$$F^{-1}_{\xi\to x}\sigma(x,\xi)(Fu)(\xi)=\sigma(x,D)u$$

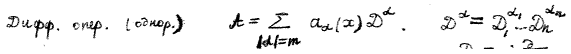


Uncertain reading. After $Au(x)=$, a crossed-out formula with $F^{-1}_{\xi\to x}$, σ, parentheses and $(Fu)(\xi)$. Interior strokes, including the first argument of σ, are covered; the reading of the entire crossed-out expression remains incomplete. A separate active formula is to the right.

Original source details: GELFAND-P000125

Differential oper. (homog.)

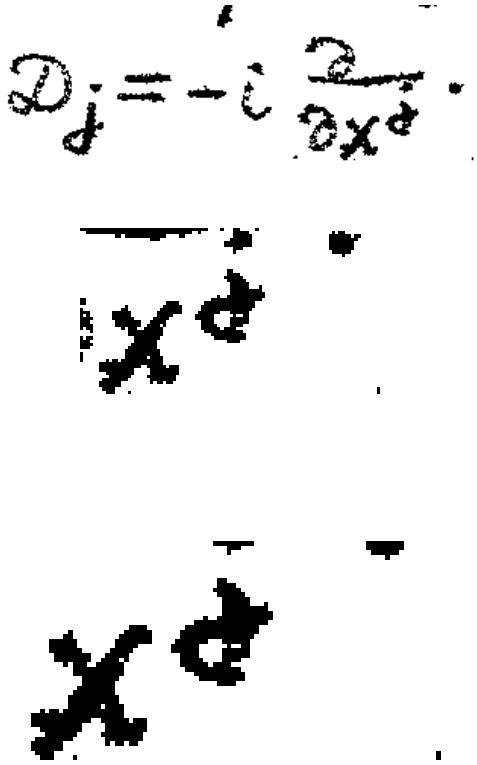
$$A = \sum_{|\alpha|=m} a_\alpha(x) D^\alpha. \qquad D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$$



Original source details: GELFAND-P000125

Diagram and typed description. The summation condition is $|\alpha| = m$. To the right of D^α are factors with subscripts $1, n$ and superscripts α_1, α_n.

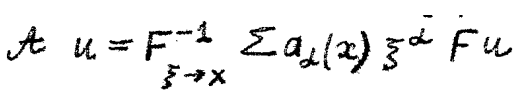
$$D_j = -i \frac{\partial}{\partial x^{\alpha[?]}}.$$



Original source details: GELFAND-P000125

Uncertain reading. The superscript of x in the denominator has a rounded counter and a hooked upper stroke: α- / a- / i- / j-like. The exact sign is undetermined.

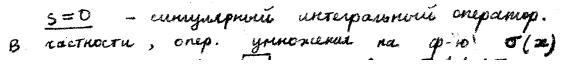
$$Au = F^{-1}_{\xi \to x} \sum a_\alpha(x) \xi^\alpha F u$$



Original source details: GELFAND-P000125

Diagram and typed description. Beneath F^{-1}, the direction $\xi \to x$. In the sum, $a_\alpha(x)$ and ξ^α, followed by Fu.

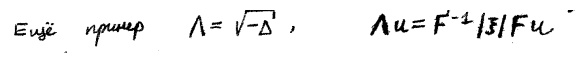
$\underline{s = 0}$ — a singular integral operator. In particular, the oper. of multiplication by the fn. $\sigma(x)$.



Original source details: GELFAND-P000125

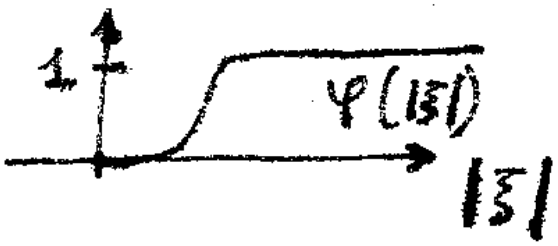
Diagram and typed description. Only $s = 0$ is underlined. The second line says multiplication by the fn. $\sigma(x)$.

Another example $\Lambda = \sqrt{-\Delta}, \qquad \Lambda u = F^{-1}|\xi| F u.$



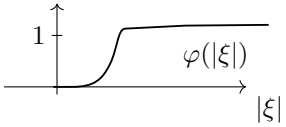
Original source details: GELFAND-P000125

Diagram and typed description. Capital Λ appears twice. Under the root are a minus sign and Δ; to the right, the modulus $|\xi|$.



Original source details: GELFAND-P000125

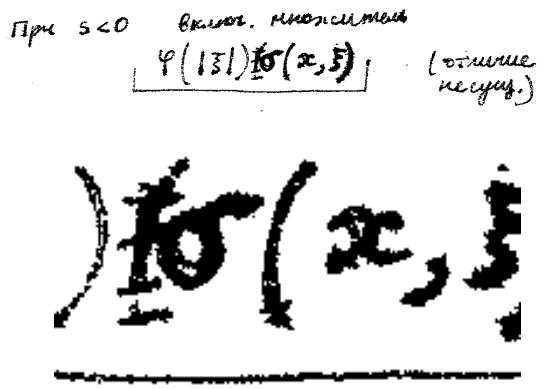
Diagram and typed description. Two axes with arrows; the horizontal one is labelled $|\xi|$. A mark 1 on the vertical axis. The curve runs along zero, rises and becomes almost horizontal; the label is $\varphi(|\xi|)$. The right end of the curve extends beyond the horizontal arrow.



For $s < 0$ include the factor [Before σ, a K-like sign is crossed out.]

$$\underline{\varphi(|\xi|)\sigma(x,\xi)}$$

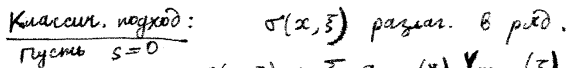
(the difference is inessential)



Original source details: GELFAND-P000125

Uncertain reading. Dense overlapping strokes at σ, including a K-like sign; the exact reading of the cancelled sign is undetermined. Beneath the product is a lower line with short vertical edges. To the right, a separate parenthesis: the difference is inessential.

Classical approach: expand $\sigma(x,\xi)$ in a series. Let $s = 0$.

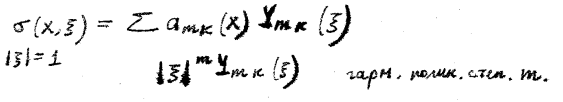


Original source details: GELFAND-P000125

Diagram and typed description. Classical approach: is underlined. To the right, expand $\sigma(x,\xi)$ in a series; beneath it, Let $s = 0$.

$$\sigma(x,\xi) = \sum a_{mk}(x) Y_{mk}(\xi)$$
$$|\xi| = 1$$

$|\xi|^m Y_{mk}(\xi)$ — a harmonic polyn. of degree m.

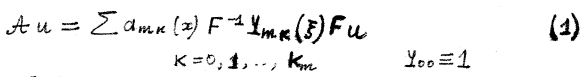


Original source details: GELFAND-P000125

Diagram and typed description. Both occurrences of Y have subscript mk. Beneath the first formula is $|\xi| = 1$; the next line has $|\xi|^m Y_{mk}(\xi)$ and harmonic polyn. of degree m.

$$Au = \sum a_{mk}(x) F^{-1} Y_{mk}(\xi) Fu \qquad (1)$$

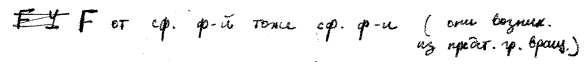
$k = 0, 1, \ldots, k_m$, $Y_{00} \equiv 1$.



Original source details: GELFAND-P000125

Diagram and typed description. The formula has number (1) at the right. Beneath it, the range $k = 0, 1, \ldots, k_m$ and a separate $Y_{00} \equiv 1$.

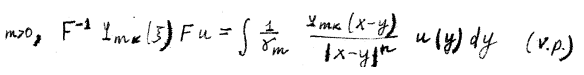
[Crossed out: FY[?].] F of spherical fns. also gives spherical fns. (they arise from reps. of the rotation grp.)

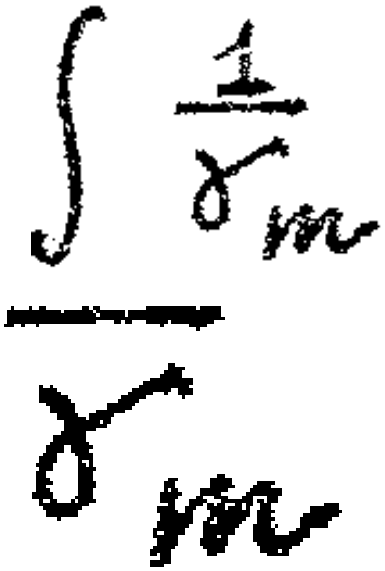


Original source details: GELFAND-P000125

Uncertain reading. Before the active F, F- and Y-like signs are crossed out. The parenthesis about representations of the rotation group is at the right of the line on spherical functions.

$$m > 0, \qquad F^{-1} Y_{mk}(\xi) Fu = \int \frac{1}{\gamma_m[?]} \frac{Y_{mk}(x-y)}{|x-y|^n} u(y)\, dy \qquad (\text{v.p.})$$





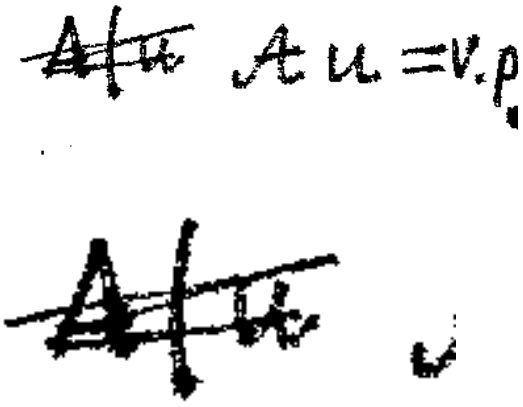
Original source details: GELFAND-P000125

Uncertain reading. In the denominator of the separate fraction $1/(\gamma_m[?])$, a looped Greek sign with an upper crossing stroke; candidates γ_m and δ_m. Subscript m is visible.

[Crossed out: Au[?].]

$$Au = \text{v.p.} \int K(x, x-y)u(y)\,dy$$

(the calculations are formal)



Uncertain reading. To the left of the active Au, an earlier Au-like beginning is crossed out with several intersecting strokes, including a tall vertical one.

Original source details: GELFAND-P000125

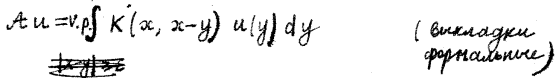


Diagram and typed description. The active Au is to the right of the cancelled beginning. The integrand is $K(x, x-y)u(y)\,dy$, preceded by v.p.; to the right, the parenthesis the calculations are formal.

Original source details: GELFAND-P000125

3rd def. Consider the fn.

[Crossed out: $x - y$[?], parentheses and a z-like sign.] $K(x, z)$ pos. homog. in z of degree $-n$.

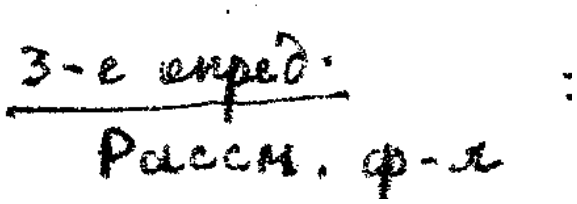
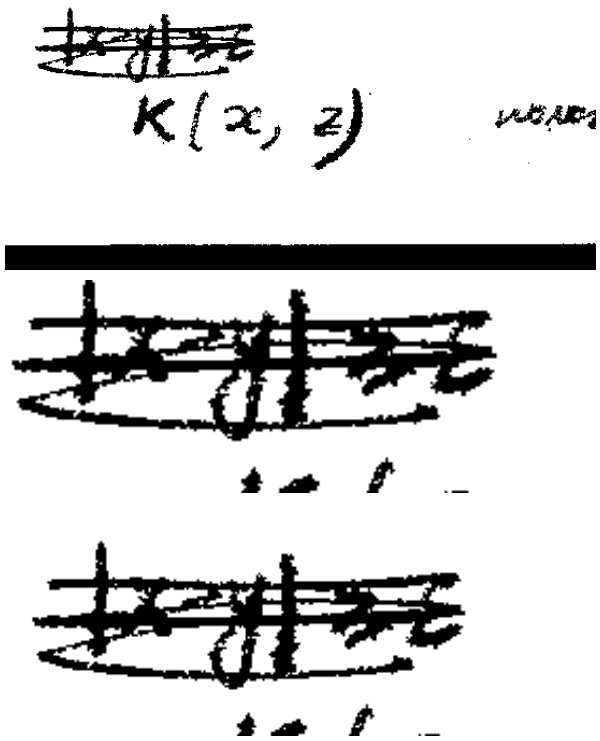


Diagram and typed description. A separate line beneath 3rd def. Then Consider the fn.

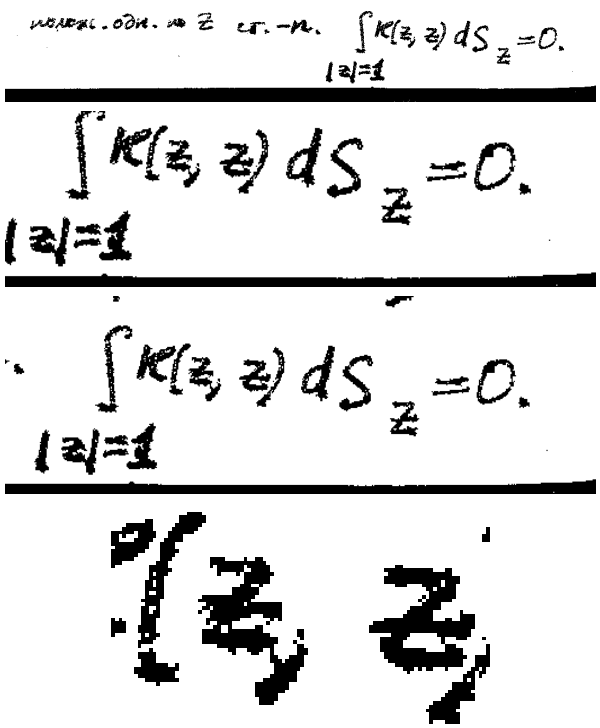
Original source details: GELFAND-P000125



Uncertain reading. Above the active $K(x, z)$, densely cancelled signs: an x-y-like portion, parentheses and a z-like sign at the right. The order and complete reading of the cancelled argument are undetermined.

Original source details: GELFAND-P000125

$$\int_{|z|=1} K(z[?], z)\, dS_z = 0.$$



Uncertain reading. The end says degree $-n$. Beneath the integral is $|z| = 1$, and dS has subscript z. The first argument of K is a z- / ξ- / x-like sign with dark horizontal strokes; its exact reading is undetermined. The second argument is z-like.

Original source details: GELFAND-P000125

Diagram and typed description. The frame around 80 has left and bottom sides.

Original source details: GELFAND-P000126

Let it stabilize as $x \to \infty$.

$$\sigma(x,\xi) = \sigma(\infty,\xi) + \sigma'(x,\xi)$$

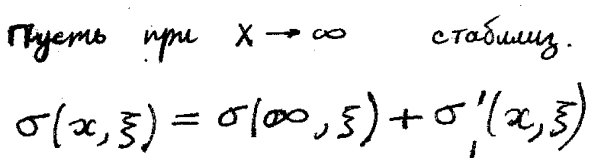
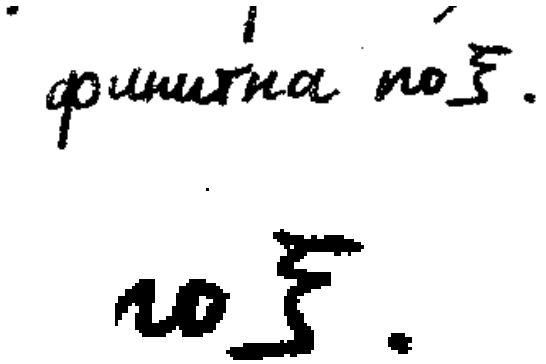


Diagram and typed description. The first line has $x \to \infty$. In the decomposition, the last σ has a prime; the first summand has argument ∞.

Original source details: GELFAND-P000126

σ' is compactly supported in ξ.



Manuscript notation. The note "compactly supported in ξ" uses the same form of ξ as the decomposition above.

Original source details: GELFAND-P000126

$$\sigma(x,D) = \sigma(\infty,D) + \sigma'(x,D)$$

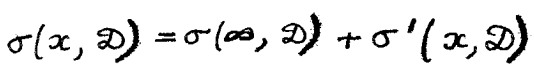
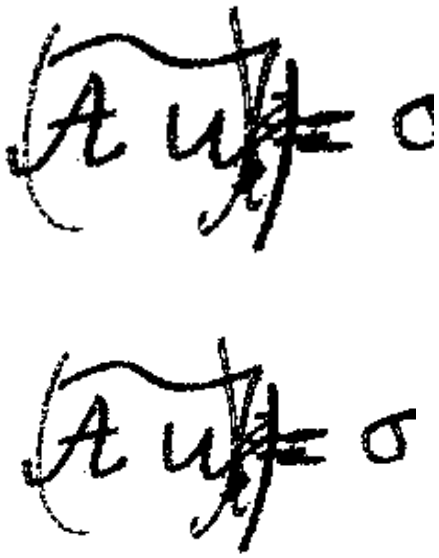


Diagram and typed description. The second argument is D in all three places. The last σ has a prime and is preceded by a plus sign.

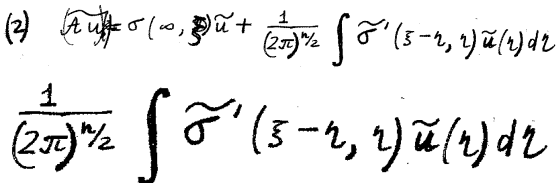
Original source details: GELFAND-P000126

$$(2)\quad \widetilde{Au}[?] = \sigma(\infty, \xi[?])\widetilde{u} + \frac{1}{(2\pi)^{n/2}} \int \widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta$$



Uncertain reading. A wavy line above Au; surrounding arcs at the left and below. The right portion at u and the equality sign is densely overwritten; the exact reading of the corrected strokes is undetermined.

Original source details: GELFAND-P000126



Uncertain reading. The second argument in $\sigma(\infty, \xi[?])$ is overwritten and ξ-like. The denominator is $(2\pi)^{n/2}$; the integrand is $\widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta$. Number (2) is at the left.

Original source details: GELFAND-P000126

Cohn and Nirenberg

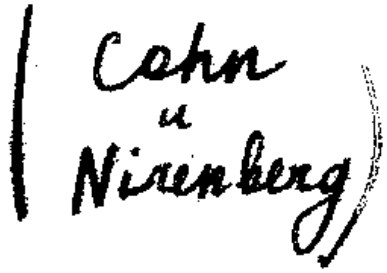


Diagram and typed description. In the side bracket, the Latin names Cohn and Nirenberg, with and between them. The left line is almost straight and vertical, and the right line is curved.

Original source details: GELFAND-P000126

$\widetilde{\sigma}$ is the Fourier transf. in the 1st variable.

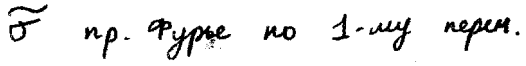
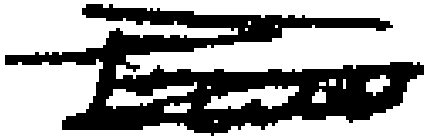


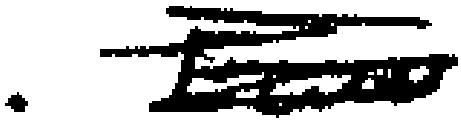
Diagram and typed description. A wavy line above σ. The end says in the 1st variable, with digit 1.

Original source details: GELFAND-P000126

To what extent are they equiv.? If the symbols are very good. [Crossed out: Was[?].]

Насколько они эквивал.? Если симв. очень хорошие. ~~Если~~





Uncertain reading. After good., a short cancelled word resembling Was; its middle strokes are covered by intersecting lines.

Original source details: GELFAND-P000126

In the general case the defs. are inequivalent (lead to different classes of oper.)

В общем случае опр. неэквив. (прив. к разн. кл. опер.)

Original source details: GELFAND-P000126

Diagram and typed description. After inequivalent, a separate parenthesis: lead to different classes of oper.

In which spaces the oper. is bounded.

$$H^\ell = W_2^{(\ell)}, \qquad \|u\|_\ell^2 = \int |\widetilde{u}|^2 (1+|\xi|^2)^\ell \, d\xi$$

В каких пр-вах опер. огран.

$H^\ell = W_2^{(\ell)}$ $\|u\|_\ell^2 = \int |\widetilde{u}|^2 (1+|\xi|^2)^\ell d\xi$

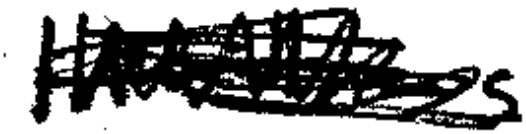
Original source details: GELFAND-P000126

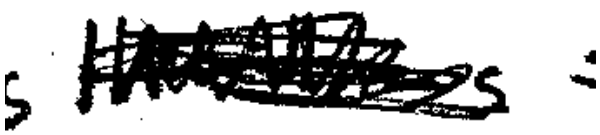
Diagram and typed description. The superscript of H is ℓ; W has subscript 2 and superscript (ℓ). Both the norm and weight have ℓ; $|\widetilde{u}|$ and $|\xi|$ are squared.

$$\|Au\|_{\ell-s} \quad \text{[Crossed out: } \|[?]\|_s\text{]} \quad \leq C\|u\|_\ell$$

for a smooth fn. σ, expansion in a series is rapid.

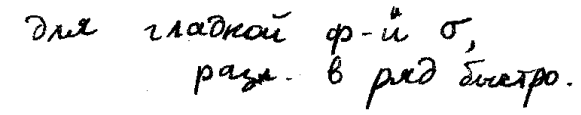
$\|Au\|_{\ell-s}$ [illegible] $\leq C\|u\|_\ell$





Uncertain reading. Between the active left norm and $\leq C\|u\|_\ell$, a densely crossed-out term. Norm-like vertical lines and an s-like sign at the lower right are visible; the interior reading is undetermined.

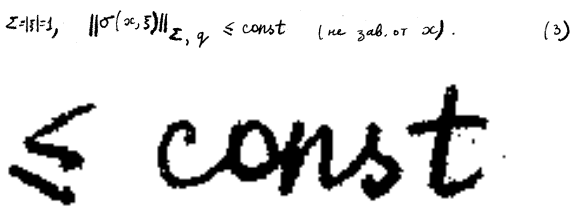
Original source details: GELFAND-P000126



Original source details: GELFAND-P000126

Diagram and typed description. Two lines at the right: for a smooth fn. σ, then expansion in a series is rapid.

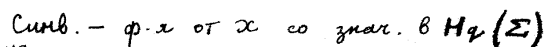
$$\Sigma : |\xi| = 1, \qquad \|\sigma(x,\xi)\|_{\Sigma,q} \leq \text{const} \quad (\text{indep. of } x). \qquad (3)$$



Original source details: GELFAND-P000126

Diagram and typed description. The norm has subscripts Σ, q. The sign before const has a lower line; at the right, the parenthesis indep. of x. Number (3) is at the right.

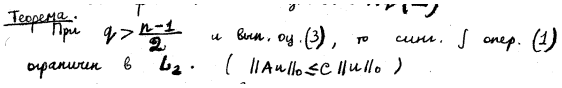
The symbol is a fn. of x with values in $H_q(\Sigma)$.



Original source details: GELFAND-P000126

Diagram and typed description. The subscript of H is q, with Σ in parentheses.

Theorem. For $q > \dfrac{n-1}{2}$ and est. (3) holding, the sing. $\int$ oper. (1) is bounded in L_2. ($\|Au\|_0 \leq C\|u\|_0$)



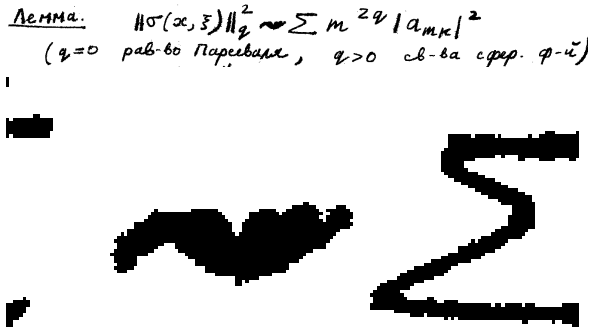
Original source details: GELFAND-P000126

Diagram and typed description. Theorem. is underlined. The condition is $q > (n-1)/2$, with references (3) and (1); the end has L_2 and a separate parenthesis with norms having subscript 0.

Lemma.

$$\|\sigma(x,\xi)\|_q^2 \sim \sum m^{2q} |a_{mk}|^2$$

($q = 0$ Parseval's equality, $q > 0$ properties of spherical fns.)

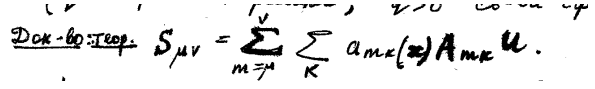


Original source details: GELFAND-P000126

Diagram and typed description. Lemma. is underlined. A dense wavy sign between the norm and sum; the exponent of m is $2q$, and a has subscripts mk. Beneath it, a separate parenthesis with $q = 0$ and $q > 0$.

Proof of the thm.

$$S_{\mu\nu} = \sum_{m=\mu}^{\nu} \sum_{k} a_{mk}(x) A_{mk} u.$$

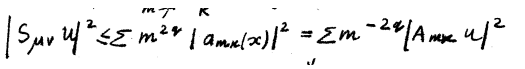


Original source details: GELFAND-P000126

Diagram and typed description. Proof of the thm. is underlined. At the left, $S_{\mu\nu}$; at the right, the first sum has limits $m = \mu$ and ν, the second has subscript k, and the end is $A_{mk}u$.

$$|S_{\mu\nu}u|^2 \le \sum m^{2q}|a_{mk}(x)|^2 = \sum m^{-2q}|A_{mk}u|^2$$

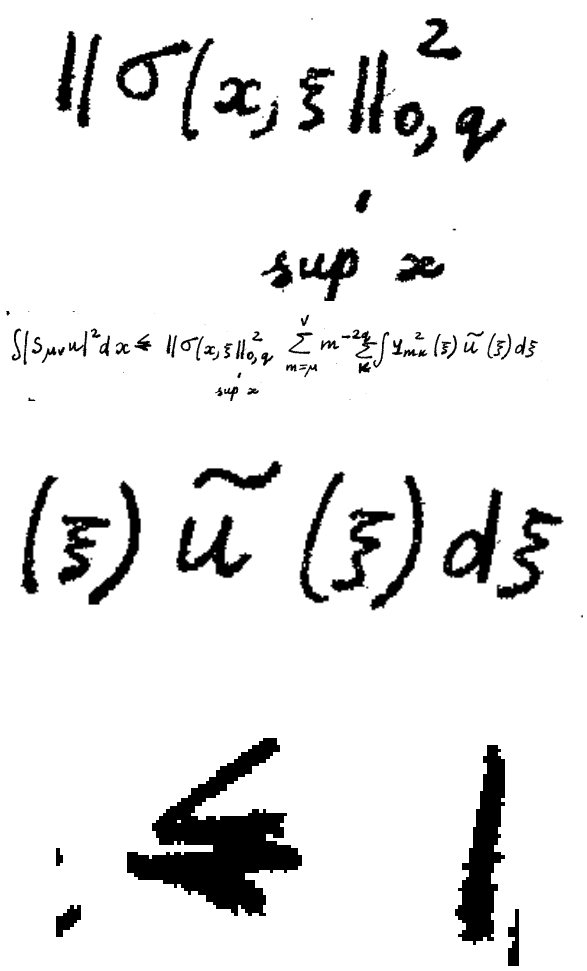
Uncertain reading. Two horizontal lines between the sums, an equality sign. The exponent of the first m is $2q$, and that of the second is $-2q$.



Original source details: GELFAND-P000126

$$\int |S_{\mu\nu}u|^2\,dx \le [?]\|\sigma(x,\xi\|^2_{0,q}$$
$$\sup_x$$
$$\sum_{m=\mu}^{\nu} m^{-2q} \sum_k \int Y^2_{mk}(\xi)\widetilde{u}(\xi)\,d\xi$$

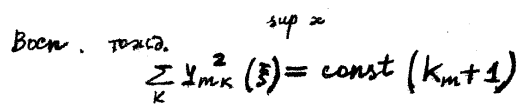
Uncertain reading. After dx, an overwritten $\le$- / $\Leftarrow$-like sign. No closing argument parenthesis is visible after ξ in the norm; sup x is written separately beneath it. Y_{mk} is squared; it is followed only by $\widetilde{u}(\xi)$, without a visible modulus or square.



Original source details: GELFAND-P000126

Use the identity

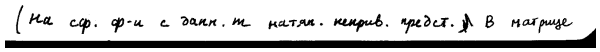
$$\sum_k Y_{mk}^2(\xi) = \text{const}\,(k_m + 1)$$



Original source details: GELFAND-P000126

Diagram and typed description. Use the identity precedes the formula. The sum has subscript k; at the right, the parenthesis $(k_m + 1)$.

(the spherical fns. for fixed m span an irred. rep.)[?] In the matrix

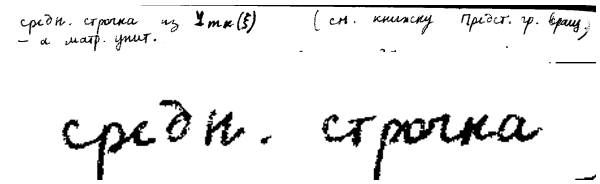


Original source details: GELFAND-P000126

Uncertain reading. The last line has «натян. неприв. предст.» (“span an irred. rep.”). Before “In the matrix” is a thick mark resembling a closing parenthesis. Its reading remains uncertain.

GELFAND-P000127

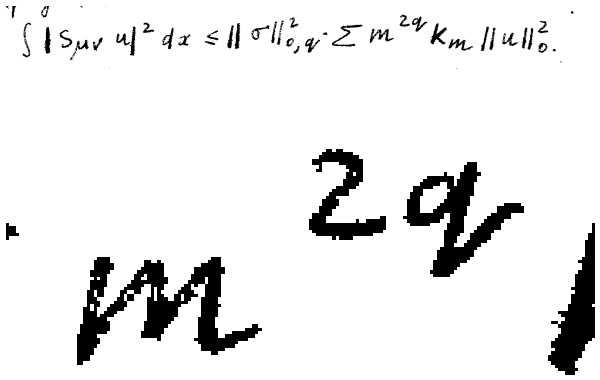
middle row of $Y_{mk}(\xi)$ (see the book Reps. of the rotation grp.) — and the matrix is unitary.



Original source details: GELFAND-P000127

Diagram and typed description. The first word is middle; then row of $Y_{mk}(\xi)$. At the right, the parenthesis see the book Reps. of the rotation grp.; beneath it, a separate line — and the matrix is unitary.

$$\int |S_{\mu\nu} u|^2 \, dx \le \|\sigma\|_{0,q}^2 \cdot \sum m^{2q} k_m \|u\|_0^2.$$

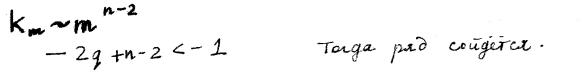


Original source details: GELFAND-P000127

Diagram and typed description. The exponent of m is $2q$, with no visible minus before 2. A separate multiplication dot follows the norm with subscripts $0, q$.

$$k_m \sim m^{n-2}, \qquad -2q + n - 2 < -1$$

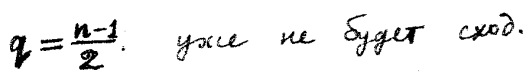
Then the series will converge.



Original source details: GELFAND-P000127

Diagram and typed description. The first formula has k_m and exponent $n - 2$. The next line begins with a minus before $2q$, then $+n - 2 < -1$. At the right, Then the series will converge.

$q = \frac{n-1}{2}$. there will no longer be conv.



Original source details: GELFAND-P000127

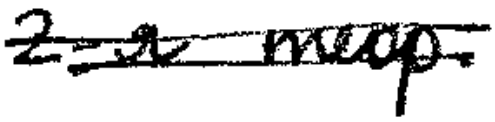
Diagram and typed description. A period follows the fraction $(n-1)/2$; then there will no longer be conv.

No smoothness in ξ is needed.

Original source details: GELFAND-P000127

Diagram and typed description. After in, a dense ξ-shaped sign; the phrase says no smoothness is needed.

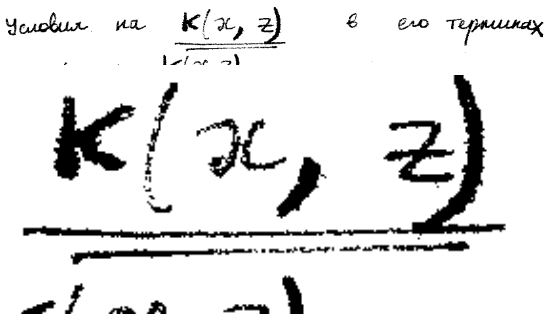
[Crossed out: 2nd thm.]



Original source details: GELFAND-P000127

Diagram and typed description. 2nd thm. is crossed out with two long lines.

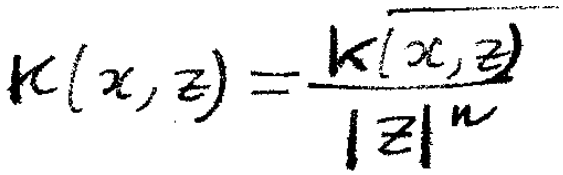
Conditions on $K(x,z)$ in its terms



Original source details: GELFAND-P000127

Diagram and typed description. Two separate lines beneath $K(x,z)$. Conditions on is at the left, and in its terms at the right.

$$K(x,z) = \frac{k(x,z)}{|z|^n}$$



Original source details: GELFAND-P000127

Diagram and typed description. At the left, $K(x,z)$; at the right, numerator $k(x,z)$ and denominator $|z|^n$.

$$\|k(X,z)\|_q \le \text{const} \qquad q > -\frac{1}{2}$$
$$(\text{Norm on } |z|=1)$$

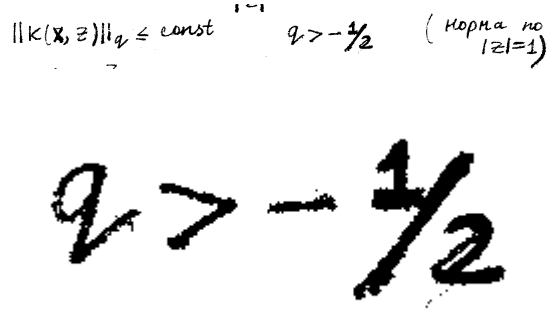


Diagram and typed description. The first argument of k is X-shaped. A short horizontal minus precedes $1/2$; a separate parenthesis says Norm on $|z| = 1$.

Original source details: GELFAND-P000127

Example ($n = 2$)

$$K(z) = \frac{z_1}{(z_1^2 + z_2^2)^{3/2}}.$$

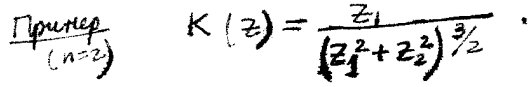


Diagram and typed description. Example is underlined, with the parenthesis $n = 2$ beneath it. The numerator is z_1; the denominator is the sum of two squares to the power $3/2$.

Original source details: GELFAND-P000127

$$\widetilde{Au}(\xi) = \sigma(\infty,\xi)\widetilde{u}(\xi) + \frac{1}{(2\pi)^{n/2}} \int \widetilde{\sigma}'(\xi-\eta,\eta)\widetilde{u}(\eta)\,d\eta$$

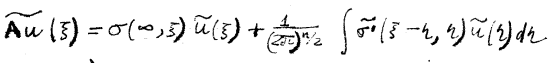


Diagram and typed description. Wavy lines above Au, u and σ'. The denominator is $(2\pi)^{n/2}$. The arguments of $\widetilde{\sigma}'$ are $\xi - \eta, \eta$.

Original source details: GELFAND-P000127

$\sigma'(x,\xi)$ is smooth in X through order $n+1$.

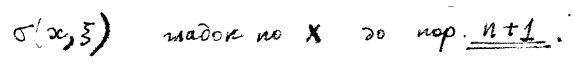
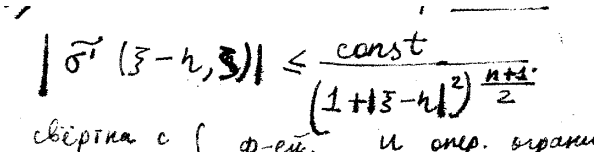


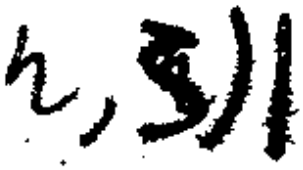
Diagram and typed description. After in, a separate X-shaped sign. Two lines beneath $n+1$.

Original source details: GELFAND-P000127

$$|\widetilde{\sigma}'(\xi-\eta,\xi[?])| \le \frac{\text{const}}{(1+|\xi-\eta|^2)^{(n+1)/2}}.$$

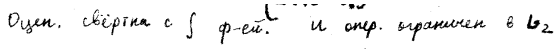
Uncertain reading. The overwritten last argument looks more like ξ. In the repeated formula above, the corresponding argument is η. The denominator is $(1+|\xi-\eta|^2)^{(n+1)/2}$.





Original source details: GELFAND-P000127

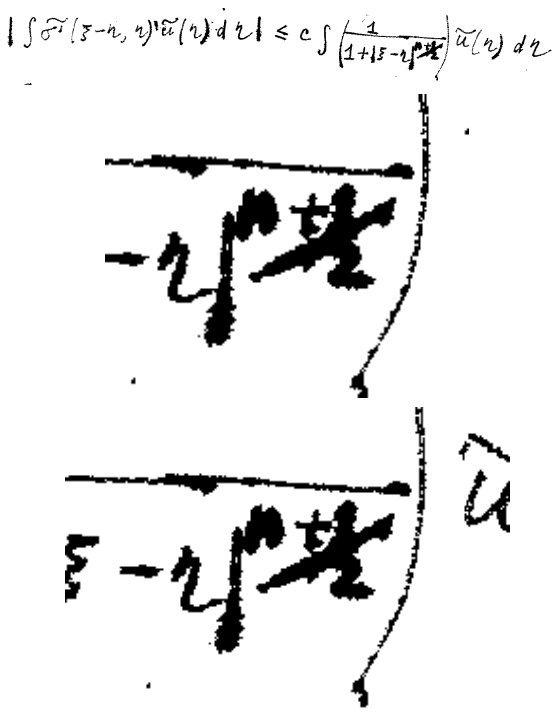
Est. by convolution with an integrable fn. And the oper. is bounded in L_2.



Original source details: GELFAND-P000127

Diagram and typed description. After “with”, the manuscript writes the symbol $\int$ before the abbreviation for “function”. The second phrase is And the oper. is bounded in L_2.

$$\left| \int \widetilde{\sigma}'(\xi-\eta,\eta)\widetilde{u}(\eta)\,d\eta \right| \le c \int \left(\frac{1}{1+|\xi-\eta|^{\text{[illegible]}}} \right) \widetilde{u}(\eta)\,d\eta$$



Uncertain reading. The power of $|\xi-\eta|$ in the right denominator is densely corrected: $n+1$ / $(n+1)/2$; its exact reading is undetermined. Large parentheses surround the fraction; they are followed by $\widetilde{u}(\eta)$ without a visible modulus.

Original source details: GELFAND-P000127

1. Commutation formula.

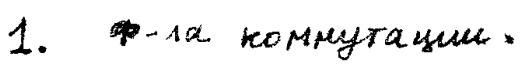


Diagram and typed description. Digit 1 with a period; then Commutation formula.

Original source details: GELFAND-P000127

2. $A \cdot B - \underset{\sigma_1\sigma_2}{A \circ B}$ — lower order.

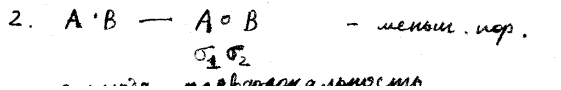


Diagram and typed description. A dot between the first A and B, and a circle in the second product. Beneath $A \circ B$, a separate $\sigma_1\sigma_2$; at the right, — lower order.

Original source details: GELFAND-P000127

Hence pseudolocality.

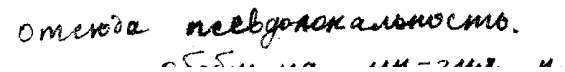
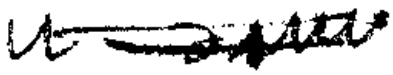


Diagram and typed description. A separate phrase Hence pseudolocality., with a period.

Original source details: GELFAND-P000127

[End of the expression crossed out: $u \longrightarrow$ [?].]

Generalization to manifolds and elliptic theory.



Uncertain reading. A u-like letter at the left, a long line with an arrow to the right and a densely cancelled endpoint. Looped and vertical strokes at the end; the complete reading is undetermined.

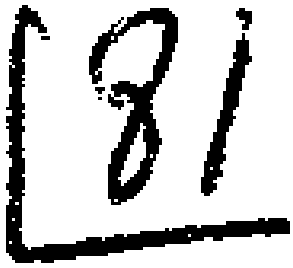
Original source details: GELFAND-P000127

Original source details: GELFAND-P000127

Diagram and typed description. The concluding phrase is Generalization to manifolds and elliptic theory. Elliptic is written with dense strokes.

GELFAND-P000128

Let us start again.

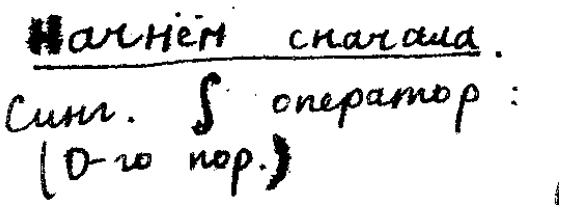


Original source details: GELFAND-P000128

Diagram and typed description. Page number 81 with the left and bottom sides of a frame.

81

Sing. $\int$ operator: (of order 0)



Original source details: GELFAND-P000128

Diagram and typed description. Let us start again. is underlined. Below, Sing. and a separate integral-like sign; of order 0 in parentheses.

$$(1)\quad \widetilde{Au}(\xi) = F_{x\to\xi}\sigma(x,\xi)u(x)$$
$$(2)\quad \widetilde{Au}(\xi) = \sigma(x,\xi)\widetilde{u}(\xi) +$$

(C. and N.) (early works)

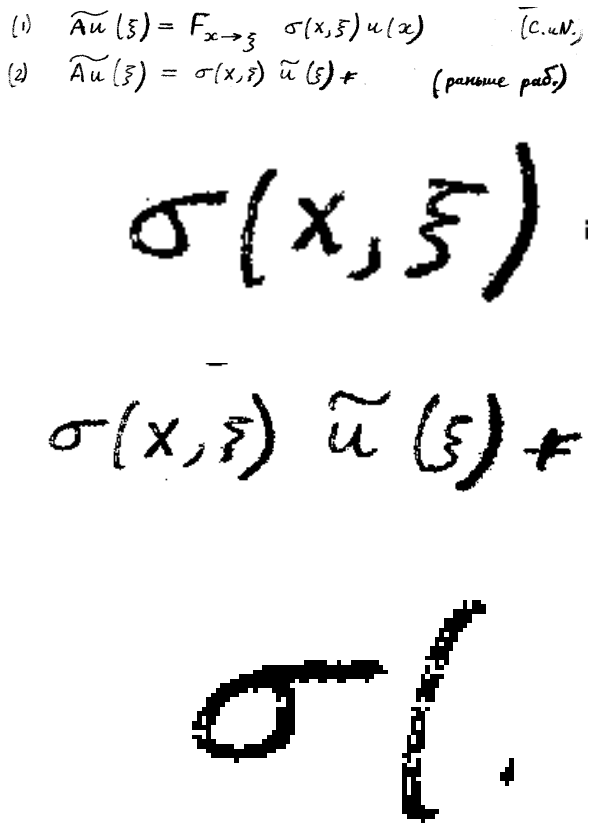


Diagram and typed description. In both initial definitions, σ has no detached prime. In the second line, a plus follows $\widetilde{u}(\xi)$; no continuation is written. At the right, separate parentheses C. and N. and early works.

Original source details: GELFAND-P000128

$$(1)\quad \widetilde{Au}(\xi) = \sigma(\infty,\xi)u(\xi) + \int \widetilde{\sigma}'(\xi-\eta,\xi)\widetilde{u}(\eta)\,d\eta$$

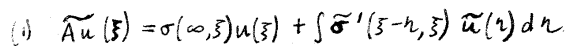
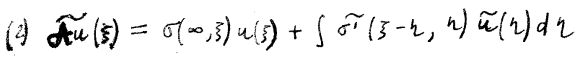


Diagram and typed description. The first term contains $u(\xi)$ with no tilde. In the integral, σ has a detached prime and a tilde; the second argument is ξ.

Original source details: GELFAND-P000128

$$(2) \quad \widetilde{\mathcal{A}[?]u}(\xi) = \sigma(\infty, \xi)u(\xi) + \int \widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta$$

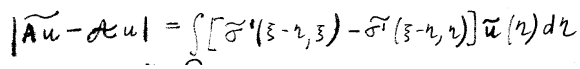


Original source details: GELFAND-P000128

Uncertain reading. The second operator sign is heavily overwritten: $\mathcal{A}$- / B-like, with a tilde above it and u. The exact overwritten letter is unresolved. The first term again has $u(\xi)$ without a tilde.

But they differ by an operator of lower order (-1).

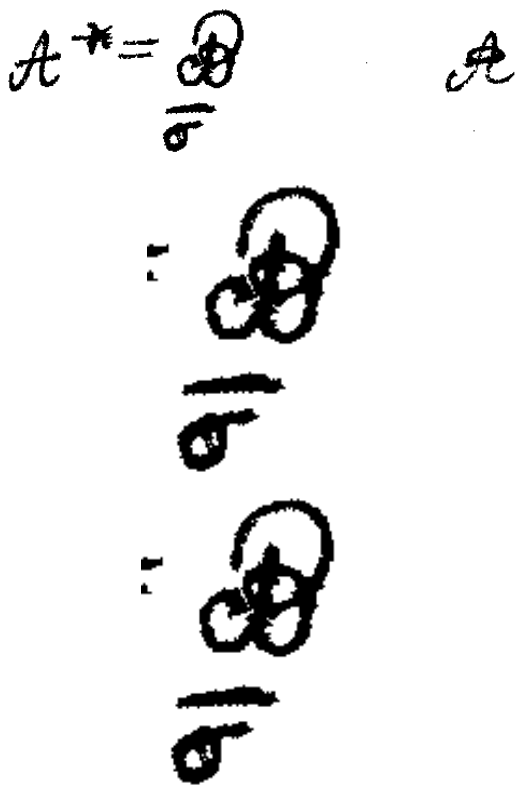
$$|\widetilde{Au} - \mathcal{A}u| = \int [\widetilde{\sigma}'(\xi - \eta, \xi) - \widetilde{\sigma}'(\xi - \eta, \eta)]\widetilde{u}(\eta)\, d\eta$$



Original source details: GELFAND-P000128

Diagram and typed description. At the left, vertical bars and a tilde only above the first Au; the second operator is calligraphic. At the right, an integral without an outer modulus, the difference in square brackets.

$$\mathcal{A}^* = B[?] \qquad \mathcal{A}$$
$\overline{\sigma}$



Uncertain reading. After calligraphic $A^* =$, a heavily overwritten looped B-like sign. Beneath it, $\overline{\sigma}$. At the right, a separate calligraphic A-like letter; its connection to the equality is unresolved.

Original source details: GELFAND-P000128

(All this is done by Taylor's formula)

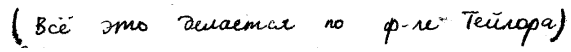
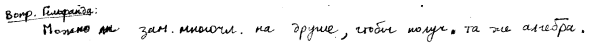


Diagram and typed description. The entire phrase is in parentheses; the source contraction for formula and Taylor's name.

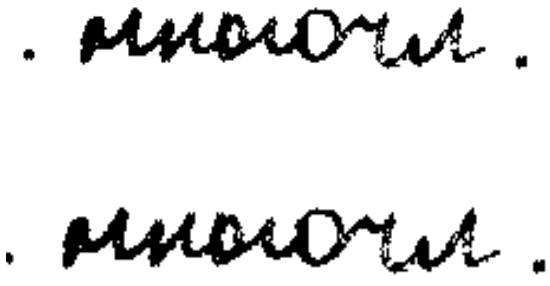
Original source details: GELFAND-P000128

Gelfand's question:

Can polyn. be replaced by others, so as to obtain the same algebra.



Manuscript notation. The manuscript abbreviates "polynomials" as «многочл.». The heading "Gelfand's question" is underlined.



Original source details: GELFAND-P000128

The symbol is defined on the cotang. bundle.

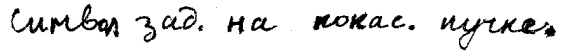
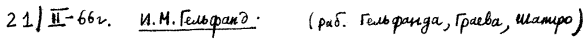


Diagram and typed description. A separate phrase: The symbol is defined on the cotang. bundle.

Original source details: GELFAND-P000128

21/II–66 I. M. Gelfand. (works of Gelfand, Graev, Shapiro)

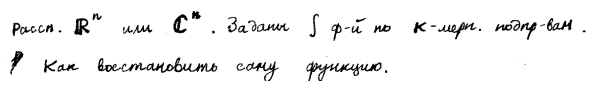


Original source details: GELFAND-P000128

Diagram and typed description. II has horizontal lines above and below. Date 21/II–66. The name I. M. Gelfand. is underlined; at the right, parentheses with three surnames.

Consider R^n or C^n. The $\int$ of functions over k-dim. subspaces are given.

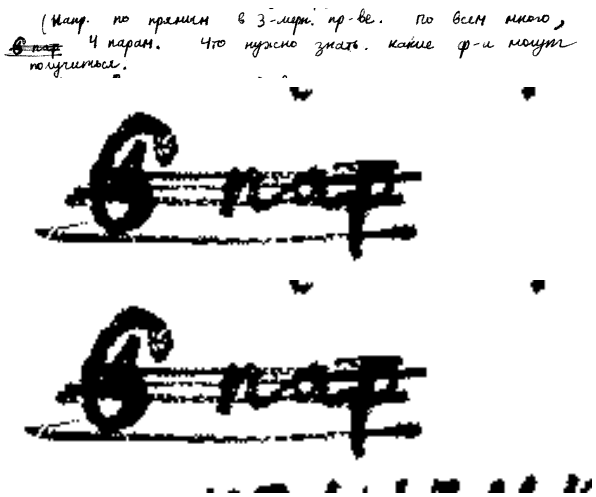
? How to recover the function itself.



Original source details: GELFAND-P000128

Diagram and typed description. At the left, a separate question mark. In the first line, R^n, C^n and a separate integral-like sign before the contraction for functions.

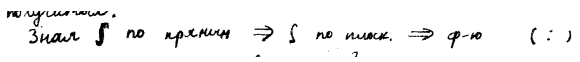
(E.g. over lines in 3-dim. space. Over all of them is too much, [Crossed out: 6 param.[?]] 4 param. What needs to be known. which functions can result.



Original source details: GELFAND-P000128

Uncertain reading. Before 4 param., 6 пар. / 6-пар. is crossed out; horizontal lines cross the contraction, and a separate hyphen is unresolved. The initial parenthesis in this passage has no visible closure.

Knowing $\int$ over lines $\Rightarrow$ $\int$ over planes $\Rightarrow$ the function (:)



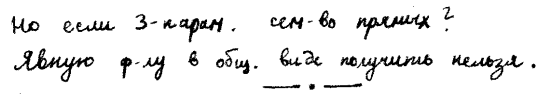
Original source details: GELFAND-P000128

Diagram and typed description. Two separate $\Rightarrow$ signs; after the function, parentheses containing a colon.

But if a 3-param. family of lines?

An explicit formula cannot be obtained in the general case.

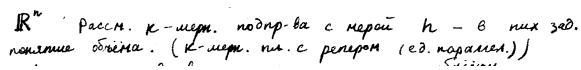
— . —



Original source details: GELFAND-P000128

Diagram and typed description. After the question, a separate sentence about an explicit formula. Below, two horizontal strokes with a dot between.

R^n Consider k-dim. subspaces with measure h — a notion of volume is given in them. (k-dim. plane with a frame (unit parallelep.))

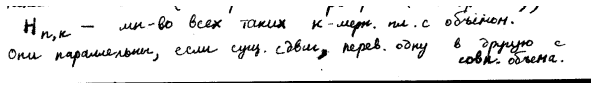


Original source details: GELFAND-P000128

Diagram and typed description. After measure, a separate h-like sign. At the end, two nested parentheses, frame and unit parallelep.

$H_{n,k}$ — the set of all such k-dim. planes with volume.

They are parallel if there exists a translation taking one into the other with the volumes made to coincide.

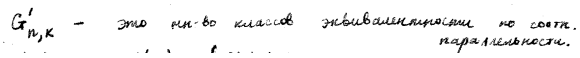


Original source details: GELFAND-P000128

Diagram and typed description. The subscript of H is n, k. The final sentence ends with the volumes made to coincide.

GELFAND-P000129

$G'_{n,k}$ — this is the set of equivalence classes under the relation of parallelism.

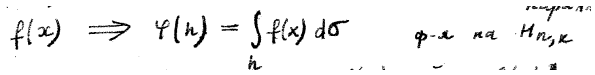


Original source details: GELFAND-P000129

Diagram and typed description. A detached prime above G, and subscripts n, k. The phrase ends with parallelism.

$$f(x) \Rightarrow \varphi(h) = \int_h f(x)\, d\sigma$$

a function on $H_{n,k}$.

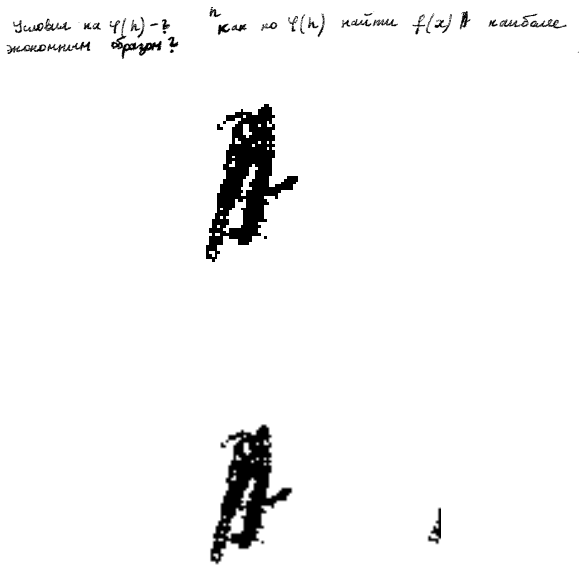


Original source details: GELFAND-P000129

Diagram and typed description. The integral has subscript h; at the right, $d\sigma$. A separate phrase: a function on $H_{n,k}$.

Conditions on $\varphi(h)$ — ?

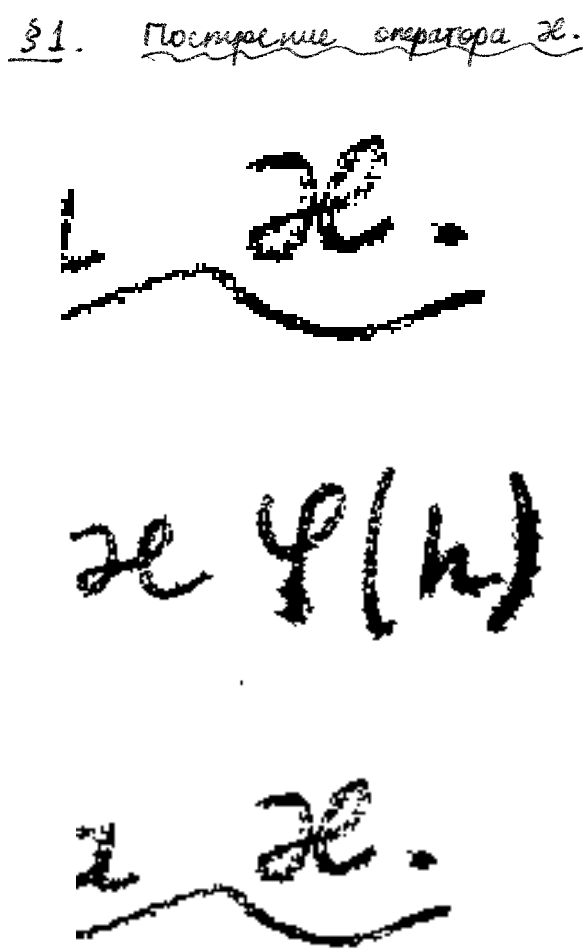
how to find $f(x)$ from $\varphi(h)$ [Crossed-out sign: illegible] in the most economical way?



Original source details: GELFAND-P000129

Uncertain reading. After $f(x)$, a dense crossed-out B-like sign; its exact reading is unresolved. The active sentence continues with most.

§1. Construction of the operator $\mathcal{X}$[?].



Uncertain reading. The operator sign is X-like / χ; the exact notation is unresolved. The same sign recurs in the concluding line. A straight underline beneath §1, a wavy one beneath the title.

Original source details: GELFAND-P000129

S_β. $\beta \in R^n$. Consider all k-dim. [Crossed out: subsp.] [Inserted above: hyperpl.], passing through β.

One parallel will be singled out.

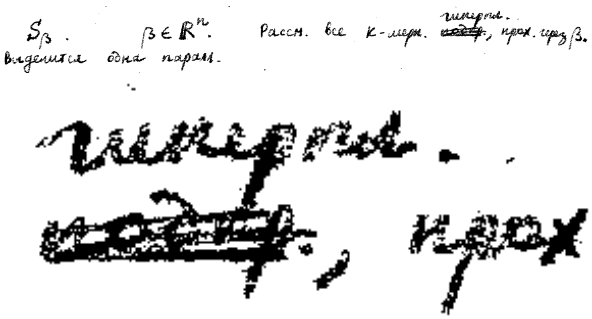


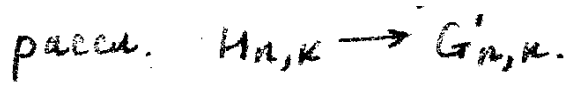
Diagram and typed description. Subsp. is crossed out, hyperpl. inserted above the line. The active line has k-dim., passing through β; below, One parallel will be singled out.

Original source details: GELFAND-P000129

$$S_\beta : G'_{n,k} \longrightarrow H_{n,k}$$

This is a global section.

bundle $H_{n,k} \to G'_{n,k}$.



Original source details: GELFAND-P000129

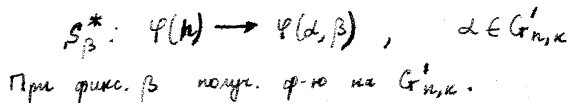
Manuscript notation. Before the reverse arrow, «рассл.» (“bundle”).

Original source details: GELFAND-P000129

Diagram and typed description. The subscript of S is β. In the first map, the arrow goes from $G'_{n,k}$ to $H_{n,k}$; in the next line, the opposite direction.

$$S^*_\beta : \quad \varphi(h) \to \varphi(\alpha, \beta), \qquad \alpha \in G'_{n,k}$$

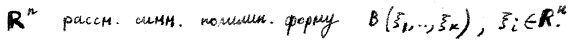
With β fixed, we obtain a function on $G'_{n,k}$.



Original source details: GELFAND-P000129

Diagram and typed description. A superscript star on S_β. An arrow after $\varphi(h)$, then $\varphi(\alpha, \beta)$; at the right, $\alpha \in G'_{n,k}$.

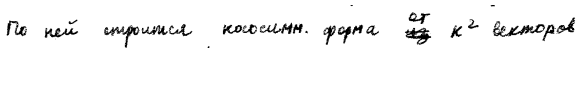
R^n consider a symm. multilinear form $B(\xi_1, \ldots, \xi_k)$, $\xi_i \in R^n$.



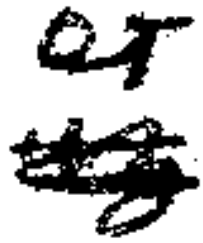
Original source details: GELFAND-P000129

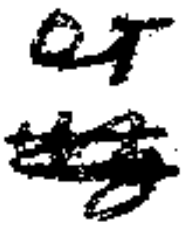
Diagram and typed description. Symm. multilinear form and $B(\xi_1, \ldots, \xi_k)$. The exponent on the last R is n.

From it is constructed a skew-symm. form [Crossed out: from[?]] [Inserted above: of] k^2 vectors



Uncertain reading. Beneath the inserted of, a short heavily crossed-out word, resembling source из; its exact reading is unresolved. Then k^2 vectors.





Original source details: GELFAND-P000129

$$\Xi = \begin{Vmatrix} \xi_{11} & \cdots & \xi_{1k} \\ \xi_{21} & \cdots & \xi_{2k} \\ \\ \xi_{k1} & \cdots & \xi_{kk} \end{Vmatrix}.$$

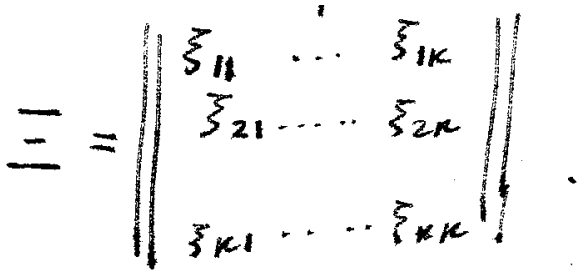
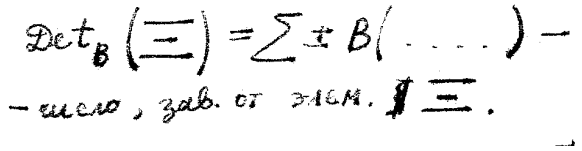


Diagram and typed description. Double vertical bars, rows with first indices 1, 2, k, and horizontal dots. There is a blank gap between the second and last rows; no vertical dots are visible.

Original source details: GELFAND-P000129

$$\mathrm{Det}_B(\Xi) = \sum \pm B(\ldots) —$$

— a number depending on the entries of [Crossed-out sign: illegible] Ξ.



Uncertain reading. Before the active Ξ, a short crossed-out vertical sign; its exact reading is unresolved. In the preceding line, $\sum \pm B(\ldots)$ and a separate stroke at the right.

Original source details: GELFAND-P000129

For vectors $\xi \in R^1$, $B = \xi_1 \cdots \xi_k$ this is the ordinary det.

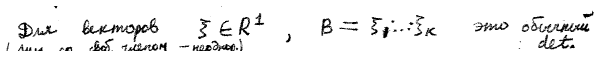
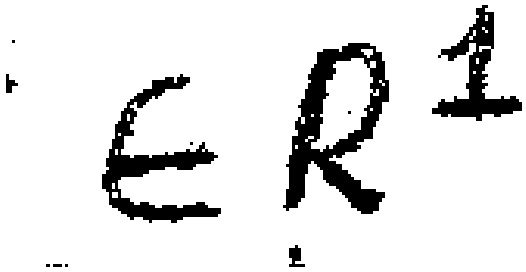
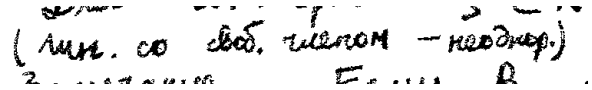


Diagram and typed description. The exponent on R is a separate digit 1. Then the product $\xi_1 \cdots \xi_k$ and ordinary det.



Original source details: GELFAND-P000129

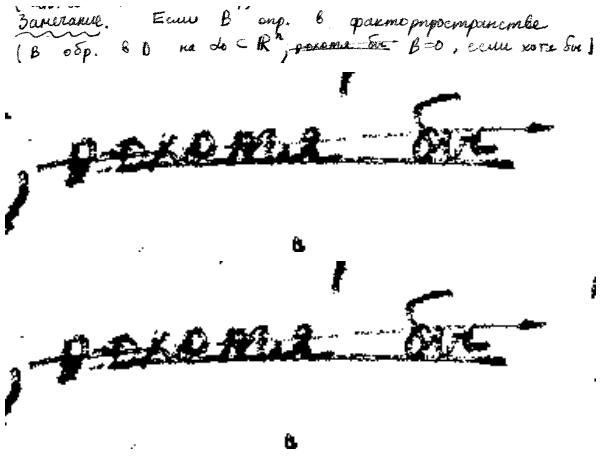
(lin. with a constant term — inhomog.)



Original source details: GELFAND-P000129

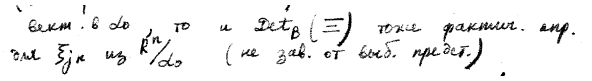
Diagram and typed description. A separate parenthesis: lin. with a constant term — inhomog.

Remark. If B is defined on a quotient space (B vanishes on $\alpha_0 \subset R^n$, [Crossed out: would have to[?]] $B = 0$ if at least 1 vector is in α_0, then $\mathrm{Det}_B(\Xi)$ too is in fact defined for ξ_{jk} from R^n/α_0 (does not depend on the choice of representative)



Original source details: GELFAND-P000129

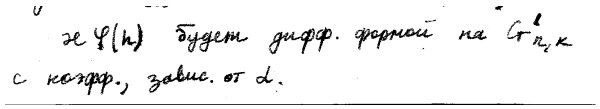
Uncertain reading. The source words resembling должны бы are crossed out; their exact reading is unresolved. The first word after the opening parenthesis is B-like, then vanishes on $\alpha_0 \subset R^n$. Remark. has a wavy underline.



Original source details: GELFAND-P000129

Diagram and typed description. After the condition $B = 0$ if at least 1 vector is in α_0, the text continues through R^n/α_0 and the parenthesis does not depend on the choice of representative. The initial parenthesis before B has no separate visible closure.

$\mathcal{X}$[?]$\varphi(h)$ will be a differential form on $G'_{n,k}$ with coefficients depending on α.

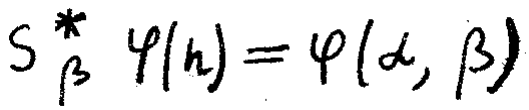


Original source details: GELFAND-P000129

Diagram and typed description. The operator sign before $\varphi(h)$ repeats the sign in the title. The last line reads with coefficients depending on α.

GELFAND-P000130

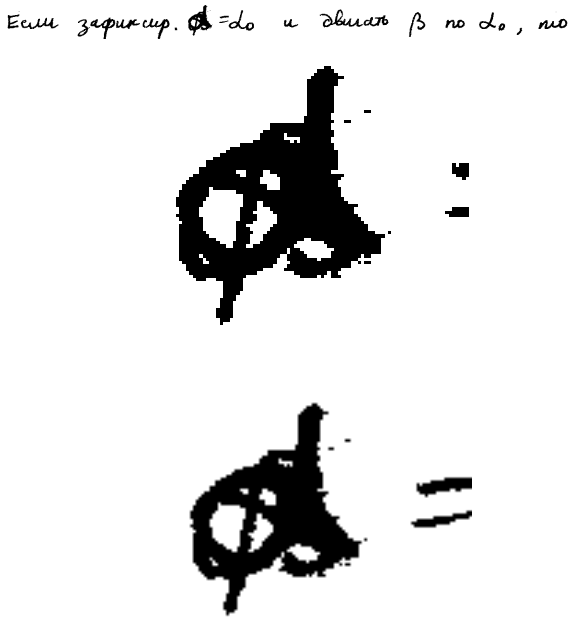
$$S^*_\beta \varphi(h) = \varphi(\alpha, \beta)$$



Original source details: GELFAND-P000130

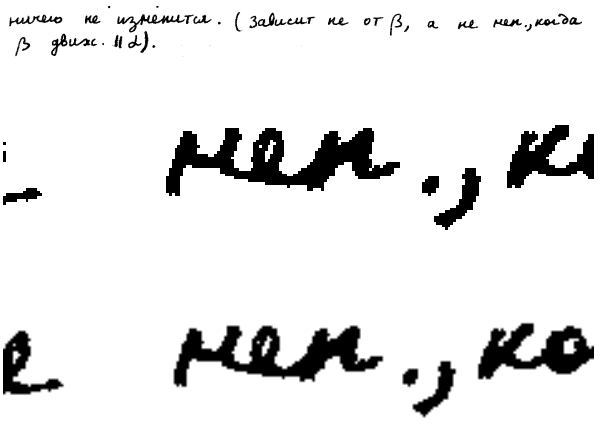
Diagram and typed description. Subscript β and superscript star on S. At the right, $\varphi(\alpha, \beta)$.

If we fix $\alpha[?] = \alpha_0$ and move β along α_0, nothing changes. (Depends not on β, but does not change when β moves $\| \alpha$).



Original source details: GELFAND-P000130

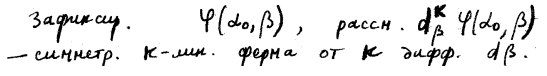
Uncertain reading. Before $= \alpha_0$, a heavily overwritten α-like letter with loops and a vertical stroke; the exact reading is unresolved.



Original source details: GELFAND-P000130

Manuscript notation. After “but does not”, the manuscript abbreviates “change” as «мен.». The parenthesis opens with “Depends not on”.

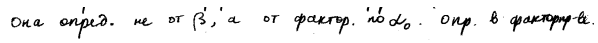
Fix $\varphi(\alpha_0, \beta)$, consider $d^k_\beta \varphi(\alpha_0, \beta)$ — — a symm. k-linear form of k differentials $d\beta$.



Original source details: GELFAND-P000130

Diagram and typed description. Superscript k and subscript β on d. A separate dash after the expression, and another at the start of the next line; then k-linear and k differentials $d\beta$.

It is defined not from β, but from the quotient by α_0. Defined on the quotient space.

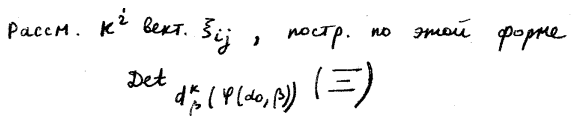


Original source details: GELFAND-P000130

Diagram and typed description. The phrase not from β, but from the quotient by α_0. Then Defined on the quotient space.

Consider k^2 vectors ξ_{ij}, construct from this form

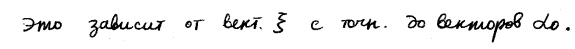
$$\mathrm{Det}_{d^k_\beta(\varphi(\alpha_0,\beta))}(\Xi)$$



Original source details: GELFAND-P000130

Diagram and typed description. The entire expression $d^k_\beta(\varphi(\alpha_0,\beta))$ is beneath Det; at the right, large parentheses containing Ξ. In the line above, k^2 vectors ξ_{ij}.

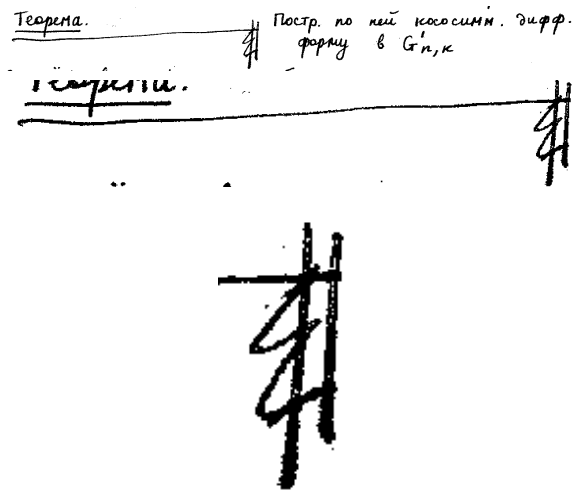
This depends on the vectors ξ up to vectors of α_0.



Original source details: GELFAND-P000130

Diagram and typed description. Up to vectors of α_0 at the end of a separate line.

Theorem. Construct from it a skew-symm. differential form in $G'_{n,k}$.

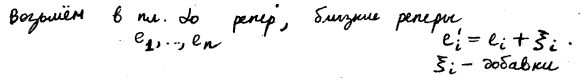


Original source details: GELFAND-P000130

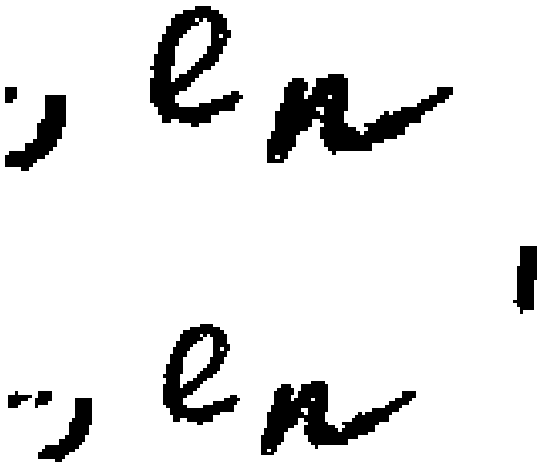
Diagram and typed description. Beneath Theorem., a separate long line to the right, two vertical strokes and two looped portions to their left. A typographic rendition of the strokes is provided alongside; there are no labels or stated geometric meaning.

Take a frame in the plane α_0, nearby frames $e_1, \dots, e_n$, $e'_i = e_i + \xi_i$.

ξ_i — increments.



Manuscript notation. The last frame index reads n. At the right, $e'_i = e_i + \xi_i$, with "increments" below.



Original source details: GELFAND-P000130

(planes up to small quantities of higher order)

They define the same plane $\Leftrightarrow \xi_i - \widetilde{\xi}_i \in \alpha_0$.

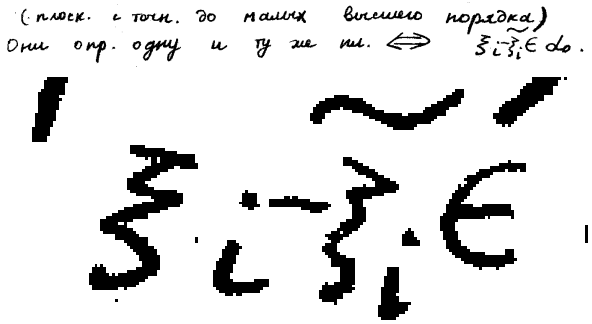


Diagram and typed description. After $\Leftrightarrow$, two ξ with subscripts i and a short minus between them; a tilde over the second. At the right, $\in \alpha_0$.

Original source details: GELFAND-P000130

k nearby planes

$$e'_{ij} = e_i + \xi_{ij} \qquad j = 1, \dots, k$$

Consider as Ξ a matrix defining k nearby planes.

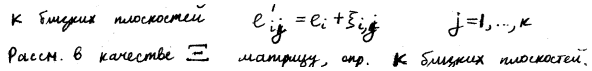


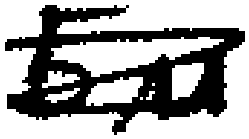
Diagram and typed description. Two subscripts i, j on e'; at the right, $j = 1, \dots, k$. The next line contains Ξ and k nearby planes.

Original source details: GELFAND-P000130

Theorem 1.

1. [Crossed-out B-like sign: illegible] $\mathrm{Det}_B(\Xi)$ depends only on the position of the planes, and not on the matrix itself.

Теорема 1. 1. Det$_B(\Xi)$ зав. только от положе. плоск., а не самой матр.



Original source details: GELFAND-P000130

Uncertain reading. Between 1. and the active Det, a dense B-like sign or short entry is crossed out; its full reading is unresolved.

2. The resulting form is a k-form (skew.) in $G'_{n,k}$.

2. Получ. форма есть к-форма (кос.) в $G'_{n,k}$.

Original source details: GELFAND-P000130

Diagram and typed description. The second item contains k-form, the parenthesis skew. and $G'_{n,k}$.

$$\sum \varphi \dots \quad d\alpha_1 \wedge \dots$$

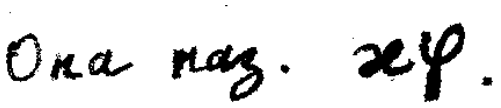
$d\alpha_i$ — some coordinates in G.

$\Sigma\varphi\ldots$ $d\alpha_1\wedge\ldots$ $d\alpha_i$ – какие-то коорд. в G.

Original source details: GELFAND-P000130

Manuscript notation. An ellipsis follows φ. The second part $d\alpha_1 \wedge \dots$ also ends in an ellipsis.

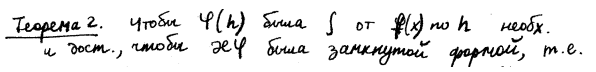
It is called $\mathcal{X}[?]\varphi$.



Original source details: GELFAND-P000130

Uncertain reading. After called, an X-like / χ-like operator sign before φ; its exact notation is unresolved.

Theorem 2. For $\varphi(h)$ to be the $\int$ of $f(x)$ over h, it is necessary and sufficient that $\mathcal{X}[?]\varphi$ be a closed form, i.e.

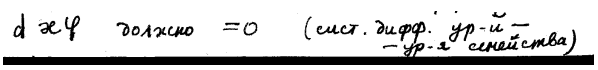


Original source details: GELFAND-P000130

Diagram and typed description. Theorem 2. is underlined. The full sentence contains necessary and sufficient, and a closed form, i.e.

$$d\mathcal{X}[?]\varphi \text{ must } = 0$$

(system of differential equations — — equations of the family)

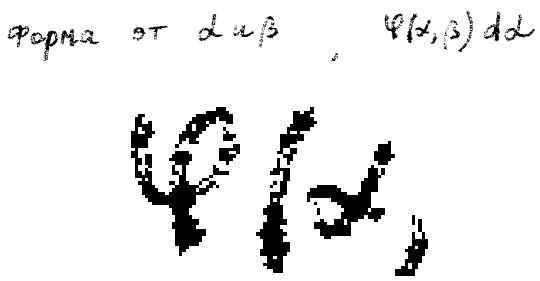


Original source details: GELFAND-P000130

Diagram and typed description. After the operator expression, the word must, an equals sign and 0. At the right, parentheses with system of differential equations, two dashes at the line junction, and equations of the family.

GELFAND-P000131

A form in α and β: $\varphi(\alpha, \beta)\, d\alpha$.



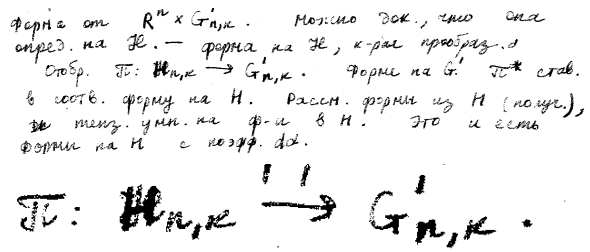
Original source details: GELFAND-P000131

Manuscript notation. The φ has the same open left branch and right loop as in the formulas below.

A form on $R^n \times G'_{n,k}$. One can prove that it is defined on H — a form on H which transforms with the map

$$\pi : H_{n,k} \longrightarrow G'_{n,k}.$$

To a form on G', π^* assigns a form on H. Consider the forms from H (obtained),



Original source details: GELFAND-P000131

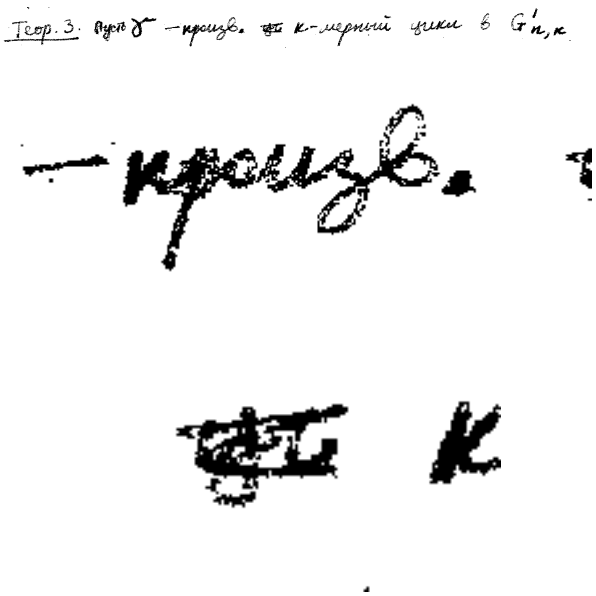
Diagram and typed description. The first product contains R^n and $G'_{n,k}$. The map goes from $H_{n,k}$ to $G'_{n,k}$; below, a separate π^*, a form on H and the parenthesis obtained.

[Crossed out: a short unreadable sign.] tensor mult. by fns. on H. These are precisely the forms on H with coeffs. $d\alpha$.

Uncertain reading. Before tensor mult., a short dense sign with loops and crossing strokes is cancelled; its full reading is unresolved.

Original source details: GELFAND-P000131

Thm. 3. Let γ be an arbitrary [Crossed out: several unreadable letters or strokes.] k-dimensional cycle in $G'_{n,k}$.



Uncertain reading. Between “arbitrary” and k-dimensional, a short entry is densely crossed out and remains unread. “Thm. 3.” is underlined.

Original source details: GELFAND-P000131

$$\int_\gamma \mathcal{X}[?]\varphi = C_\gamma f(\beta).$$

Uncertain reading. The sign after the integral has crossing loops: X-like / χ / κ; its exact notation is unresolved. Below, the same sign is called a diff. oper.

Original source details: GELFAND-P000131

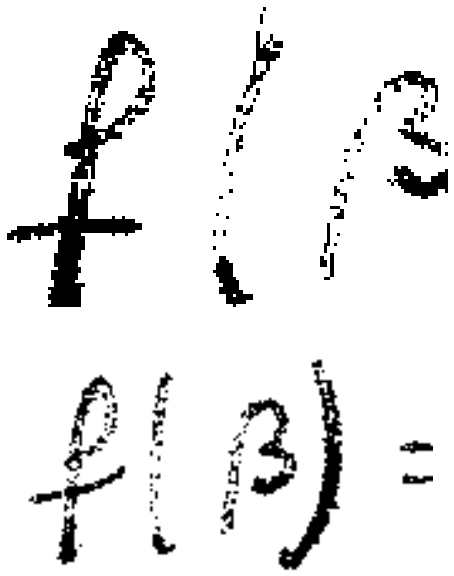


Diagram and typed description. In both occurrences of $C_\gamma f(\beta)$, a tall thin opening parenthesis follows f; no separate upper stroke on f is visible.

Original source details: GELFAND-P000131

C_γ — up to π, is the intersection index of γ with the Euler cycle (all k-dim. subspaces lying in a $(k+1)$-dimensional space).

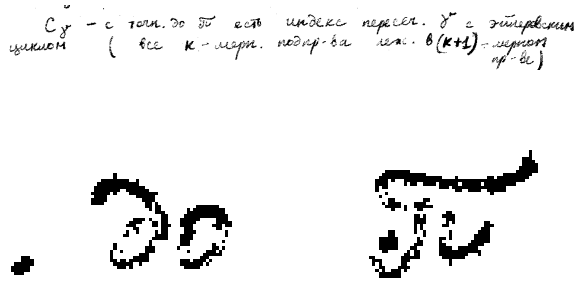
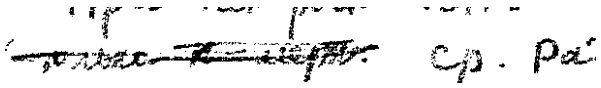


Diagram and typed description. After up to, a separate π with a long upper stroke. The parenthesis contains all k-dim. subspaces and a $(k+1)$-dimensional space; the source word for cycle continues at the beginning of the lower line.

Original source details: GELFAND-P000131

For some parities there is no local inversion formula ([Crossed out: only k-dim.[?]] cf. Radon).



Uncertain reading. In the parentheses before cf. Radon, horizontal lines cancel an entry with parts resembling only and k-dim.; the complete reading of the covered strokes is unresolved.

Original source details: GELFAND-P000131

In C^n there is an analytic formula, but there $C_\gamma \neq 0$ always. There is always an inversion formula there. One must know not only φ, but also its derivatives.

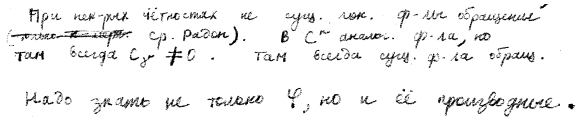


Diagram and typed description. The first phrase says parities. The next part contains C^n, then $C_\gamma \neq 0$ and a separate sentence about an inversion formula. The lower line ends with its derivatives.

Original source details: GELFAND-P000131

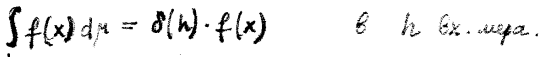
[Crossed out: a beginning resembling $\delta(h) =$.]

Original source details: GELFAND-P000131

Uncertain reading. To the left of the active integral, an earlier $\delta(h) =$ entry is crossed out by two long lines. The upper part of the opening parenthesis is thin; the covered interior strokes remain unread.

$$\int_h f(x)\, d\mu = \delta(h) \cdot f(x).$$

[At right:] in h, Eucl. measure.



Original source details: GELFAND-P000131

Diagram and typed description. The integral has subscript h; at the right, a dot between $\delta(h)$ and $f(x)$. A separate right-hand note: in h, Eucl. measure.

$$C_\gamma f(\beta) = \int_\gamma \mathcal{X}[?]\varphi(h).$$

$$C_\gamma(\delta(x-\beta), f(x)) = \int_\gamma \mathcal{X}[?](\delta(h), f(x)).$$

$$C_\gamma \delta(x-\beta) = \int_\gamma \mathcal{X}[?]\delta(h).$$

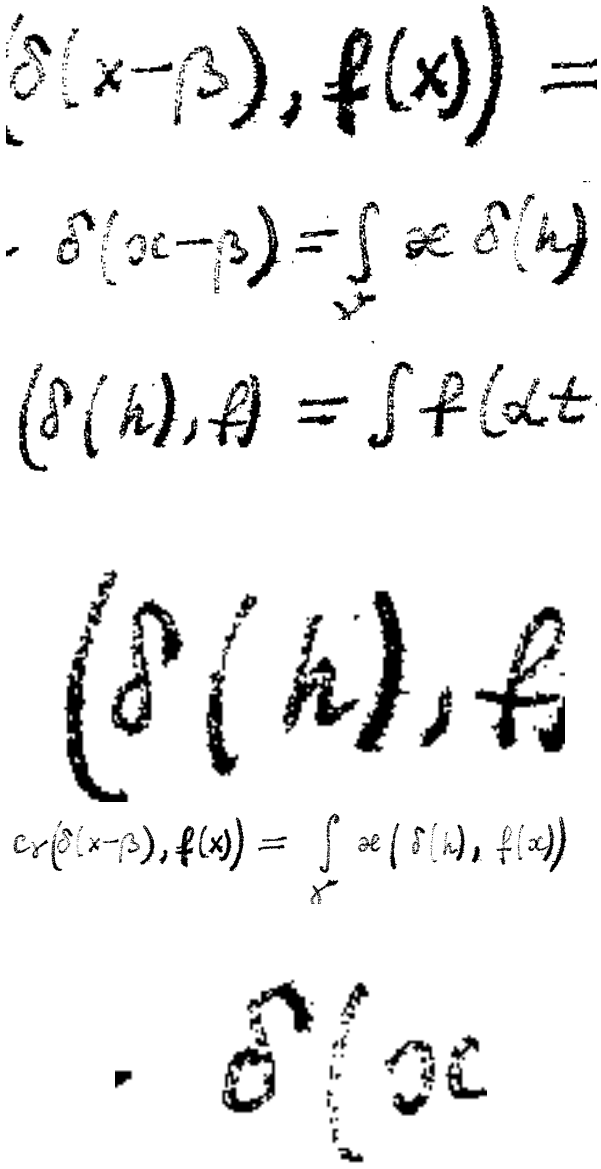


Diagram and typed description. In the formulas with $\delta(x-\beta)$ and $\delta(h)$, the tall thin strokes form opening parentheses; no separate apostrophe between the letter and parenthesis is visible. In the penultimate line, both pairings are in round brackets.

Original source details: GELFAND-P000131

[At right:] $\mathcal{X}$[?] — diff. oper.; $\int$ — over the cycle γ.

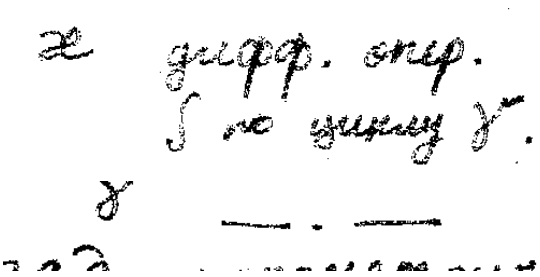


Diagram and typed description. The right-hand note contains the operator sign and diff. oper.; the next line is the integral over the cycle γ. Beneath the formula, two horizontal strokes with a dot between them.

Original source details: GELFAND-P000131

In C^n lines are given by parametric eqns.

$$x = \alpha t + \beta.$$

$$(\delta(h), f) = \int f(\alpha t + \beta)\, dt\, d\bar{t}.$$

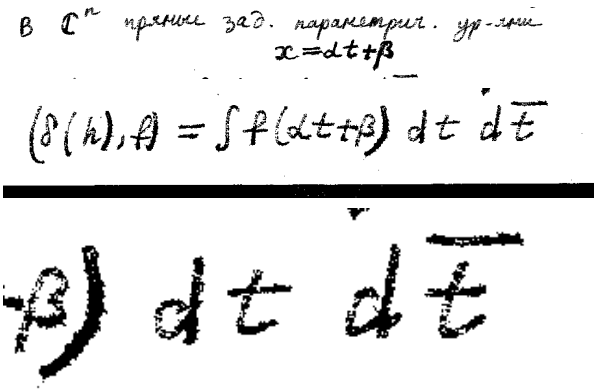
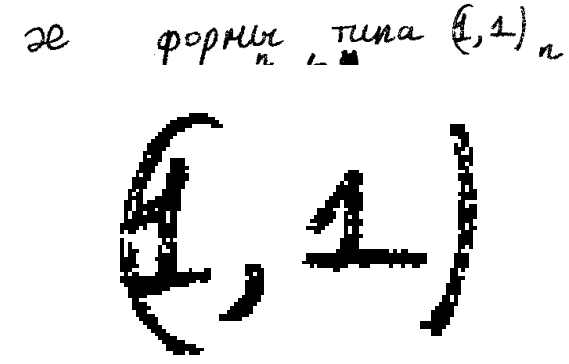


Diagram and typed description. The parametric equation has plus β. In the last line, the round pairing $\delta(h), f$ is at the left; the integrand is $f(\alpha t + \beta)$, followed by dt and $d\bar{t}$, with a separate bar only over the final t.

Original source details: GELFAND-P000131

GELFAND-P000132

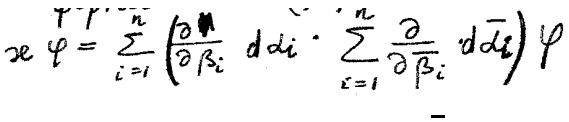
$\mathcal{X}$[?] — forms of type $(1, 1)$.



Manuscript notation. The first 1 touches the opening parenthesis. The second numeral is written separately.

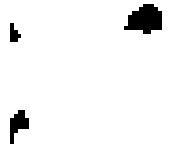
Original source details: GELFAND-P000132

$$\mathcal{X}[?]\varphi = \sum_{i=1}^{n} \left(\frac{\partial^{n[?]}}{\partial \beta_i} d\alpha_i \cdot \sum_{i=1}^{n} \frac{\partial}{\partial \overline{\beta}_i} d\overline{\alpha}_i \right) \varphi.$$



Uncertain reading. A dense raised n-like sign at the first ∂; the covered strokes have not been read.

Original source details: GELFAND-P000132



Uncertain reading. The same φ as on the preceding frame occurs on both sides. The intersecting looped operator sign remains uncertain. A dot stands between the first $d\alpha_i$ and the second sum.

Original source details: GELFAND-P000132

Final form.:

$$C_\gamma \delta(x) = \int_\gamma \left(\sum \frac{\partial}{\partial \beta_i} d\alpha_i \wedge \sum \frac{\partial}{\partial \overline{\beta}_i} d\overline{\alpha}_i \right)$$

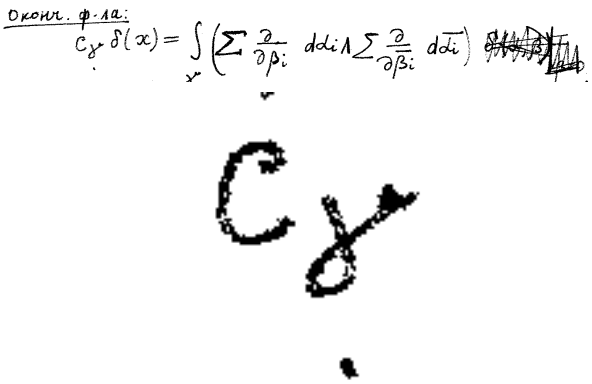


Diagram and typed description. The left side is $C_\gamma \delta(x)$; the integral has subscript γ. A $\wedge$ stands between the sums in parentheses. The heading Final form.: is underlined.

Original source details: GELFAND-P000132

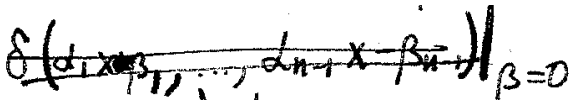
[At right, crossed out: an entry resembling $\delta(x - \beta)$ and unread strokes.]

Uncertain reading. To the right of the parenthesis, a densely cancelled entry has a δ-like beginning, parentheses and a β-like part; the complete former kernel and subsequent strokes remain unread.

Original source details: GELFAND-P000132

[Below at right, crossed out:]

$$\delta(\alpha_1 x \text{[?]} \beta_1, \ldots, \alpha_{n-1} x \text{[?]} \beta_{n-1})|_{\beta=0} .$$

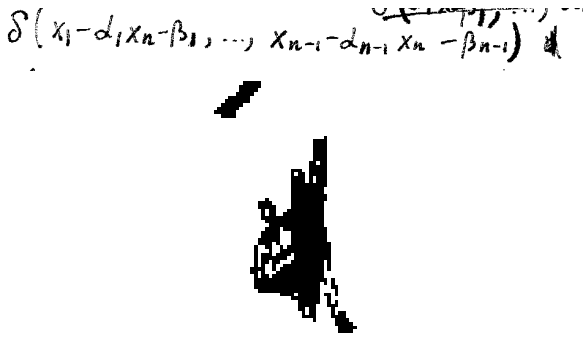


Uncertain reading. The cancelled line begins with $\delta(\alpha_1 x \ldots \beta_1, \ldots)$ and ends with $|_{\beta=0}$. Lines cross the intermediate signs by β_i; their precise reading remains uncertain.

Original source details: GELFAND-P000132

[Inserted below:]

$$\delta(x_1 - \alpha_1 x_n - \beta_1, \ldots, x_{n-1} - \alpha_{n-1} x_n - \beta_{n-1})\, d\text{[?]}$$



Uncertain reading. A dense overwritten d- / x-like sign follows the inserted kernel. Its precise reading remains uncertain.

Original source details: GELFAND-P000132

A line dep. on fewer param., not $2n$.

Original source details: GELFAND-P000132

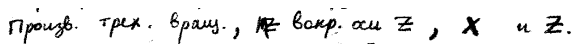
Manuscript notation. The first word is singular: “A line”. The rest reads “dep. on fewer param., not $2n$”.

Klein’s work on the top. (Karpelevich) (1896) (4 lectures at Princeton).

Original source details: GELFAND-P000132

Diagram and typed description. Klein’s work on the top. has a straight underline. The name in parentheses is followed by 1896; farther right, 4 lectures at Princeton, with the last words below.

One point fixed. Rot. of the top — a curve in the group of rot. of 3-dim. space. Coord. in this group. 2 ways (Euler angles φ, θ, ψ). Product of three rot., [Crossed out: a short dense sign.] about axes Z, X and Z.



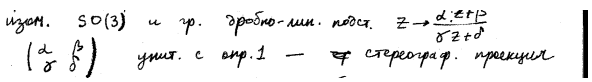
Uncertain reading. Before about axes, short dense К- / и-like strokes are crossed out. The complete reading remains uncertain.

Original source details: GELFAND-P000132

Isom. of $SO(3)$ and the gr. of fractional-lin. subst.

$$z \longrightarrow \frac{\alpha z + \beta}{\gamma z + \delta}.$$

$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ unitary with det. 1 — [Crossed out: an unread sign.] stereograph. projection.



Original source details: GELFAND-P000132

Diagram and typed description. The text reads Isom. of $SO(3)$ and the gr. of fractional-lin. subst.; at right, a fraction with $\alpha z + \beta$ and $\gamma z + \delta$. Below the line, a matrix and unitary with det. 1.

Uncertain reading. Between the dash and stereograph. projection is a dense cancelled sign resembling φ or an arrow. Its precise reading remains uncertain.

Original source details: GELFAND-P000132

(Rotate the sphere — fractional-lin. map of the plane.) (This is the 1st lecture.)
2nd lecture.

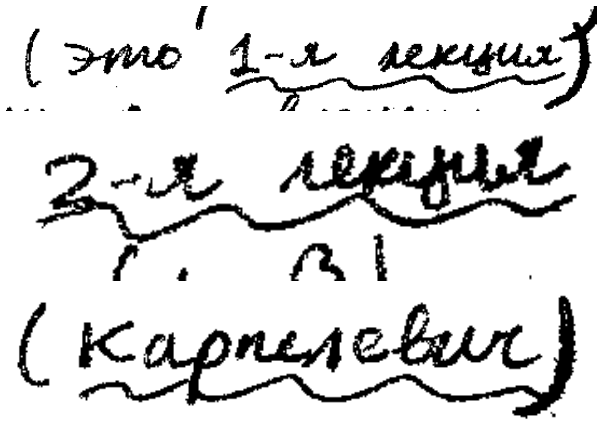


Diagram and typed description. Wavy lines under Karpelevich, under 1st lecture in parentheses, and under 2nd lecture.

Original source details: GELFAND-P000132

φ, ψ, θ sought as functions of time. $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ as functions of time are simpler. Eqs. of motion:

$$T = \frac{1}{2}(\varphi'^2 + \psi'^2 + 2\varphi'\psi' \cos\theta + \theta'^2).$$

[At right:] (All moments of inertia $= 1$ rel. to the point of attachment — sphere-top.)

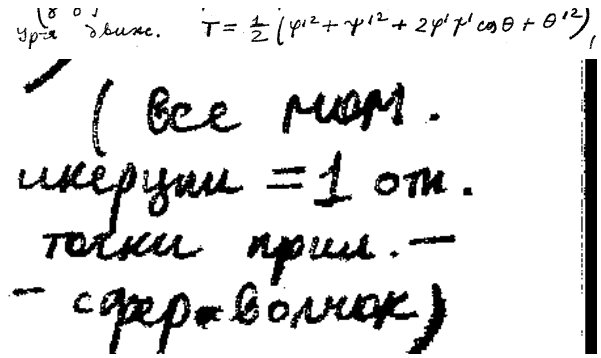


Diagram and typed description. Primes and squares at φ, ψ, θ; the mixed term is $2\varphi'\psi' \cos\theta$. A separate parenthesis at right: All moments of inertia $= 1$ rel. to the point of attachment — sphere-top.

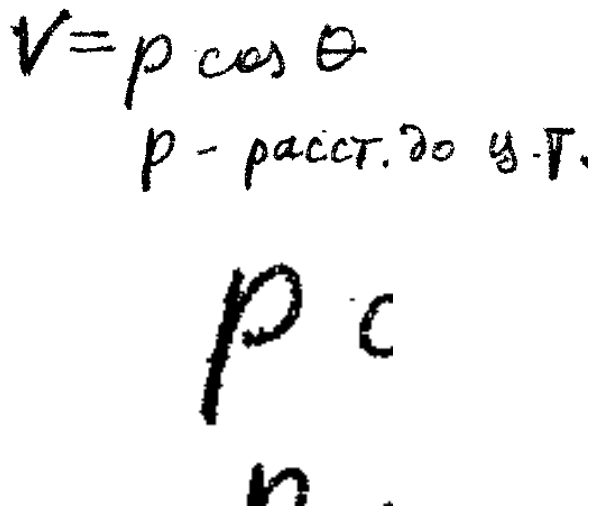
Original source details: GELFAND-P000132

Potential energy [Crossed out: a p- / ρ-like beginning.] $V = p$[?] $\cos\theta$.

Uncertain reading. Before V, a letter with a loop and a descending stroke is cancelled with long lines: p / ρ. The complete former beginning remains unread.

Original source details: GELFAND-P000132

p[?] — dist. to c. of g.



Uncertain reading. In V and in the following line, a letter p / ρ with a loop and a descending stroke. The precise symbol remains uncertain.

Original source details: GELFAND-P000132

θ — axis Z with c. of g. vert.

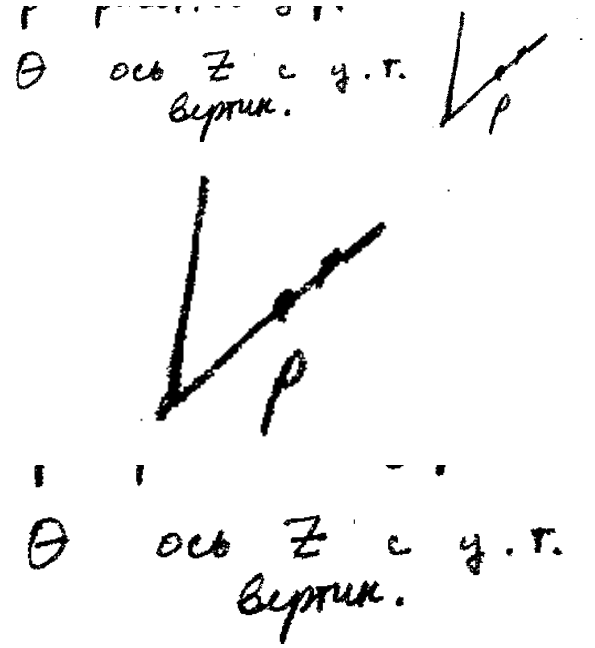
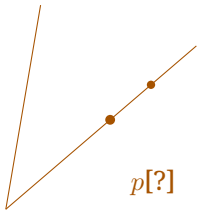


Diagram and typed description. Two segments from a common lower endpoint, one nearly vertical, the other sloping upward to the right. Two dense points lie on the oblique segment, with a p / ρ label below. The drawing has no angle arc or axis labels. Beside it: θ — axis Z with c. of g. vert.



Original source details: GELFAND-P000132

$$\frac{d}{dt}\frac{\partial T}{\partial \varphi'} = 0.$$

$$\begin{cases} \varphi' + 2\psi' \cos\theta = n, \\ \psi' + 2\varphi' \cos\theta = e, \\ T + V = h. \end{cases}$$

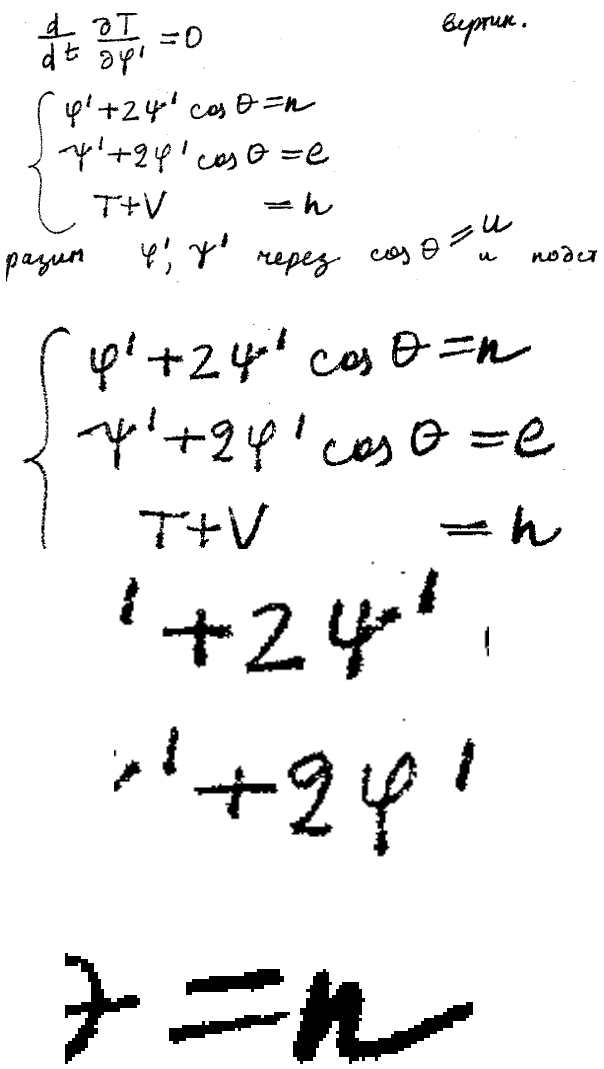


Diagram and typed description. In each of the first two lines, a clear 2 follows the plus. The right side of the first line is n, of the second e; the third line is $T + V = h$.

Original source details: GELFAND-P000132

Express φ', ψ' in terms of $\cos\theta = u$ and subst. into that eq.

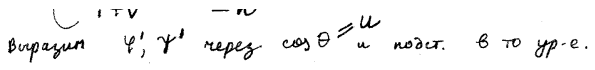
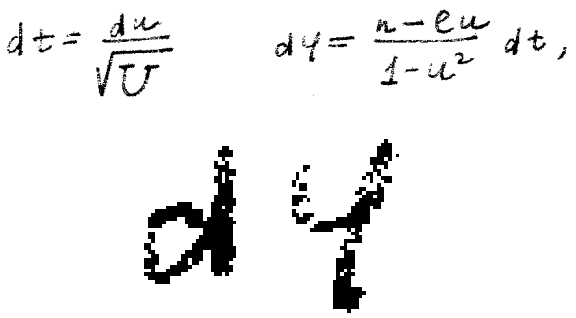


Diagram and typed description. The addition $= u$ is above the end of $\cos\theta$. The last phrase ends with into that eq.

Original source details: GELFAND-P000132

$$dt = \frac{du}{\sqrt{U}}, \qquad d\varphi[?] = \frac{n - eu}{1 - u^2}\, dt.$$

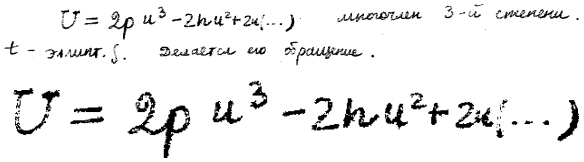


Uncertain reading. The letter in the second differential has an open branch and a descending stroke: φ / ψ. The precise symbol remains uncertain.

Original source details: GELFAND-P000133

$$U = 2p[?]u^3 - 2hu^2 + 2u(\ldots).$$

A polynomial of degree 3. t — ellipt. $\int$. Its inversion is carried out.



Uncertain reading. After 2, a letter with a loop and a descending stroke: p / ρ. The first term is $2pu^3$, then a minus before $2hu^2$ and a plus before $2u(\ldots)$; the ellipsis is in parentheses.

Original source details: GELFAND-P000133

[Crossed out: a short unread beginning.] $28/\overline{\text{II}}$.

Uncertain reading. Before $28/\text{II}$, a short looped sign is densely cancelled; its complete reading remains uncertain.

Original source details: GELFAND-P000133

Continuation of Agranovich's talk.

Diagram and typed description. The Roman II in the date has upper and lower bars. The heading Continuation of Agranovich's talk. has a wavy underline.

Original source details: GELFAND-P000133

$$Au(x) = \underset{\xi \to x}{F^{-1}_{\xi \to x}}\big(F_{x \to \xi}\, \sigma(x,\xi) u(x)\big).$$

$$\sigma(x, t\xi) = t^s \sigma(x, \xi).$$

This def. differs inessentially.

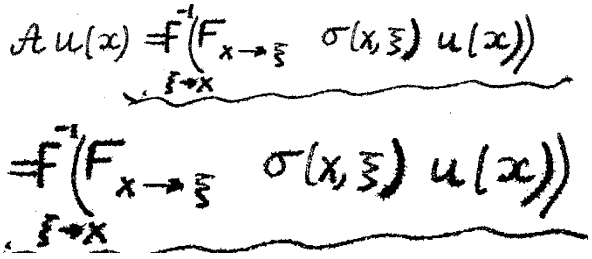
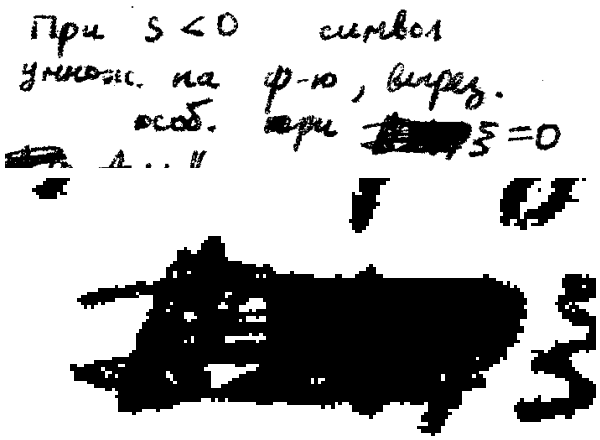


Diagram and typed description. A separate $\xi \to x$ below the outer F^{-1}; $x \to \xi$ below the inner F. The whole right side of the definition has a wavy underline. Below it, the condition with $t\xi$ and t^s.

Original source details: GELFAND-P000133

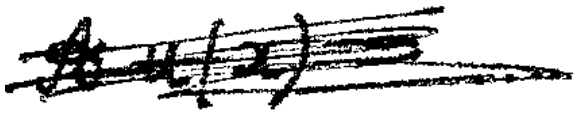
[In the right margin:] For $s < 0$, the symbol is mult. by a fn. cutting out the sing. at [Crossed out: an unread sign.] $\xi = 0$.



Uncertain reading. Between at and $\xi = 0$, a short densely cancelled entry; the covered strokes remain unread. The complete marginal note ends with $\xi = 0$.

Original source details: GELFAND-P000133

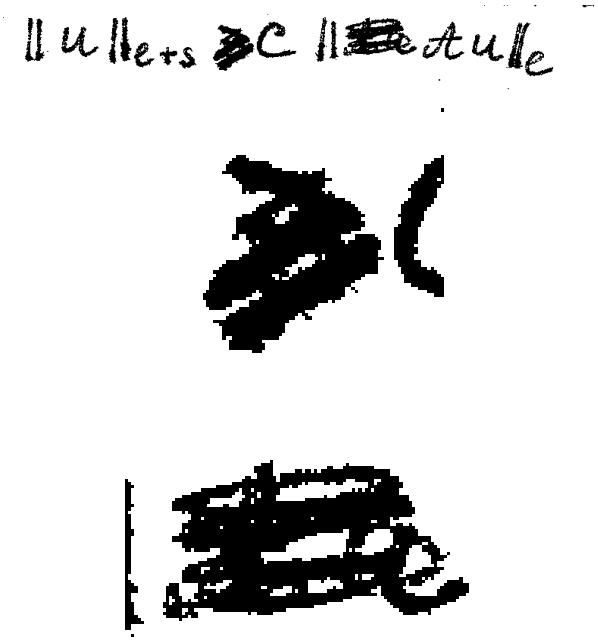
[At left, crossed out: $Au(x) =$ and an unread continuation.]



Uncertain reading. The beginning $Au(x) =$ of the cancelled line is visible; long lines and dense strokes cover its continuation.

Original source details: GELFAND-P000133

$\|u\|_{\ell+s} \geq$ [?] $C\|$[cancelled] $Au\|_{\ell}$.



Uncertain reading. The comparison sign is overwritten: $\geq$ / $\leq$. A dense sign before Au inside the right norm is cancelled. The altered strokes have not been read definitively.

Original source details: GELFAND-P000133

1) Boundedness theorem.

2) Composition theorem.

$$A_1 \cdot A_2 - A_1 \circ A_2 = T.$$
$$\sigma_1 \quad \sigma_2 \qquad\quad \sigma_1 \cdot \sigma_2$$

[At right:] ord. $s_1 + s_2 - 1$.

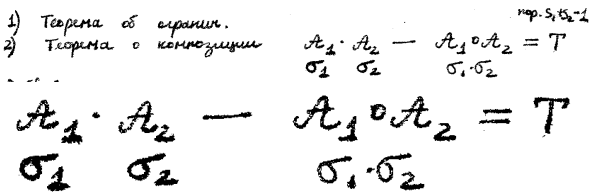
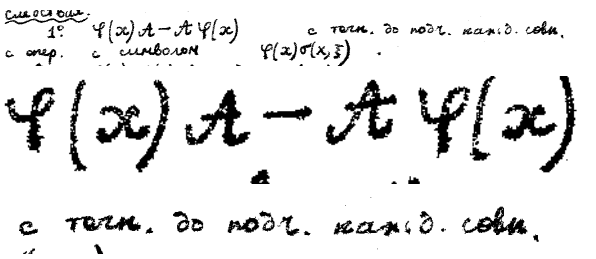


Diagram and typed description. A σ_1 below the first A_1, a σ_2 below A_2. Below $A_1 \circ A_2$, the entry $\sigma_1 \cdot \sigma_2$. Above at right, ord. $s_1 + s_2 - 1$.

Original source details: GELFAND-P000133

Corollaries.

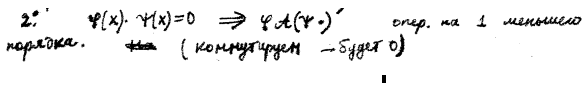
1°. $\varphi(x)A - A\varphi(x)$ up to podch. kazh.d.[?] agrees with an oper. having symbol $\varphi(x)\sigma(x,\xi)$.



Uncertain reading. After “up to”, the abbreviated reading is “podch. kazh.d.[?] agrees”. The middle abbreviation remains uncertain. A horizontal stroke stands between $\varphi(x)A$ and $A\varphi(x)$.

Original source details: GELFAND-P000133

2°. $\varphi(x)\cdot\psi(x)=0 \Rightarrow \varphi A(\psi\,\cdot)$ — an oper. of order 1 less. [Crossed out: a short unread word.] (we commute — it will be 0)



Uncertain reading. A short entry is densely crossed out before the parenthesis, which reads "we commute — it will be 0".

Original source details: GELFAND-P000133

3°. Let $s=0$. Suppose there is an estimate $\|Au\|_1 \le c\|u\|_0$ (smoothing oper.). Then the symbol is ident. 0 and the operator 0.

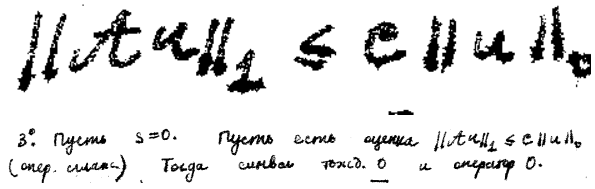
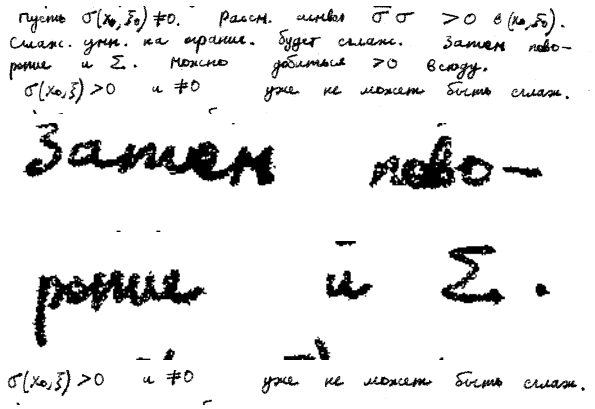


Diagram and typed description. The left norm has subscript 1, the right 0; the sign is $\le$. Then smoothing oper. in parentheses, symbol ident. 0, and operator 0.

Original source details: GELFAND-P000133

Let $\sigma(x_0,\xi_0)\neq 0$. Consider the symbol $\bar\sigma\sigma>0$ at (x_0,ξ_0). Smoothing mult. by bounded will be smoothing. Then rotations and Σ. One can obtain >0 everywhere. $\sigma(x_0,\xi)>0$ and $\neq 0$ can no longer be smoothing.



Manuscript notation. "Then rota-" at the end of one line and "tions and Σ" at the start of the next.

Original source details: GELFAND-P000133

$3)$ Coordinate transformation theorem. $x=x(y)$ — inf. smooth, nondeg.

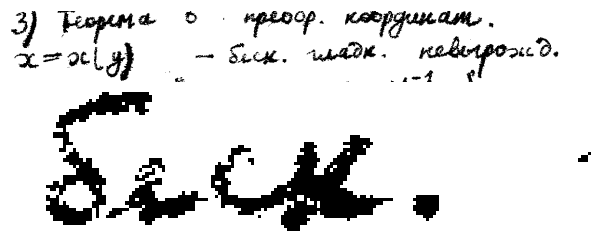
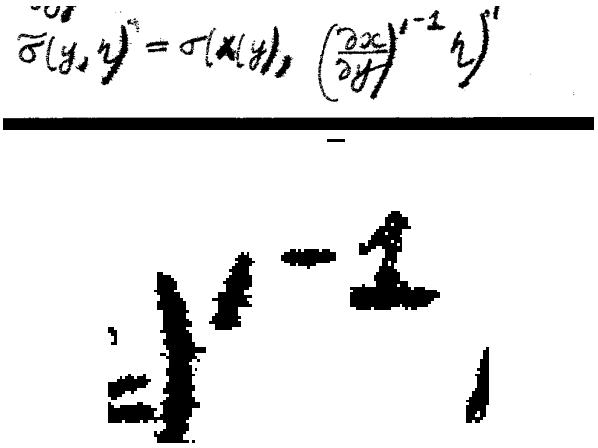


Diagram and typed description. Number $3)$, then Coordinate transformation theorem. After $x=x(y)$, the abbreviations inf. smooth, nondeg.

Original source details: GELFAND-P000133

$$\tilde{\sigma}(y,\eta) = \sigma\left(x(y), \left(\frac{\partial x}{\partial y}\right)'^{-1[?]} \eta\right).$$

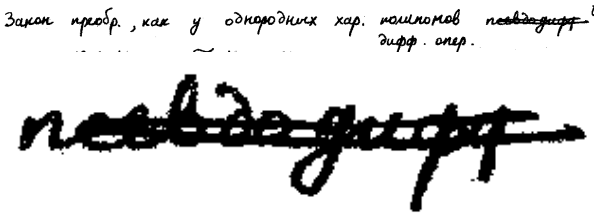


Uncertain reading. After the matrix's closing parenthesis, a separate raised apostrophe-like / t-like stroke, followed by -1. The precise superscript reading remains uncertain.

Original source details: GELFAND-P000133

GELFAND-P000134

Transformation law, as for homogeneous char. polynomials [Crossed out: pseudodiff.[?]]. [Below the line:] diff. oper.

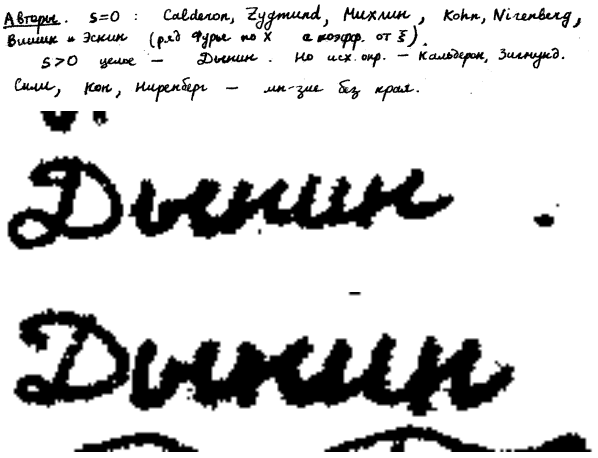


Uncertain reading. At the end of the first line, a word resembling pseudodiff. is crossed out; the covered letters remain incompletely read. Separately below: diff. oper.

Original source details: GELFAND-P000134

Consider $\varphi A\psi - \varphi\tilde{A}\psi = T$ — this is a subord. oper.

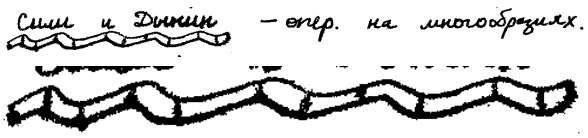
Authors. $s = 0$: Calderon, Zygmund, Mikhlin, Kohn, Nirenberg, Vishik and Eskin (Fourier series in x, with coeff. depending on ξ). $s > 0$ integer — Dynin. But the orig. def. — Calderon, Zygmund. Seeley, Kohn, Nirenberg — manifold without boundary.



Manuscript notation. Dynin is written in Cyrillic after $s > 0$ and above the wavy underline. Seeley is written Сили in Cyrillic. The names Calderon, Zygmund, Kohn, and Nirenberg in the first line are written in Latin script.

Original source details: GELFAND-P000134

Seeley and Dynin — oper. on manifolds.



Original source details: GELFAND-P000134

Diagram and typed description. Below the pair of names, a double wavy strip with short cross-strokes.

Let $\underline{s=0}$. A — sing. $\int$ oper. on the manifold V. Consider φ and ψ on V. 1) $\operatorname{supp}\varphi \cap \operatorname{supp}\psi = \varnothing \;\Rightarrow\; \varphi A(\psi u) = Tu$ $(\|Tu\|_1 \le c\|u\|_0)$.

2) Let $\operatorname{supp}\varphi$ and ψ lie in a coord. nbd. Then

$$\varphi A\psi = \varphi A_\nu \psi + Tu,$$

where A_ν — a sing. $\int$ oper., written in local coordinates.

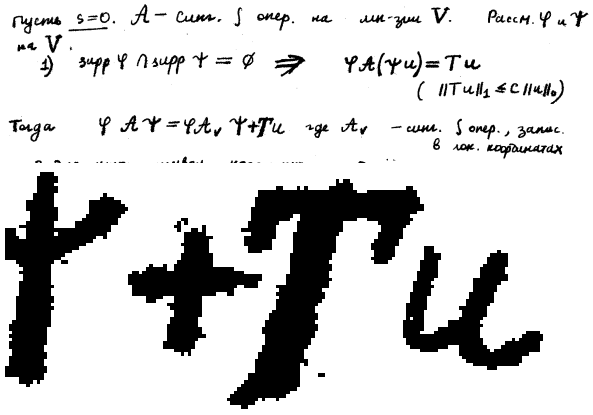


Diagram and typed description. In the first condition, two supports with empty intersection, then Tu and an estimate with indices 1 and 0. The local formula has $+Tu$ at right.

Original source details: GELFAND-P000134

The symbol is evidently well-defined as a function on the cotangent bundle. (A_ν on the overlap.) Hence, in particular, the symbol transformation formula.

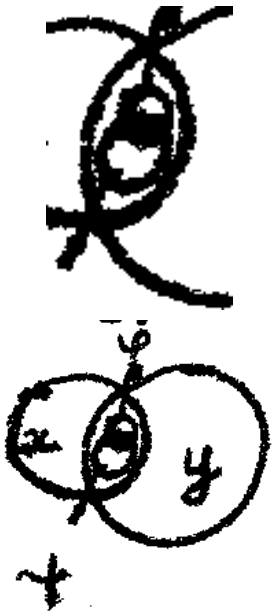
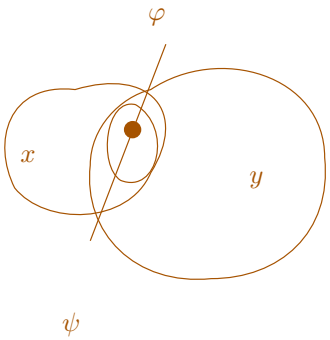


Diagram and typed description. Two overlapping rounded contours, x at left and y at right. Above, φ; below, ψ; inside the overlap, a small loop with a heavy point and an oblique stroke.



Original source details: GELFAND-P000134

$$\varphi A\psi. = \varphi A_x\psi + T_1 = \varphi A_y\psi + T_2.$$

Hence

$$\begin{array}{ll} \varphi A_x\psi & -\varphi A_y\psi \quad \text{smoothing.} \\ \| & \\ \varphi \tilde{A}_y\psi + \text{smoothing.} & \end{array}$$

[At right:] Hence smoothing symbol.

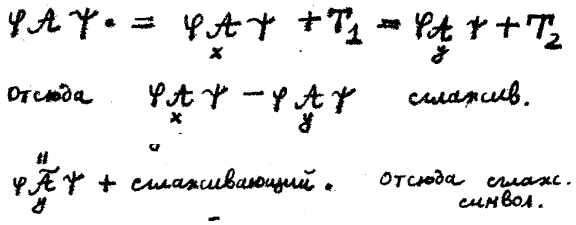


Diagram and typed description. A separate point after the first ψ and before the equals sign. Below the first term, two vertical strokes, then $\varphi\tilde{A}_y\psi+$ smoothing. Separately at right: Hence smoothing symbol.

Original source details: GELFAND-P000134

$$A = \sum \varphi_\nu A, \qquad \psi_\nu = \begin{cases} 1 & \text{in a nbd. of supp}\,\varphi_\nu, \\ 0 & \text{outside the nbd.} \end{cases}$$

$$A = \sum \varphi_\nu A\psi_\nu + \text{smoothing.} = \sum \varphi_\nu A_\nu \psi_\nu + \text{smoothing.}$$

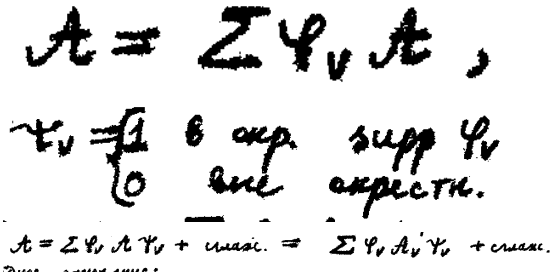
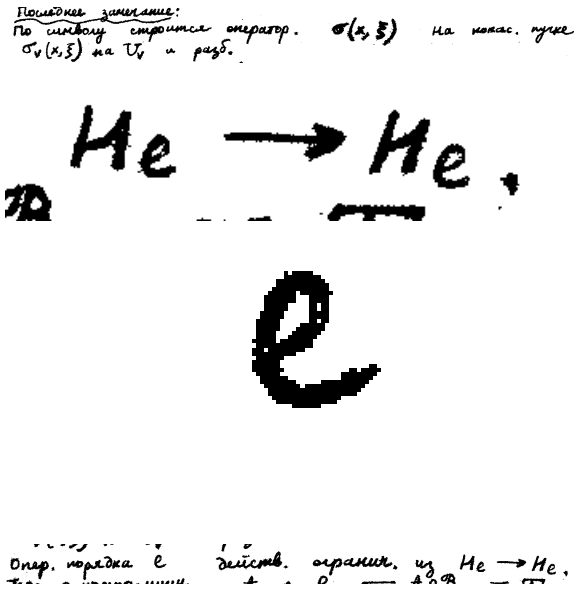


Diagram and typed description. In the first sum, only A follows φ_ν. The two cases for ψ_ν are 1 in a nbd. of the support of φ_ν and 0 outside the nbd. The lower line has two sums with ψ_ν; in the second, A also has subscript ν.

Original source details: GELFAND-P000134

Final remark: An operator is constructed from the symbol. $\sigma(x,\xi)$ on the cotang. bundle, $\sigma_\nu(x,\xi)$ on U_ν and a partition. An oper. of order ℓ[?] acts boundedly from $H_{\ell[?]} \to H_{\ell[?]}$.



Uncertain reading. The order letter and both indices have the same looped appearance: ℓ / e. Both spaces are written with the same index. Final remark: has a wavy underline.

Original source details: GELFAND-P000134

Composition thm.:

$$\underset{\sigma_1}{A} \cdot \underset{\sigma_2}{B} - \underset{\sigma_1\cdot\sigma_2}{A \circ B} = T.$$

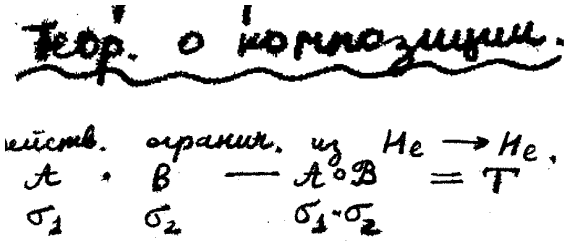


Diagram and typed description. A σ_1 below A, a σ_2 below B, and $\sigma_1 \cdot \sigma_2$ below $A \circ B$. The dot is between A and B; the upper loop above this part belongs to iz (from) in the preceding line. The heading Composition thm.: has a wavy underline.

Original source details: GELFAND-P000134

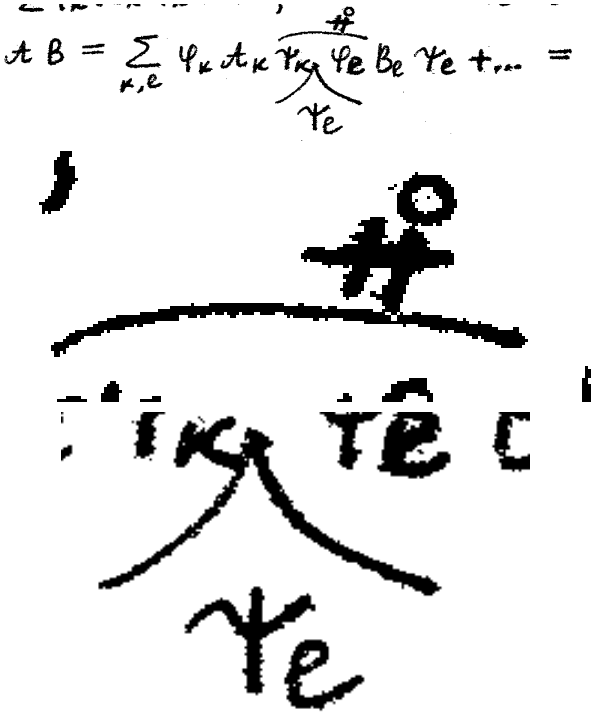
GELFAND-P000135

Proof.

$$A = \sum \varphi_k A_k \psi_k + T, \qquad B = \sum \varphi_\ell B_\ell \psi_\ell + T_2.$$

$$AB = \sum_{k,\ell} \varphi_k A_k \overbrace{\psi_k \varphi_\ell}^{A^\circ[?]} B_\ell \psi_\ell + \ldots =$$

ψ_ℓ

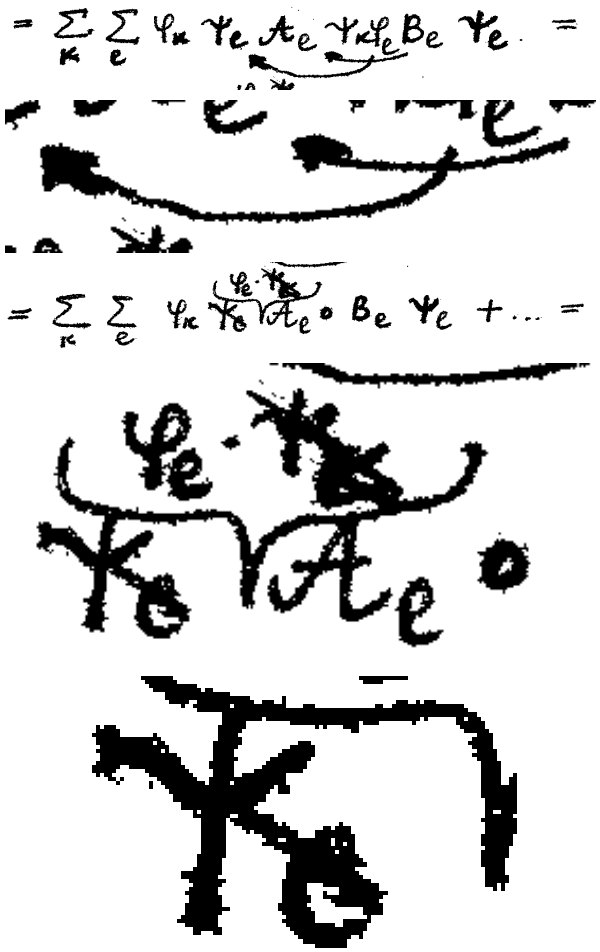


Uncertain reading. An arc above $\psi_k \varphi_\ell$ has an A°- / T°- / t°-like label; the exact reading is unsettled. Below are two converging curved strokes and the label ψ_ℓ.

Original source details: GELFAND-P000135

$$= \sum_k \sum_\ell \varphi_k \, \psi_\ell \, A_\ell \, \psi_k \, \varphi_\ell \, B_\ell \psi_\ell . \quad =$$

$$= \sum_k \sum_\ell \varphi_k \, \cancel{\psi_{\ell[?]}} \, A_\ell \circ B_\ell \psi_\ell + \ldots =$$



Uncertain reading. Below the second line are two curved arrows pointing left. In the next line, $\psi_{\ell[?]}$ before A_ℓ is crossed out; $(\varphi_\ell \cdot \psi_{k[?]})$ is inserted above, with its second part heavily crossed out. Below the insertion are joined arcs and an angular insertion mark. There is a small hollow circle between A_ℓ and B_ℓ. The covered indices remain partly unread.

Original source details: GELFAND-P000135

$$= \sum_\ell \varphi_\ell A_\ell \circ B_\ell \psi_\ell + \text{smooth.}$$

The theor. on compos. is proved.

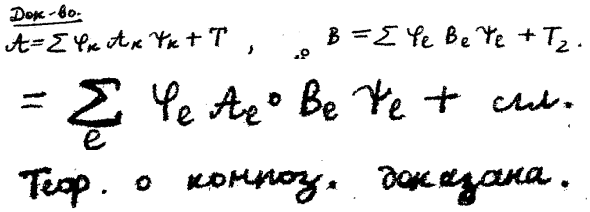
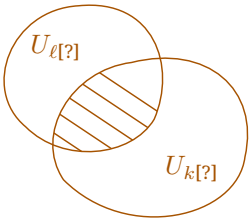


Diagram and typed description. The first three product lines also end with an equals sign; the following lines begin with an equals sign.

Original source details: GELFAND-P000135

Original source details: GELFAND-P000135

Diagram and typed description. Two intersecting contours with diagonal hatching in their overlap. The upper label is $U_{\ell[?]}$, the lower $U_{k[?]}$; the dense indices remain unsettled.



Elliptic theory. A_s, $\sigma(P,\xi_P)$; ξ_P is a cotang. vector. Elliptic operator: $\sigma(P,\xi_P)\neq 0$ for $\xi_P\neq 0$ (if a matrix, then det). Theorem 1. Ellipticity is equivalent to being Noetherian. a) Prove that an elliptic oper. is Noetherian. Consider σ^{-1}, also a symbol, homog. of degree $\underline{\underline{-s}}$, $\sigma^{-1}\leftrightarrow B$.

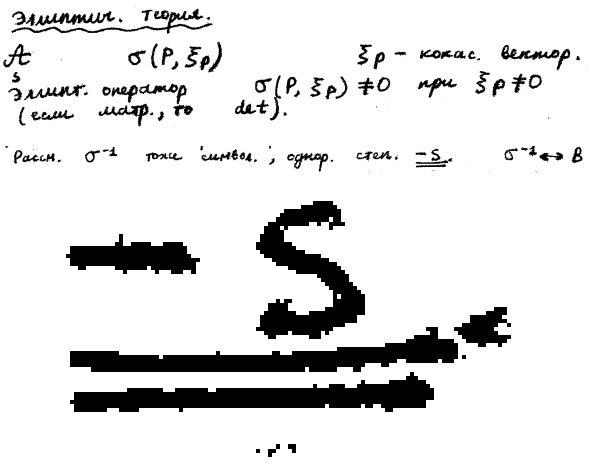
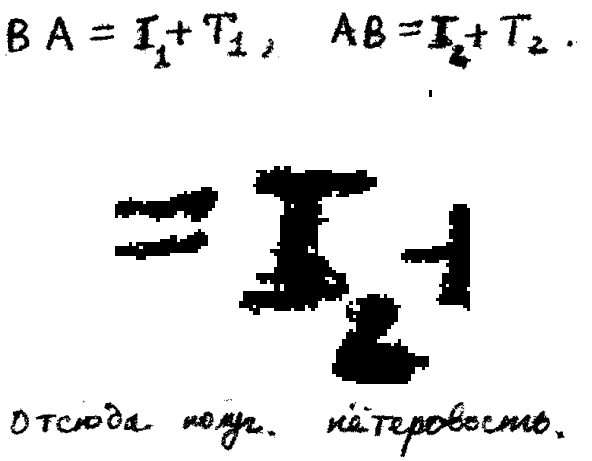


Diagram and typed description. A has the subscript s. The degree $-s$ has a double underline.

Original source details: GELFAND-P000135

$$BA = I_1 + T_1, \qquad AB = I_{2[?]} + T_2.$$

Hence we obtain Noetherianity. $d(A) < \infty, \quad \beta(A) < \infty.$



Uncertain reading. The subscript below the second I is heavily overwritten; a 2-like sign is visible, while the former stroke remains unread.

Original source details: GELFAND-P000135

b) Noeth. $\Rightarrow$ ell. $s = 1$.

$$\|u\|_1 \le C(\|Au\|_0 + \|u\|_0)$$

from Noetherianity. Passage to R^n, elim. the depend. of the symb. on x. Functions with small supports and a symb. indep. of x. $\sigma(x_0, \xi_0) = 0$ and the est. leads to a contradiction:

$$\varphi(x) e^{i\lambda\langle x, \xi_0\rangle}.$$

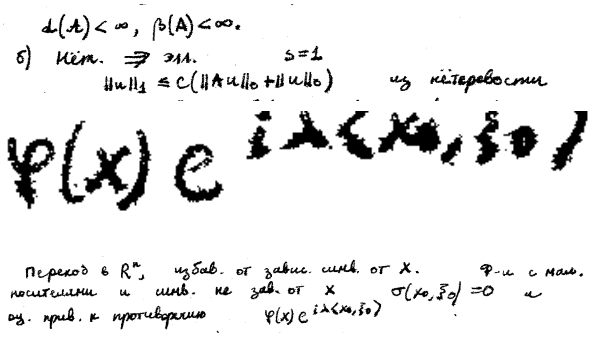


Diagram and typed description. The exponent is $i\lambda\langle x, \xi_0\rangle$. The line above has $\sigma(x_0, \xi_0) = 0$.

Original source details: GELFAND-P000135

Theorem 2 (on smoothness). $Au = f, \quad s = 0$. Let $u \in H_0,\ f \in H_1 \ \Rightarrow\ u \in H_1$.

$$Au = f, \quad u + \underset{\substack{\uparrow \\ H_1 \text{ (smooth.)}}}{T_2} f = Bf \in H_1 \ \Rightarrow\ u \in H_1.$$

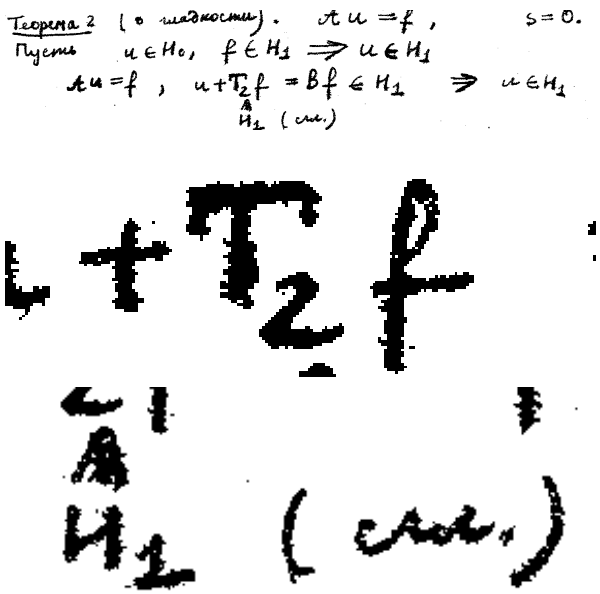


Diagram and typed description. T_2 is followed by f. Below T_2, an arrow points up from the label H_1 (smooth.).

Original source details: GELFAND-P000135

Theorem 3. Homotop. stability of the index $\chi(A) = d(A) - \beta(A)$.

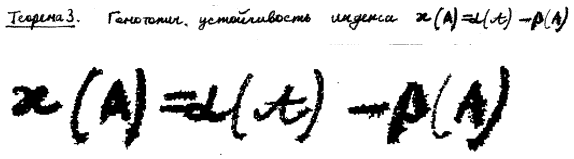


Diagram and typed description. The final formula is $\chi(A) = d(A) - \beta(A)$.

Original source details: GELFAND-P000135

GELFAND-P000136

Under small deformations of the symbol, the index does not change. A continuous deform. is approx. by a smooth deform. The oper. A with $\sigma\left(x, \frac{\xi}{|\xi|}\right)$ and B with $\underset{s}{\sigma(x,\xi)}$ have the same index.

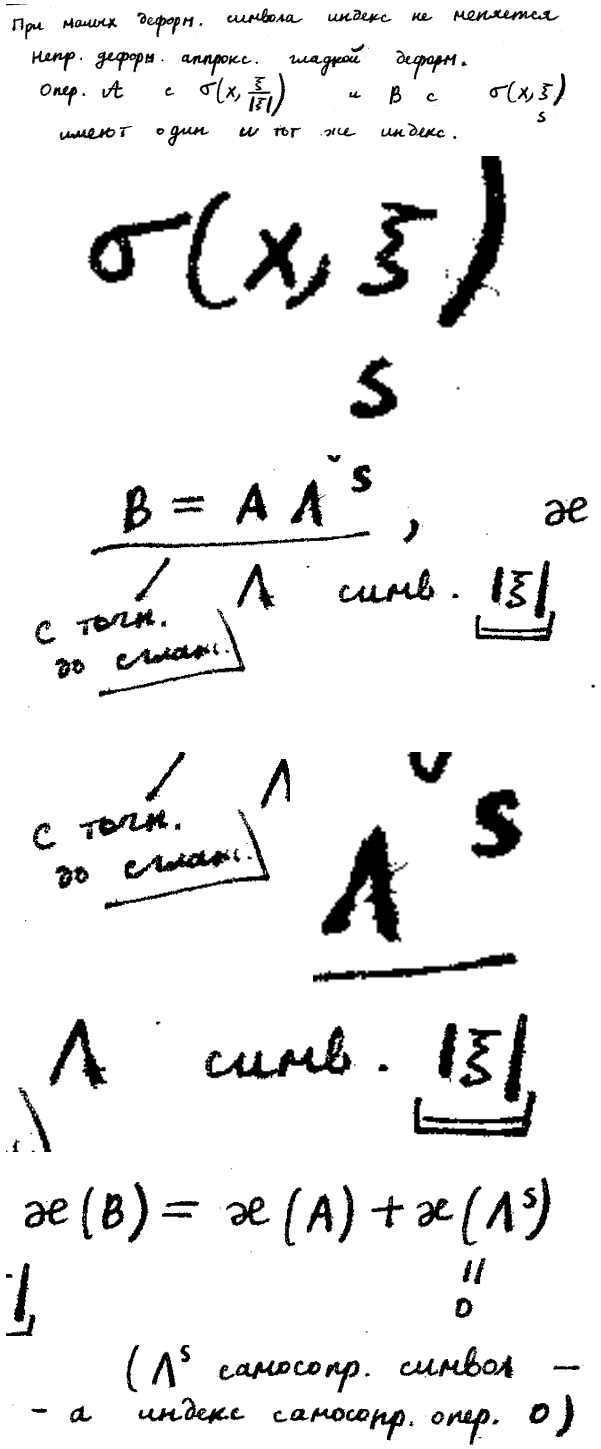


Diagram and typed description. An s is written separately beneath ξ in the second symbol. $B = A\Lambda^s$ is underlined; the margin has "up to smooth." with an oblique stroke and an angular line. Below $\chi(\Lambda^s)$ are two vertical strokes and 0. Below $|\xi|$ are two lines with short ends.

$$\underline{B = A\Lambda^s},$$

up to smooth. Λ symb. $\underline{\underline{|\xi|}}$

$$\chi(B) = \chi(A) + \underset{\|}{\chi(\Lambda^s)}.$$
$$0$$

Original source details: GELFAND-P000136

(Λ^s is a selfadj. symbol—and the index of a selfadj. oper. is 0.)

Index theorem. (Seeley) V_n, $n \times n$ systems. $\sigma(P, \xi_P)$ is a unitary matrix.

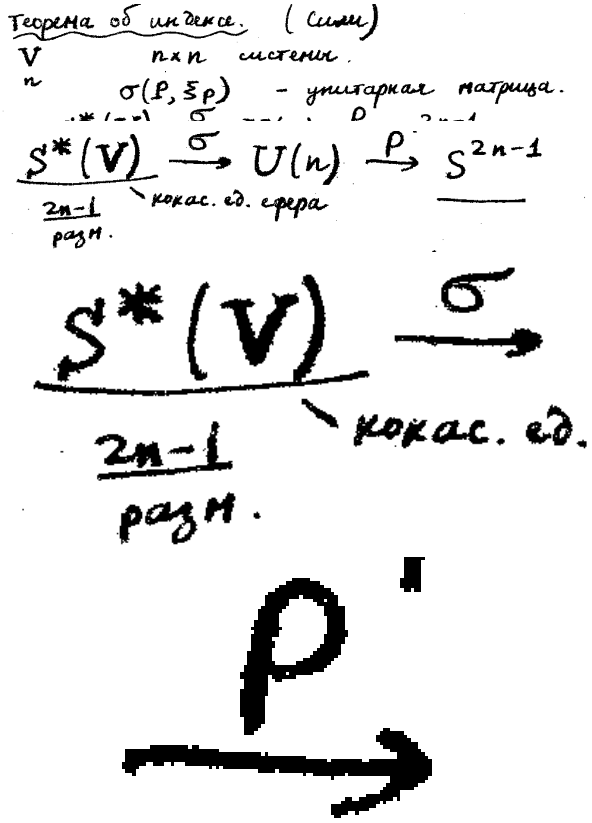


Diagram and typed description. $S^*(V)$ and the sphere on the right are underlined. Below the first space, $2n-1$ is underlined, with "dim." beneath; an oblique stroke leads to the label "unit cotang. sphere".

$$\underline{S^*(V)} \xrightarrow{\sigma} U(n) \xrightarrow{\rho} \underline{S^{2n-1}}$$

$\underline{2n-1}$ dim. — unit cotang. sphere

Original source details: GELFAND-P000136

$$\operatorname{ind} A = \frac{1}{(n-1)!} \deg(\rho \circ \sigma) \iff \operatorname{ind}_{(n-1)\times(n-1)} B = 0.$$

The operation $A\#B$ is used—$\operatorname{ind} A\#B = \operatorname{ind} A \cdot \operatorname{ind} B$. For boundaries of domains in R^{n+1}, such a formula holds. (the right side is correct) Left formula.

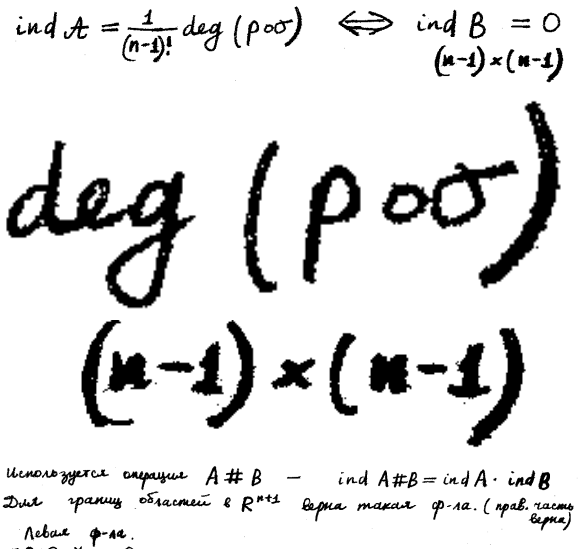
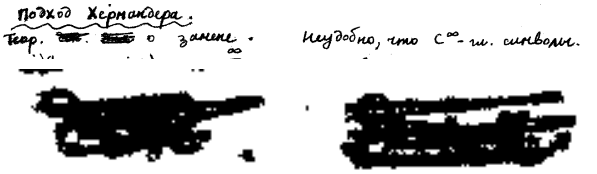


Diagram and typed description. Below B, separately, is $(n-1) \times (n-1)$. The product of indices on the right has a dot; after ind on the left is $A\#B$. The preceding line has $\rho \circ \sigma$.

Original source details: GELFAND-P000136

Hörmander's approach. Theor. [Crossed out: two unread words] on change. It is inconvenient that the symbols are C^∞.



Uncertain reading. Before "on change" are two heavily crossed-out words. The obscured letters remain unread.

Original source details: GELFAND-P000136

$$e^{-i\lambda g} A[f e^{ig\lambda}] \sim \sum_0^\infty P_j(f,g)\lambda^{s_j}, \qquad s_j \downarrow -\infty.$$

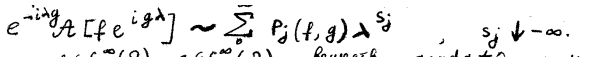
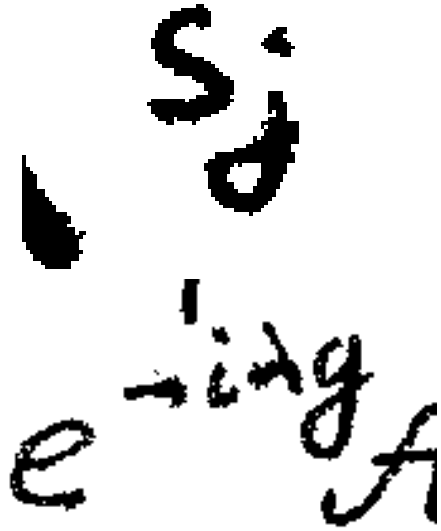


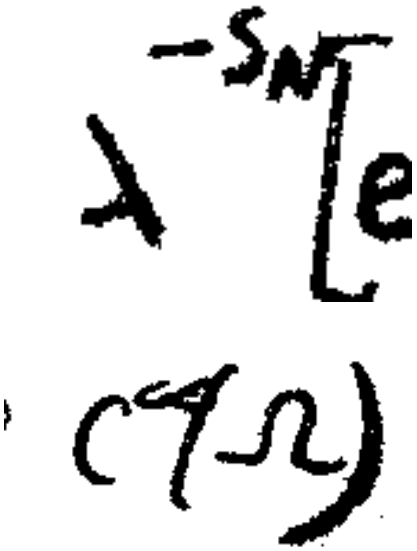
Diagram and typed description. The exponent on the left is $-i\lambda g$, and inside the first brackets it is $ig\lambda$; on the right are λ^{s_j} and $s_j \downarrow -\infty$. The lower summation limit is 0.



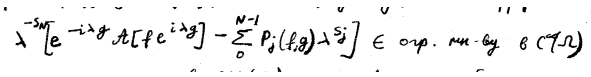
Original source details: GELFAND-P000136

$f \in C_0^\infty(\Omega), \quad g \in C^\infty(\Omega)$, real, grad $g \neq 0$ on supp f.

$$\lambda^{-s_N}\left[e^{-i\lambda g}A[fe^{i\lambda g}] - \sum_0^{N-1} P_j(f,g)\lambda^{s_j}\right] \in \text{a bounded set in } C^q(\Omega).$$



Manuscript notation. In C^q the upper loop of the exponent joins a long stroke descending below the line. Before the bracket is λ^{-s_N}, and the upper summation limit is $N-1$.



Original source details: GELFAND-P000136

$g \in$ a compact subset of $C^\infty(\Omega)$, and A must be a continuous oper. from C_0^∞ to C^∞.

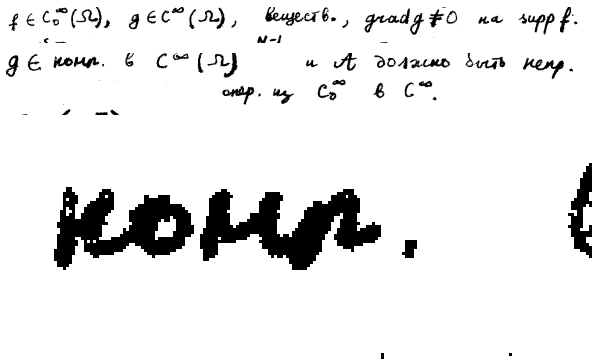


Diagram and typed description. f belongs to $C_0^\infty(\Omega)$, and g is real; the text has grad $g \neq 0$ on supp f. Below are "$g \in$ a compact subset of $C^\infty(\Omega)$" and the continuity condition on A.

Original source details: GELFAND-P000136

$\Omega \subset R^n, \quad g = \langle x, \xi \rangle$ —fund. Consider analogous diff. oper.

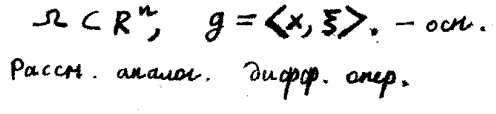
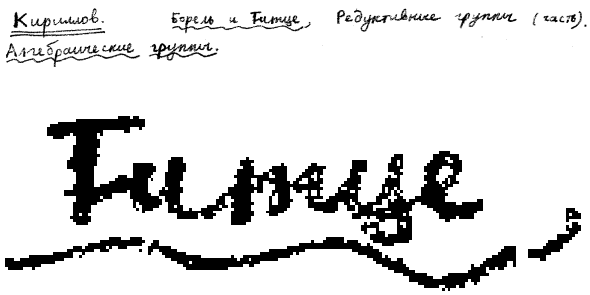


Diagram and typed description. The final lines have $\Omega \subset R^n$, the linear phase $g = \langle x, \xi \rangle$, and "Consider analogous diff. oper.".

Original source details: GELFAND-P000136

GELFAND-P000137

Kirillov. Borel and Tits, Reductive groups (part). Algebraic groups.



Manuscript notation. The manuscript writes «Борель и Титце» with a comma after the names.

Original source details: GELFAND-P000137

Algebr. variety—a locally aff. scheme. From the ring A, construct a ringed space; Spec A—the prime ideals of A. $A = C[x_1, x_2]$. A prime ideal is generated either by $\underbrace{((x_1 - a), (x_2 - b))}_{\text{(point)}}$, or by an irreduc. polynomial $I = (P(x_1, x_2))$. It is equipped with the corresponding ringed-space structure.

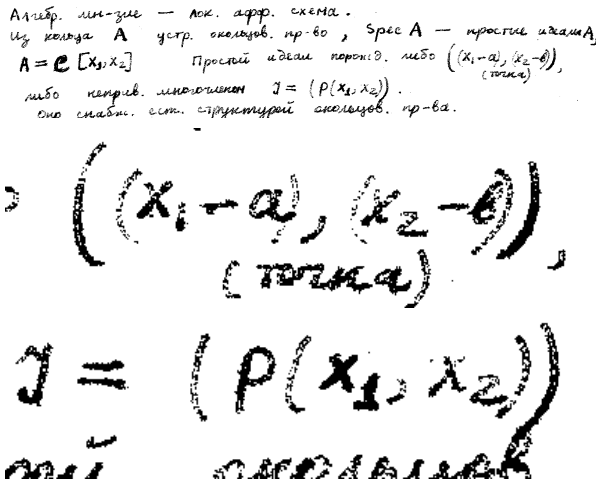


Diagram and typed description. The point example has outer parentheses and separate parentheses around $x_1 - a$ and $x_2 - b$, with "(point)" below. The second example has $I = (P(x_1, x_2))$.

Original source details: GELFAND-P000137

An algebr. group—an algebr. variety, with multipl. $G \times G \to G$, given everywhere by a regular (rat.) function. Chevalley's struct. theorem: in every alg. group G there is a max. linear group G_0 such that G/G_0 is an abelian variety.

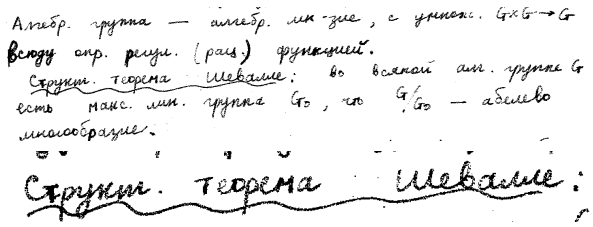
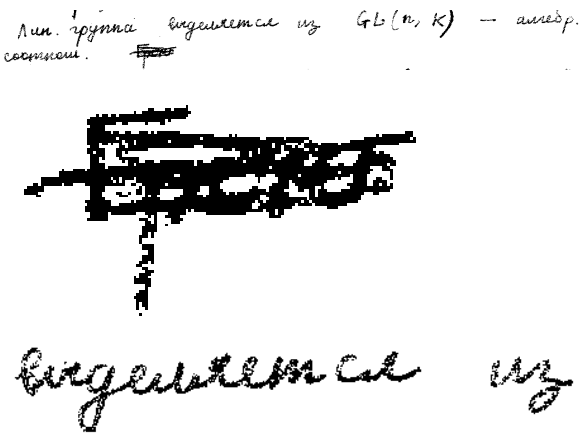


Diagram and typed description. The text has G_0 and G/G_0; the structural-theorem heading has a wavy underline.

Original source details: GELFAND-P000137

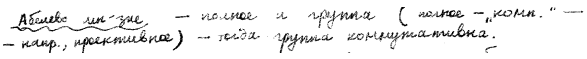
A linear group is singled out from $GL(n, K)$—algebr. relations. [Crossed out: unread word].



Original source details: GELFAND-P000137

Uncertain reading. After the line about algebr. relations, a separate word is heavily crossed out; the obscured letters remain unread.

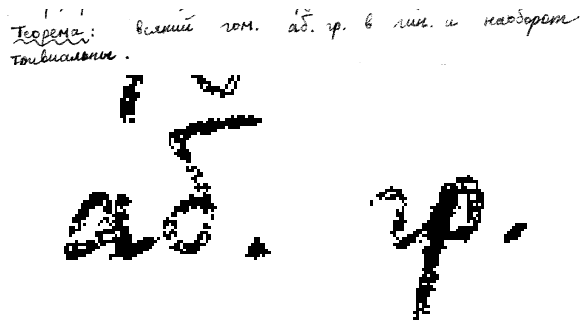
Abelian variety —complete and a group (complete—“comp.”—e.g. projective)—then the group is commutative.



Original source details: GELFAND-P000137

Diagram and typed description. “Complete” is explained in parentheses by “comp.” and the projective example; the phrase about commutativity follows.

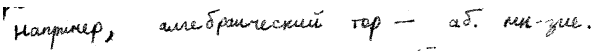
Theorem: every hom. from ab. gr. into linear, and conversely, is trivial.



Original source details: GELFAND-P000137

Manuscript notation. The abbreviation after «аб.» (ab.) is «гр.» (gr.).

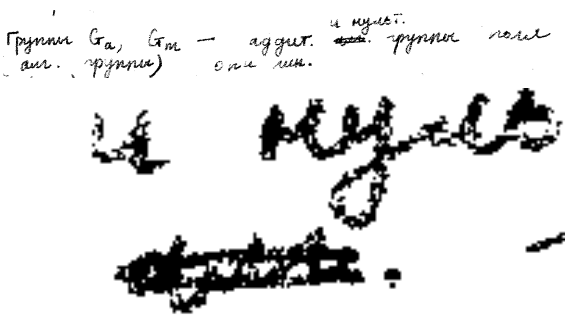
For example, an algebraic torus—an ab. variety.



Original source details: GELFAND-P000137

Diagram and typed description. The text reads “algebraic torus—an ab. variety”.

The groups G_a, G_m—additive [Crossed out: unread short word] [Inserted above: and multi.] groups of the field (alg. groups). They are linear.



Original source details: GELFAND-P000137

Uncertain reading. “And multi.” is inserted above the crossed-out short word. The obscured word remains unread.

$$\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \qquad (x).$$

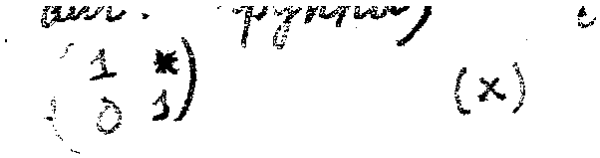


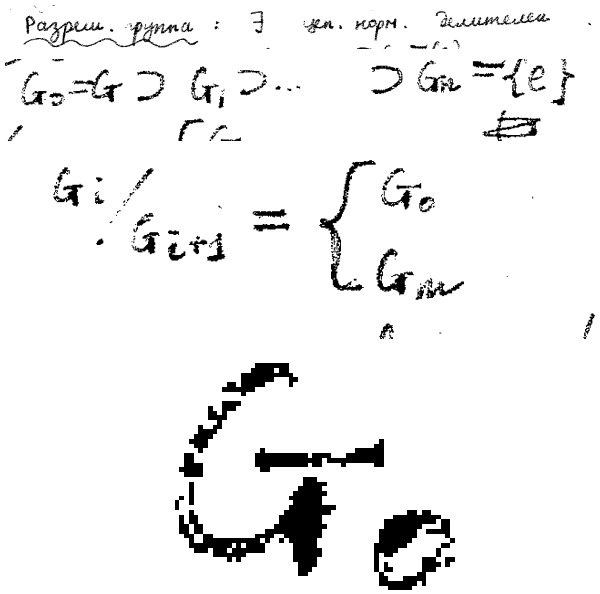
Diagram and typed description. On the left is a matrix with $1, *, 0, 1$; on the right, separately, is (x).

Original source details: GELFAND-P000137

Solvable group: $\exists$ chain of normal subgroups

$$G_0 = G \supset G_1 \supset \ldots \supset G_m = \{e\}.$$

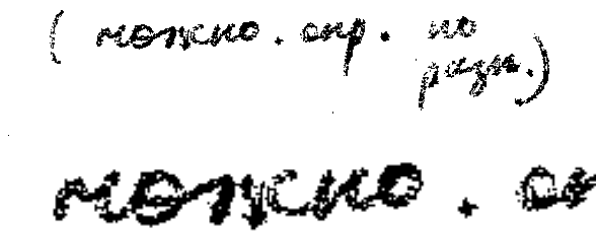
Such that $G_i/G_{i+1} = \begin{cases} G_{a[?]}, \\ G_m. \end{cases}$



Uncertain reading. In the first alternative, the subscript is a- / 0-like; the exact reading remains unsettled. The chain has $G_0 = G$, then G_1, and ends with $G_m = \{e\}$.

Original source details: GELFAND-P000137

$\Longleftrightarrow$ (imprec. def. but reason.)



Manuscript notation. The parenthesis reads “imprec. def. but reason.”.

Original source details: GELFAND-P000137

Radical—max connected (Zariski top.) [Inserted above: solvable] normal subgroup.

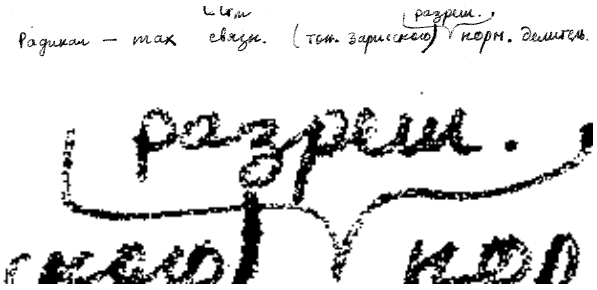
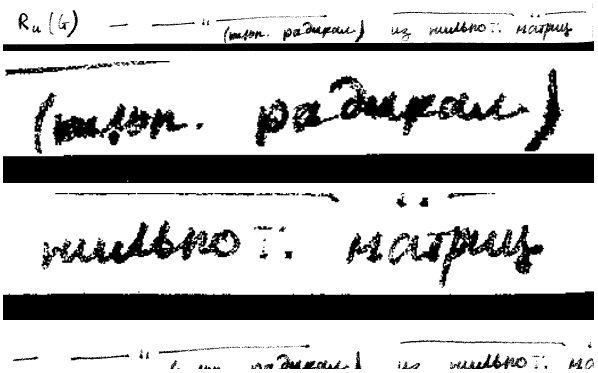


Diagram and typed description. “Solvable” is inserted above “normal”, with an arc below and an angular insertion mark.

solvable

normal

Original source details: GELFAND-P000137

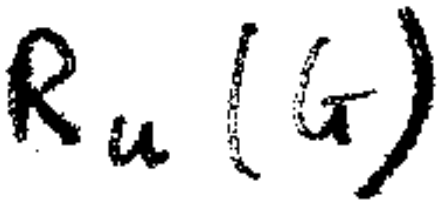


Uncertain reading. At the lower edge is $R_u(G)$, with two groups of long strokes and ditto marks; below the first is “(unip.[?] radical)”, with part of the first word obscured. On the right is “from nilpo:: [?] matrices”; the abbreviation and following marks remain unsettled.

$R_u(G)$ — —— " ———— ———— " —

(unip.[?] radical)

from nilpo:: [?] matrices



Original source details: GELFAND-P000137

GELFAND-P000138

(Nilpot. $(u-E)^k=0$.) Reductive group—its radical is a torus. (Direct product of several mult. groups of the field.)

$$G \text{ semisimple} \iff R(G)=1.$$

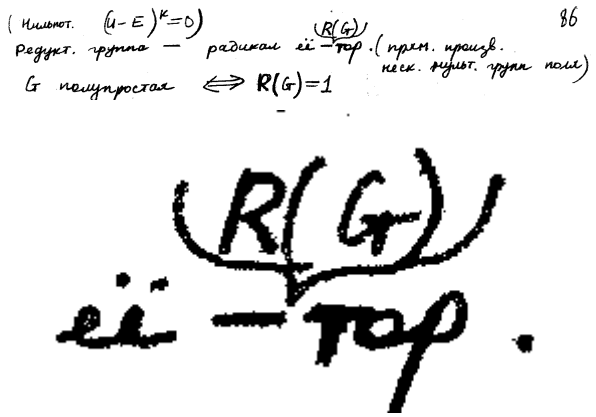


Diagram and typed description. $R(G)$ is written above “torus”, with an arc below and an angular stroke.

$R(G)$

torus

Original source details: GELFAND-P000138

$GL(n,K)$—reductive, $SGL(n,K)$—semisimple. Classif. of semisimple groups: 4 infinite series and 5 exceptional (as for Lie groups). Basic notions—max torus and a root system.

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

real matrices—isom. to the group of complex numbers under mult.—a torus. All max tori are conjugate. (A root system is associated with a torus.)

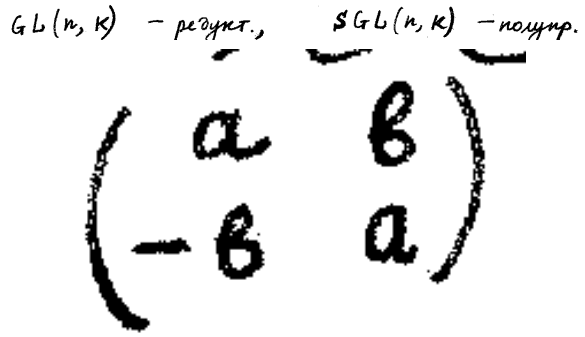
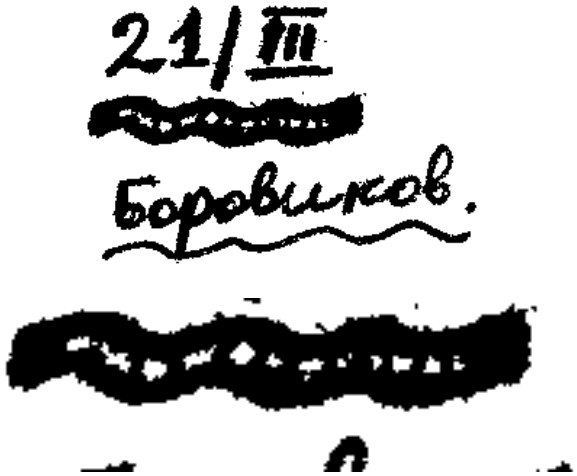


Diagram and typed description. The text has $SGL(n,K)$. The real matrix has a, b above and $-b, a$ below.

Original source details: GELFAND-P000138

21/III.



Uncertain reading. Below the date is a dense wavy band: heavy underlining / a cancelled line. There are too few distinguishable letters.

Original source details: GELFAND-P000138

Borovikov.

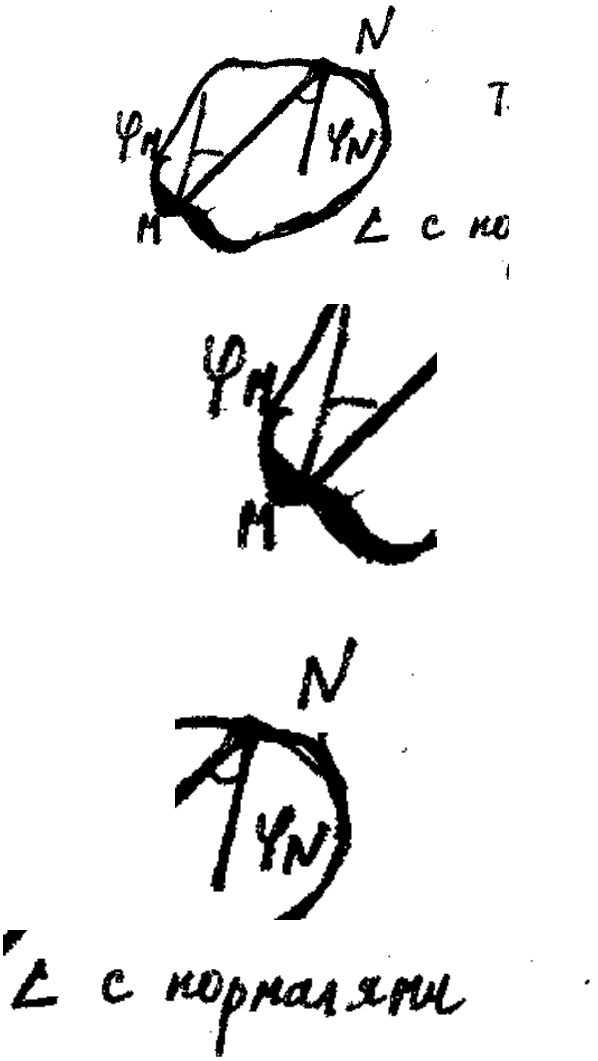
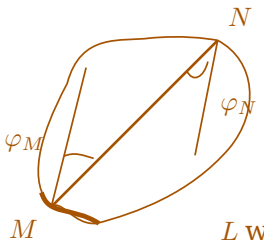


Diagram and typed description. A rounded contour, with M below left and N above right; a chord, interior strokes and angles φ_M, φ_N. The label is "L with normals". The dense lower part and exact joins remain partly unread.



L with normals

Original source details: GELFAND-P000138

Theorem.

$$\iint \cos\varphi_M \cdot \cos\varphi_N \, dM \, dN = 2\pi S.$$

(S—area.)

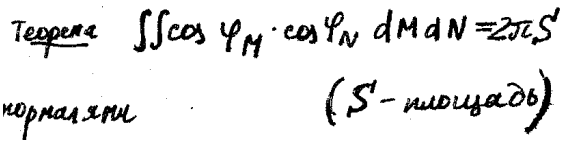
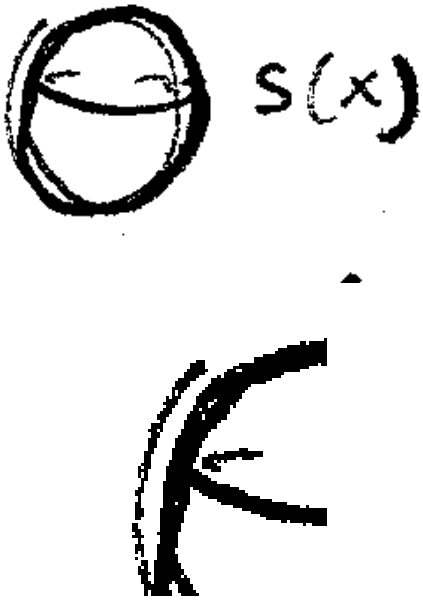


Diagram and typed description. The visible formula has $\cos\varphi_M \cdot \cos\varphi_N \, dM \, dN = 2\pi S$; no distance denominator is visible.

Original source details: GELFAND-P000138

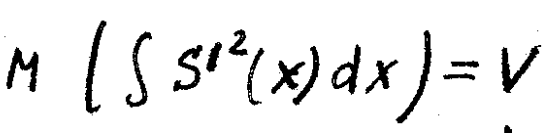
Gelfand.



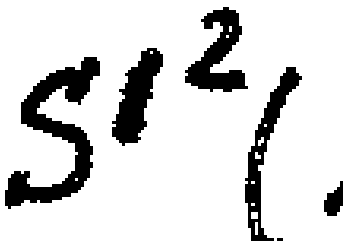
Original source details: GELFAND-P000138

Diagram and typed description. A rounded contour with a repeated left stroke, an interior section arc and the label $S(x)$ on the right. Short upper arcs are visible at the ends; the dense left join remains partly unread.

$$M\left(\int S'^{2}(x)\,dx\right) = V.$$



Manuscript notation. A separate prime follows S, with a raised 2 farther to the right.



Original source details: GELFAND-P000138

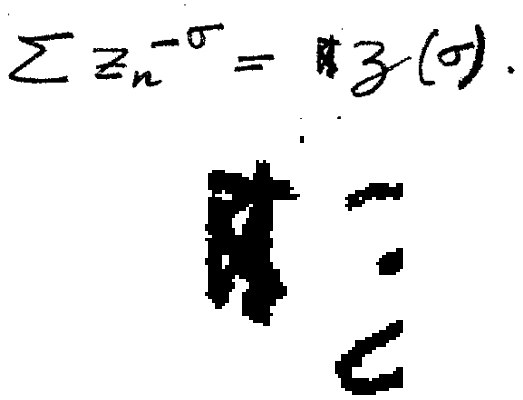
M—a universal constant (of the type of π).

Lidskii. (Viktor Borisovich.)

$$f(z) = \sum_{k=1}^{N} e^{\alpha_k z} \mathcal{P}_k(z).$$

$\mathcal{P}_k(z)$—polyn. and asympt. series in $\frac{1}{z}$.

$$\sum z_n^{-\sigma} = \text{[Crossed out: unread sign]}\ \zeta(\sigma).$$



Uncertain reading. Before $\zeta(\sigma)$ is a heavily crossed-out vertical sign; its former reading remains unsettled.

Original source details: GELFAND-P000138

$\zeta(\sigma)$ is always entire in σ. z_n are the zeros of $f(z)$.

$$\mathcal{P}_k(z) \simeq p_k(z) + \sum_{s=0}^{\infty} \frac{a_s}{z^s}.$$

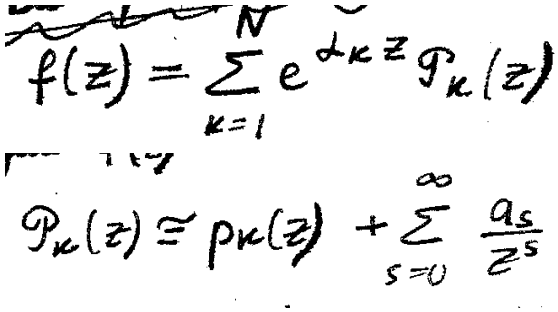
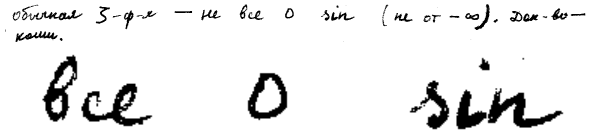


Diagram and typed description. The first sum runs from $k = 1$ to N, with exponent $\alpha_k z$ and factor $\mathcal{P}_k(z)$. The lower sum runs from $s = 0$ to ∞.

Original source details: GELFAND-P000138

The usual ζ-function—not all 0[?] sin (not from $-\infty$). Proof—
— $\int$ Cauchy.

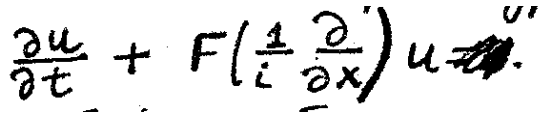


Uncertain reading. Between “all” and sin is a rounded sign: 0 / O. The contracted wording remains unsettled. In the proof note, a dash occurs both at the end of the line and before $\int$ Cauchy.

Original source details: GELFAND-P000138

Gindikin. Generalization of a parabolic eq.

$$\frac{\partial u}{\partial t} + F\left(\frac{1}{i}\frac{\partial}{\partial x}\right) u$$ [Crossed out: unread ending]



Uncertain reading. After u is a heavily crossed-out $= 0$-like ending; the exact reading remains unsettled.

Original source details: GELFAND-P000138

In place of $\frac{\partial u}{\partial t}$—any hyperb. operator. $P\left(\frac{\partial}{\partial t}\right)$. A family of oper. is associated with P, such that with $\frac{\partial u}{\partial t}$ one obtains parabolic.

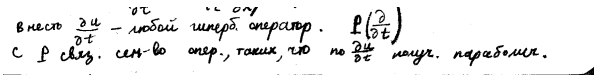


Diagram and typed description. Before $P\left(\frac{\partial}{\partial t}\right)$, the text has "hyperb. operator.". the final line associates P with a family of operators and repeats $\frac{\partial u}{\partial t}$.

Original source details: GELFAND-P000138

GELFAND-P000139

Instead of ellipt. — V-ellipt., map. in a cone, corresp. to hyperb. polyn.

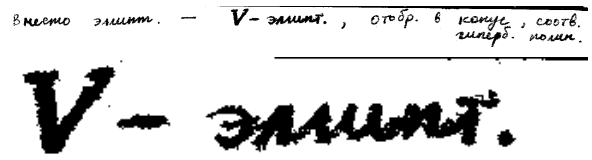


Diagram and typed description. Opening line: V-ellipt., map. in a cone, corresp. to hyperb. polyn.

Original source details: GELFAND-P000139

Walk along a line — heat eqn.

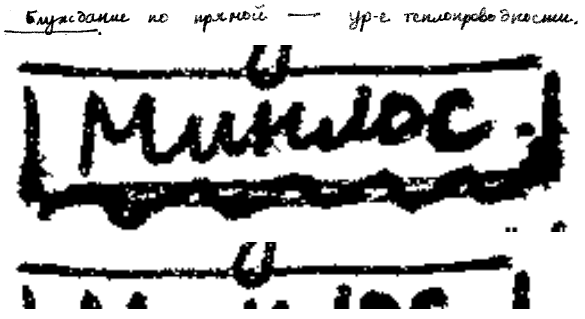
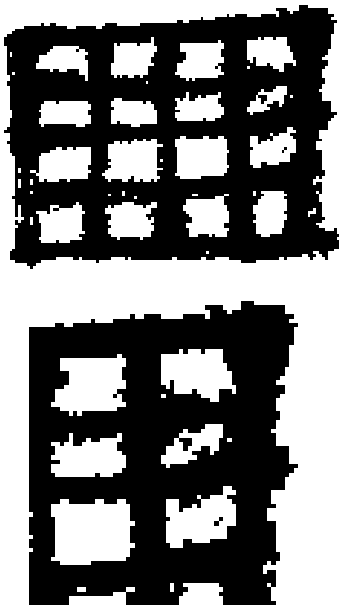


Diagram and typed description. The word "Walk" is underlined; this stroke forms the top of the name's frame. The bottom of the frame is reinforced.

Minlos.

Original source details: GELFAND-P000139

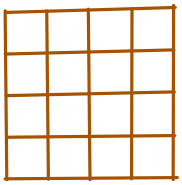
Ising model.



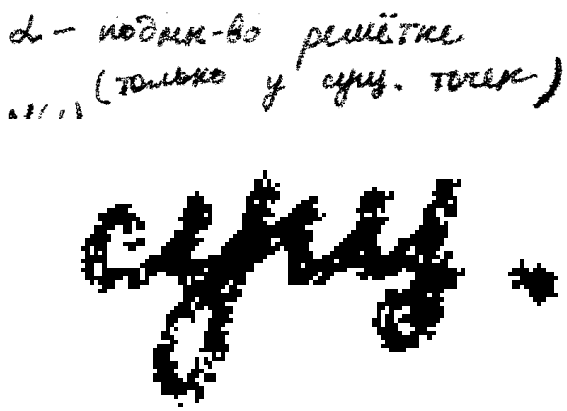
Original source details: GELFAND-P000139

L — subset of the lattice.

Diagram and typed description. Lattice: four cells in each direction. The intersections are dense; individual filled cells or state markings have not been established.



$E(L)$ — number of neighbors (only at exist. points).

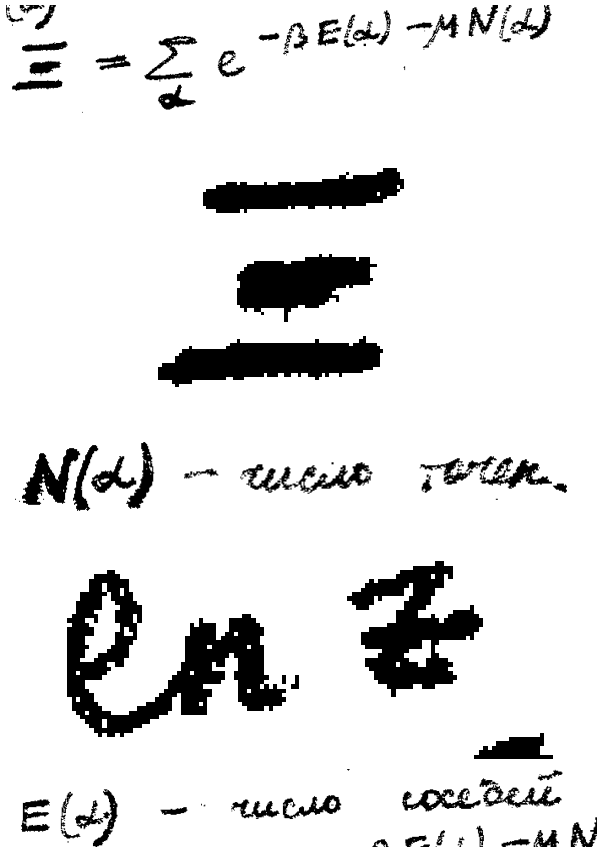


Manuscript notation. “(only at exist. points)” is written at right on the line with $E(L)$ and “number of neighbors”.

Original source details: GELFAND-P000139

$$\Xi = \sum_L e^{-\beta E(L) - \mu N(L)}.$$

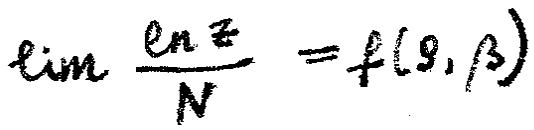
$N(L)$ — number of points.



Manuscript notation. The first sum has Ξ with three separate strokes. The next formula has Z in the logarithm.

Original source details: GELFAND-P000139

$$\lim \frac{\ln Z}{N} = f(\theta[?], \beta).$$

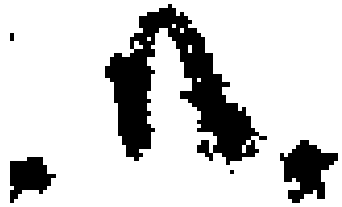


Uncertain reading. The first argument of f is a loop with a lower stroke: θ / ϑ / ρ; the exact reading is unsettled.

Original source details: GELFAND-P000139

and so on. and so forth.

Manuscript notation. After “and so on”, the letters for “and so forth” are written larger. The first letter of that abbreviation is underlined.



Original source details: GELFAND-P000139

Chapter 2

GELFAND-P000140

Gelfand seminar. $5/\overline{\text{IX}}$

1. There will be systematic stat. physics. (Minlos)
2. Status of pseudodifferential operators. (Vishik and Agranovich)
3. Algebraic groups (Kirillov)

Diagram and typed description. At the upper right, page number 5 has a line at its left and below it.

5

Original source details: GELFAND-P000140

Valuable information: Atiyah's lectures on the K-functor require no prior knowledge except the def. of a topological space and of an n-dimensional vector space (In degree of clarity it is not comparable even to Milnor)

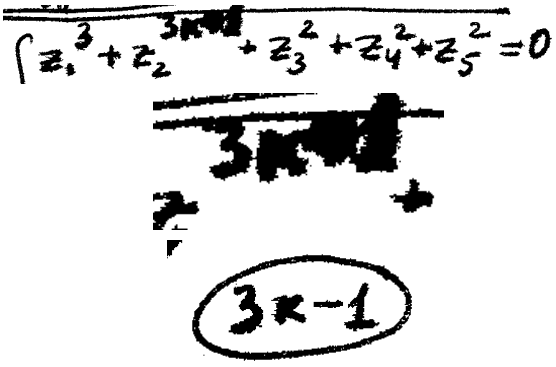
Manuscript notation. A fine stroke joins o and з in «прозрачности» (clarity).

Original source details: GELFAND-P000140

4. Deformations of every kind. (provisional name: nonlinear analysis), including Kodaira and Spencer.

Hirzebruch's result. (Gelfand)

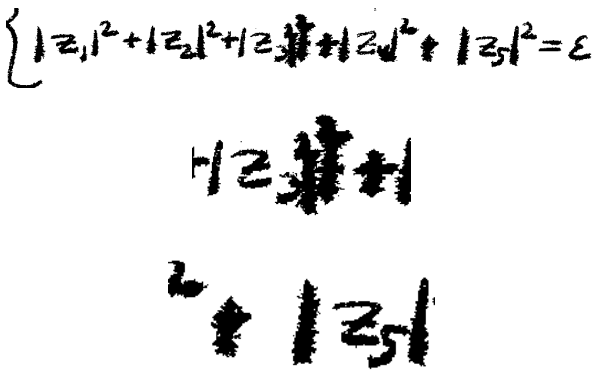
$$\begin{cases} z_1^3 + z_2^{3k-1[?]} + z_3^2 + z_4^2 + z_5^2 = 0 \\ |z_1|^2 + |z_2|^2 + |z_3|^{2[?]} + |z_4|^2 + [?]\,|z_5|^2 = \varepsilon \end{cases}$$



Uncertain reading. The exponent on z_2 begins $3k$; its ending is heavily overwritten: $3k-1$ / another overwritten ending. Separately at the right, $3k-1$ is circled with an oval. The written exponent on z_1 is 3.

$3k-1$

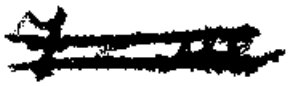
Original source details: GELFAND-P000140



Uncertain reading. In the second line, the right bar and exponent at $|z_3|$ are heavily overwritten; $|z_3|^2$ is given with this qualification. Between $|z_4|^2$ and $|z_5|^2$ the sign resembles $+$ but is dense.

Original source details: GELFAND-P000140

10 unk. and [crossed-out unread fragment] 3 eqns.



Uncertain reading. A short crossed-out fragment after "and": a 7-like sign and further dense strokes; the original content is unreadable.

Original source details: GELFAND-P000140

Milnor: this is homeomorphic to S^7

Hirzebruch: for $k = 1, 2, \ldots, 28$ there are different diff. structures there, and thereafter everything repeats periodically and these are all the Milnor spheres.

S^{11} there are 992 of them and they are obtained by such equations.

GELFAND-P000141

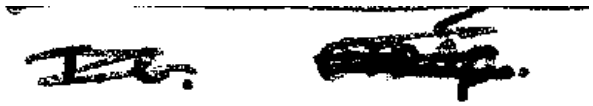
Bass (Kazhdan) rigidity of arithm. subgroups.

Manuscript notation. After Bass, the parentheses contain Каждан (Kazhdan).

Original source details: GELFAND-P000141

$SL(n, Z) \subset SL(n, R)$ — a rigid subgroup,

[A short opening and a word before "every" are crossed out.]



Uncertain reading. The initial crossed-out designation resembles "II." / Cyrillic "Те."; the following heavily cancelled word is unreadable.

Original source details: GELFAND-P000141

every deformation, (i.e. a map)

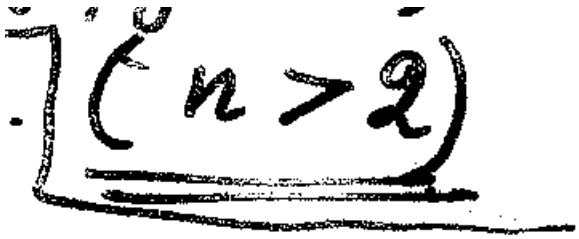


Diagram and typed description. At right, $(n > 2)$ has two underlines. The angular stroke on the left joins the long line above the condition.

$(n > 2)$

Original source details: GELFAND-P000141

$$SL(n, Z) \times [0, 1] \longrightarrow SL(n, R)$$

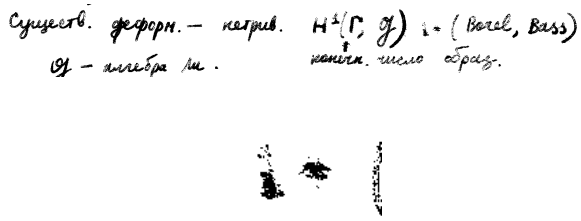
which is continuous and a homomorphism for $\forall t \in [0,1]$, at $t=0$ it is the inclusion) is trivial, i.e. it is an inner automorphism with respect to $g(t)$ (a curve in the group)

$$\varphi(h \times t) = g(t)h \cdot g(t)^{-1}, \qquad h \in SL(n, Z)$$

Weil's theorem is proved only for the case when the quotient space is compact.

The prob. reduces to a problem about congruence subgrps. over a finite field. (If there are two homomorphisms, it is unknown whether they can be joined)

Existence of deformations — nontrivial



Original source details: GELFAND-P000141

Uncertain reading. After $H^1(\Gamma, \mathfrak{g})$ are small colon-like and star-like strokes; their literal reading is not established. Below the expression is "finite no. of generators" with an upward arrow.

$H^1(\Gamma, \mathfrak{g}){:}*[?]$ (Borel, Bass)

↑

finite no. of generators

$\mathfrak{g}$ — the Lie algebra.

In $\mathfrak{g}$ there is a lattice of integer matrices L. (L — a lattice in the Lie algebra)

$$H^1(\Gamma, \mathfrak{g}) = H^1(\Gamma, L) \underset{Z}{\otimes} R$$

Prove that $H^1(\Gamma, L)$ is finite, It suffices to prove that $\exists \underset{\text{prime}}{p}$ such that multiplication by p is an epimorphism

$$H^1(\Gamma, L) \xrightarrow{p} H^1(\Gamma, L) \longrightarrow H^1(\Gamma, L/pL)$$
$$(\text{from } 0 \longrightarrow L \xrightarrow{p} L \longrightarrow L/pL \longrightarrow 0)$$

It suffices to prove that the right-hand term is 0.

One can reduce the matter to the case when Γ is finite: in place of Γ — $\Gamma/[\Gamma_p, \Gamma_p]$, Γ_p — the grp. of elems. congruent to 1 modulo p.

GELFAND-P000142

Thus, it suffices to prove that

$$H^1(\Gamma/\Gamma_p, L/pL) = 0$$

and everything reduces to fin. groups. The same Borel an incredibly difficult proof. Bass — only for split ones.

Smale (Anosov)

Diagram and typed description. Page number 6 at the upper right, with a line at its left and below it.

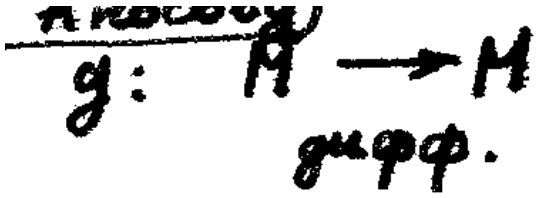
6

Original source details: GELFAND-P000142

Arnold. U-system (according to Anosov)

$$g : M \underset{\text{diff.}}{\longrightarrow} M,$$

if



Manuscript notation. Both letters M have a visible central dip. Beneath the arrow is “diff.”.

Original source details: GELFAND-P000142

there are 2 dirs.

Diagram and typed description. Two separate intersections of horizontal and nearly vertical segments; a gap in the upper part of the first vertical stroke. There are no arrows or letter labels.

Original source details: GELFAND-P000142

along the 1st it stretches, along the other it contracts ($\leq$, $\geq$ estimates on contraction and stretching)

Manuscript notation. After “along the 1st”, stretching is written as «растяг.».

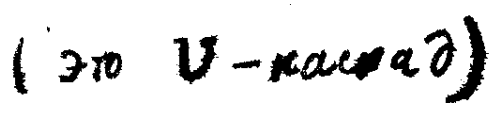
Original source details: GELFAND-P000142

> Example: autom. of the torus, nothing more was known.
> Smale asserts that he has found manifolds besides the torus and their U-automorphisms and all such automorphisms (conjecture).

Precise def.: $\exists TM_x = X_x \oplus Y_x, \forall \xi \in X_x, g^*\xi \in X_{gx}$,
a Riem. metric exists

$$|g^*\xi| < \theta|\xi|, \quad 0 < \theta < 1,$$
$$\forall \eta \in Y_x, \quad |g^*\eta| > \frac{1}{\theta}|\eta|.$$

(this is a V [?]-cascade)



Uncertain reading. The opening heading has U, whereas this sign resembles V: V-cascade / U-cascade.

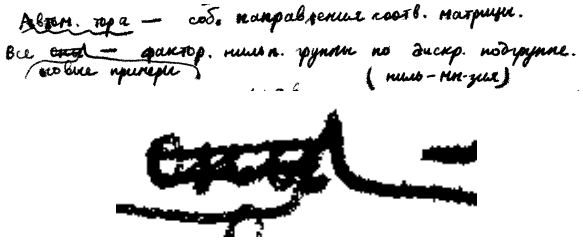
Original source details: GELFAND-P000142

Autom. of the torus — eigendirections of the corresponding matrix.

All [crossed-out unread word] — a quotient of a nilp. group by a discrete subgroup.

new examples

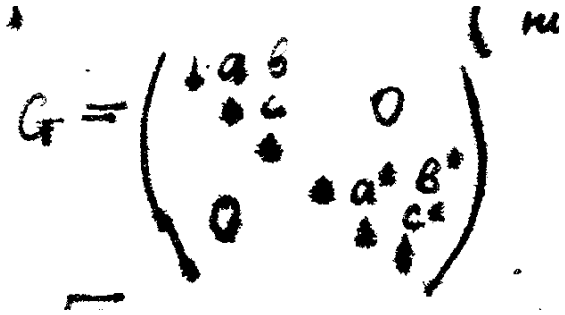
(nil-manifolds)



Original source details: GELFAND-P000142

Uncertain reading. The crossed-out word after “All” has Cyrillic с/ю/д-like remnants; the original word is unreadable. The insertion “new examples” is visible beneath a spanning curved mark.

G/Γ, $G =$



Original source details: GELFAND-P000142

Uncertain reading. A matrix in round parentheses, six diagonal positions, two upper-triangular blocks. At the top a, b, below c; in the lower block a^*, b^*, c^*; two 0 signs outside the blocks. The six dense diagonal marks are shown as dots: 1 / black dots; their exact literal reading is not established.

$$\begin{pmatrix} a & b & \\ & c & 0 \\ & & \\ 0 & & a^* \quad b^* \\ & & \quad\ c^* \\ & & \end{pmatrix}$$

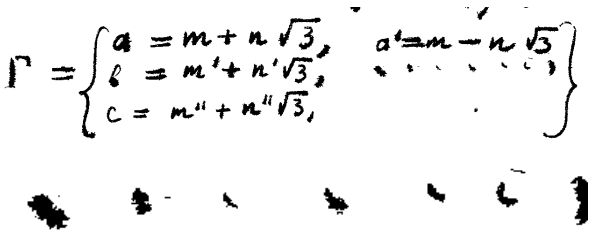
— of order 6

$$\Gamma = \left\{ \begin{array}{ll} a = m + n\sqrt{3}, & a^{*[?]} = m - n\sqrt{3} \\ b = m' + n'\sqrt{3}, & \text{`` `` `` `` `` ,} \\ c = m'' + n''\sqrt{3}, & \end{array} \right\}$$

G/Γ is compact.

Dim. $3 + 3$

contract. and expand.



Original source details: GELFAND-P000142

Uncertain reading. A short dense mark above the second a: a star / a stroke; a^* is given with this qualification. Six short repetition strokes stand at the right of the b line; their content is not expanded. An encompassing right brace. The horizontal rule beneath “G/Γ is compact.” runs above “Dim. $3 + 3$”.

GELFAND-P000143

Anosov proper.

Cascade — iterations of a diff. homeomorphism (with continuous time — a flow) Describe typical dynamical systems — cascades. (an everywhere dense set, open)

Structurally stable flow — passes to an equivalent one under a small (C^1) perturbation. (equiv. — differ by a diffeom.)

2-dimensional case: Andronov–Pontryagin thm. — structurally stable systems are described (a finite number of "stable" cycles and singular points and saddle separatrices in general position — — do not intersect, do not coincide.) Their set is open, dense.

Andronov–Pontryagin system.

1. A fin. number of equilibria and periodic trajectories, all locally structurally stable (conditions on eigenvalues and multipliers) (in the multidimensional case this is already nontriv.)

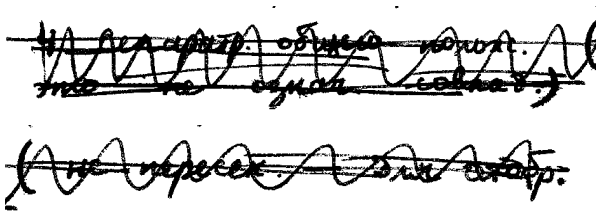
2. Separatrices in general position

(Cond.,
Conditionally stable manifolds,
Asympt. bundles,
horospheres,
horocycles)

(Manifolds of incoming and outgoing — — see Coddington and Levinson)

3. All trajectories as $t \longrightarrow \pm\infty$ tend to equilibria and periodic solutions.

[Two lines are densely cancelled with straight and wavy strokes. Surviving fragments: 4) / ∀ [?], "separatrices in general position", "(do not intersect", 2 [?], Cyrillic "собр."; below, "this does not mean[?] coincidence[?]".]



Uncertain reading. Before "separatrices in general position", the crossed-out sign resembles 4) / ∀. Then "(do not intersect" and several obscured words. The lower line tentatively reads "this does not mean coincidence". The cancellation strokes pass through the letters.

Original source details: GELFAND-P000143

4 assertions.

1. All A.–P. — structurally stable — (?)

2. All structurally stable — A.–P.
3. A.–P. are dense
~~(Cyrillic дру [?]~~
4. Structurally stable are dense

Refuted by Smale.

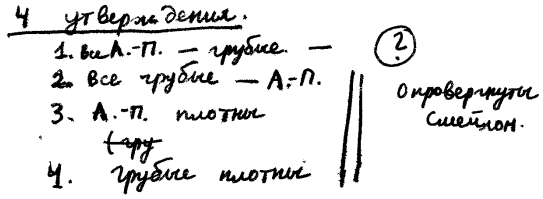
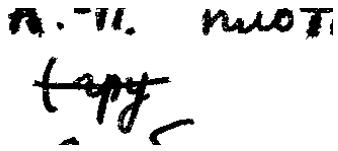


Diagram and typed description. After item 1, a hook-shaped question mark with a separate dot is circled. At the right of items 2–4 are two vertical rules and the words "Refuted by Smale".

Original source details: GELFAND-P000143



Uncertain reading. A short cancelled parenthesis beneath item 3: Cyrillic "дру" / "друг" / unreadable.

Original source details: GELFAND-P000143

GELFAND-P000144

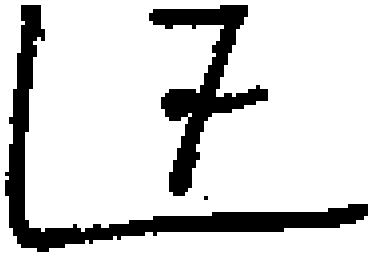


Diagram and typed description. Page number 7 at the upper right, with a line at its left and below it.

7

Original source details: GELFAND-P000144

Structurally stable system: $\dot{w} = f(w)$.

$$\forall \varepsilon\ \exists \delta, \quad \forall g, \quad \|g - f\|_{C^1} < \delta,$$

$\exists\varepsilon$-homeom., taking trajectories of f to trajectories of g.

For a cascade: commutative diagram

($\exists\varepsilon$-homeom. χ)

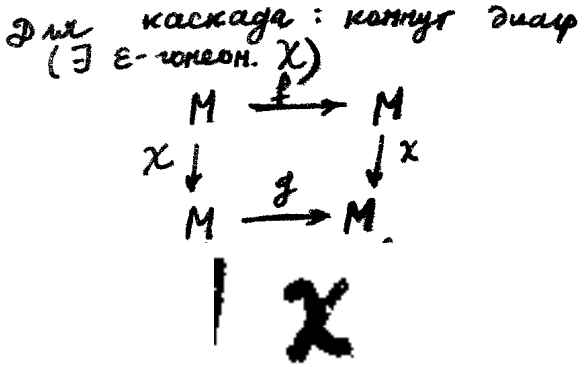
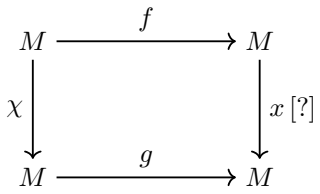


Diagram and typed description. Four M nodes; f above, g below, both horizontal arrows point right, both vertical arrows down. At the left, χ; at the right, a short x- / χ-like sign.



Original source details: GELFAND-P000144

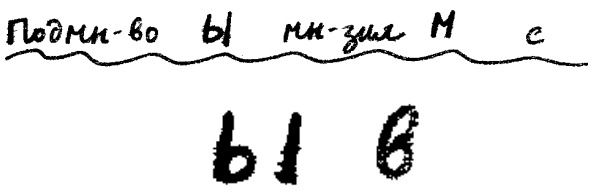
(For flows it is natural to assume that the speed may change; if this is forbidden, almost no structurally stable ones remain)

A subset $b|$ [?] of a manifold M with a smooth cascade is called a U-subset (hyperbolic subset)

compact,

invariant,

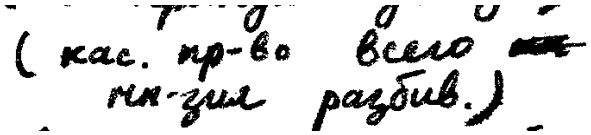
$\exists\Theta$ and a decomposition of the tangent bundle over $b|$ [?] into a direct sum, having the indicated properties. (U ...)



Uncertain reading. Subset designation: $b|$ / b_1 / another sign; a loop and a separate tall stroke. Repeated after “over”.

Original source details: GELFAND-P000144

(the tangent space of the whole [unreadable short cancelled fragment] manifold is decomposed.)



Uncertain reading. After “whole”, a short fragment is densely cancelled; the literal word cannot be read.

Original source details: GELFAND-P000144

Example: 1) a hyperbolic equilibrium (e.g. stable — — expanding — the whole space)

2) any U-system

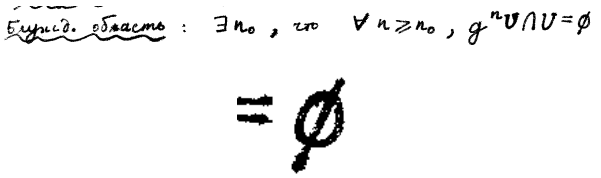
3) There are examples when such a set is a torus, a Cantor set, etc.

Smale systems.

Nonwandering point.

Wandering region: $\exists n_0$ such that

$$\forall n \geq n_0, \quad g^n U \cap U = \varnothing$$

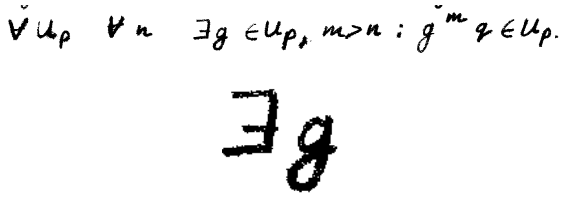


Manuscript notation. The oval to the right of the equality has a diagonal stroke extending past it: $\varnothing$.

Original source details: GELFAND-P000144

A point is called wandering if it has a wandering neighbourhood.
A nonwandering point — a point that is not wandering, i.e.

$$\forall U_p \; \forall n \quad \exists g \in U_p, \quad m > n: \quad g^m q \in U_p.$$



Manuscript notation. The g after $\exists$ has the same lower loop as the base of g^m. The letter after g^m is q.

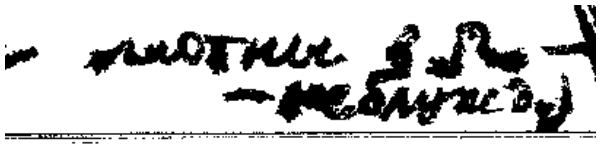
Original source details: GELFAND-P000144

The set of nonwandering points is closed. (denoted — Ω)

Smale system (def)

1. Ω — hyperbolic.

2. $\overline{\Gamma} = \Omega$ (periodic points are dense in Ω — nonwandering [?])

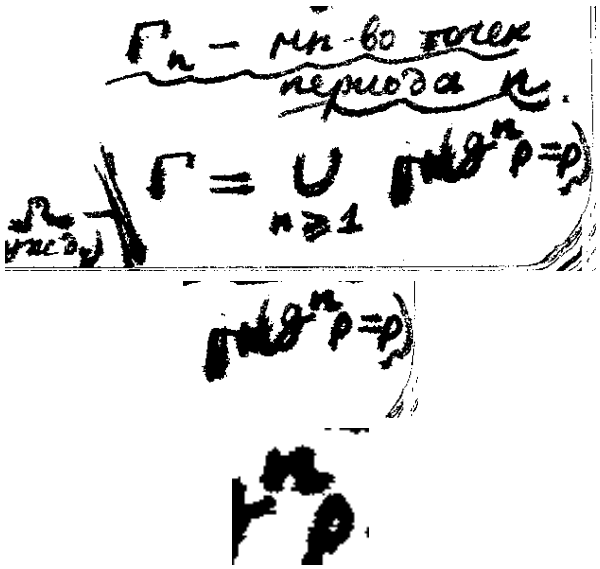


Uncertain reading. The explanation ends at the lower boundary: “— неблужд.” (nonwandering) / a similar contraction; dense strokes.

Original source details: GELFAND-P000144

Γ_n — set of points of period n.

$$\Gamma = \bigcup_{n\geq 1} \Gamma_n\,[?] \quad (g^n p = p)\,[?]$$



Uncertain reading. After the union sign, Γ_n-like signs, the opening parenthesis and g are densely overwritten; the exponent is n-like. The original overwritten signs are undetermined. At the left are two slanting strokes with a short transverse stroke.

Original source details: GELFAND-P000144

GELFAND-P000145

It can happen that all points are nonwandering, but there are no periodic ones (e.g. nil-flows [?])

Uncertain reading. Faint parenthetical entry: “ниль-потоки” (nil-flows) / “нуль-потоки” (null-flows); the middle letter is not quite clear.

Original source details: GELFAND-P000145

Conjectures.

$\{\overline{\Gamma} = \Omega\}$ — everywhere dense — proved by Pugh; 1964 (only in C^1) !!

Manuscript notation. Before “; 1964” the name Pugh is written «Пью».

Original source details: GELFAND-P000145

Smale systems are everywhere dense (?)

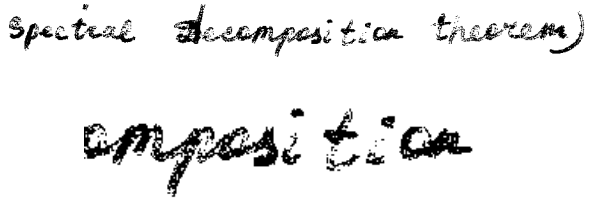
A great problem to which it is worth devoting one’s life.

Apparently 2 follows from 1 (Prob. 70%), so it suffices to prove that systems with hyperbolic Ω are dense.

Theorem S. D. T. (Smale) (Spectral Decomposition theorem)

Manuscript notation. The initials S. D. T. precede “(Smale)”.

Original source details: GELFAND-P000145



Manuscript notation. The last o and n in Decomposition are joined.

Original source details: GELFAND-P000145

on the structure of Smale systems.

$$\Omega = \bigcup_{i=1}^{k} \Omega_i$$

Ω_i — invariant, closed and topologically transitive.

All theorems follow from 1 alone.

GELFAND-P000146

12/IX Gelfand sem.

Diagram and typed description. Two wavy lines beneath the name “G. E. Shilov.”

G. E. Shilov.

Original source details: GELFAND-P000146

$$S = \{x \in H,\ |x| \leq 1\}$$

— the unit ball of a Hilbert space.

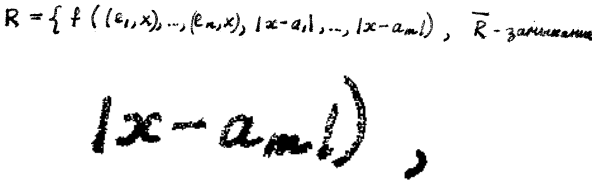
$L(S)$ — ring of all weakly continuous functions.

$$L(S) \subset \overline{R} \subset C(S)$$

$C(S)$ — strongly continuous functions.

$$R = \{f((e_1, x), \ldots, (e_n, x), |x - a_1|, \ldots, |x - a_m|),$$

$\overline{R}$ — closure



Uncertain reading. After the long formula for R, a closing round parenthesis and a comma are visible; a closing brace for the initial { is not visible.

Original source details: GELFAND-P000146

Uncertain reading. The chain $L(S) \subset R \subset \overline{R} \subset C(S)$ is cancelled; the first letters and first inclusion sign are overwritten, and the beginning is not fully established.

Original source details: GELFAND-P000146

PIM of the ring $L(S)$ — the Hilbert ball

PIM of $\overline{R}$ — a bicompact extension.

$$M_\tau(f(x), z) \qquad M^*_\tau(f(x), z)$$

Uncertain reading. The mean's subscript is τ- / r-like; the same looped sign occurs in the later limit formula and squared in its denominator.

Original source details: GELFAND-P000146

$e_1, e_2, \ldots, e_n, \ldots$ — an orthogonal basis, determining an increasing chain of subspaces $E_1 \subset E_2 \subset \ldots$

Diagram and typed description. Two ovals around "over the sphere" and "over the ball".

over the sphere over the ball

Original source details: GELFAND-P000146

The means — the limit of finite-dimensional ones. If they exist, they are equal.

The means are defined on $\overline{R}$.

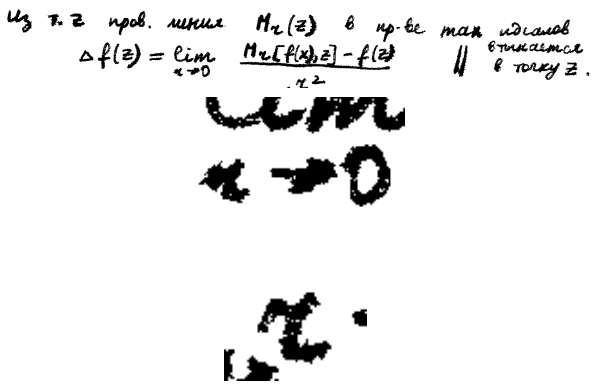
(For R, one can simply write down formulas).

M_τ — a multiplicative functional. This is a max ideal, but it is not a value at a point, but a value in a bicompact extension.

From the point z, lines $M_\tau(z)$ are drawn in the space of max ideals.

$$\Delta f(z) = \lim_{\tau \to 0} \frac{M_\tau[f(x), z] - f(z)}{\tau^2}.$$

$\|$ stick into the point z.



Uncertain reading. The looped sign in M_τ, below the limit and in τ^2 is τ / r / an undetermined letter. At the right of the formula are two vertical rules and "stick into the point z".

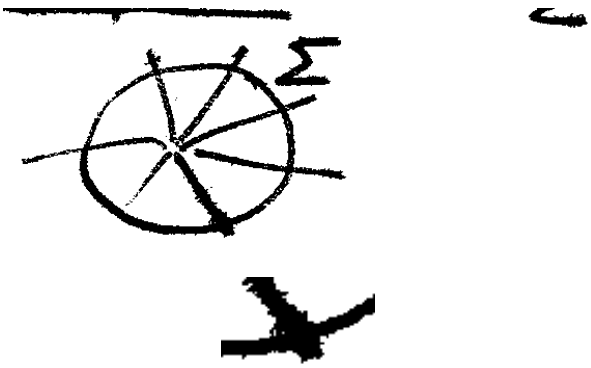
Original source details: GELFAND-P000146

This is a first-order operator!!

$$\Delta(fg) = f \cdot \Delta g + (\Delta f)g$$

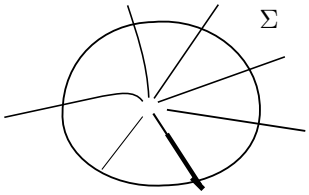
Dirichlet problem.

Σ — a closed surface (star-shaped)
(weak)



Original source details: GELFAND-P000146

Diagram and typed description. An irregular rounded closed contour with Σ at the upper right; seven radiating strokes, with small gaps at the centre. Several pass beyond the contour, the lower-left one ends inside; the left stroke bends before the centre. The lower-right stroke thickens at the contour; a separate arrowhead / point is undetermined.



$$\Delta u(x) = 0,$$
$$u(x)|_{x=\sigma} = f(\sigma)$$

GELFAND-P000147

Original source details: GELFAND-P000147

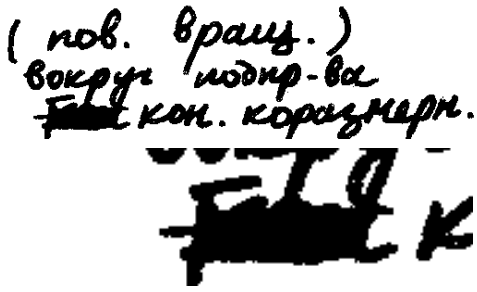
Diagram and typed description. At the upper right, page number 10 has a line at its left and below it.

10

There is uniqueness and continuous dependence for $f \in \overline{R}$ and surfaces of the form

$$\sum_{k=m+1}^{\infty} \xi_k^2 = \varphi(\xi_1, \ldots, \xi_m)$$

(surface of rotation) about a subspace [unreadable short cancelled fragment] of finite codimension.



Original source details: GELFAND-P000147

Uncertain reading. Before “of finite codimension”, a short fragment is densely cancelled; the literal word is undetermined.

and bounded curvature.

The solution formula — the angular mean of the boundary functional.

$\Delta u = g \in \overline{R}$ — also explicit

Next, $\Delta u = \varphi(x, u(x))$ is solved by the fixed-point method.

Original source details: GELFAND-P000147

Diagram and typed description. Two wavy lines with cross-strokes beneath the name “I. M. Gelfand.”

I. M. Gelfand.

$u = (u_1, \dots, u_n)$

Original source details: GELFAND-P000147

Manuscript notation. The final component is u_n. The n is shorter than the m in the formula above.

$$\frac{\partial u}{\partial t} = A(u)\frac{\partial u}{\partial x}$$

Hyperbolic (at $\forall$ point, $A(u)$ has real eigenvalues)

Is there uniqueness? Shock waves — the solution may be nonunique.

$n = 1$ — always uniqueness if $f(u)$ is convex

Hydrodynamic equations — [cancelled: “nonunique” / “may” / unreadable] uniqueness too. ($n = 2$)

Original source details: GELFAND-P000147

Uncertain reading. Before “uniqueness too”, a word is cancelled: “может” (may) / “неод.” (nonunique) / another short fragment; the cancellation crosses the letters.

In the general case it is unclear where there is uniqueness and where there is not.

Magn. hydrodyn. — uniqueness too.

Original source details: GELFAND-P000147

Manuscript notation. Before “hydrodyn.” the abbreviation is «Магн.» (Magn.).

A purely mathematical question: at $\forall$ point u, draw the eigenvectors — 3 families of lines — — characteristics.

Convexity: possibly, growth of λ along the characteristics

Better:

$$\frac{\partial u}{\partial t} = \frac{\partial f(u)}{\partial x}$$

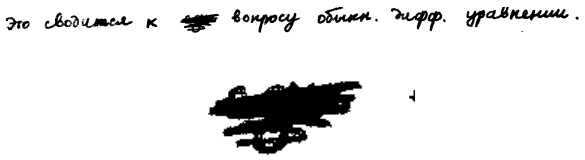
(divergence form)

There is work by Godunov, but it concerns a narrow class of systems.

When is there uniqueness upon adding viscosity:

$$\frac{\partial u}{\partial t} = A\frac{\partial u}{\partial x} + \varepsilon\frac{\partial^2 u}{\partial x^2} \quad ?$$

This reduces to the [unreadable cancelled word] question of ordinary differential equations.



Uncertain reading. In the source, between “к” (to) and “вопросу” (question), a densely cancelled word with a looped lower stroke; the literal reading is undetermined.

Original source details: GELFAND-P000147

GELFAND-P000148

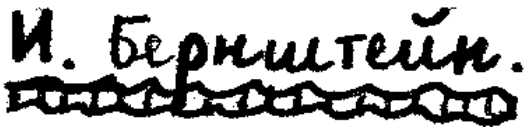


Diagram and typed description. Double wavy underline with cross-strokes under “I. Bernstein.”

I. Bernstein.

Original source details: GELFAND-P000148

Part of what Atiyah spoke about.

g — a compact Lie group. R_g — the ring of finite-dimensional representations (direct sum, tensor product and formal differences), commutative.

An embedding

$$R_g \longrightarrow K(B_g)$$

B_g — the classifying space, K — the K-functor.

A representation corresponds to a vector bundle on B_g (associated).

(I.) If the ring R_g is completed at the ideal of representations of dimension zero, there will be an isomorphism

$$\widehat{R}_g \longrightarrow k(B_g)$$

(The dimension of an element of the form of a difference of two representations — the difference of their dimensions)

Example.

$$g = S^1, \qquad R_{S^1} = \sum_{i=-\infty}^{+\infty} n_i x^i,$$

n_i — integers, finitely many nonzero.

The ideal is generated by the element $x - 1 = y$.

$$\widehat{R}_{S^1} = Z[[y]]$$

(formal power series)

Only positive powers

$$\left(\frac{1}{1-y} = 1 + y + y^2 + \dots\right)$$

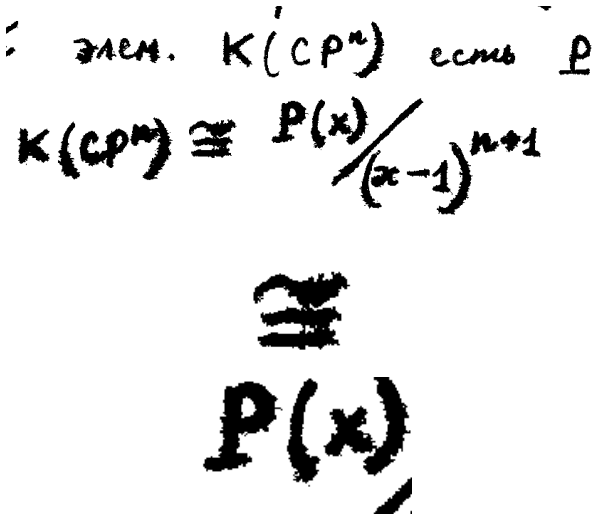
$$B_g = CP^\infty,$$
$$k(CP^\infty) = \lim K(CP^n).$$

On CP^n, the canonical bundle (fibre CP^1), total space CP^{n+1}.

Every element of $K(CP^n)$ is $P(x-1)$.

Precisely

$$K(CP^n) \cong P(x)/(x-1)^{n+1}.$$



Manuscript notation. The sign is ≅, a wave over two horizontal strokes. The numerator is $P(x)$.

Original source details: GELFAND-P000148

Next time, a lecture by Minlos on statistical physics.

GELFAND-P000149

19/IX. Gelfand sem.

I. Bernstein (continuation).

$K(B_g)$ is defined as follows:

Let

$$(B_g)_1 \longrightarrow (B_g)_2 \longrightarrow \dots$$

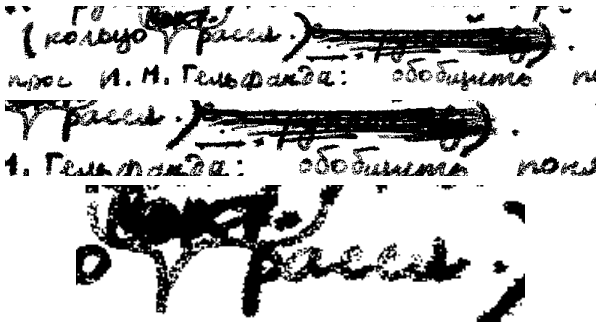
be the skeleta of the corresponding dimensions

$$K_1 \longleftarrow K_2 \longleftarrow \dots$$

One must take the projective limit.

The K-functor of a finite polyhedron is defined in the usual way (ring [overwritten: K-like sign, ?] bundl.) [~~unread long entry~~]) .

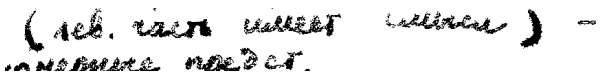
$\longrightarrow$ •



Uncertain reading. After “ring” there is an overwritten K-like sign, then the abbreviation for “bundles” and a closing parenthesis. A dense cancellation follows, with strokes below and another closing parenthesis. Under its beginning is a short rightward arrow and a dot. The cancelled letters remain unread.

Original source details: GELFAND-P000149

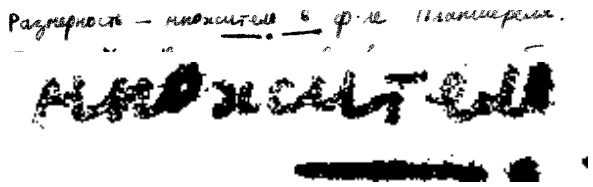
Question of I. M. Gelfand: generalize the notion of the K-functor and classifying space so that the theorem remains valid also for locally compact Lie groups. (the left-hand side makes sense) — — one must also use infinite-dimensional representations.



Original source details: GELFAND-P000149

Diagram and typed description. A wavy line below the heading. The faint parenthesis reads “the left-hand side makes sense”; the abbreviation at the right of the preceding line is “лок.”.

Dimension — a factor in the Plancherel formula.

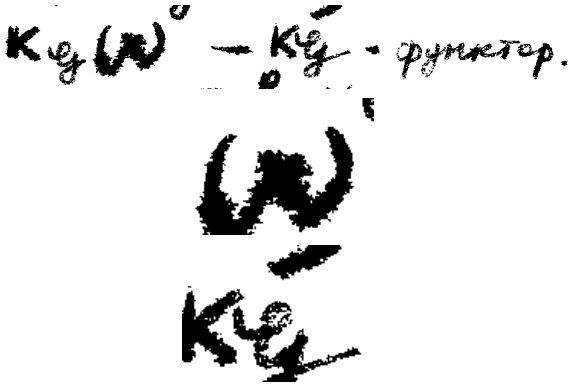


Original source details: GELFAND-P000149

Manuscript notation. After the dash the word reads “factor.” Below are two strokes with a dot between them. The second stroke rises slightly.

Let X be a g-space (i.e. there is a map $g \times X \longrightarrow X$, continuous in both arguments, and there is an action of g on X)

$K_g(X\,[?])$ — the K_g[?]-functor.



Original source details: GELFAND-P000149

Uncertain reading. At the second K, small loop-shaped strokes stand to the right, above and below: a K_g / K^g / K_g^0-like entry; a separate 0 is not established. In the parentheses of the first K_g, an X-like letter is densely overwritten.

A g-bundle $E \xrightarrow{p} X$ — a vector bundle, then E ... — a g-space, g acts compatibly with p. In the fibres g must preserve the linear structure.

In the usual way, sum and tensor product are defined and formal differences are introduced.

Example. K_g — the functor of a point — — linear spaces with representations of g, this will be exactly R_g.

~~[Cyrillic H-like beginning, ?]...~~

Original source details: GELFAND-P000149

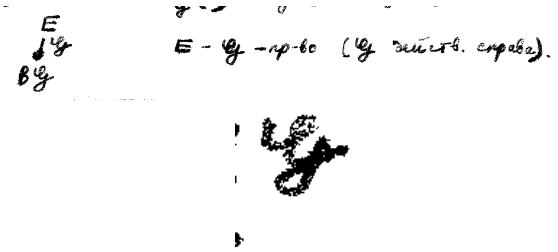
Uncertain reading. Before “Generalization”, a short cancelled fragment with a Cyrillic H-like beginning; the remaining letters are not established.

Generalization. Instead of a point — a g-space X.

The left-hand side $K_g(X)$ — already generalized.

$$\begin{array}{c} E \\ \big\downarrow g \\ B_g \end{array}$$

E — a g-space (g acts on the right).



Original source details: GELFAND-P000149

Manuscript notation. At the bottom, E is above B_g with a downward arrow. To its right is the letter g, as in the following line.

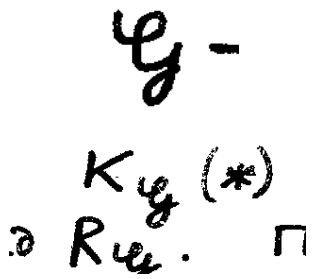
GELFAND-P000150

Diagram and typed description. At the upper right, page number 14 has a line at its left and below it.

14

Original source details: GELFAND-P000150

Let X be a G-space. Introduce the space $X_G = E \times_G X$ (in $E \times X$, the orbits of the group G are identified). If X is a point, then $X_G \cong B_G$ — Grassmann.



Manuscript notation. The looped group letter is rendered as G. In the action below, the group element is written differently, with an ordinary g.

Original source details: GELFAND-P000150

Theorem. (a generalization of the one formulated at the beginning)

$$\widehat{K}_G(X) \xrightarrow{\sim} K(X_G).$$

There is a map $K_G(*) \longrightarrow K_G(X)$, so $K_G(X)$ — — a module over R_G. Completion is taken with respect to the ideal J of the ring R_G.

An exact def. can also be given as follows:

$$\widehat{K}_G(X) = K_G(X) \underset{R_G}{\otimes} \widehat{R}_G.$$

——— • ———

1. Reduction to the unitary group U.

Let $G \subset U$. Then $K_G(X) = K_U(U \times_G X)$.

The action of G on U: $g : u \longrightarrow ug^{-1}$. On $U \times_G X$, U acts on the left. Moreover, $X_G = (U \times_G X)_U$.

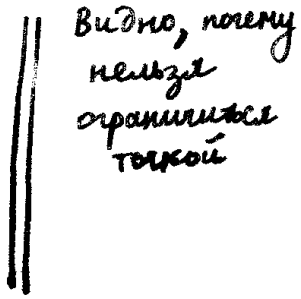


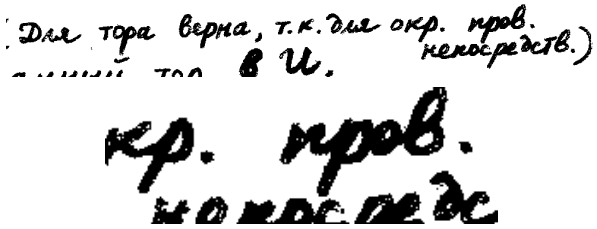
Diagram and typed description. An unfinished four-line note to the right of two vertical strokes.

It is clear why
one cannot
restrict oneself
to such

Original source details: GELFAND-P000150

2. Reduction to the torus case.

(For the torus it holds, since for the circle verif. directly.)



Original source details: GELFAND-P000150

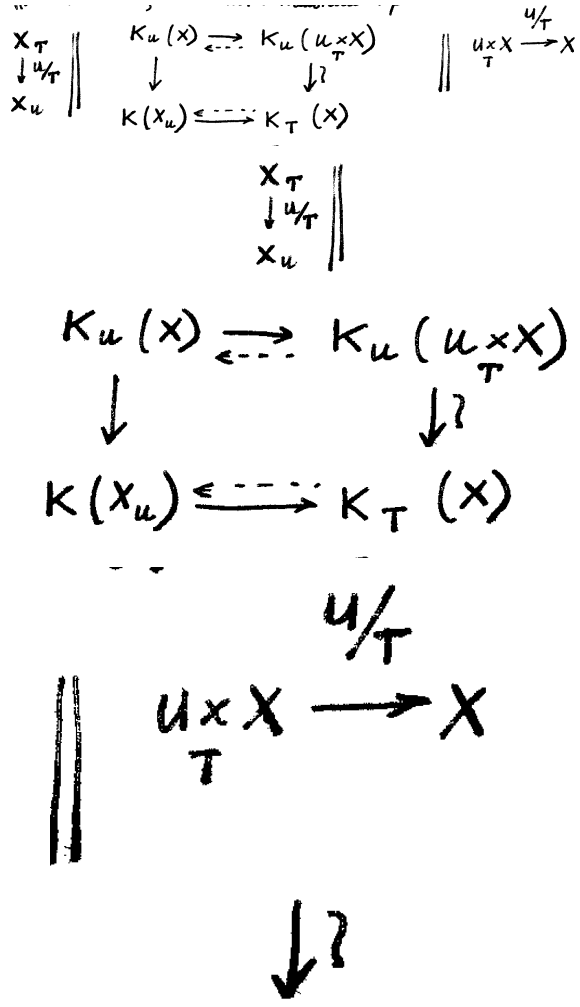
Manuscript notation. After the abbreviation for “circle” is “verif.”, with “directly” abbreviated below.

~~[unread cancelled sign]~~.

Original source details: GELFAND-P000150

Uncertain reading. Before $T \subset U$, a short number-like sign is densely cancelled; its original reading is not established. A separate dot at the right.

$T \subset U$, T is a maximal torus in U.



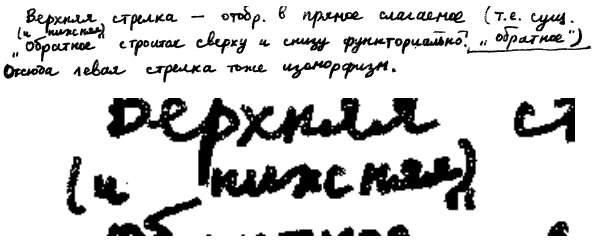
Original source details: GELFAND-P000150

Diagram and typed description. At the left, $X_T \longrightarrow X_U$ downward with U/T. Two double vertical rules separate three parts. In the square, $K_U(X)$ and $K_U(U \times_T X)$ above, $K(X_U)$ and $K_T(X)$ below; solid horizontal arrows rightward, dashed arrows backward, both vertical arrows downward. To the right of the right vertical arrow, a question-like hook with a thickened base; a separate dot cannot be distinguished. Separately at the right, $U \times_T X \longrightarrow X$ with U/T above the arrow.

$$\begin{array}{ccc} X_T & & \\ \downarrow{\scriptstyle U/T} & & \| \\ X_U & & \end{array}$$

$$\begin{array}{ccc} K_U(X) & \longrightarrow\!\!\!\!\dashleftarrow & K_U(U \times_T X) \\ \downarrow & & \downarrow \;) \\ K(X_U) & \longrightarrow\!\!\!\!\dashleftarrow & K_T(X) \end{array} \quad \| \qquad U \times_T X \xrightarrow{U/T} X$$

The upper arrow is a map into a direct summand (i.e. there exists „inverse (and the lower one) is constructed above and below functorially. „inverse“)



Original source details: GELFAND-P000150

Manuscript notation. The insertion “(and the lower one)” sits above the first “inverse”, beneath the beginning of the line “The upper arrow”.

Hence the left arrow is also an isomorphism.

GELFAND-P000151

Original source details: GELFAND-P000151

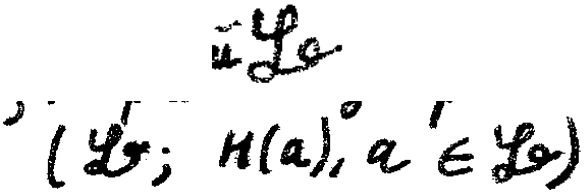
Diagram and typed description. A hatched band under the name “R. A. Minlos.”

R. A. Minlos.

A phys. system is descr. by a set of states $\mathcal{L}[?]$ and a function $H(a)$ on ~~[unreadable sign]~~, which is called the Hamiltonian fn. (a real-valued fn.). — the pair $(\mathcal{L}[?]; H(a), a \in \mathcal{L}[?])$.

Original source details: GELFAND-P000151

Uncertain reading. After “on” a letter with an upper loop and a lower stroke is densely crossed out; the original reading is not established.



Original source details: GELFAND-P000151

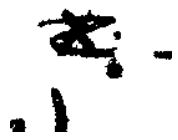
Uncertain reading. The right loop of the state-space sign joins the bottom stroke. It is provisionally set as $\mathcal{L}[?]$. A separate index on this line is unclear.

3 types.

1. Classical system — tuples

$$(p_1, \dots, p_N, x_1, \dots, x_N)$$

All p_i independently range over $\mathbb{R}^\nu$ (in physics $\mathbb{R}^3$) — — the set of momenta.
All x_i range over a region ~~x_i~~ [?] $\Omega \subset \mathbb{R}^\nu$, called the vessel (the reg. is bounded).



Original source details: GELFAND-P000151

Uncertain reading. Before Ω the crossed-out entry resembles x_i. Two separate strokes are visible below its right end.

The tuple x_i — configuration of the particles

$$(\underbrace{p_1, \dots, p_N}_{\mathbb{R}^{N\nu}}, \underbrace{x_1, \dots, x_N}_{\Omega^N})$$

Notation (will sometimes be consid.)

$$(\Omega_\delta)^N : \quad |x_i - x_j| > \delta$$

(the particles — "hard" balls)

This is the set of states.

Hamiltonian fn.:

$$H(a) = \frac{1}{2m}\sum |p_i|^2 + V(x_1, \dots, x_N), \qquad m > 0$$

V is called the potential energy.

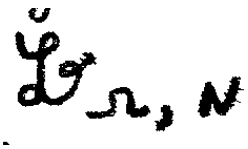
In what follows we consid. only fns. V of the form

$$V(x_1, \dots, x_N) = \sum_{i \neq j} U(|x_i - x_j|)$$

(a system with pairwise interaction)

$U(r)$ — the potential — is def. either for $r > 0$, or for $r > \delta$.

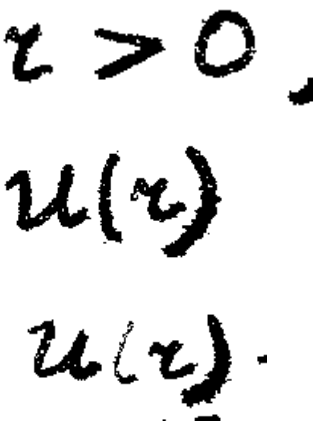
$$\{\mathcal{L}_{\Omega,N}[?], H_{m,U(r)}\}$$



Original source details: GELFAND-P000151

Uncertain reading. Inside the brace is the same connected sign, provisionally rendered as $\mathcal{L}[?]$, with a separate Ω, N below.

Ω — the vessel, N — the number of particles, m — the mass, $U(r)$ — — potential energy.



Original source details: GELFAND-P000151

Manuscript notation. The argument r in $U(r)$, $r > 0$ and $r > \delta$ has the same short upper loop and rightward tail.

Original source details: GELFAND-P000152

Diagram and typed description. Page number 15 at the upper right, with a line at its left and below it.

15

Identical sys. — differ either in the number of particles or in the vessel. (the same — mass and potential)

δ — is contained in $U(r)$.

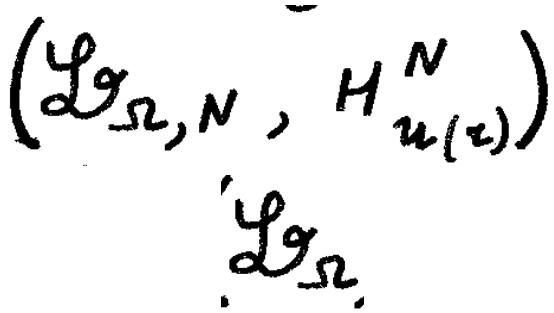
2. Configurational system.

Obt. from the preceding one. A state — $(x_1, \ldots, x_N)$ — the region Ω^N or its subregion $(\Omega_\delta)^N$.

$$H(a) = \sum_{i \neq j} U(|x_i - x_j|)$$

— without kinetic energy.

$$(\mathcal{L}_{\Omega,N}\,[?], H^N_{U(r)})$$



Original source details: GELFAND-P000152

Uncertain reading. The right loop joins the bottom of the sign provisionally rendered as $\mathcal{L}\,[?]$. Below it, Ω, N is clear.

System 1 reduces to system 2. (see below).

3. Discrete model system.

In R^ν a cubic lattice, Ω — a subset of it. A state — a tuple of N lattice sites $(x_1, \ldots, x_N)$, in which no two are equal.

$U(x)$ — an even function on the lattice.

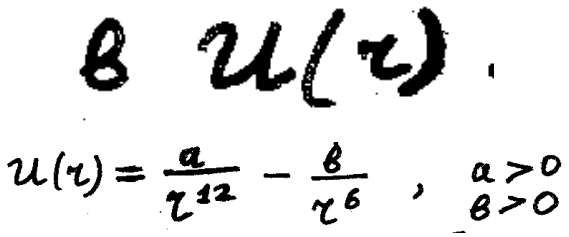
$$H(a) = \sum_{i \neq j} U(x_i - x_j)$$

(Isotropy, e.g. — spherical symm. — is not justified here)

The results of stat. physics depend only on the qualit. properties of $U(r)$, but physicists use only the potential

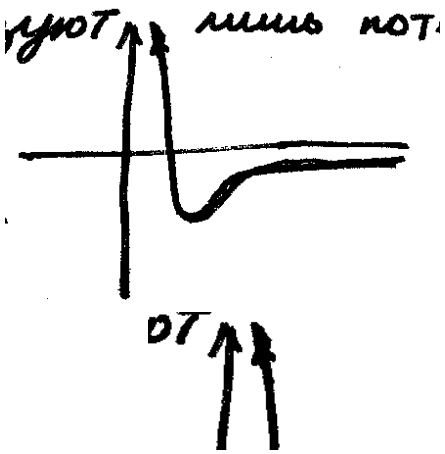
$$U(r) = \frac{a}{r^{12}} - \frac{b}{r^6}, \qquad \begin{matrix} a > 0 \\ b > 0 \end{matrix}$$

What matters to us is the growth at 0 and the decay at ∞.



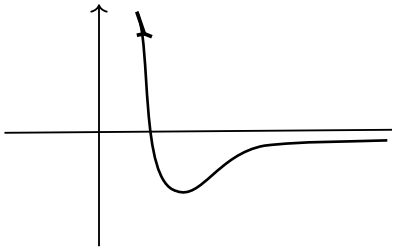
Original source details: GELFAND-P000152

Manuscript notation. The same short looped sign r recurs in $U(r)$ and both denominators.

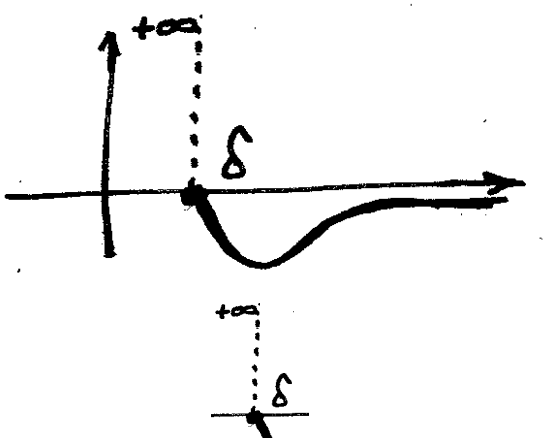


Original source details: GELFAND-P000152

Diagram and typed description. The vertical axis has an upward arrow; no horizontal arrow is established. Neither axis has letters. The branch comes from above to the right of the vertical axis, crosses the horizontal, forms a negative well and approaches the horizontal from below. The upper end of the branch is thick, with short lateral strokes; a separate arrow meaning is not established. No exact scale is given.

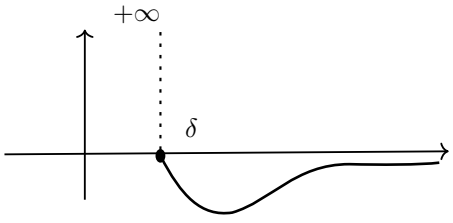


In the presence of δ:



Original source details: GELFAND-P000152

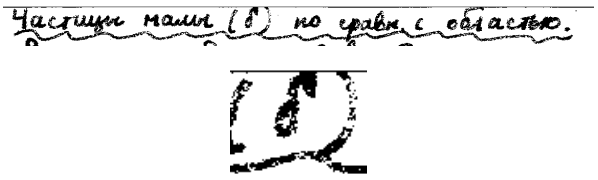
Diagram and typed description. Two axes with arrows. At the right is a dotted vertical; above it $+\infty$, a thick mark at its intersection with the horizontal, and δ to the right. The branch leaves the thick mark for a well and rises below the horizontal axis. The dense ink at the intersection is rendered without an established function value at δ or an assertion of closedness.



Sometimes only a potential of compact support is consid.

GELFAND-P000153

The particles are small (δ) compared with the region.

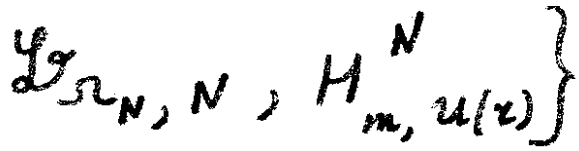


Manuscript notation. The parenthesis contains δ. The top stroke touches the frame edge.

Original source details: GELFAND-P000153

Consid. a seq. of vessels Ω_N which expands and fills the whole space (e.g. similar cubes)
Let us consider

$$\{\mathcal{L}_{\Omega_N,N}[?], H^N_{m,U(r)}\}$$



Uncertain reading. The right loop of the connected state-space sign joins its base. It is provisionally set as $\mathcal{L}[?]$. Below are Ω_N, N.

Original source details: GELFAND-P000153

Manuscript notation. The short looped r in $U(r)$ recurs from the note on the potential.

Original source details: GELFAND-P000153

$$\frac{|\Omega_N|}{N} \longrightarrow v \quad \text{(limiting specific volume)}$$

$|\Omega_N|$ — the volume of the region Ω_N

(Thermodyn. limiting transition)
In most cases everything depends only on v, and the shape of the vessel is immaterial.
Gibbs distribution.
It is assumed that a probability distr. is given on $\mathcal{L}[?]$ In the equilibrium case it is postulated (if everything is independent of time)

$$p(a) = \frac{1}{Q} e^{-\beta H(a)}, \qquad \beta > 0$$

$$Q = \int_{\mathcal{L}[?]} e^{-\beta H(a)}\, da.$$

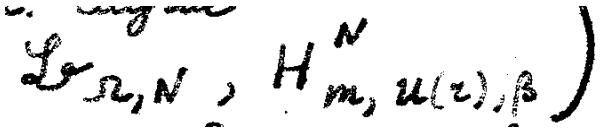
Uncertain reading. The same connected sign appears in the first line and beneath the integral. It is provisionally set as $\mathcal{L}[?]$.

Original source details: GELFAND-P000153

For lattice systems the same formula, but p now gives the probabilities themselves, rather than densities.

A phys. sys. with a given Gibbs distr. is called a small canonical ensemble

$$\begin{pmatrix}\text{in the classical case}\\ \mathcal{L}_{\Omega,N}\,[?],\, H^N_{m,U(r),\beta}\end{pmatrix}$$

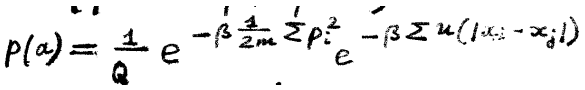


Uncertain reading. The state-space sign is provisionally set as $\mathcal{L}\,[?]$, with Ω, N below. The H has N above and $m, U(r), \beta$ below.

Original source details: GELFAND-P000153

In the thermodyn. limiting transition identical systems are always consid. (β is also fixed)

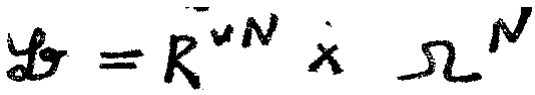
$$p(a) = \frac{1}{Q} e^{-\beta \frac{1}{2m} \sum p_i^2} e^{-\beta \sum U(|x_i - x_j|)}$$



Manuscript notation. The first exponent contains $\sum p_i^2$ and the second $\sum U(|x_i - x_j|)$. Neither sum has written limits.

Original source details: GELFAND-P000153

$$\mathcal{L}\,[?] = R^{\nu N} \times \Omega^N$$



Uncertain reading. On the left is the same connected sign, provisionally set as $\mathcal{L}\,[?]$. On the right is $R^{\nu N} \times \Omega^N$.

Original source details: GELFAND-P000153

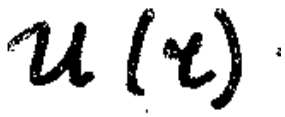
GELFAND-P000154

Diagram and typed description. Page number 16 at the upper right, with a line at its left and below it.

16

Original source details: GELFAND-P000154

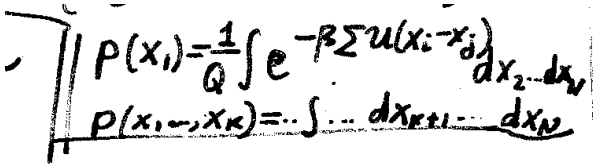
Our distr. is a product of simpler distributions and only the Gibbs distr. in the configuration space is of interest. Where $U(r) = +\infty$, the density will be 0. There are distr. fns. for a subsystem ($\int$ over the others)



Manuscript notation. The r in $U(r)$ has the same form as in the potential on the previous page.

Original source details: GELFAND-P000154

Correlation functions are convenient.

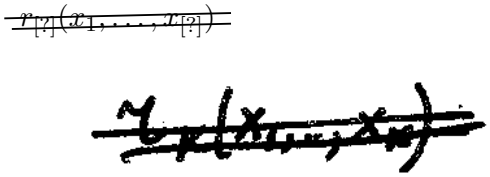


Original source details: GELFAND-P000154

Diagram and typed description. Two formulas to the right of a double rule. The long upper rule extends left over the heading.

$$p(x_1) = \frac{1}{Q}\int e^{-\beta \sum U(x_i - x_j)}\, dx_2 \dots dx_N$$

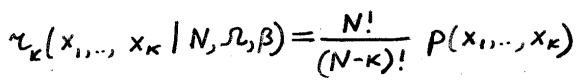
$$p(x_1, \dots, x_k) = \dots \int \dots dx_{k+1} \dots dx_N$$



Original source details: GELFAND-P000154

Uncertain reading. The opening r remains visible above two strokes. The cancelled indices are indistinct.

$$r_k(x_1, \dots, x_k \mid N, \Omega, \beta) = \frac{N!}{(N-k)!} p(x_1, \dots, x_k)$$



Manuscript notation. The r in r_k has the same upper bend and lower rightward stroke as in $U(r)$.

Original source details: GELFAND-P000154

Summatory fn. $F(a)$, $a\,(x_1, \dots, x_N)$

Of the 1st order $F(a) = f(x_1) + f(x_2) + \dots + f(x_N)$

Of the 2nd order $F(a) = \sum_{i \neq j} f(x_i, x_j)$, f — symmetric.

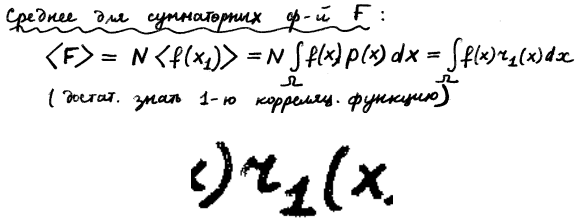
and so on.

Mean for summatory fns. F:

$$\langle F \rangle = N \langle f(x_1) \rangle = N \int_\Omega f(x) p(x)\, dx$$

$$= \int_\Omega f(x) r_1(x)\, dx$$

(it suffices to know the 1st correlation function)

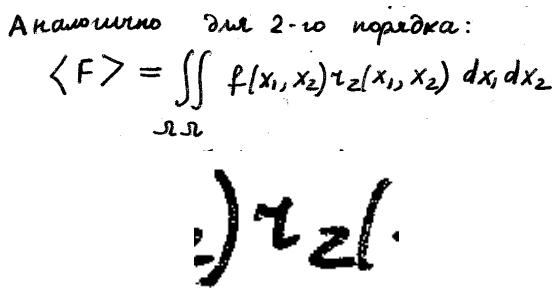


Manuscript notation. The right-hand integral contains $r_1(x)$. Both integration domains are marked Ω.

Original source details: GELFAND-P000154

Similarly for the 2nd order:

$$\langle F\rangle = \int_\Omega\int_\Omega f(x_1,x_2)r_2(x_1,x_2)\,dx_1dx_2$$



Manuscript notation. The integrand contains $r_2(x_1,x_2)$. Each of the two integrals has Ω beneath it.

Original source details: GELFAND-P000154

Probabilistic (phys.) meaning.

$C\subset\Omega$, C — a subregion, $n_C(a)$ — the number of particles that fall in the subregion, $a=(x_1,\dots,x_N)$. This is a summatory quantity of the 1st order, with

$$f(x)=\begin{cases}1, & x\in C,\\ 0, & x\notin C\end{cases}$$

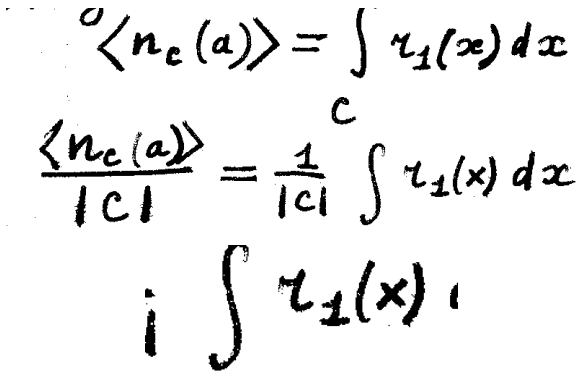
Uncertain reading. Two small angular crossed signs in the left margin; their letter reading and meaning are not established.

Original source details: GELFAND-P000154

Therefore

$$\langle n_C(a)\rangle = \int_C r_1(x)\,dx$$

$$\frac{\langle n_C(a)\rangle}{|C|} = \frac{1}{|C|}\int r_1(x)\,dx$$

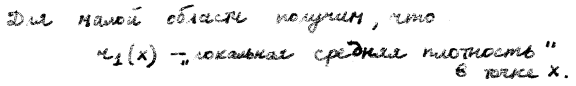


Manuscript notation. The first integral has an explicit C. The manuscript leaves the domain of the last integral unwritten.

Original source details: GELFAND-P000154

GELFAND-P000155

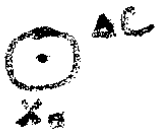
For a small region we obtain that $r_1(x)$ — the "local mean density" at the point x.



Manuscript notation. The manuscript repeats $r_1(x)$ from the previous page.

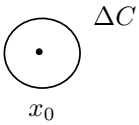
Original source details: GELFAND-P000155

Another probabil. meaning:



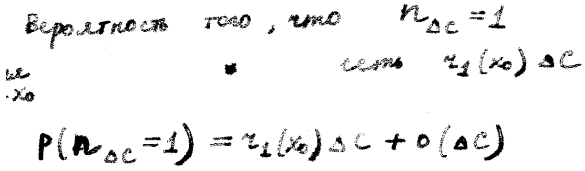
Original source details: GELFAND-P000155

Diagram and typed description. A separate approximate circle with a dot inside; ΔC at the upper right edge and x_0 below.



ΔC — a small nbhd. of x_0.
The probability that $n_{\Delta C} = 1$ is $r_1(x_0)\Delta C$.

$$p(n_{\Delta C} = 1) = r_1(x_0)\Delta C + o(\Delta C)$$



Manuscript notation. Both lines have $r_1(x_0)\Delta C$. The displayed formula ends with $o(\Delta C)$.

Original source details: GELFAND-P000155

Uncertain reading. Between the lines of the first probability explanation is a separate dense ink mark with an irregular outline; no letter reading or meaning is established.

Original source details: GELFAND-P000155

26/IX. Gelfand Sem.
R. A. Minlos. Lecture 2.

$r(x_1, x_2)\Delta x_1 \Delta x_2$ — the prob. of a particle falling in Δx_1 and a second one in Δx_2.

Manuscript notation. The same r precedes (x_1, x_2). No separate index is written on this letter.

Original source details: GELFAND-P000155

Original source details: GELFAND-P000155

Diagram and typed description. Two separate approximate outlines, with x_1, x_2 below; dense crossed strokes inside, whose separate letter reading is not established. A short stroke above the right outline. Sizes and distance are not given.

$\Omega_N \longrightarrow \infty$ — the thermodyn. limiting transition.

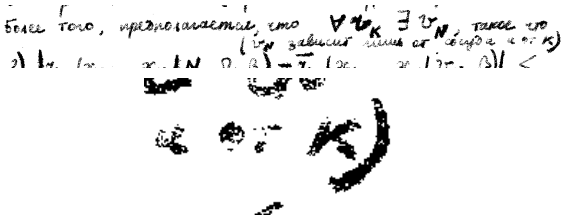
$$\frac{|\Omega_N|}{N} \longrightarrow v = \frac{1}{\rho}, \qquad \rho \text{ — density.}$$

Hypothesis: 1)

$$r_k(x_1, \ldots, x_k \mid N, \Omega, \beta) \longrightarrow \vec{r}_k(x_1, \ldots, x_k \mid v, \beta)$$

This is assumed (limiting correlation function)
Moreover, it is assumed that $\forall r_k \; \exists v_N$ such that

$$(v_N \underset{\longrightarrow}{\text{depends}} \text{ only on the vessel and on } k)$$

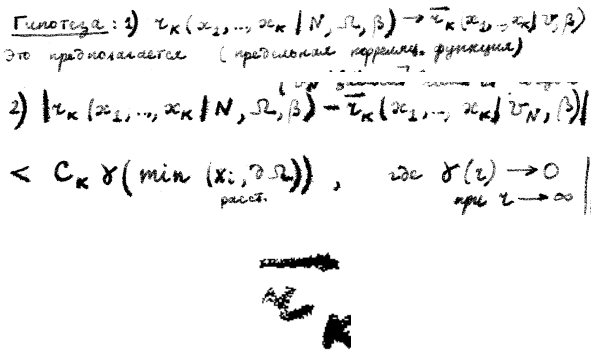


Original source details: GELFAND-P000155

Manuscript notation. Before k is “and on”. A short rightward arrow appears beneath “depends”.

$$2) \quad \left| \begin{matrix} r_k(x_1, \ldots, x_k \mid N, \Omega, \beta) \\ -\vec{r}_k(x_1, \ldots, x_k \mid v_N, \beta) \end{matrix} \right| < \\ < C_k \gamma\big(\min_{\text{dist.}}(x_i, \partial\Omega)\big),$$

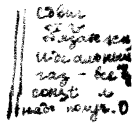
where $\gamma(r) \longrightarrow 0$ as $r \longrightarrow \infty$.

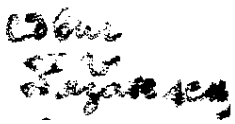


Original source details: GELFAND-P000155

Manuscript notation. A short rightward arrow stands over the limiting function’s r. The estimate has $(x_i, \partial\Omega)$ in parentheses with “dist.” beneath.

Volume [?]
and v [?]
we discuss [?]
Ideal
gas — all [?]
[dense unreadable sign]
const and
need to obtain 0

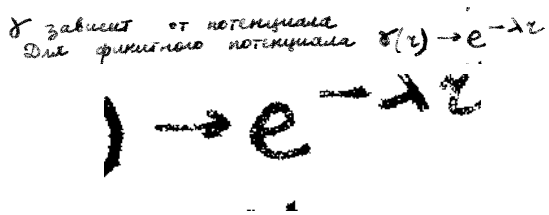




Original source details: GELFAND-P000155

Uncertain reading. Beyond two vertical lines are words resembling “Volume and v we discuss,” then “Ideal gas — all” / “… without …,” const, “and need to obtain 0.” The first connective and endings are faint; a dense unreadable sign follows the gas line. These readings are provisional.

γ depends on the potential. For a potential of compact support $\gamma(r) \longrightarrow e^{-\lambda r}$.



Original source details: GELFAND-P000155

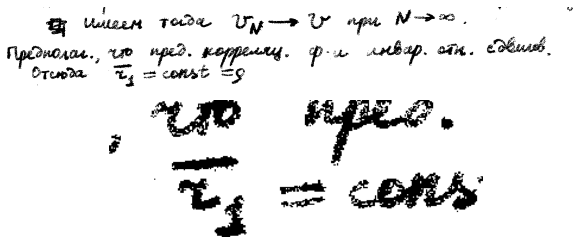
Uncertain reading. A right arrow between $\gamma(r)$ and $e^{-\lambda r}$; separate limit conditions are not repeated in this line.

~~[unreadable sign]~~

Original source details: GELFAND-P000155

Uncertain reading. Before “We then have” a short sign or number is densely crossed out; the original reading is not established.

We then have $v_N \longrightarrow v$ as $N \longrightarrow \infty$.
3) Assume that the limiting correlation fns. are invariant under translations. Hence $\vec{r}_1 = \text{const} = \rho$.



Original source details: GELFAND-P000155

Manuscript notation. The last line has $\vec{r}_1 = \text{const} = \rho$. The letters r and ρ have different forms.

GELFAND-P000156

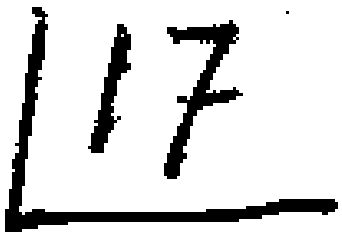


Diagram and typed description. Page number 17 at the upper right, with a line at its left and below it.

17

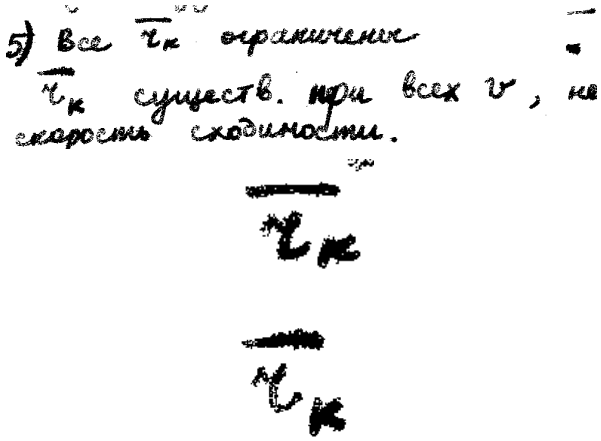
Original source details: GELFAND-P000156

4) Continuity with respect to v and β.

This theorem should hold except at exceptional values of v and β. (assumed)

Proved only for small v and β (under some assumptions). (this will be done later — — Ruelle)

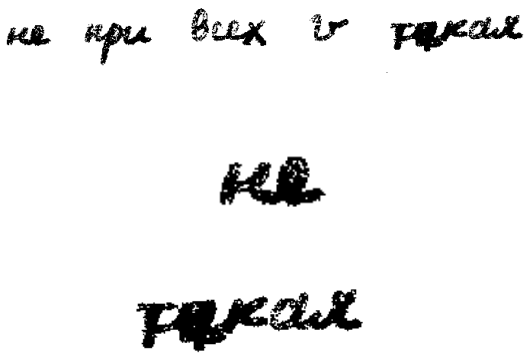
5) All $\vec{r}_k$ are bounded.



Manuscript notation. Both lines of item 5 have $\vec{r}_k$.

Original source details: GELFAND-P000156

$\vec{r}_k$ exist for all v, not for all v with such [?] a rate of convergence.



Uncertain reading. The line reads “not for all.” The opening strokes of “such” are heavily overwritten, obscuring the earlier entry.

Original source details: GELFAND-P000156

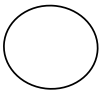
Uncertain reading. At the right beside item 5 is a separate small dense mark with an irregular outline; no letter reading is established. Below are two horizontal rules and a separate diamond-shaped central dot; the right end of the left rule is thickened.

Original source details: GELFAND-P000156

On the Gibbs distribution. (at Gelfand's request)

$$a(p_1, \dots, p_N, x_1, \dots, x_N)$$

Diagram and typed description. One unfilled approximate circle without labels or arrows to the left of the paragraph on motion; no sizes are specified.



Original source details: GELFAND-P000156

Motion on the surface of constant energy $H(a) = \text{const}$. Assume that it is ergodic and there is an invariant measure.

Perhaps any distribution, as the number of particles increases, leads to the Gibbs distribution. (Gelfand)

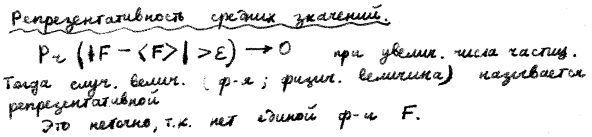
Not any: if there is no ergodic hypothesis (Arnold)

Representativity of mean values.

$$P_r\big(|F - \langle F\rangle| > \varepsilon\big) \longrightarrow 0 \quad \text{as the number of particles increases.}$$

Then the random variable (function; physical quantity) is called representative.

This is imprecise, since there is no single function F.



Manuscript notation. Below P is the same short, right-curved r used in the correlation formulas.

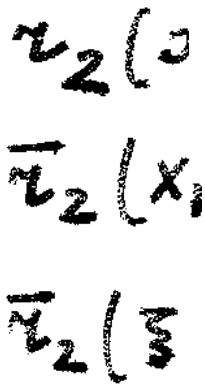
Original source details: GELFAND-P000156

The needed example. (summatory function)

$$F_N(x_1, \dots, x_N) = \sum_{i \neq j} f(x_i - x_j),$$

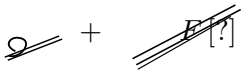
f is one and the same, $\int |f(\xi)|\, d\xi < \infty$.

$$\langle F_N \rangle = \iint r_2(x_1, x_2 \mid N, \Omega, \beta) f(x_1 - x_2)\, dx_1\, dx_2 =$$
$$= \iint \vec{r}_2(x_1 - x_2 \mid v, \beta) f(x_1 - x_2)\, dx_1\, dx_2 + o(|\Omega|) =$$
$$= |\Omega| \int \vec{r}_2(\xi \mid v, \beta) f(\xi)\, d\xi + o(|\Omega|).$$



Manuscript notation. The first line has r_2 and the next two have $\vec{r}_2$. The integration domains are unwritten. Equality signs recur at the line ends and beginnings.

Original source details: GELFAND-P000156

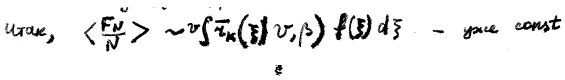


Uncertain reading. To the right of the last equality, a loop-shaped sign, a plus and an F-like form are crossed out. The first reading is not established; F is provisional. The cancellation strokes are diagonal.

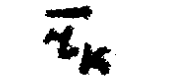
Original source details: GELFAND-P000156

Thus,

$$\left\langle \frac{F_N}{N} \right\rangle \sim v \int \vec{r}_k(\xi \mid v, \beta) f(\xi)\, d\xi \quad \text{— already const}.$$



Manuscript notation. The last formula has $\vec{r}_k$, with the divergent strokes of k. Before the integral is v.



Original source details: GELFAND-P000156

GELFAND-P000157

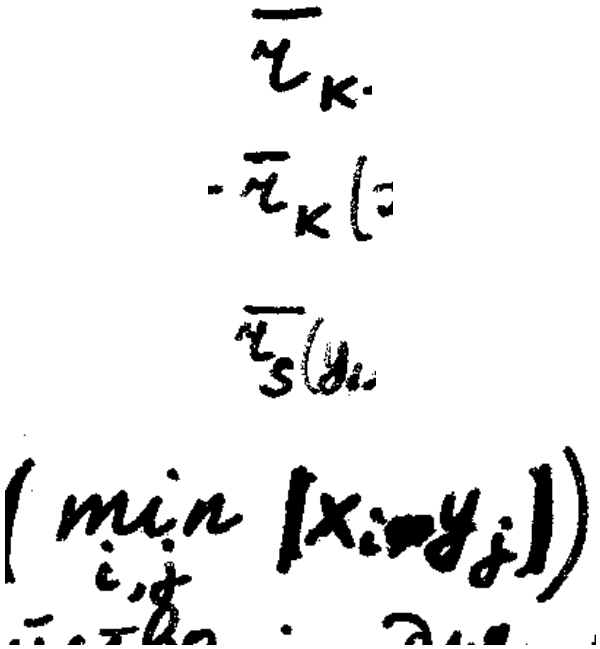
Uncertain reading. In the upper left margin is a densely cancelled loop-shaped sign resembling 6; its original reading is not established. At its right is a separate dot.

Original source details: GELFAND-P000157

One more hypothesis (but it already follows from these, though with difficulty):

$$6)\quad \left|\begin{matrix}\bar r_{k+s}(x_1,\dots,x_k,y_1,\dots,y_s\mid v,\beta)\\ -\bar r_k(x_1,\dots,x_k\mid v,\beta)\bar r_s(y_1,\dots,y_s\mid v,\beta)\end{matrix}\right|\le$$
$$\le C_{k,s}\gamma\big(\min_{i,j}|x_i-y_j|\big).$$

(multiplicative property; for the ideal gas 0).



Manuscript notation. The three functions use r with a horizontal bar. The minimum has i, j, with $x_i - y_j$ between the absolute-value bars. The manuscript repeats $\le$ across the line break.

Original source details: GELFAND-P000157

We shall prove that

$$\underset{\text{variance}}{DF_N} \sim dN, \qquad d \text{ — const}.$$

Hence $D\frac{F_N}{N} \sim \frac{d}{N}$ and by Chebyshev's inequality we already obtain ~~[unreadable word]~~. representativity of the mean.

Uncertain reading. After "obtain" a word is densely cancelled, with a loop-shaped stroke below it. Its original reading, including a possible repetition of "represent.", is not established. The following words are "representativity of the mean".

Original source details: GELFAND-P000157

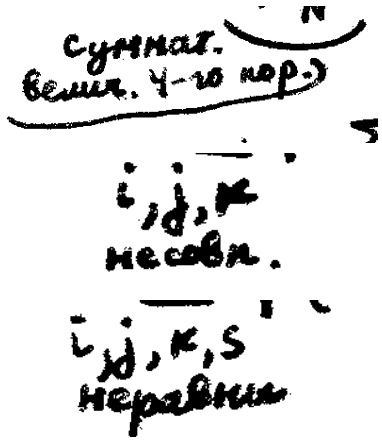
This is proved by direct calculation:

$$DF_N = \langle F_N^2\rangle - \langle F_N\rangle^2.$$

$$\overbrace{F_N^2}^{\substack{\text{summatory}\\ \text{qty. of 4th ord.}}} = 2\sum_{i\neq j} f^2(x_i - x_j)$$

$$+ 4 \sum_{\substack{i,j,k\\ \text{noncoinc.}}} f(x_i - x_j) f(x_k - x_j) +$$

$$+ \sum_{\substack{i,j,k,s\\ \text{unequal}}} f(x_i - x_j) f(x_k - x_s).$$



Original source details: GELFAND-P000157

Manuscript notation. A short arc is below F_N^2. At the left, "summatory quantity of order 4" has a longer curved stroke beneath it. Below the triple sum are i, j, k and the abbreviation for "distinct", below the quadruple sum i, j, k, s and "unequal".

We calculate

$$\langle F_N^2 \rangle = 2 \iint f^2(x_1 - x_2) r_2(x_1, x_2)\, dx_1\, dx_2 + 4 \ldots$$

and use the decay estimates.

Original source details: GELFAND-P000157

Manuscript notation. The final integral contains $r_2(x_1, x_2)$ without an overbar. Then comes $+4 \ldots$.

GELFAND-P000158

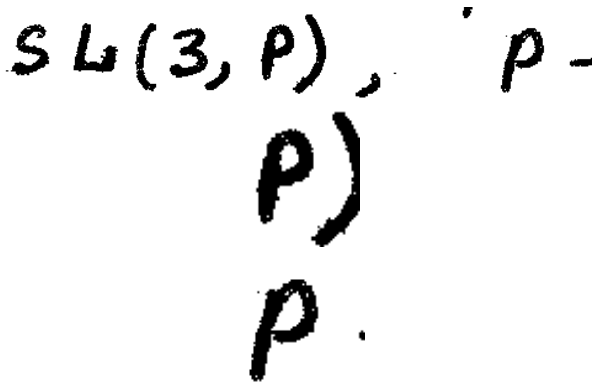
Diagram and typed description. Page number 18 at the upper right, with a line at its left and below it.

18

Original source details: GELFAND-P000158

D. Kazhdan. His results.

1) G is a simple real or p-adic Lie group containing as a subgroup $SL(3, P)$, P is the corresponding field. Γ is a discrete subgroup with a quotient space of finite volume.



Manuscript notation. In $SL(3, P)$ and before "corresponding field" is P, with an upper loop and a long downward stroke.

Original source details: GELFAND-P000158

Then a)

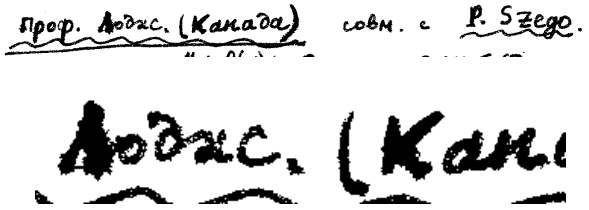
$$\Gamma / \underbrace{[\Gamma, \Gamma]}_{\text{commutator subgroup}} \quad \text{is finite.}$$

b) Γ has a finite number of generators. (even from any set of generators a finite one can be selected)

2) G is an algebraic group over the rational numbers Q, containing $SL(3, Q)$. Something about representations.

The proof uses the isolation of the trivial representation.

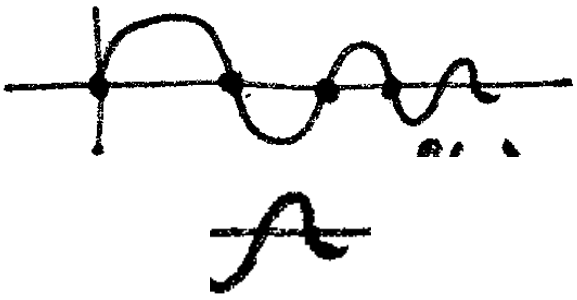
Prof. Lodzhs. (Canada) jointly with P. Szego.



Manuscript notation. The manuscript spells the name "Лоджс.", with a heavy initial stroke. The professor and Canada line has a wavy and a straight underline. P. Szego has a wavy underline.

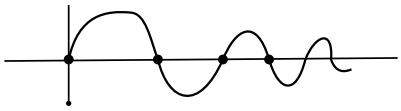
Original source details: GELFAND-P000158

$$y'' + f(x)y = 0, \qquad 0 < x < \infty.$$



Original source details: GELFAND-P000158

Diagram and typed description. A horizontal line, a short vertical at the left, four dense early crossings and a dot at the bottom of the vertical. Three upper waves alternate with lower ones, ending at the right in a short downward curl. No labels, sizes or exact coordinates are given; the contour is approximate.



Sturm: (1836) if $f(x)$ increases, the zeros get closer and conversely.

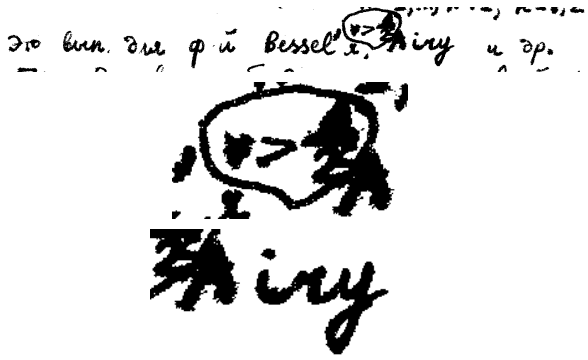
Theorem, if f has a higher degree of monotonicity.
Let

$$(-1)^j f^{(j+1)}(x) \geq 0, \qquad 0 < x < \infty, \quad j = 1, \ldots, N$$

and $f(\infty) > 0$. Then

$$(-1)^n \Delta^{n+1} x_k > 0, \qquad n = 1, \ldots, N+1; \quad k = 1, 2, \ldots$$

This holds for the Bessel functions $\nu > \frac{1}{2}$ [?], Airy and others.



Original source details: GELFAND-P000158

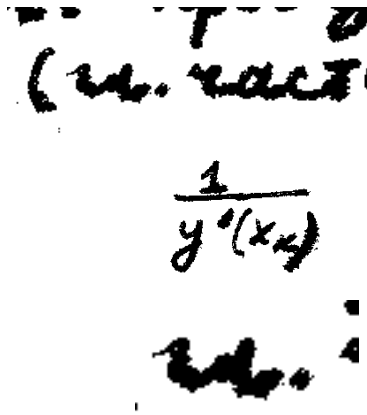
Uncertain reading. The oval insertion after Bessel reads $\nu > 1/2$ [?]. The top of the first sign and the fraction are densely overwritten. In Airy the initial letter is heavy.

The wave areas have the same property.

$$M_k = \int_{x_k}^{x_{k+1}} |y(x)|^\lambda \, dx, \qquad \lambda > -1.$$

$\lambda = 0$ — differences of zeros, $\lambda = 1$ — wave areas,

$\lambda = -1$ — diff. of derivs. (Gelfand's remark) (princ.[?] parts) $\dfrac{1}{y'(x_k)}$.



Original source details: GELFAND-P000158

Uncertain reading. Below "diff. of derivs." a parenthesis appears to read "(princ.[?] parts)". The first two strokes are joined. At the right is $1/y\prime(x_k)$.

GELFAND-P000159

I. M. Gelfand.

Repetition of I. Bernstein. (also Atiya)

G is a compact group. K_G is a functor, $K_G(X)$.

Original source details: GELFAND-P000159

Diagram and typed description. E above X, with a downward arrow between them and no additional labels.

$$\begin{array}{c} E \\ \downarrow \\ X \end{array}$$

E, X are G-spaces; a vector bundle. Fiber to fiber — linearly. Formal differences.

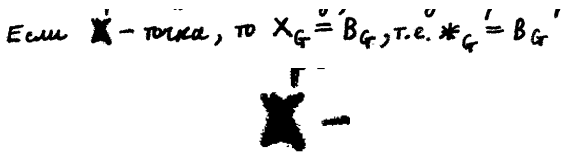
This is a contravariant functor.

Generalization of the notion of a classifying space. Consider

$$E \underset{G}{\times} X = X_G.$$

the orbits of G are contracted to a point; E is the universal space.

If X is a point, then $X_G = B_G$, i.e. $*_G = B_G$.



Uncertain reading. After "If" a dense X is written over earlier strokes. Their original reading is not established.

Original source details: GELFAND-P000159

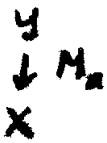
Theorem. $K_G(X)$ approximately coincides with $K(X_G)$ (already the ordinary K-functor)

$K_G(X)$ is a ring, $X \longrightarrow \bullet$, hence $R_G \longrightarrow K_G(X)$ and therefore ~~[unreadable word]~~ this is an R_G-module (a ring over a subring)

Uncertain reading. Between "therefore" and "this" a word or group of letters is densely cancelled, with a loop-shaped stroke below it; the original reading is not established.

Original source details: GELFAND-P000159

I. Bernstein. The most essential part of the proof: a construction with elliptic operators.



Original source details: GELFAND-P000159

Diagram and typed description. Y above X, a downward arrow, and M_x to its right. Beside it is the description of M as the fiber and the arrow $K(X) \longrightarrow K(Y)$.

$$\begin{array}{c} Y \\ \downarrow \\ X \end{array} \quad M_x$$

M is a manifold, the fiber; $K(X) \longrightarrow K(Y)$. Sometimes a reverse map can be constructed.

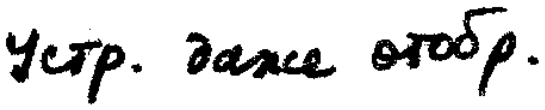
Let there be on X a family of differential operators D_x: acting in a sheaf over M_x (the fibers over x). Let there be over Y 2 sheaves N_1, N_2.

$$D_x : \Gamma(N_1|_{M_x}) \longrightarrow \Gamma(N_2|_{M_x})$$

— an elliptic operator (or a complex)

For example, the de Rham complex.

$K(Y)$ is a $K(X)$-module. Construct even a map of $K(X)$-modules.



Manuscript notation. Between the abbreviations for “construct” and “map” the manuscript reads “even”.

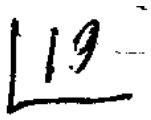
Original source details: GELFAND-P000159

V is a sheaf on Y.

$$M_x : \quad D_x \otimes 1 : \Gamma(N_1 \otimes V)_{M_x} \longrightarrow \Gamma(N_2 \otimes V)_{M_x}.$$

An elliptic operator; $\mathrm{Ker}, \mathrm{Coker} < \infty$. The difference of elements in the K-functor gives the required element.

GELFAND-P000160



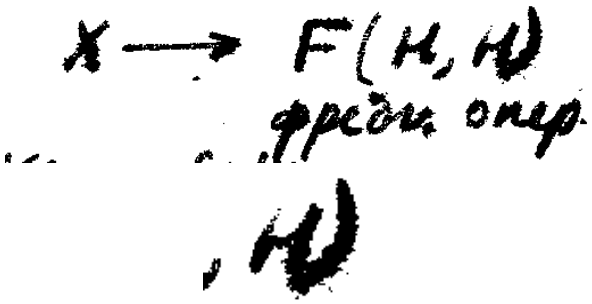
Original source details: GELFAND-P000160

Diagram and typed description. Page number 19 at the upper right, with a line at its left and below it.

19

If there is a map

$$X \longrightarrow \underset{\text{Fredholm ops.}}{F(H,H)}$$



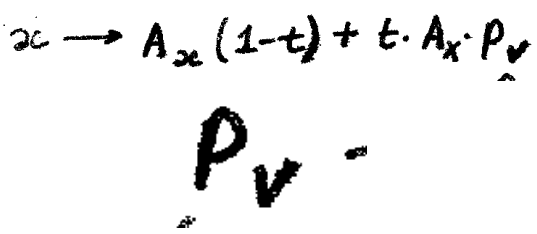
Original source details: GELFAND-P000160

Manuscript notation. The second H touches the closing parenthesis.

then one can def. Ker – Coker as a bundle over X

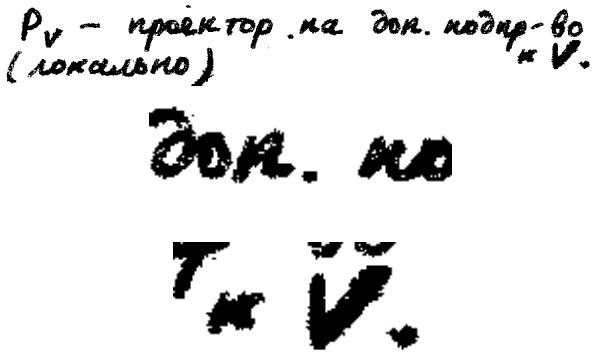
$$x \longrightarrow A_x$$
$$x \longrightarrow A_x(1-t) + t \cdot A_x \cdot P_V$$

P_V — a projector. onto a complementary subspace to V.



Original source details: GELFAND-P000160

Manuscript notation. The subscript V of P has the same angular stroke as the V at the end of the explanation.



Original source details: GELFAND-P000160

Manuscript notation. The manuscript has the abbreviated “complementary subspace,” with “to V” below.

Already const. dim. of the kernel (locally)

Bundle $\underset{\sim}{ind}\, V$.

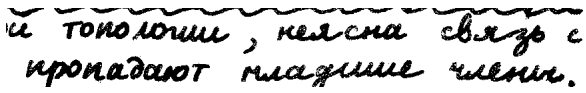
Next time, Nirenberg’s report and Minlos.

GELFAND-P000161

3/X. Gelfand's sem.

Nirenberg.

Informal expos. of Seeley and Hörmander's index f-la. The f-la is analytic, without any topology. Its connection with topology is unclear, as is how the lower-order terms disappear in it.



Original source details: GELFAND-P000161

Manuscript notation. The manuscript reads "the connection is unclear" and "lower-order terms".

However, according to Gelfand, the Atiyah–Singer f-la can be written analyt. [as the degree of a mapping, etc.] The same situation for Pontryagin classes.

L — ellipt. operator on a comp. m-fold. Consid. eigenfns.

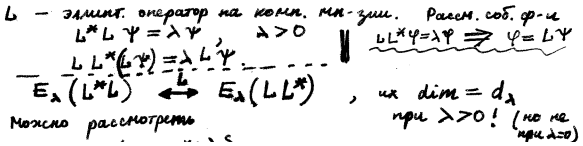
Original source details: GELFAND-P000161

Manuscript notation. Before "eigenfns." the manuscript has the abbreviation for "consider".

$$L^*L\Psi = \lambda\Psi, \qquad \lambda > 0,$$
$$LL^*(L\Psi) = \lambda L\Psi.$$

- -

$$\| \quad LL^*\varphi = \lambda\varphi \;\Rightarrow\; \varphi = L\Psi$$



Original source details: GELFAND-P000161

Diagram and typed description. Below the second formula is a dashed line. The formula on the right is separated by a double vertical line and has a wavy underline.

$$E_\lambda(L^*L) \overset{L}{\longleftrightarrow} E_\lambda(LL^*), \qquad \text{their dim} = d_\lambda \quad \begin{matrix}\text{for } \lambda>0!\\ \text{(but not for } \lambda=0)\end{matrix}$$

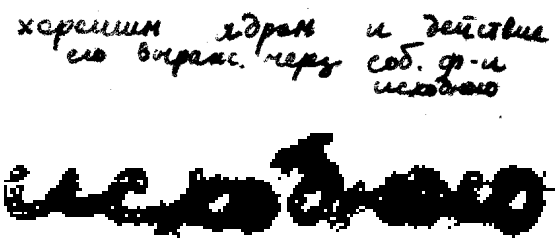
Original source details: GELFAND-P000161

Manuscript notation. Above the two-way arrow is L. Its upper stroke reaches the dashed line.

One can consider

$$(I + L^*L)^s, \qquad \operatorname{Re} s \ll 0$$

Then this is an integr. operator with a good kernel and its action is expressed through eigenfns. [Below the line: of the original [?].]



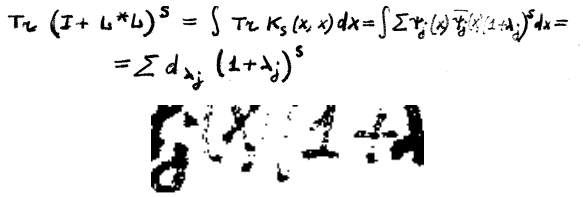
Uncertain reading. Below "eigenfns." a separate word appears to read "of the original." Dense strokes obscure its middle.

Original source details: GELFAND-P000161

$$\int K_s(x,y)u(y)\,dy =$$
$$= \int \sum_j \Psi_j(x)\overline{\Psi}_j(y)(1+\lambda_j)^s u(y)\,dy$$

Ψ_j — scalar fns. or sections of a bundle.

$$\operatorname{Tr}(I+L^*L)^s = \int \operatorname{Tr} K_s(x,x)\,dx$$
$$= \int \sum \Psi_j(x)\overline{\Psi}_j(x)(1+\lambda_j)^s\,dx =$$
$$= \sum d_{\lambda_j}(1+\lambda_j)^s$$



Manuscript notation. In the second integral the argument x and the start of $1+\lambda_j$ are faintly visible.

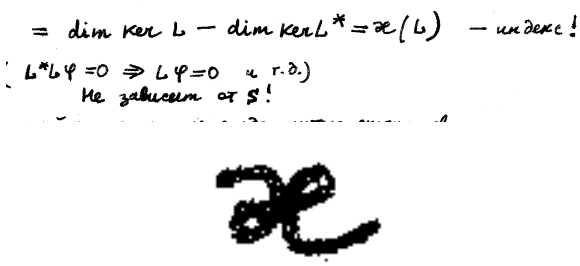
Original source details: GELFAND-P000161

Consider

$$\operatorname{Tr}(I+L^*L)^s - \operatorname{Tr}(I+LL^*)^s =$$

Everything cancels, except the terms with $\lambda = 0$

$$= \dim\ker L - \dim\ker L^* = \text{æ}\,[?](L) \qquad \text{— index!}$$



Uncertain reading. The sign before (L) has two joined loops: χ / x / an "æ"-like sign. Provisionally rendered as "æ".

Original source details: GELFAND-P000161

($L^*L\varphi = 0 \;\Rightarrow\; L\varphi = 0$ etc.)

Does not depend on s!

What follows — study of the trace of integr. operators.

GELFAND-P000162

Original source details: GELFAND-P000162

Diagram and typed description. Page number 25 at the upper right, framed on three sides.

25

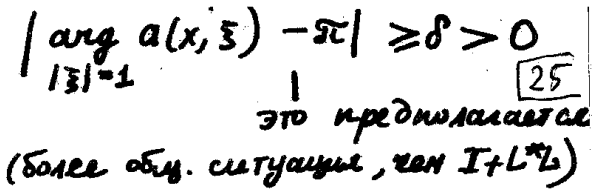
A — elliptic. — $(I + L^*L)$ —
of order $m>0$.

$$|\arg a(x,\xi) - \pi| \underset{|\xi|=1}{\geq} \delta > 0$$

|
this is assumed

(a more gen. situation than $I + L^*L$)

$$\sigma(A) = a_m(x,\xi) + a_{m-1}(x,\xi) + \dots$$



Original source details: GELFAND-P000162

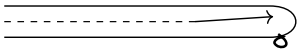
Diagram and typed description. Below "ellipt.": "of order $m > 0$.". Below the angular condition: a short vertical stroke and "this is assumed".

Take $m = 1$ (multipl. by an invertible one)
usual def.:

$$A^s = \frac{1}{2\pi i}\int \lambda^s (\lambda - A)^{-1}\, d\lambda, \qquad \|(\lambda - A)^{-1}\| \leq \frac{C}{1+|\lambda|}$$

Original source details: GELFAND-P000162

Diagram and typed description. Two long lines are open on the left and meet on the right. A dashed arrow points right inside. At the right of the lower line there is a heavy loop. The direction of traversal remains unclear.



contour, enclos. spectrum.

Thus for negat. powers. For posit. powers multiply by a positive integer power.

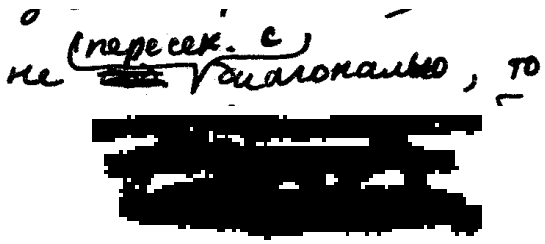
We must compute $\int \operatorname{Tr} K_s(x,x)\, dx$

$K_s(x,x)$ — kernel of the operator A^s, $\operatorname{Re} s \ll 0$.

(i) $s \longrightarrow K_s$ — holom. mapping of the region $\operatorname{Re}(ms) < -n-k$ into kernels of class C^k on $M \times M$

(smooth dep. of the kernel on s)

(ii) C — comp. in $M \times M$, not [Crossed out: unread word.] [Inserted above with a bracket: intersec. with] the diagonal, then



Uncertain reading. Between “not” and “the diagonal” a word is densely crossed out; above it, “intersec. with” is inserted with a bracket. Beneath the cancellation are separate loops without a confident literal reading.

Original source details: GELFAND-P000162

$s \longrightarrow K_s|_C$ continues [Crossed out: unread short word.] to a holom. mapping

Uncertain reading. After “continues” a short word resembling “all” / “everything” is crossed out.

Original source details: GELFAND-P000162

of the entire plane into $C^\infty(C)$

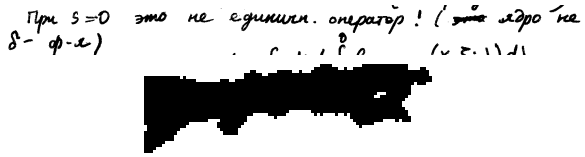
(not analyt. in x, since the coeffs. are infinitely differ.)

(iii) for every x, $K_s(x,x)$ is a meromorphic fn. on the entire plane with poles (simple), possibly only at

$$-\frac{n}{m}, \quad \frac{1-n}{m}, \quad \frac{2-n}{m}, \dots, -\frac{1}{m}, \quad \frac{1}{m}, \quad \frac{2}{m}, \dots$$

$(0$ — regul. point$)$

At $s = 0$ this is not the identity operator! ([Crossed out: unread short word.] the kernel is not a δ-fn.)



Uncertain reading. Before “the kernel is not”, short joined strokes are densely crossed out, possibly “this”; the literal reading is undetermined.

Original source details: GELFAND-P000162

$$\text{(iv)} \quad (2\pi)^n K_0(x,x) = \frac{1}{m} \int_{|\xi'|=1} d\xi' \int_{-\infty}^{0} b_{-m-n}(x,\xi;\lambda)\, d\lambda$$

K_0 has no singularities. in calc. through coeffs. [Below “coeffs.”: of oper. A.]

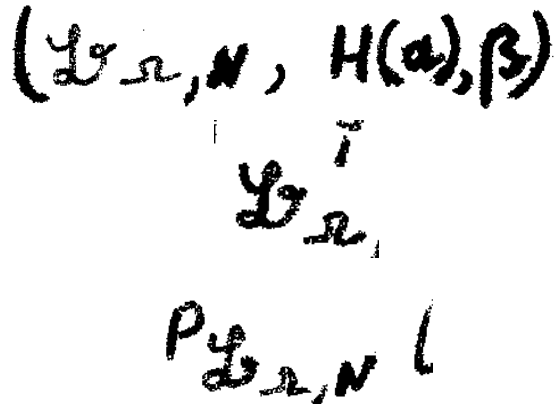
A^s — pseudodiff. oper. and its symbol is obtained in a natural way by subst. the symbol in place of A in the f-la for A^s.

The index — difference of two exprs. of type (iv).

GELFAND-P000163

Minlos. Lecture 3.

$(\mathcal{L}_{\Omega,N}\,[?], H(a), \beta)$



Uncertain reading. Before Ω, N is the same connected sign with upper and right loops as in frames 151 and 152. It is provisionally rendered as $\mathcal{L}$.

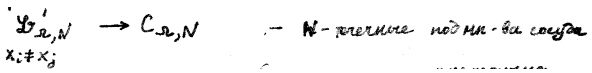
Original source details: GELFAND-P000163

Can one specify a "limiting sp." possessing asympt. props.? Probably yes, but the proof is again for low density.

Pass to indistinguishable particles

$$\mathcal{L}'_{\Omega,N} \underset{x_i \neq x_j}{\longrightarrow} C_{\Omega,N}$$

— N-point subsets of the vessel.



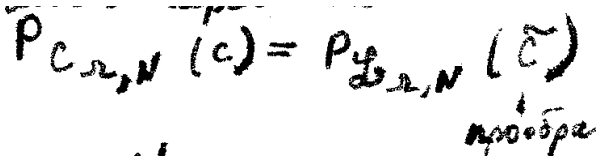
Uncertain reading. The space on the left has the same connected sign, provisionally $\mathcal{L}$, with a prime above. The explanation of N-point subsets is at the right.

Original source details: GELFAND-P000163

H can be consid. on C, since it is symmetric

The Gibbs distr. is transferred

$$P_{C_{\Omega,N}}(c) = P_{\mathcal{L}_{\Omega,N}}(\underset{\text{preimage.}}{\widetilde{c}})$$



Uncertain reading. The subscript on the right has the same connected sign, provisionally $\mathcal{L}$. Below $\widetilde{c}$ is "preimage."

Original source details: GELFAND-P000163

* the compl. of $\mathcal{L}'$ has measure 0.

C_∞ — finite or countable subsets of R^ν (having no accum. points), $d \in C_\infty$.

A Gibbs distr. can be defined on it.

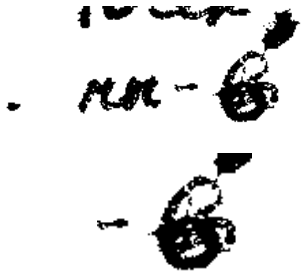
First single out the σ-algebra.

Cylindr. set. $U \subset R^\nu$

$$S_{U,k} \quad (d : n(d,U) = k)$$

the number of points falling in U equals k.

A finite coll. of nonintersect. sets [?]



Uncertain reading. The ending of the abbreviation after “nonintersect.” is densely overwritten, with an upper loop and a lower rounded knot; provisionally rendered as “sets”. The original strokes underneath the overwriting are undetermined.

Original source details: GELFAND-P000163

$$S_{U_1,U_2,\ldots,U_s;k_1,k_2,\ldots,k_s} = \{d : n(d,U_1) = k_1,\ldots\}$$

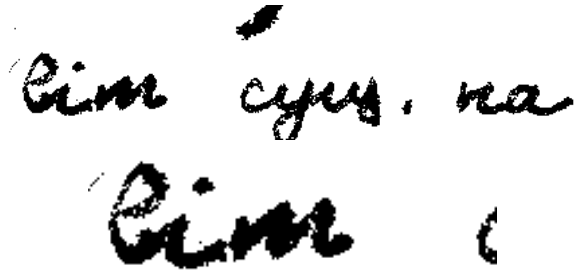
The σ-alg. generated by these cylindr. sets. The usual Gibbs distr. generates a measure on this σ-algebra. The resulting measure satisfies the condition of uniform strong mixing. this is the limiting measure. Its existence is proved for low densities.

$$\mu(S_{U_1U_2,k_1k_2}) - \mu(S_{U_1,k_1})\mu(S_{U_2,k_2}) \longrightarrow 0,$$
$$d(U_1,U_2) \longrightarrow \infty$$

(unif. strong mixing)

uniformly over all pairs U

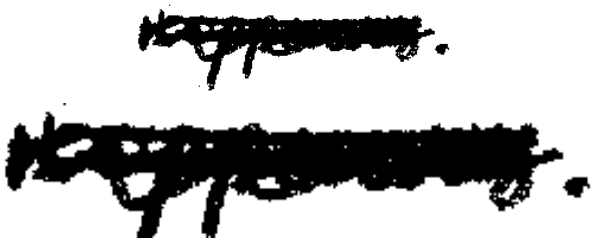
Difficulty — lim exists on



Manuscript notation. After the dash is lim, followed by the abbreviation for “exists”.

Original source details: GELFAND-P000163

[Crossed out: unread long word.] cylindr. sets

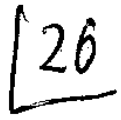


Uncertain reading. After “on”, a long word with two descenders is densely crossed out: “correl.” / another word; the readable parts do not permit a confident reconstruction.

Original source details: GELFAND-P000163

This — reformul. of props. of correlation fns.

GELFAND-P000164



Original source details: GELFAND-P000164

Diagram and typed description. Page number 26 at the upper right, with a line at its left and below it.

26

Props. following from this.

1°. For almost all d, the mean density exists —

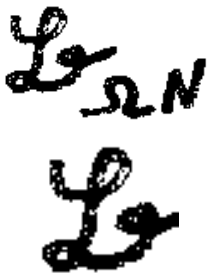
— a consequence of Birkhoff's thm. and ergodicity of the measure.

Gelfand's propos.: start with axioms on correlation functions. How to single them out?

Thermodyn. quantities.

E	P	S	F
energy	pressure	entropy	free energy

$$Z = \frac{1}{N!} \int_{\mathcal{L}_{\Omega N}\,[?]} e^{-\beta H(a)}\, da$$



Uncertain reading. The integration-domain sign matches the connected looped sign in the preceding frame. It is provisionally rendered as $\mathcal{L}$. Below it is ΩN with no visible comma.

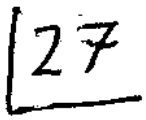
Original source details: GELFAND-P000164

— normaliz. factor.

They are all expressed through $\ln Z$

We shall prove that $\lim \frac{\ln Z}{N}$ exists

GELFAND-P000165



Original source details: GELFAND-P000165

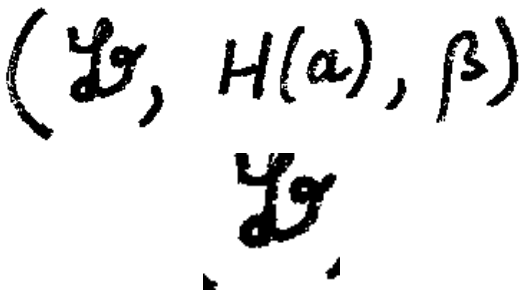
Diagram and typed description. Page number 27 at the upper right, with a line at its left and below it.

27

10/X Gelfand's sem.
Minlos Lecture 4.

E energy P pressure S entropy

$(\mathcal{L}[?], H(a), \beta)$

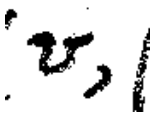


Original source details: GELFAND-P000165

Uncertain reading. The first sign in the triple is the same connected looped sign, provisionally $\mathcal{L}$.

1°. $E = \langle H(a)\rangle$. For $H(a) = \sum U(|x_i - x_j|)$ representativeness was proved, i.e.

$$\langle H(a)\rangle \approx \underset{N,\Omega\longrightarrow\infty}{Ne}, \qquad e = e(v\,[?], \beta)$$



Original source details: GELFAND-P000165

Uncertain reading. The argument in $e(v, \beta)$ is written in a v- / ν-like form; the Latin / Greek sign is undistinguished.

$$\langle H(a)\rangle = \frac{\int H(a) e^{-\beta H(a)}\, da}{Z} = -\frac{\partial}{\partial\beta} \ln Z$$

2°. $p(x) = \tau_1(x)$ [?]. $\quad P = \dfrac{1}{S}\int p(x)\,d\sigma$

(Before the lim. transition)

Uncertain reading. The sign with subscript 1 after $p(x) =$ has an upper arc and a lower hook: τ_1 / r_1; it recurs in the parentheses below.

Original source details: GELFAND-P000165

There is a hypothesis that $p(x) \longrightarrow const$ in the thermodyn. lim. transition. this does not follow from the earlier hypotheses (that $\tau_1 \longrightarrow const$, since this is on $\forall$ finite segment)

Uncertain reading. The sign with subscript 1 recurs in the parentheses. The letter is provisionally rendered as τ.

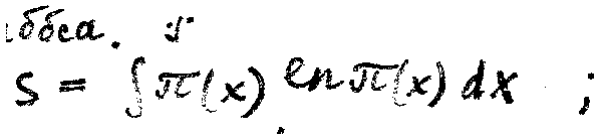
Original source details: GELFAND-P000165

$$p = \frac{\partial \ln Z}{\partial V};$$

3°. Entropy of the Gibbs distr.:

$\forall$ distr. $\pi(x)$

$$S = \int \pi(x) \ln \pi(x)\,dx; \qquad \frac{S}{N} = \frac{1}{N}\ln Z - \frac{1}{\beta}\frac{\partial}{\partial\beta}\frac{\ln Z}{N}$$



Uncertain reading. Above the start of the integral is a short hook with two separate dots. Its function is unclear.

Original source details: GELFAND-P000165

The lim. of entropy per 1 particle exists.

Diagram and typed description. Two strokes with a dot between them separate the talks.

Original source details: GELFAND-P000165

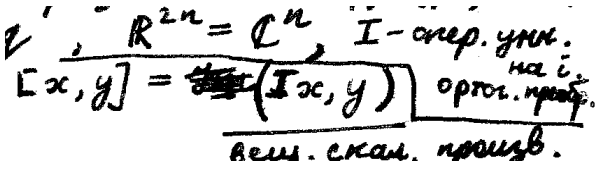
———— . ————

V. I. Arnold. On Maslov's works.

R^{2n} — phase sp. $p_1, \ldots, p_n, q_1, \ldots, q_n$, if 3 structures are introduced

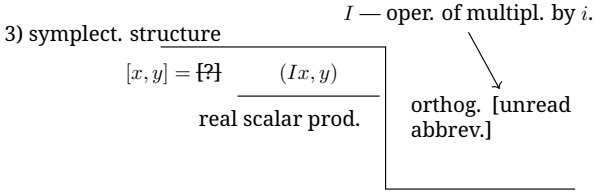
1) metric $p^2 + q^2$ and scalar prod. (Eucl. structure) (x, y)

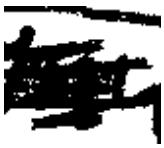
2) compl. structure $z = p + iq$, $R^{2n} = C^n$,



Original source details: GELFAND-P000165

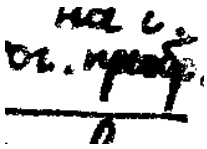
Diagram and typed description. The structures are numbered 1), 2), 3). Beside (Ix, y) is a stepped outline, with the lower caption “real scalar prod.”; on the right, an abbreviation after “orthog.” and a pointer from I.





Original source details: GELFAND-P000165

Uncertain reading. Before (Ix, y) a short mathematical sign / term is densely crossed out; the separate strokes do not permit a confident reading.

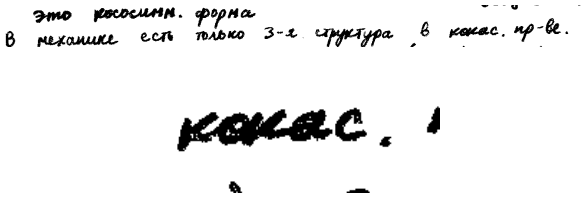


Original source details: GELFAND-P000165

Uncertain reading. After “orthog.” the abbreviation is densely overwritten: “transf.” / “prod.” / another reading; the beginning and a descender are visible.

this is a skew-sym. form

In mechanics there is only the 3rd structure in [unread abbrev.] sp.



Uncertain reading. The abbreviation before “sp.” begins with кас... (kas...). It probably refers to the tangent space.

Original source details: GELFAND-P000165

Preserv. of these structures: $O(2n)$, $GL(C, n)$, $Sp(n)$

It is easy to prove that the pairwise intersecs. equal the triple one (from [3])

this is $U(n)$.

A subsp. R^n is called Lagrangian, if the 3rd form is 0 on it $(IR^n \perp R^n)$. E.g., the real and the purely imaginary subspace.

$$[R^n, R^n] = 0$$

Example. $p = p(q)$ — Lagrangian m-fold $\iff \exists S(q)$, such that $p = \dfrac{\partial S(q)}{\partial q}$

$n = 1$ — every curve is a Lagr. m-fold

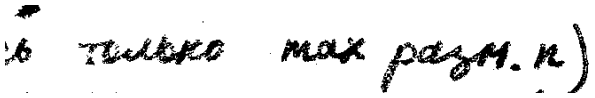
$n = 2$ — not every one.

Lagr. planes are real-like, $U(n)$ acts transitively

$O(n)$ leaves the real plane in place.

Lagr. planes $\Lambda(n) = U(n)/O(n)$ — Lagr. Grassmann m-fold

(Lagr. planes have dim. $\leq n$, but here only max. dim. n)

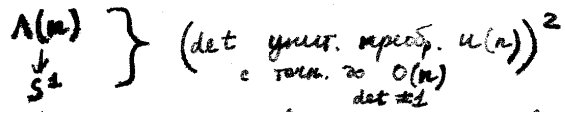


Original source details: GELFAND-P000166

Manuscript notation. Before "dim. n" the manuscript has max.

M — Lagrangian m-fold (the tang. sp. at $\forall$ point is Lagrangian)

Construct the mapping

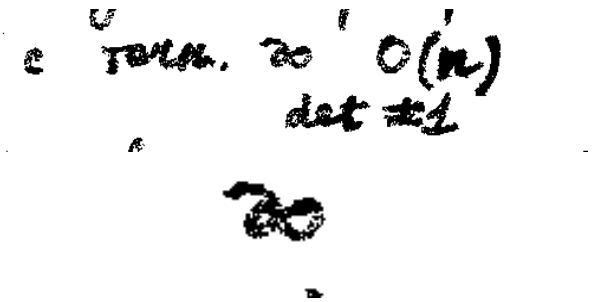


Original source details: GELFAND-P000166

Diagram and typed description. The vertical mapping $\Lambda(n) \to S^1$ is joined by a right brace to the square of the determinant.

$$\left.\begin{array}{c}\Lambda(n)\\ \downarrow\\ S^1\end{array}\right\} \quad (\text{det of a unit. transf. } U(n))^2$$

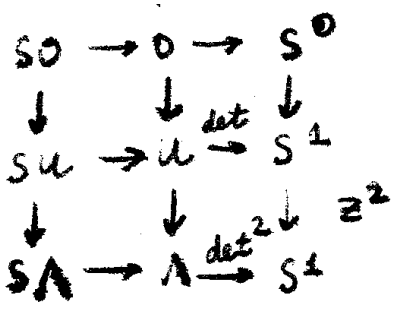
up to $O(n)$ det ± 1



Original source details: GELFAND-P000166

Manuscript notation. The manuscript has "up to $O(n)$," with det ± 1 below $O(n)$.

$U(n)$ takes the real plane to our Lagrangian one.



Original source details: GELFAND-P000166

Diagram and typed description. Three rows: SO, O, S^0; SU, U, S^1; $S\Lambda, \Lambda, S^1$. Arrow labels: det, $\det^2$, z^2.

$$\begin{array}{ccccc} SO & \longrightarrow & O & \longrightarrow & S^0 \\ \downarrow & & \downarrow & & \downarrow \\ SU & \longrightarrow & U & \xrightarrow{\det} & S^1 \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle z^2} \\ S\Lambda & \longrightarrow & \Lambda & \xrightarrow{\det^2} & S^1 \end{array}$$

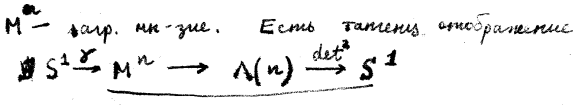
$S\Lambda$ — fibre of the corresp. bundle.

This is a diag. of bundles.

M^n — Lagr. m-fold. There is a tangent mapping
[Overwritten unread sign.]

Original source details: GELFAND-P000166

Uncertain reading. Before the chain $S^1 \longrightarrow M^n$ is a densely overwritten sign: $\forall$ / a crossed-out expression; the original sign is undetermined.

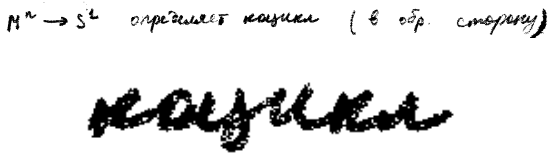


Original source details: GELFAND-P000166

Diagram and typed description. Below the chain from M^n to the last S^1 is a long line; γ is on the first arrow, $\det^2$ on the last.

$$S^1 \xrightarrow{\gamma} M^n \longrightarrow \Lambda(n) \xrightarrow{\det^2} S^1$$

$M^n \longrightarrow S^1$ defines a cocycle [?] (in the reverse direction)

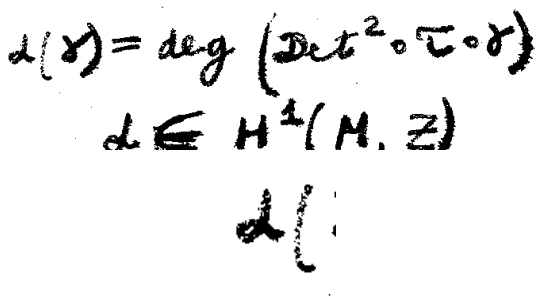


Original source details: GELFAND-P000166

Uncertain reading. The word after “defines” appears to read коцикл (cocycle). The vowel after the initial k is joined to its neighbours.

γ — a curve in M^n

$$\ell\,[?](\gamma) = \deg(\mathrm{Det}^2 \circ \tau \circ \gamma),$$
$$\ell\,[?] \in H^1(M, Z)$$



Original source details: GELFAND-P000166

Uncertain reading. The sign before (γ) and before $\in H^1$ has a tall stem and a lower loop: an l- / ℓ- / μ-like sign. Provisionally rendered as ℓ.

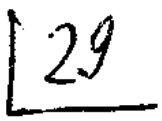
This is the Maslov index, occurring in asymptotic f-las, in particular, the Morse index is obtained (the number of conjug. points on a geod.)

There is another geom. def. of Maslov’s.

GELFAND-P000167

$17/\overline{X}$ Gelfand Sem. Minlos. Lecture 5.

Phase transitions.



Original source details: GELFAND-P000167

Diagram and typed description. Page number 29 at the upper right, with a line at its left and below it.

29

I. M. Gelfand. Tannaka—Krein duality thm.

G — compact group or finite.

Given irreduc., finite-dimensional reps. in the spaces $A_1, \ldots, A_n$

$$A_i \otimes A_k \cong \sum A_j$$

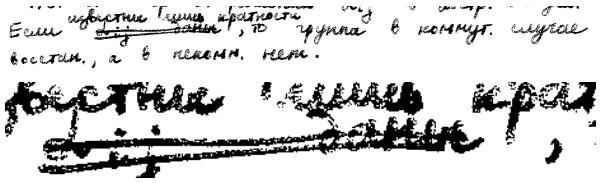
(Tensor prod. decomp. into irreducible representations)

Consider $\sum A_j$ (each 1500 times)

$$A_i \otimes A_j \longrightarrow \sum A_j;$$

i.e. there is a collection of maps d_{ij} into an abstr. module.

If [crossed out: d_{ij} are given[?]] [inserted above: only the multiplicities are known], then the group can be recov. in the commut. case, but not in the noncommut. case.



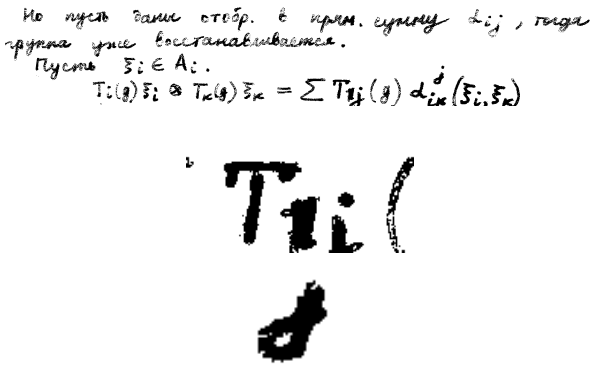
Original source details: GELFAND-P000167

Uncertain reading. Above is "only the multiplicities are known." Beneath the two strokes the entry appears to read "d_{ij} are given."

But let the maps d_{ij} into the direct sum be given, then the group can now be reconstructed.

Let $\xi_i \in A_i$.

$$T_i(g)\xi_i \otimes T_k(g)\xi_k = \sum T_{1j}\,[?](g) d^j_{ik}(\xi_i, \xi_k)$$



Original source details: GELFAND-P000167

Uncertain reading. The first stroke of the subscript on the right could be 1 or undotted i. The following j is clear. The superscript on d is j.

In place of $T_i(g)$ write arbitrary C_i — lin. transformations.

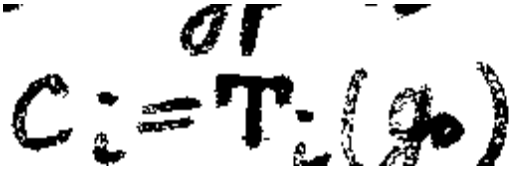
Let [?] the formula hold:

Original source details: GELFAND-P000167

Uncertain reading. The word before "the formula hold" is densely overwritten; provisional reading "Let", with the internal cancelled strokes unread.

$$C_i\xi_i \otimes C_k\xi_k = \sum C_j d_{ik}^j(\xi_i, \xi_k).$$

This eq. has solutions $C_i = T_i(g)$.
Then $C_i = T_i(g_0)$. Thus, all solutions of these eqs. will be $T_i(g)$ and everything is recov. from d_{ik}^j.



Original source details: GELFAND-P000167

Manuscript notation. In the line after "Then," g has a small subscript 0.

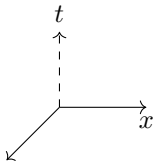
There is a generalization to $SL(2, R)$.
Nirenberg.
Varadhan.

$$\frac{\partial u}{\partial t} = \sum a^{ij}(x) u_{x^i x^j}, \qquad u(x, 0) = \delta(x - y)$$

Study the behavior of the fund. solution.

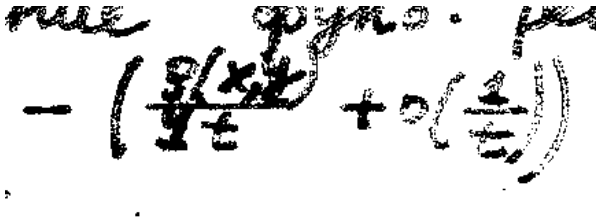
Original source details: GELFAND-P000167

Diagram and typed description. Three arrows from a common point; the vertical one is dashed with t, the right one has x, the lower-left one is unlabelled.



Then

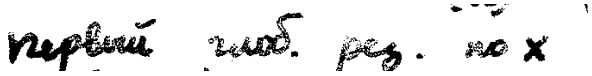
$$u(x) = e^{-\left(\frac{g(x,y)}{[?]\,t} + o(\frac{1}{t})\right)}, \quad t \longrightarrow 0$$



Uncertain reading. There is a heavy stroke to the left of t under the fraction bar, possibly 4. The denominator is provisionally rendered as t.

Original source details: GELFAND-P000167

first glob. result in x.



Manuscript notation. The line reads “first glob. result in x.”

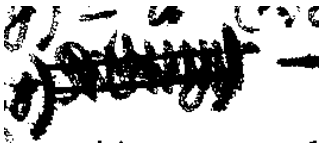
Original source details: GELFAND-P000167

For heat cond.

$$u = C\frac{e^{-\frac{(x-y)^2}{4t}}}{t^{n/2}}$$

$$g(x,y) = d^2(x,y)$$

$d(x,y)$ [crossed-out unreadable term] — dist. along a geodesic



Uncertain reading. After $d(x,y)$ a term is densely crossed out; the internal letter strokes are obscured.

Original source details: GELFAND-P000167

($t^{n/2}$ enters $o(\frac{1}{t})$)

in the metric gener. by a^{ij}.

Also asymptotics of the eigenvalues of the Laplace oper. in a domain in the same spirit.

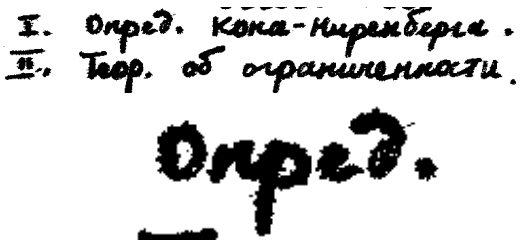
GELFAND-P000168

$24/\overline{X}$. Gelfand Sem.

Vishik. Lecture 1.

I. Defn. of Kohn—Nirenberg.

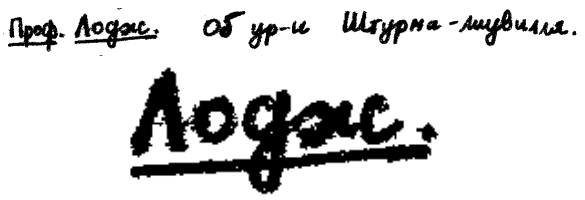
II. Thm. on boundedness.



Manuscript notation. After I is Опред. (“Defn.”). The final letter has an upper loop.

Original source details: GELFAND-P000168

Prof. Lodes. [?] On the Sturm—Liouville eq.



Original source details: GELFAND-P000168

Uncertain reading. After "Prof." the name appears to read Лодес. (Lodes.). The middle letters are joined, and the exact surname remains unclear.

31/$\overline{X}$ Gelfand Sem.
Vishik. Lecture 2.

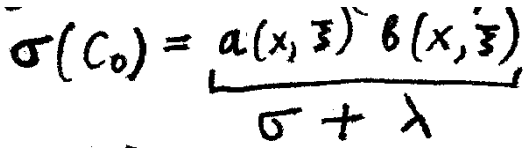
$$\sigma(A) = a(x,\xi) \longleftrightarrow A$$

$$A \cdot Bu = C_0 u + \ldots + C_{s-1}u + T_s u,$$

where

$$\sigma(C_k) = \sum_{|\alpha|=k} \frac{1}{\alpha!} \partial^\alpha a(x,\xi) D^\alpha b(x,\xi)$$

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial \xi^\alpha}, \qquad D^\alpha = \left(\frac{1}{i}\frac{\partial}{\partial x}\right)^\alpha$$



Original source details: GELFAND-P000168

Diagram and typed description. A straight bracket with short raised ends under the product $a(x,\xi)b(x,\xi)$; label $\sigma + \lambda$.

$$\sigma(C_0) = \underbrace{a(x,\xi)b(x,\xi)}_{\sigma+\lambda}$$

T_s — oper. of order $\sigma + \lambda - s$.

$$\zeta(\xi)a(x,\xi) = h(x,\xi), \qquad \zeta(\xi)b(x,\xi) = g(x,\xi)$$

$$\widetilde{Au}(\xi) = (2\pi)^{-n/2} \int \widetilde{h}(\xi-\eta,\eta)\widetilde{u}(\eta)d\eta$$
$$\widetilde{Bu}(\eta) = (2\pi)^{-n/2} \int \widetilde{g}(\eta-\tau,\tau)\widetilde{u}(\tau)d\tau$$

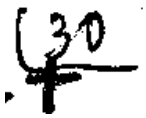
$$\widetilde{A \cdot Bu}(\xi) = (2\pi)^{-n/2} \int \widetilde{h}(\xi-\eta,\eta)\widetilde{Bu}(\eta)d\eta =$$

$$= \ldots \int \widetilde{h}(\xi-\eta,\eta)d\eta \int \widetilde{g}(\eta-\tau,\tau)\widetilde{u}(\tau)d\tau =$$
$$= \ldots \int \left(\int \widetilde{h}(\xi-\eta,\eta)\widetilde{g}(\eta-\tau,\tau)d\eta \right) \widetilde{u}(\tau)d\tau$$

Expand $\widetilde{h}$ by Taylor's formula at $\eta = \tau$

$$\widetilde{h}(\xi-\eta,\eta) = \sum_{|\alpha|\le s-1} \frac{1}{\alpha!}\partial^\alpha \widetilde{h}(\xi-\eta,\tau)(\eta-\tau)^\alpha + h_s(\xi-\eta,\eta,\tau)$$

GELFAND-P000169

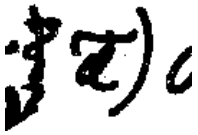


Original source details: GELFAND-P000169

Diagram and typed description. Page number 30 at the upper right. The adjacent plus sign appears beneath it.

30

$$\int \widetilde{h}(\xi-\eta,\eta)\widetilde{g}(\eta-\tau,\tau)d\eta$$
$$= \int \widetilde{h}(\xi-\eta,\tau)\widetilde{g}(\eta-\tau,\,[?]\,\tau)d\eta +$$
$$+\sum \frac{1}{\alpha!}\int \partial^\alpha \widetilde{h}(\xi-\eta,\tau)(\eta-\tau)^\alpha \widetilde{g}(\eta-\tau,\tau)d\eta$$



Original source details: GELFAND-P000169

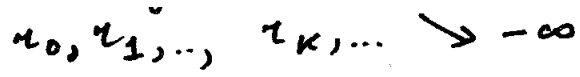
Uncertain reading. In the first right-hand integral, between $\eta-\tau$, and the final τ, is a small densely overwritten sign; a possible cancelled ξ / another sign is undetermined.

The 1st convolution corresp. to the product $h(x,\tau)g(x,\tau)$

Everything follows from this.

Defn.

$$r_0, r_1, \ldots, r_k, \ldots \searrow -\infty$$
$$A_0, A_1, \ldots, A_k, \ldots$$



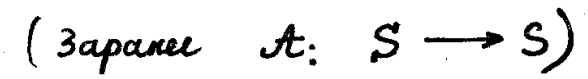
Original source details: GELFAND-P000169

Uncertain reading. The signs in the sequence are rendered as $r_0, r_1, \ldots, r_k, \ldots$.

A_i homog. of order r_i with symbols $a_i(x,\xi)$

$$A \sim \sum_{k=0}^{\infty} A_k$$

(In advance $A : S \longrightarrow S$)



Original source details: GELFAND-P000169

Manuscript notation. The parentheses contain $A : S \longrightarrow S$. There is no subscript on A.

$$A - \sum_{k=0}^{N} A_k = T_{N+1} \quad \text{— oper. of order } < r_N.$$

Symbol [crossed-out formula resembling $\sigma(A)$, reading undetermined]

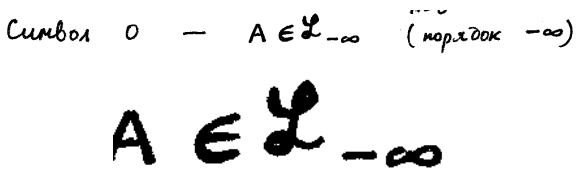
Original source details: GELFAND-P000169

Uncertain reading. After “Symbol” a formula resembling $\sigma(A)$ is crossed out; dense strokes inside the parentheses leave the reading ambiguous.

formal sum

$$\sum_{k=0}^{\infty} a_k(x,\xi) = \sigma(A)$$

Symbol 0 — $A \in \mathcal{L}_{-\infty}$ [?] (order $-\infty$)



Original source details: GELFAND-P000169

Uncertain reading. The calligraphic letter before $-\infty$ has a tall stem with upper and lower loops: $\mathcal{L}$ / $\mathcal{Y}$ / another handwritten sign. Provisionally rendered as $\mathcal{L}$.

Theorem: the symbol of an oper. is uniquely determ.
$\forall$ [crossed-out unreadable abbreviations] symbol there exists an operator.

Original source details: GELFAND-P000169

Uncertain reading. After $\forall$ two abbreviations are crossed out: “oper. exists” / another entry; lower strokes and a connecting long line are visible.

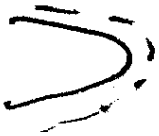
Pseudodiff. operators form an algebra.

$$\sigma(A \cdot B) = \sum \frac{1}{\alpha!} \partial^\alpha \sigma(A) D^\alpha \sigma(B)$$

O. A. Oleinik.
One particular problem in mechanics.
Viscous fluid. u, v — veloc. (plane). Navier—Stokes eqs.

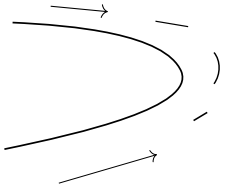
$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \nu\Delta u \\ \dfrac{\partial v}{\partial t} + u\dfrac{\partial v}{\partial x} + v\dfrac{\partial v}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial y} + \nu\Delta v \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0 \end{cases}$$

ν — viscosity (small)
$\nu = 0$ — Euler eqs. (ideal fluid).
Even for small ν these terms cannot be crossed out (Prandtl)
Ideal fluid: bdy. cond. — normal component of veloc. $= 0$
viscous — both comp. $= 0$ (sticking)

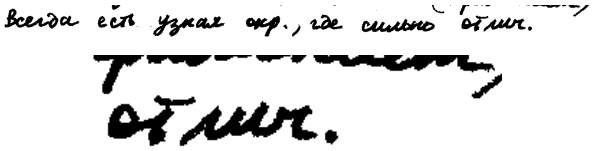


Original source details: GELFAND-P000169

Uncertain reading. Beside the boundary conditions on the left is an arc open to the left and convex to the right. Above and below it are short strokes and arrows pointing right; several separate strokes to the right. No labels; the exact directions of all short strokes are undetermined.



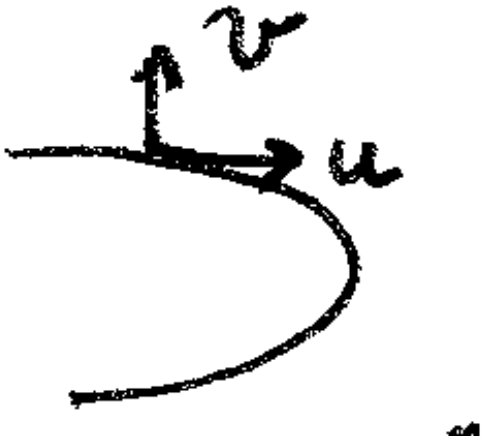
There is always a narrow neighb. where they differ markedly.



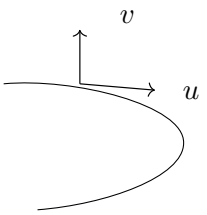
Original source details: GELFAND-P000169

Manuscript notation. The last line reads “a narrow neighb. where they differ markedly.”

GELFAND-P000170



Typeset diagram



Original source details: GELFAND-P000170

v does not change (bdy. conds. preserved).

u changes strongly.

The Prandtl eqs. remain:

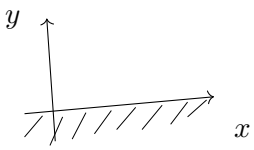
$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \nu\dfrac{\partial^2 u}{\partial y^2}, \\ \cancel{\dfrac{\partial p}{\partial y} = 0} \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0. \end{cases}$$

Manuscript notation. The middle line $\partial p/\partial y = 0$ is crossed out with two diagonal strokes.

Original source details: GELFAND-P000170

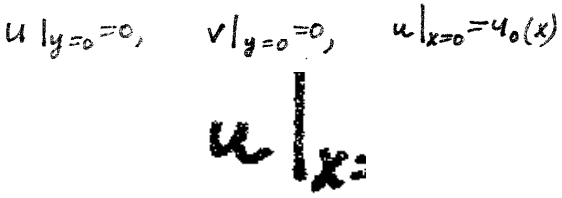
p does not change along the normal and it is consid. to be a known fn., obtained from the eqs. of an ideal fluid. $p = p(t, x)$.

Typeset diagram



Original source details: GELFAND-P000170

$$u|_{y=0} = 0, \qquad v|_{y=0} = 0, \qquad u|_{x=0} = u_0(x).$$



Manuscript notation. The third condition is written $u|_{x=0} = u_0(x)$. Both letters x have crossed strokes.

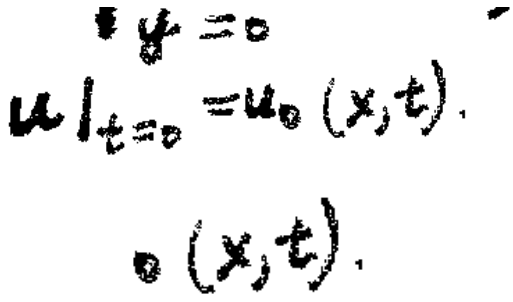
Original source details: GELFAND-P000170

$$u(t, x, y) = U(t, x) \qquad y \longrightarrow \infty.$$

$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = \nu\dfrac{\partial^2 u}{\partial y^2} - \dfrac{1}{\rho}\dfrac{\partial p(t,x)}{\partial x}, \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0. \end{cases}$$

$$u|_{y=0} = 0, \qquad v|_{y=0} = v_0(x, t), \qquad u|_{x=0} = u_0(x),$$

$$u|_{t=0} = u_0(x, t),$$



Manuscript notation. In the condition at $t = 0$, the second argument on the right is t, with a short crossbar.

Original source details: GELFAND-P000170

$$\lim_{y \to +\infty} u(t, x, y) = U(t, x).$$

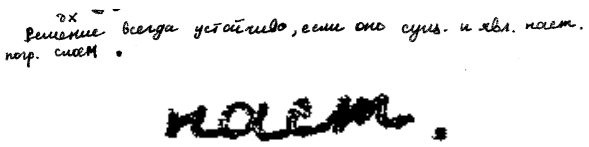
The soln. is unique, if it exists.

One can prove that it exists for an interval $0 \le t \le T$ for the whole body in the flow (separation may occur, but not instantaneously).

Near the beginning of the flow around the body, it exists for $\forall t$.

$$\frac{\partial p}{\partial x} \le 0.$$

The solution is always stable if it exists and is a genuine bdy. layer.



Manuscript notation. The first line ends with the abbreviations явл. наст. ("is a genuine"), followed by "bdy. layer".

Original source details: GELFAND-P000170

GELFAND-P000171

$14/\overline{XI}$. Sem. of Gelfand.

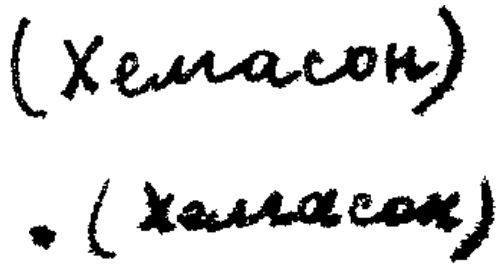
Agranovich. Lecture 3 on pseudodiff. operators.

Problems of diff. geom.

1. There is a complete Riem. manif. of neg. curvature.

$f \in C_0^\infty(\Omega)$, a ball B is given and $\int$ over any sphere containing B, $= 0$. Prove that $f = 0$ outside B.

(Helgason) Done for const. curvature. (Helgason)



Uncertain reading. Both parentheses appear to read Хелгасон (Helgason). The middle letters are joined.

Original source details: GELFAND-P000171

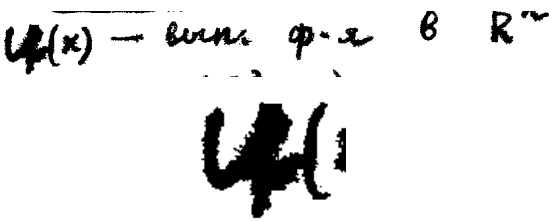
2. [Crossed out: illegible sign]

Uncertain reading. Before Calabi is a small densely cancelled sign with a vertical stroke and lower loop: a star / cancelled letter.

Original source details: GELFAND-P000171

Calabi

$U(x)$ [?] — conv. funct. in R^n.



Uncertain reading. The first function letter is reinforced and overwritten: a U- / u-like sign; the internal overlapping strokes cannot be separated.

Original source details: GELFAND-P000171

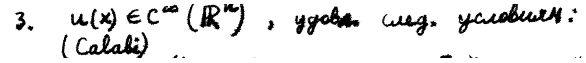
$$\det\left(\frac{\partial^2 u}{\partial x_i \partial x_j}\right) \equiv 1.$$

Prove that u is a polynomial of degree 2.

Proved for $n = 2$ and for $n \leq 5$ Calabi.

3. $u(x) \in C^\infty(R^n)$, satisf. [?] the foll. condition:

(Calabi)

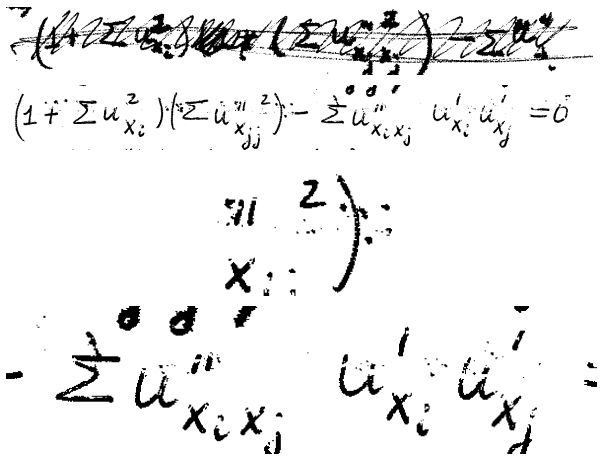


Original source details: GELFAND-P000171

Uncertain reading. The end of the abbreviation translated "satisf." is reinforced with overlapping strokes; it is followed by "the foll. condition:".

[Crossed out:]

$$\left(1 + \sum u_{x_i}^2\right) \text{ [illegible] } \left(\sum u''_{x_j j}{}^2\right)$$
$$- \sum u''_{x_i} [?]$$
$$\left(1 + \sum u_{x_i}^2\right) \cdot \left(\sum u''_{x_j j}{}^2\right) - \sum u''_{x_i x_j} u'_{x_i} u'_{x_j} = 0.$$



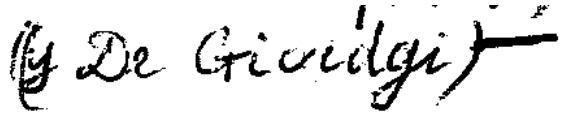
Uncertain reading. The first formula is densely crossed out. Some strokes between the brackets and at the end remain unread. In the uncancelled line there is a dot between the brackets. The second u has a double prime, a separate raised 2, and two j letters below x.

Original source details: GELFAND-P000171

Is it true that u is linear?

(These are the eqns. of min. surfaces.)

Done. (there is a reference in De Giorgi [?])



Uncertain reading. The context points to De Giorgi. The middle letters of the Latin-script surname are joined.

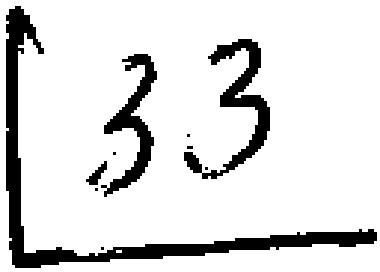
Original source details: GELFAND-P000171

4. M is a comp. Riem. manif.

$$L_1^2(M), \quad \langle \varphi, \psi \rangle = \int_M \varphi\psi \, dx + \int \langle d\varphi, d\psi \rangle \, dx.$$

GELFAND-P000172

[source number: 33]

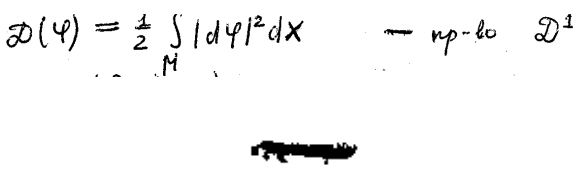


Original source details: GELFAND-P000172

Diagram and typed description. Page number 33 at the upper right, with a line at its left and below it.

33

$$D(\varphi) = \frac{1}{2}\int_M |d\varphi|^2\, dx \quad \text{— sp. } D^1\,[?].$$



Original source details: GELFAND-P000172

Uncertain reading. Before "sp. D^1" is a dense horizontal stroke: dash / arrow, with an overwritten left end; its direction is undetermined.

$$(L_1^2(M), D)$$
$$(L_1^2(M'), D').$$

There is an isometry of $L_1^2(M)$, inducing a diffeom. of M and M'. Prove that M' and M are isometric.

Invariants — eigenvalues of the Laplace oper.

[Crossed out: ~~Er~~ [?]]

Original source details: GELFAND-P000172

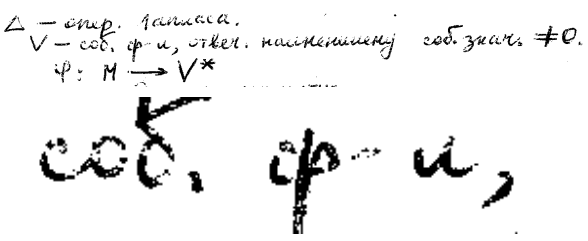
Uncertain reading. Above number 5 is a short cancelled word: "Er" / "Or"; its joined strokes are not fully distinct.

5. M is an n-dim. comp. irreduc. symm. sp.

Δ is the Laplace oper.

V — eigenfns., corresp. to the smallest eigenvalue $\neq 0$.

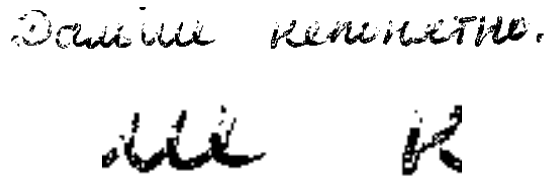
$$\varphi : M \longrightarrow V^*.$$



Original source details: GELFAND-P000172

Manuscript notation. After "eigen" the manuscript has the abbreviation for "functions." The map ends in V^*.

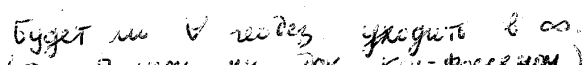
The rest is unclear.



Original source details: GELFAND-P000172

Manuscript notation. Below the map the manuscript reads “The rest is unclear.”

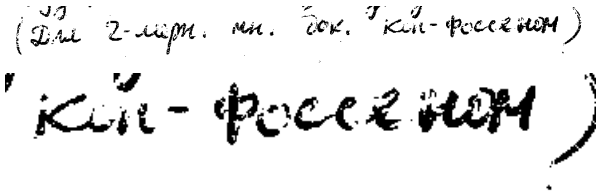
6. M is a complete noncomp. Riem. manif. (3-dim.) of pos. curvature (Ricci tensor > 0). Will $\forall$ geodes. go to ∞.



Original source details: GELFAND-P000172

Manuscript notation. After “Will” are $\forall$ and the abbreviation for “geodesic”.

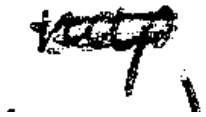
(For 2-dim. manif. prov. by Cohn-Vossen [?])



Original source details: GELFAND-P000172

Uncertain reading. The surname is “Kn-Fossen” / “Kon-Fossen”; the middle strokes and possible second “s” are joined.

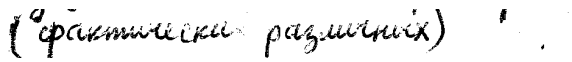
7. M is a comp. n-dimensional manif. Do ∞-many [Crossed out: ~~par~~ [?]] intersect. geodesics exist.



Original source details: GELFAND-P000172

Uncertain reading. Before “intersect.” is cancelled “par” / a short word with a descending stroke; the contraction is not expanded.

(actually [?] distinct)



Original source details: GELFAND-P000172

Uncertain reading. The first long word in parentheses is read as “actually”; some internal strokes are joined. The second word is “distinct”.

8. [?]

Original source details: GELFAND-P000172

Uncertain reading. The number after 7 is provisionally read as 8. The left edges of both loops are faint.

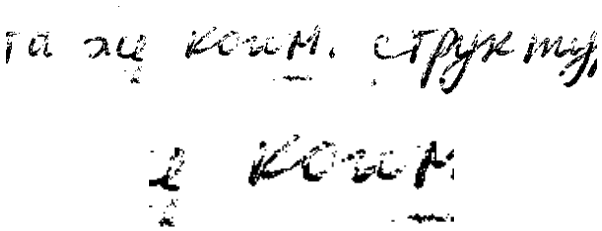
M is a comp. simply connected Riem. manif. in which all geodesics are closed and have the same length. Prove that it is diffeom. to a symm. sp. [Crossed out: $\rho =$ [?] illegible] of the 1st type. (?)

Original source details: GELFAND-P000172

Uncertain reading. Before “of the 1st type” is a cancelled $\rho =$- / $p =$-like sign with an overlapping continuation.

There is an analyt. manif., diffeom. to a sphere, but not isometr.

(Bott). Bott proved that it has the same cohom. [?] structure.



Uncertain reading. The internal strokes in the contraction before “structure” are joined; it is read as “cohom.”. Under it are two short strokes. Bott is written twice, the first time in parentheses.

Original source details: GELFAND-P000172

9. Does a contact structure exist on every comp. M^3?

GELFAND-P000173

$21/\overline{XI}$. Sem. of Gelfand.

Arnold.

Lecture 1.

Theory of singularities of diff. mappings.

$$f : M^m \longrightarrow N^n.$$

Uncertain reading. At the right of the title is an isolated short upright stroke. Its reading is unclear.

Original source details: GELFAND-P000173

$y = x^2$.

Typeset diagram

Original source details: GELFAND-P000173

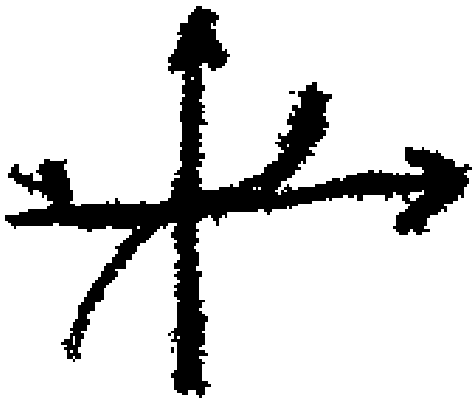
Stab. [Crossed out: illegible word] sing.

Undeciphered passage. Between “Stab.” and “sing.” is a densely cancelled short word; the letters beneath the cancellation strokes are unrecovered.

Original source details: GELFAND-P000173

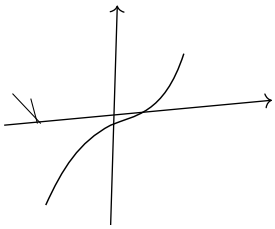
(for a nearby diff. mapping, reduced to the same form)

$y = x^3$.



Original source details: GELFAND-P000173

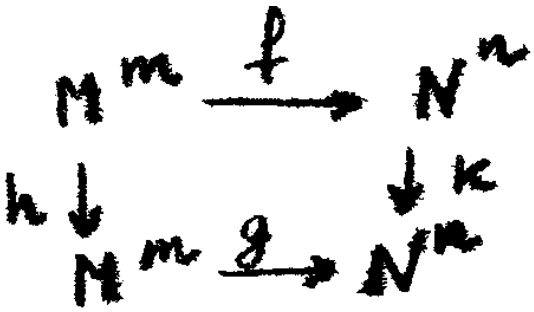
Typeset diagram



Unstab. sing.

Original source details: GELFAND-P000173

Uncertain reading. At $y = x^2$ is a U-shaped line with an almost vertical internal stroke and a horizontal line below it. At $y = x^3$ are axes with upward and rightward arrows, an S-like line from lower left to upper right, and small oblique strokes left of the crossing. The axes have no letter labels; exact contacts in the dense crossing are undetermined.



Original source details: GELFAND-P000173

Typeset diagram

$$\begin{array}{ccc} M^m & \xrightarrow{f} & N^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ M^m & \xrightarrow{g} & N^n \end{array}$$

Top. stab. $\exists$ homeom. h and k.

(diff.)

$\downarrow$

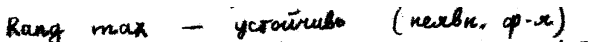
Diff. stab.

Stability at 0 — the same on neighborhoods.

(germs of a mapping)

(One can change the manifolds so that the mappings are given by the same formulas)

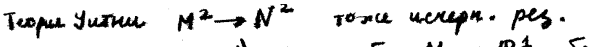
Rang max — stable (implicit funct.).



Manuscript notation. The line reads “Rang max — stable (implicit funct.)”.

Original source details: GELFAND-P000173

Morse lemma — nondeg. quadr. form ($R^n \to R^1$ stab.)
Whitney thm. [?] $M^2 \to N^2$ also an exhaustive res.



Uncertain reading. Before “Whitney” is Teop with a short joined ending. The exact expansion is unclear.

Original source details: GELFAND-P000173

Morse M is comp. 1) stab. mappings $M \to R^1$ form an open everywhere dense set.
2) Conditions for stability (necess. and suff.):

GELFAND-P000174

M.1 All crit. [Crossed out: z [?], illegible] points nondeg.

Uncertain reading. Before “points” is a cancelled short word with a “z”-like beginning; the reading “values” abbreviated / overlapping short strokes.

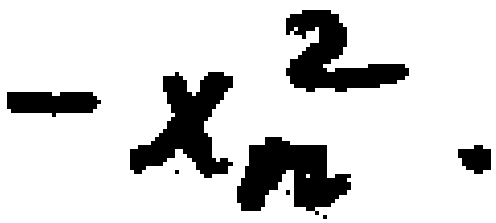
Original source details: GELFAND-P000174

M.2 All crit. values are distinct.
3) loc. assertion loc. stability $\Longleftrightarrow$ one can choose coords. so that it is written by formulas (one of the ways)

Manuscript notation. The line reads “written by formulas”.

Original source details: GELFAND-P000174

$$y = x_1$$
$$y = x_1^2 + \ldots + x_k^2 - x_{k+1}^2 - \ldots - x_n^2.$$



Manuscript notation. The last term in the quadratic form is $-x_n^2$.

Original source details: GELFAND-P000174

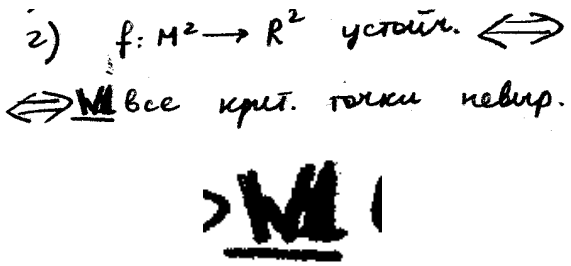
(stab. both top. and diff.)

Whitney's theorem.

1) Stab. mappings are dense.

2) $f : M^2 \to R^2$ stab. $\Longleftrightarrow$

$\Longleftrightarrow$ W1 [?] all crit. points nondeg.



Original source details: GELFAND-P000174

Uncertain reading. The first underlined number is W1 / M1; the angular strokes overlap heavily. The equivalence sign is written twice, again $\Longleftrightarrow$ before the number. The next number is W2.

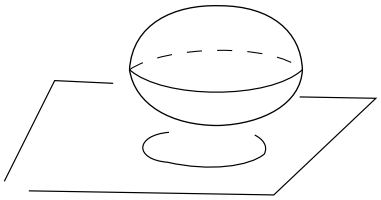
W2 crit. values are transverse (images of folds)

3) Stab. at a given point $\Longleftrightarrow$ the mapping is written by one of the formulas:

1. $y_1 = x_1, \quad y_2 = x_2,$
2. $y_1 = x_1, \quad y_2 = x_2^2$ (folds),
3. $y_1 = x_1, \quad y_2 = x_1 x_2 + x_2^3.$

Original source details: GELFAND-P000174

Typeset diagram

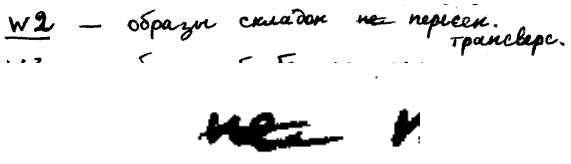


Original source details: GELFAND-P000174

Uncertain reading. A sphere-like outline with a transverse ellipse whose upper arc is dashed; below it are an open oval and a plane with gaps in the upper edge and lower left corner. There are no labels or connecting arrow; the precise projection is unspecified.

2. — fold points, these are curves where the Jacobian $= 0$.

W2 — images of folds [Crossed out: ~~not~~] intersect. transvers.

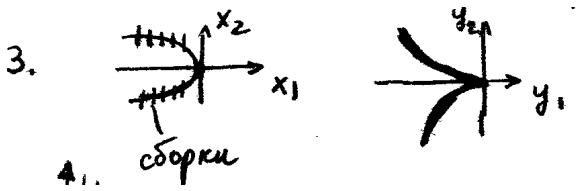


Original source details: GELFAND-P000174

Manuscript notation. Before "intersect." the manuscript has a crossed-out "not".

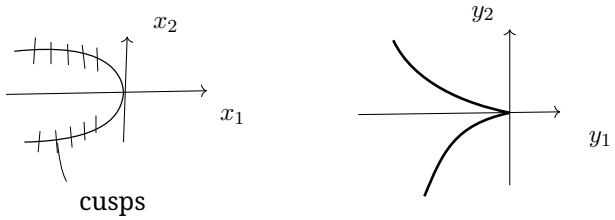
W3 — images of cusps are distinct.

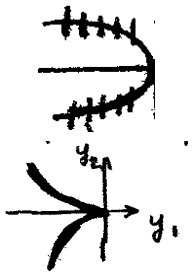
3.



Original source details: GELFAND-P000174

Typeset diagram



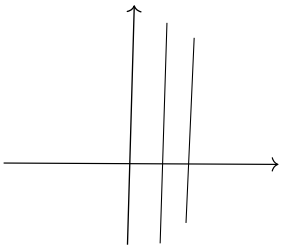


Original source details: GELFAND-P000174

Uncertain reading. At left are axes x_1, x_2, a line with two leftward branches and short strokes, and a “cusps” callout to the lower branch. At right are axes y_1, y_2 and two thick branches meeting near their crossing. Exact stroke placement and contact in the dense crossing remain uncertain.

Original source details: GELFAND-P000174

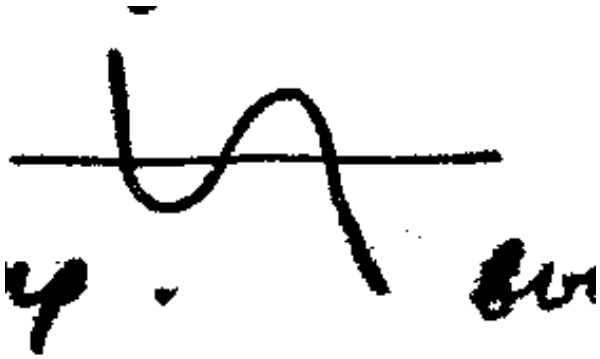
Typeset diagram



For $x_1 > 0$, monot. on vert. lines.

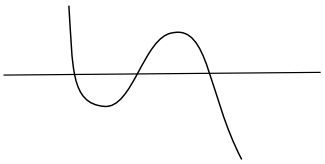
$$x_1 = \text{const} \Rightarrow y_1 = \text{const}\,.$$

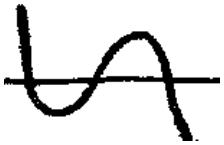
For $x_1 < 0$ there will be



Original source details: GELFAND-P000174

Typeset diagram

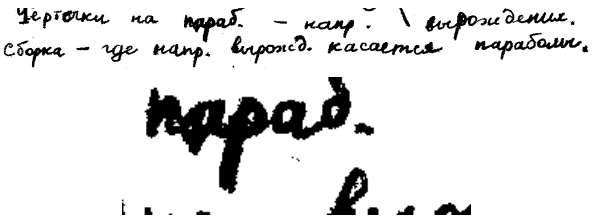




Original source details: GELFAND-P000174

Uncertain reading. An arrowless horizontal line and a curve descending from upper left, then a trough, a rise above the horizontal and a descent at right. There are no letter labels; crossing placement is specified only by the drawing.

Short strokes on the parab. [?] — directions of degeneracy.
A cusp is where the dir. of degen. is tangent to the parabola.

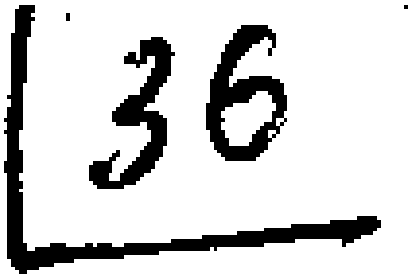


Original source details: GELFAND-P000174

Uncertain reading. The first line has “parab.” / a compressed “prab.”-like contraction with overlapping initial strokes. The last line writes “parabola” in full.

GELFAND-P000175

[source number: 36]

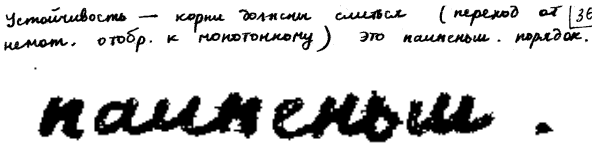


Original source details: GELFAND-P000175

Diagram and typed description. Page number 36 at the upper right, with a line at its left and below it.

36

Stability — the roots must merge (transition from a nonmonot. map to a monotone one)
This is the least order.



Manuscript notation. Before “order” the manuscript has наименьш. (“least”).

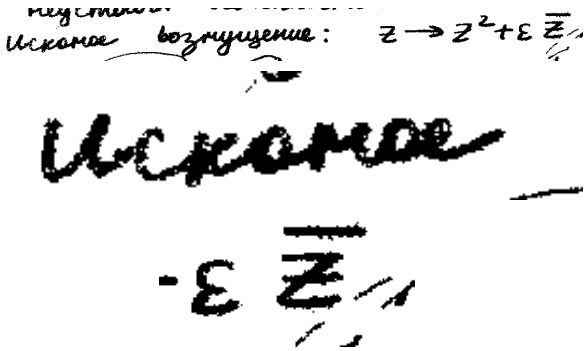
Original source details: GELFAND-P000175

$$z \longrightarrow z^2$$

Unstable singularity.

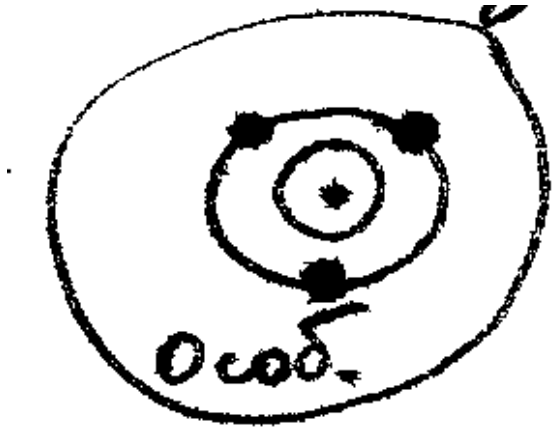
Sought perturbation:

$$z \longrightarrow z^2 + \varepsilon\bar{z}.$$

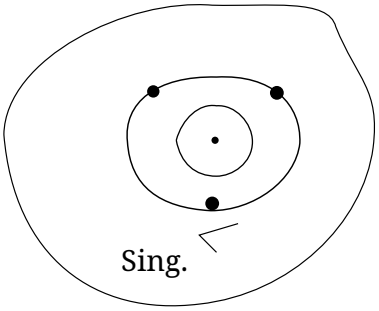


Manuscript notation. The words read “Sought perturbation.” The final z has a horizontal bar.

Original source details: GELFAND-P000175

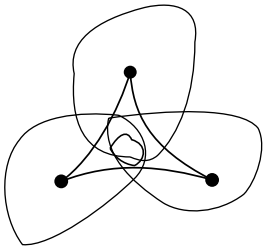


Typeset diagram

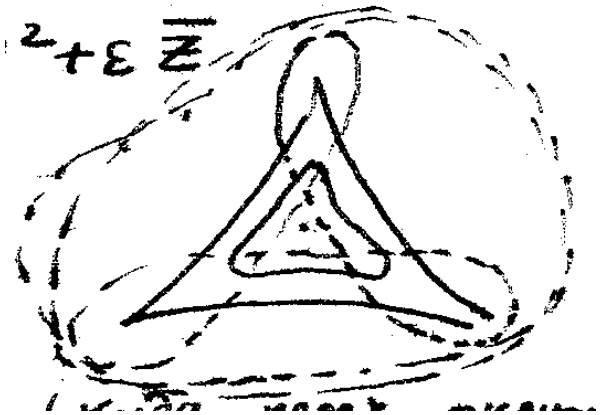


Original source details: GELFAND-P000175

Typeset diagram

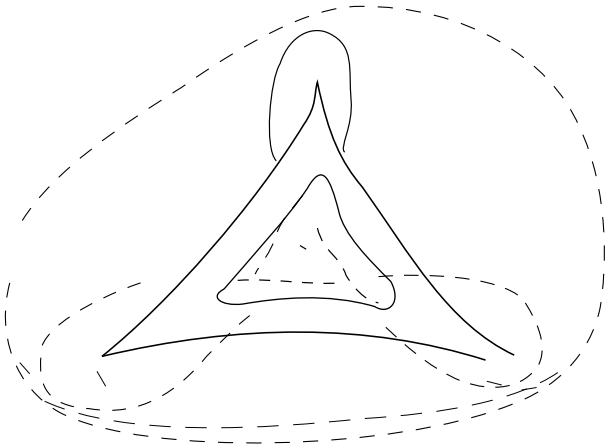


Original source details: GELFAND-P000175

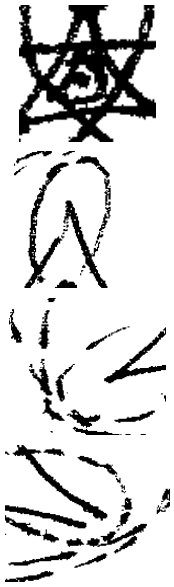


Original source details: GELFAND-P000175

Typeset diagram



Uncertain reading. Three nested circles with a central point and three points on the middle circle; callout “Sing.”. Beside them are three loop-like petals, three heavy points, connecting strokes and a small central contour. At right are concave triangular contours, an upper loop, an outer dashed arc, lower dashed loops and short oblique strokes. Exact connectivity remains uncertain at the dense crossings.

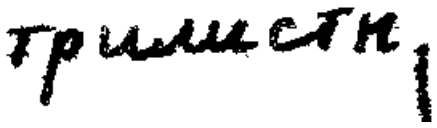


Original source details: GELFAND-P000175

(Where did the circles go?)

Large circles make 2 turns (trefoil [?] knot)

Большие окр. делают 2 оборота (трилистн, узел)



Original source details: GELFAND-P000175

Uncertain reading. “Trefoil” is abbreviated with a fused final letter; a separate descending stroke follows its dot. The number of turns is written 2.

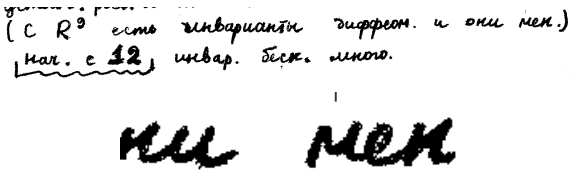
Up to dim. 5-6 there is a classific.

Theorem 1 (Thom)

For maps $R^9 \to R^9$ (and higher), everywhere density of stable germs does not hold.

(from R^9 there are diffeom. invariants and they change)

Starting from 12, there are infinitely many invariants.



Manuscript notation. The parentheses read “diffeom. invariants and they change.” “Starting from 12” has a wavy underline with raised ends.

Original source details: GELFAND-P000175

Thom:

Theorem 2.

There exists a discrete topol. classific. exists!!!

(What it looks like — is unknown)

(If from the space of all maps one discards A_∞ — infinite codim.

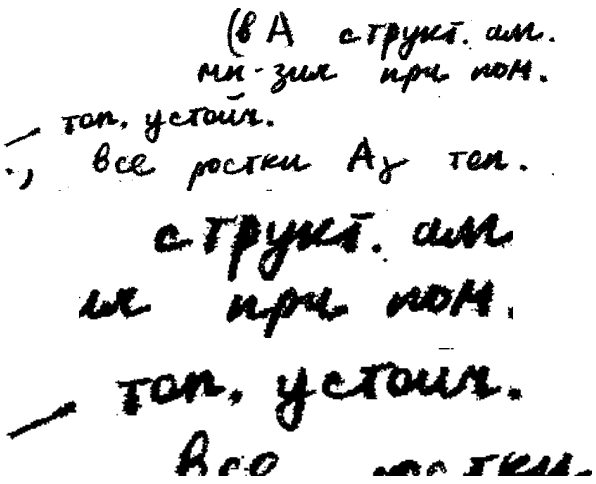
$$A = \bigcup_{\gamma \in \Gamma} A_\gamma + A_\infty.$$

Γ — a countable set.

Each A_γ — finite codim.; all germs of A_γ are topol. equiv. to one another, for different A_γ inequivalent.

(in A jet [?] alg. varieties with topol. [?]

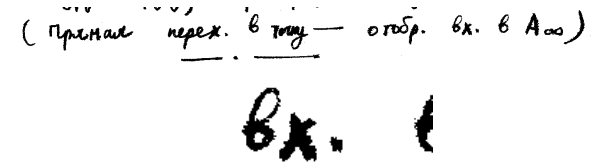
topol. stable



Uncertain reading. The right insertion has “jet” / “struct.” and “topol.” / “that”; an opening parenthesis. Below it, separately, “topol. stable” with an oblique leader from the finite-codimension line; the exact attachment of the insertion is not established.

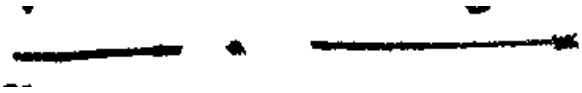
Original source details: GELFAND-P000175

(A line maps to a point — a map ent. [?] in A_∞)



Uncertain reading. The Russian contraction is “ent.” / joined “vl.”-like strokes; “maps” and “to a point” are separately underlined.

Original source details: GELFAND-P000175

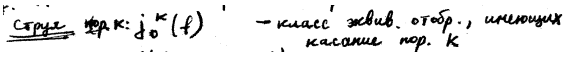


Original source details: GELFAND-P000175

Diagram and typed description. Two strokes with a dot between them precede the definition of a jet.

$f : M^m \to N^n$ a map germ.

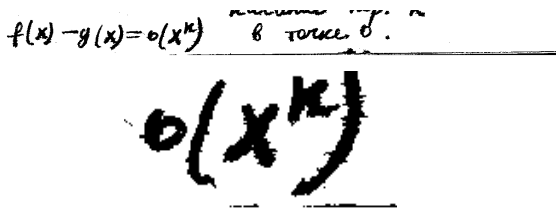
Jet of ord. [?] k: $j_0^k(f)$ — the equivalence class of maps having contact of ord. k



Uncertain reading. "Jet" is doubly underlined. The first Russian "ord." has superposed strokes; the next line has "contact of ord. k".

Original source details: GELFAND-P000175

$$f(x) - g(x) = o(x^k) \quad \text{at the point } 0.$$



Uncertain reading. The remainder sign is a small oval o / O; its argument is x^k without absolute-value bars. The line lies at the lower edge of the frame.

Original source details: GELFAND-P000175

GELFAND-P000176

Jet of order 0 — the image of a point
„ 1 — the image and its differential.

$$j_0^k(f) \quad \text{defined invariantly} \quad \text{(def. invar.)}$$

Jets of fixed ord. — an alg. variety (in coordinates — — Taylor series). The bundle of jets of maps from M to N

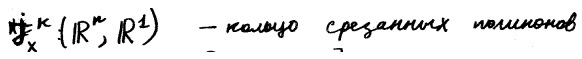
$$J^k(M, N) \quad \text{— a bundle over } M.$$

For $N = R^n$ this is a vector bundle.

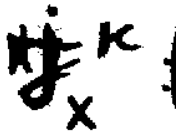
(With natural topol., diff. and alg. structure)

$$R^n \longrightarrow R^1.$$

$J_x^k(R^n, R^1)$ [?] — the ring of truncated polynomials



Uncertain reading. Initial letter J / j with dense transverse strokes through its vertical part; subscript x and superscript k are separately visible.

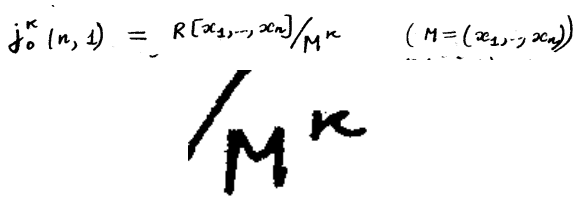


Original source details: GELFAND-P000176

$$R[x_1, \ldots, x_n]$$

$$j_x^k(f) = j_x^k(n,1)$$
$$(f : R^n \to R^1).$$

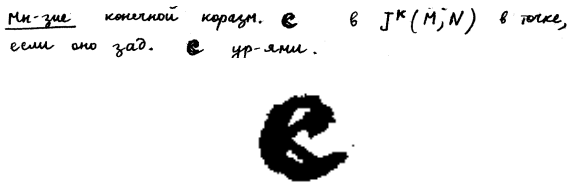
$$j_0^k(n,1) = R[x_1,\dots,x_n]/M^k \qquad (M = (x_1,\dots,x_n)).$$



Uncertain reading. The denominator visibly has M^k; no superscript $+1$ is visible. The M has a long left leg; short neighboring strokes remain unexpanded.

Original source details: GELFAND-P000176

A variety of finite codim. e [?] in $J^k(M,N)$ at a point, if it is given by e [?] equations.



Uncertain reading. The letter for codimension and number of equations is heavily retraced: e / an ε-like form. "Variety" is underlined.

Original source details: GELFAND-P000176

Natural map:

$$f \xrightarrow{j} j_x^k(f).$$

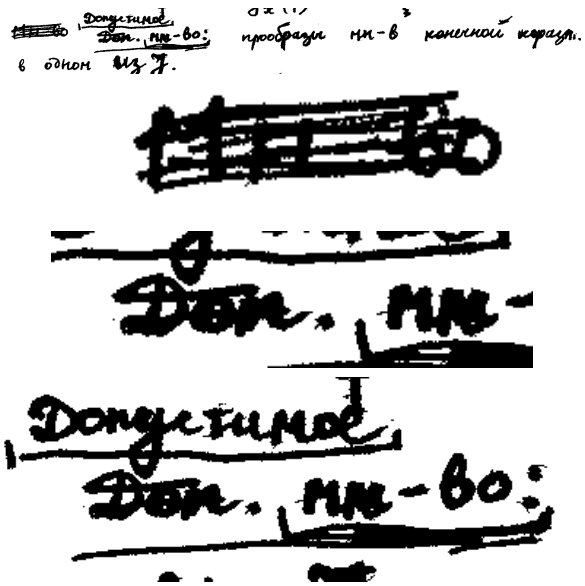
ξ [?]

Uncertain reading. At right a separate vertical three-lobed sign: ξ / flourish; no connection to the formula is shown.

Original source details: GELFAND-P000176

Admissible

[Cancelled: illegible] [Cancelled: ~~Adm.~~ [?]] set: inverse images of sets of finite codim. in one of the J.

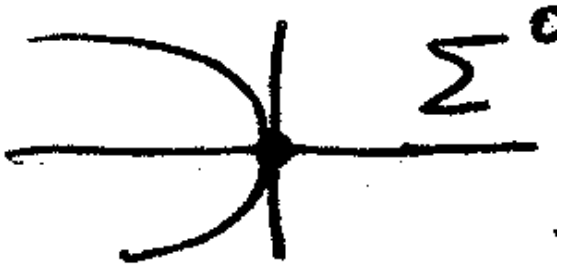


Undeciphered passage. The cancelled short word at left is unrecovered. In the next word, “Adm.” / “This”-like superposed strokes; underlined “Admissible” inserted above, followed by underlined “set:”.

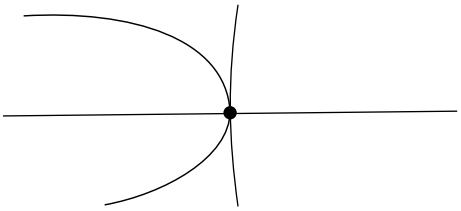
Original source details: GELFAND-P000176

Consider $f : M^m \longrightarrow N^n$.

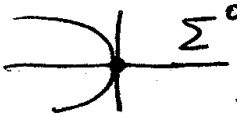
$$x \in \Sigma^i(f), \quad \text{if}$$
$$\dim \operatorname{Ker} df\big|_x = i.$$



Typeset diagram



Original source details: GELFAND-P000176



Uncertain reading. Horizontal and vertical lines without arrows or letters; a heavy point near the crossing, a rounded curve with two leftward branches. Exact contact inside the heavy point remains uncertain.

Original source details: GELFAND-P000176

Σ^0 — the whole pl. [?] without the parab.

Uncertain reading. After “the whole,” the abbreviation appears to mean “plane.” Its final strokes are joined.

Original source details: GELFAND-P000176

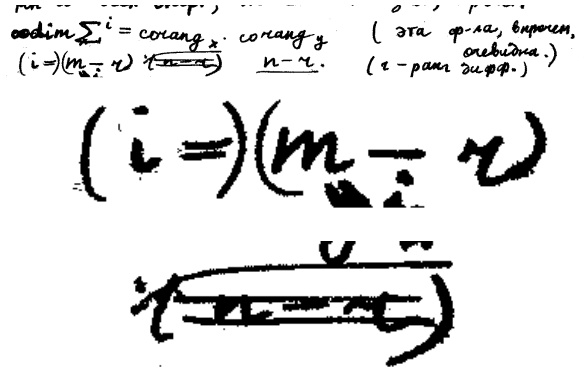
Σ^1 — the whole parabola.

Theorem.

The set of all maps where Σ^i are varieties is dense, and

$$\operatorname{codim} \Sigma^i = \underline{\operatorname{corang}_x} \cdot \operatorname{corang}_y .$$

$$\underline{(i =)(m - r)} \qquad \underline{n - r}.$$



Uncertain reading. The upper formula has a dot between coranks; the first corank is underlined. Below, separately, "$(i =)(m - r)$", a heavily crossed connecting stroke with a right arc, and underlined $n - r$. No clear product sign is recovered between the lower factors; two short heavy strokes lie under m at the edge.

Original source details: GELFAND-P000176

(r — rank of the diff.)

(This formula is, however, obvious.)

GELFAND-P000177

[source number: 37]

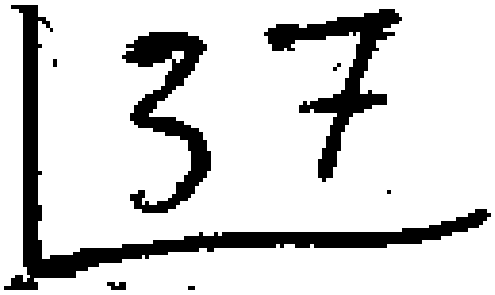


Diagram and typed description. Page number 37 with a line at its left and below it.

37

Original source details: GELFAND-P000177

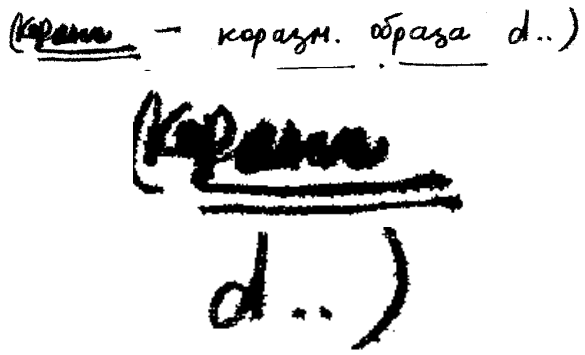
$z \longrightarrow z^2$ is already unstable.

The conditions on Σ^i — algebraic subsets. In J — the required codimensions (minors $= 0$).

In $J^1(M, N)$ the varieties $\Sigma^1, \Sigma^2, \ldots$ are defined only by germs.

An actual map $\Rightarrow$ a section.

(Corank [?] — codim. of the image d..)

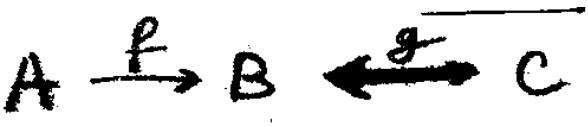


Uncertain reading. The opening "Corank" / "Codim." is heavily overwritten and doubly underlined. Two small separate strokes follow d; they are not expanded into a map name.

Original source details: GELFAND-P000177

Original source details: GELFAND-P000177

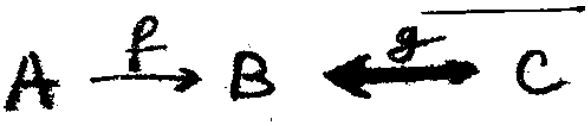
Diagram and typed description. Two strokes with a dot between them precede the map diagram.



Original source details: GELFAND-P000177

Typeset diagram

$$A \xrightarrow{f} B \xleftarrow{g} C$$



Original source details: GELFAND-P000177

Uncertain reading. At left is $A \to B$ with f. At right is a heavy line with g and two visible points, the left one larger.

f, g are transverse if $\forall a \in f^{-1}(b), c \in g^{-1}(b)$
at the point $b \in B$

Original source details: GELFAND-P000177

Manuscript notation. Below “transverse,” the manuscript says “at the point $b \in B$.”

the relation holds:

$$f_* TA_a + g_* TC_c = TB_b \qquad \text{(may intersect)}.$$

2 lines in R^3 are not transverse at even 1 point (if they intersect) (they may intersect).

If a map is transverse to a subvariety, (g — the embedding $C \subset B$), then the inverse image of the variety — is a variety.

Its Codim is the same

$$\operatorname{codim} f^{-1}C\big|_A = \operatorname{codim} C\big|_B.$$

If $f : M \longrightarrow N$ is transverse to Σ^i, then all is well.

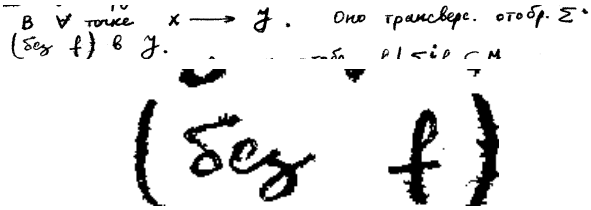
One can perturb $\forall$ map.

Further classific.

$$\Sigma^i f \subset M, \qquad f : M \longrightarrow N.$$

Suppose that the section defined by the map f is transverse to Σ^i in the jets.

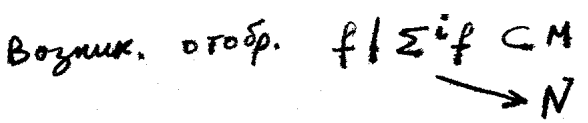
At $\forall$ point $x \longrightarrow J$. It is transverse to the map Σ^i (without f) [?] in J.



Original source details: GELFAND-P000177

Uncertain reading. After “without f”, a closing sign with several bends: a round parenthesis / curled stroke. Both letters are J; the phrase reads “map Σ^i”.

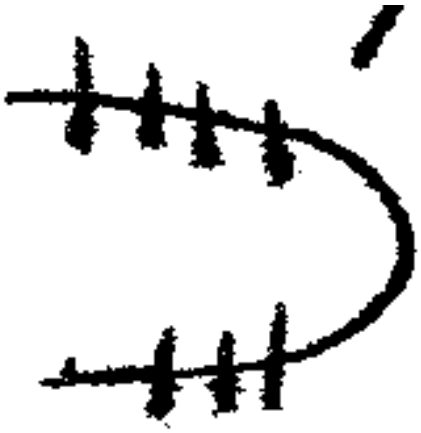
There arises a map



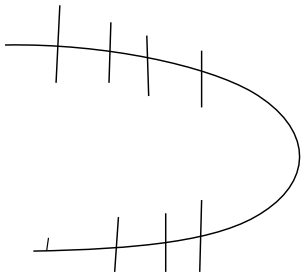
Original source details: GELFAND-P000177

Typeset diagram

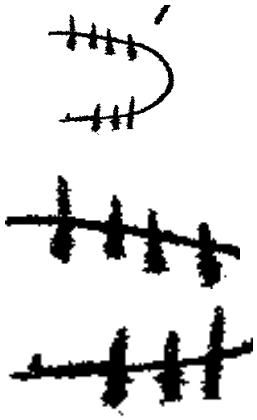
$$f|\Sigma^i f \subset M \longrightarrow N$$



Typeset diagram



Original source details: GELFAND-P000177



Uncertain reading. A rounded curve with a right vertex and two leftward branches, without axes or labels. Four long strokes on the upper branch; three long and one short stroke near the left edge on the lower branch. Exact stroke lengths and contacts remain uncertain.

Original source details: GELFAND-P000177

GELFAND-P000178

The same classific. is applied to it:

$$\Sigma^j f | \Sigma^i f$$

$x \in \Sigma^{ij}(f)$ — a subvariety (Boardman 1965).

(One can perturb) $\forall$ numbers of indices.

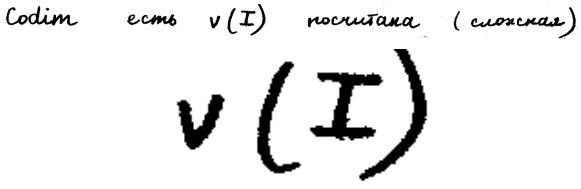
Thm.

1. $\forall k$ in $J^k(M,N)$ algebraic subvarieties are defined

$$\Sigma^I \qquad I = (i_1, \ldots, i_k)$$

(independent of f.)

The codimension $\nu(I)$ [?] has been calculated (complicated).



Uncertain reading. The function sign is ν / a v-like form, followed by argument I in parentheses. “Codim” is in Latin letters.

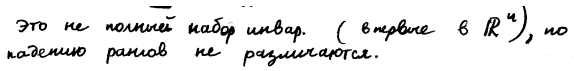
Original source details: GELFAND-P000178

2. Every map is approximated by a map transverse to all these

$$\Sigma^I(f).$$

Classific. Σ^{110} and so on.

This is not a complete set of invariants (first in R^4), they cannot be distinguished by rank drops.



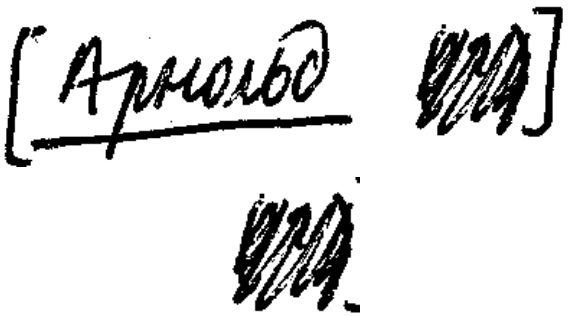
Manuscript notation. The words after the comma read “by rank drops”.

Original source details: GELFAND-P000178

GELFAND-P000179

28/XI. Sem. Gelfand.

[Arnold [Cancelled: UR [?]]]



Uncertain reading. After the surname, inside square brackets, two or three heavily cancelled signs: UR-like / Ch-like superposed strokes. The word is not established.

Original source details: GELFAND-P000179

$$f : M^m \longrightarrow N^n.$$

$$\Sigma^i(f) = \{x \in M^m : \dim \ker df_x = i\} \quad \text{— a subset.}$$

if it is a variety, then

$$\Sigma^{ij}(f) = \Sigma^j\Sigma^i(f) = \Sigma^j(f|\Sigma^i f) \quad \text{— Thom's naive def..}$$

(forget)

For every multi-index $i_1, i_2, \dots, i_k = I$ one defines

$$\Sigma^I(f),$$

if varieties are obtained at every step.

There must be

$$i_1 \geq i_2 \geq \dots$$

$$K_1 = \text{Ker}\, df_x, \qquad K_2 \subset K_1 \quad \text{— next kernel on the subsequent}$$

subvariety.

There is a subset independent of the map $\Sigma^I \subset J^k(M, N)$ (is constructed)

A — the ring of functions on a variety with coord. system $x_1, \ldots, x_m$
$B \longrightarrow$ an ideal in A.

The new ideal — the Jacobian extension of B of ord. k $\Delta_k(B)$ — — to the functions from $B \ni \{f\}$ adjoin the minors formed from them

of order k $\left|\frac{\partial f_i}{\partial x_j}\right|$ and adjoin to the ideal.

[Cancelled: ~~This~~ [?]] They are what generate $\Delta_k(B)$.

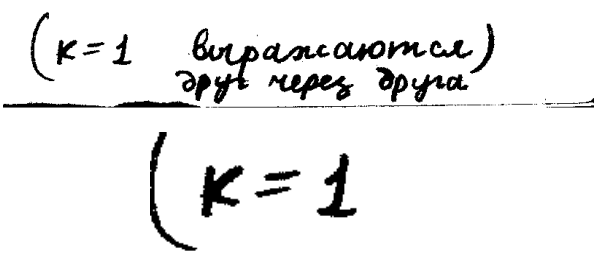
Uncertain reading. The cancelled strokes allow either “This…” or “These…”. The remainder is illegible.

Original source details: GELFAND-P000179

This does not depend on coordinates (in fact an invar. constr.)

$$\frac{\partial f}{\partial x_i'} = \sum_j \frac{\partial f}{\partial x_j} \cdot \frac{\partial x_j}{\partial x_i'}.$$

($k = 1$ they are expressed) in terms of one another



Manuscript notation. The k matches the order symbol above. “In terms of one another” is written below the line.

Original source details: GELFAND-P000179

GELFAND-P000180

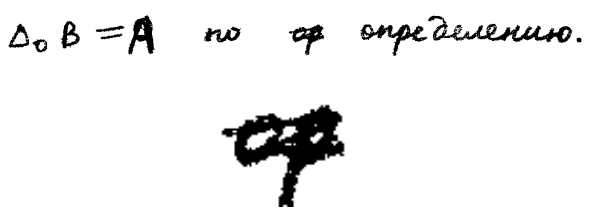
Relation of extensions of different orders.

$k \nearrow \ldots$ the ideal $\searrow \ldots$ (decreases)

The extension $\Delta_k(B)$ is critical if k is max for which the ideal is not yet all of B.

$$\Delta_{k-1}B = A, \qquad \Delta_k B \subset A$$

$\Delta_0 B = A$ by [cancelled: illegible] definition.



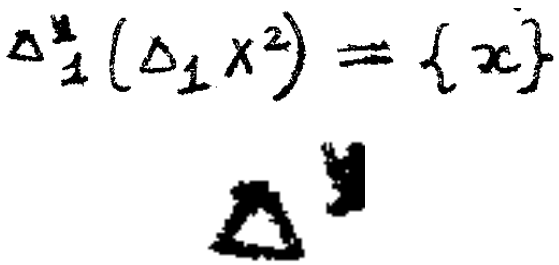
Uncertain reading. After "by" there is a loop, a cross-stroke and a long downstroke, as in a heavily cancelled abbreviation for "definition". The reading is provisional.

Original source details: GELFAND-P000180

Example: the ideal generated by x^3.

$k = 1$ $\quad \Delta_1 = \{x^2\}$ $\quad$ critical 1st

$$\Delta_1^2(\Delta_1 x^2) = \{x\}$$



Uncertain reading. The superscript on the first Δ_1 is heavily overwritten. The reading 2 is provisional. The clear exponent on x^2 is visible nearby.

Original source details: GELFAND-P000180

The extension is performed several times.

Definition: $\Delta^k = \Delta_{m-k+1}$ $\quad (f : M^m \longrightarrow N^n)$

Def. of Boardman:

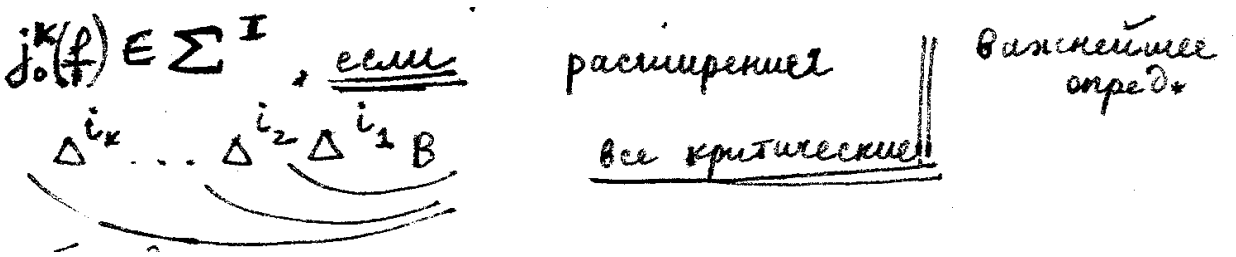
Consider a map f:

$$y_1 = f_1(x_1, \ldots, x_m)$$
$$\ldots$$
$$y_n = f_n(x_1, \ldots, x_m), \qquad f(0) = 0$$

The ring of germs of smooth functions at 0.

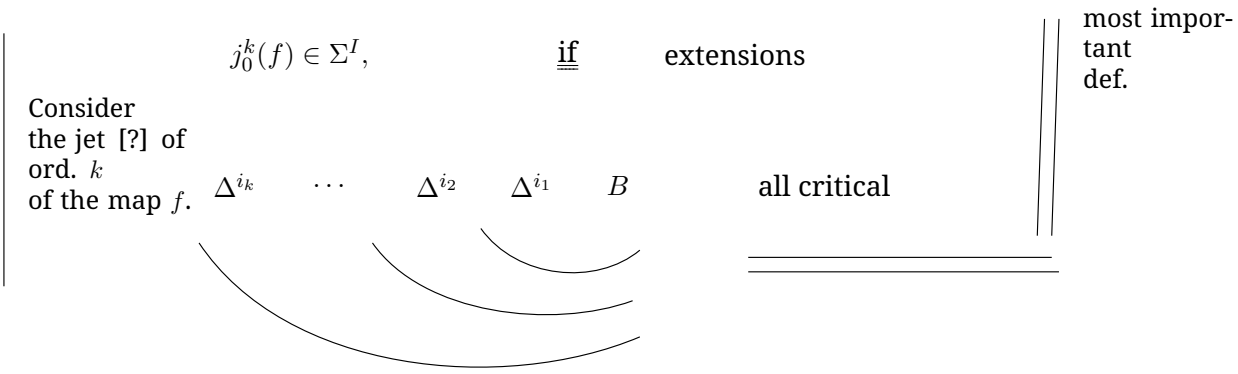
Consider the ideal generated by $f_1, \ldots, f_n$, $\quad B = \{f_i\}$

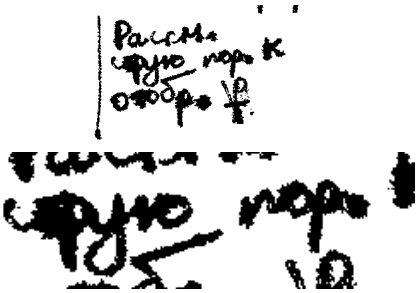
In form the def. depends on coordinates, but in fact it does not.



Original source details: GELFAND-P000180

Typeset diagram





Original source details: GELFAND-P000180

Uncertain reading. The insertion at left reads "Consider the jet of ord. k of the map f": Russian "struyu" / "struya"-like joined and reinforced strokes; the final letter and the beginning of the abbreviation for "map" are not fully separate. A vertical line separates the insertion.

It does not depend on the map f, but only on the jet of ord. k.

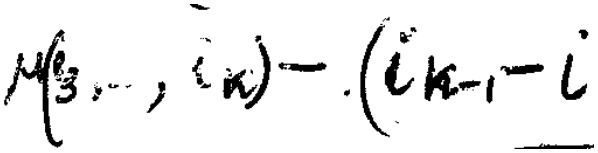
The ideal does not depend on X.

Nor does it depend on Y (needs checking).

Main theorem.

This Σ^I is an algebraic variety in the space of jets, and its codimensions have been calculated (functions $= 0$ and $\neq 0$, polynom.)

$$c(I) = (m - n + i_1)\mu(i_1, i_2, \ldots, i_k) - (i_1 - i_2)\mu(i_2, \ldots, i_k) -$$
$$-(i_2 - i_3)\mu(i_3, \ldots, i_k) - .(i_{k-1} - i_k)\mu(i_k)$$



Original source details: GELFAND-P000180

Uncertain reading. One low dot is visible between the minus sign and the final parenthesis. It may abbreviate an ellipsis.

Original source details: GELFAND-P000180

Diagram and typed description.

Three nearly horizontal traces lie below the formula. A short thick trace is at left and two thinner, offset traces are at right. Both lower traces end on the last scan row with dark pixels. They may be manuscript strokes or traces at the edge of the scan. Their purpose remains uncertain.

GELFAND-P000181

[source number: 42]

Original source details: GELFAND-P000181

Diagram and typed description. Page number 42 with a line at its left and below it.

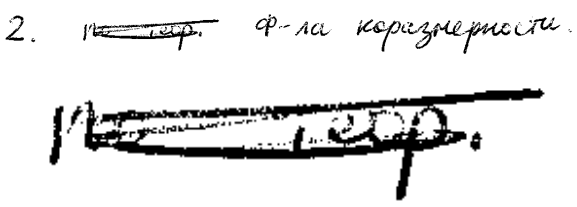
42

$\mu(i_1, \ldots, i_r)$ is the number of seq. $j_1, \ldots, j_r$ such that

$$1)\ j_1 \geq \ldots \geq j_r$$
$$2)\ i_s \geq j_s \geq 0 \quad \text{for all } 1 \leq s \leq r.$$

Theorem of Boardman

1. Σ^I is a variety in the space of jets.
2. [Cancelled: illegible] codimension formula.

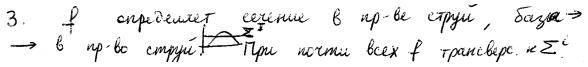


Original source details: GELFAND-P000181

Uncertain reading. Before “codimension formula” several joined letters with a downward stroke are cancelled. The literal cancelled reading is undetermined.

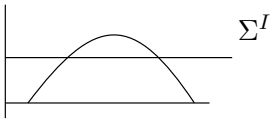
3. f determines a section in the space of jets, base $\longrightarrow$

$\longrightarrow$ into the space of jets



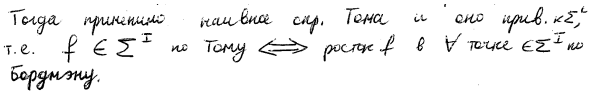
Original source details: GELFAND-P000181

Typeset diagram

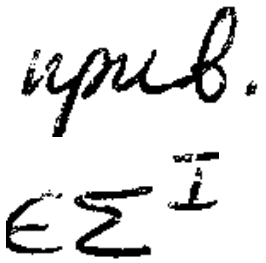


For almost all f, transv. to Σ^i

Then Thom's naive def. is applicable, and it leads to Σ^i; i.e., $f \in \Sigma^I$ in Thom's sense $\Longleftrightarrow$ the germ of f at $\forall$ point $\in \Sigma^I$ in Boardman's sense.



Manuscript notation. The first line reads "и оно прив." ("and it leads"), with the verb abbreviated.



Original source details: GELFAND-P000181

Almost all are transv. by the general theorem.

Original source details: GELFAND-P000181

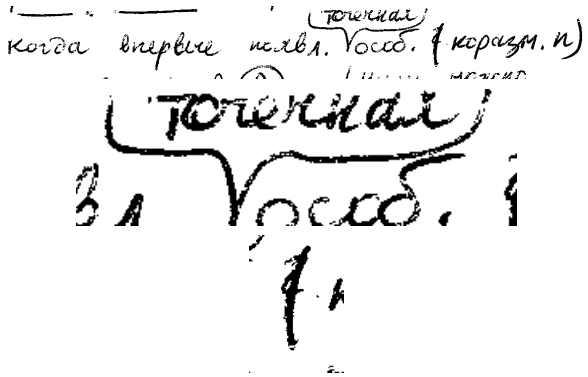
Diagram and typed description. Two strokes with a dot between them follow the general theorem.

——— . ———

$\underline{R^n \longrightarrow R^n}$.

isolated

When do sing. $\{$codim. $n)$ first appear



Original source details: GELFAND-P000181

Uncertain reading. "Точечная" ("isolated") is written above the abbreviation for "singularity". The mark before "codim." is a brace or an overwritten parenthesis. The right mark is a parenthesis.

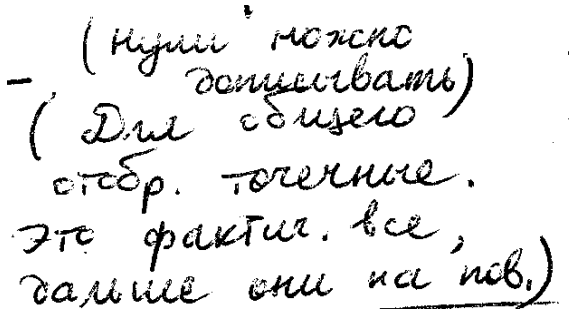
n	1	2	3	4	5	6	7	8	(9)
I	1	11	1_3	1_4	1_5	1_6	1_7	1_8	1_9
				2			21		3

x^2 $\quad$ Σ^{11}

(cusp)

(Need [?] can write out more)

(For a general map, isolated. These are in fact all, after that they are on surfaces.)

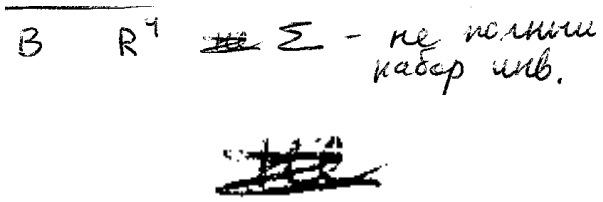


Uncertain reading. The first word in parentheses looks like an abbreviation for “need”. The second note ends “на пов.”, provisionally read as “on surfaces”.

Original source details: GELFAND-P000181

10
1_{10}
$2, 2$
$2, 1, 1$

In R^4 [cancelled: illegible] Σ is not a complete set of inv.



Uncertain reading. Between R^4 and Σ a small sign is densely cancelled; the literal reading is undetermined. The source words for “not complete” are written separately.

Original source details: GELFAND-P000181

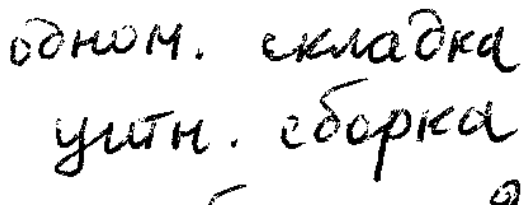
In dim. 16, sing.: 4 of them.

$n = 9$:

on M^8 — one-dim. fold

M^7 — Whitney [?] cusp

M^5 — sing. of type 2.



Uncertain reading. The abbreviation before “cusp” looks like “уитн.” (“Whitney”). The first letter has a lower loop.

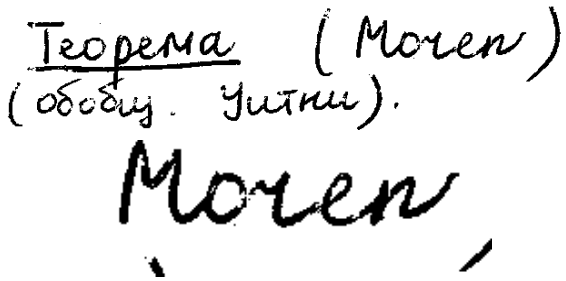
Original source details: GELFAND-P000181

The table is easily comp. using the combinat. formula.

1_9: the rank drops by 1 on M^8, then by another 1 on M^7.

GELFAND-P000182

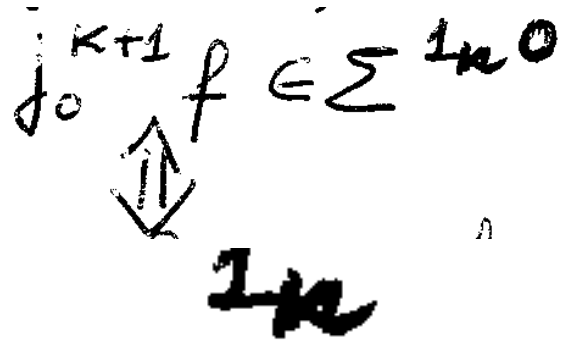
Theorem (Moren [?])
(generaliz. of Whitney).



Original source details: GELFAND-P000182

Uncertain reading. The parenthesized name: “Moren” / “Morin” / Cyrillic “Moren”; the joined third letter and following vowel are not unambiguous. “Theorem” is underlined; below, “(generaliz. of Whitney).”.

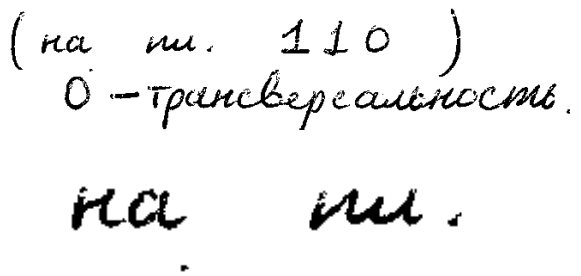
$$j_0^{k+1} f \in \Sigma^{1_k 0}$$
$$\Updownarrow$$



Original source details: GELFAND-P000182

Uncertain reading. The small subscript on 1 resembles k. The jet order at left also has k. The final 0 is separate.

(e.g. [?] 110)
0: transversality.



Original source details: GELFAND-P000182

Uncertain reading. In the parenthesis before 110, the provisional reading is “на пр.” (“e.g.”). The line below reads “0: transversality”.

$\exists$ coordinates in which

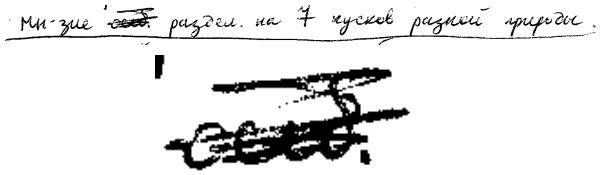
$$x_1 = y_1$$
$$\dots$$
$$x_{n-1} = y_{n-1}$$
$$y_n = x_1 x_n + x_2 x_n^2 + \dots + x_{n-1} x_n^{n-1} + x_n^{n+1}$$

(there, a cubic straight away)
(For classes 1_n, sing. det. up to diffeom.)
For type 20 the class is not a complete inv.

(up to diffeom.)

$$\Sigma^{20} \quad \text{codim. } 4$$
$$f : R^4 \longrightarrow R^4$$

The variety [cancelled: sing.?] is divided into 7 pieces of different nature.



Original source details: GELFAND-P000182

Uncertain reading. After “The variety” a cancelled contraction resembles “sing.” / another cancelled inscription. The joined letters under the strokes do not give a certain full reading. The entire line is underlined.

Quadratic differential of a map.

$f : M^m \longrightarrow N^n$

(In J^k, two diffeomorphism groups of M^m and N^n —

— orbits arise. (at singular points, what orbits))

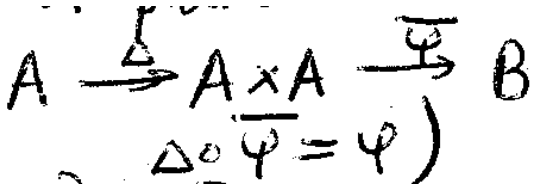
$$f_x : TM_x \longrightarrow TN_{f(x)}$$

There are $Ker\, f_x \subset TM_x$, $\quad Coker\, f_x = TN_{f(x)}/Im\, f_x$.

$$f_{xx} : Ker\, f_x \longrightarrow Coker\, f_x$$

(a quadratic map, i.e., $\psi : A \longrightarrow B$

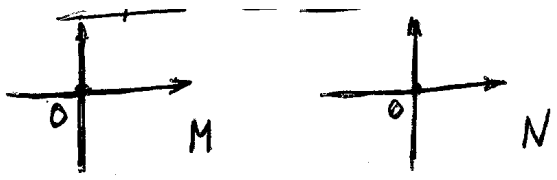
$$A \xrightarrow{\Delta} A \times A \xrightarrow{\overline{\psi}} B$$
$$\Delta \circ \overline{\psi} = \psi)$$



Original source details: GELFAND-P000182

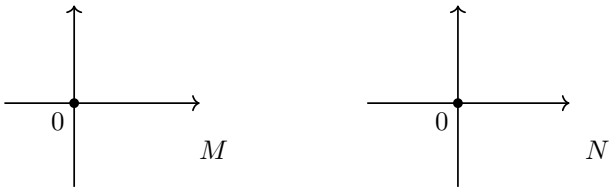
Manuscript notation. The lower line reads $\Delta \circ \overline{\psi} = \psi$.

Def. using a system of coordinates.



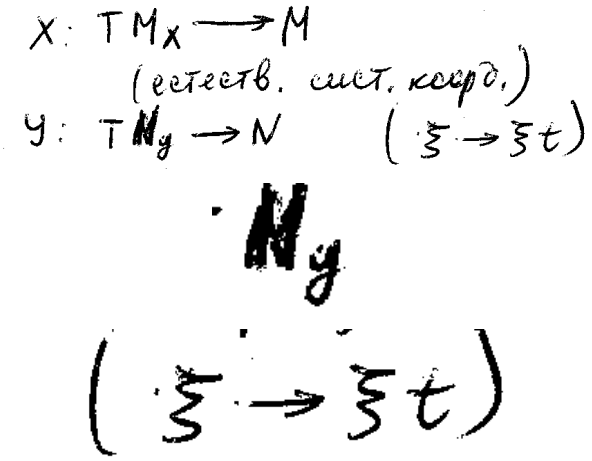
Original source details: GELFAND-P000182

Typeset diagram



$X : TM_x \longrightarrow M$ (natural coord. syst.)

$y : TN_y\,[?] \longrightarrow N \qquad (\xi \longrightarrow \xi t)$



Uncertain reading. The first letter X is capital, the second y lowercase. In TN_y, the N is densely overwritten: N / an M-like lower stroke; the exact underlying form is undetermined. At right, $\xi \longrightarrow \xi t$.

Original source details: GELFAND-P000182

GELFAND-P000183

[source number: 43]

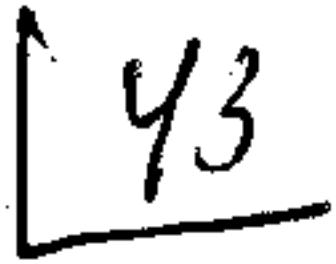


Diagram and typed description. Page number 43 with a line at its left and below it.

43

Original source details: GELFAND-P000183

$\xi \in Ker\ f_x$

$X\xi$: a point of the variety

$$\lim_{t\to 0} \frac{\left(y^{-1} f X\xi t / Im\ f_x\right)}{t^2}$$

It is easy to prove that this does not depend on the choice of coord. syst.

(The cubic diff. cannot be def. this way)

Look into this!

A bilinear map $f_{xx} \Longrightarrow$ a pencil of quadr. forms.

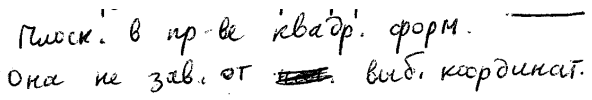
Let $\alpha \in \underbrace{Co\,Coker\, f_0}_{(Coker\, f_0)'} = C'$

αf_{00} is a quadr. form on the kernel.

(already a pure quadr. form)

A plane in the space of quadr. forms.

It does not depend on [cancelled: illegible] the choice of coordinates.



Uncertain reading. Before "the choice of coordinates" a short inscription is densely cancelled; the literal reading is undetermined.

Original source details: GELFAND-P000183

$$C' \longrightarrow \text{Qu.}_{Ker\, f_0}$$

Example. $\quad \Sigma^{20} : R^4 \longrightarrow R^4$

$$2 = dim\, Ker\, f_0,$$

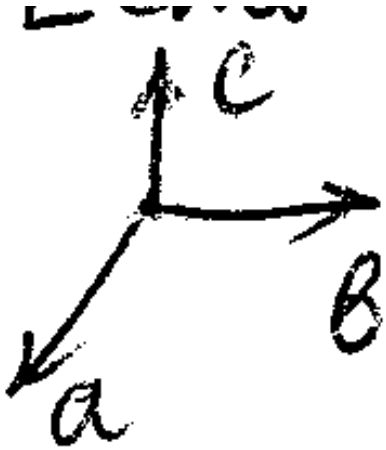
Hence $dim\, Coker\, f_0 = 2$

forms: a 3-dim. space

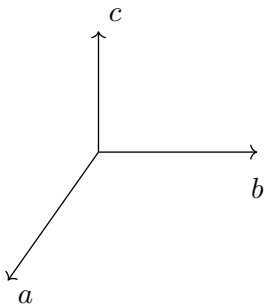
(in 2 vars.)

$$ax_1^2 + 2bx_1x_2 + cx_2^2.$$

There is a classific. by rank



Typeset diagram



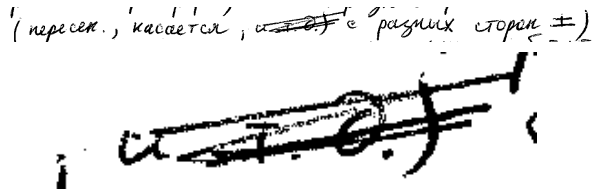
Original source details: GELFAND-P000183

$$\text{Rank} < 2 \quad \Longrightarrow \quad \text{cone } b^2 = ac$$

(inv. def.)

There is also a plane there (a map of a plane into the space of quadr. forms). It can have different positions

(intersects, is tangent, [cancelled: ~~etc.)~~ [?]] on different sides $\pm$)



Uncertain reading. Under the strokes are “and”, a short stroke with a dot, a loop and a right parenthesis. The provisional reading is “etc.)”.

Original source details: GELFAND-P000183

These positions (a total of 7 pos., including degeneracies), if different, will be different classes of maps.

And there are indeed 7 orbits.

GELFAND-P000184

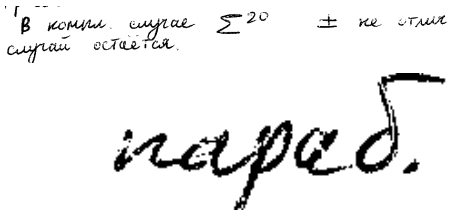
The maps can be written.

$$y_1 = x_1 x_2$$
$$y_2 = x_1^2 \pm x_2^2 + \underbrace{x_1 x_3 + x_2 x_4}_{\text{added for transversality.}}$$
$$y_3 = x_3$$
$$y_4 = x_4$$

———— · ————

In R^5 there can be a curve Σ^{20} with different sing. along the curve.

In the complex case, Σ^{20} $\pm$ do not differ, but the parabolic case remains.



Manuscript notation. The abbreviation after “but” reads параб. (“parabolic”).

Original source details: GELFAND-P000184

Σ^{20}: all sing. are stable, there is a discrete classific.

Manuscript notation. The word before “classific.” is the abbreviation for “discrete”.

Original source details: GELFAND-P000184

Σ^{30}: this is no longer so.

———— · ————

Thm. of Thom.

$$\Sigma^{30} : R^9 \longrightarrow R^9$$
$$dim\, Ker\, f_0 = 3$$
$$dim\, Coker\, f_0 = 3$$

the space of sing. is not discrete (there are moduli)

(Everything is still quadratic)

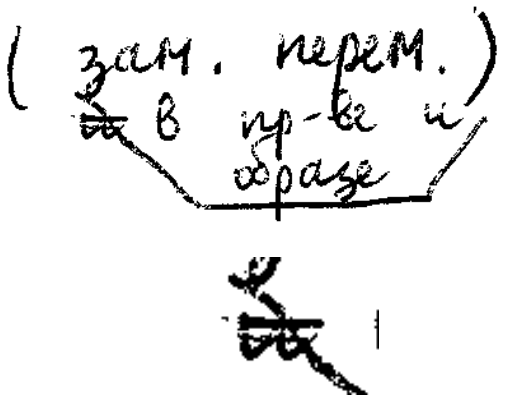
Lemma. For $n \geq 3$, the dimension of the orbit

$$GL(n,R) \times GL(n,R)$$

in $H(n,n)$: quadr. forms in n vars.

(change of vars.)

[cancelled: ?] in the space and image



Original source details: GELFAND-P000184

Uncertain reading. Beneath "(change of vars.)", before "in the space and image", a short sign with joined loops is cancelled. The original inscription is undetermined. A bent stroke connects the parenthesis and interlinear insertion.

$$codim \text{ orbits} > 0$$

(no orbits of full dim.)

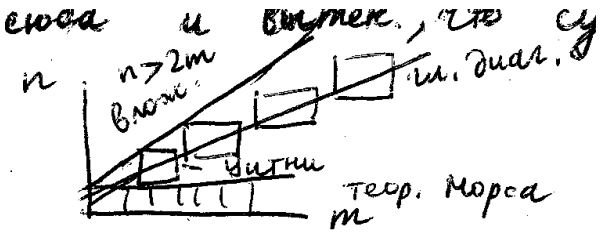
group $2n^2$, forms $\frac{n^2(n+1)}{2}$, and there is always

a one-dim. stationary subgroup ≥ 1

(mult. by a number there and there)

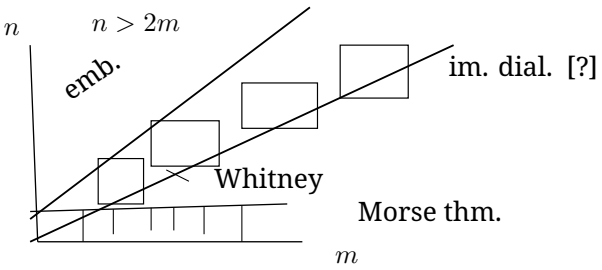
$n = 3$ and higher.

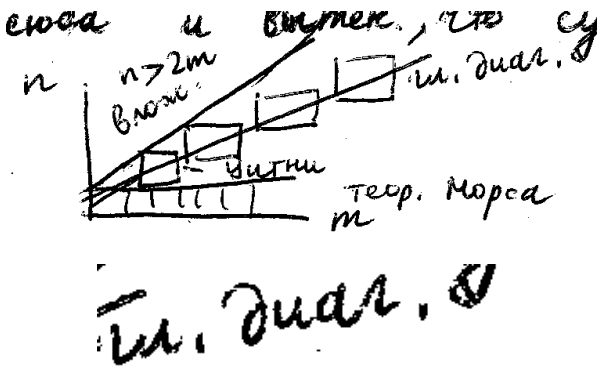
Hence it follows that moduli exist.



Original source details: GELFAND-P000184

Typeset diagram





Uncertain reading. At left a vertical axis n, below a horizontal m, two rising lines, four rectangular outlines, and a lower strip with six vertical strokes. The labels $n > 2m$, “emb.”, “Whitney”, and “Morse thm.” are legible; the small right label is “im. dial.” / “mil. diap.” / an undetermined contraction. Dense crossings at the origin and separate short strokes do not establish additional numerical boundaries.

Original source details: GELFAND-P000184

In cells $9, 9$ and $6, 9$
it has been determined where there are moduli
(*J.* Mather: announced)
He also: classific. of stable sing.

GELFAND-P000185

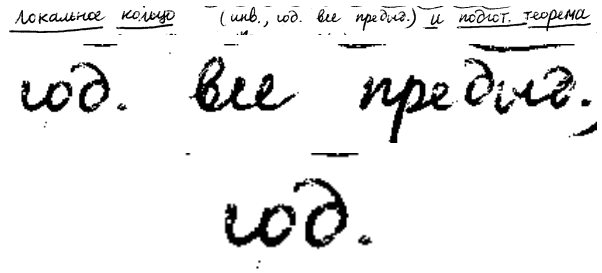
[source number: 44]

Diagram and typed description. Page number 44 with a line at its left and below it.

44

Original source details: GELFAND-P000185

The only inv. of stable sing. is the local ring.
Local ring (inv., cont. [?] all the preceding.) and prepar. theorem

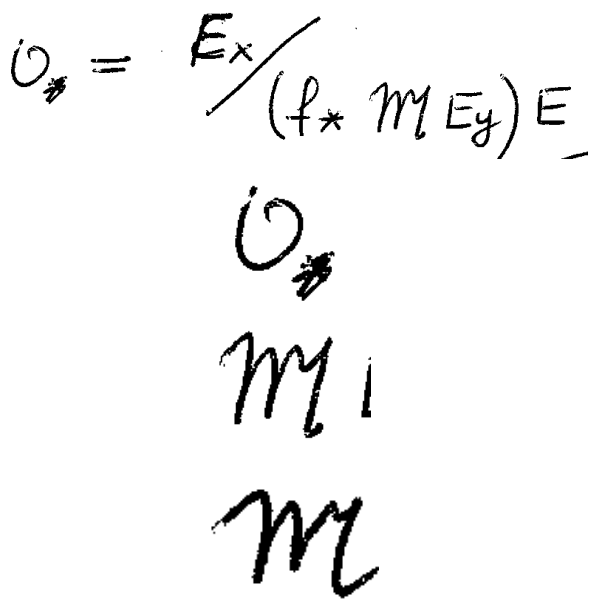


Uncertain reading. The abbreviation between “inv.” and “all the preceding” resembles сод. (possibly “contains”). The expansion remains uncertain.

Original source details: GELFAND-P000185

$$f : M^m \longrightarrow N^n, \qquad f(0) = 0.$$

$$O_* [?] = \frac{E_x}{(f_* \mathfrak{M} E_y) E}$$



Uncertain reading. At left, O has a heavily overwritten subscript, provisionally rendered as $*$. The lower stroke of $\mathfrak{M}$ recurs in the line about the maximal ideal. A separate E follows the parenthesis.

Original source details: GELFAND-P000185

E_x: the ring of germs of infinitely diff. functions on M at 0.

E_y: similarly, in y

$\mathfrak{M} E_y$: the max ideal.

f_* is def. naturally.

Meaning: "the ring of functions on the coalesced inverse image."

Example. $f(x) = x^2$,

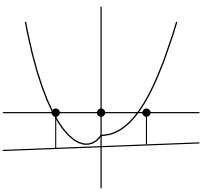
$$\mathfrak{M} E_y = y E_y = a_1 y + \dots$$
$$f_* \mathfrak{M} E_y = a_1 x^2 + a_2 x^4 + \dots$$

Mult. by E: add everything above the 2nd.

$a_1 x^2 + \dots$

Take the quotient: $a + bx$

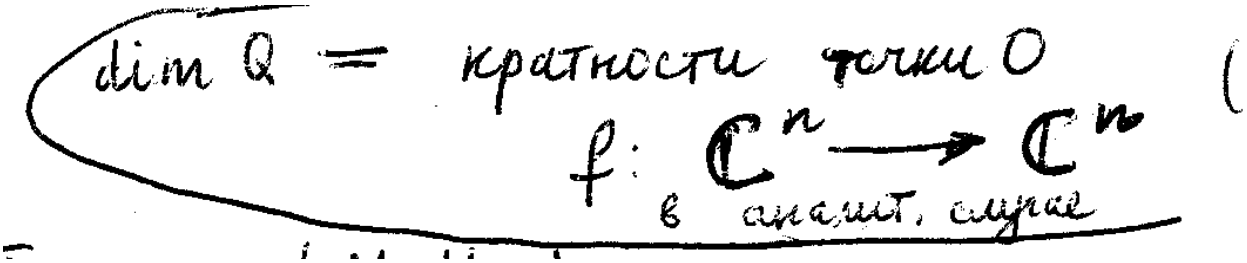
Typeset diagram



Original source details: GELFAND-P000185

the ring of functions on 2 coalesced points (an infinites. pair of points)

Let us move: it will remain so, and let $y \longrightarrow 0$.



Original source details: GELFAND-P000185

Typeset diagram

$dim\, Q =$ multiplicity of the point 0

$$f : C^n \longrightarrow C^m$$

in the analytic case

(in the general case)

Apparently so.

Theorem: (Mather)

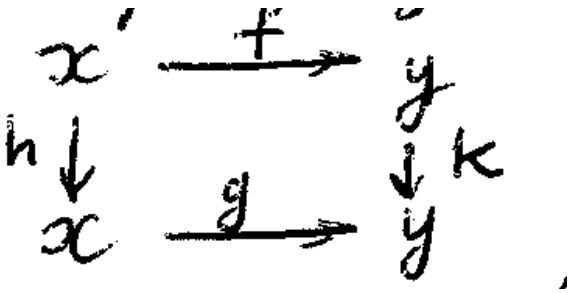
Stable sing. are equiv. $\Longleftrightarrow$ their local rings coincide.

Another theorem. (Mather)

a sing. is stable $\Longleftrightarrow$ it is infinitesim. stable.

The latter means:

At the point: $y = f(x)$

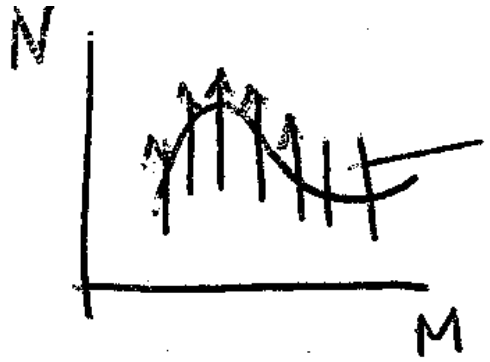


Typeset diagram

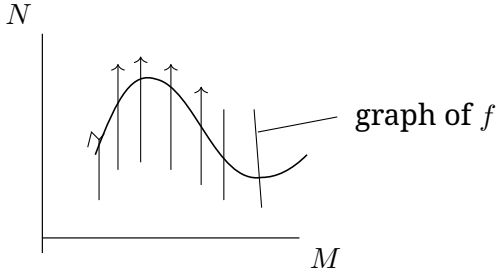
$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ x & \xrightarrow{g} & y \end{array}$$

Original source details: GELFAND-P000185

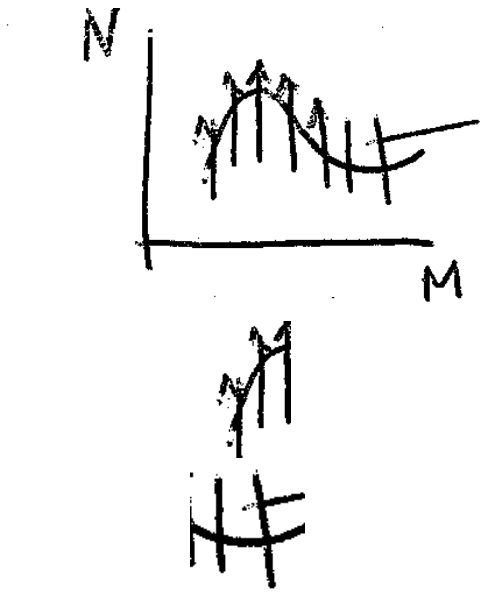
h and k must exist infinitesimally.



Typeset diagram



Original source details: GELFAND-P000185



Uncertain reading. Axes N upward and M rightward; a wavy graph with a maximum and minimum, and seven crossing segments. The four middle segments have distinguishable upper arrowheads; the densely bent far-left stroke does not give a certain direction. Separate arrowheads on the two right segments are undetermined. At right, an oblique stroke toward "graph of f"; exact attachments of dense strokes remain uncertain.

Original source details: GELFAND-P000185

A section of the bundle, parallel to N.

This is such a vect. field (a vertical vect. field).

GELFAND-P000186

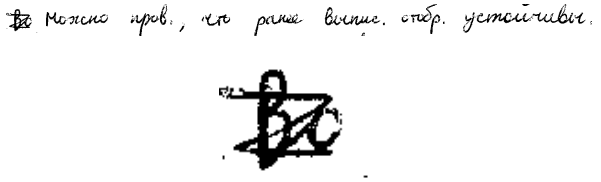
Every vertical field φ must be representable as a sum.

$$\varphi = f_*\xi + f^*\eta$$

(transfer)

$$\overset{\uparrow}{f_*}, \quad \overrightarrow{f^*}$$

[Cancelled: illegible] One can check that the maps written earlier are stable.



Uncertain reading. Before "One can check", a short sign / contraction with a high upper loop is cancelled; the literal inscription is undetermined.

Original source details: GELFAND-P000186

Weierstrass prepar. thm.

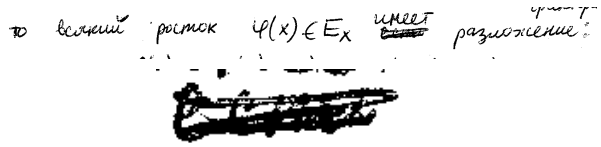
(Paris version)

If Q is finite-dimensional $(dim\, Q = r)$ and has generators

$$e_1(x), \ldots, e_r(x)$$

(functions that become generators after quotienting)

has
then every germ $\varphi(x) \in E_x$ [cancelled: ?] an expansion:



Uncertain reading. Beneath the raised insertion “has”, a joined word with an initial loop and lower teeth is cancelled. The literal reading is undetermined; then follows “an expansion:”.

Original source details: GELFAND-P000186

$$\varphi(x) = \psi_1(y)e_1(x) + \ldots + \psi_r(y)e_r(x)$$

$dim\, Q$: the correct def. of multiplicity.

Theorem (Mather)

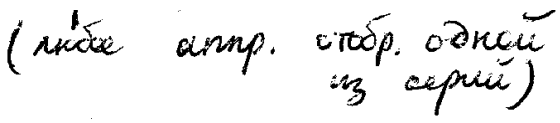
Σ^{20} is divided into parts.

1°. Five series of rings Q

2°. Remainder $codim = \infty$

$f : M^m \longrightarrow N^n$

(or [?] approx. by a map of one of the series)

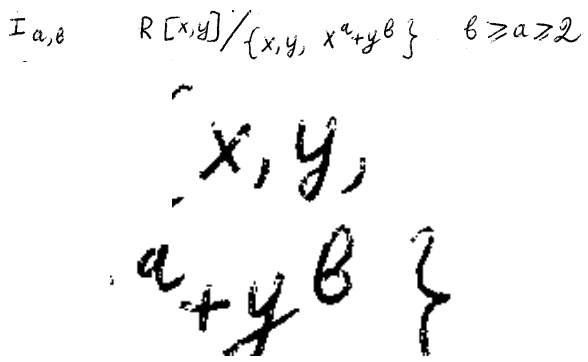


Uncertain reading. The first word resembles “либо” (“or”) or “любое” (“any”). It is followed by the abbreviated phrase “approx. by a map of one of the series”.

Original source details: GELFAND-P000186

Singularities of the five series are stable:

$$n \geq m \qquad 1. \quad \mathrm{I}_{a,b} \qquad R[x,y]/\{x,y,x^a+y^b\} \qquad b \geq a \geq 2$$



Manuscript notation. The first denominator has a comma between x and y. The second line has xy without a comma.

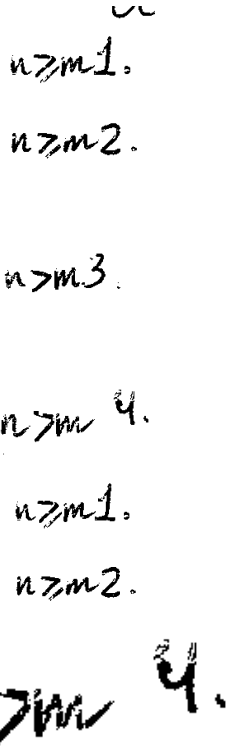
Original source details: GELFAND-P000186

$$n \geq m \qquad 2. \quad \mathrm{II}_{a,b} \qquad /\{xy, x^a - y^b\} \qquad b \geq a \geq 2 \quad \text{even}$$

$$n > m \qquad 3. \quad \mathrm{III}_{a,b} \qquad /\{x^a, y^b, xy\}$$

$$n > m \qquad 4. \quad \mathrm{IV}_a \qquad /\{x^a, xy^{a-1}, x^2+y^2\} \qquad a \geq 3$$

$$\mathrm{V}_a \qquad /\{x^a, x^2+y^2\} \qquad a \geq 3$$



Manuscript notation. The left margin gives $n \geq m$ beside rows 1 and 2, and $n > m$ beside rows 3 and 4. A gap is visible beside row 4.

Original source details: GELFAND-P000186

The codimensions have been calculated.

(for all series)

Σ^{20} itself is strat. into subvarieties

(This is the strongest res. on the classific. of singularities)

Next time $13/II$ — 1967.

GELFAND-P000187

Gelfand sem.

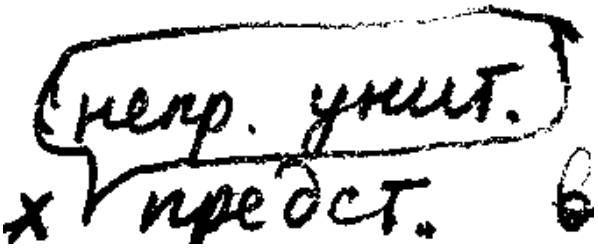
Plan.

1.	Kirillov Harish-Chandra	≈ 10 lectures.
2.	Anosov Maslov and semiclassics	2 lectures.
3.	Kazhdan A. Weil.	6–7 lectures.
4.	Dynin Index.	3 lectures.
5.	Anosov Pfaff eqs. (1st order)	1 lecture.
6.	KG-functor (Bernstein)	2 lectures.

Next time Kirillov and Anosov.

Kirillov. Lecture 1.

A series of works by H.-Ch., leading to a descr. of all irred. unit. reps. of real semisimple Lie groups.



Original source details: GELFAND-P000187

Manuscript notation. "Irred. unit." appears in a box pointing to "reps.".

Today, the scheme of the proof of one theorem (on characters).

G: a real semisimple Lie group. (reductive—for ind.)

On it, consider diff. operators commuting with both left and right translations.

This is a polynomial ring in generators $D_1, \dots, D_\ell$: the rank of the group.

$$G = SL(n), \qquad \ell = n - 1,$$

one oper. of each degree, beginning with the second.

Gelfand: E_{ik}

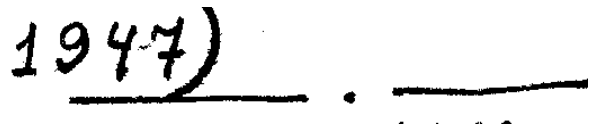
$$x = (\dots), \qquad f(x(1 + te_{ik})) = \\ = f(x) + tE_{ik}f + \dots$$

In the full group, $\sum E_{ii}$ (it is triv. in SL)

$$\sum E_{ik}E_{ki}$$
$$\sum E_{ij}E_{jk}E_{ki}$$
$$\dots$$

The rest are expressed through the first n.

(see Gelfand, 1947)



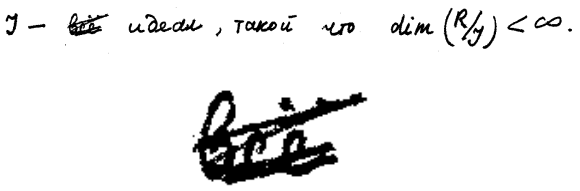
Original source details: GELFAND-P000187

Diagram and typed description. After the reference are two strokes with a dot between them.

K. Consider generalized functions on the group that are “sols. of all these eqs.”.

R: the ring of all polynomials in $D_1, \ldots, D_\ell$.

J: [cancelled: illegible] an ideal such that $dim\,(R/J) < \infty$.



Original source details: GELFAND-P000187

Uncertain reading. Before “ideal”, a short word is cancelled: a high initial loop and several crossed lower strokes. The readings “vse” / “vsyo” (“all”) are undetermined.

GELFAND-P000188

[source number: 51]

Original source details: GELFAND-P000188

Diagram and typed description. Page number 51 with a line at its left and below it.

51

Consider generalized functions T such that

$$T^x = T \quad \text{for } x \in G$$

(x: an inner autom.)—const. on classes of conj. elements.

$$\langle T^x, f\rangle = \langle T, f^{x^{-1}}\rangle, \qquad f^{x^{-1}}(y) = f(x^{-1}yx)$$

Original source details: GELFAND-P000188

Diagram and typed description. Two strokes with a dot between them.

G.: $D_1, \ldots, D_\ell$: Laplace operators on the group.
Let $D_i T = \lambda_i T$; then T is an eigenfunction. (generalized)
Among them, seek "spherically symm." (zonal) ones

$$T(y) = T(x^{-1}yx).$$

Then it is an actual function and, moreover, summable (locally)
[In the right margin: Schwarz[?]]

Original source details: GELFAND-P000188

Uncertain reading. An underlined surname in the margin: "Schwartz" / "Schwarz"; the middle joined letters are compressed, and long strokes cross and surround the inscription. Its exact spelling is undetermined.

One can consider generalized eigenfunctions:

$$(D - \lambda)^2 T = 0$$

(not R, but $JT = 0$).

$$r(D_1, \ldots, D_\ell)T = 0, \qquad r \in J.$$

For a max ideal, these are simply eigenfunctions (λ_i)
It is not necessarily continuous!

Original source details: GELFAND-P000188

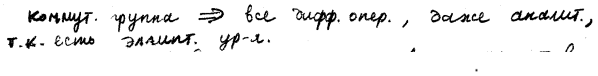
Diagram and typed description. Two strokes with a dot between them.

_____ · _____

K.

$$\begin{aligned} &1) \quad T^x = T \quad \text{for } x \in G \\ &2) \quad DT = 0 \quad \text{for } D \in J \end{aligned}$$

Then it is locally summable.
Abelian group $\Rightarrow$ all diff. operators, even analytic, since there is an elliptic eq.



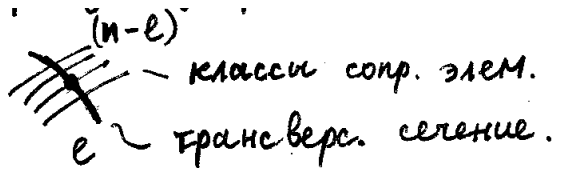
Original source details: GELFAND-P000188

Manuscript notation. The word before the abbreviation for "analytic" is "even".

General idea: ℓ oper., functions actually depending on ℓ params., satisfying ℓ eqs., construct an elliptic eq., away from singular points.
Characters become 1 at ∞—may have discontinuities.
How is this proved?

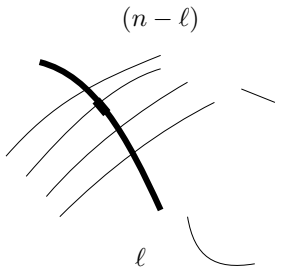
The set of classes of conj. elements is arranged thus: open subsets that are subvarieties $GL(n)$: distinct eigenvalues.
Complement of codim. 2 or 3, in turn, has subsets, etc. (2 eigenvalues coincide, etc.)—
—stratification.
GELFAND-P000189

At a reg. point $y \in G$, this is an analytic function.
In a nbhd. of y, construct a new coord. system.



Original source details: GELFAND-P000189

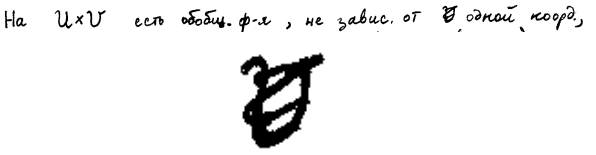
Typeset diagram



classes of conj. elements

transv. section.

On $U \times V$, there is a generalized function independent of [cancelled: illegible] one coord.,



Original source details: GELFAND-P000189

Uncertain reading. Between “of” and “one coord.”, a small sign is heavily cancelled: V / $\forall$ / another sign. Its literal reading is undetermined; the word “one” follows the cancelled sign.

reducing to a generalized function on the section. (see L. Schwarz)

$$D_1, \ldots, D_\ell$$

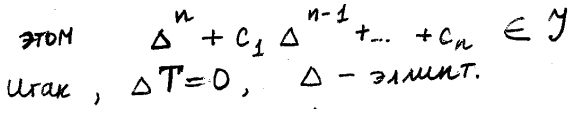
generate almost all diff. operators on the section
(already of finite codim. in the ring of all diff. operators)

$$\Delta = D_1^2 + D_2^2 + \ldots + D_\ell^2$$

is already elliptic on the section.
here

$$\Delta^n + c_1 \Delta^{n-1} + \ldots + c_n \in J$$

Thus, $\Delta T = 0$, Δ is elliptic.



Original source details: GELFAND-P000189

Manuscript notation. The line below the polynomial reads $\Delta T = 0$.

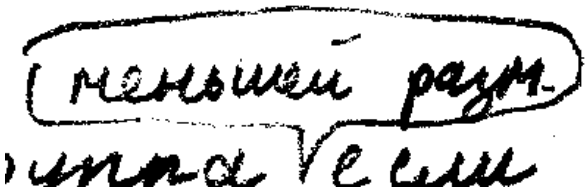
The arbitrary case reduces to this by induction on the dim. of the group.
3 different cases.

x — 1) semisimple
2) nilpotent
3) identity.

of smaller dim.

1)—reduces to the centralizer of y: a reductive group, if y is not central. (then the whole group)

Manuscript notation. The box above "if" reads "of smaller dim.".



Original source details: GELFAND-P000189

2) nilpotent—a special device

3) $x = e$: ln, exp reduce to the study of the Lie algebra—

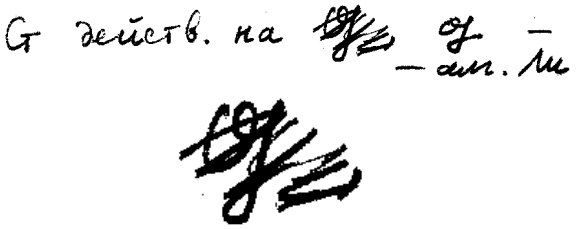
—2 separate works.

The idea for semisimple elements: passage to polar coords.

(radial part of the Laplace operator)

a: a semisimple element

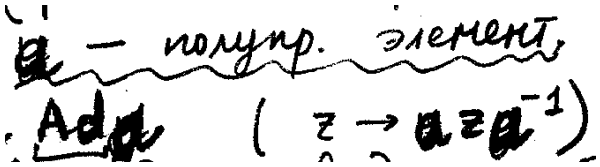
G acts on [cancelled: ?] g—
—Lie alg.



Uncertain reading. After "on", before the legible g, a symbol with loops and a long lower stroke is cancelled: g / another sign. The original inscription is undetermined.

Original source details: GELFAND-P000189

if Ad_a $(z \longrightarrow aza^{-1})$



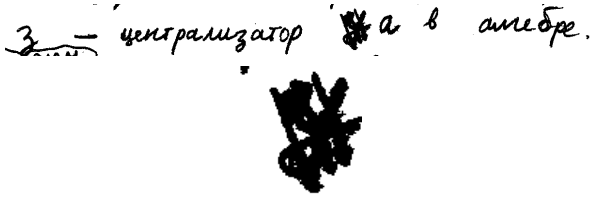
Uncertain reading. The subscript on Ad and the outer letters of aza^{-1} are heavily inked. The reading a agrees with the separate a following "centralizer".

Original source details: GELFAND-P000189

is reduced to diagonal form, i.e. without Jordan blocks

(For $SL(n)$, the matrix itself has no Jordan blocks)

z: the centralizer of [cancelled: illegible] a in the algebra.

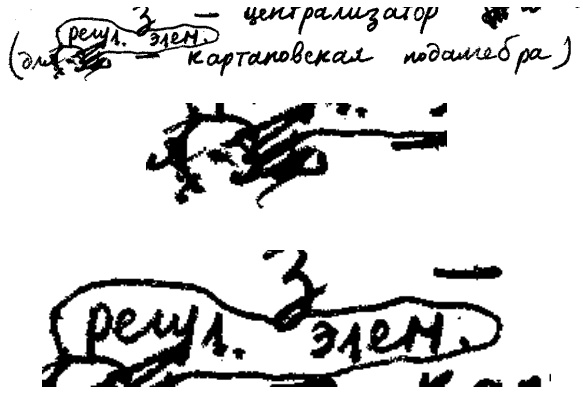


Uncertain reading. Before the separate legible a, a small symbol with several loops is heavily overwritten / cancelled. Its original reading is undetermined.

Original source details: GELFAND-P000189

reg. elem.

(for [cancelled: ?] —a Cartan subalgebra)



Uncertain reading. After “for”, beneath the boxed insertion “reg. elem.”, an inscription is cancelled: a / x_0 / another symbol. Several original strokes merge with cancellation lines; the reading is undetermined.

Original source details: GELFAND-P000189

GELFAND-P000190

[source number: 52]

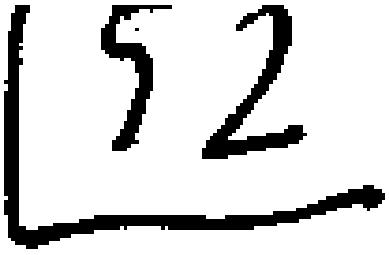


Diagram and typed description. Page number 52 with a line at its left and below it.

52

Original source details: GELFAND-P000190

Take a section (still redundant for now)

Berezin's thm. for reg. elements: restriction of operators to a Cartan subalgebra.

Ξ: the Lie group corresponding to $\mathfrak{z}$.

Construct a map

$$\begin{aligned} \varphi\colon G \times \Xi &\longrightarrow G \\ (x, y) &\longmapsto (ay)^x = xayx^{-1} \end{aligned}$$

Further notation:

$$V_a(y) = \det\bigl(Ad(ay)^{-1} - 1\bigr)_{\mathfrak{g}/\mathfrak{z}}$$

$$\Xi' = \{y \in \Xi : V_a(y) \neq 0\} \quad \text{— "reg. points" of } \Xi$$

($V_a \neq 0$, since the eigenvalue 1 is no longer present; it is on $\mathfrak{z}$)

Thm. 1 (Lemma 14)

$$\operatorname{rang} d\varphi = n = \dim G \qquad \text{on } G \times \Xi'$$

(locally this is the max of the rank: projection type)

Proof. X, Z: a pair of tangent vectors.

$$d\varphi(X, Z) = Z + \bigl(Ad(ay)^{-1} - 1\bigr)X,$$

from which everything follows.

Nondegenerate on the quotient.

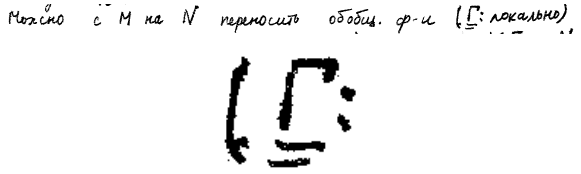
Z provides full rank on the whole space.

Result from another work.

$$d\mu \quad M^m \xrightarrow{\pi} N^n \quad d\nu \quad \text{— measures}, \qquad m \geq n$$

$\operatorname{rg} d\pi = n$.

One can transfer generalized functions from M to N (G: locally)



Manuscript notation. A doubly underlined Γ ("G") and a colon precede "locally".

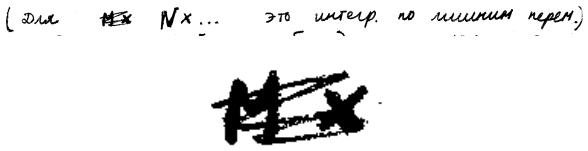
Original source details: GELFAND-P000190

$$\alpha \in C_0^\infty(M) \quad \longrightarrow \quad f_\alpha \in C_0^\infty(N),$$

so that $\forall F$ on N there will be

$$\int_M \alpha \cdot F \circ \pi \, d\mu = \int_N F f_\alpha \, d\nu$$

(For [cancelled: ~~$M\times$~~ [?]] $N \times \ldots$, this is integration over the extra variables.)



Uncertain reading. The cancellation provisionally reads $M\times$. The separate $N \times \ldots$ follows.

Original source details: GELFAND-P000190

This is done linearly, a partition of unity; locally this is a direct product.

GELFAND-P000191

Perspective: apply to φ.
Useful notation. (for illegibility)
On $M \times N$, there is a function (smooth)
D_1, D_2 —diff. operators on M, N.

$$f(x; D_1 : y; D_2) = (D_{1x} \times D_{2y} f)(x, y)$$

(apply D_1 in x and D_2 in y)
The identity operator is omitted.

$$f(x : y; D_2) \quad —$$

—explained in only one work.
Another notation.

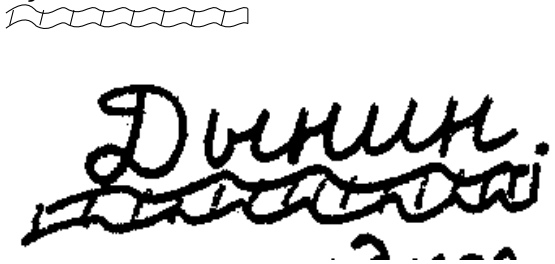
$$G, \quad f(x) \text{ on } G, \qquad f(x : y) = f(xyx^{-1})$$

Next time, generalize Berezin's theorem.
For identity operators D_1 and D_2:

$$f(x : y)$$

and everything applies to it.

Dynin.



Manuscript notation. Two wavy lines joined by short strokes underline the surname Dynin.

Original source details: GELFAND-P000191

X —a smooth compact manifold
E_1, E_2 —bundles on X

$$A\colon \Gamma(E_1) \longrightarrow \Gamma(E_2) \quad \text{— lin. diff. operator}$$

Pseudodiff. operators.

$\{U_i\}$ φ_i — subord. partition of 1

$$\psi_i \varphi_i = \varphi_i$$

1) $\psi_i A(\varphi_i \cdot)$ — an ordinary pseudodiff. operator in the space of vector-valued functions

2) $\varphi\psi = 0 \;\Rightarrow\; \psi A(\varphi \cdot)$ a smoothing operator.

GELFAND-P000192

[source number: 53]

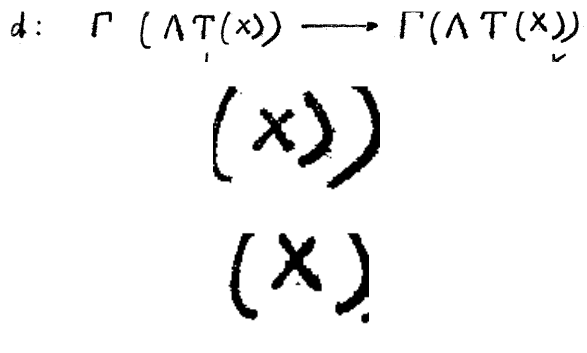
Diagram and typed description. Page number 53 with a line at its left and below it.

53

Original source details: GELFAND-P000192

Example:

$$d\colon \Gamma(\Lambda T(x)) \longrightarrow \Gamma(\Lambda T(X))$$



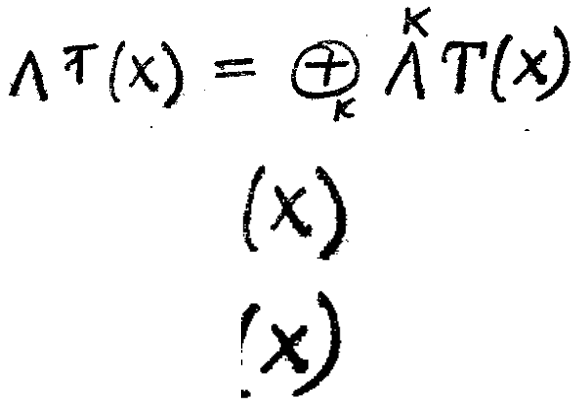
Uncertain reading. The case x/X in both arguments of T is not fully distinguishable; lower-case x at left and capital X at right are typographic candidates.

Original source details: GELFAND-P000192

!

all forms (inhomog.)

$$\Lambda T(x) = \bigoplus_k \Lambda^k T(x)$$



Manuscript notation. Both parenthesized arguments of T have a small x.

Original source details: GELFAND-P000192

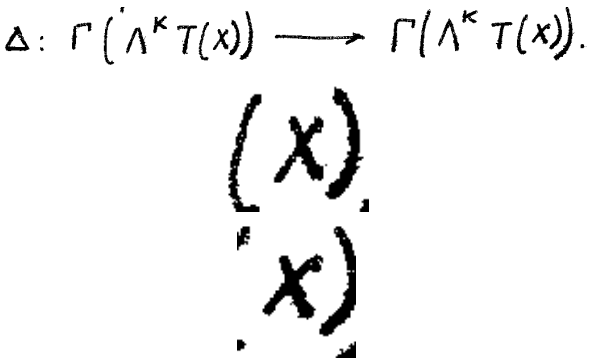
δ: —the adjoint operator (in the Riem. metric)

$$\delta = d^*.$$

$d + \delta$—the Cauchy-Riemann operator

$\Delta = (d+\delta)^2$ —the Laplace operator.

$$\Delta\colon \Gamma(\Lambda^k T(X)) \longrightarrow \Gamma(\Lambda^k T(X)).$$

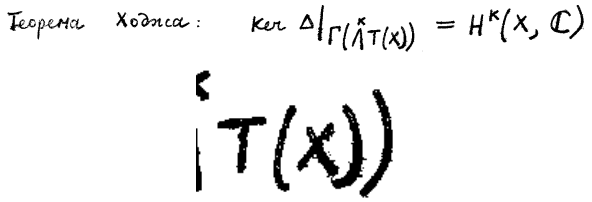


Uncertain reading. The undetermined case x/X affects both T arguments on this line; capital X signs are candidates, not established readings.

Original source details: GELFAND-P000192

Hodge's theorem:

$$\ker \Delta\big|_{\Gamma(\Lambda^k T(x))} = H^k(X, C)$$



Manuscript notation. Below the restriction bar is a small x. On the right, the X in H^k is taller.

Original source details: GELFAND-P000192

(a very simple fact, see Moren, connection with the heat eq.)

Next time.

6–7 —Kirillov

7–8 —Anosov.

Gelfand sem. $20/\overline{\text{II}}$.
Kirillov. Lecture 2.

The case where the fibre has positive dimension. (xyx^{-1})—
—y is not central.
Let a be a semisimple element.
Take a section intersecting each class of conj. elements in 1 element.

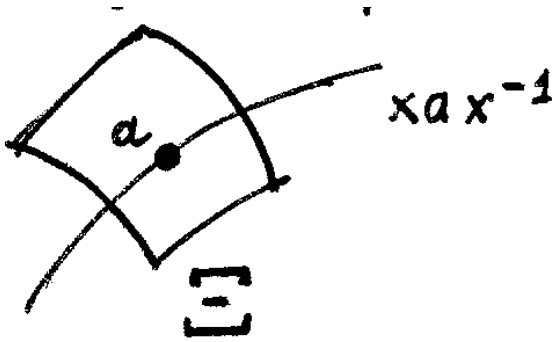
$$\mathfrak{g} \text{ — Lie alg.}, \qquad \mathfrak{z} = \{X : \operatorname{ad}_a X = X\}, \qquad \underset{\text{subgroup corresp. to } \mathfrak{z}.}{\Xi} \subset G.$$

The section is the conn. component of the centralizer of a.
Everything reduces to the group Ξ.
We shall construct a new generalized function and diff. operator.
How does one specify diff. operators on a Lie group?
Consider left-invariant diff. operators ($\exists$ a left-invariant one at $\forall$ point, agreeing with ours up to lower-order terms).
D_y: the left-invariant operator agreeing with D at y.
$D_y \in \mathfrak{g}$ describes the whole operator (on the whole group at once).

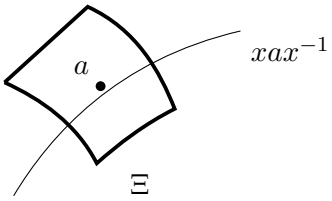
Manuscript notation. After $\in$ is a looped g with a descender, as in the earlier definition of $\mathfrak{g}$.

Original source details: GELFAND-P000193

We shall restrict a diff. operator to a subset.



Typeset diagram



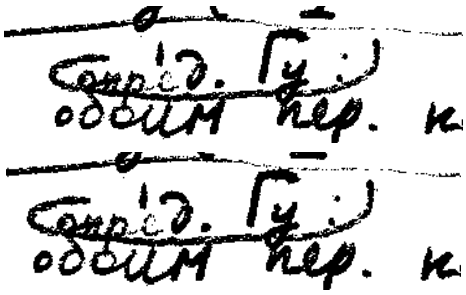
Original source details: GELFAND-P000193

We shall regard diff. operators along the fibres as zero.
Fibres close to a may have greater dimension.

$$f(x; D_1 : y; D_2) = \underline{f(x : y; \Gamma_y(D_1 \otimes D_2))}$$

(D_1 in x, D_2 in y)

def. Γ_y:

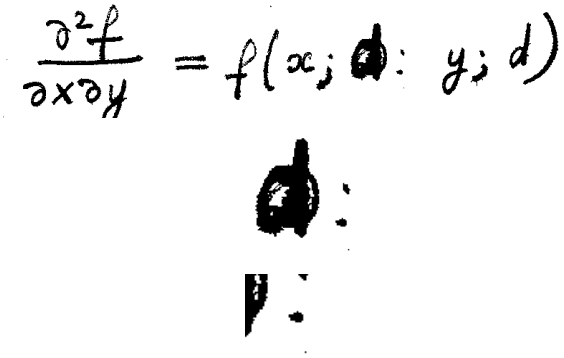


Manuscript notation. The enclosure reads “def. Γ_y”, followed by two separate dots.

Original source details: GELFAND-P000193

The explicit form of replacing operators in both variables by an operator in y.

$$\frac{\partial^2 f}{\partial x \partial y} = f(x; [d \,/\, \partial\, ?] : y; d)$$



Uncertain reading. The first sign after $x;$ is heavily overwritten and has a tall stroke. It may be d. The remaining ink does not exclude ∂.

Original source details: GELFAND-P000193

Γ_y depends bilinearly on D_1, D_2, or—linearly on $D_1 \otimes D_2$.

GELFAND-P000194

[source number: 57]

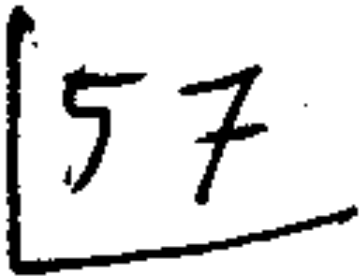


Diagram and typed description. Page number 57 with a line at its left and below it.

57

Original source details: GELFAND-P000194

Explicit expression for Γ_y. (g_1, g_2: elements of the enveloping algebra.

$$f(x; g_1 : y; g_2) = f(x : y; \Gamma_y(g_1 \otimes g_2)).$$
$$\Gamma_y(g_1 \otimes g_2) = \sigma_y(g_1) \cdot g_2,$$

where $\sigma_y(g_1)$ is a representation of the enveloping algebra, i.e.

$$\sigma_y(gh) = \\ = \sigma_y(g)\sigma_y(h).$$

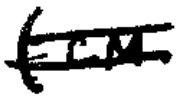
Now it suffices to calculate σ_y on the generators of the enveloping algebra.

$$\sigma_y(X) = L_{X^{y^{-1}}} - R_X \qquad (X \text{ from the Lie algebra}).$$

$$\sigma_y(X) : g \longrightarrow X^{y^{-1}} g - gX$$

Derivation.

[Cancelled: (le...?]

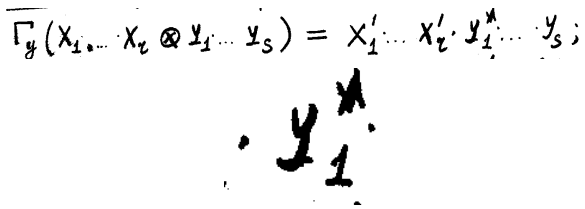


Uncertain reading. A short beginning, "(ст..." ("(le..."), beneath two cancelling lines; the further reading is undetermined.

Original source details: GELFAND-P000194

(leading term)—on a basis:

$$\Gamma_y(X_1 \cdots X_r \otimes Y_1 \cdots Y_s) = X_1' \cdots X_r' Y_1^{[*\,?]} \cdots Y_s;$$



Uncertain reading. Above Y_1, a sign with several rays: an asterisk or a cancelled prime. The brackets and question mark qualify the candidate.

Original source details: GELFAND-P000194

$$X_i' = (\operatorname{ad} y^{-1} - 1) \cdot X_i$$

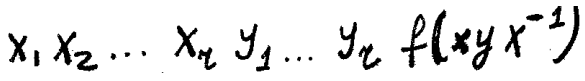
[Cancelled: $f(x; g_1$?]

Uncertain reading. Candidates for the unfinished fragment: $f(x; g_1$ / $f(x, g_1$. The sign after $f($ and the separator are crossed out; no closing parenthesis is supplied.

Original source details: GELFAND-P000194

G.

$$X_1 X_2 \cdots X_r Y_1 \cdots Y_r \, f(xyx^{-1})$$

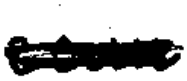


Manuscript notation. The subscripts on the last X and Y have the same r shape.

Original source details: GELFAND-P000194

The operators must be transferred to y; Y_i remains, while

[cancelled: if?]



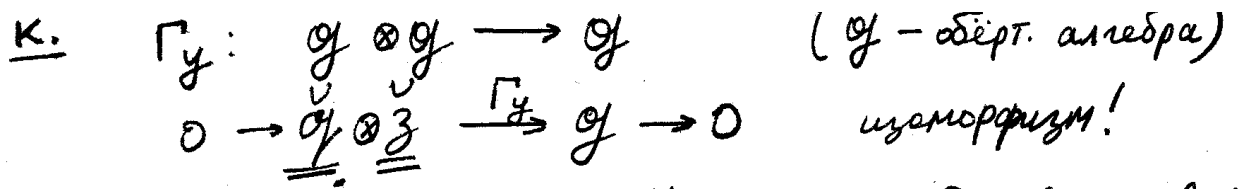
Uncertain reading. One heavily crossed-out word before the next line; candidate “if” is undetermined.

Original source details: GELFAND-P000194

if X commutes with y, nothing changes.
(It is easy to check all [inserted: this])

— · —

K.



Original source details: GELFAND-P000194

Typeset diagram

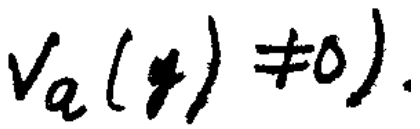
$$\begin{array}{ccccccc} \Gamma_y: & & \mathfrak{g}\otimes\mathfrak{g} & \longrightarrow & \mathfrak{g} & & (\mathfrak{g}\text{—enveloping algebra}) \\ & & \cup \quad \cup & & & & \\ 0 & \longrightarrow & \underline{\underline{\mathfrak{q}}}\otimes\underline{\underline{\mathfrak{z}}} & \xrightarrow{\Gamma_y} & \mathfrak{g} & \longrightarrow 0 & \text{isomorphism!} \end{array}$$

$\mathfrak{g} = \mathfrak{z} + \mathfrak{q}$,
[cancelled: where? / sub.?]

Uncertain reading. A short crossed-out word after the equality: “where” / “sub.”; middle and final strokes merge with the cancelling line.

Original source details: GELFAND-P000194

$\mathfrak{q}$: a complementary subspace, invariant under $\mathrm{Ad}\,\mathfrak{z}$: $[\mathfrak{z}, \mathfrak{q}] \subset \mathfrak{q}$.
$\underline{\underline{\mathfrak{z}}}$, $\underline{\underline{\mathfrak{q}}}$ are now generated by them in the enveloping algebra.
Choose y regular (i.e. $V_a(y) \neq 0$). Then the map



Manuscript notation. The condition reads $V_a(y) \neq 0$. The V matches the notation on manuscript page 52.

Original source details: GELFAND-P000194

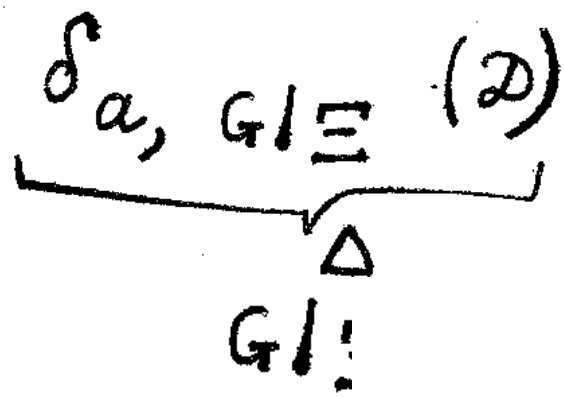
Γ_y, by the formula, is nondegenerate and unique in its leading term,

GELFAND-P000195

D: an operator on the group,

$$D \longrightarrow \underbrace{\delta_{a,\,G|\Xi}(D)}_{\Delta}$$

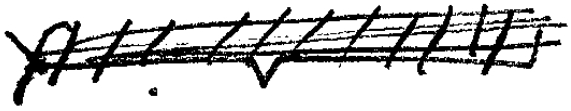
—an operator on $a\Xi$— —a submanifold.



Uncertain reading. The nearly vertical stroke is provisionally read as |. An alternative is /.

Original source details: GELFAND-P000195

The correspondence is constructed so that for functions constant on the orbits it is all the same.
[Cancelled formula: at the beginning, φ?; at the end, (D)?]



Uncertain reading. Possible separate remnants of the cancelled formula: φ at the beginning and (D) at the end. The complete formula is undetermined.

Original source details: GELFAND-P000195

Undeciphered passage. The middle and remaining undetermined part of the cancelled formula are illegible.

Original source details: GELFAND-P000195

$$\Delta_{ay} = \alpha(D_{ay})$$
$$D - \alpha(D) \in \Gamma_y(\mathfrak{q}^+ \otimes \mathfrak{z})$$

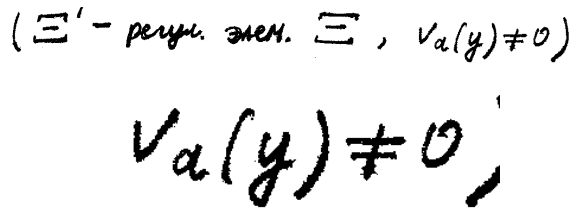
($\alpha(D)$: the constant term of D. $\mathfrak{q}^+$: elements having no constant term).
Next time, the transfer of generalized functions again.

$27/\overline{\text{II}}$ Gelfand Seminar.
Kirillov, Lecture 3.
Transfer of generalized functions.

Consider an auxiliary map:

$$\varphi : G \times \Xi' \longrightarrow G$$
$$(x, y) \longrightarrow xayx^{-1}$$

(Ξ': the regular elements of Ξ, $V_a(y) \neq 0$)



Manuscript notation. The regularity condition reads $V_a(y) \neq 0$.

Original source details: GELFAND-P000195

$$\operatorname{rank} \varphi = \dim G.$$

Now generalized functions are transferred from G to Ξ. First they are transferred to $G \times \Xi'$.

T: generalized functions on G. We must learn to transfer ordinary functions in the forward direction. Roughly speaking, this is integration (locally).

GELFAND-P000196

[source number: 58]

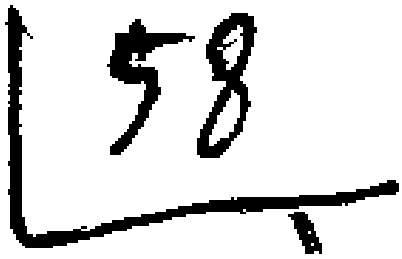
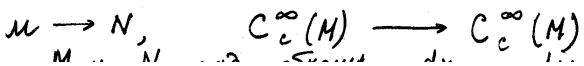


Diagram and typed description. Page number 58 with a line at its left and below it.

58

Original source details: GELFAND-P000196

$$M \longrightarrow N, \qquad C_c^\infty(M) \longrightarrow C_c^\infty(M)$$



Manuscript notation. Both function spaces are written with M in the manuscript.

Original source details: GELFAND-P000196

Volumes dx and dy are given on M and N. (differential forms)

$$\alpha(x) \longrightarrow f_\alpha(y)$$

The condition must be satisfied: $\forall F(y)$

$$\int_M \alpha(x) F(\varphi(x))\, dx = \int_N F(y) f_\alpha(y)\, dy$$

Theorem: such a map exists and is continuous in the topology of these spaces.

Example.

$$\begin{matrix} M & = & N & \times & P \\ & & dy & & dz \end{matrix} \qquad (dx = dy\, dz).$$

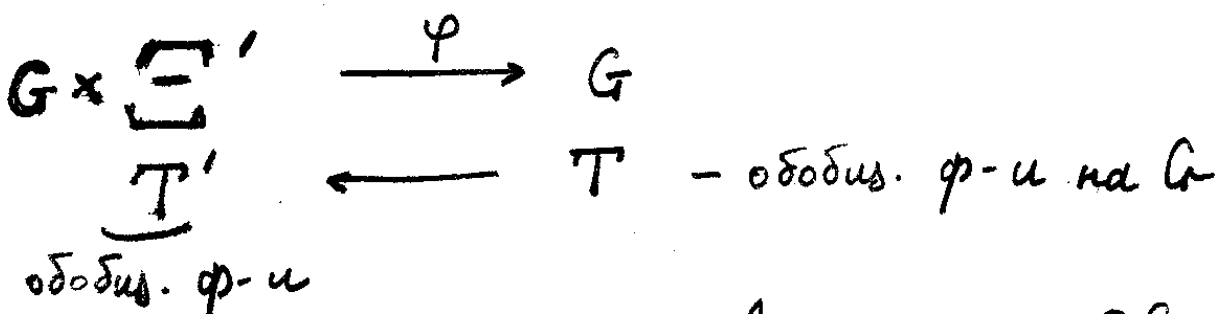
Then the map is integration with respect to z.

— · —

[Cancelled: M? / $\forall M$?]

Original source details: GELFAND-P000196

Uncertain reading. Angular strokes before $G \times \Xi'$: M or $\forall M$; the cancelling lines cannot confidently be separated from the original signs.



Original source details: GELFAND-P000196

Typeset diagram

$$G \times \Xi' \xrightarrow{\ \varphi\ } G$$

$$\underbrace{T'} \xleftarrow{\quad} T$$

—generalized functions on G

generalized functions

T' is invariant under translations in its first factor

$$\langle T', f(x,y)\rangle = \langle T', f(x'x,y)\rangle$$

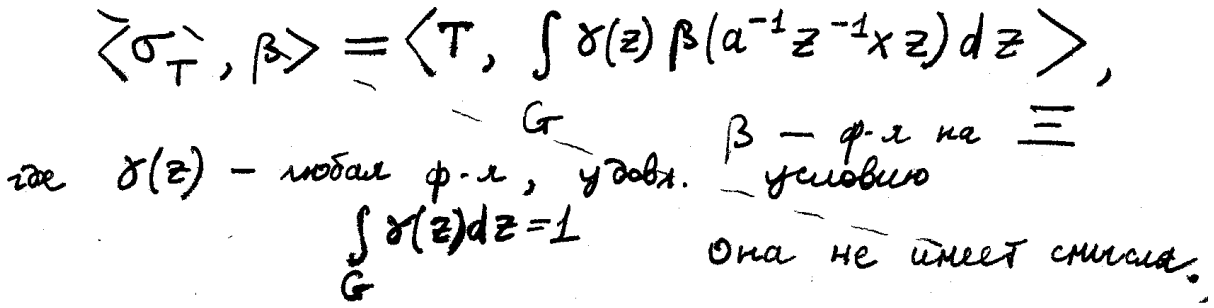
Lemma. A generalized function invariant under translations is $\int$ with respect to Haar measure (const)

— · —

T' depends only on the second argument (integrate over the first).

$$T' = 1 \otimes \sigma_T$$

An explicit formula expressing σ_T through T can be given (K.)



Original source details: GELFAND-P000196

Typeset diagram

$$\langle \sigma_T, \beta \rangle = \left\langle T, \int_G \gamma(z)\beta(a^{-1}z^{-1}xz)\,dz \right\rangle,$$

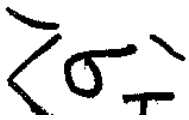
β: a function on Ξ

where $\gamma(z)$ is any function satisfying the condition

$$\int_G \gamma(z)\,dz = 1$$

It has no meaning.

Diagram and typed description. A sloping dashed line runs through the formula region toward “It has no meaning.” Its purpose—cancellation or connection with the note—is undetermined.



Original source details: GELFAND-P000196

GELFAND-P000197

Thus, we have:

$$D \longrightarrow \Delta \qquad = \delta_{a,G/\Xi}(D)$$
$$T \longrightarrow \sigma_T^{[?]}$$

Uncertain reading. The faint interrupted stroke at the upper right of σ may be a prime in σ'_T. Below, σ_T recurs.

Original source details: GELFAND-P000197

$a \in U_G \subset G$ [Cancelled: Ξ?] $\Xi \supset U_\Xi \ni 1$

Uncertain reading. The cancelled sign shows three horizontal strokes resembling the following Ξ.

Original source details: GELFAND-P000197

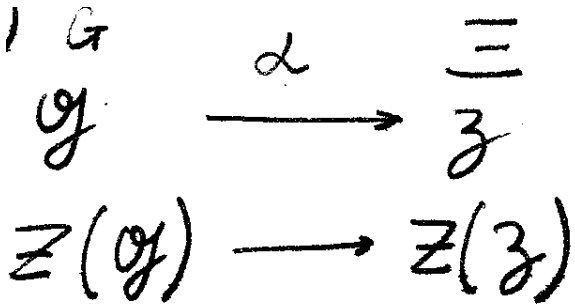
Lemma.

$$\sigma_{DT} = \Delta\sigma_T$$

(Essential use of $\Gamma_{ay} : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}$.)

Facts.

1) The operators Δ generate an ideal of finite codimension in the center of the group Ξ.
(On the Cartan subalgebra, these are operators symmetric under the Weyl group)

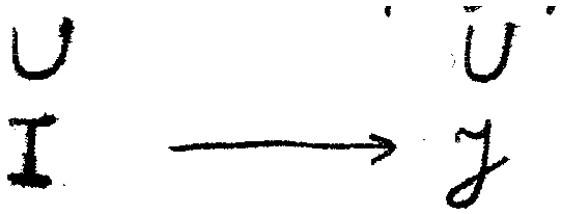


Typeset diagram

$$\begin{array}{ccc} G & & \Xi \\ & \overset{\alpha}{\longrightarrow} & \\ \mathfrak{g} & & \mathfrak{z} \\ & & \\ Z(\mathfrak{g}) & \longrightarrow & Z(\mathfrak{z}) \end{array}$$

Original source details: GELFAND-P000197

$Z(\mathfrak{g})$ is the center of the enveloping algebra; the image of $Z(\mathfrak{g})$ lies in the center $Z(\mathfrak{z})$ (can be checked). and the ideal it generates is of finite codimension.



Typeset diagram

$$\begin{array}{ccc} \cup & & \cup \\ & & \\ I & \longrightarrow & J \end{array}$$

Original source details: GELFAND-P000197

Then a complete reduction to Ξ is obtained

$$\sigma_T \text{ is locally integrable} \implies T \text{ is locally integrable too}$$

a in the center $\Longrightarrow$ reduce by translation

Undeciphered passage. The cancelled text after “reduce by translation” is completely covered by strokes and has not been read.

Original source details: GELFAND-P000197

to the case $a = e$,
there is a special technique (3 papers

Undeciphered passage. The cancelled text between “3 papers” and the name is completely covered by strokes and has not been read.

Original source details: GELFAND-P000197

by Harish-Chandra)

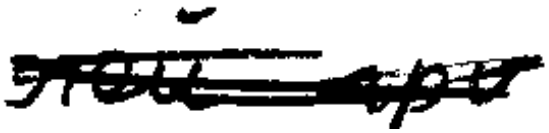
GELFAND-P000198

[source number: 59]

6/III. Vishik Seminar.
1. Continuation of Agranovich's report.
2. Continuation of G. I. Eskin's report.

Gelfand Seminar.

[Left margin] Kirillov
$x^2 + y^2 - z^2 = c$, T is invariant under
[Cancelled: this gr...? / this f...?]



Original source details: GELFAND-P000198

Uncertain reading. A cancelled word and the beginning of the next: "this gr..." / "this f..."; the continuation is undetermined.

the group preserving this function.

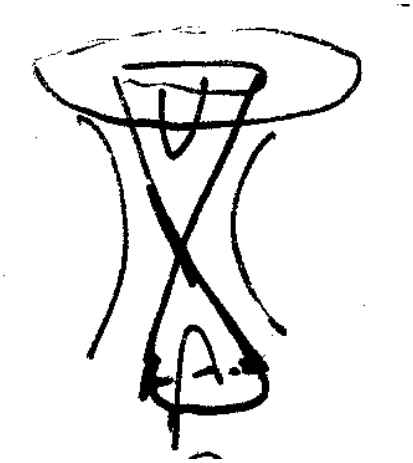
$$P\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}\right) T = 0,$$

will T be locally integrable?

Original source details: GELFAND-P000198

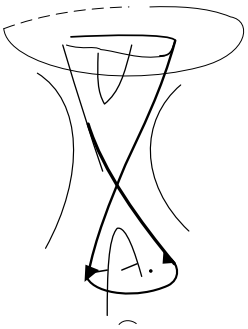
Undeciphered passage. One cancelled word before the following parenthesis is illegible; a period remains at the right.

(T: the closure of local functions constant on the fibres)



Original source details: GELFAND-P000198

Typeset diagram



Original source details: GELFAND-P000198

Uncertain reading. The large upper oval, smaller upper loop, lower oval, crossing diagonals, concave side curves and internal arcs. The purpose of the internal A-/U-shaped strokes and dark lower endpoints is undetermined; no letter labels are assigned to them.

Outside the cone, the system is elliptic.

Arnold.

C. J. Earle, J. Eells.

X^2: a two-dimensional compact manifold (oriented)

$\mathrm{Diff}_0(X)$: the group of its diffeomorphisms homotopic to the identity.

Its topology has been calculated!

$$\mathrm{Diff}_0 X = \begin{cases} X = S^2 \ [:\ ?] & \underset{\substack{\text{conformal}\\ \text{maps}}}{G} \times \underset{\text{contractible}}{\mathrm{Diff}(S^2 - a, b, c)} \quad (a, b, c \text{ fixed}) \\ X = T^2 & T^2 \times \underset{\text{contractible}}{\mathrm{Diff}(T^2 - a)} \\ \text{genus} \geq 2 & D \sim 0 \quad \text{contractible} \end{cases}$$

Original source details: GELFAND-P000198

Uncertain reading. After $X = S^2$, two dots at right and a separate sloping stroke at left. A colon with an additional stroke / a three-dot sign; the typeset colon is qualified by brackets and a question mark.

The proof is based on work by Ahlfors and Bers on quasiconformal maps, onto which any diffeomorphism contracts.

Manuscript notation. The Russian manuscript spelling is Альфорса.

Original source details: GELFAND-P000198

Diff is a Lie group in the new understanding of smooth structure.

GELFAND-P000199

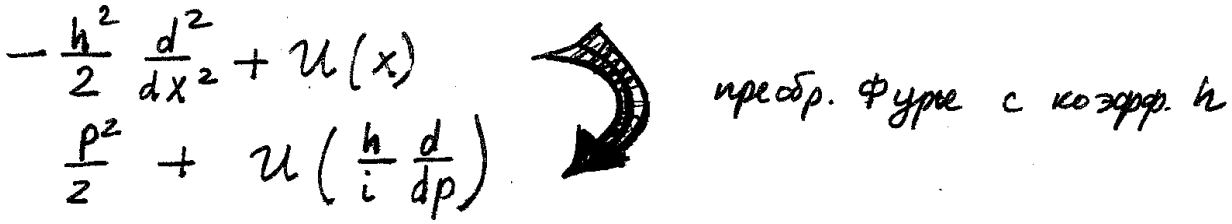
Anosov.

Statement of the problem.

$Lu = f$, $u(x) \in H$, H is a Hilbert space.

In H there is an operator A; 0 is not an eigenvalue; A is self-adjoint.

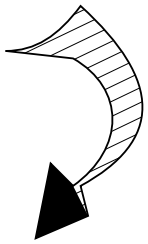
L may depend on A as well.



Original source details: GELFAND-P000199

Typeset diagram

$$-\frac{h^2}{2}\frac{d^2}{dx^2} + U(x)$$



Fourier transform with coefficient h

$$\frac{p^2}{2} + U\left(\frac{h}{i}\frac{d}{dp}\right)$$

(pseudodifferential operator)— $\overline{\mathcal{F}}_{p\to x}\,\sigma(x,p)\,\mathcal{F}_{x\to p}\,f(x)$

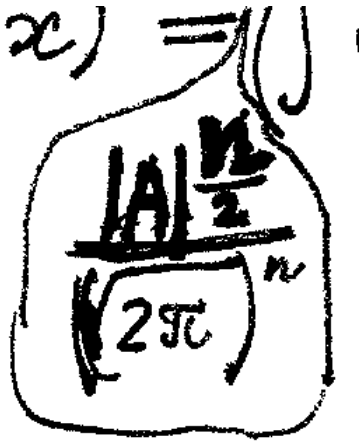
$$e^{\pm i\frac{1}{h}(p,x)} =$$

(h to make the exponent dimensionless).

According to Maslov, $\frac{1}{h}$ is an operator; since everything is a function of h

$$= e^{\pm iA(p,x)}$$

A-Fourier transform:

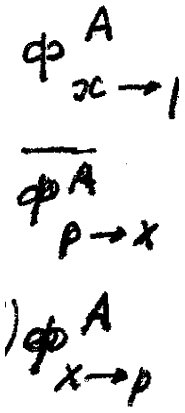


Original source details: GELFAND-P000199

Typeset diagram

$$\phi^A_{x\to p} f(x) = \left(\int e^{iA(x,p)} f(x)\, dx \right).$$

$$\frac{|A|^{n/2}}{(\sqrt{2\pi})^n}$$

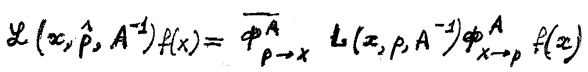


Uncertain reading. The oval sign has a long descending stroke. It is provisionally set as ϕ. Another reading is Φ.

Original source details: GELFAND-P000199

Consider

$$\mathcal{L}(x, \hat{p}, A^{-1}) f(x) = \overline{\phi^A}_{p\to x}\, L(x, p, A^{-1})\, \phi^A_{x\to p}\, f(x)$$



Manuscript notation. Both signs repeat the provisional ϕ from the preceding display. The bar belongs to the first transform.

Original source details: GELFAND-P000199

GELFAND-P000200

[source number: 60]

13/III Gelfand Seminar.

Diagram and typed description. Page number 60 with a line at its left and below it.

60

Original source details: GELFAND-P000200

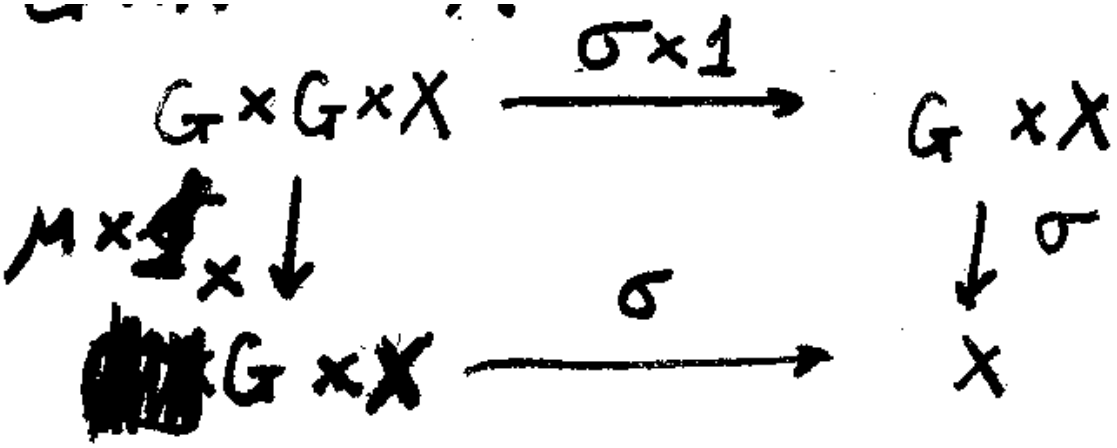
Parshin on Mumford.

Uncertain reading. A separate blackened mark to the right of the heading. Its sign and purpose are undetermined.

Original source details: GELFAND-P000200

G is an algebraic group, X is a variety, G acts regularly on X,
if

$$G \times X \longrightarrow X$$

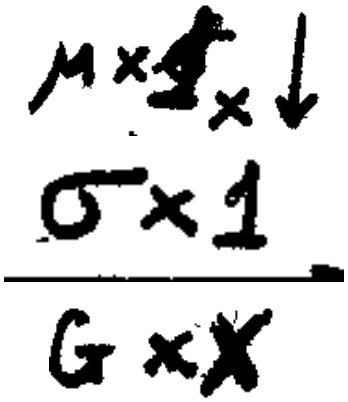


Original source details: GELFAND-P000200

Typeset diagram

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{\sigma \times 1} & G \times X \\ \downarrow{\scriptstyle \mu \times [1?]_X} & & \downarrow{\scriptstyle \sigma} \\ G \times X & \xrightarrow{\sigma} & X \end{array}$$

[cancelled: ?]



Uncertain reading. The upper label reads $\sigma \times 1$. The left label after $\mu\times$ has a heavily overwritten sign with a foot and a small X below and to its right. The reading 1_X is provisional.

Original source details: GELFAND-P000200

Undeciphered passage. The completely cancelled former entry before the lower-left $G \times X$ is illegible.

Original source details: GELFAND-P000200

Equivalent points—obtained by the action of G—an orbit.

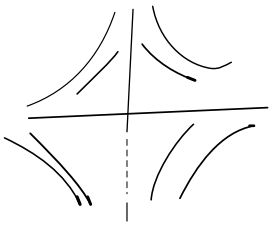
The set of orbits—? When is it an algebraic variety, and in what sense?

$$\underset{\text{orbits}}{\underline{X/G}} \longleftarrow X$$

Property of regular maps of algebraic varieties.

Original source details: GELFAND-P000200

Typeset diagram



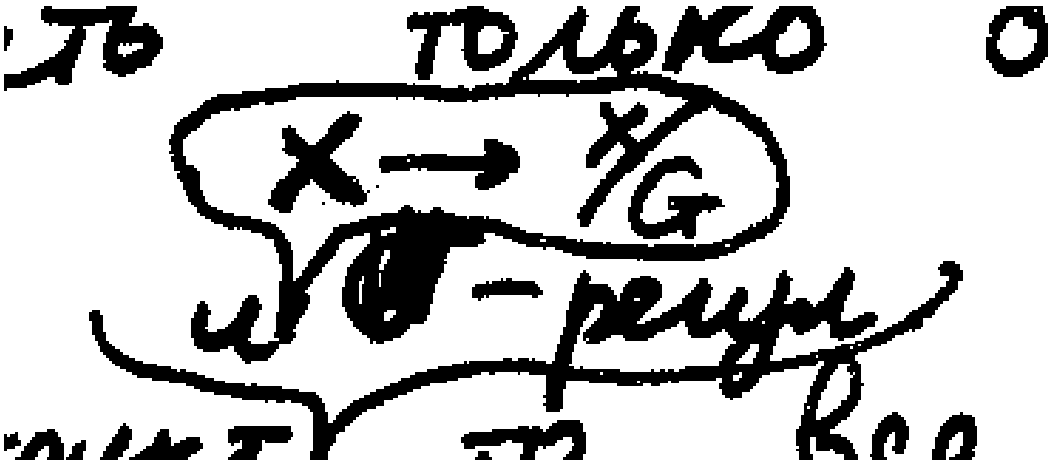
Original source details: GELFAND-P000200

Uncertain reading. Intersecting horizontal and vertical lines and two curved branches in each sector. There are no labels. Some ends are thickened and hooked; the presence and directions of possible arrows are undetermined.

A structure cannot be introduced; this is possible only after removing a point.

The dim of the stabilizer can only be larger (in comparison with the generic stabilizer).

Theorem. If X/G has an algebraic structure

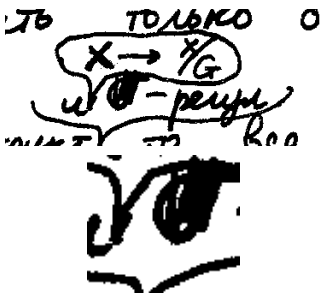


Original source details: GELFAND-P000200

Typeset diagram

$X \to X/G$

and $[\sigma?/\pi?]$ — regular.



Uncertain reading. The enclosure around $X \to X/G$ points to the overwritten sign between "and" and "—regular." The brace places the whole insertion after "algebraic structure." The overwritten sign may be σ or π.

Original source details: GELFAND-P000200

then all orbits must have the same dim. (in partic., be closed).

Reductive group (k of char. 0)

The quotient exists $\Longleftrightarrow$ all orbits are closed.

GELFAND-P000201

Dynin, Lecture 2.

$\Sigma(\mathbb{R}^n)$ from maps

$$\sigma\colon \mathbb{R}^n \times S^{n-1} \longrightarrow GL = \varinjlim GL(m, \mathbb{C})$$

with compact support

Outside $K_\sigma \times S^{n-1}$, $\sigma = 1$.

Theorem. ∃ one and only one map

$$\Sigma(\mathbb{R}^n) \xrightarrow{\text{ind}} \mathbb{Z},$$

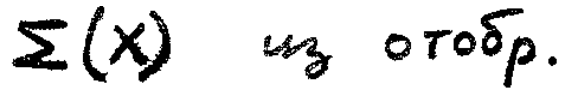
with the following properties:

(Ī) Homot. stability.

(II̅)

$$\text{ind}(\sigma_0 \oplus \sigma_1) = \text{ind}\,\sigma_0 + \text{ind}\,\sigma_1$$

$\Sigma(X)$ from maps



Original source details: GELFAND-P000201

Uncertain reading. The crossed sign is set as X, following the next lines. The isolated letterform leaves the case uncertain.

$$\sigma\colon T_0(X) \longrightarrow GL = \varinjlim GL(m, \mathbb{C}),$$

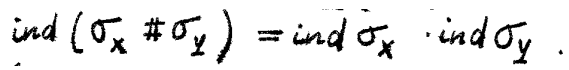
such that outside $K_\sigma \subset X$ we have $\sigma = 1$.

In the class of X whose tangent bundle is stably trivial, there exists one and only one function ind with properties Ī, II̅ and

(III̅)

$$\Sigma(X) \times \Sigma(Y) \xrightarrow{\#} \Sigma(X \times Y)$$

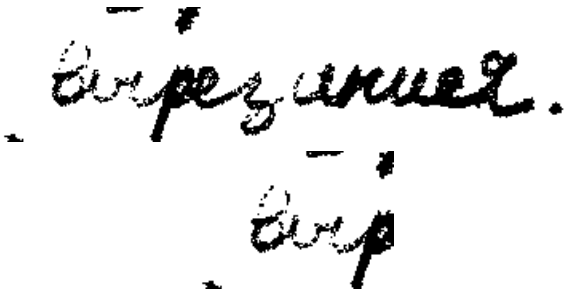
$$\text{ind}(\sigma_X \,\#\, \sigma_Y) = \text{ind}\,\sigma_X \cdot \text{ind}\,\sigma_Y.$$



Original source details: GELFAND-P000201

Uncertain reading. The subscripts match the shapes of X, Y on the product line. They are provisionally set as X, Y.

(IV̅) Invariant under excision.

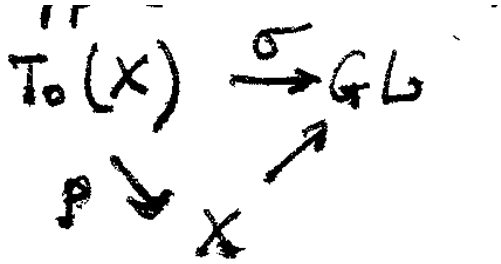


Original source details: GELFAND-P000201

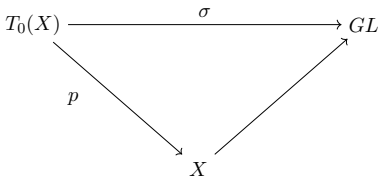
Uncertain reading. The faint opening is provisionally read as вы. The ending резания is clear.

$$\text{supp}\,\sigma \subset U \subset X \Longrightarrow \text{ind}(\sigma|U) = \text{ind}\,\sigma$$

(V)



Typeset diagram



$\Longrightarrow \quad \operatorname{ind}\sigma = 0$

Original source details: GELFAND-P000201

Uncertain reading. The label has a closed loop and a foot. The provisional reading is p. An alternative is ρ.

Original source details: GELFAND-P000201

(VI) For $X = \mathbb{R}^1$ there is the Noether-Muskhelishvili formula

$$\operatorname{ind}(\sigma) = \frac{1}{2\pi}\Delta_{\mathbb{R}^1 \times S^0} \arg\det\sigma.$$

GELFAND-P000202

20/III Continuation of Parshin's report.
The case of projective varieties.
27/III.

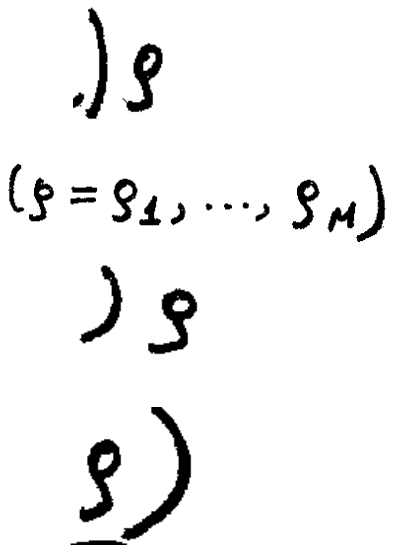
$$A(\omega,\eta_n) = A_-(\omega,\eta_n)\ M_-\ \underbrace{M_+}_{(\eta_n+i)^{n_j(\omega)}\delta_{ij}}\ A_+(\omega,\eta_n)$$

But $n_j(\omega)$ depend on ω.
And under this assumption a regularizer is constructed.

One-dimensional problem

$$\mathcal{A}(\omega, \eta_n) u_+(\eta_n) + G(\omega, \eta_n)\varrho = \ell f(\eta_n) + u_-(\eta_n)$$

$$(\varrho = \varrho_1, \dots, \varrho_M)$$



Uncertain reading. The repeated sign has a top loop and a lower hook. The provisional reading is ϱ. An alternative is s.

Original source details: GELFAND-P000202

$$\big(B(\omega, \eta_n), u_+(\,)\big)_{\eta_n} + E(\,)\varrho = g\big|_\ell.$$

Uncertain reading. The restriction stroke after g carries a looped sign followed by a point. The sign resembles ℓ on the $\ell - M$ line, with e also possible.

Original source details: GELFAND-P000202

(restr. to the bndry)

$$\ell - M = m_+(\omega) - m_-(\omega)$$

$\mathcal{A}_\omega(u_+, \varrho)$ — one-dimensional problem

Uncertain reading. The initial glyph in the lower underlined line: $\mathcal{A}_\omega(u_+, \varrho)$ / $\alpha_\omega(u_+, \varrho)$. The first candidate is in the line; the entire line has a wavy underline.

Original source details: GELFAND-P000202

It is solvable (existence and uniqueness)

$R_\omega(f, g)$ depends cont. on ω.

GELFAND-P000203

The same in the n-dimensional case: $A(\xi', \xi_n)$ also has a right and a left regularizer. (factor out $|\xi'|$).

A_+, A_- — homog. of degree 0,

ξ_-^α enters M_-.

A regul. $R^\omega(\ell f; g)$ is constructed, and then localization. (with respect to ω)

See the work of Simonenko [In the left margin: Supposedly]

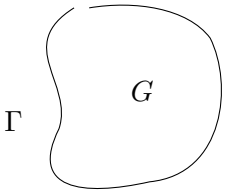
Original source details: GELFAND-P000203

Uncertain reading. Two wavy lines linked by short strokes underline the surname. A heavy vertical mark follows it. A separate dot is indistinguishable.

(in abstr. form, a disc. of inverse operators of ellipt. oper., depending on a parameter) Eskin.

Original source details: GELFAND-P000203

Typeset diagram

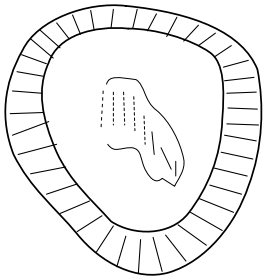


$A(x,\xi)$ — an ellipt. symbol.

A condition on the variation of $A(x,\xi)$ along the boundary.

Original source details: GELFAND-P000203

Typeset diagram



Original source details: GELFAND-P000203

Uncertain reading. Two irregular oval outlines with transverse hatching; the thin interior contour is visible with breaks. Its closure is undetermined. There are no labels or definite arrows inside.

$\Gamma_\varepsilon = \Gamma \times [0,\varepsilon)$, coord. (x', ξ', x_n, ξ_n)

$$A(x,\xi)$$

$$a_+(x') = A(x', 0, 0, 1)$$

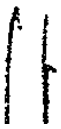
$$a_-(x') = A(x', 0, 0, -1)$$

$b(x') = a_+(x')^{-1} a_-(x')$ — measure of nonellipt.

$$\gamma(x') = \frac{1}{2\pi i} \cdot \ln b(x')$$

Typeset diagram

Original source details: GELFAND-P000203



Original source details: GELFAND-P000203

Uncertain reading. To the left of a), b) are two nearly parallel long lines with lower arrowheads. The upper left branch and right side tick are visible; whether the left branch is an upper arrowhead is undetermined.

a) Req. that the branches of ln can be chosen so that [crossed out: на / их / illegible] $\gamma_j(x')$ — are cont. on all [Γ ?]

Original source details: GELFAND-P000203

Uncertain reading. After “чтобы” (so that), approximately two letters lie beneath two thick strokes: “на” (on) / “их” (their) / illegible. The writing is crossed out.

Original source details: GELFAND-P000203

Uncertain reading. The last heavily overwritten sign in a): Γ / an overwritten illegible sign. Candidate Γ in the line is qualified by a question mark.

(locally this is always poss.)

b) $|\operatorname{Re}\gamma_i(x') - \operatorname{Re}\gamma_j(x')| < 1$ on Γ

This is the class $\widehat{\mathcal{E}}_\alpha$.

Original source details: GELFAND-P000203

Uncertain reading. The main letter beneath the hat: $\widehat{\mathcal{E}}_\alpha$ / $\widehat{\varepsilon}_\alpha$. Subscript α and upper hat are visible; the first candidate is in the line.

(For fixed x' one can — argum. in one interv. $[\lambda\,?]$ — they must not spread apart)

Original source details: GELFAND-P000203

Uncertain reading. The sign after “interv.” has a high point and a stroke curving left. It is provisionally read as λ. The sign does not establish the interval length.

$$\mathcal{A} = \begin{pmatrix} a_1(x,\xi) & & 0 \\ & \ddots & \\ 0 & & a_p(x,\xi) \end{pmatrix}$$

GELFAND-P000204

[source number: 65]

65

a: factorize — $\kappa_j(x')$.

Original source details: GELFAND-P000204

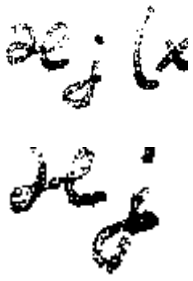
Uncertain reading. The looped glyph with subscript j and argument (x'): $\kappa_j(x')$ / $\chi_j(x')$ / $x_j(x')$. The first candidate is in the line.

γ_j will be κ_j and this class is not obtained.

Original source details: GELFAND-P000204

Uncertain reading. The sign after “будут” (will be): κ_j / χ_j / x_j. Candidate in the line: κ_j.

$$b = e^{2\pi i \kappa_j(x')}, \qquad \gamma_j = \kappa_j$$



Original source details: GELFAND-P000204

Uncertain reading. Both right-hand looped glyphs: κ_j / χ_j / x_j. Candidate in the exponent and in the right-hand equality: κ_j.

This is for oper. in ordinary spaces.

Main question: when does there globally exist $[G \in E \,/\, G \notin E \,/\, G \,\&\, E \,/\, G$ and $E\,?]$ (and B)?

Original source details: GELFAND-P000204

Uncertain reading. A heavily overwritten sign between G and E: $\in$ / $\notin$ / an ampersand / "and". All four readings remain open.

27/III Atiyah's work as presented by S. Gelfand.
(following Petrovskii)
(Report No.319 in the Bourbaki sem.).
Consider a homog. hyperb. eqn with const. coefficients

$$Pu = \sum_{|\alpha|=m} a_\alpha D^\alpha u$$

in $\mathbb{R}^n$.
a_α — real.
[Crossed out at the left: R^n / Ha(On)]

Original source details: GELFAND-P000204

Uncertain reading. An overwritten or crossed-out fragment: R^n / "Ha" (On). Neither reading is established.

Hyperbolicity: (with respect to ξ_0):

$$\sum a_\alpha \xi^\alpha = a(\xi_0, \ldots, \xi_n)$$

$\forall$ nonzero tuple $(\xi_1, \ldots, \xi_n) \neq 0$

$$a(\xi_0, \ldots, \xi_n) = 0$$

has exactly m real distinct roots.
We seek the sol. of the problem

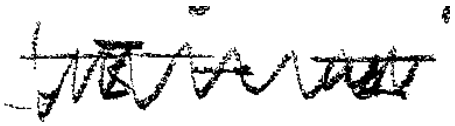
$$\begin{cases} Pu = \delta(x_0, \ldots, x_n), \\ \quad \text{with the addit. condition} \\ u = 0 \quad \text{for } x_0 < 0. \end{cases}$$

It is denoted E and is unique by the uniqueness thm for the sol. of the Cauchy problem.

It is known that it is arranged as follows:

$$\text{Cone} \qquad a(\xi_0, \ldots, \xi_n) = 0 \qquad A$$

Its dual cone B in the space $(x_0, \ldots, x_n)$.



Original source details: GELFAND-P000204

Undeciphered passage. An entire crossed-out entry is illegible. Possible candidate: “гиперпл.” (hyperpl.).

cone of hyperpl.,
tang. to A.

$$b(x_0, \ldots, x_n) = 0$$

One must take cone B, the solution is analyt. in $\forall$ connected comp. of the complement of B.

Original source details: GELFAND-P000204

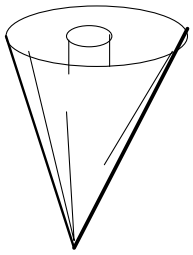
Manuscript notation. The enlarged excerpt shows дополнения, the genitive form of “complement.”

GELFAND-P000205

Domain $\overset{C}{/}$ — a lacuna if $E(x) \equiv 0$ on C.
(conn. comp.)

Original source details: GELFAND-P000205

Typeset diagram



Original source details: GELFAND-P000205

Uncertain reading. A downward-pointing cone, the upper oval, small looped contour, two short vertical strokes, and generators. The inner loop may be part of a section or a letter φ/ρ; its literal identity and the exact joins of the strokes are undetermined. No arrow directions are marked.

Stable lacuna — if it — is a lacuna for nearby eqns.

There is a nec. and suff. condition for the existence of a stable lacuna —
— certain cycles on alg. varieties are homologous to zero.
Eqn $a = 0$ — [м?]

Uncertain reading. The sign after the dash has two peaks. The provisional reading is м. Another possibility is и.

Original source details: GELFAND-P000205

$$(\xi_0, \ldots, \xi_n) \subset \mathbb{C}P^n$$

Manuscript notation. The relation has two outer strokes and no middle stroke. It reads $\subset$.

Original source details: GELFAND-P000205

$a(\xi) = 0$ — defines the variety A (complex)

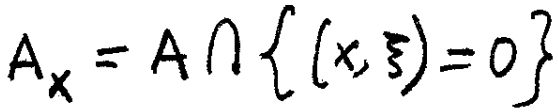
It is assumed that already in $\mathbb{C}P^n$ it is nonsingular (there are no complex singular points). This can always be achieved by a real change of coeff. (small).

$$A_{\mathbb{R}} = A \cap \mathbb{R}P(n)$$

Let there be a point $\begin{matrix} x \\ \| \\ (x_0, \ldots, x_n) \end{matrix}$. We shall construct 2 cycles from it

1.

$x_0\xi_0 + x_1\xi_1 + \ldots + x_n\xi_n = 0$ — a plane (complex)



Typeset diagram

$$\begin{matrix} A_x & = A \cap \{(x, \xi) = 0\} \\ | & \\ \text{also has} & \\ \text{a real part} & \end{matrix}$$

Original source details: GELFAND-P000205

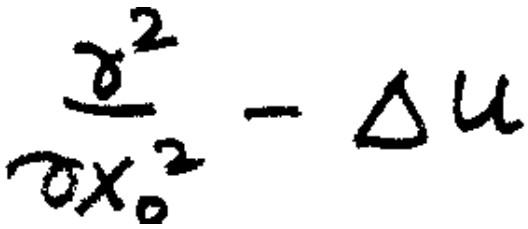
$(A_x)_{\mathbb{R}} = A_x \cap \mathbb{R}P(n)$; its dim $= n - 2$

This is a chain in homology (a cycle) — a sum of spheres:

$$\beta_x \in H_{n-2}(A_x)$$

For the wave eqn.

$$\frac{\partial^2}{\partial x_0^2} - \Delta u$$



Manuscript notation. The first fraction stands alone. After the minus sign the manuscript writes Δu.

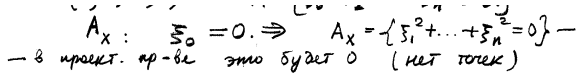
Original source details: GELFAND-P000205

$$x = (1, 0, \ldots, 0).$$

$$A = \{\xi_0^2 - \xi_1^2 - \ldots - \xi_n^2 = 0.\}$$

$$A_x : \quad \xi_0 = 0. \Rightarrow A_x = \{\xi_1^2 + \ldots + \xi_n^2 = 0\} \quad —$$

— in project. space this will be 0 (no points)



Manuscript notation. The last line says “in project. space” without specifying the field.

Original source details: GELFAND-P000205

GELFAND-P000206

[source number: 66]

2. Another cycle.

Consider the map

$$\xi \mapsto \text{sign}(x, \xi),$$
$$\xi \in A_R,$$

Typeset diagram

~~$\alpha =$ a cycle.~~

Original source details: GELFAND-P000206

Typeset diagram

~~$\partial\alpha \in (A_R)_x$~~

Original source details: GELFAND-P000206

It gives a cycle.

> Split A_R by the plane
> (x, ξ) into 2 parts
> and one with $+$, the other
> with $-$.

$$\{(x,\xi) > 0\} - \{(x,\xi) < 0\}$$
$$(a \in C_{n-1}(A))$$

We obtain an $(n-1)$-dim. chain on A $\quad (C_{n-1}(A))$

$$\partial a \in (A_x)_R \quad \text{— a cycle}$$

Thus an elem. $\alpha_x \in H_{n-1}(A, Ax)$ is defined, and

Original source details: GELFAND-P000206

Manuscript notation. In the second argument, x is on the baseline with A. The x is lowered in the following (A_x).

from the exact homology seq.:

$\partial\alpha_x \in H_{n-2}(A_x)$

[Crossed out: and / H / another sign],

Original source details: GELFAND-P000206

Uncertain reading. The densely crossed-out sign between the group and comma: “и” (and) / H / another sign.

$\partial\alpha_x = 2\beta_x$.

If x varies in a connected comp. of the cone's complement, then α_x, β_x do not change.

Original source details: GELFAND-P000206

Uncertain reading. The crossed-out parenthesis: “(in $H_{n-\ldots}$)” / “(in H_n)”; the exact index beneath the dense strokes is undetermined.

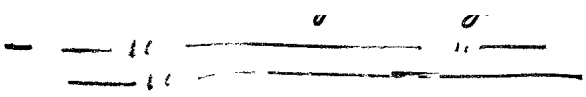
α_c, β_c — the corresp. cycles. (homology elem.).

$$\alpha_c \in H_{n-1}(A, A_c), \qquad \beta_c \in H_{n-2}(A_c)$$

Theorem. m — order of the eqn, $n+1$ — dim. of the space, m, $n+1$ are even.

a) $m \geq n+1 \Rightarrow \alpha_c = 0$ — a nec. and suff. condition for c to be a stable lacuna

b) $m < n+1 \Rightarrow \beta_c = 0$ —

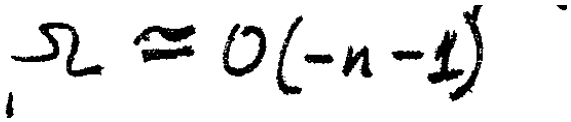


Original source details: GELFAND-P000206

Manuscript notation. Two rows of strokes and short ditto marks follow the second condition.

Proof: the explicit form of E.

On $CP(n)$ consider a form of type $(n,0)$ (holomorphic) with a single pole of order $(n+1)$ (it is unique)



Original source details: GELFAND-P000206

Typeset diagram

$$\Omega \quad \cong O(-n-1)$$

holom. forms

There are no global reg. sections. (hence uniqueness)

$$\omega = x_0\,dx_1 \wedge \ldots \wedge dx_n - x_1\,dx_0 \wedge dx_2 \wedge \ldots$$

— (in affine space) — area of the sphere.

GELFAND-P000207

Use the residue form ω_a on the surface $a = 0$.

$$\frac{\omega}{a} = \frac{da}{a} \wedge \omega_a + \Theta$$

Uncertain reading. An oval with a horizontal stroke recurs in the formula and in the parenthesis. It is provisionally read as Θ. Another reading is θ.

Original source details: GELFAND-P000207

($\Theta = 0$ on $[a = 0\,?]$)

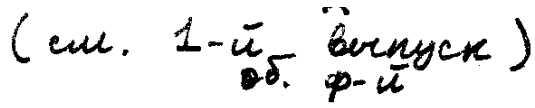
Uncertain reading. The last sign in the parenthesis is overwritten. The provisional reading is $a = 0$. An alternative is $a = \varphi$.

Original source details: GELFAND-P000207

$m \geq n+1$ — the Herglotz—Petrovskii formula

$$E(x) = c \int_{A_R} \operatorname{sgn}(x,\xi)(x,\xi)^{m-n-1}\omega_a$$

(see the 1st issue)
gen. func.



Manuscript notation. The abbreviation for “generalized functions” is written below the parenthetical reference to the first issue.

Original source details: GELFAND-P000207

Chapter 3

GELFAND-P000208

[source number: 1]

1

Year 1967. Gelfand Seminar.

18/IX. Kirillov.

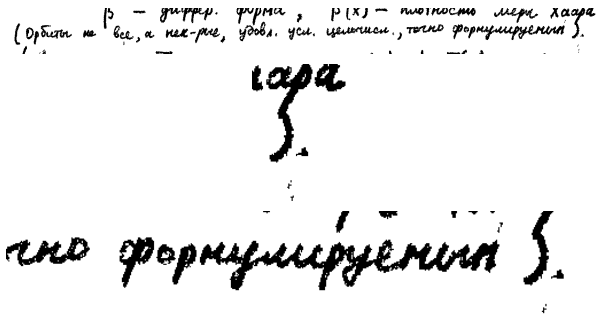
Formula (without proof or counterexamples) for characters of irreduc. repres. of Lie groups.

$$\chi(x) = p(x)^{-1/2} \int_{\Omega} e^{i\langle F,x\rangle}\beta \qquad (\beta = \beta(F))$$

$x \in \mathfrak{g}$, $F \in \mathfrak{g}^*$, Ω — an orbit in $\mathfrak{g}^*$ under G,

β — a differ. form, $p(x)$ — the density of Haar measure.

(Not all orbits, but certain ones satisfying integrality cond., precisely [formulated: ending -емый / -емым?]}.



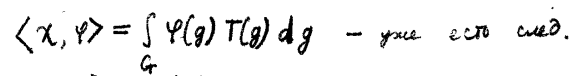
Uncertain reading. The ending appears to be -емый. The final stroke also permits -емым. A curly brace and a point follow.

Original source details: GELFAND-P000208

(G — a group, T — a unitary repres., $\chi(g) = \operatorname{tr} T(g)$ (in finite dim.)

In the general case χ — a gen. function

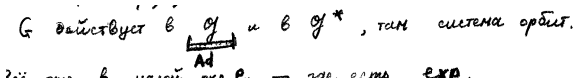
$$\langle \chi, \varphi \rangle = \int_G \varphi(g) T(g)\, dg \quad \text{— already has a trace.}$$

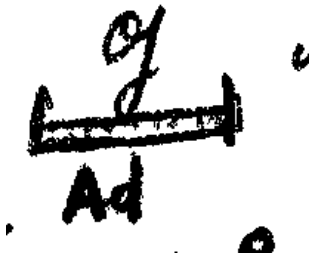


Original source details: GELFAND-P000208

Manuscript notation. The equation has no trace sign before the integral. The text after the integral says “already has a trace.”

(Such a def. of the char.).)





Original source details: GELFAND-P000208

Diagram and typed description.

G acts in $\underset{\text{Ad}}{\mathfrak{g}}$ and in $\mathfrak{g}^*$; there is a system of orbits there.

All this is in a small nbhd of e — where there is exp.

β is constructed from the struct. const. of the algebra.

[Crossed out: illegible.]

Original source details: GELFAND-P000208

Undeciphered passage. One word beneath dense repeated loops. Undetermined candidates: “Предложения” (Proposals) / “Продолжение” (Continuation). The letters beneath the strokes are illegible.

Planned.

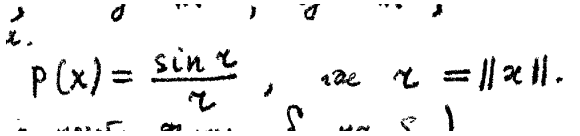
1. Expository talks on logic. (including the solution of the continuum hypothesis)
2. Lidskii “Asymptotics of eigenv. of self-adj. oper.”
3. Work of Kirillov; Kazhdan; Graev.

Example. $G = SO(3)$, $\mathfrak{g} = \mathbb{R}^3$, $\mathfrak{g}^* = \mathbb{R}^3$,

the action — rotations.

Orbit — S^2.

$$p(x) = \frac{\sin \tau}{\tau}, \qquad \text{where } \tau = \|x\|.$$

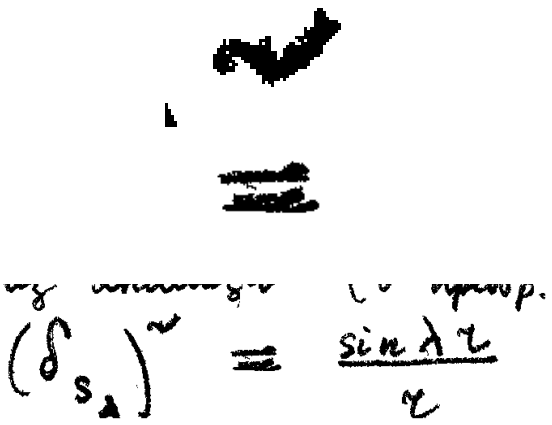


Original source details: GELFAND-P000208

Uncertain reading. The three occurrences have a hooked top and a bent stem. The provisional reading is τ. An alternative is r. The fraction has no visible power.

A fact from analysis (on the Fourier transform of δ on S_λ)

$$(\delta_{S_\lambda})^{\sim} \equiv \frac{\sin \lambda\tau}{\tau}$$



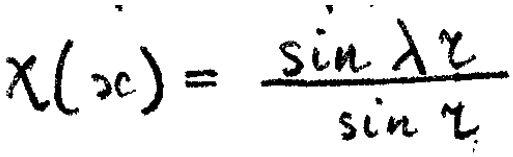
Uncertain reading. A wavy stroke $\sim$ sits above the parenthesis. The lower strokes of the equality sign run together. The provisional reading is $\equiv$. An alternative is $=$. The radial variable is the same.

Original source details: GELFAND-P000208

GELFAND-P000209

β — ordinary surface [measure] on the sphere.
Answer.

$$\chi(x) = \frac{\sin \lambda\tau}{\sin \tau}.$$



Uncertain reading. The radial grapheme is the same as on the preceding page: τ / r. Candidate in the ratio: τ; the literal letter is undetermined.

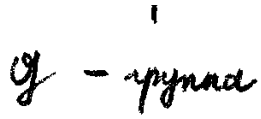
Original source details: GELFAND-P000209

All this for $\lambda = 2n + 1$ (countably many repres.)
$\beta(x)$ — $\int$ over cycles of the orbit of certain integral forms.
Work of Fuchs and Gelfand.
Fuchs. Study of principal bundles
[Crossed out: illegible.]

Undeciphered passage. A short, densely crossed-out entry between “bundles” and “with fiber”. The letters are undetermined.

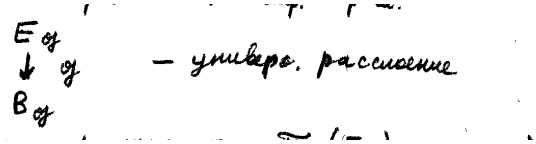
Original source details: GELFAND-P000209

with fiber G (top. group)
Types of invar. — as top. spaces and classif. spaces.
$\mathfrak{g}$ — a group



Manuscript notation. The loop and descender match $\mathfrak{g}$ on the preceding page. The earlier phrase “with fibre” uses a capital G.

Original source details: GELFAND-P000209



Original source details: GELFAND-P000209

Diagram and typed description.

$E_{\mathfrak{g}}$
$\downarrow \mathfrak{g}$ — universal bundle
$B_{\mathfrak{g}}$

($E_{\mathfrak{g}}$ [Crossed out: illegible.]

Original source details: GELFAND-P000209

Undeciphered passage. One word beneath repeated dense strokes before “aspherical”. The letters are undetermined.

aspherical, i.e. $\pi_k(E_{\mathfrak{g}}) = 0$, $k > 0$).

A char. class is determ. by specifying $\alpha \in H^q(B_{\mathfrak{g}}; A)$

(Maps of principal bundles [commute?] with the group action).

Original source details: GELFAND-P000209

Uncertain reading. The angular strokes at the end of the abbreviation are joined. The Russian reading remains перестан. or перестав.

Analogously, instead of H^q — any contravar. functor h to the category of sets.

All char. classes — $h(B_{\mathfrak{g}})$

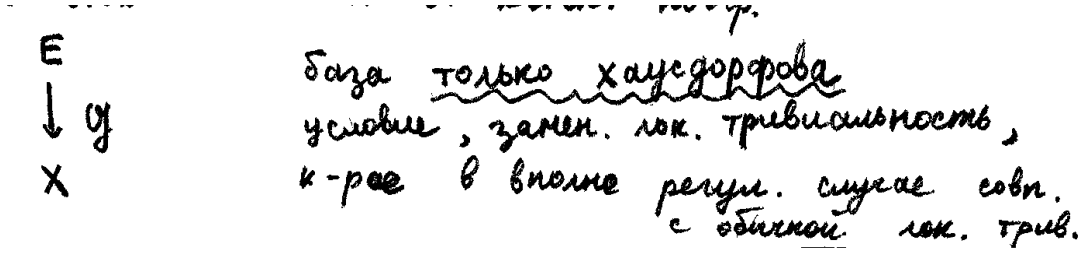
(this is if h — a homotopy functor)

$\hat{\mathfrak{g}}$ — a max. comp. subgroup

(all of them are homot. equiv. and their classif. spaces hom. equivalent.)

and equiv. to $\mathfrak{g}$.

G. Problem: find the actual classif. space of noncomp. groups, without identifying it with the max. comp. subgr.



Original source details: GELFAND-P000209

Typeset diagram

E
$\downarrow \mathfrak{g}$
X

base only Hausdorff
a condition replacing loc. triviality,
which in the completely regul. case coincides
with usual loc. triv.

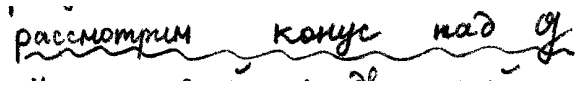
GELFAND-P000210

[source number: 2]

2

The following property is required to hold:

consider the cone over $\mathfrak{g}$

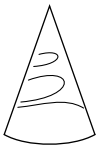


Original source details: GELFAND-P000210

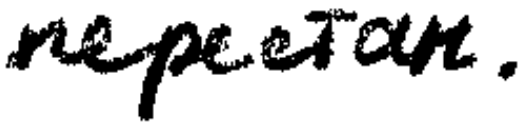
Manuscript notation. The group glyph has the same loop and descender as $\mathfrak{g}$ on the preceding page.

Original source details: GELFAND-P000210

Uncertain reading. The outer cone outline, rounded base, and internal curved paths. Closure of the paths and possible arrow directions are undetermined.



$\mathfrak{g}$ acts on it with one fixed point.
$\forall x \in X$, $\exists$ a map $E \longrightarrow C\mathfrak{g}$, commuting with the group action;



Original source details: GELFAND-P000210

Manuscript notation. The manuscript repeats перестан. before “with the group action.”

the fiber over [Crossed out: an illegible sign.]

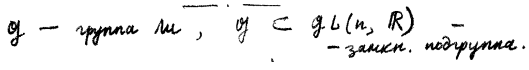
Original source details: GELFAND-P000210

Undeciphered passage. Before x a sign with a lower loop is densely crossed out. Undetermined candidates: y / g.

x maps isomorphically onto the base of the cone
(Local triviality follows from this).
(For a completely regul. base (there exists a function 1 at the point, 0 outside a given nbhd) loc. triviality $\Longrightarrow$ this property.)
(but in this case the theory becomes trivial and all computed invar. are trivial)

D. Kazhdan’s remark: then the height of the cone determines a cont. function on X.
So this def. is incorrect.

$\mathfrak{g}$ — a Lie group, $\mathfrak{g} \subset gL(n, \mathbb{R})$ — a closed subgr.

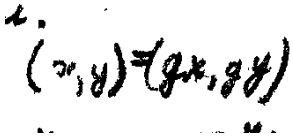


Original source details: GELFAND-P000210

Manuscript notation. The first letter before L has a loop and descender. The manuscript writes $gL(n, \mathbb{R})$.

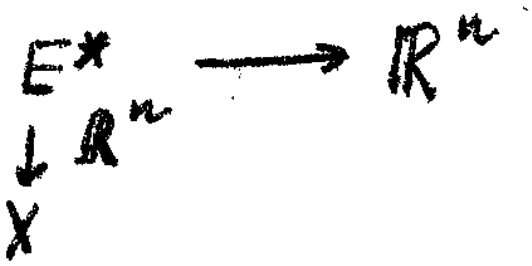
(i.e. $\mathfrak{g}$ acts on $\mathbb{R}^n$).
There exists a construction which makes a vector bundle (induced) from a principal bundle

$E^* = E \times \mathbb{R}^n/\mathfrak{g}$.
$(x, y) = (gx, gy)$



Original source details: GELFAND-P000210

Manuscript notation. Two strokes form $=$ before the second pair and touch its opening parenthesis.



Original source details: GELFAND-P000210

Diagram and typed description.

$$\begin{array}{ccc} E^* & \longrightarrow & \mathbb{R}^n \\ \downarrow & & \\ X & & \end{array} \quad \mathbb{R}^n$$

There must exist a map $E^* \longrightarrow \mathbb{R}^n$,
[Crossed out: illegible.]

Original source details: GELFAND-P000210

Undeciphered passage. A short word before “linear” is crossed out by two strokes. The letters are undetermined.

linear on the fibers and
such that at a certain point $\underset{\text{(given)}}{x_0}$
it — an isomorphism.
(No compatibility with the group action).

this is the def. of loc. triviality (for Hausd.) and this is precisely the requirement.
Kazhdan’s remark cannot be repeated.

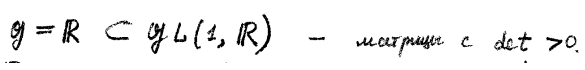
GELFAND-P000211

E is recovered from E^*.
One cannot construct a classif. space, but there is an analogue.
Example:
[An undetermined thickened sign.]

Original source details: GELFAND-P000211

Uncertain reading. Before the group notation is a separate thickened sign with a lower loop and a transverse stroke: “в” (in) / a crossed-out earlier sign. Neither candidate is established.

$\mathfrak{g} = \mathbb{R} \subset gL(1, \mathbb{R})$ — matrices with det > 0.



Original source details: GELFAND-P000211

Manuscript notation. The same looped letter with a descender as on manuscript page 2 precedes L. The manuscript writes gL.

A nontrivial $\mathbb{R}$-bundle (also "universal").
(principal)

The direct product of countably many lines in the Tych. topology.

$$(x_1, x_2, \ldots) \quad \text{— sequences}$$

$$\alpha(x_1, \ldots) = (x_1 e^{\alpha}, \ldots)$$

The quotient space S cannot be embedded in E (!) — — there are no sections.

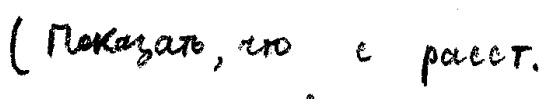
Fact: in $\mathbb{R}^{\infty}$ one cannot choose one point on each ray.

Proof.

S is Hausdorff, but every cont. function on it is const,

so such a map — to a point (coords. — cont. functions).

(Show that with the distance

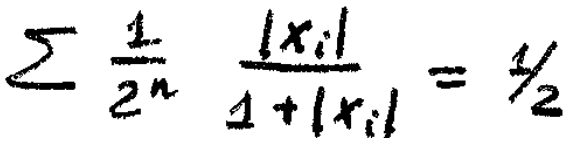


Original source details: GELFAND-P000211

Manuscript notation. The manuscript writes расст., an abbreviation for "distance." The two c letters are followed by т.

$$\sum \frac{1}{2^n} \frac{|x_i|}{1 + |x_i|} = 1/2$$

Manuscript notation. The weight is written with 2^n. Both occurrences of x have subscript i.



Original source details: GELFAND-P000211

one obtains a discontinuous map.)

Suppose there is f. (a cont. function).

$$(x_1, x_2, \ldots)$$

$$a_i = f(\underbrace{0, \ldots, 0}_{i}, 1, \ldots, 1, \ldots)$$

$$a = f(x_1, x_2, \ldots)$$

We shall prove that $a = \lim_{i \to \infty} a_i$.

$\forall \varepsilon > 0$ $\exists$ a nbhd of the point

[Crossed out: an illegible sign.]

Original source details: GELFAND-P000211

Undeciphered passage. Before the coordinate parenthesis a sign is densely crossed out. The original grapheme is undetermined.

$(x_1, x_2, \ldots) \in S$,
where the difference is $< \varepsilon$. (from a).

$$(x_1^0, x_2^0, \ldots)$$

(A neighborhood — if the preimage is open)

GELFAND-P000212

[source number: 3]

3

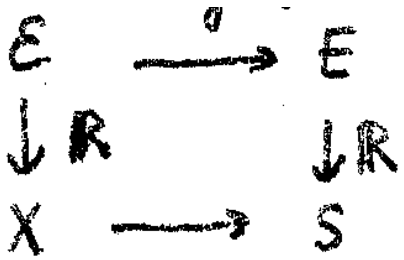
A neighborhood in E

$$|x_1 - x_1^0| < \delta$$
$$\ldots$$
$$|x_N - x_N^0| < \delta.$$

Keep these coordinates the same and make the others very large.

$$\left(\frac{x_1^0}{M}, \ldots, \frac{x_N^0}{M}, 1, 1, \ldots\right),$$

and these approximate the point $(0, \ldots, 0, 1, \ldots, 1, \ldots)$.
A map always exists (but is not unique)



Original source details: GELFAND-P000212

Uncertain reading. The upper-left vertex is rounded: calligraphic $\mathcal{E}$ / ε. Provisional reading: $\mathcal{E}$; the letter is undetermined. Both horizontal arrows point right, both vertical arrows down with labels $\mathbb{R}$.

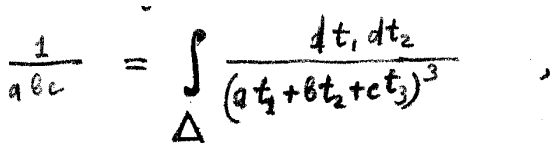
$$\begin{array}{ccc} \mathcal{E} & \longrightarrow & E \\ \downarrow{\scriptstyle \mathbb{R}} & & \downarrow{\scriptstyle \mathbb{R}} \\ X & \longrightarrow & S \end{array}$$

S is constructed from a product of $\mathbb{R}$ of any cardinality (depends on X)
$25/\overline{\text{IX}}$.
A good formula (Feynman)

$$\frac{1}{ab} = \int_0^1 \frac{dt}{(at + b(1-t))^2}$$

(decomp. into partial fractions)

$$\frac{1}{abc} = \int_{\Delta} \frac{dt_1\, dt_2}{(at_1 + bt_2 + ct_3)^3},$$



Manuscript notation. The integral immediately follows the equals sign. The manuscript shows no numerical multiplier.

Original source details: GELFAND-P000212

Δ — a simplex

$$t_1 + t_2 + t_3 = 1, \quad t_i \geq 0.$$

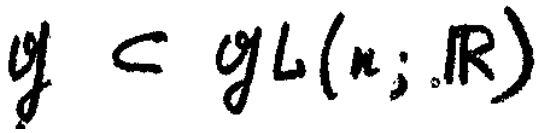
Next time Fuchs and then logic.

There is a counterexample to the hypothesis of a common fixed point of commuting cont. maps of an interval, and it is proved that if they are differentiable, there is a point fixed for one and of finite multiplicity for the other.

GELFAND-P000213

Fuchs (continuation).

$\mathfrak{g} \subset gL(n;\mathbb{R})$



Manuscript notation. Both group letters have a loop and a descender. A semicolon follows n.

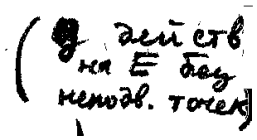
Original source details: GELFAND-P000213

principal bundles

$$(\xi) \qquad p : E \longrightarrow X$$
$$(\xi^{\#}) \qquad p^{\#} : E^{\#} \longrightarrow X$$

(X — Hausdorff)

($\mathfrak{g}$ acts on E without fixed points]



Uncertain reading. The right-hand note ends with an upright stroke that hooks inward. It is provisionally set as].

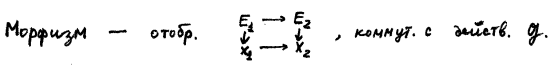
Original source details: GELFAND-P000213

(

$$xg_1 = xg_2 \Rightarrow$$
$$\Rightarrow g_1 = g_2$$

Loc. triviality: $\forall x \; \exists$ a map $E^{\#} \longrightarrow \mathbb{R}^n$, an isom. at the point x, linear on the fibers.

(One cannot require 0 outside small nbhds, since then even $X \times \mathbb{R}^n$ may fail to qualify because of nonregularity)



Original source details: GELFAND-P000213

Diagram and typed description.

A morphism — a map

$$\begin{array}{ccc} E_1 & \longrightarrow & E_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

, commuting with the action of $\mathfrak{g}$.

Examples of functors:

1) cohomology

2) H-bundles (H — another group)

correspondingly charact. classes.

$h_{\text{alg}}(\mathfrak{g})$ — the set (group) of all char. classes constructed from the functor h.

1) For any top. space

[Crossed out: an illegible sign.]

Original source details: GELFAND-P000213

Undeciphered passage. After “пр-ва” (space), before X, an additional sign is densely crossed out. The original grapheme is undetermined.

X, the sheaf $\mathbb{R}$ of germs of cont. real functions on X is defined.

Put

$$h(X) = H^q(X, \mathbb{R}).$$

(for [Crossed out: illegible.]

Original source details: GELFAND-P000213

Undeciphered passage. A fragment between “for” and “paracomp.” is crossed out; a descender is visible. The letters are undetermined.

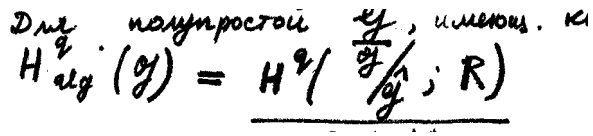
paracomp. X this is 0)

There is defined

$$H^q_{\text{alg}}(\mathfrak{g})$$

Answer. For semisimple $\mathfrak{g}$ having finitely many connected components

$$H^q_{\text{alg}}(\mathfrak{g}) = H^q(\bar{\mathfrak{g}}/\hat{\mathfrak{g}}; R)$$



Original source details: GELFAND-P000213

Manuscript notation. A straight bar marks the first group glyph and a hat marks the second. The descender on the line above touches the bar.

with coeffs. in a constant sheaf —
— real coeffs.

GELFAND-P000214

[source number: 4]

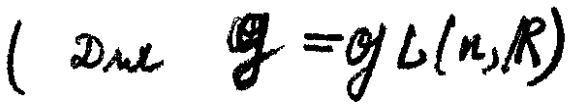
4

Here $\bar{\mathfrak{g}}$ — a compact form of $\mathfrak{g}$
$\hat{\mathfrak{g}}$ — a max. comp. subgroup
[Crossed out: $\bar{\mathfrak{g}}$?]

Original source details: GELFAND-P000214

Undeciphered passage. Before the final group sign an entry is densely crossed out. Candidate $\bar{\mathfrak{g}}$ from the visible upper bar and lower loop; the precise form beneath the strokes is undetermined.

$\mathfrak{g}$. (For $\mathfrak{g} = gL(n, R)$



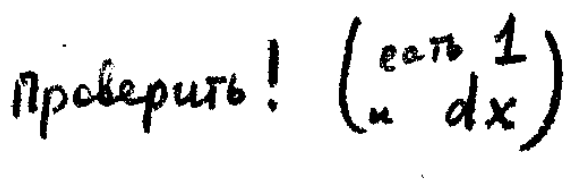
Original source details: GELFAND-P000214

Manuscript notation. The letter before L repeats the loop and descender on the preceding page. The manuscript writes gL.

there will be $\bar{\mathfrak{g}} = U(n)$, $\hat{\mathfrak{g}} = O(n)$
$\bar{\mathfrak{g}}$ is obtained as a max. comp. subgroup of the complexification of $\mathfrak{g}$)
2nd order: as for S^2.
For a compact group everything is trivial.
For a nonsemisimple one, take $\mathfrak{g}/\hat{\mathfrak{g}}$ and differ. forms on $\mathfrak{g}/\hat{\mathfrak{g}}$ invariant under $\mathfrak{g}$.
d(invar.) — invar. And the homologies of this complex are isomorphic to $H^q_{\text{alg}}(\mathfrak{g})$.

$$H^q_{\text{alg}}(\mathbb{R};\mathbb{R}) = \begin{cases} R & q = 0,1 \\ 0 & q > 1. \end{cases}$$

Check! (there are 1 and dx)



Manuscript notation. The "Check!" note sits to the right of the formula, with 1 above dx inside parentheses.

Original source details: GELFAND-P000214

2) X, H — a top. group

$EH(X)$ — the set of all loc. triv. (in the usual sense) bundles (principal) on X.

\

Theorem.

$$EH_{\text{alg}}(\mathfrak{g}) = \text{Hom}(\mathfrak{g}, H)$$

(cont. hom. [Crossed out: an illegible sign.]

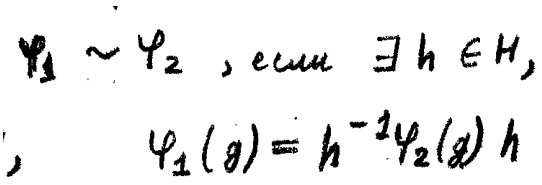
Undeciphered passage. After "cont. hom." are doubled curved and oblique strokes: a cancelled) / the sign φ. The reading is undetermined.

Original source details: GELFAND-P000214

, more precisely, classes of conjugate homom.)

The natural answer — a hom. $\mathfrak{g} \longrightarrow H$ corresponds to a char. class, naturally defined.

$$\varphi_1 \sim \varphi_2, \text{ if } \exists h \in H, \qquad \varphi_1(g) = h^{-1}\varphi_2(g)h$$

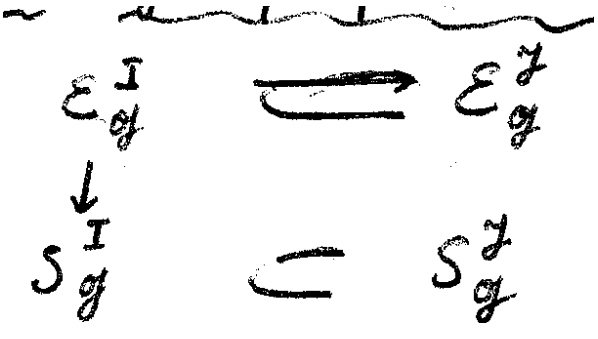


Uncertain reading. The map glyphs in the formula have a closed right loop and an open left arm. The reading φ is preferred.

Original source details: GELFAND-P000214

GELFAND-P000215

Construction of a universal bundle.



Uncertain reading. The upper stroke ends in a rightward arrowhead. The lower stroke curves upward on the left. The redraw follows both strokes as drawn. The rounded upper letters are rendered as $\mathcal{E}$.

$$\begin{array}{ccc} \mathcal{E}^I_{\mathfrak{g}} & \rightleftharpoons & \mathcal{E}^J_{\mathfrak{g}} \\ \downarrow & & \\ S^I_{\mathfrak{g}} & \subset & S^J_{\mathfrak{g}} \end{array}$$

Original source details: GELFAND-P000215

$I \subset J$ — arbitrary sets.

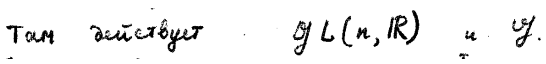
Any bundle maps into a universal one with I of sufficiently large cardinality.

Consider $\mathbb{R}^I \ni (x_i,\ i \in I)$.

The topology — Tychonoff.

and $\underset{n \text{ lin. indep. vect.}}{n\text{-frames}}$ in it — this is precisely $\mathcal{E}^I_{\mathfrak{g}}$.

$gL(n,\mathbb{R})$ and $\mathfrak{g}$ act there.



Original source details: GELFAND-P000215

Manuscript notation. The looped letter with a descender recurs before L. It is set as gL.

The quotient space is precisely our $S^I_{\mathfrak{g}}$.

Observe that $\mathcal{E}^I_{\mathfrak{g}}$ does not depend on $\mathfrak{g}$.

Lemma. $\forall \mathcal{E}^I_{\mathfrak{g}}$ and a cont. function f on it $\exists$

$J \subset I$ such that $I \setminus J$ is at most countable, and

and $f|_{\mathcal{E}^J_{\mathfrak{g}}} = \text{const}$.

GELFAND-P000216

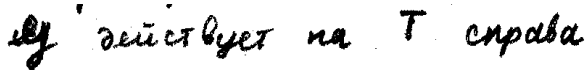
[source number: 6]

6

$2/\overline{\text{X}}$. Fuchs.

T — a space, $S \subset \text{Map}(T,T)$, compact open top.

$\mathfrak{g}$ acts on T on the right



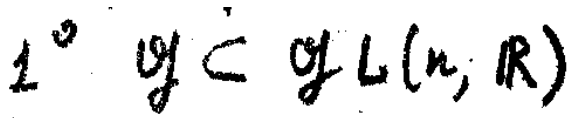
Original source details: GELFAND-P000216

Manuscript notation. The group sign has a loop and descender. Here it differs from the simpler g in the next equation.

$$f \in S, \quad g \in \mathfrak{g} \Rightarrow fg \in S,\ gf \in S.$$

Examples.

1°. $\mathfrak{g} \subset gL(n;\mathbb{R})$,



Original source details: GELFAND-P000216

Manuscript notation. The letter before L has a loop and descender. A semicolon follows n. The left stroke of $\mathbb{R}$ is doubled.

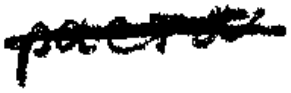
$T = \mathbb{R}^n$; S — all lin. maps (including zero maps).

2°. $\mathfrak{g}$ — a Hausdorff top. group, $T = C\mathfrak{g}$ — the cone over $\mathfrak{g}$

Original source details: GELFAND-P000216

Diagram and typed description.

$\mathfrak{g}$ acts naturally, S is generated by them and similarities [Crossed out: растя... / растяже...? (dila...?)]



Undeciphered passage. Before “contractions” a word or beginning of a word is crossed out; the opening “раст...” is distinguishable. The ending is undetermined.

Original source details: GELFAND-P000216

contractions toward the vertex (and to a point).

Axioms.

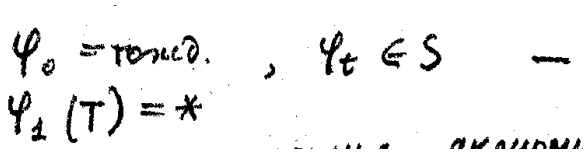
1°. T is contractible by transformations in S to a marked point $*$.

2°. $\mathfrak{g}$ is closed in the group of all homeom. of T to itself.

3°. $*$ is fixed under all transf. in S.

4°. $\varphi_t : T \longrightarrow T$

$$\varphi_0 = \text{identity}, \quad \varphi_t \in S \quad —$$
$$\varphi_1(T) = *$$



Manuscript notation. The map sign has an open left arm and a closed right loop. It reads φ.

Original source details: GELFAND-P000216

— the homotopy of axiom 1°.

Then there exist φ_t such that φ_t for $t < 1$ are topological embeddings.

5°. S contains all transf. of $\mathfrak{g}$, and maps close to maps of $\mathfrak{g}$ in S are top. embeddings.

$U(1) \subset S$ consists of embeddings.

In this situation spaces are defined

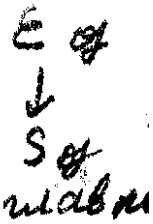


Diagram and typed description.

$$\begin{array}{c} \mathcal{E}_{\mathfrak{g}} \\ \downarrow \\ S_{\mathfrak{g}} \end{array}$$

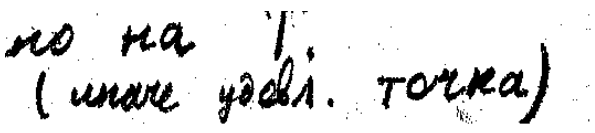
, the action of $\mathfrak{g}$ on $\mathcal{E}_{\mathfrak{g}}$ — a universal principal bundle.

Original source details: GELFAND-P000216

$\underbrace{T \times \cdots \times T}_{I \text{ times}}$ Tych. topology.

6°. $\mathfrak{g}$ acts effectively on T,

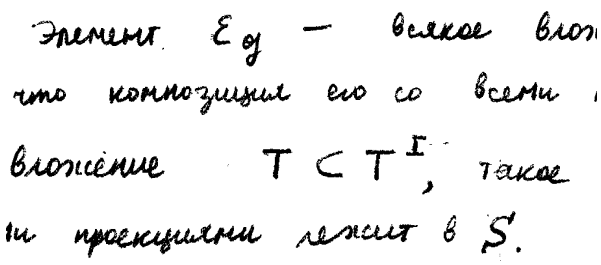
(otherwise [удвл.?] point)



Uncertain reading. The parenthesis provisionally reads “удвл.”. The expansion is undetermined.

Original source details: GELFAND-P000216

An element of $\mathcal{E}_{\mathfrak{g}}$ — any embedding $T \subset T^I$ such that its compositions with all projections lie in S.



Manuscript notation. The subscript of the rounded $\mathcal{E}$ has the same loop and descender as the group sign on the preceding page.

Original source details: GELFAND-P000217

1) $T = \mathbb{R}^n$ — all n-frames.

2) $T = C\mathfrak{g}$, $\mathcal{E}_{\mathfrak{g}} = C\mathfrak{g} \times \ldots \times C\mathfrak{g} \setminus \{*, \ldots, *\}$.

$\mathfrak{g}$ acts on $\mathcal{E}_{\mathfrak{g}}$ by multiplication by transf. in $\mathfrak{g}$.

I finite $\Rightarrow$ 1) the usual classif. space
2) Milnor's constr. with a not very substantial modification.

Diagram and typed description.

$\mathfrak{g} = \mathbb{R}$, $T =$ •————

Original source details: GELFAND-P000217

$\mathcal{E}_{\mathfrak{g}}$ — sequences of nonneg. numbers.

For each T its own definition of loc. triviality.

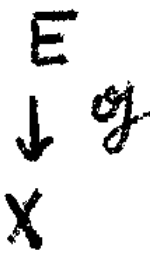


Diagram and typed description.

$$\begin{array}{c} E \\ \downarrow \mathfrak{g} \\ X \end{array}$$

Original source details: GELFAND-P000217

is loc. triv. if the induced bundle

Diagram and typed description.

$$\begin{array}{c} E^{\#} \\ \downarrow T \\ X \end{array}$$

Original source details: GELFAND-P000217

, has the property: $\forall x \; \exists$ a map $E^{\#} \longrightarrow T$, a homeom. on the fiber over x, and on $\forall$ fiber it is a transf. in S.

Theorem.

Every bundle maps into this one.

(X — all Hausdorff)

Defined $H^q_{\text{alg}}(\mathfrak{g};\mathbb{R})$

Theorem.

Whatever T and S may be, $H^q_{\text{alg}}(\mathfrak{g},\mathbb{R})$ does not depend on their choice

(Instead of $\mathbb{R}$ one may take a group A)

What will there be?

Uspenskii, Lectures on logic 2

Lidskii, Asymptotics of eigenfns. (after Hörmander)

Sinai, Conditions on diffeom. under which there is an invar. measure

Kazhdan, Discr. groups — solution of Selberg's problem

Kushnirenko, Analyt. action of a semisimple group in a nbhd. of a fixed point — linear

GELFAND-P000218

[source number: 7]

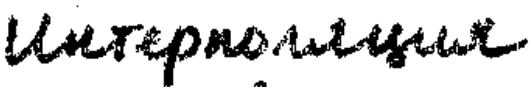
Tyurina and S. Gelfand survey of Pham's works.

Anosov and Maslov

I. Bernstein P^λ

Gelfand, Graev

S. G. Krein Interpolation



Uncertain reading. The final strokes favour the singular "Интерполяция" (Interpolation).

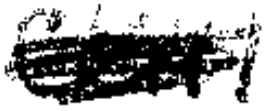
Original source details: GELFAND-P000218

Berezin
Kirillov' Grassmann analysis.

— • —

GELFAND-P000219

16/X



Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P000219

S. G. Krein.

On interpolation of linear operators.

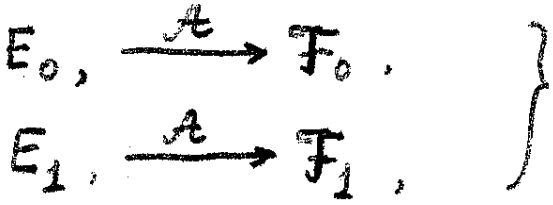
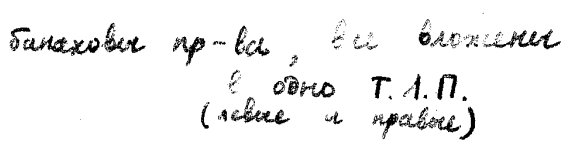


Diagram and typed description.

$$\left.\begin{array}{l} E_0, \xrightarrow{A} F_0, \\ E_1, \xrightarrow{A} F_1, \end{array}\right\}$$ Banach spaces,

Original source details: GELFAND-P000219

all are embedded into one T. L. S. (the left and the right ones)



Original source details: GELFAND-P000219

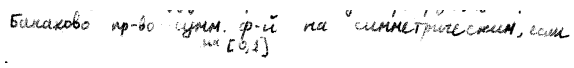
Manuscript notation. The last word reads вложены (“are embedded”).

A is bounded.

E, F — an interpolation pair if, for $\forall$ A, boundedness at 0 and 1 implies boundedness of $A : E \longrightarrow F$.

All spaces will be spaces of summable functions.

A Banach space of summable functions [Inserted below the line: on $[0, 1]$.] is called symmetric if



Original source details: GELFAND-P000219

Manuscript notation. The main line has the abbreviation на (“is called”). A separate insertion below reads “on $[0, 1]$”.

1) $y(t) \in E$

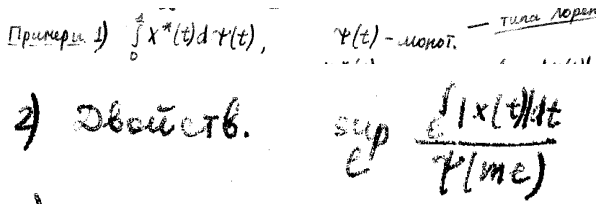
$$|x(t)| \le |y(t)| \quad \Longrightarrow \quad x \in E \qquad \text{and} \qquad \|x\| \le \|y\|$$

2) The norm does not change under transformations preserving measure (one may consider an equimeasurable monotone one)

Then $L_\infty \subset E \subset L_1$.

Examples. 1)

$$\int_0^1 x^*(t)\, d\psi(t), \qquad \psi(t) \text{ — monot.} \qquad \text{— of Lorentz type}$$



Original source details: GELFAND-P000219

Uncertain reading. The same three-armed sign occurs in the differential, the monotonicity condition and the denominator. The provisional reading is ψ. An alternative is Ψ.

$x^*(t)$ — a rearrangement of $|x(t)|$

2) Dual. $\sup\limits_e \dfrac{\int_e |x(t)|\, dt}{\psi(me)}$

3) and Orlicz spaces (these include L_p)

[At the left above the line:] 1st step.

Riesz–Thorin theorem.

GELFAND-P000220

2nd step.

Marcinkiewicz theorem.

(Less is required at the endpoints)

$$n_x(\tau) = \text{mes}\{t : |x(t)| \ge \tau\}$$

$$[n_{Ax}(\tau)]^{1/q_i} \le \frac{C}{\tau} \|x\|_{L_{p_i}}$$

— a less restrictive condition.

Most operators of analysis satisfy the Marcinkiewicz conditions at the endpoints

Matsaev and Dikarev — more precisely, from L_p into a space narrower than L_q (of Lorentz type)! — and it is already minimal (in the middle)

(Minimal for all operators under conditions both of Marcinkiewicz and of Riesz–Thorin) —

— this is for improving operators:

$$\begin{matrix} q_1 > p_1 \\ q_2 > p_2 \end{matrix} \quad \Big\| \quad \text{— as in the Marcinkiewicz theorem.}$$

$p = q$ — maps into itself — then all the other scales do too.

Let $\begin{matrix} L_1 \longrightarrow L_1 \\ L_\infty \longrightarrow L_\infty \end{matrix}$ and $\|A\| \le 1$

List all [Crossed out: illegible.] spaces for which $\|A\| \le 1$ as well.

Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P000220

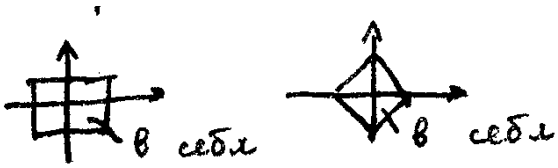
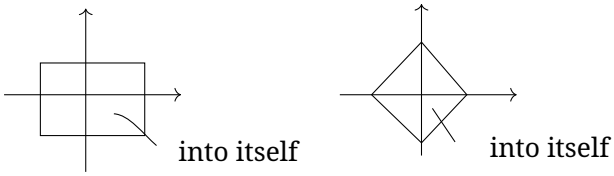


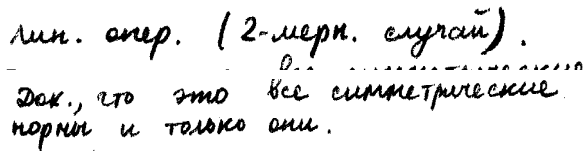
Diagram and typed description. The extra stroke curves at the rectangle and runs obliquely at the diamond. Their directions remain uncertain.



Original source details: GELFAND-P000220

[To the right of the figures:] linear oper. (2-dimensional case).

Док. [prove / proved], that these are all symmetric norms and only they.



Uncertain reading. The abbreviation “Док.” is readable; the full verb form is not written. Possible expansions: prove / proved.

Original source details: GELFAND-P000220

General theorem — B. S. Mityagin.

It is necessary and sufficient that the space be symmetric.

(also in the ∞-dimensional case)

$L_p \xrightarrow{A} L_p$ what then?

$p \neq 1, \infty$

(Marcinkiewicz case)

$$\Big\| \underset{\substack{\text{char. fn. } e \\ \uparrow}}{\chi_e} \Big\|_E = \varphi(me).$$

(symmetry is again necessary)

the resulting function $\varphi(t)$ — a characteristic of the space E.

$$\ell_E = \underline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad u_E = \overline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad 1 \le \ell_E \le u_E \le 2$$

GELFAND-P000221

[source number: 9]

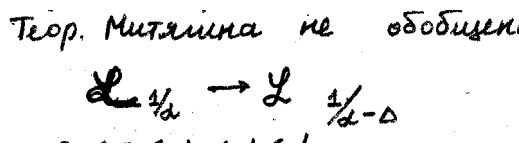
For L_p they are both equal to $2^{1/p}$

Boundedness of all such A in $E \iff 1 < \ell_E \le u_E < 2$

(Semyonov's 1st theorem)

Mityagin's theorem has not been generalized.

$$L_{1/\alpha} \longrightarrow L_{\frac{1}{\alpha-\Delta}}$$



Original source details: GELFAND-P000221

Uncertain reading. The minus sign and Δ sit on the lower level with α after the slash. The preferred reading is $1/(\alpha - \Delta)$. The manuscript has no parentheses.

$0 < \Delta \le \alpha_0 < \alpha < \alpha_1$.

How can one list the spaces in which operators of this type are bounded?

$$2^{\alpha_0} < \ell_E \le u_E < 2^{\alpha_1} \quad \Longrightarrow \quad E \longrightarrow E_\Delta,$$

where

Original source details: GELFAND-P000221

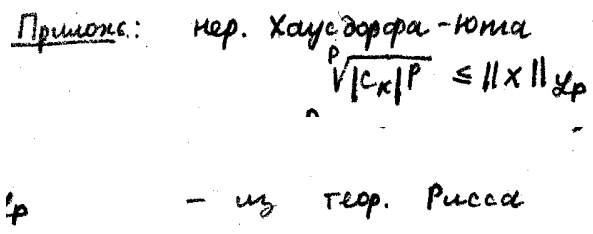
Uncertain reading. After "where" there are two dense short signs; their exact reading and whether they are crossed out are undetermined.

$$\|x\|_{E_\Delta} = \|x^*(t)t^{-\Delta}\|_E$$

(This generalizes the theorem of Dikarev and Matsaev — the best space)

Applic.: Hausdorff–Young inequality

$$\sqrt[p]{|c_k|^p} \le \|x\|_{L_p} \qquad \text{— from Riesz's theorem}$$



Original source details: GELFAND-P000221

Manuscript notation. The first radicand reads $|c_k|^p$. A summation sign is visible only under the next radical.

Poia inequality:

$$\sqrt[p]{\sum k^{p-2}c_k^{*p}} \le C\|x\|_{L_p}$$

c_k^* — the monotone rearrangement of the $|\ |$ coefficients.

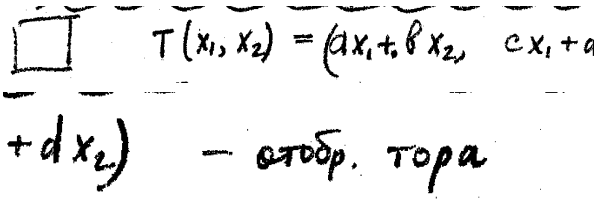
For monotone decreasing Fourier coefficients — two-sided inequalities (Hardy–Littlewood)

Between L_1 and $L_{1,5}$ there is no Hilbert interpolation space (Semyonov)

Gelfand's question: an abstract description of interpolation spaces. (it is unknown whether they are spaces of functions.)

Ya. Sinai.

Construction of invar. measures of smooth maps Markov partitions



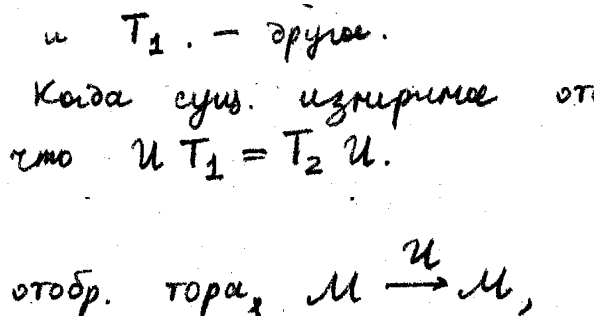
Original source details: GELFAND-P000221

Diagram and typed description.

$T(x_1, x_2) = (ax_1 + bx_2,\ cx_1 + dx_2)$ — a map of the torus

and T_1. — another.

When does there exist a measurable map of the torus, $M \xrightarrow{U} M$, such that $UT_1 = T_2U$.



Original source details: GELFAND-P000221

Uncertain reading. The description has T and T_1. The equation has T_1 and T_2. The looped glyph above the arrow recurs on both sides of the equation. The provisional reading is U.

Irrational eigenvalues $\Rightarrow$ real $\Rightarrow$ ergodic, the spectrum is Lebesgue of countable multiplicity.

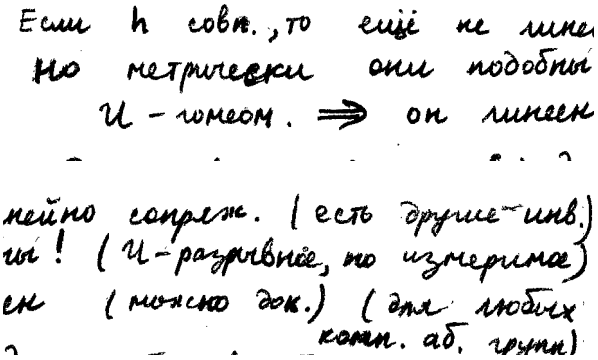
Entropy $h(T) = \ln \lambda \quad (\lambda > 1)$

GELFAND-P000222

If h agrees, this does not yet give linear conjugacy (there are other-inv.)

But metrically they are similar! (U — discontinuous, but measurable)

U — a homeom. $\Longrightarrow$ it is linear (one can prove this) (for any compact abelian groups)



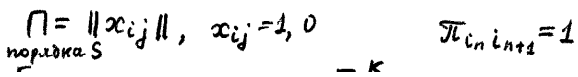
Original source details: GELFAND-P000222

Uncertain reading. The looped map sign in both lines: U / u. Candidate: U; letter case is undetermined.

Ω_{II} — the space of seq. $\omega = \{i_n\}$ — sequences infinite in both directions.

$i_n \in \{1, \dots, S\}$

$$\underset{\text{of order } S}{\Pi} = \|x_{ij}\|, \quad x_{ij} = 1, 0 \qquad \pi_{i_n i_{n+1}} = 1$$



Original source details: GELFAND-P000222

Manuscript notation. The matrix definition writes x_{ij}. The admissibility condition writes $\pi_{i_n i_{n+1}}$.

(only such sequences ω are admissible.)

We shall assume that Π^k consists of positive entries.

Consider the shift (the shift transformation on Ω).

Example. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — does not satisfy the cond., but Ω — sequences only of 1 or [Crossed out] only of 2

Original source details: GELFAND-P000222

Uncertain reading. A densely crossed-out sign follows “or”. Its reading is undetermined.

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ — anything may be put in

$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ — look into it.

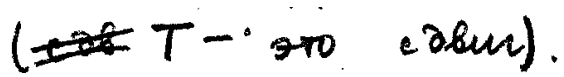
on Ω_Π we would like to specify an invariant measure μ with max entropy:

Original source details: GELFAND-P000222

Uncertain reading. [An illegible crossed-out entry to the left of the formula.]

$$h = \max_{\mu} h_\mu(T)$$

([Crossed out] T — this is the shift).

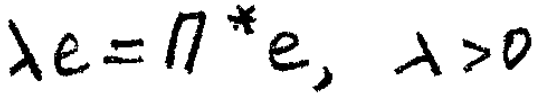


Original source details: GELFAND-P000222

Uncertain reading. A looped ending is visible beneath two diagonal cancellation strokes. The readings “where” and “shift” remain tentative.

One must consider e with positive coordinates. $\lambda e = \Pi^* e$, $\lambda > 0$

Manuscript notation. The star is at the upper right of Π.



Original source details: GELFAND-P000222

(unique.)

(G. If all entries and all minors of order 2 are > 0, then there exist 2 eigenvectors, the 1st with positive entries, the 2nd — with the 1st change of sign —

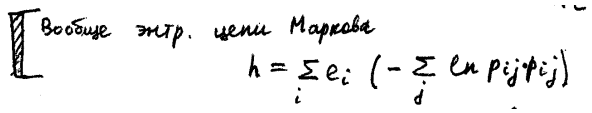
— the matrix of minors of order 2 — a representation of the matrix itself in the exterior product.)

Consider a Markov chain with probabilities $p_{ij} = \dfrac{e_j \pi_{ij}}{\lambda e_i}$ —

[At the right:] — entropy — $\ln \lambda$

In general, the entropy of a Markov chain

$$h = \sum_i e_i \left(- \sum_j \ln p_{ij} \cdot p_{ij} \right)$$



Manuscript notation. The bracket at left is hatched. The outer sum has e_i. A dot separates $\ln p_{ij}$ from p_{ij}.

Original source details: GELFAND-P000222

the measure of max entropy is unique.

The measure in Ω_Π on the Σ-alg. generated by cylindrical sets.

GELFAND-P000223

[source number: 10]

Diagram and typed description. Page number 10 with a line at its left and below it.

10

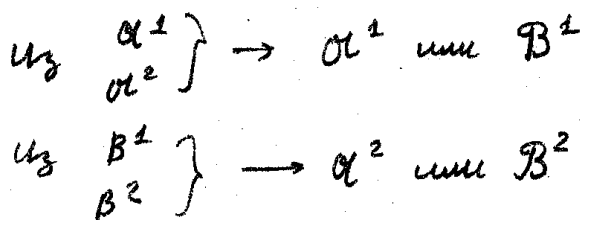
Original source details: GELFAND-P000223

a) $\{a_1^1, \ldots, a_p^1;\ a_1^2, \ldots, a_q^2;\ b_1^1, \ldots, b_r^1;\ b_1^2, \ldots, b_s^2\}$

Π

$$\text{from } \left.\begin{matrix}\alpha^1 \\ \alpha^2\end{matrix}\right\} \longrightarrow \alpha^1 \text{ or } B^1$$

$$\text{from } \left.\begin{matrix}\beta^1 \\ \beta^2\end{matrix}\right\} \longrightarrow \alpha^2 \text{ or } B^2$$



Uncertain reading. Both transitions have an uppercase B with an upper loop on the right. The left-hand forms resemble β on the second line and α on the first. Case on the left is uncertain.

Original source details: GELFAND-P000223

[In the right margin:] ‖ a partition is constructed from an integer matrix of order 2.

$$\lambda^2 - (p+s)\lambda + (ps - qr) = 0, \qquad \boxed{ps - qr = 1}$$

condition.

b) Ω_{Π_1} $\pi_{ij} = 1$ and $\pi_{N1} = 0$ (only in the 1st position)

Prop. The shifts in a) and b) are isomorphic.

⌈Everything now reduces to proving that
an algebraic automorphism ⇒
⇒ a partition.

Proof of the proposition.

Alphabet b): $A_1, \dots, A_p, \quad B_1, \dots, B_s$.

(Isomorphism of any 2 Ω_{Π} with the same entropy).

Different groups are encoded (using the symbols A_i, B_i).

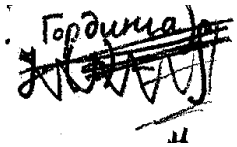
The space with the 1st max measure passes into a space with another max measure.

Next time: Markov partitions and how invariant measures are constructed with their help.

$23/\overline{\text{X}}$. Leray. Singularities of Feynman integrals.

N. Nilsson, Arkiv för Math. (Swede) 1964–1965

(student of Gårding)

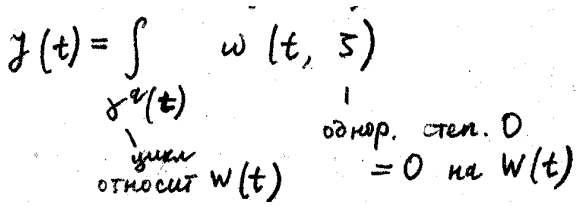


Original source details: GELFAND-P000223

Undeciphered passage. A short formula below "(student of Gårding)" is crossed out. Possible beginning: $J(\dots)$; the remaining symbols are illegible.

Original source details: GELFAND-P000223

Uncertain reading. A separate crossed-out sign below the formula: J / γ; its reading is undetermined.



Original source details: GELFAND-P000223

Uncertain reading. The cycle superscript has a loop and descender, favouring q. The same angular sign occurs in the argument of ω and in the coordinate line. The provisional reading is ξ. An alternative is ζ.

$$J(t) = \int_{\gamma^q(t)} \omega(t, \xi)$$

cycle relative to $W(t)$ — homog. deg. 0 $= 0$ on $W(t)$

$t \in T = \mathbb{C}^n$,

$\xi \in \mathbb{C}^{\ell+1}$

$\omega(t, \xi)$ on $\mathbb{P}^{\ell}$, since it is homogeneous; the cycle is there too.

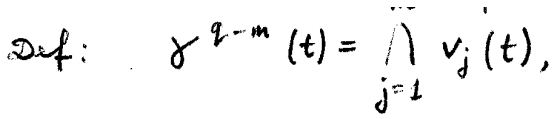
GELFAND-P000224

$W(t) = \bigcup_k W_k(t)$ — hyperplanes, algebraic functions of t, singularities may occur (not in general position)

$t \in S_0[W] \iff$ the dimension of the intersection of any number changes there.

Def:

$$\gamma^{q-m}(t) = \bigcap_{j=1}^{m} V_j(t),$$



Manuscript notation. The large sign has a curved top and descending sides. It is set as $\bigcap$.

Original source details: GELFAND-P000224

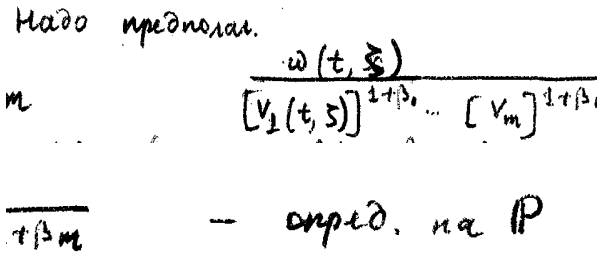
$$V_j(t) = v_j(t,x) = 0$$

)

algebraic set

One must assume

$$\frac{\omega(t,\xi)}{[v_1(t,\xi)]^{1+\beta_1}\dots[v_m]^{1+\beta_m}} \quad \text{— defined on } \mathbb{P}$$

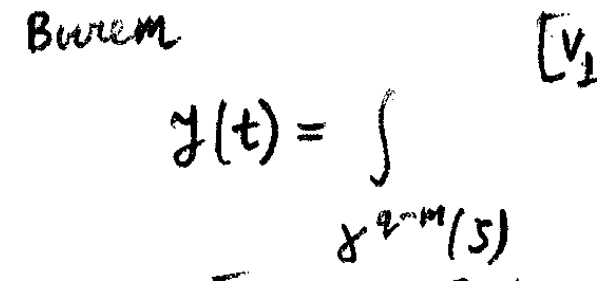


Uncertain reading. The coordinate signs in the argument of ω and first denominator: ξ / s; displayed candidate: ξ. The last factor is written $[v_m]^{1+\beta_m}$.

Original source details: GELFAND-P000224

[On the left:] Residue

$$J(t) = \int_{\gamma^{q-m}(s)} \frac{d^{\beta_1+\dots+\beta_m}\omega}{[dv_1]^{1+\beta_1}\wedge\dots\wedge[dv_m]^{1+\beta_m}}$$



Uncertain reading. The sign in parentheses has a short top stroke and a lower curve. The reading s is provisional. An alternative is ξ.

Original source details: GELFAND-P000224

The singular support is also described in the general case

(On the covering space of the complement the form will be holomorphic)

Thm. Then $J(t)$ has algebraic singular support.

Uncertain reading. Right-margin note: “of type st. algebraic” / “of type sm. algebraic”; the first word and middle abbreviation are not uniquely established.

Original source details: GELFAND-P000224

(i.e. in T there exists an algebraic set such that $J(t)$ is holomorphic on a covering of the complement.)

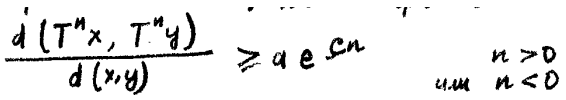
Leray described the singular support in the case of planes.

$30/\underline{\overline{\text{X}}}$ Sinai. U-diffeom.

characteristic exp growth of the distance between nearby points (separation)

either as $t \longrightarrow +\infty$, or as $t \longrightarrow -\infty$.

$$\frac{d(T^n x, T^n y)}{d(x,y)} \geq a e^{cn} \qquad \begin{matrix} n > 0 \\ \text{or } n < 0 \end{matrix} \qquad (x, y \text{ — nearby})$$



Manuscript notation. The exponent reads cn. The two lines at the right give $n > 0$ and "or $n < 0$".

Original source details: GELFAND-P000224

Consider the points $d(T^n y, T^n x) \leq a e^{-cn} d(y, x)$,

Suppose that $\forall\, x$ this is a k-dimensional subset.

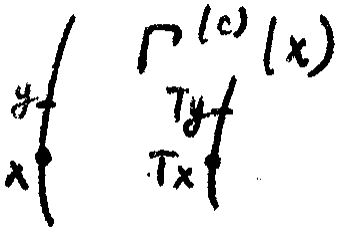
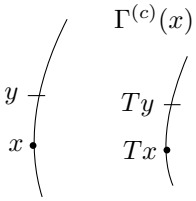


Diagram and typed description.



Original source details: GELFAND-P000224

This is a contracting subset (hence the index $\underline{\underline{c}}$)

$$d(T^n y', T^n x) \leq a e^{+cn} d(y', x) \qquad (n \longrightarrow -\infty).$$

Similarly we obtain $\Gamma^{(p)}(x)$ — an expanding subset.

(ℓ-dimensional).

Then the choice does not depend on a and c.

GELFAND-P000225

[source number: 11]

Diagram and typed description. Page number 11 with a line above, left, and below.

Original source details: GELFAND-P000225

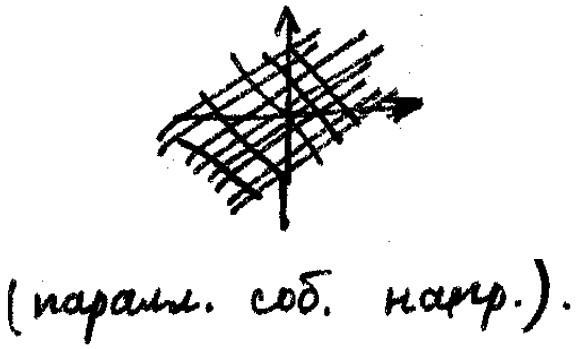
U-system: $k + \ell = n$ and the tangent spaces to the submanifolds are uniformly transverse

1. $k + \ell = n$
2. transversality.
3. Smoothness (not written out explicitly).

We obtain 2 foliations $\gamma^{(c)}$ and $\gamma^{(p)}$.

Example: linear transformation of the torus $(x, y) \longrightarrow (ax + by, cx + dy)$.

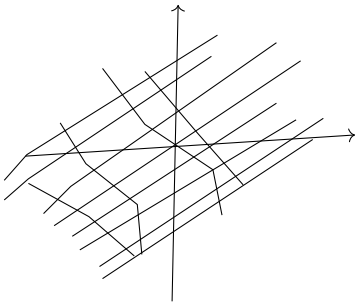
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1.$$



(параллельные соб. напр.).

Original source details: GELFAND-P000225

Uncertain reading. Two families of inclined lines and axes with arrows upwards and rightwards. The number of coinciding lines and continuations within dense crossings are not uniquely established. Individual lines have no labels.



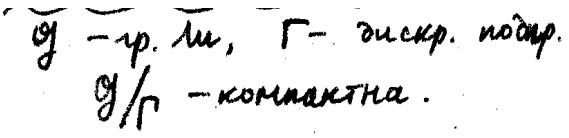
(parallel to the eigendirections).

A nonlinear perturbation (periodic and smooth) is also a U-diffeomorphism and the rotation number is the same.

Method of constructing U-diffeomorphisms.

g — a Lie group, Γ — a discrete subgroup.

g/Γ — compact.



$\mathfrak{g}$ — гр. Ли, Γ — дискр. подгр.
$\mathfrak{g}/\Gamma$ — компактна.

Original source details: GELFAND-P000225

Manuscript notation. The group sign has a loop and descender. The uppercase G below denotes the Lie algebra.

$\underbrace{T_{\text{alg}}}$ — an algebraic automorphism of g, leaving Γ fixed $\quad (T_{\text{alg}}\Gamma = \Gamma)$

automorphism as a group

Then it acts on g/Γ.

Consider G — the Lie algebra; $\hat{T}_{\text{alg}}$ — the corresponding automorphism of G; if its

[At the left edge of the following line:] eigenvalues.

Among them there are no [Crossed out] eigenvalues $= 1$ in mod $\Longleftrightarrow$

$\Longleftrightarrow$ this is a U-diffeomorphism.

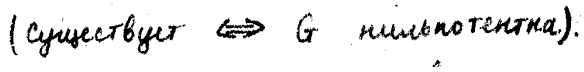
Original source details: GELFAND-P000225

Uncertain reading. Crossed out: $\Longleftrightarrow$ / $\Longrightarrow$; endpoint directions are not established.

Original source details: GELFAND-P000225

Manuscript notation. After mod there are two horizontal strokes and two arrowheads. The sign recurs at the start of the next line.

(It exists $\Longleftrightarrow$ G is nilpotent).



Original source details: GELFAND-P000225

Manuscript notation. The first word reads Существует (“It exists”). The letter before the final т is е.

Further. A small nonlinear perturbation $T = T_{\text{alg}} + \Delta T$ is also a U-diffeomorphism.

Main problem: topological similarity to the algebraic one.

For small perturbations — from Anosov’s general theorem

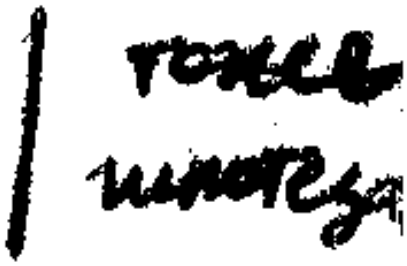
(A small perturbation of a U-diffeomorphism is topologically similar to the original).

Always.

$$M = M_1 \cup M_2$$

M_2 — wandering points

M_1 — nonwandering points. — the condition U holds on them



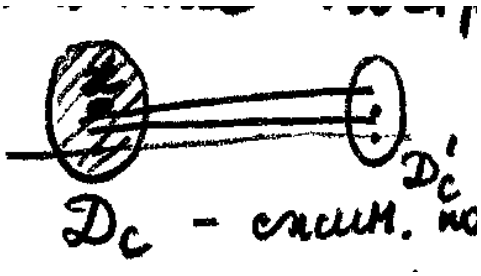
Original source details: GELFAND-P000225

Diagram and typed description. A vertical rule separates the right-margin note.

> also a
> conjecture

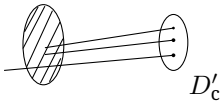
GELFAND-P000226

An important construction:



Original source details: GELFAND-P000226

Diagram and typed description.

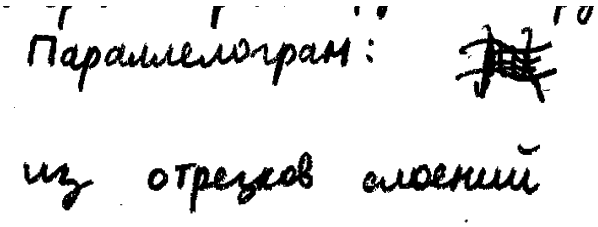


natural isomorphism of our pieces of contracting leaves (along noncontracting leaves).

D_c — a contracting subset, dim $= k$.

(π — the canonical isomorphism.)

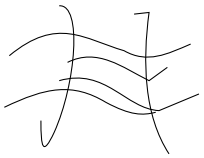
π is not always smooth (!), but it is absolutely continuous, i.e. the natural measures pass into one another.



Original source details: GELFAND-P000226

Uncertain reading. Two nearly vertical curved strokes and transverse curves; the number of coinciding strokes and their continuations at the dense middle crossings are not uniquely established. No labels or arrows are visible.

A parallelogram:



made of segments of foliations

(all its leaves are canonically isomorphic: both contracting and expanding.)

$$\Gamma(U) = \Gamma_p(U) \cup \Gamma_c(U)$$

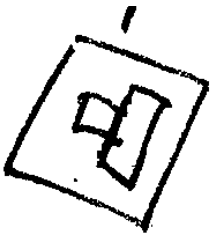
Suppose there is a system of parallelograms U_i adjoining one another.

$$M = \bigcup \overline{U}_i$$
$$U_i \cap U_j \subset \Gamma(U_i) \cap \Gamma(U_j) \qquad (\Gamma \text{ — boundaries}).$$

α — such a partition into parallelograms. Consider the boundaries:

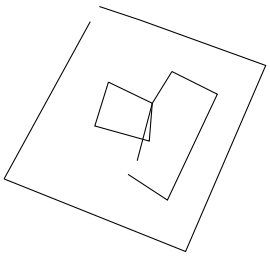
$$\Gamma_p(\alpha) = \bigcup_i \Gamma_p(U_i)$$
$$\Gamma_c(\alpha) = \bigcup_i \Gamma_c(U_i)$$

[Drawing on the right:]



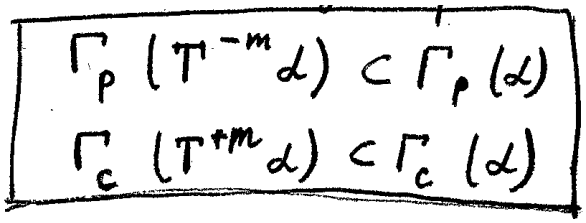
Original source details: GELFAND-P000226

Diagram and typed description.



This α is called Markov if there exists an m such that

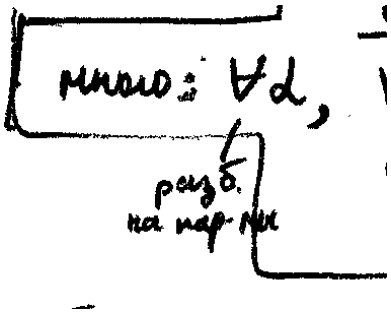
$$\boxed{\begin{array}{l}\Gamma_{\mathrm{p}}(T^{-m}\alpha) \subset \Gamma_{\mathrm{p}}(\alpha) \\ \Gamma_{\mathrm{c}}(T^{+m}\alpha) \subset \Gamma_{\mathrm{c}}(\alpha)\end{array}}$$



Manuscript notation. The upper exponent is $-m$ and the lower one is $+m$. One box surrounds both lines.

Original source details: GELFAND-P000226

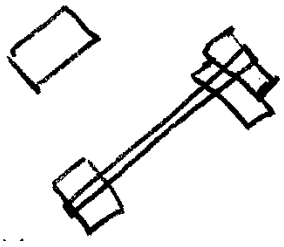
Thm. (technical) Markov partitions exist and there are many of them: $\forall$ α (partition into parallelograms), $\forall \varepsilon > 0$ $\exists m$ and a partition $\widetilde{\alpha}$, such that it will be at a distance of less than ε from the original.



Manuscript notation. The pointer links the note “partition into parallelograms” to $\forall\alpha$.

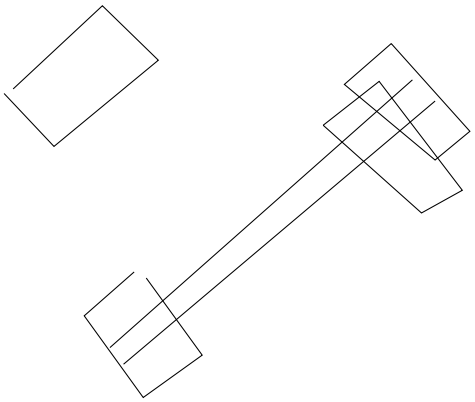
Original source details: GELFAND-P000226

[On the left:]



Uncertain reading. A separate inclined quadrilateral and a narrow inclined strip with transverse quadrilaterals at its ends. The continuations of several superposed contours at the upper right end are not uniquely established. There are no arrows.

Original source details: GELFAND-P000226



the ends cannot be in the middle.
(original parallelogram.)

— • —

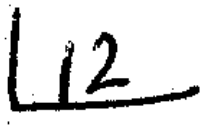
If α is a partition with m, then one can make $m = 1$, taking the intersection

$$\alpha_1 = \alpha \cdot T^{-1}\alpha \cdot \ldots \cdot T^{-m+1}\alpha \text{ —}$$

— Markov with $m = 1$

GELFAND-P000227

[source number: 12]



Original source details: GELFAND-P000227

Diagram and typed description. Page number 12 with a line at its left and below it.

12

— from

$$\Gamma_{\mathrm{p}}(\alpha_1) = \Gamma_{\mathrm{p}}(\alpha) \cup \Gamma_{\mathrm{p}}(T^{-1}\alpha) \cup \cdots \cup \Gamma_{\mathrm{p}}(T^{-m+1}\alpha)$$

Symbolic dynamics for a U-diffeomorphism.

(Suppose there is a Markov partition and with $m = 1$.)

$$\pi_{j_1 j_2} = \begin{cases} 1 & V_{j_1} \cap TV_{j_2} \neq \varnothing \\ 0 & \text{in the remaining cases} \end{cases}$$

Π

Ω_Π — the space of points

$$\omega = \{\ldots\ j_{-n}, \ldots, j_0, \ldots, j_n, \ldots\}$$
$$\pi_{j_n j_{n+1}} = 1.$$

Now associate

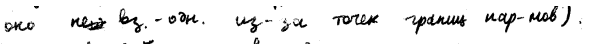
$$\omega \longrightarrow \bigcap_{-\infty}^{+\infty} T^n \overline{V}_{j_n} = x \in M \qquad (!)$$

(nonempty by the Markov property and, for a fine partition, consists of 1 point.)

We obtain a map $\varphi : \Omega_\Pi \longrightarrow M$

[A parenthesis begun on the right:] (from nonuniqueness.

it is not [Crossed out] one-to-one because of the points — boundaries of parallelograms).



Original source details: GELFAND-P000227

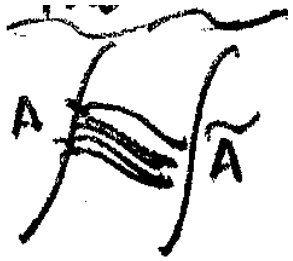
Uncertain reading. After the readable “not” a short entry is crossed out; possible “one-to-” is not established.

On a set of the 2nd category it is one-to-one.

Condition: Π^K consists of positive entries for some K.

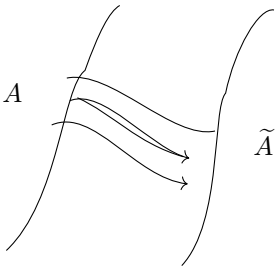
(i.e. all leaves $\Gamma^{(\mathrm{c})}$ and $\Gamma^{(\mathrm{p})}$ are dense in M — this is necessary and sufficient.)

On each leaf $\Gamma^{(c)}$ a σ-finite measure λ_c is introduced,
where
[On the left:]

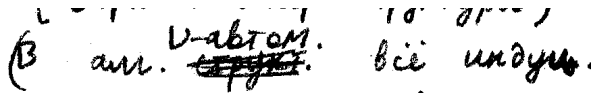


Original source details: GELFAND-P000227

Uncertain reading. The lines labelled $A, \widetilde{A}$ and connecting curves. Directed thickenings are visible at two lower right ends; the number of coinciding lower curves and character of the other ends are not uniquely established.



1) $\lambda_c(A) = \lambda_c(\widetilde{A})$
(measures of canonically isomorphic sets are equal.)
2) $\lambda_c(TA) = h\lambda_c(A) \quad (h < 1)$
(h — an algebraic number.)
— additional information.
(Reflects the homogeneous structure.)
(In an algebraic



Original source details: GELFAND-P000227

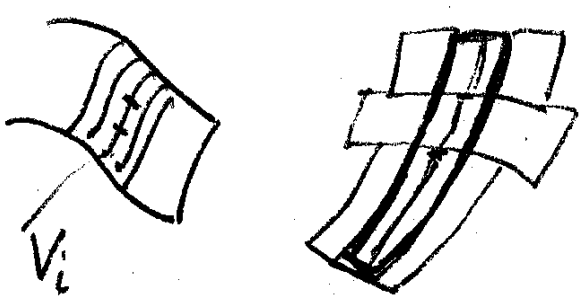
Uncertain reading. A word is crossed out, possibly “case”; its reading is not established.

[Inserted above:] U-automorphism everything is induced by Haar measure.)
Corollary: there is an invariant measure, T is [Crossed out] a finite Markov chain.

Original source details: GELFAND-P000227

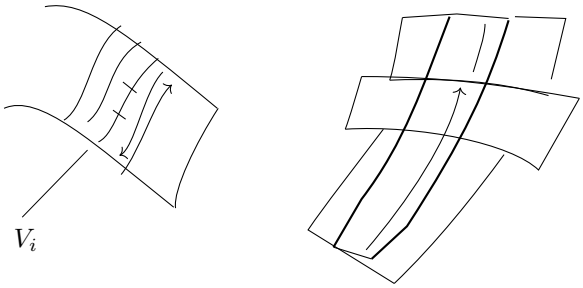
Uncertain reading. Crossed-out sign: Cyrillic “б” / “в”; its reading is not established.

Construction of this measure: $(n = 2)$

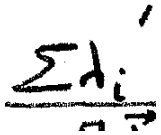


Original source details: GELFAND-P000227

Uncertain reading. Two diagrams with V_i, leader stroke, transverse marks, and visible directed strokes. Connections of thick lines at the base of the right diagram and some endpoint directions remain undetermined. There is no arrow between the diagrams.



$$\lambda_i h^{-1} = \sum \lambda'_i = \sum \pi_{ii'} \lambda_{i'}, \quad \text{i.e.}$$



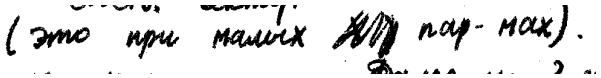
Original source details: GELFAND-P000227

Manuscript notation. In the first sum the prime sits high above the subscript. In the second sum both primes are at subscript height.

$$\boxed{h^{-1}\vec{\lambda} = \Pi\vec{\lambda}.}$$

— eigenvector.

(This is for [small?] [Crossed out] parallelograms).



Original source details: GELFAND-P000227

Uncertain reading. The letterforms favour “малых” (small), as on manuscript page 11. “Наших” (our) remains possible. The next word is crossed out.

Original source details: GELFAND-P000227

Uncertain reading. A short crossed-out entry is illegible.

The values of the measure on a segment are given. Then from the 2nd property.

[Below the last line:] (on a smaller segment.)

GELFAND-P000228

Entropy

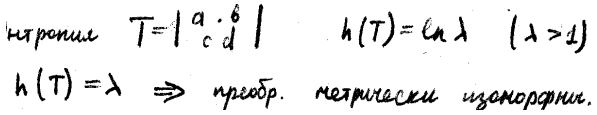
Original source details: GELFAND-P000228

Uncertain reading. The final stroke favours Энтропия (Entropy). The ending remains provisional.

$$T = \begin{vmatrix} a & \cdot & b \\ c & & d \end{vmatrix}$$

$$h(T) = \ln \lambda \quad (\lambda > 1)$$

$$h(T) = \lambda$$

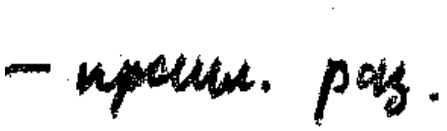


Original source details: GELFAND-P000228

Manuscript notation. The upper line writes $h(T) = \ln \lambda$. The next writes $h(T) = \lambda$. A dot stands between a and b.

$\Rightarrow$ the transformations are metrically isomorphic.

Ω_{Π_1} maps onto Ω_{Π_0} (stand.) —

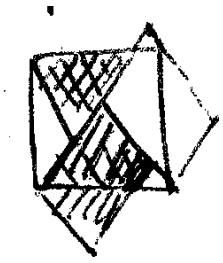


Original source details: GELFAND-P000228

Uncertain reading. The abbreviation before раз. provisionally reads прежн. (previous). The middle strokes merge.

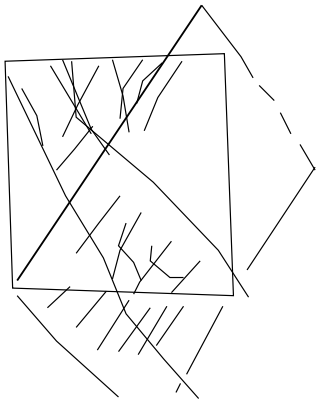
And it remains to map the torus onto Ω_{Π_1}.

Π_1 is of a certain type: 4 groups of letters and properties of transitions — see earlier.



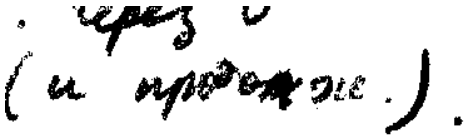
Original source details: GELFAND-P000228

Uncertain reading. A nearly rectangular frame, oblique edges projecting above, to the right and below, and hatching. Continuations of lines at dense crossings and exact boundaries of the hatched parts are not uniquely established. There are no vertex labels or arrows.



— a partition of the torus into 2 parallelograms (the sides $\parallel$ the eigendirections)

The sides lie on a line through 0 in an eigendirection (and its cont.?).

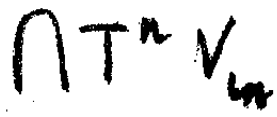


Uncertain reading. The abbreviation after "and" appears to read продолж. (continuation). The middle letters overlap.

Original source details: GELFAND-P000228

This is a Markov partition (the coarsest). It is because of it that one succeeds in solving the isomorphism problem.

$$\bigcap T^n V_{\omega_n}$$



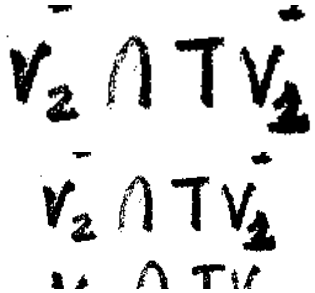
Uncertain reading. The mark below V is looped. A separate n is hard to distinguish. V_{ω_n} is provisional.

Original source details: GELFAND-P000228

may consist of a large number of points.

Consider first 4 parallelograms:

$$V_1 \cap TV_1$$
$$V_2 \cap TV_2$$
$$V_1 \cap TV_2$$
$$V_1 \cap TV_2$$



Manuscript notation. The right subscript on the second line has a foot like the 2 in the following lines. The third and fourth lines both read $V_1 \cap TV_2$.

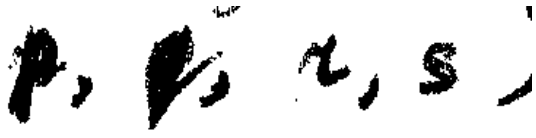
Original source details: GELFAND-P000228

(

Uncertain reading. Уже (Already); the word is readable, but whether a transverse stroke crosses it out is undetermined.

Original source details: GELFAND-P000228

a disconnected partition. Form a partition — — its connected components. Their numbers, respectively — —



Uncertain reading. p, q, r, s; the second sign is heavily retraced and q remains a candidate.

Original source details: GELFAND-P000228

)

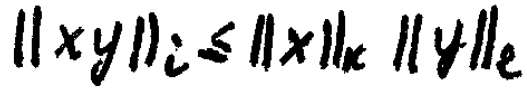
GELFAND-P000229

: Year 1968.

12/II Question (Gelfand)

R — a countably normed ring:

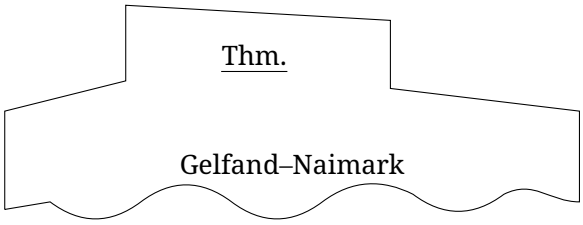
$$\|xy\|_i \le \|x\|_k \|y\|_e$$



Uncertain reading. The last subscript has a low closed loop like e. The reading e is provisional. An alternative is ℓ.

Original source details: GELFAND-P000229

Diagram and typed description.



Original source details: GELFAND-P000229

for

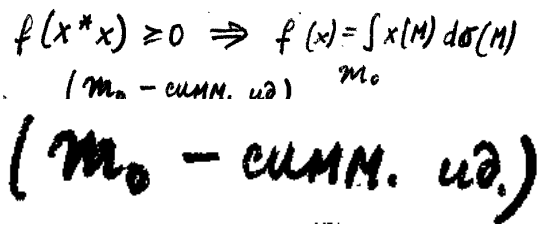
Uncertain reading. [Superscribed sign: superposed loops and strokes; its reading is undetermined.]

Original source details: GELFAND-P000229

normed rings:

$$f(x^*x) \ge 0 \Rightarrow f(x) = \int_{\mathcal{M}_0} x(M)\, d\sigma(M)$$

($\mathcal{M}_0$ — symm. ideals)



Uncertain reading. The slanted letter with a subscript zero occurs below the integral and in the parenthesis. It is provisionally set as $\mathcal{M}_0$.

Original source details: GELFAND-P000229

Extension — ? (Not always: one must characterize the class axiomatically).

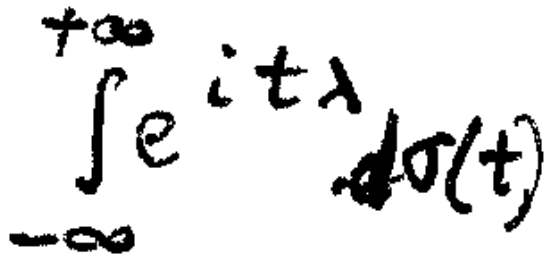
$$x^*(M) = \overline{x(M')}, \quad M = M'$$

— symm.

L^1 with convolution, $\overline{x(-t)} = x^*(t)$

⇒ Bochner's theorem

$$\int_{-\infty}^{+\infty} e^{it\lambda}\, d\sigma(t)$$



Manuscript notation. The bounds are $-\infty$ and $+\infty$. The exponent is $it\lambda$ and the differential is $d\sigma(t)$.

Original source details: GELFAND-P000229

I. Bernstein.

Original source details: GELFAND-P000229

Diagram and typed description.

Grothendieck topology.

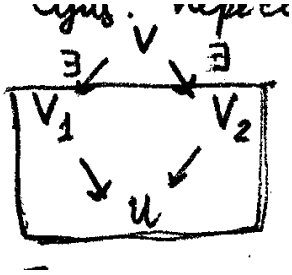
T is the following collection:

1) The category Cat T is specified (objects — open sets).

2) Certain families $\{\alpha_i : U_i \to U\}$ are specified, called coverings

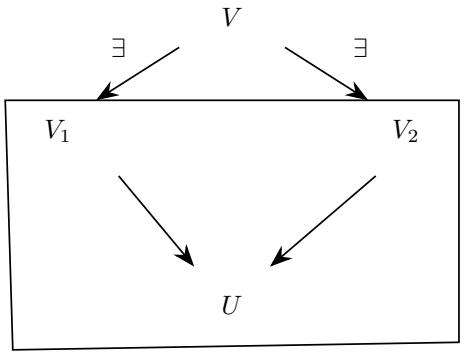
3) A system satisfying axioms is specified:

1. There exists an intersection — a product (fibered).



Original source details: GELFAND-P000229

Diagram and typed description.



(commut.).

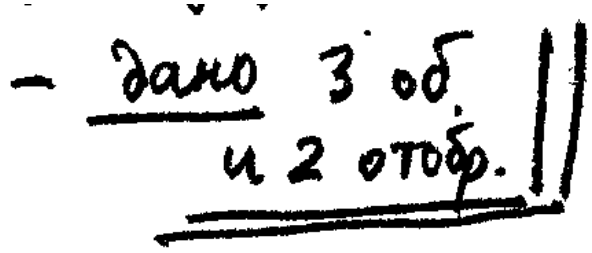


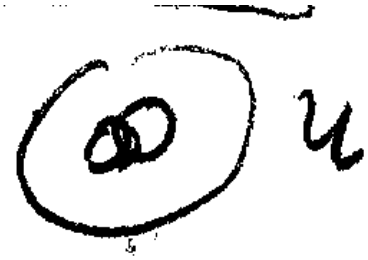
Diagram and typed description.

— given 3 objects

and 2 maps

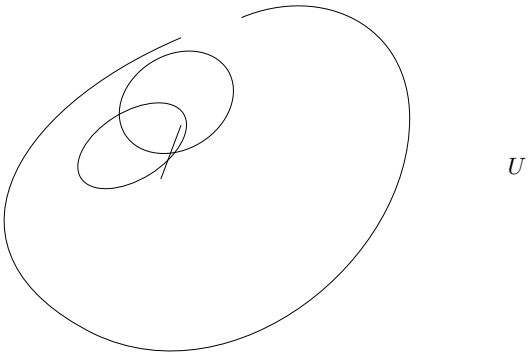
Original source details: GELFAND-P000229

$\boxed{V = V_1 \times_U V_2}$ — notation. Then there exists



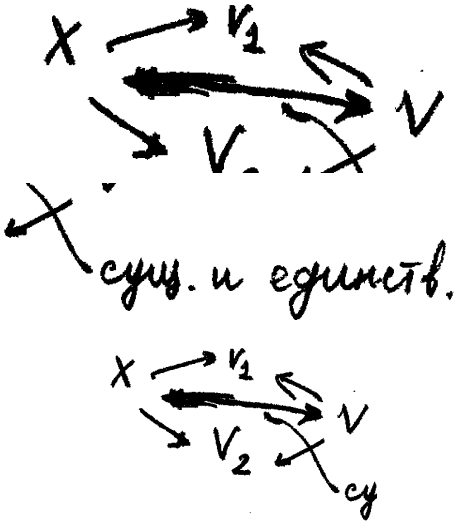
Original source details: GELFAND-P000229

Diagram and typed description.



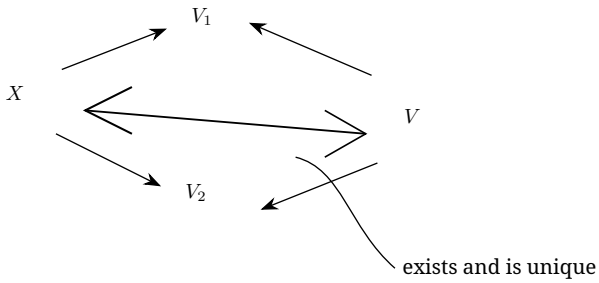
The universal property is required to hold:

$\forall X$, and $\forall$



Original source details: GELFAND-P000229

Uncertain reading. The central line has a clear arrowhead at right. The dense wedge at left may be a correction. The leader carries the note “exists and is unique”.



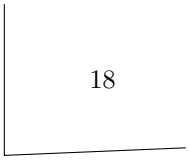
Depends essentially on U (in Grothendieck top.)

(If V exists, it is determined uniquely by the properties of the category)

GELFAND-P000230

Original source details: GELFAND-P000230

Uncertain reading. The upper right reads 18. The first digit is an upright stroke with no loop.



2. $\{U \xrightarrow{\text{id}} U\}$ — a covering.
3. $\{V_i \to V\}$ — a covering.

$\{V_{ij} \to V_i\}$ — a covering.

$\Rightarrow \{V_{ij} \to V\}$ — a covering.

4. $\{U_i \to U\}$ — a covering

$\{V \to U\}$ — a morphism

Then $\{U_i \times_U V \to V\}$

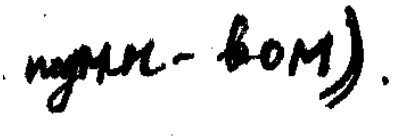
Example: unramified topology.

X — an ordinary topological space.

An open set: $U \to X$ — a local homeomorphism

(U — a topological space)

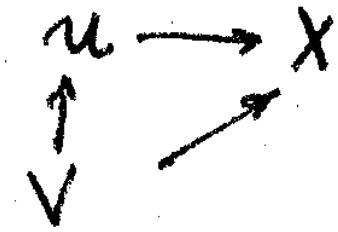
(a covering over X or its open subset).



Original source details: GELFAND-P000230

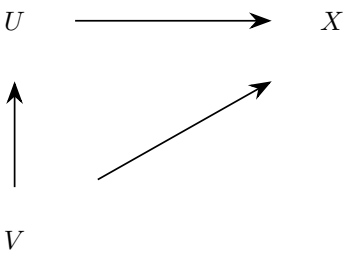
Manuscript notation. The word подмн-вом (subset) retains the opening под. No separate cancellation stroke is visible.

A map



Original source details: GELFAND-P000230

Diagram and typed description.



(any continuous one with a commutative diagram)

$\{U_i \to U\}$ — a covering, if the images

Original source details: GELFAND-P000230

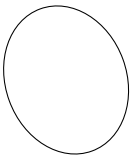
Undeciphered passage. [The crossed-out word is undeciphered.]

cover.

Axiom 1.

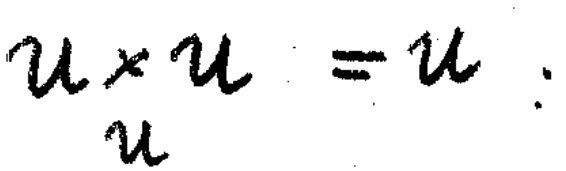
Original source details: GELFAND-P000230

Diagram and typed description.



U — a circle, U — also a circle, a map of degree 2 $U \to U$.

$$U \times_U U = U.$$

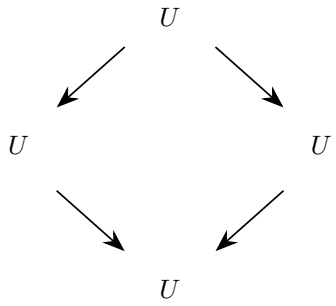


Original source details: GELFAND-P000230

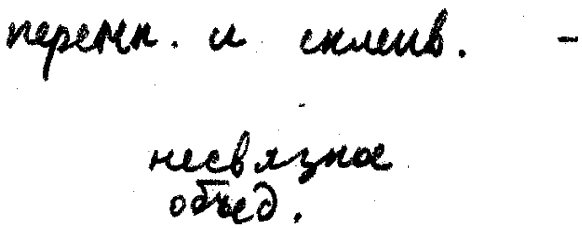
Manuscript notation. The equality has one U after the equals sign.

Original source details: GELFAND-P000230

Diagram and typed description.



$U_1 \times_X U_2$ — multipl. and gluing



Original source details: GELFAND-P000230

Manuscript notation. The first abbreviation is перемн. (multipl.), followed by и склеив. (and gluing).

— a disconnected union.

2 and 3 sheets $\Rightarrow$ 6 sheets

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \neq \mathbb{Z}_4,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \mathbb{Z}_6.$$

GELFAND-P000231

Presheaf: • a functor

$$\text{Cat}\, T \to \text{Ab} \qquad \text{(contravar.)}$$
$$u \to P(u)$$

$$v \to u \;\Rightarrow\; P(u) \to P(v)$$

A sheaf — a presheaf satisfying the axioms:

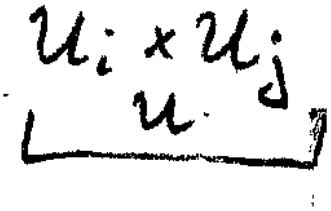
1. $\xi \in P(u)$

$\{u_i \to u\}$ — a covering

$$\xi \to \xi_i \qquad P(u) \to P(u_i)$$

ξ is uniquely determined by ξ_i

2. and is indeed determined, if ξ_i agree on intersections



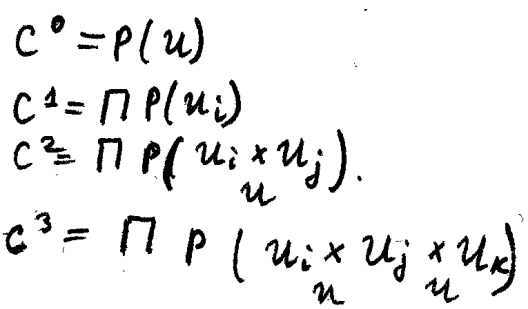
Original source details: GELFAND-P000231

Diagram and typed description.

$$u_i \times_u u_j$$

Čech cohomology is defined

$$C^0 = P(u)$$
$$C^1 = \prod P(u_i)$$
$$C^2 = \prod P(u_i \times_u u_j).$$
$$C^3 = \prod P(u_i \times_u u_j \times_u u_k)$$



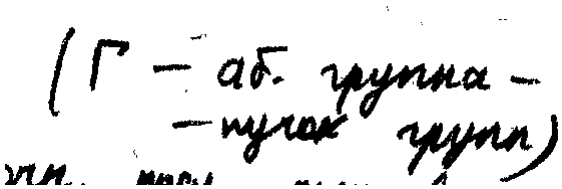
Original source details: GELFAND-P000231

Manuscript notation. The manuscript writes $C^0 = P(u)$ and begins the products at C^1. There are no indices under the product signs.

d is defined naturally

$\mathcal{F} \to \Gamma(X, \mathcal{F})$ — a functor.

(Γ — an abelian group —
— a sheaf of groups)



Original source details: GELFAND-P000231

Uncertain reading. Both lines of the parenthesis are retained. No separate symbol is visible before the dash in the second line; its implied subject is undetermined.

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0.$$

— an exact sequence of sheaves.

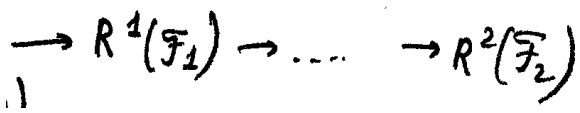
$$0 \to \Gamma(\mathcal{F}_1) \to \Gamma(\mathcal{F}_2) \to \Gamma(\mathcal{F}_3)$$

(left exact).

Derived functors exist. $R^i : S \to \text{Ab}$

S — arbit. sheaves (category).

$$\to R^1(\mathcal{F}_1) \to \ldots \to R^2(\mathcal{F}_2)$$



Original source details: GELFAND-P000231

Manuscript notation. The ellipsis is followed by $R^2(\mathcal{F}_2)$. The opening arrow and the ellipsis are as written.

(R^i — determined uniquely).

$$R^i(\mathcal{F}) \underset{\text{def}}{=} H^i(X, \mathcal{F}).$$

Finite group. Grothendieck top. from representations.

GELFAND-P000232

11/III Gelfand (and Kirillov)

$\mathfrak{g}$ — semisimple Lie algebra, (over C)

$U(\mathfrak{g})$ — its enveloping algebra

(polyn. in generators $e_1, \ldots, e_n$,

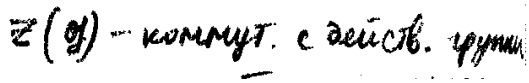
$$e_i e_k - e_k e_i = \sum C_{ik}^j e_j$$

$Z(\mathfrak{g})$ — centre of the enveloping algebra

(its elements — Laplace or Casimir operators).

Repr. of $G \Rightarrow$ diff. operators and commut.

$Z(\mathfrak{g})$ — commut. with the action of the group



Original source details: GELFAND-P000232

Manuscript notation. The right edge of the line reads группы (group).

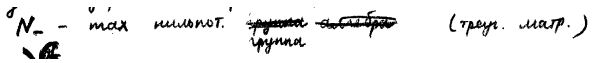
Such a repr. of the enveloping alg. by diff. operators is not necessarily faithful.

On G itself — faithful (transform.)

For $Z(\mathfrak{g})$ it makes no difference, left or right.

$\mathfrak{g}$ — semisimple Lie algebra. (over C).

N_- — max nilpot.



Original source details: GELFAND-P000232

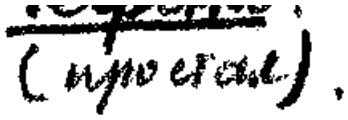
Uncertain reading. [crossed out: group algebra] [inserted below the first word: group]

(triang. matr.)

Original source details: GELFAND-P000232

Uncertain reading. Before the theorem, N_- and a diagonal stroke are visible. The upper right character is overwritten. The clearer notation in the theorem supports a provisional $N_- \backslash G$.

Theorem: in $N_-\backslash G$ the repr. of $U(\mathfrak{g})$ is faithful.
(simple?).



Uncertain reading. The word below the heading probably reads простая (simple). Its last letters are faint.

Original source details: GELFAND-P000232

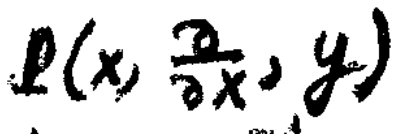
f on the group is regular if its translates — finite-dim. space
(sin on the line)
On a comp. group, functions are approx. by regular ones (completeness theorem)
Consider functions $f(g) \equiv f(n_- g)$ — invar. — ring.
Consider Spec of this ring. This is X.
$N_-\backslash G$ birat. equiv. to projective space.
Enveloping algebras are numerous, but division rings, of fractions are few

$$\underset{\substack{\text{division ring}\\ \text{of quotients}}}{D(\mathfrak{g})} \sim D_{n,k}$$

Nontriv. that embeds into a division ring (!) of quotients.

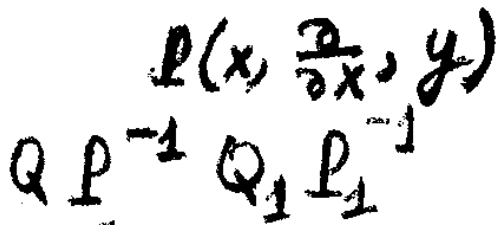
$$x_1, \dots, x_n; \quad \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \quad y_1, \dots, y_k$$

$P(x, \frac{\partial}{\partial x}, y)$



Manuscript notation. The initial P has the loop and foot seen in the denominator letter on the next line.

Original source details: GELFAND-P000232



Manuscript notation. The letter recurs in P, P^{-1} and P_1^{-1}.

Original source details: GELFAND-P000232

$\underline{\underline{QP^{-1}}}$ $Q_1 P_1^{-1}$ and so on. Everything can be done in 2 steps.
(QP^{-1} suffices)
n — number of variables, k — number of parameters — invariants.

GELFAND-P000233

[source number: 23]

Diagram and typed description.

23

Original source details: GELFAND-P000233

For a semisimple group it is almost proved that $D(\mathfrak{g}) \sim D_{n,k}$.
(matr. is valid)

Manuscript notation. The words below the formula read матр. (matr.) and верна (is valid).

Original source details: GELFAND-P000233

For nilpotent proved, and probably for algebraic solvable.

Diagram and typed description.

Adjoin to $Z(\mathfrak{g})$ (K) operators — virtual Laplace operators

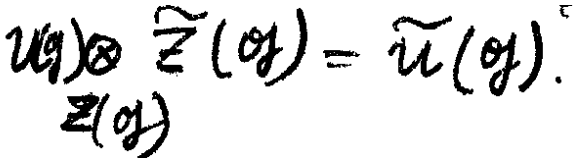
Original source details: GELFAND-P000233

New ring $\tilde{U}(\mathfrak{g})$ — with adjoined left translations

Undeciphered passage. A short word or abbreviation is crossed out; its letters beneath the heavy strokes have not been established.

Original source details: GELFAND-P000233

$$U(\mathfrak{g}) \otimes_{Z(\mathfrak{g})} \tilde{Z}(\mathfrak{g}) = \tilde{U}(\mathfrak{g}).$$

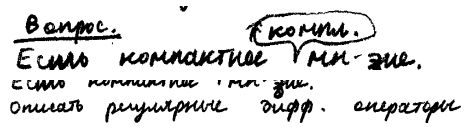


Manuscript notation. The manuscript puts $Z(\mathfrak{g})$ under $\otimes$ and writes tildes over the second factor and the U at right.

Original source details: GELFAND-P000233

$\tilde{Z}$ — Z with adjoined virt. Laplace operators.
(this is an algebraic extens., Galois group — Weyl group).
Theorem. The division ring of fractions of $\tilde{U}(\mathfrak{g})$ is $D_{n,k}$.
(For U unknown; done for unimodular).
Alg. extens. of $D_{n,k}$ — ?.

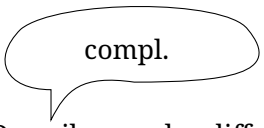
In the commut. case, alg. varieties.



Original source details: GELFAND-P000233

Diagram and typed description.

Question. There are compact



Describe regular differential operators

On the Riemann sphere $\frac{\partial}{\partial z},\ z\frac{\partial}{\partial z},\ z^2\frac{\partial}{\partial z}$

On Dixmier's work (Limanov).

$R_{1,0}$ — ring of quantum mech. — generators $x, \frac{d}{dx}$.

Diagram and typed description.

Polynomials

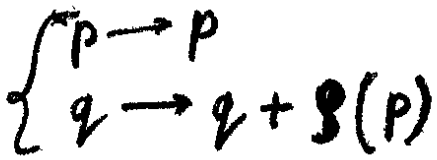
in p and q.

Original source details: GELFAND-P000233

$p, q, \quad [p, q] = 1.$

The autom. group is generated by triang. autom.

$$\begin{cases} p \to p \\ q \to q + f(p) \end{cases}$$



Uncertain reading. The letter before (p) is heavily inked. The reading f is provisional. The right-hand pair clearly has $g(q)$.

Original source details: GELFAND-P000233

and $\begin{cases} p \to p + g(q) \\ q \to q \end{cases}$

For the division ring $D_{1,0}$ unknown.

Or

Uncertain reading. A short word is crossed out. A possible reading is от, with the first letter obscured.

Original source details: GELFAND-P000233

$$\begin{cases} q \to q \\ p \to p + g(q) \end{cases}$$

and $\begin{cases} q \to p \\ p \to -q \end{cases}$

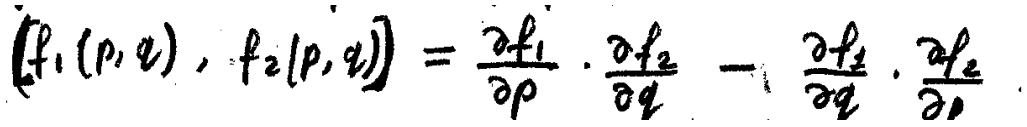
(Fourier transform)

Is generated

GELFAND-P000234

Arnold: theor.
Let there be functions $f(p,q)$

$$[f_1(p,q), f_2(p,q)] = \frac{\partial f_1}{\partial p}\cdot\frac{\partial f_2}{\partial q} - \frac{\partial f_2}{\partial q}\cdot\frac{\partial f_2}{\partial p}.$$



Original source details: GELFAND-P000234

Uncertain reading. The first numerator after the minus sign has a subscript with a foot, probably 2. The second numerator after the minus sign clearly has 2.

Consider 2 polynomials

$$P = P(p,q)$$
$$Q = Q(p,q).$$

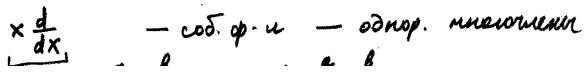
Substitute P and Q — will be 1.

They induce canon. transformations of quasiclassics.

But there is no connection, even with grading.

An element x is completely semisimple if the transformation ad X has a complete system of eigenvalues.

Then by an automorph. x is carried into $x \to pq$.



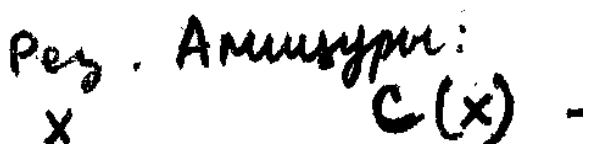
Original source details: GELFAND-P000234

Diagram and typed description.

$x\frac{d}{dx}$ — eigenfunctions — homog. polynomials.

$$p^a q^b \to (a-b)p^a q^b.$$

Result of Amitsur:



Original source details: GELFAND-P000234

Uncertain reading. The manuscript spelling probably ends -урн. An alternative is -ура.

x $C(x)$ — commuting — commut. subalgebra.
But these are not always polynomials.

$$\forall\alpha\ \exists X, Y, \qquad X^3 = Y^2 + \alpha, \qquad \alpha \in C.$$

Palamodov.

on Hörmander's work (on $\bar\partial$)

2 main results: existence theorem and approx. theorem.

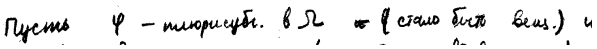
1. $\Omega \subset C^n$, $\phi \in C^2(\Omega)$

$$L(\phi) = \frac{\partial^2\phi}{\partial z_j \partial \bar z_k}\xi_j\bar\xi_k \geq 0$$

$$L_{p,q} = L_{p,q}(\Omega, \phi), \qquad \|f\|_\phi^2 = \int_\Omega |f|^2 e^{-\phi} dV$$

forms of type (p, q)

Let ϕ be plurisubh. in Ω



Original source details: GELFAND-P000234

Uncertain reading. A short dense stroke stands between Ω and the opening parenthesis. Its reading is unresolved.

(therefore real) and

$\forall c \{\phi \le c\}$ — compact. ($\Rightarrow \Omega$ — pseudoconvex)

Then $\forall f \in L_{p,q}$, $\bar\partial f = 0$

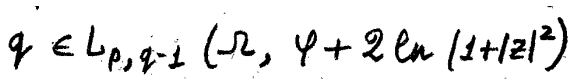
$\exists$

Original source details: GELFAND-P000234

Undeciphered passage. Crossed-out fragment; its initial symbol resembles g, the remaining letters have not been established.

g of type $p, q-1$

$$\bar\partial g = f \qquad \overset{(q\ge 1)}{\text{and}} \qquad g \in L_{p,q-1}(\Omega, \phi + 2\ln(1+|z|^2)$$



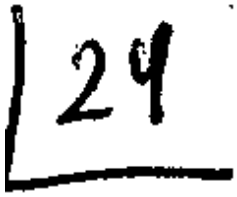
Original source details: GELFAND-P000234

Manuscript notation. One closing parenthesis is visible after $|z|^2$. The outer closing parenthesis is absent.

Here $\sqrt{q}\,\|g\| \le \|f\|$ (assuming $q \ge 1$)

GELFAND-P000235

[source number: 24]



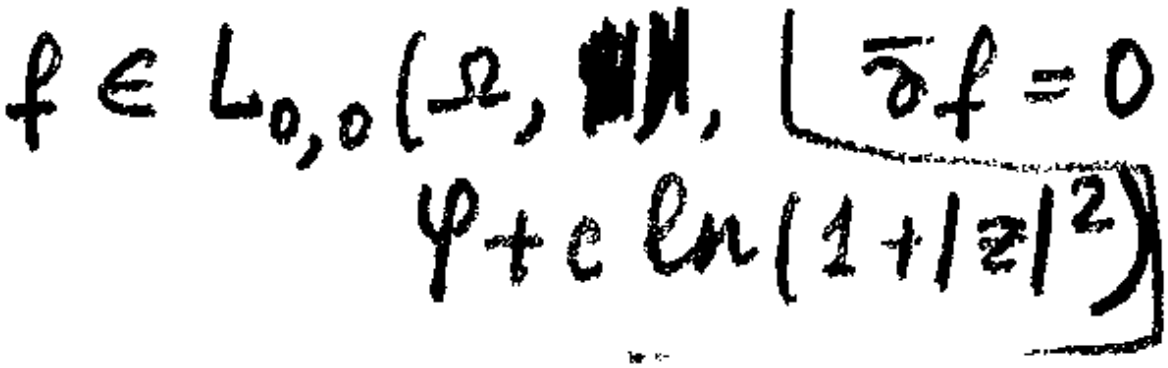
Original source details: GELFAND-P000235

Diagram and typed description.

24

2. Approx. theorem.

$\Omega = C^n$.



Original source details: GELFAND-P000235

Uncertain reading. The former second argument is crossed out: ϕ or ψ; $\phi + c\ln(1+|z|^2)$ is inserted below.

$f \in L_{0,0}(\Omega, [?])$ | $\bar\partial f = 0$ (holom. functions)

$\phi + c\ln(1+|z|^2)$

i.e. $|f| \le Ce^{\phi}(1+|z|^2)^k$, ($\exists C > 0$).

This set is nonempty (nontriv. — $\Omega = C^n$). — and dense in the space of all [holom.] entire functions with unif. convergence on $\forall$ compact set:

ϕ — homog. along radii $\Rightarrow$ indicator of an entire function.

Hence Martino: the class of indicators — all radially homog. plurisubh. harm.

Original source details: GELFAND-P000235

Uncertain reading.

$i(z) =$ | $\lim_{\tau\to\infty}$ | $\tau^{-k}\ln|f(z\tau)|$

A stroke above lim meets the left side of the box. Its function is uncertain.

18/III

If $\phi = \phi(x)$ (only on the real parts), it is plurisubharm. if and only if it is convex.

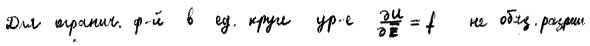
In $\forall$ one-dim. complex space subharm.

Example:

$$\frac{\partial u}{\partial \bar z} = f, \qquad |f(z)| \le Ce^{a|\operatorname{Im} z|}(1+|z|^2)^k.$$

A solution in this class — nontrivial.

For bounded functions in the unit disk, the equation $\dfrac{\partial u}{\partial z} = f$



Original source details: GELFAND-P000235

Uncertain reading. There are thick strokes between z and the fraction rule. No separate bar can be distinguished. Provisional z, possibly $\bar{z}$.

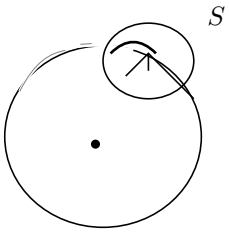
is not necessarily solvable for bounded f.

For solvability it suffices

(f — measure)

Original source details: GELFAND-P000235

Uncertain reading.



The faint upper arc and dark strokes at the intersection are transcribed without an established meaning for the strokes.

S — disk. $|f(S \cap \Omega)| \le C R_S$.

Theorem 3. More precise theorem for fixed q.

Let $\partial\Omega \in C^3$ and the domain be bounded. Local condition on the boundary:

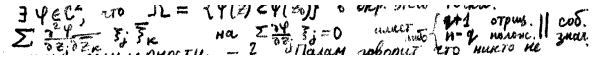
$\forall z \in \partial\Omega\ \exists \phi \in C^2$ such that $\Omega = \{\phi(z) < \phi(z_0)\}$ in a nbhd. of this point.

Condition a_q:

$$\sum \frac{\partial^2 \phi}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \quad \text{on} \quad \sum \frac{\partial \phi}{\partial z_j} \xi_j = 0$$

has —

$$\begin{cases} q+1 \text{ negative} \\ \text{or } n-q \text{ positive} \end{cases} \quad \| \quad \text{eigen-values}$$



Original source details: GELFAND-P000235

Manuscript notation. The tangent condition uses ξ_j. At right the alternatives are $q+1$ negative or $n-q$ positive, followed by two strokes.

Gelfand: domains of regularity — ?

Palam. says nobody is working on this.

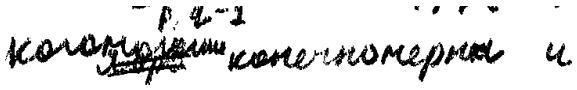
$\partial\Omega$ pseudoconvex if at $\forall$ point condition a_0 (at $\forall$ point $n-1$ positive eigenvalues)

In the tangent space, $n-q$-dim. surfaces along which the domain is pseudoconvex.

Let a_q hold at $\forall$ boundary point.

$$L_{p,q-1} \xrightarrow{\bar{\partial}} L_{p,q}(\Omega, \phi) \xrightarrow{\bar{\partial}} L_{p,q+1}$$

Cohomology



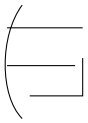
Original source details: GELFAND-P000235

Uncertain reading. [crossed out below: kernel, ядро or ядра]

is finite-dimensional and the image is closed.

Original source details: GELFAND-P000235

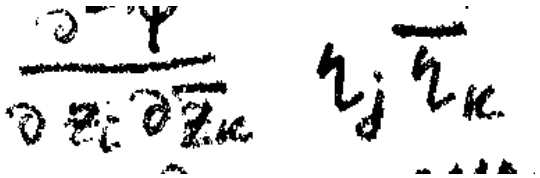
Uncertain reading.



Isolated sign: E or $\exists$; its function has not been established.

Existence of a function Ψ on the whole domain.
Ψ — real, defined inside Ω,

$$\frac{\partial^2 \Psi}{\partial z_i \partial \bar{z}_k} \eta_j \bar{\eta}_k$$



Original source details: GELFAND-P000235

Uncertain reading. The first denominator index is short with a dot, probably i. The η index has a descending loop and reads j.

satisfies all conditions

(1 more plus required, since in the whole space)

Then cohomology 0.

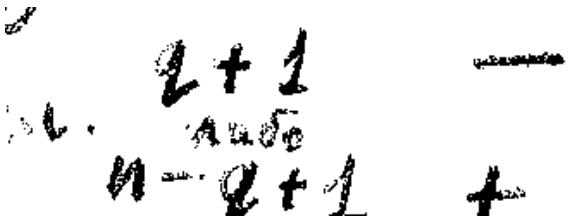
GELFAND-P000236

All these theorems for complex spaces — Andreotti and Grauert.

1. Runge-type theorem.

$\omega \subset \Omega$, $L(\Psi)$ at $\forall$ point of $\Omega \setminus \omega$ satisfies the condition

$$\begin{array}{c} q+1 \quad - \\ \text{or} \\ n-q+1 \quad + \end{array}$$

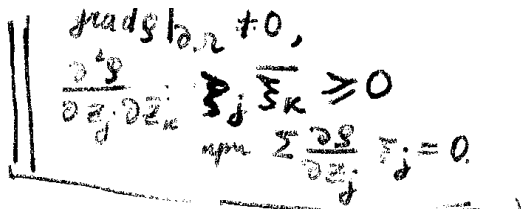


Original source details: GELFAND-P000236

Manuscript notation. The upper count is $q+1$ with a minus sign. The lower count is $n-q+1$ with a plus sign. The word between them is “or”.

$\Psi \to \infty$ near $\partial\Omega$, $\{\Psi < 0\} = \omega$

Then the cohomology of Ω restricts monomorphically to the cohom. of ω.
A closed form f in Ω and the equation $\bar\partial u = f$ solvable in $\omega \Rightarrow$ it is solvable in Ω as well.
2. Approximation theorem.
Closed forms in ω are approximated by closed forms in Ω.
Existence theorem.
Let $\partial\Omega \in C^2$ and there exist ρ, $\{\rho < 0\} = \Omega$,



Original source details: GELFAND-P000236

Diagram and typed description.

grad $\rho|_{\partial\Omega} \neq 0$,

$\frac{\partial^2 \rho}{\partial z_j \partial \bar z_k} \xi_j \bar\xi_k \geq 0$

when $\sum \frac{\partial \rho}{\partial z_j} \xi_j = 0$.

It is proved simply, using the identity.

$$L_{p,q-1} \xrightarrow{\bar\partial = T} L_{p,q} \xrightarrow{\bar\partial = S} L_{p,q+1}$$

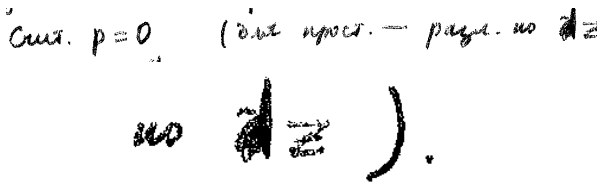
$$\|T^* f\|_\phi^2 + \|Sf\|_\phi^2 =$$

$$(u, v)_\phi = \int_\Omega u \bar v e^{-\phi} dV.$$

(adjoint with respect to this inner product)

$$= \sum_L \sum_{j,k} \left(\frac{\partial^2 \phi}{\partial z_j \partial \bar{z}_k} f_{jL}, f_{kL} \right)_\phi + \sum_J \sum_k \left\| \frac{\partial f_J}{\partial \bar{z}_k} \right\|_\phi^2 +$$

Take $p = 0$ (for simplicity — expansion in dz?).



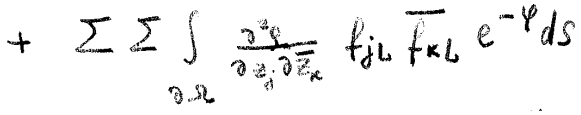
Uncertain reading. The first character is overwritten and probably reads d. A separate bar over z is uncertain. The next formula clearly has $d\bar{z}^J$.

Original source details: GELFAND-P000236

$$f = \sum f_J d\bar{z}^J, \qquad J = j_1, \ldots, j_q.$$

f_J skew-symm. in the indices, jL — long index.

$$+ \sum \sum \int_{\partial\Omega} \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} f_{jL} \overline{f_{kL}} e^{-\phi} dS.$$



Manuscript notation. Two summation signs stand before the integral without lower indices. The integral is over $\partial\Omega$ and ends in dS.

Original source details: GELFAND-P000236

The identity — integration by parts.

First — Morrey with $\phi = 0$.

grad, rot — known in mechanics.

Solvability follows from it.

Next time — Nirenberg (according to hearsay).

Chapter 4

GELFAND-P000237

Uncertain reading. The upper right has a single upright stroke. It may be the numeral 1.

Original source details: GELFAND-P000237

Gelfand Seminar. 1968-69.

1. Mumford. Moduli spaces. (Geronimus Sasha)

2. ~~Kr.~~ Kirillov. Characters. (lectures)

Manuscript notation. The short crossed-out Kp. (Kr.) precedes Kirillov on the same line.

Original source details: GELFAND-P000237

3. Kazhdan Dwork ?

4. Griffiths Mixture of diff. eqs., geom., top., complex analysis

5. Spencer resolution. (Palamodov)

6. **Gromov**

7. Buslaev on eigenvalues (incl. Maslov's work)

8. Grauert and Remmert, p-adic analysis.

(old idea of L. G. Shnirelman about the Cauchy $\int$ and the like over an arbitrary field)

9. Nilson

10. Vershik on factors $\overline{\text{II}}$ and $\overline{\text{III}}$

or Sinai (and K°?)

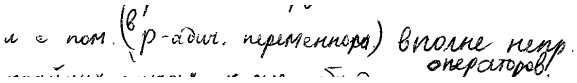
Uncertain reading. The small superscript on K is open at the top. Provisional $\circ$, possibly 0.

Original source details: GELFAND-P000237

11. Hörmander work on Lie groups and diff. eqs. (hypoell.).

Idea of **Dwork** (Kazhdan)

If some series converges near 0 in a complex variable and at every p-adic one, then it is a rational function.
idea: represent the ζ-function using ([Above the line: in] p-adic variables) completely continuous operators.



Manuscript notation. The word в (in) is written above "p-adic variables" in the parenthesis before "completely continuous operators".

Original source details: GELFAND-P000237

On Gromov: 2 extreme cases which are unified.

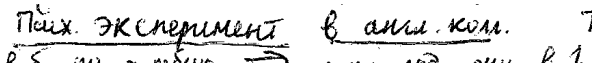
Bundle techniques are transferred to foliations (sometimes). For foliations of dimension 1 — the question of a vector field.

A foliation of codim. ≥ 1 always exists if a family of planes exists (!)

Original source details: GELFAND-P000237

Uncertain reading. The sign before 1 is heavily inked. The lower strokes favour $\geq$.

The sheaf of germs and jets — a weak homotopy equiv.
(If there is a set of coeff. glob., then one can construct a function)
Psych. experiment in an Engl. coll.



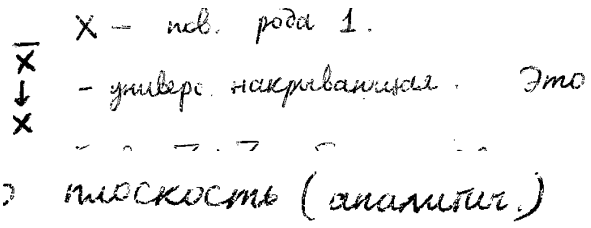
Original source details: GELFAND-P000237

Manuscript notation. The underlined words read в англ. колл. (in an Engl. coll.).

Tests $\Rightarrow$ teachers secretly told about the giftedness of 20% of the pupils, selected by lot $\Rightarrow$ after a year they are 1.5 times better than avg.! (not by marks, but by tests)

GELFAND-P000238

Geronimus on Mumford.
Object of study — compact Riem. surfaces.
$g = 0$ — all isomorphic (sphere).
$g = 1$ — complex elliptic curves.



Original source details: GELFAND-P000238

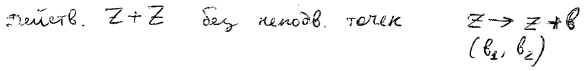
Diagram and typed description.
X — surface of genus 1.

$$\begin{array}{c} \overline{X} \\ \downarrow \\ X \end{array}$$

— universal cover. It is the plane (analytically)

$Z + Z$ acts without fixed points

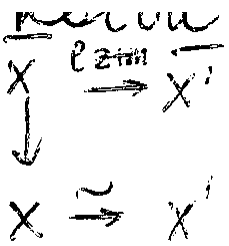
$$z \underset{(b_1,b_2)}{\to} z + b$$



Original source details: GELFAND-P000238

Manuscript notation. The pair (b_1, b_2) is written below the translation formula.

When do we obtain different surfaces?



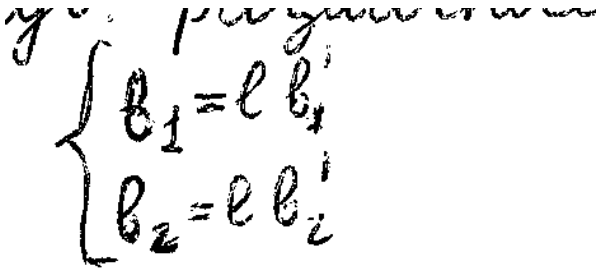
Original source details: GELFAND-P000238

Uncertain reading.

$$\begin{array}{ccc} X & \xrightarrow{\ell z + m} & X' \\ \downarrow & & \\ X & \xrightarrow{\sim} & X' \end{array}$$

Rules are visible above X and X'. The left rule is close to the underlined heading.

$$\begin{cases} b_1 = \ell b_1', \\ b_2 = \ell b_i'. \end{cases}$$



Manuscript notation. The final subscript has a dot and a lower hook and reads i. The left of the second line clearly reads b_2.

Original source details: GELFAND-P000238

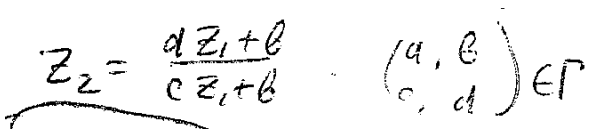
One can assume ($b_i \neq 0$) that $b_1 = 1$.

$(1, z_2)$ an automorphism of $Z + Z$ also changes nothing.

$$(1, z) \qquad \operatorname{Im} z > 0$$

$$\mathrm{SL}(2, Z) = \Gamma$$

$$z_2 = \frac{dz_1 + b}{cz_1 + b} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$$



Manuscript notation. The numerator reads $dz_1 + b$, and the denominator $cz_1 + b$. The first and last entries of the matrix are a and d.

Original source details: GELFAND-P000238

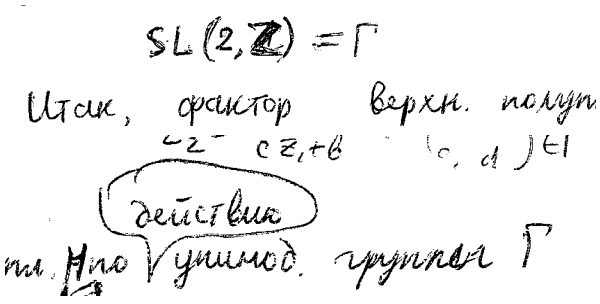
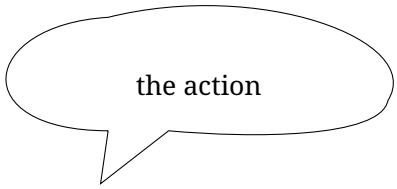


Diagram and typed description.

Thus, the quotient of the upper half-plane H by



of the unimodular group Γ

Original source details: GELFAND-P000238

$$H \longrightarrow H/\Gamma = \Pi$$

On H there is a fund. domain

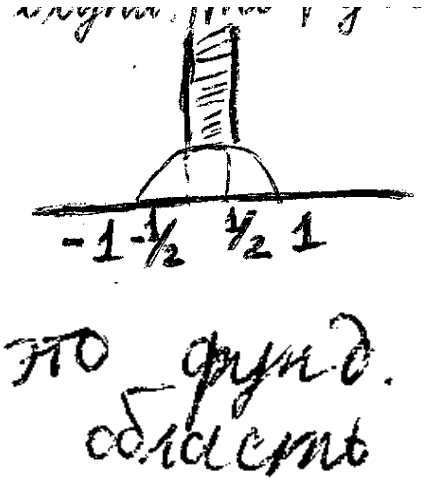


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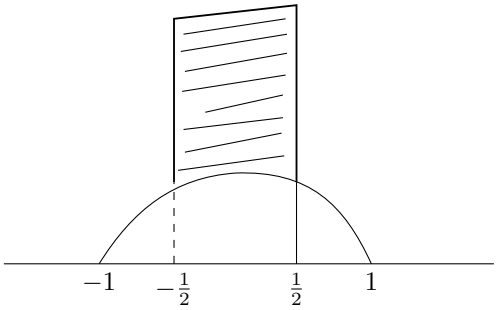
Diagram and typed description.

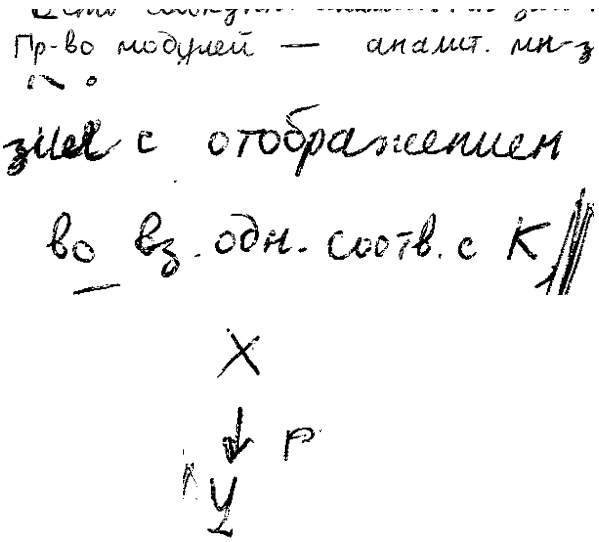


this is a fund. domain

The plane — the analytic set of all elliptic curves.

General def. 1

There is a collection of analytic sets K.



Original source details: GELFAND-P000238

Diagram and typed description.

Moduli space — an analytic set with a map

$$\begin{array}{c} X \\ \downarrow p \\ \underline{Y} \end{array}$$

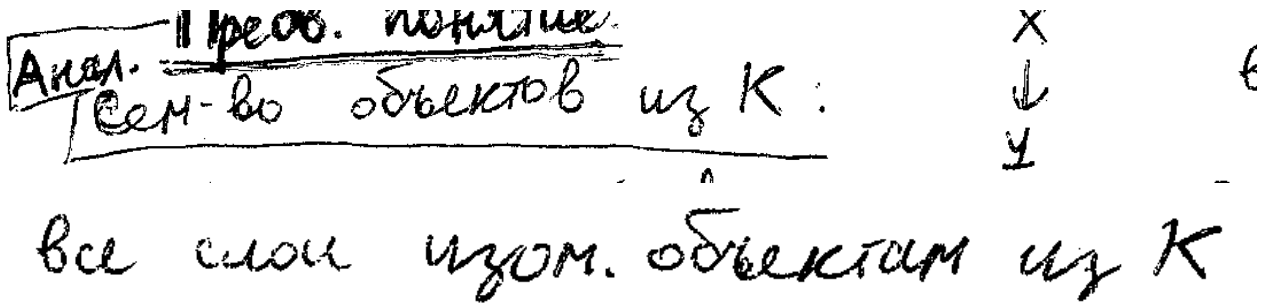
($X, \underline{Y}$ — manifolds and an analytic map). Points of $\underline{Y}$ in one-to-one corresp. with K and, in addition, $p^{-1}(y) \cong$ the corresponding object of K

(objects of K cannot be analytically isom.)

GELFAND-P000239

[source number: 2]

$Y = K$ with the structure of an analytic set
If the moduli space exists, it is unique up to analytic isom. (easy to prove).
Another def. takes account of possible automorphisms of objects of K.
Prelim. notion.



Original source details: GELFAND-P000239

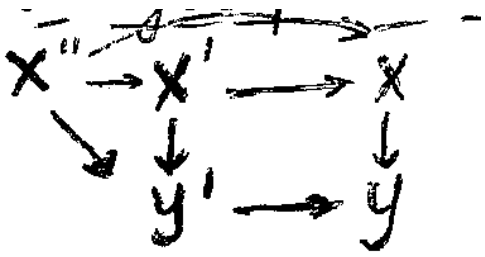
Diagram and typed description.

Anal. Family of objects of K:

$$\begin{array}{c} X \\ \downarrow \\ Y \end{array}$$

all fibers isom. to objects of K

Induced family — analogous to an induced bundle (it always exists)



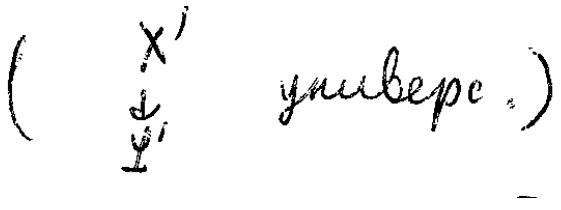
Original source details: GELFAND-P000239

Diagram and typed description.

$$\begin{array}{ccccc} X'' & \longrightarrow & X' & \longrightarrow & X \\ & \searrow & \downarrow & & \downarrow \\ & & Y' & \longrightarrow & Y \end{array}$$

This is a commut. diagram + univers.

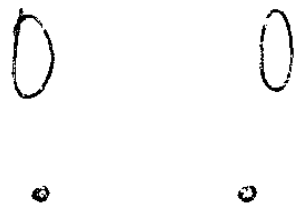
Universality



Original source details: GELFAND-P000239

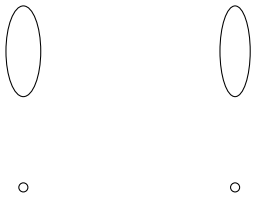
Diagram and typed description.

$$\begin{array}{ccc} & X' & \\ & \downarrow & \\ (& & \text{univers.}) \\ & \underline{Y'} & \end{array}$$

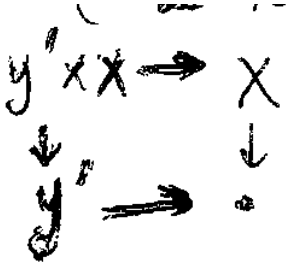


Original source details: GELFAND-P000239

Diagram and typed description.



A merely commut. diagram is insufficient (2 points and any one where a map $\exists$).



Original source details: GELFAND-P000239

Diagram and typed description.

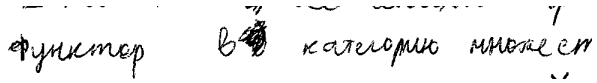
$$\begin{array}{ccc} Y' \times X & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & \bullet \end{array}$$

Kirillov (conjecture)

Representable functor. The representing object is the family of moduli

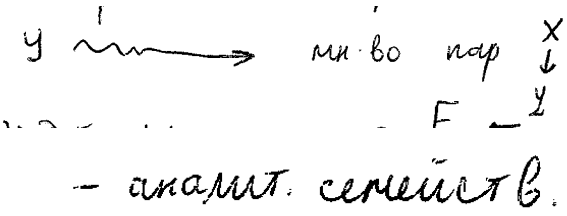
2 categ. — 1) all analytic spaces.

functor into the [Crossed out: 2)?] category of sets



Original source details: GELFAND-P000239

Uncertain reading. There is a heavily crossed-out mark just after “into”. Provisional 2), possibly 2.



Original source details: GELFAND-P000239

Diagram and typed description.

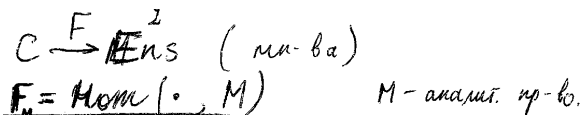
$$\begin{array}{ccc} Y & \rightsquigarrow & \text{set of pairs} \\ & & X \\ & & \downarrow \\ & & \underline{Y} \end{array}$$

— analytic families.

Repr. of the functor

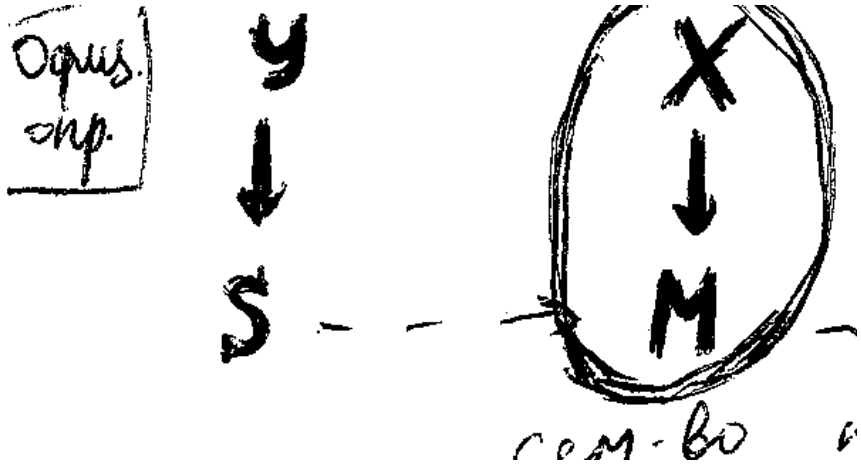
$$C \xrightarrow{F} \text{Ens} \quad (\text{sets})$$

$$F_M = \text{Hom}(\bullet, M) \qquad M \text{ — analytic space.}$$



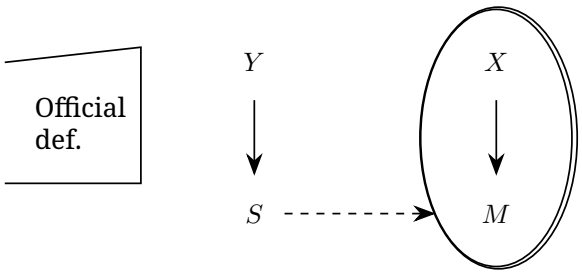
Manuscript notation. The upper line has $C \to \text{Ens}$ with F above the arrow. The first argument of Hom is a dot.

Original source details: GELFAND-P000239



Original source details: GELFAND-P000239

Diagram and typed description.



— the family is called a moduli space if $\forall$ other family is obtained as an induced one.

GELFAND-P000240

A universal family for curves of a given genus does not exist (even for elliptic ones)

Exists for polarized ones (with a chosen homology basis)

Modification — final object of the category of families

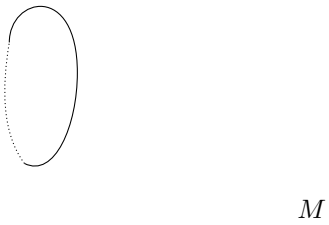
If it does not exist, adjoin it formally.

An additional structure is introduced, allowing cohom. with coeff. in a sheaf to be considered, and so on.

There is no universal object for elliptic curves

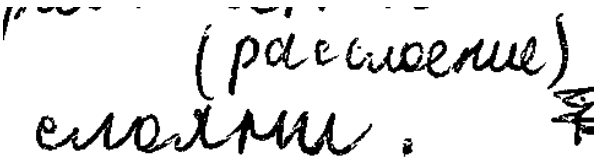
Original source details: GELFAND-P000240

Diagram and typed description.



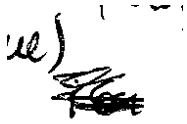
all points map unambiguously $\Rightarrow$ as a set — elliptic curves do not repeat.

Suppose there is a nontrivial family (base not a point) with identical fibers. [Above “fibers”: (bundle).]



Original source details: GELFAND-P000240

Manuscript notation. The parenthetical word расслоение (bundle) is written above слоями (fibers).

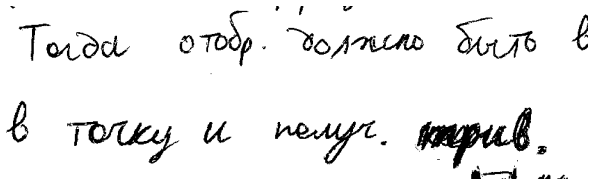


Original source details: GELFAND-P000240

Undeciphered passage. [Cancelled fragment to the right of “fibers”: reading not established.]

(because there are automorphisms)

Then the map must be into a point and one obtains triv.?



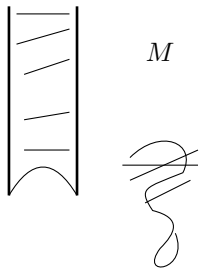
Original source details: GELFAND-P000240

Uncertain reading. The ending reads трив. (triv.). Heavy strokes precede it. A possible не (non-) is indistinct.

Exists in a weaker sense — the plane

Original source details: GELFAND-P000240

Uncertain reading.



The upper right mark resembles M. The crossed-out lower mark is illegible.

Most have one autom. ($x \to -x$) (with a marked point), and some have additional rotations.

Thus, the work consists in studying the properties of a nonexistent universal object.

Start precisely $6^{\underline{15}}$

Manuscript notation. The 15 is raised and underlined.

Original source details: GELFAND-P000240

GELFAND-P000241

[source number: 3]

$9/\underline{\text{IX}}$. **Continuation of the talk by Geronimus.**

Nonexistence of the moduli space is overcome by fixing a homology basis (a more rigid object). This is not considered.

For elliptic curves only a point is fixed (and even that not immediately).

A way (a different one) to overcome this difficulty will be indicated.

Def. of a Grothendieck topology.

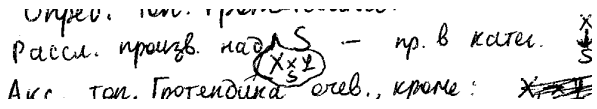
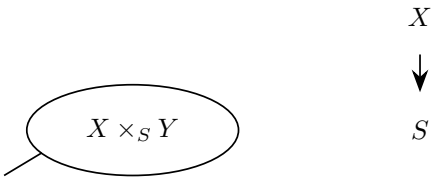


Diagram and typed description.

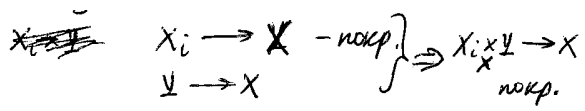
Fiber product over S — product in the categ.



Original source details: GELFAND-P000241

The axioms of a Grothendieck topology are obvious except: [Cancelled: $X_i \times Y$]

$$\left.\begin{array}{c} X_i \longrightarrow X \text{ — cover} \\ \underline{Y} \longrightarrow X \end{array}\right\} \Rightarrow \begin{array}{c} X_i \times_X \underline{Y} \longrightarrow X \\ \text{cover} \end{array}$$



Manuscript notation. The conclusion points to X. The two occurrences of Y are underlined. The crossed-out writing is to the left of the hypotheses.

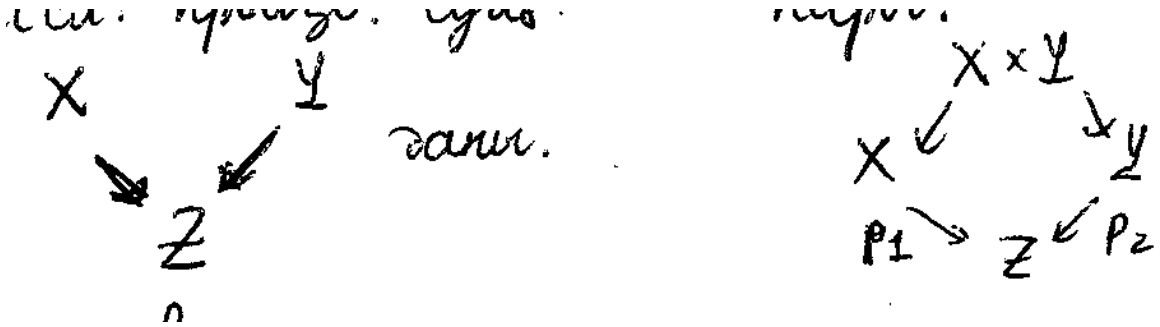
Original source details: GELFAND-P000241

A morphism of Grothendieck topologies —

— a functor $T_1 \to T_2$ preserving all structures.

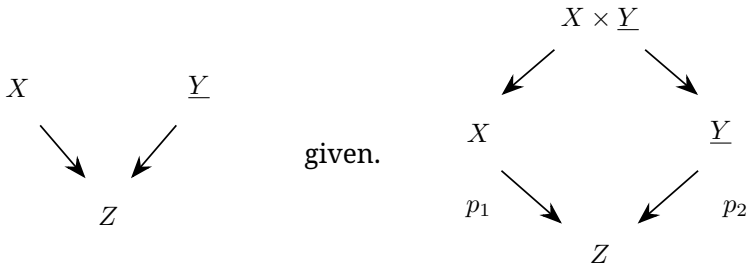
Example 1. T_Γ — Γ-sets (Γ — a group).

Fiber products exist — pairs.



Original source details: GELFAND-P000241

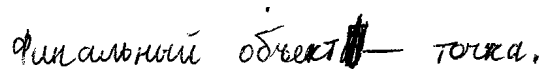
Diagram and typed description.



(in $X \times Y$ take pairs (x, y) such that $p_1(x) = p_2(y)$).

$X_i \xrightarrow{f} X$ — a covering if $\bigcup f_i(X_i) = X$.

Final object — a point.



Uncertain reading. Dense looped strokes follow "object". Any underlying reading is unresolved.

Original source details: GELFAND-P000241

Cohomology of a point with coeff. in various sheaves. $H^q(\cdot, F)$

(F — an arbitrary sheaf)

~~On~~ All sheaves are constructed as follows.

Manuscript notation. The word Ha (On) is crossed out before "All sheaves".

Original source details: GELFAND-P000241

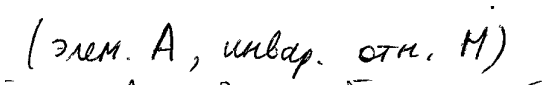
If the value $F(\Gamma) = A$ is specified (Γ acts on Γ on the left), then on all the others it is uniquely determined by the group A. (if this really is a sheaf and not a presheaf).

$\Gamma \longrightarrow \Gamma/H$ a covering.

Apply the sheaf property to the covering of the open set Γ/H. We obtain

$$F(\Gamma/H) = A^H$$

(elements of A invariant under H?)



Original source details: GELFAND-P000241

Uncertain reading. The letter in the parenthesis has a curved middle stroke. Nearby A^H supports provisional H. An alternative is M.

(Γ acts on A, because maps of sets induce maps of sections by the definition of a presheaf).

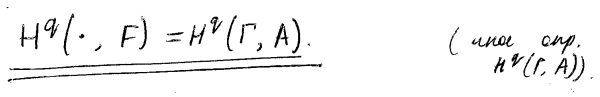
GELFAND-P000242

For an arbitrary Γ-set that is a union of disjoint orbits, one must consider the embeddings of all orbits, and there will be a direct sum.

Thus, everything is uniquely determined by the Γ-module A.
Here

$$\underline{\underline{H^q(\cdot, F) = H^q(\Gamma, A)}}.$$

(another def. of $H^q(\Gamma, A)$).

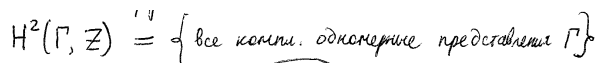


Original source details: GELFAND-P000242

Manuscript notation. The equality has two underlines. At right is the note “another def.” with $H^q(\Gamma, A)$.

A fact about group cohom.

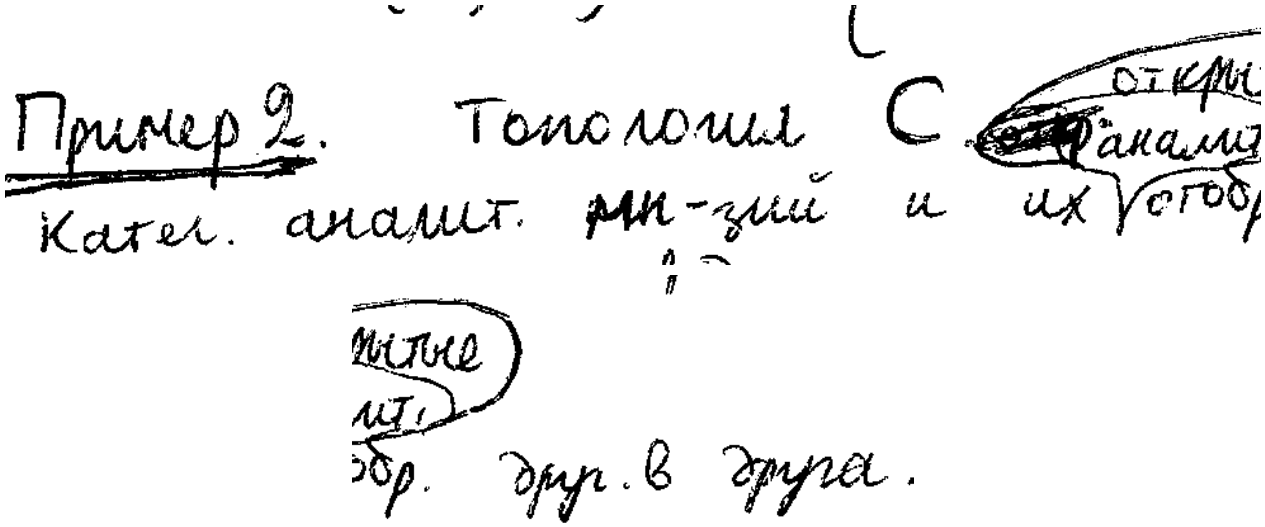
$$H^2(\Gamma, Z) = \{\text{all complex one-dimensional representations of } \Gamma\}$$



Original source details: GELFAND-P000242

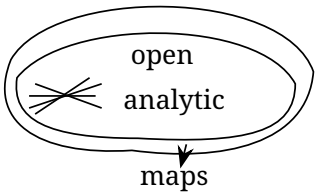
Manuscript notation. The left side is $H^2(\Gamma, Z)$. The right side gives all complex one-dimensional representations of Γ.

Example 2. Topology C



Original source details: GELFAND-P000242

Diagram and typed description.



The categ. of analytic sets and their maps to one another.

Undeciphered passage. The cancelled fragment to the left of “analytic” inside the enclosing stroke could not be deciphered.

Original source details: GELFAND-P000242

There is a fiber product.

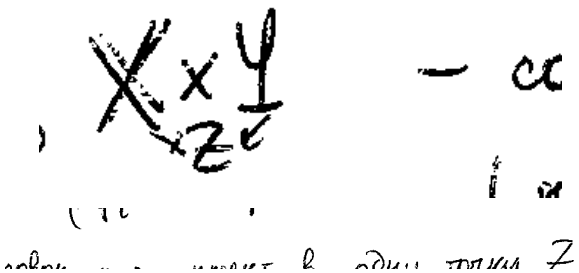
Coverings (!) X — an analytic set

$X_i \xrightarrow{f_i} X$ — a covering if

1) $\bigcup f_i(X_i) = X$

2) each f_i — an unramified covering of X_i over an open set of X

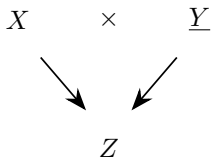
(f_i^{-1} a discrete set of points).



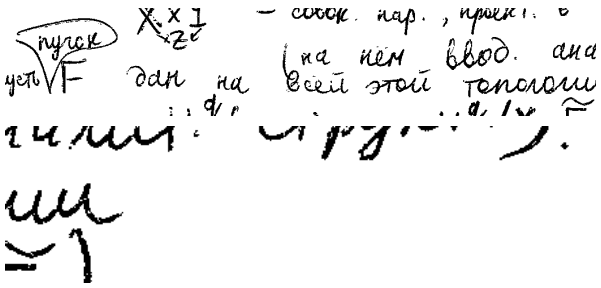
Original source details: GELFAND-P000242

(an analytic structure is introduced on it)

Diagram and typed description.

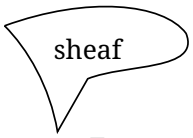


— the collection of pairs projecting to the same point of Z



Original source details: GELFAND-P000242

Diagram and typed description.



let F be given on this entire topology

$$H^q(X, F) = H^q(X, \widetilde{F})$$

$\widetilde{F}$ — a sheaf on the usual topology (restriction)

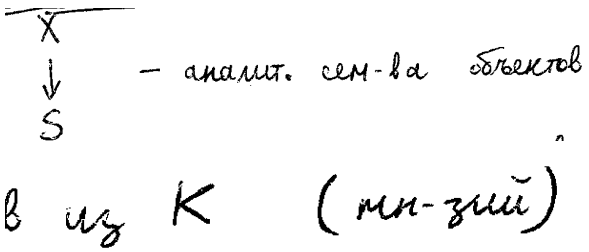
(ordinary cohomology)

(over a sufficiently small open set all coverings split).

GELFAND-P000243

[source number: 4]

Example 3. (the main one) Given K.



Original source details: GELFAND-P000243

Diagram and typed description.

$$\begin{array}{c} X \\ \downarrow \\ S \end{array}$$

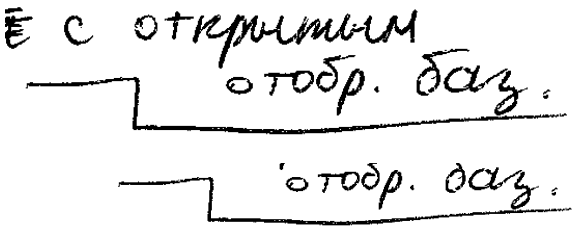
— analytic families of objects of K (sets)

A map — induction,

Original source details: GELFAND-P000243

Uncertain reading. Several strokes cross out a short word. Some loops remain visible. A possible reading is это (this).

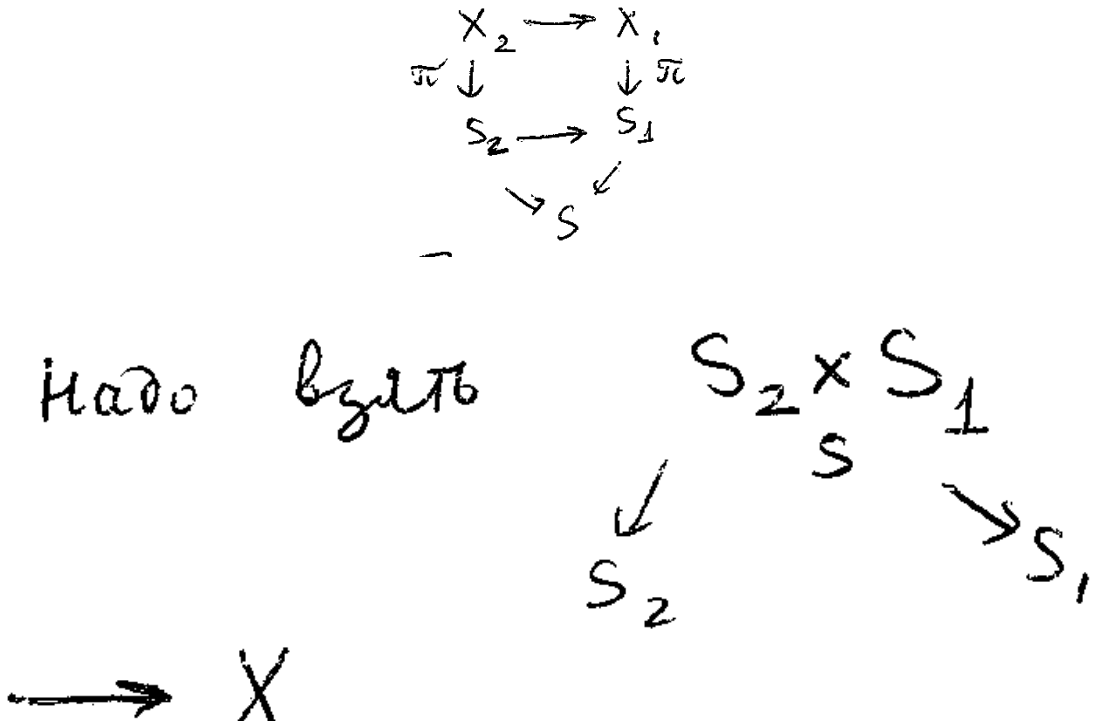
Diagram and typed description.



with an open map of bases

Original source details: GELFAND-P000243

Fiber products exist



Original source details: GELFAND-P000243

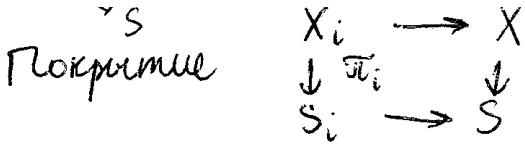
Diagram and typed description.

$$\begin{array}{ccc} X_2 & \longrightarrow & X_1 \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ S_2 & \longrightarrow & S_1 \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

One must take

$$\begin{array}{ccc} & S_2 \times_S S_1 & \\ \swarrow & & \searrow \\ S_2 & & S_1 \end{array}$$

and induce a family over it (from S_2 and S_1 one obtains the same thing)



Original source details: GELFAND-P000243

Diagram and typed description.

A covering

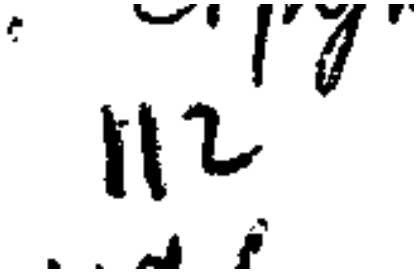
$$\begin{array}{ccc} X_i & \longrightarrow & X \\ \downarrow & \pi_i & \downarrow \\ S_i & \longrightarrow & S \end{array}$$

if the images cover all of S and each map —

— a covering (in the sense of Ex.2).

$H^q(\pi, F)$ equiv. to the preceding example $\cong$ cohom. of S in the preceding example.

(since the structures of open sets are the same and the topologies are isomorphic).



Original source details: GELFAND-P000243

Uncertain reading.

Two upright strokes and a small 2-like mark stand between the cohomology lines. The meaning of the smaller mark is uncertain.

$H^q(S, \widetilde{F})$ — ordinary cohomology.

If there is a terminal object.

Suppose there is no terminal object.

Example 4.

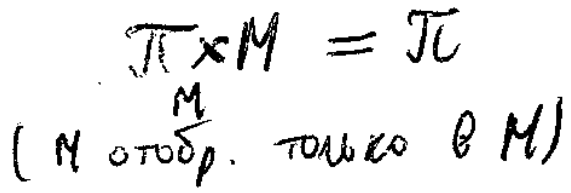
T_M — categ. — the old one + a formally adjoined terminal object M.

Its cohomology is computed as if it existed.

A fiber product exists.

$$\pi \times_M M = \pi$$

(M maps only to M)



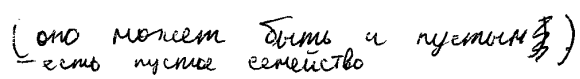
Original source details: GELFAND-P000243

Manuscript notation. The product has M below the $\times$ sign. The equation repeats π.

The fiber product $X \times_M \underline{Y}$ remains

In any category $\boldsymbol{X} \times_{\boldsymbol{M}} \underline{\boldsymbol{Y}}$ — the direct product.

(a tautological fact).



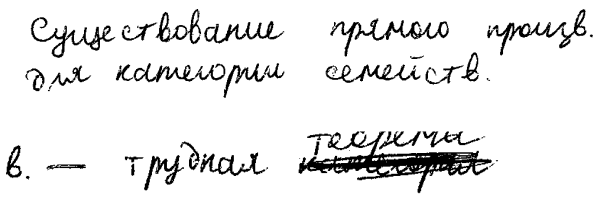
Original source details: GELFAND-P000243

Uncertain reading. (it may also be empty [a sign resembling ξ, possibly cancelled])

— there is an empty family

GELFAND-P000244

Existence of a direct product — a difficult [crossed out: category?] [inserted above: theorem] for the category of families.



Uncertain reading. The insertion above the crossed-out word reads теорема (theorem). Under the strokes, категория (category) is the likelier reading.

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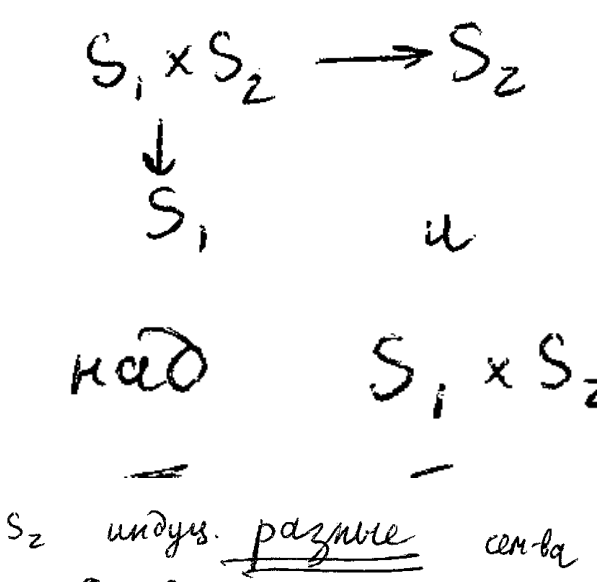

Diagram and typed description.

$$\begin{array}{ccc} S_1 \times S_2 & \longrightarrow & S_2 \\ \downarrow & & \\ S_1 & & \end{array}$$

over $S_1 \times S_2$ different families are induced

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and one cannot take $S_1 \times S_2$ as the base.

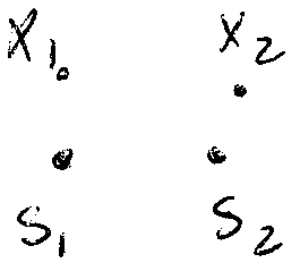
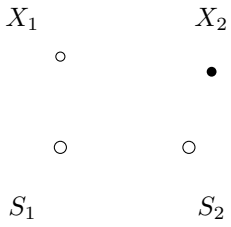


Diagram and typed description.



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If $X_1 = X_2$ then the product — itself.
If $X_1 \neq X_2$, then empty.

Manuscript notation. The crossed-out word before “If a point” reads Но (But).

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If a point, then as a rule empty.

If close to universal, then usually nonempty.

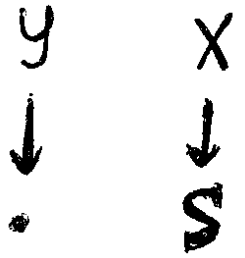
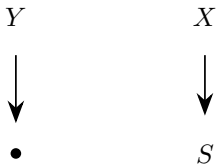


Diagram and typed description.



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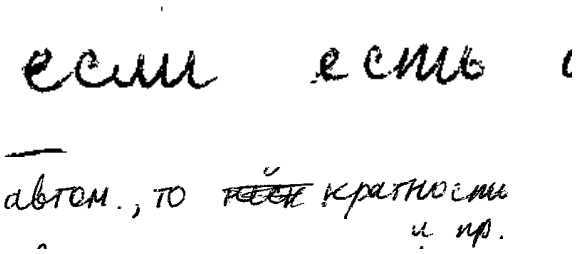
The base — a subset of S, with fibers isom. to Y.

if there are autom., then [crossed out: there is?] multiplicity etc.



Uncertain reading. A short word before “multiplicity” is crossed out, possibly есть (there is).

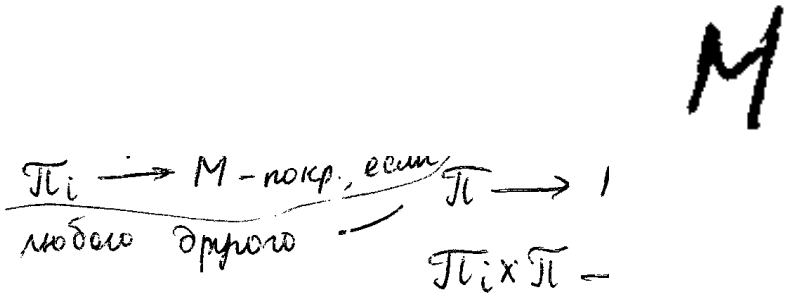
Original source details: GELFAND-P000244

Set-theoretically — the base — points in S with fibers Y, multiplicity — the number of automorphisms.

This — an analytic set! (a difficult theorem)

(the number of ways of mapping Y into this family)

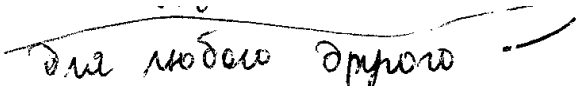
Def. of a covering of M:



Original source details: GELFAND-P000244

Diagram and typed description.

$\pi_i \longrightarrow M$ — a cover if $\pi \longrightarrow M$

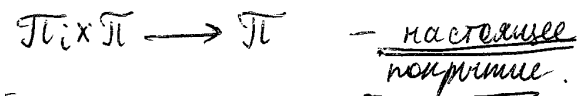


Original source details: GELFAND-P000244

Diagram and typed description.

for any other

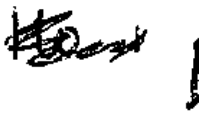
$\pi_i \times \pi \longrightarrow \pi$ — a genuine covering.



Manuscript notation. There is no index under $\times$. The words “genuine covering” are underlined twice.

Original source details: GELFAND-P000244

(one axiom is ensured by this def., the others — automatically)
All π_i — not a covering!



Uncertain reading. [Cancelled: But / To us?]

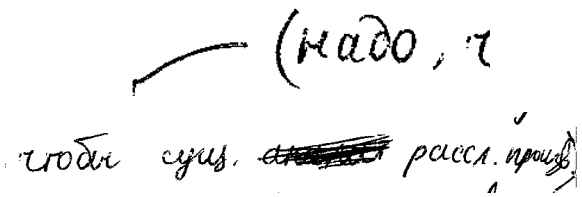
Original source details: GELFAND-P000244

Next time, continuation and or the Spencer resolvent or Dwork.

GELFAND-P000245

[source number: 5]

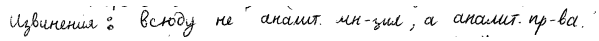
16/IX Geronimus.



Undeciphered passage. (it is necessary that there exist [a cancelled word cannot be read beneath dense strokes] fiber products)

Original source details: GELFAND-P000245

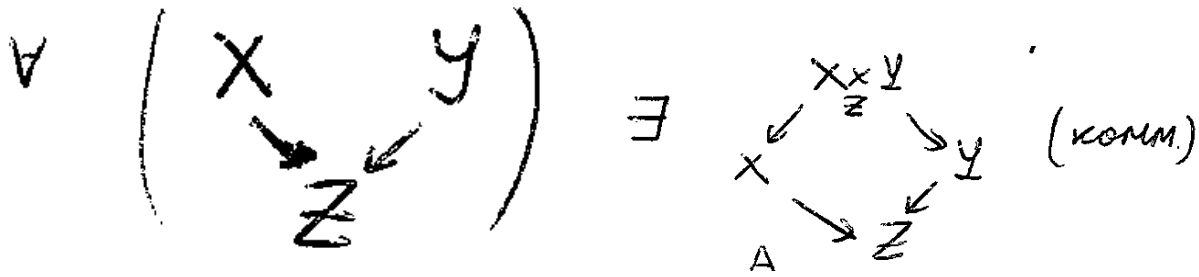
Apologies: everywhere, not analytic sets, but analytic spaces.



Manuscript notation. The two abbreviations mean “analytic sets” and “analytic spaces”.

Original source details: GELFAND-P000245

Fiber product.

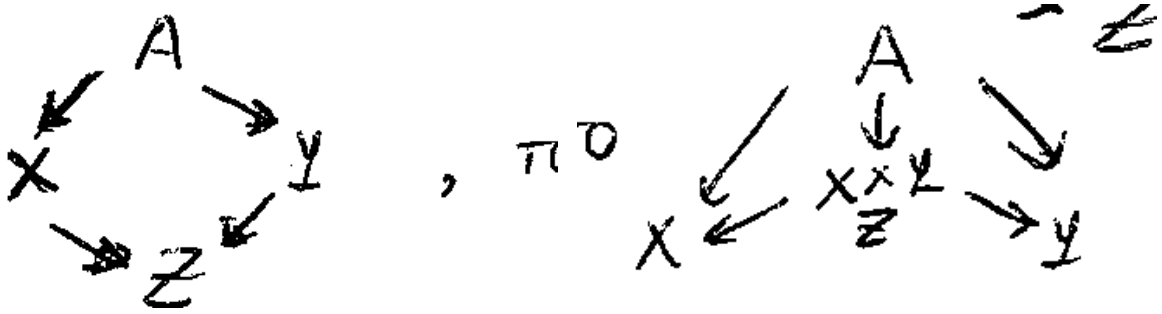


Original source details: GELFAND-P000245

Diagram and typed description.

$$\forall \left(\begin{array}{ccc} X & & Y \\ & \searrow \quad \swarrow & \\ & Z & \end{array} \right) \quad \exists \quad \begin{array}{ccccc} & & X \times_Z \underline{Y} & & \\ & \swarrow & & \searrow & \\ X & & & & \underline{Y} \\ & \searrow & & \swarrow & \\ & & Z & & \end{array} \qquad \text{(comm.)}$$

and universally, i.e. if



Original source details: GELFAND-P000245

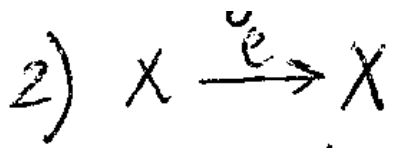
Diagram and typed description.

$$\begin{array}{ccc} & A & \\ \swarrow & & \searrow \\ X & & \underline{Y} \\ \searrow & & \swarrow \\ & Z & \end{array} \quad \text{, then} \quad \begin{array}{ccc} & A & \\ & \downarrow & \\ X & X \times_Z \underline{Y} & \underline{Y} \\ \searrow & & \swarrow \\ & Z & \end{array}$$

Covering axioms

1) $X_{ij} \longrightarrow X_i \longrightarrow X$

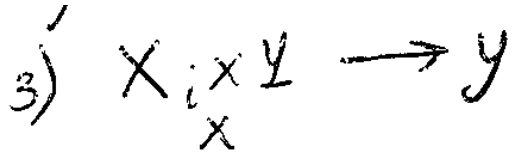
2) $X \xrightarrow{e} X$



Original source details: GELFAND-P000245

Manuscript notation. The label above the arrow reads e, with a short upper loop.

3) $X_i \times_X \underline{Y} \longrightarrow Y$



Manuscript notation. The arrow points to Y. The earlier Y is underlined and X stands below $\times$.

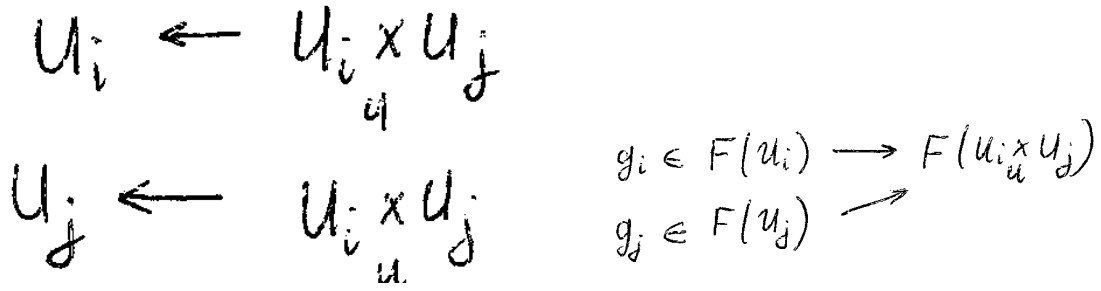
Original source details: GELFAND-P000245

Presheaf: a contravariant functor F to the categ. of groups.

Sheaf.

$U_i \longrightarrow U$ — a covering

$g_i \in F(U_i)$



Original source details: GELFAND-P000245

Diagram and typed description.

$$\begin{array}{ccccc} U_i & \longleftarrow & U_i \times_U U_j & & \\ & & & & \\ U_j & \longleftarrow & U_i \times_U U_j & & \end{array} \qquad \begin{array}{ccc} g_i \in F(U_i) & \longrightarrow & F(U_i \times_U U_j) \\ & \nearrow & \\ g_j \in F(U_j) & & \end{array}$$

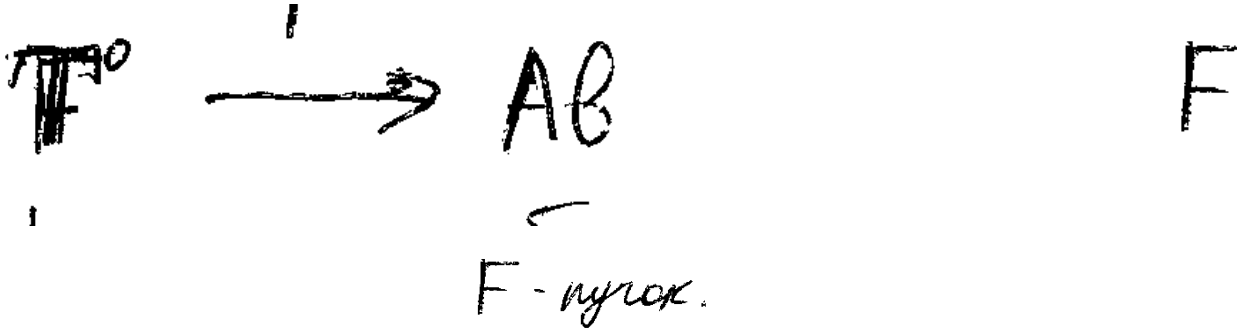
Compatible if the g_i map to the same element.

$$U_i \longrightarrow U \qquad F(U) \longrightarrow F(U_i) \quad (g_i)$$

Uncertain reading. [Cancelled: def. / exists / this?]

Original source details: GELFAND-P000245

a sheaf: if $g \in F(U)$ exists and is unique, mapping to the g_i.



Original source details: GELFAND-P000245

Uncertain reading.

$T^\circ \xrightarrow{F}$ Ab F — a sheaf.

categ. with reversed arrows. abel. groups

The base letter is overwritten and provisionally reads T. The superscript is round and is set as $\circ$.

GELFAND-P000246

Cohomology — these are groups
$H^\nu(U, F)$,

Original source details: GELFAND-P000246

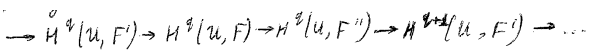
Manuscript notation. The opening degree is angular and open, ν. The exact sequence uses a looped q.

such that $H^0(U, F) = F(U)$. and if

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0, \quad \text{then}$$

(there is a procedure allowing one to construct a sheaf canonically from any presheaf — for any topology),

$$\longrightarrow H^q(U, F') \longrightarrow H^q(U, F) \longrightarrow H^q(U, F'') \longrightarrow H^{q+1}(U, F') \longrightarrow \dots$$



Original source details: GELFAND-P000246

Manuscript notation. The first three terms have q, followed by $q+1$. The final written term has one prime on F.

(they exist).

And for any injective sheaf cohom. 0

(and there are sufficiently many inj. sheaves)

Suppose there is a map compatible with all structures (a map of topologies)

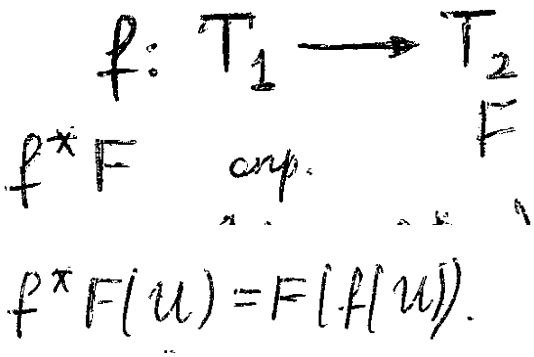


Diagram and typed description.

$$f: \quad T_1 \longrightarrow T_2$$

f^*F def. F

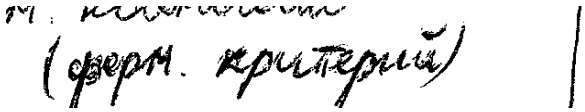
$$f^*F(U) = F(f(U)).$$

Original source details: GELFAND-P000246

$$H^q(U, f^*F) \longrightarrow H^q(f(U), F).$$

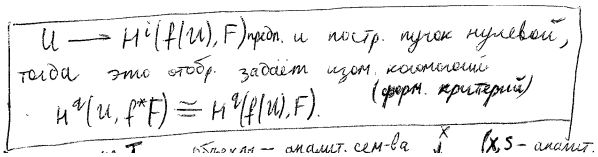
$U \longrightarrow H^i(f(U), F)$ a presheaf and the constructed sheaf is zero,
then this map gives an isom. of cohomology (form. criterion)

$$H^q(U, f^*F) \cong H^q(f(U), F).$$



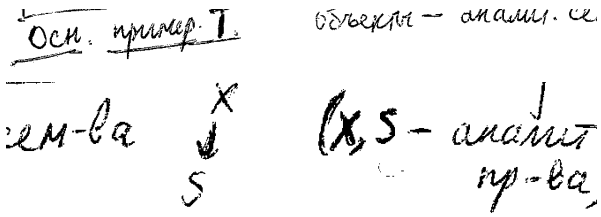
Original source details: GELFAND-P000246

Manuscript notation. The parenthesis reads форм. критерий (form. criterion). The words above it read изом. когомологий (isom. of cohomology).



Original source details: GELFAND-P000246

Manuscript notation. The box ends with $H^q(U, f^*F) \cong H^q(f(U), F)$.



Original source details: GELFAND-P000246

Diagram and typed description.

Main example T. objects — analytic families

$$\begin{array}{c} X \\ \downarrow \\ S \end{array}$$

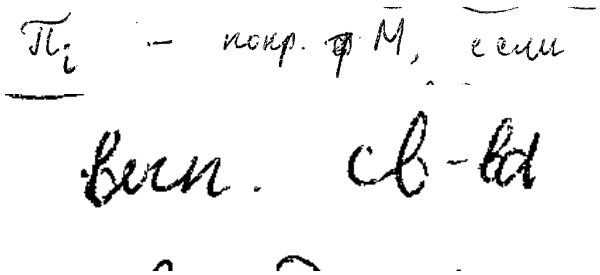
(X, S — analytic spaces)

It is necessary that

New category T_M with M. (a terminal object)

Existence of fiber products.

(topol. of the moduli space)



Uncertain reading. π_i — a covering [cancelled sign: φ / o?] of M if the properties hold

Original source details: GELFAND-P000246

1) each K occurs as a fiber in one of the families π_i

2) each K occurs only a discrete number of times

GELFAND-P000247

[source number: 6]

3) Extension axiom. $\forall X$.

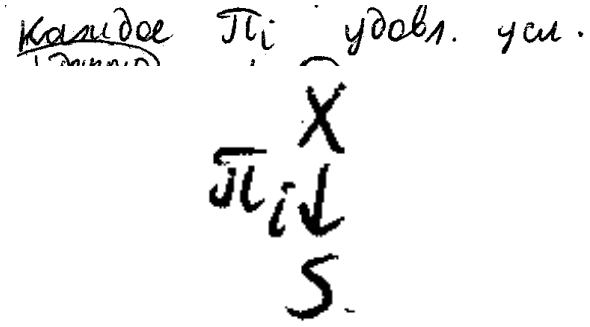


Diagram and typed description.

Each Π_i satisfies the cond.

$$\begin{array}{cc} & X \\ \pi_i & \downarrow \\ & S \end{array}$$

Original source details: GELFAND-P000247

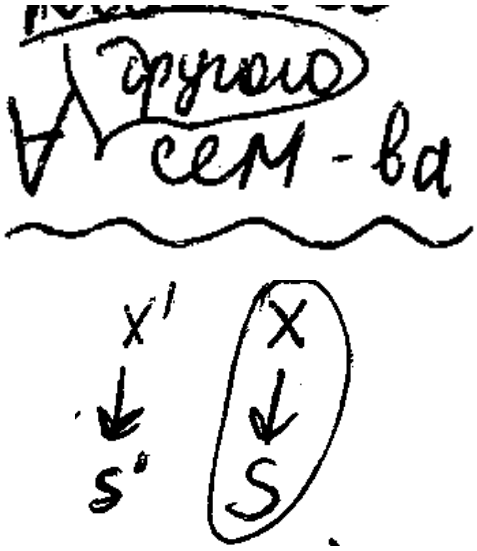
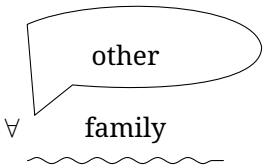


Diagram and typed description.



$$\begin{array}{cc} X' & X \\ \downarrow & \downarrow \\ S' & S \end{array}$$

Original source details: GELFAND-P000247

and a subfamily (base — a submanifold and [crossed out: i.e.?] induced restr.)

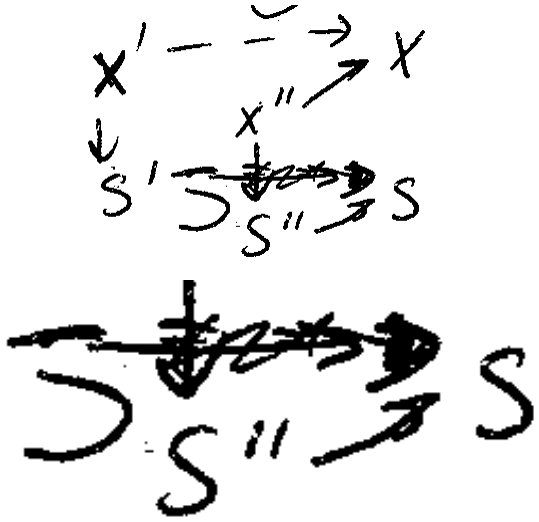
Original source details: GELFAND-P000247

Uncertain reading. A point and a low loop are visible under the strike. Provisional т.е. (i.e.), possibly т.д. (etc.).

and ∀ map of the subfamily

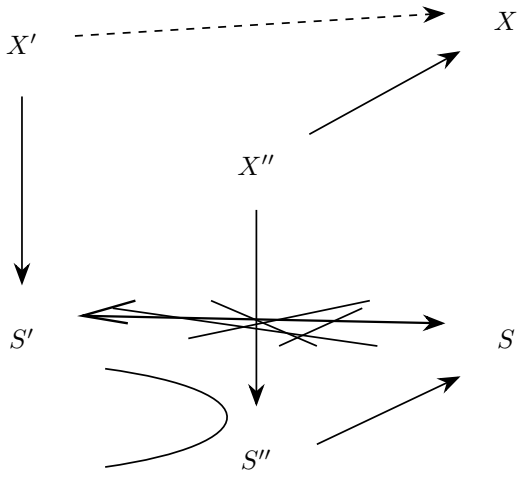
∃ its extension (local)-

— to a tubular neighborhood S''.



Original source details: GELFAND-P000247

Uncertain reading.

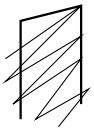


A right arrowhead and a dense left wedge survive on the crossed-over line $S' \to S$. The meaning of the curved stroke below is uncertain.

Next time — example with ellipt. curves.

Original source details: GELFAND-P000247

Diagram and typed description. Isolated hatched figure in the lower margin.



GELFAND-P000248

Palamodov.

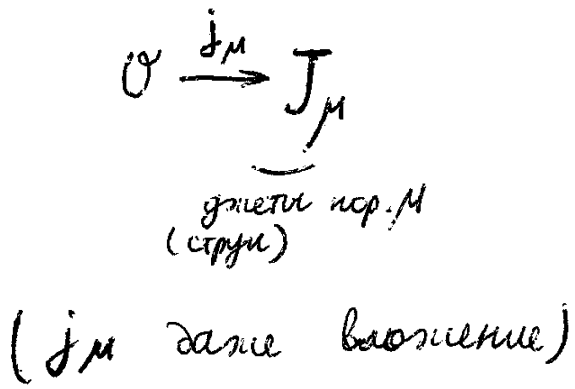
Spencer resolution.

Original source details: GELFAND-P000248

Diagram and typed description.

Working on this	Sweeny Holdschmidt Guillemin Quillen Sternberg Singer	presented by Sweeny in Acta math.

A method for solving the system $p(x, D)u = w$, where p is a matrix, u, w are vectors.



Original source details: GELFAND-P000248

Diagram and typed description.

M — a smooth manifold. $\mathcal{O}$ — the sheaf of germs of diff. fns.

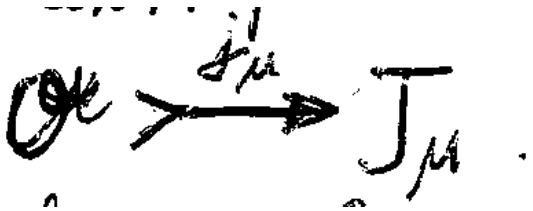
$$\mathcal{O} \xrightarrow{j_\mu} J_\mu$$

jets of ord. μ

(jets)

(j_μ is even an embedding)

J_μ — vect. bundle (or the sheaf of its germs)



Original source details: GELFAND-P000248

Uncertain reading.

$$\mathcal{O} \xrightarrow{j_\mu} J_\mu \ .$$

Retraced $\mathcal{O}$ / $\mathcal{O}$ with an upper sign; overlapping strokes not deciphered.

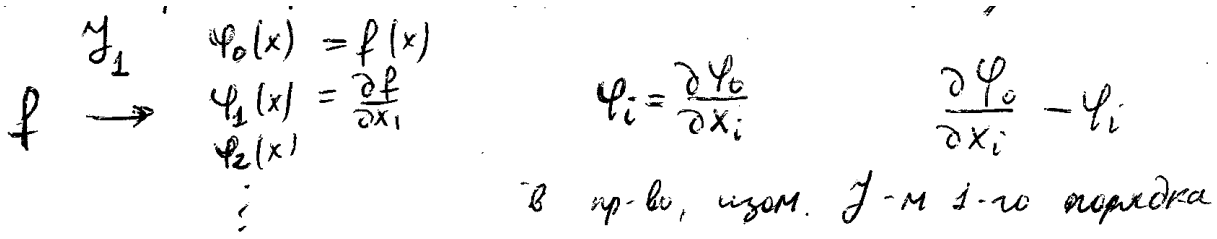
The Spencer resolution of the sheaf $\mathcal{O}$ begins with j_μ and continues so that the seq. of sheaves is exact.

$$0 \longrightarrow \mathcal{O} \xrightarrow{j_\mu} J_\mu \longrightarrow$$

$\mu = 0$ — map to 0.

A collection is given (I. M. G.'s project)

$$\varphi_i \qquad i \text{ — multi-index}, \qquad |i| \le \mu.$$



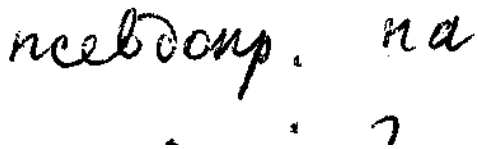
Original source details: GELFAND-P000248

Diagram and typed description.

$$f \xrightarrow{j_1} \begin{matrix} \varphi_0(x) = f(x) \\ \varphi_1(x) = \dfrac{\partial f}{\partial x_1} \\ \varphi_2(x) \\ \vdots \end{matrix} \qquad \varphi_i = \frac{\partial \varphi_0}{\partial x_i} \qquad \frac{\partial \varphi_0}{\partial x_i} - \varphi_i$$

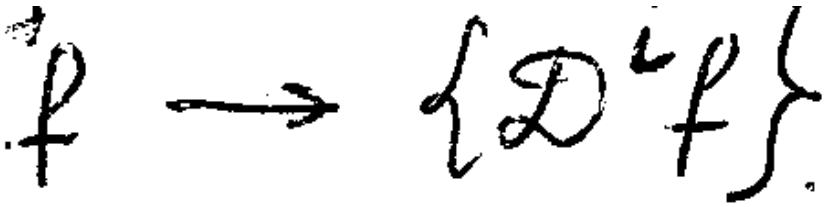
to a sp., isom. to the J's of 1st order

Cf. псевдопр.? one order higher.



Uncertain reading. Provisional reading of the abbreviation псевдопр. Its expansion is uncertain.

Original source details: GELFAND-P000248



Original source details: GELFAND-P000248

Uncertain reading.

$$f \longrightarrow \{D^i f\}$$

Above the first f are short strokes; their meaning is not established.

GELFAND-P000249

[source number: 7]

notation $1_j = (0, \ldots, \underset{j}{1}, \ldots, 0)$

cf. φ_i with φ_{i+1_j}.

Need to chk. that

$$\frac{\partial \varphi_i}{\partial x_j} = \varphi_{i+1_j}, \qquad \underset{|i| \le \mu - 1}{\forall i}, \qquad j = 1, \ldots, n.$$

Map to

Uncertain reading. Crossed out:

The first mark is heavily crossed out, possibly J or j. The start of «джеты» (jets) is rewritten.

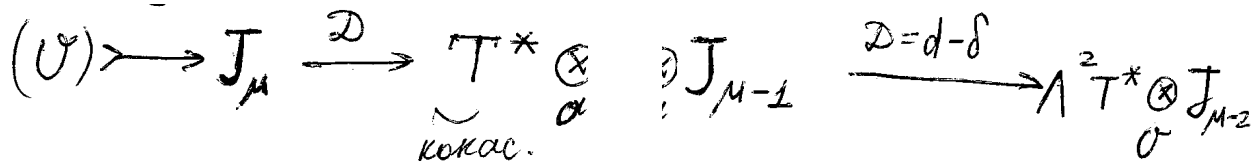
Original source details: GELFAND-P000249

jets of order lower by 1.

$$\{\varphi_i\} \longrightarrow \left\{ \frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right\}$$

$$|i| \le \mu - 1, \qquad j = 1, 2, \ldots, n.$$

$$\sum \left(\frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right) dx_j \qquad \text{(added).}$$



Original source details: GELFAND-P000249

Diagram and typed description.

$$(\mathcal{O}) \rightarrowtail J_\mu \xrightarrow{\mathcal{D}} T^* \otimes_{\mathcal{O}} J_{\mu-1} \xrightarrow{\mathcal{D}=d-\delta} \Lambda^2 T^* \otimes_{\mathcal{O}} J_{\mu-2}$$

cotang.

$$\left(\frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right) = (d-\delta)\{\varphi_i\}$$

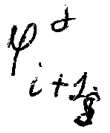
(difference of the operators of actual and formal diff.)

On $T^* \otimes J_{\mu-1}$ both ops. are also def.

d — exterior diff. in the 1st coord.

δ — formal diff.

$$\{\varphi_i^j(x)dx_j\} \overset{\mathcal{D}=d-\delta}{\longrightarrow} d(\varphi_i^j(x)dx_j) - \varphi^j_{i+1_s}\, dx_j \wedge dx_s$$



Uncertain reading. Provisional $i+1_s$. The lower s is overwritten, possibly over an earlier j.

Original source details: GELFAND-P000249

and so on.

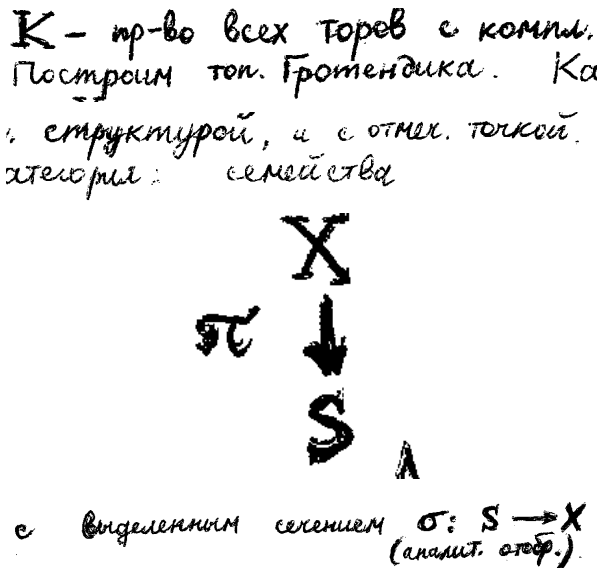
The resolution will terminate (and it is exact)

Next time Geronimus, Palamodov and Kazhdan

GELFAND-P000250

$23/\overline{\text{IX}}$

Geronimus.



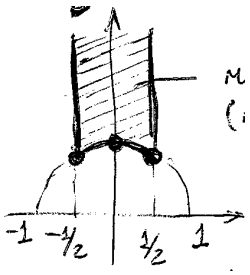
Original source details: GELFAND-P000250

Diagram and typed description.

K — sp. of all tori with a compl. structure, and with a marked point.
Let us construct Grothendieck's top.
Category: families

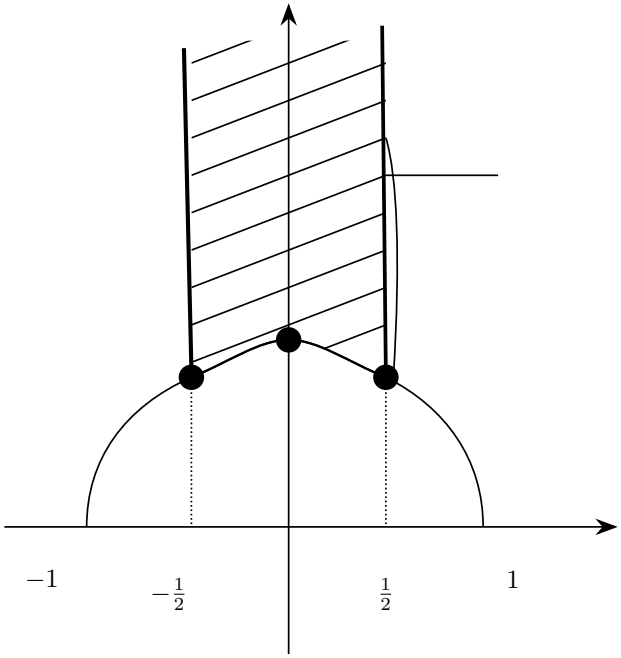
$$\begin{array}{ccc} & & X \\ & \pi & \downarrow \\ & & S \end{array}$$

with a distinguished section $\sigma\colon S \to X$ (analyt. map)



Original source details: GELFAND-P000250

Uncertain reading.



Two close strokes run along the right boundary. Their exact junction is uncertain.

set of all lattices S'

(moduli sp. in the coarse sense of the word)

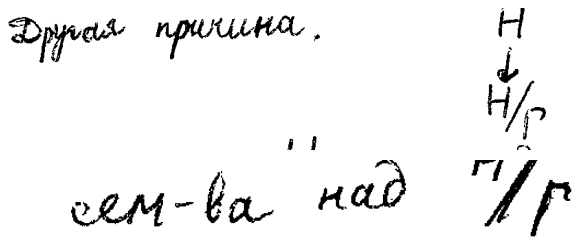
(*coarse*)

Grothendieck top. — mod. sp. in the fine (*fine*) sense of the word.

Quotient H/Γ — H — upper half-plane

There is a family with base S', but this is not a fine moduli sp.

(does not induce nontriv. bundles with const. fiber).



Original source details: GELFAND-P000250

Diagram and typed description.

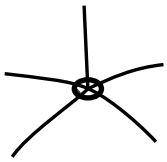
Another reason.

The family over H is not induced from a family over H/Γ.

(Under the action on H there are fixed points (3 of them)
there is branching there

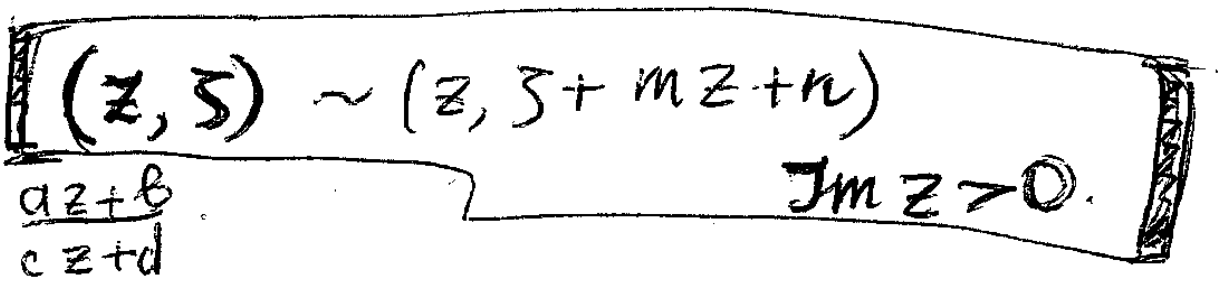
Original source details: GELFAND-P000250

Uncertain reading.



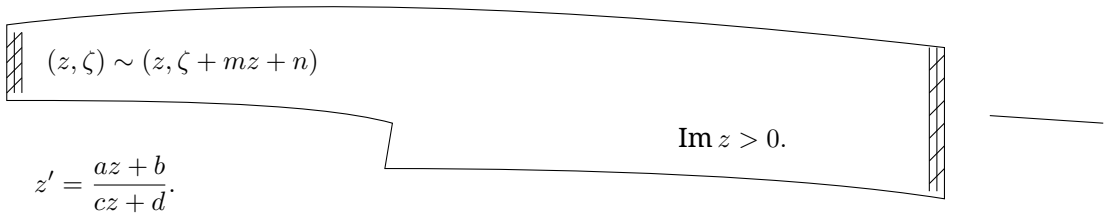
Dense retracing at the node and stroke ends; arrowheads are not established.

and different struct. of the ellipt. curve are obtained)
Family over H $\quad H \times \mathbb{C} \quad z \in H \quad (z, 1)$ —
— a lattice.



Original source details: GELFAND-P000250

Diagram and typed description.



Problem: this family,

GELFAND-P000251

[source number: 8]

and there the group acts

$$z' = \frac{az+b}{cz+d}$$
$$\zeta' = \zeta(cz+d)^{-1}$$

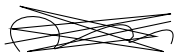
Uncertain reading. Crossed out in the upper-right margin. Provisional reading там (there).

Original source details: GELFAND-P000251

Calculation of cohomology in this case.

The topology and everything else were defined earlier.

Uncertain reading. A separate word is cancelled:



Individual strokes survive under the crossing. The word is uncertain.

Original source details: GELFAND-P000251

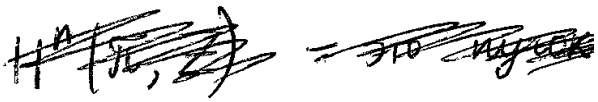


Diagram and typed description. Crossed-out formula and words.

~~$H^n(\pi, Z)$ — this is a sheaf~~

Original source details: GELFAND-P000251

$$F(\pi) = \underbrace{Z + Z + Z}_{\text{by the number of conn. comp.}}$$

(so to speak, a const. sheaf).

It is easy to see that

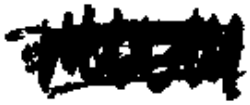
$$\underset{(\pi\text{ — ordinary})}{H^n(\pi, Z)} = \underset{\text{(Topological).}}{H^n(S, Z)}$$

Cohom. of the final object.

1) T_Γ — another top.; Γ — a group of matrices with det 1.

2) T'_Γ — those Γ-sets where Γ does not act too triv.

(if the stabilizer of $\forall$ point — integral matr. with



Original source details: GELFAND-P000251

Uncertain reading. The cancelled word has not been deciphered.

1 mod 3.

$$\sigma x = x \to \sigma \in \Gamma_3.)$$

Map $T'_\Gamma \to T$

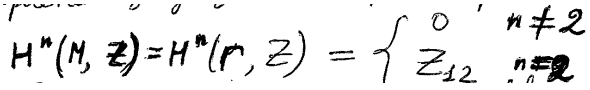
$$S \to (H \times S)/\Gamma \qquad (S \text{ — a discr. set}).$$

$$M' \to M.$$

It induces an isomorphism of cohomology.

(which is checked using the criterion stated last time).

Hence



Original source details: GELFAND-P000251

Uncertain reading.

$$H^n(M, Z) = H^n(\Gamma, Z) = \begin{cases} 0 & n \neq 2 \\ Z_{12} & n = 2 \end{cases}$$

The sign and digit in the second condition are retraced. Provisional $n = 2$.

(which, if Γ acted without fixed points, would coincide with the cohom. of the ordinary moduli sp.).

The answer is valid over any field (Char $\neq 2$).

Families have an invar.

mod 12.

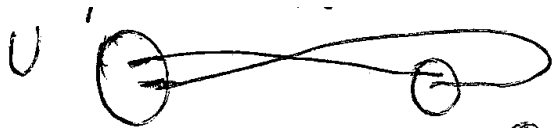
GELFAND-P000252

Shafarevich: an example of Grothendieck top.

A syst. of diff. eqns. (ordinary) is given on a set

U — transv. patch

Map



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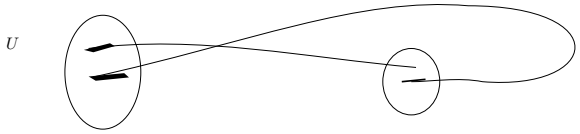
Uncertain reading.



The directions of all parts and exact joins are not established; the left-hand thickenings are retained by their shape.

$U \to V$

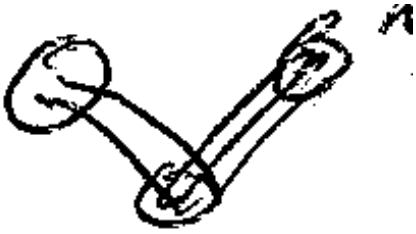
if from points of U

draw [crossed out: ...] traject. into V?.

Original source details: GELFAND-P000252

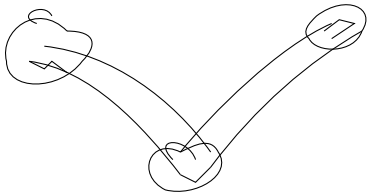
Uncertain reading. The crossed-out fragment is uncertain. The overwritten final sign is provisionally V.

there is a fiber product



Original source details: GELFAND-P000252

Uncertain reading.



Additional marks are visible within the outlines and at the upper right. Their meaning is uncertain.

If a fibration, then the cohom. of the base and the syst. are triv.

(then there is a final object).

On the plane diff. eqn. and const. sheaf of Groth. top.,

then its Euler char. — the difference [inserted above: of numbers] of lim. cycles and sing. points.

Original source details: GELFAND-P000252

Diagram and typed description. Insertion of “of numbers”.

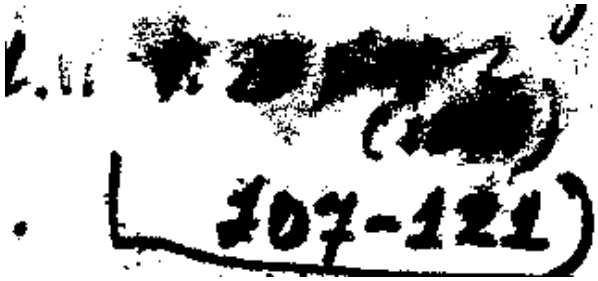
Next time Kazhdan. and Palamodov.

GELFAND-P000253

[source number: 9]

30/IX. Kazhdan.

Work of Ihara. (Jap.). (Ihara, Journ. Math. Soc. of J.



Undeciphered passage. Two short strokes follow "J." A dense cancellation follows. The range 107–121 is legible.

Original source details: GELFAND-P000253

107-121)

One-dimensional surf. over C are connected with discr. groups.

An attempt to transfer this to curves over finite fields.

$X = \Gamma\backslash M$ $\quad$ M — Lobach. pl.

Γ describes both X and unramif. coverings.

(The congruence monodromy problem).

In [crossed out: ...] $k[X, Y]$ consider the principal ideal (f)

Uncertain reading. The sign before the ring is heavily crossed out. Its reading is uncertain.

Original source details: GELFAND-P000253

$A = k[x, y]/(f)$ — ring of fns. on the curve, max. ideals — geom. points of the curve

A map $Y \longrightarrow X$ is a map of rings $A \longrightarrow B$

$$B \longleftarrow A$$

An unramif. map — unramif. at $\forall$ point.

m max. ideal. $\subset A$ $\quad$ Consider the image and B/mB

if it does not contain nilpot. elems., then it is unramified at m.

There will be no seminar next Monday.

The time after next there will be...

GELFAND-P000254

14/X Palamodov.

X — set

O — sheaf of germs of diff. fns.

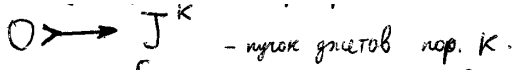


Diagram and typed description.

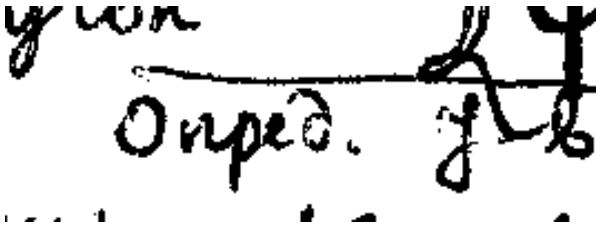
$O \rightarrowtail J^k$

— sheaf of jets of order k.

Original source details: GELFAND-P000254

presheaf $\{\varphi_i, \quad i = (i_1, \ldots, i_n), \quad |i| \leq k\}$ $\quad$ transition — as for derivatives.

Gelfand. Def. j?

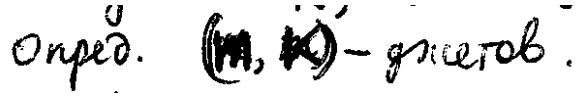


Original source details: GELFAND-P000254

Uncertain reading. After "Def." there is a small sign, provisionally j.

Let k, m be integers.

Def. of (m,k)-jets.



Original source details: GELFAND-P000254

Uncertain reading. The first letter is retraced. Provisional (m,k).

X — diff. set.

Consider fns. of the $(m+1)$-st variable.

Original source details: GELFAND-P000254

Uncertain reading. Provisional $m+1$. The letter is retraced.

(ordinary — $(1,k)$-jets).

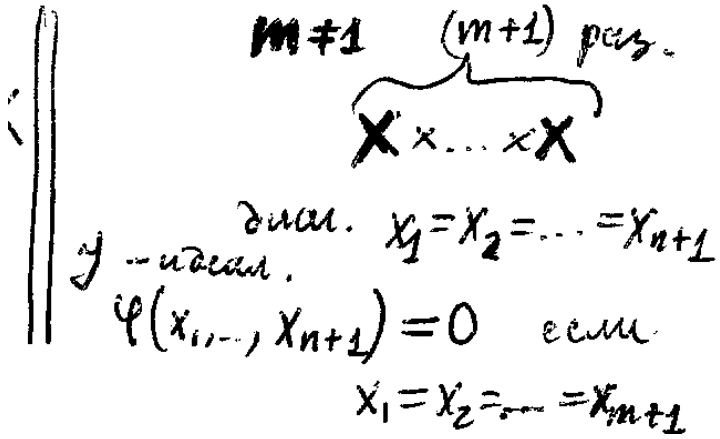
$(1,k)$-jets.

$\varphi(x,y)$ — fns. on the square $X \times X$

Consider the diagonal $x = y$

Consider the collection of fns. $= 0$ on the diagonal

This is an ideal.



Original source details: GELFAND-P000254

Diagram and typed description. Addition to the right of the two vertical strokes.

$m \neq 1 \qquad \overbrace{X \times \cdots \times X}^{(m+1) \text{ times}}$

diag. $x_1 = x_2 = \cdots = x_{n+1}$

$\mathcal{J}$ — ideal.

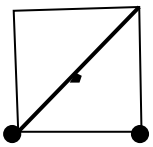
$\varphi(x_1, \ldots, x_{n+1}) = 0$ if

$x_1 = x_2 = \cdots = x_{m+1}$

Consider $\underline{\mathcal{J}^{k+1}}$ (fns. of k-th order of smallness on the diag.)

We regard $\varphi_1 \sim \varphi_2$ if $\varphi_1 - \varphi_2 \in \mathcal{J}^{k+1}$

Diagram and typed description. Unlabelled square, thick rising diagonal and two dark lower corners; central protrusion: point / small arrowhead.



Original source details: GELFAND-P000254

$\varphi(x, y)$ — [crossed out: ...]

Uncertain reading. Cancelled single sign: 0 / e.

Original source details: GELFAND-P000254

$(1, 0)$-jet. — fn. on the diagonal.

Undeciphered passage. Short cancelled entry at the left; possible beginning $(1, \ldots)$, remainder unreadable.

Original source details: GELFAND-P000254

$(1, 1)$ — jet

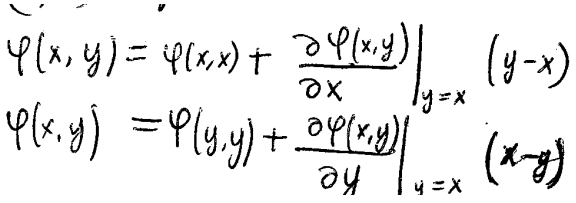
Uncertain reading. 0 / O

Original source details: GELFAND-P000254

on the diag. with 1-st derivatives.

$$\varphi(x,y) = \varphi(x,x) + \left.\frac{\partial \varphi(x,y)}{\partial x}\right|_{y=x} (y-x)$$
$$\varphi(x,y) = \varphi(y,y) + \left.\frac{\partial \varphi(x,y)}{\partial y}\right|_{y=x} (x-y)$$

Diagram and typed description. The formulas in the manuscript.



Original source details: GELFAND-P000254

There are 2 ways to define jets

$$a(x) + b(x)(y-x)$$
$$a(y) + b(y)(x-y)$$

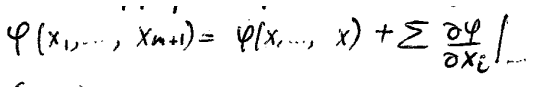
These are different ways.

GELFAND-P000255

[source number: 10]

One can differentiate — this gives a jet of lower order. 10

$$\varphi(x_1,\dots,x_{n+1}) = \varphi(x,\dots,x) + \sum \frac{\partial\varphi}{\partial x_i}\Big|_{\dots}$$



Uncertain reading. Short dots follow the lower end of the evaluation bar. The entry stops there.

Original source details: GELFAND-P000255

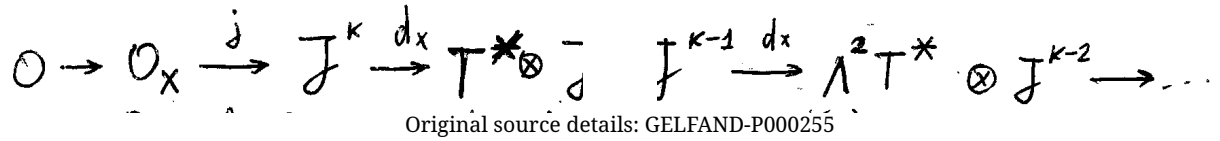
$(1,m)$-jet — scalar fn. + diff. form

Palamodov.

The case of general m reduces to $m=1$.

$$\varphi(x,y) \sim \sum_{|i|\le m} \varphi_i(x)\frac{\xi^i}{i!}$$

$$O_Y \longrightarrow O_{X\times Y}/\mathcal{J}^{k+1}$$



Original source details: GELFAND-P000255

Diagram and typed description.

$$0 \longrightarrow O_X \xrightarrow{j} J^k \xrightarrow{d_x} T^*\otimes J^{k-1} \xrightarrow{d_x} \Lambda^2 T^*\otimes J^{k-2} \longrightarrow \dots$$

Spencer resol. (triv. case)

In the gen. case — O_X is replaced by the sheaf of germs of solns. of an arbitrary lin. syst. of eqns.

Sweeny

Acta Math., 120 $N=3-4$.

and development by Malgrang

Quillen — a large article in proceedings...

Next time: Palamodov again.

GELFAND-P000256

[source number: 17]

$27/\overline{\text{X}}$. Prof. Weinstein (USA).

on operators.

4/XI?

Original source details: GELFAND-P000256

Uncertain reading. Provisional XI. The Roman numeral and the 4 are retraced.

Kirillov — Traces of operators !!

11/XI — skipped.

18/XI Kazhdan ζ-fn. following Dwork

25/XI ——— || ——— || ——— || ———

Kirillov.

$\mathcal{A}$ — operator.

M — comp. set, $\mathcal{L}_2(M, H)$ H — Hilbert.

μ — measure on it

(or a Hermitian bundle)

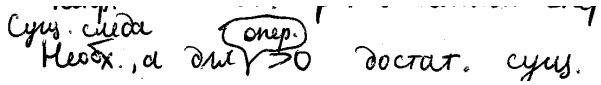
$$\mathcal{K}f(x) = \int_M K(x,y) f(y)\, d\mu(y)$$

(K — operator fn., $K(x,y)\colon H \longrightarrow H$)

jointly cont. in the weak oper. topology.

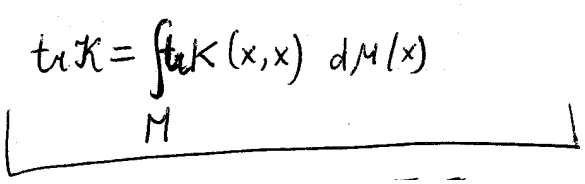
Existence of the trace

Necessary, and for [inserted above: oper.] > 0 sufficient existence.



Original source details: GELFAND-P000256

Diagram and typed description. Insertion of "oper." above the > 0 sign.



Original source details: GELFAND-P000256

Diagram and typed description. The trace formula, underlined in the manuscript.

$\operatorname{tr}\mathcal{K} = \int_M \operatorname{tr} K(x,x)\, d\mu(x)$ (tr in the usual sense).

Scalar fns.: see Gohberg and Krein (Carleman theorem)

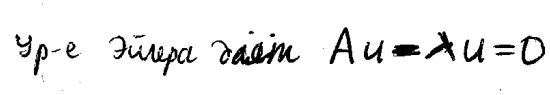
Nontriv.: if $\mathcal{K}$ has a trace, then $\operatorname{tr} K(x,x)$ exists almost everywhere.

Weinstein.

A — self-adj. oper. with discr. spectrum.

$$\mathbf{A}+\mathbf{B} \qquad \min R(u) \qquad R(u) = \frac{(Au,u)}{(u,u)}$$

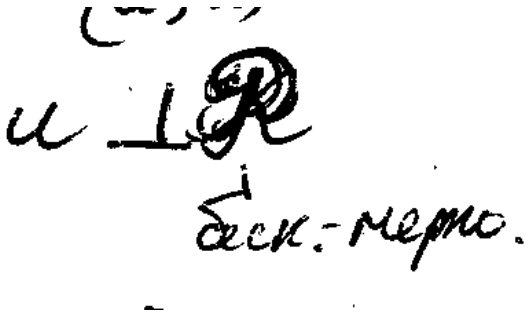
The Euler eqn. gives $Au = \lambda u = 0$



Original source details: GELFAND-P000256

Uncertain reading. The first sign after Au is retraced. Provisional $Au = \lambda u = 0$.

$u \perp \mathcal{R}$ infinite-dim.



Uncertain reading. The retraced letter has a lower-right stroke. Provisional $\mathcal{R}$, possibly $\mathcal{P}$.

Original source details: GELFAND-P000256

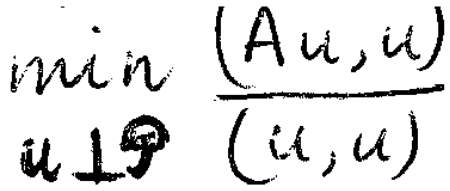
$$\mathcal{P} = \bigcup \mathcal{P}_n$$

$\min\limits_{u \perp \mathcal{P}_n} R(u)$ soln. exists: $\det \ldots = 0$

17

GELFAND-P000257

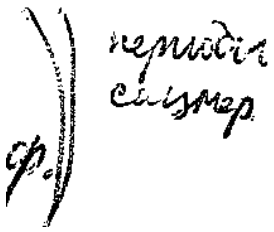
$$\min_{u \perp \mathcal{P}} \frac{(Au, u)}{(u, u)}$$



Uncertain reading. The retraced letter after $u \perp$ under the minimum is provisionally $\mathcal{P}$.

Original source details: GELFAND-P000257

exists for the Schrödinger oper.
(not only completely cont.).
In general. classical theory — estimates of eigenvals., from one side.
Weinstein and Co. — from the other side.
Someone from Poland. Rolevich.
Periodic solns. of diff. eqns. with delay.
Reduces to solving a syst. of ordinary eqns. with periodic coeffs.



Uncertain reading. Beside the double bracket, “periodic” and provisionally “case”.

periodic
case?

Original source details: GELFAND-P000257

A few more applications, etc.

Appl. to control problems.

Next will be Kirillov.

$4/\overline{\text{XI}}$ Kirillov.

G — simply conn. Lie group

T — irred. representation

$$T(\varphi) = \int_G \varphi(g) T(g)\, dg \quad \text{— operator}$$

tr as a generalized fn. sometimes exists.

$$\operatorname{tr} T(\varphi) = \int_G \varphi(g)\chi(g)\, dg$$

$\chi(g)$ — generalized fn.

In the examples it is no more complicated than a measure, but there is no general Theorem.

There are examples with no trace for any φ and even narrower than C_0^∞.

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

for it one must narrow C_0^∞ to fns. with

$$\int \varphi(a,b)\, db \underset{a}{\equiv} 0$$

(over a conjugacy class of elems. — 0)

GELFAND-P000258

[source number: 18]

18

$\mathfrak{g}$ — Lie alg. (real) $\quad \mathfrak{g}^*$ — dual sp. (real).

G acts in $\mathfrak{g}$ and $\mathfrak{g}^*$.

$\chi(g)$ resembles another generalized fn.

$\mathfrak{g}^* \supset \Omega$, $\quad \Omega$ — orbit of G.

Ω — symplectic sp. (there is a nondegen. 2-form there).

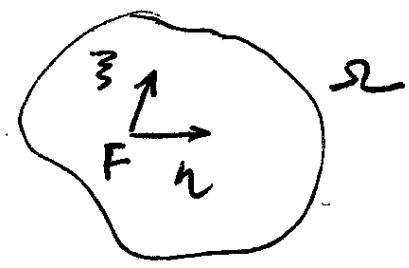
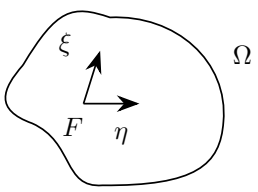


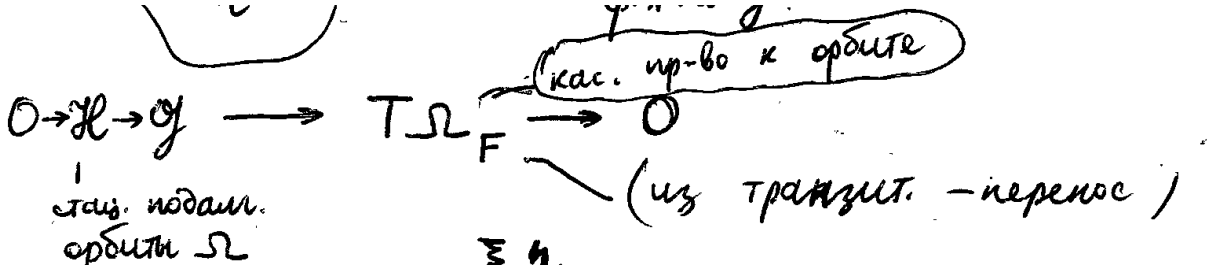
Diagram and typed description.



Original source details: GELFAND-P000258

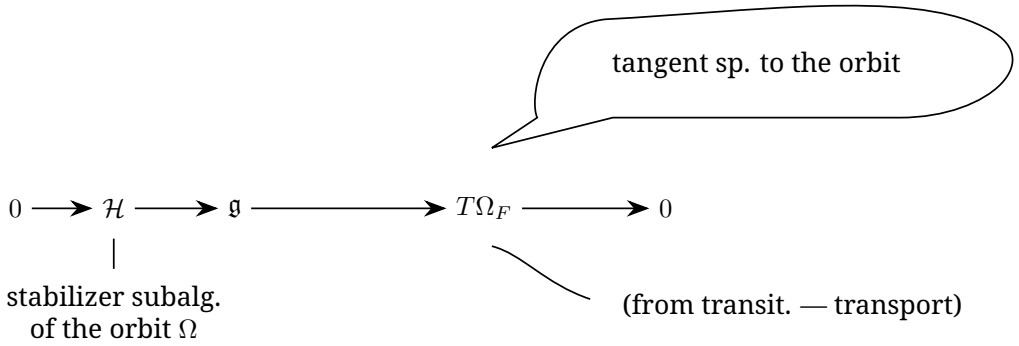
$B(\xi, \eta)$ $F \in \Omega$

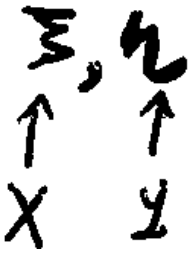
functional on $\mathfrak{g}$.



Original source details: GELFAND-P000258

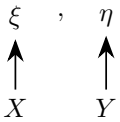
Diagram and typed description.





Original source details: GELFAND-P000258

Diagram and typed description.



One must take

$$\frac{1}{2\pi}\langle F, [X, Y]\rangle = B(\xi, \eta)$$

skew-symm.
nondegen.
form

(well-defined — independent of the choice [two signs crossed out] of representatives mod$\mathcal{H}$)

Original source details: GELFAND-P000258

Uncertain reading. Two signs after “choice” are crossed out. Their reading is uncertain.

Inhomog. form.

$$\beta = e^B = 1 + B + \frac{B \wedge B}{2!} + \cdots + \frac{B \wedge \cdots \wedge B}{k!}$$

(it terminates).

In fact, only one term is needed for integr.

Thus, a measure is defined on each orbit.

generalized fns. $\begin{cases} \mathcal{J}_\Omega(x) = \int_\Omega e^{i\langle F,X\rangle}\beta(F) & \text{(Fourier transform of the measure on the orbit)} \\ \chi(\exp X) \end{cases}$

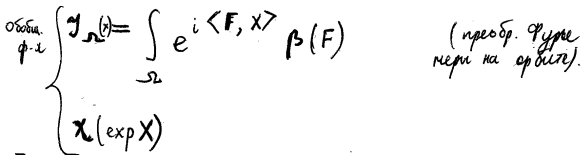


Diagram and typed description. The formulas and annotation in the manuscript.

Original source details: GELFAND-P000258

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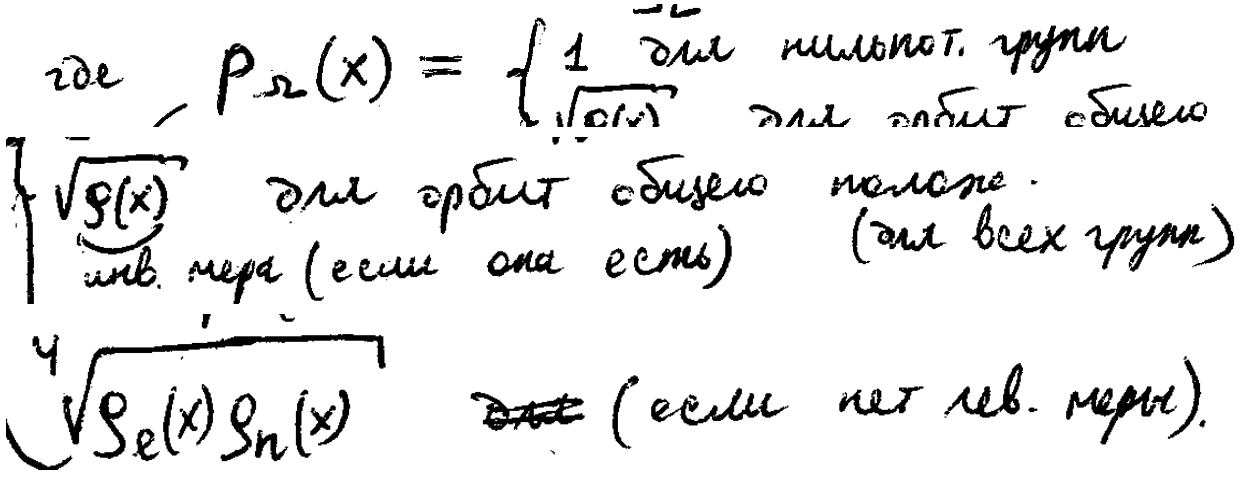
resemble one another.

the hypothesis that they coincide is false. But:

GELFAND-P000259

Their ratio $\in C^\infty$ and one can write a formula.
$\forall$ irred. repr.

$$\chi(\exp X) = p_\Omega^{-1}(x) \int_\Omega e^{i\langle F,X\rangle}\beta(F),$$

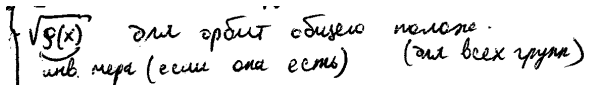


Original source details: GELFAND-P000259

Diagram and typed description.

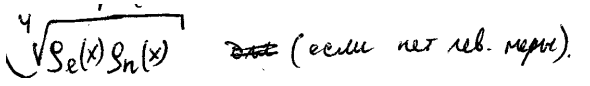
where

$$p_\Omega(x) = \begin{cases} 1 & \text{for nilpot. groups} \\ \underbrace{\sqrt{\rho(x)}}_{\text{inv. measure (if there is one)}} & \text{for orbits in general position (for all groups)} \\ \sqrt[4]{\rho_л(x)\rho_п(x)} & \text{~~for~~ (if there is no left measure)} \end{cases}$$



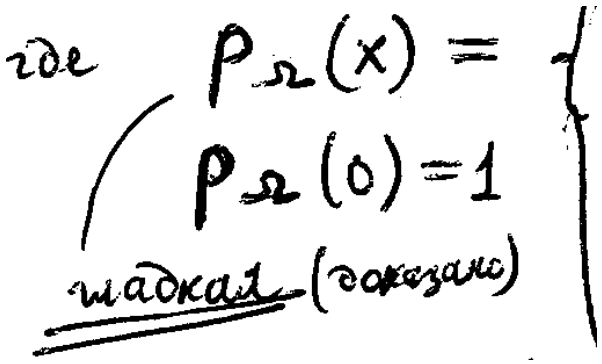
Original source details: GELFAND-P000259

Uncertain reading. Provisional $\rho(x)$ under the square root. Below it reads “inv. measure (if there is one)”.



Original source details: GELFAND-P000259

Uncertain reading. Provisional $\rho_л(x)\rho_п(x)$ under the fourth root. “For” is crossed out on the right.

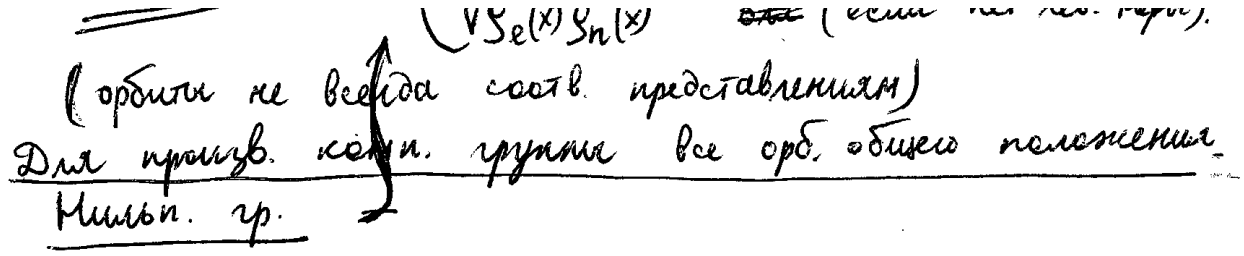


Original source details: GELFAND-P000259

Diagram and typed description.

$p_\Omega(0) = 1$

smooth (proved)



Original source details: GELFAND-P000259

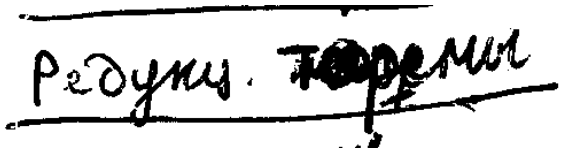
Uncertain reading. The arrow rises from beside “Nilpot. gr.” through the “comp.” line towards “always”; its reference to the formula or the orbit remark is not established.

(orbits do not always correspond to representations)

For an arbitrary comp. group all orbits are in general position.

Nilpot. gr.

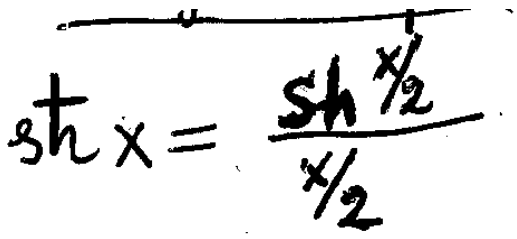
Reduction theorems



Original source details: GELFAND-P000259

Uncertain reading. Provisional reading of the retraced word: theorems (теоремы); initial strokes remain undeciphered.

$$\operatorname{sh} x = \frac{\operatorname{sh} \frac{x}{2}}{x/2}$$

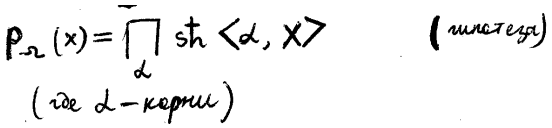


Original source details: GELFAND-P000259

Uncertain reading. A horizontal stroke crosses the ascender of the h.

$$p_\Omega(x) = \prod_\alpha \operatorname{sh}\langle \alpha, X\rangle \qquad \text{(conjecture)}$$

(where α — roots)



Original source details: GELFAND-P000259

Uncertain reading. A horizontal stroke crosses the ascender of the h.

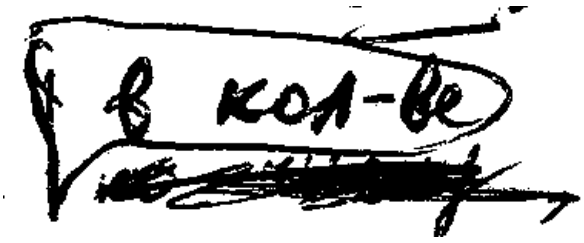
Unit. irred. reprs. of all Lie groups.

in $\mathfrak{g}^*$, orbits Ω.

Choose those for which $B \in H^2(\Omega, \mathbb{Z})$

($\int$ over $\forall$ cycle — an integer)

An integral orbit corresponds to a repr. [crossed-out entry]



Original source details: GELFAND-P000259

Undeciphered passage. The crossed-out writing is uncertain. “In a number” is inserted above it.

in a number

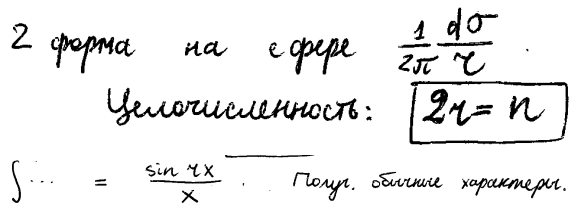
equal to the number of characters of the fund. group of the orbit.
(the fund. gr. is commutative in this case).

GELFAND-P000260

[source number: 19]

19

~~On~~ For $SU(2)$ (almost $SO(3)$)
2-form on the sphere $\frac{1}{2\pi}\frac{d\sigma}{r}$.
Integrality: $\boxed{2r = n}$ — such spheres.
$\int \ldots = \frac{\sin rx}{x}$. Obtain the usual characters.



Original source details: GELFAND-P000260

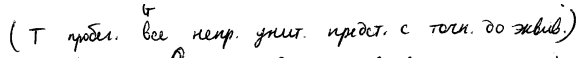
Uncertain reading. The same sign occurs in all three formulas. Provisional r, with a form that could be τ.

Universal Plancherel formula for all groups.

$$\int_G |f(g)|^2\, dg = \int \mathrm{tr}[T(f)T^*(f)]\, d\mu(T)$$

$$T(f) = \int_G f(g)T(g)\, dg$$

(T ranges over all irred.? unit. reprs. up to equiv.)

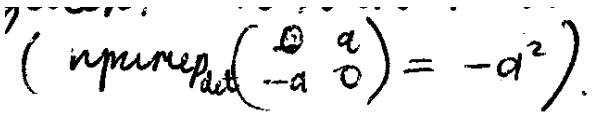


Original source details: GELFAND-P000260

Uncertain reading. The manuscript has непр., provisionally “irred.”.

Abstr. theorem — true, the question is the calculation of the measure $d\mu$.

Problem. For a skew-symm. matrix, extract a polynomial sq. root from its det.

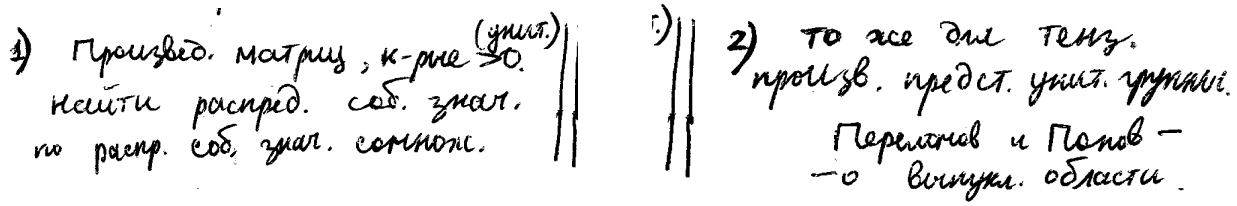


Original source details: GELFAND-P000260

Diagram and typed description. The sign before a^2 in the manuscript.

(example $\det \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} = -a^2$).

One nontriv. duality:



Original source details: GELFAND-P000260

Diagram and typed description.

1) Product of matrices which $\overset{\text{(unit.)}}{> 0}$.
Find distrib. of eigenvalues
from distrib. of eigenvalues of factors.

2) the same for tensor
product of reprs. of the unit. group.

Perelomov and Popov —
— on a convex region.

$[(x^2-1)^n]^{(n)}$ — Legendre

$[(x-1)^{m_1}(x+1)^{m_2}]^{(m_3)}$ — Jacobi polynomials

m_3 — fractional — hypergeom. function

(or $x(x-1)$)

GELFAND-P000261

$18/\overline{\text{XI}}$.

Undeciphered passage. Heavily cancelled inscription to the right of the date not deciphered; two strokes below it.

Original source details: GELFAND-P000261

Lit. on representation theory.

Minimum (0th level)

1. Kirillov lectures “Elem. theory of reprs.”, $\overline{\text{I}}, \overline{\text{II}}$
2. Gelfand, Minlos, Shapiro “Reprs. of the rotation group”
3. Zhelobenko, Lectures on theory of reprs.
4. Jacob Schwartz “Lie alg. and Lie groups” (Engl)
5. Serre — "
6. Vinberg “Compact Lie groups”.

7. Sophus Lie Seminar “Reprs. of Lie algs.* reprs. of Lie groups”

and

1. general theorems on fin. and comp. grs.

2. detailed analysis for rotation gr. and matrices of 2nd order — illustr. and concr. for 1

3. useful for physicists

4. general theory of Lie groups and class. comp. and semisimple Groups

6. ... comp. groups and Lie algebras (Schwartz, but shorter)

7. modern exposition of theory of semisimple groups

1st level.

1. Gelfand, Naimark ~~Reprs.~~ „Unit. reprs. of class. groups “

(Acad. Sci. 1950)

(do not read the chapter on the Plancherel theorem,

read the introd. and article of Gelfand and Graev in Trans. Math. Soc. vol. 4.

(Plancherel theorem).

2. Godement Matem. 5:5

Simple proof of the Harish-Chandra theorem on existence of char.

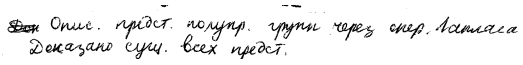
Also there, description of fin.-dim. reprs.

3 Berezin Trans. + letter to ed.

vol. 6 vol. 12

[Cancelled: For?] Description of reprs. of semisimple groups through the Laplace oper.

Existence of all reprs. proved.



Original source details: GELFAND-P000261

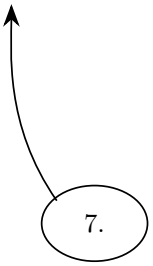
Uncertain reading. Provisional «Для» (for). The word is crossed out.

4. Kirillov Reprs. of nilpot. groups

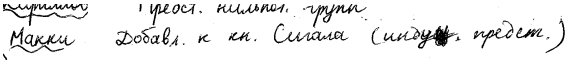
Mackey Addendum to Segal’s book (induc. reprs.)

Original source details: GELFAND-P000261

Uncertain reading. Beside “Mackey” is a circled numeral, provisionally 7. An arrow points upward at the left. Below the circle is “(in the 0th)”.



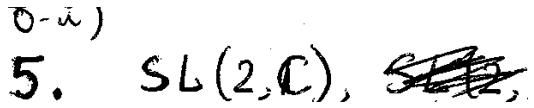
(in the 0th)



Original source details: GELFAND-P000261

Uncertain reading. The parenthesis reads инду... предст. (induc... reprs.). The end of the first word is heavily overwritten.

5. $SL(2,C)$, ~~SL_2~~,



Uncertain reading. Cancelled notation: SL_2 / $SL(2)$; position of 2 and parentheses unresolved.

Original source details: GELFAND-P000261

(5th issue of generalized functions, Naimark, Unitary repr. of the Lorentz gr.)

GELFAND-P000262

[source number: 20]

20

$SL(2,\mathbb{R})$ (5th issue of generalized functions)

$SL(2,Q_p)$ (6th issue of gen. functions) ~~or Gelfand's article)~~

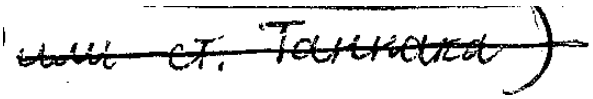


Diagram and typed description. Crossed-out reference to an article.

Original source details: GELFAND-P000262

to the next level.

1st level: no final theorems, except Berezin's work + Harish-Chandra

6. J. Schwartz, C^*-algebras *; Naimark, UMN (book)

2nd level.

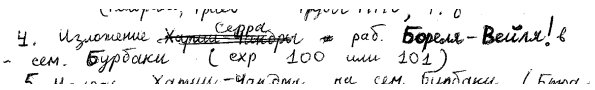
1. $SL(2,Q_p)$. Gen. functions 6.

2. Karpelevich, Gindikin (Izv. AN) ~~Method~~ Plancher. theorem for symm. spaces.

3. Method of horospheres $\rightarrow$ 1st level (Gelfand, Graev — Trans. MMO, vol. 8

Serre

4. Exposition by ~~Harish-Chandra~~ ~~o~~ of **Borel-Weil!** work in — Bourbaki sem. (exp 100 or 101)

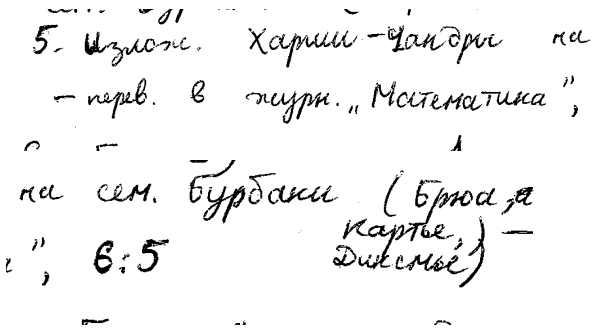


Uncertain reading. A single sign after the insertion "Serre" is heavily crossed out, possibly o.

Original source details: GELFAND-P000262

5. Exposition of Harish-Chandra at the Bourbaki sem. (Bruhat, ~~a~~ Cartier, Dixmier) —

— transl. in the journal „Matematika “, 6:5



Uncertain reading. A letter after “Bruhat” is crossed out, provisionally a.

Original source details: GELFAND-P000262

6. Groups and algs. of type 1, Glimm, Ferr, etc. —

— in Dixmier’s book „C^*-algebras and their reprs. “

7. Lie algebras and enveloping algebras.

Uncertain reading. The first numeral before the heading is overwritten. Provisional 7.

Original source details: GELFAND-P000262

7. Enveloping algebras:

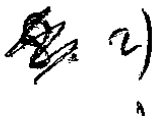
а) Sophus Lie; ch. I.

б) Harish-Chandra from 5)

в) Gelf., Kirillov; Fr. work and journal FA, vol. 3, issue 1

(Publ. Math.)

~~в)~~



Uncertain reading. A separate crossed-out sign before г) has two loops and resembles 8. Its reading is uncertain.

Original source details: GELFAND-P000262

г) Kostant, Amer. Journ., 85, pp. 327–404

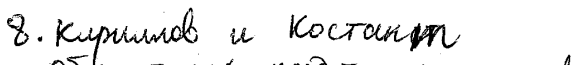
д) Zhelobenko „Enveloping algs. “

е) Nelson „Anal. vectors “ (there is a transl. in the journal Matem.)

6:3

GELFAND-P000263

8. Kirillov and Kostant

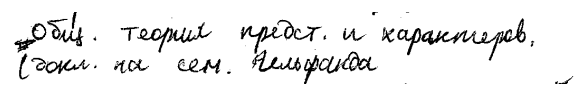


Uncertain reading. Provisional “Kostant”. The end of the surname is retraced.

Original source details: GELFAND-P000263

Gen. theory of reprs. and characters.

(talk at Gelfand’s sem.



Uncertain reading. A dense mark appears to the left of “Gen.”. Its reading is uncertain.

Original source details: GELFAND-P000263

9. Harish-Chandra — work of the latest series. (from 1963)

10. Kazhdan — func. analysis and Fr. translation at the Bourbaki sem.

vol. I, issue 1.

(Kirillov's talk)

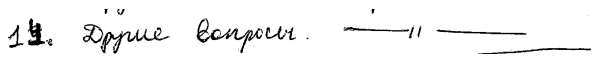
11. G/Γ — extensive lit. (conference Boulder)
Sommer Sicool?

Proceed. — translation in preparation.

Original source details: GELFAND-P000263

Uncertain reading. Below "Boulder", provisionally "Sommer Sicool". The first strokes of the second word are uncertain.

12. Other fields. Extensive lit.,
14? Other questions. — // —



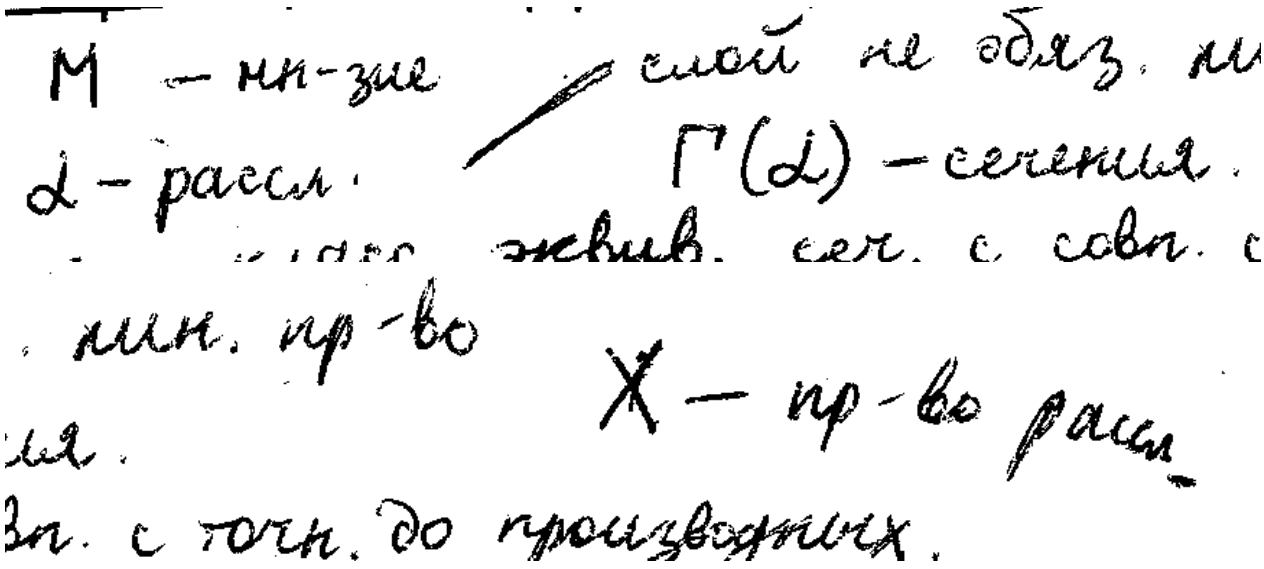
Original source details: GELFAND-P000263

Uncertain reading. The final numeral is overwritten. Provisional 14, possibly 13.

13. List of questions.
Everything will be in "Uspekhi"

Gromov.

1. Invar. def. of diff. oper.



Original source details: GELFAND-P000263

Diagram and typed description.

M — set

fibre not necessarily a linear space

$\mathcal{L}$ — bundle

$\Gamma(\mathcal{L})$ — sections.

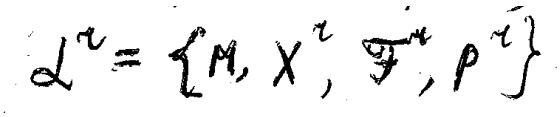
X — bundle space

Jets — equiv. class of sects. coinciding up to derivatives.
$\mathcal{L}^r$ — jet extension.

X^r — its space

Partial diff. equation —
— subset $A \subset \boldsymbol{X}^r$.
In appl. — a subset (not necessarily closed),
linear equation — a subbundle.
(add. to X of vect. spaces)

$$\mathcal{L}^r = \{M, X^r, \mathcal{F}^r, \rho^r\}$$



Uncertain reading. Last symbol of the quadruple: ρ^r / p^r; superscript r retained, the letter unresolved.

Original source details: GELFAND-P000263

Subset A in class. theory is usually closed.
But the equation $Lu \neq 0$ — an open set
$L(u) = 0$ — closed (if the func. does not satisfy, then

GELFAND-P000264

[source number: 21]

21

Or example.

$$f : \underset{g_1}{M_1} \longrightarrow \underset{g_2}{M_2}$$ with quad. forms.

smooth map, inducing forms $f^*(g_2) = g_1$.

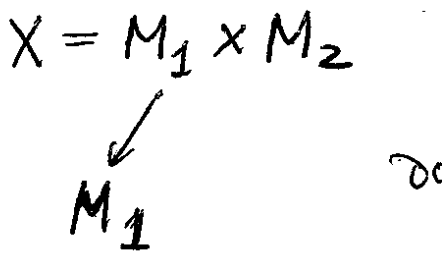


Diagram and typed description.

$X = M_1 \times M_2$

M_1

Original source details: GELFAND-P000264

map $M_1 \to M_2 \Leftrightarrow$ section
add an open cond. — so that it is an embedding.
General formulation:
solution exists or not.

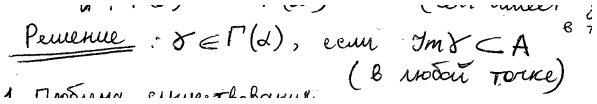
$$d^r : \Gamma(\mathcal{L}) \longrightarrow \Gamma(\mathcal{L}^r)$$

(sect. has a jet at a point).

Solution:

$$\gamma \in \Gamma(\mathcal{L}), \quad \text{if} \quad \operatorname{Im}\gamma \subset A$$

(at any point)



Uncertain reading. The same looped sign occurs twice, provisionally γ. It follows Im directly.

Original source details: GELFAND-P000264

1. Problem of existence.

Connection with others

2. Uniqueness. never occurs (almost never)
open sets

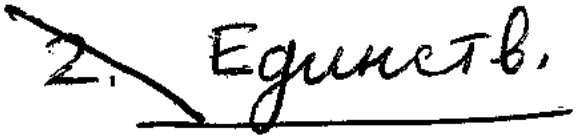


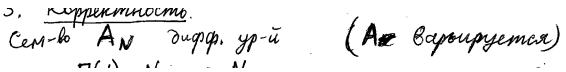
Diagram and typed description. Slanting stroke across the numeral 2.

Original source details: GELFAND-P000264

But one can speak of well-posedness.

3. Well-posedness.

A family A_N of diff. equations ($A_?$ varies)



Uncertain reading. The left occurrence is A_N. The right subscript is overwritten and uncertain.

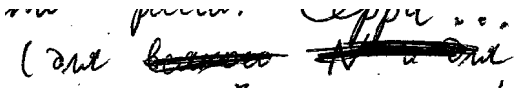
Original source details: GELFAND-P000264

$$\Gamma(\mathcal{L}) \times N \longrightarrow N$$

Well-posedness — this is a Serre fibration!!!

of a family of problems.

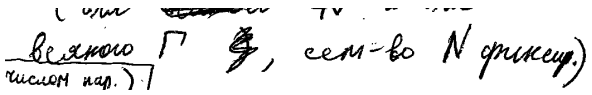
(for ~~every~~



Undeciphered passage. Following cancelled continuation unreadable; only elements resembling N and “и” are distinguishable. Complete words and number of hidden signs unknown.

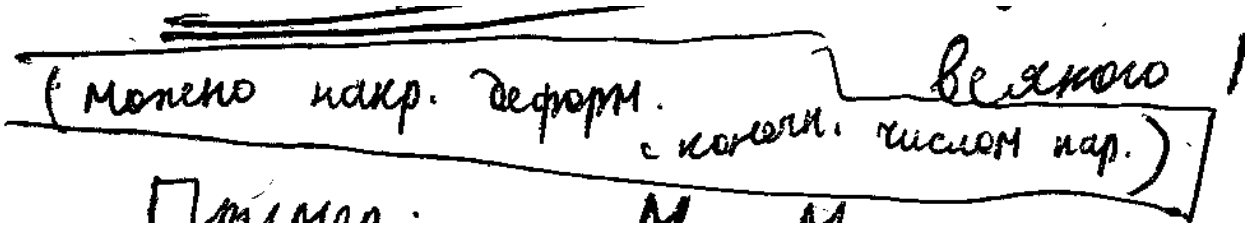
Original source details: GELFAND-P000264

every Γ, family N fixed)



Uncertain reading. After Γ, a separate cancelled looped sign is undeciphered; lower stroke and following comma visible in the complete fragment.

Original source details: GELFAND-P000264



Original source details: GELFAND-P000264

Diagram and typed description.

(one may, e.g., deform

with a finite number of param.)

Example:

$\underset{g_1}{M_1},\ \underset{g_2}{M_2}$ a pair of forms defines an eq.

All pairs of forms — N,

Idea: as a rule, one assumes that locally everything is known,

and what is global — unknown.

Local part: Nash's implicit-function theorems.

in ex.

GELFAND-P000265

Stab. well-posedness allows solving eqs.

Theorem C^1-Nash.

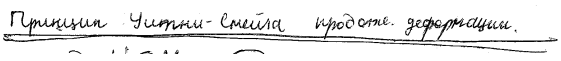
problem of C^1-imm. is posed correctly:

with a deform. of the metric one can deform the immersion.

With some metric an imm. exists — top. methods,

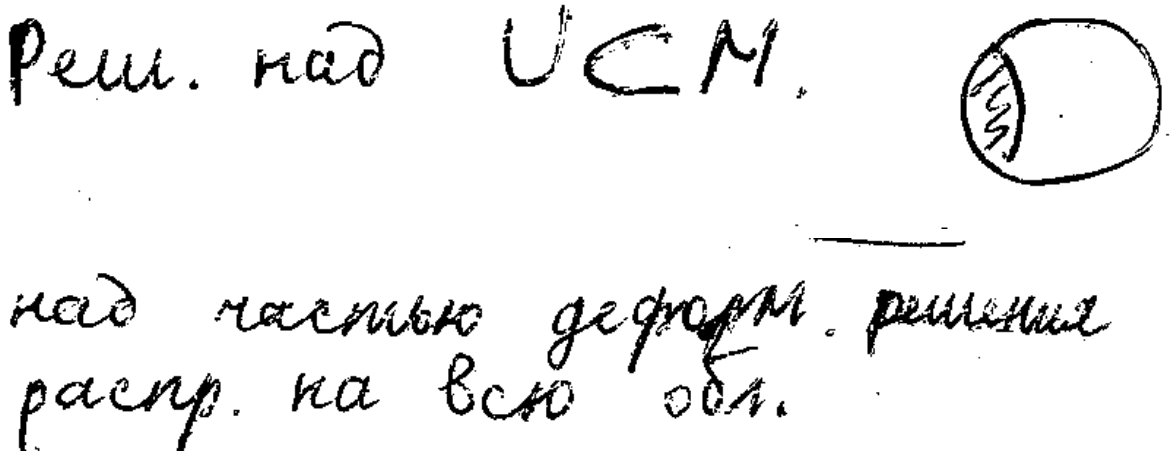
then deform it into any — obtain an emb. !!

Whitney–Smale princ. of contin. deformation.



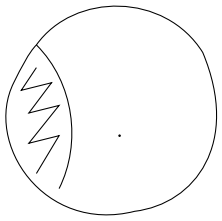
Original source details: GELFAND-P000265

Uncertain reading. The abbreviation is provisionally продолж. The upper stroke ends in a point on the right.



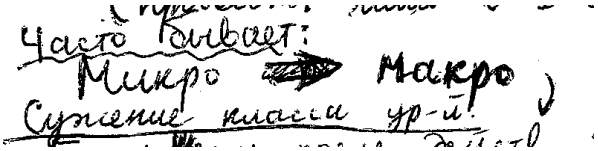
Original source details: GELFAND-P000265

Diagram and typed description.

Sol. over $U \subset M$,  deform the solution over a part

extend to the whole region

(if the boundary effect is ignored)
They proved it in a number of cases.
~~From corr. and this princ. one can~~
Microscopic realizability of these principles —
— this is the implicit-function theorem.
(contin. only at the 1st instant).



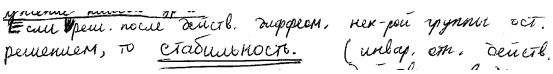
Original source details: GELFAND-P000265

Diagram and typed description.

Often:

Micro ⟹ Macro ↘

Narrowing the class of eqs.
If **[?]** a sol. after the action of diffeom. of some group remains
a solution, then Stability.



Original source details: GELFAND-P000265

Uncertain reading. A heavily overwritten sign precedes “sol.”. Its reading is uncertain.

(invar. under the action
of the group of all diff., which acts there in X —
— if this is given)
example — tang. bundle.
(maps the eq. to itself)

Original source details: GELFAND-P000265

Diagram and typed description. Crossed-out title of the problem.

~~Plateau prob~~

Example: Plateau prob.

$$R^2 \longrightarrow R^3$$ minim. films invar. under diffeom.

Then (in the stable case) micro ⇒ macro.

Another principle.

(Theorem ^{2}.)

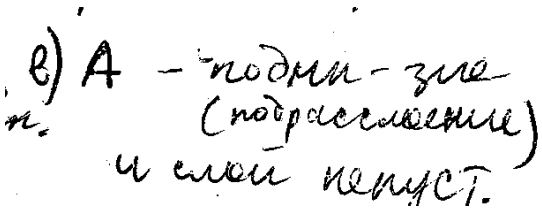
a) M parall.

noncomp. conn.

в) A — subset

(subbundle)

and fibres nonempty.



Uncertain reading. The letter before A has two loops: “в)” / “b)”. The order of the conditions does not determine its alphabet.

Original source details: GELFAND-P000265

⇒

13-04 from 16-18 Tue.; sem. Arnold

GELFAND-P000266

[source number: 22]

c) there is micro

d) there is stability

Then the eq. has a solution.

It contains Smale’s immersion theory and so on.

Noncomp.: micro ⇒ macro.

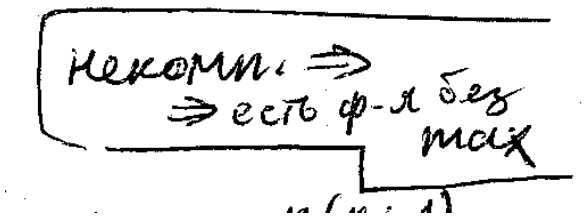


Diagram and typed description.

noncomp. ⇒

⇒ there is a fn. without max

Original source details: GELFAND-P000266

Geometric example

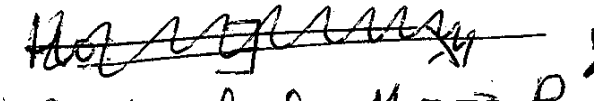
M^n — Riemannian

$M^n \subset R^s$ loc. in $s = \frac{n(n+1)}{2}$

comp. of metr. tensor

one cannot, generally speaking, fit into a lower one.

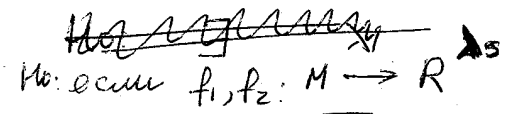
~~But, [?]~~



Original source details: GELFAND-P000266

Undeciphered passage. After “But,” the cancelled continuation is unreadable; looped and horizontal strokes are visible.

But: if $f_1, f_2 : M \to R^{?s}$.



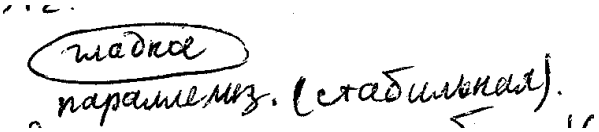
Original source details: GELFAND-P000266

Uncertain reading. The sign before the superscript s is heavily overwritten. Its final form is uncertain.

$0 < \lambda < 1$

Original source details: GELFAND-P000266

Undeciphered passage. Cancelled formula at the right; lower loop resembles g_0. Left-hand part and relation sign are unreadable.



Original source details: GELFAND-P000266

Diagram and typed description.

smooth

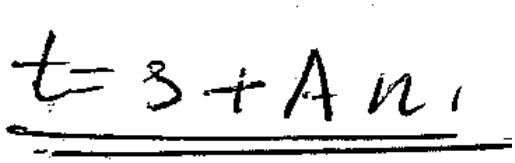
parametriz. (stable).

they are C^0-approxim. by maps φ_1, φ_2, such that the induced metrics are equal.

(Essentially there is no unique determination)

In large dim. $t \geq s$

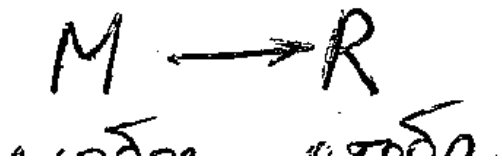
$t = 3 + An$



Original source details: GELFAND-P000266

Manuscript notation. The formula $t = 3 + An$ is doubly underlined.

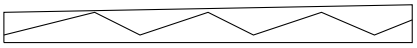
$$M \to R$$

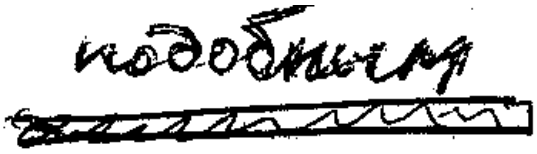


Original source details: GELFAND-P000266

Manuscript notation. The manuscript gives $M \to R$ without an exponent.

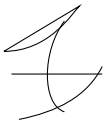
any map is approxim. by a similar one

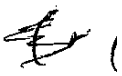




Original source details: GELFAND-P000266

Uncertain reading. Narrow hatched outline below “similar”. No letters could be identified.





Original source details: GELFAND-P000266

Uncertain reading. Separate overwritten sign: f / t. Order of the superimposed strokes undetermined.

(with induced metric λg)

All this follows from theorem 2.

Next time: Gromov

GELFAND-P000267

25/XI

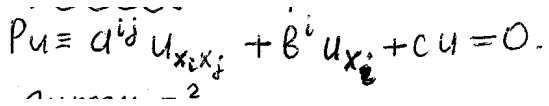
~~I?I~~

Original source details: GELFAND-P000267

Uncertain reading. Dense cancelled sign at the right of the date is unreadable.

Oleinik Radkevich’s results on eqs.

$$Pu \equiv a^{ij} u_{x_i x_j} + b^i u_{x_i} + cu = 0.$$



Original source details: GELFAND-P000267

Uncertain reading. The subscript on x is heavily overwritten. The dot and terminal curve support a provisional i.

hypoell. — ?

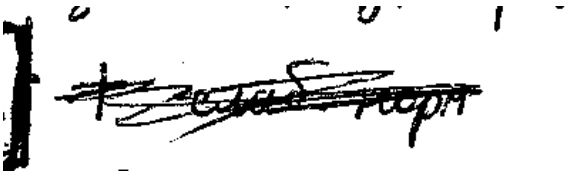
Hörmander:

$$\sum_{i=1}^{n} X_i^2 u + X_0 u + cu = f$$

if at every point there are n lin. indep. commutators, then hypoell., which is obtained from the est.:

$$\|u\|_\varepsilon \leq C\big[\|u\|_0 + \|Pu\|_0\big]$$

~~+ weak norm[?]~~

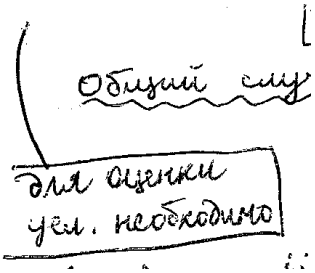


Original source details: GELFAND-P000267

Uncertain reading. After +, provisionally "weak norm". The ending of the last word is crossed out.

$\boxed{\varepsilon > 0}$ (necessity not proved)

General case: without resorting to the concrete form



Original source details: GELFAND-P000267

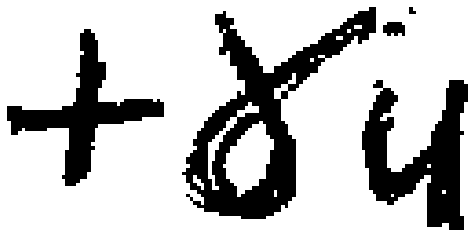
Diagram and typed description.

for the estimate

cond. necessary

Radkevich:

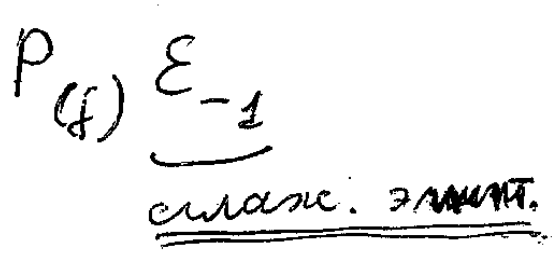
$$Lu = a^{ij} u_{x_i x_j} + \delta u$$



Original source details: GELFAND-P000267

Uncertain reading. The looped sign with crossed upper strokes before u is provisionally δ.

$$P^0 = a^{ij}u_{ij}, \qquad P^{(j)} = a^{ij}(x)D_i,$$

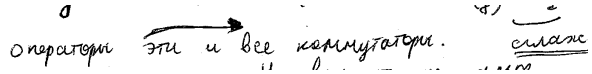


Original source details: GELFAND-P000267

Diagram and typed description.

$P_{(j)}$ $\mathcal{E}_{-1}$

smooth. ellipt.



Original source details: GELFAND-P000267

Diagram and typed description.

Operators these and all commutators.

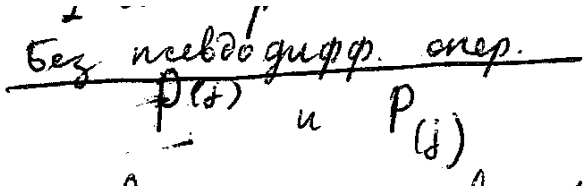
The condition is the same

($2n$ first-order oper.)

$$P_{(j)} = \sum_{i,k} \frac{\partial a_{ik}}{\partial x_j} \frac{\partial^2}{\partial x_i \partial x_k}(\mathcal{E}_{(-1)})$$

Rank n — one can select an elliptic system.

First order — the same



Original source details: GELFAND-P000267

Diagram and typed description. Short stroke below the notation on the left.

without pseudodiff. oper.

$P^{(j)}$ and $P_{(j)}$

everywhere we keep the principal parts

It is necessary to be able to obtain an ellipt. system.

There is as yet no formulation incl. Hörmander's theorem.

GELFAND-P000268

[source number: 25]

Gelfand: for applications it is important to know: when is every solution a summable function?

25

Kazhdan

Work of Dwork: rationality of the ζ-functions of all alg. varieties over a fin. field.

k — a field.

$A = k[x_1, \ldots, x_n]/\mathfrak{J}$ — an aff. scheme

Original source details: GELFAND-P000268

Uncertain reading. The ideal sign has a long descending hook. Provisionally $\mathfrak{J}$.

Points — closed points (max ideals)

Hilbert's thm.:

$$K = A/I \quad \text{— a fin. ext. of } k.$$

Prime ideals — points. $K \supset k$.

Geom. point with values in K: a homomorphism $A \to K$.

(a max ideal corresp. to it)

$$A = \frac{R[x]}{x^2 + 1}$$

2 geom. points, although 1 scheme point

i and $-i$

Alg. variety — if there are no nilpot. elements

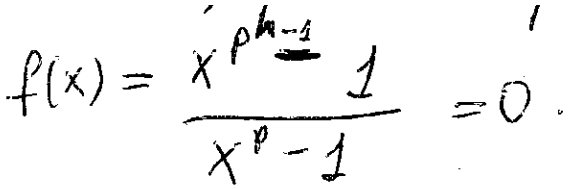
(this is the difference from a scheme)

[illegible]

Original source details: GELFAND-P000268

Uncertain reading. At the left before F_{p^n} is a separate densely cancelled unreadable sign.

F_{p^n} — a fin. field, ext. of Z_p by the eq.



Original source details: GELFAND-P000268

Uncertain reading. In the numerator $n-1$ sits above p. A heavy horizontal stroke below $n-1$ precedes a separate 1. The function of the heavy stroke is uncertain.

$$f(x) = \frac{x^{p^{n-1}} - 1}{x^p - 1} = 0.$$

Original source details: GELFAND-P000268

Diagram and typed description.

K

$\uparrow$

k

The autom. group of K is cyclic and has a canon. gen. — the Frobenius autom.

$$\sigma : x \to x^p.$$

$k = Z_p$.

A — an alg. variety. ζ-func. of it — ?

[illegible].

Uncertain reading. Interlinear mark above the following line: 3. / §.; reading and role undetermined.

Original source details: GELFAND-P000268

Consider a closed point of A — $x = I \subset A$

A/I — a finite field — an ext. of some degree.

Degree — degree of the closed point.

GELFAND-P000269

$$\zeta(t, A) = \prod_{x \in X} \frac{1}{1 - t^{\deg x}}$$

(t — a formal variable)

Theorem. (Dwork)

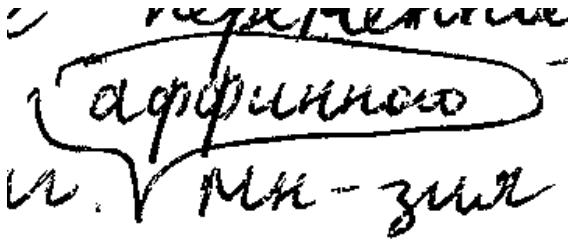


Diagram and typed description.

affine

$\forall$ alg. variety — this is a rat. func. of t

Original source details: GELFAND-P000269

(was Weil's conjecture, proved earlier for curves).

There are usually infinitely many points.

If A — the ring

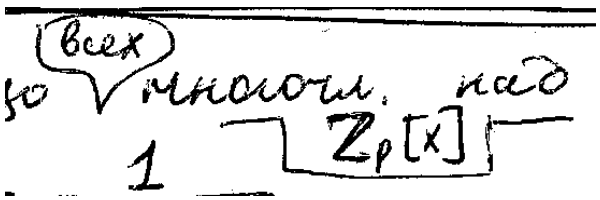


Diagram and typed description.

all

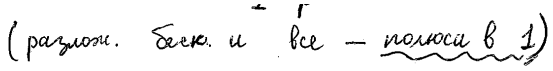
polynom. over Z_p, then

$Z_p[x]$

Original source details: GELFAND-P000269

$$\zeta(t) = \frac{1}{1 - pt}$$

(expansion inf. and all — poles at 1)



Manuscript notation. The manuscript says "and all — poles at 1", with a wavy underline under "poles at 1".

Original source details: GELFAND-P000269

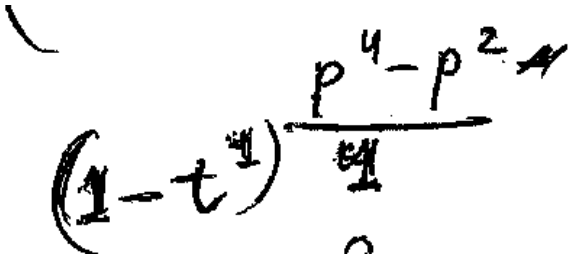
Degree 1 — exactly p points (x — Z_p).

Degree 2 — in $(p^2 - p)$ elem.

(so as not to fall among those counted
$\exists 1$ nontriv. autom.

$$\frac{1}{(1-t)^p} \quad \frac{1}{(1-t^2)^{\frac{p^2-p}{2}}} \quad \frac{1}{(1-t^3)^{\frac{p^3-p}{3}}} \quad \dots$$

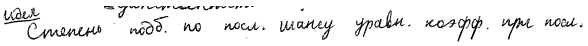
$$(1-t^4)^{\frac{p^4-p^2}{4}}$$



Uncertain reading. The fourth entry is on its own line. Two occurrences of 4 are overwritten. A short sign after p^2 is crossed out.

Original source details: GELFAND-P000269

Uniqueness — ?
Idea
Degree chosen by the last chance to equate the coeff. at the last ...



Uncertain reading. Provisionally по посл. шансу ("by the last chance"). The final abbreviation посл. is left unexpanded.

Original source details: GELFAND-P000269

Hence everything can be expanded.

GELFAND-P000270

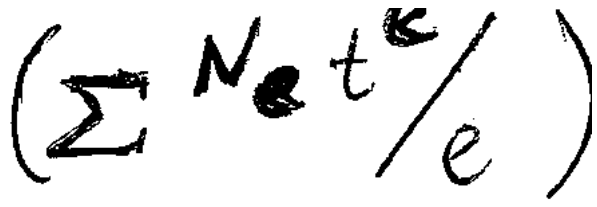
[source number: 26]

26

$$\zeta(t, A) = \sum c_k t^k \qquad c_k \text{ — integers} \qquad (\text{obv.}).$$

The equality holds.

$$\zeta(t, A) = \exp\left(\sum N_e t^e / e\right),$$



Manuscript notation. The looped letter e occurs in the indices on N and t and after the slash.

Original source details: GELFAND-P000270

where N_e — the number of geom. points with values in a field with deg $= e$.
(number of homom. from A to F_{p^e}).
This is a general equality.

But [?] it does not immediately follow that c_k are integers

Original source details: GELFAND-P000270

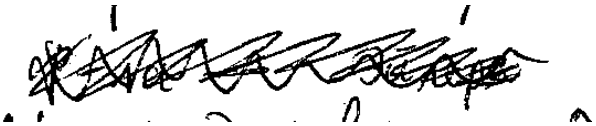
Uncertain reading. After “But” a short entry is crossed out and boxed. Its reading is uncertain.

Compute for $Z_p[x]$. $\quad N_e = p^e$.

$$\sum (pt)^e/e = -\ln(1-pt) \qquad \text{and the rest is clear.}$$

Apply at $A = Z$. The usual ζ-func. results.

[?...?]



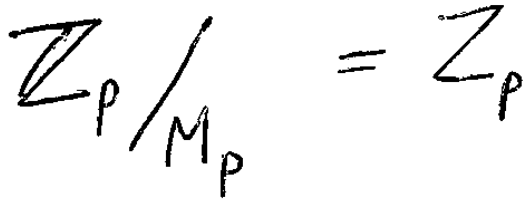
Original source details: GELFAND-P000270

Undeciphered passage. Cancelled line not fully read. Possibly Russian “Идея ... Дворк” / “... Дворк” (“Idea ... Dwork” / “... Dwork”); content beneath the hatching undetermined.

Idea of proof. The series $\zeta(t, A)$ in a compl. variable — conv. in a disc.
Consider it in a p-adic variable. It will also conv. for a small norm.
Merom. p-adic (ratio of holom. p-adic) $\Longrightarrow$
$\Longrightarrow$ autom. rational (general theorem) —
— a simple theorem

$$\mathbb{Z}_p \text{ — ring} = \{x \in Q_p;\ |x| \le 1\}$$
$$M_p = \{x \in Q_p :\ |x| < 1\} \quad \text{— max. ideal}$$

$$\mathbb{Z}_p / M_p = Z_p \quad \text{— a field of } p \text{ elem.}$$



Manuscript notation. The left Z has a doubled diagonal. The Z on the right has a single diagonal.

Original source details: GELFAND-P000270

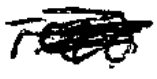
$$x_n \in Q_p. \qquad \sum_{\text{conv.}} x_n \iff x_n \to 0.$$

Banach sp. — the usual def. carries over.
Completely cont. oper.

GELFAND-P000271

Thm. In a Ban. sp. a compl. cont. oper. is nuclear,

[?]



Original source details: GELFAND-P000271

Uncertain reading. After "nuclear" there is a comma and a short crossed-out entry. Its reading is uncertain.

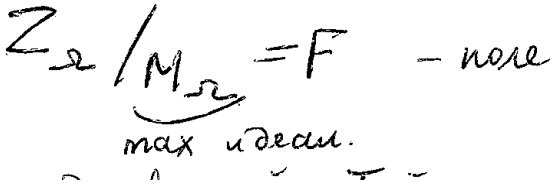
and therefore there exists $\det(1 - tA)$

— an entire p-adic func. of a p-adic variable t.

Alg. extensions of Q_p.

~~Closure~~

Ω — alg. completion of the alg. closure of Q_p



Original source details: GELFAND-P000271

Diagram and typed description.

$Z_\Omega / M_\Omega = F$ — field

max ideal

Teichmüller representatives exist.

Every elem. $f \in F$ has a canon. representative in Z_Ω.

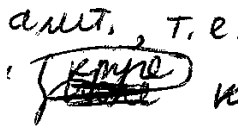
$F^* \longrightarrow Z_\Omega^*$ there is a canon. homomorphism.

Structure of analytic funcs. on the field of p-adic numbers.

$$f(t) = \sum c_k t^k, \qquad c_k \in \Omega.$$

Entire analytic, i.e. c_k sufficiently rapidly $\to 0$.

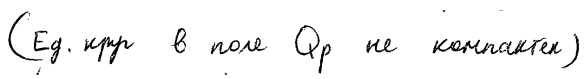
Then in every disc [?] of finite radius there are finitely many zeros.



Original source details: GELFAND-P000271

Uncertain reading. "disc" is inserted above a cancelled word and outlined with a lower point. Initial word undetermined.

(Unit disc in the field Q_p is not compact)



Original source details: GELFAND-P000271

Manuscript notation. The parenthesis reads "Unit disc in the field Q_p is not compact".

(Shnirelman)

Proof: the max principle, Schwarz lemma $\Rightarrow$ finitely many zeros.

$$f(z) = \frac{\prod(1-\alpha_i z)}{\prod(1-\beta_j z)} = 1 + \sum c_k z^k$$

c_k — integers

α_i, β_j — alg. $\Longrightarrow$ they are alg. integers.

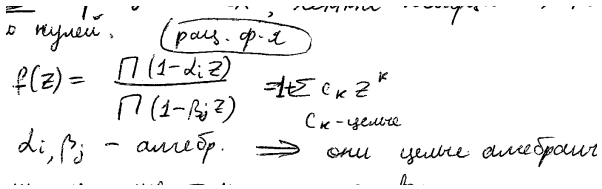
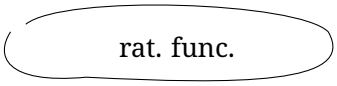


Diagram and typed description.



Original source details: GELFAND-P000271

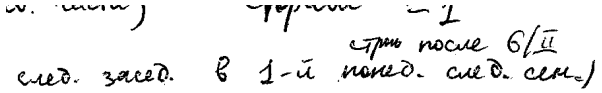
If not integers, a contradiction with the norms is obtained

$$|\ |<1 \quad \text{(on the left)} \qquad \text{on the right } =1$$

(This is an example)

str[?] after $6/\overline{\text{II}}$

next mtg. on the 1st Mon. of next sem.)



Uncertain reading. A joined entry above "Mon." and before "after 6/II", provisionally стр.

Original source details: GELFAND-P000271

GELFAND-P000272

$10/\overline{\text{II}} - 1969$ **yr.**

Margulis. Integer points in a disc on the Lobachevsky plane.

Int. points — preimages on the univers. covering.

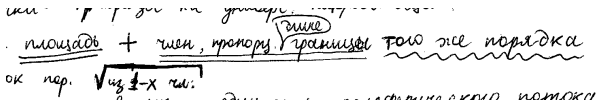
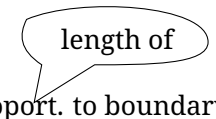


Diagram and typed description.

Obtain area +



term, proport. to boundary of the same order

Remainder of order $\sqrt{\text{of the 1st terms}}$

Original source details: GELFAND-P000272

The ergodicity of the horospherical flow is used in full.

There are works of Linnik: how many points on a sphere.

What will there be?

1. Kazhdan. 2. Margulis. 3. Kirillov.

Gelfand:

3-adic numbers (for schoolchildren)

Inf. number of digits to the left, rather than to the right.

The rules and top. are introduced analogously.

Ordinary numbers are worse because of the identification of

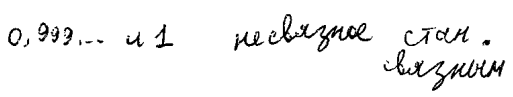
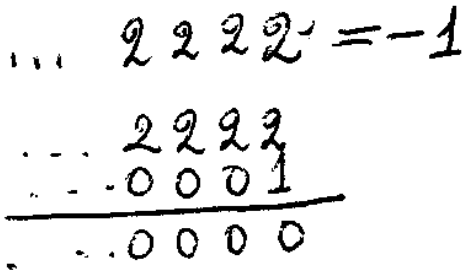


Diagram and typed description.

$0{,}999\ldots$ and 1 disconnected becomes connected

Original source details: GELFAND-P000272



Original source details: GELFAND-P000272

Diagram and typed description.

$$\ldots 2222 = -1$$

$$\begin{array}{r} \ldots 2222 \\ \ldots 0001 \\ \hline \ldots 0000 \end{array}$$

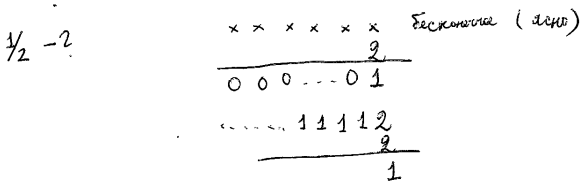


Diagram and typed description.

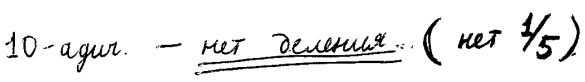
$\frac{1}{2}$ — ?

$$\begin{array}{r} \times\ \times\ \times\ \times\ \times\ \times \\ 2 \\ \hline 000\ldots01 \end{array}$$ infinite (clear)

$$\begin{array}{r} \ldots 11112 \\ 2 \\ \hline 1 \end{array}$$

Original source details: GELFAND-P000272

10-adic — no division (no $1/5$)



Manuscript notation. The last entry reads “no $1/5$”.

Original source details: GELFAND-P000272

GELFAND-P000273

$31/\overline{\text{III}}$ Bleher’s talk. On the spectr. func.

Acta Math. 121

$N = 3 - 4$, 1968 (L. Hörmander)

$$(Pu, u) \geq c(u, u),$$

$$\hat{p} = \int \lambda \, dE_\lambda, \qquad e(x, y, \lambda) \in C^\infty(\Omega \times \Omega)$$

— the spectral function of the operator.

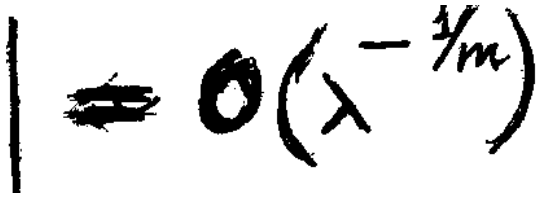
Asymptotics in λ — ?

Theorem.

$$\left| \lambda^{-n/m} e(x, x, \lambda) - (2\pi)^{-n} \int_{p^0(x,\xi)<1} d\xi \right| = O(\lambda^{-1/m})$$

Original source details: GELFAND-P000273

Uncertain reading. Second argument: final x over y / an earlier x; initial letter undetermined.



Original source details: GELFAND-P000273

Uncertain reading. Before O, two horizontal strokes lie over slanted strokes. Provisionally $=$.

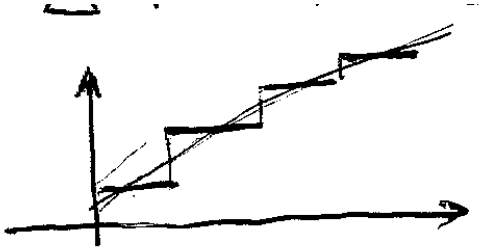
Off the diagonal 0.

Original source details: GELFAND-P000273

Uncertain reading. After “diagonal”, a round sign provisionally read as 0, possibly the letter O.

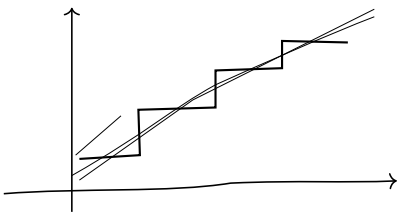
Avakumovich’s example.

Δ on $S^n \subset \mathbb{R}^{n+1}$



Original source details: GELFAND-P000273

Uncertain reading. Axes with upward and rightward arrows, four horizontal steps and three rises. Sloping strokes have no assigned formulas; their number as distinct comparison lines and their role are undetermined. No axis labels.



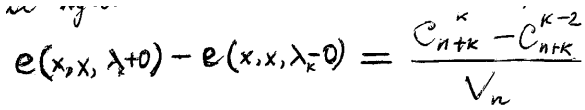
λ of large multiplicity.

the jump magnitude is of just this order.

Large multiplicities due to invariance of the sphere under rotations.

The worst situation for asymptotics is a **multiple spectrum.**

$$e(x,x,\lambda_k+0)-e(x,x,\lambda_k-0)=\frac{C^{k}_{n+k}-C^{k-2}_{n+k}}{V_n}.$$



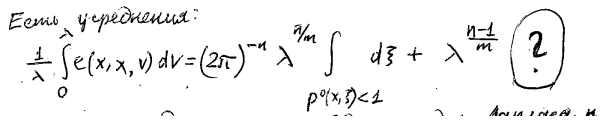
Uncertain reading. Both lower indices read $n+k$ and the second upper index reads $k-2$. The first subscript on λ is retraced, provisionally k.

Original source details: GELFAND-P000273

$$\frac{(2\pi)^{-n}}{V_n}\lambda^{n/2}$$

Jump $\sim\lambda^{(n-1)/2}$ and $C\lambda^{(n-1)/2}$ cannot be written.
There are averages:

$$\frac{1}{\lambda}\int_0^{\lambda} e(x,x,\nu)\,d\nu=(2\pi)^{-n}\lambda^{n/m}\int_{p^0(x,\xi)<1}d\xi+\lambda^{(n-1)/m}$$



Uncertain reading. The leading coefficient is written $(2\pi)^{-n}$; a question mark is outlined after the second power term.

Original source details: GELFAND-P000273

This has not been done. It has been done for the Laplacian on a square.

GELFAND-P000274

[source number: 38]

See Dikii, UMN, 1957 on traces.

and earlier in 1954 Gelfand.

38

There are 2 problems.
1) obtain the subsequent terms of the asymptotics.
2) obtain a **formal asympt. expansion** (a subtle problem).

$$A=\hat{P}^{1/m}\quad\text{— a pseudodiff. operator.}$$

$$A=a_1+a_0+a_{-1}+\cdots$$

$$a_1=p^{1/m}(x,\xi).$$

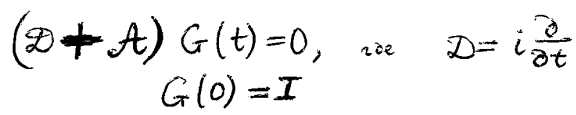
It suffices to do everything for the operator A. (pseudodiff., first order)

$$G(t)=\int e^{it\lambda}\,dE_\lambda=\exp(itA).$$

E_λ — spectr. decomp. of A: $A=\int\lambda\,dE_\lambda$.

$$(D + A)G(t) = 0, \qquad \text{where} \quad D = i\frac{\partial}{\partial t}$$

$$G(0) = I$$



Manuscript notation. A bold plus appears between D and A. At the right, $D = i\,\partial/\partial t$.

Original source details: GELFAND-P000274

This gives a hyperbolic problem, which can therefore be solved.

Generalized Huygens principle: singularities propagate at finite speed.

$$G(t)u = \iint e^{i\varphi(x,t,y,\xi)} q(x,t,y,\xi) u(y)\, dy\, d\xi + Ku,$$

$\varphi(x,t,y,\xi) = \psi(x,y,\xi) - ta_1(y,\xi).$

$$q \sim q_0 + q_1 + \cdots$$

Only this is important for the asymptotics.

We need small t. $(|t| < \varepsilon)$.

$$\psi(x,y,\xi) = (x-y,\xi) + O(|x-y|^2|\xi|).$$

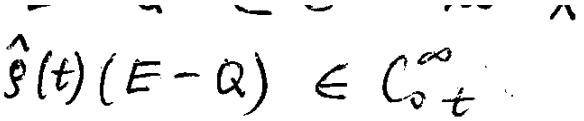
One can then refer to Marchenko's Tauberian theorem.

$E(x,y,t)$ — kernel of $G(t)$

$Q(x,y,t)$ — approx. kernel (regulariz.)

$$E - Q \in C^\infty \quad \text{in } x \text{ and } y \text{ for } |t| < \varepsilon$$

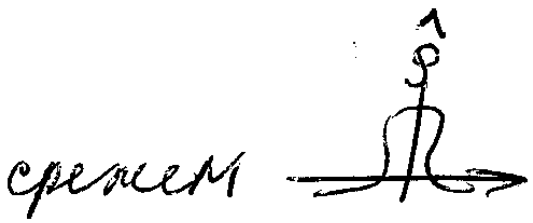
$$\hat{\varrho}(t)(E-Q) \in C^\infty_{0t}$$



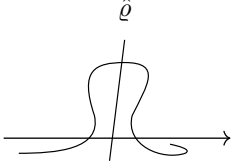
Uncertain reading. The signs 0 and t are below and to the right of C^∞; t is lower and farther right than 0. Strict index hierarchy undetermined.

Original source details: GELFAND-P000274

cut off



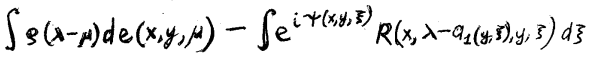
Uncertain reading. A slanted stroke marked $\hat{\varrho}$ crosses the rounded rise. Strokes continue below the horizontal axis on both sides. Their function is uncertain.



Original source details: GELFAND-P000274

GELFAND-P000275

$$\int \varrho(\lambda-\mu)\,de(x,y,\mu) - \int e^{i\psi(x,y,\xi)} R(x,\lambda-a_1(y,\xi),y,\xi)\,d\xi$$



Original source details: GELFAND-P000275

Manuscript notation. The first line is the difference of two integrals. The next line separately defines R.

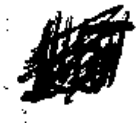
$$R(x,\lambda,y,\xi) = F^{-1}_{t\to\lambda}\big(q(x,t,y,\xi)\hat{\varrho}(t)\big)$$

Original source details: GELFAND-P000275

Diagram and typed description.

GELFAND-P000276

$14/\underline{\text{IV}} - 69.$



Original source details: GELFAND-P000276

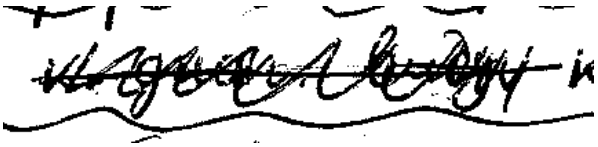
Uncertain reading. An illegible short mark is crossed out before the date.

I. M. Gelfand.

On the work of Yik-Hoi Au-Yeung (Proceed. of Amer. M. S. February 1969)

If two quadr. forms are not simult. 0, then they can be

simult. reduced [Crossed out: to ... form.]



Original source details: GELFAND-P000276

Uncertain reading. The crossed-out phrase reads “to ... form”. The middle word is uncertain.

to a sum of squares.

(Greg[?] “Lin. algebra”)

Original source details: GELFAND-P000276

Uncertain reading. The surname in the “Lin. algebra” reference is provisionally Greg. The last letter is uncertain.

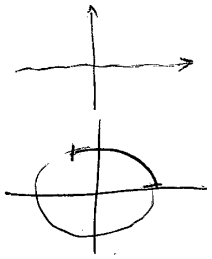
$\xi^* A\xi, \qquad \xi^* B\xi$ — corresp. quadr. forms

a pair of numbers — a vector of the plane.

[Above the line: Normalize.]

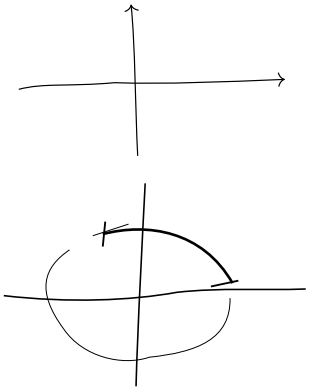
If $\forall \xi$ this is $\neq 0$, then it runs through no more than a semicirc.

After this rotate and one will be pos. definite.



Original source details: GELFAND-P000276

Uncertain reading. The upper axes have upward and rightward arrows. Below are axes without arrows and an open, nearly circular curve, heavier at the upper right. Short strokes occur at the ends of the heavy arc; their role as arrows or boundary marks is undetermined. There are no labels.



[Crossed out: in R / in R^3.]

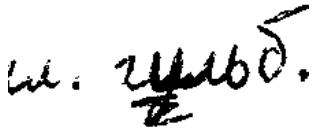
Original source details: GELFAND-P000276

Uncertain reading. A possible superscript at R is obscured by hatching; the digit 3 is undetermined.

Davis (Toronto)

Davis, Acta Sci. Math. (1968)

H — compl. Hilb.



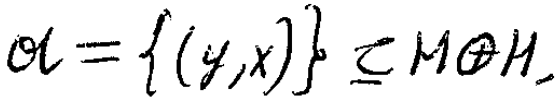
Original source details: GELFAND-P000276

Uncertain reading. Under the beginning of "Hilb." is a small correction group resembling Cyrillic "и"; its role is undetermined.

y^*x — scalar product, $(x, y) \in H \oplus H$.

$$A : H \to H, \qquad \underset{y=Ax}{(y,x)} \quad \text{— graph}$$

More gen. def. — a relation — a set $\alpha = \{(y, x)\} \subseteq M \oplus H$.



Original source details: GELFAND-P000276

Manuscript notation. The right-hand side reads $M \oplus H$.

1°

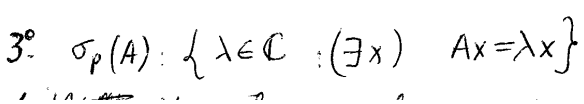
$$\|A\| \le a \quad \Rightarrow \quad \left\{ \frac{\|Ax\|^2}{\|x\|^2} : x \ne 0 \right\} \subseteq [0, a^2]$$

2°. Range of values of the form.

$$w(A) = \left\{ \frac{x^* Ax}{\|x\|^2} : x \neq 0 \right\}.$$

3°.

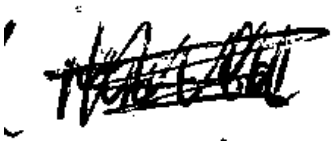
$$\sigma_p(A) : \{\lambda \in \mathbb{C} : (\exists x) \quad Ax = \lambda x\}$$



Manuscript notation. The parenthesis reads $(\exists x)$.

Original source details: GELFAND-P000276

([Crossed out: Numerical?] Numerical sets of the operator)



Uncertain reading. The cancelled word: Numerical?; some letters are obscured.

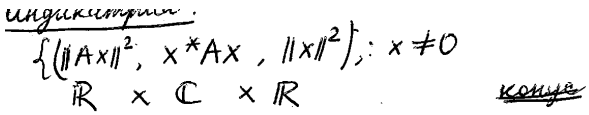
Original source details: GELFAND-P000276

Univers. inform. about the operator.
Indicatrix.

4°.

$$\{(\|Ax\|^2,\ x^* Ax,\ \|x\|^2) : x \neq 0$$

$$\mathbb{R} \times \mathbb{C} \times \mathbb{R}$$



Manuscript notation. The set is written with one opening brace. “Cone” is doubly underlined below the formula.

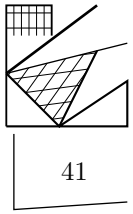
Original source details: GELFAND-P000276

cone

GELFAND-P000277

[source number: 41]

Diagram and typed description.

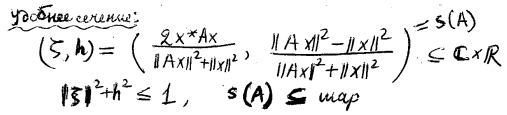


Original source details: GELFAND-P000277

A convenient section:

$$(\xi, h) = \left(\frac{2x^* Ax}{\|Ax\|^2 + \|x\|^2}, \frac{\|Ax\|^2 - \|x\|^2}{\|Ax\|^2 + \|x\|^2} \right)^{=s(A)} \subseteq \mathbb{C} \times \mathbb{R}$$

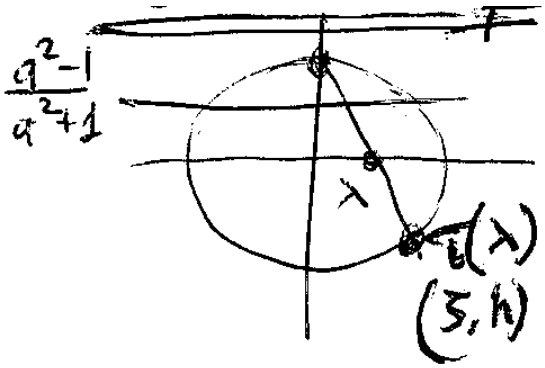
$$\|\xi\|^2 + h^2 \le 1, \qquad s(A) \subseteq \text{ball}.$$



Original source details: GELFAND-P000277

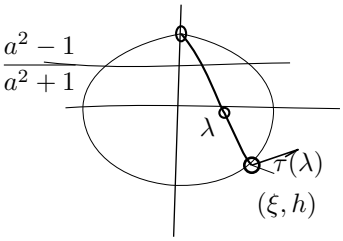
Manuscript notation. The insertion $= s(A)$ is above the right parenthesis. The main line gives $\subseteq \mathbb{C} \times \mathbb{R}$.

Eigenvector — when it belongs to the sphere.
eigenval. — stereogr. projection



Original source details: GELFAND-P000277

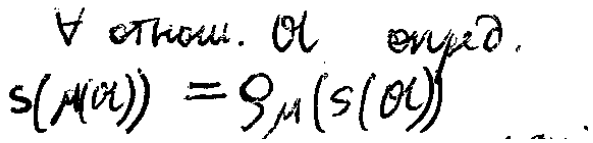
Uncertain reading. A circle, a vertical line, two horizontal lines and a sloping stroke with three marked points. Labels $\frac{a^2-1}{a^2+1}$, λ, $\tau(\lambda)$, (ξ, h). Short strokes at the lower-right point have no established arrow direction; the exact boundaries of the densely marked points are undetermined.



Norm — if it does not go above the line $\frac{a^2-1}{a^2+1}$, then a — the norm.

Th. 1. Let $\mu(\lambda) = \frac{a\lambda+b}{c\lambda+d}$. $ad - bc \neq 0$.
For each relation α, $\mu(\alpha)$ is defined,

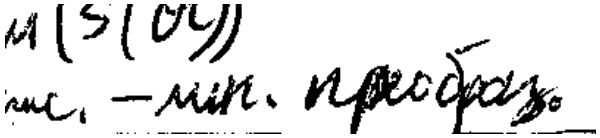
$$s(\mu(\alpha)) = \varrho_\mu(s(\alpha))$$



Original source details: GELFAND-P000277

Uncertain reading. Before the subscript μ is a loop with a curved tail. Provisionally ϱ_μ.

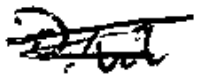
[Under the transformation symbol:]
[?]-lin. transf.



Original source details: GELFAND-P000277

Uncertain reading. The first abbreviation is uncertain. It is followed by "-lin. transf.".

[Crossed out: **[?]**]

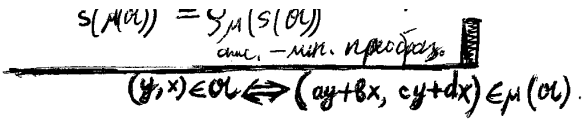


Original source details: GELFAND-P000277

Uncertain reading. A short cancelled word before “If”: Для? (For?); reading undetermined.

If A — an oper., then as usual
(if there exists $\mu(A) = (aA + b)(cA + d)^{-1}$).

$$(y, x) \in \alpha \iff (ay + bx, cy + dx) \in \mu(\alpha).$$



Original source details: GELFAND-P000277

Manuscript notation. Transformation rule below the long line.

def.:

$$X \text{ — a spectr. set of } A \iff$$

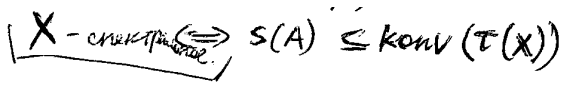
(**[Cancelled illegible mark]** $\forall$ rat. funcs. f: $f(X) \subseteq D \Rightarrow \|f(A)\| \leq 1$.)

Original source details: GELFAND-P000277

Uncertain reading. After the opening parenthesis, before $\forall$, a single mark is densely crossed out and unread.

$$D = \{\lambda : |\lambda| \leq 1\}.$$

Th. 2 description of spectr. sets
provided that $\tau(X)$ (on the sph.) — a segment (spherical).



Original source details: GELFAND-P000277

Diagram and typed description. Criterion marked by a bracket.

X — spectral $\iff$ $s(A) \subseteq \text{konv}(\tau(X))$.

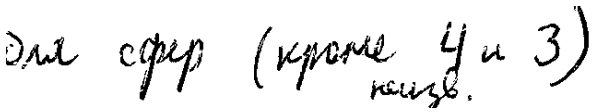
GELFAND-P000278

V. Buchstaber.

$$T^n = \underbrace{S^1 \times \ldots \times S^1}_{n}.$$

Thm. There are finitely many smooth manifolds of homotopy type torus.
(up to diffeom.)
For simply conn., it is proved: everything is determined by Pontryagin classes.
and the hom. type
(up to finitely many possibilities)

Milnor: for spheres (except $\underset{\text{unknown}}{4}$ and 3)



Manuscript notation. The note “unknown” is written below the line just to the left of 4.

Original source details: GELFAND-P000278

Novikov: For simply conn. manifolds: they are determined by the homot. type and charact. classes of the tangent bundle — then finitely many possibilities remain.

M^n — a manifold.

Smale:

Let $\partial W^{n+1} = M_1^n \cup M_2^n$

$M_1^n \subset W^{n+1}$ — a deform. retract

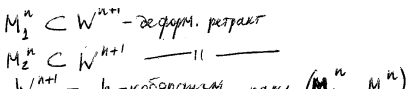


Diagram and typed description.

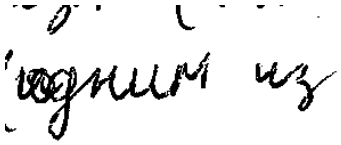
$M_2^n \subset W^{n+1}$ ——————||——————

Original source details: GELFAND-P000278

Then W^{n+1} — h-cobordism of the pair (M_1^n, M_2^n).

Smale’s theorem.

h-cobordism (film) is completely (up to diffeom.) determined by a boundary (one of M_i^n) and an invar.



Uncertain reading. The beginning of the final word “одним” (one) is overwritten over an illegible earlier entry.

Original source details: GELFAND-P000278

$$\alpha \in \underset{\text{(Whitehead gr.)}}{\mathrm{Wh}}(\pi_1(M_1^n)).$$

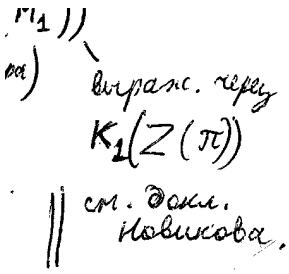


Diagram and typed description.

\

expressed through

$K_1(Z(\pi))$

||

see the talk of

Novikov.

Original source details: GELFAND-P000278

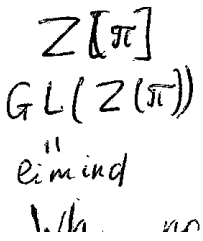


Diagram and typed description.

$Z[\pi]$

$GL(Z(\pi))$

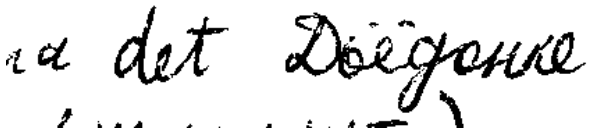
| |

lim ind

Original source details: GELFAND-P000278

group det Dieudonné $K_1(Z(\pi))$.

Uncertain reading. The name is Dieudonné. The sequence of middle strokes in the handwritten Russian spelling is uncertain.

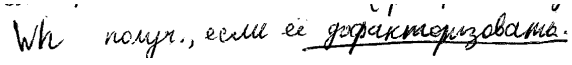


Original source details: GELFAND-P000278

(noncommut.)

(quotient by the commutator subgroup).

Wh is obtained by taking a further quotient.



Original source details: GELFAND-P000278

Manuscript notation. The underlined verb means “take a further quotient”.

The element α is obtained through Smale’s func. (nice-function).

In the Smale-Novikov case [crossed out: this?] h cobordisms $\Longleftrightarrow$ diffeom., and a cobord. is already a homot. [cancelled illegible mark] concept. (There $W^{n+1} = M_1^n \times I$)

Uncertain reading. A short word between “Smale-Novikov” and “h cobordisms” is cancelled: это? (this?); reading undetermined.

Original source details: GELFAND-P000278

Uncertain reading. Between “homot.” and “concept”, a mark or beginning of a word is cancelled; reading undetermined.

Original source details: GELFAND-P000278

GELFAND-P000279

[source number: 42]

42

Tori could be classif. because of Bass’s theorem:

$$K_1(Z + Z + \ldots + Z) = 0.$$

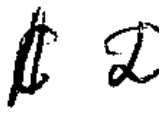
(of the group ring of the group).

(algebr. fact)

Original source details: GELFAND-P000279

Manuscript notation. The line uses K_1, ordinary plus signs and $= 0$.

For Z it is obvious.



Uncertain reading. Left of “For Z”, a narrow loop-like mark with a stroke; reading undetermined.

Original source details: GELFAND-P000279

Matrices of polynomials ~~det =~~ invertibility $\Longleftrightarrow$ det $= \pm 1$.

Uncertain reading. Cancelled: det and a following sign: $det =$ / $det \neq$; the sign is undetermined.

Original source details: GELFAND-P000279

(see Artin’s book “Geometric Algebra” — it has the Dieudonné determinant).

Whitehead det. 1 — if reducible by elem. transformations to 1

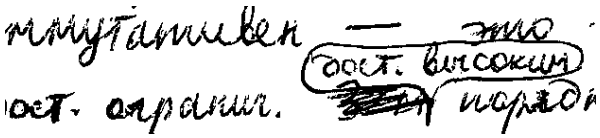
(and increase of order)

The quotient is commutative — this is a remarkable theorem! (Whitehead).

Bass — it suffices to restrict to

sufficiently high

~~[?]~~ order.

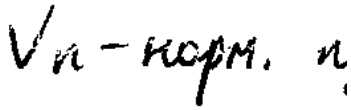


Uncertain reading. An illegible word is crossed out below the outlined “sufficiently high”.

Original source details: GELFAND-P000279

$$M^n \overset{\nu_n}{\subset} \mathbb{R}^N$$

ν_n — normal bundle



Uncertain reading. Initial letter: ν_n / V_n; the first lower index is n, and M is used below.

Original source details: GELFAND-P000279

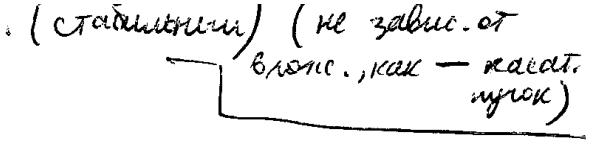


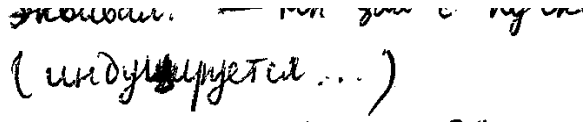
Diagram and typed description. Continuation at the right of the line.

(stable) (does not depend on embedding, like the tangent bundle)

Original source details: GELFAND-P000279

Equivalence — of manifolds with a bundle.

(is induced...)



Uncertain reading. In “индуцируется” (is induced), the middle letters are heavily overwritten; the earlier entry is undetermined.

Original source details: GELFAND-P000279

Let us calculate for the torus.

From Adams's theorem on the J-funct.

Diagram and typed description. The insertion “spher.” above “for”.

spher.

for

Original source details: GELFAND-P000279

$M^n \sim T^n$ everything is completely determined by the Pontryagin classes of M^n.

Pontryagin classes — an infinitely valued invar.

Novikov: for tori, Pontryagin classes are a homot. invariant.

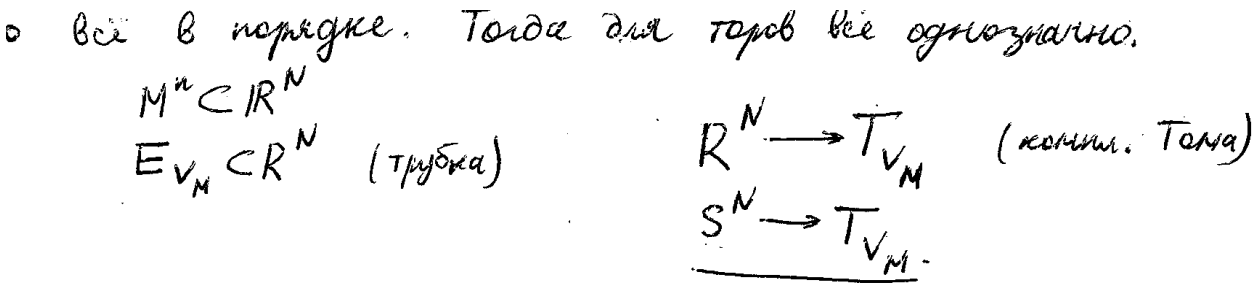
There $\nu_M = 0$.

But: on $S^2 \times S^4$ there are infinitely many smooth structures.

Again everything works for tori, since the torus is parallelizable and $\nu_M = 0$.

$$\{M_1^n\} \qquad f : M_1^n \longrightarrow M_2^n, \quad f^*\nu_{M_2} = \nu_{M_1}.$$

How many such classes? If there are finitely many here, then everything is in order. Then for tori everything is unique.



Original source details: GELFAND-P000279

Diagram and typed description.

$M^n \subset \mathbb{R}^N$

$R^N \longrightarrow T_{\nu_M}$ (Thom compl.)

$E_{\nu_M} \subset R^N$ (tube)

$S^N \longrightarrow T_{\nu_M}$.

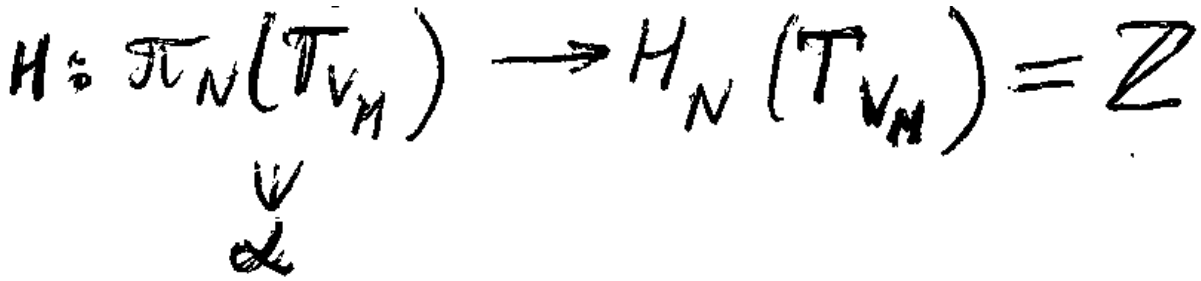
GELFAND-P000280

An elem. of $\pi_N(T_{\nu_{M_1}})$ is obtained.

The space T_{ν_M} does not depend on the manifold (!) — they are all homot. equiv., since on the base they are homot. equiv.

We obtain an element of $\pi_N(T_{\nu_M})$.

We need to consider the Hurewicz homom.



Original source details: GELFAND-P000280

Diagram and typed description.

$$H: \quad \pi_N(T_{\nu_M}) \longrightarrow H_N(T_{\nu_M}) = \mathbb{Z}$$

$$\downarrow \alpha$$

If α is realized in this way, then $H(\alpha) \longrightarrow +1$.

(on the interior of the tube it is the identity).

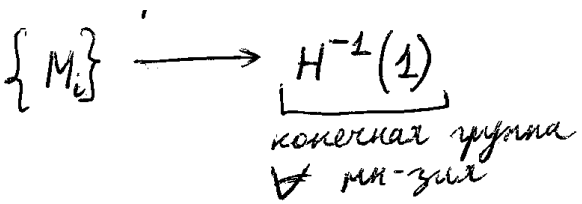
(by the Excision theorem).

There is a map

$$\{M_i\} \longrightarrow H^{-1}(1)$$

finite group

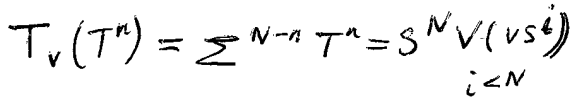
$\forall$ manifolds



Manuscript notation. The rectangular bracket is labelled “finite group $\forall$ manifolds”.

Original source details: GELFAND-P000280

$$T_\nu(T^n) = \Sigma^{N-n} T^n = S^N \vee \left(\vee_{i<N} S^i\right).$$

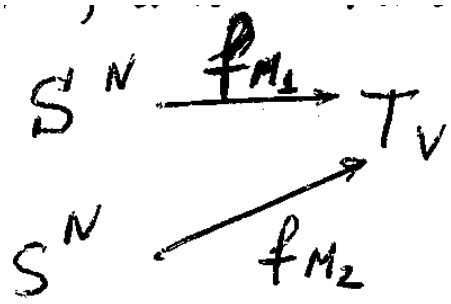


Manuscript notation. Both signs in the line are equality signs. The last parenthesis has $i < N$ below it.

Original source details: GELFAND-P000280

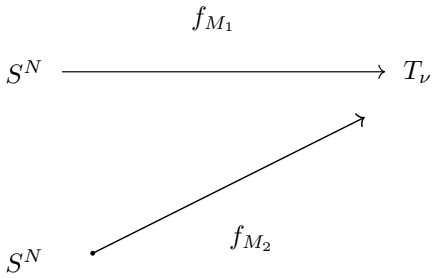
Everything follows from Serre’s finiteness theorem.

It remains to show that Ker is finite.



Original source details: GELFAND-P000280

Uncertain reading. The f_{M_2} stroke has an arrowhead on the right. The left end is thickened. A second arrowhead is indistinct.

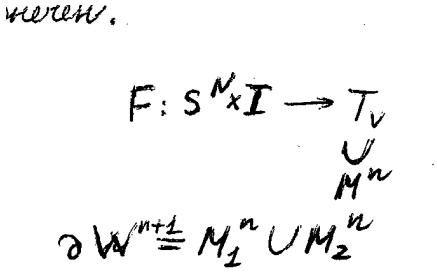


Suppose they are homotopic.

$$\partial W^{n+1} = M_1^n \cup M_2^n$$

and there exist retractions and with bundles

Everything is in order.

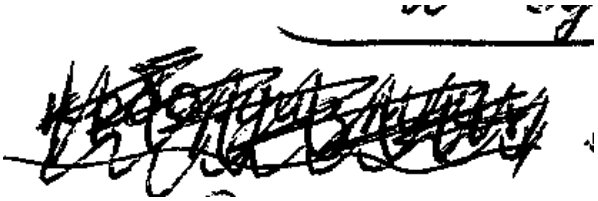


Original source details: GELFAND-P000280

Diagram and typed description.

$$\begin{array}{ccc} F : S^N \times I & \longrightarrow & T_\nu \\ & & \cup \\ & & M^n \end{array}$$

We need to classify ~~[?]~~ films which deform. retract.



Original source details: GELFAND-P000280

Uncertain reading. A possible reading under the heavy cancellation is кобордизмы (cobordisms).

A class of finite index, for which a film exists.
This is the same as saying that $i : M_1 \subset W$ induces an isom. of all homot. groups.
Obstruction theory for the modification of the film.

GELFAND-P000281

[source number: 43]

21/IV Kirillov. Borel–Weil theorem. (in a new form).

43

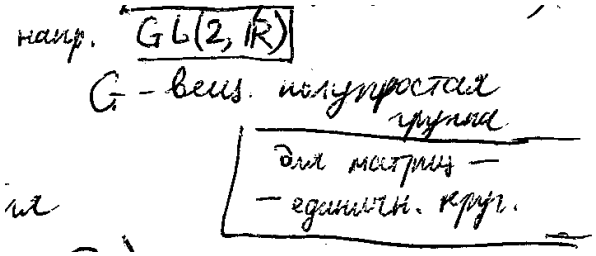
$G \supset K$ K — compact subgr. (max).
It is assumed that rank G = rank K (dimension of a max. torus).
$K \supset H$ — Cartan subgr. (common to K and G).

e.g. $GL(2,\mathbb{R})$

G — a real semisimple group.

for matrices —

— unit circle.



Original source details: GELFAND-P000281

Manuscript notation. The frame around $GL(2,\mathbb{R})$ is open on the left. The frame of the matrix note is open on the right.

$D = G/H$ — complex manifold (always)

The number of complex structures — the number of elements of the Weyl group.

$\mathcal{L}_\lambda$ — 1-dimensional bundles over D(fiber $\mathbb{C}$)

$$\mathcal{L}_\lambda \xrightarrow{\mathbb{C}} D$$

1-dimensional representation — a character of the Cartan subgroup

$A_0^i(\mathcal{L}_\lambda)$ — smooth $(0,i)$-forms with values in $\mathcal{L}_\lambda$.

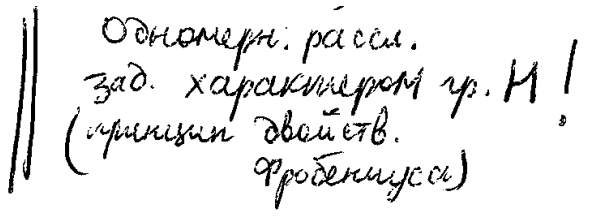
(and with compact support)

$$\bar\partial : A_0^i \longrightarrow A_0^{i+1}$$

$$\square = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial.$$

(Laplace–Beltrami)

The space of L^2-cohomology. (with coefficients in the sheaf)



Original source details: GELFAND-P000281

Diagram and typed description. Note in the right margin.

A one-dimensional bundle
is given by a character of the group H!
(the principle of Frobenius
reciprocity)

G — compact $\Rightarrow$ this is the Borel–Weil theorem.

On D there is an invariant measure $\Rightarrow$ there is [?] a Hilbert space L^2. (completion of smooth ones)

Original source details: GELFAND-P000281

Uncertain reading. Before "a Hilbert space L^2", a densely crossed-out entry in a rectangle; its reading is undetermined.

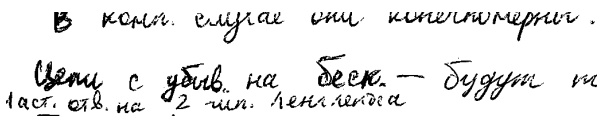
$\bar{\partial}$ on a dense set.

$$\mathrm{Ker}\,\Box = \mathcal{H}^i(\mathcal{L}_\lambda)$$

— i-dimensional L^2-cohomology.

In the compact case they are finite-dimensional. In the general case they are not.

If with decay at infinity — there will be the same cohomology.



Original source details: GELFAND-P000281

Uncertain reading. The beginning of "Если" (If) is heavily overwritten; the earlier letters are undetermined.

A partial answer to 2 conjectures of Langlands

Theorem 1. λ — a weight, $|(\lambda, \alpha)| > b \quad \forall$ simple root α.

sufficiently nondegenerate.

Then

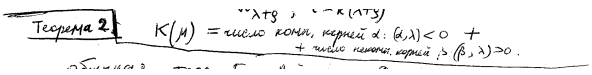
$$\mathcal{H}^i(\mathcal{L}_\lambda) = \begin{cases} 0, & i \neq k(\lambda + \rho), \\ \pi_{\lambda+\rho}, & i = k(\lambda + \rho). \end{cases}$$

Original source details: GELFAND-P000281

Uncertain reading. A looped mark after λ in the first row is struck out. It is followed by $+\rho$.

Theorem 2.

$$K(\mu) = \text{number of compact roots } \alpha : (\alpha, \lambda) < 0 \quad + \\ + \text{number of noncompact roots } \beta : (\beta, \lambda) > 0.$$



Original source details: GELFAND-P000281

Manuscript notation. The left side is $K(\mu)$ and both pairings on the right use λ. The plus sign is repeated at the line break.

For the usual Bor.–Weil theorem see Demazur

Original source details: GELFAND-P000281

Manuscript notation. The surname is written “Demazur”.

GELFAND-P000282

Original source details: GELFAND-P000282

Uncertain reading. Two loops before the surname are struck out. The letter is undetermined.

Buchstaber.

Manifolds of the homot. type of a torus are divided by h-cobordisms into a finite number of classes

Problem: to show that there are finitely many in each class

Next, the simplest films are classified by Morse modifications.

Original source details: GELFAND-P000282

Diagram and typed description. Rule at the end of the entry.

Original source details: GELFAND-P000282

Uncertain reading. A curved stroke with a dot above it is visible at the very edge of the scan. Its meaning is uncertain.

GELFAND-P000283

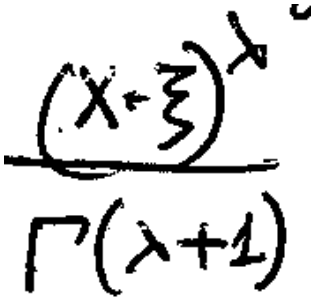
15.9.69 yr.

B. Vainberg on results of Hörmander. (see Vishik seminar). on singularities of fund. sol.
General sol. of homog. eqn.: (homog. polynomial)

$$\int_{P(\xi)=0} f[(x,\xi)]A(\xi)\,d\xi \qquad \text{(Borovikov)}$$

f, A — arbitrary functions.
Singularities in the homog. case are described by Borovikov.
For example:

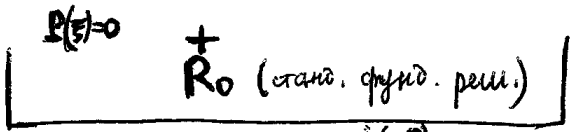
$$\int_{P(\xi)=0} \frac{(x+\xi)^\lambda}{\Gamma(\lambda+1)} A(\xi)\,d\xi$$



Original source details: GELFAND-P000283

Uncertain reading. The short crossed stroke is provisionally read as $+$. The preceding formula has (x,ξ).

— proposal of Borovikov.



Original source details: GELFAND-P000283

Diagram and typed description. Plus above R_0 within the angular rule.

$\overset{+}{R}_0$ (stand. fund. sol.)

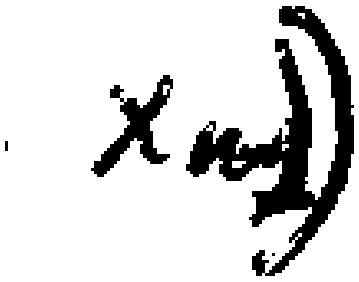
Grushin:

$$\sum_j \int \frac{\alpha_j(\xi)e^{i(x,\xi)}}{P(\xi)\pm i0}\,d\xi$$

α_j — partition of unity on the sphere.

The Bernstein problem is solved.

$$\left\{\begin{array}{l} z = f(x_1, \ldots, x_{n-1}) \text{ — min. surf.} \\ (x\text{-arbitrary}) \implies \text{it is a plane.} \end{array}\right\}$$



Uncertain reading. The final x has the provisional index $n-1$. Its end overlaps the closing parenthesis.

Original source details: GELFAND-P000283

true for $n \leq 8$; for $n \geq 9$ false.

De Giorgi, Simons and Bombieri.

Kazhdan and Bernstein.

Convergence of the spectrum as the domains expand.

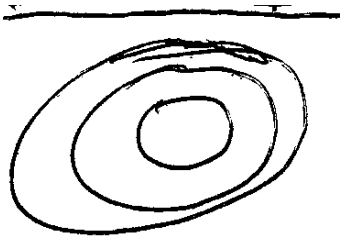
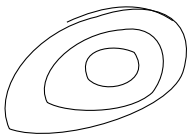


Diagram and typed description.



Original source details: GELFAND-P000283

A — self-adj. op. on the manifold X.

$\Lambda \subset \mathbb{R}$.

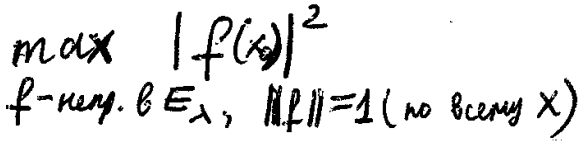
[Below Λ:] interval.

The continuous functions contained in E_Λ are dense in it.

(required) — so for elliptic op.

$x_0 \longrightarrow$ a Plancherel measure on the spectrum of A.

$$M_{x_0}(\Lambda) = \max_{\substack{f\text{-continuous in } E_\lambda,\ \|f\|=1 \\ \text{(over all of } X\text{)}}} |f(x_0)|^2.$$



Manuscript notation. Below max the manuscript has E_λ. Above it has E_Λ.

Original source details: GELFAND-P000283

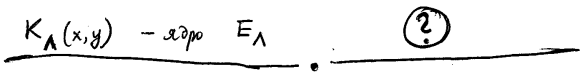
[At right:] This is a measure (possibly infinite).

GELFAND-P000284

[source number: 53]

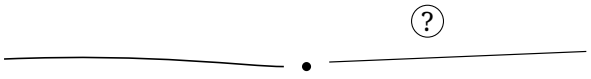
$K_\Lambda(x,y)$ — the kernel of E_Λ.

53



Original source details: GELFAND-P000284

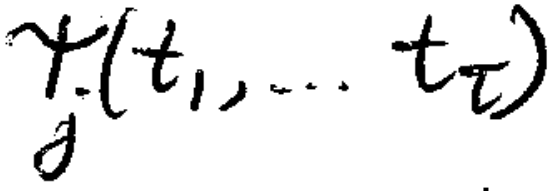
Diagram and typed description. Question mark over the right part of the rule.



Kazhdan. M. Artin in the Bourbaki seminar.

$$f_i(x_1,\dots,x_n)=0, \qquad 1\le i\le k.$$

f_i — convergent power series.
When does a sol. exist $\psi_j(t_1,\dots,t_\tau)$



Manuscript notation. The forms read ψ_j and t_τ.

Original source details: GELFAND-P000284

Answer: if and only if there is a sol. in formal power series.
(arbitrarily close to the formal one)
[Below the parenthesis:] can be completed

22.9.69 yr.
Fichera on unilateral bound. problems.

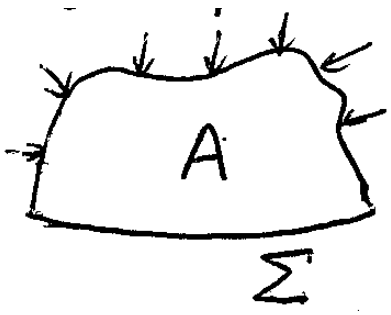
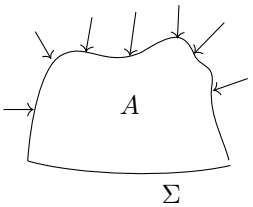


Diagram and typed description.



Original source details: GELFAND-P000284

An elastic body on a rigid, frictionless surface Σ.
And the stress tensor $\equiv 0$.
A — natural volume of the body.

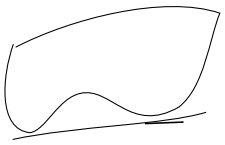
On the free part of the boundary, a force $\underbrace{\varphi}_{\text{density}}$ on $\partial A - \Sigma = \Sigma^*$.

Inside there are also certain forces f — volume forces.

Let the body move to a position of equil.

It may separate from Σ.

Uncertain reading. Contact is visible at the left. At the right, overlapping strokes and a narrow light strip; exact contact or separation and the length of contact are undetermined. There are no labels or arrows.



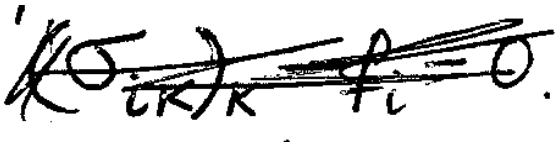
Original source details: GELFAND-P000284

This is an analytic problem (formulation A. Signorini, 1959)

σ_{ik} — the stress tensor.

Inside the body

[Crossed out:] ~~$(\sigma_{ik})_k \quad f_i = 0$~~



Manuscript notation. Earlier formula under the cancellation.

Original source details: GELFAND-P000284

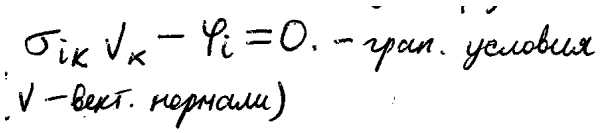
[Written at the right:]

$$\sigma_{ik/k} - f_i = 0.$$

$(\operatorname{div}\sigma = f)$.

$$\sigma_{ik}\nu_k - \varphi_i = 0.$$

— bound. conditions.
(ν — normal vector)



Uncertain reading. The sign looks like V in both lines. Provisionally ν.

Original source details: GELFAND-P000284

GELFAND-P000285

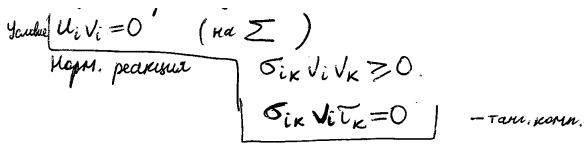
On Σ.

u_i — the displacement vector.

Condition $u_i\nu_i = 0 \quad (\text{on } \Sigma)$

Normal reaction $\sigma_{ik}\nu_i\nu_k \geq 0$

$\sigma_{ik}\nu_i\tau_k = 0$ — tang. comp.

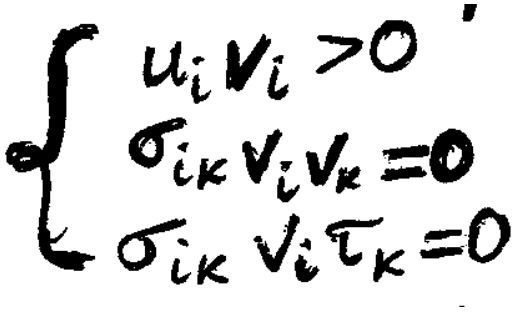


Original source details: GELFAND-P000285

Uncertain reading. The conditions are joined by a stepped rule. The normal sign looks like V and is provisionally read as ν.

At a point of the free surface:

$$\begin{cases} u_i\nu_i > 0, \\ \sigma_{ik}\nu_i\nu_k = 0, \\ \sigma_{ik}\nu_i\tau_k = 0. \end{cases}$$



Original source details: GELFAND-P000285

Manuscript notation. The first line starts with $u_i\nu_i$.

But it is unknown where which occurs.

We have not yet taken into account that the body is elastic.

$$\underline{\underline{W(x,\varepsilon)}}$$
potential

$$\varepsilon = \{\varepsilon_{ik}\} \qquad \varepsilon_{ik} = 2^{-1}(u_{i/k} + u_{k/i})$$

Strain tensor.

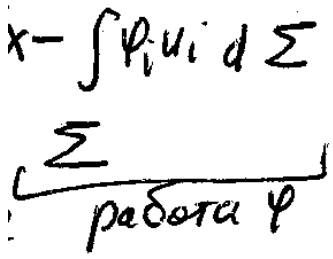
Relation:

$$\sigma_{ik} + W_{\varepsilon_{ik}} = 0 \qquad \text{on } A.$$

Problem of exist. and uniqueness.

$$I(u) = \int_A W[x, \varepsilon(u)]\, dx - \underbrace{\int_A f_i u_i\, dx}_{\text{work of } f} - \underbrace{\int_\Sigma \varphi_i u_i\, d\Sigma}_{\text{work of } \varphi}.$$

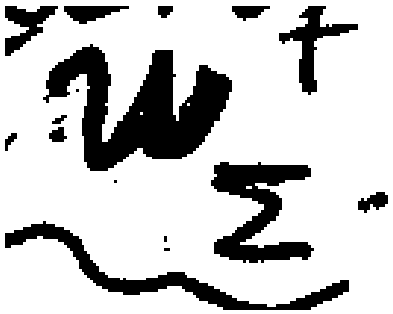
(Total energy)



Original source details: GELFAND-P000285

Manuscript notation. This integral has domain Σ. The model problem below has Σ^*.

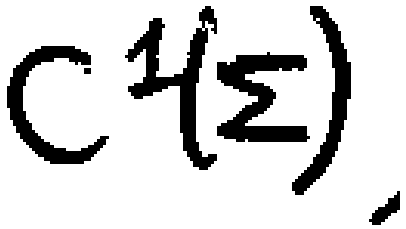
In the class of admissible displacements $\mathcal{W}_\Sigma$.



Original source details: GELFAND-P000285

Manuscript notation. The letter with two lower bends reads $\mathcal{W}_\Sigma$. The model below uses $\mathcal{U}_\Sigma$.

If a minimum exists in $C^1(\Sigma)$, then it is a solution.
conversely, a solution is a min.



Original source details: GELFAND-P000285

Manuscript notation. The line reads $C^1(\Sigma)$.

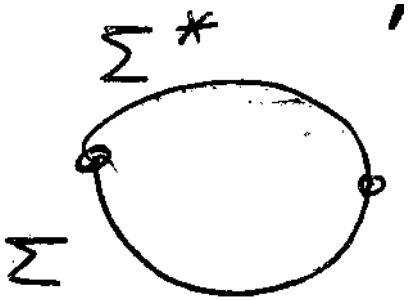
But: in this class min does not exist.

$$I(u) = \frac{1}{2}\int_A a_{nk}(x) u_{/n} u_{/k}\, dx - \int_A f u\, dx - \int_{\Sigma^*} \varphi u\, d\sigma.$$

$$a_{nk}(x)\mu_n\mu_k \geq \rho_0 \mu_n \mu_n.$$

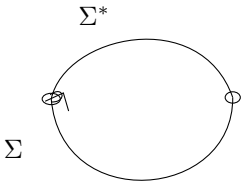
$$\mathcal{U}_\Sigma : \quad u|_\Sigma \geq 0.$$

(Model problem)



Original source details: GELFAND-P000285

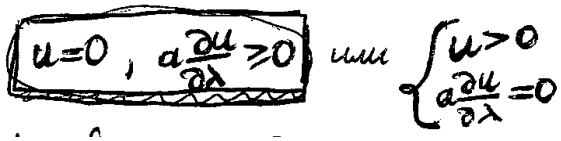
Uncertain reading. The upper arc: Σ^*; at lower left: Σ. Two ringed junction marks; the left one is retraced, with an oblique stroke. Its role and an orientation of traversal are undetermined.



GELFAND-P000286

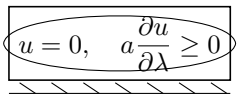
[source number: 54]

From this we obtain the problem.



Original source details: GELFAND-P000286

Diagram and typed description. Frame and brace at the two alternatives.

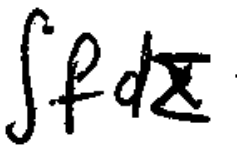


$$u = 0, \quad a\frac{\partial u}{\partial \lambda} \geq 0 \quad \text{or} \quad \begin{cases} u > 0, \\ a\dfrac{\partial u}{\partial \lambda} = 0. \end{cases}$$

$$(a_{nk}u_{/k})_{/n} = f. \qquad 54$$

Necessary:

$$\int_A f\,dx + \int_{\Sigma^*} \varphi\,d\Sigma \leq 0.$$



Original source details: GELFAND-P000286

Uncertain reading. The diagonal strokes after d overlap upper and lower horizontal lines. Provisionally dx.

Original source details: GELFAND-P000286

Uncertain reading. Below the first integral is a retraced A. The earlier strokes are illegible.

Let $f \leq 0$ on A and $\varphi < 0$ on Σ^*.

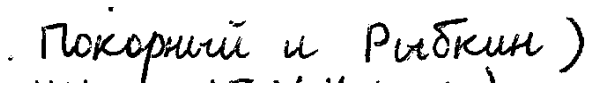
By the min principle, u cannot have a min. in A.

There is no sol. in $C^1(\bar{A})$.

(deduced from Green's formula)
an immediate contradiction — a discontinuity of the derivative is obtained.

29.9.69

Kazhdan Formulation of results of M. Artin. (1968).
(in the USSR independently Pokornii and Rybkin)



Original source details: GELFAND-P000286

Uncertain reading. The ending of the first surname provisionally reads "-ний". The second is written "Рыбкин" (Rybkin).

UMN, vol. 15, $N = 4$ (1960) p. 169
and Melamed
Sib. math. journal

$$f_i(x_1, \dots, x_k) = 0, \qquad 1 \le i \le n.$$

a collection of convergent power series (in a nbd. of 0
We try to find

$$\boxed{x_j = x_j(y_1, \dots, y_\ell)} \quad \boxed{j = 1, \dots, k.}$$

so that the equalities hold upon substitution.
It is sufficient to find such functions in formal power series.

Example.

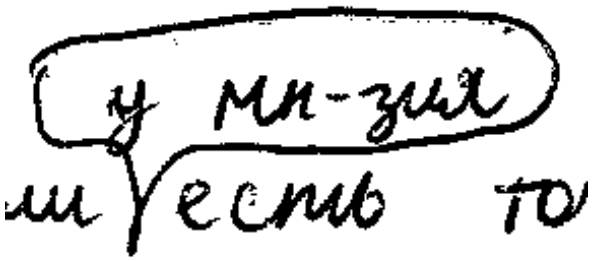
$$x_1^2 - x_2^3 = 0 \qquad \begin{aligned} x_1 &= y^3, \\ x_2 &= y^2. \end{aligned}$$

Then a sol. in analytic functions will be found, arbitrarily close. (with any number of derivatives)

the problem is handled by induction on the number of x_j.
and a more complicated ind.
Reformulation.
These are equations of an analytic set and 2 rings

$$\begin{aligned} A &= \mathbb{C}\{y_1, \dots, y_\ell\} && \text{convergent series.} \\ B &= \mathbb{C}[[y_1, \dots, y_\ell]] && \text{formal series.} \end{aligned}$$

If the manif. has a point with values in B, then there is also a point with values in A.



Manuscript notation. Insertion between “If” and “has”.

Original source details: GELFAND-P000286

GELFAND-P000287

Induction also on conn. components. (their number)

Irreduc. — a subpiece has smaller dimension.

1) If for irreduc. varieties (of the given dim.) proved, then also always proved. (given dim.)

Choose eqns. lying in the ideal, **[crossed out: ?]** corresp. to this component.

Uncertain reading. The initial Cyrillic letters appear to be ко.... The rest is obscured by the cancellation.

Original source details: GELFAND-P000287

Adding eqns. Consider a prime ideal.

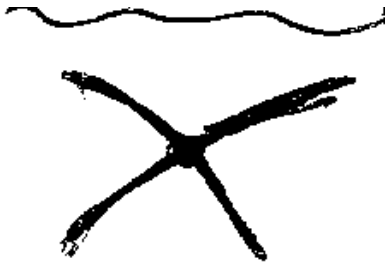
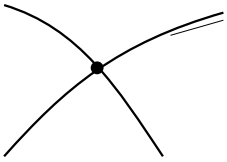


Diagram and typed description.



Original source details: GELFAND-P000287

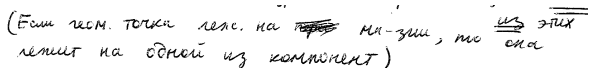
$\mathcal{J} =$ (Intersect. of primes).

Uncertain reading. The sign before the equals sign has a long lower hook. Provisionally $\mathcal{J}$.

Original source details: GELFAND-P000287

We obtain 2 systs. of eqns. Every solution satisfies one of these systems.

(If a geom. point lies on **[crossed out: ?]** a variety, then it lies on one of the components.)



Original source details: GELFAND-P000287

Uncertain reading. A short word or word beginning between “on” and “a variety” is covered by several lines; no reliable sequence of letters is established.

[Cancelled mark in the left margin: ?]

Original source details: GELFAND-P000287

Uncertain reading. A compact repeatedly crossed-out mark with loops. Whether it is a letter, number or nontextual mark is undetermined.

The rest — next time.
L. Faddeev.

On the results of a computational experiment, **[crossed out: ?]** rel. to Kirillov’s programme

descr. of reprs.

Original source details: GELFAND-P000287

Uncertain reading. The cancelled word appears to begin with the Cyrillic letters ре.... The rest is obscured.

Representation — quantization of class. mechanics.

Examples and probls. on quantization rules.

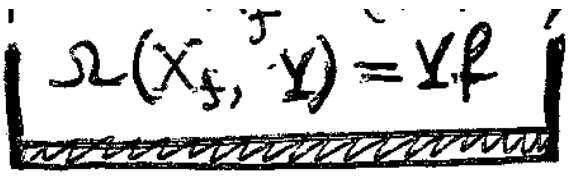
M_{2n} — a sympl. manifold, equipp. Ω — a 2-form (nondeg.). $\Omega = d\omega$.

Original source details: GELFAND-P000287

Manuscript notation. The abbreviation before the form reads обесп. (“equipp.”).

$d\mu = \Omega^n$ — a measure.

$f \longrightarrow X_f$ (a field from a fn.)



Original source details: GELFAND-P000287

Diagram and typed description.

$\Omega(X_f, Y) = Yf$

Hamiltonian fields X_f form a Lie algebra

$$[X_f, X_g] = X_{\{f,g\}},$$

where $\{f, g\}$ — the Poisson bracket (it is defined by this).
Oper. X_f is antisymm. in L^2 with measure $d\mu = \Omega^n$. (It is preserved by Liouville's theorem.)
Lie-algebra elem. $X_f \longrightarrow$ corresponding oper. in L^2.
1951 Van Hove found a 1-param. family of representations.

$$Z_f = X_f + \alpha\big(\omega(X_f) - f\big)$$

GELFAND-P000288

[source number: 55]

55

Then the representation condition holds

$$\omega(X_f) - f = \ell_f$$

Van Hove's assertion: This is again a repr., not equiv. for any α

$$[Z_f, Z_g] = Z_{\{f,g\}}$$

But this is not yet a passage to quantum mechanics, since everything is still in the space of all variables, rather than half of them.
Consider now $SO(3)$ with its nat. repr. in $\mathbb{R}^3$. On the orbits (S^2) there is a natural compl. structure.
Result of computing Z_f, when X_f — a gener. of the rotation group.

Diagram and typed description.

~~X_1, X_2, X_3~~

Original source details: GELFAND-P000288

$$x_1^2 + x_2^2 + x_3^2 = a^2 \quad \text{— orbit}$$

Functions x_1, x_2, x_3. Corresp. Van Hove opers.:

Quantization problem.

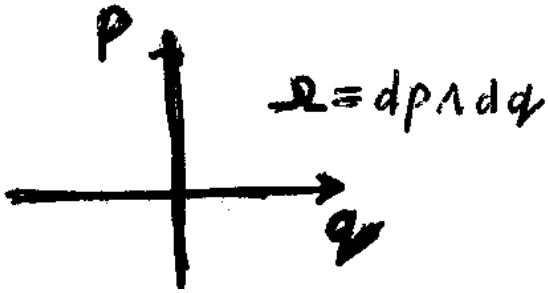
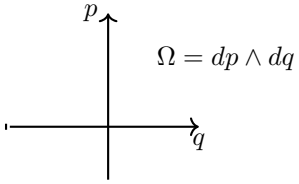


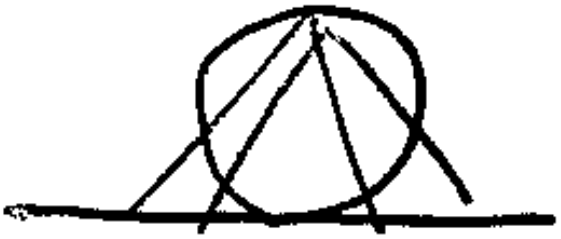
Diagram and typed description.



Original source details: GELFAND-P000288

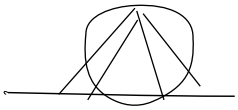
Hamiltonian $f = \dfrac{p^2}{2m}$ (free particle).
An operator must be assigned to it.

$f = x_1$.



Original source details: GELFAND-P000288

Uncertain reading. A rounded contour touches a nearly horizontal line; four sloping strokes extend from its upper region. The rightmost ends above the lower line. A small left hook may be a leftward arrow or a line irregularity; its direction and the exact upper-stroke junctions are undetermined. There are no labels.



After stereogr. projection (coord. z) — ω is not single-valued.

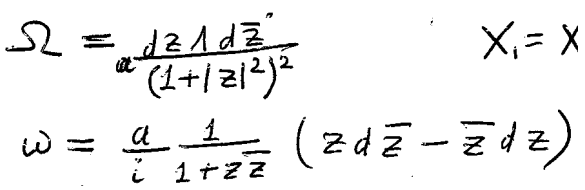
$$\Omega = a\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}$$

$$X_i = X_f \quad \text{for } f = x_i$$

Uncertain reading. The subscript on X_i has a dot. The short stroke under x_i has no distinct dot.

Original source details: GELFAND-P000288

$$\omega = \frac{a}{i}\frac{1}{1+z\bar{z}}(z\,d\bar{z} - \bar{z}\,dz)$$



Manuscript notation. The factor before the fraction in Ω is a. The entry for ω has a/i.

Original source details: GELFAND-P000288

No sing. on the plane, but a sing. at ∞ (north pole before projection).

$$X_1 = \frac{1}{2i}\left((1-z^2)\frac{\partial}{\partial z} - (1-\bar{z}^2)\frac{\partial}{\partial \bar{z}}\right)$$

$$Z_1 = X_1 + \frac{\alpha a}{2}(z+\bar{z}).$$

The analyt. and antianal. parts commute!
Therefore we take

$$Y_1 = \frac{1}{2i}(1-z^2)\frac{d}{dz} + \frac{\alpha a}{2}z$$

Y_2, Y_3 analogously. This is a basis of the Lie alg., satisfying commutation rels.

GELFAND-P000289

They give a repr. of the rotation gr.

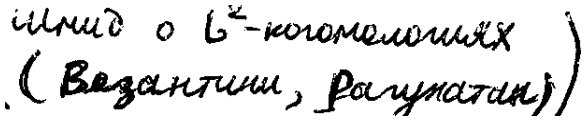
A measure where they are self-adj. exists only for integral a!

$$\frac{1}{(1+\bar{z}z)^a}$$

Square-integrable ones — a finite number — polynomials. Therefore a finite-dimensional representation.

Next time: Kazhdan;

(Schmid on L^2-cohomology (Vezantini, Raghunathan)).



Original source details: GELFAND-P000289

Uncertain reading. The first surname in parentheses provisionally reads Везантини (Vezantini).

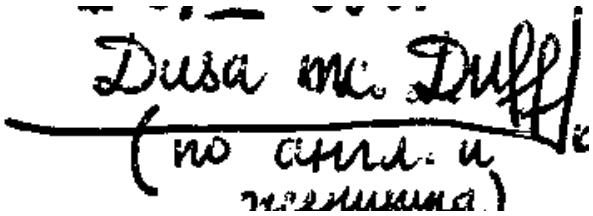
Starts at 18^{30}. (end 20^{30})

6/X-69 yr.

Original source details: GELFAND-P000289

Uncertain reading. A narrow mark before the digit 6 is heavily struck through. Its identity is uncertain.

Dusa mc Duff



Original source details: GELFAND-P000289

Manuscript notation. The name is written “Dusa mc Duff”, with lowercase “mc”.

On factors of type II_1.

(in English and a woman).

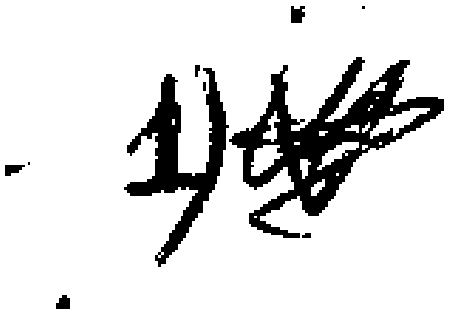
Original source details: GELFAND-P000289

Diagram and typed description. Three rules under the reference in the left margin.

see Naimark.

A subring of the ring of all opers. in Hilb. space is called a factor, if

1) [crossed out: ∀?] $X \in M \Longrightarrow X^* \in M$.



Uncertain reading. After "1)" the sign beneath slanting strokes may be the quantifier ∀; it has not been recovered.

Original source details: GELFAND-P000289

2) M is weakly closed (for ∀ seq. X_n the weak limit belongs).

3) The centre consists of opers. proportional to I.

Example. In the fin.-dimensional case:

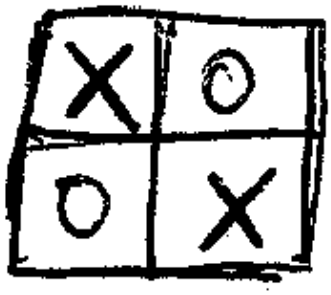


Diagram and typed description.

X	0
0	X

Original source details: GELFAND-P000289

Every factor in the fin.-dimensional case is like this: in $H_1 \otimes H_2$ consider opers. $X \otimes 1$ (X — arbitrary).

A vect. — a matrix ξ_{ik}. An oper. — acts on the 1st coordinate.

Every weakly closed ring (with involution) decomposes into a direct integral of factors.

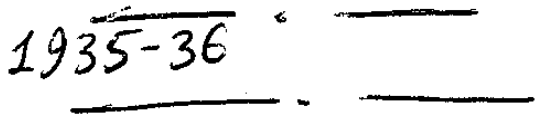


Diagram and typed description. The date with rules above and below it.

Von Neumann
1935–36

Original source details: GELFAND-P000289

Let M — a factor; then $\frac{X+X^*}{2}, \frac{X-X^*}{2i}$ generate them.

(self-adj.) $a = \int \lambda \, de_\lambda$, $e_\lambda \in$ the factor, and generate the factor.

A factor is generated by projections with $e^2 = e$, $e^* = e$.

GELFAND-P000290

[source number: 56]

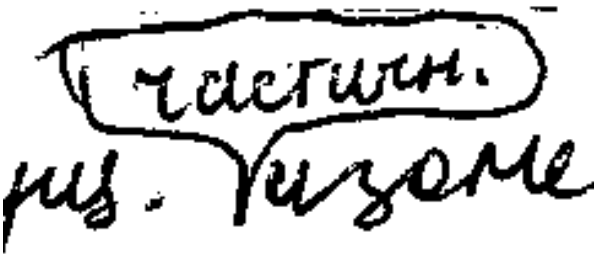
56

partial

If [crossed out: illegible] there exists an isometry from M, joining the subsp. corresponding to the projections, then the 2 opers. are called equivalent.
(subsp.)

Original source details: GELFAND-P000290

Uncertain reading. After “If” and before “there exists”, tall signs are densely crossed by diagonals; the underlying signs have not been separated or read.



Original source details: GELFAND-P000290

Manuscript notation. “Partial” is inserted above “isometry”.

If equiv. ones are identified, then an ordering results (by inclusion).

Equiv. classes of projection opers. can be added.

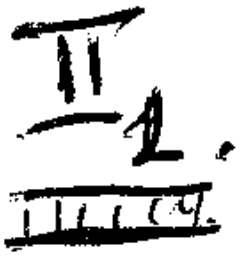
Hence subsp. can be measured (assign numbers to a subsp.).

(No Archim. axiom — ∞ is present).

Thm. If there is a min subsp., then it is isom. to the ring of all opers. in Hilb. space.

$$R = M \otimes M' \qquad (M' \text{ — commutant factor}).$$

No min $\Longrightarrow$ either II_1, II_∞ or III.
II_1 — there is never a min, but everything is meas. by numbers in $(0, 1]$ (there is a max) and the Archim. axiom.



Original source details: GELFAND-P000290

Diagram and typed description.

II_1.

[cancelled sign: ?] $\forall$ oper. has a trace and det; the minimax theory of eigenvalues holds (everything that holds for matrices).

Uncertain reading. Two nearly vertical strokes are visible under the slanted cancellation before the legible $\forall$. The cancelled sign is uncertain.

Original source details: GELFAND-P000290

$$\mathrm{Tr}(a) = \int \lambda \, d(\dim e_\lambda) \qquad (\dim \text{ from } 0 \text{ to } 1)$$

The space itself has not yet been discovered.

Which factors of type II_1?

1) hyperfinite Murray and Neumann — there is only one

2) J. Schwartz

3) Ching

4) Sakai

5–6) Dixmier, Lance

7–8) Zeller-Meier

$7, 8'$) Stepin, Golodets

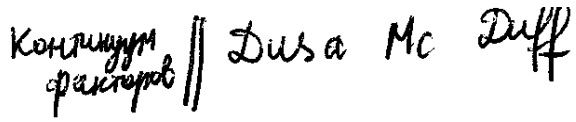


Diagram and typed description.

Continuum of factors // Dusa Mc Duff

Original source details: GELFAND-P000290

GELFAND-P000291

13/X-69 yr. Next time Minlos and Sinai.

Gelfand.

On Araki's work.

Question: Let

$$\begin{pmatrix} \alpha(t) & \beta(t) \\ -\bar{\beta}(t) & \overline{\alpha(t)} \end{pmatrix} = u(t)$$

a unit. matrix with det 1,

$$|\alpha(t)|^2 + |\beta(t)|^2 = 1, \qquad t \in [0, 1].$$

Find all irred. unit. representations of this group. This has not been done.

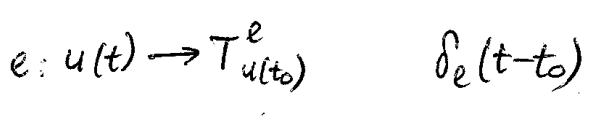
supp of repr. F — the smallest closed set for which the condition holds

$$u(t) \equiv 1 \quad \forall t \in U(F) \Longrightarrow T_{u(t)} = I$$

Problem: does an irred. unit. representation exist with supp — an interval.

supp — a point — the operator $u(t_0)$ (2-dim. repr.).
This is the repr. $\delta(t-t_0)$.

$$e:\ u(t) \longrightarrow T^e_{u(t_0)} \qquad \delta_e(t-t_0)$$

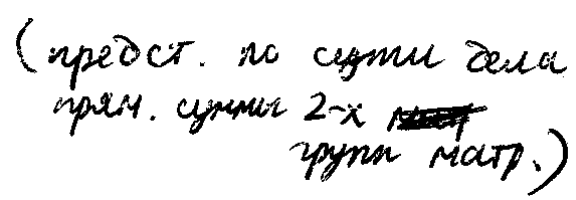


Manuscript notation. The three matching looped letters read e.

Original source details: GELFAND-P000291

$$u(t) \longrightarrow u(t_1) \otimes u(t_2)$$

an irred. repr. with supp — a pair of points.
(a repr., in effect, of a direct sum of 2 [crossed out: ?] groups of matrices).



Uncertain reading. A word after "2" is crossed out. "Groups of matrices" is written separately below.

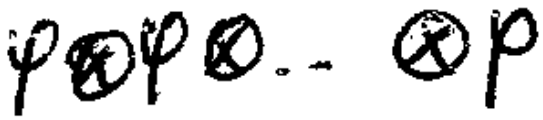
Original source details: GELFAND-P000291

One can take n points.
For a countable set this can no longer be done (it can be done for a convergent seq.).
Araki.
Idea:

$$e^K = \bigoplus_{n=0}^{\infty} S^n(K)$$

(K — Hilb. space)

$$e^{\varphi} = \bigoplus_{n=0}^{\infty} \frac{1}{\sqrt{n!}} \varphi \otimes \varphi \otimes \cdots \otimes p$$



Uncertain reading. The last factor has a right-hand loop and straight descender. Provisionally p.

Original source details: GELFAND-P000291

$\sqrt{n!}$, so that $(e^{\varphi}, e^{\psi}) = e^{(\varphi,\psi)}$.
$K \to e^K$ — a functorial passage

$$K_1 \oplus K_2 \longrightarrow e^{K_1} \otimes e^{K_2}$$

At $\forall$ point t we specify a representation in the space $K(t)$ (e.g. 2-dimensional)

$$\int_0^1 K(t)\,dt$$

GELFAND-P000292

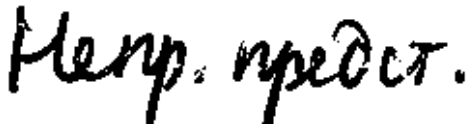
[source number: 57]

57

Consider $e^{\int_0^1 K(t)\,dt}$.
A repr. is obtained in it (from funct.) which is reducible.

———— • ————

The gr. of matrices of order 1 $\alpha(t) = e^{x(t)}$.
"Nepr." repr.



Uncertain reading. The line reads "Непр. предст." ("Nepr. repr."). The first abbreviation is unresolved.

Original source details: GELFAND-P000292

$$e^{x(t)} \longrightarrow C, \qquad e^{x_1(t)}e^{x_2(t)} \longrightarrow C_1C_2, \qquad C = e^a$$
$$x(t) \longrightarrow a$$

a — a lin. funct. of $X(t)$ — a generalized func. or distribution.

Uncertain reading. The letter before (t) is taller than the nearby x signs. Provisionally $X(t)$.

Original source details: GELFAND-P000292

$$\delta(t - t_0): \qquad e^{x(t)} \longrightarrow e^{x(t_0)}$$

$$x(t) \longrightarrow e^{\int_0^1 a(t)x(t)\,dt},$$

On the circle an integrality condition is added.

—— • ——

A repr. here is a generalization of a distribution.

GELFAND-P000293

[source number: 60]

17/XI-69 yr. Hewitt.
One result (sharp for all comp. groups)
as for the circle.
[crossed out: illegible]

Uncertain reading. Dense crossing strokes precede the heading “On s-numbers”; the underlying letters or signs have not been read.

Original source details: GELFAND-P000293

On s-numbers:

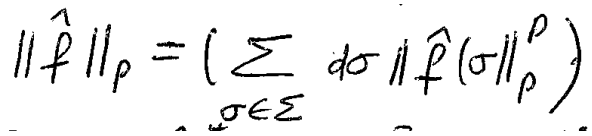
Gantmakher;
Wielandt
Berezin, Gelfand Trudy M. M. O. —
— ∀ semisimple Lie groups.
from repr. theory for a compl. func.

H. Weyl and Ky Fan.
[crossed out: illegible] All this is for an arbitrary group and a class of representations:

Uncertain reading. Before “All this”, joined vertical and slanting strokes are cancelled; the underlying sequence is undetermined.

Original source details: GELFAND-P000293

$$\|\hat{f}\|_p = (\sum_{\sigma\in\Sigma} d\sigma \|\hat{f}(\sigma\|_p^p)$$



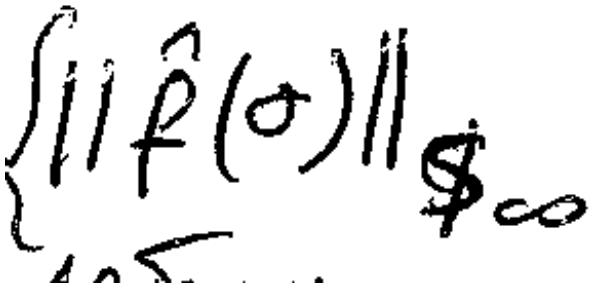
Manuscript notation. The σ next to d sits on the line. The argument σ is followed immediately by the norm bars. No exponent is legible after the outer parenthesis.

Original source details: GELFAND-P000293

Peter–Weyl–Plancherel thm.

$$\|\hat{f}\|_2^2 = \int_G |f|^2 \, d\lambda;$$

$$\|\hat{f}\|_\infty = \sup\{\|\hat{f}(\sigma)\|_{\mathfrak{S}_\infty} : \ \sigma \in \Sigma\}.$$



Uncertain reading. A vertical stroke crosses the S-shaped sign before ∞. Provisionally $\mathfrak{S}$.

Original source details: GELFAND-P000293

Riem.–Lebesgue lemma:

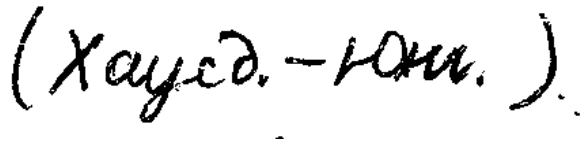
$$\forall f \in L_1(G), \quad \forall \varepsilon > 0 \quad \{\sigma : \ \|\hat{f}(\sigma)\|_{S_\infty} > \varepsilon\}$$

— a finite set.

Kunze's thm. $1 < p < 2$

$$\|\hat{f}\|_{p'} \le \|f\|_p.$$

(Hausd.–Youn.).



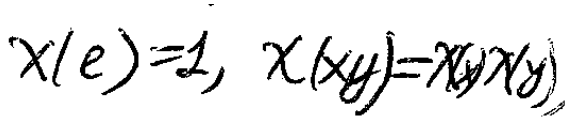
Uncertain reading. The abbreviation after "Хаусд.-" provisionally reads "Юн.". The final strokes are close together.

Original source details: GELFAND-P000293

Thm. of Hardy–Littlewood — equality only for $f(x) = \alpha e^{inx}$;

For a general group things are not so simple: there, for funcs. of the form

χ — a cont. func. on an open subgroup G_0, $\chi(e) = 1$, $\chi(xy) = \chi(x)\chi(y)$,



Uncertain reading. The first argument on the right has been retraced. Provisionally x.

Original source details: GELFAND-P000293

$\chi = 0$ outside G_0.

There are analogues of the Riesz–Thorin theorems.

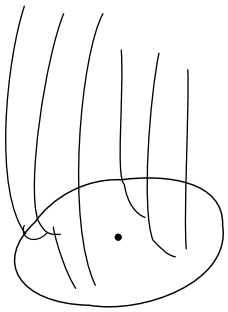
GELFAND-P000294

23.2.70 yr.

Kazhdan on Griffiths's work.

Original source details: GELFAND-P000294

Uncertain reading. On the left is an oval contour with an interior point and six long curves above it. A small loop and nearly parallel lower strokes occur at the left edge. The short-stroke connections and curve continuations are undetermined; there are no labels or definite arrows.

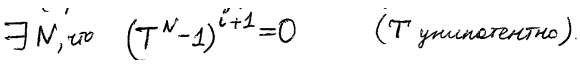


$$T:\ H^i(V_s) \longrightarrow H^i(V_s)$$

(on going around a puncture)

$$\exists N, \text{ such that } \quad (T^N - 1)^{i+1} = 0$$

(T is unipotent).



Original source details: GELFAND-P000294

Manuscript notation. To the right of $(T^N - 1)^{i+1} = 0$ the manuscript separately says "(T is unipotent)".

Original source details: GELFAND-P000294

Diagram and typed description.

$$\begin{array}{c} W \\ \downarrow \\ S \end{array}$$

an algebraic bundle

base — a 1-dim. alg. variety (a Riem. surface with punctures).

$$\pi_1(S):\ H^K(V_s) \longrightarrow H^K(V_s) \qquad (\text{coeffs. } \mathbb{C}).$$

and defines a vect. bundle [crossed out: an illegible sign] (analytic)

Uncertain reading. A narrow cancelled sign occurs between “bundle” and “(analytic)”; the underlying letter is undetermined.

Original source details: GELFAND-P000294

Diagram and typed description.

$$\begin{array}{c} E \\ \downarrow \\ S \end{array}$$

Original source details: GELFAND-P000294

$O(E)$ — the sheaf of germs of analyt. sects.

There is a canon. connection from parallel transports on the universal cover $\tilde{S}$. (There it is a direct product.)

We shall study it near a puncture.

Write the connection in the form of an eqn.

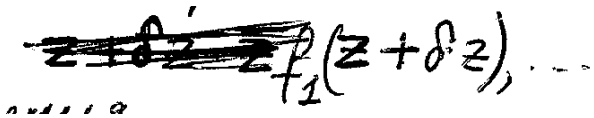
$$\begin{array}{l} f_1(z) \text{ — sections} \\ \dots \\ f_n(z) \end{array} \qquad \text{(a basis near } z\text{)}.$$

Picard–Fuchs eqn. $\dfrac{\partial}{\partial z} = A(z)$

Manuscript notation. This line reads $\partial/\partial z = A(z)$. The later equation separately includes $f(z)$.

Original source details: GELFAND-P000294

Transport $f_1(z)$ horizontally to $z + \delta z$ and expand in the basis [crossed out: $z + \delta z$, followed by illegible signs] $f_1(z + \delta z), \dots$



Uncertain reading. After “basis”, the cancelled sequence $z + \delta z$ has an illegible ending; there may be another z on the right. The uncancelled $f_1(z + \delta z), \dots$ is read separately.

Original source details: GELFAND-P000294

Consider any flat sect. of the bundle. it satisfies the eqn.

$$\frac{\partial}{\partial z} f(z) = A(z) f(z)$$

and all flat sects. are solutions of this eqn.

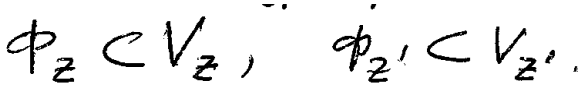
Example of flat sections.

Let Φ_z — a cycle in the fibre (K-dimensional).

Original source details: GELFAND-P000294

Uncertain reading. After "Let" a loop overlaps a heavy vertical stroke. Provisionally Φ_z.

$$\phi_z \subset V_z, \qquad \phi_{z'} \subset V_{z'}.$$



Original source details: GELFAND-P000294

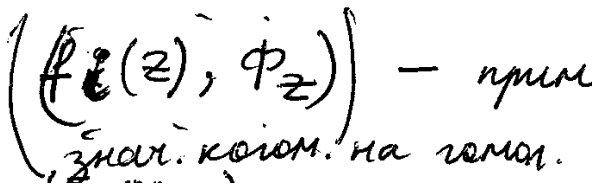
Manuscript notation. Both letters on the lower line have a thin loop.

GELFAND-P000295

[source number: 62]

62

— an example of a flat section. —



Original source details: GELFAND-P000295

Uncertain reading. The index on f is heavily retraced. Provisionally a subscript i. The outer parentheses descend toward the explanation.

$$\Big((f_i(z), \phi_z) \quad \text{value of cohom. on hom.} \Big)$$

— a vector func.

Thm. (Griffiths)

$\exists$ a basis $f_i(z)$, such that the eqn. has a reg. singular point,

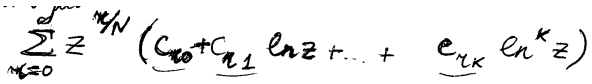
i.e. $zA(z)$ extends holomorphically to 0. (regular).

(Allows one to infer the behaviour of periods on approaching a singular point).

Corollary

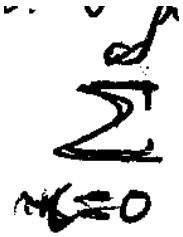
Consider any flat section — a sol. of the eqn. Then it expands in a series

$$\sum_{n\le 0} z^{n/N}\left(\underline{C_{n0}} + \underline{C_{n1}}\ln z + \ldots + \underline{c_{nk}}\ln^k z\right)$$



Original source details: GELFAND-P000295

Uncertain reading. The lower limit is provisionally $n \le 0$. The final initial resembles lowercase c or e. Three coefficients are underlined.

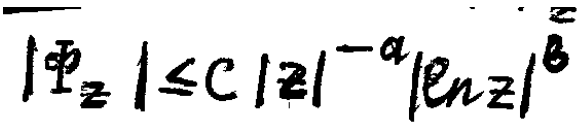


Original source details: GELFAND-P000295

Uncertain reading. A loop above the sum merges with the descender in “ряд” (“series”). The upper limit is unresolved.

Lemma $\forall$ flat sect. ϕ_z

$$|\Phi_z| \le C|z|^{-a}|\ln z|^b$$



Original source details: GELFAND-P000295

Uncertain reading. The exponent on $|\ln z|$ has two close loops. Provisionally b. The bound uses a larger Φ_z.

Original source details: GELFAND-P000295

Diagram and typed description.

. . . . . .

GELFAND-P000296

7/IX-1970 yr. Seminar of Gelfand.

Seminar programme.

1. Kazhdan on the relation between formal and algebraic geometry.
2. Arnold: Griffiths’s works.
3. Mityagin and Henkin on their works.

[cancelled number: 4.?] $1'$. Moishezon on Artin’s works.

Original source details: GELFAND-P000296

Uncertain reading. Before the separate $1'$ is a cancelled number, provisionally 4. A separate “?.” appears below.

?. K-theory of finite groups

Brown, Tate, Tomson, Swan, Katz.

(Kazhdan, Bernstein, etc.)

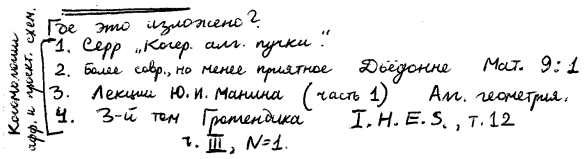
4. Hironaka.
For next semester.

ζ-func. (p-adic harm. analysis).

Finite groups.

Kazhdan.

Where is this expounded?



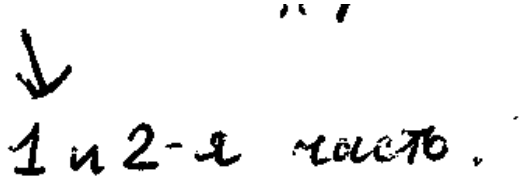
Original source details: GELFAND-P000296

Diagram and typed description.

Cohomology of aff. and proj. schemes.

1. Serre, "Coherent alg. sheaves".
2. More modern, but less pleasant Dieudonné Mat. 9:1
3. Lectures of Yu. I. Manin (part 1) Alg. geometry.
4. 3rd volume of Grothendieck I. H. E. S., vol. 12 part III, $N = 1$.

Gelfand's remark: learn Serre's books "Finite groups" and "Lie gr. and Lie alg.".
(Part of the minimum in lin. algebra)



Original source details: GELFAND-P000296

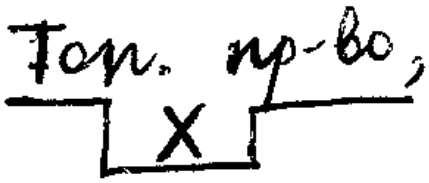
Diagram and typed description.

1 and 2nd part.

Part 1: Cohomology of coherent sheaves.

§1. Definition of a scheme.

A scheme — a ringed space of a special kind



Original source details: GELFAND-P000296

Diagram and typed description. Top. space, X equipped with a sheaf of rings. O (structure sheaf).

Affine scheme. A — a comm. ring with 1.

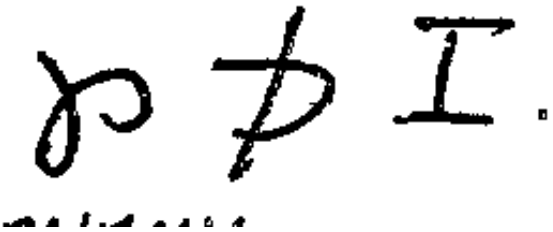
Spec A Points — prime ideals $\wp \subset A$.

Original source details: GELFAND-P000296

Uncertain reading. The looped ideal sign admits $\wp$ / $\mathfrak{p}$ / p. Both occurrences show the nearest available glyph $\wp$; the letter's identity is undetermined.

Topology: $\forall \underset{\text{ideal}}{I} \subset A$ assign the set of prime ideals $U_I \subset \operatorname{Spec} A$,
consisting of prime ideals

$$\wp \not\supset I.$$



Original source details: GELFAND-P000296

Uncertain reading. A looped ideal glyph and a crossed inclusion sign open to the left. No lower equality bar is discernible.

The sets U_I are called open.

GELFAND-P000297

[source number: 67]

67

Problem: check that a topology results.
A basis of open sets:

$$f \in A \rightsquigarrow U_f \subset \operatorname{Spec} A \quad \text{from} \quad \wp \not\ni f.$$

Original source details: GELFAND-P000297

Uncertain reading. The looped ideal sign admits $\wp$ / $\mathfrak{p}$ / p; the candidate $\wp$ is shown. The following sign has a rounded outline, a middle crossbar and a crossing stroke; $\not\ni$ is shown.

Problem: U_f — a basis.
Problem: the weakest separation ax. holds (of 2 points, 1 can be separated by a nbhd.)

Localization.

Let S — a multipl. set in A, not containing 0 (closed under mult.)

The ring A_S consists of pairs $\frac{a}{s}$:

$$\frac{a}{s} = \frac{a'}{s'} \iff \exists s'' \in S \qquad s''(s'a - sa') = 0.$$

Problem: this is an equivalence rel.

Later this will be needed for modules.

Struct. sheaf on Spec A:

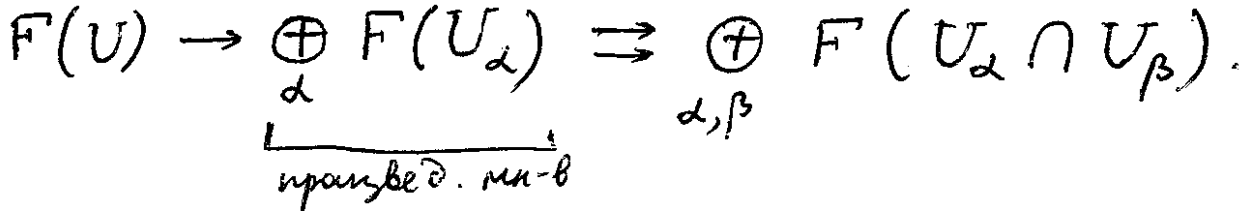
$$\Gamma(U_f, O) = A_{\substack{\{f^n\},\\ n\geq 0}}, \qquad S = \{f^n\}.$$

Reminder.

Def. of a sheaf.

A presheaf and $\forall U \subset X$ and a cover $\{U_\alpha^*\}$ of the set U the sequence is exact.

$$\{U_\alpha^*\} \subset U \qquad \left(\bigcup U_\alpha^* = U\right).$$



Original source details: GELFAND-P000297

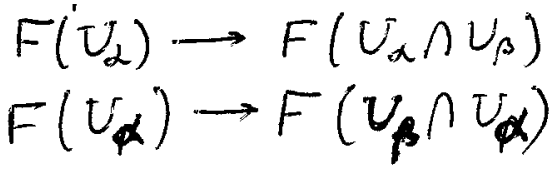
Diagram and typed description. Both signs are circled crosses. Below the first expression the note reads “product of sets”.

$$F(U) \longrightarrow \bigoplus_{\alpha} F(U_\alpha) \rightrightarrows \bigoplus_{\alpha,\beta} F(U_\alpha \cap U_\beta).$$

product of sets

$$F(U_\alpha) \longrightarrow F(U_\alpha \cap U_\beta)$$

$$F(U_\alpha) \longrightarrow F(U_\beta \cap U_\alpha)$$



Uncertain reading. The indices on the second line are overwritten. Provisionally α on the left and β, α on the right. The underlying strokes cannot be separated.

Original source details: GELFAND-P000297

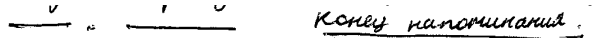
Question: why are these different arrows?

Exactness:

$$\operatorname{Im} i = \{\cdot \mid \varphi_1 = \varphi_2\}.$$

The stalk of a sheaf (presheaf) at pt. x etc.

Problem: prove that ∀ presheaf ∃ a sheaf with the same stalks (and unique)
Remark: to define a sheaf, it suffices to know the presheaf on a basis.

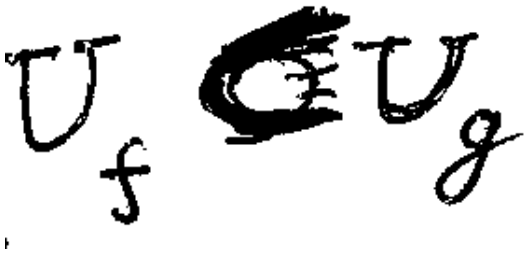


Original source details: GELFAND-P000297

Diagram and typed description.

End of reminder.

$$U_f \subset U_g \iff \exists n: \quad f^m = gh.$$



Uncertain reading. The sign between U_f and U_g is heavily overwritten; the outer contour $\subset$ is shown. The original sign is undetermined.

Original source details: GELFAND-P000297

Uncertain reading. After ∃ the letter reads n. The exponent on f provisionally reads m, with the final arches joined.

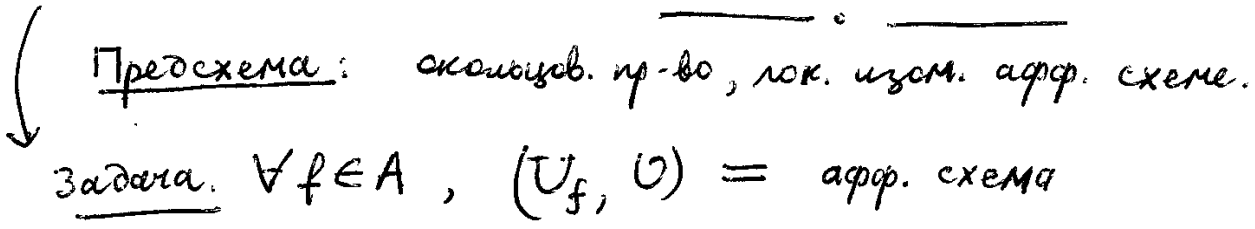
Original source details: GELFAND-P000297

(Equiv. to the thm. that ⋂ of all prime ideals is the radical)

GELFAND-P000298

$$A_{\{g^n\}} \longrightarrow A_{\{f^n\}}.$$

$$\frac{a}{g} \longrightarrow \frac{ah}{f^m}, \qquad \frac{a}{g^k} \longrightarrow \frac{ah^k}{f^{mk}}.$$



Original source details: GELFAND-P000298

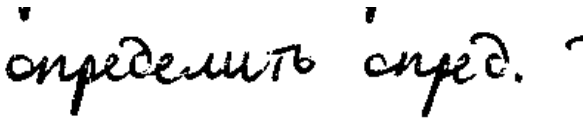
Diagram and typed description.

Prescheme: a ringed space, loc. isom. to an aff. scheme.
Problem. $\forall f \in A$, $(U_f, O) =$ an aff. scheme

Product of preschemes. Spec A, Spec B

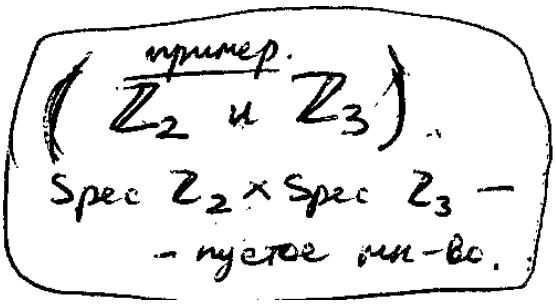
$$\text{Spec}\, A \times \text{Spec}\, B = \text{Spec}(A \otimes B)$$

Problem: determine def. of two preschemes.



Original source details: GELFAND-P000298

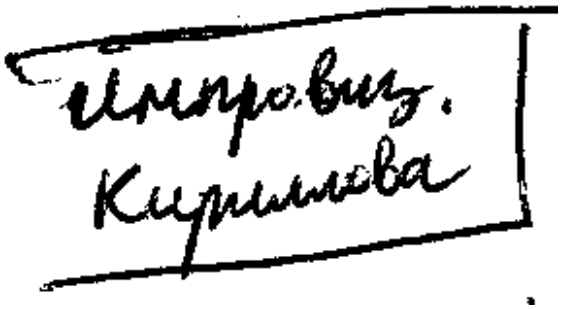
Manuscript notation. The manuscript reads “определить опред.” (“determine def.”).



Original source details: GELFAND-P000298

Diagram and typed description.

example.
(Z_2 and Z_3).
Spec Z_2 × Spec Z_3 —
— the empty set.



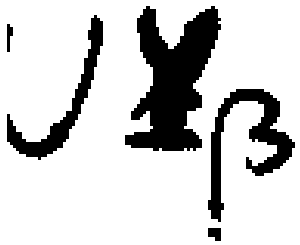
Original source details: GELFAND-P000298

Diagram and typed description.

Improvis.
of Kirillov

$$\begin{array}{cc} X, & Y \\ \| & \| \\ \bigcup X_\alpha & \bigcup Y_\beta \end{array}$$

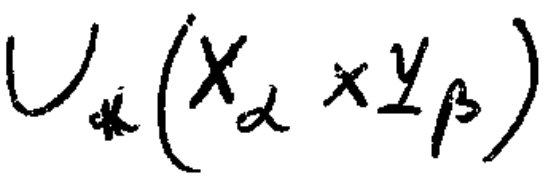
— preschemes



Original source details: GELFAND-P000298

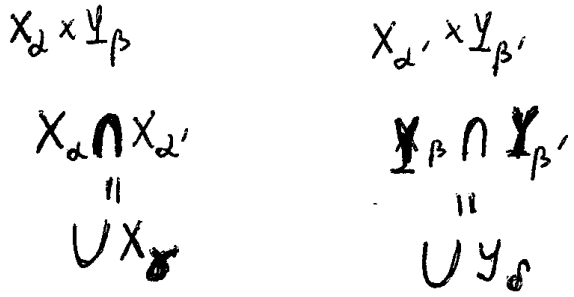
Uncertain reading. The Y above β is overwritten. The overwritten Y's in the right-hand intersection appear in the figure below.

in $\bigcup_{*}(X_\alpha \times Y_\beta)$



Original source details: GELFAND-P000298

Uncertain reading. The dark index under the union has radiating strokes, provisionally an asterisk.



Original source details: GELFAND-P000298

Diagram and typed description.

$$\begin{array}{cc} X_\alpha \times Y_\beta & X_{\alpha'} \times Y_{\beta'} \\ X_\alpha \cap X_{\alpha'} & Y_\beta \cap Y_{\beta'} \\ \| & \| \\ \bigcup X_\gamma & \bigcup Y_\delta \end{array}$$

Uncertain reading. The index in the lower left union is heavily overwritten. Provisionally γ.

Original source details: GELFAND-P000298

Lemma: $X_\alpha \subset X_{\alpha'}, Y_\beta \subset Y_{\beta'} \Rightarrow X_\alpha \times Y_\beta \subset X_{\alpha'} \times Y_{\beta'}$.

See Serre.

A prescheme X is called a scheme if $\Delta \subset X * X$ is closed.

Manuscript notation. An asterisk stands between the two X's.

Original source details: GELFAND-P000298

Problem: prove that an aff. scheme is a scheme.

Let X — an arbitr. prescheme, $\mathcal{F}$ — a sheaf of $\mathcal{O}$-modules over [cancelled sign: x?] X.

Uncertain reading. Before X is a cancelled sign with several short radiating strokes.

Original source details: GELFAND-P000298

It is called quasicoherent if it is loc. isom.

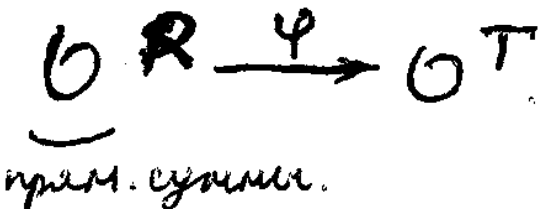
GELFAND-P000299

[source number: 68]

to the cokernel of the map:

$$\underbrace{O^R}_{\text{direct sums.}} \xrightarrow{\varphi} O^T.$$

S, T — arbitrary sets.



Uncertain reading. The first exponent resembles R, with dense strokes on the left. A separate line below reads S, T.

Original source details: GELFAND-P000299

Problem: give an example of a nonquasicoherent sheaf.

Theorem. Let A — a comm. ring and Spec A — an aff. scheme.

The map from the categ. of quasicoherent sheaves of modules

$$\mathcal{F} \longrightarrow \Gamma(\text{Spec } A, \mathcal{F})$$

to the categ. of modules is an isomorphism.

I.e. the categ. of q.-c. sheaves of mod. over the aff. scheme A is isom. to the categ. of modules over A.

(The module of glob. sections.)

(This is equivalent to saying that for aff. spaces the cohom. are zero)

(это равносильно тому, что для афф. пр-в когом. нулевые)

Original source details: GELFAND-P000299

Manuscript notation. The parenthesis is legible throughout. Degrees and coefficients are unspecified.

A quasicoherent sheaf of modules is determined by its glob. sections.

28/IX.

Direct and inv. image of sheaves.

$$X \xrightarrow{f} Y, \qquad X, Y \text{ — top. spaces,}$$

$\mathcal{F}$ — a sheaf on X,

$$\Gamma(U, f_*\mathcal{F}) := \Gamma(f^{-1}(U), \mathcal{F}), \qquad U \subset Y$$

a presheaf on Y results.

Exercise (easy) — this presheaf is a sheaf

Example: if Y — a point, then $f_*\mathcal{F}$ — the group of sections of the orig. sheaf,
($f^{-1}(U)$ may also be empty, then the set of sections is empty).

Original source details: GELFAND-P000299

Manuscript notation. The parenthesis reads "empty" twice.

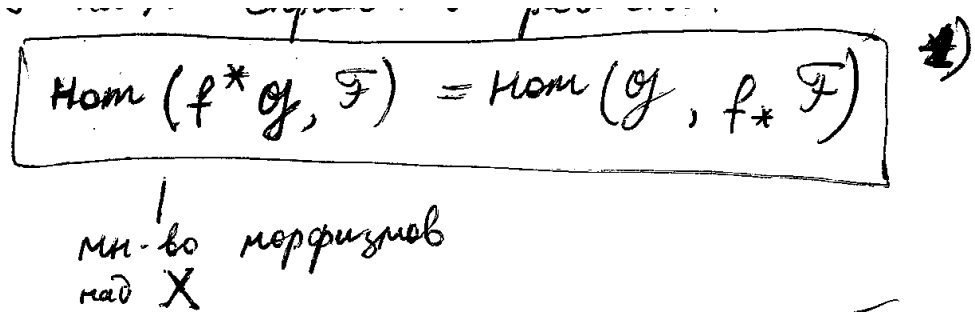
A very important concept.
Inverse image.

$$X \xrightarrow{f} Y$$

$\mathcal{G}$ — a sheaf on Y

GELFAND-P000300

The sheaf $f^*\mathcal{G}$ is defined as follows:
$\forall \mathcal{F}$ on X the equality holds:



Original source details: GELFAND-P000300

Uncertain reading. To the right of the box is a heavily overwritten number with a closing parenthesis. Provisionally 1).

$$\boxed{\mathrm{Hom}(f^*\mathcal{G}, \mathcal{F}) = \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F})}^{1)}$$

set of morphisms over X

I.e. $\forall$ sheaf $\mathcal{F}$ a one-to-one corr. between these sets is given, compatible under morphisms of $\mathcal{F}$ into one another.

(An isomorphism of functors on the cat. of sheaves).

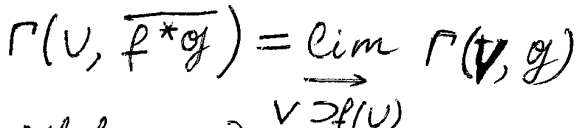
Construction, proving the existence of $f^*\mathcal{G}$

[A short expression is cancelled: ?]

Original source details: GELFAND-P000300

Uncertain reading. Beneath the diagonal strokes are an initial shaped like Γ or F and several loops. The formula is illegible.

$$\Gamma(U, \overline{f^*\mathcal{G}}) = \varinjlim_{V \supset f(U)} \Gamma(V, \mathcal{G})$$

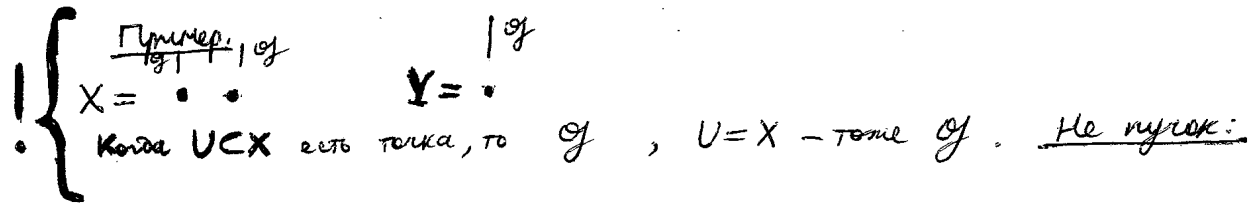


Uncertain reading. The first argument of the last Γ is heavily overwritten. The upper strokes read as V.

Original source details: GELFAND-P000300

this equality defines a presheaf $(V \subset Y)$

Exercise: the sheaf assoc. to it is the inverse image



Original source details: GELFAND-P000300

Diagram and typed description.

Example.

$X =$ • • $\mathcal{G}$ $\mathcal{G}$

$Y =$ • $\mathcal{G}$

When $U \subset X$ is a point, then $\mathcal{G}$, $U = X$ — also $\mathcal{G}$. Not a sheaf.

---

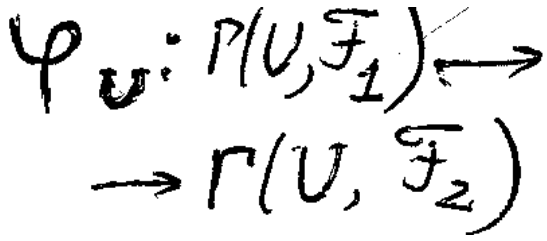
1) Definition of Hom:

A map $\mathcal{F}_1 \longrightarrow \mathcal{F}_2$ over X is called [cancelled word: ?] a collection of maps

Uncertain reading. After "is called" and before "a collection", a short word is cancelled; a lower loop and right-hand vertical strokes are visible, but no reliable letters are established.

Original source details: GELFAND-P000300

$$\varphi_U : \Gamma(U, \mathcal{F}_1) \longleftrightarrow \longrightarrow \Gamma(U, \mathcal{F}_2)$$
$$\forall U \subset X.$$



Uncertain reading. A long arrow stands at the right of the first line. The short slanted strokes at its left end may be a second head. A separate arrow points right on the following line.

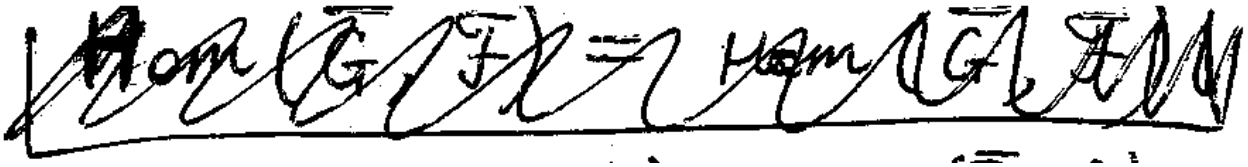
Original source details: GELFAND-P000300

and commute with the restriction opers. (A morphism of functors)

GELFAND-P000301

[source number: 69]

69



Original source details: GELFAND-P000301

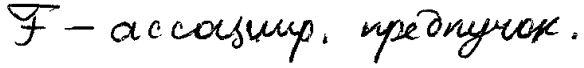
Uncertain reading. Separate bars are visible above both G signs. The upper strokes above $\mathcal{F}$ merge with the letter loops and cancellation. The bar above the second $\mathcal{F}$ is provisional.

~~$\mathrm{Hom}(\overline{G}, \mathcal{F}) = \mathrm{Hom}(\overline{G}, \mathcal{F})$~~

— def. of the sheaf corr. to a presheaf

$$\mathrm{Hom}(\mathcal{F}, G) = \mathrm{Hom}(\overline{\mathcal{F}}, G)$$

$\overline{\mathcal{F}}$ — a presheaf, $\mathcal{F}$ — the assoc. presheaf.



Manuscript notation. The entry on the right reads “the assoc. presheaf.”

Original source details: GELFAND-P000301

$\overline{\mathcal{F}} \longrightarrow \mathcal{F}$ — a map (of presheaves) must be given

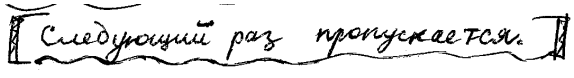


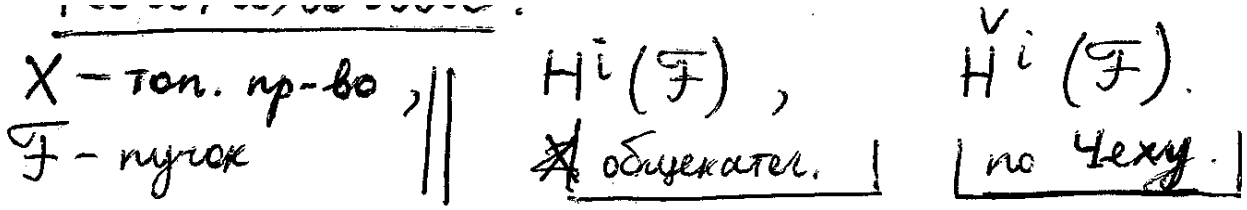
Diagram and typed description.

Next time is skipped.

Original source details: GELFAND-P000301

Reminder, § 2.

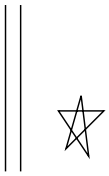
Cohomology.



Original source details: GELFAND-P000301

Uncertain reading. Two vertical rules stand at the left. The labels sit above the lower strokes of the frames.

X — a top. space, $\mathcal{F}$ — a sheaf



$H^i(\mathcal{F})$, general-categ.

$\check{H}^i(\mathcal{F})$. Čech.

Uncertain reading. Before “general-categ.” is a mark with a crossbar and a long right stroke. Whether it is a letter or an asterisk is unresolved.

Original source details: GELFAND-P000301

$H^i(\mathcal{F})$ — a deriv. functor of the sections functor.

Consider the category C of sheaves of ab. groups over the space X

Lemma 1. — this is an abelian category.

Inject. sheaves are important.

Lemma 2. There are enough inject. objects (any sheaf embeds in an inject. sheaf).

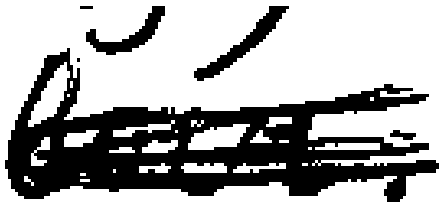
Functor to a sheaf $\longrightarrow$ sections over all of X.

$$0 \longrightarrow \Gamma(\mathcal{F}') \longrightarrow \Gamma(\mathcal{F}) \longrightarrow \Gamma(\mathcal{F}'')$$

Γ — a semiexact functor (on the left), and since there are enough inject. objects, the derived functors are defined.
Let X — a top. space $\{U\}$ — a basis of the topology in X, such that

$$\check{H}^i(\mathcal{F}, \{u_i\}) = 0, \qquad i > 0.$$

Then the actual cohom. [cancelled: calc.?] agree with the Čech calc.



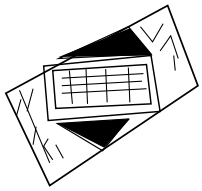
Uncertain reading. After “actual” the first stroke is a tall loop. The cancelled reading is provisionally “выч.” (“calc.”).

Original source details: GELFAND-P000301

And one can calc. them in acycl. covers.

Original source details: GELFAND-P000301

Uncertain reading. Slanted quadrilateral outlines, inner frames, dark triangular regions and short edge strokes. The reconstruction follows the visible contours; labels, exact internal-stroke junctions and mathematical purpose are undetermined.



GELFAND-P000302

An ellipt. syst. of order 2 — on a manif. without lower-order terms. Does it have solutions?

For the Laplace oper. it is known that there are none

The number of solns. $(\text{mod } 2)$.

In principle max is topology

(The Dirichlet prob. has a 0-th solution)

$\longleftarrow \cdot$

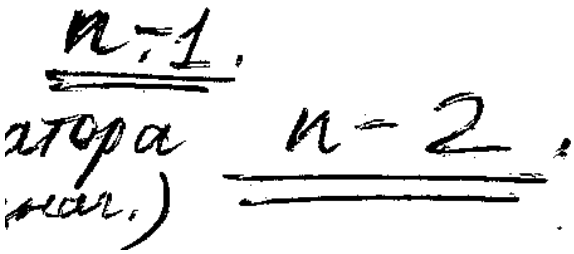
Kazhdan on one specific work.

G — a compl. semis. group $(A_n$ or $D_n)$.

A reg. elem. — if the centralizer has rank $n-1$.

Subreg. — if dim of the centralizer is $n-2$,

(two coincident eigenvals.)



Uncertain reading. $n-1$ and $n-2$ are double underlined. A separate short stroke lies below the first minus.

Original source details: GELFAND-P000302

Consider

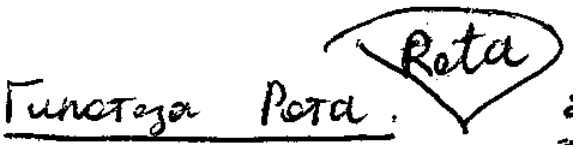
$$G \xrightarrow{\varphi} H/W \qquad \begin{array}{l}(W \text{ — the Weyl gr.}\\ H \text{ — a Cartan subgroup})\end{array}$$

An elem. [cancelled: sub / not?] is reg. $\Longleftrightarrow$ the map is reg. there.

Original source details: GELFAND-P000302

Uncertain reading. Heavily cancelled loops precede “reg.” The readings “sub” and “not” are unresolved.

12/X.



Original source details: GELFAND-P000302

Diagram and typed description.

Rota

Rota’s conjecture.

Fix a finite field F_q.

Let there be given ℓ and $k > \ell$. Then $\exists N$ [above the line: such?], that if one partitions the ℓ-dimensional subspaces in an N-dimensional subspace,

Original source details: GELFAND-P000302

Uncertain reading. Above “that” are the first letters “та” and overlapping strokes. Provisionally “такое” (“such”).

[cancelled: that?]

Original source details: GELFAND-P000302

Uncertain reading. Before “then” a short word is heavily struck through. The last letter “о” is visible.

then $\exists$ k-dimensional one, such that all subspaces of this k-dimensional one fall into one of the classes.

$$F^N \supset E \qquad \dim E = k$$

and all its ℓ-dim. subsp. are in one class.

For $\ell = 1$ it is proved.

One can use proj. terms (there a line is a point).

If P^N is partitioned into 2 classes, there is a line in one class.

GELFAND-P000303

[source number: 70]

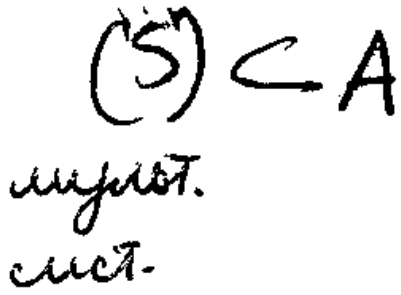
70

Kazhdan.

A quasicoherent sheaf — a sheaf loc. isom. to the cokernel of a map of free sheaves.

Example. A — a ring, M — a module, $\mathcal{F}_M$ — a sheaf of modules on Spec A.

$$\underset{\substack{\text{multipl.}\\ \text{syst.}}}{(\dot{S})} \subset A \qquad M_S \qquad \frac{m}{s} = \frac{m'}{s'} \quad \text{if } s''(s'm - sm') = 0.$$



Uncertain reading. A short separate stroke stands above S. Two other strokes touch the tops of the parentheses.

Original source details: GELFAND-P000303

$$M_f = M_{(f^n)}.$$

Sect. on U_f are M_f.

$$\Gamma(U_f, \mathcal{F}_M) = M_f.$$

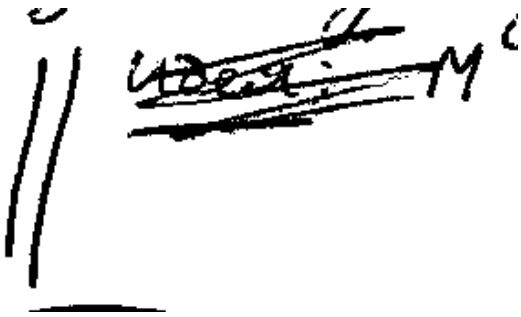
The check is left to the listeners.

Lemma: this is a sheaf.

Another def. of a quasicoherent sheaf.

Def. 2. Quasicoherent sheaves — sheaves loc. isom. to a sheaf of the form $\mathcal{F}_M$.

Problem: prove equiv. of the definitions.



Uncertain reading. Beyond two slanted rules a short word is cancelled, provisionally “idea:”. It is followed by M.

// [cancelled: idea:?] M

Original source details: GELFAND-P000303

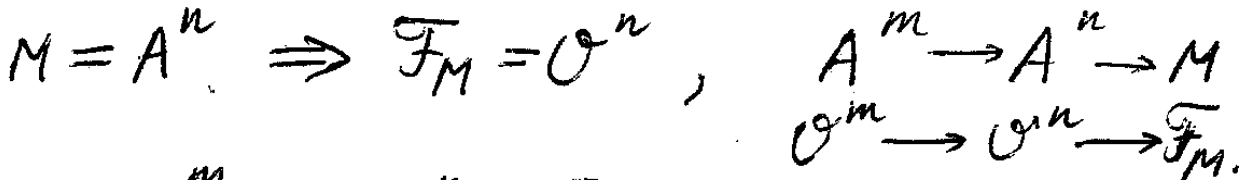
Main theorem.

$\forall$ quasicoherent sheaf $\mathcal{F}$ on an aff. scheme (Spec A) has the form $\mathcal{F}_M$, where M is an A-module.

Manuscript notation. Spec A is inserted above "aff. scheme".

Original source details: GELFAND-P000303

Problem (sol.)



Original source details: GELFAND-P000303

Diagram and typed description.

$$M = A^n \Rightarrow \mathcal{F}_M = \mathcal{O}^n, \qquad \begin{array}{l} A^m \longrightarrow A^n \longrightarrow M \\ \mathcal{O}^m \longrightarrow \mathcal{O}^n \longrightarrow \mathcal{F}_M. \end{array}$$

In the reverse direction:

$$\mathcal{O}^m \longrightarrow \mathcal{O}^n \longrightarrow \mathcal{F} \quad \text{loc.}$$

[overwritten: and?] then sect. $A^m \longrightarrow A^n \longrightarrow M$.

Uncertain reading. A densely inked, looped sign precedes "then". Provisionally "and".

Original source details: GELFAND-P000303

Proof of the theorem.

$$\mathcal{F} \qquad M = \Gamma(X, \mathcal{F}).$$

$$\forall f \in A \qquad \Gamma(U_f, \mathcal{F}) = M_f.$$

$\forall$ aff. scheme is quasicompact (no separation). From $\forall$ open cover select a finite one.

$$\mathcal{F}|_{U_{f_i}} = \mathcal{F}_{M_i}, \qquad M_i \quad A_{f_i}\text{-module.}$$

There is a nat. map

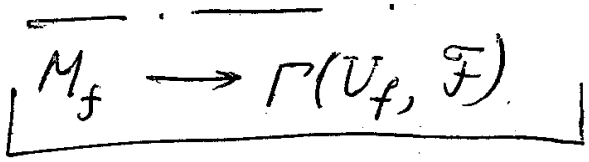


Diagram and typed description.

$M_f \longrightarrow \Gamma(U_f, \mathcal{F})$.

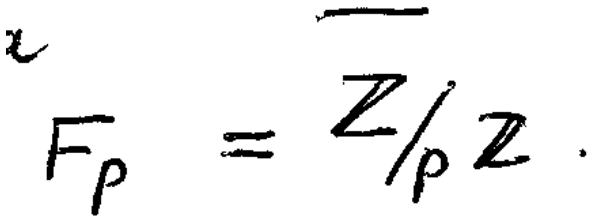
Original source details: GELFAND-P000304

$1/f \in \Gamma(U_f, \mathcal{O}_X)$, and $\mathcal{F}$ — a sheaf of modules.

Thompson as expounded by Gelfand I. M.

(G, M) — a pair.

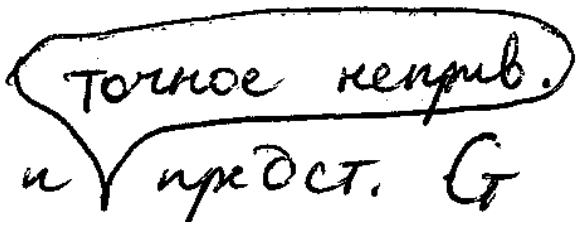
G — a group, $F_p = \overline{Z}/pZ$.



Uncertain reading. A separate horizontal stroke stands above the first Z.

Original source details: GELFAND-P000304

M — a module over F_p, and faithful irred. a repr. of G in M.



Manuscript notation. Insertion “faithful irred.” before “repr.”.

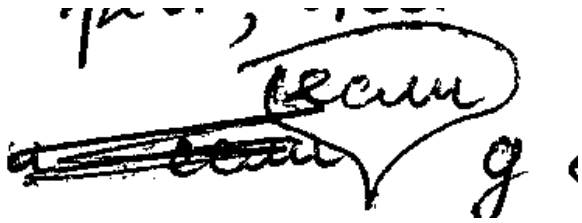
Original source details: GELFAND-P000304

Require that [cancelled sign: M?] G be generated by a collection $\langle Q \rangle$, of such elems.:

Uncertain reading. Before the clean G, a tall sign with several nearly vertical strokes is cancelled, possibly M; its letter is undetermined.

Original source details: GELFAND-P000304

~~if~~ if $g \in Q$, then $M(g-1)^2 = 0$.

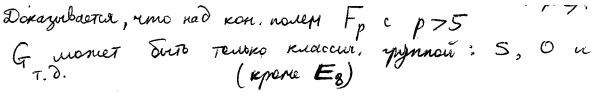


Original source details: GELFAND-P000304

Manuscript notation. "If" is inserted after the cancelled entry.

(Example: a matrix group generated by elems. with Jordan blocks of order 2).

It is proved that over the finite field F_p with $p > 5$ G can only be a class. group: S, O etc. (except E_8).



Original source details: GELFAND-P000304

Uncertain reading. The manuscript reads "$p > 5$", "class. group", and separately "(except E_8)".

Characteristics 2 and 3 remain.

Kirillov from Thompson's account.

Another finite simp. group (Conway) —

— belonging to no series.

The automorphism gr.

[margin: в]

[Cancelled: $R^{24} \supset Z^{24}$?; then an illegible cancelled word.]

Original source details: GELFAND-P000304

Uncertain reading. The cancelled line has R^{24}, an inclusion sign and Z^{24} / L^{24}, then an illegible word; the candidate Z^{24} is shown. Later loops prevent separating the original letter.

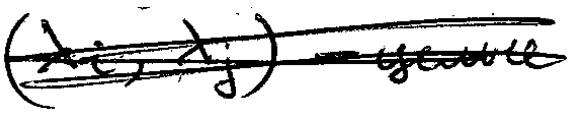
$$\mathbb{R}^{24} \supset \underset{\text{lattice}}{L}.$$

Original source details: GELFAND-P000304

Manuscript notation. The note below L reads "lattice".

1) unimodular — the fund. domain has unit volume.

2) ~~(λ_i, λ_j) — integers.~~



Original source details: GELFAND-P000304

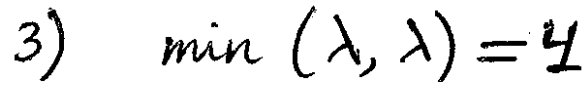
Uncertain reading. The first subscript provisionally reads i, with a loop below the cancellation. The second reads j.

(λ, μ) is always an integer, and (λ, λ) is even.

GELFAND-P000305

[source number: 71]

3) $\min(\lambda, \lambda) = 4$.



Original source details: GELFAND-P000305

Manuscript notation. The manuscript has $\min(\lambda, \lambda) = 4$ without a specified range.

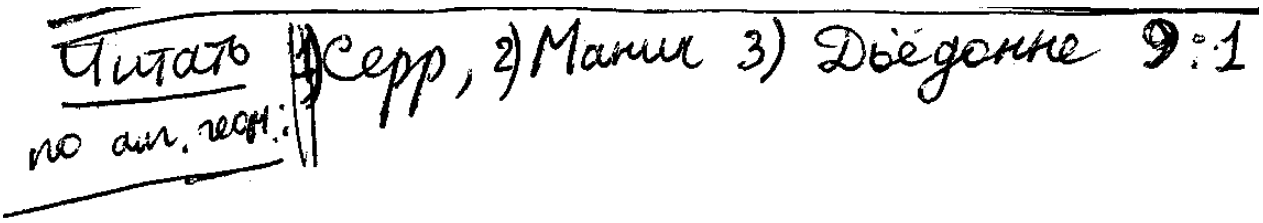
[Cancelled: 4)?]

Original source details: GELFAND-P000305

Uncertain reading. Before "Such a lattice", a short sign with a parenthesis is cancelled, possibly 4); its original and cancelling strokes cannot be separated.

Such a lattice is uniquely det. up to autom.

The signature is div. by 8 for pos. def. forms.



Original source details: GELFAND-P000305

Diagram and typed description.

To read on alg. geom. ‖ 1) Serre, 2) Manin 3) Dieudonné 9:1

Manuscript notation. After "Dieudonné" the entry reads 9:1.

Original source details: GELFAND-P000305

19/X. I. M. Gelfand.

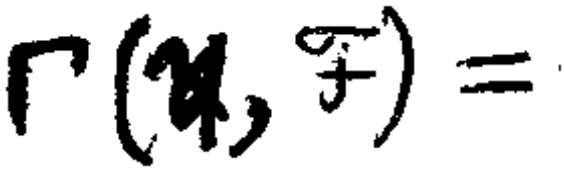
Let a sheaf of modules be given on X.

A, $X = \operatorname{Spec} A$, M — a module, $\mathcal{F}_M$ — the corresp. sheaf (sheafification of the module M).

Example of a non-quasi-coh. sheaf over Spec A:

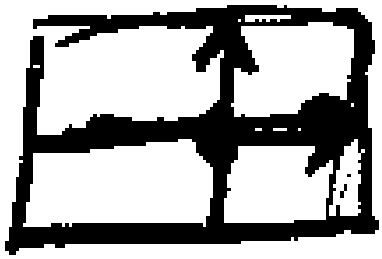
stalk A at the point x and 0 at the other points.

$$\Gamma(U,\mathcal{F}) = \begin{cases} A & \text{if } x \in U, \\ 0 & \text{if } x \notin U. \end{cases}$$

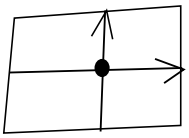


Uncertain reading. The first argument of Γ has heavily overlapping strokes. Provisionally U, from the upper strokes and the entry at right.

Original source details: GELFAND-P000305



Uncertain reading. The frame contains crossing strokes with heads at the top and right and a thickened junction. The purpose of the drawing is undetermined.



Original source details: GELFAND-P000305

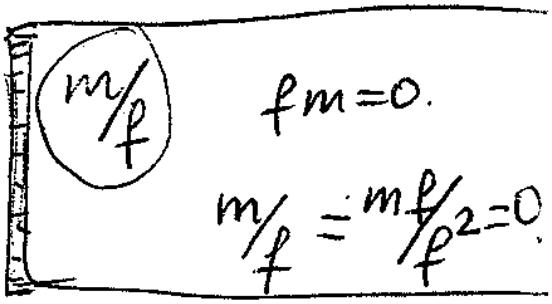
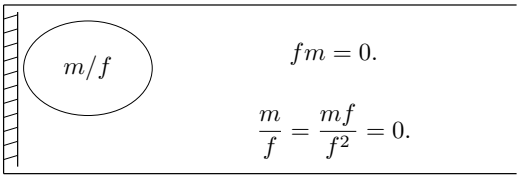


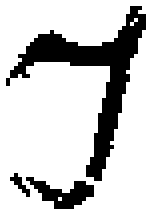
Diagram and typed description.



Original source details: GELFAND-P000305

Example (I. Bernstein).

$\mathbb{C}^n$, J — an ideal in the polynomial ring; to it there corresp. a subset, $\underset{M}{\frac{P}{Q}}$ is def. at every point of the subset.



Manuscript notation. The ideal letter hooks to the left at the foot. It reads J.

Original source details: GELFAND-P000305

Then to it there corresp. a polynomial R such that

$$\frac{P}{Q} = R \pmod{J} \quad \text{(on it)}$$

Manuscript notation. The parenthesis reads "mod J".

Original source details: GELFAND-P000305

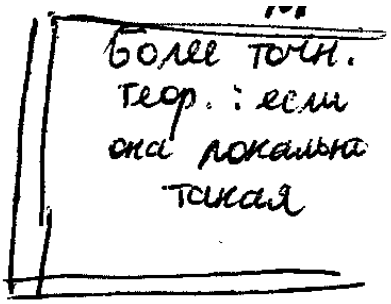


Diagram and typed description.

More prec. thm.: if it is locally such.

Original source details: GELFAND-P000305

$A = \mathbb{C}[x_1, \ldots, x_n]$, $\mathcal{O}_X$ — the struct. sheaf on $X = \operatorname{Spec} A$.

$$\operatorname{Spec} A \supset \mathbb{C}^n.$$

(not only on $\mathbb{C}^n$).

$$\underset{\text{subsheaf}}{J \subset \mathcal{O}}$$

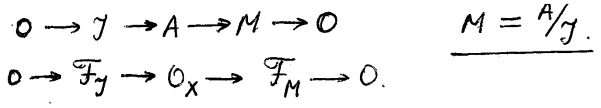
J — the ideal $(f_1, \ldots, f_n)$, a module over A.
$\mathcal{F}_J$ — the corresp. sheaf, $\mathcal{F}_J \subset \mathcal{O}_X$.

$$(J \subset A \Rightarrow \mathcal{F}_J \subset \mathcal{F}_A).$$

GELFAND-P000306

$$0 \longrightarrow J \longrightarrow A \longrightarrow M \longrightarrow 0, \qquad M = A/J.$$

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{F}_M \longrightarrow 0.$$



Manuscript notation. The letter J recurs in the exact sequences and in A/J.

Original source details: GELFAND-P000306

[Cancelled: $\mathcal{F}_M$ — rat. f-n?]

Original source details: GELFAND-P000306

Uncertain reading. After the cancelled $\mathcal{F}_M$, “рац. ф-” is legible; the final letter admits а / я / и. The candidate “ф-а?” is shown as “rat. f-n?”; the later clean “rat. f-n” does not resolve the cancelled entry.

Sections of the sheaf $\mathcal{F}_M$:

1) constr. from M

2) from the exact seq.

Let Y — the set of zeros of the ideal J.

P/Q — a rat. f-n, $Q \neq 0$ on Y.

$$P/Q \in \Gamma(X, \mathcal{F}_M) \qquad \text{from exactness of the sequence.}$$

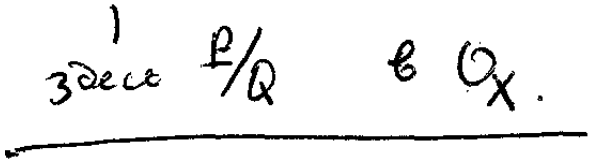
Original source details: GELFAND-P000306

Uncertain reading. The sign before Γ is heavily overwritten. Provisionally $\in$.

Consider X_Q [a small sign is cancelled] and $X \setminus Y$.

Uncertain reading. Between X_Q and “and”, a small sign with crossing strokes is cancelled; its original letter is undetermined.

Original source details: GELFAND-P000306



Original source details: GELFAND-P000306

Diagram and typed description.

here P/Q in $\mathcal{O}_X$.

Over $X \setminus Y$ consider the section $\equiv 0$.

$$0 \in \Gamma(X \setminus Y, \mathcal{F}_M).$$

$$i\left(\frac{P}{Q}\right) \in \Gamma(X_Q, \mathcal{F}_M).$$

Construct a section of the sheaf on X.

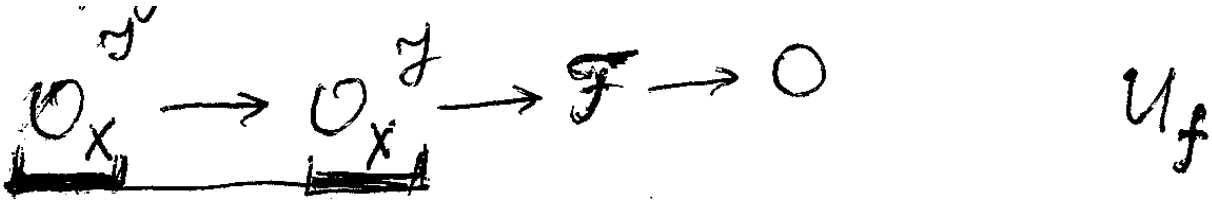
On the intersection they agree, since sections of $\mathcal{F}_M$ on $X \setminus Y$ are ident. 0, since $\mathcal{F}_M$ is supported on Y.

But! a section of this sheaf is an elem. of M — a polynomial.

[A sign is cancelled at the left: $\mathcal{O}$?]

Uncertain reading. Before the first $\mathcal{O}_X$ is a dark corrected sign with a long loop and a rising right stroke, possibly $\mathcal{O}$; the upper-right stroke cannot be separated from the cancellation.

Original source details: GELFAND-P000306

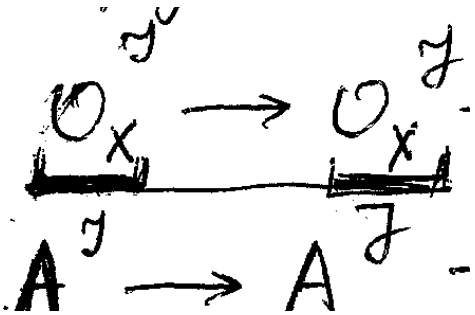


Original source details: GELFAND-P000306

Diagram and typed description.

$$\mathcal{O}_X^J \longrightarrow \mathcal{O}_X^{\mathcal{J}} \longrightarrow \mathcal{F} \longrightarrow 0 \qquad U_f$$

$$A^J \longrightarrow A^{\mathcal{J}} \longrightarrow M \longrightarrow 0.$$



Uncertain reading. In both rows the first superscript resembles a hooked J. The second has a lower loop and a cross-stroke. Provisionally J and $\mathcal{J}$.

Original source details: GELFAND-P000306

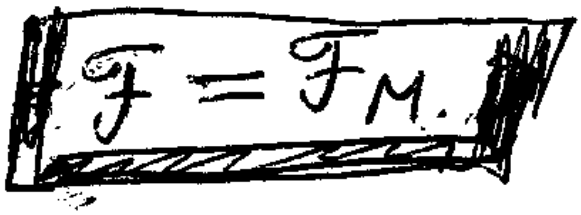
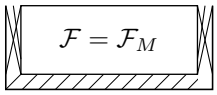


Diagram and typed description.



Original source details: GELFAND-P000306

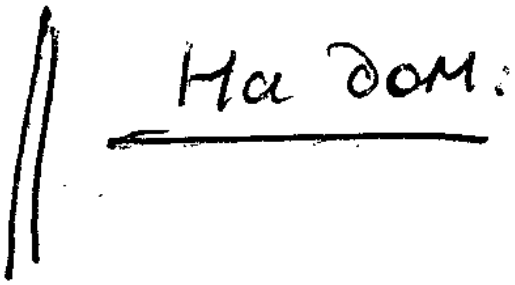


Diagram and typed description.

Homework:

Original source details: GELFAND-P000306

GELFAND-P000307

[source number: 72]

72

Ya. Sinai. Noncommut. probability theory.

H — finite-dimensional Hilbert. $\dim H = 2$ already nontriv.

$\ldots, H_{-1}, H_0, H_1, H_2, \ldots$ with a fixed isom. onto H (basis).

I — a finite subset,

$$H_I = \bigotimes_{i \in I} \text{[a sign is cancelled]}\, H_i. \qquad \dim H_I = 2^k, \quad \text{where } k = |I|.$$

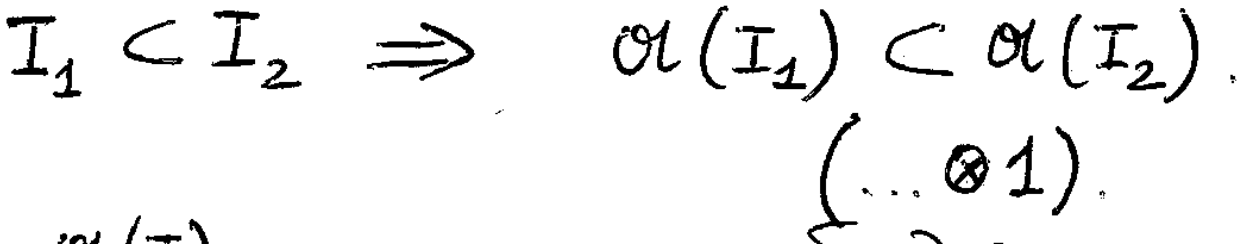
Uncertain reading. Between $\otimes$ and the clean H_i is a dark cancelled sign with several vertical strokes; an earlier H is possible, but the original entry is undetermined.

Original source details: GELFAND-P000307

$\alpha(I)$ — the algebra of matrices over H_I. [Cancelled: “Всем” (to all)?]

Uncertain reading. After the full stop following H_I, a short looped word is cancelled, resembling “Всем” / “всем” (to all); crossing strokes prevent the complete word from being established.

Original source details: GELFAND-P000307



Original source details: GELFAND-P000307

Diagram and typed description.

$I_1 \subset I_2 \implies \alpha(I_1) \subset \alpha(I_2).$

$(\ldots \otimes 1).$

Elems. of $\alpha(I)$ — local observables.

Close $\bigcup \alpha(I)$ in the uniform metric.

$$[-n, n].$$

$$\alpha([-n, n]) \subset \alpha([-n-1, n+1]).$$

closed in norm (completion) (uniform).

Obtain the algebra $\underline{\alpha}$ quasilocal observables. (with invol.).

Probability th. — Ω — a space (comp. metric), on $C(\Omega)$ positive functionals (probability distrs.).

α plays the role $C(\Omega)$.

A state φ on α [a sign is cancelled: φ?] — a positive normalized functional (probability distr.).

Uncertain reading. After α, a looped sign with a long lower stroke and several oblique strokes is cancelled; φ is possible, but original and cancelling lines cannot be separated.

Original source details: GELFAND-P000307

$$\varphi(a) \geq 0 \quad \text{when } a \geq 0; \qquad \varphi(e) = 1.$$

$\varphi(a^*)$ — real when $a^* = a$.

Uncertain reading. After a, dark crossing strokes overlap the closing parenthesis. The asterisk is provisional.

Original source details: GELFAND-P000307

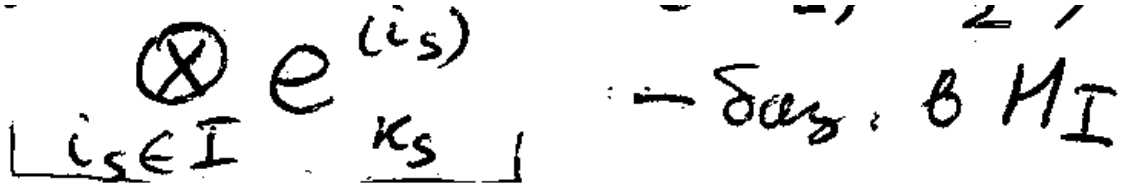
1. State of measure type.
Consider the values of the functional on $\alpha(I)$.

$$\varphi(a) = \operatorname{tr}(\rho_I \cdot a), \qquad \text{where } \rho_I \geq 0, \quad \operatorname{tr} \rho_I = 1.$$

State of measure type — the eigenv. of ρ_I agree in the following sense:
in H_i there exists a basis $(e_1^{(i)}, e_2^{(i)})$ — a basis in H_i (depends on φ).

Uncertain reading. The first subscript has heavy upper and lower strokes and a separate dot. Provisionally i.

Original source details: GELFAND-P000307



Original source details: GELFAND-P000307

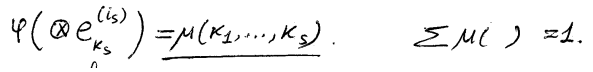
Diagram and typed description.

$\bigotimes_{i_s \in I} e_{k_s}^{(i_s)}$ — a basis in H_I

They — eigen.
for ρ_I

GELFAND-P000308

$$\varphi\left(\bigotimes e_{k_s}^{(i_s)}\right) = \mu(k_1, \ldots, k_s). \qquad \sum \mu(\,) = 1.$$

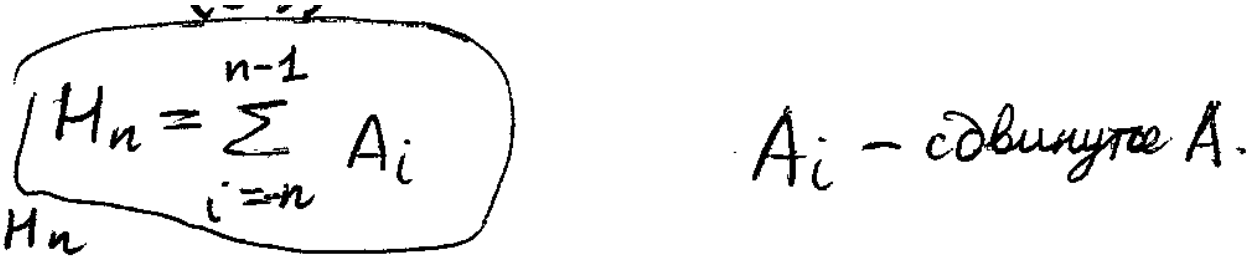


Manuscript notation. The argument of φ is written as a tensor product. After $\sum \mu$ the parentheses are empty.

Original source details: GELFAND-P000308

probability distr.
A measure on the space of seqs. taking the values 1 and 2.
2. States generated by a Hamiltonian.
$A \in \alpha((0,1)), \qquad H((I))$.

On $[-n, n]$ consider



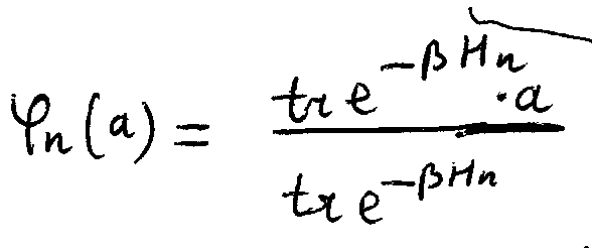
Original source details: GELFAND-P000308

Diagram and typed description.

$$H_n = \sum_{i=-n}^{n-1} A_i$$

A_i — shifted A.

$$\varphi_n(a) = \frac{\operatorname{tr} e^{-\beta H_n} \cdot a}{\operatorname{tr} e^{-\beta H_n}}.$$



Manuscript notation. The numerator has a dot before a. The expression after tr has no enclosing parentheses.

Original source details: GELFAND-P000308

$$\underline{\varphi(a) = \lim_{n\to\infty} \varphi_n(a) \text{ exists}} \quad \forall A. !$$

This is called a state, generated by the Hamiltonian A.
(from the quantum Gibbs distr.)
A noncommut. analogue of Markov chains.
Let φ — a state. A weakly closed ring of opers. in Hilbert space is constructed.
In the alg.

$$\langle a_1, a_2 \rangle = \varphi(a_2^* \cdot a_1).$$

Quotient by the ideal . . .
H_φ — Hilbert space.
[Crossed out: $h \in H_\varphi$]

Uncertain reading. Beneath the cancellation one can read $h \in H_\varphi$. The subscript extends right from the long stroke.

Original source details: GELFAND-P000308

$$h \in H_\varphi \implies L_a = ah.$$

Weak closure — M_φ.
when is it a factor?
Mainly of type 3.

GELFAND-P000309

[source number: 73]

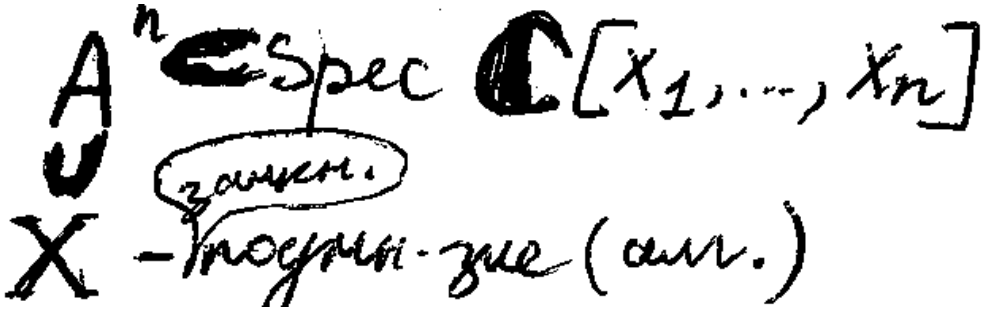
26/X. Kazhdan.

73

Triviality of the cohom. of a quasicoher. sheaf on an aff. scheme.

Uncertain reading. Between A^n and Spec is a heavily overwritten rounded sign open to the right. No middle stroke is discernible.

Original source details: GELFAND-P000309



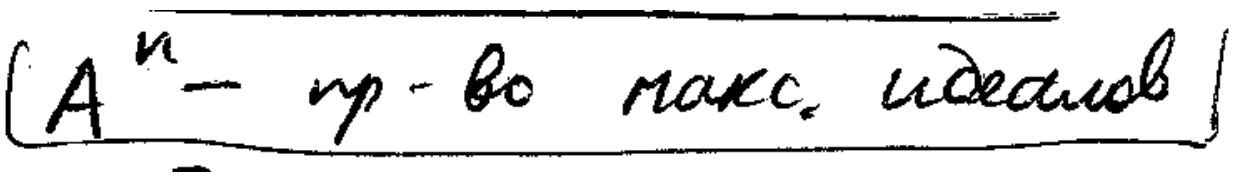
Original source details: GELFAND-P000309

Diagram and typed description.

A^n [a sign is cancelled] Spec $\mathbb{C}[x_1, \ldots, x_n]$

$\downarrow$

X closed — a subset (alg.).



Original source details: GELFAND-P000309

Diagram and typed description.

A^n — the space of max. ideals

$Q|_X \neq 0$, then P/Q — a func. on X, $\forall P$.
Then $\exists R$ — a polynomial, $(\forall P \exists R)$.

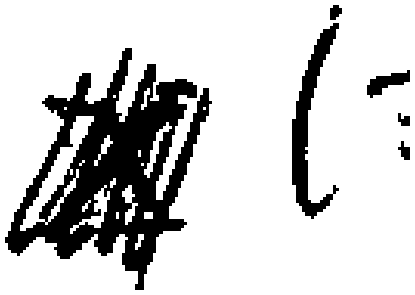
$$\left.\frac{P}{Q}\right|_X = R|_X.$$

Consider the ideal corresp. to X (0 on X),

$F_1, \ldots, F_k$ — a basis of this ideal.

Then $(F_1, \ldots, F_k, Q) = k[x_1, \ldots, x_n]$.

[A sign is cancelled: H / R?]



Original source details: GELFAND-P000309

Uncertain reading. Before the opening parenthesis is a cancelled short sign with several upright and oblique strokes; H / R are possible, but its original letter is undetermined. No closing parenthesis is visible in the following phrase.

(this ideal is not contained in any maximal

$\exists$ polynomials $L_0, \ldots, L_k$.

$$L_0 Q + L_1 F_1 + \ldots + L_k F_k = P;$$

Original source details: GELFAND-P000309

Manuscript notation. The right-hand side reads P, with an overlapping loop.

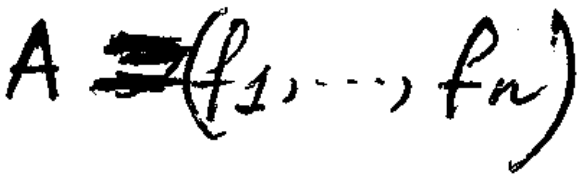
Hence one can take $R = L_0$.

(Hilbert's thm. — description of max. ideals)

This proof is analogous to the proof of triviality of cohomology

(this is triviality of 1-dimensional cohomology).

$X = \operatorname{Spec} A$, A [a sign is cancelled: $=$?] $(f_1, \ldots, f_n)$.



Original source details: GELFAND-P000309

Uncertain reading. Between A and $(f_1, \ldots, f_n)$, horizontal strokes are heavily overwritten, with an additional oblique line below. A cancelled $=$ is shown as a candidate; the exact earlier sign is undetermined.

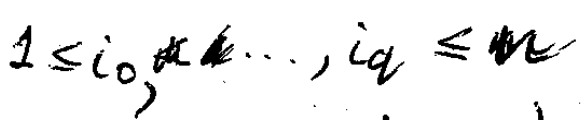
Consider $X_i = X_{f_i}$.

$$A = (f_1, \ldots, f_n) \Longleftrightarrow X = \bigcup X_i.$$

Let M — A-module, $\mathcal{F}_M$ — the corresp. sheaf.

Čech complex:

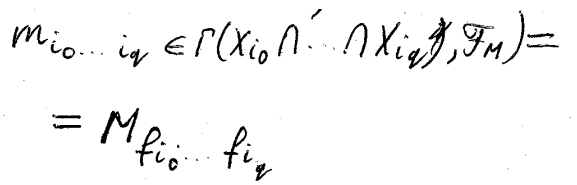
q-chain: $\forall$ index collection $1 \leq i_0$, [two overwritten signs: <?] $\ldots, i_q \leq n$ is given



Original source details: GELFAND-P000309

Uncertain reading. After i_0 are a comma and two dense marks. The final letter after $\leq$ is overwritten, provisionally n.

$$m_{i_0 \ldots i_q} \in \Gamma(X_{i_0} \cap \ldots \cap X_{i_q}), \mathcal{F}_M) = \\ = M_{f_{i_0} \ldots f_{i_q}}.$$

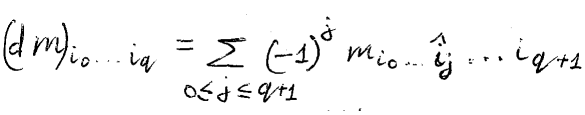


Original source details: GELFAND-P000309

Uncertain reading. After X_{i_q} is a crossed curved stroke resembling a parenthesis, followed by a comma and $\mathcal{F}_M) =$. The equals sign is repeated on the next line.

GELFAND-P000310

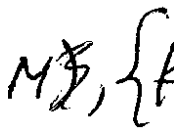
$$(dm)_{i_0 \ldots i_q} = \sum_{0 \leq j \leq q+1} (-1)^j m_{i_0 \ldots \widehat{i_j} \ldots i_{q+1}}.$$



Original source details: GELFAND-P000310

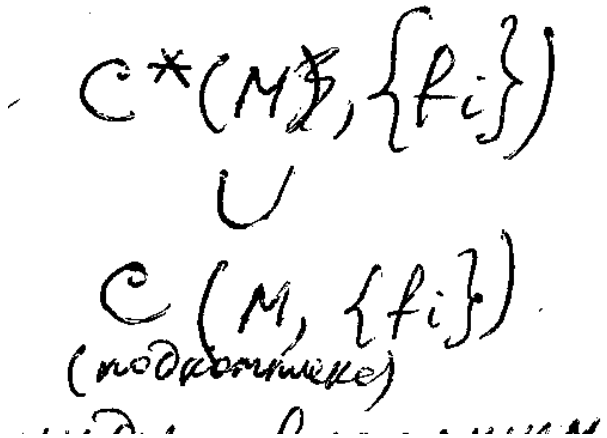
Manuscript notation. The left side has i_q, the right side i_{q+1}; the upper bound of summation is $q+1$.

$d : C^q \longrightarrow C^{q+1}$.
$d^2 = 0$ and the cochains form the Čech complex



Original source details: GELFAND-P000310

Uncertain reading. After M, before the comma, a tall looped letter is cancelled. The formula provisionally shows the candidates F/φ; the original sign is undetermined.



Original source details: GELFAND-P000310

Diagram and typed description.

$$C^*(M\,[F/\varphi?], \{f_i\})$$

$$\cup$$

antisymm. cochains — $C(M, \{f_i\})$

(subspace)

Prop.: the cohomologies coincide (the isom. is induced by inclusion).

For convenience — sometimes C^*, sometimes C.

Theorem: $H^q(\{X_i\}; \mathcal{F}_M) = 0$ for $q \neq 0$.

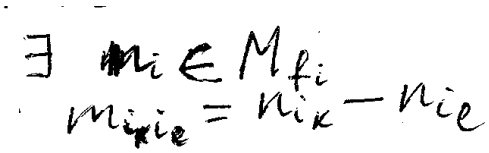
Proof: Let $m_{i_0 \dots i_q} \in M_{f_{i_0} \dots f_{i_q}}$ — cycles.

There we had $A = (f_1, \dots, f_m, Q)$,

$$m_{i_0 i_1} \in M_{f_{i_0} f_{i_1}},$$
$$(dm)_{i_0 i_1 i_2} = m_{i_0 i_1} - m_{i_0 i_2} + m_{i_1 i_2} = 0.$$

Need to prove:

$$\exists n_i \in M_{f_i}$$
$$m_{i_k i_\ell} = n_{i_k} - n_{i_\ell}.$$



Original source details: GELFAND-P000310

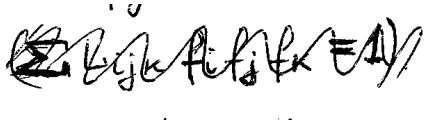
Uncertain reading. The letter before i after $\exists$ is heavily overwritten; the candidate n_i is provisional. The small subscripts on m are also overwritten; the reading $i_k i_\ell$ is not final.

Ex.: $(M_f)_g = M_{fg}$.

Original source details: GELFAND-P000310

Manuscript notation. Inside the first parenthesis, a mark after f is heavily overwritten. The original reading is undetermined.

[Cancelled; provisional reading: $(\sum L_{ijk} f_i f_j f_k = 1)$.]



Original source details: GELFAND-P000310

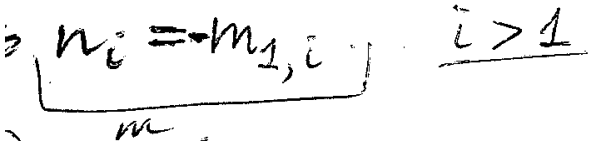
Uncertain reading. Oblique strokes cancel the sum, coefficient, and product. The candidate L_{ijk}, the small subscripts on f, and the final $= 1$ remain uncertain; the right parenthesis is visible.

$m_{1,i} = n_1 - n_i$.

Put $n_1 = 0$, [a sign is cancelled: m?]

Original source details: GELFAND-P000310

Uncertain reading. Before $n_i = -m_{1,i}$, a short sign is densely crossed out; m is possible, but the original reading is undetermined.



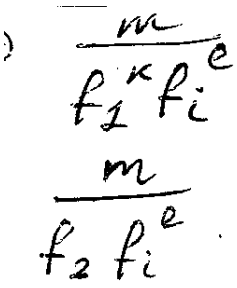
Original source details: GELFAND-P000310

Diagram and typed description.

$\underbrace{n_i = -m_{1,i}}_{m}$ $\quad i > 1$

False, since $m_{1,i}$ has the form

$$\frac{m}{f_1^k f_i^\ell},$$
$$\frac{m}{f_2 f_i^\ell},$$
$$\vdots$$



Original source details: GELFAND-P000310

Manuscript notation. In both fractions f_i has a looped exponent, provisionally ℓ. No exponent is visible on f_2.

and combine, using the fact that this is a cover.

[source number: 74]

[Cancelled: “Consider.”?] For simplicity, let $q = 1$.

Original source details: GELFAND-P000311

Uncertain reading. After the initial R-shaped letter, several horizontal strokes obscure the middle letters. The candidate “Consider.” remains provisional.

Given $m_{i_0 i_1} - m_{i_0 i_2} + m_{i_1 i_2} = 0$. $\quad 1 \le i_k \le n$.

Original source details: GELFAND-P000311

Uncertain reading. Before $m_{i_0 i_2}$ is a dark overwritten sign with two horizontal regions and a short lower extension. A minus is shown as a candidate; the earlier sign is unrecovered.

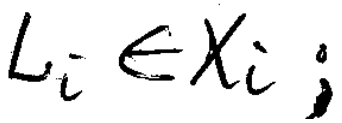
Need to find $n_i \in M_{f_i}$: $\quad m_{i_0 i_1} = n_{i_0} - n_{i_1}$.

$m_{i_0 i_1} =$ [a cancelled fraction: $\bar{m}/f_{i_0}$?] $\dfrac{\bar{m}_{i_0 i_1}}{f^N_{i_0} f^N_{i_1}}$, $\quad \bar{m} \in M$.

Original source details: GELFAND-P000311

Uncertain reading. Oblique strokes cross the old numerator and fraction bar. The reading $\bar{m}/f_{i_0}$ is provisional; a second denominator factor in the old expression is unrecovered.

Find $L_i \in X_i$;



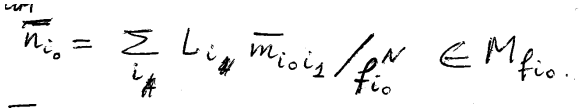
Original source details: GELFAND-P000311

Manuscript notation. After the membership sign the manuscript reads X_i.

$$\sum L_i f_i^N = 1.$$

Define

$$\bar{n}_{i_0} = \sum_{i_1} L_{i_1} \bar{m}_{i_0 i_1} / f^N_{i_0} \in M_{f_{i_0}}.$$

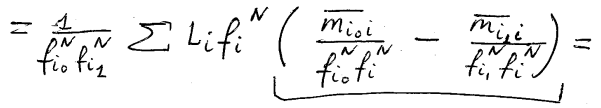


Original source details: GELFAND-P000311

Uncertain reading. The digits at the sum and on L are heavily overwritten, provisionally i_1. The second subscript on $\bar{m}$ is distinctly i_1.

$$\bar{n}_{i_0} - \bar{n}_{i_1} = \sum_i \left(\frac{L_i}{f^N_{i_0}} \bar{m}_{i_0 i} - \frac{L_i}{f^N_{i_1}} \bar{m}_{i_1 i} \right) =$$

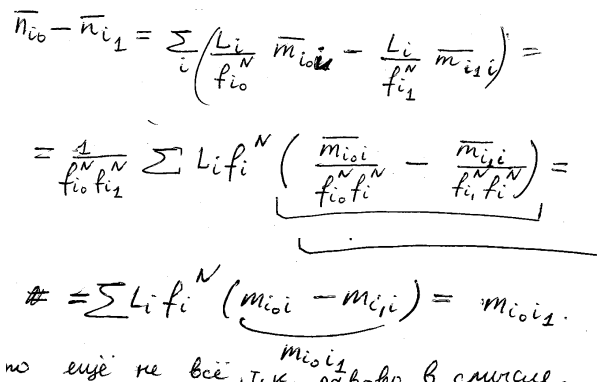
$$= \frac{1}{f^N_{i_0} f^N_{i_1}} \sum L_i f_i^N \underbrace{\left(\frac{\bar m_{i_0 i}}{f^N_{i_0} f^N_i} - \frac{\bar m_{i_1 i}}{f^N_{i_1} f^N_i} \right)} =$$



Original source details: GELFAND-P000311

Manuscript notation. The factor $\frac{1}{f^N_{i_0} f^N_{i_1}}$ is written before the sum. The denominators of both fractions above the angular bracket are also written out.

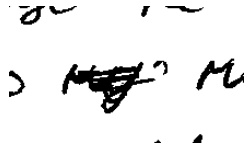
$$\text{[a sign is cancelled]} \quad = \sum L_i f_i^N \underbrace{(m_{i_0 i} - m_{i_1 i})}_{m_{i_0 i_1}} = m_{i_0 i_1}.$$



Original source details: GELFAND-P000311

Uncertain reading. In the first line, the last subscript on the first $\bar m$ is overwritten; the candidate i is provisional. A small expression before the last line's equality is cancelled. The difference of fractions has a long unlabelled angular bracket below it; the last difference has a curve labelled $m_{i_0 i_1}$.

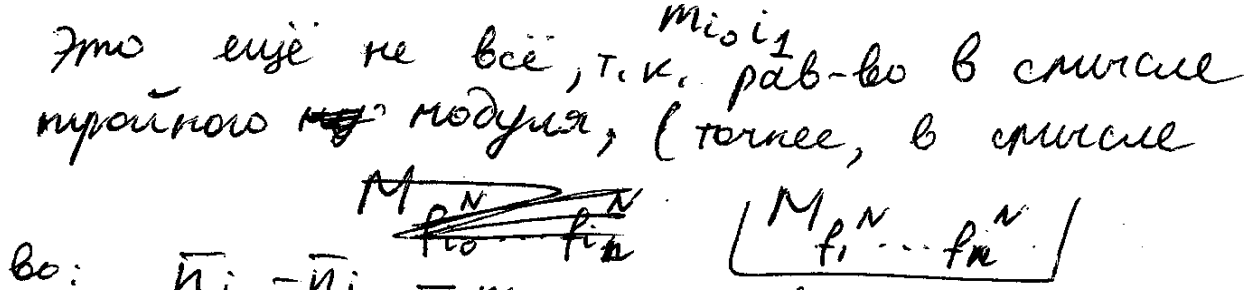
This is not all, since the eq. is in the sense of a twisted



Original source details: GELFAND-P000311

Undeciphered passage. Before "module", a short word is cancelled. An initial n-shaped stroke and a small final curve are visible; the text is unrecovered.

module, (more precisely, in the sense of
[Cancelled; candidate: $M_{f^N_{i_0} \dots f^N_{i_n}}$?]



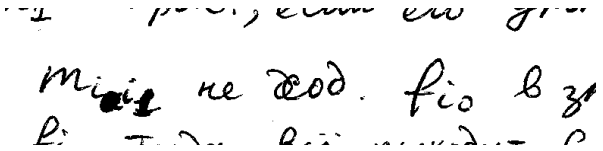
Original source details: GELFAND-P000311

Uncertain reading. Several long strokes cancel the old localization expression. The final subscript in the candidate $f^N_{i_n}$ remains uncertain; i_1 is also possible. At the right, the new expression $M_{f^N_1 \dots f^N_n}$ has lower and side strokes. No closing parenthesis is visible after it.

$M_{f^N_1 \dots f^N_n}$

The eq.: $\bar n_{i_0} - \bar n_{i_1} = m_{i_0 i_1}$ holds if it is multiplied by some power.

Bernstein's proof: let all $m_{i_0 i_1}$ not contain f_{i_0} in the denom.



Uncertain reading. The two small subscripts on m are heavily overwritten. The candidate $i_0 i_1$ is provisional.

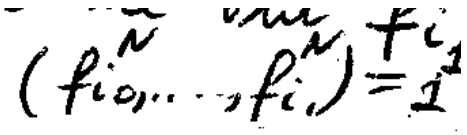
Original source details: GELFAND-P000311

and eqs. to 0 do not require multiplication by f_{i_0}. Then everything goes through.
Consequence:

$f_{i_0}^N m_{i_0 i_1}$ — a boundary. The same for $f_{i_1}^N$ and so on.

Cohom. is a module.

Since $(f_{i_0}^N, \dots, f_{i_n}^N) = 1$, everything is all right.

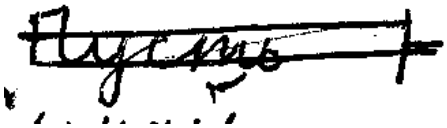


Manuscript notation. The end of the last subscript is unclear, provisionally i_n.

Original source details: GELFAND-P000311

GELFAND-P000312

[Cancelled: "Let" and a sign $\dagger$]



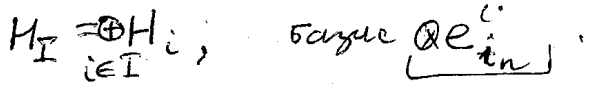
Uncertain reading. After "Let" is a tall vertical stroke with a short cross-stroke. The candidate φ is undetermined: no closed loop is visible.

Original source details: GELFAND-P000312

Sinai. H_i — 2-dimensional, $-\infty < i < +\infty$.
$I =$ any finite set.

$$H_I = \bigoplus_{i \in I} H_i;$$

basis $\bigotimes e^i_{t_n}$



Uncertain reading. The manuscript has a circled plus before H_i and a circled cross before e. The subscript on e is provisionally t_n; its first letter is heavily overwritten.

Original source details: GELFAND-P000312

$\alpha(I)$ — all matr. in H_I.
In $\alpha(I)$ introduce the operator norm in Hilb. space.

$$I \subset I_1 \Rightarrow \alpha(I) \subset \alpha(I_1). \qquad (\otimes 1).$$

$$\alpha(I_1) = \alpha(I) \otimes \alpha(I_1 \setminus I).$$

This gives a normed ring (completion) α.
(algebra of quasiloc. observables).
New operators added:

$\varphi(a)$ — a functional (lin.), $\varphi(a) \geq 0$ $\forall$ positive matr. a.
In $\alpha(I)$ we have $\varphi(a) = \mathrm{Sp}(a\rho) = \mathrm{Tr}(a\rho)$.

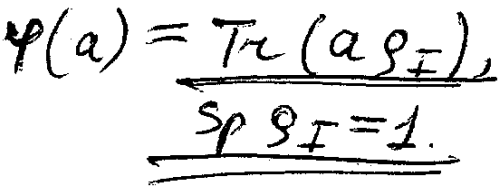
$$\underline{\underline{\varphi(a) \geq 0}} \Longleftrightarrow \rho \geq 0.$$

$\underline{\underline{\varphi(e) = 1}}$ — a noncommut. analogue of a probability distrib.; a state in QFT.

$\alpha(I)$ — a ring of matrices [cancelled: “on”?], $\varphi(a) = \underline{\underline{\mathrm{Tr}(a\rho_I)}}$, $\underline{\underline{\mathrm{Sp}\,\rho_I = 1}}$.

Original source details: GELFAND-P000312

Uncertain reading. After “matrices”, a short word is densely crossed out. The candidate “on” remains provisional.



Original source details: GELFAND-P000312

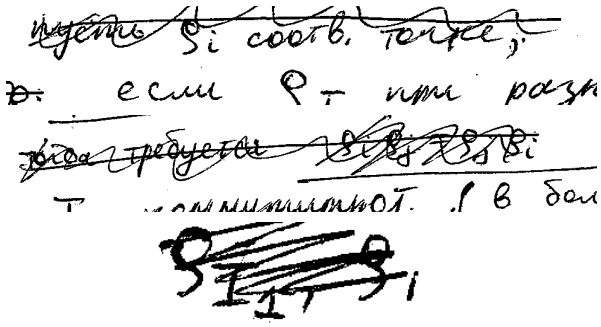
Manuscript notation. There are two strokes below $\mathrm{Tr}(a\rho_I)$ and two below $\mathrm{Sp}\,\rho_I = 1$.

On ρ_I the compatibility cond.:

$$\mathrm{Tr}_{I_1 - I}(\rho_{I_1}) = \rho_I.$$

$$\alpha(I_1) = \alpha(I) \otimes \alpha(I_1 - I).$$

$\underline{\varphi$ — a state of measure type.$}$ $\longrightarrow$ [Cancelled: let ρ_i corresp. to a point;]
[Cancelled: then require $\underline{\rho_i\rho_j = \rho_j\rho_i}$?]
if ρ_I for different I commute (in a larger space).
[At the left, two signs are cancelled; candidates $\rho_{I_1}\rho_i$?]
(one-point ρ_i and ρ_j always commute.)



Original source details: GELFAND-P000312

Uncertain reading. In the upper cancelled line, “let” is followed by a rho-shaped letter with a small subscript, then “corresp. to a point;”. In the next line, a condition with four rho-shaped letters is crossed out; the subscripts i, j are provisional. Below at the left are two separately cancelled letters whose small subscripts also remain uncertain.

GELFAND-P000313

[source number: 75]

ρ_i — a matrix in H_i,

Let e_1^i, e_2^i — a basis of eigenvectors of ρ_i,

$$e_{\alpha_1}^{i_1} \otimes e_{\alpha_2}^{i_2} \otimes \ldots \otimes e_{\alpha_n}^{i_n}$$

Then $\rho(I)$ is diagonal in it, since it commutes with all ρ_i (obvious if the spectrum is simple) ($\exists$ such a basis).

$2^{|I|}$ eigenvectors. $\mu(\alpha_1, \ldots, \alpha_n)$ — an eigenvalue.

It characterizes a distrib. of measure type. Its compatibility cond.:

$$\sum_{\alpha_n} \mu(\alpha_1, \ldots, \alpha_n) = \mu(\alpha_1, \ldots, \alpha_{n-1})$$

and so on.

Consider an infinite seq. of 0 and 1.

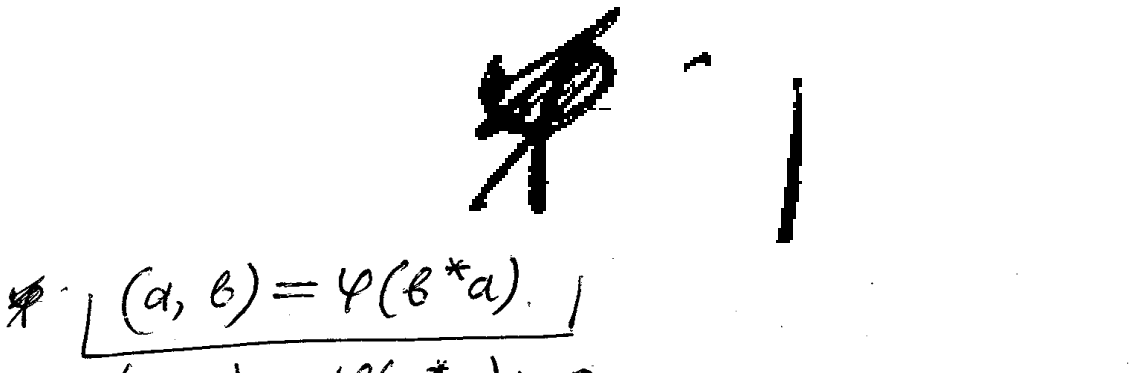
$$\ldots, \varepsilon_{-n}, \varepsilon_{-n+1}, \ldots, \varepsilon_0, \ldots$$

a group (addition mod 2). A measure is defined on cylinder sets $\alpha_1, \ldots, \alpha_n$ — fixed $\mu(\alpha_1, \ldots, \alpha_n)$.

The condition is compatibility of these measures.

$\prod \mathbb{Z}_2^i$. One can take Haar measure.

[A sign is cancelled; candidate φ?]



Original source details: GELFAND-P000313

Uncertain reading. Before (a, b), oblique strokes cross a looped letter; a lower vertical extension is visible. The candidate φ remains provisional. The uncancelled formula has a bracket with a straight lower stroke and short sides.

$(a, b) = \varphi(b^* a).$

$$(a, a) = \varphi(a^* a) \geq 0.$$

Now quotient and complete — a Hilb. space.

$xa = X_a$ — a lin. operator — continuous in this inner product.

Original source details: GELFAND-P000313

Manuscript notation. The manuscript has xa on the left and X_a on the right.

Our ring consists of lin. operators.

(see the supplement to the book by Gelfand, Shilov, Raikov)

(Naimark et al.)

The ring generated by the state φ.

Factors of type **3.** — they are important — these are shifts.

Original source details: GELFAND-P000313

Manuscript notation. The manuscript has the numeral 3 and the abbreviation «сдвиг.».

GELFAND-P000314

2/XI I. M. Gelfand.

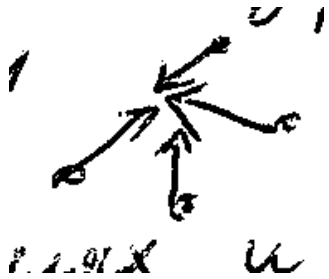
Results of the speaker and V. Ponomaryov (not the one from the Faculty of Mechanics and Mathematics).

P — a finite-dimensional space, $E_1, \dots, E_4$ — 4 subspaces (the order matters).

$$S = \{P; E_1, E_2, E_3, E_4\}.$$

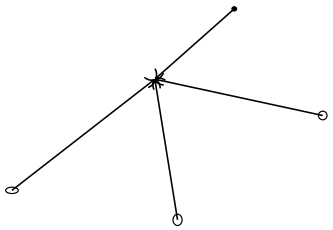
This is a category (morphisms map into).

Equivalent to studying the category of diagrams



Original source details: GELFAND-P000314

Diagram and typed description.



in the category of finite-dimensional spaces.

The methods give information about modules and submodules.

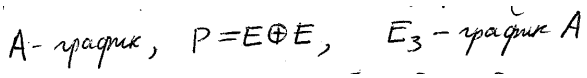
$S \sim S'$, if $\exists$ a lin. isom.

Direct sum is naturally introduced.

The permutation group acts.

Example. E; $A : E \to E$. Graph $\{(x, Ax)\} \subset E \oplus E$.

A — graph, $P = E \oplus E$, E_3 — the graph of A.



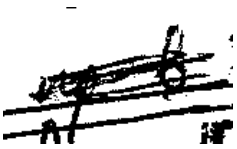
Original source details: GELFAND-P000314

Manuscript notation. This line reads "A — graph" and later "E_3 — the graph of A".

To describe the system $E \oplus E$ one must specify

$E_1 = E \oplus 0$, $E_2 = 0 \oplus E$ and the diagonal $E_4 = \{(x, x)\}$.

Pairs of maps



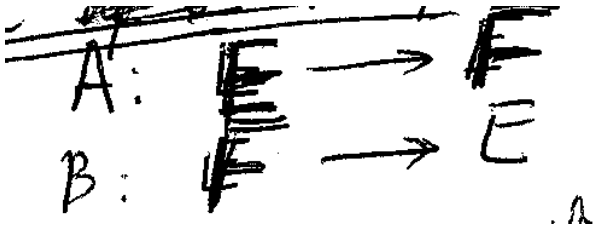
Original source details: GELFAND-P000314

Undeciphered passage. After “Pairs”, several letters before the final tall looped sign are crossed out. Their text is unrecovered.

Original source details: GELFAND-P000314

Uncertain reading. Horizontal and oblique strokes cross the final tall sign; the candidates φ / b remain unresolved.

$$A : E \longrightarrow F,$$
$$B : F \longrightarrow E.$$



Original source details: GELFAND-P000314

Uncertain reading. The starting letter of each arrow is overwritten, as is the ending letter of the first. The reading shown is $A : E \to F$, $B : F \to E$.

(problem

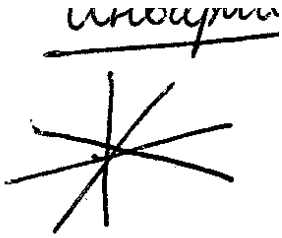
are also described.

Kronecker’s problem — in the same direction.

Relation — any graph.

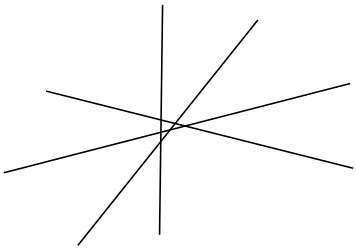
Invariants:

4 lines.



Original source details: GELFAND-P000314

Diagram and typed description.



Invariant — a number λ (one).

2 are coordinate axes.

The 3rd is diagonal.

The 4th is a graph — $y = \lambda x$.

Change the numbering: $\lambda \to 1 - \lambda$, λ^{-1}, $\lambda/(\lambda - 1)$.

Result (the proof is long).

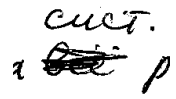
GELFAND-P000315

[source number: 76]

Undeciphered passage. Before “Let”, a short expression with several dense strokes and a large rounded final loop is cancelled. Its text is unrecovered.

Original source details: GELFAND-P000315

Let there be given P, E_1, E_2, E_3, E_4, then [cancelled: “syst.”?] [inserted above: syst.] decomposes as a direct sum of the following formations

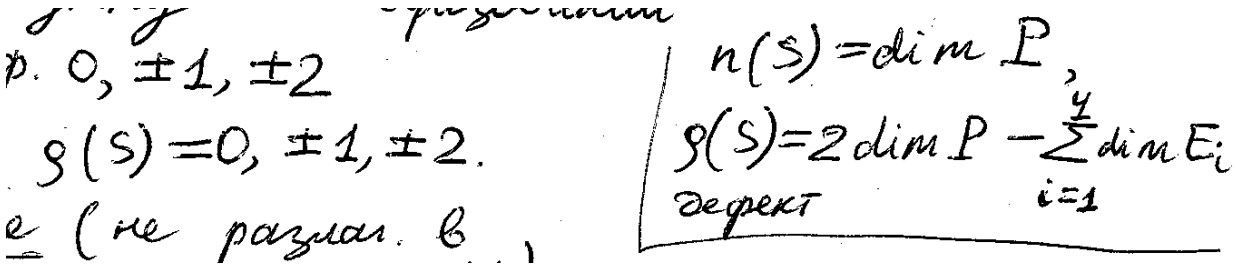


Uncertain reading. “syst.” is clearly written above the line. The earlier short expression below is densely crossed out; the candidate “syst.” remains provisional.

Original source details: GELFAND-P000315

systems with defect $0, \pm 1, \pm 2$.

$\rho(S) = 0, \pm 1, \pm 2.$



Original source details: GELFAND-P000315

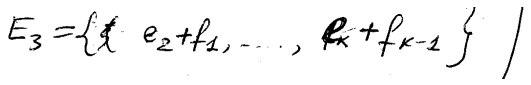
Uncertain reading. The defect symbol in the line and frame has a loop and a lower curved tail; the candidates ρ / g remain unresolved. The text provisionally uses ρ. A left and a lower rule frame the definition and “defect”, with no top or right rule.

$$n(S) = \dim P,$$

$$\rho(S) = 2\dim P - \sum_{i=1}^{4} \dim E_i$$

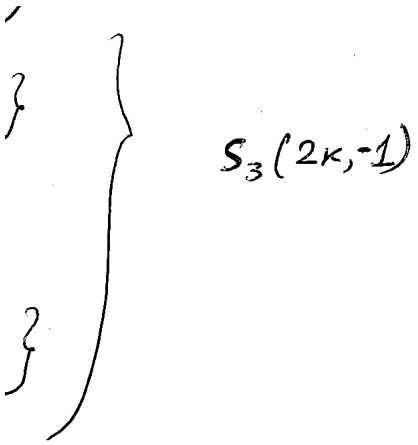
defect

Irreducible (not decomposable as a direct sum)



Original source details: GELFAND-P000315

Uncertain reading. After the opening brace of E_3, a narrow tall sign is cancelled; the candidate ξ is shown with a question mark. The last e_k is heavily overwritten; the earlier strokes are unrecovered.



Original source details: GELFAND-P000315

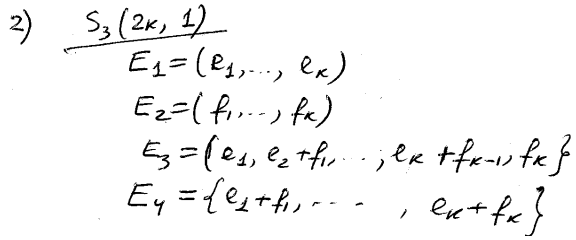
Diagram and typed description.

$$\left.\begin{array}{c} 1) \quad \text{for} \quad n(S) = 2k. \\ \text{Basis in } P \quad \{e_1 \dots e_k \; f_1 \dots f_k\}. \\ E_1 = \{(e_1, \dots, e_k)\}. \\ E_2 = \{(f_1, \dots, f_k)\}. \\ E_3 = \{[\xi?]\; e_2 + f_1, \dots, e_k + f_{k-1}\}. \\ E_4 = \{e_1 + f_1, \dots, e_k + f_k\}. \end{array}\right\} \; S_3(2k, -\underline{1})$$

---

2) $S_3(2k, 1)$.

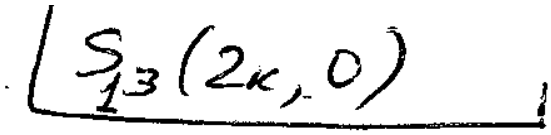
$$E_1 = (e_1, \dots, e_k).$$
$$E_2 = (f_1, \dots, f_k).$$
$$E_3 = (e_1, e_2 + f_1, \dots, e_k + f_{k-1}, f_k\}.$$
$$E_4 = \{e_1 + f_1, \dots, e_k + f_k\}.$$



Original source details: GELFAND-P000315

Manuscript notation. The series is labelled $S_3(2k, 1)$. The list for E_3 opens with a parenthesis and closes with a brace.

3) $\begin{matrix} E_1 \\ E_2 \end{matrix}$ — the same.

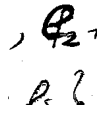


Original source details: GELFAND-P000315

Uncertain reading. The right-hand addition has 1 below S, with 3 farther right and higher, followed by $(2k, 0)$. Their positions are represented separately; the compact reading S_{13} is provisional. The frame has a left, a lower, and a short right stroke.

$$\begin{matrix} S & & \\ 1 & 3 & (2k,0) \end{matrix}$$

$$E_3 = \{e_1, e_2 + f_1, \ldots, e_k + f_{k-1}\}.$$



Original source details: GELFAND-P000315

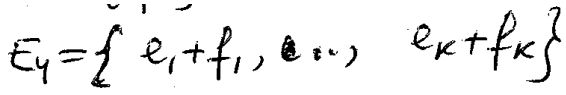
Uncertain reading. In the second position, the final e_2 is written over an earlier letter. A separate residual upright stroke resembles f, but the old letter is unrecovered.

$$E_4 = \{e_i + f_i\}.$$

GELFAND-P000316

$S(2k; 0, \lambda)$.

$$E_1 = \{e_i\},$$
$$E_2 = \{f_i\},$$
$$E_4 = \{e_1 + f_1, \ldots, e_k + f_k\},$$
$$E_3 = \{e_1 + \lambda f_1, e_2 + f_1 + \lambda f_2, \ldots, e_k + f_{k-1} + \lambda f_k\}.$$



Original source details: GELFAND-P000316

Manuscript notation. In the E_4 line, a dense e-shaped stroke precedes the ellipsis. The earlier writing is undetermined.

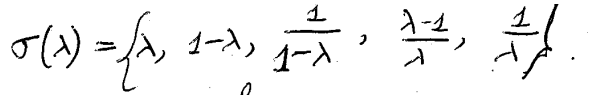
$\lambda \neq 0, \quad \lambda \neq 1.$

σ — a permutation. $\begin{matrix} 1 \longrightarrow i \\ 3 \longrightarrow k \end{matrix}$

$$\sigma(S_{13}) \xrightarrow{\text{def}} \sigma(S_{ik}).$$

$\sigma\lambda$. λ transforms as for an anharm. quadruple of points.

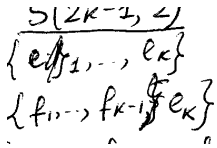
$$\sigma(\lambda) = \left\{\lambda, 1-\lambda, \frac{1}{1-\lambda}, \frac{\lambda-1}{\lambda}, \frac{1}{\lambda}\right\}.$$



Original source details: GELFAND-P000316

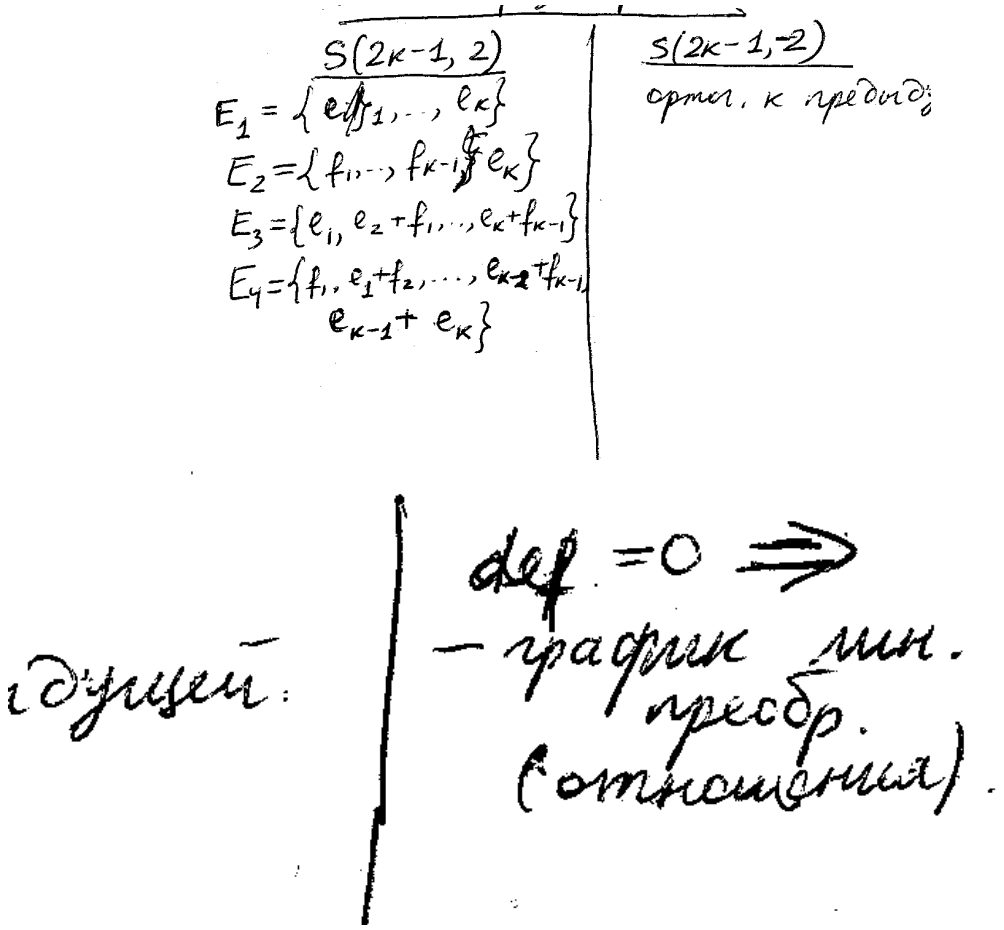
Manuscript notation. Five expressions are written inside the braces.

Odd-dimensional spaces.



Original source details: GELFAND-P000316

Uncertain reading. In the first line, near the initial subscript 1, a tall f-shaped sign is cancelled; [f?] marks it provisionally. In the second line, the old brace after f_{k-1} is heavily overwritten, with a loop above it. An added e_k stands at the right before the final brace; no separate comma is visible before it.



Original source details: GELFAND-P000316

Diagram and typed description.

$S(2k-1,2)$. $E_1 = \{e_1$ [f?]$, \dots, e_k\}$, $E_2 = \{f_1, \dots, f_{k-1}$ [overwritten] $e_k\}$, $E_3 = \{e_1, e_2 + f_1, \dots, e_k + f_{k-1}\}$, $E_4 = \{f_1, e_1 + f_2, \dots, e_{k-2} + f_{k-1}, e_{k-1} + e_k\}$.	$S(2k-1,-2)$, orthog. to the preceding one.	def. $= 0 \Rightarrow$ — the graph of a lin. transformation (relation).

def 0 does not occur in odd dimension.

def(-1) occurs for relations.

general position — without a parameter, and degenerate — with a parameter.

GELFAND-P000317

[source number: 77]

77

These trajectories.

Example. $S(5,2) \oplus S(1,2)$.

$S(5,2)$ is stable (i.e. a deform. gives an isom. syst.).

(Indecomp. systs. without parameters are always stable).

$S(5,2)$ and $S(1,2)$ are stable, but $S(5,2) \oplus S(1,2)$ is unstable and under deform. passes into the stable syst.

$$S(3,2) \oplus S(3,2).$$

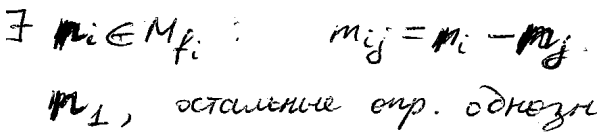
It is interesting to find a simple proof.

(The decomp. into irreducibles is unique if they are distinct, and the number is always the same)

Kazhdan.

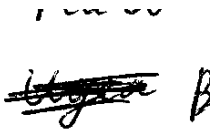
Let $m_{ij} \in M_{f_i f_j}$, $m_{ij} + m_{jk} + m_{ki} = 0 \in M_{f_i f_j f_k}$.

Need to prove: $\exists m_i \in M_{f_i}$, $m_{ij} = m_i - m_j$.



Original source details: GELFAND-P000317

Uncertain reading. The letters before i, j in the difference and on the chosen m_1 are heavily reinforced. The final reading is m; the earlier strokes cannot be separated.



Original source details: GELFAND-P000317

Uncertain reading. Before “Choose”, a short word with loops and a lower tail is crossed out. The candidate “Будем” (we shall) remains uncertain; the complete earlier reading is undetermined.

Choose m_1, the others are determined uniquely and without contradiction.

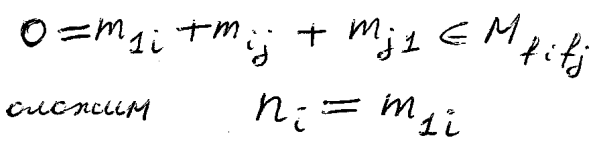
Suppose the denominators of all elems. do not contain f_1 and all rels. hold without multiplication by f_1.

Then one must put $n_1 = 0$.

Let $m_{1i} \in M_{f_i}$,

$$0 = m_{1i} + m_{ij} + m_{j1} \in M_{f_i f_j}.$$

Put $n_i = m_{1i}$.



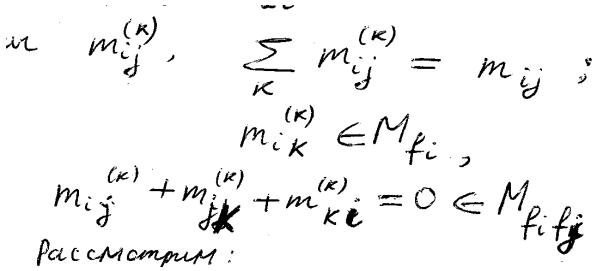
Original source details: GELFAND-P000317

Manuscript notation. The last line has n_i on the left and m_{1i} on the right, with no minus sign.

Lemma.

$$\exists \text{ cocycles } m_{ij}^{(k)}, \qquad \sum_k m_{ij}^{(k)} = m_{ij};$$

$$m_{ik}^{(k)} \in M_{f_i}, \qquad m_{ij}^{(k)} + m_{jk}^{(k)} + m_{ki}^{(k)} = 0 \in M_{f_i f_j}.$$



Uncertain reading. The k in the second subscript jk, the final i in the third term and the last j in $M_{f_i f_j}$ are reinforced. Their visible final forms are rendered provisionally; earlier strokes are unrecovered.

Original source details: GELFAND-P000317

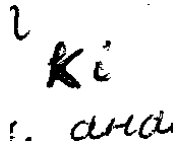
Proof. Consider:

$$m_{ij}^{(N,k)} = f_k^N m_{ij} \quad \text{— cocycles, for sufficiently large } N$$

GELFAND-P000318

satisfy all conditions.

$$\exists N: \quad m_{ki}^{(N,k)} \in M_{f_i}.$$

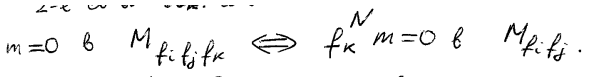


Uncertain reading. The first letter of the lower index is heavily reinforced, with an upper loop; the visible final form is rendered as ki. Earlier strokes within k cannot be separated.

Original source details: GELFAND-P000318

The 2nd property is proved similarly.

$$m = 0 \text{ in } M_{f_i f_j f_k} \quad \Longleftrightarrow \quad f_k^N m = 0 \text{ in } M_{f_i f_j}.$$



Manuscript notation. The right side reads $f_k^N m = 0$ with no separate quantifier before it.

Original source details: GELFAND-P000318

Choose N common to all k.

Let $\sum g_k f_k^N = 1$. Put

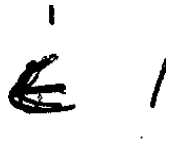
$$m_{ij}^{(k)} = g_k m_{ij}^{(N,k)};$$

they satisfy all conditions of the lemma. ■

Now it suffices to consider $m_{ij}^{(k)}$.

$$n_i^{(k)} = m_{ik}^{(k)} \in M_{f_i} \text{ by assumption.}$$

$$m_{ij}^{(k)} = n_i^{(k)} - n_j^{(k)} \in M_{f_i f_j},$$



Original source details: GELFAND-P000318

Uncertain reading. The sign before the final M has three horizontal endings and several reinforced oblique and curved strokes. The visible form is rendered as $\in$; earlier lines are unrecovered.

from the last condition of the lemma.

Example. The sheaf O^*, $X = \operatorname{Spec} A$.

Original source details: GELFAND-P000318

Manuscript notation. Below the heading, diagonal hatching fills the space between two rules.

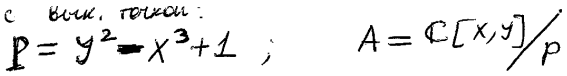
$$\Gamma(U, O^*) = \{\gamma \in \Gamma(U, O) \mid \gamma \notin p,\ \forall p \in U\}.$$

(a subsheaf of the structure sheaf)

Consider the simplest alg. variety —

— a torus with a point removed.

$$P = y^2 - x^3 + 1; \qquad A = \mathbb{C}[x, y]/p.$$

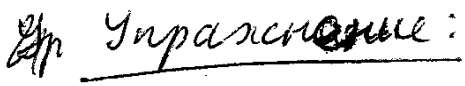


Original source details: GELFAND-P000318

Manuscript notation. A heavy minus precedes x^3. The left side uses capital P and the slash is followed by lowercase p.

[Reinforced before the heading: “Упр” (Ex.).]

Exercise: $H^1(\operatorname{Spec} A, O^*) \neq 0$.



Original source details: GELFAND-P000318

Uncertain reading. Before the full heading “Упражнение” (Exercise), the short “Упр” is superimposed on earlier loops; the earlier letters are undetermined. A dark reinforced loop also occurs within the full heading; the final word is rendered as “Упражнение”, without reconstructing earlier strokes.

$S = A^n - \{0\}$.

$$H^i(S, O) = 0 \text{ for } i \neq n-1 \text{ and } i \neq 0.$$

For $i = n - 1$ it is infinite-dimensional and easily described. $H^0 = A$.

Original source details: GELFAND-P000318

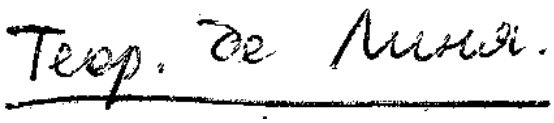
Uncertain reading. At the start of the final sentence, the letter before $= n - 1$ is heavily reinforced and has a separate dot above it. It is rendered as i; earlier strokes within the lower loop are undetermined.

GELFAND-P000319

[source number: 78]

78

Deligne's theor.



Original source details: GELFAND-P000319

Manuscript notation. The handwritten name is written as two words, «де Линя».

X — a complete variety,
Y — a subvariety, locally

Diagram and typed description.

Original source details: GELFAND-P000319

a direct prod. of transverse planes.

$f = 0$ on the variety (codim. 1).
(may)

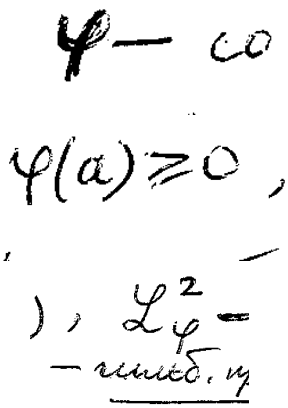
$\left\{\frac{df}{f}\right\}$ — generating diff. forms.

Resolution of sing. reduces everything to this case.
The ring does not depend on the method of resolution.

Sinai.

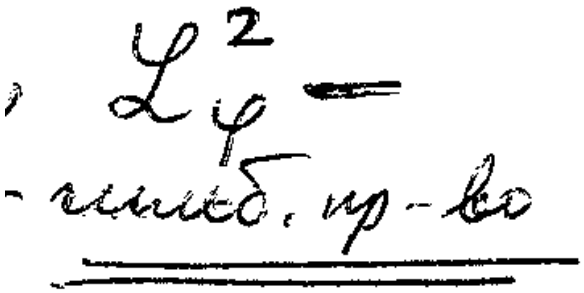
α — the alg. of quasilocal observ., φ — a state (functional),
$\mathcal{L}^2_\varphi \Longleftarrow$

— Hilb. space



Original source details: GELFAND-P000319

Uncertain reading. The reinforced initial letter has a right loop and a descending vertical stroke. Its visible form is rendered as φ; the positivity condition and the subscript on L^2 provide adjoining forms. Earlier strokes in the reinforced letter cannot be separated.



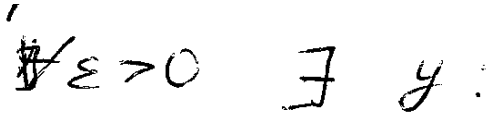
Original source details: GELFAND-P000319

Manuscript notation. Two short horizontal strokes after $\mathcal{L}^2_\varphi$ join in a heavy mark at the left, rendered as $\Longleftarrow$.

$$\varphi(a) \geq 0, \text{ if } a \geq 0, \qquad \varphi(1) = 1.$$

M — a weakly closed ring of opers.

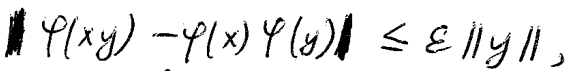
Powers's theorem. Let $\forall x \in \alpha, \quad \forall \varepsilon > 0, \quad \exists y$:



Original source details: GELFAND-P000319

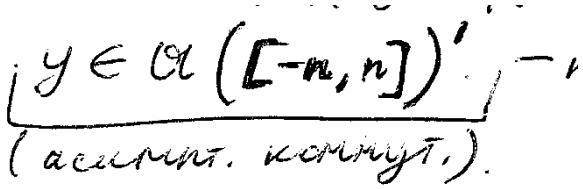
Uncertain reading. Before $\varepsilon > 0$, a $\forall$ is written over dark earlier strokes; a separate $\exists y$ is clearly written at the right. The earlier sign beneath the reinforced $\forall$ is unrecovered.

$$|\varphi(xy) - \varphi(x)\varphi(y)| \leq \varepsilon \|y\|,$$



Original source details: GELFAND-P000319

Manuscript notation. Two heavy vertical marks surround the left side. The norm on the right has double bars.



Original source details: GELFAND-P000319

Diagram and typed description.

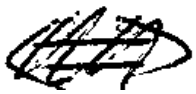
$y \in \alpha([-n, n])'$

(asympt. commut.)

— the commutant (commuting matrices).

Then M — a factor.

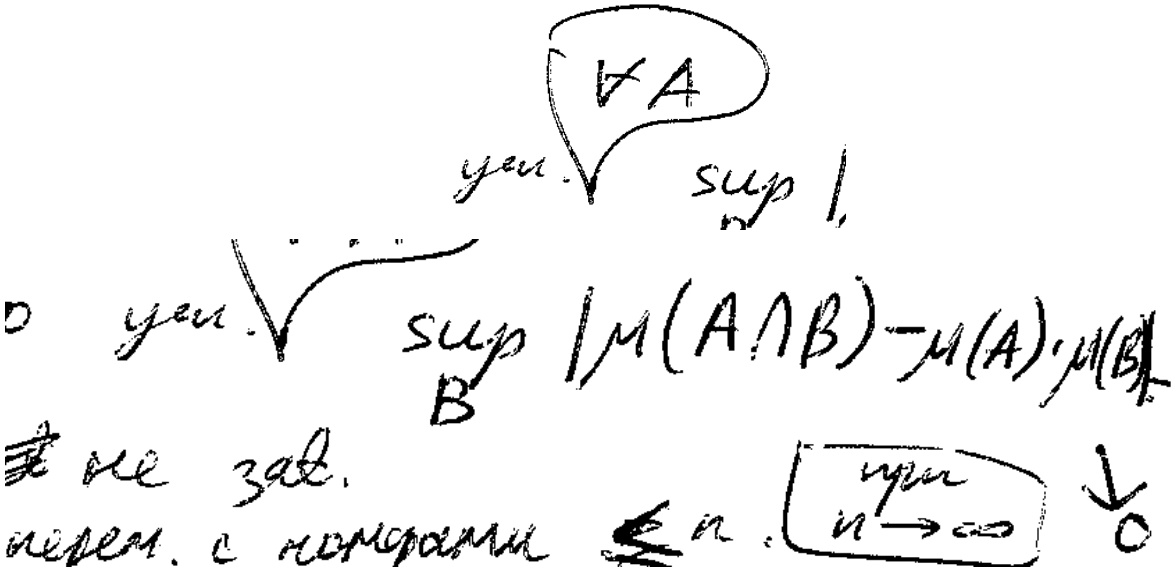
φ — of measure type



Original source details: GELFAND-P000319

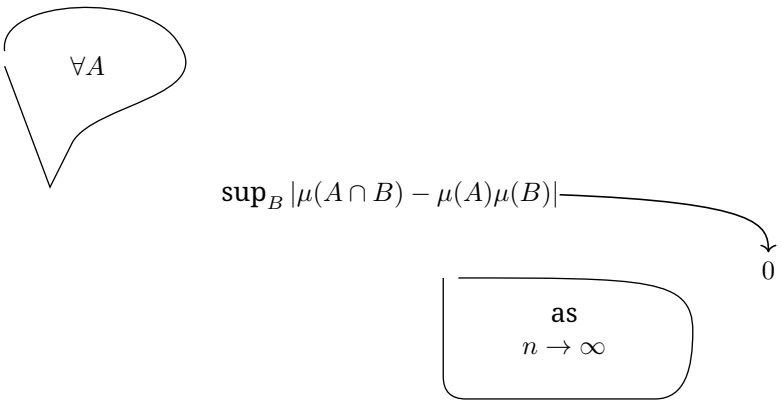
Undeciphered passage. Between “of measure type” and the dash, an entire local passage is enclosed and crossed out; its verbal or mathematical content is unrecovered.

— this is the cond.

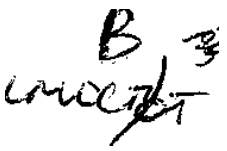


Original source details: GELFAND-P000319

Diagram and typed description.



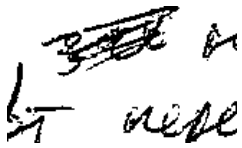
(a cond. of asympt. independence type)



Original source details: GELFAND-P000319

Uncertain reading. The ending of “независимости” (independence) has a dark overlap with a long lower stroke and an upper horizontal line. The upper-right curved stroke is provisionally rendered as a closing parenthesis; the earlier ending is unrecovered.

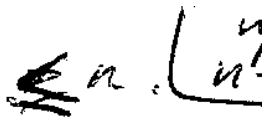
B, [a short word is cancelled; candidate “зав.” (depends)?]



Original source details: GELFAND-P000319

Uncertain reading. After B and before “не зав.” (does not depend), several horizontal and oblique strokes cross a short entry. The candidate “зав.” (depends) remains uncertain.

does not depend on variables numbered $\leq n$.



Original source details: GELFAND-P000319

Uncertain reading. The sign before n is heavily reinforced; an upper angle and two lower strokes are visible. Its final reading is rendered as $\leq$; earlier strokes are undetermined.

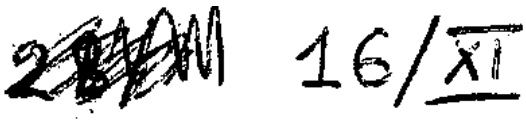
(the first n coords. take $\forall$ values)

Products of measures with prob. $\geq c > 0$ satisfy this cond.

State obtained from a Hamiltonian — a factor.

. . . . . .

GELFAND-P000320



Original source details: GELFAND-P000320

Uncertain reading. The cancelled date begins with 2, followed by dense strokes and tall loops. The day is unrecovered. A separate 16/XI is written at the right.

16/XI. Kazhdan.

Original source details: GELFAND-P000320

Uncertain reading. In the left margin below the cancelled date and to the left of “Consider”, an isolated dense mark has no established verbal reading.

Consider $A^n \setminus \{0\} = X$.

$$H^i(X, O) = ?$$

$$A^n = \text{Spec}\, A[x_1, \ldots, x_n].$$

$\{0\} = (x_1, \ldots, x_n)$ and all prime ideals containing it.

(A — a field [← / —?] it is max and, hence, only itself)

Original source details: GELFAND-P000320

Uncertain reading. Two close horizontal strokes lie between “A — a field” and “it is max”. They meet at the left, where no arrowhead is discernible.

Removing a closed subset — an open one remains.

Restr. of a sheaf to any open set is defined.

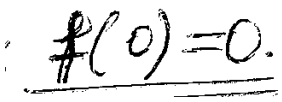
(Open sets in X — those that are open in A^n).

An open set in X — (basis)

$$U_f \cap (U_{x_1} \cup \ldots \cup U_{x_n}).$$

$(\mathbf{C}[x_1, \ldots, x_n])$.

Open sets U_f correspond to those f, for which $f(0) = 0$.

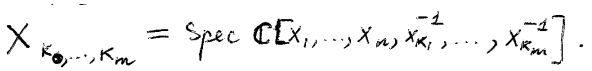


Original source details: GELFAND-P000320

Uncertain reading. The letter before $(0) = 0$ is reinforced and crossed by additional strokes; the final loop is provisionally rendered as f. Earlier strokes cannot be separated.

$$X = \bigcup_{i=1}^{n} X_i, \qquad X_i = U_{x_i} = \operatorname{Spec} \mathbf{C}[x_1, \ldots, x_n, x_i^{-1}].$$

$$X_{k_0,\ldots,k_m} = \operatorname{Spec} \mathbf{C}[x_1, \ldots, x_n, x_{k_1}^{-1}, \ldots, x_{k_m}^{-1}].$$



Original source details: GELFAND-P000320

Manuscript notation. On the left the first subscript is k_0. The first inverted element on the right is x_{k_1}.

m-dim. chains — sections on $X_{k_0,\ldots,k_m}$.

$$a_{k_0,\ldots,k_m} = \frac{P(x_1, \ldots, x_n)}{(x_{k_0} \ldots x_{k_m})^N};$$

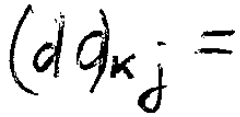
H^0:

Original source details: GELFAND-P000320

Uncertain reading. After H^0, a cancelled entry resembles a_{k_0}; the lower rounded sign and the strokes beneath the cancellation do not establish the reading conclusively.

$$a_k = \frac{P_k(x)}{x_k^{e_k}}.$$

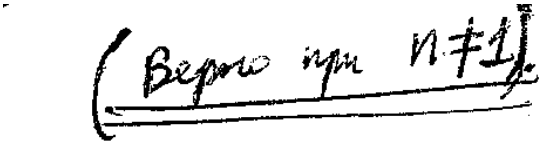
$$(da)_{kj} = \frac{P_k(x)}{x_k^{e_k}} - \frac{P_j(x)}{x_j^{e_j}} = 0 \qquad \forall k, j.$$



Original source details: GELFAND-P000320

Uncertain reading. The second sign in $(d \ldots)_{kj}$ has a loop and a right stroke close to the closing parenthesis; candidates are $(da)_{kj}$, $(dd)_{kj}$, $(dg)_{kj}$. The first candidate is provisionally shown in the formula.

Hence a_k — a polynomial for all k.



Original source details: GELFAND-P000320

Diagram and typed description.

True for $n \neq 1$!

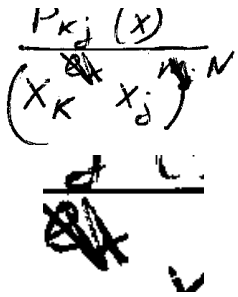
GELFAND-P000321

[source number: 79]

79

H^1:

$$a_{kj} = \frac{P_{kj}(x)}{(x_k x_j)^N}.$$



Uncertain reading. Earlier separate powers above x_k and x_j are cancelled; candidates e_k, e_j and n_k, n_j remain undetermined. The common exponent N at the right is clear.

Original source details: GELFAND-P000321

Boundary:

$$\frac{P_{kj}}{(x_k x_j)^N} + \frac{P_{js}}{(x_j x_s)^N} + \frac{P_{sk}}{(x_s x_k)^N} = 0.$$

$$\boxed{P_{kj}x_s^N + P_{js}x_k^N + P_{sk}x_j^N = 0}.$$

Bound.

$$a_{kj} = \frac{Q_k}{x_k^N} - \frac{Q_j}{x_j^N} \quad \#;$$

Manuscript notation. A short crosshatched mark precedes the semicolon, rendered as #.

Original source details: GELFAND-P000321

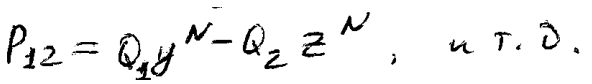
Need: $P_{kj} = Q_k x_j^N - Q_j x_k^N$.

Concretely. there are 3 polyns. in 3 vars.

$$P_{12}x^N + P_{23}y^N + P_{31}z^N = 0.$$

Then $\exists Q_1, Q_2, Q_3$

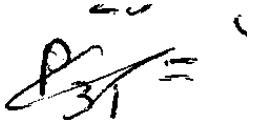
$$P_{12} = Q_1 y^N - Q_2 z^N, \quad \text{and so on.}$$



Manuscript notation. The first term has y^N, the second z^N.

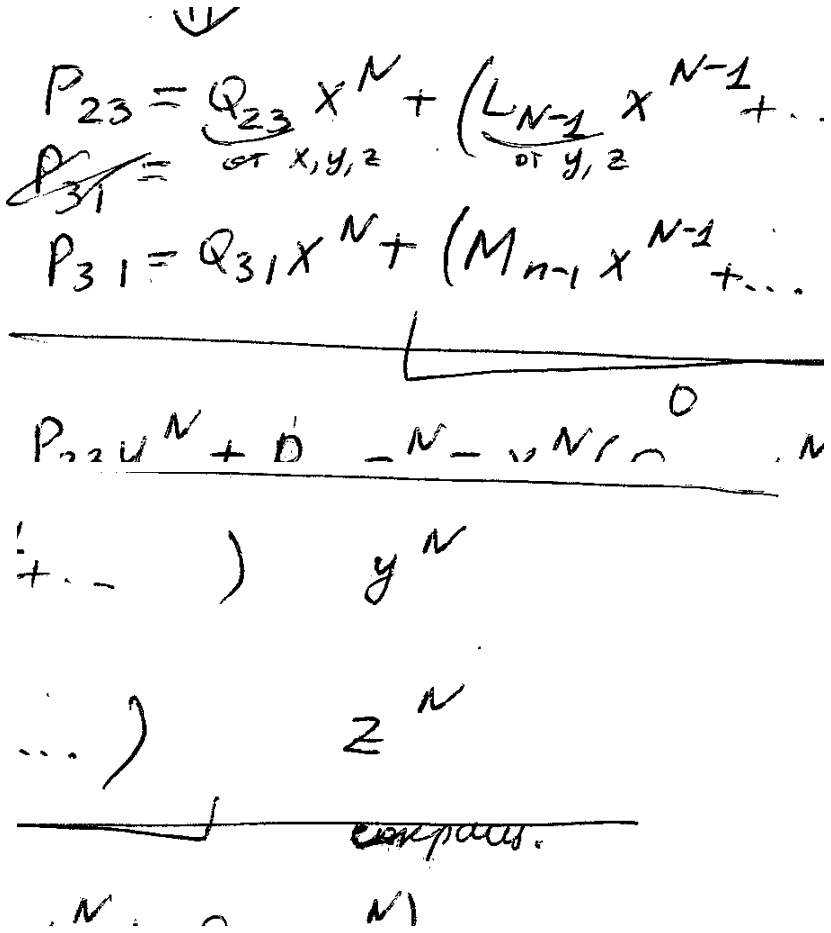
Original source details: GELFAND-P000321

⇓



Uncertain reading. Between the P_{23} and P_{31} rows, a short entry with subscript 31 and an equals sign is cancelled. The candidate $P_{31} =$ is rendered provisionally; long strokes cross its initial loop.

Original source details: GELFAND-P000321



Original source details: GELFAND-P000321

Diagram and typed description.

$P_{23} =$ $\underbrace{Q_{23}}_{\text{in } x,y,z}\, x^N + \left(\underbrace{L_{N-1}}_{\text{in } y,z}\, x^{N-1} + \ldots\right) y^N$

~~$P_{31} =$~~

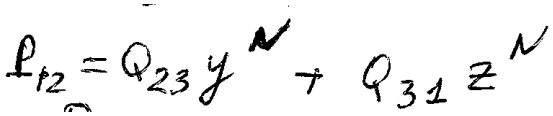
$P_{31} = Q_{31} x^N + (M_{n-1} x^{N-1} + \ldots) z^N$

cancel.

0

$$P_{23}y^N + P_{31}z^N = x^N(Q_{23}y^N + Q_{31}z^N).$$

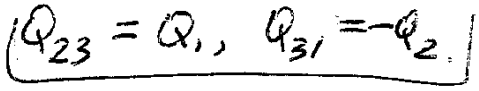
$$\Downarrow$$



Original source details: GELFAND-P000321

Uncertain reading. The first letter has a loop and a separate lower crossbar. Provisionally P_{12}, possibly f_{12}. No leading minus is visible before either term on the right.

$$P_{12} = Q_{23}y^N + Q_{31}z^N;$$

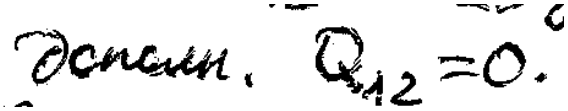


Original source details: GELFAND-P000321

Diagram and typed description.

$$Q_{23} = Q_1, \quad Q_{31} = -Q_2.$$

Complete [reading uncertain]. $Q_{12} = 0$.



Original source details: GELFAND-P000321

Manuscript notation. Before $Q_{12} = 0$ the word begins «Доп...», provisionally read as «Дополним» (“complete”).

Subtract the corresponding cobound. Cocycles with 2 terms remain.
And it now suffices to consider the problem with $\boxed{P_{12} = 0}$.

GELFAND-P000322

H^1 is also easy in the general case.
H^2 is harder. (for $n > 3$).

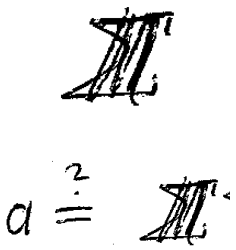
Original source details: GELFAND-P000322

Uncertain reading. An isolated reinforced letter above H^{n-1}: candidate a; additional upper and lower loops prevent recovery of the earlier letter.

H^{n-1}

$$X = \bigcup_{i=1}^{n} X_i, \qquad a = \frac{P(x)}{(x_1 \dots x_n)^N}.$$

$$a \stackrel{?}{=} \text{[cancelled sign]} \frac{\sum P_i(x) x_i^N}{\prod x_i^N}.$$

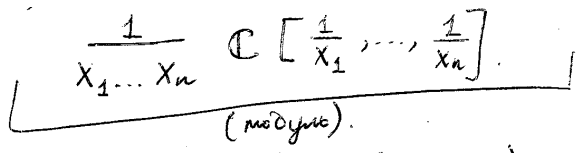


Original source details: GELFAND-P000322

Uncertain reading. Between $a \stackrel{?}{=}$ and the fraction, a large sign with two horizontal ends is cancelled: candidates $\prod$ and $\sum$ remain undetermined. Above the equality, a separate dot below the upper curved stroke forms a question mark.

$$\frac{1}{(x_1 \dots x_n)^N} \quad \text{— not representable} \quad \forall N.$$

Answer:



Original source details: GELFAND-P000322

Diagram and typed description.

$$\boxed{\frac{1}{x_1 \dots x_n} \mathbb{C}\left[\frac{1}{x_1}, \dots, \frac{1}{x_n}\right]}$$

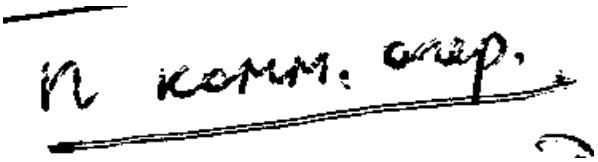
(module)

A module over $\mathbb{C}[x_1, \dots, x_n]$ (over H^0).

$$a_{1\dots n} = \frac{P(x)}{(x_1 \dots x_n)^N}.$$

---

n comm. opers. — the resolv. gives nothing (as for one) —
— localiz. in 1 var. is simple, but not here.



Original source details: GELFAND-P000322

Uncertain reading. After «n комм.» the abbreviation likely reads «опер.» (“operators”); two close strokes precede the final «р».

Original source details: GELFAND-P000322

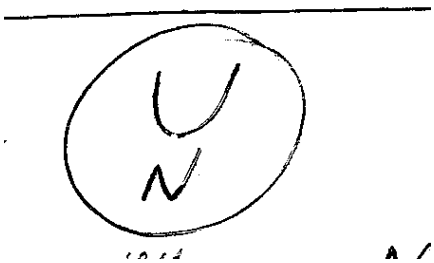
Uncertain reading. Above $\Longleftrightarrow$, a small enclosed sign is crossed by oblique strokes and the preceding paragraph’s line. The inner glyph is unrecovered; the displayed ? is only a candidate.

$a_{1\dots n}$ — bound. $\not{?}$ $\Longleftrightarrow$ [unreadable sign] $P(x) = \sum P_i(x) x_i^N$.

Uncertain reading. Before $P(x)$ is a looped mark followed by a separate short stroke. A possible reading is a with a comma; the loop is overwritten.

Original source details: GELFAND-P000322

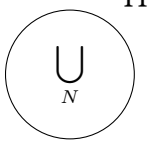
Those with degree $< N$ in every variable remain in the quotient.



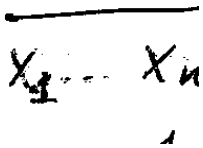
Original source details: GELFAND-P000322

Diagram and typed description.

$$\frac{C + c_i x_i + \ldots + \prod_i x_i^{N-1}}{\prod x_i^N}$$ and



There remains $\dfrac{1}{x_1 \ldots x_n}$ — the highest term. N — arbitrary.



Original source details: GELFAND-P000322

Uncertain reading. In the first lower index of the denominator $x_1 \ldots x_n$, a reinforced 1 has an additional loop and lower strokes; the final index is rendered as 1, while earlier strokes are unrecovered.

hence the answer:

$$\frac{1}{x_1 \ldots x_n} \mathrm{C}\left[\frac{1}{x_1}, \ldots, \frac{1}{x_n}\right].$$

GELFAND-P000323

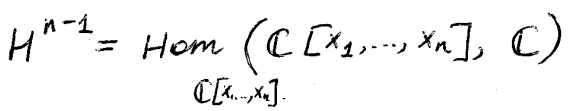
[source number: 80]

80

A module over H^0 — mult. by a polyn. and discard integer powers;

$$\left(\text{in } \mathrm{C}\left[\frac{1}{x_1}, \ldots, \frac{1}{x_n}\right]\right).$$

$$H^{n-1} = \mathrm{Hom}_{\mathrm{C}[x_1,\ldots,x_n]}(\mathrm{C}[x_1, \ldots, x_n], \mathrm{C}).$$



Manuscript notation. Under Hom the manuscript writes $\mathrm{C}[x_1, \ldots, x_n]$.

Original source details: GELFAND-P000323

C is a $\mathrm{C}[x_1, \ldots, x_n]$-module (mult. by $f(0)$).

$$\mathrm{Hom}_{H^0}(H^0, \mathrm{C}).$$

Need to construct a map

$$H^0 \times H^{n-1} \longrightarrow \mathrm{C}.$$

(product and coeff. of $\dfrac{1}{x_1 \ldots x_n}$, i.e. the residue)

$$(P, R) = \mathrm{Res}\, PR \quad \text{— — the constant term in what remains.}$$

(It no longer depends on the covering)

i

Original source details: GELFAND-P000323

Uncertain reading. Above the line on duality, an isolated vertical stroke with a dot is provisionally rendered as i; its role and final reading are undetermined.

connection — Serre's duality theorem.

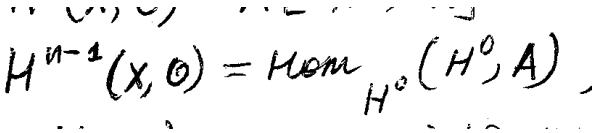
Theorem.

$$A^n = \operatorname{Spec} A[x_1, \ldots, x_n], \qquad \{0\} = (x_1, \ldots, x_n).$$

A — a module over $A[x_1, \ldots, x_n]$.

$$H^0(X, O) = A[x_1, \ldots, x_n],$$

$$H^{n-1}(X, O) = \operatorname{Hom}_{H^0}(H^0, A),$$



Original source details: GELFAND-P000323

Manuscript notation. In this formula the manuscript writes H^0 under Hom.

$$H^i(X, O) = 0 \text{ for } i \neq 0, n-1.$$

Next time — project. space.
(the Koszul complex).

Original source details: GELFAND-P000323

Manuscript notation. The vowel after the initial «К» is a closed loop, «Кошуля» (Koszul).

GELFAND-P000324

22/XI. Plan for next semester.

1) Arnold — [cancelled: new?] Griffiths.

Original source details: GELFAND-P000324

Uncertain reading. Between Arnold and Griffiths, a cancelled word resembles "новый" (new); its dense ending remains undetermined.

($\int$ of rat. funcs. on various cycles — many vars.)

$$\int \frac{P(z_1, \ldots, z_n)}{Q(z_1, \ldots, z_n)}\, dz_1 \wedge \ldots \wedge dz_n$$

Manuscript notation. A small stroke after the first dz merges with the opening of $\wedge$. The subscript 1 is provisional.

Original source details: GELFAND-P000324

over n-dim. cycles (in CP^n).

(Annals, $N = 1$ or 3 for 1969)

2) and Gabrielov — new Griffiths.

3) S. Gelfand — Brauer theory.

4) Sinai, Factors.

Uncertain reading. The labels of items 3 and 4 contain reinforcement and earlier intersecting strokes; the final digits are shown, while earlier forms are undetermined.

Original source details: GELFAND-P000324

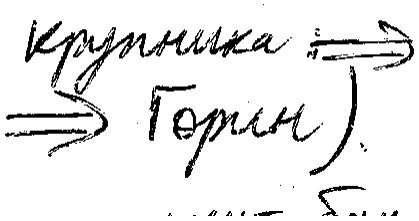
5) A new proof of Grauert's theor. (Knorr).

6) New work of Atiyah (Skew Fredholm opers. and

[cancelled: "work, opr." / "work, oper."?]

Uncertain reading. Under the cancellation, «раб.» ("work") and a final loop with a long descending stroke are visible. The continuation, possibly «опр.» or «опер.», is undetermined.

Original source details: GELFAND-P000324



Original source details: GELFAND-P000324

Diagram and typed description.

parallel $\|$ works of Gohberg and Krupnik $\Rightarrow$)

$\Rightarrow$ Gorin

Gorin.

$$\alpha \xrightarrow{\varphi} \mathbb{C} \quad \text{(hom.)};$$

α — a noncommut. Ban. algebra.

$$J = \bigcap_{\varphi} \ker \varphi,$$

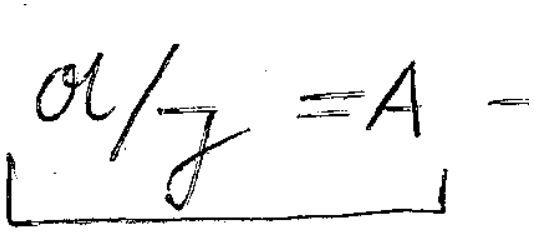


Diagram and typed description.

$\alpha/J = A$ — the algebra of symbols.

Original source details: GELFAND-P000324

Example. α — the alg. of all bounded opers. $\Rightarrow J = \alpha$ (no symbols).
$\mathcal{E}(A)$ — the gr. of invertible elements in A.

$$\alpha \xrightarrow{\psi} A, \qquad \psi^{-1}(\mathcal{E}(A)) \ni T \text{ — f-operators.}$$

$$\psi(T) \in \mathcal{E}(A), \qquad \mathcal{E}_1(A) \text{ — the conn. comp. of the identity,}$$

$$\operatorname{ind} T \in \mathcal{E}(A)/\mathcal{E}_1(A) = H^1(\mathfrak{M}_A, \mathbf{Z}).$$

(this is not a number).

GELFAND-P000325

[source number: 81]

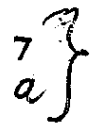
81

On works of Gohberg and Krupnik.

Example. $|\zeta| = 1$, $|\zeta| < 1$. (the $+$ ones).

$$a \longrightarrow T_a, \qquad (T_a f) = (af)_+.$$

Consider $\alpha = \{T_a\}^{?}$ generated by these operators.



Uncertain reading. At the upper end of the closing brace in $\{T_a\}$, an additional curl and separate dots are visible. The displayed superscript ? is only a candidate; an independent sign and earlier strokes remain undetermined.

Original source details: GELFAND-P000325

Δ — the unit disc. Opers. on $H_2(\Delta)$ (holom. inside).

$$\sum_{n=0}^{\infty} c_n z^n, \qquad \sum |c_n|^2 < \infty.$$

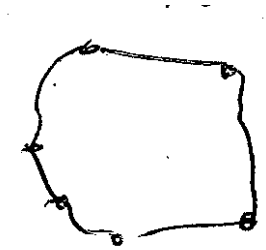
a — a piecewise cont. func.

T_a is Fredholm $\iff$ $\widetilde{a}$ never takes the value 0.

(at discontinuities, join by line segments).
The index — the winding number.

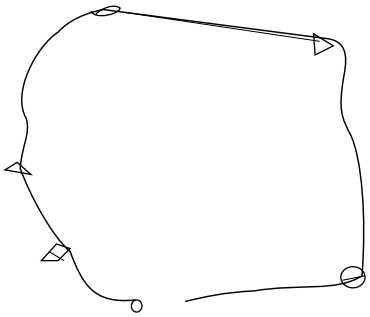
Manuscript notation. A loop is visible at the left edge of the gap, with no separate label.

Original source details: GELFAND-P000325



Original source details: GELFAND-P000325

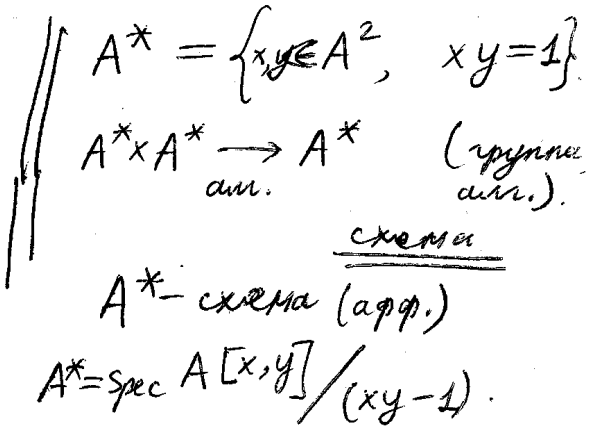
Diagram and typed description. An irregular contour with an almost straight top, curved sides and a lower gap; the round and triangular strokes are rendered by their visible shapes. A small loop is visible at the left edge of the gap; it may be the numeral 0. The directions of all small strokes are not established; the gap is not joined.



it does not
pass through
0.

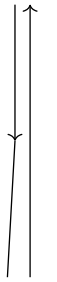
Arnold. On vector fields in general position.
Kazhdan. $A^{n+1} - \{0\}$.
A^* acts.

$$P^n = A^{n+1} - \{0\}/A^*.$$



Original source details: GELFAND-P000325

Diagram and typed description.



$$A^* = \{x, y \in A^2,\ xy = 1\}.$$

$$A^* \times A^* \underset{\text{alg.}}{\longrightarrow} A^* \quad \text{(alg. groups)}$$

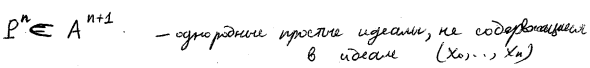
A^* — a scheme (aff.). schemes

$$A^* = \operatorname{Spec} A[x, y]/(xy - 1).$$

A precise definition of P^n.

$$A^{n+1} = \operatorname{Spec} A[x_0, \ldots, x_n].$$

$P^n \in A^{n+1}$ — homogeneous prime ideals, not contained in the ideal $(x_0, \ldots, x_n)$.



Original source details: GELFAND-P000325

Manuscript notation. The first sign after P^n has three horizontal endings. The text says “not contained in the ideal”.

Or:

Original source details: GELFAND-P000325

Uncertain reading. A long narrow mark below “Or” is hatched over, with earlier content undetermined. The sign after P^n opens to the right with no middle stroke.

$P^n \subset A^{n+1} \setminus \{0\}.$

homog. ideals. — invariant under similarity transformations.

(the set of lines — they are points, but not closed)

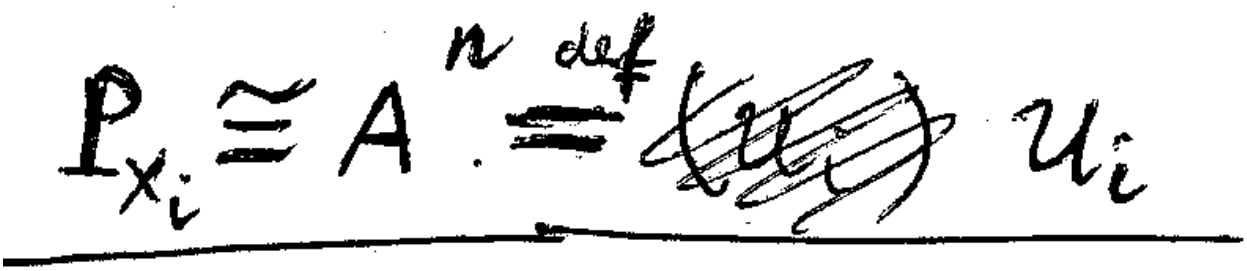
A basis of open sets: $f \in A[x_0, \ldots, x_n]$ — a homog. func.,

P_f — prime homog. ideals not containing this func.:

$\Gamma(P_f, O_P) = (A[x_0, \dots, x_n]_f)_0$ — elems. of 0 degree of homogeneity.

GELFAND-P000326

Assert:



Original source details: GELFAND-P000326

Uncertain reading. Before the final U_i, a cancelled parenthesized entry is provisionally rendered as (U_i); crossing strokes obscure the lower sign, which remains uncertain.

$$P_{x_i} \cong A^n \overset{\text{def}}{=} \cancel{(U_i)}\ U_i.$$

On P there are sheaves $O(k)$: ($k \in$ Z).

Original source details: GELFAND-P000326

Uncertain reading. In the first $O(k)$, the parenthesized letter is heavily reinforced; its final form is rendered as k, while earlier strokes cannot be separated.

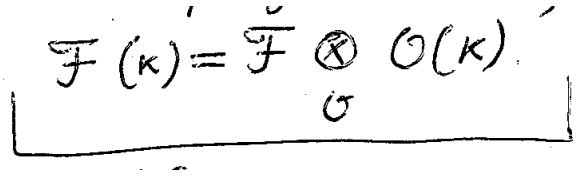
$\Gamma(P_f, O(k)) = (A[x_0, \dots, x_n]_f)_k$ — homog. degree k funcs..

$O(1)$ — the Hopf sheaf; $O(0)$ — the struct. sheaf.

A coherent sheaf: loc. finitely

$$O^J \longrightarrow O^I \longrightarrow \mathcal{F} \longrightarrow 0, \qquad J, I \text{ — finite.}$$

$\mathcal{F}$ — a coherent sheaf on P^n,
Consider



Original source details: GELFAND-P000326

Diagram and typed description.

$\mathcal{F}(k) = \mathcal{F} \otimes_O O(k)$ $\qquad O = O(0).$

Serre's theorem:

1) $\forall i \quad H^i(P, \mathcal{F})$ — A-modules of finite type;

2) (obvious). $H^i(P, \mathcal{F}) = 0$ for $i > n$;

3) $\exists k_0$, such that for $k > k_0$ $H^i(P, \mathcal{F}(k)) = 0$ for [cancelled sign] $i \neq 0$.

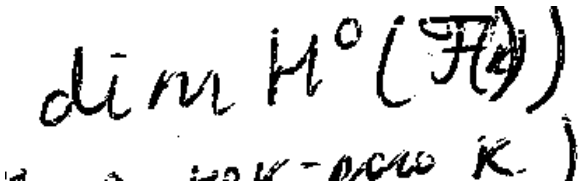
Original source details: GELFAND-P000326

Uncertain reading. Before the clear $i \neq 0$, a small sign is densely crossed by horizontal and diagonal strokes; the earlier glyph is undetermined.

Let A — a field. Define

$$\chi(F) = \sum_i (-1)^i \dim H^i(\mathcal{F}).$$

Theorem: $\forall$ coherent $\mathcal{F}$, $\chi(\mathcal{F}(k))$ — a polynomial in k.
(Serre).
(Hilbert: $\dim H^0(\mathcal{F}(k))$).

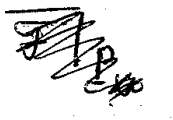


Original source details: GELFAND-P000326

Uncertain reading. After $\mathcal{F}$ are a tight loop and two right-hand curves, provisionally $\mathcal{F}(k)$. The bottom of the loop is overwritten.

(starting from certain k.)

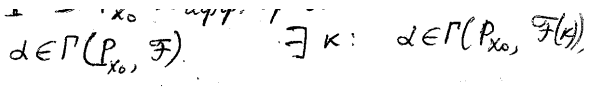
The main lemma for Serre's theor.:
$\forall \mathcal{F}\ \exists k_0$: for $k \geq k_0$ $\mathcal{F}(k)$ is generated by its global sections
(over O).
Explanation. $P^n \supset P_{x_0}$ — aff. space.
Consider [cancelled: $\mathcal{F}|_{P_{x_0}}$?].



Original source details: GELFAND-P000326

Uncertain reading. After "Consider", $\mathcal{F}$, a vertical stroke and a lower P with a dense x_0-like subscript are cancelled. The candidate $\mathcal{F}|_{P_{x_0}}$ is provisional; part of the subscript beneath the strokes cannot be separated.

$$\alpha \in \Gamma(P_{x_0}, \mathcal{F}). \qquad \exists k: \quad \alpha \in \Gamma(P_{x_0}, \mathcal{F}(k)).$$



Original source details: GELFAND-P000326

Manuscript notation. Both occurrences of Γ have P_{x_0} as their first argument.

GELFAND-P000327

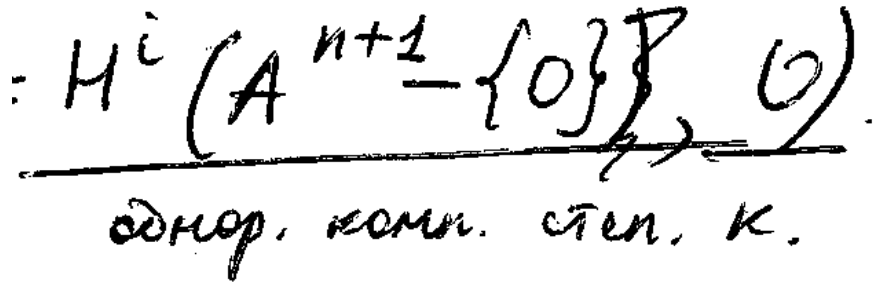
[source number: 82]

that α extends to a sect. over the whole ring.

Original source details: GELFAND-P000327

Uncertain reading. The tall stroke and closed loop resemble the α in the last line of the previous frame. The reading here is provisionally α.

Idea: the sections are simply related
sects. of $\mathcal{F}(k)$ — sects. of $\mathcal{F}$, with a pole of order k allowed.
(mult. by x_0 — extension to global sections).
Transition to the proof of Serre's theorem.
$H^i(O(k))$ have already been computed.



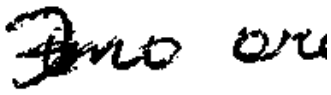
Original source details: GELFAND-P000327

Uncertain reading. Between $\{0\}$ and the comma is a tall mark with an upper loop and a descending stroke. The mark is unresolved.

$$\sum_{k\in Z} H^i(O(k)) = H^i\left(\sum_k O(k)\right) = \underline{H^i(A^{n+1} - \{0\}, O)}$$

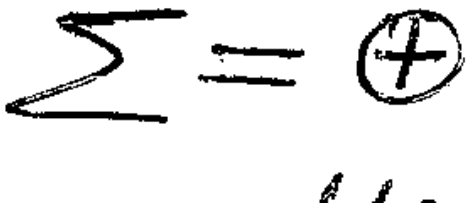
homog. comp. of degree k

This is obvious.



Original source details: GELFAND-P000327

Uncertain reading. The initial "Э" in "Это" (This) is heavily reinforced by transverse and looped strokes; the earlier short sign beneath them is undetermined.



Original source details: GELFAND-P000327

Diagram and typed description.

$\sum = \oplus$.

$P^n = \bigcup U_i$ and the Čech complex. One obtains homog. funcs. of all degrees.

$\mathcal{F}(k)$ is generated by global sects., i.e. $\exists$ an epimorphism

Original source details: GELFAND-P000327

Uncertain reading. A separate dot is visible below the start of the final arrow to 0; its punctuation value is undetermined. The dot and double underline are retained.

$$O^N \longrightarrow \mathcal{F}(k) \longrightarrow 0.$$

For O^N Serre's theorem is checked, and now $O^N \longrightarrow \mathcal{F}(k) \longrightarrow 0$.

From this follows Serre's theorem in the general case.

item 2 is obvious (there are no chains).

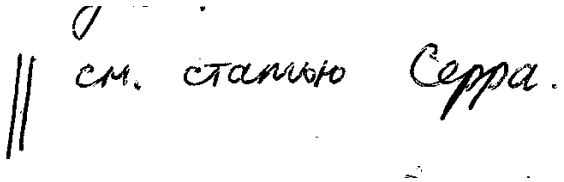


Diagram and typed description.

see Serre's article.

Original source details: GELFAND-P000327

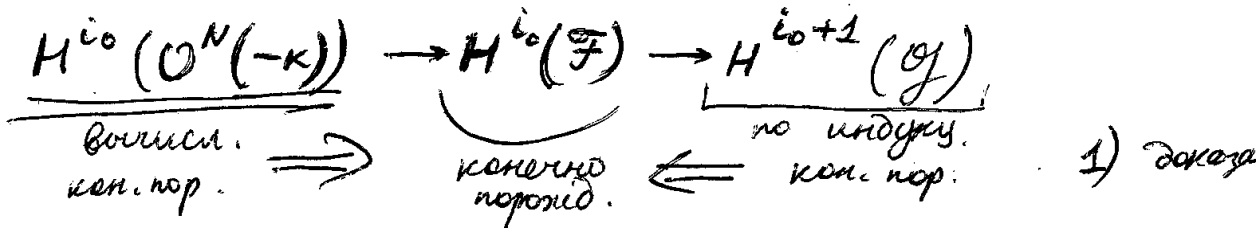
Next, induction downwards in dimensions. (simult. 1 and 3).

$i > n$ is proved.

Suppose proved for $i > i_0$.

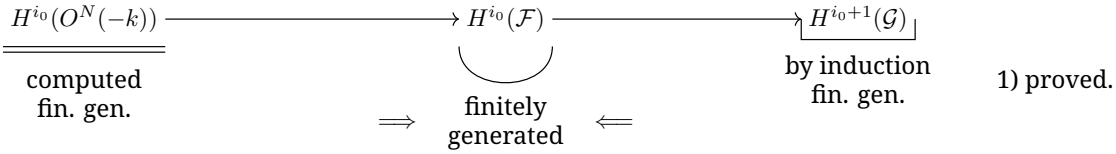
$$0 \longrightarrow \mathcal{G} \longrightarrow O^N(-k) \longrightarrow \mathcal{F} \longrightarrow 0.$$

write the exact sequence.



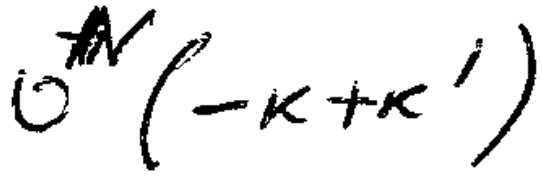
Original source details: GELFAND-P000327

Diagram and typed description.



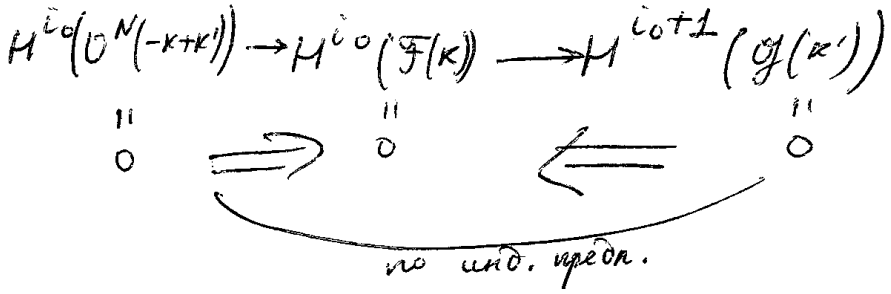
1) proved.

$$0 \longrightarrow \mathcal{G}(k') \longrightarrow O^N(-k+k') \longrightarrow \mathcal{F}(k') \longrightarrow 0.$$



Manuscript notation. The superscript on O reads N over thick strokes. The underlying form is unresolved.

Original source details: GELFAND-P000328



Original source details: GELFAND-P000328

Uncertain reading. The middle term is written $\mathcal{F}(k)$; no separate prime is visible on k, unlike k' above and at the right. The reading is retained with this uncertainty.

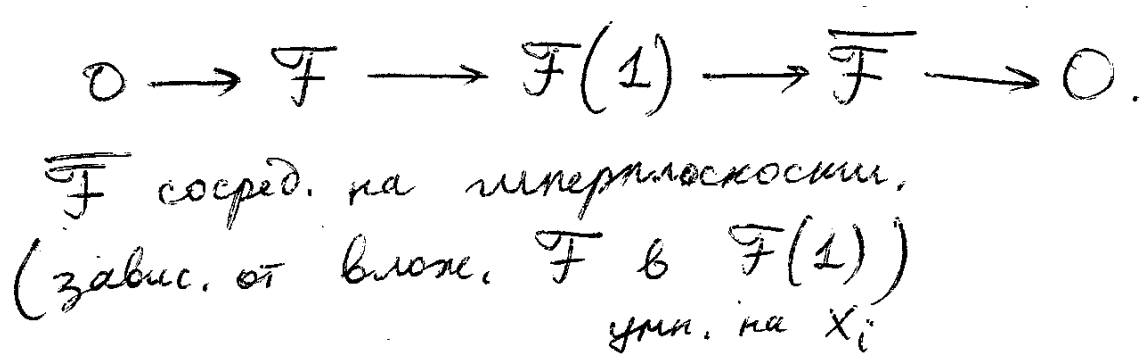
$$\begin{array}{ccccc} H^{i_0}(O^N(-k+k')) & \longrightarrow & H^{i_0}(\mathcal{F}(k)) & \longrightarrow & H^{i_0+1}(\mathcal{G}(k')) \\ \| & & \| & & \| \\ 0 & \Longrightarrow & 0 & \Longleftarrow & 0 \end{array}$$

by ind. hyp.

Corollary: ind. on dim of proj. sp.

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}(1) \longrightarrow \overline{\mathcal{F}} \longrightarrow 0.$$

$\overline{\mathcal{F}}$ concentr. on a hyperplane,



Original source details: GELFAND-P000328

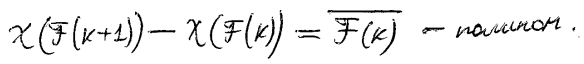
Diagram and typed description.

(depends on the emb. of $\mathcal{F}$ in $\mathcal{F}(1)$)

mult. by x_i

$$0 \longrightarrow \mathcal{F}(k) \longrightarrow \mathcal{F}(k+1) \longrightarrow \overline{\mathcal{F}}(k) \longrightarrow 0.$$

$$\chi(\mathcal{F}(k+1)) - \chi(\mathcal{F}(k)) = \overline{\mathcal{F}(k)} \quad \leftarrow \quad \text{polynomial.}$$

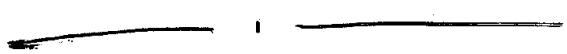


Original source details: GELFAND-P000328

Uncertain reading. After the equality there is no visible χ before $\mathcal{F}(k)$; a long upper stroke extends over this notation. The overlined notation and the left-pointing arrow from "polynomial" are retained.

hence $\chi(\mathcal{F}(k))$ — a polynomial.

For a point everything is trivial. (initial — 0th dim.)



Original source details: GELFAND-P000328

Diagram and typed description.

GELFAND-P000329

[source number: 83]

83

8/II-71 I. M. Gelfand.

Bott. Tang. bund. to X, integr. subbund. in TX;

Original source details: GELFAND-P000329

Undeciphered passage. At the left, before "Suppose", a short word with tall oblique strokes is crossed out. The text of this cancelled word is unrecovered.

Suppose there is a bundle: $\exists$ an integrable foliation homotopic to it?

Bott: example of one not homotopic to an integrable one.

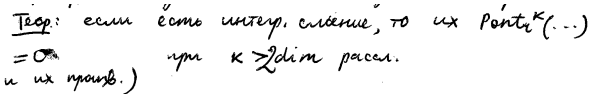
Original source details: GELFAND-P000329

Uncertain reading. In the “Bott: example” line, the provisional reading is “не гомотопного интегрируемому” (not homotopic to an integrable one). The stem “гомотопн” (homotop-) is visible, but the final word ending is heavily overwritten; earlier and latest strokes are not confidently separated. The historical alternatives “не гомотопн...” and “интегрируемых” are retained in the reading records.

There is already an example on $\mathbb{C}P^2$ — obtained using Pontr. cls.
Thm.: if there is an integr. foliation, then their

$$\text{Pontr}^k(\dots) = 0 \qquad \text{for } k > 2\,\text{dim bund.}$$

(and their products)

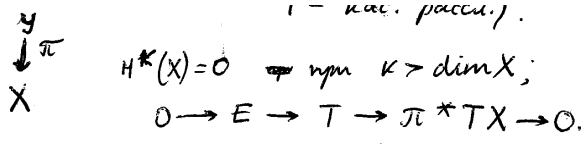


Original source details: GELFAND-P000329

Uncertain reading. The low hook in Pontr is provisionally read as part of the final letter; a separate subscript is unconfirmed. The earlier writing under the heavy strokes at the upper right of 0 is unresolved.

Idea: if the integr. syst. were not from a foliation, but a bundle,
then $\text{Pontr}^k(\ ) = 0$ for $k > \dim T/E$

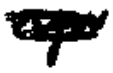
(E — our bund.,
T — tang. bund.).



Original source details: GELFAND-P000329

Diagram and typed description.

$$\begin{array}{c} Y \\ \downarrow \pi \\ X \end{array} \qquad H^k(X) = 0 \qquad \text{for } k > \dim X;$$

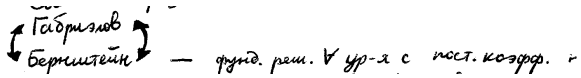


Original source details: GELFAND-P000329

Undeciphered passage. After $H^k(X) = 0$, a short looped word is crossed out. The text of this cancelled word is unrecovered.

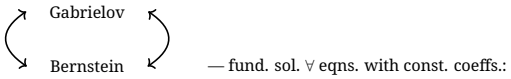
$$0 \longrightarrow E \longrightarrow T \longrightarrow \pi^* TX \longrightarrow 0.$$

and everything from naturality of Pontr. classes.
(Work in the seminar on global analysis, vol. 16).
Sinai.
Seminar plan.
Sinai 3 times.



Original source details: GELFAND-P000329

Diagram and typed description.



1) analytic.

Original source details: GELFAND-P000329

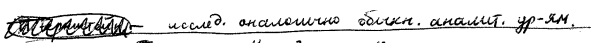
Uncertain reading. After “analytic.” is a crossed entry beginning with shapes resembling “во”; its remaining letters are covered by horizontal strokes. They are not reconstructed from the phrase’s meaning.

outside fin. codim.

2) analyt. cont. to the whole complex $\mathbb{C}^n$.

3) finite dimensionality. (branches — a fin.-dim. sp.)

GELFAND-P000330

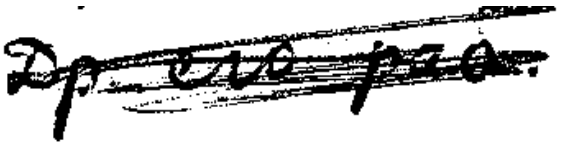


Original source details: GELFAND-P000330

Uncertain reading. The encircled, struck-out and underlined name is shown only as the candidate “Берн…штейн” (Bern…shtein). “Бернштейн” (Bernstein) and other readings of earlier strokes remain possible; historical attribution is not established.

~~Bern…shtein~~ — study analogously to ord. analyt. eqns.

~~Др.… ра~~

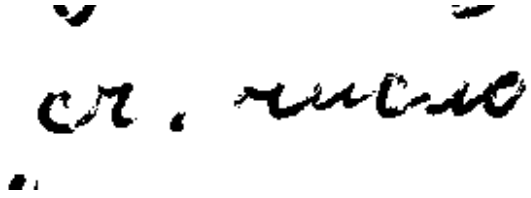


Original source details: GELFAND-P000330

Uncertain reading. A shortened entry before the equation is crossed out. The beginning “Др.” (Dr.) and ending “ра” (ra) are shown provisionally; its middle and complete reading are undetermined.

$$-y'' + x^2 y + \beta x^4 y = \lambda y.$$

As $\beta \to 0$, a [abbrev. «сл.»?] number of 2nd-order branch pts., tending to a 3rd-order branch point.



Original source details: GELFAND-P000330

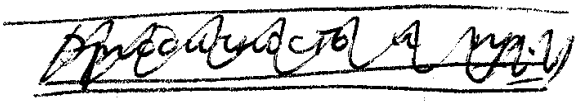
Uncertain reading. Before “number”, the second letter has a slanting leg and a short right-hand ending. Provisionally «сл.»; «сч.» is possible.

Original source details: GELFAND-P000330

Uncertain reading. At the right, in an oval, is a 2-like mark with a separate dot below. Both are shown; their combined meaning or reading as a question mark is not established.

2
.

Semiphys. work.

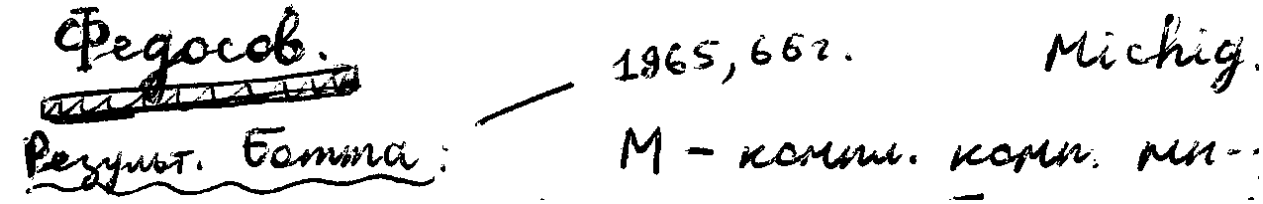


Original source details: GELFAND-P000330

Uncertain reading. The double-underlined entry provisionally reads «Преимущества ур...» (“Advantages of the eq...”). The ending of the last abbreviation is unresolved.

Advantages of the eq...

Fedosov.

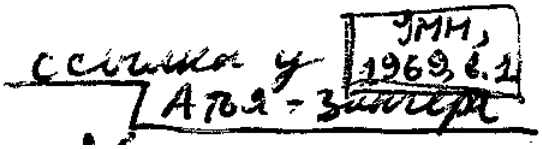


Original source details: GELFAND-P000330

Diagram and typed description.

1965, 66 Michig.

Bott’s result:



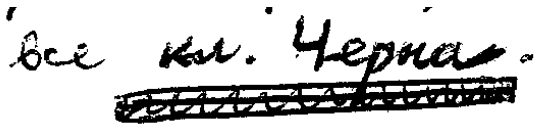
Original source details: GELFAND-P000330

Uncertain reading. In the right margin, “reference in Atiyah–Singer” is marked by an angular frame. Above is a separate box, provisionally “UMN, 1969, 6.1”; the small figures and marks after the year are not entirely certain. The bibliographical entry is not completed.

UMN,
1969, 6.1

reference in
Atiyah–Singer

M — comp. complex manif., X — holom. vect. field on M.
Let $\exists$ a fin. number of points p, where it is 0 (and nondegenerately).



Original source details: GELFAND-P000330

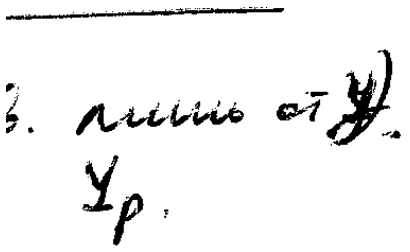
Diagram and typed description.

Using it, one can compute all Chern cls.

$$\Theta(X)Y = [X, Y].$$

Y — another vect. field.

At pt. p a lin. transformation in the fibre.



Original source details: GELFAND-P000330

Uncertain reading. After "on", the upper Y is heavily overwritten; below it is a separate Y_p. Earlier strokes at the upper letter and the value of the closing curve on the right are undetermined. The candidate Y, lower notation and curve are shown.

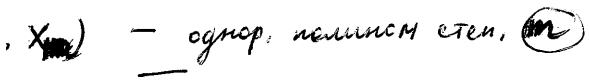
(depends only on Y)

Y_p.

Θ_p — a lin. transf. of the tang. sp.

Nondeg.: Θ_p is nondegenerate. $\forall p$.

Let dim $M = m$,



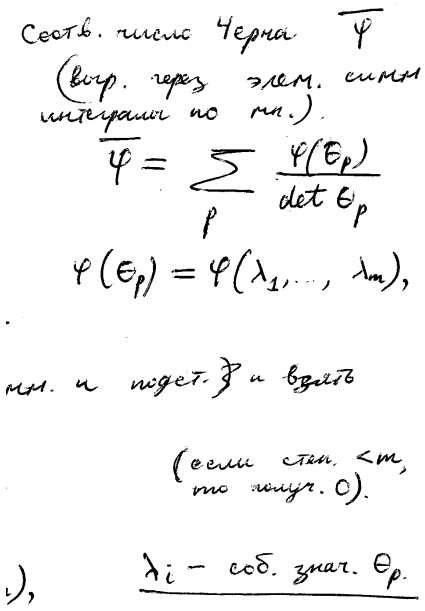
Original source details: GELFAND-P000330

Uncertain reading. The index on the final X is heavily overwritten; it is provisionally shown as m. At the right, the degree, also resembling m, is overwritten and circled. Earlier indices are unrecovered.

$\psi(X_1, \ldots, X_m)$ — a homog. polynomial of degree m

The corr. Chern number $\overline{\psi}$

(expr. through elem. symm. and subst. ξ and take integrals on "м.")



Original source details: GELFAND-P000330

Uncertain reading. After "subst." is a tall shaped sign with an upper loop and lower tail; ξ is shown provisionally, with $\mathcal{F}$ also possible. After "on" is an m-shaped sign and dot, provisionally retained as "м." without normalization to M. The full reading of these signs is undetermined.

$$\overline{\psi} = \sum_p \frac{\psi(\Theta_p)}{\det \Theta_p}.$$

(if the degree $< m$, then one gets 0).

$$\psi(\Theta_p) = \psi(\lambda_1, \ldots, \lambda_m),$$

λ_i — the eigenvalues of Θ_p.

GELFAND-P000331

[source number: 84]

This is an infinites. analogue of the Atiyah–Bott thm.

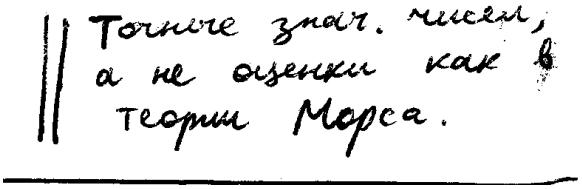
$$(e^{\text{field}} = \text{transf. of a manif. into itself})$$

~~Ka...dan~~

Original source details: GELFAND-P000331

Uncertain reading. Before “There is”, a surname is crossed out; it is provisionally shown as “Ка...дан” (Ka...dan). “Каждан” (Kazhdan) and “Каздан” (Kazdan) remain possible; its middle and ending are hidden by dense horizontal strokes.

There is an analogue for the real case.



Original source details: GELFAND-P000331

Diagram and typed description.

Exact values of the numbers, rather than estimates as in Morse theory.

Next time Sinai and the main res. of Brouwer.

15/II. I. M. Gelfand (work with Kazhdan).

Some notions of diff. geom. and their appl. to computing the cohom. of ∞ Lie algs.

Original source details: GELFAND-P000331

Uncertain reading. At the right edge are the abbreviations for computing cohomology, then a double loop provisionally read as ∞, followed by the abbreviated “Lie algs.”

Fedosov. References 1. Michigan Math. J., 1967, v.14, $N = 2$, 231–244.

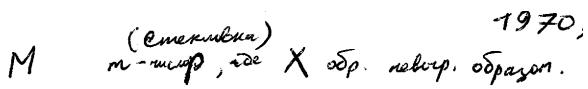
2. J. Diff. Geometry, 1967, v.1 $N = 4$, 311–330.

3. Matem., 14:2, 1970.

4. Baum, Bott, vol. dedicated to de Rham, Springer 1970, 29–47.

(Steklov)

M m-manif., where X [vanishes] nondegenerately.



Original source details: GELFAND-P000331

Uncertain reading. Between the references and the line about M, above “m-manif.”, the parenthesized entry reads “Стеклова” (Steklov). It is retained at this position; its attachment to the references or following line is undetermined.

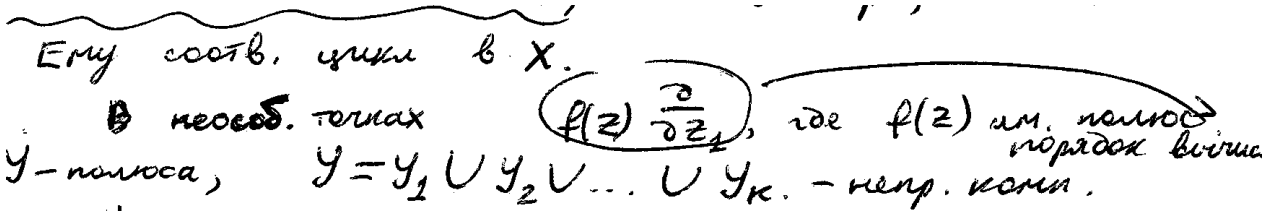
X — a holom. vect. field.

Bernstein. Formula for merom. vect. fields.

V — a merom. vect. field on a comp. analyt. manif. X.

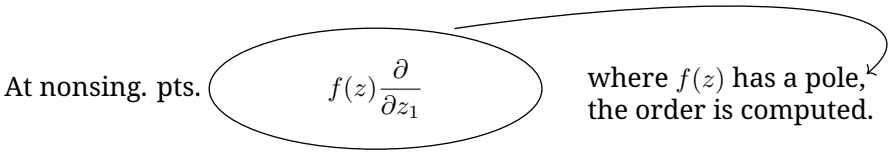
It has a fin. number of nondeg. zeros.

Poles at codim $= 1$, in general, singular.
It has a corr. cycle in X.



Original source details: GELFAND-P000331

Diagram and typed description.



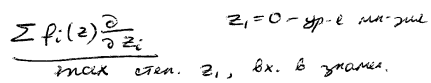
Y — the poles, $Y = Y_1 \cup Y_2 \cup \ldots \cup Y_k$ — irred. comps.
Y_i — cycles.

Cycle of the field: $\sum_{i=1}^{k} \alpha_i Y_i$, where k — the order of the pole of W.

Original source details: GELFAND-P000331

Uncertain reading. The final letter after “pole of” resembles W, possibly a rewritten V. The upper limit k, lower limit $i = 1$, coefficients α_i and following written “k is the order of the pole” are retained.

GELFAND-P000332



Original source details: GELFAND-P000332

Diagram and typed description.

$z_1 = 0$ — eqn. of the manif.

$$\frac{\sum f_i(z)\frac{\partial}{\partial z_i}}{}$$

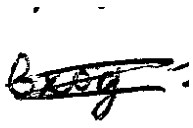
max deg. of z_1 occurring in the denom.

Cycles A_V.
p —

Original source details: GELFAND-P000332

Uncertain reading. The looped sign before the dash has a long descending stroke. p is shown provisionally; ρ is also possible.

~~ВХОД~~



Original source details: GELFAND-P000332

Uncertain reading. The first crossed-out word is provisionally shown as “вход” (vkhod, entry); dense strokes cross its visible letters. Its complete reading remains unsettled.

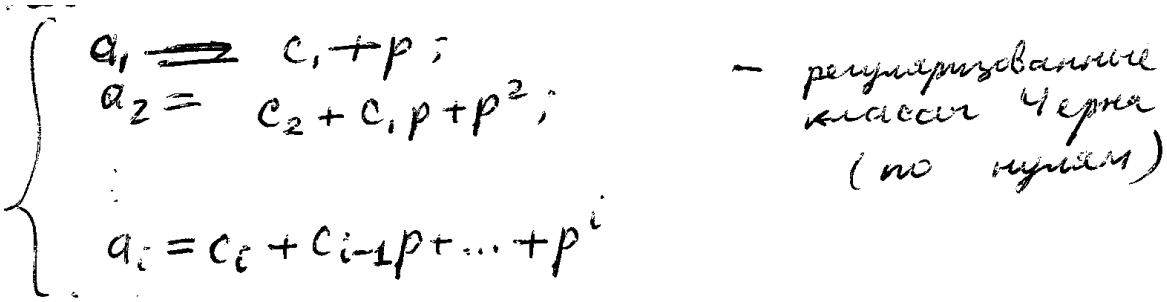
Original source details: GELFAND-P000332

Undeciphered passage. A separate second word follows, heavily crossed out; its text is unrecovered.

by Poincaré duality, 2-dim. cycles.

The right-hand side is the same.

The left-hand side:



Original source details: GELFAND-P000332

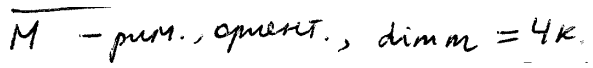
Uncertain reading. After a_1, the overwritten two-stroke sign is shown as $\Rightarrow$; a rewritten equality sign is also possible. The letter on the right is shown as p, with ρ also possible. The shared brace, all three rows and right-hand label are retained.

$$\begin{cases} a_1 \Rightarrow c_1 + p; & \\ a_2 = c_2 + c_1 p + p^2; & \text{— regularized} \\ \quad \vdots & \text{Chern classes} \\ a_i = c_i + c_{i-1}p + \ldots + p^i & \text{(from the zeros).} \end{cases}$$

$\psi(a_1, \ldots, a_m)|_X$.

Fedosov. The real case.

M — Riem., oriented, dim $m = 4k$.



Original source details: GELFAND-P000332

Uncertain reading. The initial M has a capital form; the sign after dim resembles a small m and is provisionally shown that way. The dimensional notation is retained as dim $m = 4k$.

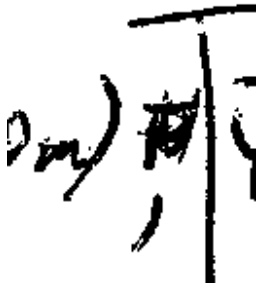
X — a vect. field; $\forall Y, Z$: (vect. fields).

$$([X, Y], Z) + (Y, [X, Z]) = X(Y, Z).$$

Displ. along traj. is isometric to ∞ small order.

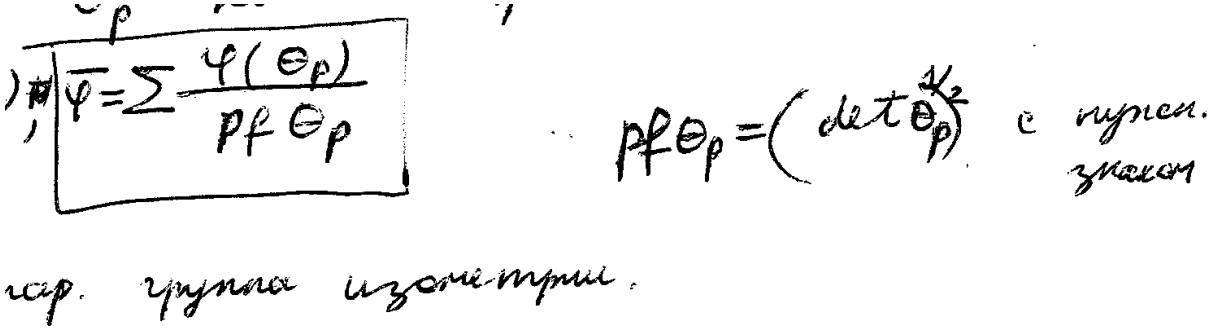
Then Θ_p is skew-symm. at sing. pts.

$\psi(p_1, \ldots, p_m)$,



Original source details: GELFAND-P000332

Uncertain reading. Between the parenthesis and the frame is a densely crossed-out mark. A lower loop and a separate comma are visible. The earlier sign is unresolved.



Original source details: GELFAND-P000332

Diagram and typed description.

$\overline{\psi} = \sum \frac{\psi(\Theta_p)}{\operatorname{pf}\Theta_p}$

$\operatorname{pf}\Theta_p = (\det\Theta_p)^{1/2}$ with the required sign.

One-param. group of isometries.

GELFAND-P000333

[source number: 85]

85

22/II Sinai.

Factors — Naimark.

Next time — Gabrielov.

Arnold.

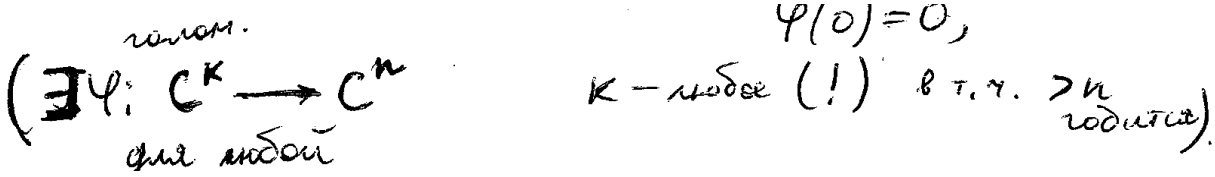
$A(\lambda)$ — a holom. family of matrices.

(or a matr. over the ring of germs of holom. fns.).

A_0 — a fixed matr., $A(\lambda)$ — its deform., $A(0) = A_0$.

Versal family: if obtained under a mapping.

$$B(\lambda) = C(\lambda)A(\varphi(\lambda))C^{-1}(\lambda); \qquad C(0) = 1, \quad \varphi(0) = 0,$$



Original source details: GELFAND-P000333

Diagram and typed description.

holom.

$(\exists\varphi : \mathbb{C}^k \longrightarrow \mathbb{C}^n$ k — any (!) incl. $> n$ works).

for any

With the least no. of parameters. (Universal, but not unique.)

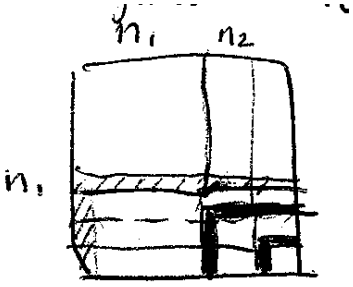
Original source details: GELFAND-P000333

Uncertain reading. The abbreviation “наим.” (least) is shown provisionally; its last letters are heavily overwritten and the earlier entry is unrecovered. The wavy underline of the whole phrase and separate right-hand parenthetical note are retained.

Answer. Versal deform. of A_0.

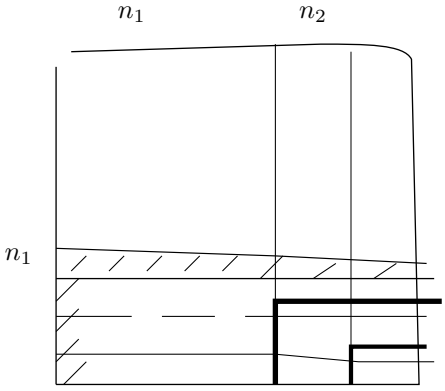
Let A_0 have 1 eigenval. (otherwise separate.)

$n_1 \geq n_2 \geq n_3 \geq \ldots$



Original source details: GELFAND-P000333

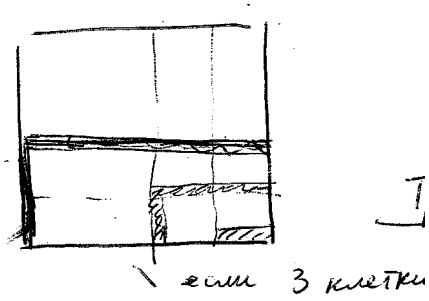
Diagram and typed description. The outer contour, two internal vertical dividers, the upper n_1, n_2 and left n_1 labels, horizontal bands, left diagonal hatching and two dense L-shaped bands at the lower right are retained. The exact cell count and earlier strokes at dense intersections are undetermined.



Insert free params. in the marked cells.

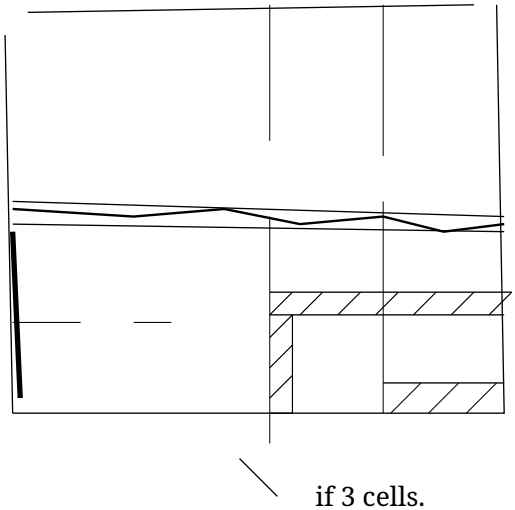
Versal deform.

Row numbered n_1, 1st column down.

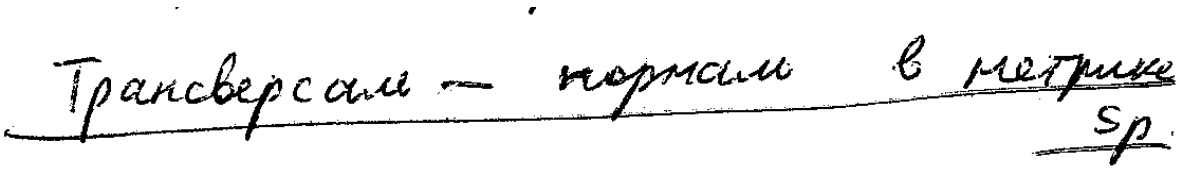


Original source details: GELFAND-P000333

Diagram and typed description. The rectangular contour, two interrupted internal vertical dividers, common dense horizontal band, two lower right hatched bands, left and internal vertical shading, slanted pointer below and label “if 3 cells” are retained. Dense intersections and the exact cell count remain unclear; no distinct arrowhead is visible on the slanted stroke.



$A_0 + \Lambda$,
Λ — free parameters.



Original source details: GELFAND-P000333

Diagram and typed description.

Transversals — normals in the metric

Sp.

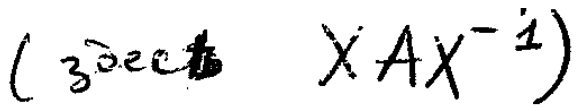
GELFAND-P000334

Sing. in a 3-param. family of matrices in general pos.

Gelfand: question:

the action of $GL(n, \mathbb{C})$ is triv., if locally — a fibration.

Let there be an action (here XAX^{-1}).



Original source details: GELFAND-P000334

Uncertain reading. The last letter in “здесь” (here) is heavily overwritten; the word is shown according to the current entry and the earlier letter is unrecovered.

Original source details: GELFAND-P000334

Uncertain reading. At the left margin, an oval encloses four slanted strokes, with a dense crossbar at the two inner strokes. The sign resembles (H) or (H); its meaning is undetermined.

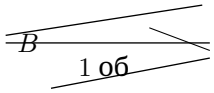
cannot split — essential.

Precisely here — a versal family

describe all types of actions (tensors of rank 15, etc.).

Original source details: GELFAND-P000334

Uncertain reading. A short entry before Ad is crossed out. The left B and small 1, “об” below are shown only as candidates from visible fragments; the full reading is undetermined. The intersecting cancellation strokes are retained.



Ad — the worst action.

Hypothesis: locally there is a fibration without sing.,

in the remainder — Ad — Jordan form.

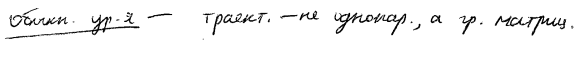
~~Near...~~

Original source details: GELFAND-P000334

Uncertain reading. After “form”, a word is crossed out; its visible beginning is “Близк...” (near...). A line crosses its middle and the ending is unrecovered. The ellipsis marks the unrecovered remainder of the word.

A nearby probl.: a generalized fn., inv. under the action of matrices.

Ordinary eqns. — traject. — not one-param., but a gr. of matrices.



Original source details: GELFAND-P000334

Manuscript notation. The middle abbreviation reads «траект.» (traject.), followed separately by «однопар.» (one-param.).

Chapter 5

GELFAND-P000335

[source number: 1]

Gelfand sem.

1/III — 71

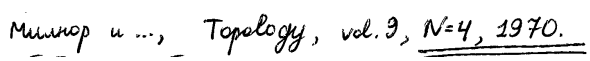
Gabrielov on the art. of Griffiths (Griffits).

On the periods of certain rational integrals

Ann. Math., v. 90, $N = 3$.

I. Bernstein.

Milnor and ..., Topology, vol. 9, $N = 4$, 1970.



Original source details: GELFAND-P000335

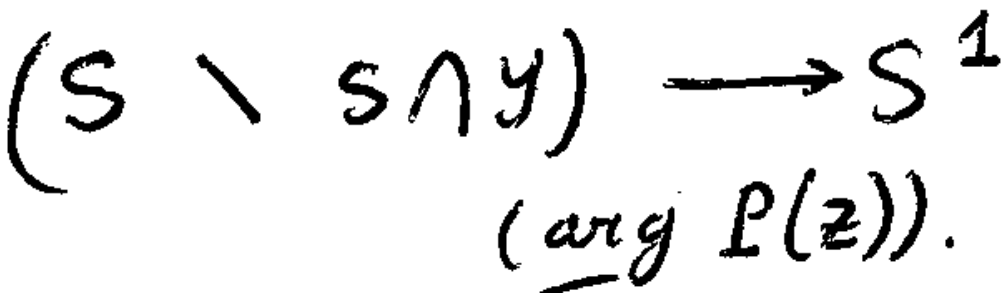
Manuscript notation. Three dots follow "Milnor and".

Generalization of Brieskorn's work.

$P(z)$ — a polynomial, $z \in \mathbb{C}^n$.

0 — a sing. point (isol.) of the surf. $\{P(z) = 0\} = Y$.

Topology of $S \setminus S \cap Y$ (S — a sphere of small rad.).



Original source details: GELFAND-P000335

Diagram and typed description.

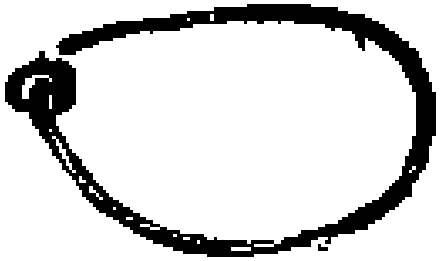
$$(S \setminus S \cap Y) \longrightarrow S^1$$

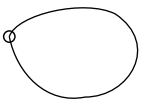
$$(\arg P(z)).$$

We study the fibres and the circuit.

(homeomorphism of the fibre: $T : F \longrightarrow F$).

Diagram and typed description.



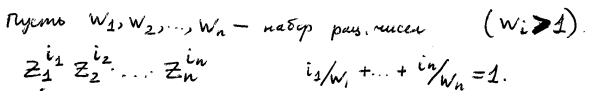


Original source details: GELFAND-P000335

Fibre F: $P(z) = \varepsilon$, ε — a small complex number.

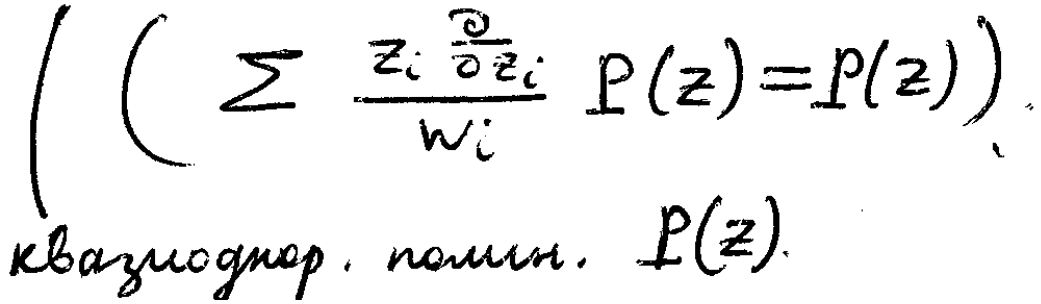
Let $w_1, w_2, \ldots, w_n$ — a collection of rational numbers ($w_i > 1$).

$$z_1^{i_1} z_2^{i_2} \dots z_n^{i_n}; \qquad \frac{i_1}{w_1} + \dots + \frac{i_n}{w_n} = 1.$$



Original source details: GELFAND-P000335

Uncertain reading. Before 1 is a heavy angular mark, provisionally $>$. The letter w in the list and denominators does not clearly distinguish upper from lower case.

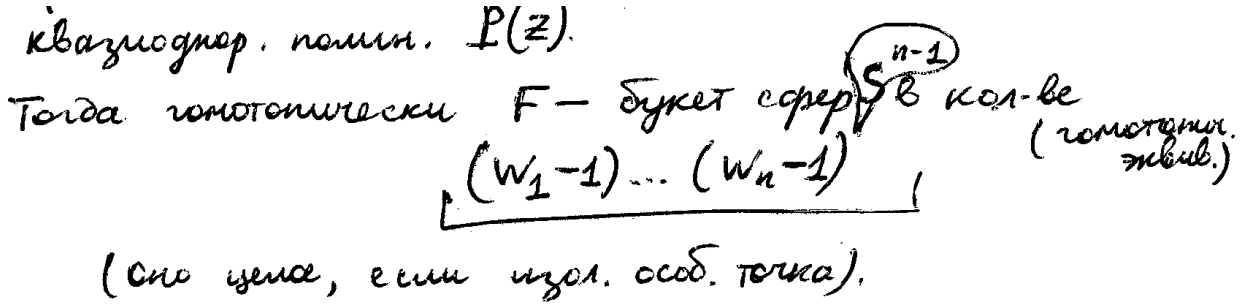


Original source details: GELFAND-P000335

Diagram and typed description.

$$\left(\left(\sum \frac{z_i \frac{\partial}{\partial z_i}}{w_i} P(z) = P(z) \right) \right)$$

quasi-homog. polyn. $P(z)$.



Original source details: GELFAND-P000335

Diagram and typed description.

Then homotopically F — a bouquet of spheres S^{n-1} in the number

$$\underbrace{(w_1 - 1) \dots (w_n - 1)}$$

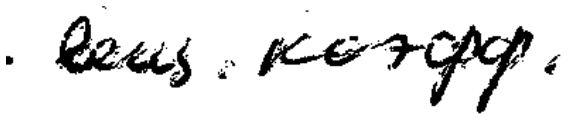
(homotopy equiv.)

(It is an integer if an isol. sing. point).

GELFAND-P000336

The characterist. polynomial of the transfor. T has been computed

(Example: $z_1^2 + \ldots + z_n^2 = \varepsilon$. — one sphere — real coord. and everything retracts onto it).



Uncertain reading. After the abbreviation for "real", the next word reads "coord." with two loops and a dot. Both words are abbreviated.

Original source details: GELFAND-P000336

Brieskorn's case:

$$z_1^{i_1} + \ldots + z_n^{i_n} = \varepsilon.$$

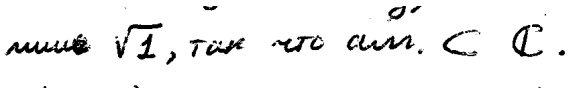
(I. Bernstein: the poles and residues of P^λ are connected with these cycles.)

$$H^{n-1}(F) = \mathbb{Z}^{(w_1-1)\ldots(w_n-1)}.$$

Consider the group ring $\mathbb{Q}(\mathbb{C}^*)$.

$$\sum r_i \langle z_i \rangle, \qquad \langle z_i \rangle \cdot \langle z_j \rangle = \langle z_i z_j \rangle.$$

Instead of $\mathbb{C}^*$ only $\sqrt{1}$ occurs, so alg. $\subset \mathbb{C}$.



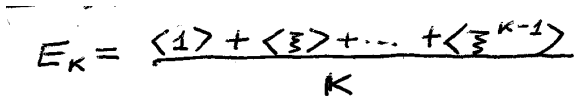
Uncertain reading. Before $\subset$ the abbreviation is provisionally "alg.". The written radical $\sqrt{1}$ has no index.

Original source details: GELFAND-P000336

$$\operatorname{div}(t - d_1)\ldots(t - d_k) = \langle d_1 \rangle + \ldots + \langle d_k \rangle \quad \text{(divisor)}.$$

Divisor of the char. polynomial $\chi(T)$

$$E_k = \frac{\langle 1 \rangle + \langle \xi \rangle + \ldots + \langle \xi^{k-1} \rangle}{k},$$



Uncertain reading. The angular denominator is larger than the subscript on E_k. The reading is provisionally k. Its case is unresolved.

Original source details: GELFAND-P000336

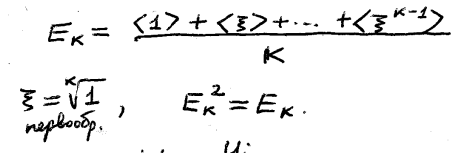


Diagram and typed description.

$\xi \underset{\text{prim.}}{=} \sqrt[k]{1}, \qquad E_k^2 = E_k.$

Original source details: GELFAND-P000336

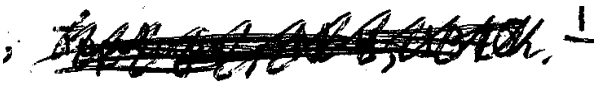
Let $w_i = \dfrac{u_i}{v_i}$ — irreduc.

$$\operatorname{div} \chi(T) = (w_1 E_{u_1} - 1)(w_2 E_{u_2} - 1)\ldots(w_n E_{u_n} - 1).$$

This gives an integral posit. divisor.

2nd work: Connection between Reidemeister tors. and ζ-funct. of the corresp. Laplace oper. on a Riem. manif.

Singer and someone else;



Original source details: GELFAND-P000336

Undeciphered passage. The crossed-out line after “Singer and someone else;” and before “Preprint”: overlapping letters and long strokes; the complete text has not been recovered.

Preprint.

GELFAND-P000337

[source number: 2]

I. M. Gelfand.

$$\alpha = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A, B, C, D \text{ — matrices.}$$

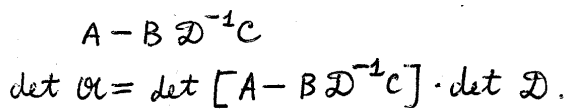
$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \text{ — ?}$$

AD cannot be mult., in diff. spaces, AB, AC do not occur in det.

$$A - B \cdot C. \qquad D \text{ only in the middle.}$$

$A - BD^{-1}C$.

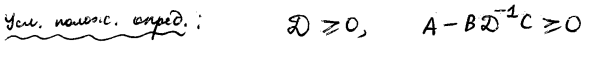
$$\det \alpha = \det[A - BD^{-1}c] \cdot \det D. \qquad (?)$$



Original source details: GELFAND-P000337

Uncertain reading. In the formula for $\det \alpha$, the letter c is shorter than the C on the line above. A separate $(?)$ stands at the right.

Cond. for posit. definiteness: $D \geq 0, \quad A - BD^{-1}C \geq 0.$



Original source details: GELFAND-P000337

Manuscript notation. Both comparisons have a lower stroke under $>$.

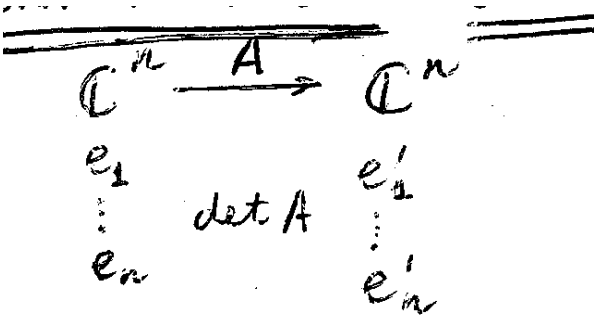
$A - BD^{-1}C$.

S. P. Novikov. On det

Original source details: GELFAND-P000337

Undeciphered passage. A short word between det and “(commut.)” is crossed out; the earlier letters overlap and have not been recovered.

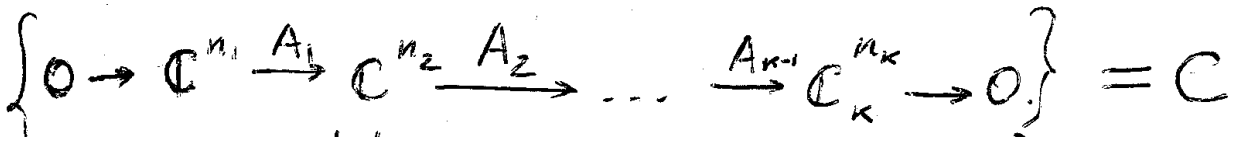
(commut.).



Original source details: GELFAND-P000337

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^n & \xrightarrow{\quad A \quad} & \mathbb{C}^n \\ e_1 & & e'_1 \\ \vdots & \det A & \vdots \\ e_n & & e'_n \end{array}$$



Original source details: GELFAND-P000337

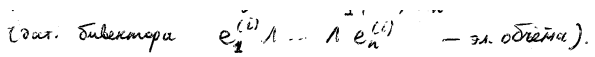
Uncertain reading. The last $\mathbb{C}$ has n_k above and a separate k-shaped sign below. Both are retained; the meaning of the lower sign is not established. Braces enclose the chain, with $= C$ at the right.

$$\left\{0 \longrightarrow \mathbb{C}^{n_1} \xrightarrow{A_1} \mathbb{C}^{n_2} \xrightarrow{A_2} \dots \xrightarrow{A_{k-1}} \mathbb{C}^{n_k}_k \longrightarrow 0.\right\} = C.$$

Classical det is defined for exact sequences.

In all $\mathbb{C}^{n_i}$ their own bases $e_1^{(i)}, \dots, e_n^{(i)}$.

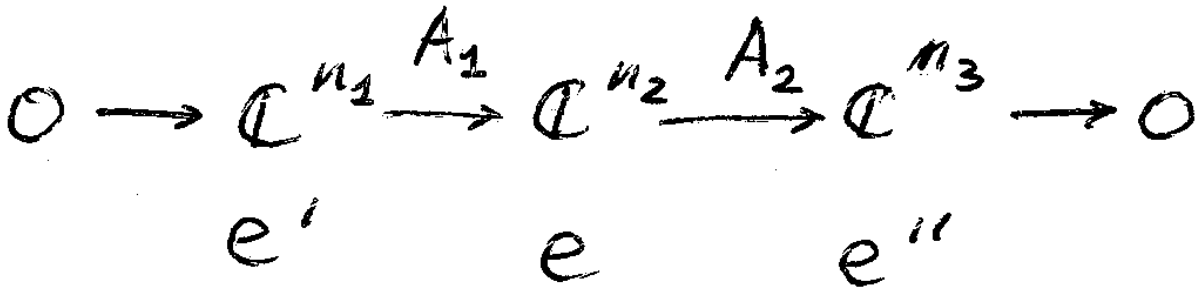
(dat. bivector $e_1^{(i)} \wedge \dots \wedge e_n^{(i)}$ — vol. elem.).



Original source details: GELFAND-P000337

Uncertain reading. The compressed first word is provisionally shown as “dat.” (Russian “дат.”); the ending of “бивектора” also permits “бивекторы”. The last $e_n^{(i)}$ has only n below, without an additional i. These provisional words and this index are retained.

$$0 \longrightarrow \mathbb{C}^n \xrightarrow{A} \mathbb{C}^n \longrightarrow 0. \qquad \det A.$$



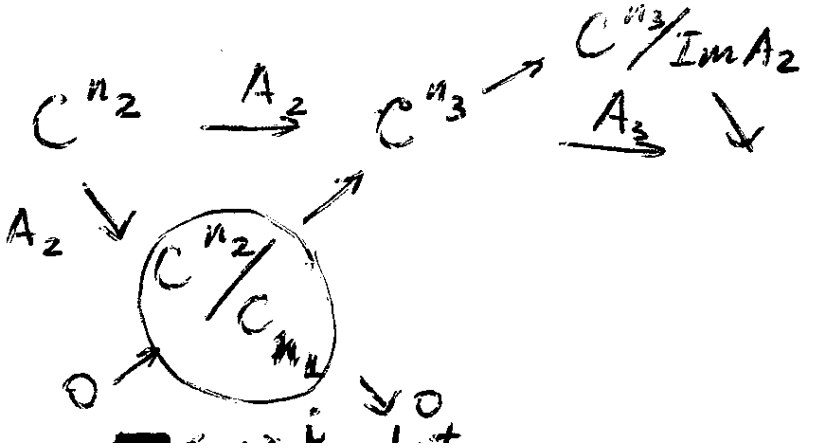
Original source details: GELFAND-P000337

Diagram and typed description.

$$0 \longrightarrow \mathbb{C}^{n_1} \xrightarrow{A_1} \mathbb{C}^{n_2} \xrightarrow{A_2} \mathbb{C}^{n_3} \longrightarrow 0$$

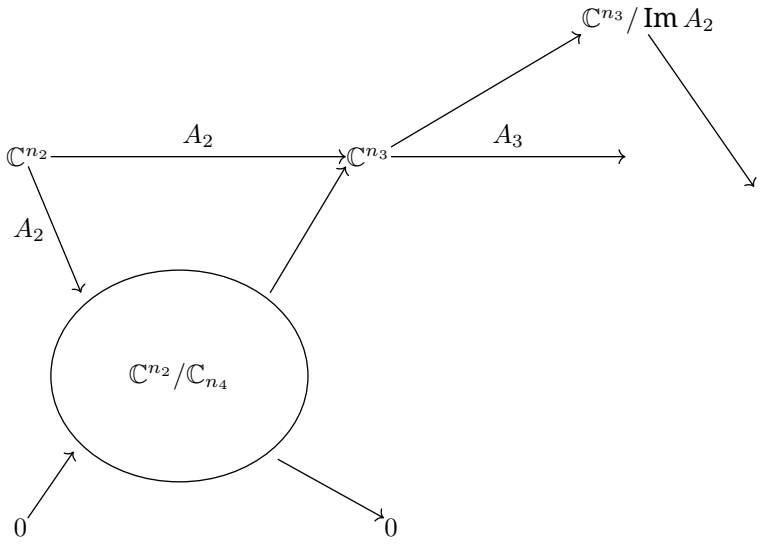
$$e' \qquad\qquad e \qquad\qquad e''$$

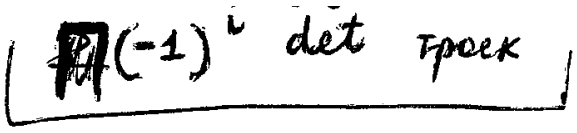
Compute the det of the transition from e to $(A_1 e', A_2^{-1} e'')$. This is $\det C$.



Original source details: GELFAND-P000338

Uncertain reading. The circled object's denominator has a dense subscript; it is provisionally shown as n_4, with n_1 also possible. Two arrows at the right end without labelled objects. The written Im A_2, both labels A_2, the label A_3, the oval and both zeros are retained; the exact index and connections are not established.





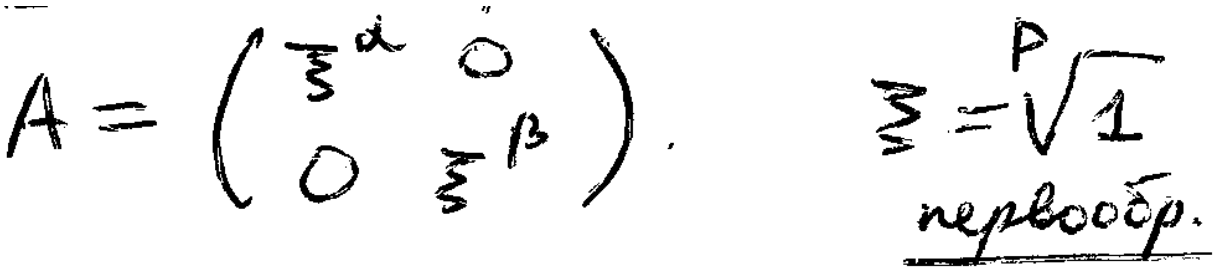
Original source details: GELFAND-P000338

Uncertain reading. The crossed-out sign before $(-1)^i$ is provisionally shown as $\prod$; two uprights and an upper stroke are visible, but the whole sign is not established. A three-sided bracket is below the entire expression.

$\not\prod (-1)^i$ det of triples

does not depend on the choice of bases in the intermediate spaces.

Lens spaces.



Original source details: GELFAND-P000338

Diagram and typed description.

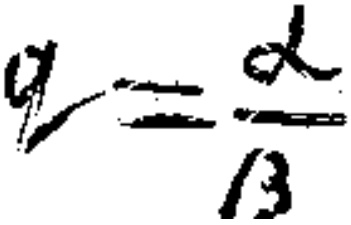
$$A = \begin{pmatrix} \xi^{\alpha} & 0 \\ 0 & \xi^{\beta} \end{pmatrix}, \qquad \xi = \sqrt[p]{1}.$$

prim.

Without fixed points.

$$q = \frac{\alpha}{\beta};$$

$A_{1,\beta}$.

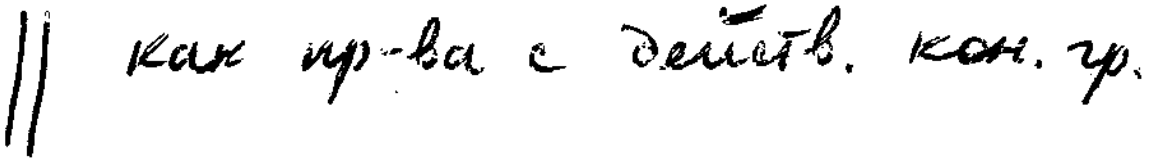


Uncertain reading. To the left of α/β is a small loop with a long downstroke and an oblique stroke, provisionally q. A separate $A_{1,\beta}$ is double-underlined below.

Original source details: GELFAND-P000338

Quotient of the sphere — lens space.

$$L^3_p = S^3/A_{1,\beta}.$$



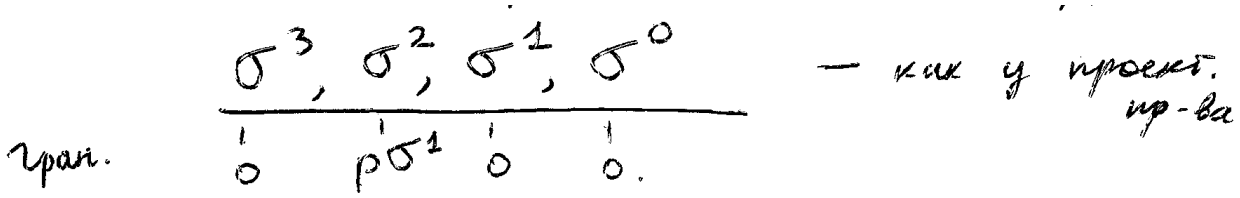
Original source details: GELFAND-P000338

Diagram and typed description.

|| as spaces with an action of a finite group.

Are they homeom. or not? The homologies are the same.

$\xi^p = 1$. The fund. gr. of a lens manif. is a cyclic group.



Original source details: GELFAND-P000338

Uncertain reading. The coefficient below σ^2 is provisionally shown as p; the form also permits ρ. A horizontal line separates the rows, with short vertical strokes above the four lower entries. The comparison with project. space is written at the right.

	σ^3,	σ^2,	σ^1,	σ^0
Bound.	0	$p\sigma^1$	0	0.

— as for project. space.

Univ. cover.

$$\xi\sigma^3$$
$$\xi^2\sigma^3$$
$$\vdots$$

Module

Over the group ring $\mathbb{Z}[\pi_1]$, $\pi_1 = \mathbb{Z}_p$.

Uncertain reading. The last $\mathbb{Z}$ has a subscript provisionally shown as p; its looped form also permits ρ.

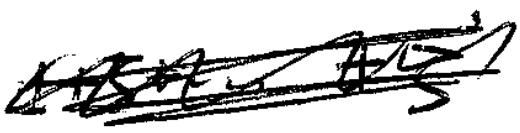
Original source details: GELFAND-P000338

GELFAND-P000339

[source number: 3]

The cochain complex is exact.

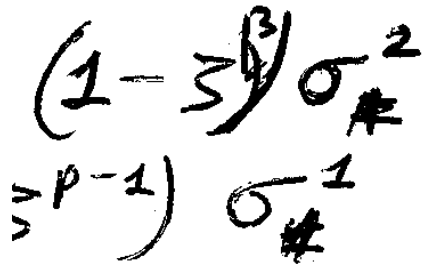
$$\sigma^3 \xrightarrow{\partial} \text{[unreadable crossed-out expression]} \quad (1-\xi^\beta)\sigma^2_{\not k}$$



Undeciphered passage. A whole expression after the first arrow and before $(1-\xi^\beta)\sigma^2$ is crossed out. Overlapping loops, sloping strokes and long horizontal strokes do not permit recovery of the entire expression.

Original source details: GELFAND-P000339

$$\sigma^2 \xrightarrow{\partial} (1+\xi+\ldots+\xi^{p-1})\sigma^1_{\not k}.$$



Original source details: GELFAND-P000339

Uncertain reading. The first ξ has an exponent with a tall stem and two right-hand loops, provisionally β. The cancelled lower marks at both occurrences of σ are provisionally shown as crossed k.

$\partial\sigma^1 = (1-\xi\ )\sigma_0$. exact.

Original source details: GELFAND-P000339

Undeciphered passage. The exponent of ξ inside $(1-\xi\ )$ is a solid dark cancelled sign. The blank preserves its position; the earlier exponent has not been recovered.

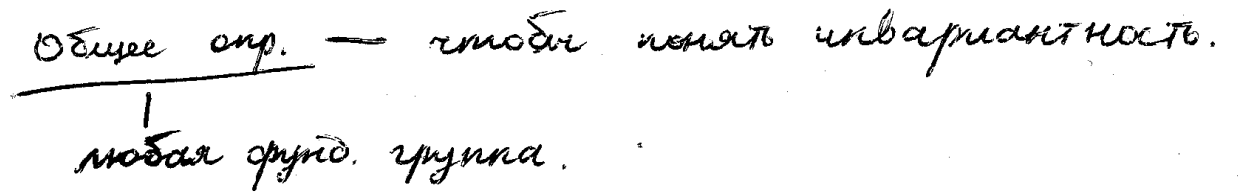
Exact sequence.

In the middle — the zero operator.

Original source details: GELFAND-P000339

Manuscript notation. The line begins «Посредине» (“In the middle”) and separately has «0-й».

$$\det = (1-\xi)(1-\xi^{\beta}).$$



Original source details: GELFAND-P000339

Diagram and typed description.

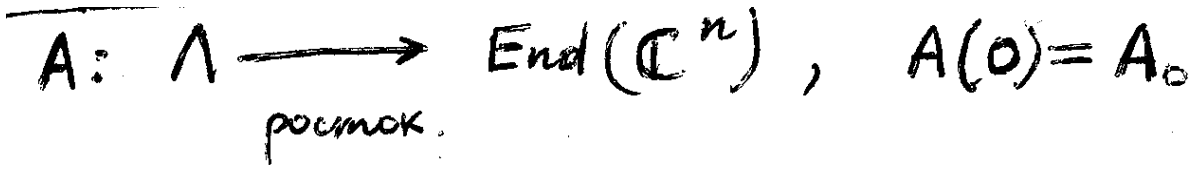
General def. — to understand invariance.

|

any fund. group.

A repres. such that the complex on the covering is exact.

V. I. Arnold. $\Lambda = \mathbb{C}^k$.



Original source details: GELFAND-P000339

Diagram and typed description.

$A : \Lambda \longrightarrow \mathrm{End}(\mathbb{C}^n)$ $\qquad A(0) = A_0$.

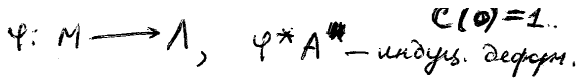
germ.

They are a category. The final object is a vers. deform.

$C(\lambda)$ — a deform. of the identity.

$$C(\lambda)A(\lambda)(C(\lambda))^{-1} = B(\lambda), \qquad C(0) = 1.$$

$\varphi : M \longrightarrow \Lambda$, $\varphi^* A$ — induced deform.



Original source details: GELFAND-P000339

Uncertain reading. The map sign is looped, with a descending stroke; it is provisionally shown as φ, with ψ also possible. The signs before the star and inside $A(\varphi(\mu))$ are also looped; the same provisional candidate is retained at all three sites.

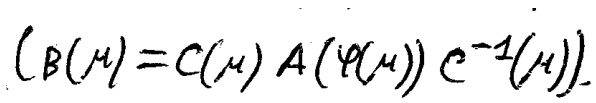
Original source details: GELFAND-P000339

Undeciphered passage. A small solid cancelled mark is above and after A. The blank preserves its position; the mark's content has not been recovered.

A versal deform. A — every deform. B of the matr. $A_0 \sim$ induced.

A.

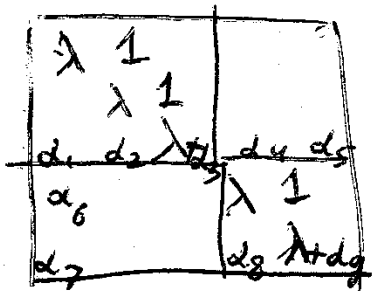
$$(B(\mu) = C(\mu)A(\varphi(\mu))c^{-1}(\mu)).$$



Original source details: GELFAND-P000339

Uncertain reading. The first letter in the final inverse factor is smaller than the neighbouring capital C and open, with a small middle curl. The provisional $c^{-1}(\mu)$ is retained; e and C are also possible.

$\exists$ a vers. fam. with 9 params.



Original source details: GELFAND-P000339

Uncertain reading. The rectangular boundary, internal vertical and horizontal lines, and the written λ, 1 and $\alpha_1, \ldots, \alpha_9$ are encoded. The label near $\lambda + \alpha_3$ intersects the lines; the exact cells and overlaps are not established. Lower labels are close to the border. Empty places remain empty, without supplied zeros.

λ	1			
	λ	1		
α_1	α_2	$\lambda + \alpha_3$	α_4	α_5
α_6			λ	1
α_7			α_8	$\lambda + \alpha_9$

α_i — free parameters.

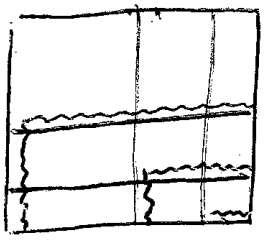
GELFAND-P000340

Dim. of the space of variations:

$$\sum_{\lambda} (n_1 + 3n_2 + 5n_3 + \ldots)$$

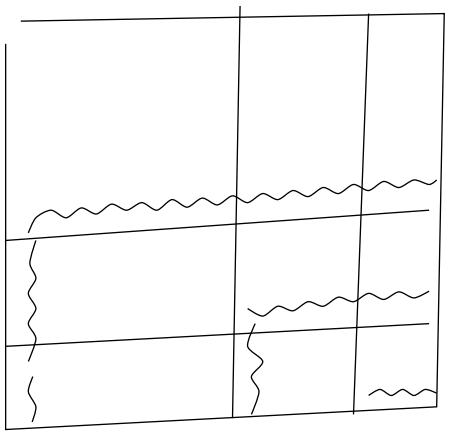
$$n_1(\lambda) \geq n_2(\lambda) \geq \ldots$$

Take a manif. through A_0 transverse to the orbit.
Deformations of the identity — n^2 params.

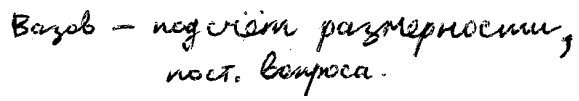


Original source details: GELFAND-P000340

Uncertain reading. An unlabelled rectangular sketch with a broad left region and two narrower right regions. Below are two horizontal divisions and several wavy bands; the outer boundary has a small gap at the upper left. The visible band positions are encoded, but the exact cells, block order and meaning of the waves are not established; no labels or entries are supplied.



Wasow [reading provisional] — dimension count, formul. of the question.

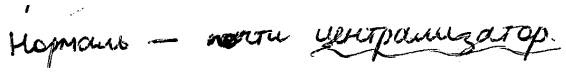


Original source details: GELFAND-P000340

Uncertain reading. The name before the dash is provisionally read «Вазов». In context it likely refers to Wolfgang Wasow.

Versality ⇔ transversality (to the orbit) —

— a normal to the orbit.
Normal — almost centralizer.



Original source details: GELFAND-P000340

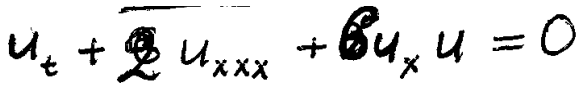
Uncertain reading. The word is provisionally read as “almost” (Russian “почти”): its beginning is overwritten by dense strokes, and the ending “ти” is readable. The earlier initial letters are not established. “Centralizer” has a wavy underline.

For quadr. forms — the canon. form of quadr. forms.

GELFAND-P000341

15/III L. Faddeev.

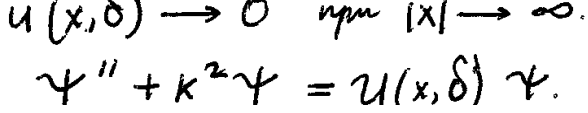
$$u_t + 2u_{xxx} + 6u_x u = 0.$$



Original source details: GELFAND-P000341

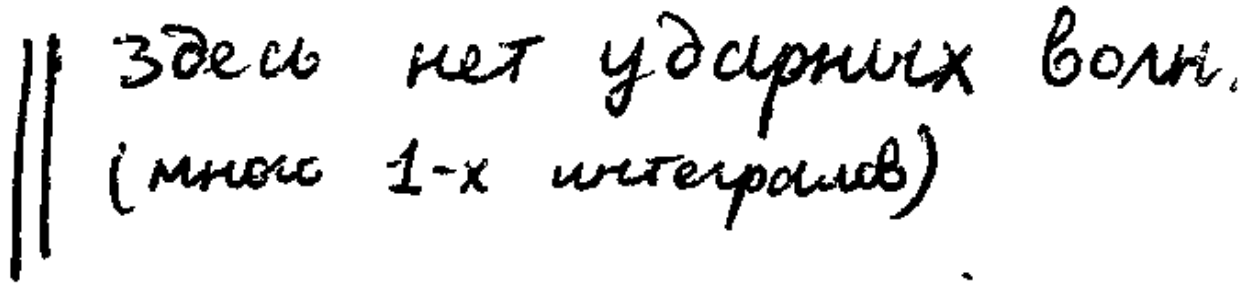
Uncertain reading. Before u_{xxx} is a visible 2 with a dense earlier looped stroke above it; before $u_x u$ is 6 with an overwritten top. The candidates 2 and 6 are retained; the earlier coefficients are not established. Both written signs before these terms are pluses.

$u(x, \delta) \longrightarrow 0$ as $|x| \longrightarrow \infty$.



Original source details: GELFAND-P000341

Uncertain reading. The second argument in both lines is rendered δ. Its upper hook and lower loop differ from the t on the lines below. The value of the argument is unspecified.



Original source details: GELFAND-P000341

Diagram and typed description.

There are no shock waves here.
(many 1st integrals).

Consider $\psi'' + k^2\psi = u(x, \delta)\psi$.

Spec. solution:

$$\psi|_{x\to-\infty} = e^{ikx} + a(k)e^{-ikx}$$
$$\psi|_{x\to+\infty} = e^{ikx}\cdot b(k).$$

One derives $|a(k)|^2 + |b(k)|^2 = 1$.

Known fact: $a(k)$ determines $u(x)$

$u(x) \longleftrightarrow a(k)$ (up to the discr. spectrum).

Recipe: $|a(k,t)| = |a(k)|$

$\arg a(k,t) = \arg a(k) + ck^3t$.

Uncertain reading. The coefficient of k^3t is provisionally read as c, with a small upper curl. A corresponding constant occurs in the formula for the phase derivative on the next leaf.

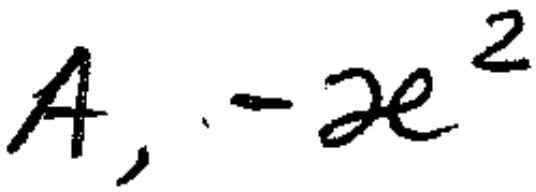
Original source details: GELFAND-P000341

$u(x,t) \longleftrightarrow a(k,t)$.

And with a discr. spectrum A, $-æ^2$ — these are solitons,

Uncertain reading. The initial sign is provisionally read as the Russian conjunction «И» (“and”). The letter u, as on the preceding line, is also possible.

Original source details: GELFAND-P000341



Uncertain reading. After A, the squared symbol made of two connected loops is rendered æ and may be handwritten $\varkappa$.

Original source details: GELFAND-P000341

when $a(k) \equiv 0$.

Uncertain reading. The letter before $(k) \equiv 0$ is dense, with an overlapping stroke; the candidate a is retained, and the earlier form is not established.

Original source details: GELFAND-P000341

GELFAND-P000342

Conclusion: resembles a Hamilton. eq. in action — angle variables.

$a(k,t)$ — a solution of a classical Hamiltonian system

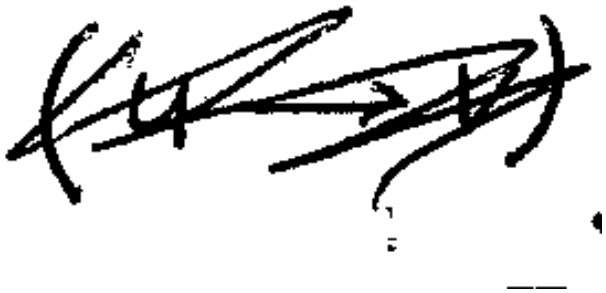
Original source details: GELFAND-P000342

Uncertain reading. At the right is a crossed-out entry with an H-shaped sign and a small k-shaped lower stroke. The crossed H_k is provisional; the exact earlier expression is not established.

$$\begin{cases} \dfrac{\partial |a|}{\partial t} = 0 \\ \dfrac{\partial \varphi}{\partial t} = Ck^3. \end{cases} \qquad \cancel{H_k}$$

Assertion: — the equation is an ∞-dimensional completely integr. Hamilton. eq. The solution of the scattering problem is its canonical transfor. to an explicitly solvable form.

Symplectic structure in the space of functions $u(x)$.



Original source details: GELFAND-P000342

Uncertain reading. Between the minus sign and the second integral are crossed parentheses, u, v and a horizontal sign. The crossed $(u \to v)$ is provisional; $(u \leftrightarrow v)$ is also possible, with the direction obscured by cancellation.

$$\langle u|v\rangle = \int_{-\infty}^{\infty} u(x)\,dx \int_{-\infty}^{x} v(y)\,dy - \cancel{(u \to v)} \int_{-\infty}^{\infty} v(x)\,dx \int_{-\infty}^{x} u(y)\,dy.$$

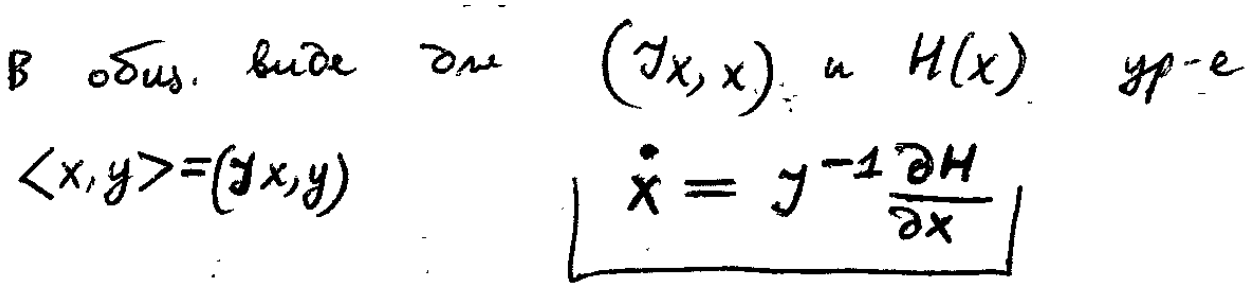
a nondeg. skew-symm. form.

Hamiltonian:

$$Q(u) = \int_{-\infty}^{\infty} (u^3 + u'^2)\,dx$$

$$\frac{\partial u}{\partial t} = \left(\frac{d}{dx}\right)\left(\frac{\delta Q}{\delta u(x)}\right).$$

In general form, for (Jx, x) and $H(x)$ the eq.

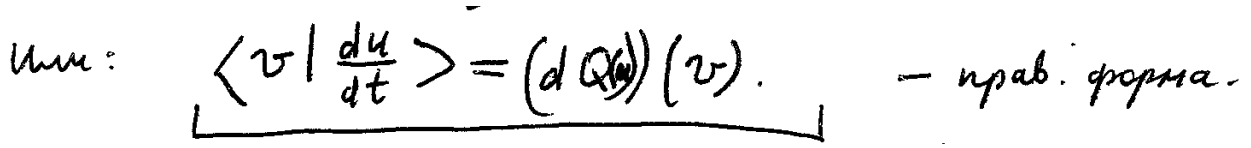


Original source details: GELFAND-P000342

Uncertain reading. Both letters in the introductory pair resemble x: (Jx, x). The next pair is (Jx, y); the difference is retained. Below the equation with J^{-1} is a line with upright sides; no upper side is visible.

$$\langle x, y\rangle = (Jx, y); \qquad \dot{x} = J^{-1}\frac{\partial H}{\partial x}.$$

Or:

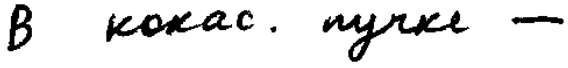


Original source details: GELFAND-P000342

Diagram and typed description.

$$\left\langle v \middle| \frac{du}{dt} \right\rangle = (dQ(u))(v). \qquad \text{— correct form.}$$

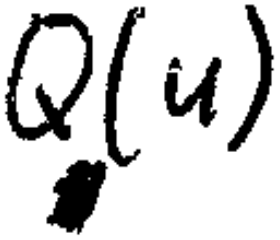
In the cotang. bundle [reading provisional] — 2 eqs. for p and q.



Uncertain reading. The manuscript abbreviation «кокас.» before “bundle” likely stands for “cotangent.”

Original source details: GELFAND-P000342

Pass from u to $a(k)$.

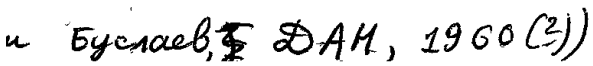


Original source details: GELFAND-P000342

Uncertain reading. Below Q in the final equality is a separate short, dense, sloping stroke. Its position is encoded; its reading and meaning are not established.

$$Q(u) = \int_{-\infty}^{\infty} (u^3 + u'^2)\,dx = \int_{-\infty}^{\infty} \ln(1 - |a|^2)k^4\,dk.$$

(Trace f-la: L. D. Faddeev and Buslaev,



Original source details: GELFAND-P000342

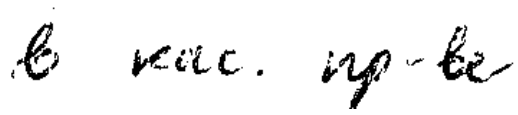
Undeciphered passage. A small sign with remnants of a loop is densely crossed out between “Buslaev,” and “DAN”; the earlier entry has not been recovered.

DAN, 1960 (2)).

GELFAND-P000343

[source number: 6]

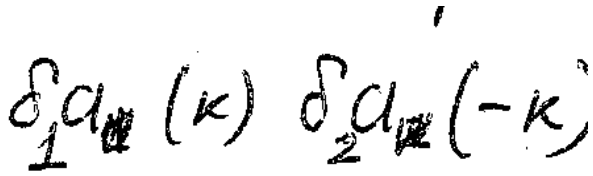
Express any $\int$ of this type (a local functional) through moments of the logarithm $\ln(1-|a|^2)$. Image of the differential form $\longrightarrow$ in the tang. space (the transition is nonlinear).



Original source details: GELFAND-P000343

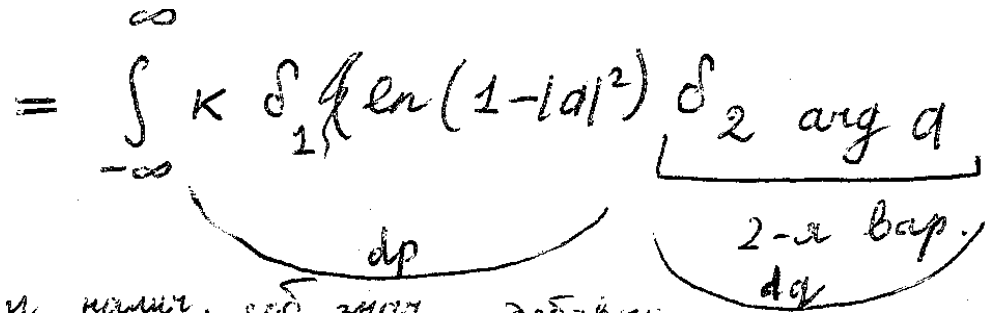
Uncertain reading. The abbreviation is provisionally read «кас.», likely short for “tangent.”

$$\langle \delta u_1, \delta u_2 \rangle_u = i \int_{-\infty}^{\infty} \frac{k}{1-|a|^2} \delta_1 a_k(k)\, \delta_2 a_k(-k)\, dk =$$



Original source details: GELFAND-P000343

Uncertain reading. Both occurrences of a have densely canceled lower signs. The overwritten sign may have been k, and the reading with no operative subscript is also possible.



Original source details: GELFAND-P000343

Uncertain reading. Before $\ln(1-|a|^2)$ is an opening brace; no closing brace or trailing dk is visible in this row. An arc labelled dp is below the left part; below $\delta_2 \arg a$ are a three-sided bracket labelled “2nd var.” and a lower arc labelled dq. Unseen completions are not supplied.

$$= \int_{-\infty}^{\infty} k\delta_1\{\ln(1-|a|^2) \qquad \delta_2 \arg a$$

dp 2nd var. dq

When eigenvalues are present, there are additional terms.

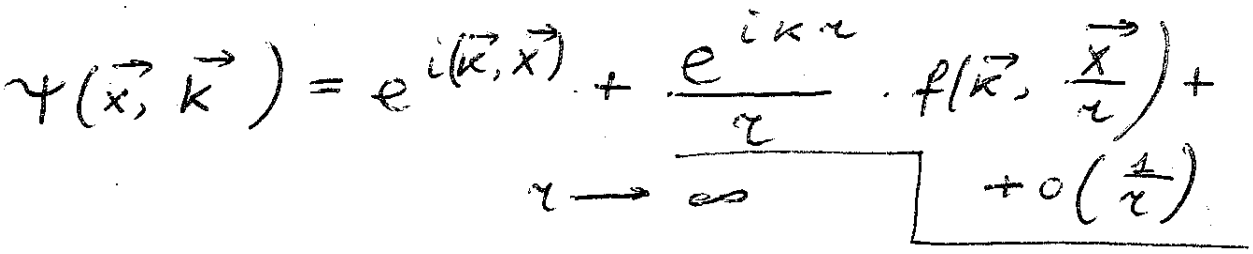
The three-dimensional inverse scattering problem.

$$-\Delta\psi + v(\vec{x})\psi = k^2\psi, \qquad \vec{x} \in \mathbb{R}^3,$$

$v(\vec{x})$ decays sufficiently rapidly as $|\vec{x}| \longrightarrow \infty$

Scattering amplitude.

The natural solution:



Original source details: GELFAND-P000343

Uncertain reading. A plus is written after f at the end of the first row, and another before $o(1/r)$ in the next row. Both are retained, and o is lowercase. A stepped line is below the fraction and formula ending, with $r \longrightarrow \infty$ at the lower left.

$$\psi(\vec{x},\vec{k}) = e^{i(\vec{k},\vec{x})} + \frac{e^{ikr}}{r}\cdot f\left(\vec{k},\frac{\vec{x}}{r}\right) +$$

$$r \longrightarrow \infty \qquad +o(1/r).$$

f — the scattering amplitude, a function of 5 variables (since $\vec{x}/r$).

Relations on the scattering amplitude have been found, eliminating 2 parameters.

Original source details: GELFAND-P000343

Uncertain reading. The crossed beginning of the line contains a tall f-shaped form and (k). The crossed $f(k)$ is provisional; the dense initial form is not definitively established.

~~$f(k)$~~

$$f\left(k\left|\frac{\vec{k}}{k},\frac{\vec{x}}{r}\right.\right) \longrightarrow f(k|\vec{\alpha},\vec{\beta})$$

in $L_2(S^2)$ a unitary operator! — this is a condition;

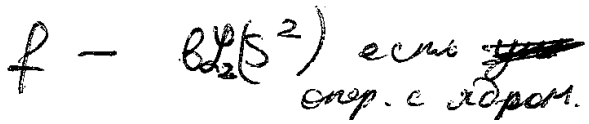
analogous.

Original source details: GELFAND-P000343

Undeciphered passage. A short expression between "analogous" and $a^2+b^2=1$ is crossed out; the overlapping letters have not been recovered.

$a^2 + b^2 = 1$.

$S(k) = I + f$, f — $\mathcal{B}L_2(S^2)$ if

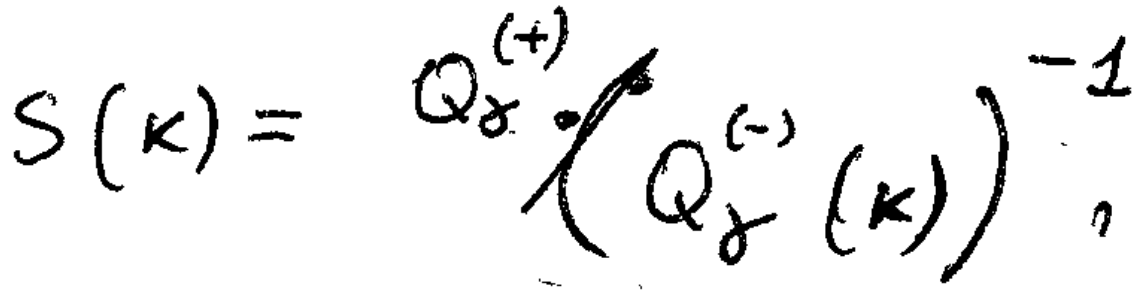


Original source details: GELFAND-P000343

Undeciphered passage. After "if" and before "operator with a kernel", a word is crossed out by overlapping strokes; the word has not been recovered. Before this is a B-shaped sign directly before the space notation, without a supplied membership sign or new parentheses.

an operator with a kernel.

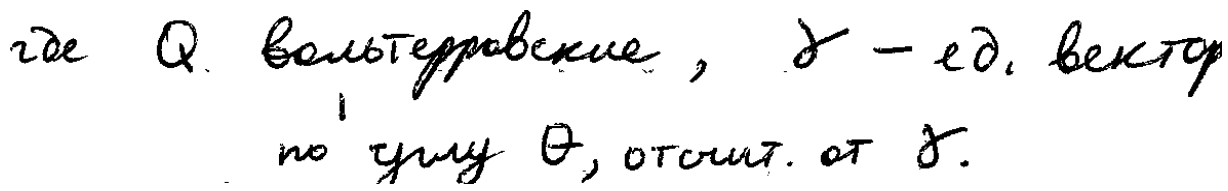
GELFAND-P000344



Original source details: GELFAND-P000344

Uncertain reading. After $Q_\gamma^{(+)}$ are a dot and a long slant near the opening of the next parenthesis. Both strokes retain their positions. A product is only a candidate; the slant's role and possible earlier division are not established. No argument k is visible on the first Q, and none is supplied.

$$S(k) = Q_\gamma^{(+)} \cdot / (Q_\gamma^{(-)}(k))^{-1},$$



Original source details: GELFAND-P000344

Diagram and typed description.

where Q are Volterra, γ — a unit vector.

|

in the angle θ, measured from γ.

$$h_\gamma(k;\alpha,\beta)\Theta[\pm(\alpha-\beta,\gamma)] = f\left(k\left|\frac{\vec{k}}{k},\frac{\vec{x}}{r}\right.\right)$$

When $(\alpha,\gamma) = (\beta,\gamma)$ there is a function analytic in k in the upper half-plane.

The Green function and factorization were written down (several years ago).

There was no inverse problem because of the discrete spectrum.

A. Gabrielov.

Griffiths's work on periods of rational integrals. Annals of Math.

$$\omega = \frac{P(z)}{Q(z)}\, dz_1 \wedge \ldots \wedge dz_n, \qquad P, Q \text{ — polynomials in } \mathbb{C}^n.$$

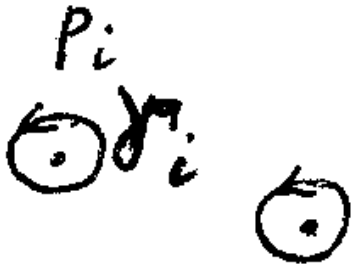
V — the zeros of $Q(z)$.

Γ — n-dimensional cycles in $\mathbb{C}^n \setminus V$.

$\int_\Gamma \omega$ — the periods.

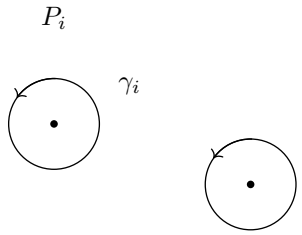
1 variable: $\omega = \dfrac{P(z)}{Q(z)}\, dz$

Γ reduces to a standard collection of paths



Original source details: GELFAND-P000344

Uncertain reading. Two contours enclose points, with both upper arrowheads pointing left. The label over the first is provisionally read P_i and may be lowercase p_i. At its right is γ_i.



$$\Gamma = \sum \mu_i \gamma_i; \qquad \int_\Gamma \omega = \sum \mu_i \int_{\gamma_i} \omega.$$

Reduction to 1st order — subtract an exact differential.

And now $\sum$ of residues.

Regard the form as defined on $\mathbb{CP}^n$.

$P_n \supset V$ — the set of poles.

There exists Q — a homogeneous form of degree m, minimal.

GELFAND-P000345

[source number: 7]

$$\Gamma \subset P_n \setminus V, \qquad \Gamma \in H_n(P_n \setminus V, \mathbb{Z}).$$

Заметим, что Γ в P_n есть 0;

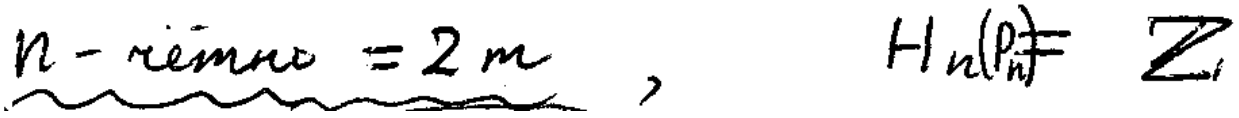
Original source details: GELFAND-P000345

Diagram and typed description.

Notice that Γ in P_n is 0;

n — odd $\Rightarrow H_n = 0$.

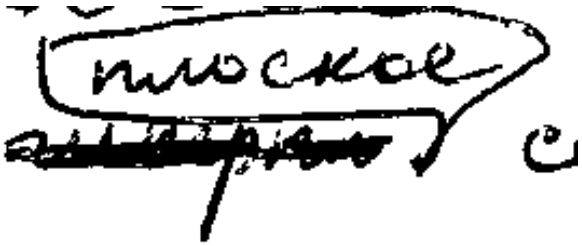
$n = 2k+1 \Rightarrow H_n(P_n) = 0$.



Original source details: GELFAND-P000345

Diagram and typed description.

n — even $= 2m$, $H_n(P_n) = \mathbb{Z}$



Original source details: GELFAND-P000345

Undeciphered passage. The earlier word after "Then" is crossed out by dense horizontal strokes; the whole word has not been recovered. The inserted "plane" is enclosed, with a downward arrow at its right towards the earlier entry. An initial loop and a long lower stroke are visible; "alg." remains only a candidate.

Then plane ~~[crossed out]~~

section P_m — the only generator. It suffices to compute the intersection index of Γ with P_m.
Consider $W = P_{m+1} \cdot V = q \cdot P_m$, q — the degree of the polyn. defining V.
This follows from $W \cdot P_m = V \cdot P_1$ (Bézout's thm.),

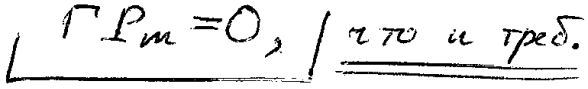
$$(z_1^2 + z_2^2 + z_3^2 = 0 \sim 2P_1).$$

Original source details: GELFAND-P000345

Uncertain reading. After ΓP_m and before the comma is a raised small crossed sign with a loop and a lower stroke. A crossed 2 is provisional; the earlier sign is not established. Its position is not expanded as a secure exponent.

$0 = \Gamma \cdot W = q\Gamma P_m$ ~~2~~ ,

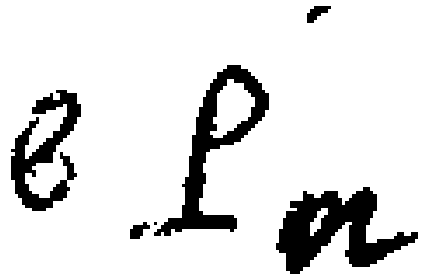
hence



Original source details: GELFAND-P000345

Diagram and typed description.

$\Gamma P_m = 0$, as required



Original source details: GELFAND-P000345

Uncertain reading. Below R is the insertion "in P_m". Its lower letter is dense, with joined strokes; m is provisionally retained, with n also possible. The insertion stays below R.

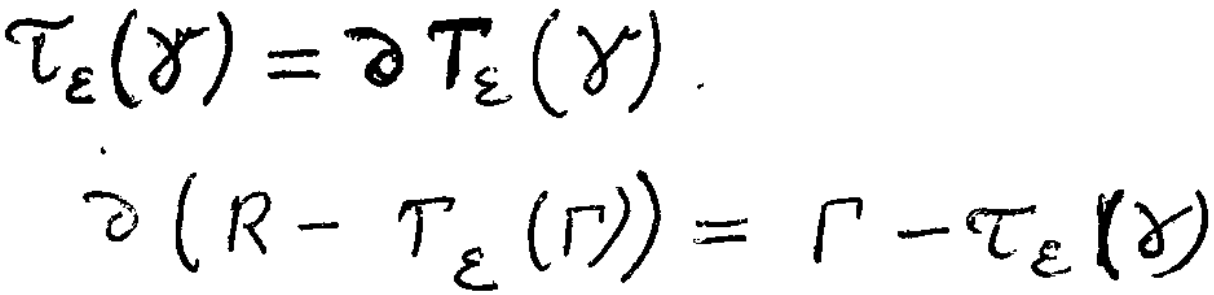
There is a chain R, such that $\partial R = \Gamma$.

in P_m

Assume R and V in general position, $\gamma = R \cdot V$, $\gamma \in H_{n-1}(V, \mathbb{Z})$.
Take $T_\varepsilon(\gamma)$ — an ε-neighbourhood of the cycle γ in P_n.

Uncertain reading. In the first argument of T_ε the letter with a lower loop is strongly reinforced and provisionally shown as γ; earlier strokes cannot be securely separated.

Original source details: GELFAND-P000345



Original source details: GELFAND-P000345

Diagram and typed description.

$$\tau_\varepsilon(\gamma) = \partial T_\varepsilon(\gamma).$$
$$\partial(R - T_\varepsilon(\Gamma)) = \Gamma - \tau_\varepsilon(\gamma)$$

Hence $\tau_\varepsilon(\gamma) \sim \Gamma$. We obtain:

$$\tau : H_{n-1}(V, \mathbb{Z}) \longrightarrow H_n(P_n \setminus V, \mathbb{Z}).$$

Manuscript notation. The upper wavy stroke corresponds to the crossbar of the handwritten τ_ε on this leaf. No separate tilde is legible.

Original source details: GELFAND-P000345

This gives a map [crossed out] of homologies.

Undeciphered passage. Between “map” and “of homologies” a word is covered by several horizontal strokes. Joined loops are visible beneath them; the whole earlier word has not been recovered, including the candidate Russian “когомол.”.

Original source details: GELFAND-P000345

$$\tau_\varepsilon(\partial\gamma) = \partial(\tau_\varepsilon(\gamma)).$$

$$\tau : H_k(V, \mathbb{Z}) \longrightarrow H_{k+1}(P_n \setminus V, \mathbb{Z}).$$

Manuscript notation. The upper wavy stroke corresponds to the crossbar of the handwritten τ_ε on this leaf. No separate tilde is legible.

Original source details: GELFAND-P000345

Original source details: GELFAND-P000345

Diagram and typed description.

GELFAND-P000346

Arnold. Let $P_n(x, y) = 0$ be a plane nonsingular alg. curve.

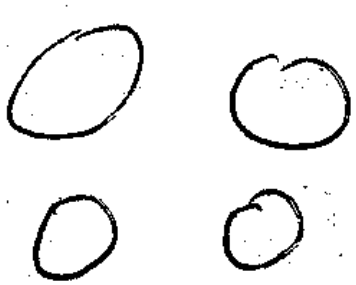
Original source details: GELFAND-P000346

Uncertain reading. The subscript on the first P is strongly reinforced; n is provisionally retained. Dense overlapping strokes do not securely separate the earlier form.

$$n = 2k.$$

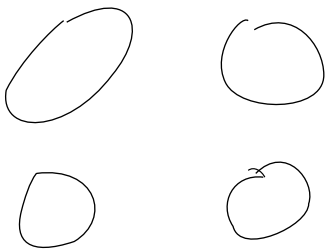
Harnack's thm.: ovals $\leq g + 1$, $g = \dfrac{(n-1)(n-2)}{2}$.

4th degree.



Original source details: GELFAND-P000346

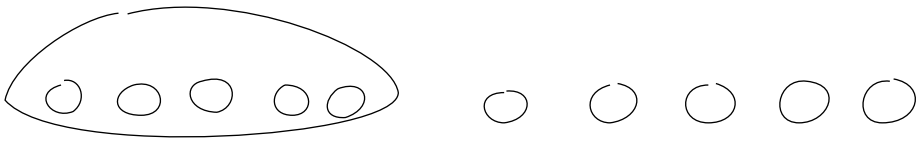
Diagram and typed description.



6th degree. — 50 cases (Gudkov). Let the max. number of ovals be 11 (for $n = 6$).

Original source details: GELFAND-P000346

Diagram and typed description.



and another 2 cases,

Posit. ovals — lie in an even number of others.

Petrovsky: $|2(p - m) - 1| \leq 3k^2 - 3k + 1$. Hence not all 11 are outside one another.

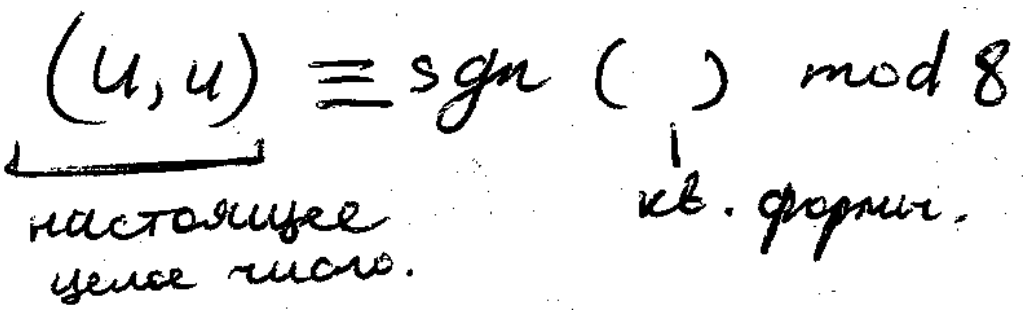
Gudkov: $p - m \equiv k^2 \pmod 8$.

Arnold and Varchenko: proof mod 4 because of integral forms.

$$(x, x) = (u, x) \pmod 2$$

— the quad. form is linear mod 2.

Lemma.



Original source details: GELFAND-P000346

Diagram and typed description.

$(u,u) \equiv \operatorname{sgn}(\ ) \bmod 8$

an actual integer.

quad. form.

Proof:

$$(u+2x, u+2x) = u^2 + 4ux + 4x^2,$$

and ux and x^2 have the same parity by the formula.

$z^2 = f(x,y)$, z, x, y; $(p-m)$ — the Euler char. of the set of smaller values.

GELFAND-P000347

[source number: 8]

22/III. A. Gabrielov.

$$\int_\Gamma \frac{P(z)}{Q(z)}\, dz_1 \wedge \cdots \wedge dz_n, \qquad \Gamma \text{ — a cycle in } P^n \setminus V.$$

$$V = \{Q(\xi) = 0\}, \qquad \deg Q = q.$$

The cycle Γ in P_n is homologous to zero.

$$\Gamma \in H_n(P_n \setminus V, \mathbb{Z}).$$

The map $H_n(P_n \setminus V, \mathbb{Z}) \longrightarrow H_n(P_n, \mathbb{Z})$ is zero.
Let $\Gamma \in H_n(P_n \setminus V, \mathbb{Z})$, $n = 2m$.

Uncertain reading. Before V in the repeated group are dense crossed oblique strokes. The resulting complement is retained as \; the earlier sign is not recovered.

Original source details: GELFAND-P000347

Intersec. no.

$$\langle P_m, P_m\rangle = 1 \text{ — direct check.}$$
$$\langle P_{m+1}\cdot V, P_m\rangle = q.$$

$$\langle P_{m+1}\cdot V; P_m\rangle = \langle V, P_{m+1}\cdot P_m\rangle = V\cdot P_1 = q.$$

(Bézout's thm.).

$$0 = \Gamma\cdot(P_{m+1}\cdot V) = q(\Gamma, P_m), \qquad \text{hence } (\Gamma, P_m) = 0.$$

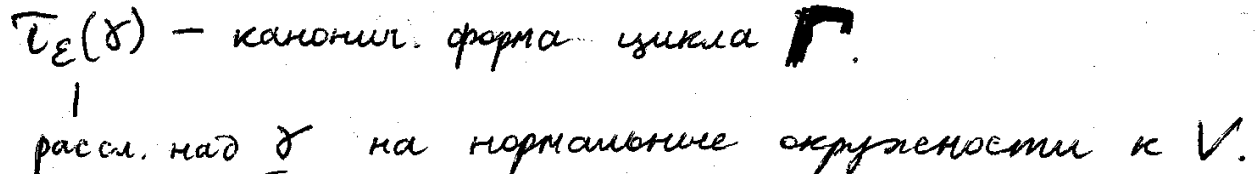
Thus, $\Gamma = \partial R$, $R\cdot V = \gamma \in H_{n-1}(V,\mathbb{Z})$.

Uncertain reading. The subscript before -1 is heavily reinforced. It is provisionally read as n, with the earlier strokes inseparable.

Original source details: GELFAND-P000347

$T_\varepsilon(\gamma)$ — fibred into small circles over γ.

$$\partial T_\varepsilon(\gamma) = \tau_\varepsilon(\gamma).$$
$$\partial(R - T_\varepsilon(\gamma)) = \Gamma - \tau_\varepsilon(\gamma).$$



Original source details: GELFAND-P000347

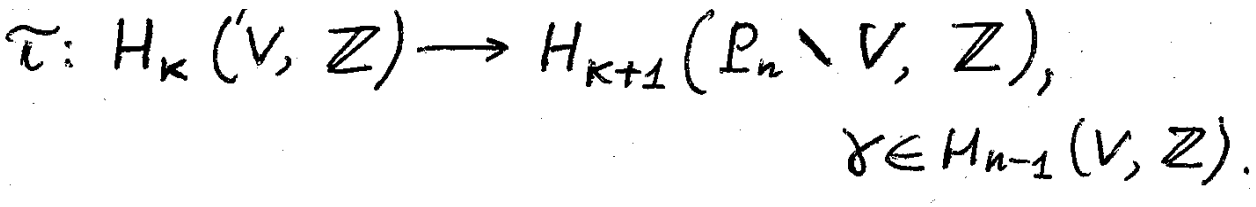
Diagram and typed description.

$\tau_\varepsilon(\gamma)$ — the canon. form of the cycle Γ.

|

fibred over γ into circles normal to V.

A map has been constructed:



Original source details: GELFAND-P000347

Uncertain reading. The map symbol has an upper wavy stroke separated from its lower part. τ is provisionally retained; a separate tilde, $\widetilde{\tau}$, is also possible.

$\tau : H_k(V,\mathbb{Z}) \longrightarrow H_{k+1}(P_n \setminus V, \mathbb{Z}),$

$\gamma \in H_{n-1}(V,\mathbb{Z}).$

How ambiguous is this?

$$\operatorname{Ker} \tau\big|_{H_k} - ?$$

Manuscript notation. The handwritten τ has a wavy crossbar, as in the τ_ε above.

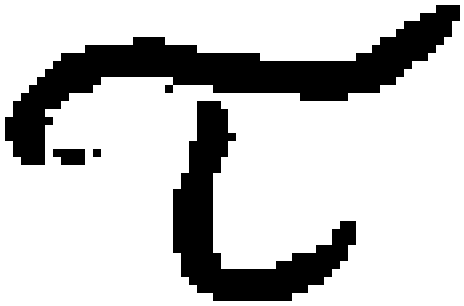
Original source details: GELFAND-P000347

GELFAND-P000348

This is part of the exact seq. of the pair:

$$H_{n+1}(P_n, \mathbb{Z}) \xrightarrow{\cdot V} H_{n-1}(V, \mathbb{Z}) \xrightarrow{\tau} H_n(P_n \setminus V, \mathbb{Z}) \xrightarrow{0} H_n(P_n, \mathbb{Z}).$$

Manuscript notation. The wavy crossbar of τ resembles that of τ_ε on the preceding leaves.

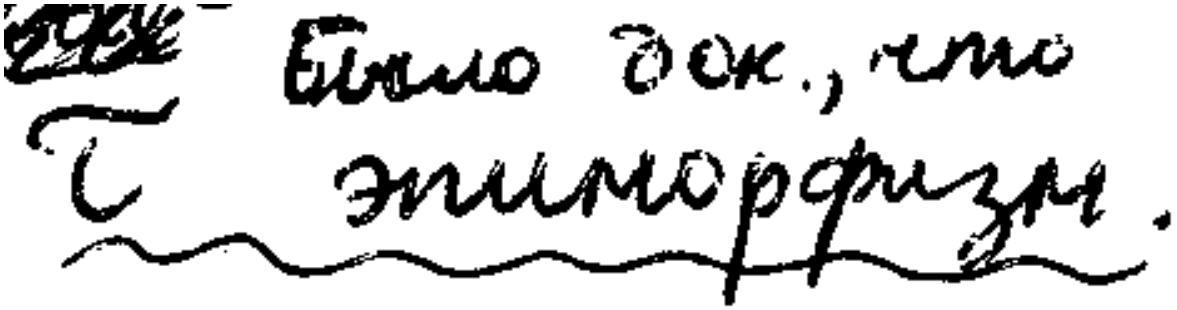


Original source details: GELFAND-P000348

Uncertain reading. A word before “It is exact” is crossed out. “otsyuda” (“hence”) remains a possible reading beneath the diagonal strokes.

Original source details: GELFAND-P000348

It is exact. Usually mod torsion, but here everything is fine.



Original source details: GELFAND-P000348

Diagram and typed description.

It was proved that

τ is an epimorphism.

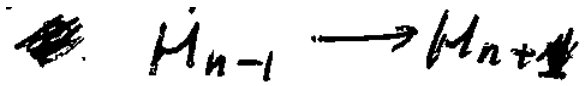
Hence exactness at that term.

From exactness: Ker τ is described.

$$\operatorname{Ker}\tau = H_{n+1}(P_n,\mathbb{Z}),\quad \begin{cases} n \text{ — even — this is } 0,\\ n \text{ — odd — } \mathbb{Z} \text{ with gen. } P_{m+1}. \end{cases}$$

Exactness is proved in the standard way.

Alternatively: the pair P_n, P_n, — a tubul. neighb. of V. Apply to the pair and the Thom isom.:

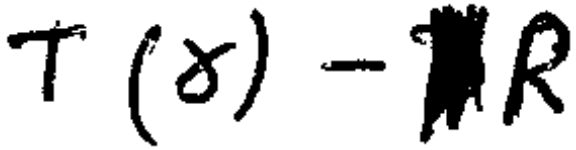


Original source details: GELFAND-P000348

Uncertain reading. A small sign after "Thom isom." is densely crossed out. In the right subscript the last digit is reinforced over other strokes; $n+1$ is provisionally retained.

$$H_{n-1} \longrightarrow H_{n+1}.$$

Proof of exactness: Let $\tau(\gamma) = \partial R$, $\partial(T(\gamma) - R) = 0$ and hence $T(\gamma) - R$ is the bdry. of something, intersect with V and precisely γ.



Original source details: GELFAND-P000348

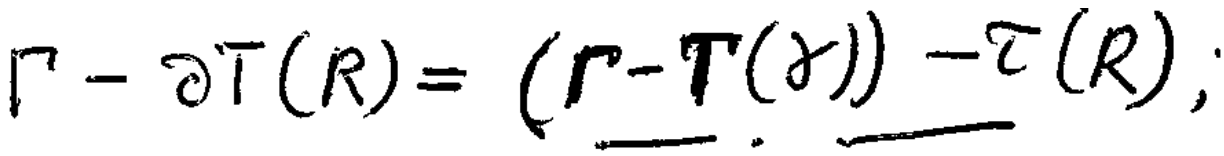
Uncertain reading. Before the final R in $T(\gamma) - R$ is a dense correction with a vertical stroke and diagonals. The earlier sign is not recovered; R is readable.

[In the left margin: $\Gamma \cdot V = \gamma = \partial R$;]

Manuscript notation. A semicolon follows R.

Original source details: GELFAND-P000348

$$\partial T(R) = \tau(R) + T(\gamma);$$



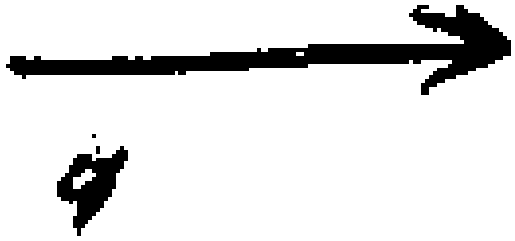
Original source details: GELFAND-P000348

Diagram and typed description.

$\Gamma - \partial T(R) = (\mathbf{\Gamma} - T(\gamma))\ -\tau(R);$

Let there be a rat. diff. form on P_n. In homogeneous coord.

$$\mathbb{C}^{n+1} \setminus \{0\} \xrightarrow[\varphi]{} P_n.$$

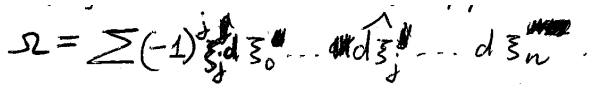


Uncertain reading. Below the arrow is a small looped sign with a descending oblique leg. φ is provisionally retained; π remains a historical candidate.

Original source details: GELFAND-P000348

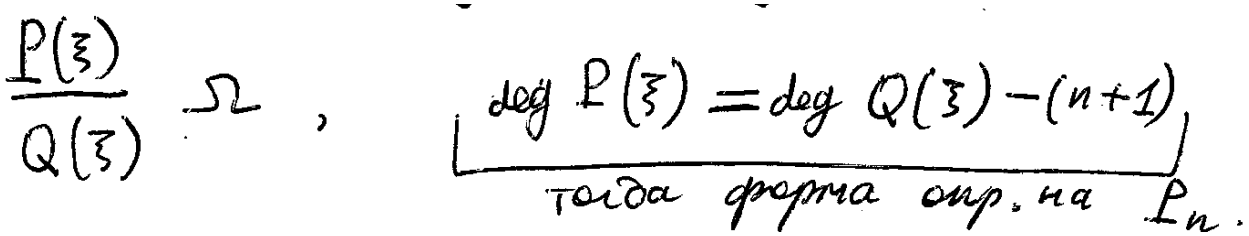
The induced form

$$\Omega = \sum (-1)^j \xi_j \, d\xi_0 \cdots \widehat{d\xi_j} \cdots d\xi_n.$$



Uncertain reading. Earlier signs above ξ_j, after $d\xi_0$, at $\widehat{d\xi_j}$ and above the final $d\xi_n$ are reinforced or crossed out. The earlier indices are not recovered.

Original source details: GELFAND-P000348



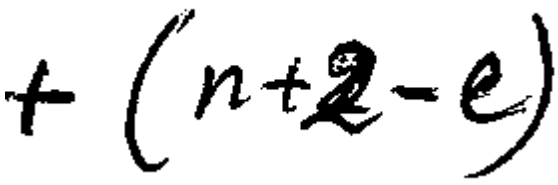
Original source details: GELFAND-P000348

Diagram and typed description.

$\frac{P(\xi)}{Q(\xi)}\Omega,$ $\quad$ $\deg P(\xi) = \deg Q(\xi) - (n+1)$

then the form is defined on P_n.

$$\varphi = \frac{1}{B(\xi)} \left\{ \sum_j (-1)^{|j|} \left[\sum_{\alpha=1}^{\ell} (-1)^{\alpha} \xi_{j_\alpha} A_{j_1 \cdots \widehat{j_\alpha} \cdots j_\ell}(\xi) \right] \right.$$
$$\left. (d\xi_0 \wedge \cdots \widehat{d\xi_{j_1}} \cdots \wedge \widehat{d\xi_{j_\ell}} \wedge \cdots \wedge d\xi_n) \right\}; \qquad \deg B = \deg A + (n + \mathbf{2} - \ell).$$



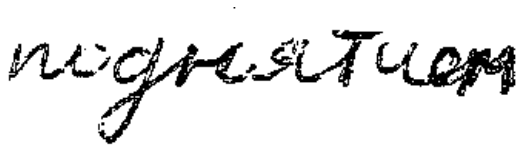
Uncertain reading. The digit before $-\ell$ is written over earlier strokes. The upper reading is provisionally 2.

Original source details: GELFAND-P000348

GELFAND-P000349

[source number: 9]

Everything is nonsingular.
This is the type of forms that are pullbacks from P_n.



Uncertain reading. The reinforced ending is provisionally read «-ием». The earlier ending is inseparable.

Original source details: GELFAND-P000349

$$\frac{P(\xi)}{Q^{\ell}(\xi)}\Omega.$$

We lower the order of poles by subtracting coboundaries.

$$\frac{1}{Q(\xi)^{\ell}}\left\{\sum_{i<j}(-1)^{i+j}[\xi_i A_j(\xi)-\xi_j A_i(\xi)]\right.$$
$$\left.(d\xi_0\wedge\cdots\wedge\widehat{d\xi_i}\wedge\cdots\wedge\widehat{d\xi_j}\cdots\wedge d\xi_n)\right\}.$$

$$P=\ell\left(\sum A_j\frac{\partial Q}{\partial\xi_j}\right)-Q\left(\sum\frac{\partial A_j}{\partial\xi_j}\right),$$
$$\deg P(\xi)=\deg Q(\xi)-(n+1).$$

Annals, v. 90, n. 3, 1969.

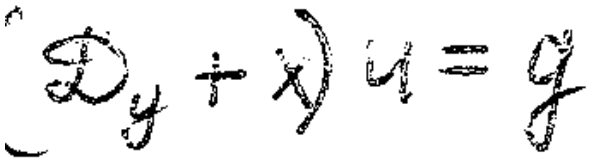
Palamodov.

On the characteristics of systems of lin. eqns.

$$\text{system}\begin{cases}a_1u=f_1\\ \vdots\\ a_tu=f_t\end{cases}\qquad a_1,\dots,a_t\text{ — diff. operators}$$

Example

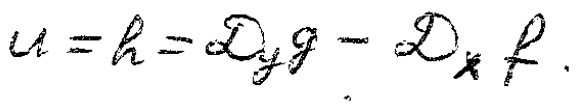
$$\begin{cases}D_xu=f,\\(D_y+x)u=g.\end{cases}\qquad\text{princ. part — ?}$$



Uncertain reading. The sign after the plus is provisionally read x. Its form also permits λ.

Original source details: GELFAND-P000349

Not $\begin{pmatrix}D_x\\D_y\end{pmatrix}$, since a consequence of the system is: $u=h=D_yg-D_xf$.



Manuscript notation. The line reads D_yg-D_xf.

Original source details: GELFAND-P000349

One must write out all consequences and deal with them.

———— . ————

Char. — ? $D_x u + D_y v = f$.

Holmgren's thm. does not apply in any direction, whereas the thm. of **C**auchy-**K**oval. in any direction.

GELFAND-P000350

29/III. Rolewicz.

A — a lin. operator. in a fin.-dim. space.

$\{x, Ax, A^2x, \ldots\}$ — closure.

Limit points: 0, $+\infty$ or

Original source details: GELFAND-P000350

Uncertain reading. The heavily reinforced mark before ∞ is provisionally read as $+$.

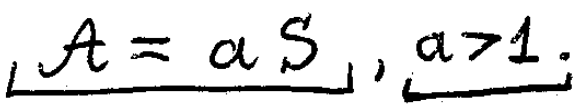
Original source details: GELFAND-P000350

Undeciphered passage. An entire crossed-out word between "or" and "a compact set" is unreadable.

a compact set not containing zero.

Let the oper. be in a Hilbert space, ℓ^p or c^0. It can be everywhere dense. — a shift with multiplication.

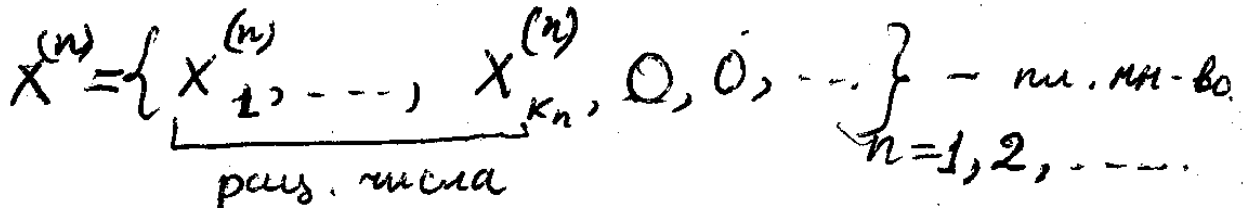
$$S(x_1, \ldots, x_n, \ldots) = (x_2, \ldots, x_n, \ldots).$$



Original source details: GELFAND-P000350

Diagram and typed description.

$A = aS$, $a > 1$.

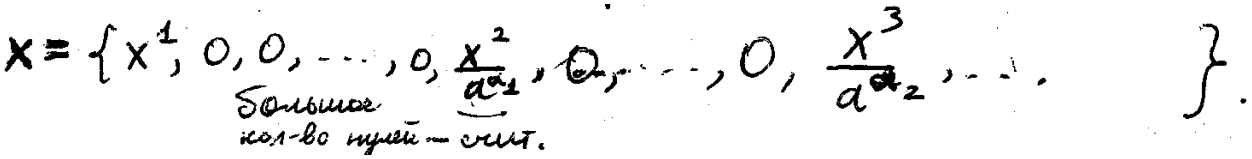


Original source details: GELFAND-P000350

Diagram and typed description.

$x^{(n)} = \{x_1^{(n)}, \ldots, x_{k_n}^{(n)}, 0, 0, \ldots\}$ — a lin. set

rat. numbers $n = 1, 2, \ldots$.



Original source details: GELFAND-P000350

Uncertain reading. After “large no. of zeros —” is a short abbreviation. “schit.” is provisionally retained; “ochit.” remains a candidate. A short arc is below the first denominator.

$$\boldsymbol{x} = \{x^1, 0, 0, \ldots, 0, \underbrace{\frac{x^2}{a^{\alpha_1}}}, 0, \ldots, 0, \frac{x^3}{a^{\alpha_2}}, \ldots\}.$$

large
no. of zeros — schit.

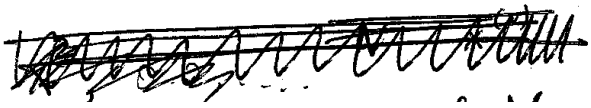
$$x^{(1)} = (x_1^{(1)}, x_2^{(1)}, \ldots, x_{k_1}^{(1)}).$$

Topol. equiv. of lin. opers.: $\exists$ a homeom. U: $A = UBU^{-1}$.

There exist inequivalent opers. — density of an orbit — a topol. prop.

Original source details: GELFAND-P000350

Uncertain reading. Before “opers. not equiv.” a word is crossed out, with “Exist.” inserted above. The earlier word is not recovered; “Eti” (“These”) remains a candidate.



Original source details: GELFAND-P000350

Undeciphered passage. An entire line between “topol. prop.” and “One can construct” is crossed out with horizontal lines and repeated loops. Its original words are unreadable.

One can construct an oper. unbounded in $\forall$ norm.

Arnold.

Milnor — the theory of unimod. quad. integral forms.

$$\varphi : \mathbb{Z}^n \times \mathbb{Z}^n \longrightarrow \mathbb{Z}.$$

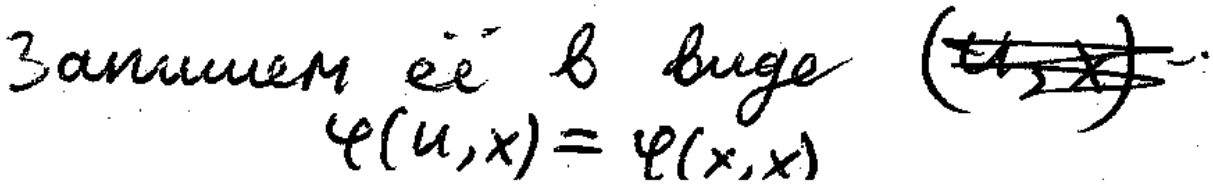
Unimod.: $\tilde{\varphi} : \mathbb{Z}^n \longrightarrow \mathbb{Z}^{n*}$ is an isomorphism.

Invariants: r — rank; sgn — signature; type, I, II.

Uncertain reading. Before "rank" is a sign with a curved upper part and a lower hook. Latin r is provisionally retained; the form also resembles τ.

Original source details: GELFAND-P000350

If the form is not definite, then the system is complete.
Consider the quad. form $\hat{\varphi}_2 : \mathbb{Z}_2^n \longrightarrow \mathbb{Z}_2$, reduced mod 2. It is linear.



Original source details: GELFAND-P000350

Diagram and typed description.

Write it in the form $\varphi(u,x)=$

$$\varphi(u,x) = \varphi(x,x).$$

GELFAND-P000351

[source number: 10]

or $\varphi(x,x) \equiv \varphi(u,x) \bmod 2 \quad \forall x$
Type I — $\exists$ an elt. whose square is odd.
Lemma: 1) $\varphi(u,u) \bmod 8$ does not depend on u. (u is defined mod $2\mathbb{Z}^n$).
2) $\varphi(u,u) \equiv \operatorname{sgn}\varphi \bmod 8$.
(from Fresnel's $\int$).
Classif.: Not definite.

$$\text{odd} \quad \sum x_i^2 - \sum y_i^2$$

Uncertain reading. After $\sum x_i^2 - \sum y_i^2$ a small term is densely crossed out with oblique and horizontal strokes. Its notation is not recovered.

Original source details: GELFAND-P000351

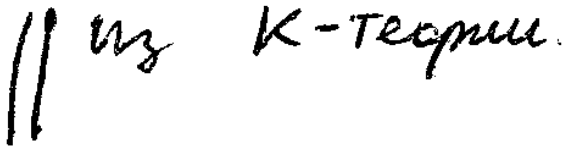


Diagram and typed description.

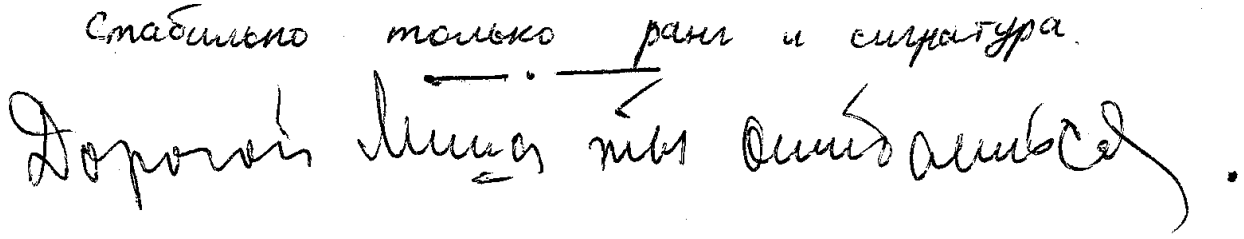
// from K-theory.

Original source details: GELFAND-P000351

even $kU \oplus \ell\Gamma_8$, Γ_8 — the Zolotarev form,
related to E_8.

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Posit. def. — a finite number in $\forall$ dimension.
Serre — a textbook of arithmetic.



Original source details: GELFAND-P000351

Diagram and typed description.

Stably only rank and signature.

Dear Misha you are wrong.

GELFAND-P000352

[source number: 12]

5/IV Kirillov Representation according to Geronimus.

Uncertain reading. The last letter of «Представление» is written over earlier strokes and is provisionally read e.

Original source details: GELFAND-P000352

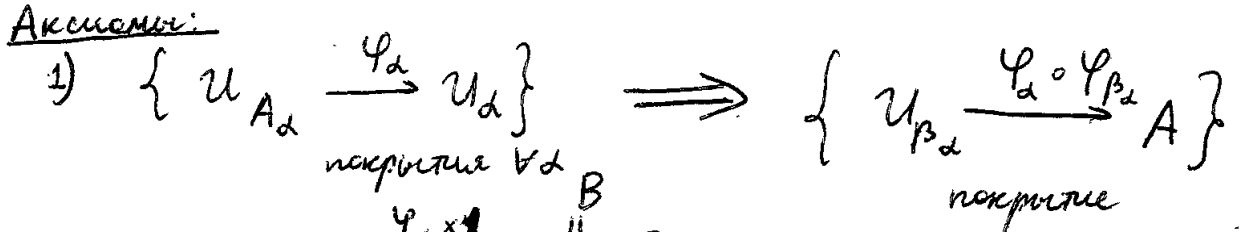
Based on the notion of a Grothendieck topology (“Topó”).

C — a category, for $\forall A \in C$ covers are specified $\{U_\alpha \xrightarrow{\varphi_\alpha} A\}$.

Uncertain reading. The loop and subscript above the arrow are heavily reinforced. The provisional reading is φ_α.

Original source details: GELFAND-P000352

Axioms:

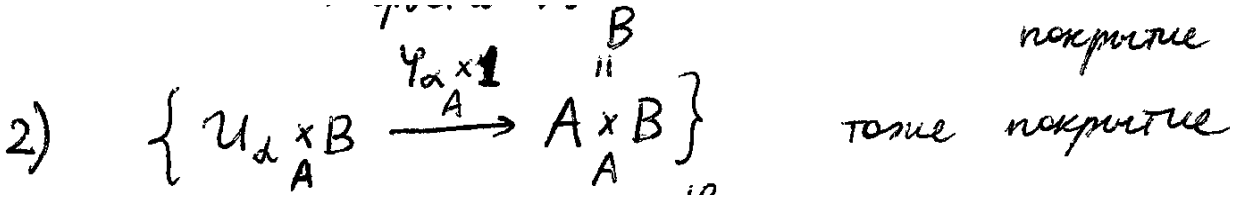


Original source details: GELFAND-P000352

Diagram and typed description.

1) $\{U_{A_\alpha} \xrightarrow{\varphi_\alpha} U_\alpha\} \quad \Longrightarrow \quad \{U_{\beta_\alpha} \xrightarrow{\varphi_\alpha \circ \varphi_{\beta_\alpha}} A\}$

covers $\forall \alpha$ — a cover



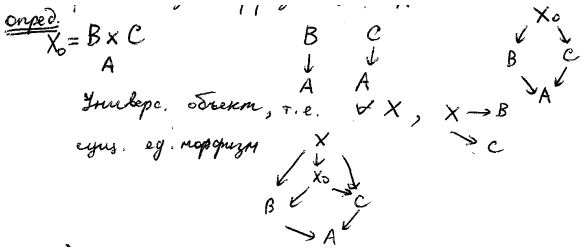
Original source details: GELFAND-P000352

Diagram and typed description.

$$\begin{array}{ccc} & & B \\ & & \| \\ \{U_\alpha \times_A B & \xrightarrow{\varphi_\alpha \times_A 1} & A \times_A B\} \end{array}$$

2) also a cover

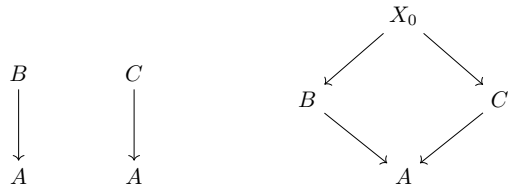
A morphism $B \xrightarrow{\varphi} A$ is given here.



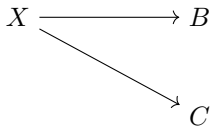
Original source details: GELFAND-P000352

Diagram and typed description.

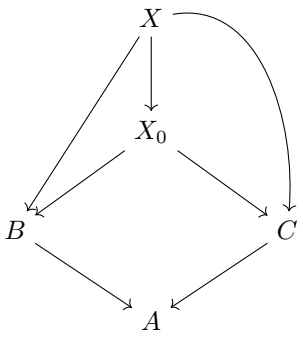
Def. $X_0 = B \times_A C$.



A universal object, i.e. $\forall X$,



there exists a unique morphism



8) $U \longrightarrow U$ a cover.

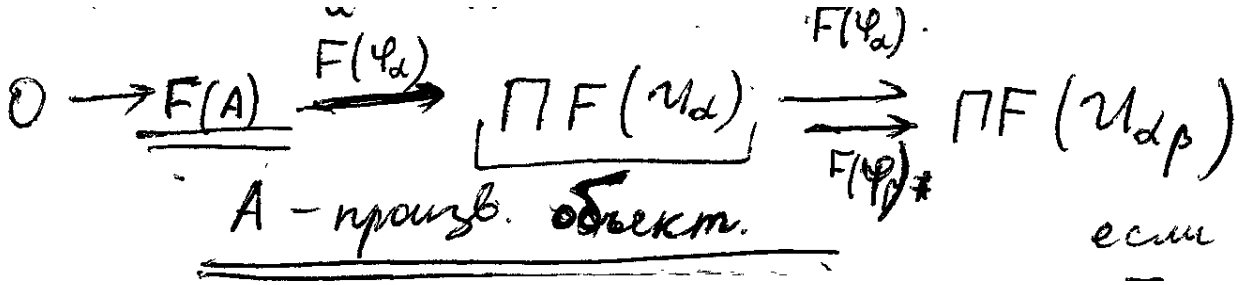
Original source details: GELFAND-P000352

Uncertain reading. Before $U \to U$ is a densely reinforced number with two loops. The visible 8 is provisionally retained; 3 remains a candidate, and the earlier number is not recovered.

(see **Mumford**.)

A sheaf — a functor $F : C \underset{\text{contravar.}}{\longrightarrow} K$ — another category. And the sheaf ax. holds.

$$U_\alpha \xrightarrow{\varphi_\alpha} A.$$



Original source details: GELFAND-P000352

Diagram and typed description. The first arrow from $F(A)$ is reinforced; beneath its strokes at the left is a remnant of a left-facing arrowhead, while the right arrowhead is clear. The order of alteration is not established. At the left of the lower parallel arrow are thickened strokes. After $F(\varphi_\beta)$ is a dense sign with two crossbars and a loop; its meaning is not established. Below $F(A)$ are two lines; below $\prod F(U_\alpha)$ is a bracket.

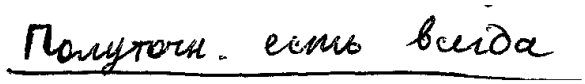
$$0 \longrightarrow F(A) \xrightarrow{F(\varphi_\alpha)} \prod F(U_\alpha) \underset{F(\varphi_\beta)}{\overset{F(\varphi_\alpha)}{\rightrightarrows}} \prod F(U_{\alpha\beta})$$

A — an arbitrary object.

If the images agree, then

$$A \longleftarrow U_\alpha \longleftarrow U_\alpha \times U_\beta.$$

Half-exactn. exists everywhere.



Original source details: GELFAND-P000352

Uncertain reading. The final abbreviation is provisionally read «Полуточн.» (“half-exactn.”).

GELFAND-P000353

$\mathcal{U} = \{U_\alpha \to A\}$. • Cohomology $H(\mathcal{U}, F)$ (Čech).

Nontriv. theorem. (!) One can construct a sheaf from a presheaf.

Original source details: GELFAND-P000353

Undeciphered passage. The entire crossed-out heading to the left of “from a group” is not read in full. “Predstavleniya” (“Representations”) remains an uncertain candidate.

from a group G construct “Topó”. and from a repr. — a sheaf over this “topó”.

Geronimus’s idea — cohomology and its relation to the old reprs.

G — a (Lie) group. C — the G-manifolds. Morphisms — G-maps, and moreover regular,

т.е. ∀ точке M $M \longrightarrow N$ кас. отобр. — эпиморфизм.

Original source details: GELFAND-P000353

Uncertain reading. Between M and "tang. map" a small sign is crossed out with several oblique strokes. Neither its letter nor its direction is established. Below M is written $M \to N$.

i.e. at $\forall$ point M the tang. map is an epimorphism.

$$M \longrightarrow N$$

2nd ex.

G — a fin. group — categ. — arbitrary G-sets. (Finite). Cat. K — the category of lin. spaces. Sheaves $\leftrightarrow$ representations. Let a repr. T_g in E be given.

[Crossed out: ~~Repr.~~ $\longrightarrow$]

Undeciphered passage. The crossed-out word at the right after the readable "Repr." and arrow is unreadable.

Original source details: GELFAND-P000353

commuting with the action of G.

Если X - G-пр-во, то надо рассм. отобр. X в E, из 5 точек.

Original source details: GELFAND-P000353

Diagram and typed description.

If X is a G-space, then one must consider maps from X to E,

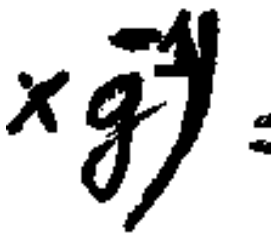
of 5 points.

$$\xi_{g(i)} = T_g \xi_i.$$

The five-tuples — a lin. space.

From X and T_g construct a space of fns. $\xi(x)$.

$$T_g \xi(xg^{-1}) = \xi(x), \qquad \xi(x) \in E.$$



Uncertain reading. The exponent on g is reinforced and is provisionally read -1.

Original source details: GELFAND-P000353

There are sheaves that are not obtained from representations.

Elementary-particle theory

Uncertain reading. Before “not constructed” is a short crossed-out word with an e-shaped first loop and following shch-shaped strokes. “eshchyo” (“yet”) and “eshch.” remain uncertain candidates; the dots over yo and the ending are not established.

Original source details: GELFAND-P000353

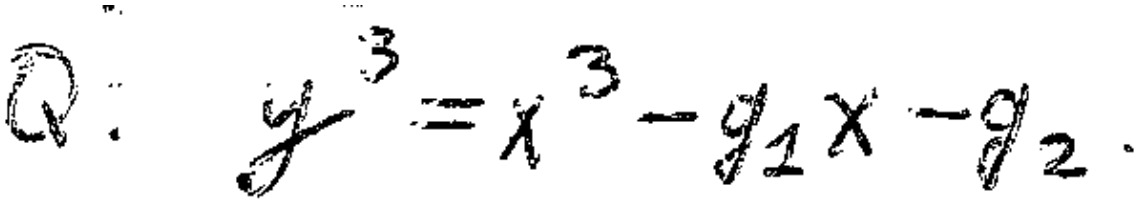
is not constructed, since not all reprs. are known. (cf. spinors and electrons.) (For a fin. group everything is obtained from here).

Sometimes one obtains the cohom. of a topol. space and cohom. as in Gelfand–Fuchs.

GELFAND-P000354

[source number: 13]

I. M. Gelfand on the Weil conjecture.



Original source details: GELFAND-P000354

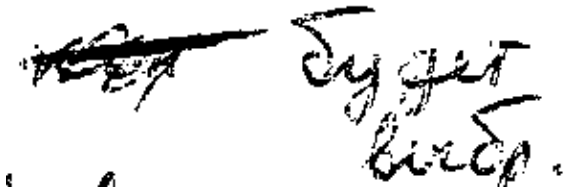
Diagram and typed description.

$$Q: \quad y^3 = x^3 - g_1 x - g_2.$$

g_1, g_2 — rat. numbers. — an ell. curve T. Over the compl. numbers.

There is a discr. group of matrices $\begin{pmatrix} \alpha & \beta \\ A\gamma & \delta \end{pmatrix}$, $\alpha, \beta, \gamma, \delta$ — integers, det $= 1$.

$A \in \mathbb{Z}$,



Uncertain reading. Between $A \in \mathbb{Z}$, and “will be chosen” a small word is crossed out with a horizontal stroke and diagonals. “pozd.” (“later”) remains an uncertain candidate.

Original source details: GELFAND-P000354

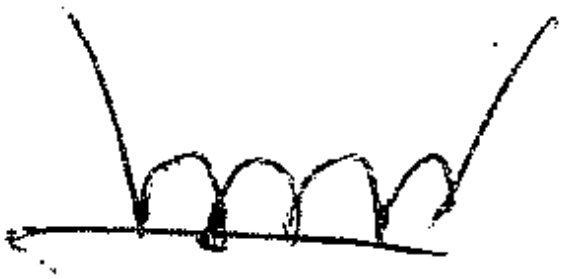
will be chosen. Group mod A.

They define fractional-lin. transf. of the upper half-plane into itself.

$$z' = \frac{\alpha z + A\gamma}{\beta z + \delta}.$$

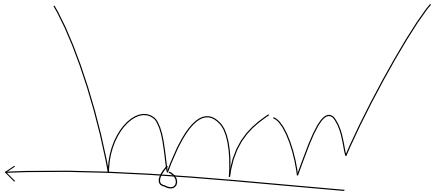
Fund. domain — the modular manifold X_A.

The group $\Gamma_0(A) = \begin{pmatrix} \alpha & \beta \\ A\gamma & \delta \end{pmatrix}$.



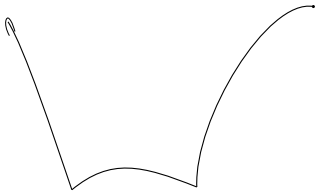
Original source details: GELFAND-P000354

Diagram and typed description. Four arcs above a slanted nearly horizontal line with a small left arrowhead. The two outer branches rise; the third arc has an upper gap, and the first inner foot has a loop. The loop is not identified as a label.

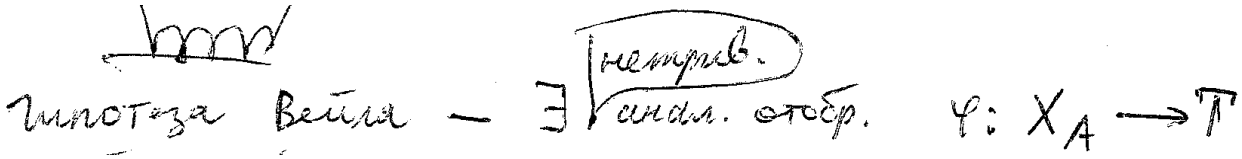


Original source details: GELFAND-P000354

Diagram and typed description.



$A = 1.$

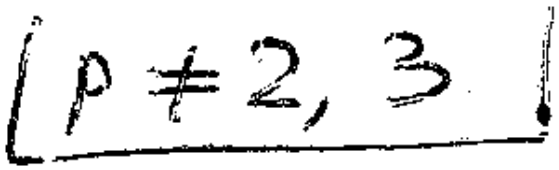


Original source details: GELFAND-P000354

Diagram and typed description.

The Weil conjecture — ∃ nontriv. an anal. map $\varphi : X_A \longrightarrow T$

(There are finitely many maps; from ∀ curve to a torus)



Original source details: GELFAND-P000354

Diagram and typed description.

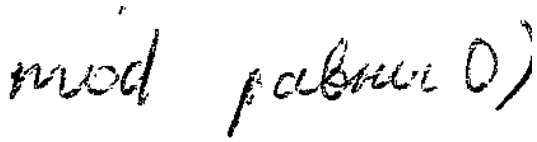
$p \neq 2, 3$

Replace T mod p. A curve over $\mathbb{F}_p$ results. It

Uncertain reading. After “It” at the start of the next line before “can” is a dense overlap of letters resembling “on” (“he/it”). It is not established whether this is a cancelled repetition of the pronoun.

Original source details: GELFAND-P000354

can become singular.
(I.e. the partial derivs. mod equal 0).



Original source details: GELFAND-P000354

Manuscript notation. After “mod” the manuscript goes directly to “equal,” with the modulus unstated.

1) No sing. points.

2) Sing. points with distinct tangents.

2a — real

Original source details: GELFAND-P000354

Undeciphered passage. An entire crossed-out word between “real” and “tang.” is unreadable.

tang.

2b — imaginary tangents.

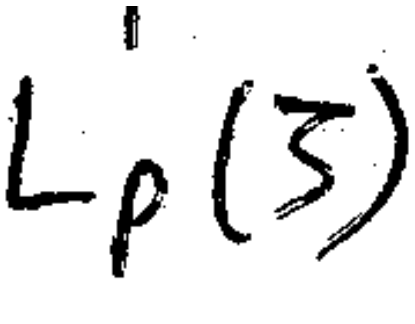
3) The tangents coincide.

$$\begin{array}{lll} A_p = 1 & p & \text{type 1} \\ A_p = p & p & \text{type 2} \\ A_p = p^2 & p & \text{type 3} \end{array}$$

GELFAND-P000355

$A = \prod_p A_p$ — the conductor of the curve.

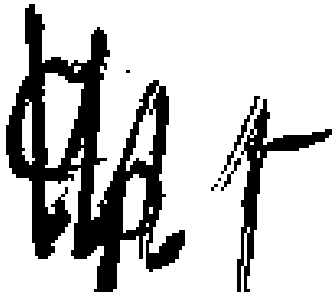
$$L_p(\zeta) = (1 - c_p p^{-s} + p^{1-2s})^{-1}, \qquad p \text{ — of type 1.}$$



Original source details: GELFAND-P000355

Manuscript notation. The first argument reads ζ, as in $\zeta(s)$ below.

[Crossed out before the sign “+”:]



Original source details: GELFAND-P000355

Undeciphered passage. The initial entry before “+ number of points of the curve” is densely overwritten and crossed out. The whole earlier sign or short entry has not been recovered.

$+$ number of points of the curve:

$$1 - c_p + p.$$

$$L_p(s) = (1 - \varepsilon_p p^{-s})^{-1}, \qquad \text{case 2)}; \qquad \begin{matrix} \varepsilon_p = +1 & \text{2a}, \\ \varepsilon_p = -1 & \text{2b}. \end{matrix}$$

$L_p(s) = 1$, case 3.

$$\zeta(s) = \prod_p L_p(s).$$

Expand $\zeta(s)$ in a series:

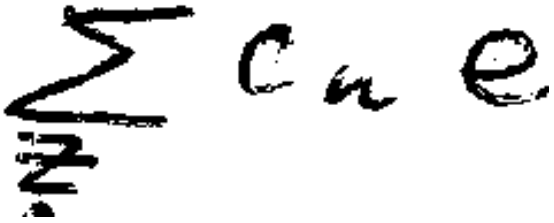
$$\zeta(s) = \sum_1^\infty \frac{c_n}{n_s}.$$

Uncertain reading. The small s in the denominator is below n. The visible n_s is retained without replacing it by a power n^s.

Original source details: GELFAND-P000355

Now consider

$$\sum_z c_n e^{inz} = \omega(z).$$

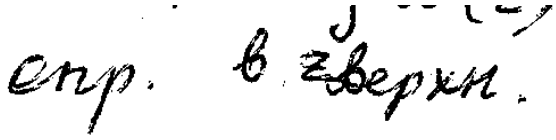


Uncertain reading. The lower summation sign has two horizontal strokes and an oblique middle, resembling z. The coefficient and exponent use n; the lower z is provisionally retained.

Original source details: GELFAND-P000355

Assertion: $\int \omega(z)\, dz$ — the sought fn..

It is str. in the upper half-plane. (and analyt.)



Uncertain reading. The abbreviation reads “str.”. Overlapping earlier strokes before “upper” are crossed out.

Original source details: GELFAND-P000355

Maps the upper half-plane into the compl. line.

The Weil conjecture:

I. $\exists$ a lattice $\Gamma \subset \mathbb{C}^1$ such that $\forall g \in \Gamma_0(A)$

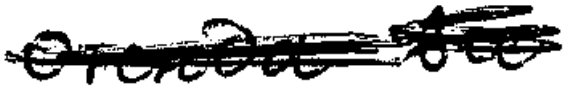
$$\int_z^{gz} \omega(z)\, dz \in \Gamma.$$

[Crossed out at lower left:]

Original source details: GELFAND-P000355

Undeciphered passage. The short entry in the lower left margin is covered by horizontal and oblique strokes. A complete reading is not established.

[Crossed-out line at the bottom:]

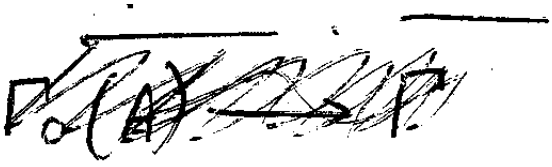


Original source details: GELFAND-P000355

Uncertain reading. The crossed line is provisionally read “otsyuda vse” (“from here all”); “vsyo” (“everything”) is also possible. The last ending and possible dots over yo are obscured by horizontal strokes.

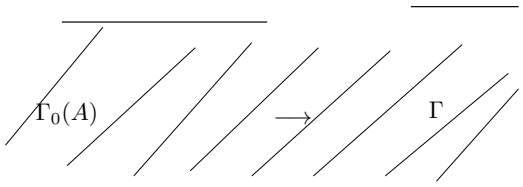
~~from here all~~

[Crossed out above the continuation of this line:]



Original source details: GELFAND-P000355

Uncertain reading. The crossed entry $\Gamma_0(A) \to \Gamma$ is shown provisionally. The letter in parentheses is reinforced over other strokes; A remains a candidate, and the earlier argument has not been recovered. The oblique strokes and two upper horizontal lines are rendered typographically.

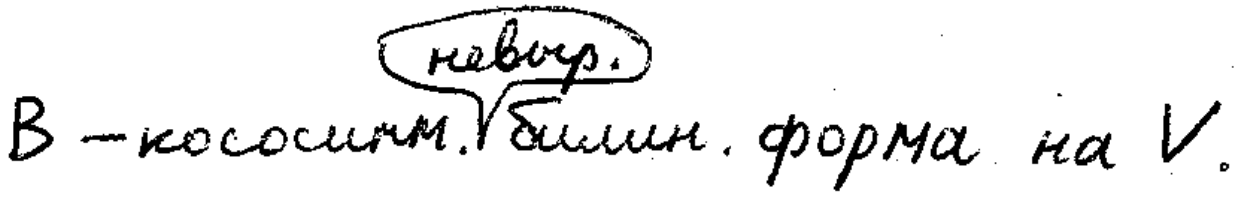


GELFAND-P000356

[source number: 14]

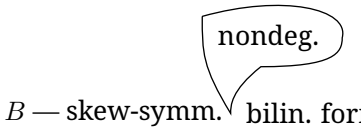
12/IV. Perelomov.

V — a $2N$-dim. real space,



Original source details: GELFAND-P000356

Diagram and typed description.

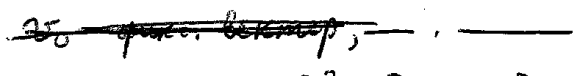


A B-lattice $L \subset V$,

Undeciphered passage. The short crossed-out entry between $L \subset V$, and the following dash has not been read in full.

Original source details: GELFAND-P000356

— on L, B takes integer values. $J^2 = -1$, J is compatible with B: $J^T B J = B$.



Uncertain reading. The crossed row is provisionally read v_0 fix. vector. Middle strokes of "fix." are covered by horizontals; the abbreviation is not expanded. After the word, a dash, comma and long horizontal stroke are retained.

~~v_0 fix. vector~~ —, ———————

Original source details: GELFAND-P000356

Gelfand on Pauli's book:

$$P^2 = P, \qquad \lambda^2 = \lambda, \qquad \lambda = 0, +1, -1.$$

-1 discarded for phys. reasons.

Diagram and typed description.

——— , ———————

Original source details: GELFAND-P000356

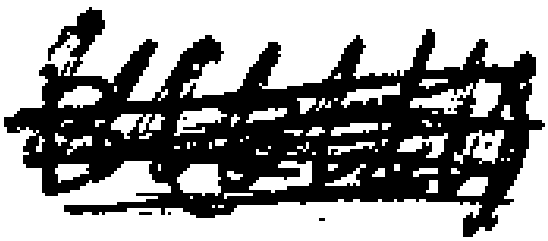
Volume of the fund. parallelotope $= 1$. (principal lattice). Let $BJ > 0$.

Uncertain reading. The sign after BJ is reinforced over earlier strokes, with an additional horizontal line. $>$ is shown provisionally; $\geq$ remains a possible reading.

Original source details: GELFAND-P000356

A compl. structure on V. $z \in \mathbb{C}^N$.

$H(z, z') = B(z, Jz') + iB(z, z')$ — a Hermitian form.



Undeciphered passage. Between the equals sign and the current $B(z, Jz')$, an earlier formula with a B-like beginning is covered by repeated cancellation. Its arguments and signs are unreadable.

Original source details: GELFAND-P000356

Consider a fn. F: — integer-valued.

$$F(\alpha+\beta) \equiv F(\alpha) + F(\beta) + B(\alpha,\beta) \pmod 2.$$

A lin. form may be added to F. Choose a basis in L.

$$v \in L \mapsto \kappa_1, \dots, \kappa_N, \ell_1, \dots, \ell_N.$$

Assume:

$$B = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Undeciphered passage. The short crossed-out word before $F(\alpha)$ has not been read in full. An initial loop and a lower stroke are visible, but a complete reading is not established.

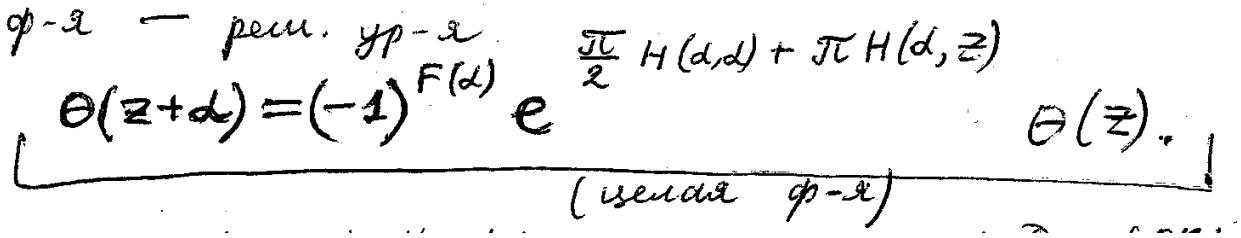
Original source details: GELFAND-P000356

$$F(\alpha) = \sum \kappa_i \ell_i + \sum \varepsilon_i \kappa_i + \sum \tilde{\varepsilon}'_i \ell_i \qquad \text{(there exist } 4^N \text{ solutions)}.$$

Uncertain reading. Above the last coefficient is a separate oblique stroke below the right part of the tilde. It is provisionally rendered as an additional prime on $\tilde{\varepsilon}_i\prime$; its exact position remains uncertain.

Original source details: GELFAND-P000356

A Θ-fn. — a solution of the eqn.



Original source details: GELFAND-P000356

Uncertain reading. The current $H(\alpha, z)$ has no separate upper bar on z; below it is a remnant of an overlapped curve, not separated into a secure sign. Below the whole formula is a three-sided bracket without a top, with "(entire fn.)" underneath.

$$\Theta(z+\alpha) = (-1)^{F(\alpha)} e^{\frac{\pi}{2} H(\alpha,\alpha) + \pi H(\alpha,z)} \Theta(z).$$

(entire fn.).

There exists only one solution of the functional eqn. up to const. (It is important that the lattice is principal — $\det B = 1$). This is written in Cartier (although known before him).

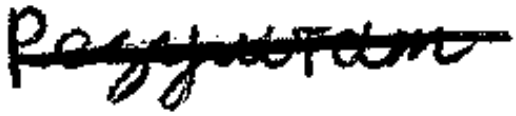
GELFAND-P000357

Phys. problem (see von Neumann). Let a — annihil. oper., $a|\alpha\rangle = \alpha|\alpha\rangle$ — eigenvectors.

$$e^{-\frac{(x-\lambda)^2}{2}}.$$

Spectrum — the whole compl. plane. The system is overcomplete. Problem: select a complete system.

~~Result~~



Manuscript notation. The crossed-out word reads “Result.”

Original source details: GELFAND-P000357

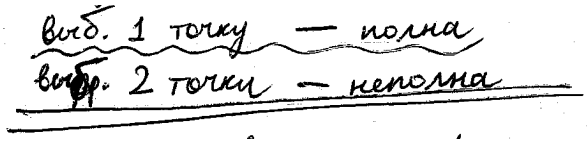
Overcomplete — if discarding $\forall$ finite number leaves a complete system. (if λ has an accumulation point, then ...) For which seq. λ_n is this a complete system? Let λ — a lattice on the compl. plane C.

cell area $> \pi$ — incomplete,
— ″ — $< \pi$ — overcomplete.

Let the cell area $= \pi$.

vyb. 1 point — complete.

vybr. 2 points — incomplete.

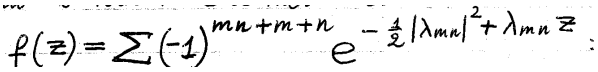


Uncertain reading. Before “2 points”, “vybr.” is shown provisionally; “vyb.” remains a candidate. After the b there is a loop and a descending r-like stroke, but the earlier entry is not securely separated. The abbreviation is left unexpanded. The first line has a wavy underline, the second two horizontal underlines.

Original source details: GELFAND-P000357

I.e., there is exactly 1 relation.

$$f(z) = \sum (-1)^{mn+m+n} e^{-\frac{1}{2}|\lambda_{mn}|^2 + \lambda_{mn} z} = 0.$$



Manuscript notation. The exponent ends in z with a crossing stroke, as in $f(z)$.

Original source details: GELFAND-P000357

$\lambda_{mn} = m\omega_1 + n\omega_2$. (already in F_2) — everything is divisible. or

$$\sum (-1)^{m\boldsymbol{n}+m+n} e^{-\frac{(x-\lambda_{mn})^2}{2}}.$$

Diagram and typed description. In the initial mn of the sign exponent, the second letter is thickened and overwritten. A separate earlier reading is not established.

Original source details: GELFAND-P000357

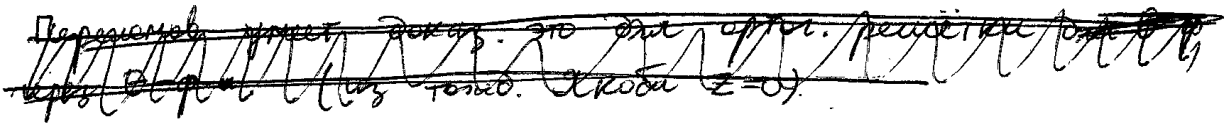
$$|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2}.$$

(Neumann and Schoenberg — pos. def. fns.). Von Neumann's question: consider a lattice with area π and fns. $|\alpha_{mn}\rangle$. In this state determine the variances of X and p. $\delta_x\,\delta_p$ usually grows for an orthonorm. system.

GELFAND-P000358

[source number: 15]

Von Neumann's question. Will $\delta_x\,\delta_p$ after orthogonalization of these vectors be bounded on this system? Gelf. considers this unlikely. The gen. question for an orthog. system is unclear.



Original source details: GELFAND-P000358

Undeciphered passage. Legible fragments of the two crossed-out lines are retained. The end of the first line and beginning of the second, including part of a parenthesis, are unrecovered.

~~Perelomov can prove this for an orthog. lattice [?]~~

~~[?] (from the Jacobi ident. $(z = 0)$).~~

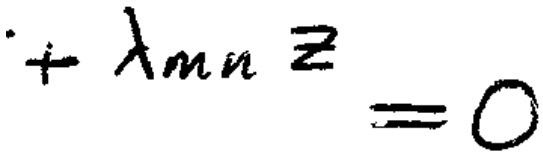
π The gen. ident. is not deduced from the Jacobi ident..

Uncertain reading. The current preposition is provisionally read "from"; it is rewritten in dark strokes. The earlier entry underneath is not securely separated.

Original source details: GELFAND-P000358

This carries over to the N-dim. case. Obtain $2^{N-1}(2^N - 1)$ identities. Thus, an elem. problem:

$$\sum (-1)^{mn+n+m} e^{-\frac{1}{2}|\lambda_{mn}|^2 + \lambda_{mn} z} = 0$$



Manuscript notation. The final z in the exponent has a stroke through the middle of the letter.

Original source details: GELFAND-P000358

if the lattice area $= \pi$. Proof: consider a Hilb. space with

$$\int \bar{f}(z) g(z) e^{-|z|^2}\, dz\, d\bar{z}.$$

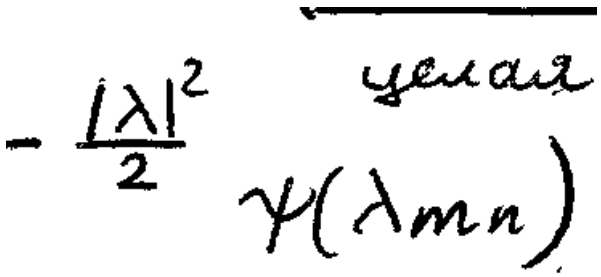
$|\boldsymbol{\lambda}_{mn}\rangle$
Consider $|\lambda\rangle$ and $|\psi\rangle \in \mathcal{H}$.

$$\langle \psi | \lambda \rangle = e^{-\frac{|\lambda|^2}{2}}\ \psi(\lambda)\cdot$$

entire analytic.

Orthog.

$$\langle \psi | \lambda_{mn} \rangle = e^{-\frac{|\lambda|^2}{2}} \psi(\lambda_{mn}).$$



Manuscript notation. The norm has λ without mn. The ket and the argument of ψ have λ_{mn}.

Original source details: GELFAND-P000358

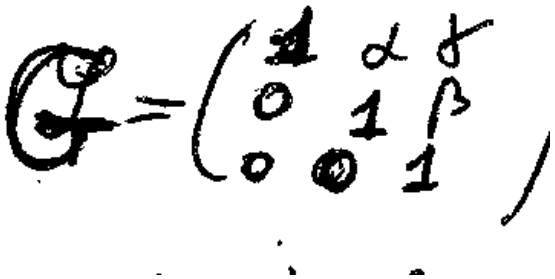
Entire with 0s at the prescribed set of points. Need to prove that it does not exist in F_2. (see Levin, standard theorem).

Undeciphered passage. The isolated crossed-out mark before G is not recovered in full.

Original source details: GELFAND-P000358

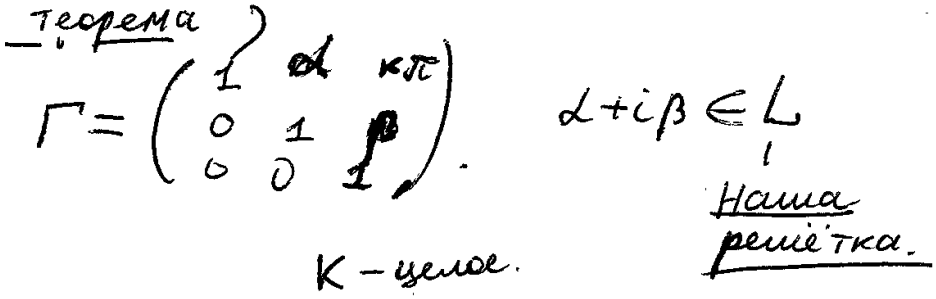
$$G = \begin{pmatrix} \mathbf{1} & \alpha & \delta \\ 0 & 1 & \beta \\ 0 & \mathbf{0} & 1 \end{pmatrix},$$

α, β, δ — real.



Manuscript notation. The upper left one and the middle zero in the bottom row are written over in heavier strokes.

Original source details: GELFAND-P000358



Original source details: GELFAND-P000358

Diagram and typed description. The current α and β in Γ are heavily overwritten; earlier strokes are unrecovered. The upper right entry is provisionally shown as $k\pi$. The note below has a capital K. Below L is a short detached vertical stroke towards the two underlined words “Our lattice”; its exact connection is not established.

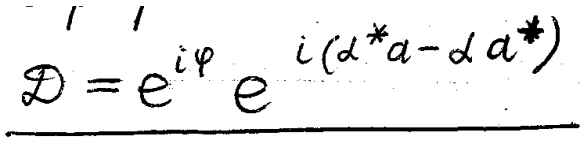
$$\Gamma = \begin{pmatrix} 1 & \boldsymbol{\alpha} & k\pi \\ 0 & 1 & \boldsymbol{\beta} \\ 0 & 0 & 1 \end{pmatrix}. \qquad \alpha + i\beta \in L$$

K — integer. Our lattice.

GELFAND-P000359

Consider the operators.

$\mathcal{D} = e^{i\varphi} e^{i(\alpha^* a - \alpha a^*)}$ — a 3-param. group.



Manuscript notation. The exponent of the second exponential has an i before the parenthesis.

Original source details: GELFAND-P000359

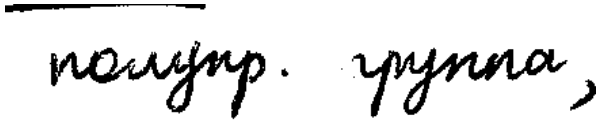
Lie alg. — a, a^+ and 1; $[a, a^+] = 1$, other commutators 0. Γ — the α-points of the lattice; φ — 0 and π. Then the repr. induced by the repr. $(-1)^{mn+m+n}$ of the discrete group Γ is irreducible. There is an extension to an arbitrary group.

Kazhdan.

Uncertain reading. “Kazhdan.” is shown provisionally; compressed middle strokes also permit the literal “Kamdan.”.

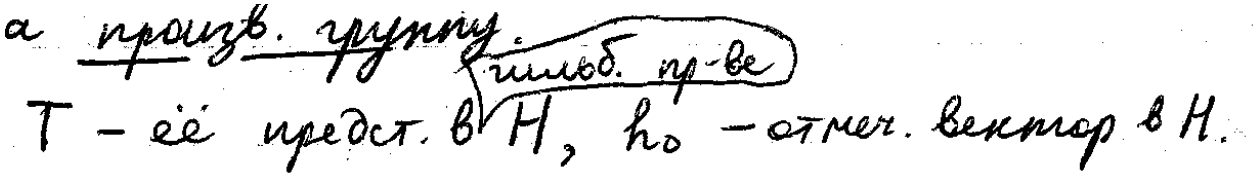
Original source details: GELFAND-P000359

G — a polupr. group,



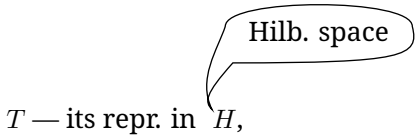
Uncertain reading. After G — “polupr.” is shown provisionally; “polugr.” remains a candidate. The abbreviation is not expanded.

Original source details: GELFAND-P000359



Original source details: GELFAND-P000359

Diagram and typed description.

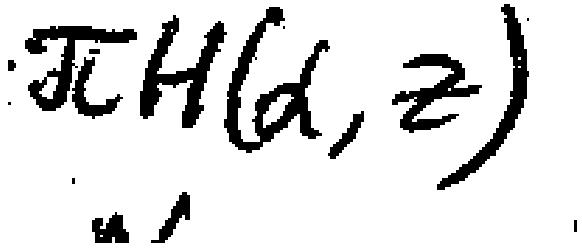


T — its repr. in H,

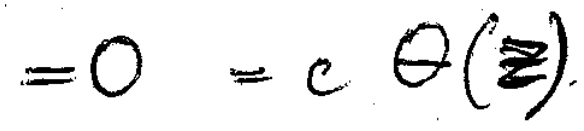
h_0 — a distinguished vector in H. $T(g)h_0 \in H$. $\Gamma \subset G$ a discrete subgroup, $T(\gamma)h_0 \in H$. For a reasonable subgroup Γ the system is complete and satisfies a finite number of relations. Here — a unipotent group.

$$f(z) = \sum_{\alpha} (-1)^{F(\alpha)} e^{-\frac{\pi}{2}H(\alpha,\alpha)+\pi H(\alpha,z)} = 0 = c\Theta(z).$$

Manuscript notation. The z in $H(\alpha, z)$ has a crossing stroke through the letter.

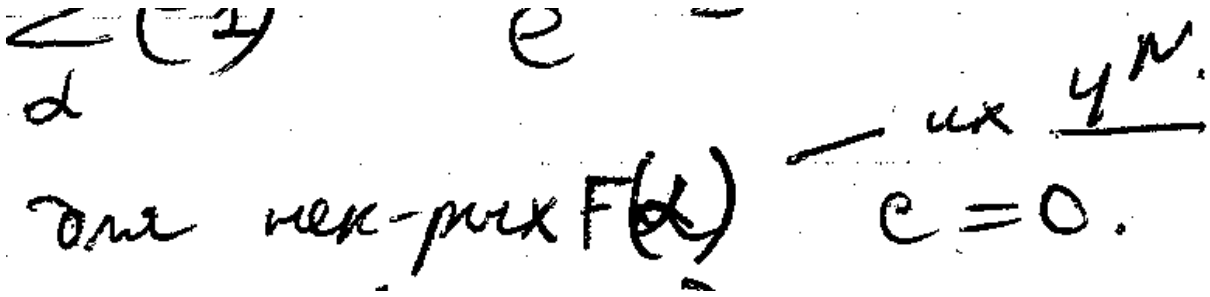


Original source details: GELFAND-P000359



Uncertain reading. The source literally writes $= 0 = c\Theta(z)$. The last argument of Θ is repeatedly retraced, with overlapping upper and lower strokes. z is shown provisionally; the earlier sign and any separate upper bar are not established.

Original source details: GELFAND-P000359



Original source details: GELFAND-P000359

Diagram and typed description.

— of them 4^N

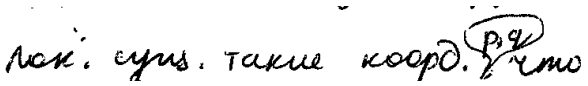
for some $F(\alpha)$ $\quad c = 0.$

$c \neq 0$ — A new repr. of the Θ-fn. For half of the $F(\alpha)$ this is 0. $\Theta(z)$ — the standard fn., $\neq 0$. Next meeting 1st Mon. of September.

GELFAND-P000360

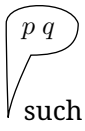
[source number: 22]

Kirillov on quantization. (Kostant etc.) Trivial part. A symplectic manifold — X^{2n} — even-dimensional with a nondeg. closed 2-form ω ($d\omega = 0$).



Original source details: GELFAND-P000360

Diagram and typed description.

Locally there exist coords.  such that

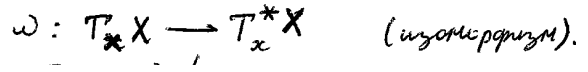
$$\omega = \sum_{i=1}^{n} dp_i \wedge dq_i.$$

(see Rashevskii, “Geom. theory of diff. eqns.”, Sternberg, “Lectures on diff. geom.”). Volume

$$\Omega = \frac{1}{n!}\omega \wedge \omega \wedge \cdots \wedge \omega = \frac{1}{n!}\omega^n \quad \text{(exterior power)}.$$

Identify tang. space with cotang. using the form ω.

$$\omega(\xi, \eta) :— \quad \omega : T_x X \to T_x^* X \quad \text{(isomorphism)}.$$



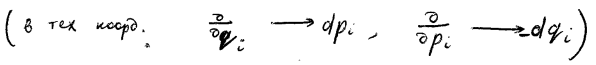
Original source details: GELFAND-P000360

Uncertain reading. The lower index of the first T is heavily overwritten. x with overlapping upper strokes is shown provisionally; $*$ remains a candidate.

$$\xi \to \alpha_\xi;$$

$$\alpha_\xi(\eta) = \omega(\xi, \eta).$$

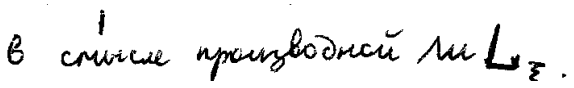
$$\left(\text{in those coords. } \frac{\partial}{\partial q_i} \to dp_i, \quad \frac{\partial}{\partial p_i} \to -dq_i\right).$$



Original source details: GELFAND-P000360

Manuscript notation. The coordinate line gives dp_i and $-dq_i$.

Closed 1-forms $\leftrightarrow$ vector fields preserving the form ω.



Original source details: GELFAND-P000360

Diagram and typed description.

|

in the sense of the Lie derivative L_ξ.

L_ξ acts on all geom. objects on the manifold. φ_t — the corresponding one-param. group, corresponding to the field ξ. ($\dot{x} = \xi(x)$).

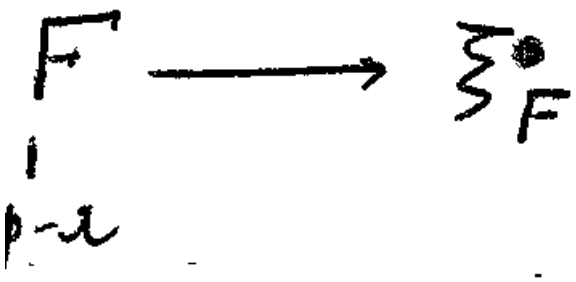
$$\left.\frac{d}{dt}\right|_{t=0} = L_\xi.$$

It commutes with differentiation of diff. forms: $L_\xi d = dL_\xi$. $L_\xi f = \xi f$, $L_\xi \eta = [\xi, \eta]$.

$$\boxed{L_\xi \omega = 0 \;\Leftrightarrow\; d\alpha_\xi = 0.}$$ — a short formulation.

Such fields — Hamiltonian, forms — gradient. ξ — loc. Hamiltonian $\Leftrightarrow \alpha_\xi = df$. f is defined up to const and is multivalued.

GELFAND-P000361



Uncertain reading. Beside the first ξ_F is a dense round overwritten mark. It is shown as a round mark without assigning a mathematical meaning; the exact earlier sign is not established.

$F \longrightarrow \xi_F^{\bullet}$

|

fn.

Original source details: GELFAND-P000361

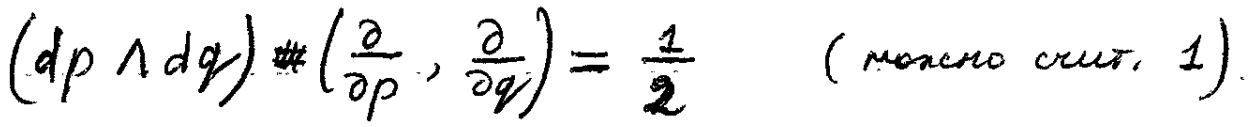
dF — a form $\longleftrightarrow$ Hamiltonian field; $dF = 2\alpha_{\xi_F}$. Poisson bracket.

$$\{F_1, F_2\} = \xi_{F_1} F_2 = -\xi_{F_2} F_1 = 2\omega(\xi_{F_1}, \xi_{F_2}).$$

Identity:

$$\xi_{\{F_1,F_2\}} = [\xi_{F_1}, \xi_{F_2}].$$

Remark:



Original source details: GELFAND-P000361

Uncertain reading. Between $(dp \wedge dq)$ and the parenthesis with vectors is a dense crossed hash-like mark. Its strokes are shown without a mathematical interpretation.

$(dp \wedge dq)$ ## $\left(\frac{\partial}{\partial p}, \frac{\partial}{\partial q}\right) = \frac{1}{2}$ (one may take 1).

$$\frac{1}{2}[\alpha_1(\xi)\alpha_2(\eta) - \alpha_2(\xi)\alpha_1(\eta)] = \alpha_1 \wedge \alpha_2.$$

A new object is constr. when $[\omega] \in H^2(X, Z)$. Constr. a complex 1-dim. bundle and a map $\mathbb{C}$.

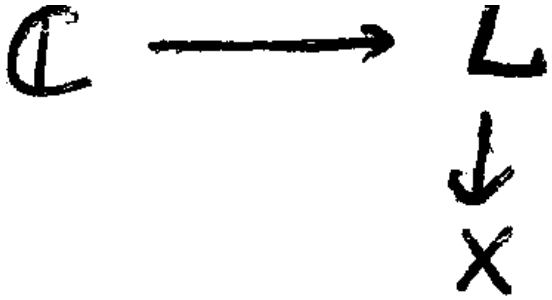
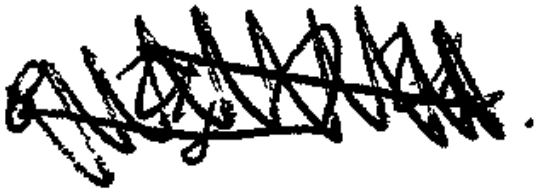


Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C} & \longrightarrow & L \\ & & \downarrow \\ & & X \end{array}$$

Original source details: GELFAND-P000361

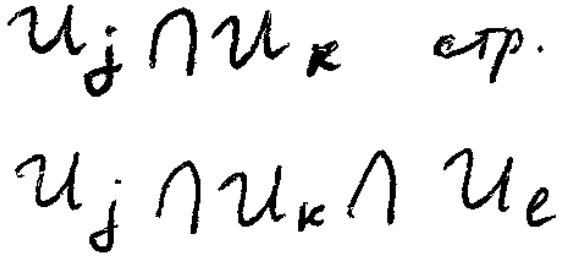
$\{U_j\}$ — a covering, $\omega = d\alpha_j$ in U_j. On $U_j \cap U_k$ constr. $\alpha_j - \alpha_k = df_{jk}$. ~~$\alpha$~~[?]



Undeciphered passage. After df_{jk}, the beginning of the crossed entry is provisionally shown as α. Small signs, subscripts and the continuation are not recovered in full.

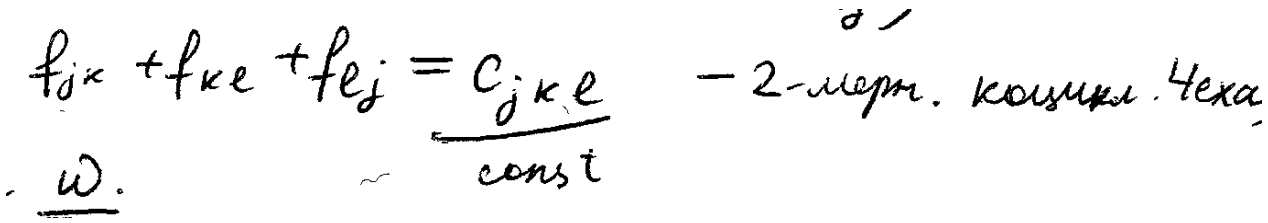
Original source details: GELFAND-P000361

on $U_j \cap U_k \wedge U_\ell$ we have $d(f_{jk} + f_{k\ell} + f_{\ell j}) = 0$ and hence



Uncertain reading. The second sign between the sets is pointed: $\wedge$ is shown provisionally, without replacing it by the usual intersection.

Original source details: GELFAND-P000361



Original source details: GELFAND-P000361

Diagram and typed description.

$f_{jk} + f_{k\ell} + f_{\ell j} = \underset{\text{const}}{C_{jk\ell}}$ — a 2-dim. Čech cocycle,

corresponding to the form $\underline{\omega}$. Integer-valued — $\exists \alpha_j, f_{jk}$ such that $C_{jk\ell}$ — integer-valued.

$g_{jk} = e^{2\pi i f_{jk}}$ — a set of bundle transition fns. (1-dim.).

$$g_{jk} \cdot g_{k\ell} \cdot g_{\ell j} = 1.$$

GELFAND-P000362

[source number: 23]

Choose other α, [?] ~~const [?]~~

Original source details: GELFAND-P000362

Undeciphered passage. In the cancelled continuation a const-like fragment is shown provisionally. Other words and signs among long looping cancellation strokes are not read in full.

The equiv. class of the constr. bundle depends on choosing an element of

$$H^1(X, \mathbb{C}^*).$$

$H^1(X, \mathbb{C}^*)$ acts simply transitively on the set of bundles. ($\mathbb{C}^*$, rather than

Undeciphered passage. The small crossed-out token before “sheaf” is wholly unreadable.

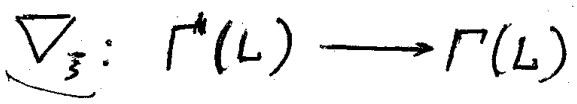
Original source details: GELFAND-P000362

the sheaf of continuous fns.) s_j in U_j, $s_j = g_{jk}s_k$ — a section of the bundle.

Original source details: GELFAND-P000362

Diagram and typed description.

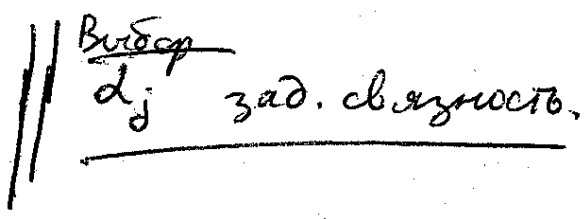
Connection — derivative of sections along a vector field.



Original source details: GELFAND-P000362

Uncertain reading. The small upper mark on the first Γ is thickened. $\Gamma^1(L)$ is shown provisionally; $\Gamma\prime(L)$ remains a candidate. Below ∇_ξ is a curved underline.

$$\nabla_\xi : \quad \Gamma^1(L) \to \Gamma(L).$$



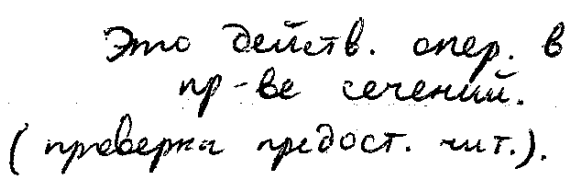
Original source details: GELFAND-P000362

Diagram and typed description.

Choice of α_j defines the connection.

$$(\nabla_\xi s)_j = \xi s_j - 2\pi i \alpha_j(\xi) s_j.$$

This acts as an oper. on the space of sections. (checking left to the reader).



Original source details: GELFAND-P000362

Uncertain reading. The first word is provisionally read “checking”; “to check” remains a candidate. The following Russian abbreviations are “predost. chit.”, with a separate o-like loop after d. The abbreviations remain unexpanded in the Russian text.

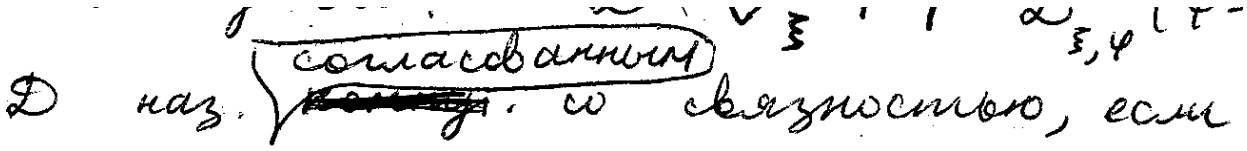
Check:

$$(\nabla_\xi s)_j = g_{jk}(\nabla_\xi s)_k.$$

Define a Hermitian bundle with a connection and a generalized Fourier transform in it. Consider an arbitrary 1st-order diff. oper. in the bundle L.

$$\mathcal{D} : \Gamma(L) \to \Gamma(L) \quad \text{(ordinary in local coords.)}.$$

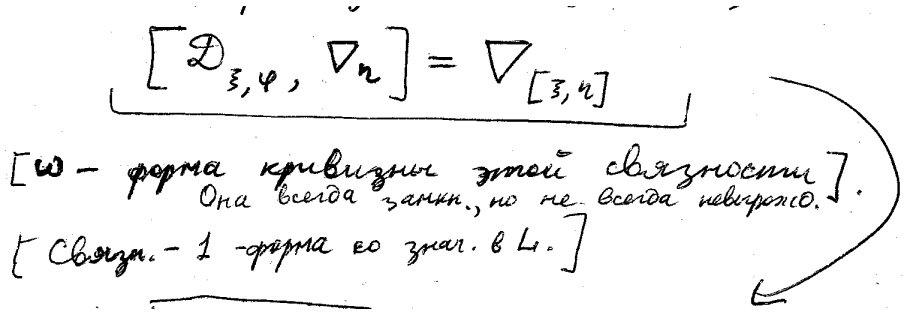
Its gen. form: $\mathcal{D} = \nabla_\xi + \varphi = \mathcal{D}_{\xi,\varphi}$ (φ — multiplication by a fn.).



Original source details: GELFAND-P000362

Undeciphered passage. The earlier adjective is crossed out and not recovered in full. Above it is the legible inserted and outlined word “compatible”.

$\mathcal{D}$ is called ~~[?]~~ compatible with the connection if



Original source details: GELFAND-P000362

Diagram and typed description. An arc starts at the right of the commutator, bows to the right beside the remarks in square brackets, and ends down-left above the lower frame. No additional mathematical relation is assigned to the arc.

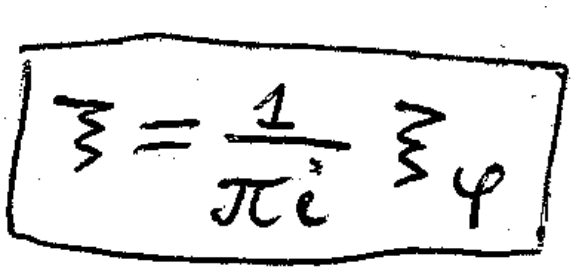
$$[\mathcal{D}_{\xi,\varphi}, \nabla_\eta] = \nabla_{[\xi,\eta]}.$$

[ω — the curvature form of this connection.
It is always closed, but not always nondeg.]

[Connection — a 1-form with values in L.]

Or

$$\boxed{\xi = \tfrac{1}{\pi i}\xi_\varphi}$$



Manuscript notation. The box reads $\xi = \frac{1}{\pi i}\xi_\varphi$.

Original source details: GELFAND-P000362

(in particular, ξ is actually Hamiltonian.)

GELFAND-P000363

~~$\mathcal{D}_\varphi$~~

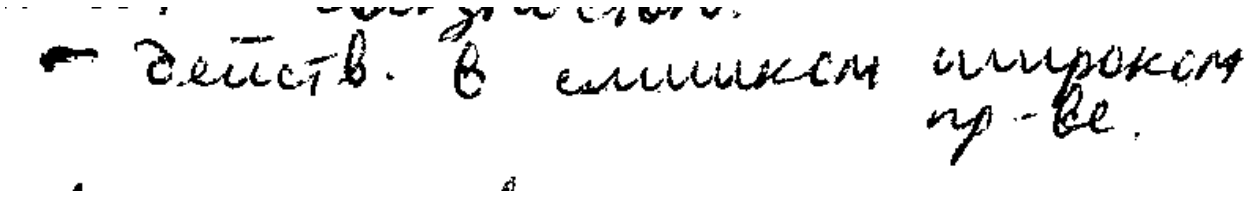
Uncertain reading. The crossed initial sign is provisionally shown as $\mathcal{D}_\varphi$; overlapping strokes do not permit a secure reading.

Original source details: GELFAND-P000363

The operator $\mathcal{D}_\varphi = \nabla_{\xi_\varphi} + \pi i\varphi : \Gamma(L) \to \Gamma(L)$. Fund. theorem:

$$[\mathcal{D}_\varphi, \mathcal{D}_\psi] = \mathcal{D}_{\{\varphi,\psi\}}.$$

(obtain a repr. of the Lie algebra) compatible with the connection. (Van Hove repr.: fns. $\to$ opers. on bundle sections) compatible with the connection. This is called prequantization.

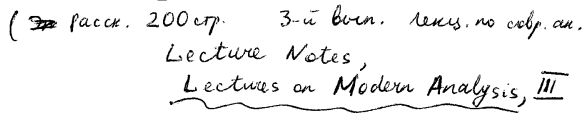


Original source details: GELFAND-P000363

Diagram and typed description. Before the note at right is a short left arrow with a shaft on its right. Its exact endpoint is not established; no additional logical connection is assigned.

$\leftarrow$ acts on too large a space.

(~~This.~~ consider p. 200 of the 3rd issue of Lectures on Modern An., Lecture Notes, Lectures on Modern Analysis, III).



Uncertain reading. Before "consider," the crossed-out word provisionally reads "This." The source puts "200" before the page abbreviation.

Original source details: GELFAND-P000363

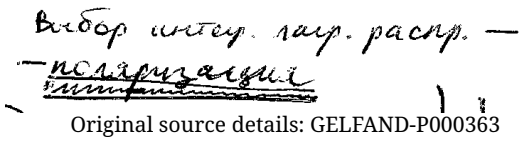
A Lagrangian distribution (isotropic space) — — at $\forall$ point a Lagr. subspace is given. $\frac{1}{2}$ the

dimension.

$$F : F_x \subset T_* X, \qquad \omega|_{F_x} = 0, \qquad \dim F_x = n = \frac{1}{2} \dim X.$$

Complex distributions will be considered:

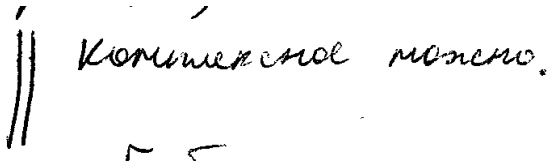
$$F_x \subset T_x^C X, \qquad \omega|_{F_x} = 0, \qquad \dim_C F_x = n = \frac{1}{2} \dim X.$$



Original source details: GELFAND-P000363

Diagram and typed description. Written vertically in the left margin.
The choice of an integr. Lagr. distrib. —
— polarization

It is considered integrable (Frobenius cond.) — see Dieudonné. (Locally exists — because of $\sum dp \wedge dq$) take the p-subspace.

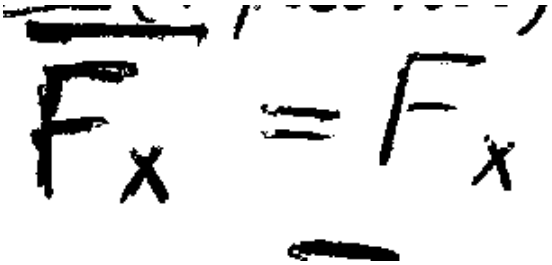


Original source details: GELFAND-P000363

Diagram and typed description.
The complex case is possible.

Example: a sphere with its usual volume. Globally impossible. Special cases (extreme).

1) $\overline{F}_x = F_x$ — a real distribution. (real polarization)



Original source details: GELFAND-P000363

Uncertain reading. The first F and its bar in the first equality are heavily thickened or rewritten. The current reading is provisionally shown as $\bar{F}_x = F_x$; earlier strokes are not separated.

2) $F_x \cap \bar{F}_x = 0$. (Kahler polarization).

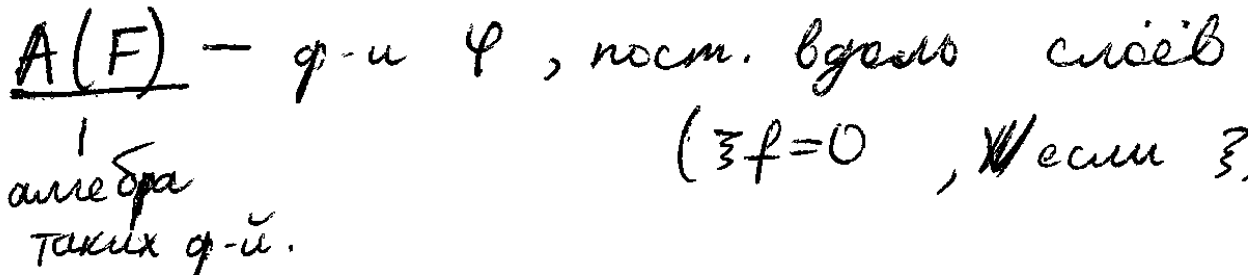
$$\dim(F_x + \bar{F}_x) = k \quad (= \text{const}) \quad (\text{and an integr. bundle}).$$

There is a mixed-mfd. structure there (a bundle with a real base and complex fibres; transitions — holom. transformations of the fibres).

GELFAND-P000364

[source number: 24]

F — a distrib., $\Gamma(L,F)$ — sections constant along the leaves, i.e., $\nabla_\xi s = 0$ if $\xi_x \in F_x$. Now consider the action of opers. on $\Gamma(L,F)$.



Original source details: GELFAND-P000364

Diagram and typed description.

$A(F)$ — fns. φ constant along the leaves.

|

algebra of
such fns.

($\xi f = 0, \forall$

Uncertain reading. The crossed sign is provisionally shown as $\forall$; overlapping diagonal strokes cross it.

Original source details: GELFAND-P000364

if $\xi_x \in F_x$). $\varphi \in A(F) \Rightarrow \mathcal{D}_\varphi$ preserves the space of sections ($\mathcal{D}_\varphi$ acts on $\Gamma(L,F)$ as multiplication by φ). $A(F)$ — a max. commutative subalg. with the Poisson bracket (locally). Now consider operators (also preserving $\Gamma(L,F)$)

$$A_1(F) = \text{normalizer of } A(F) \text{ in the Poisson-bracket alg.} =$$
$$= \{\varphi : \{\varphi, A(F)\} \subset A(F)\}.$$

Thm. Operators $\mathcal{D}_\varphi$ with $\varphi \in A_1(F)$ preserve $\Gamma(L,F)$ and act there as 1st-order opers. This is already quantization, and the fns. — quantizing. (Other quant. are already nonunique). There are many of them.

Uncertain reading. The current word “ikh” (“of them”) is readable, but its first letter is rewritten. Earlier strokes are not separated.

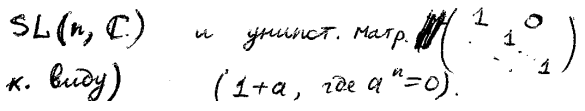
Original source details: GELFAND-P000364

p, q; $dq = 0$ — subspace, fns. — of q only. Opers. — fns. of q — multiplication opers.,
a linear fn. of p — a 1st-order oper.

Next time: Kirillov.

GELFAND-P000365

13/IX-71. I. M. Gelfand on the work of Springer (comput. char. for semisimple groups over a finite field). 1. Consider the group $SL(n, \mathbb{C})$ and unipot. matrices. [?]

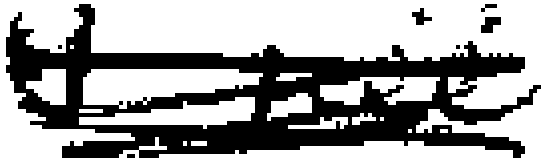


Original source details: GELFAND-P000365

Uncertain reading. A separate dark overwritten sign precedes the first matrix. Neither U nor an equals sign has been established securely.

$$\begin{pmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ \dots & & & 1 \end{pmatrix}$$

~~U~~[?]



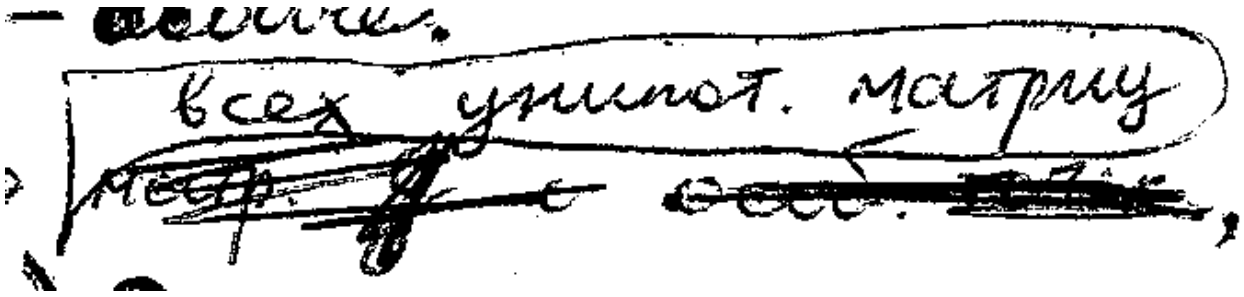
Original source details: GELFAND-P000365

Undeciphered passage. The beginning of the short cancelled entry is provisionally shown as a U-like sign. The remaining signs under horizontal strokes have not been read in full.

(reduced to this form). $(1 + a,$ where $a^n = 0)$. At 1 — a sing. point.

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $ad - bc = 1$, unipot. $a + d = 2$, in $\mathbb{R}^3$ a cone, 1 — a singularity.

Nonsingular — one Jordan block. Several Jordan blocks — **singular**. Answer: R — the set

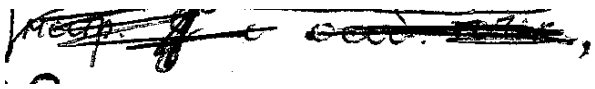


Original source details: GELFAND-P000365

Undeciphered passage. Beneath the outlined insertion “all unipot. matrices”, the earlier phrase is cancelled. Its remainder between and after the provisionally legible fragments has not been recovered.

all unipot. matrices

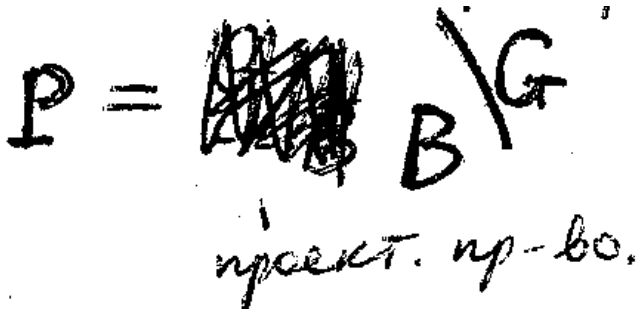
~~matr.[?] [?] sing.[?] [?],~~



Original source details: GELFAND-P000365

Uncertain reading. The contractions “matr.” and “sing.” are provisionally rendered in the cancelled line. Endings and intervening writing are covered by strokes; the contractions are not expanded.

$g \in R$.



Original source details: GELFAND-P000365

Uncertain reading. Between $P =$ and $B\backslash G$ a sign is heavily cancelled, provisionally P. The exact earlier sign has not been recovered. Beneath it is a short vertical stroke and the caption “project. space.”.

$P =$ ~~P~~[?] $B\backslash G$

project. space.

B — triangular matrices

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ \dots & & \lambda_n \end{pmatrix}.$$

Consider pairs (p, g) with conditions

$$pgp^{-1} \in U = \left\{ \begin{pmatrix} 1 & & 0 \\ & 1 & \\ & & \ddots \\ \dots & & 1 \end{pmatrix} \right\}, \qquad p \in P.$$

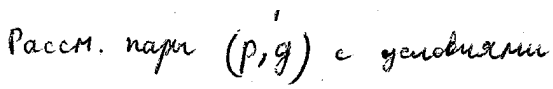
(This does not depend on the chosen representative elem. from p.) $\bar{R}$ — the set of pairs.

$$(p, g) \longrightarrow g.$$

This is $\bar{R} \to R$, a resolution of singularities (all). Over a point in gen. position one element. $\bar{R}$ — a nonsingular variety. (This is for an arbitrary reductive group over $\forall$ field.)

Uncertain reading. The Russian word “nad” (“over”) is readable, but its last letter is heavily rewritten, with a separate arc above it. Earlier strokes are not separated.

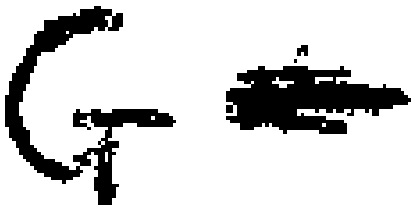
Original source details: GELFAND-P000365



Original source details: GELFAND-P000365

Manuscript notation. The words before the conditions on the pair read “with conditions.”

2. Geom. interpretation of Hall polynomials. Consider G [?]



Undeciphered passage. After G is a short overwritten and cancelled group of strokes. An equals sign remains a candidate; no secure earlier reading has been established.

Original source details: GELFAND-P000365

— the group of matrices, B — the group of triangular matrices.

$$G = GL(n, F_q), \qquad B\backslash G \text{ — flags.}$$

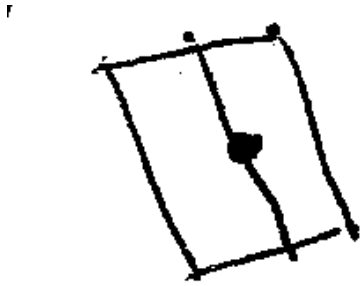
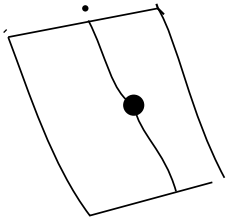


Diagram and typed description. A sloping four-sided outline has a roughly middle bent line with a large dot on it. The small detached dot above its upper middle join, short trace at upper left and thickened upper right join are encoded without assigning additional nodes or mathematical meaning.



Original source details: GELFAND-P000365

The number of flags — a polynomial in q, whose degree depends on n.

GELFAND-P000366

[source number: 25]

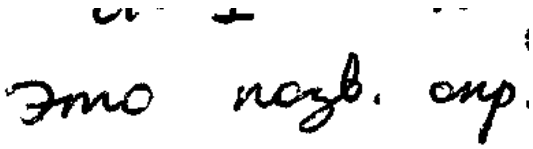
A coset class is determined by a partition $n = n_1 + \cdots + n_k$, $u \in U$ — an elem. of this class.

$$|P^u| = P_{n_1,\ldots,n_k}(q) \text{ — a Hall polynomial}$$

(the number of fixed points of this element). $u = 1$ — obtain the Poincaré polynomial. This allows one to determine the Poincaré polynomial for any split group.

Manuscript notation. The opening reads “coset class.” The Russian wording is abbreviated.

Original source details: GELFAND-P000366



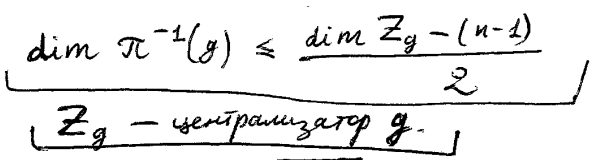
Manuscript notation. The abbreviation after “This” reads “pozv.” (“allows”).

Original source details: GELFAND-P000366

J. A. Green, Trans. A. M. S., 1955 — recommended.

There are bounds on the dim. of the variety lying over a point here.

$$\pi : \bar{R} \longrightarrow R,$$



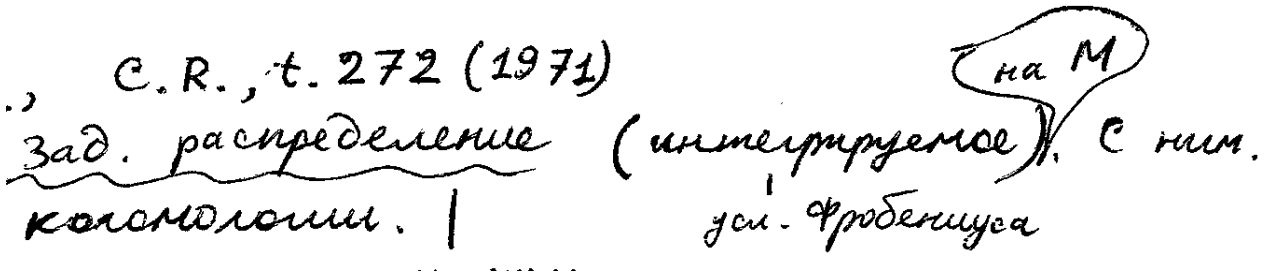
Original source details: GELFAND-P000366

Diagram and typed description.

$\dim \pi^{-1}(g) \leq \dfrac{\dim Z_g - (n-1)}{2}$

Z_g — the centralizer of g.

B. Rosenfeld. On notes by Rego, C. R., t. 272 (1971). ~~In the tang. space.~~

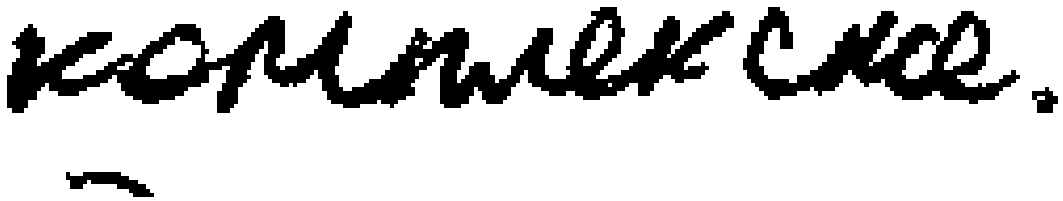


Original source details: GELFAND-P000366

Diagram and typed description.

Given a distribution (integrable) on M with it Frobenius cond.

are assoc. certain cohomologies.



Original source details: GELFAND-P000366

Diagram and typed description.

Every diff. of the graded ring complex. diff. forms on M has the form

$$d_{L_1} + i_{L_2}$$

(i_{L_2} — interior multiplication by a vector-form).

– – – – – – – – – – – – – – – – – –

$\omega(X_1, \ldots, X_q)$ — a scalar form.
$L(X_1, \ldots, X_p)$ — a form (vector-valued).
$(i_L\omega)(X_1, \ldots, X_{p+q-1})$ substitute and antisymmetrize.
$d_L\omega = (d \circ i_L + (-1)^p i_L \circ d)\omega.$

L — a form of degree 0 (vector field) — d_L — the Lie derivative. ~~$d_L^2 = 0$~~[?]

Uncertain reading. Before "integrability" a short formula is cancelled. $d_L^2 = 0$ is provisionally shown; the index and exponent on d are hidden by overlapping strokes and not established securely.

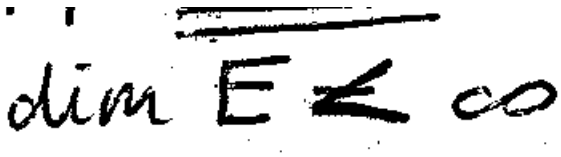
Original source details: GELFAND-P000366

Integrability. (square $= 0$)

GELFAND-P000367

Prof. Pták.

$$\dim E \leq \infty, \qquad E \text{ with a norm.}$$



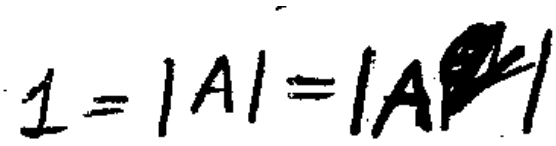
Uncertain reading. Between dim E and ∞ several oblique and horizontal strokes overlap. $\leq$ is provisionally shown; $<$ and $\neq$ remain candidates.

Original source details: GELFAND-P000367

q — an exponent (crit.) if

$$1^\circ. \quad A \in L(E), \quad 1 = |A| = |A^q| \Rightarrow |A|_\sigma = 1$$

(sp. radius)



Uncertain reading. The exponent on the second A is heavily thickened or rewritten. q is shown provisionally; the earlier exponent is unestablished.

Original source details: GELFAND-P000367

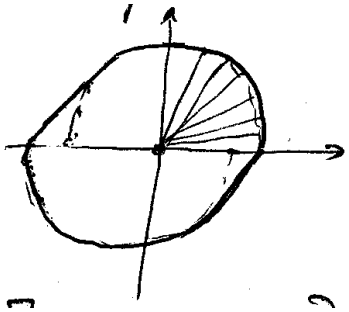
2°. $\exists B$, $\quad 1 = |B| = |B^{q-1}|$, $\quad$ but $|B|_\sigma < 1$.

$\ell^\infty \quad |x| = \max |x_i|,$

$\ell_1 \quad \sum |x_i|, \qquad q(n) = n^2 - n + 1.$

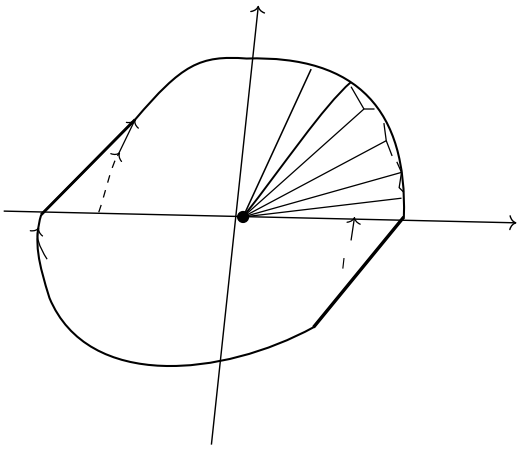
$\ell_2 \qquad q(n) = n.$

There exists a 2-dim. space in which $q = \infty$. (The norm gives poor estimates of convergence; eigenvalues are needed).



Original source details: GELFAND-P000367

Diagram and typed description.



A — rotation by a small angle (Danzer's example). ℓ_p — unknown whether it exists, except for $p = 2^m$. A useful problem for computational mathematics.

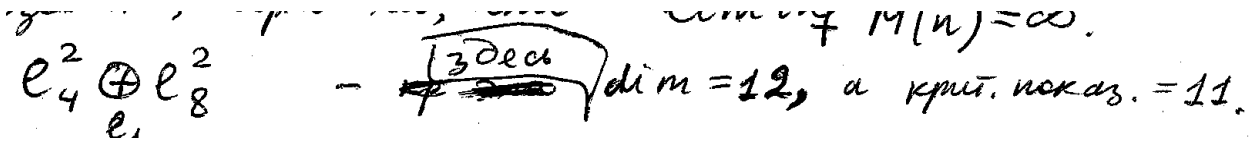
$M(n) =$ min over all ban. $\leq n$.

Original source details: GELFAND-P000367

Uncertain reading. The abbreviation after "over all" is provisionally shown as "ban.". Its middle letters are heavily thickened and rewritten; earlier strokes are not separated. The abbreviation is not expanded.

$\exists$ those for which $< n$! There exists a seq. $M(n_k) \sim \sqrt{n_k}$. It is unknown whether $\liminf M(n) = \infty$ is true.

$$\ell_4^2 \underset{\ell_1}{\oplus} \ell_8^2$$



Original source details: GELFAND-P000367

Uncertain reading. Above the cancelled earlier word is the legible outlined insertion "here". The earlier "eto" ("this") is shown provisionally; the covered letters do not establish its complete reading.

— here ~~this~~[?] dim $= 12$, and crit. exponent $= 11$.

$C(E, r, \rho) = \max\{|T^r|,\ |T| \leq 1,\ |T|_\sigma \leq \rho\}.$

Original source details: GELFAND-P000367

Diagram and typed description.

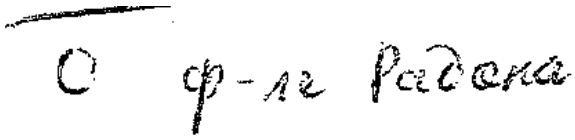
$0 < \rho < 1.$

GELFAND-P000368

[source number: 26]

I. M. Gelfand.

On the Radon formula over a finite field.



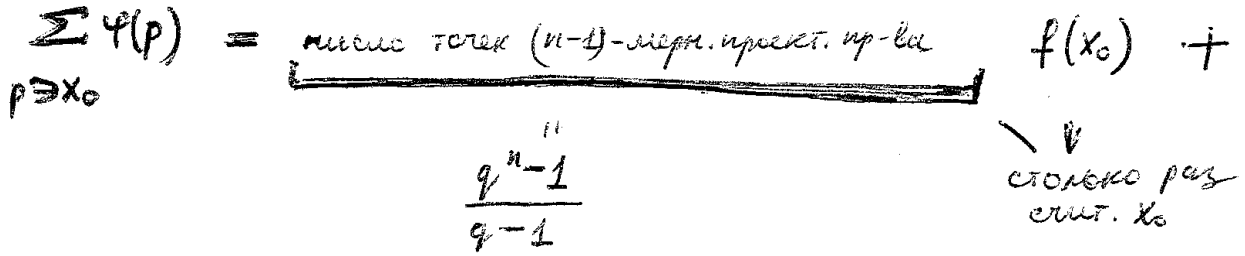
Uncertain reading. The initial glyph has a gap on the right. It is provisionally read as Russian О; its shape also allows С.

Original source details: GELFAND-P000368

Over F_q an n-dim. space. $(x_1, \ldots, x_n)$. q^n points.
Integrals over all planes are known.

$$\varphi(p) = \int_{x \in p} f(x) \quad \text{(sum)}.$$

p — a plane.

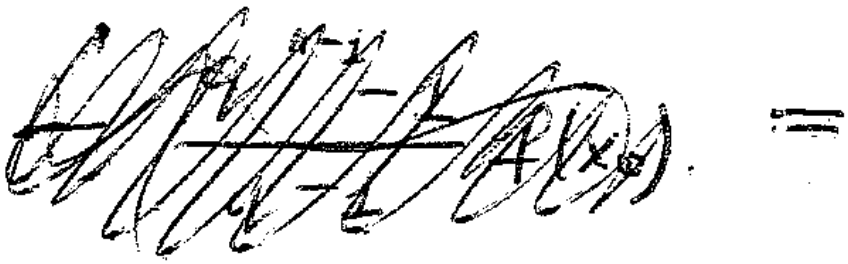


Original source details: GELFAND-P000368

Diagram and typed description.

$$\sum_{p \ni x_0} \varphi(p) = \underbrace{\text{number of points of } (n-1)\text{-dim. project. space}}_{\frac{q^n-1}{q-1}} \; \underbrace{f(x_0)+}_{\text{this many times counted } x_0}$$

$$+\frac{q^{n-1}-1}{q-1}\sum_{x\neq x_0} f(x).$$

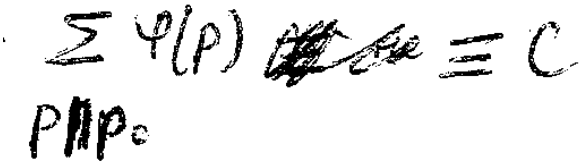


Original source details: GELFAND-P000368

Uncertain reading. In the cancelled term the angular sign before (x_0) is provisionally shown as A. f remains a candidate; the crossing cancellation strokes prevent a unique reading. The entire term is cancelled; an equals sign follows it.

~~$+\frac{q^{n-1}-1}{q-1}A(x_0)$ [?] $=$~~

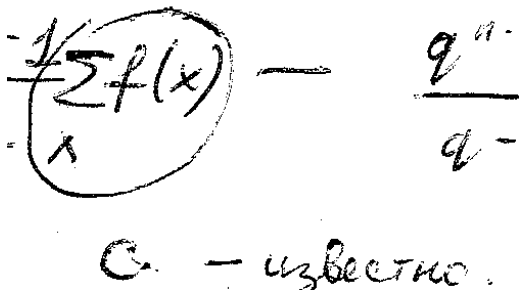
$$\sum_x f(x) = \sum_{p\|p_0} \varphi(p) \quad \text{[?]} \quad \equiv C.$$



Undeciphered passage. Between the sum over $p \parallel p_0$ and $\equiv C$ several signs are cancelled. Their reading is unestablished.

Original source details: GELFAND-P000368

(does not depend on p_0) — a cond. on $\varphi(p)$ — an analogue of the Paley–Wiener theorem.



Original source details: GELFAND-P000368

Diagram and typed description. The sum $\sum_x f(x)$ is circled; beneath it "C — known." is underlined. The lower index has overlapping strokes and a small loop; x is retained provisionally.

$$= \frac{q^n-1}{q-1}f(x_0) + \frac{q^{n-1}-1}{q-1}\left(\sum_x f(x)\right) - \frac{q^{n-1}-1}{q-1}f(x_0).$$

C — known.

Rota — as $q \to 1$ one obtains a passage to simple theorems. $C_k^n(q)$ — the number of k-dim.

subspaces of an n-dim. space.

$$\lim_{q\to 1} C_k^n(q) = C_k^n \text{ — combinations.}$$

GELFAND-P000369

Kirillov.

Consider an arbitrary generalized function a on a manifold M (e.g. $\mathbb{R}^n$). For it a wave front is defined

$$WF(A) \subset T^*M.$$

A smooth function — zero wave front (empty). Let $\xi \in T^*M$.

$$\xi \overline{\in} WF(A), \quad \text{if}$$
$$\langle A, \varphi e^{it\psi}\rangle = o(t^{-n}), \quad \forall n \quad \text{as } t \to +\infty.$$

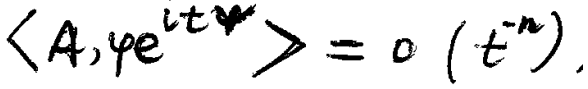
Manuscript notation. The first line uses lowercase a.

Original source details: GELFAND-P000369

Manuscript notation. A separate horizontal stroke appears above $\in$.

Original source details: GELFAND-P000369

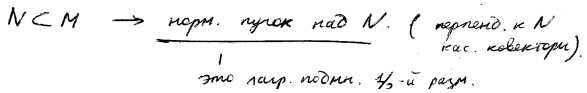
Uncertain reading. The phase sign after it is heavily thickened or rewritten. ψ is shown provisionally; the earlier strokes are not separated.



Original source details: GELFAND-P000369

(here $d\psi = \xi$ at the point π_ξ and $\varphi \in C_0^\infty$ is arbitrary).

It is convenient to study functions for which the wave front $\in$ a Lagr. manifold.



Original source details: GELFAND-P000369

Diagram and typed description.

$N \subset M \longrightarrow$ norm. bundle over N

$\Big|$

(tangent covectors perpend. to N).

This is a Lagr. subman. of $1/2$ dimension.

~~[?]~~ P. D. O. — a generalized function on $M \times M$ (kernel) with a wave front constructed in this way from the diagonal.

Original source details: GELFAND-P000369

Uncertain reading. Before “P. D. O.” a short word is heavily cancelled. “Napr.” (“e.g.”) remains a candidate; the complete reading is unestablished.

~~$A(p,q) =$~~

$$\langle A, [?]u \rangle = \int e^{i\psi(x,\xi)} a(x,\xi) u(x)\, dx\, d\xi.$$

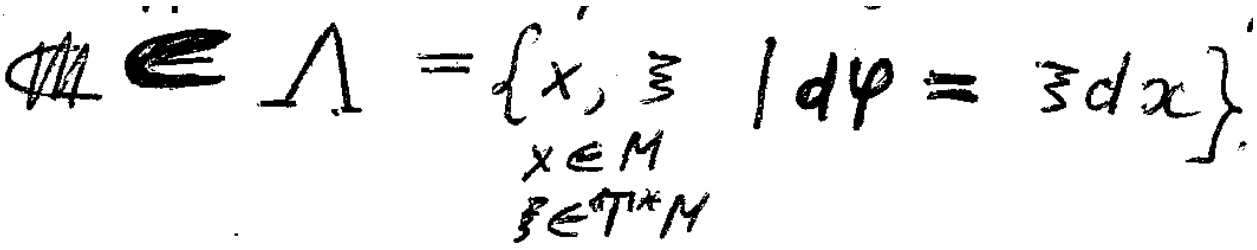
Uncertain reading. A comma is visible after A, followed by a separate thickened vertical mark before u. The mark is undeciphered. The right angle bracket is retraced.

Original source details: GELFAND-P000369

(Value of A on the test function u).
Every A ~~[?]~~ with wave front defined this way is given by this formula.

Undeciphered passage. After “Every A” a separate sign is heavily cancelled. Its reading is unestablished.

Original source details: GELFAND-P000369



Original source details: GELFAND-P000369

Uncertain reading. After “Wave front” a rounded sign is cancelled. φ is shown provisionally; another reading is not excluded. The following $\in \Lambda$ is retained separately from the cancellation.

$$\text{Wave front } \varphi[?] \in \Lambda = \left\{ \begin{matrix} x, \xi \\ x \in M \\ \xi \in T^* M \end{matrix} \;\middle|\; d\psi = \xi\, dx \right\}.$$

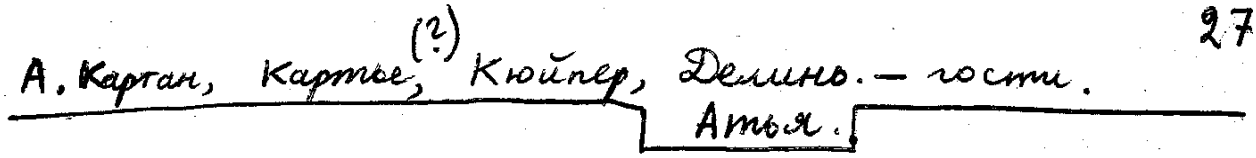
GELFAND-P000370

[source number: 27]

20/IX

Uncertain reading. To the right of 20/IX is a separate short vertical 1-like stroke. Its meaning is unestablished.

Original source details: GELFAND-P000370



Original source details: GELFAND-P000370

Diagram and typed description.

A. Cartan, Cartier (?), Kuiper, Deligne. — guests.

Atiyah.

I. Bernstein. on work of Godbillon, Vey. Comptes rendus. M — a compact manifold. On M is given a form ω (a foliation S of codim. 1), which is integrable.

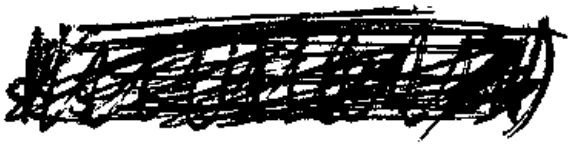
$$d\omega = \omega \wedge \omega_1.$$

— ~~[?]~~

Undeciphered passage. After the formula an earlier word is heavily cancelled; its reading is unestablished.

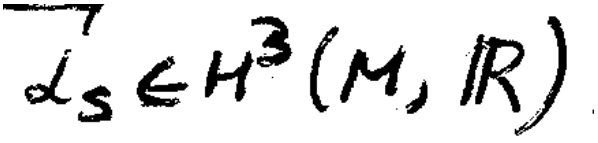
Original source details: GELFAND-P000370

~~[?]~~ To it is associated $\alpha_S \in H^3(M, \mathbb{R})$, it



Undeciphered passage. Before “To it is associated” an entire phrase is heavily cancelled; its reading is unestablished.

Original source details: GELFAND-P000370



Manuscript notation. The subscript S repeats the foliation label used above.

Original source details: GELFAND-P000370

~~[?]~~ $= 0$ if this is a fibration and is stable under deformations,

Original source details: GELFAND-P000370

Undeciphered passage. After “it” and before $= 0$ a short word is heavily cancelled. No confident reading of its letters is established.

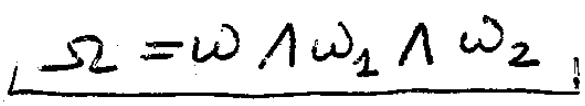
~~[?]~~ $\neq 0$.

Original source details: GELFAND-P000370

Undeciphered passage. Between “under deformations” and $\neq 0$ a short word is crossed out; its reading is unestablished.

Construction: there exist forms ω_1, ω_2:

$$d\omega = \omega \wedge \omega_1;$$
$$d\omega_1 = \omega \wedge \omega_2.$$



Original source details: GELFAND-P000370

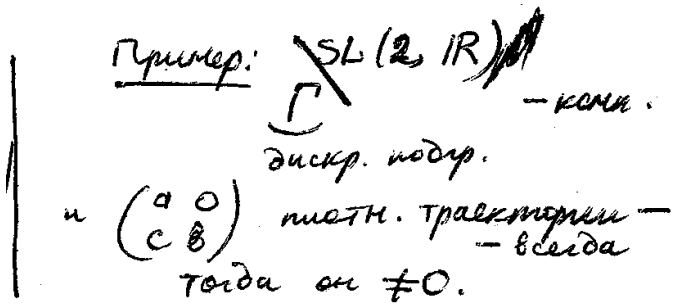
Diagram and typed description.

Consider $\Omega = \omega \wedge \omega_1 \wedge \omega_2$

— closed, and the cohom. class does not depend on the choice of ω_1 and ω_2. There is a connection with the cohomology of algebras.

Original source details: GELFAND-P000370

Uncertain reading. After $SL(2, R)$ is a reinforced M-like sign. M is shown provisionally; the overlapping strokes are unestablished.



Original source details: GELFAND-P000370

Diagram and typed description.

Example: $SL(2,R)\,M$[?]

Γ — compact.

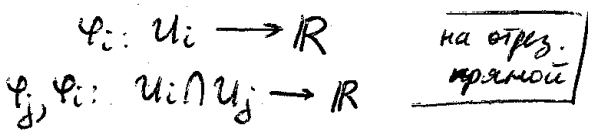
discr. subgr.

and $\begin{pmatrix} a & 0 \\ c & b \end{pmatrix}$ dense trajectories —

— always.

Then it is $\neq 0$.

Foliation:



Original source details: GELFAND-P000370

Diagram and typed description.

$\varphi_i : U_i \longrightarrow \mathbb{R}$,
$\varphi_j\varphi_i : U_i \cap U_j \longrightarrow \mathbb{R}$.

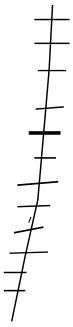
on an interval of the line

Since the leaves are the same, one obtains $\Psi_{ij} : \mathbb{R} \to \mathbb{R}$ (an interval). If $\Psi_{ij} \in$ a group, then a cocycle with coeff. in the group $\longrightarrow$ a sheaf and from it cohomology classes. But this is a pseudogroup. But consider a Lie algebra isom. to W_1 — formal vector fields on the line, $f(x)\frac{d}{dx}$. They are $= 0$, except there exists $H^3(W_1, \mathbb{R}) = \mathbb{R}$. This is precisely that class.

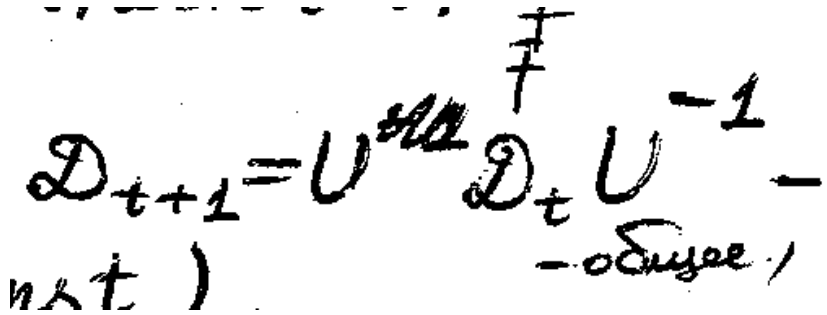
Atiyah. On the spectrum of a selfadj. elliptic operator. X — a compact manifold, D — an elliptic diff. operator, selfadj. $D^* = D$. Discr. spectrum; eigenvalues are divided into positive and negative. Let $D = D_t$, t — a param.; the spectrum changes with t.

Original source details: GELFAND-P000370

Diagram and typed description. An inclined nearly vertical line with twelve distinguishable transverse strokes of different lengths and short overlapping strokes. One stroke in the middle is thickened. Exact intervals and separation of all overlapping strokes are unestablished; there are no numerical labels.



Let $D_{t+1} = D_t$ — periodicity, or



Original source details: GELFAND-P000370

Uncertain reading. The first U stands on the main line; a small cancelled earlier sign is above and to its right. “na” remains a candidate; the exact reading is unestablished. The order of insertion is undetermined.

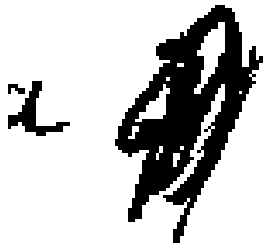
$$D_{t+1} = U \overset{\text{[?]}}{} D_t U^{-1}$$

— general case,

($U = \text{const}$).

$$\operatorname{Spec} D_{t+1} = \operatorname{Spec} D_t.$$

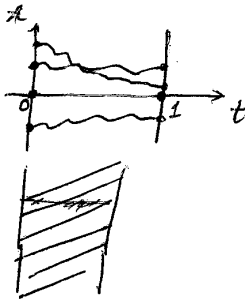
The spectrum moves (eigenvalues may shift by q[?]).



Original source details: GELFAND-P000371

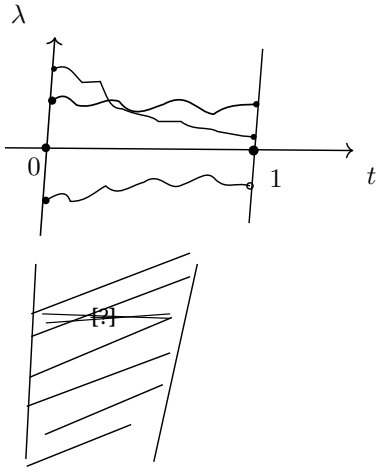
Uncertain reading. After “by” is a reinforced q-like sign. q is shown provisionally; a question mark remains a candidate.

One can define an integer characterizing this shift. (Place an observer at 0 and see what mass has passed to the left and to the right.)



Original source details: GELFAND-P000371

Diagram and typed description. Upper drawing: axes λ, t, marks $0, 1$, two verticals and three wavy curves. Thick dots occur at endpoints and on the axis; the lower right dot has a light centre. Below are six distinguishable inclined segments between nearly vertical lines. Exact separation of overlapping strokes and joins is unestablished.



Original source details: GELFAND-P000371

Undeciphered passage. In the middle of the lower drawing a short caption is cancelled, distinct from the inclined lines. No complete word or formula has been recovered.

Example. Spectral flow $\in \mathbb{Z}$.

$$X \text{ — a circle}, \qquad D_t = -i\frac{d}{dx} + t, \qquad Uf = e^{ix} f.$$

Flow $= \pm 1$ $(n \to n+1)$. This is a top. invariant under deformations of the family. Problem: compute spectral flow from the geom. data of the problem: manifold, operator, U. Connected with the index. Let the operator D be of first order (a system). Consider

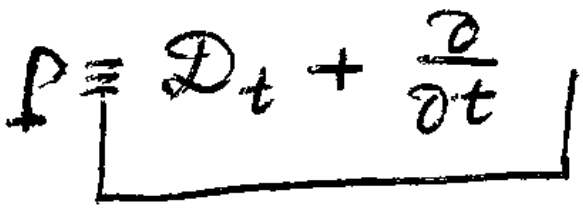
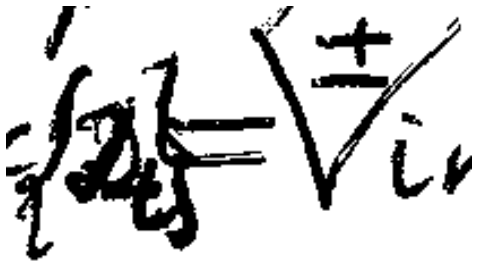


Diagram and typed description.

$$P \equiv D_t + \frac{\partial}{\partial t}$$

Original source details: GELFAND-P000371

on functions $f(x,t)$, $f(x,t+1) = f(x,t)$, $x \in X$, $(U = 1)$. (if $U \neq 1$, adjust $U \otimes 1$). This is an elliptic operator on $X \times S^1$ (or in a bundle, base S^1, fibre X). (D_t — Hermitian, $\frac{\partial}{\partial t}$ — skew-Hermitian). Theorem.



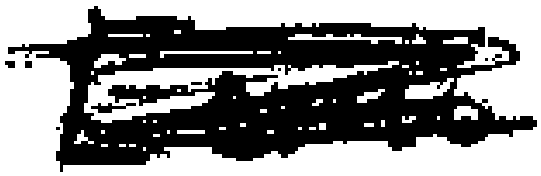
Original source details: GELFAND-P000371

Uncertain reading. Near the right family brace an upper 1-like stroke and a lower 0-like loop join the reinforced brace strokes. Possible limits 1 and 0 are not confidently separated; no definite interval is assigned.

Spectral flow $\{D_t\} = \pm$ index P

$=$ the index formula.

A direct proof is difficult. It is easier to express spectral flow by a formula. Heuristic considerations. Let D_t depend real-analytically on t. [?]



Undeciphered passage. After “real-analytic” a short word is entirely crossed out. Its reading is unestablished.

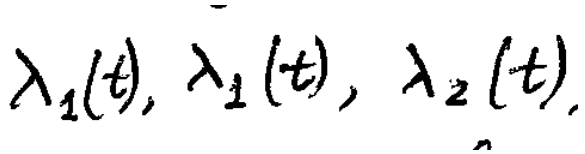
Original source details: GELFAND-P000371

[?]

Uncertain reading. At left an entry is cancelled: an initial t-like form and at right a t_0- or 0-like ending. $t \dots t_0$, $t = t_0$, $t = 0$, $t \dots 0$ remain candidates; the relation sign is unestablished.

Original source details: GELFAND-P000371

$$\lambda_1(t),\ \lambda_1(t),\ \lambda_2(t), \dots$$



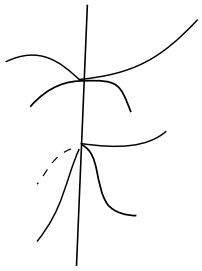
Original source details: GELFAND-P000371

Manuscript notation. The first two subscripts both read 1.

— analytic functions!, defined for all t. (Roots of char. polynomials; real $\Rightarrow$ no branching).

Original source details: GELFAND-P000371

Diagram and typed description. A nearly vertical line and two groups of branches at different levels: upper crossing curves, below a right rising branch and a downward-bent branch, and oblique strokes at left. Exact joins of thick strokes and endpoints are unestablished.



GELFAND-P000372

[source number: 28]

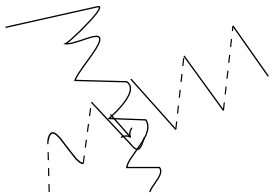
Periodicity $\Rightarrow$ a permutation σ of them.

$$-i\frac{d}{dx} + t, \qquad \sigma(n) = n + 1.$$

This σ is not arbitrary. Orbits of σ — 4 classes are possible. 1) finite — an eigenvalue that returns. 2) bounded on one side (from $-\infty$ to $-\infty$ or from $+\infty$ to $+\infty$). 3) unbounded on either side.

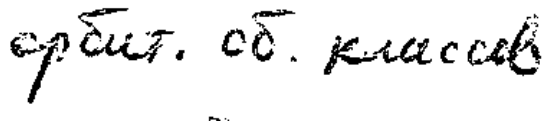
Original source details: GELFAND-P000372

Diagram and typed description. To the right of case3 is a descending zigzag and several wavy or angular strokes with solid and interrupted parts; in the middle is a downward arrow-like stroke. Exact joins and geometric meaning are unestablished.



$(|\lambda'(t)| < C|\lambda(t)|.)$

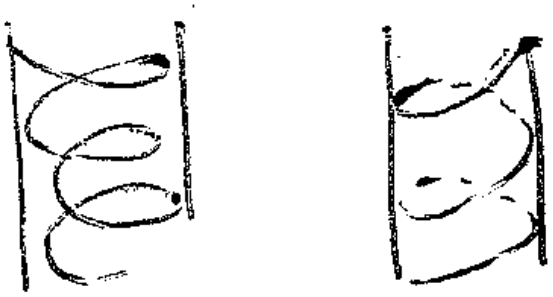
Increasing and decreasing — the numbers of orbits of ob.[?] classes p and q are finite.



Original source details: GELFAND-P000372

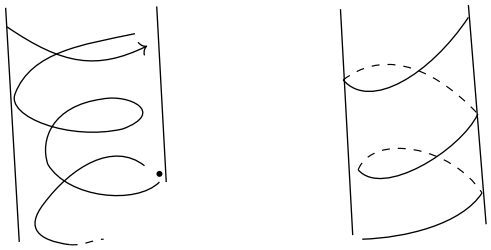
Uncertain reading. The abbreviation after “orbits” is provisionally shown as “ob.”. The first letter has a gap; “sb.” remains a candidate. The abbreviation is not expanded.

Functions $\underline{\underline{\lambda(t)}}$ on the cylinder. 3) — spirals go upward or downward.

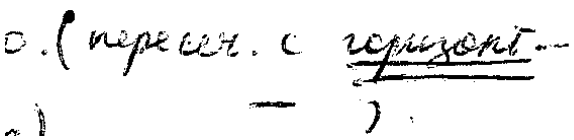


Original source details: GELFAND-P000372

Diagram and typed description. Two drawings between verticals: at left overlapping loops, solid and interrupted parts, an upper rightward arrow and a lower dot at right. At right are several arcs with solid and interrupted parts. Additional directions and hidden joins are unestablished.

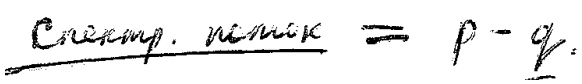


The number of orbits of class3) is finite. (intersection with a horizontal —) Spectral flow $= p - q$. (obvious from this).



Original source details: GELFAND-P000372

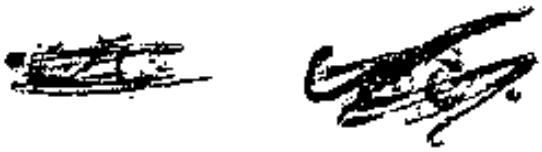
Manuscript notation. A separate horizontal stroke is drawn below the end of the line, before the closing parenthesis.



Original source details: GELFAND-P000372

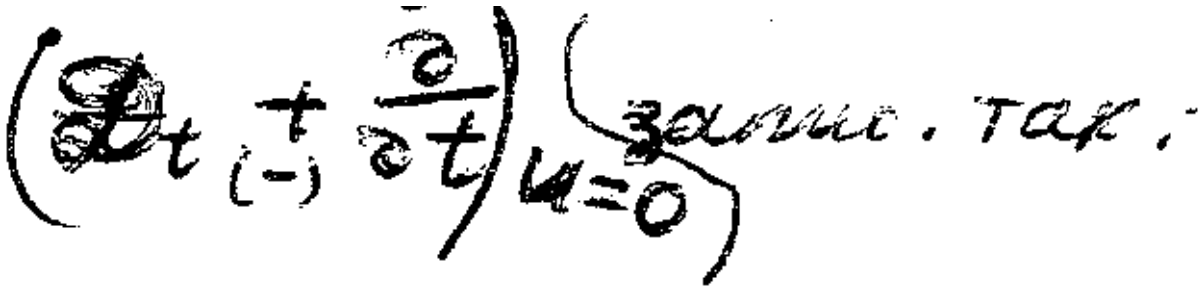
Manuscript notation. The two strokes before $p - q$ form an equals sign.

Assumption: let the eigenvectors be independent of t. Then ~~[?]~~ ~~[?]~~ exact solution



Original source details: GELFAND-P000372

Undeciphered passage. After “Then” and before “exact solution” two short words are crossed out. Neither reading is established.



Original source details: GELFAND-P000372

Uncertain reading. The first operator before the plus is reinforced over an earlier sign. D_t is shown provisionally; earlier strokes with a possible ∂ are not separated. Below the plus is a separate $(-)$; this insertion does not replace the main plus.

$\left(D_t[?] + \underset{(-)}{\frac{\partial}{\partial t}}\right) u = 0$ write it thus:

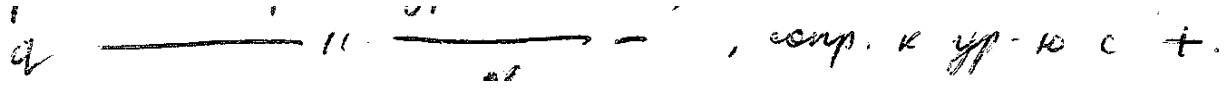
$$u_n = C_n \exp\left(-\int \lambda_n(t)\,dt\right), \qquad u = \sum u_n e_n, \quad e_n \text{ — an eigenvector.}$$

$$\lambda_n(t+1) = \lambda_{\sigma(n)}(t).$$

$$\sigma(n) \longrightarrow \infty \iff \text{finite } n \text{ and large } t.$$

$$u_n(t) = C_n \exp\left(-\int_0^t \lambda_n(s)\,ds\right).$$

p — the number of solutions of the equation with $+$,



Original source details: GELFAND-P000372

Diagram and typed description.

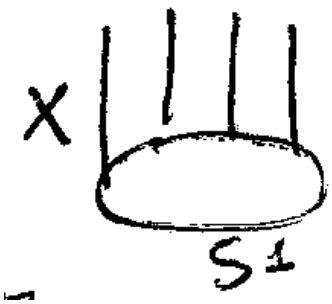
q ——————— // ——————→ $-$, adjoint to the equation with $+$.

Finite orbits $\int_0^N \lambda_n(t)\,dt = 0$, N — the period.

Index $= p - q$. In the general case (vectors move), the equation $D_t + \frac{\partial}{\partial t}$ contains an additional term expressing the motion of eigenvectors. A direct analytic proof cannot be given.

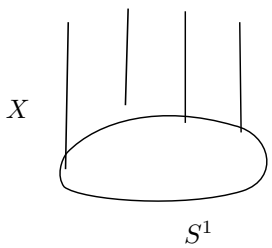
GELFAND-P000373

Examples. ①. $T : X \longrightarrow X$ — a diffeom. Define a bundle.



Original source details: GELFAND-P000373

Diagram and typed description.



Let there be a functorial situation (X, g, D), g — a Riem. metric, D — selfadjoint. (The case of a spectrum unbounded on both sides is interesting). One then obtains:

$$X_t, g_t, D_t$$

Original source details: GELFAND-P000373

Uncertain reading. After "when" a separate sign is cancelled; A remains a candidate. Above it "this" is readable. The earlier sign is not confidently recovered.

this
(E.g.: $d + \delta$, when ~~[?]~~ is selfadj.).

Spectral flow does not depend on the choice of Riem. metric. If T is an isometry, then $D_t = \text{const}$ and spectral flow $= 0$. The invariant is an obstruction to the existence of a metric preserved by T. ②. $X \times S^1$, the operator $(d + \delta)_t$ on forms. Twisted using this map. Suppose given

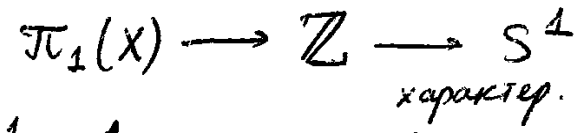


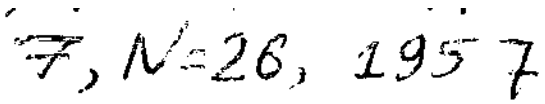
Diagram and typed description.

$\pi_1(X) \longrightarrow \mathbb{Z} \longrightarrow S^1$
character.

Original source details: GELFAND-P000373

$$X^{4k+1} \times S^1$$

— a Novikov invariant using the Hirzebruch L-polynomial. Deligne, in Manin's translation. On the theory of motives. Different cohomology theories of alg. varieties. 25/X̄-71 | S. Gelfand. Schubert cells. Horrocks, Proc. Lond. Math. Soc., 7, $N = 26$, 1957.



Manuscript notation. After 7 the manuscript reads $N = 26$.

Original source details: GELFAND-P000373

$G(N,n)$ — the complex Grassm.

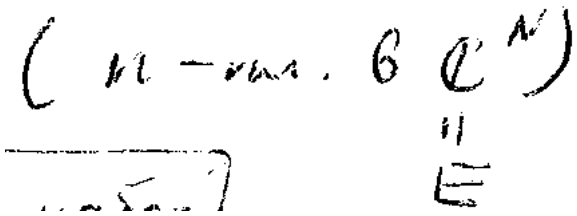
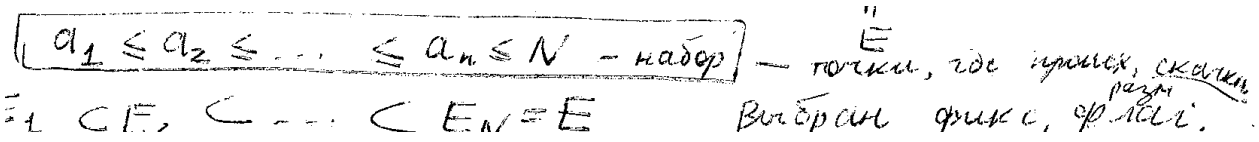


Diagram and typed description.

(n-planes in $\mathbb{C}^N$)
||
E

Original source details: GELFAND-P000373



Original source details: GELFAND-P000373

Diagram and typed description. The tuple is framed. At right a line leads to "points where jumps occur[?]"; below is the insertion "dim." joined by an oblique stroke to the end of the caption. Its exact attachment to the lower line is unestablished.

$a_1 \le a_2 \le \cdots \le a_n \le N$ — tuple — — points where jumps occur[?]

dim.

A fixed flag is chosen.

$$E_1 \subset E_2 \subset \cdots \subset E_N = E,$$
$$\dim E_i = i.$$
$$F \in \Omega(a_1, \ldots, a_n), \quad \text{[?]}$$

Undeciphered passage. After $F \in \Omega(a_1, \ldots, a_n)$ an entire word is cancelled by several thick horizontal strokes. Its reading is unestablished.

Original source details: GELFAND-P000373

$$\Updownarrow$$
$$\dim(F \cap E_{a_i+i}) = i.$$

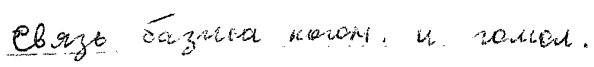
$$\varphi(i) = \dim(F \cap E_i).$$

$\underline{\underline{a_i + i}}$ — the growth points of this function.

GELFAND-P000374

[source number: 29]

This is a cell decomposition of the Grassm. manifold. Problem. Find all invariant diff. forms (Chern classes). Connection between the basis of cohom. and homol.



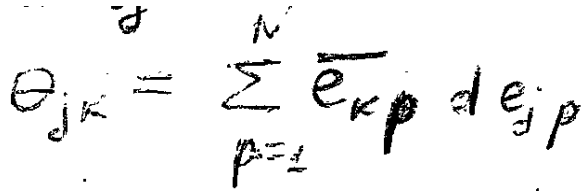
Manuscript notation. The last abbreviation reads “homol.”

Original source details: GELFAND-P000374

$$(*) \quad U(N)/(U(n)\times U(N-n)) = G(N,n) = G/F.$$

The form can be lifted to $U(N)$. Construction of forms: de_{ij}. $U=(e_{ij})$, (e_{ij}) — a matrix-element function on $U(N)$.

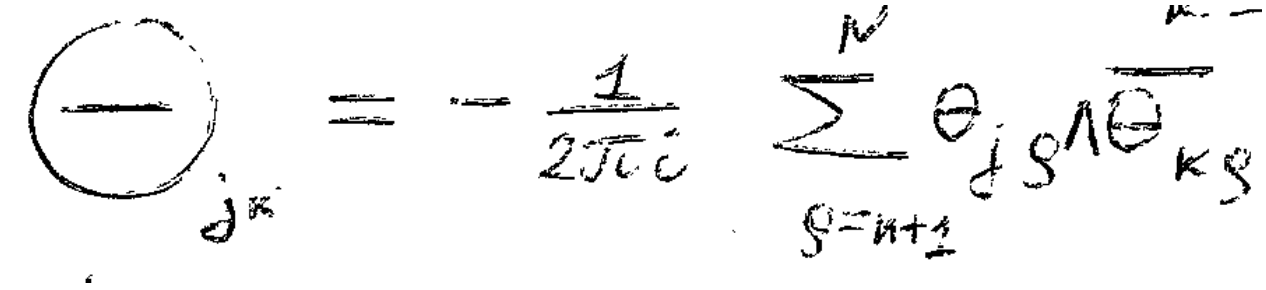
$$\theta_{jk} = \sum_{p=1}^{N} \bar{e}_{kp}\, de_{jp}, \qquad j=1,\dots,n, \quad k=n+1,\dots,N.$$



Uncertain reading. In the index on $\bar{e}$, after k, is a heavily reinforced p-like sign. p is shown provisionally; ρ remains a candidate.

Original source details: GELFAND-P000374

$j,k=1,\dots,n$.

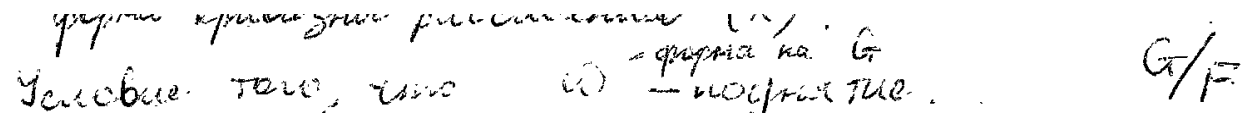


Original source details: GELFAND-P000374

Uncertain reading. The two factors use signs of the same size, smaller than Θ at left. The barred sign is provisionally read as θ; its case is uncertain.

$$\Theta_{jk} = -\frac{1}{2\pi i}\sum_{s=n+1}^{N} \theta_{js}\wedge\bar{\theta}_{ks}$$

curvature form of the bundle $(*)$.



Original source details: GELFAND-P000374

Diagram and typed description. “Form on G” is written above ω, with G/F separately at right.

ω — form on G

The condition that ω — a pullback. G/F

1) $\omega(X_1,\dots,X_k)=0$, $X_i\in T(F)$ — tangent to F. 2) ω — invariant under F. All Θ_{jk} satisfy condition 1). Invariant under $U(N-n)$; under $U(n)$ it transforms by conjugation (adjoint action).

Original source details: GELFAND-P000374

Diagram and typed description.

$$GL(n) \ni (x_{ik})_{i,k=1}^{n}.$$

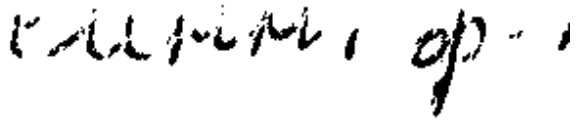
Construct polynomials in the matrix elements with the property

$$p(x) = p(gxg^{-1}).$$

Coefficients of the char. polynomial — elementary symm. functions. A basis of the linear space is needed. Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$ be integers. x — a matrix, $\alpha_1, \ldots, \alpha_n$ — eigenvalues. One must take

GELFAND-P000375

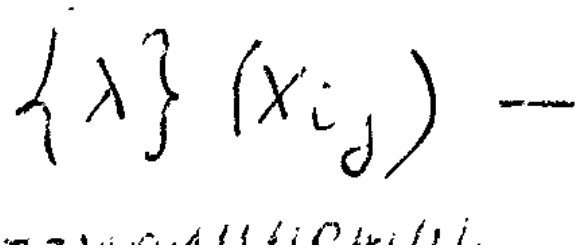
symm. fns.



Original source details: GELFAND-P000375

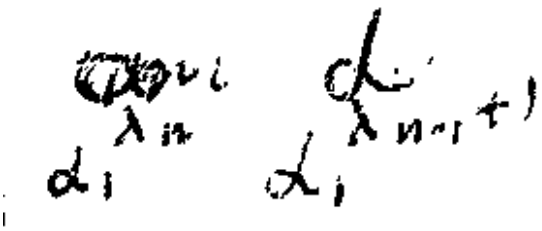
Uncertain reading. The small faint note above the first determinant is provisionally rendered "symm. fns."; a longer final contraction remains possible.

$$\{\lambda\} = \frac{\begin{vmatrix} \alpha_1^{\lambda_n} & \alpha_1^{\lambda_{n-1}+1} & \cdots & \alpha_1^{\lambda_1+n-1} \\ \vdots & \vdots & & \vdots \\ \alpha_n^{\lambda_n} & \alpha_n^{\lambda_{n-1}+1} & \cdots & \alpha_n^{\lambda_1+n-1} \end{vmatrix}}{\begin{vmatrix} 1 & \alpha_1 & \cdots & \alpha_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \cdots & \alpha_n^{n-1} \end{vmatrix}} = \{\lambda\}(X_{ij}) \text{ — a polynomial.}$$



Original source details: GELFAND-P000375

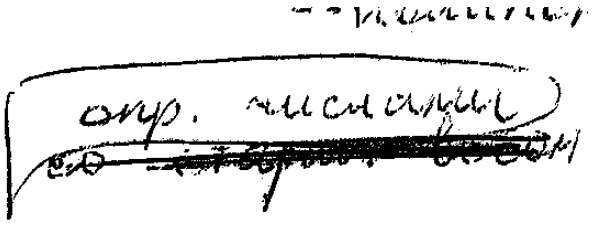
Manuscript notation. Two crossing strokes form the argument X_{ij}.



Original source details: GELFAND-P000375

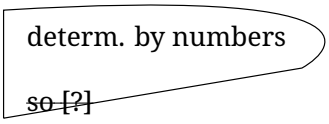
Uncertain reading. Above the first numerator entry is a separate heavily overwritten entry. An α-like form and neighboring short strokes are visible; the earlier index is not securely recovered.

This is the character of a representation.



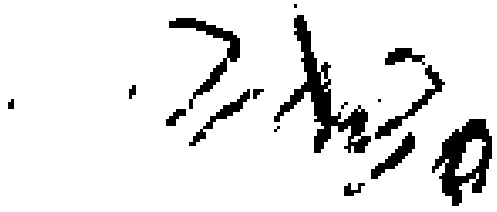
Original source details: GELFAND-P000375

Undeciphered passage. The outlined insertion “determ. by numbers” lies above the cancelled earlier phrase. Its initial Russian “so” and separate final strokes are visible; the words under the dense horizontal strokes are not established in full.



$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0[?].$$

Uncertain reading. The last small oval has an attached lower stroke. 0 is provisionally shown; a remains a candidate.



Original source details: GELFAND-P000375

$\{\lambda, \circleddash\} = \circleddash^{\{\lambda\}}$ — notation.

Uncertain reading. The elevated entry after the enclosed letter contains λ and two curly strokes. Braces $\{\lambda\}$ are provisionally rendered; not all overlapping strokes have been separated.

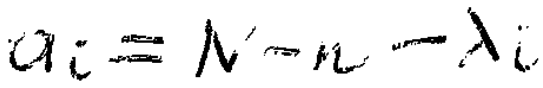
Original source details: GELFAND-P000375

$\circleddash$ — a matrix of diff. forms (they have degree 2 and commute).

Uncertain reading. The circle encloses a short crossbar and two side strokes. Its reading as a letter is uncertain.

Original source details: GELFAND-P000375

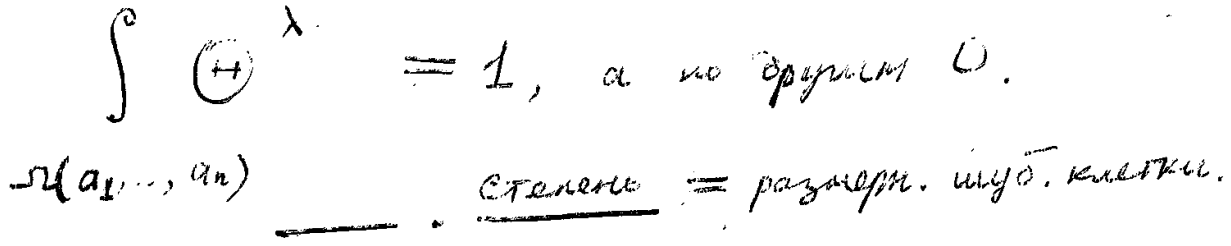
Put $a_i = N' - n - \lambda_i$.



Uncertain reading. Between N and the minus is a separate thin rising stroke above the line. N' is provisionally rendered; the meaning of the stroke is not established.

Original source details: GELFAND-P000375

Then



Original source details: GELFAND-P000375

Diagram and typed description.

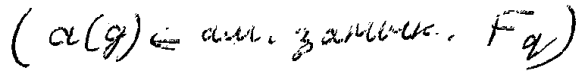
$\int \Theta^{\lambda} = 1$, and on the others 0.

$\Omega(a_1, \ldots, a_n)$ degree = dim. of the Schubert cell.

Problem (I. M. Gelfand)

$$g \in SL(2, F_q).$$

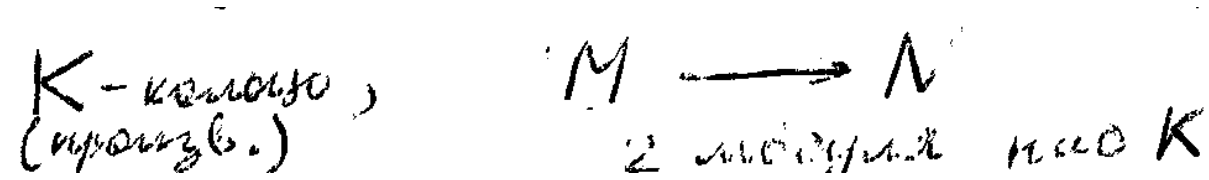
Group ring $\sum_g a(g)g$. ($a(g) \in$ alg. closure F_q).



Original source details: GELFAND-P000375

Uncertain reading. The sign after $a(g)$ is provisionally rendered as $\in$. Its curve and crossbar resemble the sign after g above, in a more compact form.

Describe it. A. Vinogradov. Global aim — to expel analysis from the theory of linear diff. operators.



Original source details: GELFAND-P000375

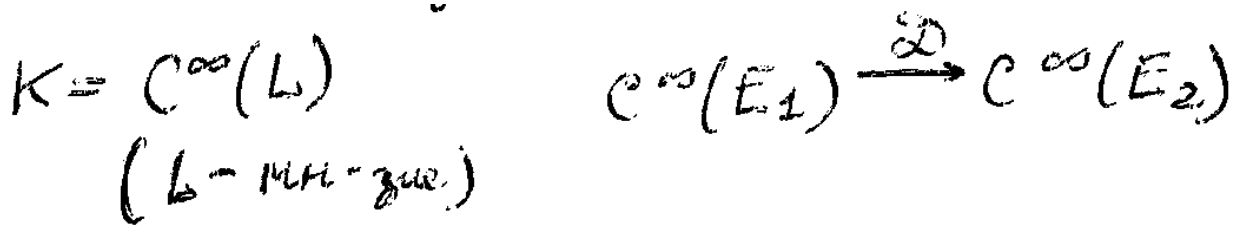
Diagram and typed description.

K — a ring, (arbitr.)

$M \longrightarrow N$

2 modules over K

Classical situation



Original source details: GELFAND-P000375

Diagram and typed description.

$K = C^{\infty}(L)$

(L — manifold)

$C^{\infty}(E_1) \xrightarrow{\mathcal{D}} C^{\infty}(E_2)$

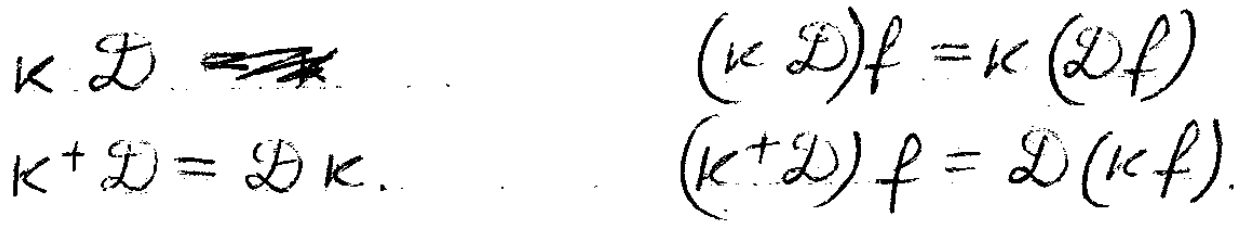
K may be noncommutative.

GELFAND-P000376

[source number: 30]

$$M \xrightarrow{\mathcal{D}} N.$$

$k \in K$,



Original source details: GELFAND-P000376

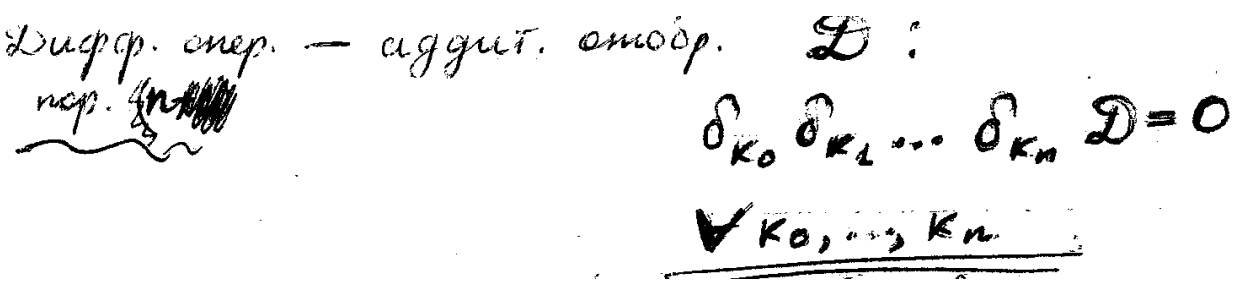
Undeciphered passage. The right-hand side after $k\mathcal{D} =$ is crossed by horizontal and oblique strokes. Its earlier signs have not been read in full.

$k\mathcal{D} =$ [?] $\qquad (k\mathcal{D})f = k(\mathcal{D}f)$

$k^{+}\mathcal{D} = \mathcal{D}k.$ $\qquad (k^{+}\mathcal{D})f = \mathcal{D}(kf).$

$$\underline{\delta_k = k^{+} - k}.$$

Basic definition. A diff. operator — an additive map



Original source details: GELFAND-P000376

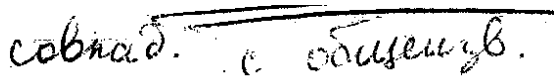
Undeciphered passage. After the visible n an earlier sign is densely cancelled. Curly strokes lie to the left of n, with a wavy line below "ord.". The complete earlier reading is not established.

$\mathcal{D}$:

ord. n [?] $\qquad \delta_{k_0}\delta_{k_1}\cdots\delta_{k_n}\mathcal{D} = 0$

$\underline{\forall k_0, \ldots, k_n}$

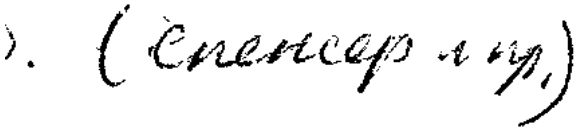
In the classical case this agrees with the customary definition.



Manuscript notation. The last word is abbreviated in the manuscript.

Original source details: GELFAND-P000376

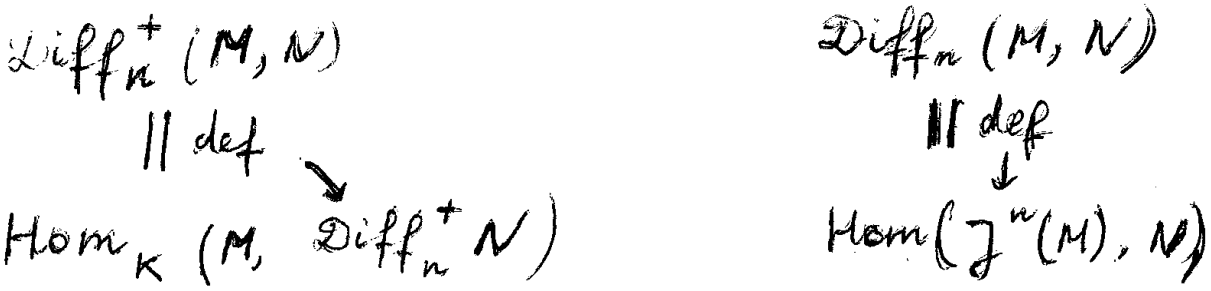
In this way one algebraically constructs the theory of de Rham homology and so on. (Spencer and oth.)



Manuscript notation. The parenthesis reads "Spencer and others," with the last word abbreviated.

Original source details: GELFAND-P000376

$\mathrm{Diff}_n^+(M,N)$ — all these operators. a module with multiplication $\underline{\underline{k^+}}$.



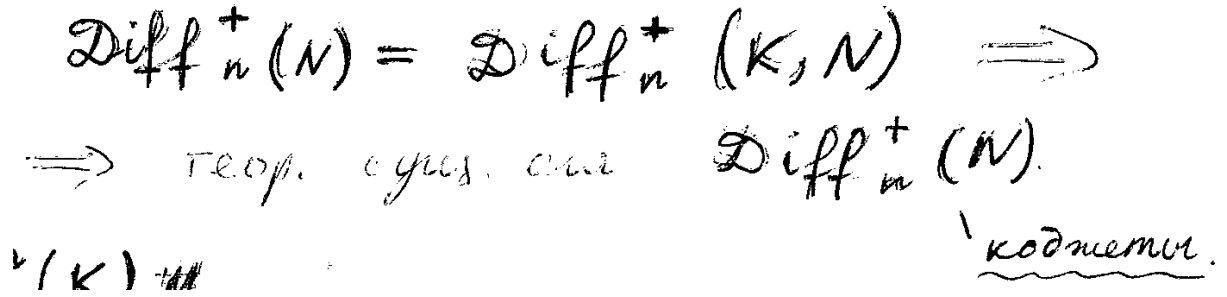
Original source details: GELFAND-P000376

Diagram and typed description.

$\mathrm{Diff}_n^+(M,N)$ $\quad\quad$ $\mathrm{Diff}_n(M,N)$

$\|\ def$ $\searrow$ $\quad\quad$ $\|\ def$ $\downarrow$

$\mathrm{Hom}_K(M,\ \mathrm{Diff}_n^+ N\ )$ $\quad\quad$ $\mathrm{Hom}(J^n(M),N)$



Original source details: GELFAND-P000376

Diagram and typed description.

$\mathrm{Diff}_n^+(N) = \mathrm{Diff}_n^+(K,N) \quad \Longrightarrow$

$\Longrightarrow$ existence theorem for $\mathrm{Diff}_n^+(N)$.

$\backslash$

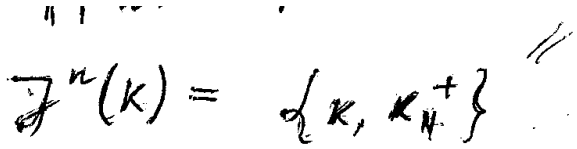
cojets.

$J^n(K)$ ~~[?]~~ — an existence theorem also exists.

Undeciphered passage. After $J^n(K)$ is a separate small sign with overlapping cancellation strokes. Its whole reading is not recovered.

Original source details: GELFAND-P000376

In Diff$_n$ K consider the algebra of operators, generated by k, k^+.



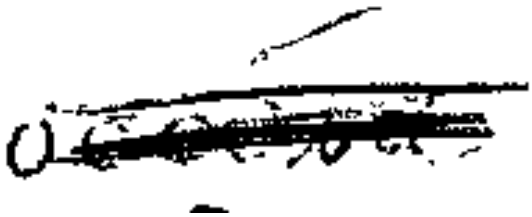
Original source details: GELFAND-P000376

Uncertain reading. Beneath the last k is a separate lower N/n-like sign. N is provisionally rendered; lowercase n and a reading without an independent index remain candidates. The upper plus is retained separately.

$$J^n(K) = \{k, k_N^+\}$$

GELFAND-P000377

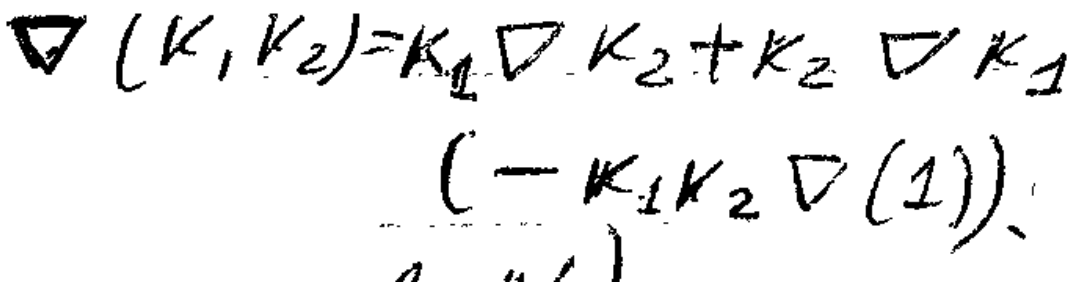
D_i — ~~[?]~~ functors $D_i(M)$ (de Rham homology).



Original source details: GELFAND-P000377

Undeciphered passage. The word before "functors" is cancelled by several horizontal strokes. It has not been recovered in full.

Define inductively. $D = D_1$ — derivations of K into N — denoted $D(N)$. $K \xrightarrow{\nabla} N$ is a derivation if



Original source details: GELFAND-P000377

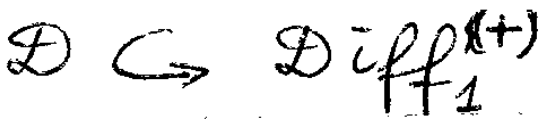
Uncertain reading. The parentheses contain k, k_2. The first low stroke reads as a comma.

$$\nabla(k, k_2) = k_1 \nabla k_2 + k_2 \nabla k_1$$

This is a diff. op. of order 1. $(-k_1 k_2 \nabla(1))$.

This is $D(N)$ (vector fields with values in N). D — ordinary vector fields.

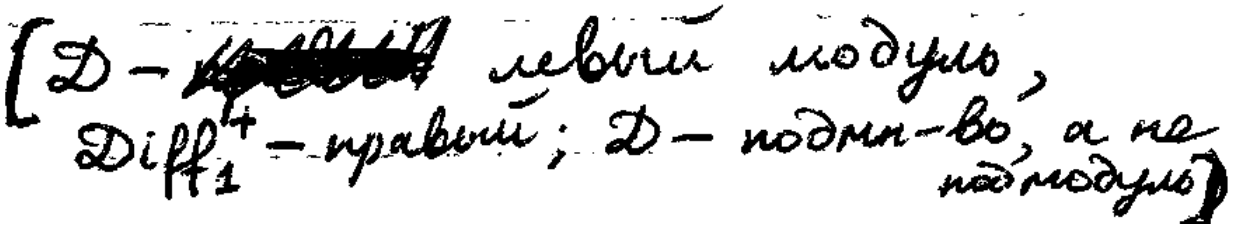
$$D_2(N) = D\{D(N) \subset \text{Diff}_1^+ N\}.$$



Original source details: GELFAND-P000377

Diagram and typed description.

$D \hookrightarrow \text{Diff}_1^{(+)}$



Original source details: GELFAND-P000377

Uncertain reading. The old adjective before “left” is densely cancelled. “right” is rendered as a candidate; the overlapping strokes do not yield an indisputable whole reading.

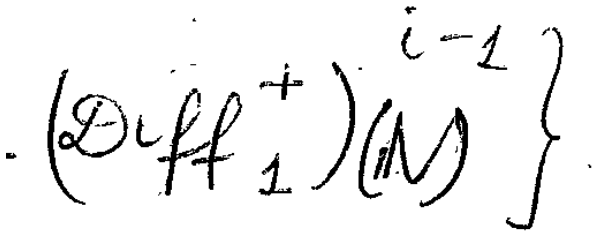
D — ~~right[?]~~ left module,

Diff_1^+ — right; D — a subset, and not a submodule

In $\mathrm{Diff}_1^{(+)}$ it is not a submodule, and the inclusion — a diff. operator of order 1. $D_2(N)$ — derivations of K into $\mathrm{Diff}_1^+ N$ which go into $D(N)$.

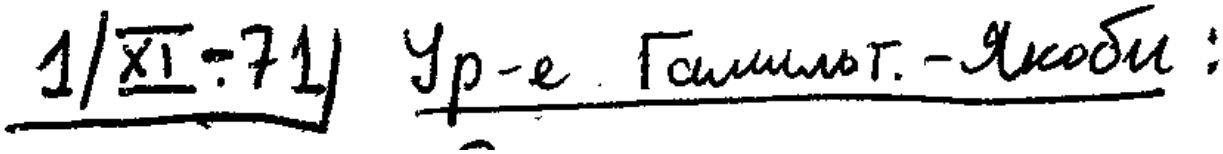
$$D_i(N) = D\{D_{i-1}(N) \subset (\mathrm{Diff}_1^+)(N)^{i-1}\}.$$

(by induction).



Manuscript notation. The exponent $i-1$ is written above (N).

Original source details: GELFAND-P000377

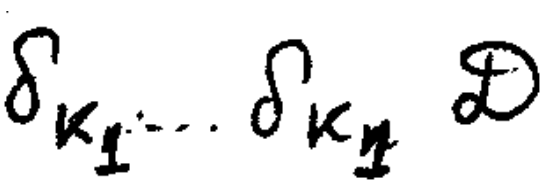


Original source details: GELFAND-P000377

Diagram and typed description.

1/XI-71 Hamilton-Jacobi equation:

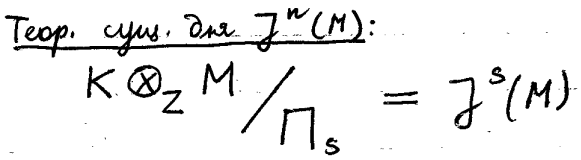
$\delta_{k_1} \cdots \delta_{k_1} \mathcal{D}$ — left side of the Hamilton-Jacobi eq.



Uncertain reading. The last lower index contains thick overlapping strokes. k_1 is provisionally rendered; k_n is retained among the candidates.

Original source details: GELFAND-P000377

(calculate) (k_1 — the same everywhere). Existence theorem for $J^n(M)$:



Uncertain reading. The heading has n; the following quotient and Π_s have s. Both source indices are retained.

Original source details: GELFAND-P000377

$$K \otimes_{\mathbb{Z}} M/\Pi_s = J^s(M).$$

Π_s — the submodule in $K \otimes_{\mathbb{Z}} M$ generated by the required relations.

$$k(k' \otimes m) = kk' \otimes m,$$
$$\bar{k}(k' \otimes m) = k' \otimes km.$$

GELFAND-P000378

[source number: 31]

$$\Delta_{k_0 \ldots k_s} = (\bar{k}_0 - k_0) \cdots (\bar{k}_s - k_s).$$

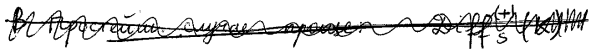
Consider the submodule generated by these Δ.

$$K \otimes_{\mathbb{Z}} M = A \supset \Delta(A) = \Pi_s \quad \text{(the usual definition of jets).}$$

Diagram and typed description.

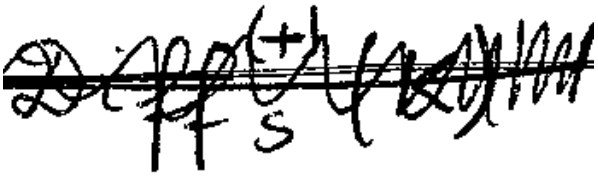
Original source details: GELFAND-P000378

~~In the simplest[?] case simply Diff$_s^{(+)}(K, N[?])$ [?]~~



Undeciphered passage. The whole line is cancelled. “In the simplest[?] case simply” and formula strokes can be distinguished. The adjective ending and final overlapping signs are not recovered in full.

Original source details: GELFAND-P000378



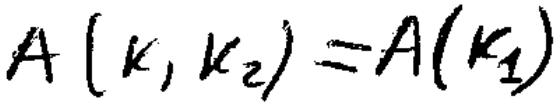
Uncertain reading. In the cancelled Diff$_s^{(+)}(K, N[?])$ the target sign is N/M-like. N is provisionally rendered; M remains a candidate.

Original source details: GELFAND-P000378

$J^s(K)$ is simpler than in the general case.

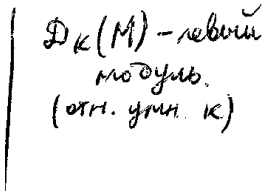
$D_1(M)$ — diff. operators without a constant term. $D_1(M) = D(M)$ (derivations of K with values in M). $D(M) \subset \mathrm{Diff}_1 M$.

$$A \in D_1(M) \overset{def}{\iff} A(k, k_2) = A(k_1)k_2 + k_1 A(k_2).$$



Manuscript notation. The argument of A reads k, k_2, as in the related formula on the previous frame.

Original source details: GELFAND-P000378



Original source details: GELFAND-P000378

Diagram and typed description.

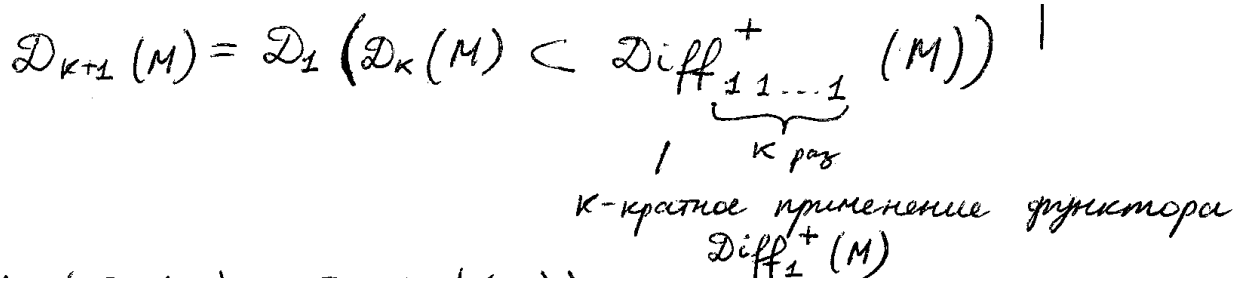
$D_0(M) = M$;

$D_k(M)$ — a left

module.

(w.r.t. multiplication by k).

$$D_2(M) = D_1(D_1(M) \subset \mathrm{Diff}_1^+(M)).$$



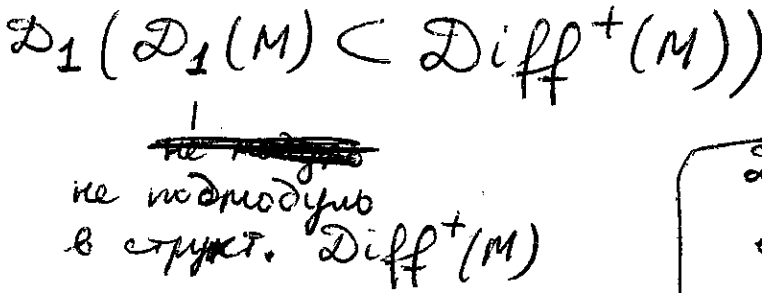
Original source details: GELFAND-P000378

Diagram and typed description.

$D_{k+1}(M) = D_1(D_k(M) \subset \underbrace{\mathrm{Diff}_{1\,1\ldots1}^+}_{k \text{ times}}(M))$

/

the k-fold application of the functor $\mathrm{Diff}_1^+(M)$.



Original source details: GELFAND-P000378

Undeciphered passage. Above the uncancelled “not a submodule” is a cancelled entry with visible parts “not” and “submod…”. The other letters are not recovered in full. Above the entry is a separate short vertical stroke.

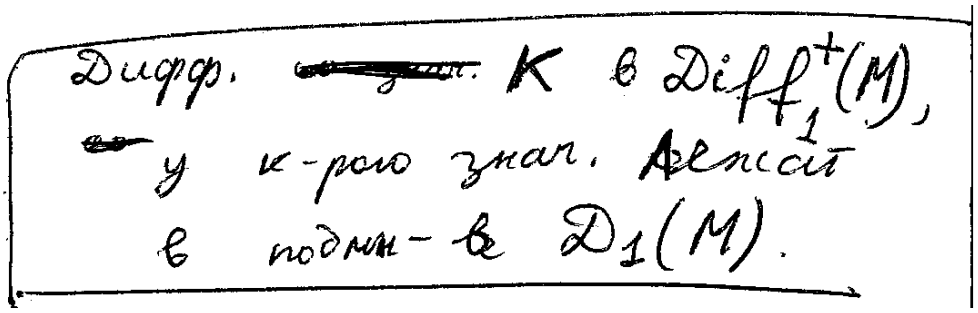
$D_1(D_1(M) \subset \text{Diff}^+(M))$.

|

~~not a submod[?]~~

not a submodule

in the structure $\text{Diff}^+(M)$.



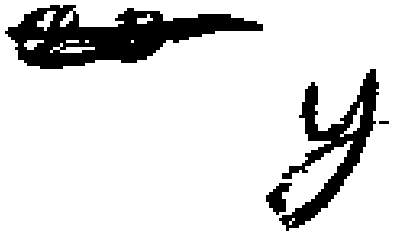
Original source details: GELFAND-P000378

Undeciphered passage. After “Diff.” before K a word is densely cancelled. It has not been read in full.

Diff. ~~[?]~~ of K into $\text{Diff}_1^+(M)$,

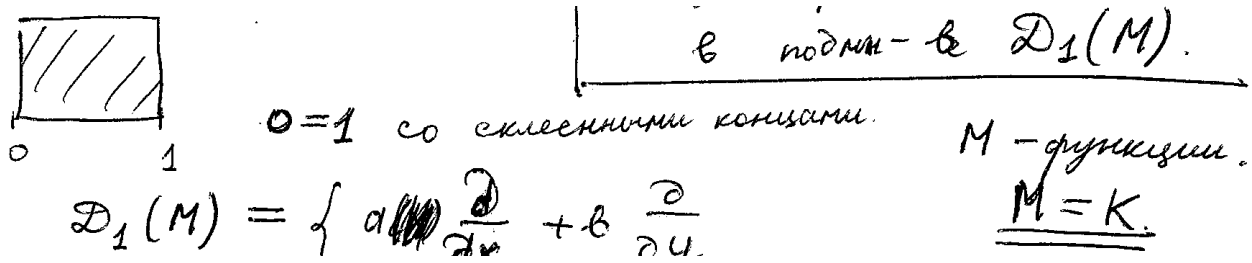
~~[?]~~ whose values lie

in the subset $D_1(M)$.



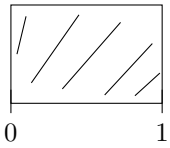
Undeciphered passage. Before “whose” is a separate short fragment beneath cancellation strokes. It has not been deciphered in full.

Original source details: GELFAND-P000378



Original source details: GELFAND-P000378

Diagram and typed description.

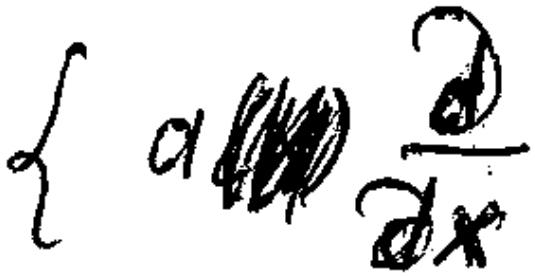


$0 = 1$ with ends identified.

M — functions.

$\underline{\underline{M = K}}$.

$$D_1(M) = \left\{ a \,[?]\, \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} \right. .$$



Undeciphered passage. After the legible a its argument is densely overwritten and cancelled. It has not been deciphered in full. An opening curly brace is visible; no closing brace has been added.

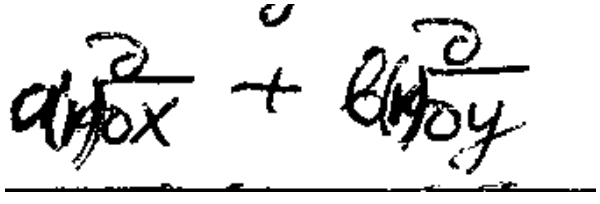
Original source details: GELFAND-P000378

$$\mathrm{Diff}_1^+ : a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c.$$

$$K(x) \longrightarrow [?]\, a_k \frac{\partial}{\partial x} + b_k \frac{\partial}{\partial y}.$$

Undeciphered passage. Immediately after the arrow a short prefix is cancelled. It cannot be read in full. The line lies at the lower edge of the frame.

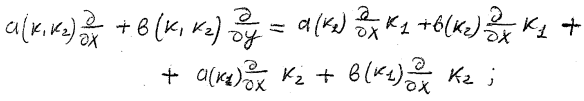
Original source details: GELFAND-P000378



Uncertain reading. Beside a and b are compact k-like strokes. Lower indices a_k, b_k are provisionally rendered; $a(k), b(k)$ remain candidates. Complete parentheses are not recovered.

Original source details: GELFAND-P000378

$$a(k,k_2)\frac{\partial}{\partial x}+b(k,k_2)\frac{\partial}{\partial y}=a(k_2)\frac{\partial}{\partial x}k_1+b(k_2)\frac{\partial}{\partial x}k_1+$$
$$+a(k_1)\frac{\partial}{\partial x}k_2+b(k_1)\frac{\partial}{\partial x}k_2;$$

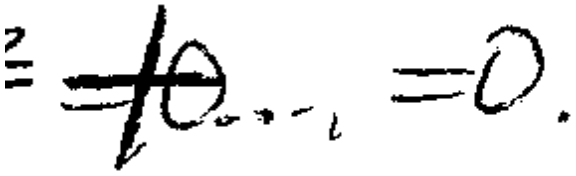


Manuscript notation. On the left, the arguments of a and b read k, k_2. All four denominators on the right read ∂x, including the terms with b.

Original source details: GELFAND-P000379

Hence $a(k,k_2)=a(k_2)k_1+a(k_1)k_2$ and b — a derivation, and also the condition:

$$a(k_2)\frac{\partial k_1}{\partial x}+a(k_1)\frac{\partial k_2}{\partial x}$$
$$+b(k_2)\frac{\partial k_1}{\partial x}+b(k_1)\frac{\partial k_2}{\partial x}\quad=0\ldots=0.$$



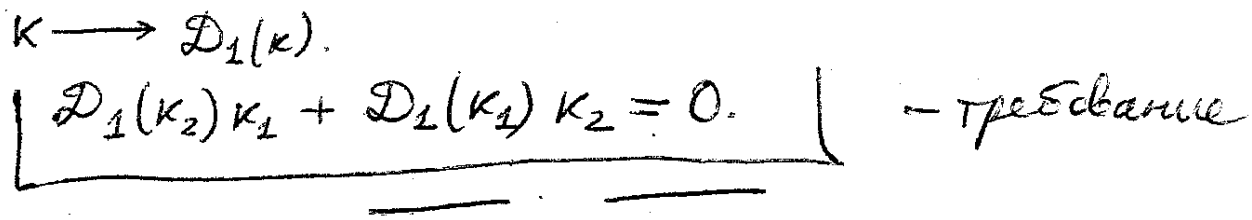
Uncertain reading. After the sum, $=0$ is crossed by a separate oblique stroke. Separate dots and a faint lower stroke follow; then there is an uncancelled $=0$.

Original source details: GELFAND-P000379

$$\left(a(k_2)\frac{\partial}{\partial x}+b(k_2)\frac{\partial}{\partial y}\right)k_1+\sim=0.$$
$$D_1(k_2)k_1+D_1(k_1)k_2=0.$$

Uncertain reading. The first lower 1 is densely overwritten. D_1 is provisionally rendered; the earlier overlapping strokes are not recovered in full.

Original source details: GELFAND-P000379



Original source details: GELFAND-P000379

Diagram and typed description.

$k \longrightarrow D_1(k)$.

$D_1(k_2)k_1 + D_1(k_1)k_2 = 0.$ — requirement.

Original source details: GELFAND-P000379

Undeciphered passage. At the left margin before $D_k(M)$ a separate sign is cancelled. It has not been deciphered in full.

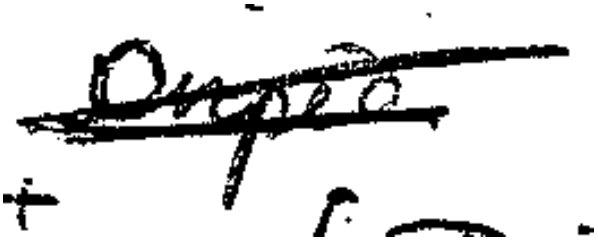
~~[?]~~ $D_k(M)$ —k-forms on the ~~[?]~~ manifold with values in M.

~~[?]~~

~~Def.-e.[?]~~

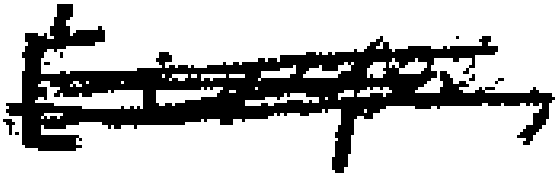
Undeciphered passage. Between “on” and “manifold” a sign is densely cancelled. It has not been recovered in full.

Original source details: GELFAND-P000379



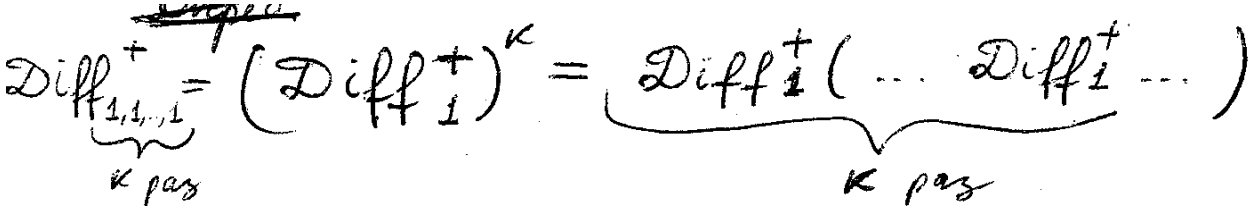
Uncertain reading. Below the line at left is a cancelled contraction, provisionally “Def.-e.”; “Def.-o.” and “Def.” are retained as alternatives. The ending is not expanded.

Original source details: GELFAND-P000379



Original source details: GELFAND-P000379

Undeciphered passage. Below “forms” is a separate cancelled entry. It has not been recovered in full.



Original source details: GELFAND-P000379

Diagram and typed description.

$$\underbrace{\mathrm{Diff}^+_{1,1,\dots,1}}_{k\text{ times}} = (\mathrm{Diff}^+_1)^k = \underbrace{\mathrm{Diff}^+_1(\cdots \mathrm{Diff}^+_1 \cdots)}_{k\text{ times}}.$$

$$(\mathrm{Diff}^+_1)^2(M) = \mathrm{Diff}_1(K, \mathrm{Diff}^+_1 M).$$

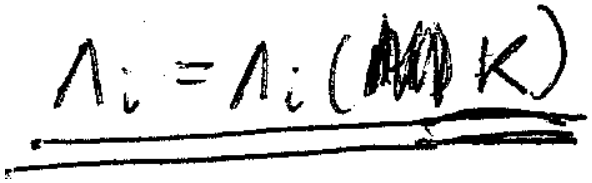
Definition of exterior differential forms. (the de Rham complex over the ring K)

Original source details: GELFAND-P000379

Diagram and typed description.

$$D_i(M) \overset{def}{=} \mathrm{Hom}_K(\Lambda_i, M);$$

$$\Lambda_i = \Lambda_i(\cancel{M}\,[?]\,K).$$



Original source details: GELFAND-P000379

Uncertain reading. In the argument before the legible K an M/N-like sign is cancelled. M is provisionally rendered; N remains a candidate.

Theorem: a representing object exists

Original source details: GELFAND-P000379

Manuscript notation. The manuscript uses the singular “a representing object exists.”

$K = \mathbb{Z}_2$ will have $D_2 \neq 0$! (the 2-dimensional de Rham complex).

GELFAND-P000380

[source number: 32]

Consider the group ring $\Gamma_2(\mathbb{Z}_2)$ (with coefficients in $\mathbb{Z}_2$).

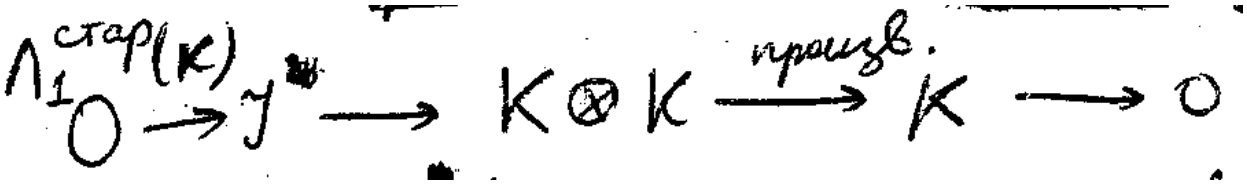
Manuscript notation. The first letter has a vertical and a top stroke, Γ, with no middle crossbar.

Original source details: GELFAND-P000380

All de Rham cohomology groups — $\mathbb{Z}_2$. For group rings, de Rham cohom. — group cohom.!

Diagram and typed description.

Original source details: GELFAND-P000380



Original source details: GELFAND-P000380

Uncertain reading. At the upper right of J in the sequence is a separate dense sign with overlapping strokes. Its whole reading and meaning are not established; it is rendered as [?].

$\Lambda_1^{\text{old}}(K)$

$$0 \longrightarrow J^{[?]} \longrightarrow K \otimes K \xrightarrow{\text{mult.}} K \longrightarrow 0$$

$$J^1/J^2 = \Lambda_1, \quad \text{the rest — exterior powers.}$$

Uncertain reading. At the upper right of the numerator are dense overwritten strokes. J^1 is provisionally rendered; J without an independent power and J with an unidentified earlier power remain candidates. The denominator J^2 is legible.

Original source details: GELFAND-P000380

For a field these cohomologies will be 0, and the Vinogradov ones $\neq 0$. This definition is unnatural in the categorical sense. They are not defined for noncommutative rings. This is the only natural definition in the category of rings. The differential d.

Original source details: GELFAND-P000380

Undeciphered passage. Below the map is a separate small sign with overlapping cancellation strokes. It has not been recovered in full.

$D_k \longrightarrow D_{k-1}(\mathrm{Diff}_1^+)$ morphism of functors.

(in other terms, d).

i.e. $D_k(M) \longrightarrow D_{k-1}(\mathrm{Diff}_1^+(M))$. natural map.

~~[?]~~

$$D_1 = D \underset{\text{incl.}}{\longrightarrow} \mathrm{Diff}_1^+ \quad (= D_0(\mathrm{Diff}_1^+)).$$

Uncertain reading. The first lower index is densely overwritten, with a loop-like component. 1 is provisionally rendered; 0 remains a visual candidate.

Original source details: GELFAND-P000380

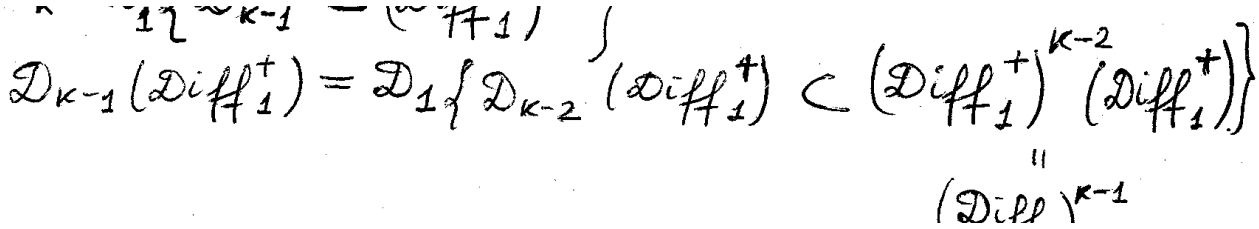
$$D_1 \longrightarrow D_0(\mathrm{Diff}_1^+).$$

(not a module inclusion, but a differential operator of order 1).

$\mathrm{Diff}_n \longrightarrow \mathrm{Diff}_n^+$
(a differential operator of order n).

Induction:

$$D_k = D_1\{D_{k-1} \subset (\mathrm{Diff}_1^+)^{k-1}\}.$$



Original source details: GELFAND-P000380

Diagram and typed description.

$$D_{k-1}(\mathrm{Diff}_1^+) = D_1\{$$

$$D_{k-2}(\mathrm{Diff}_1^+) \subset \underset{\| \|}{(\mathrm{Diff}_1^+)^{k-2}(\mathrm{Diff}_1^+)} \}.$$
$$(\mathrm{Diff}_1)^{k-1}$$

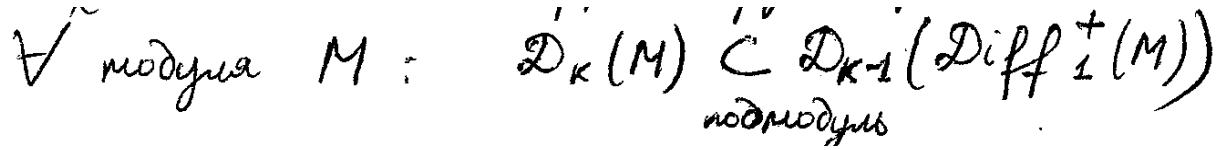
Assume inductively that $D_{k-1} \subset D_{k-2}(\mathrm{Diff}_1^+)$, and then we obtain:

GELFAND-P000381

$D_k \subset D_{k-1}(\mathrm{Diff}_1^+)$. This is an [?] injective morphism of functors, that is

Undeciphered passage. Before “injective” a short sign or fragment is cancelled. It has not been deciphered in full.

Original source details: GELFAND-P000381

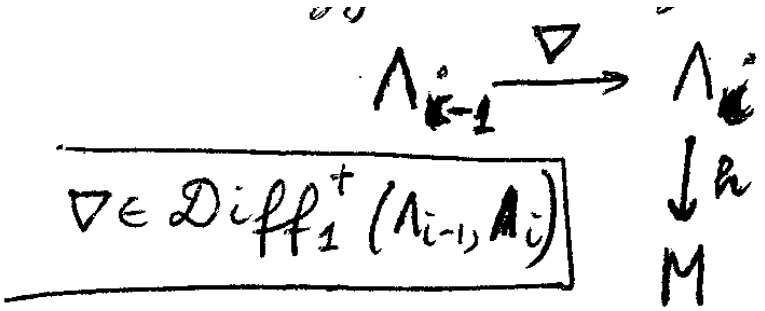


Original source details: GELFAND-P000381

Diagram and typed description.

$\forall$ module M: $D_k(M) \subset D_{k-1}(\mathrm{Diff}_1^+(M))$
submodule

Express this in the language of differential forms:



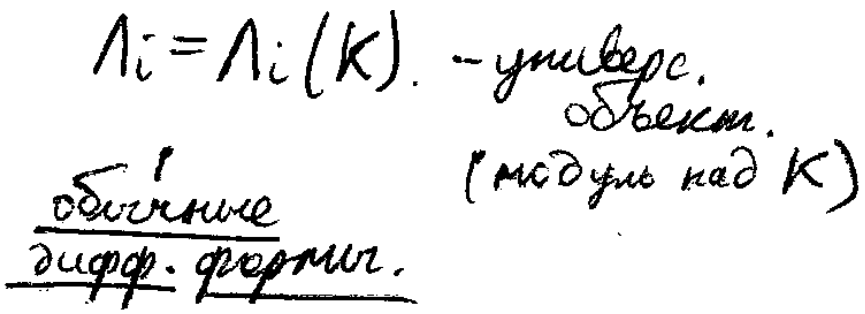
Original source details: GELFAND-P000381

Uncertain reading. In the diagram the lower $i-1$ at left and i at right are written over reinforced strokes. The current pair is provisionally rendered; the earlier lower signs are not recovered in full.

$$\begin{array}{ccc} \Lambda_{i-1} & \xrightarrow{\nabla} & \Lambda_i \\ & & \downarrow h \\ & & M \end{array}$$

$$\boxed{\nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, \Lambda_i)}$$

$$D_i(M) = \mathrm{Hom}(\Lambda_i, M).$$



Original source details: GELFAND-P000381

Uncertain reading. Above “ordinary” is a separate thick, nearly vertical stroke. An independent lower dot is not established; no complete exclamation mark has been supplied.

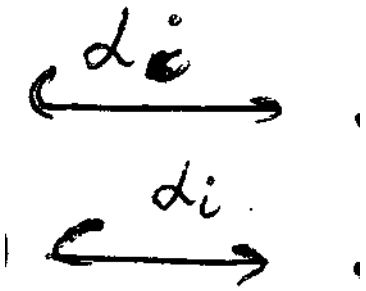
$\Lambda_i = \Lambda_i(K)$. — universal object.

(a module over K).

ordinary diff. forms.

$$D_i(M) \overset{\alpha_i}{\hookrightarrow} D_{i-1}(\mathrm{Diff}_1^+(M)).$$

$$h \in \mathrm{Hom}(\Lambda_i, M) \overset{\alpha_i}{\hookrightarrow} [?]\,\mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M)).$$



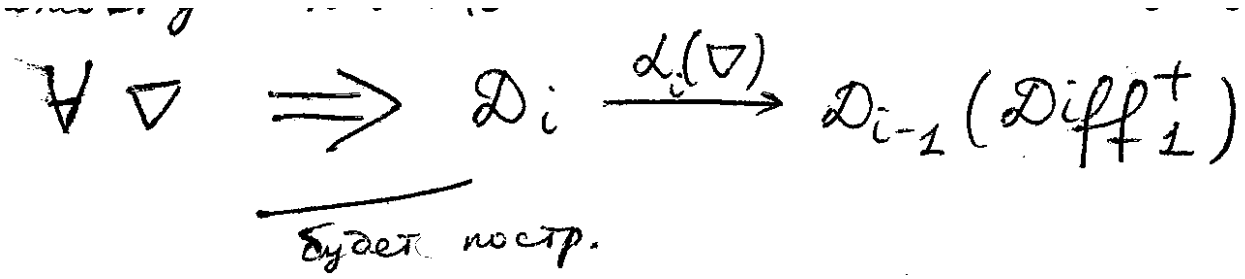
Manuscript notation. Both arrows have a hook at left and an arrowhead at right.

Original source details: GELFAND-P000381

Undeciphered passage. Before the right-hand Hom a short fragment is densely cancelled. It has not been recovered in full.

Original source details: GELFAND-P000381

Let a differential operator of order 1 be given, $\Lambda_{i-1} \overset{\nabla}{\longrightarrow} \Lambda_i$. Then to each $h : \Lambda_i \longrightarrow M$ assign the composition $h \circ \nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, M)$.

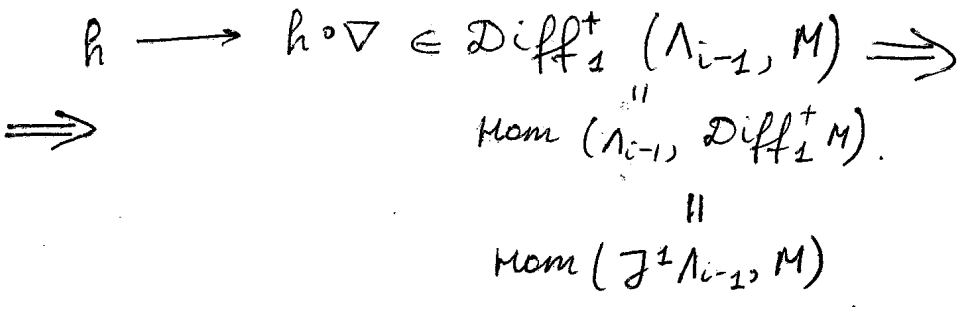


Original source details: GELFAND-P000381

Diagram and typed description.

$$\forall \nabla \Longrightarrow D_i \overset{\alpha_i(\nabla)}{\longrightarrow} D_{i-1}(\mathrm{Diff}_1^+)$$

will be constructed.



Original source details: GELFAND-P000381

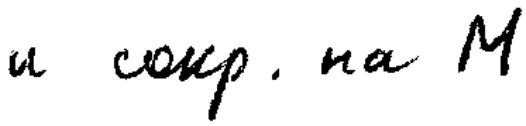
Diagram and typed description.

$$h \longrightarrow h \circ \nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, M) \quad \Longrightarrow$$

$$\Longrightarrow \qquad \mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+ M)$$

$$\mathrm{Hom}(J^1\Lambda_{i-1}, M)$$

And adj. on M



Original source details: GELFAND-P000381

Manuscript notation. The abbreviation is left unexpanded in the manuscript.

Original source details: GELFAND-P000381

Undeciphered passage. The first cancelled entry lies under the strokes of a reinforced rightward arrow; its whole reading is not recovered. At right, “diff.” is separately cancelled. Both parts and the arrow are retained.

~~[?]~~ ⟶ ~~diff.~~

a homomorphism $J^1\Lambda_{i-1} \longrightarrow \Lambda_i$, i.e. a differential operator $\Lambda_{i-1} \longrightarrow \Lambda_i$. Taking $M = \Lambda_i$ and the identity map gives what is needed.

GELFAND-P000382

[source number: 33]

$$\mathrm{Hom}(\Lambda_i, M) \xhookrightarrow{\alpha_i} \mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M)).$$

Manuscript notation. The α_i arrow has a hook at left and an arrowhead at right.

Original source details: GELFAND-P000382

$M = \Lambda_i$ and id on the left. An element of $\mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M))$ is by definition a differential operator of order 1 from Λ_{i-1} to M. (We find ∇, which exists and is unique). We must prove that

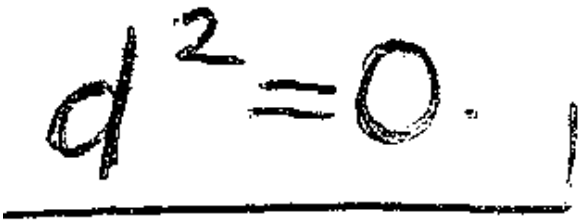
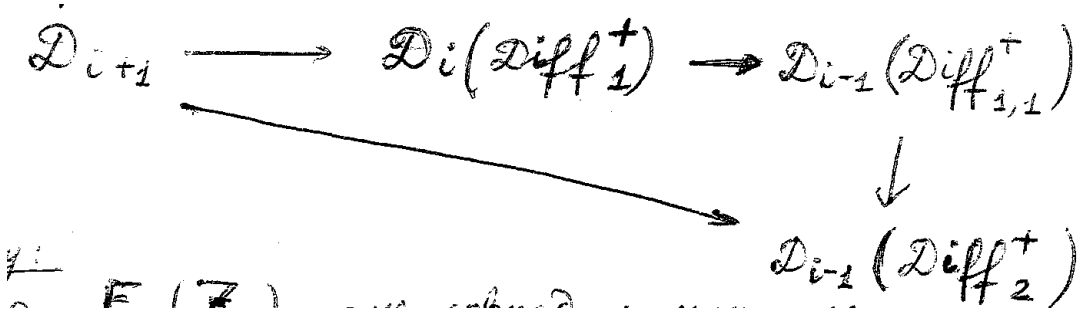


Diagram and typed description.

$$d^2 = 0.$$

Original source details: GELFAND-P000382

Consider

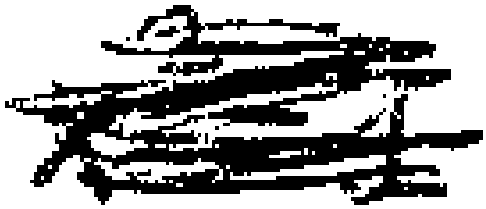


Original source details: GELFAND-P000382

Diagram and typed description.

$$\begin{array}{ccccc} D_{i+1} & \longrightarrow & D_i(\mathrm{Diff}_1^+) & \longrightarrow & D_{i-1}(\mathrm{Diff}_{1,1}^+) \\ & \searrow & & & \downarrow \\ & & & & D_{i-1}(\mathrm{Diff}_2^+) \end{array}$$

An aside: The ring $F_2(\mathbb{Z}_2)$—does not coincide with the classical ones. $0, 1, x$, ~~[?]~~, $1 + x$, $(x^2 = 1)$.



Undeciphered passage. After x and before $1 + x$ a separate element is cancelled. Its whole reading has not been recovered.

Original source details: GELFAND-P000382

$y = x + 1$, then $y^2 = 0$. $0, 1, y, 1 + y$, where $y^2 = 0$.

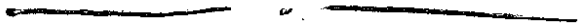
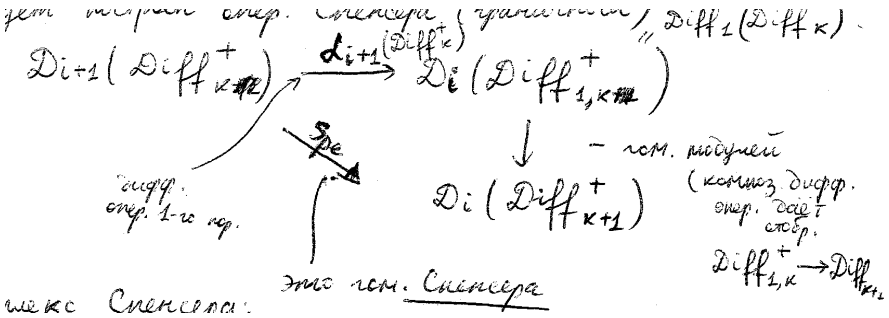


Diagram and typed description.

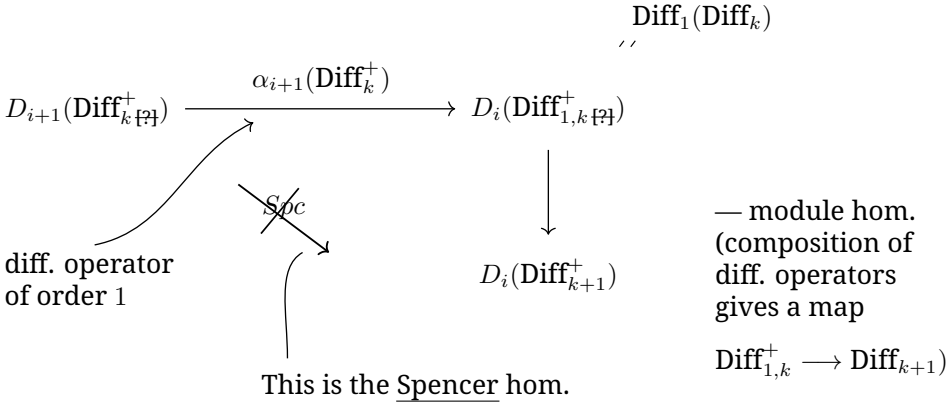
Original source details: GELFAND-P000382

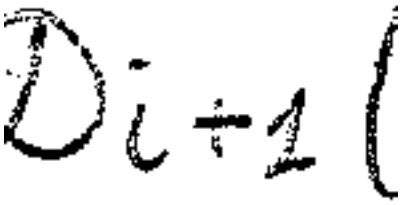
The Spencer operator (boundary) will be constructed.



Original source details: GELFAND-P000382

Diagram and typed description. At upper left is $D_{i+1}(\mathrm{Diff}_k^+)$, at upper right $D_i(\mathrm{Diff}_{1,k}^+)$, and below at right $D_i(\mathrm{Diff}_{k+1}^+)$. A horizontal arrow bears $\alpha_{i+1}(\mathrm{Diff}_k^+)$; at right is a downward arrow. A short diagonal down-right arrow stops before the lower expression; its Spc label is crossed out. A curved pointer from below at left approaches the start of the upper horizontal arrow. Under the diagonal is a curved vertical bracket to "This is the Spencer hom."; "Spencer" is underlined. Above the right expression are $\mathrm{Diff}_1(\mathrm{Diff}_k)$ and two short slanted strokes; the composition annotation is at right.





Original source details: GELFAND-P000382

Uncertain reading. The sign after i has discontinuous vertical portions. Provisionally D_{i+1}; possibly D_{i-1}.

Original source details: GELFAND-P000382

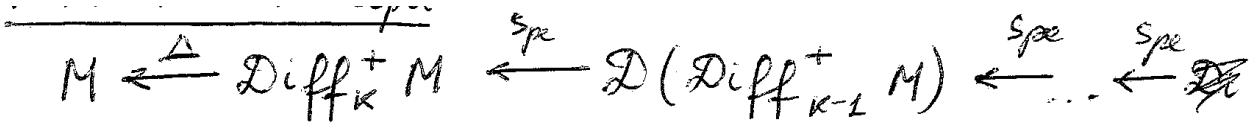
Uncertain reading. After the left k a linking mark and a 2-like sign are cancelled. The earlier reading may be -2, $+2$, or another index ending in 2.

Original source details: GELFAND-P000382

Undeciphered passage. After the right k an additional index fragment is cancelled. Its whole reading has not been recovered.

Original source details: GELFAND-P000382

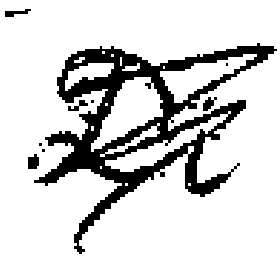
The Spencer complex:



Original source details: GELFAND-P000382

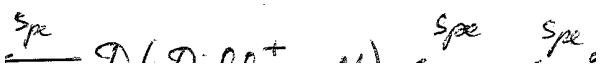
Diagram and typed description.

$$M \xleftarrow{\Delta} \mathrm{Diff}_k^+ M \xleftarrow{Spc} D(\mathrm{Diff}_{k-1}^+ M) \xleftarrow{Spc} \cdots \xleftarrow{Spc} \not{D}_i$$



Original source details: GELFAND-P000382

Uncertain reading. The cancelled expression at the far right is provisionally D_i; the lower hook also admits D_1.



Original source details: GELFAND-P000382

Uncertain reading. All four labels have a third sign resembling c/e or a lower e-like sign. Provisionally Spc; possibly Spe. The diagonal label is crossed out; the three lower labels are current.

GELFAND-P000383

$$\longleftarrow D_i(\mathrm{Diff}^+_{k-i} M) \longleftarrow \cdots$$

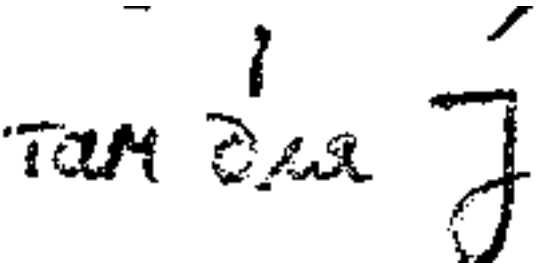
This is the Spencer resolution (coresol. Sm.)

Original source details: GELFAND-P000383

Manuscript notation. A separate dot is visible above the last sign in $k-i$.

Original source details: GELFAND-P000383

Uncertain reading. In parentheses, “coresol. Sm.”; the final two joined Cyrillic letters admit “См.” or “Сп.”. The abbreviation is left unexpanded.

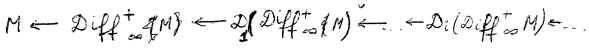


Original source details: GELFAND-P000383

Diagram and typed description. The small note reads “there for J”. A separate vertical stroke is above “for”.

there for J.

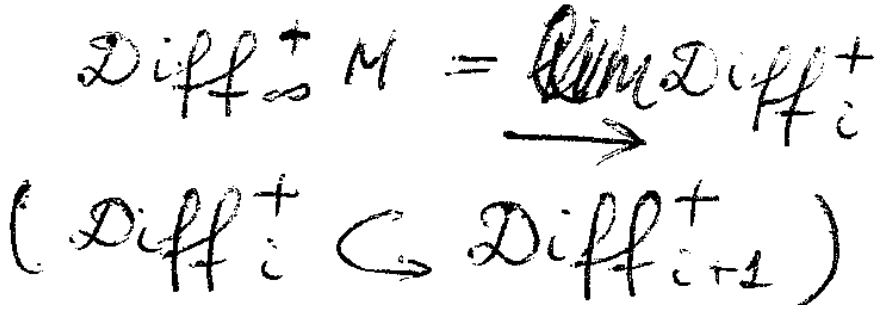
$$M \longleftarrow \mathrm{Diff}^+_\infty\{M\} \longleftarrow D_1(\mathrm{Diff}^+_\infty\{M\})$$
$$\longleftarrow \cdots \longleftarrow D_i(\mathrm{Diff}^+_\infty M) \longleftarrow \cdots$$



Original source details: GELFAND-P000383

Uncertain reading. The first M has braces with reinforced strokes. Before the second M is a brace-like sign, and after it a closing round parenthesis. Provisionally $D_1(\mathrm{Diff}^+_\infty\{M)$; the earlier form of the reinforced strokes is not established in full. All arrows point left.

$$\mathrm{Diff}^+_\infty M = \varinjlim \mathrm{Diff}^+_i .$$



Original source details: GELFAND-P000383

Diagram and typed description.

$$(\mathrm{Diff}^+_i \hookrightarrow \mathrm{Diff}^+_{i+1})$$

This is a complex. And a map (not yet defined)

$$\mathrm{Diff}^+_k M \longrightarrow M.$$

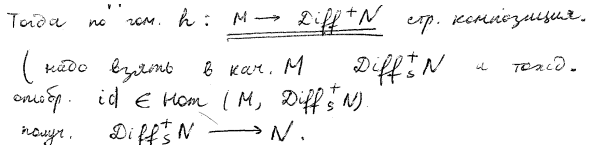
$$\mathrm{Diff}^+_s(M,N) \stackrel{def}{=} \mathrm{Hom}_K(M, \mathrm{Diff}^+_s N).$$

Let $\mathrm{Diff}^+ N \xrightarrow{\Delta} N$; Δ — an operator of order s. Then from the hom. $h : M \longrightarrow \mathrm{Diff}^+ N$ construct the composition. (we must take $\mathrm{Diff}^+_s N$ as M and the identity map

$$id \in \mathrm{Hom}(M, \mathrm{Diff}^+_s N).$$

we obtain $\mathrm{Diff}^+_s N \longrightarrow N$.

This defines the first morphism.



Original source details: GELFAND-P000383

Diagram and typed description. The map $M \longrightarrow \mathrm{Diff}^+ N$ following h is doubly underlined. Below Hom a separate thin diagonal stroke runs down to the right, without an arrowhead, towards the start of the next arrow. Before "we must" is an opening round parenthesis; no separate overall closing parenthesis is visible after the lower N.

$$\Delta \in \mathrm{Diff}^+_s(K,M) \longrightarrow \Delta(1).$$

Headings: All Spencer resolutions in non-coordinate form (a distinction from Spencer). The index over any ring. The Hamiltonian formalism for $\forall$ ring. Next time—continuation.

GELFAND-P000384

[source number: 34]

[ ]

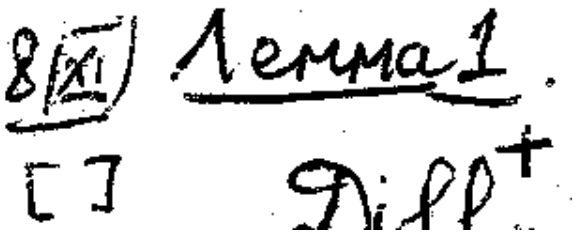


Diagram and typed description.

8 XI Lemma 1.

Original source details: GELFAND-P000384

$\mathrm{Diff}_n^+(M,N)$ for fixed N —a functor of M (contravar.).

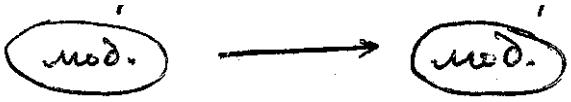


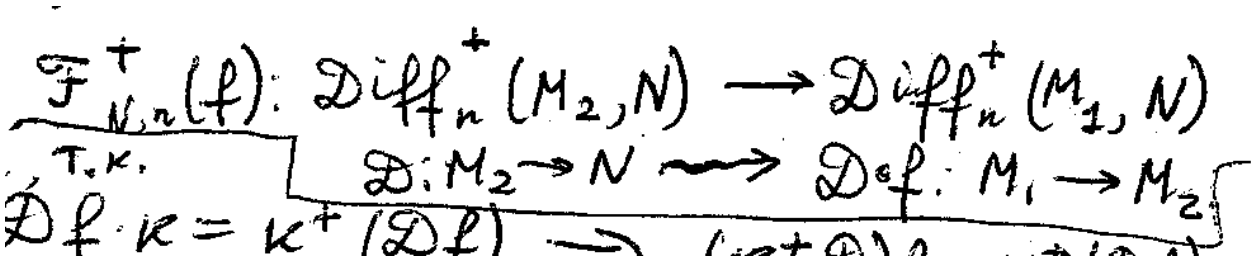
Diagram and typed description.

mod. ⟶ mod.

Original source details: GELFAND-P000384

(and if we take $\mathrm{Diff}_n(M,N)$, then this is a functor of N for fixed M (covar.)). Proof: $\mathcal{F}^+_{N,n}(M) = \mathrm{Diff}_n^+(M,N)$. If $f : M_1 \longrightarrow M_2$ a module hom. $\Longrightarrow$

$$\mathcal{F}^+_{N,n}(f) : \mathrm{Diff}_n^+(M_2,N) \longrightarrow \mathrm{Diff}_n^+(M_1,N).$$

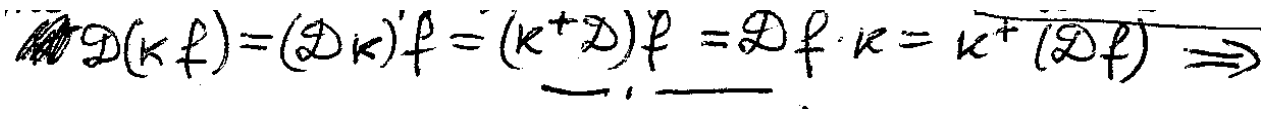


Original source details: GELFAND-P000384

Uncertain reading. The right-hand inset reads $D \circ f : M_1 \longrightarrow M_2$; the final M_2 is retained. Its frame is open above, with a long lower stroke and short side bends.

$D : M_2 \longrightarrow N \rightsquigarrow D \circ f : M_1 \longrightarrow M_2$

And this is a module homomorphism, since



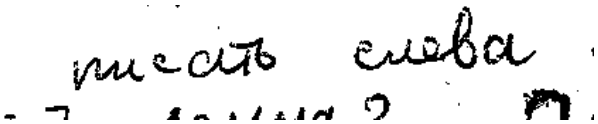
Original source details: GELFAND-P000384

Undeciphered passage. Before $D(kf)$ is a separate short, densely cancelled fragment. Its whole reading has not been recovered.

[?] $D(kf) = (Dk)f = (k^+D)f = Dfk = k^+(Df)$

$$\Longrightarrow (k^+D)f = k^+(Df).$$

Similarly, if we consider $\mathrm{Diff}_n(M,N)$ as a functor of N, then f must be written on the left, and this also gives a module homomorphism.



Original source details: GELFAND-P000384

Manuscript notation. The manuscript reads "be written on the left".

[ ] Lemma 2. Let $A \in \mathrm{Diff}_1(K,M)$. Then

$$A(k_1k_2) = k_1A(k_2) + k_2A(k_1) - k_1k_2A(1). \qquad (1)$$

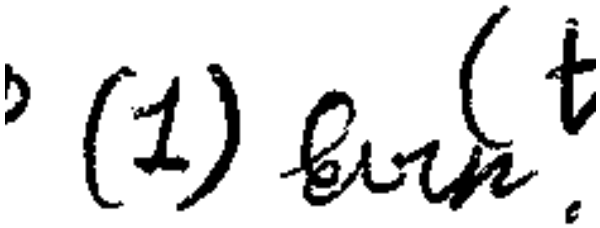
and conversely, if $A : K \longrightarrow M$ —an additive map satisfying (1), then $A \in \mathrm{Diff}_1(K,M)$.
Proof: The general situation $A \in \mathrm{Diff}_1(N,M) \Longleftrightarrow$

$$Ak_0k_1 = k_0Ak_1 + k_1Ak_0 - k_0k_1A \quad (\forall k_0, k_1). \qquad (2)$$

For $N = K$ this means that

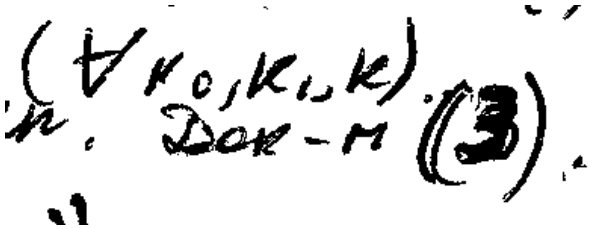
$$A(k_0k_1k) = k_0A(k_1k) + k_1A(k_0k) - k_0k_1A(k). \qquad (3)$$

Setting $k = 1$ gives (1). Conversely, let (1) hold.



Original source details: GELFAND-P000384

Manuscript notation. The manuscript abbreviates "holds" after "let (1)".

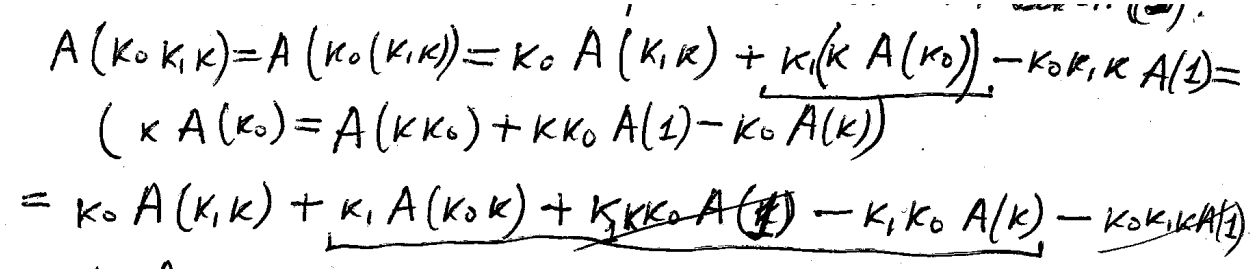


Original source details: GELFAND-P000384

Uncertain reading. The digit 3 is densely reinforced over earlier ink; an overwritten 2 is possible. At left are two close opening parentheses; at right one closing parenthesis. Above is the small insertion $(\forall k_0, k_1, k)$.

$(\forall k_0, k_1, k)$

We prove $((3)$.



Original source details: GELFAND-P000384

Uncertain reading. The whole $k_1kk_0A(1)$ is crossed out; the final argument is reinforced and also admits $A(k)$. The final term $-k_0k_1kA(1)$ is crossed out separately.

$$A(k_0k_1k) = A(k_0(k_1k)) = k_0A(k_1k)+k_1(kA(k_0))-k_0k_1kA(1) =$$

$$\big(kA(k_0) = A(kk_0) + kk_0A(1) - k_0A(k)\big)$$

$$= k_0A(k_1k)+k_1A(k_0k) + \cancel{k_1kk_0A(1)} - k_1k_0A(k)\cancel{-k_0k_1kA(1)}.$$

$$= k_0Ak_1k + k_1Ak_0k - k_0k_1Ak,$$

as required.

Corollary: $A \in \mathrm{Diff}_1(K, M) \Longrightarrow A - A(1)$ —a derivation from K to M. Lemma 3. Let $A_1 \in \mathrm{Diff}_m(M_1, M_2)$, $B \in \mathrm{Diff}_n(M_2, M_3)$.

Uncertain reading. In the hypothesis, the first A has an extra reinforced 1-like lower index. Provisionally A_1; the earlier lower sign is not established.

Original source details: GELFAND-P000384

Then $B \circ A \in \mathrm{Diff}_{m+n}(M_1, M_3)$. Proof: induction on $m+n$. For $m+n = 0 (m = n = 0)$ everything is evident. Assume it proved for $m+n \le d-1$. Let $m+n = d$.

$$\delta_k(B \circ A) = kBA - BAk = (kB - Bk)A + B(kA - Ak)$$
$$= (\delta_k B)A + B(\delta_k A) \quad \blacksquare$$

Manuscript notation. The order of the terms at the start of the line is $kBA - BAk$.

Original source details: GELFAND-P000384

GELFAND-P000385

Lemma 4. There exists a canon. diff. operator of ord. n:

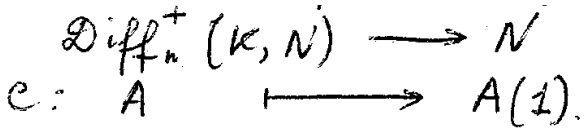


Diagram and typed description.

$$\mathrm{Diff}_n^+(K, N) \longrightarrow N$$

$$C: \quad A \longmapsto A(1)$$

Original source details: GELFAND-P000385

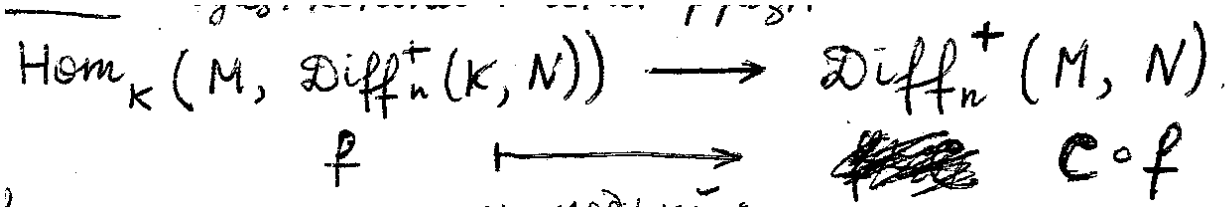
Proof.

$$(\delta_k C)(A) = (kC - Ck)(A) = kA(1) - CAk$$
$$= kA(1) - Ak(1) = (\delta_k A)(1).$$

hence

$$(\delta_{k_0}\delta_{k_1}\dots\delta_{k_n}C)(A) = (\delta_{k_0}\delta_{k_1}\dots\delta_{k_n}A)(1) = 0$$

and C is a diff. op. Consequence: a canon. homomorphism exists



Original source details: GELFAND-P000385

Diagram and typed description.

$$\mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N)) \longrightarrow \mathrm{Diff}_n^+(M,N)$$

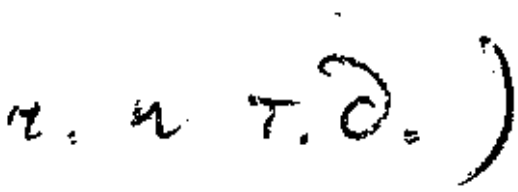
$$f \longmapsto [?] \quad C \circ f$$

Undeciphered passage. Before $C \circ f$ a formula is crossed repeatedly by horizontal and oblique strokes. Its whole earlier reading has not been recovered.

Original source details: GELFAND-P000385

(Check that this is a module hom.:

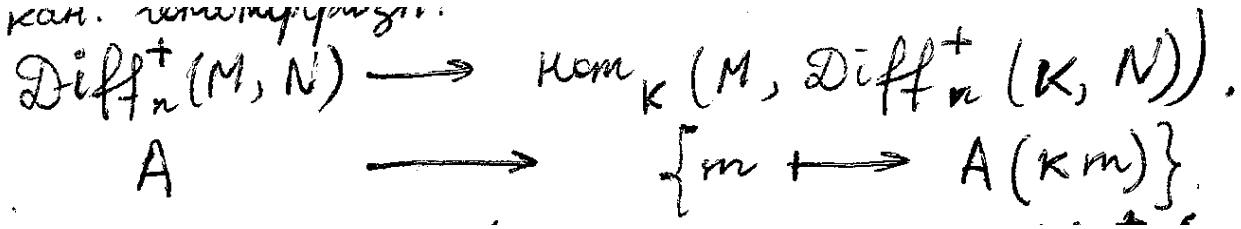
$$kf \longmapsto C \circ kf = C \circ fk = k^+(C \circ f), \quad \text{as req.})$$



Manuscript notation. The calculation ends with the customary Russian abbreviation for “as required”.

Original source details: GELFAND-P000385

Lemma 5. A canon. homomorphism exists:



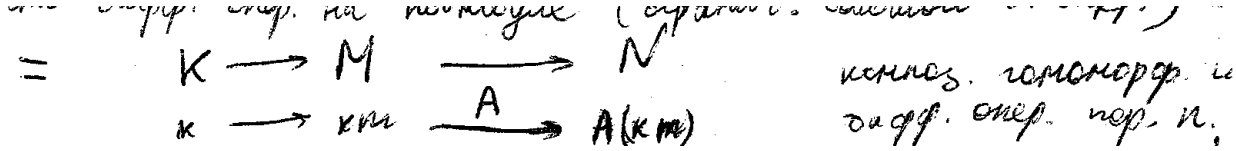
Original source details: GELFAND-P000385

Diagram and typed description.

$$\mathrm{Diff}_n^+(M,N) \longrightarrow \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$$

$$A \longrightarrow \{m \longmapsto A(km)\}$$

Proof. 1) Check that $A(km)$ (m is fixed) $\in \mathrm{Diff}_n^+(K,N)$ — — this is a diff. op. on a submodule (restrict. of the larger diff. op.) $=$



Original source details: GELFAND-P000385

Diagram and typed description.

$$K \longrightarrow M \longrightarrow N$$

$$k \longrightarrow km \xrightarrow{A} A(km)$$

canon. hom. and diff. op. of ord. n.

2) Check that $g : m \longmapsto A(km)$ is $\mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$.

Original source details: GELFAND-P000385

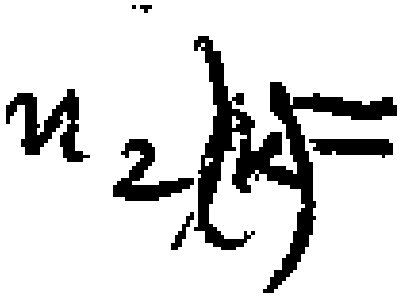
Uncertain reading. The letter before m is densely overwritten, with a loop and lower crossing strokes. Provisionally g; its earlier form is not established.

$g \ldots$

Original source details: GELFAND-P000385

Uncertain reading. Before the additivity line is a cancelled earlier prefix with a g-like beginning. Its argument strokes overlap; the complete earlier expression has not been recovered.

$$g(m_1+m_2)(k) = A(km_1+km_2) = A(km_1)+A(km_2) = g(m_1)(k)+g(m_2)(k).$$

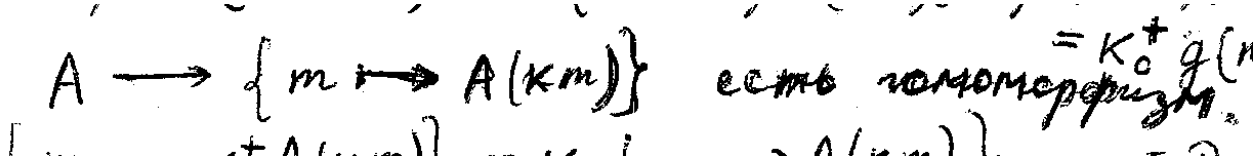


Uncertain reading. After m_2 the right argument boundary and the following (k) are densely reinforced and intersect. Provisionally $g(m_1+m_2)(k)$; possibly $g(m_1+m_2,k)$.

Original source details: GELFAND-P000385

$$g(k_0m)(k) = A(kk_0m) = A(k_0km) = (Ak_0)(km)$$
$$= (k_0^+A)(km) = k_0^+g(m).$$

3) Check that

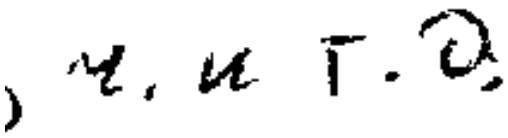


Original source details: GELFAND-P000385

Diagram and typed description.

$A \longrightarrow \{m \longmapsto A(km)\}$ is a homomorphism.

$$k^+A \longrightarrow \{m \longrightarrow k^+A(km)\} = k\{m \longrightarrow A(km)\}, \quad \text{as req.}$$



Manuscript notation. The calculation ends with the customary Russian abbreviation for “as required”.

Original source details: GELFAND-P000385

Lemma 6. The constructed maps (the consequence of lemma 4 and lemma 5) are mutually inverse and give a module isom.

$$\mathrm{Diff}_n^+(M,N) \simeq \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N)).$$

Proof. [?]

Undeciphered passage. Between “Proof” and “Let” a word is repeatedly cancelled by long horizontal strokes. The whole word has not been recovered.

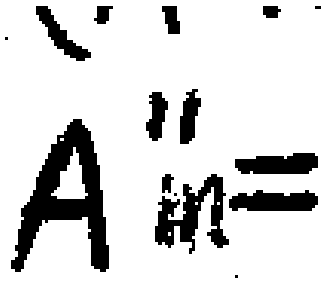
Original source details: GELFAND-P000385

Let $A \in \mathrm{Diff}_n^+(M,N)$, construct $A' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$:

$$(A'm)(k) = A(km).$$

Now from A' construct $A'' \in \mathrm{Diff}_n^+(M,N)$ by the formula

$$A''_m = C \circ A'm = (A'm)(1) = A(m), \quad \text{i.e. } A'' = A$$



Original source details: GELFAND-P000385

Uncertain reading. Below the first A'' is a small reinforced m-like entry. Provisionally a lower m; the earlier strokes are not established. The following $A'm$ is rendered without adding a lower index.

Conversely, let $A' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$. Construct $A'' \in \mathrm{Diff}_n^+(M,N)$

Original source details: GELFAND-P000385

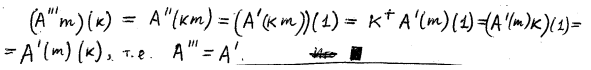
Uncertain reading. In the final $\mathrm{Diff}_n^+(M,N)$ the first argument is densely reinforced. Provisionally M; the complete earlier form underneath has not been recovered.

GELFAND-P000386

[source number: 35]

by the formula $A''m = (A'm)(1)$ and then $A''' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$ as

$$(A'''m)(k) = A''(km) = (A'(km))(1) = k^+A'(m)(1) = (A'(m)k)(1) = \\ = A'(m)(k), \quad \text{i.e. } A''' = A'.$$



Original source details: GELFAND-P000386

Manuscript notation. The equals sign is written at the end of the first line and again at the start of the second.

[?] ■

Original source details: GELFAND-P000386

Undeciphered passage. After $A''' = A'$ and before the black square, a short word is cancelled by several horizontal strokes. The whole earlier word has not been recovered.

Thus, it is proved that

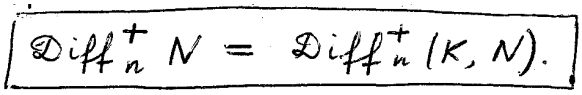


Diagram and typed description.
$\mathrm{Diff}_n^+ N = \mathrm{Diff}_n^+(K, N).$

Original source details: GELFAND-P000386

GELFAND-P000387

15/XI. I. M. Gelfand.

G is a Lie group, X a sp., G acts transitively on X (X is a homogeneous sp.).

Principal homog. sp.: G acts on X without fixed pts. ($xg = y \Rightarrow g$ is unique).

The same in terms of the Lie algebra:

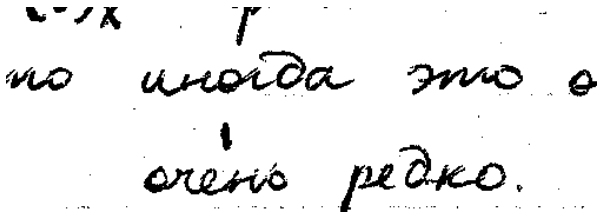
Let a Lie algebra $\mathfrak{g}$ act on X.

$$\xi \in \mathfrak{g} \rightsquigarrow \text{a vector field on } X; \quad a(\xi),$$
$$\xi_1 + \xi_2 \rightsquigarrow a(\xi_1) + a(\xi_2);$$
$$[\xi_1, \xi_2] \longrightarrow [a(\xi_1), a(\xi_2)].$$

Homogeneous sp.: if $\mathfrak{g} \longrightarrow TX_x$ epim. $\forall x$ ($a(\xi)_x$ generate all of X).

(One can prove that sometimes this means ordinary homogeneity)

very rarely.



Manuscript notation. A vertical stroke below "sometimes" leads to the annotation "very rarely".

Original source details: GELFAND-P000387

When can one extend[?] to a group:

Uncertain reading. The final cursive stroke is provisionally read as п in «доп.», probably short for "extend".

Original source details: GELFAND-P000387

when $\mathfrak{g}$ is finite-dimensional.

Principal homog. sp.: map $\mathfrak{g} \longrightarrow TX_x$ isomorphism.

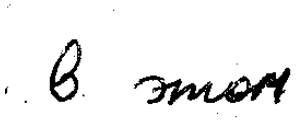
Ordinary case:

fix x_0, one gets $x_0 g = y$. $g \leftrightarrow y$, $X \cong G$ and the transf. is a right translation.

If there is no group, there may be many of them and different ones!

(All of them are candidates for the title of group)

Principal homog. sp. is now understood in this[?] sense.

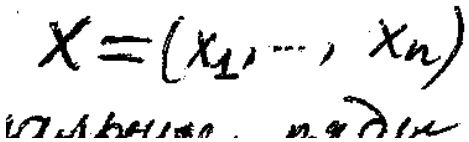


Uncertain reading. In the word before "sense," the final letters are densely reinforced and intersect earlier strokes. Provisionally "этом"(this); the whole earlier form has not been recovered.

Original source details: GELFAND-P000387

Example.
W_n — formal vect. fields;

$$\sum_{i=1}^{n} a_i(x)\frac{\partial}{\partial x_i}$$
$$X = (x_1, \ldots, x_n)$$



Original source details: GELFAND-P000387

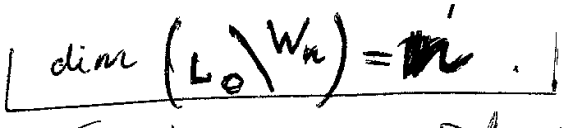
Uncertain reading. The initial cross-shaped sign on the coordinate line is noticeably larger than the components x_i. Provisionally X; possibly a large handwritten x. The case distinction is not finally established.

$a_i(x)$ — formal series (power series).

GELFAND-P000388

[source number: 36]

The group does not exist: one cannot move.
In W_n there is a subalgebra $L_0 \subset W_n$, $a_i(0) = 0$ — const. term $= 0$.



Original source details: GELFAND-P000388

Diagram and typed description.

$$\dim(L_0 \backslash W_n) = n.$$

Original source details: GELFAND-P000388

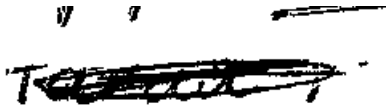
Uncertain reading. After the equals sign, the letter is densely reinforced, especially at left. Provisionally n; the complete earlier strokes have not been recovered.

The algebra L_0 can be brought to a group.

Original source details: GELFAND-P000388

Diagram and typed description. A circled mark: two roughly parallel oblique strokes and a dark spot in the middle. Its number or symbol is not established.

$y = \varphi(x)$ — formal automorphisms, 0 stays in place.
Its Lie alg. ~~then[?]~~ is L_0.



Original source details: GELFAND-P000388

Uncertain reading. Between “Its Lie alg.” and “is L_0” a word is cancelled by several strokes. Provisionally “тогда”(then); the complete earlier reading is not finally established.

Consider W_n, its gr. of automorphisms, its Lie alg. is L_0.
Thm: the Lie alg. of the automorphism gr. of W_n is L_0.

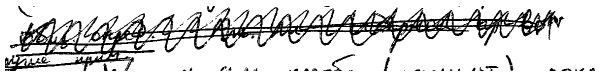
(In the classical case one gets the quotient gr. ~~[?]~~

Original source details: GELFAND-P000388

Undeciphered passage. Between “quotient gr.” and “alg.” is a short densely cancelled fragment. Its whole content has not been recovered.

of the alg. by the centre).

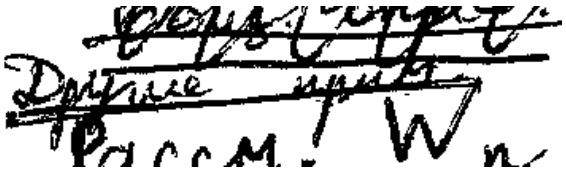
~~[?]~~



Original source details: GELFAND-P000388

Undeciphered passage. A long line above the lower cancelled heading is covered by repeated loops and horizontal strokes. The complete earlier line has not been recovered.

~~Other ex.~~[?]



Original source details: GELFAND-P000388

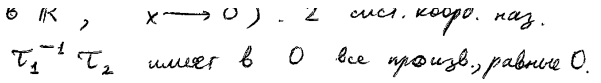
Uncertain reading. After “Other”, the abbreviation is provisionally «прим.» (ex.). The descending p-loop and adjacent strokes are visible. A cancellation line crosses some letters.

Consider W_n and sympl. transf. (Hamiltonian), volume-preserving and so on.

Construction of a principal homog. sp. for W_n.

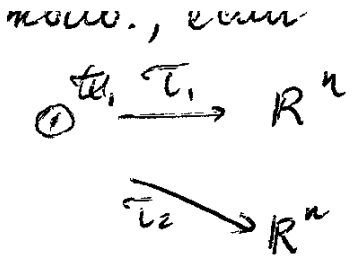
M — a sp. (arbitrary) — a finite-dimensional manifold, $x \in M$.

Consider at pt. x jets of coordinate systems (a coord. syst. is a map of a nbd. U_x into R^n, $x \longrightarrow 0$). 2 coord. systs. are called equiv. if $\tau_1^{-1}\tau_2$ has at 0 all derivs. equal to 0.



Original source details: GELFAND-P000388

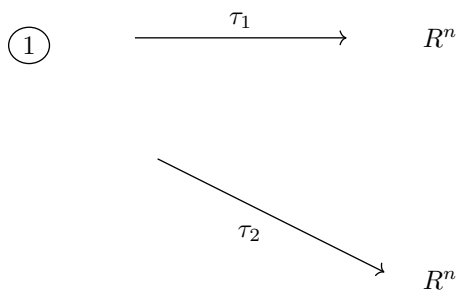
Manuscript notation. The manuscript has $\tau_1^{-1}\tau_2$ and “all derivs. equal to 0”.



Original source details: GELFAND-P000388

Diagram and typed description. Two separate arrow starts: the upper arrow points right with label τ_1 to R^n; the lower arrow slopes down and right with label τ_2 to a second R^n. Above the upper start is provisionally U_1, with a circled digit 1 below. A cursive stroke crosses the upper letter; no connection between starts is supplied.

U_1[?]



$\xi \in W_n$ acts there.

$\xi\tau(x)$ — another formal coord. syst.

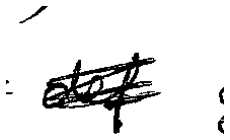
Original source details: GELFAND-P000388

Diagram and typed description. An earlier cancelled scheme: provisionally $\tau(x)$, then A, a rightward arrow and R^n. Long loops and oblique strokes cross the labels and arrow. The exact earlier beginning has not been recovered.

$X = S(M)$ — the sp. of all jets of coord. systems (at all points of M).

Theorem: $S(M)$ — a principal homog. sp. for W_n.

dim M = ~~def[?]~~ defect W_n (by how much it falls short).



Original source details: GELFAND-P000388

Uncertain reading. After dim $M =$ a short earlier beginning is cancelled by horizontal and oblique strokes. Provisionally “деф”(def); the complete earlier form has not been recovered.

Original source details: GELFAND-P000388

Uncertain reading. In the current word “дефект”(defect), the middle letters are densely overwritten. The word itself is legible; all earlier strokes underneath are not established.

It is unclear whether there are other substantive principal homog. spaces.

GELFAND-P000389

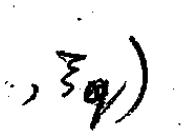
Let G act on X, ω — an invar. diff. form.

It suffices to specify it at the 1st point. Right-inv. forms correspond to forms on G.
Differential = diff. in the cohomology of the Lie algebra $\mathfrak{g}$.

W_n — a Lie alg., $X = S(M)$.

$f(\xi_1, \dots, \xi_q)$ — a multilin. skew-symm. form on $W_n \rightsquigarrow$

$\rightsquigarrow$ an invar. form on $S(M)$.

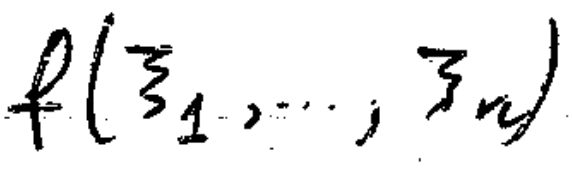


Original source details: GELFAND-P000389

Uncertain reading. The subscript of the last ξ in the first f is densely reinforced and crossed by earlier strokes. Provisionally q; possibly n. The whole earlier subscript form has not been recovered.

$x, D_1, \dots, D_q$ — vectors at the point $x \in X$.

corresponding to them are vectors $\xi_1, \dots, \xi_q \in W_n$. One gets $f(\xi_1, \dots, \xi_n)$ — a form on $S(M)$.



Original source details: GELFAND-P000389

Manuscript notation. The last subscript here is n. The earlier line provisionally reads q.

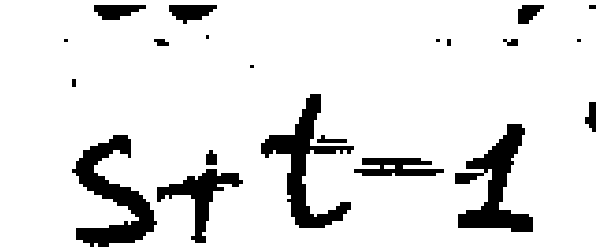
Right-invar. one can give a definition.

Original source details: GELFAND-P000389

Uncertain reading. Several initial letters in “Right-invar.” are densely overwritten. The current reading is provisionally retained; the whole earlier strokes are not established.

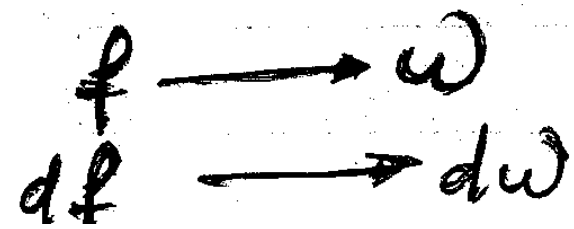
$$(df)(\xi_1,\dots,\xi_{q+1}) = \sum_{s<t}(-1)^{s+t-1}$$

$$f([\xi_s,\xi_t],\xi_1,\dots,\hat{\xi}_s,\dots,\hat{\xi}_t,\dots,\xi_{q+1}).$$



Original source details: GELFAND-P000389

Uncertain reading. The horizontal strokes nearly merge. They are provisionally read as a single retraced minus sign.



Original source details: GELFAND-P000389

Diagram and typed description.

$f \longrightarrow \omega$

$df \longrightarrow d\omega$

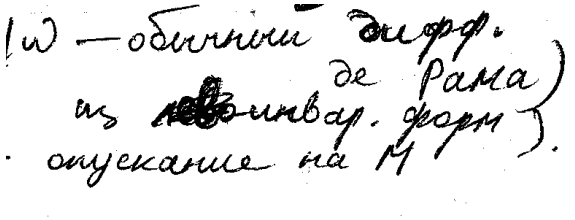
($d\omega$ — the ordinary de Rham diff.

from ~~left[?]~~ invar. forms

Tricks of diff. geometry — lowering to M.

Original source details: GELFAND-P000389

Uncertain reading. Before “invar. forms” is a cancelled joined beginning with a tall loop. Provisionally “лево”(left); the whole earlier entry is not established.



Original source details: GELFAND-P000389

Uncertain reading. At right are two joined curves: the upper is beside the de Rham line; the lower is shorter, beside the invar. forms and lowering lines. Below and right is a separate period. The exact semantic attachment of the curves is not established.

(classes and so on).

see note in DAN, I. M. Gelfand and Kazhdan.

I. Bernstein.

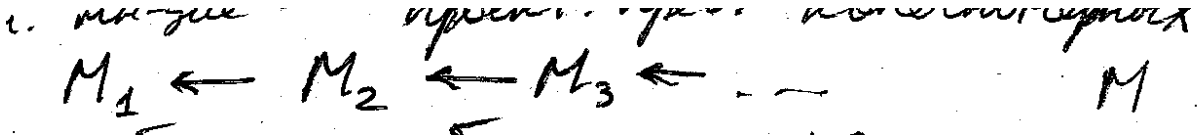
Diagram and typed description. At right in the source raster is a digital yellow comment icon with a black outline, a lower tail and three horizontal bars.

Original source details: GELFAND-P000389

Consider ∞-dimensional manifolds.

Definition:

An ∞-dim. manifold is a proj. limit of finite-dimensional ones

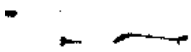


Original source details: GELFAND-P000389

Diagram and typed description.

$M_1 \longleftarrow M_2 \longleftarrow M_3 \longleftarrow \cdots \quad M$

Uncertain reading. After the last arrow are a short point-like stroke and a separate low wavy stroke. The exact continuation mark is not established; it is not normalized to three dots.



Original source details: GELFAND-P000389

All maps are onto maps.

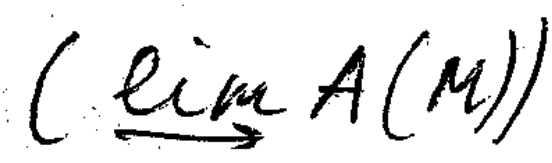
Ring of functions on M: ~~A[?]~~

Uncertain reading. After the ring-of-functions heading, a single letter-like sign is cancelled. Provisionally A; the whole earlier form is not established.

Original source details: GELFAND-P000389

$$A(M) = \bigcup A(M_i) \qquad (\varinjlim A(M))$$

Manuscript notation. A right arrow is written below lim. Its argument is M, with no subscript.

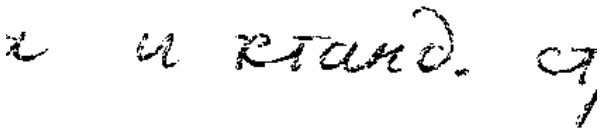


Original source details: GELFAND-P000389

GELFAND-P000390

[source number: 37]

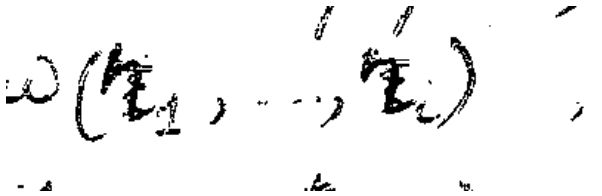
C^∞ — a vector field, C^∞ — a diff. form, and in the std.[?] way one constructs the de Rham compl.



Original source details: GELFAND-P000390

Uncertain reading. The ending of the abbreviation matches the «станд.» (std.) written lower on the page. The initial stroke is irregular, so this reading remains provisional.

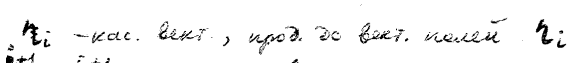
ω — a form; $\omega(\xi_1, \ldots, \xi_i)$,



Original source details: GELFAND-P000390

Uncertain reading. Two argument letters are reinforced over earlier strokes. Provisionally $\xi_1, \ldots, \xi_i$; possibly $\eta_1, \ldots, \eta_i$. The complete earlier outlines have not been recovered.

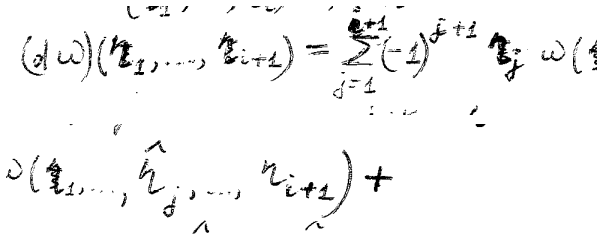
ξ_i — a tangent vect., extended to vector fields η_i.



Original source details: GELFAND-P000390

Uncertain reading. The initial letter with subscript i in the tangent-vector phrase is reinforced. Provisionally ξ_i; possibly η_i. The clean final η_i is rendered separately; the earlier strokes of the initial letter are not established.

$$(d\omega)(\xi_1, \ldots, \xi_{i+1}) = \sum_{j=1}^{q+1} (-1)^{j+1} \xi_j \omega(\xi_1, \ldots, \hat{\eta}_j, \ldots, \eta_{i+1}) + \\ + \sum_{j<k} (-1)^{j+k} \omega([\eta_j, \eta_k], \eta_1, \ldots, \hat{\eta}_j, \ldots, \hat{\eta}_k, \ldots, \eta_{i+1}). \qquad (1)$$



Original source details: GELFAND-P000390

Uncertain reading. In the first row of (1), the two letters at left, the upper summation limit, the letter before ω and the first argument of ω are reinforced. Provisionally $\xi_1, \ldots, \xi_{i+1}$ at left, $q+1$ above, then ξ_j and ξ_1; the corresponding η and $i+1$ are possible. The clean η entries in the tail and second row are retained. The complete earlier strokes have not been recovered.

Using, $S(M)$ we shall construct the Chern classes of M.

$\mathfrak{g}$ — an inf.-dimensional Lie alg. (proj. lim of finite-dimensional ones with the corresp. topol.).

X — a principal homog. sp.

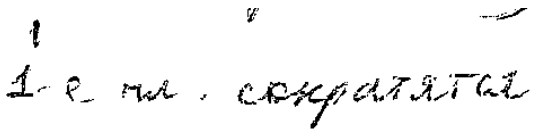
Construct the complex $C(\mathfrak{g})$

$C(\mathfrak{g}) = \{C^i(\mathfrak{g})\}$. — the standard complex of the algebra $\mathfrak{g}$.

$C(\mathfrak{g}) \longrightarrow \Omega(X)$ — the de Rham compl. of X.

A homomorphism (commuting with d) — by formula (1).

The 1st terms will cancel



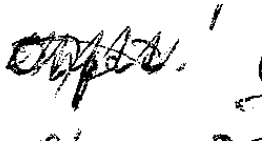
Original source details: GELFAND-P000390

Manuscript notation. The note under the short vertical stroke at right reads “The 1st terms will cancel”.

One gets a map.

$$H^i(\mathfrak{g}) \longrightarrow H^i(X)$$

How does one constr. Chern classes?
~~Def[?]...~~

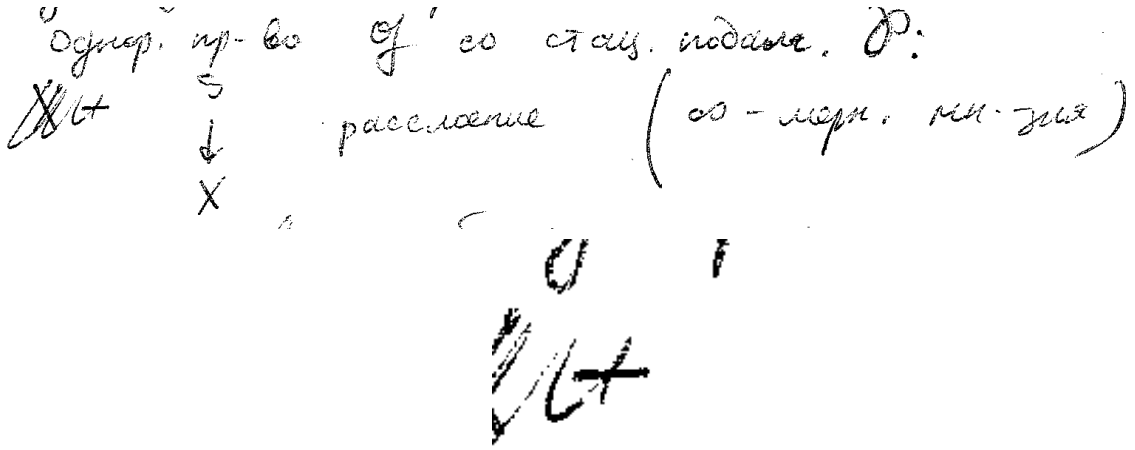


Original source details: GELFAND-P000390

Uncertain reading. Before the new heading is a cancelled earlier beginning. Provisionally “Опр...”(Def...); possibly “Однор...”(Homog...). The ending has not been recovered or completed from the following heading.

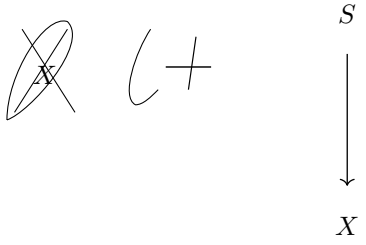
Definition.

$\mathfrak{g} \supset \mathfrak{p}$ — a subalgebra.
A homog. sp. of $\mathfrak{g}$ with isotropy subalg. $\mathfrak{p}$.



Original source details: GELFAND-P000390

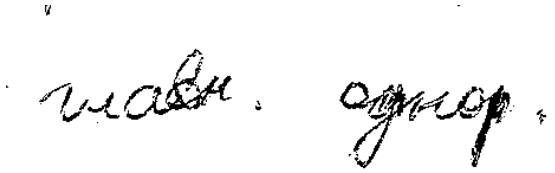
Diagram and typed description. To the left of the current vertical projection from S to X is a cancelled earlier X; beside it are an open hook-like stroke and a separate upper-right crossing resembling a plus. The exact earlier sign is not established. ι^+-, τ^+-or t^+-like forms are possible; the earlier $\mathfrak{g}/\mathfrak{p}$ hypothesis is not confirmed by a clear fraction bar.



a fibration (∞-dim. manifolds)

On tangent spaces the maps are onto

S — a princ. homog. $\mathfrak{g}$-sp. and if $s \in S$,



Uncertain reading. In the abbreviations “главн.”(princ.) and “однор.”(homog.), final and middle letters are overwritten and crossed by earlier strokes. The current reading is provisionally retained; the complete earlier outlines are not established.

Original source details: GELFAND-P000390

GELFAND-P000391

$\mathcal{F}_s$ — the fibre of S at pt. s, then $\xi \in \mathfrak{p}$ passes to vector fields, tangent to the fibre, and at the same time $\mathcal{F}_s$ — a principal homogeneous $\mathfrak{p}$-space.

Uncertain reading. The letter before “-space” is densely overwritten. Provisionally $\mathfrak{p}$; possibly $\mathfrak{g}$. The complete original outline is not established.

Original source details: GELFAND-P000391

X — a principal homog. sp.

Chain complex $C(\mathfrak{g})$

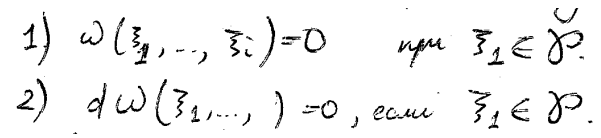
Rel. chain compl. $C(\mathfrak{g}, \mathfrak{p})$.

$$C^i(\mathfrak{g}, \mathfrak{p}) \subset C^i(\mathfrak{g})$$

$\omega(\xi_1, \ldots, \xi_i)$ — a form on $\mathfrak{g}$, belonging to $C^i(\mathfrak{g}, \mathfrak{p})$, if

1) $\omega(\xi_1, \ldots, \xi_i) = 0$ for $\xi_1 \in \mathfrak{p}$.

2) $d\omega(\xi_1, \ldots,) = 0$, if $\xi_1 \in \mathfrak{p}$.



Manuscript notation. In the second condition a comma follows the ellipsis just before the closing parenthesis.

Original source details: GELFAND-P000391

(ω invar. with respect to ~~elem[?]~~ the adjoint repr. of elem. of $\mathfrak{p}$).

Uncertain reading. The cancelled word after “invar. with respect to” is provisionally “элем”(elem.). The full earlier reading is not established.

Original source details: GELFAND-P000391

The forms ω — a complex.

Cohomology — cohom. of the quotient gr. (in the classical case).

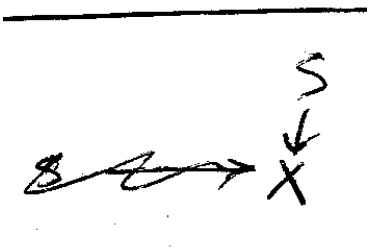
Let there be a homog. $\mathfrak{g}$-space, $\mathfrak{p}$ — an isotropy subalg.

$\exists$ a map

$$C(\mathfrak{g}, \mathfrak{p}) \longrightarrow \Omega(X).$$

Construction:

$\eta \in T_x(X)$,



Original source details: GELFAND-P000391

Diagram and typed description. The current vertical projection is from S down to X. At left is an earlier horizontal right arrow to X with its beginning and small label crossed out. The beginning resembles s; the complete earlier letter and label are not established.

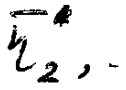
$\eta' \in T_s(S)$, $\eta' \longrightarrow \eta$.

Original source details: GELFAND-P000391

Uncertain reading. The letter before $\in T_s(S)$ is densely overwritten; the separate upper-right stroke is read as a prime. Provisionally η'; possibly ξ'. The original letter is not established.

$\omega \in C(\mathfrak{g}, \mathfrak{p}) \longrightarrow$

$$\overrightarrow{\tilde{\omega}}(\eta_1, \dots, \eta_n) = \omega(\bar{\eta}'_1, \bar{\eta}'_2, \dots, \bar{\eta}'_n)$$



Original source details: GELFAND-P000391

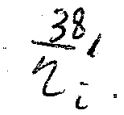
Uncertain reading. Above the second argument $\bar{\eta}_2$ is a dense crossing with a loop-like stroke. The prime in $\bar{\eta}'_2$ is provisional; $\bar{\eta}^*_2$ or a reinforced or overwritten upper stroke is possible. The exact upper mark is not established.

$\bar{\eta}'_i$ — the corresp. elements of the Lie alg. $\mathfrak{g}$.

GELFAND-P000392

[source number: 38]

There is arbitrariness, but $\tilde{\omega}$ does not depend on it — on the choice of $\bar{\eta}'_i$.

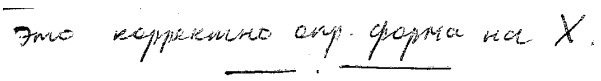


Original source details: GELFAND-P000392

Uncertain reading. The horizontal stroke immediately below page number 38 lies above η'_i. Provisionally $\bar{\eta}'_i$; possibly η'_i with only page number 38 underlined. The attachment of the horizontal stroke is not established.

It does not depend on the choice of point either.

This is a well-def. form on X.



Original source details: GELFAND-P000392

Manuscript notation. The line reads “a well-def. form on X”, in the singular.

M — a manifold, $S(M)$ — a princ. W_n-space (see above).

gl(n) — a subalg. of linear transformations

It acts on $S(M)$ and the action extends to the group

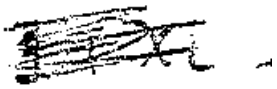
$V_x \longrightarrow V_0 \subset \mathbb{R}^n$, rotates.

Consider $S(M)/GL(n, R)$ corresp. to the pair $(W_n, \text{gl}(n))$.

This is already homotopy equiv. to M.

(in contrast to $S(M)$, which is homotopy equiv. to 1-frames).

~~[...]~~

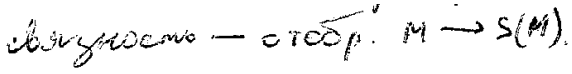


Original source details: GELFAND-P000392

Undeciphered passage. The short cancelled word before “There is a map” has not been recovered.

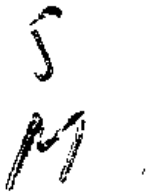
There is a map.

Connection[?] — a map $M \longrightarrow S(M)$.



Original source details: GELFAND-P000392

Uncertain reading. The word provisionally reads “Connection”, as in the lower passage. The written map is $M \longrightarrow S(M)$.



Original source details: GELFAND-P000392

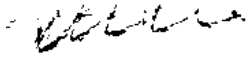
Diagram and typed description. Below the quotient is a separate curved vertical stroke and the label M. In the right margin above the formula are two vertical strokes, one with an upward arrow. Their position is retained; the mark below the quotient is not replaced by an equality assertion.

$$C(W_n, \mathfrak{gl}(n)) \longrightarrow \Omega(S(M)/GL(n,R))$$

M

Description of the cohom. of this alg:

$\psi_1, \psi_2, \ldots, \psi_n$[...ymi?] — generators of degree $2n$.

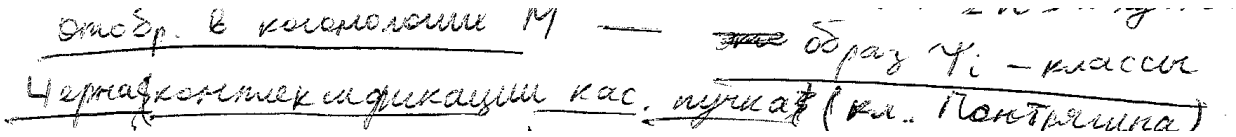


Original source details: GELFAND-P000392

Uncertain reading. After ψ_n are faint letter strokes. The earlier provisional ending “-ыми”(-ymi) is qualified; the whole word is not established.

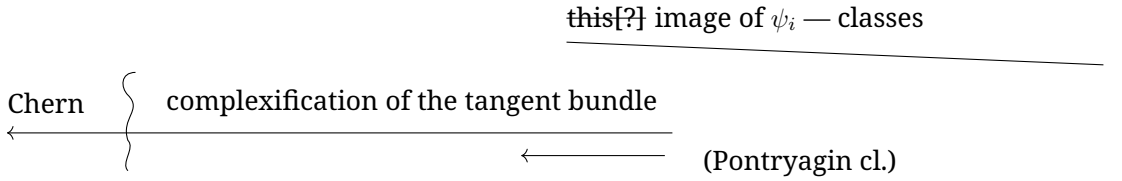
degrees $> 2n \Rightarrow$ one gets 0.

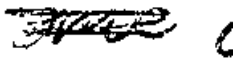
Map to the cohomology of M —



Original source details: GELFAND-P000392

Diagram and typed description. The line reads “Chern [classes] of the complexification of the tangent bundle”. A brace-like mark and arrowed lines link the parts of the entry.





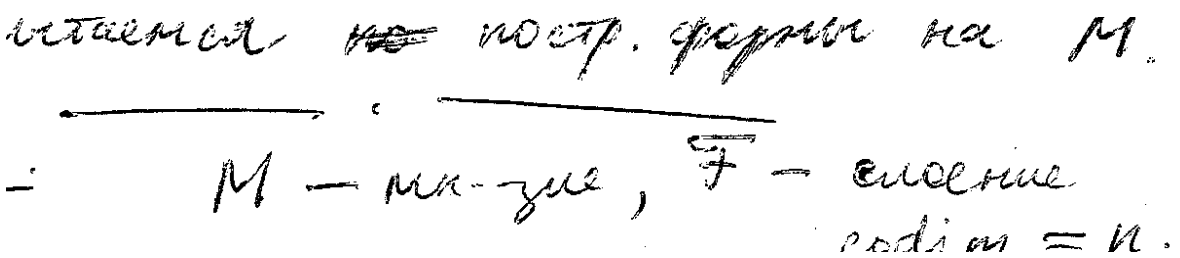
Uncertain reading. Before “image of ψ_i — classes” is a cancelled word, provisionally “это”(this). The full earlier reading is not established.

Original source details: GELFAND-P000392

A form on $S(M)/GL(n,R)$ is det. uniquely (invar.), nonuniqueness (choice of connection) appears, when we try ~~not~~[?] to constr. forms on M.

Uncertain reading. Before “constr.” is a short cancelled beginning, provisionally “не”(not). The crossings prevent recovery of the complete original reading.

Original source details: GELFAND-P000392



Original source details: GELFAND-P000392

Diagram and typed description. Above the following heading and line are a separate horizontal arrow with heads at both ends, a small open c-like stroke and a separate long sloping stroke above the foliation sign. Their meanings are not established.

Bott thm. M — a manifold, $\mathcal{F}$ — a foliation, codim $= n$.

GELFAND-P000393

$S_{\mathcal{F}}(M)$. $x \in M$, $U_x \xrightarrow{\text{form.}} U_0 \subset \mathbb{R}^n$, having the properties

1) const. on the leaves; 2) an isom. on a transverse sect. (nondeg.)

(A formal germ of a map).

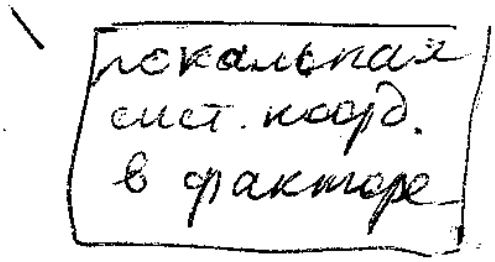


Diagram and typed description. The box reads “local coord. syst. in the quotient”. There is a short oblique stroke at left.

local
coord. syst.
in the quotient.

Original source details: GELFAND-P000393

Consider $S_{\mathcal{F}}(M)/GL(n,R)$.

On $S_{\mathcal{F}}(M)$ no algebra acts.

Consider ~~dim $M \longrightarrow N$~~ $\mathbb{R}^N_n$, in it a standard foliation of codim n.

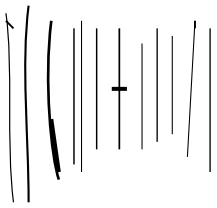
Uncertain reading. After the cancelled dim $M \longrightarrow N$ stands $\mathbb{R}^N$ with a small n lower right. The provisional positional notation is $\mathbb{R}^N_n$; the n may be separate. Its role is not established.

Original source details: GELFAND-P000393

$W_{N,n}$ — the Lie alg. of vector fields preserving the foliation

Diagram and typed description. A row of nearly vertical strokes of different lengths and thicknesses, with dark thickenings. Intertwined reinforced strokes are rendered schematically. The exact number of separate lines and the attachment of the thickenings are not established; no points or particular leaves are assigned.

Original source details: GELFAND-P000393



Linear transf.: :

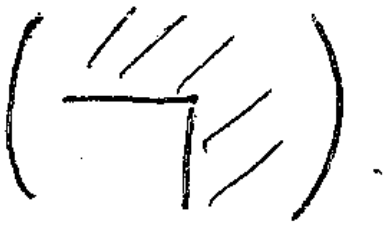
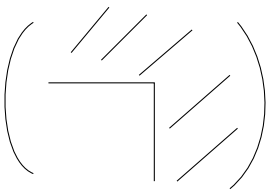


Diagram and typed description.



Original source details: GELFAND-P000393

$S'_{\mathcal{F}}(M)$, $x \in M$, $U_x \overset{\text{form.}}{\longrightarrow} U_0 \subset \mathbb{R}^N$

leaves of the foliation $\longrightarrow$ leaves of the form. foliation.

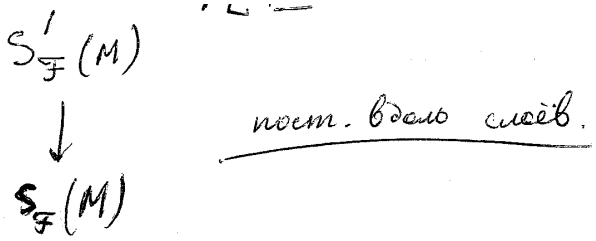
$S'_{\mathcal{F}}(M)$ — a princ. homog. $W_{N,n}$-space. !

$W_{N,n} \supset L$ — preserving each leaf

Original source details: GELFAND-P000393

Diagram and typed description. Between the quotient and W_n are two reinforced wavy upper strokes and a lower nearly straight stroke. The form is rendered as separate strokes; a standard relation is not imposed.

$$W_{N,n}/L \approx W_n.$$



Original source details: GELFAND-P000393

Diagram and typed description.

$$\begin{array}{c} S'_{\mathcal{F}}(M) \\ \downarrow \\ S_{\mathcal{F}}(M) \end{array}$$

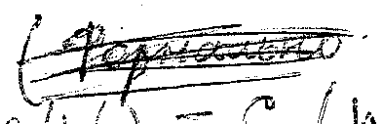
const. along the leaves.

The pair $(S'_{\mathcal{F}}(M), S_{\mathcal{F}}(M))$ — a homogeneous $(W_{N,n}, L)$-space.

GELFAND-P000394

[source number: 39]

~~(Formally[?]~~



Original source details: GELFAND-P000394

Uncertain reading. At the beginning an opening parenthesis and a word are cancelled, provisionally “Формально”(Formally). The complete earlier reading and superimposed strokes are not established.

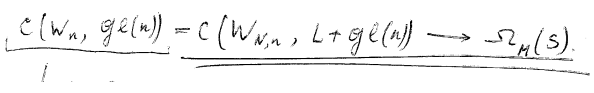
(the quotient group does not act on the quotient space)

$$C(W_n) = C(W_{N,n}, L) \longrightarrow \Omega(S_{\mathcal{F}}(M))$$

$S = S_{\mathcal{F}}(M)/GL(n, R)$ — homotopy equiv. to M.

$(S'_{\mathcal{F}}(M), S)$ — a homog. princ. sp. relative to $(W_{N,n}, L + \mathrm{gl}(n))$

$$C(W_n, \mathrm{gl}(n)) = C(W_{N,n}, L + \mathrm{gl}(n)) \longrightarrow \Omega_M(S).$$



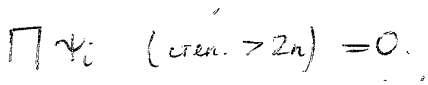
Original source details: GELFAND-P000394

Manuscript notation. The right side of the formula is doubly underlined. A short separate stroke lies below the left side.

these are known (generators) $\psi_1, \ldots, \psi_n$ — Izv. A. N., 1971, Gelfand, Fuks.

$T(M)/T(\mathcal{F})$ — Chern classes — images of the cl. $\psi_1, \dots, \psi_n$.

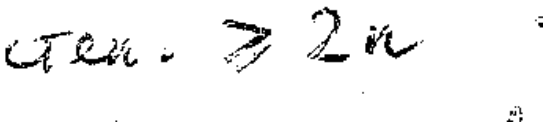
$$\prod \psi_i \quad (\text{deg. } > 2n) = 0.$$



Original source details: GELFAND-P000394

Manuscript notation. The displayed formula has deg. $> 2n$.

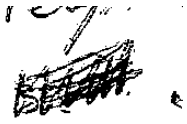
Hence for $T(M)/T(\mathcal{F})$ a product of Chern cl. deg. $\geqslant 2n$ must be 0!! — this is Bott's thm. —



Original source details: GELFAND-P000394

Uncertain reading. Below $>$ is a separate sloping stroke, provisionally rendered as $\geqslant$. The preceding display has $>$ without that stroke.

— an obstruction to being homotopically integrable [...]



Original source details: GELFAND-P000394

Undeciphered passage. The densely cancelled word after "integrable" has not been recovered.

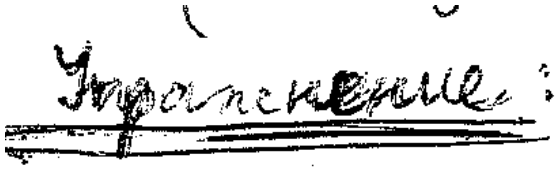
. . .

Arnold.

$$\frac{dw}{dz} = \frac{P(z,w)}{Q(z,w)} \quad \text{on } CP^2 \setminus \text{several singular points.}$$

One gets an invariant of the foliation. (on 3-dimensional complex manifold: point — vector)

Arnold's question — characterize geometrically, prove integrality (rationality) (a generalization of Poincaré rotation numbers).



Original source details: GELFAND-P000394

Uncertain reading. The current "Exercise" is reinforced over earlier strokes. The complete earlier outlines have not been recovered. The two lines beneath the heading with left ends are rendered separately.

Exercise:

define rotation numbers as such an invariant.

GELFAND-P000395

[source number: 43]

14 February.

L. Vaserstein.

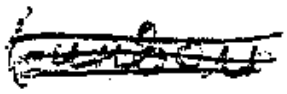
A — an assoc. ring with 1. (not necessarily commut.). V — a module over A. (right).

A quadr. form over a comm. ring — in A^n — is $A^n \longrightarrow A$, written in the form $\sum a_{ij}x_ix_j$. The matrices are det. nonuniquely.

$\exists\, 2^{-1} \Rightarrow$ there is a one-to-one corresp. with symm. matrices.

——— · ———

An involution on A: $A \longrightarrow A$.

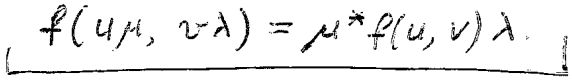


Original source details: GELFAND-P000395

Undeciphered passage. Below the arrow a short word is cancelled by horizontal and slanting strokes. Separate loops are visible; the full reading is not established.

$$(x-y)^* = x^* - y^*; \qquad (xy)^* = y^*x^*; \qquad (x^*)^* = x.$$

A sesquilin. form: $f(u,v) \in A$, $u, v \in V$; additive in all variables;



Original source details: GELFAND-P000395

Diagram and typed description.

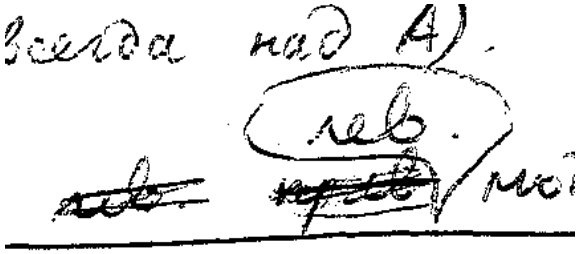
$f(u\mu, v\lambda) = \mu^* f(u,v)\lambda.$

(see Bourbaki). (There are no bilin. forms over a noncomm. ring)

$\mathrm{Hom}(V, V^*)$ — these are $1,5$-lin. forms.

$V^* = \mathrm{Hom}_A(V, A)$ (of two right modules).

(Hom is always over A).



Original source details: GELFAND-P000395

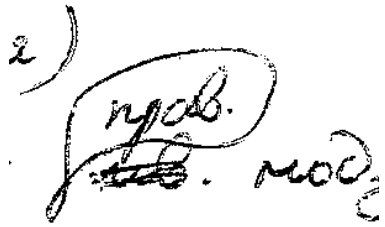
Uncertain reading. Two earlier abbreviations are cancelled. Provisionally “left.; right.”; the complete earlier strokes are not established. Above the second, “left.” is clearly inserted and circled; its pointer crosses the earlier word.

left.

V^* ~~left.[?]~~ ~~right.[?]~~ module.

(generally speaking).

But the involution makes it poss. to consider V^* as



Original source details: GELFAND-P000395

Uncertain reading. Before “module” an earlier abbreviation is cancelled, provisionally “left.”. The complete earlier strokes are not established. Above it “right.” is clearly inserted and circled, with a pointer down and left.

right.

~~left.[?]~~ module.

$$a \longmapsto f(a); \qquad \lambda f(a); \qquad a \longmapsto \lambda f(a).$$

$L \in \mathrm{Hom}(V, V^*)$ (a hom. of right A-modules).

$$f(u, v) = L(u)v.$$

GELFAND-P000396

A quadr. form — not quite that.

Undeciphered passage. At left before the ring line a word is densely cancelled. A large initial stroke and separate loops are visible; the complete word is not established.

Original source details: GELFAND-P000396

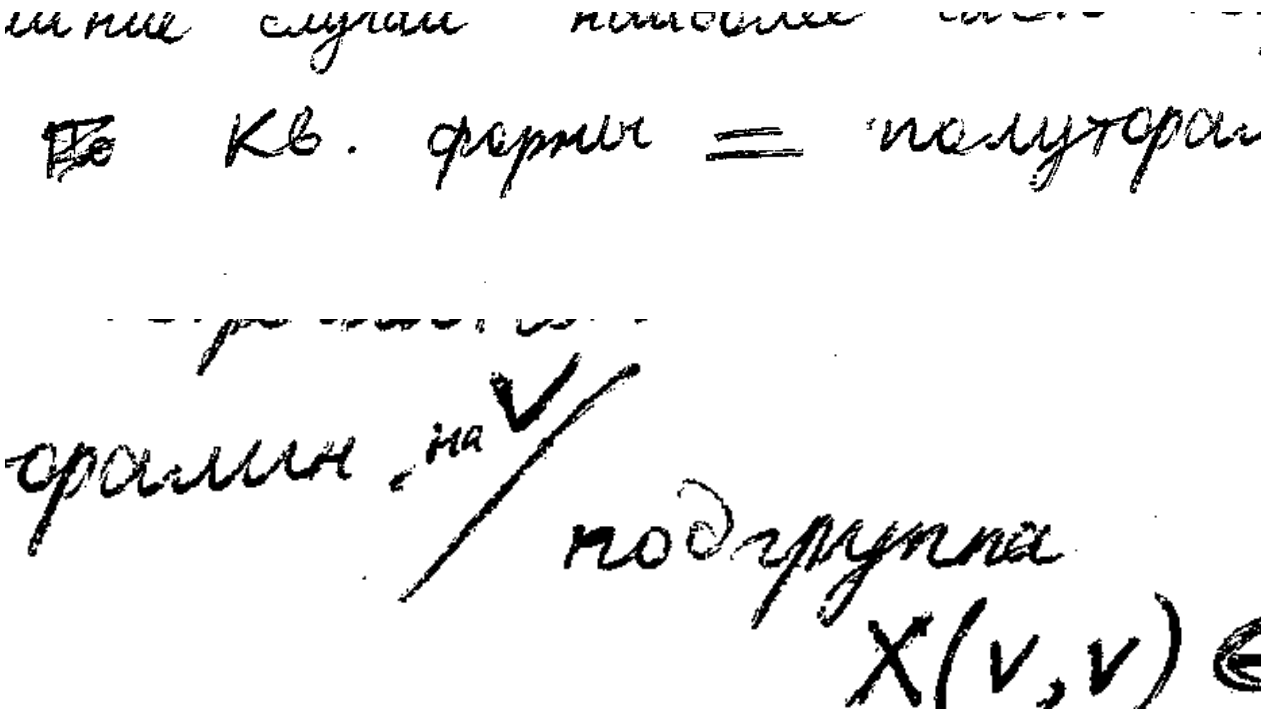
A — a ring, $*$ — an invol., $\varepsilon \in$ the centre of A, $\varepsilon \cdot \varepsilon^* = 1$. ($\varepsilon$ — the symmetry of the quadr. form). (most often one encounters $\varepsilon = 1$ or -1).
Λ — a subgroup of the additive gr. of the ring, having the properties:

$$\{x - \varepsilon x^*\}_{x \in A} \subset \Lambda \subset \{x = -\varepsilon x^*\}_{x \in A};$$
$$a^* \lambda a \in \Lambda \quad \text{for } \lambda \in \Lambda \text{ and } a \in A.$$

Original source details: GELFAND-P000396

Uncertain reading. At right below “properties” are a separate open bracket-like stroke and a small looped sign. “э” or “д” is possible; no verbal reading is established.

The extreme cases occur most often.



Original source details: GELFAND-P000396

Diagram and typed description. The slanting stroke beside "on V" descends to the left towards the conditions on X.

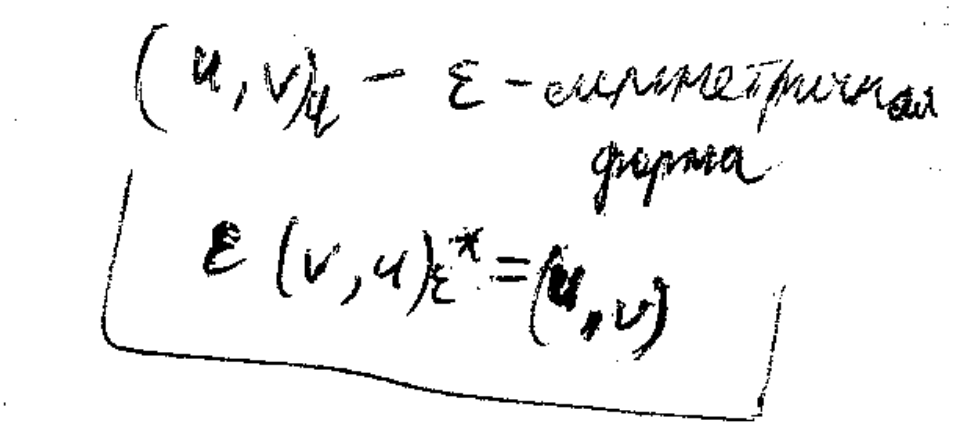
~~I.e.~~ Quadr. forms $\equiv$ sesquilin. on V / subgroup:

$$X(v,v) \in \Lambda; \qquad X = -\varepsilon X^*,$$
$$\text{i.e. } X(u,v) + \varepsilon X(v,u)^* = 0.$$

From a quadr. form q one constr. a scalar product.

$$(u,v)_q = Q(u,v) + \varepsilon Q(v,u)^* \qquad \text{(linearization)}.$$

$Q \in q$; (does not depend on the choice of representative). It is a $1/2$-lin. form.

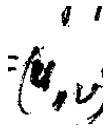


Original source details: GELFAND-P000396

Diagram and typed description. At right are two lines with a frame open at the top; in the lower line the left subscript is ε and the right pair has no subscript.

$(u,v)_q$ — an ε-symmetric form.

$$\varepsilon(v,u)^*_\varepsilon = (u,v)$$



Original source details: GELFAND-P000396

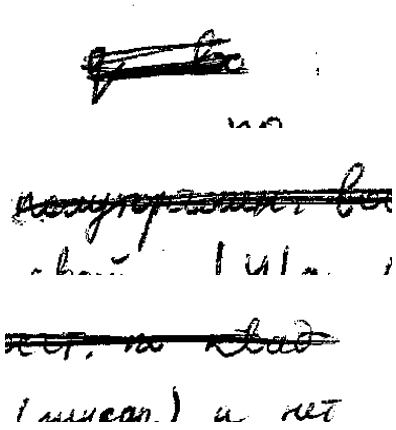
Uncertain reading. The first letter of the right pair in the lower line is reinforced by dense strokes. Provisionally u; the complete earlier strokes are not established.

Length map: $V \longrightarrow A/\Lambda$.

$$|u|_q = Q(u,u) + \Lambda.$$

(also does not depend on the representative)

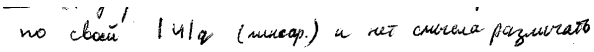
Λ max $\Rightarrow$ ~~[q?] [...] sesquilin. reconstr. from quadr[...]~~



Original source details: GELFAND-P000396

Undeciphered passage. After the arrow a short expression with a q-like sign and a phrase are cancelled. Parts reading “sesquilin.”, “reconstr.” and “from quadr...” are distinguishable; the complete earlier expression and ending have not been recovered.

quadr. is reconstr. from its $|u|_q$ (linear.) and there is no sense in distinguishing.



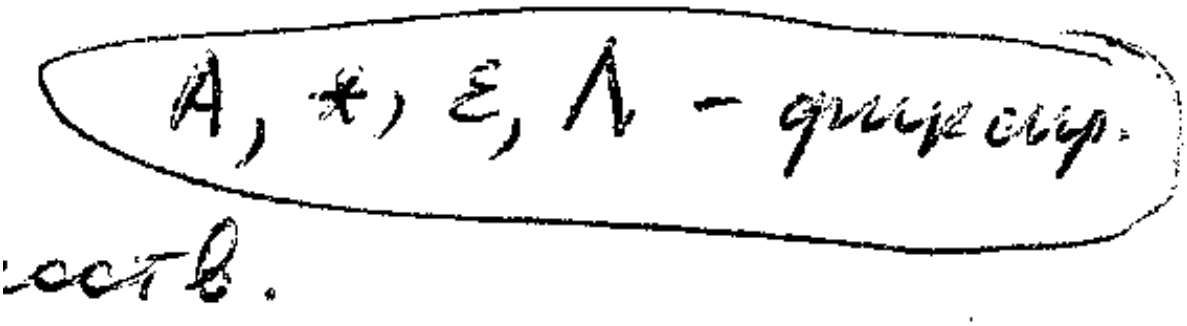
Manuscript notation. The last line has $|u|_q$.

Original source details: GELFAND-P000396

GELFAND-P000397

[source number: 44]

Example: (A Hamiltonian or hyperbolic quadr. form)



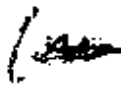
Original source details: GELFAND-P000397

Diagram and typed description. At right the parameters and the word “fixed” are circled.

$A, *, \varepsilon, \Lambda$ — fixed.

U — an A-module (any). $H(U) = U \oplus U^*$ with the quadr. form corresp. to

$$Q(u + f, u' + f') = fu'.$$



Undeciphered passage. In parentheses before “1,5 lin.” a short sign is densely cancelled. Its complete initial reading is not established.

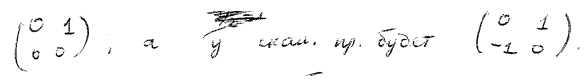
Original source details: GELFAND-P000397

(1,5 lin.).

Undeciphered passage. Below fu' a separate formula with vertical, horizontal and slanting strokes is cancelled. The complete expression is not established.

Original source details: GELFAND-P000397

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \text{and the scalar prod. will have } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$



Original source details: GELFAND-P000397

Manuscript notation. The lower-left entry of the second matrix is -1.

Witt group: from nonsing. forms: i.e. $(u,v)_q = (L(u)v)$, $L : V \longrightarrow V^*$, ~~corresp. scalar product~~, L is an isomorphism.

Original source details: GELFAND-P000397

Uncertain reading. Immediately before (u,v) a sign with a tall f-like stroke is densely overwritten. Its complete earlier outline and reading are not established; a separate initial f is not asserted.

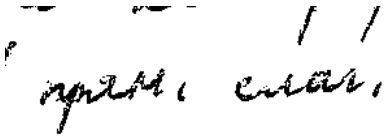
Witt gr.: $(V,q) \perp (V',q')$ — the orthog. direct sum of two modules with quadr. forms ($A, *, \varepsilon, \Lambda$ fixed).

Nonsing. quadr. forms on finitely gener.

Original source details: GELFAND-P000397

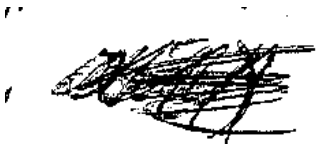
Undeciphered passage. Between “gener.” and “projective” a short fragment is densely cancelled. Its complete reading is not established.

projective modules. (dir. summ.[?],



Original source details: GELFAND-P000397

Uncertain reading. The start of the parentheses is provisionally “прям. слаг.” (dir. summ.); the joined middle letters and the ending of the second abbreviation allow other readings. The abbreviations are not expanded.



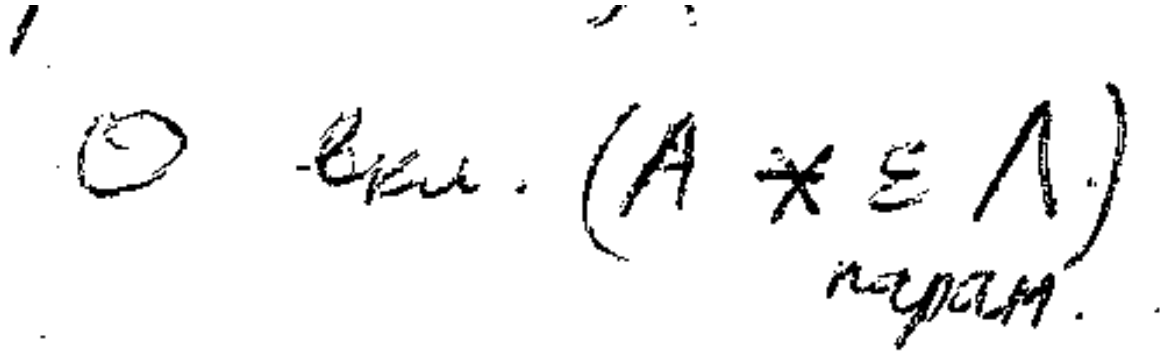
Original source details: GELFAND-P000397

Undeciphered passage. After the first two abbreviations in parentheses a word is densely cancelled. The whole earlier word has not been recovered.

finite-dimensional free ones).

They are a semigroup. From it one constr. a group (Grothendieck’s), this is the Witt gr. Over a field there is a cancellation law (Witt’s thm.).

$K_0O(A)$; O — orthog.



Original source details: GELFAND-P000397

Diagram and typed description. The line has O incl. $(A * \varepsilon\Lambda)$; below the parameters is the label "param.".

O incl. $(A * \varepsilon\Lambda)$
param.

Witt—Grothendieck. Witt, quotient by Hamilton.

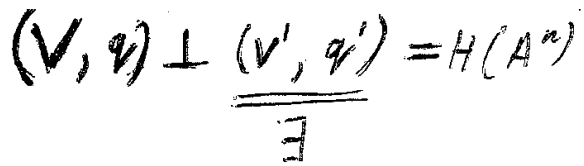
Original source details: GELFAND-P000397

Diagram and typed description. At right of these lines are two separate nearly vertical strokes.

Ex. A — a division ring $\Rightarrow$ the fund. cancellation thm.

Thm. 1. $\forall$ finitely gener. projective module with a quadr. form is a direct summand in a Hamilton. (finite-dim.).

$$(V, q) \perp \underset{\exists}{\underline{(V', q')}} = H(A^n).$$



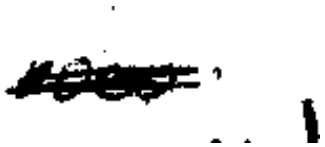
Manuscript notation. An $\exists$ sign sits under the doubly underlined pair (V', q').

Original source details: GELFAND-P000397

Remark: if cancellation by $\forall$ modules is allowed, then the Witt gr. $= 0$. $(a + b + a + b + \ldots)$.

GELFAND-P000398

Examples: 1 A — a field, $*$ the identity, $\varepsilon = 1, -1$ (the theory of symm. and

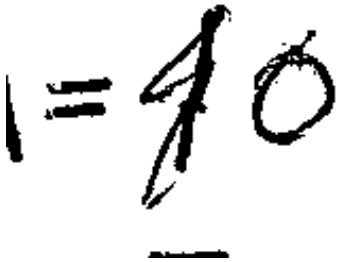


Original source details: GELFAND-P000398

Undeciphered passage. Between “and” and the following abbreviation a short word is densely cancelled. Its complete reading has not been recovered.

sympl. forms.) W.-G. gr. $= Z$.

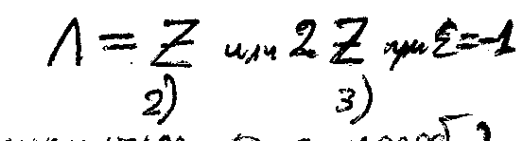
2 $A = Z$, $*$ — the identity; $\varepsilon = +1, -1$; 1) $\Lambda = 0$ for $\varepsilon = 1$;



Original source details: GELFAND-P000398

Uncertain reading. Between the equals sign and 0 is a separate tall corrected or cancelled sign: a left slanting side, a right stem and a cross-stroke. 4, a cancelled one, or an earlier A- or Λ-like letter are possible. Its reading is not established.

$\Lambda =$ 0



Original source details: GELFAND-P000398

Diagram and typed description. Below Z is the label “2)”, and below $2Z$ is “3)”.

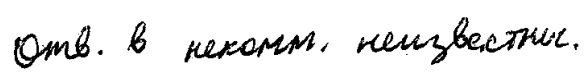
$\Lambda = \underset{2)}{Z}$ or $\underset{3)}{2Z}$ for $\varepsilon = -1$.

1) classif. of quadr. forms: invariants — rank and signature (for nonsing. ones). the cancellation thm. is false.

2) symplectic: they are isom. to Hamilton.

3) — an addit. invariant — the Arf invariant.

3 The group ring of a group. $A = Z[g]$. In the noncomm.[?] case the answers are unknown.



Original source details: GELFAND-P000398

Uncertain reading. The abbreviation is provisionally read as «некомм.» (noncomm.).

$g = Z_p \times Z_q$; K_0 is unknown.

Original source details: GELFAND-P000398

Uncertain reading. The last subscript in $Z_p \times Z_q$ is densely overwritten, with a loop and a lower tail. Provisionally q; the complete earlier strokes are not established.

This is of interest for topology (there Λ is minimal and sometimes maximal).

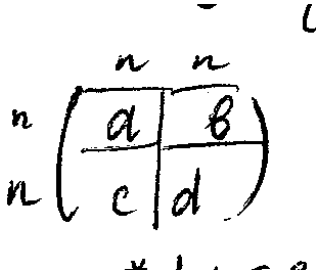
Taking Matrices over A does not change $K_0, K_1 \dots$

Remark. (there are $80 \cdot 10^6$ nonisom. quadr. forms in 32 variables) with det $= 1$.

$K_1O(A)$; $O = (A, *, \varepsilon, \Lambda)$. $O(H(A^n))$.

Original source details: GELFAND-P000398

Uncertain reading. In A below the superscript n are two parallel left strokes and a dense apex. The current A is readable; reinforcement and an earlier A- or Λ-like layer cannot be separated confidently. The earlier letter is not established.

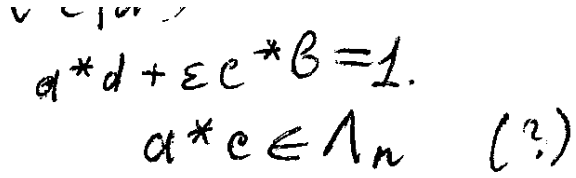


Original source details: GELFAND-P000398

Diagram and typed description. Within parentheses the blocks a, b, c, d are separated by horizontal and vertical lines; n stands above both columns and to the left of both rows.

$$\begin{array}{cc} & \begin{matrix} n & n \end{matrix} \\ \begin{matrix} n \\ n \end{matrix} & \left(\begin{array}{c|c} a & b \\ \hline c & d \end{array}\right) \end{array}$$

$$a^*d + \varepsilon c^* b = 1.$$
$$a^*c \in \Lambda_n \quad (?)$$



Manuscript notation. The second line ends with a separate $(?)$.

Original source details: GELFAND-P000398

(the scalar product $= 1$, and the length 0).

$$O(H(A^n)) \subset O(H(A^{n+m})).$$

~~$O(H(A^\infty))$.~~ $O(A)$ — their union.

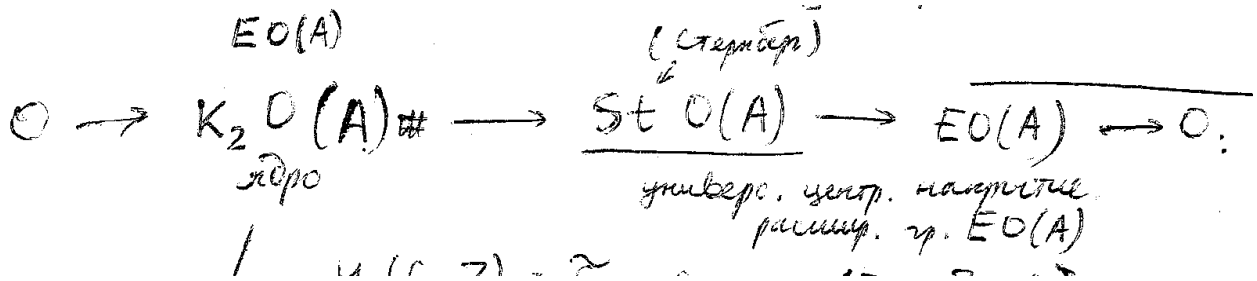
GELFAND-P000399

[source number: 45]

$$O(A)/\underline{[O(A),O(A)]} = K_1O(A).$$

Whitehead lemma $[O(A),O(A)]$ is gener. by elem. matrices

$$\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}, \qquad \begin{pmatrix}1 & 0\\ c & 1\end{pmatrix}.$$



Original source details: GELFAND-P000399

Diagram and typed description. At left is $EO(A)$; a slanting arrow from "(Steinberg)" points to the underlined $StO(A)$. Below $K_2O(A)$ is "kernel", and below $StO(A)$ are two lines of caption. The cancelled loops after the kernel are rendered separately.

$$0 \longrightarrow \underset{\text{kernel}}{K_2O(A)} \longrightarrow \underset{\substack{\text{univ. centr. cover}\\ \text{extension of the gr. } EO(A)}}{StO(A)} \longrightarrow EO(A) \longrightarrow 0$$

(Steinberg) ↙ $EO(A)$

Original source details: GELFAND-P000399

Undeciphered passage. After $K_2O(A)$ before the arrow two small loop-like signs are crossed by strokes. Their complete original expression has not been recovered.

$$H_2(G,Z) \longrightarrow \widetilde{G} \longrightarrow G \qquad ([G,G]=G).$$

Original source details: GELFAND-P000399

Uncertain reading. The first letter with subscript 2 has an M-shaped bend. The reading H_2 is provisional and the manuscript form is shown here.

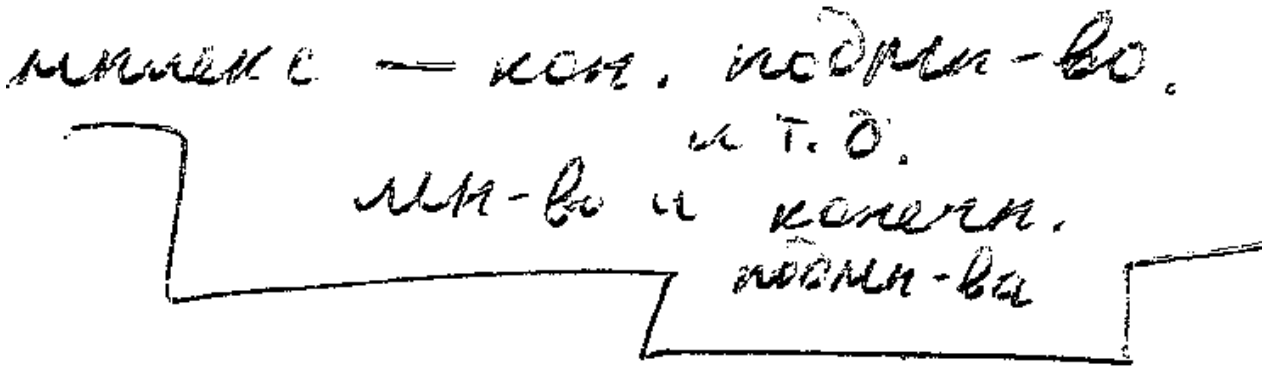
Original source details: GELFAND-P000399

Uncertain reading. The note below the line provisionally reads "nontriv. rels." The attachment of the separate slanting stroke at left is uncertain.

/

nontriv. rels.[?]

Abstr. simplicial scheme. A simplex — a finite subset. a set and finite subsets and so on.

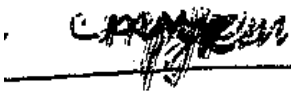


Original source details: GELFAND-P000399

Diagram and typed description. Below the words about a set and finite subsets is a stepped bracket-like line: a short upper horizontal at left, a descent, a long horizontal, a slanting descent, a lower horizontal, a rise at right and a short continuation to the right. There are no labelled vertices or arrows; the role of the steps is not established.

$$K_n O(A) = \pi_{n-1}(O(A)).$$

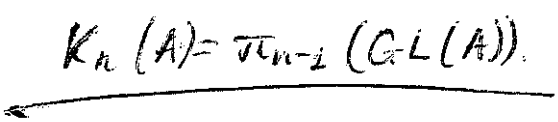
Simpl. str.[?] in $O(A)$.



Original source details: GELFAND-P000399

Uncertain reading. The abbreviation after “Simpl.” is densely overwritten. Provisionally “стр.” (str.); the longer “структ.” (strukt.) is possible. The complete earlier strokes are not established.

On finite $O(H(A^n))$.

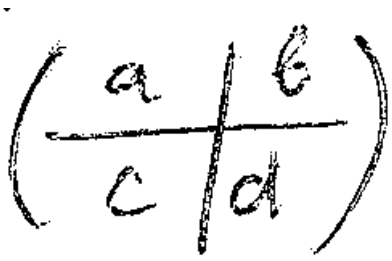


Original source details: GELFAND-P000399

Diagram and typed description. At right the formula is underlined by a long stroke with a left-pointing head at its left end.

$$K_n(A) = \pi_{n-1}(GL(A))$$

Matrices of order $(2n)$.

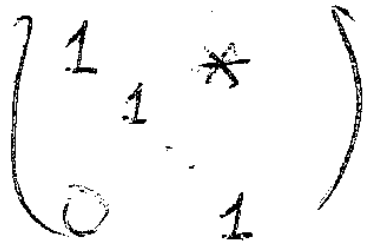


Original source details: GELFAND-P000399

Diagram and typed description. Within parentheses a, b, c, d are separated by horizontal and vertical lines.

$$\left(\begin{array}{c|c} a & b \\ \hline c & d \end{array}\right)$$

Which finite set is a simplex?



Original source details: GELFAND-P000399

Diagram and typed description. Within parentheses are three diagonal ones, $*$ at upper right, 0 at lower left and two small separate strokes between the middle and lower ones.

$$\begin{pmatrix} 1 & & * \\ & 1 & \\ & & \ddots \\ 0 & & 1 \end{pmatrix}$$

Any finite subset in this subgr. is a simplex. and right and left translations are also a simplex. (All simult. reduced to Δ form). K_1 and K_2 defined earlier agree with these.

Original source details: GELFAND-P000399

Undeciphered passage. At left before the lower line a short fragment is cancelled. The large initial stroke resembles "Э"; the whole word is not established.

Comm. relations. — relns. of elem. matrices. Triv. relns. — also conn. with π_2.

GELFAND-P000400

21/II. Kazhdan[?].

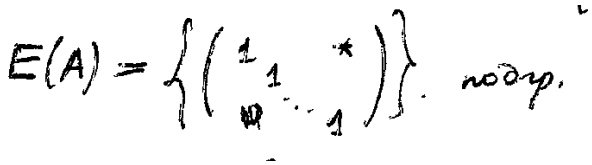
Original source details: GELFAND-P000400

Uncertain reading. The middle strokes of the surname run together. The MCCME catalogue associates the talk on 21 February 1972 with D. A. Kazhdan (record 247).

Let A be a ring. Then $K_0(A)$ is the Groth. gr. of proj. A-modules of fin. rank; $K_1(A) = GL(\infty, A)/E(A)$.

Original source details: GELFAND-P000400

Manuscript notation. The beginning of the line reads "of fin. rank".



Original source details: GELFAND-P000400

Diagram and typed description. Within braces is a matrix schematic: three diagonal ones, $*$ at upper right, three small diagonal strokes and an overwritten lower-left sign. Its provisional candidate 0 is qualified separately; no finite size is specified.

$$E(A) = \left\{ \begin{pmatrix} 1 & & * \\ & 1 & \\ & & \ddots \\ 0 & & 1 \end{pmatrix} \right\}$$

. subgr.

Original source details: GELFAND-P000400

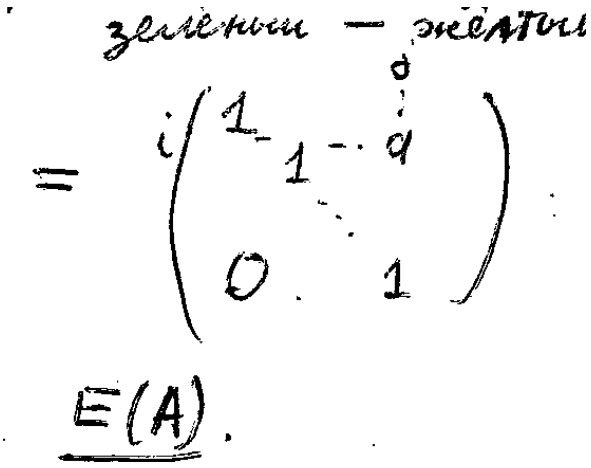
Uncertain reading. The lower-left sign is densely overwritten: a loop is crossed by vertical strokes. The candidate is 0; the earlier sign is not established.

Original source details: GELFAND-P000400

Undeciphered passage. After “subgr.” is a separate short densely cancelled fragment; its initial entry has not been recovered.

$$K_2(A) = \mathrm{Ker}\{St(A) \longrightarrow E(A)\}.$$

(Gr. of Helmholtz—Grothendieck — Helmholtz when changing colours — subtract.) green — yellow = blue.



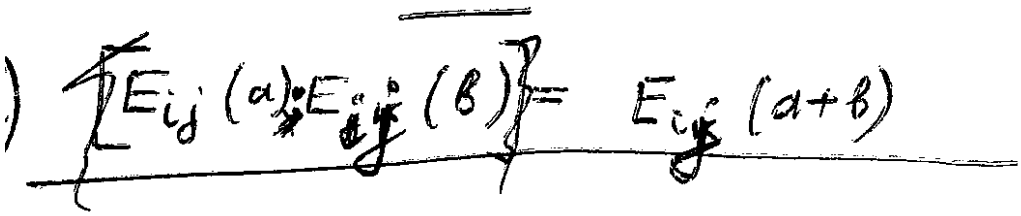
Original source details: GELFAND-P000400

Diagram and typed description. Within parentheses are three visible ones with intervening dots on a descending diagonal, 0 at lower left; a at upper right, short dotted guides to the left of and above a. At left is i, above a a small j-like label. Below $E_{ij}(a)$ is a separate $i \neq j$. The full matrix size is not specified.

$$E_{ij}(a) = \quad i \neq j \qquad \begin{matrix} & & j[?] \\ & & \vdots \\ 1 & & \\ & \ddots & -a \\ & 1 & \\ & & \ddots \\ 0 & & 1 \end{matrix}$$

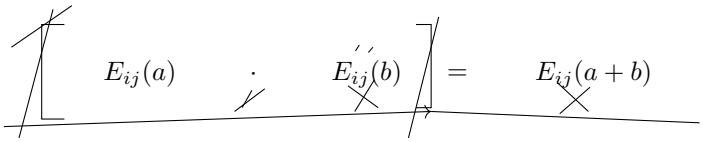
The gr. gener. by them, $\underline{\underline{E(A)}}$.

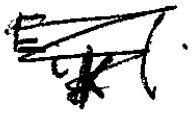
1)

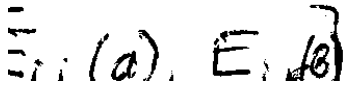


Original source details: GELFAND-P000400

Uncertain reading. The current candidate is $E_{ij}(a) \cdot E_{ij}(b) = E_{ij}(a+b)$. Earlier square brackets are crossed out, and a comma-like stroke near the dot is cancelled. The indices of the second and right-hand E are overwritten; their earlier strokes have not been recovered. Beneath the line is a long rule with a rightward head at the old right bracket and a continuation below the right-hand side.

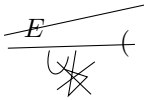




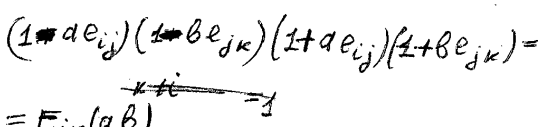


Original source details: GELFAND-P000400

Uncertain reading. Below at left a cancelled unfinished prefix E has overwritten i-/j-/k-like lower strokes and an opening parenthesis. After it is a blank interval. No written argument or closing parenthesis is visible; the exact indices are not established.

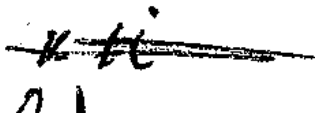


$$(1+ae_{ij})(1+be_{jk})(1+ae_{ij})(1+be_{jk}) = 1.$$



Original source details: GELFAND-P000400

Uncertain reading. In the first two factors the signs after 1 are densely overwritten; the candidates are plus and minus. Plus candidates are provisionally retained in the line; the final two plus signs are separately visible. The result 1 is written lower down at right after the equals sign.

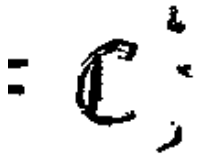


Original source details: GELFAND-P000400

Undeciphered passage. Below the product before the separately written 1 a short entry with u- or v-like and index-like strokes is cancelled; the complete earlier result has not been recovered.

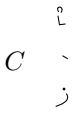
2) $[E_{ij}(a), E_{jk}(b)] = E_{ik}(ab)$. and these are all the relations for now.

In $GL(4,\mathbb{C})$ the gr. of matrices with $\det = 1$ will be gener. by elementary ones. i.e. $K_1(A) = \mathbb{C}$ for $A = \mathbb{C}$;



Original source details: GELFAND-P000400

Uncertain reading. Above and to the right of the final C is a small upper i-like or stroke-like sign, a separate dot below and further below a dot with a tail. The main C is readable; the upper sign is not identified.

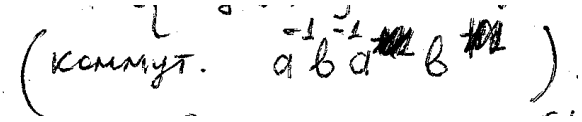


and this holds for any field.

GELFAND-P000401

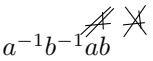
[source number: 47]

Consid. $\{E_{ij}(a)\}$ with the indicated relns. (commut. $a^{-1}b^{-1}ab$).

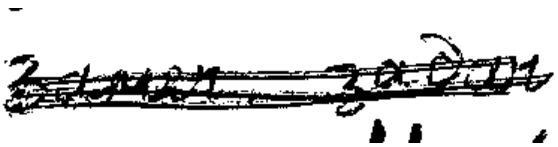


Original source details: GELFAND-P000401

Uncertain reading. Above the final a and b two exponents with minus-like and one-like strokes are cancelled. The current first two exponents $-1, -1$ are clear; the complete earlier final exponents are not established.

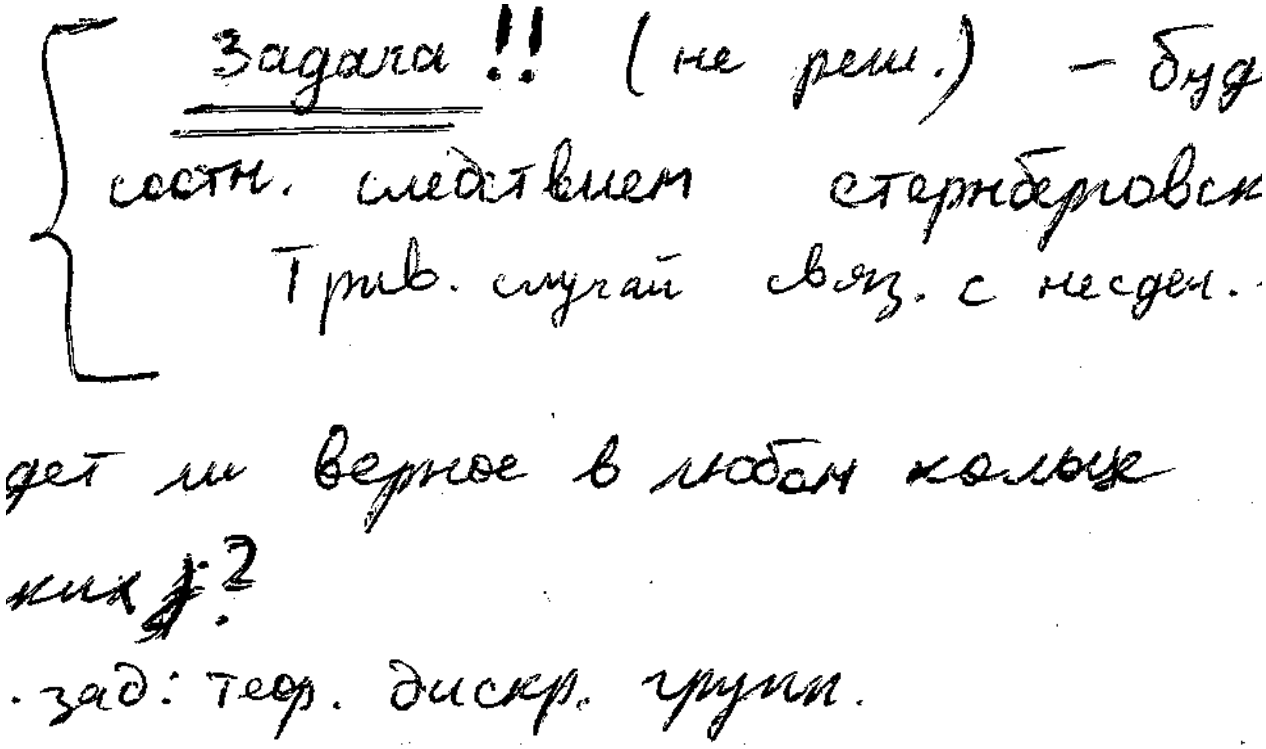


The gr. gener. by them is called $St(A)$ (an abstr. group). (Relns. in the ring, apart from the standard ones, are called $K_2(A)$) Unkn. whether these generate all relns. valid in any ring. (all Steinberg's relns.).



Original source details: GELFAND-P000401

Undeciphered passage. Before the new problem heading a line is cancelled. Parts of "Задача" (Problem) and the final word are distinguishable; the entire earlier formulation has not been recovered.



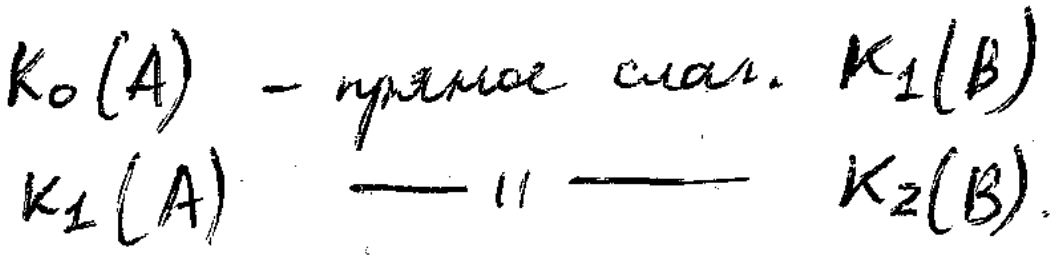
Original source details: GELFAND-P000401

Diagram and typed description. At left a brace joins the new heading, the question and the following “Triv. case”. Two rules are beneath the word “Problem”; the exclamation marks are to its right. A short stroke is cancelled before the question mark.

Problem!! (unsolv.) — will a reln. valid in any ring follow from the Steinberg ones ~~[?]~~?

The triv. case is conn. with an unsolv. problem of the theory of discrete groups.

A, $B = A[t, t^{-1}]$.



Original source details: GELFAND-P000401

Diagram and typed description. In the second line two short nearly vertical ditto strokes are between two horizontal rules. They are reproduced without expanding the repeated words.

$K_0(A)$ — direct summ. $K_1(B)$

$K_1(A)$ —— // —— $K_2(B)$.

1) $K_0(A) \longrightarrow K_1(B)$. $K_1(B)$ — autom. of a free module. Shall we specify an autom. of a proj. module? $L \in \mathrm{Aut}\, P$; $P + Q = A^n$.

Original source details: GELFAND-P000401

Uncertain reading. At right the formula $L \oplus 1 \in \mathrm{Aut}\, A^n$ is underlined. After 1 is a small dot; two more small marks are below and to the right of A^n. Their role is not established; they are rendered as separate strokes and dots.

$$L \oplus 1 \in \mathrm{Aut}\, A^n$$

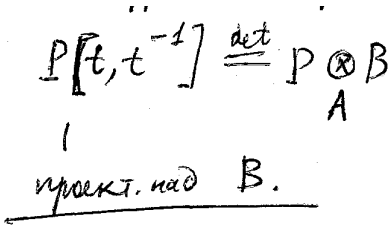
This is a well-defined def. in K_1.

$$P[t, t^{-1}] \overset{\text{def}[?]}{=} P \otimes_A B.$$

Original source details: GELFAND-P000401

Uncertain reading. Above the equals sign is a short underlined Latin label: provisionally def; the first two letters are de, the final tall crossed stroke with a lower hook is t-/f-like. Its exact letter reading is not established.

de$t$

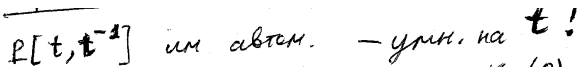


Original source details: GELFAND-P000401

Diagram and typed description. Below the equation at left is a separate short vertical stroke. The words “proj. over B.” are underlined by a long rule with a leftward head.

proj. over B.

$P[t, t^{-1}]$ has an autom. — mult. by t!



Original source details: GELFAND-P000401

Manuscript notation. The line ends “mult. by t” with an exclamation mark.

From it we construct

Original source details: GELFAND-P000401

Undeciphered passage. After “construct” a short word is crossed out. Provisionally “авто” / “авто…” (avto / avto…) can be distinguished; dense correction covers the end. The exact whole word and ending are not established.

an element of $K_1(B)$.

GELFAND-P000402

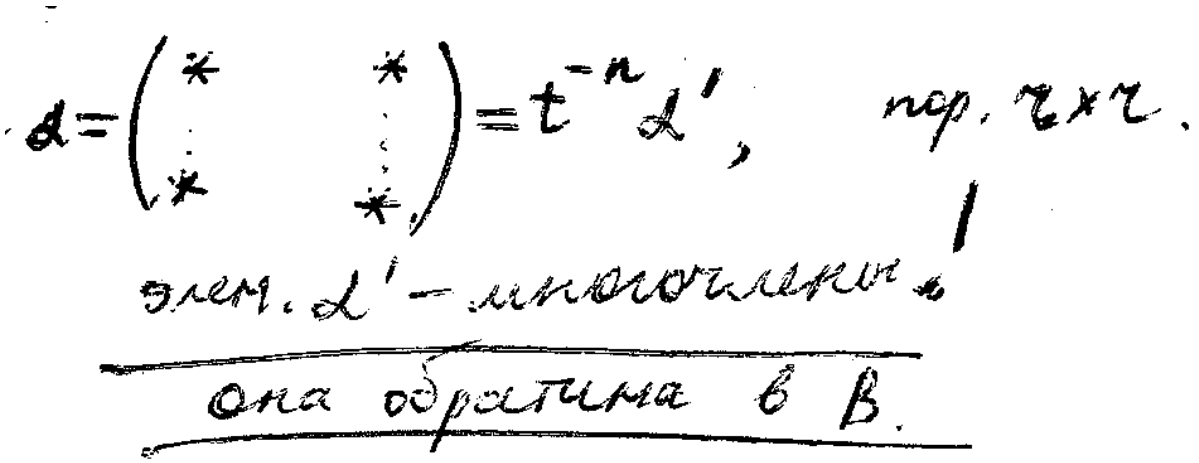
Reverse projection:

Manuscript notation. The heading "Reverse projection" has a left-pointing stroke beneath it.

Original source details: GELFAND-P000402

$$A^n \longrightarrow (B^n, t) \simeq (B, t^n).$$

An elem. of $K_1(B)$ is a matrix



Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses are four asterisks in two outer columns with dots between top and bottom. Its size is separately stated as $r \times r$; the schematic does not specify the number of intervening rows.

$$L = \begin{pmatrix} * & * \\ \vdots & \vdots \\ * & * \end{pmatrix} = t^{-n} L'$$

of order $r \times r$. The elems. of L' are polynomials! It is invertible in B.

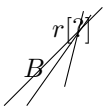
$$L' : A[t]^r \longrightarrow A[t]^r.$$

Original source details: GELFAND-P000402

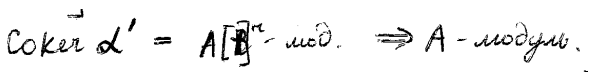
Uncertain reading. Inside the first square brackets the letter is densely overwritten, with B-like loops; provisionally t. The exponent after the bracket is r-/n-like; the line uses the candidate r. The earlier strokes are not established.

Original source details: GELFAND-P000402

Uncertain reading. Before the current right-hand $A[t]^r$ a B-like expression with a densely cancelled upper exponent is crossed out; the candidates are B^r, B^n. The drawn exponent is provisional.



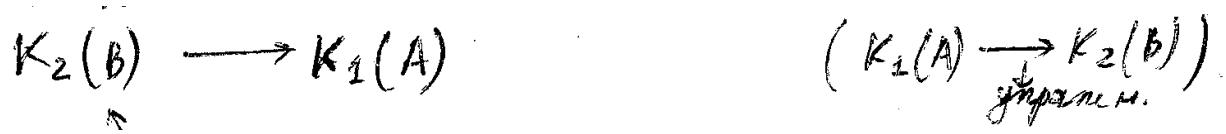
Coker $L' = A[t]^r$-mod. $\Rightarrow$ A-module.



Original source details: GELFAND-P000402

Uncertain reading. In $A[t]^r$ after Coker the letter inside the brackets is overwritten; the exponent is r-/n-like. The line provisionally uses t, r; the equals sign and abbreviation "-mod." are written in the source.

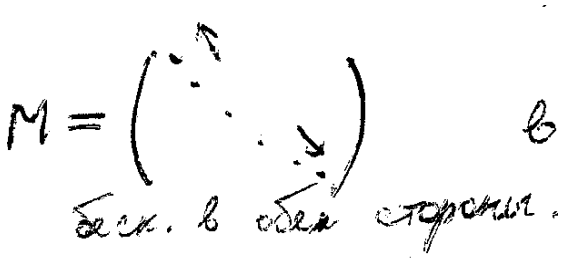
It is fin. gener. and projective over A. (like a Fredholm oper. — finite cokernels).



Original source details: GELFAND-P000402

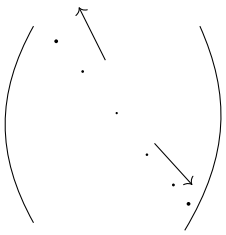
Diagram and typed description. At left is an arrow $K_2(B) \to K_1(A)$. At right in parentheses is the opposite direction $K_1(A) \to K_2(B)$; below its arrow is a short downward arrow to "exercise".

$$K_2(B) \longrightarrow K_1(A) \qquad (K_1(A) \longrightarrow K_2(B)) \downarrow \text{exercise}$$

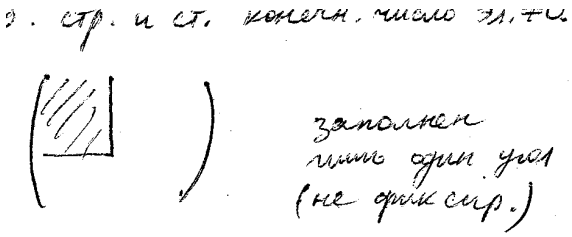


Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses is a descending diagonal row of dots; beside it are separate arrows upwards-left and downwards-right.

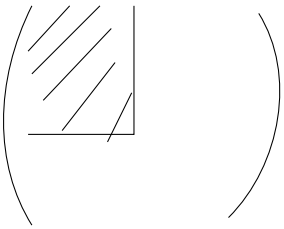
$M =$ 

infinite in both directions. In each row and col. a finite number of elems. $\neq 0$. $M^+ \subset M$, infinite in one dir.

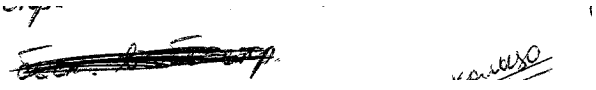


Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses the upper-left corner is filled: a vertical line is to the right of the hatching and a horizontal below it; inside are slanting strokes. The rest is unfilled, and no finite size is specified.

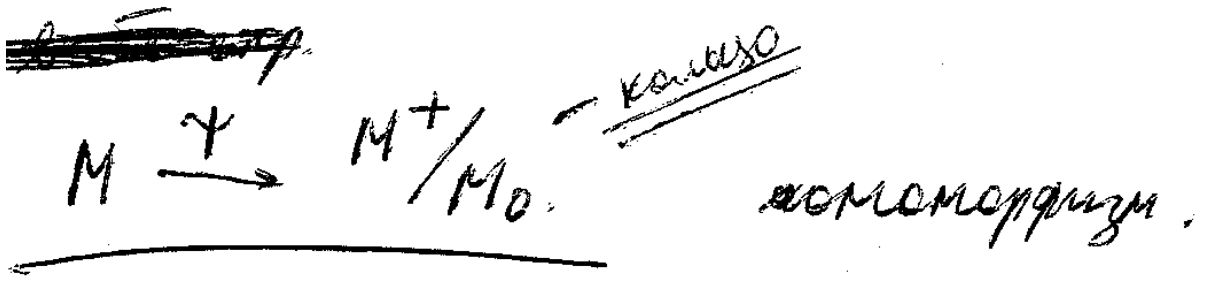


only one corner is filled (not fixed). $M_0 \subset M^+$.



Original source details: GELFAND-P000402

Undeciphered passage. After $M_0 \subset M^+$ a phrase is cancelled. An initial б-like sign and separate letter loops are visible; the whole phrase has not been recovered.



Original source details: GELFAND-P000402

Diagram and typed description. The label Ψ is above the arrow from M to M^+/M_0; "ring" is doubly underlined, and "homomorphism." is lower at right. Beneath is a long underlining rule.

$$M \xrightarrow{\Psi} M^+/M_0$$

ring,

homomorphism.

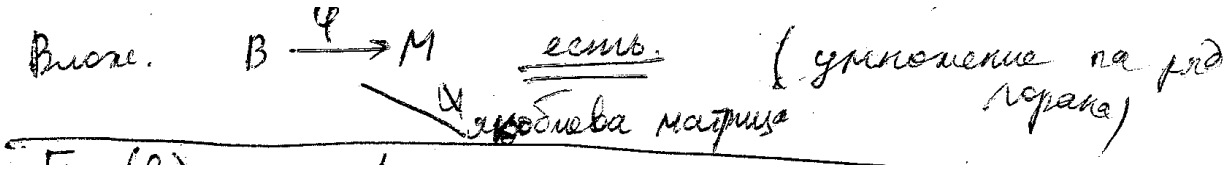
M_0 — a finite number of elems. different from 0.

Original source details: GELFAND-P000402

Uncertain reading. At M with lower 0 there is an additional densely corrected upper sign with loops, resembling 0; its precise initial position and reading are not established.

M_0

An embedding.



Original source details: GELFAND-P000402

Uncertain reading. A horizontal arrow labelled φ goes from B to M; at right "exists." is doubly underlined. From M a slanting arrow downwards-right labelled Ψ points to a caption, provisionally "якобиева матрица" (yakobieva matrix, Jacobian matrix); the beginning of the first word is densely overwritten.

$$B \xrightarrow{\varphi} M$$

exists.

Ψ

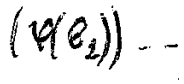
Jacobian[?] matrix

(exercise on a Laurent series)

$$E_{ij}(b_1)\ldots = 1$$

— a reln. of elem. matrices — an element of K_2 — — a certain reln. (not a Steinberg one).

$$E_{ij}(\Psi(\varphi(e_2)))\ldots = 1.$$



Original source details: GELFAND-P000402

Uncertain reading. In the last line inside φ is a looped e-like letter and a small lower sign with a bend and lower rule. The current candidate is e_2; the historical candidates e_i, b_i are not excluded. The preceding argument b_1 has not been copied here.

GELFAND-P000403

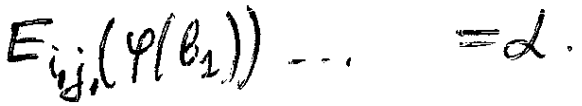
[source number: 48]

$\ldots M^+/M_0$ “all oper. po vn. nepr.”

Original source details: GELFAND-P000403

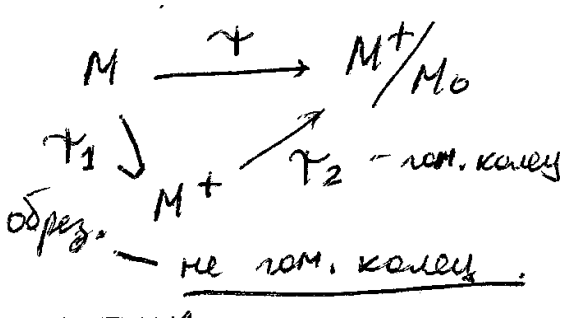
Uncertain reading. Within quotation marks is “все опер. по вн. непр.” (all oper. po vn. nepr.). The abbreviations “вн. непр.” are left unexpanded; their full sense is not established.

$$E_{i_1 j_1}(\varphi(b_1))\ldots = L.$$
$$\Psi(L) = 1.$$



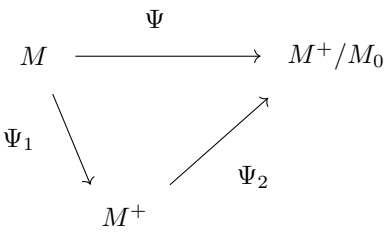
Original source details: GELFAND-P000403

Manuscript notation. Separate small ones follow i and j.

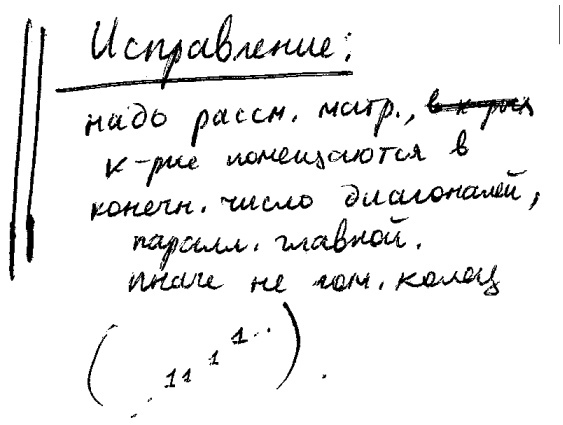


Original source details: GELFAND-P000403

Diagram and typed description. The upper arrow $M \to M^+/M_0$ is labelled Ψ; at left an arrow downwards-right labelled Ψ_1 goes to M^+, and from there an arrow upwards-right labelled Ψ_2.



Ψ_2 — a ring hom.; Ψ_1 — trunc. — not a ring hom..

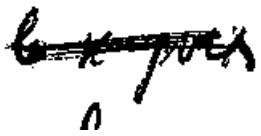


Original source details: GELFAND-P000403

Diagram and typed description. The correction at right is separated by two long nearly vertical rules; its heading is underlined.

Correction:

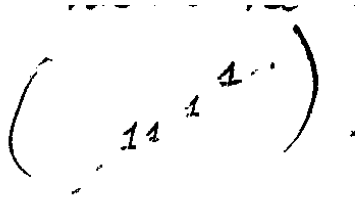
we must consid. matrices,



Original source details: GELFAND-P000403

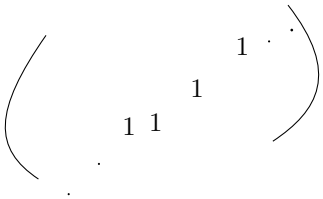
Undeciphered passage. After “matrices” a word with distinguishable в-like and p-like strokes is cancelled; its complete reading has not been recovered.

which fit in a finite number of diagonals parall. to the principal one. Otherwise not a ring hom.

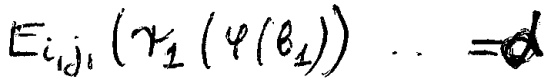


Original source details: GELFAND-P000403

Diagram and typed description. Within parentheses are four one-like signs: two beside each other below, one higher, and one still higher at right; at both ends are separate diagonally positioned dots. Full rows, band boundaries and the remaining entries are not labelled.



Consider $E_{i_1 j_1}(\Psi_1(\varphi(b_1)))\ldots = L$.



Original source details: GELFAND-P000403

Manuscript notation. Separate small ones follow i and j.

$L \neq 1$, since Ψ_1 is not a ring hom.

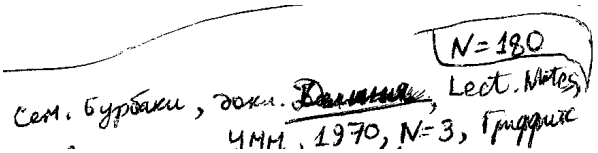
Original source details: GELFAND-P000403

Uncertain reading. The sign after the inequality is densely overwritten: a one-like stem with a lower rule and superimposed loops is visible. The current candidate is 1; the earlier sign is not established.

$L \neq$

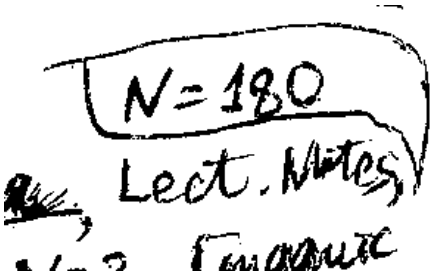
But $\Psi_2(L) = 1$. L is a matrix over M^+, hence it is over M^0 — a finite matrix, def. an elem. of $K_1(A)$.

Bourbaki Sem., report by Delinya[?];



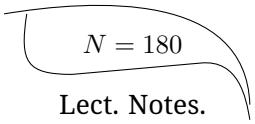
Original source details: GELFAND-P000403

Uncertain reading. The overwritten surname provisionally reads “Deligne”. The reference to Lect. Notes 180 agrees with Pierre Deligne’s “Travaux de Griffiths”, Bourbaki talk 376.



Original source details: GELFAND-P000403

Diagram and typed description. At right $N = 180$ is within a shaped frame, with Lect. Notes. below it. The frame edge bends downwards at right.



UMN, 1970, $N = 3$, Griffiths.

Gabrielov. Work of Griffiths on var. Hodge structure.

X — a nonsingular complex[?] alg. projective variety; dim $X = d$.

Original source details: GELFAND-P000403

Uncertain reading. The abbreviation after “nonsingular” is provisionally “компл.” (complex); its middle joined strokes and ending are not fully separate. The complete word is not written out here.

$$H^n(X, Z), \qquad H^n(X, C) = H^n(X, Z) \otimes_Z C.$$

X is Kähler. On it there is an integral form η of second pr.[?] of type $(1, 1)$.

Original source details: GELFAND-P000403

Uncertain reading. After “second” is a short abbreviation with п-like and p-like strokes, provisionally “пр.” (pr.); a compressed middle о in “пор.” (por., order) is possible. Its exact complete letter reading is not established.

$$H^n(X, C) = \bigoplus_{p+q=n} H^{p,q}.$$

$\eta \in H^{1,1}$, closed and def. a cohom. class. it is given a priori. η is called a polarization.

GELFAND-P000404

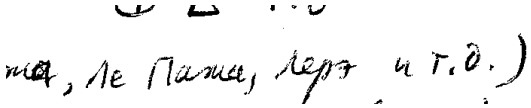
$L : H^n(X, C) \longrightarrow H^{n+2}(X, C)$ — the oper. of mult. by η.

$$\text{Put } H_0^n = \text{Ker}(L^{d-n+1}; H^n \longrightarrow H^{2d-n+2}).$$

primitive cohomology. In cohom. there is a natural scalar product. One can prove that H_0^n is the orthog. complement of the image of L.

$$H^n(X, C) = \bigoplus L^i H_0^{n-2i}.$$

(decomp. of Hodge, Le Pazha, Lert[?], and so on)

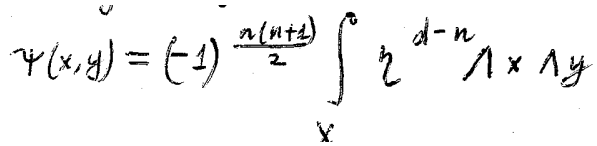


Original source details: GELFAND-P000404

Uncertain reading. After “Ходжа” (Hodge) is written “Ле Пажа” (Le Pazha), then a surname or abbreviation beginning “Лер” and ending in a t-like stroke. Provisionally “Лерт” (Lert);“Лерэ” (Lere) and “Лефт” (Left) are possible. A full name is not expanded here.

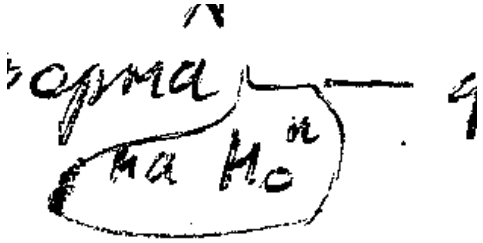
(i.e. everything decomposes into primitives). $x, y \in H_0^n$. Consid.

$$\Psi(x, y) = (-1)^{\frac{n(n+1)}{2}} \int_X \eta^{d-n} \wedge x \wedge y.$$



Original source details: GELFAND-P000404

Uncertain reading. The two tall chevrons between the factors are provisionally read as exterior products.



Original source details: GELFAND-P000404

Diagram and typed description. Below the word “form” is an insertion “on H_0^n” within a curved enclosure connected above to the line.

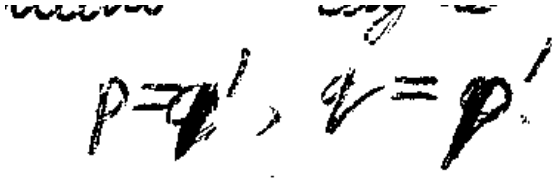
This is a bilin. form — the Hodge form.

on H_0^n

Properties:

1) $\Psi(x, y) = (-1)^n \Psi(y, x)$.

2) $x \in H_0^{p,q}$, $y \in H_0^{p',q'} \Rightarrow \Psi(x,y) \neq 0$ only when $p = q'$, $q = p'$.

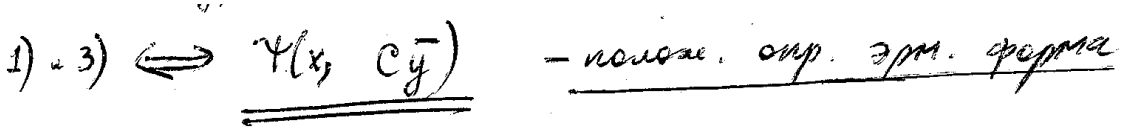


Uncertain reading. After $p =$ the q' is densely overwritten; after $q =$ the letter p' has repeated strokes. The current candidates are $p = q'$, $q = p'$; earlier letters have not been recovered.

Original source details: GELFAND-P000404

3) $0 \neq x \in H_0^{p,q} \Rightarrow i^{p-q}\Psi(x,\overline{x}) > 0$. (in cohom. there is complex conjugation, $\overline{x}$ is the complex conjugate) Let C be the Weil autom. — mult. by i^{p-q} in $H^{p,q}$. (mult. by i in the tangent space). $\Psi(x,y)$ is inv. under the Weil autom.

1) and 3) $\Longleftrightarrow$



Original source details: GELFAND-P000404

Diagram and typed description. The formula with $C\overline{y}$ is doubly underlined; the lower rule has a leftward head. At right "pos. def. Hermitian form" is underlined by one long rule.

$\Psi(x, C\overline{y})$ — pos. def. Hermitian form

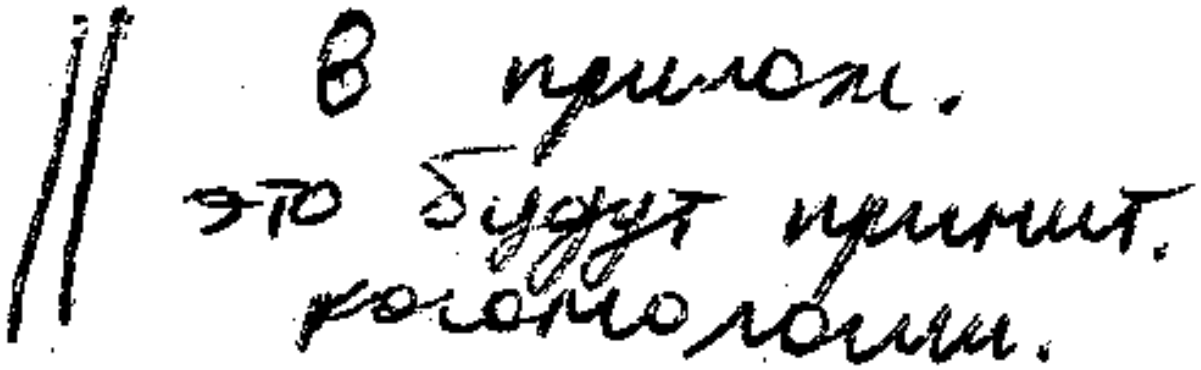
GELFAND-P000405

[source number: 49]

The basic idea of Deligne — one can forget about manif. and retain only the Hodge struct. Cohom. — lin. spaces. A problem of lin. algebra.

$H_{\mathbb{Z}}$ — a finite-type $\mathbb{Z}$-module

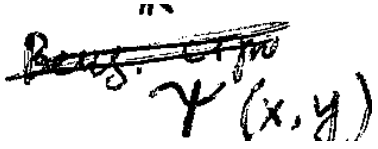
$H_{\mathbb{R}} = \mathbb{R} \otimes H_{\mathbb{Z}}$; $\quad H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$.



Original source details: GELFAND-P000405

Diagram and typed description. At right beside a double vertical rule:

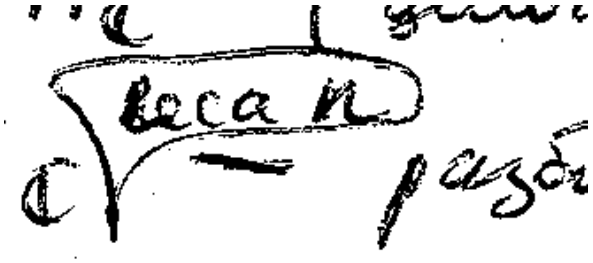
// in applic. these will be primit. cohomology.



Original source details: GELFAND-P000405

Uncertain reading. [Crossed out above the formula:] “Real struct.[?]” (“Вещ. стр.”); the ending “стро” is possible. Strokes obscure the end of the abbreviation.

$\psi(x,y)$ — a bilin. form in $H_{\mathbb{C}}$ (integral — takes integer values on $H_{\mathbb{Z}}$).



Original source details: GELFAND-P000405

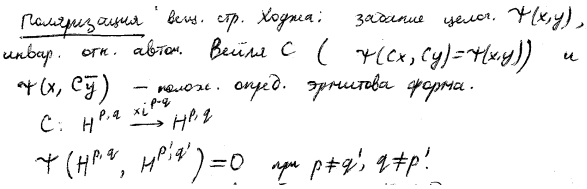
Diagram and typed description. After $H_{\mathbb{C}}$, “of weight n” is inserted within a curved enclosure.

A real Hodge struct. on $H_{\mathbb{C}}$ (of weight n) — a decomposition

$$H_{\mathbb{C}} = \bigoplus_{p+q=n} H^{p,q}, \qquad H^{p,q} = \overline{H^{q,p}}.$$

A polarization of a real Hodge struct.: specifying an integral $\psi(x,y)$, inv. under the Weil autom. C ($\psi(Cx,Cy)=\psi(x,y)$) and $\psi(x,C\bar{y})$ — a pos. def. Hermitian form.

$$C: \quad H^{p,q} \xrightarrow{\times i^{p-q}} H^{p,q},$$
$$\psi(H^{p,q}, H^{p',q'}) = 0 \quad \text{when } p \neq q',\ q \neq p'.$$

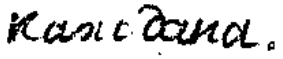


Original source details: GELFAND-P000405

Manuscript notation. The bar in $\psi(x,C\bar{y})$ is over y.

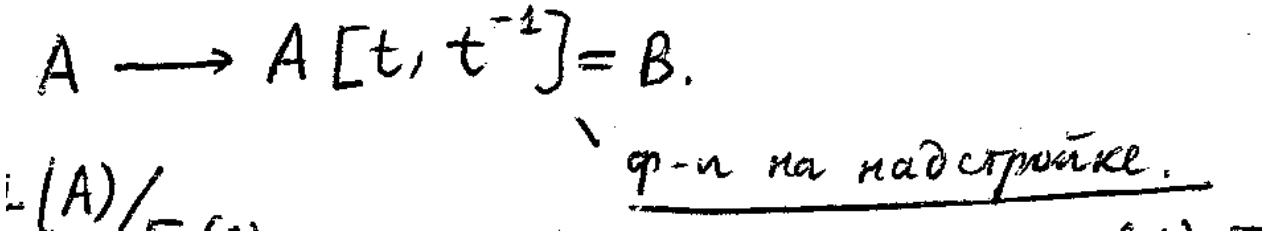
This is called a polarized Hodge structure.

27/II. Kirillov. — repetition of Kazhdan[?].



Original source details: GELFAND-P000405

Uncertain reading. After “repetition” the surname reads “Ка...дана”. “Каждана” (Kazhdan) is provisional because the middle strokes run together.



Original source details: GELFAND-P000405

Diagram and typed description. A short slanted stroke starts under the closing bracket and leads to the underlined caption. Its end is unheaded.

$A \longrightarrow A[t, t^{-1}] = B.$

fns. on the suspension.

$$K_1 = GL(A)/E(A).$$

Original source details: GELFAND-P000405

Diagram and typed description. Two separate horizontal arrows: the upper one points right, the lower one left.

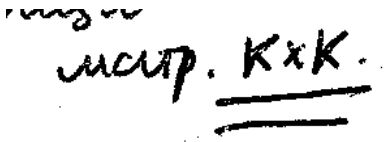
$K_0(A) \rightleftarrows K_1(B).$

$M \in GL(B)$; $\quad M = t^{-k}$[crossed-out sign]M_1,

Original source details: GELFAND-P000405

Uncertain reading. Between t^{-k} and M_1 a separate sign is densely crossed out. Its earlier shape cannot be separated; candidates are a crossed-out multiplication sign or another separate symbol.

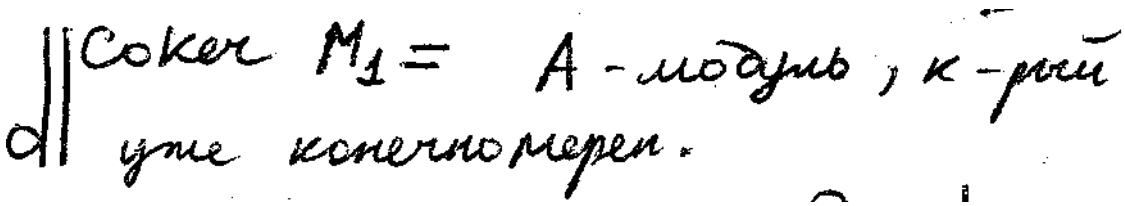
M_1 — a polyn. matrix mstr.[?] $k \times k$.



Original source details: GELFAND-P000405

Uncertain reading. Below “polyn. matrix” the abbreviation is provisionally read “мстр.” (mstr.); “матр.” (matr.) is possible. The middle letter cannot be separated reliably.

$$A[t]^k \xrightarrow{M_1} A[t]^k, \qquad \ker = 0.$$



Original source details: GELFAND-P000405

Diagram and typed description. At right of the line beside a double vertical rule:

// Coker $M_1 = A$-module, which is already finite-dimensional.

(actually one needs the difference between kernel and cokernel) Elem. transformations do not change the cokernel.

GELFAND-P000406

$$M = t^{-k}M_1 = t^{-\ell}M_2.$$

Original source details: GELFAND-P000406

Diagram and typed description. In the margin — a long lower rule and an additional short one at left:

one must subtract the excess.

Original source details: GELFAND-P000406

Diagram and typed description. Crossed out:

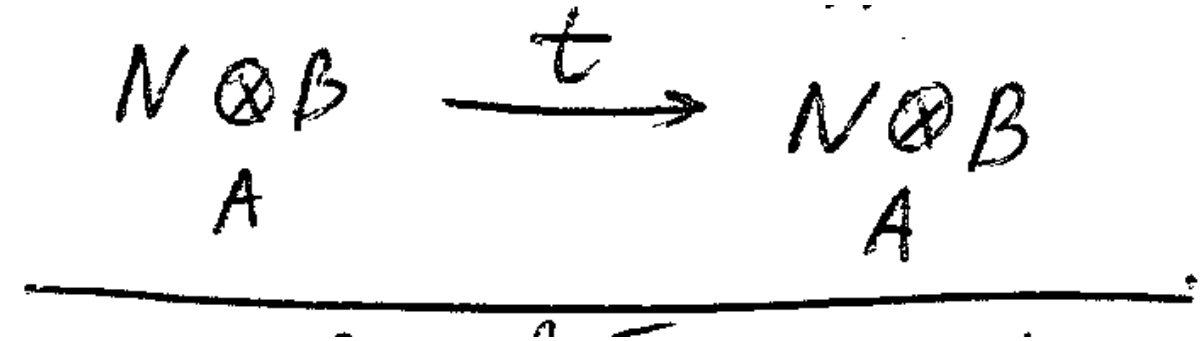
~~M~~

The inverse map:

$$N \longrightarrow N[t, t^{-1}] = N \otimes_A B.$$

Original source details: GELFAND-P000406

Uncertain reading. The sign has two horizontal bars and a separate small downward stroke; provisionally the equality sign is retained. The meaning of the lower mark is not established.



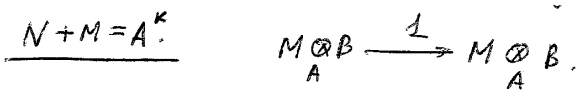
Original source details: GELFAND-P000406

Diagram and typed description. A long horizontal rule is drawn below the next line.

$$N \otimes_A B \xrightarrow{t} N \otimes_A B$$

complete to a free module (otherwise one does not obtain a matrix).

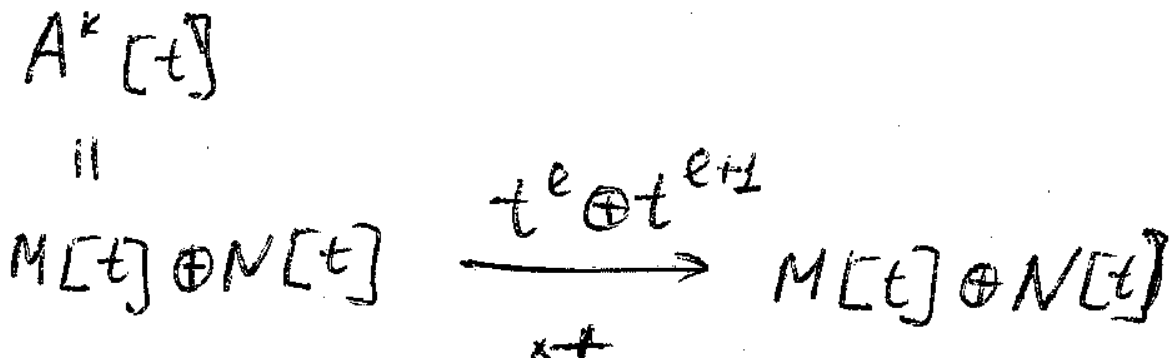
$$\underline{N + M = A^k.} \qquad M \otimes_A B \xrightarrow{1} M \otimes_A B.$$



Manuscript notation. $N + M = A^k$ is written at left. The right-hand arrow is labelled 1.

Original source details: GELFAND-P000406

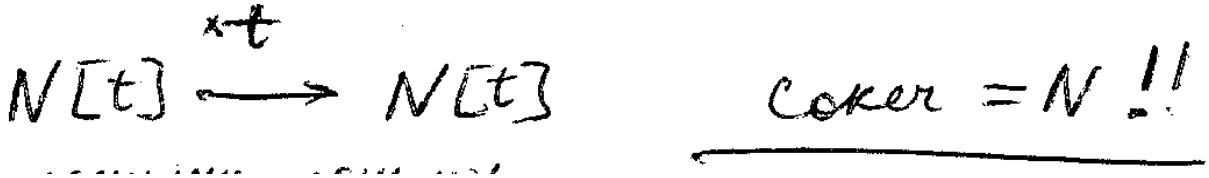
One obtains a map $B^k \longrightarrow B^k$
multiply by t^ℓ to obtain a polynomial one? [At right:] One need not multiply.



Original source details: GELFAND-P000406

Diagram and typed description. $A^k[t]$ is above the left-hand $M[t] \oplus N[t]$; between them are two vertical rules.

$$\begin{array}{ccc} A^k[t] & & \\ \| & & \\ M[t] \oplus N[t] & \xrightarrow{t^\ell \oplus t^{\ell+1}} & M[t] \oplus N[t] \end{array}$$



Original source details: GELFAND-P000406

Diagram and typed description. At right the equality is underlined and has two exclamation marks.

$$N[t] \xrightarrow{\times t} N[t] \qquad \underline{\text{Coker} = N!!}$$

And the ends have met.

$$1 \longrightarrow K_2(B) \longrightarrow St(B) \longrightarrow E(B) \longrightarrow 1.$$

Swan, Tate. Two talks in Nice from [crossed out: an unreadable word] different points of view.

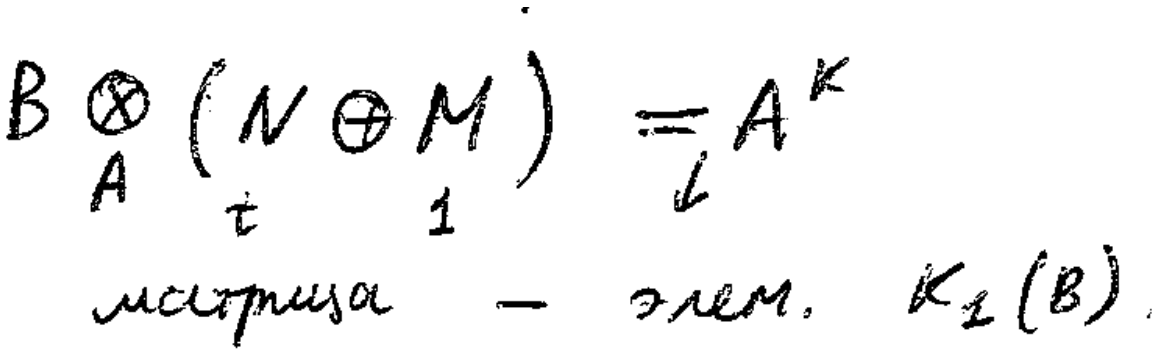
Undeciphered passage. The word after "Two talks in Nice from" is entirely covered by dense crossing-out; its reading is not established.

Original source details: GELFAND-P000406

Gelfand.

$$K_0(A) \longrightarrow K_1(B).$$

N — a proj. module.



Original source details: GELFAND-P000406

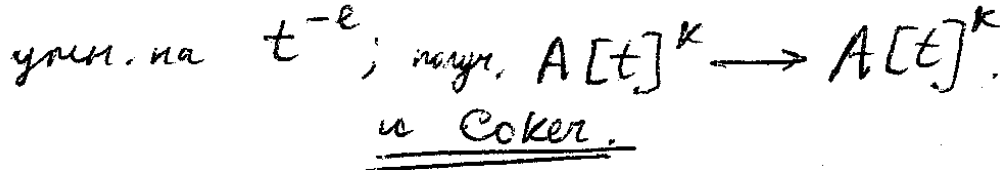
Diagram and typed description. Below N and M are written t and 1; a downward arrow leads from the equality to the caption.

$$B \otimes_A (\underset{t}{N} \oplus \underset{1}{M}) = A^k$$

$$\downarrow$$

a matrix — an elem. of $K_1(B)$.

$B^k \longrightarrow B^k$ multiply by $t^{-\ell}$; one obtains $A[t]^k \longrightarrow A[t]^k$.



Original source details: GELFAND-P000406

Diagram and typed description. The concluding words are doubly underlined.

and Coker.

GELFAND-P000407

[source number: 50]

$$0 \longrightarrow K_1 A \longrightarrow K_1 A[t] \oplus K_1 A[t^{-1}]$$
$$\longrightarrow K_1 A[t, t^{-1}] \longrightarrow K_0 A \,; \longrightarrow 0.$$

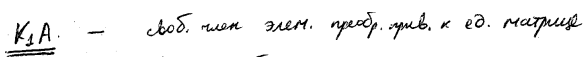
Original source details: GELFAND-P000407

Uncertain reading. Immediately after $K_0 A$ are a detached upper dot and a lower comma-like stroke; provisionally ";" in the line. The exact nature of the mark is not established.

$K_0 A$;

$$K_1A[t] = K_1A \oplus \text{Nil } A$$

$M : A[t]^k \longrightarrow A[t]^k$ an invertible matrix up to elem. transformations. For regular rings Nil $A = 0$. K_1A. — the constant term reduc. to the identity matrix by elem. transf.



Manuscript notation. The abbreviation before “to the identity matrix” reads «прив.».

Original source details: GELFAND-P000407

Now $M = 1 + tN(t)$; and invertible. one can make N independent of t; t^2 is broken up by increasing the order of the matrix (reduction to an equation of the first order).

$$1 + tN \text{ invertible in polynomials} \quad \Longrightarrow \quad N^k = 0.$$

Nil$A$ — nilpotent matrices.

Original source details: GELFAND-P000407

Diagram and typed description. Two horizontal segments with a detached dot between them.

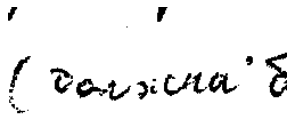
The K-functor in Quillen’s sense.

Morava. (Kolumb.[?] Univ.).

Uncertain reading. The abbreviation before “Univ.” is provisionally read «Колумб.» (Kolumb.). The first vowel could be «а».

Original source details: GELFAND-P000407

Quillen’s lemma. (it must be one of the first in topology).



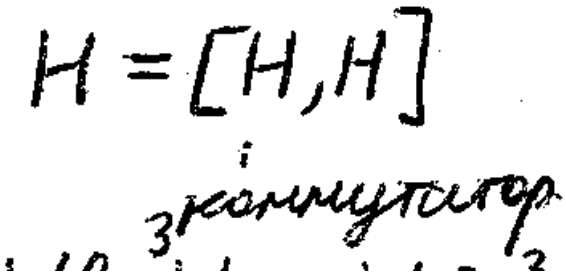
Manuscript notation. The first word in parentheses is «должна» (must).

Original source details: GELFAND-P000407

Manuscript notation. «Быть» (be) follows «должна».

Original source details: GELFAND-P000407

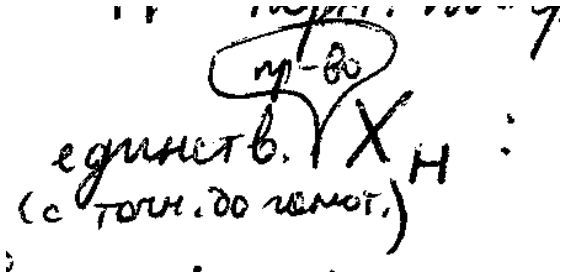
X — a space, H — a normal subgroup of $\pi_1(X)$, with



Original source details: GELFAND-P000407

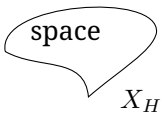
Diagram and typed description. Below the condition are two small separate vertically placed strokes/dots and a caption; no arrowhead is visible.

$H = [H, H]$
:
commutator



Original source details: GELFAND-P000407

Diagram and typed description. Between “unique” and X_H, “space” is inserted in a curved enclosure with a lower tip.

Then $\exists$ a unique 

X_H:

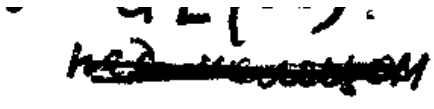
(up to homotopy),

$$X_H = X \cup e^2 \cup e^2 \cup \cdots \cup e^3 \cup \cdots \cup e^3,$$
$$\text{such that} \quad \pi_1(X_H) = \pi_1(X)/H.$$
$$\text{and} \quad H_*(X, \mathbb{Z}) \simeq H_*(X_H; \mathbb{Z}).$$

(the homology ring with integer coeffs.) Corresp. functorially in X. Proof by means of obstruction theory.

GELFAND-P000408

Let $X = BGL(R)$. — the classif. space of $GL(\infty)$. [Below $GL(\infty)$, crossed out: ned...naya.]



Original source details: GELFAND-P000408

Uncertain reading. In the crossed-out word the beginning “нед...” (ned...) and ending “...ная” (...naya) are distinguishable; the middle is covered by a line. The candidate “недискретная” (non-discrete) is not established in full.

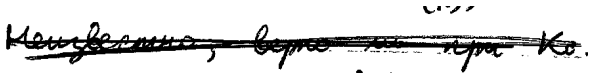
(R — any ring). everything without topology — a discr. group (has nothing in common with Grassm.).

$$E(R) = [E(R), E(R)]$$

the subgroup generated by elem. matrices.

$$\pi_1(BGL(R)) = GL(R).$$
$$\pi_1(X_{E(R)}) = GL(R)/E(R) \text{ — its fund. group } K_1(R).$$

Now $K_i(R) = \pi_i(X_{E(R)})$. [Crossed out: It is unknown whether it is true [...] for K_0.]



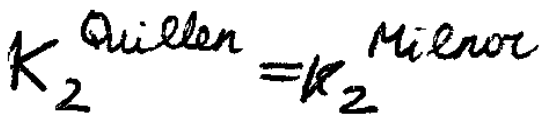
Original source details: GELFAND-P000408

Undeciphered passage. The middle words of the crossed-out line are not recovered; the surviving beginning and end are given in the line.

K_0: the def. must be changed, since $\pi_0(\ ) = \{1\}$.

Quillen: if R is a field, then

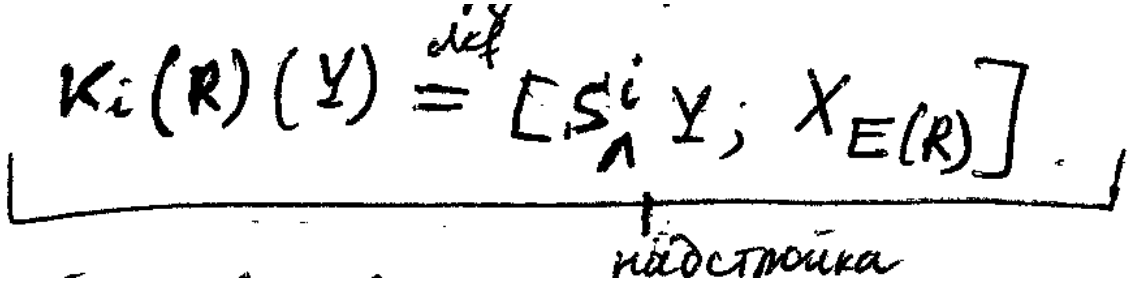
$$K_2^{\text{Quillen}} = k_2^{\text{Milnor}}.$$



Original source details: GELFAND-P000408

Uncertain reading. On the right, the letter before the subscript 2 is distinctly smaller than the left K and has a k-like form. Provisionally typeset as k_2; its case is not inferred from the expected group notation.

There is also Volodin's definition. There exists a theory, applicable to top. groups. Then for R and C one obtains the usual K-functor.



Original source details: GELFAND-P000408

Diagram and typed description. The $\wedge$ sign is lowered below the main line. Below the formula is a lower bracket with a short stroke to the caption "suspension"; the exact scope of the caption is not established.

$$K_i(R)(Y) \stackrel{\text{def}}{=} [S^i \underset{\wedge}{} Y; X_{E(R)}].$$

suspension

Novikov's problem: define the Chern char. in alg. K-theory. Here this reduces to computing $H_*(X_{E(R)}; Q)$.

GELFAND-P000409

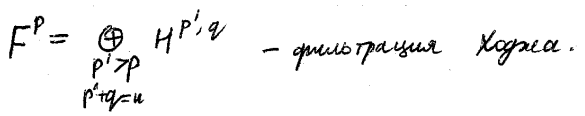
[source number: 51]

A. Gabrielov.

X — an alg. projective manifold.

$$H^n(X,\mathbb{C}) = \bigoplus_{p+q=n} H^{p,q},$$

$$F^p = \bigoplus_{\substack{p'>p \\ p'+q=n}} H^{p',q} \text{ — the Hodge filtration.}$$

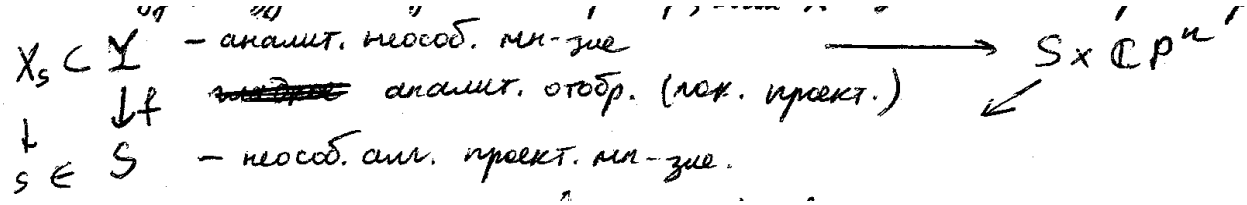


Manuscript notation. Below the direct-sum sign are $p\prime > p$ and $p\prime + q = n$.

Original source details: GELFAND-P000409

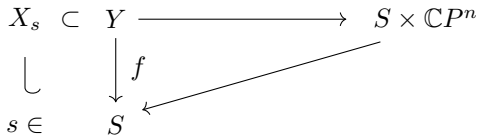
It moves holomorphically! The behaviour of this filtr. will be studied when X depends on a parameter

$X_s \subset Y$ — an analytic nonsing. manifold.



Original source details: GELFAND-P000409

Diagram and typed description. The family diagram and a separate short vertical stroke with a hook below X_s; below it, nearby, $s \in S$.

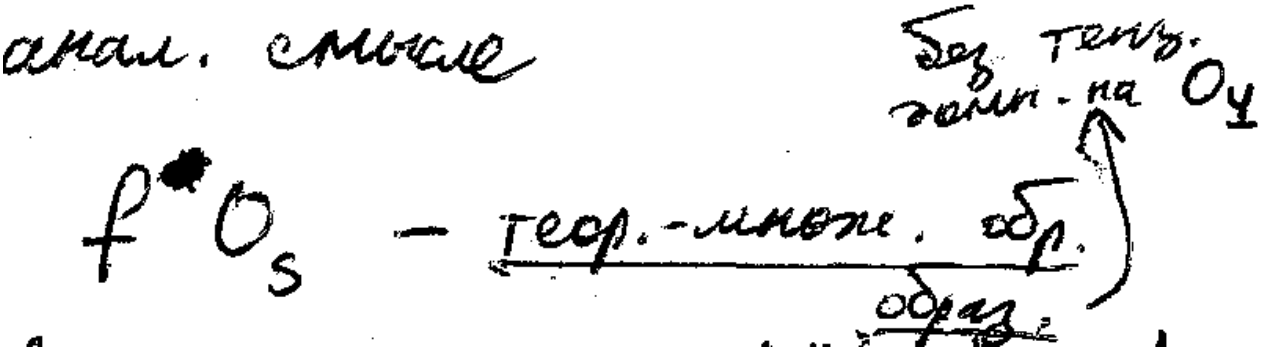


[Before the following words, crossed out:]

Undeciphered passage. The crossed-out word before “analytic map” is not recovered in full. The candidates “гладкое” (smooth) and “полное” (complete) are not fully confirmed.

Original source details: GELFAND-P000409

analytic map (loc. project.). S — a nonsing. alg. project. manifold. smooth — local triviality in the analytic sense. O_S — the sheaf of analytic functions on S;



Original source details: GELFAND-P000409

Diagram and typed description. The underlined definition, “image” inserted below, and a curved arrow to the margin.

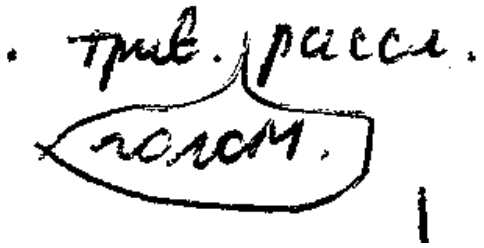
Without tensor mult. by O_Y.

f^*O_S — set-theor. inv. image.

Original source details: GELFAND-P000409

Uncertain reading. In the first f^*O_S the superscript is overwritten into a dark blot. Provisionally $*$; $\circ$ is possible. The later clear f^* and f° do not establish this written sign.

$R^n f_* f^\circ O_S$ —



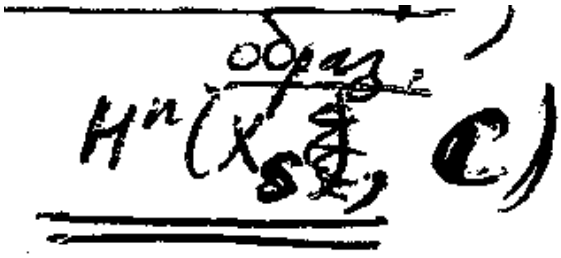
Original source details: GELFAND-P000409

Diagram and typed description. The enclosed insertion “holom.” attached to “bundle”; below to the right is a separate vertical stroke.

loc. triv. bundle

holom.

with fibre



Original source details: GELFAND-P000409

Uncertain reading. After X_s, before the separator, are crossed-out vertical and loop-like strokes. The previous signs are not read; the strokes are shown without asserting a letter Z. Below the left part of the formula is a double underline.

$H^n(X_s; \mathbb{C})$

Original source details: GELFAND-P000409

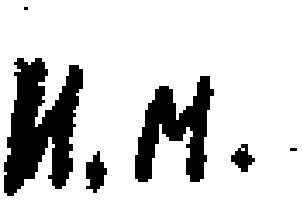
Uncertain reading. Below the formula to the right is a crossed-out fragment ending “...оем” (...oyem); “слоем” (with fibre) is possible, but not established in full. Above it is a short vertical stroke.

~~with fibre[?]~~

in the fibres, a holom. filtration by subbundles. The relative de Rham complex:

$$\Omega^*_{Y/S},$$
$$\Omega^1_{Y/S} = \Omega^1_Y / f^*\Omega^1_S; \qquad \Omega^p_{Y/S} = \bigwedge_p \Omega^1_{Y/S}.$$
$$0 \longrightarrow f^\circ O_S \longrightarrow O_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow \dots.$$

($f^*O_S = O_Y$, $f^\circ O_S$ — constant along the fibres). (a resolution of the sheaf f°). I. M.[?] $H^{p,q}$ forms in $dz, d\bar{z}$ move non-analytically, since $d\bar{z}$ transforms non-analytically.



Uncertain reading. The two letters in the left margin before $H^{p,q}$ provisionally read «И. М.» (I. M.). The first is closely written and also resembles «Н».

Original source details: GELFAND-P000409

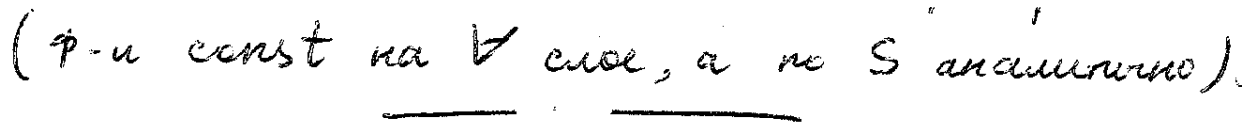
$H^{0,q}$ is clear (dz does not enter).

Original source details: GELFAND-P000409

Diagram and typed description. The last line of the leaf, with an unclosed parenthesis: $d\bar{z}$ remains, the number does not increase. (the holom. part increases.

GELFAND-P000410

S — a point, $f^\circ O_S$ — the sheaf of constants, $R^n f_* f^\circ O_S$ — cohom., depending holom. on the parameter.



Original source details: GELFAND-P000410

Diagram and typed description. Two separate horizontal segments below words in parentheses.

(functions const on $\forall$ fibre, but on S analytically).

Consider the Leray spectr. seq.:

$$E_1^{p,q} = R^q f_* \Omega^p_{Y/S} \longrightarrow R^n f_* f^\circ O_S$$

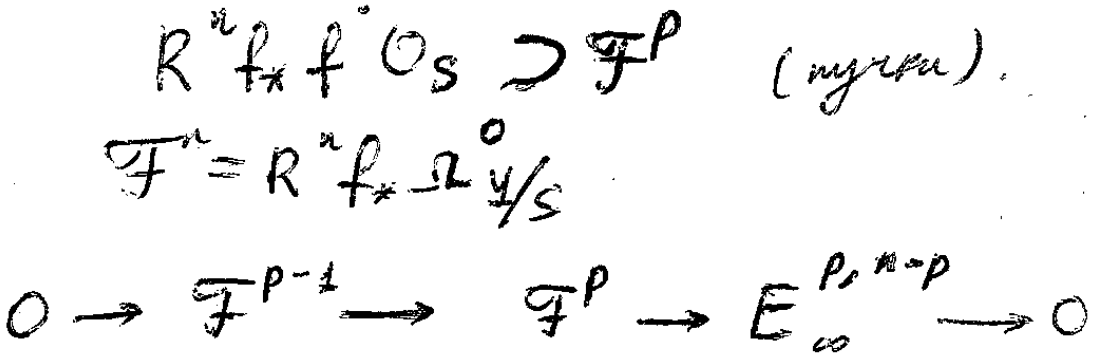
it induces a filtration

Manuscript notation. The subject is «она» (it).

она индуцирует фильтрацию

$R^n f_* f^\circ O_S \supset \mathcal{F}^p$ (пучки).

Original source details: GELFAND-P000410

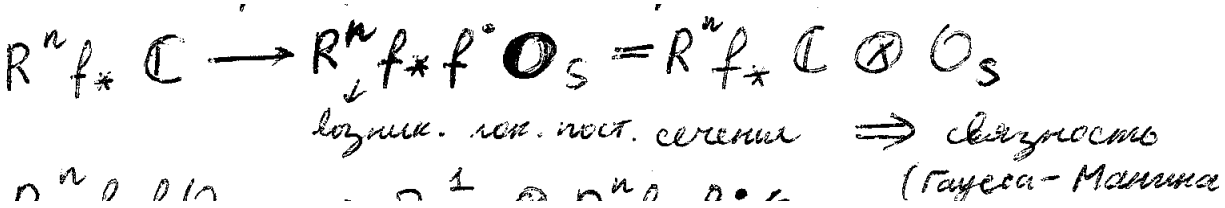


Original source details: GELFAND-P000410

Diagram and typed description. The filtration and exact sequence.

$$R^n f_* f^\circ O_S \supset \mathcal{F}^p \quad \text{(sheaves)}.$$
$$\mathcal{F}^n = R^n f_* \Omega^0_{Y/S}$$
$$0 \longrightarrow \mathcal{F}^{p-1} \longrightarrow \mathcal{F}^p \longrightarrow E_\infty^{p,n-p} \longrightarrow 0$$

The spectr. seq. degenerates (all differentials are zero) In the bundle $R^n f_* f^\circ O_S$ there is a filtration by holom. bundles with fibres F^p.

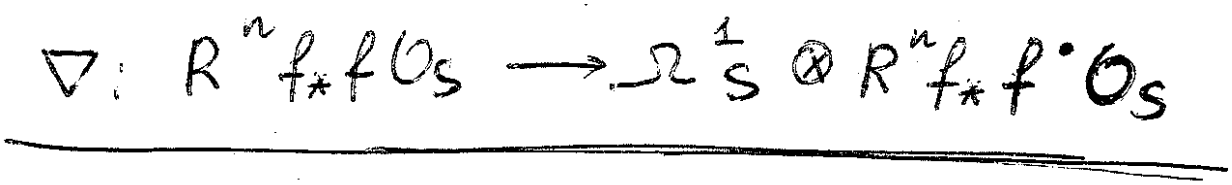


Original source details: GELFAND-P000410

Diagram and typed description. A short arrow from the formula to the lower line about sections.

$$R^n f_* \mathbb{C} \longrightarrow R^n f_* f^\circ O_S = R^n f_* \mathbb{C} \otimes O_S$$
$$\downarrow$$

loc. constant sections arise ⇒ a connection
(Gauss-Manin)



Original source details: GELFAND-P000410

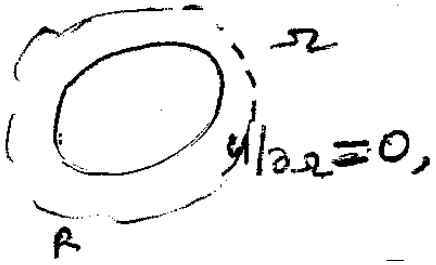
Diagram and typed description. The connection formula and several nearly coincident horizontal strokes below it.

$$\nabla : \quad R^n f_* f O_S \longrightarrow \Omega^1_S \otimes R^n f_* f^{\circ} O_S$$

GELFAND-P000411

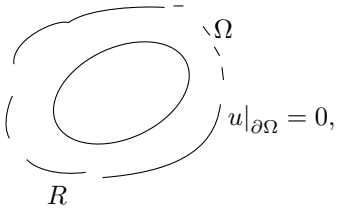
[source number: 52]

6/III. M. Sh. Birman.



Original source details: GELFAND-P000411

Diagram and typed description. A tilted solid oval inside a wider broken contour; Ω at the upper right, R at the lower left, and the condition at the right. The exact assignment of the region Ω to the contours is not established.



$u_{tt} = \Delta u.$

Original source details: GELFAND-P000411

Diagram and typed description. A phrase with a wavy underline.

Suppose Huygens' principle holds.

$\int_\Omega (|\nabla u|^2 + |u_t|^2)\, dx$ — is conserved.

Original source details: GELFAND-P000411

Diagram and typed description. Incoming and outgoing solutions — do not exhaust the subsp. with finite energy.

$K_+ \oplus K_- = \mathcal{H} \ominus K_0$; (consider the initial data of solutions). K_0 — solutions decaying exp. in time, but concentrated near the boundary. f, $\left.\frac{\partial f}{\partial t}\right|_{t=0} = f_1$. Initial data — pairs (f, f_1) in the energy metric. K_+ — an outg. wave and at $t = 0$ already concentrated outside the sphere of radius R.

Original source details: GELFAND-P000411

Diagram and typed description. A line with two separate underline segments and a dot between them.

see Lax and Phillips.

Words. (key)

1. Characteristic functions. — part 2 of the book by Gohberg and Krein.
2. Scattering,
3. “the second sheet of the resolvent”
4. S-matrix theory.
5. complex poles of

Original source details: GELFAND-P000411

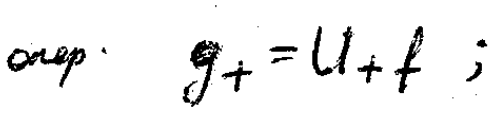
Diagram and typed description. Two separate segments below the words and a dot between them.

self-adjoint oper.

H is close to H_0 if $\forall f\ \exists g_\pm$.

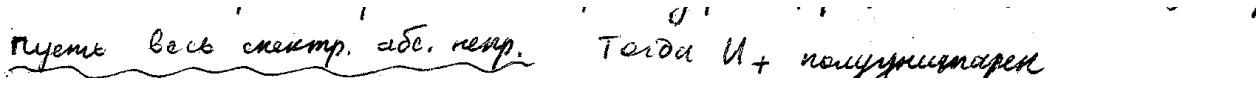
$$\|e^{-iH_0t}f - e^{-iHt}g_\pm\| \longrightarrow 0 \quad \text{as } t \longrightarrow \pm\infty.$$

On the abs. cont. spectrum there is an oper. $g_+ = U_+ f$; U_+ — a partial isometry.



Manuscript notation. Before $g_+ = U_+ f$ the abbreviation is «опер.» (oper.).

Original source details: GELFAND-P000411

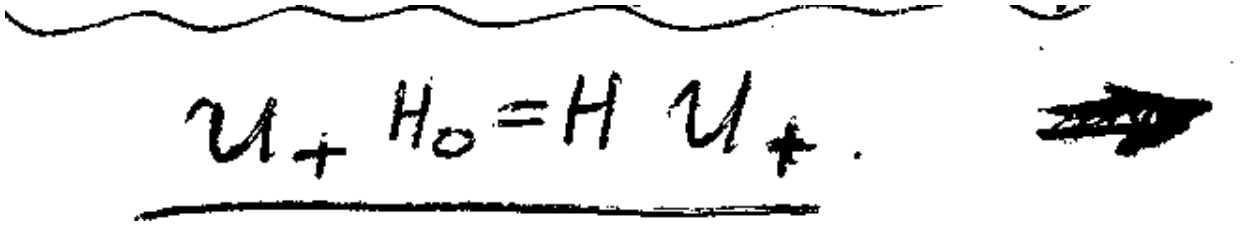


Original source details: GELFAND-P000411

Diagram and typed description. A phrase with a wavy underline.

Suppose the whole spectrum is abs. cont.

Then U_+ is semi-unitary



Original source details: GELFAND-P000411

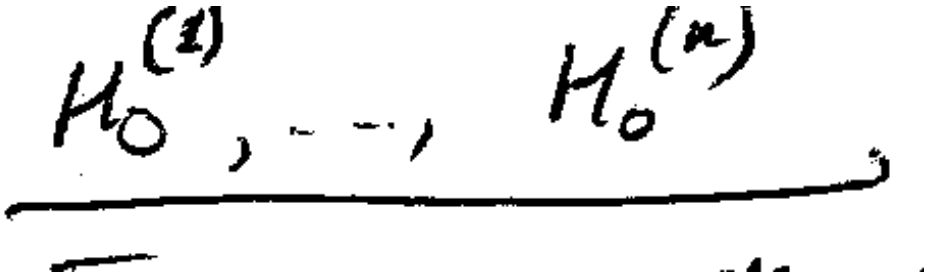
Uncertain reading. After the underlined identity is a separate densely crossed-out sign; its previous form is not established. The visible horizontal and oblique strokes are reproduced without choosing $\Rightarrow$.

$$U_+H_0 = HU_+.$$

H contains a part unitarily equiv. to H_0.

Problem: when does the range of U_+ exhaust the whole abs. cont. subsp. of H? This is single-channel scattering (in physical language). (i.e. $\exists$ complete wave operators)

multichannel: To describe H one needs



Original source details: GELFAND-P000411

Diagram and typed description. A list of operators with a lower horizontal stroke turned upward at the right.

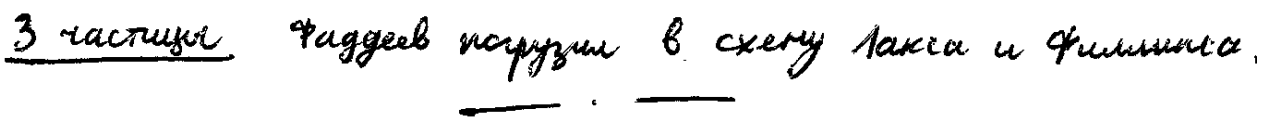
$$H_0^{(1)}, \ldots, H_0^{(n)}$$

so that the ranges of U^+ cover the whole space and are orthogonal. $H_0^{(i)}$, $i = 1, \ldots, n$ — a collection of operators; H is called admissible if the oper. $U_+^{(i)}$

such that their ranges $U_+^{(i)}$ are pairwise orthogonal. and gen.[?] the abs. cont. spectrum of H.

Original source details: GELFAND-P000412

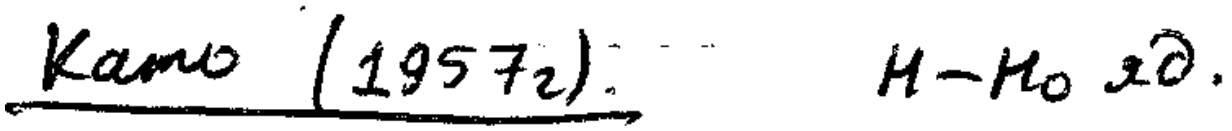
Uncertain reading. The abbreviation after "and" has a dense beginning and a distinguishable ending "...рожд." (...rozhd.). Provisionally "порожд." (gen.); it is not expanded into a full verb.



Original source details: GELFAND-P000412

Diagram and typed description. "3 particles" is underlined; below the middle of the line are two separate segments and a dot.

3 particles Faddeev embedded in the scheme of Lax and Phillips.



Original source details: GELFAND-P000412

Diagram and typed description. Below the name and date is a line ending in a point to the right.

Kato (1957). $H - H_0$ nuclear.

$\varphi(H) - \varphi(H_0) \in$ nuclear. — Birman, 1962.

Original source details: GELFAND-P000412

Uncertain reading. The second function sign, before (H_0), is heavily overwritten. Provisionally φ; a ψ-like form is not excluded. The clear signs in the later formula do not recover the previous strokes.

$S = U_+^* U_-$, — the scattering oper.

$S(\lambda)$ — the scattering matrix. (corresp. decomp. of H_0).

$S(\lambda) - I_\lambda \in$ nuclear. Birman — Krein

$$\det S(\lambda) = e^{-2\pi i \xi(\lambda)}$$

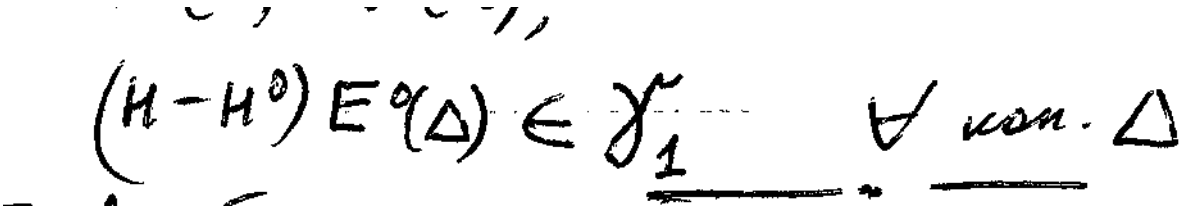
$$\mathrm{Sp}[\varphi(H) - \varphi(H_0)] = \int \frac{d\varphi(\lambda)}{d\lambda} \xi(\lambda)\, d\lambda.$$

(this defines $\xi(\lambda)$)

Original source details: GELFAND-P000412

Diagram and typed description. A dividing line.

$$D(H) = D(H_0);$$

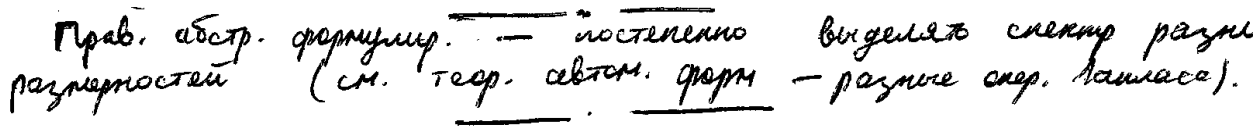


Original source details: GELFAND-P000412

Uncertain reading. A tall loop-like letter with a forked top and subscript 1. Provisionally $\mathcal{J}_1$; the candidate γ_1 remains. The two separate underline segments and dot are retained.

$(H - H^0)E^0(\Delta) \in \mathcal{J}_1 \qquad \forall$ bounded Δ

— sufficient for the existence of wave operators. The proper abstract formulation — gradually separate the spectrum of different dimensions (see



Original source details: GELFAND-P000412

Diagram and typed description. Two underline segments with a dot between them.

theory of coupled forms

— different Laplace operators).

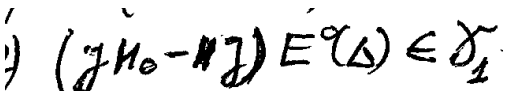
On a more general def. of wave operators.

$$U_\pm = \lim_{t \to \pm\infty} e^{iHt} J e^{-iH_0 t}$$

Corresp. to different spaces and classical scattering.

$$1) \quad D(H) = D(H_0)$$

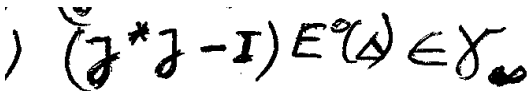
$$2) \quad (JH_0 - HJ)E^0(\Delta) \in \mathcal{J}_1$$



Original source details: GELFAND-P000412

Uncertain reading. In condition 2 is the same loop-like letter with subscript 1. Provisionally $\mathcal{J}_1$; γ_1 is possible. The letter is not established from the expected operator ideal.

$$3) \quad (J^*J - I)E^0(\Delta) \in \mathcal{J}_\infty$$



Original source details: GELFAND-P000412

Uncertain reading. In condition 3 is a loop-like letter with a lower horizontal loop. Provisionally $\mathcal{J}_\infty$; the candidate is γ_∞. The letter and reading of the lower loop remain qualified.

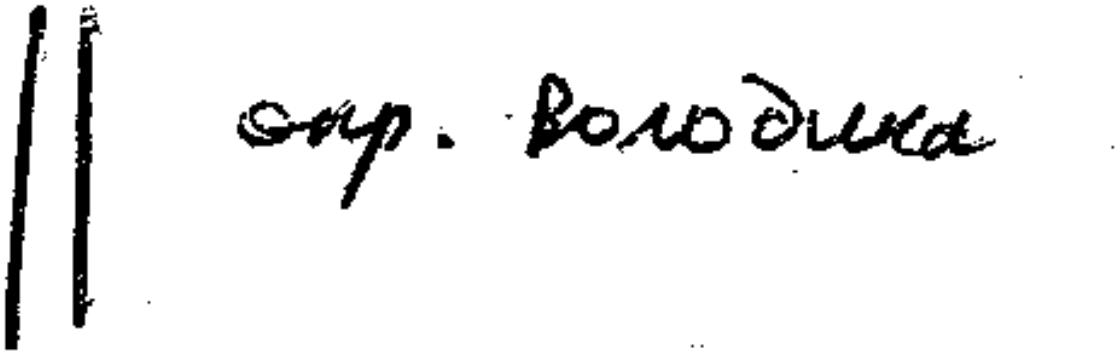
$J = U_+$ — trivially exists. One must ≈ guess U_+.

GELFAND-P000413

[source number: 53]

L. Vaserstein. Different K-theories.

1) A — an assoc. ring $\quad K_n = \pi_{n-1}(GL(A))$.



Original source details: GELFAND-P000413

Diagram and typed description. A double vertical stroke and a caption to the right.

Volodin's def.

def. of an abstr. simpl. scheme in $GL(A)$

K_1 coincides with Whitehead's; K_2 with Milnor's.

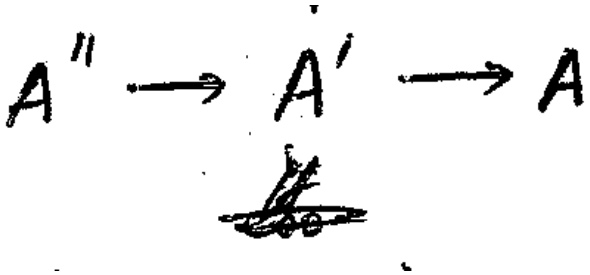
Original source details: GELFAND-P000413

Diagram and typed description. A long dividing line.

2) Derived functors (Swan; Gershtein) of GL.

$K_n^{GS} A$

Consider a free resolution of the ring A:

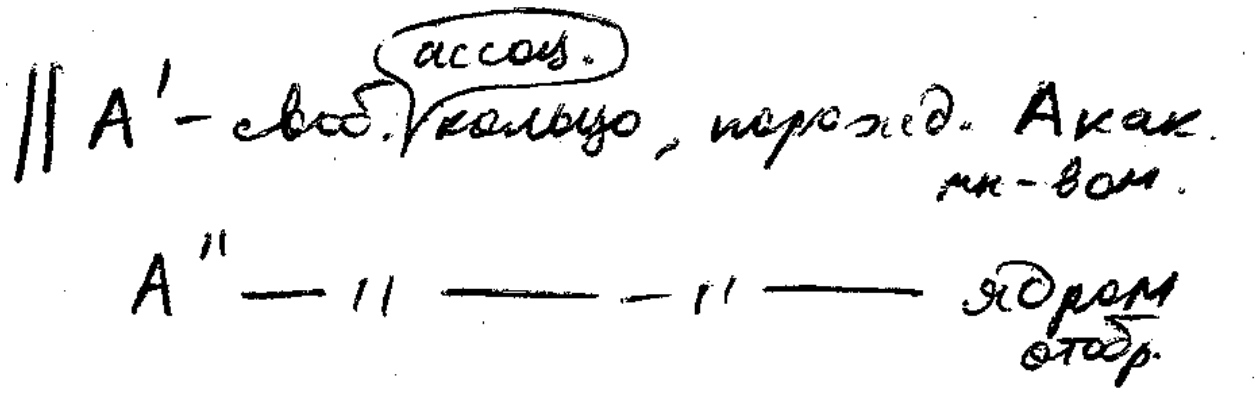


Original source details: GELFAND-P000413

Uncertain reading. Below A' is a crossed-out expression, provisionally K_0. Its meaning is not recovered from context.

$A'' \longrightarrow A' \longrightarrow A$

~~K_0~~



Original source details: GELFAND-P000413

Diagram and typed description. A double vertical stroke; “assoc.” in an enclosure with an insertion tip; ditto strokes in the second line.

assoc.

$||\ A'$ — a free ring, generated by A as a set.

A'' — $||$ — $||$ — the kernel of the map.

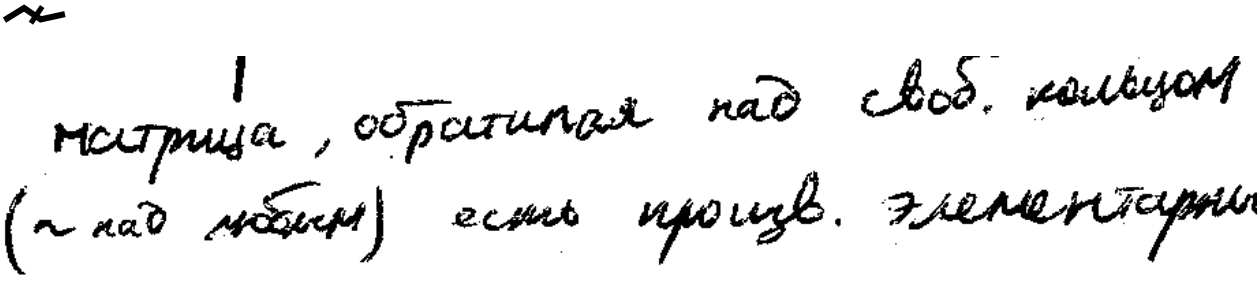
$$GL(A'') \longrightarrow GL(A') \longrightarrow GL(A)$$

is no longer exact..

K_1 coincides ([crossed-out mark] K_1 of a free ring $= 0$).

Original source details: GELFAND-P000413

Uncertain reading. Before the following K_1 is a separate densely inked mark. The previous sign is not determined.



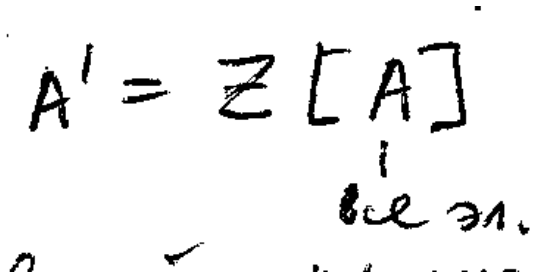
Original source details: GELFAND-P000413

Diagram and typed description. A separate vertical stroke above the marginal explanation.

|

a matrix invertible over a free ring
(∼ over any) is a product of elementary matrices.

(For simplicity the rings are taken without identity).



Original source details: GELFAND-P000413

Diagram and typed description. Below $[A]$ is a short stroke to the lower caption.

$A' = Z[A]$
|
all elem.

— noncommut. polynomials without a constant term.

The derived functor is defined using simplicial constructions.

Original source details: GELFAND-P000413

Diagram and typed description. Two horizontal segments with a detached dot between them.

$$1 \longrightarrow A'' \longrightarrow A' \longrightarrow A$$
$$K_{n+1}(A) = K_n(A'')$$

(an inductive formula): define everything through K_1.

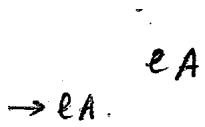
The problem of the Steinberg relations was done by L. V. and follows from the coincidence of these K-functors with Volodin's

A — a ring

MA — matrices over A.

ℓA — in $\forall$ row and column there are only finitely many nonzero elements.

$MA \longrightarrow \ell A$.

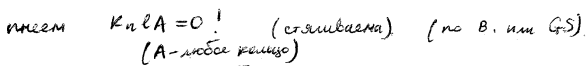


Original source details: GELFAND-P000413

Uncertain reading. In the first ring notation and in the adjacent arrow is a loop-shaped lowercase letter. Provisionally ℓ in both places; the candidate e remains. Similarity of the forms does not establish the choice of letter.

GELFAND-P000414

we have $K_n \ell A = 0$! (contractible) (according to V. or GS).

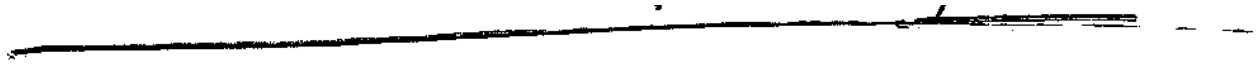


Original source details: GELFAND-P000414

Manuscript notation. The first line reads «имеем» (we have) and «стягиваема» (contractible).

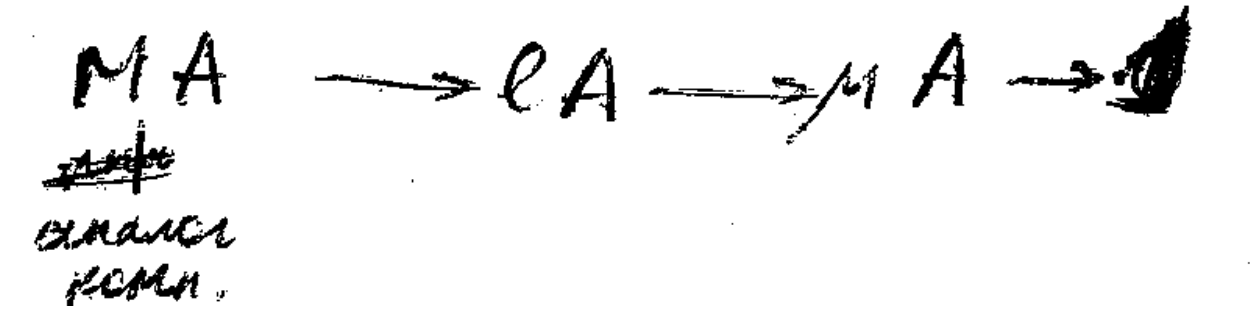
(A — any ring)

Proof $\sim$ Kuiper's theorem.



Original source details: GELFAND-P000414

Diagram and typed description. A horizontal line.



Original source details: GELFAND-P000414

Uncertain reading. At the end of the sequence is an inked sign with a rounded lower loop; 0 is only a candidate. Below MA are crossed-out strokes.

$$MA \longrightarrow \ell A \longrightarrow \mu A \longrightarrow \emptyset$$

analogue of compact.

Original source details: GELFAND-P000414

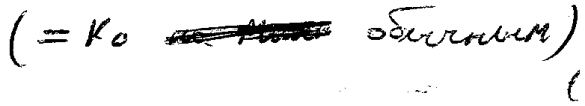
Undeciphered passage. The crossed-out word immediately below MA is not read in full. The candidates "матр." (matr.) and "комп." (compact.) are not asserted.

(matrices with finitely many nonzero elements)

$$\underline{\underline{K_n A = K_n MA.}}$$

$$\underline{\underline{K_n A = K_{n+1} \mu A.}} \quad \text{(from the exact sequence)}$$

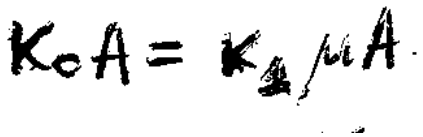
From this one obtains K_0 ($= K_0$ [crossed out: ...] ordinary) — full (a free module $\neq 0$).



Original source details: GELFAND-P000414

Undeciphered passage. The phrase before "ordinary" inside the parenthesis is not recovered in full.

$$K_0 A = K_1 \mu A.$$



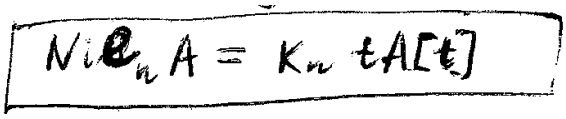
Original source details: GELFAND-P000414

Uncertain reading. The subscript on the second K is reinforced/overwritten. Provisionally 1; the former form is not established from the shift formula.

From this, K_n $(-\infty < n < \infty)$.

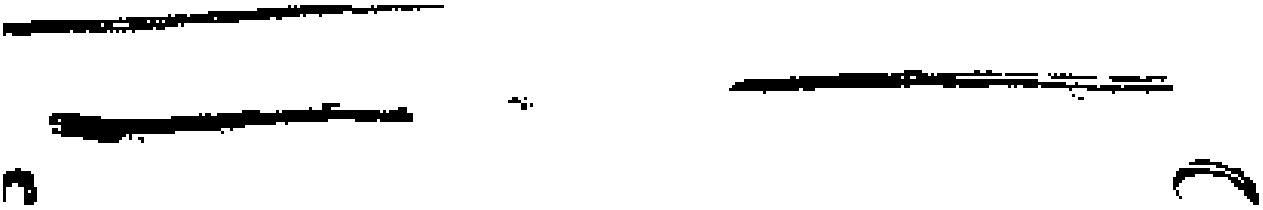
This coincides with Bass's K_n.

$$\boxed{\mathrm{Nil}_n\, A = K_n\, tA[t]}$$



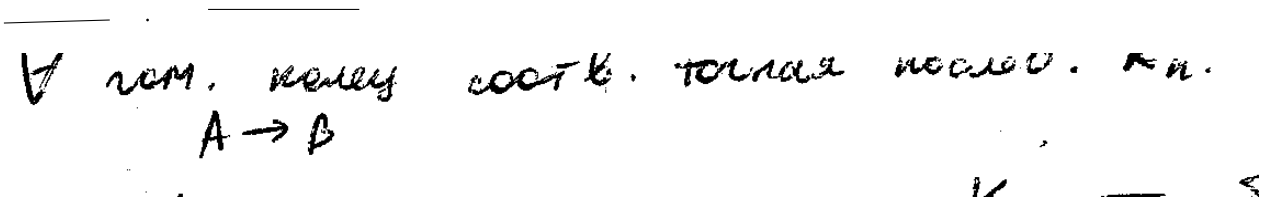
Original source details: GELFAND-P000414

Uncertain reading. The last letter of "Nil" is heavily retraced. "Nil" is retained provisionally.



Original source details: GELFAND-P000414

Diagram and typed description. Two separate horizontal segments, the right one higher, with a dot between them.



Original source details: GELFAND-P000414

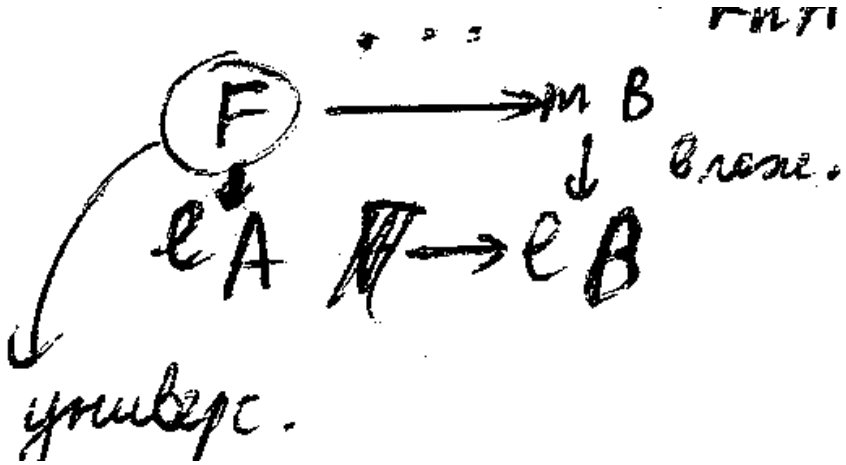
Diagram and typed description. Below "hom." is the inserted line $A \longrightarrow B$.

To $\forall$ ring hom. corresp. an exact seq. of K_n.

$A \longrightarrow B$

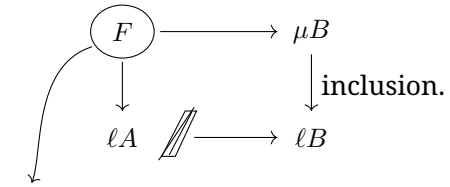
and a new ring, so that relative K_n = absolute of the new ring

$$A \xrightarrow{f} B$$
$$\cdots \longrightarrow K_n A \longrightarrow K_n B \longrightarrow K_n F \longrightarrow K_{n-1} A \longrightarrow \dots$$



Original source details: GELFAND-P000414

Diagram and typed description. A square with circled F, four arrows and an arc to "universal". Before the lower arrow is a crossed-out mark; the former sign is not recovered. The upper-right letter is μ-/m-like: provisionally μB; mB and MB are not excluded. The notation is not replaced from the expected inclusion.



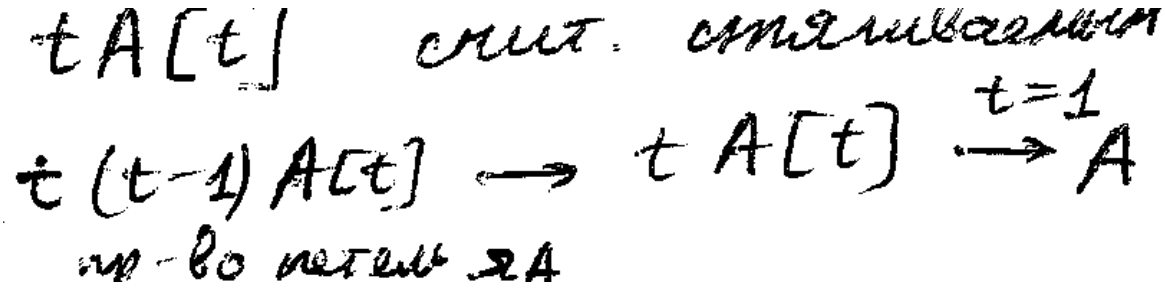
(fibred product; inverse image)

pullback.

In L. V.'s opinion, Quillen's theory coincides with these.

Karoubi's theory does not coincide.

$tA[t]$ is considered contractible



Original source details: GELFAND-P000414

Diagram and typed description. A lower caption for $t(t-1)A[t]$ and the label $t=1$ at the right arrow.

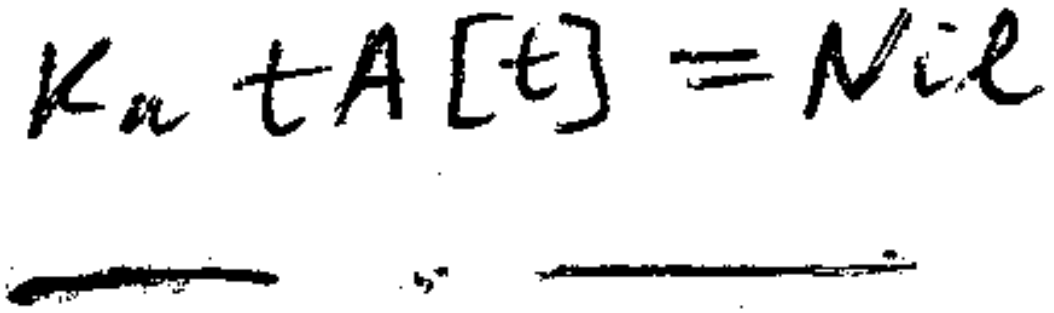
$$t(t-1)A[t] \longrightarrow tA[t] \xrightarrow{t=1} A$$

loop space ΩA

GELFAND-P000415

[source number: 54]

$K_{n+1}A = K_n\Omega A$: According to Karoubi.
In the present K-theor.



Original source details: GELFAND-P000415

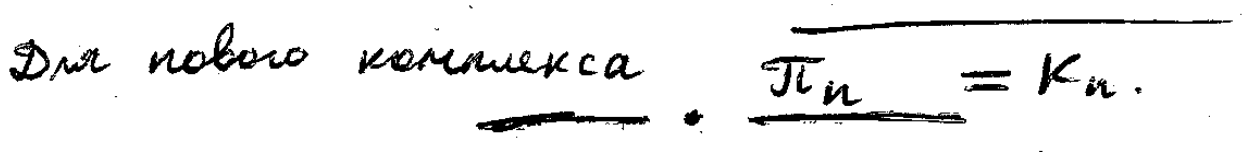
Diagram and typed description. Below the formula are two separate horizontal strokes and a small mark between them.

$K_n\, tA[t] = Nil$

On Quillen's theorem:

Let G be any group, $E = [G, G]$. and $E = [E, E]$. Consider the universal classif. space BG with a cell structure. Publ. Math., 34. $\pi_1 BG = G$, the others 0. $(BG = K(G, 1))$.

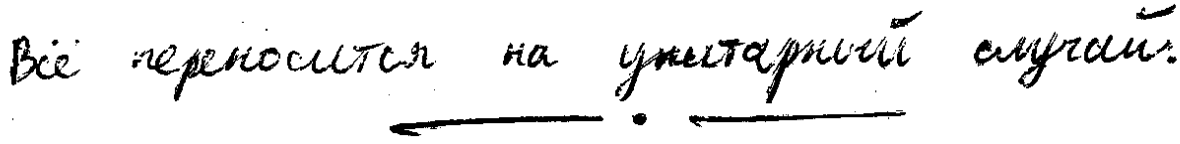
Make minimal cell attachments, turning it into a group (H-space). Homology does not change. E is killed.



Original source details: GELFAND-P000415

Diagram and typed description. Below the end of "For the new complex" is a separate stroke, then a dot and a stroke below π_n.

For the new complex $\pi_n = K_n$.



Original source details: GELFAND-P000415

Diagram and typed description. Below the line are two separate strokes and a dot between them; both strokes point left.

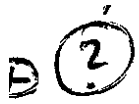
Everything carries over to the unitary case.

Theor. Valonera/Vagonera[?] on Laurent extensions.

Original source details: GELFAND-P000415

Uncertain reading. The underlined surname is provisionally “Валонера/Вагонера” (Valonera/Vagonera); the middle letters are joined. The full reading is not established.

$$K_2A[t,t^{-1}] = K_2A \oplus K_1A \oplus \textcircled{2}.$$

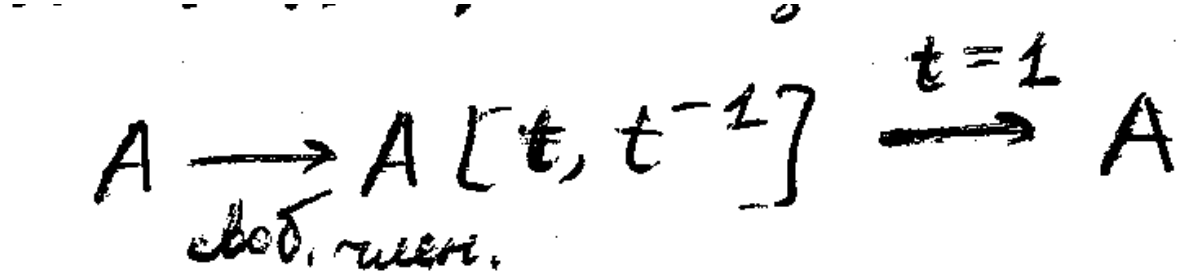


Original source details: GELFAND-P000415

Uncertain reading. The upper curve of the last circled sign is 2-like, with a separate dot below. The visible strokes are reproduced; “?” remains a candidate, and no numeric summand is asserted.

the theor. holds $\forall n$, but from this case all the others follow.

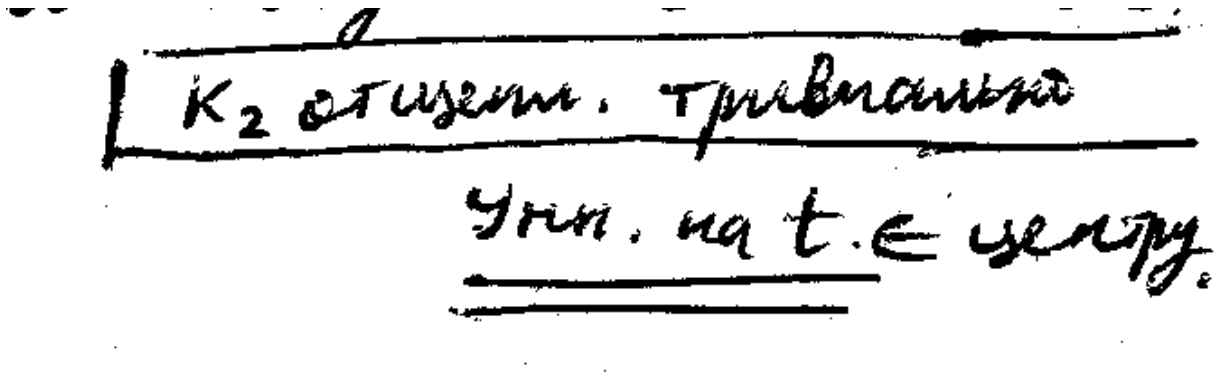
Proof.



Original source details: GELFAND-P000415

Diagram and typed description. The first caption is below its arrow, the second above its arrow.

$$A \xrightarrow[\text{constant term.}]{} A[t,t^{-1}] \xrightarrow{t=1} A$$

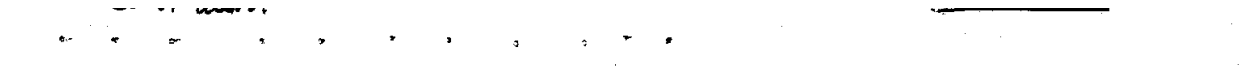


Original source details: GELFAND-P000415

Diagram and typed description. At right is an enclosure with upper, left and lower sides; the right side is open. There are two strokes below the following words.

K_2 relative is trivial

Mult. by $t \in$ the centre.

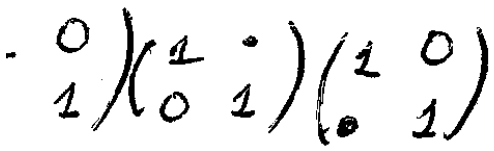


Original source details: GELFAND-P000415

Undeciphered passage. Between the diagram and the matrices is a row of eleven small separate dots/short strokes. Their purpose and any earlier inscription are not established.

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & \cdot \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \circ & 1 \end{pmatrix}.$$

(Whitehead's formula). (the dots are recovered uniquely).



Original source details: GELFAND-P000415

Uncertain reading. The upper-right entry of the third matrix is a dot. The lower-left entry of the fourth is a thick 0-like dot with a small opening; its contour is reproduced, but its value is unwritten.

g is a matrix over A. (defines an elem. of K_1)

$$\begin{pmatrix} gt & 0 \\ 0 & g^{-1}t^{-1} \end{pmatrix} \qquad \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \qquad \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}.$$

Original source details: GELFAND-P000415

Uncertain reading. The upper-left entry of the middle matrix is overwritten; provisionally t, with the earlier strokes unrecovered. There are no equality signs between the three matrices.

Now decompose them into elem.

Original source details: GELFAND-P000415

Uncertain reading. After the abbreviation "элем." (elem.) is a faint slanted chain of broken strokes with two darker spots. Its connection to the inscription and the punctuation are not established.

one obtains relations —

— an elem. of K_2

GELFAND-P000416

The converse: An elem. of $K_2A[t,t^{-1}]$.

Original source details: GELFAND-P000416

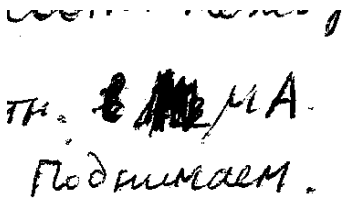
Diagram and typed description. The underlined “Laurent series” and a horizontal arrow with a wavy beginning.

Laurent series. ⇝ Mult. by it — an infinite matrix

Original source details: GELFAND-P000416

Undeciphered passage. Before “One obtains relations” is a separate densely crossed-out formula. K-like, a lower 2-like and A-like fragments are distinguishable; the full inscription is unrecovered.

One obtains relations between infinite matrices. Cut off ⇒ relations $\ell A/\mu A$. (mod. finite matrices).



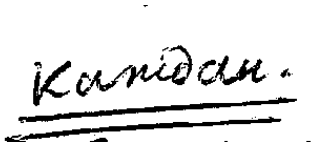
Original source details: GELFAND-P000416

Uncertain reading. The numerator is overwritten: provisionally ℓA, with MA possible beneath the dense earlier sign; the denominator is μA. The earearlier strokes are unrecovered.

~~ℓA~~$/\mu A$

We need $\ell A/\mu A$. Lift.

Kazhdan/Kazdan[?].



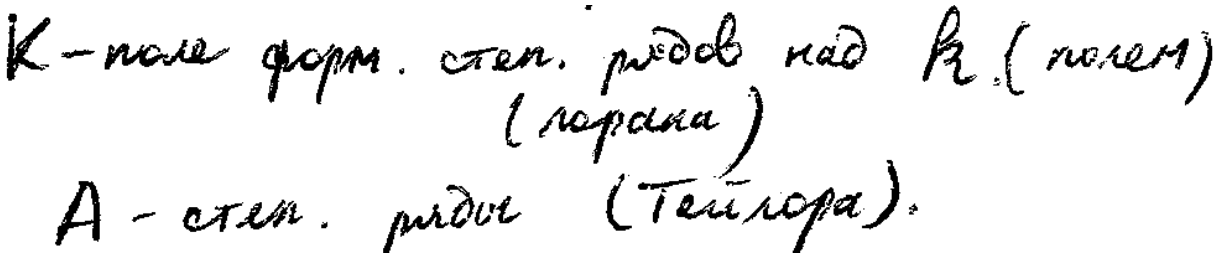
Original source details: GELFAND-P000416

Uncertain reading. The doubly underlined heading is provisionally “Каждан/Каздан” (Kazhdan/Kazdan); the join after “Ка” is ambiguous.

Original source details: GELFAND-P000416

Uncertain reading. The crossed-out heading is provisionally “Let there be[?] 2 formal power series”; “Let 2 formal power series be given” is also possible. The second word merges with the crossing stroke.

~~Let there be[?] 2 formal power series~~



Original source details: GELFAND-P000416

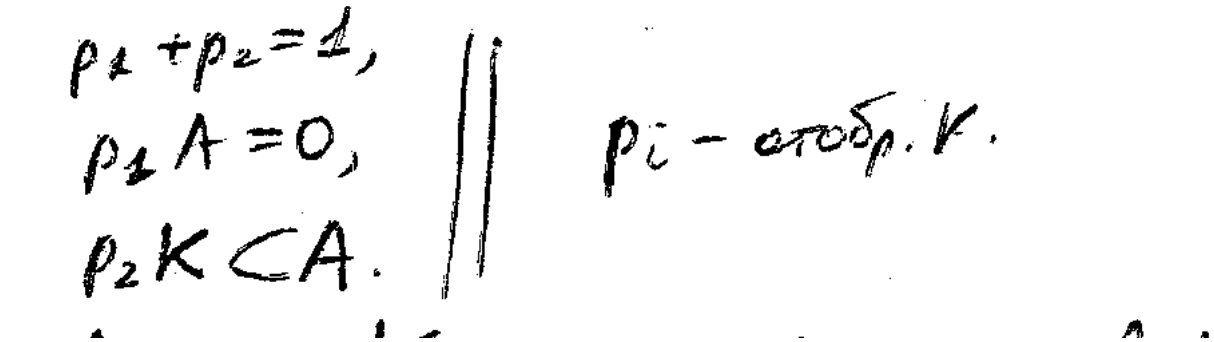
Diagram and typed description. A parenthetical caption is inserted below “power series”.

K is the field of formal power series over k (a field).

(Laurent)

A is power series (Taylor). $k \subset K \supset A$.

Let



Original source details: GELFAND-P000416

Diagram and typed description. To the right of the three conditions are two separate vertical rules and a caption.

$$\begin{aligned} p_1 + p_2 &= 1, \\ p_1 A &= 0, \\ p_2 K &\subset A. \end{aligned}$$

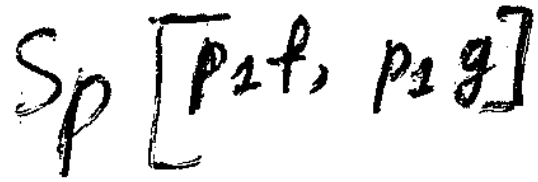
p_i — maps of K.

Let $f, g \in K$. The residue of $\underline{f\,dg}$ — ?

Original source details: GELFAND-P000416

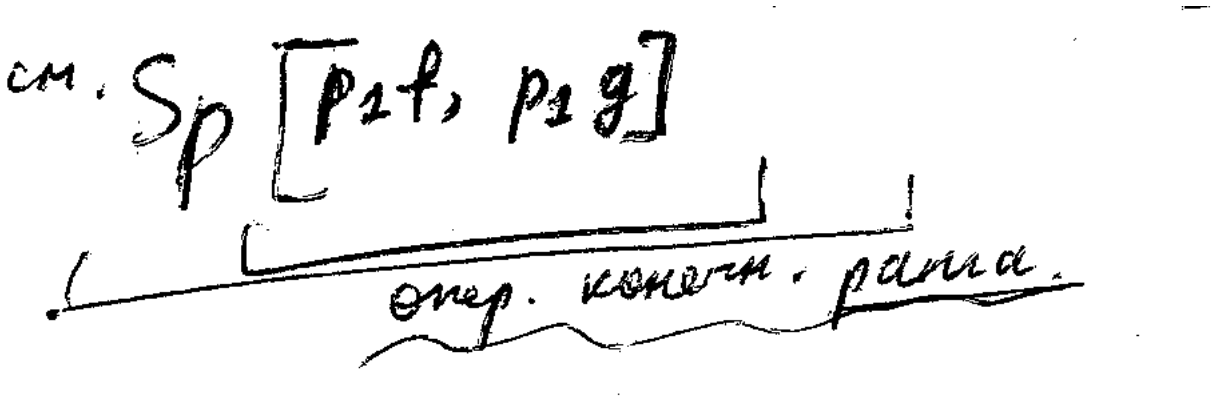
Uncertain reading. The parenthetical note is provisionally “(без парам.!)” (without param.!); the abbreviation is unexpanded. The readings “без лоран.!” (without Laurent.!) and “без полюса!” (without a pole!) are weaker candidates; the first sign of the abbreviation is closer to “п”.

Consider $Sp[p_1 f, p_1 g]$.



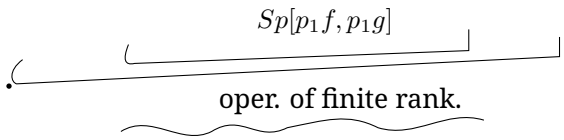
Uncertain reading. In both arguments after p are small lower signs, provisionally $1,1$; $2,1$ and $1,2$ are possible. The first sign especially merges with p.

Original source details: GELFAND-P000416



Original source details: GELFAND-P000416

Diagram and typed description. Below the commutator are two long lower contours with short raised ends. There is a separate dot at the left end of the outer contour; below is a caption with a wavy underline.



GELFAND-P000417

[source number: 55]

13/III Prof. Morava. Work by G. Segal and K°[?].

раб. G. Segal и K°.

Uncertain reading. A small circle is written above K. It could be the abbreviation for “and Co.” or the notation K^0.

Original source details: GELFAND-P000417

C is a categ., BC the simpl. set corresponding to this category. the points of the set are the objects of this category.

Гельфанд. Точки

C_0: $[n] = \{$

Original source details: GELFAND-P000417

Diagram and typed description. At the left above the beginning of the C_0 line is an underlined surname.

Gelfand.

C_0: $[n] = \{0, 1, \ldots, n\}$. — objects of an auxiliary categ.

Morphisms: $[k] \xrightarrow[\varphi(i)]{} [n]$, $\varphi(1) \leq \ldots \leq \varphi(k)$.

$[k] \xrightarrow[\varphi(i)]{} [n]$ $\quad \varphi(1) \leq \ldots \leq \varphi(k)$.

Uncertain reading. In the first argument of φ is a thick 1; an earlier overwritten 0 is possible underneath.

Original source details: GELFAND-P000417

(see Gabriel, Tsisman[?])

(см. главы. Габриэль, Цисман)

Uncertain reading. The second surname is provisionally “Цисман” (Tsisman); the joined middle letters are ambiguous, and “Зисман” (Zisman) is possible.

Original source details: GELFAND-P000417

(см. Контравар. Функт

Original source details: GELFAND-P000417

Diagram and typed description. Above the beginning of the next line, “Contravar.” is outlined; the lower contour has a narrow projecting pointer.

Contravar.

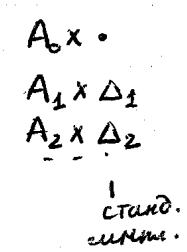
A functor A from C_0 to the categ. K is called a simplicial object in the cat. K. A_k is the image of the object $[k]$. $A_0, A_1, \ldots, A_k, \ldots$ and a system of morphisms. If K is the cat. of sets, then this is

Original source details: GELFAND-P000417

Undeciphered passage. An earlier short word is densely crossed out; its original letters are not fully recovered. A lower loop is visible.

a simplicial set.

The system A_0 is a set of points. The system A_1 — $-\|-$ of segments. and so on.



Original source details: GELFAND-P000417

Diagram and typed description. Three product lines; below are two short strokes and a dot. Under the last triangle are a short vertical stroke and a caption.

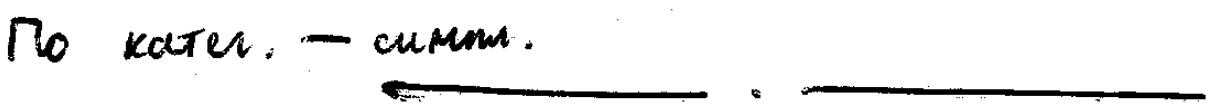
$A_0 \times \cdot$

$A_1 \times \Delta_1$

$A_2 \times \Delta_2$

stand.
simpl.

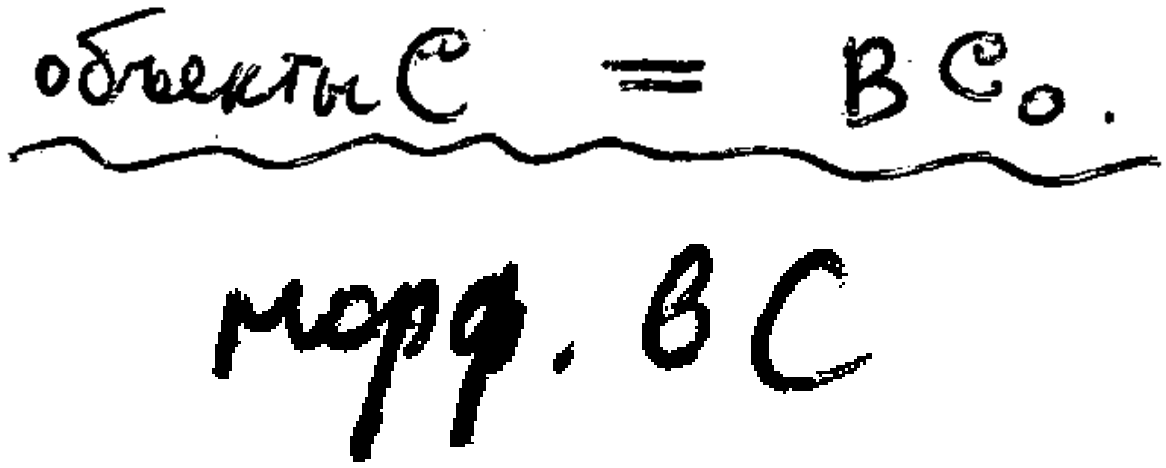
and glue in the natural way. Obtain an actual complex.



Original source details: GELFAND-P000417

Diagram and typed description. Under the line is a long leftward arrow; to the right of its left part is a gap with a separate dot, then the stroke continues.

From a categ. — a simpl.

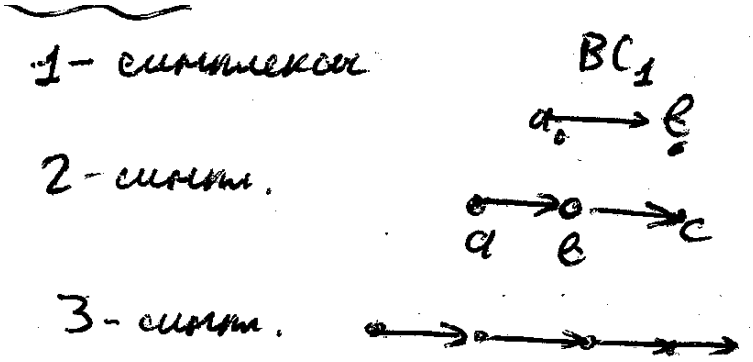


Original source details: GELFAND-P000417

Diagram and typed description. There is a wavy stroke beneath the equality. The next line is "morphisms in C".

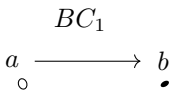
objects $C = BC_0$.

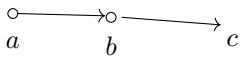
morphisms in C

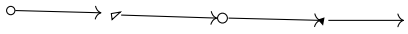


Original source details: GELFAND-P000417

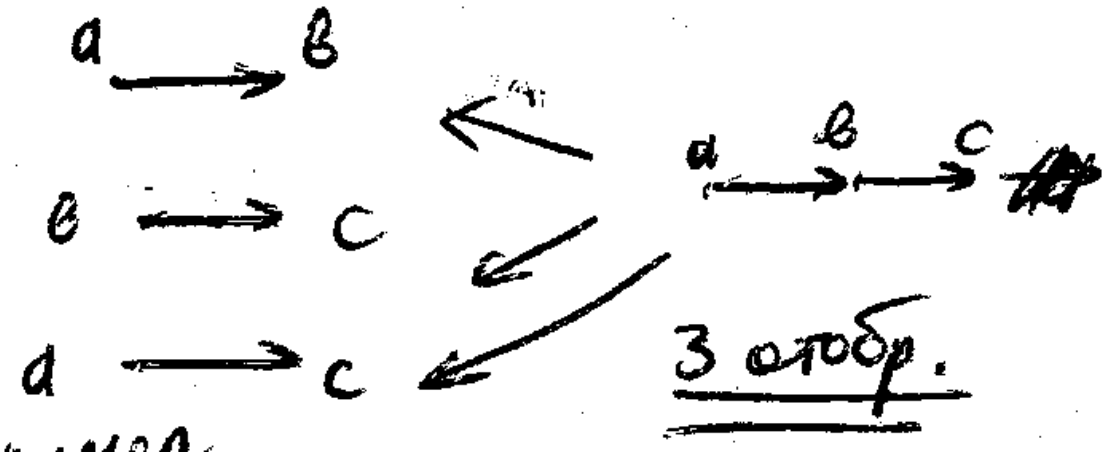
Diagram and typed description. Three lines with captions at the left. In the first, BC_1 is above the arrow; the right letter is provisionally b, possibly an e-like letter. Below a is a small closed contour, and below the right letter a separate thick stroke; the meanings of the lower signs are undetermined. In the second are two round marks and an arrow to c; the middle letter is also provisionally b, possibly e-like. In the third are four marked positions, the third and fourth merging with arrowheads; after the fourth is a further rightward arrow, with no added endpoint.

1-simplices 

2-simpl. 

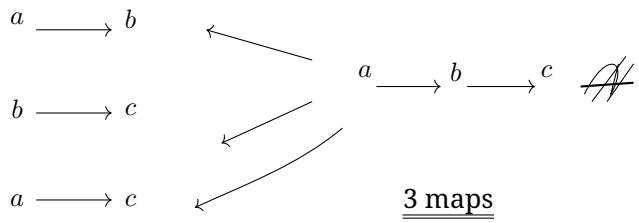
3-simpl. 

GELFAND-P000418



Original source details: GELFAND-P000418

Diagram and typed description. At the right is the chain $a \to b \to c$, with a separate densely crossed-out a-like sign after c. Three oblique arrows point left to the lines $a \to b$, $b \to c$, $a \to c$. Beneath the right chain, "3 maps" is doubly underlined. The earlier overlapping strokes of the last sign are not fully recovered.

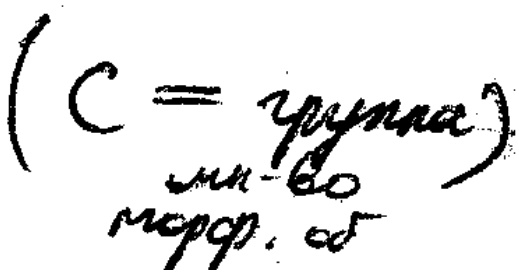


Example.

If C has one object, ~~[?]~~ $\forall$ morphism — isomorphism.

Original source details: GELFAND-P000418

Undeciphered passage. After "object" an earlier short inscription is crossed out; its letters are unrecovered. Below it an outlined e is inserted.



Original source details: GELFAND-P000418

Uncertain reading. Below the parenthetical inscription "$C =$ group" is the caption "set of morph. ob[?]". The final joined sign is provisionally read "об" (ob); the abbreviation is unexpanded and the reading is undetermined.

($C =$ group
set of
morph. ob[?])

Original source details: GELFAND-P000418

Diagram and typed description. After $BC_1 = G$, $Mor(e,e)$ is written separately. Below are three short faint marks and a separate dot.

$BC_1 = G$ $\qquad\qquad$ $Mor(e,e)$

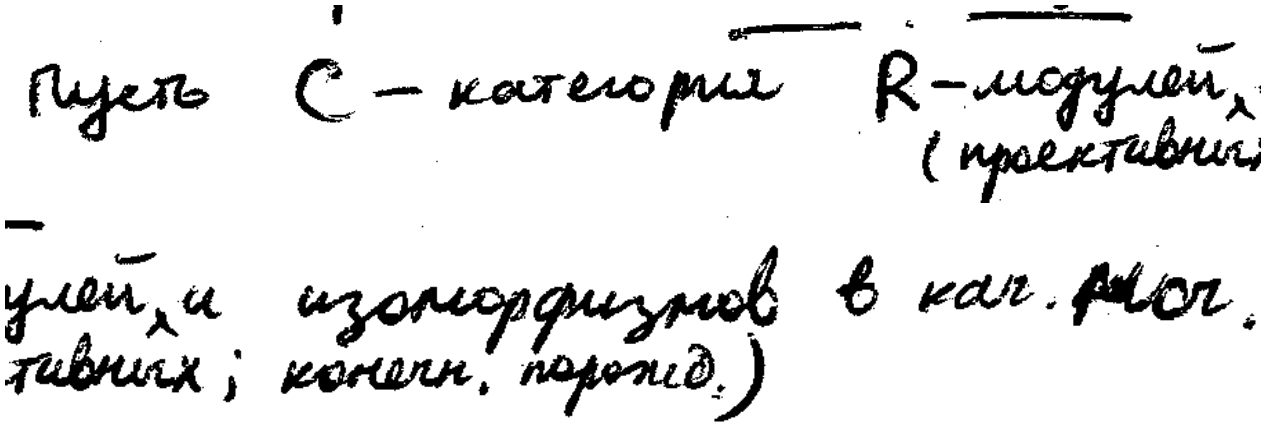
$$BC_n = G \times G \times \ldots \times G.$$

def. $\Rightarrow$ Milnor's def. of the classifying space of the group G.

Original source details: GELFAND-P000418

Undeciphered passage. Before "of the group", an earlier short word is densely crossed out and not fully recovered.

since $BC_n \times \Delta_n$ and gluing. G a top. gr. $\Rightarrow$ one may equip with a topology.



Original source details: GELFAND-P000418

Diagram and typed description. Below the R-modules is a parenthetical insertion; between "modules" and "and" is a small angular stroke.

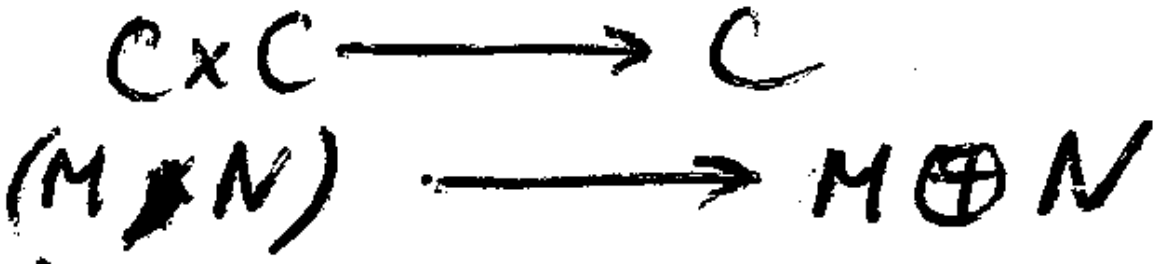
Let C be the category of R-modules and isomorphisms in the cat. Mod.

(projective; finitely generated.)

Original source details: GELFAND-P000418

Uncertain reading. The category name is provisionally "Мод." (Mod.); the initial letter merges with dense earlier strokes, and the earlier inscription is unrecovered.

BC.



Original source details: GELFAND-P000418

Diagram and typed description. Two lines of arrows; in the lower one the sign between M and N is overwritten on earlier strokes.

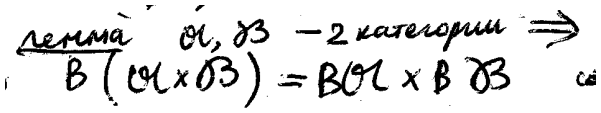
$C \times C \longrightarrow C$

$(M \times N) \longrightarrow M \oplus N$

Original source details: GELFAND-P000418

Uncertain reading. In the lower left argument the sign is provisionally $\times$; one diagonal stroke is very thick, and the earlier sign is unrecovered.

Lemma. $\mathcal{A}, \mathcal{B}$ are 2 categories $\Rightarrow B(\mathcal{A} \times \mathcal{B}) = B\mathcal{A} \times B\mathcal{B}$.



Manuscript notation. The category symbols are script capitals $\mathcal{A}$ and $\mathcal{B}$.

Original source details: GELFAND-P000418

Original source details: GELFAND-P000418

Diagram and typed description. In the left margin, “Proof” is crossed out.

~~Proof~~

Original source details: GELFAND-P000418

Undeciphered passage. At the right after the product formula an earlier phrase is densely crossed out and not fully recovered. A separate lower loop is visible beneath the stroke.

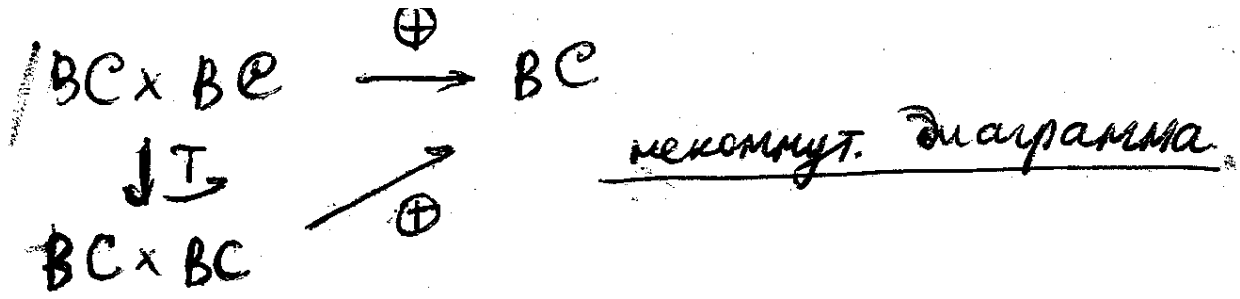
$BC \times BC \longrightarrow BC$, i.e. BC is an H-space.

Original source details: GELFAND-P000418

Undeciphered passage. After the doubly underlined heading a short inscription is densely crossed out; its letters are unrecovered. At the left is a separate angular mark.

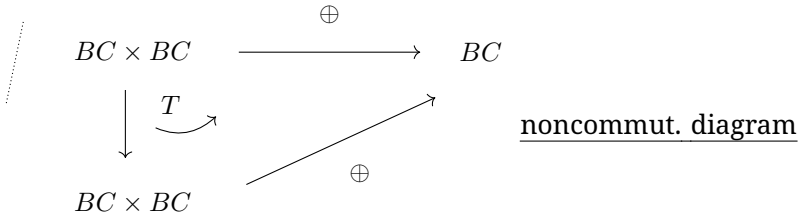
Remark:

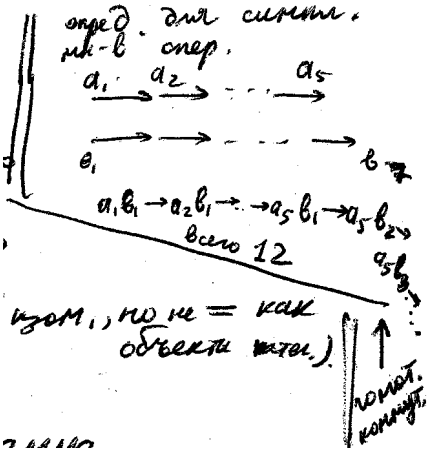
$M \oplus N \neq N \oplus M$ (isom., but not $=$ as objects of the categ.)



Original source details: GELFAND-P000418

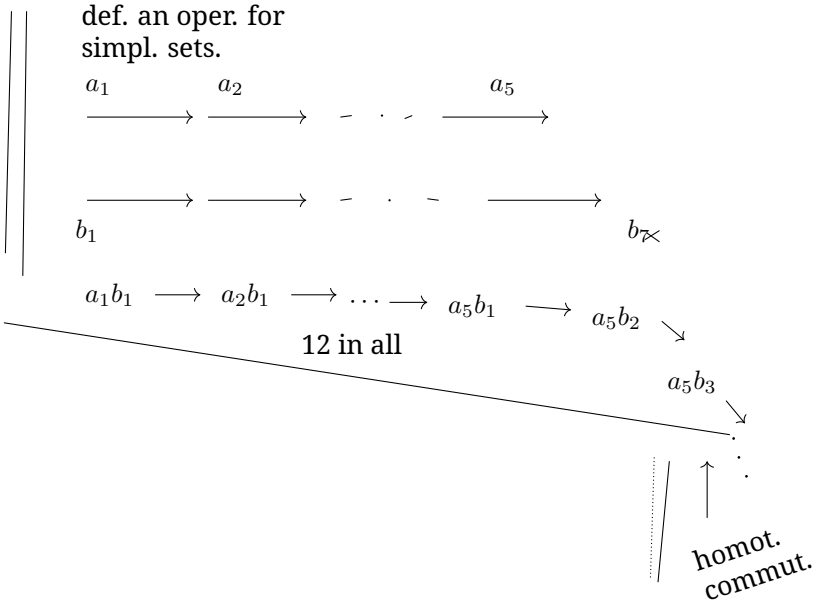
Diagram and typed description. The top and oblique arrows of the triangular diagram are labelled $\oplus$; the vertical arrow points downward and is labelled T. Below T is a separate short curved rightward arrow; at the left is a faint broken stroke.





Original source details: GELFAND-P000418

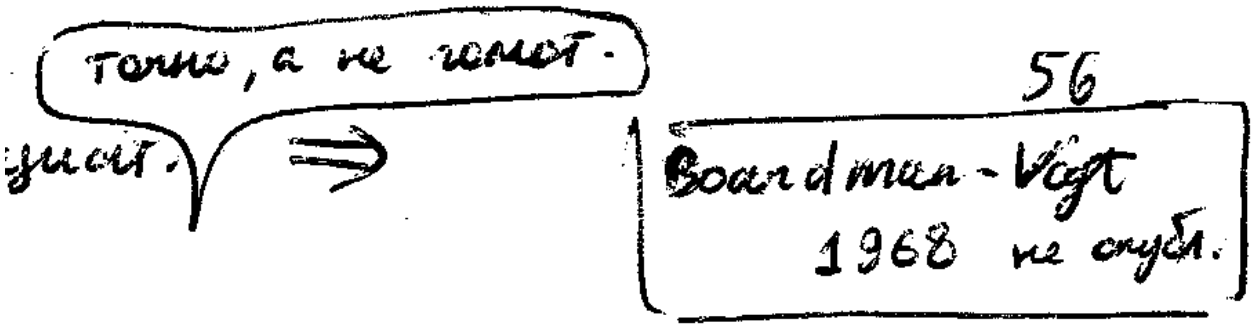
Diagram and typed description. The right-hand insertion has two left vertical rules and an oblique lower boundary. Above the arrows of the upper line are the labels a_1, a_2, a_5; below the outer arrows of the second, b_1, b_7, with the final lower index overwritten and provisionally read 7. Next is written $a_1b_1 \to a_2b_1 \to \cdots \to a_5b_1 \to a_5b_2$, with a_5b_3 below, an arrow and dots. The caption is “12 in all”. At the lower right margin are two further rules, an upward arrow and “homot. commut.”.



GELFAND-P000419

[source number: 56]

Remark. If $X \times X \longrightarrow X$ is commut. and associat.



Original source details: GELFAND-P000419

Diagram and typed description. Above the line, “exactly, and not up to homot.” is outlined with a lower projecting pointer; to the right are a separate frame and the sign ⇒.

exactly, and not up to homot.

⇒

Boardman–Vogt
1968 unpubl.

Original source details: GELFAND-P000419

Diagram and typed description. The sign between X and the product has a wavy upper stroke and two lower horizontal strokes.

$\Rightarrow X \cong \prod_n K(\pi_n, n)$

Theorem. Let $X \times X \longrightarrow X$

Original source details: GELFAND-P000419

Uncertain reading. The first word is provisionally “гомот.” (homot.); the initial letters are overwritten and the earlier strokes cannot be separated.

~~го~~mot.[?]

commut. and assoc. in a canonical way. Then ∃ a classif. space;

Original source details: GELFAND-P000419

Uncertain reading. The overwritten word is provisionally “оно” (it); “это” (this) is possible. The initial strokes merge.

~~Ait~~[?]

is also an H-space with a canonically homot. commut. and assoc. law. BX is this classif. space. $B(BX) \underset{def}{=} B^2X$.

Original source details: GELFAND-P000419

Diagram and typed description. At the left is a vertical sequence with two groups of continuation dots.

C

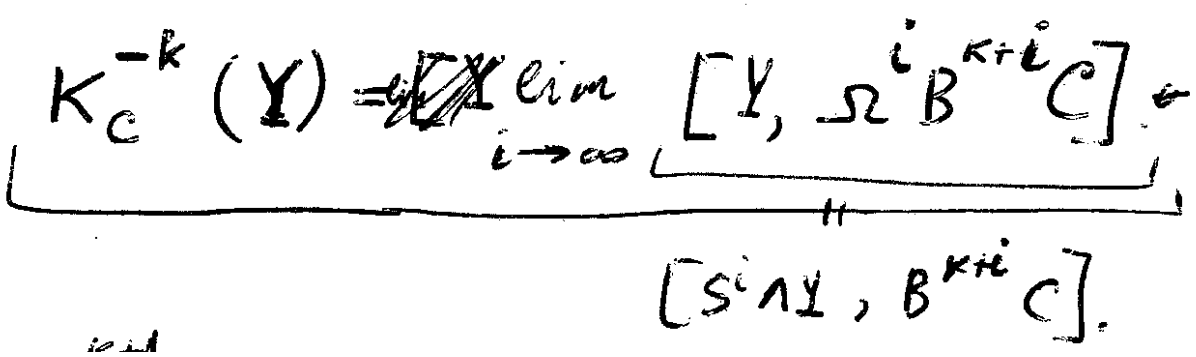
BC

B^2C

$\vdots$

B^kC

$\vdots$



Original source details: GELFAND-P000419

Diagram and typed description. The definition has two lower contours. Two short vertical strokes cross the long contour; beneath is a separate bracketed expression. The cancelled intermediate inscription is provisionally $[Y]$; the small right-hand sign is rendered by separate strokes.

$$K_C^{-k}(Y) = \cancel{[Y]} \lim_{i\to\infty} [Y, \Omega^i B^{k+i} C]$$

$$[S^i \wedge Y, B^{k+i} C]$$

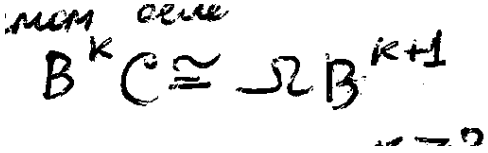
Original source details: GELFAND-P000419

Uncertain reading. Between the equality and lim is an earlier crossed-out bracketed inscription; square brackets and a Y-like letter are visible, provisionally $[Y]$. A fuller inscription is undetermined.

Original source details: GELFAND-P000419

Uncertain reading. After the right bracket is a small sign with a loop, a long rightward stroke and a separate upper dot; below is a further separate mark. An i-like inscription is possible, but its meaning and index position are undetermined.

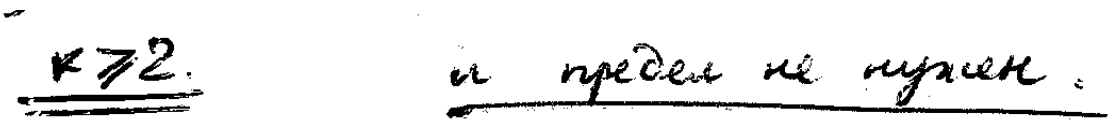
In fact



Original source details: GELFAND-P000419

Uncertain reading. At the right after the exponent $k+1$, the letter C is unwritten. The relation sign has a wavy stroke above two horizontal strokes.

$$B^k C \cong \Omega B^{k+1}$$



Original source details: GELFAND-P000419

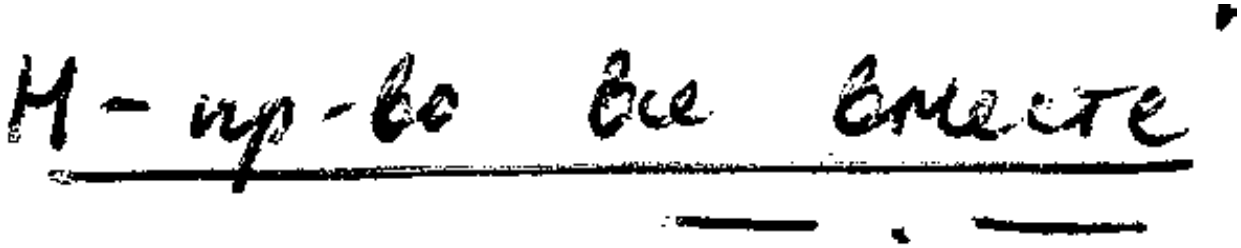
Diagram and typed description. The condition at the left is doubly underlined; at the right the entire phrase, including the initial “and”, is underlined.

$\underline{\underline{k \geq 2.}}$ $\underline{\text{and the limit is unnecessary.}}$

Example 1. $Y = \{*\}$, C the cat. of projective R-mod. $\Rightarrow$ the K-functor according to Quillen.
Example 2. C is the categ. of finite sets.

$$BC = \coprod_n B\Sigma_n$$

(disj. union of all classif. spaces of the symm. groups Σ_n).

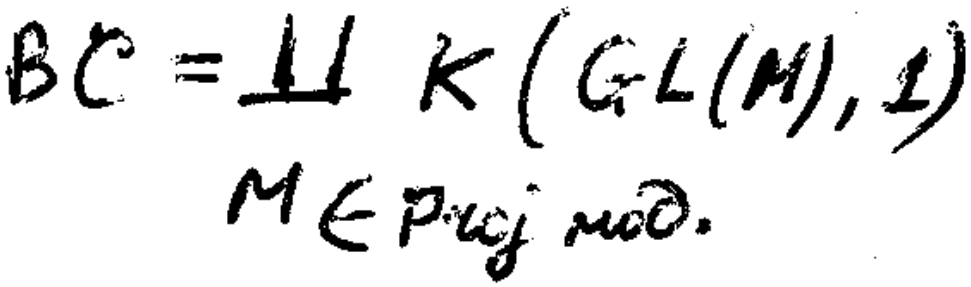


Original source details: GELFAND-P000419

Diagram and typed description. The whole line is underlined; below are two additional short strokes with a dot between them.

An H-space all together

For projective modules also



Original source details: GELFAND-P000419

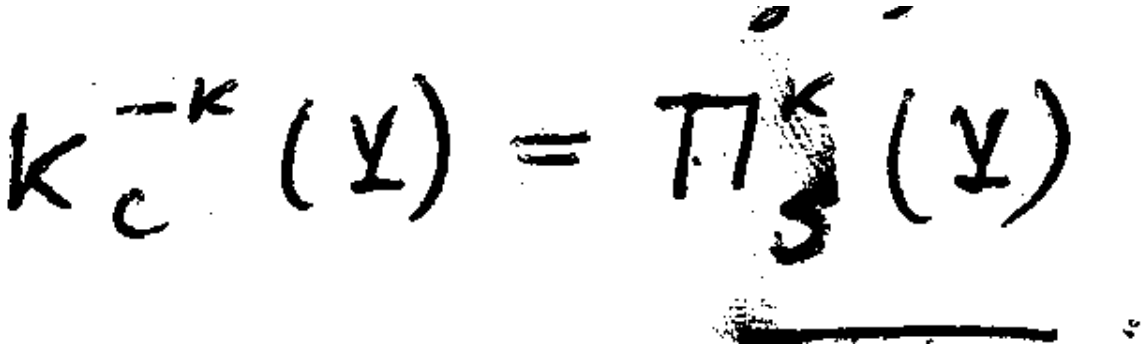
Diagram and typed description. The index $M \in$ Proj mod. is written on a separate line under the union sign.

$BC = \coprod K(GL(M), 1)$

$M \in$ Proj mod.

GELFAND-P000420

For the cat. of finite sets it is known that

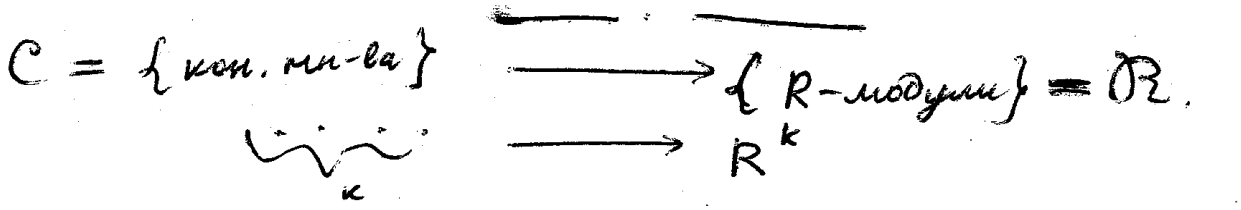


Original source details: GELFAND-P000420

Uncertain reading. The lower index at Π is overwritten and 3-like; provisionally s, possibly 3. The upper k is read separately.

$K_C^{-k}(Y) = \Pi_s^k(Y)$

(stable cohomotopy groups.)

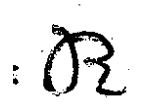


Original source details: GELFAND-P000420

Diagram and typed description. Two lines of arrows. Above the upper arrow are two separate horizontal strokes with a small mark between them. Beneath the left category are four separate small marks, a brace with a gap in its right half and the caption k; the lower arrow leads to R^k.

$C = \{\text{finite sets}\} \longrightarrow \{R\text{-modules}\} = \mathcal{R}$

$k \longrightarrow R^k$



Original source details: GELFAND-P000420

Uncertain reading. After the equality is a florid R-like letter with large left and upper loops; provisionally $\mathcal{R}$, with the precise font and overlapping strokes undetermined.

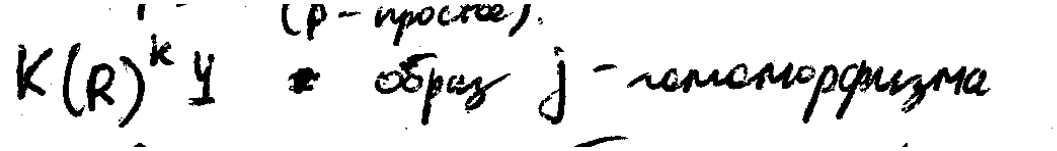
$$\Pi_s^k(Y) \longrightarrow K(R)^k(Y).$$

Y a point $\Rightarrow$ or stable homot. groups of spheres. One obtains their [?] homom. into the K-functor of any ring.

Original source details: GELFAND-P000420

Undeciphered passage. Before "homom." an earlier separate word is densely crossed out and not fully recovered.

$R = F_p$, Y a point (p prime).



Original source details: GELFAND-P000420

Uncertain reading. Between $K(R)^kY$ and "image" is a small dense sign with two widened parts and a connecting stroke; its exact reading is undetermined.

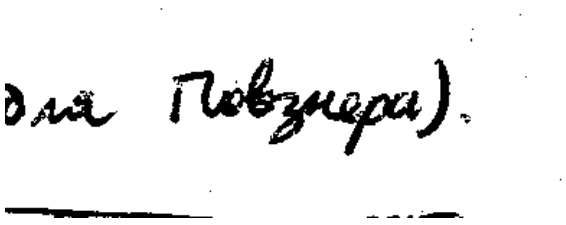
$K(R)^kY$ image of the j-homomorphism

Original source details: GELFAND-P000420

Uncertain reading. Before "-homomorphism" is a letter with a lower loop and a separate upper dot, provisionally j; the case j/J is undetermined.

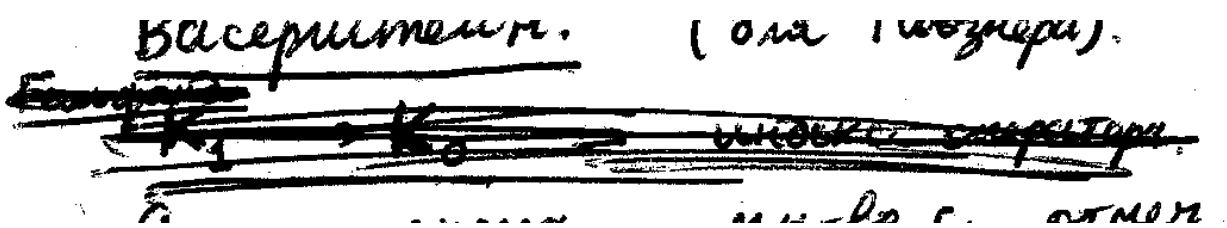
Next time — cobordisms. (in the same exposition).

Vasershtein. (for Povzner[?]).



Original source details: GELFAND-P000420

Uncertain reading. The first join after "П" (P) merges; "Повзнера" (Povzner) and "Певзнера" (Pevzner) remain candidates.



Original source details: GELFAND-P000420

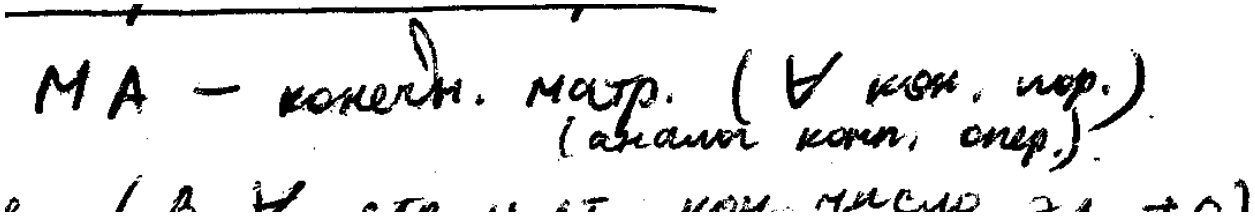
Undeciphered passage. An earlier beginning and a line with K_1, K_0 and the words “new operation” are crossed out. The initial E-like fragment is provisionally rendered “E…”; the whole earlier assertion is unrecovered.

~~E…~~

~~K_1 — K_0 … new operation~~

Simpl. scheme — a set with marked simplices (finite subsets) (together with $\forall$ set all its subsets are included). A topol. space is assigned to it. (see Godeman) homot. groups can be defined combinatorially.

A any ring; (noncommut.)



Original source details: GELFAND-P000420

Diagram and typed description. Above “finite” is a short curved stroke; the parenthetical caption is placed beneath the line.

MA — finite matrices. ($\forall$ finite order.)

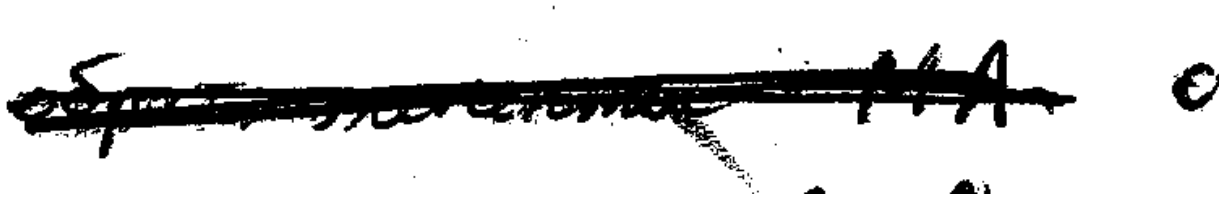
(analogue of compact oper.)

ℓA — row-finite (in $\forall$ row and col. a finite number of elements $\neq 0$) (analogue of bounded oper.)

$\mu A = \ell A / MA$ — the ring of Fredholm oper.

$K_i A$.

$GL(A)$ —



Original source details: GELFAND-P000420

Undeciphered passage. The earlier definition is crossed out; the beginning “invert…” and the final MA are visible, while the middle phrase is unrecovered. A faint oblique stroke descends from the middle part.

~~invert … MA~~

Original source details: GELFAND-P000420

Diagram and typed description. At the right is a new inscription with $1 + MA$ doubly underlined.

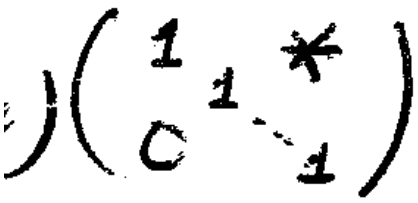
invertible elem. of the form $\underline{\underline{1 + MA}}$

A simplex — a finite subset in upper triang. matrices λ and

GELFAND-P000421

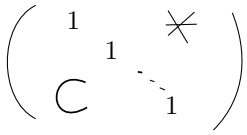
[source number: 57]

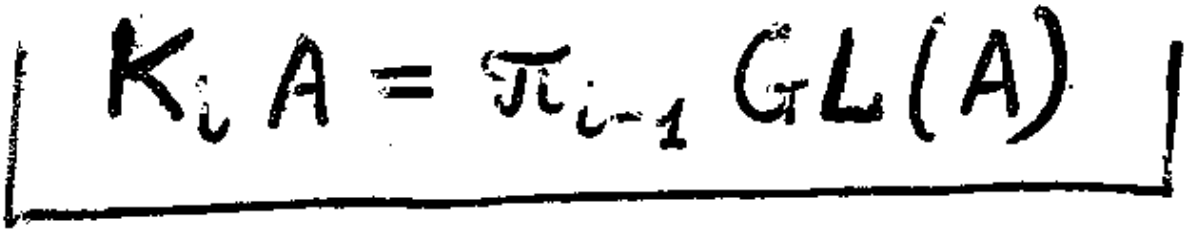
any conjugate of it (suffices by means of permutations) $a_1, \ldots, a_n$ — simpl., if $\exists f, g$ — invertible: fa_ig — upper triang. (with 1 on the diagonal).



Original source details: GELFAND-P000421

Diagram and typed description. A small matrix schematic: three 1-like diagonal entries, an upper-right star, short diagonal strokes and a lower-left $C/0$-like sign open on the right. The full size and lower-left sign are undetermined.

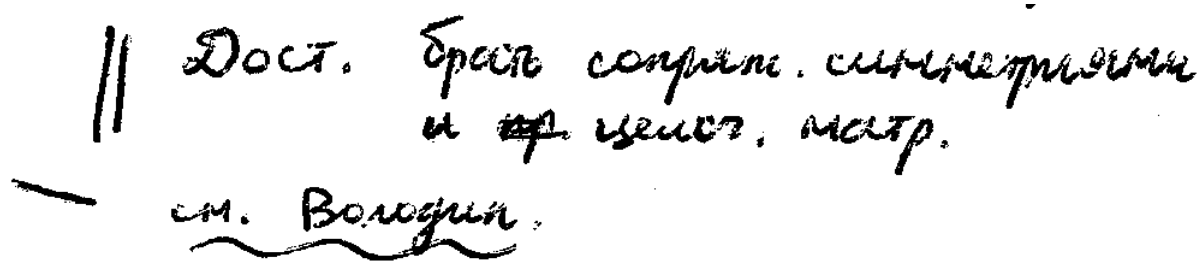




Original source details: GELFAND-P000421

Diagram and typed description. The formula is inside a three-sided enclosure with no top side.

$$K_iA = \pi_{i-1}GL(A)$$

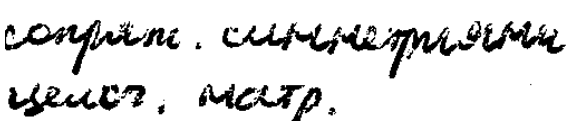


Original source details: GELFAND-P000421

Diagram and typed description. At the right are two vertical rules, two lines of a marginal note, a separate short oblique stroke lower down at the left and a wavy underline beneath “see Volodin.”.

Suffices to take conj. symmetric[?]
and integral matrices.

see Volodin.



Original source details: GELFAND-P000421

Uncertain reading. The adjective before “and” is provisionally “symmetric”; “singular” remains a candidate, and the joined middle is not confidently read.

Original source details: GELFAND-P000421

Undeciphered passage. Before “integral matrices.” a short word is densely crossed out and not fully recovered.

Exercise:

Original source details: GELFAND-P000421

Uncertain reading. find the arrows. The second word is provisional: its middle letters merge.

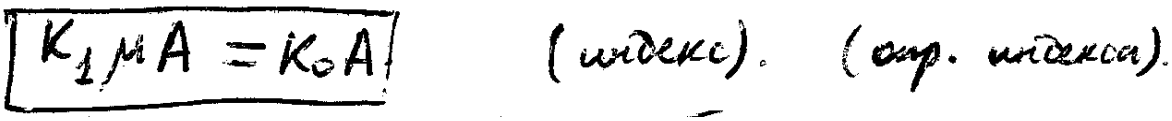
Theorem.

Original source details: GELFAND-P000421

Diagram and typed description. The formula and final period are inside the frame.

$$K_i A = K_{i+1} \mu A.$$

In particular,



Original source details: GELFAND-P000421

Diagram and typed description. The formula is framed; the explanations are outside at the right, with separate periods.

$$K_1 \mu A = K_0 A$$

(index). (def. of index).

The theor. generalizes to a Banach ring $\Rightarrow$ the usual theor. holds.

Question for Quillen's K-functor. And th

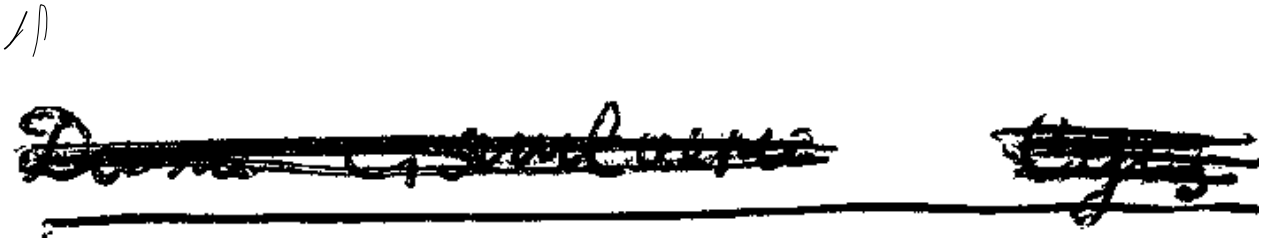
Undeciphered passage. A short unfinished beginning after the question; the continuation is unwritten.

Original source details: GELFAND-P000421

This follows from contractib. of ℓA. Sufficient is $K_i(\ell A) = 0$. Contractibility of $\ell A \Leftrightarrow$ contractib. of the complex $GL(\ell A)$. (even in the strong sense)

Original source details: GELFAND-P000421

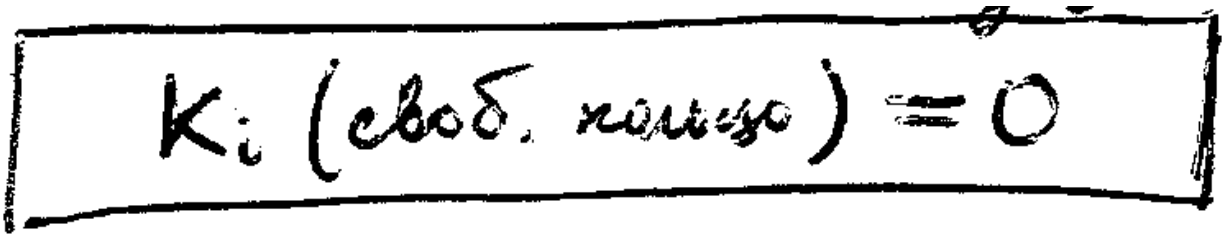
Uncertain reading. On the left margin is a separate slanting $1/i$-like mark and a tall curved stroke; the reading and function are undetermined.



Original source details: GELFAND-P000421

Undeciphered passage. "Proof of contraction" and a following separate inscription are crossed out; the last inscription is not fully recovered.

~~Proof of contraction~~



Original source details: GELFAND-P000421

Diagram and typed description. The formula is inside a closed slightly sloping frame.

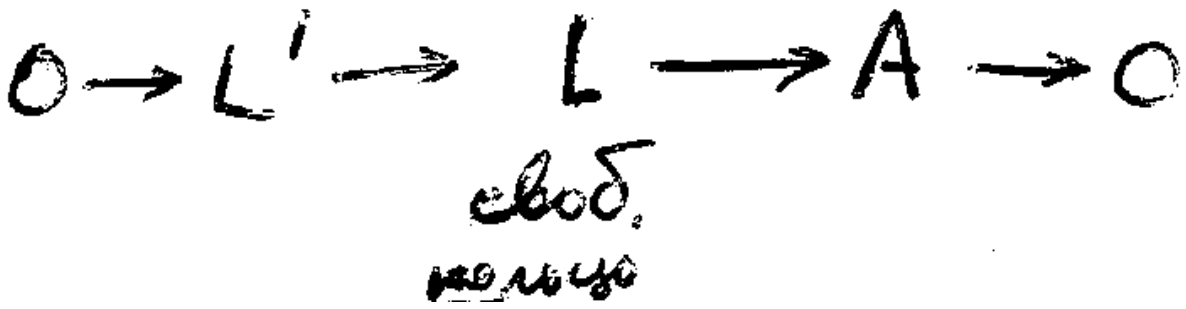
$$K_i(\text{free ring}) = 0$$

—

Original source details: GELFAND-P000421

Undeciphered passage. Before “but contractib.” is a separate short crossed inscription, not fully recovered.

but there is no contractib. of a free ring.



Original source details: GELFAND-P000421

Diagram and typed description. The arrow chain has a two-line caption below L. Its last round sign is open on the right.

$$0 \longrightarrow L' \longrightarrow L \longrightarrow A \longrightarrow C$$

free
ring.

Original source details: GELFAND-P000421

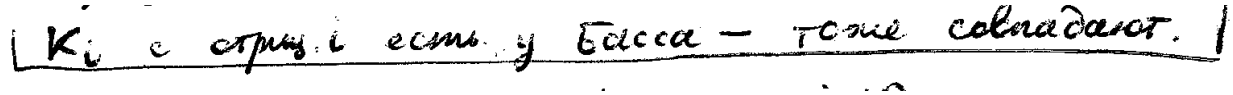
Uncertain reading. The final sign after the arrow from A has an open right edge; $0/C$ remain candidates. Its contour is rendered without selecting a value.

Original source details: GELFAND-P000421

Diagram and typed description. The formula is framed; the parenthetical explanation is at the right.

$K_i A = K_{i-1} L'$ (shift in the other direction)

There is a long exact seq. (everything follows from it) Hence follows agreement with the K-functors of Swan and Gershtein. (K_0, K_1 coincide, and further inductively).



Original source details: GELFAND-P000421

Diagram and typed description. The whole line is enclosed by a bottom rule with short vertical ends; there is no top side.

K_i with negative i occurs in Bass — also coincide.

Original source details: GELFAND-P000421

Undeciphered passage. A separate densely crossed beginning of the final line, not fully recovered.

For regular rings all $K_i = 0$ with $i < 0$.

GELFAND-P000422

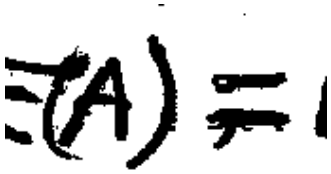
Karoubi's theor.: $i > 0$ differs from Bass

No zero divisors — Karoubi also agrees

Quillen's theor.:

Original source details: GELFAND-P000422

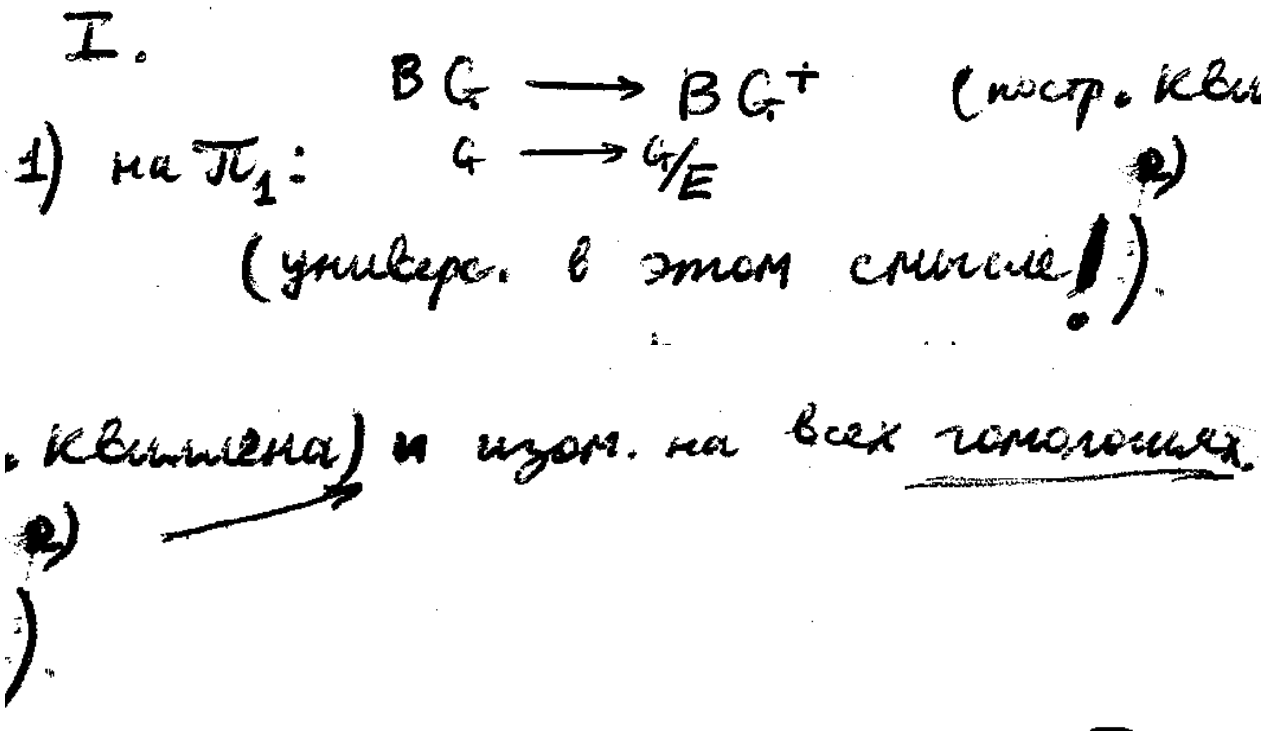
Undeciphered passage. After the heading and before G is a separate densely crossed earlier fragment, not fully recovered.



Original source details: GELFAND-P000422

Uncertain reading. After $E(A)$ are two horizontal strokes, with an oblique stroke descending from the lower one. An overwritten equality and a remnant of an earlier $\neq$ are possible; the sign is rendered as strokes without a choice.

$G = GL(A), \quad E = E(A) \quad [E, E] = [G, G].$



Original source details: GELFAND-P000422

Diagram and typed description. I and two numbered marks. The upper arrow leads from BG to BG^+; beneath it is the map on π_1. From "2)" a separate arrow points to the words about homology isomorphism; the last word is underlined.

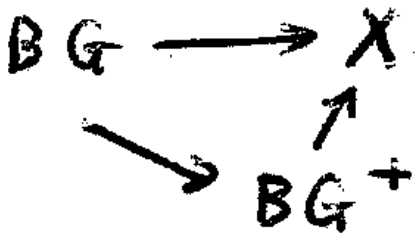
I.

$BG \longrightarrow BG^+$ (Quillen's constr.) and an isom. on all homology.

1) on π_1: $G \longrightarrow G/E$ 2)

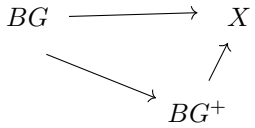
(universal in this sense!).

II. ∃ such a BG^+, which is an H-space and universal, i.e. for ∀ other map into an H-space



Original source details: GELFAND-P000422

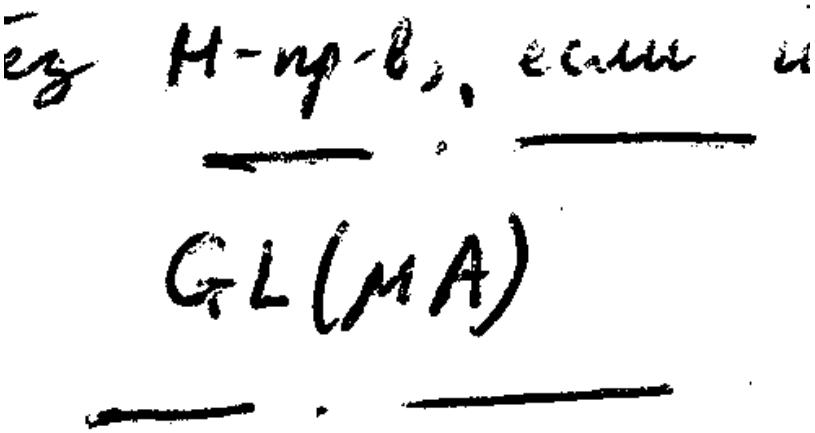
Diagram and typed description. Three arrows: $BG \to X$, $BG \to BG^+$ and from the lower BG^+ up to X.



(Morava: and even without H-spaces, if it induces an isom. [?] of homology)

Original source details: GELFAND-P000422

Undeciphered passage. Before “of homology” a short earlier word is crossed out and not fully recovered.



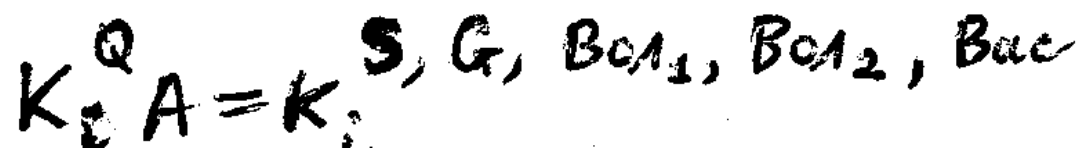
Original source details: GELFAND-P000422

Diagram and typed description. Two rows of separate horizontal strokes each have a small central mark. The upper row is below “H-spaces, if” and the lower beneath “Consider now”. The upper left stroke is pointed to the left.

Consider now $GL(\mu A)$

Quillen's constr.: $K_i^Q A = \pi_i BG^+$, $i \geq 1$.

Theorem: $K_i A = K_i^Q A$.

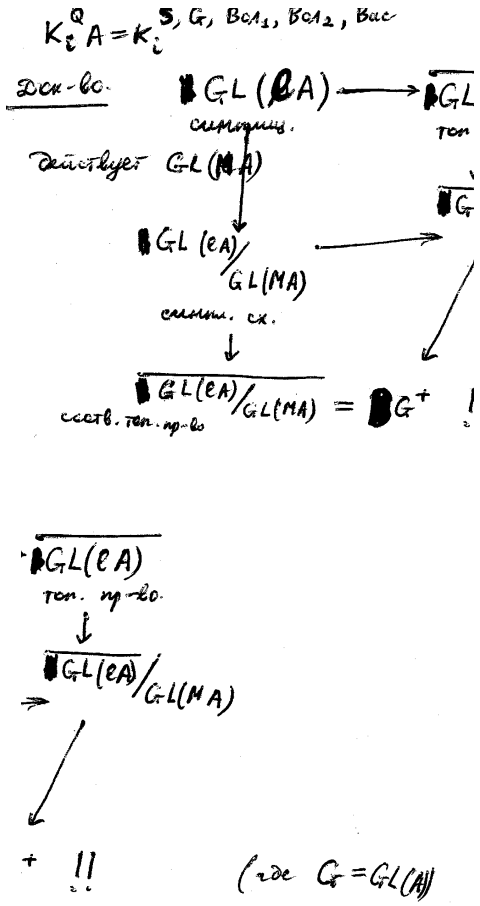


Original source details: GELFAND-P000422

Diagram and typed description. After the equality is just one K_i with the upper list S, G, "Vol" with lower $1, 2$ and "Vas"; the S is heavily inked. No additional A is written on the right.

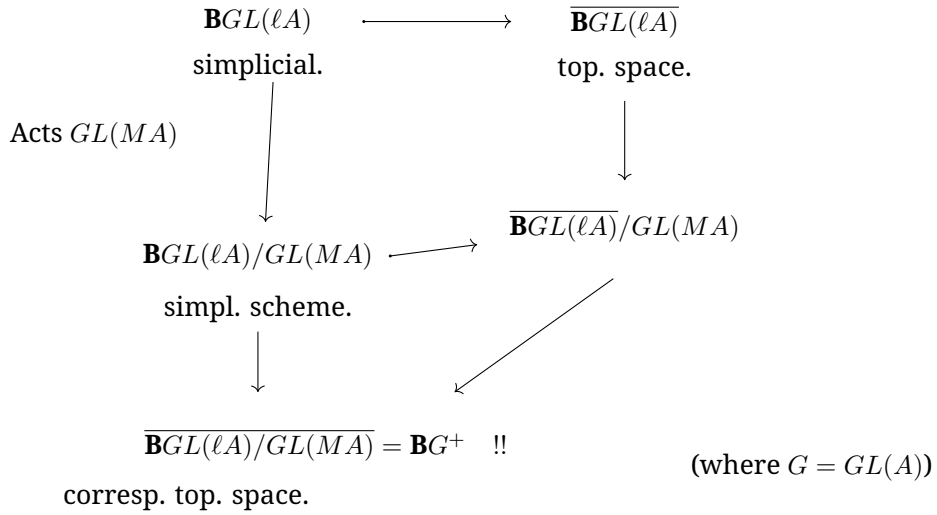
$K_i^Q A = K_i^{\mathbf{S},G,\mathrm{Vol}_1,\mathrm{Vol}_2,\mathrm{Vas}}$

Proof.



Original source details: GELFAND-P000422

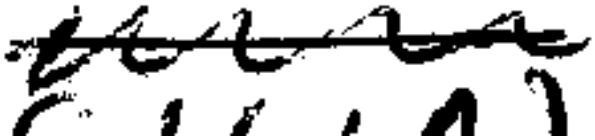
Diagram and typed description. The five nodes with BGL and the lower BG^{+} are provisional: each initial B is heavily blackened and the upper-left ℓ is overwritten; earlier signs are unrecovered. All six arrows, the $GL(MA)$ action, captions and two exclamation marks are retained. At the upper right the bar covers $BGL(\ell A)$, at the middle right only the numerator, and at the bottom the whole quotient.



GELFAND-P000423

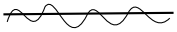
[source number: 58]

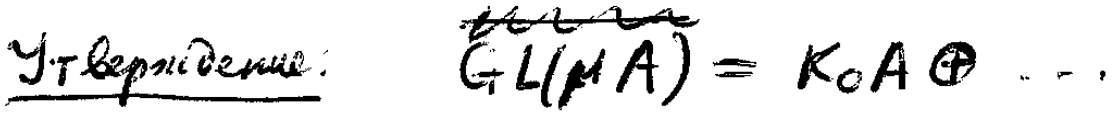
Assertion:



Original source details: GELFAND-P000423

Undeciphered passage. Above $GL(\mu A)$ is an earlier short crossed inscription; separate loops are visible, but it is not fully recovered.

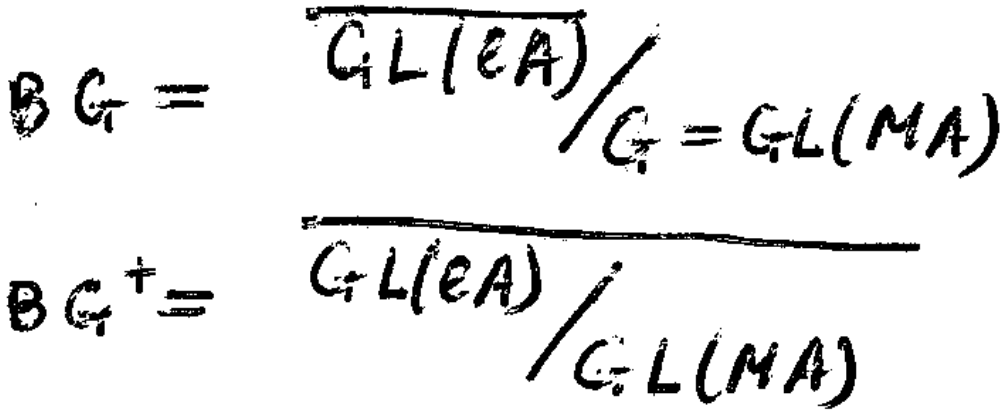




Original source details: GELFAND-P000423

Diagram and typed description. After $\oplus$ are three separate marks: a dot, a short horizontal stroke, a dot.

$GL(\mu A) = K_0 A \oplus \quad . \; - \; .$

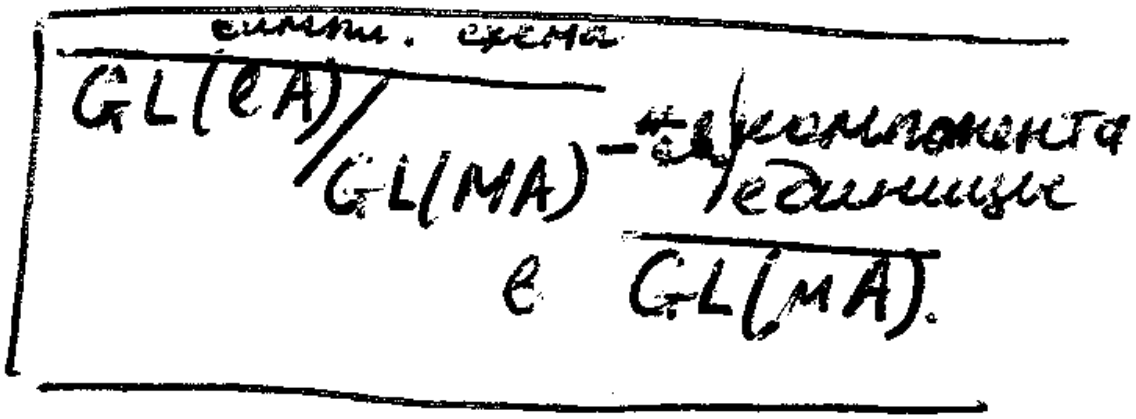


Original source details: GELFAND-P000423

Diagram and typed description. Two inscriptions with oblique quotient strokes. In the first, the upper bar covers only $GL(\ell A)$, and $G = GL(MA)$ is written beneath. In the second, the bar covers the entire quotient.

$$BG = \overline{GL(\ell A)} \,/\, G = GL(MA)$$

$$BG^{+} = \overline{GL(\ell A) \,/\, GL(MA)}$$

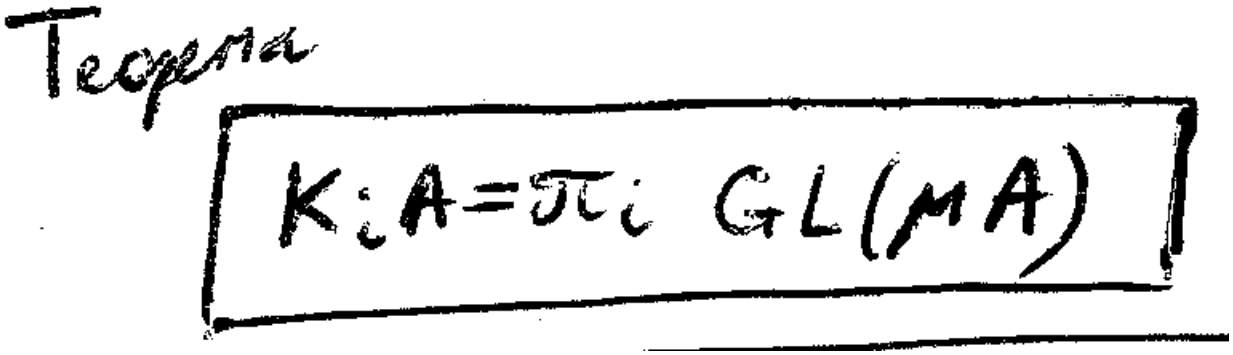


Original source details: GELFAND-P000423

Uncertain reading. At the right is a frame with no right side. The small overwritten mark before "component" has two horizontal and crossing strokes with a separate 1-like inscription; its meaning is undetermined.

simpl. scheme

$GL(\ell A)/GL(MA)$ – component of the identity in $\overline{GL(\mu A)}$.

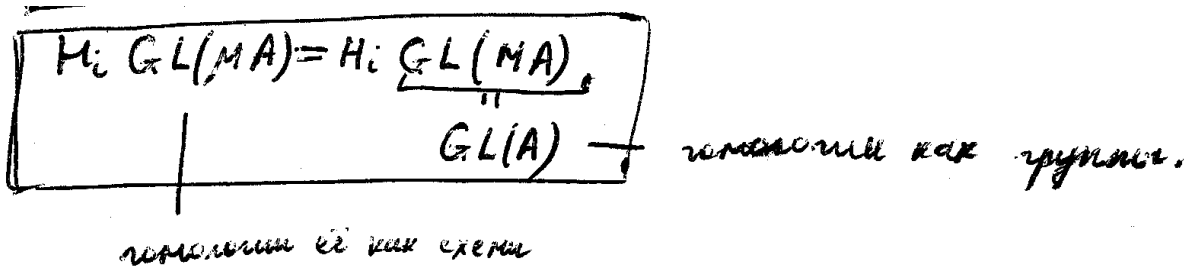


Original source details: GELFAND-P000423

Diagram and typed description. The heading is outside the sloping closed frame; the formula is inside.

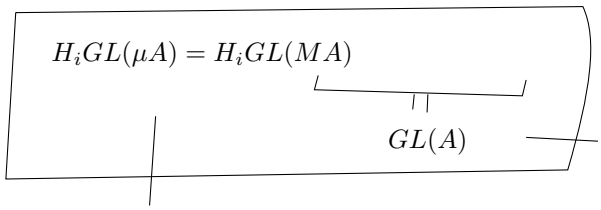
Theorem

$$K_iA = \pi_i GL(\mu A)$$



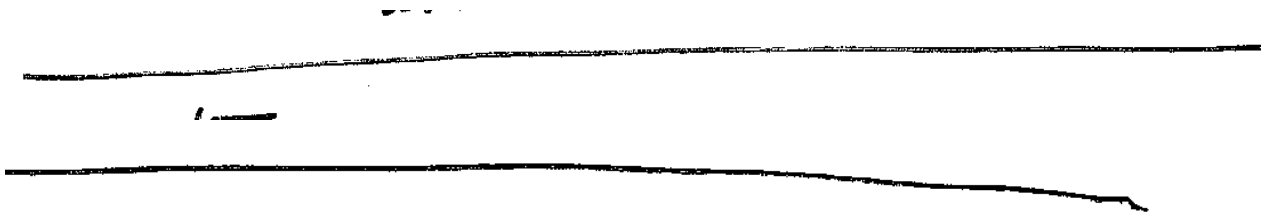
Original source details: GELFAND-P000423

Uncertain reading. Inside the frame is the upper equality; beneath $GL(MA)$ are a bracket and two short vertical strokes, with $GL(A)$ below. The link's meaning and scope of H_i are undetermined. Separate captions at the right and below have leader strokes.



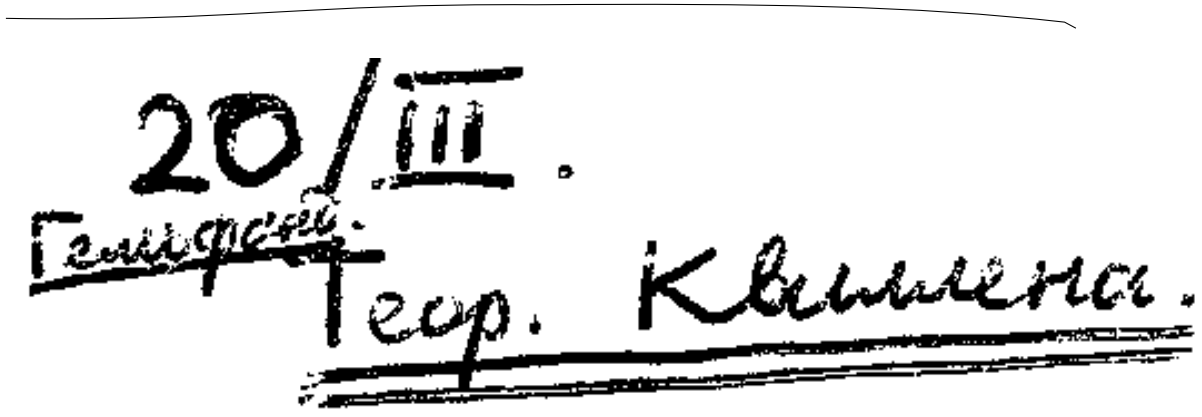
homology as a group.

its homology as a scheme



Original source details: GELFAND-P000423

Diagram and typed description. A long slightly curved transverse rule.



Original source details: GELFAND-P000423

Diagram and typed description. The date has rules above and below III; at the left is an underlined name, and below it a doubly underlined heading.

20/III.

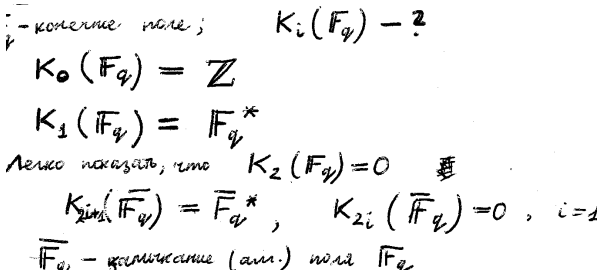
Gelfand.

Quillen's theor.

(proved nonelementarily) $\mathbb{F}_q$ — a finite field; $K_i(\mathbb{F}_q)$ — ?

$$K_0(\mathbb{F}_q) = \mathbb{Z},$$
$$K_1(\mathbb{F}_q) = \mathbb{F}_q^*.$$

It is easy to show that $K_2(\mathbb{F}_q) = 0$.

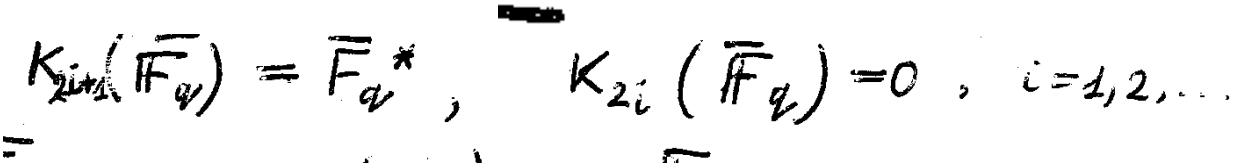


Original source details: GELFAND-P000423

Manuscript notation. The opening formulas have double-stem F. The first algebraic-closure equation has a single stem on its right-hand side.

Original source details: GELFAND-P000423

Undeciphered passage. After the equality to zero is a separate small crossed inscription, not fully recovered.



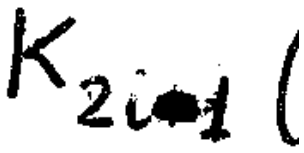
Original source details: GELFAND-P000423

Uncertain reading. The first index is provisionally $2i+1$: $2i$, a dense $+$ and 1 are visible, but an earlier overwritten layer is unrecovered. The range $i = 1, 2, \ldots$ is retained.

$$K_{2i+1}(\overline{\mathbb{F}}_q) = \overline{F}_q^*, \quad K_{2i}(\overline{\mathbb{F}}_q) = 0, \quad i = 1, 2, \ldots.$$

$\overline{\mathbb{F}}_q$ — the (alg.) closure of the field $\mathbb{F}_q$. (like the periodicity theor.)

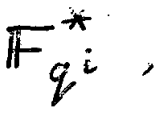
$$K_{2i}(\mathbb{F}_q) = 0,$$



Original source details: GELFAND-P000423

Uncertain reading. Between $2i$ and 1 is a dense overwritten mark, rendered by its contour; $+$ and $-$ remain candidates.

$K_{2i \bullet 1}(\mathbb{F}_q) = \mathbb{F}_q^{*}{}_i \quad , \quad i = 1, 2, \ldots .$

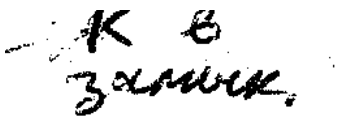


Uncertain reading. At the right of F is a lower q, a small i above and to its right, and an upper star. A candidate is $F^{*}_{q^i}$; the exact two-level placement remains uncertain.

Original source details: GELFAND-P000423

GELFAND-P000424

As i increases, the K-functor approaches elementary K in the clos.



Uncertain reading. The final abbreviation is provisionally read “замык.” (clos.); the short terminal stroke remains uncertain.

Original source details: GELFAND-P000424

Uses the Frobenius automorphism.

Kazhdan: Proof.

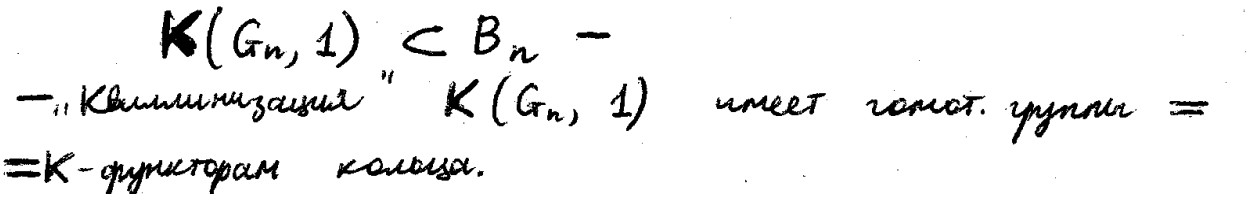
Consider

Original source details: GELFAND-P000424

Uncertain reading. The first lower index at G is overwritten; provisionally n, with the earlier sign unrecovered.

$GL(n, F_q) = G_n$

and its classifying space $\underline{K(G_n,1)}$. $GL(n,F_q) \supset SL(n,F_q) = G_n^0$.



Original source details: GELFAND-P000424

Diagram and typed description. After the inclusion is a separate dash; the next line begins with a dash and ends with an equality, with another equality at the start of the bottom line.

$$K(G_n,1) \subset B_n —$$

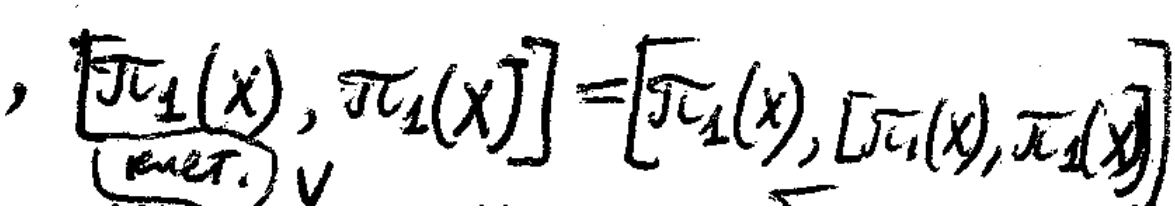
— «Quillenization» $K(G_n,1)$ has homot. groups $=$

$=$ the K-functors of the ring.

Aim: make the fund. group commutative, without changing homology groups. We fill in the commutator subgroup and the resulting 2-dimensional cycles. One can restrict to attaching 3-cycles.

Fuchs.

Existence theor. X — a cell complex,

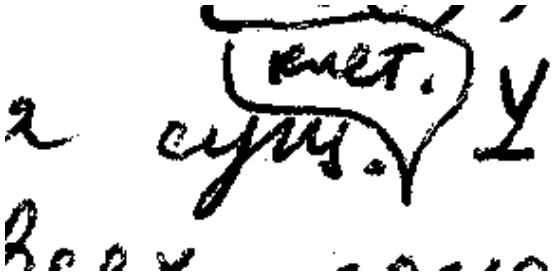


Original source details: GELFAND-P000424

Diagram and typed description. On the right, the outer first argument is $\pi_1(X)$ and the second argument is the inner commutator; all brackets are retained.

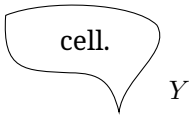
$$[\pi_1(X),\pi_1(X)] = [\pi_1(X),[\pi_1(X),\pi_1(X)]]$$

This condition is necessary and sufficient for the existence of



Original source details: GELFAND-P000424

Diagram and typed description. Above Y is the inserted “cell.” in a curved contour with a pointed lower end.



and a map $f : X \longrightarrow Y$, an isom. of all homology and

всех гомологий и
тивно. 1) $f_* : H_*(X, \mathbb{Z}) \xrightarrow{\sim} H_*(Y, \mathbb{Z})$

Original source details: GELFAND-P000424

Diagram and typed description. At the right is condition 1), a rightward arrow with a wavy stroke above it. At the left is an angular bend and below a long rule; the upper and right sides are open.

1) $f_* : H_*(X, \mathbb{Z}) \xrightarrow{\sim} H_*(Y, \mathbb{Z})$

in Y

Original source details: GELFAND-P000424

Undeciphered passage. Before "2)" a short earlier inscription is crossed out and not fully recovered.

2) $\pi_1(Y)$ is commutative. Hence $\pi_1(Y)$ is computed, namely

Отсюда $\pi_1(Y)$ вычисляется, а именно
$\pi_1(Y) = \pi_1(X) / [\pi_1(X), \pi_1(X)]$.

Original source details: GELFAND-P000424

Diagram and typed description. The formula has an oblique quotient stroke: the numerator is higher at the left, and the commutator lower at the right.

$$\pi_1(Y) = \pi_1(X) \,/\, [\pi_1(X), \pi_1(X)].$$

This is an existence theorem. Such a Y is not unique. (E.g., when $\pi_1(X)$ is commutative).

2a) let Y be simple (the fund. group acts on $\pi_n(Y)$ trivially)

Such a Y is unique ($\Rightarrow$ commutative fund. group, since π_1 acts on itself by conj.) — Whitehead's theor..

It is known that H-spaces are simple.

GELFAND-P000425

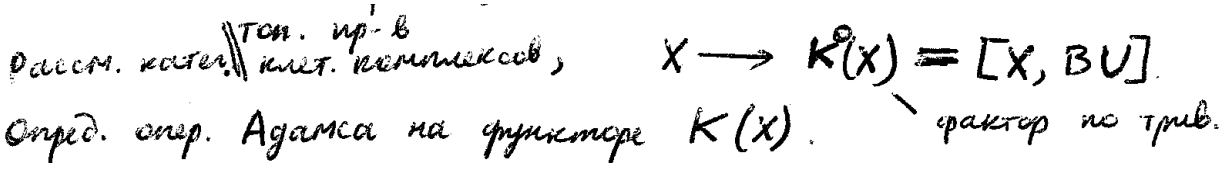
[source number: 59]

Kazhdan — B_n — H-space. (but this is not for every G, only for this one).

$$K_i(G_n) = \pi_i(B_n).$$

Quillen found a topological realization for B_n.

Consider the category \\ of topological spaces of cell complexes, $X \longrightarrow K^0(X) = [X, BU]$. \ quotient by trivial



Original source details: GELFAND-P000425

Diagram and typed description. Two slanting strokes before the inserted lines; a slanting stroke leads to the annotation under the formula.

Define the Adams operations on the functor $K(X)$.

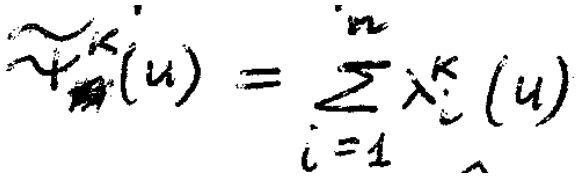
On U_n consider the function: $u \longmapsto$ eigenvalues $\lambda_1, \ldots, \lambda_n$.

Uncertain reading. The short sign between the arrow and "eigenvalues" is crossed out; the underlying letter is illegible.

Original source details: GELFAND-P000425

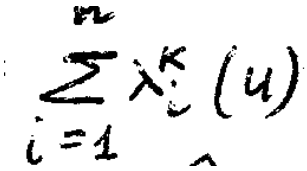
$$\widetilde{\psi}^k(u) = \sum_{i=1}^{n} \lambda_i^k(u)$$

— the character of a virtual representation (difference of characters).



Uncertain reading. Crossed-out subscript beneath the tilded ψ^k; the original index is illegible.

Original source details: GELFAND-P000425



Uncertain reading. The subscript of λ is a dot above a short curved stroke: i or j. The sum limit $i = 1$ is legible.

Original source details: GELFAND-P000425

This representation $\widehat{\psi}^k$: Sp $\widehat{\psi}^k = \widetilde{\psi}^k(u)$.

Construct, for $\forall$ bundle, another with a U action, using the representation $\widehat{\psi}^k$. This is an operation in the K-functor: $\psi^k(\alpha^\dagger)$.

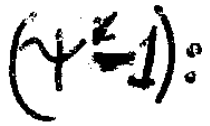
$$\exists \text{ a map } \psi^k : BU \longrightarrow BU,$$

inducing the Adams operation in K. $\alpha^\dagger \in [X, BU]$,

Original source details: GELFAND-P000425

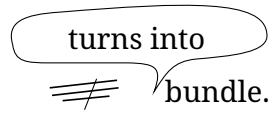
Undeciphered passage. The crossed-out writing after $[X, BU]$ has a ψ-like beginning and a rounded stroke near the end; the remainder is illegible.

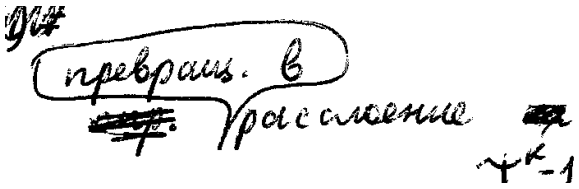
$$(\psi^k - 1) : BU \longrightarrow BU.$$



Original source details: GELFAND-P000425

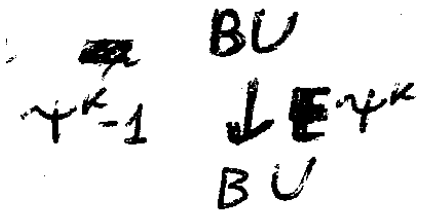
Uncertain reading. The horizontal stroke before 1 has been retraced; the lower stroke merges into it. Reading: $-$ or $=$.





Original source details: GELFAND-P000425

Undeciphered passage. Under the insertion "turns into", the crossed-out fragment before "bundle" is illegible.



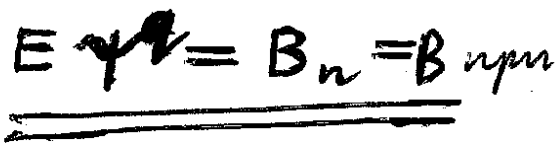
Original source details: GELFAND-P000425

Diagram and typed description. Two BU labels and a down arrow. At left $\psi^k - 1$; at right $E\psi^k$ with no separate arrow. A crossed-out mark is at the left of the upper BU.

$$\begin{array}{ccc} & BU & \\ \psi^k - 1 & \downarrow & E\psi^k \\ & BU & \end{array}$$

$E\psi^k$ — fiber of the map $\psi^k - 1$.

Claim: $E\psi^{\cdot} = B_n = B$ for large n.



Original source details: GELFAND-P000425

Uncertain reading. The exponent on $E\psi$ has been retraced: k or q. The equality is underlined twice.

Frobenius automorphism $\sim$ Adams operation in topology.

Another def.

Original source details: GELFAND-P000425

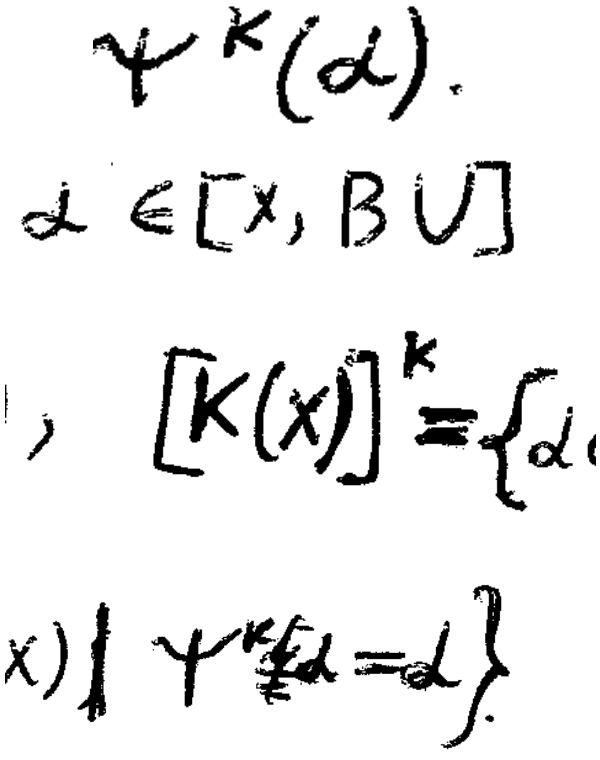
Uncertain reading. A looped mark above the line: “e” (the end of the Russian abbreviation) or a separate k.

Let $[K(X)]^k \subset K^0(X)$,

$$[K(X)]^k = \{\alpha^\dagger \in K^0(X) \mid \psi^k \epsilon \alpha^\dagger = \alpha^\dagger\}.$$

Original source details: GELFAND-P000425

Uncertain reading. Crossed-out sign after ψ^k; the original reading is undetermined.



Original source details: GELFAND-P000425

Uncertain reading. † The recurring letter has a low loop and a rising right stroke. It is provisionally read as α; d is also possible.

Original source details: GELFAND-P000425

it is representable and

$$[K(X)]^k = [X, E\psi^k]$$

easy to check.

GELFAND-P000426

$K(G,1) \subset B$. Identify B with $E\psi^q$.

[illegible]

Uncertain reading. Crossed-out traces of $[K(X)]$, a superscript (k or q), and a separate illegible sign to the right.

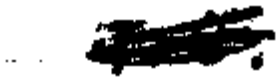
Original source details: GELFAND-P000426

Consider $GL(n, F_q)$ and fix a homomorphism $\mu : \overline{F}_q^* \longrightarrow \mathbb{C}^*$.

On $GL(n, F_q)$ construct a function — the character of a representation.

$\varphi_\mu(g) = \sum \mu(\lambda_i)$, where λ_i — eigenvalues of the matrix g.

We must construct a map $K(G_n, 1) \longrightarrow E\psi^q$ or an element of $[K(G_n, 1), E\psi^q]$, i.e. an element of the K-functor of $K(G_n, 1)$, invariant under ψ^q, [illegible]



Undeciphered passage. After the invariance clause, the short crossed-out fragment is illegible.

Original source details: GELFAND-P000426

It suffices to give a representation of the group G_n (then we obtain an element of $K(K(G_n, 1))$.

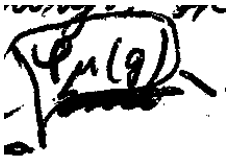
From Brauer's thm. $\varphi_\mu^\dagger(g)$ character of a virtual representation,

[illegible] G_n (тогда получ. элем. $K(K(G_n,1))$.

Из теор. Брауэра $\varphi_\mu(g)$ характер вирт. представления,

Original source details: GELFAND-P000426

Uncertain reading. † In the outlined insertion, the descending stem and the loop favor $\varphi_\mu(g)$. The outline overlaps the letter; $\psi_\mu(g)$ remains possible. A wavy line underlines "Brauer".



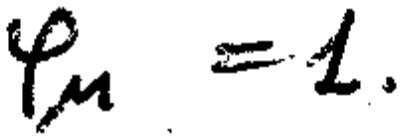
Undeciphered passage. The heavily crossed-out inscription below the insertion is illegible.

Original source details: GELFAND-P000426

The set λ_i^q — eigenvalues too — the same ones $\Rightarrow$ invariance.

On $E\psi^q$ there is an H-space structure $\Rightarrow$ the map extends to a map into B_n (onto the filled-in disks).

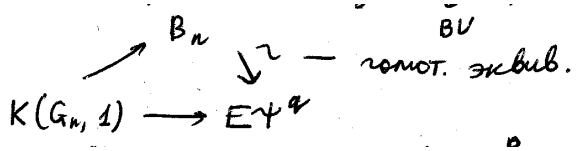
Another way: SL is generated by unipotent matrices, and these are the ones to kill $\Rightarrow$ and on them $\varphi_\mu = 1$.



Original source details: GELFAND-P000426

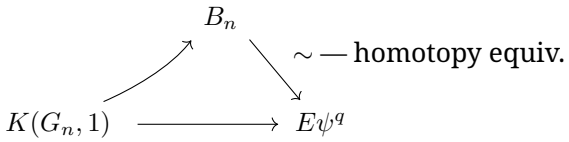
Diagram and typed description. The source reads $\varphi_\mu = 1$.

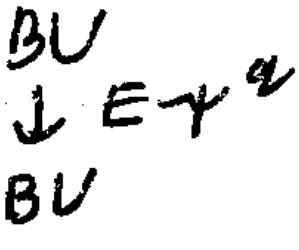
Homotopy groups of $E\psi^q$.



Original source details: GELFAND-P000426

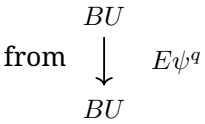
Diagram and typed description. Three arrows; the sign $\sim$ and its caption are by the descending arrow.





Original source details: GELFAND-P000426

Diagram and typed description. Two BU labels, a down arrow, and $E\psi^q$ at right without a separate connection.



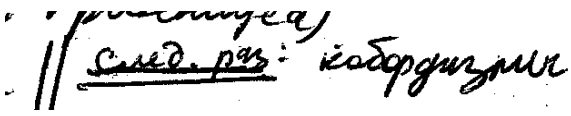
For $E\psi^q$ the homology is computed, and for B_n — ?

The homology groups of $K(G_n, 1)$ are computed. (Quillen too). using étale topology (see Annals).

$$\varphi : G \longrightarrow G$$
$$\varphi(g) = g^{-1} g^\sigma$$

(σ — Frobenius automorphism).

Now write the spectral sequence of the fibration and compute. // Next time: cobordisms.



Original source details: GELFAND-P000426

Diagram and typed description. Two near-vertical slanting strokes separate the marginal note; its opening is underlined twice.

GELFAND-P000427

[source number: 60]

M. Sh. Birman. Kato's theorem.

$H - H_0 \overset{=}{} V \in \mathfrak{S}_1$ (nuclear operator).

Original source details: GELFAND-P000427

Uncertain reading. Between $H - H_0$ and V are several overlapping horizontal strokes: $=$ or $\equiv$.

Original source details: GELFAND-P000427

Uncertain reading. An isolated mark below the page number 60 has two loops and a slanted cross stroke. Possibly f. Its connection to the formulas is undetermined.

Prove that the absolutely continuous parts of H and H_0 are unitarily equivalent. H, H_0 — self-adjoint.

Question (I. M. Gelfand): how can one determine from the collection of spectral types whether they can be realized so that the difference is a nuclear operator?

Spectrum in the quotient ring modulo the nuclear operators — ?

2 facts used:

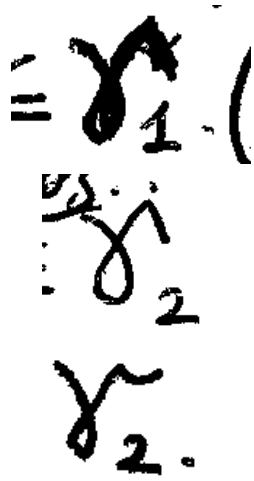
1) $D_1, D_2 \in \gamma_2$; E_λ — resolution of the identity $\Rightarrow$

$$\Rightarrow \quad \frac{d}{d\lambda} D_1 E_\lambda D_2$$

exists almost everywhere and is nuclear ($\sim$ rigging theorem).

(see Gelfand, diff. formula, strongly bounded var.)

2) $D_1 R_z D_2 \longrightarrow D_1 R_{\lambda+i0} D_2$ almost everywhere in γ_2.



Uncertain reading. Each of the three marks has a loop and two upper branches. The letter may be γ or $\mathcal{J}$. The subscripts $1, 2, 2$ are legible.

Original source details: GELFAND-P000427

Operator implementing the unitary equivalence:

$$\int_{-\infty}^{\infty} \frac{d}{d\lambda}(E_\lambda(I + VR_z^0)f, g)\Big|_{z=\lambda\pm i0} d\lambda = (W_\pm f, g).$$

(V is split into 2 Hilbert–Schmidt operators.)

Next time.

27/III Kazhdan.

Artin–Mazur theorem.

If X — set over a field k. any!! From it one can construct a complex C_X

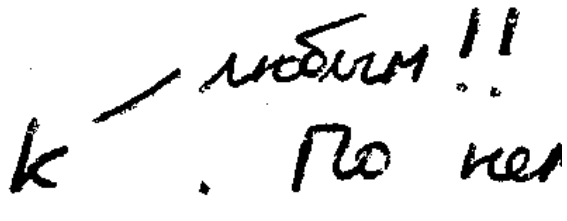


Diagram and typed description. “Any!!” is written above k, with a slanted leader.

Original source details: GELFAND-P000427

If $k = \mathbb{C} \Rightarrow C_X = \widehat{X}_{\mathbb{C}}$ — completion

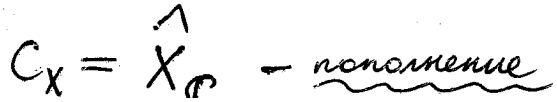
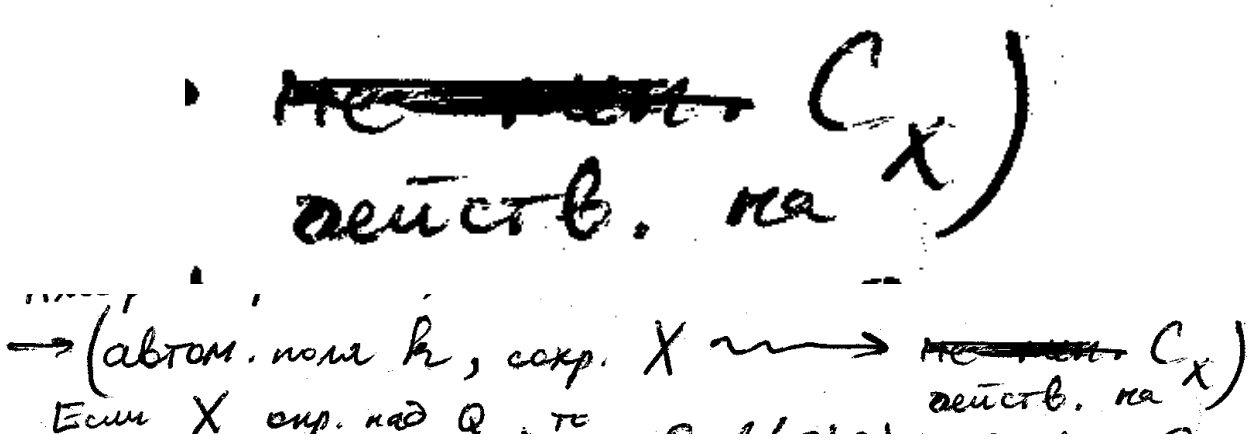


Diagram and typed description. A wavy line underlines “completion”.

Original source details: GELFAND-P000427

Algebraic construction. $\longrightarrow$

$\longrightarrow$ (autom. of the field k, preserving X $\overset{\text{act on}}{\sim\!\!\!\longrightarrow}$ C_X)



Original source details: GELFAND-P000427

Undeciphered passage. Before C_X there is a crossed-out inscription whose beginning resembles “не” (“not”). The remainder is illegible. “Act on” is inserted below.

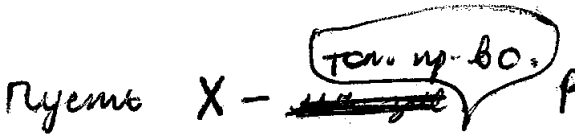
If X is defined over Q, then Gal$(\mathbb{C} \mid Q)$ acts on C_X.

Original source details: GELFAND-P000427

Uncertain reading. Before “then”, below the line, a short mark has a faint slanted continuation. It is unclear whether the continuation is part of the comma.

GELFAND-P000428

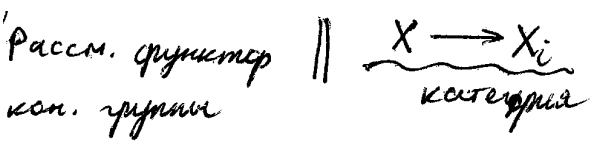
Let X — top. space.



Original source details: GELFAND-P000428

Undeciphered passage. Horizontal strokes conceal the crossed-out abbreviation. Traces resembling “мн-” or “мн-ие” remain. “Top. space.” was added in the outline above it.

Consider the functor $\| \; \underbrace{X \longrightarrow X_i}_{\text{category}}$



Original source details: GELFAND-P000428

Diagram and typed description. Two vertical strokes precede the arrow diagram. A wavy stroke and a caption are below it.

the images of all $\pi_n(X)$ — finite groups

$\forall \underline{Y}$ consider the set $\underline{Y} \longrightarrow X_i \qquad \prod_i \mathrm{Hom}(\underline{Y}, X_i)$

and compatible $\underline{h_{\widehat{X}}(Y)}$ — a functor.

It is representable and $\widehat{X}$ — the representing object.

———— · ————

I. M. Gelfand.

Standard language.

1°. Category and functor (Grothendieck, "On some questions of hom. alg." — chapter 1).

2°. Topos (topó). + theorem on descent

Uncertain reading. "Descent" is provisional. The middle strokes of the Russian word merge.

Original source details: GELFAND-P000428

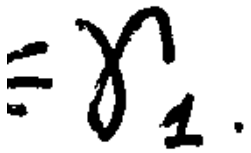
(the definition is in Manin's preface to the article by Mumford)

M. Artin — Grothendieck,

———— · ————

M. Sh. Birman. Theorem of Kato.

$\mathfrak{H}$ — Hilbert space. H, H^0 — operators, $H - H^0 \in \mathfrak{S}_1$.



Uncertain reading. The ideal sign may be γ_1 or $\mathfrak{S}_1$. The subscript 1 is legible.

Original source details: GELFAND-P000428

$$e^{iHt}e^{-iH^0t}P^0$$

P^0 — projection onto the absolutely continuous subspace of H^0

$$\underset{t\to\pm\infty}{s\text{-}\lim} e^{iHt}e^{-iH^0t}P^0 = W_\pm$$

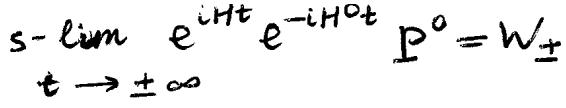


Diagram and typed description. The t under the limit is heavily retraced, followed by an arrow to $\pm\infty$.

Original source details: GELFAND-P000428

Then both W_+ and W_- give an isomorphism of the absolutely continuous spectra of H and H^0 (symmetry of the assumptions!)

$W_\pm H^0 = HW_\pm$ || $P^{0\mathfrak{H}}$ — domain of isometry of $W_\pm$ (on the complement it is 0),

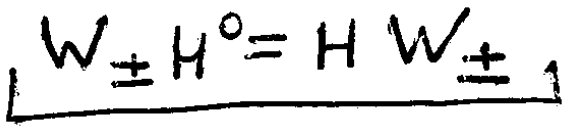


Diagram and typed description. A horizontal bracket with two short upstrokes is below the equality.

Original source details: GELFAND-P000428

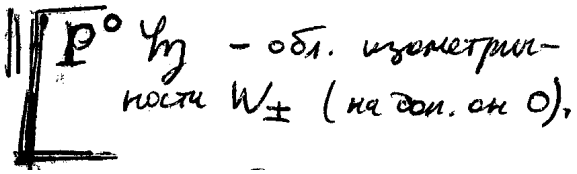


Diagram and typed description. At right, two vertical strokes and a bracket open to the right surround the isometry note.

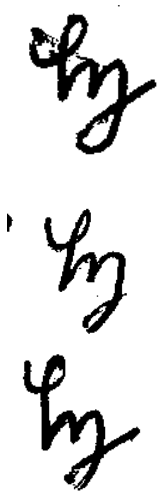
Original source details: GELFAND-P000428

Range: does it coincide with the subspace

$P^{\mathfrak{H}}$?

Diagram and typed description. There is a bracket below the subspace symbol, with the question mark outside it.

Original source details: GELFAND-P000428



Uncertain reading. All three occurrences show a looped sign with a long lower tail. $\mathfrak{H}$, $\mathcal{H}$, and Ψ are possible readings. The shape is represented by an outline.

Original source details: GELFAND-P000428

GELFAND-P000429

[source number: 61]

Prop. The range of W_+ coincides with $P^{\mathfrak{y}} \Leftrightarrow \exists$ wave oper. W_+ with the roles of H and H_0 interchanged.

Uncertain reading. The ornamental glyph after P has several loops, including a closed lower loop. $\mathfrak{H}$, $\mathcal{H}$, and Ψ are possible readings.

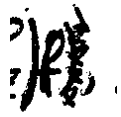
Original source details: GELFAND-P000429

Hence existence $\Rightarrow$ completeness, from the symmetric relation.

$$W_\pm(H_3, H_2)W_\pm(H_2, H_1) = W_\pm(H_3, H_1).$$

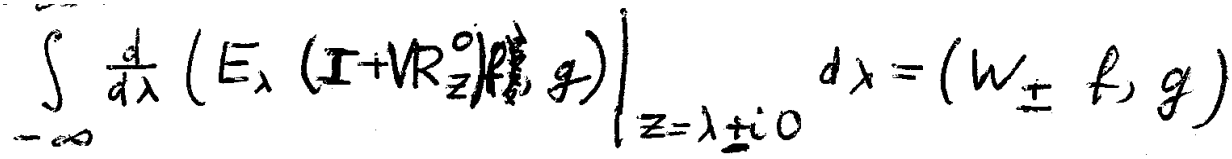
Proof of thm.

$$\int_{-\infty} \frac{d}{d\lambda}\left(E_\lambda(I + VR_z^0)f^{\natural}, g\right)\Big|_{z=\lambda\pm i0} d\lambda = (W_\pm f, g)$$



Uncertain reading. A narrow vertical crossing-out stands after f and before the comma. The underlying sign is illegible.

Original source details: GELFAND-P000429



Original source details: GELFAND-P000429

Diagram and typed description. The integral has $-\infty$ below it. No upper limit is visible.

This is the definition of the bilinear form of this operator;

One must be able to: $\frac{dE_\lambda g}{d\lambda}$, and this is possible with $\frac{dDE_\lambda g}{d\lambda}$, where D — a Hilbert–Schmidt oper.

The derivative [constitutes?] directly the absolutely continuous part of the spectrum.

Uncertain reading. The word begins with two open rounded letters. A provisional literal reading is “сставляет”. Alternatives are “составляет” (constitutes) and “оставляет” (leaves).

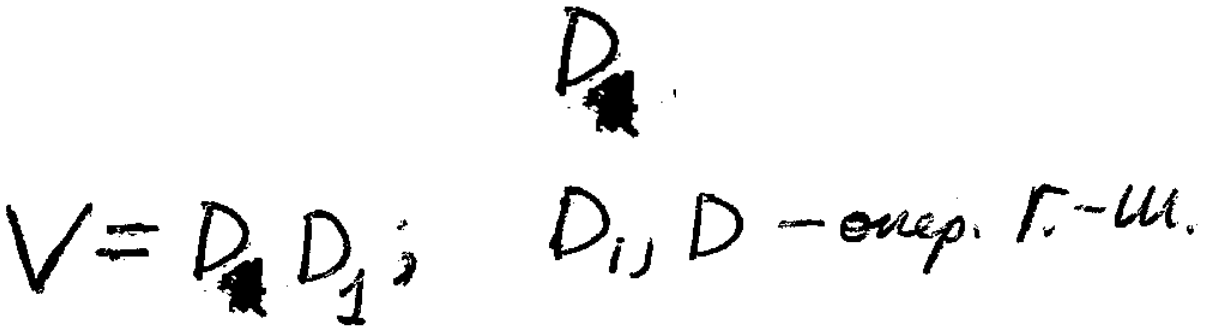
Original source details: GELFAND-P000429

Factor V into 2 Hilbert–Schmidt oper.

Almost everywhere there exists the derivative of $(E_\lambda f, g)$

$$\frac{d(E_\lambda f, g)}{d\lambda} + \left(D_1 R^0_{\lambda + i0} f, \frac{dD^* E_\lambda g}{d\lambda} \right)$$

$V = D_\bullet D_1; \qquad D_i, D$ — Hilbert–Schmidt oper.



Original source details: GELFAND-P000429

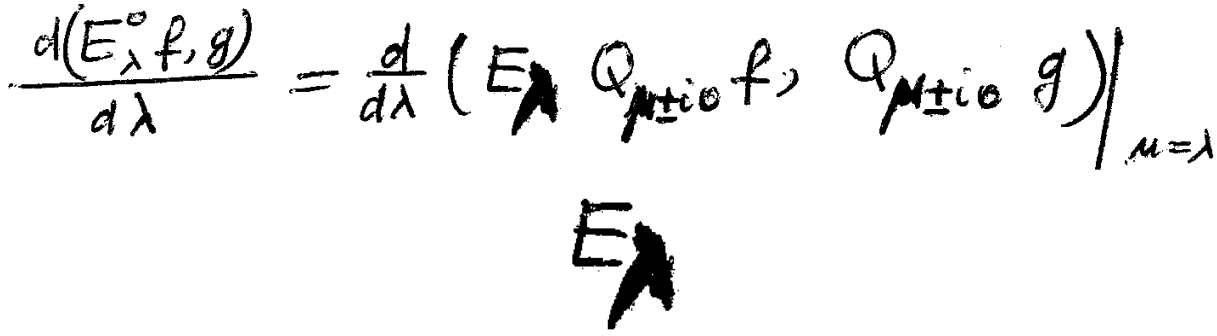
Uncertain reading. The first subscript in the product is heavily overwritten. 2 is possible. The second factor's subscript 1 is legible.

The oper. $W_\pm$ is independent of the decomposition of V as a product.

$$R^0_z - R^0_{\bar z} = Q^*_z (R_z - R_{\bar z}) Q_z; \qquad Q_z = I + V R^0_z;$$

Further,

$$\frac{d(E^0_\lambda f, g)}{d\lambda} = \frac{d}{d\lambda} \left(E_\bullet Q_{\mu \pm i0} f, Q_{\mu \pm i0} g \right) \Big|_{\mu = \lambda}$$



Original source details: GELFAND-P000429

Uncertain reading. The overwritten subscript on E at right resembles λ; μ is possible. Both Q subscripts read $\mu \pm i0$.

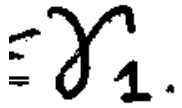
GELFAND-P000430

All D — the required riggings,
see the article in “Izvestiia”, 1967.

$$\int_{-\infty}^{\sigma} \frac{d}{d\lambda}\left(E_\lambda(I+VR_z^0)f,g\right)\Big|_{z=\lambda\pm i0} d\lambda = (E_\sigma W_\pm f, g)$$

More gen. variant:

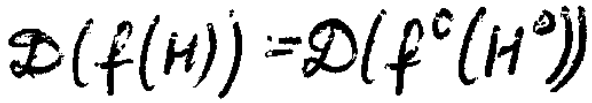
$$E(\Delta)(H-H^0)E^0(\Delta) \in \gamma_1.$$



Uncertain reading. A looped sign with subscript 1. Possible readings are γ_1 and $\mathfrak{S}_1$.

Original source details: GELFAND-P000430

$\mathcal{D}(f(H)) := \mathcal{D}(f^c(H^0))$, f, f^0 grow at infinity.



Uncertain reading. The second f has an open, c-shaped superscript. The later f has an oval superscript, set as 0. At the equality, a high dot and a low fleck sit at the left of the bars, resembling $:=$.

Original source details: GELFAND-P000430

3/IV-72 Minimum.

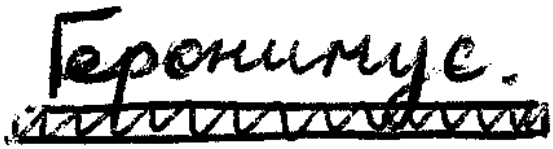
1. Grothendieck topol. and derived functors.
2. Descent.
3. Simplic.

Uncertain reading. “Simplic.” is a provisional reading of “Симплиц.” The middle strokes run together.

Original source details: GELFAND-P000430

Exam: work through Artin's article on the relation between formal and algebraic objects.

Geronimus.

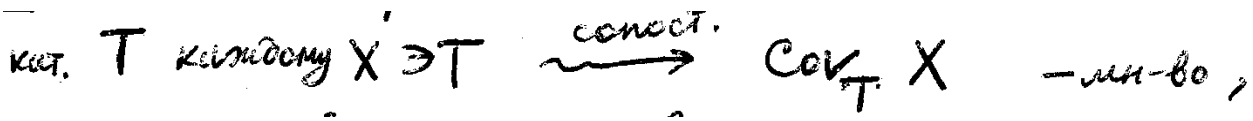


Original source details: GELFAND-P000430

Diagram and typed description. A narrow, diagonally hatched band underlines the name.

Def. Grothendieck topology. —

cat. T to each $X \ni T \overset{\text{assigns}}{\rightsquigarrow} \mathrm{Cov}_T X$ — a set,



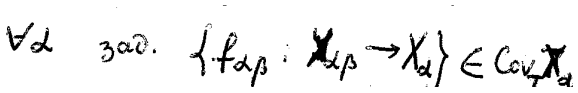
Original source details: GELFAND-P000430

Diagram and typed description. The labelled arrow begins with a wavy stroke. It follows $X \ni T$ as written.

its elements $\{f_\alpha : X_\alpha \longrightarrow X\}$. (a family of morphisms). ($\alpha \in A$ — a set).

1) $f : Y \longrightarrow X$ an isom. in cat. T. Then $\{f : Y \longrightarrow X\}$ is a covering.

2) $\{f_\alpha : X_\alpha \longrightarrow X\} \in \mathrm{Cov}_T X$ and $\forall \alpha$ given $\{f_{\alpha\beta} : \boldsymbol{X}_{\alpha\beta} \longrightarrow X_\alpha\} \in \mathrm{Cov}_T \boldsymbol{X}_\alpha$



Original source details: GELFAND-P000430

Diagram and typed description. Two X glyphs are heavily retraced, in the last family and after Cov_T.

Then

$$\{f_\alpha \circ f_{\alpha\beta} : X_{\alpha\beta} \longrightarrow X\} \in \mathrm{Cov}_T X$$

Mumford, Matematika 13:2

3) $\{f_\alpha : X_\alpha \longrightarrow X\} \in \mathrm{Cov}_T X$, $f : U \longrightarrow X$ — arbitrary morphism.

GELFAND-P000431

[source number: 62]

Then

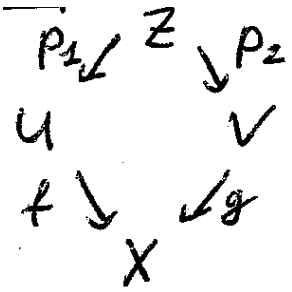
Original source details: GELFAND-P000431

Uncertain reading. Diagonal strokes cover a single sign. φ and 0 are possible readings.

3.1 $X_\alpha \underset{X}{\times} U$ exists (fibre prod.)

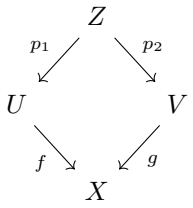
3.2 $\{p_2 : X_\alpha \underset{X}{\times} U \longrightarrow U\} \in \mathrm{Cov}_T\, U$.

Def. of fibre prod.:



Original source details: GELFAND-P000431

Diagram and typed description. The diamond has arrows p_1, p_2, f, g. The note on Z is to its right.



Z univ. — fibre prod.

T — the starting category.

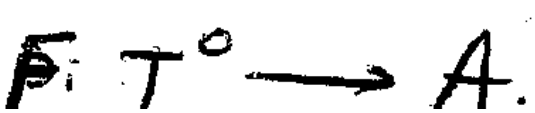
Let A be a fixed abelian category (with infinite direct sums)

Presheaf: ~~a~~ contravar. functor from T to A:

$$\boldsymbol{F} : T^0 \longrightarrow A.$$

Original source details: GELFAND-P000431

Undeciphered passage. Heavy crossing-out covers a short fragment before “contravar.”



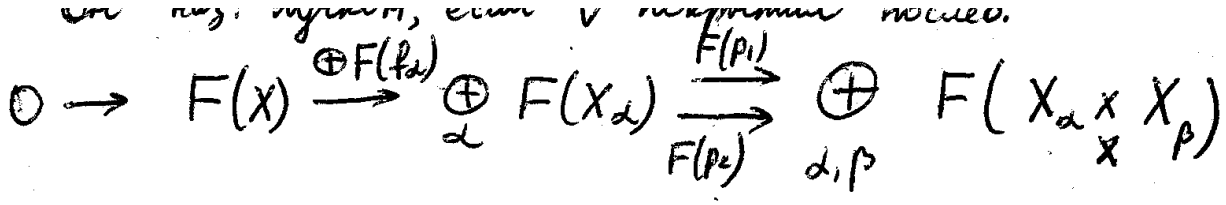
Original source details: GELFAND-P000431

Diagram and typed description. The strokes of F are reinforced. The small oval above T is set as 0.

P_T — the category of presheaves.

It is called a sheaf if $\forall$ coverings the seq.

$$0 \longrightarrow F(X) \xrightarrow{\bigoplus F(f_\alpha)} \bigoplus_\alpha F(X_\alpha) \underset{F(p_2)}{\overset{F(p_1)}{\rightrightarrows}} \bigoplus_{\alpha,\beta} F(X_\alpha \underset{X}{\times} X_\beta)$$



Original source details: GELFAND-P000431

Diagram and typed description. Two direct sums are written. The first labelled arrow reads $\bigoplus F(f_\alpha)$; the parallel arrows read $F(p_1)$ and $F(p_2)$.

S_T — the full subcat. of sheaves in the cat. of presheaves

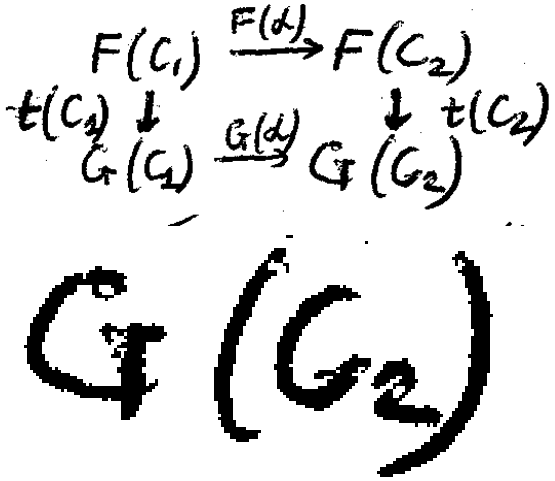
Morphism of functors

$$F : \mathcal{C} \longrightarrow \mathcal{D} \qquad F \xrightarrow{t} G$$
$$G : C \longrightarrow \mathcal{D}$$

Original source details: GELFAND-P000431

Diagram and typed description. Two rows point to $\mathcal{D}$. At right, an arrow from F to G carries the label t.

$\alpha : C_1 \longrightarrow C_2$



Original source details: GELFAND-P000431

Uncertain reading. The inner glyph at lower right has an inward stroke resembling G. The reading $G(G_2)$ is provisional. $G(C_2)$ is also possible.

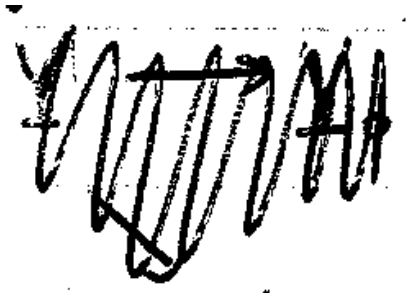
$$\begin{array}{ccc} F(C_1) & \xrightarrow{F(\alpha)} & F(C_2) \\ {\scriptstyle t(C_1)}\downarrow & & \downarrow{\scriptstyle t(C_2)} \\ G(C_1) & \xrightarrow{G(\alpha)} & G(G_2) \end{array}$$

This is a deep concept. Nontriviality: there are many different α, and there are exactly 2 of the $t(C_i)$.

GELFAND-P000432

Hence the functors themselves form a “category”.

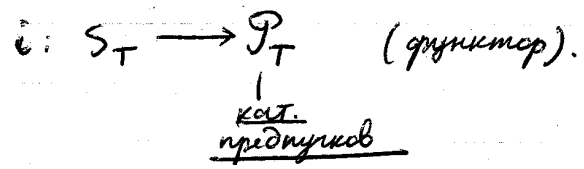
(up to the fact that morphisms may fail to form a set)



Original source details: GELFAND-P000432

Diagram and typed description. The crossing-out leaves a Y-like left label, a rightward arrow, and an A-like right label visible. A further looped stroke is at the right edge.

Theorem: the category S_T is abelian.



Original source details: GELFAND-P000432

Diagram and typed description. A vertical line without an arrowhead and two underlined rows are below $\mathcal{P}_T$.

$i : S_T \longrightarrow \mathcal{P}_T$ (functor).

cat. of presheaves

Idea of proof:

$\exists$ a functor $\# : \mathcal{P}_T \longrightarrow S_T$, adj. to i.

(associated sheaf).

Functors:

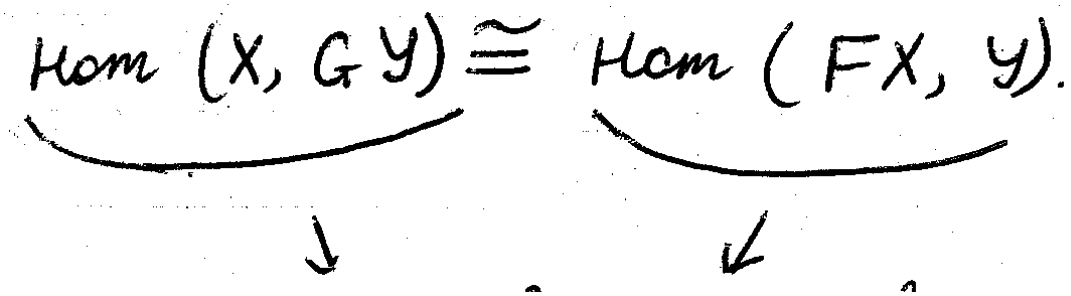
$$F : T_1 \longrightarrow T_2$$
$$G : T_2 \longrightarrow T_1$$

are called adjoint if $\forall X \in T_1$ and $\forall Y \in T_2$ a nat. isomorphism is given

$\mathrm{Hom}(X, GY) \cong \mathrm{Hom}(FX, Y)$.

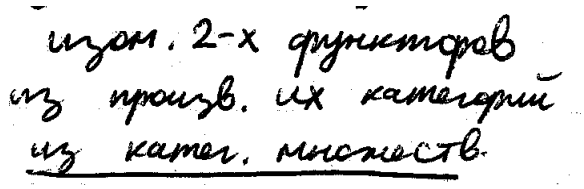
isom. of 2 functors from the product of their categories from the cat. of sets.

Def. using a univ. morphism from a presheaf to a sheaf also works



Original source details: GELFAND-P000432

Diagram and typed description. An arc and an arrow run from below each Hom expression toward the shared caption.



Original source details: GELFAND-P000432

Diagram and typed description. Both of the last two lines begin with “from” in the source. The final line is underlined.

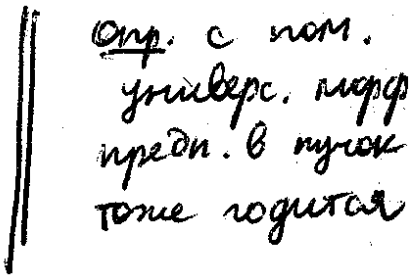
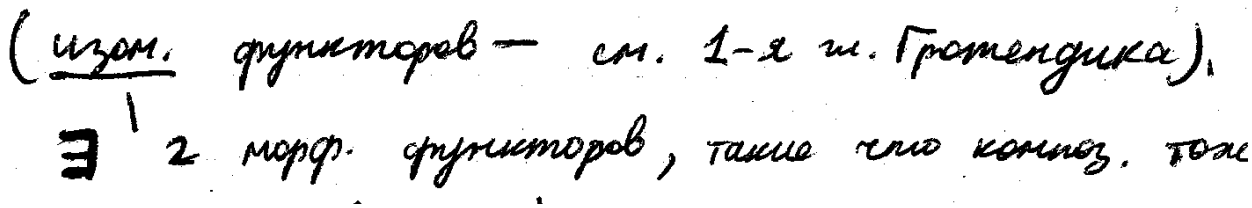


Diagram and typed description. The note is at the right of two vertical strokes.

Original source details: GELFAND-P000432

(isom. of functors — see ch. 1 of Grothendieck).

$\exists$ 2 morphisms of functors whose compositions are identities.



Original source details: GELFAND-P000432

Diagram and typed description. The abbreviation “isom.” is underlined and a separate stroke leads downward to the next line.

(for the proof see Artin).

GELFAND-P000433

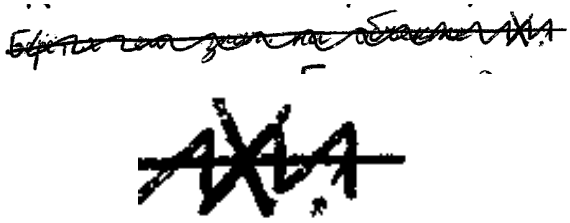
[source number: 63]

Definition. Let T, $X \in T$, $F \in S_T$

$$H^q(T, X, F) \stackrel{\text{def}}{=} R^q i(F)(X).$$

$R^q i$ — a new functor. (also from S_T to $\mathcal{P}_T$).

~~Its value is taken on the object X~~



Uncertain reading. The crossed-out sentence is provisional. There are further slanting strokes and a low mark after X, possibly “, A”.

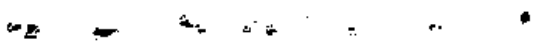
Original source details: GELFAND-P000433

Its value on F — a presheaf (i.e. also a functor)

A presheaf on X — ~~is~~

Uncertain reading. The crossed-out Russian “это” is a provisional reading.

Original source details: GELFAND-P000433



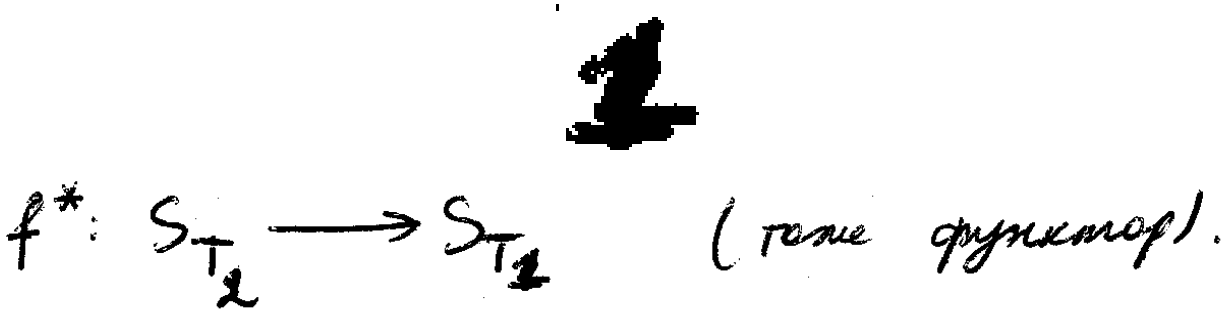
Original source details: GELFAND-P000433

Diagram and typed description. A row of dots and small strokes follows the crossing-out.

Definition.

$f : T_1 \longrightarrow T_2$ morph. of topol. $\underset{\text{def}}{\Leftrightarrow}$ covering $\longrightarrow$ cov.

$f^* : S_{T_2} \longrightarrow S_{T_1}$ (also a functor).



Original source details: GELFAND-P000433

Uncertain reading. The subscript on the right-hand T is heavily retraced. 1 and traces of 2 are possible. The shape is represented by an outline.

$$f^*F(X) = F(f(X)), \quad \text{where } X \in T_1,\ F \in S_{T_2}.$$

Easy to check that this is a sheaf.

Leray spectr. seq.

$f : T_1 \longrightarrow T_2$; $X \in T_1$; $F \in S_{T_2}$.

Then a spectr. seq. exists with second term

$$E_2^{p,q} = H^p(T_1, X, R^q f^*(F)) \Rightarrow H^*(T_2, f(X), F)$$

Let A, B, C be ab. categories

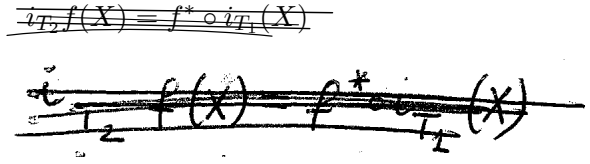
$$A \xrightarrow{F} B \xrightarrow{G} C.$$

Then (under certain cond.) there exists a spectr. seq.

$$R^pG \circ R^qF(X) \Rightarrow R^*(G \circ F)(X)$$

Leray's thm. is a special case of this situation: $\Rightarrow$

GELFAND-P000434



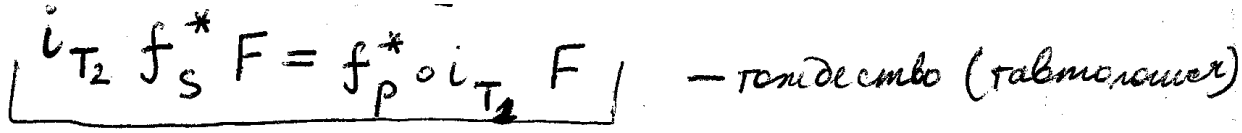
Original source details: GELFAND-P000434

Uncertain reading. The crossed-out formula is provisional. Horizontal strokes obscure the middle relation.

$i_{T_2} f_s^* F = f_p^* \circ i_{T_1} F$ — an identity (tautology)

Uncertain reading. The right-hand index on T is heavily retraced where it meets the bracket. 1 and 2 are possible. The left-hand T_2 is clear.

Original source details: GELFAND-P000434

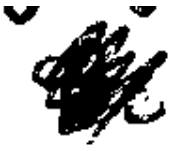


Original source details: GELFAND-P000434

Diagram and typed description. The horizontal line under the equality has two raised ends. The caption is at right.

f_p^* — the same morph. on presheaves

Leray's thm. is a special case because f_p^* is exact in the cat. of presheaves.



Uncertain reading. A separate entry is heavily crossed out. φ is possible.

Original source details: GELFAND-P000434

Classifying topology.

Definition.

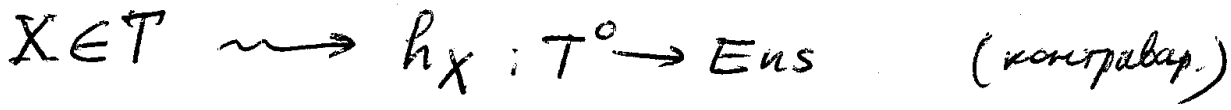
Let T be a Grothendieck topol.,

Uncertain reading. "Topol." is the provisional reading of a retraced Russian abbreviation. A wavy line runs below the phrase.

Original source details: GELFAND-P000434

G is a group in cat. T.

$$X \in T \rightsquigarrow h_X : T^0 \longrightarrow \text{Ens} \qquad \text{(contravar.)}$$



Original source details: GELFAND-P000434

Diagram and typed description. The first arrow starts with a wave. The oval above T is set as 0.

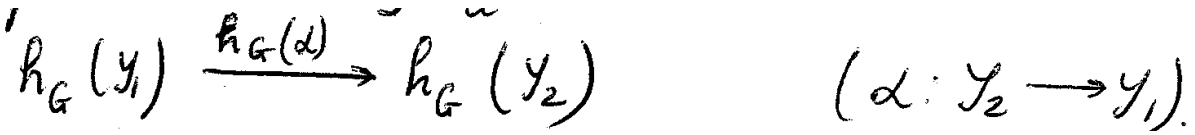
$$h_X(Y) = \mathrm{Hom}(Y, X).$$

Together with G there must be given a group structure on $h_G(Y)$,

functorial in Y and

$$h_G(Y_1) \overset{h_G(\alpha)}{\longrightarrow} h_G(Y_2) \qquad (\alpha : Y_2 \longrightarrow Y_1).$$

is a group hom.

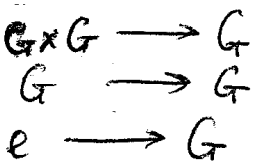


Original source details: GELFAND-P000434

Diagram and typed description. The arrow from $h_G(Y_1)$ to $h_G(Y_2)$ is labelled $h_G(\alpha)$. At right is $\alpha : Y_2 \longrightarrow Y_1$.

Another def.: a collection of 3 morphisms

$$G \times G \longrightarrow G$$
$$G \longrightarrow G$$
$$e \longrightarrow G$$



Original source details: GELFAND-P000434

Diagram and typed description. Three separate unlabelled arrows point rightward, all to G.

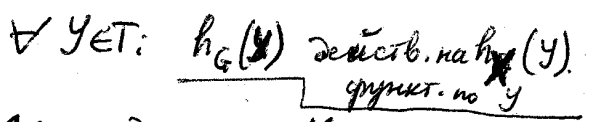
(e — terminal object in the cat.)

related by several commut. diagr.

G acts on X if $\forall Y \in T$: $h_G(Y)$ acts on $h_X(Y)$.

X — G-object. on the left

funct. in Y

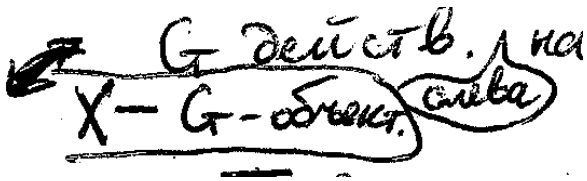


Original source details: GELFAND-P000434

Uncertain reading. Dense strokes obscure the argument of the first h and the subscript of the second. Y and X, respectively, are possible.

Original source details: GELFAND-P000434

Diagram and typed description. A stepped line runs below the formula, with the caption on the lower step.



Original source details: GELFAND-P000434

Diagram and typed description. The outline at left has a point at its lower-left end. “On the left” is in a separate outline.

TG — a new Grothendieck topol. Category —

GELFAND-P000435

[source number: 64]

— G-sets (G-objects).

$X \in T$ and on it an action of G.

Objects [...] of cat. TG — G-objects.

Undeciphered passage. The short crossed-out entry between “Objects” and “cat.” is unreadable.

Original source details: GELFAND-P000435

Morph. — commute with the action (G-morphisms).

Coverings — those cov. $\underbrace{\{f_\alpha : X_\alpha \to X\}}_{\text{in the orig. cat.}}$, which consist of G-morph.

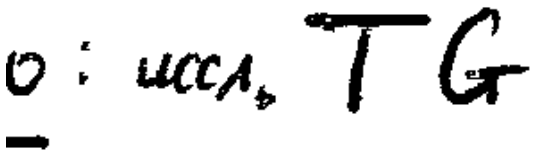
This is the classifying topology.

[...] Classif. space: final object, if it exists.

Undeciphered passage. The first sign in the cancelled entry before “Classif. space” resembles M. Two long strokes extend below it. The full reading is unresolved.

Original source details: GELFAND-P000435

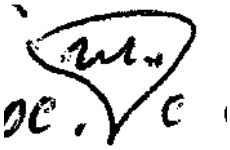
Interesting: иссл.[?] $TG \longrightarrow T$ (forgetf.).



Uncertain reading. The two curves before the last ascending stroke favour «иссл.». «исл.» and «исх.» remain possible.

Original source details: GELFAND-P000435

This is an analogue of a principal bundle (univ. гл.[?] with base the classif. space).



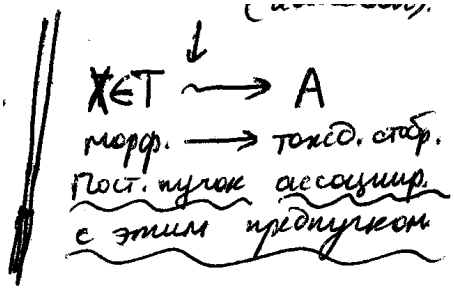
Uncertain reading. The circled insertion provisionally reads «гл.» (principal); «им.» is possible.

Original source details: GELFAND-P000435

Theorem. A const. sheaf is def. in the obvious way by a group A (abelian).

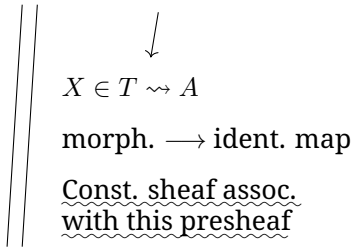
Aim: cohom. of TG from the cohom. of T.

For Quillen's theor. a const. sheaf is needed.



Original source details: GELFAND-P000435

Diagram and typed description. Two rules separate the note on the right. An arrow from the line "(abelian)" points down to the first row; its rightward arrow starts with a wavy stroke. Both closing lines have wavy underlines, with a break in the first line.

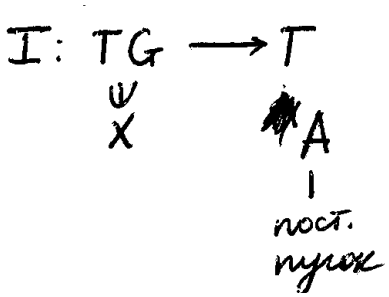


$I : TG \longrightarrow T$.

[...] The Leray spectr. seq., with some addit. restrictions, is arranged thus:

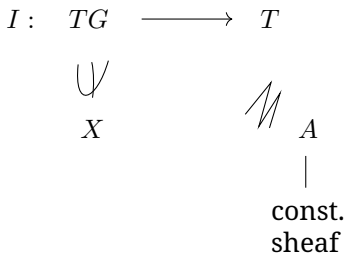
Original source details: GELFAND-P000435

Undeciphered passage. The cancelled entry before "Spectr." has a C-shaped initial curve and loops below the line. The full reading is unresolved.



Original source details: GELFAND-P000435

Diagram and typed description. Below TG is a U-shaped stroke with a middle bar, above X. A blot lies to the left of A; a short rule extends down to the two-line label. The direction of the U-shaped mark is unresolved.



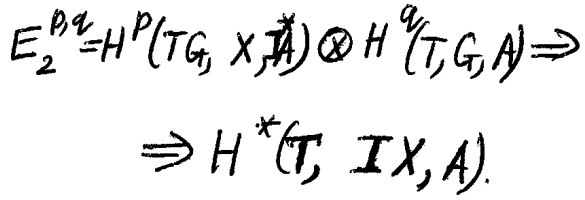
Original source details: GELFAND-P000435

Undeciphered passage. Dense ink covers the entry immediately to the left of the lower A.

$$E_2^{p,q} = H^p(TG, X, A^*) \otimes H^q(T, G, A) \Rightarrow$$
$$\Rightarrow H^*(T, IX, A).$$

Original source details: GELFAND-P000435

Uncertain reading. The coefficient is provisionally typeset as A^*. An overlapping vertical stroke with crossbars is visible below the star; the order of the layers is unresolved.



Original source details: GELFAND-P000435

Diagram and typed description. The sign $\Rightarrow$ closes the first line and is repeated at the start of the next.

X — point.

$$H^p(BG) \otimes H^q(G) \Rightarrow H^*(E).$$

GELFAND-P000436

Quillen's theor.

$G = GL(n, F_q)$

Then

$$H^*\Big(G, \underset{\substack{\uparrow \\ \text{triv. module}}}{F_\ell}\Big) = S[c_1, \ldots, c_n] \otimes \Lambda[c_1', \ldots, c_n']$$

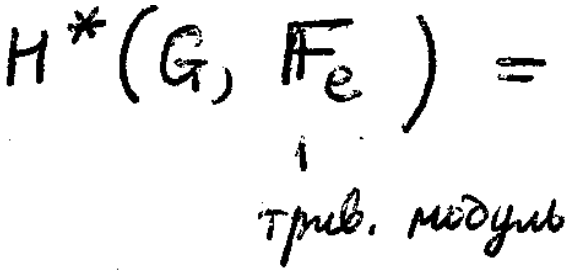


Diagram and typed description. Below the first F_ℓ is a short stroke with an upward point and the label "triv. module".

Original source details: GELFAND-P000436

$$c_i \in H^{2i}(G, F_q)$$
$$c_i' \in H^{2i-1}(G, F_\ell)$$

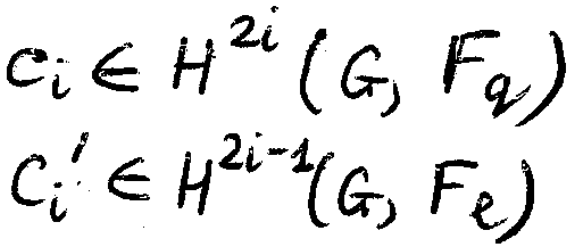
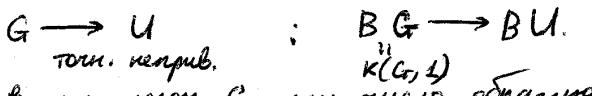


Diagram and typed description. The subscript of F reads q on the first line and ℓ on the second.

Original source details: GELFAND-P000436

Where does this come from?



Original source details: GELFAND-P000436

Diagram and typed description. Below the first arrow is “faithf. irred.”. Two short strokes above $K(G,1)$ are below BG.

If $G \longrightarrow U$; $BG \longrightarrow BU$.

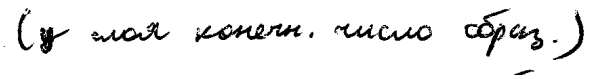
faithf. irred.

$K(G,1)$

hence in the alg. cohom. of G a fin. number of generators

(this is kn.). (from the spectr. seq.)

(I[?] have a fin. number of gen.)



Original source details: GELFAND-P000436

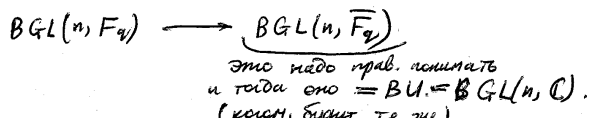
Uncertain reading. The joined strokes after «у» provisionally read «мя», a contracted form of «меня» (me). The fully written «меня» is also possible.

We need to know this map explicitly.

There is none among Lie groups!

Quillen’s rem.: take matr. over the closure of the field F_q

$GL(n, F_q)$.



Original source details: GELFAND-P000436

Diagram and typed description. An arc under the right-hand term connects it to the three lines below.

$BGL(n, F_q) \longrightarrow BGL(n, \overline{F}_q)$

This must be understood prop. and then it $= BU. = BGL(n, \mathbb{C})$. (the cohom. will be the same).

Original source details: GELFAND-P000436

Uncertain reading. The separate low mark after BU and before the reinforced equals sign is set as a dot. Its function is unresolved.

Theor. of Ma[n/m]ford — coincid. of cohom.

Original source details: GELFAND-P000436

Uncertain reading. The connected middle strokes allow «Манфорда» (Manford) and «Мамфорда» (Mamford).

Grothend. theor.

Rem. of I. Bersht./Bernsht.[?]: repres. of $\overline{F}_q$ and $\mathbb{C}$ are descr. in terms of high. weights.

Uncertain reading. After «И.» the joined strokes allow «Бершт.» (Bersht.) and «Берншт.» (Bernsht.).

Original source details: GELFAND-P000436

GELFAND-P000437

[source number: 65]

1[0/8]/IV Geronimus.

Uncertain reading. The second date digit is overwritten. Both 10/IV and 18/IV remain possible. There are rules below and to the right, with rules above and below IV.

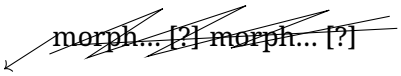
Original source details: GELFAND-P000437

Derived category.

Let A — an abelian category.

$K(A)$ — category of complexes with val. in A;

Undeciphered passage. Two fragments resembling «морф» (morph.) survive the crossing-out. The slanted arrow points down and left.



Original source details: GELFAND-P000437

$$C = \{0 \longrightarrow C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2 \xrightarrow{d_2} \dots\}$$

$$H^q(C) = \operatorname{Ker} d_q / \operatorname{Im} d_{q-1}.$$

Uncertain reading. Both subscripts of d are darkened. In the denominator the sign before -1 is provisionally q.

Original source details: GELFAND-P000437

$$D = \{0 \longrightarrow D_0 \xrightarrow{d'_0} D_1 \longrightarrow \dots\}$$

map of degree τ:

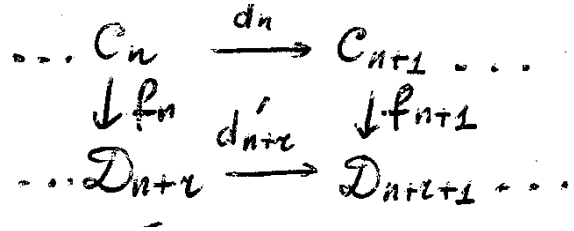


Diagram and typed description. Both vertical arrows are labelled on the right. Each horizontal arrow is labelled above.

$$\begin{array}{ccccc} \cdots & C_n & \xrightarrow{d_n} & C_{n+1} & \cdots \\ & \downarrow f_n & & \downarrow f_{n+1} & \\ \cdots & D_{n+\tau} & \xrightarrow{d'_{n+\tau}} & D_{n+\tau+1} & \cdots \end{array}$$

Original source details: GELFAND-P000437

Let f and g — 2 maps of degree τ.

They are called homotopic, if $\exists$ a map α of deg. $\tau - 1$:

$$\forall n: \quad f_n - g_n = d_n\alpha_n + \alpha_n d'_n \qquad (f - g = d\alpha + \alpha d').$$

Original source details: GELFAND-P000437

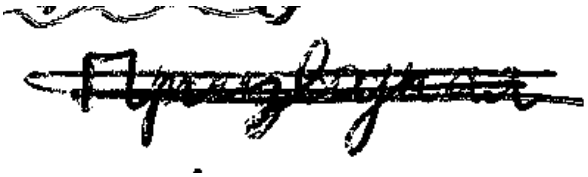
Diagram and typed description. The second α_n is more heavily retraced. Both subscripts on d and both on α read n.

Homot. maps induce the same cohom. morph.

Morphisms in $K(A)$, — morphisms of degrees τ (all) —

— direct sum — quotient by morphisms, homotopic to zero.

~~Derived~~ Category of complexes — is not abelian!



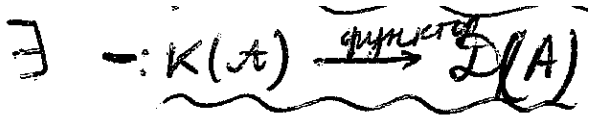
Original source details: GELFAND-P000437

Diagram and typed description. "Derived" is crossed out before "Category".

(because of homot. equival.)

Deriv. categ. for an ab. categ. A, — is the cat. $D(A)$, and

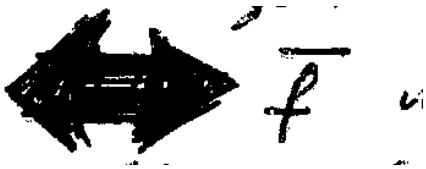
$\exists$ —: $K(A) \xrightarrow{\text{functor}} D(A)$, and $\forall$ morph. of compl. inducing a cohomology isom., is assigned an isom. in cat. $D(A)$.



Original source details: GELFAND-P000437

Diagram and typed description. "Functor" is written above the arrow. A wavy line runs below $K(A)$, the arrow and $D(A)$.

f induces a cohom. isom. $\iff$ $\bar{f}$ isom. in $D(A)$



Original source details: GELFAND-P000437

Diagram and typed description. The two-headed arrow is heavily retraced. A bar is above f.

and $D(A)$ is univ. in this sense.

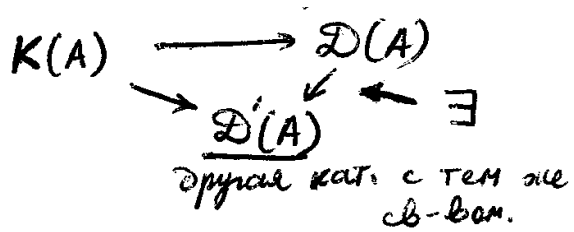
Construction:

consider morphisms the same for morph. inducing cohom.

$D(A)$ is def. by 2 prop.

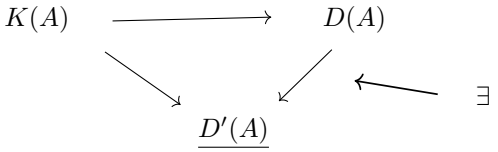
1) f^* isom. $\Rightarrow \bar{f}$ isom.

2) Let



Original source details: GELFAND-P000438

Diagram and typed description. A rule and caption sit below $D'(A)$. To the right of the diagonal arrow, a separate pointer from ∃ points left.



$D(A)$ — deriv. category for A

one can pr. (this is a theor.) that then in $D(A)$ f^* isom. $\Leftrightarrow \bar{f}$ isom.

This is not the same as the categ. C with morph. $f \sim g$, if $f^* = g^*$.

In $D(A)$ there are more morphisms.

To maps inducing cohom. isom., inverses are added formally.

It suffices to have a map $C_1 \longrightarrow C_2$, inducing a cohom. isom. — then it is an isom.

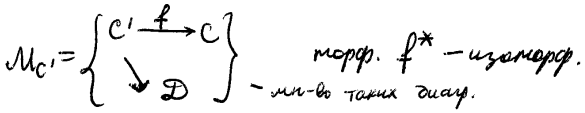
Construction. Objects — complexes

Let C, D — complexes.

~~For...~~ [?] To an object C' associate a diagr.

Original source details: GELFAND-P000438

Undeciphered passage. An opening resembling «Для» (For) and a tall loop survive before “To an object”. Crossing strokes cover the right end.



Original source details: GELFAND-P000438

Diagram and typed description. Two arrows start at C' inside the braces. Only the upper one is labelled f.

$$\mathcal{M}_{C'} = \left\{ \begin{array}{ccc} C' & \xrightarrow{f} & C \\ & \searrow & \\ & D & \end{array} \right\}$$

morph. f^* — isomorph.

— set of such diagr.

With resp. to the map C' a proj. system, indexed by objects of the category $C^{[?]}$; ...

Original source details: GELFAND-P000438

Uncertain reading. The joined strokes above C resemble an H or M. A separate prime is unresolved.

Original source details: GELFAND-P000438

Undeciphered passage. Horizontal strokes cover the entry after the category name. Separate lower loops survive.

(contravar. functor of C').

Proj. lim.it[?] of this syst. — set of maps[?] from C to D.

Original source details: GELFAND-P000438

Uncertain reading. A gap and a small dot occur before the letter д. «предел» (limit) and a contracted reading remain possible.

Original source details: GELFAND-P000438

Uncertain reading. An interrupted round stroke precedes б. «отобр.» is provisional, with «отбр.» also possible. Both are abbreviations for maps.

$$f : A \longrightarrow A'$$

Deriv. functor:

$$D(f) : D(A) \longrightarrow D(A').$$

Original source details: GELFAND-P000438

Diagram and typed description. Two separate lengthwise traces run near the right edge.

GELFAND-P000439

[source number: 66]

Ka[z/zh]dan.

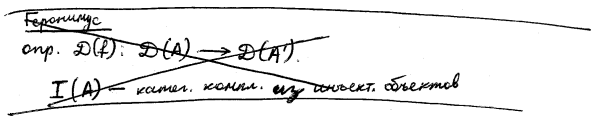
Original source details: GELFAND-P000439

Uncertain reading. The joined strokes after “Ка” allow «Каздан» (Kazdan) and «Каждан» (Kazhdan). The surname is doubly underlined.

Let A — an ab. categ. with a suff. number of inj. obj.
Consider the full subcategory of inj. objects. Inj(A)
Consider there the categ. of complexes.
Categ. of compl. in Inj(A) $\cong$ the deriv. category of A.

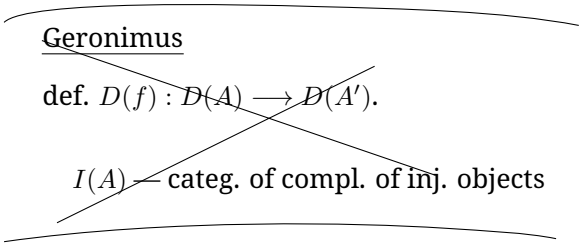
Original source details: GELFAND-P000439

Diagram and typed description. The sign between Inj(A) and "deriv." has a wave over two strokes.



Original source details: GELFAND-P000439

Diagram and typed description. Two long rules bound the three lines above and below. Two oblique strokes cross out the block.



Morava.
Cobordism.
M, N — manif.[?], compact[?], C^∞, $\partial M = \partial N = \varnothing$.

Original source details: GELFAND-P000439

Uncertain reading. The abbreviation has a loop with a descender after the hyphen. «мн-зия» is provisional, with «мн-гия» also possible.

Original source details: GELFAND-P000439

Uncertain reading. The final rounded stroke favours «компактные»; «компактны» is still possible. Both readings mean "compact".

$$\dim M = \dim N = n.$$

$$M \sim N \quad \text{cobordant} \quad \Longleftrightarrow \quad \exists\, \text{manif. } W, \quad \partial W = M \cup N$$

(W comp., C^∞).

$$N_n = \{\text{set of this cl. equiv.}\}. \quad = \{\text{cobord. cl.}\}.$$

1) $N_n = \{\text{semigroup}\}$

$$[M] + [N] = [M \cup N] \qquad \text{(disjoint)}$$
$$[M] + [\varnothing] = [M].$$

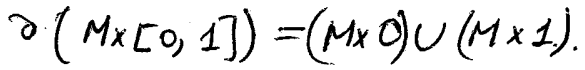
In | fact

Original source details: GELFAND-P000439

Diagram and typed description. A tall slanting rule separates “In” from “fact”.

$N_n =$ group, because $[M] + [M] = [\varnothing]$

$$\partial(M \times [0,1]) = (M \times 0) \cup (M \times 1.).$$



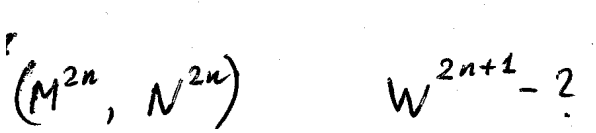
Original source details: GELFAND-P000439

Uncertain reading. A separate dot follows 1 before the last parenthesis. Its function is unresolved.

Spec. case: 1) M oriented

2) M complex

$$(M^{2n}, N^{2n}) \qquad W^{2n+1} - ?$$



Original source details: GELFAND-P000439

Uncertain reading. After the dash, the upper part of the sign resembles a 2. There is a separate dot below. “?” is a possible reading.

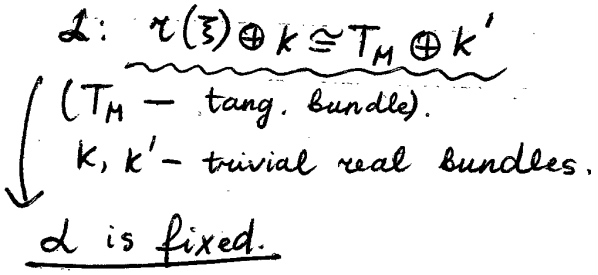
GELFAND-P000440

Weak compl. manif.

$(M, \xi) = \{M$ — manif.; ξ — compl. vect. bundle over $M\}$.

Original source details: GELFAND-P000440

Undeciphered passage. Below the pair is a short stroke and a crossed-out sign with a loop. Its lettering is unresolved.



Original source details: GELFAND-P000440

Diagram and typed description. A wavy line is below the isomorphism. At left a long curved arrow points down to the underlined condition.

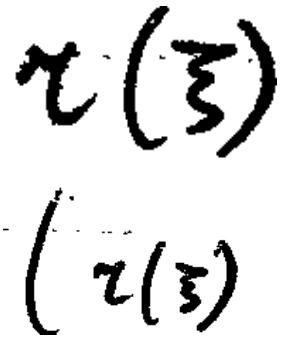
$\alpha : \tau(\xi) \oplus k \cong T_M \oplus k'$

(T_M — tang. bundle).

k, k' — trivial real bundles.

α is fixed.

($\tau(\xi)$ — real ... sous-jacent bundle).



Original source details: GELFAND-P000440

Uncertain reading. Both signs before ξ have a cross stroke and a curved stem. They are provisionally set as τ; r is possible.

Original source details: GELFAND-P000440

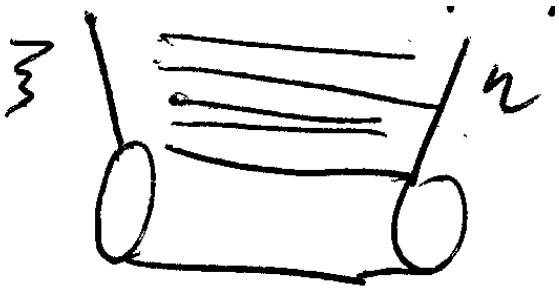
Undeciphered passage. A single sign after "real" is crossed out. An oblique stroke extends below it.

$[M, \xi, \alpha] \longrightarrow M$ ...

class of equiv. three objects

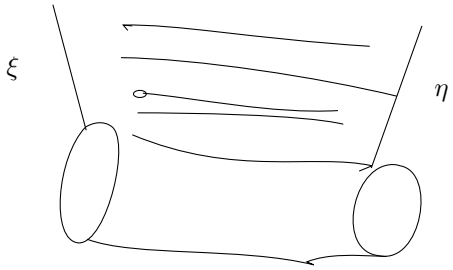
Original source details: GELFAND-P000440

Undeciphered passage. A short crossed-out entry remains to the right of M. Parts of loops show through the strokes.



Original source details: GELFAND-P000440

Diagram and typed description. The lower line connects two ovals. The slanted uprights are labelled ξ and η. Two of the five inner strokes reach the right upright. The third has a small loop at left. The role of the angular ends is unresolved.



S^1 — weak complex manif.

$T_{S^1} =$ triv.

Every W. c. m. is cobord. to compl. man.).

Ω_n^U — group | cobord. n-dim man. (w-compl.).

Film and conn. bundle. $\dots$ Σ.

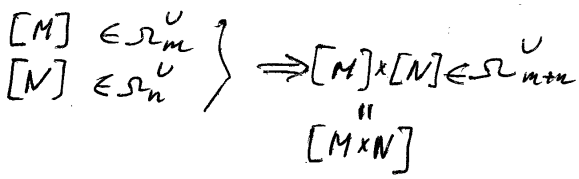
Original source details: GELFAND-P000440

Undeciphered passage. Overlapping strokes cover a loop and diagonals before Σ. The earlier sign is unreadable.

$(W, \Sigma, \dots)$ — also w. c. m.

Original source details: GELFAND-P000440

Uncertain reading. Three separate groups of strokes follow the comma after Σ. An ellipsis remains a tentative reading.



Original source details: GELFAND-P000440

Diagram and typed description. A brace gathers two lines. Two short vertical strokes sit between the product of classes and the expression below.

$$\left.\begin{array}{l}[M] \in \Omega_m^U \\ [N] \in \Omega_n^U\end{array}\right\} \Rightarrow [M] \times [N] \in \Omega_{m+n}^U$$
$$||$$
$$[M \times N]$$

GELFAND-P000441

[source number: 67]

$$\Omega_*^U = \bigoplus_{n\geq 0} \Omega_n^U$$

This|is a graded ring.

Original source details: GELFAND-P000441

Diagram and typed description. A short vertical stroke stands between “This” and “is”. The phrase has a wavy underline.

Th. $\Omega_*^U \cong \mathbb{Z}[t_0, t_1, \ldots, t_n, \ldots]$.

Original source details: GELFAND-P000441

Diagram and typed description. The enclosure is open at the top.

$\deg t_i = 2i$

(Ord$[M] \neq 2$ — compl. structure!).

Ω_n^U is group

Proof: $-[M] = [M \overset{\text{opposite}}{\text{orientation}}]$.

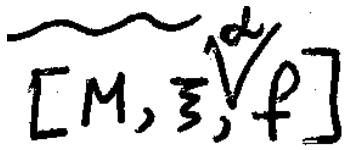
for orient cobord.

Compl. case — something similar.

X — topological space.

$X \mapsto \Omega_n^U(X)$ — group (functor of X).

$[M, \xi, \overset{\alpha}{\sqrt{}} f]$ ~~[bord...?]~~

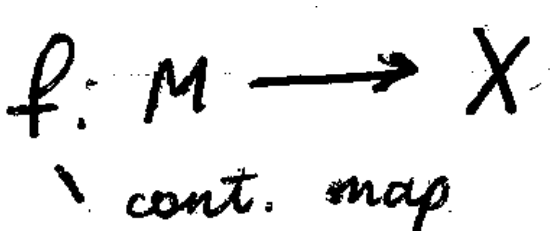


Original source details: GELFAND-P000441

Uncertain reading. The letter above the deep insertion mark reads α. The left end of the mark hooks back. A comma sits below it.

Original source details: GELFAND-P000441

Undeciphered passage. Strokes resembling “bord” or “cobord” remain under the cancellation. The full reading is unresolved.

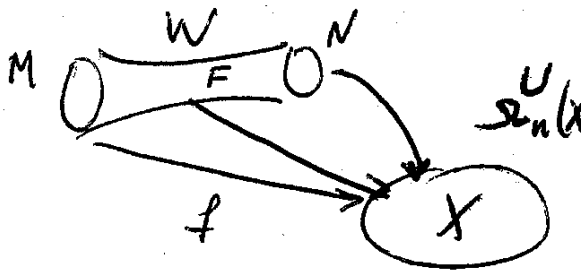


Original source details: GELFAND-P000441

Diagram and typed description. A short slanted stroke leads from the map notation to the caption below.

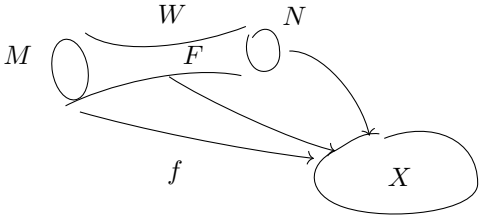
$f : M \longrightarrow X$

\ cont. map

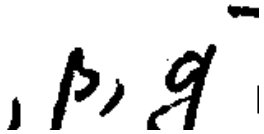


Original source details: GELFAND-P000441

Diagram and typed description. The upper line is separated from the left oval. Both lines stop before the right oval. A separate arc from the right of N points to X, as does the middle arrow. The lower arrow labelled f stops just short of the edge of the oval X.



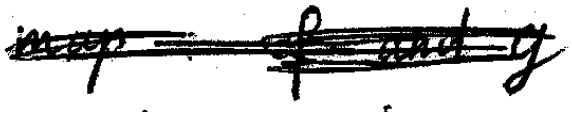
$\Omega_n^U(X)$ — equivalence classes
(group of Bordism of X).
$[M, \xi, \alpha, f] \sim [N, \eta, \rho, g]$



Original source details: GELFAND-P000441

Uncertain reading. The sign before g in the upper line has one loop and is provisionally read as ρ or p. β is possible. The β in the lower line has two bends.

iff ~~map [?] f and g~~ $[M, \xi, \alpha] \sim [N, \eta, \beta]$



Original source details: GELFAND-P000441

Uncertain reading. "map" and "f and g" are visible under the cancellation. The sign between them resembles $=$ or a dash.

and equiv. prolong. over W
$\bigoplus \Omega_n^U(X) = \Omega_*(X)$ — graded group (mult. abs.)
GELFAND-P000442

This is a Ω_*^U-module.

Original source details: GELFAND-P000442

Uncertain reading. A separate dark spot appears below "a". Its function is unresolved.

...

Original source details: GELFAND-P000442

Undeciphered passage. Horizontal and oblique strokes cover the short entry. The letters are unreadable.

Problem: describe Ω_*^U-module structure. (in Ω_*^U)

$\Omega_*^U = \mathbb{Z}[t_0, t_1, \ldots]$ = coordin. ring of a big affine space $\mathbb{A}$.

Module $\Omega_*^U(X)$ — sheaf over $\mathbb{A}$.

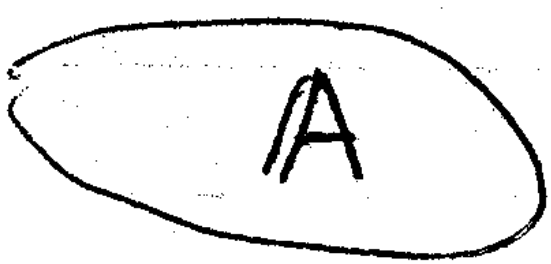
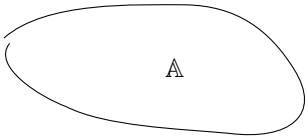


Diagram and typed description. The slanting outline almost closes on the left. Inside it is $\mathbb{A}$, with a double left stroke.

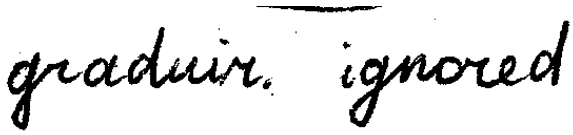


Original source details: GELFAND-P000442

graduir. ignored

$\bigoplus \Omega_{2n}^U(X)$ and $\bigoplus \Omega_{2n+1}^U(X)$ —

— from graduir. (forgets the rest)



Uncertain reading. The abbreviation in both lines reads “graduir.”.

Original source details: GELFAND-P000442

Theor. $\exists$ dens. open set in the Zar. topol. in $\mathbb{A}$, such that
Morava and Quillen,

the fibre at $\forall$ point is the gr. $K^\bullet(X)$ $\left(= K^0(X) \oplus K^1(X)\right)$!!

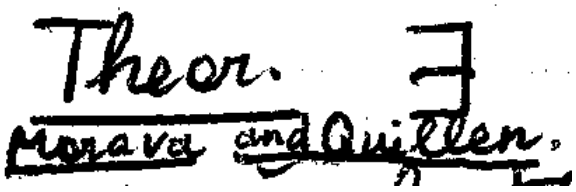


Diagram and typed description. “Morava” and “and Quillen” appear below the heading. There are two separate lines under the names.

Original source details: GELFAND-P000442

Uncertain reading. The Russian is provisionally read as “пл.” and translated as “dens.”. The second letter more closely resembles “л” than the “н” of “мн-во” to the right.

Original source details: GELFAND-P000442

Diagram and typed description. The Russian “в” is written above the line before $\mathbb{A}$, inside a loop-shaped insertion mark.

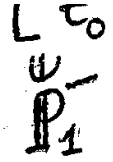
Original source details: GELFAND-P000442

Original source details: GELFAND-P000442

Uncertain reading. The heavily circled sign with intersecting strokes is provisionally set as $\oplus$. $\otimes$ is possible.

$\mathbb{Z}[t_0, \ldots, t_n, \ldots][t_1^{-1}]$ — this is that very open set.

$t_1 = \mathbb{P}_1$ — compl. proj. bundle

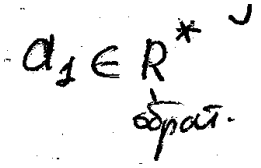


Original source details: GELFAND-P000442

Uncertain reading. A separate U-shaped mark with an inner stroke stands above $\mathbb{P}_1$. A detached rising stroke to the right may be a prime. No continuous arrow shaft is visible.

R — commutative; $\Gamma(R) = \left\{ \sum_{i=1}^{\infty} a_i t^i \;\middle|\; a_i \in R \right\}$

$a_1 \in R^*$ — invert.



Original source details: GELFAND-P000442

Diagram and typed description. A short oblique stroke lies between R^* and "invert.". No separate arrowhead is visible.

gr. under compos. of maps.

GELFAND-P000443

[source number: 68]

$\Gamma(R)$ — inf. dim alg. group. (compos. — polinomial)

Γ acts on A

Sheaf Ω_*^U is an equivariant sheaf (const on orbits).

Original source details: GELFAND-P000443

Uncertain reading. There is a dark patch of ink at the left of Ω. Any earlier writing underneath is unreadable.

Γ: {Rings} $\longrightarrow$ {groups}.

Original source details: GELFAND-P000443

Diagram and typed description. The arrowhead at right is clear. At left is a short lower hook whose role is unresolved.

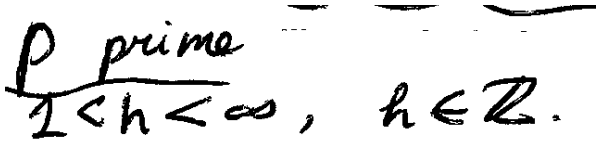
The actual theor. should consist in the descr. of the orbits.

~~Orb.[?]~~ over $\mathcal{D}$ — fibers $K^\bullet(X)$.
$\mathcal{D}_{(p,h)}$ p prime — orbits,
$1 < h < \infty, \quad h \in \mathbb{Z}$.
$\mathcal{D}_\infty$ = fibers $H^*(X, \mathbb{Z})$.

orbits.

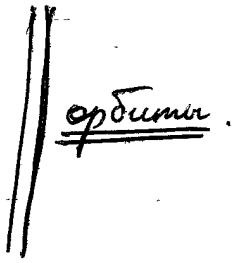
Original source details: GELFAND-P000443

Undeciphered passage. An initial loop, a tall loop and a dot show through the cancellation. “Orb.” is a provisional reading.



Original source details: GELFAND-P000443

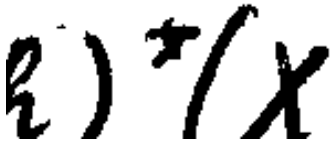
Diagram and typed description. The strict inequality $1 < h < \infty$ is written below “p prime”. The top stroke of 1 meets the wavy underline.



Original source details: GELFAND-P000443

Diagram and typed description. Two near-vertical strokes set off the doubly underlined caption on the right.

$K(h)$ — functor $K(h)^+(X, \mathbb{Z}_p)$ — $\mathbb{Z}_p$-module.



Original source details: GELFAND-P000443

Uncertain reading. The superscript of $K(h)$ has a crossbar and one slanted stroke. It is provisionally set as $+$. $*$ is possible, though the stars in Ω^U_* and H^* show extra arms.

Bott periodicity in dim $2(p^h - 1)$

$\exists$ Adams oper. $\psi^\alpha \colon K(h)$

Original source details: GELFAND-P000443

Diagram and typed description. The arc at the right of $K(h)$ runs from the lower end to the upper end. Its arrowhead points left.

$\alpha \in$ units of p-adic division algebra of ranke h^2.

Original source details: GELFAND-P000443

Uncertain reading. The sign before $\in$ resembles the superscript of ψ. It is provisionally set as α, with λ possible. The final "e" is visible in "ranke".

This is station. subgroup of a point on the orbit.

Next time — continuation

Formal groups

Original source details: GELFAND-P000443

Uncertain reading. An angular loop, a hooked stem and a separate dot lie within the outlined shape. The glyph remains unidentified.

$\mathbb{A}$ — moduli space for 1-dimensional commut. form. groups.

F — form. gr. over a ring R, if an elem. is given $F \in R[[X, Y]]$,

Original source details: GELFAND-P000443

Uncertain reading. The inner opening mark is rounded, almost like a C. Two closing brackets are visible. The double opening bracket is provisional.

1) $F(X,Y) = F(Y,X)$; 2) $F(0,X) = X$; 3) $F(X, F(Y,Z)) = F(F(X,Y),Z)$

(the inv. exists automatically). X, Y — form. var. — letters.

GELFAND-P000444

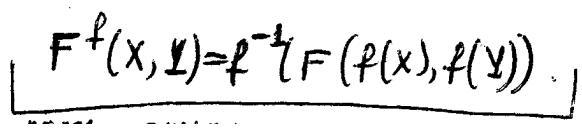
$f \in \Gamma(R)$ $\Rightarrow$ s'

Original source details: GELFAND-P000444

Uncertain reading. A separate slanting stroke sits below the right end of the underline. The small adjacent sign and its raised stroke are provisionally read as s', with $5'$ possible.

$$f = a_1 t + a_2 t^2 + \dots$$
$$a_1 \in R^*$$

Change of var. —

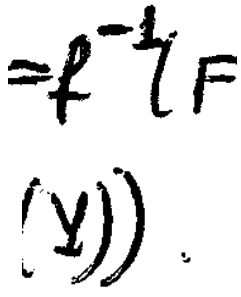


Original source details: GELFAND-P000444

Diagram and typed description. A long line with raised ends encloses the formula from below.

$F^f(X,Y) = f^{-1}\big[F\big(f(X), f(Y)\big).$

Uncertain reading. The slanted sign with a short lower hook after f^{-1} is provisionally set as [. Two round parentheses and a stop follow Y. No separate outer closing delimiter is visible.



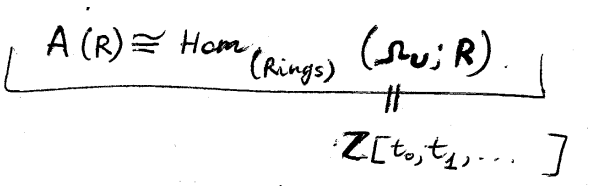
Original source details: GELFAND-P000444

f acts on the set of form. groups.

R — ring $\rightsquigarrow$ {form. groups $A(R)$ — set.}

$$A\colon (\text{Rings}) \xrightarrow{\text{Functor}} (\text{Sets})$$

it is representative



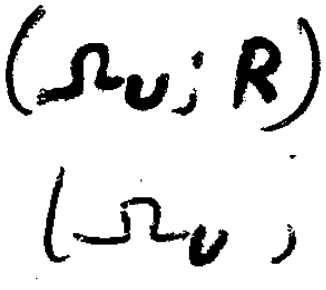
Original source details: GELFAND-P000444

Diagram and typed description. The frame is open at the top. Two vertical strokes below Ω_u cross its lower line. The polynomial ring is written below them.

$A(R) \cong \mathrm{Hom}_{(\text{Rings})}(\Omega_u; R).$

$\mathbb{Z}[t_0, t_1, \dots]$

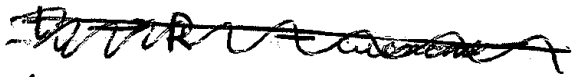
Uncertain reading. The sign after Ω is lowered in both occurrences. It is provisionally set as Ω_u. A capital U in the subscript is possible.



Original source details: GELFAND-P000444

Hom determ. by values on t_i. which form. gr.

~~$\Omega_u \longrightarrow R$[?] ...~~



Original source details: GELFAND-P000444

Undeciphered passage. At left under the cancellation are an omega-like loop and lower sign, followed by a rightward arrow-like stroke and R. The beginning may read $\Omega_u \to R$ or contain $\forall$. A C-like sign is visible farther on. The full inscription is unresolved.

$$\forall R: \quad \Gamma(R) \times A(R) \longrightarrow A(R);$$
$$\Gamma \times \mathbb{A} \longrightarrow \mathbb{A}.$$

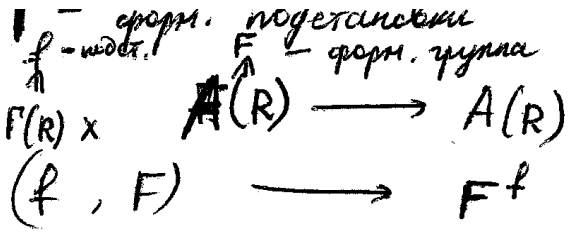
(Γ acts on A as an alg. gr. on an alg. set).

$$A(R) \underset{\text{def}}{=} \text{Hom}(\Omega_u, R).$$

Γ — form. substitutions

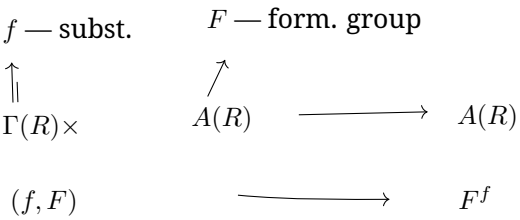
Original source details: GELFAND-P000444

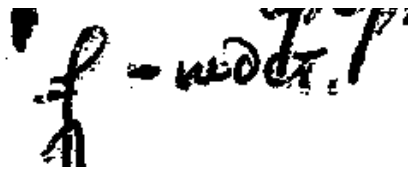
Uncertain reading. The Russian is provisionally read as “подстановки”. Joined strokes between “д” and “т” obscure the reading.



Original source details: GELFAND-P000444

Diagram and typed description. Two short arrows point up from the vicinity of $\Gamma(R)$ to f and from the first $A(R)$ to F. Both long arrows point right.





Original source details: GELFAND-P000444

Uncertain reading. The caption is provisionally read as “подст.”. It has a tall medial loop, a final cross stroke and a stop. The earlier alternatives “индив.” and “модиф.” remain possible.

Original source details: GELFAND-P000444

Uncertain reading. The first lower A is heavily overwritten. The earlier strokes cannot be separated.

GELFAND-P000445

[source number: 69]

Kazhdan.

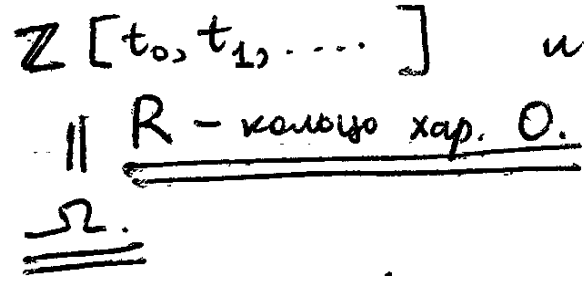


Diagram and typed description. Two vertical strokes stand between the beginning of the line with $\mathbb{Z}$ and Ω. The statement about R is to their right.

$\mathbb{Z}[t_0, t_1, \ldots]$ and form. group ~~...~~ over R.
$\|$ R — ring of char. 0.
Ω.

Original source details: GELFAND-P000445

Undeciphered passage. The short cancelled inscription before "over" is undeciphered.

Original source details: GELFAND-P000445

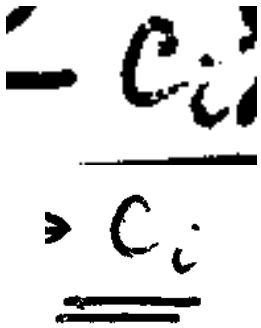
Constr. of an inv. diff. on a form. group — uniquely determ.

$$\omega = \sum c_i x^i \, dx;$$

Map: $t_i \longrightarrow C_i \in R$.

Uncertain reading. Both variable signs are heavily overwritten. They are provisionally set as x. Their case and earlier strokes are unresolved.

Original source details: GELFAND-P000445



Uncertain reading. The coefficient in the differential is set as c_i. The taller letter in the map is set as C_i. The distinction in case is provisional.

Original source details: GELFAND-P000445

Obtain a map $\Omega \longrightarrow R$.

$14/\overline{\text{IV}}$ I. M. Gelfand.

On Dyson's talk "Unrealized possibilities".

Anecdote about a tunnel.

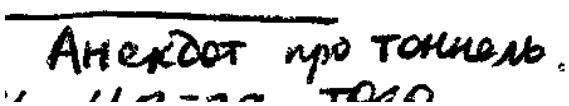


Diagram and typed description. The reference to an anecdote sits in the right margin beneath the title.

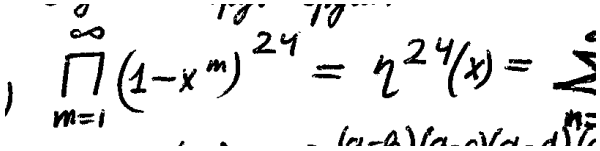
Original source details: GELFAND-P000445

Aim: to convince math. and phys. how much they lost by failing to listen to one another.

Example 1.

(Dyson's amusement)

$$\prod_{m=i}^{\infty}(1-x^m)^{24}=\eta^{24}(x)=\sum_{n=1}^{\infty}\tau(n)x^n \quad \text{(Ramanujan fn.)}$$



Original source details: GELFAND-P000445

Uncertain reading. The lower product index is provisionally read as $m=i$. A short stroke with a small mark above it follows the equals sign. The reading $m=1$ remains possible. No separate factor is visible before the product.

Identity:

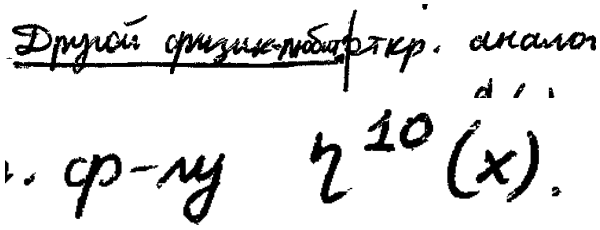
$$\tau(n)=\sum_{\substack{a,b,c,d,e\equiv 1,2,3,4,5 \pmod 5,\\ a+b+c+d+e=0\\ a^2+b^2+c^2+d^2+e^2=10n.}}\frac{(a-b)(a-c)(a-d)(a-e)(b-c)(b-d)(b-e)(c-d)(c-e)(d-e)}{1!\,2!\,3!\,4!}$$

Original source details: GELFAND-P000445

Uncertain reading. The first letter in the penultimate parenthesis is overwritten. The factor is provisionally set as $(c-e)$.

$\eta(x)$ — Dedekind fn.

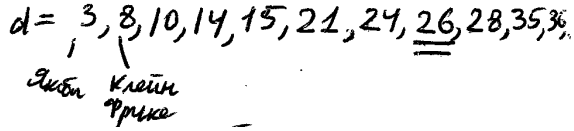
Another amateur physicist discov. an analog. fmla. $\eta^{10}(x)$.



Original source details: GELFAND-P000445

Uncertain reading. The writing is joined where the underline ends. The provisional readings are "физик-люб." ("amateur physicist", abbreviated) and "откр." ("discov."). The abbreviation before the power of η is provisionally read as "ф-лу".

Dyson: for $\eta^d(x)$,



Original source details: GELFAND-P000445

Diagram and typed description. Short slanted strokes lead from 3 and 8 toward the labels. A comma and a small stroke are visible after 36.

$d=\ 3,\ 8,10,14,15,21,24,\underline{26},28,35,36,$

Jacobi — 3; Klein, Fricke — 8

These are dimensions of simple Lie algebras.

$$\begin{array}{cccccc} A_1 & A_2 & Sp(4) & - & - & \cdot \quad A_4 = SL(5) \\ 3 & 8 & 10 & - & - & \cdot \quad 24 \end{array}$$

Prod. — over all simple roots (pos.) of gr. $SL(5)$.

Произв. — по всем простым корням (положит.) гр. $SL(5)$

Original source details: GELFAND-P000445

Uncertain reading. The wording is “простым корням (положит.)”, rendered here as “simple roots (pos.)”.

$\underline{\underline{26}}$ — drops out — unclear

GELFAND-P000446

2. $c \to 0,\ c \to \infty,\ R \to \infty$ — lim. transitions in the kinem. group.

в кинем. группе.

Original source details: GELFAND-P000446

Uncertain reading. The last word reads “группе” (“group”). Joined strokes in the middle obscure the reading slightly.

$\underline{\underline{\text{Morava.}}}$

$\underline{\text{Kirillov (recapitulation).}}$

$X \longrightarrow \Omega_n(X) \underset{[?]}{—}$ group, $\Omega_n(*)$ — module.

Original source details: GELFAND-P000446

Uncertain reading. A separate small sign with two upper strokes is visible beneath the dash after $\Omega_n(X)$. Its reading is unresolved.

~~Triple.~~ (M, ξ, α, f)

Original source details: GELFAND-P000446

Uncertain reading. “Тройка.” (“Triple.”) is the provisional reading beneath the two cancellation strokes.

(M, ξ, α) — weakly compl. set

$$\left(\cdots \quad \underbrace{\tau(M) + [k] = \xi + [k']}_{\alpha}, \quad \xi \text{ — already a compl. bundle} \right)$$

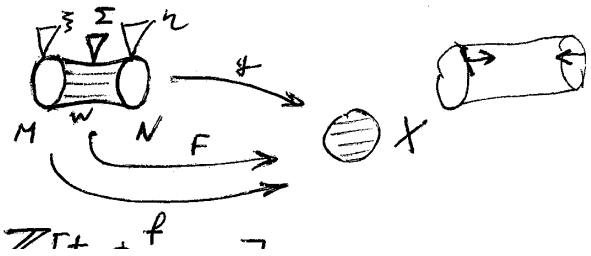
Original source details: GELFAND-P000446

Undeciphered passage. The short sign just inside the left parenthesis is cancelled. The earlier strokes are unresolved.

α — fixed decomp.

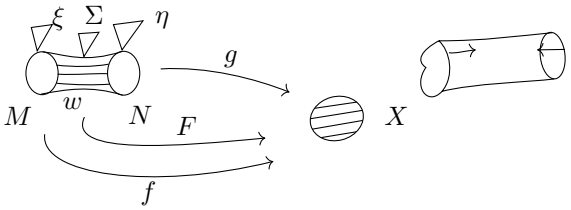
$f\colon M \longrightarrow X$

Equiv. relation.



Original source details: GELFAND-P000446

Diagram and typed description. Three arrows point right toward the hatched region X. The upper label is provisionally g, and the label beneath the middle of the left cylinder is provisionally w. Two arrows in the separate right cylinder point inward from its ends.



Theorem. $\Omega_*(*) = \mathbb{Z}[t_1, t_2, \ldots]$

$$\deg t_i = 2i$$

$t_1 \leftrightarrow CP^1$

A — cobord. ring of a point $\Omega_*(*)$.

M — module over A $(\Omega_*(X))$.

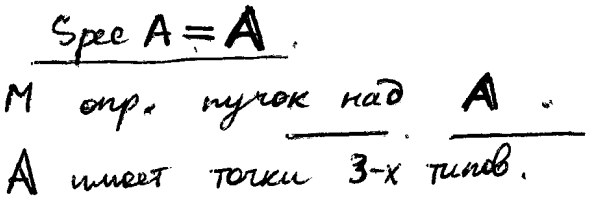
Spec $A = \mathbb{A}$.

M determ. a sheaf over $\mathbb{A}$.

Description of M_x — ? $(x \in M)$

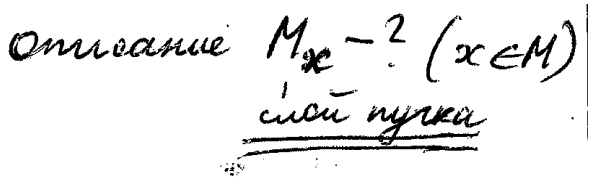
fiber of the sheaf

$\mathbb{A}$ has points of 3 types.



Original source details: GELFAND-P000446

Uncertain reading. The right slanting stroke is doubled in three signs. The sign before the equals sign is written more simply. The three signs are set as $\mathbb{A}$.



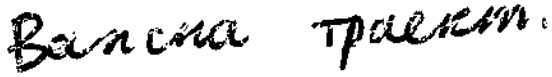
Original source details: GELFAND-P000446

Uncertain reading. The letter following the membership sign in the margin reads M. The heavily overwritten subscript of M_x is provisionally read as x.

GELFAND-P000447

[source number: 70]

What matters is the traj. of a point under the action of an infin. group.



Uncertain reading. The opening reads “Важна траект.”. Some joining strokes are faint.

Original source details: GELFAND-P000447

Orbits:

1) one open dense — gen. pos.
2) one closed
3) count. number of interm. orbits

Uncertain reading. The first letters of the abbreviation are overwritten. The provisional reading is “промежут.” (“interm.”).

Original source details: GELFAND-P000447

1) fiber $K(X)$. (at $\forall$ point) (open dense)
2) p, n (p — prime, n — natur.) $K^{\overline{n,p}}(X, \mathbb{Z}_p)$.

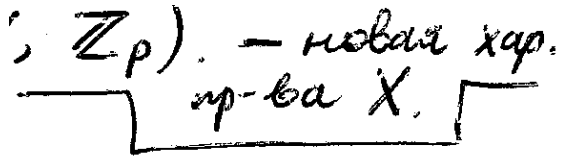
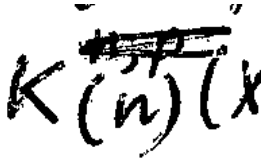


Diagram and typed description. A stepped bracket runs beneath “of the sp. X”.

— new char.
of the sp. X.

Original source details: GELFAND-P000447



Uncertain reading. A short inscription above (n), following K, is cancelled. It is provisionally read as n, p.

Original source details: GELFAND-P000447

3) $H^*(X, \mathbb{Z})$ at $\forall$ point.

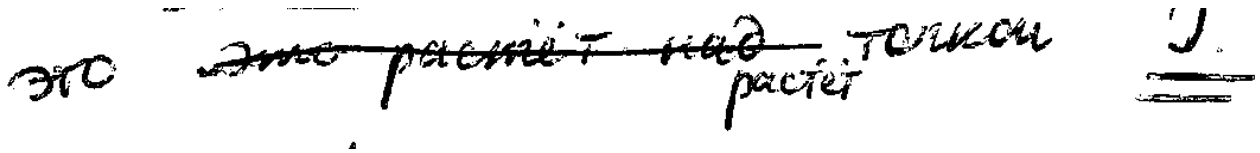
Clarification

$$(a_1, \ldots, a_n, \ldots) \in \mathbb{A}$$

max id.

$$J = (t_1 - a_1, \ldots, t_n - a_n, \ldots)A \subset \mathbb{A}$$

M/MJ — vect. sp. (more precisely A/J — module).

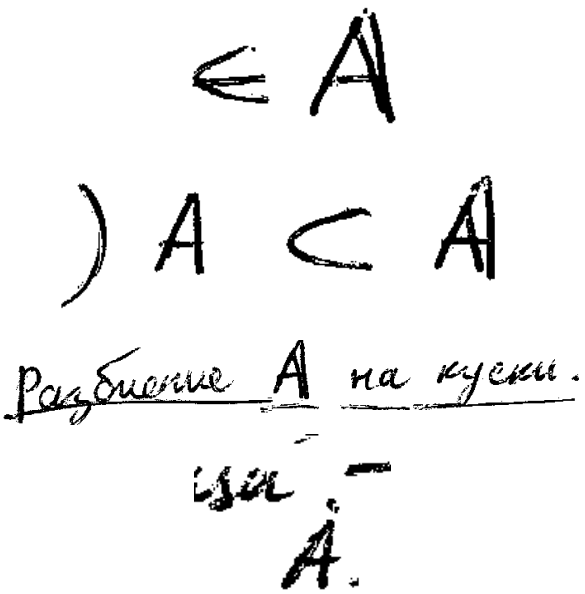


Original source details: GELFAND-P000447

Diagram and typed description. “It” is written separately on the left. “Grows” is inserted beneath the cancelled “over”.

it ~~it grows~~ ~~over~~ grows the point J.

Partition $\mathbb{A}$ into pieces.



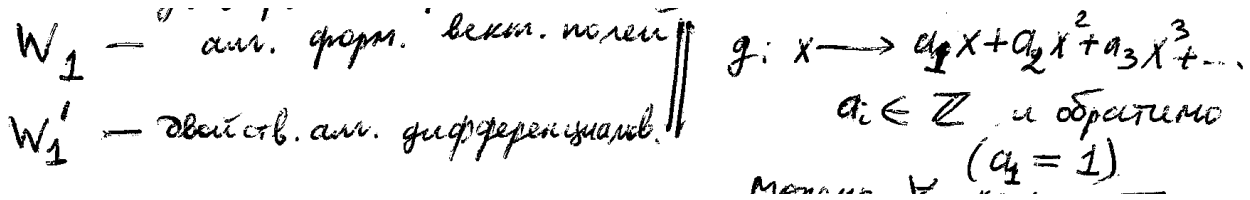
Uncertain reading. The right stroke of A is doubled after $\in$ and to the right of $\subset$. The letter in the “Partition” heading is also set as $\mathbb{A}$. The sign immediately after the closing parenthesis in the J line is plain A.

Original source details: GELFAND-P000447

~~$\mathbb{A}$ [...] one-dimension... form.~~ G — gr. of form. substitutions of variables

Undeciphered passage. Under the wavy cancellation the initial A, the stem “одномерн” (“one-dimension”), and the final “форм.” (“form.”) are visible. The reading of the horizontal stroke and the ending of the first word is unresolved.

Original source details: GELFAND-P000447

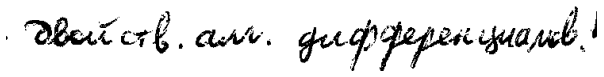


Original source details: GELFAND-P000447

Diagram and typed description. Two vertical strokes separate the W_1, W_1' entries from the substitution g. The subscript on the first a is overwritten and reads as 1.

W_1 — alg. of form. vect. fields
W_1' — dual alg. of differentials.

$g\colon x \longrightarrow$
$a_1 x + a_2 x^2 + a_3 x^3 + \dots$
$a_i \in \mathbb{Z}$ and invertible
$(a_1 = 1)$



Original source details: GELFAND-P000447

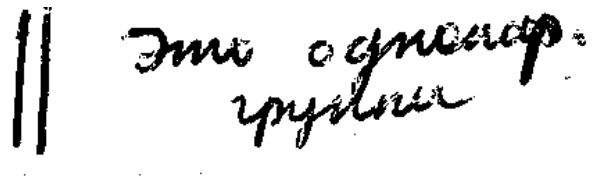
Uncertain reading. The text reads “двойств. алг. дифференциалов” (“dual alg. of differentials”). A separate stroke is visible above “й”.

Possible for $\forall$ rings. —
— in particular from A.
(a_1 — inv. el. of the ring)
Form. 1-dim. group:

$$x * y = x + y + \alpha xy + \beta x^2 y + \delta x y^2 + \dots \qquad (\alpha, \beta \in \text{the ring})$$

associativity. — sole req.
They are equiv. if one law of mult. $\longrightarrow$ the other by a form. subst. of var.

$$\varphi(x) = \lambda_1 x + \lambda_2 x^2 + \dots, \qquad \lambda_1 \text{ — inv. el. of the ring}$$

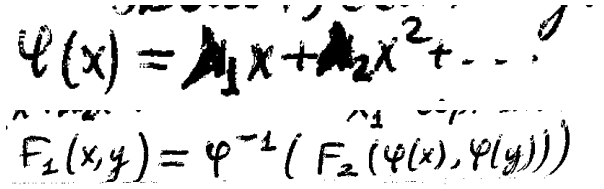


Original source details: GELFAND-P000447

Diagram and typed description. Two vertical strokes separate the right marginal note. Its abbreviation is provisionally read as “одноморф.” (“same-form”).

$F_1(x,y) = \varphi^{-1}\big(F_2(\varphi(x), \varphi(y))\big)$

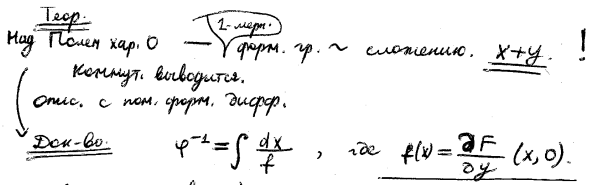
These are same-form groups



Original source details: GELFAND-P000447

Uncertain reading. The coordinate sign is set as φ in all four places. The left side of the first sign and of both signs inside the parentheses resembles ψ. The loop is clearer in the sign with exponent -1. The distinction is unresolved. The two overwritten coefficients are provisionally read as λ_1, λ_2.

GELFAND-P000448



Original source details: GELFAND-P000448

Diagram and typed description. A loop encloses the insertion “1-dim.”. The left arrow leads down to “Proof”.

Thm.

Over a Field of char. 0 — 1-dim. form. gr. $\sim$ addition. $\underline{\underline{x+y}}$. !

Commut. follows.

Descr. by means of form. diff.

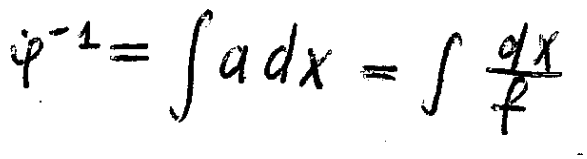
Proof. $\varphi^{-1} = \int \frac{dx}{f}$, where $\underline{f(x) = \frac{\partial F}{\partial y}(x,0)}$.

(simple check).

Outside char. 0 — the integral does not exist.

$\int x^{p-1}\,dx$ does not exist over a field of char. p.

$$\dot{\varphi}^{-1} = \int a\,dx = \int \frac{dx}{f}$$



Original source details: GELFAND-P000448

Uncertain reading. A separate dot is visible above the left part of φ in the second occurrence. It is set as a dot above the letter. It may be a spot.

$\underline{a\,dx}$ — form. diff., ~~reconstr.~~ determ. the group

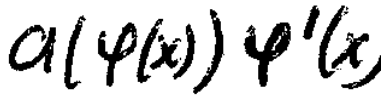
Original source details: GELFAND-P000448

Uncertain reading. The cancelled abbreviation is provisionally read as “восстан.” (“reconstr.”).

$$a(x) = 1 + a_1 x + a_2 x^2 + \dots$$

Action of the group (in diff.)

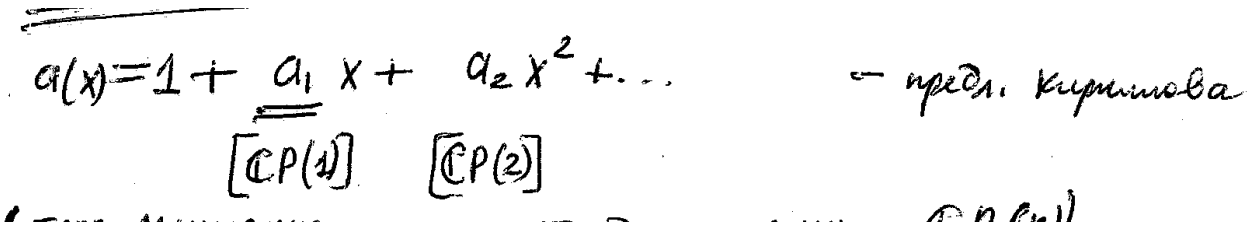
$$a(x) \longrightarrow a(\varphi(x))\varphi'(x)$$



Original source details: GELFAND-P000448

Uncertain reading. Both signs have a right loop. The left stroke also admits the reading ψ. The provisional readings are φ and φ'.

$a(x)$ — any series. To it corresp. a form. group (theorem on existence of the univ. group) Morava.



Original source details: GELFAND-P000448

Diagram and typed description. The labels under a_1 and a_2 are in square brackets. The letter C has an additional stroke inside.

$$a(x) = 1 + \underset{[\mathbb{C}P(1)]}{a_1}\, x + \underset{[\mathbb{C}P(2)]}{a_2}\, x^2 + \ldots$$

— Kirillov's prop.

(Mishchenko's thm. — ... here one obtains $\mathbb{C}P(n)$).

Undeciphered passage. The short inscription after the dash is heavily cancelled. The letters cannot be separated.

Original source details: GELFAND-P000448

Theorem, G — gr. scheme. $X_{[?]}$ — G-scheme (over $\mathbb{Z}$).

Uncertain reading. A single slanting stroke follows "Theorem". It is provisionally set as a comma.

Original source details: GELFAND-P000448

Uncertain reading. A small looped mark and an adjacent stroke appear below and to the right of X. The reading is unresolved.

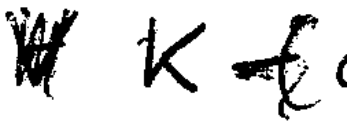
Original source details: GELFAND-P000448

Let X be smooth, G be flat and all be Noetherian

Uncertain reading. Two diagonal strokes overwrite the sign before "smooth". It is provisionally set as X. The earlier loop-like stroke cannot be separated from the overwriting.

Original source details: GELFAND-P000448

... K — (alg. closed field) and $G(K)$ acts transitively on $X(K)$



Uncertain reading. A separate sign with two slanting sides is heavily cancelled before K. One possible reading is $\forall$.

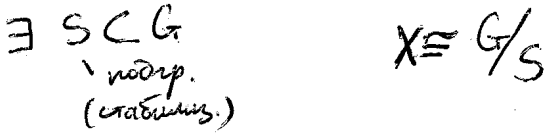
Original source details: GELFAND-P000448

Original source details: GELFAND-P000448

Uncertain reading. The letters in parentheses after G and X are smaller than the standalone K, but have a similar angular form. They are provisionally set as K.

GELFAND-P000449

[source number: 71]



Original source details: GELFAND-P000449

Diagram and typed description. A short slanted stroke links S to its label.

$\exists\ S \subset G$ $\qquad X \cong G/S$

subgr.

(stabiliz.)

Original source details: GELFAND-P000449

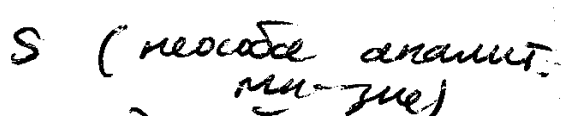
Diagram and typed description. Ten dots form a separator.

. . . . . .

GELFAND-P000450

25/IX-72 yr. Gabrielov.

Fam. of Hodge structures over S (nonsingular analyt. set)



Original source details: GELFAND-P000450

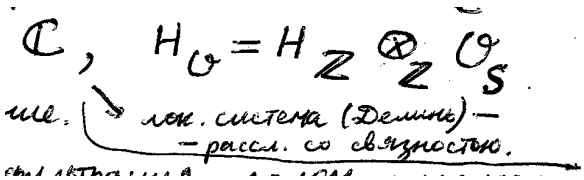
Uncertain reading. Two rounded strokes are joined after “б”. The reading is “неособое” (“nonsingular”).

1) $\mathcal{H}_{\mathbb{Z}}$ — loc. free sheaf of $\mathbb{Z}$-modules of fin. type.

~~2)~~ From it constr. a loc. triv. compl. bundle $H_{\mathbb{C}}$

Original source details: GELFAND-P000450

Uncertain reading. The cancelled beginning of the line is provisionally read as “2)”.



Original source details: GELFAND-P000450

Diagram and typed description. An arrow from the complex scalar extension points to the note. The lower line of the note turns up at its left end.

$H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C}, \quad H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathcal{O}_S.$

and $H_{\mathbb{R}}$ — real bundle. loc. system (Deligne) —

— bundle with connection.

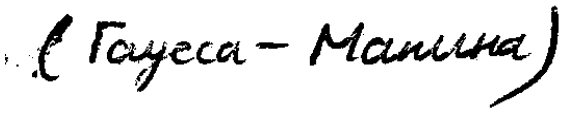
2) in $H_{\mathbb{C}}$ a decreasing filtration by holom. bundles

$$H_{\mathbb{C}} = F^0 \supset F^1 \supset \ldots \supset F^n$$

Original source details: GELFAND-P000450

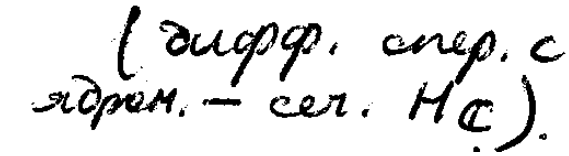
Diagram and typed description. A loop encloses the inserted “есть” (“there is”).

3) in $H_{\mathcal{O}}$ there is nat. connection. (Gauss–Manin)



Original source details: GELFAND-P000450

Uncertain reading. The middle letters of the first name are joined. The pair is read as “Гаусса—Манина” (“Gauss–Manin”).

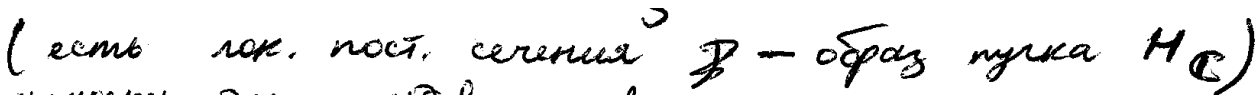


Original source details: GELFAND-P000450

Diagram and typed description. The note on the kernel is to the right of the formula. The mark before the dash is provisionally a comma.

$\nabla\colon H_{\mathcal{O}} \longrightarrow \Omega^1 \otimes_{\mathcal{O}_S} H_{\mathcal{O}}$ (diff. oper. with kernel, — sect. of $H_{\mathbb{C}}$).

(there are loc. const. sections $\mathbb{Z}$ — image of the sheaf $H_{\mathbb{C}}$)



Original source details: GELFAND-P000450

Uncertain reading. The sign between “sections” and the dash is provisionally read as $\mathbb{Z}$. It has a loop and a long lower stroke. The exact reading is unresolved.

The connection must satisfy the cond.:

$$\nabla(\mathcal{F}^p) \subset \Omega^1_S \otimes \mathcal{F}^{p-1} \qquad (\mathcal{F}^p = F^p \otimes_{\mathbb{C}} \mathcal{O}_S)$$

4) ψ — bilin., loc. const. form on $H_{\mathbb{C}}$, giving on restriction to $\forall$ fiber a genuine Hodge struct.

———— • ————

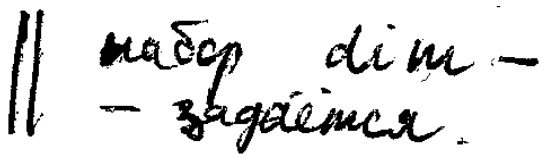
Classifying sp. of Hodge structures.
Consid.

———— • ————

$$F = \{F^0 \supset F^1 \supset \ldots \supset F^n\} \qquad \text{— flags of fixed dim.}$$

in a fixed sp., $\mathbb{C}^n$ (dim can have gaps).

———— • ————

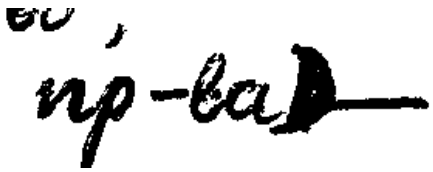


Original source details: GELFAND-P000450

Diagram and typed description. Two vertical strokes separate the right marginal note.

list of dim —
— is given.

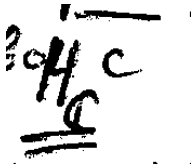
D — sp.;
Points of the sp. ~~D~~ — flags.



Original source details: GELFAND-P000450

Uncertain reading. The heavily overwritten sign before the dash resembles D. The reading of the cancelled letter is provisional.

Take a lin. sp. $H_{\mathbb{C}}$ with a bilin. form $\psi(x, y)$ (nondeg.; skew or symm.)



Original source details: GELFAND-P000450

Diagram and typed description. Two lines under $H_{\mathbb{C}}$ sit below its lowered subscript.

$\bar{D}$ — set of all flags ~~in~~ in $H_{\mathbb{C}}$ with a fixed list of dim

Original source details: GELFAND-P000450

Uncertain reading. A separate stroke is visible above the heavily retraced D.

Original source details: GELFAND-P000450

Uncertain reading. The small cancelled sign before “в” (“in”) is provisionally read as a repeated “в”.

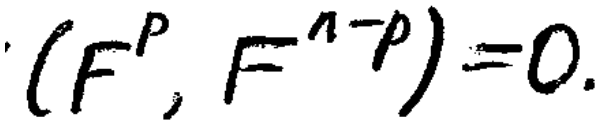
GELFAND-P000451

[source number: 72]

... with condition 1) $\psi(F^p, F^{n-p}) = 0$.

Original source details: GELFAND-P000451

Undeciphered passage. The short cancelled sign before “with” is undeciphered.



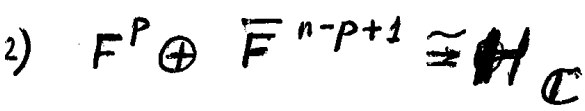
Original source details: GELFAND-P000451

Uncertain reading. The first formula clearly reads $n - p$. The next formula separately reads $n - p + 1$.

This is an algebraic subs. of $\bar{D}$.

ψ either symm. or skew-symmetric (def).

$$2) \quad F^p \oplus \bar{F}^{n-p+1} \cong H_{\mathbb{C}}$$



Original source details: GELFAND-P000451

Uncertain reading. The sign before $H_{\mathbb{C}}$ has a wave above two horizontal strokes. A slanted mark joins the lower strokes. The reading $\cong$ is provisional. The letter H is heavily retraced.

$H_{\mathbb{Z}} \subset H_{\mathbb{R}} \subset H_{\mathbb{C}}$ also given

The nondeg. flag determ. the grading and the Weil autom.

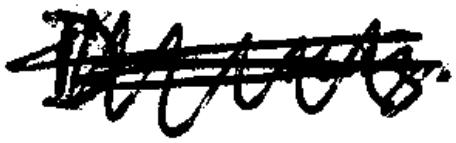
$$H^{p,n-p} = F^p \cap \bar{F}^{n-p}$$

Weil autom.:

$$C\xi = i^{2p-n}\xi \quad \text{when } \xi \in H^{p,n-p}$$

These conds. pick out in $\bar{D}$ the open subs. D.

... Another condition: $\psi(x, C\bar{x}) > 0$ when $x \neq 0$.



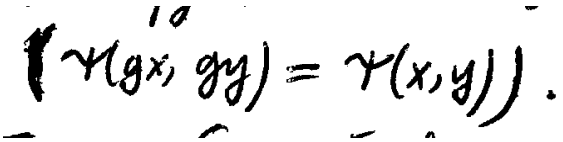
Original source details: GELFAND-P000451

Undeciphered passage. Several horizontal strokes cover the earlier writing before “Another condition”. The ending has a loop and a stop. The text is unresolved.

Define on D the structure of a homog. sp.

G — group of all real transf. of $H_{\mathbb{R}}$, preserv. ψ

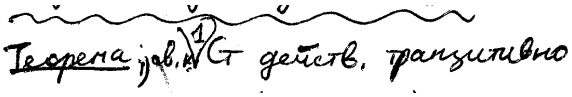
$$\big(\psi(gx, gy) = \psi(x, y)\big).$$



Original source details: GELFAND-P000451

Uncertain reading. The left outer parenthesis is heavily retraced. Two round closing parentheses are distinguishable on the right.

Then G acts on D



Original source details: GELFAND-P000451

Diagram and typed description. Before G is a short inscription provisionally read as “яв.” (“expl.”) and a heavily retraced sign. A 1 is inserted in the loop above it. The beginning of the line provisionally reads 1).

Theorem: 1) expl. ...$\overset{1}{\bigcirc}$$G$ acts transitively

2) stab. Subgroup H (preserv. the flag) is compact.

Proof: from explicit computation of G and H.

$K \subset G$ — max. comp. subgroup

$G/H \longrightarrow G/K$ it is not holomorphic

In G/K there is a compl. structure

GELFAND-P000452

Original source details: GELFAND-P000452

Undeciphered passage. Closely spaced wavy strokes cover the underlined heading. A stop is visible to the right. The letters are unresolved.

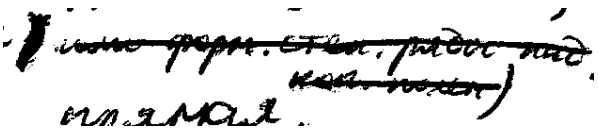
Manin’s 1st lecture on p-adic numbers. (retelling)

K — loc. field, (...

Original source details: GELFAND-P000452

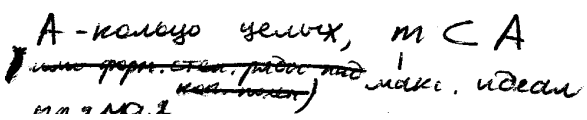
Undeciphered passage. After the comma are a left parenthesis and a short, heavily cancelled inscription. The right parenthesis in this line is unclear.

(fin. ext. of the p-adic numbers) ~~or form. pow. series over a fin. field~~)



Original source details: GELFAND-P000452

Undeciphered passage. The cancelled lines are provisionally read as “or form. pow. series over” and “a fin. field”. The closing parenthesis near “adic numbers” is heavily retraced.

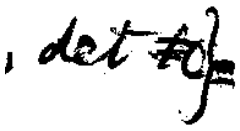


Original source details: GELFAND-P000452

Diagram and typed description. A short vertical stroke connects m with the label below.
A — ring of integers, $m \subset A$
max. ideal

$P^1(K)$ — proj. line.

$$G = \text{its motions} \quad \left\{ z \longrightarrow \frac{az+b}{cz+d} \;\middle|\; a,b,c,d \in K,\ \det \neq 0 \right\} =$$
$$= \mathrm{GL}(2,K)/K^*$$



Original source details: GELFAND-P000452

Uncertain reading. The notation before the right brace reads det $\neq 0$. The sign is heavily retraced. Two horizontal strokes immediately after the brace read as $=$. The next line also starts with $=$.

$U \subset G$; $U = \mathrm{GL}(2,A)/A^*$.

Constr. G/U — graph Δ — analog of the Lobachevsky pl.

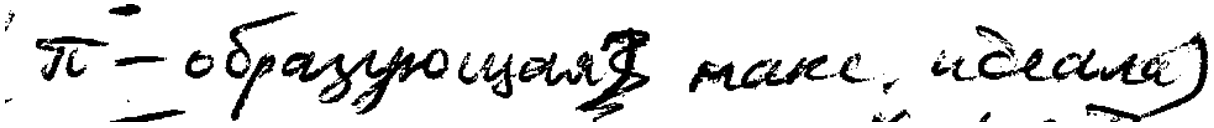
Def.: a lattice in K^2 — a free A-module with 2 gen.

$$L = Ae_1 \oplus Ae_2.$$

Lemma. If L_1 and L_2 — 2 latt., then $\exists$ a basis e_1, e_2 in K^2, such that

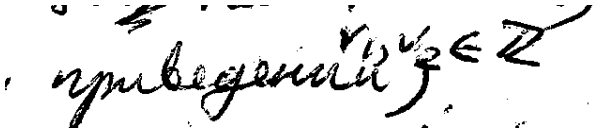
$$L_1 = Ae_1 \oplus Ae_2; \qquad L_2 = m^{\nu_1} e_1 \oplus m^{\nu_2} e_2 =$$
$$= A\pi^{\nu_1} e_1 \oplus A\pi^{\nu_2} e_2$$

(π — generator [?] of the max. ideal)



Original source details: GELFAND-P000452

Uncertain reading. A looped mark with a lengthwise stroke stands between “generator” and “max.” in the Russian. Its reading is unresolved.



Original source details: GELFAND-P000452

Diagram and typed description. The inscription $\nu_1, \nu_2 \in \mathbb{Z}$ sits at the right between the two parenthetical lines.

(analog. to the lin. alg. theor. on simult. reduction) $\nu_1, \nu_2 \in \mathbb{Z}$

Def. $L_1 \sim L_2$, if $L_1 = \alpha L_2$, where $\alpha \in K^*$.

There is an analog of distance — as in symm. sp. ... — see A. Weil

Undeciphered passage. The small sign before the dash is cancelled. Its reading is unresolved.

Original source details: GELFAND-P000452

Vertices of graph $\Delta^{(0)}$ — equiv. classes of lattices

Undeciphered passage. A small cancelled sign appears below and to the right of the first Δ. Its reading is unresolved. Above it is (0).

Original source details: GELFAND-P000452

G acts on $\Delta^{(0)}$.

GELFAND-P000453

[source number: 73]

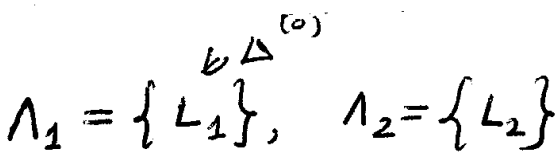


Diagram and typed description. The looped stroke by $\Delta^{(0)}$ points down and left toward the first entry. It is drawn as an arrow. A reading as "в" ("in") remains possible.

$\Delta^{(0)}$

↙

$\Lambda_1 = \{L_1\}, \quad \Lambda_2 = \{L_2\}$ — classes of lattices

Original source details: GELFAND-P000453

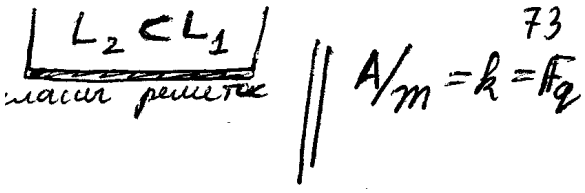


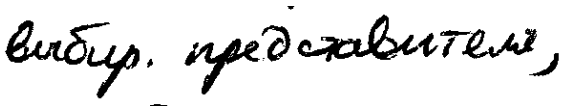
Diagram and typed description. The frame under $L_2 \subset L_1$ is open at the top. Two vertical strokes set off the residue field on the right. The left stroke of F is doubled.

$L_2 \subset L_1$ ‖ $A/m = k = \mathbb{F}_q$

Original source details: GELFAND-P000453

$$L_1 = Ae_1 \oplus Ae_2$$
$$L_2 = Ae_1 \oplus A\pi^\nu e_2$$

— thus choose the representatives, $\nu \geq 0$.



Uncertain reading. The end of the last word is faint. The reading is "представители" ("representatives").

Original source details: GELFAND-P000453

ν is called distance $\nu = d(\Lambda_1, \Lambda_2)$.

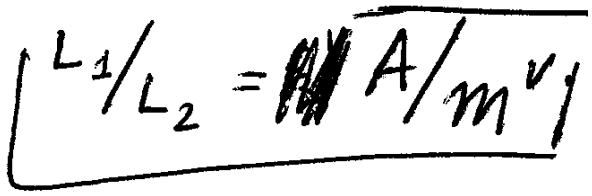


Diagram and typed description. Three strokes frame the formula below and at the sides. The inscription after the equals sign is heavily cancelled. Its reading is unresolved.

$L_1/L_2 = \dots A/m^\nu$

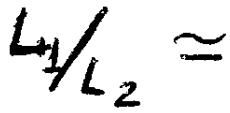
Original source details: GELFAND-P000453

Thm. This is a corr. def. and is the distance.

———— · ————

Variant:

$$L_1/L_2 \simeq A/\pi^{\nu_1}A \oplus A/\pi^{\nu_2}A$$



Uncertain reading. A single straight stroke lies under the wave. It is set as $\simeq$.

Original source details: GELFAND-P000453

$$d(\Lambda_1, \Lambda_2) = |\nu_1 - \nu_2|$$

Λ_1, Λ_2 — neighbors in Λ^0, if $d(\Lambda', \Lambda^2) = 1$.

Uncertain reading. The set has a Λ with a small raised circle. In the arguments of d, a raised stroke follows the first Λ and a raised 2 follows the second.

Original source details: GELFAND-P000453

Join them by an edge. We obtain a graph Δ.

Theorem: Δ — a tree (conn. graph without cycles), and from $\forall$ vertex there issue exactly $q+1$ edges

Δ is the Lobachevsky pl. — there acts G etc.

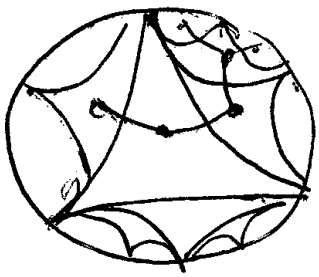
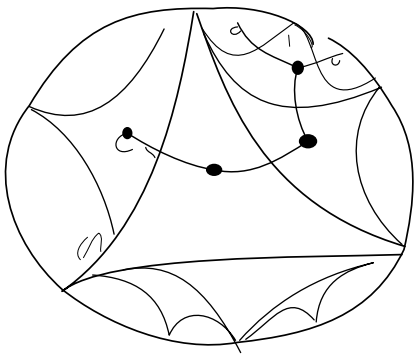


Diagram and typed description. The outer boundary has a gap at the upper right. The inner arcs and chain follow the visible strokes. A fine mark by the lower left arc resembles an oval. Some joins in the upper right remain uncertain.



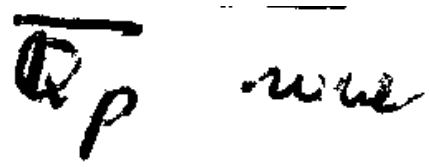
Original source details: GELFAND-P000453

———— • ————

GELFAND-P000454

D. Kazhdan. Tate's art., Matematika, 13 : 2, §3.

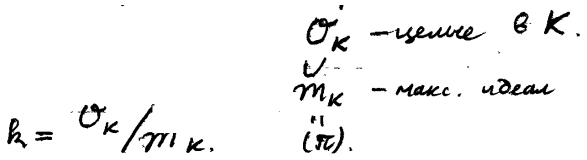
(on the field $\mathbb{C}_p$ — completion of the alg. closure $\overline{\mathbb{Q}}_p$ or $\mathbb{Q}_p$ — p-adic numbers)



Original source details: GELFAND-P000454

Uncertain reading. The connecting word reads "or."

K — finite ext. of the field $\mathbb{Q}_p$.



Original source details: GELFAND-P000454

Diagram and typed description. There is a curved stroke between the two entries. Its meaning is uncertain. Two short vertical strokes sit below the second entry.

$\mathcal{O}_K$ — integers in K.

$\cup$

m_K — max. ideal

$k = \mathcal{O}_K/m_K$. (π).

L — finite ext. of K, ... $\mathcal{O}_L$ — integers in it

Original source details: GELFAND-P000454

Uncertain reading. The crossed-out marks before $\mathcal{O}_L$ are illegible.

$m_L = (\tau)$, $\ell = \mathcal{O}_L/m_L$,

$$G = \mathrm{Gal}(L : K)$$

Introd. a filtration in G and prove that G is solvable.

On L and K ... define a norm so that $|\pi| = (\mathrm{Card}\, k)^{-1}$

Original source details: GELFAND-P000454

Uncertain reading. A crossed-out sign sits between K and "define." Its reading is uncertain.

Def. ....

Original source details: GELFAND-P000454

Undeciphered passage. A short entry is crossed out after "Def." Its reading is uncertain.

$$g \in G_i \Longleftrightarrow \forall x \in \mathcal{O}_L \quad x^g - x \in m_L^i$$

$$G = G_0 \supset G_1 \supset \ldots \supset G_n = \{e\}.$$

G_i — normal subgroup in G.

Theorem: G_i/G_{i+1} — ab. group

1) $i = 0$. $G \xrightarrow{\varphi}$ Gal$(\ell : k)$ (nat. map)

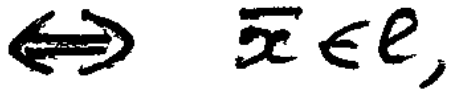
Construction of φ: $\bar{x} \in \ell,\ g \in G$

$x \in L$ $\quad \bar{x}^g \overset{\text{def}}{=} \overline{x^g}$

Uncertain reading. The manuscript reads $x \in L$.

Original source details: GELFAND-P000454

$g \in \ker \varphi \iff \bar{x} \in \ell,\ \bar{x}^g = \bar{x}$, i.e. $\bar{x}^g - \bar{x} = 0$, i.e.



Uncertain reading. The equivalence sign is followed by $\bar{x} \in \ell$. No explicit quantifier is visible.

Original source details: GELFAND-P000454

GELFAND-P000455

[source number: 74]

~~i.e. ...~~ $x^g - x \in m_L$, i.e. $g \in G_1$.

Uncertain reading. The start of the crossed-out entry provisionally reads “i.e.”.

Original source details: GELFAND-P000455

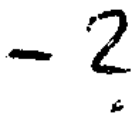
Undeciphered passage. The remainder of the crossed-out entry is illegible.

Original source details: GELFAND-P000455

$0 \longrightarrow G_0/G_1 \xrightarrow{\varphi}$ Gal$(\ell : k)$

Exercise: φ — an isomorphism.

2) G_1/G_2 — ?



Uncertain reading. A question mark follows the dash.

Original source details: GELFAND-P000455

$$G_1 \xrightarrow{\psi_1} \ell^*$$
$$g \rightsquigarrow \overline{\tau^g/\tau} \qquad \text{(indep. of the choice of } \tau\text{)}$$

Original source details: GELFAND-P000455

Uncertain reading. The first arrow is labelled ψ_1.

Its kernel is G_2.

Original source details: GELFAND-P000455

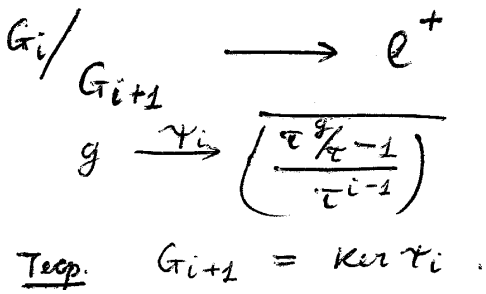
Uncertain reading. The word following "kernel" reads «ero» ("its").

$$G_i/G_{i+1} \longrightarrow \ell^+ \qquad \text{(addit. group of the field } \ell\text{)}.$$
$$g \xrightarrow{\psi_i} \overline{\left(\frac{\tau^g/\tau - 1}{\tau^{i-1}}\right)}$$

Original source details: GELFAND-P000455

Uncertain reading. The arrow from g is labelled ψ_i.

Thm. $G_{i+1} = \operatorname{Ker} \psi_i$.



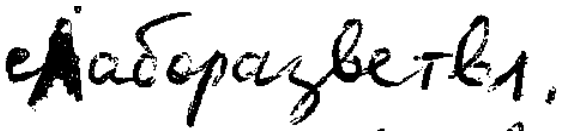
Original source details: GELFAND-P000455

Uncertain reading. The domain on the upper line is written G_i/G_{i+1}. The line below has $G_{i+1} = \operatorname{Ker} \psi_i$.

G_1 is called the inertia subgroup of the Galois grp.

Unramified extension $\underset{\text{def}}{\Longleftrightarrow} G_1 = \{e\}$;

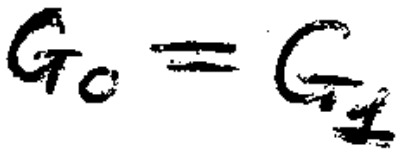
weakly ramif. $G_2 = \{e\}$.



Original source details: GELFAND-P000455

Uncertain reading. The second letter is overwritten. The abbreviation reads «слаборазветвл.» ("weakly ramif.").

purely ramified $G_0 = G_1$



Uncertain reading. The last subscript is overwritten. It provisionally reads 1.

Original source details: GELFAND-P000455

GELFAND-P000456

The seminar will start at $18^{\underline{00}}$.

2/X-72. A. Gabrielov.

S — compl. analytic set

$H_{\mathbb{Z}}$ — loc. free $\mathbb{Z}$-sheaf

($\mathbb{Z}$ — const. sheaf; a section over a conn. region is a const. integer; $H_{\mathbb{Z}}$ — loc. free sheaf of $\mathbb{Z}$-modules.

Uncertain reading. Overlapping strokes follow "modules." Their reading is uncertain.

Original source details: GELFAND-P000456

At $\forall$ point $\exists$ nbhd., in which it is isom. to $\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$).

~~H...~~

Uncertain reading. The first crossed-out sign provisionally reads H.

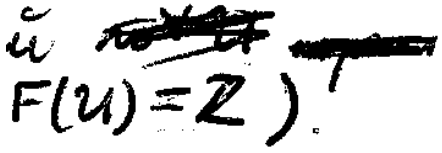
Original source details: GELFAND-P000456

Undeciphered passage. The rest of the short entry before the parenthesis is illegible.

Original source details: GELFAND-P000456

($\mathbb{Z}$-sheaf assoc. with the presheaf which ~~... ...~~

$F(U) = \mathbb{Z}$).



Undeciphered passage. Two entries after "which" are crossed out. Below them is $F(U) = \mathbb{Z}$.

Original source details: GELFAND-P000456

The space is loc. connected $\Rightarrow$ fiber $= \mathbb{Z}$.

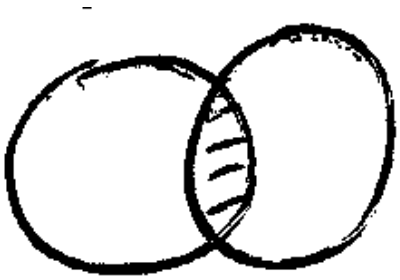
The space of the sheaf $\mathbb{Z}$ is homeom. to $S \times \mathbb{Z}$.

Loc. $H_{\mathbb{Z}}|_U \simeq U \times \mathbb{Z}^n$

$$\Gamma(U, H_{\mathbb{Z}}) \cong \mathbb{Z}^n$$

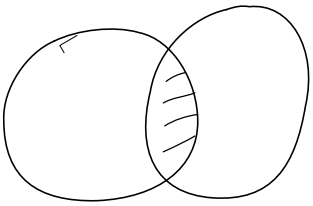
S is connected $\Rightarrow$ n is the same everywhere.

Cover by neighborhoods and fix isomorphisms over the neighborhoods (up to $GL(n, \mathbb{Z})$)



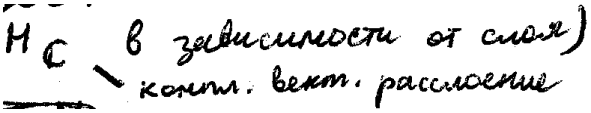
Original source details: GELFAND-P000456

Diagram and typed description. Two overlapping ovals with four strokes in their intersection. There is an angular mark on the left arc.



transition functions from $GL(n, \mathbb{Z})$.

One can construct a loc. triv. bundle with such transition functions.

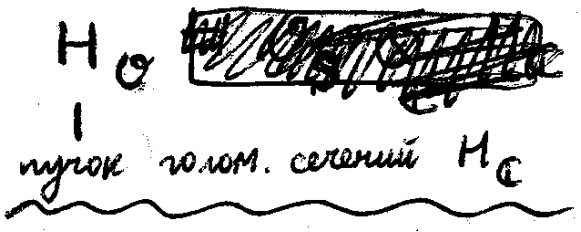


Original source details: GELFAND-P000456

Diagram and typed description. A slanted stroke links $H_{\mathbb{C}}$ to the caption.

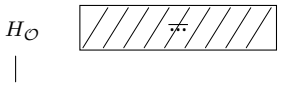
This is $H_{\mathbb{R}}$ (or $H_{\mathbb{C}}$ depending on the fiber)

compl. vector bundle



Original source details: GELFAND-P000456

Diagram and typed description. A vertical stroke leads to the wavy-underlined caption. The rectangle at right is crossed out.



sheaf of holom. sections of $H_{\mathbb{C}}$

Original source details: GELFAND-P000456

Uncertain reading. The crossed-out entry may start with H and a subscript.

Original source details: GELFAND-P000456

Undeciphered passage. The crossed-out entry inside the rectangle is illegible.

$$H_{\mathcal{O}} = H_{\mathbb{Z}} \underset{\mathbb{Z}}{\otimes} \mathcal{O}_S$$

There is a natural connection (Gauss–Manin)

$$\nabla \colon H_{\mathcal{O}} \longrightarrow \Omega^1_S \underset{\mathcal{O}_S}{\otimes} H_{\mathcal{O}} \qquad (\mathbb{C}\text{-lin. homom.})$$

Ω^1_S — holom. diff. forms of order 1

GELFAND-P000457

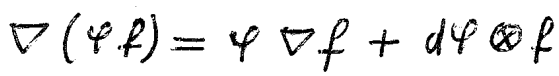
[source number: 75]

def. of connection.

1. $\mathbb{C}$-linearity.

2. $\varphi \in \mathcal{O}_{S,s}$ (germs), $f \in H_{\mathcal{O},s} \Rightarrow$

$$\Rightarrow \nabla(\varphi f) = \varphi \nabla f + d\varphi \otimes f$$



Original source details: GELFAND-P000457

Uncertain reading. Small low marks are visible just to the left of the crossbars of the first two f's in this formula. Similar strokes are visible by the f after $\otimes$ and in the next ∇f. A separate multiplication sign is uncertain.

Our connection.

Take a loc. trivialization
and a basis of const. sections (bundle $H_{\mathbb{Z}}$) u_i

$$f = \sum \varphi_i u_i, \qquad \varphi_i \in \mathcal{O}_S$$

Then

$$\nabla f = \sum d\varphi_i \otimes u_i$$

This def. is well-defined and satisfies all the axioms.

The curvature of this connection equals 0.

Uncertain reading. The last sign reads 0.

Original source details: GELFAND-P000457

Requirement on the G.-M. connection (or on the ~~...~~ filtration)

$$\nabla(\mathcal{F}^p) \subset \Omega^1_S \otimes \mathcal{F}^{p-1} \qquad \text{(Griffiths)}$$

Uncertain reading. The crossed-out sign after "or on the" is illegible.

Original source details: GELFAND-P000457

(Given a filtration $H_{\mathbb{C}} = F^0 \supset F^1 \supset \cdots$)
by holom. subbundles

$\mathcal{F}^p$ — sheaves of sections of F^p

———— . ————

There is a loc. constant bilin. form $\psi(x, y)$ on $H_{\mathbb{C}}$

Original source details: GELFAND-P000457

Uncertain reading. The letter before (x, y) reads ψ.

...

Original source details: GELFAND-P000457

Undeciphered passage. The short crossed-out entry before "and" is illegible.

and at $\forall$ point $s \in S$ this is a Hodge structure.

———— . ————

Let $0 \leq p \leq n, \quad k_p \geq k_{p+1}$ — a seq. of numbers

$$k_0 \geq k_1 \geq \cdots \geq k_n.$$

Z — space with elem. $F = \{F^0 \supset F^1 \supset \cdots \supset F^n\}$;
where dim $F^p = k_p$.

Original source details: GELFAND-P000457

Uncertain reading. The superscript p on F is overwritten.

This is a proj. alg. set.

GELFAND-P000458

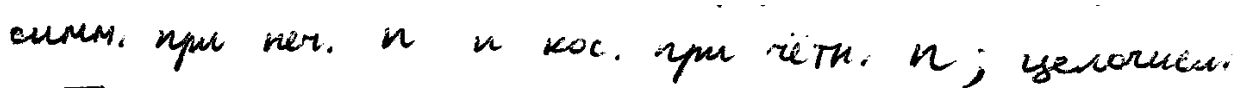
$$H_{\mathbb{C}} = F^0, \qquad H_{\mathbb{Z}} \subset H_{\mathbb{R}} \subset H_{\mathbb{C}}.$$

Let there be on $H_{\mathbb{C}}$ a given bilin. form ψ (nondeg.),

Original source details: GELFAND-P000458

Uncertain reading. The sign after "form" reads ψ.

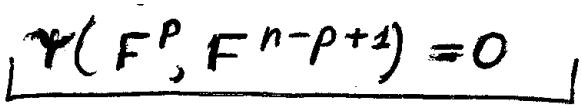
symm. for odd n and skew for even n; integral.



Original source details: GELFAND-P000458

Uncertain reading. The manuscript reads “symm. for odd n and skew for even n.”

$\overline{D}$ — subset in Z of those flags for which

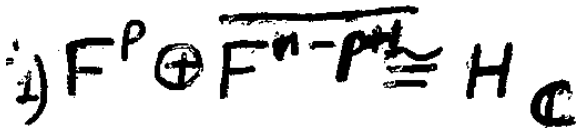


Original source details: GELFAND-P000458

Diagram and typed description. A line encloses the formula from below and at both sides.

$$\psi(F^p, F^{n-p+1}) = 0$$

Add. cond.
on the flags 1) $F^p \oplus \overline{F^{n-p+1}} \cong H_{\mathbb{C}}$



Uncertain reading. The end of the exponent $n-p+1$ is overwritten beside the sign $\cong$.

Original source details: GELFAND-P000458

2) From it are constr. the grading and the Weil autom. C and then

$\psi(x, C\bar{y})$ pos. def. Herm. form on $H_{\mathbb{C}}$

Uncertain reading. After C the manuscript reads «и тогда» (“and then”).

Original source details: GELFAND-P000458

These conditions single out $\underset{\text{(open).}}{\mathcal{D}} \subset \overline{\mathcal{D}}$.

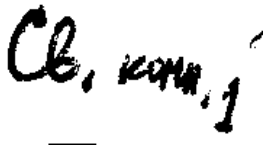
$\mathcal{D}$ is called the classif. space of Hodge structures.

Depends on the set of dimensions and the form ψ.

Next time: homogeneity.

grp. of the compl. space $H_{\mathbb{C}}$, preserv. ψ, transit. on $\overline{\mathcal{D}}$

Conn. comp. $_1$ grp. of the real space $H_{\mathbb{R}}$, preserv. ψ, transit. on $\mathcal{D}$

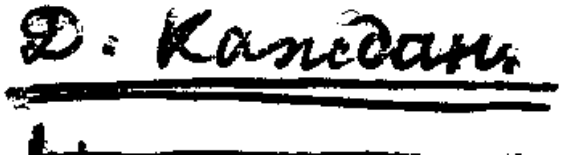


Uncertain reading. The left addition provisionally reads «Св. комп.» (“conn. comp.”) with a lowered sign resembling 1.

Original source details: GELFAND-P000458

Manin's 2nd lecture.

D: Kazhdan. More on Tate's work.

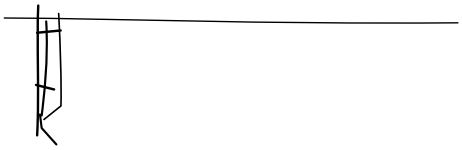


Original source details: GELFAND-P000458

Uncertain reading. Two dots sit one above the other after the initial.

Original source details: GELFAND-P000458

Diagram and typed description. A narrow mark of several strokes crosses the divider. No definite letter or arrow is visible.



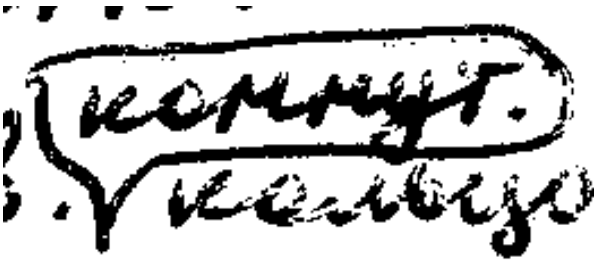
Chapter 6

GELFAND-P000459

V. Buchstaber.

Formal groups.

The def. of a formal group was given.



Original source details: GELFAND-P000459

Diagram and typed description. The abbreviation “commut.” is added in a bubble above “ring.”

commut.

A — arbitrary ring with 1;

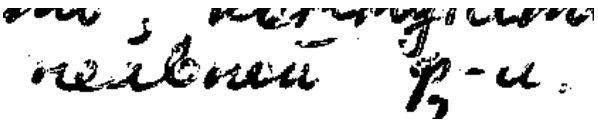
$$\sum_{i=0}^{\infty} a_i x^i, \qquad a_i \in A$$

Form. group:

$$F(x,y) = x + y + \sum_{i+k\geq 2} a_{ik} x^i y^k$$

— form. power series.

Conditions: associativity, commutativity (it follows from assoc.). Exist. of an inv. elem. from the implicit func. thm.



Original source details: GELFAND-P000459

Uncertain reading. The last words read «неявной ф-и» (“implicit func.”).

Isomorphism:

$$z = \varphi(x) = x + a_1 x^2 + \ldots; \qquad \varphi^{-1}(x) = z - a_1 z^2 + \ldots$$

Original source details: GELFAND-P000459

Uncertain reading. The sign in the first formula reads φ.

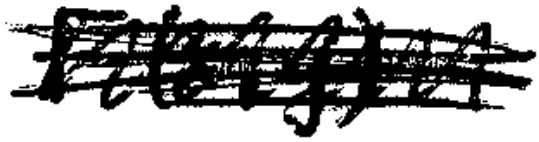
Original source details: GELFAND-P000459

Uncertain reading. The same sign φ appears before the exponent -1.

Original source details: GELFAND-P000459

Uncertain reading. The parentheses contain x. The right-hand side uses z.

... and one is obt. from the other by a change of variables.



Original source details: GELFAND-P000459

Undeciphered passage. The entry before “and one” is crossed out. A full reading is uncertain.

Form. groups will be classified (but not up to isom. — that is a separate problem)

The series $g(x) = x + \dots$ is called a logarithm of the grp. F, if

$$F(x, y) = g^{-1}(g(x) + g(y))$$

Lazard's lemma. If A is a $\mathbb{Q}$-module, then any form. group $F(x, y)$ over A has a logarithm.

Original source details: GELFAND-P000459

Uncertain reading. The name provisionally reads «Лазара» (Lazard). The vowel after «з» may be «у».

Proof. (see Mat., $13 : 6$)

GELFAND-P000460

Consider in place of A the ring $A[[t]]$; t — time.

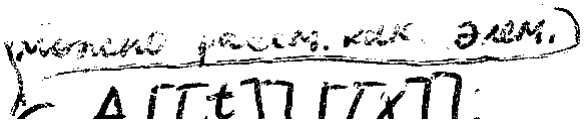
Then def. the shift $x \mapsto F(x, t)$ (shift of the form. var. by time t).

Differential:

$$\omega(x) = \psi(x)\, dx$$

Original source details: GELFAND-P000460

Uncertain reading. The first coefficient sign is set as ψ. The same letter recurs in the formulas below.



Original source details: GELFAND-P000460

Diagram and typed description. A thin line marks off the addition above the formula.

can be consid. as an elem.

$\psi(x) \in A[[t]][[x]]; \quad A \subset A[[t]].$

Inv. form (differential):

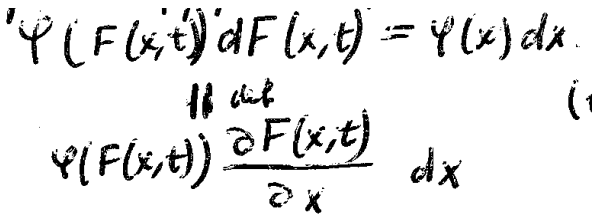


Diagram and typed description. Two vertical strokes and “or” stand between the formulas.
$\psi(F(x,t))\,dF(x,t) = \psi(x)\,dx.$

$\|$ or

$$\psi(F(x,t))\,\frac{\partial F(x,t)}{\partial x}\,dx$$

Original source details: GELFAND-P000460

(t is a parameter; the differential is taken only in x)

$$\psi(F(x,t))\,\frac{\partial F(x,t)}{\partial x} = \psi(x)$$

Set $x = 0$:

$$\psi(t)(1 + \ldots)|_{x=0} = \psi(0)$$

$$\psi(t) = \psi(0)\,\frac{1}{\left.\dfrac{\partial F(x,t)}{\partial x}\right|_{x=0}}$$

These are the series that are coeffs. in the inv. differential.

Canonical form:

$$\omega(x) = \frac{1}{\left.\dfrac{\partial F(x,y)}{\partial y}\right|_{y=0}}\,dx$$

is an ~~...~~ invar. differential (simple exercise).

Uncertain reading. A heavy vertical sign lies between the words. Its reading is uncertain.

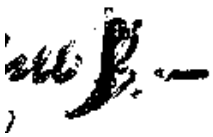
Original source details: GELFAND-P000460

This gives a characterization.

Consider the form. series

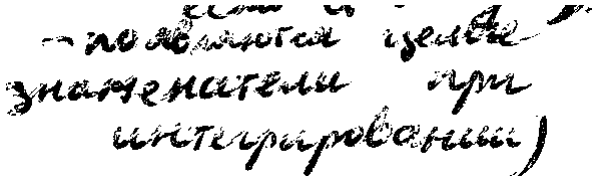
$$g(x) = \int_0^x \omega(x) = x + \ldots$$

(it is important that A is a $\mathbb{Q}$-module! — integer denominators arise during integration)



Uncertain reading. The mark after “module” provisionally reads as an exclamation mark. The strokes overlap here.

Original source details: GELFAND-P000460



Uncertain reading. The first word after the dash reads «появляются» (“arise”).

Original source details: GELFAND-P000460

Integrate the relation:

$$\psi(F(x,t))\,dF(x,t) = \psi(x)\,dx$$
$$g(F(x,y)) = g(x) + C;$$

Here integration is in x (t — param.), therefore $C = C(y)$.

Set $x = 0$. We obtain: $C = g(y)$ ...

Undeciphered passage. The entry after $C = g(y)$ is crossed out and illegible.

Original source details: GELFAND-P000460

$$g(F(x,y)) = g(x) + g(y)$$
$$F(x,y) = g^{-1}(g(x) + g(y)). \quad !! \qquad \blacksquare$$

Corollary Over $\mathbb{Q}$-mods. all form. laws are strictly isom.

Univ. law: consid. $g(x) = x + \sum g_i x^{i+1}$ — a series, its ring is $\mathbb{Q}[g_i]$.

$$F(x,y) = g^{-1}(g(x) + g(y));$$

Over $\overset{\downarrow}{\mathbb{Q}}[g_i]$ one obtains a univ. law. (for all $\mathbb{Q}$-modules)

Uncertain reading. The letter $\mathbb{Q}$ below the arrow is overwritten. The earlier strokes cannot be separated.

Original source details: GELFAND-P000460

We shall classify over arbitrary rings A (commut. and with 1)

Uncertain reading. The first word reads «Будем» (“we shall”).

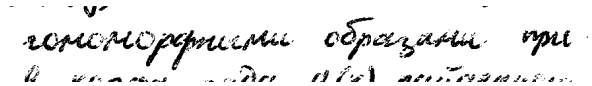
Original source details: GELFAND-P000460

Diagram and typed description. A curved arrow leads from the universal law to the bracket before “Universality.”

[Universality:

Original source details: GELFAND-P000460

any series $F(x,y)$ over the $\mathbb{Q}$-mod. A is obt. from such a series by replacing its coefficients by their homomorphic images under a ring homom. $\mathbb{Q}[g_i]$ into A, sending g_i to the coeffs. of the series $g(x)$, found above]



Uncertain reading. The two words for "homomorphic images" have plural endings.

Original source details: GELFAND-P000460

GELFAND-P000461

Now consid. the general case: A is not necess. a $\mathbb{Q}$-module.

Consid. the series $F(x,y) = x + y + \sum a_{i,j} x^i y^j$

$a_{i,j}$ — alg. indep. variables.

Consid. $\mathbb{Z}[a_{i,j}]$, $\quad \deg a_{i,j} = i + j - 1$

A form. group is obt. from it

$$F(F(x,y),z) = x + y + z + \sum a_{i,j,k} x^i y^j z^k, \quad a_{i,j,k} \in \mathbb{Z}[a_{i,j}].$$
$$F(x,F(y,z)) = x + y + z + \sum a'_{i,j,k} x^i y^j z^k \qquad a'_{i,j,k} \in \mathbb{Z}[a_{i,j}]$$

Consid. in $\overset{\mathbb{Z}[a_{i,j}]}{\overline{\ldots}}$ the ideal J, gen. by the homog. elems. $\{a_{i,j,k} - a'_{i,j,k}\}$ and by the elems. $\{a_{i,j} - a_{\overline{j,i}}\}$.

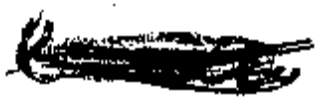
Undeciphered passage. $\mathbb{Z}[a_{i,j}]$ is inserted above the cancellation. The earlier entry is illegible.

Original source details: GELFAND-P000461

Uncertain reading. There are crossed-out strokes above the second coefficient $a_{j,i}$.

Original source details: GELFAND-P000461

Now replace $a_{i,j}$ in $F(x,y)$ with $a_{i,j}$ mod J — $\overline{\ldots}$ we obtain the series $F \in \Lambda[[x,y]]$.



Undeciphered passage. The entry before "we obtain" is crossed out and illegible.

Original source details: GELFAND-P000461

$$\Lambda = \mathbb{Z}[a_{i,j}]/J$$

This is the univ. group. (clear).

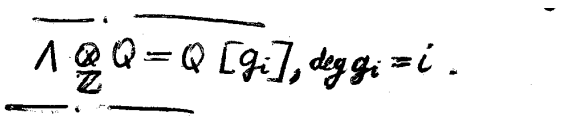
Lazard's thm., Λ — a polynomial ring on generators, one in each dimension:

$$\Lambda = \mathbb{Z}[a_i], \quad i \geq 1, \quad \deg a_i = i.$$

Original source details: GELFAND-P000461

Uncertain reading. The penultimate letter of «одной» (“one”) is heavily overwritten.

By Lazard's lemma:



Original source details: GELFAND-P000461

Diagram and typed description. Two pairs of strokes with dots lie above and below the equality.

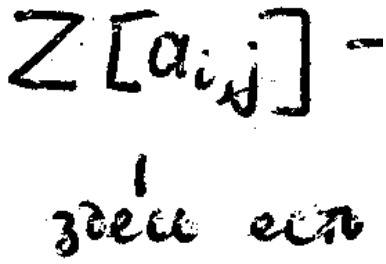
$$\Lambda \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}[g_i], \ \deg g_i = i.$$

1. How many mult. generators has Λ?

Consid. $\widetilde{\Lambda} = \Lambda/(\text{decomposable elements})$

Original source details: GELFAND-P000461

Uncertain reading. The word in the denominator provisionally reads «разложимым» (“decomposable”).



Original source details: GELFAND-P000461

Diagram and typed description. A short vertical stroke links the first ring to the explanation.

$$\mathbb{Z}[a_{i,j}] \longrightarrow \Lambda \longrightarrow 0.$$

here there is an ideal gen. by decomp. elems. (prod. of 2 of nonzero degree)

We have: $\widetilde{\Lambda} = \sum_{i \geq 0} \widetilde{\Lambda}_i$;
multiplication is zero except when one of $\dots$ is an elem. of degree 0.

Decomp. elems. are elems. of the form $a \cdot b$, where a and b *
* are noninvertible. Take the ideal they generate, and $\widetilde{\Lambda}$ — the quotient ring.
Grading $\Rightarrow$ only elems. of degree 0 are invertible.
Hence this ideal is the image of the ideal of decomp. elems. from $\mathbb{Z}[a_{i,j}]$.

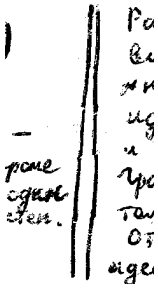
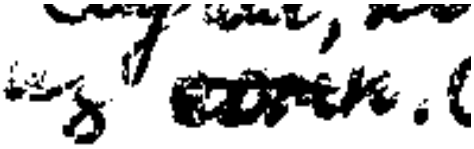


Diagram and typed description. Two vertical strokes separate the marginal note at right.

Original source details: GELFAND-P000461

Uncertain reading. The left-hand sign probably reads $\widetilde{\Lambda}$ with a joined tilde. Another possible reading is χ.

Original source details: GELFAND-P000461

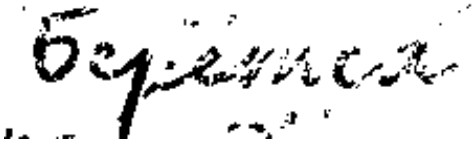


Undeciphered passage. The short entry after “of” is crossed out and illegible.

Original source details: GELFAND-P000461

Uncertain reading. The marginal note reads «необратимы» (“are noninvertible”).

Original source details: GELFAND-P000461



Uncertain reading. «Берется» (“take”) follows «необратимы».

Original source details: GELFAND-P000461

$\widetilde{\Lambda}_n$ — an abelian grp. under add. with gens. — gens. of the mult. grp. Λ in dimension n.

$$F(F(x,y),z) = F(x,y) + z + \sum a_{i,j} \dots (x+y)^i z^j$$

Undeciphered passage. The entry before $(x+y)^i$ is crossed out and illegible.

Original source details: GELFAND-P000461

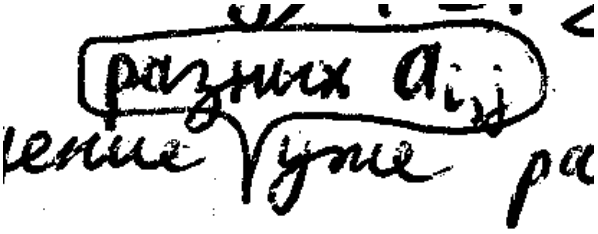


Diagram and typed description. The bubble above the line contains "of different $a_{i,j}$."

of different $a_{i,j}$

(in $\widetilde{\Lambda}$ the product already equals 0).

Original source details: GELFAND-P000461

$$F(x, F(y,z)) = x + F(y,z) + \sum a_{i,j} x^i (y+z)^j$$

$\widetilde{\Lambda}$ is obt. from the free ab. group gen. by $a_{i,j}$, by quotienting by the rels. ...

Let $h(x,y) = \sum h_{i,j} x^i y^j = F(x,y) - x - y$,

$$h(x,0) = 0, \quad h(0,y) = 0.$$

GELFAND-P000462

Lazard's rel.:

$$h(x,y) + h(x+y,z) = h(y,z) + h(x,y+z). \qquad (*)$$

It suffices to take homog. polynomials h.

In Lazard: methods of number theory.

Problem: find all solutions of $(*)$ [homog. polys.] over some ring K.

All solutions — a group L, Everything over the ring K

$$L \longrightarrow \mathrm{Ext}^{2,n}_{K[x]}(K,K) \longrightarrow 0.$$

Introduce on $K[x]$ a Hopf struct.: a mult. homomorphism

$$\Delta : K[x] \longrightarrow K[x] \otimes_K K[x] \qquad \text{(comultiplication)}.$$
$$x \longrightarrow x \otimes 1 + 1 \otimes x$$

A Hopf alg. has cohomology. The alg. is graded $\Rightarrow$

$\Rightarrow$ the cohom. is bigraded. Hence 2 indices for Ext.

Next:

$$\begin{array}{c} \text{on the left and on the right} \quad (x+y+z)^n - x^n - y^n - z^n \\ \downarrow \\ 0 \longrightarrow \{c((x+y)^n - x^n - y^n),\ a \in K\} \longrightarrow L \longrightarrow \mathrm{Ext}^{2,n}_{K[x]}(K,K) \longrightarrow 0. \end{array}$$

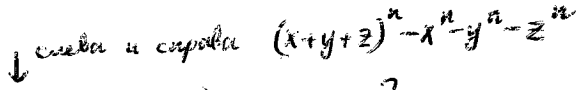


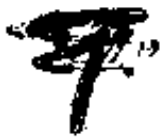
Diagram and typed description. The arrow from the addition points down toward the first term of the sequence.

Original source details: GELFAND-P000462

Uncertain reading. The sign before the parentheses provisionally reads c. The separate entry reads $a \in K$. Both signs are overwritten.

Original source details: GELFAND-P000462

$n = 2$. Polyn. of degree 2 cxy.



Uncertain reading. The sign above "Polyn." is crossed out and illegible.

Original source details: GELFAND-P000462

$n = 3$ $h(x, y) = C_1x^2y + C_2xy^2$

$$\widetilde{\Lambda} = \mathbb{Z} + \mathbb{Z}_3.$$

Uncertain reading. The line reads $\widetilde{\Lambda} = \mathbb{Z} + \mathbb{Z}_3$.

Original source details: GELFAND-P000462

1 sol. by Lazard's lemma: $c((x + y)^n - x^n - y^n)$

$h(x, y) = x^2y + xy^2$ for all rings

$h(x, y) = x^2y$ for rings that are $\mathbb{Z}_3$-modules

Uncertain reading. The subscript on $\mathbb{Z}$ in this line is 3.

Original source details: GELFAND-P000462

x^2y

Consid. comodules over a Hopf alg. A — this is an ab. grp. N, for which there is a map $N \longrightarrow A \otimes N$ with nat. conds.

We need to choose the resolution correctly: there will be all cycles without boundaries (min. resolution). ~~by~~

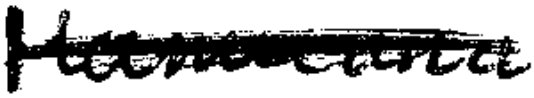
Uncertain reading. The crossed-out word after the parenthesis may read «по» (“by”).

Original source details: GELFAND-P000462

~~H...~~

Uncertain reading. The start of the crossed-out heading shows the Cyrillic «Н».

Original source details: GELFAND-P000462



Undeciphered passage. The rest of the crossed-out heading is illegible.

Original source details: GELFAND-P000462

Calculation of the group Ext^2.

1°. K — ... a comodule over the Hopf alg. $K[x]$.

Uncertain reading. A heavy rectangular mark is crossed out between the dash after K and “comodule.”

Original source details: GELFAND-P000462

... $x \longrightarrow x \otimes 1 + 1 \otimes x$

Uncertain reading. A heavily crossed-out sign appears before x in the addition.

Original source details: GELFAND-P000462

$$K \longrightarrow K \otimes K[x] \quad \overleftarrow{\ldots}$$

Uncertain reading. The beginning of the crossed-out addition resembles $\leftarrow$.

Original source details: GELFAND-P000462

Undeciphered passage. The remaining strokes after $K[x]$ are illegible.

Original source details: GELFAND-P000462

$1 \longrightarrow 1 \otimes 1$ and then by linearity.

GELFAND-P000463

Consid.

$$K_*[x] = \mathrm{Hom}_K(K[x], K).$$

("diff. operators").

$$(x_*^n, x^m) = \delta_{n,m}$$

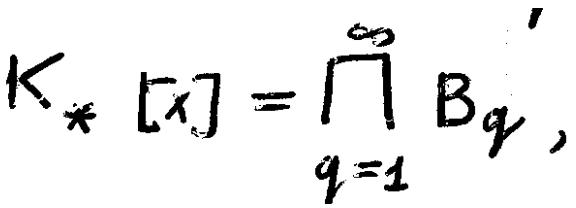
Multiplication:

$$\underline{x_*^n \cdot x_*^m = C_{n+m}^n x_*^{n+m}} \qquad \text{— } \underline{\text{table}}.$$

Here K plays no role.

It suffices to take $K = \mathbb{Z}_p$ or $\mathbb{Q}$.

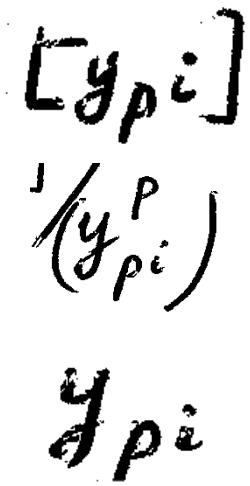
$$K_*[x] = \prod_{q=1}^{\infty} B_q, \qquad B_q = \mathbb{Z}_p[y_{p^i}]/(y_{p^i}^p)$$



Uncertain reading. The left side reads $K_*[x]$. The product has lower limit $q = 1$ and upper limit ∞.

Original source details: GELFAND-P000463

$y_1, \dots, y_i, \dots, y_{p^i}$ — generators.



Uncertain reading. In three places the i next to p is raised. This is set as y_{p^i}. In the denominator the separate outer exponent p on y is visible.

Original source details: GELFAND-P000463

GELFAND-P000464

30/X-72. Gelfand sem.

I. Bernstein

I. M. G.

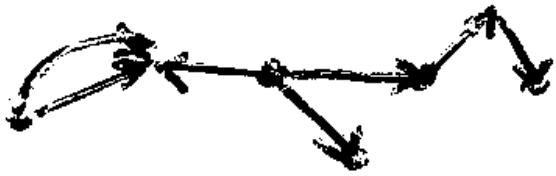
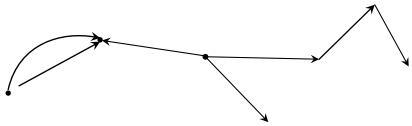


Diagram and typed description. The two arrows at left point to the same vertex. From the middle vertex arrows lead left, right and down to the right. The bent line at right has arrowheads. The small strokes at vertices have no established reading as letters.



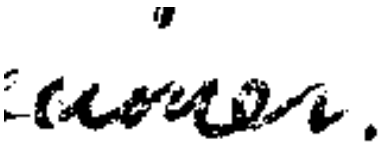
Original source details: GELFAND-P000464

There is a directed graph. This is a category.

Ꝑ Consid. some diagrams, with at the points — vect. spaces, on the arrows maps (functoriality is not required)

Uncertain reading. A loop survives at the start of the crossed-out sign. Possibly Cyrillic «Р».

Original source details: GELFAND-P000464



Uncertain reading. The words before “diagrams” provisionally read «какие л.» (“some”).

Original source details: GELFAND-P000464

Prob. — to classify.

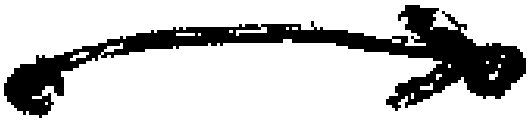


Diagram and typed description. A slightly curved rightward arrow joins the two points.

Original source details: GELFAND-P000464

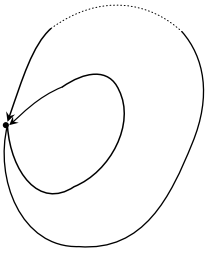
Prob. of classif. of maps. The only invs. are dim Ker and dim Coker. Reduces to an embedding.

Diagram and typed description. One loop. Its arrowhead on the left side points upward.

Original source details: GELFAND-P000464

one transformation.

Diagram and typed description. Two loops share the left vertex. The top of the outer loop is faint. Overlapping arrowhead strokes are visible near the vertex.



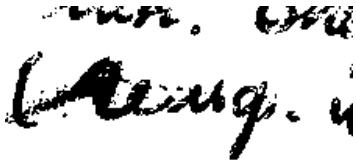
Original source details: GELFAND-P000464

— the prob. is unsolvable (a pair of arbitrary lin. operators). ...

Uncertain reading. The short entry before “it” is crossed out and illegible.

Original source details: GELFAND-P000464

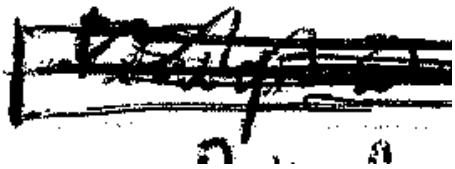
It reduces to the prob. of commuting ones (Gelf. and Ponomarev).



Uncertain reading. The first letter is overwritten. The abbreviation reads «Гельф.» (Gelf.).

Original source details: GELFAND-P000464

~~Reduc....~~



Original source details: GELFAND-P000464

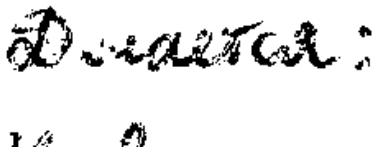
Uncertain reading. The crossed-out line may begin «Свод.» (“reduc.”).

Original source details: GELFAND-P000464

Undeciphered passage. The remainder of the crossed-out line is illegible.

Space and subspace.
Space and 2 subspaces } triv.

Can be done: Space and 3 lin. subspaces.



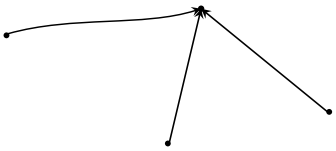
Original source details: GELFAND-P000464

Uncertain reading. The start of the line provisionally reads «Делается» (“can be done”).

There are 9 invariants, not 8!

Original source details: GELFAND-P000464

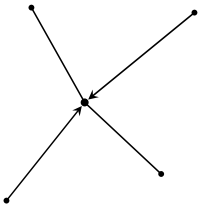
Diagram and typed description. Three arrows point toward the upper vertex. Their arrowheads partly merge.



(problem).

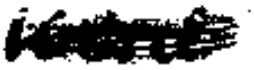
Original source details: GELFAND-P000464

Diagram and typed description. Four edges meet at the heavily inked central vertex. Overlapping strokes obscure the direction of two edges.



Space and 4 subspaces classified (Gelf. and Ponom.) — ... already an inf. number of indecomp.

Undeciphered passage. The word after the dash is crossed out and illegible.



Original source details: GELFAND-P000464

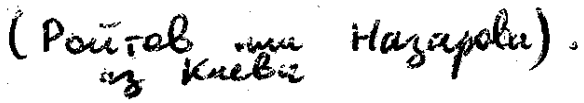
Gabriel: in which graphs are there finitely many indecomp. systems?

Answer: draw a Dynkin diagram! — these and only these are such graphs.

Exceptions: diagrams with double (generally multiple) arrows.

GELFAND-P000465

Theorem (Roytov or Nazarovy).
from Kyiv

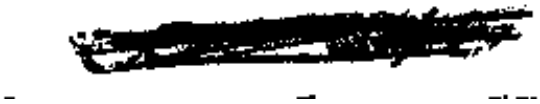


Uncertain reading. The names provisionally read «Ройтов» and «Назаровы». The addition below reads “from Kyiv.”

Original source details: GELFAND-P000465

Descr. the class of diagrams in which there is a classific. as in Gelf. and Ponomarev.

Connection ... with singularities! — see Artin, Tyurina, Brieskorn.



Undeciphered passage. The word between “Connection” and “with singularities” is crossed out and illegible.

Original source details: GELFAND-P000465

I. B. Category of vect. spaces over points and maps over arrows.

(The graph is fixed). Def. direct sum. This is Cat(Γ) (Γ — graph)

Morphisms — maps over each point with commut. diagrs.

- category of linear spaces.

Important: no need to ident. isomorphic ones.

Uncertain reading. The abbreviation before “isomorphic” provisionally reads «отожд.» (“ident.”).

Original source details: GELFAND-P000465

It is easy to see that $\forall$ object $= \sum$ indecomp. ones (uniquely up to isom.)

1) Classif. Γ, with finitely many indecomp. objects.

Restrict to connected ones:

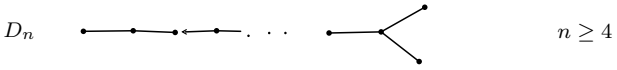
Diagram and typed description. A small angular stroke rises above the third vertex from the left.

A_n (n points)

Original source details: GELFAND-P000465

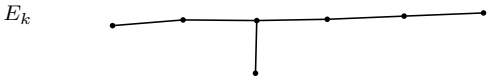
Original source details: GELFAND-P000465

Diagram and typed description. There is a small leftward arrowhead on the segment between the third and fourth vertices.



Original source details: GELFAND-P000465

Diagram and typed description. There are six vertices on the horizontal chain. A branch descends from the third vertex from the left.



$k = 6, 7, 8$ (necessarily 2 segments on the left).

Place the arrows arbitrarily!!

D_4 — space and 3 subspaces

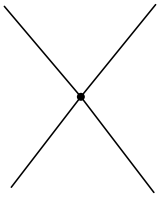
Idea: no cycles (one obt. eigenvalues), no

Original source details: GELFAND-P000465

Uncertain reading. The abbreviation in parentheses may read «получ.» (“obt.”).

Original source details: GELFAND-P000465

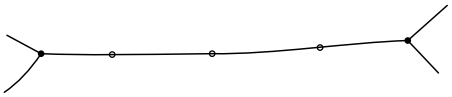
Diagram and typed description. Four segments meet at the black dot. No separate vertices are visible at the ends.



— there will be an inf. number.

Original source details: GELFAND-P000465

Diagram and typed description. The chain has a fork at each end. Five vertices are marked on the chain itself.



— also imposs., since ident.

Original source details: GELFAND-P000465

Uncertain reading. The last abbreviation provisionally reads «отожд.» (“ident.”).

2) The no. of indecomp. objs. does not depend on the dir. of the arrows.
It is characterized by the set of dimensions!!
one should consid. the root syst.
γ — pos. root, α_i — simp. roots.
To each root — an indecomp. object

$$\gamma = \sum n_i \alpha_i, \qquad n_i \text{ — dimensions!!}$$

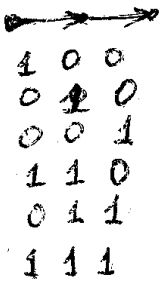
Original source details: GELFAND-P000465

Uncertain reading. Both signs have a lower loop and two upper strokes. They are provisionally set as γ.

No. = no. of simple roots

Original source details: GELFAND-P000465

Uncertain reading. The line reads «простых корней» (“simple roots”).



Original source details: GELFAND-P000465

Diagram and typed description. Six rows of dimensions stand below the chain with two rightward arrows.

1	0	0
0	1	0
0	0	1
1	1	0
0	1	1
1	1	1

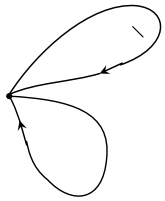
Original source details: GELFAND-P000465

Uncertain reading. The middle 1 in row two is overwritten. A remnant of a loop survives to its right.

3) Consid. those Γ for which $\mathrm{Cat}(\Gamma)$ is solvable.

Original source details: GELFAND-P000466

Diagram and typed description. The two loops share a vertex. Inside the upper loop there is a short slanted stroke separate from its arrowhead.



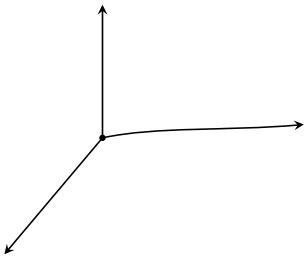
C

Unsolvable categ. — if there is an embedding $C \longrightarrow \mathrm{Cat}(\Gamma)$, taking nonisom. objects to nonisom. ones.

They are classif. and correspond to extended Dynkin diagrams!

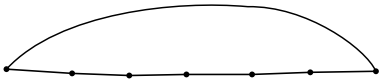
Original source details: GELFAND-P000466

Diagram and typed description. Three arrows leave one vertex, upward, rightward and down to the left.

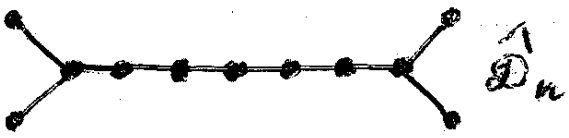


Original source details: GELFAND-P000466

Diagram and typed description. Seven points lie on a straight chain. An arc joins the endpoints.



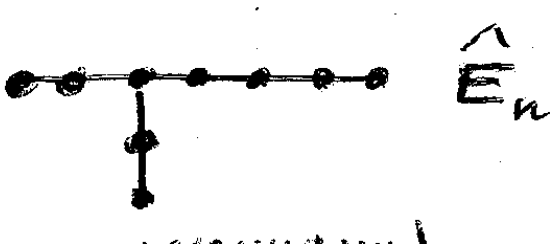
$\widehat{A}_n$



Original source details: GELFAND-P000466

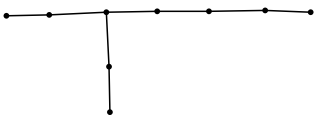
Diagram and typed description. Seven points lie along the chain, with two branches at each end.

$\widehat{D}_n$



Original source details: GELFAND-P000466

Diagram and typed description. Seven points lie along the horizontal chain. Two segments with points descend from the third point from the left.



$\widehat{E}_n$

(appear in the theory of ~~grp.~~ groups generated by reflections.)

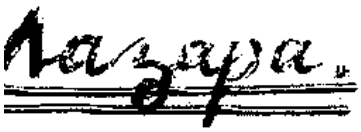
Uncertain reading. The crossed-out insertion above the line may read «гр» (“grp.”).

Original source details: GELFAND-P000466

These are all the graphs for which the prob. is solvable.

V. Buchstaber.

Lazur's thm.



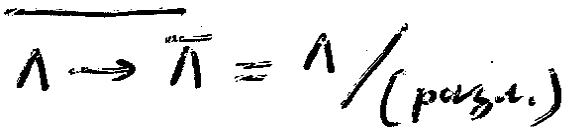
Uncertain reading. The name provisionally reads «Лазура» here. The middle vowel may be «а».

Original source details: GELFAND-P000466

$$\Lambda = \mathbb{Z}[a_1, \ldots, a_n, \ldots], \qquad \deg a_n = n \geq 1.$$

Proof.

$$\Lambda \longrightarrow \bar{\Lambda} = \Lambda/(\text{decomp.})$$



Uncertain reading. A horizontal stroke stands above the second Λ. The denominator reads «разл.» (“decomp.”).

Original source details: GELFAND-P000466

$\bar{\Lambda}$ — univ. ring for (such) polynomials $h(x, y)$,

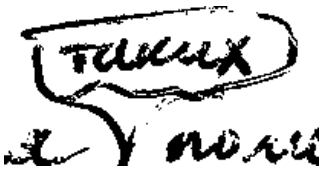


Diagram and typed description. “Such” is inserted in a bubble before “polynomials.”

Original source details: GELFAND-P000466

(that) $h(x, y) = h(y, x), \quad h(x, 0) = 0,$

$$h(y, z) + h(x, y + z) = h(x, z) + h(y, x + z).$$

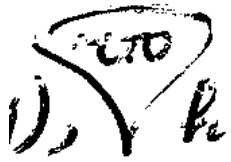
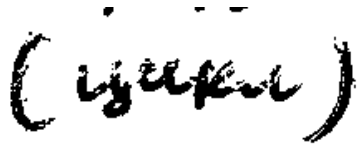


Diagram and typed description. “That” is inserted in a separate bubble before the conditions.

Original source details: GELFAND-P000466

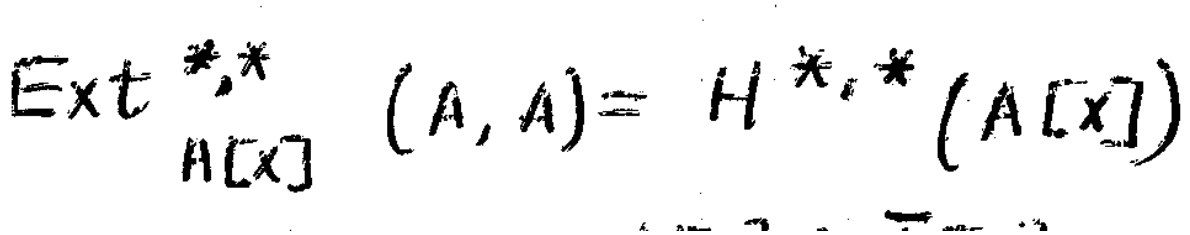
This is cohomology: $\underset{\text{(cycles)}}{dh} = 0$ (in the bar construction for polynomials)



Uncertain reading. Below dh is the word "cycles."

Original source details: GELFAND-P000466

$$\mathrm{Ext}^{*,*}_{A[x]}(A, A) = H^{*,*}(A[x])$$



Original source details: GELFAND-P000466

Uncertain reading. The subscript on Ext reads $A[x]$. There are two stars above the symbol on each side.

$$A[x] \longrightarrow A[x] \otimes \bar{A}[y]$$
$$x \longrightarrow x + y$$

Uncertain reading. Below the first arrow the manuscript reads $x \longrightarrow x + y$.

Original source details: GELFAND-P000466

Cohom. is calc. using resolutions.

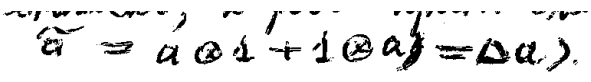
The bar constr. is one of the resolutions.

$$\mathcal{A}: \quad A[x] \xrightarrow{\Delta} A[x] \otimes \bar{A}[y] \longrightarrow A[x] \otimes \bar{A}[y] \otimes \bar{A}[z] \longrightarrow \dots$$

$\bar{A}[y]$ — polynomials with no const. term.

We obt. an acyclic complex which is used to compute cohomology

$$(\underset{h(x,y)}{\text{put everywhere}} — \tilde{a} \Rightarrow a \otimes 1 + 1 \otimes a) = \Delta a).$$



Uncertain reading. A wavy stroke stands above a. The following sign provisionally reads $\Rightarrow$. Both closing parentheses are visible.

Original source details: GELFAND-P000466

One must classif. cycles in a particular resolution.

$$0 \longrightarrow B \longrightarrow Z \longrightarrow H^2(A[x]) \longrightarrow 0$$

Original source details: GELFAND-P000466

Uncertain reading. The term after B is Z, and H has superscript 2.

$x^n \longrightarrow (x+y)^n - x^n - y^n$ (these are boundaries)

GELFAND-P000467

Passage to the obt. resol. by means of the functor $\mathrm{Hom}(\cdot, A)$

Uncertain reading. The abbreviation before "resol." reads «полученн.» ("obt.").

Original source details: GELFAND-P000467

In the acycl. resol.

$$x \longrightarrow x + y$$
$$x^n \longrightarrow (x+y)^n.$$

We must consid. $\mathrm{Hom}_{A[x]}(\mathcal{A}, A)$ (use the 1st factor to est. the $A[x]$-mod. struct.).

Uncertain reading. The subscript on Hom is $A[x]$. The arguments are $\mathcal{A}, A$.

Original source details: GELFAND-P000467

Uncertain reading. The abbreviation after "factor" reads «уст.» ("est.").

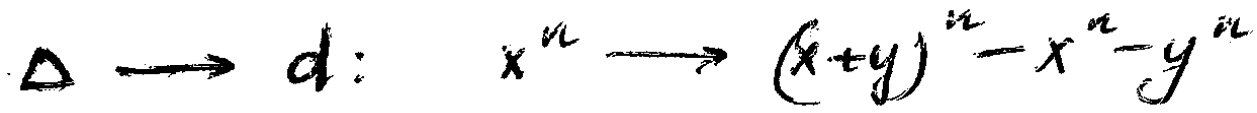
Original source details: GELFAND-P000467

$$A \xrightarrow{d} A[y] \xrightarrow{d} A[y] \otimes A[z] \xrightarrow{d} \dots$$

Original source details: GELFAND-P000467

Uncertain reading. On this line $A[y]$ and $A[z]$ have no overbars.

$$\Delta \longrightarrow d: \qquad x^n \longrightarrow (x+y)^n - x^n - y^n$$



Original source details: GELFAND-P000467

Uncertain reading. Both subtracted terms, x^n and y^n, are visible on the next line.

$$\Delta: \quad h(x,y) \longrightarrow h(\underset{\|}{\Delta x}, y) - h(x, \underset{\|}{\Delta y})$$
$$\qquad\qquad\quad x+z \qquad\qquad\quad y+z$$

Uncertain reading. There is a heavily inked horizontal sign between the terms. It is provisionally set as minus.

Original source details: GELFAND-P000467

$\mathcal{A}$ — a seq. of $A[x]$-modules,

Uncertain reading. A short low stroke at the end of the line resembles a comma.

Original source details: GELFAND-P000467

$A[x] \otimes y^k$ — submodules

Because of $\overline{A[y]}$:

$$x^n \longrightarrow (x+y)^n - x^n \qquad \text{(since there is no const. term)}$$

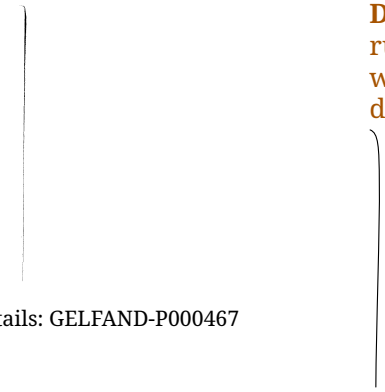


Diagram and typed description. A long line runs along the right edge of the frame. It begins with a small bend and ends just below the divider.

Original source details: GELFAND-P000467

GELFAND-P000468

13/XI-72. Gelfand sem.

L. Vaserstein.

A — assoc. ring (not necessarily with 1).

iA — ring with a formally adjoined unit.

$iA/A = \mathbb{Z}$, A — two-sided ideal in iA.

GA — mult. group of the ring

$GA \subset A$

$(a, b) \longrightarrow a + b + ab$ This is a binary assoc. oper.

$$\| \qquad (e + a)(e + b) - e = a + b + ab$$

~~iA~~ GA — set of invertible elements with respect to this operation.

Uncertain reading. The short crossed-out entry before GA may read iA.

Original source details: GELFAND-P000468

$$a \longrightarrow 1 + a, \qquad A \longrightarrow 1 + A$$

homomorphism. into the ordinary invert. elems.

$$f : A \longrightarrow B$$
$$f(GA) \subset GB$$

Uncertain reading. The sign before GB is overwritten. It is provisionally set as $\subset$.

Original source details: GELFAND-P000468

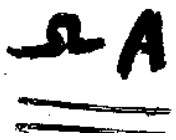
i, G — functors to the cat. of rings with 1 and the cat. of groups.

$M_k A$ — matrices $k \times k$ with ordinary multiplication.

$\widetilde{\Omega} A = \{a = (a_{ij})\}$, in $\forall$ row and column a finite no. of elems. $\neq 0$, (this is a ring)

MA — subring of finite matrices (a finite number of elems. $\neq 0$). (two-sided ideal in $\widetilde{\Omega} A$).

$\Omega A = \widetilde{\Omega} A / MA$



Uncertain reading. The letter before A on the left is overwritten. It reads Ω.

Original source details: GELFAND-P000468

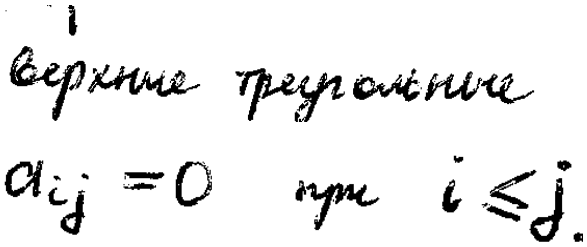
$M, \widetilde{\Omega}, \Omega$ — functors.

$$\underline{\underline{\Theta A}} \subset MA$$
$$|$$

upper triangular $a = (a_{ij})$: $a_{ij} = 0$ when $i \leq j$ ...

Diagram and typed description. A short vertical stroke is drawn below the double underline under ΘA.

Original source details: GELFAND-P000468



Uncertain reading. The manuscript has “upper triangular” and $a_{ij} = 0$ when $i \leq j$.

Original source details: GELFAND-P000468

Undeciphered passage. Several signs one above another are crossed out after the inequality.

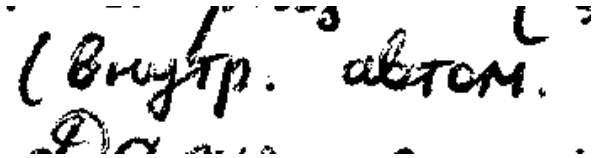
Original source details: GELFAND-P000468

Remark. If $N \xrightarrow{\sim} N$ is a bijection, there is a map of the matrix ring (~~th.~~ autom.) — simultaneous perm. of rows and columns.

Uncertain reading. The crossed-out abbreviation before “autom.” may read «эт».

Original source details: GELFAND-P000468

(inner autom. perm. of rows and columns)



Uncertain reading. The parenthesis reads «внутр. автом.» (“inner autom.”).

Original source details: GELFAND-P000468

Even if $N \longrightarrow N$ is any map, it is assigned an endomorphism.

Uncertain reading. The first word reads «Даже» (“even”).

Original source details: GELFAND-P000468

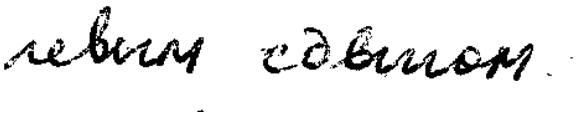
Uncertain reading. The last word is underlined and reads “endomorphism.”

Original source details: GELFAND-P000468

GMA — mult. group, MA, $GMA \subset MA$.

Simplicial structure in MA.

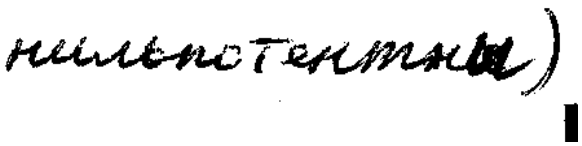
$X \subset GMA$ is called a simplex, if by a left shift and conjugation by a permutation matrix it can be taken into ΘA



Uncertain reading. The manuscript reads «левым сдвигом» (“by a left shift”).

Original source details: GELFAND-P000468

Note that $G\Theta A = \Theta A$ as a set (they are nilpotent)



Uncertain reading. The last letters are overwritten. The word reads “nilpotent.”

Original source details: GELFAND-P000468

GELFAND-P000469

Undeciphered passage. A crossed-out mark before the first line with a short stroke to its right. The reading is unresolved.

Original source details: GELFAND-P000469

In fact

Diagram and typed description. The line under the inclusion turns up at both ends.

$\llcorner GMA \supset \Theta A \lrcorner$

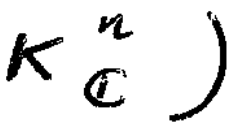
Original source details: GELFAND-P000469

Mark the vertex: 0 — the identity of the group GMA.

SA — the corresponding simplicial scheme.

(Historical remark: Volodin used SA to define the functors $K^n_{\mathbb{C}}$)

in essence



Uncertain reading. The subscript reads $\mathbb{C}$, with an extra stroke inside the letter.

Original source details: GELFAND-P000469

Suppose the ring A has a unit. Then put.

$$K_nA = \pi_{n-1}SA, \qquad n \geqslant 1.$$

For $n \geqslant 2$ there is a group structure, commutative for $n \geqslant 3$.

Another def. of the group structure (commutative).

Whitehead's lemma

Consider the spheroid $s\colon Z^n \longrightarrow SA$. The group of permutations acts on SA $f\colon N \longrightarrow N$. We obtain a new spheroid s^f:

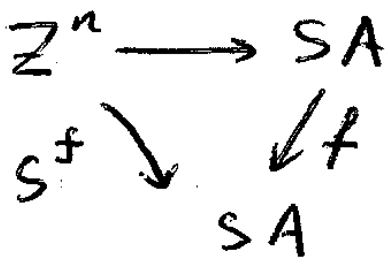
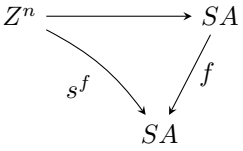


Diagram and typed description. The top arrow is unlabelled.



Original source details: GELFAND-P000469

They are always homotopic.

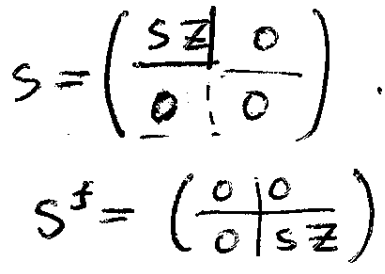
Idea of the proof:

Im $s \subset M_kA$

$$s = \left(\begin{array}{c|c} sz & 0 \\ \hline 0 & 0 \end{array}\right).$$

Everything reduces to the case

$$s^f = \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & sz \end{array}\right).$$

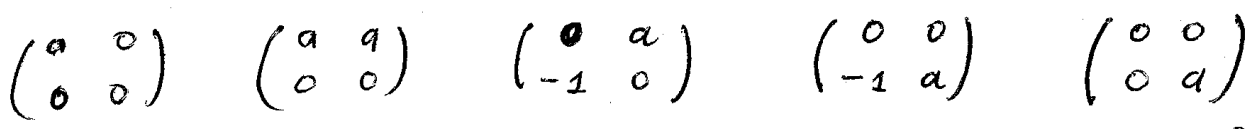


Uncertain reading. The final sign in sz has a cross-stroke and is provisionally set as a lowercase letter. Both matrices have two dividing rules.

Original source details: GELFAND-P000469

(f of a special form: $fi = i + k$, we adjust it at ∞).

$$\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} a & a \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ -1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ -1 & a \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & a \end{pmatrix}$$



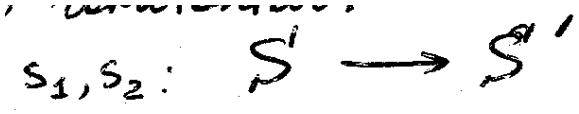
Original source details: GELFAND-P000469

Uncertain reading. The upper left zero in the third matrix is heavily retraced.

These are 5 spheroids. This is a homotopy. Each 2 neighbours are elementarily homotopic:

$$s_1, s_2 \colon S' \longrightarrow S'$$

are called elem. homotopic if $\forall$ simplices $X \subset S$, $s_1 X \cup s_2 X$. is a simplex in S'.



Uncertain reading. A short stroke is visible above the first S. The next line has $X \subset S$ without a prime.

Original source details: GELFAND-P000469

Let all a lie in one simplex X. Then $\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} a & a \\ 0 & 0 \end{pmatrix}$ lie in one simplex.

Let $a_0^{-1} \in X$. Multiply by a_0. Then they become upper triang. If all a are upper triangular, then all 5 matrices lie in one simplex.

———— · ————

The commut. group structure on $K_n A$.

$$MA \times MA \xrightarrow{\oplus} MA$$

$fi = 2i$	$f, g \colon \mathbb{N} \longrightarrow \mathbb{N}$	noncommut., nonassoc., with no
$gi = 2i + 1$	$a \oplus b = a^f + b^g$	identity elem.

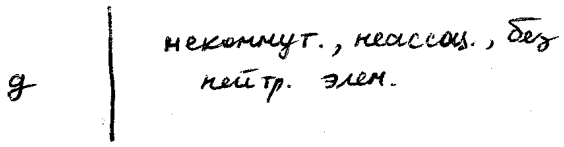


Diagram and typed description. A vertical rule separates the note about the operation.

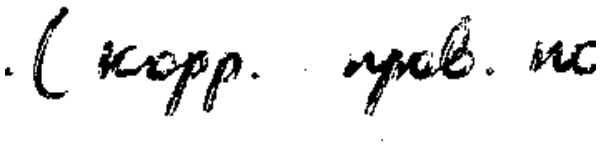
Original source details: GELFAND-P000469

Uncertain reading. Outside the left edge of the text are a slanting stroke and a mark resembling a closing parenthesis. Their purpose is unresolved.

Original source details: GELFAND-P000469

GELFAND-P000470

it induces the mapp. $SA \times SA \longrightarrow SA$, which gives operations on all K_i, $i \geqslant 0$. (corr. verif. by Whitehead's lemma)



Uncertain reading. The second abbreviation after «корр.» is provisionally read as «прв.».

Original source details: GELFAND-P000470

They are commutative and coinc. with those introduced earlier by mutual distributivity, which is verified by Whitehead's lemma.

One can define a sum — H-space. On $\widetilde{\Omega}A$ one can even add an inf. number of matrices! Hence it will have trivial K-functors.

Original source details: GELFAND-P000470

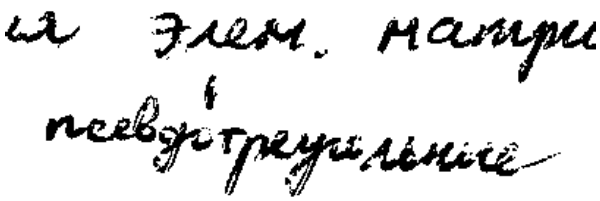
Uncertain reading. A long wavy mark is written over Ω. It is set as $\widetilde{\Omega}$.

$K_1 A = \pi_0 SA$.

This is the set of connected components in GMA.

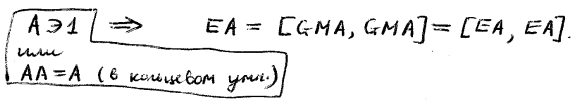
EA — the connected component of the identity. (products of elem. matrices),

pseudotriangular



Original source details: GELFAND-P000470

Diagram and typed description. "Pseudotriangular" is written under "elem." and connected to it by a stroke.



Original source details: GELFAND-P000470

Diagram and typed description. The stepped outline encloses both hypotheses. The arrow starts level with the first one.

$A \ni 1 \implies EA = [GMA, GMA] = [EA, EA]$.
or
$AA = A$ (in ring mult.)

$K_2 A$ — homot. classes of loops. Loops — a relation of elem. matrices.

ΘA has generators a^{ij}

Original source details: GELFAND-P000470

Undeciphered passage. The entry after a^{ij} is heavily crossed out. A short leftward arrow is visible below "Loops".

(at position i, j stands $a \in A$).

Relations: Steinberg

$$a^{ij} * b^{ij} = (a+b)^{ij},$$
$$[a^{ij}, b^{jk}] = (ab)^{ik}$$

and for some the commutator is 0.

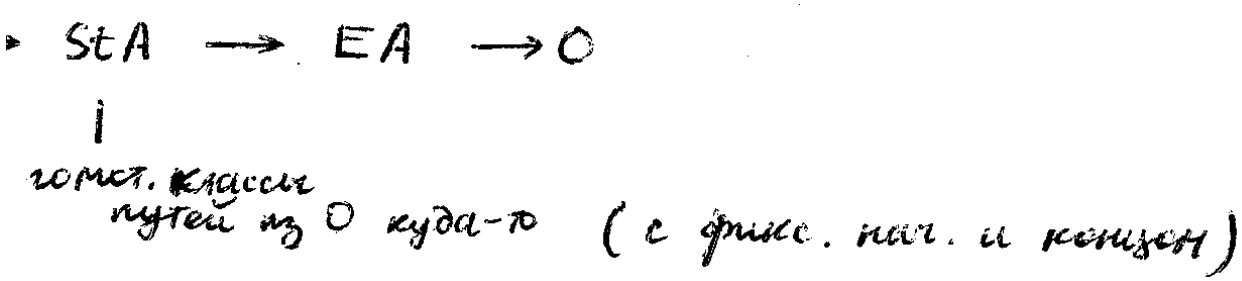
These are the defining relations.

Let StA be the group gen. by these gen. and rel.

$$0 \longrightarrow K_2 A \longrightarrow StA \longrightarrow EA \longrightarrow 0$$

$$|$$

homot. classes of paths from 0 to somewhere
(with fixed start and end)



Original source details: GELFAND-P000470

Diagram and typed description. The annotation is connected by a vertical stroke to StA.

Uncertain reading. There is a single retraced mark to the left of the formula. Its reading is unresolved.

Original source details: GELFAND-P000470

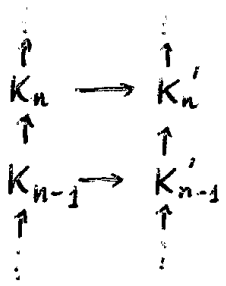
$K_2A = \underset{\substack{\text{(group}\\ \text{homology)}}}{H_2EA} \quad \sim \quad StA =$ the univ. centr. extension of EA.

Problem: suppose there are 3 matrices in GMA. All segments are a simplex. Then all 3 are a simplex.

Generalization: if any pair is a simplex, then all are a simplex. (space of tolerance)

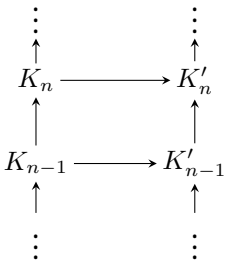
D. A. Kazhdan. Derived category.

C — an abelian category. Consider the category of complexes K_C.



Original source details: GELFAND-P000470

Diagram and typed description. The arrows point up in both columns. Two arrows point right between the columns.

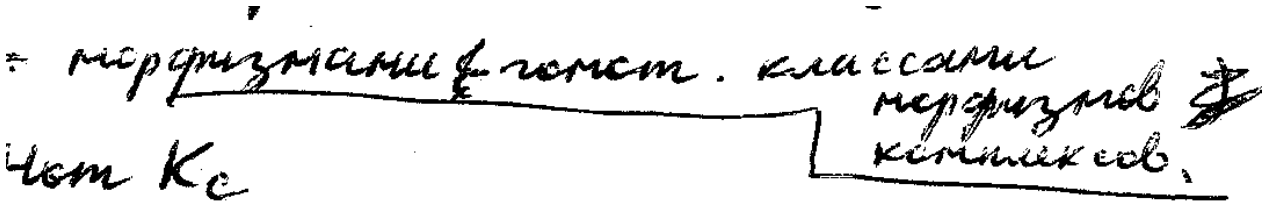


Original source details: GELFAND-P000470

Uncertain reading. A fragment before “with morphisms” is struck out. A possible reading is «с сост. об.».

Original source details: GELFAND-P000470

Uncertain reading. A single mark after “morphisms” is struck out. Its reading is unresolved.



Original source details: GELFAND-P000470

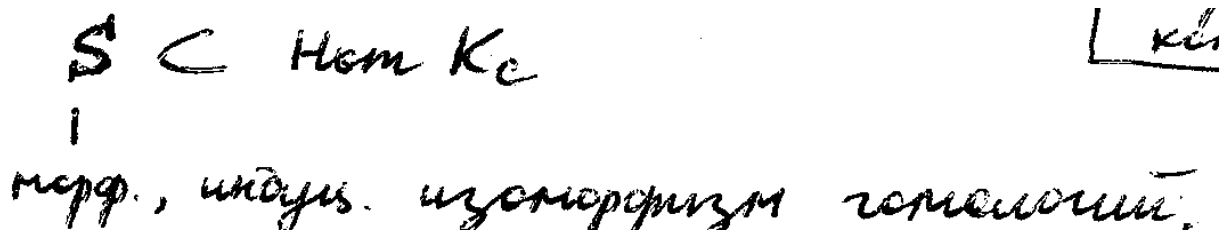
Diagram and typed description. A continuous stepped underline runs beneath the text.

with morphisms f the homot. classes of morphisms of complexes.

$S \subset \operatorname{Hom} K_C$

|

morph. inducing an isomorphism of homologies

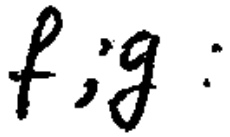


Original source details: GELFAND-P000470

Diagram and typed description. The annotation is connected by a vertical stroke to S.

GELFAND-P000471

Lemma. Let $f; g\colon K \longrightarrow K'$ be two morphisms.

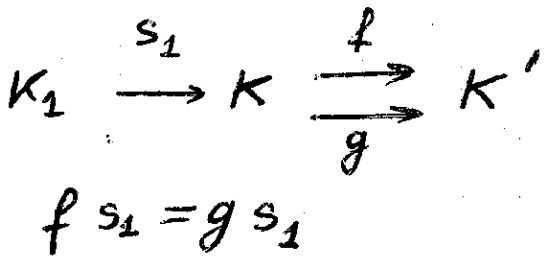


Original source details: GELFAND-P000471

Uncertain reading. Two strokes appear between f and g, here set as a semicolon.

The foll. assertions are equivalent:

1°. $\exists K_1, s_1 \in S$:



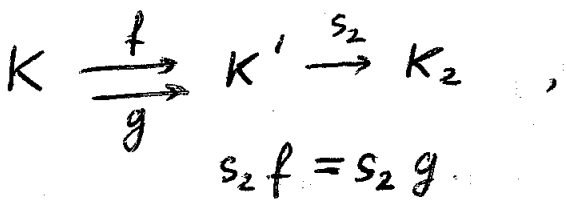
Original source details: GELFAND-P000471

Diagram and typed description. The first diagram with the equality written below it.

$$K_1 \xrightarrow{s_1} K \underset{g}{\overset{f}{\rightrightarrows}} K'$$

$$fs_1 = gs_1$$

2°. $\exists K_2, s_2 \in S$

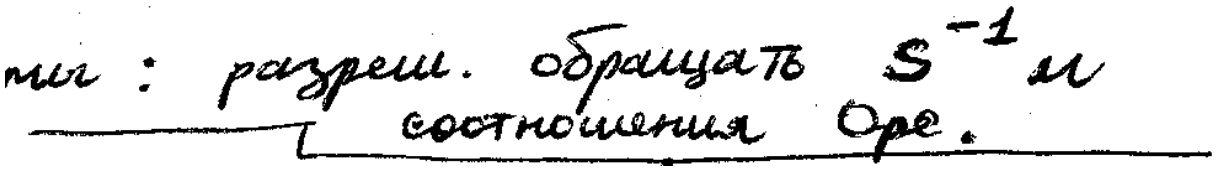


Original source details: GELFAND-P000471

Diagram and typed description. The second diagram with the equality written below it.

$$K \underset{g}{\overset{f}{\rightrightarrows}} K' \xrightarrow{s_2} K_2 \ ,$$

$$s_2 f = s_2 g.$$



Original source details: GELFAND-P000471

Diagram and typed description. “Ore relations” is written on the second line, above the lower part of the stepped rule.

L_C — the same objects, and the morphisms: allow inverting s^{-1} and Ore relations.

The lemma makes it poss. to define the product correctly.

Original source details: GELFAND-P000471

Uncertain reading. The manuscript has «возможно» immediately before «корректно».

See Hartshorne. Residues and duality.

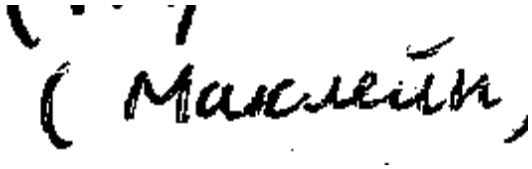
Uncertain reading. A short cross-shaped mark appears above «См.» (“See”).

Original source details: GELFAND-P000471

V. Buchstaber.

References:

	C.–E.	213–223	(Cartan and Eilenberg)
	M.	358–370	(Mac Lane)
	Adams	Mat. 3 : 4	pp. 7–19.



Uncertain reading. The parenthesis reads «Маклейн» (“Mac Lane”).

Original source details: GELFAND-P000471

$$A = \sum_{n \geqslant 0} A_n \qquad \text{— grad. ring.}$$

A_n — an A_0-module for every $n \geqslant 0$.

$$0 \longleftarrow A_0 \xleftarrow{\varepsilon} A \longleftarrow C_1 \longleftarrow C_2 \longleftarrow \cdots \longleftarrow C_s \xleftarrow{d_{s+1}} C_{s+1} \longleftarrow \cdots$$

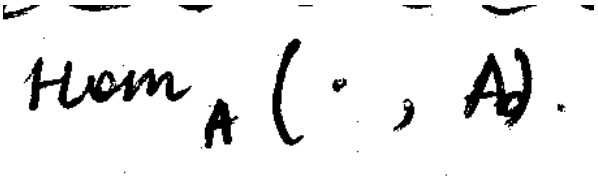
(ε assigns A zeroth comp. to each $a \in A$).

Original source details: GELFAND-P000471

Uncertain reading. A separate A is written immediately after «сопост.» (“assigns”).

Complete to a grad. free resolution.

The functor $\mathrm{Hom}_A(\cdot, A_0)$. takes d_s to $(d_s)^*$.



Uncertain reading. A small zero is visible below and to the right of the second argument A, against the parenthesis.

Original source details: GELFAND-P000471

$$H^s(A) = \mathrm{Ker}(d_{s+1})^* / \mathrm{Im}(d_s)^*$$

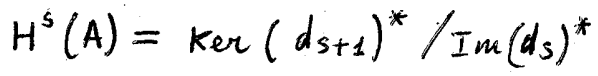


Diagram and typed description. The numerator has d_{s+1} and the denominator d_s.

Original source details: GELFAND-P000471

Theorem: 1) A free res. exists

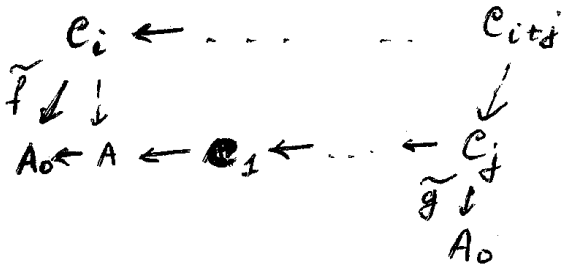
2) The groups $H^s(A)$ are independent of the resolution.

$$H(A) = \sum_{s=0}^{\infty} H^s(A) \qquad \text{— a graded ring.}$$

A def. will be given for the Yoneda mult.

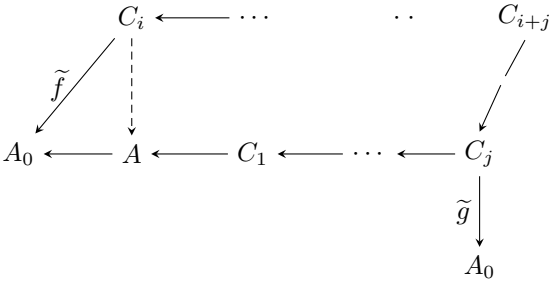
$$f \in H^i(A), \quad g \in H^j(A); \quad fg \in H^{i+j}(A).$$
$$\widetilde{f}\colon C_i \longrightarrow A_0; \qquad \widetilde{g}\colon C_j \longrightarrow A_0$$

GELFAND-P000472



Original source details: GELFAND-P000472

Diagram and typed description. The node C_1 is heavily retraced. The arrow from C_i to A is dashed, and the arrow from C_{i+j} to C_j has a single break.



We have obtained a homomorphism $C_{i+j} \longrightarrow A_0$.

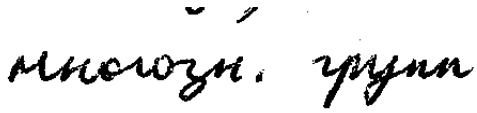
Original source details: GELFAND-P000472

Uncertain reading. The first word is read as «Получили» (“we have obtained”). Its last two letters run together.

This gives

$$A \underset{\text{assoc.}}{\rightsquigarrow} H(A) \qquad \text{(a ring)}$$

$H(A)$ is generally noncommutative (!) in a genuine sense. The classific. of multiv. groups is based on this.



Original source details: GELFAND-P000472

Uncertain reading. The manuscript abbreviates “multivalued” as «многозн.».

(So far in topology only commut. rings have occurred — there one always has Hopf algebras).

Stand. resolution or B-resolution.

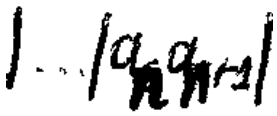
$$\overline{A} = \sum_{n>0} A_n$$

$$C_1 = A \underset{A_0}{\otimes} \overline{A} \qquad \text{Henceforth } A_0 \text{ is omitted}$$

(The init. rings are commutative!)

$$C_s = A \otimes (\overline{A})^s.$$
$$d_{s+1}\colon A \otimes (\overline{A})^{s+1} \longrightarrow A \otimes (\overline{A})^s$$

$$d_s\big(a_0[a_1|a_2|\dots|a_s]\big) = (-1)^{\varepsilon(0)} a_0 a_1 [a_2|\dots a_s] + \sum (-1)^{\varepsilon(n)} a_0[a_1|\dots|a_n a_{n+1}|\dots a_s]$$



Uncertain reading. Two indices in the product are retraced. The reading is $a_n a_{n+1}$.

Original source details: GELFAND-P000472

$\varepsilon(n)$ — certain functions of the index, ~~For even grading $\varepsilon(n) = n$.~~

$\varepsilon(n) = n$, if all elems. are even.

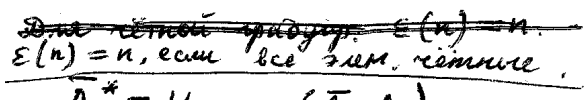


Diagram and typed description. The condition on even elements is added below the deleted line and set off by a rule.

Original source details: GELFAND-P000472

$$\mathrm{Hom}_A(C_s, A_0) \simeq (\overline{A}^*)^s, \qquad \overline{A}^* = \mathrm{Hom}_{A_0}(\overline{A}, A_0)$$
$$d_s^*\colon [\alpha_1|\alpha_2|\dots|\alpha_{s-1}] = \sum (-1)^{\varepsilon(n)} [\alpha_1|\dots|\overline{\Delta}\alpha_n|\dots|\alpha_{s-1}].$$
$$\varepsilon(n) = n, \quad \text{if } \deg \alpha_i = 2k_i$$

Def. $\overline{\Delta}$.

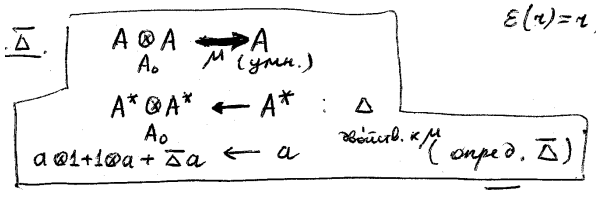
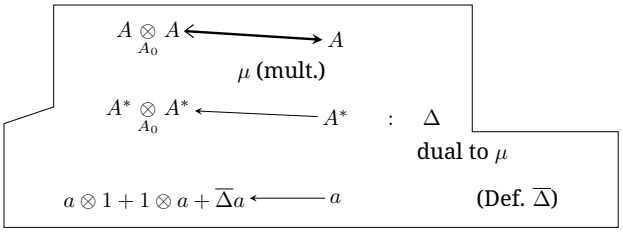


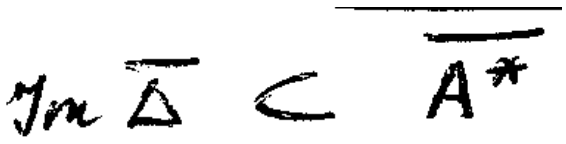
Diagram and typed description. The top arrow has a clear right head. Its retraced left end also resembles a head.



Original source details: GELFAND-P000472

In fact

$$\operatorname{Im}\overline{\Delta} \subset \overline{A^*}.$$

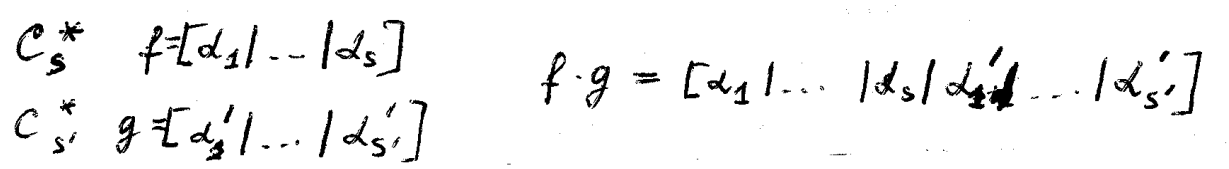


Uncertain reading. The bar on the right extends over both A and the star.

Original source details: GELFAND-P000472

This gives another definition of multiplication,

$$\begin{array}{ll} C_s^* & f = [\alpha_1|\dots|\alpha_s] \\ C_{s'}^* & g = [\alpha'_1|\dots|\alpha'_{s'}] \end{array} \qquad f \cdot g = [\alpha_1|\dots|\alpha_s|\alpha'_1|\dots|\alpha'_{s'}]$$



Original source details: GELFAND-P000472

Uncertain reading. The last index in g reads s'. The separator after the first primed term in the product is heavily retraced.

GELFAND-P000473

20/XI-72. Gelfand sem.

Kostyuchenko. One example.

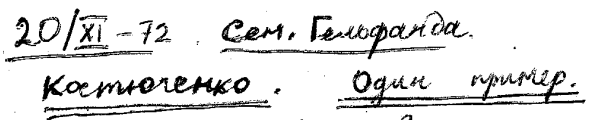


Diagram and typed description. The month is given as a Roman numeral. The name and title are double-underlined.

Original source details: GELFAND-P000473

$$\begin{array}{c} v'' + 2a\lambda v' + b\lambda^2 v = 0 \\ \underline{v(0) = v(\pi) = 0.} \end{array} \qquad \text{in } L_2[0,\pi].$$

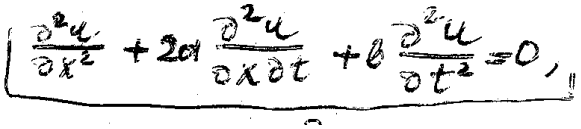


Diagram and typed description. A line under the equation turns up at both ends.

$$\underline{\left|\frac{\partial^2 u}{\partial x^2} + 2a\frac{\partial^2 u}{\partial x \partial t} + b\frac{\partial^2 u}{\partial t^2} = 0,\right.}$$

Original source details: GELFAND-P000473

$$0 < x < \pi$$
$$0 \leqslant t < +\infty$$

Uncertain reading. The second inequality sign is retraced. The provisional reading is $0 \leqslant t < +\infty$.

Original source details: GELFAND-P000473

Suppose the eq. is elliptic.

Uncertain reading. The first word reads «Пусть» (“Suppose”).

Original source details: GELFAND-P000473

$$a^2 < b; \qquad b > 0.$$

$$v_k(x) = e^{\gamma kx} \sin kx, \qquad \gamma = \frac{-a}{\sqrt{b-a^2}}; \qquad k = \pm 1,\ \pm 2,\ \ldots.$$

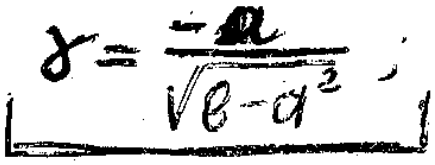


Diagram and typed description. A rule with upturned ends is drawn below the formula for γ.

Original source details: GELFAND-P000473

1. The system is complete, but it is unknown how to find the coefficients. This is the question of finding the biorthog. system.

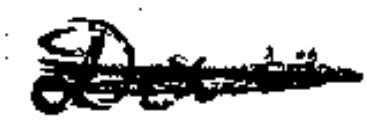
Uncertain reading. The end of “system” is retraced. A separate mark is crossed out after the period.

Original source details: GELFAND-P000473

(found in terms of the inverse fn. of the Christoffel–Schwarz integral)

(a biorthog. system exists)

2. This is not a basis! (not every vector expands into a conv. series in this system)

~~Even~~ Moreover: if there is an expansion in L^2, then the limit — an analytic fn.! — and satisfies inf. many boundary conditions (the converse holds too).

Uncertain reading. The crossed-out word may read «Даже» (“Even”).

Original source details: GELFAND-P000473

It cannot be summed by the Abel method of the first order.

if

$$\sum_{\lambda_k > 0} e_k e^{\lambda_k t} v_k(x)$$

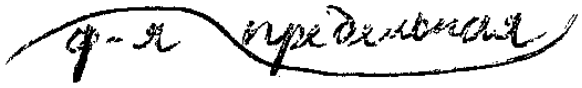
Uncertain reading. The coefficient is heavily retraced. The provisional reading is e_k.

Original source details: GELFAND-P000473

Original source details: GELFAND-P000473

Undeciphered passage. Two short entries after $v_k(x)$ are heavily struck out. Their readings are unresolved.

has a limit as $t \to 0$, then the limiting fn. is analytic.



Original source details: GELFAND-P000473

Diagram and typed description. A wavy stroke runs above “fn.” and below “limiting”.

(Discrediting the Fourier method for the elliptic eq.)

With certain nonseparated conditions one can solve it.

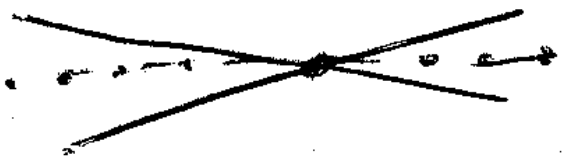
Original source details: GELFAND-P000473

Uncertain reading. The adjective before “conditions” is abbreviated as «неразд.».

Original source details: GELFAND-P000473

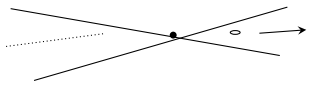
Undeciphered passage. Two marks before “Reason” are crossed out. Their readings are unresolved.

Reason for the poor behav.: the resolvent grows exp in the angles



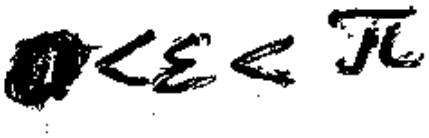
Original source details: GELFAND-P000473

Diagram and typed description. Two intersecting lines, broken strokes on the left, a dot at their crossing, and a circle and arrow on the right.



On $[0, \pi - \varepsilon]$ it is overdetermined.

On $[0, \pi + \varepsilon]$, $0 < \varepsilon < \pi$ it remains complete !!! (I. Bernstein)



Original source details: GELFAND-P000473

Uncertain reading. The lower bound is heavily retraced and is provisionally set as zero.

On $[0, 2\pi]$ many orthog. fns. arise.

A problem in the theory of entire fns.

$$\int_0^\pi f(x) e^{\gamma k x} \sin kx \, dx = 0 \quad \Longrightarrow \quad F(\lambda) = \int_0^\pi f(x) e^{\gamma \lambda x} \sin \lambda x \, dx$$

$F(\lambda)$ is an entire fn. of exp type. It is 0 at integer points $F(n) \underset{n>0}{=} 0$.

The specific features must be taken into account.

On $[0, \pi - \varepsilon]$ the type and indic. diagram work.

Uniqueness — ?

It is unknown what happens if one replaces k by λ_k.

GELFAND-P000474

I. M. Gelfand on Arnold's work.

Original source details: GELFAND-P000474

Diagram and typed description. The name and topic are double-underlined.

$$(\mathbb{R}^n, 0) \longrightarrow (\mathbb{R}, 0)$$

$f(x) = 0$ — hypersurf.

How does the hypersurf. behave in a neighb. of 0 — ?

$f \in C^\infty$; $f(0) = 0$, $df(0) = 0$. (crit. point).

(In fact consid. $\mathbb{C}^n \longrightarrow \mathbb{C}$)

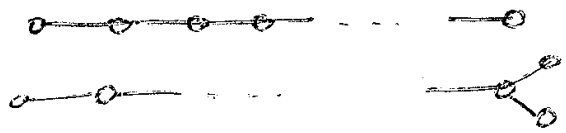
Equiv. hypersurfaces — differ by a change of variables

$$(\mathbb{R}^n, 0) \longrightarrow (\mathbb{R}^n, 0) \qquad \text{(diffeom.)}.$$

Simple hypersurf.: $f(x) = 0$

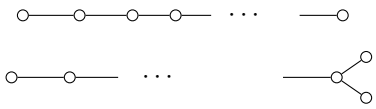
if one perturbs it, there will be finitely many orbits in jets of order N (the bound on the number of orbits does not depend on the index N).

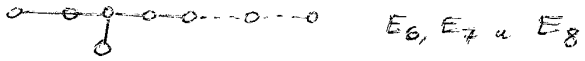
Enumeration of the simple orbits:



Original source details: GELFAND-P000474

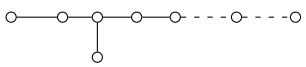
Diagram and typed description. Two chains are drawn at left. The lower chain forks at its end.





Original source details: GELFAND-P000474

Diagram and typed description. The right-hand chain has a downward branch at the third vertex. Its last links are broken.



E_6, E_7 and E_8

Next: Most orbits are simple (as a rule, the orbit is simple).

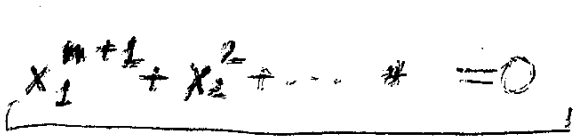
Original source details: GELFAND-P000474

Undeciphered passage. A short fragment after the opening parenthesis is struck out. Its reading is unresolved.

Nonsimple ones of codimension 6.

$$\underbrace{x_1^{n+1} + x_2^2 + \cdots * = 0}$$

— corr. to the first type.



Uncertain reading. The first letter of the exponent $n+1$ is retraced. A separate star-like mark follows the ellipsis. A rule with upturned ends is drawn below the formula.

Original source details: GELFAND-P000474

Versal deformation.

$$F(x,\lambda) = 0, \qquad F(x,0) = f(x),$$
$$\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$$

Every other def. is equiv. to a def. induced from this one as follows:

$$G(x,\mu) = F(X(x,\mu), \Psi(\mu)) \quad \text{and} \quad X(y,0) = y, \quad \Psi(0) = 0$$

Original source details: GELFAND-P000474

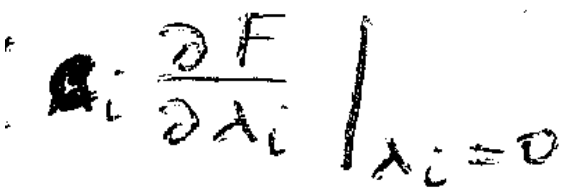
Uncertain reading. The first arguments of G and X are heavily retraced and provisionally read as x. The right side clearly has $X(y,0) = y$.

Rule for constructing a vers. family: Take an orbit and a set transverse to the orbit.

Diagram and typed description. A curved line and a short transverse segment with a point at their intersection.

Original source details: GELFAND-P000474

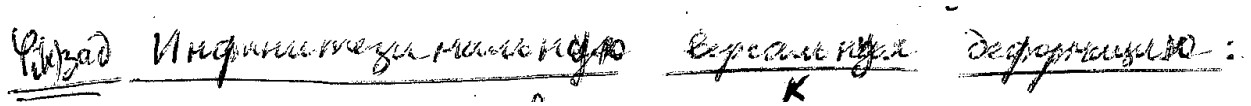
$$F(x,\lambda) = f(x) + \sum a_i \left.\frac{\partial F}{\partial \lambda_i}\right|_{\lambda_i=0} = f(x) + \sum c_i \varphi_i(x) + O(\lambda^2).$$



Uncertain reading. The multiplier in the first sum is provisionally read as a_i. The evaluation subscript is $\lambda_i = 0$.

Original source details: GELFAND-P000474

φ_i give an infinitesimal versal deformation: if $\forall$ fn. $a(x)$ is repr. in the form



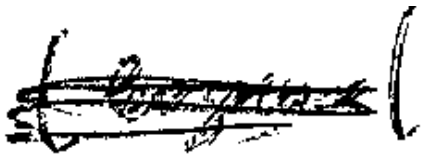
Original source details: GELFAND-P000474

Uncertain reading. The first fragment is heavily retraced and double-underlined. The provisional reading is φ_i «зад» (“give”).

$$a(x) = \sum \frac{\partial f}{\partial x_i} \cdot h_i(x) + \sum_{i=1}^{k} c_i \varphi_i(x)$$

$a(x)$ defines an infinitesim. vers. deform.

(φ_i gen. the local ring of smooth maps)



Undeciphered passage. A separate crossed-out entry precedes the parenthesis about Taylor expansion. Its reading is unresolved.

Original source details: GELFAND-P000474

(from the Taylor expans. of that fn.)

Important: the quotient ring by the Jacobian ideal $\left(\left\{\frac{\partial f}{\partial x_i}\right\}\right)$.

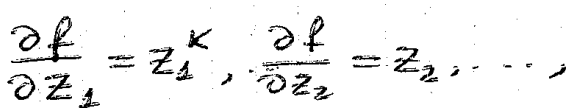
The quotient — a finite-dimensional space of dimension k.

GELFAND-P000475

$$f = z_1^{k+1} + z_2^2 + \cdots + z_n^2.$$

Then

$$\frac{\partial f}{\partial z_1} = z_1^k, \qquad \frac{\partial f}{\partial z_2} = z_2, \ \ldots,$$

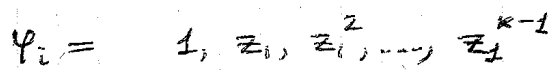


Uncertain reading. The right-hand sides are written as monomials without numerical factors.

Original source details: GELFAND-P000475

hence the quotient ring is a space of dimension k.

$$\varphi_i = 1, z_i, z_i^2, \ldots, z_1^{k-1}$$

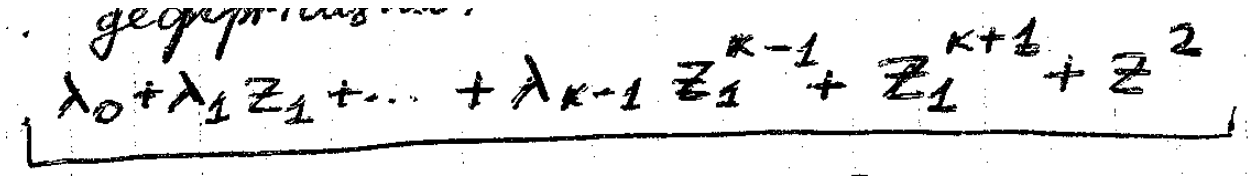


Uncertain reading. The first two powers after 1 have subscripts resembling i. The last has 1.

Original source details: GELFAND-P000475

Vers. deformation:

$$\underbrace{\lambda_0 + \lambda_1 z_1 + \cdots + \lambda_{k-1} z_1^{k-1} + z_1^{k+1} + z^2}$$ — vers. deformation



Original source details: GELFAND-P000475

Diagram and typed description. A rule with upturned ends is drawn below the polynomial. The final term is written z^2.

$F(z, \lambda) = 0$. The bifurc. diagram — the set of those λ, for which sing. points remain (in the vers. deformation).

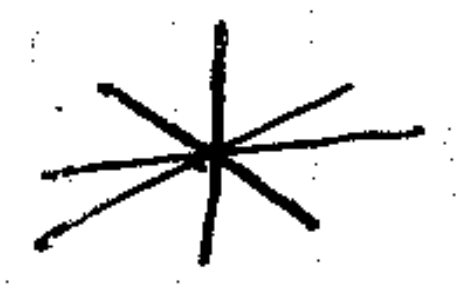
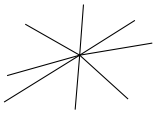


Diagram and typed description. Four segments meet at a common point to the left of the theorem.



Original source details: GELFAND-P000475

Thm.: on an open set (bifurc. diag.) the sing. points will already be simple (Morse).

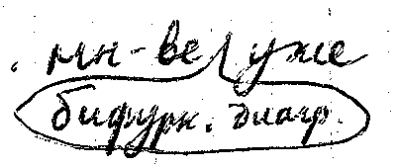


Diagram and typed description. "Bifurc. diag." is written in a bubble below "set".

Original source details: GELFAND-P000475

$z_1^{k+1} + z_2^2 = 0$ ‖ the bifurc. diag. of this point — $\longrightarrow$ Swallowtail.

$$z_1^{k+1} + z_2^2 + \lambda_{k-1} z_1^{k-1} + \cdots + \lambda_0$$

$$z_2^2 = -p_{k+1}(z_1); \qquad z_2 = \sqrt{-p_{k+1}(z_1)}.$$

sing. points $(z_2, P'(z_1)) = 0$, i.e.

$$\underbrace{P(z_1) = 0 \text{ and } P'(z_1) = 0}$$

(P has multiple roots)

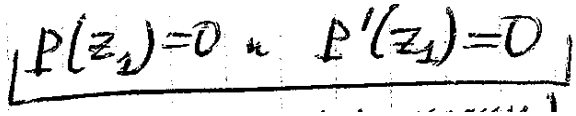


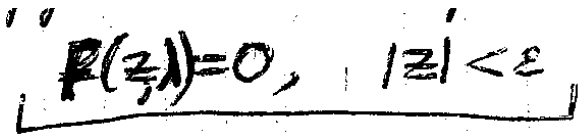
Diagram and typed description. A single rule with upturned ends surrounds the two equalities below.

Original source details: GELFAND-P000475

Consider the complement of the bifurc. diagram and its fund. group. It is the braid group of the Coxeter group.

Consider the eq.

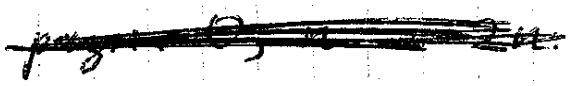
$$\underbrace{P(z,\lambda)=0, \quad |z|<\varepsilon}_{} \qquad \underline{\lambda \text{ such that there are no singularities}}$$



Original source details: GELFAND-P000475

Uncertain reading. The first letter is heavily retraced. The provisional reading is P.

the homology of this space is $\neq 0$ only in ~~dim.~~



Original source details: GELFAND-P000475

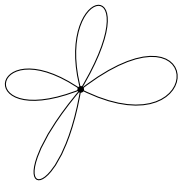
Undeciphered passage. Traces remain after the struck-out “dim.”. A full reading is unresolved.

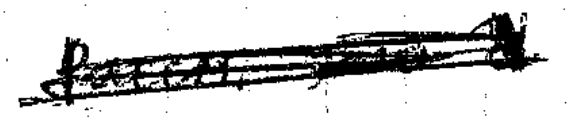
half-dimension $n-1$.

It is a bouquet of μ spheres.

Original source details: GELFAND-P000475

Diagram and typed description. Four loops with a common point depict the bouquet.



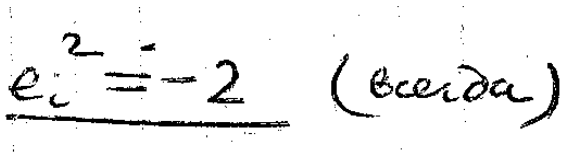


Original source details: GELFAND-P000475

Undeciphered passage. The entry to the left of H^{n-1} is heavily struck out. Its reading is unresolved.

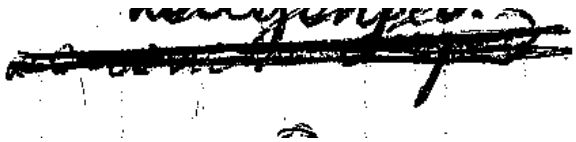
H^{n-1} has generators $e_1\ e_2 \dots\ e_\mu$.
The intersection index defines a scalar prod. (e_i, e_k).

$$n-1 \equiv 2 \pmod 4 \quad \Big\| \quad \begin{matrix} \text{neg. def.} \\ \text{semidef.} \\ \text{indef.} \end{matrix} \qquad \underline{e_i^2 = -2} \quad \text{(always)}$$



Uncertain reading. A separate short horizontal stroke appears above the equals sign. Its meaning is unresolved.

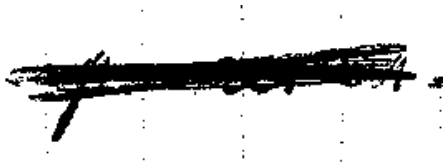
Original source details: GELFAND-P000475



Undeciphered passage. One line between “semidef.” and “indef.” is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000475

monodromy: rep. of the fund. gr. in



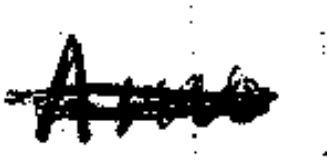
Undeciphered passage. An entry after “in” is crossed out and followed by an ellipsis. Its reading is unresolved.

Original source details: GELFAND-P000475

...

———— • ————

~~Atiyah~~



Uncertain reading. The struck-out name may read «Атья» (“Atiyah”).

Original source details: GELFAND-P000475

There are works by Atiyah, Bott,...: how to get rid of derivatives in Sing.’s calculations. How can one study the topology of the world while sitting at one point? Finding explicit asymptotics for eqs. of the 2nd order (considerations of I. M. Gelfand, V. I. Arnold and others)

(will be in the next semester)

A. Shnirelman on Maslov’s work.

GELFAND-P000476

V. M. Buchstaber. (continuation).

$$A \overset{\text{addit.}}{=} \Lambda[[X]], \qquad X_n \text{ — addit. generators}$$
$$X_n \cdot X_m = C^n_{n+m} X_{n+m}$$

Uncertain reading. Two short parallel strokes under "addit." read as an equals sign.

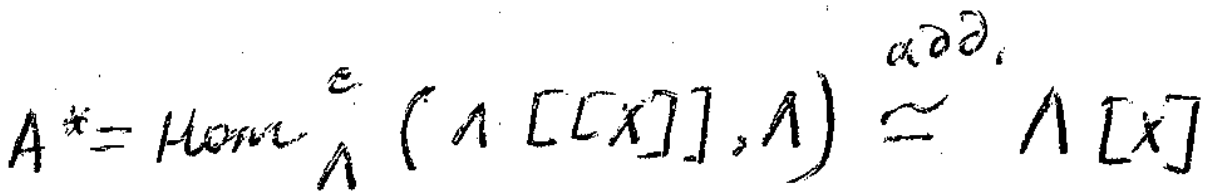
Original source details: GELFAND-P000476

$$\begin{array}{c} \operatorname{Ext}^{2,*}_A(\Lambda,\Lambda) \\ \| \\ \operatorname{Ker} d_3^* / \operatorname{Im} d_2^* \end{array}$$ in terms of the B-construction

1°.

$$d_3^* \colon \overline{A}^* \otimes \overline{A}^* \longrightarrow \overline{A}^* \otimes \overline{A}^* \otimes \overline{A}^*$$

$$A^* = \operatorname{Hom}^c_\Lambda(\Lambda[[X]], \Lambda) \overset{\text{add.}}{\simeq} \Lambda[x]$$



Original source details: GELFAND-P000476

Uncertain reading. The upper index of Hom reads c and the lower one Λ.

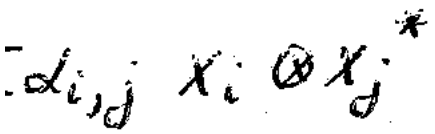
|

compact. (nonzero val. on a finite number).

Diagram and typed description. "Compact" is written below Hom and connected to it by a stroke.

Original source details: GELFAND-P000476

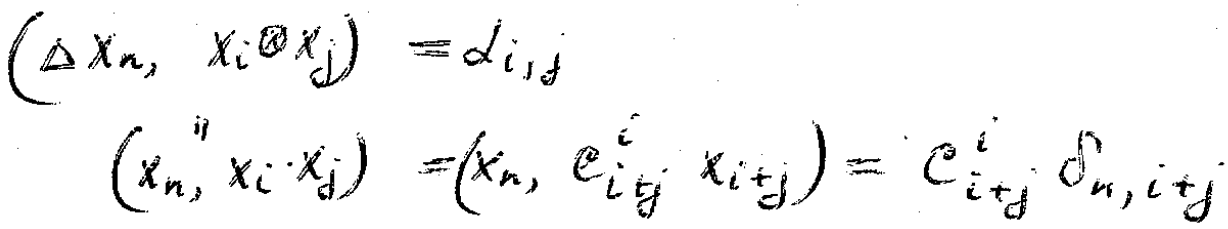
$$\Delta \colon A^* \longrightarrow A^* \otimes A^*$$
$$\Delta x_n \longrightarrow x_n \otimes 1 + 1 \otimes x_n + \sum d_{i,j} x_i \otimes x_j^*$$



Uncertain reading. A star is visible after x_j. It is absent from the later formula for Δx_n.

Original source details: GELFAND-P000476

$$(\Delta x_n, x_i \otimes x_j) = d_{i,j}$$
$$\|$$
$$(x_n, x_i \cdot x_j) = (x_n, C^i_{i+j} x_{i+j}) = C^i_{i+j} \delta_{n,i+j}$$



Original source details: GELFAND-P000476

Diagram and typed description. A vertical equals sign joins the two lines of the pairing. The case of the x-shaped glyphs is uncertain. Lowercase is used provisionally.

$$\Delta x_n = x_n \otimes 1 + 1 \otimes x_n + \sum C^i_n x_i \otimes x_{n-i}$$

And put $x_n = x^n$

$$\Delta x^n = (x \otimes 1 + 1 \otimes x)^n$$
$$d_3^*\colon \overline{\Lambda[x,y]} \longrightarrow \overline{\Lambda[x,y,z]}$$

$$d_3 \left(\begin{array}{c} h(x,y) \\ \| \\ \sum h_{i,j} x^i y^j \end{array} \right) = \sum h_{ij} (\overline{\Delta} x^i) z^j - \sum h_{i,j} x^i \overline{\Delta} y^j =$$

Uncertain reading. The calculation has d_3 without a star before $h(x,y)$. The preceding line has d_3^*.

Original source details: GELFAND-P000476

$$\overline{\Delta} x^n = \begin{array}{c} \Delta x^n \\ \| \\ (\Delta(x))^n \\ \| \\ (x+y)^n \end{array} - x^n - y^n.$$

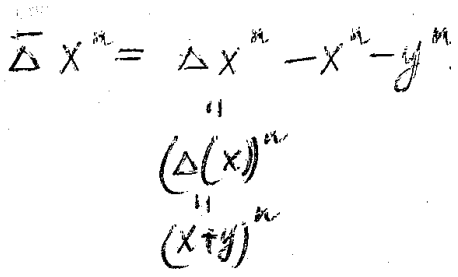
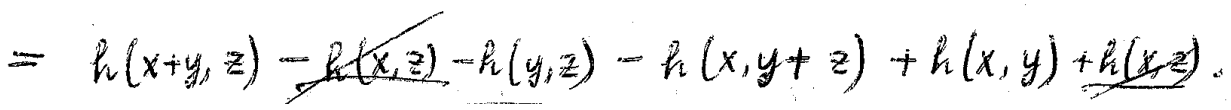


Diagram and typed description. Below Δx^n, vertical equals signs connect $(\Delta(x))^n$ and $(x+y)^n$.

Original source details: GELFAND-P000476

$$= h(x+y,z) - \cancel{h(x,z)} - h(y,z) - h(x,y+z) + h(x,y) + \cancel{h(x,z)}.$$



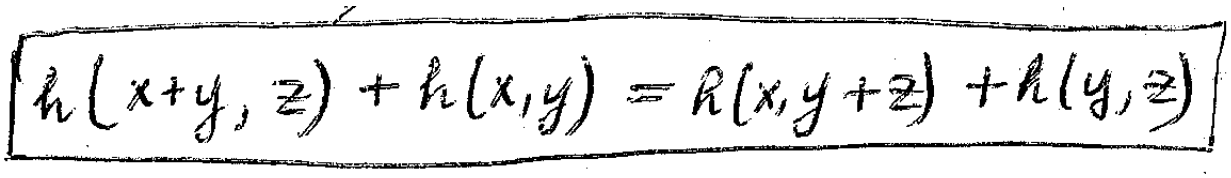
Original source details: GELFAND-P000476

Diagram and typed description. Both instances of $h(x,z)$ are crossed out. The minus and plus signs in front remain visible.

Ker d_2^*:

$$\boxed{h(x+y,z) + h(x,y) = h(x,y+z) + h(y,z)}$$

Denote these cycles by L.



Original source details: GELFAND-P000476

Diagram and typed description. The equality is enclosed in a rectangular box.

This is precisely the Lazard relation.

Original source details: GELFAND-P000476

Uncertain reading. The adjective before “relation” reads «лазаровское» (“Lazard”).

$$\underline{\operatorname{Im} d_2^*} = \Lambda\{d_2^* x^n\}$$

$$0 \longrightarrow \operatorname{Im} d_2^* \longrightarrow L \longrightarrow \operatorname{Ext}_A^{2,n}(\Lambda,\Lambda) \longrightarrow 0$$

Calculation of the group Ext.

The eq. does not involve Λ. Hence it suffices to calc. 2 cases:

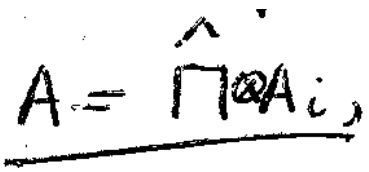
1) Λ — Z_p-module (p — prime)

2) $\Lambda = Q$

GELFAND-P000477

there remains X_{p^i}

$$\underline{A = \widehat{\prod} \otimes A_i}, \qquad A_i = Z_p[X_{p^i}]/(X_{p^i}^p) \underset{Z_p}{\otimes} \Lambda$$

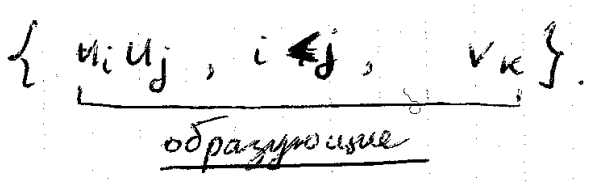


Original source details: GELFAND-P000477

Uncertain reading. The product sign has a hat. A tensor sign is written immediately to its right.

$$H^*(A_i) = \Lambda_p(u_i) \otimes Z_p[v_i], \qquad \deg u_i = p^i, \quad \deg v_i = p^{i+1}$$
$$u_i \in \mathrm{Ext}^{1,p^i}; \qquad v_i \in \mathrm{Ext}^{2,p^{i+1}}$$

$$\mathrm{Ext}_A^{2,*} = \left\{ \underbrace{u_i u_j, \quad i < j, \quad v_k}_{\text{generators}} \right\}.$$



Uncertain reading. The sign between i and j is heavily retraced. It is provisionally set as $i < j$.

Original source details: GELFAND-P000477

~~This~~ But: symmetric ones are needed. $h(x,y) = h(y,x)$

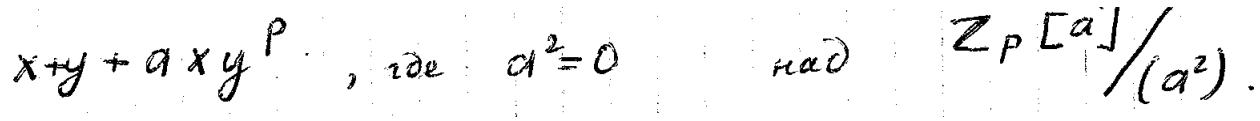
Uncertain reading. The struck-out word before "But" may read «Это» ("This").

Original source details: GELFAND-P000477

There are noncommut. one-dimensional groups!! (if there is torsion)

(corr. to the cycles $u_i u_j$)

Example: $x + y + axy^p$, where $a^2 = 0$ over $Z_p[a]/(a^2)$.



Original source details: GELFAND-P000477

Diagram and typed description. The term axy^p and the condition $a^2 = 0$ occur on the same line.

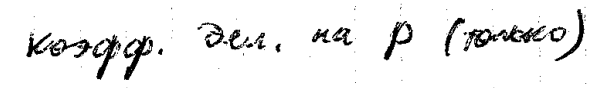
This is a noncommut. form. group!!

There are as many of them as there are cycles.

v_k corr. to commut. groups

$$h(x,y) = (x+y)^{p^{k+1}} - x^{p^{k+1}} - y^{p^{k+1}}$$

coeff. divisible by p (only)



Uncertain reading. The word in parentheses reads «только» ("only").

Original source details: GELFAND-P000477

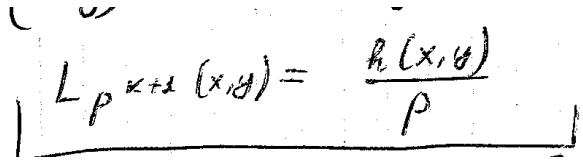


Diagram and typed description. A rule with upturned ends is drawn below the formula.

$$L_{p^{k+1}}(x,y) = \frac{h(x,y)}{p}$$

Original source details: GELFAND-P000477

(in $\forall$ dimension one generator each)

That is all for this semester.

(в $\forall$ размерности по 1-й образующей)

В этом семестре конец.

Original source details: GELFAND-P000477

Diagram and typed description. The last line has both a wavy underline and a broken straight underline.

GELFAND-P000478

12/II-73. Gelfand sem.

Plan: 1. Buchstaber, Mishchenko: Topology (Sullivan's work)

2. Works by Deligne and Langlands

V. Buchstaber. On Sullivan's 1st paper.

M^n — manif.

A smoothing of M^n — a sm. manif. N^n and a homot. equiv. $N^n \xrightarrow{f} M^n$

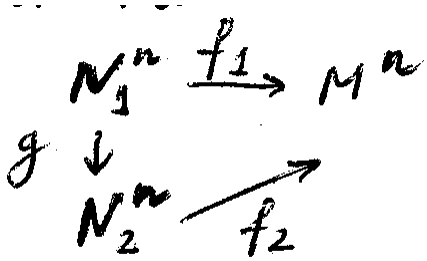


Diagram and typed description. Three arrows connect the two smooth manifolds and their common target.

$$\begin{array}{ccc} N_1^n & \xrightarrow{f_1} & M^n \\ {\scriptstyle g}\downarrow & \nearrow_{f_2} & \\ N_2^n & & \end{array}$$

Original source details: GELFAND-P000478

Equiv. $\Longleftrightarrow \exists$ a diffeom. g, such that this square commutes (up to homotopy).

$S(M^n)$ — the set of classes of equiv. smoothings.

Undeciphered passage. A short entry above "Milnor" is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000478

Milnor: Card $S'(S^7) = 28$.

Card $S'(S^7) = 28$.

Uncertain reading. A short stroke is visible between the first S and the parenthesis. It is set as S'.

Original source details: GELFAND-P000478

Description of $S(M^n)$ — ?

Sullivan considers in M^n a fixed piecewise-linear structure.

and solved the prob. in this formulation: (M^n — a polyhedron)

(in place of a diffeom. — a piecewise-lin.

Original source details: GELFAND-P000478

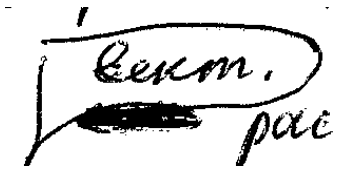
Undeciphered passage. A heavily crossed-out entry follows “piecewise-lin.”. Its reading is unresolved.

homeomorphism)

Obtain $PL(M^n)$.

Here always $PL(S^n) = \{S^n\}$ — where the Poincaré prob. has been solved (for $n \geqslant 5$).

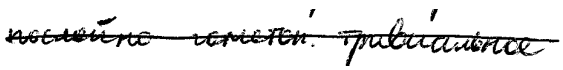
Consider on M^n (Vect.) a bundle ξ, stable (of large dim.).



Original source details: GELFAND-P000478

Undeciphered passage. The old entry before “bundle” is inked over. “Vect.” is inserted above it.

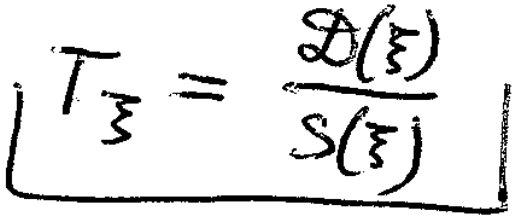
~~trivial in several degrees~~



Original source details: GELFAND-P000478

Uncertain reading. The struck-out phrase is provisionally read as «несколько степен. тривиальное».

T_ξ — the Thom complex of the bundle ξ,



Original source details: GELFAND-P000478

Diagram and typed description. The formula has a bottom rule with upturned ends.

$$T_\xi = \frac{D(\xi)}{S(\xi)}$$

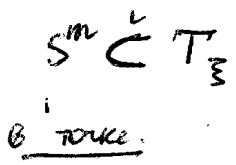
D — bundle of balls,

S — bundle of spheres.

dim $\xi = m$.

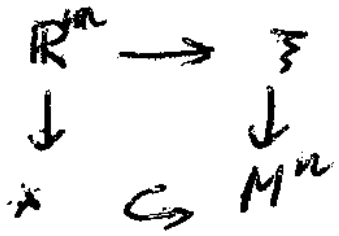
$$S^m \overset{i}{\subset} T_\xi$$

at a point.



Original source details: GELFAND-P000478

Diagram and typed description. A stroke connects “at a point” to the left-hand sphere.

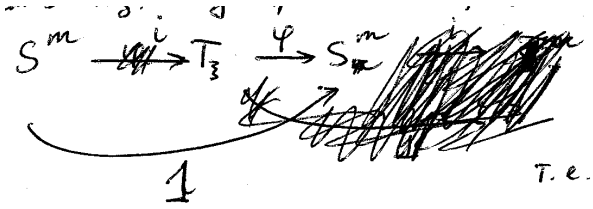


Original source details: GELFAND-P000478

Diagram and typed description. In the diagram at right, the point maps by inclusion into the base.

$$\begin{array}{ccc} \mathbb{R}^m & \longrightarrow & \xi \\ \downarrow & & \downarrow \\ * & \hookrightarrow & M^n \end{array}$$

The bundle ξ is called J-trivial, if $\exists$ a map φ; such that



Original source details: GELFAND-P000478

Diagram and typed description. The first arrow is labelled i; part of its shaft is crossed out. A separate cancellation is visible below T_ξ near the arc. The mark at the right-hand sphere and the large crossed-out entry to its right remain unresolved. The arc from the left sphere to the right sphere is labelled 1.

$$\begin{array}{ccccc} S^m & \xrightarrow{\ i\ } & T_\xi & \xrightarrow{\ \varphi\ } & S^m \\ & \searrow & & \swarrow & \\ & & 1 & & \end{array}$$

(homot.)

i.e. $\deg(\varphi \cdot i) = 1$

Original source details: GELFAND-P000478

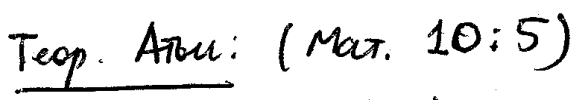
Uncertain reading. A dot is visible between φ and i.

(the choice of $*$ does not matter since M^n is assumed connected).

$K_J(M^n)$ — the group of stab. bundles over M^n, each of which is J-trivial

GELFAND-P000479

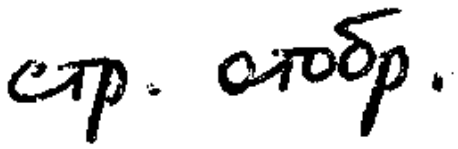
Atiyah's thm.: (Mat. $10:5$)



Original source details: GELFAND-P000479

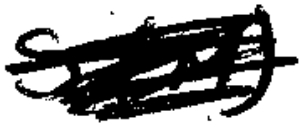
Diagram and typed description. A short reference follows the name, «Мат. 10:5».

стр. отобр. ~~$S\ldots)$~~



Original source details: GELFAND-P000479

Uncertain reading. «стр. отобр.» may abbreviate “a map is constructed”.



Original source details: GELFAND-P000479

Undeciphered passage. An S and a closing parenthesis are visible under the crossing-out. $S(M^n)$ is a possible reading.

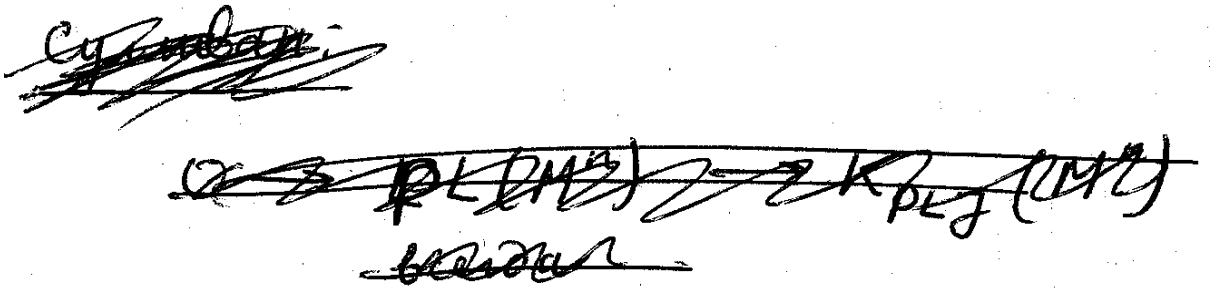
$$\underline{S(M^n) \longrightarrow K_J(M^n).}$$

$$N^n \longrightarrow M^n \qquad \underline{\text{difference}}.$$

~~Sullivan:~~

$$\sout{0 \longrightarrow PL(M^n) \longrightarrow K_{PLJ}(M^n)}$$

~~always~~



Original source details: GELFAND-P000479

Diagram and typed description. The name, sequence and word “always” are crossed out and remain legible underneath.

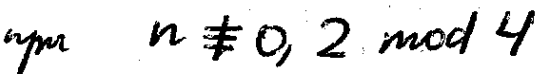
and

$$0 \longrightarrow PL(M^n) \longrightarrow K_{PLJ}(M^n)$$

Original source details: GELFAND-P000479

Undeciphered passage. An entry after the second $K_{PLJ}(M^n)$ is crossed out. It may read $\to 0$.

for $n \not\equiv 0, 2 \bmod 4$



Original source details: GELFAND-P000479

Uncertain reading. The comparison sign has three horizontal strokes crossed by one stroke.

$n \equiv 0 \bmod 4$ cokernel Z
$n \equiv 2$ —//— —//— Z_2.

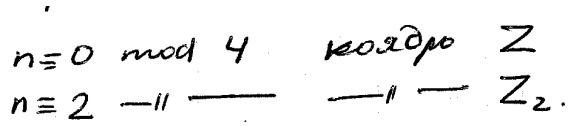


Diagram and typed description. The second row has ditto marks under “mod 4” and “cokernel”.

Original source details: GELFAND-P000479

Sullivan calc. the homot. groups of the repr. space $K_{PLJ}(M^n)$.

Uncertain reading. The loop on the descending stroke is retraced. The final strokes merge. The manuscript may read «груп» or «групп».

Original source details: GELFAND-P000479

For $K_J(M^n)$ it has not been done. A seq. for $S(M^n)$ exists too.

GELFAND-P000480

19/II-73. R. L. Dobrushin.

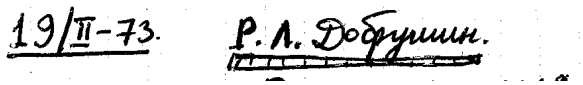


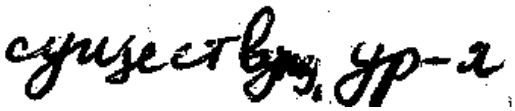
Diagram and typed description. The month is written as a Roman numeral. Two lines with cross-strokes are drawn below the name.

Original source details: GELFAND-P000480

On field theory.

Gelfand: 2 problems: — · —

1) quantize existing eqs.



Uncertain reading. The end of the abbreviation is provisionally read as «существующ.» (“existing”).

Original source details: GELFAND-P000480

2) write eqs. that actually «ответ.» (phys.)

Uncertain reading. The abbreviation «ответ.» after “actually” may mean “corresponding to” or “responsible for”.

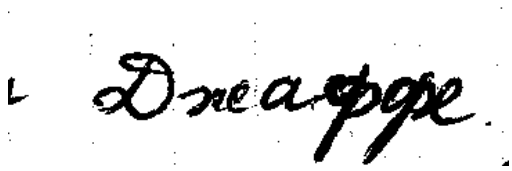
Original source details: GELFAND-P000480

— · —

A method of constr. objects satisfying the Wightman axioms (probab. objects).

Principal author — Nelson.

One can (almost) obtain the res. of Glimm and Jaffe.



Uncertain reading. The manuscript writes Jaffe as «Джеаффе», with «е» before «а».

Original source details: GELFAND-P000480

Random variable: ξ_t, $t \in \mathbb{R}^{v+1}$

Uncertain reading. The exponent begins with a v-shaped sign. Greek ν is also possible.

Original source details: GELFAND-P000480

1) it is Eucl. invari.

2) Mark. prop.

GELFAND-P000481

26/II-73 V. M. Buchstaber. $hs(M^n)$ — (homotop.) smoothings.

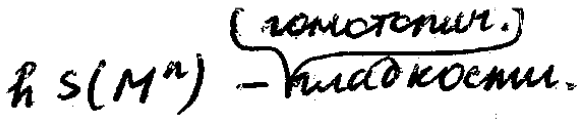


Diagram and typed description. "(Homotop.)" is inserted above "smoothings", with a pointer to the word.

Original source details: GELFAND-P000481

$ht(M^n)$ — homot. triangulations.
is described.

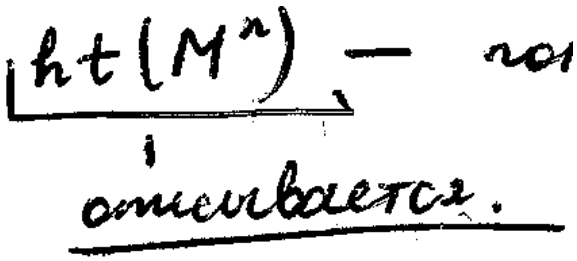


Diagram and typed description. "Is described." is connected by a stroke to the bracket below $ht(M^n)$.

Original source details: GELFAND-P000481

Hauptvermutung — a homeomorphism (of manif.) can be replaced by a piecewise-lin. homeom. (if M is simply conn. and there is in H^3 no 2-torsion)

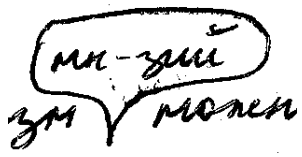
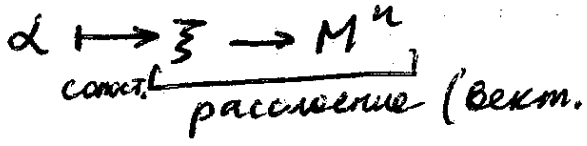


Diagram and typed description. "Of manifolds" is inserted in a callout after "a homeomorphism".

Original source details: GELFAND-P000481

Let $\alpha \in hs(M^n)$ (hom. smoothing).

$$\alpha \underset{\text{сопст.}}{\mapsto} \underbrace{\xi \longrightarrow M^n}_{\text{bundle (vect.)}} \qquad \text{smooth.}$$



Uncertain reading. The abbreviation under the arrow appears to read «сопст.». «Сопост.» ("associate") remains possible.

Original source details: GELFAND-P000481

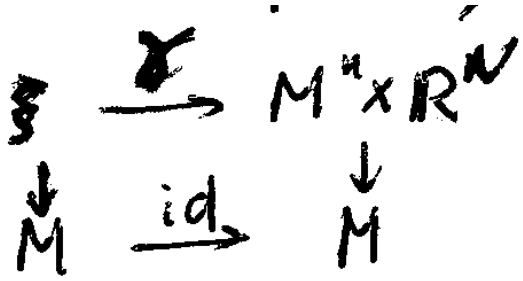
$$f\colon N^n \longrightarrow M^n;$$
$$\exists\, g\colon M^n \longrightarrow N^n; \qquad f\circ g \sim 1, \quad g\circ f \sim 1.$$
$$\tau_{M^n} + g^* v_{N^n} \qquad (v_{N^n} = -\tau_{N^n} \text{ mod triv.})$$

Original source details: GELFAND-P000481

Uncertain reading. The normal-bundle sign is reproduced as a v-shaped letter. Greek ν is possible.

Ex. Prove that 2 equiv. smoothings induce isom. bundles ξ.

The resulting smooth bundle ξ is J-trivial, i.e.



Original source details: GELFAND-P000481

Diagram and typed description. The square has an arrow labelled γ with a heavily retraced sign.

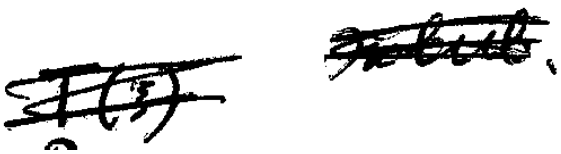
$$\begin{array}{ccc} \xi & \xrightarrow{\gamma} & M^n \times R^N \\ \downarrow & & \downarrow \\ M & \xrightarrow{\mathrm{id}} & M \end{array}$$

and γ is a proper map and a homot. equiv.

Original source details: GELFAND-P000481

Uncertain reading. The sign before “is” is heavily retraced. It is provisionally set as γ.

~~$T(\xi)$ equiv.,~~

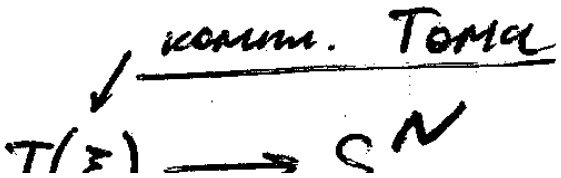


Original source details: GELFAND-P000481

Uncertain reading. After the crossed-out $T(\xi)$, «эквив.» (“equiv.”) is a provisional reading. «трив.» (“triv.”) remains possible.

Thom comp.

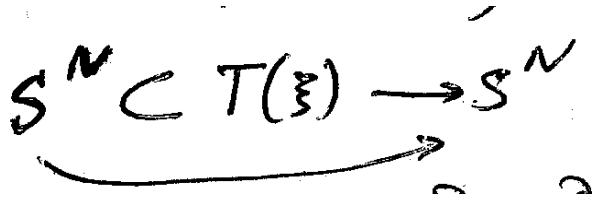
2nd def. — $\exists$ a map $T(\xi) \longrightarrow S^N$, such that



Original source details: GELFAND-P000481

Diagram and typed description. The underlined «комп. Тома» (“Thom comp.”) points to $T(\xi)$. The abbreviation may stand for “complex”.

the composite



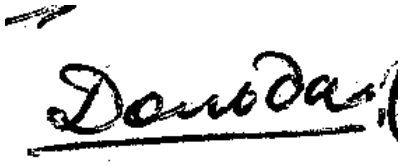
Original source details: GELFAND-P000481

Diagram and typed description. The lower arrow denotes the composite.

$$S^N \subset T(\xi) \longrightarrow S^N$$

is a homot. equiv.

Equivalence — thm. of Dold (technical, trivial in idea)



Original source details: GELFAND-P000481

Diagram and typed description. Dold's name is underlined.

GELFAND-P000482

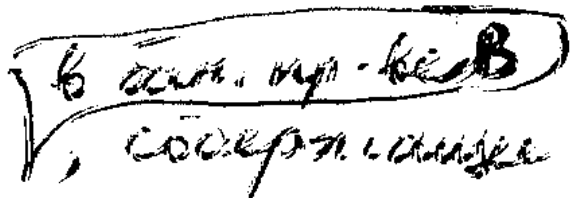
5/III-73 Gelfand's Sem.

Theorem. ~~F. A. Berezin~~ Lomonosov

Original source details: GELFAND-P000482

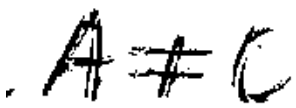
Uncertain reading. The heavily crossed-out name is provisionally read as "F. A. Berezin".

R — a family of pairwise commut. (in Banach space B) op., containing a completely cont. op. $A \neq C$.



Original source details: GELFAND-P000482

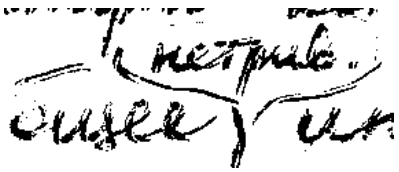
Diagram and typed description. "In Banach space B" is written in a callout after "op.".



Original source details: GELFAND-P000482

Uncertain reading. The open-right sign after $\neq$ is reproduced as C. The context suggests 0; the sign below in $Ax_0' \neq 0$ is closed.

Then R has a common (nontriv.) invariant subsp.



Original source details: GELFAND-P000482

Diagram and typed description. "Nontriv." is inserted between "common" and "invariant".

(generalization of the Aronszajn–Smith theorem)

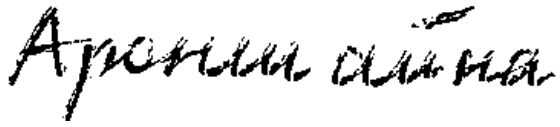


Diagram and typed description. The first surname reads «Ароншайна» in Russian.

Original source details: GELFAND-P000482

The proof works in the case of an LCS, short and simple.

Uncertain reading. A short sign before “short” is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000482

Proof. S' — a closed ball, $r = 1$ centered at x_0, such that
(in B)

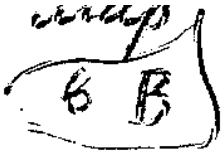


Diagram and typed description. “In B” is inserted below “ball” in a callout.

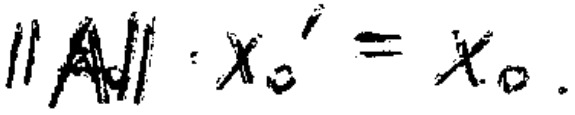
Original source details: GELFAND-P000482

Diagram and typed description. The first occurrence of the ball symbol S' is primed. The manuscript subsequently writes S.

Original source details: GELFAND-P000482

$$\inf_{x \in S} \|Ax\| > 1/2$$

$$Ax_0' \neq 0 \qquad \|x_0'\| = 1 \qquad \text{and we take} \qquad \|A\| \cdot x_0' = x_0.$$



Uncertain reading. The letter between the norm bars is heavily retraced. A provisional reading is $\|A\|$.

Original source details: GELFAND-P000482

$\overline{AS}$ — a compact set that does not contain zero.

Consid. the $\overset{\mathcal{R}}{\text{algebra}}$ with 1, generated by the family R (without closure).

Diagram and typed description. The symbol inserted above “algebra” is provisionally set as $\mathcal{R}$.

Original source details: GELFAND-P000482

$$\forall y \in B, \qquad \mathcal{R}y \quad \text{— lin. set.}$$

It is easy to see that $\mathcal{R}y$ is invar. under $\mathcal{R}$.

$\overline{\mathcal{R}y}$ — an inv. subsp.

We must prove that it can be nontriv.

Let $y \in \overline{AS}$. We must check that $\exists y$; $\mathcal{R}y$ is not dense,

Suppose otherwise, then

Uncertain reading. The crossed-out word after "then" is undeciphered. «Оно» ("it") is a possible reading.

Original source details: GELFAND-P000482

$$\forall y \in \overline{AS} \quad \exists T \in \mathcal{R}, \qquad \|Ty - x_0\| < 1.$$

From the compactness of $\overline{AS}$ it follows that $\exists T_1, \ldots, T_n$

$$\forall y \; \exists i \qquad \|T_i y - x_0\| < 1,$$

$$\forall y \in AS \quad \{T_1 y, \ldots, T_n y\} \cap S \neq \varnothing.$$

Let $f(t) \in C(0, +\infty)$

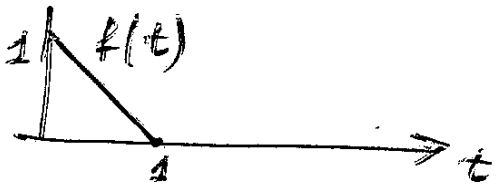
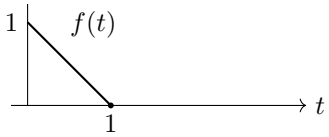


Diagram and typed description. The graph of $f(t)$, the marks 1 on both axes and the point on the t-axis.

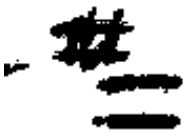


Original source details: GELFAND-P000482

GELFAND-P000483

Consid.

$$\Phi y \overset{\#}{=} \frac{\sum_1^n f(\|T_i y - x_0\|) T_i y}{\sum_j f(\|T_j y - x_0\|)} \qquad \text{(convex comb.)}$$



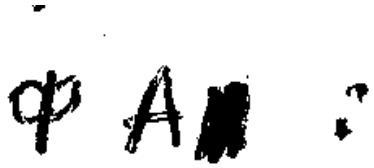
Uncertain reading. A sign resembling # is written above and to the left of the equals sign.

Original source details: GELFAND-P000483

$$\Phi\colon \overline{AS} \longrightarrow S$$

This is a cont. function on $\overline{AS}$.

Consid. $\Phi A\blacksquare\colon S \longrightarrow S$,



Uncertain reading. The solid mark stands for the heavily overwritten sign after ΦA. A possible reading is x.

Original source details: GELFAND-P000483

where the image of ΦA is compact and it is cont.

By Schauder's thm. $\exists$ a fixed point, i.e.,

$$\sum_1^n \alpha_i T_i A x = x, \qquad \alpha_j = \frac{f(\|T_j y - x_0\|}{\sum_i f(\|T_i y - x_0\|)}$$

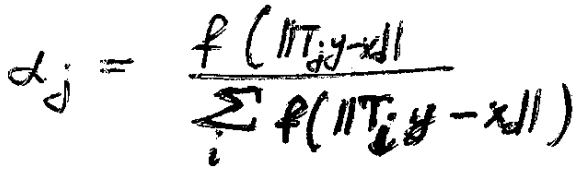
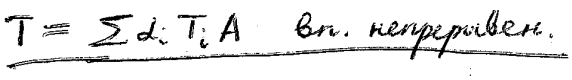


Diagram and typed description. Both arguments show x_0. The closing parenthesis after the norm is absent in the numerator.

Original source details: GELFAND-P000483

$T \equiv \sum \alpha_i T_i A$ is completely continuous.



Uncertain reading. Three horizontal strokes precede the sum.

Original source details: GELFAND-P000483

$\exists x\colon Tx = x$. Hence an eigenvect. with eigenval. 1 — a nontriv. invar. subsp. (even finite-dimensional). Hence $\mathcal{R}$ has an invar. subsp.

Diagram and typed description. End mark with an oval above two crossing strokes and a dot to the right.

Original source details: GELFAND-P000483

(This does not mean there is a finite-dimensional one in general).

Corollary. If a comm. (Banach) algebra has an elem.,

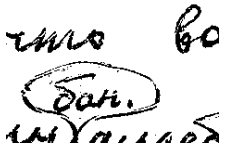
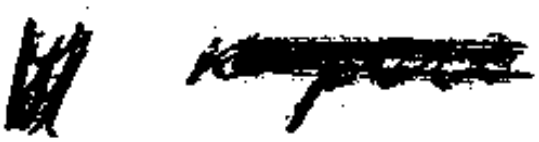


Diagram and typed description. "Banach" is inserted after "comm.".

Original source details: GELFAND-P000483

~~и ...~~



Uncertain reading. A sign resembling «и» (“and”) and an undeciphered word are crossed out after “elem.”.

Original source details: GELFAND-P000483

(by which)
mult. is completely cont., then ∃ a nontriv. ideal.

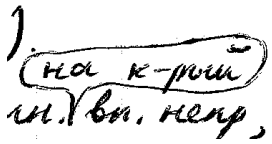


Diagram and typed description. “By which” is inserted after “mult.”.

Original source details: GELFAND-P000483

Henkin. (announcement)

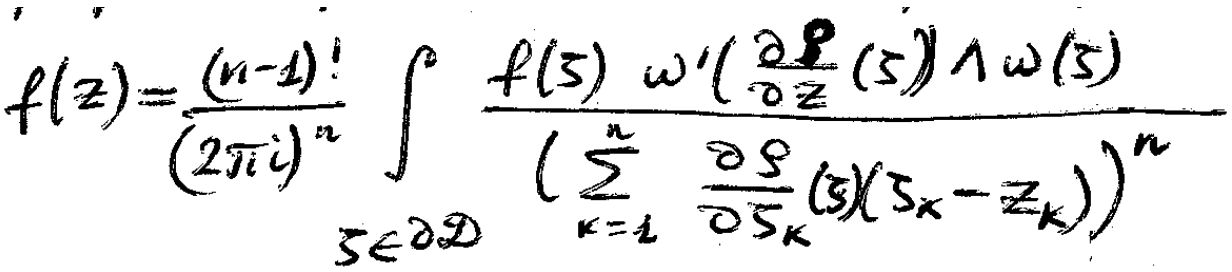
Let there be in $\mathbb{C}^n$ a convex dom. $D \subset \mathbb{C}^n$, given by the eq. $\varrho(z) < 0$.

Uncertain reading. The letter D before the inclusion sign is heavily retraced.

Original source details: GELFAND-P000483

Leray integral formula for bounded holom. functions:

$$f(z) = \frac{(n-1)!}{(2\pi i)^n} \int_{\zeta \in \partial D} \frac{f(\zeta)\, \omega'\left(\frac{\partial \varrho}{\partial z}(\zeta)\right) \wedge \omega(\zeta)}{\left(\sum_{k=1}^{n} \frac{\partial \varrho}{\partial \zeta_k}(\zeta)(\zeta_k - z_k)\right)^n}$$



Original source details: GELFAND-P000483

Uncertain reading. The sign ϱ in the numerator of the derivative in ω' is heavily retraced.

1) Leray-type integral:

$$f_g(z) = \cdot\ \cdot \int^{g(\zeta)\cdots}_{\cdots} \qquad g(\zeta) \text{ — a function on the boundary.}$$

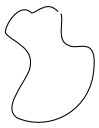
$$f \longrightarrow [f(\zeta)g(\zeta) - f_g(\zeta)] \quad \text{— a completely cont. operator}$$

and a Hölder condition with exp. $1/2$

2) $\varrho(z)$ — strictly plurisubharm.

Original source details: GELFAND-P000483

Diagram and typed description. Contour to the left of “Then”.



Then D — ~~dom.~~ is a closed subset of a strictly convex dom., approaching the boundary transversely.

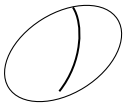
Original source details: GELFAND-P000483

Uncertain reading. The crossed-out word is provisionally read as «обл.» (“dom.”), possibly «область» (“domain”).

(follows from Oka’s thm.)

Original source details: GELFAND-P000483

Diagram and typed description. A tilted oval with an interior arc. The arc reaches the upper edge and ends near the lower edge.



3) in item 2) write out the extension from the subset with estimates! (in the class of bounded functions)

The converse is also true: such a subset is ~~a domain~~ a bounded domain of holomorphy.

Original source details: GELFAND-P000483

Uncertain reading. The crossed-out word is provisionally read as «область» (“domain”), possibly «обл.» (“dom.”).

4) Corollary: for functions anal. inside and cont. in the closure, the space of maximal ideals $=$ the closure of the domain.

GELFAND-P000484

F. A. Berezin.

Quantization problem.

M — sympl. manifold (ω — a form ■ with matrix ω_{ik}; ω^{ik} — inverse matrix)

Uncertain reading. The solid mark reproduces the heavily retraced sign after “form”. Possible readings are 2 and ξ.

Original source details: GELFAND-P000484

$$[f, g] = \sum \omega^{ik} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^k}$$

Quantization: a family of assoc. rings A_h with elements $f(x)$ (functions on M)

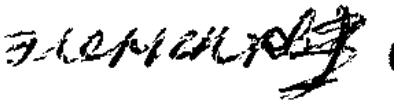
$$f_1 * f_2 = g(x \mid h)$$

$$\left.\begin{array}{ll} 1) & g(x \mid 0) = f_1(x) \cdot f_2(x) \\ 2) & \dfrac{\partial}{\partial h}(f_1 * f_2 - f_2 * f_1)\Big|_{h=0} = [f_1, f_2] \end{array}\right\} \quad \text{(correspondence principle)}.$$

Several further reqs. (less obligatory)

3) all A_h — algebras with 1, corresp. $f_0 \equiv 1$.

4) A_h — an algebra with trace (def. on a distinguished set of elements in A_h).



Uncertain reading. The ending of the word “elements” was rewritten. The earlier reading is unresolved.

Original source details: GELFAND-P000484

$$\mathrm{Sp}(f_1 * f_2) = \mathrm{Sp}(f_2 * f_1)$$

and

$$\mathrm{Sp}\, f = c \int f \, d\mu,$$

where $d\mu$ — the nat. measure on the manifold (a power of ω).

Diagram and typed description. The parenthetical reads «степень ω» (“a power of ω”).

Original source details: GELFAND-P000484

Req. of math. naturalness — functoriality with respect to canonical Diffeomorphisms (with respect to some family of maps — some fixed category).

A solution is known when $M \in$ a certain class (2 classes of Kähler manifolds).

1) dom. in $\mathbb{C}^n$
2) compact symm. manifolds,

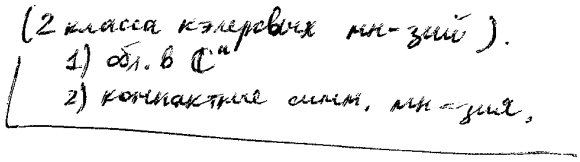


Diagram and typed description. A common stroke runs to the left of both items and below them.

Original source details: GELFAND-P000484

Fock space.

Uncertain reading. A sign above the start of the heading is heavily blacked out. Its reading is unresolved.

Original source details: GELFAND-P000484

$$M = \mathbb{C}^n \qquad \omega = \sum dz_i \wedge d\bar{z}_i.$$

Functions — entire functions $f(z, \bar{z})$ (expan. in x, y in a series converg. everywhere)

$$(f_1 * f_2)(z, \bar{z}) = \frac{1}{h^n} \int f_1(z, \bar{v}) f_2(v, \bar{z}) e^{-\frac{1}{h}(z-v)(\bar{z}-\bar{v})}\, dv\, d\bar{v}$$

Important: the structure of lin. rep. of this ring.

Let $\varphi(z)$ — ordinary entire functions satisfying

$$(\varphi, \varphi) = \frac{1}{h^n} \int |\varphi(z)|^2 e^{-\frac{z\bar{z}}{h}}\, dz\, d\bar{z} < +\infty \qquad \text{(space } F_2\text{)}$$

There acts a lin. rep. of the ring A_h:

$$(\hat{f}\varphi)(z) = \frac{1}{h^n} \int f(z, v) \overline{\varphi(v)}\, e^{-\frac{1}{h}(v-z)\bar{v}}\, dv\, d\bar{v},$$

f — Wick symbol.

Diagram and typed description. The bar is over $\varphi(v)$. The preceding factor $f(z, v)$ has no bar over v.

Original source details: GELFAND-P000484

One more ring of operators:

$$(\hat{f}\varphi)(z) = \frac{1}{h^n}\int \mathring{f}(v,\bar{v})\varphi(v)e^{-\frac{1}{h}(v-z)\bar{v}}\,dv\,d\bar{v},$$

$\mathring{f}$ — anti-Wick symbol.

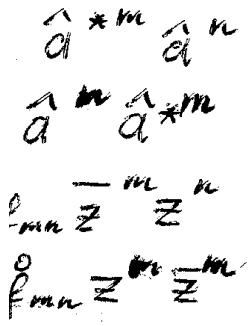
$$(\hat{a}_k\varphi)(z) = \frac{\partial\varphi}{\partial z_k}, \qquad (\hat{a}_k^*\varphi)(z) = z_k\varphi.$$

$$\hat{f}^{\blacksquare} = \sum f_{mn}\hat{a}^{*m}\hat{a}^n \quad\longleftrightarrow\quad f(z,\bar{z}) = \sum f_{mn}\bar{z}^m z^n$$

$$\hat{f} = \sum \mathring{f}_{mn}\hat{a}^n\hat{a}^{*m} \quad\longleftrightarrow\quad \mathring{f}(z,\bar{z}) = \sum \mathring{f}_{mn}z^n\bar{z}^m.$$

Original source details: GELFAND-P000484

Uncertain reading. The solid superscript at the first $\hat{f}$ reproduces the overwritten mark. A possible reading is $*$.



Original source details: GELFAND-P000484

Uncertain reading. The exponents on the first $\hat{a}$ and the first z of the lower row are heavily retraced. Both are provisionally read as n.

GELFAND-P000485

Consid. $e_{\bar{v}}(z) = e^{\frac{1}{h}\bar{v}z}$ (coherent states or Poisson vectors).

~~$f(\dots$~~

Original source details: GELFAND-P000485

Uncertain reading. An f and the beginning of a parenthesis are visible under the crossing-out. The remaining strokes are undeciphered.

Consid. now

$$\hat{f} \longleftrightarrow f(v,\bar{v}) = \frac{(\hat{f}e_{\bar{v}}, e_v)}{(e_{\bar{v}}, e_v)} \quad \text{(extension)} =$$
$$= \mathrm{Sp}(\hat{f}P_{\bar{v}}).$$

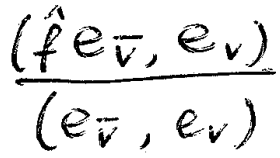


Diagram and typed description. In both rows of the fraction the first e has subscript $\bar{v}$ and the second e has subscript v.

Original source details: GELFAND-P000485

Uncertain reading. A short stroke above the projector index in the trace is provisionally set as an overbar. P_v remains possible.

Original source details: GELFAND-P000485

(P_v — orthog. proj. onto e_v)

$$\hat{f} = \int \mathring{f}(v,\bar{v}) P_v \, dv \, d\bar{v} \qquad \text{(ext. of anti-Wick symbols).}$$

Duality:

1.

Wick	↑	op. ⟶ symbol
anti-Wick		symb. ⟶ operator

Diagram and typed description. A heavily retraced upward arrow before “op.” points toward the wavy underline. Its connection with the two rows is unresolved.

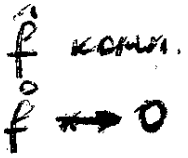
Original source details: GELFAND-P000485

2.

$$\hat{f} \text{ — bd.} \Longrightarrow f \text{ bd.}$$
$$\mathring{f} \text{ — bd.} \Longrightarrow \hat{f} \text{ bd.}$$

3.

$$\hat{f} \text{ comp.} \Longrightarrow f \to 0$$
$$\mathring{f} \leftrightarrow 0 \Longrightarrow \hat{f} \text{ comp.}$$



Uncertain reading. The heavy arrow in the second row has a right point and strokes at the left. A double-headed arrow preserves this outline; $\to$ is possible.

Original source details: GELFAND-P000485

4. Nuclearity (analog.)

$$\operatorname{Sp} \hat{f} = \frac{1}{h^n} \int f(z, \bar{z})\, dz\, d\bar{z} = \frac{1}{h^n} \int \mathring{f}(z, \bar{z})\, dz\, d\bar{z}.$$

Conversion formula:

$$f(z, \bar{z}) = \frac{1}{h^n} \int \mathring{f}(v, \bar{v}) e^{-\frac{1}{h}(z-v)(\bar{z}-\bar{v})}\, dv\, d\bar{v}.$$

At $h/2$ one obtains the Weyl symbol. (Gelfand's remark: (these are symbols) of Poincaré .)

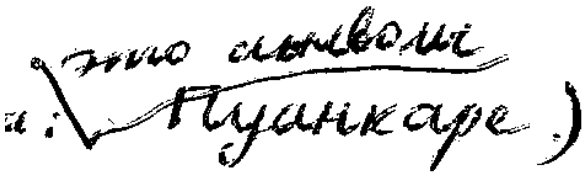


Diagram and typed description. "These are symbols" is inserted above, before "of Poincaré".

Original source details: GELFAND-P000485

General construction.

H — Hilbert space, M — set with a measure, $\alpha \in M$, measure $d\alpha$.

e_α — overcompl. system, if

$$(f, f) = \int |(f, e_\alpha)|^2\, d\alpha.$$

Consid. then the correspondence:

$$\hat{A} \longrightarrow \frac{(\hat{A}e_\alpha, e_\alpha)}{(e_\alpha, e_\alpha)} = \underset{\operatorname{Sp}(\hat{A}\hat{P}_\alpha)}{A(\alpha)} \qquad \text{(covar. symbol)}.$$

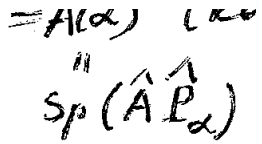


Diagram and typed description. Two strokes and the trace expression appear below $A(\alpha)$.

Original source details: GELFAND-P000485

contravar. symbol.

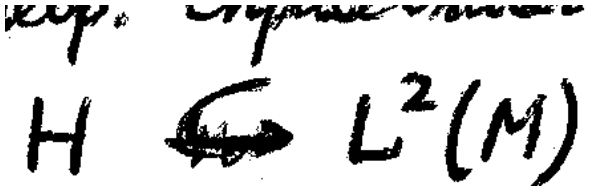
$$\hat{A} = \int \mathring{A}(\alpha) \hat{P}_\alpha\, d\alpha.$$

Earlier: $e_\alpha = e^{\bar{v}z}$, $\alpha = (v, \bar{v})$.

All thms. hold.

Another method:

$H \hookrightarrow L^2(M)$, $\hat{T}$ — orthog. projector from $L^2(M)$ into H.



Uncertain reading. Between H and $L^2(M)$ there is a hook on the left and a point on the right. The lower strokes are heavily retraced. Possible readings are $\hookrightarrow$ and $\to$ with a retraced or cancelled initial stroke.

Original source details: GELFAND-P000485

$$(\hat{B}f)(\alpha) = \mathring{A}(\alpha) f(\alpha) \quad \text{— operator in } L^2(M).$$

Then $\hat{A} = \hat{T}\hat{B}\hat{T}.$

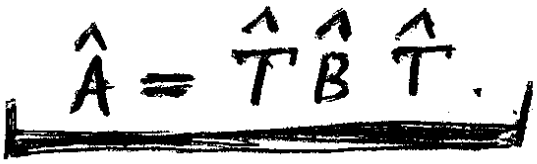


Diagram and typed description. A stroke runs below and to either side of the formula.

Original source details: GELFAND-P000485

GELFAND-P000486

Example: $e^{i(p,x)}$ in bounded domains.

Let $\varphi(x)$ be an arbit. convex (downward) function of a real var. x.

Diagram and typed description. “Downward” is inserted above, after “convex”. An arc is drawn beneath the word.

Original source details: GELFAND-P000486

Then

$$\int \varphi(A)\, d\mu \leqslant \operatorname{Sp} \varphi(\hat{A}) \leqslant \int \varphi(\mathring{A})\, d\alpha$$

Uncertain reading. The left differential is provisionally read as $d\mu$. Another possible reading is $d\alpha$.

Original source details: GELFAND-P000486

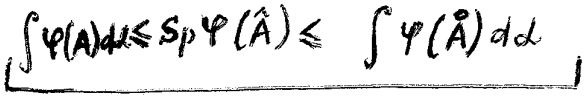


Diagram and typed description. A stroke runs below and to either side of the inequality.

Original source details: GELFAND-P000486

Left side — a consequence of Jensen’s ineq.

Right side — $\rightarrow$

$$\boxed{\operatorname{Sp}(P\varphi(A)P) \geqslant \operatorname{Sp} \varphi(PAP)}$$

Uncertain reading. The sign before the box is provisionally set as a right arrow. It may be crossed out.

Original source details: GELFAND-P000486

Original source details: GELFAND-P000486

Uncertain reading. On the right side the parenthesis merges with adjacent strokes. A provisional reading is Sp $\varphi(PAP)$.

Consid. the domain $\underline{G \subset \mathbb{C}^n}$,
in it there exists a nat. | Kähler metric.

Original source details: GELFAND-P000486

Diagram and typed description. A vertical stroke precedes “Kähler”.

We construct a ring ~~of~~ from real-anal. functions $f(z, \bar{z})$

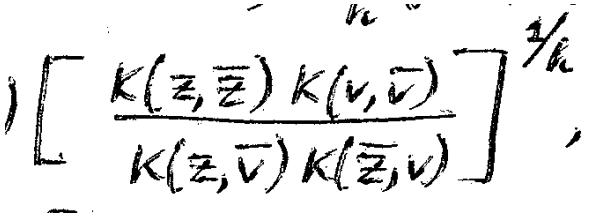
Original source details: GELFAND-P000486

Uncertain reading. A short sign after “ring” is crossed out. The provisional reading is «от» (“of”).

$$f_1 * f_2 = \int f_1(z, \bar{v}) f_2(v, \bar{z}) G_h(v, \bar{v}, z, \bar{z})\, dv\, d\bar{v}$$

$$G_h = c(h) \left[\frac{K(z, \bar{z}) K(v, \bar{v})}{K(z, \bar{v}) K(\bar{z}, v)} \right]^{1/h},$$

(one can prove that $K(z, \bar{v}) \neq 0$)
where K — the Bergman kernel fn.



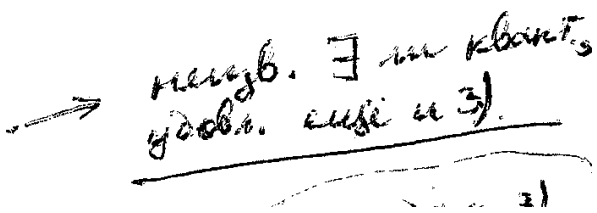
Original source details: GELFAND-P000486

Diagram and typed description. The diagonal factors are in the numerator. The second denominator factor is $K(\bar{z}, v)$. The exponent is $1/h$.

$$c(h) = h^{-\dim G} c_1(h),$$

$c_1(h)$ is cont. as $h \to 0$. (arbitrary)

$\longrightarrow$ $\underline{\text{unknown if } \exists \text{ a quantiz. also satisfying 3).}}$



Original source details: GELFAND-P000486

Diagram and typed description. The rightward arrow from “(arbitrary)” points to the underlined marginal note.

Then all reqs. are met except 3).

on the basis of 3).
in a nat. way.
If G — a homog. dom., this too is met and $c_1(h)$ is specified

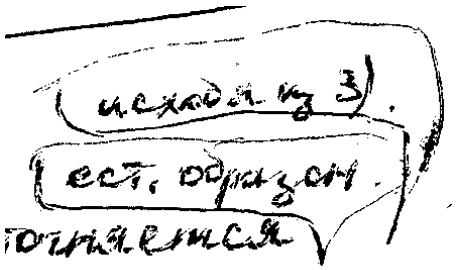


Diagram and typed description. Two callouts, one above the other, appear on the right above "is specified".

Original source details: GELFAND-P000486

If G is even symmetric, and motions — autom. of the ring, then only this.
Overcompl. system here (for a homog. dom.)

$$M = G \subset \mathbb{C}^n = \mathbb{R}^{2n}$$

$e_v(z) = (K(z, \bar{v}))^{1/h}$ (for a homog. domain).

Uncertain reading. A sign below the exponent $1/h$ is crossed out. It may be an earlier exponent.

Original source details: GELFAND-P000486

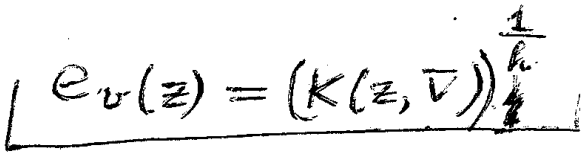
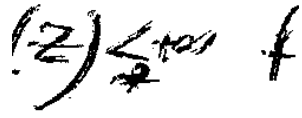


Diagram and typed description. A stroke runs below and to either side of the formula.

Original source details: GELFAND-P000486

Hilbert space:

$$H = \{f : \int |f(z)|^2 K^{1/h}(z, \bar{z})\, d\mu(z) < +\infty \quad f \text{ (holom.)}\}.$$



Uncertain reading. A separate heavy stroke sits below the angle $<$. It may be a crossed-out "2" or the lower stroke of $\leqslant$.

Original source details: GELFAND-P000486

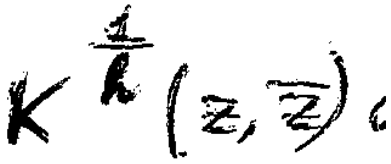


Diagram and typed description. The weight has exponent $1/h$. There is no visible minus before the exponent.

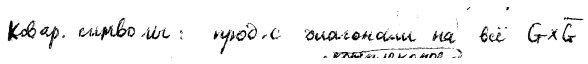
Original source details: GELFAND-P000486

$\mu(z)$ — an invar. measure generated by the Bergman metric.

A larger space — all functions.

In the gen. case it is unknown whether $K(z, \bar{v})$ can become 0.

Covar. symbols: cont. from the diagonal to all of $G \times \bar{G}$

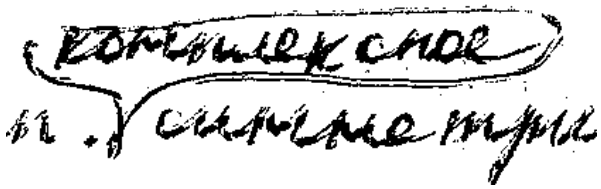


Original source details: GELFAND-P000486

Diagram and typed description. The line on covar. symbols reads «прод. с диагонали» (“cont. from the diagonal”).

(complex)

Compact case. Comp. symmetric space.



Original source details: GELFAND-P000486

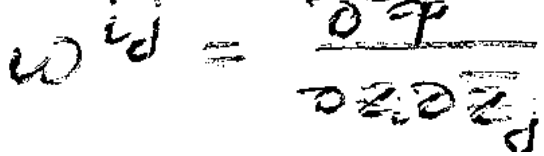
Diagram and typed description. “Complex” is inserted in a callout after “Comp.”.

Consid. the potential of the Kähler metric, i.e. $\Phi(z, \bar{z})$

GELFAND-P000487

Consid.

$$e^{\frac{1}{h}\Phi(z,\bar{z})} = K_h(z, \bar{z}) \qquad \left(\Phi \text{ is def. by the cond. } \omega^{ij} = \frac{\partial^2 \Phi}{\partial z_i \partial \bar{z}_j}\right).$$



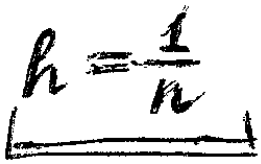
Original source details: GELFAND-P000487

Diagram and typed description. The two indices i, j on ω are written above.

At $h = \frac{1}{n}$ $\Phi(z, \bar{z})$ extends and there exists a single-valued fn.

$$K_h(z, \bar{v})$$

and the whole preceding theory.



Original source details: GELFAND-P000487

Diagram and typed description. The formula $h = 1/n$ has a double stroke underneath with turned-up ends.

Then the Hilbert space H_α is finite-dimensional and dim is easily computed.

Original source details: GELFAND-P000487

Uncertain reading. The small retraced mark is beside H. It is set as a small baseline α to reproduce its position. Possible readings are subscripts α and h.

Lobachevsky plane:

$$K^{1/h} = (1 - z\bar{z})^{+2/h}$$

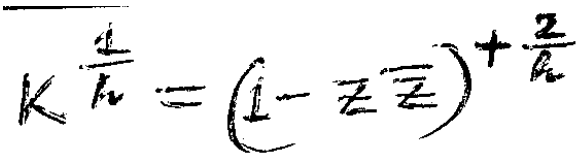


Diagram and typed description. A plus sign is written before $2/h$.

Original source details: GELFAND-P000487

Autom. of the space generate autom. of the ring.

(autom. of)

One obtains a projective rep. of the Lobachevsky plane.

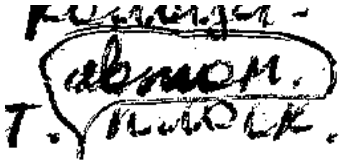


Diagram and typed description. “Autom.” is inserted in a callout after “rep.”.

Original source details: GELFAND-P000487

GELFAND-P000488

12/III-73 Gelfand’s Sem.

V. M. Buchstaber.

$S(M^n)$ — set of smoothings on M^n
$hS(M^n)$ — homot. smoothings
$t(M^n)$ — triangulations
$ht(M^n)$ — hom. triangul.

— · —

Consid. the set $hS(M^n)$.

On it, consid. smooth vect. bundles.

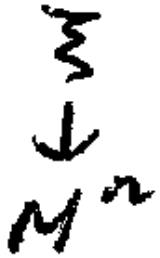


Diagram and typed description. An arrow runs down from ξ to M^n.

$$\begin{array}{c} \xi \\ \downarrow \\ M^n \end{array}$$

Original source details: GELFAND-P000488

J-triv. bundles («псл.» homot. triv.), i.e.

Uncertain reading. The first abbreviation appears to read «псл.», possibly «посл.», which may stand for “fibrewise”.

Original source details: GELFAND-P000488

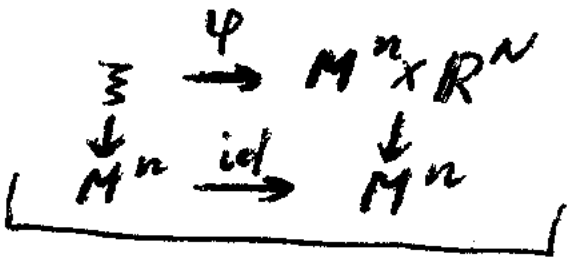


Diagram and typed description. Square with an underline whose ends turn upwards.

$$\begin{array}{ccc} \xi & \xrightarrow{\varphi} & M^n \times \mathbb{R}^N \\ \downarrow & & \downarrow \\ M^n & \xrightarrow{\mathrm{id}} & M^n \end{array}$$

Original source details: GELFAND-P000488

$\underline{\underline{N = \dim \xi}}$

Diagram and typed description. The equality has a double underline.

Original source details: GELFAND-P000488

φ — a proper homotop. equivalence.

$\Phi(M^n)$ — set of smooth J-triv. bundles on M^n. (up to trivial ones).

Let

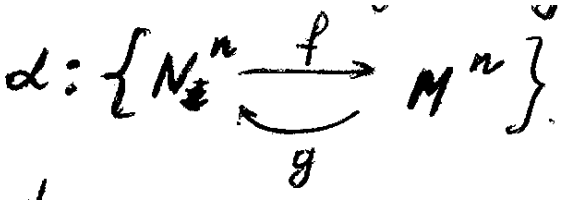


Diagram and typed description. The heavily retraced subscript on N is provisionally set as ξ. Another possible reading is ζ.

$$\alpha : \{ \quad N^n_\xi \underset{g}{\overset{f}{\rightleftarrows}} M^n \quad \}$$

Original source details: GELFAND-P000488

g — a homot. inverse.

From α we construct an elem. of the group $\Phi(M^n)$

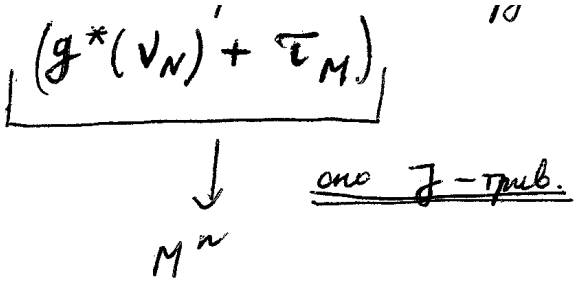


Diagram and typed description. A stroke runs below the sum. An arrow points from it to the base.

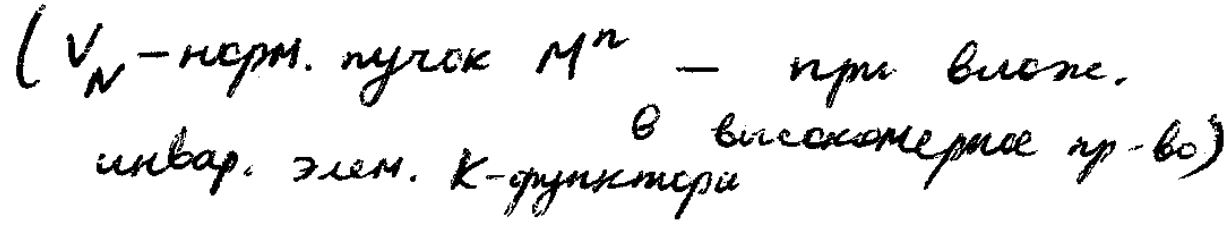
$$\underline{(g^*(v_N) + \tau_M)}$$
$$\downarrow$$
$$M^n$$

it is J-triv.

Original source details: GELFAND-P000488

(v_N — norm. bundle of M^n — for an embedd. in a high-dimensional space)

invar. elem. of the K-functor.



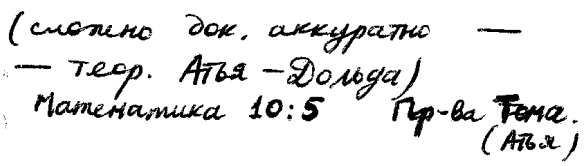
Original source details: GELFAND-P000488

Uncertain reading. The explanation for v_N reads M^n. The normal-bundle sign is v-shaped and could be Greek ν.

$$g \text{ — ident.} \Longrightarrow v_N + \tau_M \underset{\text{def}}{=} \text{triv.}$$

$$g \text{ — hom. equiv.} \Longrightarrow \dots$$

(difficult to prove carefully —
— Atiyah–Dold thm.)
Matematika 10:5 Thom spaces. (Atiyah)



Original source details: GELFAND-P000488

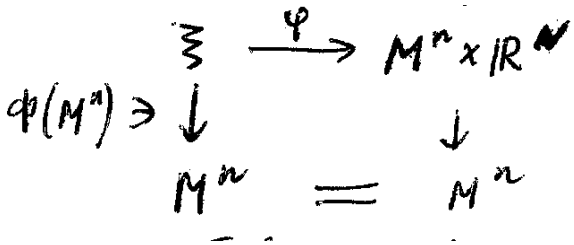
Diagram and typed description. The note and reference are to the right of the calculation.

Let $\pi_1(M^n) = 0$.

Exact seq.

$$hS(M^n) \longrightarrow \Phi(M^n) \longrightarrow P_n$$

$$P_n = \begin{cases} 0, & n \neq 4k,\ 4k+2 \\ \mathbb{Z}, & n = 4k \\ \mathbb{Z}_2, & n = 4k+2 \end{cases}$$

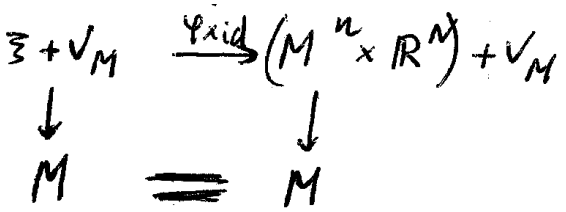


Original source details: GELFAND-P000488

Diagram and typed description. The bottom horizontal connection is written as an equals sign. The lower-case sign at the left is set as ϕ.

$$\begin{array}{ccc} \xi & \xrightarrow{\varphi} & M^n \times \mathbb{R}^N \\ \phi(M^n) \ni \downarrow & & \downarrow \\ M^n & = & M^n \end{array}$$

Let us add v_M:



Original source details: GELFAND-P000488

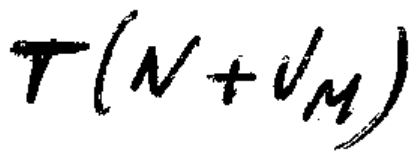
Diagram and typed description. The normal bundle is added to both upper entries.

$$\begin{array}{ccc} \xi + v_M & \xrightarrow{\varphi \times \mathrm{id}} & (M^n \times \mathbb{R}^N) + v_M \\ \downarrow & & \downarrow \\ M & = & M \end{array}$$

$\varphi \times \mathrm{id}$ — a proper homot. equiv.

Hence

$$T(\xi + v_M) \longrightarrow T(N + v_M) \qquad (*)$$



Original source details: GELFAND-P000488

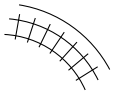
Diagram and typed description. The argument on the right reads $N + v_M$.

GELFAND-P000489

Remark on the props. of the Thom complex, norm. bundle.

$M^n \subset \mathbb{R}^L$.

Diagram and typed description. Three arcs and transverse strokes.



Original source details: GELFAND-P000489

The space of the tube is the space of the norm. bundle.

There is a nat. map

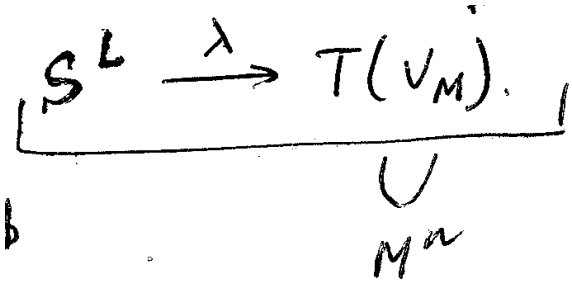


Diagram and typed description. The arrow λ and its endpoints have a stroke underneath. Below $T(\nu_M)$ are $\cup$ and M^n.

$$\underbrace{S^L \xrightarrow{\lambda} T(\nu_M).}_{\cup\ M^n}$$

Original source details: GELFAND-P000489

Uncertain reading. A sign below the left part is crossed out. Possible readings are β and φ.

Original source details: GELFAND-P000489

The map λ is transv. regular on M^n. Its preimage is M^n. Only this will be used.

$\exists$ a map $S^L \xrightarrow{\lambda} T(\nu_M)$

Uncertain reading. An entry following the second formula is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000489

and an embedding $M \subset T(\nu_M)$, which is transv. reg. on M.

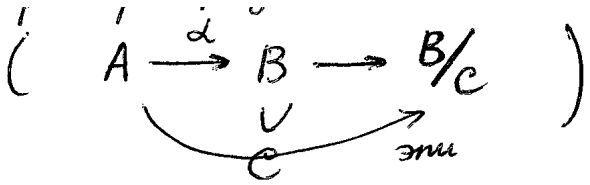


Diagram and typed description. The lower arc goes from A to B/C. Its endpoint is labelled "epi".

$$\left(A \xrightarrow{\alpha} B \longrightarrow B/C \right), \quad C \cup \text{epi}$$

Original source details: GELFAND-P000489

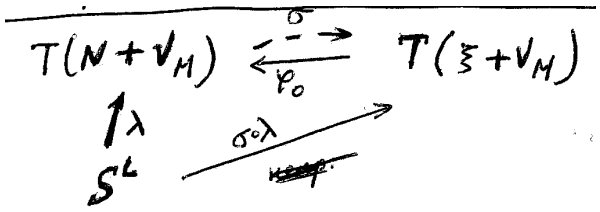
Original source details: GELFAND-P000489

Diagram and typed description. A left-pointed stroke, a dot and a right stroke.

The map $\phi(M^n) \longrightarrow P_n$.

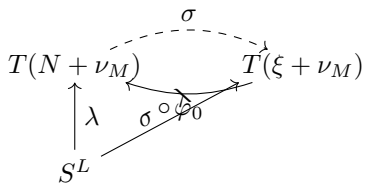
Original source details: GELFAND-P000489

Uncertain reading. The sign before (M^n) is provisionally set as ϕ.



Original source details: GELFAND-P000489

Diagram and typed description. The reverse arrow σ from $T(N+\nu_M)$ is dashed.



~~cont.~~

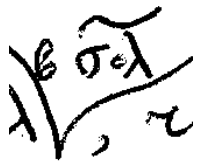
Original source details: GELFAND-P000489

Uncertain reading. The crossed-out abbreviation below the diagonal arrow is provisionally read as «непр.» (“cont.”).

φ_0 is a homot. equiv. by constr. σ — a hom. inverse.

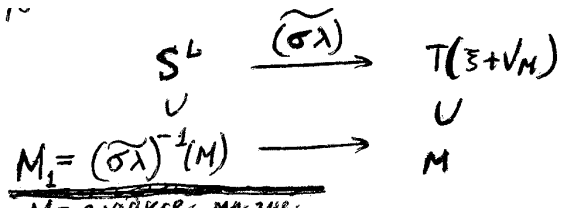
$\sigma \circ \lambda$ — a cont. map. $\quad M \subset T(\xi + \nu_M)$

We homotope $\sigma \circ \lambda$ (to $\widetilde{\sigma \circ \lambda}$), so that on M it becomes transv. reg.



Original source details: GELFAND-P000489

Diagram and typed description. The callout after $\sigma \circ \lambda$ reads “to” followed by the same composite with a tilde.



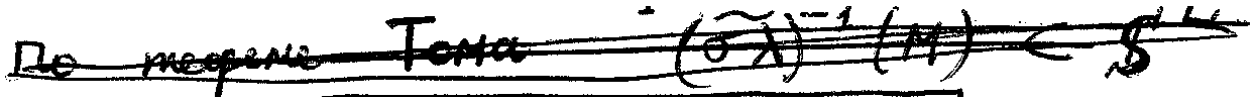
Original source details: GELFAND-P000489

Diagram and typed description. The preimage M_1 is below S^L. The two rows have $\cup$ signs between them.

$$\begin{array}{ccc} S^L & \xrightarrow{\widetilde{(\sigma\lambda)}} & T(\xi+\nu_M) \\ \cup & & \cup \\ M_1 = (\widetilde{\sigma\lambda})^{-1}(M) & \longrightarrow & M \end{array}$$

M_1 — a smooth manifold. dim M_1 = dim M.

~~By Thom's theorem $(\widetilde{\sigma\lambda})^{-1}(M) \subset S^L$.~~



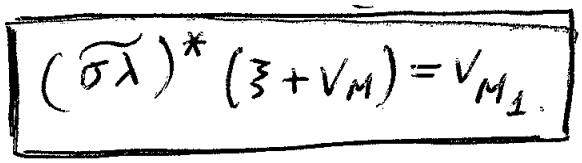
Original source details: GELFAND-P000489

Diagram and typed description. The entire line is crossed out.

Important:

$$\boxed{(\widetilde{\sigma\lambda})^*(\xi + \nu_M) = \nu_{M_1}.}$$

(reformul. of transverse regularity)

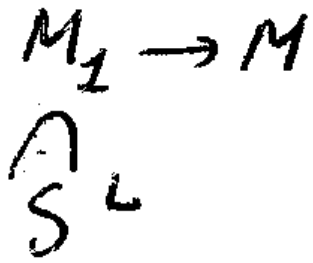


Original source details: GELFAND-P000489

Diagram and typed description. The equality is enclosed in a rectangle.

$\widetilde{\sigma\lambda} : M_1 \longrightarrow M$.

Thus from the bundle ξ we have constr. a manifold M_1 and its cont. map



Original source details: GELFAND-P000489

Diagram and typed description. Below M_1 are $\cap$ and S^L.

$$\begin{array}{ccc} M_1 & \longrightarrow & M \\ \cap & & \\ S^L & & \end{array}$$

M_1 with norm. bundle $\nu_{M_1} = (\widetilde{\sigma\lambda})^*(\xi + \nu_M)$. (depends on ξ)

There arose a ~~problem~~ surgery situation.

Original source details: GELFAND-P000489

Uncertain reading. The crossed-out word is provisionally read as «задача» (“problem”).

A problem in smooth topology: how uniquely is M_1 determined.

Milnor–Kervaire thm. All M_1 are obtained from one another by Morse surgeries.
(necess. and suff.)
Is there a manifold there homot. equiv. to M.
P_n — the obstruction group

Uncertain reading. A word after "obstructions" is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000490

to surgeries as far as homotop. equiv.

$$\sigma_0 : M_1 \longrightarrow M$$

The seq. is exact.

$$? \longrightarrow hS(M^n).$$

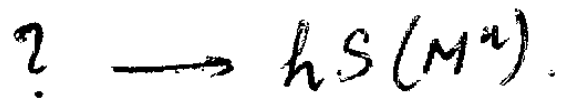


Diagram and typed description. A question mark is written in place of the first term.

Original source details: GELFAND-P000490

$$\alpha_1. \quad \{N_1 \xrightarrow{f_1} M\},$$
$$\alpha_2 : \quad \{N_2 \xrightarrow{f_2} M\}.$$

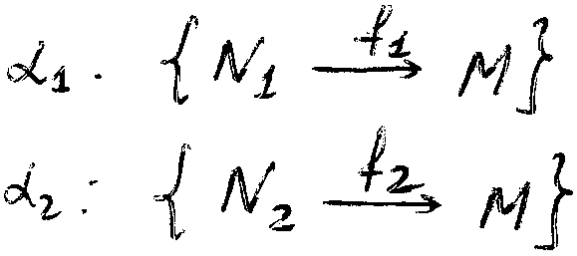


Diagram and typed description. A period follows α_1, and a colon follows α_2.

Original source details: GELFAND-P000490

$\xi_1 \in \phi(M)$ is constr. from α_1.

Diagram and typed description. The sign before (M) is set as ϕ.

Original source details: GELFAND-P000490

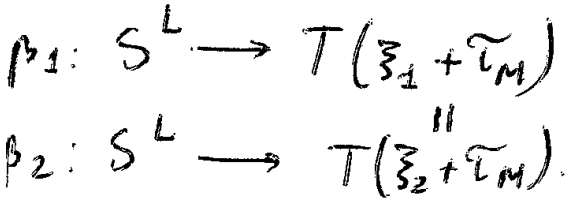
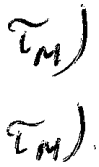


Diagram and typed description. Two arrows point to Thom complexes. Two vertical strokes stand between the targets.

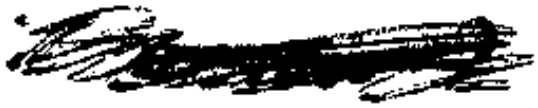
$$\beta_1 : S^L \longrightarrow T(\xi_1 + \widetilde{\tau}_M)$$
$$\| \qquad\qquad\qquad\



Uncertain reading. A separate wavy stroke is visible above τ. Set as $\widetilde{\tau}_M$.

Original source details: GELFAND-P000490

We obtain the problem



Uncertain reading. An entry between "problem" and "diss." is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000490

Novikov's diss.:

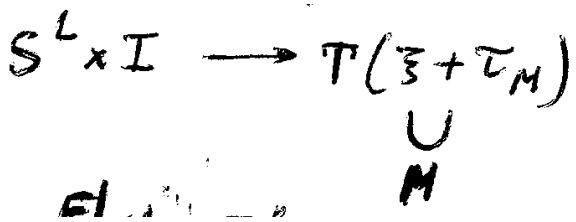


Diagram and typed description. Below the target are $\cup$ and M.

$$\begin{array}{ccc} S^L \times I & \longrightarrow & T(\xi + \widetilde{\tau}_M) \\ & & \cup \\ & & M \end{array}$$

Original source details: GELFAND-P000490

Film:

$$F|_{S^L \times 0} = \beta_1$$
$$F|_{S^L \times 1} = \beta_2$$

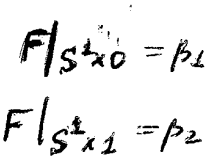


Diagram and typed description. The two restrictions of F at 0 and 1.

Original source details: GELFAND-P000490

Uncertain reading. The sign before the quantifier is overwritten. Its reading is unresolved.

Original source details: GELFAND-P000490

$\exists$ among the films h-cobordisms?

GELFAND-P000491

19/III-73 Gelfand's Sem.

Uncertain reading. A short entry before the date is crossed out. It may contain digits.

Original source details: GELFAND-P000491

Arnold. Consid. the germ of a fn. $f : (\mathbb{C}^n, 0) \longrightarrow (\mathbb{C}, 0)$. 0 — a crit. point.

Diagram and typed description. A yellow callout icon with three black strokes is visible above the line. Its text is unavailable in this frame.

Original source details: GELFAND-P000491

Consid. the space of k-jets. The group of k-jets of diffs. acts.

$j^k f$ k-suff. (the subsequent ones have no effect)

m — number of moduli. Thm. $m = 0 \Longrightarrow$ the list

$$\begin{array}{ll} A_k & x^{k+1} \\ D_k & x^2 y + y^{k-1} \\ E_6 & x^3 + y^4 \\ E_7 & x^3 + xy^3 \\ E_8 & x^3 + y^5 \end{array}$$

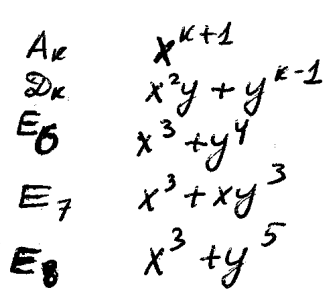


Diagram and typed description. Several strokes in the list are retraced, including those in E_6 and E_8.

Original source details: GELFAND-P000491

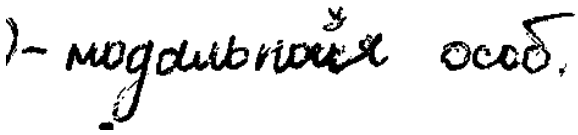
Stab. equiv. $\oplus$ with nondeg. quadr. forms.

The quadr. form is pos. definite

GELFAND-P000492

An analyt. fn., up to an analyt. change of var., uniquely det. the change of var.

0-модульно[...]я sing. — after changing the coeffs. $\exists$ only a finite number of variants.



Uncertain reading. The visible sequence is «модульно», then overwritten strokes with a й-like mark above, and final «я». The ending may have been changed from «модульной» to «модульная» or have other superposed strokes. The earlier reading «модульных» remains doubtful. The word concerns modality.

Original source details: GELFAND-P000492

$f = \varepsilon$ level line: Intersection with a ball — an even-dimensional manifold with boundary.

$$f^{-1}(\varepsilon) \cap B_\delta$$

Diagram and typed description. Two closed curves are joined by inward-curving sides.

Original source details: GELFAND-P000492

Intersection index

Uncertain reading. An entry after "intersection" is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000492

of real homology in the middle dim. (integral)

(neg. def.)

It is definite in sign exactly in the listed cases, provided $n \equiv 3 \pmod 4$. (otherwise add extra variables) — Tyurin's thm.

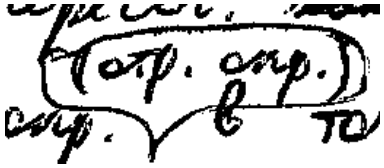


Diagram and typed description. "(Neg. def.)" is inserted in a callout after "definite in sign".

Original source details: GELFAND-P000492

$$f^{-1}(\varepsilon) \cap B_\delta \approx \underbrace{S^{n-1} \vee \ldots \vee S^{n-1}}_{\mu}.$$

$\mu =$ dim of the local ring.

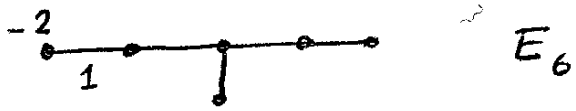
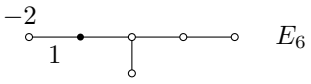


Diagram and typed description. Diagram of E_6. The leftmost vertex is marked -2 and the first edge is marked 1.



Original source details: GELFAND-P000492

Uncertain reading. An entry to the right of E_6 is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000492

The quadr. form is given by the Dynkin diag.

$\boxed{m = 1.}$ — one also finds a list (long).

First: $x_1^3 + x_2^3 + x_3^3$.

$$\mu = \mu_+ + \mu_0 + \mu_-$$

$\mu_+ \leqslant 1. \Longrightarrow$ always $m = 1$.

Uncertain reading. The sign before 1 is retraced. It is provisionally set as $\leqslant$. The conclusion reads $m = 1$.

Original source details: GELFAND-P000492

For all of them

$$\mu_+ \leqslant 1 \Longrightarrow \mu_0 = 1 \qquad \text{(always degen.)}$$

$\mu_+ = 1.$ $axyz + x^k + y^\ell + z^m$ — three-index series $T_{k\ell m}$.

A. S. Mishchenko.

Sullivan: localization, periodicity and ... Galois-

$$K(X)/J\text{-equiv.} = J(X) \quad \text{— a finite group.}$$

Adams: it is described in terms of the cohom. ops. of K-theory ψ^k.

$$\psi^k(\xi) = \xi^k \quad \text{and additive} \quad \Longrightarrow \quad \text{this determines them uniquely}$$

$(\dim \xi = 1) \qquad k = 2, 3, \ldots$

Thus $J(X)$ is determined by the group $K(X)$ as a module over these ops.

J-equiv. — fibrewise homot. equiv.

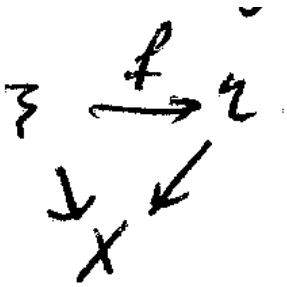


Diagram and typed description. The arrow f joins ξ and η. Both letters have arrows to X.

$$\begin{array}{ccc} \xi & \xrightarrow{f} & \eta \\ & \searrow \quad \swarrow & \\ & X & \end{array}$$

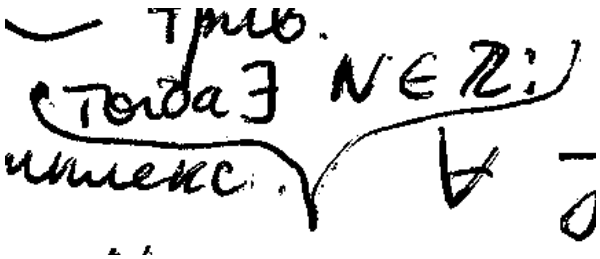
Original source details: GELFAND-P000492

f — proper. $\deg f = 1$ on fibres.

It was proved (Adams, Atiyah):

$$\xi \overset{J}{\sim} \text{triv.}$$

X — a finite complex. (then $\exists N \in \mathbb{Z}$:)



Original source details: GELFAND-P000492

Diagram and typed description. "Then $\exists N \in \mathbb{Z}$:" is inserted after "complex".

$\forall$ J-triv. bundles ξ $(\forall k)$ $\exists$

Original source details: GELFAND-P000492

Diagram and typed description. The quantifier $\forall k$ is inserted between ξ and $\exists$.

Original source details: GELFAND-P000492

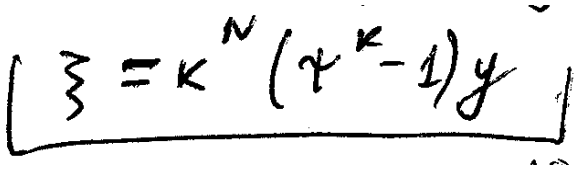
Uncertain reading. A word before "bundle" is crossed out. Its reading is unresolved.

a bundle y:

Original source details: GELFAND-P000492

Uncertain reading. An entry after the colon is crossed out. Its reading is unresolved.

$$\underline{\xi = k^N(\psi^k - 1)y}$$



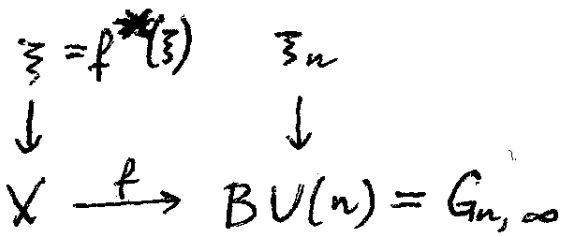
Original source details: GELFAND-P000492

Diagram and typed description. A stroke runs below and to either side of the equality.

The converse — the Adams conjecture: $\forall$ finite compl. X $\exists N$; $\forall k$

$$k^N(\psi^k - 1)\xi \overset{J}{\sim} 0.$$

Idea.



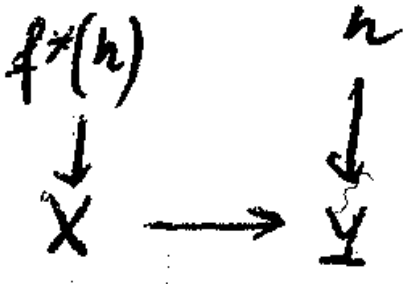
Original source details: GELFAND-P000493

Diagram and typed description. The left side reads $\xi = f^*(\xi)$ and the right side ξ_n.

$$\begin{array}{ccc} \xi = f^*(\xi) & & \xi_n \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & BU(n) = G_{n,\infty} \end{array}$$

Original source details: GELFAND-P000493

Uncertain reading. The sign after f is heavily retraced and provisionally set as a star.



Original source details: GELFAND-P000493

Diagram and typed description. The second diagram shows $f^*(\eta)$ and η.

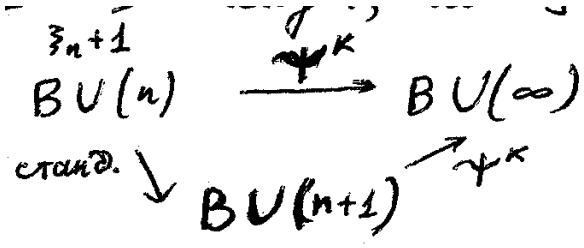
$$\begin{array}{ccc} f^*(\eta) & & \eta \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

$$\psi^k(f^*(\eta)) = f^*\psi^k(\eta).$$

Hence the op. ψ^k is det. by specifying a bundle on $\underline{\underline{BU(n)}}$. $(\psi^k(\xi_n))$.

From ψ^k one constructs a map $BU(n) \longrightarrow BU(\infty)$ (unknown dimension).

$\psi^k(1) = 1 \Longrightarrow$ it follows that $\exists$ a sequence of maps.



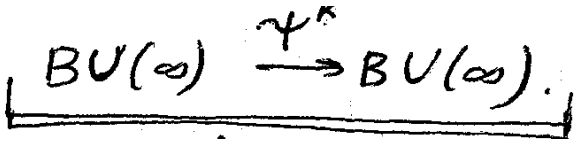
Original source details: GELFAND-P000493

Diagram and typed description. Above $BU(n)$ is $\xi_n + 1$.

$$\begin{array}{ccc} \xi_n + 1 & & \\ BU(n) & \xrightarrow{\psi^k} & BU(\infty) \\ & \searrow^{\text{std.}} \quad \nearrow_{\psi^k} & \\ & BU(n+1) & \end{array}$$

Thus ψ^k is given by a single map.

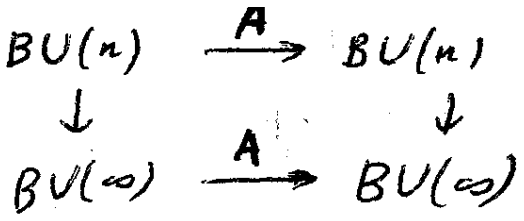
$$\underline{\underline{BU(\infty) \xrightarrow{\psi^k} BU(\infty).}}$$



Original source details: GELFAND-P000493

Diagram and typed description. A double stroke runs below the arrow, with turned-up ends.

Suppose there is a cohom. op. $BU(\infty) \xrightarrow{A} BU(\infty)$, and suppose it preserves dim.

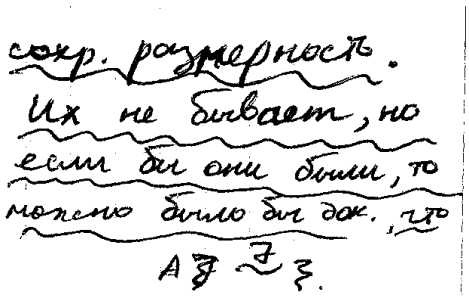


Original source details: GELFAND-P000493

Diagram and typed description. The square has A on both horizontal arrows.

$$\begin{array}{ccc} BU(n) & \xrightarrow{A} & BU(n) \\ \downarrow & & \downarrow \\ BU(\infty) & \xrightarrow{A} & BU(\infty) \end{array}$$

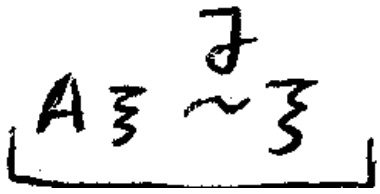
preserves dimension. They do not occur, but if they did, one could prove that $A\xi \overset{J}{\sim} \xi$.



Original source details: GELFAND-P000493

Diagram and typed description. The note on the right has a wavy underline.

Then $A\xi \overset{J}{\sim} \xi$



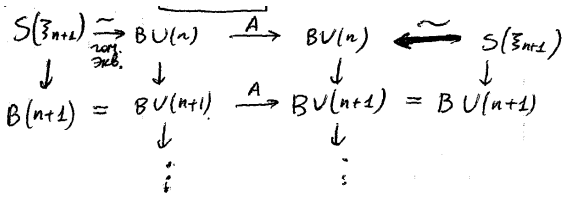
Original source details: GELFAND-P000493

Diagram and typed description. An open-top bracket beneath the formula.

spheres

Original source details: GELFAND-P000493

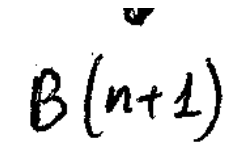
Diagram and typed description. The frame around “spheres” is open on the left.



Original source details: GELFAND-P000493

Diagram and typed description. The diagram has two levels and vertical ellipses.

$$\begin{array}{ccccccc} S(\xi_{n+1}) & \xrightarrow[\text{hom. equiv.}]{\sim} & BU(n) & \xrightarrow{A} & BU(n) & \xleftarrow{\sim} & S(\xi_{n+1}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ B(n+1) & = & BU(n+1) & \xrightarrow{A} & BU(n+1) & = & BU(n+1) \\ & & \downarrow & & \downarrow & & \\ & & \vdots & & \vdots & & \end{array}$$



Original source details: GELFAND-P000493

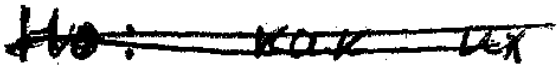
Diagram and typed description. The lower-left node reads $B(n+1)$.

Original source details: GELFAND-P000493

Uncertain reading. The heavily retraced stroke has a clear point on the left and a short oblique stroke on the right. The right stroke may be the remnant of another arrowhead.

A fibrewise homot. equiv. is obtained.

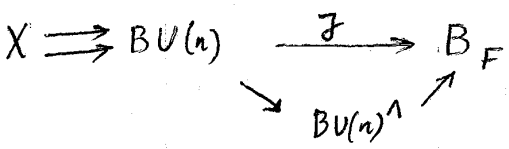
~~But: how [to ...] them~~



Original source details: GELFAND-P000493

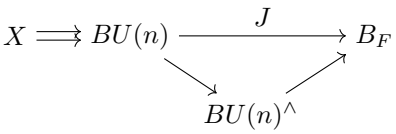
Diagram and typed description. "But: how [to ...] them" is crossed out.

There are none, but how do we "find" them?

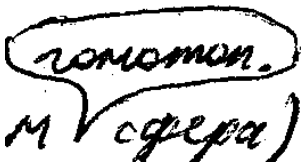


Original source details: GELFAND-P000493

Diagram and typed description. Two arrows go from X to $BU(n)$.



(classif. space of bundles with fibre a (homotopy) sphere)

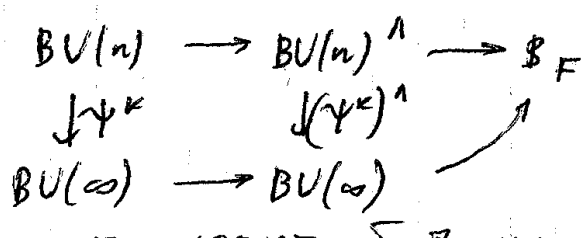


Original source details: GELFAND-P000493

Diagram and typed description. "Homotopy" is inserted before "sphere".

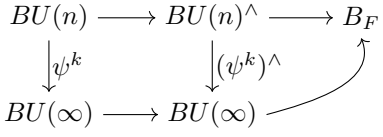
J — a functor.

Suppose $\exists BU(n)^\wedge$, to which the Adams operations transfer, i.e. $\exists$ an op. $(\psi^k)^\wedge$, for which the diag. commutes.



Original source details: GELFAND-P000493

Diagram and typed description. The lower-right arrow bends up toward B_F.



Perhaps, it may be that $(\psi^k)^\wedge$ preserves dimension.

(orientations — dim.-preserving maps)

These spaces were constr. by Sullivan.

$BU(n)^\wedge$ will be the compl. Grassmannian.

On

Original source details: GELFAND-P000494

Uncertain reading. A short entry after "On" is crossed out. Its reading is unresolved.

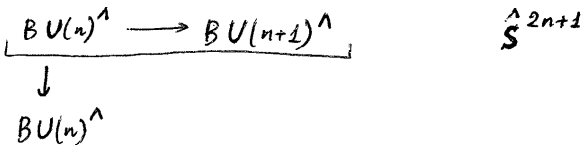
$S^{2n\cdots 1}$ the group $U(n)$.

Original source details: GELFAND-P000494

Uncertain reading. Strokes between $2n$ and 1 are heavily retraced. Possible readings are $2n-1$ and $2n+1$. The similar expression immediately below has a clear $2n-1$.

We obtain bundles (spher). on $BU(n)$.

Sullivan: another space $\widehat{S}^{2n-1}$ and the group G_n. Consid. the bundle and the classif. space.



Original source details: GELFAND-P000494

Diagram and typed description. A stroke runs below the horizontal arrow. The separate entry on the right is $\widehat{S}^{2n+1}$.

$$\begin{array}{ccc} BU(n)^\wedge & \longrightarrow & BU(n+1)^\wedge \\ \downarrow & & \\ BU(n)^\wedge & & \end{array} \qquad \widehat{S}^{2n+1}$$

and the fibrewise homot. type will be preserved.

Original source details: GELFAND-P000494

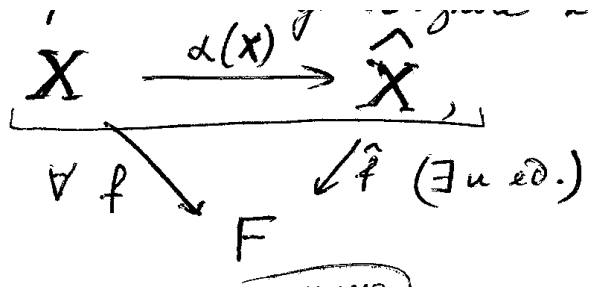
Undeciphered passage. The line is heavily crossed out. A continuous reading is unresolved.

Simply connected compls. with fin. gen. homot. grps. are considered.

Claim. $\begin{matrix}\forall X \\ \exists\end{matrix}$

Original source details: GELFAND-P000494

Diagram and typed description. $\forall X$ is inserted above $\exists$.



Original source details: GELFAND-P000494

Diagram and typed description. A stroke runs under the upper arrow. Two arrows point to the bottom node.

$$\begin{array}{ccc} X & \xrightarrow{\alpha(X)} & \widehat{X}, \\ & \searrow^{\forall f} \quad \swarrow_{f\ (\exists \text{ and unique})} & \\ & F & \end{array}$$

(functor)

$\alpha(X)$ — a transformation of the ident. functor.

($\forall F$, (for which) $\pi_i(F)$ are finite).

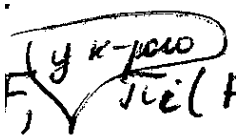


Diagram and typed description. “For which” is inserted after F.

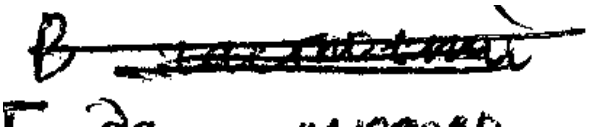
Original source details: GELFAND-P000494

Uncertain reading. An entry before “are finite” is blacked out. Its reading is unresolved.

Original source details: GELFAND-P000494

$\pi_i(\widehat{X})$ are profinite ($\varprojlim$ fin. groups). (in partic., compact).

~~In particular~~



Uncertain reading. The crossed-out words are provisionally read as “In particular”.

Original source details: GELFAND-P000494

Then one can compute:

$$\pi_i(\widehat{X}) = \widehat{\pi_i(X)}$$

(profin. completion). Neighb. — subgroups of fin. index and their limit.

$Z^\wedge$ is descr. by (integral) p-adic numbers.

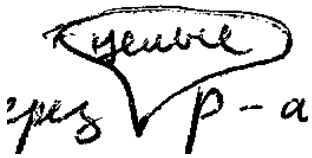


Diagram and typed description. “Integral” is inserted before p.

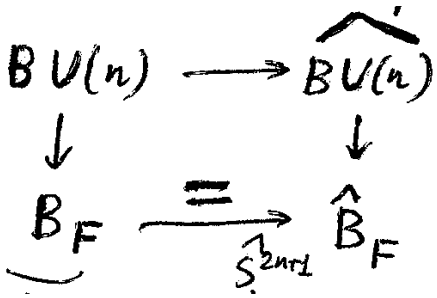
Original source details: GELFAND-P000494

The homot. groups of $\widehat{X}$ behave like integral p-ad. numbers.

~~Action of the Galois grp.~~

Diagram and typed description. The heading is crossed out.

Original source details: GELFAND-P000494

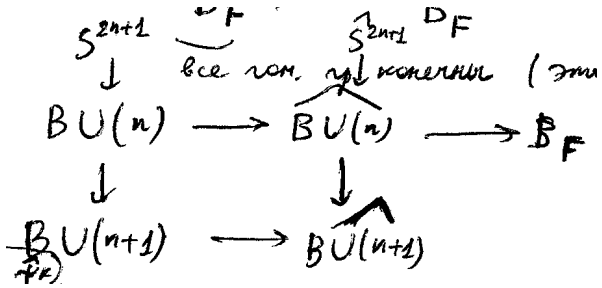


Original source details: GELFAND-P000494

Diagram and typed description. The lower arrow points right and has an equals sign above it.

$$\begin{array}{ccc} BU(n) & \longrightarrow & \widehat{BU(n)} \\ \downarrow & & \downarrow \\ B_F & \xrightarrow{=} & \widehat{B}_F \end{array}$$

all homot. grps. are finite (these are stable homot. groups of spheres)



Original source details: GELFAND-P000494

Diagram and typed description. The lower horizontal arrow points right. There is an additional stroke at its left end. The arrow from $\widehat{S}^{2n+1}$ crosses the neighbouring writing.

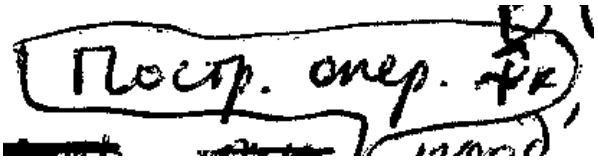
$$\begin{array}{ccccc} S^{2n+1} & & \widehat{S}^{2n+1} & & \\ \downarrow & & \downarrow & & \\ BU(n) & \longrightarrow & \widehat{BU(n)} & \longrightarrow & B_F \\ \downarrow & & \downarrow & & \\ BU(n+1) & \longrightarrow & \widehat{BU(n+1)} & & \end{array}$$

~~Suppose~~

Original source details: GELFAND-P000494

Uncertain reading. The following short word is crossed out. Its reading is unresolved.

(The constr. op. $\widehat{\psi}_k$) extends the Adams ops. and preserves dimension.

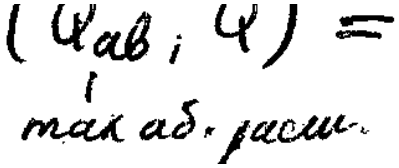


Original source details: GELFAND-P000494

Uncertain reading. A small k stands beside ψ. Angular strokes above ψ intersect the neighbouring writing. Possible readings include ψ_k and $\widehat{\psi}^k$.

Consid. the Grassmannians: $G_{N,n}$. There are finitely many aff. alg. charts, with eqs. over Z (scheme over Z). — an alg. set over Q. $\Rightarrow$ the Galois grp. acts there $\mathrm{Gal}(\mathbb{C} : Q)$ — acts on $G_{N,n}$. It is discontinuous in the usual sense, but continuous in the Zariski top. Consid. Étale top. ...

$$\mathrm{Gal}(\mathbb{C} : Q) \longrightarrow \mathrm{Gal}\Big(\underset{\text{max. ab. ext.}}{Q_{ab}} ; Q \Big) = \widehat{Z}^*$$



Original source details: GELFAND-P000494

Diagram and typed description. Below Q_{ab} is the label “max. ab. ext.”.

$$H_i(V_c, A) \approx H_i(V_{et}, A)$$

$$\widehat{V}_c \sim \varprojlim N(V_{et}), \qquad N \text{ — nerve.}$$

GELFAND-P000495

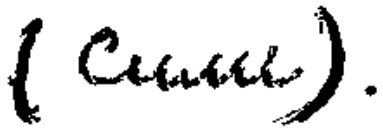
26.3.73

Statement of the res. of Atiyah, Patodi, Singer. (I. M. Gelfand).

Let A — a self-adj. ell. op.

$$\zeta_A(s) = \operatorname{Tr} A^{-s} = \sum \lambda_i^{-s}$$

at $s = 0$ has no pole (Seeley).



Uncertain reading. The Cyrillic spelling of the name is provisionally read as «Сили», possibly «Силли».

Original source details: GELFAND-P000495

$\zeta_A(0)$ — Atiyah and others.

Consid. In place of the ζ-fn., L-series.

Undeciphered passage. The crossed-out phrase to the right may begin “Not”. The rest is illegible.

Original source details: GELFAND-P000495

Let A — not pos. def. (self-adj. ellip.)

$$\eta(s) = \sum_{\lambda \neq 0} (\operatorname{sign} \lambda_i)\, |\lambda_i|^{-s}$$

Difficult theor.: $s = 0$ — no pole !!

Diagram and typed description. An open-top bracket runs under the statement about the pole.

Original source details: GELFAND-P000495

Important example — 1st-order op.

General prob.: no pos. def. Consid. $\sum$ (mult. characters of λ_i) (L-series). On compl. numbers there are more complicated mult. characters.

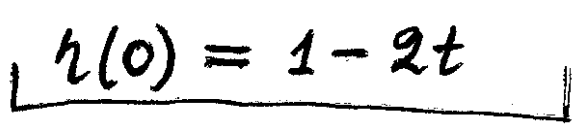
Example.

Uncertain reading. The crossed-out entry before A partly overlaps A.

Original source details: GELFAND-P000495

$$A = i\frac{d}{dx} + t \qquad \text{on } [0, 2\pi] \qquad \text{periodic}$$

$$\underline{\eta(0) = 1 - 2t}$$



Manuscript excerpt. An open-topped bracket runs under the formula.

Original source details: GELFAND-P000495

Eigenvals. $n + t$, $n = -2, -1, 0, 1, \ldots$

$\eta(0)$ — the "asymmetry" of the operator:

$$\sum_{\lambda_i > 0} 1 - \sum_{\lambda_i < 0} 1.$$

S. I. Gelfand. Account of Manin's last lecture:

p-adic integration (Mazur, Swinnerton — Dyer)

Uncertain reading. The middle letters of the first part of the name run together. Provisionally read as Swinnerton.

Original source details: GELFAND-P000495

GELFAND-P000496

Z_p — integral p-adic numbers, a fn. is defined on them. It is to be integrated.
(p-adic val.)
lit. L. G. Schnirelmann, Dwork, Grauert, Krasner, Tate
$Z_p \supset A_i = p^i Z_p$ — syst. of neighb. of 0. The measure is def. on sets $a + A_i$
$\mu(a + A_i)$ — a p-adic number (not nec. integral).

$$a \in Z_p \quad \text{or} \quad \underline{a \in Z_p/p^i}$$

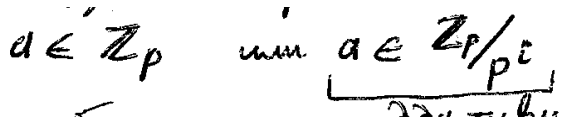


Diagram and typed description. An open-top bracket runs under the second membership expression.

Original source details: GELFAND-P000496

There must be finite additivity; reduces to the cond.:

$$\mu(a + A_i) = \sum_{t=0}^{p-1} \mu(a + tp^i + A_{i+1})$$

2 examples: 1) invar. measure: indep. of a, but dep. on i. Then

$$\mu(a + A_i) = p^{-i} \qquad \text{(only possibility).}$$

The measure of a small set is large! — far from 0

$$\left(\left\| \frac{a}{b} \right\| = p^{-n}, \quad \text{where } \frac{a}{b} = p^n \frac{a_1}{b_1}, \quad a_1, b_1 \text{ rel. prime to } p \right).$$

2) More important example. (Leopoldt — Kubota measures).

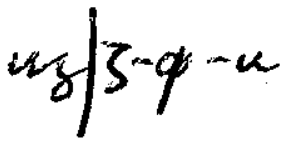
2 parameters n and a

$$(n, p) = 1, \qquad 0 \le a \le p^i - 1$$

$$\mu_n(a + A_i) = \left[\frac{na}{p^i} \right] - \frac{n-1}{2}$$

$$\mu_2(a + A_i) = \begin{cases} +1/2, & \text{if } a > \dfrac{p^i}{2} \\ -1/2, & \text{if } a < \dfrac{p^i}{2} \end{cases}$$

It is addit. !! The measure of $\forall$ set is bounded
(it is obtained by the Mellin transf. from the | ζ-fn. of Leopoldt — Iwasawa)



Uncertain reading. A vertical stroke appears between “from” and ζ. Its function is unresolved.

Original source details: GELFAND-P000496

$$Q_p \ni f(x), \quad x \in Z_p \qquad \int_{Z_p} f(x)\mu(x) \qquad \text{in Riemann's sense}$$

R_m — representatives of $Z_p/p^m Z_p$ (1 from $\forall$ class)

This is a partition. Form

$$S(R_m) = \sum_{a \in R_m} f(a)\mu(a + A_m)$$

they converge under conds. on the measure and the fn.

Moderately growing measure — if it grows more slowly than the invariant one

$$\lim_{m\to\infty} \max |\mu(a + A_m)|p^{-m} = 0$$

f satisfies the Lipschitz cond. if

$$|f(a) - f(b)| < Cp^{-m}, \quad \text{if } |b - a| < p^{-m}$$

(in partic., if given by a conv. power series)

If the measure grows moderately and the fn. satisfies the Lipschitz cond., then $\exists$ a limit, called the integral.

GELFAND-P000497

If we consid. the ordinary $\zeta(s) = \sum 1/n^s$. Then

$\zeta(1 - 2k)$ — a rat. number $\forall k > 0$ (expr. through Bern. numbers).

Uncertain reading. The crossed-out entry in the left margin resembles a sum. An s is visible on the right. The full reading is unresolved.

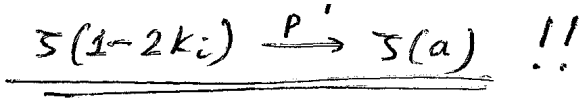
Original source details: GELFAND-P000497

Let k_i — ~~p-adic numbers~~ be integers and in the p-adic sense

$$k_i \longrightarrow a \in Z_p.$$

Iwasawa — Leopoldt:

$$\zeta(1 - 2k_i) \xrightarrow{p} \zeta(a) \quad !!$$



Manuscript excerpt. The right-hand side reads $\zeta(a)$, with p over the arrow.

Original source details: GELFAND-P000497

We obtain a cont. fn. on Z_p —

Uncertain reading. A short word is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000497

p-adic var. It is the Mellin transf. of the measure of example 2.

A. Gabrielov. Dynkin diagr. of isol. sing. of fns.

$f(x)$ — complex-analytic in a neighb. of 0, 0 — an isol. sing.

Ball of rad. r_0 (small). Consid. $f(x) = \delta, \quad |x| < r_0$

Choose also δ.

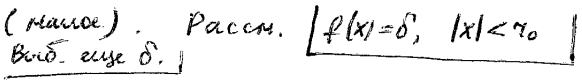
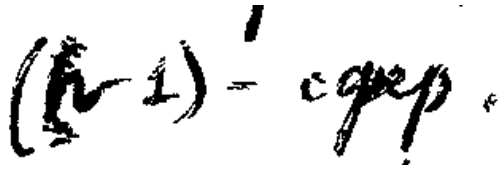


Diagram and typed description. Separate brackets mark the formula and the note "Choose also δ."

Original source details: GELFAND-P000497

For small δ this is a complex set of dim. $(n-1)$, homot. equiv. to a bouquet of $(\ldots - 1)$-spheres.



Uncertain reading. The first letter in the sphere dimension is heavily retraced. Possible readings are $(n-1)$ and $(k-1)$. The preceding manifold dimension reads $(n-1)$.

Original source details: GELFAND-P000497

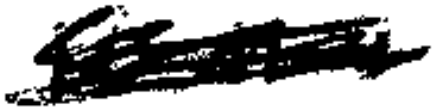
и van. cycles.

Uncertain reading. The heavily retraced initial mark resembles «и» ("and").

Original source details: GELFAND-P000497

GELFAND-P000498

9/IV-73 yr.



Uncertain reading. A short entry to the right of the date is crossed out. Possibly «Сем.» ("Sem.").

Original source details: GELFAND-P000498

S. I. Gelfand. (account of lectures by Yu. I. Manin)

$Q_p \supset Z_p$,

{A measure μ on Z_p with vals. in the fin.-dim. space W over Q_p.}

$$\mu : U \longmapsto \mu(U) \in W, \qquad U \subset Z_p$$

$$\mu(U_1 \cup U_2) = \mu(U_1) + \mu(U_2).$$

Original source details: GELFAND-P000498

Manuscript excerpt. The additivity formula in the manuscript.

In Q_p a basis of open-and-closed sets $A_n = p^n Z_p$. $\quad U = (a + A_n)$.

Original source details: GELFAND-P000498

Uncertain reading. The heavily retraced letter before the subscript p has a rounded part and a descending stroke. Provisionally read as Q_p.

Riemann integral.

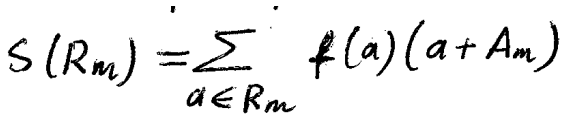
f — a fn. on Z_p with vals. in Q_p.

$m \geqslant 1$,

R_m — representatives of $Z_p / p^m Z_p$. (a finite set).

Riemann sum

$$S(R_m) = \sum_{a \in R_m} f(a)(a + A_m)$$



Original source details: GELFAND-P000498

Manuscript excerpt. The entry immediately after $f(a)$ is $(a + A_m)$.

Theor.: as $m \to \infty$ under certain conds. on the fn. and measure $\exists$ a limit

$$\lim_{m \to \infty} S(R_m) = \int_{Z_p} f(a)\mu(a) \in W$$

Conditions: μ is bounded, and the fn. is loc. constant

Leopoldt — Kubota measures:

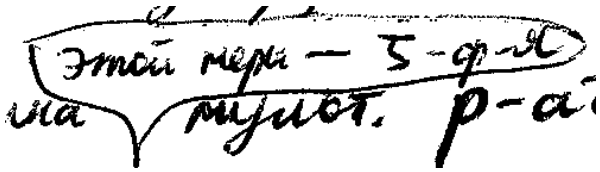
k — an integer, $(k, p) = 1$,

$$\mu_k(a + A_n) = \left[\frac{ka}{p^n}\right] - \frac{k-1}{2}$$

$0 \leq a \leq p^n - 1$

Easy to check (useful) that it is finitely addit.

(Mellin transf.



Original source details: GELFAND-P000498

Diagram and typed description. A callout between “Mellin” and “mult.”.

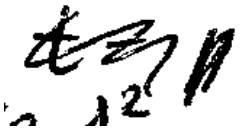
Of this measure — ζ-fn.

mult. p-adic char. on Z_p)

Mult. char.: p-adic number to a p-adic power.

$$\chi_t(a) = e^{t \ln a}, \qquad a = 1 \bmod p.$$

~~t ...~~

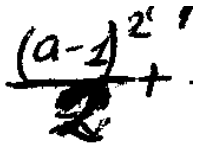


Uncertain reading. The crossed-out entry begins with t. The following signs are illegible.

Original source details: GELFAND-P000498

$t \in Z_p$ (for convergence)

$$\ln a = (a-1) - \frac{(a-1)^2}{2} + \dots$$



Uncertain reading. A 2 is visible in the denominator. The other strokes below the fraction bar are illegible.

Original source details: GELFAND-P000498

2nd cl. of measures using modular forms:

$$H = (z \mid \operatorname{Im} z > 0)$$

Acts

$$SL(2,R)/(\pm 1) \supset \Gamma = SL(2,Z)/(\pm 1)$$

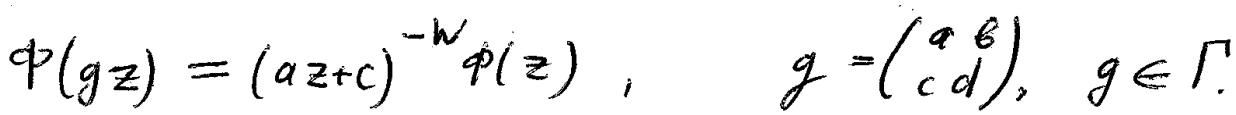
Consid.

$$\varphi(z) = \sum_{n=1}^{\infty} \lambda_n e^{2\pi i n z}$$

(var. invar. decr. fns.)

Autom. form of weight w

$$\varphi(gz) = (az+c)^{-w}\varphi(z), \qquad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad g \in \Gamma.$$



Original source details: GELFAND-P000498

Manuscript excerpt. The factor is written $(az+c)^{-w}$.

GELFAND-P000499

Further condition: φ eigenfn. with respect to

Hecke operators.

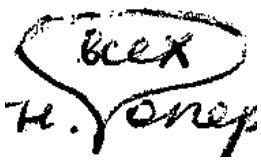
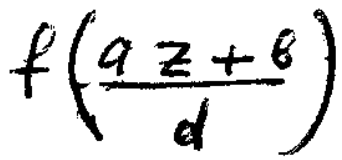


Diagram and typed description. "All" is inserted above the line in an oval callout with a downward point.

Original source details: GELFAND-P000499

$$(T_k\varphi)(z) = \sum_{\substack{ad=k \\ 0\le b\le d-1}} f\left(\frac{az+b}{d}\right)$$



Original source details: GELFAND-P000499

Manuscript excerpt. The summand has f. The left-hand side has φ.

(a, b, d, k — integers)

$T_k\varphi$ — also a modular form

Let $\lambda_1 = 1$, Then

$$T_k\varphi = \lambda_k\varphi \quad _____ \ . \ _____ \quad (\lambda_k \text{ — coeff. of } e^{2\pi ikz}).$$

$$p: \qquad \chi : (\mathbb{Z}/p^n\mathbb{Z})^* \to \mathbb{C}^*$$

(primitive character) (p^n — conductor of the char.)

(on $p^{n\bullet 1}\mathbb{Z}/p^n\mathbb{Z}$ nontriv.)

Original source details: GELFAND-P000499

Uncertain reading. A solid ink patch between n and 1 is represented by a dot. The sign underneath is obscured. Possible readings are $n-1$ and $n+1$.

$$\varphi_\chi(z) = \sum \lambda_n \chi(n) e^{2\pi inz} \quad \text{— also modular.}$$

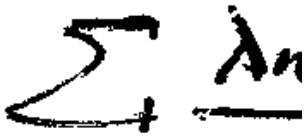
$$L(\varphi, \chi, 1) = C \int_0^{i\infty} \varphi_\chi(z)\, dz$$

Original source details: GELFAND-P000499

Uncertain reading. Before the integral are an arc, a solid patch above, and small loops and strokes below. The entry may be C or a longer expression. The drawing follows the visible strokes.

Has the interpretation

$$L(\Phi,\chi,s)=C\int_0^{i\infty}\Phi_\chi(z)z^{s-1}\,dz=\sum{}_{\!\prime}\frac{\lambda_n\chi(n)}{n^s}\qquad(L\text{-fn.})$$



Uncertain reading. A short cross-stroke touches the lower-right end of the summation sign. It may be a pen terminal or a separate mark. No index can be read.

Original source details: GELFAND-P000499

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p-adic part.

χ — also a loc. const. fn. on integral p-adic numbers

(on the invertible ones as before, elsewhere 0).

Let $\chi(-1)=1$

Aim: $L(\varphi,\chi,1)$ repr. by $\int\chi$ on p-adic numbers.

$$\int\chi(a)\mu(a)=\textcircled{?}\,L(\varphi,\chi,1).$$

Manuscript excerpt. The question mark is circled in the manuscript.

Original source details: GELFAND-P000499

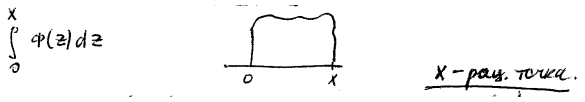
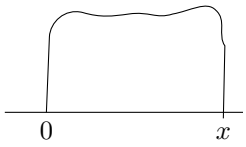


Diagram and typed description. The line from 0 to x runs above the baseline. There are no arrows.

Original source details: GELFAND-P000499

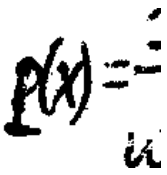
$\int_0^x\varphi(z)\,dz$ 

x — rat. point.

(essentially an $\int$ over a cycle in the quotient space)

$$\mathcal{P}(x)=\frac{1}{\omega}\operatorname{Re}\int_0^x\Phi(z)\,dz\text{ — an alg. integer}\qquad\exists\omega\text{ (transcend.)}$$

from a fixed ext. of Q.



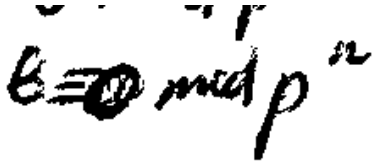
Uncertain reading. The mark before (x) has a round upper loop, long stem and flat foot. Its outline is drawn. Possible readings are P and ℓ.

Original source details: GELFAND-P000499

(in many cases)

$$T^2 - \lambda_p T + p^{w-1} = 0, \qquad \pi, \bar{\pi} \text{ — roots}, \qquad \lambda_p \neq 0 \bmod p.$$

$$\mu(a + A_n) = \lim_{\substack{m\to\infty \\ m\geq n}} \frac{1}{\pi^m + \bar{\pi}^m} \sum_{\substack{b \bmod p^m \\ b\equiv a \bmod p^n}} \left(\frac{b}{p^m}\right)$$



Original source details: GELFAND-P000499

Uncertain reading. The three strokes of $\equiv$ are followed by a heavily retraced loop. Its outline is drawn. Possible readings are 0 and a.

Original source details: GELFAND-P000499

Uncertain reading. The mark before the fraction repeats the upper loop and flat foot. Its outline is drawn. Possible readings are P and ℓ.

lim exists.

$$\text{Final result:} \quad \pi^{k-1} \int_{\mathbb{Z}_p} \chi^{-1}(a)\mu(a) = -\frac{p^k}{\omega g(\chi)} L(\Phi, \chi, 1)$$

GELFAND-P000500

~~Numbers~~

Original source details: GELFAND-P000500

Uncertain reading. The crossed-out word is provisionally read as «Числа» (“Numbers”).

Original source details: GELFAND-P000500

Diagram and typed description. Two lines with transverse hatching run under the name.

I. M. Gelfand.

$$1^k + 2^k + \ldots + n^k = P_{k+1}(n) \quad \text{— Bernoulli polynomial.}$$

$$\sum_{i=1}^{n} f(i), \qquad F'(x) = f(x).$$

$$F(n+1) - F(n) = f(n) + \frac{f'(n)}{2!} + \frac{f''(n)}{3!} + \ldots$$
$$F(n) - F(n-1) = f(n-1) + \frac{f'(n-1)}{2!} + \frac{f''(n-1)}{3!} + \ldots$$
$$F(2) - F(1) = f(1) + \frac{f'(1)}{2!} + \ldots$$

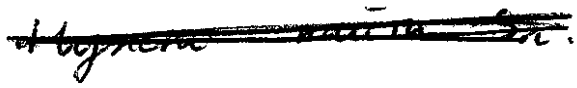
Original source details: GELFAND-P000500

Uncertain reading. The right margin has retraced loops, a tall stroke at the right, and a separate short stroke below and to the left. The outline is drawn. It could be “36”, “38”, or a non-numerical pen mark.

Hence

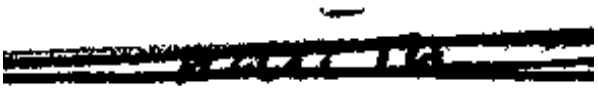
$$F(n+1) - F(1) = S_n + \frac{S'_n}{2!} + \frac{S''_n}{3!} + \ldots \quad \Big\| \quad S_n = \sum_{i=1}^{n} f(i)$$
$$S'_n = \sum_{i=1}^{n} f'(i)$$
$$- \quad - \quad - \quad -$$
$$S_n^{(k)} = \sum_{i=1}^{n} f^{(k)}(i)$$

Similarly:



Original source details: GELFAND-P000500

Uncertain reading. After “Similarly” an initial stroke (“and”?) and a word with a descending loop remain visible (“take” or “we shall”?). At the right are an S- or Σ-like shape and a lower stroke, possibly i.



Original source details: GELFAND-P000500

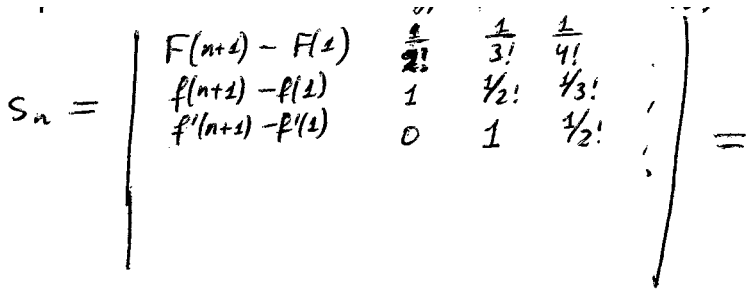
Undeciphered passage. Pen strokes cover the middle of the crossed-out entry. Its reading is unresolved.

$$f(n+1) - f(1) = S'_n + \frac{S''_n}{2!} + \dots$$
$$f'(n+1) - f'(1) = S''_n + \dots$$

We shall consid. this as a syst. of eqs. in $S_n, S'_n, S''_n, \dots$

Hence

$$S_n = \begin{vmatrix} F(n+1) - F(1) & \frac{1}{2!} & \frac{1}{3!} & \frac{1}{4!} & \cdot \\ f(n+1) - f(1) & 1 & 1/2! & 1/3! & \cdot \\ f'(n+1) - f'(1) & 0 & 1 & 1/2! & \cdot \\ & & & & \cdot \end{vmatrix} =$$



Original source details: GELFAND-P000500

Manuscript excerpt. Three rows are written. The dots descend at the right and the bars extend below the written rows.

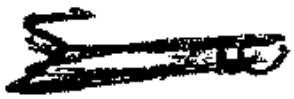
Expand along the 1st column:

$$= B_0\big(F(n+1) - F(1)\big) + B_1\big(f(n+1) - f(1)\big) + B_2\big(f'(n+1) - f'(1)\big) + \dots$$

But: B do not depend on the form of the fn.!

Take $f(x) = e^{tx}$. Then $f'(x) = te^{tx}$. Geom. progr.

~~$\sum$ [...]~~



Original source details: GELFAND-P000500

Undeciphered passage. At the left, $\sum$ and a continuation ending in a loop are crossed out. The strokes after $\sum$ and any limits are illegible.

$$S_n = \frac{e^{t(n+1)} - e^t}{e^t - 1}$$ expand in t

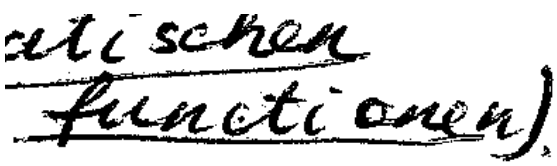
in a power series — we get the coefficients.

Bernoulli numbers in arbitrary fields.

Math. Z., 1898 Hurwitz (Analogue of the Bernoulli numbers for Gaussian numbers)

Hurwitz, Math. Ann., 1899, pp. 196–226

(Über die Entwicklungskoeffizienten der Lemniskatischen functionen).



Uncertain reading. The last word starts on the next line and is spelled with “c”. Its looped, descending initial may be lowercase “f” or uppercase “F”.

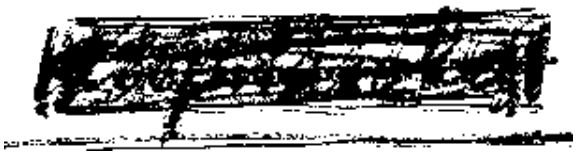
Original source details: GELFAND-P000500

Chapter 7

GELFAND-P000501

[source number: 1]

3/IX-73. Sem. of



Undeciphered passage. The entry between “Sem.” and “Gelfand” is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000501

Gelfand.

Plan:

1. Derived category (Kirillov, Bernstein).

2. Ahlfors — Nevanlinna theory for many variables (according to Grauert). (Henkin, Shabat).

3. K-theory (simplicial version) according to Quillen (Fuchs).

Uncertain reading. The faint words are provisionally read as “simplicial version”.

Original source details: GELFAND-P000501

4. Lichtenbaum’s conjectures on the connection between the ζ-fn. at integers and higher K-theories (Manin).

5. Weil: the ideas of «Герстенцуса».

Uncertain reading. The Cyrillic spelling is provisionally read as «Герстенцуса».

Original source details: GELFAND-P000501

Interlude:

Uncertain reading. The number at the left is heavily retraced, possibly 6 or 8.

Original source details: GELFAND-P000501

2 problems of Serre:

1) $k[t]$, k — finite; $SL_n(k[t])$, $n \geq 3$; $SL_n(k[t, t^{-1}])$, $n \geq 2$

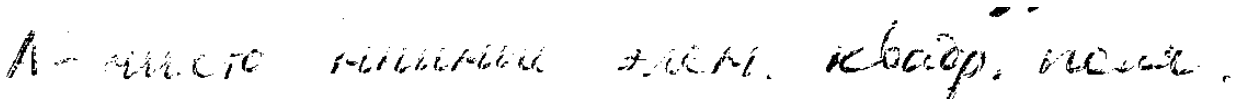
These groups have finitely many generators. Are they finitely presented? (Do there exist generators with finitely many rels. among them.).

2) Coher. group: $\forall$ subgrp. with finitely many gens. is finitely presented. Free and nilpot. are coherent.

Original source details: GELFAND-P000501

Uncertain reading. The faint abbreviation reads “nilpot.”.

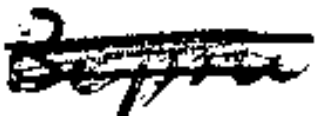
The fund. group of 3-dimensional manifolds is coherent (G. P. Scott, J. Lond. Math. Soc., (2)6 (1973), 437–440). Article $SL_2(A)$ is coher., where Λ — purely imaginary elem. of a quadr. field.



Original source details: GELFAND-P000501

Uncertain reading. The letter in the formula has the crossbar of A. In the explanation its crossbar is indistinct, so the letter is provisionally set as Λ. The rest provisionally reads “purely imaginary elem. of a quadr. field”.

~~also~~

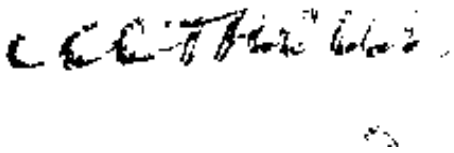


Original source details: GELFAND-P000501

Uncertain reading. The crossed-out word provisionally reads “also”.

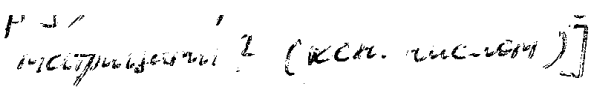
Are $SL_3(\mathbb{Z})$ or $SL_2(\mathbb{Z}[\frac{1}{p}])$ coherent? $GL_n(\mathbb{Q})$?

[are there many rels. among distinct irreducibles? (a fin. number)]



Original source details: GELFAND-P000501

Uncertain reading. The faint abbreviation is provisionally read as “rels.”.



Original source details: GELFAND-P000501

Uncertain reading. The word is provisionally read as “irreducibles”.

[Which fund. groups occur?]

6. Smale: optimization of several functions.

Original source details: GELFAND-P000501

Uncertain reading. The digit is heavily retraced. Its outlines and the adjacent numbering support 6.

7. Brown, Douglas, Fillmore Proper K-theory of several variables (Kirillov).

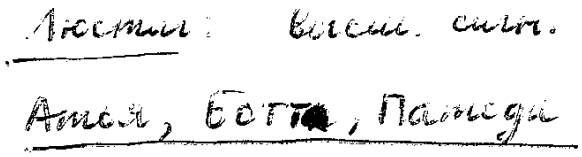
8. Mishchenko: representations in his sense.

9. Lusztig: higher sign. of Novikov J. Diff. Geom. 197(1)

Uncertain reading. The faint name is provisionally read as “Lusztig”.

Original source details: GELFAND-P000501

10. Atiyah, Bott, Patodi on L-functions associated with eqs. (spectral asymmetry, etc.) (Shubin)



Uncertain reading. A heavily retraced round mark follows “Bott”. Its reading is unresolved.

Original source details: GELFAND-P000501

11. Donovan, Freilich a (V. Ponomarev) — a quadruple of subspaces, etc.

Uncertain reading. The small mark is provisionally read as «a».

Original source details: GELFAND-P000501

Modules, etc.

12. Yu. I. Manin.

13. Vershik, Gelfand, Graev — 6 variations on SL_2.

Yu. I. Manin. p-adic measures

Setting: $\mathbb{Q}_p$; K — a fin. ext. X — a compact p-adic analytic set over K. Example: integers in K, their direct prod., etc.

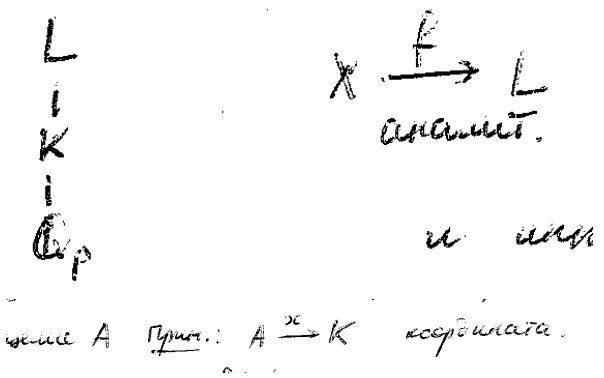


Diagram and typed description. The field tower at the left has no arrows. Beside it are an analytic map and an example of a coordinate.

$$\begin{array}{c} L \\ | \\ K \\ | \\ \mathbb{Q}_p \end{array} \qquad X \xrightarrow[\text{analyt.}]{f} L \qquad \text{integers } A \qquad \underline{\text{Ex.}}: \quad A \xrightarrow{x} K \quad \text{coordinate.}$$

Original source details: GELFAND-P000501

Problem: define $\int f\,d\mu$, i.e. determine a reasonable and inter. class of measures of p-adic type on X.

Uncertain reading. The abbreviation is provisionally read as “inter.”.

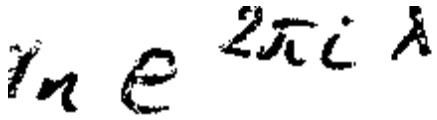
Original source details: GELFAND-P000501

GELFAND-P000502

Motivation: a) there are ζ-fns. $L(s)$ $\quad \mathbb{C} \longrightarrow \mathbb{C}$

$$\sum n^{-s}; \qquad \sum \chi(n) n^{-s}; \qquad \sum a_n n^{-s},$$

where $\sum a_n e^{2\pi i \lambda}$ is an autom. form.



Uncertain reading. The last sign in the exponent is provisionally set as λ.

Original source details: GELFAND-P000502

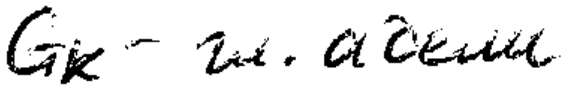
For them, a class of int. formulas with adelic integration

$$L(\chi) = \int_{G_A/G_K} \chi\, d\mu$$

Uncertain reading. The denominator is provisionally set as G_K. There is an additional loop at the base of G.

Original source details: GELFAND-P000502

G_K — princ. adeles

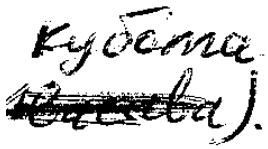


Uncertain reading. The faint note is provisionally read as “G_K — princ. adeles”.

Original source details: GELFAND-P000502

In fact, the argument is a character $L(\chi)$

b) there are p-adic analogues of $L(\chi)$ and $\zeta(s)$ (Leopoldt — ~~Iwasawa~~ Kubota).



Uncertain reading. “Iwasawa” is crossed out. “Kubota” is inserted above.

Original source details: GELFAND-P000502

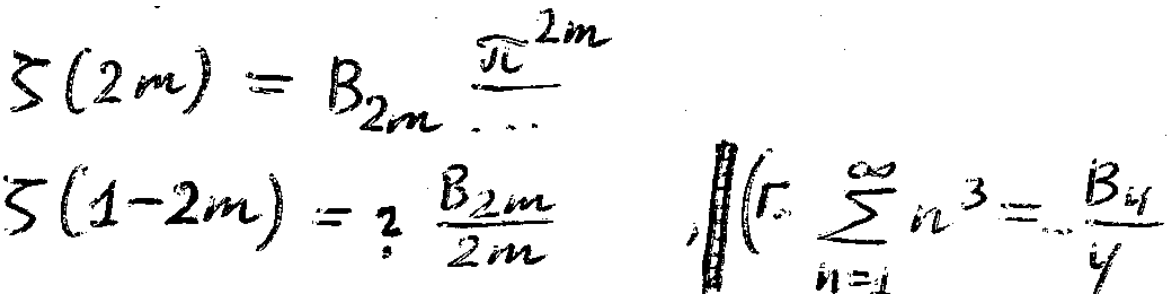
$$\zeta(2m) = B_{2m}\frac{\pi^{2m}}{\dots}$$

$$\zeta(1-2m) = ?\,\frac{B_{2m}}{2m},$$

$$\| \quad \left(\text{G.}\sum_{n=1}^{\infty} n^3 = \frac{B_4}{4}\right.$$

Uncertain reading. The letter after the parenthesis is provisionally read as «Г.». A hatched stroke is at the left.

Original source details: GELFAND-P000502



Original source details: GELFAND-P000502

Uncertain reading. There are faint marks below the line between the equals sign and $\frac{B_4}{4}$. Their reading is unresolved.

even and odd differ:

$$-y'' + p(x)y = \lambda y, \qquad \sum \lambda_n^k$$

integers and half-integers differ)

Still unknown why such a $\sum n^3$ is expr. through B .)

The point of the next plan — pres. of Kronecker's formulas.

Uncertain reading. The abbreviation before "formulas" is provisionally read as «пред.».

Original source details: GELFAND-P000502

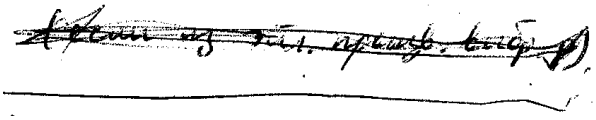
Choose p from the Euler prod.

$$1-2m \longrightarrow \frac{B_{2m}}{2m}(1-p^{2m-1}) \qquad \text{Fix } p!$$

p-adic cont. and even analytic !!! ~~(from the Euler prod. we removed the chosen p)~~.

Uncertain reading. The ending of the last word is provisionally read as «аналитичне».

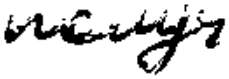
Original source details: GELFAND-P000502



Original source details: GELFAND-P000502

Uncertain reading. The crossed-out entry is provisionally reconstructed. A long stroke runs below the line in the manuscript.

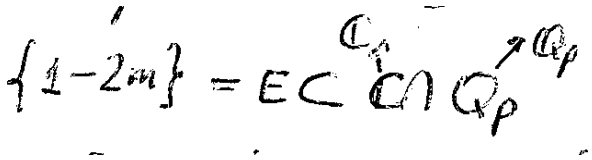
$$\zeta(s)(1-p^{-s}) \qquad \text{between.} \qquad \zeta^{(p)}(s_p)$$
$$\mathbb{C} \longrightarrow \mathbb{C} \qquad\qquad \mathbb{Z}_p \longrightarrow \mathbb{Q}_p.$$



Original source details: GELFAND-P000502

Uncertain reading. The faint word between the two displays is provisionally read as “between”.

there is an intersection of $\mathbb{C}$ and $\mathbb{Z}_p$ (integers), the values there are the same.



Original source details: GELFAND-P000502

Diagram and typed description. The lower left letter is provisionally set as $\mathbb{C}$. A short arrow leads upward from it. The right arrow goes up and to the right.

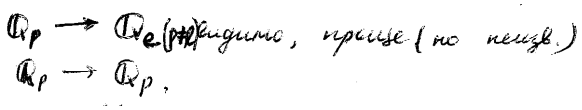
$$\{1-2m\} = E \subset \begin{array}{ccc} \mathbb{C} & & \mathbb{Q}_p \\ \uparrow & \nearrow & \\ \mathbb{C} \cap \mathbb{Q}_p & & \end{array}$$

and they are analytic

In fact, the correct argument is apparently adelic.

$$\mathbb{Q}_p \longrightarrow \mathbb{Q}_\ell \ (p \neq \ell) \quad \text{apparently simpler (but unknown)}$$

Difficult $\mathbb{Q}_p \longrightarrow \mathbb{Q}_p$,



Original source details: GELFAND-P000502

Uncertain reading. The subscript on the right-hand field and the heavily retraced condition in the first line are provisionally set as ℓ and $(p \neq \ell)$.

Was begun by Mazur.

Measures on compact p-adic spaces.

K — fin. ext. of $\mathbb{Q}_p$; A — integers in K; $A \supset \overline{m}$ (max. ideal)

(roots of eqs. with integral coeffs.)

$$\text{Card}(A/\overline{m}) \underset{\text{def}}{=} q.$$

Compact anal. set over K.

$$A^n = \underbrace{A \times \ldots \times A}_{n} \quad \text{— } n\text{-dimensional ball}$$

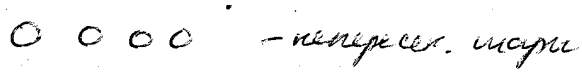
Anal. set: space and a sheaf of fns., loc. structure — a ball.

Theorem (Serre): X — a comp. an. set over K of dim. n. Then

a) $X \simeq rA^n$ (analytic isom.), $r \in \mathbb{N}$

Original source details: GELFAND-P000502

Uncertain reading. The number of balls is provisionally set as r, possibly τ.



Original source details: GELFAND-P000502

Diagram and typed description. Four ovals with a label at the right.

◯ ◯ ◯ ◯ — disjoint balls

b) ~~if~~

$$rA^n \simeq r'A^n \quad \Leftrightarrow \quad r \equiv r' \ \text{mod}(q-1)$$

Original source details: GELFAND-P000502

Uncertain reading. The crossed-out word is provisionally read as “if”. The equivalence sign is encircled.

GELFAND-P000503

[source number: 2]

$$A = \{a \in K \mid a \text{ is integral}\}. \qquad A = \bigcup_{a_i \bmod m} \{a \equiv a_i \bmod m\}.$$

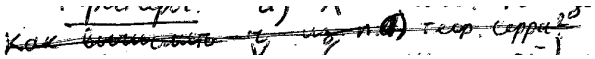
How to compute r?

Uncertain reading. The letter is provisionally set as r. It may be τ.

Original source details: GELFAND-P000503

Examples: a) X — an alg. set over A (a smooth scheme over Spec A).

~~How to compute r from pt. a) of Serre's theor.?~~



Uncertain reading. The crossed-out question is read provisionally. The strokes over the part number are particularly heavy.

Original source details: GELFAND-P000503

Reduction X_0 (closed fibre) — an alg. set over $A/\bar{m}$
over Spec A.

$$\text{Card}\, X_0(A/\bar{m}) = r(X).$$

Example of a scheme: $X = P^1$ or a circle. 2 aff. lines glued by $x = y^{-1}$
Reduction: proj. line over a fin. field of q elems. — there are $q+1$ elems.

$$q + 1 = 2 \bmod (q-1).$$

So 2 balls:

Diagram and typed description. The top of the left contour is blacked out. The arrows of the right contour point towards its upper gap. At the right are "and" and a small round sign.

Original source details: GELFAND-P000503

$$P^1(K) = \mathbb{Q}_p \cup \infty, \qquad pA \cup A^{-1}.$$

b) X — p-adic comp. Lie grp.; it is profinite, $\forall$ factor has an order.
Order: $2^2\, 3^\infty\, 5^1\, 7^0\, 11^\infty \ldots$ — a supernatural number (since they divide one another).

Manuscript excerpt. The stroke left of the exponent belongs to the digit 5. The exponent reads 1.

Original source details: GELFAND-P000503

It turns out ord $X = p^\infty \cdot N$, where $(p, N) = 1$, N — an ord. integer.

$$\underline{\underline{N = r(X)}}$$

SL_2 — $(p-1)$ balls $(= p^2 - 1 \bmod p)$

Original source details: GELFAND-P000503

Manuscript excerpt. The manuscript reads $p^2 - 1 \bmod p$.

Diagram and typed description. The mark between L and K is blacked out. Its shape and direction are unresolved.

L

K

Original source details: GELFAND-P000503

Definition: L-measure on X — an L-valued finitely addit. fn. on open-and-closed subsets of X.

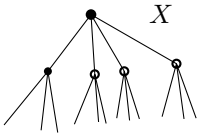
(X — a comp. anal. set).

Uncertain reading. The start of “anal.” is heavily overwritten.

Original source details: GELFAND-P000503

Examples: a) $X = \mathbb{Z}_p$

Diagram and typed description. Four branches descend from the vertex X. Some lower strokes merge.



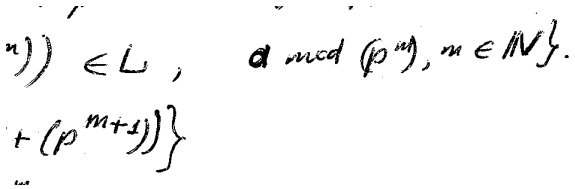
Original source details: GELFAND-P000503

The vertices are open-and-closed subsets.

Construct these measures successively. We obtain a fund. syst. of neighbourhoods:

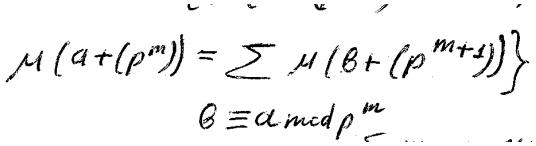
$$\text{Measure on } X \leftrightarrow \{\mu(a+(p^m)) \in L; \quad a \bmod (p^m),\ m \in \mathbb{N}\}.$$

$$\mu(a+(p^m)) = \sum_{\beta \equiv a \bmod p^m} \mu(\beta+(p^{m+1}))\}$$



Manuscript excerpt. Each of the two written lines ends in a closing brace.

Original source details: GELFAND-P000503



Uncertain reading. The summation index has two loops. It is provisionally set as β. A Latin b is possible.

Original source details: GELFAND-P000503

And this applies to any compact subset.

$$\text{b) } \mu(a+(p^m)) = \underbrace{\text{const.}\, p^{-m}}_{\in L} \quad \text{— invar. measures.}$$

There is on L a metric (p-adic absolute value).

$$|\text{const.}\, p^{-m}|_L \longrightarrow \infty \quad \text{as } m \longrightarrow \infty.$$

The smaller the sets, the larger the measure!!

It cannot tend to zero! It would follow from the non-Archim. property that the measure of the whole equals 0.

Invar. measures are bad.

The slowest-growing is bounded.

$$\text{c) } \mu(a+(p^m)) = 1 \text{ if } a < \frac{p^m}{2}, \qquad -1 \text{ if } a > p^m/2$$

Original source details: GELFAND-P000503

Manuscript excerpt. The left value is 1 and the right value is -1. The first threshold uses a stacked fraction and the second a slash.

$0 < a < p^m, \quad (a,p) = 1$ measure on $\mathbb{Z}_p^*$

Undeciphered passage. A short entry between $0 < a < p^m$ and $(a,p) = 1$ is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000503

GELFAND-P000504

Exercise: check additivity. Interference occurs.

Integr. against it — ζ-fns. of Leopoldt — Kubota.

Mazur devised this measure.

Integrals. X, μ — a measure with vals. in L.

Diagram and typed description. In the right margin an arrow points down from L to K.

$$\begin{array}{c} L \\ \downarrow \\ K. \end{array}$$

Original source details: GELFAND-P000504

$X \xrightarrow{f} L$, analyt. fn.

$$\int_X f\,d\mu$$

$$R = \{(U_i), \quad \underset{\text{open-and-closed}}{U_i} \text{ — a partition of } X, \quad a_i \in U_i\}$$

$$S(f,R) = \sum f(a_i)\mu(U_i).$$

$$\int_X f\,d\mu = \lim_R S(f;R)$$

if the limit exists under refinement

Example. $X = \mathbb{Z}_p$, μ — invar. $(\mu(a + (p^m)) = p^{-m}.)$,

$L = \mathbb{Q}_p$. $f = 1$ is integrable

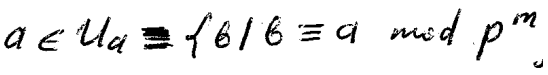
$f = x$ — simple nonconst.

$$\int f\,d\mu \quad \text{does not exist!}$$

$$R = \{0 \le a \le p^{m-1} - 1, \quad a \in U_a \equiv \{\beta \mid \beta \equiv a \bmod p^m\}.$$

Original source details: GELFAND-P000504

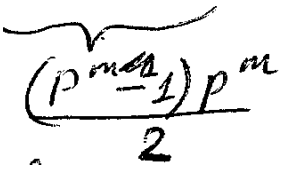
Uncertain reading. An entry resembling -1 is crossed out after the exponent m. The next -1 is part of the bound itself.



Original source details: GELFAND-P000504

Uncertain reading. The two-loop letter in the description of U_a is provisionally set as β. A Latin b is possible.

$$S(f,R) = \frac{1}{p^m} \underbrace{\sum_{a=0}^{p^m-1} a}_{\frac{(p^{m-1}-1)p^m}{2}} \longrightarrow -\frac{1}{2}$$



Original source details: GELFAND-P000504

Uncertain reading. Below the brace, an entry resembling -1 is crossed out after the exponent m of the first p.

But this is not the only limit point.

$$R' = \{p^m \in U_0 \quad \text{the rest as in } R\}.$$

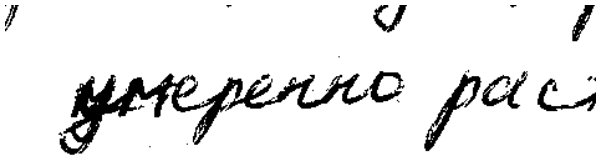
The limit will be different: variation of the fn. on the interval $\approx$ measure of the interv.

hence a cond. on the measure: under refinement the ratio of the measure to the invar. measure must $\longrightarrow 0$ (this is in the case $n = 1$).

Theorem: let X be one-dimensional over $\mathbb{Q}_p$

1) μ — a measure on X with vals. in L, moderately growing, i.e.

$$U \longrightarrow \underset{\text{point}}{\bullet} \Longrightarrow \left|\frac{\mu(U)}{\mu_{\text{inv}}(U)}\right|_L \longrightarrow 0$$



Uncertain reading. The beginning of "moderately" is heavily retraced.

Original source details: GELFAND-P000504

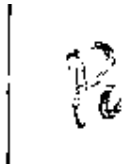
(μ_{inv} — the usual inv. meas. on $\mathbb{Z}_p$).

2) $f : X \longrightarrow L$ satisfies the Lipschitz cond. with exponent 1

Then $\int f\, d\mu$ exists.

|| Next time: concrete measures and mod. forms.

GELFAND-P000505



Uncertain reading. The entry beyond the right border is cut off by the edge of the scan. Its reading is unresolved.

Original source details: GELFAND-P000505

10.9.73. Gelfand Sem.

Manin (continued).

X — 1-dimensional p-adic space;

μ — K-measure on X (K — a fin. ext. of $\mathbb{Q}_p$)

$f : X \longrightarrow K$

$$R: \quad \text{Riemann data: } \{X = \bigcup_{i=1}^{n} U_i\}, \quad a_i \in U_i\}.$$

Uncertain reading. A crossed-out closing brace is visible after U_i.

Original source details: GELFAND-P000505

$$S(f,R)=\sum \mu(U_i)f(a_i).$$

Theorem:

$\exists \int_X f\,d\mu$, if

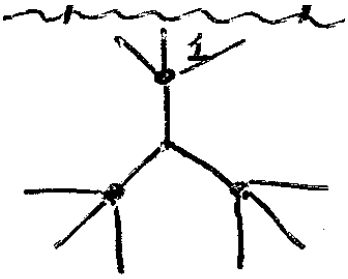
a) f is Lipschitz with exp. 1

b) $(U \mapsto \text{point}) \Longrightarrow \left|\frac{\mu(U)}{\mu_{\text{inv}}(U)}\right| \longrightarrow 0.$

Example. $X=\mathbb{Z}_p$;

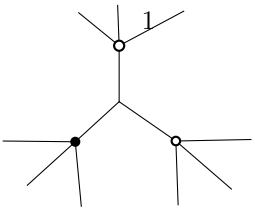
$$\mu(a+(p^m))=\begin{cases}-1 & \text{if } 0\le a<\dfrac{p^m}{2}\\ +1 & \text{if } \dfrac{p^m}{2}<a<p^m.\end{cases}$$

Tree of open-and-closed subsets:

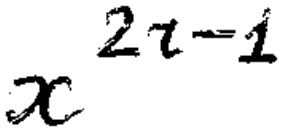


Original source details: GELFAND-P000505

Diagram and typed description. Three strokes extend upward from the top circle. There are two three-ray forks below. A 1 is written at the upper branch.



$f(x)=x^{2r-1}$ (for $f(x)=x^{2r}$ one gets 0).



Original source details: GELFAND-P000505

Uncertain reading. The letter resembles τ and is provisionally set as r.

$$\int_{\mathbb{Z}_p} x^{2r-1}\,d\mu=(-1)^r(1-2^{-r})\frac{B_{r-1}}{2r}$$

$$\mathbb{Z}_p=\bigcup_{0\le a<p^{m-1}}\{a+p^m\mathbb{Z}_p\}.$$

Original source details: GELFAND-P000505

Manuscript excerpt. In the upper bound, -1 is written at the level of the exponent m.

R

Original source details: GELFAND-P000505

Uncertain reading. The sign in the left margin is heavily retraced. Provisionally read as R.

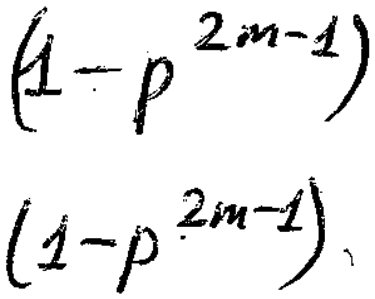
$$S(f,R) = -\sum_{0\le a<\frac{p^m}{2}} a^{2r-1} + \sum_{p^m>a>\frac{p^m}{2}} a^{2r-1}$$

There are Voronoi formulas interpreted as evaluations of integrals of this kind,

$$\int_{\mathbb{Z}_p^*} = \int_{\mathbb{Z}_p} - \int_{p\mathbb{Z}_p}$$

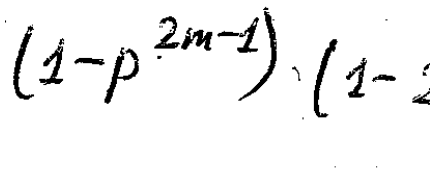
$$\int_{\mathbb{Z}_p^*} x^{2r-1}\,d\mu = (1-p^{2m-1}) \int_{\mathbb{Z}^p}$$

$$= (-1)^r(1-p^{2m-1})\,,(1-2^{-r})\frac{B_{r-1}}{2r} =$$



Manuscript excerpt. The exponent is written as $2m-1$ in both factors.

Original source details: GELFAND-P000505



Uncertain reading. A faint curved stroke between $(1-p^{2m-1})$ and $(1-2^{-r})$ is provisionally set as a comma.

Original source details: GELFAND-P000505

Manuscript excerpt. The p is written to the upper right of $\mathbb{Z}$.

Original source details: GELFAND-P000505

(computed by the Mellin transf. of characters)

$$= (\ldots)\zeta_{(p)}(1-2r)$$

GELFAND-P000506

[source number: 5]

Consider $X = \mathrm{Hom}(\mathbb{Z}_p^*; K^*)$.

Uncertain reading. The mark above K merges with the parenthesis. Provisionally read as a star.

Original source details: GELFAND-P000506

$$\mathbb{Z}_p^* = (\mathbb{Z}/p\mathbb{Z})^* \times (1+p)^{\mathbb{Z}_p}.$$

$\chi \in X \Longrightarrow \chi = \chi_0 \cdot \chi_1$, where

$$\chi_0 = 1 \quad \text{on } (1+p)^{\mathbb{Z}_p}.$$
$$\chi_1 = 1 \quad \text{on } (\mathbb{Z}/p\mathbb{Z})^*.$$

$$\forall u \in \mathbb{Z}_p^*: \quad u = \varepsilon \cdot (1+p)^a; \qquad \text{where } \varepsilon^{p-1} = 1, \quad a \in \mathbb{Z}_p$$

Therefore

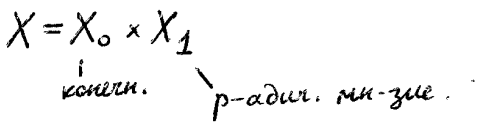


Diagram and typed description.
$X = X_0 \times X_1$
finite
p-adic set.

Original source details: GELFAND-P000506

Thus: X — p-adic set

$$\chi_1(1+p) = 1+t, \qquad t \in \overline{m}. \qquad \text{(notation: } \chi_{(t)})$$

Hence $X_1 \cong \overline{m}$ (but with a different law of composition)

Manuscript excerpt. The first letter of the Russian word for “but” is heavily retraced.

Original source details: GELFAND-P000506

Theorem. If μ grows moderately (more slowly than an invari. one), then

$$\int_{\mathbb{Z}_p^*} \chi \, d\mu \quad \text{is analytic in } \chi.$$

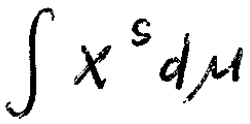
In more detail:

$$\int \chi_{(t)} \, d\mu = F(t) \in K[[t]]$$

converges for $|t| < 1$.

Next: $\int x^s \, d\mu$ is analytic in s for $s \equiv 0 \ (\mathrm{mod}(p-1))$ and for $s = k \,\mathrm{mod}(p-1)$

Manuscript excerpt. The integrand reads x^s.



Original source details: GELFAND-P000506

Modular forms. $\Gamma = SL(2,\mathbb{Z})/(\pm 1)$ acts on

$$H = \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}, \qquad w \equiv 0\ (2), \qquad w \geq 0$$

P_w: Parab. forms of weight $w+2$ on H:

a) $\varphi(z)(dz)^{\frac{w+2}{2}}$ Γ-invariant

b) $\varphi(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z}$

1st nontriv. fn. at $w+2 = 12$ with gaps up to 26 (1-dimensional); then a fin.-dim. space throughout [Crossed out:

Undeciphered passage. The reading under the heavy strikeout is unresolved. A small two-loop stroke sits just below its left end.

Original source details: GELFAND-P000506

]

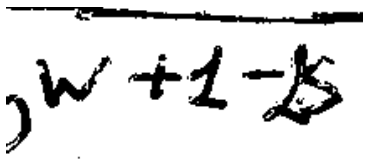
$$w+2 = 12 \quad \Longrightarrow \quad e^{2\pi i z} \prod_{n=1}^{\infty} (1 - e^{2\pi i n z})^{24} = \sum_{n=1}^{\infty} \tau(n) e^{2\pi i n z}$$

In P_w there exists a basis of forms with algebraic integer coefficients

GELFAND-P000507

and such that if $\sum_{1}^{\infty} a_n e^{2\pi i n z}$ — an elem. of this basis, then

$$\sum a_n n^{-s} = \prod_p \frac{1}{1 - a_p p^{-s} + p^{w+1-[2s?]}}.$$



Uncertain reading. The end of the exponent in the first product is overwritten. The $2s$ is provisional. The final mark also resembles k.

Original source details: GELFAND-P000507

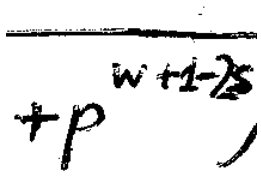
(For each [Crossed out: p-adic.; above the line, circled: parab.] form we will construct a par. measure below)

Manuscript excerpt. The circled insertion "parab." points to the crossed-out word.

Original source details: GELFAND-P000507

Hecke op.: let us see what the values of a are, if

$$\sum a_n n^{-s} = \frac{1}{\left(1 - \lambda_p p^{-s} + p^{w+1-[2s?]}\right)} \left(\sum \dots\right)$$



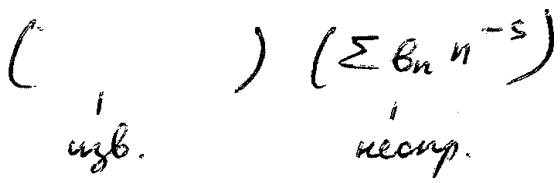
Uncertain reading. The end of the exponent is heavily retraced and is provisionally set as $2s$. The subsequent expansion of this factor shows a separate 2 and s.

Original source details: GELFAND-P000507

i.e.

$$\sum a_n n^{-s} = (\quad)\Big(\sum b_n n^{-s}\Big)$$

known | undet.



Manuscript excerpt. The empty parentheses are labelled “known” and the series “undet.”.

Original source details: GELFAND-P000507

$$1 - \lambda p^{-s} + p^{w+1-2s} = (1 - \alpha_1 p^{-s})(1 - \alpha_2 p^{-s}).$$

We get a lin. condition on the coeffs. a_n.
All together — an infin. matrix.

Uncertain reading. The mark after the period is heavily overwritten. Its reading is unresolved.

Original source details: GELFAND-P000507

This cond. $\Longleftrightarrow$ The coeff. vector is an eigenvector of the Hecke operator.
Let us write its action on the fn. instead of on the Fourier coeffs.

$$(\Phi \mid T_p)(z) \underset{\text{def}}{=} p^{w+1}\Phi(pz) + p^{-1}\sum_{k=0}^{p-1} \Phi\left(\frac{z+k}{p}\right).$$

Eigenfns. $\Longleftrightarrow$ extract the p-factor from the corr. Dirichlet series.

$$(P_w \mid T_p) = P_w.$$

Measures: on $\mathbb{Z}_p^*$, assoc. with φ, $\varphi \mid T_p = \lambda_p \varphi \quad \forall p$.
[Crossed out:

Uncertain reading. Possibly $\mathbb{Q}_p$. The reading under the strikeout is unresolved.

Original source details: GELFAND-P000507

] with values in $\{\mathbb{Q}_p(\lambda_1, \ldots, \lambda_p, \ldots)\}_p$

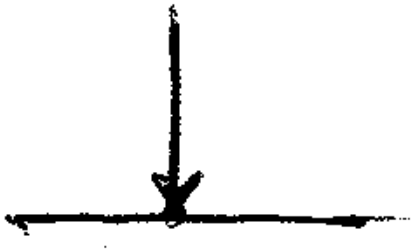
Half-measures with values in [Crossed out:

Original source details: GELFAND-P000507

Uncertain reading. Possibly $\mathbb{Q}$. The reading under the strikeout is unresolved.

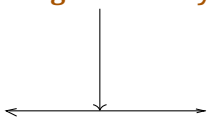
] $\mathbb{C}$.

$x \in \mathbb{Q}$. $\quad P(x) = \int_x^{i\infty} \Phi(z)\, dz$



Original source details: GELFAND-P000507

Diagram and typed description.



Convergence: obvious at the upper end ($e^{2\pi i n z} \to 0$), and unclear at the lower end. From the funct. eq.: it converges at the lower end too, but the dependence on x is discontinuous (from subtle arithmetic). At irr. points it generally diverges. $P(x+1) = P(x)$.

$\lambda_p P(x) = \int_x^{i\infty} \Phi(z) \mid T_p\, dz$ we substitute and get a funct. eq.

$$\lambda_p P(x) = p^w P(px) + \sum_{k=0}^{p-1} \overset{[P?]}{P}\Big(\frac{x+k}{p}\Big)$$

Original source details: GELFAND-P000507

Uncertain reading. A separate sign stands over the argument in the right-hand summand. It is provisionally set as P. Lowercase p is possible.

$$\frac{a}{p^m} \sim a + p^m \mathbb{Z}_p.$$

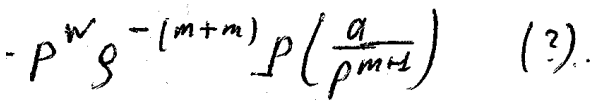
The measure $P\left(\frac{a}{p^m}\right)$ gives almost fin. additivity.

GELFAND-P000508

[source number: 6]

Correction:

$$\mu_\Phi(a + p^m\mathbb{Z}_p) = \varrho^{-m} P\left(\frac{a}{p^m}\right) - p^w \varrho^{-(m+m)} P\left(\frac{a}{p^{m+1}}\right) \qquad (?).$$

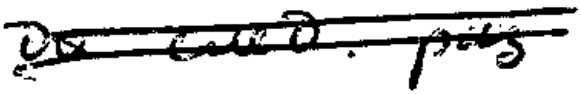


Original source details: GELFAND-P000508

Manuscript excerpt. The second exponent on ϱ reads $-(m+m)$. The last denominator reads p^{m+1}. The $(?)$ is in the manuscript. The first P is heavily retraced.

We get a measure on $\mathbb{Z}_p^*$.

~~until next time~~



Original source details: GELFAND-P000508

Manuscript excerpt. Two strokes cross out the phrase. The last word reads “раз”.

GELFAND-P000509

17.9.73. Gelfand Sem.

Original source details: GELFAND-P000509

Uncertain reading. The second digit of the day is overwritten. A 7 is visible over the earlier strokes.

Yu. I. Manin (continued)

$$P(x) = \int_x^{i\infty} \Phi(z)\, dz$$

Consider the eq. $\varrho^2 - \lambda_p \varrho + p^{w+1} = 0$. and ϱ is any root of it

Proposition. Put

$$\mu_\Phi\big(a + (p^{m+1})\big) = \varrho^{-m} P\left(\frac{a}{p^{m+1}}\right) - p^w \varrho^{-(m+1)} P\left(\frac{a}{p^m}\right).$$

Then this is a half-measure (on $\mathbb{Z}_p^*$ taking values in $\mathbb{C}$ and fin.-additive).

Consider the periods:

of the form

$$\tau_k(\varphi) = \int_0^{i\infty} \varphi(z) z^k\, dz, \qquad 0 \le k \le w.$$

$$L_\varphi(s) = \sum_{n=1}^{\infty} \lambda_n n^{-s} = \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty \varphi(iy) y^{s-1}\, dy$$

$$\tau_k(\varphi) = \frac{k!\, i^{k+1}}{(2\pi)^{k+1}} L_\varphi(k+1)$$

$[1, w+1]$ — crit. strip (funct. eq. relative to it)

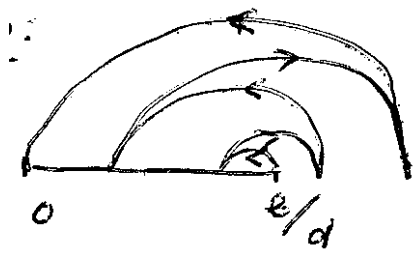
Proposition. $\forall x \in \mathbb{Q},\ P(x) \in \sum_{k=0}^{w} \mathbb{Z} \cdot \tau_k(\Phi)$

Re and Im have a common transcend. factor, dividing by which gives algebraic

$$\int_{b/d}^{i\infty} \Phi(z)\, dz = \int_0^{i\infty} - \int_0^{b/d}$$

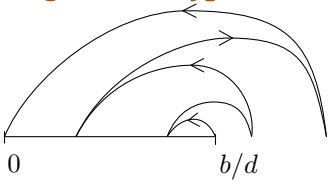
(in neighbourhoods of points of the axis, integrate along geodesics).

Contour:



Original source details: GELFAND-P000509

Diagram and typed description.



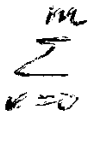
The inner arcs partly overlap. The direction of the fourth arc from the top is unresolved.

from the cont. fraction exp.

$$\frac{b}{d} = \frac{b_m}{d_m}, \quad \frac{b_{m-1}}{d_{m-1}}, \quad \dots, \quad \frac{b_0}{d_0} = \frac{0}{1}$$

$$g_k = \begin{pmatrix} b_k & (-1)^{k-1} b_{k-1} \\ d_k & (-1)^{k-1} d_{k-1} \end{pmatrix} \in SL(2, \mathbb{Z}). \qquad \text{(choose signs).}$$

$$\begin{aligned} \int_0^{b/d} \varphi(z)\, dz &= \sum_{k=0}^{m} \int_{b_{k-1}/d_{k-1}}^{b_k/d_k} \Phi(z)\, dz \\ &= \sum_{k=1}^{m} \int_{g_k(0)}^{g_k(i\infty)} \varphi(z)\, dz \\ &= \sum_{k=1}^{m} \int_0^{i\infty} \varphi(z) \big(d_k z + (-1)^{k-1} d_{k-1}\big)^w\, dz. \end{aligned}$$



Manuscript excerpt. The first sum reads $k = 0$. The next two read $k = 1$.

Original source details: GELFAND-P000509

This is an integer lin. comb. of $\tau_k(\varphi)$.

Summary:

$$P\left(\frac{b}{d}\right) = \sum F(d_k, d_{k-1})$$

F — polynomial (form) of deg. w

Hence the arithmetical char.

$$P\left(\frac{b}{d}\right) = \sum_{k=0}^{w} A_k \left(\frac{b}{d}\right) \tau_k(\varphi)$$

— separating out the irrationalities

In this expr. addit. is unclear...

Manuscript excerpt. The abbreviation before "addit." reads "выраж." in Russian.

Original source details: GELFAND-P000509

k — even $\Rightarrow$ $\tau_k(\varphi)$ purely imaginary

k — odd $\Rightarrow$ $\tau_k(\varphi)$ real

GELFAND-P000510

Uncertain reading. In the upper-right corner, two strokes and a faint joining stroke sit in an open angle. The mark may read "II" or Cyrillic "И".

Original source details: GELFAND-P000510

Proposition.

$$\exists\, \omega^+, \omega^- \in \mathbb{C}, \qquad \left(\frac{\tau_0}{\omega^+}, \frac{\tau_2}{\omega^+}, \ldots, \frac{\tau_w}{\omega^+}\right), \quad \left(\frac{\tau_1}{\omega^-}, \ldots, \frac{\tau_{w-1}}{\omega^-}\right) \quad -$$

— algebraic (from the field containing the coeffs. of Φ)

Proof.

Linear eqs. for $\tau_k(\Phi)$.

a) Eichler — Shimura; depend only on w:

$$SL(2,\mathbb{Z})/(\pm 1). \qquad \text{Consider } s = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{(generators).}$$

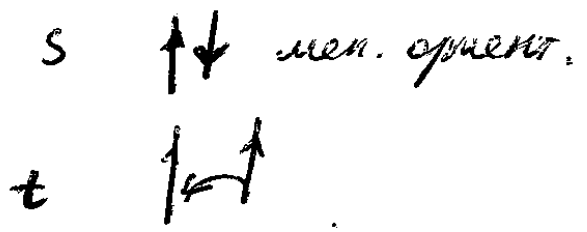
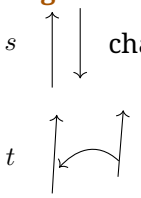


Diagram and typed description.



Original source details: GELFAND-P000510

$$\left(\int_0^{i\infty} + \int_{s(0)}^{s(i\infty)}\right) \Phi(z)\, dz = 0 = \left(\int_0^{i\infty} + \int_{t(0)}^{\ldots} + \ldots\right)$$

$$\underline{\tau_k(\Phi) + (-1)^k \tau_{w-k}(\varphi) = 0} \qquad \text{(also follows from the funct. eq.)}$$

$$\int_0^{i\infty} \varphi(z) \left[z^k + z^{w-k}(z-1)^k + (-1)^k (z-1)^{w-k}\right] dz = 0.$$

This gives the sought lin. eq. for τ_k.

$S_{w+2} = R$-space of forms φ with real coeffs.

$$S_{w+2} \longrightarrow R$$

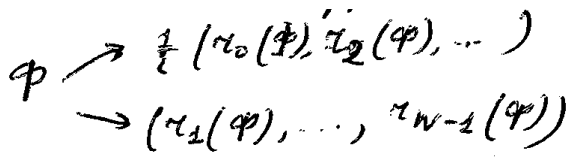
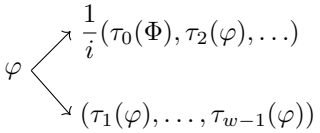
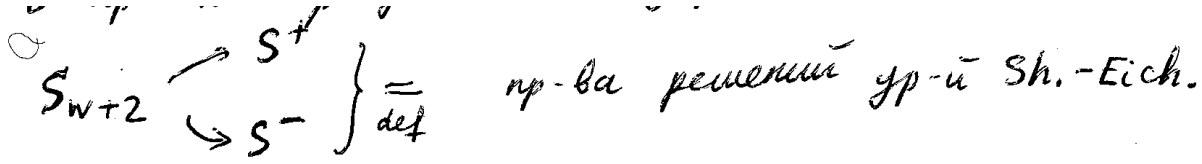


Diagram and typed description.



Original source details: GELFAND-P000510

In the period spaces take the subspaces given by the eqs.



Original source details: GELFAND-P000510

Diagram and typed description. $S_{w+2} \nearrow S^+ \searrow S^- \Big\} \underset{\text{def}}{=}$ spaces of solutions of the eqs. of Sh.-Eich.

Uncertain reading. A faint loop with a short tail is visible at the upper left of the diagram. Its significance is unresolved.

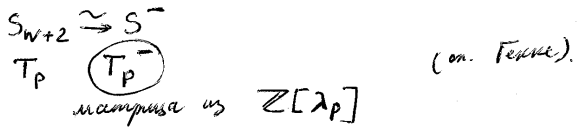
Original source details: GELFAND-P000510

Theorem.

a) $S_{w+2} \xrightarrow{\sim} S^-$

b) $S_{w+2} \hookrightarrow S^+$ embedding codim 1.

Theorem.



Original source details: GELFAND-P000510

Diagram and typed description.

$S_{w+2} \xrightarrow{\sim} S^-$

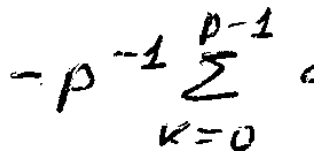
T_p $\quad$ T_p^- (Hecke op.).

matrix over $\mathbb{Z}[\lambda_p]$

Here

$$(\Phi \mid T_p)(z) = p^{w+1}\varphi(pz) - p^{-1}\sum_{k=0}^{p-1}\varphi\left(\frac{z+k}{p}\right)$$

Manuscript excerpt. A minus sign is written before $p^{-1}\sum$.



Original source details: GELFAND-P000510

Eigenfns. of the Hecke op. $\Leftrightarrow$ [Crossed out:

Undeciphered passage. The reading under the heavy strikeout is unresolved.

Original source details: GELFAND-P000510

] factor into an Euler prod.

T_p^- — expression in terms of periods (the lin. form is uniquely det.)

Application.

$$w = 10, \qquad S_{12} = R \cdot e^{2\pi i z} \prod_{n=1}^{\infty} (1 - e^{2\pi i n z})^{24}$$
$$= R \left(\sum_{n=1}^{\infty} \tau(n) e^{2\pi i n z} \right)$$

(Ramanujan fn.)

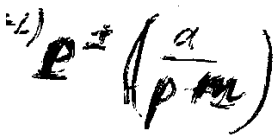
Ramanujan identity

$$\tau(n) \equiv \sum_{d|n} d^{11} \pmod{691}$$

GELFAND-P000511

Measures.

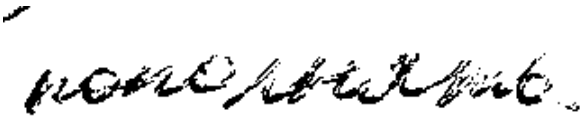
$$P^+(x) = \frac{i}{\omega^+} \operatorname{Im} \int_x^{i\infty} \varphi(z)\, dz,$$
$$P^-(x) = \frac{1}{\omega^-} \operatorname{Re} \int_x^{i\infty} \Phi(z)\, dz$$
$$\mu_\varphi^\pm \left(a + (p^{m+1})\right) \underset{\text{def}}{=} \varrho^{-m} \underline{P}^\pm \left(\frac{a}{p^{m+1}}\right) - p^w \varrho^{-(m+1)} \underline{P}^\pm \left(\frac{a}{p^m}\right)$$



Uncertain reading. Both occurrences of P on the right are underlined. The second P and the exponent on p are heavily retraced. The last term provisionally reads with $\underline{P}^\pm(a/p^m)$.

Original source details: GELFAND-P000511

measures with alg. values, and one can complete.



Uncertain reading. The last Russian verb is provisionally read “пополнять”. Another possible reading is “пополнить”.

Original source details: GELFAND-P000511

Growth. $P^\pm$ have bounded denominators. They can turn up because of ϱ.

$$\varrho^2 - \lambda_p \varrho + p^{w+1} = 0$$

$$\begin{cases} \varrho_1 \cdot \varrho_2 = p^{w+1} \\ \varrho_1 + \varrho_2 = \lambda_p. \end{cases}$$

Typ. case: λ_p copr. to $p \Rightarrow$ one root gives a bounded measure, the other a growing one!
Unknown relation [crossed out:

Undeciphered passage. The reading between “relation” and the next symbol is unresolved.

Original source details: GELFAND-P000511

] λ_p and p.

Integrals (Mellin transf. of p-adic char.)

$$\dots \int_{\mathbb{Z}_p^*} \chi \, d\mu_\Phi^\pm$$

Manin can do it only for a char. of finite ord. $\chi : \mathbb{Z}_p^* \to K^*$, $\chi^m = 1$ and primitive.

Theorem

$$\dots \int_{\mathbb{Z}_p^*} \chi \, d\mu_\varphi^\pm = \dots L_\varphi(1, \chi),$$

where

$$L_\Phi(s, \chi) = \sum_{n=1}^{\infty} \chi(n) a_n n^{-s}$$

$$\dots \dot{L}_\varphi(1, \chi) = \dots \int_{R_+^*} \varphi_\chi(z) \, dz, \qquad \text{where } \Phi_\chi = \sum \chi(n) a_n e^{2\pi i n z}.$$

Uncertain reading. A separate dot is visible above L in the manuscript. Its meaning is unresolved.

Original source details: GELFAND-P000511

— · —

On Lichtenbaum's conjectures: on the connection of K-groups, alg. fields with ζ-fns. at int. points.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^3} = 2\pi^2 |K_4\mathbb{Z}| \cdot e_5(\alpha_5)$$

For the first time there is a conjecture

α_5 — a generator of $K_5\mathbb{Z}$. (of the free part)

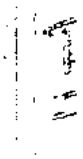
e_5 — a homom. into the real numbers (a certain one)

It is unproved that $\sum_1^\infty \frac{1}{n^3}$ is transcend. Euler's const. also unproved.

Undeciphered passage. The text inside the rectangle is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000511

GELFAND-P000512

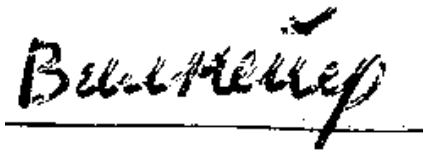


Uncertain reading. Clipped strokes are visible at the upper right. The reading is unresolved.

Original source details: GELFAND-P000512

24.9.73. Gelfand Sem.

Pincus; Douglas, Coburn, Vilneyer and Singer; Helton and Roger Howe. (Howe)



Original source details: GELFAND-P000512

Uncertain reading. The surname after Coburn is provisionally transcribed “Вилнейер” and romanized “Vilneyer”. The middle letters run together.

(interesting work in funct. analysis).

Theorem (Helton, How). close to the work of Krein, Gohberg, Simonenko, etc.

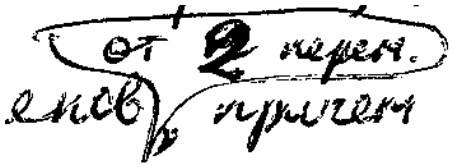
Let B — a bilin. form on 2 polynomials [inserted above the line: of 2 var.], with

$$B(P_1(Q), P_2(Q)) = 0,$$

if $P_1(z), P_2(z)$ — polyn. in one var. z, and Q — a polyn. in any number of var. Then

$$B(R, S) = L(\{R, S\}), \qquad \{R, S\} \text{ — Poisson bracket}$$

L — lin. form.



Original source details: GELFAND-P000512

Manuscript excerpt. The insertion “of 2 var.” is circled and points to the place after “polynomials”.

— · —

Let U, V — self-adj. and commuting mod nuclear operators. Then

$$\mathrm{Tr}\{[R(U,V), S(U,V)]\} = \iint \{R, S\}\, d\sigma$$

R, S — polynomials. The order of U and V in them is immaterial.

— · —

Book by Douglas Banach algebras technique.
$\Rightarrow$ in the p-adic case suitably chosen normed rings are needed in place of diff. operators.

— · —

Kirillov.
L. Brown, R. Douglas, P. Fillmore.
Extensions of Banach algebras (of a nontraditional type).
$\mathcal{K}$ — alg. of comp. operators on a sep. Hilbert space.
Def. 1.

$$0 \to \mathcal{K} \to \mathcal{A} \to C(X) \to 0, \qquad X \text{ — compact}$$

Exts. form a group $\mathrm{Ext}(X) = \mathrm{Ext}_1(X)$.
Another definition gives the same group:
Def. 2.

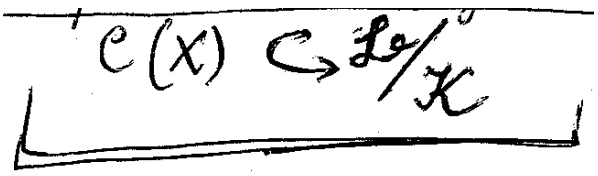


Diagram and typed description.

$$C(X) \hookrightarrow \mathcal{L}/\mathcal{K}$$

Original source details: GELFAND-P000512

1) consider all embeddings.
2) identify embeddings differing by a unit. op. (unit. eq. UAU^{-1}).
3) Embeddings of the form $C(X) \to \mathcal{L} \to \mathcal{L}/\mathcal{K}$ are regarded as triv.
Equiv. classes then form a group.

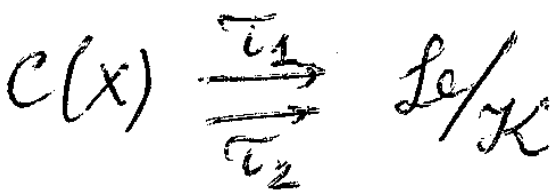


Diagram and typed description.

$$C(X) \underset{\tau_2}{\overset{\tau_1}{\rightrightarrows}} \mathcal{L}/\mathcal{K}$$

Original source details: GELFAND-P000512

$\tau_1 \oplus \tau_2$ already in $H_1 \oplus H_2 \cong H$

Functor

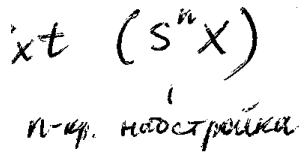
Diagram and typed description.

$$\underline{\text{Top}} \rightsquigarrow \underline{\text{Ab}}.$$

Original source details: GELFAND-P000512

Main result: the functor $\text{Ell}(X)$ reduces to $\text{Ext}(X)$ (answer to a question of Atiyah).

$$\text{Ext}_{1-n}(X) = \text{Ext}\left(\begin{array}{c} S^n X \\ | \\ n\text{-fold suspension} \end{array}\right)$$



Manuscript excerpt. A short stroke under $S^n X$ points to the label.

Original source details: GELFAND-P000512

GELFAND-P000513

[source number: 14]

Th. 1. Exact seq.

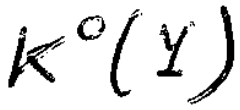
$$\text{Ext}_1(A) \to \text{Ext}_1(X) \to \text{Ext}_1(X/A) \to \text{Ext}_0(A) \to \dots$$

$A \subset X$

Th. 2. There is a pairing

$$K^1 \times \text{Ext}_1 \to \text{Ext}_0$$

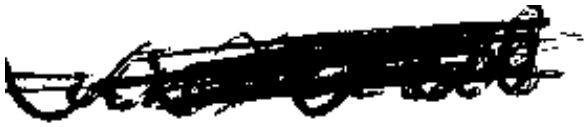
$$K^1(X \times Y) \times \text{Ext}_1(X) \to K^\circ(Y)$$



Uncertain reading. A small oval appears above K before (Y), represented by $\circ$. It may be the superscript zero of $K^0(Y)$.

Original source details: GELFAND-P000513

~~The functor Ext_1 is computed for all~~



Undeciphered passage. The end of the crossed-out line is illegible.

Original source details: GELFAND-P000513

Uncertain reading. The crossed-out verb is provisionally read as “computed”.

Original source details: GELFAND-P000513

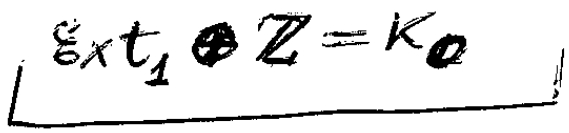


Diagram and typed description.
$\mathrm{Ext}_1 \oplus Z = K_0$

Original source details: GELFAND-P000513

for simpl. complexes.

Uncertain reading. The sign before Z is heavily retraced. Provisionally set as $\oplus$. It could be $\otimes$.

Original source details: GELFAND-P000513

Example of an extension.

$$0 \to \mathcal{K} \to C^*(\text{„}ax + b\text{”}) \to C(\infty) \to 0$$

$$\mathrm{Ext}(\infty) = Z + Z.$$

What is this pair of numbers for a specific op.?

— · —

~~$X = S^1$~~

Manuscript excerpt. $X = S^1$ is crossed out on the left.

Original source details: GELFAND-P000513

$$X = \mathbb{T} = S^1$$

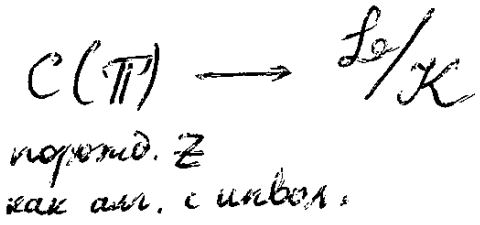


Diagram and typed description.
$C(\mathbb{T}) \longrightarrow \mathcal{L}/\mathcal{K}$
gen. by z
as an alg. with invol.

Original source details: GELFAND-P000513

Let T — the image of z.

$$TT^* = 1 \bmod \mathcal{K}$$

up to unit. equiv. and mod $1 + \mathcal{K}$.

Answer: index (classific.).

Diagram and typed description. ○ A fine loop with a short side stroke appears to the right of the parenthesis. Its meaning is unresolved.

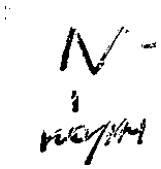
Original source details: GELFAND-P000513

— · —

Th.

$$\left\{ \begin{matrix} N \\ | \\ \text{norm.} \end{matrix} \quad + \mathcal{K} \right\}$$

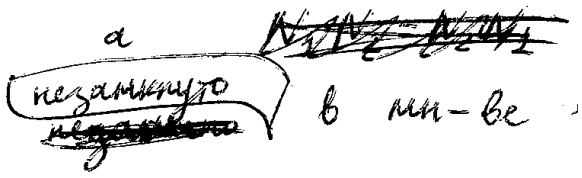
is norm-closed among all op.,



Manuscript excerpt. A short stroke connects N to the label “norm.” below.

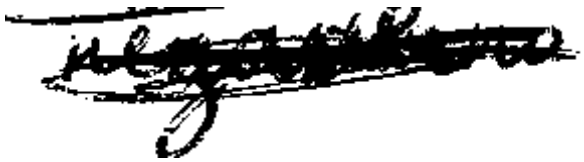
Original source details: GELFAND-P000513

but ~~$N_1N_2 = N_2N_1$~~ pairs $\{N_1 + K_1,\ N_2 + K_2\}$ with cond. $N_1N_2 = N_2N_1$ [inserted and circled at the left: not closed] in the set of all pairs of this kind.



Manuscript excerpt. The equality is crossed out before “pairs”. The insertion at the left is circled.

Original source details: GELFAND-P000513



Uncertain reading. The crossed-out word below the circle is provisionally read as “not closed”.

Original source details: GELFAND-P000513

Next time: Fuchs's talk; pres. on Mishchenko.

Fuchs. Quillen Higher algebraic K-theory I

(Proc. of symp. on K-theory)

Introduction. M — a category (abelian or spec. additive). The $K_i(M)$ are defined.

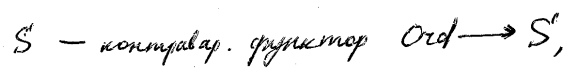
Constructions

QM — cat. of quotients $\Rightarrow$ classif. space BQM $\Rightarrow$

$$\Rightarrow \quad K_i(M) = \pi_{i+1}(BQM).$$

Essential part.

A simpl. object in a category S' — a contravar. functor Ord $\to S'$, where Ord — the can. category of nonnegative integers,

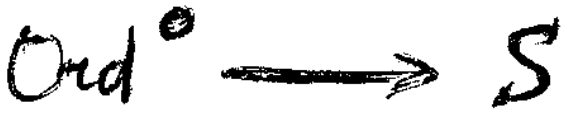


Original source details: GELFAND-P000513

Uncertain reading. Both S glyphs have raised terminal strokes, provisionally represented as primes in S'. They may be flourishes of the letter.

GELFAND-P000514

$$\text{Ord}^{\circ} \longrightarrow S$$



Original source details: GELFAND-P000514

Uncertain reading. There is a small oval above Ord. Provisionally set as $\circ$. It could also be o or 0.

(Functor always means covar. functor).

Ord objects — $0, 1, 2, 3, \ldots$

Morphism [crossed out:

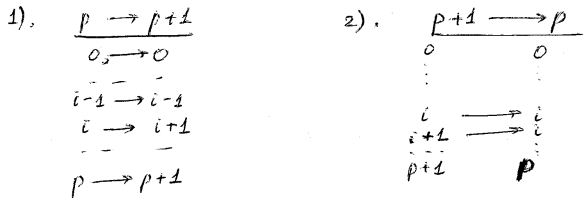
Original source details: GELFAND-P000514

Undeciphered passage. The reading under the strikeout is unresolved.

] $p \to q$ is called a monotonically nondecr. map

$$(0, 1, \ldots, p) \longrightarrow (0, 1, \ldots, q)$$

All are repr. as compositions of morphisms of 2 kinds:



Original source details: GELFAND-P000514

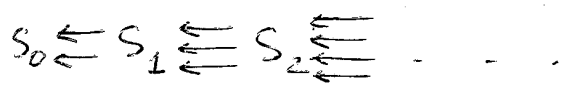
Diagram and typed description.

$$\begin{array}{cccc} & \underline{p \longrightarrow p+1} & & \underline{p+1 \longrightarrow p} \\ & 0, \longrightarrow 0 & & 0 \quad\quad 0 \\ & - \quad - \quad - & & \vdots \quad\quad \vdots \\ 1). & i-1 \longrightarrow i & 2) & i \longrightarrow i \\ & i \longrightarrow i+1 & & i+1 \longrightarrow i \\ & - \quad - \quad - & & \\ & p \longrightarrow p+1 & & \ldots \quad\quad \vdots \\ & & & p+1 \quad\quad p \end{array}$$

To specify a simpl. obj. it suffices to give a seq. of objects

$$S_0, S_1, S_2, \ldots$$

and morphisms



Original source details: GELFAND-P000514

Diagram and typed description.

$S_0 \longleftarrow S_1 \;\substack{\longleftarrow\\ \longleftarrow} S_2 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow} \cdots$

(correspond to embeddings of type 1))

and

Diagram and typed description.

$$S_0 \longrightarrow S_1 \;\substack{\longrightarrow\\ \longrightarrow}\; S_2 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; S_3 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots$$

Original source details: GELFAND-P000514

(corr. to maps of type 2))

(The relations must be satisfied;).

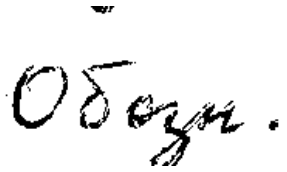
they are easy to write down

S — either the cat. of sets or the cat. of top. spaces.

But one can use groups; smooth manifolds, etc.

Important example: sing. complex of a top. space. S — cat. of sets.

X — top. space. Denote by S_q — all sing. simplices of X

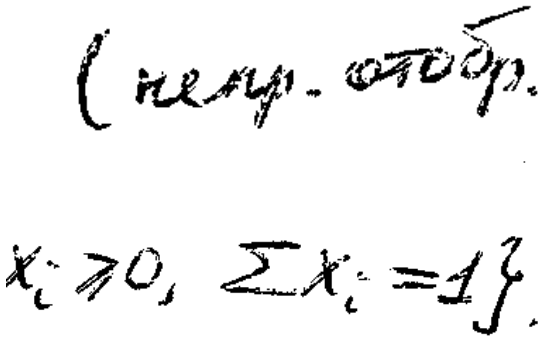


Manuscript excerpt. The Russian abbreviation reads “Обозн.”.

Original source details: GELFAND-P000514

(cont. maps $\Delta^q \to X$, where

$$\Delta^q = \{(x_1, \ldots, x_{q+1}) \in R^{q+1}, \quad x_i \geq 0, \quad \sum x_i = 1\}.$$



Manuscript excerpt. An opening parenthesis precedes “cont. maps”. There is no visible matching parenthesis before the next formula.

Original source details: GELFAND-P000514

$$\exists\ \delta_i : \Delta^{q-1} \longrightarrow \Delta^q$$

canon. map onto the i-th face (vertices ordered)

Uncertain reading. The letter with index i is provisionally set as δ_i. It could be γ_i.

Original source details: GELFAND-P000514

Sing. simplex — composition

$$\Delta^{q-1} \longrightarrow \Delta^q \longrightarrow X$$

Uncertain reading. The initial Russian abbreviation is provisionally read as “Син.”. Its last letter is uncertain.

Original source details: GELFAND-P000514

We get $q+1$ maps

Diagram and typed description.
$S_q \;\overset{\longrightarrow}{\underset{\longrightarrow}{\cdots}}\; S_{q-1}$ (faces).

Original source details: GELFAND-P000514

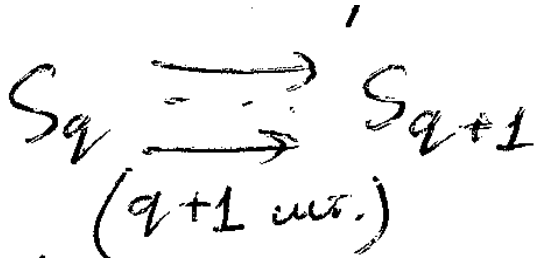


Diagram and typed description.
$S_q \;\overset{\longrightarrow}{\underset{\longrightarrow}{\cdots}}\; S_{q+1}$
($q+1$ in all)

Original source details: GELFAND-P000514

from $\Delta^{q+1} \longrightarrow \Delta^q$ affine projections (i-th identifies the i-th vertex with the $(i+1)$-st).

—·—

GELFAND-P000515

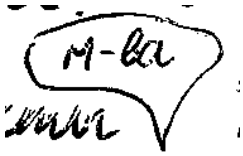
[source number: 15]

Realization.

Let S — the cat. of sets or top. spaces.

A top. space is associated with a simpl. obj. in the cat. of sets or top. spaces.

Elements of [inserted above the line, circled: the set] S_i are called simplices



Manuscript excerpt. The circled abbreviation for “set” points to the place after “Elements”.

Original source details: GELFAND-P000515

The images of arrows from left to right are called degenerate simplices.

The others are called nondegenerate. Arrows from right to left take nondeg. to nondegenerate simplices.

Original source details: GELFAND-P000515

Manuscript excerpt. The manuscript states that the right-to-left arrows take nondeg. to nondegenerate simplices.

Consider $\Delta^q \times S^0_q$

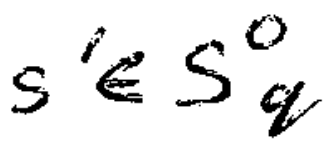
nondeg. simpl. from S_q. (S^0_q — a discrete top. space).

In $\coprod_q \Delta^q \times S^0_q$ (disjoint union) an identification.

$$\Delta^{q-1} \times s, \qquad s \in S^0_{q-1}$$

is identified with part of $\Delta^q \times s'$, $s' \in S^0_q$, when s — the image of s' under the map (i-th)

$$S_{q-1} \xleftarrow{i} S_q \qquad (i\text{-th face})$$



Manuscript excerpt. A prime is visible in the membership condition: $s' \in S^0_q$.

Original source details: GELFAND-P000515

We get a top. space. This is a cell space (the interiors are left unglued).

But: boundary points can be glued together (so it is not a simplicial space).

The same ex. if S — top. spaces. A top. struct. no longer arises, since in $\Delta^q \times S^0_q$ there is the direct product top.

Original source details: GELFAND-P000515

Manuscript excerpt. The manuscript reads “A top. struct. no longer arises”.

Ex.: for a good top. space the realization of the sing. [crossed out:

Undeciphered passage. The reading under the strikeout is unresolved.

Original source details: GELFAND-P000515

] complex is homotopy equiv. to X.

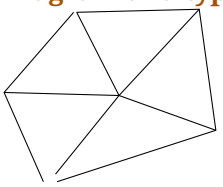
To reconstr. a polyhedron, take for S_i the aff. maps of simplices into it

Manuscript excerpt. The Russian abbreviation before “polyhedron” reads “восстан.”.

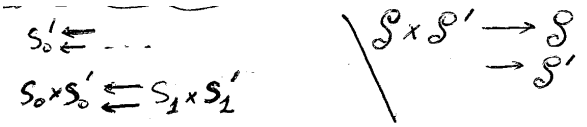
Original source details: GELFAND-P000515

Original source details: GELFAND-P000515

Diagram and typed description.



Realization commutes with multiplication.

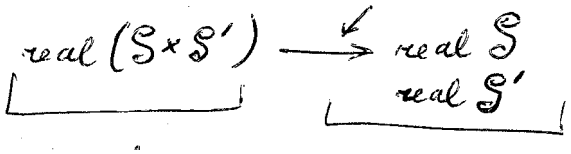


Original source details: GELFAND-P000515

Diagram and typed description.

$$S_0' \longleftarrow \dots \qquad \mathcal{S}\times\mathcal{S}' \longrightarrow \mathcal{S}$$
$$S_0\times S_0' \longleftarrow S_1\times S_1' \qquad \longrightarrow \mathcal{S}'$$

hence

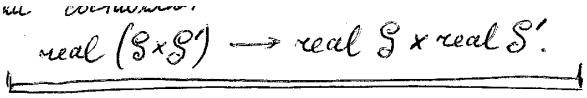


Original source details: GELFAND-P000515

Diagram and typed description.

$$\begin{array}{ccc} \operatorname{real}(\mathcal{S}\times\mathcal{S}') & \longrightarrow & \operatorname{real}\mathcal{S} \\ & & \operatorname{real}\mathcal{S}' \end{array}$$

and together they form



Original source details: GELFAND-P000515

Diagram and typed description.

$$\operatorname{real}(\mathcal{S}\times\mathcal{S}') \longrightarrow \operatorname{real}\mathcal{S}\times\operatorname{real}\mathcal{S}'.$$

Prop. It turns out to be — a homeomorphism.

Example.

Original source details: GELFAND-P000515

Diagram and typed description. ⟼

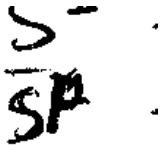
Ordered segment.

S^0 — 2 elements

S^1 — 3 elem. (counting degenerate ones)

$\overline{S^p} = (p+2)$ elem. i-th:

$$[0, i-1] \longrightarrow 0, \qquad [i, p] \longrightarrow 1$$



Original source details: GELFAND-P000515

Uncertain reading. The short horizontal stroke above S is shown as a bar. It may belong to the preceding line. The heavily retraced raised sign at the right is provisionally read as p.

$S_p \times S_p$

Original source details: GELFAND-P000515

Uncertain reading. Faint detached strokes follow the product. Their reading is unresolved.

(next time.)

GELFAND-P000516

§2. Classif. space in a category.

C — a category (small). Associated to it is its nerve NC (simpl. object of the cat. of sets).

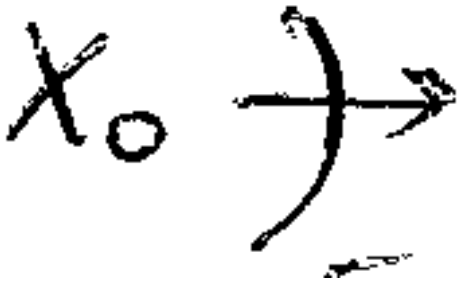
$$NC_q = \{\bullet \quad X_0 \longrightarrow \cdots \longrightarrow X_q\} \qquad \text{set of such diagrams}$$

Original source details: GELFAND-P000516

Uncertain reading. A small heavily retraced mark appears after the opening brace before X_0. The drawing follows its outline. It may be a cancelled sign or an asterisk-like stroke. Its reading is unresolved.

$$NC_q \longrightarrow NC_{q-1}$$

0th

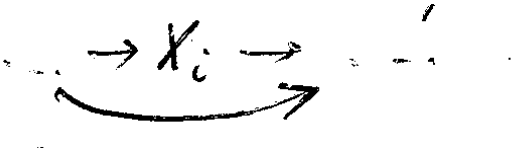


Original source details: GELFAND-P000516

Diagram and typed description. X_0 (discard it).

qth — discard X_q

ith — compose the ith and $(i+1)$st arrows



Original source details: GELFAND-P000516

Diagram and typed description.

$$\cdots \longrightarrow X_i \longrightarrow \cdots$$

$$NC_q \longrightarrow NC_{q+1}$$

one of the X is doubled and the id. morphism is inserted.

~~Nerve~~ Real $NC = BC$ — classif. space of the category.

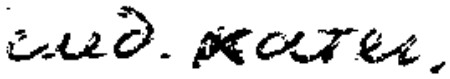
Original source details: GELFAND-P000516

Uncertain reading. The crossed-out word is provisionally read as “Нерв.” (“Nerve”).

Example: the classif. space of the group G is obtained from the following cat.:

Undeciphered passage. The reading under the strikeout after the colon is unresolved.

Original source details: GELFAND-P000516



Manuscript excerpt. The Russian abbreviations after “from” read “след. катег.”.

Original source details: GELFAND-P000516

1 object, with morphisms the elements of G.

$$NC_q = \underbrace{G \times \cdots \times G}_{q}$$

— · —

$G = Z_2$. Simplices:

Nondeg. simpl. — do not contain 1.

One nondeg. simpl. of each dimension.

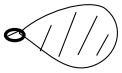


Diagram and typed description. Three long strokes and one short stroke in an oval. Dark overlapping marks at the left edge.

Original source details: GELFAND-P000516

Ex. Wind the 2-dimensional one twice. One gets RP_∞.

Manuscript excerpt. The manuscript reads “2-мерную намот. дважды”.

Original source details: GELFAND-P000516

Manuscript excerpt. The phrase ends with “дважды” (“twice”).

Original source details: GELFAND-P000516

— · —

Let M — a category that is either abelian or a full subcat. of an abelian cat., closed under extensions. Suppose the isom. classes of objects form a set. Then it is equivalent to a small category. Category QM. $\mathrm{Ob}(QM) = \mathrm{Ob}(M)$.

$$\mathrm{Hom}_{QM}(X', X'') \Rightarrow \text{diagrams}$$

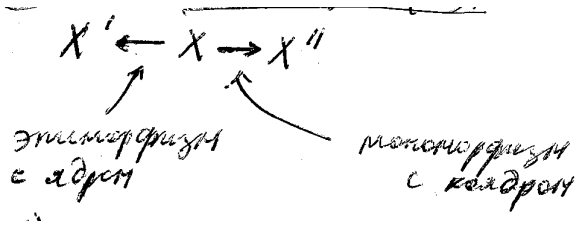


Diagram and typed description.

$X' \longleftarrow X \longrightarrow X''$

epimorphism with kernel

monomorphism with cokernel

Original source details: GELFAND-P000516

This is a category.

Definition.

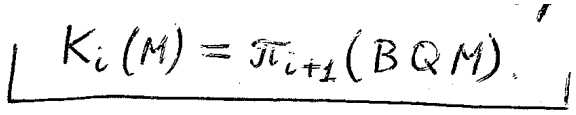


Diagram and typed description.

$K_i(M) = \pi_{i+1}(BQM).$

Original source details: GELFAND-P000516

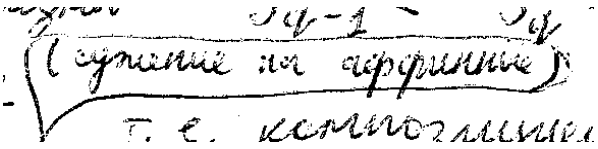
GELFAND-P000517

Continuation of Fuchs's talk (following Marchenko).

Example of realization

S_q consists of $q+2$ elements. Interpret it as the set of aff. maps from Δ^q into the segment Δ^1, compatible with the ordering of the vertices of Δ^q and Δ^1.

The morphisms $S_{q-1} \leftarrow S_q \rightarrow S_{q+1}$ are def. as for the singular simplices of the space $I = \Delta^1$ [inserted above the line: (restriction to affine maps)], i.e. by composition with an embedding and restrictions.



Manuscript excerpt. The circled insertion "restriction to affine maps" points to the place after the formula.

Original source details: GELFAND-P000517

Nondegenerate simplices — 2 0-dimensional $* \rightarrow I$ and 1 1-dimensional one.

$S_q \times S_q$ when is an element of $S_q \times S_q$ nondeg.?

Elem. — a pair (φ, ψ), it is nondeg. when there are no 2 vertices in Δ^q, mapped by φ and ψ [crossed out:

Undeciphered passage. The reading of the crossed-out sign is unresolved.

Original source details: GELFAND-P000517

] into one [crossed out:

Undeciphered passage. The reading of the crossed-out word is unresolved.

Original source details: GELFAND-P000517

] vertex.

$$S_0 \times S_0 \quad \text{— 4 elems.} \qquad * \to (0,0),(0,1),(1,0),(1,1)$$

$$S_1 \times S_1 \qquad \begin{pmatrix}1\\0\end{pmatrix} \to \begin{matrix}\varphi & \psi\\ 0 & 0\\ 0 & 1\end{matrix} \qquad \left(\begin{matrix}0 & 1\\ 0 & 1\end{matrix} \text{ degenerate}\right)$$

$$\begin{pmatrix}0&0\\1&1\end{pmatrix}\begin{pmatrix}0&0\\1&0\end{pmatrix}\begin{pmatrix}0&1\\1&1\end{pmatrix}\begin{pmatrix}1&0\\1&1\end{pmatrix}$$

Uncertain reading. A stroke resembling a closing parenthesis is crossed out after the second column, before “degenerate”.

Original source details: GELFAND-P000517

$$S_2 \times S_2 \qquad \Delta \to \begin{matrix}0&0&0&0\\1&0&0&1\\1&1&1&1\end{matrix}$$

Original source details: GELFAND-P000517

Manuscript excerpt. The manuscript has four columns with no dividers.

Realization of $S \times S$.

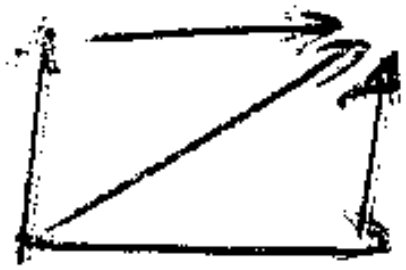
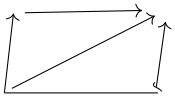


Diagram and typed description. 

Strokes overlap at the right end of the lower side. Its direction is unresolved.

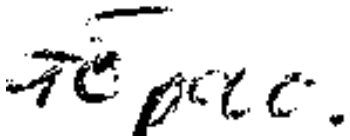
Original source details: GELFAND-P000517

Classif. space of a category.

C — a small cat., NC — the nerve of the category

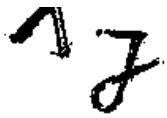
$(NC)_q$ $(= S_q)$ — all diagrams in C of the form $X_0 \to \ldots \to X_q$

the maps — taking compositions, discarding endpoints and inserting id X_j.



Original source details: GELFAND-P000517

Uncertain reading. The Russian abbreviation before “endpoints” is provisionally read “отбрс.”. It could be “отброс.”.



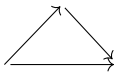
Original source details: GELFAND-P000517

Uncertain reading. The subscript on X is provisionally set as j. It could be a capital J.

$$BC = \text{Real}\, NC.$$

The vertices of BC — objects of C; 1-dimensional simplices are nonidentity morphisms. Δ are stretched over commutative diagrams

Original source details: GELFAND-P000517

Diagram and typed description. 

and so on.

Barycentr. subdiv. of a polyhedron of high deg. is the nerve of some category.

Original source details: GELFAND-P000517

Uncertain reading. The Russian abbreviation is provisionally read “подразд.”. The last letter could be “б”.

Properties:

$$B(C \times C') \cong BC \times BC'$$

$$f : C \rightsquigarrow C' \quad \Rightarrow \quad Bf : BC \to BC'$$

If there is also $g : C \to C'$ and a nat. map from f to g, then $Bg : BC \to BC'$ is connected to Bf by a canonical homotopy.

Let $\{0, 1\}$ — a cat. of 2 objects with one morphism $0 \to 1$

A morphism of the functors f and g — a functor from $C \times \{0, 1\}$ to C'.

GELFAND-P000518

[source number: 21]

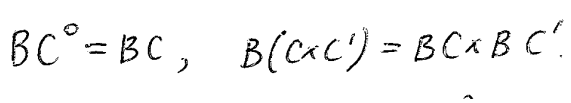
$F : C \times \{0,1\} \to C'$. At the endpoints f, g; morph. $0, 1$ — nat. map.
It determines a map, a homotopy connecting f with g

$$BC \times I = B(C \times I) \to BC'.$$

If C has a terminal or an initial object, then BC is contractible —
— it is a cone.

If $f : C \rightsquigarrow C'$ has a left or right adjoint, then Bf — a homotopy equivalence.

$$BC^{\circ} = BC, \qquad B(C \times C') = BC \times BC'.$$



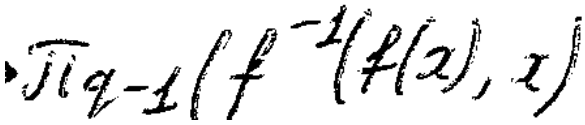
Original source details: GELFAND-P000518

Uncertain reading. A small closed oval is visible above C, represented by $\circ$. A raised o or 0 is also possible.

Analogue of the homotopy sequence of a fibration.

Let $f : X \to Y$ — a Serre fibration. Then

$$\ldots \to \pi_q(X, x) \to \pi_q(Y, f(x)) \to \pi_{q-1}(f^{-1}(f(x), x) \\ \to \pi_{q-1}(X, x) \to \ldots$$



Original source details: GELFAND-P000518

Manuscript excerpt. The parentheses follow the manuscript. One closing parenthesis is visible before the comma.

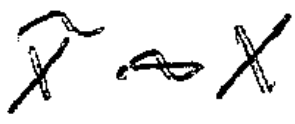
If f does not satisfy the homotopy lifting axiom, then after replacement by an equivalent map it will satisfy it and we get the sequence. Here is the construction:

Let $\widetilde{X} = \{(x, s),\ x \in X,\ s : I \to Y,\ s(0) = f(x)\}$.

$\widetilde{X} \underset{\sim}{\supset} X$. The map $\widetilde{f} : \widetilde{X} \to Y$ is constructed as follows:

$\widetilde{f}(x, s) = s(1), \qquad \widetilde{f}|_X = f$ but now the homotopy lifting axiom holds.

Because $\widetilde{X} \rightsquigarrow X$ we have an exact sequence.



Original source details: GELFAND-P000518

Uncertain reading. Between $\widetilde{X}$ and X a wavy stroke crosses a straight one with a small point at the right. Possible readings are an overlapped $\simeq$ or a retraced wave.

$$\ldots \to \pi_q(X) \to \pi_q(Y) \to \pi_{q-1}(\text{homot. fibre}) \to \pi_{q-1}(X) \to \ldots$$

Now let $g : C \rightsquigarrow C'$, then we have

$Bg : BC \rightsquigarrow BC'$. We would like to construct a "homotopy fibre" of Bg.

This has not been done. There is a surrogate.

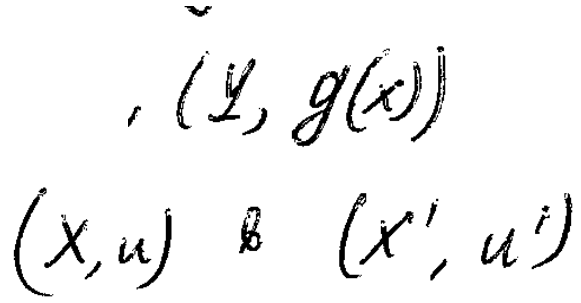
The constr. also depends on some object $Y \in \mathrm{Ob}\, C'$.

Def. the category $Y\backslash g$. $\mathrm{Ob}(Y\backslash g)$ — the pairs (X, u), where

$$X \in \mathrm{Ob}\, C, \qquad u \in \mathrm{Mor}_{C'}(Y, g(x)).$$

A morphism (X, u) to (X', u') — a morphism $\varphi : X \to X'$, compat. with u and u'. There is a nat. map

$$B(Y\backslash g) \to \text{homot. fibre } Bg : BC \to BC'$$



Manuscript excerpt. The argument of $g(x)$ shows a lowercase x. The letters in the pairs in the description of the morphism read as uppercase X and X'.

Original source details: GELFAND-P000518

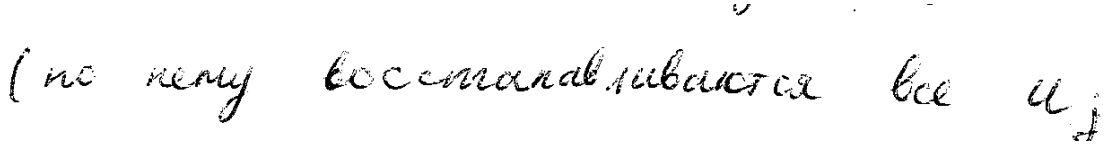
Theorem. Let the functor g be such that

$\forall v \in \mathrm{Mor}_{C'}(Y, Y')$ the resulting map is $Y'\backslash g \xrightarrow{v\backslash g} Y\backslash g$ and $B(v\backslash g) : B(Y'\backslash g) \to B(Y\backslash g)$ is a homot. equivalence (which in general is false). Then $B(Y\backslash g)$ — homot. fibre $BC \overset{Bg}{\rightsquigarrow} BC'$.

GELFAND-P000519

Construct the map $B(Y\backslash g)$ — homot. fibre of Bg.

If there is $(X_0, u_0) \leftarrow \cdots \leftarrow (X_\tau, u_\tau)$ (a simplex of $B(Y\backslash g)$) then there is also $X_0 \leftarrow \cdots \leftarrow X_\tau$ and one only needs to know $Y \to g(X_\tau)$ (from it all the u_j are recovered



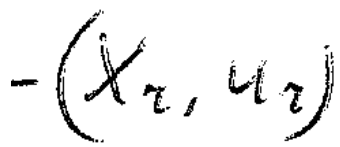
Original source details: GELFAND-P000519

Manuscript excerpt. The manuscript has no closing parenthesis after this line.

$X_0 \leftarrow \cdots \leftarrow X_\tau$ — a simplex of BC of dimension τ. Together with u_τ it defines a simplex

$$g(X_0) \leftarrow \cdots \leftarrow g(X_\tau) \overset{u}{\leftarrow} Y$$

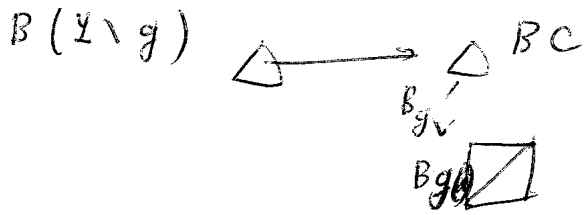
in BC'.



Uncertain reading. The index is provisionally set as τ. A reading t is possible. The same shape appears in the dimensions.

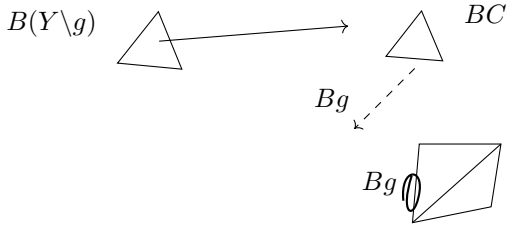
Original source details: GELFAND-P000519

Thus, from a simplex in $B(Y\backslash g)$ we have defined 2 simplices, in BC and in BC' of dimensions τ and $\tau+1$ resp. A simplex from $B(Y\backslash g)$ is mapped as follows: the point of the space BC corresponding to that point (in barycentric coord.) and the straight path in the simplex of BC', connecting Bg of this point to Y:



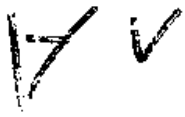
Original source details: GELFAND-P000519

Diagram and typed description.



Overlapping strokes follow Bg on the left side of the quadrilateral. Their reading is unresolved.

This is a homot. equiv. $\forall v$ iff the conditions of the theorem hold.



Original source details: GELFAND-P000519

Manuscript excerpt. The beginning of the second line to the right of the sketch reads $\forall v$.

Remark by Fuchs: probably $Y\backslash g$ should be def. as the inverse limit over the spectrum of all $Y\backslash g$ with different Y.

Mishchenko. (Representations according to Mishchenko)

Λ — a ring with involution, H — a Hilbert space and 2 reprs. T_1 and T_2 of the ring Λ into the algebra of bounded op. in H, F — a Fredholm operator in H and let

$$\forall \lambda \in \Lambda \qquad FT_1(\lambda) - T_2(\lambda)F \in \mathrm{Comp}(H)$$

This system $\{T_1, T_2, F\}$ is called a Fredholm repr. of Λ.

Atiyah considered $\Lambda = C(M)$, M — a manifold (symposium in Tokyo).

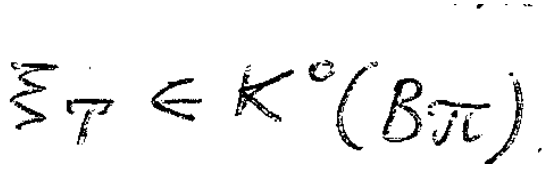
Original source details: GELFAND-P000519

Manuscript excerpt. The parenthetical in the manuscript reads “symposium in Tokyo”.

Constr. with these representations

1) Atiyah (1970) $\Lambda = C(M)$ $\forall$ repr., assign index $T \in K_0(M)$

2) $\Lambda = \mathbb{Z}[\pi] \quad \Rightarrow \quad \forall$ repr., construct $\xi_T \in K^\circ(B\pi)$.



Uncertain reading. In item 2 a small oval above K is represented by $\circ$. It may be the superscript zero of $K^0(B\pi)$.

Original source details: GELFAND-P000519

3) Λ — arbitrary, $A = (a_{ij})$ — a quadr. form on Λ^n $(a_{ij} \in \Lambda)$

$$A^* = A, \quad \exists A^{-1} \quad \Rightarrow \quad \text{Construct } \operatorname{sign}_T A \in \mathbb{Z}.$$

~~4)~~

Undeciphered passage. The crossed-out number before "1)" is provisionally read as "4)". Strokes cross the first digit.

Original source details: GELFAND-P000519

1) $\Lambda = C(M)$, T — a Fredh. repr. of Λ in H, $T = (T_1, T_2, F)$.

H_i — modules over Λ. ind F — a number. Let η — a bundle, $\Gamma(\eta)$ — a proj. module over $C(M)$. Consider

Undeciphered passage. An isolated small mark at the lower edge below the last line. Its reading is unresolved.

Original source details: GELFAND-P000519

GELFAND-P000520

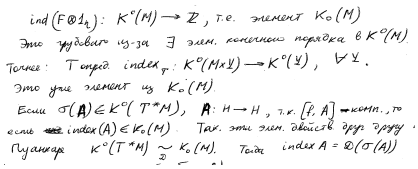
[source number: 22]

$$H_1 \otimes_{C(M)} \Gamma(\eta) \xrightarrow{F\otimes 1_\eta} H_2 \otimes_{C(M)} \Gamma(\eta)$$

$$\operatorname{ind}(F \otimes 1_\eta) : K^\circ(M) \to \mathbb{Z}, \quad \text{i.e. an element of } K_0(M)$$

This is somewhat crude because $\exists$ elements of finite order in $K^\circ(M)$.

More precisely: T determines $\operatorname{index}_T : K^\circ(M \times Y) \to K^\circ(Y)$; $\forall Y$. This is already an element of $K_0(M)$.



Original source details: GELFAND-P000520

Uncertain reading. The small raised ovals over K in these formulas are represented by $\circ$. A possible reading is superscript zero, K^0. The subscript zeros are retained separately.

If $\sigma(A) \in K^\circ(T^*M)$, $A : H \to H$, since $[f, A] \to$ compact, there is

Original source details: GELFAND-P000520

Undeciphered passage. The short crossed-out group of signs before "index" cannot be read.

$\operatorname{index}(A) \in K_0(M)$.

Original source details: GELFAND-P000520

Uncertain reading. The retraced stroke after the commutator has a pointed right end. It is provisionally represented by $\to$. It may be a retraced dash.

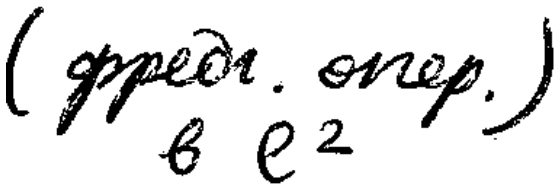
Thus these elems. are Poincaré dual to one another $K^\circ(T^*M) \underset{\mathcal{D}}{\sim} K_0(M)$. Then

index $A = \mathcal{D}(\sigma(A))$

(NB one can constr. the direct image of A)

2) π — an arbitrary discr. group, $\Lambda = \mathbb{Z}[\pi]$ — its group ring, B_π —
— the classif. space. T — a Fredh. repr.: $T = (T_1, F, T_2)$. We need to constr. $\xi_T \in K(B_\pi)$.

$K(B_\pi)$ is understood as $[B_\pi, BU]$, but $BU \simeq \mathcal{F}$ (Fredh. op. [inserted below: in ℓ^2])



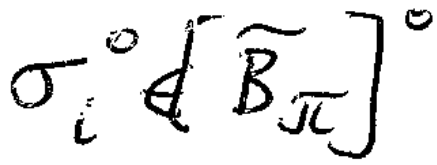
Original source details: GELFAND-P000520

Manuscript excerpt. "In ℓ^2" is written below "Fredh. op.".

$$B_\pi \ni x \to \Phi_x \in \mathcal{F}.$$

Let. $\varphi_x : H_1 \to H_2$ where F from T acted, $\widetilde{B}_\pi$ — the universal cover of B_π, on which there acts π. Take $\widetilde{B}_\pi \times H_i$ with π acting on $\widetilde{B}_\pi$ in the standard way, and on H_i via T_i. Let $\widetilde{B}_\pi$ — cell complexes (even simplicial ones).

Constr. $\varphi_x : \widetilde{B}_\pi \times H_1 \to \widetilde{B}_\pi \times H_2$. Let $[\widetilde{B}_\pi]^k$ — the k-dim. skeleton of $\widetilde{B}_\pi$. Choose one $\sigma_i^0 \in [\widetilde{B}_\pi]^0$ in any orbit of the action of π and set $\Phi_{\sigma_i^0} \overset{\text{def}}{=} F$.



Original source details: GELFAND-P000520

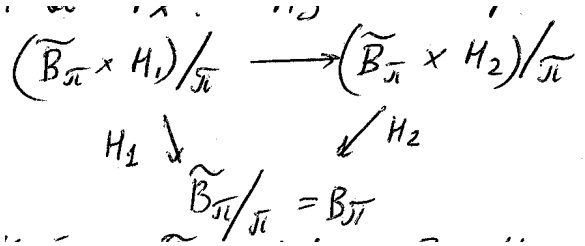
Manuscript excerpt. The $\in$ sign touches the left bracket of the skeleton.

We have:

a) equivariance

b) $\forall x, y \in [\widetilde{B}_\pi]^0 \quad \Rightarrow \quad \varphi_x - \varphi_y \in \text{Comp}$

Extend this to $[\widetilde{B}_\pi]^1$; by the simplicial action of π we can choose one simplex in $\forall$ orbit and extend over the simplex as $p\varphi_1 + q\varphi_2$, $p + q = 1$ with equivariance satisfied. By ind. we obtain the whole family φ_x. From equivariance we have

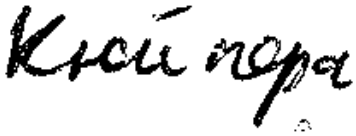


Original source details: GELFAND-P000520

Diagram and typed description.

$$\begin{array}{ccc} (\widetilde{B}_\pi \times H_1)/\pi & \longrightarrow & (\widetilde{B}_\pi \times H_2)/\pi \\ & {\scriptstyle H_1} \searrow \quad \swarrow {\scriptstyle H_2} & \\ & \widetilde{B}_\pi/\pi = B_\pi & \end{array}$$

But by Kuiper $(\widetilde{B}_\pi \times H_i)/\pi \approx B_\pi \times H_i$.



Original source details: GELFAND-P000520

Manuscript excerpt. The manuscript gives the Russian genitive “Кюйпера” for Kuiper.

After identification we have $B_\pi \times H_1 \to B_\pi \times H_2$, which defines $\varphi_x : H_1 \to H_2$.

GELFAND-P000521

15/$\overline{X}$-73 Gelfand's seminar.

Rudakov. On representations of infinite-dimensional Lie algebras.

Consider W_n — the Lie alg. of form. vector fields ([inserted below: cont.] deriv. of the ring $k[[x_1, \ldots, x_n]]$).

Original source details: GELFAND-P000521

Manuscript excerpt. The circled “cont.” is inserted from below before “deriv.”.

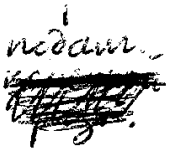
$$\left(\sum_{i=1}^{n} a_i(x_1, \ldots, x_n)\frac{\partial}{\partial x_i}\right)$$

$W_n = \mathcal{L} \supset$ [crossed out] $M \supset M'$.

Uncertain reading. Lower loops are visible under the crossing out before M. This could also be M.

Original source details: GELFAND-P000521

[under M: subalg.]



Undeciphered passage. The crossed-out words under "subalg." cannot be read.

Original source details: GELFAND-P000521

[under M':] an ideal of finite codim. in M.

Suppose we are given a repr. of

$$M/M' \longrightarrow \text{repr. of } M. \underset{\text{induc.}}{\longrightarrow} \text{representation of } \mathcal{L}.$$

Thm.: all irred. reprs. of W_n are obtained this way, with finitely many exceptions.

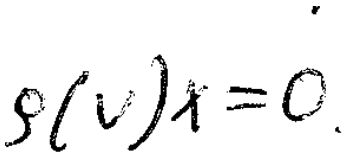
Spaces countable-dimensional discrete (without top.). E

[under «countable-dimensional»: (vect. spaces)]

Repr. $\varrho : \mathcal{L} \longrightarrow \operatorname{Aut} E$, with

$$x \in E,\ \exists V \subset \mathcal{L} \qquad \varrho(v)x = 0. \qquad \text{(loc. const. funcs.).}$$

[under V: nbhd. of 0]



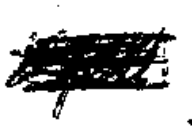
Manuscript excerpt. The argument of ϱ is a lowercase v. The condition on the left uses a capital V.

Original source details: GELFAND-P000521

Filtrations of nbhds. of zero

$$\mathcal{L} = \mathcal{L}_{-1} \supset \mathcal{L}_0 \supset \ldots \supset \mathcal{L}_k \supset \ldots$$

$$\mathcal{L}_k = \left\{ \sum a_i \frac{\partial}{\partial x_i} \text{ [crossed out]}, \text{ deg. } a_i \geqslant k+1 \right\}.$$



Undeciphered passage. The crossed-out notation between the derivative and the comma cannot be read.

Original source details: GELFAND-P000521

Assertion: consider cont. reprs. of $\mathcal{L}_0/\mathcal{L}_k$ and irred. reprs. (and inducing representations).
Exceptions: in the dual spaces to spaces of closed forms
Suppose there is a repr. of the alg. $\mathcal{L}$.

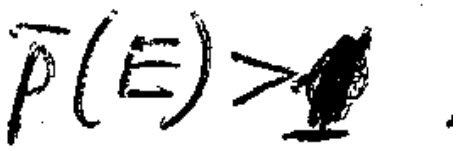
$$x \in E; \quad p(x) = \{\min p : \mathcal{L}_p x = 0\}$$

($\exists$ a bounded cont. repr.?)
Consider

$$\bar{p}(E) = \min_{x \in E,\ x \neq 0} p(x).$$

$$E_0 = \{x \in E,\ p(x) = \bar{p}(E)\} \text{ — } \mathcal{L}_0\text{-submodule}.$$

If $\bar{p}(E) > 1$, then the repr. is induced: $E = \mathrm{Ind}(E_0)$



Uncertain reading. A heavily retraced sign after $>$ has a short detached horizontal stroke at the foot. The stroke supports 1, with the interior shape of the digit obscured.

Original source details: GELFAND-P000521

— · —

Exceptions:

Ω^p — form. forms of deg. p, $\quad d\Omega^p$ — exact forms,

$(d\Omega^p)^*$ — the dual space (with the natural topology)

— · —

There are corresponding results for Hamiltonian ones as well.

— · —

GELFAND-P000522

[source number: 23]

Kirillov. A similar result about the group
Unproved, but well known
M — a manifold, Diff M — the group of diffeomorphisms.

Uncertain reading. The Russian abbreviation is “мн-зие”, probably short for “manifold”.

Original source details: GELFAND-P000522

describe those reprs. that have finite funct. dimension.

$$\mathrm{Diff}\, M \supset H_{x_1,\dots,x_n}$$

[under $H_{x_1,\dots,x_n}$:] keeps fixed $x_1, \dots, x_n$.
Consider cont. reprs. of H/H_0 and induce.
[on the right:] H_0 preserves infin. nbhds. of order n.
Fuchs. (continuation).
$f : C \longrightarrow C'$ a functor
[under C, C':] small cats.

$$Y \in \mathrm{Ob}\, C', \qquad Y\backslash f$$

$$\mathrm{Ob}(Y\backslash f) = \{(X, u),\ X \in \mathrm{Ob}\, C,\ u : Y \longrightarrow f(X)\}.$$

$$v : Y \longrightarrow Y' \quad \rightsquigarrow \quad v\backslash f : Y'\backslash f \longrightarrow Y\backslash f$$

$\exists$ a nat. map

$$B(Y\backslash f) \longrightarrow \text{homot. fibre of the map } Bf : BC \longrightarrow BC'$$

over the point $Y \in BC'$.

$$\longleftarrow \cdot \longrightarrow$$

new cat. of pairs (X, u), where $X \in \mathrm{Ob}\, C$, $u : f(X) \longrightarrow Y$

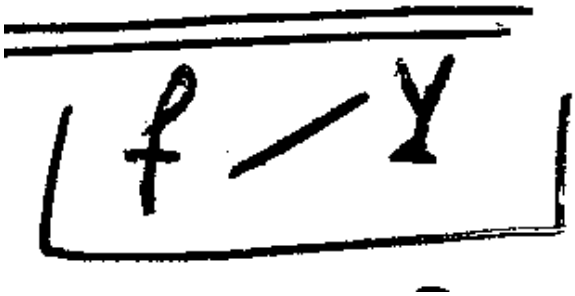
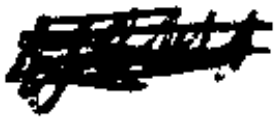


Diagram and typed description. f/Y

Original source details: GELFAND-P000522

another model of the homot. fibre.
The thm. also holds (dual)
Theorem A. This condition holds if all cats. $Y\backslash f$ are contractible (i.e.



Undeciphered passage. The word after "i.e." is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000522

have contractible classif. spaces). Then the conclusion also holds (in particular, the exact homot. seq.) $\Rightarrow Bf : BC \longrightarrow BC'$ induces isos. of homot. groups.

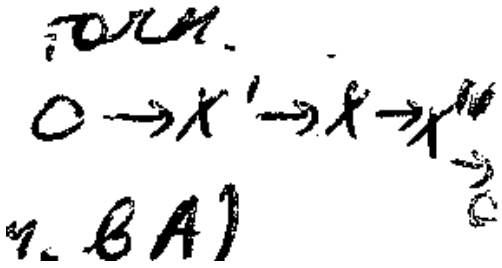
— · —

$\underline{\underline{K_i(M)}}$

M — a full subcat. of the ab. cat. A, closed under extensions, i.e. from

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0, \quad X', X'' \in \mathrm{Ob}\, M \quad \Rightarrow \quad X \in \mathrm{Ob}\, M.$$

If a morphism $i : X' \longrightarrow X$ (from M) is included in a short exact seq., it is called an admissible monomorphism.



Original source details: GELFAND-P000522

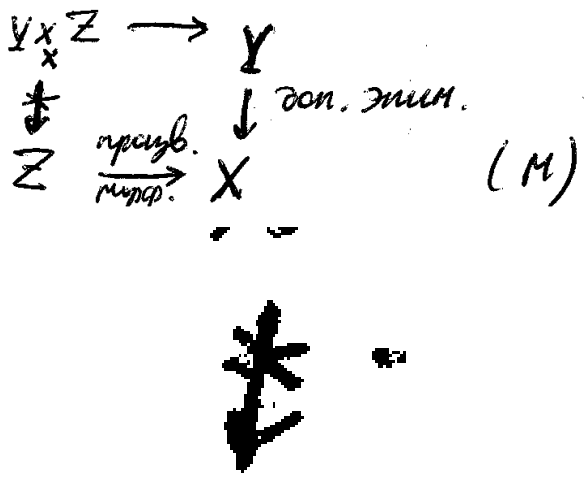
Diagram and typed description.

$0 \to X' \to X \to X''$ with a 0 below X''

In the right-hand insertion the final arrow bends down to an open rounded sign with a short stroke. The sign is provisionally read as 0.

(exactness is understood in A)

Dual def. if a morph. $p : X \longrightarrow X''$ is included in a short exact seq., it is called an adm. epimorphism.



Original source details: GELFAND-P000522

Diagram and typed description.

$$\begin{array}{ccc} Y \times_X Z & \longrightarrow & Y \\ \downarrow & & \downarrow \text{adm. epim.} \\ Z & \xrightarrow[\text{morph.}]{\text{arbitr.}} & X \end{array} \qquad (M)$$

Then $Y \times_X Z \in \mathrm{Ob}\, M$ and $\downarrow$ — adm. epim.

GELFAND-P000523

From M we construct the cat. QM:

$$\mathrm{Ob}\, M = \mathrm{Ob}\, QM.$$

$$\mathrm{Mor}_{QM}(X_1, X_2) = \left\{ X_1 \xleftarrow{\text{a.e.}} X \xrightarrow{\text{a.m.}} X_2 \right\}.$$

Def. equivalence in the obvious way.

([crossed out] X_1 — an admiss. subquotient of X_2)

Undeciphered passage. Two rounded strokes are visible under the cancellation before X_1. The letters or signs remain unresolved.

Original source details: GELFAND-P000523

Def. of composition:

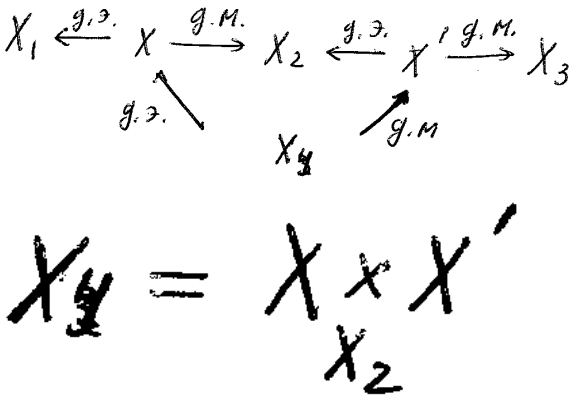
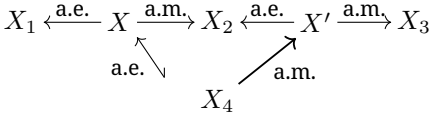


Diagram and typed description.



$X_4 = X \underset{X_2}{\times} X'$.

The subscript 4 in the lower vertex and the product formula overwrites an earlier sign. The earlier sign is unreadable. One short spur branches from the lower end of the left diagonal arrow. It may be part of a second arrowhead.

Original source details: GELFAND-P000523

and one must take

$$X_1 \xleftarrow{\text{a.e.}} X_4 \xrightarrow{\text{a.m.}} X_3.$$

By definition

$$K_i(M) = \pi_{i+1}(BQM, 0),$$

where 0 — any of the zero objects of the category M.

Isom. classes of objects of M must form a set.

Undeciphered passage. The crossed-out entry before “Special case” cannot be read.

Original source details: GELFAND-P000523

Special case: (the main one).

A — an arbitrary ring, PA — the cat. of finitely generated projective left A-modules.

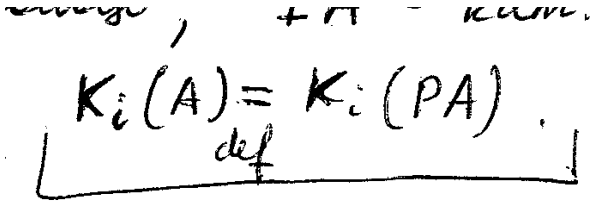


Diagram and typed description.

$K_i(A) = K_i(PA)$.
def

Original source details: GELFAND-P000523

~~Properties of a certain K-functor.~~

Original source details: GELFAND-P000523

Uncertain reading. The crossed-out abbreviation between “Properties” and K is provisionally read as “некотор.” (“a certain”).

$$\varinjlim \mathrm{GL}(n, A) = \mathrm{GL}(A).$$

$$B\,\mathrm{GL}(A)$$

We construct $B_+\,\mathrm{GL}(A)$ with the properties

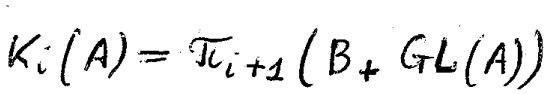
1) it is an H-space,

2) $B\,\mathrm{GL}(A) \longrightarrow B_+\,\mathrm{GL}(A)$

3) the homot. fibre is acyclic

4) surj. map on π_1.

$K_i(A) = \pi_{i+1}(B_+\,\mathrm{GL}(A))$ — the old def. (Quillen’s)



Manuscript excerpt. The subscript $i+1$ in this formula follows the manuscript.

Original source details: GELFAND-P000523

Uncertain reading. A crossed-out entry before “Apparently” may be the beginning of “По-видимому”.

Original source details: GELFAND-P000523

Apparently $BQPA$ is $B_+ \mathrm{GL}(A)$. This is false, but there is a map.

$$B\,\mathrm{GL}(A) \longrightarrow BQPA.$$

Let S_n — a free A-mod. with n gens.

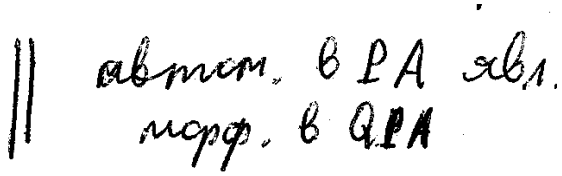
$$S_n \in \mathrm{Ob}(PA) = \mathrm{Ob}(QPA).$$

Elems. of the group $\mathrm{GL}(n, A)$ — the automs. $S_n \longrightarrow S_n$. This lets us def. the functor

$$\mathrm{GL}(n, A) \rightsquigarrow PA.$$

[under $\mathrm{GL}(n, A)$:] corr. cat. (a single obj.)

$$\text{obj.} \longrightarrow S_n$$



Manuscript excerpt. ‖ automs. in PA are morphs. in QPA

Original source details: GELFAND-P000523

Functor $\mathrm{GL}(n, A) \longrightarrow QPA$.

GELFAND-P000524

[source number: 24]

Hence we obtain a map of classif. spaces

$$B\,\mathrm{GL}(n, A) \longrightarrow BQPA$$

It remains to pass to the limit (they are incompatible).

$$\begin{array}{c} QPA \longrightarrow QPA \\ X \longrightarrow X \oplus A \end{array}$$ is a homotopy equivalence.

Diagr.

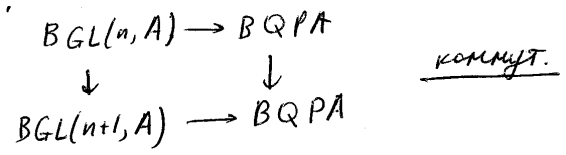


Diagram and typed description.

$$\begin{array}{ccc} B\,\mathrm{GL}(n, A) & \longrightarrow & BQPA \\ \downarrow & & \downarrow \\ B\,\mathrm{GL}(n+1, A) & \longrightarrow & BQPA \end{array}$$ commutes.

Original source details: GELFAND-P000524

and hence the limit and the map

$$B\,\mathrm{GL}(A) \longrightarrow BQPA$$

GELFAND-P000525

22.10.73 Gelfand's seminar.

Undeciphered passage. The crossed-out entry before "Gelfand" cannot be read.

Original source details: GELFAND-P000525

Gelfand. Publicizing work on nonlin. scattering theory. (Straus)

Original source details: GELFAND-P000525

Manuscript excerpt. The surname in parentheses is written “Straus”.

$$U(t)U(s) = U(t+s),$$

Relation for $U(t)f$ as $t \longrightarrow \pm\infty$

Typical features: $E(U(t)f) = E(f)$; E — some funct.

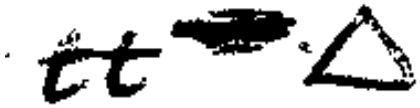
$\exists$ a simpler group $U_0(t)$, such that

$$E\big(U(t)f - U_0(t)f_\pm\big) \underset{t\to\pm\infty}{\longrightarrow} 0 \qquad (\exists f_\pm)$$

S — the relation between f_+ and f_-.

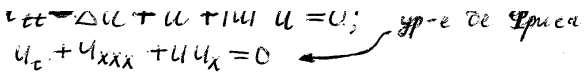
Examples: wave eq.

$$u_{tt} - \Delta u + u + |u|^2 u = 0;$$



Original source details: GELFAND-P000525

Uncertain reading. A thick horizontal trace precedes the Laplacian and the first u is heavily retraced. The horizontal shape supports a provisional minus sign.



Original source details: GELFAND-P000525

Diagram and typed description.

de Vries eq.

$$u_\tau + u_{xxx} + uu_x = 0$$

The first subscript has a short top bar and a curved foot. It is provisionally set as τ, with a single t also possible.

— • —

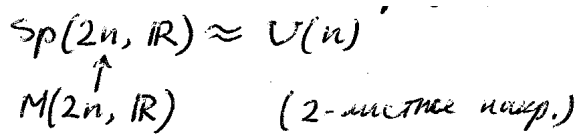
Kirillov.

1. Kostant’s work.

M — a symplectic manifold (a nondeg. closed 2-form ω is given).

TM — a bundle over M with struct. group $\mathrm{Sp}(2n,\mathbb{R})$. It is not simply connected.

Topol.



Original source details: GELFAND-P000525

Diagram and typed description.

$\mathrm{Sp}(2n,\mathbb{R}) \approx U(n)$

$\uparrow$

$M(2n,\mathbb{R})$ (2-sheeted cover)

Sometimes the struct. group lifts to

Undeciphered passage. The crossed-out sign before $M(2n, \mathbb{R})$ cannot be read.

Original source details: GELFAND-P000525

$M(2n, \mathbb{R})$.

Then on M one constructs a spinor bundle (if C_1 is divisible by 2).

M has a special

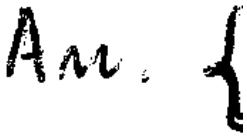
Undeciphered passage. The crossed-out word before "repr. T" cannot be read.

Original source details: GELFAND-P000525

repr. T, called the spinor repr. (or the Weyl repr.)

Alg. $\{p_1, \ldots, p_n, q_1, \ldots, q_n\}$ and the rel.

$$p_i q_j - q_j p_i = \delta_{ij}$$



Manuscript excerpt. The abbreviation at the start of the line reads "Ал.".

Original source details: GELFAND-P000525

(if $+$, then the spin repr. of the orthog. group).

$\mathrm{Sp}(2n, \mathbb{R})$ linearly changes p and q.

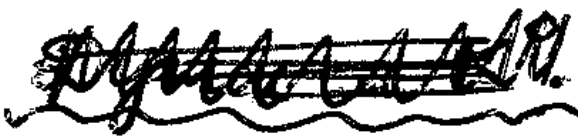
Realiz. $p_i \longrightarrow \frac{\partial}{\partial x_i}$, $q_i \longrightarrow x_i$ in $L^2(\mathbb{R}^n)$.

Automs. of the alg. $\Longrightarrow$ automs. of the alg. of bounded operators. But they are all inner: $A \longrightarrow UAU^{-1}$ (U up to a factor) $\Rightarrow$ a projective representation. The factors reduce to ± 1. The repr. becomes single-valued on the 2-sheeted cover.

GELFAND-P000526

[source number: 28]

2. [crossed out] p-adic measures.



Undeciphered passage. The crossed-out part of the heading cannot be read.

Original source details: GELFAND-P000526

p-adic measures; $\mathbb{Z}_p$ — integers;

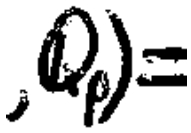
Undeciphered passage. The crossed-out entry after “integers” cannot be read.

Original source details: GELFAND-P000526

Consider $C(\mathbb{Z}_p, K)$, K — a finite ext. of the p-adic field
(cont. funcs. with values in K)

$$\mathrm{Hom}\big(C(\mathbb{Z}_p, K), \mathbb{Q}_p\big) = C(\mathbb{Z}_p, K)' = M(\mathbb{Z}_p, K)$$

(measures with p-adic values)



Manuscript excerpt. The second argument of Hom reads $\mathbb{Q}_p$.

Original source details: GELFAND-P000526

There is a convolution on measures

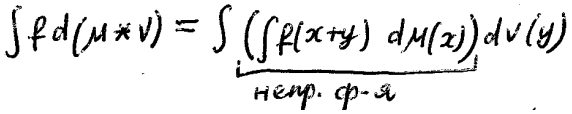
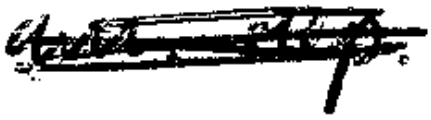


Diagram and typed description.

$$\int f\, d(\mu * \nu) = \int \underbrace{\left(\int f(x+y)\, d\mu(x) \right)}_{\text{cont. func.}} d\nu(y)$$

Original source details: GELFAND-P000526

Theorem 1.



Undeciphered passage. The crossed-out entry after “Theorem 1” cannot be read.

Original source details: GELFAND-P000526

K-algebra of measures $M(\mathbb{Z}_p, K) = K[[T]]$.

$$\left\| \sum a_k T^k \right\| = \sup_k |a_k| \qquad \text{(series of finite norm)}$$

Form. series are obtained from unbounded measures.

Theorem 2. Consider $M(\mathbb{Z}_p^*, K) = (p-1)K[[T]]$.

Unbounded measures are finitely add. on elem. sets (dual to loc. const. funcs.).

Proof of Thm. 1.

$$\mu \longrightarrow \sum a_k T^k, \qquad \text{where } a_k = \int_{\mathbb{Z}_p} \frac{x(x-1)\cdots(x-k+1)}{k!}\, d\mu(x)$$

$$\sum a_k T^k = \int_{\mathbb{Z}_p} (1+T)^x \, d\mu(x).$$

it follows that convolution of measures $\longrightarrow$ product of series.

Original source details: GELFAND-P000526

Diagram and typed description.
A tilted shaded quadrilateral between the formulas. Its purpose is unclear.

$$1 + T = e^{\tau} \qquad (\tau \text{ — form. var.}) \qquad \int_{\mathbb{Z}_p} e^{\tau x}\, d\mu(x) = \widetilde{\mu}(\tau)$$

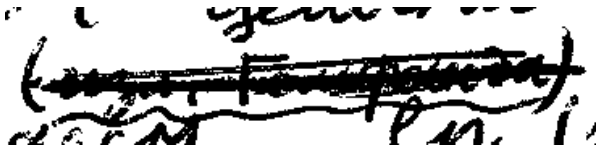
(rewrite the analysis!)

τ can be treated as a number.

$$\tau = \ln(1+T) = T - \frac{1}{2}T^2 + \frac{1}{3}T^3 - \dots$$

It is unclear what range of funcs. will result (after substitution one should get a series with integer coeffs.)

Example ([crossed out]) (Cartier) as presented by Gelfand

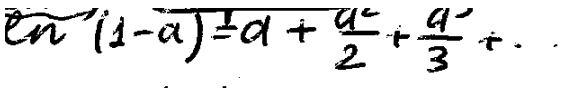


Original source details: GELFAND-P000526

Uncertain reading. The first parentheses after "Example" contain a crossed-out fragment. Its latter part resembles "Гельфанда" ("Gelfand's"). The beginning remains unreadable.

Consider

$$\ln(1-a) = a + \frac{a^2}{2} + \frac{a^3}{3} + \dots \qquad \text{and } a = 2 \in \mathbb{Z}_2.$$



Original source details: GELFAND-P000526

Manuscript excerpt. The manuscript has a after the equals sign, followed by plus signs.

The series converges [inserted below: and equals] $\ln(-1)$.

Original source details: GELFAND-P000526

Manuscript excerpt. The circled insertion “and equals” points to the line above.

$$\ln[(-1)^2] = 2\ln(-1) = 0 \quad \Rightarrow \quad \ln(-1) = 0.$$

hence the series converges to 0

$$2 + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \frac{2^5}{5} + \ldots + \frac{2^n}{n} + \ldots = 0$$

$$S_n = 2 + \frac{2^2}{2} + \frac{2^3}{3} + \ldots + \frac{2^n}{n} \qquad \text{for large } n \text{ is divisible by } 2^k$$

2, 4, $\frac{20}{3}$, $\frac{32}{3}$ and so on.

— · —

Theorem.

$$\|\mu\| = \sup_k |a_k| \qquad (\approx \text{ Plancherel})$$

$$\|\mu\| \underset{\text{def}}{=} \sup_{|f|\le 1} \left| \int f \, d\mu \right|$$

From the formula $\|\mu\| \ge \sup_k |a_k|$

We need to check

$$|f| \le 1 \quad \Rightarrow \quad f(x) = \sum c_k P_k(x), \qquad |c_k| \le 1,$$

$$P_k(x) = \frac{x(x-1)\cdots(x-k+1)}{k!}$$

(Proof: approx. by polynomials and apply the thm.)

$$c_k = \sum_{i=0}^{k} (-1)^{i+k} C_k^i f(i) = \Delta^k f(0) \qquad \text{(Taylor formula)}$$

$$X^{\textcircled{k}} = x(x-1)\cdots(x-k+1),$$

$$\Delta X^{\textcircled{k}} = kX^{\textcircled{k-1}}, \qquad \text{where } \Delta f(x) = f(x+1) - f(x).$$

$c = 0$

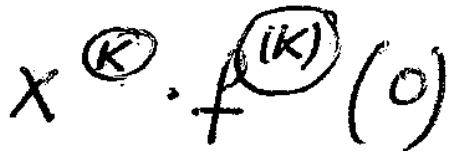
$X^{\textcircled{k}} = x(x-1) \ldots (x-k+1)$,

$\Delta X^{\textcircled{k}} = kX^{\textcircled{k-1}}$, where $\Delta f(x) = f(x+1) - f(x)$.

Original source details: GELFAND-P000526

Uncertain reading. The three letters with circled exponents have similar heights, including the one after the factor k. They are provisionally set as X. Lowercase x remains possible for the last letter.

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} X^{(k)} \cdot f^{(k)}(0), \qquad \text{where } f^{(k)}(x) = (\Delta^k f)(x)$$



Original source details: GELFAND-P000527

Manuscript excerpt. The letter with a circled k in the first product reads as X. Both exponents on f are circled.

Original source details: GELFAND-P000527

Uncertain reading. A short stroke and a dot precede the first k. They could be the opening parenthesis with a separate dot or the letter i in (ik). The stroke and dot are drawn separately.

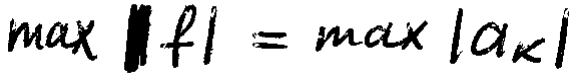
Assn.: $\forall$ cont. funcs.

Original source details: GELFAND-P000527

Undeciphered passage. The crossed-out word before "series" cannot be read.

the Maclaurin series converges uniformly, and,

$$\max |f| = \max |a_k|.$$



Original source details: GELFAND-P000527

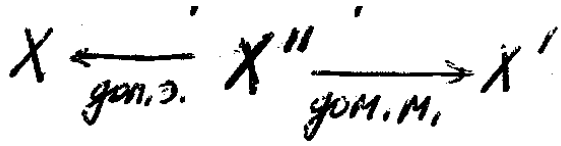
Uncertain reading. There is a thick stroke before f in the manuscript. It is provisionally set as one absolute-value bar.

Next time Kirillov (3 more min.) and Mishchenko.

Fuchs. (continuation).

M — an add. full subcat. of an abelian category, closed under extensions

$$QM: \qquad \mathrm{Ob}\, QM = \mathrm{Ob}\, M.$$



Original source details: GELFAND-P000527

Diagram and typed description.
$X \xleftarrow[\text{adm. e.}]{} X'' \xrightarrow[\text{adm. m.}]{} X'$

$$\pi_i(BQM)$$

BQM is homol. equiv. to $B\,\mathrm{GL}(\infty, A)$

Def. Grothendieck gr. of the category M. Gens. — the objects of M, rels.

$$0 \to X' \to X \to X'' \to 0, \qquad \Rightarrow \quad X = X' + X''$$

The gr. is commutative, since $\exists\ X = X' \oplus X''$.

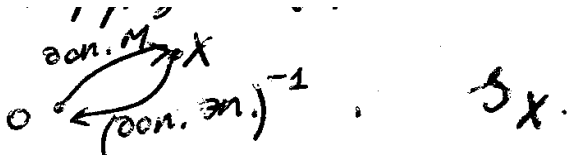
Theorem:

$$\underset{\pi_1(BQM,0)}{K_0(M)} \cong \text{Grothendieck gr. of the cat. } M.$$

Zero-dim. skeleton $= \mathrm{Ob}(M)$

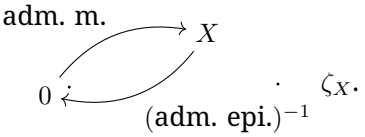
1-dim. skeleton — the morphisms of QM

$\forall X$ a loop is def.



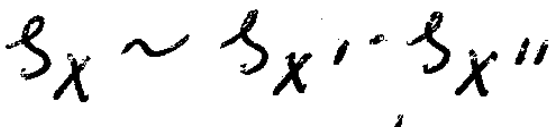
Original source details: GELFAND-P000527

Diagram and typed description.



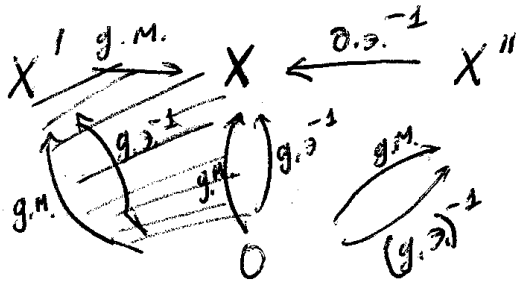
If in M there is an exact seq. (short) $0 \to X' \to X \to X'' \to 0$, then

$$\zeta_X \sim \zeta_{X'} \cdot \zeta_{X''}.$$



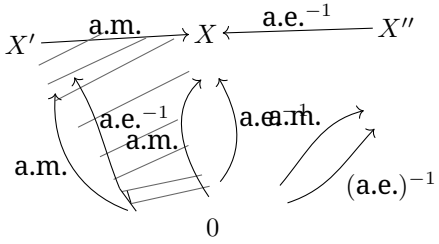
Original source details: GELFAND-P000527

Uncertain reading. The letter denoting the loop is provisionally set as ζ.



Original source details: GELFAND-P000527

Diagram and typed description.



The lower ends of the arcs are left separate as in the sketch. The thin lines represent the diagonal hatching. The lower end of the second arc from the left has an angular spur, possibly part of a second arrowhead.

and recall the 2-simplices.

— • —

GELFAND-P000528

29/$\overline{X}$-73. Gelfand's seminar.

A. A. Kirillov.

There is a symplectic manifold M

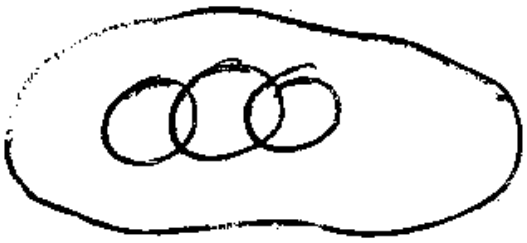
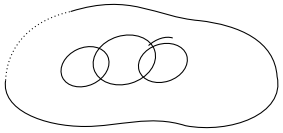


Diagram and typed description.



Original source details: GELFAND-P000528

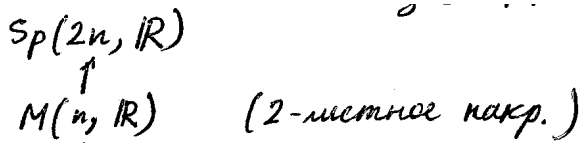


Diagram and typed description.

$\mathrm{Sp}(2n,\mathbb{R})$

$\uparrow$

$M(n,\mathbb{R})$ (2-sheeted cover)

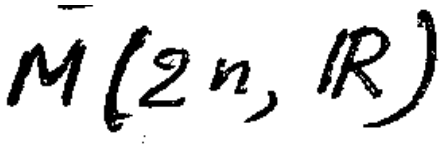
Original source details: GELFAND-P000528

loc. we may take

$$\omega = \sum_{k=1}^{n} dp_k \wedge dq_k$$

Transition funcs. (in TM) $g_{ij} \in \mathrm{Sp}(2n,\mathbb{R})$

Suppose they are in the 2-sheeted cover $M(2n,\mathbb{R})$



Manuscript excerpt. The line reads $M(2n,\mathbb{R})$. The diagram above and the insertion below read $M(n,\mathbb{R})$.

Original source details: GELFAND-P000528

The spinor representation [inserted below: $M(n,\mathbb{R})$] in Hilb. space

Undeciphered passage. The crossed-out sign after "space" cannot be read.

Original source details: GELFAND-P000528

induces a Hilbert bundle HM over M

Manuscript excerpt. The circled insertion $M(n,\mathbb{R})$ points to "representation".

Original source details: GELFAND-P000528

$$SM \subset HM \subset \cancel{H} S'M$$

Uncertain reading. A heavily cancelled sign with two slanted strokes occurs between the second $\subset$ and $S'M$. It is provisionally set as a struck-through H.

Original source details: GELFAND-P000528

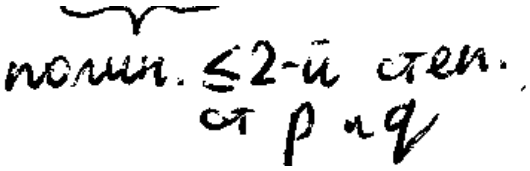
[under SM: rapidly decreasing]

Symplectic transf. $\rightsquigarrow$ op. in Hilb. space (can. transf. with p_i, q_i)

We obtain a projective representation

$$\underbrace{GM} = TM \oplus \mathbb{R} \times M$$

[under the brace:] polyn. of deg. ≤ 2 in p and q (Heisenberg Lie alg.)



Manuscript excerpt. The sign before 2 reads $\leq$.

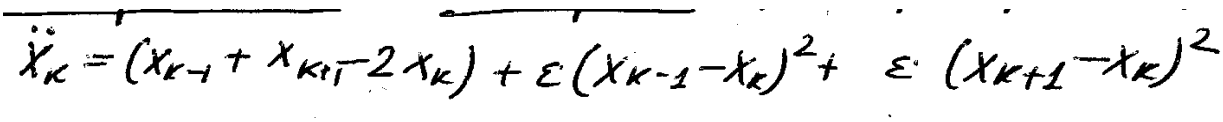
Original source details: GELFAND-P000528

GM acts on HM compat. with the bundle structure.

RM, R_x — Lagrangian subspaces in T_xM (max. dim.)

A. Shnirelman. On Fermi: equipartition of energy among degrees of freedom.

$$\ddot{x}_k = (x_{k-1} + x_{k+1} - 2x_k) + \varepsilon(x_{k-1} - x_k)^2 + \varepsilon(x_{k+1} - x_k)^2$$

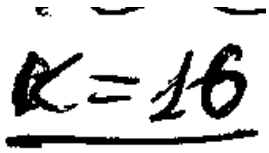


Original source details: GELFAND-P000528

Manuscript excerpt. Both quadratic terms have plus signs in the manuscript.

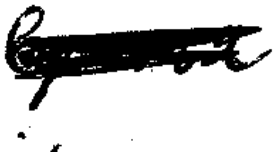
Solved num. on a computer

$k = 16$ after fin. ~~time~~ time intervals they examined the Fourier transf.



Uncertain reading. The number 16 is legible. The letter has a heavily retraced upright. It is provisionally set as k, with capital K also possible.

Original source details: GELFAND-P000528



Uncertain reading. The crossed-out word is provisionally read as “time”.

Original source details: GELFAND-P000528

Init. cond. — sin (all the energy in the 1st harm.)

By the estimate $t = 10^{-35}$ in this state.

Manuscript excerpt. The exponent on the line reads -35.

Original source details: GELFAND-P000528

Calculations:

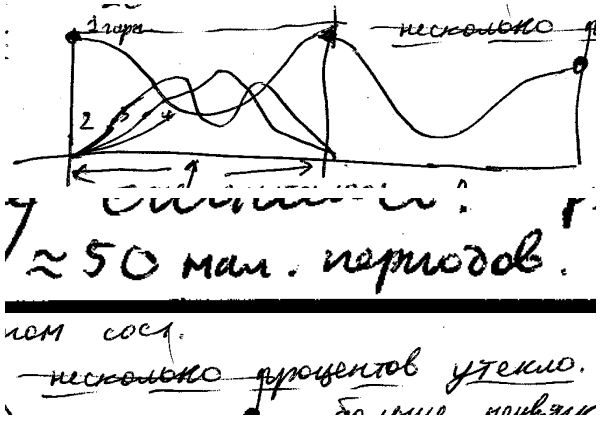
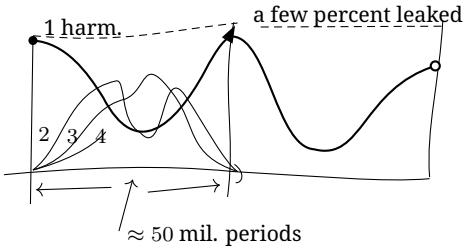


Diagram and typed description.



The numerals are placed by the curves as in the manuscript. Their precise assignment and the continuation of the short rising line among the crossings are unresolved.

Original source details: GELFAND-P000528

larger residual

10 steps — per. $1/3$

Another 10 steps — again
the same thing!!

Cubic nonlin. was also computed. The same result was obtained.

GELFAND-P000529

Zakharov: approx. by a diff. system with first integrals. (Possible cause)

A string with sin — likewise.

There are init. data for which uniform distribution occurs.

$$u_{tt} = u_{xx} \qquad \text{on } S^1 \quad \text{all traj. are closed}$$

$$u_{tt} = u_{xx} + \varepsilon f'(u)$$

First-order drift of the trajectories: also periodic for $u_{tt} = u_{xx} - \varepsilon u^3$.

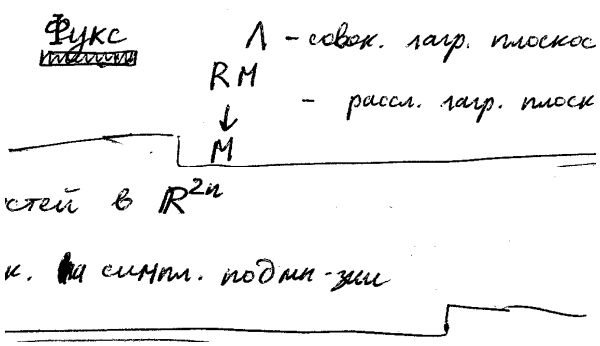
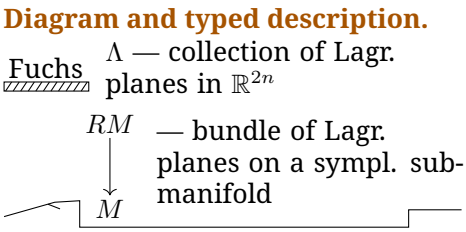


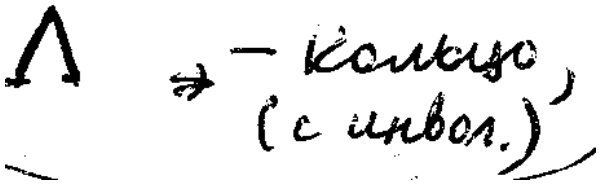
Diagram and typed description.



The first stroke in “on” is heavily retraced. The Russian abbreviation reads “подмн-зии”.

Original source details: GELFAND-P000529

Mishchenko.

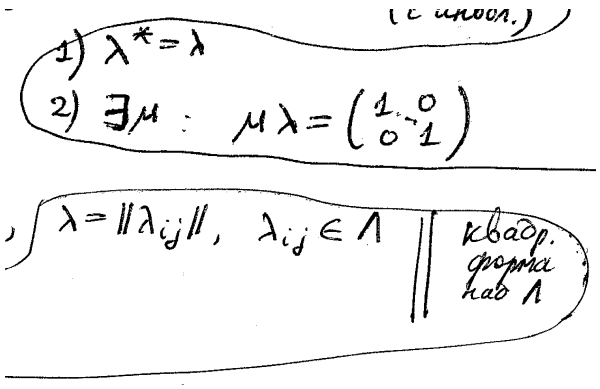


Original source details: GELFAND-P000529

Diagram and typed description.

Λ — a ring,
$\Rightarrow$ (with invol.)

Two short, closely spaced strokes below and to the right of Λ end in a right-pointing head. A possible reading is $\Rightarrow$.



Original source details: GELFAND-P000529

Diagram and typed description.

$\lambda = \|\lambda_{ij}\|, \quad \lambda_{ij} \in \Lambda$

1) $\lambda^* = \lambda$
2) $\exists \mu : \quad \mu\lambda = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$

quadr. form over Λ

The short diagonal trace between the rows of the identity matrix is rendered as $\ddots$.

A Fredholm rep. is a triple $T = \{T_1, F, T_2\}$;
T_i — a rep. of Λ in $\mathcal{L}(H)$ (with involution), where

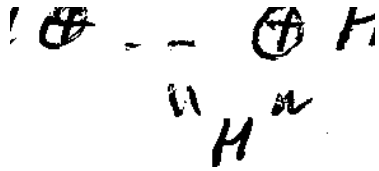
F — a Fredholm operator

$$FT_1(x) - T_2(x)F \quad \in \text{comp} \qquad \forall x \in \Lambda.$$

Original source details: GELFAND-P000529

Undeciphered passage. After the second F, before $\in$, a parenthesized fragment is heavily crossed out.

Construct $\lambda \in \mathcal{L}(H \oplus \cdots \oplus H)$ from the quadr. form.



Original source details: GELFAND-P000529

Manuscript excerpt. H^n

Below the direct sum are two slanted strokes and H^n. A rotated equality sign represents the strokes.

$$\lambda_{T_1} = (T_1(\lambda_{ij})), \qquad \lambda_{T_2} = (T_2(\lambda_{ij}))$$

[at right:] — invertible and self-adj.

They are bounded, invertible, self-adjoint.

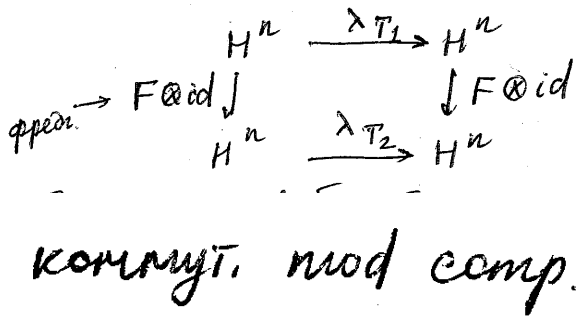


Diagram and typed description.

$$\begin{array}{ccc} H^n & \xrightarrow{\lambda_{T_1}} & H^n \\ {\scriptstyle F\otimes id}\downarrow & & \downarrow{\scriptstyle F\otimes id} \\ H^n & \xrightarrow{\lambda_{T_2}} & H^n \end{array}$$

Fredh. commutes mod comp.

Original source details: GELFAND-P000529

Choose bases so that

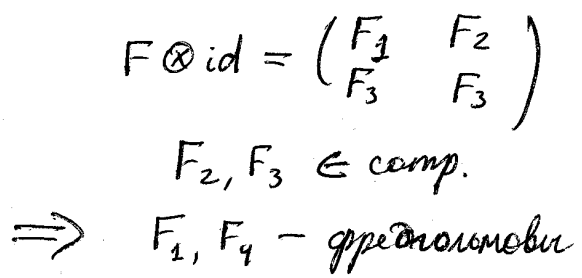
$$\lambda_{T_1} = \begin{pmatrix} A & 0 \\ 0 & -B \end{pmatrix} \qquad \lambda_{T_2} = \begin{pmatrix} A' & 0 \\ 0 & -B' \end{pmatrix}$$

A, B — inv. and self-adj. and > 0.

$$F \otimes id = \begin{pmatrix} F_1 & F_2 \\ F_3 & F_3 \end{pmatrix}$$

$F_2, F_3 \in$ comp.

$\Rightarrow$ F_1, F_4 — Fredholm



Manuscript excerpt. The lower-right entry of the matrix reads F_3. The final line reads F_4.

Original source details: GELFAND-P000529

GELFAND-P000530

[source number: 32]

Definition. $\operatorname{sign}_T \lambda = \operatorname{index} F_1 - \operatorname{index} F_4$.

Original source details: GELFAND-P000530

Manuscript excerpt. The definition gives the difference of the indices of F_1 and F_4.

It is independent of the presentation as a direct sum and of the representative in the class of equiv. quadr. forms.

Diagram and typed description.

$\longleftarrow \cdot \longrightarrow$

Original source details: GELFAND-P000530

Let M — a manifold (comp.); $\pi_1(M) = \pi$, $\dim M = 4K$; $\Lambda = \mathbb{Z}[\pi]$.

Uncertain reading. An angular letter follows 4. The provisional setting is K, with lowercase k also possible.

Original source details: GELFAND-P000530

There is a quadr. form $\sigma(M)$ over Λ. Formula:

$$\operatorname{sign}_T(\sigma(M)) = \langle \operatorname{ch} \xi_T \cdot L(M), [M] \rangle$$

ξ_T — a bundle on M

Hirzebruch proved it for $(\operatorname{ch} \xi_T = 1.)$

Finite-dimensional reps. are insufficient.

Diagram and typed description.

Original source details: GELFAND-P000530

Uncertain reading. A short isolated slanted stroke in the blank right margin. Its reading as the numeral “1” remains unresolved.

Original source details: GELFAND-P000530

GELFAND-P000531

[source number: 37]

12.11.73. Gelfand's sem..

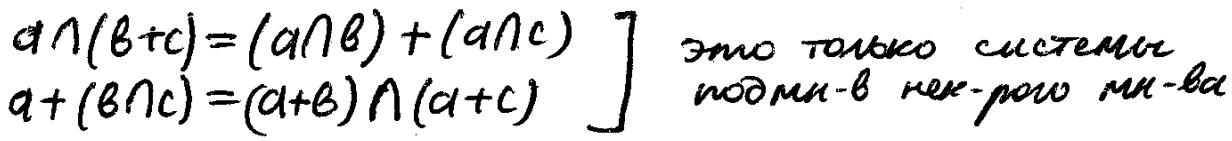
V. Ponomarev.

Suppose the set has 2 operations $\cap$ and $+$ with the natural properties (assoc., commut., $x \cap x = x$, $x + x = x$, $a + (a \cap b) = a$, $a \cap (a + b) = a$) this is a lattice.

Distributive lattice:

$$\left.\begin{aligned} a \cap (b + c) &= (a \cap b) + (a \cap c) \\ a + (b \cap c) &= (a + b) \cap (a + c) \end{aligned}\right]$$

[at right:] these are only systems of subsets of some set.



Original source details: GELFAND-P000531

Manuscript excerpt. The right square bracket joins the two identities to the note.

example: the lattice of subspaces with intersection and sum.

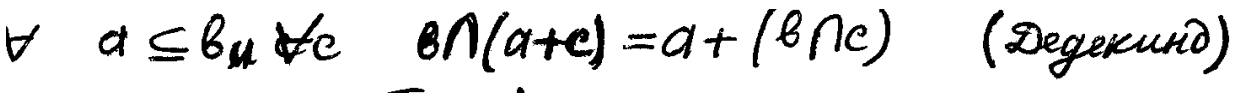
order relation in the lattice: $a \leq b \iff a \cap b = a$

Uncertain reading. The heavily retraced sign between a and b is provisionally set as $\leq$.

Original source details: GELFAND-P000531

Modular latt.:

$$\forall\, a \subseteq b \text{ and } \forall c \qquad b \cap (a + c) = a + (b \cap c) \qquad \text{(Dedekind)}$$



Original source details: GELFAND-P000531

Manuscript excerpt. The condition in the modular law reads $a \subseteq b$.

($\forall$ distributive is modular, but not conversely).

Uncertain reading. The heavily retraced sign before "distributive" resembles $\forall$.

Original source details: GELFAND-P000531

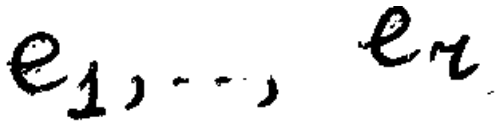
the lattice of subspaces is modular, but not distributive.

the lattice of submodules of a module is modular too.

Question: are there more identities or is this all? (Conjecture: $\forall$ modular latt. is the latt. of submod. of a module).

For latt. of subspaces there are more!!

Let $e_1, \ldots, e_r$ — be letters.



Manuscript excerpt. The first subscript has the same shape as r in the second list and in the notation for generators below.

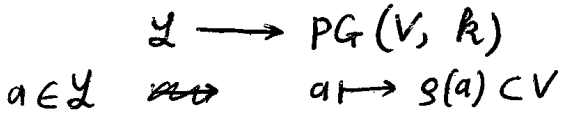
Original source details: GELFAND-P000531

Consider the free modular lattice generated by them (polyn. in $e_1, \ldots, e_r$ with relations) linearly equiv. polynomials $P(e_1, \ldots, e_r) \sim Q(e_1, \ldots, e_r)$, if it holds on substitution of subspaces.

V — a fin.-dim. space over the field k.

A representation of the latt. is a map into the lattice of subspaces of V (which is denoted $PG(V, k)$)

$$\begin{array}{cc} \mathcal{L} \longrightarrow PG(V,k) & ab \mapsto s(a)s(b) \\ a \in \mathcal{L} \quad \longrightarrow \quad a \mapsto s(a) \subset V & a + b \mapsto s(a) + s(b) \end{array} \Big\|$$

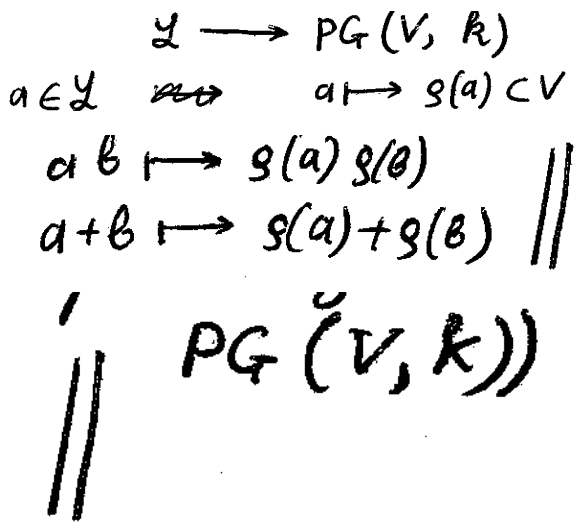


Manuscript excerpt. The arrows are written in two rows. A double vertical stroke follows the right column.

Original source details: GELFAND-P000531

Manuscript excerpt. The arrow after $a \in \mathcal{L}$ is overwritten or struck through with slanted loops.

Original source details: GELFAND-P000531



Uncertain reading. The looped letter is provisionally set as s throughout the notation for representations.

Original source details: GELFAND-P000531

$\mathcal{D}^r$ — the free modular lattice on r generators $e_1, \ldots, e_r$.

A rep. — is V and $s(e_1), \ldots, s(e_r)$ — a collection of subspaces $E_1^0, \ldots, E_r^0$, and conversely such a collection determines a representation.

Undeciphered passage. A heavily crossed-out fragment lies between "A rep. —" and "is V".

Original source details: GELFAND-P000531

Problem: describe all reps. of $\mathcal{D}^r$.

~~An elem. a is called perfect if~~

Original source details: GELFAND-P000531

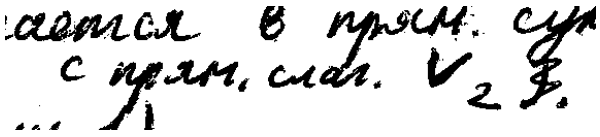
Manuscript excerpt. A single stroke crosses out the beginning of the definition.

s_1, V_1; s_2, V_2

$$s_1 \oplus s_2 \text{ — a rep. in } V_1 \oplus V_2 : \qquad (s_1 \oplus s_2)(a) = s_1(a) \oplus s_2(a)$$

Decomposable representation is defined naturally.

An elem. $a \in \mathcal{L}$ (inserted above: modular latt.) is called perfect, if for $\forall$ rep. s in a fin.-dim. space V over $\forall$ field k the subspace $s(a)$ can be complemented by a subspace V_2 and the rep. decomposes as a direct sum [inserted below:] with direct summand V_2.



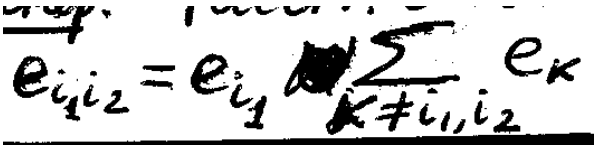
Uncertain reading. After V_2 a curl with a dot adjoins the descender from the line above.

Original source details: GELFAND-P000531

(Or: in an indecomp. rep. it maps to 0 or 1). If $a \sim b$ (lin.), then both are perfect at the same time.

Example. Consider the free mod. latt. on r generators $e_1, \dots, e_r$. Consider the elements

$$e_{i_1 i_2} = e_{i_1} \text{ [overwritten sign] } \sum_{k \neq i_1, i_2} e_k$$

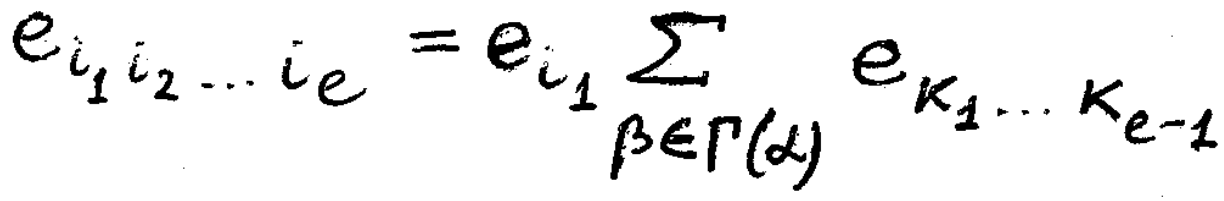


Uncertain reading. The sign between e_{i_1} and the sum is heavily overwritten. Its reading remains unresolved.

Original source details: GELFAND-P000531

GELFAND-P000532

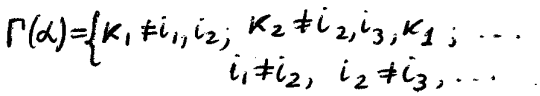
$$e_{i_1 i_2 \dots i_e} = e_{i_1} \sum_{\beta \in \Gamma(\alpha)} e_{k_1 \dots k_{e-1}}$$



Original source details: GELFAND-P000532

Diagram and typed description. Both indices use e. There is no separate sign between e_{i_1} and the sum.

$$\alpha = (i_1, \dots, i_e), \qquad \beta = (k_1, \dots, k_{e-1})$$
$$\Gamma(\alpha) = \{\, k_1 \neq i_1, i_2; \quad k_2 \neq i_2, i_3, k_1; \quad \dots$$
$$i_1 \neq i_2, \quad i_2 \neq i_3, \quad \dots$$



Manuscript excerpt. The line of conditions on i_1, i_2, i_3 is indented. There is no closing brace.

Original source details: GELFAND-P000532

. . . . . .

Triv. perfect elements: $m_t = \sum_{i \neq t} e_i$ — perfect elements.

m_t generate a Boolean alg. with r generators.

S. Gelfand.

Bruhat–Tits complex.

G — a p-adic group, K — a max. comp. subgroup.

B.–T. cplx. — description of G/K, analogue of a symmetric space.

К Б.-Т. — описание G/K, аналог симметрич. пр-ва.

Original source details: GELFAND-P000532

Manuscript excerpt. The manuscript abbreviation reads “К Б.-Т.”.

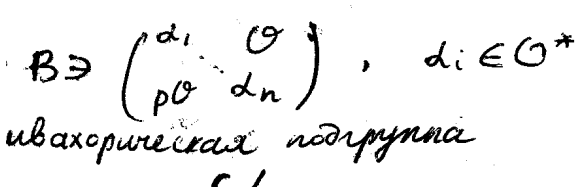
K is open (contains a nbhd. of 1) $\Rightarrow G/K$ is discrete.

example: $G = \mathrm{GL}(2, Q_p)/C$ (C — the centre).

$$K = \mathrm{GL}(2, \mathcal{O}) \qquad \mathcal{O} \text{ — integers}$$

$$\mathrm{GL}(n, Q_p) \supset B, \qquad B \ni \begin{pmatrix} \alpha_1 & & \mathcal{O} \\ & \ddots & \\ p\mathcal{O} & & \alpha_n \end{pmatrix}, \qquad \alpha_i \in \mathcal{O}^*$$

[under the matrix:] Iwahori subgroup.



Manuscript excerpt. The diagonal letter matches the shape of α at the start of the frame. At upper right is $\mathcal{O}$ and at lower left is $p\mathcal{O}$.

Original source details: GELFAND-P000532

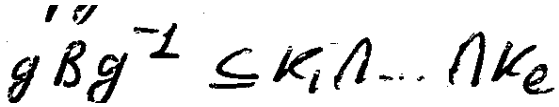
$x_1, \ldots, x_e$ — points of G/K

$K_1, \ldots, K_e \leftarrow$ the same ones, subgroups conjugate to K

Manuscript excerpt. A short leftward arrow precedes “the same ones”.

Original source details: GELFAND-P000532

They span a simplex if $gBg^{-1} \subseteq K_1 \cap \ldots \cap K_e$



Manuscript excerpt. The sign between gBg^{-1} and the intersection reads $\subseteq$.

Original source details: GELFAND-P000532

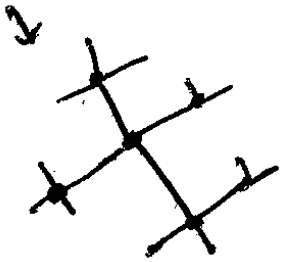
$n = 2 \Rightarrow$ it is contractible.

Uncertain reading. The heavily overwritten letter before $= 2$ is provisionally set as n.

Original source details: GELFAND-P000532

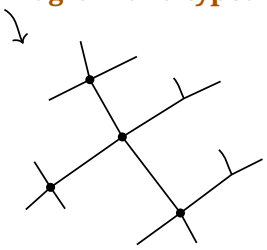
Undeciphered passage. A heavily struck-out rectangular fragment sits to the left of the tree sketch. The writing inside is illegible.

Original source details: GELFAND-P000532



Original source details: GELFAND-P000532

Diagram and typed description.

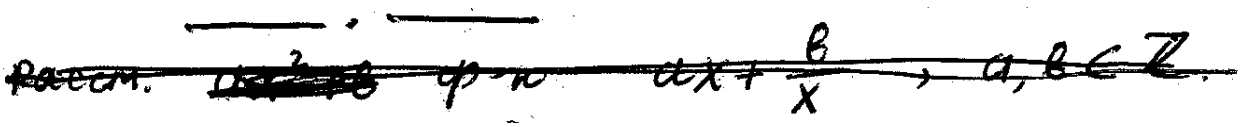


The slanted arrow at left is separate from the upper node. Two hooked strokes appear on the right branches.

Consider $V_3(F_q)$. Points of type 1 — lines in V_3, 2 — planes in V_3.
Incidence $\Rightarrow$ spans a simplex.

D. Kazhdan.

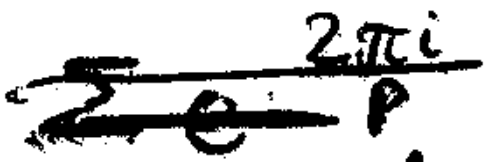
~~Consider the func.~~ $\sim\!\!\sim ax + \frac{b}{x},\ a, b \in \mathbb{Z}. \sim\!\!\sim$



Original source details: GELFAND-P000532

Undeciphered passage. In the crossed-out line between “Consider” and “the func.” there is a raised 2. The writing below it remains undeciphered.

[in the left margin, crossed out:] $\sum e^{\frac{2\pi i}{p}}$.

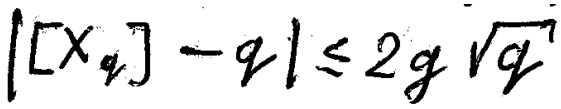


Original source details: GELFAND-P000532

Manuscript excerpt. In the crossed-out expression the numerator reads $2\pi i$ and the denominator p.

X — a curve over F_q. How many points are on it?

$$\left|[X_q] - q\right| \leq 2g\sqrt{q}$$

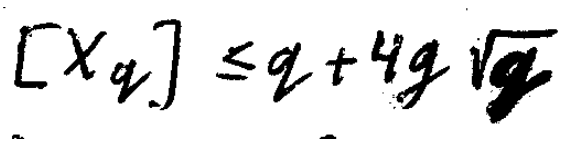


Original source details: GELFAND-P000532

Manuscript excerpt. Inside the absolute-value signs the manuscript reads $[X_q] - q$.

(g — genus of the curve, $[X_q]$ number of points of X over F_q).
This is equiv. to the Weil conjecture.
Stepanov's proof (in Bombieri's presentation at the Bourbaki sem.). He proved

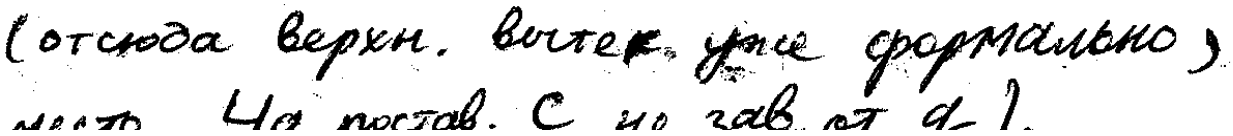
$$[X_q] \leq q + 4g\sqrt{q}$$



Manuscript excerpt. The heavily retraced letter under the radical reads q.

Original source details: GELFAND-P000532

(from this the upper bound follows formally).



Original source details: GELFAND-P000532

Uncertain reading. The final stroke in the Russian abbreviation for "follows" is retraced. The reading is provisional.

(It even suffices, instead of $4g$ to put C, independent of q).
X — a curve, on it a divisor D. Number of funcs. divisible by the divisor D:

$$L(D) = \deg D - g + 1 \qquad \text{when } \deg D > 2g$$

(if there are many more poles than zeros).

Original source details: GELFAND-P000532

Manuscript excerpt. The left side is $L(D)$. The condition has a strict $>$ before $2g$.

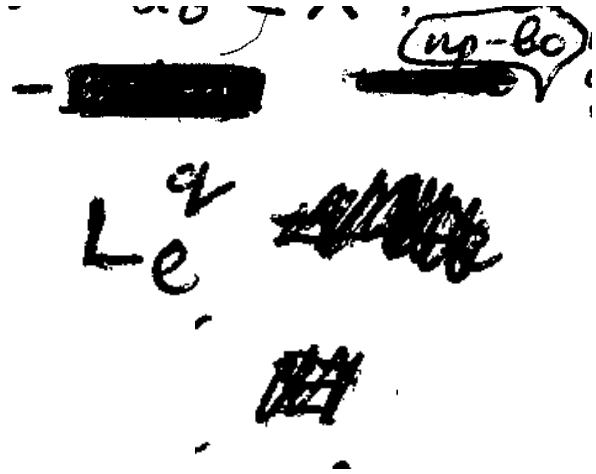
GELFAND-P000533

[source number: 38]

Let $x_0 \in X$, fixed.

L_e — [inserted above:] space of funcs. with a pole of mult. $\leq e$ at pt. x_0.

L_e^q has the same number of elements as L_e.

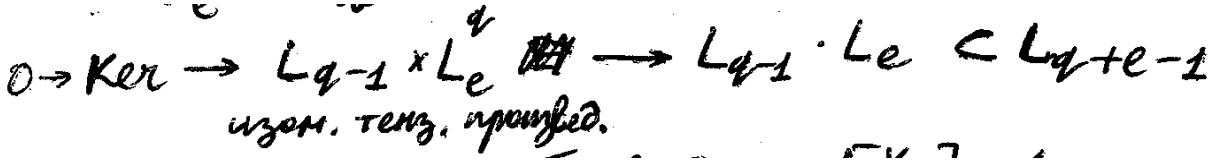


Undeciphered passage. Two heavily crossed-out entries follow the dash on the L_e line. One follows L_e^q, and one precedes the product arrow. "Space" is circled above the second cancelled entry. The cancelled entries are illegible.

Original source details: GELFAND-P000533

$$0 \to \text{Ker} \to L_{q-1} \times L_e^q \longrightarrow L_{q-1} \cdot L_e \subset L_{q+e-1}$$

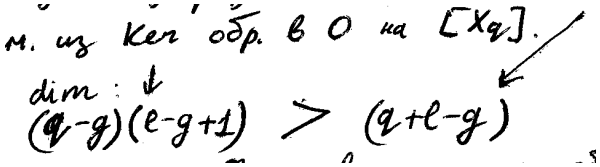
[under the middle term:] isom. tensor prod..



Original source details: GELFAND-P000533

Manuscript excerpt. The middle term uses $\times$. Below it is the note "isom. tensor prod." The subscript reads e.

Elems. of Ker are 0 on $[X_q]$.



Original source details: GELFAND-P000533

Diagram and typed description.

dim :

$(q-g)(e-g+1) \; > \; (q+e-g)$

A short downward arrow points to the first product, and a longer diagonal arrow points to the right side.

And there is a kernel. Funcs. of low deg. are 0 on $[X_q]$.

Original source details: GELFAND-P000533

Diagram and typed description.

GELFAND-P000534

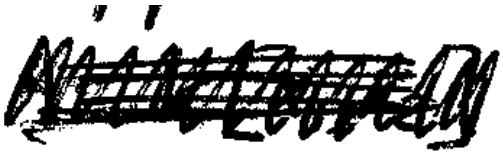
19/XI-73. Gelfand's sem.. (last meeting this semester).

Manuscript excerpt. The Roman XI has a rule above and below.

Original source details: GELFAND-P000534

Serre's problem on polyn. in 3 variables has been solved.

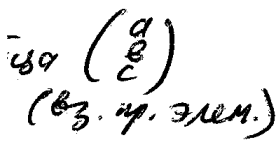
Vaserstein.



Undeciphered passage. An elongated entry to the right of the surname is crossed out with horizontal and slanting strokes. Its reading remains unresolved.

Original source details: GELFAND-P000534

Def. The stab. rank of the ring $A \leq 2$, if $\forall$ unimod. column $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ (rel. prime elems.)

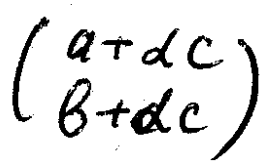


Manuscript excerpt. The parenthesis "(rel. prime elems.)" is below the three-entry column.

Original source details: GELFAND-P000534

$\exists \alpha, \beta \in A$, such that

$$\begin{pmatrix} a+\alpha c \\ b+\alpha c \end{pmatrix} \quad \text{is unimodular.}$$

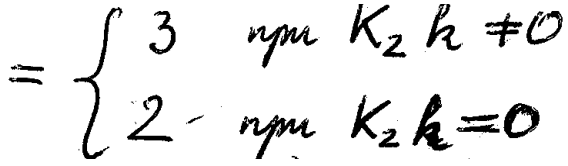


Manuscript excerpt. Both entries of the short column contain α. The condition above gives α, β.

Original source details: GELFAND-P000534

Thm.

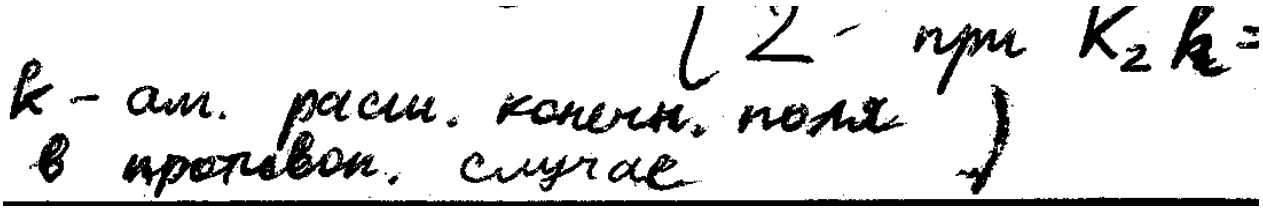
$$\text{st. rank } k[x_1, x_2] = \begin{cases} 3 & \text{when } K_2 k \neq 0 \\ 2 — & \text{when } K_2 k = 0 \end{cases}$$



Manuscript excerpt. There is a short stroke after 2 before "when".

Original source details: GELFAND-P000534

$$\left(K_2 k = \begin{cases} 0 & k \text{ — alg. ext. of a finite field} \\ \neq 0 & \text{otherwise} \end{cases}\right)$$



Original source details: GELFAND-P000534

Diagram and typed description. The manuscript reads “ал.” before the abbreviation for “ext.”.

GELFAND-P000535

[source number: 45]

Example over $\mathbb{R}$.

Undeciphered passage. A long entry after $\mathbb{R}$ is crossed out with slanting and horizontal strokes.

Original source details: GELFAND-P000535

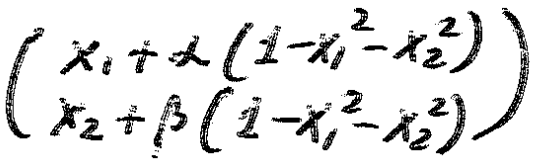
$$\begin{pmatrix} x_1 \\ x_2 \\ 1 - x_1^2 - x_2^2 \end{pmatrix}$$

there do not exist $\alpha, \beta \in \mathbb{R}$, such that

$$\begin{pmatrix} x_1 + \alpha(1 - x_1^2 - x_2^2) \\ x_2 + \beta(1 - x_1^2 - x_2^2) \end{pmatrix}$$

Undeciphered passage. There is a dense cancellation between “not” and “exist”. The underlying writing is illegible.

Original source details: GELFAND-P000535



Manuscript excerpt. The second entry of the short column here contains β.

Original source details: GELFAND-P000535

(by degree).

Manuscript excerpt. Below the line with “do not exist” is the note “(by degree)”.

Original source details: GELFAND-P000535

There is a curve $C(x_1, x_2) \equiv C = 0$

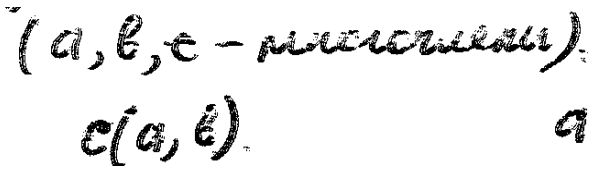
Winding index with respect to the set $a(x_1, x_2) = 0$, $b(x_1, x_2) = 0$

Original source details: GELFAND-P000535

Manuscript excerpt. The abbreviation before the equations reads “set”.

(a, b, c — polynomials).

$c(a, b)$. $a = x_1$, $b = x_2$ — number of turns about the origin.



Original source details: GELFAND-P000535

Diagram and typed description. The letter before (a, b) is c.

B — a ring

$K_1 \mathrm{Sp}\, B =$ Quotient of symplectic matrices of order $2n$ by the elementary ones.

Question: $z_1^2 + z_2^2 = 1$ index 0 or non 0?

Original source details: GELFAND-P000535

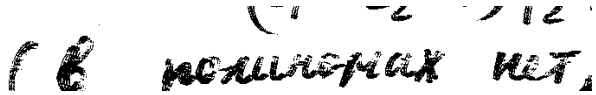
Manuscript excerpt. Both signs in the question are read as 0. A faint closing stroke is visible at the right of each oval.

$\Longleftrightarrow \exists$ are there 2 funcs. $f_1(z_1, z_2)$ and $f_2(z_1, z_2)$, such that

(entire, entire of exp. growth, etc.)

$$\begin{matrix}(z_1^2 + z_2^2 - 1) f_1 + z_1 \\ (z_1^2 + z_2^2 - 1) f_2 + z_2\end{matrix}$$ have no common div.

(no for polynomials, yes for cont. funcs.).



Original source details: GELFAND-P000535

Manuscript excerpt. The first phrase reads “for polynomials”.

Kazhdan. Derived category.

A — a ring, ~~$\forall$ — modules over A.~~

Original source details: GELFAND-P000535

Uncertain reading. The crossed-out initial sign is provisionally set as $\forall$. “Modul...” and “over A” are visible under the cancellation.

~~$\mathrm{Hom}(M, N)$ — functor~~

Original source details: GELFAND-P000535

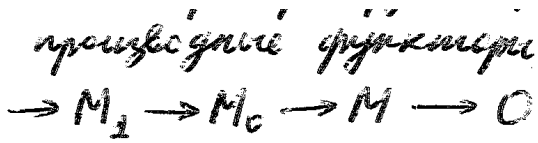
Uncertain reading. The word after the crossed-out $\mathrm{Hom}(M, N)$ is provisionally read as “functor”.

Consider the category of modules over A.

(Everything will be illustrated by the functor $\mathrm{Hom}(M, N)$).

Define the derived functors $\mathrm{Ext}^i(M, N)$:

$$\ldots \to M_1 \to M_0 \to M \to 0 \qquad (M_i \text{ — free})$$



Original source details: GELFAND-P000535

Uncertain reading. The subscript before M is open at right. It is provisionally read as 0.

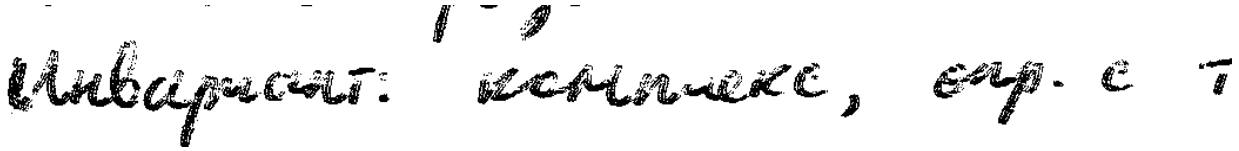
Consider the complex $C^i = \mathrm{Hom}(M_i, N)$

$$0 \to C^0 \to \ldots \to C^{i-1} \to C^i \to \ldots$$

Its cohomology grs. $\mathrm{Ext}^i(M, N)$. $\mathrm{Ext}^0(M, N) = \mathrm{Hom}(M, N)$.

Any 2 resol. are homotopic.

Invariant: complex, def. up to homotopy.



Original source details: GELFAND-P000535

Manuscript excerpt. The first word on the line reads “Invariant”.

There are complexes with isom. homologies that are not homotopic. (see topology).

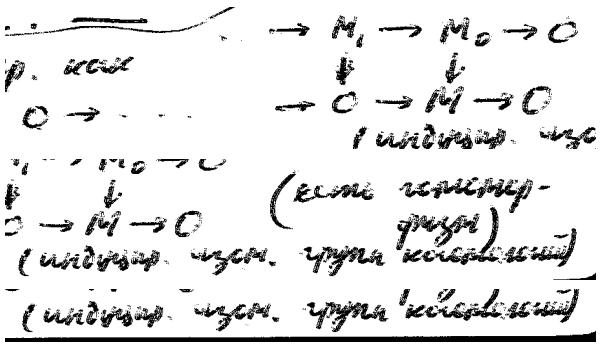
The natural object is a complex.

Let $\dot{M}$ — a complex $0 \to M_1 \to \ldots \to M_i \to 0$

$\dot{N}$ — — " — $0 \to N_1 \to \ldots \to N_j \to 0$

$\mathrm{Hom}(\dot{M}, \dot{N})$

(!) M is interpr. as



Original source details: GELFAND-P000535

Diagram and typed description.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & M_1 & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow \cdots \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

(if a homomorphism)

(induc. an isom. of cohomology groups)

The lower letter beneath M_0 is provisionally set as M. The small stroke cannot securely be separated from the arrow.

GELFAND-P000536

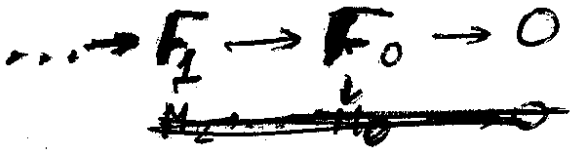
M is replaced by a compl. of free modules, for which $\to 0 \to M \to 0 \to \dots$ is mapped with an induced isom. of cohomology grs..

(quasi-isomorphic complexes)

Resolution — replacement of M by a quasi-isom. compl. of free modules.

$\mathrm{Hom}(\dot{M}, \dot{N})$ in the derived category.

Replace $\dot{M}$ by a quasi-isom. complex of free modules



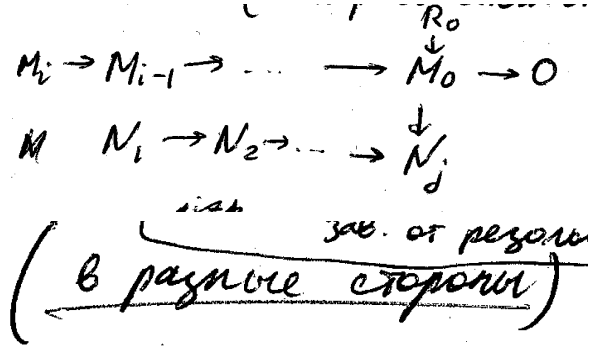
Original source details: GELFAND-P000536

Diagram and typed description.

$$\cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0$$
$$\downarrow$$
$$\cancel{M_2} \longrightarrow \cancel{M_0} \longrightarrow \cancel{0}$$

The first crossed-out subscript is provisionally read as 2. A downward arrow is visible below F_0. Below F_1 a separate arrow cannot be distinguished from the subscript stroke.

Construct the complex $C^i = \mathrm{Hom}^i(R, N)$ — derived functor. (maps with shift by i)



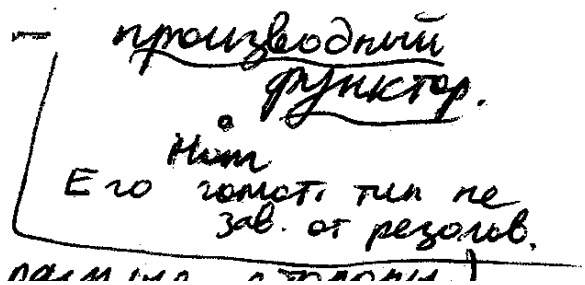
Original source details: GELFAND-P000536

Diagram and typed description.

$$\begin{array}{ccccccccc} & & & & & & R_0 & & \\ & & & & & & \downarrow & & \\ M_i & \longrightarrow & M_{i-1} & \longrightarrow & \cdots & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & & & & & \downarrow & & \\ M & & N_i & \longrightarrow & N_2 & \longrightarrow & N_j & & \end{array}$$

(in opposite directions)

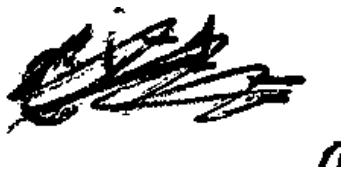
The first subscript below is provisionally set as i. An isolated M stands to the left of the lower chain. A faint leftward arrow is visible under the note at right.



Original source details: GELFAND-P000536

Diagram and typed description.

Its [inserted above: $\overset{\circ}{\mathrm{Hom}}$] homot. type is independent of the resol.



Original source details: GELFAND-P000536

Undeciphered passage. A short, crossed-out fragment lies below the chains. The underlying writing is illegible.

2 complexes of free modules are quasi-isom. $\Longleftrightarrow$ they are homot. equiv.

Original source details: GELFAND-P000536

Manuscript excerpt. Two heads are visible on the retraced arrow, rendered as $\Longleftrightarrow$.

$K(A)$ — the categ. of complexes. Morphisms — homot. classes of morphisms.
The category $D(A)$. $\mathrm{Ob}\,\mathcal{D}(A) = \mathrm{Ob}(K(A))$, and $\mathrm{Mor}(\mathcal{D}(A))$ — a seq.

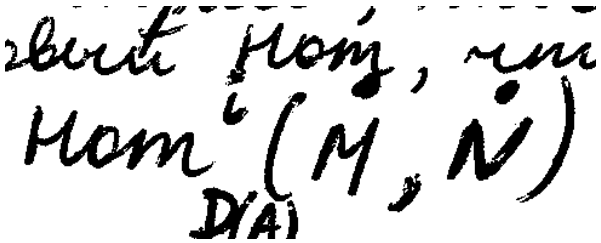
$$\dot{M} \to \dot{M}_1 \to \ldots \to \dot{M}_k \to \dot{N} = \mathrm{Mor}(\dot{M}, \dot{N})$$

where $\to$ — a morph. to the right or a quasi-isom. to the left.
We want: to introduce a new Hom, so that it becomes a complex.
~~$\mathrm{Hom}_{D(A)}$~~

Uncertain reading. The crossed-out fragment shows Hom. The subscript resembles $D(A)$. The rest of the entry remains undeciphered.

Original source details: GELFAND-P000536

$$\mathrm{Hom}^i_{D(A)}(\dot{M}, \dot{N})$$



Uncertain reading. There is a dot over N. The mark over M touches a letter on the line above and is provisionally rendered as a dot.

Original source details: GELFAND-P000536

. . . . . .

F — a contravar. functor; $F : \underset{\text{modules over } A}{M(A)} \to \underset{\text{modules over } B}{M(B)}$

$\dot{M} \in D(A)$

It is bad to take $F(\dot{M})$, since quasi-isoms. will not go to quasi-isoms.

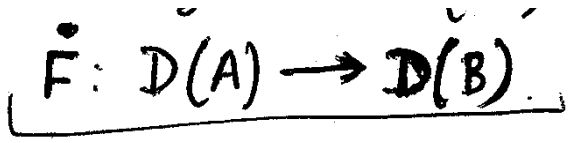


Diagram and typed description.

$\dot{F} : D(A) \longrightarrow D(B).$

Original source details: GELFAND-P000536

(Its cohom. are the ordinary derived functors)

$\dot{M} \in D(A)$. Choose a representative of free modules $\dot{M}$ and apply to it F — we obtain a new complex. Regard it as an element of $D(B)$. Well-definedness follows from the stated theorem on homotopy.

More gen. def. $F : A \to B$, $\dot{F} : D(A) \to D(B)$

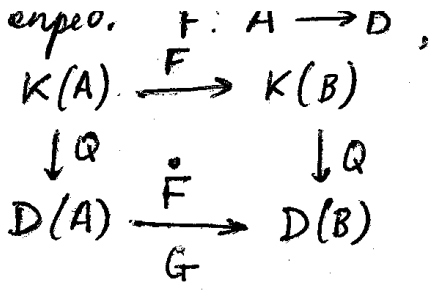
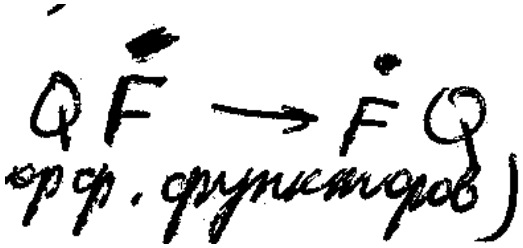


Diagram and typed description.

$$\begin{array}{ccc} K(A) & \xrightarrow{F} & K(B) \\ \downarrow Q & & \downarrow Q \\ D(A) & \xrightarrow[G]{\dot{F}} & D(B) \end{array}$$

Original source details: GELFAND-P000536

Req.: $\exists$ a map $Q\overline{F} \to \dot{F}Q$ (morph. of functors)



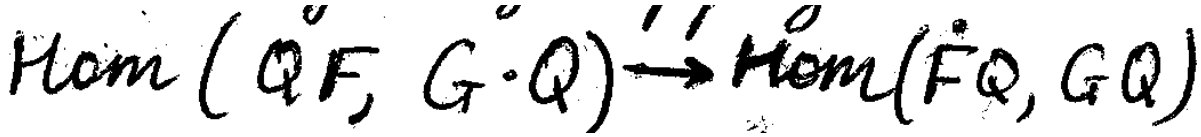
Uncertain reading. The first superscript mark is a heavy slanted stroke, provisionally set as a bar. The second F has a dot.

Original source details: GELFAND-P000536

$\forall G$ we obtain an isomorphism

$$\mathrm{Hom}(QF, G \cdot Q) \longrightarrow \mathrm{Hom}(\dot{F}Q, GQ)$$

would be an isomorphism for $\forall G$.



Original source details: GELFAND-P000536

Manuscript excerpt. The left side reads QF without a superscript mark, and the right side $\dot{F}Q$. The arrow points right.

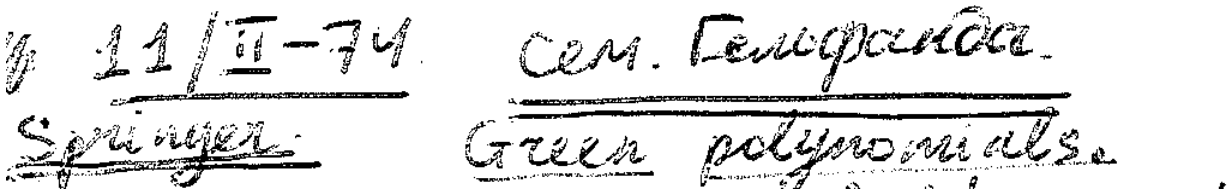
GELFAND-P000537

11/II-74. Gelfand's sem..

Uncertain reading. A crossed-out entry lies to the left of 11/II-74. Its first stroke resembles 1. The full entry remains unreadable.

Original source details: GELFAND-P000537

Springer. Green polynomials.



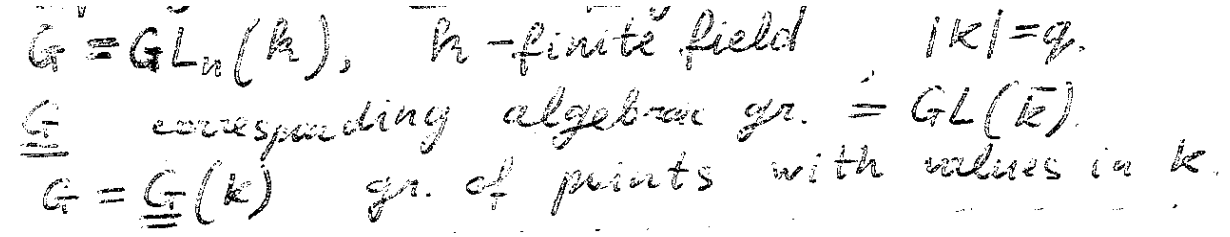
Original source details: GELFAND-P000537

Manuscript excerpt. The title and the lines about the group are written in English in the manuscript.

$$G = \mathrm{GL}_n(k), \qquad k \text{ — finite field}, \qquad |k| = q.$$

G corresponding algeb-raic gr. $= \mathrm{GL}(\bar{k})$.

$G = G(k)$ gr. of points with values in k.



Original source details: GELFAND-P000537

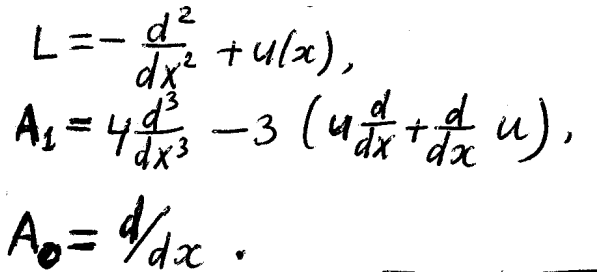
Manuscript excerpt. Two rules below G distinguish the algebraic group from the plain G. After "gr." the manuscript gives $\mathrm{GL}(\bar{k})$ without the subscript n.

GELFAND-P000538

25/II-74. Gelfand's sem.

S. P. Novikov. (continued).

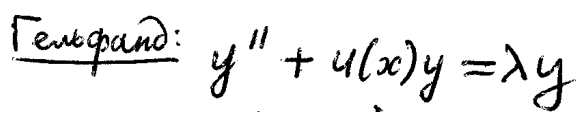
$$L = -\frac{d^2}{dx^2} + u(x),$$
$$A_1 = 4\frac{d^3}{dx^3} - 3\left(u\frac{d}{dx} + \frac{d}{dx}u\right),$$
$$A_0 = d/dx.$$



Manuscript excerpt. The definition of L has a minus sign before the second derivative.

Original source details: GELFAND-P000538

Gelfand: $y'' + u(x)y = \lambda y$



Manuscript excerpt. There is no separate minus sign before y'' on this line.

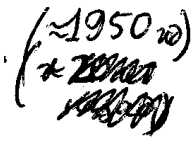
Original source details: GELFAND-P000538

Original source details: GELFAND-P000538

Undeciphered passage. A small entry to the right of λ_y lies under a heavy cancellation.

$$u(x+T) = u(x)$$

structure of the spectrum — shortest proof in Gelfand DAN (≈ 1950).



Original source details: GELFAND-P000538

Undeciphered passage. Two crossed-out entries lie below the year. A sign resembling $\approx$ precedes the upper one.

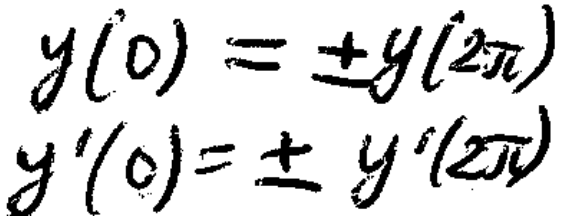
$u(x)$ — real and its closure (from C_0^∞) is self-adj.

Another case:

Original source details: GELFAND-P000538

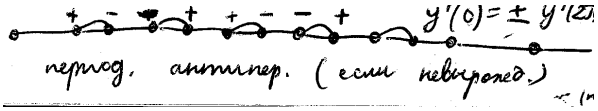
Diagram and typed description.

$$y(0) = \pm y(2\pi), \qquad y'(0) = \pm y'(2\pi)$$



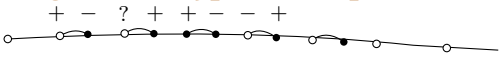
Original source details: GELFAND-P000538

Manuscript excerpt. Both boundary conditions carry a $\pm$ sign.



Original source details: GELFAND-P000538

Diagram and typed description.



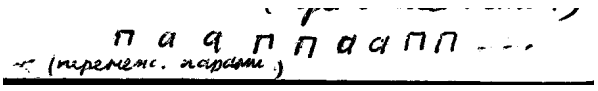
periodic, antiper. (if nondegenerate)

The question mark stands for the overwritten label above the second arc. The fill of partially retraced circles follows the visible center.

(alternation!)

п а а п п а а п п ...

$\sim$ (altern. in pairs)



Original source details: GELFAND-P000538

Uncertain reading. The short wavy stroke below the letters is provisionally rendered as $\sim$.

GELFAND-P000539

[source number: 59]

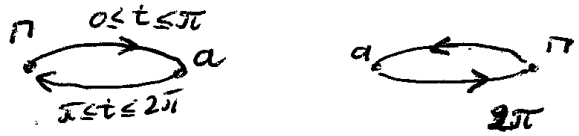
On the line, consid. the following bound. conditions:

$$y(x+T) = e^{it} y(x)$$
$$y'(x+T) = e^{it} y'(x)$$

Original source details: GELFAND-P000539

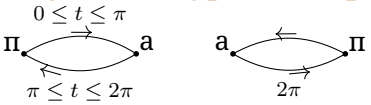
Uncertain reading. The sign after i in the first exponent is heavily overwritten. The reading t is provisional. The second exponent it is independently legible.

$\forall t$ self-adj. prob. with discr. spectrum.



Original source details: GELFAND-P000539

Diagram and typed description.

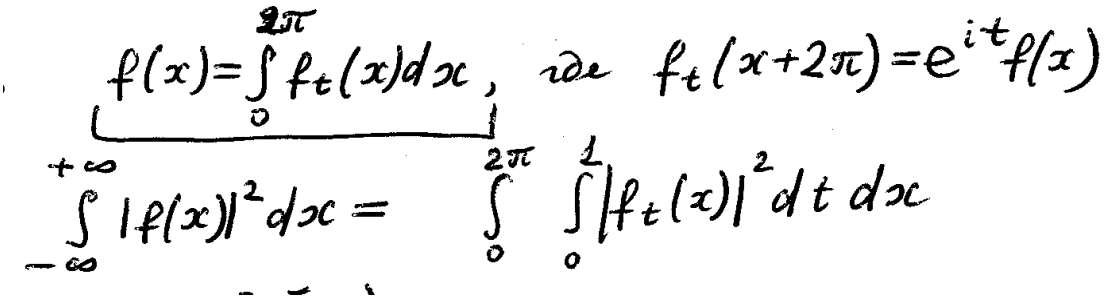


Lemma. $\forall f(x)$,

$$f(x) = \int_0^{2\pi} f_t(x)\,dx, \qquad \text{where } f_t(x+2\pi) = e^{it} f(x)$$

At the same time

$$\int_{-\infty}^{+\infty} |f(x)|^2\,dx = \int_0^{2\pi} \int_0^1 |f_t(x)|^2\,dt\,dx$$



Original source details: GELFAND-P000539

Manuscript excerpt. The first integral has dx, and the right side of the condition has $f(x)$ without a subscript. In the second equality the upper limits are 2π and 1, with differentials $dt\,dx$.

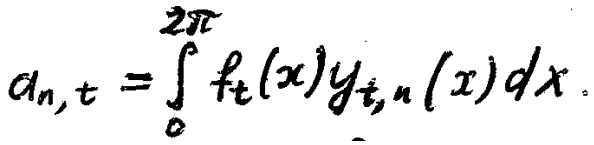
(reduc. to a comp. group — Weyl)

$0 \le x \le 2\pi \Rightarrow$ expand in a Fourier series in the eigenfunctions of the problem.

Obtain an integral over the whole line now $\| \; f_t(x) = \sum_n a_{n,t} y_{t,n}(x)$

$y_{t,n}$ — Bloch funcs.

$$a_{n,t} = \int_0^{2\pi} f_t(x) y_{t,n}(x)\, dx.$$



Manuscript excerpt. There is no bar over $y_{t,n}(x)$ in the integral.

Original source details: GELFAND-P000539

$\rightleftarrows$ is calculated $\dfrac{d\lambda}{dt}$.

S.P.N.

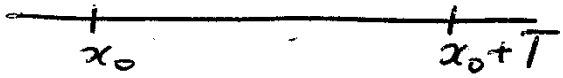


Diagram and typed description.

Original source details: GELFAND-P000539

Consid. $y(x_0, x)$

$Ly = \lambda y, \quad \lambda = k^2$

$y|_{x=x_0} = 1$
$y'|_{x=x_0} = ik$ — one eigenfunc. (closer to e^{ikx}).

Undeciphered passage. A heavily crossed-out entry before “and”.

Original source details: GELFAND-P000539

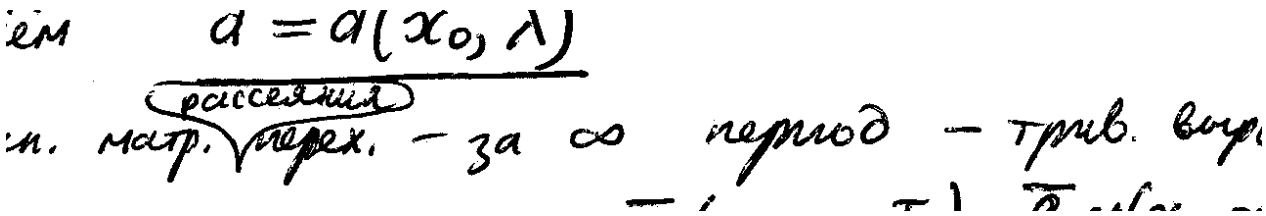
and $\bar{y}(x_0, x)$ — another eigenfunc.

Monodromy matr.

$$y(x_0, x+T) = a y(x_0, x) + b\bar{y}(x_0, x),$$

moreover $a = a(x_0, \lambda)$.

(ord. transfer matr. [above the line: scattering] — over an ∞ period — triv. degener. case).



Original source details: GELFAND-P000539

Manuscript excerpt. “Scattering” is circled and linked by a stroke to the line at “transfer”.

$$\bar{y}(x_0, x+T) = \bar{b}y(x_0, x) + \bar{a}\bar{y}(x_0, x).$$

$$T = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \text{ — monodromy matr.} \qquad T = T(x_0, \lambda)$$

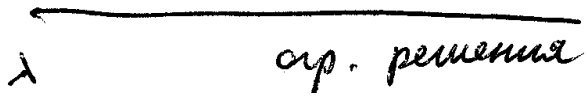
$\det = 1$ (the Wronskian is constant)

$$|a|^2 - |b|^2 = 1;$$
$$\operatorname{Sp} T = 2a_R = a + \bar{a};$$

Eigenvalues

$$\alpha_\pm(\lambda) = a_R \pm \sqrt{a_R^2 - 1}$$

consid. the graph in λ



Original source details: GELFAND-P000539

Diagram and typed description.

λ

There is a small hook at the left of the line.

bounded solutions $\iff |a_R| \le 1 \implies$ obtain $e^{\pm it}$
unbounded solutions $\iff |a_R| > 1 \implies$ obtain $e^{\pm t}$

GELFAND-P000540

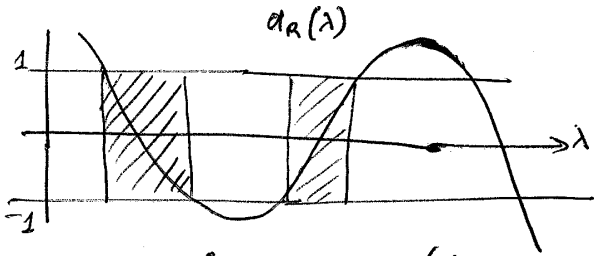
$(y, y') = (1, ik)$

Relation to the spectrum.

Original source details: GELFAND-P000540

Uncertain reading. Before “For” is a dense mark with two lower ends. The sign is undeciphered.

For $|a_R| \le 1$ there is Bloch spectrum.



Original source details: GELFAND-P000540

Diagram and typed description.

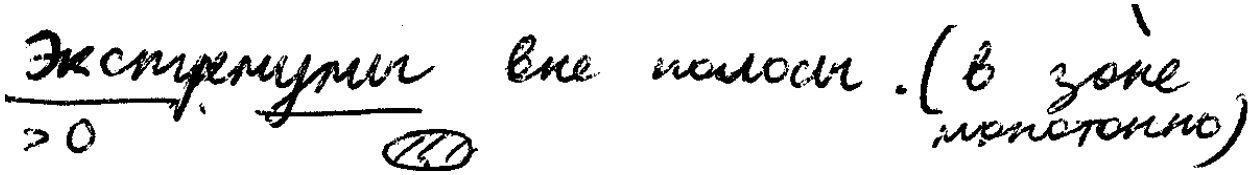
$a_R(\lambda)$

1

λ

−1

Extrema outside the strip. (monotonically in the band)



Original source details: GELFAND-P000540

Manuscript excerpt. “Extrema” is underlined.

G. $u(x) = \text{const}$ — spectrum > 0

$u(x) < 0$ is possible.

stability bands extend into the neg. region

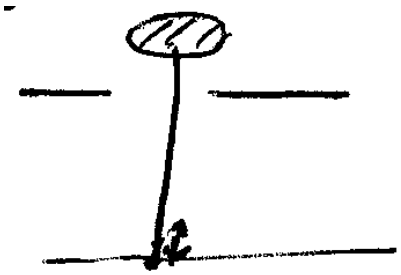
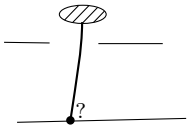


Diagram and typed description.



The question mark at the lower point stands for the undeciphered mark.

Original source details: GELFAND-P000540

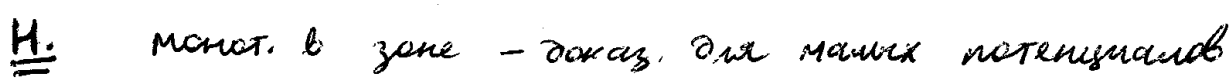
$$-y'' + (a + b\cos 2x)y = 0$$

bands of stability and instab.

1st stab. band when $a + b\cos 2x < 0$

(does not occur for constant ones)

N. monot. in the band — proved for small potentials



Original source details: GELFAND-P000540

Manuscript excerpt. The Russian abbreviations read “монот.” and “доказ.”.

G. Contest for a proof (for next time)

N. Periodic and antiperiodic $|a_R| = 1$ — band edges.

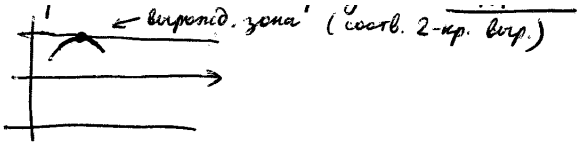
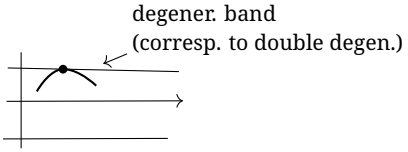


Diagram and typed description.

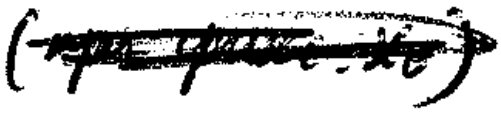


Original source details: GELFAND-P000540

For a const. or zero one it is always thus:

$$\begin{pmatrix} e^{iT} & 0 \\ 0 & e^{-iT} \end{pmatrix} \qquad a_R = \cos T$$

On inverse problems: $a_R(k) = a_R(x_0, k)$ does not determine the potential



Undeciphered passage. After “potential” there is a crossed-out entry in parentheses. Its reading is unresolved.

Original source details: GELFAND-P000540

(≈ inform. about the spectrum).

1) nondegener. $\lambda_n : b(x_0, \lambda) \neq 0$

2) doubly degener. $\lambda_n : b(x_0, \lambda_n) \equiv 0$.

Uncertain reading. A separate heavily overwritten sign precedes “doubly degener.”.

Original source details: GELFAND-P000540

What scattering data determine $u(x)$ (apart from the spectrum). Dubrovin and Faddeev's students are working on it. _____ · _____

If from some index onward $\forall$ eigenvalues are doubly degener., then the forbidden bands contract to a point. _____ · _____

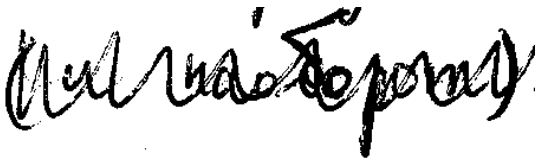
Theorem.

1) Consid. $\sum_{i=0}^{n} Q_i(u, \dots, u^{(k_i)}) \frac{d^i}{dx^i}$ and let

$$[A, L] = P(u, \dots, u^{(\ell)}) \qquad \text{(multiplication oper.)}$$

and consid. the diff. eq. $P(u, \dots, u^{(\ell)}) = 0$ and suppose it is periodic. Then it is a finite-gap potential.

~~or conversely~~ (in fact, the converse is true too).



Manuscript excerpt. “Or conversely” is crossed out with a wavy stroke.

Original source details: GELFAND-P000540

GELFAND-P000541

[source number: 60]

one of them at each odd order.
(and any lin. combination)
2) They are all Hamilton. syst. with n degrees of freedom. _____ · _____
N.'s question: can any potential be approx. by finite-gap ones? _____ · _____

$$P_0 = u', \qquad P_1 = 6uu' - u''' + cu', \dots$$

2) ∃ a matrix of order 2

$$\Lambda(u, \dots, u^{(k)}, \delta) = \begin{pmatrix} \lambda & \mu \\ \bar{\mu} & \bar{\lambda} \end{pmatrix}$$

polynomial of degree $2k+1$ in δ

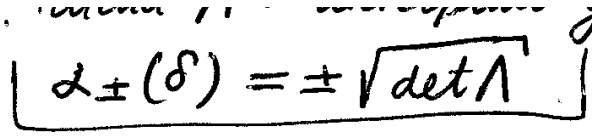
$$\lambda + \bar{\lambda} = 0.$$

and another matrix

$$Q(u, \delta) = i\delta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \frac{iu}{2\delta} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix},$$

such that $P(u, \dots, u^{(2k+1)}) = 0$ is equiv. to the eq. $\Lambda' = [\Lambda, Q]$
(in partic., the eigenvalues of Λ — integrals of the eq.)

$\alpha_\pm(\delta) = \pm\sqrt{\det \Lambda}$ — a polynomial in δ.

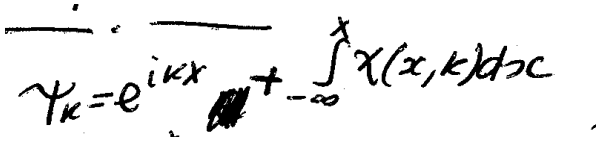


Manuscript excerpt. The symbol reads $\alpha_\pm(\delta)$. The radicand is written $\det \Lambda$.

Original source details: GELFAND-P000541

Commuting integrals — max set. _____ · _____
Consid. an eigenfunc.:

$$\psi_k = e^{ikx + \int_{-\infty}^{x} \chi(x,k)\,dx}, \qquad L\psi = k^2\psi.$$



Undeciphered passage. Below the exponent line after ikx there is a heavy cancellation. The plus and the integral are at exponent height.

Original source details: GELFAND-P000541

Undeciphered passage. A separate crossed-out mark precedes the parenthesis. There is an extra heavy stroke at the end of the word for "correction".

Original source details: GELFAND-P000541

(correction to the phase $\int_{-\infty}^{x} \chi(x,k)\,dx$)

$I = \int \chi(x,k)\,dx$ conserved

$$\chi(x,k) = \sum_{n\geq 1} \frac{\chi_n(x)}{k^n}, \qquad \chi_n(x) \text{ — a polynomial}$$

$$I_n = \int_{-\infty}^{\infty} \chi_n(x)\,dx$$

as a func. of u — int. of the KdV eq.

Let $(*)$ $\underline{\dot{u} = [A,L]}$ — mult. op.;

Undeciphered passage. A crossed-out word precedes L. Its reading is unresolved.

Original source details: GELFAND-P000541

L — Sturm–Liouville oper..

ψ_k — its eigenfunc. (evolution of ψ_k — according to the potential u).

$$\dot{\psi}_k = A\psi_k + \lambda\psi_k + \mu\bar{\psi}_k, \qquad \Lambda = \begin{pmatrix} \lambda & \mu \\ \bar{\mu} & \bar{\lambda} \end{pmatrix}.$$

(tail of the time dep. of the eigenfuncs.)

Manuscript excerpt. The initial letter is retraced. The word reads “tail”.

Original source details: GELFAND-P000541

Apply d/dx — obtain an eq.

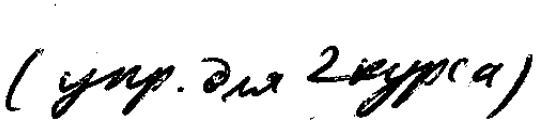
$\psi_k(x_0, x)$

At $x = x_0$ $(1, ik)$

$\underline{\dot{T} = [\Lambda, T]}$ — evolution of the monodromy matr. T

|| *Novikov's thm.*

$$\frac{dT}{dx_0} = [Q,T], \qquad Q \text{ — stand. matrix (exercise for the second year)}$$



Manuscript excerpt. The parenthesis reads “exercise for the second year”.

Original source details: GELFAND-P000541

$\Lambda' = [\Lambda, Q]$. Band edges are the roots of a polynomial. (for stat. solns.)

Yakubovich, Starzhinskii: "Diff. eqs. with periodic coefficients".

Original source details: GELFAND-P000541

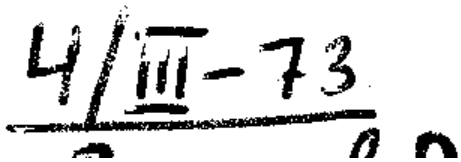
Manuscript excerpt. The upper rule touches the surname "Yakubovich".

GELFAND-P000542

4/III-73 Gelfand's sem..

Zakharov V.

Manuscript excerpt. The year is written "73".

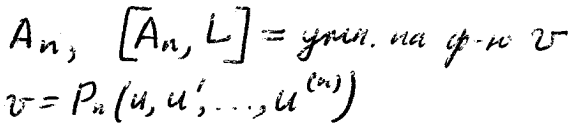


Original source details: GELFAND-P000542

G. $L = -\dfrac{d^2}{dx^2} + u(x)$. Then $\exists$ oper. A_n, $[A_n, L] =$ mult. by the func. v

$$v = P_n(u, u', \ldots, u^{(n)})$$

Uncertain reading. The same form appears in both lines. The provisional reading is v, with ν also possible.



Original source details: GELFAND-P000542

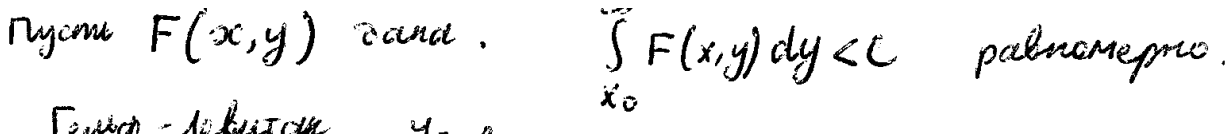
$$A_n = \frac{d^{2n+1}}{dx^{2n+1}} + \ldots \qquad \text{lower terms}$$

How to find A_n? How are A_n and P_n related?

Relation to the trace formula? Other eqs?

3. Results of A. B. Shabat and Zakharov.

Let $F(x, y)$ be given. $\int_{x_0}^{\infty} F(x, y)\, dy < C$ uniformly.



Original source details: GELFAND-P000542

Manuscript excerpt. The word after $F(x, y)$ reads "given".

Gelf.—Levitan. Eq.

$$K(x,y) + F(x,y) + \int_x^\infty K(x,s)F(s,y)\,ds = 0 \tag{1}$$

x — parameter.

We shall consid. them as oper. kernels.

$$1 + \hat{F} = (1 + \hat{K}_+)^{-1}(1 + \hat{K}_-), \qquad \hat{K}_\pm \text{ — Volterra.}$$

$$\begin{aligned} K_+(x,y) &= 0 \quad \text{when } y < x \\ K_-(x,y) &= 0 \quad \text{when } y > x \end{aligned}$$

This is the same thing,

$$K + F + K * F = 0 \qquad \text{form (1)}$$

Differentiate in t (parameter)

$$D_tK + D_tF + D_tK * F + K * D_tF = 0 \tag{2}$$

Let D and $\tilde{D}$ be other operators: D acts on F, $\tilde{D}$ acts on K. We seek to have

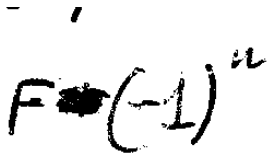
$$\tilde{D}K + DF + \tilde{D}K * F + K * DF = 0 \tag{3}$$

Let

$$DF = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}, \qquad \tilde{D}K = \frac{\partial K}{\partial x} + \frac{\partial K}{\partial y}.$$

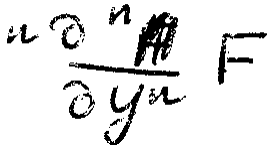
Then all is in order (direct calculation) shows that identity (3) holds.

$$DF = \frac{\partial^n}{\partial x^n}F \,?\, (-1)^n \frac{\partial^n}{\partial y^n}F \tag{4}$$



Original source details: GELFAND-P000542

Uncertain reading. The sign between the first F and $(-1)^n$ is heavily obliterated.



Original source details: GELFAND-P000542

Undeciphered passage. In the numerator of the second differential operator, to the right of ∂^n, there is a crossed-out sign.

For it one can construct $\tilde{D}$.

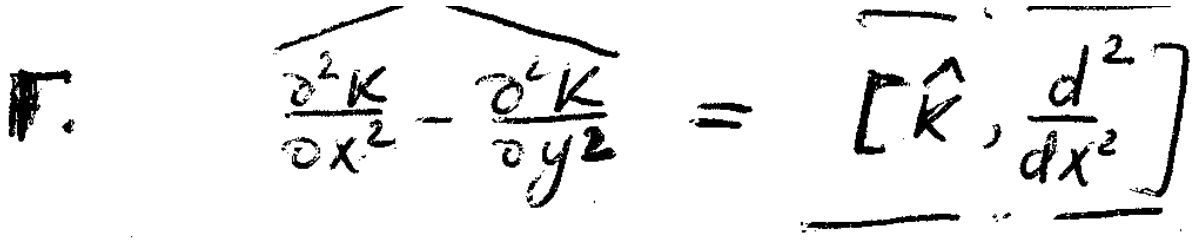
$DF = [D_n\, F]$, more precisely $DF =$ the kernel of $[\hat{D}_n, \hat{F}]$, where $\hat{D}_n = \dfrac{\partial^n}{\partial x^n}$.

Namely

$$\tilde{D}K = L_n K + (-1)^n \frac{\partial^n}{\partial y^n} K, \tag{5}$$

G.

$$\widehat{\frac{\partial^2 K}{\partial x^2} - \frac{\partial^2 K}{\partial y^2}} = \left[\hat{K}, \frac{d^2}{dx^2}\right]$$



Original source details: GELFAND-P000542

Manuscript excerpt. An arch spans the difference of derivatives. On the right $\hat{K}$ is first in the brackets.

where $L_n = \dfrac{\partial^n}{\partial x^n} + \ldots$ (lower terms)

K, F can also be matrices.

GELFAND-P000543

[source number: 61]

$$L_n = \frac{\partial^n}{\partial x^n} + n \underbrace{\left(\frac{d}{dx} K(x,x)\right)}_{u_0(x)} \frac{\partial^{n-2}}{\partial x^{n-2}} + \ldots \tag{6}$$

There are recurr. formulas for the lower terms.

L_n — nonlinear operator in K, but its coeffs. (all of them) are made from the quantities

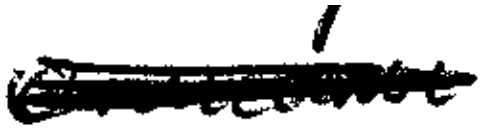
$$\xi_n(x) = \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right)^n K\Big|_{y=x}$$

$$u_0 = \frac{d\xi_0}{dx}$$

The coeffs. are already constant (of these).

These are not all the operators that differ. this convolution.

These D — linear opers. with const. coeffs. Variable ones are possible too. This is a Lie algebra



Undeciphered passage. A crossed-out entry follows the opening parenthesis. Its reading is unresolved.

Original source details: GELFAND-P000543

(all D, that can be extended (A. Shabat) — they are described).

Suppose that (1) is uniquely solvable for K (F given). And let F be such that $\exists D\colon DF = 0$. Hence

$$\tilde{D}K = 0 \tag{7}$$

(from (2) and unique solvability of the eq.)

Hence

$$DF = 0 \longleftrightarrow \tilde{D}K = 0. \tag{8}$$

Obtain some linearizable eqs. But they are uninteresting.

Let

$$D_1F = 0 \quad \text{and} \quad D_2F = 0 \tag{9}$$

Uncertain reading. A separate loop-like stroke stands to the left of “Let”. Its meaning is unresolved.

Original source details: GELFAND-P000543

Uncertain reading. The crossed-out word provisionally reads “where”.

Original source details: GELFAND-P000543

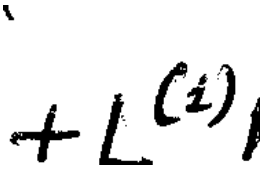
2 parameters are given t, z; $\quad F = F(x, y; t, z)$ and

$$D_1F = \alpha\frac{\partial F}{\partial t} + ([\hat{L}_0^{(1)}, \hat{F}])_{\text{kernel}} \tag{10}$$

$$D_2F = \beta\frac{\partial F}{\partial z} + ([\hat{L}_0^{(2)}, \hat{F}])_{\text{kernel}} \tag{11}$$

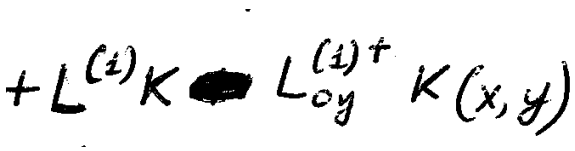
$$\hat{L}_0^{(1,2)} = \sum C^{(n)}\frac{\partial^n}{\partial x^n}$$

$$\tilde{D}_1K = \alpha\frac{\partial K}{\partial t} + L^{(i)}K\ ?\ L_{0y}^{(1)+}K(x,y) \tag{12}$$



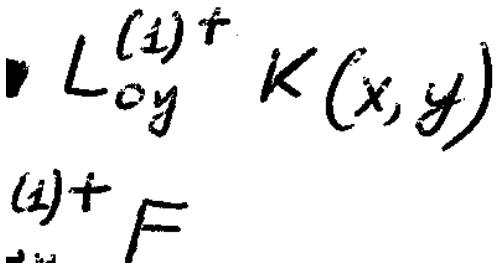
Manuscript excerpt. The index above the first L has a distinct dot and reads i.

Original source details: GELFAND-P000543



Uncertain reading. The sign between $L^{(i)}K$ and the next term is heavily obliterated.

Original source details: GELFAND-P000543



Manuscript excerpt. The last operator in (12) and on the following line carries a superscript $+$. The sign is absent in (13).

Original source details: GELFAND-P000543

$$[\hat{L}^{(1)}, \hat{F}] = L_x^{(1)} F - L_y^{(1)+} F$$

$$\tilde{D}_2 K = \beta \frac{\partial K}{\partial z} + L^{(2)} K - L_{0y}^{(2)} K \tag{13}$$

$$\boxed{D_1 F = 0,\ D_2 F = 0 \Longrightarrow \tilde{D}_1 K = 0,\ \tilde{D}_2 K = 0.} \tag{14}$$

Theorem.

$$\boxed{\beta \frac{\partial L^{(1)}}{\partial z} - \alpha \frac{\partial L^{(2)}}{\partial t} = [L^{(1)}, L^{(2)}]} \tag{15}$$

$L^{(1)}, L^{(2)}$ — linear operators (for given K).

In $L^{(1)}$ and $L^{(2)}$ the coeffs. are expressed in terms of

$$\xi_n(x) = \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y} \right)^n K(x,y) \Big|_{y=x}$$

GELFAND-P000544

This is a syst. of nonlin. diff. eqs. for $\xi_n(x, y, t, z)$.

Uncertain reading. The crossed-out argument after x provisionally reads y.

Original source details: GELFAND-P000544

Undeciphered passage. A short crossed-out entry follows the first line. Its reading is unresolved.

Original source details: GELFAND-P000544

The syst. for K reduces to the syst. for F. using the G.–L. eq.

Manuscript excerpt. ~~from F construct $K \Rightarrow \xi_n$~~

Original source details: GELFAND-P000544

Undeciphered passage. There is a heavily crossed-out fragment after the arrow at the end of the line.

Original source details: GELFAND-P000544

Give a pair of lin. diff. ops. with const. coeffs. $L_0^{(1)}$ and $L_0^{(2)}$
from the pair construct the syst. of eqs. (15). Let $D_1F = D_2F = 0$, where
$F = F(x, y; t, z)$

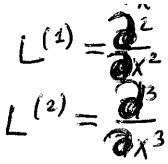
From F construct K, and from K construct a soln. of the system.
Obtain a solution of the system.
1st nontriv. — eq. Kadomtsev–Petriashvili (?).

Original source details: GELFAND-P000544

Manuscript excerpt. The manuscript reads "Kadomtsev–Petriashvili" with a question mark in parentheses.

$$L_0^{(1)} = \frac{\partial^2}{\partial x^2}, \qquad L_0^{(2)} = \frac{\partial^3}{\partial x^3} + \lambda\frac{\partial}{\partial x}.$$

$$L^{(1)} = \frac{\partial^2}{\partial x^2} + u(x), \qquad \text{where } u(x) = 2\frac{d}{dx}K(x,x),$$
$$L^{(2)} = \frac{\partial^3}{\partial x^3} + \frac{3}{4}\left(u\frac{\partial}{\partial x} + \frac{\partial}{\partial x}u\right) + \lambda\frac{\partial}{\partial x} + w,$$



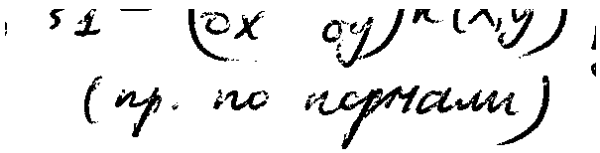
Original source details: GELFAND-P000544

Manuscript excerpt. The ∂ signs in the leading terms of both operators are retraced.

where

$$w = \frac{3}{2}\frac{d}{dx}(\xi_1 + \xi_0^2)$$
$$\xi_0 = K(x,x), \qquad \xi_1 = \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right)K(x,y)\Big|_{y=x}$$

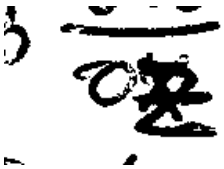
(deriv. in the normal direction)



Original source details: GELFAND-P000544

Manuscript excerpt. The parenthesis reads "deriv. in the normal direction".

$$\begin{cases} \frac{3}{4}\beta\frac{\partial u}{\partial z} = -\frac{\partial w}{\partial x}; \\ \beta\frac{\partial w}{\partial y} = \frac{1}{4}\left(u_{xxx} + 3uu_x\right) + \lambda u_x + \alpha\frac{\partial}{\partial t}u \end{cases}$$

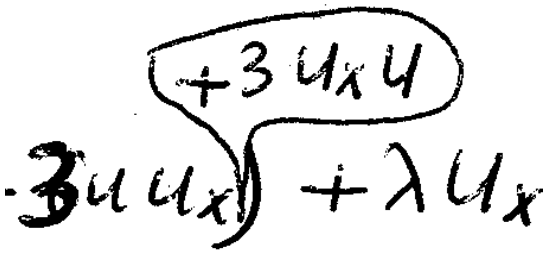


Original source details: GELFAND-P000544

Manuscript excerpt. In the denominator of the derivative in the first equation, z is written over earlier strokes.

Manuscript excerpt. In the second equation the variable below the derivative sign clearly reads y.

Original source details: GELFAND-P000544



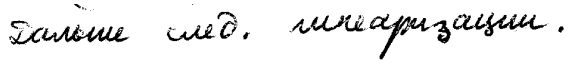
Manuscript excerpt. The coefficient 3 of uu_x is retraced. A circled insertion $\boxed{+3u_xu}$ is attached to the closing parenthesis.

Original source details: GELFAND-P000544

u can be matrix-valued

$$\begin{pmatrix} u_0 & u_1 \\ 0 & u_0 \end{pmatrix}$$ for u_1 obtain a linearized eq. for u_0.

Next come the following linearizations.



Manuscript excerpt. The abbreviation before "linearizations" reads "след.".

Original source details: GELFAND-P000544

β real or imaginary, λ...

All have a phys. meaning.

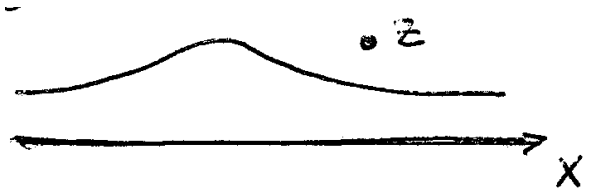
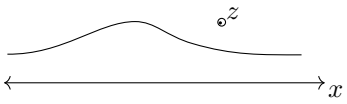
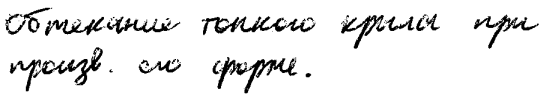


Diagram and typed description.



Original source details: GELFAND-P000544

Flow around a thin wing of arbitrary shape.



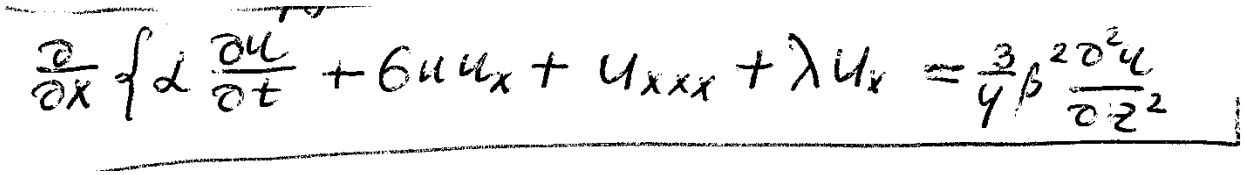
Manuscript excerpt. The manuscript describes a wing of arbitrary shape.

Original source details: GELFAND-P000544

KdV if independent of z.

Nonlin. string.

$$\frac{\partial}{\partial x}\left\{\alpha\frac{\partial u}{\partial t}+6uu_x+u_{xxx}+\lambda u_x=\frac{3}{4}\beta^2\frac{\partial^2 u}{\partial z^2}.\right.$$



Original source details: GELFAND-P000544

Manuscript excerpt. The sum starts with an opening brace. The manuscript has no closing brace before the equals sign.

A medium with sound dispersion and shallow water.

Karpman's book — see the derivations there. (book on nonlin. waves)

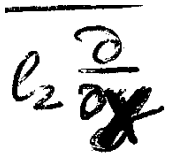
Uncertain reading. The middle letter of the surname is overwritten. The reading "Karpman" is provisional.

Original source details: GELFAND-P000544

$$L_0^{(1)} = \ell_1 \frac{\partial}{\partial x} + \ell_2 \frac{\partial}{\partial y}, \qquad [\ell_1, \ell_2] = 0, \qquad \ell_i \text{ — matrices (diagonal)},$$

$$L^{(1)} = \ell_1 \frac{\partial}{\partial x} + [\ell_1, \xi_0], \qquad \xi_0(x) = K(x,x), \qquad K \text{ — a matrix},$$

$$L^{(2)} = \ell_2 \frac{\partial}{\partial x} + [\ell_2, \xi_0].$$



Uncertain reading. The variable in the denominator of the second term of $L_0^{(1)}$ is overwritten. The reading y is provisional.

Original source details: GELFAND-P000544

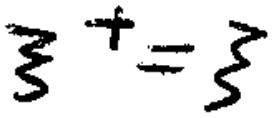
GELFAND-P000545

[source number: 62]

$$\ell_1 = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}; \qquad \ell_2 = \begin{pmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{pmatrix}.$$

The 1st nontriv. one arises in a matrix of size 3×3

$$\frac{\partial \xi}{\partial t} + \vec{V} \nabla \xi \qquad u_1 = \xi_{13}, \quad u_2 = \xi_{12}, \quad u_3 = \xi_{23}. \qquad \xi^+ = \xi$$



Manuscript excerpt. The superscript on ξ is clearly a $+$.

Original source details: GELFAND-P000545

$$\begin{cases} \dfrac{\partial u_1}{\partial t} + \vec{V}_1 \nabla u_1 = i\lambda u_2 u_3, \\ \dfrac{\partial u_2}{\partial t} + \vec{V}_2 \nabla u_2 = i\lambda u_1 u_3^*; \\ \dfrac{\partial u_3}{\partial t} + \vec{V}_3 \nabla u_3 = i\lambda u_1 u_2^* \end{cases}$$

Manuscript excerpt. The rewritten subscript of the last factor reads 2.

Original source details: GELFAND-P000545

$u_i(t,x,z)$, ∇ — 2-dimensional, $\vec{V}_i$ — arb. vectors

Describes the interaction of waves of 3 types in a nonlin. medium.

Dissipation cannot be introduced (ever). _____ · _____

Conjecture: they are all Hamiltonian. From one L one constructs the symplect. form, and from the other, the Hamiltonian. _____ · _____

$u_{tt} - u_{xx} = \sin u$ also comes from the matrices

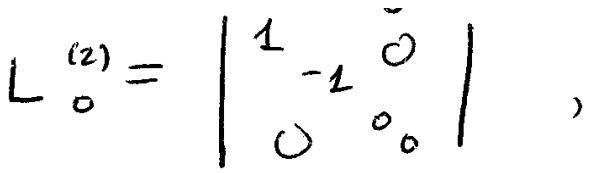


Diagram and typed description.

$$L_0^{(2)} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

Original source details: GELFAND-P000545

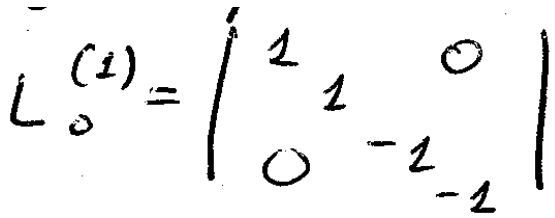
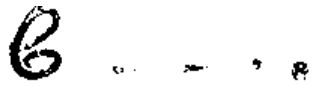


Diagram and typed description.

$$L_0^{(1)} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix}$$

Original source details: GELFAND-P000545

and an eq. for propag. of a laser pulse in . . .



Manuscript excerpt. The manuscript has a series of dots after “in”.

Original source details: GELFAND-P000545

_____ · _____

Bloch eq. for density matrices (quant. mech. in a 2-level system). _____ · _____

Matrix ops. of high orders.

$$L_0^{(1)} = \ell_1 \frac{\partial}{\partial x}, \qquad L_0^{(2)} = \frac{\partial^2}{\partial x^2}.$$

One obtains

$$i\frac{\partial \xi}{\partial t} + \xi_{xx} \pm |\xi|^2 \xi = 0 \qquad \text{(nonlin. Schrödinger eq.)}$$

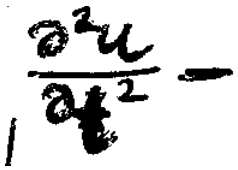
there is a scheme for solving the Cauchy prob. _____ · _____

11/III-74. Gelfand's sem..

Diagram and typed description.
Povzner. Witham.

Original source details: GELFAND-P000546

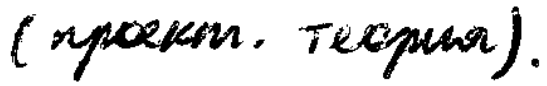
$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + T'(u) = 0$$



Manuscript excerpt. The variable t in the denominator of the first term is retraced.

Original source details: GELFAND-P000546

1) a wave travels $x \sim t$ (pract. theory).



Manuscript excerpt. The parenthesis reads "pract. theory".

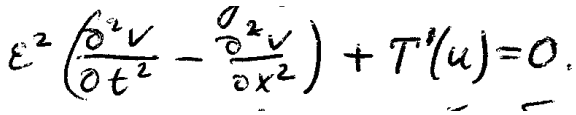
Original source details: GELFAND-P000546

and the asympt. of what happens far away

$$x \longrightarrow \frac{x}{\epsilon}, \qquad t \longrightarrow \frac{t}{\epsilon}$$

the asympt. is now sought in a finite region and as $\epsilon \to 0$.
$f(x - vt)$, v depends little on t and x (varies slowly in the sense of ϵ).
The method consists of the following:

$$\epsilon^2 \left(\frac{\partial^2 v}{\partial t^2} - \frac{\partial^2 v}{\partial x^2} \right) + T'(u) = 0.$$



Manuscript excerpt. Both numerators carry v, and the last term is $T'(u)$.

Original source details: GELFAND-P000546

study the asympt. in a finite region.

Add. assumption (justif. by scientific experience):

$$u(x,t) = f\left(\frac{\Theta_0(x,t)}{\epsilon}; x, t\right)$$

$$f(x_{\text{old}} - ct_{\text{old}}) = f\left(\frac{x-ct}{\epsilon}\right)$$

— exact trav. waves are a special case with $\Theta_0(x,t) = x - ct$

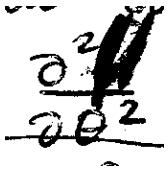
Manuscript excerpt. Both coordinate subscripts read “cr”, translated as “old”.

Original source details: GELFAND-P000546

$$\frac{\Theta_0(x,t)}{\epsilon} = \Theta \qquad f(\Theta; x, t).$$

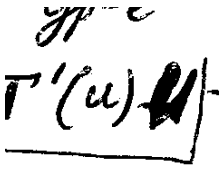
hence one obtains a diff. eq.

$$\epsilon^2(\omega_2^2 - \omega_1^2)\frac{\partial^2 u}{\partial \Theta^2} + T'(u)u + [\epsilon(u_x, u_t)\ldots] = 0$$



Uncertain reading. The letter in the derivative numerator is heavily overwritten. The reading u is provisional.

Original source details: GELFAND-P000546



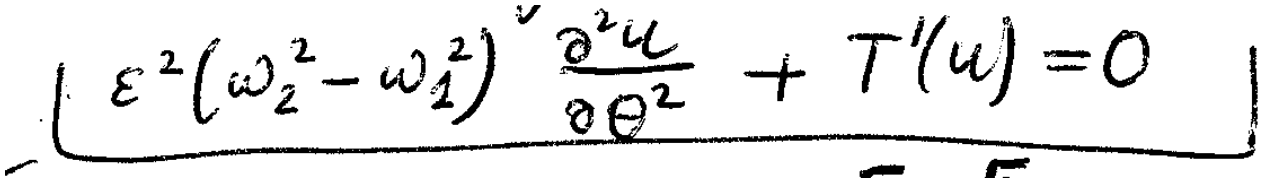
Uncertain reading. Immediately after $T'(u)$ is a heavily overwritten mark. The reading u is provisional.

Original source details: GELFAND-P000546

where $\omega_2 = \Theta_{0t}, \quad \omega_1 = \Theta_{0x}$.

As $\epsilon \to 0$ one obtains

$$\epsilon^2(\omega_2^2 - \omega_1^2)\frac{\partial^2 u}{\partial \Theta^2} + T'(u) = 0$$



Original source details: GELFAND-P000546

Manuscript excerpt. The factor ϵ^2 remains in the displayed equation after $\epsilon \to 0$.

ordinary eq. of 2nd order, E_1, E_2 — arb. consts.

$$u = u(E_1, E_2; x, t, \Theta)$$

Equation for Θ.

Now substitute $\Theta = \dfrac{\Theta_0(x,t)}{\epsilon}$.

$$u = u\left(E_1, E_2; \frac{\Theta_0(x,t)}{\sqrt{\omega_2^2 - \omega_1^2} \cdot \varphi(E) \cdot \epsilon}\right)$$

(the solution is obtained in this form)

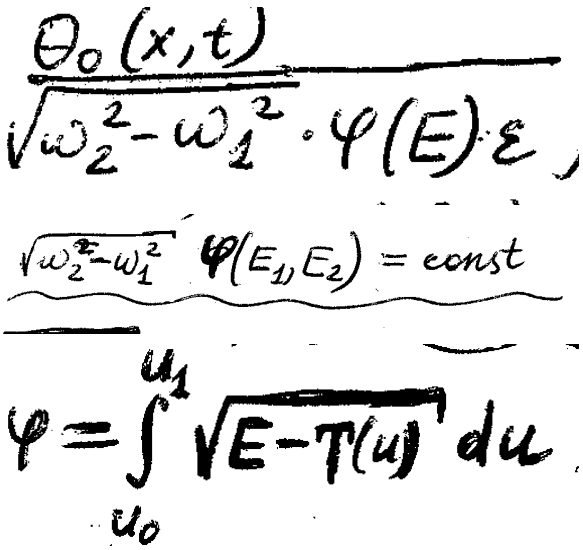
Add. eq.:

Uncertain reading. A crossed-out symbol resembling Θ_0 precedes the square root.

Original source details: GELFAND-P000546

$$\sqrt{\omega_2^2 - \omega_1^2}\,\varphi(E_1, E_2) = \text{const}$$

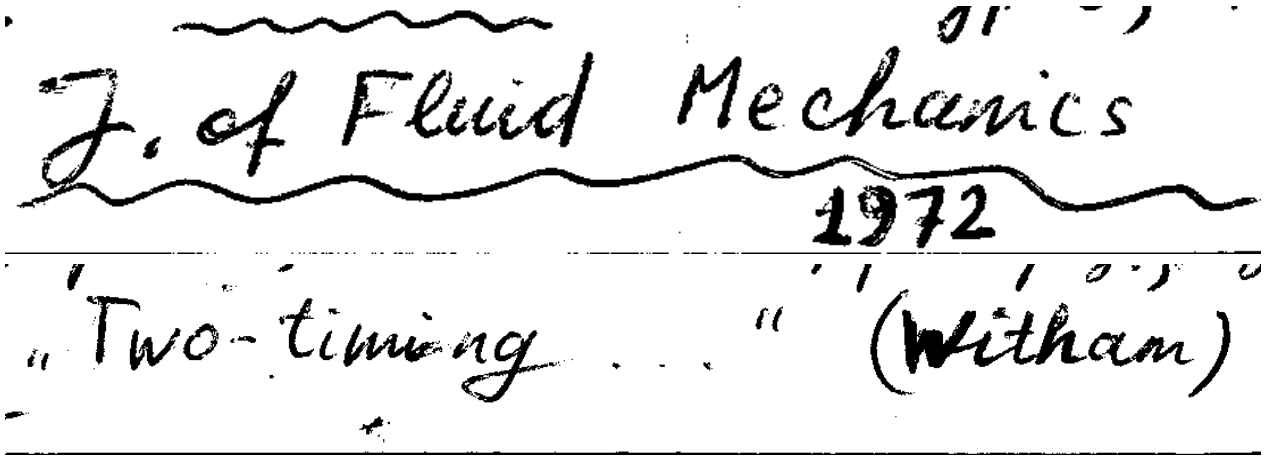
$$\varphi = \int_{u_0}^{u_1} \sqrt{E - T(u)}\, du$$



Manuscript excerpt. The letter φ has a closed right loop in all three places. The middle occurrence is retraced.

Original source details: GELFAND-P000546

Ref.: Witham. This eq., Korteweg, heavy fluid, radiophys., lasers.
J. of Fluid Mechanics 1972 "Two-timing ..." (Witham)



Original source details: GELFAND-P000546

Manuscript excerpt. The parenthesized name and the year follow the manuscript. "1972" is written below the journal title.

GELFAND-P000547

[source number: 66]

V. N. Tulovsky. On Hörmander's work.
Wave fronts.

$$u \in \mathcal{D}'(\Omega), \qquad \Omega \subset \mathbb{R}^n.$$

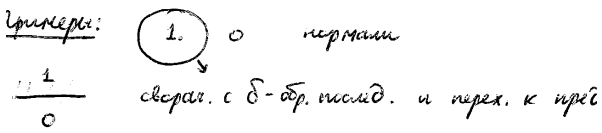
$(x_0, \xi_0) \notin WF(u)$, if $\exists \varphi \in C_0^\infty(\Omega), \quad \varphi(x_0) \neq 0$

$$\left|\langle u, \varphi e^{-i\langle x,\xi\rangle}\rangle\right| \leq C_N(1+|\xi|)^{-N}$$

in a small conic neighborhood of the point ξ_0:

$$\left|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}\right| < \epsilon.$$

Examples:



Original source details: GELFAND-P000547

Diagram and typed description.

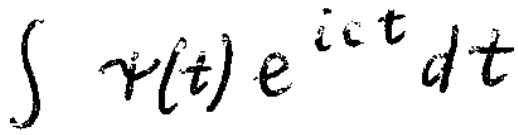
normals
convol. with a δ-like seq. and pass. to the limit.

G.

$$\int \varphi(x_1, x_2) e^{i(x_1\xi_1 + x_2\xi_2)}\, dx_1\, dx_2$$

stat. int. over the plane.

$$\int \psi(t) e^{ict}\, dt$$



Manuscript excerpt. A faint c is written between i and t.

Original source details: GELFAND-P000547

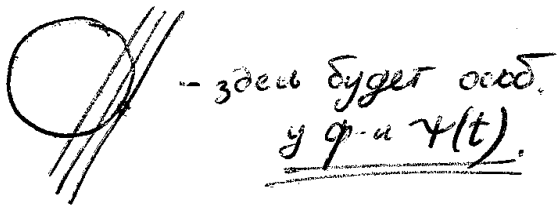
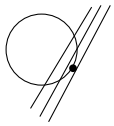


Diagram and typed description.



— there will be a sing. here of the func. $\psi(t)$.

Original source details: GELFAND-P000547

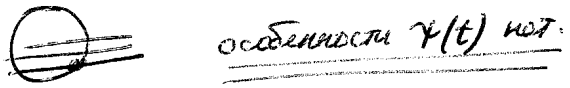


Diagram and typed description.

no singularity of $\psi(t)$.

Original source details: GELFAND-P000547

Tul. $\pi WF(u) = \operatorname{sing\,supp} u.$

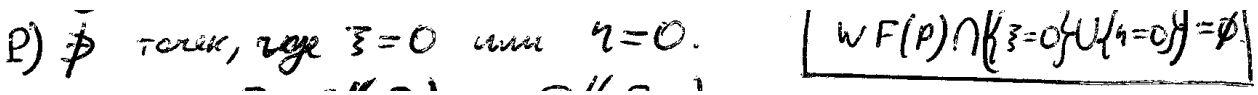
$$\pi(x, \xi) = x.$$

$$P : C_0^\infty(\Omega) \longrightarrow C^\infty(\Omega_1), \qquad \exists K_P \in \mathcal{D}'(\Omega \times \Omega_1),$$
$$\langle K_P, u(x)v(y)\rangle = \langle Pu, v\rangle$$
$$WF(K_P) \underset{\text{def}}{=} WF(P)$$

In the space (x, y, ξ, η).

Suppose $WF(P) \not\ni$ points where $\xi = 0$ or $\eta = 0$.

$$\boxed{WF(P) \cap (\{\xi = 0\} \cup \{\eta = 0\}) = \varnothing}$$



Original source details: GELFAND-P000547

Manuscript excerpt. A slashed $\ni$ precedes "points".

Then one can extend $P : \mathcal{E}'(\Omega) \to \mathcal{D}'(\Omega_1)$.

Def.

$$WF'(P) = \{(x, y, \xi, \eta) : (x, y, \xi, -\eta) \in WF(P)\}.$$

Def.

$$WF'(P) \circ WF(u) = \{(y, \eta) : \exists (x, y, \xi, \eta) \in WF'(P),\ (x, \xi) \in WF(u)\}.$$

Theorem.

$$WF(Pu) \subset WF'(P) \circ WF(u).$$

Suppose there are 2 operators:

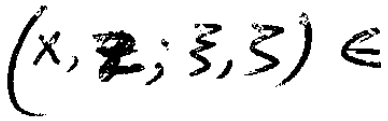
$$Q : \underset{y}{\mathcal{E}'(\Omega_1)} \longrightarrow \underset{z}{\mathcal{D}'(\Omega_2)},$$
$$P : \underset{x}{\mathcal{E}'(\Omega)} \longrightarrow \underset{y}{\mathcal{E}'(\Omega_1)}$$

$$WF'(Q \circ P) \subseteq WF'(P) \circ WF'(Q).$$

Original source details: GELFAND-P000547

Manuscript excerpt. The left side has $Q \circ P$ in parentheses. On the right $WF'(P)$ comes first.

$$(x, z; \xi, \zeta) \in WF'(P) \circ WF'(Q) \underset{\text{def}}{\Longleftrightarrow} \exists\, (x, y, \xi, \eta) \in WF'(P),$$
$$(y, z, \eta, \zeta) \in WF'(Q).$$



Manuscript excerpt. In the second coordinate on the left, z is written over earlier strokes.

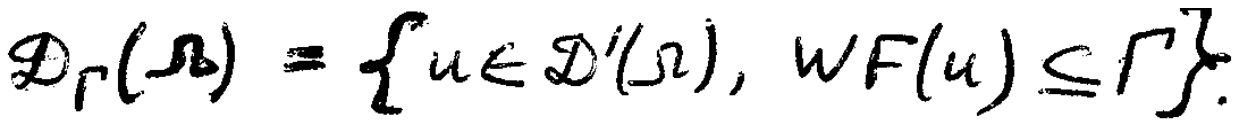
Original source details: GELFAND-P000547

GELFAND-P000548

Product of generalized funcs..

$\Gamma \subseteq T^*(\Omega) \setminus 0$, Γ — a conic closed set.

$$\mathcal{D}_\Gamma(\Omega) = \{u \in \mathcal{D}'(\Omega),\ WF(u) \subseteq \Gamma\}.$$



Original source details: GELFAND-P000548

Manuscript excerpt. The first $\mathcal{D}_\Gamma$ is written without a prime. The inner $\mathcal{D}'$ carries one.

Theorem: let Γ_1, Γ_2 — 2 closed conic sets in $T^*(\Omega) \setminus 0$ and suppose

$$\Gamma_1 + \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$$

$$(x,\xi) \qquad (x,\eta) \qquad \Gamma_1 + \Gamma_2 = \{(x, \xi+\eta)\}, \qquad \xi + \eta \neq 0$$

Then there exists a cont. mapping

$$\mathcal{D}_{\Gamma_1}(\Omega) \times \mathcal{D}_{\Gamma_2}(\Omega) \longrightarrow \mathcal{D}_{\Gamma_1 \cup \Gamma_2 \cup (\Gamma_1+\Gamma_2)}(\Omega),$$

extending the product of ordinary funcs.
(and the product of a generalized func. by a smooth one)
(It is unique owing to the topology consid.)

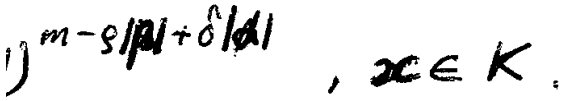
Def. of Fourier integral ops.

$\varphi(x,\Theta), \quad (x,\Theta) \in \Omega \times (\mathbb{R}^N \setminus 0)$ — a phase func., if it is real, homogeneous of order 1 [in Θ] and $\varphi'_{x,\Theta} \neq 0$.

Original source details: GELFAND-P000548

Manuscript excerpt. The circled insertion "in Θ" is attached to "homogeneous".

$$S^m_{\rho,\delta}(X \times \mathbb{R}^N) \ni a(x,\Theta),$$
$$\left|\partial_x^\alpha \partial_\Theta^\beta a(x,\Theta)\right| \leq C_{\alpha,\beta}(1+|\Theta|)^{m-\rho|\beta|+\delta|\alpha|}, \qquad x \in K.$$



Original source details: GELFAND-P000548

Uncertain reading. Two indices in the exponent and the variable before $\in K$ are retraced. The reading is $m - \rho|\beta| + \delta|\alpha|$ and $x \in K$.

$$\langle A, u\rangle = \int e^{i\varphi(x,\Theta)} a(x,\Theta) u(x)\, dx\, d\Theta$$
$$= \lim_{\epsilon \to 0} \int e^{i\varphi(x,\Theta)} a(x,\Theta) \chi(\epsilon\Theta) u(x)\, dx\, d\Theta$$

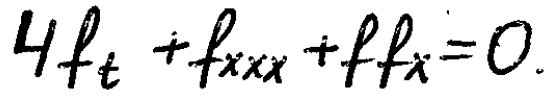
where $\chi \in C_0^\infty(\mathbb{R}^N)$, $\chi = 1$ when $|\Theta| \leq 1$.

Original source details: GELFAND-P000548

Manuscript excerpt. The last condition clearly reads $|\Theta| \leq 1$.

18/III-74. Gelfand's sem.. A. B. Shabat.

$$4f_t + f_{xxx} + ff_x = 0.$$



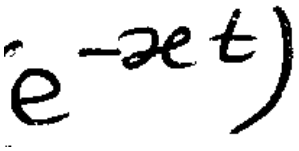
Manuscript excerpt. The plus sign is followed immediately by ff_x.

Original source details: GELFAND-P000549

$$f(x,t) = g(x-ct) = \frac{2\lambda^2}{\mathrm{ch}^2[\lambda(x-\lambda^2 t - h)]};$$

2-solit. solution

$$f(t,x) = \frac{\lambda_1}{\cdot} + \frac{\lambda_2}{\cdot} + O(e^{-\varkappa t}) \quad \text{as } t \to +\infty.$$



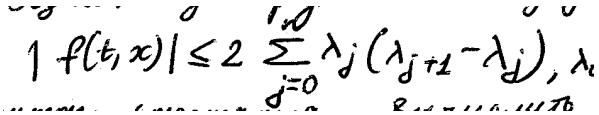
Uncertain reading. The letter in the exponent has two loops. The reading $\varkappa$ is provisional.

Original source details: GELFAND-P000549

one can prove that it is unique.

1) Estimate the total amplitude at the instant of collision.

$$|f(t,x)| \le 2\sum_{j=0}^{n} \lambda_j(\lambda_{j+1} - \lambda_j), \qquad \lambda_0 = 0$$



Uncertain reading. A small sign above the sum touches a descender from the adjacent line. The reading n is provisional.

Original source details: GELFAND-P000549

(the amplit. do not add as in the lin. case)

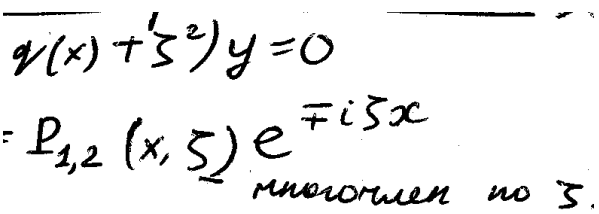
2) The centers will shift. Calculate the shift. The answer is stand. — scattering.

Bargmann potential $q(x)$, if $\exists 2$ solns. of the eq.

$$y'' + (q(x) + \zeta^2)y = 0$$

of the form

$$y_{1,2} = P_{1,2}(x,\zeta)e^{\mp i\zeta x} \qquad \text{polynomial in } \zeta.$$



Manuscript excerpt. The spectral letter reads ζ.

Original source details: GELFAND-P000549

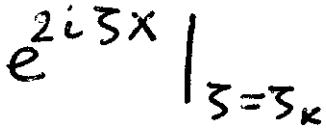
Undeciphered passage. A neighboring entry is cut off at the left edge. The closing parenthesis and equals sign are visible $\ldots) = \ldots$

Original source details: GELFAND-P000549

GELFAND-P000550

Let P, Q be such that:

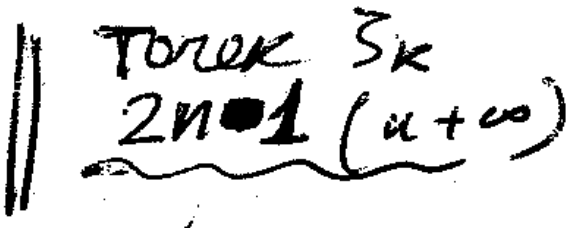
$$c_k e^{2i\zeta x}\Big|_{\zeta=\zeta_k} = \frac{P(x,\zeta)}{Q(x,\zeta)}\Big|_{\zeta=\zeta_k}, \qquad \frac{P}{Q}\Big|_{\zeta=\infty} = 1$$



Manuscript excerpt. The spectral letter reads ζ.

Original source details: GELFAND-P000550

|| Points ζ_k $2n\,?\,1$ (and $+\infty$)



Uncertain reading. Ink obscures the sign between $2n$ and 1.

Original source details: GELFAND-P000550

For any choice of parameters c_k, ζ_k one obtains a fund. syst. of solutions of the S.–L. oper. with Bargmann pot.

$$y_1 = P(x,\zeta)e^{i\zeta x}$$
$$y_2 = Q(x,\zeta)e^{-i\zeta x}$$

1.

$$P'_x Q - Q'_x P - 2i\zeta PQ = \prod(\zeta - \zeta_k).$$

Diff. with respect to x.

$$\frac{d}{dx}\frac{P}{Q}\Big|_{\zeta=\zeta_k} = \frac{P'_x Q - Q'_x P}{Q^2}\Big|_{\zeta=\zeta_k} = 2i\zeta\,\frac{P}{Q}\Big|_{\zeta=\zeta_k}$$

is a rat. func. in ζ with zeros at the [crossed out] giv. points

Uncertain reading. The short crossed-out entry before “giv.” is illegible.

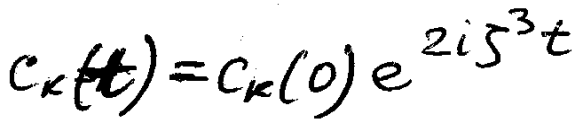
Original source details: GELFAND-P000550

$$\left.\frac{P_xQ - Q_xP - 2i\zeta PQ}{Q^2}\right|_{\zeta=\zeta_k} = 0.$$

and one cond. at ∞.

Now let c_k depend on t (the parameter):

$$c_k(t) = c_k(0)e^{2i\zeta^3 t}$$



Uncertain reading. The argument of $c_k(t)$ reads t over earlier strokes. The underlying entry is illegible.

Original source details: GELFAND-P000550

$$\left.\frac{P}{Q}\right|_{\zeta=\zeta_k} = \left. c_k(t)e^{2i\zeta x}\right|_{\zeta=\zeta_k}$$

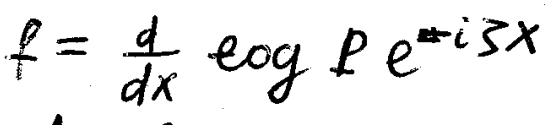
Then also

$$\left(4\frac{\partial}{\partial t} + \frac{\partial^3}{\partial x^3} - 3q\frac{\partial}{\partial x}\right)P = (\qquad)Q = 0$$

Manuscript excerpt. The manuscript leaves the parentheses before Q empty.

Original source details: GELFAND-P000550

Then $f = \frac{d}{dx}\log Pe^{?i\zeta x}$ satisf.



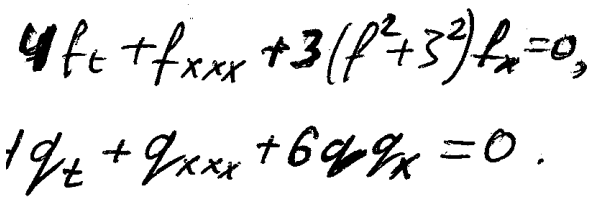
Uncertain reading. The sign before $i\zeta x$ is overwritten and its reading remains unresolved.

Original source details: GELFAND-P000550

$$4f_t + f_{xxx} + 3(f^2 + \zeta^2)f_x = 0,$$

while $f' + f^2 = q$ satisf. the eq.

$$4q_t + q_{xxx} + 6qq_x = 0.$$



Manuscript excerpt. In the first equation the 4, the $+$ sign and the 3 are retraced. In the second, qq_x is retraced.

Original source details: GELFAND-P000550

GELFAND-P000551

25/III-74. Gelfand's sem..

Kirillov. On inf. groups. Descr. of repres. of the fol. groups

$$G_1 = \text{Diff}\, M, \qquad G_2 = H^M = C^\infty(M, H) \qquad (H \text{ is a group}).$$

Uncertain reading. A separate two-stroke glyph stands to the left of the diagram. Its reading remains unresolved.

Original source details: GELFAND-P000551

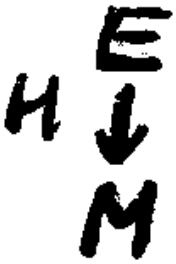


Diagram and typed description.

$$\begin{array}{cc} & E \\ H & \downarrow \\ & M \end{array}$$

Original source details: GELFAND-P000551

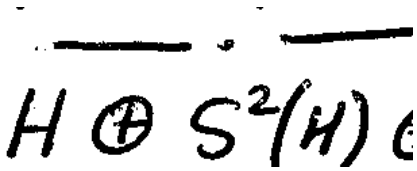
principal bundle and its automorphism group

It projects onto G_1 with fiber G_2

Undeciphered passage. The rectangular entry below the diagram is densely crossed out and illegible.

Original source details: GELFAND-P000551

$$e^H = H_0 \oplus H \oplus S^2(H) \oplus S^3(H) \oplus \dots$$

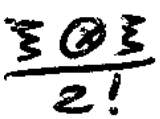


Uncertain reading. Two horizontal strokes and a small loop between them stand above $H, \oplus, S^2(H)$. Their meaning and attachment remain unresolved.

Original source details: GELFAND-P000551

(H_0 is one-dimensional; $S^k(H)$ is a symm. power

$$1 \qquad \xi \qquad \frac{\xi \otimes \xi}{2!} \qquad\qquad |\xi|^2 = \sum \frac{|\xi_n|^2}{n!}$$



Uncertain reading. The circled sign in the numerator provisionally reads $\otimes$.

Original source details: GELFAND-P000551

Gurevich. On the K.–d.–V. eq. by Whitham's method.

Uncertain reading. The ending of the abbreviated word for "equation" is overwritten in the Russian. Its exact reading remains unresolved.

Original source details: GELFAND-P000551

GELFAND-P000552

1/*IV*-74. Gelfand's sem..

Yura Koblents. On the asymptotics of the eigenvalues of the Laplace oper..

(from Leningrad)

M — closed Riem. manif., Δ — Beltrami–Laplace oper..

$$\lambda_0 = 0, \qquad 0 \le \lambda_1 \le \lambda_2 \le \dots \qquad \text{— spectrum.}$$

$$Z(s) = \operatorname{Tr}\exp(-s\Delta) = \sum_i e^{-\lambda_i s}$$

$$\Delta f = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i}\left(\sqrt{g}\, g^{ij} \frac{\partial f}{\partial x^j}\right), \qquad \text{where } g = \det \|g_{ij}\|,$$

Undeciphered passage. There is a densely crossed-out entry between the equals sign and $1/\sqrt{g}$. A stroke on the right cannot confidently be separated from the cancellation.

Original source details: GELFAND-P000552

$$E(m, m', t) = \sum e^{-\lambda_i t} \varphi_i(m) \varphi_i(m').$$

Both factors use φ.

Original source details: GELFAND-P000552

$$Z(s) = \int E(m, m, s)\, dm$$

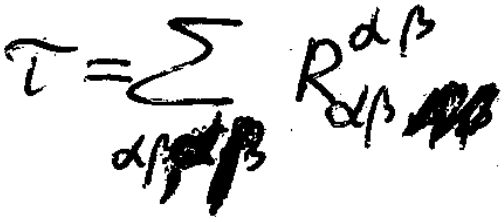
Minakshisundaram, 1949

$$Z(t) = (4\pi t)^{-n/2}(a_0 + a_1 t + a_2 t^2 + \ldots), \qquad t \to 0.$$

$$n = \dim M, \qquad a_0 = \operatorname{vol} M, \qquad \operatorname{Re} t > 0.$$

$$a_1 = \int_M \frac{\tau}{6}\, dV, \qquad \text{where } \tau \text{ — scal. curvature}$$

$$\tau = \sum_{\alpha\beta[?]\beta} R^{\alpha\beta}_{\alpha\beta[\ldots]}$$



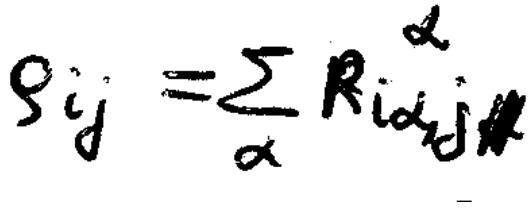
Uncertain reading. The sum index shows an initial $\alpha\beta$ and a final β, with overwritten strokes between them. Two signs after the lower $\alpha\beta$ of R are crossed out.

Original source details: GELFAND-P000552

$$a_2 = \frac{1}{360} \int \left(2|R|^2 - 2|\rho|^2 + 5\tau^2\right)$$

R — curvature tensor; ρ — Ricci tensor,

$$\rho_{ij} = \sum_{\alpha} R^{\alpha}_{i\alpha j}$$



Uncertain reading. There is a heavily overwritten mark after the lower j of R. The original stroke is illegible.

Original source details: GELFAND-P000552

$$|R|^2 = \sum R^2_{\alpha\beta\gamma\delta} \qquad \text{in norm. coord.} \qquad |R|^2 = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$$

$$|\rho|^2 = \rho_{\alpha\beta}\rho^{\alpha\beta}$$

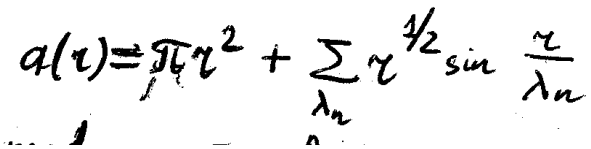
$$E(t, m, m') = (4\pi t)^{-n/2}\left(u_0(m, m') + u_1(m, m')t + u_2(m, m')t^2 + \ldots\right)$$

in a neighb. of the diagonal.

a_3 — Sakai and the speaker (independently).

G. Number of int. points in a disk:

$$a(r) = \pi r^2 + \sum_{\lambda_n} \tau^{1/2} \sin \frac{\tau}{\lambda_n}$$



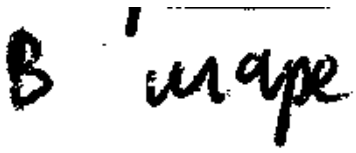
Uncertain reading. The letter at the start of the formula provisionally reads r. Both occurrences in the summand have a broad wavy top, provisionally read as τ.

Original source details: GELFAND-P000552

(λ_n — zeros of the Riemann ζ-func.) mod the Riemann hypothesis

Schnirelmann.

In the ball λ_n — lengths of closed billiard paths.



Manuscript excerpt. The first letter of “шаре” has three strokes.

Original source details: GELFAND-P000552

$$\sum \delta(\lambda - \lambda_k) * \frac{\varepsilon}{\lambda^2 + \varepsilon^2} = f(\lambda)$$

GELFAND-P000553

[source number: 83]

$$f(\lambda) = f_0(\lambda) + \sum_k c_k \lambda^{\alpha_k} \sin(\ell_k \lambda + \omega_k)$$

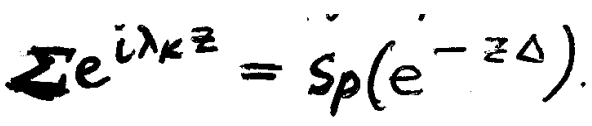
There is exper. confirmation (Balian and Bloch)

Uncertain reading. The end of the surname Balian is retraced. The reading of the final n is provisional.

Original source details: GELFAND-P000553

(and even a precise statement).

$$\sum e^{i\lambda_k z} = \mathrm{Sp}(e^{-z\Delta}).$$



Manuscript excerpt. The exponent on the left is $i\lambda_k z$, and the one on the right is $-z\Delta$.

Original source details: GELFAND-P000553

contin. downward (from the upper half-plane)

Twin probl.:

$$\lambda_{k+1} - \lambda_k > c\lambda_k^{-\nu} \qquad \text{for general operators}$$

Uncertain reading. The letter after the minus has two converging strokes and provisionally reads ν.

Original source details: GELFAND-P000553

but on the circle they are double. (stably asympt. double spectrum).

T^2 $\quad g_{ij} \sim \delta_{ij}$ $\quad$ the theorem holds

Theorem.

$$\min(\lambda_{k+1} - \lambda_k, \lambda_k - \lambda_{k-1}) < C_N \cdot k^{-N}, \qquad \forall N$$

(Lazutkin, Schnirelmann)

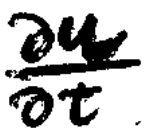
Uncertain reading. The strokes leading into the last letters of "Schnirelmann" are overwritten. The underlying strokes are illegible.

Original source details: GELFAND-P000553

(a small but finite perturbation) — stab.

For T^3 this is no longer so. (apparently).

$$\frac{\partial u}{\partial t} = i\sqrt{-\Delta}\, u; \qquad u|_{t=0} = \delta_x$$

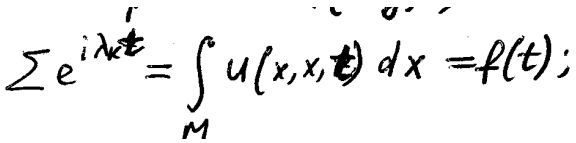


Uncertain reading. The overwritten letter after ∂ provisionally reads u.

Original source details: GELFAND-P000553

Its soln. $\quad u(x, y, t)$

$$\sum e^{i\lambda_k t} = \int_M u(x, x, t)\, dx = f(t);$$



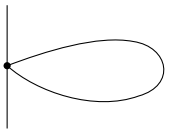
Uncertain reading. The exponent in the summand and the last argument of the integrand both contain t. The earlier strokes beneath the retracing are indistinct.

Original source details: GELFAND-P000553

singul. of $u(x, y, t)$ where $|x - y| = t$

Original source details: GELFAND-P000553

Diagram and typed description.



for some t bifurcations occ.

Original source details: GELFAND-P000553

Uncertain reading. The Russian ending of “bifurcations” is overwritten. The reading is provisional.

singul. move at speed 1

Chapter 8

GELFAND-P000554

2/IX-74. Gelfand's sem. (32nd year of the seminar)

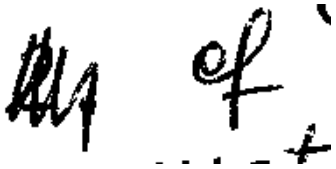
Giusti.

1. Singularities [crossed out] and regularity

Uncertain reading. Crossed-out strokes lie between "Singularities" and "and". The original entry is illegible.

Original source details: GELFAND-P000554

[crossed out] of solutions of non-linear elliptic (and parabolic) systems.



Uncertain reading. A separate crossed-out entry precedes "of solutions". Its reading remains unresolved.

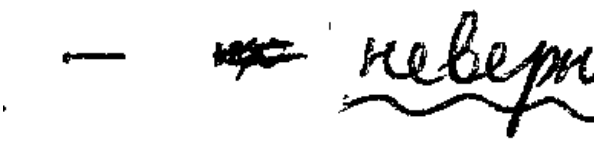
Original source details: GELFAND-P000554

Nash, De giorgi, Moser. — Regularity: 1 equation linear ~~di...~~ with discont. coeff. (Höld. coeff.) ⇒ regul. for non linear with smooth coeff.

Uncertain reading. There are crossed-out letters before "with". The opening provisionally reads "di".

Original source details: GELFAND-P000554

Systems — [crossed out] false. There are [?] coeff. and [?] solutions



Uncertain reading. A heavily crossed-out sign follows the dash before "false".

Original source details: GELFAND-P000554

Original source details: GELFAND-P000554

Uncertain reading. The Russian abbreviations before "coefficients" and "solutions" provisionally read "Быв. ср." and "несгр.".

1968 Miranda, Giusti — syst. with analytic coeff. with discont. solutions

Uncertain reading. The beginning of the surname Giusti is overwritten and the underlying strokes are illegible.

Original source details: GELFAND-P000554

Σ — singular set.

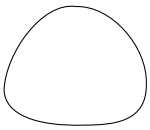
(1) Σ closed
(2) $\dim \Sigma < n - 2$

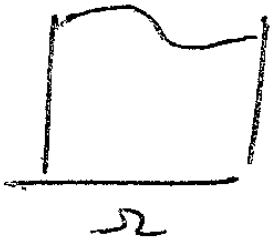
2. Boundary value problems. for surfaces of prescribed mean curvature.

$u : \Omega \longrightarrow \mathbb{R}$ graph $u = S$

Original source details: GELFAND-P000554

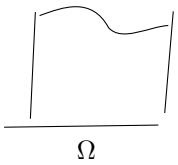
Diagram and typed description.





Original source details: GELFAND-P000554

Diagram and typed description.



$$H(x) = \frac{1}{n} \sum \frac{\partial}{\partial x_i} \left(\frac{\partial u / \partial x_i}{\sqrt{1 + |\operatorname{grad} u|^2}} \right)$$

H is not very large (otherwise unsolvable). Depends on the geometry of the domain, the func. $H(x)$ and so on

That is what will come next time.

S. P. Novikov. (and Dubrovin).

Abelian var. $\xi_1, \ldots, \xi_{2n}$ — basis in $\mathbb{C}^n$ (over $\mathbb{R}$), T^{2n} — corresp. torus. Let the Abelian var. (algebraic var.).

How to describe all tori?

$n = 2$ already nontrivial.

$y^2 = P_{2n+1}(x)$ Riem. surface $\Rightarrow$ torus.

Undeciphered passage. A crossed-out entry before H^1 provisionally begins “Вид...”. The rest is illegible.

Original source details: GELFAND-P000554

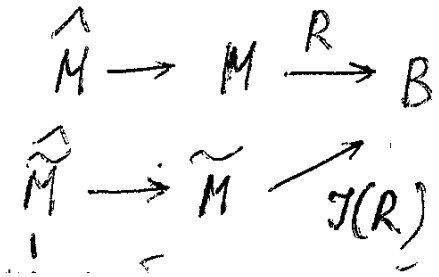
$$\underbrace{H^1(R,\mathbb{C})/H^1(R,\mathbb{Z})}_{\text{the torus is like this}} \qquad R \longrightarrow J^{2n}(R). \quad \text{(torus)}$$

$$k = 0, \dots, n-1, \qquad \boxed{\omega_k = \frac{x^k\,dx}{\sqrt{P_{2n+1}(x)}}} \quad \longleftarrow \quad \text{everywhere regular differentials.}$$

Basis — the forms ω_k.

$$P_{2n+1}(x) = \prod (x - x_\alpha).$$

Everything is det. by the set of roots.



Original source details: GELFAND-P000554

Diagram and typed description.

$$\begin{array}{ccccc} \widehat{M} & \longrightarrow & M & \xrightarrow{R} & B \\ & & & \nearrow & \\ \widehat{\widetilde{M}} & \longrightarrow & \widetilde{M} & & J(R) \\ | & & & & \end{array}$$

The lower diagonal arrow ends below and to the left of the upper B. Its exact connection remains unresolved.

M — collection of Riem. surfaces

B — Siegel upper half-space

(cover of the moduli set).

from the set of roots one constructs the polynomial, and from it the torus.

GELFAND-P000555

$$p_1, p_2, q_1, q_2, c, d. \qquad \mathbb{C}^6 \text{ or } \mathbb{C}^4_{c,d}$$

$$J_1 = p_1 p_2 + V(q_1, q_2)$$

$$J_2 = p_1^2 - 2q_1 p_1 p_2 + 2(q_2 - c)p_2^2 + \mathcal{D}(q_1, q_2)$$

$$\Omega = \sum_{\alpha=1}^{2} dp_\alpha \wedge dq_\alpha.$$

$\langle J_1, J_2 \rangle \equiv 0.$ Poisson br.

$$V = -\left(\frac{q_2^2}{2} + \frac{5}{2}q_1^2 q_2 + \frac{5}{2}q_1^4\right) + c_1 q_1^2 - d_1 q_1$$

Original source details: GELFAND-P000555

Uncertain reading. The sign before the equals sign is overwritten. It has two upper strokes and provisionally reads V.

$$\mathcal{D} = q_1^5 + 2c_1 q_1^3 + d_1 q_1^2 - 4q_1 q_2^2 + 4c_1 q_1 q_2 - 2d_1 q_2$$

$[X_{J_1}, X_{J_2}] = 0.$ Obtain 2 commut. fields.

The corresp. orbits — an Abelian var.

$$\left.\begin{array}{l} J_1 = \text{const} \\ J_2 = \text{const} \end{array}\right\} \quad \text{torus.}$$

Uncertain reading. A crossed-out entry left of the equations starts with a J-like form. The rest is illegible.

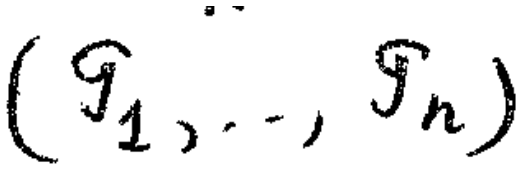
Original source details: GELFAND-P000555

Corollary: the coll. of them all — a unirat. variety.

$$R: \quad y^2 = x^5 + \frac{c_1}{4}x^3 - \frac{d_1}{16}x^2 + \left(\frac{J_1}{32} + \frac{c_1^2}{4}\right)x + \frac{J_2}{2^8} + \frac{c_1 d_1}{2^7}$$

R — Riem. surface, $S^n R$ — symm. power of R.

$(J_1, \ldots, J_n)$ and symmetrization.



Uncertain reading. Both letters in the row before “and symmetrization” have a hook. The reading J is provisional.

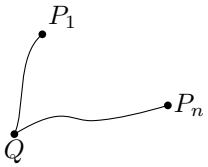
Original source details: GELFAND-P000555

$$S^n R \longrightarrow J(R).$$

$(P_1, \ldots, P_n) \xrightarrow{A} (\alpha_1, \ldots, \alpha_n)$ Abel map.

(Abel’s theor. — it is an isomorphism).

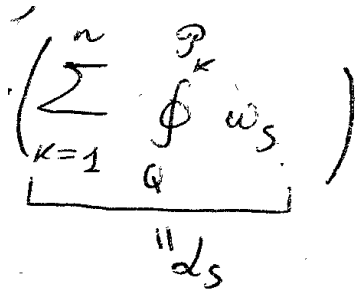
Diagram and typed description.



The upper curve thickens at P_1. An arrowhead cannot be distinguished.

Original source details: GELFAND-P000555

$$(P_1, \dots, P_n) \longrightarrow \left(\underbrace{\sum_{k=1}^{n} \oint_Q^{P_k} \omega_s}_{\| \ \alpha_s} \right)$$



Manuscript excerpt. A loop is visible on the integral sign. Two short strokes stand above α_s below the bracket.

Original source details: GELFAND-P000555

$$L = -\frac{d^2}{dx^2} + u(x),$$

$u(x+T) = u(x)$ or quasiperiodic.

Eigenfunc. $L\Psi(x, E) = E\Psi,$

$$\Psi(x+T) = \lambda\Psi(x), \qquad |\lambda| = 1, \qquad \lambda = e^{ip(E)}$$

(Bloch eigenfuncs.)

For almost periodic [crossed out] the collection of such eigenfuncs. does not form a complete system.

Undeciphered passage. A short densely crossed-out entry follows the Russian abbreviation for “periodic”.

Original source details: GELFAND-P000555

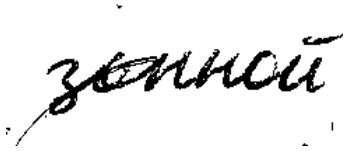
$\Psi|_{x_0} = 1$ — normalization

$\Psi(x, E, x_0)$ $(\Psi, \overline{\Psi})$ — 2 eigenfuncs.

branches in E at the endpoints of the spectral intervals. Obtain a Riem. surface of inf. genus. 2 sheets, since there are 2 eigenfuncs.

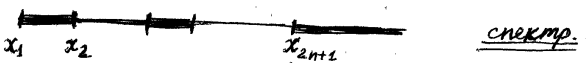
GELFAND-P000556

The set of pot. with given band structure — the Jacobian var. of this Riem. surface (there are formulas)



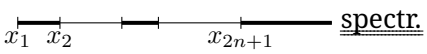
Uncertain reading. The first strokes in the Russian word for “band” are overwritten.

Original source details: GELFAND-P000556



Original source details: GELFAND-P000556

Diagram and typed description.



Pot. with given band structure — the corresp. torus (Jacobian var.). .....

9/IX-74 . Gelfand's sem..

Bessaga (Warsaw). Anderson's Z-sets and their applications. (R. D. Anderson)

X — a complete metriz. space. $A \subset X$, $\overline{A} = A$.

A is called a Z-set if $\forall$ cont. maps

$$f : \underbrace{I^\infty}_{\text{Hilb. cube}} \longrightarrow X, \qquad \forall \varepsilon > 0, \qquad \exists g : I^\infty \longrightarrow X \setminus A, \qquad \rho(f,g) < \varepsilon.$$

$$Q = [-1,1]^\infty, \qquad P = (-1,1)^\infty. \qquad (Q \setminus P) \text{ — } Z\text{-set}$$

Prop.: 1°. If A, B are Z-sets (in Q), then $f : A \xrightarrow{\sim} B$ extends to a ~~homeo~~ homeom. of the entire space.

Original source details: GELFAND-P000556

Uncertain reading. The numeral of 1° is heavily overwritten. The small circle and period are distinguishable.

Original source details: GELFAND-P000556

Uncertain reading. The crossed-out abbreviation before "homeom." provisionally reads "homeo".

Z_σ-set — a countable union of Z-sets.

2°. There exists a maximal Z_σ-set _____ · _____

Gelfand

Consid. $\mathbb{R}^\infty$ in the Tych. topology and the space S of rays [crossed out] issuing from 0 in the nat. topology. Prove that on S every cont. func. is [crossed out] const

Original source details: GELFAND-P000556

Undeciphered passage. There is a densely crossed-out entry between "rays" and "issuing".

Original source details: GELFAND-P000556

Uncertain reading. A small crossed-out mark precedes "const". Its reading remains unresolved.

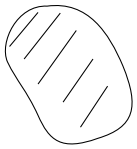
(Gelf., Fuks. Funct. Anal. I, 4). _____ · _____

Bessaga. All separab. Fréchet spaces are homeomorphic.

Giusti. $\Omega \subset \mathbb{R}^n$

Original source details: GELFAND-P000556

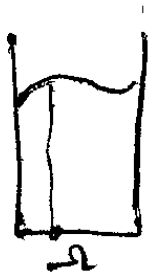
Diagram and typed description.



$$\begin{cases} \sum_{i=1}^{n} D_i \left\{ \dfrac{D_i u}{\sqrt{1+|Du|^2}} \right\} = -H(x,u) \quad \text{in } \Omega, \\ u = \phi \quad \text{on } \partial\Omega \end{cases} \qquad D_i = \frac{\partial}{\partial x_i}.$$

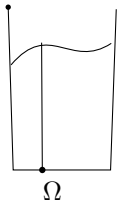
n times the mean curvature $=$ the left-hand side of the eq.

$H \equiv 0 \quad \Rightarrow \quad$ minimal surface.



Original source details: GELFAND-P000556

Diagram and typed description.

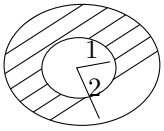


A dark mark sits at the lower end of the inner vertical stroke. Its shape is indistinct.

Example 1. $\Omega, \quad n = 2, \quad H = 0$

Original source details: GELFAND-P000556

Diagram and typed description.

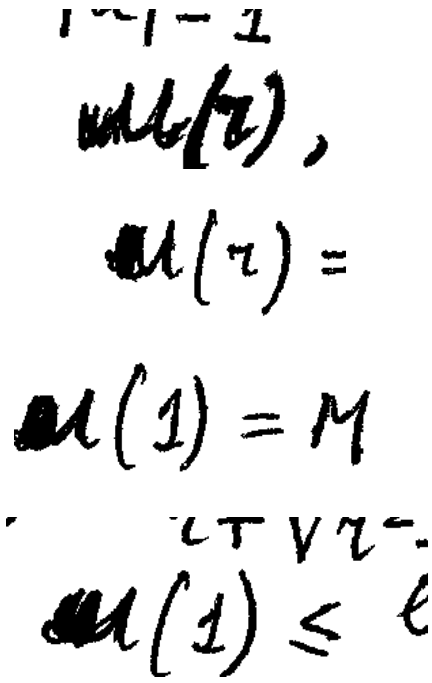


circular annulus. The labels 1 and 2 stand beside the two radial strokes.

$$\phi = \begin{cases} 0 & |x| = 2 \\ M & |x| = 1 \end{cases}$$

Solution: $u(r)$, $r = |x|$.

$$u(r) = c \log \frac{2 + \sqrt{4 - c^2}}{r + \sqrt{r^2 - c^2}}, \qquad 0 \le c \le 1,$$
$$u(1) = M \quad \Rightarrow \quad u(1) \le \log(2 + \sqrt{3}) = M_0.$$



Uncertain reading. The opening strokes of $u(r)$ and $u(1)$ are heavily overwritten. The underlying forms are illegible.

Original source details: GELFAND-P000556

GELFAND-P000557

For $M > M_0$ there are no solutions.

Exper. check: soap bubble

we carefully lift the stick steak.

Manuscript excerpt. Below the Russian word for “stick” the manuscript reads “steak.” with a wavy underline.

Original source details: GELFAND-P000557

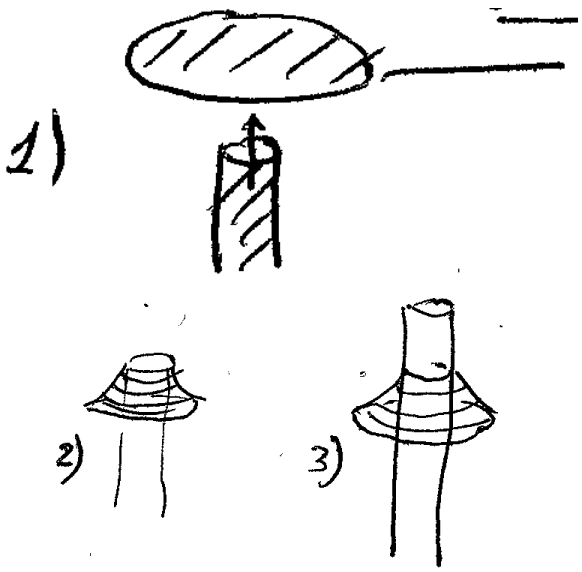
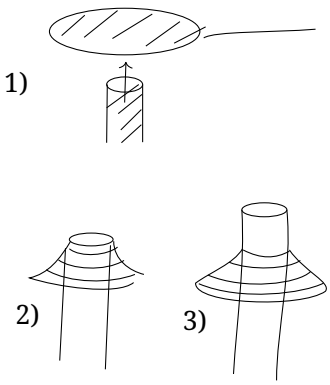


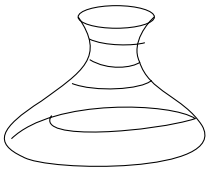
Diagram and typed description.



Original source details: GELFAND-P000557

if it is a wire ring, then

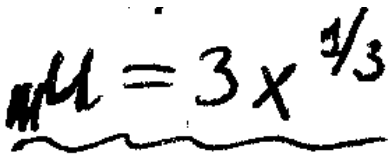
Diagram and typed description.



Original source details: GELFAND-P000557

Example 2. $n = 1, \quad \Omega = (-1, 1)$

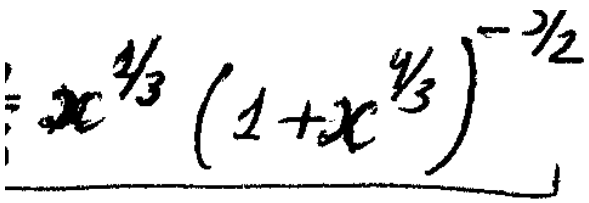
$u = 3x^{1/3}$



Uncertain reading. The initial letter is overwritten. Its final outline provisionally reads u.

Original source details: GELFAND-P000557

$$\left(\frac{u'}{\sqrt{1+u'^2}}\right)' = \underbrace{-\frac{2}{3}x^{1/3}\left(1+x^{4/3}\right)^{-3/2}}_{\text{cont.}}$$



Manuscript excerpt. The exponent on the first x is $1/3$. The exponent on the second x is $4/3$.

Original source details: GELFAND-P000557

[crossed out] u — nondiff.

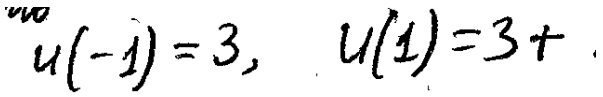
Uncertain reading. The crossed-out Russian entry may begin “об…”. Its end is illegible.

Original source details: GELFAND-P000557

$$u(-1) = -3, \qquad u(1) = 3$$

Let

$$u(-1) = 3, \qquad u(1) = 3 + A$$



Manuscript excerpt. After “Let” the manuscript reads $u(-1) = 3$.

Original source details: GELFAND-P000557

$$\frac{u'}{\sqrt{1+u'^2}} = (1+x^{4/3})^{-1/2}$$

$$u' \le x^{-2/3}$$

$$u(x) \le -3 + \int_{-1}^{x} t^{-2/3}\,dt = 3x^{1/3}$$

_____ · _____

$$\mathcal{F}(u) = \int_\Omega \sqrt{1+|Du|^2}\,dx + \int_\Omega \lambda(x,u)\,dx$$

Need to find min.

$$\lambda(x,t) = \int_0^t H(x,s)\,ds$$

Under the cond. $|u'| \le K$ there is a min. (otherwise none).

u_k — soln. Let $k \to +\infty$. $\quad u_k \to u_0$? _____ · _____

u — Lip-contin. solut. $\quad A \subset \Omega$

$$\int_A H(x)\,dx = \int_{\partial A} n_i \frac{D_i u}{\sqrt{1+|Du|^2}}\,dH_{n-1}$$

$$\left|\int_A H(x)\,dx\right| \le (1-\varepsilon_0) H_{n-1}(\partial A)$$

$$\boxed{1-\varepsilon_0 = \sup_\Omega \frac{|Du|}{\sqrt{1+|Du|^2}}.}$$

Therefore H is not very large. _____ · _____

GELFAND-P000558

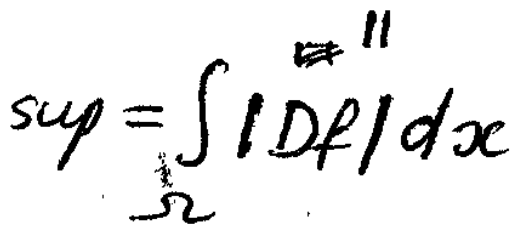
① $H^{1,1} = W_1^1$ ‖

$$\mathcal{F}(u) = \int_\Omega \sqrt{1+|Du|^2}\,dx + \int_\Omega \lambda(x,u)\,dx + \int_{\partial\Omega} |u-\phi|\,dH_{n-1}$$

$BV(\Omega) = L^1$ funcs., whose first derivs. are measures (Radon).

$$\text{i.e.} \quad f \in BV \iff f \in L^1,$$
$$\sup\left\{\int_\Omega f \operatorname{div}\vec{g}\,dx, \quad \vec{g} \in C_0^1(\Omega),\ |g| \le 1\right\} < +\infty$$

$$\sup = \int_\Omega |\overset{||}{D}f|\,dx$$

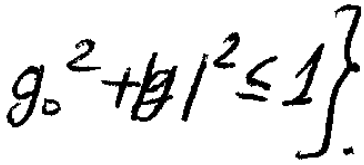


Uncertain reading. Two separate strokes stand above Df. The crossed-out mark to their left is illegible.

Original source details: GELFAND-P000558

Area $f \in BV$.

$$\mathcal{A}(f) = \int \sqrt{1+|Df|^2}\,dx \underset{\text{def}}{=}$$
$$\sup\left\{\int_\Omega (g_0 + f \operatorname{div}\vec{g})\,dx, \quad g_0, g \in C_0^1, \quad g_0^2 + |g|^2 \le 1\right\}.$$



Both vertical bars are visible at g. The expression reads $|g|^2$.

Original source details: GELFAND-P000558

area.

$(\mathcal{A}f)$ lower semicont. with respect to L^1 ~~cont...y~~.

Uncertain reading. The crossed-out English word provisionally reads “cont...y”. The full word remains unresolved.

Original source details: GELFAND-P000558

② Bounded sets in BV are relatively compact in L_1

. . . . . .

Dubrovin.

$$L = -\frac{d^2}{dx^2} + u(x), \qquad u(x+T) = u(x)$$

E — spectral parameter.

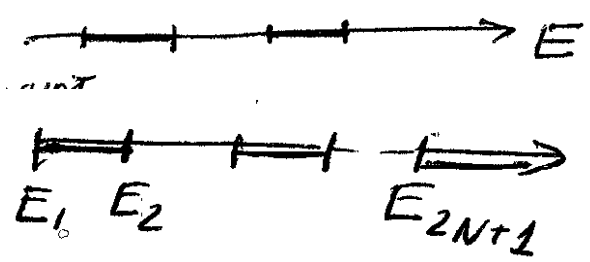
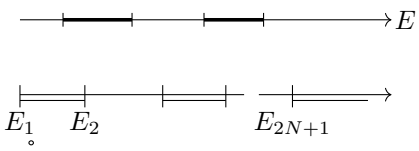


Diagram and typed description.



A small unfilled loop stands below E_1. Its meaning remains unresolved.

Original source details: GELFAND-P000558

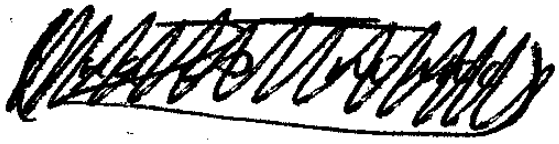
Sometimes the gaps disappear.

Let $u(x)$ — a finite-gap potential

Let $R(E) = \prod(E - E_i)$

Consid. the Riem. surface of the func. $\sqrt{R(E)}$ — denote by Γ_N.

Bloch eigenfuncs. continue analyt. to $\Gamma_N \setminus \infty$ [crossed out]



Undeciphered passage. The parenthetical entry at the right is densely crossed out and illegible.

Original source details: GELFAND-P000558

(meromorphically)

N — genus.

and has N poles and N zeros

Bloch eigenfunc.: $\Psi(x, E, x_0)$

Poles $\gamma_i(x_0)$. || there is a natural proj.

$$\Gamma_N \longrightarrow \mathbb{C}_E.$$

2-sheeted covering

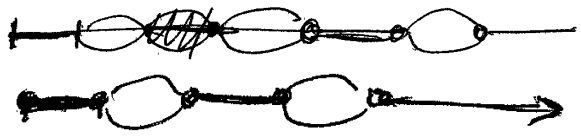
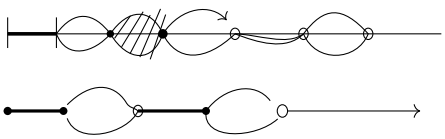


Diagram and typed description.



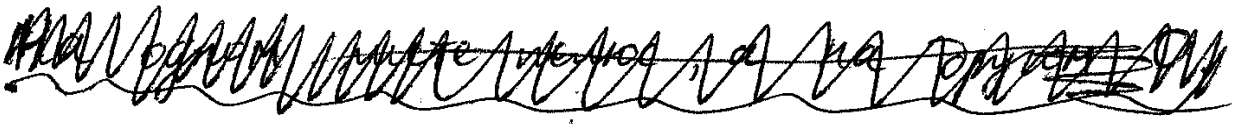
Original source details: GELFAND-P000558

An oval projects to a gap: on each of them lies one pole. The loc. of the poles depends on the normalization: x_0.

~~On one sheet [...], and on the other [...]~~

Uncertain reading. The crossed-out line provisionally begins “On one sheet”.

Original source details: GELFAND-P000558



Original source details: GELFAND-P000558

Undeciphered passage. There are isolated strokes after “sheet”. The next legible words provisionally read “and on the other”. The full reading remains unresolved.

On each oval also one zero.

Next time 18^{30}

GELFAND-P000559

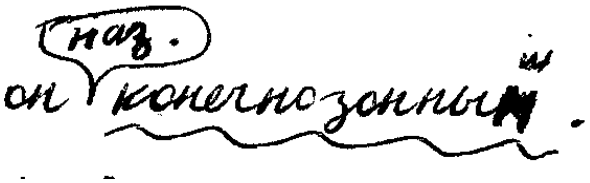
16.9.74. Gelfand Sem. Dubrovin.

Uncertain reading. The second digit of the date has been written over with a heavy loop. The current reading “6” is provisional. The earlier mark is illegible.

Original source details: GELFAND-P000559

$$\text{Let} \quad L = -\frac{d^2}{dx^2} + u(x), \qquad u(x+T) = u(x).$$

The spectrum L — a set of intervals. If there are finitely many of them, it is [inserted above: called] finite-gap.



Uncertain reading. A letter at the end of the Russian word is heavily crossed out. The insertion above provisionally reads “й”. The earlier ending is illegible.

Original source details: GELFAND-P000559

The spectrum — those λ for which there is a bounded soln. of the eq. $Ly = \lambda y$.

$$y_1(x+T) = \alpha_{11} y_1(x) + \alpha_{12} y_2(x)$$
$$y_2(x+T) = \alpha_{21} y_1(x) + \alpha_{22} y_2(x).$$

The matrix $(\alpha_{ij}) = \widehat{T}$ (of monodromy).

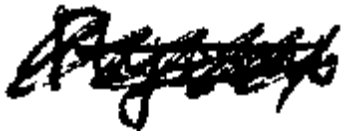
We need $\widehat{T}^n y$ to be bounded, i.e. $\exists$ an eigenvalue with $|\lambda| = 1$

but: $\det \widehat{T} = 1$ (Wronskian); $\lambda_1 \lambda_2 = 1$.

$$\lambda^2 - (Sp\,\widehat{T})\lambda + 1 = 0$$

GELFAND-P000560

~~Roots~~



Original source details: GELFAND-P000560

Uncertain reading. The crossed-out word provisionally reads “Roots”. Crossing strokes obscure some letters.

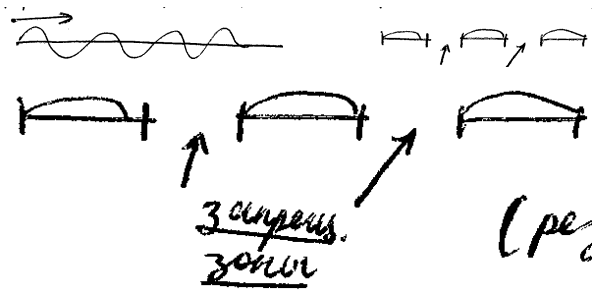
$$|\alpha_{11} + \alpha_{22}| < 2 \quad \Rightarrow \quad \text{complex roots.}$$
$$|\alpha_{11} + \alpha_{22}| > 2 \quad \Rightarrow \quad \text{real roots.}$$

The roots run in one direction (can be proved).

Phys. interpret.: if there is a bounded eigenfunc., the particle will pass through, if not, it will ~~not~~ get trapped

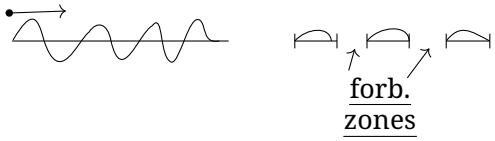
Original source details: GELFAND-P000560

Uncertain reading. The short crossed-out word before the boxed word provisionally reads “not”.



Original source details: GELFAND-P000560

Diagram and typed description.



exp decrease forb. zones. (resonance of particle and field)

Original source details: GELFAND-P000560

Uncertain reading. At the start of the Russian word for “decrease”, there is a high stroke above overwritten marks. The earlier spelling is illegible.

Ince, Akhiezer, Hochstadt.

$$\text{Bloch func.} - \boxed{u(x+T) = \nu u(x)}, \qquad |\nu| = 1. \quad (\text{eigenvector } \widehat{T}).$$

$$\nu = e^{ip}, \qquad p \text{ — “quasimomentum”.}$$

Bounded eigenfuncs. — if $p \in \mathbb{R}$.

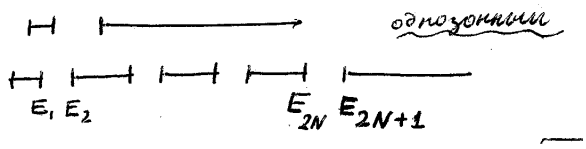
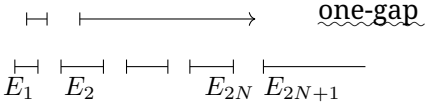


Diagram and typed description.



Original source details: GELFAND-P000560

$$\Gamma_N \text{ — Riemann surface of } \sqrt{R(E)}, \qquad \text{where} \quad R(E) = \prod_1^{2N+1} (E - E_i)$$

G. Let $y(x, E)$ — Bloch func. for complex E.

$$y(x+T) = \nu y(x)$$

$y(x_0, x, E)$ normalization: $y(x_0, x_0, E) = 1$.

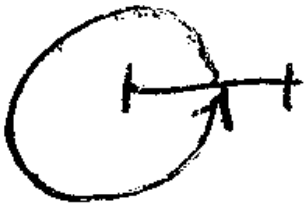
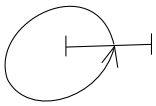


Diagram and typed description.



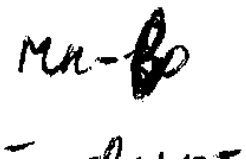
Two short diverging strokes are visible where the loop meets the segment. The mark provisionally points upward.

Original source details: GELFAND-P000560

D. Th. 1. Bloch eigenfuncs. continue analyt. to Γ_N.

Th. 0. Any finite-gap pot. — a stat. soln. of a higher KdV eq..

Th. 2 The set of pots. with a given spectrum is $J(\Gamma_N)$ — the Jacobi manifold.



Uncertain reading. In the Russian abbreviation for “set”, the letter after the hyphen was written over. The earlier strokes are illegible.

Original source details: GELFAND-P000560

KdV (dyn. syst.) and its higher analogues act there.
On solns. $Ly = Ey$ acts $(\widehat{T}y)(x) = y(x+T)$

$$\begin{matrix} c=1, & c'=0 \\ s=0, & s'=1 \end{matrix} \quad \text{(at pt. } x_0) \qquad c(x, x_0, E)$$

In this basis the monodromy matr. $\alpha_{ij}(E, x_0)$.
Everything is det. by the matr. trace $\tau = \frac{1}{2} Sp\, \alpha_{ij}$

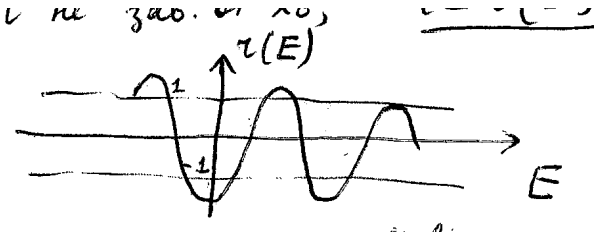
$$|\tau| \le 1 \quad \Rightarrow \quad \text{spectral bands}$$

Uncertain reading. The ending of the Russian word for "bands" is overwritten. The earlier strokes are illegible.

Original source details: GELFAND-P000560

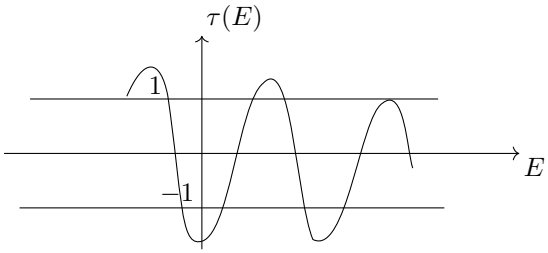
τ is indep. of x_0, $\qquad \tau = \tau(E)$.

$$\lambda^2 - 2\tau\lambda + 1 = 0$$



Original source details: GELFAND-P000560

Diagram and typed description.



it is known that the graph looks like this — max and $min \gtrless 1$ — prob.

The sign before 1 is $\gtrless$.

Original source details: GELFAND-P000560

GELFAND-P000561

for the 2^* year There are always infinitely many of them (max and min).

Uncertain reading. A small crossed mark follows "2". The visible shape is retained. The ending remains unresolved.

Original source details: GELFAND-P000561

merger of 2 forb. zones — there is no allowed zone

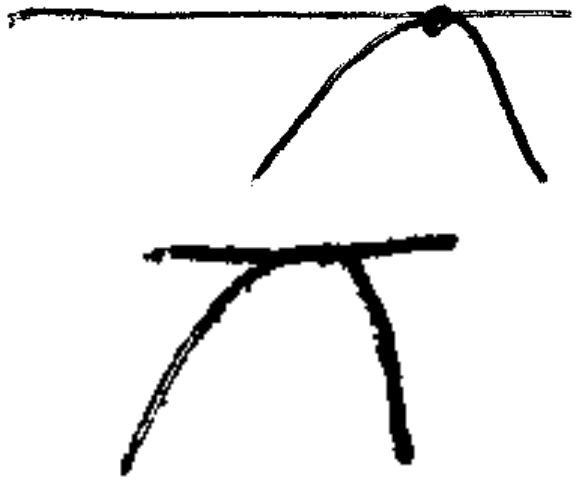
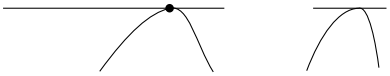


Diagram and typed description.



Original source details: GELFAND-P000561

From the eq. $\tau(E) = \pm 1$ one must select the simple roots to det. the band edges.
The band edges give a Jordan block. If diagonal, we get the degenerate case.
See the work of Gelfand and Lidskii (1955) on systems with periodic coeffs..
Def. $\Psi(x, x_0, E), \qquad \Psi(x+T) = e^{ip}\Psi(x), \qquad \Psi(x_0) = 1.$
The other is complex conjugate (when $|\tau(E)| < 1$).

$$\text{Let} \qquad \chi = -i\frac{\Psi'}{\Psi}$$

1. $\chi = \chi(x, E)$ (indep. of x_0).

Uncertain reading. A detached ", and" stands to the right between the first and second items. Its attachment remains unclear.

Original source details: GELFAND-P000561

2. $-i\chi' + \chi^2 + u - E = 0$ (Riccati eq.).

3. $\chi_I = \frac{1}{2}\frac{\chi_R'}{\chi_R}, \qquad \chi = \chi_R + i\chi_I.$

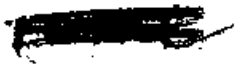
4. $\chi(x, E) \sim k + \sum_{n=1}^{\infty} \frac{\chi_n(x)}{(2k)^n}, \qquad$ where $k = \sqrt{E}$

Uncertain reading. A wavy stroke lies over earlier strokes before k. The reading $\sim$ is provisional.

Original source details: GELFAND-P000561

5. $\chi(x+T)=\chi(x)$. (asympt. expansion).

$$2 \Rightarrow 3, \qquad 2 \Rightarrow 4$$



Undeciphered passage. A group of signs after $2 \Rightarrow 4$ is heavily crossed out. The surviving strokes are insufficient to read it.

Original source details: GELFAND-P000561

Uncertain reading. A short group is crossed out before χ_R. It provisionally reads χ with a subscript, possibly R.

Original source details: GELFAND-P000561

$$\chi_R(x,E) = k + \sum_{n=0}^{\infty} \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$

$$\Psi = \exp\left(i\int_{x_0}^{x} \chi\, dx\right) = \sqrt{\frac{\chi_R(x_0)}{\chi_R(x)}}\exp\left(i\int_{x_0}^{x} \chi_R\, dx\right).$$

1. $|\Psi|^2 = \dfrac{\chi_R(x_0)}{\chi_R(x)}$

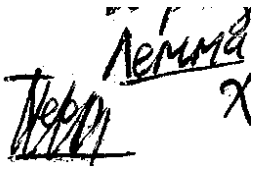
2. $p(E) = \displaystyle\int_0^T \chi_R(x,E)\, dx + \pi n$

Uncertain reading. A heavy slanted mark precedes πn. A separate factor remains unresolved.

Original source details: GELFAND-P000561

Express χ in terms of the monodromy matr.

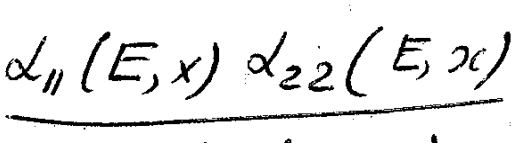
~~Th.~~ Lemma



Uncertain reading. The underlined “Th.” is crossed out. “Lemma” is inserted above and to its right.

Original source details: GELFAND-P000561

$$\chi(x,E) = \frac{\sqrt{1-\tau^2(E)}}{\alpha_{21}(E,x)} + \frac{i}{2}\frac{\alpha_{11}(E,x)\alpha_{22}(E,x)}{\alpha_{21}(E,x)} \tag{1}$$



Uncertain reading. A separate sign is indistinguishable between $\alpha_{11}(E,x)$ and $\alpha_{22}(E,x)$. The groups are typeset side by side.

Original source details: GELFAND-P000561

Proof. $\Psi(x, x_0, E)$ — an eigenvector of the monodr. matr. Use the rules of lin. algebra:

$$\Psi(x, x_0, E) = c(x, x_0, E) + i \begin{pmatrix} \text{right-hand side of} \\ \text{formula (1) at pt. } x_0 \end{pmatrix} s(x, x_0, E).$$

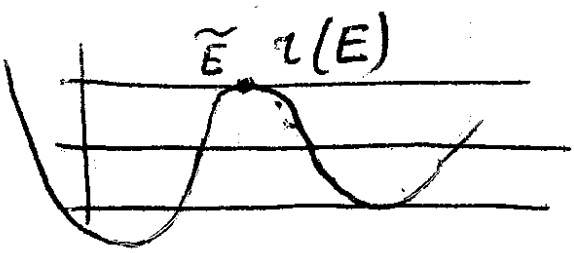
GELFAND-P000562

Find the Wronskian det. of this func. and c (it is indep. of x).

$$\{c, \Psi(x, x_0, E)\}|_{x_0} = \quad \text{everything works out.}$$
$$i\chi(x_0, E) = i(\dots)$$
$$\{c, \Psi\}|_{x_0} = i\Psi'(x_0) = i\chi(x_0).$$

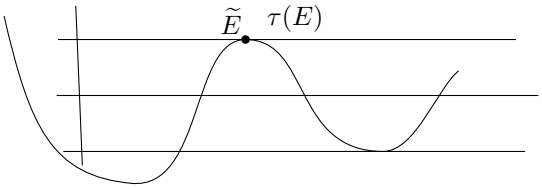
Original source details: GELFAND-P000562

Uncertain reading. An overwritten mark with a prime follows i. The current reading Ψ' is provisional.



Original source details: GELFAND-P000562

Diagram and typed description.



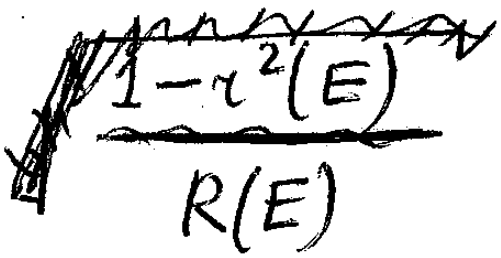
If $(1 - \tau(E))$ has a multiple zero, then the monodromy matr. is diagonal, i.e.

Original source details: GELFAND-P000562

Uncertain reading. A short mark resembling "π" is crossed out before the equalities. Its reading remains unresolved.

$$\alpha_{12}(\widetilde{E}, x_0) = 0, \qquad \alpha_{21}(\widetilde{E}, x_0) = 0.$$

$\dfrac{1 - \tau^2(E)}{R(E)}$ has only double degen. zeros.



Original source details: GELFAND-P000562

Uncertain reading. An oblique stroke and an upper line are heavily crossed out to the left of and above the fraction. They may be a deleted radical sign.

$\sqrt{\dfrac{1-\tau^2(E)}{R(E)}}$ has only simple zeros. (see Titchmarsh)

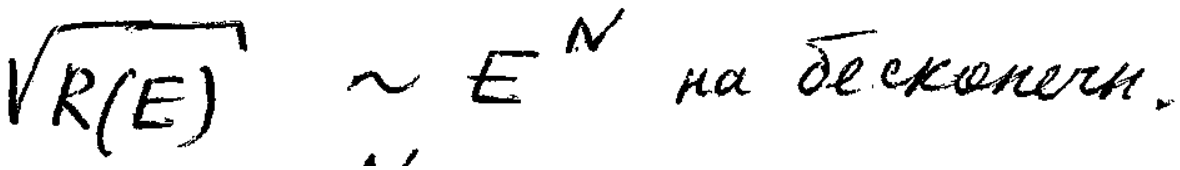
$\alpha_{21}(\widetilde{E}, x_0) = 0$ too

$$\alpha_{21} = \widetilde{\alpha}_{21} \cdot \sqrt{\frac{1-\tau^2(E)}{R(E)}} \qquad \text{(divided)}.$$

Substitute into the formula for χ_R:

$$\chi_R = \frac{\sqrt{R(E)}}{\widetilde{\alpha}_{21}(E,x)}, \qquad \text{where } \widetilde{\alpha}_{21}(E,x) \text{ — an entire anal. func. of } E.$$

$$\chi_R = k + O\left(\frac{1}{k}\right), \qquad \sqrt{R(E)} \sim E^N \quad \text{at infinity.}$$



Original source details: GELFAND-P000562

This line reads E^N. The next fraction reads kE^N/E^N.

$$\chi_R(x,E) \sim \frac{kE^N}{E^N} \quad \Rightarrow \quad \widetilde{\alpha}_{21}(E,x) \text{ — a polynomial in } E.$$

$P(E,x)$

$\chi_R(x,E)$ — 2-valued analyt. func. (analyt. continuation from real E in a band)

$$\chi_R(x,E) = \frac{\sqrt{R(E)}}{P(E,x)}$$

$$P(E,x) = \prod (E - \gamma_i(x)), \qquad \gamma_i(x) \text{ — roots of the polynomial}$$

Hence : $\boxed{\chi(x,E) = \left(\sqrt{R} - \frac{i}{2}\frac{d}{dx}P\right)/P.}$

$$\chi_R \sim k + \sum \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$

Uncertain reading. The letter n in the subscript $2n+1$ has been written over. The earlier strokes are illegible.

Original source details: GELFAND-P000562

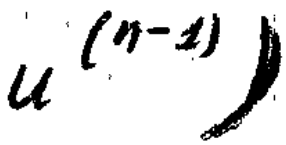
Construct the functionals

$$I_n\{u\} = \int_0^T \chi_{2n+3}(x)\,dx$$

GELFAND-P000563

$$\underbrace{-i\chi' + \chi^2 + u - E = 0} \quad \Rightarrow \quad \text{asymptotics.}$$

and in this case $\chi_{2n+1} = Q(u, u', \ldots, u^{(n-1)})$ — a polynomial.



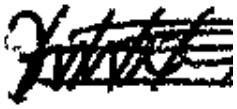
The exponent on the last u is $(n-1)$.

Original source details: GELFAND-P000563

Uncertain reading. A group with vertical strokes and a loop is crossed out at the left. The visible sign resembles "δ" or "$\int$". The earlier writing remains unresolved.

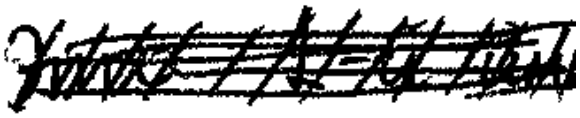
Original source details: GELFAND-P000563

~~This [...]~~



Uncertain reading. The beginning of the crossed-out line provisionally reads "This".

Original source details: GELFAND-P000563



Undeciphered passage. The remaining words in the crossed-out line are illegible.

Original source details: GELFAND-P000563

$$\frac{\delta}{\delta u}\left\{\sum c_n I_n\right\} = d \quad (= const).$$

. . . . . .

GELFAND-P000564

———— ——

23/IX – 74 Gelfand Sem.

On the work of Gabrielov, I. Gelfand and Losik on combinatorial topology.

It is interesting to do it analytically. ~~$\int$ [...]~~

Uncertain reading. A mark with two loops begins the crossed-out group. It provisionally reads $\int$.

Original source details: GELFAND-P000564

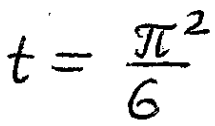
Undeciphered passage. Oblique strokes obscure the remaining marks. They are illegible.

Original source details: GELFAND-P000564

$$\int_{1-x}^{x} \left[\frac{\log|t|}{1-t} + \frac{\log|1-t|}{t} \right] dt$$

cannot be evaluated in elem. funcs.. But:

$$\int_0^1 \frac{\log|1-t|}{t}\, dt = \frac{\pi^2}{6} \quad \left(= \int_0^1 \sum_{n=1}^{\infty} \frac{t^{n-1}}{n} = \sum_{n=1}^{\infty} \frac{1}{n^2} \right).$$

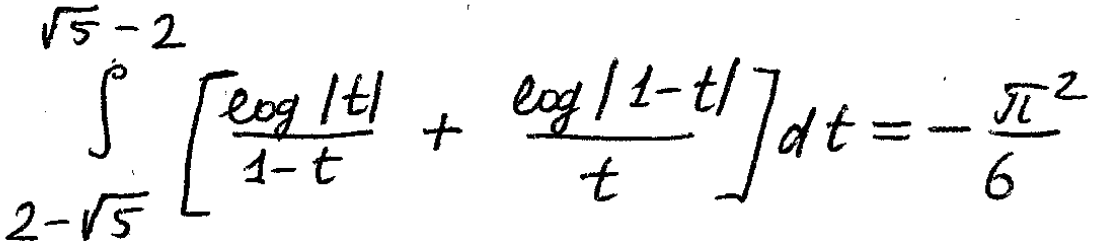


Uncertain reading. There is no minus sign before $\pi^2/6$. The middle term has no written differential.

Original source details: GELFAND-P000564

One can evaluate

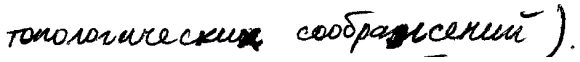
$$\int_{2-\sqrt{5}}^{\sqrt{5}-2} \left[\frac{\log|t|}{1-t} + \frac{\log|1-t|}{t} \right] dt = -\frac{\pi^2}{6}$$



Original source details: GELFAND-P000564

Uncertain reading. This line shows a minus sign before $\pi^2/6$. The limits read $2-\sqrt{5}$ and $\sqrt{5}-2$.

(from topological considerations).



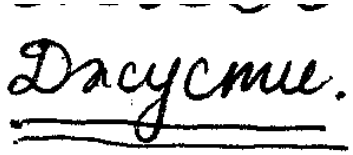
Uncertain reading. Both Russian words contain overwritten letters. The earlier strokes are illegible.

Original source details: GELFAND-P000564

They occur in a finite combinatorial problem.

Graeme Segal — all of topology.

Giusti. On singularities. (On regularity of ~~[...]~~ solv. of nonlin. syst.).



Uncertain reading. The name in the Russian handwriting reads “Джусти”.

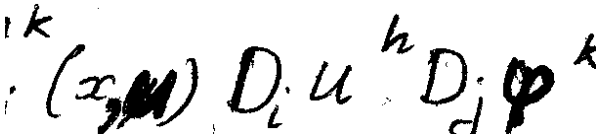
Original source details: GELFAND-P000564

Undeciphered passage. A short entry before “solv.” in the English parenthesis is heavily crossed out and illegible.

Original source details: GELFAND-P000564

Symplest case:

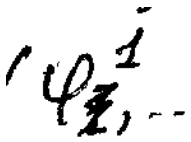
$$\sum_{i,j=1}^{n}\sum_{h,k=1}^{N}\int a_{ij}^{hk}(x,u)D_iu^hD_j\varphi^k\,dx=0$$



Uncertain reading. There are overwritten marks in the argument (x,u) and before the superscript k. The current readings u and φ are provisional.

Original source details: GELFAND-P000564

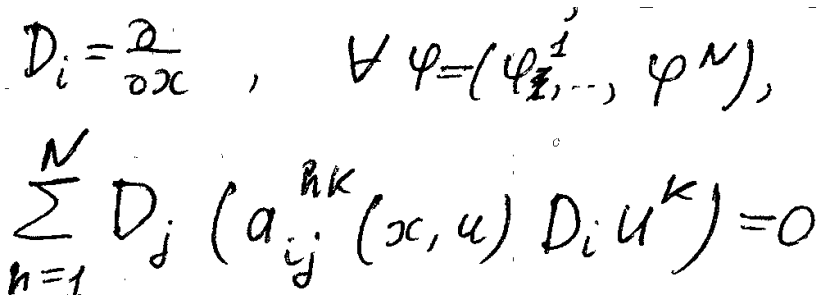
$$D_i=\frac{\partial}{\partial x},\qquad \forall\varphi=(\varphi^1,\dots,\varphi^N),\qquad \varphi_i\in W_0^{1,2}(\Omega)$$



Uncertain reading. There is a heavy subscript mark below φ^1. It may be 2, i or z. Its reading remains unresolved.

Original source details: GELFAND-P000564

$$\sum_{i,j=1}^{n}\sum_{h=1}^{N}D_j\big(a_{ij}^{hk}(x,u)D_iu^k\big)=0\qquad(k=1,2,\dots,N)$$



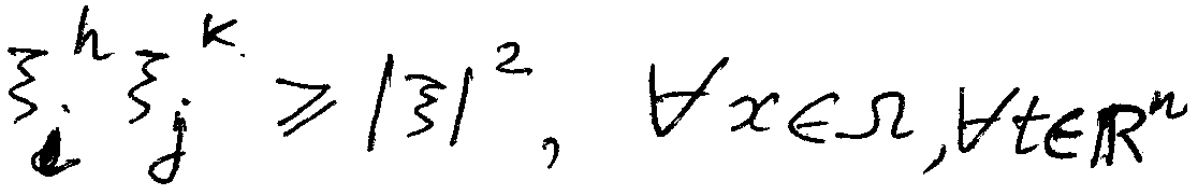
Original source details: GELFAND-P000564

Uncertain reading. The written denominator of D_i is ∂x. In the sum over h the visible superscript of u is k.

$$u = (u^1, \dots, u^N).$$

Ellipticity:

$$\sum_{ijhk} a_{ij}^{hk}(x,t)\xi_i^h \xi_j^k \geqslant |\xi|^2, \qquad \forall x \in \Omega,\ \forall t \in \mathbb{R}^n$$



Original source details: GELFAND-P000564

Uncertain reading. The first subscript on ξ provisionally reads i. The final dimension reads n.

GELFAND-P000565

$$\sum_{i,j=1}^{n} D_j\big(A_{ij}(x,u)D_i u\big) = 0, \qquad \begin{matrix} u \text{ — a vector} \\ A_{ij} \text{ — a matrix.} \end{matrix}$$

$N = 1 \Rightarrow \forall$ solution is regular

or $n \leqslant 2$ $\quad \forall u \in W^{1,1}$ is regular (depende on reg. of A-coefficients)

Uncertain reading. The sign between n and 2 has been written over. The current reading $\leqslant$ is provisional.

Original source details: GELFAND-P000565

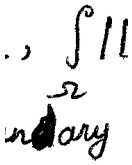
Uncertain reading. The mark before "coefficients" has been written over. It provisionally reads "A". The earlier strokes are illegible.

Original source details: GELFAND-P000565

$$A \in C^\infty \quad \Rightarrow \quad u \in C^\infty \quad \text{etc.}$$

$$W^{1,2}, \qquad u \in L_2, \qquad \int_\Omega |Du|^2\, dx < +\infty$$

$W_0^{1,2}$ — 0 boundary values.



Uncertain reading. The "d" in "boundary" has been written over. Above the word, Ω is below the integral in the previous line.

Original source details: GELFAND-P000565

$n = N \geqslant 3$ falls.

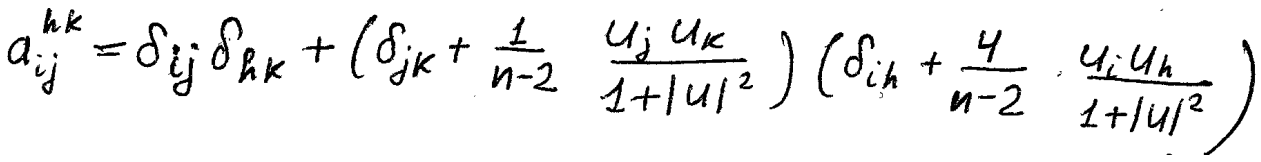
Uncertain reading. The digit “3” has been written over. The earlier strokes beneath it remain unresolved.

Original source details: GELFAND-P000565

Uncertain reading. The English word has two tall loops with no dot over the first. It is transcribed as “falls”.

Original source details: GELFAND-P000565

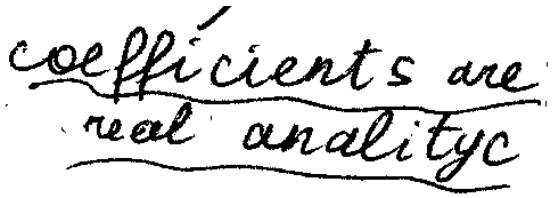
$$a_{ij}^{hk} = \delta_{ij}\delta_{hk} + \left(\delta_{jk} + \frac{1}{n-2}\frac{u_j u_k}{1+|u|^2}\right)\left(\delta_{ih} + \frac{4}{n-2}\frac{u_i u_h}{1+|u|^2}\right)$$



Original source details: GELFAND-P000565

Uncertain reading. The written factors preceding the fractions are $1/(n-2)$ and $4/(n-2)$, in that order.

$u_i = x_i/|x|$ — solution. coefficients are real analityc



Uncertain reading. The original English spelling “analityc” is retained.

Original source details: GELFAND-P000565

$n = 2$ $\forall$ sol. is regular

$n = 3,\ N = 2$ — there is an example of non regular solution.

For Dirichlet problem. $u_i = x_i$ when $|x| = 1$. $\exists!$ solution and it is not regular!

Suppose a_{ij}^{hk} are continuous

Then (1) $\forall$ solution u of system is regular in an open set $\Omega_0 \subset \Omega$

Uncertain reading. An entry in parentheses is crossed out before “(2)”. The earlier number is illegible.

Original source details: GELFAND-P000565

(2). $K = \Omega - \Omega_0$ (K is closed in Ω) ~~a[...]~~

$$\Rightarrow \underset{H_{n-2}}{\text{meas}} K = 0$$

Uncertain reading. An initial “a” is visible in the crossed-out English word. The remaining letters are unresolved.

Original source details: GELFAND-P000565

$$\underline{\dim K < n - 2.}$$

Uncertain reading. The lower stroke of the sign in the dim K line is thickened. The visible outline is transcribed as $<$.

Original source details: GELFAND-P000565

(for order $m > 2$ $\quad$ $\dim K \leqslant n - 2$).

~~There~~ Are ~~also~~ [inserted above] there variational problems with nonregular solutions?

Uncertain reading. The crossed-out word before “Are” reads “There”. The crossed-out word after it provisionally reads “also”. The inserted “there” is enclosed above the line.

Original source details: GELFAND-P000565

Problem of Gelfand. Devise an analogue of Hironaka’s resol. of sing. for elliptic eqs..

GELFAND-P000566

I. M. Gelfand recap of Dubrovin’s previous talk.

Diagram and typed description.

There is a small detached slanted stroke above the left end.

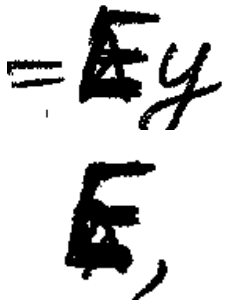
Original source details: GELFAND-P000566

$$-y'' + u(x)y = Ey$$
$$u(x + T) = u(y)$$

Uncertain reading. The right-hand side clearly reads $u(y)$.

Original source details: GELFAND-P000566

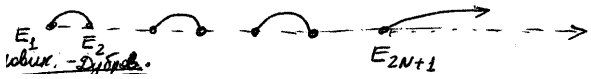
The spectrum — those E, for which there are bounded eigenfuncs.



Original source details: GELFAND-P000566

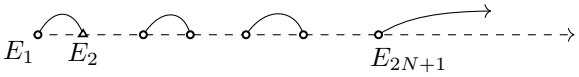
Uncertain reading. Both letters E are written over with heavy strokes. The earlier signs remain unresolved.

Novikov's problem: when are there finitely many forbidden zones?

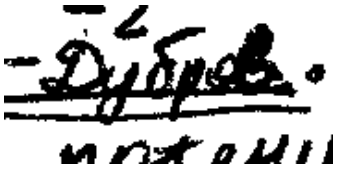


Original source details: GELFAND-P000566

Diagram and typed description.



Th. Novik.–Dubrov.



Original source details: GELFAND-P000566

Uncertain reading. A heavy stroke is written over the end of the abbreviated name "Dubrov." The earlier letter remains unresolved.

These are the potentials $\Leftrightarrow$ satisf. the nonlin. eq.

$$K(u, u', \dots, u^{(2N)}) = 0$$

(Higher Korteweg–de Vries eq.)

$$\begin{aligned} c(x, x_0, E): &\qquad c(x_0) = 1, \quad c'(x_0) = 0 \\ s(x, x_0, E): &\qquad s(x_0) = 0, \quad s'(x_0) = 1. \end{aligned}$$

These are analyt. funcs. of the parameter E.

$$\widehat{T}\Psi = \Psi(x+T), \qquad \widehat{T} = \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix} \qquad \det \|\widehat{T}\| = 1.$$

Bloch func.: $\Psi(x+T) = \lambda\Psi(x)$, $\qquad \Psi(x_0) = 1$

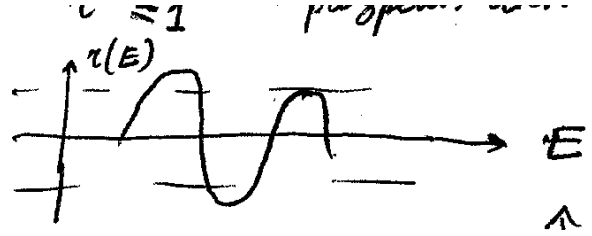
This is an eigenvector for $\widehat{T}$.

$$\lambda^2 - (\alpha_{11} + \alpha_{22})\lambda + 1 = 0,$$

$$\lambda^2 - 2\tau\lambda + 1 = 0, \qquad \tau(E) = \frac{\alpha_{11} + \alpha_{22}}{2}.$$

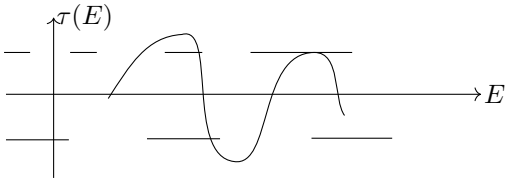
$E \in$ an allowed interv. $\Leftrightarrow \exists$ a Bloch func. with $|\lambda| = 1$.

$\tau^2 \leqslant 1$ — allowed int., $\tau^2 > 1$ — forbidden



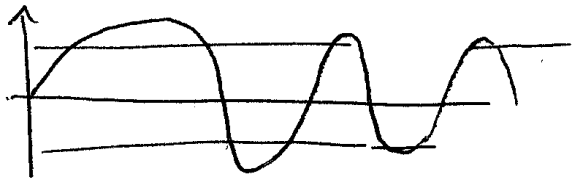
Original source details: GELFAND-P000566

Diagram and typed description.



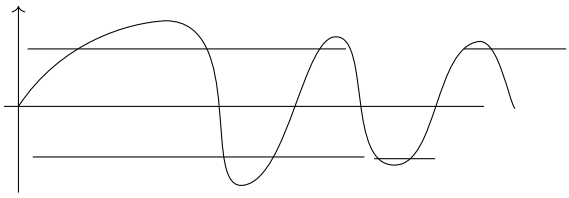
all $max \geqslant 1$ (this shape).

Correct fig.:



Original source details: GELFAND-P000566

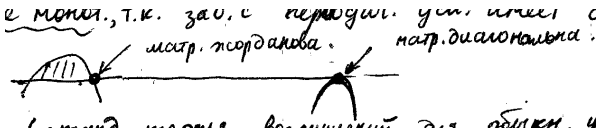
Diagram and typed description.



The forb. zones shrink.

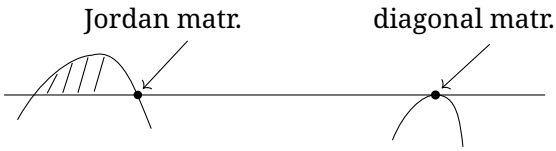
Important conclusion: if all the zones have merged, then, max and min are equal from some point on to 1.

They cannot, since the problem with periodic conds. has infinite spectrum (there $\tau = 1$).



Original source details: GELFAND-P000566

Diagram and typed description.



(stand. perturbation theory for ordinary eqs.).

$$\tau^2(E) = 1 \quad \text{has a mult. root} \quad \Rightarrow \quad \alpha_{12} = \alpha_{21} = 0.$$

Uncertain reading. The first digit of the subscript on α_{12} has been written over. The visible reading is 1.

Original source details: GELFAND-P000566

$\alpha_{12}(E)$ and $\alpha_{21}(E)$ have a zero at the same points where $\tau^2(E) - 1$ has multiple zeros.
Consid. the Bloch func. $\Psi(x, x_0, E)$: $\Psi(x, x_0, E)$, $\Psi(x_0) = 1$.
there are 2 Bloch funcs. $\forall E$ (since the eq. is quadr. in λ).
it depends analyt. on E (two-valued).

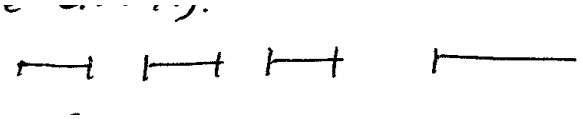


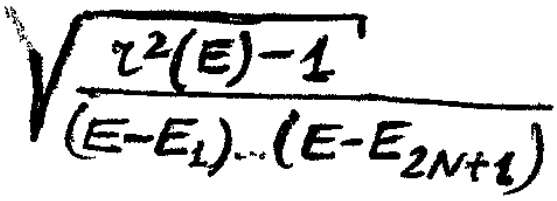
Diagram and typed description.

Original source details: GELFAND-P000566

$\tau^2(E) - 1$ has a zero at $E_1, \ldots, E_{2N+1}$. The rest are double.

GELFAND-P000567

$$\sqrt{\frac{\tau^2(E) - 1}{(E - E_1) \cdots (E - E_{2N+1})}} \quad \text{— single-valued func.}$$



Uncertain reading. The top stroke of the radical ends over the numerator. The bottom of the radical descends to the denominator. Its scope remains ambiguous.

Original source details: GELFAND-P000567

Consid. the eq.

$$-\Psi'' + u(x)\Psi = E\Psi, \qquad \Psi(x+T) = \lambda\Psi(x)$$
$$u(x+T) = u(x)$$

Let $\chi = -i\,\Psi'/\Psi$.

1°. χ does not depend on x_0, $\quad \chi = \chi(x, E)$.

2°. $\chi(x+T) = \chi(x)$.

3°. $-i\chi' + \chi^2 + u - E = 0$.

$\chi = \chi_R + i\chi_I$ in an allowed band — then analyt. continuation in E.

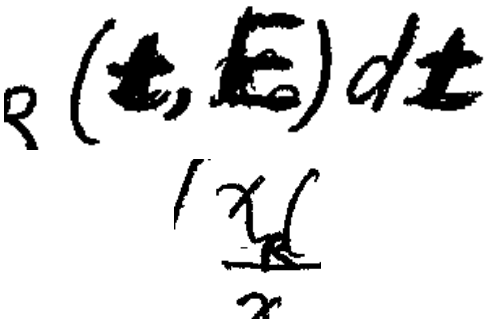
$$4^\circ. \quad \chi_I = \frac{1}{2}\,\chi_R'/\chi_R.$$

$$5^\circ. \quad \chi(x, E) \sim k + \sum_{n=1}^{\infty} \frac{\chi_n(x)}{(2k)^n}, \qquad E = k^2$$
$$\chi_n = P_n(u, u', \ldots, u^{(n)}) \text{ — a polynomial}$$

(det. successively from the asympt. equality)

$$4^\circ \Rightarrow 6^\circ. \quad \chi_R(x, E) \sim k + \sum_{n=0}^{\infty} \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$

$$7^\circ. \quad \Psi(x, x_0, E) = \sqrt{\frac{\chi_R(x_0, E)}{\chi_R(x, E)}} \cdot \exp\left[i \int_{x_0}^{x} \chi_R(t, E)\, dt\right].$$



Uncertain reading. The subscript of χ_R in the numerator has been written over. In the exponent, t, E and the final t of dt have been written over. The earlier strokes remain unresolved.

Original source details: GELFAND-P000567

$$7^\circ \Rightarrow 8^\circ. \quad |\Psi|^2 = \chi_R(x_0, E)/\chi_R(x, E) \qquad \underline{\text{in an allowed band}}$$

9°. ~~[...]~~

Uncertain reading. An entry before λ is heavily crossed out. Some strokes resemble an equals sign and $p(E)$. The exact reading remains unresolved.

Original source details: GELFAND-P000567

$$\lambda = e^{ip(E)} \quad \Rightarrow \quad p(E) = \int_{x_0}^{x_0+T} \chi_R(x,E)\,dx + 2\pi n.$$

The integral — a functional of u with an asympt. expansion. Its Euler eq. — KdV eqs., Allowed and forb. zones — where $|\lambda| = 1$

Uncertain reading. An earlier number with a superscript circle is crossed out to the left of "10°". A vertical stroke and a loop are visible. The exact digits remain unresolved.

Original source details: GELFAND-P000567

$$10^\circ. \quad \frac{\delta p(E)}{\delta u(x)} = -\frac{1}{2}\frac{1}{\chi_R(x,E)}$$

(left-hand side of the Euler eq. for KdV).

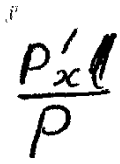
$$11^\circ. \quad \chi(x,E) = \frac{\sqrt{1-\tau^2(E)}}{\alpha_{21}(x,E)} + \frac{i}{2}\frac{\alpha_{11}(x,E)-\alpha_{22}(x,E)}{\alpha_{21}(x,E)}$$

Rough idea: eigenvalue obtained from the quadr. eq.

$$12^\circ. \quad \text{Let} \quad R(E) = \prod_1^{2N+1}(E-E_j). \qquad \text{Then}$$

$$\chi(x,E) = \frac{\sqrt{R(E)}}{P(E,x)} - \frac{i}{2}\frac{P'_x}{P},$$

where P — a polynomial in E of degree N.



Uncertain reading. A heavy stroke follows P'_x in the numerator. It resembles an opening parenthesis. Its meaning remains unresolved.

Original source details: GELFAND-P000567

GELFAND-P000568

$$P(E,x)=\prod_{j=1}^{N}(E-\gamma_j(x)).$$

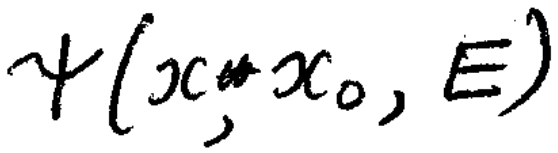
G.'s problem. Let R and P — entire funcs. of the 1st order of growth. Then corresponding to them are potentials $u(x)$. Which ones?

class of potentials (not necessarily periodic). ____ · ____

Dubrovin.

Γ_N — Riem. surf. of $\sqrt{R(E)}$, $\chi(x,E)$ — an algebr. func. on this surface

Theorem: $\Psi(x,x_0,E)$ a single-valued func. on Γ_N, merom. on $\Gamma_N\setminus\infty$, has N zeros and N poles.



Uncertain reading. A comma follows the first x. A small mark is heavily crossed out above it. The earlier reading remains unresolved.

Original source details: GELFAND-P000568

Poles at $E=\gamma_i(x_0)$, zeros above $E=\gamma_i(x)$.

GELFAND-P000569

30/IX – 74. Gelfand Sem.

Polyakov. On elementary particles. Yang–Mills fields. (difference analogue).

Graeme Segal. A work of Dusa Mc Duff on config. space.

$$\mathbb{R}^n,\qquad C_k(\mathbb{R}^n)=\{\text{subsets of }\mathbb{R}^n\text{ with }k\text{ points (no ordered)}\}.$$

From a top. point of view it is not quite trivial.

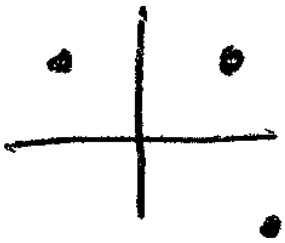
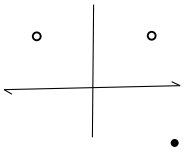


Diagram and typed description.



Original source details: GELFAND-P000569

~~[...] f_S~~

Undeciphered passage. The left part of the crossed-out entry is illegible.

Original source details: GELFAND-P000569

Original source details: GELFAND-P000569

Uncertain reading. The end of the crossed-out entry provisionally reads f_S.

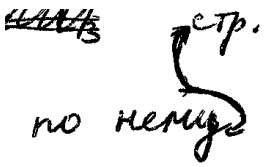
constr. a map $f_S : \mathbb{R}^n \longrightarrow \mathbb{R}^n \cup \infty$

$S \subset \mathbb{R}^n$ from it

finite.

Original source details: GELFAND-P000569

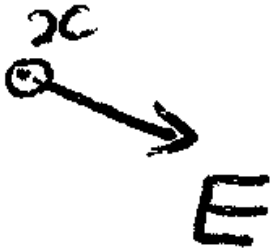
Uncertain reading. The abbreviation before "map" reads "стр." The curved arrow has an upper head beside the abbreviation and a lower head beside the note below.



Original source details: GELFAND-P000569

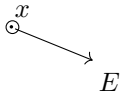
Uncertain reading. The end of the word merges with the arrow. It provisionally reads "from it", possibly "after the German".

E — field strength (electr.), produced by unit charges at points $x \in S$. We get a vector $E \in \mathbb{R}^n \cup \infty$



Original source details: GELFAND-P000569

Diagram and typed description.



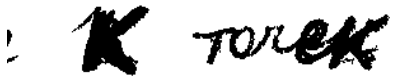
At the points $x \in S$ it will be ∞. $\infty \longrightarrow 0$

$$f_S : \mathbb{R}^n \cup \infty \longrightarrow \mathbb{R}^n \cup \infty$$

We get a map

$$C_k(\mathbb{R}^n) \longrightarrow \mathrm{Map}_k(S^n, S^n). \qquad (\text{map of degree } k).$$

(When far away, all k points are perceived as one).



Original source details: GELFAND-P000569

Uncertain reading. The k and the end of the Russian word for "points" are written over. The earlier strokes are illegible.

A point corresp. to a long vector — near one of the k charges
— and there are such points.

Uncertain reading. A letter in the middle of the Russian word for "such" is heavily overwritten. The earlier strokes are illegible.

Original source details: GELFAND-P000569

$C_k(\mathbb{R}^n)$ — a good approx. of $\mathrm{Map}_k(S^n, S^n)$!! (in a certain sense)
A good approx. for large k.

$$H_i(C_k(\mathbb{R}^n)) \underset{\cong}{\longrightarrow} H_i(\mathrm{Map}_k)$$

for $i \ll k$ (there are many proofs, including Segal's)

Uncertain reading. Below the arrow is a wave over two horizontal strokes, in the form ≅.

Original source details: GELFAND-P000569

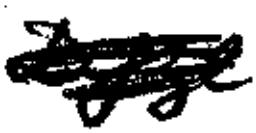
Inventiones

Uncertain reading. The underlined word reads "Inventiones". The final letters are joined.

Original source details: GELFAND-P000569

GELFAND-P000570

A generalization of this thm. will be described (due to ~~[...]~~ Dusa Mc Duff)



Undeciphered passage. A short entry before the name "Dusa Mc Duff" is crossed out and illegible.

Original source details: GELFAND-P000570

Fuchs. $C_k(\mathbb{R}^n)$ gives various classifying spaces.

Uncertain reading. The initial letter of "Fuchs" is written over with a heavy stroke. The earlier strokes remain unresolved.

Original source details: GELFAND-P000570

2 kinds of questions arise:

The right side is indep. of k ($\approx$), on the left pass to the limit in k.

Original source details: GELFAND-P000570

Uncertain reading. The middle of the Russian word for “limit” is written over. The earlier strokes are illegible.

(homot. eq.)

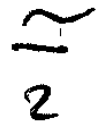
$C_k^{\pm}(\mathbb{R}^n)$ finite subsets with charges of diff. signs

topology — if 2 oppositely charged particles approach, they disappear.

Many components (total charge), $-\infty < k < +\infty$

Does not depend strongly on k.

$$C_k^{\pm}(\mathbb{R}^n) \underset{2}{\simeq} \mathrm{Map}_k(S^n, S^n)$$



Original source details: GELFAND-P000570

Uncertain reading. A separate mark in the shape of 2 is below $\simeq$. Its meaning remains unresolved.

a very good approx. (hypothesis). or

Original source details: GELFAND-P000570

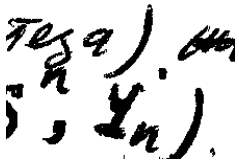
Uncertain reading. Heavy crossing strokes follow “hypothesis”. The provisional reading is “or”.

There is a space Y_n, $\dim Y_n = 2n$,

$$C_k^{\pm}(\mathbb{R}^n) \simeq \mathrm{Map}_k(S^n, Y_n).$$

Original source details: GELFAND-P000570

Uncertain reading. The upper stroke of the relation is written over. The visible wave over a line is transcribed as $\simeq$.



Original source details: GELFAND-P000570

Uncertain reading. The Y in the final Y_n is written over. The earlier strokes are illegible.

$C_k(M)$ M — a manifold

What in place of Map(S^n, S^n)?

$$C_k(M) \longrightarrow \Gamma(T_M^{\text{comp}}),$$

∞ is added in each fibre. (all fibres compactified).

At [...] infinity one can take 0 (with comp. supp.)

Uncertain reading. A small loop-shaped mark is crossed out before "infinity". Its reading remains unresolved.

Original source details: GELFAND-P000570

$$C_k(M) \longrightarrow \Gamma_k^{\text{cpt}}(T_M^{\text{comp}})$$

induces a homol. isom. for $i \ll k$.

(The electr. field is taken to have finite range — then everything is constr.)

The proof method is of greater interest than the assertion.

$U \subset M$ a small piece of M. $C_k(U)$ [...] $C_k(M)$.

Uncertain reading. A sign between $C_k(U)$ and $C_k(M)$ is heavily crossed out. Its reading remains unresolved.

Original source details: GELFAND-P000570

We would like to obtain $C_k(M)$, by joining $C_k(U)$.

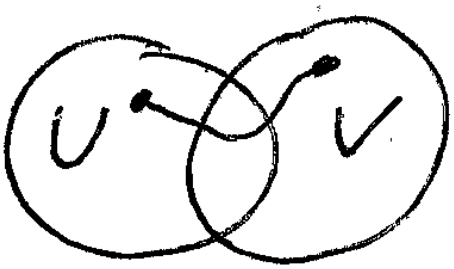
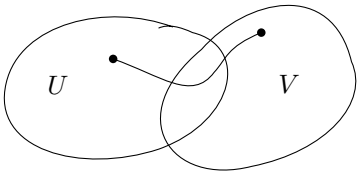


Diagram and typed description.



Original source details: GELFAND-P000570

$$C(M) \underset{\text{discontinuous}}{\longrightarrow} C(U) \times C(V)$$

Uncertain reading. "Discontinuous" is written below the formula and above the next line.

Original source details: GELFAND-P000570

Let U, V — manifolds with boundary.

$\overline{C}(U)$ — the quotient space of $C(U)$

$$S_1 \sim S_2 \quad \Leftrightarrow \quad S_1 \cap \overset{\circ}{U} = S_2 \cap \overset{\circ}{U}$$

(points on the boundary disappear)

Thm. U — a manif. with boundary (nonempty)

$$\Rightarrow \quad \overline{C}(U) \underset{\simeq}{\longrightarrow} \Gamma(TU^{\text{comp}}) \qquad \text{(homotopic equiv.)}.$$

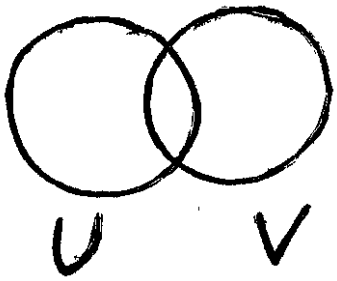
Proof. Case 1. $U =$ the unit disk.

Left side: $C_\varepsilon(U)$ — configurs. with distance $> \varepsilon$. Configurs. of $\leqslant 1$ particles remain.

right side $\quad \text{Map}(U, S^n) \simeq S^n$

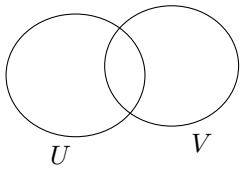
Radial stretch. $\longrightarrow S^n$ (pushed out to the boundary)

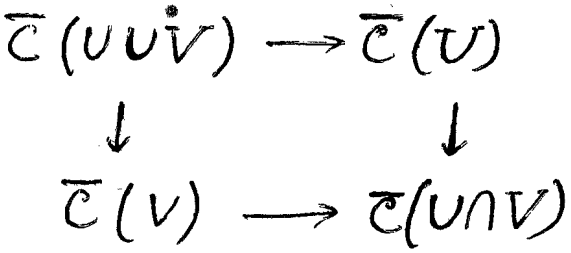
GELFAND-P000571



Original source details: GELFAND-P000571

Diagram and typed description.





Original source details: GELFAND-P000571

Diagram and typed description.

$$\begin{array}{ccc} \overline{C}(U \cup \dot{V}) & \longrightarrow & \overline{C}(U) \\ \downarrow & & \downarrow \\ \overline{C}(V) & \longrightarrow & \overline{C}(U \cap V) \end{array}$$

A dot stands above V in the upper left. Its meaning remains unresolved. The lower arrow points right.

there are well-defined maps and if 2 points from $\overline{C}(U)$ and $\overline{C}(V)$ det. one in $\overline{C}(U \cap V)$, then ~~this~~ det. a point in $\overline{C}(U \cup V)$.

Uncertain reading. "This" is crossed out after "then".

Original source details: GELFAND-P000571

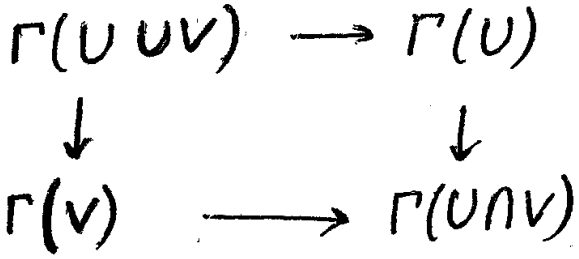
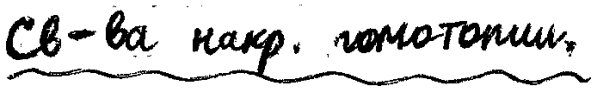


Diagram and typed description.

$$\begin{array}{ccc} \Gamma(U \cup V) & \longrightarrow & \Gamma(U) \\ \downarrow & & \downarrow \\ \Gamma(V) & \longrightarrow & \Gamma(U \cap V) \end{array}$$

Original source details: GELFAND-P000571

Props. of covering homotopy.



Uncertain reading. The middle Russian abbreviation reads "накр.".

Original source details: GELFAND-P000571

the analogous thm. on $C_k(M)$ is omitted.

Next time: Polyakov or Dubrovin.

GELFAND-P000572

7/X–74. Gelfand Sem.

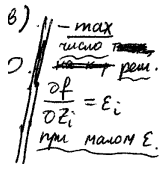
Kushnirenko. $f(z_1, \dots, z_n) = \sum a_m z^m$

0 — sing. point. (series without const. and linear terms).

Milnor number $\mu(f)$ = multiplicity of the zero sol.

$$\frac{\partial f}{\partial z_1} = 0, \dots, \frac{\partial f}{\partial z_n} = 0.$$

— max number [...] [...] sols.
$\frac{\partial f}{\partial z_i} = \varepsilon_i$
for small ε.

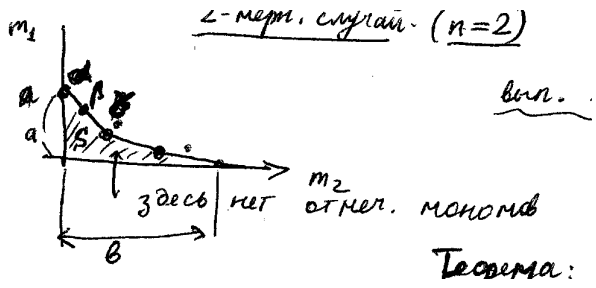


Original source details: GELFAND-P000572

Undeciphered passage. Two groups in the right margin are crossed out after “number” and before “sols.”. Remnants of strokes are visible. The full reading remains unresolved.

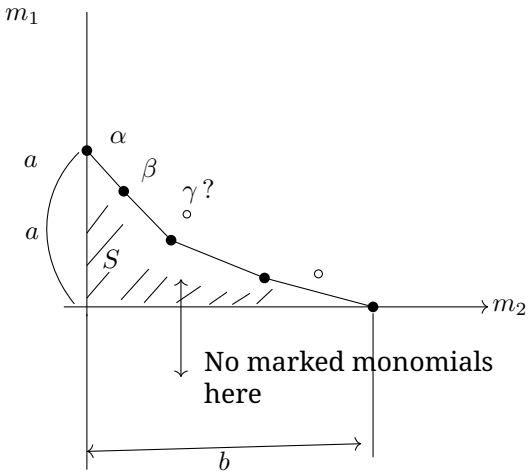
Problem: compute $\mu(f)$ from the coeffs. of the Taylor series.

2-dim. case. ($n = 2$)



Original source details: GELFAND-P000572

Diagram and typed description.



The third label is written over. Its shape and the formula below suggest γ. The reading remains provisional. The two small marks above the polygonal line are retained.

conv. polygonal line

$$\nu(f) = 2S - a - b + 1.$$

Theorem: 1) $\mu(f) \geqslant \nu(f)$

2) $\mu(f) = \nu(f)$, if certain nondegeneracy conds. hold

$\beta^2 \neq 4\alpha\gamma$ (α, β, γ — coefficients of the resp. monomials).

(in the gen. case $\neq 0$ discriminant of the left edge.)

↘ on the edge essentially a polynomial of 1 var.

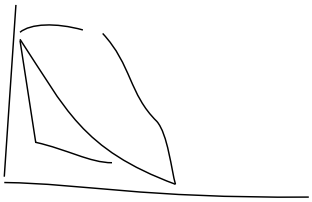
$n = 3$.

$$\nu(f) = 3!V - 2!(S_{12} + S_{13} + S_{23}) + 1!(L_1 + L_2 + L_3) - 1.$$

S_{12} — area obtained on the plane.

Original source details: GELFAND-P000572

Diagram and typed description.



The contours and their spatial relation are schematic in the manuscript.

Original source details: GELFAND-P000572

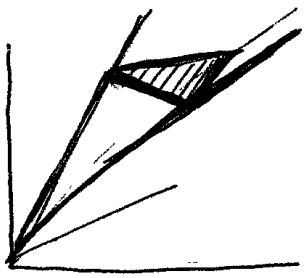
Uncertain reading. A short detached oblique stroke is visible between the formula for ν and the drawing. Its function is unresolved.

Theorem: 1) $\mu(f) \geqslant \nu(f)$

2) nondegen. cond. $\Rightarrow$ equal-ty.

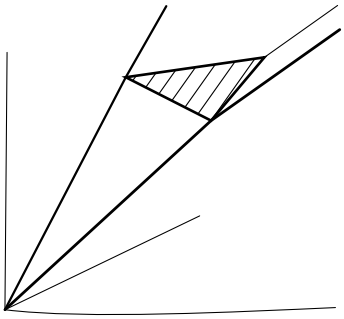
(holds almost always)

GELFAND-P000573



Original source details: GELFAND-P000573

Diagram and typed description.



The hatched region lies between the upper rays. Heavy and retraced strokes are drawn schematically. The precise directions of the extensions are unresolved.

$$z_i \frac{\partial f_\Delta}{\partial z_i}$$

f_Δ — corresp. to the monomials of the face.

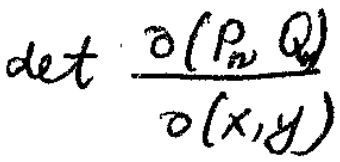
$$\dim_{\mathbb{C}} \mathbb{C}[P(\Delta)] / \left(z_i \frac{\partial f}{\partial z_i} \right) < \infty.$$

nondegen. cond.

Proof: commut. algebra. Use that $\mathbb{C}[P(\Delta)]$ — a Cohen–Macaulay ring (Hochster).

Arnold.

$$\sum_{x,y:\begin{cases} P_n(x,y)=0 \\ Q_n(x,y)=0 \end{cases}}^{n^2} \frac{R_{n-4}(x,y)}{\det \frac{\partial (P_n\, Q)}{\partial (x,y)}} = 0.$$



Original source details: GELFAND-P000573

Uncertain reading. The numerator of the Jacobian shows Q after P_n. No distinct comma is visible, and any index on Q is obscured by the tail and parenthesis. The summation conditions separately show Q_n.

$P_n(x,y)$, $Q_n(x,y)$ — 2 polynomials.

(R — an arb. polynomial of deg. $\leq n-4$)

$\sum$ over all zeros.

Polyakov. Consider $\mathbb{Z}^2$ and a fn. of an oriented edge connecting adjacent vertices.

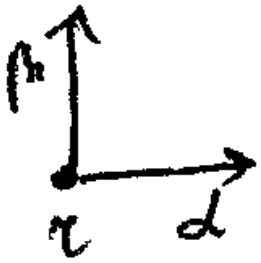
Source raster detail: GELFAND-P000573

Icon visible in the source raster. A yellow icon with three strokes is visible above the word for “oriented” in the raster. Its text is absent from the image and its origin is unresolved.

Yang–Mills fields.

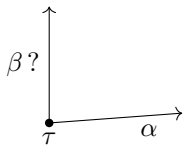
$$B_{\tau,\alpha}, \quad \tau \in \mathbb{Z}^2, \quad \alpha \text{ — dir.}$$

a fn. with vals. in SU(2) (or another compact gr.).



Original source details: GELFAND-P000573

Diagram and typed description.



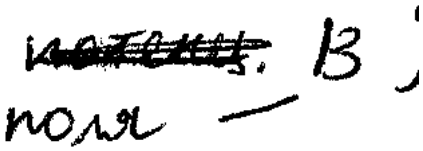
The label by the vertical arrow resembles β, but its shape is ambiguous. The horizontal arrow is labelled α.

$$F_{\vec{\tau},\alpha\beta} = \quad \text{circul.} = B_{\tau\alpha} B_{\tau+a_\alpha,\beta} B^{-1}_{\tau+a_\beta,\alpha} B^{-1}_{\tau\beta}.$$

Manuscript excerpt. The top, left, bottom and right arrows point left, down, right and up, respectively.

Original source details: GELFAND-P000573

(field strength of ~~[...]~~ B). $\hookleftarrow$ field

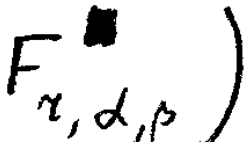


Uncertain reading. The earlier word before B is crossed out, provisionally "of the contour". The inserted word "field" appears below with a connecting stroke.

Original source details: GELFAND-P000573

Energy

$$E = \sum_{\tau,\alpha,\beta} \chi(F_{\tau,\alpha,\beta}) \qquad Z = \int \prod dB_{\tau,\alpha}\, e^{-\frac{1}{T}E[B]}.$$



Uncertain reading. A dense dark patch is visible above F. The underlying sign and its function are unresolved.

Original source details: GELFAND-P000573

χ — trace.

Correl. fns.:

$$\Pi(C) = Z^{-1} \int \prod dB_{\tau\alpha}\, \chi \left[\prod_{(C)} B_{\tau\lambda} \right] e^{-\frac{1}{T}E(B)}.$$

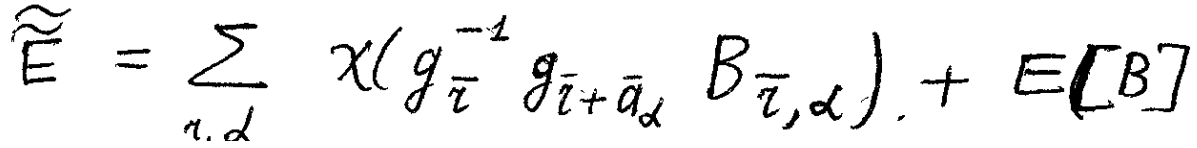
$\alpha, \beta = 1$ or 2 C — contour.

Lattice of size $N \times N$.

The planar prob. is solved exactly.

$$\widetilde{E} = \sum_{\tau,\alpha} \chi(g^{-1}_{\vec{\tau}} g_{\vec{\tau}+\vec{a}_\alpha})$$

$$\widetilde{\widetilde{E}} = \sum_{\tau,\alpha} \chi(g^{-1}_{\bar{\tau}} g_{\bar{\tau}+\bar{a}_\alpha} B_{\bar{\tau},\alpha}). + E[B].$$



Original source details: GELFAND-P000573

Uncertain reading. In the double-tilde line B follows the second g within the parentheses with no visible comma between them. The τ index on B has a bar. There is a small separate stroke near the baseline before the plus sign.

Field th. = stat. mech. in $\mathbb{R}^n$ (Euclidean).

(Fourier transf. in time)

Sinai.

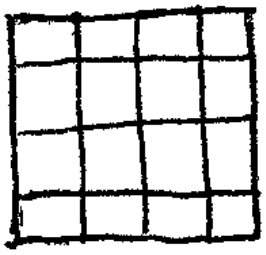
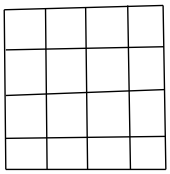


Diagram and typed description.



Original source details: GELFAND-P000573

V — finite volume.

$$E(\{B_{\tau\alpha}\}) = \sum \operatorname{tr}\left(B_{\tau\alpha} B_{\tau+a_\alpha,\beta} B^{-1}_{\tau+a_\beta,\alpha} B^{-1}_{\tau,\beta}\right)$$

$$\underline{\underline{\mathcal{M} e^{-\beta E} = Z(\beta, V)}}$$

Calculated for the 2-dim. case.

$$\ln Z(\beta, V) = V \ln I_0(\beta) \qquad (I_0(\beta) \text{ — Bessel fn.})$$

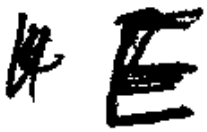
GELFAND-P000574

$$B_{\tau\alpha} = e^{iA_{\tau\alpha}}$$

Uncertain reading. The first subscript on B and on A has the same hooked shape. The reading τ is tentative.

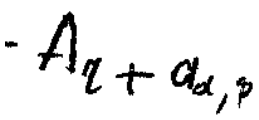
Original source details: GELFAND-P000574

[...] $E = \sum \cos(A_{\tau\alpha} + A_{\tau+a_\alpha,\beta} - \ldots)$



Uncertain reading. A short sign is densely crossed out before E. φ is possible, but the reading is unresolved.

Original source details: GELFAND-P000574



The subscript on the second A under the cosine is $\tau + a_\alpha, \beta$. The a has a separate subscript α. The reading τ is tentative.

Original source details: GELFAND-P000574

$$\lim_{V\to\infty} \frac{\ln Z(\beta, V)}{V} = f(\beta)$$

does not depend on the choice of boundary conds.

$f(\beta)$ may have singularities — they are called phase transitions.

(≈ a rearrangement of typical configs. occurs).

$$p(\{B_{\tau,\alpha}\}) = Z^{-1} e^{-\beta E(\{B_{\tau\alpha}\})} \qquad \text{— Gibbs distrib.}$$

Correl. fn. — mean value of the circulation around the contour.
Phase trans. — in one phase ∼ area
in the other ∼ length.

GELFAND-P000575

~~[...]~~ 14/X–74. Gelfand Sem.

Uncertain reading. A short group is crossed out before the date. The Russian letters “Пo” are possible, but the strokes overlap.

Original source details: GELFAND-P000575

Graeme Segal.
Classifying space of category.
$\mathcal{C}$ — category, $|\mathcal{C}|$ is top. space
Objects — points
Morphisms — 1-dim symplex.
Compositions — ?

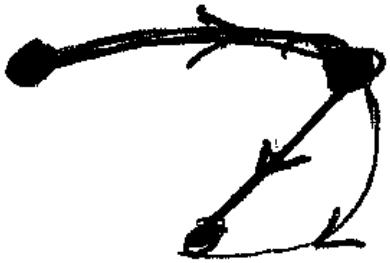
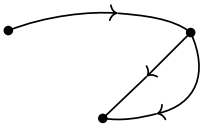


Diagram and typed description.



Original source details: GELFAND-P000575

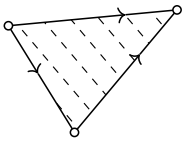
we have a graph.
It describe not cat. [?]

Uncertain reading. A low loop-like sign follows “cat.”. The reading $\mathcal{C}$ is tentative. Its shape differs from the clearly written letter above.

Original source details: GELFAND-P000575

(composition!)

Diagram and typed description.



Original source details: GELFAND-P000575

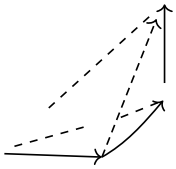
two-symplex — composition of two morphisms.

3 morph., and all compos. — 3-dim symplex.

which can be composed ↑

Original source details: GELFAND-P000575

Diagram and typed description.



The solid arrows and discernible thin strokes are drawn schematically. Some overlapping edges have unresolved directions.

etc.

We define $|\mathcal{C}|$.

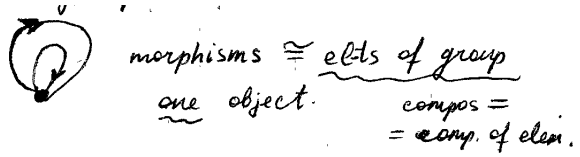
Diametr. opposed examples.

Partially ↓

Ordered sets.

≡ categ. with at most one morph. between two objects.

groups



Original source details: GELFAND-P000575

Diagram and typed description.

morphisms ≅ elts of group
one object.
compos =
= comp. of elem.

X top space, $\{U_\alpha\}_{\alpha\in S}$ — covering $U_\alpha \subset X$

open sets

We suppose:

Property:

$$U_\alpha \cap U_\beta = \bigcup_{\text{certain } \gamma} U_\gamma.$$

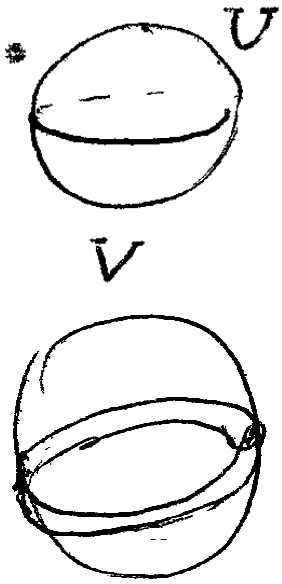
$$\alpha \le \beta \underset{\text{def}}{\iff} U_\alpha \subset U_\beta.$$

If all sets are contractible then $|\mathcal{C}| \underset{\text{hom. equiv.}}{\simeq} X$.

Add. intersections.

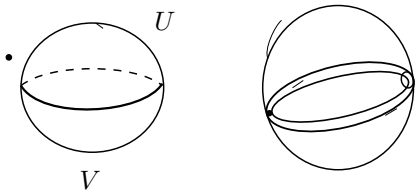
GELFAND-P000576

Example. $X = S^2$. $w = U \cap V$.

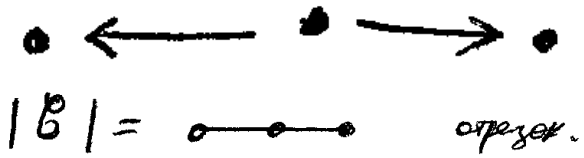


Original source details: GELFAND-P000576

Diagram and typed description.



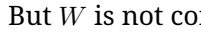
A small mark is visible to the left of the upper outline. The band lines and side marks overlap in parts of the lower drawing.



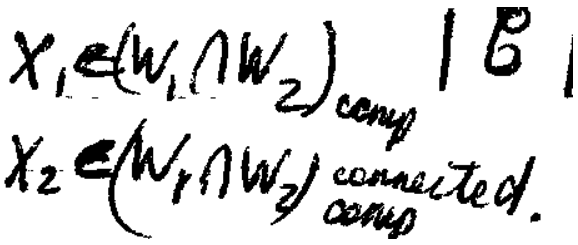
Original source details: GELFAND-P000576

Diagram and typed description.

But W is not contractible.

$|\mathcal{C}| =$  segment.

$$w = W_1 \cup W_2$$
$$x_1 \in (W_1 \cap W_2)_{\text{comp}}$$
$$x_2 \in (W_1 \cap W_2)_{\substack{\text{connected} \\ \text{comp}}}.$$



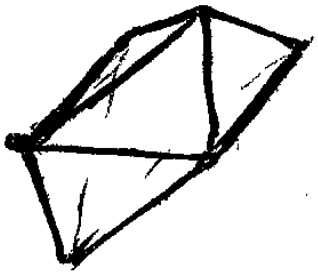
Original source details: GELFAND-P000576

The first parenthesis is followed by “comp”. The second is followed by “connected”, with “comp” below.

~~oc?~~ — surface of octahedron.

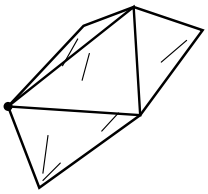
Original source details: GELFAND-P000576

Uncertain reading. Two loop-like letters are visible under the cancellation, tentatively “oc”.



Original source details: GELFAND-P000576

Diagram and typed description.



The redraw follows the clear outline and inner strokes. Some thin and superimposed edges are only partly discernible.

Ex. 2

V — finite dimensional vect. space ($\forall$ field).

$\mathcal{C}$ — cat of proper subspaces and inclusions.

$|\mathcal{C}|$ = "building" of V building

"immeuble"

Thm. (Solomon — Tits).

~~[...]~~ $|\mathcal{C}| \simeq$ wedge (bouquet) of spheres of dim $(n-2)$

$(n = \dim V)$,

Original source details: GELFAND-P000576

Uncertain reading. A cancelled group with a vertical and loops precedes $|\mathcal{C}|$. An earlier $|\mathcal{C}|$ is possible.

(quantity of spheres depends on field — infinite if the field is infinite).

Proof.

1. If ord. set has max elt then $|\mathcal{C}|$ is contractible.

Original source details: GELFAND-P000576

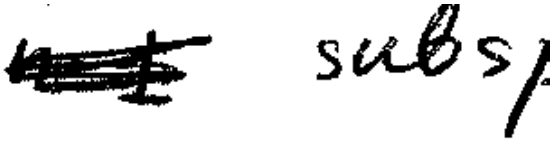
Diagram and typed description.

evidente.

max elements are ~~[...]~~ subsp. of codimension 1.

Original source details: GELFAND-P000576

Uncertain reading. A short abbreviation, possibly "vect", is crossed out before "subsp.".

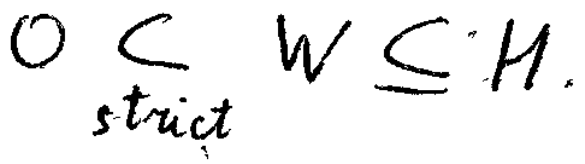


Original source details: GELFAND-P000576

Undeciphered passage. Multiple horizontal strokes conceal the middle of the cancelled abbreviation.

$$|\mathcal{C}| = \bigcup_{H\text{—hyperplanes}} |\mathcal{C}_H|.$$

$\mathcal{C}_H$ — is the cat of subspace W, $0 \underset{\text{strict}}{\subset} W \subseteq H$.



The first inclusion sign $\subset$ is labelled “strict”. The second has a separate lower stroke, $\subseteq$.

Original source details: GELFAND-P000576

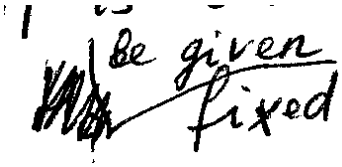
$|\mathcal{C}_H|$ is contractible.

be given ↙

Let ~~[...]~~ fixed ~~hyperplane~~ line L.

Uncertain reading. A group between “Let” and “fixed” is crossed out, possibly “We”.

Original source details: GELFAND-P000576

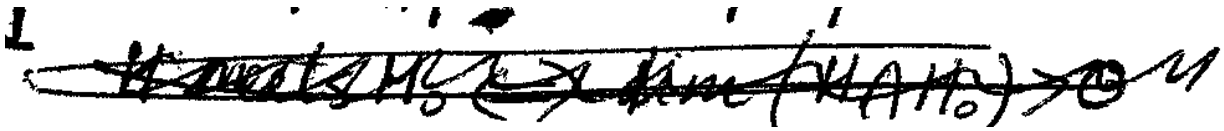


Undeciphered passage. Dense slanted strokes obscure the full group. “be given” is written separately above the line.

Original source details: GELFAND-P000576

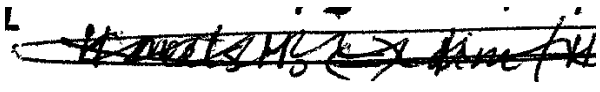
$$\text{Let} \quad Y = \bigcup_{H \supset L} |\mathcal{C}_H| \subset |\mathcal{C}|.$$

~~[...] $\Rightarrow$ dim$(W \cap H_0) > 0$ [...]~~



Original source details: GELFAND-P000576

Uncertain reading. The end of the cancelled line reads dim$(W \cap H_0) > 0$. The sign before “dim” resembles an implication arrow, and the final stroke may be Cyrillic “и”.



Undeciphered passage. The opening and intervening parts of the cancelled line remain unresolved.

Original source details: GELFAND-P000576

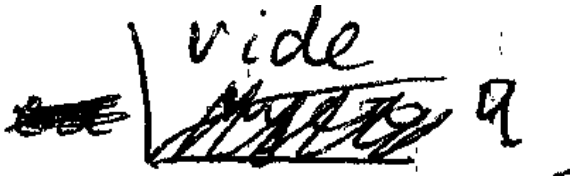
Y is contractible because

$$|\mathcal{C}_{H_1}| \underset{\text{contr.,}}{\cap} |\mathcal{C}_{H_2}| \underset{\text{contr.}}{=} |\mathcal{C}_{H_1 \cap H_2}| \underset{\text{also contractible}}{.}$$

GELFAND-P000577

If Y is covered by pieces Y_i such that all $Y_{i_1} \cap \ldots \cap Y_{i_k}$

is contractible (and not ~~[...]~~ vide ~~empty?~~ [?]) $\Rightarrow Y$ contractible.



Original source details: GELFAND-P000577

Uncertain reading. “vide” is above the crossed-out word, tentatively Russian “пусто”. A loop-like sign with a descender is to the right. Its reading is unresolved.

Original source details: GELFAND-P000577

Undeciphered passage. The first short word after “not” is crossed out. “et” is possible, but the full lettering is indistinct.

~~[...]~~

Original source details: GELFAND-P000577

Uncertain reading. The short cancellation below “is” begins with strokes that resemble “te”.

Original source details: GELFAND-P000577

Undeciphered passage. Dense crossing strokes obscure the ending.

in partial.

$$|\mathcal{C}| = Y \cup \bigcup_{H \not\supset L} |\mathcal{C}_H|.$$

Original source details: GELFAND-P000577

The union is indexed by $H \not\supset L$. The slashed inclusion arc has no separate lower stroke.

Lemma 1.

If $X = Y \cup Y_1 \cup \ldots \cup Y_k$

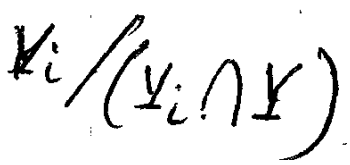
and $Y_?, Y_1, \ldots, Y_k$ contractible.

Original source details: GELFAND-P000577

Uncertain reading. A subscript on the first Y is crossed out. Its reading is unresolved.

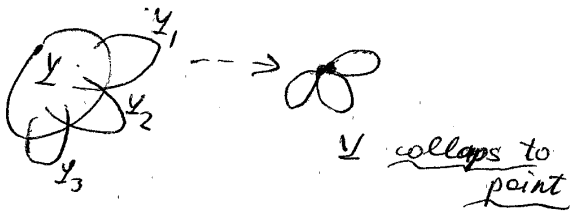
and $Y_i \cap Y_j \subset Y$, then

$$X \simeq \bigvee_{\text{wedge}} Y_i/(Y_i \cap Y)$$



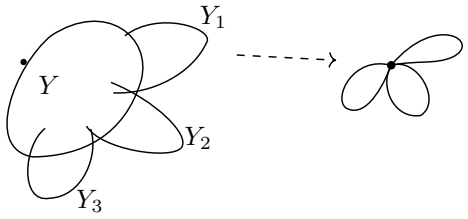
Original source details: GELFAND-P000577

The numerator reads Y_i. The short foot on the Y also appears on both occurrences of Y in the denominator.



Original source details: GELFAND-P000577

Diagram and typed description.



Y collaps to point

Lemma 2.

Also if Z is contractible and $Z_0 \subset Z$ then

$$Z/Z_0 \simeq \underset{\text{suspension}}{SZ_0}$$

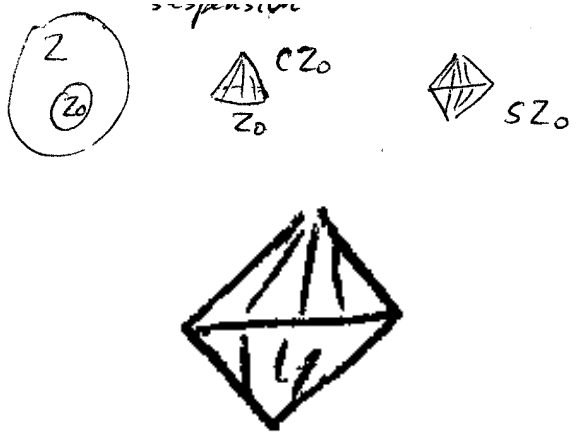
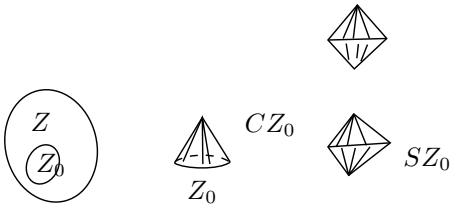


Diagram and typed description.



Original source details: GELFAND-P000577

$$\Rightarrow X \simeq \bigvee S(Y_i \cap Y)$$

Lemma 3.

$|\mathcal{C}_H| \cap Y =$ "building" of H.

and reduction by dimension.

dim $= 2$. (first step of induction).

Original source details: GELFAND-P000577

Diagram and typed description.

Lines

wedge of !! points ~~[...]~~

Original source details: GELFAND-P000577

Uncertain reading. "of" is visible in an outline after "wedge", with two vertical marks nearby. Their function is uncertain.

~~[...]~~

Original source details: GELFAND-P000577

Uncertain reading. Some strokes in the lower cancelled group resemble "wedge".

Original source details: GELFAND-P000577

Undeciphered passage. The lower inscription and the cancelled group after "points" are incompletely legible.

Suspension of wedge of spheres is also a wedge of spheres.

$H_{n-2}(|\mathcal{C}|;\mathbb{Z}) =$ Steinberg module of V.

Original source details: GELFAND-P000577

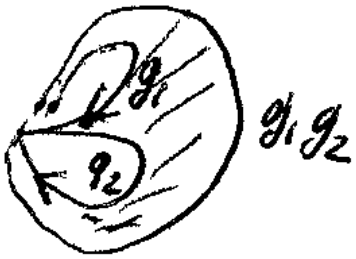
Uncertain reading. Typeset as $=$. The lower stroke is retraced, so $\equiv$ is a possible reading.

(repres. in subgrps. of a fin. group)

Other extreme case.

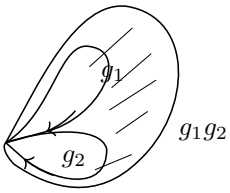
$\mathcal{C}$ of group G. $\quad |\mathcal{C}| = BG$ — classifying space of group

$\pi_1(|\mathcal{C}|) = G$. $\quad$ (evidently)



Original source details: GELFAND-P000577

Diagram and typed description.



The outline and loops follow the discernible strokes. Short, superimposed marks near the outline do not establish further directions.

Universal covering is contractible.

Let $\widetilde{\mathcal{C}}$ $\quad$ objects $=$ elts of G.

$\exists\, 1$ morph. $g \to g'$ for each pair of elts.

GELFAND-P000578

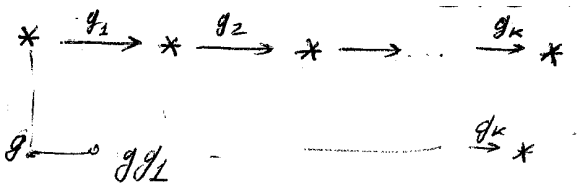
$|\widetilde{\mathcal{C}}|$ is contractible.

($\forall$ el-t is connected by line with unity 1).

$\widetilde{\mathcal{C}} \xrightarrow{\text{functor}} \mathcal{C}$ universal covering space.

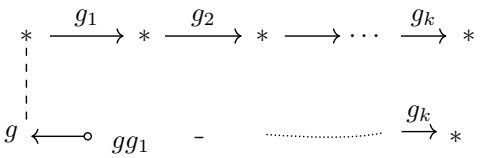
$g \mapsto *$

$g \to g' \mapsto g^{-1}g'$



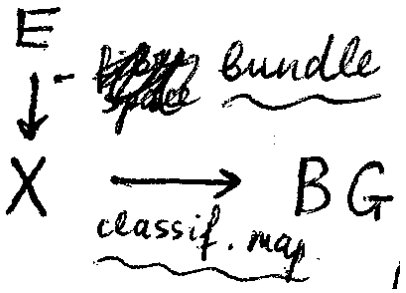
Original source details: GELFAND-P000578

Diagram and typed description.



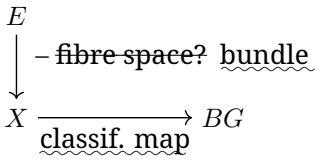
The short lower-left arrow points from a small loop towards g. An asterisk is visible at the right.

$|\widetilde{\mathcal{C}}| \to |\mathcal{C}|$ universal covering.



Original source details: GELFAND-P000578

Diagram and typed description.



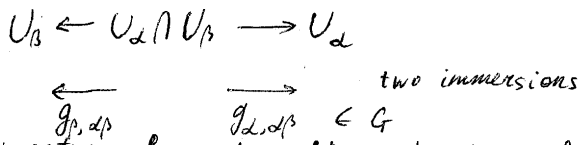
Original source details: GELFAND-P000578

Uncertain reading. The crossed-out expression next to “bundle” tentatively reads “fibre space”.

$\{U_\alpha\}$ — covering, on U_α bundle is trivial.

Original source details: GELFAND-P000578

Uncertain reading. A heavy superimposed stroke follows “covering” before the comma. A separate character is uncertain.



Original source details: GELFAND-P000578

Diagram and typed description.

$U_\beta \longleftarrow U_\alpha \cap U_\beta \longrightarrow U_\alpha$

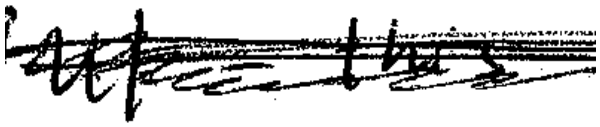
$\overleftarrow{g_{\beta,\alpha\beta}}$ $\overrightarrow{g_{\alpha,\alpha\beta}}$ two immersions $\in G$

Let us consider categ. of sets U_α and their finite intersections

~~let ... this cat. of sets~~

Original source details: GELFAND-P000578

Uncertain reading. The crossed-out line shows "let", tentatively "this", and "cat. of sets".



Original source details: GELFAND-P000578

Undeciphered passage. The strokes after "let" may include "us". The full passage is indistinct.

GELFAND-P000579

21/X-74. Gelfand Sem. Kushnirenko.

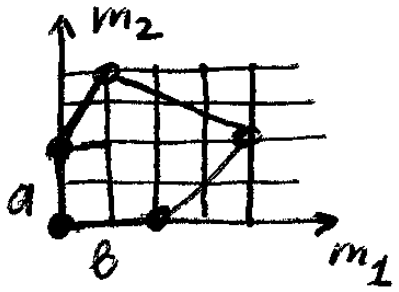
Melvin Hochster.

$P\colon \mathbb{C}^n \to \mathbb{C}$, P — a polynomial, $P = \sum a_m z^m$.

Singular points: $\{z : \frac{\partial P}{\partial z_i} = 0\}$. Consider all the sing. points. Suppose they are isolated.

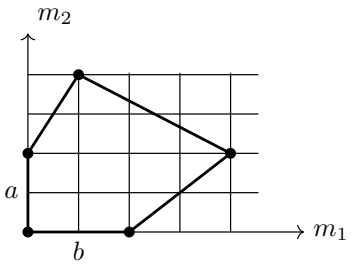
How do we find them, counting multiplicity?

Consider the convex hull of 0 and all $m \in \mathbb{Z}^n$ for which $a_m \neq 0$.



Original source details: GELFAND-P000579

Diagram and typed description.



$$2S - a - b + 1.$$

1) the number of isol. sing. points, counting mult., $\leq 2S - a - b + 1$.

2) if the nondegeneracy cond. holds $= 2S - a - b + 1$.

Idea: consider the degree of the map $S^{2n-1} \to S^{2n-1}$.

$$\underset{z}{|z|=R} \longmapsto \frac{\left(\frac{\partial P}{\partial z_1}, \ldots, \frac{\partial P}{\partial z_n}\right)}{|\text{grad } P|}.$$

This was known in the homogeneous or quasihomog. case.

$\sum$ of Milnor numbers = degree (abs. fact)

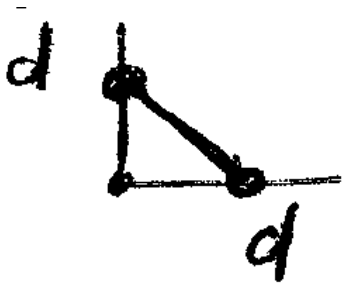
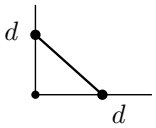


Diagram and typed description.



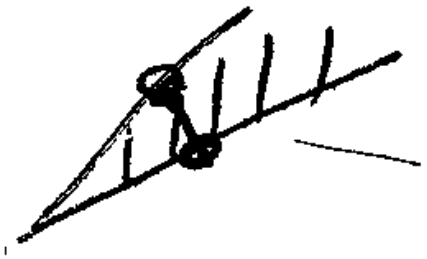
Original source details: GELFAND-P000579

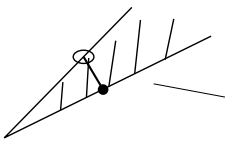
Hochster: consider the ring of monomials with exponents m in the angle.

The heavily traced word before the Russian for "angle" reads "в" ("in").

Original source details: GELFAND-P000579

Diagram and typed description.





Original source details: GELFAND-P000579

This is a Cohen–Macaulay ring.

Mumford — toric embeddings.

Polyakov:

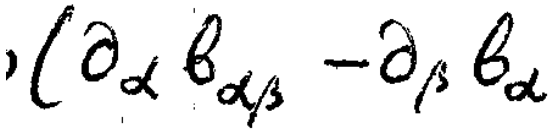
Yang–Mills field.

$$S \dots = \int d^4x \left\{ \frac{1}{e^2} \operatorname{Sp}(\partial_\alpha b_{\alpha\beta} - \partial_\beta b_\alpha - [b_\alpha, b_\beta]^2) \right.$$

⟵ action
– Lagrangian
(under the integral sign)

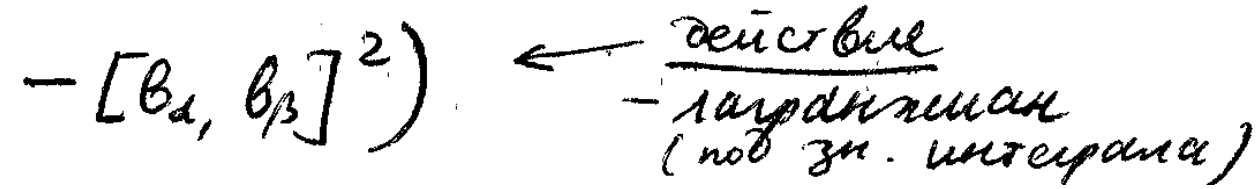
Uncertain reading. A group of marks after S is crossed out. Its reading is obscured by dense strokes.

Original source details: GELFAND-P000579



In the first term the two subscripts on b are $\alpha\beta$.

Original source details: GELFAND-P000579



Original source details: GELFAND-P000579

The exponent 2 immediately follows the square bracket. The brace after d^4x remains open.

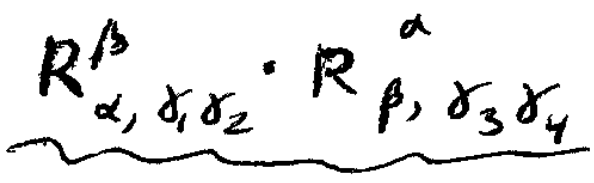
b_α — elem. of a Lie algebra, given at any point.

There is a vect. bundle over $\mathbb{R}^4$ with fibre a Lie algebra (preferably $SU(2)$)

If $(b_\alpha)^b_a$ — the matrix Γ_α, we obtain $R^b_{\alpha\beta a}$ under the integral sign.

Elem. of a Lie alg. at $\forall$ point ⟷ connection.

Pontryagin class: $R^\beta_{\alpha,\delta_1\delta_2} \cdot R^\alpha_{\beta,\delta_3\delta_4}$.



The lower pairs read $\delta_1\delta_2$ and $\delta_3\delta_4$.

Original source details: GELFAND-P000579

GELFAND-P000580

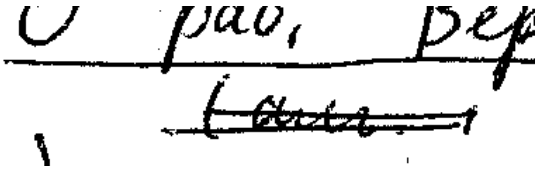
28/X-74. Gelfand Sem.

M. I. Graev. On work by Vershik, Graev and Gelfand

Uncertain reading. The second initial has been heavily retraced. Cyrillic “И.” is probable, with “Н.” also possible.

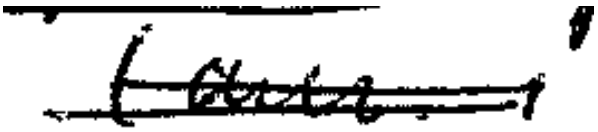
Original source details: GELFAND-P000580

~~(Alg.?)~~



Uncertain reading. The cancelled parenthetical abbreviation below the heading tentatively reads “Алг.” (“Alg.”).

Original source details: GELFAND-P000580



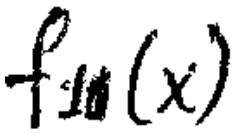
Undeciphered passage. The crossing strokes obscure some of the letters in the abbreviation.

Original source details: GELFAND-P000580

On current algebra for physicists.

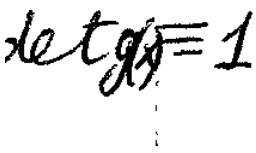
(Funct. Anal., vol. 2, $N = 1$)

$$g(x) = \begin{pmatrix} f_{11}(x) & f_{12}(x) \\ f_{21}(x) & f_{22}(x) \end{pmatrix}, \qquad x \in [0,1], \qquad \det g(x) = 1$$



Uncertain reading. The upper-left entry tentatively has subscript 11. The second digit is overwritten.

Original source details: GELFAND-P000580



Uncertain reading. The strokes at the argument of g overlap. The probable reading is $\det g(x) = 1$.

Original source details: GELFAND-P000580

$f_{ij} : [0,1] \to \mathbb{R}$ continuous functions.

A method for constructing irreducible representations is given, analogous to taking an integral.

The end of the line reads “взятию интеграла” (“taking an integral”).

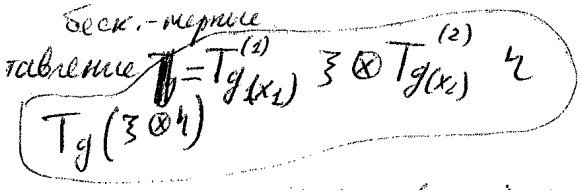
Original source details: GELFAND-P000580

GELFAND-P000581

G. Replace the interval by a set of points. Suppose there are 2 points

$$\begin{matrix} x_1 & x_2 \\ g_1 & g_2 \\ H & \end{matrix} \qquad T^{(1)}_{g_1}\xi \quad T^{(2)}_{g_2}\xi$$

Consider $H_1 \otimes H_2$ and the representation



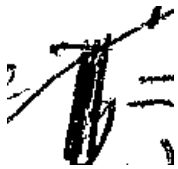
Original source details: GELFAND-P000581

Diagram and typed description.

infinite-dim.

$T = T^{(1)}_{g(x_1)}\xi \otimes T^{(2)}_{g(x_2)}\eta$

$T_g(\xi \otimes \eta)$



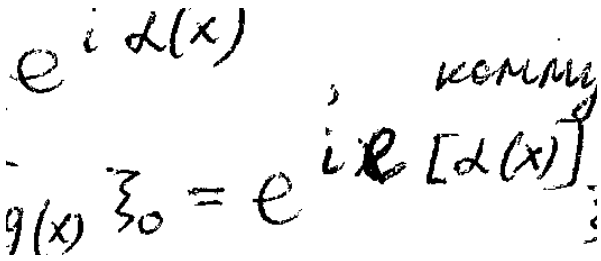
Original source details: GELFAND-P000581

Uncertain reading. The sign before the equals sign resembles T, with heavily retraced lower strokes. T_g is possible. Both operators in the upper pair act on ξ.

Example: $e^{i\alpha(x)}$, commut. group. Rep. (irred.) is 1-dimensional

$$T_{g(x)}\xi_0 = e^{i\ell[\alpha(x)]}\xi_0, \qquad \ell \text{ — lin. functional} \qquad \int \alpha(x)\,dx$$

$\alpha \in C^\infty \Rightarrow \ell$ — a generalized function



Original source details: GELFAND-P000581

The first exponent contains $\alpha(x)$. The letter α recurs in the exponent below.

Original source details: GELFAND-P000581

Uncertain reading. The Russian abbreviation “непр.” is tentatively expanded as “irreducible”.

matrix

Reps. of groups of order 1 $\Longleftrightarrow$ generalized functions

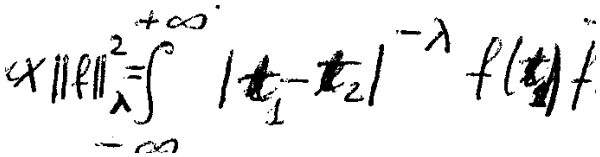
Is there an irred. rep. that is structured in the same way everywhere.

Grp. $SL_2(\mathbb{R})$ representations T_λ in spaces H_λ

H_λ — those $f(t)$ for which

$$\|f\|_\lambda^2 = \int_{-\infty}^{+\infty} |t_1 - t_2|^{-\lambda} f(t_1)\overline{f(t_2)}\, dt_1\, dt_2 < \infty$$

$0 < \lambda < 1$ for pos. definiteness

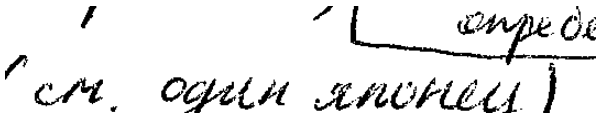


The norm has one integral sign and two differentials, $dt_1\ dt_2$. The strokes of the variables have been retraced.

Original source details: GELFAND-P000581

G. There are representations close to 1 (but infinite-dimensional!!)

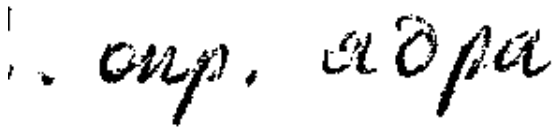
Problem. There is a dimension in Kolmogorov's sense (see one Japanese person).



The words in parentheses read "один японец" ("one Japanese man").

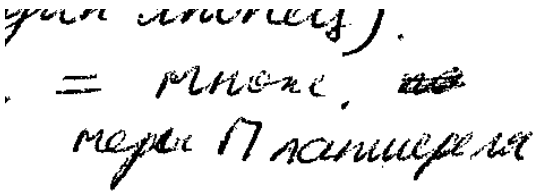
Original source details: GELFAND-P000581

With a proper def. of the kernel we can consider dim in Kolmogorov's sense $=$ mult. $\dots$ of the Plancherel measure.



The word after "опр." reads "ядра" ("of the kernel").

Original source details: GELFAND-P000581



Uncertain reading. The abbreviation after the equals sign tentatively reads "множ." ("mult."). The final strokes overlap.

Original source details: GELFAND-P000581

Uncertain reading. A short group of strokes to the right of the abbreviation is crossed out.

Original source details: GELFAND-P000581

Undeciphered passage. The entire crossed-out group remains unread.

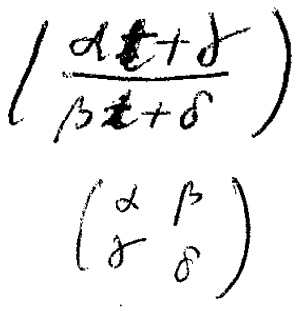
Original source details: GELFAND-P000581

Rep. of $SL_2(\mathbb{R})$,

$$f(t) \longrightarrow f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{\rho-1}, \qquad \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}$$

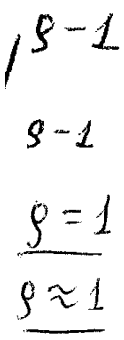
$(\dots)^{\rho-1}$

$\underline{\rho=1}$ — identity rep.
$\underline{\rho\approx 1}$ — close to the identity



The numerator of the fraction has γ. Its long rightward stroke matches the lower-left letter in the matrix.

Original source details: GELFAND-P000581



Uncertain reading. The looped sign in both exponents and both following lines is rendered as ρ. s is possible.

Original source details: GELFAND-P000581

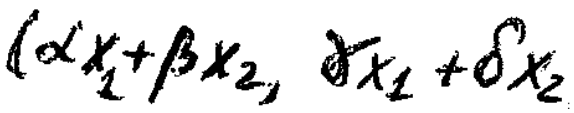
$$f(x_1,x_2) \longrightarrow f(\alpha x_1+\beta x_2, \gamma x_1+\delta x_2)$$

on homog. polynomials

$$x_2^n f(x_1/x_2,1)$$

n — weight
$\approx$ dim.

The second argument has γ before x_1 and δ before x_2.

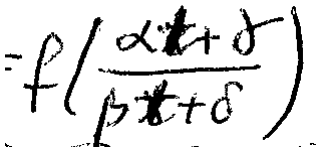


Original source details: GELFAND-P000581

Gr.

$$(T_\lambda^{(g)} f)(t) = f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{\lambda-2}, \qquad g=\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}$$

$$0<\lambda<1$$



The numerator has γ, and the denominator has δ.

Original source details: GELFAND-P000581

$\lambda = 0 \Rightarrow$ the inner prod. degenerates to

$$\left|\int_{-\infty}^{\infty} f(x)\,dx\right|^2 = \|f\|^2.$$

Analyt. cont.:

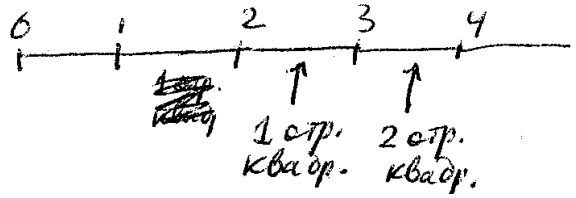


Diagram and typed description.

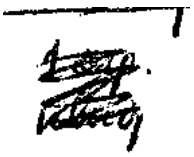
0 1 2 3 4

~~1 neg. quadr.?~~ 1 neg. quadr. 2 neg. quadr.

Original source details: GELFAND-P000581

Uncertain reading. A two-line inscription below the interval 1–2 is crossed out. It tentatively reads “1 neg. quadr.”

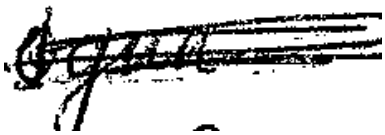
Original source details: GELFAND-P000581



Undeciphered passage. Both lines are only partially legible beneath the cancellation.

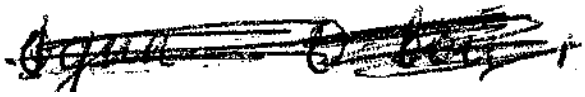
Original source details: GELFAND-P000581

$\lambda = 0$ — ~~func.? ...~~ obtain ~~...~~ a rep. in a subspace



Uncertain reading. The crossed-out passage after the equality begins with what appears to be “функц.” (“func.”).

Original source details: GELFAND-P000581



Undeciphered passage. The remaining crossed-out strokes in the phrase are unresolved.

Original source details: GELFAND-P000581

Uncertain reading. A further word is crossed out between “obtain” and “rep.”

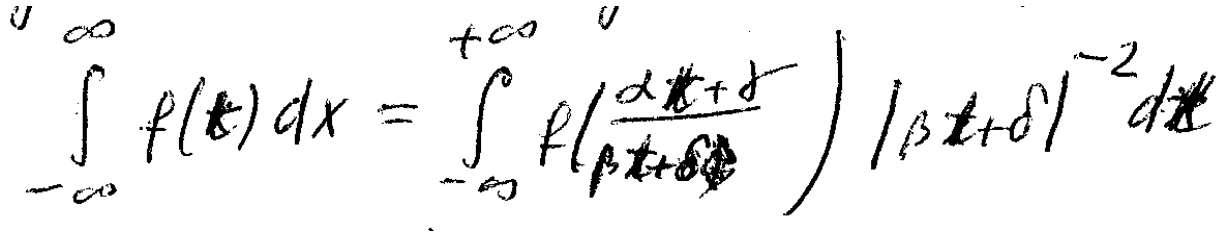
Original source details: GELFAND-P000581

Undeciphered passage. The crossed-out word remains unread.

Original source details: GELFAND-P000581

$$\int_{-\infty}^{\infty} f(t)\,dx = \int_{-\infty}^{+\infty} f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{-2}\,dt.$$

($\int$ — invariant)



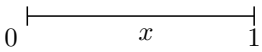
Original source details: GELFAND-P000581

Uncertain reading. The left side has $f(t)\,dx$ and the right has dt. The looped numerator letter is read as γ. Heavy strokes near the denominator’s δ obscure the earlier writing.

GELFAND-P000582

Fix λ, $(0 < \lambda < 1)$ and consider const. matrices $g(x) \equiv \text{const} = g$.

Diagram and typed description.



Original source details: GELFAND-P000582

This is $SL_2(\mathbb{R}) = G$.

Consider the rep. T_λ of this group in H_λ.

Divide the interval into 2 equal parts and take functions const. on the left and right halves

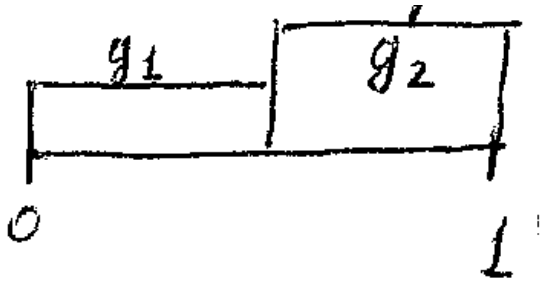
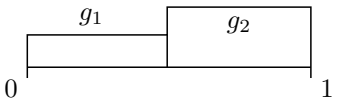


Diagram and typed description.



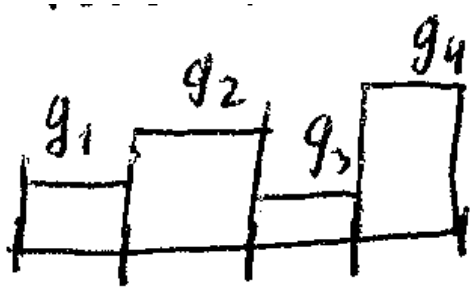
Original source details: GELFAND-P000582

$\dots$ This is $G \times G$.

Original source details: GELFAND-P000582

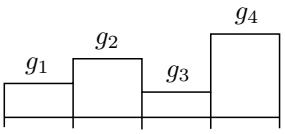
Uncertain reading. A sign before "This is" is heavily crossed out. Its reading is unresolved.

Consider $H_{\lambda/2} \otimes H_{\lambda/2}$ and define there a rep. of $G \times G$



Original source details: GELFAND-P000582

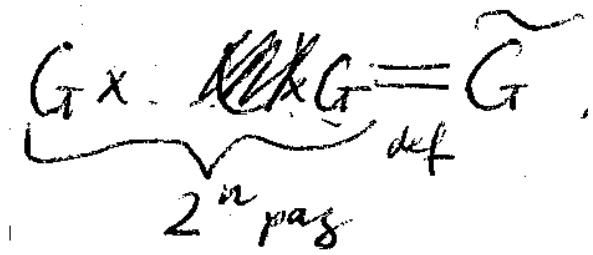
Diagram and typed description.



$$G \times G \times G \times G \longrightarrow \text{rep. in } H_{\lambda/4} \otimes H_{\lambda/4} \otimes H_{\lambda/4} \otimes H_{\lambda/4}.$$

and so on

$$G \subset G \times G \subset G \times G \times G \times G \subset \dots \qquad \varinjlim_{n \to +\infty} \underbrace{G \times \dots \times G}_{2^n \text{ times}} = \widetilde{G}.$$



Original source details: GELFAND-P000582

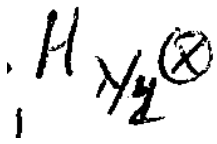
Uncertain reading. A group of symbols between the outer factors is crossed out. The brace reads "2^n times".

Representation of $\widetilde{G}$?

We need to embed $H_\lambda \subset H_{\lambda/2} \otimes H_{\lambda/2}$ compatibly with the group action.

Decomp. into irreducibles — in the discrete part enters H_λ once, and in the complement, the rest, and so on. This lets us embed

$$H_\lambda \hookrightarrow H_{\lambda/2} \otimes H_{\lambda/2} \hookrightarrow \underbrace{H_{\lambda/4} \otimes \dots \otimes H_{\lambda/4}}_{4 \text{ times}} \hookrightarrow .$$



Original source details: GELFAND-P000582

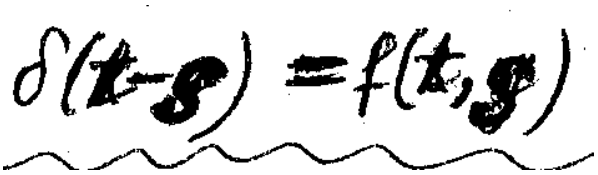
Uncertain reading. The denominator of the first subscript under the brace is overwritten. The probable reading is $\lambda/4$.

$$H_{\lambda/2} \otimes H_{\lambda/2} = H_\lambda \oplus \text{the rest}.$$

The embedding of H_λ is determined (up to a factor)

$$H_\lambda \hookrightarrow H_{\lambda/2} \otimes H_{\lambda/2}$$

$$f(t) \longrightarrow f(t)\delta(t-s) = f(t,s) \qquad \text{(unit. embedding)}$$



Uncertain reading. Both occurrences of the second argument are overwritten with a looped sign. s is a tentative reading.

Original source details: GELFAND-P000582

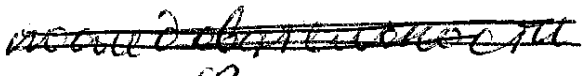
Uncertain reading. The last letter of "вложение" ("embedding") is heavily retraced. The ending of "вложения" may underlie it.

Original source details: GELFAND-P000582

$$\varinjlim H_{\lambda/2^n} \otimes \ldots \otimes H_{\lambda/2^n}$$ and take the completion

the group of step functions acts there. The rep. extends to a rep. of the group of cont. functions.

Elements of the space — ~~sequences~~ functions $f(x,t)$, $-\infty < t < +\infty$, $0 < x < 1$



The crossed-out word reads "последовательности" ("sequences").

Original source details: GELFAND-P000582

~~$f(t_0)f(t)$~~? $\int_{-\infty}^{\infty} f(x,t)\,dt = 1$

Uncertain reading. The cancellation to the left of the integral resembles $f(t_0)f(t)$. The complete reading is unresolved.

Original source details: GELFAND-P000582

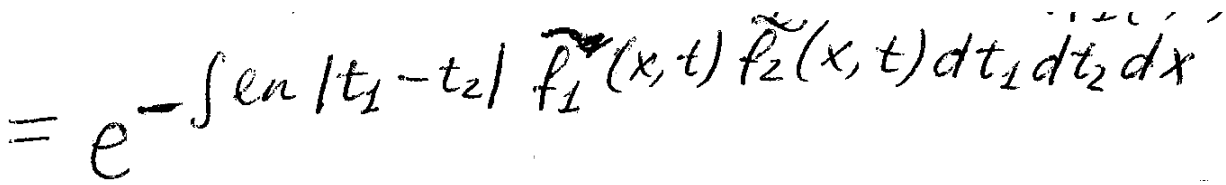
Consider formal lin. combs. $\sum \lambda_i \cdot f_i(x,t)$.

$\sum \lambda_i \cdot 0 \neq 0$ (!).

Inner prod.

$$\langle f_1(x,t), f_2(x,t)\rangle = e^{-\int \ln|t_1-t_2|\, \widetilde{f}_1(x,t)\widetilde{f}_2(x,t)\,dt_1\,dt_2\,dx}$$

and by linearity



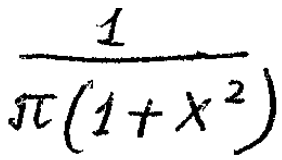
Original source details: GELFAND-P000582

The exponent has one integral sign. Both occurrences of $\widetilde{f}_i$ have the argument (x,t). The tilde above the first f is retraced.

$f_1(x,t)$ — a curve in a Hilb. space. Inner prod. of δ-functions on them.

GELFAND-P000583

$$\widetilde{f}(x,t) = f(x,t) - \frac{1}{\pi(1+x^2)}.$$

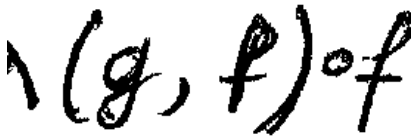


The denominator of the subtracted fraction has x^2.

Original source details: GELFAND-P000583

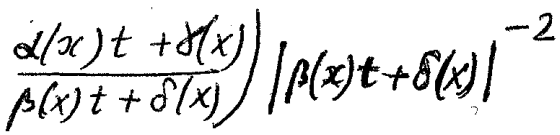
Representation:

$$U(g(x))f(x,t) = \lambda(g,f) \circ f\left(x, \frac{\alpha(x)t+\gamma(x)}{\beta(x)t+\delta(x)}\right)|\beta(x)t+\delta(x)|^{-2}$$



An open circle $\circ$ appears between $\lambda(g,f)$ and f.

Original source details: GELFAND-P000583



The numerator has $\gamma(x)$, and the denominator has $\delta(x)$.

Original source details: GELFAND-P000583

GELFAND-P000584

17/II-75 Gelfand Sem.
Tulovskii FIO — I and II.

$$\Omega \subset \mathbb{R}^n, \qquad u \in \mathcal{D}'(\Omega), \qquad \mathrm{WF}(u) \subseteq \Omega \times (\mathbb{R}^n \setminus 0).$$

~~$u \in \mathcal{D}'(\Omega)$~~

Uncertain reading. The cancelled formula resembles $u \in \mathcal{D}'(\Omega)$.

Original source details: GELFAND-P000584

Definition. $(x_0, \xi_0) \notin \mathrm{WF}(u)$, if $\exists\, \varphi(x) \in C_0^\infty(\Omega)$, $\varphi(x_0) \neq 0$

$$|\widetilde{u\varphi}(\xi)| \leq C_N(1+|\xi|)^{-N} \quad \text{when} \quad |\dot{\xi} - \dot{\xi}_0| < \varepsilon \qquad \left(\dot{\xi} = \frac{\xi}{|\xi|}\right)$$

G.

Original source details: GELFAND-P000584

Diagram and typed description.

Crossed-out diagram. The cancellation partly covers the upper arcs.

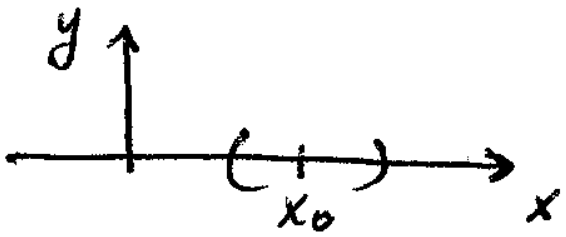
Smoothness of a function of one var., obtained as an $\int$ over parallel hyperplanes — $\sim$ membership of the covector in the wavefront.

Original source details: GELFAND-P000584

Uncertain reading. The initial letter of “фронту” (“front”) is heavily overwritten. The earlier layer is unresolved.

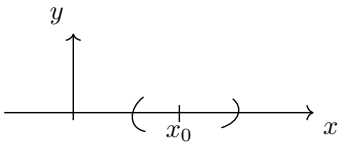
— · —

1 variable.



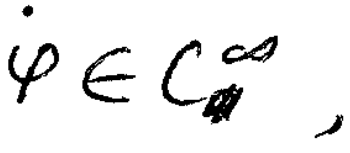
Original source details: GELFAND-P000584

Diagram and typed description.



$$(x_0, \xi_0) \notin \mathrm{WF}(u), \quad (\xi_0 > 0), \quad \text{if} \quad u = \varphi(x) + u_1 \quad \text{(locally)}$$

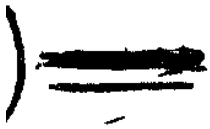
$\varphi \in C_0^\infty$, u_1 analytically continues into the upper half-plane.



Original source details: GELFAND-P000584

Uncertain reading. The overwritten subscript in C_0^∞ resembles 0. A separate dot is visible above φ.

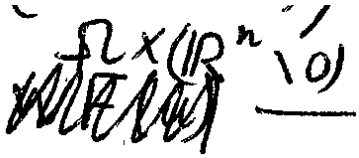
$$\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp} u. \qquad (\pi : T^*\Omega \to \Omega).$$



Uncertain reading. The lines read as $=$ overlap a heavier, arrow-like stroke.

Original source details: GELFAND-P000584

$$\pi' : \overset{\Omega\times(\mathbb{R}^n\setminus 0)}{\cancel{\mathrm{WF}(u)}} \longrightarrow \mathbb{R}^n \setminus 0.$$



Uncertain reading. The domain $\mathrm{WF}(u)$ is crossed out. $\Omega \times (\mathbb{R}^n \setminus 0)$ is inserted above it.

Original source details: GELFAND-P000584

$\pi'\mathrm{WF}(u)$ — the smallest closed cone outside which the Fourier transform decays rapidly.

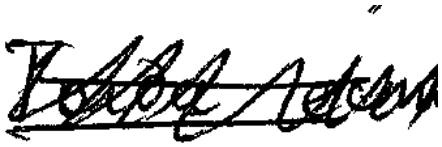
Uncertain reading. The initial letter of “Фурье” (“Fourier”) is retraced, with short strokes above and below.

Original source details: GELFAND-P000584

Let $\Gamma \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$ — an open conic set,

$$\begin{aligned} &\Gamma_1, \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0), \\ &\Gamma_1 + \Gamma_2 = \{(x; \xi + \eta) : (x, \xi) \in \Gamma_1,\ (x, \eta) \in \Gamma_2\}, \\ &\mathcal{D}'_\Gamma(\Omega) = \{u \in \mathcal{D}'(\Omega) : \mathrm{WF}(u) \subseteq \Gamma\}. \end{aligned}$$

~~Then ...~~ let $\Gamma_1 + \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$.



Undeciphered passage. Before “let”, “Then” and a further word are crossed out. The second word remains partly unread.

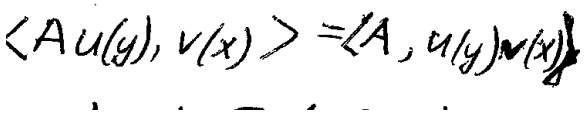
Original source details: GELFAND-P000584

Then $\exists$ a bilin. map:

$$\mathcal{D}'_{\Gamma_1}(\Omega) \times \mathcal{D}'_{\Gamma_2}(\Omega) \longrightarrow \mathcal{D}'_{\Gamma_1+\Gamma_2\cup\Gamma_1\cup\Gamma_2}(\Omega).$$

—— · ——

$$\begin{aligned} &\Omega_x \subseteq \mathbb{R}^{n_x}, \\ &\Omega_y \subseteq \mathbb{R}^{n_y}, \qquad A \in \mathcal{D}'(\Omega_x \times \Omega_y). \\ A : C_0^\infty(\Omega_y) \to \mathcal{D}'(\Omega_x), \qquad &\langle Au(y), v(x)\rangle = \langle A, u(y)v(x)\rangle \end{aligned}$$



Uncertain reading. The strokes near v and the closing sign on the right are heavily retraced.

Original source details: GELFAND-P000584

(x, ξ, y, η) — $\mathrm{WF}(\text{of the operator}) \underset{\text{def}}{=} \mathrm{WF}(\text{of the kernel})$

Thm. Suppose for simplicity that in $\mathrm{WF}(A)$ there are no points where $\xi = 0$ or $\eta = 0$
Then $A : C_0^\infty(\Omega_y) \to C^\infty(\Omega_x)$ and extends to a map $A : \mathcal{E}'(\Omega_y) \to \mathcal{D}'(\Omega_x)$.
Moreover: if $u \in \mathcal{E}'(\Omega_y)$, then $\mathrm{WF}(Au) \subseteq \mathrm{WF}'(A) \circ \mathrm{WF}(u)$,

GELFAND-P000585

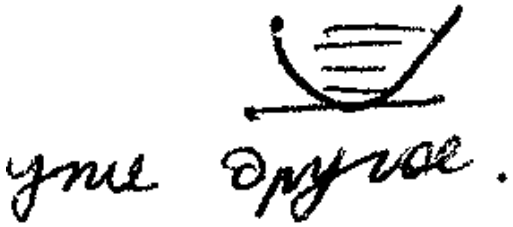
where

$$\mathrm{WF}'(A) = \{(x, \xi, y, -\eta), \text{ where } (x, \xi, y, \eta) \in \mathrm{WF}(A)\}.$$

$$(x, \xi) \in \mathrm{WF}'(A) \circ \mathrm{WF}(u) \iff \exists\, (x, \xi, y, \eta) \in \mathrm{WF}'(A), \quad (y, \eta) \in \mathrm{WF}(u).$$

— · —

G.'s idea — integrate both lin. functions and polynomials.



Original source details: GELFAND-P000585

Diagram and typed description.

already different.

Original source details: GELFAND-P000585

Uncertain reading. The Russian caption reads "уже другое" and ends with a full stop.

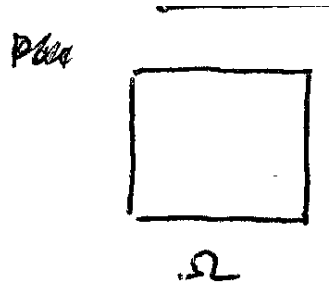
— · —

$$\frac{A : \mathcal{E}'(\Omega_y) \to \mathcal{D}'(\Omega_x) \qquad B : \mathcal{E}'(\Omega_z) \to \mathcal{E}'(\Omega_y)}{\mathrm{WF}'(A \circ B) \subseteq \mathrm{WF}'(A) \circ \mathrm{WF}'(B)}$$

Original source details: GELFAND-P000585

Uncertain reading. Both occurrences of WF on the right of the inclusion carry a prime.

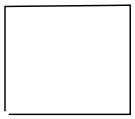
Sinai. Stat. physics (from the end)
Gibbs random field.



Original source details: GELFAND-P000585

Diagram and typed description.

Fig.?



Original source details: GELFAND-P000585

Uncertain reading. Crossed-out vertical strokes lie to the left of the outline. “II” and “III” are possible.

Original source details: GELFAND-P000585

Uncertain reading. The upper-left label resembles Russian “Рис” (“Fig.”). $\mathbb{R}^n$ is also possible.

G. Consider a measure μ on countable subsets of $\mathbb{R}^n$ without accumulation points (configurations denoted by X).

Measurable sets — the smallest σ-algebra generated by cylinder sets:

$$\{K : |X \cap K| = n\} = \text{Ц}_{K,n} \qquad (K \text{ — compact})$$

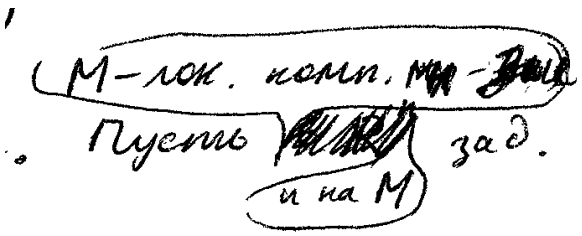
The measure is countably additive.

Main example: Poisson distribution.

(M — loc. comp. ...)
Let ... be given

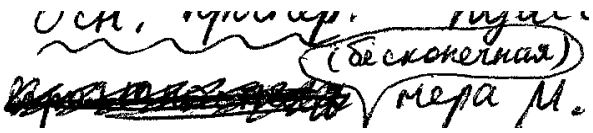
and on M

(infinite)
... measure μ. Then



Undeciphered passage. Strokes overlap at the end of the upper insertion and in the notation after “Let”. The lower insertion reads “and on M”.

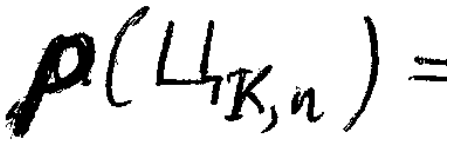
Original source details: GELFAND-P000585



Original source details: GELFAND-P000585

Undeciphered passage. A phrase before “measure” is heavily crossed out. “Infinite” is inserted in an outline above it.

$$\rho(Ш_{K,n}) = \frac{m(K)^n}{n!} e^{-m(K)}$$



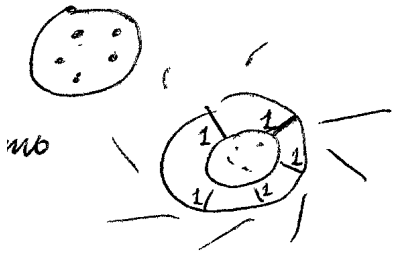
Original source details: GELFAND-P000585

Uncertain reading. The first symbol is heavily retraced and resembles ρ. The numerator and exponent use m.

— · —

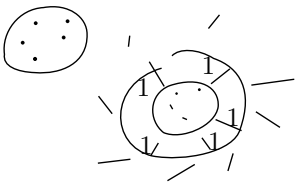
Gibbs distributions:

“Interaction is finite”: independence.



Original source details: GELFAND-P000585

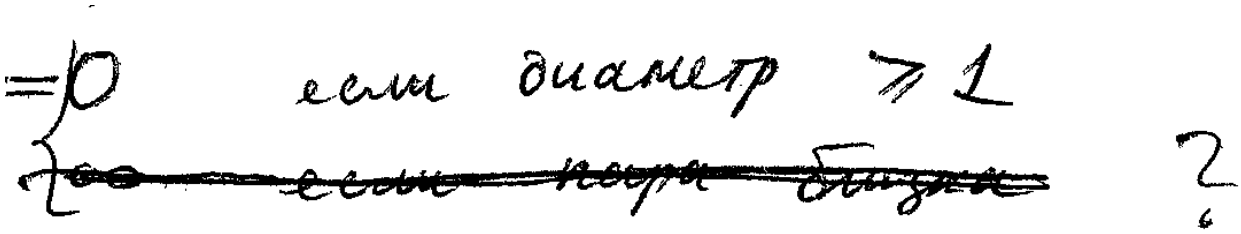
Diagram and typed description.



The upper contour encloses five dots. Some of the five segments in the lower drawing stop short of the inner contour.

Consider $U(x_1, x_2)$, $U(x_1, x_2, x_3)$, …

$$U(x_1, \dots, x_n) = \begin{cases} 0 & \text{if the diameter is } \geqslant 1 \\ \infty \text{ if} & \text{... close?} \quad ? \end{cases}$$



Original source details: GELFAND-P000585

Undeciphered passage. The second case is cancelled. The infinity sign and “if” are visible, followed by what appears to end in “close”. The intervening strokes are obscured.

The seq. breaks off, since it becomes $\equiv +\infty$.

Original source details: GELFAND-P000585

Uncertain reading. The word after “since” reads “становится” (“becomes”).

Consider countable configurations.

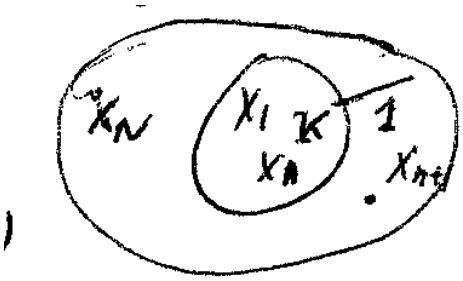
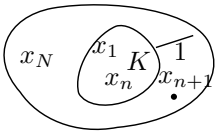


Diagram and typed description.

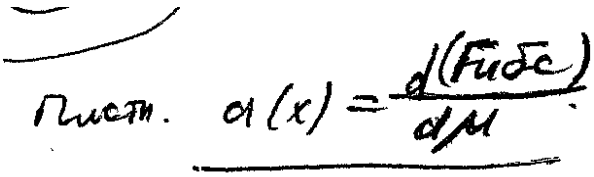


The right subscript tentatively reads $n+1$. The final strokes meet the outline.

Original source details: GELFAND-P000585

$$d(X) = e^{-\sum U_k(x_1,\ldots,x_k)}$$

Density $d(x) = \dfrac{d(\text{Gibbs})}{d\mu}$.



Uncertain reading. The abbreviation to the left of the fraction reads “Плотн.” (“density”).

Original source details: GELFAND-P000585

GELFAND-P000586

S. Consider $\underbrace{K \times \ldots \times K}_{n \text{ times}}$ with measure $m \times \ldots \times m$.

$$\frac{dG(X)}{dm} = d(X)$$

- -

GELFAND-P000587

24?/II–75. Gelfand’s sem.

Uncertain reading. The second digit of the day is overwritten. The reading 24 is tentative.

Original source details: GELFAND-P000587

Ya. Sinai. Let $X = \{x\}$ be an infinite discrete set in $\mathbb{R}^2$,

Undeciphered passage. A short mark after the comma is crossed out.

Original source details: GELFAND-P000587

m — a measure in $\mathbb{R}^2$. $\mathcal{M}$ — all such sets.

Poisson measure on $\mathcal{M}$: let O be a region.

$$\mu_0(|X \cap O| = k) = e^{-m(O)}[m(O)]^k/k! \qquad (1)$$

$m(O)$ — expected number of particles found in O.

More generally: (M, m) — a space with a measure [inserted below: "and a metric"]
$m(M) < \infty$ — uninteresting case

Interesting case: $m(M) = \infty$ and certain homogeneity properties.

Original source details: GELFAND-P000587

Undeciphered passage. A short mark is crossed out above "certain".

Let 2 functions be given: $U_1(x)$ [below: "bounded"] and

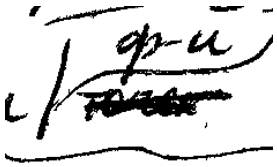
$$U_2(x,y) = \begin{cases} \infty & \text{if } \operatorname{dist}(x,y) < \varepsilon, \\ 0 & \text{if } \operatorname{dist}(x,y) \geq r_0. \end{cases}$$

Original source details: GELFAND-P000587

Uncertain reading. The overwritten sign in the second row reads as 0.

($U_2(x,y)$ is symmetric — a function on a pair of points.)

A Gibbs measure is constructed from a pair of [inserted: "functions"] U_1, U_2.



Original source details: GELFAND-P000587

Undeciphered passage. The word beneath the insertion "functions" is crossed out.

Def. by means of conditional probabilities.

Let there be a measure μ on the space of configurations $\mathcal{M}$

$$X = O \cap \ldots, \quad \underline{Y} = O' \cap \ldots$$

Original source details: GELFAND-P000587

Uncertain reading. The overwritten sign before $\cap$ reads as O'.

x — inside, y — outside (in the given configuration)

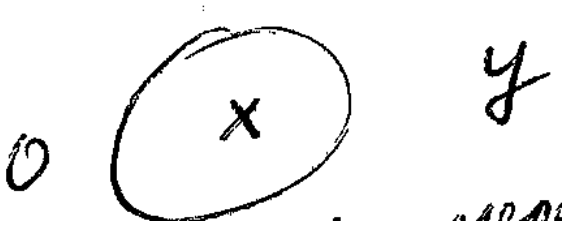
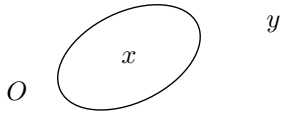


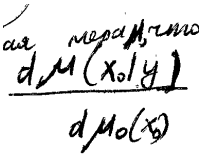
Diagram and typed description.



Original source details: GELFAND-P000587

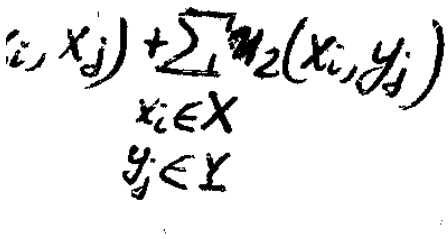
A Gibbs measure is a measure such that:

$$\frac{d\mu(x_0 \mid y)}{d\mu_0(x_0)} = \frac{1}{\Xi(y;O)} e^{-\left[\sum_{x_i \in X} U_1(x_i) + \sum_{(x_i,x_j)\in X} U_2(x_i,x_j) + \sum_{\substack{x_i \in X \\ y_j \in Y}} U_2(x_i,y_j)\right.}$$



Uncertain reading. A subscript 0 is visible after x in the denominator.

Original source details: GELFAND-P000587



Uncertain reading. The first symbol in the final summand is overwritten and tentatively reads U_2. The closing square bracket is not visible.

Original source details: GELFAND-P000587

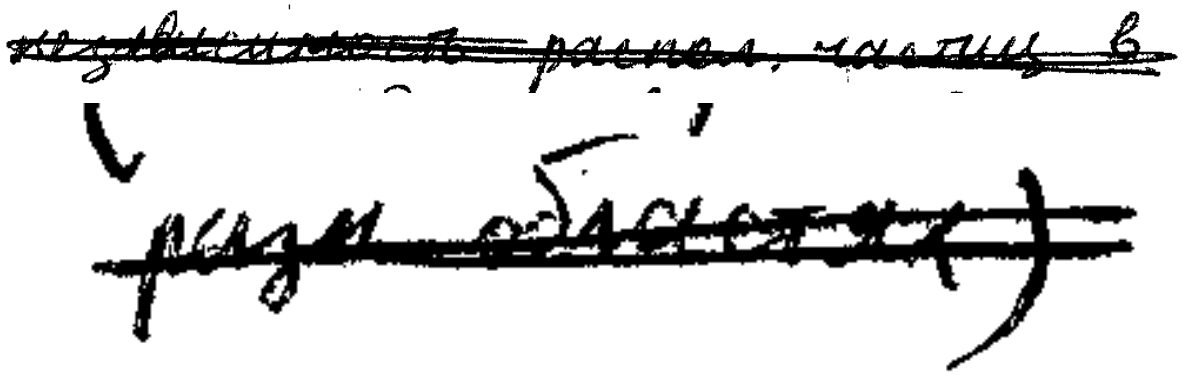
$$x = (x_1, \ldots, x_k) \qquad X = \{X_0 = X \cap O,\ Y = X \setminus X_0\}$$

All this depends only on the r_0-neighbourhood of O (on those points y, that fell within it).

$\mu(x \mid y)$ — a measure on configurations in O.

$\mu_0(x)$ — a measure —"— O.

(In def. (1) it is understood, ~~the independence of different particles in different regions)~~ that we take the σ-algebra generated by the corresponding cylinder sets.)



Original source details: GELFAND-P000587

Uncertain reading. The first word in the crossed-out clause tentatively reads “independence”.

[G. $\mu(x \mid y)$ is defined because there is a measurable partition (Fubini's theorem — general) The measure on Y is induced by the measure on the whole space, by means of preimages.] Let there be 2 regions.

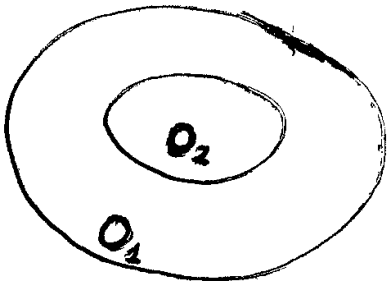
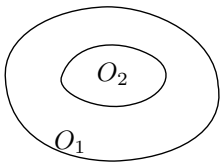


Diagram and typed description.

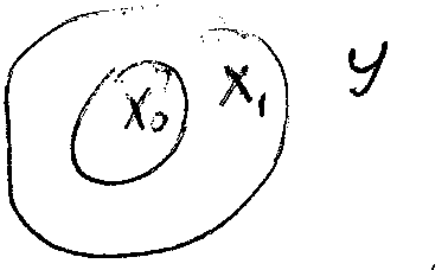


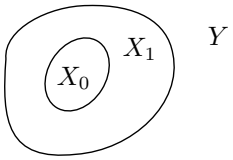
Original source details: GELFAND-P000587

From the measure μ on X a measure is constructed on Y (complete in all possible ways with the set X_0).

GELFAND-P000588

Diagram and typed description.





Original source details: GELFAND-P000588

Fix X_0, X_1 and Y. Then the (cond.) measure is calculated in two ways:

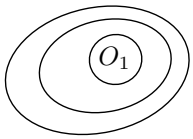
Theorem. ∃ at least one Gibbs distribution.

Undeciphered passage. A short mark before ∃ is crossed out.

Original source details: GELFAND-P000588

Proof.

Diagram and typed description.



Original source details: GELFAND-P000588

$O_1 \subset O_2 \subset O_3 \subset \ldots$ and the boundary conditions are fixed.

~~Let~~ The space of config. in O_1 is compact. $\exists$ a subseq. having a limit

and so on ...

Ex. 1 M — a line, m — Lebesgue measure $\Rightarrow$ there is only one Gibbs distribution.

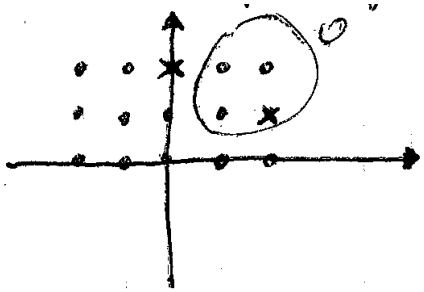
Ex. 2 $M = \mathbb{R}^k$, $U_1 \equiv 0$, $m(O) \leq c\ell(O)$ (ℓ — Lebesgue measure)

For small c the Gibbs distrib. is unique.

— · —

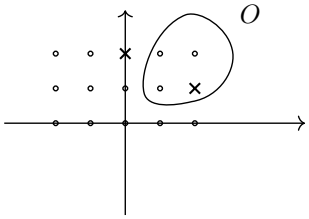
Lattice Gibbs distrib.

Consider the lattice in $M = \mathbb{Z}^d$, $d \geq 2$.



Original source details: GELFAND-P000588

Diagram and typed description.



The lattice has two crosses and round marks. Region O encloses four sites. The drawing leaves the meaning of the marks unlabelled.

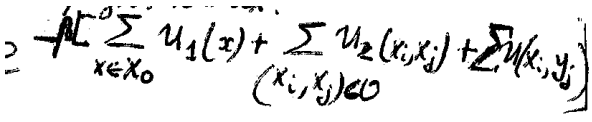
X — infinite subsets of lattice points

$\mathcal{M}$

μ_0 prob. that a particle is present $\frac{1}{2}$, a particle is absent $\frac{1}{2}$, and independence at different sites.

O — a finite region.

$$\mu(X_0 \mid Y) = \frac{1}{\Xi(Y,O)} e^{-\beta\left[\sum_{x\in X_0} U_1(x) + \sum_{(x_i,x_j)\in O} U_2(x_i,x_j) + \sum U(x_i,y_j)\right]}$$



Original source details: GELFAND-P000588

Uncertain reading. No subscript is visible on U in the final sum. The symbol before the bracket reads β.

U_2 also vanishes for large $|x-y|$.

$U_1 \equiv 0$, U_2 — small $\Rightarrow$ one Gibbs distrib.

(perturbation, prob. joint distrib. of independent random variables).

$$U_1(x) = \alpha \qquad U_2(x_i, x_j) = \cancel{U_2(|x_i - x_j|)}\, U_2(x_i - x_j)$$

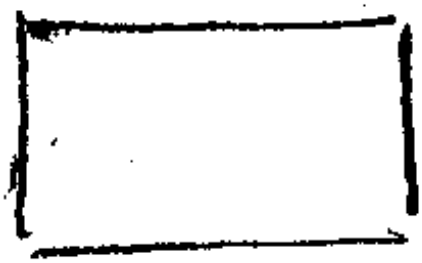
$$U_2(-x) = U_2(x).$$

Original source details: GELFAND-P000588

Uncertain reading. A horizontal stroke between x_i and x_j is crossed by a diagonal cancellation stroke. Both $+$ and $-$ are possible. The vertical strokes and outer parentheses are tentative.

1. Periodic ground state.

X — a periodic configuration (repetition of a rectangle)

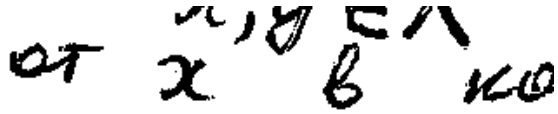


Original source details: GELFAND-P000588

Diagram and typed description.

$$H(X) = \sum_{x \in X} U_1(x) + \sum_{x,y \in X} U_2(x,y) \qquad \text{(formal expression)}$$

Let X' be nonperiodic and differ from x in finitely many places. Write $H(X')$ (also formally).

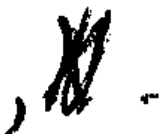


Original source details: GELFAND-P000588

Uncertain reading. The symbol after "from" is a lowercase x.

If $\underbrace{H(X') - H(X) \geq 0}_{\text{already def.}}$ $\forall\, X'$, then X is called a ground state.

1) Let $(X_1, X_2, \dots, X_k) = G(U)$ — the set of periodic ground states corresponding to U.



Original source details: GELFAND-P000588

Undeciphered passage. One mark before the ellipsis is crossed out.

GELFAND-P000589

2) Peierls condition (nondegeneracy of ground states):

(so that ground states — the main contribution to $e^{-\beta H}$).

Undeciphered passage. One mark between the dash and “main” is crossed out.

Original source details: GELFAND-P000589

$$H(X') - H(X) > \dots$$

Undeciphered passage. A short entry before “Let” is heavily crossed out.

Original source details: GELFAND-P000589

Let $\bar{X}$ — an arbitrary configuration.

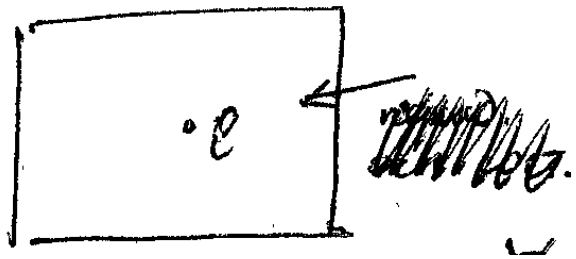
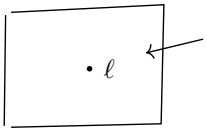


Diagram and typed description.



Original source details: GELFAND-P000589

Undeciphered passage. The label by the arrow is crossed out.

Original source details: GELFAND-P000589

ℓ is called a boundary point of $\bar{X}$, if in a neighbourhood of ℓ $\bar{X}$ does not agree with any of the periodic ground states.

Suppose $\forall$ config. X', differing from X in finitely many points,

Peierls cond.:

$$H(X') - H(X) \geq g\,\Gamma(X') \qquad (g > 0).$$

$\Gamma(X') \longleftarrow$ number of boundary points in X'.

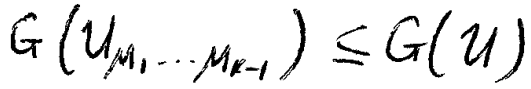
Undeciphered passage. A crossed-out entry stands between g and $\Gamma(X')$.

Original source details: GELFAND-P000589

— · —

Include $\underset{\text{(pair } U_1, U_2)}{U}$ in a $(k-1)$-parameter family $U_{\mu_1,\dots,\mu_{k-1}}$, $U_{0,\dots,0} = U$ (there are k ground states — here the same k).

$$G(U_{\mu_1,\dots,\mu_{k-1}}) \le G(U) \qquad \text{for small } \mu$$



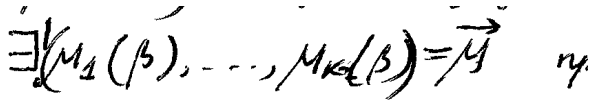
Uncertain reading. The sign between the two G expressions reads as $\le$.

Original source details: GELFAND-P000589

(ground states can only disappear). Let the perturbations be “lin. indep.”.

Then $\exists \beta_0, \forall \beta > \beta_0$

$$\exists!\big(\mu_1(\beta), \dots, \mu_{k-1}(\beta)\big) = \vec{\mu}$$

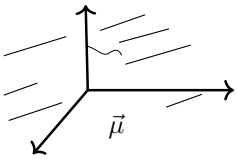


Uncertain reading. An exclamation mark is visible after $\exists$. The last subscript reads as $k-1$, with its end touching the parenthesis.

Original source details: GELFAND-P000589

for which there are k Gibbs distributions. Through it pass k curves, on which there are $(k-1)$ Gibbs distributions, C_k^2 surfaces, on which there are $(k-2)$ Gibbs distributions and so on.

Diagram and typed description.

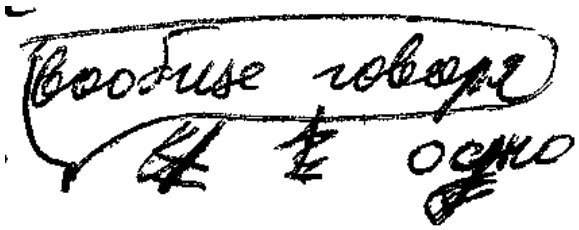


The grouping of the short strokes is approximate.

Original source details: GELFAND-P000589

With shifts one obtains, “as a rule”, one ground state.

Nondegeneracy: as $\mu_1, \dots, \mu_k$ vary there is [inserted: “generally speaking”] one ground state, on the curves $\le (k-1)$ and so on.



Uncertain reading. Two altered signs precede “one”.

Original source details: GELFAND-P000589

Then the same holds for Gibbs distributions.

Gelfand’s question: Gibbs distrib. in the p-adic case.

GELFAND-P000590

[The upper part of the frame contains no text.]

—— · ——

3/III–75 Gelfand's sem.

V. N. Tulovskii. WF. $\Omega \subset \mathbb{R}^n$, $u \in \mathcal{D}'(\Omega)$.

$(x_0, \xi_0) \notin \mathrm{WF}(u)$, if $(x_0, \xi_0) \in \Omega \times (\mathbb{R}^n \setminus 0)$ and from $\phi'(x_0) = \xi_0$ it follows that

$$\left|\langle u, \varphi e^{i\lambda\phi}\rangle\right| \le C_N \lambda^{-N}, \qquad \forall N$$

(the linearity of ϕ does not matter)

—— · ——

Let $\Omega \subset \mathbb{R}^n$, $\theta \in \mathbb{R}^N$, $a(x,\theta) \in S^m_{1,0}(\Omega \times \mathbb{R}^N)$.

Φ — phase function.

$$\langle A, u\rangle = \int a(x,\theta) e^{i\phi(x,\theta)} u(x)\, dx\, d\theta =$$

GELFAND-P000591

$$= \lim_{\varepsilon \to 0} \int a(x,\theta) e^{i\phi(x,\theta)} u(x) \chi(\varepsilon\theta)\, dx\, d\theta$$

Φ - фазовая ф-я $\quad \langle A, u\rangle = \int a(x,\theta)\, e^{i\,\phi(x,\theta)}\, u(x)\, dx\, d\theta =$

Original source details: GELFAND-P000590

Uncertain reading. An uppercase Φ precedes "phase", and a lowercase ϕ appears in the exponent.

Uncertain reading. The overwritten sign before (x) tentatively reads u.

Original source details: GELFAND-P000590

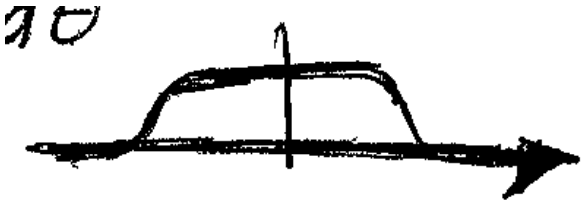
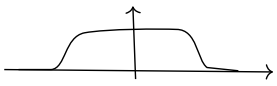


Diagram and typed description.



Original source details: GELFAND-P000591

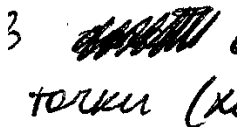
Let $\phi'_\theta(x_0, \theta_0) \neq 0$. Write the decomposition $a = a_1 + a_2$,

(example: $a = 1$, $\phi = \langle x, \theta\rangle$, $\langle A, u\rangle = u(0)$, i.e. $A(x) = \delta(x)$.).

Undeciphered passage. A short entry before "i.e." is crossed out.

Original source details: GELFAND-P000591

where a_2 is supported in a conic neighbourhood of (x_0, θ_0), and $a_1(x, \theta) = 0$ in a neighbourhood of the point (x_0, θ_0).



Original source details: GELFAND-P000591

Undeciphered passage. An entry after "in" is crossed out. "Point" is inserted below the line.

$$\langle A_2, u\rangle = \int a_2(x,\theta) e^{i\phi(x,\theta)} u(x)\, dx\, d\theta, \qquad \text{here } \phi'_\theta \neq 0 \Rightarrow A_2 \in C^\infty.$$

$$C_\phi = \{(x,\theta) : \phi'_\theta = 0\}$$

Theorem. ~~WF~~ Consider the map

$$\lambda_\phi : C_\phi \longrightarrow \Omega \times (\mathbb{R}^n \setminus 0),$$
$$(x,\theta) \longrightarrow (x, \phi'_x).$$

Denote $\lambda_\phi(C_\phi) = \Lambda_\Phi$.

Then $\mathrm{WF}(A) \subset \Lambda_\Phi$.

Original source details: GELFAND-P000591

Uncertain reading. The sign following WF(A) is retraced and reads as $\subseteq$.

—— · ——

Nondegenerate phase functions — if $\mathrm{rank}(\phi''_{\theta x}, \phi''_{\theta\theta}) = N$ on C_ϕ.

Then λ_ϕ — a local diffeom. and Λ_ϕ — a conic Lagrangian manifold, i.e. $\langle \xi, dx\rangle|_{\Lambda_\phi} = 0$.

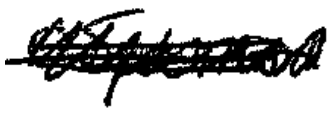
Original source details: GELFAND-P000591

Undeciphered passage. A short mark between "and" and Λ_ϕ is crossed out.

Conversely: $\forall$ conic Lagr. manifold can locally be obtained from a nondegenerate phase function (and one can take $N = n$).

—— · ——

~~Let there be 2 nondegenerate phase functions.~~



Original source details: GELFAND-P000591

Undeciphered passage. The crossed-out ending tentatively begins "Обра..." ("Converse...").

Check of the Lagrangian property:

$$\phi'_\theta(x,\theta) = 0 \;\Rightarrow\; \phi(x,\theta) = 0$$

$$\phi'_\theta(x+dx,\theta+d\theta) = 0, \qquad \phi'_x\,dx + \underbrace{\phi'_\theta\,d\theta}_{\text{“0”}} = 0.$$

hence $\phi'_x\,dx = 0$ on Λ_ϕ, or $\xi\,dx = 0$ on Λ_ϕ. ▷.

—— · ——

Let there be 2 phase functions ϕ_1 and ϕ_2,

$$\lambda_{\phi_1} : C_{\phi_1} \to \Lambda_\phi, \qquad \lambda_{\phi_2} : C_{\phi_2} \to \Lambda$$

Uncertain reading. A subscript is visible on the first target Λ and tentatively reads ϕ.

Original source details: GELFAND-P000591

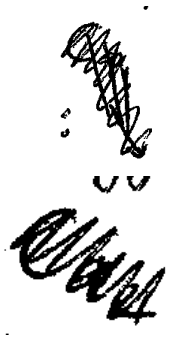
(Λ is the same).

$$(x_0,\theta_1^0) \longrightarrow (x_0,\xi_0),$$
$$(x_0,\theta_2^0) \longrightarrow (x_0,\xi_0).$$

The class of functions A will be the same for ϕ_1 and for ϕ_2.

—— · ——

FIO:



Undeciphered passage. Two entries by X, Y are heavily crossed out.

Original source details: GELFAND-P000591

X, Y — 2 manifolds, Λ — a closed conic Lagr. manifold without singularities in $T^*(X \times Y) \setminus 0$.

Then locally Λ is given by means of a phase function $\phi(x,y,\theta)$. (In a neighbourhood of the point (x_0,ξ_0,y_0,η_0))

Undeciphered passage. A mark before “point” is heavily crossed out.

Original source details: GELFAND-P000591

and choose an amplitude a and define the operator by means of the kernel

$$\int a(x,y,\theta)e^{i\phi(x,y,\theta)}u(x,y)\,dx\,dy\,d\theta$$

Undeciphered passage. A short entry to the right of the integral is crossed out.

Original source details: GELFAND-P000591

GELFAND-P000592

[source number: 4]

Kirillov. Fact: M — a sympl. manifold, ω — the corresponding form.

Undeciphered passage. A short entry after "form" is crossed out.

Original source details: GELFAND-P000592

on M there is a Lagrangian foliation.

Diagram and typed description.

Original source details: GELFAND-P000592

Then there is an invariantly defined affine structure on its leaves.

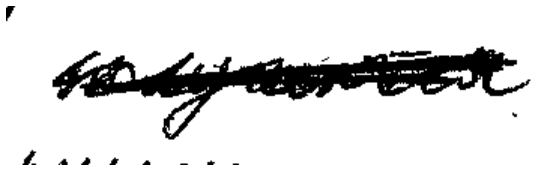
Uncertain reading. A mark before "If" is crossed out.

Original source details: GELFAND-P000592

If the functions on M are constant on the leaves of the foliation, then their gradients (vectors corresponding to the covectors $d\varphi$) define parallel transport there. The corresponding vector fields commute.

$$\varphi \longrightarrow d\varphi \longrightarrow \xi_\varphi.$$

— these vector fields ~~commute~~ are called parallel.



Uncertain reading. The crossed-out word tentatively repeats “commute”.

Original source details: GELFAND-P000592

Corollary: the leaves — $\mathbb{R}^n$ or tori.

“Kokas.” construction — an example of a Lagr. foliation.

Uncertain reading. The visible layer of the overwritten abbreviation reads “Кокас.” (“Kokas.”).

Original source details: GELFAND-P000592

— · —

Tulovskii.

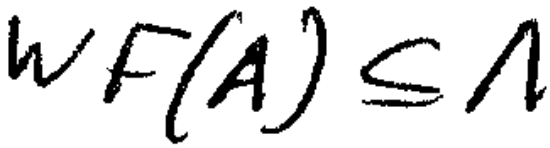
In the general case:

$$\sum_j \int a_j(x,y,\theta_j) e^{i\phi_j(x,y,\theta_j)} u(x,y)\, dx\, dy\, d\theta_j$$

— this is an FIO.

~~Theorem.~~

This is a class of operators for which $\mathrm{WF}(A) \subseteq \Lambda$ and which act well on rapidly oscillating exponentials.



Uncertain reading. The sign between $\mathrm{WF}(A)$ and Λ reads as $\subseteq$.

Original source details: GELFAND-P000592

Uncertain reading. The overwritten word reads as “which”.

Original source details: GELFAND-P000592

Let (x, Ψ'_x) be transverse to Λ.

$$A(e^{it\Psi}\varphi) \sim e^{it\phi} \sum_{j=0}^{\infty} b_j(x) t^{-\lambda_j}$$

Then this is an FIO.

— · —

GELFAND-P000593

[The upper part of the frame contains no text.]

10/III–75. Gelfand's sem.

Tulovskii. (continued) $\Omega_x \subseteq \mathbb{R}^{n_x}$, $\Omega_y \subseteq \mathbb{R}^{n_y}$.

$$Au(x) = \int a(x,y,\theta) e^{i\phi(x,y,\theta)} u(y)\, dy\, d\theta$$

$$A : \mathcal{E}'(\Omega_y) \longrightarrow \mathcal{D}'(\Omega_x)$$

With Φ is associated a Lagr. manifold

$$\Lambda \subset (\Omega_x \times \Omega_y) \times (\mathbb{R}^{n_x+n_y} \setminus 0)$$

Original source details: GELFAND-P000593

Uncertain reading. The overwritten sign after Λ reads as $\subset$.

PDO case: $\Lambda = (x, \xi, x, -\xi)$.

Original source details: GELFAND-P000593

Undeciphered passage. A short entry to the right of the formula is crossed out.

Question from G.: it would be good to construct this in the p-adic case.

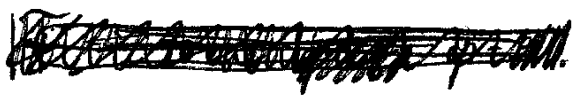
When is the kernel $A(x,y)$ — the kernel of an FIO?

Tulovskii's answer:

1) $\mathrm{WF}(A) \subset \Lambda$ (Λ — some Lagr. manifold)

2) $\forall \Psi : \{(x, \Psi'_x, y, \Psi'_y)\}$ transverse to $\Lambda \Rightarrow$

$$\langle A, e^{it\Psi}\varphi(x,y)\rangle \sim \sum e^{it\tilde{\Psi}} a_j(t,\ldots)$$



Original source details: GELFAND-P000593

Undeciphered passage. A line beneath the asymptotic expression is heavily crossed out.

and $\widetilde{\Psi}$ is related to Ψ through Λ.

Thm. 1. $P(x, D_x)$ — a first-order PDO with real principal part $p(x, \xi)$, $p(x_0, \xi_0) = 0$,

$$p'_{x,\xi}(x_0, \xi_0) \neq 0 \quad \Rightarrow \quad \exists\, 2 \text{ FIOs}$$

$$A : \mathcal{E}'(\Omega_x) \longrightarrow \mathcal{E}'(\mathbb{R}^n),$$
$$B : \mathcal{E}'(\mathbb{R}^n) \longrightarrow \mathcal{E}'(\Omega_x)$$

$$\mathrm{WF}\left[P - B\frac{\partial}{\partial X_n}A\right] \not\ni (x_0, \xi_0, x_0, -\xi_0)$$

Uncertain reading. There are crossed-out strokes immediately after the first ξ_0.

Original source details: GELFAND-P000593

Moreover

$$\mathrm{WF}(B \circ A - I) \not\ni (x_0, \xi_0; x_0, -\xi_0)$$

Uncertain reading. The overwritten letters read as B and A. A small circle is visible between them.

Original source details: GELFAND-P000593

~~Theorem~~

Thm. 2 $P(x, D_x)u = f$,

$$H_p = \sum \left(\frac{\partial p}{\partial x_j}\frac{\partial}{\partial \xi_j} - \frac{\partial p}{\partial \xi_j}\frac{\partial}{\partial x_j} \right)$$

Undeciphered passage. A short mark between f and H_p is crossed out.

Original source details: GELFAND-P000593

$$\mathrm{WF}(u) \setminus \mathrm{WF}(f) \subset \{p(x, \xi) = 0\}$$

and invar. under the Ham. field H_p.

next time R. L. Dobrushin.

GELFAND-P000594

I. Bernstein. WF of a left ideal of diff. operators with polynomial coeffs.

Uncertain reading. The crossed-out abbreviation after “ideal” tentatively reads “oper.”.

Original source details: GELFAND-P000594

$$= \{(x_0, \xi_0) : \dddot{-} p(x_0, \xi_0) = 0,\ \forall p\}$$

Uncertain reading. The initial sign before p is heavily overwritten. The ending $\forall p$ is clear.

Original source details: GELFAND-P000594

Thm. $\dim \mathrm{WF} \geq \dim X$.

GELFAND-P000595

[The upper part of the frame contains no text.]

17/III–75. Gelfand's sem.

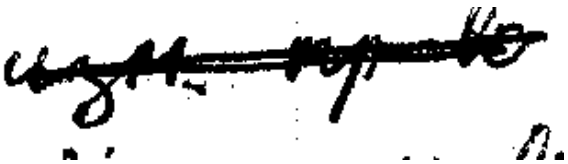
Seminar programme:

Dobrushin R. L.

Objects

1) random fields with a discrete argument

X — ~~measurable space~~ a finite set



Uncertain reading. The crossed-out expression reads "measurable space".

Original source details: GELFAND-P000595

T — a countable set

$$X^T = (x_t,\ t \in T)$$

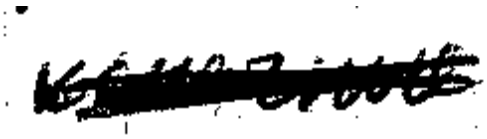
A random field — a prob. measure on X^T.

Physical interp. — spin systems

2) point random fields

T — a continuous space $\mathbb{R}^\nu$

Realizations — ~~finite~~ subsets of $\mathbb{R}^\nu$, $\cap$ of which with $\forall$ finite volume is finite.



Uncertain reading. The crossed-out word tentatively reads "finite".

Original source details: GELFAND-P000595

A measure on realizations — a point random field.

$$3) \qquad \underset{\substack{\downarrow \\ \text{functions (real)}}}{X(t)} \quad , \quad t \in \mathbb{R}^\nu \qquad S^*(\mathbb{R}^\nu)$$

a random field — a measure on S^*

$\Rightarrow$ quantum field theory.

GELFAND-P000596

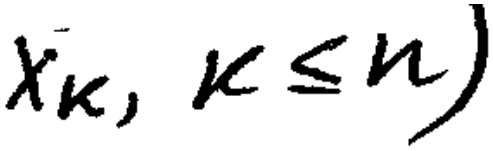
Problem: description of random fields

$$T = \mathbb{Z}, \qquad X^{\mathbb{Z}}\ (x_t,\ t \in \mathbb{Z})$$

Def

$$\Pr(x_n \mid x_k,\ k \le n) \underset{\text{Markov}}{=} \Pr(x_n \mid x_{n-1}) \qquad \text{(Markov chain)}$$

Cond. prob. is known.



Uncertain reading. The condition on the left reads $k \le n$.

Original source details: GELFAND-P000596

Task: first specify the conditional probability, then see which measures have these conditional probabilities.

—— · ——

$$V \subset \mathbb{Z}^\nu, \quad \text{specify} \quad \Pr\{\xi_n,\ n \in V \mid \xi_k,\ k \in \mathbb{Z}^\nu \setminus V\}.$$

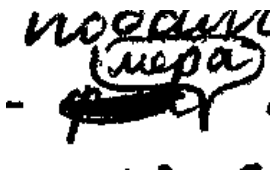
—— · ——

Conditional probability: $(X, \mathcal{B}, \mu)$ — a prob. space.

$$\mu(A \mid B) = \frac{\mu(A \cap B)}{\mu(B)} \qquad \text{— insufficient}$$

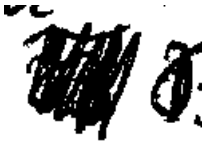
Let $\widetilde{\mathcal{B}}$ — be a subalgebra in $\mathcal{B}$

Conditional probability — measure on $\mathcal{B}$, depending on $x \in X$



Uncertain reading. The crossed-out word under the insertion “measure” tentatively reads “function”.

Original source details: GELFAND-P000596



Uncertain reading. An entry before $\mathcal{B}$ is crossed out.

Original source details: GELFAND-P000596

$$\mu(\cdot \mid x), \quad x \in X. \qquad\qquad \mu(B \mid x), \quad B \in \mathcal{B},\ x \in X.$$

1) $\forall B:\ \mu(B \mid x)$ is measurable on $\widetilde{\mathcal{B}}$

2)

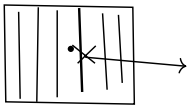
$$\int \psi(x)\varphi(x)\mu(dx) = \int \psi(x)\mu(dx \mid y)\varphi(y)\mu(dy)$$

(φ is measurable with respect to $\widetilde{\mathcal{B}}$)

Then the corresponding μ is called the conditional probability distribution.

In a broad class of cases $\exists$ and uniqueness..

Diagram and typed description.



The exact shape of the crossing marks on the middle segment is unclear.

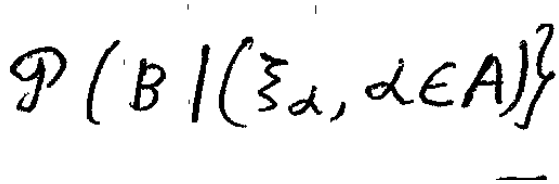
Original source details: GELFAND-P000596

here the measure $\overset{\mu(B|x)}{\uparrow}$ sits on the segment. One must take Borel sets (rather than Lebesgue-measurable ones).

In this example:

$$\int \psi(x_1, x_2)\varphi(x_2)\,dx_1\,dx_2 = \int \left(\int \psi(x_1, x_2)\,dx_1 \right) \varphi(x_2)\,dx_2$$

$$\mathcal{P}\big(B \mid (\xi_\alpha, \alpha \in A)\big\}. \qquad \mathcal{P}_{\mathcal{B}(\xi_\alpha, \alpha \in A)}(B \mid X)$$



Uncertain reading. The first expression uses round parentheses and ends with a closing brace.

Original source details: GELFAND-P000596

$$\xi_\alpha : X \underset{\text{functions}}{\longrightarrow} F_\alpha \qquad \text{or one} \quad \xi : X \to F \quad \text{(into the direct product)}$$

points of F — layers

preimages of sets in F — $\widetilde{\mathcal{B}}$

↓ there the σ-algebra $\mathcal{B}_F$.

— · —

GELFAND-P000597

Gelfand Sem. 31/III-75

Polyakov (Sasha) On the idea of [inserted above the line and outlined: “spontaneously”] broken symmetries

“Bubbles in the ether of quant. field theory are particles”

Skyrme; Nielsen, Olsen, Faddeev, Polyakov, ’t Hooft.

Next time — Nevanlinna theory (following Griffiths)

GELFAND-P000598

7/IV-75. Gelfand Sem.

King (?) Nevanlinna theory following Griffiths.

Uncertain reading. The manuscript places a question mark in parentheses after “King”.

Original source details: GELFAND-P000598

2 ideas. 1) hyperbolic analysis

$$\underset{\text{holomorphic}}{f} : X \to V \qquad \text{negatively curved}$$

$\Rightarrow$ there are some restr. on behaviour of f.

2) “counting” of volume of divisers of analytic sets by residue calculus.

First 2)

g — holom. function, $D = \{g^{-1}(0)\}$

Diagram and typed description.

In the left margin two curved strokes cross a nearly vertical line. Their direction and exact closure are unclear.

Original source details: GELFAND-P000598

integral over a neighbourhood of a form (of lower order).

Let η — any form of deg. r on $\mathbb{C}^n \setminus D$

[outlined above the line] $\leftarrow$ this too is a form

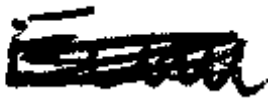
$$\eta = \frac{dg}{2\pi i g} \wedge \zeta + \xi, \qquad \zeta, \xi \text{ — forms, [crossed out] then}$$

Uncertain reading. An entry before “then” is crossed out.

Original source details: GELFAND-P000598

$$\lim_{\varepsilon\to 0}\int_{C(\varepsilon)}\eta=\zeta|_D$$

η,ζ — currents (current) (functionals on forms): ~~so~~ the easiest way to define them:



Original source details: GELFAND-P000598

Uncertain reading. A short entry before η,ζ is crossed out.

Current:

$$T: A_c^k(V)\to\mathbb{C}$$

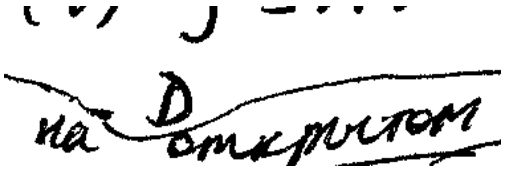
$$\lambda\in L^1_{\text{loc}}\ r\text{-form}\quad\Rightarrow\quad[\lambda](\varphi)=\int_X\lambda\wedge\varphi,$$

$$dT(\varphi)=\pm T(d\varphi).$$

The form $\zeta|_D$ defines a current:

$$[\zeta|_D](\varphi)=\int\zeta\wedge\varphi.$$

Def. Let η — L^1_{loc} form, $\in C^\infty$ on ^{D} an open dense set

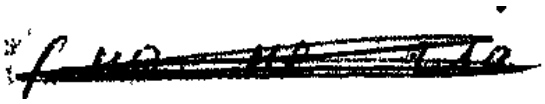


Original source details: GELFAND-P000598

Uncertain reading. D is written above the line. The oval below it is the first letter of the Russian word for “open”.

$$\operatorname{Res}\eta=[\zeta|_D]:=d[\eta]-[d\eta]$$

[at right below the outlined line] η — an ordinary form



Original source details: GELFAND-P000598

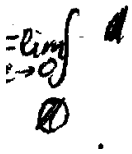
Uncertain reading. An entry below “ordinary form” is crossed out. It tentatively begins “(on”.

$$d[\eta](\varphi) \underset{\text{def}}{=} [\eta](d\varphi) = \int_V \eta \wedge d\varphi$$
$$= \lim_{\varepsilon \to 0} \int_{V - U_\varepsilon(D)} \eta \wedge d\varphi =$$
$$= \int_V d\eta \wedge \varphi - \lim_{\varepsilon \to 0} \int_{\partial U_\varepsilon(D)}$$
$$= \int_V d\eta \wedge \varphi - \int_D \dots$$

Example:

$$\eta = \frac{dz}{z}, \qquad [\eta](\varphi) = \int \frac{dz \wedge \varphi}{z}$$

$$d[\eta](\varphi) - [\eta](d\varphi) = \lim_{\varepsilon \to 0} \int_{\text{[overwritten]}} \varphi \, dz = \varphi(0).$$

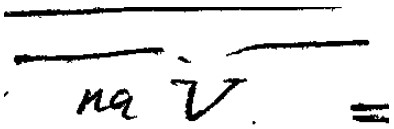


Uncertain reading. The round mark below the integral is overwritten and may be O. A short stroke above it is also crossed out.

Original source details: GELFAND-P000598

Res η — a current.!

g — holom. function on $\bar{V}$.

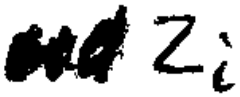


Uncertain reading. There are separate strokes above V, tentatively read as an overbar.

Original source details: GELFAND-P000598

$$\Rightarrow \quad \operatorname{Res}\left[\frac{dg}{2\pi i g}\right] = [Z],$$

$$Z = \{g = 0\}, \quad Z = \bigcup Z_i, \quad \operatorname{ord} g = m_i \text{ [overwritten] } Z_i$$



Uncertain reading. The word before Z_i is overwritten and may read “on”.

Original source details: GELFAND-P000598

$$Z = \sum m_i \int_{Z_i} \qquad \text{(as a current)}$$

GELFAND-P000599

Let $X \subset V$; Codim $X = 1$.

$$X \cap U_i = \{g_i = 0\} \qquad f_{ij} = \frac{g_i}{g_j} \text{ in } U_i \cap U_j$$

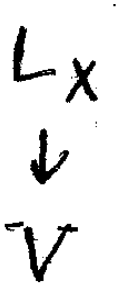


Diagram and typed description.

$$\begin{array}{c} L_X \\ \downarrow \\ V \end{array}$$

Original source details: GELFAND-P000599

$$\sigma = \{g_i\}, \quad \sigma \in \Gamma(L_X).$$

$\sigma = 0$ on X.

—— · ——

Let L have a Herm. metric (L — a 1-dim. bundle)
We can consider $|\sigma|^2 \in C^\infty$ on V and $= 0$ on X.

$$\eta = \frac{1}{2\pi i}\partial \log |\sigma|^2, \qquad \partial \log |z|^2 = \frac{dz}{z}.$$

Then Res $\eta = [X]$.

$$[d\eta] = \frac{\bar\partial\partial}{2\pi i} \log |\sigma|^2 = \frac{\bar\partial\partial}{2\pi i} \log h_i = c(L_D).$$

(Chern class)

where $|\sigma|^2 = h_i |g_i|^2$

$$\partial\bar\partial \log \bar g_i = 0$$

Uncertain reading. The separate line has a bar over the second ∂ and over g_i.

Original source details: GELFAND-P000599

De Rham's thm. holds! (in currents):

$$X = d[\eta] + [c(L_D)]$$

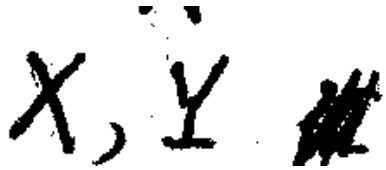
(the Chern class is cohomologous to the zero set of any section)!

—— · ——

Let ψ — a closed form. [inserted above the line and outlined: Then]

$$\int_X \psi = \int_V c(L_D) \wedge \psi.$$

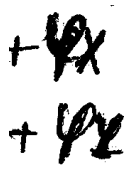
Bézout-type theorems. Let X, Y [crossed out] — two curves in $\mathbb{CP}^2$.



Original source details: GELFAND-P000599

Uncertain reading. A short mark after X, Y is crossed out.

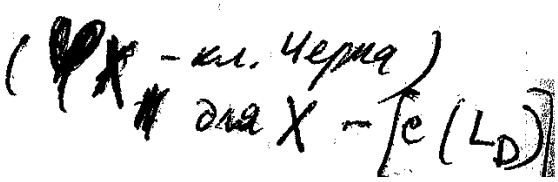
$$X = d[\eta_X] + \varphi_X$$
$$Y = d[\eta_Y] + \varphi_Y$$



Original source details: GELFAND-P000599

Uncertain reading. The last term of each equation is heavily inked and reads φ_X, φ_Y.

(φ_X — Chern cl.)
[crossed out] for X — $[c(L_D)]$



Original source details: GELFAND-P000599

Uncertain reading. A small mark before "for" is crossed out.

Consider

$$\int_V \varphi_X \wedge \varphi_Y = \int_X \varphi_Y = [X \wedge Y](1)$$

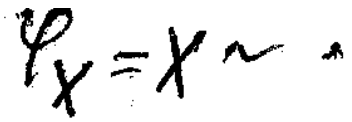
(number of intersection points =
= prod. of Chern classes)

Example: $V = \mathbb{P}^2, \quad X, Y = H$. (hyperpl.)

$$d_X = \deg X = \#(X \cap H) = \int_X \omega$$

(ω — std. form dual to the hyperpl. section)

$$\varphi_X = X \sim \text{[small mark]}$$



Uncertain reading. The short mark after $\sim$ resembles a small angle.

Original source details: GELFAND-P000599

$$\int_{\mathbb{P}^2} \varphi_X \wedge \varphi_Y = d_X d_Y \int_{\mathbb{P}^2} \omega^2 = d_X \cdot d_Y$$

— this is the proof of Bézout's thm. on the number of intersection points of curves.

GELFAND-P000600

X with singularities $\Rightarrow$ the thm. still holds (Lelong, 1953)

—— · ——

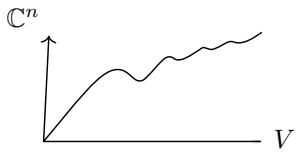
1) idea.

$$f : \mathbb{C}^n \xrightarrow[\text{holom.}]{} V$$

[below $\mathbb{C}^n$:] $\varphi \leftarrow$ Eucl. Kähler
[arrow to V:] compact. Kaehler ([inserted above the line and outlined: with] form ω).
Let $B(r)$ — be the ball of rad. r.

$$\text{vol}(f(B(r))) = \int_{B(r)} f^*(\omega^n)$$

Diagram and typed description.



The right end of the horizontal axis is thickened. An arrowhead cannot be established.

Original source details: GELFAND-P000600

[at right] Γ [overwritten subscript] [inserted above the line and outlined: part] of the graph of f. over $B(r)$

Uncertain reading. The subscript of Γ is heavily overwritten and may be f or r.

Original source details: GELFAND-P000600

$$\int_{\Gamma(r)} (\varphi+\omega)^n = \sum \binom{n}{k} \int_{B(r)} \varphi^k \wedge f^*\omega^{n-k}$$

Def.

$$T_k(f,r) \underset{\text{def}}{=} \int_{B(r)} \varphi^{n-k} \wedge f^*(\omega^k)$$

In many cases $T_k = O(T_1)$ (growth in r).

[below the line] of the same order.

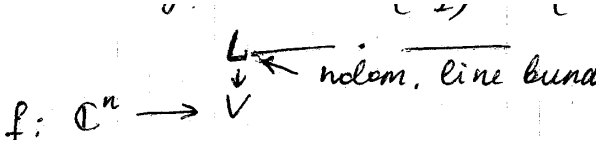


Diagram and typed description.

$$\begin{array}{ccc} L & \longleftarrow & \text{holom. line bundle.} \\ \downarrow & & \\ f:\ \mathbb{C}^n & \longrightarrow & V \end{array}$$

Original source details: GELFAND-P000600

Let $0 < \omega = c(L)$ — its Chern class

Let $\sigma \in O(L), \quad D = \{\sigma = 0\}$.

$$d[\eta] - [\omega] = [D]; \qquad D_f = f^{-1}D.$$

$$d[f^*\eta] - [f^*\omega] = [f^{-1}D] \quad \text{(on } \mathbb{C}^n\text{)}.$$

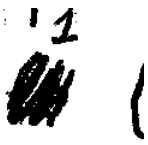
Mult. by $\chi_{B(r)}\varphi^{n-1}$:

$$\underbrace{\int_{S(r)} \varphi^{n-1} \wedge f^*\omega}_{\text{“}T_1} = \int_{D_f} \chi_{B(r)}\varphi^{n-1}.$$

Uncertain reading. The integral subscript reads $S(r)$.

Original source details: GELFAND-P000600

$$\eta = \underbrace{\frac{1}{4\pi} \text{[crossed out]}\ (\bar\partial - \partial) \log|\sigma|^2}_{d^c}.$$



Uncertain reading. A mark before $(\bar\partial - \partial)$ is crossed out and may be i.

Original source details: GELFAND-P000600

$$\int_{D_f \cap B(r)} \varphi^{n-1}$$

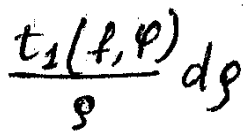
(volume of the zero set, meas. using the std. φ).

Uncertain reading. The abbreviation after "zero set" reads "помер." (meas.).

Original source details: GELFAND-P000600

$$T_1 = \int_0^r \frac{t_1(f,\varphi)}{\rho}\, d\rho$$

— $t_1 \leftrightarrow T_1$ in ordinary Nevanlinna th..



Uncertain reading. The argument of t_1 has φ. The denominator and differential have ρ.

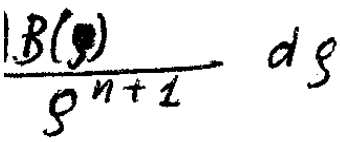
Original source details: GELFAND-P000600

GELFAND-P000601

Uncertain reading. The beginning of a formula $N(\dots)$ is heavily crossed out.

Original source details: GELFAND-P000601

$$N(D_f, r) = \int_0^r \frac{\int_{D_f \cap B(\rho)} \varphi^{n-1}}{\rho^{n+1}}\, d\rho$$



Uncertain reading. The argument of $B(\rho)$ is overwritten and reads ρ.

Original source details: GELFAND-P000601

$$N(D_f, r) \le T(f, r) + O(1) \qquad \text{(1st Nevanlinna inequality)}$$

GELFAND-P000602

14/IV-75. Gelfand Sem. King (continued).

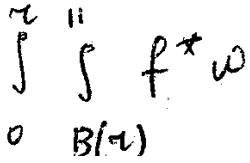
$$\mathbb{C}^n \xrightarrow{f} \underset{\text{projective}}{V^n}, \qquad \underset{\text{divisor.}}{D} \subset V^n, \qquad D \sim c(L_D) = \underset{\text{Kähler form}}{\omega}$$

$$N(D_f, r) \le T(f, r)$$

[below N:] volume of the preimage of the divisor ("number of zeros")
[below T:]

$$\int_0^r \int_{B(r)}'' f^*\omega \wedge \psi^{n-1} \frac{dt}{t}$$

("order of growth")



Uncertain reading. Two short vertical strokes are written above the second integral.

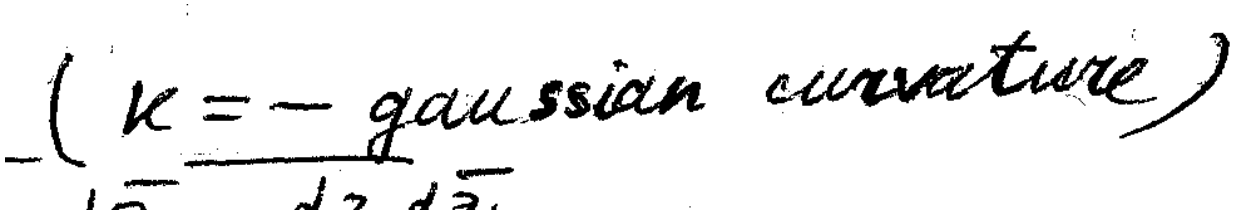
Original source details: GELFAND-P000602

[at right] (1st Nevanlinna thm. — triv.)
uses the residue thm.

$$\operatorname{Ric}\Omega = i\partial\bar\partial \log g \quad \Leftarrow \quad \left(\Omega = \frac{i}{2} g\, dz \wedge d\bar z\right).$$

(Ricci tensor).

$$n = 1 \quad \Rightarrow \quad \operatorname{Ric}\Omega = K\Omega \quad (K = -\text{gaussian curvature})$$



Original source details: GELFAND-P000602

Uncertain reading. The long stroke after $K =$ is read as a minus sign.

Ω — an (n, n) volume form

$$a(\cdot)\, dz_1\, d\bar{z}_1 \ldots dz_n\, d\bar{z}_n$$

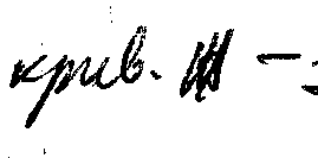
$$\operatorname{Ric}\Omega = \frac{i}{2\pi}\partial\bar{\partial}\log a$$

1) $\operatorname{Ric}\Omega > 0$

2) $(\operatorname{Ric}\Omega)^n \geq \Omega$ (vol. form).

Ahlfors Lemma.

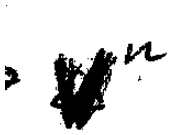
Let B^n or Δ^n have a Bergman form with curv. [overwritten] -1



Uncertain reading. A heavily overwritten mark before -1 resembles H or K.

Original source details: GELFAND-P000602

$\Rightarrow$ $\forall$ holom. $f : B^n \to V^n$; $\quad \Omega_V$ — volume form with neg. curv.



Uncertain reading. The overwritten letter before the superscript n reads V.

Original source details: GELFAND-P000602

$f^*\Omega_V \leq \Phi_B$ — volume form

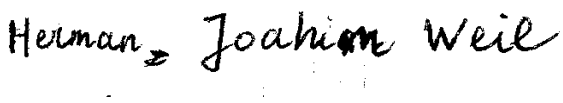
$\ldots$ ___ $\cdot$ ___ $\ldots$

~~Acta~~ Acta Math.; Griffits, King, [crossed out]

Uncertain reading. A small mark after "King," is crossed out and may read "a".

Original source details: GELFAND-P000602

Herman $\geq$ Joahim Weil



Uncertain reading. The manuscript reads "Herman" followed by a mark shaped like $\geq$. The end of "Joahim" is heavily inked.

Original source details: GELFAND-P000602

A nontriv. formulation is needed in the theory of eigenfunctions.

Griffits, Cowen

Chapter 9

GELFAND-P000603

15.9.75. Gelfand Sem.

There will be a series of lectures by Polyakov on physics.

Kirillov. Thm. J. Vey on deformation of the Poisson bracket.

Uncertain reading. The initial reads “J.”.

Original source details: GELFAND-P000603

(M, ω) — a sympl. manifold, $[u, v]$ — the Poisson bracket. There are essentially no other Lie alg. structures (Kirillov et al.). But if diff. oper. of order ∞ are allowed, the structure can be deformed.

Thm.: $\exists$ a formal deformation of the Poisson bracket, inequivalent to it.

$$[u, v]_t = \sum_{k=0}^{\infty} t^k [u, v]^{(k)}$$

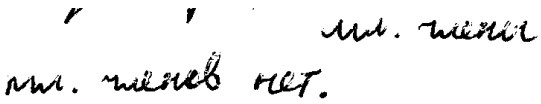
[stroke from $[u, v]^{(k)}$:] oper. of order $(2k + 1)$ in u and v.

nontriviality: a change of coordinates does not reduce it to the original Poisson bracket.

$$[u, v]^{(k)} = \frac{1}{(2k+1)!} \left(\frac{\partial^{2k+1} u}{\partial p^{2k+1}} \frac{\partial^{2k+1} v}{\partial q^{2k+1}} - \frac{\partial^{2k+1} u}{\partial q^{2k+1}} \cdot \frac{\partial^{2k+1} v}{\partial p^{2k+1}} \right) + \dots$$

lower terms

On the plane (or torus) there are no lower terms.



Uncertain reading. The abbreviation in both places reads “мл.” (lower).

Original source details: GELFAND-P000603

Defined globally on M provided that $H^3(M, R) = 0$.

$$? \qquad pq - qp = t$$

GELFAND-P000604

Polyakov. (Alexander Markovich).

Elementary particle physics.

γ — photon. — a separate group

Leptons

$$\left.\begin{array}{ll} e^- & m = 0{,}5 \\ \mu^- & m = 105 \\ \nu_e & m = 0 \\ \nu_\mu & m = 0 \end{array}\right\} \text{ neutrinos}$$

Hadrons.

$$\begin{array}{ll} p & m = 940 \\ n & \text{— ‖ —} \\ \pi & m = 140 \end{array}$$

etc.

Uncertain reading. The top stroke of π is separate from the lower strokes.

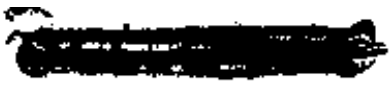
Original source details: GELFAND-P000604

Interactions: Intensity in units $x = \dfrac{E_{\text{int}}}{m}$

Gr	(gravit.)	$x \sim 10^{-38}$
W	(weak)	$x \sim 10^{-5}$
EM	(electromagn.)	$x \sim 10^{-2}$
S	(strong)	$x \sim 1$

The weaker the interact., the more accommodating it is (it violates fewer conservation laws).

They are distinguished only by symmetry (the strong from the weak, since both are [crossed out] short-range).



Undeciphered passage. The word after “both” is heavily crossed out.

Original source details: GELFAND-P000604

Leptons — all interact. except the strong.

Hadrons — all 4 interactions.

Quantum electrodynamics describes the electromagn. interact. of leptons. (the others are weaker and are ignored for now).

A particle is described by its own field. One must specify the action:

$$S = S[e(x), \mu(x), \dots]$$

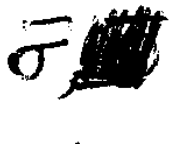
Invariance: rot., transl., Lorentz transf..

Exper. fact: under rot. the electron transf. according to the spinor representation

$$\begin{pmatrix} e_1' \\ e_2' \end{pmatrix} = \exp\left(\frac{i}{2}(\bar{\sigma}\,\bar{\Theta})\right)\begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

($e_j \in \mathbb{C}$ — components of the spinor)

$\bar{\sigma}$ [crossed out] — Pauli matrices (a set of 3 such matrices); $\bar{\Theta} \in \mathbb{R}^3$;



Uncertain reading. The entry after $\bar{\sigma}$ is heavily crossed out.

Original source details: GELFAND-P000604

$|\bar{\Theta}|$ — angle of rot., dir. of Θ — dir. of the axis of rotation.

GELFAND-P000605

$$\sigma_x = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Uncertain reading. The upper-right entry in σ_x reads -1.

Original source details: GELFAND-P000605

Calc. also the transf. under the action of the Lorentz group. 2 possibilities:

Uncertain reading. The initial abbreviation reads “Вычис.” (calc.).

Original source details: GELFAND-P000605

$$e'_L = \exp\left(\frac{\bar\sigma\bar\psi}{2}\right) e_L,$$

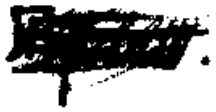
where $\bar\psi$ — a vector collinear with the velocity vect.:

$$v = \operatorname{th}|\bar\psi|, \qquad \frac{\vec v}{|\bar v|} = \frac{\bar\psi}{|\bar\psi|} \qquad (\bar\psi \text{ — also a 3-vector})$$

[below the line] “rapidity”

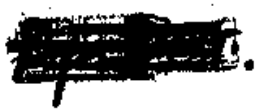
$$e'_R = \exp\left(-\frac{\bar\sigma\bar\psi}{2}\right) e_R$$

Explan. (G.) [crossed out]. $SO(3) \longrightarrow$ [crossed out]. $SU(2)$.



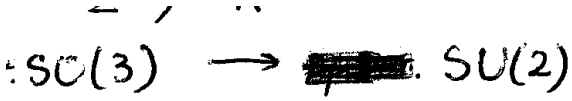
Original source details: GELFAND-P000605

Undeciphered passage. An entry before $SO(3)$ is crossed out.



Original source details: GELFAND-P000605

Undeciphered passage. An entry before $SU(2)$ is crossed out.



Original source details: GELFAND-P000605

Uncertain reading. The handwritten arrow points from $SO(3)$ to $SU(2)$.

$$\bar\Theta \text{ — rot. in } \mathbb{R}^3 \Longrightarrow \text{matr.} \in SU(2)$$

(representation of $SU(2)$)

This rep. extends to a rep. of the Lorentz group.

Lorentz transf. — the velocity vector $\vec v$ (rotation is omitted) (proper Lorentz transf.)

Rapidity $\bar\psi$ — introduced in place of $\bar v$

P.

$$\begin{cases} t' = t\,\mathrm{ch}\,\psi + z\,\mathrm{sh}\,\psi \\ z' = z\,\mathrm{ch}\,\psi + t\,\mathrm{sh}\,\psi \end{cases}$$

Original source details: GELFAND-P000605

Uncertain reading. In the first line z is written over an earlier mark.

[at right] — Lorentz transf. written in terms of ψ. (proper Lorentz transf. — with no added rot. or translation).

The electron is either e_L or e_R.

But there is a symmetry $\vec{x} \longrightarrow -\vec{x}$, $\vec{v} \longrightarrow -\vec{v}$, they turn into each other.

If reflection is adjoined, the rep. is only obtained in combined form:

$$\hat{p}\begin{pmatrix} e_L \\ e_R \end{pmatrix} = \begin{pmatrix} e_R \\ e_L \end{pmatrix}.$$

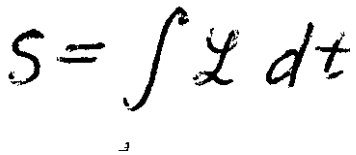
We need to introd. a bispinor: $\begin{pmatrix} e_L \\ e_R \end{pmatrix}$.

Then the Lagrangian density:

$$\mathcal{L}' = e_L^+(x)\left\{\frac{\partial}{\partial t} - \left(\bar{\sigma}\frac{\partial}{\partial \bar{x}}\right)\right\} e_L(x) + e_R^+(x)\left\{\frac{\partial}{\partial t} + \left(\bar{\sigma}\frac{\partial}{\partial \bar{x}}\right)\right\} e_R(x)$$

$(z^+ = \bar{z})$.

Action: $S = \int \mathcal{L}\, dt\, d^3\bar{x}$.



Original source details: GELFAND-P000605

Uncertain reading. No prime is visible on $\mathcal{L}$ in the action formula.

Invariance is easy to check.

Masslessness Lagr. eq.

$$\left(\frac{\partial}{\partial t} - \bar{\sigma}\frac{\partial}{\partial \bar{x}}\right) e_L = 0$$

Take the Fourier transf.: $(E - \bar{\sigma}\bar{p})e_L = 0$ and hence $E^2 = p^2$.

GELFAND-P000606

Hence E is arbitrarily small (for small p) $\Rightarrow m = 0$.

We must add a term with the mass. The inv. way is unique

e_L [crossed out] ~~me~~

Original source details: GELFAND-P000606

Uncertain reading. Under the crossing-out, the mark between e_L and me resembles a right-pointing arrow.

$$m(e_L^+ e_R + e_R^+ e_L).$$

Thus, the final form of the Lagrangian:

$$\mathcal{L} = \mathcal{L}' + m(e_L^+ e_R + e_R^+ e_L).$$

Equiv. form: let $e = \begin{pmatrix} e_L \\ e_R \end{pmatrix}$. (bispinor).

$$\mathcal{L} = \bar{e}\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) e$$

$$\gamma_\mu \frac{\partial}{\partial x_\mu} = \gamma_0 \frac{\partial}{\partial t} - \bar{\gamma}\frac{\partial}{\partial \bar{x}}, \qquad \bar{e} = e^+\gamma_0$$

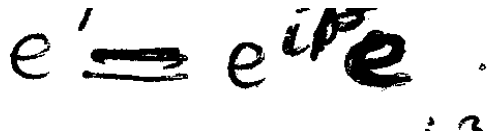
γ_μ — Dirac matr., $\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}$,

$$g_{\mu\nu} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix}$$ — Minkowski metr..

γ_μ are written in terms of the matrices σ
(References: Landau, Berestetskii, Lifshitz, Pitaevskii — Relativistic Quant. Theory, pt. 1).
Spectrum: $E = \sqrt{p^2 + m^2}$.
(In the Dirac eq. $\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) e = 0$ we pass to the Fourier transf. and obtain this res.)

—— · ——

Gauge group $e' \Rightarrow e^{i\beta}e$. $(\beta \in \mathbb{R})$.



Uncertain reading. There are two horizontal strokes between $e\prime$ and $e^{i\beta}e$. The upper stroke has a point at the right.

Original source details: GELFAND-P000606

$\mathcal{L}$ is also invariant with resp. to this group $e' = (2,7\ldots)^{i\beta} e$

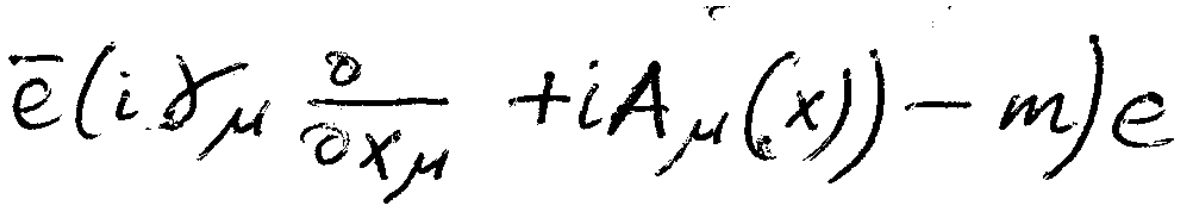
Hypothesis: vary the phase differently at different points and require covar. with resp. to these transformations (extreme locality)

$$e' = (2,7)^{i\beta(x)} e$$

Invar. is now lost:

$$\bar{e}\left(i\gamma_\mu\left(\frac{\partial}{\partial x_\mu} + i\partial_\mu\beta\right) - m\right)e.$$

$$\mathcal{L}' = \bar{e}\left(i\gamma_\mu\frac{\partial}{\partial x_\mu} + iA_\mu(x)\right) - m\Big)e$$



Original source details: GELFAND-P000606

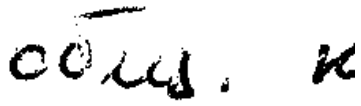
Uncertain reading. A closing parenthesis follows $A_\mu(x)$, and another follows $-m$. No factor γ_μ is visible before $iA_\mu(x)$.

GELFAND-P000607

($A_\mu(x)$ — a connection).

$$A'_\mu(x) = A_\mu - \partial_\mu\beta.$$
$$\nabla_\mu = \partial_\mu + iA_\mu.$$

If we do this, general covariance will be restored.



Uncertain reading. The abbreviation before “covariance” reads “общ.” (general).

Original source details: GELFAND-P000607

From $\mathcal{L}'$ no eqs. arise for A_μ. We need the curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

This is a tensor, and in [crossed out] $\mathcal{L}$ we introd. its square.

Uncertain reading. An entry before $\mathcal{L}$ is crossed out.

Original source details: GELFAND-P000607

$$\mathcal{L} = \text{[crossed out]}\ \mathcal{L}' + \left(\frac{1}{4e^2}\right) F_{\mu\nu}^2$$

Uncertain reading. A sign before $\mathcal{L}'$ on the right-hand side is crossed out.

Original source details: GELFAND-P000607

(e — the electron charge)

—— · ——

Scattering. $S = S[\varphi(x)]$.

B $\varphi_{\text{in}}(x) \longrightarrow \varphi_{\text{out}}(x)$

Uncertain reading. A separate mark to the left of the arrow resembles a B with a vertical stroke.

Original source details: GELFAND-P000607

$$\varphi_{\text{in}}(x) = \lim_{t\to-\infty} \varphi(\bar{x}, t), \qquad \varphi_{\text{out}}(x) = \lim_{t\to+\infty} \varphi(\bar{x}, t).$$

$$\int_{\Phi_{\text{in}}}^{\Phi_{\text{out}}} D\varphi(x)\, e^{iS[\varphi]} = M \text{ — transition amplitude}$$

(functional integration, $D\varphi$ — the natural measure, [below the line] (Lebesgue), integral over all $\Phi(x,t)$, $\Phi(x,+\infty) = \Phi_{\text{out}}$, $\varphi(x,-\infty) = \varphi_{\text{in}}$)

There are rules for calculation.

[crossed out] [inserted above the line and outlined] Without A_μ — a Gaussian integral (Fresnel)

Undeciphered passage. An entry before the insertion and the words “Gaussian integral” is heavily crossed out.

Original source details: GELFAND-P000607

Then — perturbation theory (take A_μ small); $eA_\mu e$ — a cubic term

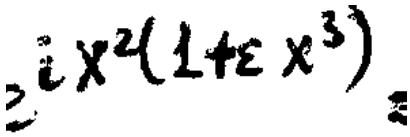
---

G. $\int e^{i(x^2+\varepsilon x^3)}\,dx$ — your theory

Important:

$$\int e^{-\sum a_{ik}x_ix_k+\varepsilon\sum x_i^3}\,dx$$

$$\int e^{ix^2(1+\varepsilon x^3)} = \sum c_k \int x^k e^{ix^2}\,dx$$

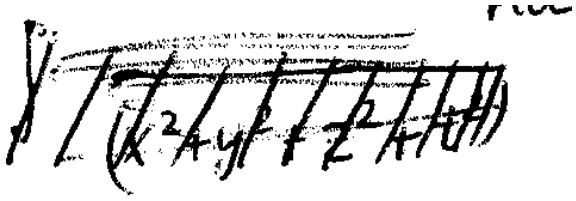


Uncertain reading. The exponent reads $ix^2(1+\varepsilon x^3)$.

Original source details: GELFAND-P000607

For this we use Feynman diagrs.

[crossed out at left] $\int \frac{}{(x^2+y^2+z^2+t\ldots)}$

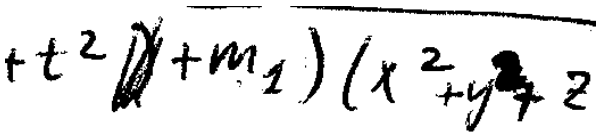


Undeciphered passage. In the crossed-out entry the integral sign, fraction bar, and part of the denominator are legible.

Original source details: GELFAND-P000607

E.g.

$$\int \frac{dx\,dy\,dz\,dt}{(x^2+y^2+z^2+t^2+m_1)(x^2+y^2+z^2+t^2+m_2)}$$



Uncertain reading. A parenthesis is crossed out after the first t^2. The exponent on y in the second factor is overwritten.

Original source details: GELFAND-P000607

(stolen by F. from Schwinger)

$$\frac{1}{ab} = \int_0^1 \frac{ds}{(sa+(1-s)b)^2}. \qquad \frac{1}{abc} \text{ analogously.}$$

GELFAND-P000608

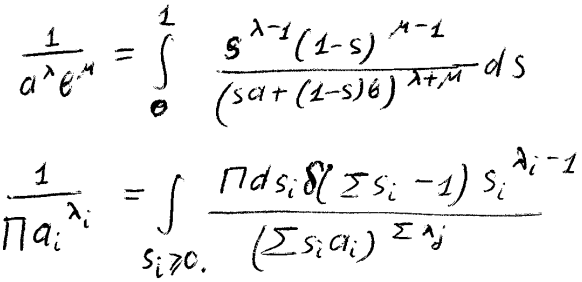
P. one can also do $\frac{1}{a^\lambda b^\mu c^\gamma}$

Uncertain reading. The exponent on a is overwritten and tentatively reads λ.

Original source details: GELFAND-P000608

$$\frac{1}{a^\lambda b^\mu} = \int_0^1 \frac{s^{\lambda-1}(1-s)^{\mu-1}}{(sa+(1-s)b)^{\lambda+\mu}}\, ds$$

$$\frac{1}{\prod a_i^{\lambda_i}} = \int_{s_i \geq 0.} \frac{\prod ds_i\, \delta(\sum s_i - 1)\, s_i^{\lambda_i - 1}}{(\sum s_i a_i)^{\sum \lambda_j}}$$



Original source details: GELFAND-P000608

Uncertain reading. Both formulas are written without a separate normalization factor.

— · —

We calculate by perturbation theory:

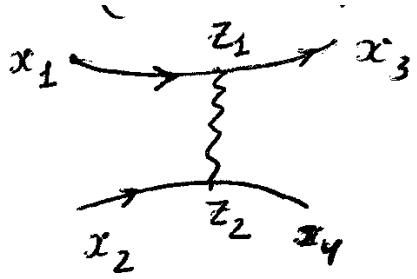
$x,\ x'$

Amplitude of trans. from x to x': $K(x,x')$ according to Maxwell

$$\left(i\gamma_\mu \frac{\partial}{\partial x_\mu} - m\right) K(x,x') = \delta(x-x').$$

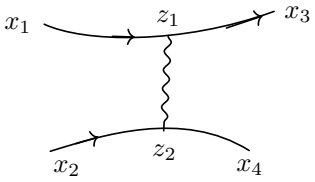
Uncertain reading. The γ is heavily inked and may be overwritten.

Original source details: GELFAND-P000608



Original source details: GELFAND-P000608

Diagram and typed description.

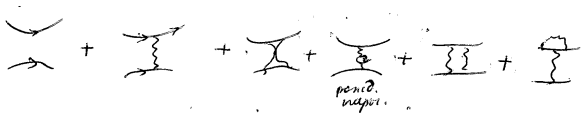


Emission and absorption of a photon are accomp. by a factor e.

$$\int K(x_3 - z_1)K(z_1 - x_1)K(x_4 - z_2)K(z_2 - x_2)$$
$$D(z_2 - z_1)\, d^4 z_1\, d^4 z_2$$

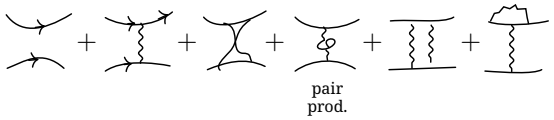
D — fund. soln. of Maxwell's eqs.

Other pictures:



Original source details: GELFAND-P000608

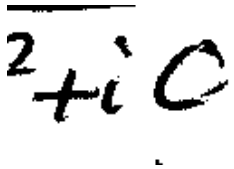
Diagram and typed description.



The meeting of the strokes in the third sketch and the direction of the loop in the fourth are unclear.

$$(\gamma_\mu p_\mu - m)K(p) = 1$$
$$K(p) = \frac{\gamma_\mu p_\mu + m}{\underline{p}^2 - m^2}$$
$$K(x, x') = \int K(p) e^{ip(x-x')}\, d^4 p$$

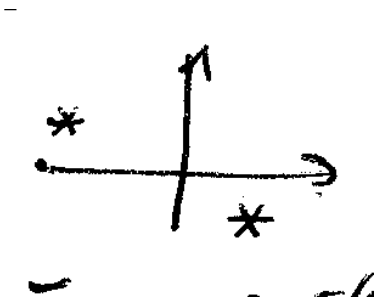
The pole is bypassed if we take $K(p) = \dfrac{\gamma_\mu p_\mu + m}{\underline{p}^2 - m^2 + i0}$



Original source details: GELFAND-P000608

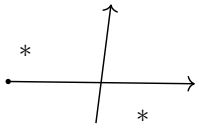
Uncertain reading. The last mark is open and is read as zero.

$$\underline{p}^2 = E^2 - \vec{p}^2.$$

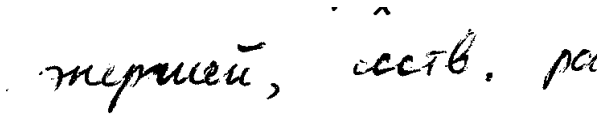


Original source details: GELFAND-P000608

Diagram and typed description.



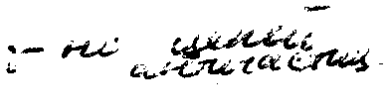
There are levels with neg. energy, nat. propagate backwards in time (pred. the existence of antiparticles).



Original source details: GELFAND-P000608

Uncertain reading. The abbreviation after "energy" tentatively reads "ество." (nat.).

There is a unique choice of bypass leading to a consistent theory: — nor have antiparticles



Uncertain reading. The last two lines are close together. The reading “nor have antiparticles” is tentative.

Original source details: GELFAND-P000608

GELFAND-P000609

Charge normalization.

e — is this the obs. electron charge?

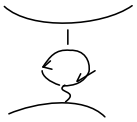
Scattering: electrons exch. photons, force $\sim e^2$

Diagram and typed description.

Original source details: GELFAND-P000609

Correction: $\sim e^4$.

Diagram and typed description.



The directions of the two arrow-like strokes on the loop are unclear. The visible gaps are retained.

Original source details: GELFAND-P000609

Obs. charge $e^2_{\text{phys}} < e^2$.

In strong interact. theory the reverse holds $g^2_{\text{phys}} > g^2$.

—— · ——

GELFAND-P000610

22.9.75. Gelfand Sem.

Diagram and typed description.

Yellow comment icon in the upper right margin of the digitized frame.

Original source details: GELFAND-P000610

Gelfand on Atiyah's res.

$\mathfrak{A}$ — all oper. in Hilb. sp., $\mathcal{K}$ — comp. oper.

$$\mathfrak{A}/\mathcal{K} = \mathfrak{B}.$$

When are $B_1, B_2 \in \mathfrak{B}$ unitarily equivalent?

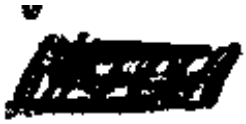
This is [crossed out] Brown, Douglas, Fillmore (Lect. Notes, $N = \ldots$)

Original source details: GELFAND-P000610

Uncertain reading. An entry before the surnames is crossed out. It may read "Все" (all).

a) spectra coincide (ess.)

b) indices coincide in [crossed out] $\forall$ conn. comp. of the complement

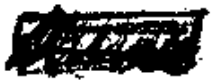


Original source details: GELFAND-P000610

Undeciphered passage. An entry before the quantifier is crossed out.

Thm. 2. [crossed out] Let X — be a compact space,

$$\underset{\substack{\uparrow \\ \text{set of embeddings of } C(X) \text{ in } \mathfrak{B}.}}{\mathrm{Ext}(X)} \qquad \cong K_1(X)$$



Original source details: GELFAND-P000610

Undeciphered passage. An entry before "Let" is crossed out.

—— · ——

Moser — Korteweg–de Vries

—— · ——

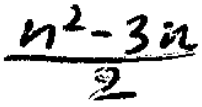
Kostant

$$\psi(x) = \prod_{n=1}^{\infty}(1 - x^n); \qquad \psi(x)^{-1} = \sum_{n=0}^{\infty} p(n)x^n$$

($p(n)$ — number of part. of n).

This is

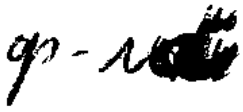
$$\psi(x) = \sum (-1)^n x^{\frac{n^2-3n}{2}}$$



Original source details: GELFAND-P000610

Uncertain reading. The exponent is written $(n^2 - 3n)/2$.

- “The most remarkable formula in all of mathematics” (Jacobi).



Uncertain reading. The heavily overwritten abbreviation after “remarkable” tentatively reads “ф-ла” (formula).

Original source details: GELFAND-P000610

$$(\psi(x))^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) x^{\frac{n(n+1)}{2}}$$

$$\eta(x) = x^{1/24} \cdot \psi(x).$$

$$\eta^{24}(x) = \sum \tau(n) x^n, \qquad \tau(n) \text{ — Ramanujan's function}$$

$\psi^{10}(x)$ exists too

Manin: if 6 different fns., obt. from different considerations

$$\eta(x)^{\square} = \sum \dots$$

Macdonald found powers for which it makes sense: These are the dim of compact Lie groups.

$$\psi(x)^{\dim K} = \sum_{\lambda} \text{[crossed out]}\ \chi_\lambda(a)\ \dim V_\lambda \cdot x^{(\text{depends on } \lambda)}$$

Undeciphered passage. An entry after the summation sign and before $\chi_\lambda(a)$ is crossed out.

Original source details: GELFAND-P000610

$a \in K$, $\chi_\lambda(a) = \pm 1$ or 0

Kostant: $\mathrm{Sp}(-\Delta + a)^{-s}$ etc.

Internal [ч-ие] G.: $\psi(x)^z$ — ? $\qquad (z \in \mathbb{C})$

Uncertain reading. The middle abbreviation tentatively reads “ч-ие”.

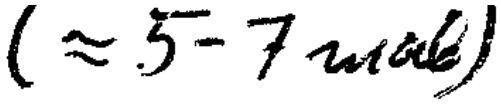
Original source details: GELFAND-P000610

— · —

Polyakov. (continued).

References:

quant. mech.
1) Landau and Lifshitz (quant. mechanics). (≈ 5–7 chapters)
2) Dirac
3) Feynman (for schoolchildren)
4) Feynman. Path integrals



Uncertain reading. The word after “5–7” reads “глав” (chapters).

Original source details: GELFAND-P000610

Quant. field theory: 1st part of vol. IV Berestetskii, Lifshitz, Pitaevskii

GELFAND-P000611

$e(x)$ — field (bispinors)

$$\mathcal{L}_{\text{free}} = \bar{e}\, i\gamma_\mu \partial_\mu e - m\bar{e}e + \frac{1}{4e^2} F_{\mu\nu}^2$$

$$\mathcal{L}_{\text{int}} = \ldots \bar{e}\gamma_\mu e A_\mu \ldots \qquad \text{cont. 3-... terms}$$

Uncertain reading. The abbreviation after “3-” is uncertain.

Original source details: GELFAND-P000611

— · —

Localization of large int. symmetries — weak interaction.

Uncertain reading. “б-их вн.” tentatively reads “large internal”.

Original source details: GELFAND-P000611

$$\mathcal{L} = \bar{e}_L i\gamma_\mu \partial_\mu e_L + \bar{e}_R i\gamma_\mu \partial_\mu e_R + \bar{\nu}_L i\gamma_\mu \partial_\mu \nu_L +$$
$$+ \begin{pmatrix} e \to \mu \\ \nu_L \to \nu'_L \end{pmatrix} + m(\bar{e}_L e_R + \bar{e}_R e_L).$$

(all leptons) $\downarrow$

ν'_L — μ-meson neutrino

$$\mathcal{L} = \bar{e}_L i\gamma_\mu \partial_\mu e_L + \bar{\nu}_L i\gamma_\mu \partial_\mu \nu_L + \bar{e}_R i\gamma_\mu \partial_\mu e_R$$

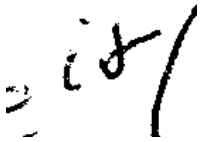
$$\begin{cases} \begin{pmatrix} e'_L \\ \nu'_L \end{pmatrix} = U \begin{pmatrix} e_L \\ \nu_L \end{pmatrix} \qquad SU_L(2) \\ e'_R = e_R \end{cases}$$

•

A new symmetry gr. $SU(2)$ arises, and also $U(1)$

$$F = SU_L(2) \times U_R(1) \times U_L(1) \quad \text{— overall lepton group.}$$

$$\begin{pmatrix} e_L \\ \nu_L \end{pmatrix} \overset{U_L(1)}{\longrightarrow} e^{i\gamma} \begin{pmatrix} e_L \\ \nu_L \end{pmatrix}$$



Original source details: GELFAND-P000611

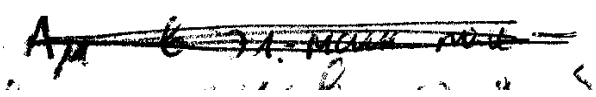
Uncertain reading. The letter in the phase reads γ.

Localization

Original source details: GELFAND-P000611

Undeciphered passage. The crossed-out entry in a box in the left margin is illegible.

[Crossed out: ~~A_μ ... electromagn. field~~.]

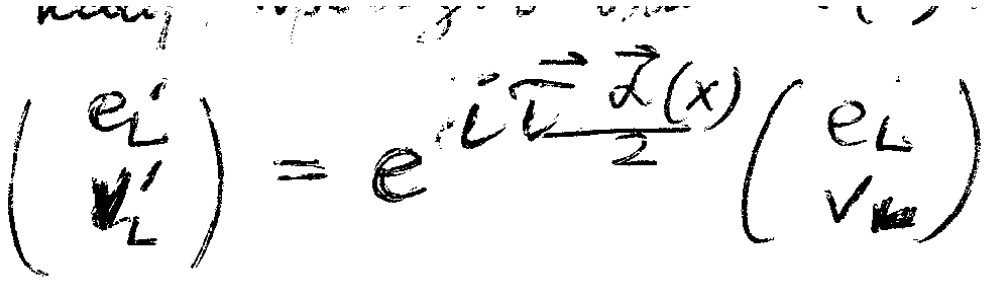


Original source details: GELFAND-P000611

Uncertain reading. A mark between A_μ and "electromagn. field" is obscured by the crossing-out.

Covar. derivative for $SU(2)$. $\bar{\tau}$ — Pauli matr. (3 in all)

$$\begin{pmatrix} e'_L \\ \nu'_L \end{pmatrix} = e^{i\frac{\bar{\tau}\vec{\alpha}(x)}{2}} \begin{pmatrix} e_L \\ \nu_L \end{pmatrix}, \qquad E' = e^{i\frac{\bar{\tau}\vec{\alpha}(x)}{2}} E$$

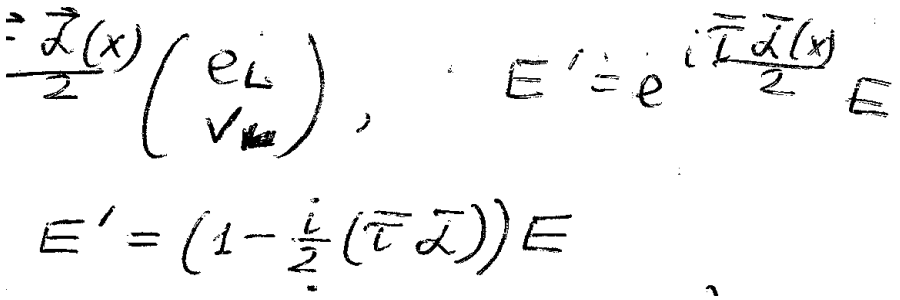


Original source details: GELFAND-P000611

Uncertain reading. The lower letter on the left and the lower subscript on the right are overwritten.

Infinitesim.:

$$E' = \left(1 - \frac{i}{2}(\bar{\tau}\bar{\alpha})\right) E$$



Original source details: GELFAND-P000611

Uncertain reading. The finite exponent is written with an implicit plus sign. A minus sign is visible in the infinitesimal expression.

$$\partial_\mu E' = \left(1 - \frac{i}{2}(\bar{\tau}\bar{\alpha})\right) \partial_\mu E - \frac{i}{2}(\bar{\tau}\partial_\mu\bar{\alpha})E$$

$$\boxed{\nabla_\mu E = \partial_\mu E + \frac{i}{2}(\vec{\tau}\vec{A}_\mu)E}$$

$$E' = e^{i\frac{\bar{\tau}\bar{\alpha}(x)}{2}} E$$

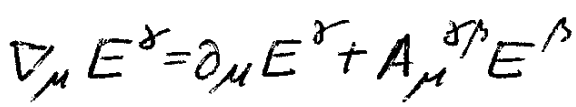
$$\nabla_\mu E' = e^{i\frac{\bar{\tau}\bar{\alpha}}{2}} \nabla_\mu E$$

Here 3 massless particles arise

GELFAND-P000612

one can write directly

$$\nabla_\mu E^\gamma = \partial_\mu E^\gamma + A_\mu^{\gamma\beta} E^\beta$$

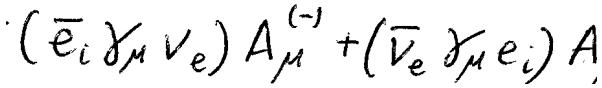


Original source details: GELFAND-P000612

Uncertain reading. The superscript on E and the first superscript on A read γ.

and we look at how A transforms so that $\nabla_\mu E^\gamma$ transforms like E^γ.

$$\mathcal{L}_{\text{int}} = (\bar{e}_i \gamma_\mu \nu_e) A_\mu^{(-)} + (\bar{\nu}_e \gamma_\mu e_i) A_\mu^{(+)} + (\bar{e}_L \gamma_\mu e_L - \bar{\nu}_L \gamma_\mu \nu_L) A_\mu^3$$



Uncertain reading. The electron index in the first two terms is written with a dotted i.

Original source details: GELFAND-P000612

where

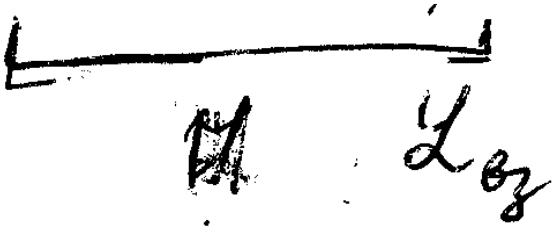
$$A_\mu^\pm = A_\mu^1 \pm i A_\mu^2$$

Example of weak interaction: $\mu^- \to e^-, \bar{\nu}_e, \nu_\mu$.

$$\mathcal{L} = \bar{E}\left(i\gamma_\mu \frac{\partial E}{\partial x_\mu}\right) + \underbrace{\bar{E}\gamma_\mu \vec{\tau} E \vec{A}_\mu}_{\mathcal{L}_{\text{int}}} + \bar{e}_R i \gamma_\mu \partial_\mu e_R$$

Uncertain reading. There is a separate short stroke with dots to the left of $\mathcal{L}$. Its meaning is uncertain.

Original source details: GELFAND-P000612



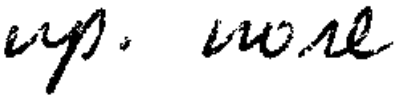
Undeciphered passage. An entry beneath the left side of the bracket is crossed out.

Original source details: GELFAND-P000612

Undeciphered passage. A separate crossed-out entry in the right margin is illegible.

Original source details: GELFAND-P000612

We also localize the r. field



Uncertain reading. The abbreviation “пр.” tentatively means “right”.

Original source details: GELFAND-P000612

$$\bar{E}\left(i\gamma_\mu \frac{\partial E}{\partial x_\mu}\right) + \bar{E}\gamma_\mu \vec{\tau} E \vec{A}_\mu + \bar{e}_R i\gamma_\mu \partial_\mu e_R$$
$$+ \bar{e}_R \gamma_\mu e_R B_\mu$$
$$A_\mu = A_\mu^3 \cos\theta + B_\mu \sin\theta$$
. . . . . _ . . . . .

Next time — ≈ in 3 weeks.

12 quarks: 4 flavors and 3 colors.

Diagram and typed description.

Yellow comment icon to the right of the quark entry in the digitized frame.

Original source details: GELFAND-P000612

—— · ——

GELFAND-P000613

29.9.75. Gelfand Sem.

Yu. I. Manin.

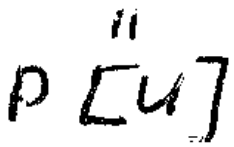
$$x\prod_{n=1}^{\infty}(1-x^n)^{24} = \sum_{n=1}^{\infty}\tau(n)x^n; \qquad \tau \text{ — Ramanujan's function.}$$

I. M. Gelfand. (and Dikii)

∞-dimensional Lie algebra, more remarkable than all the Cartan ones.

Elements — polynomials $P(u, u', \ldots, u^{(n)})$ in any number of variables.

$\overset{''}{P}[u]$, where $u = \{u, u', \ldots, u^{(n)}, \cdot\}$



Uncertain reading. Two strokes stand above the left square bracket.

Original source details: GELFAND-P000613

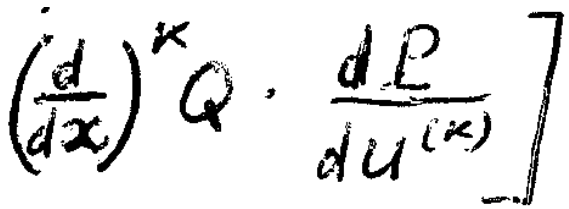
There is a notion of

$$\frac{d}{dx}P(u,u',\dots,u^{(n)}) = \sum_{i=0}^{\infty}\frac{\partial P}{\partial u^{(i)}}\cdot u^{(i+1)}$$

There is multiplication (algebr.)

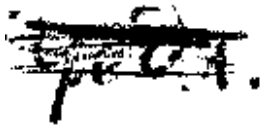
Unexpectedly, there is a notion of the commutator:

$$[P,Q] = \sum_{k=0}^{\infty}\left[\left(\frac{d}{dx}\right)^k P\cdot\frac{\partial Q}{\partial u^{(k)}} - \left(\frac{d}{dx}\right)^k Q\cdot\frac{dP}{du^{(k)}}\right]$$



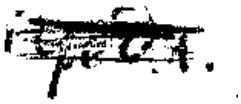
Original source details: GELFAND-P000613

Uncertain reading. The mark after Q reads as a multiplication dot. The last fraction uses d.



Original source details: GELFAND-P000613

Uncertain reading. A short entry before the surname is crossed out.



Original source details: GELFAND-P000613

Undeciphered passage. The crossed-out entry cannot be read in full.

Povzner. This alg. is connected with the eq. $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}$. Can be connected with $\forall$ eq.

$$\frac{\partial u}{\partial t} = a(u,u_x,\dots).$$

Consider fns. $F(u,u_x,\dots)$ and the operator

$$a[F(u,u_x,\dots)] = aF_u + \left(\frac{d}{dx}a\right)F_{u_x} + \frac{d^2a}{dx^2}F_{u_{xx}} + \dots$$

The operator is det. by the fn. a and an algebra is built from these oper.

$[a,b] = a[b] - b[a]$ — the same algebra.

— · —

Here the natural repr. of this algebra is indicated.

Original source details: GELFAND-P000613

Uncertain reading. A mark to the left of “Here” is blacked out.

Manin. 6 expressions for Ramanujan's fn. $\tau(n)$ (not obt. by a finite algorithm).

⊕ ① Definition ⇒

$$\tau(n) = \sum (-1)^k$$

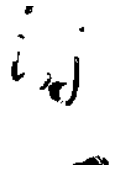
$$n - 1 = n_1 + \ldots + n_k, \qquad n_i \geq 1$$

$$n_1 \leq n_2 \leq \ldots \leq n_k$$

in a row ≤ 24 equality signs.

⊕ ② Macdonald (Dyson)

$$\tau(n) = \frac{1}{1!2!3!4!} \sum \prod_{i<j} (u_i - u_j)$$



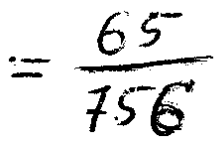
Uncertain reading. The index condition below the product tentatively reads $i < j$.

Original source details: GELFAND-P000613

$$10n = \sum_{i=1}^{5} u_i^2, \qquad \sum u_i = 0, \qquad u_i \equiv i \bmod 5$$

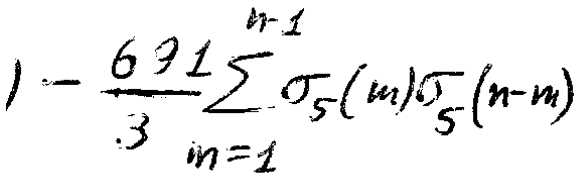
$\binom{+}{+}$ ③ Ramanujan

$$\tau(n) = \frac{65}{756}\sigma_{11}(n) + \frac{691}{756}\sigma_5(n) - \frac{691}{3}\sum_{m=1}^{n-1} \sigma_5(m)\sigma_5(n-m)$$



Uncertain reading. The last digit of the first denominator is overwritten and reads as 6.

Original source details: GELFAND-P000613



Uncertain reading. The argument in the final factor reads $n - m$.

Original source details: GELFAND-P000613

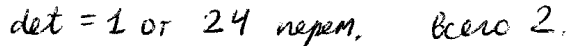
$\sigma_k(n)$ — $\sum$ of the kth powers of the divisors of n.

Corollary: $\tau(n) \equiv \sigma_{11}(n) \bmod 691$. (guessed by Ramanujan).

GELFAND-P000614

④ Int. quad. forms with det $= 1$ in 24 variables. 2 in all.

$\binom{+}{+}$ Let Q_{24} — one of them.



Uncertain reading. The line reads “2 in all”.

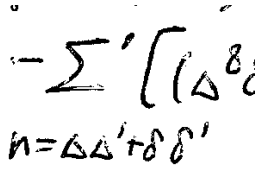
Original source details: GELFAND-P000614

$$\tau(n) = \alpha \left(\sum_{Q_{24}=n} 1 \right) - \beta \sigma_{11}(n)$$

(α, β computed from $n = 1$ and 2).

⑤ (Manin) $\binom{+}{+}$

$$\tau(n) = \sigma_{11}(n) - {\sum_{n=\Delta\Delta'+\delta\delta'}}' \left[(\Delta^8\delta^2 - \Delta^2\delta^8) + \frac{691}{6}(\Delta^6\delta^4 - \Delta^4\delta^6) \right],$$



Uncertain reading. There is a prime on the summation sign.

Original source details: GELFAND-P000614

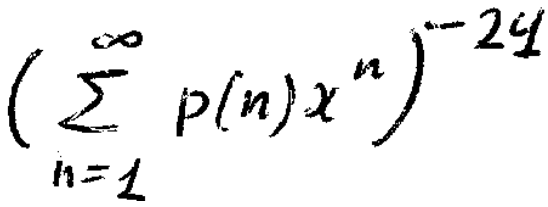
$$\Delta > \delta > 0, \quad \text{and } [\, \Delta' > \delta' > 0 \quad (\text{integers})$$
$$\text{or } \Delta \mid n, \quad \Delta' = \frac{n}{\Delta}, \quad \delta' = 0, \quad 0 < \frac{\delta}{\Delta} \le \frac{1}{2}\,].$$

⑥ $\binom{+}{+}$

$$\sum p(n)x^n = \prod_{n=1}^{\infty} (1 - x^n)^{-1}$$

($p(n)$ — number of partitions of n into any number of summands).

$$\Rightarrow \quad \sum \tau(n)x^n = x \left(\sum_{n=1}^{\infty} p(n)x^n \right)^{-24}$$



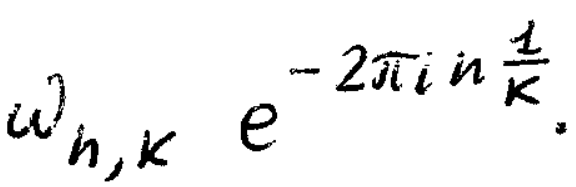
Uncertain reading. The lower limit of the inner sum is written $n = 1$.

Original source details: GELFAND-P000614

Using $p(n)$, find $\tau(n)$ recursively.

Hardy–Ramanujan–Rademacher formula

$$p(n) = \frac{1}{\sqrt{3}} \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \sum_{\substack{h \bmod k \\ (h,k)=1}} \omega_{h,k} e^{-2\pi i n \frac{1}{k}} \cdot \frac{d}{dn} \frac{\operatorname{sh} \frac{\pi}{6k} \sqrt{24n-1}}{\frac{\pi}{6k} \sqrt{24n-1}}$$



Uncertain reading. The exponent is written with $1/k$.

Original source details: GELFAND-P000614

$\omega_{h,k}^{24} = 1$ there are also explicit formulas for $\omega_{h,k}$.

Hardy–Ramanuj. wrote [crossed out] sh $\to e^{\cdots}$.

Undeciphered passage. A short entry before "sh" is crossed out.

Original source details: GELFAND-P000614

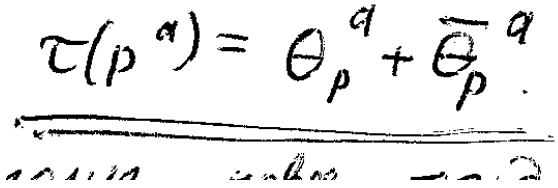
[Below sh $\to e^{\cdots}$: diverges.] but if one truncates at the $\sqrt{n}$ term, it approximates well. One has to stop at $\sqrt{n}$ terms. Each of these formulas is part of an infinite series of formulas.

⑦ $\binom{+}{+}$ $\underline{\underline{\tau(mn) = \tau(m)\tau(n)}}$, $(m,n) = 1$

(not apparent from the formulas).

$$\tau(p) = \theta_p + \bar{\theta}_p, \qquad \theta_p \bar{\theta}_p = p^{11} \quad \Rightarrow$$

$$\underline{\underline{\tau(p^a) = \theta_p^a + \bar{\theta}_p^a.}}$$

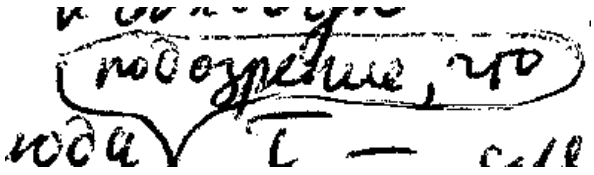


Uncertain reading. The doubly underlined formula is given as written.

Original source details: GELFAND-P000614

One can substitute in each formula and obtain new identities.

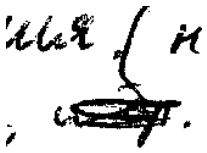
⑧ $\begin{pmatrix} + \\ + \\ + \\ + \end{pmatrix}$ From this [circled insertion: the suspicion that] τ — trace of a representation (much harder) — Serre, Deligne.



Uncertain reading. The circled insertion reads "the suspicion that".

Original source details: GELFAND-P000614

Grothendieck, Weil, Kuga, Sato, [crossed out] Rankin.



Undeciphered passage. The crossed-out entry before Rankin is illegible.

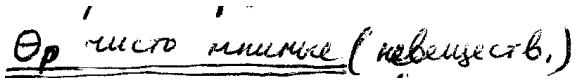
Original source details: GELFAND-P000614

An ℓ-adic repr.

$$\mathrm{Gal}_{\{\ell\}}(\bar{Q}/Q) \longrightarrow Q_\ell^2$$

(ramif. only at ℓ). The trace is taken of the Frobenius autom. gen. by p.

From this it was possible to prove that θ_p are purely imaginary (nonreal)



Uncertain reading. The manuscript reads "purely imaginary".

Original source details: GELFAND-P000614

— The longest thm. in modern mathematics (Deligne's estimate: 2,000 pages).

GELFAND-P000615

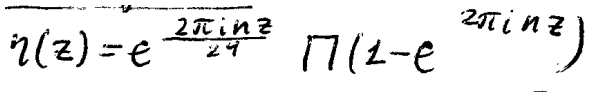
Levels of difficulty.

$+$	combinatorics
$++$	modular forms
$+++$	complex analysis
$++++$	alg. geom. and the like, étale cohom. and so on

—— · ——

How it is obtained.

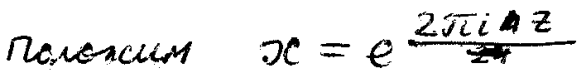
$$\eta(z) = e^{\frac{2\pi i n z}{24}} \prod (1 - e^{2\pi i n z})$$



Uncertain reading. The prefactor exponent is written $2\pi inz/24$.

Original source details: GELFAND-P000615

Set $x = e^{\frac{2\pi i n z}{24}}$.



Uncertain reading. The heavily inked mark between i and z tentatively reads n. The denominator is overwritten.

Original source details: GELFAND-P000615

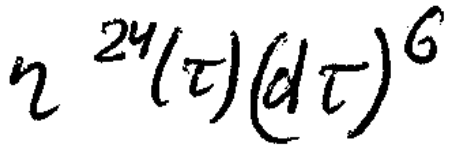
The fn. behaves [crossed out] well under the substitution $z \to \dfrac{az+b}{cz+d}$, $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1$.

Undeciphered passage. Short strokes after the verb are crossed out.

Original source details: GELFAND-P000615

$$\eta(z+1) = e^{\pi i/12}\eta(z),$$
$$\eta(-1/z) = \sqrt{z/i}\,\eta(z).$$

or $\eta^{24}(\tau)(d\tau)^6$ — inv. under frac.-lin. subst..

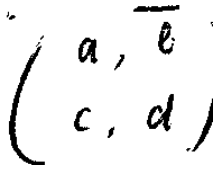


Uncertain reading. The exponent on $d\tau$ is written 6.

Original source details: GELFAND-P000615

Loss of inf. — $\sqrt[24]{\ldots} = \omega_{c,d}$ — the same ones.

$$\begin{pmatrix} a, & \bar{b} \\ c, & d \end{pmatrix}$$



Uncertain reading. In the matrix below "Loss of inf." there is a separate stroke over b.

Original source details: GELFAND-P000615

Fns. of this kind (modular) form a ring with 2 generators, and one must write them in terms of the generators, [crossed out ~~or~~] Identity in the ring ⇒ equality of Fourier coeffs. (⇒ ③, ④).

Uncertain reading. A word above the line is crossed out. It tentatively reads "or".

Original source details: GELFAND-P000615

Pure algebra:

$$\left\{\varphi(z) \,\middle|\, \sum_{n=0}^{\infty} a_n e^{2\pi i n z}\right\}, \qquad \exists k: \quad \varphi(z)dz^k \,\middle|\, \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \varphi(z)dz^k.$$

$\longleftarrow$ converg. series for $\operatorname{Im} z > 0$.

Thm.: the ring $\Phi = C[E_4, E_6]$.

$$\left.\begin{aligned} a_n|_{E_4} &= \sigma_3(n) \\ a_n|_{E_6} &= \sigma_5(n) \end{aligned}\right\| \qquad \sigma_3 \text{ — sums of cubes of divisors.}$$

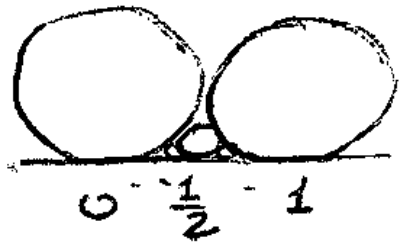
—— · ——

H. R. R. (formula 6).

$$\sum p(n) e^{2\pi i n z} = \frac{1}{\prod(1 - e^{2\pi i z n})}.$$

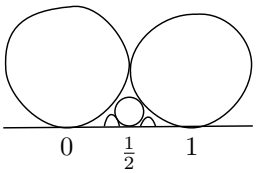
Sing. points on the boundary of convergence (poles of order ∞ at rat. points).

Rademacher's integr. contour: at $\forall$ rat. point, circles of a given radius.



Original source details: GELFAND-P000615

Diagram and typed description.

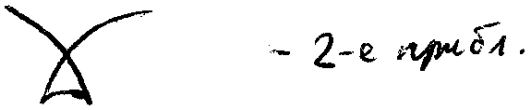


The exact shape of the short arcs near the small circle is uncertain.

Original source details: GELFAND-P000615

Diagram and typed description.

— 1st approx.



Original source details: GELFAND-P000615

Diagram and typed description.

— 2nd approx.

etc.

Special case of Macdonald's formulas in the form transformed by Kostant:

$$\varphi(x) = \prod_{n=1}^{\infty}(1 - x^n), \qquad \varphi(x) = \sum_{\lambda \in D} \chi_\lambda(a) \dim V_\lambda \, x^{c(\lambda)}.$$

Original source details: GELFAND-P000615

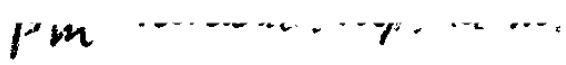
Undeciphered passage. A short entry beneath the product is heavily crossed out.

[under $\lambda \in D$: highest weights of irred. reps.; under $\chi_\lambda(a)$: $0, \pm 1$.]

$$c(\lambda) = (\lambda + \rho, \lambda + \rho) - (\rho, \rho).$$

GELFAND-P000616

$6/\overline{X} - 75$ Gelfand Sem.



Undeciphered passage. Only the lower parts of a line of strokes are visible at right, level with the date.

Original source details: GELFAND-P000616

Leray.

Undeciphered passage. A hatched strip is visible under the name. Any letters beneath it are illegible.

Original source details: GELFAND-P000616

“5 years ago Arnold suggested that I understand one of Maslov’s books.”

“Now I am beginning to understand where this book leads.”

Uncertain reading. “мне” (to me) is overwritten. The next word reads “понять” (understand).

Original source details: GELFAND-P000616

Algebraic prelim. to Maslov’s work,
in part. Buslaev’s metaplectic group.

Maslov: $X = \mathbb{R}^\ell$, Buslaev: X — any set.

$$X = R^\ell: \quad S(X),\ S'(X),\ \mathcal{H}(X) \text{ — intermed. Hilb. sp..}$$

$$a\left(x, \frac{1}{\nu}\frac{\partial}{\partial x}\right), \qquad \frac{\nu}{i} \in \mathbb{R}_+.$$

Assume a — lin. [crossed out] in both arguments, to begin with.

Undeciphered passage. An entry after “lin.” is heavily crossed out.

Original source details: GELFAND-P000616

We seek a gr. of automorph. of $S'(X)$.

$$S: f' \longrightarrow Sf'.$$

$$SaS^{-1} \in A$$

(A — [crossed out] the class of oper. linear in x and $\partial/\partial x$.)

Undeciphered passage. An entry between A and “class of oper.” is crossed out.

Original source details: GELFAND-P000616

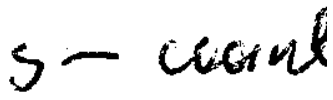
Example: $\overline{\mathcal{F}}$ $\overline{\mathcal{F}}^{-1}$ and $\overline{\mathcal{F}}$ in several coord..

lin. change of var. in x-sp..

Original source details: GELFAND-P000616

Uncertain reading. The first bar over $\mathcal{F}$ has an arched edge. The next two are closer to horizontal.

S — symb. autom. A (inner).



Original source details: GELFAND-P000616

Uncertain reading. The first abbreviation reads "симв." (symbol).

$$a : X \oplus X^* \longrightarrow \mathbb{C}$$

$$a(x, p)$$

Original source details: GELFAND-P000616

Uncertain reading. A short blackened stroke is visible between X^* and the line below.

gr. of autom. — sympl. group

$$\langle p', x\rangle - \langle p, x'\rangle.$$

$$X \oplus X^* = Z(\ell).$$

$G(\ell)$ — gr. of autom. in the sp. of distr. $\Rightarrow$ obtain its repr.

$$G(\ell) \longrightarrow \mathrm{Sp}(\ell), \qquad \text{kernel: } \dot{\mathbb{C}}$$

Original source details: GELFAND-P000616

Undeciphered passage. A short entry to the left of "kernel" cannot be read with confidence.

$$G(\ell)/\dot{\mathbb{C}} = \mathrm{Sp}(\ell)$$

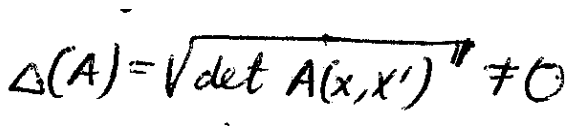
$$c \in \dot{\mathbb{C}} = \mathbb{C} \setminus 0$$

In particular, $G(\ell)$ — a Lie gr..

GELFAND-P000617

Many elem. are known — full and partial Fourier transf..

$$A : X \oplus X \longrightarrow \mathbb{R} \quad \text{real quadr.}, \qquad \Delta(A) = \sqrt{\det A(x, x')''} \neq 0$$



Original source details: GELFAND-P000617

Uncertain reading. Two primes follow the parenthesis at the end of the radical: $A(x, x')''$.

$$(S_A f')(x) = c\left(\frac{|\nu|}{2\pi i}\right)^{\ell/2} \Delta(A) \int_X e^{\nu A(x,x')} f'(x')\, d^\ell x'$$

$$(x,p) = s_A(x',p') \quad \Leftrightarrow \quad p = A_x,\ p' = -A_{x'}$$

— the corr. sympl. transformation.

S' — a unit. operator.

Uncertain reading. The short stroke at the top of S is the prime in S'.

Original source details: GELFAND-P000617

Unitarity $\Rightarrow$ metapl. group $M_p(\ell)$

I. Segal ; A. Weil ; K. Siegel

Math. phys. arbitr. field $\longrightarrow$ to understand Siegel's work

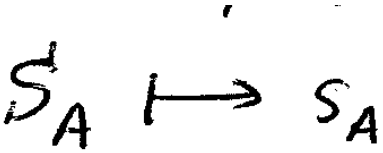
Uncertain reading. The note beneath Weil reads "arbitr. field". The arrow leads to "to understand Siegel's work".

Original source details: GELFAND-P000617

$$M_p(\ell) \ni \text{[crossed out]} \quad S'_A \longrightarrow s_A$$

Undeciphered passage. The entry after the membership sign is heavily crossed out.

Original source details: GELFAND-P000617



Uncertain reading. The stroke at the first S_A is the prime in S'_A.

Original source details: GELFAND-P000617

this is the formula with $|c| = 1$.

Take $c = \pm 1$

$$S_A \cdot S_{A'} = S_{A''}$$

$$\text{[crossed out]} \quad \underline{\underline{S_A S_{A'} S_{A''} = \pm E}}$$

$$\Uparrow \quad \overset{\text{sign}}{\sim} \text{ Maslov ind.}$$

$$s_A s_{A'} s_{A''} = e$$

Undeciphered passage. An entry to the left of the doubly underlined equality is crossed out.

Original source details: GELFAND-P000617

Decompose:

$$A(x,x') = \langle Px, x\rangle - \langle Lx, x'\rangle + \langle Qx', x'\rangle, \qquad P, Q, L : X \longrightarrow X'.$$

. . . . . .

Uncertain reading. A crossed stroke is visible to the left of the row of dots.

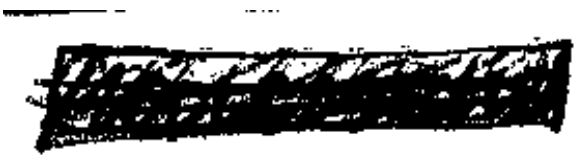
Original source details: GELFAND-P000617

GELFAND-P000618

—— · ——

13.10.75. Gelfand Sem.

Fradkin. (Efim [crossed out] Samoilovich).



Undeciphered passage. An entry before "Samoilovich" is crossed out.

Original source details: GELFAND-P000618

What does one need to know?

1. Quant. mechanics: Landau
2. Field theory Bogoliubov, Shirkov
   Akhiezer and Berestetskii
   Pauli — old.

Fradkin, Proceedings of FIAN, (field theory and stat. from the funct. approach)
vol. 29 (old) ↓ briefly. S-matr. and so on..

3. Theory of group reps. Gelfand, Minlos, Shapiro

Uncertain reading. The middle surname reads "Minlos".

Original source details: GELFAND-P000618

4. Anticommuting vars. (Fermi vars.) Berezin; methods of second quant..

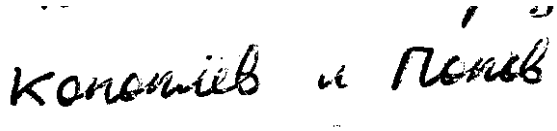
5. Abrikosov, Gor'kov, Dzyaloshinskii, "Methods of Field Theory in Stat. Physics" (book).

Original source details: GELFAND-P000618

Diagram and typed description.

The digitized frame's yellow icon obscures part of the name "Gor'kov" and nearby strokes.

6. "Gauge Fields" Konoplev and Popov.



Original source details: GELFAND-P000618

Uncertain reading. The surnames in the manuscript tentatively read "Коноплёв и Попов".

7. Reviews by Weinberg, Lee and others in

Perelomov. On completely int. Ham. systems. ($\exists n$ first int. in involution)

$$H = \frac{1}{2}\sum_{k=1}^{n} p_k^2 + U(q_1, \ldots, q_n)$$

G. Liouville's thm.: the syst. is then solved by quadratures.

$$H = \tfrac{1}{2}p^2 + \sin q$$

2nd int. — elliptic fn..

0th example — [crossed out] chain [crossed out] Toda.

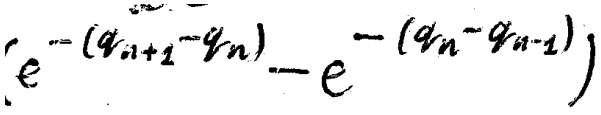
Original source details: GELFAND-P000618

Undeciphered passage. An entry before "chain" is crossed out.

Original source details: GELFAND-P000618

Undeciphered passage. An entry before "Toda" is crossed out.

$$U = g^2 \sum_{k=1}^{n} \left(e^{-(q_{n+1}-q_n)} - e^{-(q_n-q_{n-1})}\right), \qquad q_{k+n} = q_k.$$



Original source details: GELFAND-P000618

Uncertain reading. The summation starts at $k = 1$. The exponents use $n+1, n, n-1$, and a minus separates the exponentials.

[crossed out] S. Manakov ξ 1974; Toda 1970 (Prog. Th. Ph.);

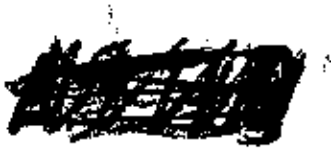
Original source details: GELFAND-P000618

Undeciphered passage. An entry before “S. Manakov” is crossed out.

Original source details: GELFAND-P000618

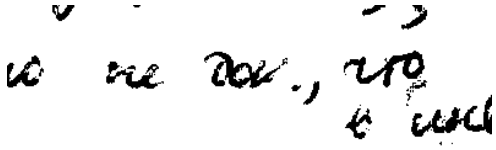
Uncertain reading. A heavily inked mark resembling ξ appears between “Manakov” and the date.

1973 Hénon, Flaschka [crossed out] 1974 ⟶ there were int., but it was not proved that they were in invol..



Original source details: GELFAND-P000618

Undeciphered passage. An entry after “Flaschka” is crossed out.

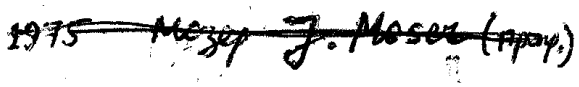


Original source details: GELFAND-P000618

Uncertain reading. The abbreviation after “not” reads “док.” (proved).

M. Kac, van Moerbeke 1975.

[crossed out ~~1975 Moser J. Moser (prepr.)~~]

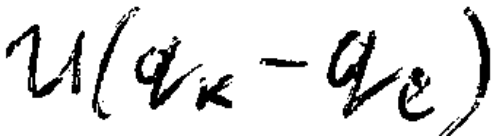


Original source details: GELFAND-P000618

Uncertain reading. The Moser reference is crossed out. The last abbreviation tentatively reads “prepr.”.

A:

$$U = g^2 \sum_{k<\ell} U(q_k - q_\ell).$$



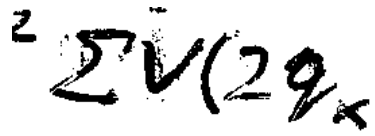
Original source details: GELFAND-P000618

Uncertain reading. The letter under the summation sign in A is U.

G. Problems: a) find the int. b) prove invol. c) solve the inv. problem d) express the int. by quadratures.

B.

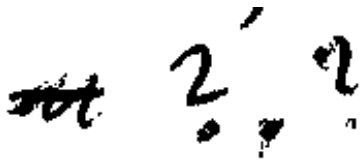
$$U = g^2 \sum_{k<\ell} [V(q_k - q_\ell) + \varepsilon V(q_k + q_\ell)] + g_1^2 \sum V(q_k) + g_2^2 \sum V(2q_k$$



Undeciphered passage. The closing parenthesis is not visible at the right edge of the frame.

Original source details: GELFAND-P000618

	$V(q)$	I_k	$\{I_k, I_\ell\} = 0$	I.P. (inv. problem)	$x(q)$
1	q^{-2}	M, OP	M, OP	?	q^{-1}
2	$a^2 \operatorname{sh}^{-2} aq$	CMR, OP	CMR, OP	?	$a \operatorname{cth} aq$
3	$a^2 \sin^{-2} aq$	M, OP	CMR, OP	?	$a \operatorname{ctg} aq$
4	$a^2 \wp(aq \mid \omega_1, \omega_2)$	C, OP	?, ?	?	$a \frac{\operatorname{cn} aq}{\operatorname{sn} aq}$
5	$q^{-2} + a^2 q^2$	P, P	ϕ, ϕ	?	



Undeciphered passage. In row 4 a short entry before the two question marks is crossed out.

Original source details: GELFAND-P000618

Uncertain reading. The involution column in row 5 reads ϕ, ϕ.

Original source details: GELFAND-P000618

$\wp$ — Weierstrass fn.. 2, 3 — limiting cases of 4.

Undeciphered passage. Separate strokes before the note on the columns are crossed out.

Original source details: GELFAND-P000618

1 column — A

2 —"— — B

GELFAND-P000619

1975 J. Moser prepr.

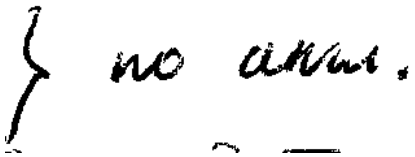
F. Calogero, M., R., (3 authors) Lett. Notes

Uncertain reading. The manuscript reads “Lett.” before “Notes”.

Original source details: GELFAND-P000619

Calogero

OP Olshanetsky, Perelomov
P Perelomov
} by name



Uncertain reading. The words beside the brace read “по имен.” (by name).

Original source details: GELFAND-P000619

Coleman, Schroer — connection [crossed out ~~with the mod.~~] of the Thirring mod. with sin-Gordon.

Uncertain reading. The crossed-out entry before “Thirring mod.” reads “с мод.” (with the mod.).

Original source details: GELFAND-P000619

0 — inv. problem — Stieltjes.

Uncertain reading. The surname after “inv. problem” reads “Стильтьес” (Stieltjes).

Original source details: GELFAND-P000619

$$0. \qquad g^2 \sum_{k=1}^{n} e^{-(q_{k+1}-q_k)}$$

The inverse problem is solved nowhere (except case 0).

[left margin || types 1–4.] The int. always have the form $I_k = \mathrm{Sp}(L^k)$, where the explicit form of L is written out (L-matrix).

$$L = P + iX, \qquad P_{k\ell} = p_k \delta_{k\ell}, \qquad X_{k\ell} = (1 - \delta_{k\ell}) x(q_k - q_\ell),$$

[under $P_{k\ell}$: diag. matr..] where $x(q)$ is written in the table.

The inv. problem here is to find matr. entries from the traces of powers. Idea: from Lax in KdV.

$$N = 5 \quad I_k = \mathrm{Sp}(N^k), \quad N = L^2 + \omega^2 Q^2, \qquad \text{where } Q_{ij} = q_j \delta_{ij}$$

Uncertain reading. $N = 5$ is written separately to the left of the formula.

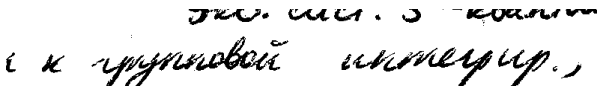
Original source details: GELFAND-P000619

Manakov — finite-dim. analogue of higher KdV. Treated the first two. [under Manakov: (dissert.)] Quant. syst. have also been treated and are also of interest.

$$p = -i\frac{\partial}{\partial q}.$$

Then $H =$ Laplace oper. on a symm. sp. (radial part). Equiv. to syst. 3 — the quantized one.

Gelfand's question: can everything be reduced to group integrability, but in a larger sp..



Uncertain reading. The abbreviation after "group" reads "интегрир." (integrability).

Original source details: GELFAND-P000619

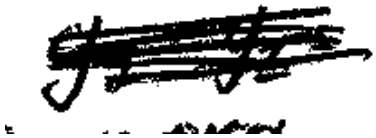
— · —

Connection with Dynkin diag.: $\{\alpha\}$ — roots, $q_\alpha = (\alpha, q)$.

$$U = \sum_{\alpha>0} g_\alpha^2 V(q_\alpha), \qquad g_\alpha = g_\beta, \quad \text{if } \alpha \text{ and } \beta \text{ have equal lengths}$$

$\alpha_{ij} = e_i - e_j$ — one obtains Moser's syst.. A — root syst. of type A_n. B — at diff. values of const. B, C, BC, D etc..

[crossed out] There are also all the exceptional groups.



Undeciphered passage. A short entry before "There are" is crossed out.

Original source details: GELFAND-P000619

It is unclear as yet whether the syst. for exceptional groups are completely integrable.

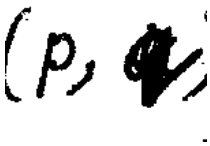
Fradkin. E. S. What is new in the last 20 years?

1. Consider an arbitr. system. It is given by a Lagr. or a Hamiltonian $H(p, q)$ (classical for now).

p, q can be Fermi and Bose (a mixture).

p, q — either Bose or Grassmann (Fermi)
$\downarrow$
$pq + qp = 1.$
(see Berezin).

There are also constraints $\varphi_k(p, q) = 0$.

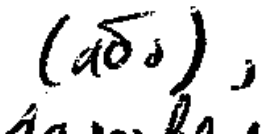


Uncertain reading. The overwritten second argument reads q.

Original source details: GELFAND-P000619

(distinguish between Bose and Fermi vars.).

Example: gauge fields (electrodynamics (Ab.), Yang–Mills, gravit. field, Einstein eqs.), grad. inv. of Maxwell's eqs..



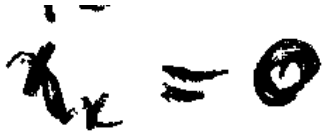
Original source details: GELFAND-P000619

Uncertain reading. The parenthesis after "electrodynamics" reads "аб." (Ab.).

and light is in 2 states, rather than 3, the extra one is eliminated by gauge inv. all fields with higher spins.

$\varphi_k = 0$ Bose constraints

$\chi_k = 0$ Fermi constraints



Original source details: GELFAND-P000619

Uncertain reading. The overwritten first sign in the last line reads χ.

GELFAND-P000620

[top left, crossed out: $\prod(d\varphi, d\pi, d\psi, d\pi^{(\psi)})$.]

Original source details: GELFAND-P000620

Uncertain reading. Parts of a product of differentials are legible. The formula is crossed out with diagonal strokes.

Lagrangian $L(q, \dot{q})$, $p = \dfrac{\partial L}{\partial^r \dot{q}}$. Poisson bracket: ($r$ — denotes the right derivative).

Original source details: GELFAND-P000620

Uncertain reading. The heading uses the singular "bracket".

$$\{A, B\} = \frac{\partial A}{\partial^r q}\frac{\partial B}{\partial^\ell p} - (-1)^{n_A n_B}\frac{\partial B}{\partial^r q}\frac{\partial A}{\partial^\ell p}.$$

$n_A = 0$, if the number of Fermi vars. is even

$n_A = 1$ —"— odd

(in A).

Nondegeneracy: $\dot{q} = f(q, p)$ (can be expressed). Sometimes it cannot: when $\det \dfrac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} = 0$.

Then addit. constraints arise.

One can proceed in the same spirit and obtain (see Dirac's slim book). all the constraints. [crossed out] Close under the Poiss. bracket (to a Lie alg.).

Original source details: GELFAND-P000620

Undeciphered passage. A parenthesized entry after "all the constraints" is crossed out.

ψ_k — addit. conditions obtained in this way.

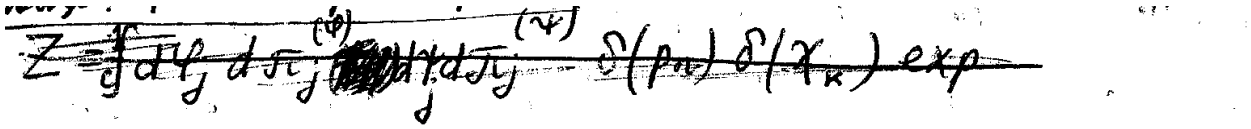
Dirac: [circled important] second-class constraints, between which $\{\varphi_k, \psi_j\} \neq 0$ taking the addit. conditions into account. Discard constraints belonging to the center, etc.

Uncertain reading. The first abbreviation reads “Выбрас.” (discard).

Original source details: GELFAND-P000620

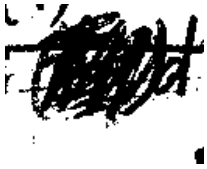
Generating fn. for all Green’s fns.

[crossed out] ~~$Z = \frac{1}{C}\int d\varphi_j\, d\pi_j^{(\varphi)} \ldots d\psi_j\, d\pi_j^{(\psi)}\, \delta(\varphi_k)\delta(\chi_k)$ exp~~



Original source details: GELFAND-P000620

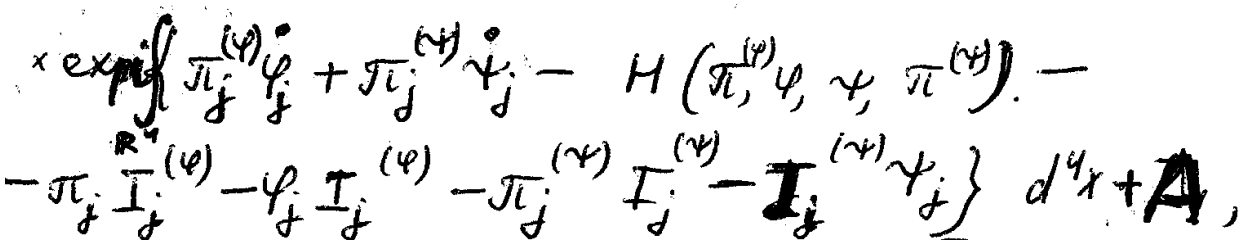
Uncertain reading. The whole preceding line is crossed out. The legible parts are typeset.



Undeciphered passage. A measure factor in the crossed-out line is obscured by heavy ink.

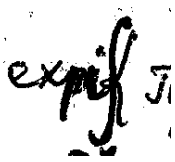
Original source details: GELFAND-P000620

$$Z = \frac{1}{C}\int \prod_x d\varphi_j(x)\, d\pi_j^{(\varphi)}(x)\, d\psi_j(x)\, d\pi_j^{(\psi)}(x)\, \delta(\varphi_k)\delta(\chi_k)\times$$
$$\times \exp i \int_{\mathbb{R}^4} \Big\{ \pi_j^{(\varphi)}\dot{\varphi}_j + \pi_j^{(\psi)}\dot{\psi}_j - H(\pi^{(\varphi)}, \varphi, \psi, \pi^{(\psi)}) -$$
$$- \pi_j \bar{I}_j^{(\varphi)} - \varphi_j I_j^{(\varphi)} - \pi_j^{(\psi)} I_j^{(\psi)} - \bar{I}_j^{(\psi)}\psi_j \Big\}\, d^4x + A,$$



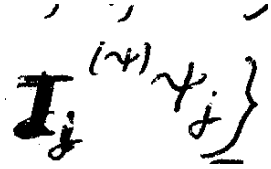
Original source details: GELFAND-P000620

Uncertain reading. A minus is visible at the end of the H line and at the start of the next. The closing brace precedes $d^4x + A$.



Uncertain reading. The overwritten sign between exp and the integral reads i.

Original source details: GELFAND-P000620

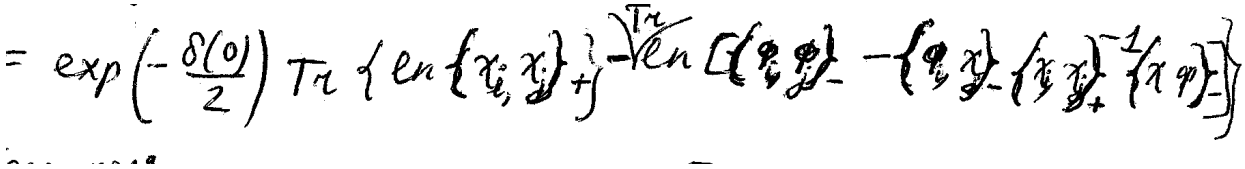


Original source details: GELFAND-P000620

Uncertain reading. The subscript on the final I reads j. The upper stroke merges with the letter.

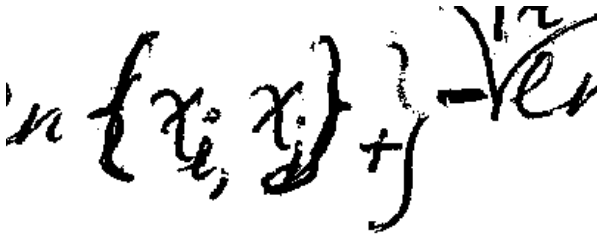
where

$$A = \exp\left(-\frac{\delta(0)}{2}\right) \operatorname{Tr}\{\ln\{\chi_i, \chi_j\}_+\} \\ - \operatorname{Tr}\ln\left[\{\varphi_i, \varphi_j\}_- - \{\varphi_i, \chi_j\}_-\{\chi_i, \chi_j\}_+^{-1}\{\chi\varphi\}_-\right]\Big\}.$$



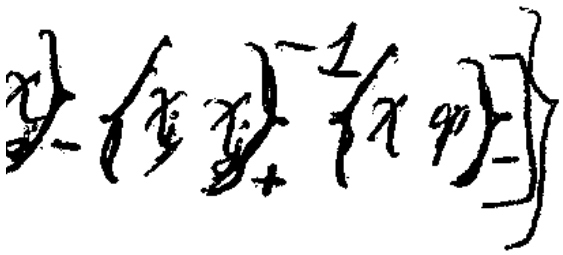
Original source details: GELFAND-P000620

Uncertain reading. The grouping after $\delta(0)/2$ and at the end follows the manuscript.



Original source details: GELFAND-P000620

Uncertain reading. An additional closing brace follows the first bracket with a plus sign. The second Tr is inserted above the line.



Original source details: GELFAND-P000620

Uncertain reading. The last bracket contains χ and φ without explicit indices.

φ — Bose fields
ψ — Fermi fields

Tr of matrices

(general formula for all quantum theories).

φ, χ — constraints (only noncommuting ones).

I — "sources" φ and ψ — elem. of the conj. space.

One obtains an analogue of the Fourier transform.

A — a term associated with the constraints.

$\delta(0)$ then cancels (in the S-matrix)

$\{\chi, \chi\}_\pm$ — Poisson brackets (Bose and Fermi)

—— · ——

GELFAND-P000621

$20/\overline{X} - 75.$ Gelfand Sem.

I. M. Gelfand:

$$L = -\frac{d^2}{dx^2} + u(x), \qquad A = \frac{d^3}{dx^3} + \dots, \qquad [L, A] = K(u).$$

$A = L^{3/2}$ with the nondifferential terms cut off.

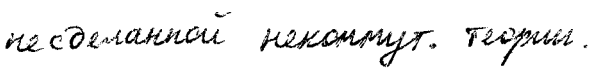
1-zone potential: $K(u) = 0$. The point (hypothesis): for u the operators L and A commute.

Higher eqs.. $A = L^{5/2}, L^{7/2}, \dots$ almost commute — the remaining oper. is multiplication by a fn.

[crossed out] Hypothesis: this is a piece of an unworked-out noncommut. theory.

Original source details: GELFAND-P000621

Undeciphered passage. A short entry before "Hypothesis" is crossed out.



Original source details: GELFAND-P000621

Uncertain reading. The adjective before "noncommut. theory" reads "несделанной" (unworked-out).

Krichever: in fact started from this.

[crossed out] Krichever. Igor.

Original source details: GELFAND-P000621

Undeciphered passage. An entry to the left of the name is crossed out.

Consider the operator eq. $[L_n, L_m] = 0$, where

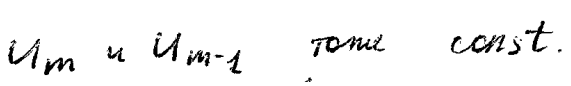
$$L_n = \sum_{i=0}^{n} u_i(x) \frac{d^i}{dx^i}, \qquad u_n(x) \equiv c_0, \quad u_{n-1} \equiv c_1$$

$$L_m = \sum_{i=0}^{m} v_i(x) \frac{d^i}{dx^i},$$

[crossed out ~~$u_m(x) \equiv$~~] u_m and u_{m-1} are also const.

Original source details: GELFAND-P000621

Uncertain reading. The crossed-out expression tentatively reads $u_m(x) \equiv$.



Original source details: GELFAND-P000621

Uncertain reading. The two constant coefficients in this line are written with u.

(operators of this kind are always considered).

Write the syst. of eqs.:

$$0 = \sum_{k,\ell} u_k C_k^{k+\ell-s} v_\ell^{(k+\ell-s)} - \sum_{k,\ell} v_k C_k^{\ell+k-s} u_\ell^{(\ell+k-s)}$$
$$0 \le s \le m+n-3.$$

Number of eqs. $=$ number of unknowns.

$$mu'_{n-2} - nv'_{m-2} = 0 \quad \Rightarrow \quad v_{m-2} = \frac{m}{n} u_{n-2} + h_2 \qquad (h_2 = \text{const}).$$

It is easy to check that the first [above the line $(m-1)$] eqs. (for $n-1 \le s \le m+n-3$) allow one to express all v_i as polynomials in u_i (and their derivatives). Also [crossed out] constants of integration arise.

Undeciphered passage. A short entry after "Also" is crossed out.

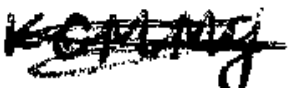
Original source details: GELFAND-P000621

Substituting them in the eqs. $0 \le s \le n-2$, we obtain eqs. for the [crossed out] coeffs. of L_n.

Undeciphered passage. An entry before "coeffs." is crossed out.

Original source details: GELFAND-P000621

for $L = -\dfrac{d^2}{dx^2} + u(x)$ one obtains for $m = 2k+1$ a set of oper., [crossed out] whose commutator is multiplication by a fn.



Undeciphered passage. A word before "whose" is crossed out.

Original source details: GELFAND-P000621

For u one eq. remains, the higher KdV eq.. The family [crossed out] depends on the const. of integration.

Undeciphered passage. An entry between "The family" and "depends" is crossed out.

Original source details: GELFAND-P000621

Theorem (A. B. Shabat). If [crossed out] for L_n there exists an oper. L_m:

Undeciphered passage. Part of the hypothesis before "for" is crossed out.

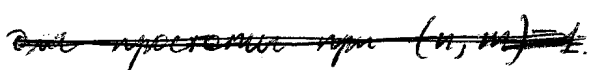
Original source details: GELFAND-P000621

$$[L_n, L_m] = 0, \quad \text{then} \quad \exists \text{ polynomial } P(x,y): \qquad P(L_n, L_m) = 0$$

(commutative).

GELFAND-P000622

Proof. [crossed out ~~all simple when $(n,m)=1$.~~]



Uncertain reading. The crossed-out phrase tentatively reads “all simple when $(n,m)=1$”.

Original source details: GELFAND-P000622

Let y — a solution of the eq.:

$$L_n y = Ey, \qquad y_j(x,E) \text{ — basis of solns. (entire fns. of } E\text{).}$$

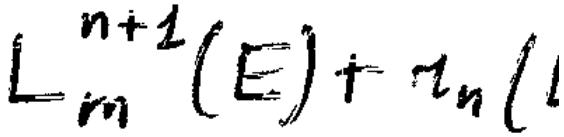
$$y_i^{(j)}(x_0, E) = \delta_{ij}$$

Uncertain reading. The equals sign meets the overwritten parenthesis.

Original source details: GELFAND-P000622

L_m gives a lin. oper. in the sp. of solutions of $L_n y = Ey$ $(\forall E)$. Its char. polynomial

$$0 = L_m^{n+1}(E) + r_n(E) L_m^n(E) + \ldots + r_0(E).$$



Uncertain reading. The first exponent is written $n+1$.

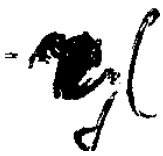
Original source details: GELFAND-P000622

Coeffs. — entire fns. of E. Prove that they are polynomials in E. This follows from the asymptotics of $y_i(E)$ in E (as $E \to \infty$).

$$y_i(x,E) \sim \sum_{j=0}^{n-1} a_{ij} e^{\varepsilon_j k x}, \qquad \varepsilon_j^n = 1, \quad k^n = E.$$

Hence the growth order of $r_j(E) \leq$ a polyn. in $k \Rightarrow$ they are themselves polynomials.

$$r_j(E) y = r_j(L_n) y$$



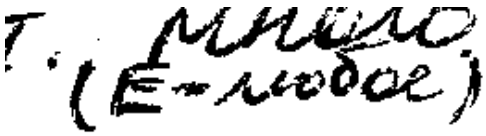
Uncertain reading. The r on the right is overwritten.

Original source details: GELFAND-P000622

Hence the char. polynomial

$$P(L_n, L_m)y = 0.$$

But there are sufficiently many solutions (E — arbitrary).



Uncertain reading. Below “many” the manuscript reads “E arbitrary”.

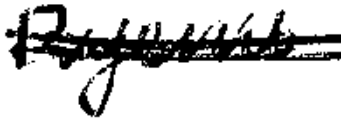
Original source details: GELFAND-P000622

[circled inf.-dimensional sp.]

$$\Rightarrow P(L_n, L_m) = 0. \qquad \triangleright$$

I. Bernstein: for polynomial coeffs. it was in F. A., vol. 1, no. 4.

[crossed out ~~Take $\psi_i(x, E)$ — eigenfn.~~]



Uncertain reading. The opening word is crossed out. The first letter looks like “В”, and the whole word tentatively reads “Возьмём” (take).

Original source details: GELFAND-P000622

Uncertain reading. The crossed-out line contains $\psi_i(x, E)$ and “соб.” (eigenfn.).

Original source details: GELFAND-P000622

Let $\psi_i(x, E)$ be an eigenfn. for the pair of op.:

$$L_n\psi_i(x, E) = E\psi_i(x, E)$$
$$L_m\psi_i(x, E) = E_i\psi_i(x, E).$$

Then consider the surf. [left, crossed out ~~$\psi(x, P)$~~.]

Uncertain reading. The entry under the crossing-out tentatively reads $\psi(x, P)$.

Original source details: GELFAND-P000622

$P(x, y) = 0$, x, y — eigenvalues of L_n and L_m.

$\Rightarrow E_i$ alg. fns. of E,

$$\psi|_{x_0} = 1 \text{ (norm.)}.$$

$\Psi(x, P)$ — a fn. on this surf. [crossed out ~~BAKER~~.]

Uncertain reading. The hatched entry tentatively reads “BAKER”. Some letters are obscured.

Original source details: GELFAND-P000622

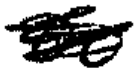
$$P \in X.$$

—— · ——

Consider $\forall$ nonsing. alg. curve X with marked point $P_0 \in X$.

The following properties uniquely determine the fn. $\Psi(x, P)$, $x \in (a, b)$, $P \in X$. (Akhiezer).

1°. $\psi(x, P)$ is meromorphic on [crossed out ~~x_0~~] $X \setminus P_0$, has polar divisor D with $\deg D = g$, and D is nonspecial and independent of x.



Uncertain reading. The entry crossed out before X tentatively reads x_0.

Original source details: GELFAND-P000622

2°.

$$\psi(x, P) = e^{k(x-x_0)} \underbrace{\left(1 + \sum_{i=1}^{\infty} \frac{\xi_i(x)}{k^i}\right)}_{\text{regular fn. near } P_0}, \qquad \frac{1}{k} = z \text{ — loc. parameter near } x_0.$$

Uncertain reading. The overwritten letter in the numerator tentatively reads ξ.

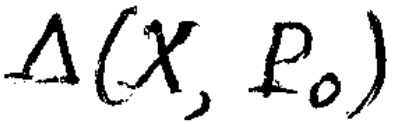
Original source details: GELFAND-P000622

Uncertain reading. The note on the right reads "near x_0". The label under the bracket reads "near P_0".

Original source details: GELFAND-P000622

GELFAND-P000623

Introduce the ring $\Delta(X, P_0)$ of fns. rat. on the curve X with a single sing. at P_0.



Uncertain reading. The first sign resembles Δ with a thin lower stroke. Below, the ring is written with Λ.

Original source details: GELFAND-P000623

Theorem. $\forall E \in \Lambda(X, P_0)$, $\exists!$ diff. oper. L, such that

$$L\Psi(x, P) = E(P)\psi(x, P)$$

[crossed out] Here if

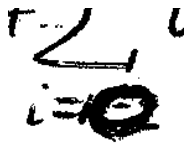
$$E(P) = c_n k^n + c_{n-1} k^{n-1} + \ldots + c_0 + \ldots$$

Undeciphered passage. An entry before "Here if" is crossed out. It may read "Д-во" (proof).

Original source details: GELFAND-P000623

($1/k$ — loc. param. near P_0), then

$$L = c_n \frac{d^n}{dx^n} + c_{n-1} \frac{d^{n-1}}{dx^{n-1}} + \sum_{i=0}^{n-2} u_i(x) \frac{d^i}{dx^i},$$



Uncertain reading. The lower limit is overwritten and reads $i = 0$.

Original source details: GELFAND-P000623

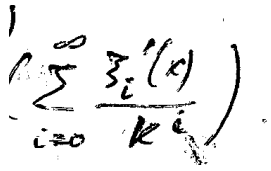
where u_i — certain fns..

Proof.

$$\psi(x, P) = e^{k(x-x_0)} \left(1 + \sum_{i=1}^{\infty} \frac{\xi_i(x)}{k^i} \right)$$

Differentiate in x: the poles are the same (the same divisor), and near P_0 (apply L) obtain

$$\frac{d}{dx}\psi(x, P) = k\Psi + e^{k(x-x_0)} \left(\sum_{i=0}^{\infty} \frac{\xi_i'(x)}{k^i} \right).$$



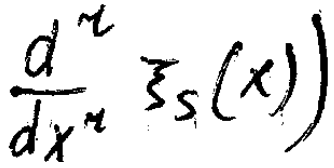
Manuscript excerpt. The lower limit in the differentiated sum is written $i = 0$.

Original source details: GELFAND-P000623

i.e. one obtains a new pole at P_0.

Seek L by requiring the sing. parts to coincide for $L\psi$ and $E\psi$. We obtain:

$$u_i(x) = Q\left(c_j, \frac{d^r}{dx^r}\xi_s(x) \right), \qquad \begin{array}{l} 0 \le s \le i-1, \\ 0 \le r \le i-s, \end{array}$$



Manuscript excerpt. The derivative order in the numerator and denominator reads r.

Original source details: GELFAND-P000623

i.e. u_i is found explicitly from the asymptotics.

Mapping $E \longrightarrow L$.

Uncertain reading. The letter at the end of the arrow is overwritten. It tentatively reads L.

Original source details: GELFAND-P000623

This is a ring homomorphism

$$\varphi : \Lambda(X, P_0) \longrightarrow \mathcal{L} \qquad \text{(diff. oper.)}$$

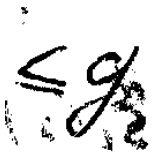
The image is a commut. ring. [left] Let n — the least possible, Let $\mu_1, \ldots, \mu_{n-1}$ — be numbers such that $\exists \psi$ with a pole of ord. $s + \mu_s n$.

$$\mu_1 + \ldots + \mu_{n-1} = g.$$

At a point in gen. position — g algebraically independent commuting operators.

Def. A diff. op. L_n is called algebraic of genus g, if it arises in this construction.

Theorem. L_n — alg. of genus $\leq g \Longleftrightarrow \exists$ operators $L^{(s)}$, $1 \leq s \leq n-1$, such that



Uncertain reading. There is a separate dark trace below g. An index cannot be read with confidence.

Original source details: GELFAND-P000623

$$[L_n, L^{(s)}] = 0, \quad \text{and} \quad [L^{(s)}, L^{(1)}] = 0, \qquad \text{ord}\, L^{(s)} = s + \mu_s n,$$

Uncertain reading. The second superscript in the commutator reads (1).

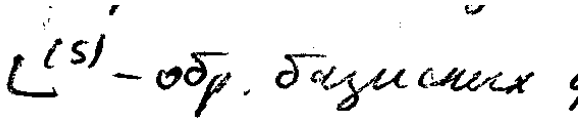
Original source details: GELFAND-P000623

where μ_i are such that $\mu_1 + \ldots + \mu_{n-1} = g$.

GELFAND-P000624

Remark. If op. commute with a pair, they commute with one another.

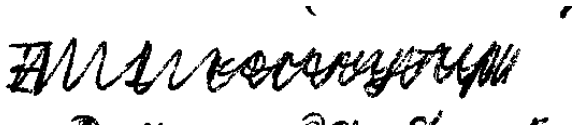
Proof: a) necessity is obvious ($L^{(s)}$ — "обр." of basis fns.) from the preceding.



Uncertain reading. The abbreviation reads "обр.".

Original source details: GELFAND-P000624

b) sufficiency — [crossed out: constr...]



Undeciphered passage. The crossed-out word before "construct" begins "constr...". The ending is illegible.

Original source details: GELFAND-P000624

from the eigenvalues construct fns. on the curve; suff. for genus g.

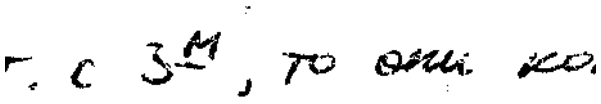
Uncertain reading. A separate closing mark appears after g.

Original source details: GELFAND-P000624

Limanov: if 2 oper. with polyn. coeffs. commute with a third, then they commute with one another (oper. in 1 variable).

Original source details: GELFAND-P000624

Manuscript excerpt. The surname is written “Лиманов” (Limanov).



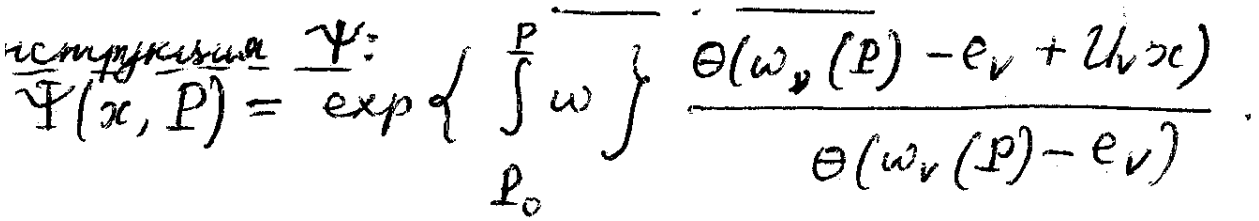
Original source details: GELFAND-P000624

Manuscript excerpt. “With a third” is written as 3 with a raised Cyrillic “м” and a short stroke.

—·—

Construction of Ψ:

$$\Psi(x,P)=\exp\left\{\int_{P_0}^{P}\omega\right\}\frac{\Theta\big(\omega_\nu(P)-e_\nu+U_\nu x\big)}{\Theta\big(\omega_\nu(P)-e_\nu\big)}.$$

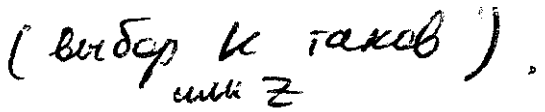


Original source details: GELFAND-P000624

Uncertain reading. The two strokes and dot above the formula form a divider. No factor x is written in the exponent.

$$\omega_2=\frac{dz}{z^2}\qquad\text{(the choice of } k \text{ is such)}.$$

[Beneath k: or z.]

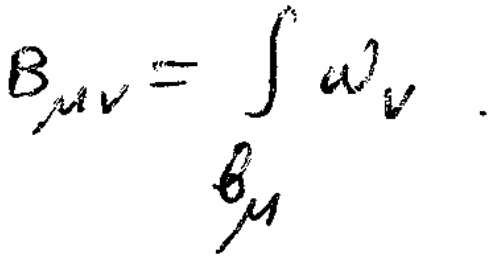


Uncertain reading. “or z” is written below the line.

Original source details: GELFAND-P000624

$$\omega_\nu(P) = \int_{P_0}^{P} \omega_\nu, \qquad \omega_\nu \text{ — Ab. diff. on the curve.}$$

$$\omega_\nu(P) = \int_{P_0}^{P} \omega_\nu, \qquad \int_{a_i} \omega_\nu = \delta_{i\nu}, \qquad B_{\mu\nu} = \int_{b_\mu} \omega_\nu.$$



Manuscript excerpt. The curved stroke at right forms part of the letter ω.

Original source details: GELFAND-P000624

$$\Theta(u_1, \dots, u_n) = \sum_{m_1,\dots,m_n} \exp\left\{2\pi i \,[\text{crossed out}] \sum_{\mu,\nu} B_{\mu\nu} m_\mu m_\nu + \sum_\nu m_\nu u_\nu\right\}.$$

Undeciphered passage. The mark after $2\pi i$ is heavily crossed out.

Original source details: GELFAND-P000624

—·—

GELFAND-P000625

27/X-75. Gelfand Sem.

Dubrovin.

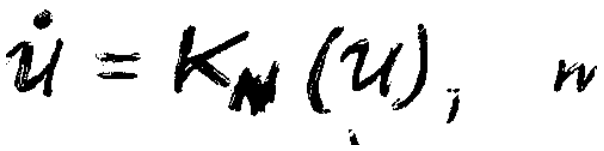
$\forall$ oper. L $\exists$ a series of eqs. of KdV type.

$$L = \frac{d}{dx} + U(x), \qquad U(x) \text{ — matr. } n \times n. \qquad (u_i^j(x)) \tag{1}$$

The eq. will be

$$\dot{U} = K_N(U); \qquad m = 1, 2, \dots. \tag{2}$$

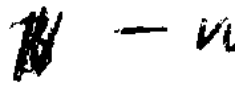
[Below $K_N(U)$:] nonlin. diff. op. in x.



Uncertain reading. The overwritten index on K tentatively reads N. To the right is $m = 1, 2, \dots$.

Original source details: GELFAND-P000625

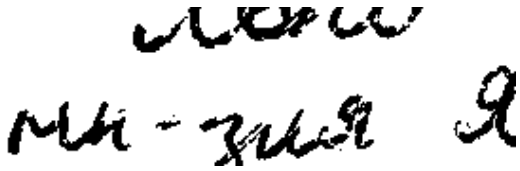
$K_m(U) = 0$ — a completely int. Ham. syst. which is solved explicitly. It admits a commut. repr. of Lax type on matrices $\Lambda' = [\Lambda, Q]$. (matr. entries of Λ — polynomials in $U, U', \ldots$).



Uncertain reading. The symbol before the dash is overwritten. It tentatively reads Λ.

Original source details: GELFAND-P000625

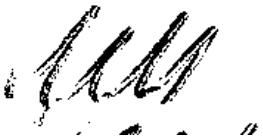
Eigenvalues — integrals of ord. eqs.. It is solved explicitly!! Introduce an extra par. E, alg. geom. arises: Jacobi varieties of Riemann surfaces.



Manuscript excerpt. The abbreviation before "Jacobi" reads "мн-зия" (varieties).

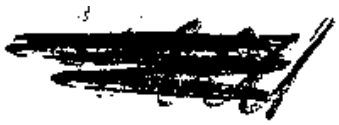
Original source details: GELFAND-P000625

Consider the spectral problem $L\Psi = EA\Psi$, $L = \dfrac{d}{dx} + U(x)$; [Upper right: uncertain entry] [crossed out].



Uncertain reading. Slanted strokes are visible above the crossed-out entry. No reading is secure.

Original source details: GELFAND-P000625



Undeciphered passage. The entry to the right of the spectral equation is heavily crossed out.

Original source details: GELFAND-P000625

$$A = (a_i \delta_i^j), \qquad a_i \neq a_j.$$

Let next $U_i^i = 0$ (this can always be achieved).

From it we construct eqs. preserving the spectrum of this problem.

$$Q = U - EA.$$

Consider the eq. $\lambda' = [\lambda, Q]$ and its soln. as a formal series

$$\lambda_B = B + \frac{\lambda_{1,B}}{E} + \frac{\lambda_{2,B}}{E^2} + \cdots, \qquad \lambda_B = \lambda_B(x, E). \tag{3}$$

B — const. diag. matrix, $= b_i \delta_i^j$; $\lambda_{j,B}$ — polyn. in the matr. entries of [crossed out] U.

Uncertain reading. A short entry before U is crossed out.

Original source details: GELFAND-P000625

Consider the eq.

$$\dot{U} = [A, \lambda_{N+1,B}] \tag{4}$$

or their lin. comb..

This is an eq. of KdV type. All these eqs. leave the spectrum of L, unchanged, since there is a Lax-type repr.

$$\dot{L} = [L, M_N] \tag{5}$$

Now seek all stat. solns.

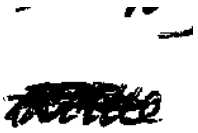
$$K_N(U) = 0. \tag{6}$$

Hence $[L, M_N] = 0$ for this U.

Lemma. The eq. $K_N(U) = 0$ [crossed out] admits a Lax-type repr. [crossed out].

Original source details: GELFAND-P000625

Manuscript excerpt. The right parenthesis in $K_N(U)$ touches the sign, which reads as an equality.



Original source details: GELFAND-P000625

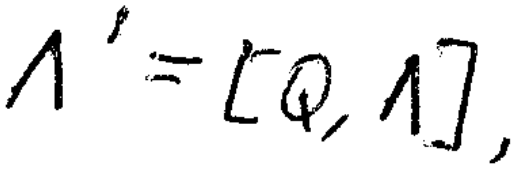
Undeciphered passage. A word immediately after the zero is heavily crossed out.

Original source details: GELFAND-P000625

Uncertain reading. A short entry after “Lax” is crossed out.

$$\Lambda' = [Q, \Lambda], \qquad \text{where } \Lambda \text{ — a polynomial in } E \tag{7}$$

(and in $U, U', \ldots$).



Original source details: GELFAND-P000625

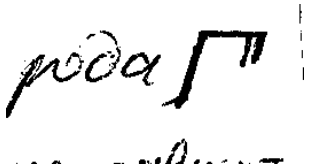
Uncertain reading. In (7), Q comes first in the commutator.

Original source details: GELFAND-P000625

Manuscript excerpt. The overwritten equation number reads (7).

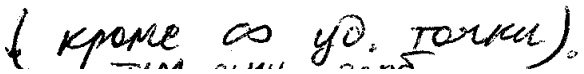
Proof. The eq. $L\Psi = EA\Psi$ has a basis of solns. $y_1, \dots, y_n$, M_N determines there (in the soln. space) a lin. transf. — a matrix, which we denote by Λ. We then have $[L, \Lambda] = 0$, i.e. $\Lambda' = [Q, \Lambda]$. ▷.

Finite-gap lin. operator. $\mathcal{L}$ — a lin. diff. operator, depending on par. E [inserted: (polynomially)] and the problem considered is $\mathcal{L}_E\Psi = 0$. $\mathcal{L}$ is called finite-gap, if the problem $\mathcal{L}_E\Psi = 0$ has an eigenfn., merom. in E on a Riemann surf. of finite genus Γ covering the E-plane (except for ∞ the distant point). (Γ is independent of x).



Original source details: GELFAND-P000626

Uncertain reading. Two short strokes descend at the right of the top bar of Γ.



Original source details: GELFAND-P000626

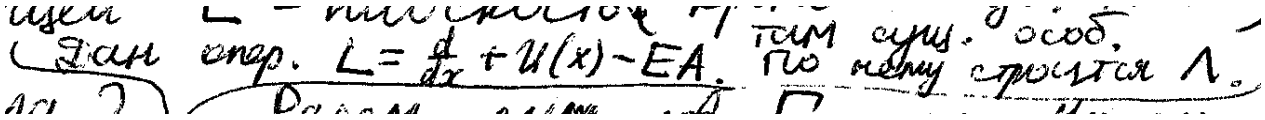
Uncertain reading. After ∞ the manuscript reads “уд. точки” (distant point).

[Vertically in the left margin: Γ is called the spec. of the oper.]

Original source details: GELFAND-P000626

Uncertain reading. The underlined entry tentatively reads “наз. спек. оп-ра”. Additional strokes are visible below the line.

Lemma 2. [inserted: Given oper. $L = \frac{d}{dx} + U(x) - EA$. From it we construct Λ. [above the insertion: there is an essent. sing. there]]

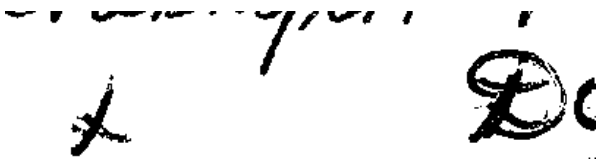


Original source details: GELFAND-P000626

Uncertain reading. The insertion begins “Given oper.” Above “From it” is “there is an essent. sing. there”.

Consider the Riem. surf. Γ, defined by the eq. $R(z, E) = 0$ ($R(z, E) = \det(z \cdot 1 - \Lambda(E))$). We then have L — finite-gap with spectrum Γ.

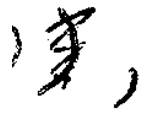
Proof.



Original source details: GELFAND-P000626

Uncertain reading. The initial letter is overwritten. There is a separate mark to the left of the heading.

Construct an eigenfn. Ψ simultaneously for L and Λ. ($L\Psi = 0$ [crossed out], $\Lambda\Psi = z\Psi$, z — the same z, as above).



Uncertain reading. A sign resembling a closing parenthesis is crossed out after the zero.

Original source details: GELFAND-P000626

Remark 1. $U(x+T) = U(x) \Longrightarrow \Psi$ [crossed out] [inserted: turns out] to be a Bloch fn..

Undeciphered passage. The word before "Bloch" is crossed out. "получ." (turns out) is inserted above it.

Original source details: GELFAND-P000626

(eigenfn. of translation by a period). (One can show that $[\Lambda, \widehat{T}] = 0$ [crossed out], and deduce the Bloch property from this).

Uncertain reading. A sign resembling a parenthesis is crossed out after the zero.

Original source details: GELFAND-P000626

Remark 2. $[L, M_N] = 0 \Longrightarrow \Psi$ — an eigenfn. of this pair.

—·—

Consider the eq. $\Lambda' = [Q, \Lambda]$. Coeffs. of the polynomial $R(z, E)$ — integrals. One can verify that the eqs. are Hamiltonian, the integr. are in involution.

Original source details: GELFAND-P000626

Uncertain reading. The final word is abbreviated "инволют.".

$$\sum_{i,j} I_{ij} \text{ [crossed out] } z^i E^j = R(z, E).$$

Uncertain reading. A separate vertical stroke stands to the left of the summation.

Original source details: GELFAND-P000626

Uncertain reading. A short sign is crossed out between I_{ij} and z^i.

Original source details: GELFAND-P000626

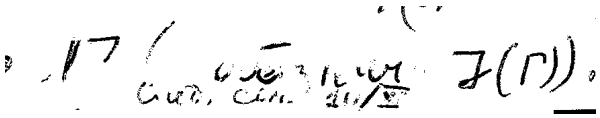
We then have $I_{ij} = I^0_{ij}$ — integrals.

Problem: determine the finite-gap oper. L with the same spectrum Γ.

Theorem. All finite-gap oper. with given spectrum Γ — the space of a principal [inserted: π-]bundle, where

$$\pi = (\text{diag. matr.})/(\text{scalar matr.})$$

over the Jacobian variety of the complete Γ ([faint entry] $J(\Gamma)$).

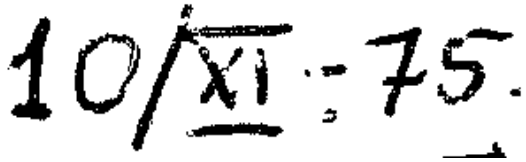


Original source details: GELFAND-P000626

Undeciphered passage. There is a separate angular stroke near Γ. Before $J(\Gamma)$ is a faint two-line entry. Parts of “см.” and of a fraction bar are visible in the lower line.

GELFAND-P000627

10/XI-75. Gelfand Sem.



Original source details: GELFAND-P000627

Uncertain reading. The month has a bar above and below. Two small marks are visible near the dash.

Original source details: GELFAND-P000627

Undeciphered passage. There is a hatched strip beneath the heading. No letters can be read with confidence.

Talk for Polyakov.

G. Torsion for schoolchildren.

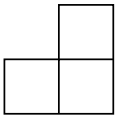
Original source details: GELFAND-P000627

Uncertain reading. The doubly underlined initial “Г.” stands in the left margin before “Torsion”.

An ant lives on a cube.

Original source details: GELFAND-P000627

Diagram and typed description.



Three faces unfolded into the plane, with no labels.

[To the right of the figure:] go round a closed path in the plane corresp. path on the cube: we will arrive at another point (torsion) and turn (curvature).

It is unclear what the basic geom. things look like on polyhedra (n-dimensional case). Char. classes and so on. [crossed out]

Undeciphered passage. A short entry after "and so on" is heavily crossed out.

Original source details: GELFAND-P000627

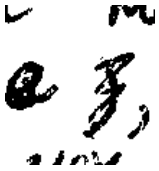
Problem: make a general theory of relativity on lattices.

A list of what one needs to know from physics (for a sensible mathematician).

1. Feynman, Hibbs «Quant. Mech. and Path Ints.».

2. Feynman's lectures for schoolchildren.

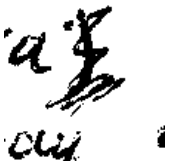
3. Quantum mechanics: Landau, Lifshitz (perturbation theory, quasiclassical approximation [crossed out], Lie groups, basic concepts of scattering theory).



Uncertain reading. A sign resembling a parenthesis is crossed out after "approximation". A comma is visible to its right.

Original source details: GELFAND-P000627

4. Relativ. quant. mech. — the beginning of Berestetskii, Lifshitz, Pitaevskii; second quantization (Dirac's book [crossed out]; Feynman «Stat. Mechanics»).



Uncertain reading. A sign resembling a parenthesis is crossed out after "Dirac". A semicolon is visible to its right.

Original source details: GELFAND-P000627

Stat. physics — Landau and Lifshitz.

GELFAND-P000628

What mathematics should physicists know?

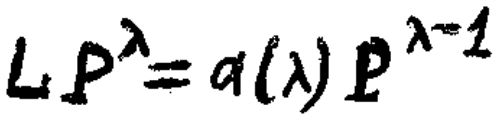
1. Group reprs.

2. Pseudodiff. operators.

3. Fourier–Hörmander integral oper..

4. P^λ (Bernstein polynomials).

$$LP^\lambda = a(\lambda)P^{\lambda-1}.$$



Uncertain reading. The coefficient on the right reads $a(\lambda)$.

Original source details: GELFAND-P000628

5. Riemann–Roch theorem.

6. Relation of sing. to top. cycles.

Polyakov.

Undeciphered passage. There is a heavily hatched strip below the surname.

Original source details: GELFAND-P000628

[Right:] TMF, the next-to-last issue; art. on the Yang–Mills field — worked out in detail. Yang, Mills — trans. in the coll. «Elem. Particles and Compensating Fields» (1964).

Uncertain reading. The abbreviation before “в сб.” (in the collection) reads “перев.” (translation).

Original source details: GELFAND-P000628

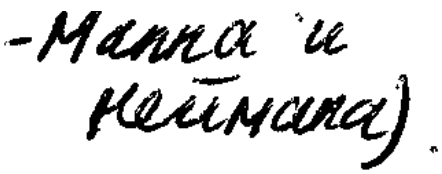
There are quarks $q \simeq \begin{pmatrix} u \\ d \\ s \end{pmatrix}$. quark spin $\frac{1}{2}$, $B \sim$ spin $\frac{1}{2}$ (baryon), M — spin 0 (meson).

$$M \sim \bar{q}q \quad \text{(quark and antiquark)},$$
$$B \sim qqq.$$

Where from: Grouping into multiplets (8 at a time). (the idea of Gell-Mann and Ne’eman).

Uncertain reading. The entry reads “мультиплеты” (multiplets).

Original source details: GELFAND-P000628



Uncertain reading. The second surname is written “Неймана”.

Original source details: GELFAND-P000628

$\Rightarrow$ Symmetries: the most suitable gr. $SU(3)$, reprs. $8, 10, 27$ (various).

Lowest representations? [crossed out] scalar, stand., and tens. products.

Uncertain reading. There is a short crossed-out mark before “scalar”.

Original source details: GELFAND-P000628

Let these be real particles! remarkable idea:

$$3^* \times 3 = 1 + 8,$$
$$3 \times 3 \times 3 = 1 + 8 + 8 + 10.$$

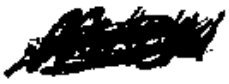
$\Rightarrow$ one cannot build mesons as 3 quarks; there are 8 of them, and $3 \times 3 = 3^* + 6$.

We scatter:

$$\frac{\sigma(BB)}{\sigma(BM)} = \frac{3}{2} \qquad \text{(cross sections)}.$$

— that is how it comes out (accur. 10%). High en. — scattering of a [crossed out] particle on a particle as of a part. on an antiparticle.

Undeciphered passage. A short word before "particle" is heavily crossed out.



Original source details: GELFAND-P000628

$qqqq\bar{q}$ — absent in experiment (called exotic).

Many particles — speeds of rot. of quarks (like the electron in the hydrogen atom). but: unstable (resonance). Trans. to the ground state.

Uncertain reading. The abbreviation before "вращ." (rotation) reads "скор.".

Original source details: GELFAND-P000628

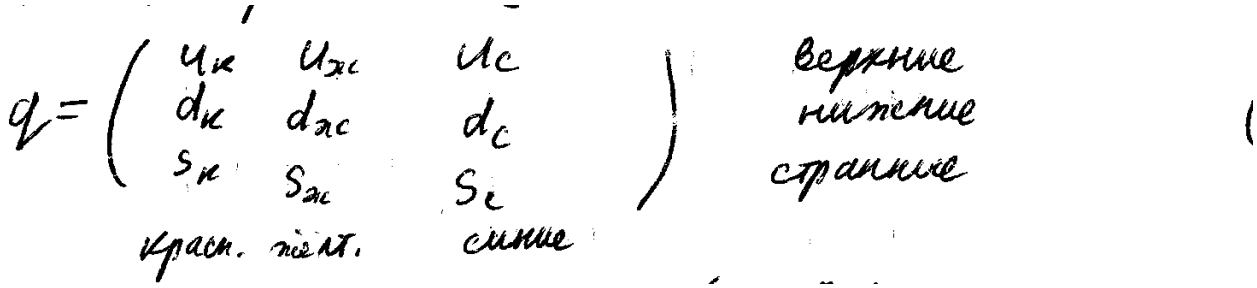
Baryon wave fn.:

$$\Delta^{++} \simeq u(x_1)u(x_2)u(x_3),$$

but antisymm. is needed.

Propose 3 kinds and 3 colors.

$$q = \begin{pmatrix} u_{\text{к}} & u_{\text{ж}} & u_{\text{с}} \\ d_{\text{к}} & d_{\text{ж}} & d_{\text{с}} \\ s_{\text{к}} & s_{\text{ж}} & s_{\text{с}} \\ \text{red} & \text{yellow} & \text{blue} \end{pmatrix} \begin{matrix} \text{up} \\ \text{down} \\ \text{strange} \end{matrix} \qquad \text{(Gell-Mann)}.$$



Original source details: GELFAND-P000628

Uncertain reading. The matrix retains the source indices "к", "ж", "с". The color names are written under the corresponding columns.

In experiment quarks do not exist (90%). $\mathcal{L}$ cannot be invar. under

$$SU(3) \otimes SU_c(3) \qquad \text{color (change of color)}.$$

GELFAND-P000629

Assume that only colorless part. have masses that are not very large, colored ones — ∞.

$$M \sim \overline{q}_{\text{к}} q_{\text{к}} + \overline{q}_{\text{жс}} \overline{q}_{\text{жс}} + \overline{q}_{\text{с}} q_{\text{с}}.$$

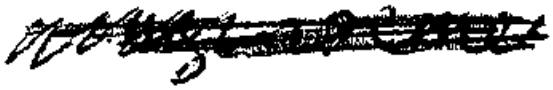
Manuscript excerpt. Both qs in the middle term are barred. Both subscripts read “жс”.

Original source details: GELFAND-P000629

The Lagrangian must prohibit the appearance of colored particles (they must have inf. masses). Let the interact. be carried by colored Yang–Mills fields (gluons).

$SU(3)$ — approx. symmetry. (20–30%). Let $SU_c(3)$ — be the exact group. (Possibly there is a fourth kind (charmed), but there are certainly 3 colors).

Of mesons (see the fmla.) there should be 9. But there are more. Hence a fourth quark is necessary. (4 years ago). [crossed out] $SU(4) \otimes SU_c(3)$.



Undeciphered passage. The entry after “4 years ago” is heavily crossed out.

Original source details: GELFAND-P000629

$$\mathcal{L} = \overline{q}(i\gamma_\mu \nabla_\mu - m)q + \frac{1}{4e^2} \operatorname{Sp}(F_{\mu\nu}^2).$$

There are 8 Yang–Mills fields. (Gluons — from glu — glue — stick quarks together). They are massless. 8 colored gluons.

Conserved currents $\Longleftrightarrow$ presence of symmetry. We want to show that if the gr. is nonabelian $\Rightarrow$ interact. energy $\sim$ [crossed out] R (dist.) ($\Rightarrow$ the string joining the quarks does not break) (in the abelian case $\sim R^{-1}$, Maxwell eqs.).

Uncertain reading. A short entry with a horizontal stroke is crossed out before R.

Original source details: GELFAND-P000629

$$\mathcal{L} = \frac{1}{4e^2} \operatorname{tr}(F_{\mu\nu}^2).$$

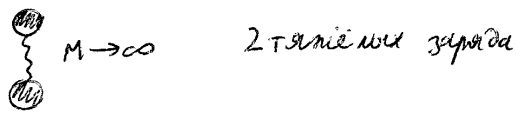


Diagram and typed description.

$M \to \infty$
2 heavy charges

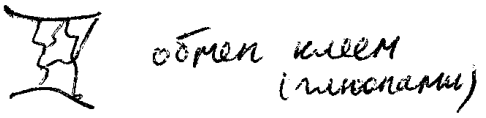
Two shaded circles are joined by a wavy line.

Original source details: GELFAND-P000629

G. Classical vers.: noncommut. electrical engineering.

Original source details: GELFAND-P000629

Manuscript excerpt. After “Классич.” (classical) the abbreviation reads “вар.” (version).

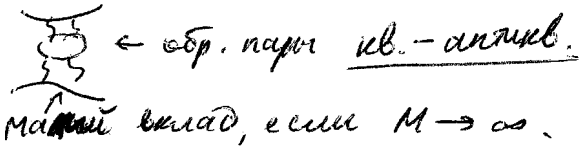


Original source details: GELFAND-P000629

Diagram and typed description.

exchange of glue
(gluons)

Two boundaries and connecting strokes. The exact joins in the middle are uncertain.



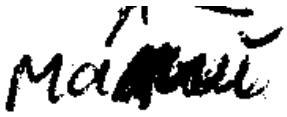
Original source details: GELFAND-P000629

Diagram and typed description.

← formation of a pair
q.–antiquark

Small contribution if $M \to \infty$

An oval between two boundaries. Arrows lead to the oval and to the lower line.



Original source details: GELFAND-P000629

Uncertain reading. The first word beneath the sketch is overwritten and tentatively reads “Малый” (small).

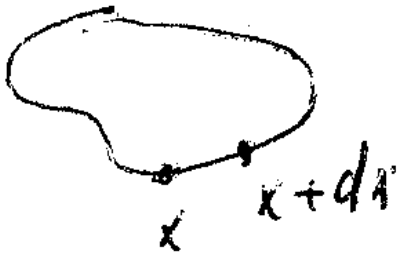
$$\frac{\int DA_\mu\, e^{i\int \mathcal{L}\, d^4x}\ \mathrm{tr}\, e^{i\oint A_\mu\, dx_\mu}}{\int DA_\mu\, e^{i\int \mathcal{L}\, d^4x}} = \left\langle \mathrm{tr}\, e^{i\oint_C A_\mu\, dx_\mu} \right\rangle = e^{iW(C)}.$$

Original source details: GELFAND-P000629

Uncertain reading. The last character after d^4 in the numerator is overwritten. It tentatively reads x.

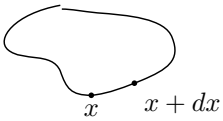
Original source details: GELFAND-P000629

Manuscript excerpt. The x after d in the numerator contour integral is overwritten. No subscript C is visible on that integral.



Original source details: GELFAND-P000629

Diagram and typed description.

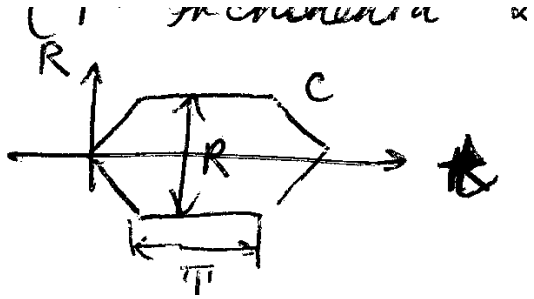


Two points are marked on the contour. A short stroke follows $x + dx$. No index can be read with confidence.

$$\operatorname{tr} \prod_x (1 + iA_\mu(x)\, dx_\mu)$$

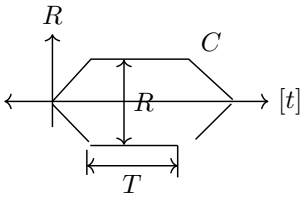
— contour integral. (this ordering). (T — Dyson exponential).

Assertion:



Original source details: GELFAND-P000629

Diagram and typed description.



Contour C, the R axis, and spans R and T. There is a gap at the right of the lower side of the contour.

Uncertain reading. The overwritten horizontal axis label tentatively reads t.

Original source details: GELFAND-P000629

Consider such a contour. Here $W(C) \xrightarrow[T\to\infty]{} E(R)T$, where $E(R)$ — the interaction energy of the charges.

GELFAND-P000630

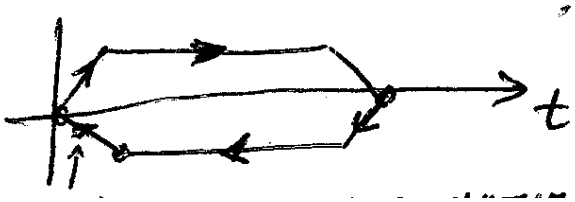
Explanation: if the syst. is in a stat. state, the trans. amplitude in time T: $F = e^{-iET}$ (E — energy).

Original source details: GELFAND-P000630

Uncertain reading. The exponent is written $-iET$. A separate stroke is visible above the first F.

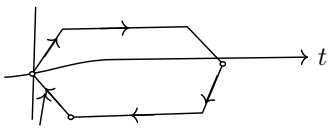
On the other hand

$$F = e^{-iET} = \sum_{\text{(paths)}} e^{iS}.$$



Original source details: GELFAND-P000630

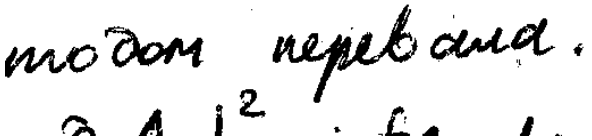
Diagram and typed description.



Arrows point right on the upper side and left on the lower side. A separate arrow points up toward the start.

[Beneath the figure:] creation of a part.–antipart. pair (superheavy); we hold it for a long time.

Classical action $S = \int_{(x)}^{(x')} A_\mu \, dx_\mu$. In the el.-magn. field. Antipart. $\sim$ a part. going backward in time. Hence $W(C) \sim S_C$ (area inside the contour). $\sim RT \to E \sim R$. It is calculated by the saddle-point method.



Uncertain reading. The last word is written “перевала” (saddle point).

Original source details: GELFAND-P000630

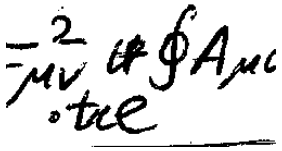
$$\int DA_\mu \, e^{i\int (\partial_\mu A_\nu - \partial_\nu A_\mu)^2} e^{i \oint A_\mu \, dx_\mu} = e^{-\oint\oint dx\, dx'\, D(x-x')},$$

where $D = \dfrac{1}{(x-x')^2 + i0}$ (Green’s fn. of the d’Alembert eq.).

$$W = \text{const} \cdot T + \frac{e^2 T}{R} + \dots$$

Next time:

$$\frac{\int DA \, e^{\frac{1}{4e^2}\int F_{\mu\nu}^2} \text{[crossed out]} \cdot \operatorname{tr} e^{\oint A_\mu \, dx_\mu}}{N}$$ — we will calculate.



Uncertain reading. A short sign is crossed out before the contour integral. The insertion below reads $\cdot \operatorname{tr} e$.

Original source details: GELFAND-P000630

non-Gaussian, since $F_{\mu\nu}$ has the quadr. term $[A_\mu, A_\nu]$.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

(In the el.-magn. field the commutator is 0). El.-dyn. with a compact gauge group — nonuniqueness of extremals already arises. Quarks in Yang–Mills have yet to be worked out.

Uncertain reading. The last word reads “сделана”, with the feminine ending.

Original source details: GELFAND-P000630

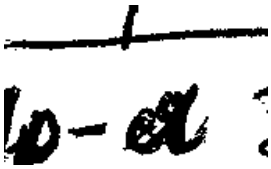
—·—

9/II-76. Gelfand Sem.

Uncertain reading. The digit of the day is heavily overwritten and reads “9”.

Original source details: GELFAND-P000631

Krichever. Zakharov–Shabat eqs.



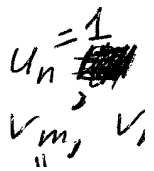
Uncertain reading. The ending of the abbreviation before “Zakharov” is heavily overwritten. It may read “-a”.

Original source details: GELFAND-P000631

$$\left[L_1 - \frac{\partial}{\partial y}, L_2 - \frac{\partial}{\partial t}\right] = 0, \tag{1}$$

$$L_1 = \sum_{i=0}^{n} u_i(x,y,t)\frac{\partial^i}{\partial x^i}, \qquad L_2 = \sum_{i=0}^{m} v_i(x,y,t)\frac{\partial^i}{\partial x^i},$$

$$\left.\begin{matrix} u_n = 1, & u_{n-1} \\ v_m = 1, & v_{m-1} \end{matrix}\right\} = \text{const.}$$



Undeciphered passage. An entry after u_n is crossed out, with $= 1$ above it. A 1 is inserted below v_m.

Original source details: GELFAND-P000631

Consider the formal series

$$\Psi = e^{kx+k^n y}\left(\sum_{i=0}^{\infty} \xi_i(x,y,t)\frac{1}{k^i}\right), \qquad \xi_0 = 1. \tag{2}$$

Let $\left(L - \frac{\partial}{\partial y}\right)\Psi = 0$, where $L = L_1$. The coeffs. are computed recursively.

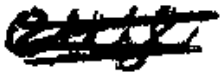
$$\sum_{i=0}^{n} u_i \sum_{\ell=0}^{i} C_i^{\ell}\, \xi_{\ell-s}^{(i-\ell)} = \xi_{n-s} + \frac{\partial}{\partial y}\xi_{-s}, \quad s = n, n-1, \ldots \text{ down to } -\infty. \tag{3}$$

$\xi_{-s} = 0$ for $s > 0$.

Uncertain reading. The sign before 0 is heavily overwritten. It tentatively reads $>$.

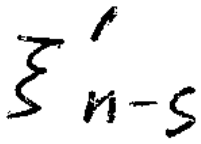
Original source details: GELFAND-P000631

[crossed out] determines $\xi'_{n-s} \Rightarrow$ const. of integr. are needed.



Undeciphered passage. A short word before "determines" is crossed out.

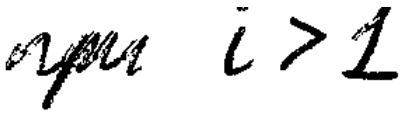
Original source details: GELFAND-P000631



Manuscript excerpt. The subscript of ξ' reads $n - s$.

Original source details: GELFAND-P000631

$\xi_0 = 1, \quad \xi_i(0, y, t) = 0$ for $i > 1$ — normalization.



Uncertain reading. The bound in the normalization is visibly written $i > 1$.

Original source details: GELFAND-P000631

Conversely, $\forall$ series $\exists$ an oper., for which everything holds to accuracy k^{-1}.

Algebraic operator. — if $\exists$ an anal. continuation in $1/k$ to a compact alg. curve (merom. on the complement of the point, [below the line: where there is an essent. sing.]).

G. Zakharov–Shabat eq.

$$\frac{\partial K_2}{\partial y} - \frac{\partial K_1}{\partial t} = [K_1, K_2],$$

K_1, K_2 — diff. operators. K_1, K_2 — matr. in y and t or oper. in other variables. There are many ways leading to this eq.. y, t — parameters. [Left, crossed out: On]

Manuscript excerpt. The short crossed-out entry reads "Ha" (on).

Original source details: GELFAND-P000631

Put on an equal footing x, y, t. Then it is written in the form (1).

Problem for schoolchildren: find oper. of the form

$$-\frac{d^2}{dx^2} + u(x) = L_1, \qquad \frac{d^3}{dx^3} + v_1\frac{d}{dx} + v_2 = L_2,$$

which commute. This is done via ellipt. fns. or ellipt. curves. If we have only $\frac{d^n}{dx^n}$, then

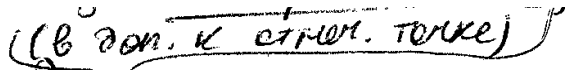
$$\frac{d^n}{dx^n}\Psi - \frac{\partial \Psi}{\partial y} = 0 \quad \Rightarrow \quad \Psi = e^{kx + k^n y}.$$

More precisely: the method of Fourier integral oper..

$$e^{kx+k^n y}\left(1+\frac{\xi_1(x,y,t)}{k}+\dots\right).$$

Kr. Associate to each oper. L — the formal series (2). The recurrence relation (3) recovers it uniquely up to normalization.

Definition. L is called algebraic, if the formal series (2) has a merom. continuation [inserted: (in the complement of the marked point)] to a compact alg. curve, with poles independent of x, y, t.



Original source details: GELFAND-P000632

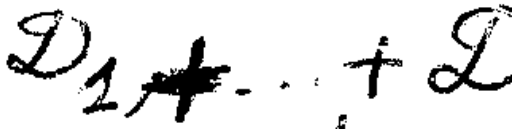
Uncertain reading. "In the complement of the marked point" is inserted above the line.

$L \longrightarrow$ [crossed out] $\mathcal{R}$

Original source details: GELFAND-P000632

Undeciphered passage. An entry after the arrow and before $\mathcal{R}$ is heavily crossed out.

$\mathcal{R}$ — alg. curve with marked point P_0 and set of poles $D_1 + \dots + D_g$, g — genus of the curve — these are called scattering data for L.



Original source details: GELFAND-P000632

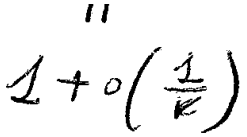
Manuscript excerpt. The first sign after D_1 is overwritten and reads $+$.

P_0 — point where there is an essent. sing.. $1/k$ — loc. param. near P_0.

G. An asympt. expansion as a rule does not converge. The def. assumes convergence. K. To an alg. operator $L \rightsquigarrow$ alg. curve with a marked point and a set of poles. G. C^∞ — Taylor fmla. asympt. $\Rightarrow$ analytic — converg.. Analogue of quasianalyticity?

K. Thm. (Akhiezer) $\forall$ nonsing. alg. curve $\mathcal{R}$ with marked point P_0 and a fixed loc. param. near P_0 and a marked set of poles $\exists$ a fn. determined uniquely up to proportionality $\Psi(x,y,P)$ with the given set of poles and

$$\underbrace{\Psi(x,y,P)e^{-kx-k^n y}}_{1+o(1/k)} \quad \text{— reg. near } P_0.$$



Original source details: GELFAND-P000632

Uncertain reading. Below the bracket is $1+o(1/k)$, with a small o.

Diagram and typed description.

Unlabelled rectangle.

Original source details: GELFAND-P000632

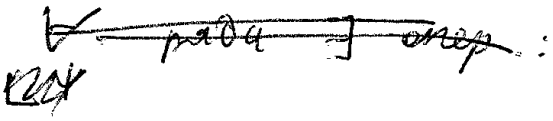
$e^{\zeta(P)x}$ [arrow to ζ:] Weierstrass (Kamke).

Proof: explicit fmla. (will be presented later). Thm. The eq. is algebraic $\Rightarrow \exists$ a fn. $\Psi \Rightarrow$ we know all the $\xi \Rightarrow$ we find all the u_i (uniquely) — why do they exist? From [crossed out] Ψ one finds the u_i!!

Uncertain reading. A short sign is crossed out before Ψ.

Original source details: GELFAND-P000632

[Lower right, crossed out: $\forall$ series $\exists$ an oper.:]



Undeciphered passage. A second short crossed-out entry stands beneath the left end of the cancelled line.

Original source details: GELFAND-P000632

GELFAND-P000633

$\forall$ series $\exists$ an oper.:

$$L\Psi - \frac{\partial}{\partial y}\Psi = \text{[crossed out]} \left(e^{kx+k^n y} o(1/k)\right).$$

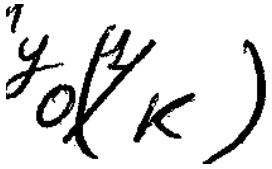
(first $n-2$ eqs.)

Uncertain reading. The first line reads “ряда” (series).

Original source details: GELFAND-P000633

Uncertain reading. A short entry after the equals sign is heavily crossed out.

Original source details: GELFAND-P000633



Uncertain reading. The manuscript reads $o(1/k)$.

Original source details: GELFAND-P000633

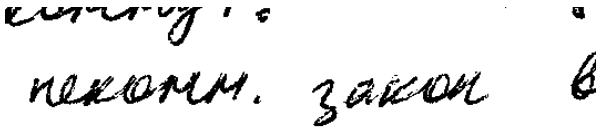
$L\Psi - \dfrac{\partial \Psi}{\partial y}$ meets all the requirements of Akhiezer's thm. $\Rightarrow$ it $\equiv 0$.

Drinfeld.

An attempt at algebraizing Krichever's previous talk

Description of commut. subrings in the ring of diff. operators.

G. The goal is a noncommut. reciprocity law.



Uncertain reading. The abbreviation before "law" reads "некомм." (noncommutative).

Original source details: GELFAND-P000633

B — [crossed out] a comm. ring with 1, equipped with a deriv. [crossed out] $b \mapsto b'$ [crossed out]. ($\mathbb{C}$-algebra)

Undeciphered passage. The entry after B and the dash is heavily crossed out.

Original source details: GELFAND-P000633

Uncertain reading. A short sign is crossed out after "дифф." (deriv.).

Original source details: GELFAND-P000633

Uncertain reading. A short sign is crossed out after b'.

Original source details: GELFAND-P000633

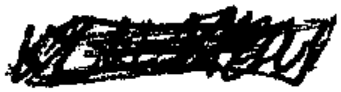
$$B[\mathcal{D}], \qquad \underbrace{\sum b_i \mathcal{D}^i}_{\text{polyn.}}, \qquad \underbrace{Db - bD = b'}_{\text{multiplication rule}} .$$

(b' — derivative in B)

X — a compact Riem. surf. over $\mathbb{C}$, P — a fixed point. Denote by A [crossed out] the ring of merom. fns. on X [crossed out] with a pole only at P.

Uncertain reading. A sign after A is heavily crossed out.

Original source details: GELFAND-P000633



Original source details: GELFAND-P000633

Undeciphered passage. The word after X is heavily crossed out.

We describe the homomorphisms $A \to B[\mathcal{D}]$. (in fact there are no other comm. subrings — Shabat's thm.)

G. The real problem: find all extensions of Gelfand–Kirillov skew fields.

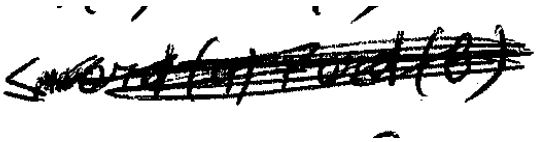
Polynomiality gets in the way.

$$\sum_{i=0}^{k} b_i D^i, \qquad \text{ellipticity}: \quad \exists b_k^{-1}.$$

Req.: the image of $\forall$ nonzero elem. of A is elliptic.

Rank. $a \neq 0, \quad a \in A$.

$$\left.\begin{array}{c} \operatorname{ord}(a) \text{ — order of the image of } a \\ \operatorname{ord}(ab) = \operatorname{ord}(a) + \operatorname{ord}(b) \\ \operatorname{ord}(a+b) \leq \text{[crossed out]} \max(\operatorname{ord}(a), \operatorname{ord}(b)) \\ \operatorname{ord}\lambda = 0 \quad \text{for } \lambda \in \mathbb{C} \end{array}\right] \quad \text{valuation}.$$



Original source details: GELFAND-P000633

Uncertain reading. An entry after the inequality sign is crossed out. It tentatively reads $\operatorname{ord}(a) + \operatorname{ord}(b)$.

$\deg a$ — order of the pole of a at the point P.

Then $\operatorname{ord}(a) = n \deg(a)$. (clear for polynomials)

n — rank of the homomorphism.

Krichever — $n = 1$ (coprimality of the orders of the oper.).

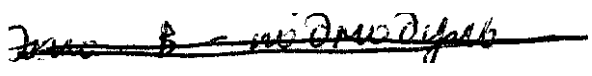
—·—

$M = B[\mathcal{D}]$, M has the structure of a B-module; $\varphi : A \to B[\mathcal{D}]$.

GELFAND-P000634

M_n — oper. of order $\leq n$.

[crossed out: This is a B-submodule.]



Original source details: GELFAND-P000634

Uncertain reading. The cancelled line reads "This is a B-submodule".

M — this is an $A \otimes B$-module, i.e. an A-module and B-module and mult. by A and B commute.

$$bx = bx \quad \text{(mult. on the left)},$$
$$ax = x\varphi(a),$$
$$a \in A \Rightarrow aM_i \subset M_{i+\deg a \cdot n}.$$

$\nabla : M \to M$ — mult. on the left by D.

$$\nabla(x) = Dx.$$

∇ commutes with the action of A, and $\nabla b - b\nabla = b'$.

Conversely, if there is an $A \otimes B$-mod. M with a filtr. and all these relations, then $\exists$ a homom. $\varphi : A \to B[\mathcal{D}]$ of rank n.

We can forget φ and consider the problem of $A \otimes B$-modules.

$\to$ Thus, M — a module over $A \otimes B$, equipped [crossed out] with the filtration

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots$$

Undeciphered passage. A short entry before "filtration" is crossed out.

Original source details: GELFAND-P000634

[Under A: merom. fns. on X with a single pole. Under B: a ring ($\mathbb{C}$-alg.) with deriv. $b \mapsto b'$.] and with the operator $\nabla : M \to M$, such that

1) $\nabla a = a\nabla$.

2) $\nabla b - b\nabla = b'$.

3) $\nabla(M_i) \subset M_{i+1}$, $\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i$ for $i \geqslant 0$.

Uncertain reading. A separate slanted stroke lies below $>$. The bound reads $i \geqslant 0$.

Original source details: GELFAND-P000634

4) $bM_i \subset M_i$, M_i/M_{i-1} [crossed out: $\to$] a free B-module with one generator.

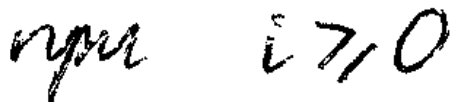
Uncertain reading. A rightward arrow is crossed out before "free".

Original source details: GELFAND-P000634

5) $aM_i \subset M_{i+n\deg a}$,

$$a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+n\deg a}/M_{i+n\deg a-1} \qquad \text{for } i \geqslant 0.$$

Uncertain reading. Item 5 repeats the sign with a separate stroke. The bound reads $i \geqslant 0$.



Original source details: GELFAND-P000634

$\sim$ the data of a sheaf over $X \otimes B$ with certain filtrations.

$\nabla \sim$ a connection with a sing. at ∞.

—·—

GELFAND-P000635

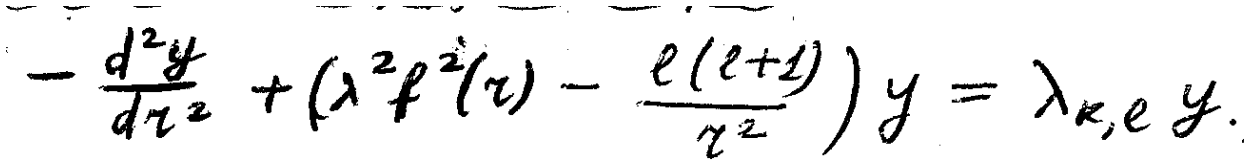
23/II-76. Gelfand Sem.

S. P. Novikov on the work of M. Brodsky. (Minsk)

Some cases of explicitly solvable problems unrelated to KdV.

Problem of vibrations of an elastic sphere.

$$-\frac{d^2y}{dr^2} + \left(\lambda^2 f^2(r) - \frac{\ell(\ell+1)}{r^2}\right) y = \lambda_{k,\ell} y.$$



Original source details: GELFAND-P000635

Uncertain reading. The right-hand side is written $\lambda_{k,\ell} y$. A minus sign precedes the fraction in parentheses.

Inverse problem — recover the par. of the elastic medium from the set of frequencies $\lambda_{k,\ell}$. The problem is overdetermined. Find when it is solvable and how.

Let $rf(r) = u(r)$ be a monotone fn..

Consider

$$m(\tau) = \int_{H(\tau)}^{1} \left(\tau^{-2} f^2(r) - \frac{1}{r^2}\right)^{1/2} dr \qquad \text{(Bohr quant. rule)}.$$

($H(\tau)$ — a zero of the expression under the radical)

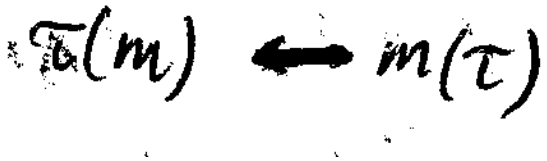
Consider

$$\tau^*_{k\ell} = \frac{\sqrt{\ell(\ell+1)}}{\lambda_{k\ell}}, \qquad m_{k\ell} = \frac{k + \frac{1}{4} + C}{\sqrt{\ell(\ell+1)}}.$$

$\tau^*(m_{k\ell}) = \tau^*_{k\ell}$ establishes a one-to-one corresp.

$m_{k\ell}$ are dense everywhere; nec. cond. — $\tau_{k\ell}$ dense.

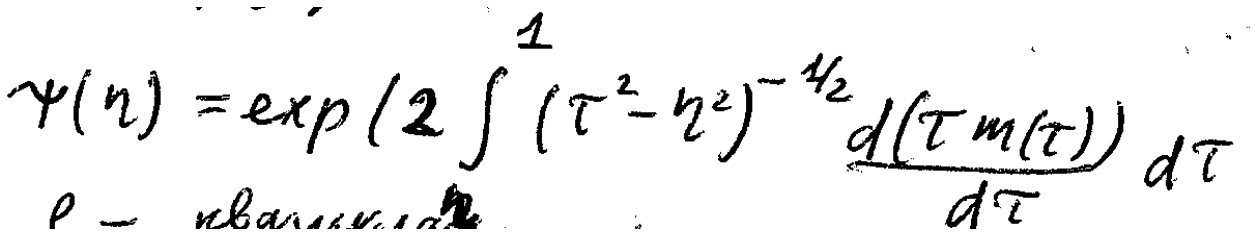
$$(rf(r))^{-1} = \psi(\eta). \qquad \tau(m) \longleftrightarrow m(\tau)$$



Uncertain reading. The arrow between the expressions has heavy strokes at both ends.

Original source details: GELFAND-P000635

$$\psi(\eta) = \exp\Big(2\int^{1} (\tau^2 - \eta^2)^{-1/2} \frac{d(\tau m(\tau))}{d\tau}\, d\tau$$



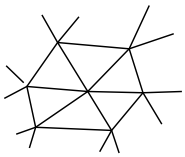
Original source details: GELFAND-P000635

Uncertain reading. The upper limit of the integral is written 1. The lower limit and the closing parenthesis are not visible.

Large ℓ — quasiclassics.

Fuchs. There is a triang. manifold.

Diagram and typed description.



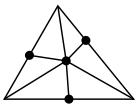
A triangular mesh with edges extending beyond it.

Original source details: GELFAND-P000635

Consider $\sum$ of all simplices of dim. 1 with coeff. 1 mod 2. It is not a cycle.
For the first barycentr. subdivision 0.

Original source details: GELFAND-P000635

Diagram and typed description.



A triangle with three internal lines and four marked points.

all the centers, 0 mod 2 (even).
Stief. class — is Poincaré dual to it.

Original source details: GELFAND-P000635

Uncertain reading. The entry reads “Штиф. класс” (Stief. class).

$$\sum \text{number} \underset{\text{mod } 2}{=} \sum (-1)^i \text{ of numbers} = \text{Euler char.}.$$

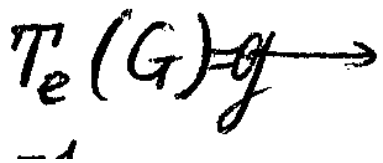
Losik (Mark).
Connection. $E \to X$ vector bundle with fiber $\mathbb{R}^n$; $p : P \to X$ assoc. principal bundle.
$x \in X$, E_x — fiber of E at x.
$P_x \ni z$, $z = (e_{i,x})$, $i = 1, \dots, n$, z — a basis in E_x.
Consider $G = GL(n, \mathbb{R})$ and define its action on P.

$$A \in G \rightsquigarrow R_A : P \to P, \qquad R_A(z) = zA = \left(\sum_{j=1}^{n} e_{j,x} A_i^j \right), \quad A = (A_i^j), \quad i, j = 1, \dots, n.$$

This action is smooth.
GELFAND-P000636

1. $(zA)B = z(AB)$.
2. G preserves the fibers and acts simply transitively.

$z \in P_x$, V_z — vertical tangent space ($\subset T_z P$). $V_z = T_z(P_x)$.
define the map $\ell_z : G \to P_x$, $\ell_z(A) = zA$
and the induced map $\alpha_z : T_e(G) = \mathfrak{g} \to V_z$ (isomorphism).



Original source details: GELFAND-P000636

Uncertain reading. In $T_e(G) = \mathfrak{g} \to V_z$, the lettering of $\mathfrak{g}$ overlaps the strokes of the equals sign and arrow.

$\beta_z = \alpha_z^{-1} : V_z \to \mathfrak{g}$ (a linear form on V_z with values in $\mathfrak{g}$).

A connection on a principal bundle — a 1-form on P with values in $\mathfrak{g}$, such that

1. $\omega|_{V_z} = \beta_z$,

2. $R_A^* \omega = \operatorname{ad} A^{-1} \omega$ (invariance condition).

For the tangent bundle — an affine connection.

Consider the equation $\omega = 0$ at the point z,

$$H_z = \{\xi \in T_z P : \omega(\xi) = 0\}.$$

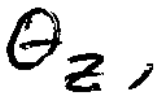
H_z is called the horizontal space. $T_z = H_z \oplus V_z$. H_z is preserved under the action of G.

—·—

A bundle will be constructed whose sections are in one-to-one correspondence with connections on P.

Consider $T_z^* \otimes \mathfrak{g}$ — 1-forms at z with values in $\mathfrak{g}$.

Define $\Theta_z \subset T_z^* \otimes \mathfrak{g}$ as the subspace of those Θ_z, such that $\theta_z|_{V_z} = 0$.



Uncertain reading. After "those", the manuscript uses uppercase Θ_z. The restriction condition reads lowercase θ_z.

Original source details: GELFAND-P000636

$$C_z \subset T_z^* \otimes \mathfrak{g}, \qquad C_z \ni \omega_z : \omega_z|_{V_z} = \beta_z.$$
$$C_z - C_z \subset \Theta_z, \qquad \omega_{1,z} - \omega_{2,z} \in \Theta_z, \qquad \omega_z + \Theta_z \in C_z.$$

Uncertain reading. The last relation has the sign $\in$ before C_z.

Original source details: GELFAND-P000636

i.e. C_z — a plane parallel to Θ_z, which does not pass through 0.

This gives a bundle of affine spaces («the difference of two Christoffel symbols is a tensor»).

Over P consider the bundle with fiber $T_z^* \otimes \mathfrak{g}$.

There are maps:

$$\widetilde{r}_{z,A} : T_z^* \otimes \mathfrak{g} \to T_{zA}^* \otimes \mathfrak{g},$$

induced by the maps $R_{A^{-1}}^* : T_z^* \to T_{zA}^*$ and $\operatorname{ad} A^{-1} : \mathfrak{g} \to \mathfrak{g}$. [At right: Tensor product of maps.]

GELFAND-P000637

C_z goes to the corresponding plane at zA under this map.

$$C(P) = \bigcup_{z \in P} C_z$$

— a natural smooth bundle over P.

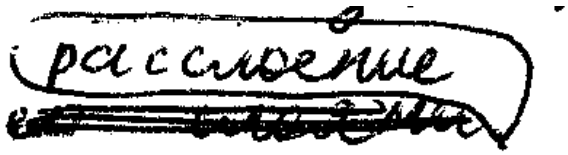
G acts on $C(P)$.

[Crossed out: C?] Invariant sections $\longleftrightarrow$ connections.

Uncertain reading. The crossed-out entry before "Invariant" begins with a mark resembling C.

Original source details: GELFAND-P000637

$C(P)/G = B(P)$, ["bundle" inserted and circled, with an arrow to π] $\pi : B(P) \to X$.



Undeciphered passage. The words below the circled "bundle" are heavily crossed out.

Original source details: GELFAND-P000637

$C(P)$ — a principal bundle with base P, there is a canonical connection there.

Gabrielov. [Crossed out: The connection γ gives an isomorphism $T^*X \otimes \mathfrak{g} \to B(P)$.]

Uncertain reading. After "The connection", γ is crossed out.

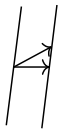
Original source details: GELFAND-P000637

(on $\forall$ fiber — an affine transformation)

1 conn. chosen $\to$ another conn. determines a transformation of the original space.

($\Longleftrightarrow$ the difference of connections is a tensor)

Diagram and typed description.



Two slanted lines and two arrows pointing right in the margin.

Original source details: GELFAND-P000637

$$\lambda \in T_x X \rightsquigarrow E_x \to E_x$$

i.e. a form with values in homomorphisms.

$T^*X \otimes \mathrm{Hom}(E, E)$.

—·—

GELFAND-P000638

1/III-76. Gelfand Sem.

Original source details: GELFAND-P000638

Diagram and typed description.

Hatched strip with a wavy upper edge below the heading.

Drinfeld. (Volodya).

$\overline{X}$ — a smooth proj. curve over $\mathbb{C}$, P — a fixed point on X,

$A \sim$ the ring of rat. fns. on X with a pole at P (reg. on $X \setminus P$).

Original source details: GELFAND-P000638

Undeciphered passage. A separate word below the definition of the ring is obscured by heavy hatching.

G. Example. $y^2 = x(x-1)(x-2)$ (elliptic curve).

$$P = \infty, \qquad A = \mathbb{C}[x,y]/(y^2 - x(x-1)(x-2)).$$

Original source details: GELFAND-P000638

Uncertain reading. The field symbol in the first quotient was overwritten. Tentatively $\mathbb{C}$.

There is no fn. with a pole of order 1. — since [crossed out] it is a torus.

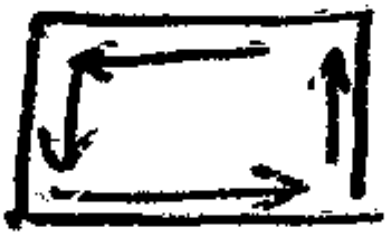
Original source details: GELFAND-P000638

Undeciphered passage. A short word between “since” and “it is a torus” is heavily crossed out.

doubly per.

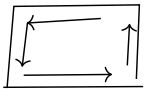
Original source details: GELFAND-P000638

Uncertain reading. The abbreviation reads “двоякопер.” (doubly periodic).



Original source details: GELFAND-P000638

Diagram and typed description.



$\int = 0$, residue $= 0$.

Four arrows run counterclockwise along the contour.

Pole of order 2 — Weierstrass $\wp$-fn. (see Hurwitz, Courant).

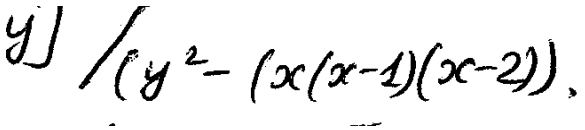
[crossed out] $y^2 = x(x-1)$ of genus 0 — Euler subst..

Original source details: GELFAND-P000638

Undeciphered passage. A short word before the equation is heavily crossed out.

A for a curve of genus 0 — polynomials, gen. z.

A for an ell. curve — $\mathbb{C}[x,y]/(y^2 - (x(x-1)(x-2))$.



Original source details: GELFAND-P000638

Uncertain reading. The manuscript lacks a further closing parenthesis after the product.

x has a pole of order 2. It goes to the Weierstrass fn. $x = \wp(z)$,

y —"— order 3. $y = \wp'(z)$.

The ring A uniquely determines the curve and the point.

How to describe these rings?

—·—

Krichever.

almost $\forall$ points P least pole order $g+1$.

Sometimes it can be lower — at Weierstrass points.

$$y^2 = \prod_{i=1}^{2g+1} (z - z_i)$$

(hyperelliptic curve) — at a branch point there is a fn. with a pole of order 2.

n — least possible pole mult. at $P \Rightarrow$ there are poles of mult. $2n, 3n, \ldots$.

Let μ_s be such that $\mu_s n + s$ — pole multiplicity, $s = 1, \ldots, n-1$.

(μ_s — the least).

Then $\mu_1 + \cdots + \mu_{n-1} = g$. ‖ all generators are among them. (Weierstrass theorem).

GELFAND-P000639

P in general position — the ring A has g generators — with poles of $g+1, g+2, \ldots, 2g+1$ order.

Original source details: GELFAND-P000639

Uncertain reading. The manuscript gives g before "generators" and lists orders from $g+1$ to $2g+1$.

B — an algebra over $\mathbb{C}$ with a derivation $\mathcal{D}$.
There is a ring $B[\mathcal{D}] = \{\sum b_i D^i\}$, $\mathcal{D}b - b\mathcal{D} = b'$.
An elem. is called elliptic if its lead. coeff. is invertible.
We describe homs. $A \to B[\mathcal{D}]$, such that the image of $\forall$ elem. is elliptic.
The rank of such a homomorphism is a (nat.) number n, such that

$$\forall a \in A, \quad \operatorname{ord} \varphi(a) = n \cdot \deg a.$$

$\deg a$ — pole order of a [crossed out] at the point P.

Undeciphered passage. The entry after a is covered by heavy cross-out strokes.

Original source details: GELFAND-P000639

(G. There are diff. opers. of orders 2 and 3 that commute with each other.)
[crossed out: M?] homoms. $\longleftrightarrow$ modules with additional structures.

Uncertain reading. The crossed-out mark resembles M.

Original source details: GELFAND-P000639

Krichever thm.
The leading coefficient of the homomorphism $\varphi : A \to B[\mathcal{D}]$.
Let z — a local coordinate at P. Let $a = z^{-k} + \cdots$ (Laurent).
[crossed out: Lead.?] $\varphi(a) = b^k D^{nk} + \cdots$,

Uncertain reading. The crossed-out entry tentatively reads “Стар” (lead.).

Original source details: GELFAND-P000639

where $\underset{B}{b}$ is independent of k and of the choice of loc. parameter.
(there are poles of $\forall$ orders from a certain order onward. One has to divide)
b is called the leading coeff. of the homomorphism φ.
def. [crossed out: This is?] $\varphi_1 \sim \varphi_2$, if $\exists b \in B^*$, $\varphi_1(a) = b\varphi_2(a)b^{-1}$.

Uncertain reading. The crossed-out entry above the start of the definition tentatively reads “Это” (this is).

Original source details: GELFAND-P000639

$\frac{d}{dx} \sim \frac{b'}{b} + \frac{d}{dx}$. Among germs, b'/b — any fn..

Krichever thm. B [crossed out] — germs of smooth fns. on the line.

Undeciphered passage. The formula after B is crossed out. Delimiters are visible, and the reading remains unresolved.

Original source details: GELFAND-P000639

homom. φ of rank 1 (up to equiv.!) $\longleftrightarrow$ the leading coeff. and a point on the Jacobian of the curve (Jacobian variety), J_X.

S. Gelf. 1-dimensional vect. bundles of degree 0 ($\sim$ divisor up to equivalence)

$$\begin{matrix} Q_1, \dots, Q_\ell \\ n_1, \dots, n_\ell \\ \text{integers} \end{matrix} \qquad \text{degree } n_1 + \dots + n_\ell = 0.$$

equivalence: if they differ by the divisor of a (meromorphic) fn..

Equiv. — bundles given by global merom. sections.

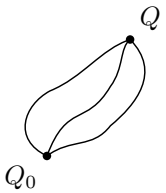
GELFAND-P000640

G. Analytic def. of the Jacobian variety: on $\overline{X}$ there exist differentials of the first kind (without sing.) there are g of them.

$y^2 = x(x-a)(x-b)$,

$\| \frac{dx}{y}$ has no sing. (at $x = 0, a, b$ there are none, since x is no loc. param.).

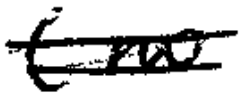
Diagram and typed description.



Three paths between Q_0 and Q.

Original source details: GELFAND-P000640

$z_i = \int_{Q_0}^{Q} \omega_i$. [crossed out: This is?] ($\int$ over homol. classes. Basis — g numbers)



Uncertain reading. The crossed-out entry after the integral tentatively reads “Это” (this is).

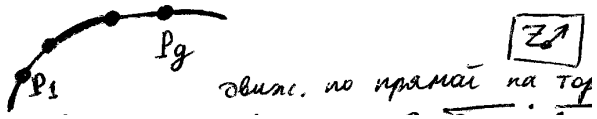
Original source details: GELFAND-P000640

the form determines a set of int. over periods.

up to periods — $\mathbb{C}^g/\underline{\text{lattice}}$. $n_1\alpha_1 + \cdots + n_{2g}\alpha_{2g}$.

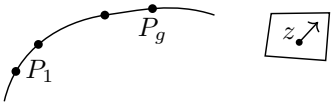
Only some tori arise.

A divisor is a lin. comb. of points. Points of the torus can be added.



Original source details: GELFAND-P000640

Diagram and typed description.



Points on the curve and an arrow from z in a rectangle.

motion along a straight line on the torus $\to$ motion of $P_1, \ldots, P_g$ [crossed out] along the curve.

Original source details: GELFAND-P000640

Undeciphered passage. The word between P_g and "along the curve" is heavily crossed out.

—·—

Giving a hom. from A into the ring $B[\mathcal{D}]$ is equiv. to giving the following.

M — a module over $A \otimes B$, $\nabla : M \to M$, $\nabla(am) = a\nabla(m)$, $\nabla(bm) = b\nabla(m) + b'm$.

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots, \qquad \dim_B M_i/M_{i+1} = 1 \quad \text{for } i \geqslant 0.$$

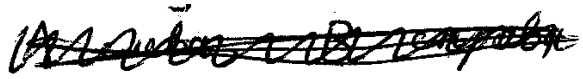
Original source details: GELFAND-P000640

Uncertain reading. The manuscript writes M_i/M_{i+1} in the dimension statement.

$$\begin{aligned} BM_i &\subset M_i, \\ \nabla M_i &\subset M_{i+1}, \qquad \nabla_* : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i, \\ aM_i &\subset M_{i+\deg a}, \\ a : M_i/M_{i-1} &\xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1}. \end{aligned}$$

—

$\underline{\varphi : A \to B[\mathcal{D}]}$, $\Rightarrow B[\mathcal{D}] = M$. [crossed out: $A \otimes B$-module?]

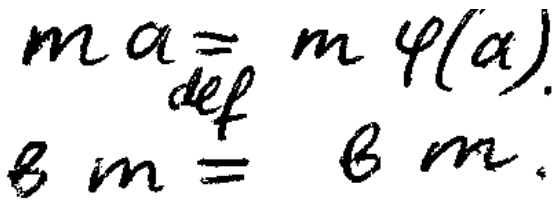


Original source details: GELFAND-P000640

Uncertain reading. The crossed-out line tentatively reads $A \otimes B$-module.

Mult. by B on the left and by A on the right,

$$\begin{array}{rcl} ma & = & m\varphi(a). \\ & \overset{\text{def}}{} & \\ bm & = & bm. \end{array}$$



Uncertain reading. The mark def is written between the two equals signs.

Original source details: GELFAND-P000640

$\nabla : M \to M$ mult. by $\mathcal{D}$ on the left.

Filtr. — order.

last cond. — ellipticity.

—·—

How, from M to construct φ?

[crossed out: M?] $M_{-1} = 0$,

Uncertain reading. The crossed-out mark resembles M.

Original source details: GELFAND-P000640

$\nabla : M_0 \xrightarrow{\sim} M_1/M_0$. in M_0 there is a generator e (over B).

put $D = \nabla(e)$.

$\varphi(a)$ is clear since there is an A-module, etc..

—·—

Next step: translate into the language of sheaves.

[crossed out: on the curve] $M_{-1} = 0 \subset M_0 \subset M_1 \subset \cdots$

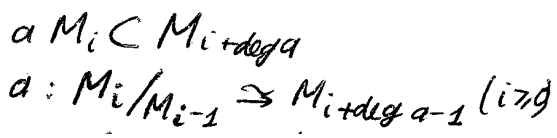
Uncertain reading. “On the curve” is crossed out before the filtration.

Original source details: GELFAND-P000640

$\sim$ a vect. bundle [inserted: 1-dimensional] on X with 0 cohom. (0-dimensional and 1-dimensional) of the sheaf of sections.

[At right]

$$aM_i \subset M_{i+\deg a}, \qquad a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a-1} \quad (i \geqslant g$$



Uncertain reading. One filtration term is written at the target of the arrow in the right margin. An opening parenthesis precedes the condition $i \geqslant g$.

Original source details: GELFAND-P000640

GELFAND-P000641

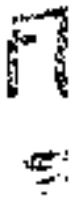


Diagram and typed description.

A three-sided mark at upper right with several detached strokes below.

Original source details: GELFAND-P000641

L — a 1-dimensional vect. bundle over $\overline{X}$, $H^0(X, \underline{L}) = H^1(X, \underline{L}) = 0$.

($H^0(X, \underline{L}) = 0$ and $\deg L = g - 1$).

$M = H^0(X - P, \underline{L})$ (algebraic sects., holom. with a single pole).

Uncertain reading. The abbreviation after “algebraic sects.” reads “голом.” (holom.).

Original source details: GELFAND-P000641

M_i [crossed out: $-$?] with a pole of order $i + 1$.

Uncertain reading. A short horizontal mark after M_i is crossed out.

Original source details: GELFAND-P000641

$M_{-1} = 0$ — first condition, ($H^0 = 0$).

$$\dim M_i/M_{i-1} = 1, \qquad M_0 = H^0(X, \quad \underset{\uparrow}{\underline{L}(P)} \quad).$$
with a pole at P

$$0 \to \underline{L} \to \underline{L}(P) \to \underline{L}(P)/\underline{L} \to 0.$$

exact cohom. seq. $\Rightarrow \dim_{\mathbb{C}} M_i/M_{i-1} = 1$.

$=$

Conversely, given a seq. of modules, construct a sheaf

$$0 = M_{-1} \subset M_0 \subset M_1 \subset M_2 \subset \cdots, \qquad \dim_{\mathbb{C}} M_i/M_{i-1} = 1.$$

M — a module over A.

$$aM_i \subset M_{i+\deg a}$$
$$a_* : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1}, \qquad i \geqslant 0.$$

Uncertain reading. The subscript on a in the map resembles an asterisk.

Original source details: GELFAND-P000641

We give sects. of the sheaf on Zariski open sets.

The sheaf is quasi-coherent. It suffices to give it on open affine subsets.

$a_0 \in A$, $a_0 \neq$ const $\|$ cover

$$X = (X \setminus P) \cup (X \setminus \text{zeros of } a_0).$$

Sects. over $X \setminus P$, A-module — M.

Over $X \setminus$ zeros of a_0 — m/a_0^k, $m \in M_{k \cdot \deg a_0 - 1}$.

Next time: Drinfeld 20 min., Shnirelman.

Original source details: GELFAND-P000641

Undeciphered passage. Only isolated upper strokes are discernible below the last line.

GELFAND-P000642

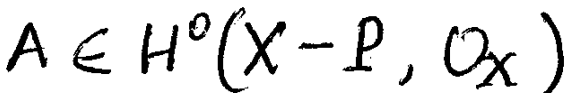
. —·—

15/III-76 Gelfand Sem. Drinfeld.

$\overline{X}$ — a smooth proj. curve over $\mathbb{C}$

$P \in \overline{X}$,

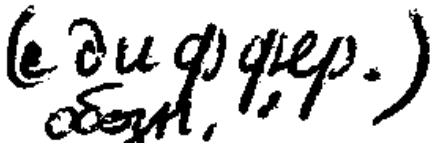
$A \in H^0(X - P, \mathcal{O}_X)$ — rat. fns. on $\overline{X}$ with a pole at P.



Uncertain reading. The sign between A and H^0 is $\in$.

Original source details: GELFAND-P000642

B — an algebra over $\mathbb{C}$ (with a deriv. [below: notation]), z — a loc. coord. on $\overline{X}$ near P.



Original source details: GELFAND-P000642

Uncertain reading. Below "with a deriv." the abbreviation reads "обозн." (notation). The neighboring short strokes are unclear.

We seek homomorphisms $\varphi : A \to B[\mathcal{D}]$ of rank 1,

i.e. $\forall a \in A$, $a = z^{-m} + \cdots \Rightarrow \varphi(a) = b^m D^m + \cdots$

b — the leading coefficient of φ.

They are considered up to conjugation by inv. elements of B.

[crossed out: B?] Giving φ is then equivalent to giving an $A \otimes B$-module M

Original source details: GELFAND-P000642

Uncertain reading. The mark crossed out before "Giving" resembles B.

[crossed out: e?] and a map $\nabla : M \to M$, such that

Original source details: GELFAND-P000642

Uncertain reading. The short cancelled mark before "and a map" resembles "e".

$\nabla(am) = a\nabla(m)$, [crossed out: and?] $\nabla(bm) = b\nabla(m) + b'm$, $M = \bigcup M_i$,
(filtration)

Original source details: GELFAND-P000642

Uncertain reading. The mark crossed out between the formulas for the map resembles "и" (and).

$BM_i \subset M_i$, [crossed out: BM_i?] $M_{-1} = 0$, $M_i/M_{i-1} = B$ for $i \geqslant 0$.

Original source details: GELFAND-P000642

Uncertain reading. The crossed-out entry resembles BM_i.

$$\forall a \in A, \qquad aM_i \subset M_{i+\deg a},$$
$$a : M_i/M_{i-1} \xrightarrow{\sim} M_{i+\deg a}/M_{i+\deg a-1},$$
$$\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i, \qquad i \geqslant 0.$$

G. corollary: $\exists$ an oper. whose powers, except for the 1st — are diff. operators (2nd, 3rd etc.).

$\overline{X}$ — an alg. curve of genus g, P — a point.

n — the least possible pole multiplicity at P (the sole pole).

μ_i — the least numbers for which there is a pole of mult. $i + \mu_i n$, $i = 1, \dots, n-1$.

Then $\sum \mu_i = g$.

GELFAND-P000643

[crossed out: There exist fn...?]

Uncertain reading. The crossed-out opening tentatively reads “Сущ. ф...” (there exist fn...).

Original source details: GELFAND-P000643

(There are g omitted pole multiplicities*)

[Arrow from the asterisk] this [crossed out] also conversely implies the statement formulated by Krichever.

Undeciphered passage. The word after “this” is heavily crossed out.

Original source details: GELFAND-P000643

D. Into the language of sheaves.

Everything except ∇, $\sim$ specifying a vect. bundle L on $\overline{X}$, for which $H^0(X, \underline{L}) = H^1(X, \underline{L}) = 0$.

$\sim$ specifying a 1-dimensional bundle L over $X \otimes B$,

$$H^0(X \otimes B, \underline{L}) = H^1(X \otimes B, \underline{L}) = 0.$$

$$X = U_1 \cup \dots \cup U_k, \qquad U_i = \text{Spec}\, R_i.$$

Glue the aff. schemes Spec [crossed out] $R_i \otimes B$.

Undeciphered passage. Marks between Spec and the expression on the right are heavily crossed out.

Original source details: GELFAND-P000643

Intuitively: a germ of a smooth family of vect. bundles.

$$\nabla : \underline{L} \longrightarrow \underset{\text{with a first-order sing. at } P}{\underline{L}(P)} \qquad \underline{L} \longrightarrow \underline{L}(P)/\underline{L} \longrightarrow 0.$$

[At right] $H^0(\underline{L}(P)/\underline{L})$ is free (the restriction to P is trivial).

$\nabla b - b\nabla = b'$, ∇ — O_X-linear.

leading coeff. — ?

Trivialize near P, Let e — a basis.

$$\frac{\nabla(e)}{e} = \frac{1}{b_0 z} + \cdots \quad \Rightarrow \quad b_0 \text{ — leading coeff..}$$

Krichever thm. Homom. of rank 1 up to conjugation by inv. elements of $B \Longleftrightarrow$ a point on the Jacobian of $\overline{X}$ in general position.

1. Given φ. $\Rightarrow$ a 1-dimensional bundle is given over $X \otimes B$. We construct a point on the Jacobian.

Let $\mathbb{C} \leftarrow B$ and the corresp. embedding $\overline{X} \subset X \otimes B$, since there is a natural map $B \to \mathbb{C}$ (value at 0).

Restrict L to $\overline{X}$, i.e. obtain a vect. bundle on $\overline{X}$.

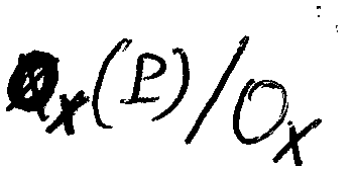
2. Let there be given L_0 — a bundle over $\overline{X}$ and let there be given B. We construct everything else.

J — the Jacobian of $\overline{X}$, its points are vect. bundles of the same degree.

Tangent space to the Jacob. $\cong H^1(X, O_X)$ (the Jacobian itself $H^1(X, O_X^*)$. $\Rightarrow$ the tangent space is obtained by removing $*$).

Consider the exact seq.

$$0 \longrightarrow O_X \longrightarrow O_X(P) \longrightarrow O_X(P)/O_X \longrightarrow 0.$$



Original source details: GELFAND-P000643

Uncertain reading. The first letter of the third term has been overwritten. Tentatively $O_X(P)/O_X$.

Original source details: GELFAND-P000643

Undeciphered passage. A crossed-out entry is visible below the third term of the sequence.

[Upward arrow] a sheaf over a point.

Consider

$$\frac{1}{[\text{crossed out}]z} \in H^0(O_X(P)/O_X) \xrightarrow[\text{boundary hom.}]{} H^1(X, O_X).$$

Undeciphered passage. The mark before z in the denominator is obscured by ink.

Original source details: GELFAND-P000643

(z — a loc. parameter).

The image of $1/z$ in the tangent space to the Jacobian is denoted by $\eta_P \in H^1(X, O_X)$.

GELFAND-P000644

ℓ_0 — the corresp. point on the Jacobian. As we vary x we get a smooth path on the Jacobian (a point of the Jacobian with values in B).

$$\underline{\ell'(x) = \frac{\eta_P}{b_0(x)}}$$

(a straight line with changing parameter).

Uncertain reading. The looped sign in both entries resembles ℓ or e.

Original source details: GELFAND-P000644

Again: curve X, point P, loc. param..

$\Rightarrow$ a line in the tangent space to the Jacobian.

$x \in \overline{X}$ [crossed out] $x \mapsto \overline{x - P}$ — divisor.

Undeciphered passage. A group of looped signs is crossed out after the membership expression.

Original source details: GELFAND-P000644

We obtain a map from $\overline{X}$ into the Jacobian J.

The curve has at P a tangent space. The image — a 1-dimensional space tangent to the Jacobian.

—·—

A. Polyakov. Gauge fields. Quantization of the Yang–Mills eqs.

Lagrangian

$$S = \int \mathcal{L}(x) d^4x, \qquad \mathcal{L}(x) = \frac{1}{4} \operatorname{tr} F_{\mu\nu}^2.$$

For the group SU(2) we have $F_{\mu\nu}$ — an elem. of the Lie algebra.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

[Outside the right rule]

Uncertain reading. The isolated mark resembles “ɜ” or “3”.

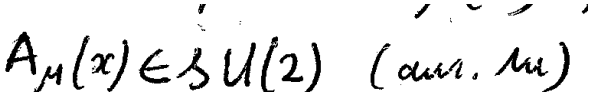
Original source details: GELFAND-P000644

A_μ, $\mu = 0, 1, 2, 3$ — anti-Hermitian with trace 0 (2×2) (“traceless”).

The fields A_μ — dynamical variables.

Elem. fields — fns. $A_\mu(x)$, $\mu = 0, 1, 2, 3$.

$$A_\mu(x) \in \mathrm{SU}(2) \quad \text{(Lie alg.)}.$$



Uncertain reading. The line reads $SU(2)$, followed by “Lie alg.” in parentheses.

Original source details: GELFAND-P000644

$$F_{\mu\nu}^2 = \sum_{\mu,\nu} F_{\mu\nu} F_{\mu\nu}.$$

Undeciphered passage. Several signs to the right of the sum are crossed out.

Original source details: GELFAND-P000644

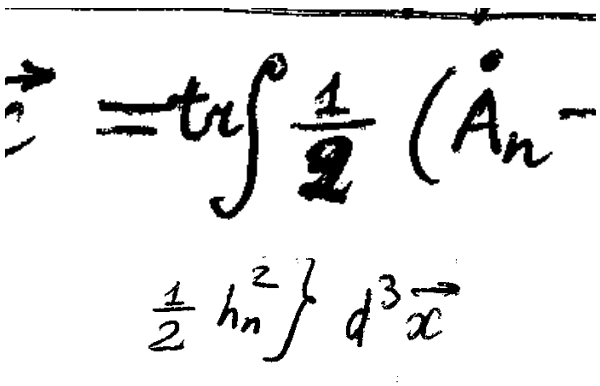
A_μ is called the vect. potential,

$F_{\mu\nu}$ —"— the field-strength tensor.

Passage to the Hamiltonian formalism. $n = 1, 2, 3.$

$$L = \int \mathcal{L} d^3\vec{x} = \text{tr} \int \frac{1}{2}(\dot{A}_n - \nabla_n A_0)^2 - \frac{1}{2} h_n^2 \Big\} d^3\vec{x}$$

(we single out time)



Uncertain reading. The manuscript has a closing brace. No opening brace is visible after the integral sign.

Original source details: GELFAND-P000644

Here $\nabla_n A_0 = \partial_n A_0 + g[A_n, A_0]$ (and we call it the covariant derivative).

$$h_n = \frac{1}{2}\varepsilon_{nm\ell}(\partial_m A_\ell - \partial_\ell A_m + g[A_m, A_\ell]),$$
$$F_{0n} = \partial_0 A_n - \partial_n A_0 + g[A_0, A_n].$$

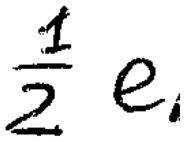
$$\nabla_\mu X \underset{\text{def}}{=} \partial_\mu X + g[A_\mu, X].$$

GELFAND-P000645

L does not contain $\dot{A}_0$, $e_n = \dfrac{\partial \mathcal{L}}{\partial \dot{A}_n} = \dot{A}_n - \nabla_n A_0$ — canon. momentum.

Hamiltonian.

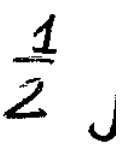
$$\mathcal{H} = e_n \dot{A}_n - \mathcal{L} = \frac{1}{2} e_n (e_n + \nabla_n A_0) + \frac{1}{2} h_n^2.$$



Uncertain reading. The manuscript has $\frac{1}{2}$ before $e_n(e_n + \nabla_n A_0)$.

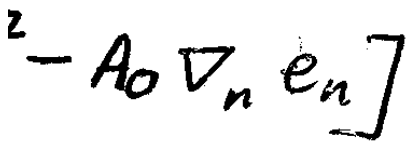
Original source details: GELFAND-P000645

$$H = \frac{1}{2} \int d^3\vec{x}\ \text{tr}[e_n^2 + h_n^2 - A_0 \nabla_n e_n].$$



Uncertain reading. The prefactor of the integral reads one half.

Original source details: GELFAND-P000645



Original source details: GELFAND-P000645

Uncertain reading. A minus sign stands before $A_0 \nabla_n e_n$, with no separate numerical factor.

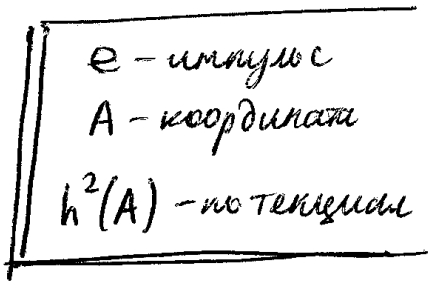
$$-\frac{\delta H}{\delta A_0} = \dot{e}_0 = 0.$$

Hamilton eqs. $\leftrightarrow$ Lagrange eqs..

Hence let us impose the cond.

$$\nabla_n e_n = 0 \qquad (*)$$

$$\widehat{H} = \frac{1}{2}\int d^3\vec{x}\,[e_n^2 + h_n^2].$$



Original source details: GELFAND-P000645

Diagram and typed description.

e — momentum
A — coordinates
$h^2(A)$ — potential

A note in the right margin with an open-sided frame.

Will the cond. hold at the next instant $(*)$?

Denote $\nabla_n e_n = \rho(\vec{x})$.

This is the condition $\{\rho(x), H\} = 0$, where

$$\{A, B\} = \int d^3\vec{x} \sum_n \left(\frac{\delta A}{\delta e_n} \cdot \frac{\delta B}{\delta A_n} - \frac{\delta A}{\delta A_n}\frac{\delta B}{\delta e_n} \right).$$

$$\begin{cases} \dot{A}_n = \dfrac{\delta H}{\delta e_n}, \\ \dot{e}_n = -\dfrac{\delta H}{\delta A_n}, \\ \nabla_n e_n = 0. \end{cases}$$

[At right] These eqs. $\sim$ the Lagrange eqs. This generalizes the usual result..

(generalization of Maxwell's eqs. from $U(1)$ to any [crossed out] compact group — and even a semisimple one).

Original source details: GELFAND-P000645

Undeciphered passage. A word between "any" and "compact" is crossed out. Lower loops are visible.

—·—

$\exists$ a comb. of Y.–M. fields on which the potential energy does not depend h^2.

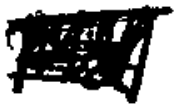
$$h_n = \frac{1}{2}\varepsilon_{nm\ell}F_{m\ell},$$
$$F_{m\ell} = \partial_m A_\ell - \partial_\ell A_m + g[A_m, A_\ell].$$

$$\widetilde{A}_n(x) = \Omega^+ A_n \Omega + \frac{1}{g}\Omega^+\partial_n\Omega(\vec{x}),$$

$\Omega(x)$ — arbitrary matrix, $\Omega(x) \in SU(2)$.

$\widetilde{h}_n = \Omega^+ h_n \Omega$ — checked directly.

$\Rightarrow$ the pot. en. [crossed out] invar. (Sp h^2 does not change).



Undeciphered passage. A short entry before "invar." is heavily crossed out.

Original source details: GELFAND-P000645

Introduce the field a_n. consider fields with cond. $\partial_n a_n = 0$.

Set

$$\begin{cases} A_n(x) = \Omega^+(x) a_n \Omega(x) + \frac{1}{g}\Omega^+\partial_n\Omega(x), \\ \partial_n a_n = 0. \end{cases}$$

[At right] $\|$ a new set of variables.

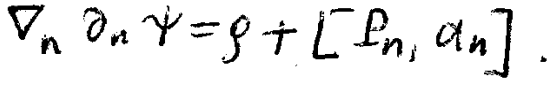
GELFAND-P000646

Canon. momentum split into two parts. P_n, $\rho(x)$, $\partial_n P_n = 0$.

$$e_n = P_n + \partial_n\Psi, \qquad H = \frac{1}{2}\int (P_n^2 + h_n^2(a) + (\partial_n\Psi)^2)d^3\vec{x}.$$

To find Ψ we use that $\nabla_n e_n \underset{\text{def}}{=} \rho(\vec{x})$.

$$\nabla_n P_n + \nabla_n\partial_n\Psi = \rho,$$
$$\nabla_n\partial_n\Psi = \rho + [P_n, a_n].$$



Uncertain reading. The first formula has $[P_n, a_n]$ with no factor g.

Original source details: GELFAND-P000646

$(\partial_n \Psi)^2$ is called Coulomb energy.

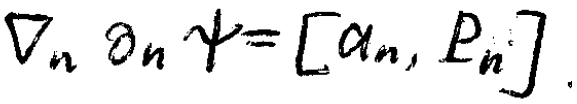
—·—

$$H = \int (e_n^2 + h_n^2(A)),$$
$$A_n \longrightarrow \Omega^+ a_n \Omega + \Omega^+ \partial_n \Omega,$$
$$\partial_n a_n = 0.$$

$$\begin{cases} H = \int (P_n^2 + (\partial_n \Psi)^2 + h^2), \\ \nabla_n \partial_n \Psi = [a_n, P_n]. \end{cases} \qquad \text{(Schwinger)}$$

Uncertain reading. The parenthesis at right reads “швингер” (Schwinger).

Original source details: GELFAND-P000646



Uncertain reading. Here the bracket reads $[a_n, P_n]$.

Original source details: GELFAND-P000646

Here $h_n(a) = \frac{1}{2}\varepsilon_{nm\ell}(\partial_m a_\ell - \partial_\ell a_m + g[a_m, a_\ell])$. Ψ is found from the last eq..

$$\nabla_n e_n = \rho(\vec{x}) \quad \Rightarrow \quad \begin{cases} \partial^2 \Psi = \rho(x) \quad \text{(in the scalar case)}, \\ \int (\partial \Psi)^2 dx = \int \dfrac{\rho(x)\rho(y)}{|x-y|}\, dx\, dy. \end{cases}$$

— Coulomb interaction.

Uncertain reading. Above the equality the entry reads “в скал. случае” (in the scalar case).

Original source details: GELFAND-P000646

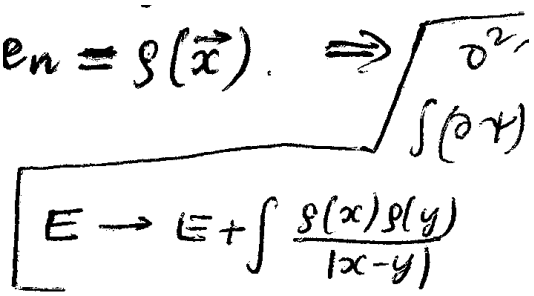


Diagram and typed description.

A bent line connects the energy expression to the preceding formulas.

Original source details: GELFAND-P000646

Non-Ab. case. First by perturbation theory (first Khriplovich, then Gross, etc.).

Classical case: $g = 0 \Rightarrow$ the problem is Abelian.

We obtain

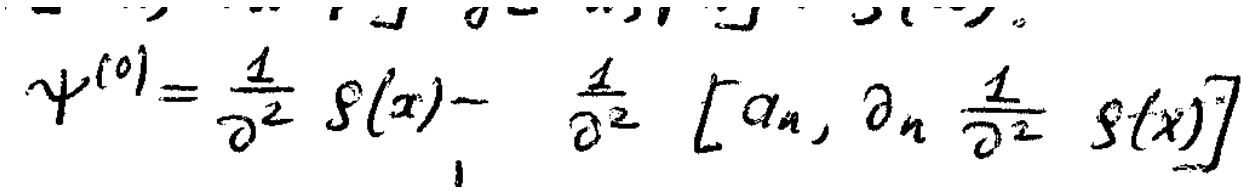
$$\partial^2 \Psi + g[a_n, \partial_n \Psi] = g[a_n, P_n] + \rho(x).$$

Uncertain reading. The right-hand side reads $g[a_n, P_n] + \rho(x)$.

Original source details: GELFAND-P000646

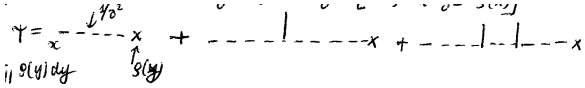
Approximations:

$$\Psi^{(0)} = \frac{1}{\partial^2}\rho(x) - \frac{1}{\partial^2}\left[a_n, \partial_n \frac{1}{\partial^2}\rho(x)\right].$$



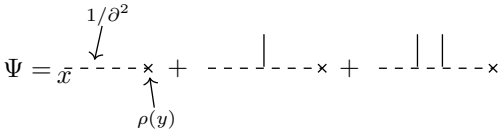
Original source details: GELFAND-P000646

Uncertain reading. The second term has prefactor $-1/\partial^2$. No separate factor g is visible.



Original source details: GELFAND-P000646

Diagram and typed description.



Three dashed diagrams with zero, one and two vertical branches.

[Below the first diagram] $\int \frac{1}{|x-y|}\rho(y)dy$.

Branch — a_n.

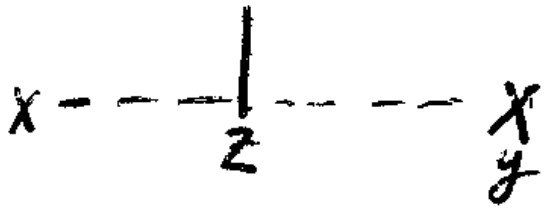
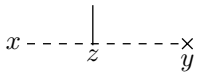


Diagram and typed description.

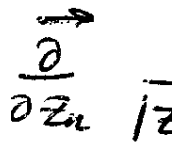


The lower diagram has a branch at z and a cross above y.

Original source details: GELFAND-P000646

[At right]

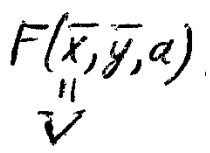
$$= \int \frac{1}{|x-z|} a_n(z) \frac{\vec{\partial}}{\partial z_n} \frac{1}{|z-y|} \rho(y)\, d\vec{y}\, d\vec{z}.$$



Uncertain reading. An arrow is written over the derivative sign in the numerator.

Original source details: GELFAND-P000646

$$\mathcal{H} \to \mathcal{H} + \int \rho(\bar{x}) F(\bar{x}, \bar{y}, a) \rho(\bar{y})\, d\bar{x}\, d\bar{y}.$$
$$\Big\Vert$$
$$V$$



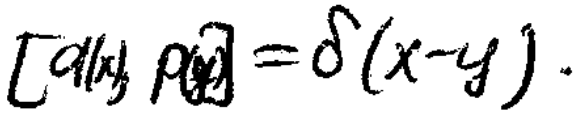
Uncertain reading. Below the arguments of F are two short strokes and a sign resembling V.

Original source details: GELFAND-P000646

Previously $F = \dfrac{1}{|x-y|}$. now F — an operator.

GELFAND-P000647

Passage to quant. mech. — a, p — opers., $[a(x)\ p(y)] = \delta(x-y)$.

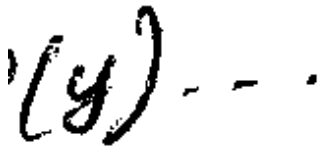


Uncertain reading. The second entry in the bracket reads $p(y)$. The marks between entries are crowded, and a separate comma is indistinct.

Original source details: GELFAND-P000647

$$\delta E = \langle V \rangle = \int \rho(x) \langle F(x,y,a) \rangle \rho(y) \dots$$

(averaging)



Uncertain reading. Three short strokes follow $\rho(y)$. The integration measure is unwritten.

Original source details: GELFAND-P000647

5 April — continuation.

GELFAND-P000648

22/III-76. Gelfand Sem.

G. A prob. distrib. arising from biology.

$$P_n = e^{-x_0 - \alpha n} x_0 \frac{(x_0 + \alpha n)^{n-1}}{n!}, \qquad 0 \leqslant \alpha < 1.$$

Uncertain reading. The initial letter appears to be uppercase P.

Original source details: GELFAND-P000648

For $\alpha = 0$ — Poisson distrib..

Exercise. check that the sum equals 1.

“This is the only case where math. makes sense”. There is one more case in the literature.

Uncertain reading. The end of the sentence reads “в лит-ре есть” (there is one in the literature).

Original source details: GELFAND-P000648

Roiter.

Λ — a finite-dimensional algebra over the field k.

There may be finitely or infinitely many indecomp. reps..

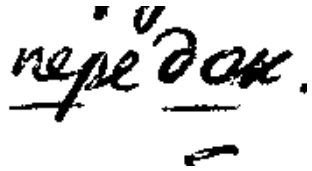
1968 Roiter: if there are infinitely many, there are indecomp. reps. of arbitrarily large dimension. Gabriel — a simpler proof.

(Maurice) Auslander: 1) generalized to Artinian rings.

2) if Λ has infinitely many indecomp. reps., there are infinite chains of modules embedded in one another.

$$A_1 \to A_2 \to A_3 \to \ldots \to A_n \to \ldots$$

Roiter reproved this with additional information.



Uncertain reading. The abbreviation before "with additional information" reads "передок." (reproved).

Original source details: GELFAND-P000648

Λ — a finite-dimensional algebra, A — an indecomp. module over Λ.

$\ell(A)$ — length of the mod. A — max. length

$$0 \subset A_1 \subset A_2 \subset \cdots \subset A_n = A.$$

$\bar{\ell}(A)$ — max. length [inserted with arrow: of such a] chain of indecomp. modules.

We have $\ell(A) \geqslant \bar{\ell}(A)$.

Theorem $\forall \Lambda$ and $\forall$ nat. N, $\exists n$:

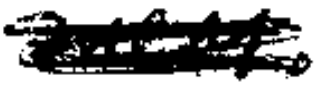
$$\bar{\ell}(A) \leqslant n \Longrightarrow \ell(A) \leqslant N.$$

Uncertain reading. The arrow goes from $\bar{\ell}(A) \leqslant n$ to $\ell(A) \leqslant N$.

Original source details: GELFAND-P000648

—

There is an extended Dynkin diagram with n [crossed out] vertices.



Undeciphered passage. The word between n and "vertices" is heavily crossed out.

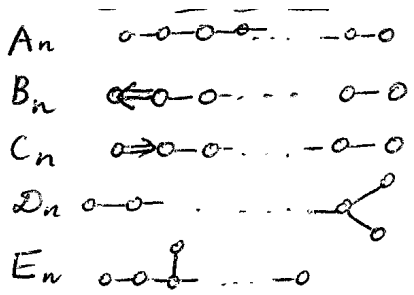
Original source details: GELFAND-P000648

It has an associated quadr. form

$$f(x_1, \dots, x_n) = \sum_{i=1}^{n} a_i x_i^2 + \cdots$$

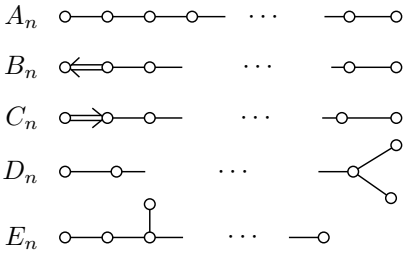
[At left] Kirillov.

Dynkin diagram one of the pictures.

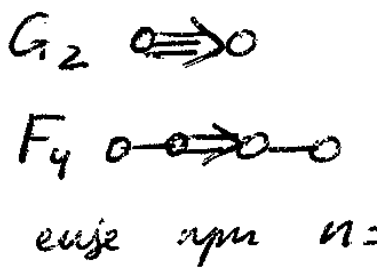


Original source details: GELFAND-P000648

Diagram and typed description.

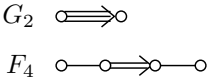


Left column with the diagrams A_n, B_n, C_n, D_n, E_n.



Original source details: GELFAND-P000648

Diagram and typed description.



Right column with the diagrams G_2, F_4.

[At right] and also for $n = 6, 7, 8$.

$\longrightarrow$

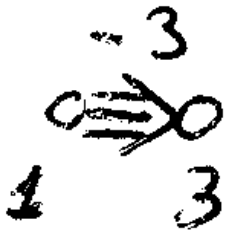
[At left] Roiter.

Dynkin diag. — an integer-valued quadr. form

$$\sum a_i x_i^2 + \sum_{i<j} b_{ij} x_i x_j.$$

additional conditions 1) positive definiteness. 2) $a_i > 0$. 3) $b_{ij} \leqslant 0$. 4) $a_i \mid b_{ij}$ (divides).

[At right] || one obtains precisely the Dynkin diagrams.



Original source details: GELFAND-P000648

Diagram and typed description.

The lower diagram has 1 and 3 below the circles. The stroke before the upper 3 resembles a minus sign.

|| Extended diag. — if the condition is replaced by nonnegative definiteness.

GELFAND-P000649

A reflection is a change of one coord. of the vector that leaves the quadr. form unchanged. (lin. transf.).

Generate the group by all the refl.

Simple root — $(0, 0, \dots, 1, \dots, 0)$.

Root: a vector obtained from a simple one by several reflections.

Given a vector $(x_1, \dots, x_n)$. When is it a root?

If all the strokes are single, then root $\iff$ the quadr. form on it equals 1.

Original source details: GELFAND-P000649

Uncertain reading. The abbreviation before “are single” reads “чёрт.” (strokes).

[Crossed out: If they are not all single] If they are not all single?

Original source details: GELFAND-P000649

Uncertain reading. The first “If they are not all single” is crossed out with several strokes.

Thm.

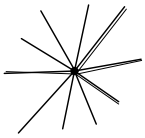
$$(x_1, \dots, x_n) \text{ — root} \quad \iff \quad \forall i: \ f(x) \mid a_i x_i.$$

Exercise (for first-year students).

$$\sum_{i=1}^{n} a_i x_i^2 - \sum_{i<j} b_{ij} x_i x_j, \qquad \frac{b_{ij}}{a_i} \in \mathbb{Z}, \quad a_i > 0.$$

Original source details: GELFAND-P000649

Diagram and typed description.

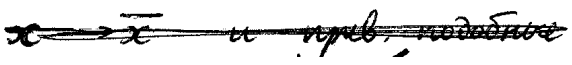


Rays at the left of the following formula. Some strokes are doubled.

$$x_0^2 + \dots + x_n^2 - x_0 x_1 - x_0 x_2 - \dots - x_0 x_n. \quad \text{unclear.}$$

Refl. — a lin. transf. that leaves $f(x)$ unchanged and changes only one coord.

[Crossed out: $x \to \bar{x}$ and пр...в., similar.]



Original source details: GELFAND-P000649

Undeciphered passage. The interior letters of the crossed-out abbreviation are only partly legible.

describe all the roots! (obtained by refl. from standard vectors) (ВОЛК→КОЗА).

[At right, column]

ВОЛК (wolf)
ПОЛК (regiment)
ПОЛА (coat flap)
ПОЗА (pose)
КОЗА (goat)

— · —

Kirillov on Berezin's talk.

$$x_i x_k = x_k x_i, \qquad x_1, \dots, x_p$$
$$\xi_i \xi_k = -\xi_k \xi_i, \qquad \xi_1, \dots, \xi_q$$

Supermanifold: Berezin, Leites.

[In the left margin] Following Rudakov.

Consider the manifold M^p and the bundle $E \to M^p$, whose fibre is a Grassmann algebra (with q generators) (the struct. is specific to $\forall$ point).

Morphisms are described in a more complicated way.

A fn. on a superman. is a smooth sect. of the bundle.

Integration rules. w.r.t. x as usual, and

$$\int d\xi = 0, \qquad \int \xi \, d\xi = 1.$$

Reps. of a 64-dimensional Lie superalgebra describe elem. particles better than reps. of $SU(3)$.

Change of var.:

$$\begin{cases} x \to x + \xi_1 \xi_2, \\ \xi_1 \to \xi_1 + x \xi_2, \\ \xi_2 \to \xi_2 + \xi_1 \xi_2 \xi_3 \end{cases}$$

$$\int P(\xi_1, \xi_2, \xi_3) \, d\xi_1 \, d\xi_2 \, d\xi_3 = \text{coeff. of the highest term.}$$

Lie superalgebra: $\mathbb{Z}_2$-graded, and

$$[x, y] = -(-1)^{\deg x \cdot \deg y} [y, x].$$

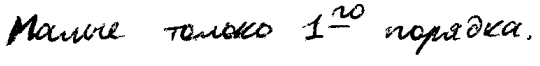
Jacobi id. $[x, \cdot]$ — a homogeneous derivation.

more conv. def.: a superalg. is obtained from a graded assoc. algebra by the rule

$$[x; y] = xy - (-1)^{\deg x \cdot \deg y} yx.$$

Lie thm. — Berezin and Kac from Kyiv.

G.'s remark. $p(t + \xi) = p(t) + p'(t)\xi$. Small quantities only of first order.



Uncertain reading. The first word reads "Малые" (small quantities).

Original source details: GELFAND-P000649

GELFAND-P000650

Fns. on a superman. form a superalgebra.

There is a Poisson bracket.

Find deformations in the direction of the Poisson bracket.

Sinai. Theory of phase transitions — spectra of matrices.

It was possible to calc. one of the interesting examples.

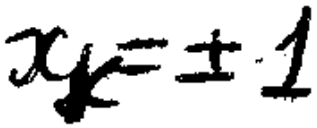
Uncertain reading. The abbreviation after "of" reads "инт." (interesting).

Original source details: GELFAND-P000650

Dimension of the space 2^n.

$$a_{ij} \qquad a(\underbrace{x_1, \dots, x_n}_{i}, \underbrace{y_1, \dots, y_n}_{j})$$
$$= e^{\beta(J_1 \sum_{i=1}^{n} x_i x_{i-1} + J_2 \sum_{i=1}^{n} x_i y_i)}.$$

$x_k = \pm 1$, $x_0 = x_n$.



Uncertain reading. The subscript on x is heavily overwritten and resembles k.

Original source details: GELFAND-P000650

Physical problem: one needs to study $\frac{e^{-\beta H}}{Z}$. Here [crossed out]

Undeciphered passage. A short passage after "Here" is heavily crossed out.

Original source details: GELFAND-P000650

$$\frac{\ln Z}{R^2} \underset{R \to \infty}{\longrightarrow} f(\beta).$$

The fns. $f(\beta)$ can have singularities.

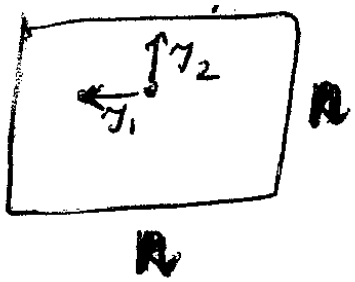
Study $f(\beta)$ — [crossed out]

Undeciphered passage. A short word before "there is" is heavily crossed out.

Original source details: GELFAND-P000650

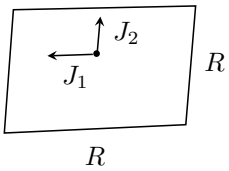
there is a transfer matrix (transfer-matrix).

Matrix with positive entries.



Original source details: GELFAND-P000650

Diagram and typed description.



The arrows point left and up. The labels read as J_1 and J_2.

$$Z = \sum e^{-\beta \sum_{x',x''} H(x',x'')}, \qquad \underline{Z = \operatorname{tr} A^R}.$$

Original source details: GELFAND-P000650

Uncertain reading. The superscript is heavily overwritten and reads as R.

A^R — transition matrix from layer to layer.

(Onsager's 2-dimensional problem).

Onsager computed all the eigenvalues of this matrix:

Anticomm. variables help here: there are anticomm. matrices A_i, A_j, such that

$$A = e^{\sum c_{ij} A_i A_j}$$

and the integrals can be calculated.

[In the right margin] ‖ see Huang "Stat. Mech.".

Lieb similarly calculated the entropy of square ice.

— · —

GELFAND-P000651

29/III-76. Gelfand Sem.

Krichever: problem: solve the eq. $[L_1, L_2] = 0$. If ord L_1 and ord L_2 are coprime, the answer is a curve and a bundle (1-dimensional).

The orders are not coprime — multidimensional sheaves arise.

E.g. 4th and 6th order — not done. Reduced to 2-dimensional bundles.

$$\left.\begin{aligned} L_1 y = E_1 y \\ L_2 y = E_2 y \end{aligned}\right\} \quad P(E_1, E_2) = 0 \quad \text{— curve.}$$

Original source details: GELFAND-P000651

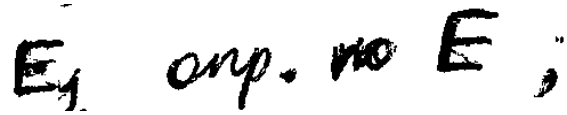
Undeciphered passage. A heavily crossed-out passage below the brace of the simultaneous equations.

Joint solution — a 2-dimensional bundle over the curve.

Coprime — the basis is 1-dimensional. (from the asymptotics).

$(m, n) = 1 \Longrightarrow$ the number of common eigenfns. equals n. (and the eigenvalues are distinct)

E_1 is determ. by E; y a uniquely determ. fn. on a Riemann surf.



Original source details: GELFAND-P000651

Uncertain reading. The subscript on the first E tentatively reads 1. The word “по” (by) is heavily overwritten.

Inverse problem: construct commuting diff. ops. from the bundle — not done.

Drinfeld.

$d = (m, n) \Longrightarrow$ specifying a pair of comm. ops. $\Longleftrightarrow$ specifying $(d-1)$ arbitrary fns.

$B^{-1}LB$ can be solved — 1 arbitrary fn. remains — the leading coeff.

Orders 4 and 6 — if square and cube — already 2 arbitrary fns.

Shnirelman. M — an n-dimensional compact analytic Riemannian manifold,

Δ — Laplace op. on M;

$$\Delta u_k + \lambda_k^2 u_k = 0; \qquad (1)$$

def. of Δ:

Original source details: GELFAND-P000651

Diagram and typed description.

sphere of rad. τ;

Circle with a dot at the centre.

$$\frac{\text{avg. over sphere of rad. } \tau \text{ — value at centre}}{\tau^2} \longrightarrow \Delta f(x).$$

Growth as k: $\lambda_k \sim ck^{1/n}$, [crossed out: $a \ldots$]

Original source details: GELFAND-P000651

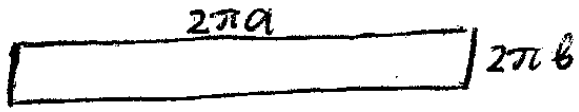
Undeciphered passage. A sign resembling a is visible at the start of the cancellation.

Let $a > 0$. How is the following distributed $a \cdot \lambda_k \bmod 2\pi$

(There are often arithmetic progressions).

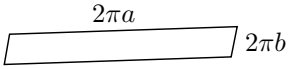
— subseq.

Torus.



Original source details: GELFAND-P000651

Diagram and typed description.



Rectangle under the heading “Torus”.

$$\Delta u = \lambda^2 u.$$

$\sin \alpha x \cdot \sin \beta y$ [crossed out: cos . . .]

Original source details: GELFAND-P000651

Undeciphered passage. The start of the crossed-out entry resembles cos. The remaining signs are illegible.

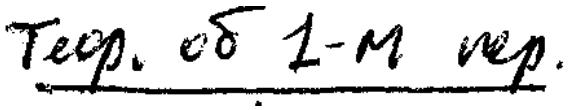
or $\cos \alpha x \cdot \sin \beta y$ etc.

$$\lambda^2 = \frac{n^2}{a^2} + \frac{m^2}{b^2}, \qquad \lambda = \sqrt{\frac{n^2}{a^2} + \frac{m^2}{b^2}}.$$

Series — for different n and m (for fixed).

Which 2 indices — unclear.

Thm. for 1 dim.



Original source details: GELFAND-P000651

Uncertain reading. The last abbreviation tentatively reads “мер.” (dimension).

$$-y'' + u(x)y = \lambda y \quad \Longrightarrow \quad \text{on}$$

Original source details: GELFAND-P000651

Diagram and typed description.

⟝

Horizontal line with two end ticks.

$$\sum_{i=0}^{n-1} c_i y_i \quad \text{has} \leq n \text{ zeros} \quad \Longrightarrow \quad \text{behaves like a polyn. of degree 1.}$$

Hence 2 dim. — a polynomial in n variables.

Original source details: GELFAND-P000651

Uncertain reading. The abbreviation after 2 tentatively reads “мер.” (dimension).

Lowest — positive everywhere.

1st — like linear fns.

Next — a curve of 2nd order.

follow the lin. combin.

Original source details: GELFAND-P000651

Diagram and typed description.

Two ovals. The sloping line in the left one runs from an interior point to the upper boundary. In the right one it runs from the left boundary to an interior point.

Rayleigh — theory of sound (2 volumes).

Arnold: transfer Harnack's thm. on the number of ovals (unknown for lin. combs.).
[In the right margin] ‖ For the nth eigenfn. the zeros divide the region $\leq$ into n parts.

GELFAND-P000652

Consider

$$\Phi(\lambda, a) = \lim_{\varepsilon \to 0} \varepsilon \left(\sum \delta((\lambda - a\lambda_k) \bmod 2\pi) e^{-\varepsilon \lambda_k} - \sum_k \frac{e^{-\varepsilon \lambda_k}}{2\pi} \right).$$

$$\mu_k = a\lambda_k \bmod 2\pi.$$

μ — param. on the circle.

Uncertain reading. The heavily retraced letter tentatively reads μ. The following words read "param. on the circle".

Original source details: GELFAND-P000652

$$\Phi(\mu, a) = \lim_{\varepsilon \to 0} \varepsilon \left(\sum_{k=1}^{\infty} \delta(\mu - \mu_k) e^{-\varepsilon \lambda_k} - \sum e^{-\varepsilon \lambda_k} / 2\pi \right).$$

[Arrow to the subtracted term] subtract so that the avg. value is 0. constant.

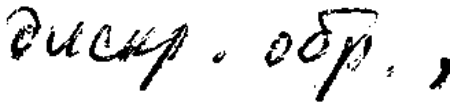
$\Phi^{\varepsilon}(\mu, a)$ — pre-limit expression.

Theorem. Let $\gamma_1, \gamma_2, \ldots$ — closed geodesics on M, $\ell(\gamma_1), \ell(\gamma_2), \ldots$ — their lengths. Let невд. расп.,

Uncertain reading. The first Russian abbreviation resembles "невд.". The next reads "расп.".

Original source details: GELFAND-P000652

in a discr. manner,



Uncertain reading. The Russian abbreviation reads "дискр." (discr.).

Original source details: GELFAND-P000652

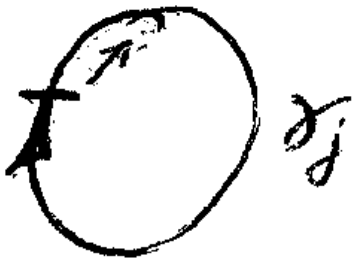
the lengths are all distinct and incommensurable. (general pos.).

$$\Phi(\mu, a) \begin{cases} \equiv 0 & \text{if } a \neq \frac{p}{q} \ell(\gamma_j), \\ \not\equiv 0 & \text{[crossed out] if } a = \frac{p}{q} \ell(\gamma_j). \end{cases} \qquad (p, q \in \mathbb{Z})$$

Undeciphered passage. The word before the second "if" is heavily crossed out.

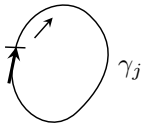
Original source details: GELFAND-P000652

[In the left margin]



Original source details: GELFAND-P000652

Diagram and typed description.



Oval and arrows. The heavy strokes on the left edge overlap.

consider the geodesic flow. ($\in SM$).

S — a plane transv. to the closed geod. (dim. $2n-2$).

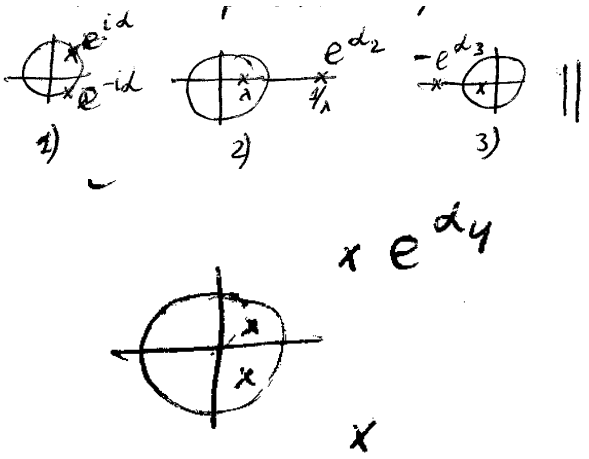
return map $\Psi : S \to S$.

$S \cap \gamma_j$ — fixed point.

$$d\Psi\big|_{S\cap\gamma_j} = \mathcal{M} \quad \text{— monodromy op.} \quad (\underline{\text{monodromy matrix}}).$$

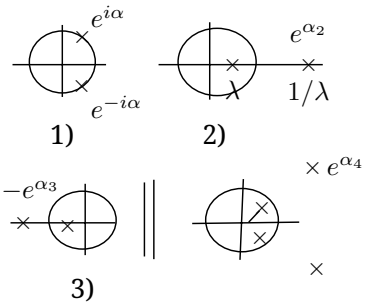
This transf. is symplectic, since the flow is Hamiltonian.

The eigenvalues fall into pairs and quadruples.



Original source details: GELFAND-P000652

Diagram and typed description.



Three numbered sketches and an unnumbered fourth. No legible label is visible at the inner cross in the third sketch.

[At the right of the fourth sketch] — this only in 3-dimensional space.
Consider the Fourier coeffs.:

$$\varphi_n(a) = \int_0^{2\pi} \varphi(\mu, a) e^{-i\mu n}\, d\mu.$$

$$1)\quad \varphi_n(a) = \frac{c}{\sin \frac{n\alpha_1}{2}},$$
$$2)\quad \varphi_n(a) = \frac{c}{\operatorname{sh} \frac{n\alpha_2}{2}},$$
$$3)\quad \varphi_n(a) = (-1)^n \frac{c}{\operatorname{sh} \frac{n\alpha}{2}}.$$

c depends on the dimension.
[A long enclosing line runs from the three formulas to the entry at right.]
1) α_1 is poorly approx. by rationals (required) $\Longrightarrow$

Uncertain reading. The word before “approx.” reads “плохо” (poorly).

Original source details: GELFAND-P000652

Uncertain reading. The parenthesis reads “треб.” (required).

Original source details: GELFAND-P000652

$$\text{A)}\quad \varphi(\mu, a) = \sum_{n=0}^{\infty} \delta\left(\mu - \left(n + \frac{1}{2}\right)\alpha - \frac{1}{2\pi}\right).$$

(exactly in the reg. sense

$$\varphi(\mu, a) = \lim_{\varepsilon \to 0} \sum_{n=0}^{\infty} \delta\left(\mu - \left(n + \frac{1}{2}\right)\alpha e^{-n\varepsilon} - \frac{1}{2\pi}\right).$$

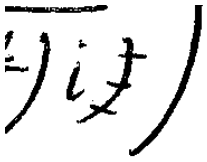
[In the lower left margin] 2 quantum numbers — 2 indices.
[In the lower left margin] [heavy mark] Infinitely many arithm. progressions.

Undeciphered passage. Before the line is a heavy mark resembling the letter A.

Original source details: GELFAND-P000652

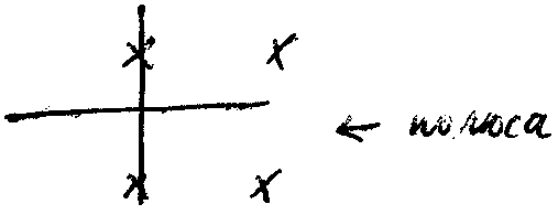
$$2) \quad \varphi(\mu, a) = c \sum_{m,n=-\infty}^{\infty} \left(\frac{1}{\mu - n \cdot 2\pi - (m + \frac{1}{2})i} + \frac{1}{n \cdot 2\pi + (m + \frac{1}{2})i \text{ [cancelled mark]}} \right)$$
$$= c \int_0^{\mu} \wp(\xi)\, d\xi,$$

$\wp$ — Weierstrass fn.



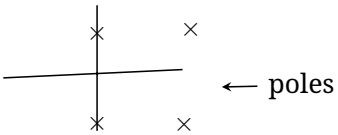
Original source details: GELFAND-P000653

Uncertain reading. A stroke with several crossbars follows i in the second denominator. The cancelled letter is undeciphered.

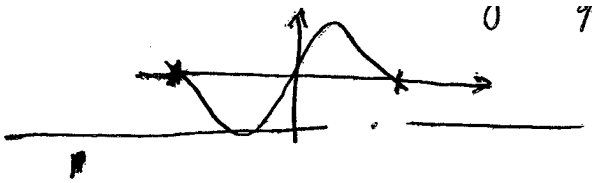


Original source details: GELFAND-P000653

Diagram and typed description.

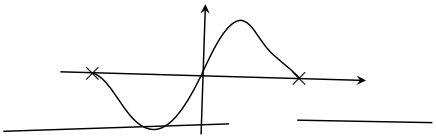


Crosses on the vertical axis and to its right.



Original source details: GELFAND-P000653

Diagram and typed description.



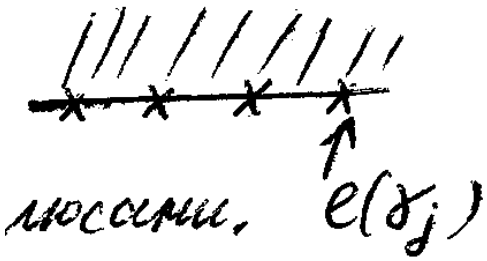
Curve with a minimum and a maximum, two crosses on the axis, and a broken lower line. A heavy illegible mark lies below the left part of the line.

Consider the fn.

$$f(z) = \sum_{k=1}^{\infty} e^{i\lambda_k z}.$$

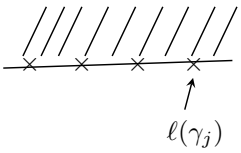
Continues analytically downwards with simple poles.

[At right]



Original source details: GELFAND-P000653

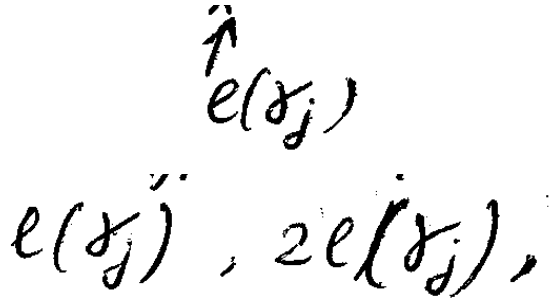
Diagram and typed description.



Four crosses on a hatched horizontal line. The arrow from below points to the rightmost cross.

Fund. sol. of the wave eq. Continuat.

Residues at $\ell(\gamma_j)$, $2\ell(\gamma_j)$, ... — Fourier coeffs., etc.



Uncertain reading. The letter with subscript j in the label and residue line reads γ, as in the geodesic notation on the preceding frame.

Original source details: GELFAND-P000653

Uncertain reading. The abbreviation before "Fourier" is heavily overwritten and reads "коэфф." (coeffs.).

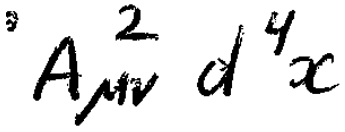
Original source details: GELFAND-P000653

GELFAND-P000654

5/IV-76. Gelfand Sem. Polyakov.

Action

$$\begin{cases} S = \dfrac{1}{4e^2}\,\mathrm{tr}\int A_{\mu\nu}^2\, d^4x \\ F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \end{cases}$$



Uncertain reading. The integrand is written $A_{\mu\nu}^2$. The next line defines $F_{\mu\nu}$.

Original source details: GELFAND-P000654

$A_\mu^+ = -A_\mu$, A_μ — a skew-Herm. matr. 2×2, which is a fn. on $\mathbb{R}^4$.

[Below A_μ] $\mu = 0, 1, 2, 3$.

A_μ — the field. S — a functional of the field.

Classical eqs. of motion — stationarity of S.

For numerical A_μ — electromagn. field.

Analogue for discrete x and t.

$$x = (t, \vec{x}).$$

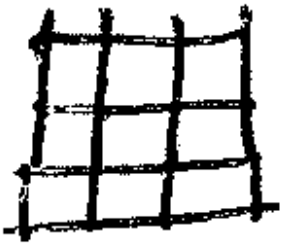
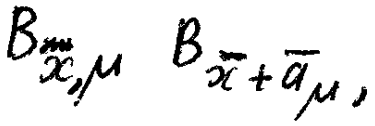


Diagram and typed description.

Four roughly horizontal lines and four vertical lines.

Original source details: GELFAND-P000654

$$S = \frac{1}{4e_0^2} \sum_{\bar{x}} \chi \left(B_{\bar{x},\mu} B_{\bar{x}+\bar{a}_\mu,\nu} B^{-1}_{\bar{x}+\bar{a}_\nu,\mu} B^{-1}_{\bar{x},\nu} \right).$$



Original source details: GELFAND-P000654

Uncertain reading. Short bars appear over x and a in the product. The bar over the first x is heavily retraced.

a_μ — frame, $\quad B_{x,\mu} \in SU(2)$.

χ — any character (crossed out)

Original source details: GELFAND-P000654

Undeciphered passage. The crossed-out word before "fn." is illegible.

(a fn. constant on classes of conj. elements — e.g. trace).

$$Z = \int \prod_{\vec{x},\mu} dB_{\vec{x},\mu}\, e^{-S(B)}.$$

$12n$ indep. variables (n — the number of lattice sites, at $\forall$ — 4 matrices).

$B_{\vec{x},\mu}$ — indep. vars. $\Longrightarrow dB_{\vec{x},\mu}$ — inv. measure ($4n$ integrals with inv. measures).

$$t \to it \quad \Longrightarrow \quad e^{iS} \to e^{-S}.$$

In the continuous case — a functional integral.

Correl. fns.

$\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots$

GELFAND-P000655

12/IV-76. Gelfand Sem.

Fuchs. Use of computers in theor. mathematics.

Comput. of the cohom. of the alg. of formal divergence-free vect. fields

Given $\mathbb{R}^n$. consider $(\mathbb{R}^n)'$ and polynomials

$$p(\xi_1, \dots, \xi_q; \eta_1, \dots, \eta_q), \qquad \xi_i \in \mathbb{R}^n, \quad \eta_i \in (\mathbb{R}^n)'.$$

They have the property

a) if the pair ξ_i, η_i is put in the place of ξ_j, η_j, the polynomial changes sign.

$dP = P_1$ given by the formula

$$P_1(\xi_1, \dots, \xi_{q+1}, \eta_1, \dots, \eta_{q+1}) = \sum_{i<j} (-1)^{i+j-1} \big[(\xi_i, \eta_j) - (\xi_j, \eta_i)\big] \\ P(\xi_i + \xi_j, \xi_1, \dots, \widehat{\xi_i}, \dots, \widehat{\xi_j}, \dots, \xi_{q+1}, \\ \eta_i + \eta_j, \eta_1, \dots, \widehat{\eta_i}, \dots, \widehat{\eta_j}, \dots, \eta_{q+1}).$$

problem: find the cohomology of this complex (still undone).

A computer helped find nontriv. cycles.

Consider those linear in η; then this is the cohom. complex of the algebra W_n.

Undone even for $n = 1$.

All results obtained concern this case.

complex = the sum of many finite-dimensional ones.

Total degree in ξ: $m_1 + q$, in η: $m_2 + q$. [At right] ‖ subcomplex K_{m_1,m_2}.

(q varies)

polys. in $\xi_1, \dots, \xi_q, \eta_1, \dots, \eta_q$.

Manuscript excerpt. The first abbreviation reads “мн-ны” (polynomials).

Original source details: GELFAND-P000655

Each of them is finite-dimensional.

$m_1 \neq m_2 \Longrightarrow$ it is acyclic.

In fact, $\sum$ homol. of a countable number of subcompls.

$$H(K) = \bigoplus_m H(K_{m,m}).$$

(for now n is arbitrary)

$n = 1$.

$m = 1, 2, 3$ are acyclic.

$m = 4$ nontriv. in dim. 7 and 10 — computed on a machine (Gelfand, Fuchs, Kalinin).

Uncertain reading. The third surname reads “Калинин” (Kalinin).

Original source details: GELFAND-P000655

$m = 5$ — “work area overflow $N = 2$”.

Euler char. does not depend on the differential.

Perchik (Bott’s grad. student) computed up to $m = 35$.

Uncertain reading. The manuscript spells the surname “Першик”.

Original source details: GELFAND-P000655

$$K_{m,m} \supset K'_{m,m} \qquad H(K_{m,m}) = H(K'_{m,m}) \otimes H(S^3).$$

$$\chi^*_m \text{ — ?} \qquad (\text{for } K'_{m,m}).$$

Perchik found the generating fn. for χ_m.

Uncertain reading. After “for” the manuscript has χ_m without a star.

Original source details: GELFAND-P000655

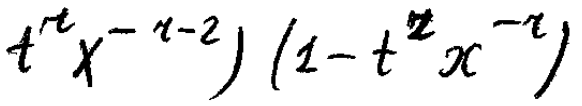
Interesting: prove that for arbitrarily large m there are $\chi_m \neq 0$.

Uncertain reading. The middle letters are retraced. The word reads “Интересно” (interesting).

Original source details: GELFAND-P000655

$$\tfrac{1}{2}(1 - t^{-1}x^{-1})(1 - t^{-1}x)$$
$$(1 - x^{-2}) \qquad (1 - x^2)$$
$$(1 - tx^{-3})(1 - tx^{-1})(1 - tx)(1 - tx^3)$$
$$\dots$$

$$(1-t^{\ell}x^{-\ell-2})(1-t^{2}x^{-\ell})\dots(1-t^{\ell}x^{\ell+2}).$$



Uncertain reading. The middle exponent of t is heavily retraced and resembles 2. The outer exponents read ℓ.

Original source details: GELFAND-P000656

Take the product of all of them. Discard everything that contains x. Then one obtains

$$\sum \chi_m t^{2m} = t^{-2} + 2 - t^{8} - t^{14} - t^{22} + \dots + 10t^{70} + \dots$$

Wrote the expansion through t^{70}.

(Existence proved of $\geqslant 57$ nontriv. [inserted above: classes] of cohomology.)

Buchstaber on Adams's work on Buchstaber's work.

X — top. space. Consider $X \times X \to X$ for a group. $\Longrightarrow$

[At left, with arrow to "Consider"] If X — an actual group, then,

$$C(X \times X) \longleftarrow C(X) \quad \text{— coalgebra struct..}$$

Roughly:

$$M(f(x \cdot y)) \longleftarrow f(x)$$

(expected value of the values).

— · —

S. P. Novikov General theory of relativity.

Read: Landau and Lifshitz (together with Riem. geometry) — Theory of Fields.

$(t, \vec{x})$ — space of events. $(t, \vec{x}) \to (x^0, \vec{x}), \quad x^0 = ct.$

$$ds^2 = g_{ij}\, dx^i\, dx^j, \qquad g_{ij} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix}.$$

Proper time — length of a curve.

Let M^4; given g_{ij}. U — coord. system x^0, x^1, x^2, x^3.

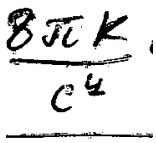
Uncertain reading. The symbol before "coord. system" tentatively reads U. Its shape resembles Cyrillic И.

Original source details: GELFAND-P000656

Einstein: the metric is called the gravit. field (1913) (+ Grossmann)

Einstein's eq. — independently Einstein and Hilbert (1916)

$$\underbrace{R_{ij} - \frac{1}{2} R g_{ij} = \frac{8\pi K}{c^4} T_{ij}}\,.$$



Uncertain reading. The numerator has uppercase K.

Original source details: GELFAND-P000656

Here $R_{j\ell} = \sum_i R^i_{ji\ell}$ — Ricci tensor.

[At right] $\| \ (R^i_j)_{k\ell}$ — 2-form (skew-symm. in k, ℓ) with values in the Lie alg. of the group $O(n)$.

T_{ij} — matter energy-momentum tens. (external mass).

[At right] T_{ij} accounts for everything except the gravit. field.

$$R = \sum_{j,\ell} g^{j\ell} R_{j\ell}.$$

In any dimension

$$\underbrace{R_{ij} - \frac{1}{2} R g_{ij} = T_{ij}}\,.$$

Let the phys. quantities be determined by the Lagrangian L.

$$S = \int L(q, g_{ij}) \underbrace{\sqrt{g}\, d^4x}_{d\sigma}\,.$$

$d\sigma$ — inv. volume elem..

Then

$$\sqrt{g}\, T_{ij} = \frac{\delta S}{\delta g_{ij}}.$$

Uncertain reading. This fraction has g_{ij}, with lower indices.

Original source details: GELFAND-P000656

Electromagn.

$$L_{\text{elm.}} = F_{\mu\nu} F^{\mu\nu}, \quad \text{where} \quad F_{\mu\nu} dx^\nu \wedge dx^\mu = d(\vec{A}).$$

[Arrow to $F^{\mu\nu}$] the metric enters.

No forces:

$$\Longrightarrow \quad \frac{\delta S}{\delta g^{ij}} = 0, \qquad S = \int R\sqrt{g}\, d^4x.$$

GELFAND-P000657

Einstein's eq.: in the grav. field there is no energy-mom. apart from that in other forms of energy.

G. Is it quantized? If so, there is no metric (geom. structure of space).

N. But this affects only 10^{-33} cm (whereas nuclear interactions at 10^{-23} cm).

As long as space makes sense, Einstein's eq. holds.

Quantum mech. is limited only by times of 10^{-43} sec.

$$T_{00} \text{ — energy density} = \varepsilon > 0.$$

[heading crossed out]

Undeciphered passage. Dense hatching obscures the heading before "Problems".

Original source details: GELFAND-P000657

Problems

1) continuum approx.: hydrodyn. form with Pascal's law

$$T_{ij} = (p+\varepsilon)u_i u_j - p g_{ij}.$$

[At right] (p — press., ε — energy density

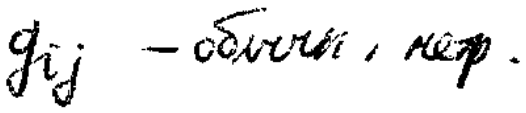
$\varepsilon, p, \vec{u}$ — fns. of the point.

$$\varepsilon = \varepsilon(p) \text{ — given} \qquad \text{or} \quad p = p(\varepsilon).$$

Problem: find (M^4, g_{ij}). Dust-like fluid $p = 0$.

or $p = \dfrac{\varepsilon}{3}$ — ultrarelativistic fluid (light

Hydrodyn. — g_{ij} — ordinary metr.



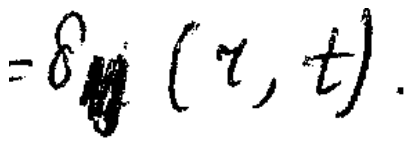
Manuscript excerpt. The final abbreviation reads "метр." (metric).

Original source details: GELFAND-P000657

$$\underline{\sum_k \nabla_k T_i^k = 0} \qquad \Longrightarrow \qquad \text{laws of hydrodyn.}$$

Field of a point mass

$$R_{ij} - \tfrac{1}{2} R g_{ij} = \delta_{ij}(r,t).$$



Uncertain reading. The heavily retraced signs on δ tentatively read ij.

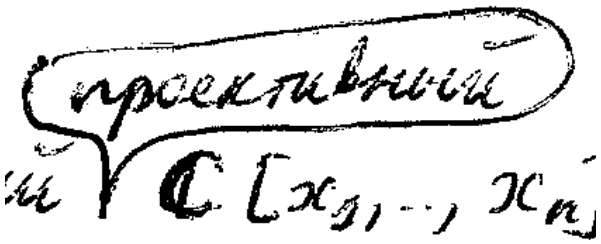
Original source details: GELFAND-P000657

symmetry grp. $SO(3)$ — Schwarzschild soln.. First M^4 given by Kruskal in 1960.
Spherical coords. describe one chart (and not the whole manifold).
r in preferred coords. has saddles (gravit. radii).
Next time Novikov.

Original source details: GELFAND-P000657

Uncertain reading. The letters before "time" read "След." (next).

19/IV-76. Gelfand Sem.
Serre's problem. Is every [inserted in a callout: projective] $\mathbb{C}[x_1, \ldots, x_n]$-module free?



Original source details: GELFAND-P000657

Diagram and typed description.

projective

The word "projective" in an outlined callout before the ring.

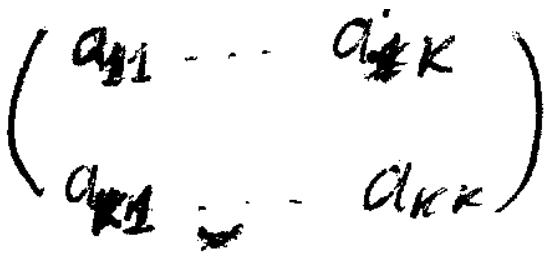
— ∘ —

Gelfand, following Vaserštein's notes

$$a_1, \ldots, a_k \in \mathbb{C}[x_1, \ldots, x_n] = A.$$

is called unimodular if $\exists p_1, \ldots, p_k : \sum p_i a_i = 1$.
Is there a matrix with entries in A, and with det $= 1$

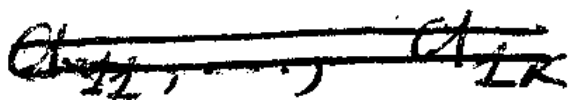
$$\begin{pmatrix} a_{11} & \ldots & a_{1k} \\ a_{k1} & \ldots & a_{kk} \end{pmatrix},$$



Original source details: GELFAND-P000657

Uncertain reading. The indices in a_{11}, a_{1k}, a_{k1} are retraced or corrected.

and with first row [crossed out: $a_{11}, \ldots, a_{1k}$] $a_1, \ldots, a_k$?



Original source details: GELFAND-P000657

Uncertain reading. The indexed row before the corrected first row is crossed out. The last digit on the first a resembles 2. A reading of 1 is possible.

Drinfeld on the same topic.

B — a ring

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \quad \text{— a unimod. column in } B[x]$$

(x — one var.).

leading coeff. of $f_1 = 1$ Then $f(x) \sim f(0)$ (transformed by a nonsing. matrix over $B[x]$).

Main tool — resultant:

$$\prod (\xi_i - \eta_k)$$

[At right] ξ_i, η_k — roots of 2 polys. in 1 var.

GELFAND-P000658

Main lemma. Let $c = \mathrm{Res}(f_1, f_2)$. Then if f — any column, then,

$$f(Y_1 + cY_2) \sim f(Y_1) \quad \text{over } B[Y_1, Y_2].$$

Uncertain reading. The relation sign has several retraced wavy strokes. The reading $\sim$ is tentative.

Original source details: GELFAND-P000658

Proof: by an explicit construction (Suslin)

Need: $c = 1$ and set $Y_2 = -Y_1$.

Reduction to this lemma.

A change of var. reduces to the case where 1 of the leading coeffs. is 1.

Remove all vars. in turn.

S. P. Novikov.

M^n, g_{ij}, sign. $(1, -1, -1, -1)$

$$R_{ij} - \frac{1}{2} R g_{ij} = T_{ij}.$$

[At right] || Spher. symm. problem: to completion in 1960, Kruskal.

Phys. formulation of the problem — hypotheses on T_{ij}.

Continuum

$$T_{ij} = (p + \varepsilon) u_i u_j - p g_{ij}, \quad \text{where } p = p(\varepsilon);$$
$$\vec{u} = (u_0, u_1, u_2, u_3); \qquad 1 = \langle u, u \rangle.$$

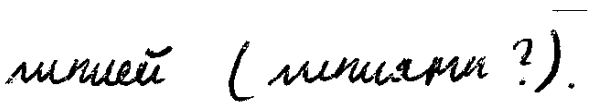
Dust: $p = 0$. Speed of sound:

$$c^2_{\text{sound}} = \frac{\partial p}{\partial \varepsilon} \geqslant 0,$$
$$c^2_{\text{sound}} \leqslant c^2 = 1.$$

There are too few eqs. But there are gauge transfs.

Hilbert wrote down the eqs. but gave them up as incorrect.

There is the metric of Gödel (that very one) with a closed timelike line (lines?).



Uncertain reading. The parenthesis reads “линиями?” (lines?) after the singular “линией”.

Original source details: GELFAND-P000658

Chapter 10

GELFAND-P000659

6/IX-76. Gelfand Sem.

What should one know from physics? [At right] (Following Sinai?)

Manuscript excerpt. The note at right reads "Following Sinai?" with a question mark.

Original source details: GELFAND-P000659

1. Classical mechanics and Ham. formalism.
   Arnold (≤ 30 pp.), Landau and Lifshitz.
2. Quantum mechanics and scattering theory.
   Landau and Lifshitz, Quantum mechanics (minus the first two chapters).
   Fermi, Quantum mechanics.
3. Field theory and relativity.
   Landau and Lifshitz, Weinberg, Cosmology and the theory of gravitation (chapters 1–4).
4. Quantum field theory.

[crossed out]

Undeciphered passage. The line under item 4 is heavily crossed out.

Original source details: GELFAND-P000659

For next time, consult the physicists and type it up.

Quantum field theory

1. Second quantization (Bose and Fermi). [Inserted below: statistics] — Landau and Lifshitz.
2. Writing the equations [inserted in callout: of the field] using a functional integral,

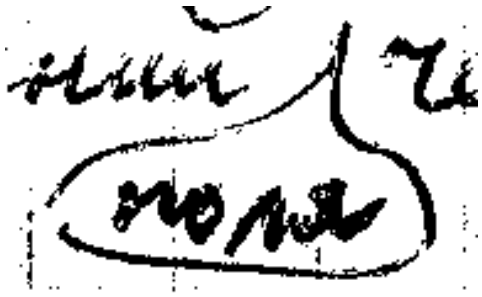


Diagram and typed description.

of the field

The callout after "equations" reads "поля" (of the field).

Original source details: GELFAND-P000659

perturbation series, Feynman diagrams.

3. Renormalizations. — Hepp.

4. Yang–Mills fields — there are surveys.

Gindikin (and Gelfand). All about $SL(2,\mathbb{R})$.

$$G = SL(2,\mathbb{R})$$

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \det(\,) = 1$$

(hyperboloid in $\mathbb{R}^4$)

$$\alpha\delta - \beta\gamma = 1$$

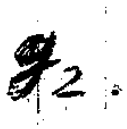
It corresponds to a form with 2 pluses and 2 minuses:

$$x^2 + y^2 - z^2 - t^2 = 1.$$

The group acts on the hyperboloid $G \times G$ by left and right translations

$$T_{(g_1,g_2)}g = g_1^{-1} g g_2 \,.$$

Uncertain reading. The final g_2 is heavily retraced.



Original source details: GELFAND-P000659

Any lin. transf. preserving the hyperboloid can be written in this form.

The stabilizer of the point $e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ consists of transfs. [crossed out]

Undeciphered passage. A parenthesis is visible. Its contents are heavily crossed out.

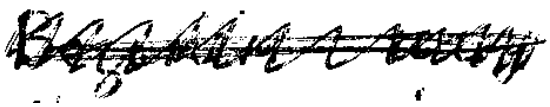
Original source details: GELFAND-P000659

of the form (g, g).

GELFAND-P000660

[Line crossed out at top]

Undeciphered passage. The line is heavily crossed out. Individual letter strokes remain visible.



Original source details: GELFAND-P000660

Exercise: compute the stabilizer subgrps. of other points.

Consider fns. on the hyperboloid.

For $\beta \neq 0$ convenient coordinates are [crossed out: α, β, δ] α, δ, β.

Uncertain reading. The first sign of the crossed-out triple is retraced and the reading is tentative.

Original source details: GELFAND-P000660

Fns. on the hyperboloid can be regarded as fns. $\psi(\alpha, \delta, \beta)$.

The functions are to be integrated against the invariant measure:

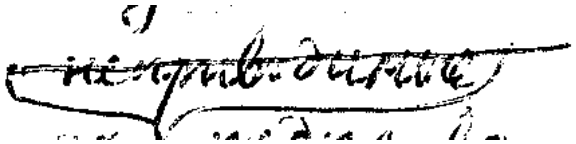
$$dg = \frac{d\alpha\, d\beta\, d\delta}{|\beta|}$$

(inv. under motions of the hyperboloid).

We will consider the space $\mathcal{H} = L^2(G)$ with respect to this measure.

$$\|\psi\|^2 = \int |\psi(g)|^2\, dg.$$

Problem: decompose the fns. into [circled and crossed out: irreducible] subspaces inv. under



Manuscript excerpt. The enclosed insertion "irreducible" above the line is crossed out.

Original source details: GELFAND-P000660

$$T_{(g_1,g_2)}\psi(g) = \psi(g_1^{-1} g g_2).$$

In $\mathbb{R}^4$ we consider hyperplanes

$$\xi\alpha + \eta\delta - p\beta - q\gamma = \zeta.$$

G. Philosophy: divide into series (an algebraic problem)

(Bernstein, Gelfand, Gelfand)

Manuscript excerpt. The surname "Gelfand" is written twice in succession.

Original source details: GELFAND-P000660

Series into representations — a different (transcend.) problem.

This well-known example (Bargman, Gelfand and Graev, Molchanov, Strichartz, Vilenkin et al.) — everything about hyperboloids is already known.

Manuscript excerpt. The first surname is written "Bargman" in Latin script.

Original source details: GELFAND-P000660

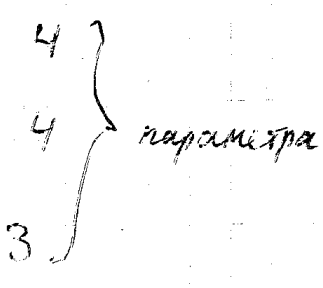
Into irreducibles — no need.

The following types of sections are possible.

$$\text{I.}\quad \Delta = \begin{vmatrix} \xi & p \\ q & \eta \end{vmatrix} < 0 \quad \text{one-sheeted hyperboloid} \qquad 4 \text{ parameters}$$

$$\text{II.}\quad \Delta = \begin{vmatrix} \xi & p \\ q & \eta \end{vmatrix} > 0$$

a)	$\lvert\zeta\rvert > 2\sqrt{\Delta}$	one-sheeted hyperboloid	4	} parameters
b)	$\lvert\zeta\rvert < 2\sqrt{\Delta}$	two-sheeted hyperbol.	4	
c)	$\lvert\zeta\rvert = 2\sqrt{\Delta}$	cone	3	



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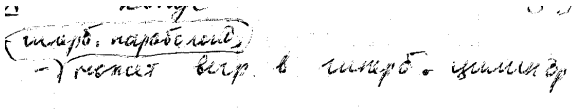
Manuscript excerpt. The numbers 4, 4 and 3 are written at right and grouped with a brace.

III. $\Delta = 0$ — flat thing in hyperbol. cylinder 2 parameters.



Original source details: GELFAND-P000660

Diagram and typed description.

hyperbol. paraboloid

"Hyperbol. paraboloid" is enclosed above the line. The single sign before "hyperbol. cylinder" resembles "в" (in).

GELFAND-P000661

A — one-sheeted hyperboloid (*cases* I, IIa), 4-param..

B — two-sheeted hyperboloid (—»— IIb), 4-param..

[Single letter crossed out: A?]

Original source details: GELFAND-P000661

Uncertain reading. The crossed-out letter between B and D resembles A.

D — cones, 3-param..

E — hyperbolic paraboloid (case III), 3-param..

$$\psi \in \mathcal{H}.$$

$$\widehat{\psi}(\xi, \eta, p, q, \zeta) = \int \psi(g)\delta(\xi\alpha + \eta\delta - p\beta - q\gamma - \zeta)\, dg.$$

(Radon transform of the fn. in the whole space)

(G. The technique goes back to that of the 1947 paper by G. and Naimark.)

$\mathcal{H}_A, \mathcal{H}_B, \mathcal{H}_D, \mathcal{H}_E$ — fns. for which the integrals are 0 [sign crossed out: φ?] over the hyperplanes of the given family.

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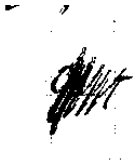
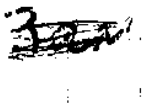
Uncertain reading. The third subscript is overwritten. Its outline reads D.



Original source details: GELFAND-P000661

Uncertain reading. A loop and a descending stroke in the crossed-out sign resemble φ.

[Crossed out: Problem [abbr.]?] How does one describe which fns. of 4 and 3 variables appear?

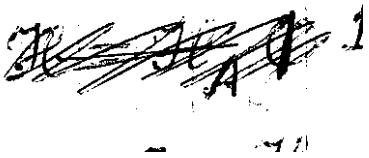


Original source details: GELFAND-P000661

Uncertain reading. The crossed-out opening tentatively reads “Зад.” (Problem).

Main theorem 1.

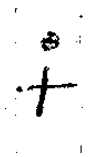
[Formula crossed out: $\mathcal{H} = \mathcal{H}_A \ldots$]



Original source details: GELFAND-P000661

Uncertain reading. The crossed-out formula shows $\mathcal{H}$, an equals sign and $\mathcal{H}$ with subscript A. The final sign is unclear.

1. $\mathcal{H} = \mathcal{H}_A \overset{\circ}{+} \mathcal{H}_B.$



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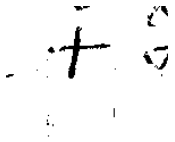
Uncertain reading. There is a small retraced circle above the plus sign.



Uncertain reading. A separate dot is also visible to the left of the plus sign.

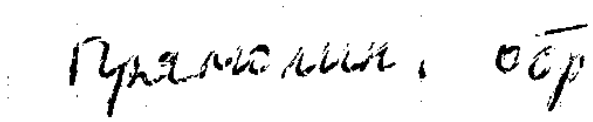
Original source details: GELFAND-P000661

$$2. \quad \mathcal{H}_A = \mathcal{H}_D = \mathcal{H}_E.$$

The fn. $\widehat{\psi}$ satisfies a second-order ultrahyperbolic eq. that can be written down.

The families A, D, E are minimal.

Straight-line rulings — a 3-param. family C.



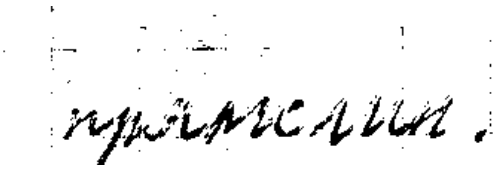
Manuscript excerpt. The abbreviation reads “Прямолин.” (straight-line).

Original source details: GELFAND-P000661

$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}$ — an example of a straight line, $\begin{pmatrix} h & z \\ 0 & h^{-1} \end{pmatrix}$ (h — fixed)

or

$$g_1^{-1} \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} g_2 \quad \text{(general form of a straight-line ruling)}.$$



Manuscript excerpt. The same abbreviation for “straight-line” is repeated in parentheses.

Original source details: GELFAND-P000661

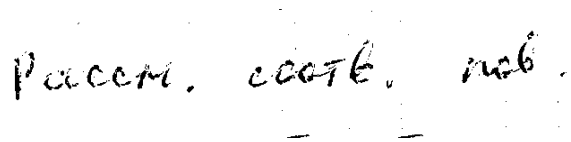
C — straight-line rulings.

$$2'. \quad \mathcal{H}_A = \mathcal{H}_D = \mathcal{H}_E = \mathcal{H}_C. \qquad \text{(discrete series)}$$

Explicit specification of projectors and description of these spaces.

$$\alpha\delta - \beta\gamma = 1.$$

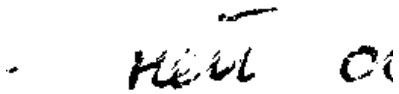
Consider the corresponding surface in $\mathbb{C}^4$ with the same eq..



Manuscript excerpt. The abbreviation before “in” reads “пов.” (surface).

Original source details: GELFAND-P000661

and on it, the domains:



Manuscript excerpt. The pronoun is “ней”, the feminine form used with “surface”.

Original source details: GELFAND-P000661

$$\underline{(\Im\alpha\bar{\beta})(\Im\delta\bar{\beta}) > (\Im\beta)^2} \qquad (1)$$

Consider it in $\mathbb{C}^3$.

Manuscript excerpt. The pronoun after "Consider" reads "её" (it).

Original source details: GELFAND-P000662

It splits into 2 domains whose distinguished boundary is the real space $\mathbb{R}^3$.

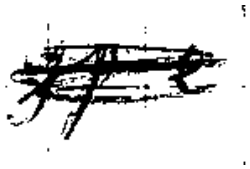
The inequality is inv. under fractional-linear transfs. (one of the Siegel domains).

Distinguished boundary $G = SL(2, \mathbb{R})$.

The domains are separated by the sign of the parentheses.

(The distinguished boundary of a domain D is the smallest set in ∂D such that a fn. holom. in D and cont. on $\bar{D}$ is uniquely recovered from its restriction to that set.)

Lemma. In $\mathbb{C}^3$ with coords. (α, β, δ) [text crossed out] inequality (1) singles out 2 domains, each with distinguished boundary $\mathbb{R}^3$.



Undeciphered passage. The short entry after the coordinate triple is heavily crossed out.

Original source details: GELFAND-P000662

Manuscript excerpt. Before the 2 the writing reads "выд." (singles out).

Original source details: GELFAND-P000662

Theorem. On the surface $\alpha\delta - \beta\gamma = 1$ in $\mathbb{C}^4$ there exist 2 domains inv. under real motions, with distinguished boundaries given by the hyperboloid [sign crossed out] $\alpha\delta - \beta\gamma = 1$ in $\mathbb{R}^4$.

Undeciphered passage. The initial sign to the left of the hyperboloid equation is crossed out.

Original source details: GELFAND-P000662

Denote these domains by D_+ and D_-.

Main thm. 2.

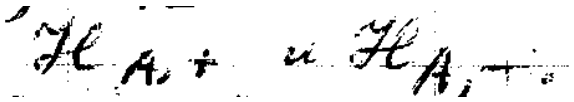
$$\psi \in \mathcal{H}_A \iff \psi = \psi_+ + \psi_-, \quad \text{where}$$

ψ_+ continues analyt. into D_+, ψ_- continues analyt. into D_-.

[Above the line: Denote [sign crossed out] the subspaces by $\mathcal{H}_{A,+}$ and $\mathcal{H}_{A,-}$.]

Original source details: GELFAND-P000662

Undeciphered passage. A sign with diverging strokes between "Denote" and "the subspaces" is crossed out.



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Uncertain reading. A thin vertical stroke crosses the long bar in the second subscript. The reading as a minus is tentative.

The projector onto $\mathcal{H}_{A,+}$ is given by the fn.

$$\int_G K(g_1, g_2)\psi(g_2)\, dg_2,$$

where

$$K(g_1, g_2) = c\,\frac{1}{[-\alpha_1\delta_2 - \delta_1\alpha_2 + \gamma_1\beta_2 + \beta_1\gamma_2 + 2]^{3/2}}.$$

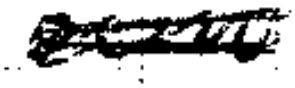
$$g_1 = \begin{pmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & \delta_1 \end{pmatrix}, \qquad g_1 \in D_+.$$

The 2nd projector is analogous.

$$\mathcal{H}_{A,+} \cap \mathcal{H}_{A,-} = \{0\},$$

since [text crossed out] everything is in L^2.

Undeciphered passage. The word after "since" is heavily crossed out.



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The submerged part: Fourier-integral technique.

$K(g_1, g_2)$ — analogue of the Cauchy kernel.

Arnold's conjecture. D_+ — those that map the disc into itself,

D_- — those that map the exterior of the disc [text crossed out] into itself,

Original source details: GELFAND-P000662

Undeciphered passage. A short entry after "disc" is heavily crossed out.

$SL(2, \mathbb{R})$ — preserves the circle.

GELFAND-P000663

13/IX-76 Gelfand Sem..

A. A. Kirillov. Superalgebras, supergroups and supermanifolds.

D. B. Fuchs. All about the Weil algebra.

$\mathfrak{g}$ — Lie algebra. Construct from it the assoc. Weil algebra:

$$W(\mathfrak{g}) = \underset{\text{exterior algebra}}{\Lambda^*(\mathfrak{g}')} \overset{\otimes}{\underset{\text{II}}{}} \underset{\text{symmetric algebra}}{S^*(\mathfrak{g}')}.$$

Uncertain reading. Two vertical strokes with shared crossbars appear below the tensor product sign.

Original source details: GELFAND-P000663

If $\mathfrak{g}$ has basis $e_1, \dots, e_n$, then the elements of $W(\mathfrak{g})$ — functions of $2n$ variables, super

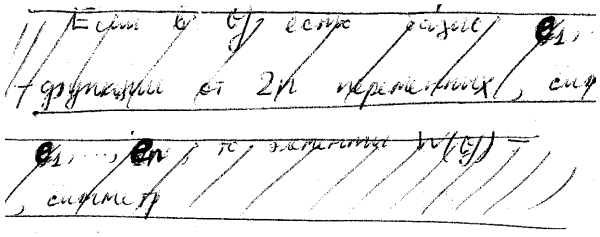
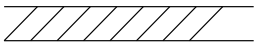


Diagram and typed description.



Diagonal strokes cross the two lines between the horizontal rules. The last word tentatively reads “super”.

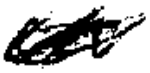
Original source details: GELFAND-P000663

Grading: $\Lambda^p(\mathfrak{g}') \otimes S^q(\mathfrak{g}')$ has degree $p + 2q$.

Differential: $d : W(\mathfrak{g}) \to W(\mathfrak{g})$ with properties

1) $d^2 = 0$, 2) acts in the usual way on [entry crossed out] the product.

Undeciphered passage. The short entry before “the product” is heavily crossed out.



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Let $h \in \underset{\Lambda^1(\mathfrak{g}')}{\mathfrak{g}'}$. Consider $h \otimes 1 \in W(\mathfrak{g})$.

$$d : \Lambda^1(\mathfrak{g}') \otimes S^0(\mathfrak{g}') \to \Lambda^2(\mathfrak{g}') \otimes S^0(\mathfrak{g}') \oplus \Lambda^0(\mathfrak{g}') \otimes S^1(\mathfrak{g}').$$

$$h \otimes 1 \text{ [}h\text{? crossed out]} \mapsto \{(g_1, g_2) \mapsto h([g_1, g_2])\} \otimes 1; \qquad 1 \otimes h.$$

Uncertain reading. The crossed-out sign before the arrow resembles h.

Original source details: GELFAND-P000663

This extends uniquely to a diff. with properties 1) and 2).

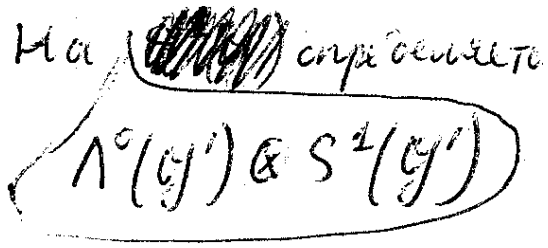
Property 2).

$$\xi \in W^k(\mathfrak{g}), \quad \eta \in W^\ell(\mathfrak{g}) \implies d(\xi \cdot \eta) = d\xi \cdot \eta + (-1)^k \xi \cdot d\eta.$$

On [entry crossed out] it is determined by the condition $d^2 = 0$.

Original source details: GELFAND-P000663

Undeciphered passage. The entry after "On" is crossed out with diagonal strokes.



Original source details: GELFAND-P000663

Diagram and typed description.

$\Lambda^0(\mathfrak{g}') \otimes S^1(\mathfrak{g}')$

The enclosed insertion is connected to the place after "On".

$$d : W^k(\mathfrak{g}) \to W^{k+1}(\mathfrak{g}).$$

$$d(h \otimes 1) = (dh \otimes 1) \oplus (1 \otimes h),$$

where $(dh)(g_1, g_2) \doteq h([g_1, g_2])$.

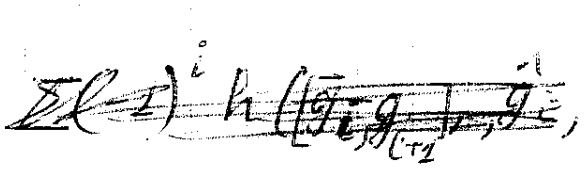
$\doteq$ means $\overset{\text{def}}{=}$.

$$(dh)(g_1, \ldots, g_{\ell+1}) =$$

Manuscript excerpt. The subscript on the last g reads $\ell + 1$.

Original source details: GELFAND-P000663

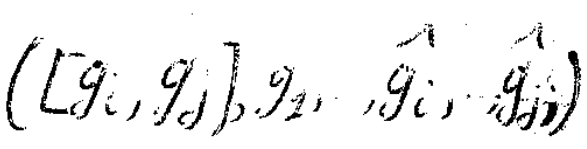
[First sum crossed out: $\sum(-1)^i h(\ldots)$.]



Undeciphered passage. The sum sign, the sign with superscript i, h, brackets and several instances of g are visible. The full entry cannot be read.

Original source details: GELFAND-P000663

$$\sum_{i<j} (-1)^{i+j-1} h([g_i, g_j], g_1, \ldots, \widehat{g_i}, \ldots, \widehat{g_j}).$$



Uncertain reading. The first subscript after the commutator reads 1. The last two gs carry hats. An extra stroke is visible at the end.

Original source details: GELFAND-P000663

$$dh = \sum a_i h_{i_1} \wedge h_{i_2} = \sum a_i (h_{i_1} \otimes 1)(h_{i_2} \otimes 1).$$

Differentiate by the Leibniz rule.

A nonzero element appears in $\Lambda^1 \otimes S^1$.

Let $e_1, \dots, e_n$ — a basis in $\mathfrak{g}$, also a basis of linear fns. on $\mathfrak{g}'$.

$h^1, \dots, h^n$ — the dual basis in $\mathfrak{g}'$.

Then

$$d(1 \otimes h^k) = c_{ij}^k (h^i \otimes h^j).$$

Main theorem: The homologies of the complex $\dots \to W^k \xrightarrow{d} W^{k+1} \to \dots$ are equal to 0.

Next time: how it works in char. classes.

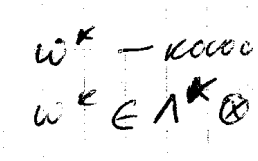
A. A. Kirillov. Devised to compute the homologies of simple compact Lie groups.

The easy part is the inv. forms.

Symmetric ones are easy to find. Skew-symmetric ones?

ω^k — a skew-symm. inv. diff. form.

$$\omega^k \in \Lambda^k \otimes S^0 + \dots; \qquad d\omega^k = 0.$$



Manuscript excerpt. The retraced superscript on ω in the membership line reads k.

Original source details: GELFAND-P000664

Then ω^k — the differential of a symm. expr..

Conclusion: as many skew-symm. [inserted above: inv.] of deg. k as symm. of deg. $\frac{k+1}{2}$.

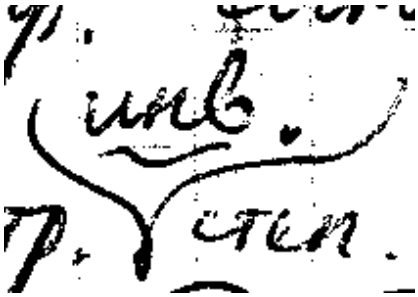


Diagram and typed description.

inv.

The abbreviation “inv.” is inserted above the line between “skew-symm.” and “deg.”.

Original source details: GELFAND-P000664

—”— 3 —”— 2.

Arnold. On works from the end of the last century on 2 questions.

1) $y'' = F(x, y, y')$ (e.g., $y'' = 0$).

Tresse (a student of Sophus Lie) 1896, Acta Math. 1894.

$y'' = 0$ the graphs are straight lines.

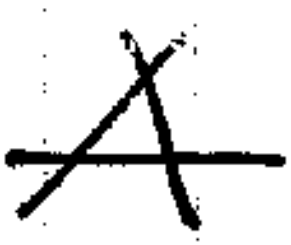
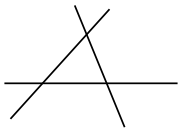


Diagram and typed description.



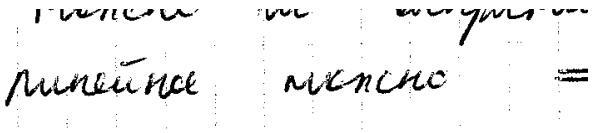
Three straight lines at right intersect pairwise to form a triangle.

Original source details: GELFAND-P000664

There are configuration theorems (Pappus, Desargues, Pascal, etc.).

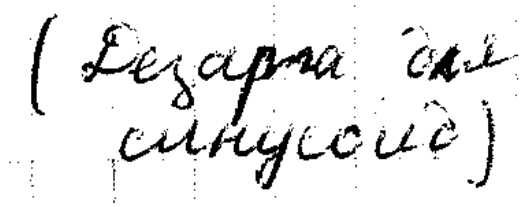
Can the solns. of the general eq. be straightened?

Linear menu $\Longrightarrow$ all the config. thms. hold (Desargues for sinusoids).



Manuscript excerpt. The second word reads “меню” (menu).

Original source details: GELFAND-P000664



Manuscript excerpt. The word between “Desargues” and “sinusoids” reads “для” (for).

Original source details: GELFAND-P000664

$$y'' + a(x)y' + b(x)y = 0$$

is reduced by a diffeom. of the plane (x, y), linear in y:

$$x = f(\tilde{x})$$
$$y = g(\tilde{x})\tilde{y}$$

Locally it is reduced to $y'' = 0$.

GELFAND-P000665

Consider an arbitrary family.

The infinitesimal theory is fairly well known.

E. Cartan in 1924–25, — projective connection —

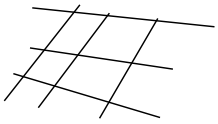
— the solutions are geodesics with respect to this connection.

Projective connection — a connection in a G-bundle, where G — the projective group acting in the tangent space.

Space of line elements (x, y, y').

2) In $\mathbb{R}^3$ there are 2 direction fields.

Diagram and typed description.



Two families of three strokes each cross to form a slanted grid.

Original source details: GELFAND-P000665

The diffeom. group of $\mathbb{R}^3$ acts.

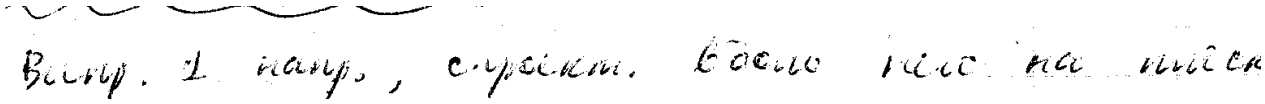
[retraced sign] To what form can one reduce it?

Uncertain reading. The isolated, heavily retraced sign before the question is undeciphered.

Original source details: GELFAND-P000665

These problems are equivalent.

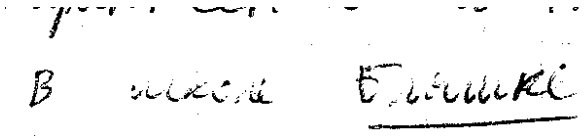
Choose 1 dir., project along it onto the plane. There is a 2-param. family there, etc.



Original source details: GELFAND-P000665

Manuscript excerpt. The abbreviation after “1 dir.” reads “спроект.” (project).

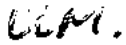
In his lifetime Blaschke worked on it.



Manuscript excerpt. The word before “Blaschke” reads “жизни” (life).

Original source details: GELFAND-P000665

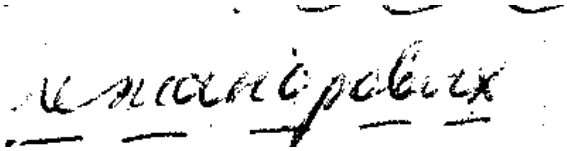
In $\mathbb{R}^3$ there is a contact structure. (Planes spanned by tangent vectors).
The curves have locally the structure of a projective line.



Manuscript excerpt. The abbreviation after “curves” reads “им.” (have).

Original source details: GELFAND-P000665

(from the general theory of multidimensional bundles)



Uncertain reading. The adjective before “bundles” tentatively reads “многомерных” (multidimensional).

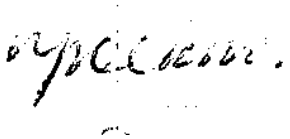
Original source details: GELFAND-P000665

On the circle there are many inequivalent structures of a locally projective space.

$$1)\quad \mathbb{RP}^1 \qquad 2)\quad S^1 \xrightarrow[\text{deg. } 2]{} \mathbb{RP}^1$$

2) inequivalent structure.

Proof of inequivalence: we will follow the vect. fields (infinitesim. automs. of this structure).

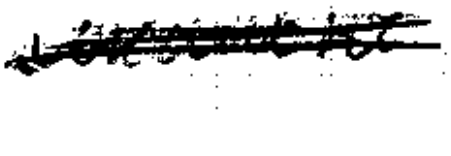


Uncertain reading. The verb before “vect. fields” tentatively reads “проследим” (we will follow).

Original source details: GELFAND-P000665

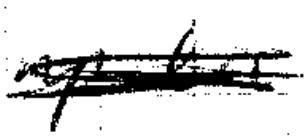
$$\text{on } \mathbb{RP}^1 \quad \frac{\partial}{\partial x},\quad x\frac{\partial}{\partial x},\quad x^2\frac{\partial}{\partial x}.$$

Problem: how [entry crossed out] the equation [entry crossed out] globally a locally projective space?



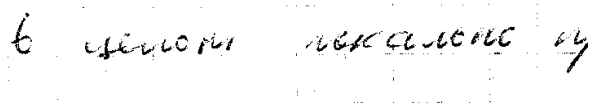
Undeciphered passage. The word between “how” and “the equation” is heavily crossed out.

Original source details: GELFAND-P000665



Undeciphered passage. The word after “the equation” is crossed out with several strokes.

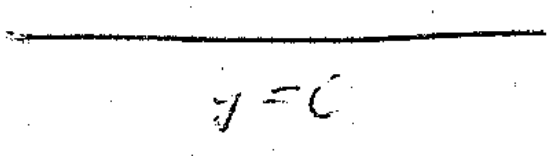
Original source details: GELFAND-P000665



Manuscript excerpt. After the crossed-out word the writing reads “в целом локально” (globally, locally).

Original source details: GELFAND-P000665

Take one soln. as a straight line.



Original source details: GELFAND-P000665

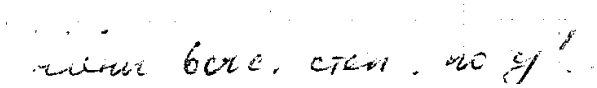
Diagram and typed description.

$y = 0$

The line at right is labelled $y = 0$.

$$y'' = a(x)y + b(x)p + \dots$$

[At right: terms of higher deg. in y'.]

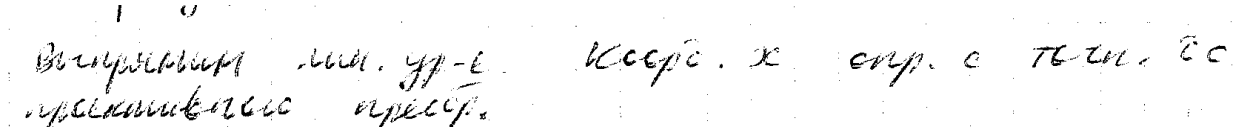


Original source details: GELFAND-P000665

Manuscript excerpt. The first words of the note read “члены выс. степ.” (terms of higher deg.).

$$p = y'.$$

Straighten the lin. eq.. The coord. x is associated with points of its projective space.



Original source details: GELFAND-P000665

Uncertain reading. After “with” the abbreviation tentatively reads “т-ми” (points). An additional stroke is visible before the dash.

$$\begin{cases} \tilde{x} = f(x), \\ \tilde{y} = g(x)y. \end{cases}$$

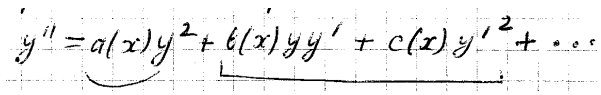
(f satisfies an eq. of Schwarz type).

GELFAND-P000666

Hence a projective structure arises on $\forall$ solns..

Now start with the quadr. part.

$$y'' = a(x)y^2 + b(x)yy' + c(x)y'^2 + \dots$$



Original source details: GELFAND-P000666

Diagram and typed description.

$a(x)y^2 + b(x)yy' + c(x)y'^2$

An arc is drawn below the first term. A bracket runs under the next terms through $c(x)y'^2$.

Now diffeom. of the plane of the form:

$$\begin{cases} \tilde{x} = x + yf(x,y), \\ \tilde{y} = y(1 + yg(x,y)). \end{cases}$$

Kill b and c with these diffeom..

There remains a, determined up to a mult. constant.

$\underline{\underline{a(x)(dx)^{5/2}}}$ determined up to a mult. constant.

This is a 5/2-form along the solution, determined up to a mult. const. depending on the solution.

Theorem. If $a(x)(dx)^{5/2} \equiv 0$, then the eq. can be straightened.

(in formal series)

Further diffeoms. lead to $\underline{\underline{y'' = y'^2 A(x,y,y')}}$.

Consider an eq. of the form $\underline{y'' = P_3(y')}$ — a 3rd-degree poly. in y'.

This class is inv. under diffeomorphisms.

The occurrence of these eqs. was discovered by R. Liouville.

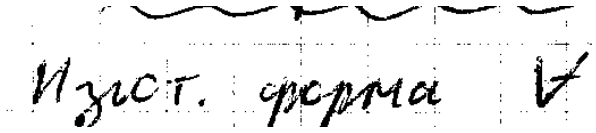
Original source details: GELFAND-P000666

Manuscript excerpt. The verb before the name reads “открыл” (discovered).

Scalar invariant.

From this, a form of $\forall$ other degs. (in partic. a vect. field).

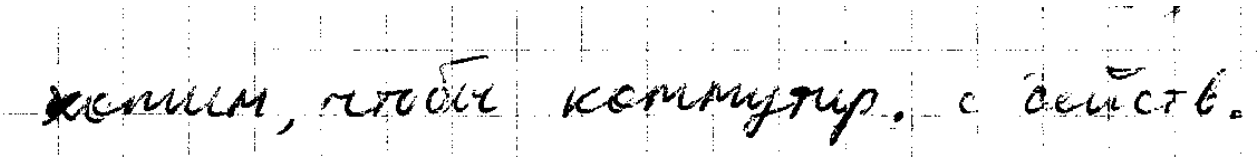
$$\underline{v(x) = (a(x))^{-2/5} \frac{\partial}{\partial x}.}$$



Original source details: GELFAND-P000666

Manuscript excerpt. The opening reads “Из эт.” (from this).

Now construct a scalar: we want it to commute with the action of the projective group.



Original source details: GELFAND-P000666

Manuscript excerpt. After “we want” the entry reads “чтобы коммутир.” (to commute).

We obtain invariants:

$I_2, I_3, \ldots$ — invariants.

Consider the 2-jet of the field. On 2-jets the projective group acts by the adjoint

representation.

$$I_2 = 2v''v - v'^2$$
$$I_3 = vI_2'$$
$$— \quad — \quad —$$
$$I_n = vI_{n-1}'$$

Gelfand conjecture: connected with Gelfand–Fuchs cohomology.

Pommaret worked on diff. invariants.

GELFAND-P000667

20/IX-76 Gelfand Sem..

S. P. Novikov. Superalgebras in physics.

[Crossed out: Rev.?] Ogievetsky's review in UFN and Zumino's review.

Uncertain reading. The crossed-out opening tentatively reads "Обз." (Rev.).

Original source details: GELFAND-P000667

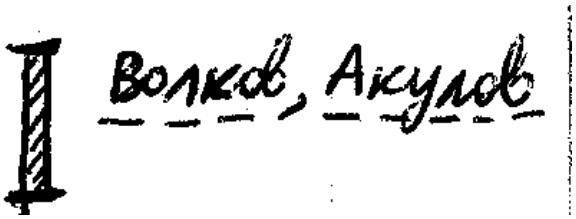


Diagram and typed description.

Volkov, Akulov

The surnames are underlined with broken strokes in the right margin. A narrow hatched mark stands to their left.

Original source details: GELFAND-P000667

Superalgebra [Crossed out: Lie?] — a direct sum $L = L_{\text{even}} \oplus L_{\text{odd}} = L_0 \oplus L_1$.

Uncertain reading. The strokes of the short crossed-out word resemble "Ли" (Lie).

Original source details: GELFAND-P000667

$[a, b]$ defines the struct. of a graded Lie algebra.

$$[a, b] = (-1)^{da \cdot db + 1}[b, a], \qquad da = \dim a.$$

Manuscript excerpt. The definition of da reads "dim a".

Original source details: GELFAND-P000667

The Jacobi id. also holds with signs.

Let L_0 and L_1 be embedded in an assoc. algebra (e.g., of matrices).

L — a subalgebra of the matrix alg., closed under the operation

$$1) \quad a, b \in L_0 \implies ab - ba = [a, b] \in L_0,$$
$$2) \quad a, b \in L_1 \implies ab + ba = [a, b] \in L_0.$$

$$3) \quad a \in L_0,\ b \in L_1 \implies ab - ba = [a, b] \subset L_1.$$

Or

$$[a, b] = (-1)^{\varepsilon_1 \varepsilon_2 + 1}[b, a],$$
$$\varepsilon_1 = \deg a, \qquad \varepsilon_2 = \deg b.$$

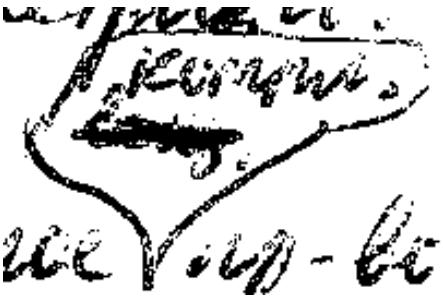
Concrete superalgebras were first considered in the work of Gelfand and Likhtman.

$$\mu = 0, 1, 2, 3.$$

Everything in $\mathbb{R}^4$ Minkowski.

$$\eta_{\nu,\mu} = \begin{pmatrix} -1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & 1 \end{pmatrix} \quad \text{— metric.}$$

space of spinors — 4-dimensional space [inserted above: real. Word crossed out.] (with basis) Q_j, $j = 1, 2, 3, 4$.

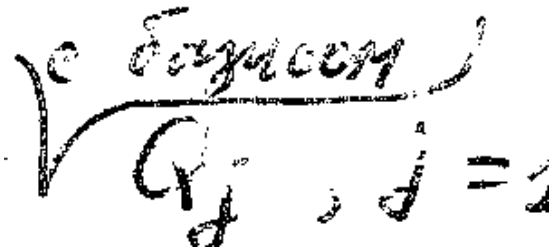


Original source details: GELFAND-P000668

Diagram and typed description.

real
[crossed out]

"Real" is written in the enclosed insertion. The lower word is crossed out.



Original source details: GELFAND-P000668

Diagram and typed description.

with basis

The second outlined insertion, "with basis", points to the start of the list.

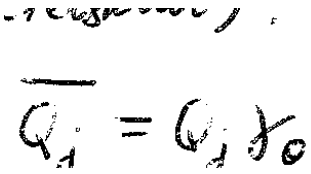
$$\alpha = \sum \alpha^j Q_j.$$

Matrix algebra generated by matrices γ_μ (real) — Majorana basis.

$$\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\nu\mu}. \qquad (\approx \text{ quaternions})$$

Choose 4 matrices γ_μ, real and forming a Majorana basis
(see Akhiezer, Berestetskii).
Set

$$\bar{Q}_j = Q_j\gamma_0;$$
$$\bar{\alpha} = \alpha\gamma_0.$$



Manuscript excerpt. The letter Q has a descending diagonal stroke.

Original source details: GELFAND-P000668

L_0 — alg. of [crossed out: Poincaré] translations.

Manuscript excerpt. “Poincaré” is crossed out before “translations”.

Original source details: GELFAND-P000668

L_1 — space of spinors.

$$1) \quad [L_1, L_0] = 0,$$
$$2) \quad [L_1, L_1] \text{ — ?}$$

$$[\bar{Q}_i, \bar{Q}_j] = -2(\gamma^\mu)_{ij} P_\mu.$$

P_μ — Poincaré translation oper., μ — index.

G. Problem. extend the Poincaré alg. to a superalgebra.

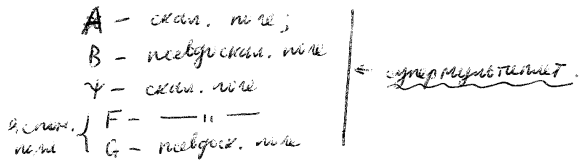


Diagram and typed description.

A — scalar field
B — pseudoscalar field
ψ — scalar field
aux. fields: F — —”—
aux. fields: G — pseudoscal. field
← supermultiplet

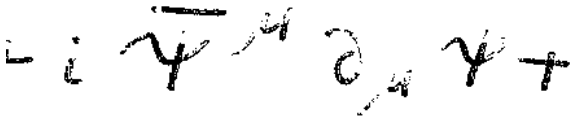
The brace at left groups F, G. The label at right points to the vertical line beside the list.

Original source details: GELFAND-P000668

Manuscript excerpt. The label beside ψ also reads “скал. поле” (scalar field).

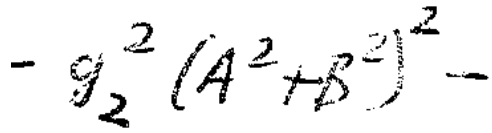
Original source details: GELFAND-P000668

$$
\begin{aligned}
L = & -\tfrac{1}{2}\big((\partial_\nu A)^2 + (\partial_\mu B)^2 + i\bar\psi\gamma^\mu\partial_\mu\psi \\
& \quad + m^2A^2 + m^2B^2 + im\bar\psi\psi\big) \\
& + gmA(A^2+B^2) - g_2^2(A^2+B^2)^2 \\
& - ig\,[im\text{? crossed out}]\,\bar\psi(A-\gamma_5 B)\psi.
\end{aligned}
$$



Original source details: GELFAND-P000668

Uncertain reading. Faint strokes before the raised μ after $\bar\psi$ tentatively read γ.



Original source details: GELFAND-P000668

Manuscript excerpt. Two instances of 2 are visible, one above g and one below. No fraction bar is distinguishable.

Original source details: GELFAND-P000668

Uncertain reading. The crossed-out strokes after $-ig$ tentatively read im.

[crossed out: (supersymm. Lagrangian?)]

Original source details: GELFAND-P000668

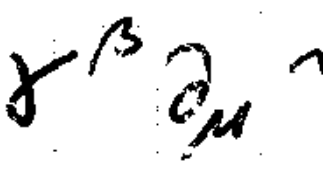
Uncertain reading. The final word in the crossed-out parenthesis tentatively reads “лагранжиан” (Lagrangian).

GELFAND-P000669

The opers. from L_0 and L_1 act on the Lagrangian L (a rep. is constructed) and it is invariant.

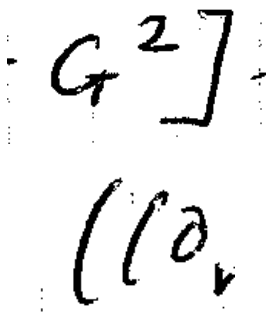
The Wess–Zumino supersymm. Lagrangian.

$$
\begin{aligned}
L = & -\tfrac{1}{2}\big((\partial_\nu A)^2 + (\partial_\mu B)^2 + i\bar\Psi\gamma^\beta\partial_\mu\Psi - F^2 - G^2\big] \\
& + m(FA + GB - i\bar\Psi\Psi) \\
& + g(FA^2 - FB^2 + 2GAB - i\bar\Psi\Psi A + i\bar\Psi\gamma_5\Psi B).
\end{aligned}
$$



Original source details: GELFAND-P000669

Manuscript excerpt. The upper index of γ is β. The lower index of ∂ is μ.



Manuscript excerpt. The outer opening bracket is round and the closing bracket is square.

Original source details: GELFAND-P000669

No derivs. in F and $G \Rightarrow$ they are eliminated by the Euler eqs. $\Rightarrow$

$\Rightarrow$ the preceding Lagrangian.

Action of the supergenerators.

Manuscript excerpt. The heading reads “суперобразующих” (of the supergenerators).

Original source details: GELFAND-P000669

$$L = L_0 + L_1.$$

$$\alpha \in L_1, \quad \alpha = \sum \alpha_j Q_j, \qquad \bar{\alpha} = \alpha\gamma_0.$$

1) $[\bar{\alpha}, A] = \delta A = i\bar{\alpha}\Psi$;
2) $\delta B =$ [crossed out] $i\bar{\alpha}\gamma_5\Psi$;
3) $\delta\Psi = \partial_\mu(A - \gamma_5 B)\gamma^\mu\alpha + (F + \gamma_5 G)\alpha$;
4) $\delta F = i\bar{\alpha}\gamma^\mu\partial_\mu\Psi$;
5) $\delta G = i\bar{\alpha}\gamma_5\gamma^\mu\partial_\mu\Psi$.

Undeciphered passage. The entry after $\delta B =$ is crossed out. Individual signs are visible.

Original source details: GELFAND-P000669

Migdal (Sasha). On gauge fields.

They arose from attempts to describe exp. data (on nucleons). These say that nucleons are made up of quarks.

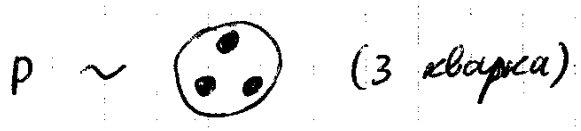


Diagram and typed description.

$p \sim$ (3 quarks)

After $p \sim$, an oval with three dots and the label “3 quarks”.

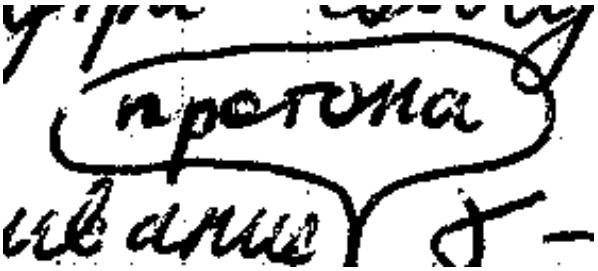
Original source details: GELFAND-P000669

They are free inside, and it is impossible to take them out.

[crossed out: They.] Quarks are visible — probing [inserted above: the proton] γ-quanta (as in a cellophane bag).

Original source details: GELFAND-P000669

Manuscript excerpt. “They” is crossed out before “Quarks”.



Original source details: GELFAND-P000669

Diagram and typed description.

the proton

“The proton” is inserted before the quanta.

In the theory the forces must increase with distance!

[crossed out] By the uncertainty princ. they cannot merge.

Original source details: GELFAND-P000669

Undeciphered passage. The short entry before “By the principle” is crossed out.

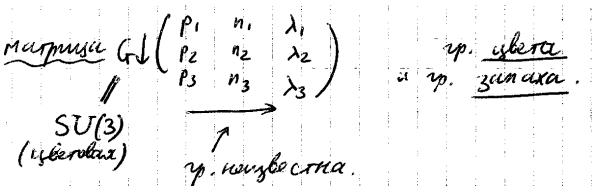
Relativ. inv. hinders specifying a law of forces (the interaction potential is relativ. noninvariant).

Locality is confirmed by probing — quarks are nearly pointlike.

one must construct Lagrangians.

Gauge theory gives such a law of forces.

Quark — matrix.



Original source details: GELFAND-P000669

Diagram and typed description.

$G \downarrow \begin{pmatrix} p_1 & n_1 & \lambda_1 \\ p_2 & n_2 & \lambda_2 \\ p_3 & n_3 & \lambda_3 \end{pmatrix}$ gr. of colour and gr. of flavour.

$SU(3)$ (colour)

group unknown.

At left G and a downward arrow. Below the matrix a rightward arrow with an annotation. The labels at right name colour and flavour.

There are 9 quarks.

Forces that prevent breaking apart are connected with colour.

To the colour gr. corresponds [crossed out] a much stronger symm. — gauge.

Undeciphered passage. The isolated sign after “corresponds” is heavily crossed out.

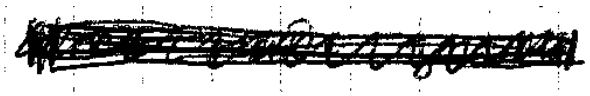
Original source details: GELFAND-P000669

Mesons are constructed from a quark and an antiquark.

GELFAND-P000670

18/X-76. Gelfand Sem..

[crossed out] Write down the basic notions and equations one needs to know. And the problems.



Undeciphered passage. The long entry before “Write down” is heavily crossed out.

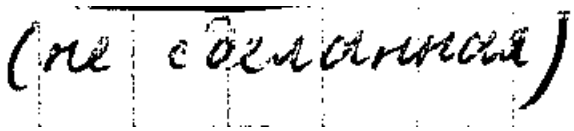
Original source details: GELFAND-P000670

Problem: express II_1 factors using matrices of continuous order.

Manuscript excerpt. The subscript is heavily retraced. Its outline reads 1.

Original source details: GELFAND-P000670

(undone)



Manuscript excerpt. The parenthetical reads “не сделанная” (undone).

Original source details: GELFAND-P000670

I. Bernstein. (continued).

Superspace — informally, there are 2 sets of variables:

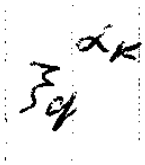
$$\left\| \begin{array}{l} x_1, \ldots, x_p, \\ \xi_1, \ldots, \xi_q \end{array} \right.$$

Formally

$$C^\infty(\mathbb{R}^p) \otimes \Lambda(\xi_1, \quad, \xi_q)$$

An element:

$$\sum f_{\alpha_1 \ldots \quad \alpha_q}(x) \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_k}, \qquad \alpha_i = 0, 1.$$



Manuscript excerpt. The last ξ_q has exponent α_k. The coefficient ends in subscript α_q.

Original source details: GELFAND-P000670

All f commute with ξ, and $\xi_i\xi_j = -\xi_j\xi_i,\ \xi_i^2 = 0$.

$\mathbb{Z}_2$ — Grading. Degree — sum of α_i mod 2. ||

Another def. [crossed out] Ringed space.

Original source details: GELFAND-P000670

Undeciphered passage. The narrow sign before "Ringed" is blacked out.

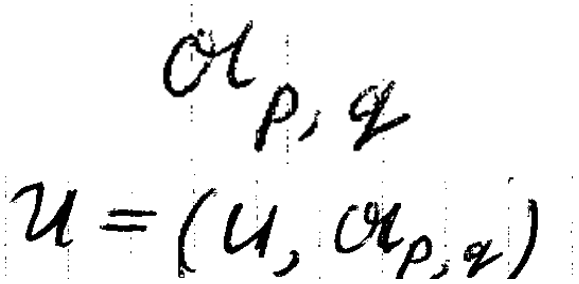
$\mathbb{R}^p$ with the sheaf $\mathcal{A}_{p,q}$.

$$U \subset \mathbb{R}^p \quad \longmapsto \quad \text{ring } C^\infty(U) \otimes \Lambda(\xi_1, \quad , \xi_q)$$

$U \subset V \quad \Rightarrow \quad$ there is a natural restriction map.

Superdomain. This is a topological space $U \subset \mathbb{R}^p$ with the same sheaf

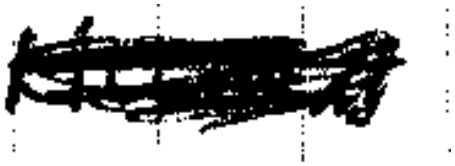
$$\mathcal{U} = (U, \mathcal{A}_{p,q}).$$



Original source details: GELFAND-P000670

Manuscript excerpt. The symbol is repeated in the ordered pair and reads as a script $\mathcal{A}$.

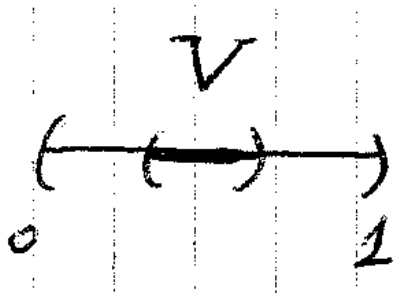
[Crossed out above the heading]



Original source details: GELFAND-P000670

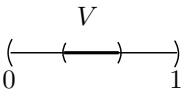
Undeciphered passage. The short word above "Example" is heavily crossed out.

Example. $u, \xi_1, \xi_2, \qquad 0 < u < 1.$



Original source details: GELFAND-P000670

Diagram and typed description.



The inner interval V is marked by a heavier line between parentheses.

functions on $V = C^\infty(V) \otimes \Lambda(\xi_1, \xi_2)$.

$$f_{00}(u) + [\text{crossed out}]\, f_{10}(u)\xi_1 + f_{01}\xi_2 + f_{11}(u)\xi_1\xi_2.$$

Original source details: GELFAND-P000670

Undeciphered passage. A group of signs after $f_{00}(u)+$ is crossed out.

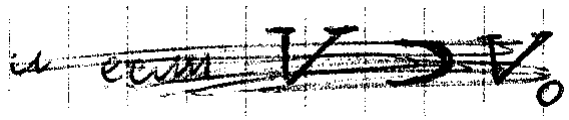
$C^\infty(\widetilde{U})$ — notation for the space of fns. on the superdomain $\widetilde{U}$.

$$\|$$
$$C^\infty(U) \otimes \Lambda(\xi_1, \ldots, \xi_q)$$

Map of superdomains $\varphi : \widetilde{U} \longrightarrow \widetilde{V}$
this is a collection of objects: a map $\varphi' : U \to V$
and a map

$$C^\infty(\widetilde{V}) \longrightarrow C^\infty(\widetilde{U})$$

a ring hom. covering φ' (with unit) and preserving parity!!
[crossed out: and if $V \supset V_0$]



Original source details: GELFAND-P000670

Manuscript excerpt. The crossed-out line has $\supset$ between V and V_0.

Example.

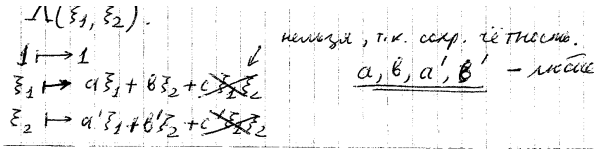
Original source details: GELFAND-P000670

Diagram and typed description.

$U \quad \circ^{1} \quad \xi_1, \xi_2$

Beside the point are the labels U, 1 and ξ_1, ξ_2.

$p = 0,\ q = 2.$ Superdomain of dimension $(0, 2)$.
Geometrically, a point. a map of the superdomain to itself is an endomorphism.
Rings $\Lambda(\xi_1, \xi_2)$.



Original source details: GELFAND-P000670

Diagram and typed description.

$1 \mapsto 1$

$\xi_1 \mapsto a\xi_1 + b\xi_2 + c\xi_1\xi_2$

$\xi_2 \mapsto a'\xi_1 + b'\xi_2 + c'\xi_1\xi_2$

The arrowed note reads “disallowed, since it preserves parity”. The coefficients before the crossed-out products are c and c', respectively.

a, b, a', b' — arbitrary.

GELFAND-P000671

The algebra $\Lambda(\xi_1, \xi_2)$ is commutative in the super sense.

$$xy = (-1)^{p(x)p(y)}yx.$$

Example 2.

$$\begin{matrix} u, \xi_1, \xi_2 \\ v, \eta_1, \eta_2 \end{matrix} \quad \left\| \begin{matrix} v = u + \xi_1\xi_2 \\ \eta_1 = \xi_1 \\ \eta_2 = \xi_2. \end{matrix} \right.$$

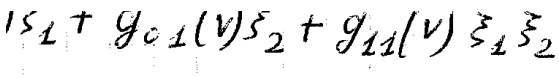
Problem: what is the image of $\sin v$?

An element of the left ring has the form

$$f_{00}(u) + f_{10}(u)\xi_1 + f_{01}(u)\xi_2 + f_{11}(u)\xi_1\xi_2$$

An elem. of the 2nd ring

$$g_{00}(v) + g_{10}(v)\xi_1 + g_{01}(v)\xi_2 + g_{11}(v)\xi_1\xi_2$$



Original source details: GELFAND-P000671

Manuscript excerpt. The 2nd ring is written with ξ_1, ξ_2, as in the 1st.

map

$$\widetilde{U} \longrightarrow \widetilde{V}$$
$$C^\infty(\widetilde{V}) \longrightarrow C^\infty(\widetilde{U})$$
$$g_{00}(v) \mapsto g_{00}(u + \xi_1\xi_2) = g_{00}(u) + \xi_1\xi_2 g'_{00}(u).$$

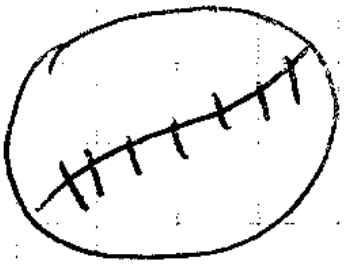
$x = (u_1, \ldots, u_p, \xi_1, \ldots, \xi_q)$ — a coordinate system.

$$\frac{\partial}{\partial u_i} f\xi_1^{\alpha_1} \ldots \xi_q^{\alpha_q} = \frac{\partial f}{\partial u_i}\xi_1^{\alpha_1} \ldots \xi_q^{\alpha_q}$$

Stokes' theorem — ?

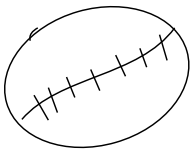
The integral of a superfunction over a superdomain.

Analogue in ordinary analysis



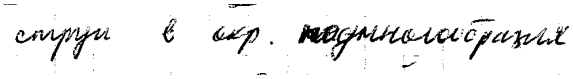
Original source details: GELFAND-P000671

Diagram and typed description.



A rising curve and seven visible cross-strokes lie inside the oval. A short mark meets the outline at the upper left.

jets in a nbhd. of a submanifold.



Original source details: GELFAND-P000671

Manuscript excerpt. The first words read "струи в окр." (jets in a nbhd.). The start of "submanifold" is retraced.

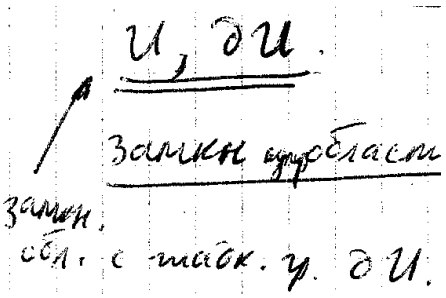
The integral for fns. on a manifold with boundary is noninv. under changes of variables.

Difference: here one cannot integrate forms. One can integrate densities (under change of coords. mult. by the Berezinian).

What does one integrate over the boundary?

$U, \partial U$. $f = 0$, where $f \in C^\infty(U)$ — a superfunction.

Closed superdomain. u, ξ_1, ξ_2



Original source details: GELFAND-P000671

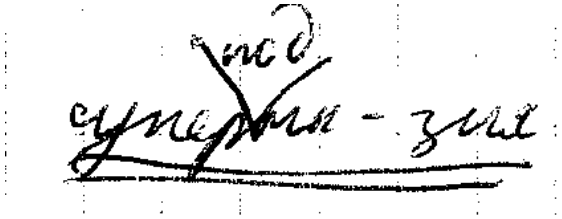
Diagram and typed description.

$U, \partial U$

closed domain with smooth bdry. ∂U

The arrow points from the note towards $U, \partial U$.

On ∂U there is the struct. of a [inserted above: sub] supermanifold.



Original source details: GELFAND-P000671

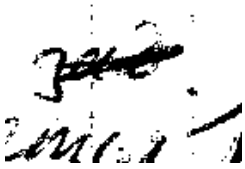
Diagram and typed description.

sub

supermanifold

The “sub” insertion points after “super”. The Russian abbreviation reads “супермн-зия”.

(by an even eq. [crossed out: this?] is cut out)



Original source details: GELFAND-P000671

Uncertain reading. The crossed-out word tentatively reads “это” (this).

$[0, \infty)$ u, ξ_1, ξ_2.

Different superdomains

1. $u = 0$
2. $u + c\xi_1\xi_2 = 0$

Original source details: GELFAND-P000671

Diagram and typed description.

even eq.

A narrow slanted quadrilateral with heavy lengthwise strokes. The words at right label the figure.

GELFAND-P000672

Submanifold: in the ring choose an ideal of fns..

Boundary — even.

$C^\infty(\widetilde{U})$ Boundary — eq. $f(\widetilde{U}) = 0$

$$\int \text{of a compactly supported fn.} = \int f_{11\ldots1}\, du_1 \ldots du_p$$

Different for closed domains.

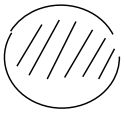
Let the half-space be $u_1 \geq 0$

Then the integr. is given by the same formula.

It is preserved if it maps into itself as a supermanifold.

$u_1^2 + u_2^2 \leq 1$

Diagram and typed description.



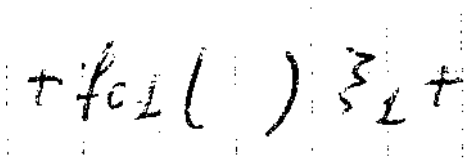
Slanted strokes fill the disc.

Original source details: GELFAND-P000672

$$u_1^2 + u_2^2 - 1 = 0 \quad (1) \quad \text{— 2 boundaries}$$
$$u_1^2 + u_2^2 - 1 - \xi_1\xi_2 = 0 \quad (2)$$

Integral over the domain with boundary (2).

$$\int [f_{00}(u) + f_{01}(\,)\xi_1 + \ldots]$$



Manuscript excerpt. After f_{01} the parentheses are empty, followed by ξ_1.

Original source details: GELFAND-P000672

Roughly

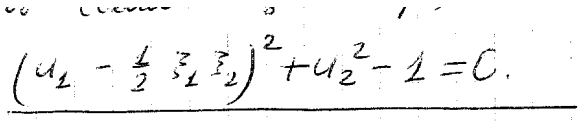
$$\int \xi_1 \xi_2 \, d\xi_1 \, d\xi_2 = 1$$

— For the ordinary boundary —

$$\int_{u_1^2+u_2^2<1} f_{11}(u) \, du_1 du_2$$

Check of invariance — the boundary eq. must stay the same.
If the boundary is $u_1^2 + u_2^2 - 1 - \xi_1\xi_2 = 0$
change variables so the boundary goes to the normal one

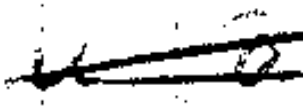
$$\left(u_1 - \tfrac{1}{2}\xi_1\xi_2\right)^2 + u_2^2 - 1 = 0.$$



Original source details: GELFAND-P000672

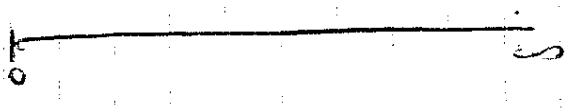
Manuscript excerpt. Inside the parentheses the term after u_1 is $-\frac{1}{2}\xi_1\xi_2$.

[crossed out: $u = 0$?]



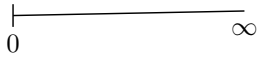
Original source details: GELFAND-P000672

Uncertain reading. The crossed-out formula above the line tentatively reads $u = 0$.



Original source details: GELFAND-P000672

Diagram and typed description.



A tick and 0 mark the left end. The right end is marked ∞.

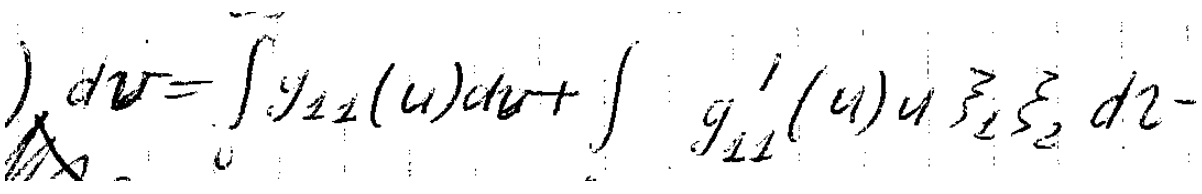
$$\int_0^\infty f_{11}(u) \, du \qquad u, \xi_1, \xi_2$$

bdry. $u = 0$.

$v = u + u\xi_1\xi_2$.

The boundary will be $v = 0$

$$\begin{aligned}\int_0^\infty g_{11}(v) \, dv &= \int_0^\infty g_{11}(u + u\xi_1\xi_2) \, dv \\ &= \int_0^\infty g_{11}(u) \, dv + \int_0^\infty g'_{11}(u) u \xi_1 \xi_2 \, dv\end{aligned}$$



Original source details: GELFAND-P000672

Manuscript excerpt. The third integral has differential dv.

Original source details: GELFAND-P000672

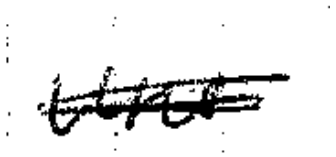
Diagram and typed description.

An open descending curve with diagonal and looping strokes. Some of their junctions are indistinct.

$$\frac{Dv}{Du}$$

Integral over a domain with boundary — — Bernstein and Leites

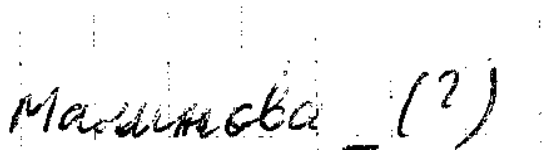
[crossed out]



Original source details: GELFAND-P000672

Undeciphered passage. The word before “There is work” is heavily crossed out.

There is work by Berezin and Marinov (?)



Original source details: GELFAND-P000672

Manuscript excerpt. The surname reads “Маринова” (Marinov). The question mark is in the manuscript.

class. mech. in Fermi and Bose variables

Problem: Stokes’ thm. for the circle $u_1^2 + u_2^2 + c\xi_1\xi_2 - 1 = 0$.

GELFAND-P000673

Notes on I. Bernstein’s previous super talks (from I. Dorfman’s notebook)

Original source details: GELFAND-P000673

Manuscript excerpt. The name in parentheses reads “И. Дорфман” (I. Dorfman).

Def. Superring

$$A = \underset{\text{even}}{A_0} \oplus \underset{\text{odd}}{A_1} \qquad \begin{aligned} A_0A_0 &\subset A_0 \\ A_0A_1 &\subset A_1 \\ A_1A_0 &\subset A_1 \\ A_1A_1 &\subset A_0. \end{aligned}$$

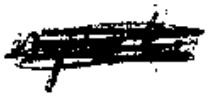
It is called commutative, if $ab = ba(-1)^{\deg a \cdot \deg b}$

Linear algebra.

Undeciphered passage. Crossed-out mark before the bracket.

Original source details: GELFAND-P000673

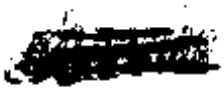
[Over a commut. superring consider



Undeciphered passage. Crossed-out word after "consider".

Original source details: GELFAND-P000673

any A-module A^n and its

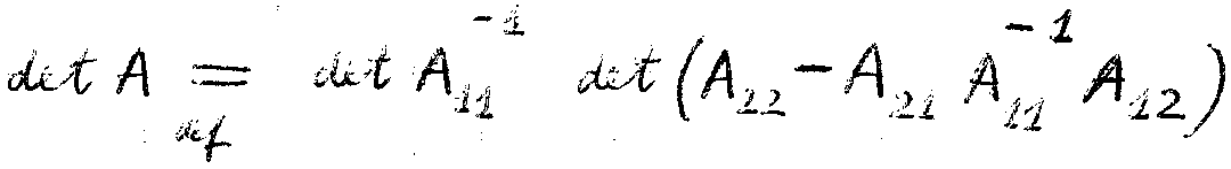


Undeciphered passage. Crossed-out word after "its".

Original source details: GELFAND-P000673

endomorphism given by a block matrix]

$$[A =] \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \qquad \det A \underset{\text{def}}{=} \det A_{11}^{-1} \det(A_{22} - A_{21}A_{11}^{-1}A_{12})$$



Original source details: GELFAND-P000673

Manuscript excerpt. The faint superscript on the first A_{11} reads -1.

$$\operatorname{str} A = \operatorname{tr} A_{11} - \operatorname{tr} A_{22}.$$
(supertrace)

Properties:

$$\det XY = \det X \cdot \det Y.$$
$$\det X = \exp \operatorname{str} \ln X$$

Uncertain reading. After X there is a dark mark below the line. It may be a comma or a crossed-out sign.

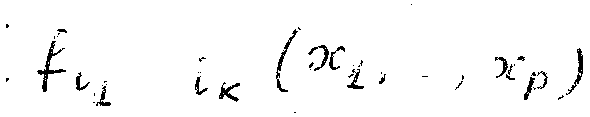
Original source details: GELFAND-P000673

$$\operatorname{str}(XY) = \operatorname{str}(YX)$$
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^* = \begin{pmatrix} A^* & C^* \\ -B^* & D^* \end{pmatrix}$$

Superanalysis.

Consider

$$\sum f_{i_1 \quad i_k}(x_1, -, x_p)\, \xi_1^{i_1} \dots \xi_k^{i_k}$$



Manuscript excerpt. A faint short stroke appears between the commas after x_1.

Original source details: GELFAND-P000673

Rules of differentiation

$$\frac{\partial}{\partial\xi_1}(\xi_1\xi_2)=\xi_2,\qquad \frac{\partial}{\partial\xi_2}(\xi_1\xi_2)=-\xi_1,$$
$$\frac{\partial}{\partial\xi_1}\xi_2=0,\qquad \frac{\partial}{\partial\xi_1}\xi_1=1,\qquad \frac{\partial}{\partial\xi_1}1=0.$$

_____ · _____

Consider block matrices $\begin{pmatrix}A & B\\ C & D\end{pmatrix}$

$$A=A_{\bar 0}\oplus A_{\bar 1}$$
$\mathbb{Z}/2\mathbb{Z}$ — grad. elem. $A,D\in A_{\bar 0}$
$\bar 0,\bar 1$ — elem. elem. $B,C\in A_{\bar 1}$
$p(a)$ — parity

$$\operatorname{Ber} J=(\det D)^{-1}\cdot\det(A-BD^{-1}C)\in A_{\bar 0}$$
(Berezinian)

Assert

$$\operatorname{Ber} J_1J_2=\operatorname{Ber} J_1\cdot\operatorname{Ber} J_2.$$
$$\operatorname{str} J=\operatorname{tr} A-\operatorname{tr} D$$

Assert.

$$\operatorname{str}(J_1J_2)=\operatorname{str}(J_2J_1).$$

_____ · _____

In $\mathbb{R}^p$ consider $C^\infty(\mathbb{R}^p)\otimes\Lambda(\xi_1,\text{—},\xi_q)$

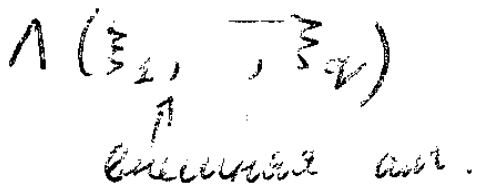


Diagram and typed description.
$\Lambda(\xi_1,\text{—},\xi_q)$
↑
exterior alg.
The arrow from the note points up towards the comma after ξ_1.

Original source details: GELFAND-P000673

$p(\xi_i)=1,\qquad \xi_i\xi_j=-\xi_j\xi_i.$

$$\Lambda(\xi_1, \dots, \xi_q) \ni \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}, \qquad \alpha_i = 0, 1.$$
$$f = \sum f_{\alpha_1 \alpha_2 \dots \alpha_q} \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}.$$

Def.

$$\mathbb{R}^{p,q} = (\mathbb{R}^p, C^\infty(\mathbb{R}^p) \otimes \Lambda(\xi_1, \dots, \xi_q))$$

— superspace of dim. (p, q).

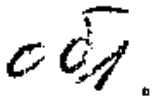
Def.

$$U = \left(u, \underbrace{C^\infty(u) \otimes \Lambda(\xi_1, \dots, \xi_q)}_{(\text{def})\, C^\infty(U)} \right)$$

superdomain.

(ringed sp.!)

— ring of fns. on the dom.



Manuscript excerpt. The abbreviation after "on" reads "обл." (dom.).

Original source details: GELFAND-P000674

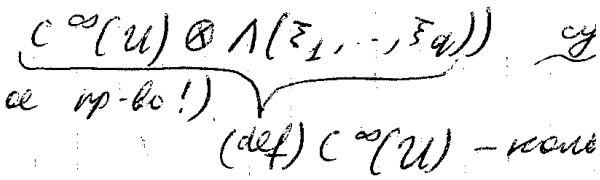


Diagram and typed description.
$\underbrace{C^\infty(u) \otimes \Lambda(\xi_1, \dots, \xi_q)}_{(\text{def})\, C^\infty(U)}$
The brace under the tensor product points to $(\text{def})\, C^\infty(U)$.

Original source details: GELFAND-P000674

map $\widetilde{\psi} : U \to V$

Uncertain reading. The isolated mark before the dash is undeciphered.

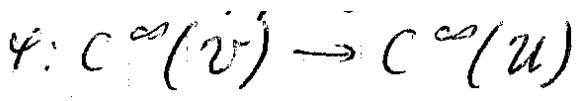
Original source details: GELFAND-P000674

— morphism of ringed sp. — this is a map

Undeciphered passage. Crossed-out writing before ψ.

Original source details: GELFAND-P000674

$$\psi : C^\infty(V) \to C^\infty(U).$$



Manuscript excerpt. The function rings use V and U, as in the notation $U \to V$.

Original source details: GELFAND-P000674

Coord. notation. Let a tuple be chosen in the sp. U

$$(u_1, \dots, u_p, \xi_1, \dots, \xi_q)$$

— " — V — " — $(v_1, \dots, v_m, \eta_1, \dots, \eta_n)$.

It turns out that to specify the map $\psi : U \to V$

Undeciphered passage. Crossed-out word after $U \to V$.

Original source details: GELFAND-P000674

it suffices to specify

$$v_1^*, \dots, v_m^*, \eta_1^*, \dots, \eta_n^*.$$

Example.

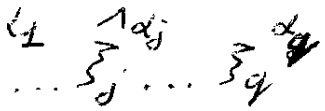
$$v = u + \xi_1 \xi_2$$
$$\eta_1 = u\xi_1$$
$$\eta_2 = \xi_2.$$

Problem. What is the form of $\sin v$?

Assert.

$$\frac{\partial}{\partial u_i}(f\xi^\alpha) = \frac{\partial f}{\partial u_i} \cdot \xi^\alpha,$$

$$\frac{\partial}{\partial \xi_j}(f\xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}) = \alpha_j (-1)^{\alpha_1 + \dots + \alpha_{j-1}} f\xi_1^{\alpha_1} \cdots \widehat{\xi_j^{\alpha_j}} \cdots \xi_q^{\alpha_q}.$$



Manuscript excerpt. The omission mark is over ξ_j, with α_j still written at upper right. The final q is retraced.

Original source details: GELFAND-P000674

Let U be a superdomain

$$x = (u, \xi)$$
$$y = (v, \eta)$$

$$J = (J_{ij}), \qquad J_{ij} = \frac{\partial y_j}{\partial x_i}.$$

$$\operatorname{Ber} J \underset{\text{def}}{=} D(y)/D(x).$$

Def. U, $x = (u, \xi)$.

The expr. $\rho = fD(x)$ is called a volume form.

$$fD(x) = f\big(D(x)/D(y)\big) \cdot D(y).$$

Definition.

$$\int_u fD(x) \underset{\text{def}}{=} \int_u f_{11\dots1}(u)\, du_1 \dots du_p$$

(integral of a volume form)

Theorem (Berezin) This expr. is independent of the coordinate system.

Here one integrates a compactly supported fn.

GELFAND-P000675

25/X-76. Gelfand seminar.

Gelfand.

What is a connection? (for first-year students and Polyakov).

— see Nomizu, Sternberg, Bishop and Crittenden, Landau and Lifshitz.

Popov.

The founders' paper (Yang, Mills — translation in the coll. "Elem. particles and compensating fields").

Consider the Dirac eq.

without mass

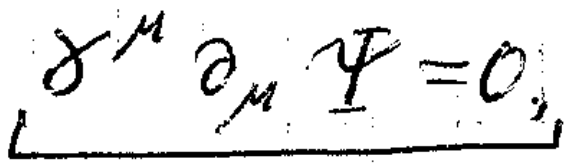


Diagram and typed description.

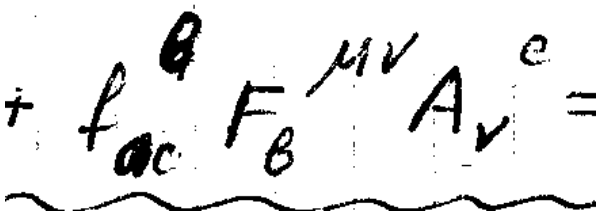
$\gamma^\mu \partial_\mu \Psi = 0,$

Original source details: GELFAND-P000675

$\Psi = (\Psi_\alpha)$ — ordinary 4-comp. spinor.

Yang–Mills equations:

$$\partial_\nu F_a^{\mu\nu} + f_{ac}^b F_b^{\mu\nu} A_\nu^c = 0,$$



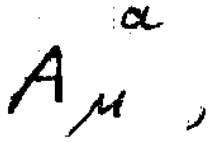
Manuscript excerpt. The retraced indices on the coefficient read f_{ac}^b.

Original source details: GELFAND-P000675

where

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + \tfrac{1}{2} f_{bc}^a (A_\mu^b A_\nu^c - A_\nu^b A_\mu^c)$$

on $\mathbb{R}^4$ consider the Minkowski metric $g_{\mu\nu}$, A_μ^α, $a = 1, \ldots, \dim G$



Manuscript excerpt. The superscript on A_μ is α. The range immediately after it uses a.

Original source details: GELFAND-P000675

(G — some group, usually taken to be semisimple).
lowering of indices in the eq. with the aid of

$$K_{ab} = f^d_{ac} f^c_{bd},$$

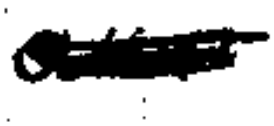
where f^d_{ac} — struct. const. of the Lie algebra of the group.

——— · ———

G. On quarks see Zel'dovich's article in UFN, "Quarks for pedestrians".

——— · ———

The Yang–Mills eqs. describe quarks.
$\mu, \nu = 1, \dots, 4$;



Undeciphered passage. Crossed-out writing before $F^{\mu\nu}_a$.

Original source details: GELFAND-P000675

$F^{\mu\nu}_a$ — tensor with values in the Lie algebra.
A — analogue of the vect. potential

$$\partial_\nu = \partial/\partial x^\nu.$$

F — analogue of the field-strength tensor.
Example.
For $G = U(1)$ one obtains the eqs. of electrodynamics.
Equations:

$$\partial_\mu F^{\mu\nu} = 0$$
$$F^{\mu\nu} \underset{\text{def}}{=} \partial_\mu A_\nu - \partial_\nu A_\mu$$

We get Maxwell's eqs.

——— · ———

Gauge transformations.
In the el. and magn. field:

$$\begin{pmatrix} 0 & H_{12} & H_{13} & E_1 \\ -H_{12} & 0 & H_{23} & E_2 \\ -H_{13} & -H_{23} & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix} = F^{\mu\nu};$$

2-form

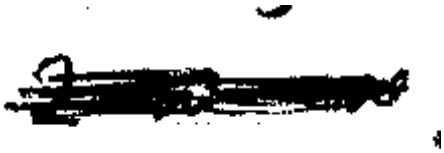
Manuscript excerpt. The third diagonal entry is retraced and reads as zero.

Original source details: GELFAND-P000675

Maxwell's equations:

$$F^{\mu\nu} = \frac{\partial}{\partial x^\mu} A_\nu - \frac{\partial}{\partial x^\nu} A_\mu.$$

— tensor



Undeciphered passage. Crossed-out writing after "tensor".

Original source details: GELFAND-P000675

of 2nd ord.

Transformations of A that leave F unchanged are called gradient transformations.

$$A_\mu \to A_\mu + \partial_\mu \alpha$$

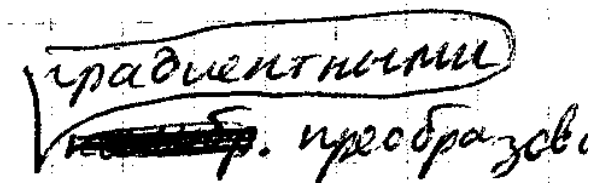


Diagram and typed description.

gradient

~~gauge~~

The outline of "gradient" points to the crossed-out abbreviation, tentatively "калибр." (gauge).

Original source details: GELFAND-P000675

GELFAND-P000676

Let (T_a) be a basis in the Lie algebra $\mathcal{L}(G)$.

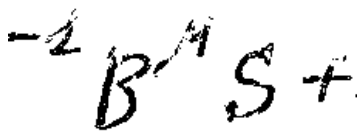
Consider

$$B_\mu = T_a A^a_\mu \quad \text{— field of matrices.}$$

$$F_{\mu\nu} = T_a F^a_{\mu\nu} \quad \text{—} \quad \text{— " —.}$$

Gauge transf. on fields of matrices:

$$B_\mu \to B'_\mu = S^{-1} B^\mu S + S^{-1} \partial_\mu S,$$



Manuscript excerpt. Between S^{-1} and S, μ is written as a superscript on B.

Original source details: GELFAND-P000676

$$F^{\text{[crossed out]}}_{\mu\nu} \to F'_{\mu\nu} = S^{-1} F_{\mu\nu} S,$$

Undeciphered passage. Crossed-out mark above the first F.

Original source details: GELFAND-P000676

where $S : \mathbb{R}^4 \to G$.

Direct calculation readily shows that the eqs. are inv. under these transformations.

Unknowns — A.

Nonlinearity due to the noncommut. group.

(F — curvature of the connection)

Yang–Mills equations.

Let (P, M, G) be a principal bundle

(M — manifold

G — group

Undeciphered passage. Crossed-out writing after “group”.

Original source details: GELFAND-P000676

— arbitrary)

on M there is a pseudo-Riemannian metric g.

In the principal bundle a connection is given by the form $\Omega \in \mathcal{L}(G) \otimes \Lambda^1(P)$.

The eq. will be for this form.

Fix an arbitrary (nondegenerate) quadratic form K on the Lie algebra $\mathcal{L}(G)$.

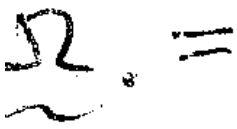
From these data a natural metric is constructed on P.

$$h_p(Y_p, Y'_p) = g_{\pi(p)}(d\pi Y_p, d\pi Y'_p) + K\big(\Omega(Y_p), \Omega(Y'_p)\big)$$

in general the metric is noninvariant.

Curvature of the connection.

$$\Phi = H\, d\Omega\,, \; = d\Omega + \tfrac{1}{2}[\Omega, \Omega]$$



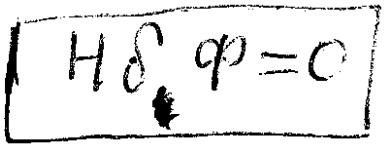
Uncertain reading. The low mark after $d\Omega$ before the second equality is rendered as a comma.

Original source details: GELFAND-P000676

H — operator taking the horiz. part of a form, constructed from the connection.

Yang–Mills equation for the free field

$$\boxed{H\delta_{\text{[crossed out]}}\varphi = 0}$$



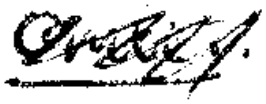
Undeciphered passage. Dark crossed-out mark below δ inside the box.

Original source details: GELFAND-P000676

δ — adjoint operator for the metric h.

This is the eq. on the principal bundle, and above is the eq. on the base. How do we identify them?

It turns out the eqs. are independent of the choice of metric K!



Manuscript excerpt. The retraced word reads “Оказ.” (it turns out).

Original source details: GELFAND-P000676

(on equiv. horiz. forms the operator $H \circ \delta$ is independent of the metric;

on arbitrary forms with values in a vector sp.

horiz. form — $\equiv 0$, if one of the vectors is vertical.

We must show that $H\delta\Phi$ — an equiv. horiz. form.

GELFAND-P000677

Consider a nbhd. $U \subset N$ and there take a trivialization ζ of the principal bundle (arbitrary section).

Manuscript excerpt. The letter after $\subset$ reads N.

Original source details: GELFAND-P000677

Uncertain reading. The section symbol is rendered as ζ. A mark to its lower right is crossed out.

Original source details: GELFAND-P000677

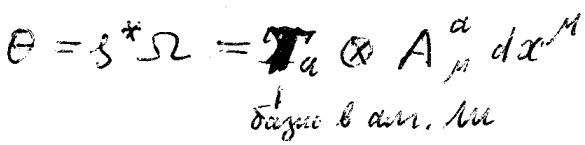


Diagram and typed description.

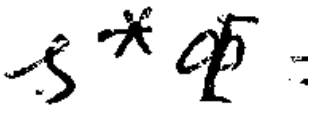
$\theta = \zeta^*\Omega = T_a \otimes A^a_\mu \, dx^\mu$

basis in the Lie alg.

The letter T is retraced. The stroke underneath marks the label.

Original source details: GELFAND-P000677

$$\zeta^*\Phi = T_a \otimes F^a_{\mu\nu} \, dx^\mu \wedge dx^\nu.$$

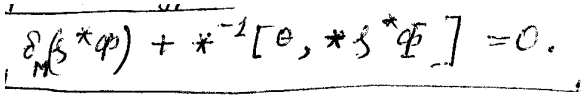


Uncertain reading. The symbol before the star is the same sign rendered as ζ.

Original source details: GELFAND-P000677

Projection of the eq.:

$$\delta_M(\zeta^*\varphi) + *^{-1}[\theta, *\zeta^*\Phi] = 0.$$



Original source details: GELFAND-P000677

Uncertain reading. The first section symbol is followed by φ, the second by Φ. Two small strokes lie at the lower edge of the bracket.

δ_M — the op. δ on the base.

Direct calc. shows this is the same eq.

It is the same eq. if the group has a 2-sided inv. metric.

The vacuum Einstein eqs. are here (Ricci tensor$= 0$)

$G =$ affine transformations of $\mathbb{R}^4$

(eq. in the bundle of aff. frames)

On the aff. frames there are 2 natural connections ind. by the base metric.

On the class of Cartan connections ind. by metrics.

Original source details: GELFAND-P000677

Uncertain reading. The retraced sign on the left resembles a 4.

There is a unique conn.

Original source details: GELFAND-P000677

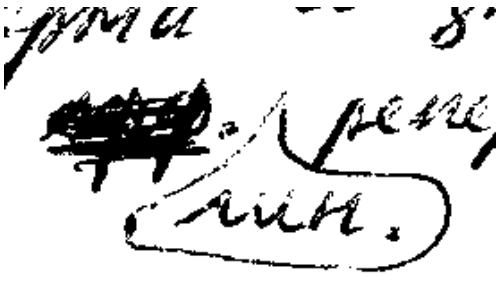
Undeciphered passage. Crossed-out mark before $\bar{\Omega}$.

$\bar{\Omega}$ in the bundle of aff. frames, such that

$$\text{if} \quad \underset{\text{lin.}}{L} \overset{i}{\hookrightarrow} \underset{\text{aff.}}{AL} \qquad i^*\bar{\Omega} = (\Omega^g, \chi);$$

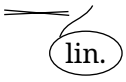
χ — displacement form

(nat. form with values in $\mathbb{R}^4$ on the bundle of lin. frames)



Original source details: GELFAND-P000677

Diagram and typed description.



The outlined “lin.” below points to the position before “frames”. The earlier writing is crossed out.

Ω^g — metric connection

_____ . _____

def. of a Cartan connection — see Bishop and Crittenden.

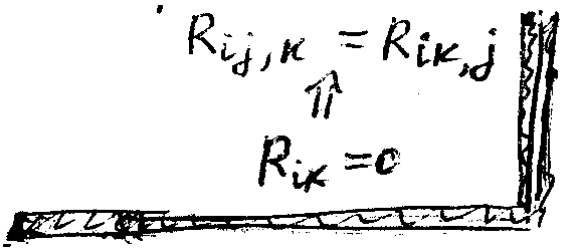
on these connections $H\delta\varphi = 0$ — Einstein's equation. (1st order in the connection!! — unlike for general Yang–Mills fields)

2nd order eqs. —

Original source details: GELFAND-P000677

Undeciphered passage. Crossed-out writing before "consequence".

a consequence of the 1st order eqs.



Original source details: GELFAND-P000677

Diagram and typed description.

$$R_{ij,k} = R_{ik,j}$$
$$\Uparrow$$
$$R_{ik} = 0$$

The double arrow points up. A hatched corner runs along the bottom and right.

GELFAND-P000678

Gelfand seminar.

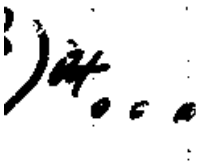
I. N. Bernstein.

Kostant.

$$\varphi(x) = \prod_{n=1}^{\infty}(1-x^n) = (1-x)(1-x^2)(1-x^3)\times\dots$$
$$= \sum_{n=-\infty}^{\infty}(-1)^n x^{(3n^2+n)/2}.$$

Original source details: GELFAND-P000678

Uncertain reading. The retraced lower limit of the product tentatively reads $n = 1$.



Original source details: GELFAND-P000678

Uncertain reading. The dark mark between the third factor and the ellipsis resembles $\times$.

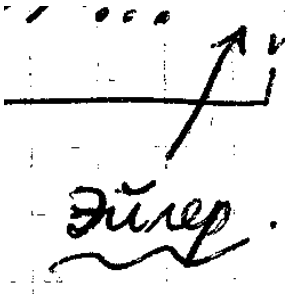


Diagram and typed description.

$= \sum$

Euler.

The arrow from the name points to the equality with the sum.

Original source details: GELFAND-P000678

Jacobi:

$$\varphi(x)^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) x^{n(n+1)/2}$$

(from the Θ-function).

Uncertain reading. The oval crossed by a horizontal stroke is rendered as uppercase Θ.

Original source details: GELFAND-P000678

Others have since been found (Dyson et al.)

Macdonald found a group-theoretic basis.

It is easy to check that

$$\varphi(x)^{-1} = \sum p(n) x^n,$$

$p(n)$ — number of partitions of n into a sum of nonnegative summands

$$x\varphi(x)^{24} = \sum \tau(n) x^n,$$

$\tau(n)$ — Ramanujan's function

this is its definition

$\tau(n)$ has a combinatorial definition

Undeciphered passage. Crossed-out writing above the line beginning "via".

Original source details: GELFAND-P000678

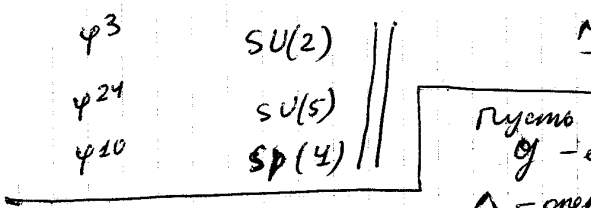
(via Dedekind's Θ-fn.)

Uncertain reading. The glyph before "fn." is rendered as Θ, as in the line about Jacobi's formula.

Original source details: GELFAND-P000678

one can write

$$\varphi(x)^k, \quad \text{where } k \text{ — dimension of the Lie group (semisimple)}$$



Original source details: GELFAND-P000678

Diagram and typed description.

φ^3	$SU(2)$
φ^{24}	$SU(5)$
φ^{10}	$Sp(4)$

A double rule stands to the right of the three rows. The stepped line below rises to the next paragraph.

Macdonald: writes out the formula explicitly.
Let there be a compact semisimple group

Original source details: GELFAND-P000678

Undeciphered passage. Two crossed-out pieces of writing between “group” and its symbol.

G

g — its Lie algebra, on g there is the Killing form B

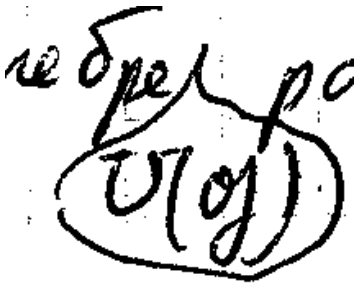
Δ — the Laplace operator on

Original source details: GELFAND-P000678

Undeciphered passage. Two crossed-out pieces of writing after “on”.

G

$X_1, \ldots, X_k$ — orthonorm. basis in g.
In the enveloping algebra



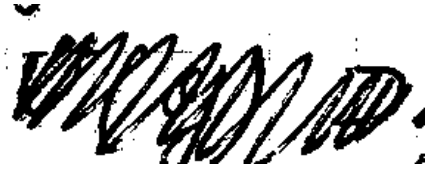
Original source details: GELFAND-P000678

Diagram and typed description.
algebra

$U(g)$

$U(g)$ is outlined below “algebra”.

consider the element



Original source details: GELFAND-P000678

Undeciphered passage. Crossed-out formula before the element.

$\Delta = \sum X_i^2$. it is 2-sided invariant.

$$\varphi(x)^k = \sum_{\text{irred. reps. } V} c_V \cdot \dim V \cdot x^{\Delta|_V}$$

Undeciphered passage. Crossed-out writing in the exponent before $\Delta|_V$.

Original source details: GELFAND-P000679

($\Delta|_V$ is a multiple of the identity; eigenvalue $\Delta|_V$.

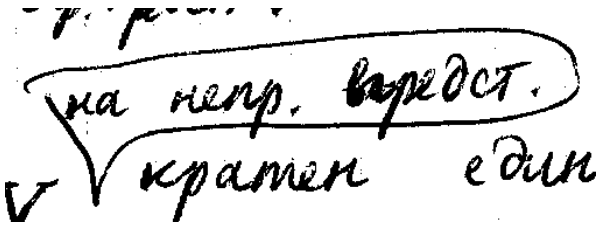


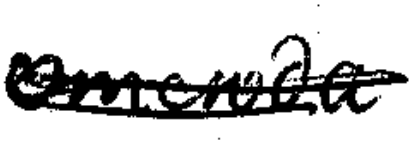
Diagram and typed description.

on an irred. rep.

The insertion above the line points after $\Delta|_V$.

Original source details: GELFAND-P000679

~~Hence~~ The asymptotics depend on the dimension.



Uncertain reading. The crossed-out word tentatively reads “Отсюда” (hence).

Original source details: GELFAND-P000679

Groups vary, and so do the formulas.

$$c_V = \chi_V(a), \qquad \text{where} \quad \underset{\text{a certain element}}{a \in G}$$

~~For the series.~~ For $SU(n)$ there is a unique reg. element. (class of conjugate elements;) i.e., one whose eigenvalues are distinct and whose

Undeciphered passage. Crossed-out mark before the nth power.

Original source details: GELFAND-P000679

nth power is scalar.

Undeciphered passage. Crossed-out mark left of the matrix bracket.

Original source details: GELFAND-P000679

$$\begin{pmatrix} \xi & & & & 0 \\ & \xi^2 & & & \\ & & \xi^3 & & \\ & & & \ddots & \\ 0 & & & & \xi^n \end{pmatrix} c, \qquad c \quad \text{chosen so that } \det = 1.$$

Consider the ζ-function of the Laplace op.

V. Kac — proof via ∞-dimensional Lie algebras.

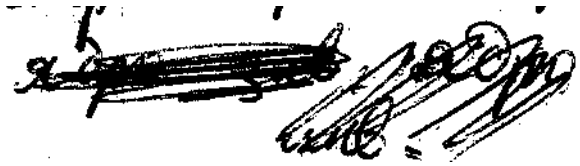
Consider $-\Delta + \dfrac{\dim g}{24}$ and set

$$\zeta(s) \underset{\text{def}}{=} \left(-\Delta + \frac{\dim g}{24}\right)^{-s}\Big|_a$$

Undeciphered passage. Crossed-out writing after the defining equality.

Original source details: GELFAND-P000679

(op. of convolution with a fn.,



Uncertain reading. Two crossed-out passages. The second tentatively contains “ядро” (kernel).

Original source details: GELFAND-P000679

constant on classes of conj. elements, its value is taken at a)

$K(x,y)$ — kernel. In fact $K(xy^{-1})$

About ~~$\zeta(s)$~~ one can say something

Uncertain reading. The crossed-out expression tentatively reads $\zeta(s)$.

Original source details: GELFAND-P000679

Theorem.

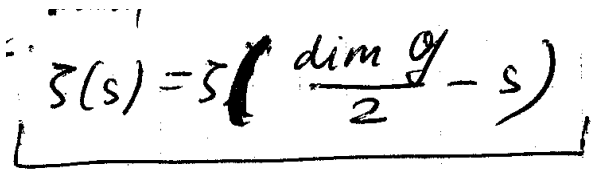


Diagram and typed description.

$$\underbrace{\zeta(s) = \zeta\left(\frac{\dim g}{2} - s\right)}$$

The line below the formula has upturned ends.

Original source details: GELFAND-P000679

Original source details: GELFAND-P000679

Uncertain reading. The letter after dim is rendered as g in all three formulas.

The theorem is equiv. to an

Original source details: GELFAND-P000679

Undeciphered passage. Crossed-out word after “equiv.”.

identity for $\varphi(x)^k$.

There are also φ^{39}, φ^{45} Van Esh

Original source details: GELFAND-P000679

Manuscript excerpt. The name is written in Latin script as “Van Esh”.

see Garland, Lepowsky, Inv. math., 34 (1976), 37–76

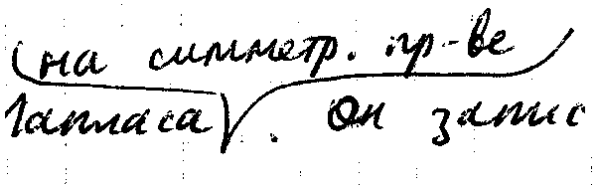
———— · ————

Original source details: GELFAND-P000679

Undeciphered passage. The underlined word before the surname is crossed out.

Olshanetsky.

Δ^0 — rad. part of the Laplace op.



Original source details: GELFAND-P000679

Diagram and typed description.

on the symmetric sp.

The insertion stands above the line after “Laplace”.

It is written in the form

$$\longrightarrow \frac{1}{2}\frac{\partial^2}{\partial t_k^2} + U(t_1, \ldots, t_n)$$

$$U = g \sum_{k>\ell} \frac{1}{\sin^2(t_k - t_\ell)}$$

And there is a complete system of observables in the quantum sense.

I. Bernstein. on V. G. Kac's work.

Construction of a Lie algebra from a Cartan matrix.

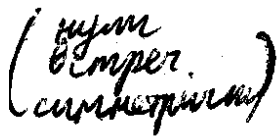
$(a_{ij})_{n\times n}$ — matrix with integer entries, for which

$$a_{ij} < 0 \quad \text{when } i \neq j \qquad \| \qquad \begin{array}{c} a_{ii} = 2, \\ a_{ij} \neq 0 \Rightarrow a_{ji} \neq 0 \end{array}$$

(zeros occur symmetrically)

Manuscript excerpt. The sign between a_{ij} and zero is written without a lower stroke.

Original source details: GELFAND-P000679



Uncertain reading. The ending of the last word runs into the bracket. It tentatively reads "симметрично" (symmetrically).

Original source details: GELFAND-P000679

GELFAND-P000680

Diagram and typed description. Vertical rule along the left margin.

Original source details: GELFAND-P000680

A Lie alg. is constructed from such a matrix.

Consider h_i, e_i, f_i, free generators (n of them), and write the relations

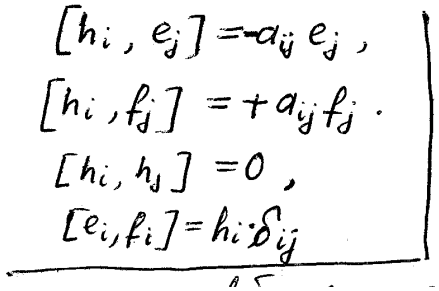
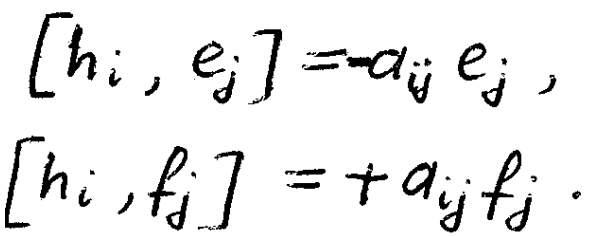


Diagram and typed description.

$$[h_i, e_j] = -a_{ij} e_j,$$
$$[h_i, f_j] = +a_{ij} f_j.$$
$$[h_i, h_j] = 0,$$
$$[e_i, f_i] = h_i \cdot \delta_{ij}$$

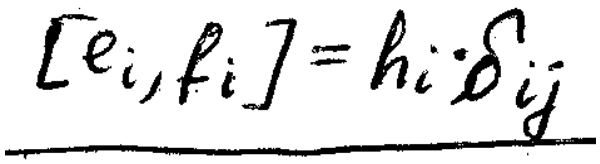
The outline runs below and to the right of the relations.

Original source details: GELFAND-P000680



Manuscript excerpt. The first line has a minus sign before a_{ij}, and the second has a plus sign.

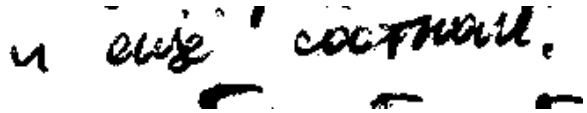
Original source details: GELFAND-P000680



Manuscript excerpt. The last line appears to give both e and f the same subscript i.

Original source details: GELFAND-P000680

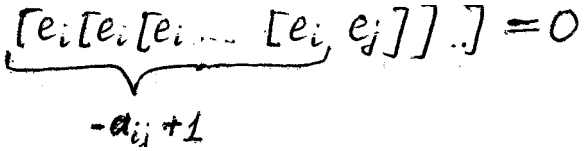
(take the quotient of the free alg. by these relations) and take the quotient of this alg. by the max normal subalgebra [Inserted at left above the formula: and further rels.]



Manuscript excerpt. The last word is abbreviated in the manuscript as “соотнош.” (relations).

Original source details: GELFAND-P000680

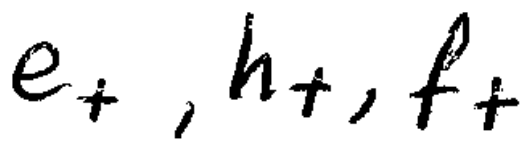
$$\underbrace{[e_i[e_i[e_i \ldots [e_i}_{-a_{ij}+1}, e_j]]..] = 0$$



Manuscript excerpt. The underbrace ends before the comma. After e_j there are two closing brackets, two dots and a further closing bracket.

Original source details: GELFAND-P000680

For certain matrices, classical groups arise. $SU(2)$ e_+, h_+, f_+ matrix $1 \times 1 \| (2)$



Manuscript excerpt. All three letters have a plus subscript.

Original source details: GELFAND-P000680

$$[h,e]=2e,\qquad [h,f]=-2f,\qquad [e,f]=h,$$

$SU(3)$ is obtained for the matrix $\begin{pmatrix}2&-1\\-1&2\end{pmatrix}$.

On roots: Serre "Lie Groups and Lie Algebras" — read only this section.

If $a_{ij}=a_{ji}$, then the following holds:

The group is finite-dimensional $\Leftrightarrow$ the matrix is positive definite.

$$\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}$$ — infinite-dimensional extension of $SU(3)$.

V. G. Kac. : the quotient of this alg. by its one-dimensional centre is

$$SU(3)\otimes\mathbb{C}[x,x^{-1}].$$

$$[x^k g, x^\ell g']=x^{k+\ell}[g,g'].$$

all the non-strictly def. ones are obtained this way.

Original source details: GELFAND-P000680

Manuscript excerpt. The two abbreviated words read "нестрог. опред." (non-strictly definite).

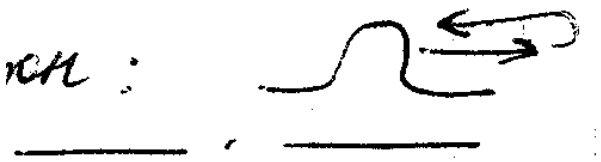
Happy New Year!

GELFAND-P000681

14/II-77. Gelfand Sem.

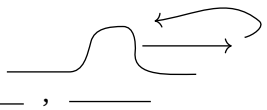
Yu. I. Manin.

Calogero: boomeron:

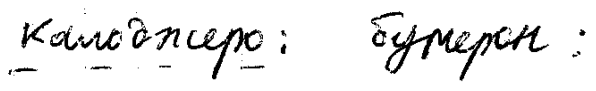


Original source details: GELFAND-P000681

Diagram and typed description.



The upper arrow hooks down at the right. A gap is visible between the hook and the point of the lower arrow.

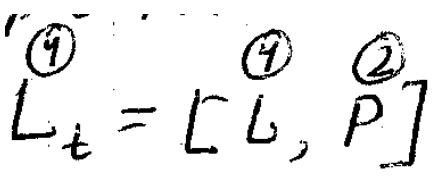


Original source details: GELFAND-P000681

Manuscript excerpt. The underline under "Calogero" is broken. The next word reads "бумерон" (boomeron).

Equation of Lax type

$$\overset{④}{L}_t=[\overset{④}{L},\overset{②}{P}]$$



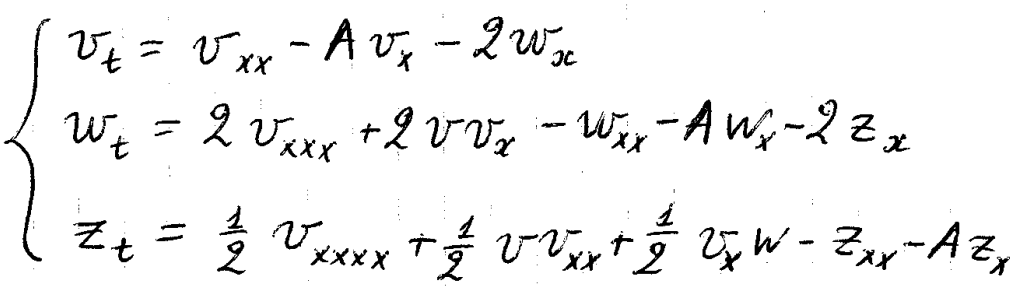
Original source details: GELFAND-P000681

Manuscript excerpt. A circled 4 stands over each L, and a circled 2 over P.

$$L = \partial^4 + v\partial^2 + w\partial + z, \qquad \partial = \partial/\partial x.$$
$$P = \partial^2 + A\partial + u$$

u is expressed in terms of v, w, z.

$$\begin{cases} v_t = v_{xx} - Av_x - 2w_x \\ w_t = 2v_{xxx} + 2vv_x - w_{xx} - Aw_x - 2z_x \\ z_t = \frac{1}{2}v_{xxxx} + \frac{1}{2}vv_{xx} + \frac{1}{2}v_x w - z_{xx} - Az_x \end{cases}$$

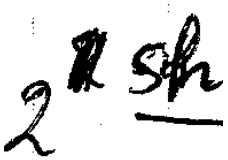


Original source details: GELFAND-P000681

Manuscript excerpt. The first line has no written coefficient before v_{xx}. The second has a 2 before vv_x.

1-soliton solution

$$\mu_1 = 2\frac{\operatorname{ch} 2\xi - \cos 2\xi + 2\sin \sigma t}{\operatorname{ch} 2\xi + \cos 2\xi - 2\cos \sigma t}$$
$$\mu_2 = 2\frac{\operatorname{sh} 2\xi - \sin 2\xi}{\operatorname{ch} 2\xi + \cos 2\xi - 2\cos \sigma t}$$

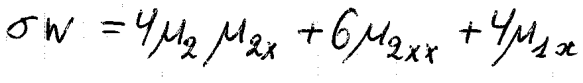


Uncertain reading. A thickened mark above and to the right of the factor 2, before sh, may be a crossed-out digit.

Original source details: GELFAND-P000681

$$\xi = x + ct \qquad \begin{matrix} c \in \mathbb{R} \\ \sigma \in \mathbb{R} \\ \sigma \leftrightarrow A \quad \text{(is expressed).} \end{matrix}$$

$$\begin{cases} v = -4\mu_{2x} \\ \sigma w = 4\mu_2\mu_{2x} + 6\mu_{2xx} + 4\mu_{1x} \\ z = \dots \quad \text{(inessential)} \end{cases}$$



Manuscript excerpt. The left-hand side of the second line reads σw.

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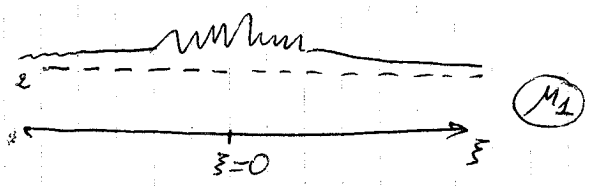
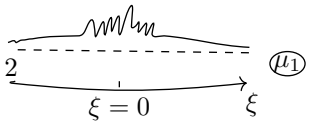


Original source details: GELFAND-P000681

Diagram and typed description.



At the left end of the axis there is a hook and a set of small illegible marks.

[To the right of the graph:] every $\frac{2\pi}{\sigma}$ periodically

Undeciphered passage. A sign or short passage is crossed out at the end of the line.

Original source details: GELFAND-P000681

it spits (geyseron, spiton, ...) explosive instability.

Algebraic method.

a) construct an aux. stationary eq. $[L_1, P_1] = 0$ then introduce t.

b)

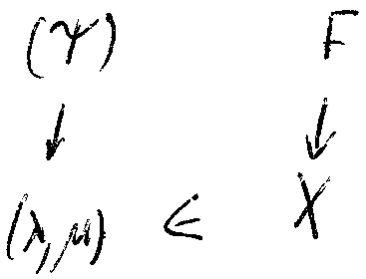
$$\begin{cases} L_1\psi = \lambda\psi \\ P_1\psi = \mu\psi \end{cases} \quad (\lambda, \mu) \in X \quad \text{— aff. alg. curve}$$

the case studied was: dim(space of ψ) $= 1$.

Krichever did not state it explicitly, but it followed from the constraints.

Drinfeld — did not specify the solution explicitly, but indicated the difficulties.

GELFAND-P000682

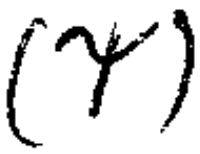


Original source details: GELFAND-P000682

Diagram and typed description.

$$\begin{array}{ccc} (\psi) & & F \\ \downarrow & & \downarrow \\ (\lambda, \mu) & \in & X \end{array}$$

Both arrows point down. The mark between the lower entries is rendered as $\in$.



Original source details: GELFAND-P000682

Manuscript excerpt. The sign in parentheses reads as ψ. There is an additional fine stroke in its centre.

The bundle extends to the completion.

Original source details: GELFAND-P000682

Manuscript excerpt. The middle word is abbreviated “продолж.” (extends).

We obtain a bundle on the curve

$$t \longrightarrow \text{path on the Jacobian.}$$

$$\underset{\text{Moduli of the system;}}{\underline{r > 1}} \quad \Rightarrow \quad \text{path in the moduli of bundles.}$$

Hyperell. curve $\longrightarrow$ intersection of 2 quadrics in $\mathbb{C}^5$.

Manuscript excerpt. The space is written $\mathbb{C}^5$.

Original source details: GELFAND-P000682

Analyt. paths?

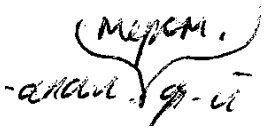
A soliton is a solution of the nonstationary eq. obtained by taking a curve with one double point with distinct tangents in the stationary eq.

Notation. k — field of char. 0.

$$\underset{\text{in the analytic setting,}}{(k = \mathbb{R} \text{ or } \mathbb{C})}$$

$\mathcal{B}$ — k-alg.

$$\begin{pmatrix} \mathcal{B} = \underset{\text{matrices}}{M_\ell}\ (\mathcal{B}_0), \\ \\ \mathcal{B}_0 = \text{field of rat. } k\text{-anal. } \overset{\text{merom.}}{\text{functions of}} \begin{matrix} x \\ x, t \\ x, t, z \end{matrix} \end{pmatrix}$$



Manuscript excerpt. The abbreviation “мером.” (meromorphic) is inserted above the space before “ф-й” (functions) with a curved callout.

Original source details: GELFAND-P000682

$$\underset{\text{derivations}}{\partial, \partial_1, \partial_2 : \mathcal{B} \longrightarrow \mathcal{B}}, \qquad [\partial, \partial_i] = [\partial_i, \partial_j] = 0$$

(in the anal. setting $\partial = \partial/\partial x$, $\partial_1 = \partial/\partial t$, $\partial_2 = \partial/\partial z$).

$\mathcal{B}[\partial]$

(L, P) — pair of ops. satisfying one of the 3 eqs.

$$\begin{cases} [L, P] = 0 & \text{stationary Lax} \\ \partial_1 L = [L, P] & \text{nonstationary Lax} \\ \partial_1 L + \partial_2 P = [L, P] & \text{Zakh.–Shabat} \end{cases}$$

① Krich.–Drinfeld bimodule $\mathcal{M}$.

Manuscript excerpt. A circled 1 stands to the left of the heading.

Original source details: GELFAND-P000682

Structures.

a) $\mathcal{M}$ — k-lin. space, left $\mathcal{B}$-module, right $\mathcal{O}$-module, where

$\mathcal{O}$ — ring of functions on an aff. curve over k; irreducible over $\bar{k}$.

b) distinguished element $\mathbf{1} \in \mathcal{M}$

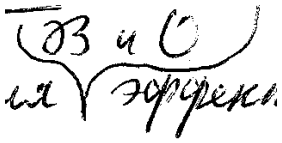
c) increasing filtration

$$\mathcal{M}_{-1} \subset \mathcal{M}_0 \subset \mathcal{M}_1 \subset \mathcal{M}_2 \subset \ldots, \qquad \bigcup_{i=0}^{\infty} \mathcal{M}_i = \mathcal{M}$$

d) $\nabla : \mathcal{M} \longrightarrow \mathcal{M}$ $\quad$ k-lin. operator.

Axioms.

e) the actions of $\mathcal{B}$ and $\mathcal{O}$ are faithful and commute



Manuscript excerpt. "$\mathcal{B}$ and $\mathcal{O}$" is inserted above the space after "actions" with a curved callout.

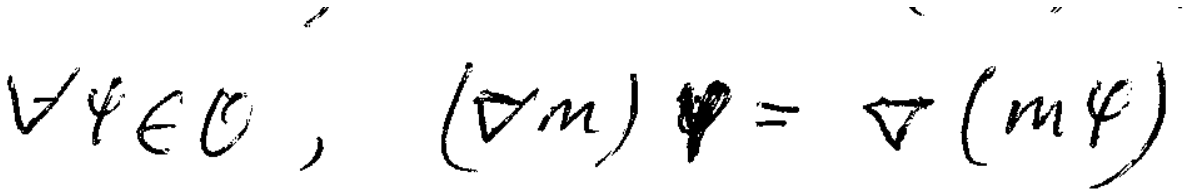
Original source details: GELFAND-P000682

f) $\mathcal{M}_{-1} = 0, \qquad \mathcal{B}\mathcal{M}_i \subset \mathcal{M}_i$

g) $\mathcal{M}_i/\mathcal{M}_{i-1}$ — free $\mathcal{B}$-module of rank 1; $\qquad \mathcal{M}_0 = \mathcal{B}\mathbf{1}$.

h)

$$\forall b \in \mathcal{B},\ m \in \mathcal{M}: \quad \nabla(bm) = (\partial b)m + b\nabla m,$$
$$\forall \varphi \in \mathcal{O}, \qquad (\nabla m)\varphi = \nabla(m\varphi)$$



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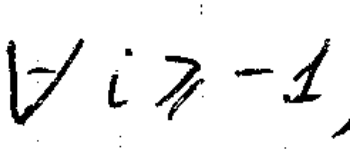
Manuscript excerpt. The letter φ in $(\nabla m)\varphi$ is retraced.

GELFAND-P000683

("nabla" is the name of an old musical instrument of the lyre type).

i)

$$\forall i \geq -1, \qquad \nabla\mathcal{M}_i \subset \mathcal{M}_{i+1},$$
$$\nabla : \mathcal{M}_i/\mathcal{M}_{i-1} \xrightarrow{\sim} \mathcal{M}_{i+1}/\mathcal{M}_i$$



Manuscript excerpt. The start of the line reads $i \geq -1$.

Original source details: GELFAND-P000683

② Construction of the bimodule from a pair $L, P \in \mathcal{B}[\partial]$ with $[L,P]=0$ and $\bar{k}[L,P]$ without zero divisors.

Set

$$\mathcal{M} = \mathcal{B}[\partial]; \qquad \mathcal{O} = k[L,P].$$

— ring of polys. in L and P.

action of $\mathcal{B}$ — mult. on the left, action of $\mathcal{O}$ — mult. on the right

1 = identity operator

$$\mathcal{M}_i = \left\{ \sum_{k \le i} b_k \partial^k \mid b_k \in \mathcal{B} \right\}. \qquad \nabla m = \partial \circ m$$

The axioms are verified trivially.

Theorem 1. If there is a $(\mathcal{B},\mathcal{O})$-bimodule $\mathcal{M}$, then it gives an embedding $i : \mathcal{O} \longrightarrow \mathcal{B}[\partial]$, such that the reverse construction from i gives $\mathcal{M}$.

Construction.

a)

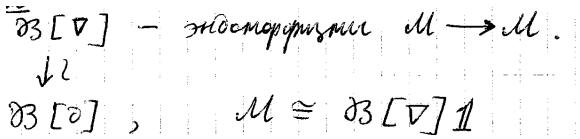


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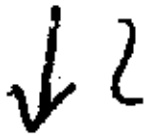
Diagram and typed description.

$\mathcal{B}[\nabla]$ — endomorphisms $\mathcal{M} \longrightarrow \mathcal{M}$.

$\downarrow$

$\mathcal{B}[\partial]$ $\quad \mathcal{M} \cong \mathcal{B}[\nabla]\mathbf{1}$

Manuscript excerpt. A separate vertical wavy stroke stands to the right of the down arrow.



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This step did not use multiplication by $\mathcal{O}$.

b)

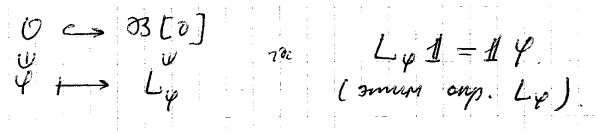

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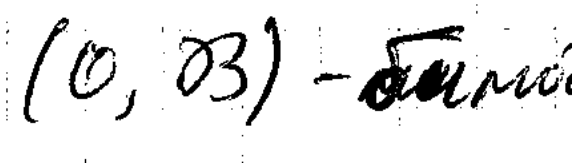
Diagram and typed description.

$$\begin{array}{ccc} \mathcal{O} & \hookrightarrow & \mathcal{B}[\partial] \\ \Psi & & \Psi \\ \varphi & \longmapsto & L_\varphi \end{array}$$

such that $L_\varphi \mathbf{1} = \mathbf{1}\varphi$.
(this determines L_φ).

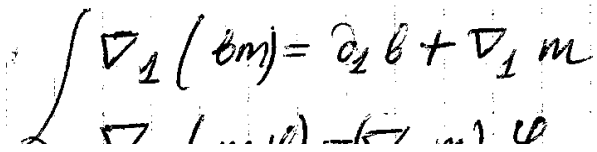
Theorem 2. Let there be an $(\mathcal{O},\mathcal{B})$-bimodule $\mathcal{M}$ and a deriv. $\partial_1 : \mathcal{B} \longrightarrow \mathcal{B}$ (in the anal. setting $\partial/\partial t$), and suppose there is also $\nabla_1 : \mathcal{M} \longrightarrow \mathcal{M}$



Original source details: GELFAND-P000683

Manuscript excerpt. Theorem 2 writes $\mathcal{O}$ first and then $\mathcal{B}$.

$$\begin{cases} \nabla_1(bm) = \partial_1 b + \nabla_1 m \\ \nabla_1(m\varphi) = (\nabla_1 m)\varphi \\ [\nabla_1, \nabla] = 0 \end{cases}$$



Manuscript excerpt. The right-hand side shows $\partial_1 b$ and $\nabla_1 m$, with a plus sign between them.

Original source details: GELFAND-P000683

$[\mathcal{M}, \nabla_1] \longrightarrow$ family of solns. of nonstationary Lax eqs.

$$\nabla_1 \mathbf{1} = P\mathbf{1}, \qquad P \in B[\partial].$$
$$\mathbf{1}\varphi = L_\varphi \mathbf{1}$$

Then

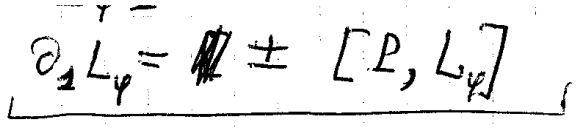


Diagram and typed description.
$\partial_1 L_\varphi =$ [crossed out] $\pm [P, L_\varphi]$

Original source details: GELFAND-P000683

Undeciphered passage. A sign is crossed out between the equality sign and $\pm$.

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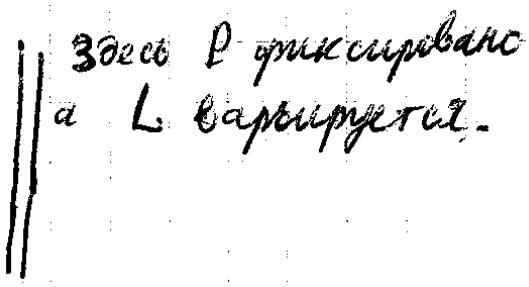


Diagram and typed description.
Here P is fixed
and L varies.

Original source details: GELFAND-P000683

GELFAND-P000684

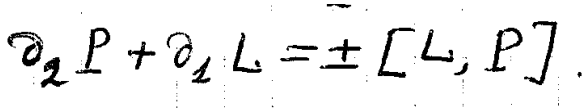
Theorem 3. A module is given $\mathcal{M}$ with no mention of $\mathcal{O}$.

Manuscript excerpt. The wavy underline under “with no mention of” ends before $\mathcal{O}$.

Original source details: GELFAND-P000684

$\nabla, \nabla_1, \nabla_2$ commute pairwise. Then we obtain solns. of the Zakh.–Shabat eqs.:

$$\nabla_1 \mathbf{1} = P\mathbf{1}$$
$$\nabla_2 \mathbf{1} = L\mathbf{1}$$
$$\partial_2 P + \partial_1 L = \pm[L, P].$$



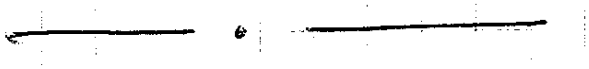
Original source details: GELFAND-P000684

Manuscript excerpt. There is a plus sign between the two derivatives and a $\pm$ before the commutator.

Next time Duistermaat.

Original source details: GELFAND-P000684

Manuscript excerpt. The underlined surname is written “Дюйстермаат” (Duistermaat).



Original source details: GELFAND-P000684

Diagram and typed description.

There is a small hook at the left end of the shorter rule. A dot separates the two rules.

GELFAND-P000685

21/II-77 Gelfand Sem.

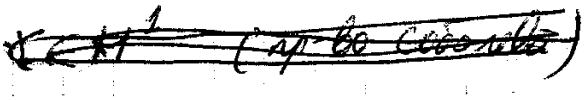
Duistermaat.

Original source details: GELFAND-P000685

Manuscript excerpt. The surname is written “Дейстермаат” here.

On the Morse index.

$I(\gamma)$ — functional, γ — curve in $\mathbb{R}^n$, ~~$\gamma \in H^1$ (Sobolev sp.)~~



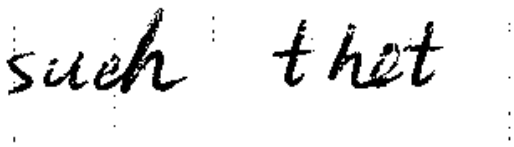
Original source details: GELFAND-P000685

Manuscript excerpt. The line crosses out $\gamma \in H^1$ and the parenthetical gloss.

$$I(\gamma) = \int_0^T f(\gamma(t), \dot{\gamma}(t))\, dt$$

Variational problem.

$$\gamma \in \text{space of curves such thet } (\gamma(0), \gamma(T)) \in R$$



Original source details: GELFAND-P000685

Manuscript excerpt. The line is in English in the manuscript. The word after “such” is “thet”.

R — subset of $\mathbb{R}^n \times \mathbb{R}^n$.

Examples; a) $R = (0,0) \in \mathbb{R}^n \times \mathbb{R}^n$ — fixed endpoints

b) $R = \{(x,x)\}$ — $\gamma(0) = \gamma(T)$ — periodic condition

Uncertain reading. The thick horizontal mark between the formulas is rendered as a dash.

Original source details: GELFAND-P000685

Extremal — critical point in H^1.

$$\text{If } \frac{\partial^2 f}{\partial \dot\gamma^2} > 0, \qquad \text{then ind } \frac{\partial^2 I}{\partial \gamma^2} < \infty$$

(positive def.) (finitely many neg. squares)

GELFAND-P000686

$$\text{ind } \frac{\partial^2 I}{\partial \gamma^2} = \text{Morse index for the boundary cond.} = \text{ind}_R(\gamma).$$

There will be a formula for the Morse index in terms of the Arnold index of a curve in the Lagr. set.

Theorem.

$$\text{Morse index}_R(\gamma) = \text{ind}(\varphi) + i(I, \rho, \varphi(T)) + \dim \pi(\rho \cap I) - 2n.$$

Original source details: GELFAND-P000686

Manuscript excerpt. The right-hand side reads $i(I, \rho, \varphi(T))$ and $\dim \pi(\rho \cap I) - 2n$.

Eq. for stat. curves (Euler–Lagrange) $\underset{\text{Legendre tr.}}{\longrightarrow}$ Hamilton eq.

$$(\gamma, \dot\gamma) \underset{\text{Legendre transform}}{\mapsto}$$

The Hamilton eq. determines a flow of sympl. transformations.

Consider the eq. in variations along the extremal. We obtain a 1-param. family of lin. symplectic transformations.

$\Phi(t) =$ derivative of the Hamiltonian flow along the extremal curve

this is a lin. transformation:

$$\underset{\text{phase space}}{\mathbb{R}^{2n}_\sigma} \longrightarrow \mathbb{R}^{2n}_\sigma$$

Uncertain reading. The first $\mathbb{R}^{2n}$ has a heavily retraced subscript, tentatively read as σ.

Original source details: GELFAND-P000686

Original source details: GELFAND-P000686

Uncertain reading. The second $\mathbb{R}^{2n}$ also has a heavily retraced subscript, tentatively read as σ.

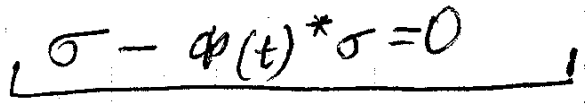
$\varphi(t) = \text{graph}\,\varphi(t) \subset \underline{\mathbb{R}^{2n}_{\sigma} \times \mathbb{R}^{2n}_{-\sigma}}$ (sympl. structure — difference)

Original source details: GELFAND-P000686

Manuscript excerpt. After "graph" there is a lowercase φ.

$$\Phi(t)^*\sigma = \sigma. \qquad t \text{ fixed.}$$

$$\Updownarrow$$



Original source details: GELFAND-P000686

Diagram and typed description.

$\underbrace{\sigma - \varphi(t)^*\sigma = 0}$

The formula has a ruled bracket below. The letter before (t) is transcribed as lowercase φ.

or

$$\underline{\sigma \oplus (-\sigma) = 0 \text{ on } \varphi(t)}$$

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Undeciphered passage. The passage before the parenthesis is heavily crossed out.

(Lagrangian subspace)

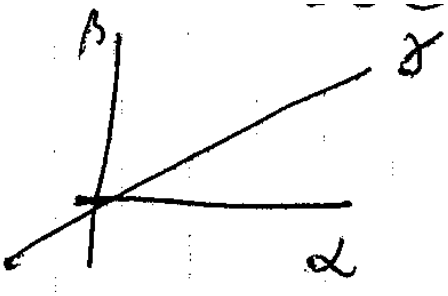

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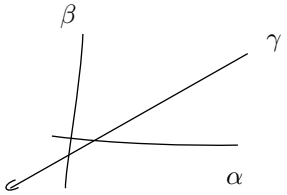
Diagram and typed description.



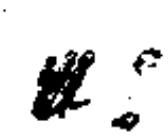
Three strokes labelled α, β, γ meet near a single point.

α, β, γ — 3 Lagr. planes

$$\alpha \cap \beta = 0$$
$$\gamma \cap \beta = 0$$

Then we can take

$$\gamma = \text{graph}\,\Gamma, \qquad \Gamma \text{ linear } : \alpha \to \beta$$



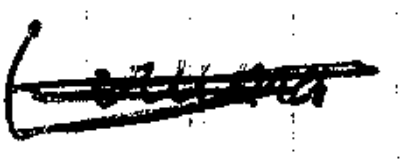
Undeciphered passage. A mark after “linear” and before the colon is overwritten.

Original source details: GELFAND-P000686

$$Q(u, v) = \sigma(\Gamma u, v), \qquad u, v \in \alpha$$

We obtain a bilin. form on α

$$\sigma = 0 \text{ on } \Gamma \quad \Leftrightarrow \quad Q \text{ is symmetric.}$$



Undeciphered passage. The passage before $Q = Q(\alpha, \beta, \gamma)$ is heavily crossed out.

Original source details: GELFAND-P000686

$$Q = Q(\alpha, \beta, \gamma)$$

If $\gamma = \gamma(t)$ (curve of Lagr. planes), then at some point γ may intersect α in a 1-dim. subspace.

At the moment of intersection take the difference of indices (it is independent of the choice of β).

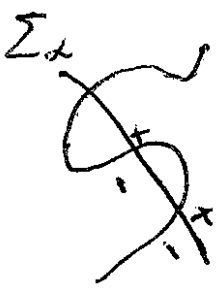
$[\gamma : \alpha]$ — intersection index of the curve $\gamma(t)$ with the hypersurf.

$$\Sigma_\alpha = \{\gamma \mid \gamma \cap \alpha \neq 0\}.$$

Manuscript excerpt. The abbreviation “гиперповерхн.” (hypersurface) is retraced in places.

Original source details: GELFAND-P000686

GELFAND-P000687



Original source details: GELFAND-P000687

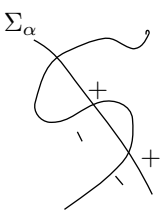


Diagram and typed description. The curved line crosses Σ_α three times. Two crossings have a plus sign and a short stroke each. A small loop marks the upper end of the curve.

$\Rightarrow$ this is a homotopy inv. with fixed endpoints.

Σ_α is embedded two-sidedly

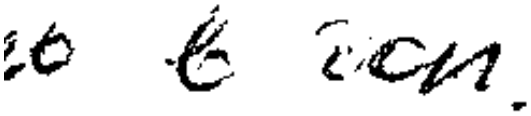
Undeciphered passage. A passage after an opening parenthesis is crossed out.

Original source details: GELFAND-P000687

Hence $\pi_1(\text{Lagr. planes}) = \mathbb{Z}$

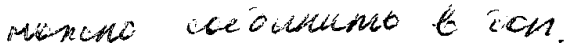
$$\text{since } (\text{Lagr. planes}) = \Sigma_\alpha \cup (\text{transverse planes})$$
$$\| $$
$$(\text{quadratic forms}) \quad \text{— homot. trivial}$$

Initial and final points outside Σ_α, can be joined in the compl. and the homot. type is independent of any choice.



Manuscript excerpt. The line ends “в доп.” (in the compl.).

Original source details: GELFAND-P000687



Manuscript excerpt. The words before “в доп.” read “можно соединить” (can be joined).

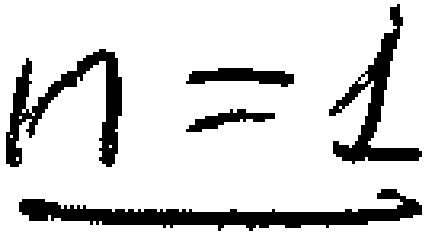
Original source details: GELFAND-P000687

For an open curve:

$$\text{ind}(\gamma) = [\gamma : \alpha] + \text{ind}\, Q\{\gamma(T), \alpha, \gamma(0)\}$$

(to make it independent of α).

Diagram and typed description. $\xrightarrow{n=1}$



Original source details: GELFAND-P000687

Lagr. planes form a circle

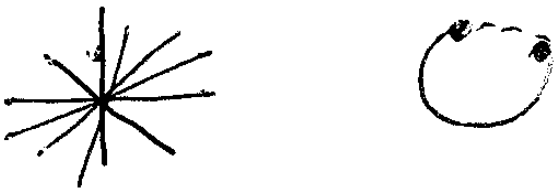
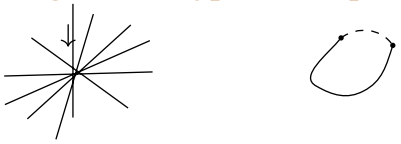


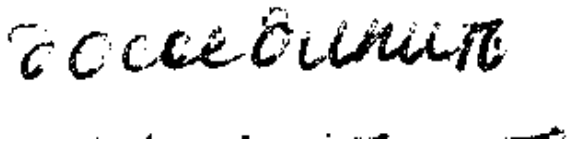
Diagram and typed description.



The pencil of lines has a downward arrow. The upper arc of the circle is dashed. Thick marks appear at its ends.

Original source details: GELFAND-P000687

[To the right of the sketches:] join up the open curve clockwise, from the initial point to the final one.



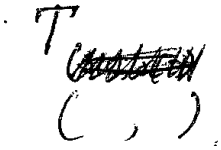
Manuscript excerpt. The verb reads “досоединить” (join up), with the prefix “до”.

Original source details: GELFAND-P000687

$\mathrm{ind}(\gamma)$ — resulting number of turns.

R — set of bdy. conditions

$$R^{\perp} = \text{normal bundle}$$
$$= \left\{ \underset{\mathbb{R}^{2n}}{(x,\xi)}, \underset{\mathbb{R}^{2n}}{(y,-\eta)} \;\middle|\; (\xi,\eta) \perp T_{(x,y)}R \right\}.$$
$$\rho = T_{(\ ,\ )}(R^{\perp})$$



Undeciphered passage. The first subscript at T is crossed out. Parentheses with a comma and empty arguments appear below.

Original source details: GELFAND-P000687

— fibre of the normal bundle over a pt. of the extremal

Uncertain reading. The last word reads “экстремали” (of the extremal). Some strokes may cross it out.

Original source details: GELFAND-P000687

this is a Lagrangian plane

$$i(I,\rho,\varphi(T)) = (\mathrm{ind} + \mathrm{null})Q(I,\rho,\varphi(T))$$

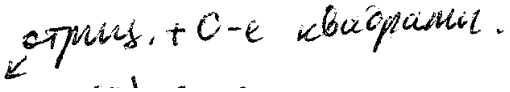


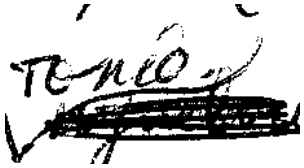
Diagram and typed description.
neg. $+0$ squares.

Original source details: GELFAND-P000687

I — graph of the [inserted above: ident.] map.

Undeciphered passage. The word between “graph” and “map” is crossed out.

Original source details: GELFAND-P000687



Manuscript excerpt. The insertion above the crossing-out reads “тожд.” (ident.).

Original source details: GELFAND-P000687

π — projection to the base.

Selberg trace formula. in the simplest case:

G — semisimple connected real Lie group, K — max compact subgrp., $K\backslash G$

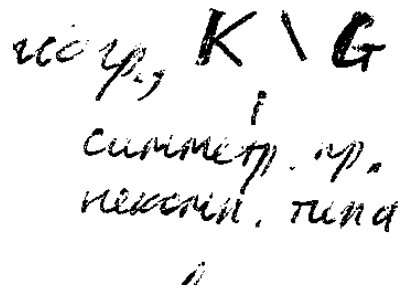


Diagram and typed description.

$K\backslash G$
symm. sp.
of noncompact type

Original source details: GELFAND-P000687

Γ — discrete subgrp. of G, acting on $M = K\backslash G$ on the right.

$M/\Gamma = X$ — compact set.

We will consider all ops. $C^\infty(M) \to C^\infty(M)$, commuting with the action of G.

$$f \longmapsto \varphi * f, \qquad (\varphi * f)(x) = \int_G \varphi(xy^{-1}) f(y)\, dy$$

$$\varphi \in \mathcal{E}'(K\backslash G)$$

GELFAND-P000688

f invar. $\Rightarrow$ we may take $\varphi \in \mathcal{E}'(K\backslash G/K)$ (two-sided invar.) — spherical fns.

The operator on X has a Schwartz kernel.

For φ supported at $e \in G$ we obtain a diff. operator.

The formula is for the spectrum of a diff. op. on M, descended to X.

Set

$$K_\varphi(x, y) = \sum_{\gamma \in \Gamma} \varphi(x\gamma y^{-1})$$

It follows from invariance that $K_\varphi \in \mathcal{E}'(X \times X)$.

Now consider an op. with smooth φ (compactly supported).

Γ — discr. $\Rightarrow$ [overwritten] the sum for fixed x, y is finite.

Undeciphered passage. A mark before “the sum” is heavily overwritten.

Original source details: GELFAND-P000688

$$\varphi \in C_c^\infty(K\backslash G/K) \quad \Rightarrow \quad K_\varphi \in C^\infty(X \times X).$$

We have:

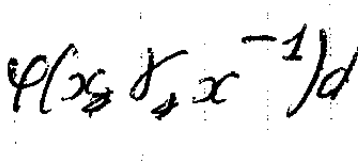
$$K_\varphi \circ K_\psi = K_{\varphi * \psi}$$
$$\varphi * \psi = \psi * \varphi \qquad \text{on spherical fns.}$$

$\Rightarrow$ we obtain a commut. algebra of ops. K_φ with involution. They are all compact $\Rightarrow$
$\Rightarrow$ $\exists$ orthon. basis of common eigenvectors $e_j \in L^2(X)$

$$K_\varphi(e_j) = \lambda_j(\varphi) e_j, \qquad \forall \varphi \in C_c^\infty(K\backslash G/K)$$

We have:

$$\sum_j \lambda_j(\varphi) = \operatorname{Sp} K_\varphi = \int_X K_\varphi(x,x)\,dx$$
$$= \sum_{\gamma \in \Gamma} \int_X \varphi\big(x_? \gamma_? x^{-1}\big)\,dx$$



Uncertain reading. Two similar small marks appear below the first x and below γ. Their readings remain uncertain.

Original source details: GELFAND-P000688

$$\big(C_c^\infty(K\backslash G/K), *\big) \xrightarrow{\sim} C_c^\infty(A)^W, \qquad W \text{ — Weyl group}$$

Uncertain reading. The delimiter after the final K and before the comma is heavily retraced.

Original source details: GELFAND-P000688

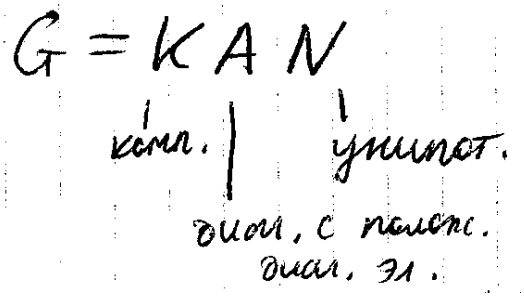


Diagram and typed description.

$G =$ K (compact) A (diag. with positive diag. entries) N (unipot.)

— Iwasawa decomposition.

Original source details: GELFAND-P000688

Manuscript excerpt. The manuscript spells the surname “Ивасавы”.

Original source details: GELFAND-P000688

$$f(a) = e^{\langle \rho, a \rangle} \int_N \varphi(an)\, dn.$$

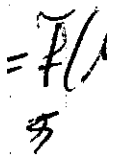
(horospherical map $\varphi \mapsto f$).

We have:

$$\lambda_j(\varphi * \psi) = \lambda_j(\varphi) \cdot \lambda_j(\psi).$$

Hence

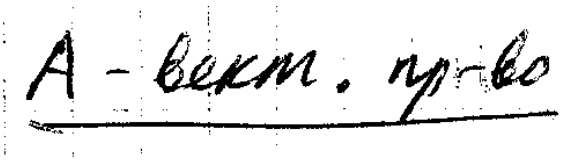
$$\lambda_j(\varphi) = \langle f, e^{i\mu_j} \rangle = \widetilde{f}_?(\mu_j), \qquad \mu_j \in A^*_{\mathbb{C}}.$$



Original source details: GELFAND-P000688

Uncertain reading. There is a separate undeciphered mark below $\widetilde{f}$.

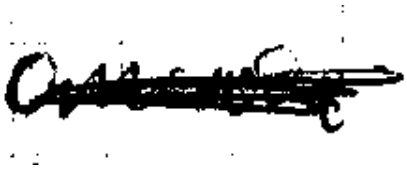
[Right margin:] A — vector sp.



Original source details: GELFAND-P000688

Manuscript excerpt. The note about A stands in the right margin and is underlined.

Here $\rho = \frac{1}{2} \sum$ pos. roots



Original source details: GELFAND-P000688

Undeciphered passage. A short word above "We have" is crossed out.

We have:

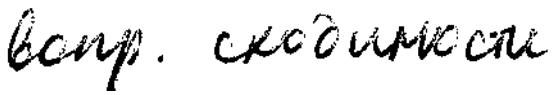
$$\sum_j \lambda_j(\varphi) = \sum_{\gamma \in \Gamma} \langle T_\gamma, f \rangle,$$

(we rewrote each summand)

T_γ — locally finite family of distributions.

GELFAND-P000689

All the brackets encountered mysteriously disappear



Original source details: GELFAND-P000689

Uncertain reading. The noun tentatively reads "скобочки" (brackets).

$$\sum_j \widetilde{f}(\mu_j) = \sum_{\gamma\in\Gamma} \langle T_\gamma, f\rangle, \qquad f \in C_c^\infty(A)^W$$

where $T_\gamma \in \mathcal{D}'(A)^W$

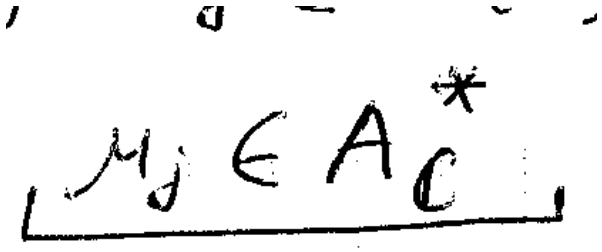


Diagram and typed description. $\mu_j \in A_{\mathbb{C}}^*$

Original source details: GELFAND-P000689

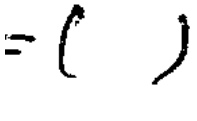
— a countable set of points called the spectrum [crossed out] Spec.

Undeciphered passage. Short marks before "Spec" are heavily crossed out.

Original source details: GELFAND-P000689

Next, information about μ_j from T_γ, given by explicit integrals.

$$T_\gamma(f) = \int_X \varphi(x\gamma x^{-1})\,dx, \qquad f(a) = (\ ) \int \varphi(an)\,dn$$



Uncertain reading. There are empty parentheses before the integral in the formula for $f(a)$.

Original source details: GELFAND-P000689

$$T_e(f) = \varphi(e) \cdot \mathrm{vol}(X)$$

Undeciphered passage. A short passage before the surname is crossed out.

Original source details: GELFAND-P000689

Harish-Chandra

$$T_e = \mathrm{vol}(X) \cdot \widetilde{\beta}, \qquad \beta \text{ — Plancherel measure.}$$

$\ell(\gamma)$ — length of the closed geodesic corresponding to a class in the fund. group.

Theorem.

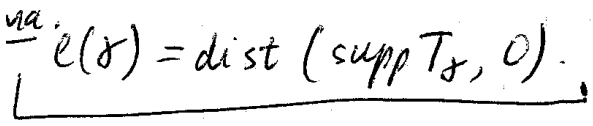


Diagram and typed description. $\ell(\gamma) = \mathrm{dist}(\mathrm{supp}\, T_\gamma, 0)$.

Original source details: GELFAND-P000689

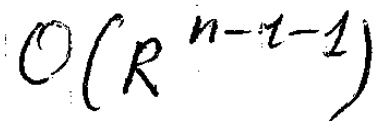
Thus,

$$\sum_j \widetilde{f}(\mu_j) = \langle \widetilde{f}, \beta \rangle \operatorname{vol}(X) + \sum_{\gamma \neq e} \langle T_\gamma, f \rangle.$$

Diagram and typed description.

Original source details: GELFAND-P000689

$$\#\{j : |\mu_j| \leq R\} = \operatorname{vol}(X) \int_{\text{ball}(R)} \beta + O(R^{n-r-1})$$



Manuscript excerpt. The middle sign in the exponent reads r.

Original source details: GELFAND-P000689

GELFAND-P000690

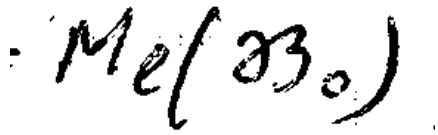
28/II-77 Gelfand Sem.

Manin (continued).

$k = \mathbb{R}$ or $\mathbb{C}$,

$\mathcal{B}_0$ — alg. of germs of k-merom. functions of x or x, t or x, t, y

$$\mathcal{B} = M_\ell(\mathcal{B}_0), \qquad \partial_x, \partial_y, \partial_t : \mathcal{B} \to \mathcal{B}$$



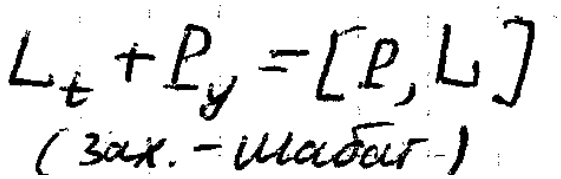
Manuscript excerpt. The subscript of M in the matrix algebra reads ℓ.

Original source details: GELFAND-P000690

$$\mathcal{B}[\partial_x]$$

Aim: find pairs $L, P \in \mathcal{B}[\partial_x]$, satisfying one of the eqs.

$[L, P] = 0$	or	$L_t = [P, L]$	or	$L_t + P_y = [P, L]$
(stationary Lax)		(nonstationary Lax)		(Zakh.–Shabat)



Manuscript excerpt. The third equation writes the commutator in the order $[P, L]$.

Original source details: GELFAND-P000690

$\mathcal{O}$ — ring of functions on an affine k-curve. Suppose we have constructed a k-space M, $(\mathcal{B}, \mathcal{O})$-bimodule,

$$0 \subset M_0 \subset M_1 \subset \cdots \quad \text{— filtration of } \mathcal{B}\text{-submodules}$$

$$\mathbf{1} \in M, \qquad \nabla : M \to M$$

Undeciphered passage. The formula below $\nabla : M \to M$ is crossed out. Parentheses and bm are visible.

Original source details: GELFAND-P000690

$$\nabla(bm) = (\partial_x b) \cdot m + b \cdot (\nabla m)$$

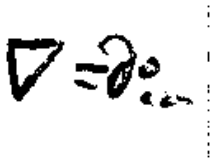
Further axioms [crossed out] see above.

Undeciphered passage. A short word before "see" is crossed out.

Original source details: GELFAND-P000690

Example (basic)

$$\mathcal{O} = k[L, P], \quad M = \mathcal{B}[\partial_x], \quad \mathbf{1} = \mathrm{id}, \quad \nabla = \partial\,[?]$$



Uncertain reading. There is a loop to the right of ∂ and several strokes below. The lower mark cannot be confirmed as x.

Original source details: GELFAND-P000690

L, P — solve the stationary problem.

~~Definition.~~

Uncertain reading. The crossed-out heading tentatively reads "Определение" (Definition).

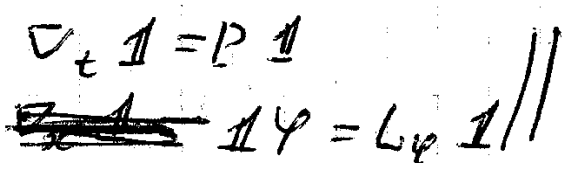
Original source details: GELFAND-P000690

Lemma. M — $\mathcal{B}[\partial_x]$-free module of rank 1 with gen. $\mathbf{1}$
(∂_x acts through ∇)
I. stationary Lax:

$$\mathcal{O} \ni \varphi \mapsto L_\varphi \in \mathcal{B}[\partial_x], \quad \mathbf{1}\varphi = L_\varphi \mathbf{1} \quad \Rightarrow \quad [L_\varphi, L_\psi] = 0$$

II. nonstationary Lax:

$$\nabla_t : M \to M, \qquad [\nabla_x, \nabla_t] = 0,$$



Original source details: GELFAND-P000690

Diagram and typed description.

$$\nabla_t \mathbf{1} = P\mathbf{1}$$

[crossed out] $\mathbf{1}\varphi = L_\varphi \mathbf{1}$ //

Original source details: GELFAND-P000690

Undeciphered passage. The short formula to the left of $\mathbf{1}\varphi = L_\varphi \mathbf{1}$ is crossed out.

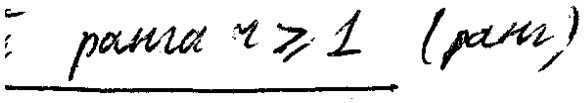
$$\Rightarrow L_t = [P, L]$$

GELFAND-P000691

III. $\nabla_y : M \to M$

$$\nabla_t, \nabla_y, \nabla_x \quad \Rightarrow \quad \nabla_t \mathbf{1} = P\mathbf{1}, \qquad \nabla_y \mathbf{1} = L\mathbf{1}$$

Explicit construction of a class of bimodules of rank $r \geq 1$ (rank)



Original source details: GELFAND-P000691

Manuscript excerpt. “(rank)” follows the underlined $r \geq 1$.

$r = 1 \Rightarrow N$-soliton solns. of KdV and higher KdV.
Fix $N \geq 1$ (number of solitons).

$$\alpha_1, \ldots, \alpha_N \in k, \qquad \alpha_i \neq \alpha_j$$
$$\beta_1, \ldots, \beta_N$$

$\alpha_i = \overline{\beta_i}$ when $k =$ [crossed out] $\mathbb{R}$. (meaning — the curve is real)

Original source details: GELFAND-P000691

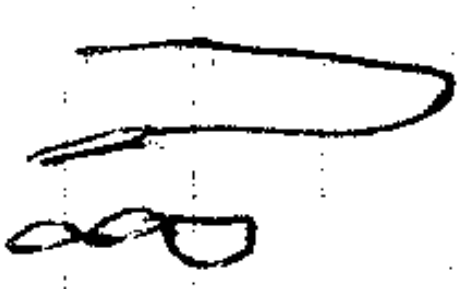
Undeciphered passage. A mark between $k =$ and $\mathbb{R}$ is crossed out.

Original source details: GELFAND-P000691

Manuscript excerpt. The first word in the parentheses reads “смысл” (meaning).

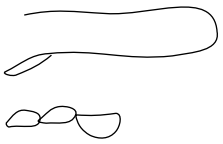
$$\mathcal{O} = \{f \in k[\lambda] \mid f(\alpha_i) = f(\beta_i), \quad \forall i = 1, \dots, N\}.$$

The module consists of sections of a bundle, trivial when lifted.



Original source details: GELFAND-P000691

Diagram and typed description.



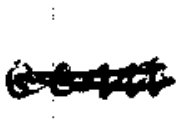
Two unlabelled sketches. The long upper line is open on the left. The lower line forms three loops.

Set

$$N = \mathcal{B}[\lambda]^r \quad \text{and} \quad N \supset M$$

All structures on M are induced by those on N. Introduce on N

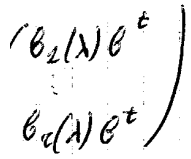
$$\mathcal{B} \times N \to N, \quad \text{[crossed out]} \quad N \times k[\lambda] \to N$$



Original source details: GELFAND-P000691

Undeciphered passage. A short word between the two actions is crossed out.

$$b \begin{pmatrix} b_1(\lambda) \\ \vdots \\ b_r(\lambda) \end{pmatrix} = \begin{pmatrix} b_1(\lambda)b^t \\ \vdots \\ b_r(\lambda)b^t \end{pmatrix}$$



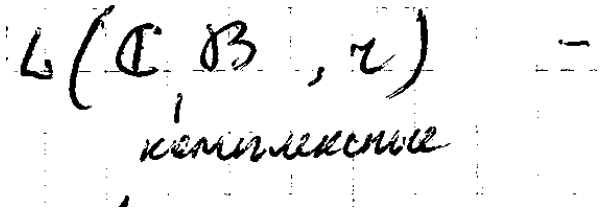
Original source details: GELFAND-P000691

Manuscript excerpt. Both visible components of the right-hand column have b^t.

$$\mathrm{GL}(\mathcal{B}[\lambda], r) \times N \to N$$

$$M = \{m(\lambda) \in N \mid m(\alpha_i) = g_i m(\beta_i), \quad \forall i = 1, \dots, N\}$$

(nontrivial gluing cond., as distinct from functions)



Original source details: GELFAND-P000691

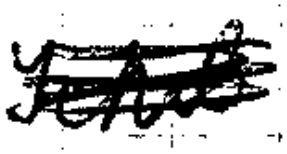
Diagram and typed description.

$g_i \in \mathrm{GL}(\mathbb{C}\,\mathcal{B}, r)$

complex

There is no clear operation sign between $\mathbb{C}$ and $\mathcal{B}$.

— parameters (gluing matrices, giving rise to nontrivial variants).



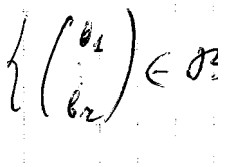
Undeciphered passage. A short passage before the conjugation condition is crossed out.

Original source details: GELFAND-P000691

$$\overline{g_i} = g_i^{-1} \quad \text{when } k = \mathbb{R}$$

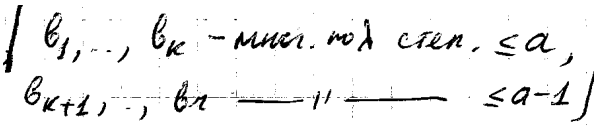
Filtration on N:

$$N_{\substack{ar+k \\ 0\le k\le r-1}} = \left\{ \begin{pmatrix} b_1 \\ \vdots \\ b_r \end{pmatrix} \in \mathcal{B}[\lambda]^r \;\middle|\; \begin{array}{r} b_1, \dots, b_k \text{ — polys. in } \lambda \text{ of deg. } \le a, \\ b_{k+1}, \dots, b_r \text{ — " — } \le a-1 \end{array} \right\}$$



Manuscript excerpt. The column in the braces starts with b_1 and ends with b_r.

Original source details: GELFAND-P000691



Manuscript excerpt. The second line uses two rules and two short slanted strokes as a ditto mark.

Original source details: GELFAND-P000691

$$\mathcal{B}N_i \subset N_i,$$

M will start at the wrong place (there is a system of lin. eqs.).

$$M_{(i)} = M \cap N_i \quad \text{— still not what we want}$$

Lemma. Consider the matrix (block)

$$(\alpha_i^j \operatorname{Id}_r - \beta_i^j g_i)_{\substack{i=1,\dots,N \\ j=0,\dots,N-1}}$$

Suppose

$$\det(\alpha_i^j \operatorname{Id}_r - \beta_i^j g_i) \neq 0.$$

GELFAND-P000692

Then $M_{(rN)} = \{0\}$. For every $i > rN$ there is a canonical inclusion

$$M_{(i)}/M_{(i-1)} \xrightarrow{\sim} N_i/N_{i-1}$$

which is an isomorphism.

Def.: $M_i = M_{(i-rN)}$.

Manuscript excerpt. The index in parentheses reads $i - rN$.

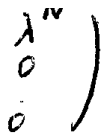
Original source details: GELFAND-P000692

Find the explicit form of **1** from a system of linear equations. We have:

$$\mathbf{1} = \sum_{j=0}^{N-1} m_j \lambda^j + \begin{pmatrix} \lambda^N \\ 0 \\ \cdot \\ 0 \end{pmatrix} \qquad \text{(by def. of } M_1\text{)}.$$

Undeciphered passage. A short mark below the equals sign is crossed out.

Original source details: GELFAND-P000692



Manuscript excerpt. A single dot is visible between the two zeros in the column.

Original source details: GELFAND-P000692

The coeff. of $\lambda^N = 1$ [crossed out] — normalization condition.

Undeciphered passage. A mark after the one is crossed out.

Original source details: GELFAND-P000692

m_j satisfy a system of linear equations

$$\sum_{j=0}^{N-1} (\alpha_i^j E_r - \beta_i^j g_i) m_i + (\alpha_i^N - \beta_i^N g_i) \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = 0,$$

$$i = 1, \ldots, N.$$

Manuscript excerpt. The first parenthesis is followed by m_i.

Original source details: GELFAND-P000692

This uniquely determines m_i.

Variant: $\nabla_x \stackrel{?}{=} \partial_x$. The gluing conditions fail.

Add an endomorphism of the module M.

$$\nabla_x = \partial_x + \underbrace{d_x(\lambda)}_{\text{matrix}},$$

where

$$d_x(\lambda) \in M_r(\mathcal{B}[\lambda]) = \mathrm{End}_{\mathcal{B}}(N = \mathcal{B}[\lambda]^r).$$

Undeciphered passage. A mark before M_r is crossed out.

Original source details: GELFAND-P000692

Lemma. $\nabla_x M \subset M$, if g_i satisfy the system of diff. eqs.

$$\partial_x g_i = g_i d_x(\beta_i) - d_x(\alpha_i) g_i, \qquad i = 1, \dots, N.$$

d_x — connection matrix. (include in the geom. structure in place of g_i).

d_x — arbitrary $\qquad$ g_i — to be found.

Simplest solutions for g_i:

Let $d_x(\lambda) \in M_r(M_\ell(k))[\lambda]$. Then (with fixed $\lambda = \mathrm{const}$)

$$g_i(x) = \exp(-x d_x(\alpha_i)) g_i(0) \exp(x d_x(\beta_i)).$$

Additional params.: the matrices $d_x(\lambda)$ with constant coefficients.

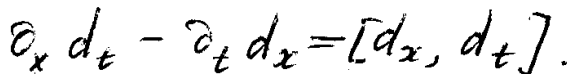
$r = 1 \Rightarrow$ ordinary exp.

Similarly $\rightarrow$ write equations for ∇_t.

$$\nabla_t = \partial_t + d_t(\lambda)$$
$$\nabla_y = \partial_y + d_y(\lambda).$$

Commutation ?

$$\partial_x d_t - \partial_t d_x = [d_x, d_t].$$



Manuscript excerpt. The commutator on the right has no minus sign.

Original source details: GELFAND-P000692

we need to be able to solve matrix equations. When $d_x = d_t =$ const we obtain

$$[d_x, d_t] = 0.$$

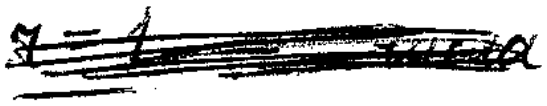
Manuscript excerpt. The last word on the line reads “получим” (we obtain).

Original source details: GELFAND-P000692

The simplest choice is to take $d_x \in k[d_t]$.

[crossed out: $r = 1$ ...]

Undeciphered passage. The rest of the line after $r = 1$ is obscured by the crossing-out.



Original source details: GELFAND-P000692

Undeciphered passage. A separate mark to the right of the line is crossed out.

Original source details: GELFAND-P000692

We obtain solns. of uncontrolled order. Reduce the orders where possible.

Consider the nonstationary Lax eq. for def.

Diagram and typed description.

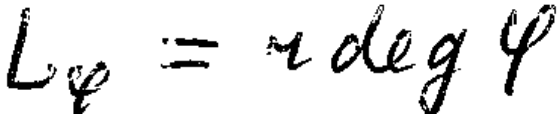
An upright flourish at lower right. No label is discernible.

Original source details: GELFAND-P000692

GELFAND-P000693

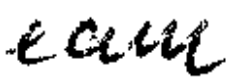
Lemma. ord $L_\varphi = r \deg \varphi$, if

Manuscript excerpt. The subscript of L and the symbol after deg are read as φ.



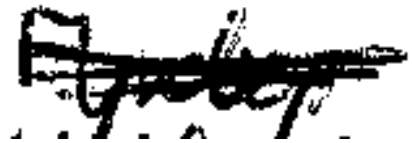
Original source details: GELFAND-P000693

Undeciphered passage. The line ends with “if.”



Original source details: GELFAND-P000693

[crossed out]



Original source details: GELFAND-P000693

Uncertain reading. The crossed-out word tentatively reads “Пример” (Example).

On the preceding discussion:

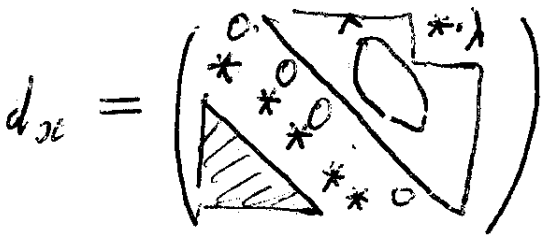
From

$$\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i$$

we obtain an add. condition on d_x:

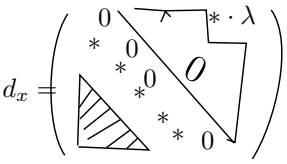
Original source details: GELFAND-P000693

Manuscript excerpt. The abbreviation before “condition” reads “доп.” (additional).



Original source details: GELFAND-P000693

Diagram and typed description.



Schematic with five asterisks below the diagonal, a hatched triangle and a stepped outline around the zero region.

$*$ — invertible elements.

Example:

$$\begin{pmatrix} 0 & \lambda \\ b & 0 \end{pmatrix}.$$

Original source details: GELFAND-P000693

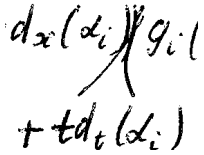
Uncertain reading. The lower left entry tentatively reads b. Another possible reading is β.

Remark. $\exists \psi$, $\deg \psi = 2$ in $\mathcal{O}$, if $\alpha_i + \beta_i$ is independent of i. (degenerate hyperelliptic curve)

$$\Rightarrow \quad \exists \psi : \operatorname{ord} L_\psi = 2r \qquad \text{(in particular one can take } 4.)$$

$$d_t = cE_2 + \dots$$

$$g_i(x) = \exp(-x d_x(\alpha_i) + t d_t(\alpha_i))\, g_i(0,0)$$
$$\exp(x d_x(\beta_i) + t d_t(\beta_i)).$$



Original source details: GELFAND-P000693

Manuscript excerpt. The term $+td_t(\alpha_i)$ is inserted from below. The left side reads $g_i(x)$.

Question: behavior of the N-soliton solution?

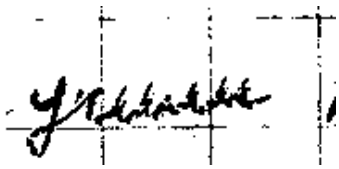
GELFAND-P000694

14/III-77. Gelfand Sem.

Gabrielov. “Combinatorial formulas for Pontryagin classes”

Gelfand:

MacPherson, Sullivan, Deligne learned from the work of G.–G.–L.



Original source details: GELFAND-P000694

Uncertain reading. The verb after “Deligne” tentatively reads “учились” (learned).

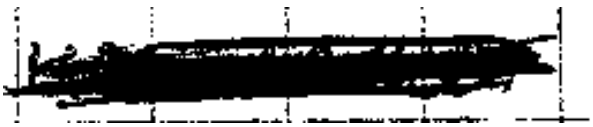
Original source details: GELFAND-P000694

Uncertain reading. The short word before “work” is overwritten. It tentatively reads “на,” with “по” another possible reading.

Elementary geometry of higher-dimensional polyhedra has yet to be developed.

An n-dimensional polyhedron is a complex (a collection of simplices with boundaries specified).

Each $(n-1)$-dimensional simplex [crossed out] — is the boundary of exactly two n-dimensional ones.

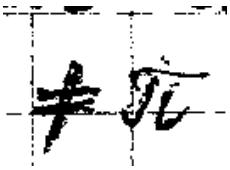


Original source details: GELFAND-P000694

Undeciphered passage. A short passage is obscured by dense horizontal strokes.

How do we assign a number to each vertex to compute the Euler characteristic?

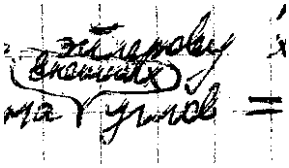
On going around a vertex, the sum of exterior angles is $\neq \pi$.



Original source details: GELFAND-P000694

Uncertain reading. Two transverse strokes and a slanting stroke with a loop appear before π. The sign tentatively reads $\neq$. The slanting stroke may be a cancellation.

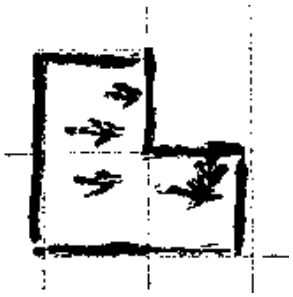
Sum of [inserted: exterior] angles $= \sum$ vertex defects.



Original source details: GELFAND-P000694

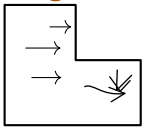
Manuscript excerpt. “Exterior” is circled between the lines, with a pointer before “angles.”

In three dimensions it is much more complicated. Going around a vertex of a square gives a turn of $90°$.



Original source details: GELFAND-P000694

Diagram and typed description.



Stepped outline with three rightward arrows at left. Strokes overlap in the lower right and their individual directions are uncertain.

$SO(3)$ is already noncommutative. The geometry of three-dimensional polyhedra is unclear — there is no substantive theory.

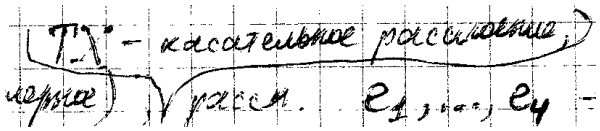
In two dimensions, commutativity reduces a contour to going around vertices.

For four-dimensional polyhedra there are two invariants (the Euler characteristic and one more). For smooth ones this has long been known.

MacPherson's exposition is in the Bourbaki Seminar (February 1977)

Gabrielov.

X — a manifold (four-dimensional). [above the line: TX — tangent bundle.]



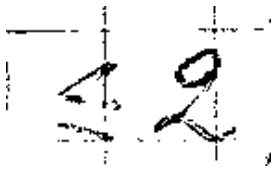
Original source details: GELFAND-P000694

Manuscript excerpt. The insertion is above the line inside a curved bracket pointing to the space before "consider."

consider $e_1, \dots, e_4$ — vector fields on X

$$Z = \{x : \mathrm{rk}(e_1, \dots, e_4) \le 2\}, \qquad \mathrm{Codim}\, Z = 4$$

for fields in general position.



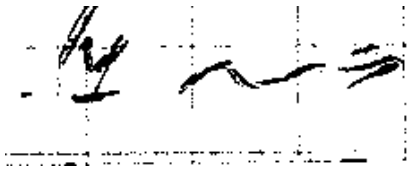
Original source details: GELFAND-P000694

Manuscript excerpt. The lower stroke of $\le$ is separate from the angle above it.

Z — a set of points.

A cooriented subset — an orientation of the normal bundle is specified

(orient. $Y \rightsquigarrow$ orient. X)



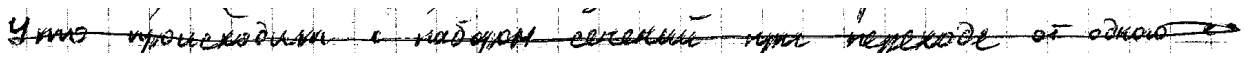
Original source details: GELFAND-P000694

Manuscript excerpt. Between the two orientations there is a wavy stroke and an arrowhead pointing right.

GELFAND-P000695

Let $L_{m,n} = \mathrm{Hom}(\mathbb{R}^m, \mathbb{R}^n)$. A point is specified by a tuple $(e_1, \dots, e_m)$, $e_j \in \mathbb{R}^n$. The set Z has a canonical coorientation (coorient. of the point x — orient. of T_xX). We obtain a set of points with signs $\pm$. The corresponding homology class is called the first Pontryagin class. (Number of plus signs) $-$ (number of minus signs) is independent of the choice of fields.

[crossed out with a rightward arrow: What happens to the collection of sections when passing from one]



Original source details: GELFAND-P000695

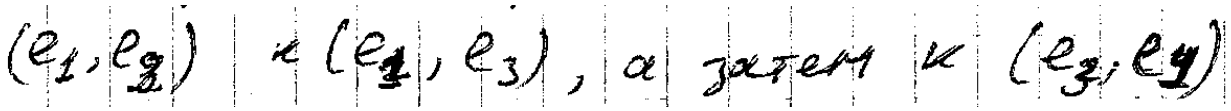
Manuscript excerpt. A horizontal stroke through the line ends in an arrowhead at right.

How does the cycle change when we replace the collection?

Let the bundle E be two-dimensional, $e_1(x), e_2(x)$ — sections.

$$Z = \{x \in X \mid e_1(x) = e_2(x) = 0\}$$

— a set of points. There is a canonical coorientation. This is the first Pontryagin class of the two-dimensional bundle. Pass from (e_1, e_2) to (e_1, e_3), and then to (e_3, e_4).



Original source details: GELFAND-P000695

Uncertain reading. The second subscript of the first pair and the first subscript of the second pair are overwritten. The last pair reads (e_3, e_4).

[crossed out: also?]

Uncertain reading. The short crossed-out word tentatively reads “ещё” (also).

Original source details: GELFAND-P000695

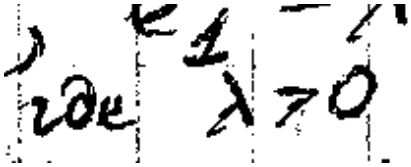
we can pass by elementary operations (changing one vector at a time). We have:

$$Z(e_2, e_3) - Z(e_1, e_3) = \partial Z^1(e_1, e_2; e_3), \tag{1}$$

where

$$Z^1 = \{x \mid e_3 = 0,\ e_1 = \lambda e_2\}, \qquad \mathrm{Codim}\, Z^1 = 3.$$

[inserted below $e_1 = \lambda e_2$: where $\lambda > 0$.]



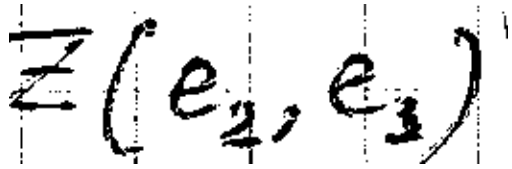
Manuscript excerpt. The inequality is inserted below the equation.

Original source details: GELFAND-P000695

The sign in the formula is easy to check. The boundary occurs at $\lambda = 0$ and $\lambda = \infty$. There is also $Z^1(e_1, e_3; e_2)$.

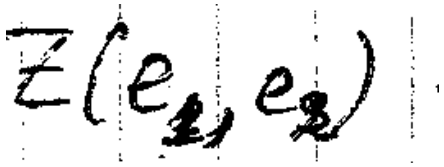
$$Z(e_1, e_2) - Z(e_2, e_3) = \partial Z^1(e_1, e_3; e_2) \tag{2}$$

$$Z(e_1, e_3) - Z(e_1, e_2) = \partial Z^1(e_2, e_3; e_1) \tag{3}$$



Original source details: GELFAND-P000695

Manuscript excerpt. The first subscript of the second term in (2) is read as 2.

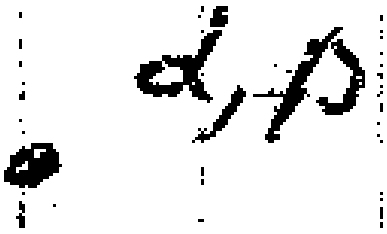


Original source details: GELFAND-P000695

Uncertain reading. Both subscripts of the second term in (3) are overwritten. The tentative reading is $1, 2$.

Add (1), (2) and (3). We obtain a closed curve. It is the boundary of the surface $e_1 = \lambda e_2 = \mu e_3,\ \lambda > 0,\ \mu > 0$.

0-dimensional hypersimplex



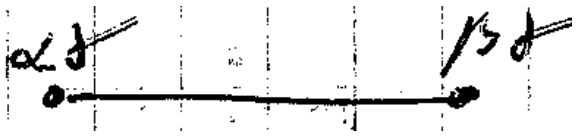
Original source details: GELFAND-P000695

Diagram and typed description.

α, β

(a point with two indices)

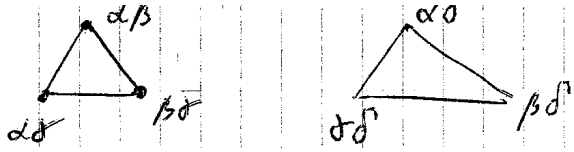
1-dimensional hypersimplex



Original source details: GELFAND-P000695

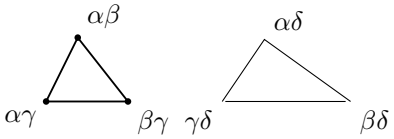
Diagram and typed description.

2 — $\|$ —



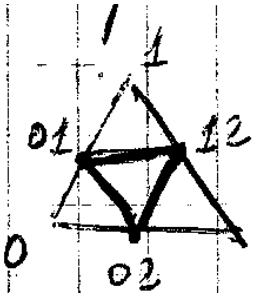
Original source details: GELFAND-P000695

Diagram and typed description.



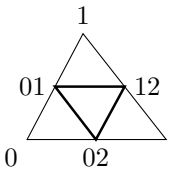
there are two types.

An n-dimensional hypersimplex is a polytope whose vertices are 0-dimensional hypersimplices, whose edges are 1-dimensional hypersimplices, and so on.

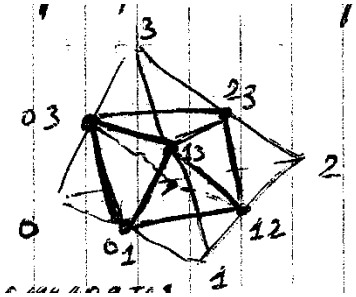


Original source details: GELFAND-P000695

Diagram and typed description.

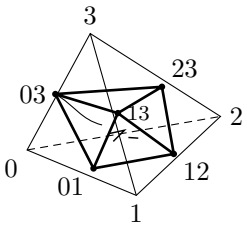


There is no discernible numeral at the lower right outer vertex.



Original source details: GELFAND-P000695

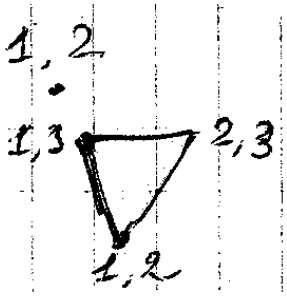
Diagram and typed description.



The heavy edges run through vertices labelled 03, 23, 13, 01, 12. A further mark and several overlapping strokes in the interior are unclear.

more complicated strata

[to the right of the drawings, beyond the two rules:] There are two types of triangles in the octahedron. Convex hull of the set of centres of the faces of the simplex. At each vertex are indices corresponding to the indices of the vertices of the corresponding face. One can also append the same index everywhere.

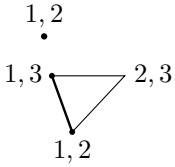


Original source details: GELFAND-P000695

Diagram and typed description.

$Z(e_1, e_2)$ — corresponds

$Z^1(e_1, e_2, e_3)$. corresponds

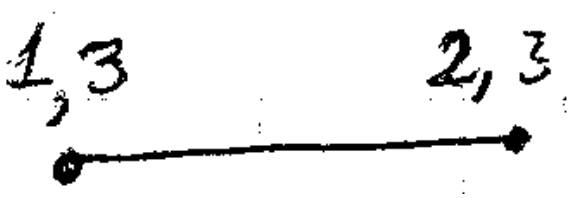


Problem: construct chains for each hypersimplex with the required properties.

GELFAND-P000696

A chain in the space of tuples of vectors in the plane. To the point 12 — in the space of pairs of vectors $e_1 = e_2 = 0$

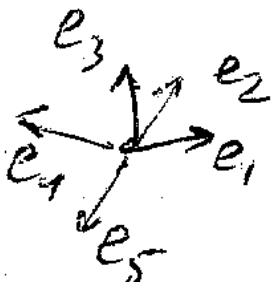
To the segment



Original source details: GELFAND-P000696

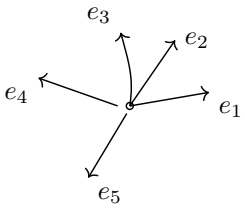
Diagram and typed description.

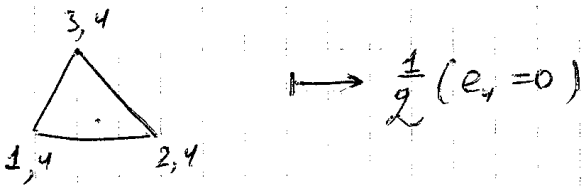
there corresponds $(e_1 = \lambda e_2,\ e_3 = 0)$. (3 — superfluous!)



Original source details: GELFAND-P000696

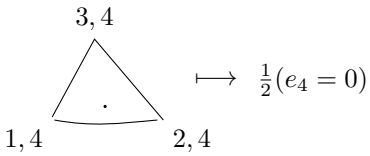
Diagram and typed description.

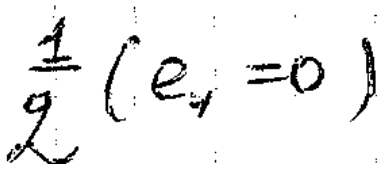




Original source details: GELFAND-P000696

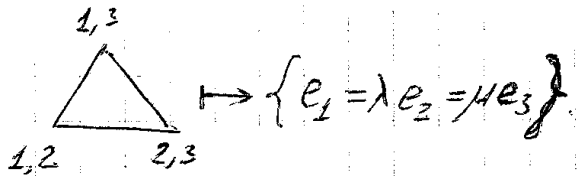
Diagram and typed description.





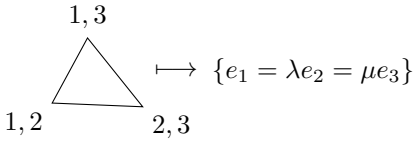
Original source details: GELFAND-P000696

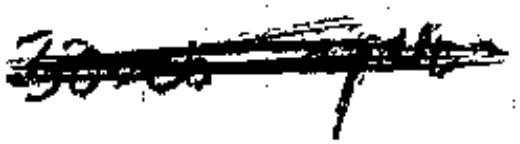
Manuscript excerpt. The coefficient reads $\frac{1}{2}$.



Original source details: GELFAND-P000696

Diagram and typed description.





Original source details: GELFAND-P000696

Undeciphered passage. Several strokes cross out a short passage under the left triangle.

Original source details: GELFAND-P000696

Diagram and typed description.

Two long strokes with a separate dot between them. The forms of their left ends are uncertain.

GELFAND-P000697

21/III — 77 Gelfand Sem.

M. I. Graev. Consider $\mathbb{C}^3$ and a fn. on it $f(x) = f(x_1, x_2, x_3)$. Consider the line

$$\begin{cases} x_1 = \alpha_1 x_3 + \beta_1 \\ x_2 = \alpha_2 x_3 + \beta_2 \end{cases}$$

$$\int f(\alpha_1 x_3 + \beta_1 \text{ [crossed out]}, \alpha_2 x_3 + \beta_2, x_3)\, dx_3\, d\bar{x}_3$$
$$= \varphi(\alpha_1, \alpha_2; \beta_1, \beta_2)$$

Undeciphered passage. A mark after β_1 in the first argument is crossed out.

Original source details: GELFAND-P000697

How can f be recovered from φ? The problem is overdetermined and trivial.

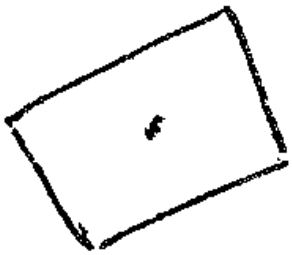
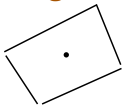


Diagram and typed description.



Original source details: GELFAND-P000697

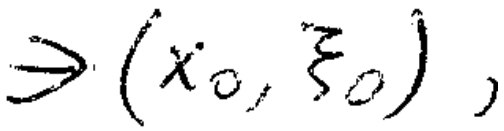
Draw a plane. This gives the classical Radon problem. Its soln. is well known. Solution. Write a $(1,1)$-form on the set of lines in the plane. Its integral gives the answer.

In the real case the equation is

$$\frac{\partial^2 \varphi}{\partial \alpha_1 \partial \beta_2} = \frac{\partial^2 \varphi}{\partial \alpha_2 \partial \beta_1}.$$

One can prove that it is sufficient.

Idea $WF(u) \ni (x_0, \xi_0)$, if the integral over the system of parallel planes with direction ξ_0 has a singularity.



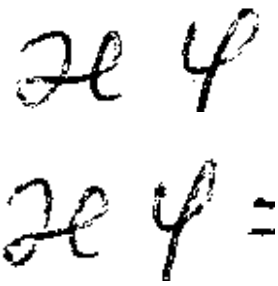
Manuscript excerpt. The sign after $WF(u)$ is read as $\ni$.

Original source details: GELFAND-P000697

Construct a differential form

$$\mathcal{H}\varphi = \sum_{i,j=1}^{2} \frac{\partial^2 \varphi}{\partial\beta_i\, \partial\bar{\beta}_j}\, d\alpha_i \wedge d\bar{\alpha}_j \quad (\mathbb{C}^3)$$

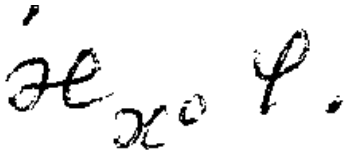
$$\mathcal{H}\varphi = \frac{\partial\varphi}{\partial\beta_1} d\alpha_1 + \frac{\partial\varphi}{\partial\beta_2} d\alpha_2 \qquad (\mathbb{R}^3)$$



Original source details: GELFAND-P000697

Uncertain reading. The handwritten sign before φ is rendered as $\mathcal{H}$.

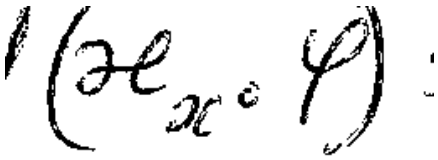
Let there be a point (x_1^0, x_2^0, x_3^0). Substitute $\beta_i = x_i^0 - \alpha_i x_3^0$. We obtain the restriction of the form to the lines through this point. Denote this by $\mathcal{H}_{x^0}\varphi$.



Original source details: GELFAND-P000697

Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.

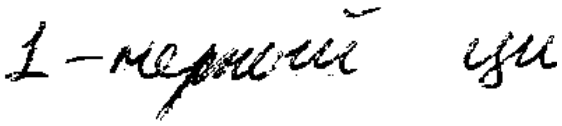
Theorem. φ is representable in terms of $f \iff d(\mathcal{H}_{x^0}\varphi) = 0,\ \forall x_0$.



Original source details: GELFAND-P000697

Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.

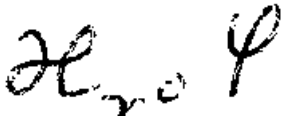
Inversion formula in $\mathbb{C}^3$ (in $\mathbb{R}^3$ it is already nonlocal). Among the lines through x_0 choose a one-dimensional cycle γ and then



Original source details: GELFAND-P000697

Manuscript excerpt. The manuscript reads “one-dimensional.”

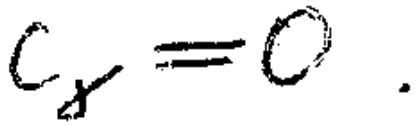
$$\int_\gamma \mathcal{H}_{x^0}\varphi = c_\gamma f(x^0)$$



Original source details: GELFAND-P000697

Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.

In the real case $C_\gamma = 0$.



Uncertain reading. The letter in this line is rendered as C, and in the preceding formula as c.

Original source details: GELFAND-P000697

A complex K — a three-dimensional set of lines.

When, given $\mathcal{H}\varphi$ on K, can we recover φ?

Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.

Original source details: GELFAND-P000697

All complexes K, with this property are of the following kinds:

a) all lines tangent to a surface

b) — $\|$ —, meeting a curve

Linear complex — a complex of lines in $\mathbb{C}P^3$ ($\mathbb{R}P^3$). all its lines through a given point — a plane.

GELFAND-P000698

$$\begin{matrix} x \\ y \end{matrix} \quad \left\| \begin{matrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{matrix} \right\|$$

Plücker coordinates $p_{ij} = x_i y_j - x_j y_i$ — coordinates of the plane.

Manuscript excerpt. The surname is written “Плюккеровы” (Plücker).

Original source details: GELFAND-P000698

Lin. complex: $\sum c_{ij} p_{ij} = 0$.

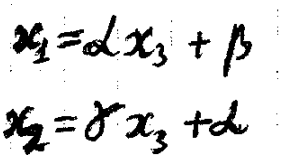
Lin. complex. a plane is associated with $\forall$ points of the space (it contains all the lines through that point)

a line is associated with $\forall$ lines, meeting all lines that pass through the given line.

There is a Legendre transform.

Example of a complex.

$$x_1 = \alpha x_3 + \beta$$
$$x_2 = \gamma x_3 + \alpha$$



Manuscript excerpt. α is the coefficient of x_3 in the first equation and the constant term in the second.

Original source details: GELFAND-P000698

Any other is isomorphic to it.

Uncertain reading. The final letter of “Любой” (any) is written over earlier strokes.

Original source details: GELFAND-P000698

References: Chapters 1 and 2 of vol. 5 of Generalized Functions.

F. A., vol. 1, $N = 1$, paper by Gelfand, Graev, Shapiro.

Manuscript excerpt. The notation after the volume number reads $N = 1$.

Original source details: GELFAND-P000698

GELFAND-P000699

28/III-77 Gelfand Sem.

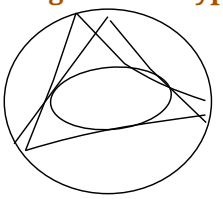
I. M. Gelfand. Problem (from Griffiths’s paper [crossed out]: Variations on a theorem of Abel.)

Undeciphered passage. A small mark between the surname and the colon is crossed out.

Original source details: GELFAND-P000699

An ellipse inside a circle.

Diagram and typed description.



A circle, an elongated oval, and overlapping sides. Some of the joins are uncertain.

Original source details: GELFAND-P000699

A polygon, inscribed in one curve of the second degree and circumscribed about another. When does this hold? (When can one inscribe it this way?)

Poncelet's thm.: if it closes from one point, then from $\forall$ other point it closes after the same number of steps.

Jacobi: a relation in elliptic integrals

From Poncelet's theorem $\Rightarrow \exists$ an invariant measure and an isomorphism to a rotation of the circle.

Latest issue of Compositia Math. — the simplest proof of the theorem of Dvoretzky.

Uncertain reading. The manuscript title reads "Compositia Math."

Original source details: GELFAND-P000699

(Figiel)

A. M. Vershik. Probability theory and the symm. group

S_n — group of permutations of a set of n elements.

S_∞ — group of finitary bijective permutations of the natural numbers, affecting only [crossed out] finitely many elements

Undeciphered passage. Several crossed-out strokes precede "finitely."

Original source details: GELFAND-P000699

Asymptotic problems related to S_n — asymptotics of Plancherel measures.

Young tableau. A collection of rows beginning at a vertical line and decreasing in length

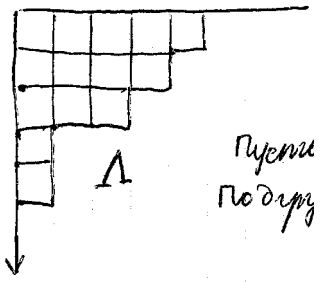
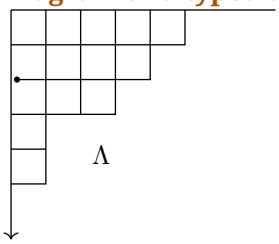


Diagram and typed description.



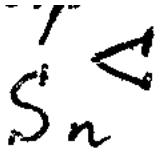
Original source details: GELFAND-P000699

(see the end of van der Waerden's book
Curtis, Reiner,
Kirillov

Let there be n cells;

Subgroup

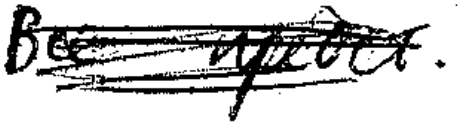
G_1, | preserves partition into rows $S_n^<$ — $G_2 \subset S_n$ | preserves partition into columns S_n^Λ — permutes elements within a column.



Uncertain reading. The first S_n has a superscript angle with a short cross-stroke at each tip. It is provisionally set as $<$.

Original source details: GELFAND-P000699

[crossed out: All reprs.]



Manuscript excerpt. The crossed-out words read "Все предст." ("All reprs.").

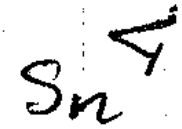
Original source details: GELFAND-P000699

GELFAND-P000700

Consider the trivial representation of $S_n^<$ and a one-dimensional one of S_n^Λ.

Manuscript excerpt. The word after "Consid." reads "единичное" ("unit").

Original source details: GELFAND-P000700



Uncertain reading. The first S_n has a superscript angle with a short cross-stroke at each tip. It is provisionally set as $<$.

Original source details: GELFAND-P000700

$$\begin{pmatrix} \text{even} — \text{identity} & 1 \\ \text{odd} — & (-1) \end{pmatrix} \longleftarrow \text{sign representation}$$

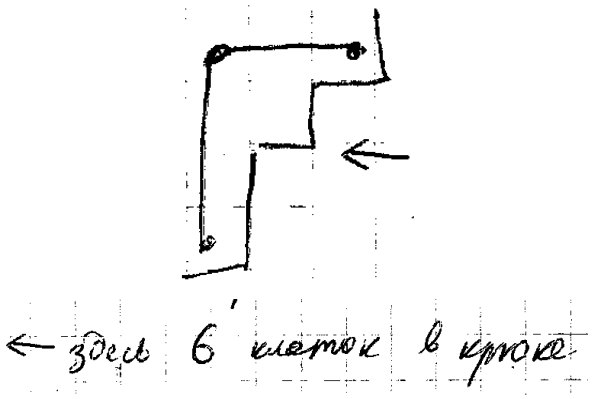
From each such representation induce a representation of S_n.

They are reducible, but have a unique common irreducible (theorem!) — this is the one assigned.

1955. hook formula (Robinson et al.)

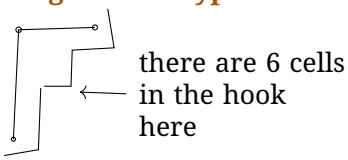
hook.

A hook is the total number of cells to the right of and below the given one —



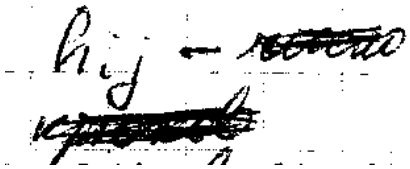
Original source details: GELFAND-P000700

Diagram and typed description.



$$\frac{n!}{\prod h_{ij}} = \dim \Lambda$$

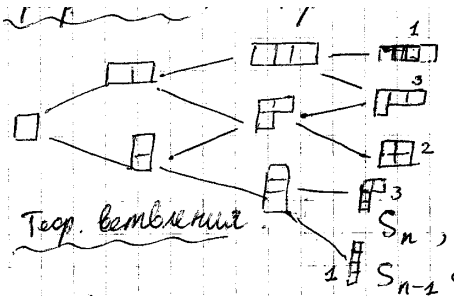
h_{ij} — [crossed out] (— number of elements in the hook



Original source details: GELFAND-P000700

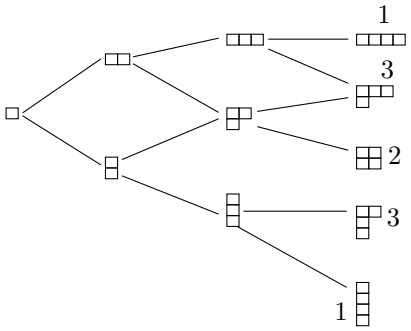
Uncertain reading. Two words after h_{ij} are crossed out. The first looks like “число” (number).

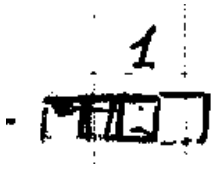
[overwritten] Young graph. The vertices are Young tableaux. In column n there are tableaux of n cells. There is an edge when one can obtain the next tableau by adding a single cell.



Original source details: GELFAND-P000700

Diagram and typed description.





Original source details: GELFAND-P000700

Uncertain reading. The upper right row of cells is heavily retraced. There is a 1 above it.

Original source details: GELFAND-P000700

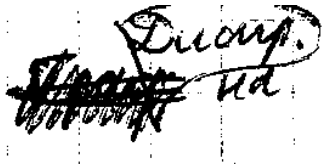
Undeciphered passage. The mark before the heading is heavily shaded over.

Branching thm. S_n, π — an irred. repr. of S_n

$S_{n-1} \subset S_n$

In $\pi|_{S_{n-1}}$ each irred. repr. occurs with multiplicity 1.

[crossed out] [inserted and circled: Diagr.] at step n decomposes into precisely the preceding ones.



Original source details: GELFAND-P000700

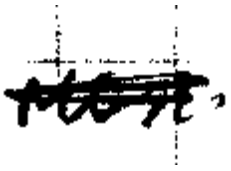
Undeciphered passage. The original word below “Diagr.” is crossed out.

— There are as many one-dimensional subspaces as there are paths in the graph leading to 1.

— Thus the count is a combinatorial problem that is unsolved (only via Frobenius’s formula and so on).

Gelfand–Tsetlin basis

Question: $S_m \subset S_n$. [crossed out] how many paths reach a fixed Young tableau?



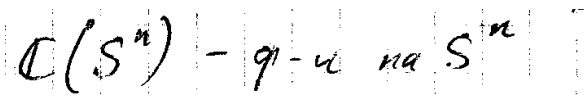
Original source details: GELFAND-P000700

Undeciphered passage. A short word after the inclusion is crossed out.

Length of the first row

$$\ell_1(\Lambda) = \max\{m : \ \pi|_{S_m} \supset \mathrm{id}\}$$

Regular repr. $C(S^n)$ — fns. on S^n.



Original source details: GELFAND-P000700

Uncertain reading. The manuscript writes n as a superscript in both instances of S^n.

$$(U_g f)(h) = f(g^{-1}h)$$

Plancherel measure on Young tableaux

It is known that

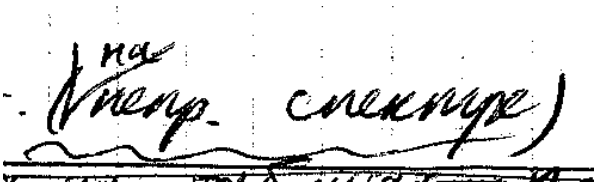
$$n! \underset{\text{dim. of the regular repr.}}{=} \sum_{\Lambda} (\dim \Lambda)^2$$

Plancherel measure

$$\mu(\Lambda) = \frac{(\dim \Lambda)^2}{n!}$$

Gelfand's problem.

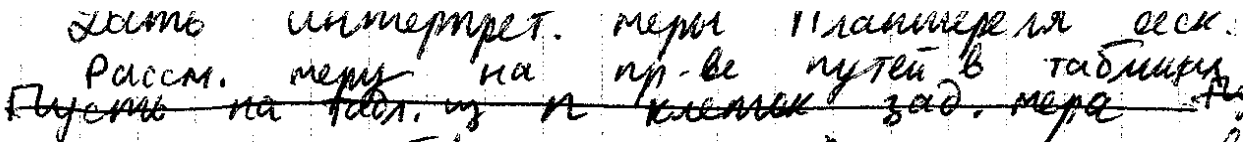
Give an interpretation of the Plancherel measure of an infinite-dimensional repr. ([“on” inserted above] the continuous spectrum)



Uncertain reading. “On” is inserted above the start of the phrase in parentheses.

Original source details: GELFAND-P000700

[written above the crossed-out line] Consider a measure on the space of paths in tableaux

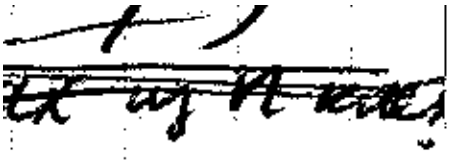


Original source details: GELFAND-P000700

Uncertain reading. The new phrase stands immediately above the horizontally crossed-out line.

[crossed out] Suppose that on tableaux with n cells a measure is given

[crossed out] Suppose there is a measure on tableaux with n cell...



Uncertain reading. The end of the crossed-out phrase is obscured by strokes at the right edge.

Original source details: GELFAND-P000700

Suppose cells are added. Consider the conditional probability that a cell appears given the history

GELFAND-P000701

Theorem

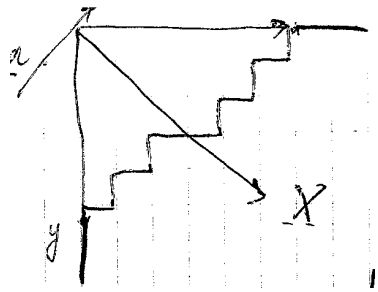
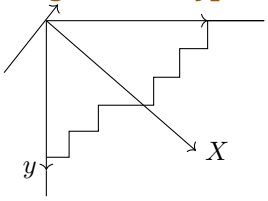


Diagram and typed description.



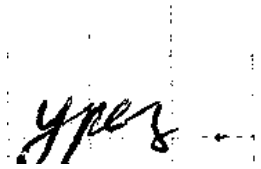
Stepped outline with the labels y and X. Strokes at the horizontal arrowhead may form x.

Original source details: GELFAND-P000701

Consider the coord. system

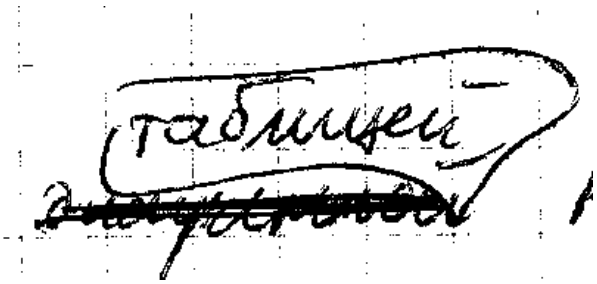
$$\left\| \begin{array}{l} X = \dfrac{x+y}{2}; \\ Y = \dfrac{x-y}{2}; \end{array} \right.$$

Consider the graph of the fn. $|X|$, trunc. [crossed out: diagram] [inserted above: tableau] of Young.



Original source details: GELFAND-P000701

Manuscript excerpt. The abbreviation reads “упез.” (“truncated”).



Original source details: GELFAND-P000701

Uncertain reading. “Tableau” is circled above the crossed-out word, which tentatively reads “diagram.”

The tableau is identified with the fn. $Y = \widetilde{\Lambda}(X)$

$$\mu_n \left\{ \widetilde{\Lambda} : \sup \left| \frac{1}{\sqrt{n}} \widetilde{\Lambda}(\sqrt{n}\, X) - \Omega(X) \right| < \varepsilon \right\} \longrightarrow 1, \qquad \forall \varepsilon > 0$$

compress by a factor of $\sqrt{n}$. Then all are near the graph of the fn.

$$\Omega(X) = \frac{2}{\pi} \left(X \arcsin X + \sqrt{1 - X^2} \right) \qquad \left(\Omega'(X) = \frac{2}{\pi} \arcsin X \right)$$

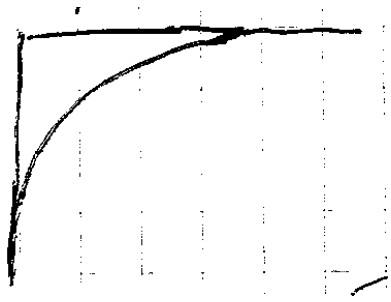
This is conv. in measure, [crossed out]

Original source details: GELFAND-P000701

Undeciphered passage. A short passage after the comma is crossed out.

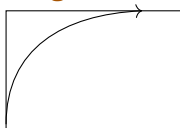
There is a thm. on conv. almost everywhere. (for a seq. of tableaux)

A diagram is an infinite path. It converges for almost all diagrams.



Original source details: GELFAND-P000701

Diagram and typed description.



Corollary:

$$\mu_n\left\{\Lambda : \left|\frac{\ell_1(\Lambda)}{2\sqrt{n}} - 1\right| < \varepsilon\right\}$$

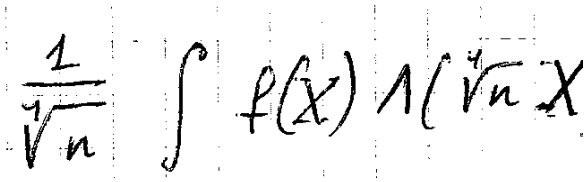
(with prob. close to 1 there is an inv. vector)

(soln. of Ulam's problem)

This is the law of large numbers. Deviations give white noise.

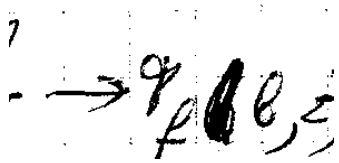
For the functional $\int f(X)\Lambda(X)\,dX$ there is a central limit thm.

$$\mu_n\left\{\Lambda : \frac{1}{\sqrt[4]{n}}\int f(X)\Lambda(\sqrt[4]{n}\,X)\,dX - \int f(X)\Omega(X)\,dX < \ell\right\}$$
$$\longrightarrow \varphi_f(\ell,\varepsilon)$$



Original source details: GELFAND-P000701

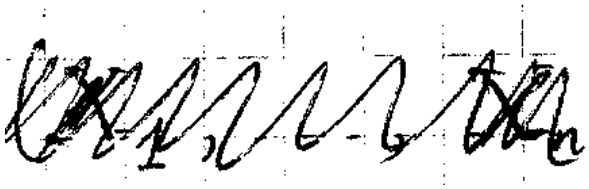
Uncertain reading. An index 4 is visible at both radicals.



Original source details: GELFAND-P000701

Uncertain reading. The right-hand side tentatively reads $\varphi_f(\ell,\varepsilon)$. The mark before ℓ is heavily overwritten.

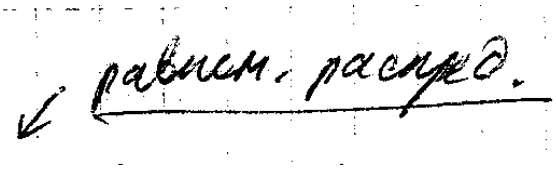
the Markov property and stationarity can be checked.



Original source details: GELFAND-P000701

Undeciphered passage. Crossed-out writing before $(x_1,\dots,x_n)$. A parenthesis, x_1 and a final index n are visible.

$$(x_1,\dots,x_n), \qquad \underset{\swarrow}{\underline{\text{uniformly distrib.}}}\ 0 \le x_i \le 1 \quad ; \quad i = 1,\dots,n$$



Original source details: GELFAND-P000701

Uncertain reading. Underlined annotation with an arrow: "uniformly distrib."

Consider a monot. incr. subseq. (not necessarily consecutive) What is its length?

$$\max\{r;\ \exists\, i_1 < \cdots < i_n, \quad X_{i_1} \le X_{i_2} \le \cdots \le X_{i_n})$$

Original source details: GELFAND-P000701

Manuscript excerpt. The letter after the opening brace is r. The final subscripts use n. The closing bracket is round.

grows like $\underline{\underline{2\sqrt{n}}}$ (with respect to Lebesgue measure on the unit cube)

1970, Knuth, Pac. Journ.

Original source details: GELFAND-P000701

Uncertain reading. The journal abbreviation is written “Pac. Journ.”

1961, [crossed out] Shenstead

Original source details: GELFAND-P000701

Undeciphered passage. Several letter strokes before the surname are crossed out. The manuscript spells the name “Shenstead.”

> There is a map into diagrams
> (bumping)

GELFAND-P000702

4/IV-77 Gelfand Sem.

A. Tyurin. FG_g — a group with $2g$ gen. $a_1, \ldots, a_g, b_1, \ldots, b_g$ and rel.

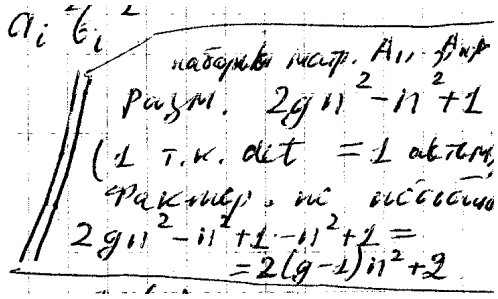
$$\prod_i [a_i, b_i] = 1, \qquad [a_i, b_i] = a_i b_i a_i^{-1} b_i^{-1}$$

Original source details: GELFAND-P000702

Uncertain reading. The day of the month is heavily retraced. It reads as 4.

R_C — set of classes of reprs. in $GL(n, C)$

an alg. set with singularities.



Original source details: GELFAND-P000702

Diagram and typed description. [Right margin, in a frame with a double left rule:]

tuples of matrices $A_1, \ldots, A_{2g}$
Dim. $2gn^2 - n^2 + 1$
(1 since $\det = 1$ automatically,
Quotient by similarity
$2gn^2 - n^2 + 1 - n^2 + 1 =$
$= 2(g-1)n^2 + 2$

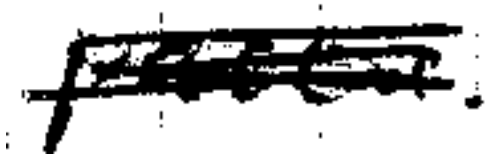
Original source details: GELFAND-P000702

Uncertain reading. The subscript on the last matrix is overwritten. It tentatively reads $2g$.

$$\pi_1(X) \longleftrightarrow FG_g.$$

Repr. $\varrho \rightsquigarrow$ bundle E_ϱ on X

Analyt. [crossed out] equiv. $\underset{\text{def}}{\Longleftrightarrow}$ $E_{\varrho_1} \cong E_{\varrho_2}$

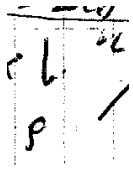


Original source details: GELFAND-P000702

Uncertain reading. The crossed-out abbreviation between “Analyt.” and “equiv.” tentatively reads “предст.” (repr.).

[At right, below the frame:]

$$\underset{\pi_1}{U(X)} \times \underset{\varrho}{V^n}/\pi_1 \longrightarrow X$$



Original source details: GELFAND-P000702

Uncertain reading. The letter with superscript n is nearly vertical in the right-hand formula. It is set as V^n following the formula below.

Packet of representations

$$\underset{\text{aff. space of dim. } (n^2-1)(g-1)}{\uparrow H(X,E)} \subset R_C$$

For a fixed curve X and a fixed bundle E.

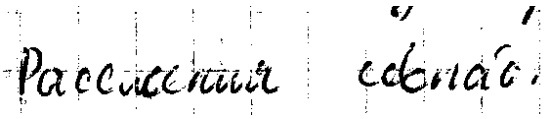
The space of reprs. has dimension $2((g-1)n^2+1)$

A repr. in general position will be irreducible.

$$E_\varrho = \underset{\text{universal cover}}{\uparrow U(X)} \times V^n/\pi_1 \qquad x \in U(X)$$

$$(x,\xi) \sim (xg, \varrho(g)\xi)$$

Bundles are conn. $\Rightarrow$ what can one say about reprs.?



Manuscript excerpt. The word after "Bundles" reads "связ." ("connected", abbr.).

Original source details: GELFAND-P000702

Any two irred. reprs. are equiv. with respect to some Riem. surface (belong to some packet).

Uncertain reading. The middle of the abbreviation after "Any two" is overwritten. It tentatively reads "неприв." (irred.).

Original source details: GELFAND-P000702

Uncertain reading. The opening strokes of "эквивал." (equiv.) are written over earlier strokes.

Original source details: GELFAND-P000702

Fix $X \Rightarrow$

$$R_C \xrightarrow{\text{affine bundle}} S_n(X) \qquad \text{(moduli of bundles)}$$

GELFAND-P000703

Typical problem: consider packets of reprs. corresp. to a one-dim. bundle L.

How a description of the whole set $H(X,L) \subset \mathbb{C}^{*2g}$ — space of characters.

Uncertain reading. The word after "description" tentatively reads "всё" (whole).

Original source details: GELFAND-P000703

L — a one-dimensional bundle.

L: given by transition fns. $\{\alpha_{ij}\} \in H^1(\{U_i\}, \mathcal{O}_X^*)$.

L: obt. from [crossed out] a repr. of $\pi_1 \Leftrightarrow \{\alpha_{ij}\} \in H^1(X, \mathbb{C}^*)$ (one can take $\alpha_{ij} = \text{const}$).

Undeciphered passage. Two rules obscure the word after “from.” Two lower strokes remain visible.

Original source details: GELFAND-P000703

Sections: f_i on U_j.

$$f_i = \alpha_{ij} f_j.$$

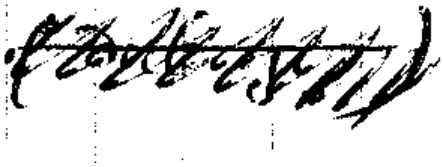
Uncertain reading. The lower stroke in the symbol after f_i resembles j. Readings i and ij are also possible.

Original source details: GELFAND-P000703

$$\alpha_{ij} = \text{const} \quad \Rightarrow \quad \frac{df_i}{f_i} = \omega\big|_{U_i}$$

— one differential (meromorphic).

Singularities of ω: poles of order ≤ 1 [crossed out: (divisor?)].



Uncertain reading. The crossed-out word in parentheses tentatively reads “divisor.”

Original source details: GELFAND-P000703

The residues are integers.

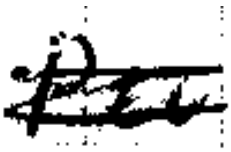
Principal parts — divisor

$$[\omega] = \sum \operatorname{Res}_x \omega \cdot x$$

(zeros and poles of the fns. f_i).

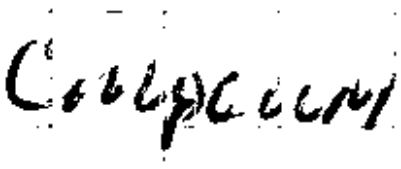
2 other representatives $\Rightarrow$
$\Rightarrow$ equiv. divisor.

Fix the divisor. [crossed out] Construct ω with the given set of principal parts.



Undeciphered passage. Two rules cross out a short passage before the verb.

Original source details: GELFAND-P000703



Uncertain reading. The verb tentatively reads “Соорудим” (construct). The middle letters run together.

Original source details: GELFAND-P000703

Solve the eq. $\frac{df}{f} = \omega + \omega_0$ globally on $U(X)$.

ω_0 — a holom. differential on X.

$$f(g(u)) = \underset{\text{character!}}{\chi_g} \quad f(u).$$

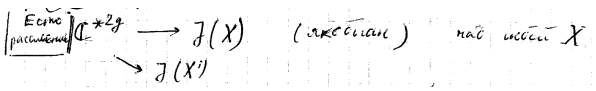
Here we obt.

All characters corresp. to the packet.

Corollary: the packet of reprs. uniquely recovers X from Terelli's thm.

Original source details: GELFAND-P000703

Uncertain reading. The manuscript spells the name "Терелли" (Terelli), referring to Torelli's theorem.



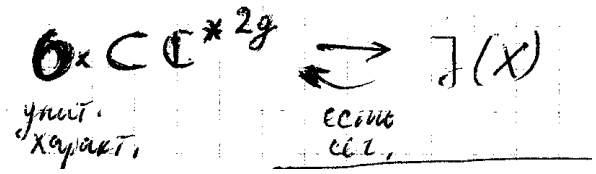
Original source details: GELFAND-P000703

Diagram and typed description.

There is a fibration $\mathbb{C}^{*2g} \longrightarrow J(X)$, $\mathbb{C}^{*2g} \searrow J(X')$

(Jacobian) over all X.

0th fibre — exponentials of primitives of holom. differentials.



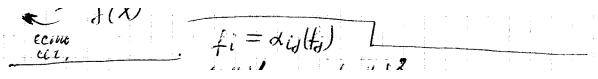
Original source details: GELFAND-P000703

Diagram and typed description.

$[O_x?] \subset \mathbb{C}^{*2g} \rightleftarrows J(X)$

unit. chars. — there is a sect.

The overwritten round symbol with subscript x may be O_x or Θ_x.



Original source details: GELFAND-P000703

Diagram and typed description.

$f_i = \alpha_{ij}(f_j)$

Schwarzian derivative:

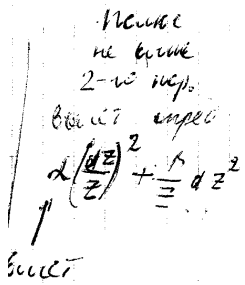
$$S_z^f = \left(\frac{f''}{f'}\right)' - \frac{1}{2}\left(\frac{f''}{f'}\right)^2$$

$$S_t^{f(g(t))}\,dt^2 = S_z^f\,dz^2 + \underbrace{S_t^g\,dt^2}_{\substack{\text{for fractional-linear}\\ \text{this is 0}}} .$$

We obt. general quadr. differentials.

[In the right margin, beyond the rule:] Pole of order at most 2. Residue is def.

$$\underset{\substack{\uparrow\\ \text{residue}}}{\alpha}\left(\frac{dz}{z}\right)^2 + \frac{r}{z}\,dz^2$$



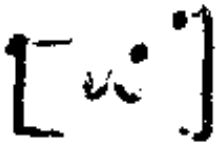
Uncertain reading. The first coefficient tentatively reads α, possibly λ. The numerator of the fraction reads r. The arrow marked "residue" points to the first coefficient.

Original source details: GELFAND-P000703

GELFAND-P000704

$$S_{z_i}^{f_i}\,dz_i^2$$

$[\omega]$ — principal part of the quadr. differential.



Uncertain reading. The sign in square brackets resembles ω. A dot or short stroke is visible above it at right, with two more dots by the right bracket.

Original source details: GELFAND-P000704

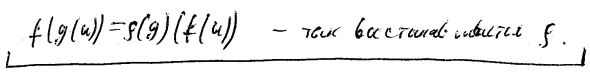


Diagram and typed description.

$$f(g(u)) = \varrho(g)(f(u))$$

— this is how one recovers ϱ.

Original source details: GELFAND-P000704

Chapter 11

GELFAND-P000705

5/IV-77 Gelfand Sem. (34th year of activity).

Original source details: GELFAND-P000705

Manuscript excerpt. The Roman marks between the rules read IV.

Possible seminar topic: noncommutative algebraic geometry (drawing on recent work by I. N. Bernstein, I. M. Gelfand, S. I. Gelfand

V. Drinfeld. On Atiyah's work on the Yang–Mills eqs.

A 4-dimensional sphere is given S^4, with a smooth complex vector bundle ξ. A connection ∇ on this bundle.

The pair satisfies the eqs. of duality, if

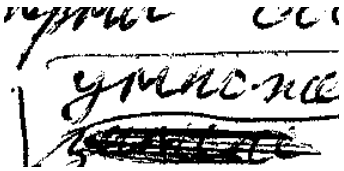
$$*K = K \tag{1}$$

(K — the curvature form of this connection)

K — a form with values in endomorphisms of the bundle

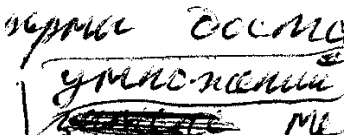
$*$ — is constructed from the standard metric on S^4; either orientation may be fixed.

To construct $*$ from 2-forms to 2-forms, it is enough to have a conformal structure (on [crossed out: replacing?] [inserted above: multiplying] the metric



Original source details: GELFAND-P000705

Undeciphered passage. The crossed-out word begins with a stroke like "з." One possible reading is "замене" (replacing).



Original source details: GELFAND-P000705

Manuscript excerpt. "Multiplying" is enclosed above the line. The pointer leads to the start of the cancelled writing after "on."

[source number: 40]

by a number leaves it unchanged)

The fibres also have a metric (Hermitian). The connection preserves the metric on the fibre.

The connection is a classical field, whose quantum analogue describes the gluons that are supposed to hold the quarks together.

What does one do next with a solution? Quantization and so on. To the pair (ξ, ∇) corresponds a linear operator, from the spectrum, extract a number (renormalized determinant). Then integrate over the set of solutions. This is the denominator.

Uncertain reading. The abbreviation before "number" tentatively reads "извл." (extract).

Original source details: GELFAND-P000706

There is a one-to-one corresp. between solutions and certain alg. bundles on $\mathbb{C}P^3$. Consider the projection $\mathbb{C}P^3 \to S^4$.

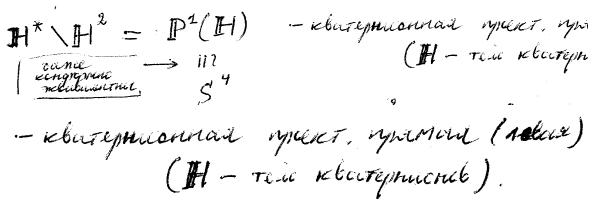


Diagram and typed description.

$\mathbb{H}^*\backslash\mathbb{H}^2 =$ $\mathbb{P}^1(\mathbb{H})$

they are conformally equivalent $\longrightarrow$ S^4

— the quaternionic projective line (left) ($\mathbb{H}$ — the division ring of quaternions).

Original source details: GELFAND-P000706

$\mathbb{C} \subset \mathbb{H}$, hence a surj. $\mathbb{C}^*\backslash\mathbb{H}^2 = \mathbb{C}P^3$ and we have a map $\mathbb{C}P^3 \to S^4$ (its fibres are two-dimensional).

Conf. structure on $\mathbb{P}^1(\mathbb{H})$ — with a point removed, it is $\mathbb{R}^4 \cong \mathbb{H}$ with the standard conf. structure.

Manuscript excerpt. The superscript before the parenthesis reads 1. Its foot touches the parenthesis.

Original source details: GELFAND-P000706

Conf. structure: at $\forall$ points a metric is given up to a positive multiplier. The fibres of $\mathbb{C}P^3 \to S^4$ are $\mathbb{C}P^1$.

Atiyah's thm. establishes an isomorphism between

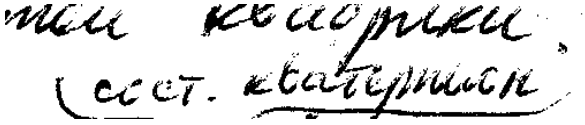
1) pairs ξ, ∇, satisfying the eqs. of duality

2) alg. vect. bundles on $\mathbb{C}P^3$, trivial on each fibre of the map $\mathbb{C}P^3 \xrightarrow{f} S^4$.

Establishing the isomorphism:

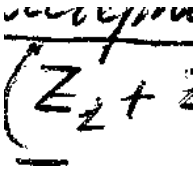
All lines (proj. complex) in $\mathbb{C}P^3$ form a quadric, the fibres of the map f — the real points of this quadric. In $\mathbb{C}P^3$ there is an involution (z_1, z_2, z_3, z_4);

form a quaternion
$$(z_1 + z_2 j,\ z_3 + z_4 j)$$



Original source details: GELFAND-P000706

Manuscript excerpt. "сост. кватернион" (form a quaternion) is written above the pair, with a brace beneath it.



Original source details: GELFAND-P000706

Manuscript excerpt. The first subscript in the pair reads 1. The following subscript 2 has a broad upper curve.

Multiply on the left by j:
$$\bar{z}_1 j - \bar{z}_2, \qquad \bar{z}_3 j - \bar{z}_4$$

GELFAND-P000707

Involution:
$$(z_1, z_2, z_3, z_4) \mapsto (-\bar{z}_2, \bar{z}_1, -\bar{z}_4, \bar{z}_3)$$

Take the complex lines invariant under this involution. This gives S^4.

Triviality is understood in the alg. ($\sim$ complex) sense.

The bundles are pulled back from S^4, but on S^4 there is no complex structure.

1) $\Rightarrow$ 2)
$$\xi \mapsto f^* \xi$$

Details next time.

I. Krichever. On constructing exact solutions of other nonlinear equations.

$$\left[\frac{\partial}{\partial y} - L_1, \frac{\partial}{\partial t} - L_2\right] = 0 \tag{1}$$

$$L_1 = \sum_{i=0}^{n} u_i(x, y, t) \frac{d^i}{dx^i} \tag{2}$$

Original source details: GELFAND-P000707

Undeciphered passage. Two vertical strokes with transverse traces are visible in the right margin beside (2).

$$L_2 = \sum_{j=0}^{m} v_j(x,y,t)\frac{d^j}{dx^j} \tag{3}$$

The coefficients may also be matrices.

There are solutions in general position.

Separatrix family — multisoliton [crossed out] solutions, of the eq.

$$u_t = 6uu_x + u_{xxx}.$$

Undeciphered passage. A short passage between "multisoliton" and "solutions" is heavily crossed out.

Original source details: GELFAND-P000707

There are parameters. They are almost always conditionally periodic.

For certain vals. of the parameter — rational.

Manin. Solutions of alg. type (without y) — those that are common solutions of (1) and

$$[L_1, L_3] = 0 \tag{4}$$

Undeciphered passage. An isolated T-shaped mark is visible to the right of (4).

Original source details: GELFAND-P000707

(L_3 — aux. operator)

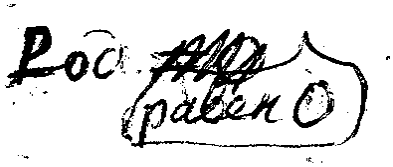
Solns. [crossed out] of alg. type — such that $\exists L_3 \notin \mathbb{C}[L_1, L_2]$, such that cond. is satisfied (4).

Undeciphered passage. A short passage between "Solns." and "of alg. type" is crossed out.

Original source details: GELFAND-P000707

Degenerate solutions: in the lin. space of lin. combinations of coeffs. of fractional powers, to each corresponds an alg. curve.

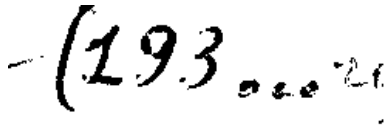
Its genus is constant almost everywhere. Degeneration — where the genus changes. The genus [crossed out $\neq 0$?] [boxed below the line: equals 0] where there are multisoliton solutions and something else.



Uncertain reading. The crossed-out writing resembles $\neq 0$. "Equals 0" is enclosed below.

Original source details: GELFAND-P000707

There is alg. dependence of L_1 and L_2 — Jacobson's thm. (193...



Uncertain reading. Three low marks resembling an ellipsis follow "193," with faint strokes at the right edge.

Original source details: GELFAND-P000707

GELFAND-P000708

[source number: 41]

Moser's work: [above the line: (Mc-Kean and Airault)] seek solns. rational in x of KdV. In t one obt. ordinary eqs. that can be solved explicitly.

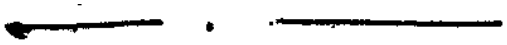


Diagram and typed description.

$\longleftarrow \bullet \longrightarrow$

Original source details: GELFAND-P000708

The rationality cond. for KdV replaces the 3rd operator.

Apparently the same holds for $\forall$ of the Lax eqs.:

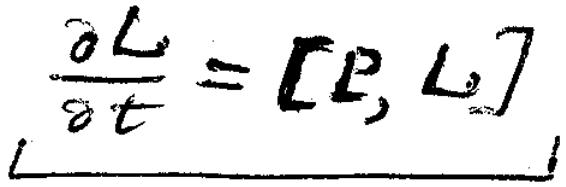


Diagram and typed description.

$$\underbrace{\frac{\partial L}{\partial t} = [P, L]}$$

Original source details: GELFAND-P000708

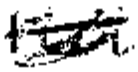
GELFAND-P000709

12/IX-77 Gelfand Sem.

G. Margulis.

G — a semisimple connected Lie group, for simplicity let its centre be trivial, $G \supset \Gamma$ — a discrete subgroup, such that [crossed out]

$$\nu(G/\Gamma) < +\infty \qquad (\nu \text{ — volume induced by Haar measure})$$

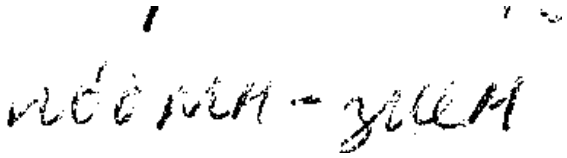


Undeciphered passage. A short passage to the left of the formula is crossed out.

Original source details: GELFAND-P000709

Such Γ are called lattices.

An algebraic group G — a subgroup of $GL(n, \mathbb{C})$, that is an alg. variety in $GL(n, \mathbb{C})$.



Uncertain reading. The abbreviation after "alg." tentatively reads "много-зием" (variety).

Original source details: GELFAND-P000709

k-group — a group def. over the field k (given by eqs. with coeffs. in k). (here $k \subset \mathbb{C}$ a subfield) (Otherwise: it is called k-closed).

Let G — $\mathbb{Q}$-group.

$G(k)$ — set of matrices with coeffs. in k, $\in$ our set.

One can take points over a subring of $\mathbb{C}$

$$G(\mathbb{R}) \supset G(\mathbb{Z})$$

$G(\mathbb{Z})$ is a group if [crossed out: $G(\mathbb{R})$] $G(\mathbb{Z}) \subset SL(n, \mathbb{Z})$.

Original source details: GELFAND-P000709

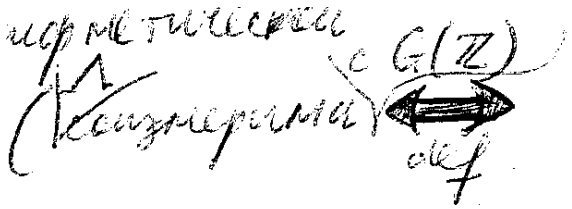
Manuscript excerpt. $G(\mathbb{R})$ is crossed out before $G(\mathbb{Z}) \subset SL(n, \mathbb{Z})$.

1) $G(\mathbb{Z})$ is called an arithm. subgroup.

2) $\Lambda \subset G(\mathbb{R})$

if Λ is commensurable with $G(\mathbb{Z})$, then Λ is also called arithmetic.

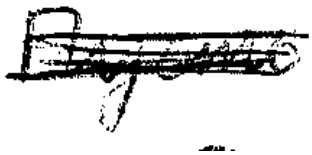
([above the line: Λ] commensurable [above the line: with $G(\mathbb{Z})$] $\underset{\text{def}}{\Longleftrightarrow}$ $\Lambda \cap G(\mathbb{Z})$ has finite index in Λ and in $G(\mathbb{Z})$)



Original source details: GELFAND-P000709

Manuscript excerpt. Λ and "with $G(\mathbb{Z})$" are inserted above the line. "def" stands under the double arrow.

[crossed out: Question]

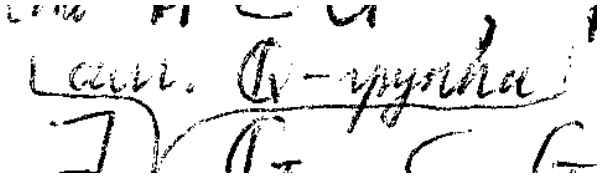


Original source details: GELFAND-P000709

Uncertain reading. Several rules cross out a word that tentatively reads "Question."

Let $H \subset G$, H is called an arithmetic subgroup, if [above the line with a pointer: alg. $\mathbb{Q}$-group]

$$\exists G \subset GL(n, \mathbb{C}) \quad \text{and} \quad f : G(\mathbb{R}) \longrightarrow G,$$



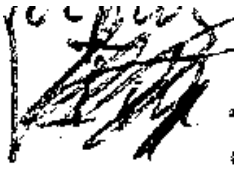
Original source details: GELFAND-P000709

Manuscript excerpt. "alg. $\mathbb{Q}$-group" is written above with a pointer toward the following G.

such that

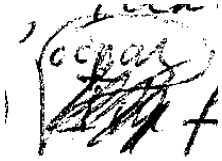
1) f is an epimorphism, 2) Ker f is compact.

3) [above the line with a pointer: (image)] [crossed out] $f(G(\mathbb{Z}))$ is commensurable with H.



Original source details: GELFAND-P000709

Undeciphered passage. Writing before $f(G(\mathbb{Z}))$ is crossed out with slanting strokes.



Original source details: GELFAND-P000709

Manuscript excerpt. "Image" is inserted above the crossed-out writing.

GELFAND-P000710

Property: if H — an arithm. subgroup, then $\nu(G/H) < \infty$.

(Borel–Harish-Chandra theorem

Example: $SL(n,\mathbb{R}) \supset SL(n,\mathbb{Z})$

$$\nu\big(SL(n,\mathbb{R})/SL(n,\mathbb{Z})\big) < +\infty$$

(Minkowski) [crossed out: or Hermite?].

Original source details: GELFAND-P000710

Uncertain reading. Two words after "Minkowski" are crossed out. They tentatively read "or Hermite."

Exercise: do it for $n = 2$ — this is equiv. to finding the area of a triangle in the Lobachevsky plane.

Original source details: GELFAND-P000710

Diagram and typed description.

Theorem: under certain additional conds. the converse holds, i.e., every lattice is arithmetic. Proof is in the appendix to the book by Raghunathan.

For $SL(2,\mathbb{R})$ it is false — there are parameters for tiling the Lobachevsky plane, and only countably many arithm. subgroups.

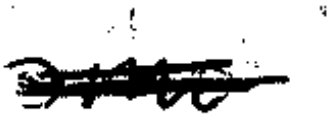
Selberg's conjecture: it is true for the remaining groups Siegel also worked on this.

$$G = \prod G_i, \qquad G_i \text{ — simple Lie groups}$$

A latt. Γ is called reducible, if $G = G_1 \times G_2$, $\dim G_i > 0$, and $(G_1 \cap \Gamma) \times (G_2 \cap \Gamma)$ has finite index in Γ.

We study irreducible lattices.

Conjecture: for an irred. subgroup in $G \neq SL(2,\mathbb{R})/\{\pm E\}$ [crossed out: this?] $\forall$ lattice is arithmetic.



Uncertain reading. The crossed-out word before $\forall$ tentatively reads "это" (this).

Original source details: GELFAND-P000710

Makarov, Vinberg: found nonarithm. subgrps. in groups of motions of Lobachevsky spaces $SO(3,1)$, $SO(4,1)$, $SO(5,1)$).

Answer: if

$$\operatorname{rank} G \geq 2, \quad \text{then the theorem holds.}$$

There is a parameter only in $SL(2,\mathbb{R})$. In the remaining cases rigidity. (Selberg, A. Weil)

In the other groups of rank 1 it is unknown.

Corollary of the rigidity thm. (Selberg): $\forall$ lattice [crossed out] in $SL(n,$ [crossed out] $\mathbb{R})$ is conjugate to a lattice with algebraic coefficients.

Undeciphered passage. A short passage after "lattice" is heavily crossed out.

Original source details: GELFAND-P000710

Undeciphered passage. An isolated sign is crossed out before $\mathbb{R}$ in $SL(n,\mathbb{R})$.

Original source details: GELFAND-P000710

GELFAND-P000711

[source number: 44]

Main lemma.

Let G — a semisimple Lie group, [enclosed insertion above the line: rank $G \geq 2$], Γ — [enclosed insertion above the line: an irreducible] lattice in G, [inserted between the lines: G has no compact factors], k — a local field of char. 0, i.e. $\mathbb{C}$, $\mathbb{R}$ or a finite extension of $\mathbb{Q}_p$, $n \in \mathbb{N}$ and let there be given a homomorphism $T : \Gamma \longrightarrow GL(n, k)$. Suppose

1) $\overline{T(\Gamma)}$ is connected and simple. (Zariski closure)

2) $T(\Gamma)$ is not rel. compact

Then

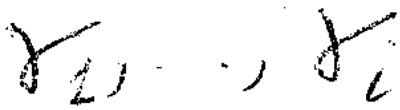
I. $k = \mathbb{C}$ or $\mathbb{R} \Rightarrow T$ extends to a repr. of the group G

II. $k = \mathbb{Q}_p \Rightarrow$ conditions 1) and 2) are incompatible

Deduction of the theorem from the lemma.

$G = G(\mathbb{R})$, K — a field of alg. numbers, we may assume $\Gamma \subset G(K)$ (easily proved [inserted above the line: by Selberg's thm.]).

Let $\gamma_1, \ldots, \gamma_i$ be generators.



Uncertain reading. The last subscript is tentatively i, possibly l.

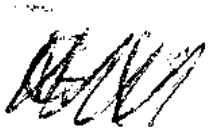
Original source details: GELFAND-P000711

Consid. the denominators of the corresp. matrices and the prime factors occurring in their factorization $p_1, \ldots, p_m$.

Uncertain reading. The last subscript in the list of prime factors is overwritten. It tentatively reads m.

Original source details: GELFAND-P000711

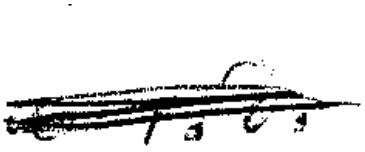
It suff. to prove that the image of



Undeciphered passage. A short passage before Γ is crossed out.

Original source details: GELFAND-P000711

Γ in $G(\mathbb{Q}_p)$ is relat. compact. Then the denominators are bounded



Undeciphered passage. A word after "bounded" is crossed out.

Original source details: GELFAND-P000711

jointly (from the structure of the topology in $\mathbb{Q}_p$).

Compactness follows from assertion II of the lemma. ■

GELFAND-P000712

19/IX—77 Gelfand Sem..

V. Drinfeld. Consid. S^4 with the conformal structure induced by the standard metric.

ξ — a smooth complex vector bundle over S^4.

Uncertain reading. The crossed-out text before the bracket tentatively reads “Кир” (Kir).

Original source details: GELFAND-P000712

[A physicist is present: Kirzhnits, David Abramovich].

Duality equations. A_μ — a matrix-valued fn. of 4 vars.,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad \mu = 1, 2, 3, 4. \tag{1}$$

Manuscript excerpt. F_{ijk} is crossed out before equation (2).

Original source details: GELFAND-P000712

$$F_{ij} = \frac{1}{2}\varepsilon_{ijk\ell}F_{k\ell} \qquad \text{(eqns. of duality)} \tag{2}$$

$$[\partial_\mu + A_\mu, F_{\mu\nu}] = 0 \qquad \text{(eqns. of Yang–Mills)} \tag{3}$$

GELFAND-P000713

Kirzhnits David Abramovich (from FIAN).

On certain characteristic ideas of theor. physics.

Manuscript excerpt. A separate “4” is written before “Frontier.”

Original source details: GELFAND-P000713

4 Frontier: 1) elem. particles 2) cosmology 3) transition from dyn. systems to statist.

All this is now closely connected!

Undeciphered passage. A heavily overwritten or crossed-out mark precedes the heading.

Original source details: GELFAND-P000713

Theory of elementary particles.

Arose ≈ 50 years ago.

For the first 20 years QFT ideas predominated, i.e. as in quant. mechanics.

Late 1950s: doubts.

Landau: "The Hamilton. method is a corpse. Our task is to bury it with all the honors due to it!"

1960s — group approach.

Now a renaissance of the field approach. The idea responsible:

1. Spontaneously broken symmetry.

Sensitivity to macroscopic effects (symmetry breaking).

2. Macroscopic effects.

3. History of the universe — an example of $\overbrace{\text{breaking}}^{\text{of such an influence}}$ of spont. symm.

Cosmology

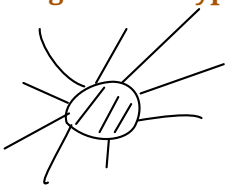
The universe is hot. Why?

4. "Dynamics — statistics".

First, on pt. 4. A few years ago something fundamentally new was said: Hawking, Bekenstein

a region in phase space spreads out (owing to instability)

Diagram and typed description.



Original source details: GELFAND-P000713

— was known. (e.g. Sinai's billiard)

a syst. with positive entropy.

The language of information (Szilard, 1927, then mathematicians) worked here.

It spread out — we lost information

$$\Delta I = \text{Information} = -\ln \frac{P_f \ (= \text{no. of poss. answers to the question at the end})}{P_i \ (= \text{—}\,''\,\text{— at the beginning})}$$

Entropy

The macrostate is determ. by a small number of parameters. With these, different particle distributions are possible (degeneracy).

Entropy $S = \ln(N_{\text{microscopic}})$

I. M. G.: Entropy: 27 coins, 1 counterf. lighter than the others — 3 weighings.

N coins, 1 counterf. differs in weight. For what N can we find it in 3 weighings:

Solution: we know half as much (13).

N weighings $(3^N - 1)/2$.

GELFAND-P000714

[source number: 47]

Algorithm.

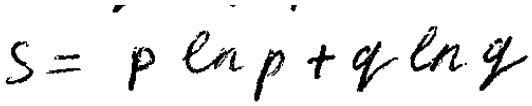
The coins must be divided so that the piles have equal entropy!!

(Divide the entropy into 3 equal parts).

No need for exhaustive search!

coin p — prob. of heads
q — prob. of tails

$$S = p \ln p + q \ln q$$



Manuscript excerpt. The manuscript reads $S = p \ln p + q \ln q$.

Original source details: GELFAND-P000714

Additivity makes it reasonable.

N outcomes $\quad -\sum p_i \ln p_i \qquad$ If $p_i = \frac{1}{N}$, we get $S = \ln N$

Inform. = change in entropy.

Soln. of problem: look at the change in probabilities.

Start: $4 + 4, 5$

$$\Delta I = -\ln \frac{N_f}{N_i}$$

K. D. A. $\quad \Delta S = -\Delta I.$

Example: gas

Diagram and typed description.

initial state final

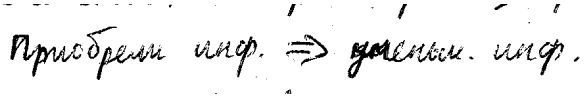
Original source details: GELFAND-P000714

$$S = \ln \frac{V}{N} \qquad \text{(per molecule)}.$$

$$\boxed{\Delta S = \ln 2.}$$

Consequence: loss of inf. $\Rightarrow$ increase in entropy

Gained inf. $\Rightarrow$ decrease in inf.



Uncertain reading. The start of “уменьш.” (decrease) is overwritten. The next abbrev. reads “инф.” (inf.).

Original source details: GELFAND-P000714

Maxwell's demon lowers entropy (keeps track of particles)

Are there any other ways of losing inf.?

Perhaps there are

Undeciphered passage. A word after "there are" is heavily crossed out.

Original source details: GELFAND-P000714

forces retaining information

Undeciphered passage. An isolated mark before "in a g. volume" is crossed out.

Original source details: GELFAND-P000714

in a g. volume.

A known situation where inform. is cut off by phys. means

— inf. does not get out.

There is such a force: gravitation! (and it is the only one)

Long-range and of one sign (!)

"Black hole" — a body that does not let information out.

Laplace thought of it: Speed of light = 2nd cosmic speed.

$$\frac{mv^2}{2} - \gamma\frac{mM}{R} = 0$$

$$\frac{mc^2}{2} = \gamma\frac{mM}{R},$$

$$\boxed{R = \frac{2\gamma M}{c^2} = r_g}$$

At this radius light will not leave the body (nor any signal)

— it turns out to hold in exact science too

GELFAND-P000715

$$R_{\text{Sun}} = 10^6 \text{ km}$$

Uncertain reading. An isolated sign before "For" is heavily crossed out.

Original source details: GELFAND-P000715

For its mass the required rad. is 7 km.

In principle we can know nothing about a black hole except a few params. — — mass, charge,

Original source details: GELFAND-P000715

Undeciphered passage. The entry after "charge" is heavily crossed out.

radius.

As with a gas. An enormous number of microstates. In fact it is a thermodyn. object!!

Hawking: the black hole's surf. S (!) increases in all processes. Merger of 2 black holes.
(earlier)
Geom. clearly increases (from Laplace's formula)

The letter S played a remarkable role.

$$\boxed{\frac{1}{T} = \frac{\partial S}{\partial E}} \quad \Rightarrow \quad T \text{ — the black hole's "temp.".}$$

Bekenstein — removed the quotation marks! Counted the information lost in forming
(Hawking's pupil)
black holes!! Agrees with informational entropy.

Let us throw a hot body into a black hole. It increases the surf. more than a cold one, in exact corresp. with the entropy increase.

[crossed out: Hawking.] Hawking. Merger of pairs of particles. The black hole radiates and its spectrum is Planckian!

$$\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1}$$

T — that temperature.

Inf. is lost not by stochasticization!!, but goes into black holes. (mechanically cut off)

Other ways

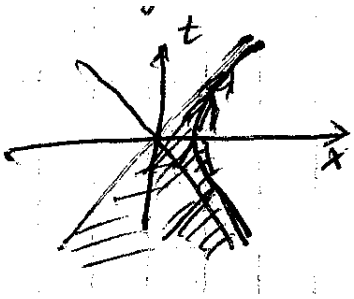
Uhnru

Original source details: GELFAND-P000715

Manuscript excerpt. The manuscript spells the surname "Uhnru." The passage refers to W. G. Unruh.

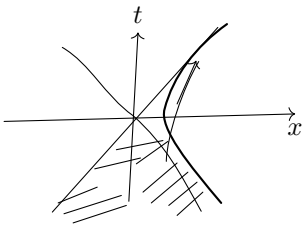
Homog. gravitational field.

Let a particle move with uniform acceleration in vacuum. Light from some places will not reach it.



Original source details: GELFAND-P000715

Diagram and typed description.

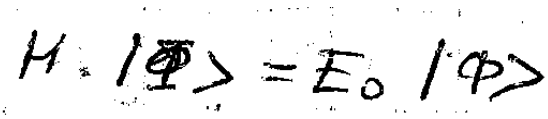


The exact joins of the overlapping strokes and curves at right are indeterminate.

It turns out that light reaches it only from half of space. The obs. will see (calculation) a Planck distribution. The zero-point oscillations look distorted to him. This is a difficult calculation.

H — Ham. of the electromagn. field.

$$H|\Phi\rangle = E_0|\varphi\rangle$$



Original source details: GELFAND-P000715

Manuscript excerpt. The left state letter is a heavily retraced Φ. The right state letter is φ.

The state vector from the point of view of a non-inert. obs. $\to$ Calc. prob. of $1^{\underline{\text{st}}}$, $2^{\underline{\text{nd}}}$ and more

Uncertain reading. Two clipped groups of strokes are visible beyond the right page border.

Original source details: GELFAND-P000715

GELFAND-P000716

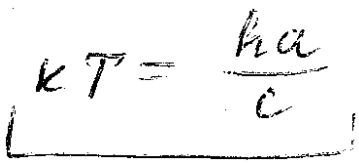
[source number: 48?]

Original source details: GELFAND-P000716

Uncertain reading. The last digit of the folio number tentatively reads 8, possibly 2.

photons.

$$\boxed{kT = \frac{ha}{c}} \quad \Rightarrow \quad \text{temperature}$$



Original source details: GELFAND-P000716

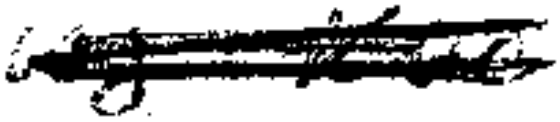
Manuscript excerpt. The first sign in the numerator is written h.

Heating of the gas at the finish.

Why is the universe hot?

[At left] Experiment — it is cooling now ||

[At right] $T = 3°\mathrm{K}$ — from the distrib. of



Undeciphered passage. A word before “Planck” is heavily crossed out.

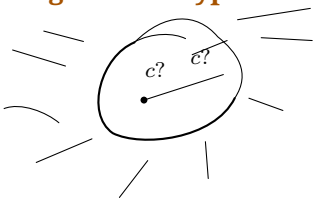
Original source details: GELFAND-P000716

Planck for rad. from space. [below the formula: (relic radiation)]

There was a region of the unknown. It arose at some point in time. Before this it makes no sense [below: to ask].

There is an event horizon.

Diagram and typed description.



The ends of the interior strokes and the two small signs are only partly legible.

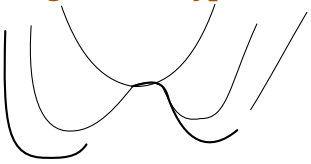
Original source details: GELFAND-P000716

closed region (light has had no time to arrive from there). (⇒ hence this temp. for us.)

—·—

Spontaneous symm. breaking.

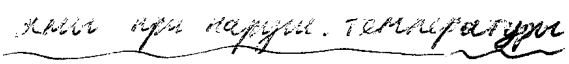
Diagram and typed description.



The precise joins where the lines overlap are indeterminate.

Original source details: GELFAND-P000716

evolution of the well upon a disruption of temperature



Manuscript excerpt. The caption reads “evolution of the well upon a disruption of temperature.”

Original source details: GELFAND-P000716

—·—

An idea of Senya Vishik, on why the Gelfand sem. is unsterilized: “To integrate successfully over a large domain one must use the Monte Carlo method.”

GELFAND-P000717

[source number: 55]

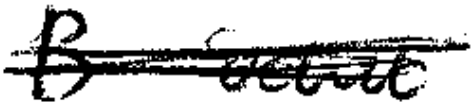
3/X—77 Gelfand Sem..

To be treated systematically: “noncommutative algebraic geometry”

L. A. Dikii. On new Lie algebras.

(Gelf., Dikii, Funct. Anal., 11:2).

Def. 1. The algebra A.



Uncertain reading. The crossed-out text at right begins “В” (In), followed by a word tentatively read as “себе.”

Original source details: GELFAND-P000717

$$u_k^{(i)}, \qquad k = 0, 1, \dots, n; \qquad i = 0, 1, 2, \dots$$

$$A = \left\{ \sum a\, u_{k_1}^{(i_1)} u_{k_2}^{(i_2)} u_{k_3}^{(i_3)} \cdots u_{k_N}^{(i_N)} \right\}, \qquad a \in \underset{\text{(field)}}{K},$$

A — the alg. of polynomials in the generators $u_k^{(i)}$

$$\boxed{A = K[u_k^{(i)}].}$$

$$\xi = \sum a_{k,i} \frac{\partial}{\partial u_k^{(i)}}, \qquad a_{k,i} \in A.$$

(the sum is infinite) ξ is called a derivation

$$\frac{d}{dx} = \sum_{k,i} u_k^{(i+1)} \frac{\partial}{\partial u_k^{(i)}}$$

$$\frac{d}{dx} u_k^{(i)} = u_k^{(i+1)}; \qquad u_k^{(i)} = \left(\frac{d}{dx}\right)^i u_k; \qquad u_k \underset{\text{def}}{=} u_k^{(0)}$$

Def. 2. The space $\widetilde{A}$ (space of functionals)

From now on, always $u_n = 1$, $u_{n-1} = 0$.

$$\left\{ f : \exists g,\ f = \frac{d}{dx} g \right\} = \frac{d}{dx} A;$$

$$\widetilde{A} = A \Big/ \frac{d}{dx} A$$

$$f \longmapsto f \bmod \frac{d}{dx} A = \widetilde{f} \in \widetilde{A}$$

$$A \underset{\int \cdots dx}{\longrightarrow} \widetilde{A} \qquad \widetilde{f} = \int f\, dx$$

Problem.

$$f[u] \in \mathbb{C}[u_k^{(i)}];$$

$$\int_{-\infty}^{\infty} f[u(x)]\,dx = 0, \quad \forall u \in S(\mathbb{R}^1) \quad \Longleftrightarrow \quad f \in \frac{d}{dx}\mathbb{C}[u].$$

Def. 3. $A^{n-1} =$ space of tuples $(f_0, f_1, \ldots, f_{n-2})$, $f_k \in A$

[in the left margin: $\frac{\delta}{\delta u}$.]

$$\frac{\delta}{\delta u} : \widetilde{A} \longrightarrow A^{n-1}$$

$$\left(\frac{\delta \widetilde{f}}{\delta u}\right)_k = \sum_{\ell=0}^{\infty} \left(-\frac{d}{dx}\right)^{\ell} \frac{\partial f}{\partial u_k^{(\ell)}} = \frac{\delta \widetilde{f}}{\delta u_k}$$

Manuscript excerpt. The letter f in the numerator under the sum is heavily retraced.

Original source details: GELFAND-P000717

(here $\widetilde{f} = f \bmod \frac{d}{dx}A$)

GELFAND-P000718

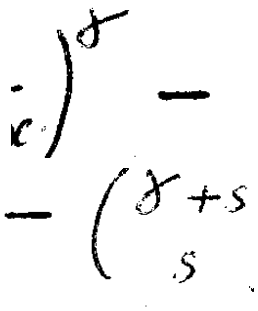
Def. 4. $\ell : A^{n-1} \longrightarrow A^{n-1}$

$$f = (f_0, \ldots, f_{n-2})$$

$$(\ell f)_r = \sum_s \ell_{rs} f_s$$

$$\ell_{rs} = \sum_{\gamma=0}^{n-1-r-s} \left[\binom{\gamma + r}{r} u_{r+s+\gamma+1} \left(-i\frac{d}{dx}\right)^{\gamma} - \right.$$

$$\left. - \binom{\gamma + s}{s} \left(i\frac{d}{dx}\right)^{\gamma} u_{r+s+\gamma+1} \right]$$



Manuscript excerpt. A minus is written at the end of the first line and again at the start of the second.

Original source details: GELFAND-P000718

$n = 2, 3$ easy; $n = 4$ — 1st substantial example.

Def. 5. $\forall f \in A^{n-1}$ we construct

$$\partial_f = \sum_{k,\ell} f_k^{(\ell)} \frac{\partial}{\partial u_k^{(\ell)}} \qquad \text{(a derivation)}$$

It is easy to check that

$$\partial_f \frac{d}{dx} = \frac{d}{dx} \partial_f$$

The operation is therefore defined $\partial_f : \widetilde{A} \longrightarrow \widetilde{A}$.

Def. 6. Let $f, g \in A^{n-1}$. Put

$$([f,g])_r = \partial_f g_r - \partial_g f_r$$

It is easy to check that

$$[\partial_f, \partial_g] = \partial_{[f,g]}.$$

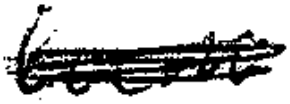
Hence the commutator $[\cdot,\cdot]$ defines on A^{n-1} the structure of a Lie algebra.

Def. 7. def. of the commutator in $\widetilde{A}$.

$$\{\widetilde{f}, \widetilde{g}\} = \int \sum_{r,s} \frac{\delta \widetilde{f}}{\delta u_r} \cdot \ell_{rs} \frac{\delta \widetilde{g}}{\delta u_s}\, dx$$

Theorem:

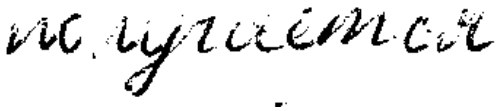
Undeciphered passage. A word after "Theorem" is heavily crossed out.



Original source details: GELFAND-P000718

we get a Lie algebra

Manuscript excerpt. Before "a Lie algebra," the verb reads "получается" ("we get").



Original source details: GELFAND-P000718

(i.e. the Jacobi identity is satisfied)

Problem: prove it more simply.

S. P. Novikov $A = \frac{d^3}{dx^3} + \left(u\frac{d}{dx} + \frac{d}{dx}u\right)$

also defines a Lie algebra.

(and the KdV eq. can be written with its help).

GELFAND-P000719

[source number: 56]

S. I. Gelfand. on the work of I. M. Gelf., I. N. Bernsh. and his own?:

Uncertain reading. The last word tentatively reads "свои." (his own). The final strokes run together.

Original source details: GELFAND-P000719

“The local structure of the category of Harish–Chandra modules”
or “Noncommutative algebraic geometry I”.

G — a real Lie group, semisimple

$K \subset G$, K — a maximal compact subgroup

$\mathfrak{g}$ — the Lie algebra of the group G

Definition. A Harish–Chandra module M — is a linear space over $\mathbb{C}$ with actions of $\mathfrak{g}$ and K and with conditions (besides the natural conds. for the actions and compatibility of the actions of $\mathfrak{g}$ and K).

1) the actions of [crossed out: the Lie algebra of the group K] $\mathfrak{g}$ and K are compatible:

a) the Lie alg. of K acts on M as a subalgebra in $\mathfrak{g}$.

b) [crossed out: a)?]

Uncertain reading. A sign after “б)” is crossed out. It tentatively reads “а)”.

Original source details: GELFAND-P000719

$$T\big(\underset{\uparrow K}{k}\ \underset{\uparrow \mathfrak{g}}{X}\ k^{-1}\big) = T_1(k)T(X)T_1(k^{-1}).$$

T — the action of $\mathfrak{g}$, T_1 — the action of K

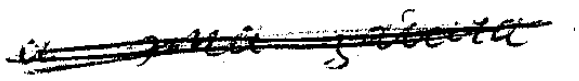
2) M decomposes with respect to K into a direct sum of [inserted above: irred.] representations and each [inserted below: irred.] repr. occurs a finite number of times.

H.–C. mods. form a category. Morphisms are defined in the natural way.

Problem: description of this category.

Idea: there is a noncommut. algebra responsible for this.

Gelfand, Ponomarev: for the Lorentz group [inserted below: $SL(2,\mathbb{C})$] it reduces to classifying pairs of matrices A, B, such that $AB = BA = 0$, [crossed out: and]



Undeciphered passage. Two more words after “and” are crossed out.

Original source details: GELFAND-P000719

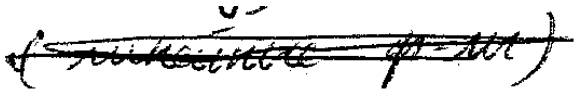
$U(\mathfrak{g})$ — the enveloping alg. of $\mathfrak{g}$

$Z(\mathfrak{g})$ — its centre

$$M \rightsquigarrow J_M = \{z \in Z(\mathfrak{g}) \mid zM = 0\} \qquad \text{(an ideal) in } Z(\mathfrak{g}).$$

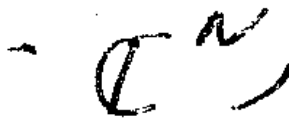
Θ — the set of characters $\theta : Z(\mathfrak{g}) \to \mathbb{C}$. (homomorphisms)

[crossed out beneath the formula: (linear formulas?)] $(\subset \mathbb{C}^N)$.



Uncertain reading. The crossed-out abbreviation after “linear” tentatively reads “ф-лы” (formulas).

Original source details: GELFAND-P000719



Uncertain reading. The case of the exponent in $\mathbb{C}^N$ is unclear, N or n.

Original source details: GELFAND-P000719

$$\operatorname{supp} M \subset \Theta$$
$$\operatorname{supp} M = \{\theta \mid J_M \subset \ker \theta\}$$

$\operatorname{supp} M$ — a finite set.

Let $D \subset \Theta$,

$$C_D = \{M \mid \operatorname{supp} M \subset D\}.$$

GELFAND-P000720

We can consid. H.–C. modules with supp. at a point $C_{\{\theta\}}$

Theorem (easy).

$$C_D = \bigoplus_{\theta \in D} C_{\{\theta\}}$$

Undeciphered passage. A mark between the equals sign and $\bigoplus$ is crossed out.

Original source details: GELFAND-P000720

in a certain sense,

(they decompose and there are no homomorphisms).

1st main theorem. Let D — a bounded domain in Θ (in the complex sense). Then $\exists$ $Z(\mathfrak{g})$-algebra Q

Uncertain reading. A small mark at the lower right of Q is crossed out. It may be a subscript.

Original source details: GELFAND-P000720

[crossed out: such] with a finite number a of additive $Z(\mathfrak{g})$-generators, [inserted above: such that] [crossed out: Then?] C_D is equivalent to the category of Q-modules that are finite-dimensional over $\mathbb{C}$.

Uncertain reading. The crossed-out word before C_D tentatively reads “Тогда” (Then).

Original source details: GELFAND-P000720

(∃ a functor realizing the equivalence)

Q — $Z(\mathfrak{g})$

— · —

"Izvestia" and "Socialist Industry" [crossed out: dated the 23rd]

$23/\overline{\text{IX}}\text{-}77$ $22/\overline{\text{IX}}\text{-}77$

a report from Petrozavodsk of an unidentified flying object

— · —

Example.

$$G = SL_2(\mathbb{R})/\{\pm 1\}$$
$$Z(\mathfrak{g}) \cong \mathbb{C}[z]$$
$$\Theta = \mathbb{C}^1$$

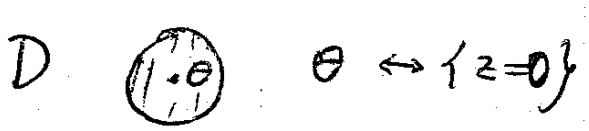


Diagram and typed description.

D $\theta \leftrightarrow \{z = 0\}$

Original source details: GELFAND-P000720

We will construct the algebra over analytic fns.

$$Q = (a_{ij}(z))_{i,j=1}^{3} \text{ — matr. } 3 \times 3, \qquad a_{ij}(z) \text{ — entire fns.}$$

$$(a_{ij}(0)) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$

GELFAND-P000721

10/X—77 Gelfand Sem..

S. Gelfand.

G — a semisimple real Lie group

K — a maximal compact subgroup

$\mathfrak{g}$ — the Lie algebra of the group G

$U(\mathfrak{g})$ — the enveloping algebra.

$Z(\mathfrak{g})$ — the centre of $U(\mathfrak{g})$

Harish–Chandra module (HCM) V, if

[H.–C. thm.: in an irreduc. repr. of a semisimple group, every [inserted above: irred.] repr. of the max. comp. group K occ. a finite no. of times]

In V there are given reprs. T_1 of the alg. $U(\mathfrak{g})$, T_2 — a repr. of the group K.

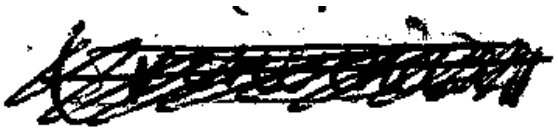
1) V — a finitely gen. $U(\mathfrak{g})$-module.

2) $T_1|_{\mathfrak{k}} = dT_2$,

$$T_2(k)T_1(X)T_2(k^{-1}) = T_1(kXk^{-1})$$

3) $T_2 = \bigoplus$ irred. cont. reprs. of K, and each occurs a finite number of times.

(direct sum in the literal sense).

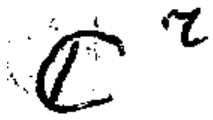


Undeciphered passage. A short passage before "Result" is heavily crossed out.

Original source details: GELFAND-P000721

Result: let

$$\Theta = \{\theta : Z(\mathfrak{g}) \to \mathbb{C}\} = \mathbb{C}^r$$



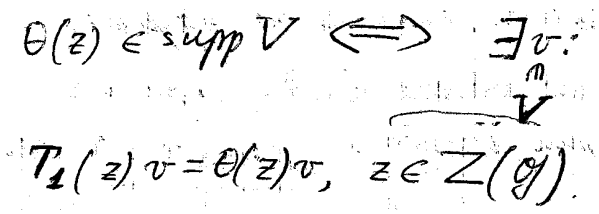
Uncertain reading. The exponent tentatively reads r, possibly τ.

Original source details: GELFAND-P000721

(ring homomorphisms)

$$V \mapsto \operatorname{supp} V \subset \Theta$$

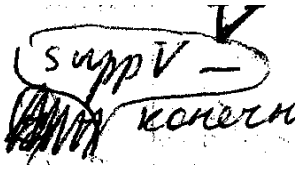
$$\theta(z) \in \operatorname{supp} V \iff \exists \underset{V}{\cap} v : T_1(z)v = \theta(z)v, \quad z \in Z(\mathfrak{g}).$$



Manuscript excerpt. The manuscript has $\theta(z)$ on the left and marks membership in V with a rotated sign below v. The condition $v \neq 0$ is absent.

Original source details: GELFAND-P000721

In fact



Original source details: GELFAND-P000721

Undeciphered passage. A short passage under the insertion “supp V —” is heavily crossed out.

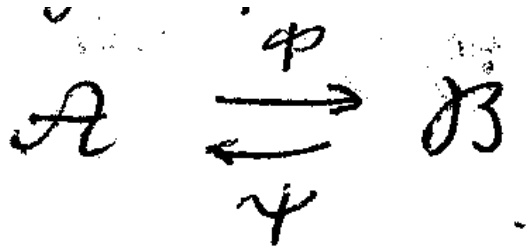
[inserted above: supp V —] a finite set.

Let there be a bounded domain $D \subset \Theta$,

$$C_D = \{V : \operatorname{supp} V \subset D\}.$$

For disjoint sets D_1 and D_2 we get the direct sum of categories.

Equivalence of categories:



Original source details: GELFAND-P000721

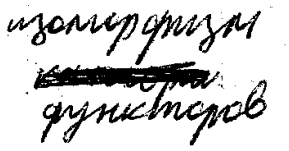
Diagram and typed description.

$$\mathcal{A} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{B}$$

φ, ψ — functors

$$\varphi \circ \psi \underset{\uparrow}{\sim} \mathrm{id}_B, \qquad \psi \circ \varphi \underset{\nearrow}{\sim} \mathrm{id}_A$$

[arrows from the inscription to both signs] isomorphism



Original source details: GELFAND-P000721

Undeciphered passage. A word between “isomorphism” and “of functors” is heavily crossed out.

of functors

GELFAND-P000722

[source number: 61]

Isomorphism of functors:

$$\forall a: \qquad \psi\varphi(a) \longleftrightarrow a$$

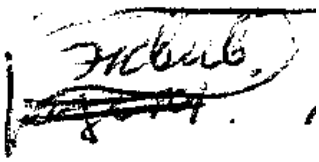
(do not coincide, but are isomorphic!) and all diags. commute.

Example of equiv. but non-isom. categories:

1) categ. of finite-dimensional lin. spaces over $\mathbb{C}$

2) — ″ — of finite-dimensional modules over $M(2,\mathbb{C})$.

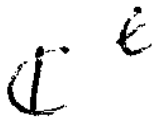
The category of HCMs [enclosed insertion above: is equiv. to] a certain category of finite-dimensional modules.



Original source details: GELFAND-P000722

Undeciphered passage. A word below the insertion “эквив.” (equiv.) is heavily crossed out.

$\underset{\text{open}}{E} \subset \mathbb{C}^{\ell}$, B — a direct sum of several matrix algebras.



Original source details: GELFAND-P000722

Uncertain reading. The exponent on $\mathbb{C}$ tentatively reads ℓ.

$B(E)$ — analytic fns. on E with values in B

Def. A subalg. $R \subset B(E)$ is called nice, if

1) it cont. all scalar fns.:

$$f(z)\mathbf{1}_B \in R$$

2) $\{f(z), f \in R\} = B$ for almost all $z \in E$

(i.e. for all except an analytic subset of smaller dimension)

[crossed out: Theorem]

Original source details: GELFAND-P000722

Manuscript excerpt. The crossed-out heading reads “Теорема” (“Theorem”).

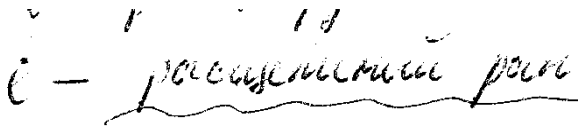
Mod R — categ. of R-modules that are finite-dimensional lin. spaces.

Theorem. $C_D \sim \text{Mod}\, R$

for a certain nice alg. R.

r — rank of the group

ℓ — split rank of the group.



Original source details: GELFAND-P000722

Uncertain reading. The sign before "split rank" tentatively reads ℓ.

— · —

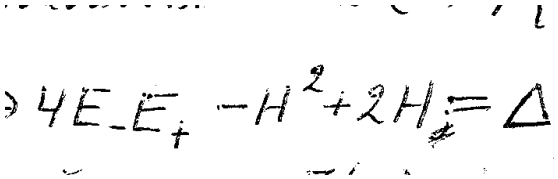
The rest will be on an example.

$$G = [\text{crossed out } SL(2,\mathbb{R})] \quad SL_2(\mathbb{R})/\{\pm 1\}, \qquad \underset{\text{Lie alg.}}{\mathfrak{sl}_2(\mathbb{R})} = \{E_+, E_-, H\}$$

Original source details: GELFAND-P000722

Manuscript excerpt. The crossed-out expression after $G =$ reads $SL(2,\mathbb{R})$.

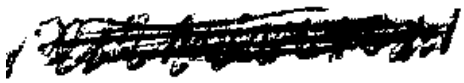
$$Z(\mathfrak{g}) \ni 4E_-E_+ - H^2 + 2H := \Delta \qquad E_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Original source details: GELFAND-P000722

Uncertain reading. The manuscript has a minus before H^2. The crossed-out strokes below the defining sign for Δ are illegible.

Every other element of $Z(\mathfrak{g})$ is a polynomial



Original source details: GELFAND-P000722

Undeciphered passage. A passage after "polynomial" is heavily crossed out.

in Δ.

$\Theta = \mathbb{C}^1$, $D =$ a neighbourhood of the point 0, suff. small

Original source details: GELFAND-P000722

Uncertain reading. The sign after "point" is open at the right and tentatively reads 0.

A nice subalgebra:

$$E = D, \qquad B = M_3(\mathbb{C}), \qquad R \subset \begin{pmatrix} a_{11}(z) & a_{12}(z) & a_{13}(z) \\ \cdot & - & - \\ \cdot & - & - \end{pmatrix} = (a_{ij}(z))$$

GELFAND-P000723

The condition singling out R:

$$a_{ij}(0) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$

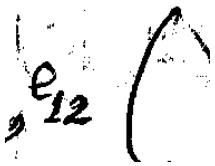
There is a basis of 9 elements over the fns.

$$e_{11} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_{12} \begin{pmatrix} 0 & z & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and so on.

Original source details: GELFAND-P000723

Manuscript excerpt. The first entry of the matrix by e_{12} is a heavily retraced zero.



Original source details: GELFAND-P000723

Manuscript excerpt. The opening parenthesis of the matrix follows e_{12} directly.

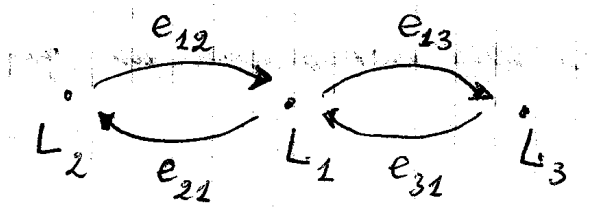
$$e_{21}e_{12} = ze_{22}$$

We want to find finite-dimensional modules for which mult. by z is a nilpot. operator.

$$e_{ii}^2 = e_{ii}.$$

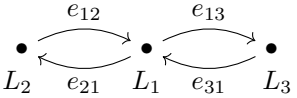
$$L_i = \{m : \ e_{ii}m = m\}.$$

$$M = L_1 \oplus L_2 \oplus L_3$$



Original source details: GELFAND-P000723

Diagram and typed description.



In the manuscript, e_{13} points from L_1 to L_3, and e_{31} from L_3 to L_1.

$e_{21}e_{12} = ze_{22}$ is nilpotent

$$e_{12}e_{21} = ze_{11} = e_{13}e_{31}$$

hence the categ. of modules over this ring $\cong \ldots$

For $SL(3,\mathbb{R})$ the answer is as follows:

$$E = \{z_1, z_2\} \qquad \text{in a nbhd. of zero:}$$

$$B = \underset{a_{ij}}{M_4} \oplus \underset{b_{ij}}{M_3} \oplus \underset{c_{ij}}{M_3} \oplus \underset{d}{M_1}$$

Conditions.

1) At $z_1 = 0$ $\quad\Big\|\quad$ $a_{13} = a_{14} = a_{23} = a_{24} = b_{12} = b_{13} = 0$, $a_{33} = b_{22}, \quad a_{34} = b_{23}, \quad a_{43} = b_{32}, \quad a_{44} = b_{33}$

GELFAND-P000724

[source number: 62]

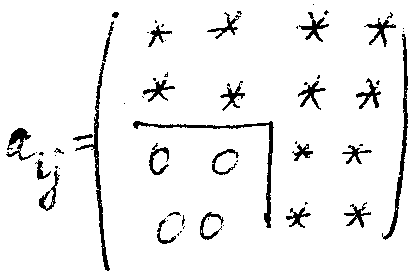


Diagram and typed description.

$$a_{ij} = \left(\begin{array}{cc|cc} * & * & * & * \\ * & * & * & * \\ \hline 0 & 0 & * & * \\ 0 & 0 & * & * \end{array}\right)$$

Original source details: GELFAND-P000724

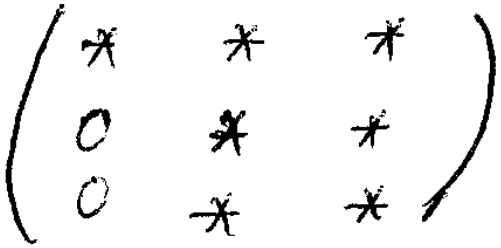
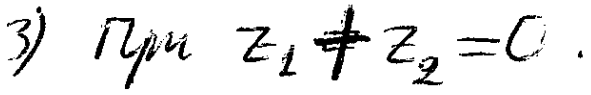


Diagram and typed description. $\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$

Original source details: GELFAND-P000724

2) When $z_2 = 0$ $\quad\Big\|\quad$ $a_{12} = a_{14} = a_{34} = c_{12} = c_{13} = a_{32} = 0$, $a_{22} = c_{22}, \quad a_{24} = c_{23}, \quad a_{42} = c_{32}, \quad a_{44} = c_{33}$

3) When $z_1 + z_2 = 0$. $\quad\Big\|\quad$ $b_{13} = b_{23} = c_{13} = c_{23} = 0$; $b_{33} = c_{33}$.



Uncertain reading. The overwritten sign between z_1 and z_2 is tentatively read as $+$.

Original source details: GELFAND-P000724

GELFAND-P000725

24/X–77

S. I. Gelfand. Harish-Chandra Modules

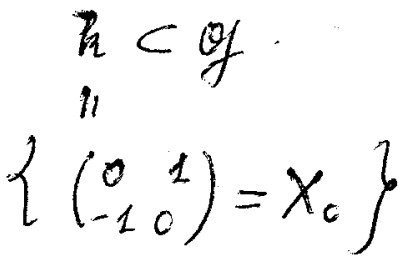
HC modules for

$$G = SL(2,\mathbb{R})/(\pm 1)$$
$$K = SO_2/(\pm 1)$$
(max. comp. subgroup)

$\mathfrak{g}$ — Lie algebra

$$\underset{\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}}{E_+}, \quad \underset{\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}}{E_-}, \quad \underset{\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}}{H}$$ — generators of $\mathfrak{g}$

$$\begin{array}{l}\mathfrak{k} \subset \mathfrak{g}.\\ \|\\ \left\{\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} = X_c\right\}\end{array}$$



Uncertain reading. The subscript on X reads as an open c.

Original source details: GELFAND-P000725

GELFAND-P000726

[source number: 67]

A module H.–C. — is a (infinite-dimensional) repr. in [crossed out] V, such that

Uncertain reading. The crossed-out fragment before V is illegible.

Original source details: GELFAND-P000726

1) X_0 has only integer eigenvalues, and even ones at that

2) [crossed out] each eigenvalue [inserted above: of the operator X_0] has finite multiplicity

Uncertain reading. The heavily crossed-out mark after 2) is illegible.

Original source details: GELFAND-P000726

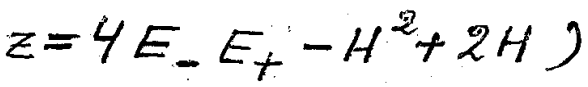
3) V is finitely generated (over the envelop. algebra)

($\Leftrightarrow$ the eigenvalue multiplicities are bounded).

$v_1, \dots, v_n$ — generators

> *[In the right margin beyond the double line]* 1) — so that one can integrate to a group repr. (e^{inx} etc.)

Definition. $Z(\mathfrak{g})$ — the centre of the envelop. algebra $U(\mathfrak{g})$
(gen. by the element $z = 4E_-E_+ - H^2 + 2H$)



Manuscript excerpt. The manuscript has a minus before H^2 and a plus before $2H$.

Original source details: GELFAND-P000726

supp $V =$ the set of eigenvalues of the operator z in V

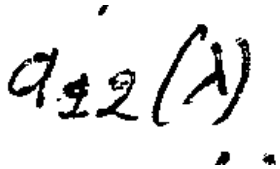
Theorem: this is a finite set for any V.

Definition. Let D — a domain in $\mathbb{C}$.

$$C_D = \{V,\ \operatorname{supp} V \subset D\}$$

Let D — a small neighbourhood of zero (disc of radius $< 1/2$).

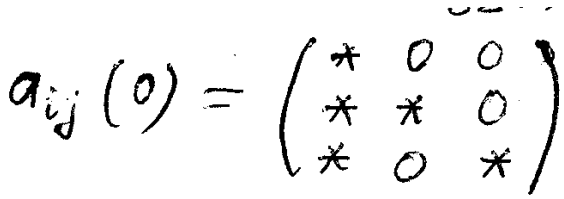
$$R = \left\{ \begin{pmatrix} a_{11}(\lambda) & a_{12}(\lambda) & a_{13}(\lambda) \\ a_{21}(\lambda) & a_{22}(\lambda) & a_{23}(\lambda) \\ a_{31}(\lambda) & a_{32}(\lambda) & a_{33}(\lambda) \end{pmatrix} \right., \quad a_{ij}(\lambda) \text{ — analytic fns. in } D$$



Manuscript excerpt. The first digit of the subscript on $a_{12}(\lambda)$ is overwritten. The visible digit is 1.

Original source details: GELFAND-P000726

$$a_{ij}(0) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$



Manuscript excerpt. The left-hand side is written $a_{ij}(0)$.

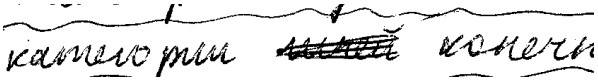
Original source details: GELFAND-P000726

$$\operatorname{Mod} R = \left\{ \underset{R\text{-module}}{M} \;:\; \dim_{\mathbb{C}} M < \infty \right\}$$

Theorem. The category C_D is equivalent to the category Mod R.

Example.

The category of modules over the matrix ring is equivalent to the category of [crossed out: linear...?] finite-dimensional linear spaces



Original source details: GELFAND-P000726

Uncertain reading. The crossed-out word tentatively begins “linear”; the ending is uncertain.

($\sim$ a repr. of the matrix alg. is a direct sum)

V — the stand. repr. of the alg. M_ℓ

Original source details: GELFAND-P000726

Uncertain reading. The subscript on M is tentatively read as ℓ; the mark also resembles e.

$$E \longmapsto V \underset{\mathbb{C}}{\otimes} E, \qquad M_\ell \text{ acts on the first variable}$$

$$W \text{ — } M_\ell\text{-module} \quad \longmapsto \quad E = \operatorname{Hom}_{M_\ell}(V, W).$$

(the 2 functors giving the equivalence are written out)

Old formulation: $\forall\, M_\ell$ module decomposes as a direct sum.

GELFAND-P000727

Proof of the theorem.

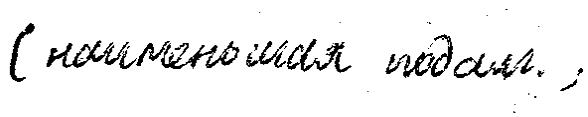
I. Let a list of numbers be given $n_1, \ldots, n_k$. Construct from it the ring $Q_{n_1,\ldots,n_k}$

a) $J_{n_1 \ldots n_k}$ — the left ideal in $U(\mathfrak{g})$, generated by the element

$$(X_0 - 2in_1)(X_0 - 2in_2) \cdots (X_0 - 2in_k)$$

b) $N_{n_1,\ldots,n_k} = \{u \in U(\mathfrak{g}) \mid J_{n_1,\ldots,n_k} u \subset J_{n_1,\ldots,n_k}\}$

(the smallest subalg., in which J — a 2-sided ideal



Original source details: GELFAND-P000727

Manuscript excerpt. The manuscript reads “smallest”.

$$Q_{n_1 \dots n_k} = N_{n_1,\dots,n_k}/J_{n_1,\dots,n_k}$$

Def. V — an H.–C. module

$$\tau_{n_1,\dots,n_k}(V) \underset{\text{def}}{=} \{v : \ (X_0 - 2in_1)\cdots(X_0 - 2in_k)v = 0\}$$

This is a finite-dimensional subspace

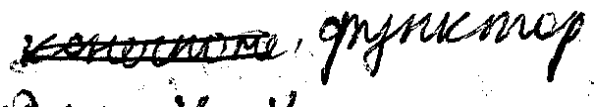
Dimension $=$ sum of the multiplicities

N maps it into itself

J —"— to zero

$$\Rightarrow \quad Q_{n_1,\dots,n_k} \text{ acts in } \tau_{n_1,\dots,n_k}(V)$$

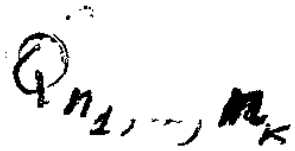
This is [crossed out: construction?], a functor



Original source details: GELFAND-P000727

Uncertain reading. The crossed-out word is tentatively read as "construction".

H.–C. modules $\longrightarrow$ finite-dimensional modules over $Q_{n_1,\dots,n_k}$



Original source details: GELFAND-P000727

Manuscript excerpt. The first mark in the last subscript of Q is retraced. The last subscript reads n_k.

It is clear that

$$z \in N_{n_1,\dots,n_k}$$

$$\operatorname{supp} M = \{\text{eigenvalues of } z\}$$

$$(\operatorname{Mod} Q_{n_1,\dots,n_k})_D$$

(finite-dimensional modules with support in D).

Original source details: GELFAND-P000727

Diagram and typed description.

$\longmapsto$

The short arrow points left and has a crossbar at the right. Its connection to the adjacent line is uncertain.

Theorem 1. Let D — be any bounded domain in $\mathbb{C}^1$. Then there exist $n_1, \dots, n_k$, such that C_D and $(\operatorname{Mod} Q_{n_1,\dots,n_k})_D$ are equiv.

GELFAND-P000728

[source number: 68]

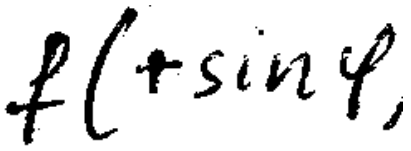
II. Definition of the reprs. of the principal series.

from s construct the module $T^{(s)} = V$

$$V = \{ \underset{\substack{\text{functions}\\ \text{on the plane}}}{f(x,y)} \}$$

1) $f(tx, ty) = |t|^s f(x, y)$

2) $f(+\sin\varphi, \cos\varphi) = P(\sin\varphi, \cos\varphi)$



Manuscript excerpt. The first argument has a $+$ before $\sin\varphi$.

Original source details: GELFAND-P000728

P — a polynomial.

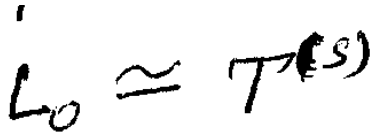
The Lie algebra acts in the natural way (diff. operators)

(the group acts on the argument)

$$E_+ \sim x\frac{\partial}{\partial y}; \quad E_- \sim y\frac{\partial}{\partial x}; \quad H \sim x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

Restr. to $S^1 \cong$ the space of polynomials (even on the circle)

$$L_0 \simeq T^{(s)} \qquad \text{(for any } s\text{)}.$$



Manuscript excerpt. The opening parenthesis in the superscript (s) is heavily retraced.

Original source details: GELFAND-P000728

Lemma 1. Let $X \in U(\mathfrak{g})$. The action of X depends polynomially on s.

(Matrix entries in the monomial basis are polynomials in s)

$$\tau_{n_1,\dots,n_k}(T^{(s)}) = L_{n_1,\dots,n_k}$$

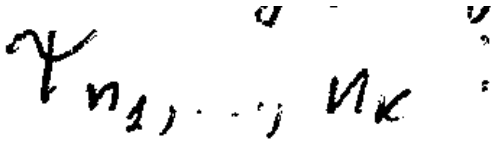
Lemma 2. Let $q \in Q_{n_1,\dots,n_k}$. The action of q depends polynomially on s.

Theorem 2. Set

$$B_{n_1,\dots,n_k} = \mathrm{End}(L_{n_1,\dots,n_k})$$

Then there exists a map

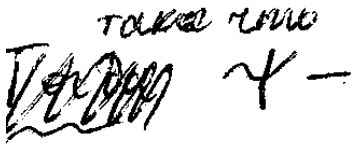
$$\psi_{n_1,\dots,n_k} : Q_{n_1,\dots,n_k} \longrightarrow B_{n_1,\dots,n_k}[s],$$



Uncertain reading. The symbol for the map is tentatively read as ψ.

Original source details: GELFAND-P000728

[crossed out: Thm.?] [inserted above: such that] ψ — an embedding



Uncertain reading. The crossed-out heading tentatively reads “Teop.” (Thm.).

Original source details: GELFAND-P000728

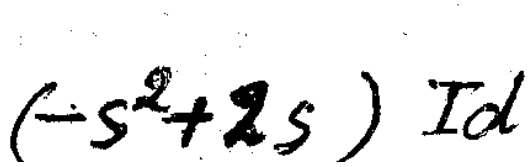
(i.e. the principal series contains as sub- and quotient reprs. $\forall$ HC modules)

III. Combining stages I and II.

D — a small nbhd. of zero

Lemma 0.

On $T^{(s)}$ the operator z is $(-s^2+2s)\,\mathrm{Id}$



Manuscript excerpt. The manuscript gives $(-s^2+2s)\,\mathrm{Id}$.

Original source details: GELFAND-P000728

$z=0$ for $s=0$ and $s=2$.

Lemma 1. Let s be near the point 0.

1) Then T^s, $s\neq 0$ is irreducible

$$0 \longrightarrow \mathrm{Id} \longrightarrow T^{(0)} \longrightarrow T_+ \oplus T_- \longrightarrow 0$$

Manuscript excerpt. The second letter of Id is heavily retraced.

Original source details: GELFAND-P000728

and the exact seq. does not split

GELFAND-P000729

$$2)\quad T^{(s)}\big|_K = \bigoplus_{j=-\infty}^{\infty} \delta_j.$$

$\delta_k =$ eigensp. of X_0 with eigenval. $2ik$.

(1-dimensional repr.)

$$I\big|_K = \delta_0, \qquad T_+\big|_K = \bigoplus_{j>0} \delta_j, \qquad T_-\big|_K = \bigoplus_{j<0} \delta_j.$$

Uncertain reading. A narrow mark resembling I is visible beyond the right border. Its connection to the text is uncertain.

Original source details: GELFAND-P000729

[In the left margin: B] Lemma 2. Near $s = 2$

Manuscript excerpt. A B-shaped letter in the left margin is heavily retraced.

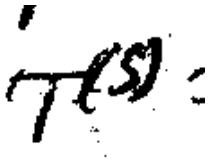
Original source details: GELFAND-P000729

$$0 \longrightarrow T_+ \oplus T_- \longrightarrow T^{(2)} \longrightarrow I \longrightarrow 0$$

$T^{(s)}$, $s \neq 2$ is irreducible

Lemma 3. $s \neq 0$ $\quad T^{(s)} \simeq T^{(2-s)}$

s near zero



Manuscript excerpt. The opening parenthesis in the superscript of $T^{(s)}$ is retraced.

Original source details: GELFAND-P000729

Consider the numbers $(n_1, n_2, n_3) = (-1, 0, 1)$ and the ring

$$Q_{(-1,0,1)}$$

D — a suff. small nbhd. of zero $\quad \Rightarrow \quad C_D \sim Q_{(-1,0,1)}$

(explanation: in T_+ there is a repr. with $+1$, in T_- — " — with -1 etc.)

$$\tau_{-1,0,1}(T^{(s)}) = \{e_{-1}, e_0, e_1\}$$

$q \in Q_{-1,0,1}$

. . . . . .

Done: $SL(2,\mathbb{R})/(\pm 1)$, $SL(2,\mathbb{R})$, $PGL(2,\mathbb{R})$,
$SL(2,\mathbb{C})$, $SL(3,\mathbb{R})$

Probably possible $SL(3,\mathbb{C})$ and $SU(2,1)$

Recommended, calc. $SO(4,1)$ (easy), $Sp(4,\mathbb{C})$ (easy)

Difficult: $SL(4,\mathbb{R})/(\pm 1)$, $SL(4,\mathbb{C})$.

Read the works of Hirai, Zhelobenko

Gelfand, Graev

Krichever

2 minutes for a sensational announcement.

[In the left margin: On eqn.]

$$[L_1, L_2] = 0$$

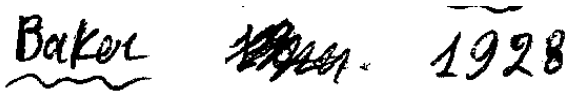
Manuscript excerpt. The underlined Russian abbreviation in the left margin reads "Об ур-и".

Original source details: GELFAND-P000729

Coddington pointed to the works of

Burchnal, Chaundy 1922, 1928, 1931

and Baker [crossed out] 1928



Uncertain reading. The writing between "Baker" and "1928" is heavily crossed out.

Original source details: GELFAND-P000729

[To the right of the list of works:] — they objected to Baker, who recommended using alg. geom.

The problem of describing commuting ops. with scalar coeffs. of relatively prime ord. was posed and solved.

the answer uses no Θ-fns. and is insufficiently effective

the 1st was found via Lamé fns. (as in Akhiezer). they did not see the straight-line behav.

GELFAND-P000730

[source number: 69]

path on the Jacobian (and even conditional periodicity).

Forgotten — in the 1939 paper there are no longer any references.

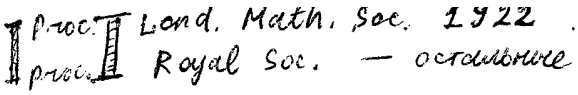


Diagram and typed description.
Proc. Lond. Math. Soc. 1922.
Proc. Royal Soc. — the others

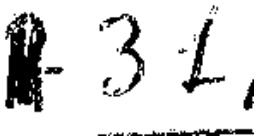
Original source details: GELFAND-P000730

almost commuting [crossed out] and Lax pairs exist.

Uncertain reading. A small mark after "almost commuting" is crossed out.

Original source details: GELFAND-P000730

[crossed out] 31/X–77 Gelfand Sem. S. P. Novikov. Instantons.



Uncertain reading. The heavily crossed-out mark before the date is illegible.

Original source details: GELFAND-P000730

$$\mathbb{R}^4 \subset S^4 \qquad (|x| \to \infty)$$

$A =$ [crossed out] $A_\mu \, dx^\mu$ connection form

Uncertain reading. The crossed-out expression after $A =$ appears to contain A and d. The indices are illegible.

Original source details: GELFAND-P000730

If the curvature is 0, then

$$(1) \qquad A_\mu \longrightarrow \frac{\partial g}{\partial x^\mu}\, g^{-1}(x), \qquad g(x) \in G$$

$$A_\mu(x) \in L_G \qquad \text{(Lie algebra)}$$

Curvature form:

$$(2) \qquad F_{\nu\mu} = dA + [A, A] = \mathcal{D}A$$

(covariant derivative)

Most often $G = SU_2$ (Yang–Mills).

$[A, A]$ — exterior prod., but mult. of coeffs. is in the Lie alg..

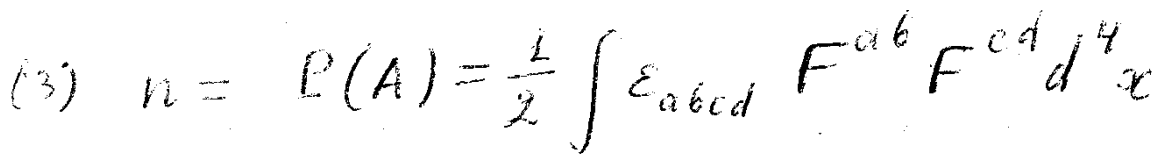
Other notation: $A \wedge A = [A, A]$, $\mathcal{D}A = \nabla_A A$

The requirement that everything be on S^4 — is slightly stronger than finite energy.

(Rem. Nonlinear (chiral) fields $\leftrightarrow$ connection with zero curvature; given by fns. with values in the group.)

Set

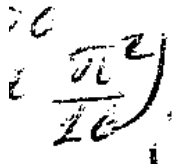
$$(3) \qquad n = P(A) = \frac{1}{2}\int \varepsilon_{abcd} F^{ab} F^{cd}\, d^4x$$



Original source details: GELFAND-P000730

Manuscript excerpt. Equation (3) has no written trace symbol before the product $F^{ab}F^{cd}$.

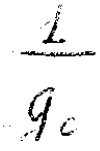
(char. Chern class — top. invariant — an integer up to a factor $\frac{\pi^2}{16}$).



Original source details: GELFAND-P000730

Manuscript excerpt. The denominator of the fraction reads 16.

$$(4) \qquad S(A) = \frac{1}{g_c}\int_{\mathbb{R}^4} \mathrm{Sp}(F_{\mu\nu}F^{\mu\nu})\, d^4x$$

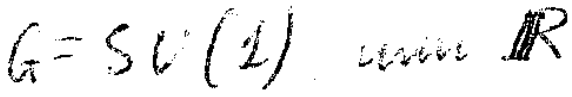


Original source details: GELFAND-P000730

Uncertain reading. The open subscript on g tentatively reads c, possibly 0.

(Yang–Mills functional)

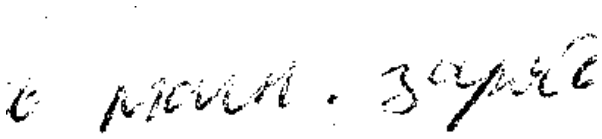
Class. electrodynamics — $G = SU(1)$ or $\mathbb{R}$



Original source details: GELFAND-P000730

Manuscript excerpt. The manuscript reads $SU(1)$.

depending on whether there are real charges or not.



Manuscript excerpt. The abbreviation before “charges” reads “реал.” (real).

Original source details: GELFAND-P000730

Yang–Mills eqns.:

$$(5) \qquad \delta S = 0.$$

Class. problem: find all such A, that $S(A) < +\infty$.

Rem. The metric of S^4 is conformally equiv. to the metric of $\mathbb{R}^4$,

$$g_{\mu\nu}(y) = g(y)\delta_{\mu\nu}$$

The Maxwell and Yang–Mills eqns. are conformally invariant.

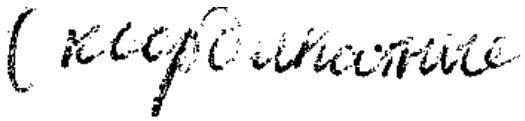
Grp. of conformal transfs. of Euclidean sp. and the sphere $\quad \| \quad O(5,1) = K_4$

GELFAND-P000731

Solutions are mult. by a scalar under the action of the conformal group:

$$f : X \longrightarrow X, \qquad f^*(g_{ij}) = g(x)g_{ij}.$$

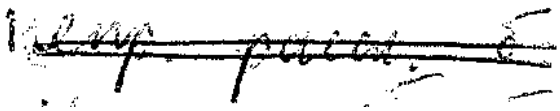
(continuous conf. transfs.)



Uncertain reading. The first word in parentheses tentatively reads “continuous”.

Original source details: GELFAND-P000731

[crossed out: Cont. bund. into?]



Uncertain reading. The crossed-out line tentatively reads “Непр. рассл. в” (Cont. bund. into).

Original source details: GELFAND-P000731

A cont. map of bases takes a bundle to a bundle and a connection to a connection. Then the new connection again satisfies the Yang–Mills eqn..

$O(5,1)$

On the sphere S^4 there is $O(5)$

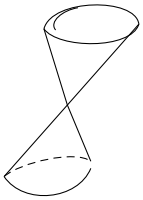
Stereogr. transf. S^4 onto $\mathbb{R}^4$ and in $\mathbb{R}^4$ take similarity transfs..

Where does $O(5,1)$?

We must consider the cone

$$x_1^2 + x_2^2 + \cdots + x_5^2 - x_6^2 = 0.$$

Diagram and typed description.



Original source details: GELFAND-P000731

[undeciphered mark] and lin. transfs. preserving the cone.

Uncertain reading. The isolated mark before "and" is illegible.

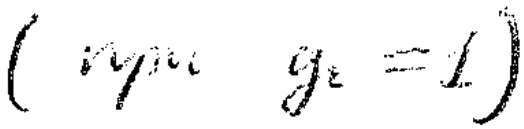
Original source details: GELFAND-P000731

The rays of the cone — S^4. The transfs. are conformal on S^4.

The conformal group preserves the functional $S(A)$

Belavin–Polyakov instantons.

$$(6) \qquad \min_{P(A)=n} S(A) = n. \qquad \text{(when } g_c = 1\text{)}$$



Uncertain reading. The subscript on g is tentatively read as c, possibly 0.

Original source details: GELFAND-P000731

and in fact the minima satisfy the eqn. [crossed out]

Uncertain reading. The crossed-out expression resembles $F = *F$; the middle signs are uncertain.

Original source details: GELFAND-P000731

$$(7) \qquad \underline{F = *F}$$

(duality eqns.)

(Belavin and Polyakov)

And conversely, solns. of these eqns. with given n — realize a strict minimum in the given class.

A number of authors gave explicit solns. (for $\forall\, n$ at least one solution exists).

[crossed out: Ger?] t' Hooft found explicit fmlas. for $\underline{n = 2}$. (all solutions!)

Uncertain reading. The crossed-out fragment before the surname tentatively reads "Гер" (Ger).

Original source details: GELFAND-P000731

by means of an elem. substitution.

The solutions form an $(8n-3)$-dimensional manifold (Schwarz, from the Atiyah–Singer thm.). This is M^{8n-3}.

Atiyah–Singer 3 months later.

t' Hooft found [crossed out] a manifold of dimension $\underline{\underline{5n+4}}$. ($n \geq 3$)

Uncertain reading. The short crossed-out entry after "found" is illegible.

Original source details: GELFAND-P000731

For $n = 2$ they coincide. It is unclear whether there is another connected component.

Proof of Schwarz's thm.

Linearize eqns. (7):

$$(A_\mu) \longrightarrow A_\mu + a_\mu = A + \delta A$$

linearized operator $\quad a_\mu \longrightarrow Ta_\mu$

Uncertain reading. An isolated, undeciphered trace is visible to the right of Ta_μ.

Original source details: GELFAND-P000731

GELFAND-P000732

[source number: 70]

$$B(A) = F - *F = 0,$$
$$B(A + \delta A) = T(\delta A) + O((\delta A)^2)$$

Counting the index: cn — a linear fn. of n; compute the vals. at 0 and 1.

Cokernel $= 0$ [crossed out] $\Leftarrow$ Hopf considerations.

Uncertain reading. A passage after "Cokernel = 0" is heavily crossed out. A left-pointing arrow is visible at right.

Original source details: GELFAND-P000732

Obtain $8n$: 3 parameters redundant due to gauge invariance.

Use of the conf. group $\Rightarrow$ For $n = 1$ all solns. are spher. symmetric.

$O(5,1)$ acts on the set of solns.

For $n = 1$ the set of solns. is 5-dim. [At right:] $\dim O(5,1) = 15$.

on $\forall$ conn. comp. acts $O(5,1)$.

Hence the stab. subgroup has dim. ≥ 10. But > 10 is impossible, since the stab. subgroup is compact (there is a metric)

$\Rightarrow$ they are homog. spaces for $SO(5)$, i.e. spher. symmetric.

(spher. symm. up to gauge invariance).

2-instant. — still incomplete.

Method of Atiyah and Zakharov–Belavin

Cond. $F = *F$.

we want to express it as a cond. of commutativity of diff. operators

$$\left(\text{e.g., } \left[L - \frac{\partial}{\partial t},\ A - \frac{\partial}{\partial y}\right] = 0\right)$$

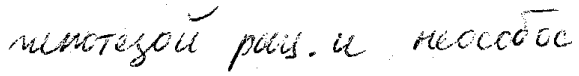
Theorem

$$(8) \quad [\widehat{B}_1(\lambda), \widehat{B}_2(\lambda)] = 0,\ \forall\lambda \quad \Longleftrightarrow \quad \text{duality eqns. (7).}$$

(1st order in λ).

[At right, ‖] (Zakharov, Belavin, Yang).

Single out solns. by assuming rationality and nonsingularity.



Manuscript excerpt. After "Single out solns." the Russian reads "гипотезой рац." (by assuming rationality).

Original source details: GELFAND-P000732

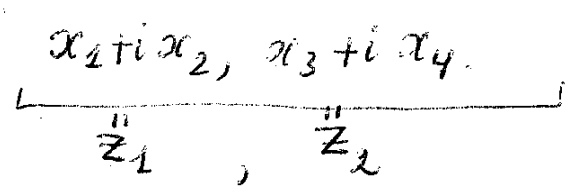
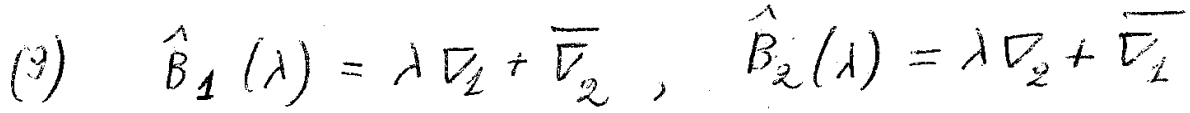


Diagram and typed description.

$$\underbrace{x_1 + ix_2,\quad x_3 + ix_4}_{\substack{\| \\ z_1}\quad\quad\quad

$B_j(\lambda)$ themselves write as covar. diff. operators.

$$(9)\quad \widehat{B}_1(\lambda) = \lambda\nabla_1 + \overline{\nabla}_2, \qquad \widehat{B}_2(\lambda) = \lambda\nabla_2 + \overline{\nabla}_1.$$



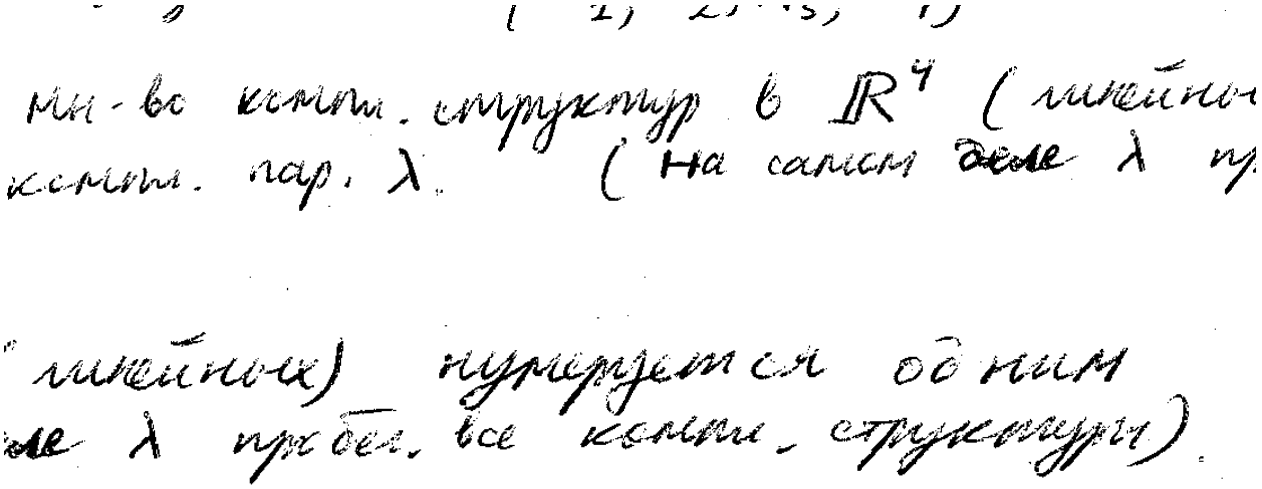
Original source details: GELFAND-P000732

Manuscript excerpt. Both operators in (9) have a written + before the conjugate derivative.

$$(10)\quad \nabla_1 = A_1 + iA_2 + \frac{\partial}{\partial z_1}, \qquad \nabla_2 = A_3 + iA_4 + \frac{\partial}{\partial z_2}.$$

Connection $= \{A_1, A_2, A_3, A_4\}$.

The set of complex structures in $\mathbb{R}^4$ (linear) is indexed by one complex param. λ. (In fact λ runs through all complex structures).



Original source details: GELFAND-P000732

Manuscript excerpt. The parentheses say “all complex structures”. No further restriction is written here.

GELFAND-P000733

Atiyah.

(approach, did more)

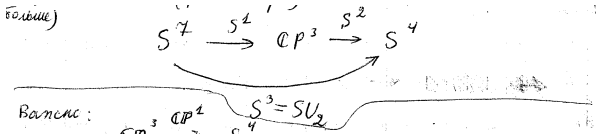
Original source details: GELFAND-P000733

Diagram and typed description.

$\mathbb{C}P^3 \xleftarrow{S^2} S^4$

The arrow in the upper line points left.

(Hopf fibr.)



Original source details: GELFAND-P000733

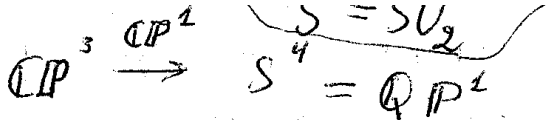
Diagram and typed description.

$$S^7 \xrightarrow{S^1} \mathbb{C}P^3 \xrightarrow{S^2} S^4, \qquad S^3 = SU_2$$

Important:

Original source details: GELFAND-P000733

Manuscript excerpt. The heading reads “Важно” (Important).



Original source details: GELFAND-P000733

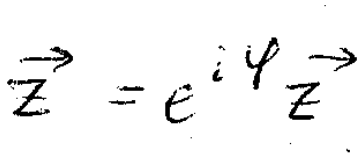
Diagram and typed description.

$$\mathbb{C}P^3 \xrightarrow{\mathbb{C}P^1} S^4 = \mathbb{Q}P^1$$

q_1, q_2 — quaternions.

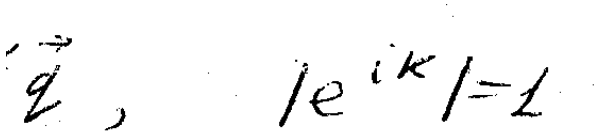
$$q_1 = (z_1, z_2), \qquad q_2 = (z_3, z_4).$$

$$\vec{z} = e^{i\varphi}\vec{z}, \qquad \vec{q} \to e^{ik}\vec{q}, \qquad |e^{ik}| = 1.$$



Original source details: GELFAND-P000733

Manuscript excerpt. There is an equals sign between $\vec{z}$ and the exponential.



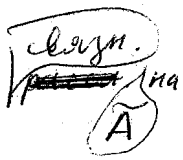
Original source details: GELFAND-P000733

Uncertain reading. After i in the exponent, the sign is tentatively k. It also resembles κ.

(unit quaternion; they form the grp. SU_2).

$$\left(\begin{array}{c|c} \underline{z_1, z_2} & \underline{z_3, z_4} \\ \text{quatern.} & \text{quatern.} \end{array} \right).$$

The conn. A on S^4 we lift to [crossed out: bund.?] [inserted above in a circle: conn.] [inserted below in a circle: $\widetilde{A}$] on $\mathbb{C}P^3$.

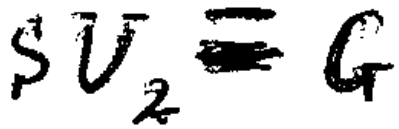


Original source details: GELFAND-P000733

Uncertain reading. The crossed-out word is tentatively “рассл.” (bund.). The insertions are circled.

Connection with grp. SU_2.

$$SU_2 = G$$



Uncertain reading. The sign before G tentatively reads $=$. The lower stroke is heavily retraced.

Original source details: GELFAND-P000733

The connection gives rise to an almost complex structure.

(the fibres are compl., and in the horiz. sps. it is lifted from the base).

Fibre $\mathbb{C}^2$.

Integrability of the almost compl. struct. $\Longleftrightarrow$ duality eqns..

$\Longleftrightarrow$ the ops. $\dfrac{\partial}{\partial z}$ commute. (repr. as a commuting pair)

Recapitulation.

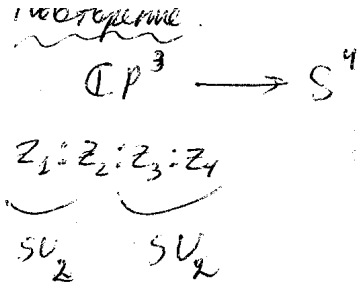


Diagram and typed description.

$$\mathbb{C}P^3 \longrightarrow S^4$$

$$\underbrace{z_1 : z_2}_{SU_2} : \underbrace{z_3 : z_4}_{SU_2}$$

Original source details: GELFAND-P000733

we act here and there by the same transfs. from SU_2.

[crossed out: $SU_2 \ldots \mathbb{C}P^3$?]

Uncertain reading. The crossed-out expression tentatively contains SU_2 and $\mathbb{C}P^3$. The sign between them is uncertain.

Original source details: GELFAND-P000733

$$\text{Fibre} = SU_2/(\text{diag. matrices}) = \mathbb{C}P^1.$$

There is a complex structure in the base and in the fibre.

Suppose on M^{2n} there is an almost compl. structure

(a way of making $\forall$ tangent sp. complex)

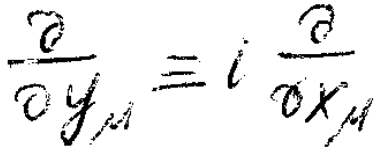
$$\partial_1, \ldots, \partial_n \text{ — a basis of vectors in the tangent sp.}$$
$$\partial_{n+1}, \ldots, \partial_{2n}$$

GELFAND-P000734

[source number: 71]

Let $\nabla_j = \partial_j + i\partial_{n+j}$

$$\frac{\partial}{\partial y_\mu} \equiv i \frac{\partial}{\partial x_\mu}$$



Manuscript excerpt. The manuscript uses a three-bar sign ≡ between the derivatives.

Original source details: GELFAND-P000734

Req. that they commute.

This is precisely integrability of the almost compl. structure.

Manuscript excerpt. The middle letters of the Russian abbreviation “квазикомпл.” (almost complex) are heavily retraced.

Original source details: GELFAND-P000734

GELFAND-P000735

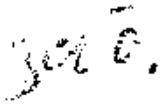
15/XI-77. Gelfand Sem.

B. Sz.-Nagy.

A number of operator problems reduce to constructing contraction operators on Hardy spaces.

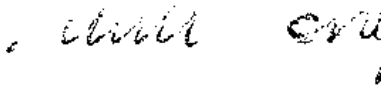
Uncertain reading. The faint first word tentatively reads “Ряд” (a number of).

Original source details: GELFAND-P000735



Uncertain reading. The second word tentatively reads “зад.” (problems).

Original source details: GELFAND-P000735



Uncertain reading. The abbreviation following “constructing” tentatively reads “сжат.” (contraction).

Original source details: GELFAND-P000735

H^∞ — bounded analytic functions in the disk.

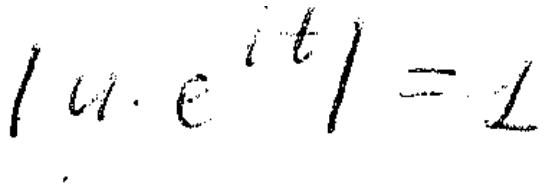
[Crossed out] inner function

$$\underset{\text{def}}{=} \quad |u \cdot e^{it}| = 1$$

almost everywhere

Uncertain reading. The crossed-out signs before “inner function” are illegible.

Original source details: GELFAND-P000735



Uncertain reading. The manuscript places a dot between u and e^{it}.

Original source details: GELFAND-P000735

$$\bigwedge_{\alpha} u_\alpha$$

— greatest inner divisor of a family of functions

Uncertain reading. A short curved stroke in the right margin. Its purpose is unclear.

Original source details: GELFAND-P000735

Blaschke decompos.:

$$\forall \quad \text{[Crossed out]} \quad u = u_i u_o$$

(Smirnov thm.)

Uncertain reading. The crossed-out signs between $\forall$ and u are illegible.

Original source details: GELFAND-P000735

Uncertain reading. The last subscript tentatively reads o.

Original source details: GELFAND-P000735

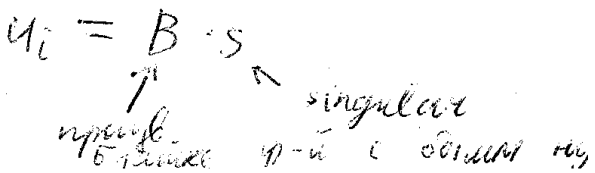


Diagram and typed description.

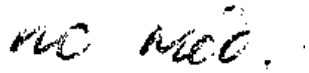
$u_i = B \cdot s$

Blaschke product

singular

Original source details: GELFAND-P000735

of functions with a common zero and by meas. [Crossed out].



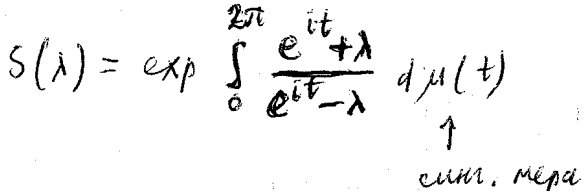
Original source details: GELFAND-P000735

Uncertain reading. The last Russian abbreviation tentatively reads “мас.”. Its expansion is uncertain.

Original source details: GELFAND-P000735

Uncertain reading. The crossed-out group at the end of the line is illegible.

GELFAND-P000736



Original source details: GELFAND-P000736

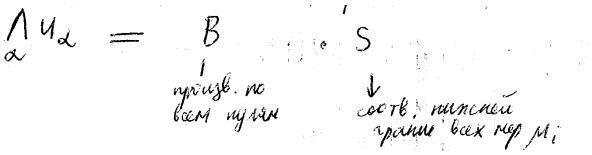
Diagram and typed description.

$$s(\lambda) = \exp \int_0^{2\pi} \frac{e^{it}+\lambda}{e^{it}-\lambda} \quad d\mu(t)$$

↑ sing. measure

Original source details: GELFAND-P000736

Uncertain reading. The initial sign tentatively reads s. Its case is uncertain.



Original source details: GELFAND-P000736

Diagram and typed description.

$$\bigwedge_\alpha u_\alpha = \quad B \quad \cdot \quad S$$

	↓	↓
	product over all zeros	corr. to the infimum of all measures μ_i

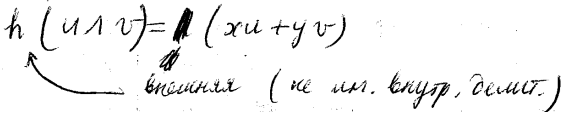
Original source details: GELFAND-P000736

Uncertain reading. The second factor tentatively reads capital S.

In the Hardy alg. it is false that

$$u \wedge v = xu + yv.$$

It is false even that



Original source details: GELFAND-P000736

Diagram and typed description.

$$h \quad (u \wedge v) = \quad \square \quad (xu + yv)$$

outer (having no inner divisor)

Uncertain reading. The sign before the parentheses is blacked out. The stroke directly below it is also heavily crossed out.

Original source details: GELFAND-P000736

Uncertain reading. A broken curved stroke appears in the right margin. Its purpose is unclear.

Original source details: GELFAND-P000736

Fix an inner fn. ω. Then $\forall u, v\ \exists x, y$:

$$xu + yv = h_\omega(u \wedge v), \quad \text{where } h_\omega \wedge \omega = 1$$

(but h_ω need not be outer)

Uncertain reading. The last word tentatively reads “внешний” (outer). The middle letters are heavily retraced.

Original source details: GELFAND-P000736

In fact, one can choose $x, y = \text{const}$

$$u + tv = h_t(u \wedge v)$$

M. Sherman (beautiful proofs of the lemma)

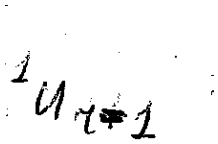
Generalization:

[crossed out] $\forall u_1, \dots, u_r,$

Undeciphered passage. The overwritten group before the quantifier is illegible.

Original source details: GELFAND-P000736

$$u_1 + tu_2 + \cdots + t^{r-1}u_{r+1} = h_t\Big(\bigwedge_k u_k\Big); \quad h_t \wedge \omega = 1$$



Manuscript excerpt. The last subscript reads $r + 1$.

Original source details: GELFAND-P000736

Undeciphered passage. The earlier strokes beneath the heavy mark in the index are illegible.

Original source details: GELFAND-P000736

(Except for a countable set of values of t)
and likewise for a finite number of rows
Then some elementary matrix transformations.
I. Krichever.
Rat. solutions of the Kadomtsev–Petviashvili equation:

$$\frac{3}{4}u_{yy} + \frac{\partial}{\partial x}\left\{[\text{crossed out}]\frac{\partial u}{\partial t} + \frac{1}{4}(6uu_x + u_{xxx})\right\} = 0 \tag{1}$$

Undeciphered passage. The crossed-out sign before the time derivative is illegible.

Original source details: GELFAND-P000736

GELFAND-P000737

[source number: 78]

Theorem. Seek solns. rational in x $u(x, y, t)$, decaying as $|x| \to \infty$. Then it is easy to see that

Uncertain reading. The word before “see” reads “легко” (easy). Its last letter is heavily retraced.

Original source details: GELFAND-P000737

Undeciphered passage. The earlier strokes of the last letter are illegible beneath the heavy retracing.

Original source details: GELFAND-P000737

$$u(x, y, t) = \sum \frac{-2}{\big(x - x_j(y, t)\big)^2}$$

Uncertain reading. The lower loop identifies the first argument of x_j as y.

Original source details: GELFAND-P000737

Undeciphered passage. The earlier strokes of the argument are illegible beneath the correction.

Original source details: GELFAND-P000737

1) The poles move in y acc. to Moser's syst. with Hamiltonian

$$H = \frac{1}{2}\sum p_j^2 + \sum_{i<j} \frac{1}{(x_i - x_j)^2}$$

(t fixed)

2) The poles move in t [crossed out] according to Hamiltonian F_3, one of the integrals of Moser's syst.

Undeciphered passage. The crossed-out mark after the abbreviation for "move" is illegible.

Original source details: GELFAND-P000737

We obtain 2 commuting flows determining [inserted above: (rat.)] solns. of eq. (1).

Diagram and typed description.

Original source details: GELFAND-P000737

$$(1) \Longleftrightarrow \left[\frac{\partial}{\partial y} - L_1, \frac{\partial}{\partial t} - L_2\right] = 0,$$

$$L_1 = \frac{d^2}{dx^2} + u, \qquad L_2 = \frac{d^3}{dx^3} + \frac{3}{2}u\frac{d}{dx} + u$$

Undeciphered passage. The coefficient reads $3/2$. The earlier strokes under the heavy retracing are illegible.

Original source details: GELFAND-P000737

Uncertain reading. The final term is written $+u$.

Original source details: GELFAND-P000737

We obtain commutative subalgebras in $\mathcal{D}_n$ (division ring of fractions of the ring of diff. ops. with polyn. coeffs. in n variables)

Next time Yu. I. Manin

Manuscript excerpt. The abbreviation before "раз" (time) reads "След." (next).

Original source details: GELFAND-P000737

Chapter 12

GELFAND-P000738

[source number: 6]

21/XI-77. Gelfand Sem.

Diagram and typed description.
Yu. I. Manin.

Original source details: GELFAND-P000738

(on work with V. Drinfeld)

Description of self-dual solutions of the Yang–Mills eqs.

$\mathbb{R}^4 \ni (x_1, x_2, x_3, x_4)$. Unknown functions:

$$A_\mu(x) \in u(r), \qquad r \geq 2$$

(Hermitian or anti-Hermitian matrices; in the Lie algebra)

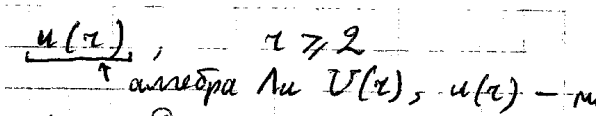


Diagram and typed description.
$\underbrace{u(r)}$
Lie algebra of $U(r)$

Original source details: GELFAND-P000738

$u(r)$ — the set of skew-Hermitian $r \times r$ matrices.

$$F_{\mu\nu}(x) \underset{\text{def}}{=} \frac{\partial}{\partial x_\mu} A_\nu - \frac{\partial}{\partial x_\nu} A_\mu + [A_\mu, A_\nu] \tag{1}$$

1) Duality equations:

$\rho\sigma$ — complementary pair of indices to $\mu\nu$, $F_{\mu\nu} = \pm F_{\rho\sigma}$; more precisely:

$$F_{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \tag{2}$$

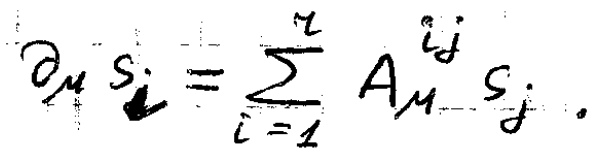
($\varepsilon^{\mu\nu\rho\sigma}$ — the sign of the permutation).

2) Regularity conditions: A) at finite points B) at infinity

A) singularities at finite points can be removed by a gauge transformation.

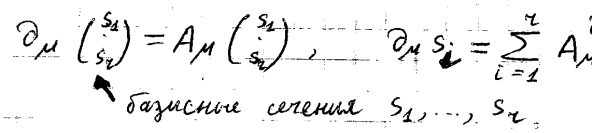
More precisely, consid. the bundle $\mathbb{C}^r \times \mathbb{R}^4$ and a basis of sections in it; the covariant differential, determined by the matrices A_μ, is given by the formula:

$$\partial_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix} = A_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}, \qquad \partial_\mu s_i = \sum_{i=1}^{r} A_\mu^{ij} s_j. \tag{3}$$



Uncertain reading. The lower limit of summation reads $i = 1$. The i in s_i on the left is heavily retraced.

Original source details: GELFAND-P000738



Original source details: GELFAND-P000738

Diagram and typed description.

$$\partial_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}$$

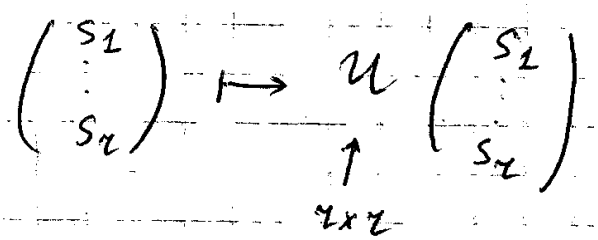
basis sections $s_1, \dots, s_r$

Pass to another basis. We obtain a transformation (gauge transformation).

$$A_\mu \mapsto u^{-1}\partial_\mu u + u^{-1}A_\mu u = A'_\mu \tag{4}$$

Regularity condition at a finite point: there is a genuine connection without singularities, i.e. the sing. in A_μ can be removed by a gauge transformation.

Transition:



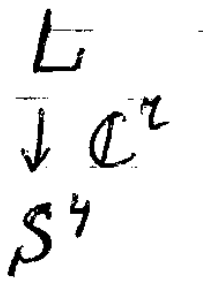
Original source details: GELFAND-P000738

Diagram and typed description.

$$\begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix} \mapsto \underset{r \times r}{u} \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}$$

B) Conditions at $P = \infty$.

The bundle on $\mathbb{R}^4$ extends to a bund. over $S^4 = \mathbb{R}^4 \cup \{\infty\}$.



Original source details: GELFAND-P000738

Diagram and typed description.

$$\begin{array}{c} L \\ \downarrow \mathbb{C}^r \\ S^4 \end{array}$$

L — the space of our bundle.

Regularity cond.: The connection ∂_μ extends to a connection in L over S^4 (the bundle L need no longer be trivial)

This also means that a twist at infinity can remove the singularity.

An essential invariant of the solution is the bundle L. This is a discrete invariant. On L there is a unitary bundle structure.

Classification of L.

Gluing the trivializations on the equator gives a map $S^3 \to U(r)$, i.e. an element of $\pi_3(U(r)) = \mathbb{Z}$.

($SU(2) \subset U(r)$, $SU(2) \cong S^3$; hence an integer-valued sufficient invariant = degree of the map n = Pontryagin number)

Uncertain reading. The word after "integer-valued" reads "достаточный" (sufficient). Its first letters are heavily retraced.

Original source details: GELFAND-P000739

the embedding $SU(2) \subset U(r)$ is taken in the form

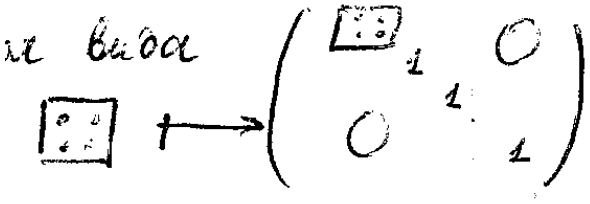


Diagram and typed description.

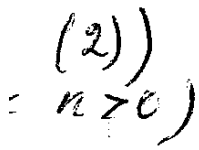
$$\begin{array}{|c|} \hline \cdot\,\cdot \\ \hline \cdot\,\cdot \\ \hline \end{array} \longmapsto \begin{pmatrix} \begin{array}{|c|} \hline \cdot\,\cdot \\ \hline \end{array} & 0 \\ 1\;1 & \\ 0 & 1 \end{pmatrix}$$

Original source details: GELFAND-P000739

We shall assume $n \geq 0$ and the duality conditions

($n < 0 \Rightarrow$ we can change the sign in eq. (2))

— anti-self-dual solns. — reduce to $n > 0$)

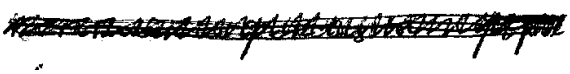


Uncertain reading. Closing parentheses are visible after "(2)" and at the end of the following line.

Original source details: GELFAND-P000739

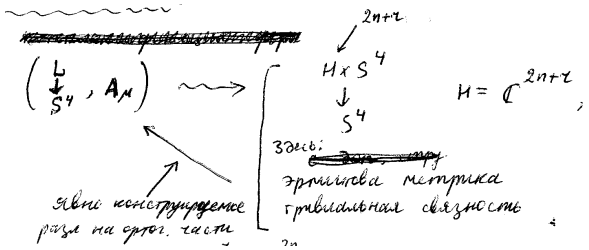
Plan for what follows:

[Crossed-out line]



Undeciphered passage. The line below the heading is heavily crossed out and illegible.

Original source details: GELFAND-P000739



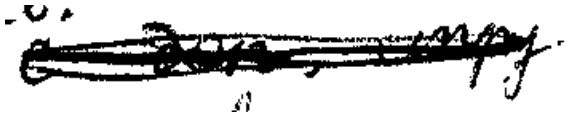
Original source details: GELFAND-P000739

Diagram and typed description.

$$\begin{pmatrix} L \\ \downarrow \\ S^4 \end{pmatrix}, A_\mu \rightsquigarrow \left[\begin{array}{l} \overset{2n+r}{\downarrow} \\ H \times S^4 \\ \downarrow \\ S^4 \end{array}\right. \qquad H = \mathbb{C}^{2n+r},$$

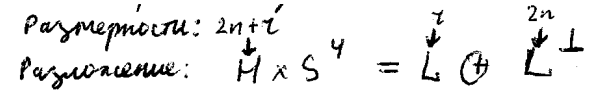
Here:
[crossed out]
Hermitian metric
trivial connection

Explicitly constructed
decomp. into orthog. parts



Original source details: GELFAND-P000739

Undeciphered passage. A short phrase is crossed out inside the large bracket. The first letter resembles Cyrillic “c”.



Original source details: GELFAND-P000739

Diagram and typed description.

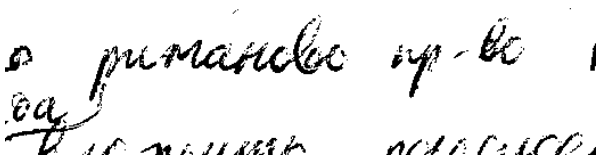
$$\begin{array}{lccccc} \text{Dimensions:} & 2n+r & & r & & 2n \\ & \downarrow & & \downarrow & & \downarrow \\ \text{Decomposition:} & H \times S^4 & = & L & \oplus & L^\perp \end{array}$$

$\{A_\mu\}$ — a direct summand of the trivial connection in $H \times S^4$

(Differentiate and project onto L).

Unitarity of the connection is proved.

[This generalizes the fact that a Riemannian space can be embedded in Euclidean space; can we [inserted above: always] embed a bundle with connection?]



Original source details: GELFAND-P000739

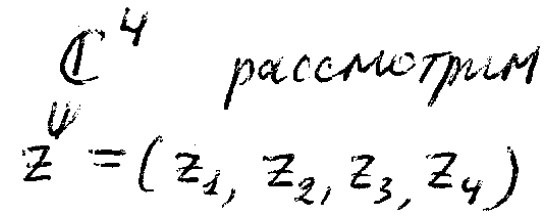
Uncertain reading. The abbreviation after “Riemannian” reads “пр-во” (space).

Which decompositions give self-dual fields?

We must pass from the base S^4 to its complexification.

In $\mathbb{C}^4$ consider the map

$$\sigma : \; z \mapsto (-\overline{z}_2, \overline{z}_1, -\overline{z}_4, \overline{z}_3).$$



Original source details: GELFAND-P000739

Diagram and typed description.

$$\begin{array}{l} \mathbb{C}^4 \\ \downarrow \\ z = (z_1, z_2, z_3, z_4) \end{array}$$

(in quaternions, multiplication by i)

For σ there are no fixed pts. (except 0) and no fixed lines, but there are invariant planes

$$\mathbb{C}z + \mathbb{C}\sigma(z) \subset \mathbb{C}^4 \tag{5}$$

GELFAND-P000740

[source number: 7]

We shall call them real (or σ-invariant) (σ — involution).

$$\{\sigma\text{-invar. planes}\} = S^4 \tag{6}$$

(proof omitted)

(in $\mathbb{C}P^3$ this is the set of σ-invar. lines).

$E^4 \subset$ [crossed out] $S^4 = E^4 \cup \{\infty\}$.

Undeciphered passage. A mark between the inclusion sign and S^4 is heavily crossed out.

Original source details: GELFAND-P000740

Hence there is a map

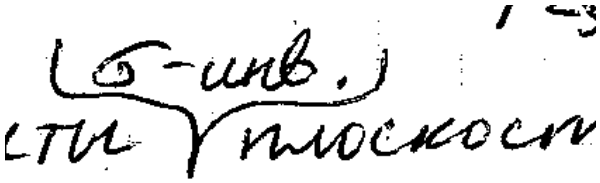
$$\mathbb{C}^4 \setminus \{0\} \xrightarrow{D} S^4 \supset E^4. \tag{7}$$

$$z \longmapsto \sigma\text{-invar. plane through } z.$$

Write D in coordinates (in E^4 the coordinates are (x_i)). The normalization is as in Zakharov–Belavin.

$$(z_1, z_2, z_3, z_4) \longmapsto \begin{cases} x_2 + ix_1 = \dfrac{z_1\bar{z}_3 + \bar{z}_2 z_4}{|z_3|^2 + |z_4|^2}, \\ x_4 + ix_3 = \dfrac{\bar{z}_2 z_3 - z_1\bar{z}_4}{|z_3|^2 + |z_4|^2}. \end{cases}$$

(Plücker coordinates of the [inserted above: σ-inv.] plane through pt. z)



Uncertain reading. The abbreviation "σ-инв." is inserted above the line with a mark indicating its position.

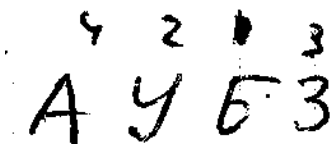
Original source details: GELFAND-P000740

This is the map

$$\mathbb{C}^4 \setminus \{z_3 = z_4 = 0\} \longrightarrow E^4.$$

Remark: to recover a σ-inv. plane, 3 coordinates suffice.

Lemma 1 ($^{4}_{\text{A}}\,^{2}_{\text{W}}\,^{1?}_{\text{B}}\,^{3}_{\text{Z}}$ — Atiyah, Ward, Belavin, Zakharov).



Uncertain reading. Above "Б" is a heavy vertical mark, tentatively 1. The other numbers read 4, 2 and 3.

Original source details: GELFAND-P000740

Original source details: GELFAND-P000740

Uncertain reading. The beginning of “Belavin” is heavily retraced.

a)

$$D^*(dx_\mu \wedge dx_\nu + \varepsilon^{\mu\nu\rho\sigma} dx_\rho \wedge dx_\sigma) \quad \text{— a form of type } (1,1).$$

(Consider such forms and lift them to $\mathbb{C}^4 \setminus 0$ using D)

b)

$$\left[D^* \begin{pmatrix} dx_1 \wedge dx_2, \\ dx_1 \wedge dx_3, \\ dx_1 \wedge dx_4 \end{pmatrix} \right]^{2,0} \quad \text{linearly independent.}$$

[crossed out]

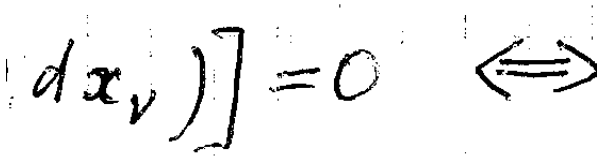
Original source details: GELFAND-P000740

Undeciphered passage. A short fragment after “linearly independent” is heavily crossed out.

$(2,0)$-components.

Corollary 1 (AW).

$$\left[D^* \left(\sum F_{\mu\nu}\, dx_\mu \wedge dx_\nu \right) \right] = 0 \quad \Longleftrightarrow \quad F_{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \tag{8}$$

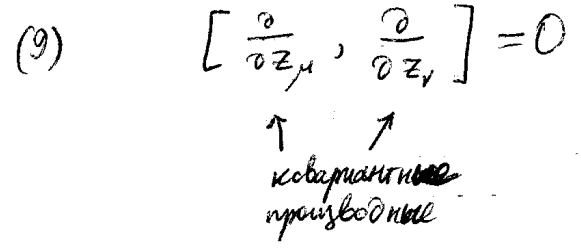


Original source details: GELFAND-P000740

Uncertain reading. The closing square bracket in (8) has no written component index.

GELFAND-P000741

Corollary 1′ (BZ)



Original source details: GELFAND-P000741

Diagram and typed description.

$$(9) \qquad \left[\frac{\partial}{\partial z_\mu}, \frac{\partial}{\partial z_\nu} \right] = 0$$

↑ ↑

covariant derivatives

Original source details: GELFAND-P000741

Uncertain reading. The ending of the Russian word for “covariant” is heavily retraced.

Belavin. Or:

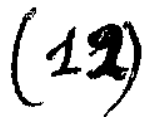
$$[\nabla_\mu, \nabla_\nu] = 0, \tag{10}$$

where

$$\nabla_\mu = \frac{\partial}{\partial z_\mu} + C_\mu, \tag{11}$$

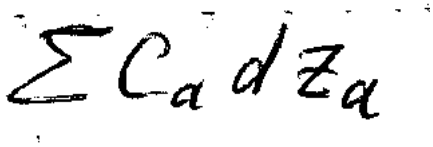
$$\left[D^*\left(\sum A_\mu\, dx_\mu\right)\right]^{1,0} = \sum C_a\, dz_a \tag{12}$$

Uncertain reading. The digit 2 in equation number (12) is heavily retraced.



Original source details: GELFAND-P000741

Uncertain reading. The indices on the right of (12) tentatively read a, possibly α.

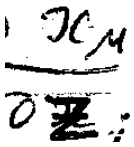


Original source details: GELFAND-P000741

Or: set

$$C_i(z) = \frac{\partial x_\mu}{\partial z_i} A_\mu(x) \tag{13}$$

Uncertain reading. The z in the denominator of (13) is heavily retraced.



Original source details: GELFAND-P000741

Then

[crossed out] $$\frac{\partial C_k}{\partial z_i} - \frac{\partial C_i}{\partial z_k} + [C_i, C_k] = 0 \tag{14}$$

Undeciphered passage. The first sign of the crossed-out group resembles ∂. The other signs are illegible.

Original source details: GELFAND-P000741

by virtue of the duality equations.

Or:

$$[L_1, L_2] = 0, \tag{15}$$

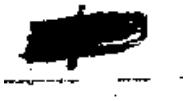
where

$$L_1 = \lambda(\nabla_1 - i\nabla_2) + (\nabla_3 + i\nabla_4), \tag{16′}$$

Uncertain reading. The sign after the minus in (16′) is heavily retraced and tentatively reads i.

Original source details: GELFAND-P000741

$$L_2 = \lambda(\nabla_3 - i\nabla_4) - (\nabla_1 + i\nabla_2), \tag{16″}$$

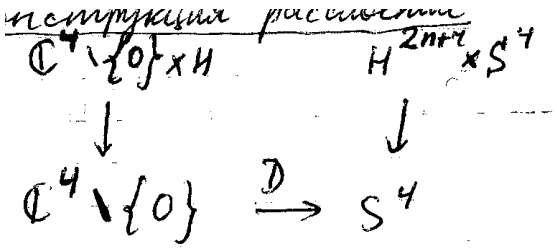


Original source details: GELFAND-P000741

Uncertain reading. Between the parentheses in (16″) there is a heavy horizontal stroke and short vertical traces. It is tentatively read as a minus.

$$\nabla_\mu \equiv \partial_\mu + A_\mu \tag{17}$$

M. Construction of bundles. They will be constructed on the left.



Original source details: GELFAND-P000741

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^4 \setminus \{0\} \times H & & H^{2n+r} \times S^4 \\ \downarrow & & \downarrow \\ \mathbb{C}^4 \setminus \{0\} & \xrightarrow{\quad D \quad} & S^4 \end{array}$$

Objects of linear algebra $\Rightarrow$ direct summand in $H \times S^4$.

H — $(2n + r)$-dimensional Hermitian space,
form Q.

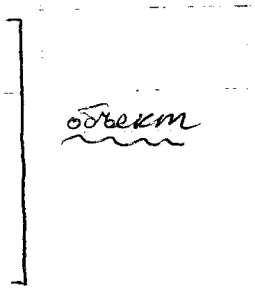
Consider [crossed out]

$$I \subset H \otimes_{\mathbb{C}} \mathbb{C}^4 \qquad (*) \tag{18}$$

Original source details: GELFAND-P000741

Undeciphered passage. A separate small crossed-out mark appears before (18).

I — n-dimensional subspace



Original source details: GELFAND-P000741

Diagram and typed description.

object

The right bracket covers the passage from the definition of H to the line about the subspace I.

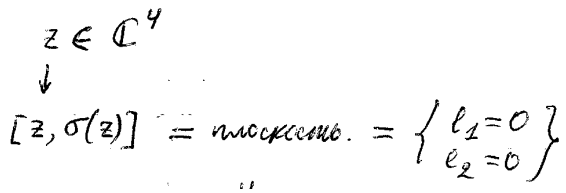
GELFAND-P000742

[source number: 8]

Conditions.

From the object we construct something like a bundle.

$L(*) =$ “system of linear spaces over S^4”.

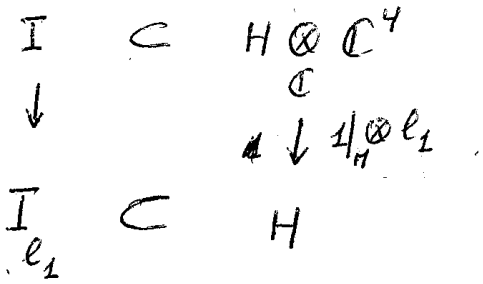


Original source details: GELFAND-P000742

Diagram and typed description.

$$z \in \mathbb{C}^4 \;\downarrow\; [z, \sigma(z)] \quad = \text{plane.} = \left\{ \begin{array}{l} \ell_1 = 0 \\ \ell_2 = 0 \end{array} \right\}$$

$\ell_{1,2} : \mathbb{C}^4 \to \mathbb{C}$ — linear forms.



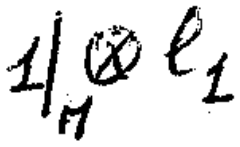
Original source details: GELFAND-P000742

Diagram and typed description.

$$\begin{array}{ccc} I & \subset & H \otimes_{\mathbb{C}} \mathbb{C}^4 \\ \downarrow & & \square \downarrow 1|_H \otimes \ell_1 \\ I_{\ell_1} & \subset & H \end{array}$$

Original source details: GELFAND-P000742

Undeciphered passage. The heavily retraced mark to the left of the right-hand arrow is illegible.



Original source details: GELFAND-P000742

Uncertain reading. To the right of the arrow the label reads $1|_H \otimes \ell_1$.

Construct analogously I_{ℓ_2}.

Thus we have constructed 2 subspaces $I_{\ell_1}, I_{\ell_2} \subset H$

It turns out that

$$I_{\ell_1} + I_{\ell_2} \quad \text{[crossed out: depends?]}$$

Original source details: GELFAND-P000742

Uncertain reading. The crossed-out word to the right of the sum tentatively reads “зависит” (depends).

depends only on the σ-invar. plane $[z, \sigma(z)]$, i.e. a point of S^4

$$(I_{\ell_1} + I_{\ell_2})^{\perp} = \text{lin. subspace in } H, \text{ corresponding to } [z, \sigma(z)].$$

Condition 1. A bundle must result (the dimensions are maximal, i.e. the right ones), namely

$$\dim(I_{\ell_1} + I_{\ell_2})^{\perp} = r \tag{19}$$

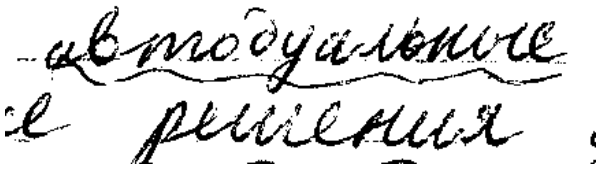
Condition 2. Hermitian property of the bundle.

The form $Q|_{I_{\ell_1}+I_{\ell_2}} > 0$

(a priori Q is not assumed to be > 0, but it turns out to do so under this condition).

A direct summand results. The connection is obtained by projecting the trivial one.

Theorem 1. This gives all [inserted above: self-dual] solutions of the Y.–M. equation, nonisom. bunds. correspond to inequivalent solutions.



Original source details: GELFAND-P000742

Uncertain reading. "Self-dual" is inserted above "solutions" and wavy-underlined.

One more condition has been omitted.

Take the eqs.:

$$\left\{ \begin{array}{l} \ell(z) = 0 \\ \ell(\sigma z) = 0 \end{array} \right\}$$

GELFAND-P000743

Condition 0.

$$Q(I_{\ell}, I_{\ell \circ \sigma}) = 0 \tag{20}$$

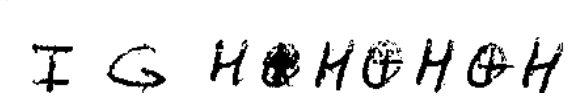
i.e. I_{ℓ} and $I_{\ell \circ \sigma}$ are orthogonal.

This gives a problem in linear algebra.

G. it is contained in an unpubl. work of Gelfand and Ponomarev on quadruples with an automorphism.

Ponomarev:

$$I \hookrightarrow H \oplus H \oplus H \oplus H \longrightarrow A \longrightarrow 0$$



Original source details: GELFAND-P000743

Uncertain reading. The first inclusion has an arrowhead. The first direct-sum sign is heavily retraced.

The images of the components H in A give 4 subspaces.

The complex invariants of the quadruple have been described.

Hermitian property: classify the quadruples invariant under the operation $\perp$

Uncertain reading. The abbreviation before “operation” tentatively reads “отн.” (with respect to). The middle strokes are heavily retraced.

Original source details: GELFAND-P000743

a) Lagrangian subspaces, etc.

Diagram and typed description.

Original source details: GELFAND-P000743

M. Turning all this into numbers.

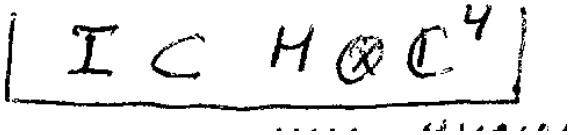


Diagram and typed description.

$$I \subset H \otimes \mathbb{C}^4$$

Original source details: GELFAND-P000743

— parameters of a multi-instanton solution

isomorphism of diagrams gives identical solutions.

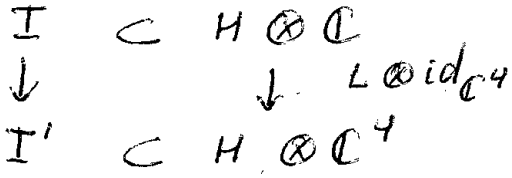


Diagram and typed description.

$$\begin{array}{ccc} I & \subset & H \otimes \mathbb{C} \\ \downarrow & & \downarrow L \otimes \mathrm{id}_{\mathbb{C}^4} \\ I' & \subset & H \otimes \mathbb{C}^4 \end{array}$$

Original source details: GELFAND-P000743

Uncertain reading. The upper row reads $H \otimes \mathbb{C}$, the lower row $H \otimes \mathbb{C}^4$.

Original source details: GELFAND-P000743

such isomorphisms of diagrams are allowed

L preserves the Hermitian form.

Scheme for computing $A_\mu(x)$.

$$(I \subset H \otimes \mathbb{C}^4, \quad e_1, \dots, e_r \text{ — a basis of } I) \longmapsto 4n \text{ vectors in } H$$
$$(e_{ij}), \quad i = 1, \dots, n; \quad j = 1, \dots, 4.$$
$$\dim H = 2n + r.$$

Uncertain reading. The last subscript in the list of basis vectors tentatively reads r.

Original source details: GELFAND-P000743

From them construct the matrix $\widetilde{Q}$

of $(n \times n)$ blocks of size 4×4

$(k\ell)$-th block at position (rs) has

$$\underset{\text{numerical matrix}}{Q(e_{kr}, e_{\ell s})}$$

Uncertain reading. The second index of the first argument is heavily retraced. The reading is tentatively e_{kr}.

Original source details: GELFAND-P000743

The matrix $\widetilde{Q}$ determines the whole situation

Assumption: $\operatorname{rk} \widetilde{Q} = 2n + r$ ($\Leftrightarrow L$ has no nontrivial horizontal sections) — we omit such degeneracies for simplicity.

GELFAND-P000744

[source number: 9]

Which $\widetilde{Q}$ are obtained:

a) $\widetilde{Q}$ is Hermitian;

b) $\operatorname{rk} \widetilde{Q} = 2n + r$;

c) “Condition 0”. This is a condition on all 4×4 blocks

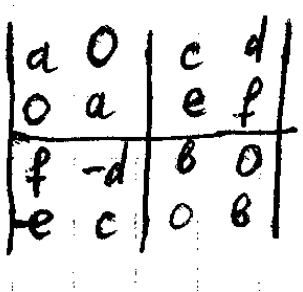


Diagram and typed description.

a	0	c	d
0	a	e	f
f	$-d$	b	0
$-e$	c	0	b

Original source details: GELFAND-P000744

scalar on the diag.;
symmetric with resp. to the diag.

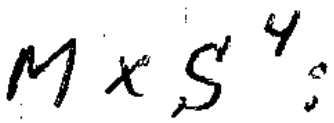
Uncertain reading. The crossed-out word tentatively reads “блоки” (blocks). Its middle is obscured by strokes.

Original source details: GELFAND-P000744

are related as written.
Conditions 1 and 2 are omitted (they too can be written out).
Next, the following are used: e_{ij}, but they do not actually enter the equations.

$$H \times S^4 = L \oplus L^{\perp}$$

$L^{\perp}$ is generated globally by $2n$ sections of the bundle $M \times S^4$;



Uncertain reading. This line reads $M \times S^4$.

Original source details: GELFAND-P000744

$$(21)_1 \qquad h_{k1} = (x_2 + ix_1)e_{k1} + (x_4 - ix_3)e_{k2} + e_{k3}$$
$$(21)_2 \qquad h_{k2} = -(x_4 + ix_3)e_{k1} + (x_2 - ix_1)e_{k2} + e_{k4}$$

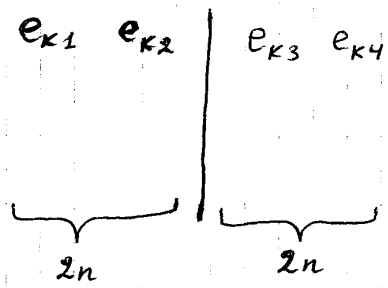


Diagram and typed description.

e_{k1} e_{k2} | e_{k3} e_{k4}

$\underbrace{\phantom{e_{k1}\ e_{k2}}}_{2n}$ $\underbrace{\phantom{e_{k3}\ e_{k4}}}_{2n}$

Original source details: GELFAND-P000744

At infinity everything becomes trivial.

“An instanton lives only a short while, and in imaginary time at that —
— a humanoid formation”

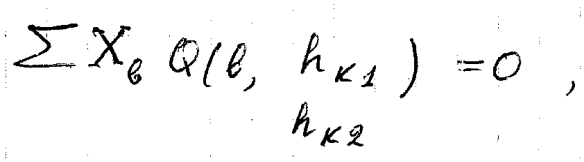
Choose from among e_{ij} a basis of H, $\quad B \subset (e_{ij})$

$\Leftrightarrow (2n+r)$-diagonal minor in $\widetilde{Q}$

(one such minor)

Using it, expand the system of generating sections of L.

$$(22) \qquad \sum_{\substack{h_{k1}\\ h_{k2}}} X_b\, Q\big(b, h_{k1}\big) = 0, \qquad k = 1, \dots, n, \qquad b \in B.$$



Uncertain reading. The manuscript places h_{k2} below h_{k1}, beneath the parentheses.

Original source details: GELFAND-P000744

this is a system of $2n$ eqs. for $2n+r$ unknowns X_b.

Fund. system of solutions $X^{(i)}$, $i = 1, \dots, r$.

Orthonormalize it to obtain $[Y^{(i)}]$ and now write down the answer:

GELFAND-P000745

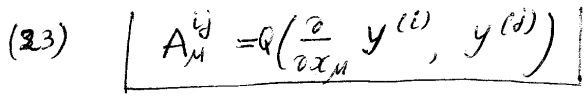
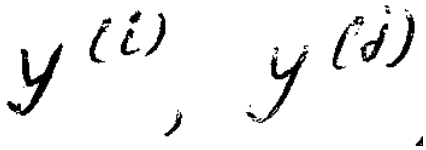


Diagram and typed description.

$$(23) \qquad \boxed{A_\mu^{ij} = Q\left(\frac{\partial}{\partial x_\mu} y^{(i)}, y^{(j)}\right)}$$

Original source details: GELFAND-P000745



Uncertain reading. Both letters in (23) have descenders and read as lowercase y.

Original source details: GELFAND-P000745

$X^{(i)}$ — rational functions with values in H;

roots of rational functions arise in orthonormalization.

Geometric theorem.

The diagrams $I \subset H \otimes \mathbb{C}^4$ with certain cond. ...

classify bundles over $\mathbb{C}P^3$ with conditions.

1) $H^i(L(-2)) = 0$ and likewise for L^*;

Uncertain reading. The dot of the superscript i is visible along the thin horizontal streak in the scan.

Original source details: GELFAND-P000745

2) $H^2(L(-1)) = 0$;

3) $H^0(L(-1)) = 0$.

GELFAND-P000746

[source number: 15]

28/XI-77. Gelfand Sem.

L. D. Faddeev. Solitons as particles.

I. $u(x,t)$ — a field, $x \in \mathbb{R}^d$, $d = 1, 2, 3$.

$$\Box = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}$$

$$\Box u + m^2 u + \varepsilon V'(u) = 0 \tag{1}$$

$$\mathcal{L} = \int \left[\tfrac{1}{2}\,\partial u \cdot \partial u - \tfrac{1}{2} m^2 u^2 - \varepsilon V(u)\right] dx \tag{2}$$

$$\partial_0 u(x,t) \underset{\text{def}}{=} \pi(x,t)$$

$$\{u(x,t), \pi(y,t)\} = \delta(x-y) \tag{3}$$

$$H = \int \left[\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla u)^2 + \tfrac{1}{2} m^2 u^2 + \varepsilon V(u)\right] dx \tag{4}$$

Free field: $\varepsilon = 0$

In quantum theory u, π — operators.

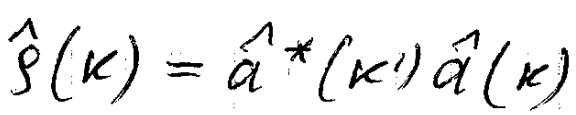
Fourier transf. for $\varepsilon = 0$:

$$H = \int \sqrt{k^2 + m^2}\, a^*(k) a(k)\, dk \tag{5}$$

$$\{a^*(k), a(k')\} = i\delta(k - k') \tag{6}$$

Quantization: let a, a^* — operators, the operator

$$\hat{\rho}(k) = \hat{a}^*(k')\hat{a}(k) \tag{7}$$

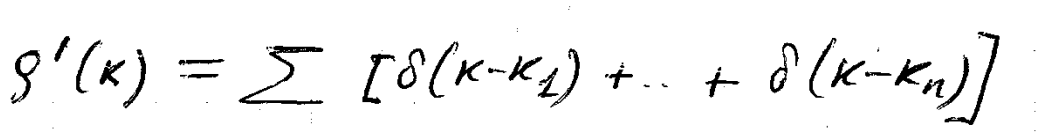


Uncertain reading. A prime is visible on k' in the first factor. The k in the second factor has no prime.

Original source details: GELFAND-P000746

has eigenvalues:

$$\rho'(k) = \sum \left[\delta(k - k_1) + \cdots + \delta(k - k_n)\right] \tag{8}$$



Original source details: GELFAND-P000746

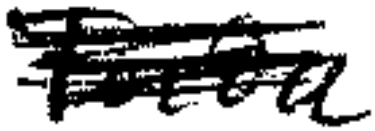
Uncertain reading. After the equals sign comes $\sum$ without limits, then a square bracket.

The spectrum of the Hamiltonian is interpreted in terms of particles.

$$E(k_1, \ldots, k_n) = \omega(k_1) + \cdots + \omega(k_n), \qquad \omega = \sqrt{k^2 + m^2} \tag{9}$$

(see: analogy with the oscillator)

In classical terms: if a finite-energy soln. of eq. (1) asymptotically approaches at ∞ a soln. of the free eq., then this picture is natural.



Uncertain reading. The crossed-out word tentatively reads “Тогда” (then).

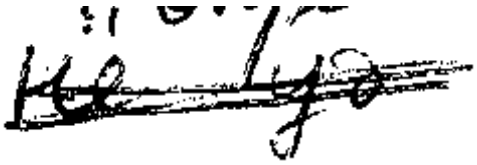
Original source details: GELFAND-P000747

Then $H(u) = H(u_0)$, since H — an integral.

Solitons are a counterexample to this hypothesis.

Solitons: “everything that fails to satisfy the condition $u \to u_0$ as $|t| \to \infty$”

“Definition”: a soliton is something that



Uncertain reading. The crossed-out fragment tentatively reads “не уд” (beginning of “does not satisfy”).

Original source details: GELFAND-P000747

does not spread out at infinity while having finite energy.

Example: a solution of the stationary eq..

$u_s(x)$ — one solution $\Rightarrow$ a Lorentz transf. gives

Uncertain reading. A heavily crossed-out entry precedes the left parenthesis. u_s is written above it.

Original source details: GELFAND-P000747

$$u_s\left(\frac{x - x_0 - vt}{\sqrt{1 - v^2}}\right) = u_s(x, t \mid x_0, v) \tag{10}$$

Uncertain reading. There is a small crossed-out mark at the right end of the fraction bar. The parenthesis is retraced and extended downward.

Original source details: GELFAND-P000747

already a soliton, depending on 2 parameters.

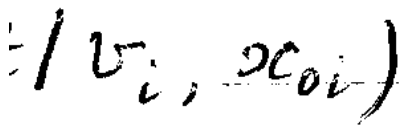
At large times: if $v_1 \neq v_2$ then

$$u(x,t \mid x_{01}, v_1) + u(x,t \mid x_{02}, v_2) \tag{11}$$

is asymptotically a solution.

Then the asymptotics: as $|t| \to \infty$

$$u \to u_0 + \sum u(x,t \mid v_i, x_{0i}) \tag{12}$$



Uncertain reading. After the vertical bar the manuscript reads v_i, x_{0i}.

Original source details: GELFAND-P000747

(a hypothesis in general; a theorem in models)

As $t \to +\infty$ and as $t \to -\infty$ they differ.

Energy

$$H(u) = H(u_0) + \sum_{i=1}^{n} H_s(v_i) \tag{13}$$

It is clear from the relativistic formula that

$$H_s(v) = \frac{M}{\sqrt{1-v^2}} = \sqrt{p^2 + M^2} \qquad \text{(“soliton energy”)} \tag{14}$$

M — the energy of a static soliton (“mass” by definition).

(14) — the definition of p.

This gives 2 parts of phase space. Energy:

$$\int \omega(k)\rho(k)\,dk + \sum_{i}^{n} \sqrt{p_i^2 + M^2} \tag{15}$$

Solitons contributed already in the classical theory. In quantum mechanics there enters $\frac{M}{\overline{h}c}$; solitons persist as $\overline{h} \to +0$.

GELFAND-P000748

[source number: 16]

In quantum mechanics the spectrum will be

$$\sum_{i=1}^{N} \sqrt{k_i^2 + m^2} + \sum_{i}^{n} \sqrt{p_i^2 + M^2} \tag{16}$$

From one field, 2 kinds of particles (semiclassically)

Periodic solution.

$$u_p(x, t \mid x_0, \bar{v}, \beta, \alpha) \tag{17}$$

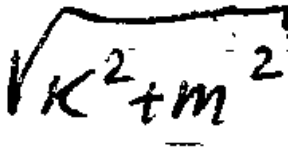
Uncertain reading. The retraced sign tentatively reads β.

Original source details: GELFAND-P000748

β is related to T (period), α (initial phase) are parameters

In this case

$$\sum_{i=1}^{N} \sqrt{k^2 + m^2} + \sum_{i}^{n} \sqrt{p_i^2 + M^2} + \sum_{k}^{\tau} \sqrt{p_k^2 + M(\alpha_k)} \tag{18}$$

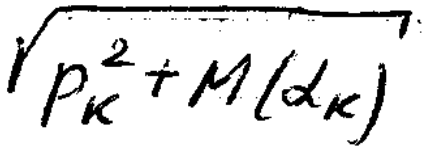


Uncertain reading. The first radicand reads k^2, with no visible subscript.

Original source details: GELFAND-P000748

Uncertain reading. The last upper summation limit tentatively reads τ. Another possible reading is r.

Original source details: GELFAND-P000748



Uncertain reading. The last radicand has $M(\alpha_k)$, with no visible exponent 2.

Original source details: GELFAND-P000748

The main model problem:

$$v(u) = 1 - \cos u - \frac{u^2}{2} \tag{19}$$

Uncertain reading. The initial sign before (u) is rendered as v. It may be υ.

Original source details: GELFAND-P000748

The eq. takes the form:

$$\Box u + m^2 \sin u = 0 \tag{20}$$

(the $(\sin\varphi)_2$)

One more parameter: in the Poisson bracket put

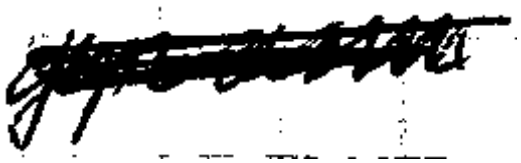
$$\{\partial_0 u, u\} = \gamma\delta(\ldots) \tag{21}$$

Then it turns out that

$$M = \frac{8m}{\gamma} \qquad \text{(“heavy particle”)} \tag{22}$$

$$M_2(\alpha) = \frac{16m}{\gamma}\sin\alpha, \qquad 0 < \alpha \leq \frac{\pi}{2}. \tag{23}$$

These



Original source details: GELFAND-P000748

Uncertain reading. The crossed-out word tentatively reads “двумя” (two).

solutions determine everything.

Here one must take

$$\alpha = \frac{\gamma}{16}n, \quad n = 1, 2, \ldots \tag{24}$$

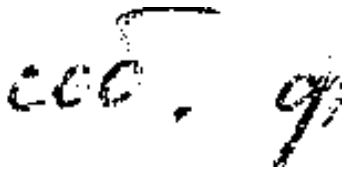
(Takhtajan, Faddeev, ≈ 1974)

A detailed review (in English) should appear shortly.

GELFAND-P000749

II. Consistent quantization and the S-matrix.

It has not yet been possible to find the actual eigenfunctions.



Original source details: GELFAND-P000749

Uncertain reading. The abbreviation in the first sentence reads “соб.” (eigenfunctions).

But the S-matrix has been computed! (by an indirect route: the path integral)
Definitions: write

$$H = \int \left[\frac{\pi^2}{2} + \frac{(\nabla u)^2}{2} + \frac{m^2}{2} u^2 + V(u) \right] \tag{25}$$

and take π and u to be operators with the relation

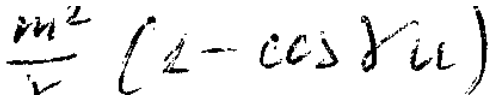
$$[\hat{\pi}(x), \hat{u}(y)] = i\delta(x - y) \tag{26}$$

There is no contradiction, but one must begin with renormalizations.
Glimm, Jaffe:
Frohlich

Original source details: GELFAND-P000749

Uncertain reading. The middle letters of the surname are unclear. The tentative reading is "Frohlich".

$$\begin{bmatrix} \frac{m^2}{\gamma}(1 - \cos \gamma u) & \text{for small } \gamma & \text{constructed} \\ \text{the Hamiltonian rigorously} & (\text{for } x \in \mathbb{R}^1). \end{bmatrix}$$



Original source details: GELFAND-P000749

Uncertain reading. The denominator and the factor before u read γ, with no visible exponent 2.

Lemma 1 (from ordinary quantum mechanics). Consider the eq.

$$i\frac{\partial \Psi}{\partial t} = H\Psi \tag{27}$$

$$H = -\frac{\Delta}{2m} + v(x), \quad v \text{ decays rapidly} \tag{28}$$

Original source details: GELFAND-P000749

Uncertain reading. The rounded sign before (x) and later in the line is rendered as v. It may be v.

Then the limit exists

$$S = \lim_{\substack{t'' \to \infty \\ t' \to -\infty}} e^{iH_0 t''} e^{-iH(t''-t')} e^{-iH_0 t'} \tag{29}$$

(S-matrix) and S — a unitary operator.
It is better to work in the momentum representation: the kernel is denoted by $S(p'', p')$. It suffices to know the kernel of the evolution op. $e^{-iH(t''-t')} = K$

$$K(x'', t'' \mid x', t')$$

One must take

$$\begin{cases} x'' = \frac{p''}{m} t'' \\ x' = \frac{p'}{m} t' \end{cases} \tag{30}$$

(by the classical formulas with the free H_0). Then

$$S(p'',p') = \lim_{\substack{t''\to+\infty \\ t'\to-\infty}} \frac{K\left(\frac{p''}{m}t'',t'' \mid \frac{p'}{m}t',t'\right)}{K_0\left(\frac{p''}{m}t'',t'' \mid 0,0\right) K_0\left(0,0 \mid \frac{p'}{m}t',t'\right)} \tag{31}$$

(analogous to (29))

GELFAND-P000750

[source number: 17]

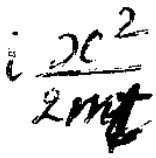
The right-hand side of (31) is computed by a path integral over trajectories with classical asymptotics.

Comment on (31): apply it to Ψ, considering a wave packet

$$\Psi(x,t) = \left[\frac{1}{2\pi}\int \Psi(p) e^{ipx} e^{-i\frac{p^2}{2m}t}\, dp\right] \underset{\substack{\uparrow \\ \text{stationary-phase} \\ \text{method}}}{\cong} \widetilde{\Psi}\left(\frac{mx}{t}\right) e^{i\frac{x^2}{2mt}} \tag{32}$$

Uncertain reading. A short mark appears above the Ψ under the integral. A tilde is uncertain.

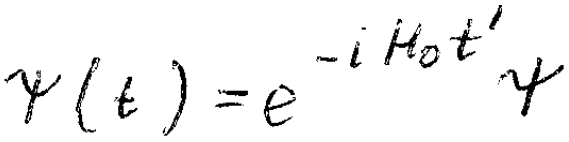
Original source details: GELFAND-P000750



Uncertain reading. The last denominator symbol in the exponential is heavily retraced and tentatively reads t.

Original source details: GELFAND-P000750

$$\Psi(t) = e^{-iH_0t'}\Psi$$



Uncertain reading. The left side reads $\Psi(t)$, while the exponential on the right has t'.

Original source details: GELFAND-P000750

We obtain

$$K(\ldots \mid \tfrac{p}{m}t,t)\,\widetilde{\Psi}(p)\sqrt{\frac{m}{t}}\,dp \tag{33}$$

with factors (in each part we pass to its own coordinate system).

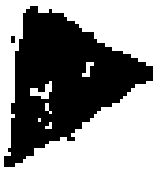
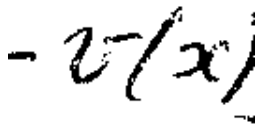


Diagram and typed description. ▶

Original source details: GELFAND-P000750

Path integral. First for quantum mechanics

$$K(x'', t'' \mid x', t') = \frac{1}{N} \int e^{i \int_{t'}^{t''} \left[\frac{m\dot{x}^2}{2} - v(x)\right] dt} \prod_{t' \le t \le t''} dx(t), \tag{34}$$



Uncertain reading. The first sign in $v(x)$ is rounded. It may be υ.

Original source details: GELFAND-P000750

over trajectories such that

$$x(t') = x', \quad x(t'') = x'' \tag{35}$$

N — a normalization factor (depending on time).

Now one must take $x(t') = \frac{pt'}{m}$, $x(t'') = \frac{pt''}{m}$.

In field theory one must write

$$K(u''(x), t'' \mid u'(x), t') = \frac{1}{N} \int \exp\left\{ i \int \mathcal{L}(x)\, dx \right\} \prod_{\substack{t' \le t \le t'' \\ -\infty < x < \infty}} du\, [\text{crossed out}](x, t) \tag{36}$$

Undeciphered passage. There is a crossed-out fragment after du and before (x, t). Its reading is undetermined.

Original source details: GELFAND-P000750

over u, such that

$$\begin{cases} u(x, t'') = u''(x) \\ u(x, t') = u'(x) \end{cases} \tag{37}$$

For the initial and final configurations one must take sums of solitons. This gives transition probabilities.

There is as yet no good quantum version of the inverse problem.

Uncertain reading. A thin diagonal stroke crosses the end of “Хорошего” (good) and continues downward to the right.

Original source details: GELFAND-P000750

For the S-matrix of sine-Gordon, an exact formula has been guessed (Zamolodchikov from ITEP; see Comm. on Math. Physics)

Undeciphered passage. Crossed-out strokes and part of a rectangular box precede “Гипотеза” (hypothesis). Their reading is undetermined.

Original source details: GELFAND-P000750

Hypothesis: all classically solvable problems are also solvable in the quantum version

$$\langle u''(x) \mid e^{-iH(t''-t')} \mid u'(x)\rangle \tag{38}$$

— a matrix element given by the path integral. Now one must divide and let t tend to $\pm\infty$. Corrections to the

Undeciphered passage. A short fragment between “к” (to) and “классической” (classical) is crossed out and undeciphered.

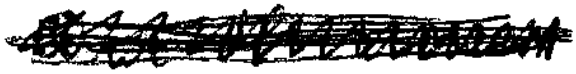
Original source details: GELFAND-P000751

classical mass arise already for a single soliton.

Problems arising in the computation of the integral.

The stationary-phase method.

$$u(x,t) = U_{cl}(x,t) + \sqrt{h}\,\varphi(x,t) + \dots \tag{39}$$



Undeciphered passage. A phrase preceding “Возникает оператор” (the operator arises) is heavily crossed out. Its reading is undetermined.

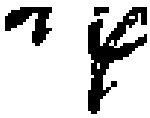
Original source details: GELFAND-P000751

The operator arises

$$A\varphi = \Box\varphi + m^2\varphi + V''(u_\varphi)\varphi \tag{40}$$

Uncertain reading. The heavily retraced signs after m^2 tentatively read $\varphi+$.

Original source details: GELFAND-P000751



Uncertain reading. The subscript on u tentatively reads φ.

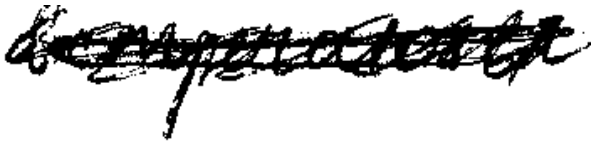
Original source details: GELFAND-P000751

Corrections are computed through $\det A$.

It is expressed through the S-matrix for an auxiliary linear equation. Loops arise next. One needs the inverse operator $R = A^{-1}$ and the kernel $R(x,y)$, namely

$$\int R^3(x,x)\,d^2x \tag{41}$$

Hopes: it is possible that such functionals can be computed explicitly.



Uncertain reading. The crossed-out phrase tentatively reads “в квадратурах” (by quadratures).

Original source details: GELFAND-P000751

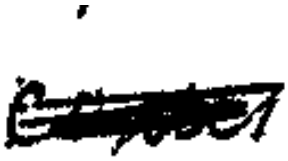
Let u_1 — a one-soliton solution (stationary),

$$V''(u_1(x)) = q(x). \tag{42}$$

Computing

$$\det\left(\frac{d^2}{dt^2} - \frac{d^2}{dx^2} + q(x)\right) \tag{43}$$

reduces to the spectral density of the stationary op. (the variables separate). One obtains $q(x) = \frac{1}{\mathrm{ch}^2\, x}$ and all eigenfunctions are found explicitly (the simplest Bargmann potential).
$\det A$ is expr. through the eigenfunction with zero eigenvalue.
Corrections to the mass of the



Original source details: GELFAND-P000751

Undeciphered passage. A short word before “солитона” (soliton) is heavily crossed out and undeciphered.

soliton:

$$M = \frac{8m}{\gamma} - \frac{m}{\pi} + O(\gamma) \tag{44}$$

Conjecture: for sine-Gordon

$$M = \frac{8m}{\gamma} - \frac{m}{\pi}, \tag{45}$$

i.e. there are no higher corrections. (justif. by Zamolodchikov)

Original source details: GELFAND-P000751

Uncertain reading. The abbreviation before the surname tentatively reads “оправд.” (justified). Its first letter may be “с”.

For $\gamma \leq 8\pi$ everything is fine, and beyond this the soliton disappears.

GELFAND-P000752

[source number: 18]

Formulas for the S-matrix of sine-Gordon.
One must compute the abbreviated action for the 2-soliton solution.

$$S_{sa}(p_1'', p_2''; p_1', p_2') = \delta(p_1'' - p_1')\delta(p_2'' - p_2')e^{i\Theta(p_1, p_2)} \tag{46}$$

Original source details: GELFAND-P000752

Uncertain reading. The separator between the arguments is heavily retraced. It is tentatively rendered as a semicolon.

(There is a soliton and an antisoliton)

The action (classical)

$$\Theta = \frac{8}{\gamma}\int_0^{\pi} d\alpha \left[\ln \frac{e^{i(\beta-\alpha)}+1}{e^{i\beta}+e^{-i\alpha}}\right] \tag{47}$$

[Crossed out: For $\gamma = \frac{8\pi}{N}$, N — an integer]

Original source details: GELFAND-P000752

Undeciphered passage. The end of the crossed-out line following “целое” (integer) is undeciphered.

(the integral cannot be evaluated).

One must take $\gamma = \frac{8\pi}{N}$, N — an integer

$$\Theta \longrightarrow \sum \ln(\dots) \tag{48}$$

Zamolodchikov took this to hold for $\gamma = \frac{8\pi}{N}$ and interpolated.

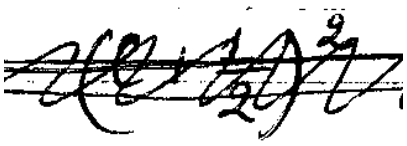
As a result sine-Gordon turned out to resemble a quantum-mechanical system [Crossed out: and] with

$$H = -\frac{d^2}{dx^2} + \frac{a}{\mathrm{ch}^2\,\alpha x} \tag{49}$$

Uncertain reading. The numerator tentatively reads a, possibly q.

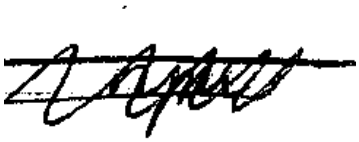
Original source details: GELFAND-P000752

[Crossed out: Eigenvalue $\alpha(\ell+\frac{1}{2})^2$]



Uncertain reading. The crossed-out formula tentatively reads $\alpha(\ell+\frac{1}{2})^2$.

Original source details: GELFAND-P000752



Undeciphered passage. The end of the crossed-out line is undeciphered. A fragment tentatively reads “с пр...”.

Original source details: GELFAND-P000752

A remarkable dynamical system

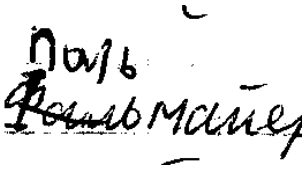
is associated with a field (chiral)

$$n(x) : R^2 \longrightarrow S^n \subset R^{n+1} \tag{50}$$

$$\mathcal{L} = \frac{1}{2} \int \partial n \cdot \partial n \, dx \tag{51}$$

it is nontrivial because of its values (there is nonlinearity).

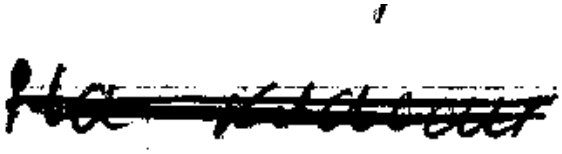
L-A pair has been written down ([Crossed out: Fol; written above: Pol]meyer (?); Zakharov and Mikhailov)



Original source details: GELFAND-P000752

Uncertain reading. The beginning of the surname is corrected from tentative “Фоль” (Fol) to “Поль” (Pol).

[Crossed out: Na]



Original source details: GELFAND-P000752

Undeciphered passage. The remainder of the crossed-out phrase before “Квантование” (quantization) is undeciphered.

Quantization is weakly connected with the classical theory.

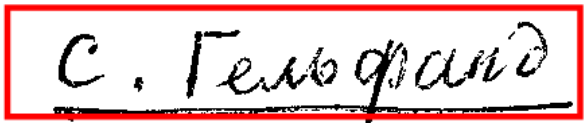
Massive particles appear as solitons.

There will be a talk on this at the seminar named after I. G. Petrovsky.

GELFAND-P000753

13/II-78. Gelfand Seminar.

I. M. G. there are 2 primary concepts in category theory: 1) Topos 2) Derived category.



Original source details: GELFAND-P000753

Diagram and typed description. S. Gelfand

Problem: describe all algebraic bundles over

Original source details: GELFAND-P000753

Undeciphered passage. A short sign before the projective-space notation is heavily crossed out.

$\mathbb{P}^n$. (n-dimensional projective space; for example, over $\mathbb{C}$).

Ξ — affine space.

$$\xi \sim \alpha\xi, \quad \alpha \in \mathbb{C}^*, \quad (\xi \neq 0); \qquad \text{Then } \Xi/\sim = \mathbb{P}^n.$$

Original source details: GELFAND-P000753

Uncertain reading. The multiplier and the following sign are tentatively read as α.

Λ — the Grassmann algebra in $(n+1)$-th variable (

Original source details: GELFAND-P000753

Undeciphered passage. An abbreviation at the beginning of the parentheses is heavily crossed out; its reading is unresolved.

algebra over Ξ).

$$\Lambda = \{\xi_0, \dots, \xi_n \mid \xi_i\xi_j + \xi_j\xi_i = 0\}.$$

$$\dim \Lambda = 2^{n+1}.$$

Introduce a grading on Λ:

$$\Lambda = \Lambda_0 \oplus \Lambda_{-1} \oplus \cdots \oplus \Lambda_{-(n+1)}, \qquad \text{[Crossed out: } \Lambda_0 \text{ — the field of constants]} \tag{1}$$

where

$$\Lambda_{-k} \text{ [Crossed out: } \Lambda_k?] = \{\xi_{i_1} \cdots \xi_{i_k}\}, \qquad \Lambda_0 \text{ — the field of constants.}$$

Original source details: GELFAND-P000753

Uncertain reading. The cancelled expression before the equality sign tentatively reads Λ_k.

Objects: a graded finite-dimensional representation of Λ or a graded Λ-module:

$$V = \bigoplus V_j, \qquad \dim V < \infty. \tag{2}$$

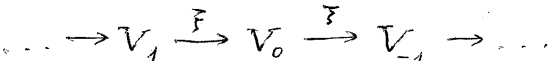
$$\Lambda_{-k} V_j \subset V_{j-k}. \tag{3}$$

Let $\xi_0, \dots, \xi_n$ — be a basis in Ξ, $\xi \in \Xi$.

$$\xi = c_0\xi_0 + \cdots + c_n\xi_n. \tag{4}$$

$(-k)$ — for historical reasons (from Serre's paper).

Then $\xi : V_j \longrightarrow V_{j-1}$. Hence, since $\deg \xi = -1$, we have



Original source details: GELFAND-P000753

Diagram and typed description.

$$\cdots \longrightarrow V_1 \xrightarrow{\xi} V_0 \xrightarrow{\xi} V_{-1} \longrightarrow \cdots$$

$\xi^2 = 0 \Rightarrow$ we have obtained a complex.
Thus, from an element $\xi \neq 0$ and a module V one constructs the complex

$$L_\xi(V) = \{\cdots \longrightarrow V_1 \xrightarrow{\xi} V_0 \xrightarrow{\xi} V_{-1} \longrightarrow \cdots\}. \tag{5}$$

Definition. V is called proper, if

$$\forall \xi \neq 0, \qquad H_i(L_\xi(V)) = 0 \quad \text{for } i \neq 0. \tag{6}$$

Here:

$$H_i \text{ [Crossed out: ?] } (L_\xi(V)) = \ker(V_i \longrightarrow V_{i-1}) / \operatorname{Im}(V_{i+1} \longrightarrow V_i) \tag{7}$$

Original source details: GELFAND-P000753

Undeciphered passage. A sign between the index and the parenthesis in (7) is heavily crossed out.

Original source details: GELFAND-P000753

Uncertain reading. The equation number is heavily retraced; it tentatively reads (7).

— [Crossed out: quotient spaces?]

Original source details: GELFAND-P000753

Uncertain reading. The crossed-out word tentatively reads “факторпространства” (quotient spaces); its beginning is obscured by dense strokes.

homology of the complex.

GELFAND-P000754

[source number: 43]

In this case

$$\dim H_0^?(L_\xi(V)) = \text{const}$$

Original source details: GELFAND-P000754

Uncertain reading. A dense mark appears above H_0; its internal strokes cannot be distinguished. A question mark represents it in the typeset formula.

(does not depend on ξ)

by Euler's formula

(left side — the Euler characteristic ξ?

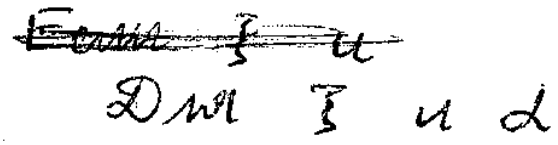
Original source details: GELFAND-P000754

Uncertain reading. The overwritten mark after "характеристика" (characteristic) may be ξ.

equal to $\sum(-1)^i \dim V_i$.)

Bundle: $\xi \longmapsto H_0(L_\xi(V))$.

[Crossed out: If ξ and]



Original source details: GELFAND-P000754

Uncertain reading. The short crossed-out line tentatively reads "Если ξ и" (If ξ and).

For ξ and $\alpha\xi$ it is the same (there is a natural isomorphism).

Example. $V = \Lambda$ (the regular representation).

$V_j = \Lambda_j$ (the same grading as before).

Answer: $H_i(L_\xi(V)) = 0, \quad \forall i, \quad \forall \xi \neq 0$.

We obtain the zero bundle.

Direct sums give the same result.

$V_j = \Lambda_{j+\ell} \quad (\ell$ — fixed$)$

$\Rightarrow$ we again obtain a trivial bundle.

Hence free modules lead to trivial bundles.

Theorem 1 Every bundle is obtained by such a construction, and V is determined uniquely up to adding a free module.

(Bernstein, S. Gelfand, I. Gelfand; independently a similar result by Beilinson, a 3rd-year student)

Refinement of the bundle construction.

V — an arbitrary Λ-module.

$O(k)$ — the standard bundles (their sections are homogeneous rational functions of degree $k \in \mathbb{Z}$), one-dimensional.

Consider

$$\mathcal{L}_j = V_{-j} \otimes O(j). \tag{8}$$

— a bundle over $\mathbb{P}^n$; each fiber is multiplied by V_{-j}.

$O(1)$ — the one that has a section.

Construct

$$d : \mathcal{L}_j \longrightarrow \mathcal{L}_{j+1}. \tag{9}$$

$f(\xi)$ — a section of $\mathcal{L}_j$ over an open set

a homogeneous function of degree j with values in V_{-j}

$$d(f(\xi)) = \xi(f(\xi)). \tag{10}$$

(on the right is the action of ξ on V_{-j}, giving V_{-j-1}).

GELFAND-P000755

$O(1)$ — the Hopf bundle. (or $O(-1)$). We obtain the complex

$$\{\ldots \longrightarrow \mathcal{L}_{-1} \xrightarrow{d} \mathcal{L}_0 \xrightarrow{d} \mathcal{L}_1 \longrightarrow \ldots\} = \mathcal{L}(V)\,\xi?. \tag{11}$$

Uncertain reading. The overwritten mark after $\mathcal{L}(V)$ tentatively reads ξ; it may be a cancellation.

Original source details: GELFAND-P000755

over each point ξ there is the same complex as before (easy to check)

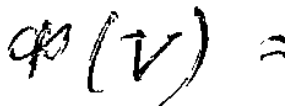
Assertion: if V — proper, then $H^0(\mathcal{L}(V))$ — a bundle and $H^i(\mathcal{L}(V)) = 0$ for $i \neq 0$.

Set

$$\Phi(V) = H^0(\mathcal{L}(V)). \tag{12}$$

Theorem $1'$. 1) Every bundle is $\Phi(V)$ for some proper V

2) $\varphi(V) \approx \Phi(V') \Leftrightarrow \exists$ free P, P': $V \oplus P \cong V' \oplus P'$.



Uncertain reading. The left side of part 2 has φ, while the right side has Φ. Both letters in (15) have the form φ.

Original source details: GELFAND-P000755

The inverse map.

From a bundle one must construct a Λ-module V.

$$\underset{\text{bundle}}{\mathcal{E}} \rightsquigarrow \mathcal{E} \otimes O(1) \tag{13}$$

what does this correspond to in Λ-modules?

Suppose there is an exact sequence of Λ-modules

$$0 \longrightarrow \underset{\text{proper}}{V} \longrightarrow \underset{\text{free}}{P} \longrightarrow V' \longrightarrow 0 \tag{14}$$

Construct $W \circ ?$; $W_j = V'_{j+1}$.

Uncertain reading. The small separate mark after W resembles $\circ$ crossed by a horizontal stroke. Its function is unresolved.

Original source details: GELFAND-P000755

$$\underline{\text{Assert.}}: \quad W \text{ — proper and } \varphi(W) = \varphi(V) \otimes O(1) \tag{15}$$

Let $\mathcal{E}$ — a bundle; consider $\mathcal{E} \otimes O(N)$ for sufficiently large N. Serre's theorem: a bundle $\mathcal{E} \otimes O(N)$ has many sections ("an ample bundle"), they pass through any point in any fiber.

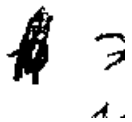
Consider $C_N = \Gamma(\mathbb{P}^n, \mathcal{E} \otimes O(N))$;

$$\dim \Gamma(\mathbb{P}^n, O(1)) = n + 1 \tag{16}$$

(linear functions of ξ)

$\xi_0, \ldots, \xi_n$ — a basis in Ξ

$x_0, \ldots, x_n$ — the dual basis in $\Gamma(\mathbb{P}^n, O(1))$ ([crossed out: v?]



Original source details: GELFAND-P000755

Uncertain reading. A heavily cancelled mark before "это пр-во" (this is the space) may be the letter "в".

this is the space of linear functions)

(coefficients of the expansion in the preceding basis)

There are operators

$$X_i : C_N \longrightarrow C_{N+1}, \quad i = 0, 1, \ldots, n. \tag{17}$$

(multiplication by x_i)

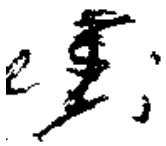
GELFAND-P000756

[source number: 44]

Now construct 2 Λ-modules

$$V = C_N \otimes \Lambda \tag{18}$$

(on C_N the action is trivial Ξ?;



Original source details: GELFAND-P000756

Uncertain reading. The sign following "действие" (action) is heavily retraced; the reading Ξ is tentative.

on Λ the usual action)

$$V' = C_{N+1} \otimes \Lambda. \tag{18'}$$

We have

$$\partial : V \longrightarrow V' \tag{19}$$

$$\partial(c \otimes \lambda) = \sum_{i=0}^{n} X_i(c) \otimes \xi_i \lambda \tag{20}$$

The map ∂ does not depend on the basis. Set $W = \operatorname{Ker} \partial$.

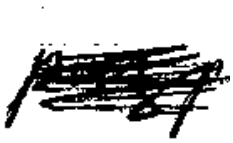
This is a Λ-module. Now one must shift back N times

$$0 \longrightarrow W' \longrightarrow P \longrightarrow W \longrightarrow 0$$

$W \longmapsto W'$ perform N times (with a shift of grading).

This is precisely that representation (that same Λ-module)

I.M.G. Classification of bundles — not quite the right problem (they do not form an abelian category). One must allow the fibers to change



Undeciphered passage. A short word between “менять” (change) and “размерность” (dimension) is heavily crossed out.

Original source details: GELFAND-P000756

dimension — introduce sheaves. The work being presented develops Serre’s theory.

Serre’s theorem

$$\{\text{Sheaves on } \mathbb{C}P^n\} \Leftrightarrow \left\{ \begin{array}{l} \text{finitely generated } \overset{\text{(graded)}}{\text{modules}} \\ \text{over } \mathbb{C}[x_0, \ldots, x_n] \end{array} \Big/ \begin{array}{l} \text{up to} \\ \text{a finite} \\ \text{number of homogeneous} \\ \text{components} \end{array} \right\}$$

ξ?

Uncertain reading. The separate overwritten sign to the left of the reference may be ξ; its function is unresolved.

Original source details: GELFAND-P000756

(an article in the collection “Fiber spaces”; 1953).

Undeciphered passage. A short fragment after the collection reference is crossed out.

Original source details: GELFAND-P000756

Construct a sheaf from the module M.

Define its sections over $E_i = \{x.\ x_i \neq 0\}$

[Crossed out: $\Gamma(E_i, M)$?]

Uncertain reading. The cancelled formula tentatively reads $\Gamma(E_i, M)$; diagonal strokes obscure the arguments.

Original source details: GELFAND-P000756

$$\Gamma(E_i, \underset{\text{our sheaf}}{L}) = \left\{ \begin{matrix} \dfrac{m}{x_i^s}, & \deg m = s \\ & m \in M \end{matrix} \right\}.$$

$O(N)$ arises from $M = \mathbb{C}[x_0, \ldots, x_n]$ with shifted grading

$$M_i = \{\mathbb{C}[x_0, \ldots, x_n]\}_{i+N}.$$

GELFAND-P000757

27/II-78. Gelfand Seminar.

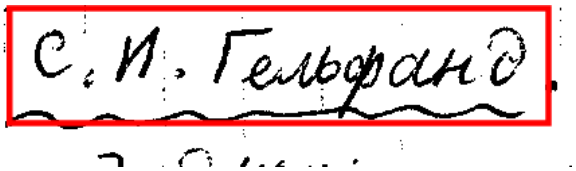


Diagram and typed description. S. I. Gelfand

Original source details: GELFAND-P000757

Problem: describe algebraic vector bundles over projective space.

Ξ — $(n+1)$-dimensional space (vector).

$\mathbb{P} = \mathbb{P}(\Xi)$ — projective n-dimensional space

[Crossed out]

Undeciphered passage. A short word before “Выберем базис” (choose a basis) is crossed out; a complete reading is unresolved.

Original source details: GELFAND-P000757

Choose a basis of Ξ: $\xi_0, \ldots, \xi_n$

and let

$\Lambda = \Lambda(\Xi)$ — the exterior algebra of Ξ

$$\Lambda = \bigoplus_{i=-n-1}^{0} \Lambda_i$$

Λ has generators $\xi_0, \ldots, \xi_n$ and relations $\xi_i\xi_j + \xi_j\xi_i = 0, \quad \xi_i^2 = 0;$

$$\Lambda_{-k} = \{\xi_{i_1} \ldots \xi_{i_k}\}.$$

$$\dim \Lambda_{-k} = \binom{n+1}{k}$$

Remark: let $\xi \in \Lambda_{-1}$. Then $\xi^2 = 0$. (and this is equivalent to all the relations).

Object: a graded finite-dimensional Λ-module

$$V = \bigoplus_j V_j, \qquad \lambda \in \Lambda_i,\ v \in V_j \Rightarrow \lambda v \in V_{i+j}$$

(j may be arbitrary integers).

From each V construct a collection of vector bundles over $\mathbb{P}(\Xi)$ and their maps (a complex of bundles).

$\forall j$ construct the bundles $O(j)$

3 definitions of $O(j)$

1) transition functions $(z_k/z_\ell)^j$;

2) sections over an open set $=$ homogeneous rational functions of degree j;

3) j-th tensor power of the anti-Hopf bundle.

The corresponding sheaf can be defined as follows:

$$\Gamma(O(j), U) = \{\text{rational functions of degree } j \text{ without singularities in } \bar{U}\}$$

$U \subset \mathbb{P}^n$, $\bar{U}$ — the preimage of U in Ξ

Problem: why does a one-dimensional bundle result?

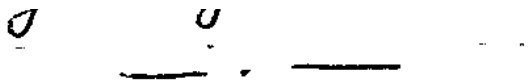
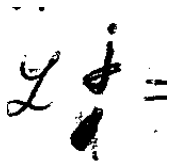


Diagram and typed description. —— · ——

Original source details: GELFAND-P000758

Set

$$\mathcal{L}^j = V_{-j} \otimes O(j) \qquad \text{— a bundle.}$$

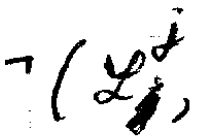


Uncertain reading. The subscript of $\mathcal{L}$ is blotted out; j is written above. The cancelled subscript tentatively reads j.

Original source details: GELFAND-P000758

V_{-j} means the constant bundle with fiber V_{-j}

$$\Gamma(\mathcal{L}^j, U) = \left\{ \begin{matrix} \text{rational functions with values in } V_{-j}, \\ \text{homogeneous degree } j, \text{ nonsingular in } \bar{U} \end{matrix} \right\}.$$



Uncertain reading. In $\Gamma(\mathcal{L}^j, U)$ the subscript is also blotted out; j is written above. The cancelled subscript tentatively reads j.

Original source details: GELFAND-P000758

Construct the maps:

$$d : \mathcal{L}^j \longrightarrow \mathcal{L}^{j+1}$$

On sections: let $f(\xi) \in \Gamma(\mathcal{L}^j, U)$,

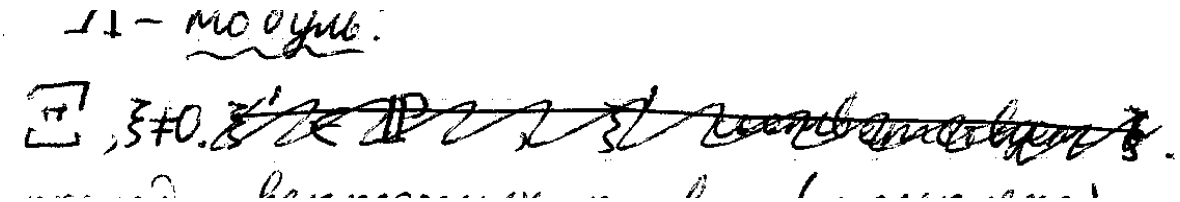
$$\xi \in \Lambda_{-1} = \Xi$$

$$(df)(\xi) = \xi[f(\xi)]$$

(on the right is the action of ξ as an element of the exterior algebra on V_{-j}).
A complex of bundles results:

$$\mathcal{L}(V) = \{\mathcal{L}^j, d\}, \qquad d^2 = 0.$$

Proper Λ-module:
let $\xi \in \Xi, \quad \xi \neq 0$. [Crossed out: $\xi' \in \mathbb{P}$ — ξ' corresponding to ξ.]



Original source details: GELFAND-P000758

Uncertain reading. The cancelled projective-point statement tentatively reads $\xi' \in \mathbb{P}$ — ξ' corresponding to ξ. Some crossed-out signs remain unclear.

Consider the sequence of vector spaces (complex)

$$\{\ldots \longrightarrow V_{-j+1} \xrightarrow{\xi} V_{-j} \xrightarrow{\xi} V_{-j-1} \longrightarrow \ldots\} = L_\xi.$$

GELFAND-P000759

[source number: 56]

Definition. V proper, if

$$H^i(L_\xi) = 0, \quad \forall i \neq 0, \quad \forall \xi \neq 0.$$

L_ξ — the complex obtained from $\mathcal{L}(V)$ at the point ξ.
For proper V

$$\dim H^0(L_\xi) = \text{const}$$

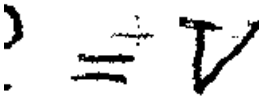
The spaces $H^0(L_\xi)$ glue into a vector bundle. The precise definition is as follows: set

$$\begin{aligned}\Phi(V) &= H^0(\mathcal{L}(V)) \\ &= \operatorname{Ker}(d : \mathcal{L}^0 \to \mathcal{L}^1) / \operatorname{Im}(d : \mathcal{L}^{-1} \to \mathcal{L}^0).\end{aligned}$$

Theorem 1. 1) Every algebraic vector bundle $\mathcal{E}$ has the form $\Phi(V)$ for some proper V.
2)

$$\Phi(V) \cong \Phi(V') \Longleftrightarrow V \oplus P \cong V' \oplus P'$$

for free P, P'.



Uncertain reading. In the second relation there is a faint stroke above the two horizontal lines: tentatively $\cong$, possibly $=$.

Original source details: GELFAND-P000759

The inverse map.

Coherent sheaves of modules on $\mathbb{P}^n$

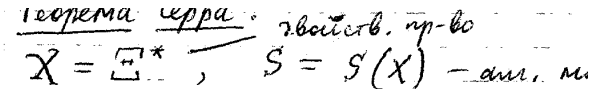
(= quotient sheaves of the sheaves of sections of bundles).

This is more convenient, since one can take images, cokernels, etc..

(~ adjoining differences).

Serre's theorem:

$$X = \Xi^*, \qquad S = S(X) \text{ — the algebra of polynomials.}$$



Original source details: GELFAND-P000759

Diagram and typed description.

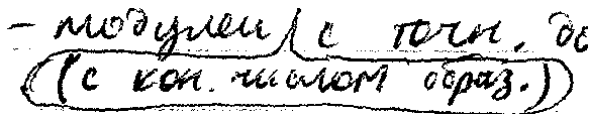
Ξ^* / dual space

The category of sheaves on $\mathbb{P}^n$ (coherent) [crossed out: po?]

Uncertain reading. The cancelled word before "эквивал." (equivalent) tentatively reads "по", possibly "до".

Original source details: GELFAND-P000759

is equivalent to the category of graded $S(X)$-modules up to finite-dimensional ones.



Original source details: GELFAND-P000759

Diagram and typed description.

modules

(with finitely many generators)

$$\mathcal{E} \longmapsto M = \bigoplus M_j, \qquad M_j = \Gamma(\mathcal{E} \otimes O(j), \mathbb{P}^n)$$

$$S_k = \Gamma(O(k), \mathbb{P}^n)$$

The module structure is introduced in the obvious way.

Original source details: GELFAND-P000759

Diagram and typed description.

Proof of Theorem 1.

1°. From a Λ-module V construct a complex of S-modules $W = (W^k, d)$:

$$W^k = V_{-k} \otimes_{\mathbb{C}} S(X)$$

Let $x_0, \ldots, x_n$ — a basis in X, dual to the basis in Ξ. set

$$d(v \otimes s) = \sum \xi_i v \otimes x_i s.$$

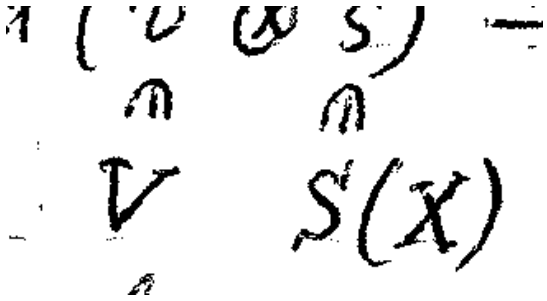


Diagram and typed description.

$$\begin{array}{cc} v & s \\ \cup & \cup \\ V & S(X) \end{array}$$

Original source details: GELFAND-P000759

It is easy to check that $d : W^k \to W^{k+1}$ and $d^2 = 0$.

In each W^k define the grading:

$$(W^k)_j = V_{-k} \otimes S_{j+k}$$

d — a morphism of graded modules (it does not change degree).

GELFAND-P000760

(This is the algebraic version of the geometric construction given above)

$V \longmapsto$ the complex of bundles $\mathcal{L}(V)$

$\overline{W}$ — This is the algebraic analogue of $\mathcal{L}(V)$.

Uncertain reading. A short detached stroke over the left of W is represented by an overbar.

Original source details: GELFAND-P000760

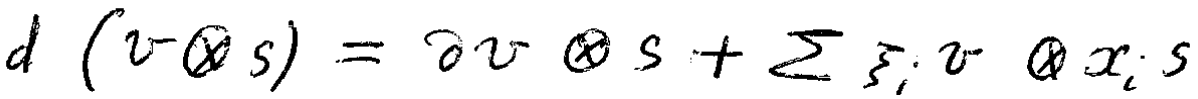
Generalization: consider a complex of Λ-modules

$$\ldots \xrightarrow{\partial} V^i \xrightarrow{\partial} V^{i+1} \xrightarrow{\partial} V^{i+2} \xrightarrow{\partial} \ldots$$

Imprecisely: construct $W^{i,k}$ (bicomplex) and sum over the diagonals
(the complex from the bicomplex)

$$V^i = \bigoplus (V^i)_j; \qquad W^k_j = \bigoplus_\ell V^{\ell+k}_{j-\ell} \otimes S_\ell$$

$$d(v \otimes s) = \partial v \otimes s + \sum \xi_i v \otimes x_i s$$



Original source details: GELFAND-P000760

Uncertain reading. There is a plus sign between the terms. No additional sign factor is visible before the sum.

It does not change [crossed out: its?]

Uncertain reading. The short cancelled entry may read “ero” (its) or j.

Original source details: GELFAND-P000760

the lower index, and increases the upper one by 1.

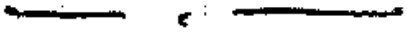


Diagram and typed description.

——— • ———

Original source details: GELFAND-P000760

2°. Complex of graded S-modules $\longmapsto$ complex of graded Λ-modules

$$\ldots \longrightarrow W^k \xrightarrow{d} W^{k+1} \xrightarrow{d} \ldots$$

Set $\overline{W} = \bigoplus W^k$

Uncertain reading. Here W is underlined and also has a short stroke over its left side.

Original source details: GELFAND-P000760

and $\underline{V} = \mathrm{Hom}_{\mathbb{C}}(\Lambda, W)$

Introduce a grading on $\underline{V}$:

$$v \in V_j^i \Longleftrightarrow v(\Lambda_k) \subset W_{j+k}^{i-j-k} \quad \text{for all } k.$$

Set:

$$\partial : V^i \longrightarrow V^{i+1}$$

$$\partial(v)(\lambda) = d(v(\lambda)) + \sum x_i v(\xi_i \lambda)$$

This map is in a certain sense inverse to the preceding one.

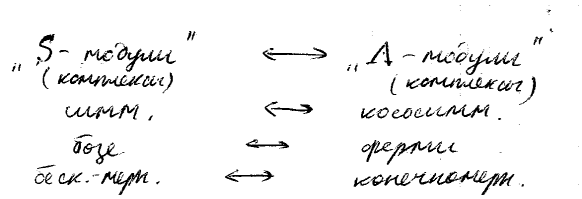


Diagram and typed description.

"S-modules" (complexes)	$\longleftrightarrow$	"Λ-modules" (complexes)
symm.	$\longleftrightarrow$	skew-symm.
Bose	$\longleftrightarrow$	Fermi
infinite-dim.	$\longleftrightarrow$	finite-dim.

Original source details: GELFAND-P000760

2°. $C^b(\Lambda)$ — the category of finite-length complexes of Λ-modules.

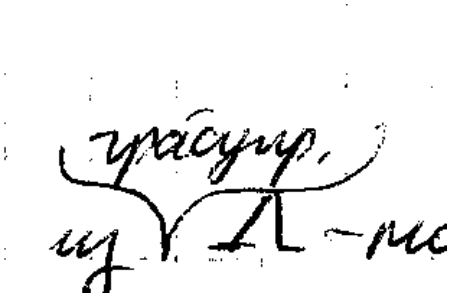


Diagram and typed description.

graded

of Λ-modules

Original source details: GELFAND-P000760

$C^b(S)$ — __ " ___ of finitely generated graded S-modules

(the differential preserves the grading)

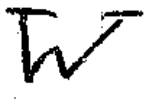
GELFAND-P000761

[source number: 57]

$C(\Lambda)$ — the category of infinite complexes of graded Λ-modules (but infinite-dimensional ones)

The functors have been constructed:

$$F : \ C^b(\Lambda) \longrightarrow C^b(S)$$
$$V \longmapsto W \qquad \text{(see item } 1^\circ\text{)}.$$



Uncertain reading. The upper left stroke touches W and is rendered as part of the letter.

Original source details: GELFAND-P000761

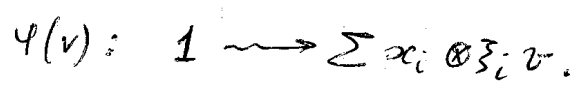
$$G:\ C^b(S) \longrightarrow C(\Lambda) \qquad \text{(see item } 2^\circ\text{)}.$$

Main technical theorem.

Consider the natural inclusion $i : C^b(\Lambda) \longrightarrow C(\Lambda)$. There is a morphism of functors $i \longrightarrow G \circ F$, inducing an isomorphism in cohomology.

This means that there is a natural map

$$\varphi : V \rightsquigarrow \mathrm{Hom}(\Lambda, S \otimes V)$$
$$\varphi(v) : \quad 1 \rightsquigarrow \sum x_i \otimes \xi_i v.$$



Uncertain reading. The manuscript gives the value at 1 as $\sum x_i \otimes \xi_i v$.

Original source details: GELFAND-P000761

($\forall V$ one must construct a homomorphism

Undeciphered passage. A sign after the abbreviation for homomorphism is heavily crossed out.

Original source details: GELFAND-P000761

$V \longrightarrow G \circ F(V)$)

V — a one-dimensional trivial module $\Rightarrow$ on the right is the Koszul complex.

Diagram and typed description.

——— • ———

Original source details: GELFAND-P000761

A — a category (abelian), $C(A)$ — the category of complexes over A (infinite ones)

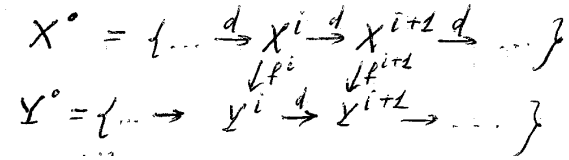


Diagram and typed description.

$$X^\bullet = \{\dots \xrightarrow{d} X^i \xrightarrow{d} X^{i+1} \xrightarrow{d} \dots\}$$
$$\downarrow f^i \qquad \downarrow f^{i+1}$$
$$Y^\bullet = \{\dots \longrightarrow Y^i \xrightarrow{d} Y^{i+1} \longrightarrow \dots\}$$

Original source details: GELFAND-P000761

morphism $\{f^i\}$ — the diagram commutes.

Cohomology:

$$H^i(X^\bullet) = \ker d / \operatorname{Im} d$$

map

$$X^\bullet \xrightarrow{f} Y^\bullet \rightsquigarrow f_i^* : H^i(X^\bullet) \longrightarrow H^i(Y^\bullet)$$

f is called a quasi-isomorphism, if it induces an isomorphism on all cohomology groups.

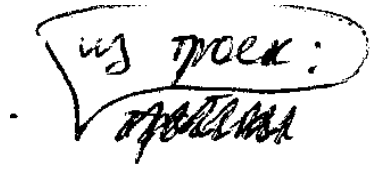
Roughly: the derived category is obtained by inverting quasi-isomorphisms.

Precise definition.

S — the set of quasi-isomorphisms. ("ensemble" in Verdier)

$$\mathrm{Ob}(D(A)) = \mathrm{Ob}(C(A))$$

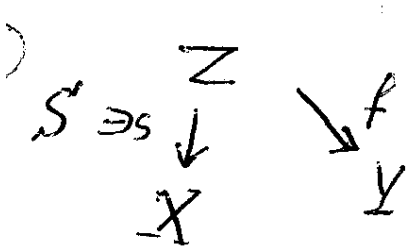
$\mathrm{Hom}_{D(A)}(X, Y)$ — [crossed out: triples?]



Original source details: GELFAND-P000761

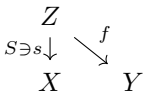
Uncertain reading. The crossed-out word tentatively reads "тройки" (triples). Above it the outlined insertion reads "из троек:" (from triples).

[inserted above in an outline: from triples:]



Original source details: GELFAND-P000761

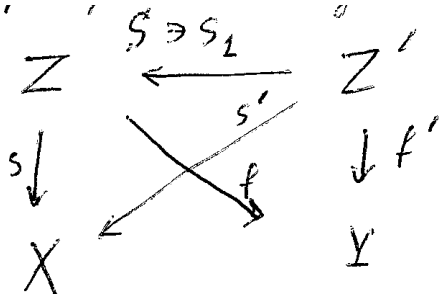
Diagram and typed description.



GELFAND-P000762

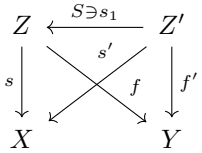
[Crossed out: 2 triples are called equivalent if]

Map of triples: a commutative diagram.



Original source details: GELFAND-P000762

Diagram and typed description.



$\mathrm{Hom}_{D(A)}(X, Y)$ — the inductive limit over such triples.

Element:

$$Z \longleftarrow Z' \longleftarrow Z'' \longleftarrow \cdots$$

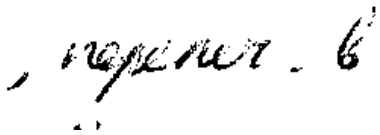
identified if they agree from some point onward

2 places where this is set out:

1) Hartshorne 20 vol. L.N.

Residues and duality

2) Verdier dissertation abstract (1963), reprinted?



Original source details: GELFAND-P000762

Uncertain reading. The abbreviation between the date and the volume number tentatively reads “перепеч.” (reprinted).

in 569 vol. Lect.Notes

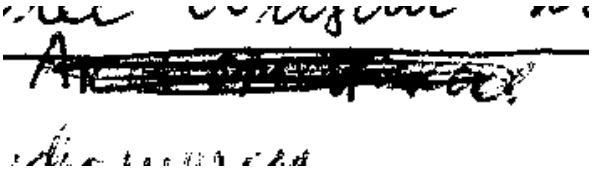
(there are no proofs here)

More general construction (axiomatic)

Original source details: GELFAND-P000762

Diagram and typed description. —— . ——

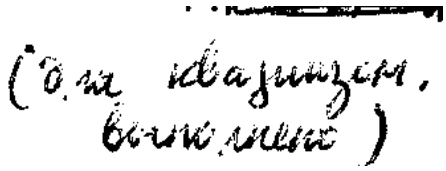
[Crossed out: A...]



Original source details: GELFAND-P000762

Undeciphered passage. The initial A is visible in the cancelled heading. Dense strokes obscure the remainder.

[In the left margin: (for quasi-isomorphisms they hold)]



Original source details: GELFAND-P000762

Uncertain reading. The end of the abbreviated word “квазиизм.” (quasi-isomorphisms) in the margin is tentative.

1)

$$\left.\begin{array}{l} s_1 \in S, \quad s_2 \in S \\ s_1 s_2 \text{ — exists} \end{array}\right\} \Rightarrow s_1 s_2 \in S.$$

2)

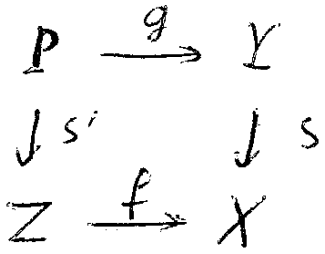


Diagram and typed description.

$$\begin{array}{ccc} P & \xrightarrow{g} & Y \\ \downarrow s' & & \downarrow s \\ Z & \xrightarrow{f} & X \end{array}$$

Original source details: GELFAND-P000762

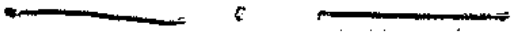
$$\forall\, Z, X, Y,\ f, s$$
$$\exists\, P, g, s'$$

3) Let $f : X \longrightarrow Y$. Then the following properties are equivalent

(i) $\exists\, s \in S :\ sf = 0$

(ii) $\exists\, s' \in S :\ fs' = 0$

Diagram and typed description.



Original source details: GELFAND-P000762

What is needed to construct the morphism of functors?

(the one we need).

1) It is constructed.

2) One must check that for one-dimensional V everything is in order (simple calculations with the Koszul complex)

3) Introduce filtrations, the factors are standard.

Derived category: reduction to the factors.

GELFAND-P000763

[source number: 62]

6/III–78 Gelfand seminar.

[I. M. G.] Sheaf of C^∞ functions on $(-1, 1)$, vanishing at 0 at the point 0 — a sheaf of sections of a vector bundle?

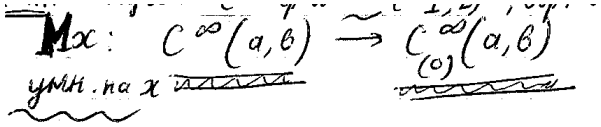


Diagram and typed description.

$$\begin{array}{ccc} Mx: & C^\infty(a,b) & \longrightarrow & C^\infty_{(0)}(a,b) \\ & \text{mult. by } x & & \end{array}$$

Original source details: GELFAND-P000763

isomorphism of $C^\infty(a,b)$-modules.

Diagram and typed description. Aomoto

Original source details: GELFAND-P000763

Exactly solvable models in neutral massive field theories. (on $\mathbb{R}^2$)

Uncertain reading. The final letter x of «нейтральных» (neutral) is retraced.

Original source details: GELFAND-P000763

3 types of fields.

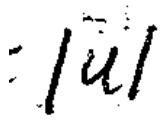
1°. Auxiliary fermions (on $\mathbb{R}$)

on $\mathbb{R}$; $u \in \mathbb{R}$; fields covariant with respect to the action of $\mathbb{R}^+$ on $\mathbb{R}$;

$\psi(u)^+$ and $\psi(u)$ — creation and annihilation operators;

$$\psi(-u) = \psi^+(u); \tag{1}$$

$$[\psi(u), \psi(u')]_+ = 2\pi|u|\delta(u+u'); \tag{2}$$



Uncertain reading. Short separate strokes are visible above the u in the absolute value. Their status as a distinct sign is uncertain.

Original source details: GELFAND-P000763

left-hand side $\underset{\text{def}}{=} \psi(u)\psi(u') + \psi(u')\psi(u)$,

2°. $\varphi_F(x)$, $\varphi^F(x)$. (on $\mathbb{R}^2$)

Consider a pair of functions:

$$w(x) = (w_+(x), w_-(x))$$

$$\frac{\partial w_\pm}{\partial x^\pm} \mp m w_\mp(x) = 0, \tag{3}$$

(Dirac equation) (2 equations), m — a number.

where

$$x^\pm = \frac{x^0 \pm x^1}{2}, \qquad p^\pm = p^0 \pm p^1.$$

$$\langle w, w' \rangle = \int_{-\infty}^{\infty} dx^1 \big(w_+(x) w'_+(x) + w_-(x) w'_-(x)\big) \tag{4}$$

$$\psi_\pm(x) = \int_{-\infty}^{\infty} \underline{du}\, \sqrt{0 + iu}^{\pm} \psi(u) \exp\left(-im\left(x^- u + \frac{x^+}{u}\right)\right), \tag{5}$$

Uncertain reading. A small oval is visible above the right parenthesis. Its meaning is uncertain.

Original source details: GELFAND-P000763

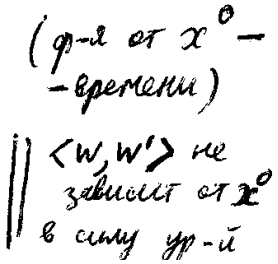


Diagram and typed description.
(a function of x^0 — time)
$\| \langle w, w' \rangle$ is independent of x^0 by the equations.

Original source details: GELFAND-P000763

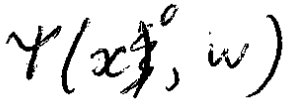
Here

$$\underline{du} = \frac{du}{2\pi|u|}$$

GELFAND-P000764

$\pm$ above the radical — choice of branch $+0$ or -0

$$w(x) \longleftrightarrow \int_{-\infty}^{\infty} dx^1 (w_+\psi_+ + w_-\psi_-) = \psi(x^0, w) \tag{6}$$



Uncertain reading. The first argument contains a dense correction after x, with a visible 0 above it. The reading x^0 is tentative.

Original source details: GELFAND-P000764

This is already an operator

There is an orthogonal decomposition of the solution space:

$$W = \left\{ w \;\middle|\; \frac{\partial w_{\pm}}{\partial x^{\pm}} \mp m w_{\mp} = 0 \right\} = W_x^+ \oplus W_x^-, \tag{7}$$

where

$$W_x^+ = \{ w \in W \mid w = 0 \text{ in the hatched region} \} \tag{8}_1$$

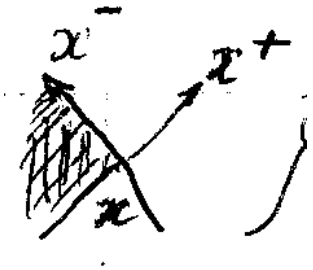
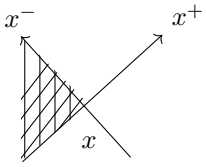


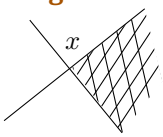
Diagram and typed description.



Original source details: GELFAND-P000764

$$W_x^- = \{ \dots \mid w = 0 \text{ in the hatched region} \}. \tag{8}_2$$

Diagram and typed description.



Original source details: GELFAND-P000764

In the Bose case instead of the Fermi case, the functions are analogous to a Lagrangian decomposition into p and q (into 2 Lagrangian planes).

$A(W)$ — [Crossed out: ...] Clifford algebra over W

Undeciphered passage. Slanted strokes obscure the short crossed-out entry before “Clifford”.

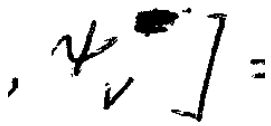
Original source details: GELFAND-P000764

$\{\psi_\mu^\pm\}$ — bases in $W_x^\pm$

The Clifford algebra is generated by them with relations

$$\psi_\mu^+\psi_\nu^- + \psi_\nu^-\psi_\mu^+ \underset{\text{def}}{=} [\psi_\mu^+, \psi_\nu^-] = \delta_{\mu\nu} \tag{9}_1$$

$$[\psi_\mu^+, \psi_\nu^+] = [\psi_\mu^-, \psi_\nu^-] = 0 \tag{9}_2$$



Uncertain reading. The superscript of the final ψ_ν is heavily crossed over. The minus is tentative.

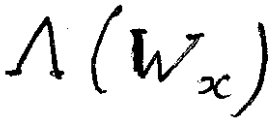
Original source details: GELFAND-P000764

Notation:

$$\underline{\underline{\psi_\mu^- = \psi_\mu}}$$

Relation:

$$A(W) = \Lambda(W_x^+) \cdot \Lambda(W_x) \tag{10}$$



Uncertain reading. The second factor $\Lambda(W_x)$ shows the subscript x and has no distinct superscript.

Original source details: GELFAND-P000764

The bases are orthogonal with respect to the scalar product (4).

There is a map:

$$A(W) \longrightarrow \Lambda W_x^+ \wedge \Lambda W_x^- \qquad \text{isomorphism}$$

$$\lambda \cdot \mu \overset{N}{\longrightarrow} \lambda \wedge \mu$$

inverse map:

$$:\varphi: \longleftarrow \varphi$$

(Wick product)

In the Bose case this is the “symbol of the operator”.

GELFAND-P000765

[source number: 63]

Definition.

$$\varphi_F(x) \in A(W)$$

$$\text{Spin}(W_x) = \{g \in A(W_x) \mid gW_xg^{-1} \subset W_x\}$$

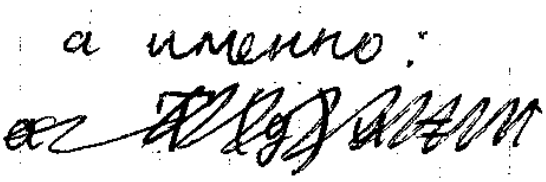
In fact

$$\varphi_F(x) \in \text{Spin}(W)$$

There is a map:

$$\text{Spin}(W) \xrightarrow{T} O(W); \qquad \text{namely} \quad T_g w = gwg^{-1}; \tag{11}$$

[Crossed out: and ...]



Undeciphered passage. Between $O(W)$ and $T_g w$, "a" (and) and a mark possibly reading $T(g)$ are visible. Dense crossings obscure the rest. "Namely" is written above.

Original source details: GELFAND-P000765

W has a scalar product and hence we have a definition of $O(W)$.

$$T_{\varphi_F(x)}(w^+ + w^-) \underset{\text{def}}{=} w^+ - w^- \tag{12}$$

(this is the definition of $\varphi_F(x)$).

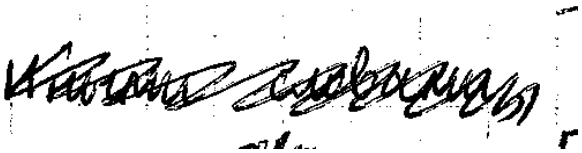
T fails to be a monomorphism, so the definition is still ill-defined. More precisely,

$$\varphi_F(x) = :\exp L_F(x): \tag{13}$$

where

$$\Lambda W \ni L_F(x) = \frac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \underline{du}\,\underline{du'}\,\frac{-i(u-u')}{u+u'-i0}\cdot\Psi(u)\Psi(u')$$
$$\exp\left(-im\left(x^-(u+u') + \right.\right. \tag{14}$$
$$\left.\left. + x^+\left(\frac{1}{u}+\frac{1}{u'}\right)\right)\right).$$

[Crossed out: One can speak?]

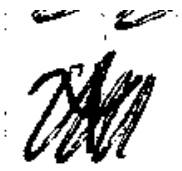


Uncertain reading. Two words beneath (14) are heavily crossed out. "Можно говорить" (one can speak) is tentative.

Original source details: GELFAND-P000765

Next,

[Crossed out: ...]



Undeciphered passage. Dense slanted strokes obscure the mark above and to the left of (15).

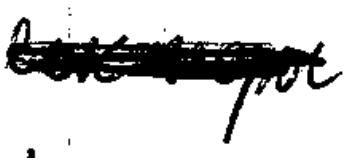
Original source details: GELFAND-P000765

$$\varphi^F(x) = \Psi_0(x) \exp L_F(x), \tag{15}$$

where

$$W \ni \Psi_0(x) = \int_{-\infty}^{\infty} \underline{du}\, \Psi(u) \exp\left(-im\left(x^- u + \frac{x^+}{u}\right)\right) \tag{16}$$

There are [Crossed out: vectors?] 2 objects:



Original source details: GELFAND-P000765

Uncertain reading. The crossed-out word before “2 objects” tentatively reads “векторы” (vectors).

$$\langle \text{vac}|, \qquad |\text{vac}\rangle$$

$$A(W)/A(W)W_x^- = |\text{vac}\rangle \tag{17}_1$$

$$W_x^+ A(W)\backslash A(W) = \langle \text{vac}| \tag{17}_2$$

Properties:

$$W_x^-|\text{vac}\rangle = 0, \qquad \langle \text{vac}|W_x^+ = 0 \tag{18}$$

(natural for the vacuum).

Consider the function

$$\langle \text{vac}|\Psi(x_1)\dots\Psi(x_k)\varphi_F(a_1)\dots\varphi_F(a_m)|\text{vac}\rangle; \tag{19}$$

($\Psi(x)$ from formula (6)).

defined on $(\mathbb{R}^2)^k$

All a_ν are fixed.

(2-dimensional Ising model?)

$k = 1 \quad \Rightarrow$

$$\langle \text{vac}|\Psi(x)\varphi_F(a_1)\cdot\varphi^F(a_i)\dots\varphi_F(a_n)|\text{vac}\rangle \underset{\text{def}}{=} W^i_{F,n}(x) \tag{20}$$

A system of equations is obtained; put

$$M_F = z\partial_z - \bar{z}\partial_{\bar{z}} + \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \tag{21}$$

The 1st equation has the form:

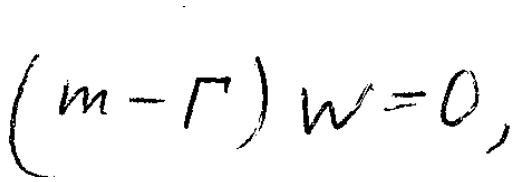
$$M_F w = (Bm^{-1}\partial_z - \bar{B}m^{-1}\partial_{\bar{z}} + E)w \tag{22}$$

$$z = (x^0 + x^1), \qquad \bar{z} = (x^0 - x^1)$$

B, E — constant matrices

The 2nd equation

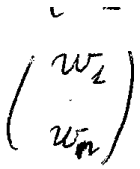
$$(m - \Gamma)w = 0, \qquad \Gamma = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \tag{23}$$



Uncertain reading. The manuscript writes $m-$ before Γ in (23).

Original source details: GELFAND-P000766

$$w_i = \begin{pmatrix}w_{i+}\\ w_{i-}\end{pmatrix}, \qquad \begin{pmatrix}w_1\\ w_n\end{pmatrix}$$



Uncertain reading. The subscript of the lower entry in the right column reads n.

Original source details: GELFAND-P000766

(22)–(23) — a maximally overdetermined system
(in fact this is the Gauss–Manin connection).
Consider also

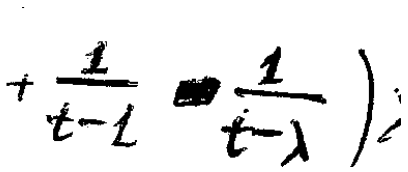
$$\langle \text{vac}|\varphi_F(a_1)\dots\varphi_F(a_n)|\text{vac}\rangle = \tau \tag{24}$$

as a function of a. (τ-function)
For $n = 2$ this is a Painlevé function of the 3rd kind.
[Written beneath “Пенлеве”: (Painleve)]
Wu, Mc Coy, Tracy, Phys. Rev. B 13 (1976)
[Beneath the names] something about Painlevé functions.
I. M. G.
Garnier — a memoir on Painlevé functions.
6 types of equations, 2nd order, fixed singular points. The most general

$$\begin{aligned}\frac{1}{y}y'' = {} & \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{c}{(x-t)^2} + \frac{d}{x(x-1)} + \frac{3}{4(x-\lambda)^2} \\ & + \frac{\alpha}{x(x-1)(x-t)} + \frac{\beta}{x(x-1)(x-\lambda)}\end{aligned} \tag{А}$$

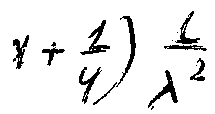
The periods (monodromy) are independent of the parameters $\Leftrightarrow$

$$\begin{aligned}\Leftrightarrow \quad \lambda'' = {} & \frac{1}{2}\left(\frac{1}{\lambda} + \frac{1}{\lambda-1} + \frac{1}{\lambda-t}\right)(\lambda')^2 \\ & - \left(\frac{1}{t} + \frac{1}{t-1}\,[?]\,\frac{1}{t-\lambda}\right)\lambda' + \\ & + 2\frac{\lambda(\lambda-1)(\lambda-t)}{t^2(t-1)^2}\left[a+b+c+d+1 - \left(a+\frac{1}{4}\right)\frac{1}{\lambda^2}\right. \\ & \left. + \left(b+\frac{1}{4}\right)\frac{t-1}{(\lambda-1)^2} - \frac{ct(t-1)}{(\lambda-t)^2}\right]\end{aligned} \tag{Б}$$



Original source details: GELFAND-P000766

Uncertain reading. A dense correction obscures the sign between $1/(t-1)$ and $1/(t-\lambda)$.



Original source details: GELFAND-P000766

Uncertain reading. The manuscript writes 1 over the first λ^2.

Only these have fixed singular points.
Constancy of the matrix monodromy (А) $\Leftrightarrow$ fixedness of the singular points of (Б)
t, λ vary in accordance with (Б)

GELFAND-P000767

Diagram and typed description. 13/III – 78 Gelfand Seminar

Original source details: GELFAND-P000767

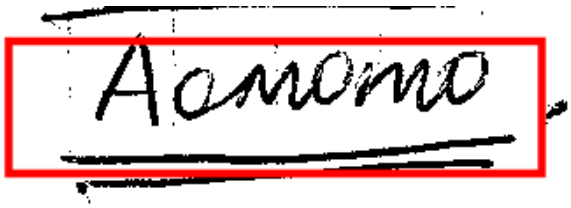


Diagram and typed description. Aomoto.

Original source details: GELFAND-P000767

On generalizing the classical hypergeometric function using integrals of the type of Bernstein (I.)

Uncertain reading. "И." (I.) appears in parentheses after "Bernstein". The initial is left unexpanded.

Original source details: GELFAND-P000767

Appell function.
The case of 2 variables.

I. M. G.
Classical hypergeometric function:

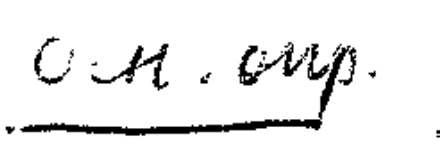
$$\sum_{n=0}^{\infty} x^n \frac{\alpha(\alpha+1)\dots(\alpha+n-1)\,\beta(\beta+1)\dots(\beta+n-1)}{\gamma(\gamma+1)\dots(\gamma+n-1)\,n!}$$

GELFAND-P000768

$$F_1 = \sum_{m,n\geq 0} \frac{(\alpha, m+n)(\beta, m)(\beta', n)}{(\gamma, m+n)(1, m)(1, n)} x^m y^n \tag{1}$$

— · —

I. M. G. One? definition of hypergeometric functions:



Uncertain reading. The first abbreviation before "опр." (definition) tentatively reads "Од." (one).

Original source details: GELFAND-P000768

$$\left((x^2-1)^n\right)^{(n)} \qquad \text{(Legendre polynomials)}$$
$$\|$$
$$\left[((x-1)(x+1))^n\right]^{(n)}$$

Generalization:

$$\left[(x-1)^{\alpha-1}(x+1)^{\beta-1}\right]^{(\gamma)}$$

Integrals of order α.

$$I_\alpha f = \int_0^x \underbrace{\frac{(x-t)^{\alpha-1}}{(\alpha-1)!}}_{\Gamma(\alpha)} f(t)\,dt, \qquad D^\alpha = I_{-\alpha} \quad \text{(analytic continuation)}$$

$$\left[x^{\alpha-1}(1-x)^{\beta-1}\right]^{(\gamma)}$$

Hypergeometric function: 1) good series expansion 2) good differential equation 3) good integral.

$$\int_0^x t^{\alpha-1}(1-t)^{\beta-1}(x-t)^{\gamma-1}\,dt$$

B-functions are simpler.

— · —

$$(\alpha, n) = \alpha(\alpha+1)\dots(\alpha+n-1)$$
$$F_4 = \sum_{m,n\geq 0} \frac{(\alpha, m+n)(\beta, m+n)}{(\gamma, m)(\gamma', n)(1, m)(1, n)} \tag{2}$$

+n) (β, m+n)

(γ', n) (1, m) (1, n)

Uncertain reading. The factor $x^m y^n$ is absent after the fraction in (2).

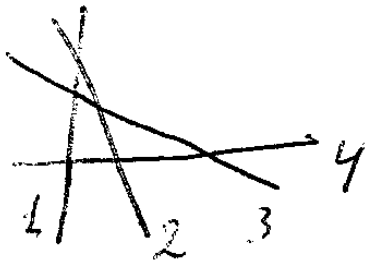
Original source details: GELFAND-P000768

There is an integral representation [Crossed out: ...] for F_1

Undeciphered passage. Slanted strokes obscure the short mark before "for F_1".

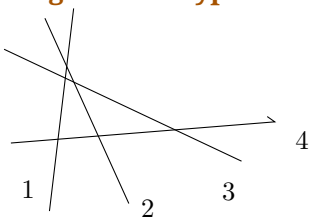
Original source details: GELFAND-P000768

(3)



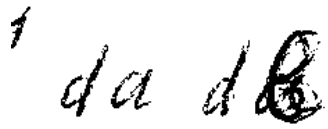
Original source details: GELFAND-P000768

Diagram and typed description.



Line 4 has a short hook at its right end. Its function is uncertain.

$$\int f_1^{\lambda_1} f_2^{\lambda_2} f_3^{\lambda_3} f_4^{\lambda_4}\, da\, db$$



Original source details: GELFAND-P000768

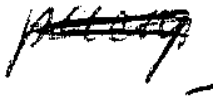
Uncertain reading. The loops in db overlap. The b is tentative.

f_j — linear forms; x, y are among the coefficients.

3 lines are reduced to a simple form (affine transformation) and fixed

The 4th is given by parameters a, b.

F_2, F_3 are obtained similarly, but with a different [Crossed out: arrangement?] arrangement of lines, (out of general position):



Original source details: GELFAND-P000768

Uncertain reading. The crossed-out word tentatively begins “расп” (arrangement).

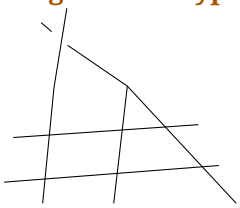
[Crossed out: F_1?]

Original source details: GELFAND-P000768

Uncertain reading. The crossed-out mark at lower left resembles F with a short stroke at its lower right. F_1 is tentative.

Original source details: GELFAND-P000768

Diagram and typed description.

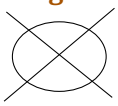


GELFAND-P000769

[source number: 68]

F_4 :

Diagram and typed description.



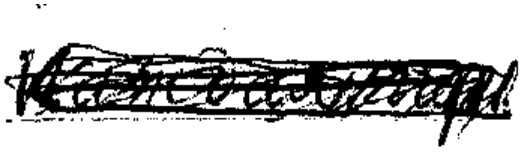
Original source details: GELFAND-P000769

a quadratic form is used in place of 2 lines.

The number of linearly independent solutions of the corresponding differential equations = the number of bounded connected components of the corresponding picture.

General remark on the coefficients: the ratio of neighbors = a rational function of m, n.

[Crossed out: ...]



Undeciphered passage. Dense horizontal and slanted strokes obscure the opening words before “An analogous problem”.

Original source details: GELFAND-P000769

An analogous problem can be considered with any number of hyperplanes

$$f_i = \sum_{\nu=1}^{n} a_{i\nu} x_\nu + a_{i0} \tag{4}$$

$$1 \le i \le m$$

$$f_{m+1} = 1 \qquad \text{Usually } m \ge n.$$

Consider the matrix formed from the coefficients of the forms

$$A = \begin{pmatrix} a_{10} & \dots & & a_{1n} \\ \vdots & & & \\ a_{m0} & \dots & & a_{mn} \\ 1 & \dots & 0 & 0 \end{pmatrix} \in \mathbb{C}^{m(n+1)} \tag{5}$$

The affine group in n variables acts there
(on x it acts)
— Change of variable in x
Let

$$\mathcal{M} \subset \mathbb{C}^{m(n+1)} \times \mathbb{C}^n$$

consist of pairs

$$\{(A, x) : f_j(x) \neq 0\} \qquad \text{(none of the equations is satisfied)}$$

(complement of all the hyperplanes)

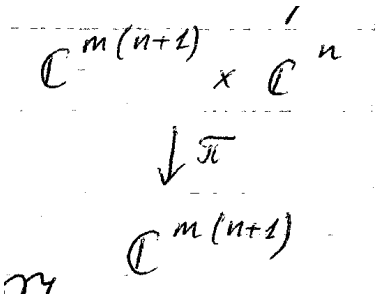


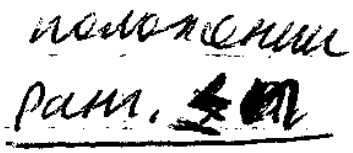
Diagram and typed description.

$$\begin{array}{c} \mathbb{C}^{m(n+1)} \times \mathbb{C}^n \\ \downarrow \pi \\ \mathbb{C}^{m(n+1)} \end{array}$$

Original source details: GELFAND-P000769

$$\gamma \subset \mathcal{M}$$

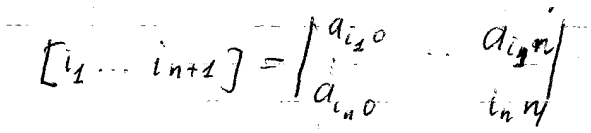
the set of such equations for which the hyperplanes are out of general position
rank $\le n$



Uncertain reading. The inequality after “rank” is densely overwritten. The reading $\le n$ is tentative.

Original source details: GELFAND-P000769

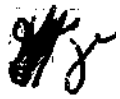
$$[i_1 \dots i_{n+1}] = \begin{vmatrix} a_{i_1 0} & \dots & a_{i_1 n} \\ \vdots & & \\ a_{i_n 0} & & i_n n \end{vmatrix} \qquad \text{(minor obtained from rows } i_1, \dots, i_n\text{)}$$



Uncertain reading. The left bracket has i_{n+1}, while the last row and explanation have i_n. The lower right entry is written $i_n n$.

Original source details: GELFAND-P000769

The projection of [Crossed out: γ?] γ is given by conditions on minors:



Uncertain reading. The looped crossed-out sign before the clean γ may be γ.

Original source details: GELFAND-P000769

$$Y = \{A \mid [i_1, \dots, i_{n+1}] = 0\} \tag{6}$$

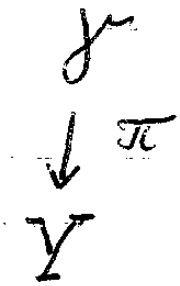


Diagram and typed description.

$$\begin{array}{c} \gamma \\ \downarrow \pi \\ Y \end{array}$$

Original source details: GELFAND-P000769

Uncertain reading. A separate densely retraced sign beneath (6), at the left, tentatively reads γ.

Original source details: GELFAND-P000769

GELFAND-P000770

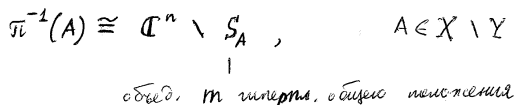


Diagram and typed description.

$$\pi^{-1}(A) \cong \mathbb{C}^n \setminus S_A \,, \qquad A \in X \setminus Y$$

union of m hyperplanes in general position

Original source details: GELFAND-P000770

$\Omega_A^\bullet$ — the complex of rational forms in $\mathbb{C}^n$ with poles on S_A
(exterior differentiation)
$\forall \varphi \in \Omega_A$ there is a formula

$$U = f_1^{\lambda_1} \dots f_m^{\lambda_m},$$

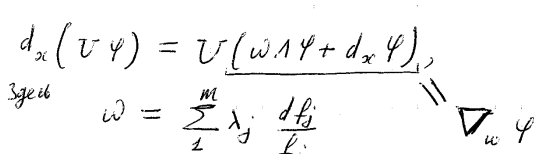


Diagram and typed description.

$$d_x(U\varphi) = U\underbrace{(\omega \wedge \varphi + d_x\varphi)}_{=\nabla_\omega \varphi},$$

Original source details: GELFAND-P000770

Here

$$\omega = \sum_1^m \lambda_j \frac{df_j}{f_j} \qquad (\nabla_\omega \text{ — a connection}).$$

ω — the connection form

$$\nabla_\omega : \Omega_A^p \longrightarrow \Omega_A^{p+1}$$
$$\nabla_\omega \varphi = \omega \wedge \varphi + d_x \varphi$$

(a connection in the trivial bundle)

We have:

$$d\omega + \omega \wedge \omega = 0$$

(integrability condition)

It follows that $\nabla_\omega^2 = 0$ and we obtain a complex.

Deligne, Grothendieck: a theory of rational de Rham hypercohomology arose.

Assumption.

$$\lambda_{m+1} = -\lambda_1 - \ldots - \lambda_m$$

$\forall$ collections

$$1 \le i_1 < \ldots < i_\nu \le m+1, \qquad 1 \le \nu \le n.$$

Condition:

$$\lambda_{i_1} + \ldots + \lambda_{i_\nu} \notin \mathbb{Z}^+ \qquad \mathbb{Z}^+ = \{n \in \mathbb{Z},\ n > 0\}$$

Proposition.

$$H^p(\Omega_A, \nabla_\omega) = 0, \quad p \ne n$$
$$H^n(\Omega_A, \nabla_\omega) = \chi(\mathbb{C}^n \setminus S_A)$$
$$\left\{ \frac{df_{i_1}}{f_{i_1}} \wedge \ldots \wedge \frac{df_{i_n}}{f_{i_n}} \right\}$$
$$\cong \frac{\Omega^n(\log\langle S\rangle)}{\omega \wedge \Omega^{n-1}(\log\langle S\rangle)} \quad = \text{def}$$

$H^n(\Omega_A, \nabla_\omega) = \chi(\mathbb{C}^n \setminus S_A$

$) \cong \left\{ \frac{df_{i_1}}{f_{i_1}} \wedge \ldots \wedge \frac{df_{i_n}}{f_{i_n}} \right\} \Big/ \omega \wedge \Omega^{n-1}(\log\langle S\rangle)$

$\Omega^n/\log\langle S\rangle$ // def

Original source details: GELFAND-P000770

Uncertain reading. The manuscript writes $H^n(\Omega_A, \nabla_\omega) = \chi(\mathbb{C}^n \setminus S_A)$, with no dim before H^n.

GELFAND-P000771

[source number: 69]

Explanation:

$$\Omega^1(\log\langle S\rangle) = \left\{\frac{df_i}{f_i}\right\}_{i\le m}$$

$$\bigwedge \Omega^1(\log\langle S\rangle)/\sim\ = \Omega^\bullet(\log\langle S\rangle)$$

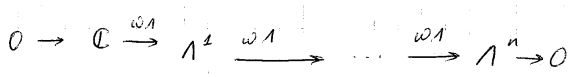


Diagram and typed description.
$0 \longrightarrow \mathbb{C} \overset{\omega\wedge}{\longrightarrow} \Lambda^1 \overset{\omega\wedge}{\longrightarrow} \cdots \overset{\omega\wedge}{\longrightarrow} \Lambda^n \longrightarrow 0$

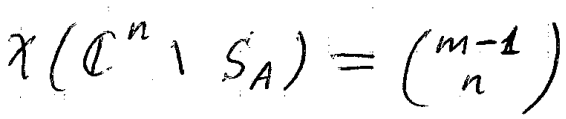
Original source details: GELFAND-P000771

(They are closed: $\nabla_\omega = \omega\wedge$ for such forms)

$$\chi(\mathbb{C}^n \setminus S_A) = \binom{m-1}{n}$$

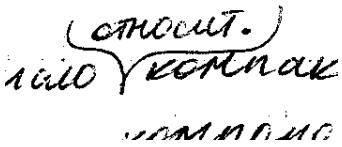
$$= \text{the number of (relatively) compact connected}$$

components of $\mathbb{R}^n \setminus S_A$



Uncertain reading. The manuscript has no additional sign factor before $\binom{m-1}{n}$.

Original source details: GELFAND-P000771



Uncertain reading. “(относит.)” (relatively) is inserted above the line before “compact”.

Original source details: GELFAND-P000771

—, —

$\int U\varphi$ [Crossed out: ...] Duality

$\varphi \in \Omega_A$

Undeciphered passage. The short entry between the integral and “Duality” is heavily crossed out.

Original source details: GELFAND-P000771

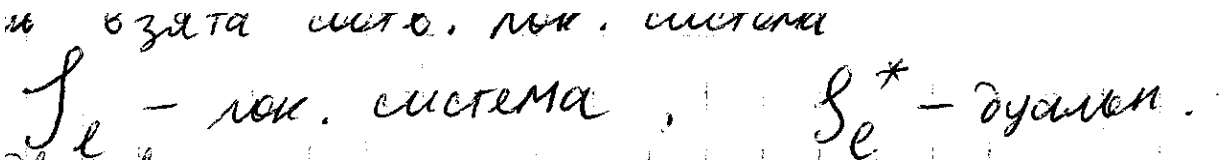
U is multivalued $\Rightarrow$ continuation around loops gives a representation of the fundamental group:

$$\ell_U : \pi_1(\mathbb{C}^n - S_A) \longrightarrow \mathbb{C}^*$$

(continuation multiplies it by a nonzero complex number)

Take the corresponding local system

$$\zeta_\ell \text{ — local system,} \qquad \zeta_\ell^* \text{ — the dual.}$$



Original source details: GELFAND-P000771

Uncertain reading. The looped local system symbol with subscript ℓ is tentatively rendered as ζ_ℓ.

There is a duality

$$H^n(\mathbb{C}^n \setminus S_A, \zeta_\ell) \times H_n(\mathbb{C}^n - S_A, \zeta_\ell^*) \longrightarrow \mathbb{C}$$

$\forall$ compact connected component $\longrightarrow$ element $\in$

$$H_n(\mathbb{C}^n - S_A, \zeta_\ell)$$

Original source details: GELFAND-P000771

Uncertain reading. The final local system symbol is heavily retraced. A separate star is indistinct, unlike in the preceding pairing.

GELFAND-P000772

20/III–78 Gelfand Seminar

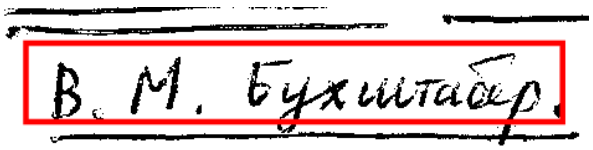
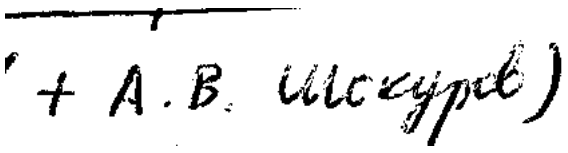


Diagram and typed description.

V. M. Buchstaber.

Original source details: GELFAND-P000772

(+ A. V. Shokurov)



Uncertain reading. The surname after “A. B.” (A. V.) tentatively reads “Шокуров” (Shokurov).

Original source details: GELFAND-P000772

Landweber–Novikov algebras and formal vector fields on the line. [Crossed out: ...]

Undeciphered passage. Dense strokes obscure the small entry beneath the title.

Original source details: GELFAND-P000772

$\mathbb{R}^1$, Diff_1 — formal diffeomorphisms of the line with the additional condition: $\varphi'(0) = 1$

$$\varphi(0) = 0, \qquad \varphi'(0) = 1$$

The group $= \varprojlim$ finite-dimensional groups

(in jets of order n)

Lie algebra — the Lie algebra of formal vector fields on $\mathbb{R}^1$ — this is L_1.

$$t + a_1 t^2 + \ldots + a_n t^{n+1} + \ldots$$

a_j — coordinates.

Ring of functions: $\underline{P}$ (polynomials in all the a_j).

Uncertain reading. The ring is denoted by P with a lower stroke.

Original source details: GELFAND-P000772

Hirzebruch genus: M — a complex or stably complex manifold

[Crossed out: ...]

Undeciphered passage. A heavily crossed-out sign precedes M.

Original source details: GELFAND-P000772

$M \rightsquigarrow M(\varphi)$ — a polynomial on Diff_1.

$$\frac{t}{\varphi(t)} = \psi(t)$$

$$\prod_{j=1}^{N} \psi(\beta_j t), \qquad N > \dim_{\mathbb{C}} M$$

The coefficient of t^n = a symmetric function of β_j =

$= \Omega(\sigma_1, \ldots, \sigma_N)$, σ_j — elementary symmetric polynomials

GELFAND-P000773

[source number: 75]

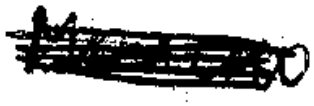
$\sigma_j \rightsquigarrow j$-th Chern class

Consider

$$\langle \Omega(\sigma_1, \ldots, \sigma_N), M \rangle$$

this is precisely the function on Diff_1.

[Crossed out: ...]



Uncertain reading. Dense strokes obscure several words before the line about operations.

Original source details: GELFAND-P000773

Operations on manifolds $\rightsquigarrow$ operations on polynomials.

left-invariant differential operators — ?

1) $M \times M_1$

2)

Diagram and typed description.

$$\begin{array}{c} \tau \\ \downarrow \\ M \end{array}$$

Original source details: GELFAND-P000773

— canonical bundle.

$c_1(\tau) \rightsquigarrow$ the cycle dual to it $M_1 \subset M$

$$\operatorname{Codim}_{\mathbb{C}} M_1 = 1.$$

(zeros of sections of the canonical bundle).

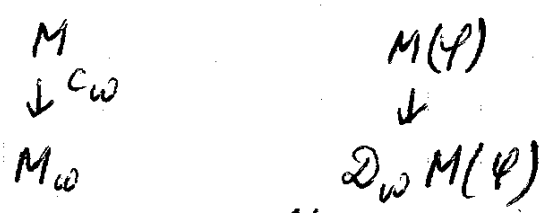
Analogue: dual to $\forall$ polynomial in the Chern classes.

Landweber–Novikov; there is nothing else. (these are all natural cohomological operations)

Theorem 1.

Diagram and typed description.

$$\begin{array}{ccc} M & & M(\varphi) \\ \downarrow{\scriptstyle c_\omega} & & \downarrow \\ M_\omega & & D_\omega M(\varphi) \end{array}$$



Original source details: GELFAND-P000773

c_ω — a polynomial in the Chern classes

D_ω — a left-invariant differential operator.

$\Longleftrightarrow$ a one-to-one correspondence

$M(\varphi)$ — integer polynomials

When do polynomials correspond to manifolds?

Answer:

Thm. 2.

$\mathrm{Diff}_1(\mathbb{Z})$ — formal diffeomorphisms with integer coordinates.

Todd genus $\varphi = 1 - \exp(-t)$ (diffeomorphism)

The polynomial must take integer values on the right coset of this φ [Crossed out: in?]

Uncertain reading. The heavily overwritten and cancelled letter tentatively reads “в” (in).

Original source details: GELFAND-P000773

$\mathrm{Diff}_1(\mathbb{Z})$.

[Crossed out: ...]

Uncertain reading. A separate crossed-out entry precedes the theorem reference.

Original source details: GELFAND-P000773

(∼ Thm. Stong–Hattori)

— · —

S — the set of all operations of Landweber–Novikov; this is an algebra.

Addition: disjoint union, multiplication — composition.

The cohomology of this algebra is important. All homotopy invariants of continuous maps are expressed through it. (Novikov)

Thm. 3

$$\mathrm{Ext}_S^*(\mathbb{Q},\mathbb{Q}) = H^*(L_1)$$

Cohomology of the algebra = cohomology of the enveloping algebra.

$H^*(L_1)$ was calculated in the works of Gelfand, Fuks, Goncharova

Problem: calculate the cohomology of the group $\mathrm{Diff}_1(\mathbb{Z})$.

GELFAND-P000774

Calogero. (and A. De gasperis).

A class of nonlinear evolution equations solvable by a spectral transform associated **with** Schrödinger.

Uncertain reading. The preposition is heavily retraced; it reads “c” (with).

Original source details: GELFAND-P000774

$$u_t(x,t) = \alpha(L,t)u_x(x,t) \tag{1}$$

$$Lf(x) = f_{xx} - 4u(x,t)f(x) + 2u_x(x,t)\int_x^\infty dx'\, f(x') \tag{2}$$

Sol. [Above the line, with an insertion mark: lin.] of the Cauchy problem reduces to a linear problem for the Fourier transform

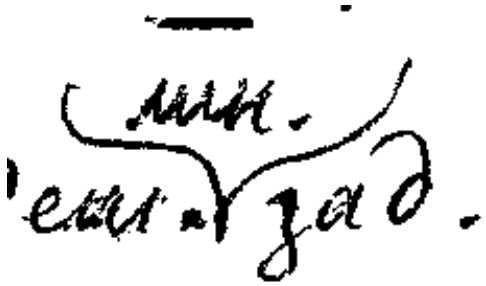


Diagram and typed description.

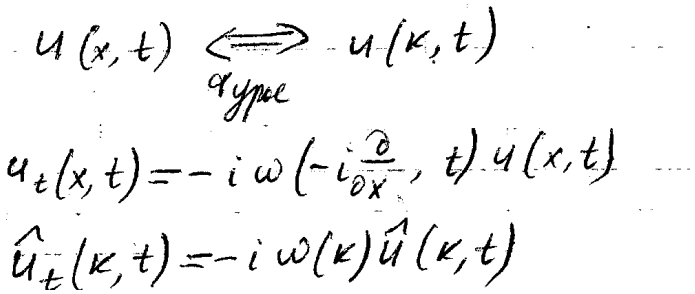
lin.
Sol.↓ problem

Original source details: GELFAND-P000774

$$u(x,t) \underset{\text{Fourier}}{\Longleftrightarrow} u(k,t)$$

$$u_t(x,t) = -i\omega\left(-i\frac{\partial}{\partial x}, t\right)u(x,t)$$

$$\hat{u}_t(k,t) = -i\omega(k)\hat{u}(k,t)$$



Original source details: GELFAND-P000774

Uncertain reading. The first right-hand $u(k,t)$ has no visible hat; the last line has $\omega(k)$ without the argument t.

A wider class:

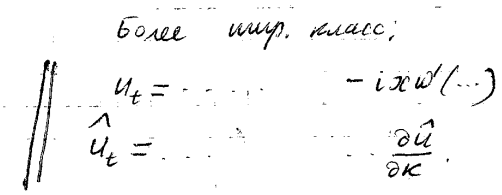


Diagram and typed description.

$$\begin{cases} u_t = \cdots - ix\omega'(\dots) \\ \hat{u}_t = \cdots \quad \cdots \dfrac{\partial \hat{u}}{\partial k} \end{cases}$$

Original source details: GELFAND-P000774

one can solve a wider class of equations:

$$\begin{aligned} u_t(x,t) = {} & \alpha(\underline{L},t)u_x(x,t) \\ & + \beta(\underline{L},t)[xu_x(x,t) + 2u(x,t)] \end{aligned} \tag{3}$$

α — a function of the operator

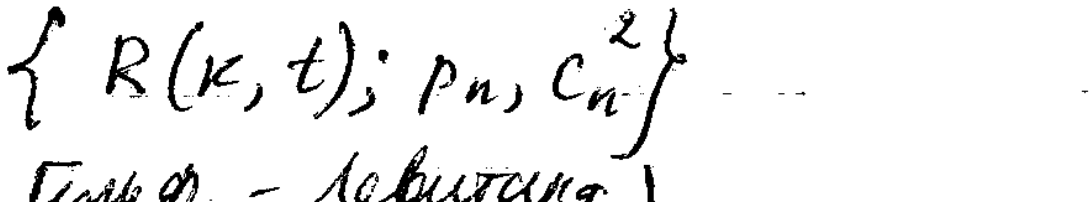
β — likewise

$$\alpha(\underline{L},t) = -\underline{L} \quad \Rightarrow \quad \text{KdV} \quad (\text{when } \beta = 0)$$

This includes the higher KdV equations: $\alpha(z,t) = z^k$.

For (3) there is a linear spectral problem describing the evolution of u.

$$u(x,t) \Longleftrightarrow \{R(k,t); p_n, c_n^2\}$$



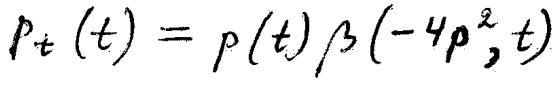
Original source details: GELFAND-P000774

Uncertain reading. p_n or ρ_n; the same sign recurs in (4').

(solution of the Gelfand–Levitan equation)

$$\begin{aligned} R_t(k,t) = {} & 2ik\alpha(-4k^2,t)R(k,t) \\ & - k\beta(-4k^2,t)R_k(k,t) \end{aligned} \tag{4}$$

$$p_t(t) = p(t)\beta(-4p^2,t) \tag{4'}$$



Uncertain reading. The shape transcribed as p at all three positions also permits the reading ρ.

Original source details: GELFAND-P000774

for c the equation is as for R (not written out)

The velocity of the solitons is not conserved, nor is the amplitude.

Conservation laws.

$$c^{(n)} = \int_{-\infty}^{\infty} dx\, \underline{L}^n[xu_x(x,t) + 2u(x,t)] \tag{5}$$

Theorem 1. $\beta \equiv 0 \quad \Rightarrow \quad (c^{(n)})_t = 0$
(in particular: KdV).

GELFAND-P000775

[source number: 76]

AKNS
(Ablowitz, Kaup, Newell, —)
— such a form of the higher KdV equations.
KdV is written for two Hamiltonian structures.
Poisson brackets — infinitely many.

A form of higher KdV with the integral: $\int_{-\infty}^{\infty} u^2 dx$.

(Faddeev et al. — the Leningrad group)
and a complicated Hamiltonian structure.

$$I_0 = \int u^2 dx;$$

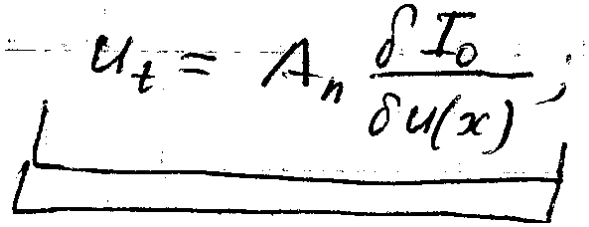


Diagram and typed description.

$$u_t = A_n \frac{\delta I_0}{\delta u(x)};$$

Original source details: GELFAND-P000775

these are the higher KdV equations

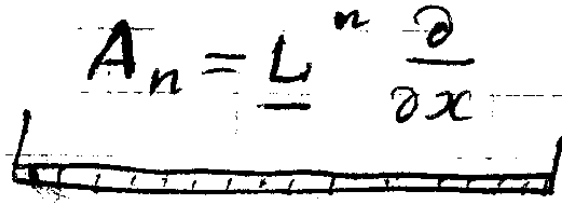


Diagram and typed description.

$$A_n = L^n \frac{\partial}{\partial x}$$

Original source details: GELFAND-P000775

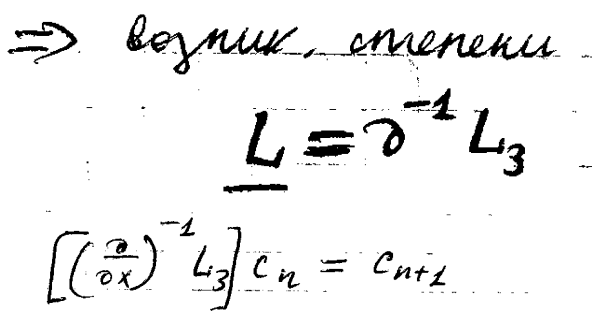
The transformation laws are invariant under shifts in x, and the integral contains x.

Novikov:

$$L_3 c_n = \frac{\partial}{\partial x} c_{n+1} \quad \Rightarrow \quad \left[\left(\frac{\partial}{\partial x}\right)^{-1} L_3\right] c_n = c_{n+1}$$

$\Rightarrow$ powers of the operator arise.

$$\underline{L} = \partial^{-1} L_3$$



Uncertain reading. In both entries the inverse derivative precedes L_3; the factor order is retained.

Original source details: GELFAND-P000775

Consider the Schrödinger equation.

$$\Psi_{xx} = [u(x) - k^2]\Psi \tag{6}$$

$$\Psi(k,x) \underset{x\to-\infty}{\longrightarrow} T(k)e^{-ikx}$$

$$\Psi(k,x) \underset{x\to+\infty}{\longrightarrow} e^{-ikx} + R(k)e^{ikx}$$

$$T(k) = |T(k)|e^{2i\varphi(k)}$$

$$T(\pm k) = |T(k)|e^{\pm 2i\varphi(k)}$$

$$\varphi(k) = \sum_{n=0}^{N} k^{-(2n+1)}\varphi^{(n)} + O(k^{-(2N+3)}). \tag{7}$$

Then

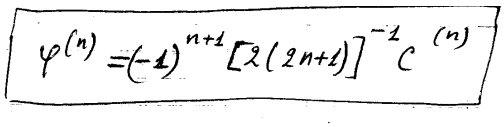


Diagram and typed description.

$$\varphi^{(n)} = (-1)^{n+1}[2(2n+1)]^{-1}c^{(n)} \tag{8}$$

Original source details: GELFAND-P000775

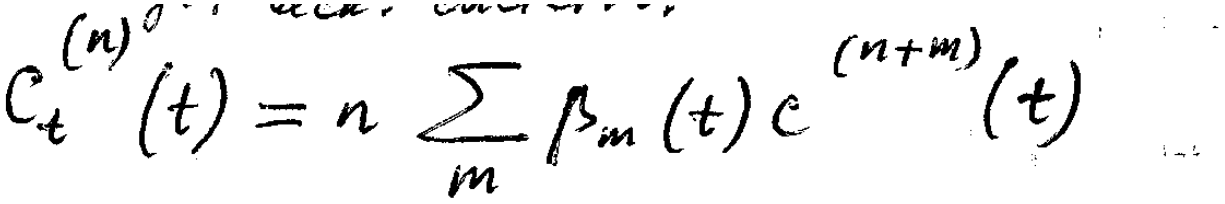
$$\begin{aligned} u_t(x,t) = {} & \alpha(\underline{L},t)u_x(x,t) \\ & + \beta(\underline{L},t)[xu_x(x,t) + 2u(x,t)] \end{aligned} \tag{9}$$

Let

$$\beta(z,t) = \sum \beta_n(t) z^n \tag{10}$$

Then one obtains an infinite system:

$$C_t^{(n)}(t) = n \sum_m \beta_m(t) c^{(n+m)}(t) \tag{11}$$



Original source details: GELFAND-P000776

Uncertain reading. The case of C/c in (11): a larger C on the left, a smaller c on the right.

β — a polynomial in $z \Rightarrow$ the system is triangular.

$\beta_0 = i\omega \Rightarrow$ all quantities are periodic in t.

(a consequence of the structure)

Next time — on the Bäcklund transform

Diagram and typed description. 21/III–78.

Original source details: GELFAND-P000776

V. P. Palamodov at the Society.

Lacunas, sharp fronts and deformations of singularities.

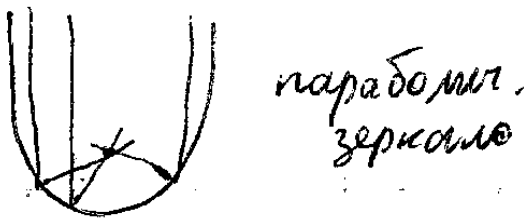
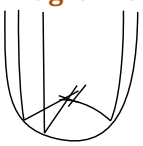


Diagram and typed description.

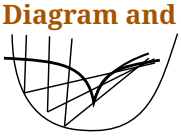


parabolic mirror

Short crossing strokes occur above the bottom; arrowheads cannot be distinguished.

Original source details: GELFAND-P000776

Diagram and typed description.



spherical mirror

Some crossings and junctions of the heavy lines cannot be distinguished.

Original source details: GELFAND-P000776

Caustic — “burning”, “hot” (in Greek)

Rays reflected from a spherical mirror are tangent to a curve with a singularity —
— the caustic.

Dynamic picture: wave fronts

Original source details: GELFAND-P000776

Diagram and typed description.

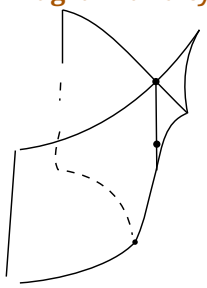
Some thin branches cross and merge with the heavy strokes; not every junction is established.

at the instant the caustic forms, a singularity $y = x^{4/3}$

There is a surface whose horizontal sections coincide with the positions of the front

Original source details: GELFAND-P000776

Diagram and typed description.



Gaps, crossings and possible overlaps of the surface are not always distinguishable; there are no labels.

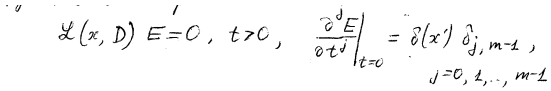
Consider a hyperbolic equation

$$\mathcal{L}(x, D)u = 0, \tag{1}$$

for example

$$\left(\frac{\partial^2}{\partial t^2} - \frac{1}{c^2}\Delta\right) u = 0 \tag{2}$$

Fundamental solution:



Original source details: GELFAND-P000776

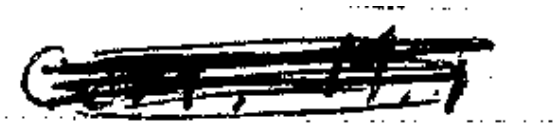
Diagram and typed description.

$$\mathcal{L}(x, D)E = 0, \quad t > 0,$$
$$\left.\frac{\partial^j E}{\partial t^j}\right|_{t=0} = \delta(x')\delta_{j,m-1}, \qquad j = 0, 1, \dots, m-1 \tag{3}$$

Chapter 13

GELFAND-P000777

27/III–78. [Crossed out: C...]



Undeciphered passage. An initial C remains between the date and the heading; dense cancellation strokes obscure the rest.

Original source details: GELFAND-P000777

Gelfand Seminar.

F. Calogero. On the Bäcklund transform.

(all of this jointly with A. Degasperis)

$$\Psi_{xx}(x,k) = [q(x)\ [?]\ k^2]\Psi(x,k), \qquad -\infty < x < +\infty \tag{1}$$

Uncertain reading. Heavy overwriting obscures the sign between $q(x)$ and k^2; its reading is unresolved.

Original source details: GELFAND-P000777

$$q(x) \to 0 \quad \text{as } |x| \to \infty \quad \text{sufficiently rapidly} \tag{2}$$

Let Ψ — a solution such that

$$\begin{cases} \Psi(x,k) \underset{x\to+\infty}{\longrightarrow} e^{-ikx} + R(k)e^{ikx}, \\ \Psi(x,k) \underset{x\to-\infty}{\longrightarrow} T(k)e^{-ikx} \end{cases} \tag{3}$$

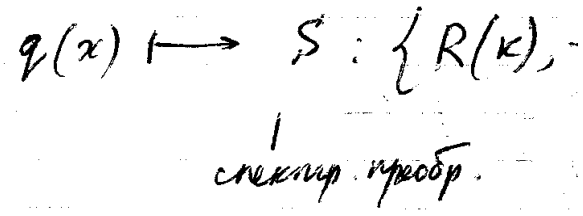


Diagram and typed description.

$q(x) \longmapsto S \quad : \{R(k),\ -\infty < k < +\infty;$

spectral transform

and some discrete spectrum information}.

Original source details: GELFAND-P000777

For a wide class of potentials, the discrete spectrum = the poles of $R(k)$:

$$R(k) \sim \frac{ic_n^2}{k - i\rho_n} \tag{4}$$

The potential q is determined by $R(k)$.

[Crossed out:]

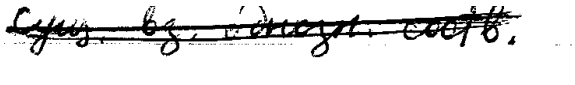


Diagram and typed description.

~~There exists a one-to-one correspondence.~~

Original source details: GELFAND-P000777

Let [crossed out: $q(x)$]

Uncertain reading. The heavily cancelled expression tentatively reads $q(x)$.

Original source details: GELFAND-P000777

$q = q(x,t)$, $R = R(k,t)$.

In which cases

R changes simply, while q changes interestingly?

[At the left edge of the frame, beyond the margin line: ...a))]

Uncertain reading. The image edge clips the entry; the end of an a-like letter and two curved strokes remain visible.

Original source details: GELFAND-P000777

Technique: since $R(k)$ is determined by asymptotics, it is natural to use Wronskians.

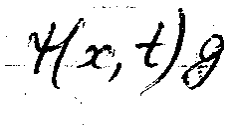
2 equations. Let there be $q, q' \longleftrightarrow R, R'$.

$$2ikf(-4k^2)[R'(k) - R(k)]$$

$$= \int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k) f(\Lambda)[q'(x) - q(x)] \quad (5)$$

$$(2ik)^2 g(-4k^2)[R'(k) + R(k)]$$

$$= \int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,t) g(\Lambda)\Gamma \cdot 1 \quad (6)$$

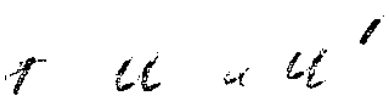


Uncertain reading. The second factor in (6) visibly reads $\Psi(x,t)$, although the first has argument k.

Original source details: GELFAND-P000777

GELFAND-P000778

Λ — an integro-differential operator depending on u and u'



Uncertain reading. The opening definition has u and u', unlike the q, q' in (7), (8).

Original source details: GELFAND-P000778

$\underline{\Gamma}$ — also an operator

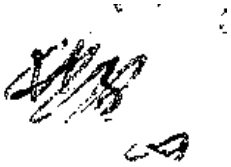
Here

$$\underline{\Lambda}F(x) = F_{xx}(x) - 2[q'(x) + q(x)]F(x) + \underline{\Gamma}\int_x^\infty dx'\, F(x'); \tag{7}$$

$$\underline{\Gamma}G(x) = [q'_x(x) + q_x(x)]G(x) + [q'(x) - q(x)]\int_x^\infty dx'[q'(x') - q(x')]G(x'); \tag{8}$$

f, g — arbitrary functions.

[Crossed out: ...]



Uncertain reading. Slanting strokes cancel the short entry before “Property”.

Original source details: GELFAND-P000778

Property of $\underline{\Lambda}$:

$$\int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k)\underline{\Lambda}F(x) = -4k^2\int_{-\infty}^{+\infty} dx\, \Psi'(x,k)\Psi(x,k)F(x). \tag{9}$$

Let q and q' be distinct (a limit will be taken later).

If q and q' are such that

$$f(\underline{\Lambda})[q'(x) - q(x)] + g(\underline{\Lambda})\underline{\Gamma}\,1 = 0, \tag{10}$$

then

$$R'(k) = \frac{f(-4k^2) - 2ikg(-4k^2)}{f(-4k^2) + 2ikg(-4k^2)}R(k) \tag{11}$$

[Crossed out: ...]

Uncertain reading. The short entry to the left of the parenthesis is cancelled; its reading is unresolved.

Original source details: GELFAND-P000778

(this follows from formulas (5), (6))

All of this is for the stationary Schrödinger equation.

Now one must take infinitesimal deformations

$$q = q(x,t), \qquad q' = q(x, t+dt)$$

$$\begin{aligned} R' - R &\longmapsto R_t(k,t) \\ q' - q &\longmapsto q_t(x,t) \\ \underline{\Lambda} &\longmapsto \underline{L} \qquad (= \underline{\Lambda}|_{q'=q}) \end{aligned}$$

$\underline{\Gamma}$ [crossed out: and?]

Uncertain reading. The cancelled entry between the operator sign and the arrow tentatively reads “и” (and).

Original source details: GELFAND-P000778

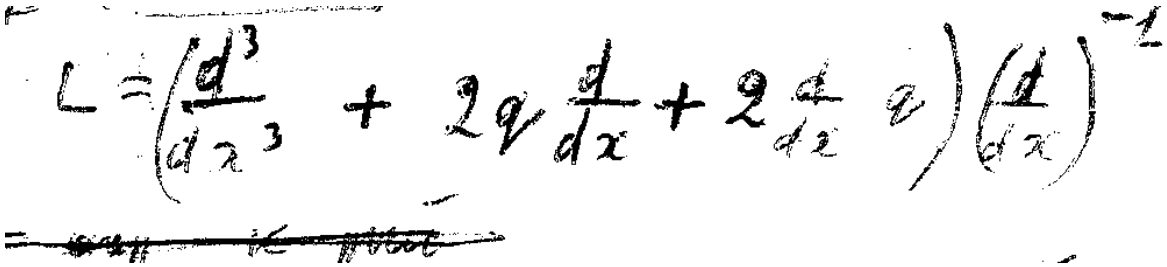
$\longmapsto$ multiplication by $2q_x(x)$

We obtain:

$$\begin{aligned} \underline{L}F(x) = F_{xx}(x) - 4q(x,t)F(x) \\ + 2q_x(x,t)\int_x^\infty dx'\, F(x') \end{aligned} \tag{12}$$

Gelf. (and Dickey):

$$L = \left(\frac{d^3}{dx^3} + 2q\frac{d}{dx} + 2\frac{d}{dx}q\right)\left(\frac{d}{dx}\right)^{-1}$$



Original source details: GELFAND-P000778

Uncertain reading. In this entry both signs before the terms containing q are “+”.

[Crossed out:]

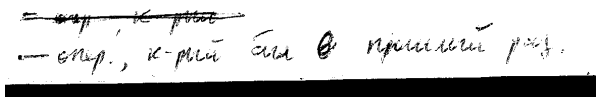


Diagram and typed description.

~~= the operator which~~

Original source details: GELFAND-P000778

— the operator which was present last time.

GELFAND-P000779

[source number: 5]

appears in many places.

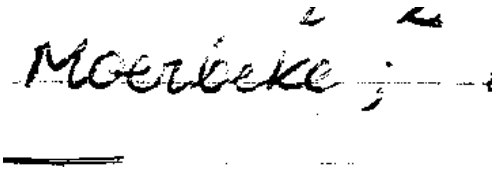
a) [Crossed out: and?]

Uncertain reading. The short cancelled sign after a) tentatively reads “и” (and).

Original source details: GELFAND-P000779

Ψ_1, Ψ_2 satisfy a second-order equation $\Rightarrow$ $\Psi_1\Psi_2$ satisfies a third-order equation.

Gelf. Dickey; Lax; Marchenko; Mose...ka; et al.



Uncertain reading. The surname after Marchenko starts with “Moce...” and ends with ka-like letters; the complete reading is unresolved.

Original source details: GELFAND-P000779

Property of $\underline{\Lambda}$.

$$\int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k)(\underline{\Lambda} + 4k^2)F(x) = 0 \tag{13}$$

$\Psi = \Psi' \Rightarrow \underline{\Lambda}$ is replaced by $\underline{L}$.

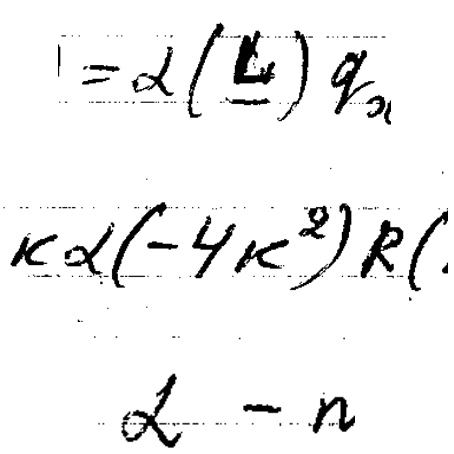
[Above the following line: a row of dots.]

. . . . . .

For $f(\underline{L}) = \underline{L}$ we have

$$\begin{gathered} q_t(x,t) = \alpha(\underline{L})q_x(x,t) \\ \Longleftrightarrow R_t(k,t) = 2ik\alpha(-4k^2)R(k,t) \end{gathered} \tag{14}$$

α — a polynomial $\Rightarrow$ KdV and higher KdV



Uncertain reading. The sign is transcribed as α in all three places; its shape also permits a calligraphic L.

Original source details: GELFAND-P000779

$R' \leftrightarrow R$ by formula (11) and R — a solution of the linear equation $(14)_2$, then R' is also a solution.

Then q', constructed from q, satisfies equation $(14)_1$ together with q.

$(14)_1$ — higher KdV in the form $AKNS$

The simplest case: let [crossed out: $f, g =$ const] $g = 1$, $f = 2\rho$.

Original source details: GELFAND-P000779

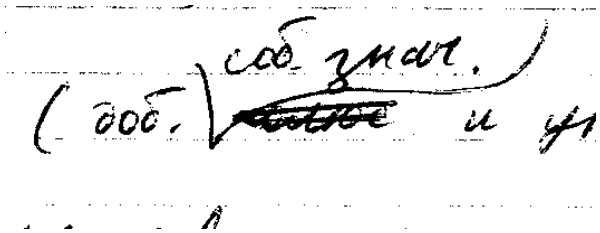
Diagram and typed description.

~~$f, g =$ const~~

One obtains the simplest transformation:

$$R'(k,t) = -\frac{k+i\rho}{k-i\rho} R(k,t) \tag{15}$$

This adds a soliton (adds [crossed out: ...; inserted above with a connecting line: eigenvalue] and multiplies by a coefficient)



Original source details: GELFAND-P000779

Diagram and typed description.

eigenvalue

...

Heavy cancellation strokes obscure the word beneath the insertion; its reading is unresolved.

The relation between the potentials is written out:

let

$$Q(x,t) = \int_x^\infty dx'\, q(x',t) \tag{16}$$

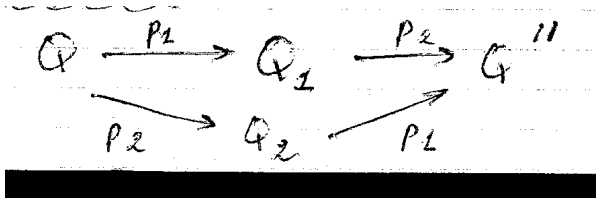
$$\begin{aligned} Q'_x(x,t) + Q_x(x,t) = & -\tfrac{1}{2}[Q'(x,t) - Q(x,t)] \\ & \cdot [4\rho + Q'(x,t) - Q(x,t)] \end{aligned} \tag{17}$$

$Q \equiv 0 \Rightarrow Q'$ — a one-soliton solution, (for $q_t = q_{xxx} - 6q_x q$)

(17) — a Riccati equation (with respect to Q' or with respect to Q)

All integrals are obtained by successive application of the Bäcklund transform from the trivial $\int u^2\, dx$

Another application



Original source details: GELFAND-P000779

Diagram and typed description.

$$\begin{array}{ccccc} Q & \xrightarrow{\rho_1} & Q_1 & \xrightarrow{\rho_2} & Q'' \\ & \searrow^{\rho_2} & & \nearrow_{\rho_1} & \\ & & Q_2 & & \end{array} \tag{18}$$

commutativity!

This is immediately visible; for the functions R it is simply multiplication. For the functions q it is nontrivial. One obtains a principle of nonlinear superposition:

$$Q''(x,t) = Q(x,t) - \frac{(\rho_1+\rho_2)[Q_1(x,t)-Q_2(x,t)]}{\{\rho_1-\rho_2+\frac{1}{2}[Q_1(x,t)\ [?]\ Q_2(x,t)]\}} \tag{19}$$

[In the manuscript the fraction in (19) is written with a slash at the end of the first line and the start of the second.]

Uncertain reading. Heavy overwriting obscures the sign between $Q_1(x,t)$ and $Q_2(x,t)$; $+$ and $-$ are possible.

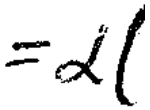
Original source details: GELFAND-P000780

— · —

f and g may depend on t! The relation between (10) and (11) remains valid.

$$q_t(x,t) = \alpha(\underline{L})q_x(x,t) \tag{20}$$

$$R(k,t) = e^{2ik\alpha(-4k^2)(t-t_0)}R(k,t_0) \tag{21}$$



Uncertain reading. The sign is provisionally rendered α in (20), (21); the handwriting also permits $\mathcal{L}$.

Original source details: GELFAND-P000780

Choose f and g so that the fraction in (11) coincides with the displayed exponential in (21). Then we obtain a correspondence between $q(x,t)$ and $q(x,t_0)$ (formula (10)).

Evolution in t — also a Bäcklund transform!

There is an analogy with the Fourier transform:

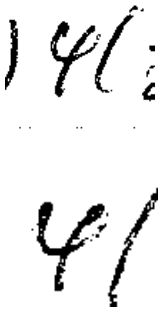
$$u(x,t) = e^{(t-t_0)\psi(\frac{\partial}{\partial x})}u(x,t_0),$$

Uncertain reading. The first u overwrites an earlier unreadable sign.

Original source details: GELFAND-P000780

when

$$u_t(x,t) = \psi\left(\frac{\partial}{\partial x}\right) u(x,t).$$



Uncertain reading. Both lines provisionally have ψ; another possible reading is φ.

Original source details: GELFAND-P000780

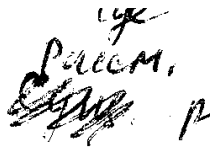
— · —

Consider the equation

$$\Psi_{tt} + \tfrac{1}{2}\Psi_{xx} - x\Psi_x + n\Psi = 0, \tag{22}$$

where $n \in \mathbb{Z}$

[Crossed out, tentatively: Its. Inserted above: Consider.]



Uncertain reading. The cancelled word may read “Его” or “Её” (its).

Original source details: GELFAND-P000780

a solution (of the form):

$$\Psi(x,t) = \prod_{j=1}^{n}(x - x_j(t)) \tag{23}$$

[Crossed out to the right of (23):]

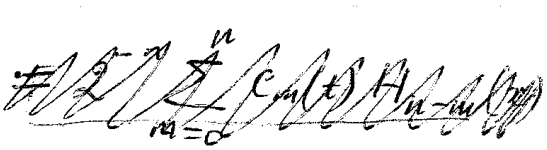


Diagram and typed description.

~~$= 2^{-n} \sum_{m=0}^{n} c_m(t) H_{n-m}(x)$~~

Original source details: GELFAND-P000780

Then

$$\ddot{x}_j = -x_j(t) + \sum_{k=1}^{n}{}' \frac{1 + 2\dot{x}_j(t)\dot{x}_k(t)}{x_j(t) - x_k(t)} \tag{24}$$

(Moser) [A word is crossed out at right; only a possible initial M or H is distinguishable.]

Undeciphered passage. The crossed-out word to the right of “Мозер” (Moser) cannot be read.

Original source details: GELFAND-P000780

[Inserted beneath, one below the other, with a brace at right:]

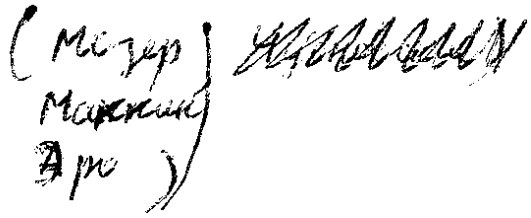


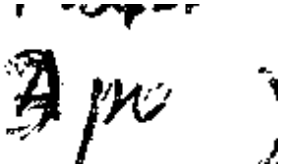
Diagram and typed description.

(Moser)
Makkin
Ero

Original source details: GELFAND-P000780

Uncertain reading. “Маккин” (Makkin) is tentative.

Original source details: GELFAND-P000780



Uncertain reading. “Эро” (Ero) is tentative.

Original source details: GELFAND-P000780

GELFAND-P000781

[source number: 6]

Another form:

$$\Psi(x,t) = 2^{-n} \sum_{m=0}^{n} c_m(t) H_{n-m}(x) \tag{25}$$

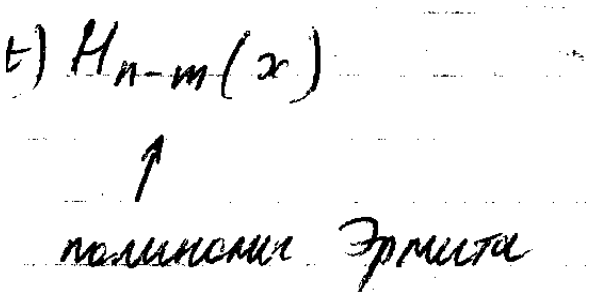


Diagram and typed description.

$H_{n-m}(x)$
↑
Hermite polynomials

Original source details: GELFAND-P000781

Then

$$\ddot{c}_m(t) + m c_m(t) = 0 \tag{26}$$

Nonlinear correspondence:

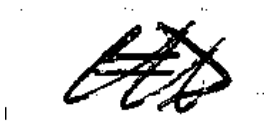
$$\{x_j\} \leftrightarrow \{c_m\}$$

From roots to coefficients — one obtains a linear system of differential equations.

Equilibrium solutions of (24)

$$\bar{x}_j = \sum_{k=1}^{n}{}'(\bar{x}_j - \bar{x}_k)^{-1} \tag{27}$$

In this case $c_m \equiv 0$, [Crossed out: ...]



Uncertain reading. The short mathematical entry is heavily cancelled; it may read $\forall t$.

Original source details: GELFAND-P000781

$m = 1, 2, \ldots, n$; $c_0 = 1$

the equilibrium positions are the zeros of the Hermite polynomial of order n

Corollary:

the zeros of the Hermite polynomial satisfy (27).

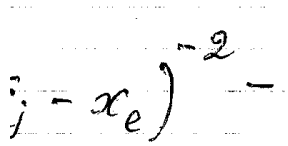
(Stieltjes; 1885)

Without Ψ_{tt} equation (22) — the equation for Hermite polynomials.

Small oscillations about the equilibrium positions:

$$x_j = \bar{x}_j + \varepsilon_j(t) \tag{28}$$

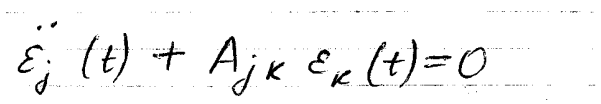
$$A_{jk} = \delta_{jk} \sum_{\ell=1}^{n}{}'(\bar{x}_j - x_\ell)^{-2} - (1 - \delta_{jk})(\bar{x}_j - \bar{x}_k)^{-2}$$



Uncertain reading. The first term has x_ℓ without a visible bar; the other term has visible bars.

Original source details: GELFAND-P000781

$$\ddot{\varepsilon}_j(t) + A_{jk}\varepsilon_k(t) = 0$$



Uncertain reading. The equation has $A_{jk}\varepsilon_k(t)$, with no separate term $\varepsilon_j(t)$.

Original source details: GELFAND-P000781

The eigenvalues are integers —

$$Av^{(s)} = a_s v^{(s)}, \qquad a_s = s - 1, \quad s = 1, 2, \ldots, n.$$

— · —

GELFAND-P000782

3/IV-78 Gelfand Sem.

D. B. Fuks (and I. M. Gelfand)

Computation of the cohomology of various Lie algebras.

g — a real Lie algebra;

$C_q(g)$ — the group of q-dimensional chains

$$C_q(g) \underset{\text{def}}{=} \Lambda^q g \qquad \text{(exterior power)}$$

$$d : C_q(g) \longrightarrow C_{q-1}(g) \tag{1}$$

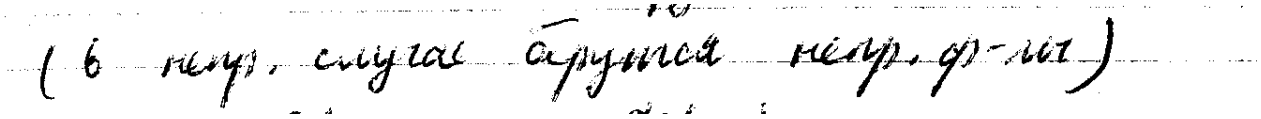
$$d(g_1 \wedge \ldots \wedge g_q) = \sum_{1 \le s < t \le q} (-1)^{s+t-1} [g_s, g_t] \wedge g_1 \wedge \ldots \wedge \hat{g}_s \wedge \ldots \wedge \hat{g}_t \wedge \ldots \wedge g_q \tag{1'}$$

GELFAND-P000783

[source number: 11]

It is easy to see that $d^2 = 0$ and this gives rise to homology, denoted by $H_q(g)$.

The dual construction gives cohomology (in the continuous case one takes continuous



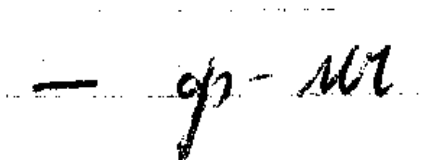
Original source details: GELFAND-P000783

Uncertain reading. Abbreviation: “ф-лы” / “ф-ии” (functionals / functions).

)

$$\delta : C^{q-1}(g) \longrightarrow C^q(g) \tag{2}$$

$C^q(g)$ —



Uncertain reading. Abbreviation: “ф-лы” / “ф-ли” (functionals).

Original source details: GELFAND-P000783

on $C_q(g)$

$H^q(g)$ — the homology of the dual complex with differential δ;

An inner product lets us identify C^q and C_q. We obtain:

$$d : C_q(g) \longrightarrow C_{q-1}(g)$$
$$\delta : C_q(g) \longrightarrow C_{q+1}(g)$$

Put

$$\Delta = d\delta + \delta d \tag{3}$$

(Laplace operator)

Δ commutes with d and δ.

It is easy to show that each homology class has exactly one harmonic element.

$$\Delta a = 0 \quad \Longleftrightarrow \quad da = 0 \text{ and } \delta a = 0 \tag{4}$$

Consider the Lie algebra [inserted above the line: (formal)] of vector fields on $\mathbb{R}$ of the form:

$$f(x)\frac{d}{dx}, \tag{5}$$

where

$$f(x) = \sum_{k\geq 2} a_k x^k. \tag{6}$$

Put

$$e_n = x^{n+1}\frac{d}{dx} \qquad \text{(element of "dimension" } n\text{)}$$

The cohomology was computed by L. Goncharova, but since then there have been further attempts to understand this.

The algebra W_1 does not even have a local Lie group

[Below W_1 is written $\{a(x)\frac{d}{dx}\}$; two short vertical strokes link the inscription to W_1.]

[Crossed out: ...]

Uncertain reading. A short cancelled word; possibly "Но" or "Ново".

Original source details: GELFAND-P000783

The group of formal diffeomorphisms has Lie algebra: $\{a(x)\frac{d}{dx},\ a(0)=0\}$

$$L_0 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a(0)=0\right\} \tag{7}$$

— it is the Lie algebra of the formal group

$$W_1 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a \text{ — a formal series}\right\} \tag{8}$$

$$L_1 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a(0)=0,\ a'(0)=0\right\} \tag{9}$$

[At the lower left, beyond the edge of the sheet:

Uncertain reading. a cropped fragment of a sign

Original source details: GELFAND-P000783

.]

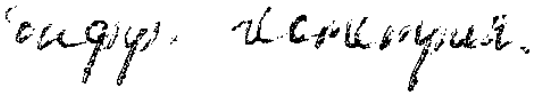
GELFAND-P000784

$$L_0 \subset W_1, \qquad \dim W_1/L_0 = 1.$$

L_0 integrates to a group

W_1 does not integrate to a group.

This is why there is this whole



Uncertain reading. “следу.” (abbreviation, probably “following”).

Original source details: GELFAND-P000784

story.

Uncertain reading. “Спрв.” / “Справ.” (abbreviated heading) 1.

Original source details: GELFAND-P000784

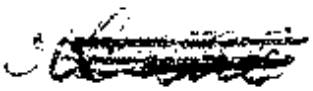
A Lie alg. is called an algebra of finite defect, if it has a subalgebra of finite codimension integrating to a Lie group. The minimum codimension is called the defect.

How it began: automorphism groups.

Uncertain reading. “Спрв.” / “Справ.” (abbreviated heading) 2.

Original source details: GELFAND-P000784

Suppose there is no centre; then the automorphism group [crossed out:



Uncertain reading. A short word, possibly “имеет” (“has”).

Original source details: GELFAND-P000784

] has a Lie algebra that is an

Uncertain reading. overalgebra / subalgebra

Original source details: GELFAND-P000784

of the algebra itself.

The cohomology of W_1 and L_0 is easily found (Gelfand, Fuks); but that of L_1 is harder.

[A short underlined inscription with diagonal hatching:

Uncertain reading. “Фукс;” / “Факт;” (Fuks; / Fact;)

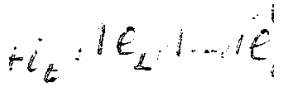
Original source details: GELFAND-P000784

.]

L_1 is generated by the elements $e_1, e_2, \ldots$

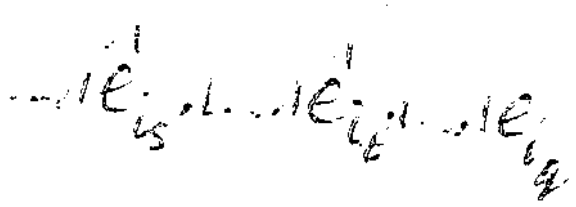
$$[e_n, e_m] = (m-n)e_{m+n} \tag{10}$$

$$d(e_{i_1} \wedge \cdots \wedge e_{i_q}) = \sum_{s<t} (-1)^{s+t-1}(i_t - i_s)e_{i_s+i_t} \wedge e_1 \wedge \cdots \wedge \overset{\prime}{e}_{i_s} \wedge \cdots \wedge \overset{\prime}{e}_{i_t} \wedge \cdots \wedge e_{i_q} \tag{11}$$



Uncertain reading. After $e_{i_s+i_t}$ the source has e_1, with no visible i in the subscript.

Original source details: GELFAND-P000784



Uncertain reading. Above e_{i_s} and e_{i_t} there are short almost vertical strokes; the shape of the marks is not entirely clear.

Original source details: GELFAND-P000784

We see that the sum of the indices does not change! We obtain a

Uncertain reading. “расслоение” / “разложение” (splitting / decomposition)

Original source details: GELFAND-P000784

into a direct sum of finite-dimensional complexes.

Theorem 1 (L. Goncharova)

a) $\forall q \geq 1 \quad \Rightarrow \quad \dim H_q(L_1) = 2$

b) «weight» of the generators

$$\frac{3q^2 \pm q}{2}$$

(sum of the indices)

— · —

$C_{q,n}$ — q-chains of weight n. There is a Laplace operator:

$$\Delta : C_{q,n} \longrightarrow C_{q,n} \tag{12}$$

[Introduce another structure]

Uncertain reading. The line has diagonal hatching; whether this is emphasis or cancellation is unclear.

Original source details: GELFAND-P000784

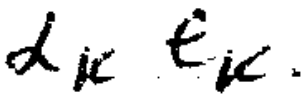
δ is the adjoint with respect to the inner product in which the monomials $e_{i_1} \wedge \cdots \wedge e_{i_q}$ $_{i_1<\cdots<i_q}$ form an orthonormal basis.

GELFAND-P000785

[source number: 12]

3 candidates for an orthonormal basis:

$$e_k, \qquad \frac{e_k}{k!}, \qquad \alpha_k e_k.$$



Uncertain reading. The coefficient in the third candidate: α_k / λ_k.

Original source details: GELFAND-P000785

Here we use e_k

This carries over to the exterior powers.

We obtain the Laplace operator (12).

Facts

Original source details: GELFAND-P000785

Uncertain reading. After the heading there is a tall hatched sign, possibly III.

A. All] eigenvalues — integers!

Original source details: GELFAND-P000785

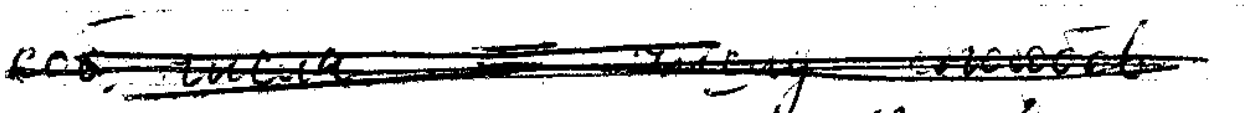
Uncertain reading. After “All” there is a tall sign resembling a closing square bracket.

B. they can be enumerated!

In the space

$$C_{*,n} = \bigoplus_q C_{q,n}$$

[Crossed out: eigenvalues ...



Original source details: GELFAND-P000785

Undeciphered passage. The middle of the phrase is illegible; the right end may read “способ...”.

]

The number of distinct eigenvalues $=$
$=$ the number of partitions of n into a sum

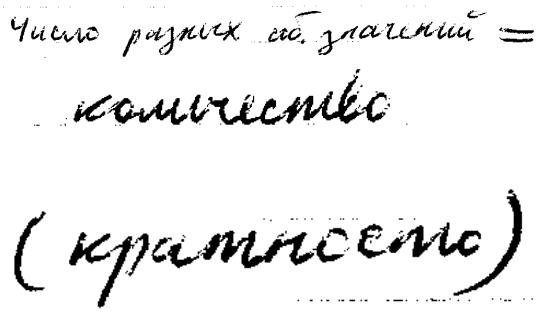
$$n = i_1 + \cdots + i_q \tag{13}$$

$$\begin{cases} i_1 \geq 1 \\ i_2 - i_1 \geq 3 \\ \cdots \\ i_q - i_{q-1} \geq 3 \end{cases} \tag{14}$$

To the partition (13) there corresponds the quantity

$$C_{i_1}^3 + \cdots + C_{i_q}^3 - \sum_{s<t} i_s i_t \tag{15}$$

of eigenvalues (multiplicity)



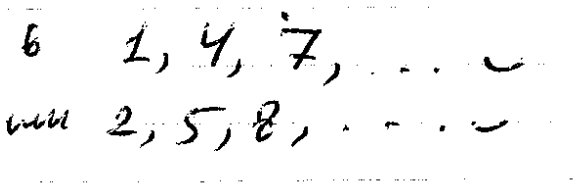
Uncertain reading. The words “quantity” and “of eigenvalues (multiplicity)” translate the literal source wording.

Original source details: GELFAND-P000785

Exercise:

the number (15) ≥ 0 and $= 0$

$$\iff \begin{array}{lll} \text{in } 1,4,7,\ldots & \smile & \Sigma = \dfrac{3q^2 - q}{2} \\ \text{or } 2,5,8,\ldots & \smile & \Sigma = \dfrac{3q^2 + q}{2} \end{array}$$



Uncertain reading. After each sequence there is a short upward-opening arc; the meaning of the marks is unclear.

Original source details: GELFAND-P000785

The eigenvectors can be written down

Symbolic calculus:

$$z = e_1, \quad e_2 = z^2, \quad e_3 = z^3; \ \ldots$$

Description of an eigenvector:

$$\left(\prod_{1 \leq i < j \leq q} (z_i - z_j) \right)^3$$

multiply by

$$\prod z_i \qquad \text{or} \qquad \prod z_i^2$$

One obtains an antisymmetric polynomial. It is represented as a combination

$$z_1^{i_1} \cdots z_q^{i_q} + \underbrace{\cdots}_{\text{with the required signs}}$$

Replace by

$$e_{i_1} \wedge \cdots \wedge e_{i_q}$$

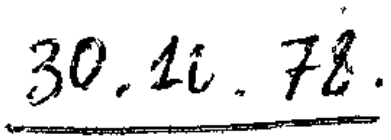
Next meeting 4/IV-78

Chapter 14

GELFAND-P000786

[source number: 4]

30.10.78. Gelfand Seminar



Uncertain reading. The month looks more like 10. A break in the second digit leaves 11 possible.

Original source details: GELFAND-P000786

Kruskal

Surreal numbers

there are 2 books: John Conway — (“combinatorial games”).

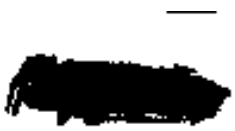
{Knuth D. “[crossed out: ...] — novel”}

Undeciphered passage. Dense hatching obscures the entry before “— novel”.

Original source details: GELFAND-P000786

Def.

Number = map from a well-ordered set to a set of 2 elements ↑ and ↓.



Undeciphered passage. A heavily inked horizontal mark lies below the wavy line. No letters can be distinguished.

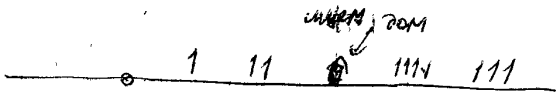
Original source details: GELFAND-P000786

$x \neq y$ — compare the arrows (lexicographically).

$$\widehat{\uparrow} = \uparrow\uparrow\uparrow \ldots = \omega$$

$\widehat{\ } \leftrightarrow$ (periodic)

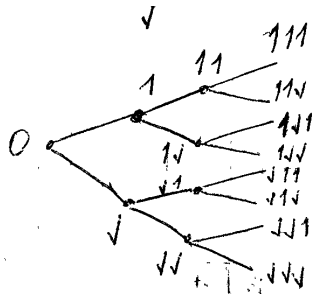
$\widehat{\uparrow\downarrow}$ $\widehat{\downarrow}$



Original source details: GELFAND-P000786

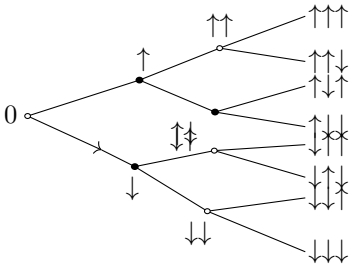
Diagram and typed description.

The crossed-out word before “house” is illegible.

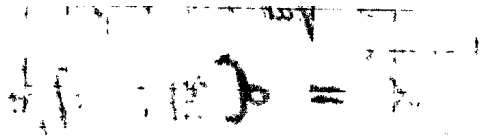


Original source details: GELFAND-P000786

Diagram and typed description.



$\ldots\} = \ldots$



Original source details: GELFAND-P000786

Undeciphered passage. A right brace and an equals sign are visible under the number line. The other faint strokes do not establish the expression.

$$x \longleftarrow y$$
earlier
(earlier)

(i.e. x is on the path from 0 to y)

$y]$ — an element of the set of numbers that come before y

Truncation operation — x has fewer arrows

$y\rceil$ a number earlier than y, but greater than it

$y\rfloor$ —"— and less than it

$x < y$ is also defined

$$\{y]\} = \{y\rfloor\} \cup \{y\rceil\},$$
$$\{y\rceil\} \cap \{y\rfloor\} = \Lambda$$

Example:

Main def.:

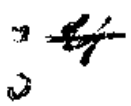
GELFAND-P000787

Thm. 1

(simple)

A, B — sets of numbers

$\forall a \in A, b \in B$; [crossed out: ...] $a < b$



Original source details: GELFAND-P000787

Uncertain reading. A heavily crossed-out sign lies between the semicolon and the inequality.

Then $\exists x: \quad a < x < b$.

and there exists a unique earliest x.

Proof: transfinite ind., we look at $x = 0$ and if we have not found x up to a certain point, we look further.

Notation:

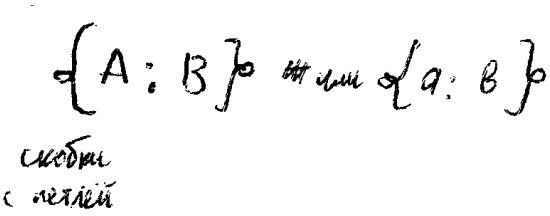
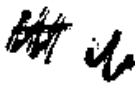


Diagram and typed description.
$\{A : B\}$ ... or $\{a : b\}$

braces with a loop

Original source details: GELFAND-P000787

Uncertain reading. The cancelled sign before "or" resembles Russian «и» ("and").



Original source details: GELFAND-P000787

Addition.

by transfinite induction.

Theorem 2

$$x = \{x\rfloor : x\rceil\}$$

(obvious)

Def. of addition.

$$x + y = \{x\rfloor + y,\ x + y\rfloor :\ x\rceil + y,\ x + y\rceil\}$$

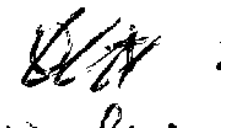
Theorem 3.

a) $0 + 0 = \{:\} = 0$.

[Below the formula:] there are no conditions, so the earliest of all

b) [crossed out: ...] $x + 0 = x$ by induction

Uncertain reading. The cancelled strokes resemble $\forall x$.



Original source details: GELFAND-P000787

Proof:

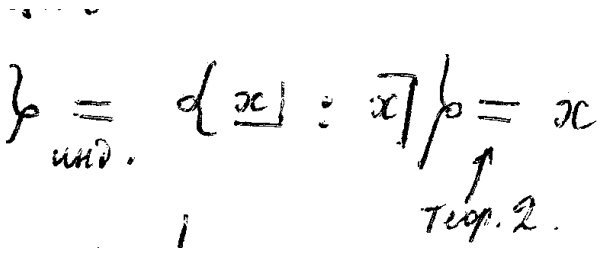


Diagram and typed description.
$x + 0 = \{x\rfloor + 0 : x\rceil + 0\}$
$\underset{\text{ind.}}{=} \{x\rfloor : x\rceil\} \underset{\text{Thm. 2.}}{=} x$

Original source details: GELFAND-P000787

In the ordinary theory there is no notion of earlier!

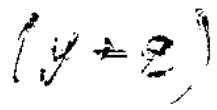
$x + y$ — the simplest number, monotone in each argument

a) monotone in x, y

b) the earliest of all such numbers.

The usual properties hold

$$x + y = y + x$$
$$(x + y) + z = x + (y\ [?]\ z)$$



Uncertain reading. The sign between y and z resembles $\geq$. The reading $+$ is also possible.

Original source details: GELFAND-P000787

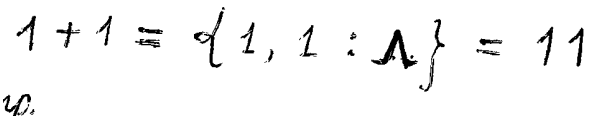
There is an inverse operation.

The opposite of x — reversal of the arrows $\quad x + \mathrm{opp}\, x = 0.$

GELFAND-P000788

[source number: 5]

$$\uparrow + \uparrow = \{1, 1 : \Lambda\} = \uparrow\uparrow$$



Uncertain reading. The initial signs on either side of the plus resemble up arrows. “1+1” is also a possible reading. The ones within the braces have feet.

Original source details: GELFAND-P000788

Ex.

$$\uparrow\downarrow + \uparrow\downarrow = ?$$
$$\hat{\uparrow} + \downarrow\downarrow\downarrow = ?$$

Multiplication.

Basic inequality of mult’n.

$$r < s, \qquad u < v$$

Uncertain reading. The sign before $< s$ is provisionally typeset as r. Its shape also admits η.

Original source details: GELFAND-P000788

Then

$$ru + sv \text{ [?] } rv + su$$

Original source details: GELFAND-P000788

Uncertain reading. The first letters in ru and rv have been retraced. The sign after sv is overwritten and resembles $>$.

xy — any of the four [?] $\Rightarrow$ 4 inequalities

Uncertain reading. The abbreviation after "four" may be «слаг.» (summands) or «оп.» (options).

Original source details: GELFAND-P000788

$$\begin{aligned} x \cdot y = \{&\lfloor x \rfloor \cdot y + x \cdot y \rfloor - \lfloor x \rfloor \cdot \lfloor y \rfloor, \\ &\lceil x \rceil \cdot y + x \cdot y \rceil - \lceil x \rceil \cdot \lceil y \rceil : \\ &\lfloor x \rfloor \cdot y + x \cdot y \rceil - \lfloor x \rfloor \cdot \lceil y \rceil, \\ &\lceil x \rceil \cdot y + x \cdot y \rfloor - \lceil x \rceil \cdot \lfloor y \rfloor\} \end{aligned}$$

Uncertain reading. A heavy cross-shaped mark stands above the comma at the end of the third line. Its purpose is unclear.

Original source details: GELFAND-P000788

The natural properties of multiplication hold.

$$x \cdot 0 = \{:\} = 0, \qquad x \cdot 1 = x$$

$$\begin{aligned} x \cdot 1 &= \{\lfloor x \rfloor \cdot 1 + x \cdot 0 - x \rfloor \cdot 0 : \\ &\quad \lceil x \rceil \cdot 1 + x \cdot 0 - x \rceil \cdot 0\} \\ &= \{\lfloor x \rfloor + 0 - 0 : x \rceil + 0 - 0\} = x \end{aligned}$$

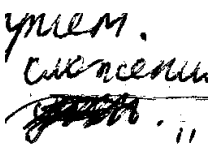
Associat., commut., distribut. hold.

Prove at home.

Advice. Technical device.

Prove that the def. [crossed out: ?] [inserted above the line with a caret:] of addition is "indifferent to brackets"

bracket indifferent



Uncertain reading. The heavily crossed-out abbreviation may read «умн.» (mult.).

Original source details: GELFAND-P000788

$$x = \{a : b\}$$
$$y = \{c : d\}$$
$$x + y = \{a + y, x + c : b + y, x + d\}$$

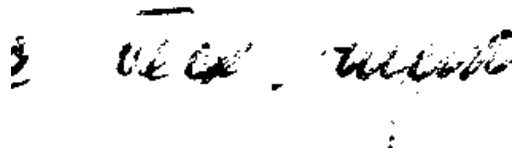
Multiplication also has this property.

Associat.

There exist inf. large and inf. small numbers.

GELFAND-P000789

$\uparrow\uparrow\uparrow \ldots = \widehat{\uparrow} = \omega$ — the simplest inf. number.



Uncertain reading. The abbreviation before «число» (“number”) provisionally reads «беск.» (inf.).

Original source details: GELFAND-P000789

$\uparrow\downarrow\downarrow \ldots = \uparrow \widehat{\downarrow} = \varepsilon$

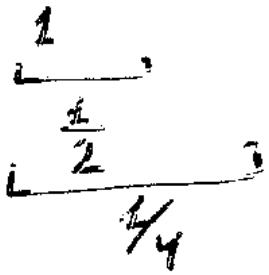
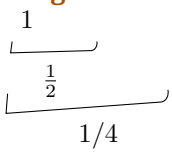


Diagram and typed description.



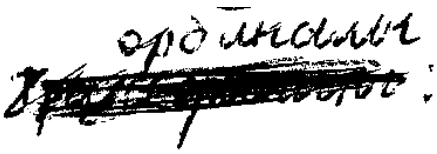
Original source details: GELFAND-P000789

$$\omega \cdot \varepsilon = 1$$
$$\omega + \varepsilon = \widehat{\uparrow} \uparrow \widehat{\downarrow}$$

Sometimes one can shorten it, but sometimes one cannot.

Ordinal arithmetic.

[crossed out: ...] [inserted above:] ordinals: numbers with arrows ↑ (without arrows ↓).



Undeciphered passage. The word below the insertion “ordinals” is heavily crossed out and illegible.

Original source details: GELFAND-P000789

The arithmetic is different from the ordinary one!

The ordinary one is not commutative!

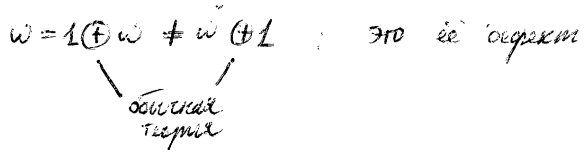


Diagram and typed description.
$\omega = 1 \oplus \omega \neq \omega \oplus 1$ This is its defect
ordinary theory

Original source details: GELFAND-P000789

Hessenburg devis. a natural def. of add. and mult. of transfin. numbers.

Uncertain reading. The last letters of the surname provisionally read “burg”.

Original source details: GELFAND-P000789

Uncertain reading. The abbreviation after the name provisionally reads «прид.» (devis.).

Original source details: GELFAND-P000789

It turned out that it coincid. with this one

$\forall$ number, except 0, has an inverse $\frac{1}{x}$, $x \neq 0$.

$\frac{y}{x}$ exists

$\forall x > 0$ there is $\sqrt{x}$ — by inf. ind.

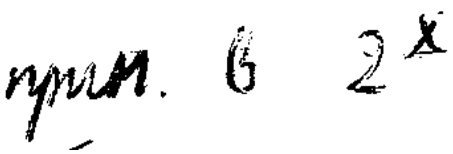
Uncertain reading. The variable after $\forall$ is overwritten. The upper writing reads x.

Original source details: GELFAND-P000789

Uncertain reading. The abbreviations after “by” provisionally read «инд. беск.» (inf. ind.).

Original source details: GELFAND-P000789

$\forall$ polynomial taking vals. of different signs at 2^x points becomes 0 somewhere between them; but this is inf. without contin. — has gaps, e.g., between finite and inf. numbers.



Uncertain reading. The abbreviation after “polynomial” provisionally reads «прин.» (“taking”). The numeral 2 is followed by a superscript Cyrillic x.

Original source details: GELFAND-P000789

[crossed out on the left: ?] $\overset{\ell}{\uparrow}$ — “the largest” ⟶ [inserted above:] already not a surreal number.

Uncertain reading. A loop at the top and several intersecting strokes are visible in the crossed-out sign.

Original source details: GELFAND-P000789

Uncertain reading. The looped sign above the arrow provisionally reads ℓ.

Original source details: GELFAND-P000789

$$\widehat{\uparrow\downarrow} = \frac{2}{3} \qquad \uparrow\downarrow\uparrow = \frac{3}{4}$$

$$\widehat{\uparrow}\overset{\ell}{\downarrow} \quad \text{— gap}$$

Uncertain reading. The looped sign above the down arrow provisionally reads ℓ.

Original source details: GELFAND-P000789

there are exp x; x^y; log

GELFAND-P000790

[source number: 6]

If x is pos. inf., then

$$2^x > x^n \qquad \text{for any } n.$$

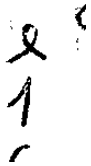
x — inf. small ⇒

$$\exp x = \sum_{n \in \text{fin.}} \frac{x^n}{n!}$$

x — inf. small, n — real ⇒ the series $\sum x^n a_n$ conv. $\forall a_n$

One possible appl. of the theory:

$$\int_x^{\overset{\ell}{\uparrow}} e^{-\xi^2}\, d\xi = \sum \frac{n!}{x^n}$$



Uncertain reading. The looped sign above the arrow at the integral’s upper limit provisionally reads ℓ.

Original source details: GELFAND-P000790

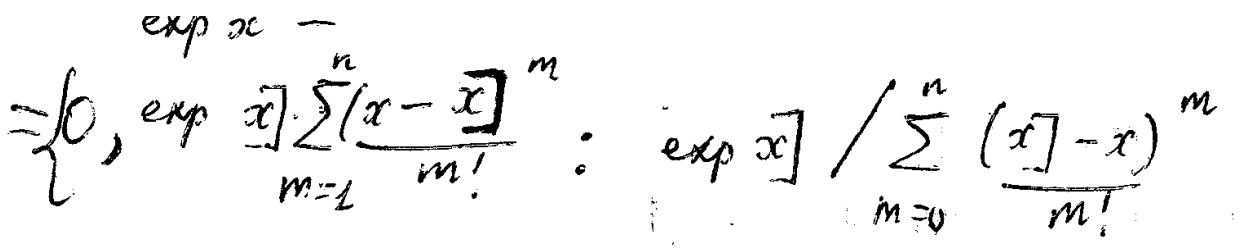
Every number = ratio of 2 integers $(= \frac{m}{n})$.

$$\widehat{\uparrow\downarrow} = \omega - 1$$

$$|x] - x| = 1 \qquad \text{— def. of an integer}$$

$$\exp x =$$

$$= \{0,\ \exp x] \cdot \sum_{m=1}^{n} \frac{(x-x]^m}{m!} :\ \exp x] \Big/ \sum_{m=0}^{n} \frac{(x]-x)^m}{m!}$$



Original source details: GELFAND-P000790

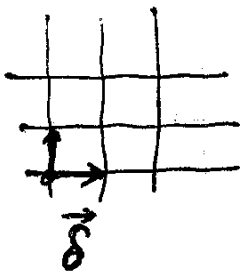
Uncertain reading. No closing parenthesis is visible after the square bracket in the first sum. The final closing brace is unwritten.

GELFAND-P000791

13.11.78. Gelfand Sem.

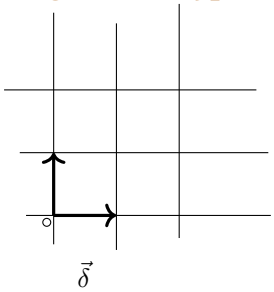
A. Polyakov 2-dimensional discr. system.

$$(1) \qquad S = \mathrm{Tr} \sum_{\vec{x},\vec{\delta}} g_{\vec{x}}^{-1} g_{\vec{x}+\vec{\delta}}$$



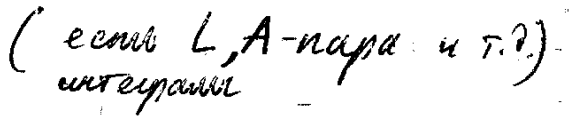
Original source details: GELFAND-P000791

Diagram and typed description.



$$(2) \qquad g_{\vec{x}} \in GL(n,\mathbb{R})$$
$$(3) \qquad \partial_\mu F_{\mu\nu}(x) + [A_\mu, F_{\mu\nu}] = 0$$
$$(4) \qquad \sum_\delta \left\{ \underbrace{g_{\vec{x}+\vec{\delta}} g_{\vec{x}}^{-1}}_{A_{\vec{x}\delta}} - g_{\vec{x}} g_{\vec{x}-\vec{\delta}}^{-1} \right\} = 0$$

The syst. is completely integr. and there is an inf. set of currents. (there is an L, A-pair and so on) [under “there is” is written: integrals]



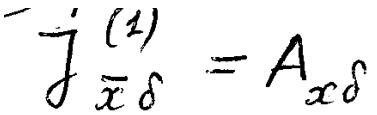
Uncertain reading. The final abbreviation provisionally reads «т. д.» (“so on”).

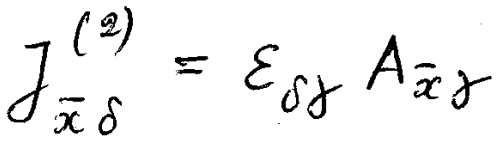
Original source details: GELFAND-P000791

$$(5) \qquad \sum_\delta \left(J^{(k)}_{\vec{x}\delta} - J^{(k)}_{\vec{x}-\vec{\delta},\delta} \right) = 0$$

Examples of currents:

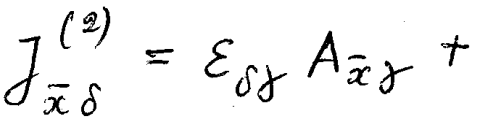
$$(6)_1 \qquad J^{(1)}_{\bar{x}\delta} = A_{x\delta}$$
$$(6)_2 \qquad J^{(2)}_{\bar{x}\delta} = \varepsilon_{\delta\gamma} A_{\bar{x}\gamma} +$$





Uncertain reading. Short bars are visible above x in the subscripts of $(6)_1$ and $(6)_2$. There is no bar over the x in $A_{x\delta}$.

Original source details: GELFAND-P000791



Undeciphered passage. The entry $(6)_2$ breaks off after the plus sign.

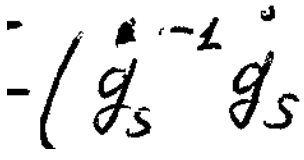
Original source details: GELFAND-P000791

For an ab. group and in the limit (cont.) in one var. — the Toda chain.

Here discr. time. and the nonab. case

Toda chain:

$$(7) \qquad \mathcal{L} = \mathrm{Tr} \sum_s \left(\dot{g}_s^{-1} \dot{g}_s + g_s^{-1} g_{s+1} \right)$$



Uncertain reading. A dot appears above the first g_s in (7), to the left of -1. The order of inversion and differentiation is unparenthesized.

Original source details: GELFAND-P000791

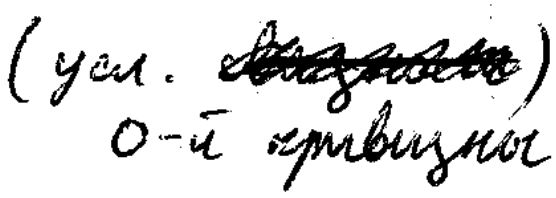
Let

$$(8) \qquad \begin{cases} A_s = g_s^{-1}\dot{g}_s \\ B_s = g_s^{-1}g_{s+1} \end{cases}$$

Zero-curvature conditions:

$$(9) \qquad \begin{cases} \dot{B}_s + A_sB_s - B_sA_{s+1} = 0 \\ \dot{A}_s = B_s - B_{s-1} \end{cases}$$

[to the right of the first line: (cond. of ~~connectedness?~~); inserted below: zero curvature]



Uncertain reading. The crossed-out word provisionally reads «связности» (“connectedness”).

Original source details: GELFAND-P000791

Find L_s, M_s, also satisf. the zero-curv. condition.

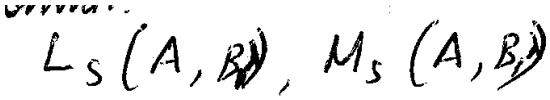
$$(10) \qquad \dot{L}_s + M_sL_s - L_sM_{s+1} = 0$$

Uncertain reading. The sign after $\dot{L}_s$ is overwritten and provisionally reads as a plus.

Original source details: GELFAND-P000791

We want:

$$L_s(A, B, \lambda), \quad M_s(A, B, \lambda)$$



Uncertain reading. The signs by the closing parentheses provisionally read λ. Some strokes touch.

Original source details: GELFAND-P000791

such that $(10) \Rightarrow (9)$ and conversely.

Then

GELFAND-P000792

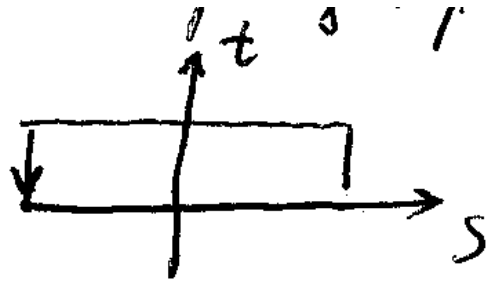
[source number: 17]

$$(11) \qquad \frac{d}{dt}\left\{\prod_{s=-\infty}^{\infty} L_s\right\} = 0$$

Uncertain reading. A short isolated dotted stroke under the product resembles i.

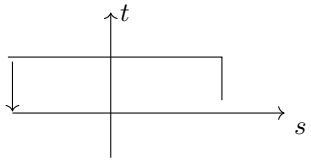
Original source details: GELFAND-P000792

generating fn. of all the integrals



Original source details: GELFAND-P000792

Diagram and typed description.



L, M — connections

$$I(0) \underset{\text{def}}{=} \prod L_s(0)$$

Since

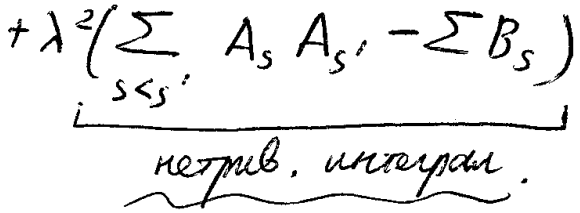
$$I(0)I^{-1}(\delta t) = 1,$$

then $I(\delta t) = I(0)$ and $I(t)$ — an integral of motion.
"Dumb way" of finding an L, A-pair: iterations in λ
answer.

$$(12) \qquad L_s = 1 + \lambda A_s - \lambda^2 B_s L_{s+1}^{-1}$$

Iter. in λ

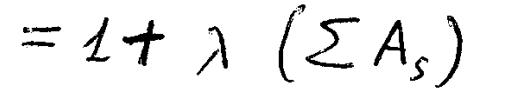
$$\prod_s (1 + \lambda A_s) = 1 + \lambda \left(\sum A_s \right)$$

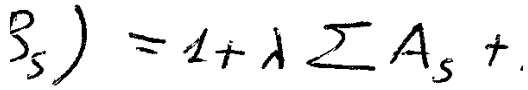


Original source details: GELFAND-P000792

Diagram and typed description.

$$(13) \quad \prod (1 + \lambda A_s - \lambda^2 B_s)$$
$$= 1 + \lambda \sum A_s + \lambda^2 \underbrace{\left(\sum_{s<s'} A_s A_{s'} - \sum B_s \right)}_{\text{nontriv. integral.}}$$

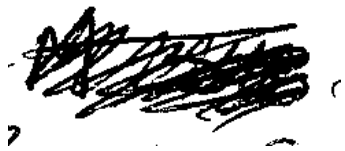




Original source details: GELFAND-P000792

Uncertain reading. Both expansions show equality signs. The subsequent powers are unwritten.

$$(14) \qquad \begin{cases} L_s = e^{-\partial} + A_s + B_s e^{\partial} \\ \dots \qquad M_s = B_s e^{\partial} \end{cases}$$



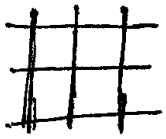
Original source details: GELFAND-P000792

Undeciphered passage. The entry at the beginning of the second line of (14) is heavily crossed out. Its symbols are illegible.

Here e^{∂} — shift by 1 on the lattice

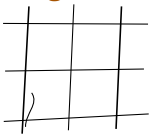
L_s, M_s — difference operators (Jacobi matr.)

Gauge theory.



Original source details: GELFAND-P000792

Diagram and typed description.



$B_{\vec{x}\alpha}$

$\vec{x}$ — point, α — direction

($\alpha = 1, \dots, d$ —

— d — dimension)

2-dim. case — 2 dir.

(only positive ones are taken)

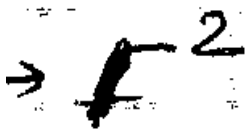
GELFAND-P000793

$$(15) \qquad F_{\bar{x}\alpha\beta} = B_{\bar{x},\alpha} B_{\bar{x}+\bar{\alpha},\beta} B^{-1}_{\bar{x}+\bar{\beta},\alpha} B^{-1}_{\bar{x},\beta}$$

Action:

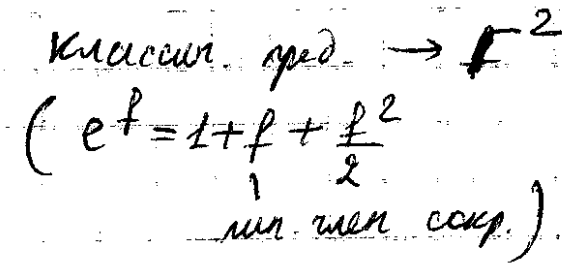
$$(16) \qquad S = \sum_{\bar{x},\alpha,\beta} \operatorname{Tr} F_{\vec{x}\alpha\beta}$$

[on the right:] Class. limit $\longrightarrow F^2$



Original source details: GELFAND-P000793

Uncertain reading. The sign before the raised 2 provisionally reads capital F.

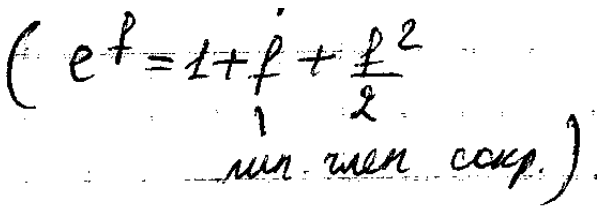


Original source details: GELFAND-P000793

Diagram and typed description.

($e^f = 1 + f + \frac{f^2}{2}$

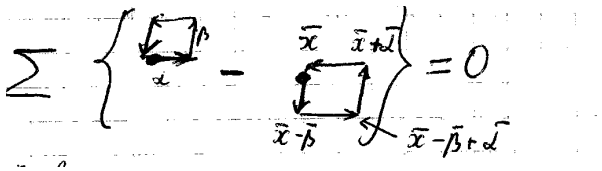
lin. term canc.).



Original source details: GELFAND-P000793

Uncertain reading. An equality sign is written in the parenthesis. The series ends after $f^2/2$.

Eqns. of motion:



Original source details: GELFAND-P000793

Diagram and typed description.
(17)

$$\sum \Big\{ \square_{\alpha}^{\beta} - \square_{\bar{x}}^{\bar{x}+\bar{\alpha}} \Big\} = 0$$

$\bar{x}-\bar{\beta}$ $\quad$ $\bar{x}-\bar{\beta}+\bar{\alpha}$

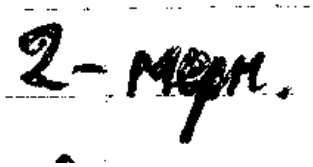
The dot marks the start

$\sum_\beta$, α — fixed (Cf.: $\nabla_\beta F_{\alpha\beta} = 0$)

Original source details: GELFAND-P000793

Uncertain reading. The overwritten sign provisionally reads ∇.

In the 2-dim. case there is no summation.



Original source details: GELFAND-P000793

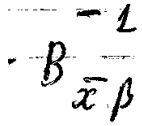
Uncertain reading. The abbreviation for “dim.” is retraced. The earlier strokes are indistinct.

(17) means:

$$(17)_2 \quad \sum \{ B_{\bar{x},\alpha} B_{\bar{x}+\alpha,\beta} B^{-1}_{\bar{x}+\bar{\beta},\alpha} B^{-1}_{\bar{x},\beta} - B^{-1}_{\bar{x}\beta} B_{\bar{x}-\bar{\beta},\alpha} B_{\bar{x}-\bar{\beta}+\bar{\alpha},\beta} B^{-1}_{\bar{x}\alpha} \} = 0$$

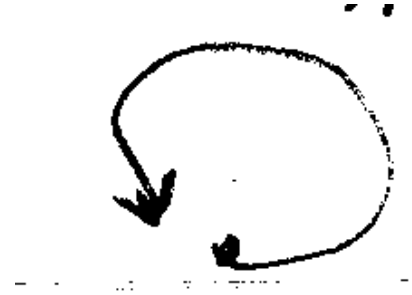
Original source details: GELFAND-P000793

Uncertain reading. The $\bar{x}$ in the last factor of the first product is retraced.



Original source details: GELFAND-P000793

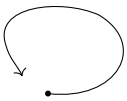
Uncertain reading. The index in the first inverse factor of the second product reads $\bar{x}\beta$.



Original source details: GELFAND-P000793

Diagram and typed description.

(18) $\Psi(c_{\bar{y}}) = \prod_{C_{\bar{y}}} B_{\bar{x}\beta}$



Original source details: GELFAND-P000793

Uncertain reading. The letter in the argument provisionally reads lowercase c. Its $\bar{y}$ subscript is retraced.

Original source details: GELFAND-P000793

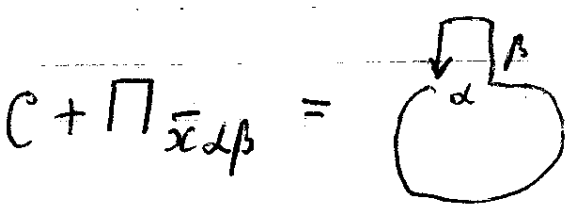
Uncertain reading. The subscript below the circled product sign provisionally reads $C_{\bar{y}}$.

Original source details: GELFAND-P000793

Uncertain reading. The overwritten sign in the subscript of B provisionally reads $\bar{x}$.

Consider

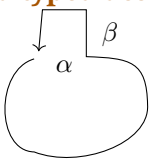
$$\Psi(c_y + \Pi_{\bar{x}\alpha\beta})$$



Original source details: GELFAND-P000793

Diagram and typed description.

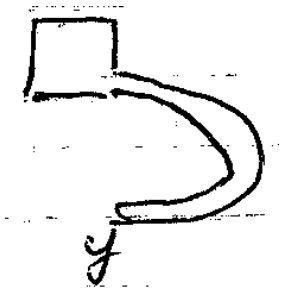
$C + \Pi_{\bar{x}\alpha\beta} =$



a small square has been attached

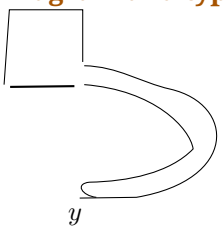
Now

$$\Psi(c+\Pi_{\bar{x}\alpha\beta})\Psi^{-1}(C)= \tag{19}$$



Original source details: GELFAND-P000793

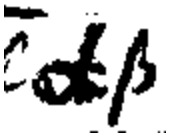
Diagram and typed description.



The gaps near the square and the thickened ends are shown. The directions at these points are unclear.

Original source details: GELFAND-P000793

Uncertain reading. The first sign resembles a looped lowercase c. The inverse function has argument C.



Original source details: GELFAND-P000793

Uncertain reading. The retraced subscript provisionally reads α.

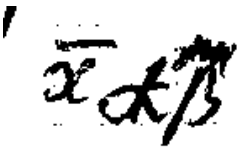
dir. at pt. $\bar{x}, \alpha$, transported to point y

The eqns. are written in the form:

$$\sum_{\beta}\{\Psi(c+\Pi_{\bar{x}\alpha\beta})\Psi^{-1}(C) - \Psi(C)\Psi^{-1}(C-\Pi_{\bar{x}\alpha\beta})\}=0 \tag{20}$$

Original source details: GELFAND-P000793

Uncertain reading. The first contour is provisionally labelled with a lowercase c.



Original source details: GELFAND-P000793

Uncertain reading. The retraced pair of indices provisionally reads $\alpha\beta$.

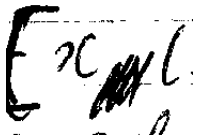
We get something like a connection (trivial) in the space of contours. (i.e. [inserted above the line:] matrix function of the contour)

(cf. (4))

Cont. case.

$\Psi[x_\mu(s)]$ — a functional of the curve (indep. of parametrization)

[on the right:] $x(s) = (x_\mu(s)); \quad \mu = 1, 2, 3, 4.$



Uncertain reading. The retraced subscript of x provisionally reads μ.

Original source details: GELFAND-P000793

Current

$$(21) \qquad \mathcal{F}_\mu(s, C) = \frac{\delta\Psi(x(s))}{\delta x_\mu(s)}\Psi^{-1}$$

GELFAND-P000794

[source number: 18]

Eqn.:

$$(22)_1 \qquad \frac{\delta\mathcal{F}_\mu(s, C)}{\delta x_\mu(s)} = 0$$

[on the right]

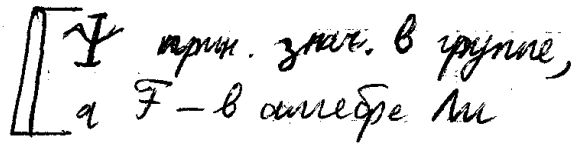


Diagram and typed description.

Ψ takes val. in the group,

and $\mathcal{F}$ — in the Lie algebra

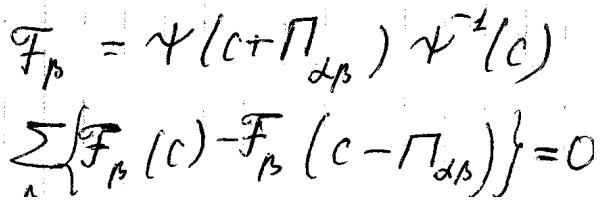
Original source details: GELFAND-P000794

Ψ is inv. with resp. to parametrization: this is written as

$$(23) \qquad \frac{dx_\mu}{ds}\frac{\delta\Psi}{\delta x_\mu(s)} = 0$$

$$\mathcal{F}_\beta = \Psi(c + \Pi_{\alpha\beta})\Psi^{-1}(c)$$

$$\sum_\beta \{\mathcal{F}_\beta(c) - \mathcal{F}_\beta(c - \Pi_{\alpha\beta})\} = 0$$



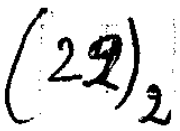
Uncertain reading. The letters in the arguments provisionally read lowercase c.

Original source details: GELFAND-P000794

the cont. limit gives (22).

$$\Psi = p\, e^{\oint A_\mu\, dx_\mu}$$

$$(22)_2 \qquad \frac{\delta \mathcal{F}_\mu(s)}{\delta x_\nu(s')} - \frac{\delta \mathcal{F}_\nu(s')}{\delta x_\mu(s)} + [\mathcal{F}_\mu(s), \mathcal{F}_\nu(s')] = 0$$

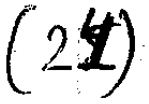


Uncertain reading. The second digit is overwritten. The upper writing provisionally reads $(22)_2$.

Original source details: GELFAND-P000794

Action

$$(24) \qquad S' = \int \left(\frac{\delta \Psi^{-1}}{\delta x_\mu(s)} \frac{\delta \Psi}{\delta x_\mu(s)} \right) (D x_\mu(s))$$



Uncertain reading. The second digit combines strokes resembling 4 and 1. Their order is unclear.

Original source details: GELFAND-P000794

Uncertain reading. The raised stroke to the right of S is typeset as a prime, S'.

Original source details: GELFAND-P000794

It is unclear how to define it

Carefully: 3 dimensions.

(in $4^{\underline{x}}$ dimensions it has yet to be checked in full).

The eqns. $(22)_1$ — $(22)_2$ are solved, with $\mathcal{F}$ obtained from formula (21) and Ψ satisf. cond. (23).

A Y.–M. field is a chiral field on the space of loops.

On the loops there is a fn. with val. in a finite-dimensional group. For it the eqns. of the principal chir. field (Ψ is such a field) are (22).

[inserted above the line on the left:] If Ψ comes from a connection ($\Leftrightarrow$ Ψ — homom. of loops into the group) $\Rightarrow$ one gets the Yang–Mills eqns..

Currents: ($\sim$ integrals of motion):

$$(25) \qquad \frac{\delta \mathcal{J}_\mu^{(k)}(s, C)}{\delta x_\mu(s)} = 0$$

$$\mathcal{J}_\mu^{(1)} = \mathcal{F}_\mu(s, C) \tag{26}$$

$$\mathcal{J}_\mu^{(2)} = \varepsilon_{\mu\nu\lambda} t_\nu(s) \mathcal{F}_\lambda(s, C) + [\mathcal{F}_\mu(s), \chi] \tag{27}$$

t — tang. vector to the curve

[$\mathcal{F}$] χ we shall now find

Uncertain reading. The crossed-out sign resembles $\mathcal{F}$.

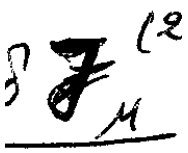
Original source details: GELFAND-P000795

We have:

$$\frac{\delta \mathcal{J}_\mu^{(2)}}{\delta x_\mu} = \varepsilon_{\mu\nu\lambda} t_\nu(s) \frac{\delta \mathcal{F}_\lambda(s, c)}{\delta x_\mu(s)} \tag{28}$$

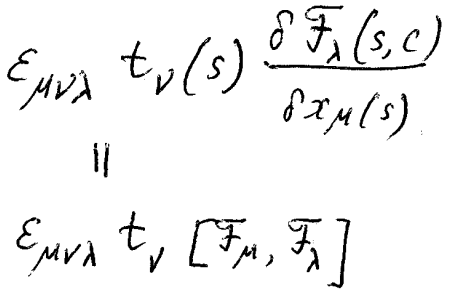
$$\| \varepsilon_{\mu\nu\lambda} t_\nu [\mathcal{F}_\mu, \mathcal{F}_\lambda]$$

$$+ \left[\mathcal{F}_\mu(s, c), \frac{\delta \chi}{\delta x_\mu(s)} \right]$$



Uncertain reading. The letter in the numerator on the left is overwritten. It may read $\mathcal{J}$ or $\mathcal{F}$.

Original source details: GELFAND-P000795



Uncertain reading. Two vertical strokes connect the first summand with the entry below it. No separate coefficient is visible before ε on the lower line.

Original source details: GELFAND-P000795

Let

$$\frac{\delta\chi}{\delta x_\mu} = -\varepsilon_{\mu\nu\lambda} t_\nu \mathcal{F}_\lambda \tag{29}$$

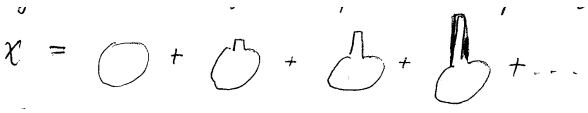
Then

$$\mathcal{F}_\mu = \varepsilon_{\mu\nu\lambda} t_\nu \frac{\delta\chi}{\delta x_\lambda} \tag{30}$$

This is possible if the following holds $(22)_1$
We obtain a nonlocal conservation law. Expr. χ:

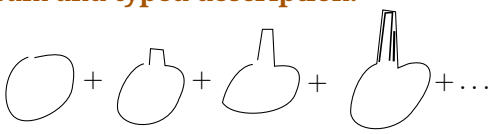
Original source details: GELFAND-P000795

Uncertain reading. The abbreviation before χ provisionally reads «Выраж-е.» (“expr.”). The dividing stroke is unclear.



Original source details: GELFAND-P000795

Diagram and typed description.

$\chi =$  $+\ldots$

The short gaps and thickened strokes on the last protrusion are shown. Arrowheads are indistinct.

(a sq. is added in one direction)
The nonlocality is 1-dimensional.
Contours: 1. S. Mandelstam, Phys. Rev., 1968.
2. Minlos, Sinai
[on the right, alongside item 1:] quant. Y.–M. in the spirit of Faddeev–Popov
γ-pencil (instead of the L, A-pair)

Original source details: GELFAND-P000795

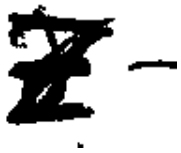
Uncertain reading. The overwritten sign in the heading provisionally reads γ. λ is also possible.

$$\frac{\delta\varphi}{\delta x_\mu(s)} + Z\,\varepsilon_{\mu\nu\lambda} t_\nu \frac{\delta\varphi}{\delta x_\lambda} = \mathcal{F}_\mu(s, C)\varphi(s, C) \tag{31}$$

Original source details: GELFAND-P000795

Uncertain reading. The overwritten parameter resembles a barred Z. Its letter case is unclear.

Z — arbitrary parameter



Uncertain reading. The letter in the explanation has a similar shape. Its case is unclear.

Original source details: GELFAND-P000795

The following has an analog. form:

Std. chiral field: (Pohlmayer, Zakharov, Mikhailov):

$$(32) \qquad A_\mu = g^{-1}\partial_\mu g(x);$$
$$[\partial_\mu + Z\,\varepsilon_{\mu\nu}\partial_\nu]\,\varphi = A_\mu\varphi, \qquad \mu = 1, 2, \ldots$$

Uncertain reading. The letter before ε provisionally reads Z. Its case is unclear.

Original source details: GELFAND-P000795

GELFAND-P000796

[source number: 19]

(31) — a system of eqns., Z — spectral param.

Uncertain reading. The letter in the explanation is retraced and resembles a barred Z. Its case is unclear.

Original source details: GELFAND-P000796

Compatibility of $(31) \Rightarrow$ eqns. (22).

$$Z = 0 \quad \Rightarrow \quad \varphi = \Psi$$

Uncertain reading. The letter before $= 0$ is retraced and resembles a barred Z. In the following formulas the letter is written smaller.

Original source details: GELFAND-P000796

Integrals — ?

In model (32) let us see how the integrals are obtained.

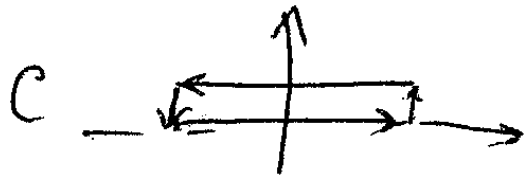
$$(32) \Leftrightarrow$$

$$(33) \qquad \partial_\mu \varphi = \frac{A_\mu + z\varepsilon_{\mu\nu}A_\nu}{1-z^2}\,\varphi$$

$$(34) \qquad P\exp \oint_C \frac{A_\mu + z\varepsilon_{\mu\nu}A_\nu}{1-z^2}\,dx^\mu = 1$$

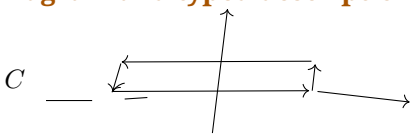
(ord. expon)

Spec. choice of contour:



Original source details: GELFAND-P000796

Diagram and typed description.



Additional strokes appear at the lower left corner. Their purpose is unclear.

$$(35) \qquad \frac{d}{dt}\,P\exp \int_{-\infty}^{\infty} dx_1 \left\{\frac{A_1 + zA_2}{1-z^2}\right\} = 0$$

This is the generating fn. of the integrals of motion.

The usual relations carry over to “contourology.”

There are Feynman diagrs. with surfaces in place of lines.

For free field theory

Original source details: GELFAND-P000796

Diagram and typed description.

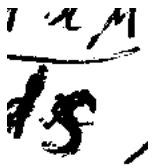
the Green fn. $G(x,x')$ satisfies the K.–G. eqn.

$$(36) \qquad (\partial_x^2 - m^2)G(x,x') = \delta(x-x')$$

$$(37) \qquad G(x,x') \underset{|x-x'|\to\infty}{\sim} e^{-m|x-x'|}$$

Here on contours

$$(38) \qquad \left\{\frac{\delta^2}{\delta x_\mu^2(s)} - m^4\left(\frac{dx_\mu}{ds}\right)^2\right\} G(C) = 0$$



Original source details: GELFAND-P000796

Uncertain reading. The overwritten denominator provisionally reads ds.

$$(39) \qquad G(C) = \langle \Psi(C) \rangle$$

How does one find the contour Green fn.?

Math. problem: (38) — the "string" K.–G. eqn.
Boundary cond.: decay for large contours:

$$(40) \qquad G(C) \sim e^{-m^2 A(C)}$$

($A(C)$ — min. area spanning the contour C)

GELFAND-P000797

Let G depend only on the tensors

$$(41) \qquad \sigma_{\alpha\beta} = \oint_C x_\alpha \, dx_\beta$$

$$(42) \qquad \frac{\delta \sigma_{\alpha\beta}}{\delta x_\mu(s)} = \delta_{\alpha\mu} \frac{dx_\beta}{ds} - \delta_{\beta\mu} \frac{dx_\alpha}{ds}$$

$$t_\mu(s) = \frac{dx_\mu}{ds}$$

$$(43) \qquad \dddot{\dots} \, t_\alpha t_\beta \left[\frac{\partial^2}{\partial \sigma_{\alpha\gamma} \partial \sigma_{\beta\gamma}} - m^4 \delta_{\alpha\beta} \right] G = 0$$

Uncertain reading. The crossed-out entry before $t_\alpha t_\beta$ resembles a summation sign with subscripts.

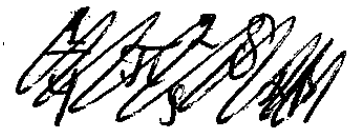
Original source details: GELFAND-P000797

Let

$$\pi_{\alpha\beta} = \frac{\partial}{\partial \sigma_{\alpha\beta}}$$

Then we obtain

$$(44) \qquad \pi_{\alpha\beta}\pi_{\beta\gamma} = \frac{1}{4}\pi^2 \delta_{\dots} \quad C\delta_{\alpha\beta};$$



Uncertain reading. The crossed-out signs after the equals sign resemble $\frac{1}{4}$, π^2 and δ. Their subscripts are unclear.

Original source details: GELFAND-P000797

Uncertain reading. The subscripts of the uncrossed δ on the right read $\alpha\beta$.

Original source details: GELFAND-P000797

This is possible only when

$$(45) \qquad \pi_{\alpha\beta} = \pm {}^*\pi_{\alpha\beta} = \pm \varepsilon_{\alpha\beta\gamma\delta}\pi_{\gamma\delta}$$

Hence a fn. only of

$$(46) \qquad U_\pm = \sqrt{\sigma_{\alpha\beta}^2 \pm \sigma_{\alpha\beta}\, {}^*\sigma_{\alpha\beta}}$$

Explicit solution:

$$G(U_+, U_-)$$

expressed via Bessel fns.

[between the lines:] Look at infinity:

Min. area

$$(47) \qquad A_{\min}(C) = \sqrt{\sigma^2 \pm \sigma\, {}^*\sigma}$$

σ — projection onto coord. planes

$$\sigma_{\alpha\beta} = \oint x_\alpha \, dx_\beta$$

$$A = \int \sqrt{\dot{x}^2 x'^2 - (\dot{x}x')^2}\, dt\, ds$$

The Euler eqn. gives

$$A_{\min}(x(s))$$

Uncertain reading. The signs of the argument on the last line are retraced. They provisionally read $x(s)$.

Original source details: GELFAND-P000797

GELFAND-P000798

[source number: 20]

Ham.–Jacobi eqn.

$$\left(\frac{\delta A_{\min}}{\delta x_\mu(s)}\right)^2 = \left(\frac{dx_\mu}{ds}\right)^2$$

We find a solution. (47) — satisfies the Ham.–Jacobi eqn.

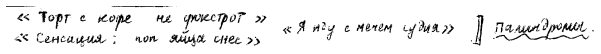
For contours that are not too [...] twisted, (47) — a solution.

Undeciphered passage. A short entry between “too” and “twisted” is crossed out. The letters are indistinct.

Original source details: GELFAND-P000798

Sinai asserts that Courant has something.

20.11.78. Gelfand Sem.



Original source details: GELFAND-P000799

Diagram and typed description.
«Торт с кофе не фокстрот»
«Сенсация: поп яйца снес»
«Я иду с мечем судия»
Palindromes.

Cake with coffee is no foxtrot. Sensation: a priest laid eggs. I go with a sword, O judge.

Krichever: instantons without bundles.

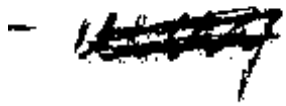
A. S. Schwarz.

1. Quantum [inserted above the line: field theory and] stat. phys. — the science of computing the integral (partition function)

$$Z = \int e^{-S(A)}\, dA \tag{1}$$

A ranges over the space of trajectories or histories: $A = A(x, t)$.

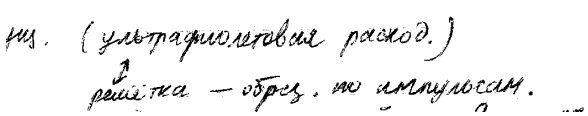
Meaning: we replace it by a finite-dimensional one (lattice approx.) and pass to the limit; a — [...] lattice spacing.



Original source details: GELFAND-P000799

Uncertain reading. A short word before “lattice spacing” is crossed out. It may read «шаг» (“step”).

QFT: the limit as $a \to 0$ does not exist.



Original source details: GELFAND-P000799

Diagram and typed description.
(ultraviolet divergences)
↙
lattice — momentum cutoff.

One way out is to take the limit for correl. fns., i.e. to compute

$$\frac{\int \varphi(A) e^{-S(A)}\, dA}{\int e^{-S(A)}\, dA} \tag{2}$$

(Green fns. or correlation fns.).

Important: the limit of the ratio already makes sense.

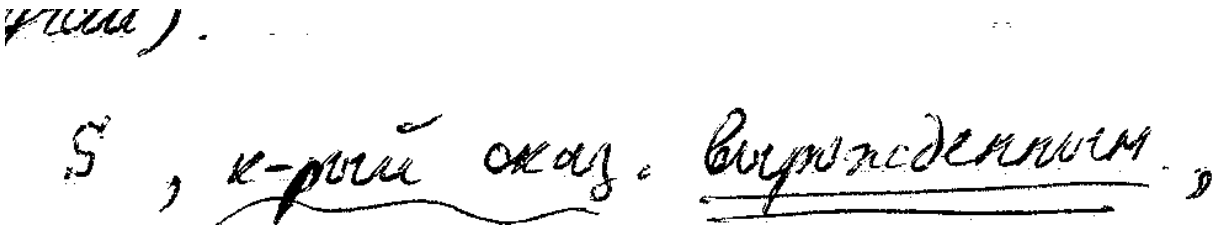
Also: when removing the cutoff one must do “renormalization”, i.e. the parameter λ_a must be varied as a function of a when passing to the limit.

There are 2 methods for computing cont. integrals:

perturbation theory and semiclassics

(in fact the 1st is a special case).

An important class of functionals S, which turns out to be degenerate, i.e. admitting an ∞-dimensional symmetry group:



Original source details: GELFAND-P000799

Uncertain reading. The abbreviation before «вырожденным» (“degenerate”) reads «оказ.». A faint dot is visible before the comma after the word.

$$g \in G \quad \rightsquigarrow \quad A \mapsto A^g$$
$$S(A^g) = S(A) \tag{3}$$

Physical examples: gauge theories.

Pure Y.–M. theory; $A = (A_\mu(x))$, $A_\mu \in$ Lie alg. of a finite-dimensional Lie group.

$$S(A) = \int \langle F_{\mu\nu}, F^{\mu\nu} \rangle\, dV, \tag{4}$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \tag{5}$$

Original source details: GELFAND-P000799

Uncertain reading. The subscript of the second ∂ is heavily retraced. It provisionally reads ν.

GELFAND-P000800

[source number: 26]

Symmetry group:
if $g(x)$ — a fn. with values in the Lie group, then it generates a transformation of the fields

$$A_\mu \mapsto g A_\mu g^{-1} + g\,\partial_\mu(g^{-1}) \tag{6}$$

[crossed out above the next line: ...]

Original source details: GELFAND-P000800

Undeciphered passage. A short entry above «Способ» (“Method”) is obscured by heavy hatching.

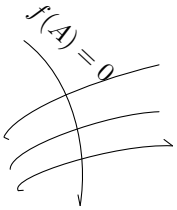
Method for dealing with degeneracy. In the case of the electromagn. field.
“The potentials have no meaning”, let us impose a gauge condition

$$\partial_\mu A_\mu = 0 \tag{7}$$

Difficulties: it is hard to construct a perturbation theory.
2. Semiclassics $\sim$ stationary phase or Laplace method.
a) A problem solved long ago: how to write the cont. integral (Faddeev and Popov)
[below the line:] — at the physicist’s level of rigor.
b) Math. theory — when S is quadratic.
c) Application to computing the instanton contribution.
d) Ray–Singer invariants.
a) Quantization of degen. Lagrangians.
Difficulty: the integral diverges, since the volume of the orbits is infinite.
moral: choose a gauge condition — a transversal to the orbits — and integrate over it.

Original source details: GELFAND-P000800

Diagram and typed description.



Angled strokes are visible at the right end of the lowest curve and the bottom of the transverse curve. The redraw shows the visible sides of those strokes.

but: one must insert the factor $\delta(f(A))$.

In addition, one must insert the Jacobian $\Delta_f(A)$

$$\Delta_f(A)\delta(f(A)) \tag{8}$$

The F.–P. method is to compute $\Delta_f(A)$. (he does not go into this)

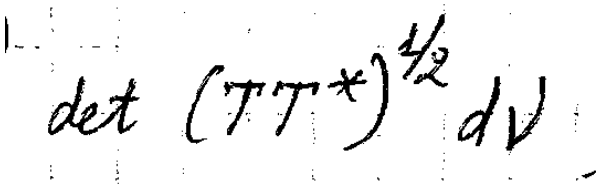
$f(A) = 0$ — eqn. of the transversal.

For finite-fold integrals, here — a theorem.

[added on the left:] Another method.

Let us consider the integral again (in the finite-fold case)

$$\int_M e^{-S(A)}\, dA = \int_{M/G} e^{-S(A)} \det(TT^*)^{1/2}\, dV, \tag{9}$$



Uncertain reading. Equation (9) reads $\det(TT^*)^{1/2}$. No prime is visible on det.

Original source details: GELFAND-P000800

where dV — the measure on the set of orbits, obtained from the std. Riem. metric on it.

T — the operator obtained as follows: take (assuming the action free):

$$T: \underset{\text{Lie alg.}}{\mathfrak{g}} \xrightarrow{T_x} T_x(M)$$

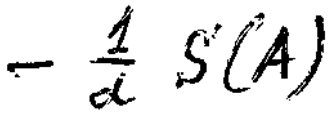
[on the right:] map obtained from the group action.

Assume that G acts by isometries on M.

GELFAND-P000801

In physics the problem arises of computing by Laplace's method, i.e. of finding the asymptotics of

$$\int e^{-\frac{1}{\alpha}S'(A)}\, dA \qquad \text{as } \alpha \to 0 \tag{10}$$



Uncertain reading. A separate short stroke follows S in the exponent of (10). It is provisionally read as S'.

Original source details: GELFAND-P000801

One needs to know the stat. points of $S'(A)$, called instantons.

Uncertain reading. The short stroke before the opening parenthesis is provisionally read as a prime on S.

Original source details: GELFAND-P000801

The math. problem is as follows. There is a set of stat. points (even several manifolds), and even ∞-dimensional ones. There is also strong degeneracy.

Examples:

1) Y.–M. field. $G = SU(2)$

$\forall q$ there is a family of $8q-3$ instantons apart from the ∞-dimensional symmetry.

$$2) \qquad S(\vec{n}) = \frac{1}{f_0}\int (\partial_\mu \vec{n})^2 \dots d^2x, \quad \mu = 0,1; \quad x = (x_0, x_1) \tag{11}$$

Undeciphered passage. An entry between $(\partial_\mu \vec{n})^2$ and d^2x is heavily crossed out. A loop is visible at the left. The entry itself is illegible.

Original source details: GELFAND-P000801

One has to compute Gaussian integrals, and there arises det.

A degen. Lagrangian gives one more det — from Faddeev–Popov — it is $\det(TT^*)^{1/2}$.

The determinants are ∞-dimensional. They require analytic continuation or renormalization.

And they turn out to depend on the choice of metric.

But one can trace the change in det under variation of the metric.

In (11): $\vec{n} = (n_1, n_2, n_3) \in S^2, \quad |\vec{n}| = 1.$

The extremals in (11) were found by Belavin and Polyakov.

Let us pass to stereogr. variables on the plane — to the sphere.

We obtain $w(z)$.

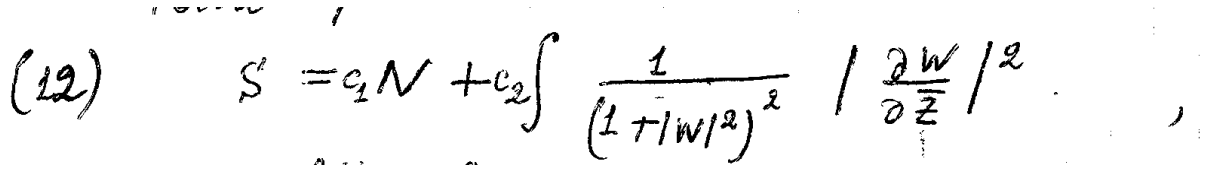
Answer: extremals $\leftrightarrow w(z)$ — rational fns. of z!

Explanation: there is a top. number N — the degree of the map

($\sim$ integral of the Jacobian).

Then a simple calculation shows

$$S' = c_1 N + c_2 \int \frac{1}{(1+|w|^2)^2} \left|\frac{\partial w}{\partial \bar{z}}\right|^2, \tag{12}$$



Original source details: GELFAND-P000801

Uncertain reading. A short stroke is visible beside S on the left of (12). It is provisionally read as S'. No differential is visible after the squared modulus.

from which everything follows.

One must compute det.

There is no gauge group here and there is one det. The asympt. of (10) is given by the

integral

$$\int \det{}^{-1/2}(\dots)\, d\mu_0, \tag{13}$$

where μ_0 — some measure on instantons.

~~It turns out to be convenient~~

The operator is constructed from $\forall$ rational fn. as the [inserted above the line: op. of the quadr. form corresp. to] the principal quadr. part of the f'nal

$$z = x_0 + ix_1$$

$$z \xrightarrow{w} \mathrm{CP}(1) \tag{14}$$

GELFAND-P000802

[source number: 27]

$T\mathrm{CP}(1)$ induces a bundle on the z-plane (by this map).

[below $T\mathrm{CP}(1)$:] (holomorphic)

[on the right:] w — rat'l.

$$\Delta = \bar\partial^* \bar\partial, \qquad \bar\partial = \frac{\partial}{\partial \bar z} : \Lambda^{0,0} \to \Lambda^{0,1} \tag{15}$$

[under the right-hand part:] on forms with values in the bundle.

The metric also depends on w.

(induced by the std. metric of CP^1)

w actually enters only in the $*$.

One has to compute det. This is done using the general theory.

[crossed-out block: ...]

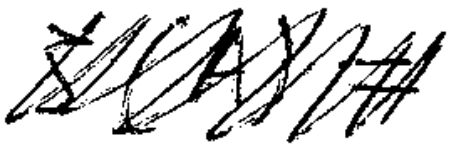
Undeciphered passage. Dense horizontal and diagonal strokes fill the outlined block. The underlying entry is illegible.

Original source details: GELFAND-P000802

det is regularized, simultaneously renormalizing f_0 (coupling const.) in (11).

One way to regularize is to consider

[crossed out: ~~$\zeta(A) =$~~]



Uncertain reading. The crossed-out entry is provisionally read as $\zeta(A) =$. Cancellation strokes intersect its ending.

Original source details: GELFAND-P000802

$$\zeta(s) = \sum_{\lambda_i \neq 0} \lambda_i^{-s} \tag{16}$$

$s = 0$ — point of analyticity.

$$\ln\det \underset{\text{def}}{=} -\zeta'(0). \tag{17}$$

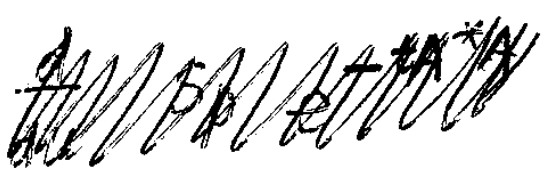
The op. $\bar{\partial}$ is “one and the same” for all instantons with the same top. number N.

General statement: — when the metric changes, one can trace the change in

$$\det(A^*A)$$

idea:

[$\frac{d}{du}\operatorname{Sp} e^{-tA^*A}$]



Uncertain reading. The crossed-out entry is provisionally read as $\frac{d}{du}\operatorname{Sp} e^{-tA^*A}$. Diagonal strokes intersect the exponent and denominator.

Original source details: GELFAND-P000802

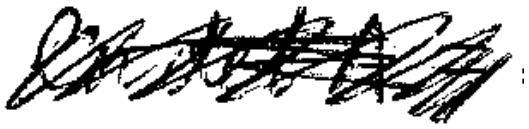
$$\frac{d}{du}A^*A = BA^*A + A^*CA \tag{18}$$

Here B arises from the formula giving the variation of the scalar product:

$$(u, v) = \langle u, v\rangle + \langle Bu, v\rangle. \tag{19}$$

The variation of $\det A^*A$ will be expressed in terms of $\operatorname{Sp} e^{-tA^*A}$ using the Frullani integral.

[crossed out above the next formula: ln det A =]



Uncertain reading. The crossed-out line is provisionally read as $\ln\det A =$. The final strokes are obscured.

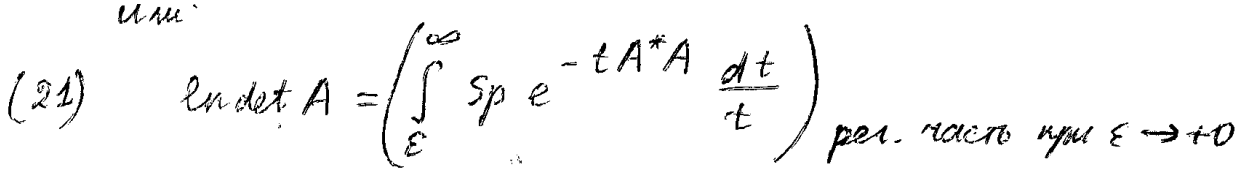
Original source details: GELFAND-P000802

$$\int_0^\infty \frac{e^{-ta} - e^{-tb}}{t}\, dt = \ln\frac{b}{a} \tag{20}$$

(Frullani integral).

or:

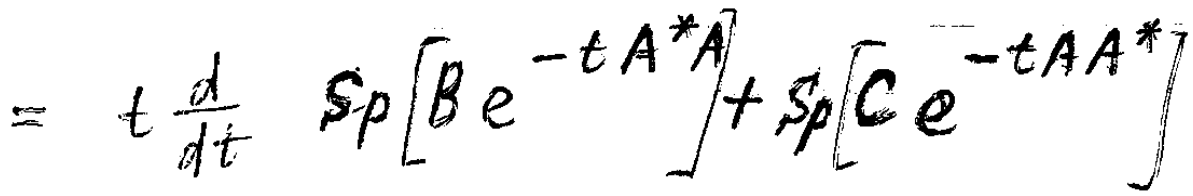
$$\ln\det A = \left(\int_\varepsilon^\infty \operatorname{Sp} e^{-tA^*A}\frac{dt}{t}\right)_{\text{reg. part as } \varepsilon\to+0} \tag{21}$$



Original source details: GELFAND-P000802

Uncertain reading. The left side of (21) reads ln det A. No minus sign or numerical factor is visible before the integral.

$$\begin{aligned}\frac{d}{du}\operatorname{Sp} e^{-tA^*A} &= -t\operatorname{Sp}\big[(BA^*A + A^*CA)e^{-tA^*A}\big] = \\ &= t\frac{d}{dt}\operatorname{Sp}\big[Be^{-tA^*A}\big] + \operatorname{Sp}\big[Ce^{-tAA^*}\big]\end{aligned} \tag{22}$$



Original source details: GELFAND-P000803

Uncertain reading. In the second line of (22) the bracket after the trace with B closes before the plus sign. The trace with C is written separately, with no visible $t\,d/dt$.

t — cancels with $\frac{1}{t}$ in (21)

and integrals come out, the corresp. coeffs. of the Seeley asympt.

$$\ln\det A = \int_\varepsilon^\infty \operatorname{Sp} e^{-tA^*A}\frac{dt}{t} + \sum a_i\varepsilon^i + a_0\ln\varepsilon.$$

What matters is a_0.

$t\to\infty \quad\Rightarrow\quad$ zero modes.

this part is left unevaluated.

One has to compute the integral with respect to $d\mu_0$.

$d\mu_0$ has to be computed too! — it is an n-dimensional det

One has to compute the $(4q+2)$-dimensional det of the metric tensor.

The metric on rat'l fns. comes from the std. metric on all fns. (in their tangent space) $=\sum$ squares of variations.

In fact the **2** unevaluated parts (from the modes and from $d\mu_0$) cancel!!

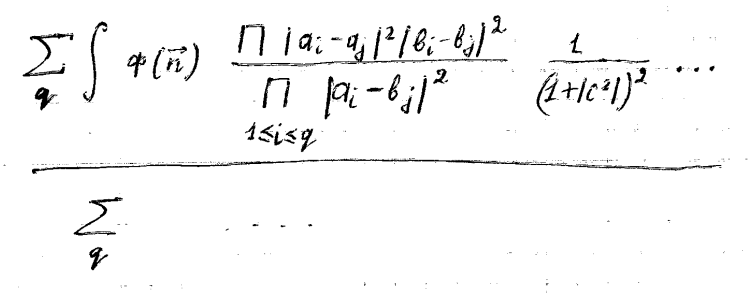
One still has to compute

$$\int \Phi(\vec n)\,\overset{-1/2}{\det}(\dots)\,d\mu_0.$$

Specify a rat'l fn. by its zeros and poles.

$$\begin{array}{l|l} a_1\dots a_n & \text{— zeros} \\ b_1\dots b_n & \text{— poles} \end{array} \quad c \text{ — asympt. at } \infty.$$

$$\frac{\sum_q \int \varphi(\vec n)\dfrac{\prod|a_i-a_j|^2|b_i-b_j|^2}{\prod_{1\le i\le q}|a_i-b_j|^2}\dfrac{1}{(1+|c^2|)^2}\cdots}{\sum_q \cdots}$$



Original source details: GELFAND-P000803

Uncertain reading. The upper product has no written limits. The lower one has $1\le i\le q$ beneath it, while both i and j occur in the factor. The exponent in $|c^2|$ is inside the modulus bars.

$$\varphi(\vec{n}) = n_\alpha(z')n_\beta(z'')$$

The denominator is the partition function of a Coulomb gas in the plane.

At poles — $+$ charges

at zeros — $-$ charges.

[on the right:]

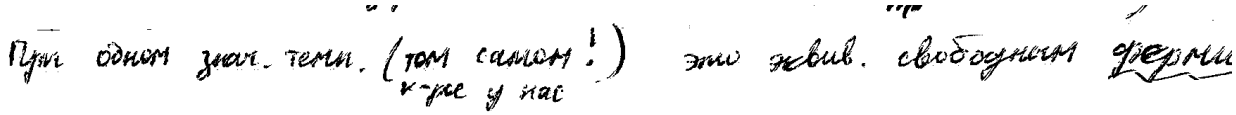
$$\left(\prod_{\dots} = \dots e^{\sum \ln \dots}\right)$$

Undeciphered passage. The sign before e is heavily crossed out. Angled strokes remain visible and the sign is illegible.

Original source details: GELFAND-P000803

At one value of the temp. (that very one!) this is equiv. to free fermions.

[inserted below: the one we have]



Original source details: GELFAND-P000803

Uncertain reading. The abbreviation in the note below the parenthesis is provisionally read as «к-рое» (“which”, referring to the value).

GELFAND-P000804

27.11.78. Gelfand Sem.

Lev Nikolaevich Lipatov

on the asympt. of diagrams.

Zakharov.

For integrable systems “in general position” the var. principle will hold.

(all the known ones are out of general position)

$$\begin{cases} \Psi_\xi = U\Psi \\ \Psi_\eta = V\Psi \end{cases} \tag{1}$$

$$\left.\begin{aligned} U &= U(\xi,\eta,\lambda) \\ V &= V(\xi,\eta,\lambda) \end{aligned}\right\} \quad \text{rat'l fns. in } \lambda$$

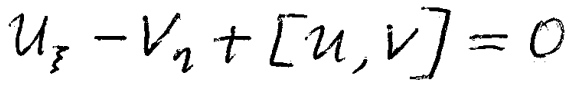
$$\begin{cases} U = U_0 + \sum\limits_{n=1}^{N_1} \dfrac{U_n}{\lambda - \lambda_n}, \\ V = V_0 + \sum\limits_{n=1}^{N_2} \dfrac{V_n}{\lambda - \mu_n} \end{cases} \tag{2}$$

$$\lambda_i \neq \lambda_j; \qquad \mu_i \neq \mu_j; \qquad \lambda_n \neq \mu_m$$

Ψ, U, V — matrices

Int. system:

$$U_\xi - V_\eta + [U,V] = 0 \tag{3}$$



Uncertain reading. In (3) the subscript on U is ξ, and the one on V is η.

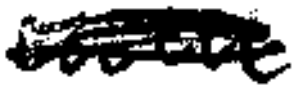
Original source details: GELFAND-P000804

Lax pairs: U has a simple pole at the 1^{st} point, V has a pole of any order at that same point

GELFAND-P000805

[source number: 37]

[crossed out at the start of the line: ...]



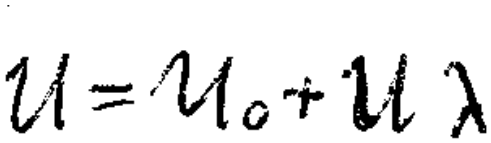
Undeciphered passage. Several horizontal strokes cover a short word. Its letters are indistinct.

Original source details: GELFAND-P000805

Then U, V can be made into polynomials:

$$U = U_0 + U\lambda$$

$$V = P_n(\lambda)$$



Uncertain reading. No subscript is visible on the U before λ.

Original source details: GELFAND-P000805

In more detail, (3) has the form:

$$\begin{cases} U_{0\eta} - V_{0\xi} + [U_0, V_0] = 0 \\ U_{n\eta} = [U_n, A_n] \\ V_{n\xi} = [V_n, B_n] \end{cases} \tag{4}$$

Here

$$A_n = V\big|_{\lambda=\lambda_n}, \qquad B_n = U\big|_{\lambda=\mu_n}. \tag{5}$$

This is a syst. of $N_1 + N_2 + 1$ eqns. for $N_1 + N_2 + 2$ quantities.

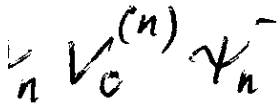
Let us integrate partially. Let $U_n^{(0)}$ and $V_n^{(0)}$ — Jordan forms, then

$$U_n^{(0)} = U_n^{(0)}(\xi), \qquad V_n^{(0)} = V_n^{(0)}(\eta) \qquad \text{(by } (4)_{2,3}\text{)} \tag{6}$$

$$\begin{cases} U_n = \varphi_n U_n^{(0)} \varphi_n^{-1}, \\ V_n = \psi_n V_0^{(n)} \psi_n^{-1} \end{cases} \tag{7}$$

Uncertain reading. The heavily retraced number provisionally reads (7). Earlier strokes within the digit are indistinct.

Original source details: GELFAND-P000805



Uncertain reading. The middle V on the second line of (7) has subscript 0 and superscript (n).

Original source details: GELFAND-P000805

We take the matrices in (6) as given.
The syst. will be integrated if we find φ_n and ψ_n in (7).
φ_n, ψ_n are determined nonuniquely. One representative is needed.
Set:

$$\nabla_\eta = \frac{\partial}{\partial \eta} - V_0, \qquad \nabla_\xi = \frac{\partial}{\partial \xi} - U_0. \tag{8}$$

Then the eqns. take the form

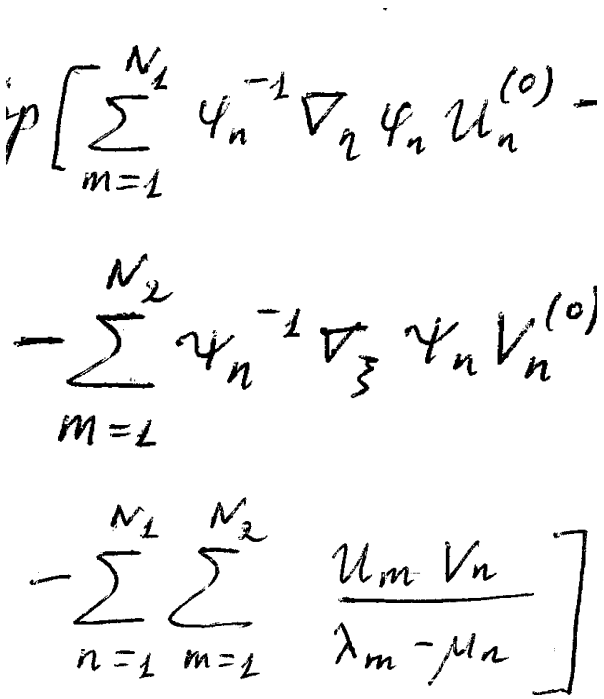
$$\begin{cases} \left(\nabla_\eta - \sum\limits_{m=1}^{N_2} \dfrac{V_m}{\lambda_n - \mu_m} \right) \varphi_n = 0 \\ \left(\nabla_\xi - \sum\limits_{m=1}^{N_1} \dfrac{U_m}{\mu_n - \lambda_m} \right) \psi_n = 0 \end{cases} \tag{9}$$

[on the right:] (it is understood that V_m, U_m are expr. by (7))
Here we have class. spinor models (for physicists). Action:

$$S' = \int d\xi \, d\eta \ \mathrm{Sp} \Big[\sum_{m=1}^{N_1} \varphi_n^{-1} \nabla_\eta \varphi_n U_n^{(0)} - \sum_{m=1}^{N_2} \psi_n^{-1} \nabla_\xi \psi_n V_n^{(0)} = - \sum_{n=1}^{N_1} \sum_{m=1}^{N_2} \frac{U_m V_n}{\lambda_m - \mu_n} \Big] \tag{10}$$

Uncertain reading. A raised stroke beside S in (10) is provisionally read as a prime, S'.

Original source details: GELFAND-P000805



Uncertain reading. The first two sums start at $m = 1$, while the factors use index n. The double sum has $n = 1$ beneath N_1 and $m = 1$ beneath N_2. The fraction reads $U_m V_n / (\lambda_m - \mu_n)$.

Original source details: GELFAND-P000805

Uncertain reading. Two strokes at the end of the first line are provisionally read as =. The next line begins with a minus sign.

Original source details: GELFAND-P000805

GELFAND-P000806

Hence series of conservation laws (related to the number of poles)

$N_1 + N_2$ series.

Simplest syst. — sine–Gord. — poles at 0 and ∞.

Conjecture: there are many Ham. structures.

(as for KdV — an infinite number of Ham. structures)

Lipatov

ZhETF 72, 411, 1977.

$$\alpha = \frac{e^2}{4\pi\hbar c} = \frac{1}{137}$$

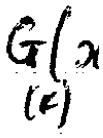
$$\mathcal{L} = \int d^3x\, dt \left[\frac{[\partial_\mu \varphi]^2}{2} - \frac{m_0^2 \varphi^2}{2} - \lambda_0 \frac{\varphi^4}{4!} \right] \tag{1}$$

Problem: [...] computing Green fns. (in the Euclidean region):

Undeciphered passage. A heavily inked sign follows «Задача» (“Problem”). Its form is illegible.

Original source details: GELFAND-P000806

$$G_{(c)}(x_1, \ldots, x_n) = Z_0^{-1} \int \prod_x d\varphi(x)\, e^{-H(\varphi)} \prod_{i=1}^{n} \varphi(x_i) \tag{2}$$



Uncertain reading. The parenthesized subscript is provisionally read as (c). It has an angled stroke open at the right.

Original source details: GELFAND-P000806

$$H(\varphi) = \int d^4x \left[\frac{(\partial_\sigma \varphi)^2}{2} + \frac{\lambda_0 \varphi^4}{4!} \right] + \frac{m_0^2 \varphi^2}{2}$$

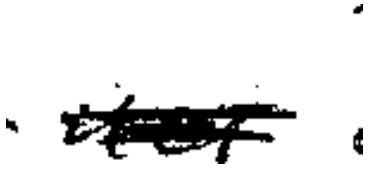
Uncertain reading. The closing square bracket after the fraction with λ_0 is crossed out. A further bracket stands after the final summand.

Original source details: GELFAND-P000806

(= $\mathcal{L}$ in the Euclidean region)

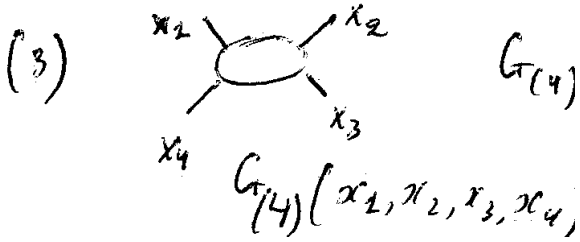
Integral (2) diverges. But if we make λ_0, m_0 dep. on the cutoff parameters, the answer will be finite (renormalizations!)

m_c^2 — pole of the Green fn. (m_c — [...] phys. mass).



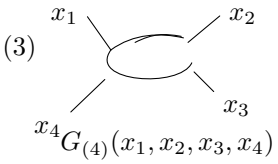
Original source details: GELFAND-P000806

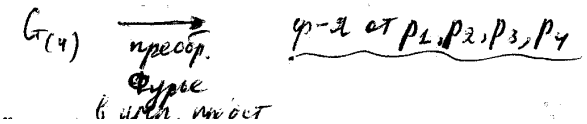
Undeciphered passage. A short entry before "phys. mass" is crossed by several strokes. Its letters are indistinct.



Original source details: GELFAND-P000806

Diagram and typed description.





Original source details: GELFAND-P000806

Diagram and typed description.

$G_{(4)} \xrightarrow[\text{transf.}]{\text{Fourier}}$ fn. of p_1, p_2, p_3, p_4
in mom. space

(∼ scattering of 4 particles)

Pole = [...] m_c^2

Original source details: GELFAND-P000806

Uncertain reading. The whole crossed-out expression before m_c^2 is illegible. Its initial strokes may belong to m_0.

fn. (3) depends only on p_j^2 and $p_i \cdot p_j$ (by Lorentz invariance)

and contains $\delta(p_1 + p_2 - p_3 - p_4)$ (by momentum conservation)

Vertex Green fn.:

$$\Gamma = \frac{G(p_1, p_2, p_3, p_4)}{G(p_1)G(p_2)\dots G(p_4)} \tag{4}$$

(after factoring out the δ-fn.)

By perturbation theory (with $\lambda_0 = 0$) we obtain

$$G^0(p) = \frac{1}{p^2 + m^2}. \tag{5}$$

GELFAND-P000807

[source number: 38]

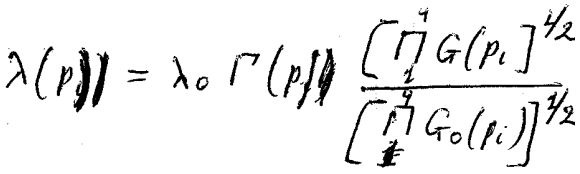
[crossed out: ...]

$$\frac{[G(p_1)\dots G(p_4)]^{1/2}}{[G^0(p_1)\quad G^0(p_4)]^{1/2}}$$

Undeciphered passage. A heavily crossed-out entry before the upper fraction may read $\Gamma(p)$. The strokes inside the parentheses are indistinct.

Original source details: GELFAND-P000807

$$\lambda(p_i)) = \lambda_0\Gamma(p_i))\frac{[\prod_1^4 G(p_i]^{1/2}}{[\prod_1^4 G_0(p_i)]^{1/2}} \quad \text{— charge (invariant)}$$



Uncertain reading. The arguments of λ and Γ are provisionally read as p_i with two closing-parenthesis strokes. In the numerator a square bracket follows p_i with no separate closing parenthesis visible.

Original source details: GELFAND-P000807

$$p = (p_1, p_2, p_3, p_4)$$

λ_0 is chosen so that the phys. charge is finite (and all the Green fns. too)

The paradox of Landau and Pomeranchuk: if we set perturb. theory aside and keep the leading ln, then we get

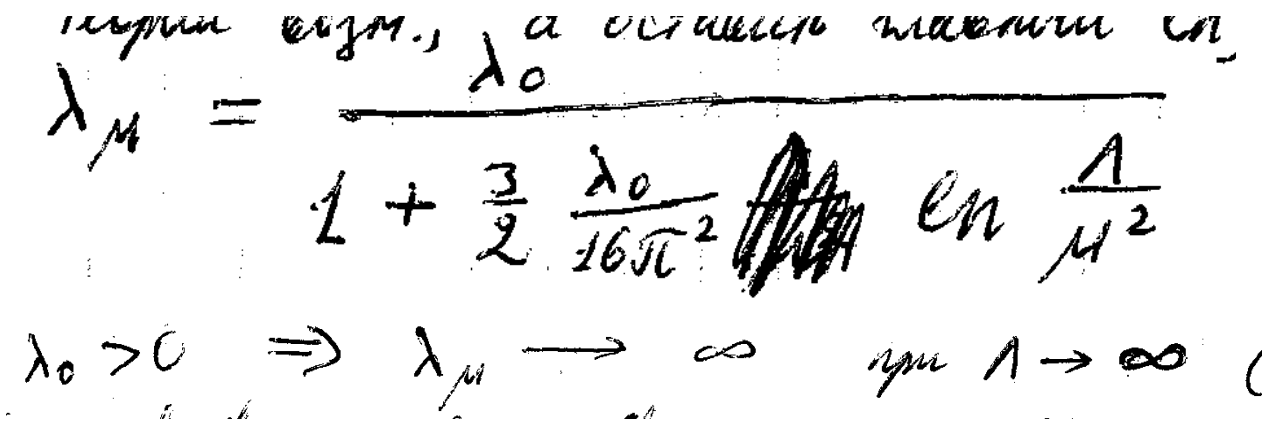
$$\lambda_\mu = \frac{\lambda_0}{1 + \frac{3}{2}\frac{\lambda_0}{16\pi^2} \dots \ln\frac{\Lambda}{\mu^2}}.$$

Undeciphered passage. The entry before ln is heavily crossed out. Its symbols are indistinct.

Original source details: GELFAND-P000807

[to the right of the formula:] μ — defining point: $p_i p_j = \mu^2$, $p_i^2 = (\dots)\mu^2$.
$(\cdot)$ from $(\sum p_i)^2 = 0$.

$$\lambda_0 > 0 \quad \Rightarrow \quad \lambda_\mu \longrightarrow \infty \quad \text{as } \Lambda \longrightarrow \infty \quad \text{(i.e. the cutoff is removed).}$$



Original source details: GELFAND-P000807

Uncertain reading. The fraction has a plus sign after 1. Below, $\lambda_\mu \to \infty$ is written for $\Lambda \to \infty$.

The same in quant. electrodyn. [~~...~~] (for no bare λ_0 does one obtain $\frac{1}{137}$)

Undeciphered passage. Diagonal strokes cross out an entry between "electrodyn." and the opening parenthesis. It is illegible.

Original source details: GELFAND-P000807

One needs the asymptotics of the coeffs. of the perturbation series arising for

$$\psi = \frac{d\lambda_\mu}{d \ln \frac{\Lambda}{\mu^2}} = a_2 \lambda_\mu^2 + a_3 \lambda_\mu^3 + \dots$$

the asympt. in k as $k \to +\infty$ exists
The first 4 have been found explicitly.
Idea: Laplace's method for the continuum integral
Sobolev inequality:

$$\left[\int (\partial_\sigma \varphi)^2 \, d^4x \right]^2 \geq \frac{32\pi^2}{3} \int \varphi^4(x) \, d^4x$$

equality is attained at

$$\varphi = \frac{c_1}{(x - x_0)^2 + \lambda^2}$$

[Zakharov's insertion $\Delta\varphi + \varphi^3 = 0$ — obtained from the Y.–M. eqns.]

GELFAND-P000808

4.12.78 Gelfand Sem.

A. Guichardet

Extensions of reps. of top. groups.

1. Exts. of modules over discr. groups.

E, F — 2 G modules.

Uncertain reading. The heavily retraced digit between the dash and G provisionally reads 2.

Original source details: GELFAND-P000808

An ext. of E by F — is a third G-module H, such that $H \supset F$ and $H/F \cong E$ or an exact seq.

$$0 \longrightarrow F \longrightarrow H \longrightarrow E \longrightarrow 0 \tag{1}$$

2 exts. are equiv. if $\exists$ an isom. such that the diag. commutes.

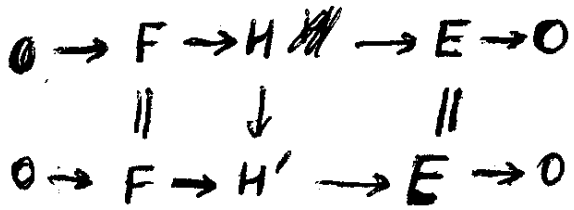


Diagram and typed description.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F & \longrightarrow & H & \cdots\!\!\longrightarrow & E & \longrightarrow & 0 \\ & & \| & & \downarrow & & \| & & \\ 0 & \longrightarrow & F & \longrightarrow & H' & \longrightarrow & E & \longrightarrow & 0 \end{array}$$

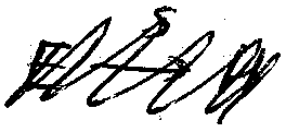
An entry immediately to the right of the upper H is crossed out. It may be another H.

Original source details: GELFAND-P000808

Trivial extension:

$$H = F \oplus E$$

Strong extension: (1) — is strong if $\exists$ a section [~~$E \overset{s}{\to} H$~~]



Uncertain reading. The crossed-out arrow is provisionally read as $E \overset{s}{\to} H$. Diagonal strokes cross the right-hand letter.

Original source details: GELFAND-P000808

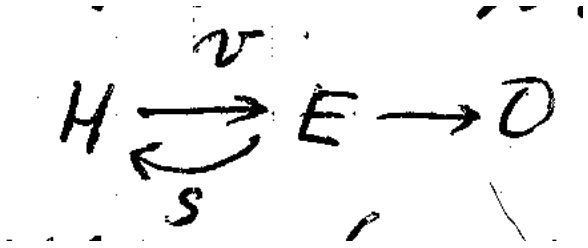
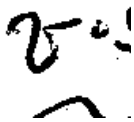


Diagram and typed description.

$$H \underset{s}{\overset{v}{\rightleftarrows}} E \longrightarrow 0$$

Original source details: GELFAND-P000808

$v \cdot s =$ id; s — a morphism of groups (and not of G-modules)



Original source details: GELFAND-P000808

Uncertain reading. The sign in $v \cdot s =$ id and over the arrow provisionally reads v.

[inserted below "groups": (abelian).]

(if s — a morph. of G-mods., the extension is trivial)

Set of equiv. classes of [inserted above the line: strong] extensions: $\mathrm{Ext}^1(E, F)$

Method: cohomology

$\mathrm{Hom}(E, F)$ — morph. of groups

this is a G-module. The action is given by

$$gu = g \cdot u \cdot g^{-1}$$

GELFAND-P000809

[source number: 43]

Def.

$$H^1(G, \mathrm{Hom}(E, F))$$

1-cocycles

$$f : G \longrightarrow \mathrm{Hom}(E, F)$$

$$f(gg') = f(g) + g \circ f(g') \circ g^{-1}. \tag{2}$$

Original source details: GELFAND-P000809

Uncertain reading. The lower loop of the first g after the plus sign in (2) is retraced. It reads g, with an indistinct adjoining stroke.

Z^1 — all cocycles.

1-coboundaries (B^1).

$$\exists\, u \in \mathrm{Hom}(E,F): \quad f(g) = g \circ u \circ g^{-1} - u$$

$$H^1 \underset{\text{def}}{=} Z^1/B^1.$$

There is a correspondence (one-to-one)

$$H^1 \cong \mathrm{Ext}^1(E,F)$$

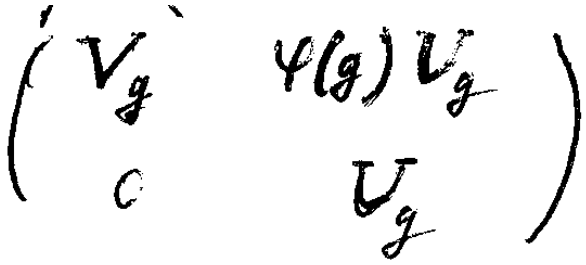
From the cocycle φ construct an extension:

$$H = F \oplus E$$

$$g(f,e) = (gf + \varphi(g)g \cdot e, \quad g \cdot e). \tag{3}$$

or in matrices:

$$\begin{pmatrix} V_g & \varphi(g)U_g \\ 0 & U_g \end{pmatrix}$$ — this is how the action is constructed.



Uncertain reading. The last letter at the top right is provisionally read as U_g, as on the diagonal. The lower-left sign is open at the right and provisionally reads 0.

Original source details: GELFAND-P000809

V_g, U_g — [...] reps. of G in the corresp. spaces.

Uncertain reading. An overwritten or crossed-out entry before “reps.” may be the numeral 2.

Original source details: GELFAND-P000809

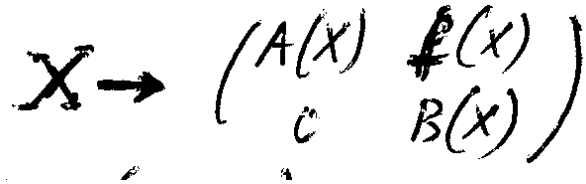
From this one can obtain the def. of a cocycle.

2 reps. equiv. $\Longleftrightarrow$ the cocycles are homologous

2. Extensions of modules over Lie algebras.

[on the right:]

$$X \longrightarrow \begin{pmatrix} A(X) & f(X) \\ 0 & B(X) \end{pmatrix}$$



Uncertain reading. The retraced letter at the top right provisionally reads f. The lower-left sign is open at the right and is rendered as 0.

Original source details: GELFAND-P000809

a cocycle is a linear map $f : \mathfrak{g} \longrightarrow \mathrm{Hom}(E,F)$

the def. of a cocycle has the form:

$$f([X,Y]) = X \circ f(Y) - f(Y) \circ X - Y \circ f(X) + f(X) \circ Y. \tag{4}$$

f — a coboundary if $\exists\, u$

$$f(X) = X \circ u - u \circ X. \tag{5}$$

Lemma 1. E, F have [inserted above the line: distinct] infinitesim. characters $\Rightarrow$

$$\Rightarrow \quad \mathrm{Ext}^1(E,F) = 0$$

$\mathfrak{g} \longmapsto U(\mathfrak{g})$ univ. enveloping algebra, $Z(\mathfrak{g})$ — its center

GELFAND-P000810

Let $u \in Z(\mathfrak{g})$ act in E by mult. by a scalar

$$u \cdot e = \chi(u)e$$

Then $\chi \in \mathrm{Hom}(Z(\mathfrak{g}), \mathbb{C})$

this is an infinitesim. character.

$H^1(\mathfrak{g}, \mathrm{Hom}(E,F))$ — is

$H^1(\mathfrak{g}, \mathfrak{h}, \mathrm{Hom}(E,F))$, $\mathfrak{h} \subset \mathfrak{g}$

[In the right margin: ....]

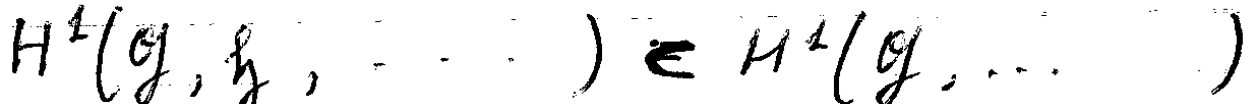
Uncertain reading. Isolated illegible mark

Original source details: GELFAND-P000810

(describes extensions trivial from the point of view of $\mathfrak{h}$).

$$Z^1(\mathfrak{g}, \mathfrak{h}, \mathrm{Hom}(E,F)) = \{f \in Z^1(\mathfrak{g}, \ldots), \quad f|_{\mathfrak{h}} = 0\}$$

$$H^1(\mathfrak{g}, \mathfrak{h}, \ldots) \in H^1(\mathfrak{g}, \ldots)$$



Original source details: GELFAND-P000810

Uncertain reading. The manuscript places the sign $\in$ between the groups.

Comments:

One can use the differential:

$$(df)(x,y) = f([x,y]) - xf(y) + yf(x)$$

$xf(y)$ — action Lie alg.

(above, written using the commutator)

3. Extensions of modules over Lie groups.

G-module — top. vect. space, G acts continuously.

$\mathfrak{g}$ — Lie alg.

$K \subset G$

|

max. comp. subgp.,

$\mathfrak{k}$ — corresp. alg.,

E, F — t.v.s.

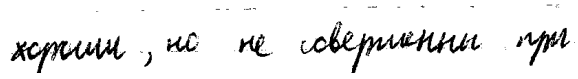
1st idea. cocycles

$$f : G \longrightarrow \mathrm{Hom}(E,F)$$

$(e,g) \longmapsto f(g)e$ cont. in g and e.

2nd idea: in place of E, F — the space of diff. vectors E_∞, F_∞

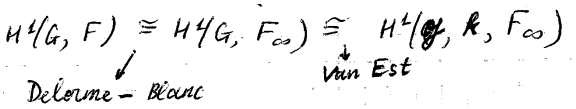
good, but imperfect when $E = \mathbb{C}$. Suppose so.



Original source details: GELFAND-P000810

Uncertain reading. The manuscript leaves the sense of "imperfect" unspecified.

Theorem:



Original source details: GELFAND-P000810

Diagram and typed description.

$H^1(G,F) \cong H^1(G,F_\infty) \cong H^1(\mathfrak{g},\mathfrak{k},F_\infty)$

↙ ↓

Delorme — Blanc Van Est

Original source details: GELFAND-P000810

Uncertain reading. The first algebra symbol in the right-hand group is retraced and provisionally reads $\mathfrak{g}$.

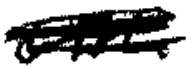
For $\mathbb{R}$ nontriv.

$$\begin{pmatrix} 1 & at \\ 0 & 1 \end{pmatrix}$$

— an extension of the triv. $\mathbb{R}$-module by the trivial one

there are no corresp. reps. of groups and algebras

(in cohom.: one must take cohom. triv. [crossed out: . . .] on the compact subgroup).



Undeciphered passage. word obscured by strokes

Original source details: GELFAND-P000810

GELFAND-P000811

[source number: 44]

Imperfection:
even if F — an irred. unit. rep., then it may be that

$$\dim H^1(G, F) \stackrel{?}{=} +\infty$$

Original source details: GELFAND-P000811

Uncertain reading. The sign has two nearly horizontal strokes with a superimposed angle. Its shape could be $\geq$. Equality agrees with the following example.

Example G — the aff. group of the line $ax + b$

Original source details: GELFAND-P000811

Uncertain reading. The mark resembles R, with some strokes possibly crossing it out.

B^1 is everywhere dense in Z^1, but $\dim Z^1/B^1 = +\infty$

Reason: F weakly contains the trivial rep.

For semisimple groups all is well, [crossed out: . . .]

Original source details: GELFAND-P000811

Undeciphered passage. the original signs are illegible

Let G — semisimple;

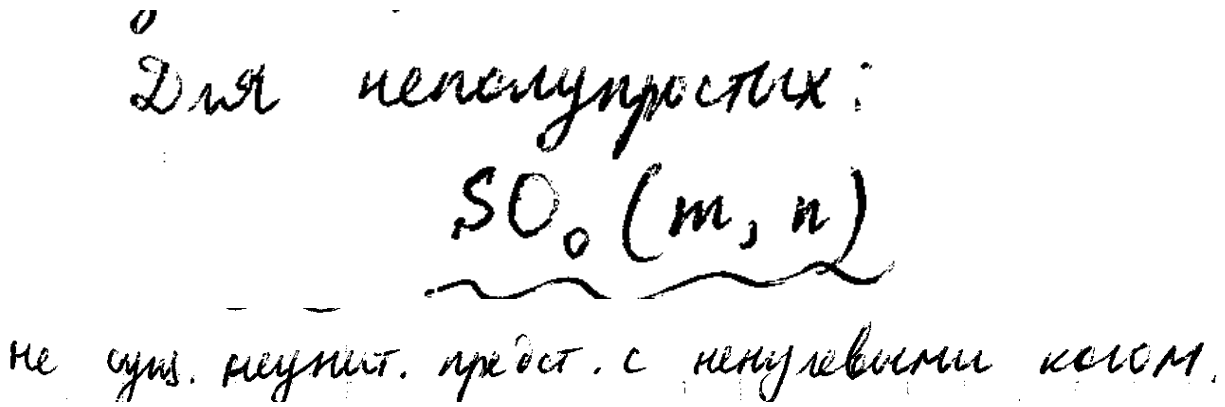
unit. irred. rep.

It is known when $H^1 = 0$.

For nonsemisimple groups:

$$SO_0(m,n)$$

there are no nonunit. reps. with nonzero cohom.



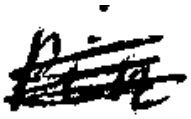
Original source details: GELFAND-P000811

Uncertain reading. The manuscript gives the heading “For nonsemisimple groups” and the abbreviation “nonunit.” alongside $SO_0(m,n)$.

Add. drawback of the 1st idea.

The lemma about infinitesim. char. is false!

There is a counterexample [crossed out: . . .] for $SL(2,\mathbb{R})$



Uncertain reading. Possibly “all”, with several letters obscured by transverse strokes.

Original source details: GELFAND-P000811

2nd idea: the lemma about infinitesim. chars. is true, but

[In the left margin, crossed out: . . .]

Uncertain reading. Possibly a repeated “but”.

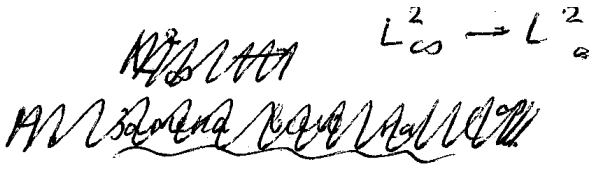
Original source details: GELFAND-P000811

$$H^1(G, \mathrm{Hom}(E_\infty, F_\infty))$$

is hard to compute.

$$L^2_\infty \longrightarrow L^2_\infty$$

[To the left and on the next line, crossed out: ~~$L^2_\infty \longrightarrow \ldots$~~; ~~Problem …~~]



Original source details: GELFAND-P000811

Undeciphered passage. The start of the upper entry provisionally reads L^2_∞ and an arrow is visible farther right. In the lower line, strokes obscure the words and signs following "Problem".

3rd idea.

$$E, F \rightsquigarrow E_K, F_K$$

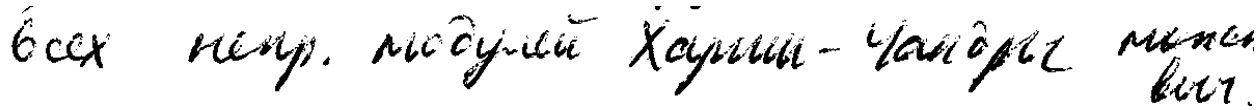
(vectors incl. in finite-dimensional reps. of K).

Original source details: GELFAND-P000811

Undeciphered passage. Two text fragments are crossed out. Isolated loops are visible, with the words obscured.

K-finite; K — max. comp. subgroup

$SL(2, \mathbb{R})$ — for all irred. Harish-Chandra modules one can compute



Original source details: GELFAND-P000811

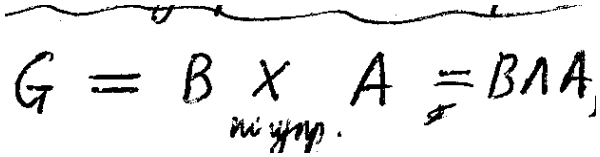
Uncertain reading. The abbreviation «непр.» is provisionally expanded as "irreducible".

$$H^1(\mathfrak{g}, \mathfrak{k}, \mathrm{Hom}(E_K, F_K))$$

algebra without topology.

4. The case of a semidirect product.

$$G = B \underset{\text{semidir.}}{\times}_{\sigma} A = B \wedge A, \qquad B = \mathbb{R}^n, \quad A \text{ — Lie group}$$



Original source details: GELFAND-P000811

Uncertain reading. The sign below the equality provisionally reads σ, possibly g. The sign between B and A is $\wedge$.

(e.g., the aff. group of the line).

B^* — dual space, $\langle b, x \rangle$ — duality.

$X = A$-orbit in B^*

(the action of A in B is precisely the def. of the semidirect prod.)

GELFAND-P000812

[To the left of the date, crossed out: . . ..]

Undeciphered passage. the word or digits are illegible

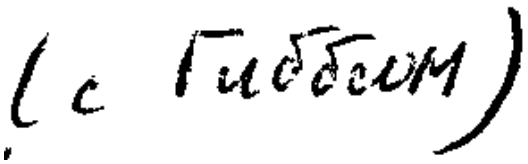
Original source details: GELFAND-P000812

12/II-79 Gelfand Sem.

What does one need to know? (according to A. A. Belavin):

I. Feynman.

1. Course for housewives, chs. 8, 9.
2. Statistical physics
3. Path integrals (with Gibbs)



Uncertain reading. The manuscript reads «Гиббсом» (Gibbs). The probable reference is to A. R. Hibbs, Feynman's coauthor on path integrals.

Original source details: GELFAND-P000812

II. Huang "Statist. Mechanics" Chapter: Onsager's solution.

III. F. Klein "Lectures on the History of Mathematics in the XIX Century"

[Crossed-out number: . . .]

Uncertain reading. possibly IV

Original source details: GELFAND-P000812

[4. The little books with blue cubes on the cover (against a white background) (Feynman: "Fund. Laws of Physics").]

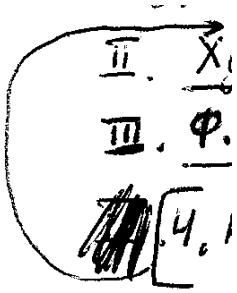
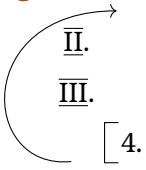


Diagram and typed description.



The arrow in the left margin runs from the crossed-out numeral to the space before II.

Original source details: GELFAND-P000812

IV. Dirac Article in the book “Dirac’s Monopole”

books. “Lectures on Quantum Mechanics”

“[ditto marks for ‘Lectures on’] Field Theory”

“General Relativity”

V. Akhiezer, Berestetskii “Quantum Electrodynamics”

(one should know the contents, but needn’t read it)

VI. Collection “Particles and Compensating Fields”, Article by Yang and Mills.

VII. Konopleva, Popov: “Gauge Fields” (the part written by Popov)

Belavin

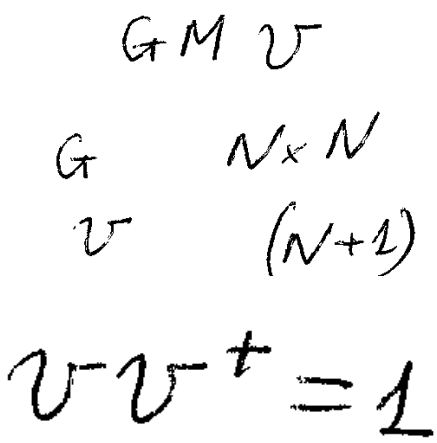
Problem put fwd. for a competition by the Chernogolovka Royal Phys. Soc.: or

What remains unsolv. about instantons?

A, B — 2 quaternionic matrices $N \times (N+1)$

One must find a parametrization [inserted above the line: “of such”] of pairs of matrices A and B, such that if $M = xA + B$, then MM^+ is real (for every quaternion x), up to the group action

$$\begin{array}{cc} & GMV \\ G & N \times N \\ V & (N+1) \times (N+1) \qquad VV^+ = 1 \end{array}$$



Uncertain reading. The same handwritten symbol is provisionally set as V throughout. Its shape could also be U or υ.

Original source details: GELFAND-P000812

There should be $8N - 3$ parameters in all.

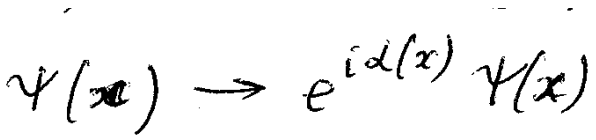
One must find a beautiful parametrization of such matrices.

On some exactly solvable 2-dimensional QFT problems.

A naive argument about non-escape. (Wilson)

$\psi(x)$ $\quad S[\psi, \ldots]$ — action, $\quad \delta S = 0$ — eqs. of motion

$\psi(x) \longrightarrow e^{i\alpha(x)}\psi(x)$ — gauge invariance



Uncertain reading. On the left of the formula, x is written over earlier strokes. The original sign is illegible.

Original source details: GELFAND-P000812

GELFAND-P000813

[source number: 63]

$$\langle \psi(x)\overline{\psi(y)}\rangle = e^{i(\alpha(x)-\alpha(y))}\langle \dots \rangle$$

$\Rightarrow$ correl. fns. $0 \Rightarrow$ impossibility [crossed out: ...] for charged objects.

Undeciphered passage. letters obscured by strokes

Original source details: GELFAND-P000813

Talk on hidden symmetry.

Passing from a cont. int. to a finite multiple integral can break the symmetry.

1) There is a theory

$$\Box\varphi = \sin\varphi, \qquad \varphi(x,t), \qquad \Box = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}$$

a quasiclass. spectrum has been obtained.

2) Next, there is a Ham. (Baxter):

$$H = \sum_{n=1}^{N}\sum_{\alpha=1}^{3} J_\alpha \sigma_n^\alpha \sigma_{n+1}^\alpha \tag{1}$$

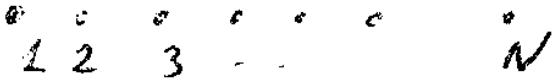


Diagram and typed description.

1 2 3 · · · N

Original source details: GELFAND-P000813

$$\sigma_n^\alpha = 1 \otimes 1 \otimes \dots \otimes \sigma^\alpha \otimes 1 \otimes \dots \otimes 1.$$
$$\alpha = 1,2,3$$

Uncertain reading. Below σ^α is a crossed-out mark, possibly n.

Original source details: GELFAND-P000813

[On the right] (1-dimensional quant. chain)

$\boxed{N+1=1}$ — periodicity condition.

$$\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \qquad J_\alpha \in \mathbb{R}$$

H — a matrix $2^N \times 2^N$

Problem: find the eigenvecs. and eigenvalues.

In meaning, these models are analogous to 4-dim. gauge theories.

In the limit, 2) yields sin-Gordon.

Baxter solved syst. (1)

The Hamiltonian itself is Heisenberg's. (anisotropic, because of J_α — Heisenberg ferromagnet)

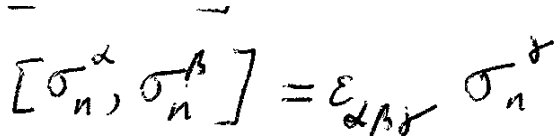
(or XYZ model)

Luther reduced (1) to the Thirring model.

The relations hold:

$$\begin{cases} [\sigma_n^\alpha, \sigma_m^\beta] = 0, & n \neq m \\ [\sigma_n^\alpha, \sigma_n^\beta] = \varepsilon_{\alpha\beta\gamma}\, \sigma_n^\gamma \\ [\sigma_n^1, \sigma_n^2]_+ = 0 \end{cases} \tag{2}$$

$[\,,\,]_+$ — anticommutator



Original source details: GELFAND-P000813

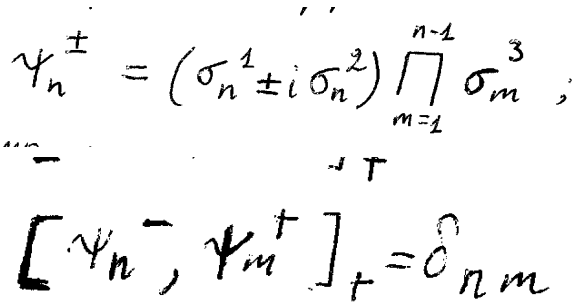
Uncertain reading. The middle line of (2) shows no factor before $\varepsilon_{\alpha\beta\gamma}$.

Let us make lattice fermions.

$$\psi_n^\pm = (\sigma_n^1 \pm i\sigma_n^2) \prod_{m=1}^{n-1} \sigma_m^3; \tag{3}$$

these are fermions, i.e.

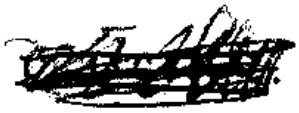
$$\begin{cases} [\psi_n^+, \psi_m^+]_+ = 0 \\ [\psi_n^-, \psi_m^+]_+ = \delta_{nm} \end{cases} \tag{4}$$



Original source details: GELFAND-P000813

Uncertain reading. In (3) there is no factor before the parentheses. In (4) the right-hand side is δ_{nm}.

In [crossed out: . . .] fact — this is how Onsager solved the Ising model:



Undeciphered passage. the word is obscured by strokes

Original source details: GELFAND-P000813

express σ in terms of ψ: H will become quadr. and everything is diagonalized

GELFAND-P000814

Let us take a Fourier transf.

$$\widetilde{\psi}_k^{\pm} = \sum_n \psi_n^{\pm} e^{ikn} \tag{5}$$

[crossed out: . . .]

Uncertain reading. Possibly “I will formulate” or “we will formulate”. Cancellation strokes obscure the ending.

Original source details: GELFAND-P000814

Assertion.

$$H = \sum_n (\sigma_n^1 \sigma_{n+1}^1 + \Delta \sigma_n^3), \qquad \Delta \text{ — a number} \tag{6}$$

$$\dot{\sigma}_n(t) = i[\sigma_n, H] \tag{7}$$

— eqs. of motion; for the solutions $\forall t$ the commut. rels. hold (2)

$$\sigma_n(0) = \sigma_n.$$

After substitution the eqs. become linear (in ψ)

$\Rightarrow$ this is equiv. to finding eigenvectors.

(σ can be expressed in terms of ψ)

We obtain the eqs.

$$\dot{\psi}_n = \psi_{n+1} - \psi_{n-1} \tag{8}$$

Expr. for the Hamiltonian in the orig. problem.

$$H \sim \sum_n \left(\psi_n \psi_{n+1}^+ - \psi_n^+ \psi_{n+1} + \psi_n^+ \psi_n \psi_{n+1}^+ \psi_{n+1}\right) \tag{9}$$

[crossed out: ~~σ_1 and σ_2~~.]

Uncertain reading. The cancelled symbols provisionally read σ_1 and σ_2. The indices may be superscripts.

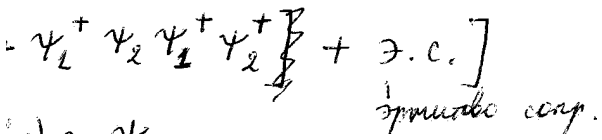
Original source details: GELFAND-P000814

σ_n is expr. in terms of ψ by induction.

(9) — Ham. of the lattice Thirring model (already nonlin.)

Passing to the continuous case, we obtain:

$$\int dx \left[\psi_1^+(x) \frac{\partial \psi_2(x)}{\partial x} + \psi_1^+ \psi_2 \psi_1^+ \psi_2^+ \right] + \underset{\text{Hermitian conj.}}{\text{h.c.}} \tag{10}$$



Uncertain reading. The subscript on the third factor is retraced and reads 1. The first closing bracket is crossed out.

Original source details: GELFAND-P000814

$$x = na, \qquad \psi(x) \approx \psi_n$$

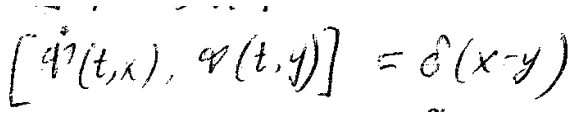
ψ_1, ψ_2 — from different even and odd lattice sites

1975 Mandelstam and...: quant. Thirring model $\sim$ quant. mod. sin-Gordon

quant. mod. sin-Gord

$$\Box\varphi = \sin\varphi \tag{11}$$

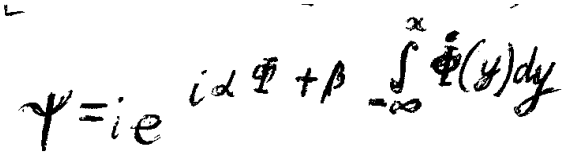
$$[\dot{\varphi}(t,x), \varphi(t,y)] = \delta(x-y) \tag{12}$$



Uncertain reading. The right-hand side of (12) shows $\delta(x-y)$ without a factor i.

Original source details: GELFAND-P000814

$$\psi = ie^{i\alpha\Phi + \beta \int_{-\infty}^{x} \dot{\Phi}(y)\, dy} \tag{13}$$



Uncertain reading. The field in (13) is set as Φ, possibly φ. There is an i before e and no additional i is visible before β.

Original source details: GELFAND-P000814

$\Downarrow$ (when $\alpha\beta = 4\pi$)

$$[\psi(x), \psi^+(y)]_+ = \delta(x-y) \tag{14}$$

End of the introduction.

Sasha Zamolodchikov. Zamolodchikov algebra.

[Under the name:] (Moscow)

$$\underset{\text{symbols}}{A_i(\theta)}\,, \qquad 1 \le i < M$$

GELFAND-P000815

[source number: 64]

[At the upper left: . . .]

Uncertain reading. An isolated short vertical stroke with a rightward hook at its top. Its meaning is unclear.

Original source details: GELFAND-P000815

One can write products

$$A_i(\theta_1)A_\alpha(\theta_2) = S_{ik}^{\alpha\beta}(\theta_1 - \theta_2)A_\beta(\theta_2)A_k(\theta_1) \tag{15}$$

$$1 \le \beta < M, \qquad 1 \le \alpha < M.$$

Uncertain reading. Both upper bounds in the manuscript use strict $<$. The example below uses A_2 with $M = 2$.

Original source details: GELFAND-P000815

[crossed out: . . .]

Undeciphered passage. the original mark is illegible

Original source details: GELFAND-P000815

$M = 2$

$$\begin{cases} A_1 A_1 = a A_1 A_1 + d A_2 A_2 \\ A_1 A_2 = b A_2 A_1 + c A_1 A_2 \end{cases} \tag{16}$$

and 2 more rels. obt. by replacing 1 with 2.

This still has to be made compatible with associativity.

On S the corresp. condition is imposed

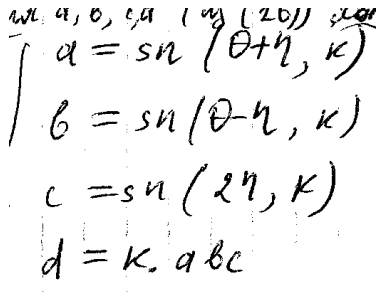
$$\big(A_1(\theta_1) A_1(\theta_2)\big) A_2(\theta_3) = A_1(\theta_1) \big(A_1(\theta_2) A_2(\theta_3)\big)$$

The eqs. lead to factorization of multiparticle S-matrices

Thus he found the S-matrix of sin-Gordon etc.

The eqs. for a, b, c, d (from (16)) are solved explicitly (!)

$$\begin{cases} a = \mathrm{sn}(\theta + \eta, k) \\ b = \mathrm{sn}(\theta - \eta, k) \\ c = \mathrm{sn}(2\eta, k) \\ d = k \cdot abc \end{cases} \tag{17}$$



Uncertain reading. The second argument of sn and the factor in d are provisionally set as k, possibly κ.

Original source details: GELFAND-P000815

These are all solns. up to a factor.

Baxter used such objects

$$T^{\alpha_1, \beta_1, \ldots, \alpha_N, \beta_N}(\theta) = S^{\alpha_1 \beta_1}_{i_1 i_2} S^{\alpha_2 \beta_2}_{i_2 i_3} \ldots S^{\alpha_N \beta_N}_{i_N i_1} \tag{18}$$

The result is a $2^N \times 2^N$ matrix. It turns out that

[To the left of (19) crossed out: ...]

Undeciphered passage. the original mark is illegible

Original source details: GELFAND-P000815

$$[T(\theta), H] = 0 \tag{19}$$

(⇒ infin. set of integrals of motion)

$$[T(\theta_1), T(\theta_2)] = 0 \tag{20}$$

(⇒ in involution)

In fact he solved it exactly.

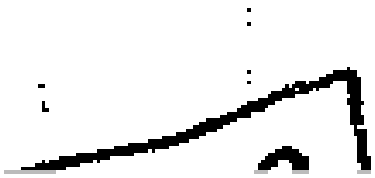


Diagram and typed description.

Angular mark at the lower left. Its strokes end at the same horizontal boundary in the source image. The full shape and purpose are unclear.

Original source details: GELFAND-P000815

GELFAND-P000816

19/II-79 Gelfand Sem.

Belavin.

$$A_i(\theta), \qquad 1 \le i \le M, \qquad \text{— symbols}$$

θ — a parameter

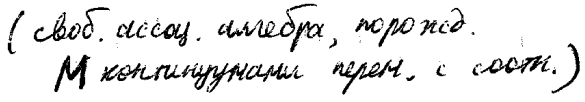
Zamolodchikov algebra.

$$A_i(\theta_1)A_\alpha(\theta_2) = S_{ik}^{\alpha\beta}(\theta_1 - \theta_2)A_\beta(\theta_2)A_k(\theta_1) \tag{1}$$

Associativity is required too. Then cubic rels. arise.

$$A_i(\theta_1)A_j(\theta_2)A_k(\theta_3)$$

[On the right:] (free assoc. algebra gen. by M continua of vars. with rels.)



Uncertain reading. The marginal phrase reads «континуумами перем.» (continua of vars.).

Original source details: GELFAND-P000816

Interpretation (physical)

Diagram and typed description.

Original source details: GELFAND-P000816

pos. — particles, θ_i — velocities

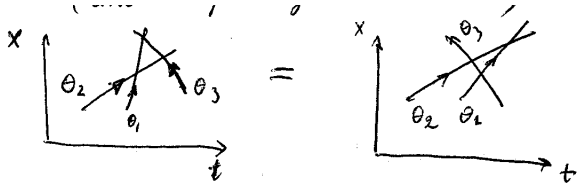
Original source details: GELFAND-P000816

Uncertain reading. The abbreviation below the drawing provisionally reads «полож.» (positions).

$A_i A_\alpha$ — the ith part. is farther left.

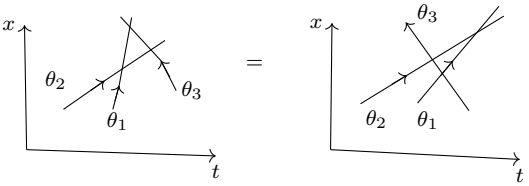
$\theta_1 > \theta_2 \Rightarrow$ the ith catches up.

They can catch up in different orders, but the beginning and end matter (what happens in between does not matter)



Original source details: GELFAND-P000816

Diagram and typed description.

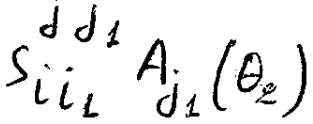


The arrowhead on the descending θ_3 line on the left is retraced. Its direction is provisionally drawn up and to the left.

$$
\begin{aligned}
&\big(A_i(\theta_1)A_j(\theta_2)\big)A_k(\theta_3)\\
&= S^{jj_1}_{ii_1}A_{j_1}(\theta_2)\big(A_{i_1}(\theta_1)A_k(\theta_3)\big) =\\
&= S^{jj_1}_{ii_1}(\theta_{12})S^{kk_1}_{i_1i_2}(\theta_{13})\big(A_{j_1}(\theta_2)A_{k_1}(\theta_3)\big)A_{i_2}(\theta_1) =\\
&= S^{jj_1}_{ii_1}(\theta_{12})S^{kk_1}_{i_1i_2}(\theta_{13})S^{k_1k_2}_{j_1j_2}(\theta_{23})\\
&\quad A_{k_2}(\theta_3)A_{j_2}(\theta_2)A_i(\theta)
\end{aligned}
$$

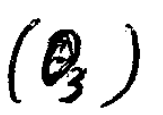
Original source details: GELFAND-P000816

Uncertain reading. The letter A in $A_j(\theta_2)$ in the first product is retraced. The original stroke is illegible.



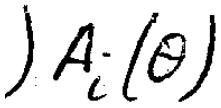
Original source details: GELFAND-P000816

Uncertain reading. There is no written argument for $S^{jj_1}_{ii_1}$ in the first equality.



Original source details: GELFAND-P000816

Uncertain reading. The θ in θ_3 after A_{k_2} in the final product is retraced. The original stroke is illegible.



Uncertain reading. The last factor is written $A_i(\theta)$ without further numerical indices.

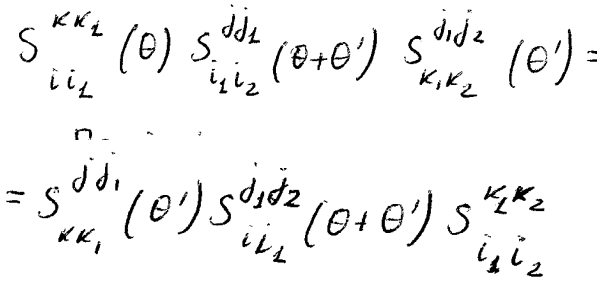
Original source details: GELFAND-P000816

GELFAND-P000817

[source number: 67]

The eqs. take the form:

$$S_{ii_1}^{kk_1}(\theta)S_{i_1i_2}^{jj_1}(\theta+\theta')S_{k_1k_2}^{j_1j_2}(\theta') = S_{kk_1}^{jj_1}(\theta')S_{ii_1}^{j_1j_2}(\theta+\theta')S_{i_1i_2}^{k_1k_2} \tag{2}$$



Uncertain reading. The indices in (2) follow the manuscript. The last S has no written argument.

Original source details: GELFAND-P000817

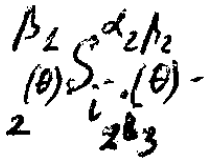
Over i_1, j_1, k_1 sum.

In all: M^6 eqs. for M^4 fns. $S_{ik}^{\alpha\beta}(\theta)$

S plays the role of an S-matrix.

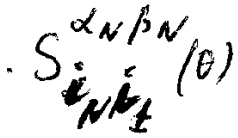
Put

$$T_{\vec{\alpha},\vec{\beta}}(\theta) = S_{i_1i_2}^{\alpha_1\beta_1}(\theta)S_{i_2i_3}^{\alpha_2\beta_2}(\theta)\cdot\ \cdot\ \cdot S_{i_Ni_1}^{\alpha_N\beta_N}(\theta) \tag{3}$$



Uncertain reading. The retraced lower indices of the middle factor provisionally read i_2i_3.

Original source details: GELFAND-P000817



Uncertain reading. The retraced lower indices of the last factor provisionally read i_Ni_1, possibly i_Nk_1.

Original source details: GELFAND-P000817

The result will be a matrix $M^N \times M^N$ (“transfer matrix”)

[On the right:] $(\alpha_1, \ldots, \alpha_N)$ — indices.

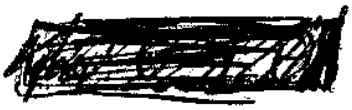
$(\beta_1, \ldots, \beta_N)$ — 2nd set of inds.

It is asserted that

$$[T(\theta), T(\theta')] = 0 \tag{4}$$

[To the right of zero, crossed out: . . .]

Undeciphered passage. the entry is illegible



Original source details: GELFAND-P000817

(by virtue of the factorization eqs.).

[On the left, an underlined entry is crossed out: . . .]

Undeciphered passage. the entry is illegible

Original source details: GELFAND-P000817

Factoriz. of the S-matr. is a consequence of a fin. number of conserv. laws.

Explicit form of the eqs. for $M = 2$

(case of Maxwell’s eqs.)

$$\begin{cases} A_1 A_1 = a\, A_1 A_1 + d\, A_2 A_2 \\ A_1 A_2 = b\, A_2 A_1 + c\, A_1 A_2 \end{cases} \qquad \begin{pmatrix} A_1(\theta_1),\ A_2(\theta_2) \\ a(\theta_1 - \theta_2) \end{pmatrix} \tag{5}$$

[On the right:] and 2 more eqs. with 1 replaced by 2 (with the same coeffs.) (see $(5)_2$)

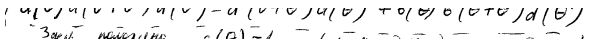
There are 4 fns. a, b, c, d.

There could be 16 fns. In fact 5 or 6 fns. (from the eqs.).

Here 4 are retained

The factorization eqs. give:

$$\begin{cases} a(\theta)a(\theta+\theta')d(\theta') = d(\theta+\theta')a(\theta') \\ \qquad\qquad\qquad\qquad + b(\theta)b(\theta+\theta')d(\theta') \end{cases} \tag{6}$$



Original source details: GELFAND-P000817

Diagram and typed description.

A row of short strokes beneath the written line of (6).

Here we have set $c(\theta) \equiv 1$ (the eqs. are homogeneous — let us divide).

Uncertain reading. The verb provisionally reads «положено» (has been set), possibly «положим» (let us set).

Original source details: GELFAND-P000817

[On the right:] (6 funct. eqs. in all)

$$\begin{cases} A_2A_2 = a\,A_2A_2 + d\,A_1A_1 \\ A_2A_1 = b\,A_1A_2 + c\,A_2A_1 \end{cases} \qquad (5)_2$$

Uncertain reading. The original mark under the retraced numeral 5 in $(5)_2$ is illegible.

Original source details: GELFAND-P000817

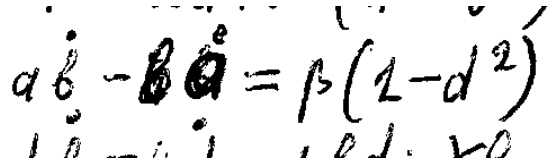
$$S_{ik}^{\alpha\beta}(0) = \delta_{i\beta}\delta_{\alpha k} \qquad (7)$$

— init. condition

We obtain diff. eqs. from (6):

$$\begin{cases} \dot a = \alpha a - \beta b + \gamma ad & \\ \dot d = \alpha d + \gamma(a^2 - b^2) & \dot a(0) = \alpha \\ a\dot b - b\dot a = \beta(1 - d^2) & \dot b(0) = \beta \\ d\dot b - b\dot d = \alpha bd + \gamma b - \beta ad & \dot d(0) = \gamma \\ \text{and 2 more eqs.} & \end{cases} \qquad (8)$$

Uncertain reading. The $b\dot a$ in the third line of (8) is retraced. The original strokes are illegible.



Original source details: GELFAND-P000817

GELFAND-P000818

All the eqs. are independent.

3 algebr. relations arise, but one follows from the other 2.

Elliptic fns. are obtained (Baxter)

A noncommut., nonassoc. algebra arises, of quadruples of fns.

Uncertain reading. The word reads «четверок» (quadruples). Strokes above «е» may indicate «ё».

Original source details: GELFAND-P000818

It would be interesting to find an explicit realiz. (e.g., on matrices)

Yu. I. Manin's idea: take points on an ell. curve, of finite order

Let us expand

$$T(\theta) = \sum_{\ell=0}^{\infty} I_\ell \theta^\ell \tag{9}$$

Then

$$[I_n, I_m] = 0 \tag{10}$$

and in the example ($M = 2$):

$$I_0^{-1}\,\cancel{(0)}\,I_1\,\cancel{(0)} = \sum_{n=1}^{N}\sum_{\alpha=1}^{3} J_\alpha \sigma_n^\alpha \sigma_{n+1}^\alpha \quad (= T^{-1}(0)\dot{T}(0)) = H \tag{11}$$

Original source details: GELFAND-P000818

Uncertain reading. A mark after I_0^{-1} is crossed out. It may read (0).

Original source details: GELFAND-P000818

Uncertain reading. A separate mark after I_1 is crossed out. It too may read (0).

Here

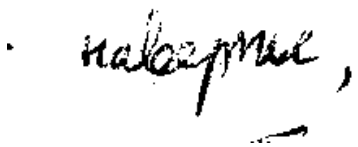
$$\sigma_n^\alpha = 1 \otimes \ldots \otimes \sigma^\alpha \otimes \ldots \otimes 1, \qquad \sigma^\alpha \text{ — the Pauli matrices.}$$

$$[I_\ell, H] = 0, \quad \ell = 0, 1, 2, \ldots \tag{12}$$

The transfer matr. is written in terms of S, i.e., in terms of ellipt. fns.

I_2 contains 3 spins, and so on.

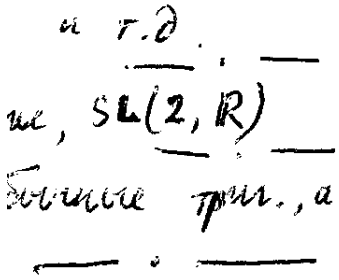
I. M. G. a, b, c, d — probably, $SL(2, R)$



Original source details: GELFAND-P000818

Uncertain reading. The word provisionally reads «наверное», possibly «наверно» (probably).

$d = 0 \Rightarrow$ we obtain ordinary trig. fns. instead of ellipt. fns.



Original source details: GELFAND-P000818

Diagram and typed description.

Three rows of separate strokes beneath the entries about spins, $SL(2, R)$, and trig. fns.

General assertion (now M is arbitrary):

Put

$$\dddot{\overline{H}} = \sum_{n=1}^{N} \dot{S}^{\alpha\beta}_{ik}(0)\sigma^{i\beta}_{n}\sigma^{\alpha k}_{n+1} = I_0^{-1} I_1 \qquad \text{(see (9))}. \tag{13}$$

Undeciphered passage. A mark above H is crossed out. The original sign is illegible.

Original source details: GELFAND-P000818

σ — a matrix of size $M \times M$.

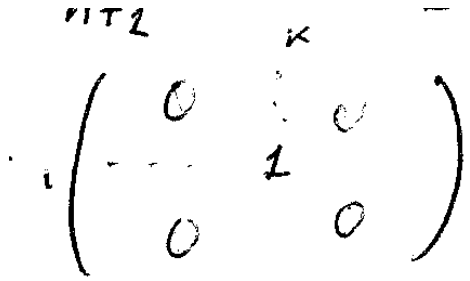


Diagram and typed description.

$$\sigma^{ik} = i \begin{pmatrix} & & k & \\ & 0 & \vdots & 0 \\ \cdots & & 1 & \\ & 0 & & 0 \end{pmatrix}$$

basis in the space of matrices (1 in position i, k).

Original source details: GELFAND-P000818

$$\sigma^{ik}_{n} = 1 \otimes \ldots \otimes \sigma^{ik}_{n} \otimes \ldots \otimes 1$$

We have a loc. Hamiltonian.

Uncertain reading. The abbreviation before "Hamiltonian" provisionally reads «лок.» (loc.).

Original source details: GELFAND-P000818

Zamolodchikov obtained particular solutions (for different M)

Let us use them for $M = 3$, We obtain

$$H = \sum_{n} \dddot{(}(\vec{S}_n \vec{S}_{n+1})^2 - 2\vec{S}_n \vec{S}_{n+1} \tag{14}$$

Undeciphered passage. An entry between the summation sign and the opening parenthesis is crossed out. The original signs are illegible.

Original source details: GELFAND-P000818

$M = 2$ — isotr. Heisenberg ferromagnet

$$H = \sum_{n} \vec{\sigma}_n \vec{\sigma}_{n+1} \tag{15}$$

spin $1/2$ — there is nothing else inv.

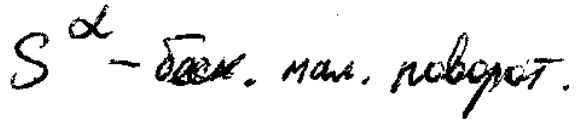
GELFAND-P000819

[source number: 68]

In (14): S^α are such that

$$[S^\alpha, S^\beta] = \varepsilon_{\alpha\beta\gamma} S^\gamma, \tag{16}$$

S^α — inf. small rotation.



Uncertain reading. The first abbreviation before “small rotation” provisionally reads «беск.» (inf.). Some strokes overlap.

Original source details: GELFAND-P000819

(rep. of the group $SO(3)$)

adj. = standard

There are no parameters here

$M = 3 \Rightarrow$ there is no general soln.

(!) It is unknown how to add parameters to Ham. (14) so as to fit Zamolodchikov's scheme.

This scheme cont. all fully integr. quant. systems (2-dimensional)

($\Rightarrow$ classical ones too after taking the quasiclass. limit).

$$H = \sum_{n=1}^{N} \dot{S}^{\alpha\beta}_{ik}(0) \sigma^{i\beta}_n \sigma^{\alpha k}_{n+1}; \tag{17}$$

I.M.G.'s question: rels. on S — of Maurer–Cartan type

(17) — the most general form of Ham. with nearest-neighbor interact.

It always has a nontriv. integral, since the diff. eqs. extend to the 2nd neighborhood.

$$S = S(0) + S'(0)\theta + \frac{S''(0)\theta^2}{2}$$

S'' is recovered.

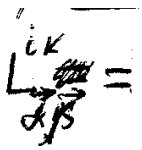
$A_i(\theta)$ are matrices $\Rightarrow$ there is assoc. automatically

There is a way to multiply algebras.

Suppose we found $S^{\alpha\beta}_{ik}$. Let us construct

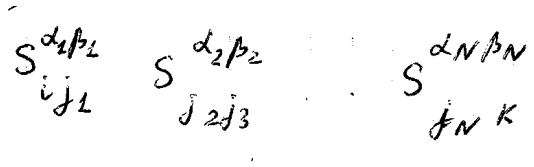
(e.g. $M = 2$)

$$L^{ik}_{\vec{\alpha}\vec{\beta}} = S^{\alpha_1\beta_1}_{ij_1} S^{\alpha_2\beta_2}_{j_2 j_3} \ldots S^{\alpha_N\beta_N}_{j_N k} \quad \text{(we did not take the trace)} \tag{18}$$



Uncertain reading. The lower indices of L are set as $\vec{\alpha}\vec{\beta}$. There is a crossed-out entry alongside them.

Original source details: GELFAND-P000819



Uncertain reading. The second factor has lower indices $j_2 j_3$, while the first ends with j_1. Two dots separate the second and last factors.

Original source details: GELFAND-P000819

We get the matrix

$$L = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} = (L^{\alpha\beta}) \tag{19}$$

its elems. — are matrices too.

[crossed out: ~~...~~] Rels. of type (2) $--$ hold for the matrix L

Undeciphered passage. The crossed-out sign before "Rels." is illegible.

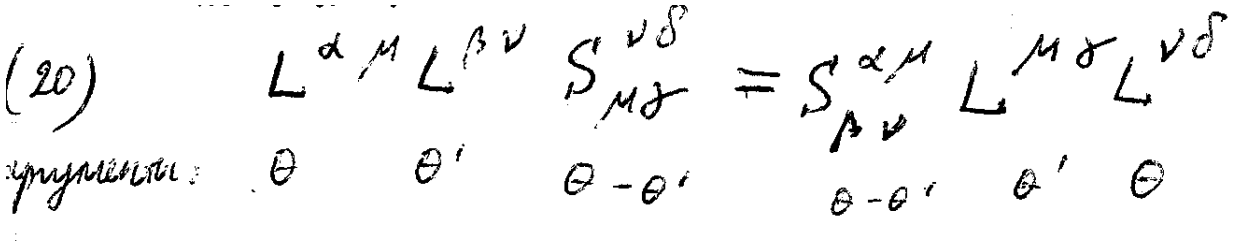
Original source details: GELFAND-P000819

$$S^{\alpha\mu}_{ik} S^{\beta\nu}_{kj} S^{\nu\delta}_{\mu\gamma} = S^{\alpha\mu}_{\beta\nu} S^{\mu\gamma}_{ik} S^{\nu\delta}_{kj} \tag{2}'$$

L^{ik} — matr. $M^N \times M^N$ for all i, k.

There are relations:

$$\begin{array}{lccccccc} & L^{\alpha\mu} & L^{\beta\nu} & S^{\nu\delta}_{\mu\gamma} & = & S^{\alpha\mu}_{\beta\nu} & L^{\mu\gamma} & L^{\nu\delta}. \\ \text{arguments:} & \theta & \theta' & \theta-\theta' & & \theta-\theta' & \theta' & \theta \end{array} \tag{20}$$



Original source details: GELFAND-P000819

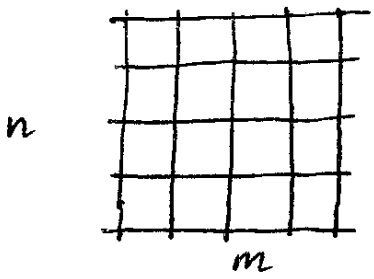
Uncertain reading. The arguments are written under their corresponding factors. On the right, the two L factors are marked first θ', then θ.

GELFAND-P000820

Now suppose

$$A_{\alpha\mu} = L^{\alpha\mu} \tag{21}$$

We get a new algebra of higher order



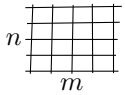
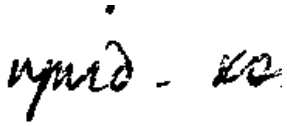


Diagram and typed description. 2-dim. lattice rolled into a torus.

Original source details: GELFAND-P000820

Onsager in ’44 — 2-dim. Ising model

—| |— ’65 — devis. commut. transfer matrix.



Uncertain reading. The abbreviation before “commut.” provisionally reads «прид.» (devis.). «пред.» (propos.) is also possible.

Original source details: GELFAND-P000820

$$\sigma_{n,m} = \pm 1$$

We need to calc. the sum:

$$Z = \sum_{\{\sigma_{n,m}\}} e^{J_1 \sum \sigma_{n,m}\sigma_{n,m+1} + J_2 \sum \sigma_{n,m}\sigma_{n+1,m} \cdots} \tag{22}$$

Undeciphered passage. The entry at the end of the exponent is crossed out. Its signs are illegible.

Original source details: GELFAND-P000820

(nearest-neighbor interact.)

Idea:

$$Z = \operatorname{Tr} T^N \tag{23}$$

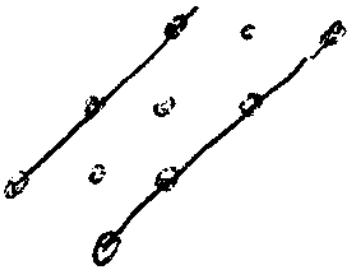
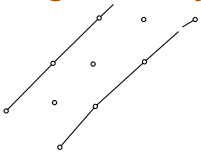


Diagram and typed description.



The strokes meet at the top circle on the left line. A separate arrowhead cannot be distinguished.

Original source details: GELFAND-P000820

[At right:] Onsager: we go along the diagonals (we sum over the spins on the interm. diag.)

$$T_{\vec{\alpha}\,\vec{\beta}} = S_1 \dots S_n, \tag{24}$$

where

Diagram and typed description.

(25) $S_{ik}^{\alpha\beta} =$

Original source details: GELFAND-P000820

We get the tr. matr. $T(J_1, J_2)$

Prop.

$T(J_1, J_2)$ commut. wh

Undeciphered passage. The last visible letters are «пр», possibly the start of «при» (“when”). The entry breaks off.

Original source details: GELFAND-P000820

Chapter 15

GELFAND-P000821

[At the upper left, crossed out: ~~22...~~]

Uncertain reading. The start of the crossed-out entry provisionally reads “22”. The ending is illegible.

Original source details: GELFAND-P000821

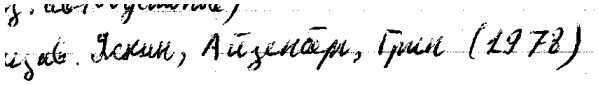
26.2.79 Gelfand Sem.

S. Gelfand from Manin’s lecture

This concerns the relation of Y.–M. solns to alg.-geom. objects

(solns need not be self-dual)

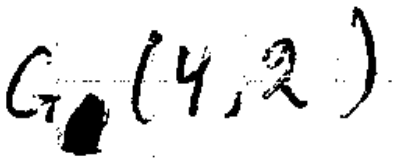
— Witten; indep. Yasskin, Isenberg, Green (1978)



Uncertain reading. The names are written «Яскин, Айзенберг, Грин».

Original source details: GELFAND-P000821

$\widehat{\mathbb{C}M} = G(4,2)$ (Grassmannian set of 2-dim. compl. planes in $\mathbb{C}^4$)



Undeciphered passage. There is a heavily inked-out entry at the lower right of G. Its sign is illegible.

Original source details: GELFAND-P000821

[Below $G(4,2)$:] (compl. Grassmannian set)

[Below the formula at left, a vertical line to the label:] complexif. Minkowski space

(and compactif.)

(in $G(4,2)$ there is an open subset isom. to $\mathbb{C}^4$).

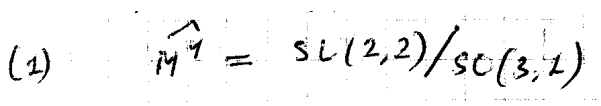
S. Gelfand

M^4 (Mink. space) with conf. metric is compactif. i.e. light cone

(at infinity). All conf. transf. are also $SU(2,2)$

We get the stat. group $SO(3,1)$

$$\widehat{M^4} = SU(2,2)/SO(3,1) \tag{1}$$



Uncertain reading. The denominator in the manuscript is $SO(3,1)$.

Original source details: GELFAND-P000821

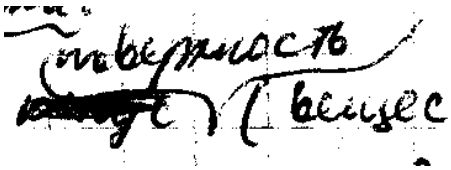
Now complexify $SU(2,2)$ and $SO(3,1)$ and take the quotient

$$(SU(2,2) = SO(4,2)) \tag{2}$$

$SU(2,2)$ — conf. group for M^4

Geometric picture.

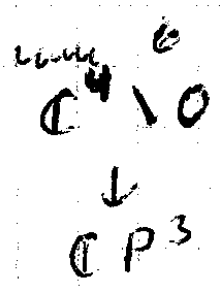
In $\mathbb{C}P^3$ consid. [crossed out: ~~cone~~] [inserted above: surface] (real 5-dimensional)



Original source details: GELFAND-P000821

Uncertain reading. The crossed-out word provisionally reads “cone”. “Surface” is written above it.

$$R = \{z : |z_1|^2 + |z_2|^2 - |z_3|^2 - |z_4|^2 = 0\} \subset \mathbb{C}P^3 \tag{3}$$



Original source details: GELFAND-P000821

Diagram and typed description.

$$(4) \quad \begin{array}{c} \text{or in } \mathbb{C}^4\backslash 0 \\ \downarrow \\ \mathbb{C}P^3 \end{array}$$

acts $GL(4,\mathbb{C})$; surf. (3) is preserved by the group $SU(2,2)$

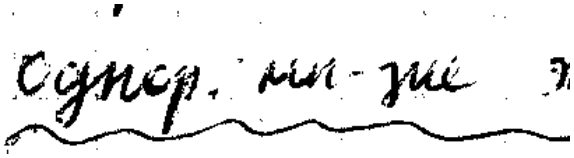
$G(4,2)$ — set of lines in $\mathbb{C}P^3$;

(or 2-dim. subspaces in $\mathbb{C}^4$)

On surf. (3) there lies wholly a set of lines of real dimension 4.

$SU(2,2)$ carries over to this set.

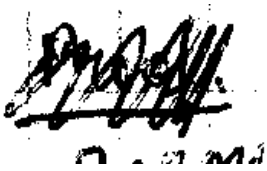
Theorem 1. Contain. set of these lines — Minkowski space.



Original source details: GELFAND-P000821

Uncertain reading. The first abbreviation provisionally reads «Содерж.» (“contain.”), possibly «Соотв.» (“correspond.”).

[crossed out: ~~Proof~~]



Original source details: GELFAND-P000821

Uncertain reading. The cancelled heading provisionally reads «Доказ.» (“Proof”).

The lines on surf. (3) are parametriz. by unit. matrices of order 2

$$(z_3, z_4) = U(z_1, z_2), \qquad U \in U(2)$$

Minkowski metric.

Consid. a fixed line L_0 on R. Consid. all other lines on R, that intersect it. We get a 3-dim. cone K_{L_0}. It is preserved under all transf. preserving L_0. We get $SO(3,1)$ and a cone called

the light cone.

The points of surf. R can be interpreted as rays of light.

One can collect all points of intersection of $L \subset R$ with L_0. This gives a generator of the cone (light cone).

GELFAND-P000822

[source number: 2]

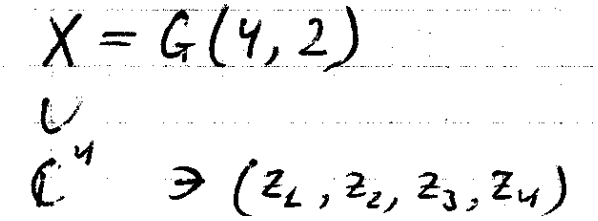
Cf.: the quaternionic version (it was discussed at some point).

Stat. subgroup:

$$GL(2,\mathbb{C}) \times \mathbb{R}^3 \qquad (?)$$

The conf. group is obtained by adding inversion $\frac{1}{\tau}$ (τ — Minkowski dist.) to the motions (the grp. of Poincaré).

$$\begin{array}{l} X = G(4,2) \\ \cup \\ \mathbb{C}^4 \ni (z_1, z_2, z_3, z_4) \end{array}$$



Uncertain reading. The sign below X resembles $\cup$. It is set in the same vertical position.

Original source details: GELFAND-P000822

There we consid. holom. connections satisfying the Y.–M. eqs.

G — a compl. Lie grp.; $\mathfrak{g}$ — its Lie algebra.

$(A_1(z), A_2(z), A_3(z), A_4(z))$ — holom. fns. with values in $\mathfrak{g}$

(holom. in an open region)

$$F_{\mu\nu}(z) = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad \partial_\mu = \frac{\partial}{\partial z_\mu}. \tag{5}$$

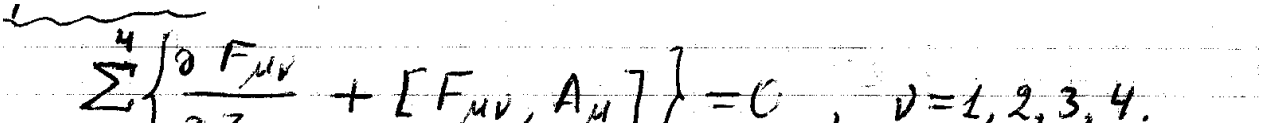
Connection form:

$$\omega = \sum A_\mu(z)\, dz_\mu$$

F — the curvature form of this connection (again with values in the Lie alg.)

Y.–M. eqs.

$$\sum_{\mu=1}^{4} \left\{ \frac{\partial F_{\mu\nu}}{\partial z_\mu} + [F_{\mu\nu}, A_\mu] \right\} = 0, \qquad \nu = 1,2,3,4. \tag{6}$$



Original source details: GELFAND-P000822

Uncertain reading. In (6) the manuscript has $[F_{\mu\nu}, A_\mu]$ after the plus sign.

I.M.G. Landau: “the Lagrangian is a corpse” thought that one need only know the S-matrix, and that is all.

The corresp. real eqs. are obtained from the Lagrangian:

$$L = \operatorname{tr} \int F_{\mu\nu} F^{\mu\nu}\, d^4x \tag{7}$$

Solutions of (6) are related to certain objects of compl. analytic geometry.

From $\mathbb{C}^4$ we construct

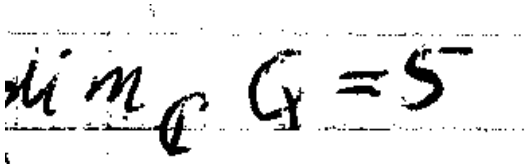
$$\underset{\text{lines in } \mathbb{C}^4}{\mathbb{C}P^3} \qquad \text{and} \qquad \underset{\text{hyperpl. in } \mathbb{C}^4}{\mathbb{C}P^3}$$

[Crossed out before the formula: ...]

Undeciphered passage. A separate sign is heavily crossed out. Its numeral cannot be established.

Original source details: GELFAND-P000822

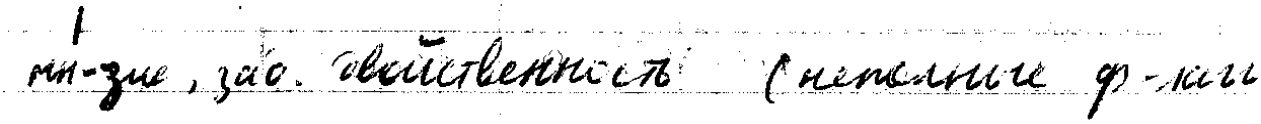
$$\mathbb{C}P^3 \times \mathbb{C}P^3 \supset Q = \{(\ell, \pi) \mid \ell \subset \pi\}, \qquad \dim_{\mathbb{C}} G = 5 \tag{8}$$



Uncertain reading. The letter after $\dim_{\mathbb{C}}$ provisionally reads G, possibly Q. The first Q on the same line has a closed upper loop.

Original source details: GELFAND-P000822

[Vertical stroke down from Q:] the set defin. duality (partial fl-ags)



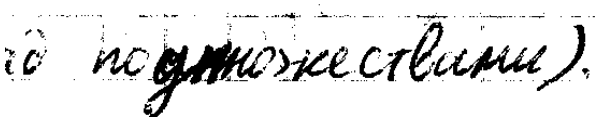
Original source details: GELFAND-P000822

Uncertain reading. The abbreviation before “duality” is provisionally set as «зад.» (“defin.”). The initial descender and final loop admit another reading.

2nd object.

(9) Holomorphic bundles on Q with structure group G, which extend holom. to the 3rd infinitesim. neighborhood of Q in $\mathbb{C}P^3 \times \mathbb{C}P^3$.

(In fact everything is over subsets).



Uncertain reading. Several letters in “subsets” are overwritten. The earlier form is illegible.

Original source details: GELFAND-P000822

All connections $\Longleftrightarrow$ all bundles (not necessarily extending to the 3rd infinit. neighborhood)

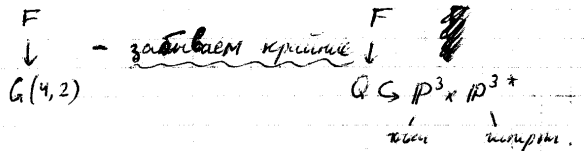
GELFAND-P000823

A. Beilinson

on $G(4,2)$ we consid. the full flag space F (line, plane, hyperpl.)

Diagram and typed description.
The upper and right edges are broken. The slanted strokes have thickened parts.

Original source details: GELFAND-P000823



Original source details: GELFAND-P000823

Diagram and typed description.

$$\begin{array}{ccc} F & & F \\ \downarrow & & \downarrow \\ G(4,2) & & Q \hookrightarrow P^3 \times P^{3*} \\ & & \text{points} \quad \text{hyperpl.} \end{array}$$

— forget the outer ones

In the right diagram a hooked arrow points right from Q.

[Crossed out to the right of the right-hand F: ...]

Undeciphered passage. Strokes cover a vertical entry. The original sign is illegible.

Original source details: GELFAND-P000823

Diagram and typed description.

There is a separate point between the figures. A short upper-right stroke has an unclear relation to the outline.

Original source details: GELFAND-P000823

(E, ∇)
bundle with connection
on $G(4,2)$

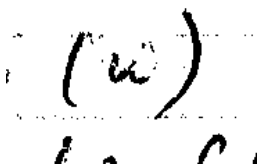
The fiber of the right bundle is isom. to $\mathbb{C}P^1$

Under the left projection it goes to $\mathbb{C}P^1$

on $\mathbb{C}P^1$ the bundle trivializes (canonically), since the curvature $= 0$ on account of the dimension being one

(fund. group $= 0$).

The connection (ω) we lift to F and try to descend to Q.



Original source details: GELFAND-P000823

Uncertain reading. The symbol in parentheses provisionally reads ω. Its shape differs from ∇ in the pair (E, ∇).

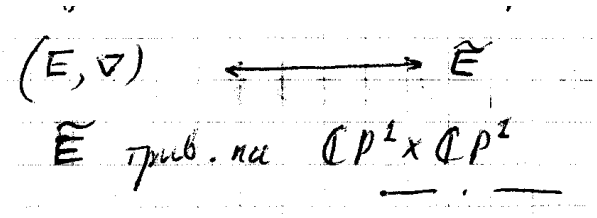
[Below the line] (from $G(4,2)$)

It is on $\forall$ fibers trivial $\Rightarrow$ can be carried over to Q

We get $\widetilde{E}$ — a holom. vect. bundle on Q

Original source details: GELFAND-P000823

Uncertain reading. The first E has a thick, almost horizontal stroke above it. The wavy mark is clearer lower down.



Original source details: GELFAND-P000823

Diagram and typed description. $(E, \nabla) \longleftrightarrow \widetilde{E}$
$\widetilde{E}$ is triv. on $\mathbb{C}P^1 \times \mathbb{C}P^1$

Two strokes and a dot between them appear under the final line.

GELFAND-P000824

5/III-79 Gelfand Sem.

D. B. Fuchs.

On ∞-dimensional Lie algebras

(construction and classification).

Graded Lie alg.:

$$\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i, \qquad [\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$$

hence $\mathfrak{g}_0$ — a subalg., and all the $\mathfrak{g}_i$ — $\mathfrak{g}_0$-modules.

Cartan's construction:

$$\mathfrak{g}_i = 0 \quad \text{for } i < i_0. \tag{1}$$

Transitivity: there is a homom.

$$\mathfrak{g}_i \to \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_{i-1}) \tag{2}$$

$$\mathfrak{g}_i \to \mathrm{Hom}(\mathfrak{g}_1, \mathfrak{g}_{i+1}) \tag{3}$$

$$\begin{cases} i \geq 0 \ \Rightarrow \ (2) \text{ monomorphic} \\ i \leq 0 \ \Rightarrow \ (3) \text{ monomorphic} \end{cases} \tag{4}$$

Then there is Cartan's classification thm.: there are 4 series.

1) W_n — polynomial vect. fields on $\mathbb{C}^n$:

$$\sum p_i(x_1, \ldots, x_n) \frac{\partial}{\partial x_i} \tag{5}$$

deg $\frac{\partial}{\partial x_i} = -1$, [crossed out: ...] deg $x_i = 1$.

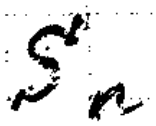
Original source details: GELFAND-P000824

Undeciphered passage. A sign between the two degrees is crossed out. Its original form is illegible.

Here

$\mathfrak{g}_0$ — matrix alg.(!) (gen. by the fields $x_j \frac{\partial}{\partial x_i}$).

2) divergence-free vect. fields S_n



Original source details: GELFAND-P000824

Uncertain reading. A short rising stroke joins the upper-right edge of S. It may be the end of the letter or a prime mark.

3) Hamiltonian H_n

4) contact K_n pres. the cont. struct. in $\mathbb{C}^{2n+1}$

(description of the grading is omitted)

(all are polynomial)

Idea of the proof:

Cart. prolongation.

For simplicity assume $i_0 = -1$ (in (1))

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \cdots \tag{6}$$

From $\mathfrak{g}_0$ and $\mathfrak{g}_{-1}$ everything is recovered

GELFAND-P000825

[source number: 10]

We have:

$$\mathfrak{g}_1 \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0) \tag{7}$$

$$f \in \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0), \quad \alpha_1, \alpha_2 \in \mathfrak{g}_{-1} \quad \Rightarrow$$

$$\Rightarrow$$

$$f(\alpha_1)\alpha_2 = f(\alpha_2)\alpha_1 \tag{8}$$

(from the Jacobi id.)

Uncertain reading. The subscript on the second α on the left is retraced. Its visible form reads 2.

Original source details: GELFAND-P000825

Let

$$\mathfrak{g}_1 = \{f \in \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0) \mid f(\alpha_1)\alpha_2 = f(\alpha_2)\alpha_1\} \tag{9}$$

$$\mathfrak{g}_2 \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_1) \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0)) = \\ = \mathrm{Hom}(\mathfrak{g}_{-1} \otimes \mathfrak{g}_{-1}, \mathfrak{g}_0) \tag{10}$$

Jacobi id. $\Rightarrow$

$$f(\alpha_1 \otimes \alpha_2)\alpha_3 \text{ — symm. in all 3 vars.} \tag{11}$$

Put

$$\mathfrak{g}_2 = \left\{ \begin{matrix} f \in \mathrm{Hom}(\mathfrak{g}_{-1} \otimes \mathfrak{g}_{-1}, \mathfrak{g}_0) \mid \\ f(\alpha_1 \otimes \alpha_2)\alpha_3 \text{ symm. in all vars.} \end{matrix} \right\} \tag{12}$$

and then

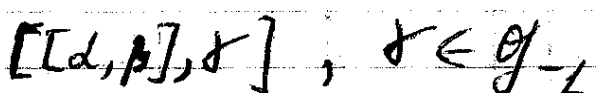
$$\mathfrak{g}_k = \left\{ \begin{matrix} f \in \mathrm{Hom}(\bigotimes^k \mathfrak{g}_{-1}, \mathfrak{g}_0) \mid \\ f(\alpha_1 \otimes \cdots \otimes \alpha_k)\alpha_{k+1} \text{ is symmetric} \end{matrix} \right\} \tag{13}$$

The commutator in $[\bigoplus \mathfrak{g}_k]$ — ?

with $\mathfrak{g}_{-1}$ the commut. is already defined, with $\mathfrak{g}_0$ — likewise.

$$\alpha \in \mathfrak{g}_k, \quad \beta \in \mathfrak{g}_\ell \quad \Rightarrow \quad \text{to def. } [\alpha, \beta] \text{ one needs to specify}$$

$$[[\alpha, \beta], \delta], \qquad \delta \in \mathfrak{g}_{-1}$$

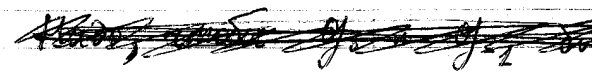


Uncertain reading. The same letter in both places provisionally reads δ. γ is also possible.

Original source details: GELFAND-P000825

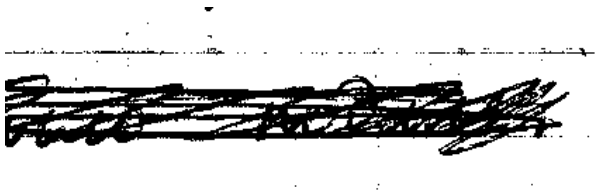
Jacobi id. $\rightsquigarrow$ into terms def. by induction.

[Crossed out: ~~We need $\mathfrak{g}_0$ and $\mathfrak{g}_{-1}$ to be ...~~]



Original source details: GELFAND-P000825

Uncertain reading. The start of the crossed-out line is provisional. Strokes cross the connector between the two $\mathfrak{g}$ terms and the sign before the index 1.



Original source details: GELFAND-P000825

Undeciphered passage. The end of the crossed-out line is illegible.

$\mathfrak{g}_0, \mathfrak{g}_{-1}$ can be arbitrary. Simplicity is an addl. condition.

But as a rule we get a finite-dimensional algebra!!

∞-dimensionality — luck.

Another constr.

Let $\mathfrak{g}_{-1}, \mathfrak{g}_0$ be given, $\quad \mathfrak{g}_0 \subset \mathrm{End}(\mathfrak{g}_{-1})$.

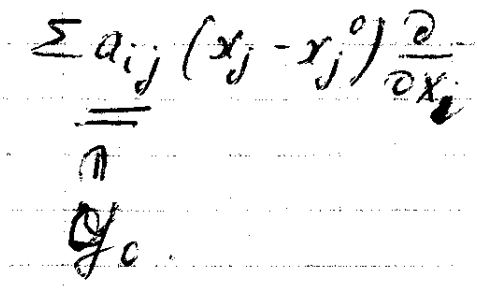
GELFAND-P000826

In $\mathfrak{g}_{-1}$ we consid. poly. vect. fields. A subalgebra is specified in it.

$$\sum_i p_i \frac{\partial}{\partial x_i} \in \mathfrak{g}, \quad \text{if } \forall x^0 \in \mathfrak{g}_{-1},$$

the lin. part of the Tayl. expan. at pt. x^0 belongs to $\mathfrak{g}_0$.

$$\sum a_i \frac{\partial}{\partial x_i} + \sum \underset{\underset{\mathfrak{g}_0}{\cap}}{\underline{\underline{a_{ij}}}} (x_j - x_j^0) \frac{\partial}{\partial x_i} \tag{14}$$



Original source details: GELFAND-P000826

Uncertain reading. Below the doubly underlined a_{ij} the manuscript has a rotated membership sign, then $\mathfrak{g}_0$.

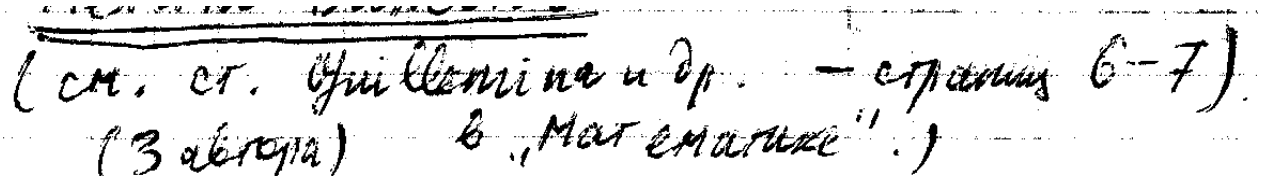
Natural grading — by degrees.

This coincides with the Lie alg. constructed above

Wilson's lemma

(see the paper by Guillemina et al. — pages 6–7).

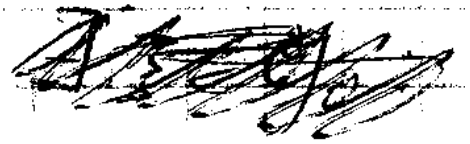
(3 authors) in "Matematika".)



Original source details: GELFAND-P000826

Uncertain reading. The name provisionally reads "Guillemina". The final «a» may be a Russian case ending. The title on the lower line is "Matematika".

Cart. prolong. leads to an ∞-dim. Lie alg. $\Longleftrightarrow$ [crossed out: $\exists \xi \in \mathfrak{g}_0$]



Original source details: GELFAND-P000826

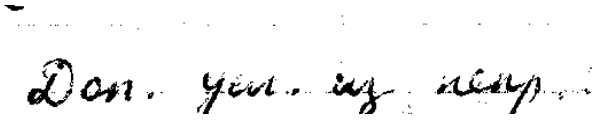
Uncertain reading. The crossed-out formula provisionally reads $\exists \xi \in \mathfrak{g}_0$. Several strokes overlap.

$$\exists \xi \underset{\mathfrak{g}_0}{\cap} : \quad \xi : \mathfrak{g}_{-1} \to \mathfrak{g}_{-1} \quad \text{has rank } 1 \tag{15}$$

(this has nothing to do with simplicity)

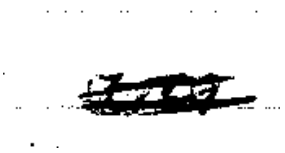
Addl. cond. from the lit. [crossed out: ...] something must be reductive,

—"———"— irreducible



Original source details: GELFAND-P000826

Uncertain reading. The abbreviation after "from" is provisionally set as «литр.» ("lit.").

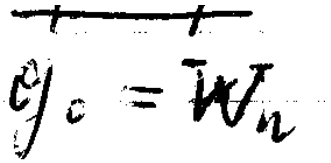


Original source details: GELFAND-P000826

Undeciphered passage. A short word is crossed out. Its original letters are illegible.

Example.

$\mathfrak{g}_0 = W_n, \qquad \mathfrak{g}_{-1} = \text{polynomials} = \mathbb{C}[x_1, \ldots, x_n]$



Original source details: GELFAND-P000826

Uncertain reading. There is a short separate stroke above the left edge of W. It may be a serif or a bar above the letter.

Cart. expans. (1st method) $\Rightarrow$ an alg. barely different from Ham. fields

(alg. of Poisson — there is a certain bracket)

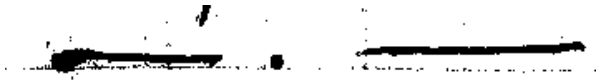


Diagram and typed description. ___ . ___

Original source details: GELFAND-P000826

Another method, leads to the algebras Kac-Moody [«Каца-» is written above “Kac”]

($\Rightarrow$ Kac–Macdonald id.)

A sq. (Cartan) matrix is needed

$$A = (n_{\alpha\beta}) \quad \text{of ord. } N, \quad \text{integer entries} \tag{16}$$

$$\begin{cases} n_{\alpha\alpha} = 2, \quad \text{off diag. — integers } \leq 0, \\ n_{\alpha\beta} \text{ and } n_{\beta\alpha} \text{ become } 0 \text{ only simultaneously.} \end{cases} \tag{17}$$

(18) Symmetrizability: bring to a symm. matrix in the ambient space by dilations along the coord. axes:

$$n_{\alpha\beta} \mapsto n_{\alpha\beta} \frac{\lambda_\alpha}{\lambda_\beta}$$

GELFAND-P000827

[source number: 11]

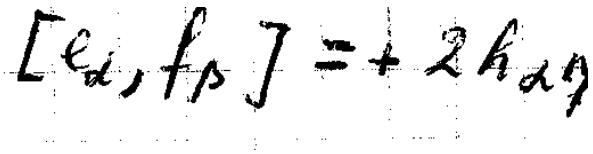
($\Rightarrow \exists$ inv. form)

A Lie alg. is def. by gen. and rels.

(19) Generators: $f_\alpha, h_\alpha, e_\alpha, \quad \alpha = 1, \ldots, N$

Relations:

$$\begin{cases} [h_\alpha, h_\beta] = 0, \quad [h_\beta, e_\alpha] = n_{\alpha\beta} e_\alpha, \quad [h_\beta, f_\alpha] = -n_{\alpha\beta} f_\alpha, \\ [e_\alpha, f_\beta] = 0 \text{ for } \alpha \neq \beta, \quad [e_\alpha, f_\beta] = +2h_\alpha \text{ for } \alpha = \beta, \end{cases} \tag{20}$$

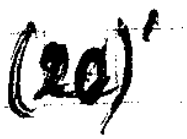


Uncertain reading. The final relation in (20) reads $+2h_\alpha$.

Original source details: GELFAND-P000827

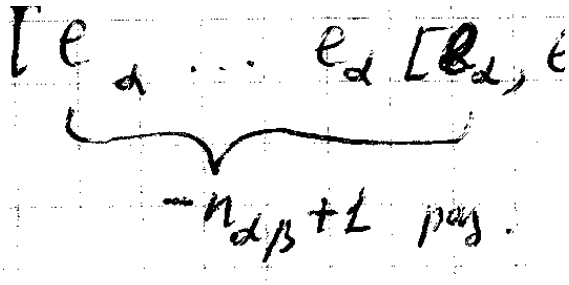
$$[\underbrace{e_\alpha \ldots e_\alpha [e_\alpha}_{-n_{\alpha\beta}+1 \text{ times}}, e_\beta] \ldots] = 0 \tag{20}'$$

[At right:] and similar conds. for f



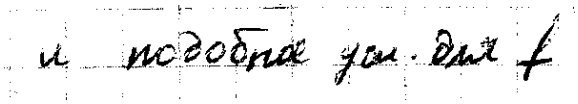
Uncertain reading. The number is retraced and reads $(20)'$. The preceding strokes are indistinct.

Original source details: GELFAND-P000827



Original source details: GELFAND-P000827

Uncertain reading. The innermost e_α is retraced. Below the brace is "$-n_{\alpha\beta}+1$ times".



Original source details: GELFAND-P000827

Uncertain reading. Before "for" the abbreviation provisionally reads «усл.» ("conds.").

$f_\alpha, h_\alpha, e_\alpha$ are lin. indep..

There are 2 gradings.

Fine

$$\Gamma = \mathbb{Z}\Big[\underset{\text{symbol}}{\alpha}\Big] \quad \text{— lattice gen. by the symbols } \alpha.$$

$$\begin{cases} \mathfrak{g} = \bigoplus\limits_{\gamma\in\Gamma} \mathfrak{g}_\gamma \\ f_\alpha, h_\alpha, e_\alpha \quad \text{homog. of degrees} \\ -\alpha, 0, \alpha \end{cases} \tag{21}$$

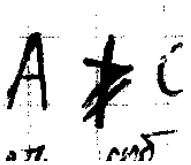
and the grading then extends uniquely.

$$\begin{cases} \underline{\text{Coarse}}: \quad \mathfrak{g} = \bigoplus\limits_{i\in\mathbb{Z}} \mathfrak{g}_i \\ \begin{array}{ccc} f_\alpha & h_\alpha & e_\alpha \\ -1 & 0 & 1 \end{array} \end{cases} \tag{22}$$

$$\underset{(\text{eigenv.} > 0)}{A} \gg 0 \quad \Rightarrow \quad \text{the alg. is finite-dimensional } (!) \tag{23}$$

$$\underset{\substack{(\text{there is an eigenv.} = 0,\\ \text{all} \geq 0)}}{A} \geq 0 \quad \Rightarrow \quad \begin{array}{c}\text{the alg. is } \infty\text{-dimensional, but growth 1} \\ (\dim \mathfrak{g}_i \text{ are bounded collectively})\end{array} \tag{24}$$

$$\underset{(\text{there is an eigenvalue} < 0)}{A} \not\geq 0 \quad \Rightarrow \quad \text{has exp. growth} \tag{25}$$



Original source details: GELFAND-P000827

Uncertain reading. A descending stroke crosses the inequality sign. It is provisionally set as $\not\geq$. A retraced $\neq$ is also possible.

For finite-dimensional algebras, Cartan matrs. also arise.

$$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & & 0 \\ & & \ddots & -1 \\ 0 & & -1 & 2 \end{pmatrix} \quad - \quad sl(n+1)$$

GELFAND-P000828

Roots of Kac–Moody algs.:

$$\gamma \neq 0, \quad \mathfrak{g}_\gamma \neq 0 \quad \Rightarrow \quad \gamma \text{ — a root}$$

Roots, e.g., the α — simple pos. roots

$$\text{pos. roots} — \sum z_\alpha \alpha, \quad z_\alpha \geq 0$$

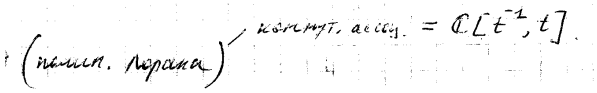
Most interesting.
degenerate $A \geq 0$ —
— then there is a center.

$$\begin{aligned} &\sum k_\beta n_{\alpha\beta} = 0 \quad \Rightarrow \\ &\Rightarrow \quad \sum k_\beta h_\beta \in \text{ the center} \end{aligned} \tag{26}$$

Example. Matrs. of ord. 2.

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \tag{27}$$

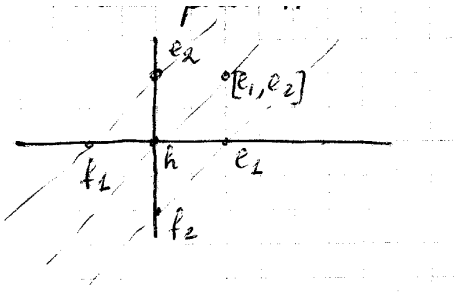
$$\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix} \quad \text{and its symmetric counterpart.} \tag{28}$$



Original source details: GELFAND-P000828

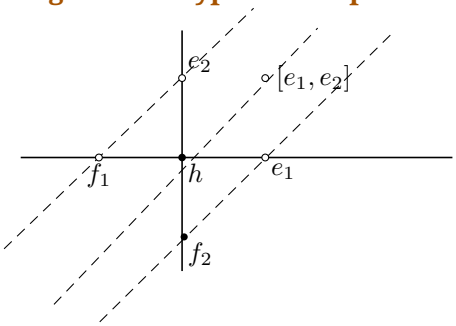
Diagram and typed description.
commut., assoc. $= \mathbb{C}[t^{-1}, t]$.
$(27) \Rightarrow sl(2) \otimes$ (Laurent polyn.)

Γ — a lattice in the plane.



Original source details: GELFAND-P000828

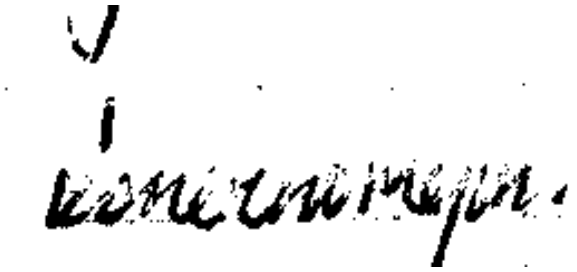
Diagram and typed description.



The broken diagonals are drawn without arrowheads. The point f_2 merges with the vertical line.

e, h, f multipl. in all poss. powers.

In general



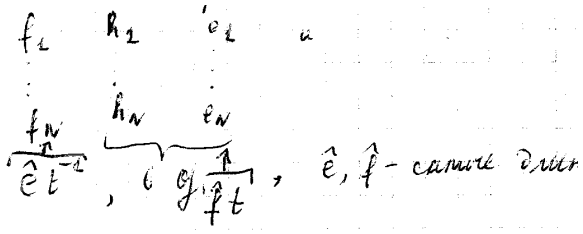
Original source details: GELFAND-P000828

Diagram and typed description.

$\mathfrak{g} \otimes \mathbb{C}[t^{-1}, t]$ — a Kac–Moody alg..

|

finite-dimension.



Original source details: GELFAND-P000828

Diagram and typed description.

$f_1 \quad h_1 \quad e_1$ and

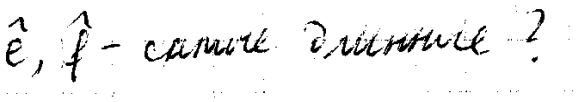
$\vdots \quad \vdots \quad \vdots$

$f_N \quad h_N \quad e_N$

$\uparrow \quad \underbrace{\qquad\qquad}_{\text{in } \mathfrak{g}}$

$\hat{e}t^{-1} \qquad \hat{f}t$

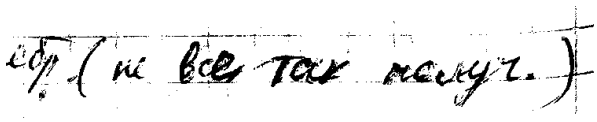
$\hat{e}, \hat{f}$ — the longest ?



Original source details: GELFAND-P000828

Uncertain reading. The manuscript has a question mark immediately after “longest”.

There are also other ways of constr. these algebras (not all are obtained this way)



Original source details: GELFAND-P000828

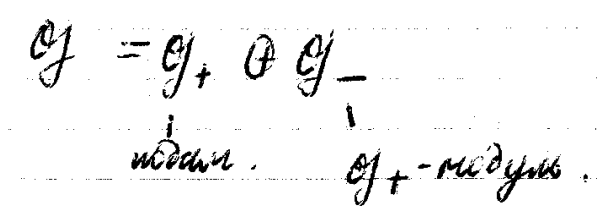
Uncertain reading. The opening words in parentheses read «не все» (“not all”). «все так» is retraced.

GELFAND-P000829

[source number: 12]

2nd constr. Suppose

There is an involution in $\mathfrak{g}$: $\tau : \mathfrak{g} \to \mathfrak{g}$. Then



Original source details: GELFAND-P000829

Diagram and typed description.

$\mathfrak{g} = \mathfrak{g}_+ \oplus \mathfrak{g}_-$

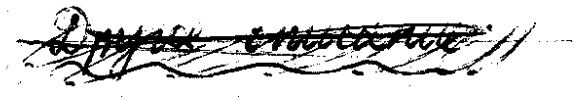
| \

subalg. $\mathfrak{g}_+$-module.

The algebra is a subalgebra of $\mathfrak{g} \otimes \mathbb{C}[t, t^{-1}]$
of the form

$$\ldots \oplus \mathfrak{g}_- t^{-1} \oplus \mathfrak{g}_+ \oplus \mathfrak{g}_- t \oplus \mathfrak{g}_+ t^2 \oplus \ldots$$

$\mathfrak{g} = sl(3)$, τ : refl. in the diag. (−transpose)
[Crossed out: ~~Other algebras~~]



Original source details: GELFAND-P000829

Uncertain reading. The crossed-out heading provisionally reads “Other algebras”. A wavy line is drawn underneath.

[At left, crossed out: $\mathfrak{g}$]

Original source details: GELFAND-P000829

Uncertain reading. A lower loop is visible under the strokes. The sign resembles $\mathfrak{g}$.

Other Lie algebras.
gen. $\varepsilon_i,\ i \in \mathbb{Z}, \quad [\varepsilon_i, \varepsilon_j] = a_{ij}\varepsilon_{i+j}$
there are 3 simple ones
1st:

$$a_{ij} = \begin{cases} 1, 0, -1 \\ \equiv j - i \mod 3 \end{cases}$$

These 2 constrs. exhaust everything up to grading.

GELFAND-P000830

19/III - 79 Gelfand Sem.
Ogievetsky (Viktor Isaakovich)
Supergravity
(= gravity + supersymmetry)
$P_m,\ L_{mn}$ — generators of the Poincaré grp.
[At right:]

P_m	— 4 of them
L_{mn}	— 6 of them
total	10 of them

$Q_\alpha, \overline{Q}_\alpha$ — spinor generators

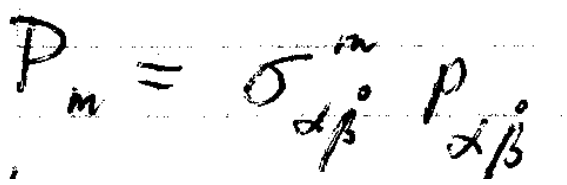
$$\alpha = 1, 2$$

Original source details: GELFAND-P000830

Uncertain reading. The initial letters of “spinor” are retraced. The earlier strokes are illegible.

$$\{Q_\alpha, \overline{Q}_{\dot\alpha}\} = 2\sigma^m_{\alpha\dot\alpha} P_m; \tag{1}$$

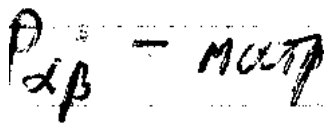
$$P_m = \sigma^m_{\alpha\dot\beta} P_{\alpha\dot\beta}$$



Uncertain reading. The upper index on σ provisionally reads m, possibly n. The short stroke over α on the second P resembles a detached tip of the letter. A dot is also possible.

Original source details: GELFAND-P000830

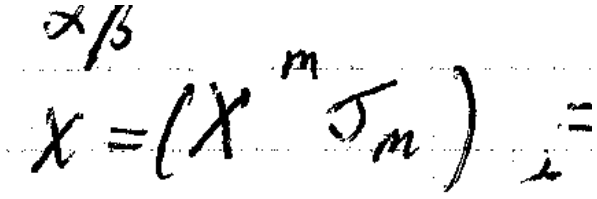
[At right:] $P_{\alpha\dot\beta}$ — matr. of ord. 2, corresp. to the vector P.



Uncertain reading. Faint marks above β in the right-hand annotation may be the trace of a dot. A separate dot above α cannot be distinguished.

Original source details: GELFAND-P000830

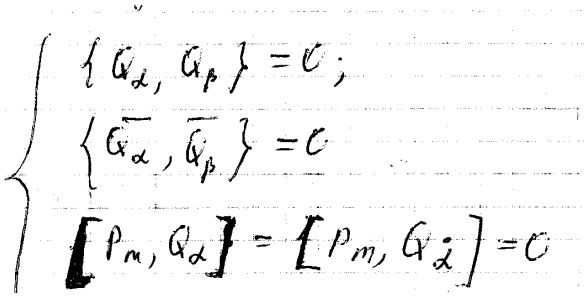
$$\underbrace{\mathbf{1}, \sigma_i}_{\substack{\text{matrs.}\\ \text{Pauli}}}; \qquad x = (X^m \sigma_m), = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}$$



Uncertain reading. The first letter provisionally reads x, possibly χ. A short mark after the closing parenthesis is provisionally set as a comma.

Original source details: GELFAND-P000830

$$\begin{cases} \{Q_\alpha, Q_\beta\} = 0; \\ \{\overline{Q}_\alpha, \overline{Q}_\beta\} = 0 \\ [P_m, Q_\alpha] = [P_m, Q_{\dot\alpha}] = 0 \end{cases} \tag{1}$$



Original source details: GELFAND-P000830

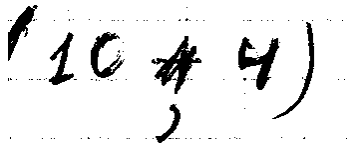
Uncertain reading. The number (1) is repeated. There are no visible dots on the indices in the middle line and no visible bar over $Q_{\dot\alpha}$ in the last line.

Thus a Lie superalgebra has been constructed whose even part is the Poincaré algebra.
Take 2 reps. of the Lorentz grp. $\left(\frac{1}{2}, 0\right),\ \left(0, \frac{1}{2}\right)$ (2-dimensional)
(it is $SL(2, \mathbb{C})$ 2 reps. — stand. and conjugate)
(Lorentz grp.: [crossed out: A] $A \to X^{+} A X, \quad X \in SL(2, \mathbb{C})$
A — Hermitian 2×2, det — Lorentz metric.
we get a map into the Lorentz grp.)

Uncertain reading. The crossed-out sign before the map resembles A.

Original source details: GELFAND-P000830

Thus, 14-dimensional $(10, 4)$ — a superalgebra



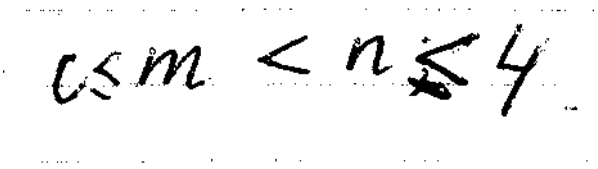
Uncertain reading. The sign between 10 and 4 is crossed out and resembles a plus. The comma is inserted below.

Original source details: GELFAND-P000830

L_{mn} skew-symm., there are [crossed out: ...] for $0 \leq m < n < 4$. [Below L_{mn}:] (since these are rotations)

Undeciphered passage. The entry after "there are" is crossed out. Its original signs are illegible.

Original source details: GELFAND-P000830



Uncertain reading. The lower stroke in the last inequality is crossed out. The provisional reading is < 4. ≤ 4 is also possible.

Original source details: GELFAND-P000830

Suppose there are several spinor operators
Precisely: there are $\mathfrak{g}_0$ and $\mathfrak{g}_1$
$\mathfrak{g}_0$ — a Lie alg.; $\mathfrak{g}_1$ — $\mathfrak{g}_0$-module.
We want to constr. $\mathfrak{g}_0 \oplus \mathfrak{g}_1$. Write down the relations: in $\mathfrak{g}_1$ one needs to introduce.

GELFAND-P000831

[source number: 13]

In our case it is unique.

Now take several:

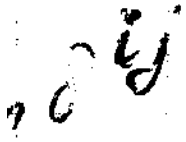
$$\left(\frac{1}{2},\frac{1}{2}\right) \qquad (1,0)+(0,1) \qquad \text{[crossed out: } \cancel{\left(\tfrac{1}{2},0\right)}\text{]},$$

Uncertain reading. The crossed-out pair at the right provisionally reads $\left(\frac{1}{2},0\right)$.

Original source details: GELFAND-P000831

One can modify system (1).

$$\{Q^i_\alpha, \bar{Q}^j_{\dot\alpha}\} = 2\sigma^m_{\alpha\dot\alpha} P_m \delta^{ij} \tag{2}$$



Uncertain reading. The last factor is provisionally set as δ^{ij}. The upper stroke is detached from the lower loop.

Original source details: GELFAND-P000831

i, j — indices of internal symmetry.

We get a nontrivial mixing of internal symmetry with the space group —

proton + antiproton = momentum

(impossible in ordinary Lie algebras — there is a theorem)

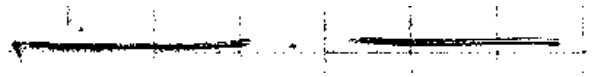


Diagram and typed description.

Original source details: GELFAND-P000831

Consider [crossed out: ...] supergravity — alg. (1).

Undeciphered passage. A short entry before “supergravity” is crossed out with dense diagonal strokes. Its letters are illegible.

Original source details: GELFAND-P000831

Realization (in Minkowski space):

$$P_m = -i\frac{\partial}{\partial x^m} \tag{3}_1$$

$$L_{mn} = -i(x_m\partial_n - x_n\partial_m), \qquad \partial_n = \frac{\partial}{\partial x^n}. \tag{3}_2$$

Q is not realized in ordinary space

We want to extend the space. New variables:

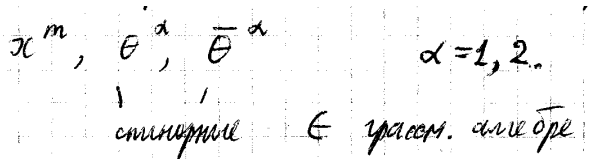


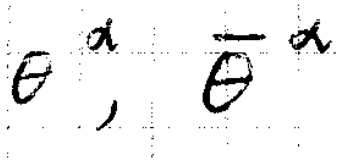
Diagram and typed description.

x^m, θ^α, $\bar\theta^\alpha$ $\alpha = 1, 2.$

spinorial $\in$ the Grassmann algebra

Original source details: GELFAND-P000831

$$\{\theta, \theta\} = 0$$



Uncertain reading. There is no discernible dot over the α on $\bar\theta$ in the list of variables.

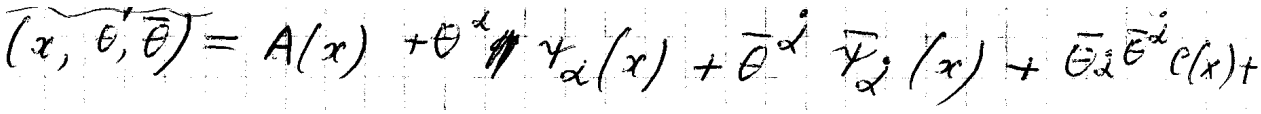
Original source details: GELFAND-P000831

Functions on this space:

$$\begin{aligned}\varphi(x,\theta,\bar\theta) = A(x) + \theta^\alpha \dots \psi_\alpha(x) + \bar\theta^{\dot\alpha}\bar\psi_{\dot\alpha}(x) + \bar\theta_{\dot\alpha}\bar\theta^{\dot\alpha}C(x) + \\ + \theta^\alpha\theta_\alpha B(x) + \theta\sigma^\alpha\bar\theta h_\alpha(x) + \dots + \theta^\alpha\theta_\alpha\bar\theta_{\dot\beta}\bar\theta^{\dot\beta}D(x)\end{aligned} \tag{4}$$

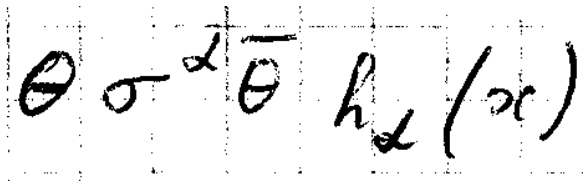
Undeciphered passage. A sign is crossed out between θ^α and $\psi_\alpha(x)$. It may be φ or ψ.

Original source details: GELFAND-P000831



Original source details: GELFAND-P000831

Uncertain reading. A + occurs both at the end of the first manuscript line and at the start of the second.



Uncertain reading. Both indices in $\theta\sigma^\alpha\bar\theta h_\alpha(x)$ are provisionally read as α. The Latin a is possible.

Original source details: GELFAND-P000831

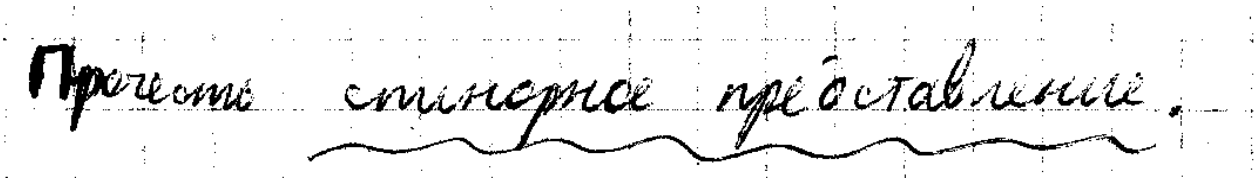
Original source details: GELFAND-P000831

Uncertain reading. The ellipsis before the last term is inserted from below with a caret.

There are 4 variables θ.

Metric tensor $\varepsilon_{\alpha\beta}$: $\varepsilon_{11} = \varepsilon_{22} = 0, \qquad \varepsilon_{12} = -\varepsilon_{21} = 1$

Simplest spinor representation.



Original source details: GELFAND-P000831

Uncertain reading. The Russian word for “simplest” may be abbreviated. The wavy underline begins at “spinor”.

$$\begin{cases} Q_\alpha = -i\dfrac{\partial}{\partial\theta^\alpha} - (\sigma^m\bar\theta)_\alpha\overline{\cdots}\partial_m \\ \bar Q_{\dot\alpha} = i\bar\partial_{\dot\alpha} + (\theta\sigma^m)_{\dot\alpha}\partial_m \end{cases} \tag{5}$$

Original source details: GELFAND-P000831

Undeciphered passage. Writing is visible between $(\sigma^m\bar\theta)_\alpha$ and ∂_m under a vertical zigzag. The original sign is illegible.

$\bar\partial_{\dot\alpha}$ — differentiation with respect to a Grassmann variable

GELFAND-P000832

This realizes the system of commutation relations (1).

Action of the group in superspace itself.

(the action on functions is given)

$$x'^m = x^m + c^m$$

under supertranslations:

$$\begin{cases} x^{m'} = x^m + i(\theta\sigma^m\bar\lambda - \lambda\sigma^m\bar\theta) \\ \theta^{\alpha'} = \theta^\alpha + \lambda^\alpha \\ \bar\theta^{\dot\alpha'} = \bar\theta^{\dot\alpha} + \bar\lambda^{\dot\alpha} \end{cases} \tag{6}$$

Original source details: GELFAND-P000832

Uncertain reading. The first θ on the right in the line for $x^{m'}$ has been retraced. It reads θ. Any earlier sign is indiscernible.

$\lambda, \bar{\lambda}$ — infinitesimal shifts of the Grassmann variables.

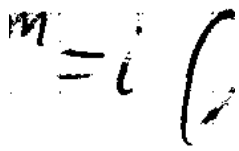
Perform the same transformation a second time with different parameters. We get a pure translation in x^m.

Set:

$$\delta_\lambda x^m = x^{m'} - x^m = i(\theta\sigma^m\bar{\lambda} - \lambda\sigma^m\bar{\theta})$$

Then

$$(\delta_\lambda\delta_\mu - \delta_\mu\delta_\lambda)x^m = i(\mu\sigma^m\bar{\lambda} - \lambda\sigma^m\bar{\mu}) \tag{7}$$



Uncertain reading. The right-hand side has i, with no visible 2 preceding it.

Original source details: GELFAND-P000832

(i.e., a pure translation)

Transformation law of the superfield

$\psi'(x') = \psi(x)$ — the ordinary transformation law of a scalar field

For supertranslations of the (scalar) superfield:

$$\varphi'(x', \theta', \bar{\theta}') = \varphi(x, \theta, \bar{\theta}) \tag{8}$$

For the fields $A(x), \ldots$ from the expansion (4) this gives complicated formulas.

Consider some further new variables:

$$\begin{cases} x_L^m = x^m + i\theta\sigma^m\bar{\theta} \\ \delta x_L^m = 2i\theta\sigma^m\bar{\lambda} \end{cases} \qquad \delta\theta^\alpha = \lambda^\alpha \tag{9}$$

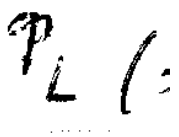
(there is only θ and no $\bar{\theta}$)

Special fields (**chiral**)

Uncertain reading. The middle letters in the Russian word for “chiral” are retraced. The earlier strokes there cannot be distinguished.

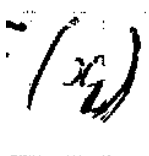
Original source details: GELFAND-P000832

$$\varphi_L(x_L, \theta) = A(x_L) + \theta^\alpha\psi_\alpha(x_L) + \theta^\alpha\theta_\alpha F(x_L) \tag{10}$$



Uncertain reading. The field symbol is provisionally set as φ_L. It may be Φ_L.

Original source details: GELFAND-P000832



Uncertain reading. The final x provisionally has subscript L. Its lower stroke touches the closing parenthesis.

Original source details: GELFAND-P000832

$$\begin{cases} x_R^m = x^m - i\theta\sigma^m\bar{\theta} \\ \delta x_{\underline{\underline{R}}}^m = -2i\lambda\sigma^m\bar{\theta} \end{cases} \qquad \delta\bar{\theta}^{\dot{\alpha}} = \bar{\lambda}^{\dot{\alpha}} \tag{11}$$

Uncertain reading. The R in the line for δx_R^m is underlined twice.

Original source details: GELFAND-P000832

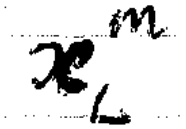
GELFAND-P000833

[source number: 14]

We obtain.

$$x^m = \frac{1}{2}(x_L^m + x_R^m)$$

$$x_L^m - x_R^m = 2i\theta\sigma^m\bar{\theta}$$



Uncertain reading. The initial x in the second formula has been retraced. The earlier strokes are indiscernible.

Original source details: GELFAND-P000833

Uncertain reading. A faint trace to the right of $\bar{\theta}$ lies along the paper ruling. It may be a full stop.

Original source details: GELFAND-P000833

Diagram and typed description.

. . . . . .

Divider of 26 dots. The fine horizontal stroke above the dots coincides with the paper ruling. Its relation to the divider is unclear.

Original source details: GELFAND-P000833

GELFAND-P000834

26.3.79 Gelfand Sem.

Diagram and typed description.

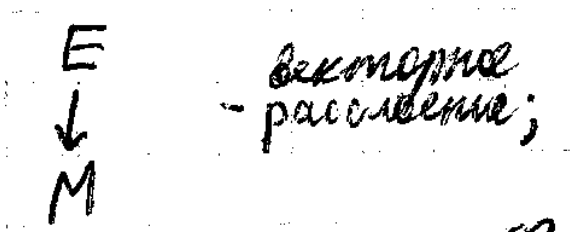
Overlapping horizontal and angular strokes appear below the date. No individual letters can be made out.

Original source details: GELFAND-P000834

Lebedev (Mitya)

On results of Ritt and Kirillov in differential algebra.

M — a smooth manifold, dim $M = n$



Original source details: GELFAND-P000834

Diagram and typed description.

E — vector bundle; M

$\Gamma^\infty(E)$ — smooth sections.

$$s_1, s_2 \in \Gamma^\infty(E)$$

Introduce a bilinear operation: $[\cdot, \cdot]$:

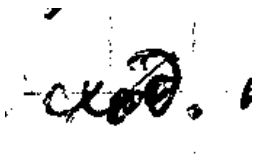
$$[s_1, s_2] \in \Gamma^\infty(E)$$

Original source details: GELFAND-P000834

Uncertain reading. The sign before $\Gamma^\infty(E)$ is retraced and provisionally read as $\in$.

a) bilin., skew-symm., Jacobi id.

b) cont. in the topol. of unif. conv. of derivs. on compact sets

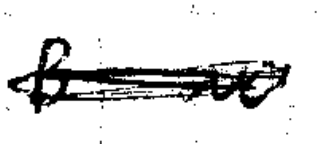


Original source details: GELFAND-P000834

Uncertain reading. The Russian abbreviation for convergence provisionally reads «схд.». A separate «о» may be present between «х» and the retraced «д».

c) $\operatorname{supp}[s_1, s_2] \subset \operatorname{supp} s_1 \cap \operatorname{supp} s_2$ (locality)

[crossed out: ...] This is called a differential Lie algebra.



Original source details: GELFAND-P000834

Undeciphered passage. The beginning of the crossed-out fragment has a loop resembling «В». The remaining letters are illegible.

GELFAND-P000835

[source number: 20]

The same definition in loc. coordinates

Let F — the fiber; dim $F = 1$

$x_1, \dots, x_n$ — coords. on the base.

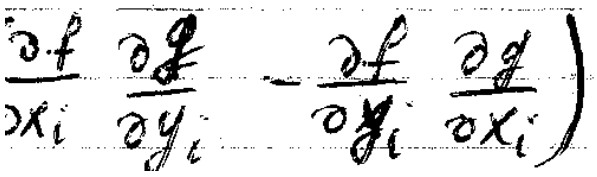
Then locally

$$[f, g] = \sum c_{k\ell}(x) \partial^k f \cdot \partial^\ell g, \qquad k, \ell \text{ — multi-indices} \tag{1}$$

Kirillov's main result: classification of such algebras.

Theorem 1. All loc. algebras with a fiber of dim. 1 are exhausted by two (essentially)

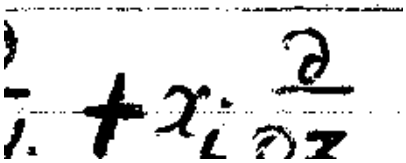
$$\text{a)} \quad M = \mathbb{R}^{2n}, \qquad [f, g] = \sum \left(\frac{\partial f}{\partial x_i} \frac{\partial g}{\partial y_i} - \frac{\partial f}{\partial y_i} \frac{\partial g}{\partial x_i} \right) \tag{2}$$



Uncertain reading. Overlapping strokes appear in the first numerator with g and the second denominator with y_i. The earlier letter forms cannot be distinguished.

Original source details: GELFAND-P000835

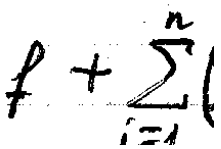
$$\text{b)} \quad M = \mathbb{R}^{2n+1}, \quad X_i = \frac{\partial}{\partial x_i}, \quad Y_i = \frac{\partial}{\partial y_i} + x_i \frac{\partial}{\partial z}, \quad Z = \frac{\partial}{\partial z} \text{ — vect. fields} \tag{3}$$



Uncertain reading. A plus is visible before $x_i \partial/\partial z$ in the entry for Y_i.

Original source details: GELFAND-P000835

$$[f, g] = fZg - gZf + \sum_{i=1}^{n} (X_i f \cdot Y_i g - X_i g \cdot Y_i f) \tag{3}_1$$



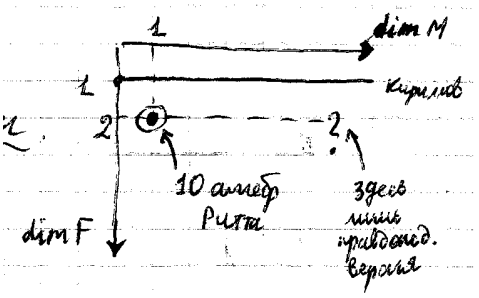
Uncertain reading. A plus is visible before the sum in $(3)_1$.

Original source details: GELFAND-P000835

All the others are formally isomorphic to them.

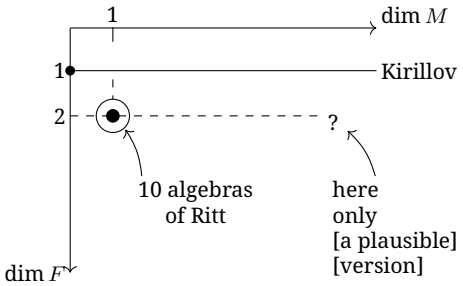
Ritt.

Differential Lie group and diff. Lie algebra.



Original source details: GELFAND-P000835

Diagram and typed description.



At the right margin. The last two lines at the right are provisionally read as «правдопод. / версия» (“plausible / version”).

Consider series

$$\sigma_1 = \sum_{i\geq 1} a_i\, a\, \xi^{-(i+1)} \tag{4}$$

Original source details: GELFAND-P000835

Uncertain reading. A separate dark sign resembling a occurs between a_i and ξ.

[crossed out: ...] and the commutator

Original source details: GELFAND-P000835

Undeciphered passage. A short entry at the left is crossed out. The underlying marks are illegible.

$$[\sigma_1, \sigma_2] = \sigma_1 \cdot \sigma_2 - \sigma_2 \cdot \sigma_1 \tag{5}$$

(here $\dim M = 1$, $\dim F = \infty$).

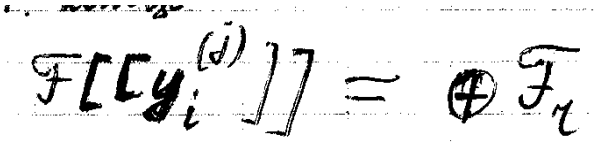
$[\xi, x] = 1$ — this determines everything.

Original source details: GELFAND-P000835

Diagram and typed description.

$\mathcal{F}$ — diff. field of char. 0, “$\partial = \frac{\partial}{\partial x}$”
$y_1(x), \dots, y_n(x)$ — diff. independent
Consider the ring

$$\mathcal{F}[[y_i^{(j)}]] \simeq \bigoplus \mathcal{F}_\tau, \qquad \tau \text{ — “order” — degree — number of factors} \tag{6}$$

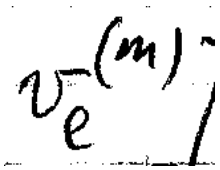


Original source details: GELFAND-P000835

Uncertain reading. The relation sign has a wavy upper stroke and is provisionally set as $\simeq$. The index is provisionally read as τ.

Formal diff. Lie group ≡ form. group law ≡ set of fns.

$$U_i \in \mathcal{F}[[u_i^{(j)}, v_\ell^{(m)}]] \tag{7}$$



Original source details: GELFAND-P000835

Uncertain reading. A short horizontal stroke at the upper right of v is joined to the letter. It may be a separate bar.

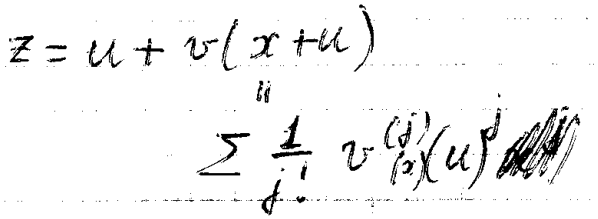
Satisfies 2 axioms:

a) $U_i(\bar{u}, 0) = \bar{u}_i, \qquad U_i(0, \bar{v}) = \bar{v}_i$

b) associativity.

GELFAND-P000836

Example.



Original source details: GELFAND-P000836

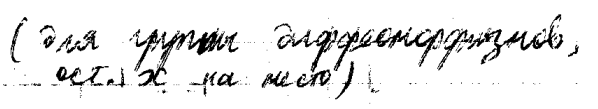
Diagram and typed description.
$z = u + v(x+u)$
$\sum \frac{1}{j!} v^{(j)}_{(x)} (u)^j \cdots$

Below $v(x+u)$, two vertical strokes stand between the formula and the series.

Original source details: GELFAND-P000836

Uncertain reading. A further entry after $(u)^j$ is crossed out. It may read u^j.

(for the group of diffeomorphisms, leav. $\downarrow x$ in place)



Original source details: GELFAND-P000836

Uncertain reading. A short vertical stroke occurs between the Russian abbreviation «ост.» and x. It is provisionally rendered as a downward arrow.

Form. series:

$$U = \sum_{k=0}^{\infty} P_k,$$

$$P_k = \text{homog. polyn. of deg. } k \text{ in } u_1, \ldots, u_n, u_1^{(1)}, \ldots, u_n^{(1)},$$

over a diff. field (a field of functions with diff.)

Uncertain reading. The start of the Russian word for “functions” is overwritten. Its earlier form is illegible.

Original source details: GELFAND-P000836

Theorem 2

Classific. of diff. groups with a fiber of dim. 2

A^h.

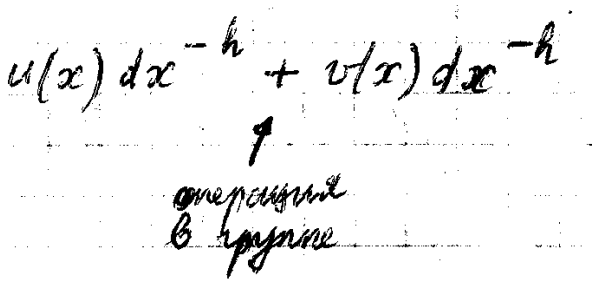


Diagram and typed description.

$u(x)\,dx^{-h} + v(x)\,dx^{-h}$

↑

operation
in the group

Original source details: GELFAND-P000836

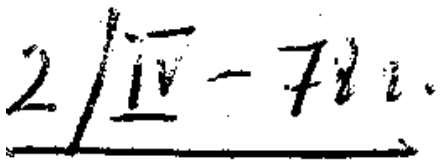
$$u \circ v = u + v$$

$$\Pi: \qquad u \circ v = u + v(x + u)$$

All groups are ext. of Π by A^h.

GELFAND-P000837

2/IV–78 Gelfand Sem.



Uncertain reading. The last digit of the year is provisionally read as 8. It may be 9.

Original source details: GELFAND-P000837

K. Shabat. From the work of Griffiths and Chern.

Uncertain reading. The initial is provisionally read as the Russian «К.». It may be «Ю.».

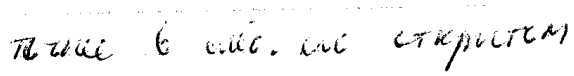
Original source details: GELFAND-P000837

A web — a finite number of foliations with transv. leaves. It is constructed on a manifold.
(net) — Web

$$X \subset P^n, \qquad \dim X = m, \qquad \deg X = d.$$

[Below X:] proj. set (algebr.)

A web in $\mathrm{Gr}(m, n-m)$, more precisely in some open subset of it:



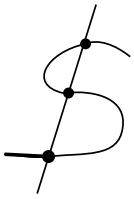
Original source details: GELFAND-P000837

Uncertain reading. The Russian phrase between “in” and “open” provisionally reads «нек. его» (“some … of it”). The middle is faint.

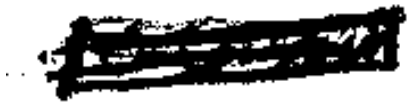
$$U = \{H \mid H \cdot X = d \text{ distinct points}\} \quad \text{— an open set} \tag{1}$$

Original source details: GELFAND-P000837

Diagram and typed description.



([crossed out: ~~…~~] when there is multiplicity, discard it)



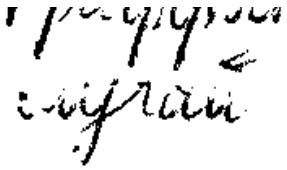
Original source details: GELFAND-P000837

Undeciphered passage. The first word inside the parentheses is crossed out. Its letters are indiscernible.

I.M.G.

Crofton’s formula: length of a curve = average number of points at which a line intersects it (average over lines with respect to an invariant measure).

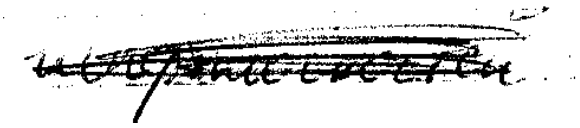
Griffiths and Chern, inspired by Penrose’s work, generalized this to the multidimensional [inserted below at left: ← case].



Original source details: GELFAND-P000837

Uncertain reading. The word “case” is below the line at the left. A short arrow above it points left.

The leaf passing through the point $H_0 \in U$, consists of [crossed out: ~~surfaces~~].

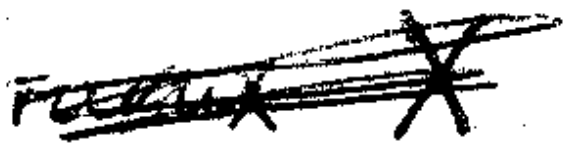


Original source details: GELFAND-P000837

Uncertain reading. The crossed-out Russian word is provisionally read as «поверхностей» (“surfaces”).

[crossed out: ~~points of X~~] planes

$$\{H \in U \mid \exists i \in \{1, 2, \ldots, d\};\ P_i \in H\}.$$

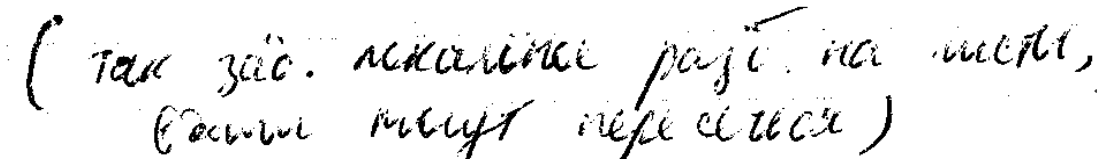


Original source details: GELFAND-P000837

Uncertain reading. "Points of X" is crossed out at the start of the line. There is a short stroke below the first P_i.

$$H_0 \cdot X = P_1 + \ldots + P_d$$
$$\{H \in U \mid \exists i \in \{1, 2, \ldots, d\},\ P_i \in H\} \tag{2}$$

This is a foliation of codimension 1. (so here the planes are div. into leaves, they themselves can intersect)



Original source details: GELFAND-P000837

Uncertain reading. The Russian abbreviations provisionally read «зд.» ("here") and «разд.» ("div.").

I.M.G. [crossed out: ~~Questions~~]

Original source details: GELFAND-P000837

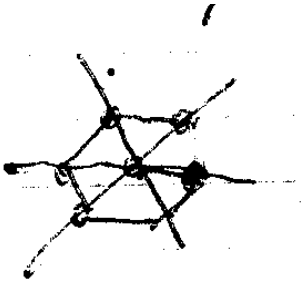
Uncertain reading. The crossed-out word provisionally reads «Вопросы» ("Questions").

Problem: there are 3 foliations (transv.) When can they be straightened? [Inserted below "there are 3 foliations":] in the plane.

(2 always!!)

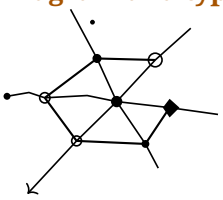
3 can be straightened $\Leftrightarrow$ it is hexagonal, i.e.

≈ 1927



Original source details: GELFAND-P000837

Diagram and typed description.



The redraw follows the visible lines, small circles and thickened nodes. The order of traversal and the direction of some small arrows are indistinct.

— such a path (infinitesimal) must be closed. — something like curvature.

Diagram and typed description.

Original source details: GELFAND-P000837

GELFAND-P000838

[source number: 29]

Now $m = 1$.

$$X \subset P^n$$
$$|$$
proj. curve

$\check{P}^n$ — dual proj. space.

$$\check{\mathbb{P}}^n \supset U = \{H \mid H \circ X = d \text{ distinct points}\} \tag{3}$$

Uncertain reading. A small oval is visible between H and X, provisionally set as $\circ$.

Original source details: GELFAND-P000838

On U we get a web of codimension 1.

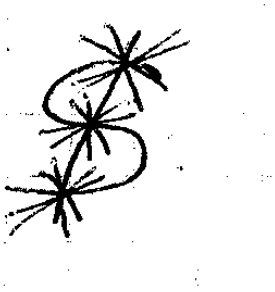
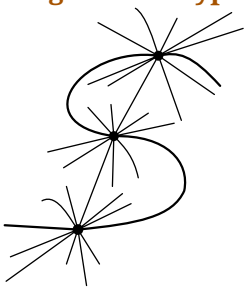


Diagram and typed description.



Original source details: GELFAND-P000838

The redraw follows the three centers and the discernible strokes. Some strokes merge, so their exact number is uncertain.

Abelian eq. for the web.

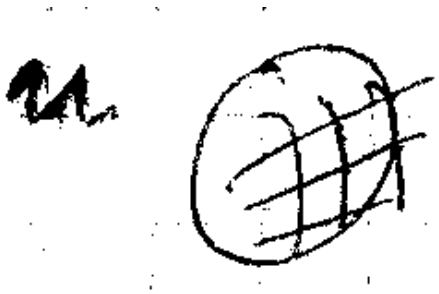
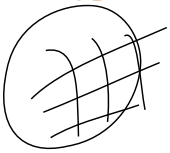


Diagram and typed description.

$[u]$



Original source details: GELFAND-P000838

Uncertain reading. The sign to the left of the oval is provisionally set as u. It may be U.

Original source details: GELFAND-P000838

$$\begin{cases} u_1 = \text{const.} \\ \vdots \\ u_n = \text{const} \end{cases} \tag{4}$$

The Ab. eq. for the web has the form:

$$\sum_{i=1}^{d} f_i(u_i)\, du_i = 0, \qquad \sum_{i=1}^{d} f_i(u_i)\, du_i \in \Omega^1(U) \tag{5}$$

There is a dependence among the normal vectors (cotang. vectors), and the coefficients have the form $f_i(u_i)$.

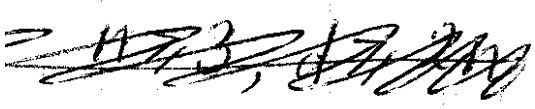
Rank of the web $\underset{\text{def}}{=}$ max. number of lin. independent Abelian eqs., that it satisfies.

Prop. 1 The rank of the web constructed from a curve of genus g, equals g.

[crossed out: ~~Another theorem 1 — Under the conditions~~ $\sim\!\!\sim n \geq 3,\ d \geq 2n \sim\!\!\sim$]

Uncertain reading. The first crossed-out word provisionally reads «Еще» (“Another”).

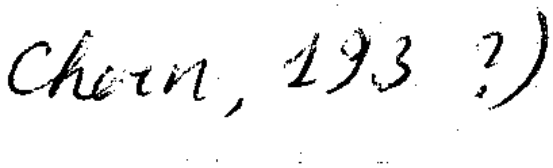
Original source details: GELFAND-P000838



Uncertain reading. The crossed-out conditions provisionally read $n \geq 3,\ d \geq 2n$.

Original source details: GELFAND-P000838

Theorem 1 (Chern, 193 ?)



Uncertain reading. A question mark follows 193 in the manuscript.

Original source details: GELFAND-P000838

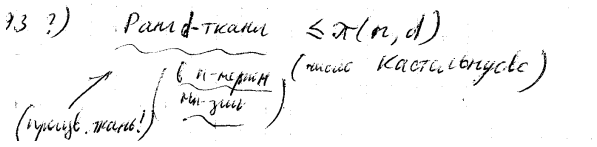


Diagram and typed description.

Rank of the d-web $\leq \pi(n,d)$ (Castelnuovo number)

(in an n-dim. set)

(arbitr. web!)

Original source details: GELFAND-P000838

$$\pi(n,d) \underset{\text{def}}{=} \text{max. genus of a curve of degree } d \text{ in } \mathbb{P}^n, \qquad n, d \geq 2. \tag{6}$$

There is a short formula

$$\pi(n,d) = m\left(d - \frac{1}{2}(m+1)(n-1) - 1\right), \quad \text{where } m = \left[\frac{d-1}{n-1}\right]. \tag{7}$$

GELFAND-P000839

Theorem 2.

(main)

If $n \geq 3,\ d \geq 2n$ and the d-web has rank $\pi(n,d)$,

then

1) It can be linearized

2) It extends to a web obtained from $X \subset \mathbb{P}^n$.

The connection between such eqs. and Abelian differentials is explained.

Reference:

(Sonderdruck aus) Jber. d. Dt. Math.-Verein, 80 (1978), 13–110

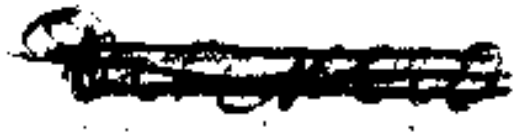
[Under «Sonderdruck aus»: offprint from.]

(Jahresberichte der Deutsche[...] Math. Vereinigung)

Original source details: GELFAND-P000839

Uncertain reading. The ending after “Deutsche” is uncertain. “Deutschen” and “Deutscher” are possible.

[crossed out: ...]



Original source details: GELFAND-P000839

Undeciphered passage. A word below the reference is crossed out. Its initial loop is visible, and the remaining letters are obscured.

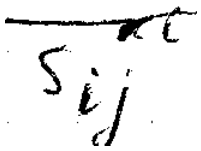
A geometry is being developed for

Original source details: GELFAND-P000839

Diagram and typed description.
projective connections.

The left stroke has a left-pointing arrow. A separate stroke runs under the words at right.

Zamolodchikov A. (Sasha) Suppose there is $S_{ij}^{k\ell}(\lambda)$



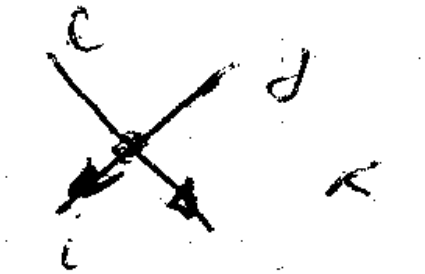
Original source details: GELFAND-P000839

Uncertain reading. The upper indices are provisionally read as k, ℓ. They touch the stroke from the preceding line.

Suppose there is also a picture:
L lines, angles $\lambda_1, \dots, \lambda_L$
λ_i in increasing order
(from some line
[crossed out: ~~Then~~]

Uncertain reading. The crossed-out word provisionally reads «То» (“Then”).

Original source details: GELFAND-P000839



Original source details: GELFAND-P000839

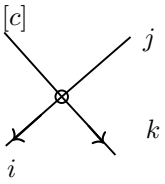
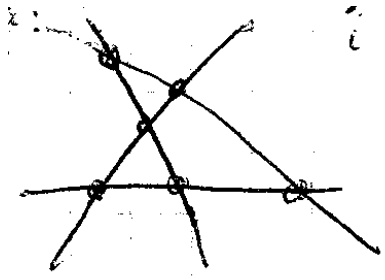


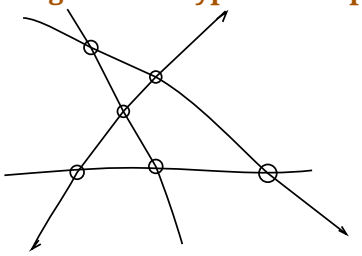
Diagram and typed description. The upper-left sign resembles c and may be ℓ.

indices $= 1, 2, \dots, N$



Original source details: GELFAND-P000839

Diagram and typed description.

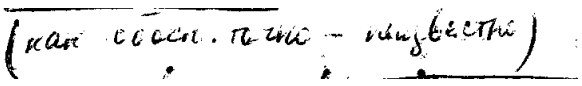


There are six circled intersections. Short strokes at the ends are drawn as visible. Some arrow directions are unclear.

To a node — a matrix; to tails — indices, I sum over the links.
We get a matrix (large). What is its behavior as $L \to \infty$?

$$Z^{j_1 \dots j_L}_{i_1 \dots i_L}(\lambda_1, \dots, \lambda_L).$$

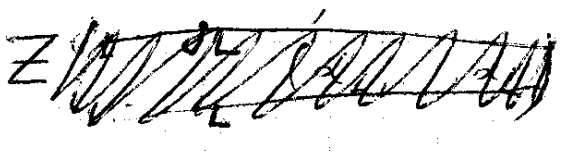
Problem 1. There is an asymptotic formula
(how to calc. it exactly — unknown)



Original source details: GELFAND-P000839

Uncertain reading. The Russian words provisionally read «сосч. точно» (“calculate exactly”). «сосч. её» (“calculate it”) is possible.

$$\ln Z_{\dots} \sim \text{proport. to the number of nodes} \tag{1}$$



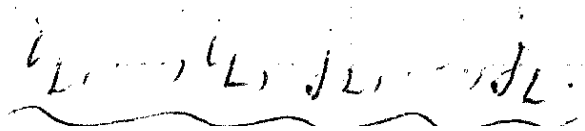
Uncertain reading. After Z the hatching covers signs resembling the indices $j_1 \dots j_L$, $i_1 \dots i_L$ and arguments $(\lambda_1, \dots, \lambda_L)$. Some of the signs cannot be made out.

Original source details: GELFAND-P000839

$$Z \underset{\text{df}}{=} Z^{i_1 \dots i_L}_{i_1 \dots i_L} \qquad (\text{“trace”}) \qquad (\text{sum over repeated indices}) \tag{2}$$

Each line $\rightsquigarrow$ parameter λ_j

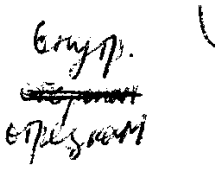
(the rays are numbered clockwise $i_1, \dots, i_L, j_1, \dots, j_L$.)



Uncertain reading. The first subscripts on i and j are provisionally set as the digit 1. Their shapes also resemble L.

Original source details: GELFAND-P000839

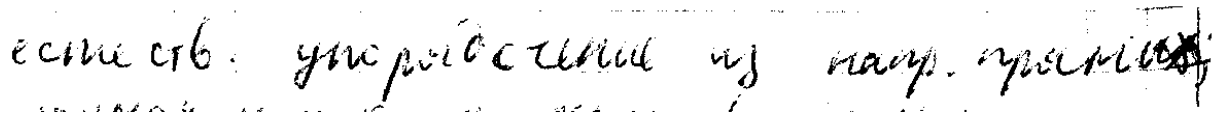
$$Z^{j_1 \dots j_L}_{i_1 \dots i_L} = \sum_{\substack{\text{internal} \\ \text{segments}}} \left(\prod_{\substack{\text{points of} \\ \text{intersection}}} S^{**}_{**}(\lambda_a - \lambda_b) \right) \tag{3}$$



Undeciphered passage. The middle line below the sum is crossed out. The word is illegible.

Original source details: GELFAND-P000839

[At right below the formula:] natural ordering from the dir. of the lines



Original source details: GELFAND-P000839

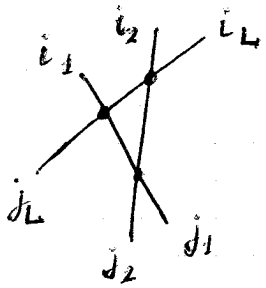
Uncertain reading. The ending of the Russian word for “lines” is overwritten. The original strokes are indiscernible.

there is a horiz. line that they cross at the very bottom.

There are fns. that do not change under parallel translation.

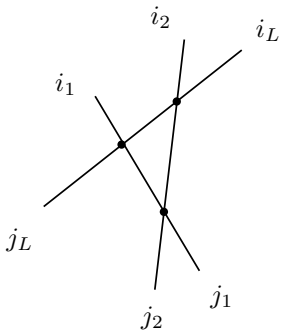
GELFAND-P000840

[source number: 30]

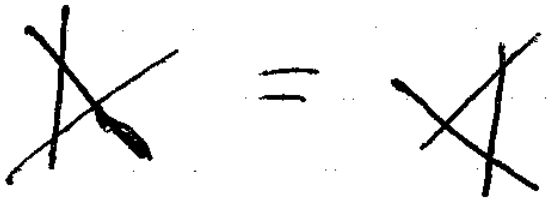


Original source details: GELFAND-P000840

Diagram and typed description.

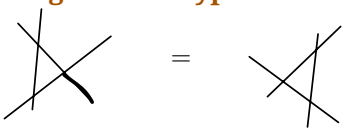


Associat.: Z does not depend on the relative arrangement of the lines.
It suffices for it to hold for 3 lines:



Original source details: GELFAND-P000840

Diagram and typed description.



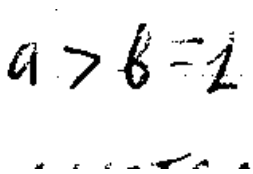
The thickened part of the descending line at left may be a retraced stroke or an arrow.

Prop. 2. such matrices exist.
On asymptotics:

$$\ln Z^{j_1\dots j_L}_{i_1\dots i_L} \sim L^2 \qquad \text{as } L \to \infty \tag{4}$$

(if without associativity) and the asympt. does not depend on the choice of indices i and j.
[Under «asympt.»: (leading term)]
Prop. 3. Asymptotics as $L \to \infty$

$$\ln Z(\lambda_1, \dots, \lambda_L) = \sum_{a>b=1}^{L} f(\lambda_a - \lambda_b) + \text{remainder term} \tag{5}$$

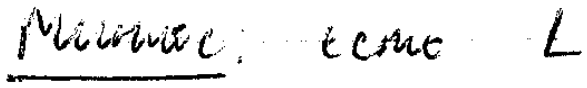


Uncertain reading. The lower limit reads "$a > b = 1$." The upper limit is L.

Original source details: GELFAND-P000840

(free en. of a large system = sum of pairwise

Opinion: there are L kinds of particles; they move along straight lines and collide; on colliding they retain their velocities, but change color



Uncertain reading. The heading provisionally reads «Мнение» (“Opinion”).

Original source details: GELFAND-P000840

Uncertain reading. The opening strokes of the Russian word for “velocities” are overwritten. The earlier letterform is indistinguishable.

Original source details: GELFAND-P000840

[Under «color»: (kind)]

Here Z — S-matrix;

There is an interpr. in which Z — the partition function

The weights of S — exp of the potential. (probabilities?)

There are initial and final colorings $i_1, \dots, i_L$

$$j_1, \dots, j_L.$$

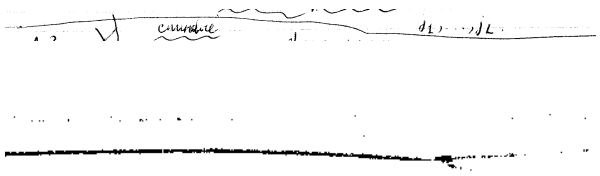
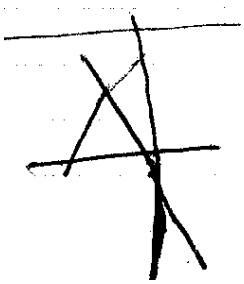


Diagram and typed description.

Original source details: GELFAND-P000840

A. Z.



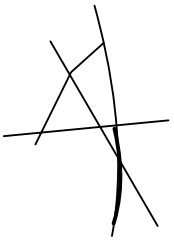
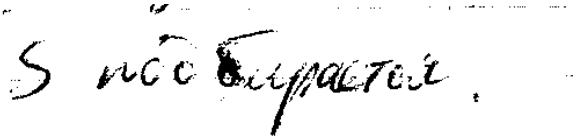


Diagram and typed description. Two crossings near the horizontal line merge in a thickened stroke. The long divider is visible on the excerpt at left.

Original source details: GELFAND-P000840

spin variables — in polygons; nearest-neighbor interaction;

S is chosen.



Uncertain reading. The short note provisionally reads «подбирается» (“is chosen”). The opening letters and the heavy stroke remain unclear.

Original source details: GELFAND-P000840

$$Z = \sum_{\{\sigma\}} e^{-\beta H} \qquad H = \sum h(\sigma_A, \sigma_B)$$

Exactly calculable partition functions — all of this kind!

For the Ising model:

$$S_{ij}^{k\ell} = \frac{ijk\ell + 1}{2} e^{-\frac{\beta}{2}(i+j+k+\ell)}$$

(after passing to the dual lattice)

GELFAND-P000841

24.9.79 Gelfand Sem.

Renata Ernestovna KALLOSH on the conf. in Cambridge

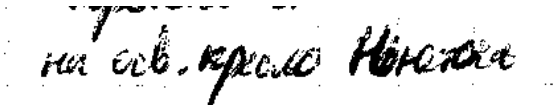
(physicist)

(month-long working meeting on quantum grav., approx. 40 people)

20 people from the USA; including Chern, Singer, Milnor, Penrose;

Hawking, De Witt, Sciama, Coleman, Jackiw

[Below Hawking, with a short vertical stroke:] candidate for Newton's vacant chair



Uncertain reading. The abbreviation preceding "chair" is provisionally read «св.». The translation takes it as "vacant."

Original source details: GELFAND-P000841

Gibbons — physicist, works with Hawking — talk on quantum grav.

Uncertain reading. The abbreviation before "on quantum grav." provisionally reads «докл.» ("talk"). «сем.» ("seminar") is possible.

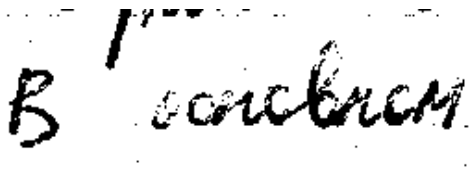
Original source details: GELFAND-P000841

Idea: one must take account of the effects of nontriv. topology of space/time on small scales.

(many little "handles").

The theory has yet to be constructed, but there has been signif. progress.

Mainly Hawking:

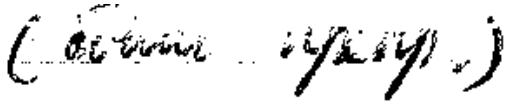


Uncertain reading. The Russian before Hawking provisionally reads «В основном» ("Mainly"). The opening strokes of the second word are faint.

Original source details: GELFAND-P000841

1) Space-time foam (foam) — published in Nuclear Physics

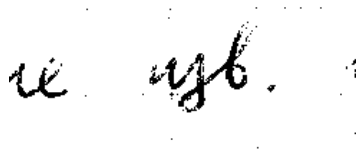
(ordinary journ.)



Uncertain reading. The note below the journal title is provisionally read «обычн. журн.» ("ordinary journ.").

Original source details: GELFAND-P000841

2) Page, Atiyah something in the field of known gravitons

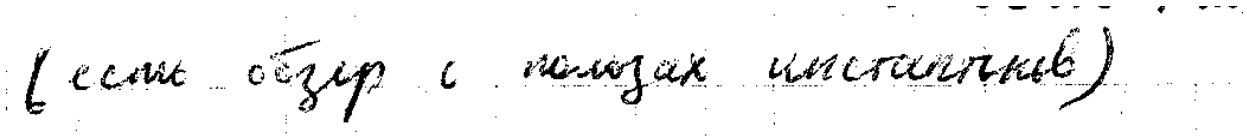


Uncertain reading. The abbreviation provisionally reads «изв.» (“known”). «изл.» (“emitted”) is possible.

Original source details: GELFAND-P000841

(from the group of Hawking)

3) Coleman, Jackiw — on grav. taking topol. numbers into account —
on Yang–Mills.
(there is a survey on instanton methods)



Original source details: GELFAND-P000841

Uncertain reading. The preposition before “methods” provisionally reads «о» (“on”). The letter is open and resembles «с».

4) Milnor — classif. of 4-dimensional manifolds
Someone on bundles on 4-manifolds.
5) Singer — on the signature of a 4-manifold.
On supermanifolds
Considered a route to quantization
De Witt is writing a book on supermanifolds
Singer is giving lectures

Diagram and typed description.

Original source details: GELFAND-P000841

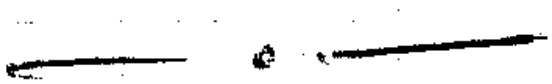
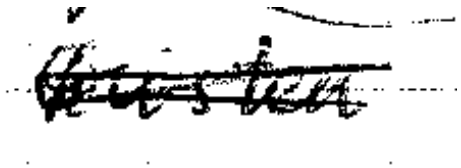


Diagram and typed description.

Original source details: GELFAND-P000841

Something obtained independently by the speaker and by someone from the conf.
[At right:] [Crossed out: ~~Christen[…]~~] Christensen, Duff;



Uncertain reading. The crossed-out name provisionally reads “Christen[…].”

Original source details: GELFAND-P000841

Kallosh — the speaker

Consider the eq.

$$\Delta[A,B]\,\Phi[A,B]=0 \tag{1}$$

Here

$$\phi[A,B]=\Psi_{\alpha_1\ldots\alpha_{2A},\dot\alpha_1,\ldots,\dot\alpha_{2B}}(x)$$

— a multispinor, transforms under the group $SO(4)$

Original source details: GELFAND-P000841

Uncertain reading. The second letter of the group name is open. It is provisionally set as $SO(4)$.

$x\in$ [Crossed out: ~~...~~] R^4.

Original source details: GELFAND-P000841

Undeciphered passage. One sign is crossed out before R^4. Its original form is indistinguishable.

Δ — some operator; at one time from Hodge–de Rham.

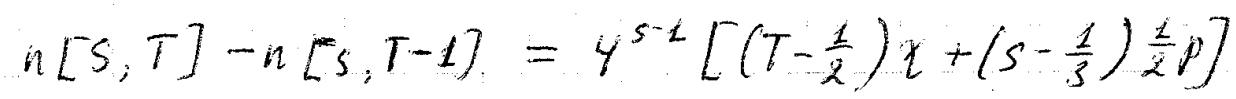
there is also the half Dirac here

Original source details: GELFAND-P000841

Uncertain reading. The abbreviation before “Dirac” is provisionally read «пол.» and rendered “half.”

$$S=A+B,\qquad T=A-B.$$

$$n[S;T]-n[s;T-1]=4^{s-1}\left[\left(T-\frac{1}{2}\right)\chi+\left(s-\frac{1}{3}\right)\frac{1}{2}p\right] \tag{2}$$



Original source details: GELFAND-P000841

Uncertain reading. The S/s forms in (2) vary in size. Their case is uncertain in places.

[Below n, with a short vertical stroke:] number of zero modes

(dim. of the solution space, with spin A and B);

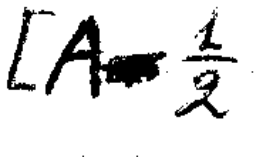
χ,p — Euler and Pontryagin numbers

GELFAND-P000842

[source number: 66]

Here there are the Gauss–Bonnet and Atiyah–Singer theorems, and so on.

$$n[A,B]-2n\left[A\overset{?}{-}\frac{1}{2},B\right]=4^{A+B-1}\left(A-\frac{7}{12}\right)\left(\chi+\frac{1}{2}p\right). \tag{3}$$

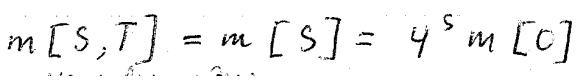


Uncertain reading. The sign before $1/2$ is heavily inked over. A minus is a possible reading.

Original source details: GELFAND-P000842

On nonzero modes:

$$m[S;T]=m[S]=4^{s}m[0] \tag{4}$$



Uncertain reading. The case of s/S in the exponent is unclear. The last argument is provisionally set as 0.

Original source details: GELFAND-P000842

[Under $m[S;T]$:] nonzero modes:

$$\Delta\Psi_i=\lambda_i\Psi_i \tag{5}$$

In supergravity the nonzero modes cancel and only the contribution of the zero ones remains.

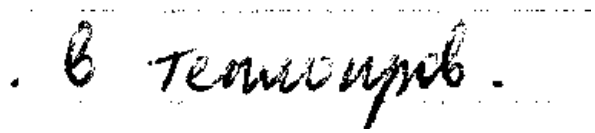
(3) [Crossed out: ~~...~~] only if the manifold is self-dual.

Undeciphered passage. The word is obscured by heavy crossing-out.

Original source details: GELFAND-P000842

Hawking proved something earlier.

Coeff. in heat prop.



Uncertain reading. The Russian abbreviation is provisionally read «теплопров.», rendered “heat prop.”

Original source details: GELFAND-P000842

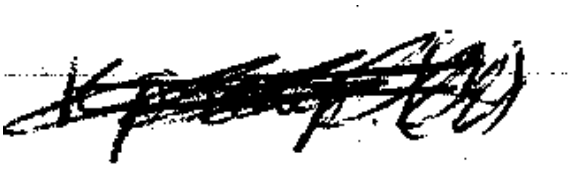
$$B_4 = n + m$$

They proved that the funct. integral has no divergences.
(in gravity)
The speaker did something too.
Interesting problem: the same on a supermanifold.
Singer is working on this now.

Diagram and typed description.

Original source details: GELFAND-P000842

The group $SO(8)$, came up, something was wrong.
French group: [Crossed out: ... (8)] symmetry E_7 and



Undeciphered passage. The crossed-out letters before the visible "(8)" cannot be read confidently.

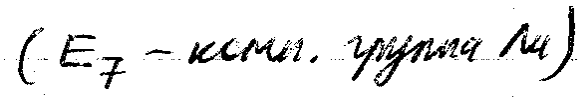
Original source details: GELFAND-P000842

$$E_7 \longrightarrow SU(8)$$

then ext. supergravity

$$\dim E_7 = 133$$

(E_7 — comp. Lie group)



Uncertain reading. The abbreviation before "Lie group" provisionally reads «комп.» ("comp.").

Original source details: GELFAND-P000842

Interesting
[Crossed out: ...] to make a super index theorem

Undeciphered passage. A mark before "to make" is inked out. Its original form is indistinguishable.

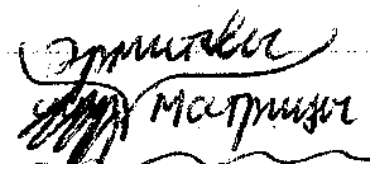
Original source details: GELFAND-P000842

Remarks:

F_4

[Inserted above the line:] Hermitian

[Crossed out: ...] matrices of order 3 made of octonions; in them one constructs a Jordan add..



Original source details: GELFAND-P000842

Undeciphered passage. A passage before “matrices” is crossed out. “Hermitian” is inserted above it.

Original source details: GELFAND-P000842

Uncertain reading. The last abbreviation is provisionally read «слож.» (“add.”).

product $\frac{1}{2}(AB + BA)$.

one must take the automorphism group of this Jord. algebra. We get F_4

Chern — adviser to the Chin. president — works in Chicago.

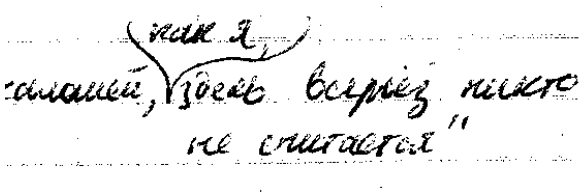
Yang gives lectures in China to an audience of a thousand.

I. M. G. — what physicists said about Einstein.

a) E. and N. Bohr are the greatest phys. of the century

b) E. did splendid work in his youth, but then spent 30–40 years on nonsense (tried to create a unified field theory).

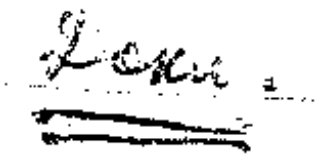
c) E. to Born: “I want to invite you to Princeton, but an old nag, [Inserted above the line: like me,] no one takes seriously here.”



Original source details: GELFAND-P000842

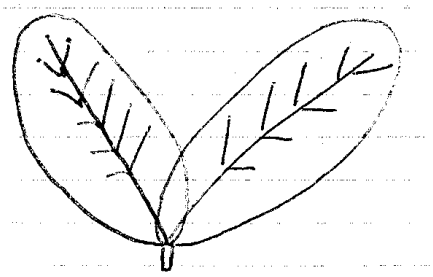
Uncertain reading. The end of the quotation reads «никто не считается». An angled mark leads to the insertion “like me.”

[In the left margin:] Blackboard.



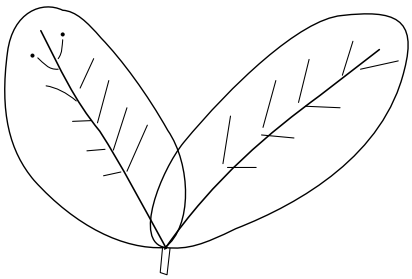
Original source details: GELFAND-P000842

Uncertain reading. The last letter of the Russian word is provisionally read «а».



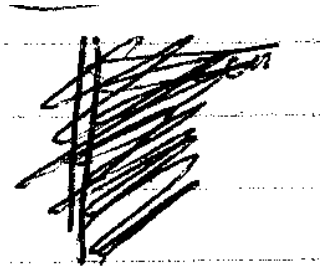
Original source details: GELFAND-P000842

Diagram and typed description.



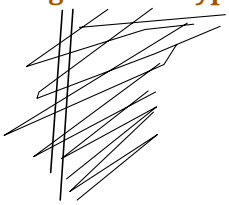
The two outlines meet at the base. At right, some short branches are detached from the main line.

connection of theor. physics with mathematics:
in fundamental matters
(Yang's drawing)



Original source details: GELFAND-P000842

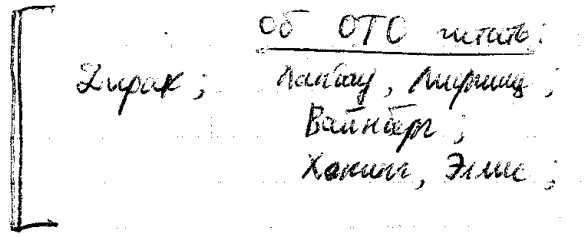
Diagram and typed description.



Near-vertical lines and diagonal hatching. The relationship to the drawing at left is unclear.

GELFAND-P000843

V. E. Zakharov.



Original source details: GELFAND-P000843

Diagram and typed description.

On general relativity, read:

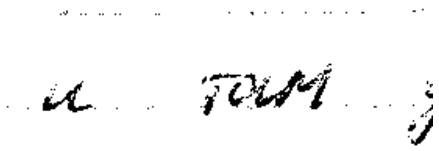
Dirac; Landau, Lifshitz;
Weinberg;
Hawking, Ellis;

In the right margin. A left bracket groups the list.

Mathematical problem:

$$k = (k_1, \ldots, k_N)$$

Consider the fn. $\omega(k)$ (real) and the space of pairs (k_1, k_2) of dimension $2N$ and there the eq.



Original source details: GELFAND-P000843

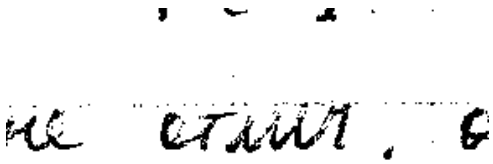
Uncertain reading. The short Russian word is provisionally read «там» ("there").

$$\omega(k_1 + k_2) = \omega(k_1) + \omega(k_2) \tag{1}$$

ω — nonlin. $\Rightarrow$ we get a surface Γ in R^{2n}.
Is there another fn. f, satisfying the same eq. on Γ:

$$f(k_1 + k_2) = f(k_1) + f(k_2)$$

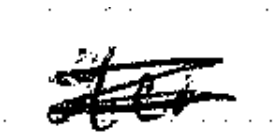
and which does not differ from ω by a linear one?



Original source details: GELFAND-P000843

Uncertain reading. The abbreviated verb after the negation is provisionally read «отлич.» (“differ”).

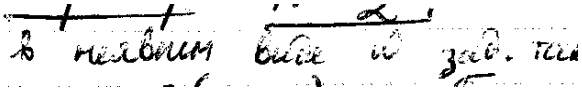
[Crossed out: ~~This~~]



Original source details: GELFAND-P000843

Uncertain reading. The crossed-out word provisionally reads «Это» (“This”).

This sit. is exceptional because the dimension is large.
Idea of Zakh. and Shulman:
how to get all eqs. having infinitely many first integrals.
Example: $N = 2$.
We will specify it by means of a parametrization, $\vec{k} = (p, q)$;
[In small writing below the line:] implicitly ω is given as follows:



Original source details: GELFAND-P000843

Uncertain reading. The small Russian words provisionally read «в неявном виде» (“implicitly”).

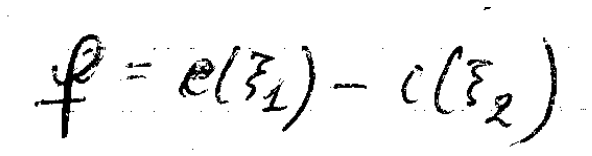
$$\omega(p, q), \qquad \begin{cases} p = \xi_1 - \xi_2 \\ q = a(\xi_1) - a(\xi_2) \\ \omega = b(\xi_1) - b(\xi_2) \end{cases} \tag{2}$$

Another [Crossed out: ~~...~~] fn. f:

Original source details: GELFAND-P000843

Undeciphered passage. A short entry before “fn.” is crossed out. The complete word is illegible.

$$\begin{cases} p = \xi_1 - \xi_2 \\ q = a(\xi_1) - a(\xi_2) \\ f = c(\xi_1) - c(\xi_2) \end{cases} \qquad c \text{ — another fn. (arbitrary)} \tag{3}$$

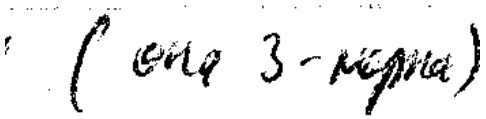


Uncertain reading. Provisionally set as $f = c(\xi_1) - c(\xi_2)$. The letter at left also resembles φ, and the first at right resembles e.

Original source details: GELFAND-P000843

Question: is this a general construction?

Idea: we will give a parametrization of Γ (it is 3-dimensional)



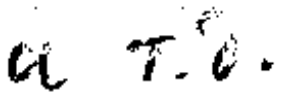
Uncertain reading. The Russian endings provisionally read «она 3-мерна» (“it is 3-dimensional”).

Original source details: GELFAND-P000843

$$\begin{cases} p_1 = \xi_1 - \xi_2 \\ q_1 = a(\xi_1) - a(\xi_2) \\ p_2 = \xi_2 - \xi_3 \\ q_2 = a(\xi_2) - a(\xi_3) \end{cases} \tag{4}$$

$$\underbrace{p_1 + p_2}_{p_3} \qquad \underbrace{q_1 + q_2}_{q_3} \qquad p_3 = \xi_1 - \xi_3$$

and so on.



Uncertain reading. The opening sign is provisionally read «и» (“and”).

Original source details: GELFAND-P000843

GELFAND-P000844

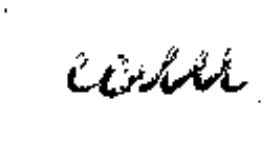
[source number: 67]

[Above the first line crossed out: ~~...~~]

Undeciphered passage. A short entry above the first line is inked out. The letters are indistinguishable.

Original source details: GELFAND-P000844

Hence if degenerate, then serious (functional parameter).



Uncertain reading. The short Russian word is provisionally read «если» (“if”).

Original source details: GELFAND-P000844

There is a method of constr. systems with a Lax repr. (Zakharov–Shabat)

$$\frac{\partial L}{\partial t} - \frac{\partial A}{\partial y} + [L, A] = 0 \tag{5}$$

$$L = \frac{\partial^n}{\partial x^n} + \dots$$
$$A = \frac{\partial^m}{\partial x^m} + \dots$$

(5) — nonlin. syst. of eqs. for the coeffs.

Consider its linearization.

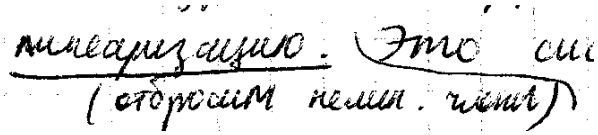


Diagram and typed description.
linearization. This

(discard the nonlin. terms)

A note below the line. The connecting stroke reaches “This.”

Original source details: GELFAND-P000844

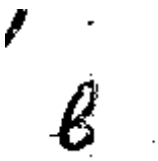
This is a syst. with const. coeffs., it is solved by the method of Fourier.

$$u_j \sim \exp(-\omega t + qy + px)$$
$$\omega = \omega(p, q)$$

Those syst. look as in

Uncertain reading. The opening Russian word provisionally reads «Те» (“Those”).

Original source details: GELFAND-P000844



Uncertain reading. After “as” there is a separate «в» (“in”). The sentence breaks off before the symbols.

Original source details: GELFAND-P000844

$a(\xi)$ — symbol of the oper. L

$b(\xi)$ — ″ — A

[In the right margin:]

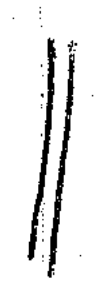


Diagram and typed description.

$$L = L_0 + \delta L$$
$$A = A_0 + \delta A$$

Original source details: GELFAND-P000844

$\delta L, \delta A$ — everything that contains variable coeffs.

Degenerate disp. laws are connected with the inverse problem.

$$p = \xi_1 - \xi_2$$
$$q = \ell(\xi_1) - \ell(\xi_2)$$
$$\omega = a(\xi_1) - a(\xi_2)$$

a, ℓ — symbols of A_0, L_0.

Consider the G.–L.–M. eq.:

$$F + K + K \oplus F = 0$$

Uncertain reading. A circled plus is visible between K and F.

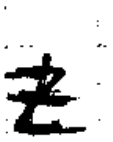
Original source details: GELFAND-P000844

$$\frac{\partial F}{\partial t} + [L_0, F] = 0$$
$$\frac{\partial F}{\partial y} + [A_0, F] = 0$$

Uncertain reading. The plus on the second line is retraced and still visible.

Original source details: GELFAND-P000844

$$K = -F$$
$$F = \exp\big(\xi_1 x - \xi_2 z + [\ell(\xi_1) - \ell(\xi_2)]y + [a(\xi_1) - a(\xi_2)]t\big)$$



Uncertain reading. The letter after ξ_2 is heavily overwritten. It provisionally reads z.

Original source details: GELFAND-P000844

Here the connection is estab. when ℓ and a — polynomials. One can estab. a connection for rat. functions.

Uncertain reading. The letter ℓ before “and” has been retraced.

Original source details: GELFAND-P000844

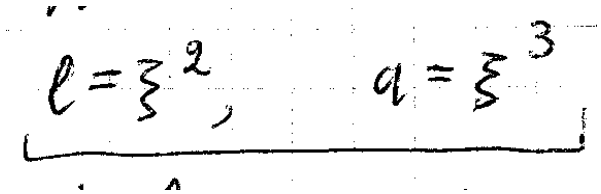


Diagram and typed description.
$\ell = \xi^2, \qquad a = \xi^3$

Original source details: GELFAND-P000844

L, A are to be found — we get KdV.

Uncertain reading. The Russian verb provisionally reads «ищутся» (“are to be found”).

Original source details: GELFAND-P000844

GELFAND-P000845

Integr. by the inverse problem $\Rightarrow$ the disp. law is degenerate.

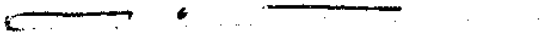


Diagram and typed description.

Original source details: GELFAND-P000845

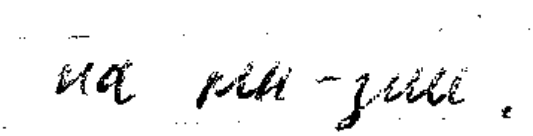
Matrices L and A: no longer necessarily

[In the left margin:] Altern.

Uncertain reading. The underlined marginal word is provisionally read «Альтерн.» (“Altern.”).

Original source details: GELFAND-P000845

If the eq. is integr. and the disp. law is nondeg., then the interaction $\equiv 0$ on the manifold.



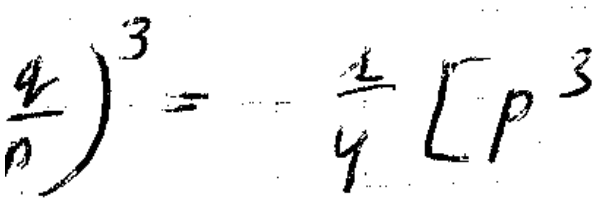
Uncertain reading. The final Russian abbreviation provisionally reads «мн-зии» (“manifold”).

Original source details: GELFAND-P000845

Diagram and typed description.

Original source details: GELFAND-P000845

$$\omega = \frac{1}{8}\left[(p+\tfrac{q}{p})^3 - (-p+\tfrac{q}{p})^3 = \frac{1}{4}\left[p^3 + \tfrac{3q^2}{p}\right]\right.$$



Uncertain reading. No closing square bracket is visible before the second equals sign.

Original source details: GELFAND-P000845

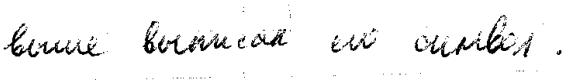
Kad.–Petv. eq.:

$$u_t \overset{?}{+} \frac{1}{4}\left(u_{xxx} + 6uu_x - 3\int_{-\infty}^{x} u_{yy}\,dx\right)$$

Uncertain reading. A vertical stroke crosses two horizontal strokes. A plus is provisional. The earlier sign is unclear.

Original source details: GELFAND-P000845

I wrote out its symbol above.



Uncertain reading. The Russian verb provisionally reads «выписал» (“I wrote out”). «выписан» (“is written out”) is possible.

Original source details: GELFAND-P000845

ξ^2 and ξ^3 we dress — we get the K.–P. eq..

That is, K.–P. has exactly this disp. law (as in the example)

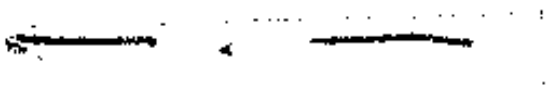


Diagram and typed description. —— . ——

Original source details: GELFAND-P000845

Experim. fact: we dress a pair

$$L_0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{\partial}{\partial x}, \qquad A_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{\partial^2}{\partial x^2}$$

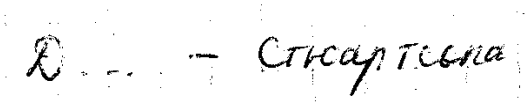
We get

$$L = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{\partial}{\partial x} + \begin{bmatrix} 0 & \psi \\ \bar{\psi} & 0 \end{bmatrix}$$

and the eq.

$$\begin{cases} i\dfrac{\partial \psi}{\partial t} - \dfrac{\partial^2 \psi}{\partial x^2} + \dfrac{\partial^2 \psi}{\partial y^2} + u\psi = 0 \\ \left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}\right) u = \left(\dfrac{\partial^2}{\partial x^2} - \dfrac{\partial^2}{\partial y^2}\right) |\psi|^2 \end{cases} \tag{6}$$

D...–Stewartson syst. (see hydrodyn., theory of surface waves)



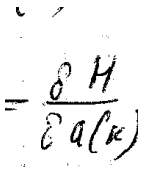
Uncertain reading. The first name begins with the Russian «Д» (“D”). Its continuation is indistinct.

Original source details: GELFAND-P000845

Take the Fourier trans. We get a syst. of the form:

$$\psi \to a(k)$$

$$i\frac{\partial a(k)}{\partial t} = \frac{\delta H}{\delta a(k)}$$



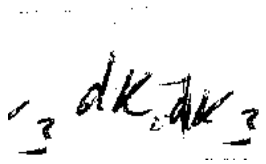
Uncertain reading. There is no visible bar over $a(k)$ in the denominator of the variational derivative.

Original source details: GELFAND-P000845

$$H = \int \omega_k |a_k|^2 \, dk + \int w^{[\Gamma]}_{kk_1k_2k_3} \bar{a}_k \bar{a}_{k_1} a_{k_2} a_{k_3} \delta_{k+k_1-k_2-k_3} \, dk_2 \, dk_3$$

Uncertain reading. The coefficient is provisionally read w. A separate mark at upper right resembles Γ.

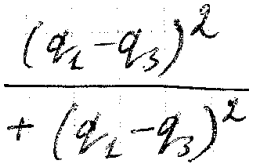
Original source details: GELFAND-P000845



Uncertain reading. Two differentials are visible at the frame edge. The first subscript is provisionally read as 2.

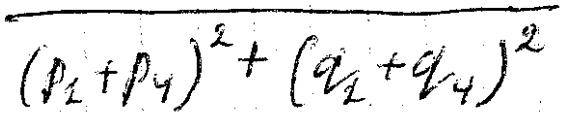
Original source details: GELFAND-P000845

$$w = \frac{(p_1 - p_3)^2 - (q_1 - q_3)^2}{(p_1 - p_3)^2 + (q_1 - q_3)^2} + \frac{(p_2 - p_4)^2 - (q_2 - q_4)^2}{(p_2 + p_4)^2 + (q_2 + q_4)^2}$$



Uncertain reading. The subscript on the first q on both lines of the first fraction is provisionally read as 1.

Original source details: GELFAND-P000845



Uncertain reading. Plus signs are visible inside both parentheses of the second denominator.

Original source details: GELFAND-P000845

GELFAND-P000846

[source number: 68]

In the space of k_1, k_2, k_3, k_4

Uncertain reading. The letter with subscript 4 is retraced. It is provisionally set as k_4.

Original source details: GELFAND-P000846

consider the eq.

$$\omega(k_1) + \omega(k_2) = \omega(k_3) + \omega(k_4);$$
$$k_1 + k_2 = k_3 + k_4$$

[Crossed out: ...]

Undeciphered passage. A short entry and a long line below the equations are heavily crossed out. The text is illegible.

Original source details: GELFAND-P000846

It turns out that $w|_\Gamma = 0$.

Arnold's conj.: This is Poincaré's theor. on integrability. (when there is a first int. indep. of the en.)

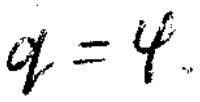
Poincaré's cond.: the ampl. of reson. harmonics must be 0

(see Poincaré, cel. mech., or Arnold: suppl. chapters)

Poincaré: Hamiltonian

$$H = H_0(p) + \varepsilon \sum h_k(p) e^{ikq}$$

$p = I, \quad q = \psi.$



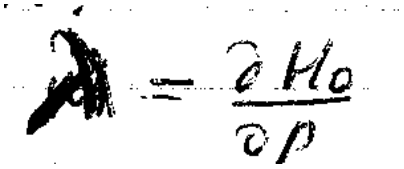
Uncertain reading. The angle sign is provisionally read ψ. φ is possible.

Original source details: GELFAND-P000846

When $\exists$ an int. $F(p, q, \varepsilon)$ (everything is periodic in q),

[Under F:] indep. of H.

Nec. condition: form $\lambda = \frac{\partial H_0}{\partial p}$ and impose a nondeg. condition on it.



Original source details: GELFAND-P000846

Uncertain reading. The sign to the left of the derivative is heavily retraced. It is provisionally set as λ.

$$\det \left| \frac{\partial \lambda}{\partial p} \right| \neq 0$$

Original source details: GELFAND-P000846

Uncertain reading. The sign in the numerator is also retraced. It is provisionally set as λ.

— one of the nondeg. conditions. (there are 2 in all)

There are no integrals if

$$\Gamma: \quad \langle k, \lambda(p) \rangle = 0$$

Original source details: GELFAND-P000846

Uncertain reading. The sign before (p) is retraced. It is provisionally set as λ.

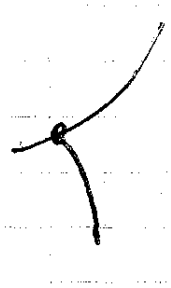
Poincaré's cond.:

$$h_k|_{\Gamma_k} = 0, \qquad k \in \mathbb{Z}^n.$$

if it is violated infinitely often, there is no integral

Idea: $\lambda \leftrightarrow \omega$.

Arn. formulas (2) mean there is a transl. surface.



Original source details: GELFAND-P000846

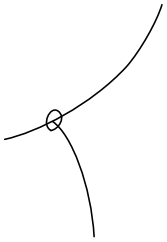


Diagram and typed description. The curves meet near a small loop. The thickening at left and the loop are drawn as visible. No arrows can be distinguished confidently.

we move the 2nd curve along the 1st,
translating it in parallel
Curves:

$$\begin{aligned} p &= \xi_1; & p &= -\xi_2 \\ q &= a(\xi_1); & q &= -a(\xi_2) \\ \omega &= b(\xi_1); & \omega &= -b(\xi_2) \end{aligned}$$

The curve moves by itself.
Zakharov.

$$\omega_1(k_1 + k_2) = \omega_2(k_1) + \omega_3(k_2)$$

we get general transl. surfaces.

Diagram and typed description.

Original source details: GELFAND-P000846

Procedure. We check whether the disp. law is degen., then see.
0 or not 0 there, the Ham. interaction.

Uncertain reading. The abbreviation before “interaction” provisionally reads «гам.» (“Ham.”).

Original source details: GELFAND-P000846

What if the integrals start directly at higher orders?

GELFAND-P000847

Arn. Birkhoff.

Uncertain reading. The opening abbreviation reads «Арн.» (“Arn.”). The full stop after the following word is unclear.

Original source details: GELFAND-P000847

$$H = \sum \omega_k z_k \bar{z}_k + \dots$$

[crossed out: ~~$H =$~~]

Uncertain reading. H can be read under the strokes. An equals sign may follow it.

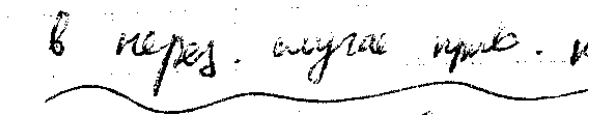
Original source details: GELFAND-P000847

$$\sum \omega_k (p_k^2 + q_k^2) + \dots$$

[At right:]

$$z = p + iq, \qquad \Pi_k = z_k \bar{z}_k$$

Birkhoff: in the nonres. case, red. to $H(\Pi_k)$.



Original source details: GELFAND-P000847

Uncertain reading. The abbreviation after “case” is provisionally read «прив.» (“red.”).

(if the ω_k are independent)

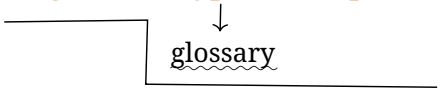
[At right:] (at the level of form. series)

I. M. G. Short list of what one needs to know.

1. Modern abstr. language (see e.g. the 1st part of the book by Kirillov)

Original source details: GELFAND-P000847

Diagram and typed description.

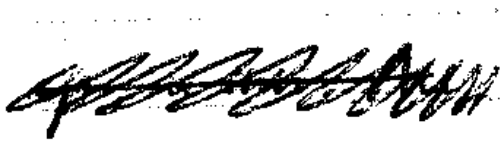


A stepped stroke and an arrow pointing to “glossary” are below the reference.

2. Arnold: diff. eqs. + suppl. chapters

(to get used to analysis)

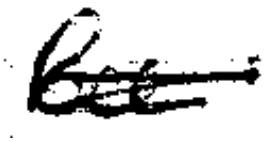
3. On algebra: 2 chapters of the book by Serre on [crossed out: ~~representations~~] reprs. of finite groups.



Original source details: GELFAND-P000847

Uncertain reading. The crossed-out word provisionally reads «представлениям» (“representations”).

[Above the line crossed out: ~~all~~]

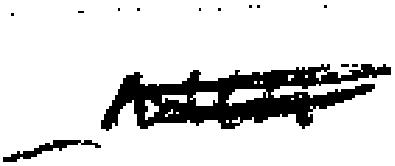


Original source details: GELFAND-P000847

Uncertain reading. The crossed-out word provisionally reads «все» (“all”).

all except ch. 1 of the book by Serre on Lie grps. and algebras (especially the last chapter)

[Above the line crossed out: ~~...~~]

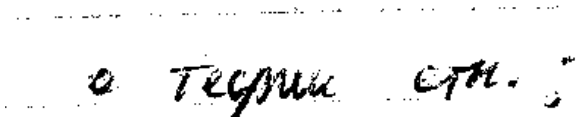


Original source details: GELFAND-P000847

Undeciphered passage. A short word above “Lin. algebra” is crossed out. It is illegible.

Lin. algebra: Gelfand; Maltsev

4. Diff. geometry in 2 vers.: the old tensor version (see Dirac on the theory of rel.; and Landau and Lifshitz)



Original source details: GELFAND-P000847

Uncertain reading. The abbreviation after “theory” provisionally reads «отн.» (“rel.”).

modern vers.: vector bundles and connections.

5. Representations: H. Weyl

lectures of Raka Racah.

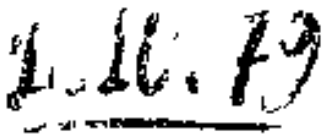
Gelfand, Minlos, Shapiro

Chapter 16

GELFAND-P000848

[source number: 1]

1.10.79 Gelfand Sem.



Original source details: GELFAND-P000848

Uncertain reading. The month is provisionally read as “10.” The second digit is incomplete in the scan.

[Crossed-out heading: ~~KALLOSH?~~]

Original source details: GELFAND-P000848

Uncertain reading. «КАЛЛОШ» (“KALLOSH”) is a possible reading of the crossed-out heading.

I. M. G. What is a spinor?

Pauli’s book on quant. mech. — the best expos. of Dirac’s theory (Dirac eq.)

Original source details: GELFAND-P000848

Uncertain reading. The overwritten abbreviation is provisionally read «квант.» (“quant.”).

His objections to Dirac’s theor.:

a) exist. of positrons, predicted by Dirac’s theor.

b) if they exist, then on colliding with a positr. one gets 2 photons

Original source details: GELFAND-P000848

Uncertain reading. The abbreviation after “with” is provisionally read «позитр.» (“positr.”).

c) according to this theory 2 photons can produce an electr./positron pair

Original source details: GELFAND-P000848

Uncertain reading. The abbreviation before the slash is tentatively read «эл-н» (“electr.”).

Dirac refus., so that they would not make him “Sir” (?)

Original source details: GELFAND-P000848

Uncertain reading. The abbreviation after “Dirac” is provisionally read «отказ.» (“refus.”).

[The lines are grouped by a bracket on the left.]

Once someone told him that he did not understand the form. in the lower left corner of the board.

Dirac does not answer.

The chairman asks him to answer

[Crossed out: ...] Dirac: “there was no question, there was a statement.”

Original source details: GELFAND-P000848

Undeciphered passage. The crossed-out fragment before “Dirac” is illegible.

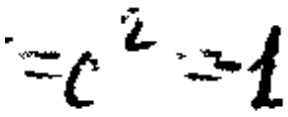
Original source details: GELFAND-P000848

Diagram and typed description. Bracket to the left of the preceding four lines.

Hamilton: wanted to extract the root of $x^2 + y^2 + z^2$, and obtain a “lin. form”

$$-(x^2 + y^2 + z^2) = (ax + by + cz)^2$$

$$a^2 = b^2 = c^2 = -1$$



Original source details: GELFAND-P000848

Uncertain reading. The overwritten sign after c^2 is provisionally read $= -1$.

$$(ab + ba)\,[?]\,xy + (ac + ca)xz + (bc + cb)yz$$
$$+ a^2x^2 + b^2y^2 + c^2z^2 = -(x^2 + y^2 + z^2).$$

Original source details: GELFAND-P000848

Uncertain reading. There is an unresolved dark stroke between the first parenthesis and xy.

hence the relations

$$ab + ba = 0 \quad \text{etc.}$$

Add the unit — get quaternions.

Consider the transformation

$$z' = \frac{az + b}{cz + d}$$

or

$$\begin{cases} z_1' = az_1 + bz_2 \\ z_2' = cz_1 + dz_2 \end{cases} \qquad z' = \frac{z_1'}{z_2'}, \quad z = \frac{z_1}{z_2}.$$

Spinors — a pair of numbers [below "numbers": (compl.)], transf. according to this law

Compl. conj. — conj. spinor.

Diagram and typed description.

Two short strokes with a dot between them.

Original source details: GELFAND-P000848

Gravitation should be read in Landau and Lifshitz (vol. 2)

chs. 1, 2, 10, 11.

Renata E. Kallosh.

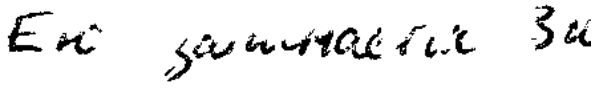
Program.

1. Two-component spinors.
2. Self-duality.
3. Diff. operator $\Delta(A, B)$, descr. the eq. for particles of higher spins.
4. Heat eq., zero and nonzero modes.
5. Index theorems.
6. Supertheor. in a grav. field
7. Superself-duality.

(8. Index supertheor.! Singer is working on it, but in fact there is no such theorem)

Uncertain reading. The numeral after the opening parenthesis is tentatively read as 8.

Original source details: GELFAND-P000848



Uncertain reading. The opening phrase is provisionally read «Ею занимается» ("is working on it").

Original source details: GELFAND-P000848

GELFAND-P000849

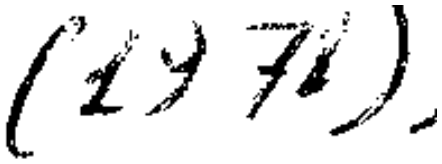
Refs.

1. [Crossed out: ...] S. W. Hawking, C. N. Pope

Undeciphered passage. A short fragment before “S. W. Hawking” is crossed out and illegible.

Original source details: GELFAND-P000849

Nucl. Phys., B 146 (1978), 381



Uncertain reading. The final digit of the year is provisionally read as 8. The strokes could also be read as 1.

Original source details: GELFAND-P000849

2. R. E. Kallosh, JETP Letters,

29 (1979), 192; 493

3. S. M. Christensen, M. J. Duff.

Nucl. Phys. (1979), 301

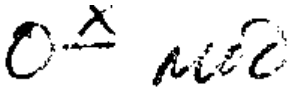
4. — „ —, — „ — Deser, Gr[?]

Uncertain reading. The name after “Deser” is provisionally read “Gr” with an unclear ending. The first letter could be read as “C.”

Original source details: GELFAND-P000849

Phys Lett. 84 B (1979), 411

[1] — main contrib. in supergrav. (in the 1-loop approx.) — from $0^{\underline{x}}$ modes



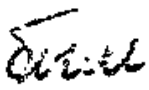
Uncertain reading. There is a small underlined superscript «x» (the Russian ordinal ending) at the zero.

Original source details: GELFAND-P000849

(in Y.-M. — from all modes!)

[2], [3] indep. — in supergrav. (for self-dual) for higher orders

Before this it was so in the 1st and 2nd ord. supergrav. did not cont. diverg.



Uncertain reading. The word after “Before this” is provisionally read «было» (“it was”). The last letter is unclear.

Original source details: GELFAND-P000849

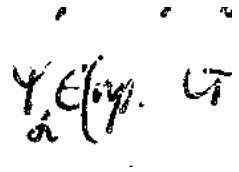
[3] based on the ordinary index theor.

Two-comp. van der Waerden formalism.

$$\Psi_\alpha \longrightarrow \Psi_\beta \Omega_\alpha^\beta \tag{1}$$

[At right:] Transformation law

$\Psi_\alpha \in$ (repr. stand. 2-dim. repr. $SL(2,C)) = V$;

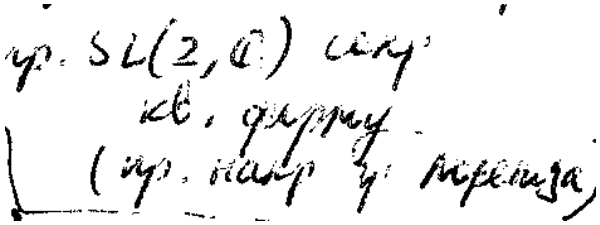


Original source details: GELFAND-P000849

Uncertain reading. The first abbreviation is provisionally read «пр.» (“repr.”). Its initial letter could be «и».

$\Psi_{\dot\alpha} \in \bar V$, $V \otimes \bar V$ has a real str. and the gr. $SL(2,C)$ pres. the quad. form.

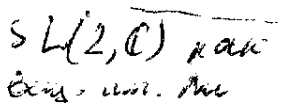
(gr. e.g. in Penrose)



Original source details: GELFAND-P000849

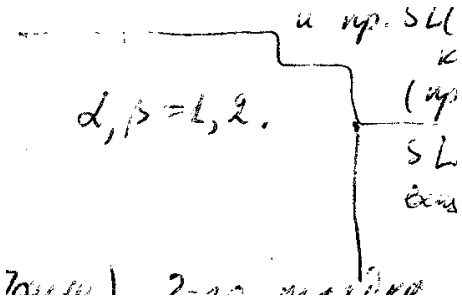
Uncertain reading. The abbreviation in parentheses is provisionally read «напр.» (“e.g.”).

$SL(2,C)$ as a real Lie man.



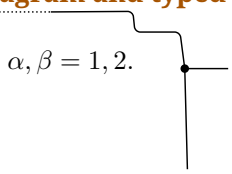
Original source details: GELFAND-P000849

Uncertain reading. The abbreviation after “real” is provisionally read «мн.» (“man.”). «гр.» (“gr.”) is possible.



Original source details: GELFAND-P000849

Diagram and typed description.



A stepped line between the index formula and the note at right. A short line extends right from the dot. The correspondence between strokes and lines of text is unclear.

[Main text:] If the Lorentz gr. or $SO(4)$

$$\Omega_\alpha^\beta = (\omega^{ab}\Sigma_{ab})_\alpha^\beta \tag{2}$$

$\alpha, \beta = 1, 2.$

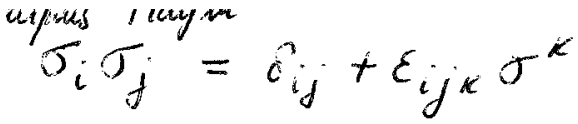
(transf. law of a two-comp. spinor)

Σ_{ab} — fixed matrices (like the Pauli matr.) of order 2

ω^{ab} — parameters

For the Pauli matrices

$$\sigma_i\sigma_j = \delta_{ij} + \epsilon_{ijk}\sigma^k \tag{3}$$



Uncertain reading. The letter ϵ is visible after the plus sign in (3). No separate factor precedes it.

Original source details: GELFAND-P000849

$$\sigma^a = (I, i\vec{\sigma}), \qquad a = 1, 2, 3, 4.$$

$$\Sigma_{ab} = \sigma_a\bar{\sigma}_b - \bar{\sigma}_b\sigma_a \tag{4}$$

Now about quantities in spinor notation.

Penrose formulated the theory of gravitation in this way.

Let there be a vector x_a; then

$$X_{\alpha\dot{\alpha}} = \sigma^a_{\alpha\dot{\alpha}}x_a \quad \text{— spinor} \tag{5}$$

$$\Psi_{\alpha_1\ldots\alpha_{2A},\dot{\alpha}_1\ldots\dot{\alpha}_{2B}} = \Psi(A, B) \tag{6}$$

Uncertain reading. The first dotted index is provisionally set as $\dot{\alpha}_1$. The numeral can also be read as 2.

Original source details: GELFAND-P000849

GELFAND-P000850

[source number: 2]

and if one fully symmetrizes, one gets an irred. representation.
Incidentally, about the Pauli matrices: they are as follows.

$$\begin{cases} \sigma^1_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \\ \sigma^2_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \\ \sigma^3_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \\ \sigma^4_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases} \tag{7}$$

Uncertain reading. The first lower index of σ^1 is overwritten. The upper layer reads α. The earlier sign is illegible.

Original source details: GELFAND-P000850

There is also $\epsilon^{\alpha\beta}$

$$\epsilon^{12} = 1, \qquad \epsilon_{\alpha\beta} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \tag{8}$$

Lowering indices:

$$\Psi_\alpha = \epsilon_{\alpha\beta}\Psi^\beta \tag{9}$$

I. M. G.
In coord. $\dot\alpha$ the transf. is complex conjugate.
Form. (5) — constr. of a spinor from a vector

Diagram and typed description. (!)
Circled exclamation mark to the left of the line about formula (5).

Original source details: GELFAND-P000850

Diagram and typed description. — . ——

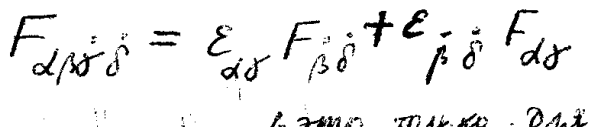
Original source details: GELFAND-P000850

K. Consid. [crossed out: ...] now antisymmetr. tensors

Undeciphered passage. The word crossed out before “now” is illegible.

Original source details: GELFAND-P000850

$$F_{[ab]} \longmapsto F_{\alpha\beta\dot\gamma\dot\delta} = \epsilon_{\alpha\gamma} F_{\dot\beta\dot\delta} + \epsilon_{\dot\beta\dot\delta} F_{\alpha\gamma}, \tag{10}$$

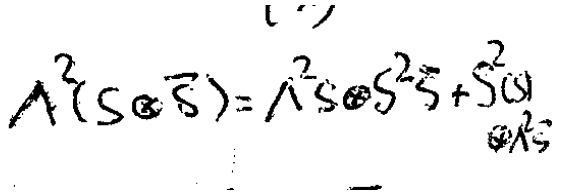


Uncertain reading. On the left of the equals sign the dots are visible above the third and fourth indices. On the right the dotted indices are provisionally read as β, δ.

Original source details: GELFAND-P000850

[Below $F_{[ab]}$:] antisymm. tensor
(this is only for antisymm.)
— easy to check
[At right:] $F_{\alpha\gamma}, F_{\dot\beta\dot\delta}$ — symm. tensors
[In the right margin:]

$$\Lambda^2(S \otimes \bar S) = \Lambda^2 S \otimes S^2 \bar S + S^2(S) \otimes \Lambda^2 \bar S \tag{*}$$



Uncertain reading. The second summand continues on a new line. The S after S^2 appears to be in parentheses.

Original source details: GELFAND-P000850

eigenval. of oper. $*$ — ± 1
Self-dual tensor: def. oper. $*$:

$${}^*F_{ab} = \frac{1}{2}\epsilon_{abcd} F^{cd} \tag{11}$$

self-duality:

$$F_{ab} = \pm {}^*F_{ab}; \quad \text{[crossed out: ...]} \quad \{F_{\alpha\beta} = 0 \quad \text{or} \quad F_{\dot\alpha\dot\beta} = 0\} \tag{12}$$

Undeciphered passage. The crossed-out passage before the brace is illegible.

Original source details: GELFAND-P000850

On gravitation

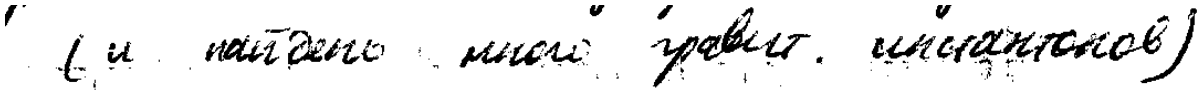
Curvature tensor: R_{abcd}. A metric is assumed to exist
and definition of the operation $*$:

$$ {}^*R_{abcd} = \frac{1}{2}\epsilon_{ab}{}^{c'd'} R_{c'd'cd} \tag{13} $$

Undeciphered passage. Faint marks are visible to the right of (13). They do not yield a reading.

Original source details: GELFAND-P000850

Spinor equivalent. Einste. eqs. follow from the self-duality cond.
(and many grav. instantons have been found)



Original source details: GELFAND-P000850

Uncertain reading. The Russian word after “and” is provisionally read «найдено» (“have been found”).

$$ R_{\alpha\beta\gamma\delta} \quad \text{and} \quad R_{\dot\alpha\dot\beta\dot\gamma\dot\delta} $$
$$ R_+ = R + {}^*R = 0 \quad \text{— see eq. } (*). $$

[At right:] see Hawking’s lect., 1978 with a reference to Penrose

Diagram and typed description. Vertical mark with transverse strokes to the left of the reference to the lectures.

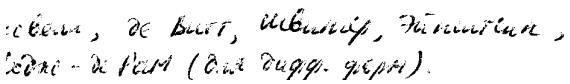
Original source details: GELFAND-P000850

$$ \Phi(A,B) = \Psi_{\alpha_1,\ldots,\alpha_{2A},\dot\alpha_1,\ldots,\dot\alpha_{2B}} \tag{14} $$

Uncertain reading. The letter in (14) is set as Φ. It is unclear whether the top stroke is a separate bar.

Original source details: GELFAND-P000850

The eqs. cont. a bunch of different eqs. — Maxwell, de Wit, Schwinger, Einstein, Hodge–de Rham (for diff. forms).



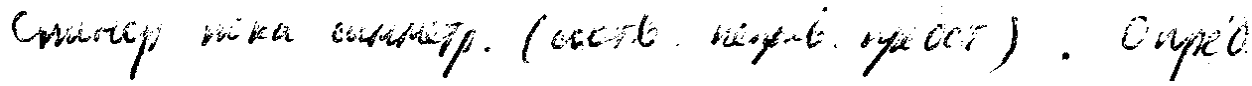
Uncertain reading. The surname after “de Wit” is provisionally read «Швингер» (“Schwinger”).

Original source details: GELFAND-P000850

(see [3])

GELFAND-P000851

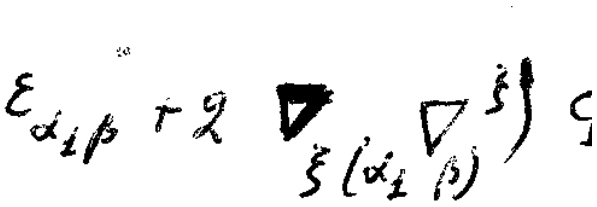
The spinor is symmetr. for now (corresp. irred. repr.). Definition of Δ.



Original source details: GELFAND-P000851

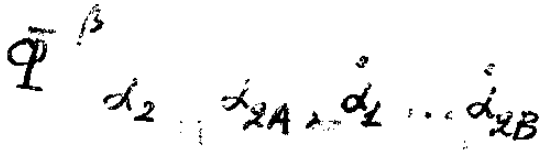
Uncertain reading. The abbreviation before “irred.” is provisionally read «соотв.» (“corresp.”).

$$(\Delta\phi)_{\alpha_1\dots\alpha_{2A},\dot\alpha_1\dots\dot\alpha_{2B}} = \big(-\Box\epsilon_{\alpha_1\beta} + 2\nabla_{\dot\xi(\alpha_1}\nabla^{\dot\xi}_{\beta)}\big) \bar\phi^{\beta}{}_{\alpha_2\dots\alpha_{2A},\dot\alpha_1\dots\dot\alpha_{2B}}. \tag{15}$$



Original source details: GELFAND-P000851

Uncertain reading. The small dots in the contraction are provisionally assigned to the two ξ indices. Parentheses enclose the lower α_1, β.



Original source details: GELFAND-P000851

Uncertain reading. The right-hand side of (15) is provisionally set with $\bar\phi^{\beta}$. A short bar is visible above the letter.

Properties.

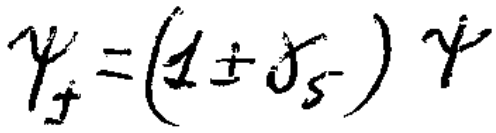
$$1) \qquad (\phi, \Delta\phi) = (\Delta\phi, \phi) \tag{16}$$

$$2) \qquad \Delta\phi(0,0) = -\Box\phi, \qquad \text{here} \quad \Box = \nabla_{\alpha\dot\alpha}\nabla^{\alpha\dot\alpha}, \tag{17}$$

$$\nabla_{\alpha\dot\alpha} = \sigma^a_{\alpha\dot\alpha}\mathcal{D}_a; \qquad \mathcal{D}_a = \partial_a + B_a$$

3) Dirac $\phi(1/2, 0)$ (in curved sp.)

$$\Psi_\pm = (1 \pm \gamma_5)\Psi, \tag{18}$$

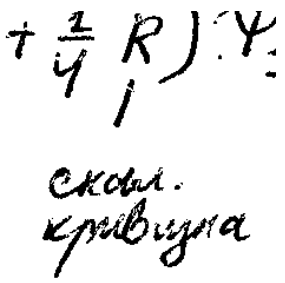


Original source details: GELFAND-P000851

Uncertain reading. A short horizontal stroke follows 5. Its function is unclear.

Ψ — 4-compon. Dirac spinor

$$\Delta\Psi_{\pm} = \left(-\Box + \frac{1}{4}R\right)\Psi_{\pm}$$



R / scal. curvature

Diagram and typed description. A stroke beneath R points to the label.

Original source details: GELFAND-P000851

4) Maxwell in harm. gauge

Uncertain reading. The abbreviation before “gauge” is provisionally read «гарм.» (“harm.”).

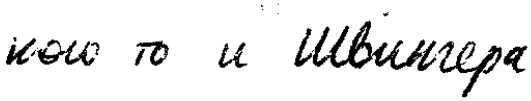
Original source details: GELFAND-P000851

$$\Delta A_a = (-\Box g_{ab} + R_{ab})A^b, \qquad \mathcal{D}^a A_a = 0 \tag{19}$$

Uncertain reading. The operator before A_a is heavily overwritten. It is provisionally set as $\mathcal{D}$. ∇ is possible.

Original source details: GELFAND-P000851

5) $\underline{s = \frac{3}{2}}$ somebody and Schwinger (20)



Uncertain reading. The opening words are provisionally read «кого то» (“somebody”). The name before “and Schwinger” is left unspecified.

Original source details: GELFAND-P000851

6) $\underline{s = 2}$ Einstein (21)

$$7) \qquad w = \frac{1}{p!}\omega_{\mu_1\ldots\mu_p}\, dx^{\mu_1} \wedge \ldots \wedge dx^{\mu_p} \tag{22}$$

Uncertain reading. The sign left of the equals sign resembles Latin w. The indexed symbol at right is ω.

Original source details: GELFAND-P000851

$$\phi(0,0), \quad \phi(1/2,1/2), \quad \phi(1,0), \quad \phi(0,1)$$

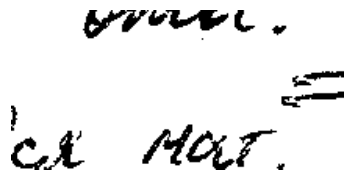
$$\Delta = \delta d + d\delta$$

[crossed out: ...] Stand. theory of diff. forms descr. bosons.

Undeciphered passage. A sign before “Stand.” is crossed out. Its original form is illegible.

Original source details: GELFAND-P000851

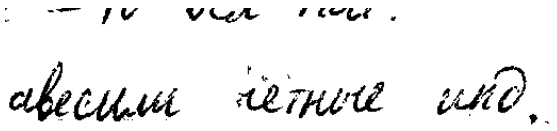
If there were odd ones — then all of math.



Uncertain reading. The line ends with «мат.» (“math.”) and breaks off.

Original source details: GELFAND-P000851

[3] — they attached even ind. to the form.



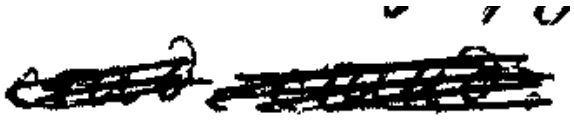
Uncertain reading. The last adjective is provisionally read «чётные» (“even”). Its opening strokes are faint.

Original source details: GELFAND-P000851

Needed: make anticommuting dx.

[At right:] Berezin did part of it.

There are as yet no [crossed out: ... ...]



Undeciphered passage. The two crossed-out words after “no” are illegible.

Original source details: GELFAND-P000851

diff. forms cont. among the dx comm. and anticomm. ones

introduce the operator Δ and construct an index theor.

The boson field Ψ_a gives the form $\Psi_a dx^a$.

For spin $\frac{5}{2}$ none of the eqs. will do (one cannot construct the interaction)

— „ — $\frac{3}{2}$ it has just been done (supergravity)

Idea of I. M. G. possibly, it need not be $d\delta + \delta d$.

GELFAND-P000852

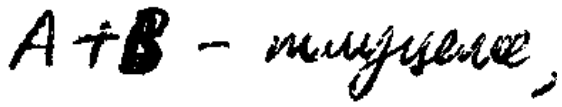
[source number: 3]

Equivalent form of the expression for Δ:

$$\Delta = -\Box + X, \tag{23}$$

$$X = -\Sigma^{ab} R^{cd}_{ab} \Sigma_{cd}, \qquad A + B \text{ — integer} \tag{24}$$

$$X = -\frac{1}{A} \Sigma^{ab} R^{cd}_{+ab} \Sigma_{cd}, \qquad A + B \text{ — half-integer}, \quad A > B. \tag{25}$$



Uncertain reading. The letter B in $A + B$ is heavily retraced. Its underlying form is indistinguishable.

Original source details: GELFAND-P000852

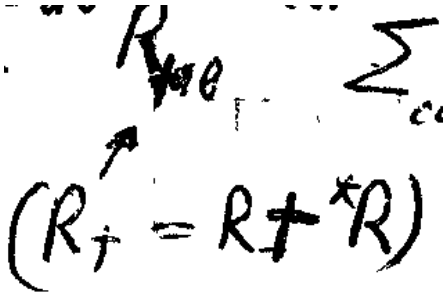


Diagram and typed description.

R^{cd}_{+ab}

$\uparrow$

$(R_+ = R + {}^*R)$

The arrow runs from the insertion to the overwritten lower sign at R in (25).

Original source details: GELFAND-P000852

φ is symmetric here in $\alpha_1 \ldots \alpha_{2A}$ and in $\dot\alpha_1 \ldots \dot\alpha_{2B}$.

$$\Delta\varphi = \lambda\varphi \tag{26}$$

Uncertain reading. The sign after Δ in (26) is overwritten. It is provisionally set as φ.

Original source details: GELFAND-P000852

The compatibility condition will follow from this.

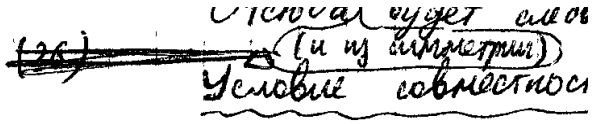


Diagram and typed description.

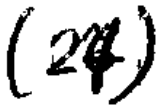
~~(26)~~ → (and from symmetry)

The stroke through the crossed-out number (26) becomes an arrow to the circled insertion.

Original source details: GELFAND-P000852

Compatibility condition:

$$\nabla^{(\alpha_1}_{\dot\alpha} \nabla^{\alpha_2)\dot\alpha} \varphi_{\alpha_1\alpha_2\ldots\alpha_{2A},\dot\alpha_1\ldots\dot\alpha_{2B}} = 0 \tag{27}$$



Uncertain reading. The last digit is overwritten. The label is provisionally set as (27). (28) is possible.

Original source details: GELFAND-P000852

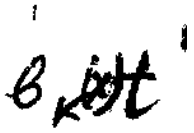
$A \le 1$, $B \le 1 \Rightarrow$ the case of all phys. fields

Consider the heat eq.

$$\frac{d}{dt}F(x,x',t) + \Delta F(x,x',t) = 0, \qquad F\big|_{t=0} = \delta(x-x'). \tag{28}$$

on a compact man.

$$\operatorname{Tr} F(x,x,t) \underset{t\to+0}{\sim} \sum_{k=0}^{\infty} b_k\, t^{(k-4)/2} \tag{29}$$



Uncertain reading. There is an overwritten mark before t, possibly a crossed-out (x).

Original source details: GELFAND-P000852

Undeciphered passage. The earlier writing before the exponent $(k-4)/2$ is crossed out and illegible.

Original source details: GELFAND-P000852

$$B_k = \int d^4x\sqrt{g}\, b_k, \tag{30}$$

Then

$$B_4 = n + m. \tag{31}$$

[At right:] B_4 — the first nondivergent one.

n — number of zero eigenval. (26)

m — [crossed out: ~~number of nonzero eigenval.~~] [Inserted below:] multiplicity of a nonzero (any) eigenval.

[At right:] [their common multiplicity — all multiplicities are the same!!]

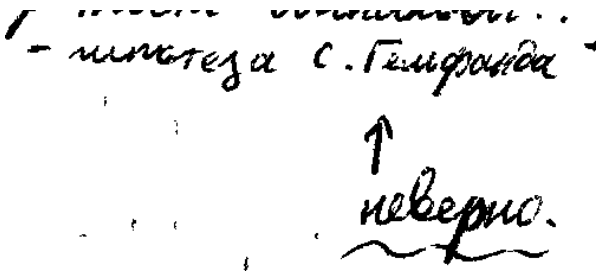


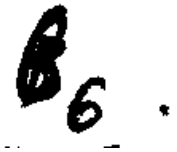
Diagram and typed description.
— S. Gelfand's conjecture
↑
incorrect.

An arrow from the underlined "incorrect" points upward to the conjecture.

Original source details: GELFAND-P000852

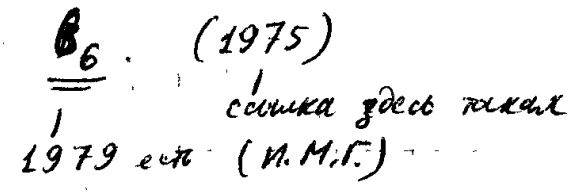
Aim — expr. in terms of top. invariants.

Gilkey got as far as B_6. (1975)



Uncertain reading. The overwritten letter is provisionally set as B. b is possible.

Original source details: GELFAND-P000852



Original source details: GELFAND-P000852

Diagram and typed description.

B_6 (1975)

the reference here is this one

1979 exists (I. M. G.)

Two short strokes link the upper symbols to the notes beneath them.

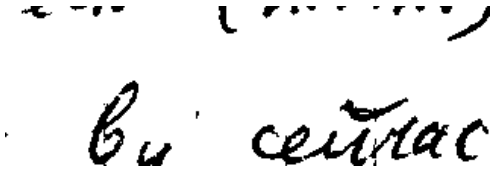
Original source details: GELFAND-P000852

Uncertain reading. The word after 1979 is provisionally read «есть» (“exists”). «ещё» (“also”) is possible.

Original source details: GELFAND-P000852

Uncertain reading. The last word after “the reference here” is provisionally read «такая» (“this one”).

If there is a boundary, then likewise b_4 now [crossed out: ...] recently written up; the question is consid. difficult.



Original source details: GELFAND-P000852

Uncertain reading. The coefficient on the line about the boundary is provisionally read as b_4.

Original source details: GELFAND-P000852

Undeciphered passage. The word before “recently” is crossed out and illegible.

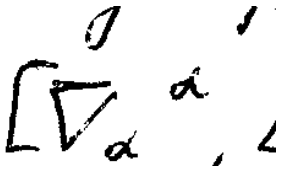
Consid. reducible reprs.

$$\varphi_{\alpha\beta} = \varphi_{(\alpha\beta)} + \frac{1}{2}\varepsilon_{\alpha\beta}\varphi^{\alpha}_{\alpha} \tag{32}$$

$$[1,0] = (1,0) \oplus (0,0)$$

By the compatibility conditions

$$[\nabla^{\dot\alpha}_{\alpha}, \Delta]\varphi[A,B] = 0 \tag{33}$$



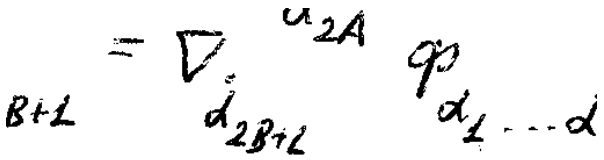
Original source details: GELFAND-P000852

Uncertain reading. There is a small stroke above the upper α. It is provisionally taken as a dot.

$$(\Delta - \lambda)\varphi_{\alpha_1 \ldots} = 0 \tag{34}$$

Consid.

$$\Psi_{\alpha_1 \ldots \alpha_{2A-1} \dot\alpha_1 \ldots \dot\alpha_{2B+1}} = \nabla^{\alpha_{2A}}_{\dot\alpha_{2B+1}} \varphi_{\alpha_1 \ldots \alpha_{2A} \dot\alpha_1 \ldots \dot\alpha_{2B}} \tag{35}$$



Uncertain reading. In (35) the dot belongs to the lower derivative index. The upper index is provisionally read as α_{2A}.

Original source details: GELFAND-P000853

$\Psi \neq 0$ when $\varphi \neq 0$ (otherwise $\overset{\circ}{\times}$ with (34)).

Uncertain reading. The small oval above the diagonal cross is provisionally set as a circle. The meaning of the mark is unclear.

Original source details: GELFAND-P000853

$$m[A, B] = m[A - \tfrac{1}{2}, B + \tfrac{1}{2}] \tag{36}$$

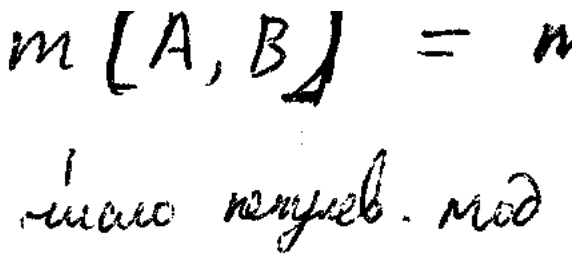


Diagram and typed description.
$m[A, B]$
|
number of nonzero modes
A short stroke below m points to the label.

Original source details: GELFAND-P000853

$$A + B = S, \qquad A - B = T.$$

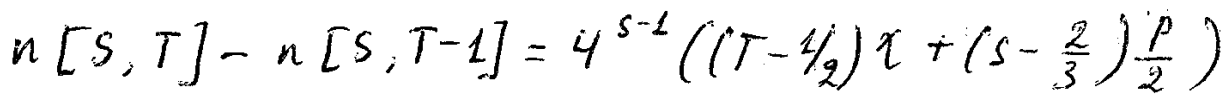
Then we get

$$m[S, T] = m[S] \tag{37}$$

Hence

$$n[S, T] - n[S, T'] = B_4[S, T] - B_4[S, T'] \tag{38}$$

$$n[S, T] - n[S, T - 1] = 4^{s-1}\big((T - \tfrac{1}{2})\chi + (s - \tfrac{2}{3})\tfrac{p}{2}\big) \tag{39}$$

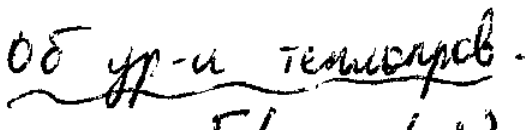


Original source details: GELFAND-P000853

Uncertain reading. The two letters at right in (39) are provisionally read as lowercase s. The notation at left looks like uppercase S.

χ — Euler number, p — Pontryagin number

On the heat eq.



Original source details: GELFAND-P000853

Uncertain reading. The end of the underlined heading is provisionally read «теплопров.» (“heat”).

$$F(x,x',t)=\sum_{i=0}^{\infty}\varphi_i(x)\varphi_i(x')e^{-\lambda_i t} \tag{40}$$

φ_i — orthonorm. syst.

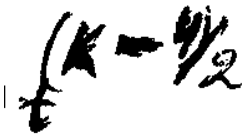
$$F(x,x',0)=\delta(x,x') \tag{41}$$

$$\int_M d^4x\sqrt{g}\,F(x,x,t)=\sum_{i=0}^{\infty}e^{-\lambda_i t}=n+\sum_{i=n+1}^{\infty}e^{-\lambda_i t} \tag{42}$$

Original source details: GELFAND-P000853

Uncertain reading. There is a small blacked-out mark above the second x, possibly a prime.

$$F(x,x,t)\sim\sum_{k=0}^{\infty}b_k(x)t^{(k-4/2} \tag{43}$$



Original source details: GELFAND-P000853

Uncertain reading. The 4 is overwritten. A separate closing parenthesis before the fraction slash cannot be distinguished.

m is determ. from this!! — by discarding the diverg. terms of $\sum e^{-\lambda_i t}$ and n contributes.

m — regulariz. sum

Original source details: GELFAND-P000853

Uncertain reading. The abbreviation in the underlined note is provisionally read «регуляриз.» (“regulariz.”).

“m — is an integer (!) on deep top. grounds ”

These are the stand. one-loop diverg. of the cont. integral.

Original source details: GELFAND-P000853

Uncertain reading. The Russian end of the line is provisionally read «контин. интеграла». The last word looks like a genitive form.

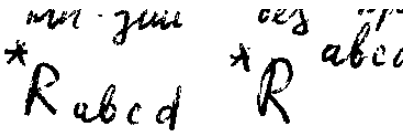
[At right: в?]

Uncertain reading. The isolated dark sign may be the letter «в». Its reading is uncertain.

Original source details: GELFAND-P000853

Def. χ and p for manifs. without boundary

$$\chi = \int {}^*R_{abcd}\, {}^*R^{abcd}\, d^4x\sqrt{g} \tag{44}$$



Uncertain reading. Stars are visible before both occurrences of R in (44).

Original source details: GELFAND-P000853

$$p = \int {}^*R_{abcd} R^{abcd}\, d^4x\, \sqrt{g} \tag{45}$$

GELFAND-P000854

[source number: 4]

8.10.79. Gelfand Sem.

Draft theor. minimum, needed in any specialization. (for 1st-year students)

1. Algebra

van der Waerden, chs. 1–5, 12.

Gelfand lectures on lin. algebra

Atiyah, Macdonald

2. Analysis

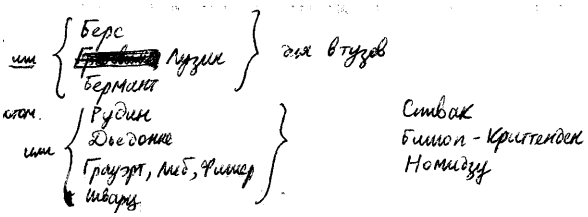


Diagram and typed description.

or { Bers / [crossed out: ...] Luzin / Berman } for technical institutes.

then.

or { Rudin / Dieudonné / Grauert, Lieb, Fischer / Schwartz }

[At right:]
Spivak
Bishop–Crittenden
Nomizu

Two pairs of braces group the lists. Three separate lines stand to the right of the second pair.

Original source details: GELFAND-P000854

Original source details: GELFAND-P000854

Uncertain reading. A surname is crossed out before “Luzin.” “Berman” is a possible reading.

Check question: calculate approximately

$$\int_{-\infty}^{\infty} e^{-x^2} \cos 1000x \, dx, \qquad \int_{-\infty}^{\infty} e^{-1000x^2} \cos x \, dx$$

(and comp. with tables)

$$\int_{-\infty}^{\infty} e^{-1000x^4} \cos x \, dx, \qquad \int_{-\infty}^{\infty} e^{-x^4} \cos 1000x \, dx$$

3. Arnold I, II, III.

4. Kirillov $\frac{1}{2} + \frac{1}{2}$

Serre lin. reprs. of finite groups

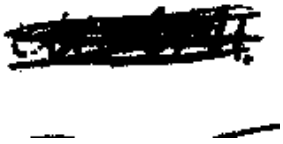
(he does not understand the 3rd chapter)

Generalized fns. I issue, chs. 1–2

5. Know the partial differential equations. (there is nothing to read).

Check problem in lin. algebra:

find the can. form of a self-adj. oper. in a [crossed out: ...] symplect. space.



Original source details: GELFAND-P000854

Undeciphered passage. The word after “in” is crossed out and illegible.

Original source details: GELFAND-P000854

Uncertain reading. The abbreviation after the crossed-out word is provisionally read «симплект.» (“symplect.”).

6. [blacked out] Spec. theory of rel. + GR

Original source details: GELFAND-P000854

Uncertain reading. A single sign before “Spec.” is blacked out. It may be the letter «А».

+ geometry

[At right:] Landau, Lifshitz (vol. 2)

jointly with Nomizu

Original source details: GELFAND-P000854

Uncertain reading. The abbreviation before “with Nomizu” is provisionally read «совм.» (“jointly”).

S. Gindikin. Beginnings of twistor theory.

To be presented in the language of integr. geometry.

[At right:] Twistors were introduced by Penrose.

see one of the recent issues of BAMS,

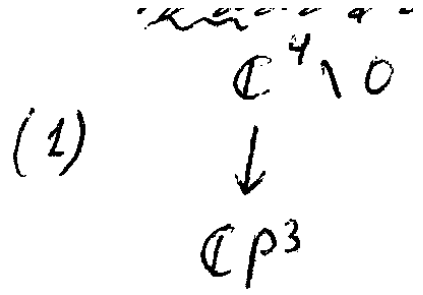
1979, vol. 1 (new series),

art. by Wells (Wells.)

[crossed out: ~~... let us consider?~~]

Original source details: GELFAND-P000854

Undeciphered passage. The crossed-out line before (1) cannot be read in full. The last word is provisionally read «рассмотрим» (“let us consider”).



Original source details: GELFAND-P000854

Diagram and typed description.

$$(1) \qquad \begin{array}{c} \mathbb{C}^4 \setminus 0 \\ \big\downarrow \\ \mathbb{C}P^3 \end{array}$$

In $\mathbb{C}^4$ fix a Hermitian form [blacked out] $(++--)$. $\langle\ ,\ \rangle$.

Original source details: GELFAND-P000854

Undeciphered passage. A sign is blacked out before the parentheses with the signature. It is illegible.

We obtain α-twistors: elements of $\mathbb{C}P^3$ or corresp. elems. of $\mathbb{C}^4$.

$(\mathbb{C}P^3)'$ — dual sp.. A form arises there too, and these are — β-twistors.

$$\langle z, z\rangle > 0 \Rightarrow \text{pos. twistor}$$
$$\langle z, z\rangle = 0 \Rightarrow \text{null} — „ —$$
$$\langle z, z\rangle < 0 \Rightarrow \text{neg.} — „ —$$

In $\mathbb{C}P^3$ consid. lines (2-dim. subsp. in $\mathbb{C}^4$). Their set: $G = G_{4,2} = G^{\mathbb{C}}_{4,2}$.

Over G there is a natural princ. bundle of 2-frames in $\mathbb{C}^4$:

Original source details: GELFAND-P000855

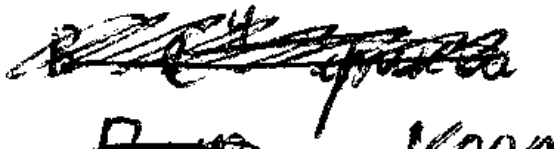
Uncertain reading. The word before “bundle” is provisionally read «натурельное» (“natural”). Its middle and final strokes run together.

Original source details: GELFAND-P000855

(2)

Diagram and typed description.

[Crossed out above the heading: ~~In $\mathbb{C}^4$ ...~~]



Original source details: GELFAND-P000855

Undeciphered passage. The crossed-out line may begin «В $\mathbb{C}^4$» (“In $\mathbb{C}^4$”). The full line is illegible.

[Crossed out at left: ~~...~~] Coordinates in G.

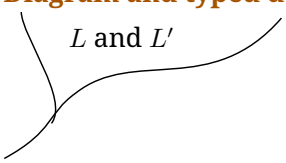
Original source details: GELFAND-P000855

Undeciphered passage. A separate short entry at the left of the heading is crossed out and illegible.

Choose in $\mathbb{C}^4$ 2 planes (transvers.) L and L', then any other is a graph, the matrix of the corresp. transf. gives coordinates.

Original source details: GELFAND-P000855

Diagram and typed description.



The insertion mark leads to the plane labels above the line.

Choose in [crossed out: ...] L basis $e_0\ e_1$; In L' basis e_3, e_4.

Original source details: GELFAND-P000855

Undeciphered passage. The entry before L is crossed out and illegible.

Then if $\widetilde{L} \in G_{4,2}$, then in $\widetilde{L}$ there is a basis of the form

Original source details: GELFAND-P000855

Uncertain reading. The repeated plane symbol is heavily retraced. It is provisionally set as $\widetilde{L}$.

$$\begin{cases} f_0 = e_0 + u_{03}e_3 + u_{04}e_4 \\ f_1 = e_1 + u_{13}e_3 + u_{14}e_4 \end{cases} \tag{3}$$

Original source details: GELFAND-P000855

Uncertain reading. The first subscript of u before e_3 in the second line is provisionally read as 1. The numeral has been retraced.

u_{ij} — coordinates.

Let L, L' be transvers. and isotropic with resp. to $\langle \cdot, \cdot \rangle$. Problem. Such L and L' exist.

$$\{e_A,\ A = 0, 1\} \text{ — basis in } L. \tag{4}$$

In L' introduce a basis

$$\{e^{A'},\ A' = 0', 1'\} \text{ — basis in } L' \tag{5}$$

[crossed out: ...] [above the line: Let [crossed out: ...]] the form [above the line: $\langle\ ,\ \rangle$] in the basis $(e_0, e_1, e^{0'}, e^{1'})$ have [crossed out: ...] the matrix

Original source details: GELFAND-P000855

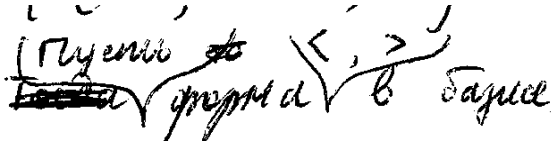
Undeciphered passage. The entry before "the form" on the main line is crossed out and illegible.

Original source details: GELFAND-P000855

Undeciphered passage. A short mark after the inserted "Let" is crossed out and illegible.

Original source details: GELFAND-P000855

Undeciphered passage. The word between “have” and “the matrix” is crossed out and illegible.



Original source details: GELFAND-P000855

Diagram and typed description.

Let ... $\langle\ ,\ \rangle$

... the form in the basis

Two marks give the insertion points for “Let” and the angle brackets.

$$\begin{pmatrix} 0 & iE \\ -iE & 0 \end{pmatrix}; \tag{6}$$

i.e.

$$\langle e_0, e^{0'} \rangle = \langle e_1, e^{1'} \rangle = i, \qquad \text{the other scal. prod.} = 0 \tag{7}$$

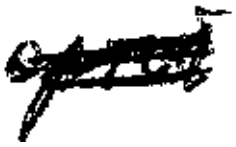
Make a change of variables in L:

$$\widetilde{e}_B = g^A_B e_A \tag{8}$$

Then it is easy to check that

$$e^{A'} \text{ transf. by means of } (g^*)^{-1} \tag{9}$$

(follows from the relat. of [crossed out: ...] orthogonality)



Original source details: GELFAND-P000855

Undeciphered passage. The entry before “orthogonality” is crossed out and illegible.

There are other transf. laws: [Crossed out at left: ...]

Original source details: GELFAND-P000855

Undeciphered passage. The mark before the lines with indices is crossed out and illegible.

Original source details: GELFAND-P000855

Diagram and typed description.

A' by $\bar{g}$ $\qquad$ B' by $(g^*)^{-1}$

B' — „ — $(g^T)^{-1}$

The arc is above A' and below B and B'. The meaning of the arcs is unclear.

GELFAND-P000856

[source number: 5]

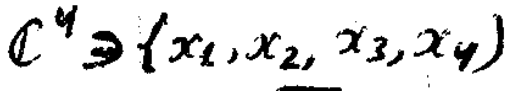
Problem. Prove that $U(2,2)$ has a subgroup isom. to $GL(2,\mathbb{C})$ (hint: use the preceding result).

Coordinates:

$$\{z^A, z_{A'}\} \qquad \text{in the basis } \{e_A, e^{A'}\} \tag{10}$$

$$z = z^A e_A + z_{A'} e^{A'} \in \mathbb{C}^4 \tag{11}$$

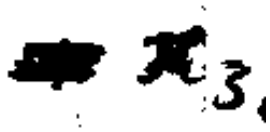
I.M.G. $\mathbb{C}^4 \ni \{x_1, x_2, x_3, x_4\}$.



Original source details: GELFAND-P000856

Uncertain reading. The opening mark resembles a brace. The closing mark is a parenthesis.

$$\langle x, y\rangle = x_1\bar{y}_1 + x_2\bar{y}_2 - x_3\bar{y}_3 - x_4\bar{y}_4 \tag{12}$$



Original source details: GELFAND-P000856

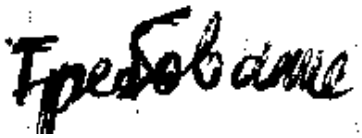
Uncertain reading. The mark before the third term is provisionally read as a minus sign. The stroke has been overwritten.

L, L' — 2 isotropic subsp. and 2 bases $\{e_0, e_1\}$; $\{e^{0'}, e^{1'}\}$. Then

$$z = z^0 e_0 + z^1 e_1 + z_{0'} e^{0'} + z_{1'} e^{1'} \tag{13}$$

$$\{z^A, z_{A'}\} \text{ — spinor coordinates in } \mathbb{C}^4 \tag{14}$$

Requirement on the basis:



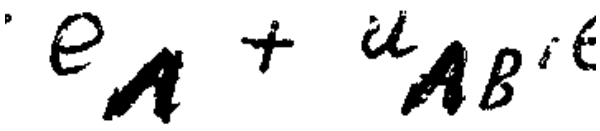
Original source details: GELFAND-P000856

Uncertain reading. The initial letter of «Требование» (“Requirement”) is retraced.

$$\langle e_0, e^{0'}\rangle = \langle e_1, e^{1'}\rangle = i, \qquad \text{the other scal. prod.} = 0. \tag{15}$$

S.G. Let $\widetilde{L} \in G$; $\widetilde{L} \cap L' = \{0\}$. Then choose a basis as follows:

$$f_A = e_A + u_{AB'} e^{B'} \tag{16}$$



Original source details: GELFAND-P000856

Uncertain reading. The two indices on the left and in the first term on the right are retraced and read as A.

The spinor indices themselves show how they transform. (worth writing out!)

(Twistor theory in curved space has yet to be worked out, though there are already papers

on it). Consider groups acting in $\mathbb{C}^4$. The action of $GL(4,\mathbb{C})$ carries over to $G_{4,2}$.

$$\begin{aligned} T_+ &: \quad \langle z,z\rangle > 0 \\ T_- &: \quad \langle z,z\rangle < 0 \\ N &: \quad \langle z,z\rangle = 0 \qquad \text{— 7-dimensional surface in } \mathbb{C}^4. \end{aligned}$$

S_+ — 2-dimensional planes $\subset T_+$ (except 0)
S_- — — „ — $\subset T_-$ (— „ —)
M — — „ — $\subset N$

S_+, S_- — lie entirely within the range of the coord. $u_{AB'}$.

$$\begin{cases} S_+ = \{u : \ \frac{1}{i}(u-u^*) \gg 0\} \\ S_- = \{u : \ \frac{1}{i}(u-u^*) \ll 0\} \end{cases} \tag{17}$$

GELFAND-P000857

M no longer lies wholly in the coord. neighborhood. Let M_0 — be the part that does lie in it.

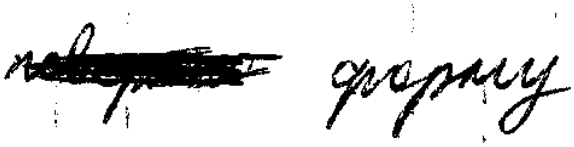
$$M_0 = \{u : \ u = u^*\} \tag{18}$$

$$\dim_{\mathbb{R}} M_0 = 4, \qquad \dim_{\mathbb{C}} S^{\pm} = 4.$$

M_0 — the common skeleton of $S^{\pm}$.

$$\partial S_+ = \{\text{supporting planes to } T_+\}$$

Now consider in $\mathbb{C}^4$ the group $U(2,2)$, preserving [crossed out: ...] the form



Undeciphered passage. The entry before “the form” is crossed out and illegible.

Original source details: GELFAND-P000857

(and, hence, the surface N). Then it is clear that $U(2,2)$ preserves all [crossed out: ...] the domains $S_\pm, M$

Undeciphered passage. The entry before “domains” is crossed out and illegible.

Original source details: GELFAND-P000857

(canon. construction) and acts on each transitively!! (A problem in lin. algebra!)

On $(\mathbb{C}^4, \langle\ ,\ \rangle)$ acts $U(2,2)$ and preserves the whole construction. $g \in U(2,2)$ is written in a basis of the kind descr. above:

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{19}$$

Action on u:

$$g:\ u \mapsto (Au + B)(Cu + D)^{-1} \tag{20}$$

The action makes M into the conformal compactification of Minkowski space: the Poincaré gr. $\subset SU(2,2)$ (addition of conformal transf.).

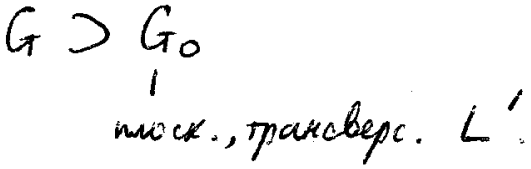


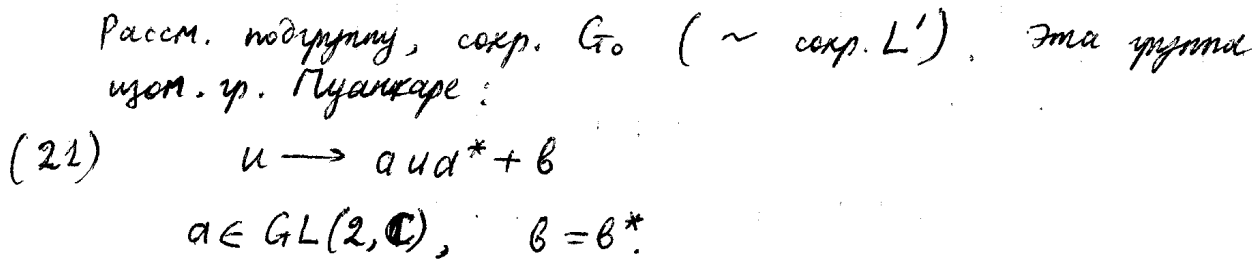
Diagram and typed description.
$G \supset G_0$
|
planes transv. to L'.

Original source details: GELFAND-P000857

Consider the subgroup preserving G_0 ($\sim$ preserving L'). This group is isom. to the Poincaré gr.:

$$u \to aua^* + b \tag{21}$$

$a \in GL(2,\mathbb{C})$, $b = b^*$.



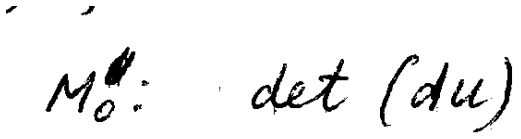
Original source details: GELFAND-P000857

Manuscript excerpt. The manuscript reads “This group is isom. to the Poincaré gr.” Equation (21) and its parameters follow.

In the notation of (19) here:

$$A = a, \quad C = 0, \quad D = (a^*)^{-1}, \quad BD^{-1} = b. \tag{22}$$

The Minkowski metric on $M_0^{\cdots}$: $\det(du)$ (23)

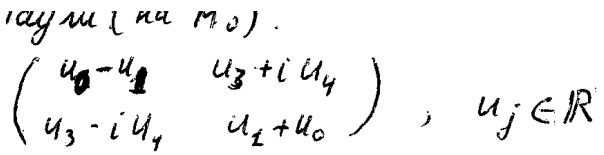


Uncertain reading. The stroke above M_0 is obscured. The superscript is illegible.

Original source details: GELFAND-P000857

Pauli coord. (on M_0):

$$\begin{pmatrix} u_0 - u_1 & u_3 + iu_4 \\ u_3 - iu_4 & u_1 + u_0 \end{pmatrix}, \qquad u_j \in \mathbb{R} \tag{24}$$



Uncertain reading. The two indices in the upper left entry are retraced and read as 0 and 1.

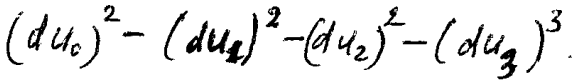
Original source details: GELFAND-P000857

GELFAND-P000858

[source number: 6]

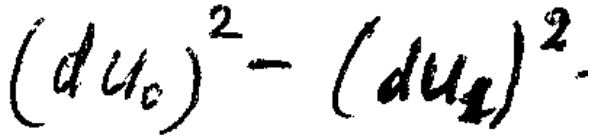
Then the metric takes the form

$$(du_0)^2 - (du_1)^2 - (du_2)^2 - (du_3)^3. \tag{25}$$



Uncertain reading. The last term has exponent 3 in the manuscript.

Original source details: GELFAND-P000858



Uncertain reading. The first term reads 0. The retraced subscript in the second term reads 1.

Original source details: GELFAND-P000858

$$M_0 = M \cap G_0 \underset{\text{def}}{=} \text{ordinary Minkowski space}. \tag{26}$$

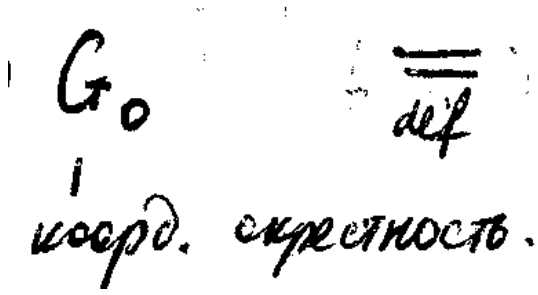


Diagram and typed description.

G_0

|

coord. neighborhood.

Original source details: GELFAND-P000858

A point of M — is a point of G or a line in $\mathbb{C}P^3$. Restrict attention to M_0.

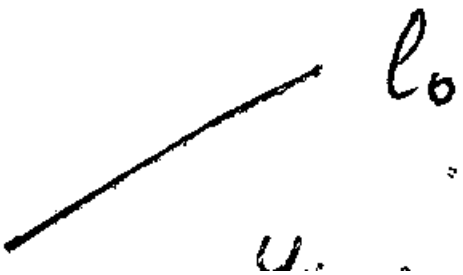


Diagram and typed description.

ℓ_0

Original source details: GELFAND-P000858

Consider all lines intersecting ℓ_0. Their eq. in coordinates:

$$\det(u^0 - u) = 0. \tag{27}$$

It turns out that this is [crossed out: ...] the Minkowski isotropy cone (the cone of null rays).

Original source details: GELFAND-P000858

Undeciphered passage. The word before "the Minkowski isotropy cone" is crossed out and illegible.

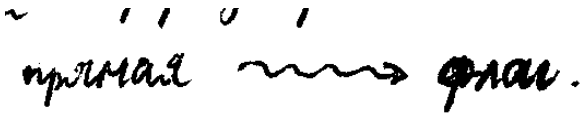
$$(27) \cap M = \{\text{light cone}\}.$$

There is a one-to-one corresp. between points of M and rays on N.

Original source details: GELFAND-P000858

Diagram and typed description.
Light rays on M
$=$ isotropic points in $\mathbb{C}P^3$

Contest: formulate a higher-dimensional generalization of this situation.



Original source details: GELFAND-P000858

Diagram and typed description.
line $\rightsquigarrow$ flag.

— · —

GELFAND-P000859

15.10.79. Gelfand Sem.

F. A. Berezin. Supermathematics.

$\mathbb{K}$ — a field ($\mathbb{R}$ or $\mathbb{C}$).

$\mathbb{Z}_2$-graded space $L = L_0 \oplus L_1$.

The elements of L_0 and L_1 are called homogeneous.

Parity

$$\alpha(x) = \begin{cases} 0, & x \in L_0 \\ 1, & x \in L_1. \end{cases}$$

Sums and prod. are taken mod2.

Grassm. algebra Λ over $\mathbb{K}$ — assoc. alg. with generators

$$1\ (\in \mathbb{K}), \quad \xi_1, \ldots, \xi_N$$

and relat.

$$\xi_i \xi_j + \xi_j \xi_i = 0, \qquad \forall i, j.$$

$$\{1,\ \xi_{i_1} \ldots \xi_{i_k},\ i_1 < i_2 < \ldots < i_k\} \quad \text{— a basis in } \Lambda \text{ over } \mathbb{K}.$$

$\Lambda = \Lambda_0 \oplus \Lambda_1$, Λ_0 is gen. by 1 and prod. of an even number of ξ_i.

GELFAND-P000860

[source number: 10]

$$\dim_{\mathbb{K}} \Lambda = 2^N.$$

For homogeneous f, g

$$f \cdot g = (-1)^{\alpha(f)\cdot\alpha(g)} gf. \tag{1}$$

Grassmann envelope of the lin. space L:

[crossed out: ~~(2) $L(\Lambda) = L \otimes_{\mathbb{K}} \Lambda$~~]

Original source details: GELFAND-P000860

Uncertain reading. The hatched-out entry provisionally reads (2) $L(\Lambda) = L \otimes_{\mathbb{K}} \Lambda$.

e_i — a basis in L of homog. elements ($L = L_0 \oplus L_1$).

Now consider

$$\left\{ \sum a_i e_i,\ a_i \in \Lambda,\ \alpha(a_i) = \alpha(e_i) \right\} \underset{\text{def}}{=} L(\Lambda). \tag{2}$$

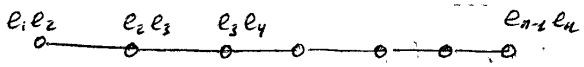
This is a left Λ_0-module. Equivalently:

$$L(\Lambda) = \left(\Lambda \underset{\mathbb{K}}{\otimes} L \right)_{\bar{0}}. \tag{2}'$$

Linear operators.

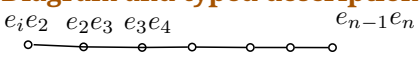
I.M.G. These alg. arise in the theory of quiver repr. (cf. FA, 1979; $N = 3$).

Conj.: $\forall$ Dynkin diagram there should be an analogue of the even part of the Grassm. algebra. (so that the ordinary one corresp. to the diagram A_n



Original source details: GELFAND-P000860

Diagram and typed description.



The dotted first subscript is read as i. No arrowheads can be made out on the chain.

if joined by an arrow, then the prod. is $= 0$, if unjoined, then there is a new element. Idea: there are other alg. of Grassmann type.

— · —

operator in L:

Original source details: GELFAND-P000860

Uncertain reading. There is a dark stroke at the right end of L. A separate subscript cannot be established.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}. \tag{3}$$

Grading:

[crossed out: ...]

Undeciphered passage. The sign before the first matrix is crossed out. It may be A.

Original source details: GELFAND-P000860

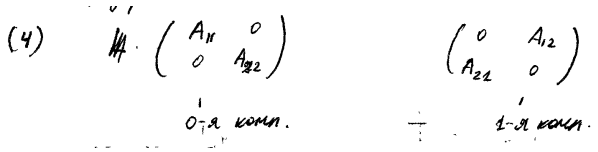
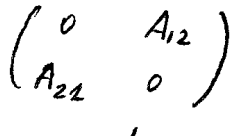


Diagram and typed description.

(4) $\begin{pmatrix} A_{11} & 0 \\ 0 & A_{22} \end{pmatrix}$ $\begin{pmatrix} 0 & A_{12} \\ A_{21} & 0 \end{pmatrix}$

0th comp. 1st comp.

Original source details: GELFAND-P000860



Uncertain reading. The lower left block in the second matrix is read as A_{21}.

Original source details: GELFAND-P000860

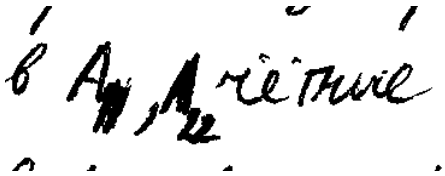
The space $L_{p,q}$ — all such oper., $p = \dim L_0$, $q = \dim L_1$.

$L_{p,q}(\Lambda)$ — Grassm. envelope:

Remark: a_i and e_i are always taken to commute.

$L_{p,q}(\Lambda)$ — also matrices of Grassm. numbers:

in A_{11}, A_{22} even



Uncertain reading. The first pair of subscripts is heavily retraced and provisionally reads 11. The second reads 22.

Original source details: GELFAND-P000860

in A_{12}, A_{21} odd.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in L_{p,q}(\Lambda).$$

$L_{p,q}(\Lambda)$ — is an algebra (and a Λ_0-module).

Supertrace

$$\operatorname{str} A = \operatorname{tr} A_{11} - \operatorname{tr} A_{22}. \tag{5}$$

Prop.:

$$\operatorname{str}(AB) = \operatorname{str}(BA) \tag{6}$$

Superdeterminant. (Berezinian):

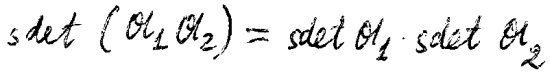
$$\operatorname{sdet} = \det(A_{11} - A_{12}A_{22}^{-1}A_{21})(\det A_{22})^{-1} \tag{7}$$

(for numerical matrices there is a fmla. [crossed out: ...] of the same form, but with $\det A_{22}$ in place of $(\det A_{22})^{-1}$).

Undeciphered passage. The entry after "fmla." is crossed out and illegible.

Original source details: GELFAND-P000861

$$\operatorname{sdet}(\alpha_1\alpha_2) = \operatorname{sdet}\alpha_1 \cdot \operatorname{sdet}\alpha_2 \tag{8}$$



Uncertain reading. The symbols in (8) are provisionally set as α_1, α_2. They could be cursive capitals A_1, A_2.

Original source details: GELFAND-P000861

Exercise. Do this if $\dim L_0 = \dim L_1 = 1$, i.e. for matrices of order 2.

Explanation. Consider matrices with anticomm. entries:

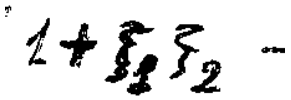
$$\operatorname{sdet}\begin{pmatrix} 1 & \xi_1 \\ \xi_2 & 1 \end{pmatrix} = 1 - \xi_1\xi_2.$$

$$\begin{pmatrix} 1 & \xi_1 \\ \xi_2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1+\xi_1\xi_2 & 2\xi_1 \\ 2\xi_2 & 1+\xi_2\xi_1 \end{pmatrix}.$$

Diagram and typed description. A small hash-like sign sits to the right of the matrix. Its purpose is unclear.

Original source details: GELFAND-P000861

$$\text{sdet} = (1 + \xi_1\xi_2 - 2\xi_1(1 + \xi_1\xi_2)2\xi_2)(1 + \xi_1\xi_2) =$$
$$= (1 - 3\xi_1\xi_2)(1 + \xi_1\xi_2) = 1 - 2\xi_1\xi_2 = (1 - \xi_1\xi_2)^2.$$



Uncertain reading. In the first product after $1+$, the first subscript is retraced and read as 1.

Original source details: GELFAND-P000861

Reduction of matrices to canon. form.

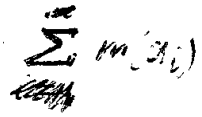
$L(\Lambda)$

Uncertain reading. The separate faded entry is provisionally read as $L(\Lambda)$.

Original source details: GELFAND-P000861

There is a mapping

$$m. \quad L(\Lambda) \longrightarrow L_0$$
$$\sum a_i e_i \longmapsto \left(\sum a_i e_i\right)\Big|_{\xi=0} = \sum_i m(a_i)e_i,$$
$$m(a_i) = a_i|_{\xi=0}.$$



Undeciphered passage. The lower entry at the last summation is crossed out. An undeciphered stroke is visible above the sign.

Original source details: GELFAND-P000861

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in L_{p,q}(\Lambda).$$

Uncertain reading. The stroke of L in the second matrix line is broken. The expression is provisionally read as $L_{p,q}(\Lambda)$.

Original source details: GELFAND-P000861

$$m(A) = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}. \tag{9}$$

Let $\mathbb{K} = \mathbb{C}$.

Theorem.

Let $m(A)$ have distinct eigenv.. Then A also reduces to diag. form:

$$BAB^{-1} = \begin{pmatrix} c_1 & & 0 \\ & \ddots & \\ 0 & & c_{p+q} \end{pmatrix}. \tag{10}$$

GELFAND-P000862

[source number: 11]

Analogues of Jordan blocks are unknown here.

Problem 1. Do this (analogue of normal forms)!

A problem related to instantons:

Problem 2. Consid. the matrix $A = (a_{ik})_{i,k=1}^{n}$ with entries $a_{ik} \in \Lambda_1$ (odd Grassm. numbers).

Describe all $B = (b_{ik})_{i,k=1}^{n}$, with entr. $b_{ik} \in \Lambda_0$ such that

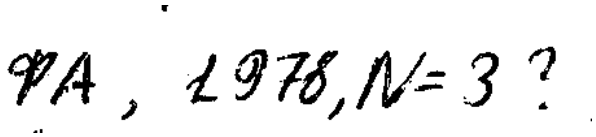
$$AB = BA.$$

If Λ has 4 generators, then it has a relation to Y.-M. If all entr. of A are independent (in Λ), then it can be solved.

— · —

S.G. Connection with the work of Beilinson and Bernst.–Gelf.–Gelf.

(see FA, 1978, $N = 3$?; or Drinfeld, Manin; Nucl. Physics, $N = 6$, 1979: the constr. is given explicitly there, translated into the language of solutions).

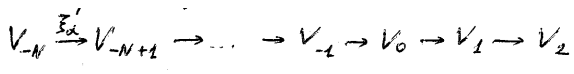


Original source details: GELFAND-P000862

Uncertain reading. The question mark after $N = 3$ is in the manuscript.

classif. of vect. bundles on $\mathbb{C}P^3$. (holom.)

Objects — graded modules over $\Lambda = \Lambda(4)$:



Original source details: GELFAND-P000862

Diagram and typed description.

$V_{-N} \xrightarrow{\xi_\alpha} V_{-N+1} \longrightarrow \cdots \longrightarrow V_{-1} \to V_0 \to V_1 \to V_2$

$\alpha = 1, 2, 3, 4$. Repr. of the Grassm. algebra:

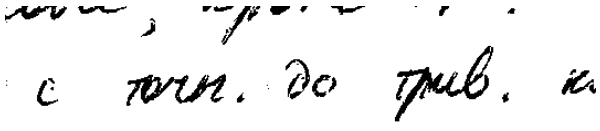
$$\xi_i \xi_j + \xi_j \xi_i = 0.$$

$a_1\xi_1 + a_2\xi_2 + a_3\xi_3 + a_4\xi_4$ — an operator from V_k to V_{k+1}

and we get a complex;

the cohom. vanish except for H^0.

Such modules up to triv. classify instantons.



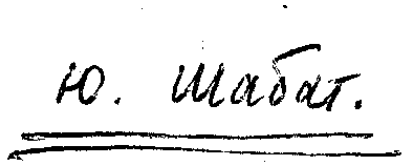
Original source details: GELFAND-P000862

Uncertain reading. The abbreviation after “up to” provisionally reads «трив.» (“triv.”).

Problem: give an explicit construction. It has yet to be done (the spaces are constructed, but the mappings are missing).

Problem: give the classif. for the Grassmannian!!

Yu. Shabat.



Original source details: GELFAND-P000862

Uncertain reading. The manuscript gives the initial «Ю.» (“Yu.”). MCCME catalogue record 469 gives the name as “G.B. Shabat”.

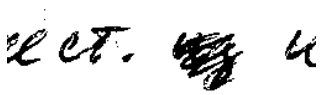
Lie–Wirtinger theorem (classical)

$\Gamma \subset \mathbb{C}^n$ — a germ of an analytic hypersurface. It is called a double translation surf., if it can be represented in 2 ways as a sum

$$\Gamma = \gamma_1 + \ldots + \gamma_{n-1}, \qquad \Gamma = \delta_1 + \ldots + \delta_{n-1}$$

γ_i, δ_i — germs of analytic curves.

$\{\gamma_i\}$ and $\{\delta_i\}$ cannot be obtained from one another by perm. [crossed out: ...] and translations.



Original source details: GELFAND-P000862

Undeciphered passage. The entry after “perm.” is crossed out and illegible.

Example: X — a smooth curve of genus n.

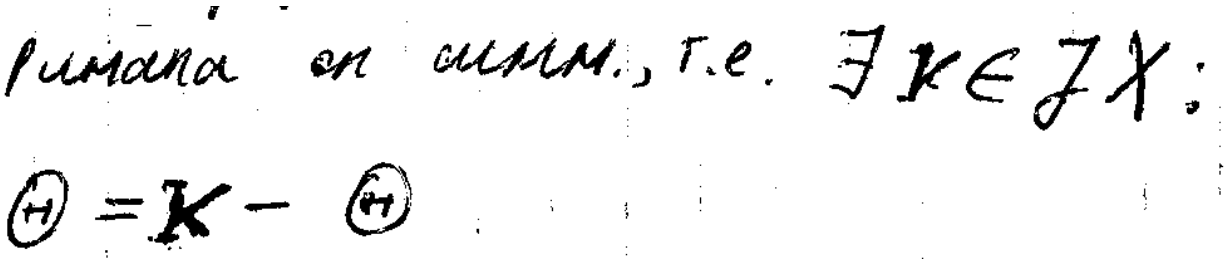
JX — the Jacobian of the curve X; $\quad j : X \longrightarrow JX$.

GELFAND-P000863

$$\Theta = \underbrace{j(X) + \ldots + j(X)}_{n-1 \text{ times}}, \qquad (\Theta \text{ — Poincaré divisor}).$$

By Riemann's thm. it is symm., i.e. $\exists K \in JX$:

$$\Theta = K - \Theta.$$



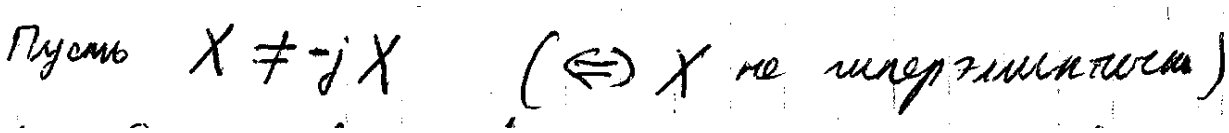
Original source details: GELFAND-P000863

Uncertain reading. The sign is set as K in both places. Another possible reading is κ.

Therefore up to translation

$$\Theta \sim \underbrace{-jX - \ldots - jX}_{n-1}.$$

Let $X \neq -jX$ ($\Leftrightarrow X$ is nonhyperelliptic).



Original source details: GELFAND-P000863

Uncertain reading. The manuscript has X on the left with no preceding j.

Assertion: all translation surf. are these surf. and their degenerations
(if Γ — a double translation surf., then a small deform. of Γ — a germ of $\Theta \subset JX$).
Conjecture of I.M.G.: 1) All integr. systems have this theorem as their basis.
2) This thm. is the first in a series of coloss. theorems of this kind.

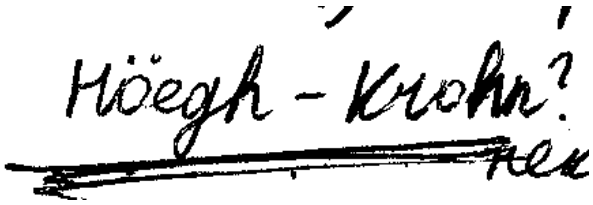
Uncertain reading. The abbreviation «колосал.» is a provisional reading. The middle strokes run together.

Original source details: GELFAND-P000863

1) — the reason for the appearance of Θ-funcs..

— · —

Höegh-Krohn? someone from Norway. (interpreted by Gelfand).



Uncertain reading. The manuscript puts a question mark after the name. MCCME catalogue record 470 gives “Raphael Hoegh-Krohn”.

Original source details: GELFAND-P000863

On mapping grps.
(Ismagilov, Graev, Vershik; Albeverio)
[below the list: and ...]
X — a Riem. manifold
G — a compact Lie gr. (simple)

$$\varphi \in C_0^\infty(X, G) \equiv G^X.$$

Uncertain reading. The open-loop sign is provisionally set as φ. Another possible reading is ψ.

Original source details: GELFAND-P000863

We study unit. representations.

g — the Lie alg. of G.

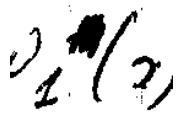
[crossed out at left: ...] $\Omega(X, g)$ — 1-forms with values in the Lie alg..

Undeciphered passage. The sign before $\Omega(X, g)$ is crossed out and illegible.

Original source details: GELFAND-P000863

$$\omega_i \in \Omega(X, g)$$

$$(\omega_1, \omega_2) = \int_X \operatorname{tr}\big(\omega_1(x)\omega_2^*(x)\big)\, dx. \tag{1}$$



Uncertain reading. The dark stroke above ω_1 may be a crossed-out star. Its reading is uncertain.

Original source details: GELFAND-P000863

$$\omega_i(x) : T_x \longrightarrow g$$

there is an inner prod. in both spaces (on the right, Killing)

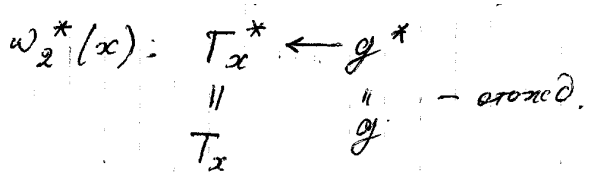


Diagram and typed description.

$$\omega_2^*(x): \quad T_x^* \longleftarrow g^* \qquad \text{— identif.}$$
$$\quad\;\; \| \qquad\quad \| \qquad\quad \text{— atomic.}$$
$$\quad\;\; T_x \qquad\; g$$

Original source details: GELFAND-P000863

GELFAND-P000864

[source number: 12]

$$\omega_1\omega_2^* : g \longrightarrow g$$

the trace and the integr. can now be taken

$\mathcal{H}$ — a Hilbert sp., completion of $\Omega(X, g)$

$$L_2(\mathcal{H}, \mu)$$

$$\dim \mathcal{H} < \infty \quad \Rightarrow \quad d\mu \sim e^{-\frac{1}{2}|x|^2}\, dx$$

Uncertain reading. The ∞ sign is retraced. A separate stroke is visible beneath it.

Original source details: GELFAND-P000864

(and analog. in an [crossed out: ~~finite~~] ∞-dim. sp.)

Uncertain reading. The crossed-out word provisionally reads «конечно» (“finite”).

Original source details: GELFAND-P000864

$T \in E(\mathcal{H})$ — Euclidean transf. (motions) of the space $\mathcal{H}$,

$$Tx = Ox + a, \qquad x \in H, \quad a \in H$$

μ is inv. under O and quasi-inv. under translations.

$g \in E(\mathcal{H})$, $g \to U_g$ unitary on $L_2(\mathcal{H}, \mu)$

hence [«ср.»] a unit. repr. of $E(\mathcal{H})$ in $L_2(\mathcal{H}, \mu)$

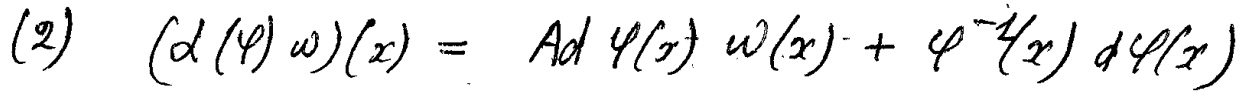
Uncertain reading. The abbreviation is provisionally read «ср.». The second sign has a long descending stroke.

Original source details: GELFAND-P000864

$$(\alpha(\varphi)\omega)(x) = \mathrm{Ad}\,\varphi(x)\,\omega(x) + \varphi^{-1}(x)\,d\varphi(x). \tag{2}$$

Uncertain reading. The open-loop sign is provisionally set as φ. Another possible reading is ψ.

Original source details: GELFAND-P000864



Original source details: GELFAND-P000864

Uncertain reading. The manuscript shows the order $\varphi^{-1}(x)\,d\varphi(x)$ and the exponent -1.

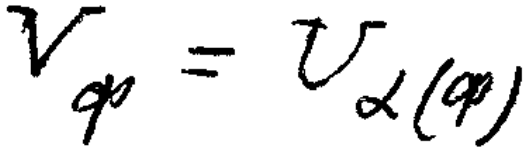
$$\alpha : C_0^\infty(X, G) \longrightarrow E(\Omega(X))$$

— this is a group homomorphism:

$$\alpha(\varphi_1\varphi_2) = \alpha(\varphi_1)\alpha(\varphi_2).$$

Take the comp. with U_g;

$\underline{\underline{V_\phi = U_{\alpha(\phi)}}}$ — we get a unit. repr.



Uncertain reading. The loops in the underlined equation have a different shape. They are set here as ϕ.

Original source details: GELFAND-P000864

$$\varphi \longrightarrow V_\varphi.$$

Theorem 1. It is irreducible if $\dim X \geq 3$.

≥ 5 — Ismagilov
≥ 4 — Gelf., Graev, Vershik
≥ 3 — Albeverio and Høegh-Krohn.

Theorem 2. $\dim X = 1 \quad \Rightarrow$ the repr. is reducible

$$\dim X = 2 \qquad \text{— ?}$$

$b(t)$ — Brownian mot. in g. (standard; with resp. to the Killing form)
Consid. the eq.

$$\eta^{-1}(t)\, d\eta(t) = db(t), \qquad \eta \in C(\mathbb{R}^+, G). \tag{3}$$

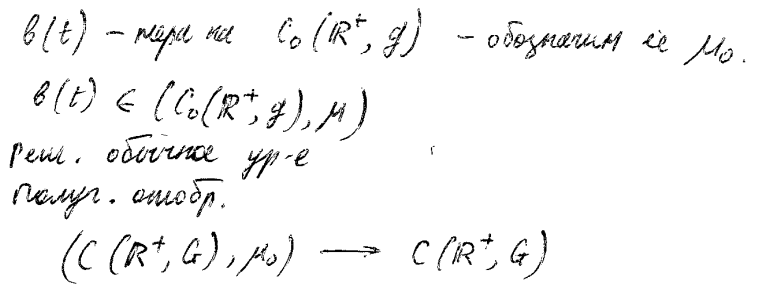
GELFAND-P000865

Explanation:
$b(t)$ — a measure on $C_0(\mathbb{R}^+, g)$ — denote it by μ_0.

$$b(t) \in \big(C_0(\mathbb{R}^+, g), \mu\big)$$

Solve the ordinary eq.
We obtain the map

$$\big(C(\mathbb{R}^+, G), \mu_0\big) \longrightarrow C(\mathbb{R}^+, G)$$

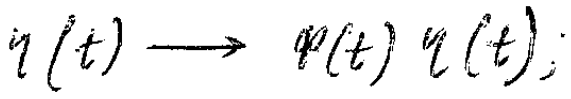


Original source details: GELFAND-P000865

Uncertain reading. The first formula has $C_0(\mathbb{R}^+, g)$ with μ. The domain of the map has $C(\mathbb{R}^+, G)$ with μ_0.

Stochastic — solve and transfer the measure.

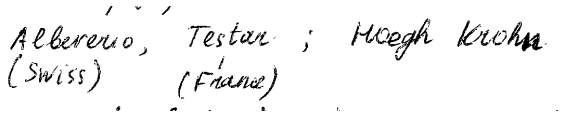
$$\eta(t) \longrightarrow \phi(t)\eta(t);$$



Uncertain reading. The closed-loop letter in these two formulas is provisionally set as ϕ.

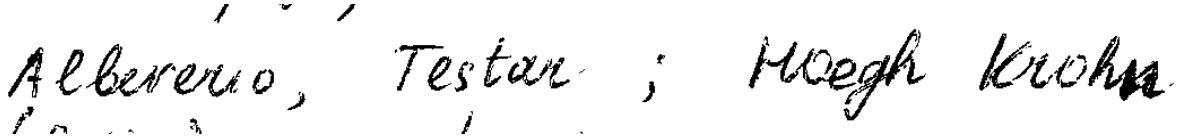
Original source details: GELFAND-P000865

$$\frac{d\mu(\phi\eta)}{d\mu(\eta)}$$



Original source details: GELFAND-P000865

Diagram and typed description.
Albeverio, Testar ; Hoegh Krohn.
(Swiss) (France)



Original source details: GELFAND-P000865

Uncertain reading. The first name is provisionally read as “Albeverio”. The second reads “Testar”. In the later 1981 article the names are printed as S. Albeverio, R. Høegh-Krohn and D. Testard.

$$L_2\big(C(\mathbb{R}^+, G), \mu\big), \qquad \varphi \in C^\infty(\mathbb{R}^+, G)$$

Uncertain reading. The open loop in the line with C^∞ is set as φ.

Original source details: GELFAND-P000865

$$\eta \longrightarrow \varphi\eta$$

$\varphi \longrightarrow W_\varphi$ we get a unit. repr. on $L_2(C(\mathbb{R}^+, G), \mu)$

Theorem $W_\varphi \cong V_\varphi$ equivalent.

The corresp. cocycle — the Maurer–Cartan form.

Factors:

$$[L_\infty \cup W_\varphi]'' \qquad \text{— a factor of type 3.}$$

GELFAND-P000866

22.10.79 Gelfand Sem.

Someone. (Baturin)

Uncertain reading. The handwritten name is provisionally read «Батурин» (“Baturin”). MCCME catalogue record 471 gives “I.A. Batalin”.

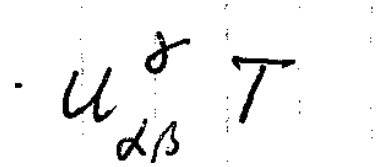
Original source details: GELFAND-P000866

On constraints.

First-class constraints

$$\begin{matrix} H \text{ — Hamiltonian}, & T_\alpha \text{ — constraints} \\ H, T_\alpha \text{ — funcs. of } q \text{ and } p. \end{matrix} \tag{1}$$

$$\{T_\alpha, T_\beta\} = u^\gamma_{\alpha\beta} T, \qquad g^a = \begin{pmatrix} q^i \\ p_i \end{pmatrix}. \tag{2}$$



Original source details: GELFAND-P000866

Uncertain reading. No subscript is visible on T after $u^\gamma_{\alpha\beta}$.

$$\{A, B\} = \frac{\partial A}{\partial g^a} \varepsilon^{ab} \frac{\partial B}{\partial g^b}, \qquad \text{here } -\varepsilon^{ba} = \varepsilon^{ab} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{3}$$

First-class constraints — if condition (2) is satisfied, $u^\gamma_{\alpha\beta} = u^\gamma_{\alpha\beta}(p, q)$.

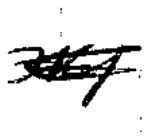
Question: can changes of var. be made so that $u^\gamma_{\alpha\beta} = \text{const}$. (4)

Rem. on the motivation: gauge transf. are gener. only by first-class constraints

Action:

$$S = \int dt \left[\frac{1}{2} g^a \varepsilon_{ab} \dot{g}^b - H - \lambda^\alpha T_\alpha \right]. \tag{5}$$

(λ^α — ordinary variables; [crossed out: ~~...~~] the extremum is now unconstrained)

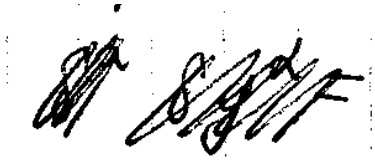


Original source details: GELFAND-P000866

Undeciphered passage. The entry before "the extremum" is crossed out and illegible.

Gauge transf. are gener. by the constraints

[crossed out below: ~~$g^a \quad \delta g^a = \dots$~~]



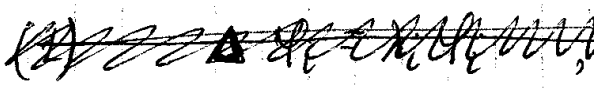
Original source details: GELFAND-P000866

Uncertain reading. The strokes under the hatching provisionally read g^a and $\delta g^a =$. The ending is illegible.

— • —

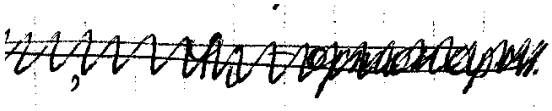
R. Kallosh. Quantum field theory

[line crossed out: ~~(1) $\Delta\phi_l = \lambda_l \phi_l, \dots$~~]



Original source details: GELFAND-P000866

Uncertain reading. The left part of the crossed-out line provisionally reads (1) $\Delta\phi_l = \lambda_l \phi_l$. The subscripts are indistinct.



Undeciphered passage. The right part after the comma is heavily hatched and illegible.

Original source details: GELFAND-P000866

New Nobel laureates

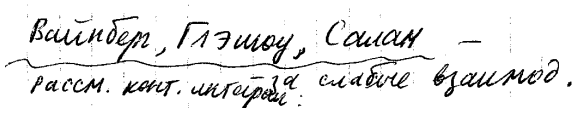
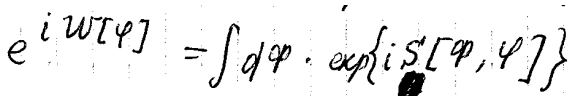


Diagram and typed description.
Weinberg, Glashow, Salam —
for weak interact.

Original source details: GELFAND-P000866

Symmetry underlies everything. Consid. the cont. integral.

$$e^{iW[\varphi]} = \int d\phi \cdot \exp\{iS[\phi, \varphi]\}. \tag{1}$$



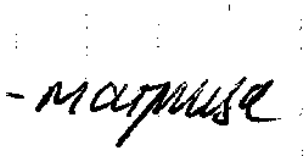
Uncertain reading. The open loop is set as φ (possibly ψ). The closed loop of the integration variable is set as ϕ.

Original source details: GELFAND-P000866

Undeciphered passage. The sign beneath S is blackened and illegible.

Original source details: GELFAND-P000866

S is constructed from the class. action
from (1) much is determ.: probabilities, S-matrix.



Uncertain reading. The last letter in «матрице» (“matrix”) provisionally reads «е».

Original source details: GELFAND-P000866

ϕ — quantum fluct. (one averages over them)
S-matrix

$$S[\varphi_{in}] = e^{iW[\tilde{\varphi}[\varphi_{in}]]}, \tag{2}$$

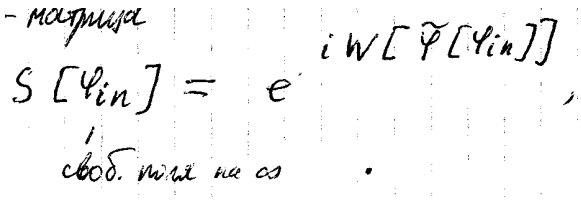
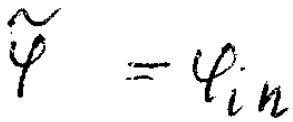


Diagram and typed description.
φ_{in}
/
free fields at ∞.

Original source details: GELFAND-P000866

$$\widetilde{\varphi} = \varphi_{in} + \int D^c(x - x') \frac{\delta \mathcal{L}_{int}}{\delta \widetilde{\varphi}(x')} \, d^4x'. \tag{3}$$



Uncertain reading. The looped sign in (3) and in the diagram is read as φ (possibly ψ).

Original source details: GELFAND-P000867

(Yang–Feldman eq.)

Iterative sol. of (3) gives trees.

D^c — Green's fn. for the free Klein–Gordon eq.

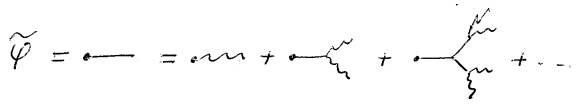


Diagram and typed description.

$\widetilde{\varphi} =$ —— $=$ ∿∿ $+$ —< $+$ >— $+ \cdots$

Original source details: GELFAND-P000867

Formula (2) contains amplitudes for prod. of any number of particles.

For gravity

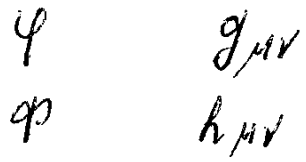
Uncertain reading. A sign, possibly $\mathcal{L}$, is cancelled before $S_{\text{кл}}$.

Original source details: GELFAND-P000867

$$S_{\text{кл}} = \int \mathcal{L} \, d^4x, \qquad \mathcal{L} = \sqrt{g} \, R[g_{\mu\nu}]. \tag{4}$$

S in (1) differs by quantum corrections

$$\begin{array}{ll} \varphi & g_{\mu\nu} \\ \phi & h_{\mu\nu} \end{array} \tag{5}$$



Uncertain reading. The upper sign is read as φ (possibly ψ), and the lower is rendered as ϕ.

Original source details: GELFAND-P000867

For gravity.

$$e^{iW[g_{\mu\nu}]} = \int dh \cdot e^{i\{S_{\text{кл}}[g_{\mu\nu}+h_{\mu\nu}]+S_{\text{кал}}+S_{\text{ф.ч.}}\}}\, d\bar{c}\, dc. \tag{6}$$

ф. ч. — fictitious particles.

$S_{\text{кл}}$ is given by formula (4) and is a functional of the metric g.

$h_{\mu\nu}$ — metric fluctuations.

$$S_{\text{кал.}} = \int \frac{1}{2}\left(D_\mu[g]h^\mu_\nu - \frac{1}{2}D_\nu[g]h^\mu_\mu\right)^2 \sqrt{g}\, d^4x. \tag{7}$$

Original source details: GELFAND-P000867

Uncertain reading. The argument of the second D runs into the beginning of h and is read as $[g]$.

D_μ — covariant deriv. for metric g

Raise and lower indices using g.

$$S_{\text{ф.ч.}} = \int \sqrt{g}\, \bar{c}^\nu D_\mu[g] D^\mu[g+h] c_\nu\, d^4x. \tag{8}$$

$S_{\text{ф.ч.}}$ is constructed from $S_{\text{кал}}$ so that the answer does not depend on $S_{\text{кал}}$
(principle of quantum relativity).

Observed by Feynman: without $S_{\text{кал}}$ there is no [cancelled] unitarity (total prob. $= 1$)

Original source details: GELFAND-P000867

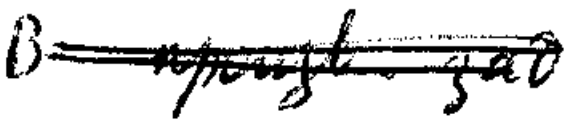
Uncertain reading. The short cancellation before “unitarity” is illegible.

Original source details: GELFAND-P000867

Manuscript excerpt. The manuscript remark reads $S_{\text{кал}}$.

He [did so] in 1st approx.

De Witt, Popov, Faddeev — further.



Original source details: GELFAND-P000867

Uncertain reading. Cancelled Russian text: “В … зад”. The middle abbreviation is illegible, possibly “прибл.” (approx.).

In supergrav. one further term suffices.

GELFAND-P000868

[source number: 26]

In an arbitrary theory — infinitely many terms.

$$\underset{\text{(Einst.)}}{S_{\text{кл}}[g]} \qquad \text{inv. under transformations} \tag{9}$$

$$\delta g_{\mu\nu} = D_\mu \xi_\nu + D_\nu \xi_\mu, \qquad \xi_\mu \quad \begin{matrix}\text{— arbitrary vector} \\ \text{(local fn. of } x\text{).}\end{matrix} \tag{10}$$

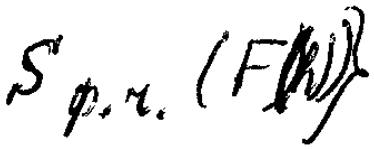
The quantum problem has an invariance:

$$W[g_{\mu\nu}] = W[g_{\mu\nu} + \delta g_{\mu\nu}], \qquad \delta g_{\mu\nu} \text{ taken from (10)} \tag{11}$$

In fact (6) needs further specification.

Another invariance law. Consider:

$$e^{iW_{\text{рад}}} \underset{\text{def}}{=} \int dh \cdot \exp i\Big\{ S_{\text{кл}}[g+h] - S_{\text{кл}}[g] - \frac{\delta S[g]}{\delta g_{\mu\nu}} h_{\mu\nu} + \\ + S_{\text{кал}}\big(F(h)\big) + S_{\text{ф.ч.}}\big(F(h)\big)\Big\}. \tag{12}$$



Uncertain reading. The stroke before h has been retraced. The last argument is read as $F(h)$, possibly Fh.

Original source details: GELFAND-P000868

F — a certain function satisfying certain conditions.

Gauge transf. [cancelled] preserve things with phys. meaning.

Uncertain reading. The cancelled word is illegible.

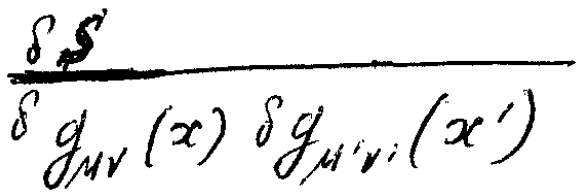
Original source details: GELFAND-P000868

(The space remains isometric to the old one; same space, but I change the coord.).
Adding the terms $S_{\text{кал}}$, $S_{\text{ф.ч.}}$ is needed to handle degen. in the stationary-phase expansion.

$$W_{\text{рад}}[g_{\mu\nu}]_F = W_{F+\delta F}[g] - \frac{\delta S[g]}{\delta g_{\mu\nu}}\langle \delta h_{\mu\nu}\rangle \tag{13}$$

$S_{\text{кал}}$ is added for convergence (correcting the degeneracy)

$$\frac{\delta S}{\delta g_{\mu\nu}(x)\,\delta g_{\mu'\nu'}(x')}$$

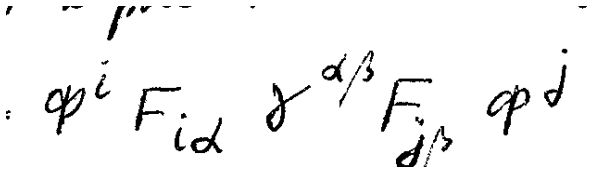


Uncertain reading. The numerator shows δS. No power on δ is discernible.

Original source details: GELFAND-P000868

Form of the terms that can be added:

$$S[\Phi] + \phi^i F_{i\alpha}\gamma^{\alpha\beta}F_{j\beta}\phi^j \tag{14}$$

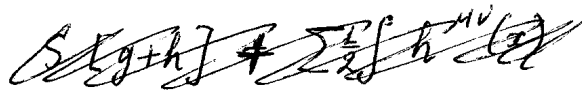


Uncertain reading. The two lowercase signs have two loops and are rendered as ϕ, as is the lower sign in (5).

Original source details: GELFAND-P000868

Let the external metric be flat

$$S[\Phi] + \frac{1}{2}\Phi^i F_{ij}\Phi^j \tag{15}$$



Uncertain reading. Cancelled line, read as $S[g+h] + \sum \frac{1}{2}\int h^{\mu\nu}(x)$. A cancellation stroke crosses the sign that resembles $\sum$.

Original source details: GELFAND-P000868

$$S[g+h] + \frac{1}{2}\int h^{\mu\nu}(x) F_{\mu\nu\mu'\nu'}(x,x',g) h^{\mu'\nu'}(x')\, d^4x\, d^4x' \tag{16}$$

F — a sufficiently arbitrary fn. of the points and the metric, for example:

$$\frac{1}{2}\left(D_\mu h^\mu_\nu - \frac{\beta}{2} D_\nu h\right)^2 \qquad \text{or} \qquad h^\mu h^\nu_\mu \tag{17}$$

Uncertain reading. After “or” the manuscript gives $h^\mu h^\nu_\mu$. No lower index on the first h is visible.

Original source details: GELFAND-P000869

Principle of relativity — it is independent of F. The result is a quadratic form

$$\frac{1}{2}\frac{\delta^2 S[g+h]}{\delta h_{\mu\nu}(x)\, \delta h_{\mu'\nu'}(x')} + F_{\mu\nu\mu'\nu'}(x,x') \tag{18}$$

The quadratic form must be nondegenerate. This is a condition on F.

The addition in (16) is the gauge-fixing action.

Read: Faddeev, Popov — review in UFN.

Uncertain reading. Two cancelled entries. The first may read “Теперь” (now). The lower one is illegible.

Original source details: GELFAND-P000869

In (13):

$\langle \delta h_{\mu\nu} \rangle$ — the change in the integral brought about by the given change of gauge.

$$\langle \delta h_{\mu\nu} \rangle = \int dh \cdot \exp i\{\widetilde{S}\}$$

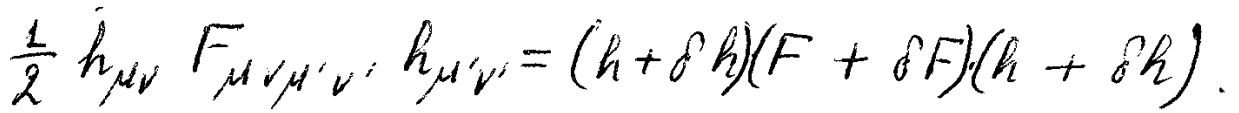
Uncertain reading. The right-hand side ends with $\exp i\{\widetilde{S}\}$.

Original source details: GELFAND-P000869

$$\delta h_{\mu\nu} = D_\mu \xi_\nu + D_\nu \xi_\mu, \qquad \xi_\mu[\delta F] \tag{19}$$

Choose F such that

$$\frac{1}{2} h_{\mu\nu} F_{\mu\nu\mu'\nu'} h_{\mu'\nu'} = (h + \delta h)(F + \delta F) \cdot (h + \delta h). \tag{20}$$



Original source details: GELFAND-P000869

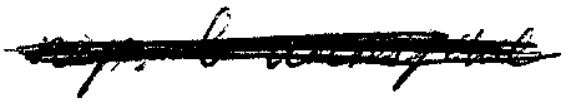
Uncertain reading. The factor $\frac{1}{2}$ is visible on the left-hand side only.

By construction:

$$W[g]_F = W[g + \delta g] \tag{21}$$

Another condition besides that. Correct construction of $S_{\text{ф.ч.}}$.

(13) is proved using a change of variables [cancelled] and precisely because of the correct choice of the action $S_{\text{ф.ч.}}$.



Original source details: GELFAND-P000869

Uncertain reading. The cancellation after "change" may read "пер. в интеграле" (variables in the integral).

In quantum electrodynamics.

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 \tag{22}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{23}$$

$$\delta A_\mu = \partial_\mu \xi, \qquad \xi = \xi(x) \tag{24}$$

Original source details: GELFAND-P000869

Uncertain reading. The ξ after ∂_μ has been retraced. The underlying strokes are illegible.

Original source details: GELFAND-P000869

Uncertain reading. An equality is cancelled between the parts of (24), possibly $\Lambda = \Lambda(x)$ or $\lambda = \lambda(x)$.

Nothing changes under this — $F_{\mu\nu}$ and $\mathcal{L}$ are unchanged.

$$\int dA_\mu \cdot \exp i\{S_{\text{кл}}[A] + S_{\text{кал}}[A] + S_{\text{ф.ч.}}\} \tag{25}$$

GELFAND-P000870

[source number: 27]

Choice of gauge-fixing term

$$S_{\text{кал}} = \frac{1}{2\alpha}(\partial_\mu A_\mu)^2 \tag{26}$$

or

$$\frac{1}{2\alpha}(\partial_\mu A_\mu + \beta n_\mu A_\mu)^2 \tag{27}$$

Uncertain reading. The second digit has been heavily retraced and is read as 7. The underlying strokes are illegible.

Original source details: GELFAND-P000870

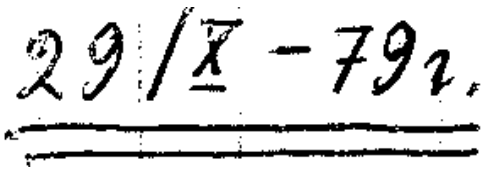
Quadr. form in the classical action:

$$\frac{1}{2}A_\mu(\Box g_{\mu\nu} - \partial_\mu\partial_\nu)A_\nu \tag{28}$$

GELFAND-P000871

[source number: 31]

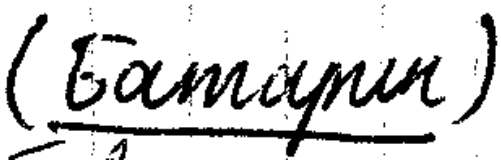
29/X-79 Gelfand Sem.



Uncertain reading. The manuscript gives 29/X-79. The MCCME catalogue dates record 473 “29-Sep-79”.

Original source details: GELFAND-P000871

Physicist (the same one). (Baturin)



Uncertain reading. The manuscript appears to read “Батурин” (Baturin). The MCCME catalogue gives I.A. Batalin.

Original source details: GELFAND-P000871

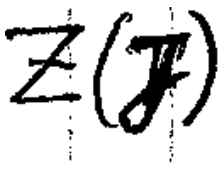
On quantizing gauge theories by means of functional integrals.

Consider the integral

$$Z(J) = c\int e^{iW(\varphi)+i\int J(x)\varphi(x)\,d^4x}\,d\mu(\varphi) \tag{1}$$

(J — “source”).

(Fourier transf. of a functional.)



Uncertain reading. The retraced argument of Z is read as J.

Original source details: GELFAND-P000871

Original source details: GELFAND-P000871

Uncertain reading. The open-loop sign is rendered as φ (possibly ψ).

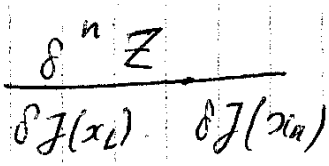
Original source details: GELFAND-P000871

Uncertain reading. The retraced argument of μ is read as φ. The underlying strokes are illegible.

(derivative of the functional)

$$\frac{\delta^n Z}{\delta J(x_1) \quad \delta J(x_n)} = G(x_1, \ldots, x_n) \quad \text{— Green's fn.} \tag{2}$$

(everything is expressed through it.)

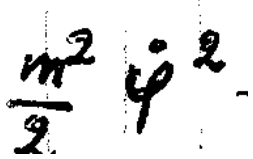


Original source details: GELFAND-P000871

Uncertain reading. The denominator shows $\delta J(x_1)$ and $\delta J(x_n)$, with a space between them.

$$W(\varphi) = \text{classical action of the field} = \int \mathcal{L}\, d^4x \tag{3}$$

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \varphi)^2 - \frac{m^2}{2}\dot{\varphi}^2 - \lambda\varphi^4 \tag{4}$$



Original source details: GELFAND-P000871

Uncertain reading. A separate dot is visible above the letter in the second term. It is set as $\dot{\varphi}^2$. The meaning of the dot is unclear.

(for example)

(1) — over all fields φ with certain decay at ∞.

Interpretation of (1): a) expansion in λ into a perturbation series

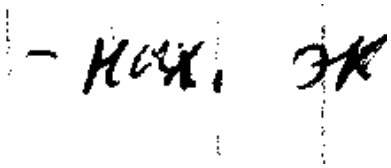
(in grav. — small departures from flat space)

b) evaluation by stationary phase near a classical extremal.

a) is currently inapplicable to grav..

b) better — accounts for more complicated classical fields at once.

b) — certain extremals of the full action.



Original source details: GELFAND-P000871

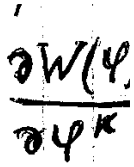
Uncertain reading. The abbreviation before "extremals" is provisionally read as "нек." (certain).

Stationary phase gives the asymptotics of

$$\int e^{\frac{i}{\hbar}W(\varphi)}(d^n\varphi) \tag{5}$$

as $\overline{h} \to 0$

$$\frac{\partial W(\varphi)}{\partial \varphi^\kappa} = 0 \quad \text{— stationary points.} \tag{6}$$



Uncertain reading. The superscript is rendered as κ, possibly Latin k.

Original source details: GELFAND-P000871

At the stationary point $\bar{\varphi}$, expand by Taylor's formula

$$\text{Leading term:} \qquad e^{\frac{i}{\hbar}W(\bar{\varphi})} \frac{1}{\sqrt{\operatorname{Det}\left\{\dfrac{\partial^2 W(\bar{\varphi})}{\partial\varphi\,\partial\varphi}\right\}}}(1 + o(\overline{h})) \tag{7}$$

Then everything reduces to a Gaussian integral, answer $\dfrac{1}{\sqrt{\det A}}$.

A — degenerate $\Rightarrow$ fails. This is what arises in gauge theories.

GELFAND-P000872

Example:

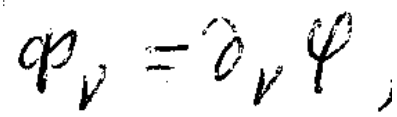
in electrodynamics

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{8}$$

Quadr. form:

$$\frac{1}{2}A_\mu \underbrace{(\Box\delta_{\mu\nu} - \partial_\mu\partial_\nu)}_{K_{\mu\nu}} A_\nu \tag{9}$$

$$\text{if} \quad \phi_\nu = \partial_\nu\varphi, \quad \text{then} \quad K_{\mu\nu}\phi_\nu = 0 \quad \text{and cannot be inverted.} \tag{10}$$

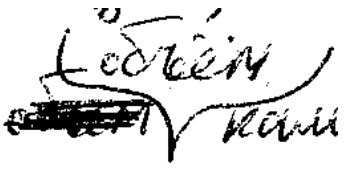


Uncertain reading. The two letter shapes in (10) are distinguished. The form with two loops is set as ϕ. The form with an open left loop is set as φ (possibly ψ).

Original source details: GELFAND-P000872

The same in gravity, Y.–M., etc.

We have to factor out [cancelled] [above the line: the volume] of the gauge group.

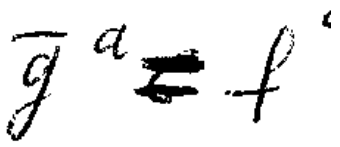


Uncertain reading. The cancelled word before the inserted "volume" is illegible. It may be the same word.

Original source details: GELFAND-P000872

$g^a \in M$ — space
and let a group act

$$g^a \longrightarrow \bar{g}^a = f^a(g,\theta), \qquad \theta \text{ — group parameters.} \tag{11}$$



Uncertain reading. The heavily retraced sign between $\bar{g}^a$ and f^a is read as equality.

Original source details: GELFAND-P000872

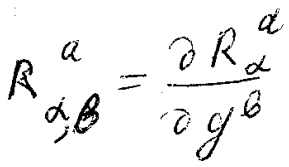
$$\int e^{iW(g)}[dg] \tag{12}$$

$W(g)$ is invariant
Consider a transf. (infinitesimal)

$$\delta g^a = R^a_\alpha(g)\theta^\alpha, \qquad \theta^\alpha \to 0 \tag{13}$$

$$R^a_\alpha(g) = \left.\frac{\partial f^a(g,\theta)}{\partial \theta^\alpha}\right|_{\theta=0} \tag{14}$$

$$R^a_{\alpha,b} = \frac{\partial R^a_\alpha}{\partial g^b} \tag{15}$$



Uncertain reading. The final lower index on the left has been retraced and is read as b, following the index in the denominator.

Original source details: GELFAND-P000872

Condition:

$$R^a_{\alpha,a} = 0 \qquad \text{(unimodularity} \Rightarrow \text{left and right measures coincide)} \tag{16}$$

Lie eqs.

$$\frac{\partial \bar{g}^a}{\partial \theta^\beta} = R^a_\alpha(\bar{g}) \underbrace{\lambda^\alpha_\beta(\theta)}_{\text{group frame}}, \tag{17}$$

there are also $t^\alpha_{\beta\gamma}$ — structure consts. of the Lie alg.
Group composition law

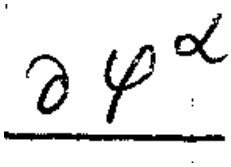
$$f^a(f(g,\theta),\theta') = f^a(g,\varphi(\theta,\theta')) \tag{18}$$

GELFAND-P000873

[source number: 32]

$$\lambda = \mu^{-1}, \quad \text{where} \tag{19}$$

$$\mu^{\alpha}_{\beta}(\theta) = \left.\frac{\partial \varphi^{\alpha}}{\partial \theta'^{\beta}}\right|_{\theta'=0}. \tag{19}'$$



Uncertain reading. The sign in the numerator is rendered as φ, as in (18). The manuscript uses a distinct shape for ψ in (20) and the following equations.

Original source details: GELFAND-P000873

Consider a hypersurface in M.

$$\psi^{\alpha}(f(g,\theta)) = 0 \tag{20}$$

We want $\forall g\ \exists!$ sol. θ of eq. (20). (cross-section!)

The class of admissible gauges is where this condition holds. We have:

$$1 = \int [d\theta]\, \delta(\psi(f(g,\theta))\, \mathrm{Det}\left\{\frac{\partial \psi(f(g,\theta))}{\partial \theta}\right\} \tag{21}$$

Uncertain reading. Two closing parentheses follow the argument of f, with three opening parentheses in the expression.

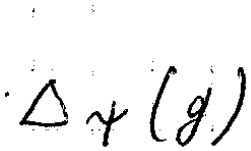
Original source details: GELFAND-P000873

Compute the Jacobian from the Lie eqs.:

$$\frac{\partial \psi^{\alpha}(f(g,\theta))}{\partial \theta^{\gamma}} = \psi^{\alpha}_{,a}(f(g,\theta)) R^{a}_{\beta}(f(g,\theta)) \lambda^{\beta}_{\gamma}(\theta) \tag{22}$$

Hence

$$\mathrm{Det}\left\{\frac{\partial \psi}{\partial \theta}\right\} = \Delta_{\psi}(g)\, \mathrm{Det}\{\lambda(\theta)\} \tag{23}$$



Uncertain reading. Equation (23) gives $\Delta_{\psi}(g)$ with argument g.

Original source details: GELFAND-P000873

Here

$$\Delta_\psi(g) = \operatorname{Det} \mathcal{D}^{-1} \tag{24}$$

$$\mathcal{D}^{-1}{}^{\alpha}_{\beta}(g) = \psi^{\alpha}_{,a}(g) R^{a}_{\beta}(g) \tag{25}$$

Substitute the Jacobian in formula (21). We obtain:

$$1 = \int \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta)) \underbrace{d\sigma(\theta)}_{\substack{\text{invariant measure} \\ \text{on the group}}} \tag{26}$$

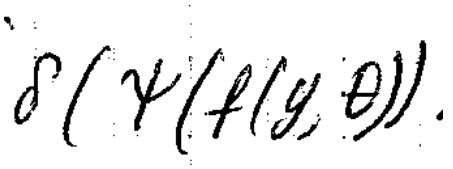
Original source details: GELFAND-P000873

Uncertain reading. In (26), two closing parentheses follow the argument of f, as in (21).

$$d\sigma(\theta) = \operatorname{Det}\{\lambda(\theta)\}[d\theta] \quad \text{— invariant measure on the group.} \tag{27}$$

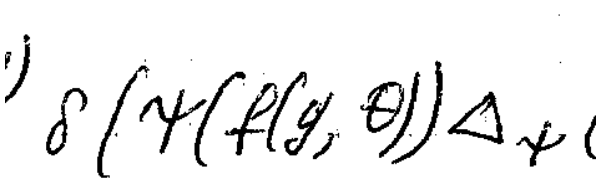
Return to the integral (12)

$$\begin{aligned} &\int e^{iW(g)}[dg] \\ &\qquad = \int e^{iW(g)} \int \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta))\, d\sigma(\theta)[dg] = \\ &\qquad = \int d\sigma(\theta) \int e^{iW(g)} \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta))[dg] \end{aligned} \tag{28}$$



Original source details: GELFAND-P000873

Uncertain reading. The upper line of (28) shows two closing parentheses after the argument of f in the delta expression.



Original source details: GELFAND-P000873

Uncertain reading. The lower line of (28) uses the same notation.

Change of variable;

$$\underset{\text{old var.}}{g^a} \longrightarrow \underset{\text{new var.}}{f^{-1\,a}(g,\theta)} \tag{29}$$

θ is fixed here (the same one over which we integrate)

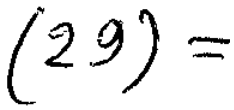
$W(g)$ inv.; $[d\sigma]$ inv. $\Rightarrow$ they will stay the same.

Uncertain reading. The manuscript gives $[d\sigma]$ as invariant after $W(g)$.

Original source details: GELFAND-P000874

$$\psi(f(g,\theta)) \rightsquigarrow \psi(g)$$
$$\Delta_\psi(f(g,\theta)) \rightsquigarrow \Delta_\psi(g)$$

$$(29) = \int d\sigma(\theta) \int e^{iW(g)} \delta(\psi(g)) \Delta_\psi(g) [dg] \qquad (30):$$



Uncertain reading. The left-hand side of (30) reads (29).

Original source details: GELFAND-P000874

Now we divide by $\int d\sigma(\theta)$ (drop it)

On the hypersurf. [cancelled] $\psi(g) = 0$ the quadr. form is already nonsingular.

Uncertain reading. The cancellation is illegible. It may be $\psi(g)$.

Original source details: GELFAND-P000874

the cross-section was

$$\psi^\alpha(g) = 0 \tag{31}$$

Now replace

$$\psi^\alpha(g) \longrightarrow \psi^\alpha(g) - c^\alpha \tag{32}$$

and obtain a cross-section.

$$\psi^\alpha(g) = c^\alpha. \tag{33}$$

This gives

$$\int e^{iW(g)} \delta(\psi(g) - c) \Delta_\psi(g)[dg] \tag{34}$$

and consider

$$\int \Big(\int e^{iW(g)} \delta(\psi(g) - c) \Delta_\psi(g)[dg]\ e^{i\Omega(c)}[dc] \Big) \tag{35}$$

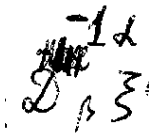
By construction it is independent of ψ and therefore of $c \Rightarrow$ integration changes nothing $\Omega(c)$ — any fn. of c. Hence

$$(35) = \int e^{iW(g) + i\Omega(\psi(g))} \Delta_\psi(g)[dg] \tag{36}$$

(we have integrated over c)

We have:

$$\Delta_\psi(g) = \mathrm{Det}\, \mathcal{D}^{-1}(g) = \int [d\bar{\xi}][d\xi] e^{i\bar{\xi}_\alpha \mathcal{D}^{-1}{}^\alpha_\beta \xi^\beta} = \tag{37}$$



Original source details: GELFAND-P000874

Uncertain reading. Above $\mathcal{D}$, superscripts are cancelled and $-1\,\alpha$ is inserted. The cancelled marks are illegible.

Original source details: GELFAND-P000874

Uncertain reading. After the last equality, $\int [d\bar{\xi}]$ and a second factor are cancelled. The second factor may be $[d\xi]$ or $[dg]$.

GELFAND-P000875

[source number: 33]

$$= \int [d\bar{\xi}][d\xi] e^{i\bar{\xi}_\alpha \psi^\alpha_{,a} R^a_\beta \xi^\beta}$$

ξ, $\bar{\xi}$ — anticommuting Grassmann variables

Now substitute in (36):

$$(36) = \int e^{iW_{\text{eff}}(g,\bar{\xi},\xi)} [dg][d\bar{\xi}][d\xi], \tag{38}$$

$$W_{\text{eff}} = W(g) + \Omega(\psi(g)) + \bar{\xi}_\alpha \psi^\alpha_{,a}(g) R^a_\beta(g) \xi^\beta \tag{39}$$

It is in fact independent of ψ by construction.

It does depend on Ω. But

$$\frac{(36)}{\int e^{i\Omega(c)} dc} \qquad \text{is independent of } \Omega. \tag{40}$$

W_{eff} is, of course, now noninvariant.

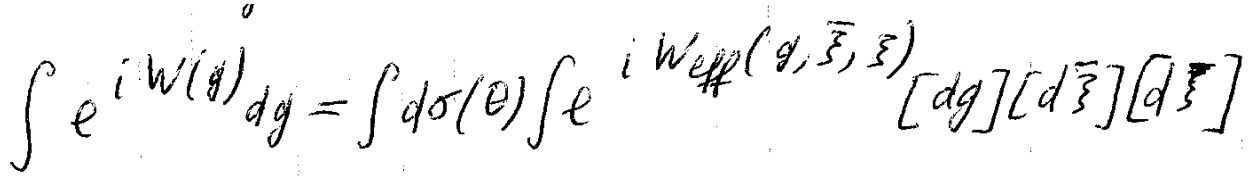
But it exhibits deeper invariance properties

(BRS — invariance)

or BRS — symmetry

Thus: it has been proved that

$$\int e^{iW(g)} dg = \int d\sigma(\theta) \int e^{iW_{\text{eff}}(g,\bar{\xi},\xi)} [dg][d\bar{\xi}][d\xi] \tag{41}$$



Original source details: GELFAND-P000875

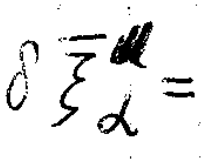
Uncertain reading. The right-hand side of (41) has no denominator from (40).

Reminder. We had

$$\delta g^a = R^a_\alpha(g) \theta^\alpha, \quad \theta^\alpha \to 0, \tag{42}$$

Forget that. Instead:

$$\begin{cases} \delta g^a = R^a_\alpha(g) \xi^\alpha \mu \\ \delta \xi^\alpha = \frac{1}{2} t^\alpha_{\beta\gamma} \xi^\beta \xi^\gamma \mu \\ \delta \bar{\xi}_\alpha = \Omega_{,\alpha}(\psi(g)) \mu \end{cases} \tag{43}$$



Uncertain reading. The superscript on $\bar{\xi}$ is cancelled, possibly α or a. The subscript is α.

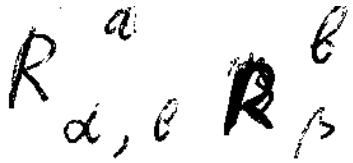
Original source details: GELFAND-P000875

μ — anticommuting Grassmann param.;

$$\mu^2 = 0, \quad \mu\xi = -\xi\mu, \quad \mu\bar{\xi} = -\bar{\xi}\mu, \quad \xi\bar{\xi} = -\bar{\xi}\xi. \tag{44}$$

To check invariance, use the ordinary Lie relations for R:

$$R^a_{\alpha,b} R^b_\beta - (\alpha \leftrightarrow \beta) = -t^\gamma_{\alpha\beta} R^a_\gamma \tag{45}$$



Uncertain reading. The last lower index on the first R resembles e. It is read as b, matching the superscript of the second R.

Original source details: GELFAND-P000875

GELFAND-P000876

We can now work directly with expr. (38), bearing in mind that it has BRS symmetry.
In modern theories, conversely, W_{eff} is sought from considerations of BRS symmetry.
ξ-fields — ghosts; they have opposite statistics (Bose $\leftrightarrow$ Fermi and vice versa).
Independence of ψ and Ω — quantum gauge invariance.

Undeciphered passage. The small cancelled sign above (46) is illegible.

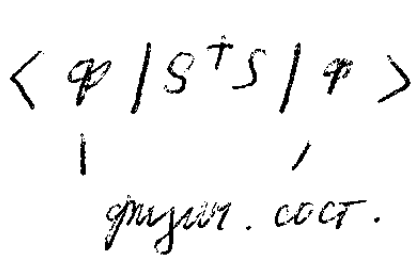
Original source details: GELFAND-P000876

$$W_{\text{eff}} \rightsquigarrow W_{\text{eff}} + J_a g^a \tag{46}$$

gives the generating functional. ~~This~~. S-matrix only in the g sector
S-matrix turns out to be unitary
Ghosts never appear externally — there are no sources.
Add sources for the ghosts.
S^+S unit. on the full space.
In $\langle \phi \mid S^+S \mid \phi \rangle$ the contributions from nonphysical degrees of freedom of gauge [cancelled] fields and ghosts cancel.

Uncertain reading. The entry after “кал.” is cancelled. It may read “по”.

Original source details: GELFAND-P000876



$\langle \phi | S^+S | \phi \rangle$ — phys. states

Diagram and typed description.

Original source details: GELFAND-P000876

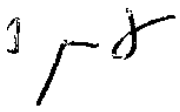
Rem. on the Lie alg.

$$\Gamma_\alpha = R^a_\alpha \frac{\partial}{\partial g^a}, \tag{47}$$

$$[\Gamma_\alpha, \Gamma_\beta] = t^\gamma_{\alpha\beta}\Gamma_\gamma \tag{48}$$

hence

$$2(\Gamma\xi)(\Gamma\xi) = t^\gamma_{\alpha\beta}\xi^\alpha\xi^\beta\Gamma^\gamma \tag{49}$$



Uncertain reading. At the end of (49), γ is written as a superscript on Γ.

Original source details: GELFAND-P000876

$[dg][d\bar{\xi}][d\xi]$ — is invariant under the BRS transf..

How do we check independence of ψ?

Choose μ in a special way:

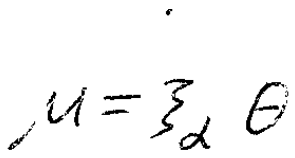
$$\mu = \bar{\xi}_\alpha \delta\psi^\alpha \tag{50}$$

and perform a BRS transformation.

The integral's independence of coordinates gives independence of ψ.

For

$$\mu = \xi_\alpha\theta \tag{51}$$



Uncertain reading. In (51), ξ has no bar and θ has no index.

Original source details: GELFAND-P000876

we obtain Ward identities, the eqs. of renormalization theory.

Bilinearity in the ghosts follows because the gauge transf. form a group.

GELFAND-P000877

[source number: 34]

In supergravity, gauge transf. do not form a group.

Yang–Mills.

Let a semisimple compact Lie group be given

t^{abc} — struct. constants of the Lie alg.

and

$$t^{abc} = (t^b)^{ac}, \quad t^b \text{ — matrix}, \quad t^b \text{ — generator of the adjoint rep.} \tag{52}$$

$$[t^a, t^b] = t^{abc} t^c \qquad \text{(assume an orthogonal basis)} \tag{53}$$

$$A_\mu(x) = A^a_\mu(x) t^a \tag{54}$$

g^a — includes μ, A and x.

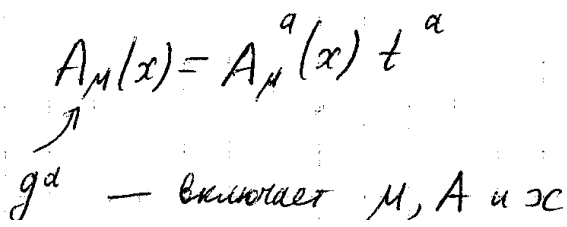


Diagram and typed description. $A_\mu(x)$ over g^a

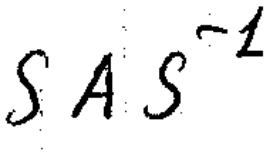
Original source details: GELFAND-P000877

Y.–M. Lagr.

$$\mathcal{L}_{\text{Y.M.}} = -\frac{1}{4} G^a_{\mu\nu} G^a_{\mu\nu} \tag{55}$$

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g t^{abc} A^b_\mu A^c_\nu \tag{56}$$

$$\bar{A}_\mu = S A S^{-1} - \frac{1}{g} (\partial_\mu S) S^{-1} \tag{57}$$



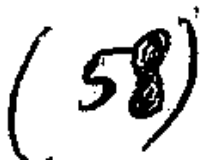
Uncertain reading. The first term of (57) reads SAS^{-1}, with no index on A.

Original source details: GELFAND-P000877

$S(x) = S$ — matrix from the adjoint rep. of the group

infinitesimal gauge transf.

$$\delta A^a_\mu(x) = \nabla^{ab}_\mu(A) \theta^b(x) \tag{58}$$

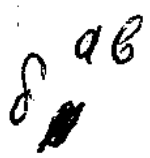


Uncertain reading. The 8 in equation number (58) is retraced.

Original source details: GELFAND-P000877

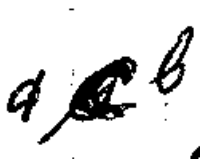
Here

$$\nabla^{ab}_{\mu}(A) = \partial_{\mu}\delta^{ab} + gt^{acb}A^{c}_{\mu} \quad \text{— covariant derivative.} \tag{59}$$



Uncertain reading. There is an illegible cancelled mark below δ^{ab}. It may be μ.

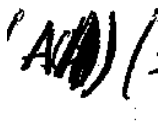
Original source details: GELFAND-P000877



Uncertain reading. The middle superscript on t is overwritten. It may be c as in t^{acb}. The underlying mark is illegible.

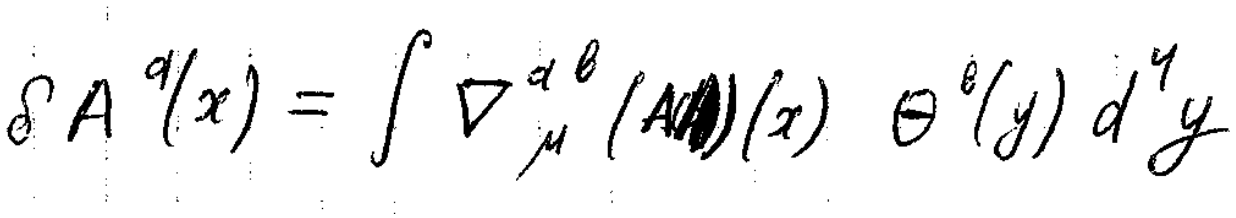
Original source details: GELFAND-P000877

$$\delta A^{a}(x) = \int \underset{\text{analogue of } R}{\nabla^{ab}_{\mu}(A)(x)}\ \theta^{b}(y)d^{4}y \tag{60}$$



Uncertain reading. There is a cancelled entry in parentheses after A. It may be (x) or (y).

Original source details: GELFAND-P000877



Original source details: GELFAND-P000877

Uncertain reading. On the left of (60), A^{a} has no subscript μ. No separate factor connecting x and y is written.

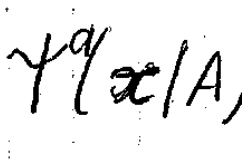
correspondence with the old notation

$$\begin{cases} \theta \text{ — parameter} \\ a \text{ — is } \mu, A, x. \\ \alpha \text{ — is } y \text{ and } b. \end{cases}$$

Now write out:

W_{eff}

$$\psi^{a}(x \mid A) = \partial_{\mu}A^{a}_{\mu}(x) \tag{61}$$



Uncertain reading. The x in the left-hand argument of (61) is retraced.

Original source details: GELFAND-P000877

$$\Omega(\psi) = -\frac{\alpha}{2}\int (\psi^{a})^{2}d^{4}x \tag{62}$$

Eff. Lagrangian

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{Y.M.}} - \frac{\alpha}{2}(\partial_\mu A^a_\mu)^2 - (\partial_\mu \bar{\xi}^a)\nabla^{ab}_\mu(A)\xi^b. \tag{63}$$

Next time 12 November.

GELFAND-P000878

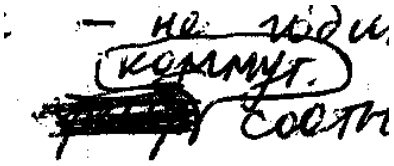
12/XI-79. Gelfand Sem..

I. M. G. Following Hawking

There are 3 approaches to quantizing GR

1) operators — unsuitable, since one must lower and raise ind.; noncomm.

2) canonical commut. rel. — poor, since space has to be foliated — against the spirit of GR



Original source details: GELFAND-P000878

Uncertain reading. "commut." is inserted above an illegible cancelled word and circled.

— and take $M^4 = M^3 \times R$, whereas all topologies are needed.

Original source details: GELFAND-P000878

Uncertain reading. There is a cancelled sign before the last line. It may read "3)".

Original source details: GELFAND-P000878

Uncertain reading. There is a cancelled word before "Points". The first letter may be "K".

Points $\to$ uncertainty princ.

GELFAND-P000879

[source number: 38]

Consider

$$\frac{1}{16\pi G}\int (R - 2\Lambda)(-g)^{1/2}\, d^4x + \int L_m(-g)^{1/2}\, d^4x \tag{1}$$

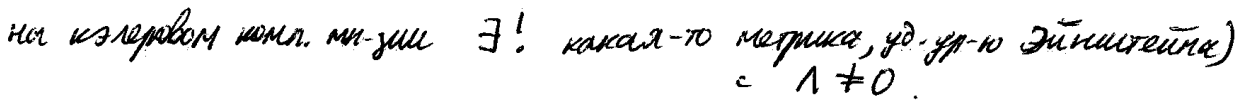
Λ — cosmological term, $\Lambda = \text{const}$.

([possibly "naïv.:"] Yau's thm.: on a compact Kähler manif. $\exists!$ some metric satisfying Einstein's eq.)

[Added below the line: with $\Lambda \neq 0$.]

Original source details: GELFAND-P000879

Uncertain reading. The word before "Yau's thm." is unclear. It may read "наивн:" ("naïv.:").



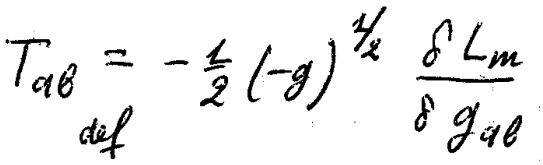
Original source details: GELFAND-P000879

Uncertain reading. The manuscript gives $\exists!$ and $\Lambda \neq 0$. The second condition is added below the line.

L_m — matter-field Lagrangian
(functional of the metric and extra fields)

$$R_{ab} - \frac{1}{2} g_{ab} R + \Lambda g_{ab} = 8\pi G T_{ab}, \qquad \text{— Einstein's eq.} \tag{2}$$

$$T_{ab} \underset{\text{def}}{=} -\frac{1}{2}(-g)^{1/2} \frac{\delta L_m}{\delta g_{ab}} \qquad \text{— stress–energy tensor} \tag{3}$$



Uncertain reading. Equation (3) has coefficient $-1/2$, exponent $1/2$ and lower indices on g_{ab} in the manuscript.

Original source details: GELFAND-P000879

(the most correct method,)
devised by Hilbert
(4) Boundary conditions: something $= 0$ on the boundary.
Gibbons, Hawking (1977) added a boundary term

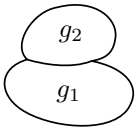
$$\frac{1}{8\pi G} \int K (\pm h)^{1/2} \, d^3x + C \tag{5}$$

K — trace of the 2nd quadr. form of the surface,
C depends only on the boundary metric h (induced)
and is independent of g at interior points
(see also Landau and Lifshitz)
We need:

$$I[g_1 + g_2] = I[g_1] + I[g_2] \tag{6}$$

Original source details: GELFAND-P000879

Diagram and typed description.

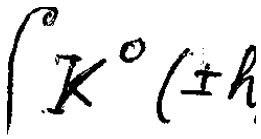


This will hold if we add term (5).

awkward — clumsy (неуклюжий)

Then

$$C = -\frac{1}{8\pi G}\int K^0(\pm h)^{1/2}\,d^3x \tag{7}$$



Uncertain reading. A small oval is written above K in (7). It may be 0 or $\circ$.

Original source details: GELFAND-P000879

Uncertain reading. There is a cancelled entry below (7). The initial sign may be Z. The full reading remains uncertain.

Original source details: GELFAND-P000879

GELFAND-P000880

Batalin.

Quantization of systems with constraints.

2 classes of constraints: 1st and 2nd class.

Eff. they are not solved.

Uncertain reading. The preferred reading is “решаются” (plural). The singular “решается” is also possible.

Original source details: GELFAND-P000880

One must write a Hamiltonian path integral with constraints

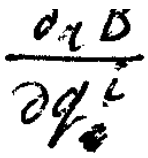
$$q^i, p_i \text{ — can. coordinates}, \qquad M \text{ — phase space} \tag{1}$$

$i = 1, \dots, n$.

q^i, p_i can be even or odd. There is a bracket

$$H_0(q,p) \text{ — Hamiltonian}, \quad T_\alpha(q,p) \text{ — constraints}, \quad \alpha = 1, \dots, 2m. \tag{2}$$

$$\{A,B\} = \frac{\partial_r A}{\partial q^i}\frac{\partial_\ell B}{\partial p_i} - (-1)^{n_A n_B}\frac{\partial_r B}{\partial q^i}\frac{\partial_\ell A}{\partial p_i} \tag{3}$$



Uncertain reading. A subscript on the second q^i is cancelled. It may be i.

Original source details: GELFAND-P000880

n_A, n_B — parities.

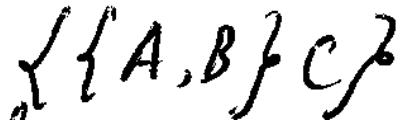
$$\{B, A\} = -(-1)^{n_A n_B}\{A, B\} \tag{4}$$

hence

$$\{A, A\} = -(-1)^{n_A}\{A, A\} \tag{5}$$

BRS — symmetry
(Becchi, Rouet, Stora).

$$(-1)^{n_A n_C}\{\{A, B\}C\} + \text{cycl. } (A, B, C) = 0 \tag{6}$$



Uncertain reading. In (6) no separate comma is visible between the inner brace and C.

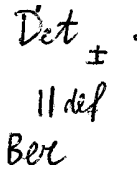
Original source details: GELFAND-P000880

Upon quantization:

$$\{A, B\} \mapsto [A, B] = AB - (-1)^{n_A n_B} BA \tag{7}$$

Definition

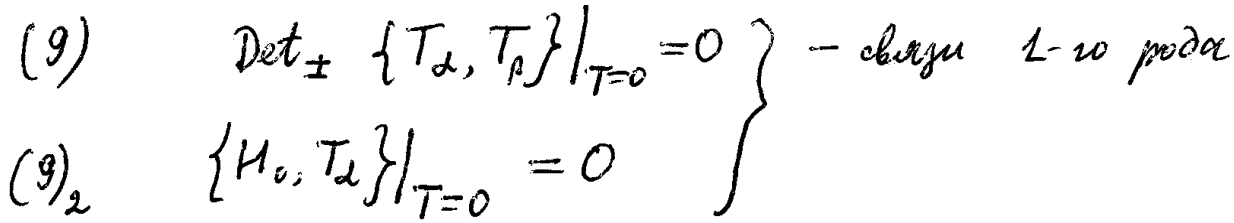
$$\underset{\substack{\| \text{ def} \\ \text{Ber}}}{\text{Det}_{\pm}\{T_\alpha, T_\beta\}}\Big|_{T=0} \neq 0 \quad \text{— 2nd-class constraints} \tag{8}$$



Uncertain reading. In (8), $\|$ def is added below $\text{Det}_{\pm}$, with Ber below it.

Original source details: GELFAND-P000880

$$\left.\begin{array}{ll} (9) & \text{Det}_{\pm}\{T_\alpha, T_\beta\}|_{T=0} = 0 \\ (9)_2 & \{H_0; T_\alpha\}|_{T=0} = 0 \end{array}\right\} \quad \text{— 1st-class constraints}$$



Original source details: GELFAND-P000880

Uncertain reading. The first line of (9) sets the determinant to zero. The second line is labelled $(9)_2$.

quant. of 1st-class constraints reduces to 2nd-class constraints (see below)

Consider 2nd-class constraints.

Class. dynamics — finding a constrained extremum of the class. action

$$S[q, p] = \int dt\,(p_i \dot{q}^i - H_0)\Big|_{T=0} \tag{10}$$

This is equiv. to an unconstrained extremum problem:

GELFAND-P000881

[source number: 39]

$$S[q,p,\lambda] = \int dt\,(p_i\dot{q}^i - H_0 - T_\mu\lambda^\mu) \tag{11}$$

Dirac bracket:

$$\{A,B\}_D \underset{\text{def}}{=} \{A,B\} - \{A,T_\alpha\}(Q^{-1})^{\alpha\beta}\{T_\beta,B\} \tag{12}$$

Here:

$$Q_{\alpha\beta} = \{T_\alpha, T_\beta\} \tag{13}$$

(I. M. G.: this is the bracket constr. from the restriction of the sympl. form to the submanif.). Consider the eq. for extremals of this action:

$$\begin{cases} \dot{q}^i = \{q^i, H_0\}_D \\ \dot{p}_i = \{p_i, H_0\}_D \\ \lambda^\alpha = -(Q^{-1})^{\alpha\beta}\{T_\beta, H_0\} \end{cases} \tag{14}$$

Original source details: GELFAND-P000881

Uncertain reading. A subscript is inked out after the final bracket in (14). It may be D.

One obtains the stand. Hamilton eqs. The Dirac bracket plays the role of the bracket. Hence it is precisely this bracket that is to be [with] replaced with the commutator.

Original source details: GELFAND-P000881

Uncertain reading. The manuscript gives an extra “на” (“with”) before “заменять” (“replace”).

From eq. [cancelled]

$$T_\alpha(q,p) = 0 \tag{15}$$

Original source details: GELFAND-P000881

Uncertain reading. A sign is cancelled between “Из” (“From”) and “ур-я” (“eq.”). It may be T.

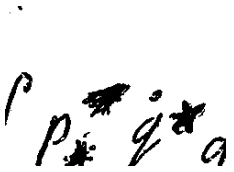
express

$$q^i = q^i(q^*, p^*), \quad p_i = p_i(q^*, p^*) \tag{16}$$

$* = 1, \ldots, n - m$

We want q^*, p_* to be canonical, i.e.

$$\int p_i \dot{q}^i \, dt \Big|_{T=0} = \int p_* \dot{q}^* \, dt \tag{17}$$



Original source details: GELFAND-P000881

Uncertain reading. On the right in (17), a superscript on p is cancelled, possibly an asterisk. The subscript asterisk is retained.

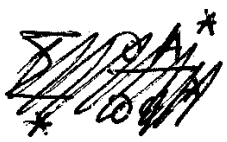
The eq. will take the form:

$$\dot{q}^* = \{q^*, H^{\text{phys}}\} \qquad \text{and likewise for } p. \tag{18}$$

$$H^{\text{phys}}(q^*, p_*) = H_0(q, p)|_{T=0} \tag{19}$$

This follows from the usual prop. of the Poisson bracket

$$\{A, B\}_D|_{T=0} = \text{[cancelled]} \quad \{A^*, B^*\}, \tag{20}$$



Original source details: GELFAND-P000881

Uncertain reading. After the equals sign a formula is cancelled, tentatively $\sum_* \frac{\partial A^*}{\partial q^*}$.

where

$$A^*(q^*, p^*) = A\big(q(q^*, p^*), p(q^*, p^*)\big) \tag{21}$$

GELFAND-P000882

Quantization

Without constraints. Let the Hamiltonian $H(q, p)$ and the action be given

$$S(q, p) = \int dt \, (p\dot{q} - H) \tag{22}$$

Original source details: GELFAND-P000882

Uncertain reading. There is a cancelled entry before $S(q, p)$. It may read $H(q, p)$.

Eqs.:

$$\dot{q} = \{q, H\} \qquad \text{and likewise for } p \tag{23}$$

Quantization: quantities are repl. by operators, and the eqs. hold

$$\dot{q} = \frac{1}{i}[q, H], \quad \dots \tag{24}$$

Original source details: GELFAND-P000882

Uncertain reading. The denominator sign in (24) is heavily retraced. It is read as i.

and the commut. rel.

$$\frac{1}{i}[q^i, p_j] = \{q^i, p_j\} = \delta^i_j \tag{25}$$

Original source details: GELFAND-P000882

Uncertain reading. A sign is cancelled below q^i in the first bracket in (25). It may be a subscript i.

Consider:

$$Z(J, I) = \langle 0 | T e^{-i \int H_{\text{ext}}\, dt} | 0 \rangle \tag{26}$$

$\langle 0|$ — ground-state vector of the syst. at the initial instant

(arb. initial vector

T — stand. T-product (arrang. the opers. in increasing time order).

Here:

$$H_{\text{ext}} = -J_i q^i - p_i I^i \tag{27}$$

Original source details: GELFAND-P000882

Uncertain reading. There is a cancelled entry at the end of (27). The final sign resembles q. The full reading remains uncertain.

(here the operators evolve in time by means of eqs. (24) —
— this is how the equation involves H).

Source eqs.. (follow from this)

Original source details: GELFAND-P000882

Uncertain reading. Above (28), a cancelled line contains $\dot{q}$ and $\{q, H\}$. The sign between them may be $=$ or $-$.

$$\left(\dot{q} - : \{q, H + H_{\text{ext}}\} :\right)\Big|_{\substack{q=\frac{\delta_r}{i\delta J} \\ p=\frac{\delta_\ell}{i\delta I}}} Z = 0 \tag{28}$$

$: \dots :$ — all sources are placed to the left of the differentiation opers.

[Below “all sources”: I and J.]

Eq. (28) is solved using a Fourier transf.:

$$Z = N \int [dq][dp] \exp\left\{ i \int dt \, (p\dot{q} - H - H_{\text{ext}}) \right\} \tag{29}$$

N — normalizing constant.

[In the right margin:] T-prod. is underdefined (at the same instant it does not distinguish opers.)

With 2nd-class constraints.

Dirac bracket $\rightsquigarrow$ commutator.

one must specify equal-time commut. rel.

GELFAND-P000883

[source number: 40]

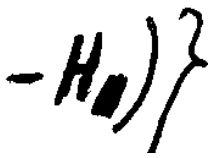
$$\frac{1}{i}[q^i, p_j] = \{q^i, p_j\}_D = \delta^i_j - \{q^i, T_\alpha\}(Q^{-1})^{\alpha\beta}\{T_\beta, p_j\} \tag{30}$$

(31) Eq. (28) is the same, with the Dirac bracket in it

It is solved again by the Fourier method:

Answer:

$$Z = \int [dq][dp] \exp\left\{ i \int dt (p\dot{q} - H_0) \right\} \prod_t \delta(T(q,p)) \big(\text{Det}_{\pm}\{T,T\}\big)^{1/2}, \tag{32}$$



Original source details: GELFAND-P000883

Uncertain reading. The subscript on H in (32) is inked over. It may be 0.

Here:

[On the right:] δ — $(2m)$-dimensional δ-fn.

$$H = H_0 + H_{\text{ext}} \tag{33}$$

On 1st-class constraints.

Let there be constraints T'_α, $\alpha = 1, \ldots, 2m$, of the 1st class

T'' — 2nd-class constraints.

The action

$$S = \int dt(p\dot{q} - H)\Big|_{T=0} \tag{34}$$

is invariant under transformations

$$\delta^F q = [q, T'_\alpha]F^\alpha \tag{35}$$

$F^\alpha = F^\alpha(t)$ — parameter!

This is gauge inv. in the Ham. formalism.

The constraints do not uniquely fix the dynamics.

Choose a repr. from each class — gauge fixing!

Add new constraints:

$$T'_\alpha = 0, \qquad \Phi_\alpha = 0 \tag{36}$$

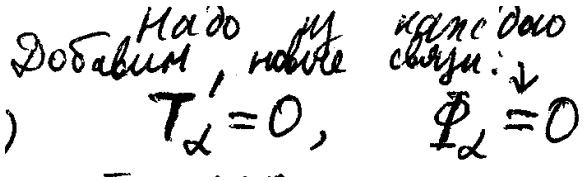


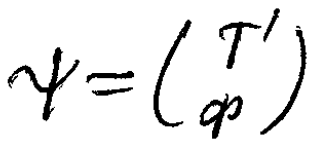
Diagram and typed description.
new constraints:
↓
$\Phi_\alpha = 0$

Original source details: GELFAND-P000883

such that

$$\text{Det}\{\Phi, T'\} \neq 0 \tag{37}$$

$\psi = \begin{pmatrix} T' \\ \varphi \end{pmatrix}$ — 2nd-class constraint!

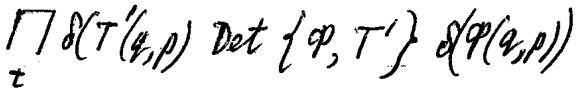


Uncertain reading. A psi sign precedes the column. Both ψ and Ψ are possible.

Original source details: GELFAND-P000883

The answer is already written down

$$Z' = \int [dq][dp] \exp\left\{ i \int dt (p\dot{q} - H) \right\} \prod_t \delta(T'(q,p)) \operatorname{Det}\{\varphi, T'\} \delta(\varphi(q,p))$$



Uncertain reading. One closing parenthesis is visible after $T'(q,p)$ and before Det.

Original source details: GELFAND-P000883

Independence of φ is traced using BRS symmetry.

GELFAND-P000884

19.11.79. Gelfand Sem..

Where do dilogarithms occur?

1) K-theory (Bloch, Beilinson.)

2) combinat. theory of char. classes (G.-G.-L.)

3) volume of a non-Euclidean tetrahedron (Gauss, --)

4) $\zeta(3)$ (Apéry)

5) Webs (Bol et al.)

[To the right of item 1:] 6) Scattering theory

Dilogarithm:

Uncertain reading. The end of the heading is heavily retraced. It may be singular or plural.

Original source details: GELFAND-P000884

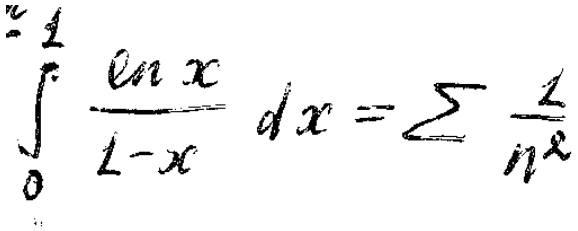
$$\int_0^{\xi} \frac{\log x}{1-x}\, dx \tag{1}$$

$$\sum_{j=1}^{5} \left(\frac{\log u_j}{1-u_j} + \frac{\log(1-u_j)}{u_j} \right) du_j = 0 \tag{2}$$

GELFAND-P000885

Beilinson

$$\int_0^1 \frac{\ln x}{1-x}\, dx = \sum \frac{1}{n^2} \tag{3}$$



Uncertain reading. No minus sign is visible before the sum in (3).

Original source details: GELFAND-P000885

Generalization to curves

(Bloch)

On the right $\zeta(2) \rightsquigarrow \zeta$ is repl. by the curve's ζ-fn.

X — Riem. surface over $\mathbb{C}$ (not necess. compact)

f, g — invertible holom. fns. on X. Consider the form:

$$\omega_{f,g} = \ln|f|\, d(\arg g) - \ln|g|\, d(\arg f) \tag{4}$$

Prop.:

$$1)\ d\omega_{f,g} = 0 \Rightarrow \text{integration along any path is well-defined} \tag{5}$$

$$2)\ \omega_{f_1 f_2, g} = \omega_{f_1, g} + \omega_{f_2, g} \tag{6}$$

$$3)\ \omega_{f,g} + \omega_{g,f} = 0 \tag{7}$$

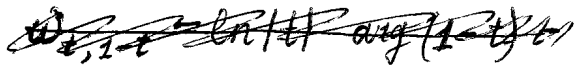
$$4)\ \omega_{f,1-f} \quad \text{is exact if } f \text{ and } 1-f \text{ are never equal to } 0 \tag{8}$$

(Steinberg ident.)

$$X = \mathbb{C}P^1 \setminus \{0, 1, \infty\}$$

$f = t$ — stand. parameter.

Then



Uncertain reading. The cancelled line reads $\omega_{t,1-t} = \ln|t| \arg(1-t)$, followed by illegible signs.

Original source details: GELFAND-P000885

$$\int_0^\xi \omega_{t,1-t} = \ln|\xi| \arg(1-\xi) - \operatorname{Im} \int_0^\xi \ln(1-t)\, d(\ln t) \tag{9}$$

The right-hand side is a single-valued fn.!! — one can show this

D. Wigner devised it and connected it to cont. cohomology of $SL(2)$.

(son)

5) Consider $\circledcirc$, going around one of the punctures x. Then

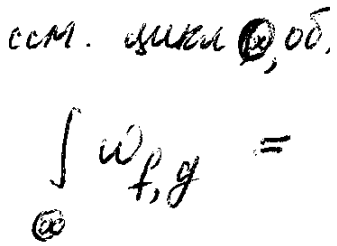
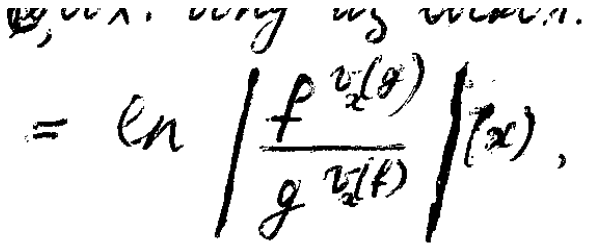


Diagram and typed description.

The interior mark resembles x. The orientation is unclear.

Original source details: GELFAND-P000885

$$\int_{\circledcirc} \omega_{f,g} = \ln \left| \frac{f^{v_x(g)}}{g^{v_x(f)}} \right| (x), \tag{10}$$



Uncertain reading. No factor is visible before the logarithm in (10).

Original source details: GELFAND-P000885

$v_x(g)$ — order of the zero of g at the point x.

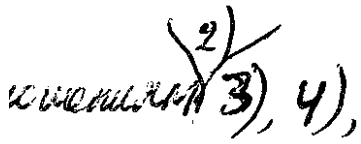
(an alg. curve is compactified uniquely. E.g., embed it into proj. space and take the closure)

This gives a map

$$K_2(X) \to H^1(X, \mathbb{R}) \tag{11}$$

f, g — fns. on X, vanishing nowhere.

Consider $\{f, g\}$ — as a free group and quotient by the relations 2), 3), 4),



Uncertain reading. "2)" is inserted above the line with an insertion mark before "3)".

Original source details: GELFAND-P000885

GELFAND-P000886

[source number: 47]

with the relations of bimultiplicativity, etc.

$\omega_{f,1-f} = 0$ in $H^1(X, \mathbb{R})$, i.e. $\{f, 1-f\} = 0$.

This is $K_2(X)$ of the field of rat. fns. on the curve.

$K_2(X)$ Define using the exact sequence:

$$\begin{aligned} K_2(X) \to K_2(\mathbb{C}(X)) &\to \bigoplus_{x \in X} \mathbb{C}^* \\ \{f, g\} &\longrightarrow \sum_{x \in X} (-1)^{v_x(f) v_x(g)} \frac{f^{v_x(g)}}{g^{v_x(f)}} \end{aligned} \tag{12}$$

From identity (10) it follows that there is a unique well-defined map (11). Consider the lin. combination

Uncertain reading. An illegible sign is cancelled before the sum in (13). It may be φ.

Original source details: GELFAND-P000886

$$\sum n_i\{f_i, g_i\} \in K_2(\mathbb{C}(X)) \tag{13}$$

Discard the zeros and poles. The integrals around the loops $= 0$

$\Rightarrow$ we obt. a cohomology class in $H^1(X, \mathbb{R})$ after taking the differ.

$$\left\{\sum n_i \omega_{f_i, g_i}\right\}. \tag{14}$$

There is a construction

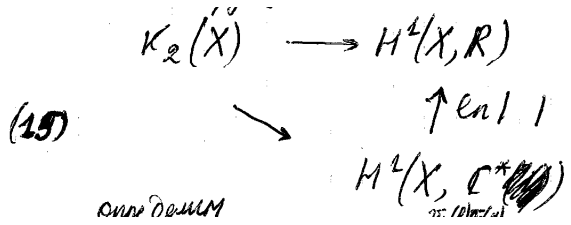


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Diagram and typed description.

$$\begin{array}{ccc} K_2(X) & \longrightarrow & H^1(X, \mathbb{R}) \\ & \searrow & \uparrow \ln | \, | \\ & & H^1(X, \mathbb{C}^* [\ldots]) \end{array} \tag{15}$$

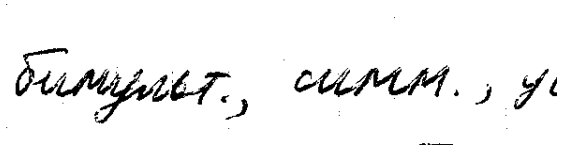
The signs after $\mathbb{C}^*$ are cancelled and illegible.

Define

$$\int_\gamma \{f, g\} = \begin{cases} (-1)^{v_\gamma(f) v_\gamma(g)} \exp \frac{1}{2\pi i} \int_\gamma \left(\ln f - \frac{v_\gamma(f)}{v_\gamma(g)} \ln g\right) d \ln g, & v_\gamma(g) \neq 0 \\ \exp \frac{1}{2\pi i} \int - \ln g \, d(\ln f) \end{cases} \tag{16}$$

(Beilinson)

This symbol is bimult., symm., satisfies the Steinberg ident., and gives the preceding one upon integration.



Uncertain reading. The manuscript reads “симм.” (“symm.”).

Original source details: GELFAND-P000886

X — aff. curve $\Rightarrow$

$$K_2(X) = H_2(SL_n(\text{alg. fns. on } X), \mathbb{Z}). \tag{17}$$

$$1 \to K_2(X) \to St^X \to SL^X \to 1 \tag{18}$$

(see Milnor and Steinberg).

— · —

GELFAND-P000887

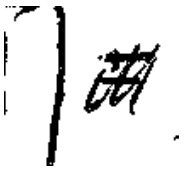
Batalin. On quantization of systems with constraints in a relativistic gauge.

$$S = \int (p\dot{q} - H_0 - \lambda T)\, dt \tag{1}$$

$$\varphi = \begin{pmatrix} q \\ \lambda \end{pmatrix} \quad \text{— relativ. field;} \tag{2}$$

In electrodyn.

$$\begin{pmatrix} \vec{A} \\ A \end{pmatrix} \quad \text{— where } A \text{ — } \lambda\text{-multiplier of the constraint.} \tag{3}$$



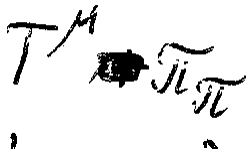
Original source details: GELFAND-P000887

Uncertain reading. An illegible sign is cancelled at the upper right of the column in (3).

In gravitation $\varphi^\alpha(q,p)$, $g^{0\alpha}$.

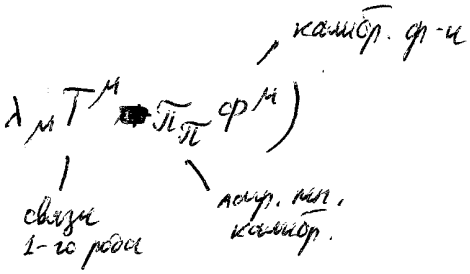
Now

$$S = \int (p\dot{q} - H_0 - \lambda_\mu T^\mu \text{ [?] } \tilde{\pi}_{\tilde{\pi}} \varphi^\mu) \tag{4}$$



Original source details: GELFAND-P000887

Uncertain reading. The sign before $\tilde{\pi}_{\tilde{\pi}}$ is heavily overwritten. It may be $-$ or $+$.



Original source details: GELFAND-P000887

Diagram and typed description.

gauge fns.

$\lambda_\mu T^\mu$ [?] $\tilde{\pi}_{\tilde{\pi}}$ φ^μ

1st-class constraints

Lagr. mult. for the gauge

$$\Phi = \Phi(p,q)$$

Relativ. inv. gauge condition

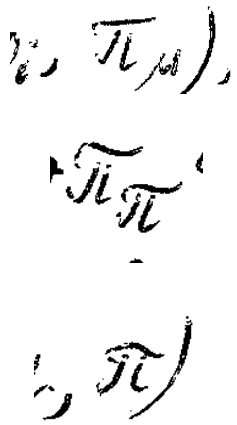
$$\Phi^\mu(p,q,\lambda,\dot{\lambda}) = -\dot{\lambda}^\mu + \chi^\mu(\lambda,p,q,\tilde{\pi}) \tag{5}$$

Original source details: GELFAND-P000887

Uncertain reading. The first argument of χ^μ in (5) is retraced. It may be λ.

$$\begin{cases} q^A = \begin{pmatrix} q^i \\ \lambda^\mu \end{pmatrix}, \quad p_A = (p_i, \bar{\pi}_\mu), \quad A = 1, \dots, n+m; \\ G_a = (\bar{\pi}_\mu, T_\mu), \quad a = 1, \dots, 2m; \quad \chi^a = -\begin{pmatrix} \chi^\mu \\ \lambda^\mu \end{pmatrix} \end{cases} \tag{6}$$

Uncertain reading. In (4) the subscript on $\tilde{\pi}$ resembles a second $\tilde{\pi}$. In (5) the stroke over π is wavy. In (6) the stroke over π is flatter.



Original source details: GELFAND-P000887

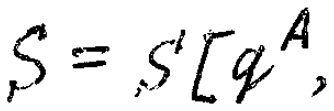
This is relativ. phase space. Then

Uncertain reading. An illegible sign is cancelled before “This”.

Original source details: GELFAND-P000887

$$\begin{aligned} S = S'[q^A, p_A] = \int dt(p_A \dot{q}^A - H_{n+m}); \\ H_{n+m} = H_0 - G_a \chi^a. \end{aligned} \tag{7}$$

Uncertain reading. A short detached stroke follows the second S in (7). It is read as the prime in S'.



Original source details: GELFAND-P000887

Here the increase in the number of variables is explicit
(but in fact $(n-m)$)

$$n - m = (n+m) - 2m$$

We must suppress $2m$ extra degrees of freedom.

$$Z = \int \left(\prod_{t,A,a} dq^A \, dp_A \, d\eta^a \, d\mathcal{P}_a \right) \exp\left\{ i \int dt \left(p_A \dot{q}^A + \mathcal{P}_a \dot{\eta}^a - H_{\text{complete}} \right) \right\} \tag{8}$$

Uncertain reading. The subscript of $\mathcal{P}$ in the exponent in (8) is retraced. It is read as a.

Original source details: GELFAND-P000887

where

$$H_{\text{complete}} = H_1 - \{\Psi, \Omega\}. \tag{9}$$

GELFAND-P000888

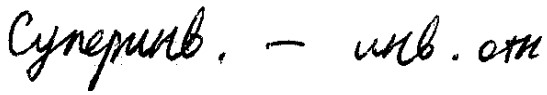
[source number: 48]

Here H_1 and Ω are found from the eqs.:

$$\{\Omega, \Omega\} = 0, \qquad \{H_1, \Omega\} = 0 \tag{10}$$

$$\Psi = \mathcal{P}_a \chi^a \tag{11}$$

Ψ, Ω — fermionic var.
If the constraints are 1st class, the eqs. are solved as series in powers of η^a and $\mathcal{P}_a$. In supergravit. these tend to terminate.
Superinv. — inv. under [cancelled] BRS-transf.:



Uncertain reading. The line begins with "Супериив." ("Superinv.").

Original source details: GELFAND-P000888

Uncertain reading. There is an illegible cancelled entry after "отн." ("under").

Original source details: GELFAND-P000888

$$\delta X = \{X, \Omega\}\mu, \qquad \mu \text{ — odd var. (constant).} \tag{12}$$

Ω — fermion
choose:

$$\mu = i \int dt \, \mathcal{P}_a \cdot \delta\chi^a \tag{13}$$

This gives BRS-inv.

One Monday is skipped.

GELFAND-P000889

3.12.79. Gelfand Sem.

Physicist. On modern theory of elem. particles.

QCD — strong 1972 (final form) Gell-Mann

$$\underbrace{\text{QED} + \text{weak}}_{\text{quant. electrodyn.}} \Longrightarrow \text{QFD}$$

1967 (Weinberg, Salam) Glashow

GR Exper. facts — the Lagrangians are correct

Faddeev, Slavnov — wrote down the Lagrangians

Original source details: GELFAND-P000889

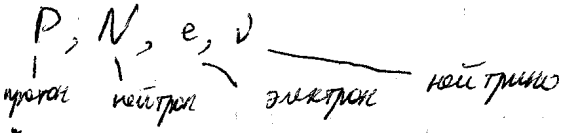
Uncertain reading. After the names the manuscript reads “выписали лагранжианы” (“wrote down the Lagrangians”).

We need:

1) group reprs.

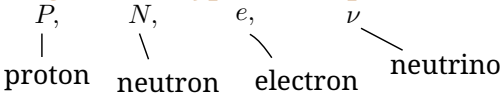
2) Feynman diag. (see Bjorken, Drell, vol. 1)

The path from the Lagrangians to exp. data is short !!



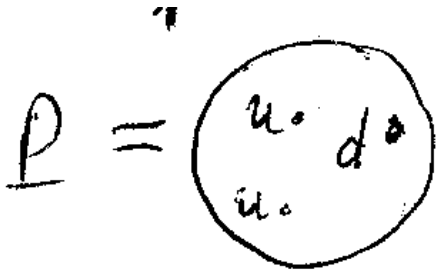
Original source details: GELFAND-P000889

Diagram and typed description.



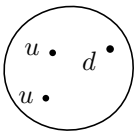
Original source details: GELFAND-P000889

Uncertain reading. A sign, possibly P, is cancelled below “proton”.



Original source details: GELFAND-P000889

Diagram and typed description.

$\underline{P} =$ 

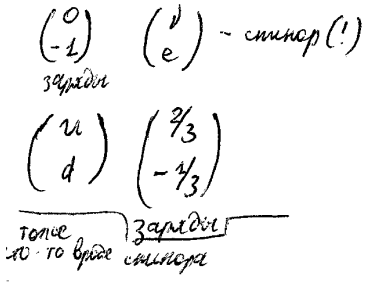


Diagram and typed description.

$\binom{0}{-1}$ $\binom{\nu}{e}$ — spinor (!)

charges

$\binom{u}{d}$ $\binom{\frac{2}{3}}{-\frac{1}{3}}$

also charges

something like a spinor

Original source details: GELFAND-P000889

GELFAND-P000890

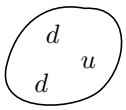
Fractions cannot be avoided (make the charges the unit matrix)

$$\binom{\nu}{e} \quad \binom{u}{d} \text{ — basic elem. particles.}$$

[On the right:] (1st generation) of particles

They suffice to construct the surrounding world

Diagram and typed description.

$N =$ 

Original source details: GELFAND-P000890

There are other generations

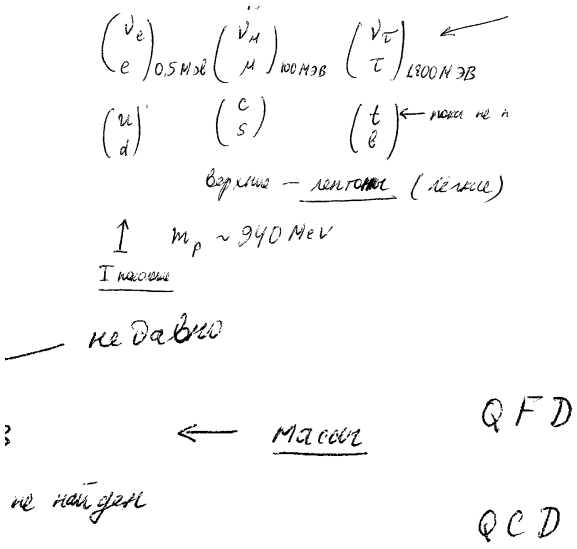


Diagram and typed description.

recently

$\binom{\nu_e}{e}$ $\binom{\nu_\mu}{\mu}$ $\binom{\nu_\tau}{\tau}$ QFD

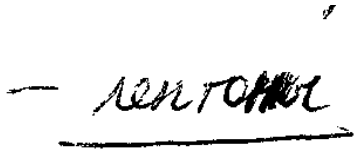
0,5 MeV 100 MeV 1800 MeV ← masses

$\binom{u}{d}'$ $\binom{c}{s}$ $\binom{t}{b}$ ← not yet found QCD

1st generation

Original source details: GELFAND-P000890

The upper ones — leptons (light)



Original source details: GELFAND-P000890

Uncertain reading. The end of “лептоны” (“leptons”) is overwritten.

$$m_p \sim 940\,\text{MeV}$$

The next generations, once produced, disappear quickly.

Leptons take part only in flavour dyn. — el.-magn. and weak interac.

(flavour dinamics)

Each quark is tripled:

u^1, u^2, u^3 and likewise for all quarks

d^1, d^2, d^3 (c, s, t, b)

Corresponding symmetry:

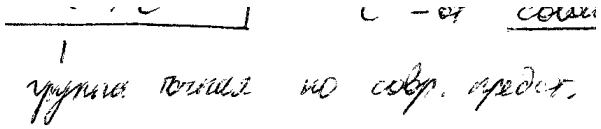
each quark is a representation of $SU(3)$

Original source details: GELFAND-P000890

Uncertain reading. The end of “представление” (“representation”) is overwritten.

$\boxed{u^1, u^2, u^3 \qquad SU(3)_c}$ C — from colour (Russian «цвет»)

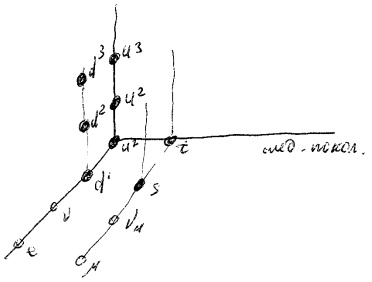
[Line from $SU(3)_c$] the group [possibly “also”] by the adj. repr.



Original source details: GELFAND-P000890

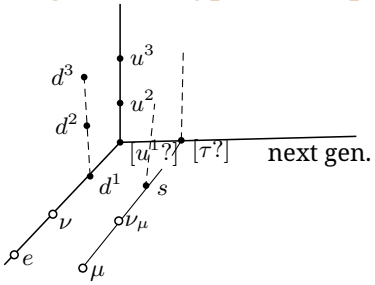
Uncertain reading. The word after “группа” (“group”) may read “тоже” (“also”).

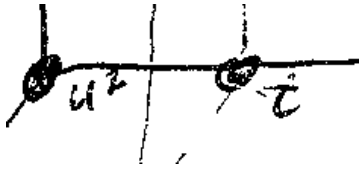
Draw the generations in a 3-dimensional diagram



Original source details: GELFAND-P000890

Diagram and typed description.





Original source details: GELFAND-P000890

Uncertain reading. The superscript on u at the vertex and the label of the right-hand point are uncertain. The right-hand label resembles τ with a detached stroke above.

[To the right of the diagram:]

$SU(3)_c$ changes the vert. comp.
(handles all verticals at once)
leptons are not tripled
(only quarks)
GELFAND-P000891

[source number: 59]

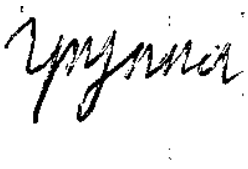
Action of the group.

$$|d^{1\prime}\rangle = U_{11}|d^1\rangle + U_{12}|d^2\rangle + U_{13}|d^3\rangle$$

U is a unitary matrix.
Or.

$$|d^{1\prime}\rangle = U_1^1|d^1\rangle + U_2^1|d^2\rangle + U_3^1|d^3\rangle$$

The leptons are permuted too!! — approximately (the masses differ)
Permutation group $SU(2)_w$ (Weinberg group)



Uncertain reading. The Russian word «Группа» is singular.

Original source details: GELFAND-P000891

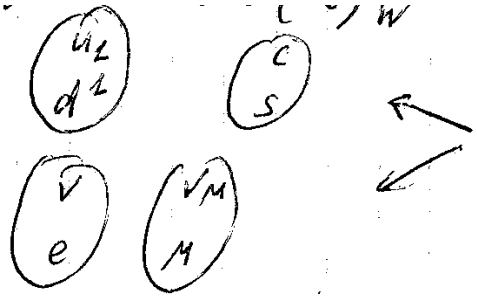
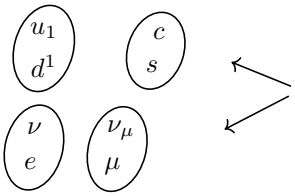
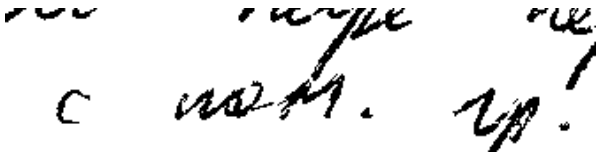


Diagram and typed description. Ovals enclosing the weak doublets, with two arrows



Original source details: GELFAND-P000891

the particles are permuted in sync within each pair by means of the group $SU(2)$



Uncertain reading. The abbreviation reads «с пом.» (by means of).

Original source details: GELFAND-P000891

We still need $U(1)$

$$[SU(2) \times U(1)]_w$$

[Crossed out:

Uncertain reading. illegible entry

Original source details: GELFAND-P000891

.] ($U(1)$ twists the phases)

Presumably e ought to have had mass 0. A transition took place as the Universe cooled.

Fermions (all those mentioned above) are described by the Dirac equation

$$(i\hat{\partial} - m)\Psi = 0 \tag{1}$$

$$\Psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix}, \qquad i = 0, 1, 2, 3$$

$$\hat{\partial} = \partial^i \gamma_i = \partial_i \gamma^i, \tag{2}$$

[After the second equals sign, crossed out:

Uncertain reading. $\gamma_i \partial^i$?

Original source details: GELFAND-P000891

.]

where the γ^i are the Dirac matrices,

$$\gamma^i \gamma^k + \gamma^k \gamma^i = 2g^{ik} \tag{3}$$

$$g^{ik} = \text{diag}(1, -1, -1, -1). \tag{4}$$

Eq. (1) is Lorentz invariant

$$\partial_t = \partial_0, \quad \partial_x = \partial_1, \quad \partial_y = \partial_2, \quad \partial_z = \partial_3.$$

Degeneracy of the Dirac particle:

$$\gamma^5 = \alpha \gamma^0 \gamma^1 \gamma^2 \gamma^3 \tag{5}$$

Consider the massless eq. (1): $m = 0$

Choose α so that in the representation

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^\alpha = \begin{pmatrix} 0 & \sigma^\alpha \\ -\sigma^\alpha & 0 \end{pmatrix} \tag{6}$$

the σ^α are the Pauli matrices ($\frac{1}{2}\sigma^\alpha$ are the generators of $SU(2)$)

we get

$$\gamma^5 = -\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \tag{7}$$

Original source details: GELFAND-P000891

Diagram and typed description. Momentum, spin, and the screw sense of the neutrino

$\longleftarrow$ p momentum

— spin

neutrino left-handed screw

antineutrino — — right-handed screw

GELFAND-P000892

$$\Psi_\nu = \underbrace{\frac{1+\gamma^5}{2}\Psi_\nu}_{\text{left-handed screw}} + \underbrace{\frac{1-\gamma^5}{2}\Psi_\nu}_{\text{right-handed screw}}$$

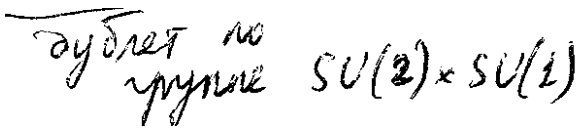
$\frac{1\pm\gamma^5}{2}$ are projectors onto invariant subspaces

for the neutrino only $\frac{1+\gamma^5}{2}\Psi_\nu$ occurs

for the electron, both

$$\boxed{\begin{array}{c}\frac{1+\gamma^5}{2}\Psi_e \\ \text{left electron} \\ \frac{1+\gamma^5}{2}\Psi_\nu\end{array}} \qquad \boxed{\begin{array}{c}\frac{1-\gamma^5}{2}\Psi_e \\ \text{right electron}\end{array}}$$

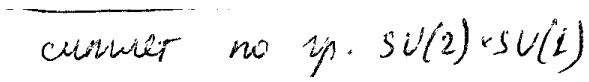
[Under the left box:] doublet under the group $SU(2) \times SU(1)$



Uncertain reading. The final factor is written $SU(1)$.

Original source details: GELFAND-P000892

[Under the right box:] singlet under the group $SU(2) \times SU(1)$



Uncertain reading. The sign between the groups looks like an incomplete $\times$ or like $<$.

Original source details: GELFAND-P000892

[Formula crossed out:

Uncertain reading. $\bar{e}\gamma^0(1+\gamma^5)\nu_e$? The superscript on the first γ and the final strokes are indistinct.

Original source details: GELFAND-P000892

]

QFD: $\gamma,\ Z^0,\ W^+,\ W^-$

Z^0, W^+, W^- are massive

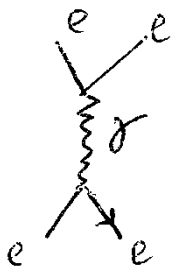
γ is light

QCD: $SU_c(3)$ 8 gluons

adjoint repr. in the Lie algebra

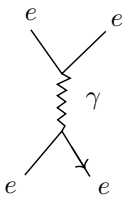
Lagrangian.

First for vector fields V



Original source details: GELFAND-P000892

Diagram and typed description. Feynman sketch with a photon



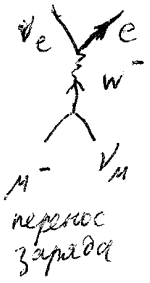
Feynman graph

γ is the quantum [crossed out:

Original source details: GELFAND-P000892

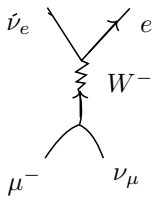
Uncertain reading. «несущ.» (carrying)?

] that transfers momentum



Original source details: GELFAND-P000892

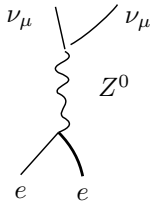
Diagram and typed description. Sketch with W^-

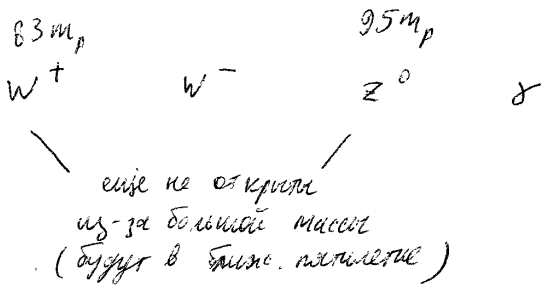


[The upper-left label resembles ν_e with an extra stroke. The mark on the adjacent line is unclear.]

Original source details: GELFAND-P000892

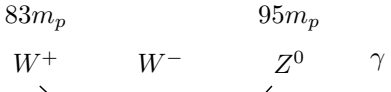
Diagram and typed description. Sketch with Z^0





Original source details: GELFAND-P000892

Diagram and typed description. Mass labels and handwritten prediction



still undiscovered
because of their large mass
(will be found in the next five years)

Charge is an eigenvalue.

pairs $\begin{pmatrix} e \\ \nu \end{pmatrix}$ are eigenvectors

GELFAND-P000893

[source number: 60]

A_a^i, a is the field number in the adjoint repr. (8)

$$F_a^{ik} = \partial^i A_a^k - \partial^k A_a^i - g f_{abc} A_b^i A_c^k \tag{9}$$

$$\mathrm{Sp}(I_a I_b) = \frac{1}{2}\delta_{ab} \tag{10}$$

I_a is an oper. in the adjoint repr. of G (basis)

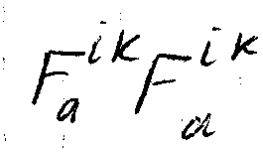
Uncertain reading. The Russian abbreviation reads «опер.».

Original source details: GELFAND-P000893

Lagrangian:

$$-\frac{1}{4}F_a^{ik}F_a^{ik} + \bar{\Psi}(i\hat{D})\Psi \tag{11}$$

Uncertain reading. Both F factors are written with ik above and a below.



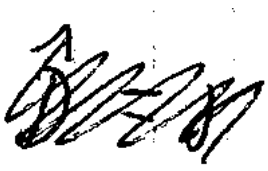
Original source details: GELFAND-P000893

Here

$$D_i = \partial_i + igI^a A_i^a \tag{12}$$

$$\hat{D} = \gamma^i D_i \tag{12}_2$$

[Crossed out at right:



Uncertain reading. $\hat{D} = \gamma$?

Original source details: GELFAND-P000893

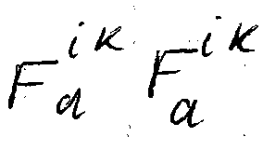
.]

Field (11) is massless

We also need scalar fields Φ (Higgs fields), transforming as a multiplet

We need:

$$-\frac{1}{4}F_a^{ik}F_a^{ik} + \bar{\Psi}(i\hat{D})\Psi + \frac{1}{2}(D\varphi)^2 - P(\varphi) \tag{13}$$



Uncertain reading. Both F factors are written with ik above and a below.

Original source details: GELFAND-P000893

Required

$$P(\varphi) = P(g\varphi) \tag{14}$$

Example: $G = U(1)$

$$P(\varphi) = \lambda^2(\varphi_1^2 + \varphi_2^2 - \eta^2)^2 \tag{15}$$

Uncertain reading. A short sloping stroke of uncertain function precedes $P(\varphi)$ in (15).

Original source details: GELFAND-P000893

(the second power is the max compatible with renormalizability)

Classical vacuum: solutions of the eqs. with min energy, i.e. those giving min $P(\varphi)$

GELFAND-P000894

10.12.79. Gelfand Sem.

The same physicist. There were 2 Lagrangians

(Igor Yuryevich Kobzar'- (?)).

Uncertain reading. The ending before the dash resembles «рь».

Original source details: GELFAND-P000894

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}F_a^{ik}F_a^{ik}, \qquad SU_c(3) \tag{1}$$

$\mathcal{L}_{\text{FD}}$ similarly with $SU(2)\times$ [crossed out:

Uncertain reading. illegible letter

Original source details: GELFAND-P000894

] $U(1)$.

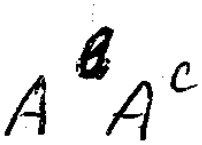
$$F_a^{ik} = \partial^i A_a^k - \partial^k A_a^i - f_{abc} A^b A^c \tag{2}$$

[Written separately below the last term:]

$$A^a$$

Uncertain reading. There is an additional loop at the upper stroke of $=$.

Original source details: GELFAND-P000894



Uncertain reading. The index b in $A^b A^c$ is heavily traced over.

Original source details: GELFAND-P000894

From this it is easy to compute

$$\int E^2 \, dV \sim \frac{e^2}{r} \tag{3}$$

Uncertain reading. The sign in the denominator is read as r.

Original source details: GELFAND-P000894

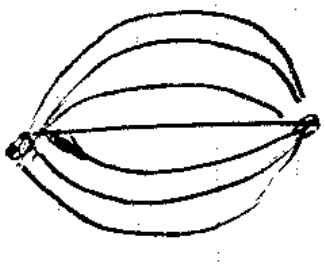
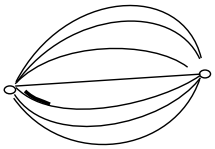


Diagram and typed description. Field lines at two small circles



Original source details: GELFAND-P000894

[A thickened mark near the left end of the inner lower arc resembles an arrowhead.]

Nonlinearity of the medium ⇒ the field lines stick together:

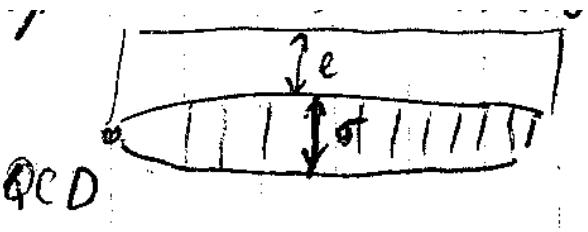
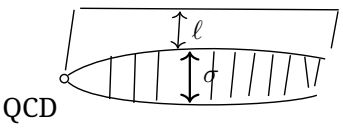


Diagram and typed description. QCD tube with a bracket and two vertical arrows



Original source details: GELFAND-P000894

[The upper label is read as ℓ.]

[To the right of the drawing:] Hypothesis: the cross-section (or diameter) of the tube is independent of the distance

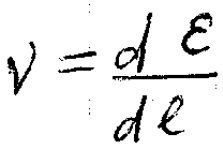
min., energy density is constant $\Rightarrow$ a string results

Uncertain reading. The initial abbreviation resembles «мин,».

Original source details: GELFAND-P000894

$$u^1, u^2, u^3$$

$$\nu = \frac{d\varepsilon}{d\ell} \sim E^2\sigma$$



Uncertain reading. The sign to the left of the fraction is read as ν.

Original source details: GELFAND-P000894

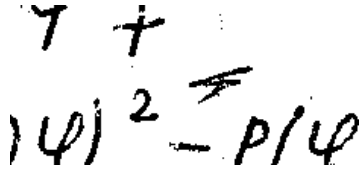
Manin's remark: when quantizing Y.–M. in the semiclassical approximation.

(and there is no exact model yet)

Weak Lagrangian:

$$\mathcal{L}_{\text{FD}} = -\frac{1}{4}F_a^{ik}F_a^{ik} + \sum_{\text{rep}} \bar{\Psi}(i\hat{D})\Psi + \\ + \frac{1}{2}(D\varphi)^2 - P(\varphi); \quad \big| (SU(2)\times U(1))_{w,s} \tag{4}$$

[Crossed out above the minus sign before $P(\varphi)$:



Uncertain reading. illegible sign

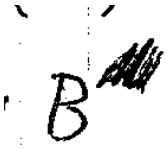
Original source details: GELFAND-P000894

.]

Here

$$D_i = \partial_i + igt^\alpha W_i^\alpha + ig'yB \tag{5}$$

[Crossed out above and to the right of B:



Uncertain reading. illegible index

Original source details: GELFAND-P000894

.]

t^1, t^2, t^3 are generators of the repr. of $SU(2)$

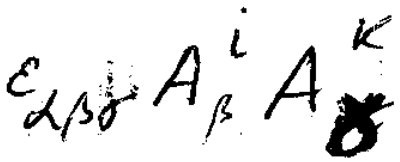
GELFAND-P000895

[source number: 66]

$$-\frac{1}{4}F_\alpha^{ik}F_\alpha^{ik}-\frac{1}{4}H^{ik}H^{ik}+\sum_{\text{rep}}\bar{\Psi}(i\hat{D})\Psi \\ +\frac{1}{2}(D\varphi)^2-P(\varphi) \tag{6}$$

[Below the first term:] $\alpha=1,2,3.$

$$F_\alpha^{ik}=\partial^i A_\alpha^k-\partial^k A_\alpha^i-\varepsilon_{\alpha\beta\delta}A_\beta^i A_\delta^k \tag{7}$$



Uncertain reading. The third index on ε and the subscript on the final A read δ. The latter is traced over.

Original source details: GELFAND-P000895

[A separate sign above (7) on the right is crossed out:

Uncertain reading. illegible

Original source details: GELFAND-P000895

.]

$$W^{ik}=\partial^i B^k-\partial^k B^i,\qquad A_i^\alpha=W_i^\alpha \tag{8}$$

We get invariance under a local group. This was the Yang–Mills idea.

t^α, y are operators (in formulas (4))

A_α^k is a connection with values in the Lie algebra

α — $SU(2)$

k is the 1-form index

B^k is a connection with values in $U(1)$

$$\hat{D}=\gamma_i D^i \quad \text{is the standard notation} \tag{9}$$

Ψ is a 4-component spinor

$SU(2)$ is a 3-dimensional (adjoint) repr., trivial on a 1-dimensional space.

Let us examine [crossed out] the situation without vector fields: the scalar part

Uncertain reading. An illegible entry is crossed out after «Исследуем» (let us examine).

Original source details: GELFAND-P000895

$$\mathcal{L}_\Phi = [?]\,\frac{1}{2}(\partial\varphi)^2 - P(\varphi) \tag{10}$$

Uncertain reading. A small sloping sign precedes $\frac{1}{2}$. Its reading is uncertain.

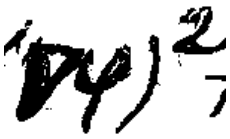
Original source details: GELFAND-P000895

φ has multiple components with respect to the group G, $\varphi = \varphi^\alpha$, $\alpha = 1, \ldots, N$
N is the dimension of the repr. corresponding to the scalar quantities.
We can assume that φ is real (if not, increase the number of components).

$$P(g\varphi) = P(\varphi) \tag{11}$$

$$T_{00} = \overset{\text{kinetic energy}}{\frac{1}{2}(\partial_t\varphi)^2} + \frac{1}{2}(\nabla\varphi)^2 + P(\varphi) \tag{12}$$

[At right:] positive def.



Uncertain reading. The sign before φ in the second term is traced over and reads ∇.

Original source details: GELFAND-P000895

min is always attained at $\varphi = \text{const}$

$$\frac{\partial P}{\partial \varphi^\alpha} = 0 \tag{13}$$

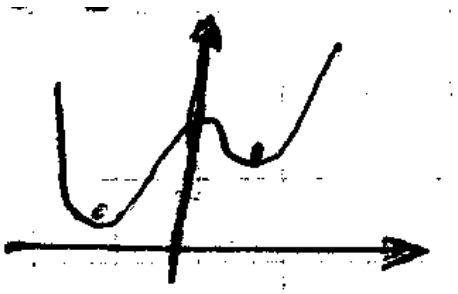
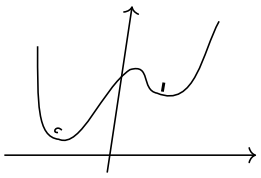


Diagram and typed description. Graph with two minima



Original source details: GELFAND-P000895

[The small marks at the minima are redrawn as strokes. The left mark resembles a letter or a point marker.]

[To the right of the drawing:] we will call any minimum (local) a vacuum.

$$\varphi^{\alpha}_{\text{vac}} = \lambda^{\alpha} \tag{14}$$

We will consider small departures from the vacuum

$$\varphi'^{\alpha} = \lambda^{\alpha} + \varphi^{\alpha} \tag{15}$$

We get the wave equation

Uncertain reading. The ending of «волновое» (wave) is written over.

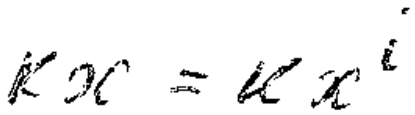
Original source details: GELFAND-P000895

$$\Box\varphi^{\alpha} + \frac{\partial^2 P}{\partial\varphi^{\alpha}\partial\varphi^{\beta}}\varphi^{\beta} = 0. \tag{16}$$

GELFAND-P000896

$$\varphi^{\alpha} = c^{\alpha} e^{-ikx}, \qquad kx = kx^{i} \tag{17}$$

Uncertain reading. The second k has no written subscript.



Original source details: GELFAND-P000896

$\Rightarrow$

$$m^2 c^{\alpha} = \frac{1}{2}\frac{\partial^2 P}{\partial\varphi^{\alpha}\partial\varphi^{\beta}} c^{\beta}, \qquad m^2 = k^2 \tag{18}$$

Uncertain reading. The manuscript writes the factor $\frac{1}{2}$.

Original source details: GELFAND-P000896

Symmetry (11) $\Rightarrow$ we always get N oscillations with $m^2 = 0$

Uncertain reading. The overwritten letter in the main line resembles N. The note at right has n.

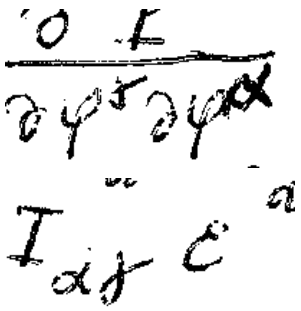
Original source details: GELFAND-P000896

[At right:] ($n = \dim G$, the number of generators, and not the dimension of the repr.)
Write the condition for invariance under infinitesimal transformations.

$$g = 1 + iI^a\varepsilon^a \tag{19}$$

$$\frac{\partial P}{\partial \varphi^\alpha} I^a_{\alpha\beta}\varphi^\beta\varepsilon^a = 0 \tag{20}$$

$$\frac{\partial^2 P}{\partial\varphi^\delta\partial\varphi^\alpha} I^a_{\alpha\beta}\varphi^\beta\varepsilon^a + \frac{\partial P}{\partial\varphi^\alpha} I^a_{\alpha\delta}\varepsilon^a = 0 \tag{21}$$

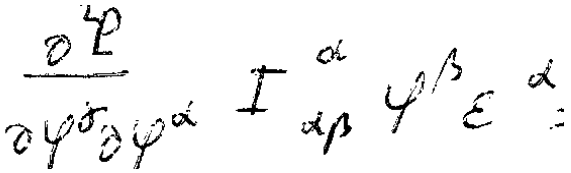


Uncertain reading. The first superscript on φ and the second subscript on I in the final term read δ.

Original source details: GELFAND-P000896

Hence.

$$\frac{\partial^2 P}{\partial\varphi^\delta\partial\varphi^\alpha} I^\alpha_{\alpha\beta}\varphi^\beta\varepsilon^\alpha = 0 \tag{22}$$



Uncertain reading. The first index in the denominator reads δ. The superscripts on I and ε are shaped like α.

Original source details: GELFAND-P000896

$$c^a_\alpha = I^a_{\alpha\beta}\varphi^\beta \tag{23}$$

Uncertain reading. The index α on c is written over.

Original source details: GELFAND-P000896

is the vector corresponding to the zero mode. (Goldstone theorem)
Some can turn out to be dependent, in general (depending on the repr.)

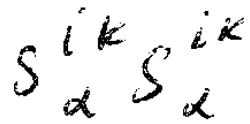
Uncertain reading. The initial word reads «Часть» (some).

Original source details: GELFAND-P000896

$$\tau^a I^a \lambda = 0 \tag{24}$$

Consider Y.–M. on an arbitrary group G. Lagrangian.

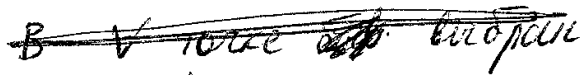
$$-\frac{1}{4} S^{ik}_\alpha S^{ik}_\alpha - \mathcal{L}(\varphi) \tag{25}$$



Uncertain reading. Both letters in the quadratic term are written as S.

Original source details: GELFAND-P000896

[Crossed out: At $\forall$ points



Uncertain reading. illegible sign

Original source details: GELFAND-P000896

we chose.]

φ^α is a scalar field (repr. of G)

$\alpha = 1, \ldots, N$

Example

$$P(\varphi) = \frac{1}{2}(\varphi_1^2 + \varphi_2^2 - \eta^2)^2, \qquad G = O(2) \quad \text{in the real repr.} \tag{26}$$

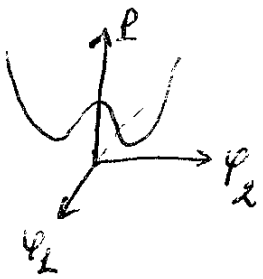
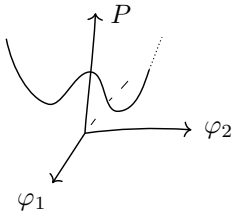


Diagram and typed description. Potential profile with three axes



Original source details: GELFAND-P000896

"bottle"

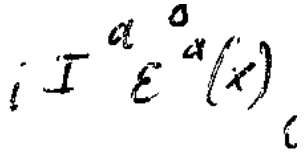
the vacuum is any state at the bottom of the bottle.

Disappearance of the Goldstone modes — a gauge is constructed.

Gauge: choice of a basis of sections, in which the connection is represented by a function

The basis of sections is constructed from the Goldstone modes. We construct rotations

$$\Phi \longrightarrow e^{iI^a \varepsilon^\alpha(x)} \Phi, \tag{27}$$



Uncertain reading. The superscript on ε resembles α.

Original source details: GELFAND-P000896

[In small writing below the formula:] and use them to arrange for Φ to be orthogonal to all the Goldstone modes.

N is the dimension of Φ

n zero modes

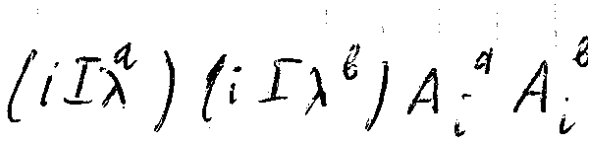
φ can be changed so that at $\forall$ points it is orthogonal to the Goldstone modes (without destroying all the properties)

GELFAND-P000897

[source number: 67]

In this procedure the particles acquired masses.

$$-\frac{1}{4}F_\alpha^{ik}F_\alpha^{ik}+\frac{1}{2}(iI\lambda^a)(iI\lambda^b)A_i^aA_i^b \tag{28}$$

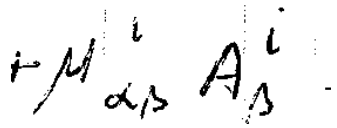


Uncertain reading. The indices a, b are written above and to the right of λ.

Original source details: GELFAND-P000897

We get

$$\frac{\partial F_\alpha^{ik}}{\partial x^i}+M_{\alpha\beta}^iA_\beta^i=0 \tag{29}$$



Uncertain reading. Both M and A have superscript i as written.

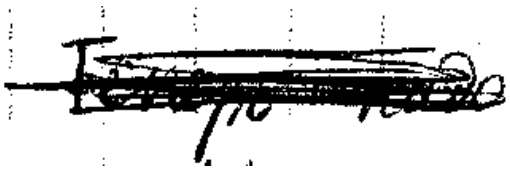
Original source details: GELFAND-P000897

(since

$$D\varphi=\partial\varphi+iI^aA^a\varphi) \tag{30}$$

This was the work of Higgs.

[Crossed out:



Uncertain reading. illegible line

Original source details: GELFAND-P000897

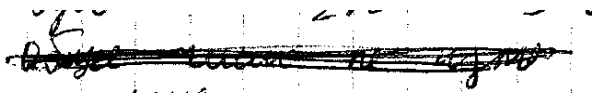
.]

The number of degrees of freedom increased to $3n$ (3 polarization dir.).

Goldstone $N \to N-n$

Here $2n \to 3n$

[Crossed out:



Uncertain reading. The start resembles «Общее» (total). The rest is illegible.

Original source details: GELFAND-P000897

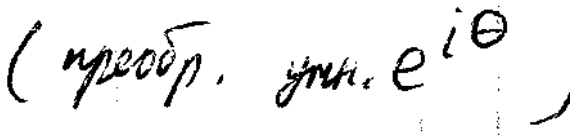
]

Exercise. eliminate the Goldstone modes for [crossed out] example (26)

Original source details: GELFAND-P000897

Uncertain reading. The word before «примера» (example) is crossed out and illegible.

$$A \to A + \text{const} \cdot \partial\theta \qquad \text{(transf. by mult. by } e^{i\theta}\text{)}.$$



Original source details: GELFAND-P000897

Uncertain reading. The abbreviation before $e^{i\theta}$ reads «ум.» (mult.).

Return to Lagrangian (6), fields W, B

$$\varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \quad \text{— spinor} \qquad \text{(4 degrees of freedom in all)} \tag{31}$$

Generators of $SU(2) \times U(1)$

$$I'_1 = gt_1, \quad I'_2 = gt_2, \quad I'_3 = gt_3, \quad I'_4 = g'y \tag{32}$$

Gauge rotations take spinor (31) to $\begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$

$$t_3 \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$$

$$y \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix} = p \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$$

Original source details: GELFAND-P000897

Uncertain reading. The coefficient and the letter in the denominators are read as p.

Hence

$$\left(2t_3 + \frac{y}{p}\right)\begin{pmatrix}0\\ \varphi_2\end{pmatrix} = 0$$

Thus $\exists$ a lin. comb. of operators that is $\equiv 0$ (depends on the vector)
(i.e. for the given state one of the Goldstone modes does not arise)

$$t_3 + \frac{y}{2p} = t_3 - y'; \qquad +y' = -\frac{y}{2p}$$

1 degree of freedom remains.
γ quantum — a vector gauging this freedom.

Uncertain reading. The heavily overwritten word reads «эту» (this).

Original source details: GELFAND-P000897

GELFAND-P000898

The mass matrix for fields $\alpha = 1, 2, 3, 4$ has the form

$$\begin{vmatrix} a^2 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & a^2 & ab \\ 0 & 0 & ab & b^2 \end{vmatrix}, \tag{33}$$

where

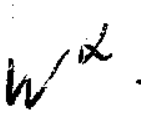
$$\begin{cases} a = g\,\text{const} \\ b = g' \cdot \text{const} \end{cases} \tag{34}$$

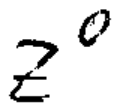
$$\frac{1}{2}\left(igt^\alpha \lambda W^\alpha + ig' y \lambda B\right)^2 \tag{35}$$

Weinberg.

$$\frac{g'}{g} = \operatorname{tg}\theta_w \tag{36}$$

$$m_z^0 = \frac{m_w}{\cos\theta_w}, \qquad Z^0 \text{ is the direction}$$





Uncertain reading. The field letters are set as W, Z. Their handwritten case is sometimes indistinct. The small indices w, z are retained.

Original source details: GELFAND-P000898

We obtain:

$$\mathcal{L}_{\text{int}} = -\frac{g}{\sqrt{2}} \left(\bar{e}_L \gamma^i \nu_L\right) W^- + \text{Hermitian conjugate} - \tag{37}$$

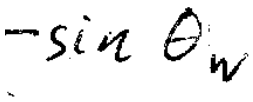
$$(W^- \text{ are new coord.})$$

$$- g \sin\theta_w \left(\bar{\Psi} \gamma^i q \Psi\right) A_i$$

$$- \frac{q}{\cos\theta_w} \left(\bar{\Psi}_L t_3 \gamma^i \Psi_L - \sin\theta_w \, \bar{\Psi} q \gamma^i \Psi\right) Z^0$$

Original source details: GELFAND-P000898

Uncertain reading. The numerator before the final parentheses is read as q.



Original source details: GELFAND-P000898

Uncertain reading. The final parentheses have $\sin\theta_w$ with no written exponent.

Original source details: GELFAND-P000898

Uncertain reading. One minus sign ends the first line of (37). Another begins its continuation below the annotation.

Original source details: GELFAND-P000898

q is the charge operator, $\quad q = t_3 - y$

The amplitudes of all the vertices can be read off from this, and the probability follows from it.

$$\Psi = \Psi_L + \Psi_R.$$

$$\Psi_L = \frac{1+\gamma^5}{2}\Psi, \qquad \frac{1-\gamma^5}{2}\Psi = \Psi_R$$

Original source details: GELFAND-P000898

Uncertain reading. The spinor symbol is rendered as Ψ. Its handwritten case is indistinct.

$\Psi \in$ tensor [inserted above: prod.] repr. of 2 spin reprs. $\times$ stand. repr. of $SU(2)$

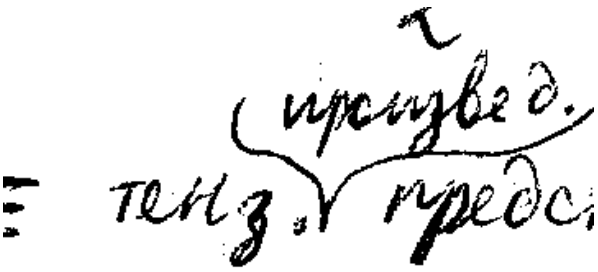
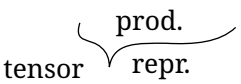


Diagram and typed description. Insertion of “prod.” between “tensor” and “repr.”



Original source details: GELFAND-P000898

In (37) there are currents:
1st parentheses J^-, 2nd — J^+
(charged)
3rd J^{el} — electromagn., 4th — J^0
Consider the left lepton ℓ^α_L
the right one $\bar{e}_R$
[At right:] λ_α is the vacuum expectation value of the Higgs field

$$f\,(\bar{e}_R \ell^\alpha_L \lambda_\alpha) + \text{Herm. conj.}$$

GELFAND-P000899

[source number: 68]

For a Spinor of the form $\begin{pmatrix} 0 \\ \lambda \end{pmatrix}$ we get

$$f(\bar{e}_R e_L)\lambda + \text{Herm. conj.}$$

φ — There were 4 degrees of freedom.
3 turned into Goldstones and were eaten by the vector [crossed out] mesons

Uncertain reading. The ending of «голдстоны» (Goldstones) is overwritten. A descender is crossed out below it.

Original source details: GELFAND-P000899

Uncertain reading. A word resembling «бозоны» (bosons) is crossed out before “mesons.”

Original source details: GELFAND-P000899

To read:

a) Faddeev, Slavnov

b) Taylor (more physical)

c) Weinberg's talk at the conf. on elementary particle physics (1974, London).

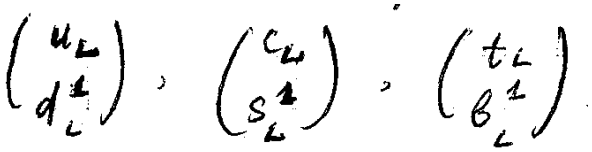
Bjorken, Drell — old.

— · —

$$\begin{pmatrix}\nu\\ e\end{pmatrix}, \qquad \begin{pmatrix}\nu_\mu\\ \mu\end{pmatrix}, \qquad \begin{pmatrix}\nu_\tau\\ \tau\end{pmatrix}$$

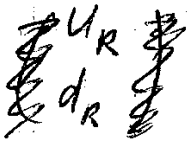
How do we write the Lagr. for quarks?

$$\begin{pmatrix}u_L\\ d_L^1\end{pmatrix}, \qquad \begin{pmatrix}c_L\\ s_L^1\end{pmatrix}, \qquad \begin{pmatrix}t_L\\ b_L^1\end{pmatrix}; \qquad \begin{matrix}u_R\\ d_R\end{matrix}$$



Uncertain reading. The three doublets have superscript 1 on d, s, b. The mixing formula at right has primes.

Original source details: GELFAND-P000899

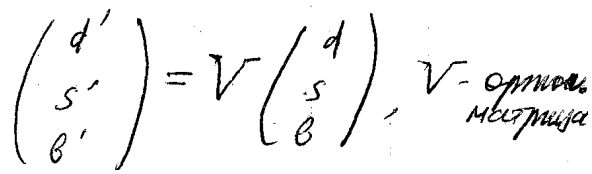


Uncertain reading. Both parentheses around u_R, d_R are crossed out.

Original source details: GELFAND-P000899

[At right:]

$$\begin{pmatrix}d'\\ s'\\ b'\end{pmatrix} = V \begin{pmatrix}d\\ s\\ b\end{pmatrix}, \qquad V \text{ is an orthog. matrix}$$



Uncertain reading. The label reads "orthog. matrix."

Original source details: GELFAND-P000899

the left ones will be doublets

the right ones are singlets

Next time: **21** January.

Uncertain reading. The digit 1 in 21 is heavily traced over.

Original source details: GELFAND-P000899

Chapter 17

GELFAND-P000900

[source number: 3]

21.1.80. Gelfand Sem.

A. Migdal. On a new approach to gauge theories.

This is a new language, more adequate for describing strong interactions.

Old language: $SU(N)$

fields

$$\underset{\text{(connection)}}{A_\mu(x)} = \sum_{a=1}^{N^2-1} T_a A_\mu^a, \qquad T_a \text{ are the generators of } SU(N). \tag{1}$$

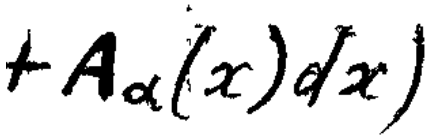
The language would be adequate if the fields were close to 0 or to another classical solution (if they did not fluctuate so strongly).

We need to consider.

$$\Psi(c) = \operatorname{Tr} p \exp\left(\oint A\,dx\right), \tag{2}$$

where c is a loop,

$$p \exp\left(\oint A\,dx\right) = \prod (1 + A_\alpha(x)\,dx) \tag{3}$$



Uncertain reading. Equation (3) reads $A_\alpha(x)\,dx$, with no visible index on dx.

Original source details: GELFAND-P000900

Without the trace in (2), a [crossed out] covariant quantity.

Uncertain reading. The crossed-out entry before «ковар.» (covar.) may begin «ков.».

Original source details: GELFAND-P000900

It turns out that we can restrict ourselves to what results when we take the trace.
Thus we need loop space.
Noninvariant quantities come out equal to 0 in quantum theory.

— • —

I.M.G. The point is that a repr. of the group of paths is determined by its character.

— · —

Here we are speaking of the lowest character.
Parallel transport:

$$\frac{dF}{dx} = AF \quad \Rightarrow \quad F = Ce^{\int A\,dx}$$

for matrices the integral is ordered.
Group of paths.

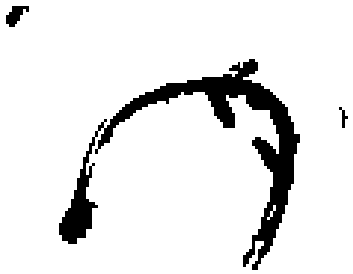


Diagram and typed description. An arc with two arrowheads. The strokes on the right branch are close together.

Original source details: GELFAND-P000900

we cancel the paths going out and back.

— · —

Let A_μ satisfy the classical equations of motion:

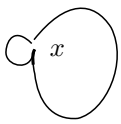
$$\nabla_\mu F_{\mu\nu} = 0, \tag{4}$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \tag{5}$$

(curvature of the connection)
Let us write the condition satisfied in this case by the function $\Psi(c)$.

Diagram and typed description. A small loop at x, with a thickened stroke at the join.

(6) 

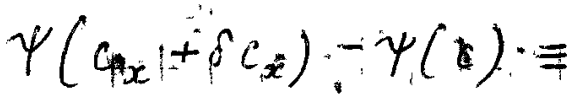
Original source details: GELFAND-P000900

$c + \delta\sigma_{\mu\nu}$, $\delta\sigma_{\mu\nu}$ is the area (we have attached a small loop to the contour c)

$$\frac{\delta\Psi}{\delta\sigma_{\mu\nu}} = \mathrm{Tr}\left(\hat{p}F_{\mu\nu}\exp\left(\oint A\,dx\right)\right) \tag{7}$$

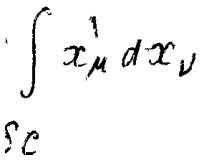
Let us prove (7). We have

$$\Psi(c_x + \delta c_x) - \Psi(c) \equiv \frac{\delta\Psi}{\delta\sigma_{\mu\nu}}\int_{\delta c} x'_\mu\,dx_\nu \tag{8}$$



Original source details: GELFAND-P000900

Uncertain reading. The first c_x is heavily overwritten. The argument of the second Ψ resembles a corrected c or δ.



Original source details: GELFAND-P000900

Uncertain reading. The detached stroke above x_μ is rendered as a prime on x.

Stokes-type functional. if the variation is proportional to the area (this is not always so, e.g. for length)

GELFAND-P000901

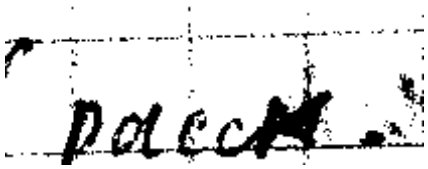
Original source details: GELFAND-P000901

Uncertain reading. Unclear mark in the upper-left margin, resembling a question mark.

In the Abelian case this is Stokes's theorem.

$$\oint_{\delta C} A_\alpha\,dx_\alpha = \int_{\delta S} F_{\alpha\beta}\,d\sigma_{\alpha\beta}. \tag{9}$$

(discusses work with Makeenko and uses work by the authors Mandelstam, Nambu, Polyakov)



Original source details: GELFAND-P000901

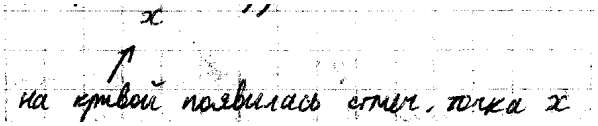
Uncertain reading. The first abbreviation may mean 'discusses' or 'considers'.

$\hat{p}$ in (7) acts on both $F_{\mu\nu}$ and exp.

$F_{\mu\nu}(x)$ is placed where the point x is

Or:

$$\frac{\delta\Psi}{\delta\sigma_{\mu\nu}} = \mathrm{Tr}\left(F_{\mu\nu}(x)\, p \exp\left(\int_x^x A\, dx\right)\right) \tag{10}$$



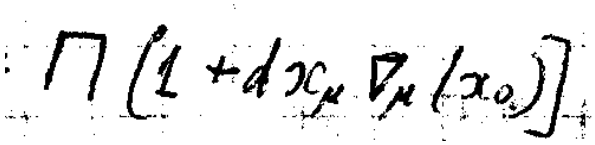
Original source details: GELFAND-P000901

Diagram and typed description. Annotation with an arrow beneath the integral in (10). The abbreviation before 'point' is read tentatively as 'marked'.

↗ a marked point x has appeared on the curve

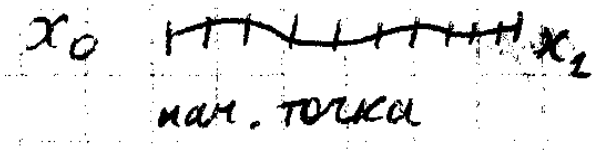
Proof of (7):

$$\begin{aligned}\prod(1 + A(x)\, dx) &= \prod\left(1 + dx_\mu\left(A_\mu(x_0) + \frac{\partial}{\partial x_\mu}\right)\right) \\ &= \prod(1 + dx_\mu \nabla_\mu(x_0)]\end{aligned} \tag{11}$$



Original source details: GELFAND-P000901

Uncertain reading. The last factor in (11) opens with a parenthesis and ends with a square bracket.

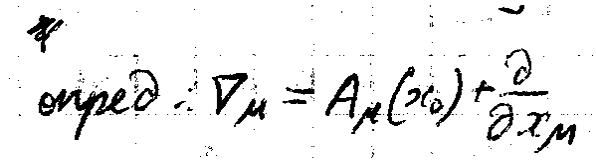


Original source details: GELFAND-P000901

Diagram and typed description. Subdivision of the curve beneath (11).

x_0 ⊢⊣⊣⊣⊣⊣⊣⊣⊣ x_1

initial point

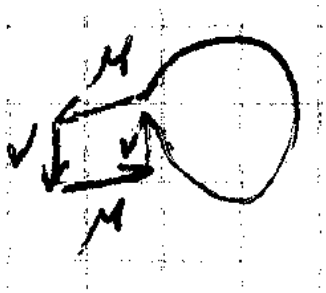


Original source details: GELFAND-P000901

Diagram and typed description. Definition of ∇_μ with an arrow toward (11).

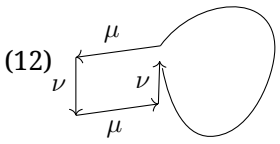
↗ def. $\nabla_\mu = A_\mu(x_0) + \frac{\partial}{\partial x_\mu}$

$dx_\mu \frac{\partial}{\partial x_\mu}$ shifts the argument



Original source details: GELFAND-P000901

Diagram and typed description. Loop with the rectangular circuit (12).



— consider the following variation

$$(1 + dx_\mu \nabla_\mu)(1 + dx_\nu \nabla_\nu)(1 - dx_\mu \nabla_\mu)(1 - dx_\nu \nabla_\nu) = \\ = 1 + dx_\mu \wedge dx_\nu [\nabla_\mu, \nabla_\nu] \tag{13}$$

(here all ∇_μ refer to the initial point)

Take two segments:

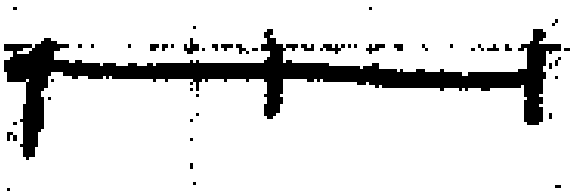
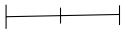
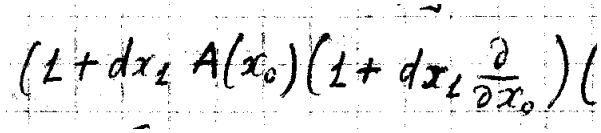


Diagram and typed description. Two consecutive segments.



Original source details: GELFAND-P000901

$$(1 + A(x_0)\, dx_1)(1 + A(x_0 + dx_1)\, dx_2) = \\ = \left[1 + dx_1 \left(A(x_0) + \frac{\partial}{\partial x_0}\right)\right](1 + A(x_0)\, dx_2) = \\ = \left(1 + dx_1 A(x_0)\left(1 + dx_1 \frac{\partial}{\partial x_0}\right)(\dots)\right. \tag{14}$$



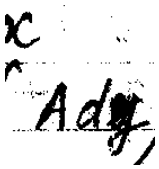
Uncertain reading. No separate parenthesis closing the first factor is visible after $A(x_0)$ in the last line of (14).

Original source details: GELFAND-P000901

(∼ transport around a closed contour is curvature)

Differentiate (10) with respect to x (without thinking, for now):

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\Big(\{\partial_\mu F_{\mu\nu}(x) + [A_\mu(x), F_{\mu\nu}]\} \\ p \exp \int_x^x A\, dy\Big) \tag{15}$$



Uncertain reading. The differential after A in (15) is overwritten. dy is the preferred reading, with dx also possible.

Original source details: GELFAND-P000901

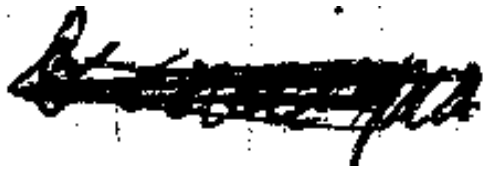
commutator due to cyclic invariance of the trace

GELFAND-P000902

[source number: 4]

Geometrically:

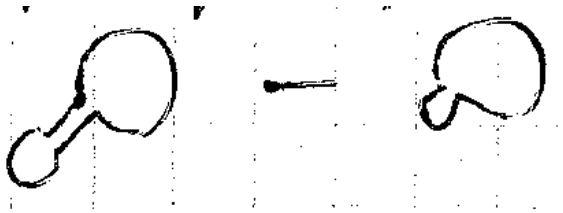
we shift the attached loop



Original source details: GELFAND-P000902

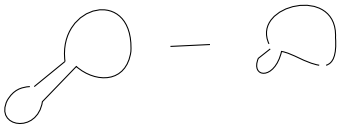
Uncertain reading. The word before ‘by an amount’ is crossed out. Individual strokes are visible and the reading is unresolved.

by an amount dx.



Original source details: GELFAND-P000902

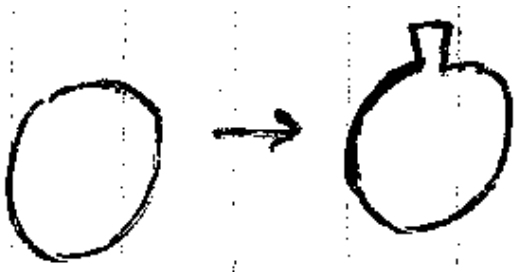
Diagram and typed description. Two positions of the attached loop.



I.M.G. A repr. of the group of paths is called a connection.

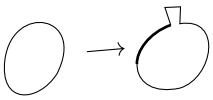
Lie alg.?

Curvature — something like a Lie algebra.

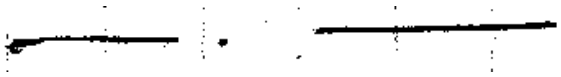


Original source details: GELFAND-P000902

Diagram and typed description. An arrow from a loop to a loop with a protrusion.



— like a shift by an element of the Lie algebra.

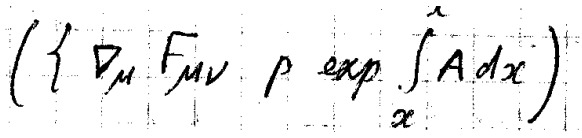


Original source details: GELFAND-P000902

Diagram and typed description. Manuscript divider.

A.M. The braces $\{\ldots\}$ (in (15)) vanish by the classical eqs. of motion.

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\left(\{ \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dx \right) \tag{15'}$$



Original source details: GELFAND-P000902

Uncertain reading. In (15′) a brace opens before the derivative. The expression ends with a parenthesis.

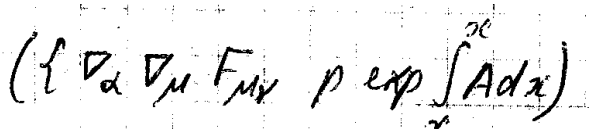
In fact

Original source details: GELFAND-P000902

Uncertain reading. The mark before ‘one can’ is crossed out. Part of its upper arc is visible and the reading is unresolved.

one can further add:

$$\partial_\alpha \partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \mathrm{Tr}\left(\{ \nabla_\alpha \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dx \right) \tag{16}$$



Original source details: GELFAND-P000902

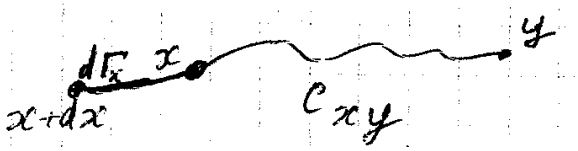
Uncertain reading. In (16) a brace opens before the derivatives. The expression ends with a parenthesis.

The left-hand side is noncommutative!! (since it is a functional of the point and also of the path)

$$\frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = \hat{\varphi}_{\mu\nu}(C_x) \tag{17}$$

— a function on contours passing through the point x

The left-hand side of (15) needs a definition:

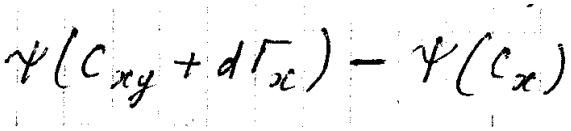


Original source details: GELFAND-P000902

Diagram and typed description. The contour C_{xy} and the short segment $d\Gamma_x$.

$d\Gamma_x$ — the contour going from x to $x + dx$.

$$\Psi(C_{xy} + d\Gamma_x) - \Psi(C_x) = dx_\mu \frac{\partial \Psi}{\partial x_\mu} \tag{18}$$



Original source details: GELFAND-P000902

Uncertain reading. The first argument contains C_{xy}, the second C_x.

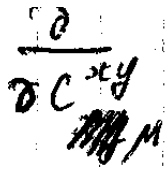
— this defines the right-hand side (and hence $\frac{\partial \Psi}{\partial x_\mu}$).

The contour passes through the point x twice: we shift both the beginning and the end

For C_{xx}, the beginning and the end are shifted in turn.

Translate the contour:

$$\frac{\partial}{\partial C_\mu^{xy}} = \frac{\partial}{\partial x_\mu} + \frac{\partial}{\partial y_\mu} + \int_x^y dz_\alpha \frac{\delta}{\delta \sigma_{\alpha\mu}(z)} \tag{19}$$



Uncertain reading. An index is crossed out before μ in the left denominator of (19).

Original source details: GELFAND-P000902

(translation of the contour)

For transl. invariant functionals, we get:

$$\partial_\mu = \oint dz_\nu \frac{\delta}{\delta \sigma_{\mu\nu}(z)}. \tag{20}$$

(for closed contours)

GELFAND-P000903

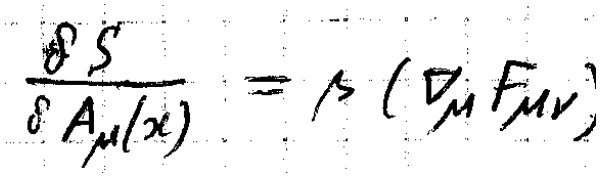
In the classical case we obtain

$$\mathcal{L} = \beta \operatorname{Tr} F_{\mu\nu}^2 \tag{21}$$

$$S = \beta \int \operatorname{Tr} F_{\mu\nu}^2 \, d^4x \tag{22}$$

As is known, we have:

$$\frac{\delta S}{\delta A_\mu(x)} = \beta(\nabla_\mu F_{\mu\nu}) \tag{23}$$



Uncertain reading. The denominator at left reads $A_\mu(x)$ and the right-hand side has $\nabla_\mu F_{\mu\nu}$.

Original source details: GELFAND-P000903

Therefore (15) $\Rightarrow$

$$\partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}(x)} = 0 \tag{24}$$

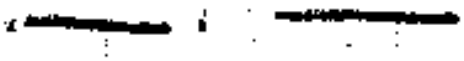


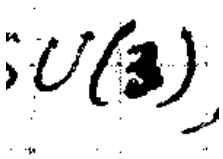
Diagram and typed description. Manuscript divider.

——— · ———

Original source details: GELFAND-P000903

The $SU(\infty)$ theory is close to an exact solution

(expansion in powers of $\frac{1}{n}$; in reality $SU(3)$)

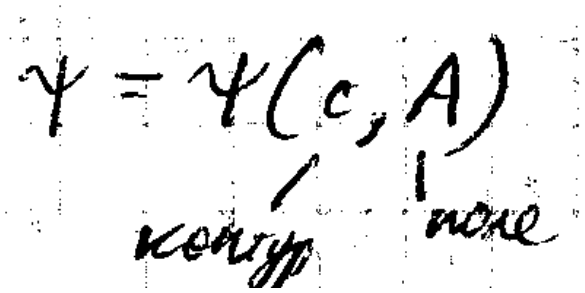


Original source details: GELFAND-P000903

Uncertain reading. The digit in $SU(3)$ is heavily traced over. It reads as 3.

Let us derive the quantum eq. of motion.

QFT Consider



Original source details: GELFAND-P000903

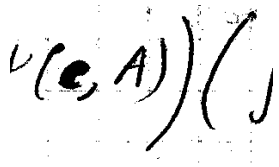
Diagram and typed description. Labels on the two arguments.

$\Psi = \Psi($ c, $A)$

contour field

and now consider

$$\Psi(c) = \left(\int \mathcal{D}A_\mu \, \exp(-S)\, \Psi(c, A)\right)\left(\int \mathcal{D}A_\mu \, \exp(-S)\right) \tag{25}$$



Original source details: GELFAND-P000903

Uncertain reading. In (25) the two parenthesized groups are written next to each other. No separate division sign is visible between them. The c in $\Psi(c, A)$ is heavily traced over.

Quantum eq. of motion: a property of the integral

$$\int \mathcal{D}A_\mu(x) \frac{\delta}{\delta A_\mu(x)} \{e^{-S} Q[A]\} \equiv 0 \tag{26}$$

$Q[A]$ — an arbitrary functional.

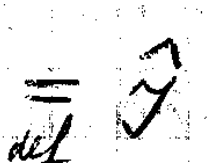
From (26) it follows that

$$\left\langle \frac{\delta S}{\delta A_\mu(x)} Q \right\rangle = \left\langle \frac{\delta Q}{\delta A_\mu(x)} \right\rangle \tag{27}$$

(Schwinger eq.)

From this, omitting Q and introducing $\overline{h}$, we obtain:

$$\frac{\delta S}{\delta A_\mu(x)} = \overline{h} \frac{\delta}{\delta A_\mu(x)} \underset{\text{def}}{=} \hat{\mathcal{J}} \tag{28}$$

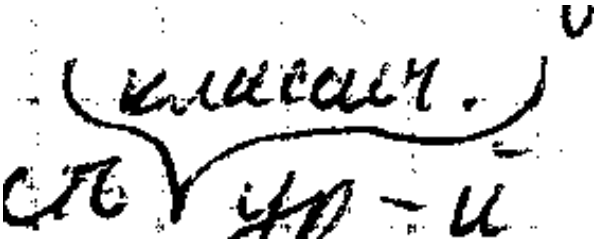


Original source details: GELFAND-P000903

Uncertain reading. The handwritten sign under the hat is read tentatively as $\mathcal{J}$.

Or:

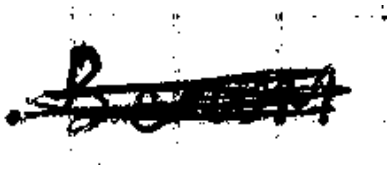
the left-hand side of the [inserted above: classical] eqs. of motion is replaced by variation with respect to the field.



Original source details: GELFAND-P000903

Diagram and typed description. The insertion ‘classical’ above ‘eqs.’

$$\underbrace{\text{classical}}$$



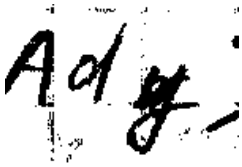
Original source details: GELFAND-P000903

Uncertain reading. The crossed-out word may begin with the Cyrillic letter «В». The remainder is unresolved.

The fields Ψ are not independent (much redundant information).

$(15') \Rightarrow$

$$\begin{aligned} \beta\, \partial_\mu \frac{\delta \Psi}{\delta \sigma_{\mu\nu}} &= \beta \left\langle \operatorname{Tr} \nabla_\mu F_{\mu\nu}\, p \exp \int_x^x A\, dy \right\rangle = \\ &= \left\langle \operatorname{Tr} \frac{\delta}{\delta A_\mu(x)}\, p \exp\left(\int_x^x A\, dy \right) \right\rangle, \qquad \beta = \overline{h}^{-1}. \end{aligned} \tag{29}$$



Original source details: GELFAND-P000903

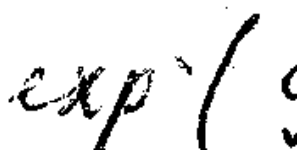
Uncertain reading. In the first integral in (29), the last letter in dy is overwritten. The second integral clearly reads dy.

GELFAND-P000904

[source number: 5]

Here

$$\Psi(c, A) = \operatorname{Tr} p \exp\left(\oint_c A\, dx\right) \tag{30}$$



Original source details: GELFAND-P000904

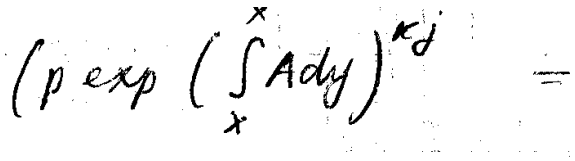
Uncertain reading. A separate short stroke of uncertain significance is visible between exp and the opening parenthesis.

[Crossed out: $\oint_c dy$.]

Original source details: GELFAND-P000904

Uncertain reading. The crossed-out entry reads as a contour integral with a lower label and a differential.

$$\frac{\delta}{\delta A_\mu^{jk}(x)}\left(p \exp\left(\int_x^x A\, dy\right)^{kj} = \right.$$
$$= \oint_c dy_\nu\, U^{ii}(c_{xy}) U^{kk}(c_{yx}) \delta^4(x-y), \tag{31}$$

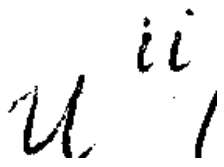


Original source details: GELFAND-P000904

Uncertain reading. In the first line of (31), the outer opening parenthesis is unpaired. An equals sign follows the superscripts.

Original source details: GELFAND-P000904

Uncertain reading. A short entry is crossed out above the integral sign. Its reading is unresolved.



Original source details: GELFAND-P000904

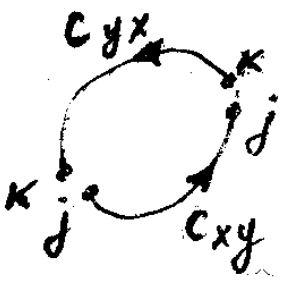
Uncertain reading. The first matrix U on the right of (31) has two superscript i's, each with a separate dot.

(32) where U is the transport matrix

Original source details: GELFAND-P000904

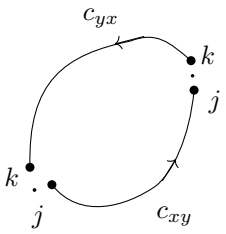
Uncertain reading. A short crossed-out mark appears between 'transport' and the formula for U.

$$U(C_{xy}) = P \exp \int_x^y A\, dx \qquad \delta^4(x-y) = \frac{\delta A(y)}{\delta A(x)}$$



Original source details: GELFAND-P000904

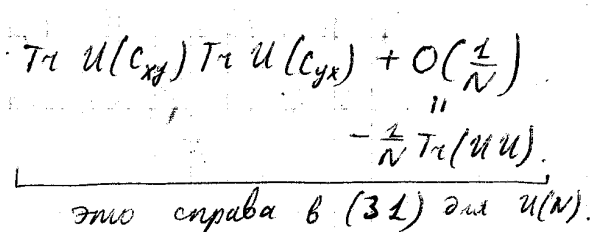
Diagram and typed description. Arcs c_{yx} and c_{xy} with their directions and the labels j, k.



For an arbitrary (semisimple) group:

$$\sum_a \operatorname{Tr}\big(T^a U(c_{xy}) T^a U(c_{yx})\big) = \operatorname{Tr} U(c_{xy}) \operatorname{Tr} U(c_{yx}) + \underbrace{O\left(\frac{1}{N}\right)}_{\| \; -\frac{1}{N}\operatorname{Tr}(UU).} \tag{33}$$

this is on the right in (31) for $U(N)$.



Uncertain reading. A straight bracket underlines the right-hand side of (33). Two vertical strokes appear above the term $-\frac{1}{N}\operatorname{Tr}(UU)$.

Original source details: GELFAND-P000904

It follows that

$$\beta\, \partial_\mu \frac{\delta\langle\Psi\rangle}{\delta\sigma_{\mu\nu}} = \oint dy_\nu\, \delta^4(x-y) \langle \Psi(c_{xy}) \Psi(c_{yx}) \rangle \tag{34}$$

We do not obtain closed equations.

c_{xy}, c_{yx} are closed because of $\delta^4(x-y)$

In fact, this is the first eq. of a chain of multiloop eqs. (one loop is insufficient) for Green functions.

Uncertain reading. The crossed-out entry before ‘We need’ may begin with «Не» (‘not’).

Original source details: GELFAND-P000904

We need to consider self-intersections of the contour as well

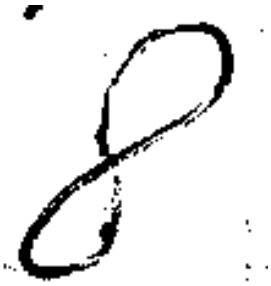
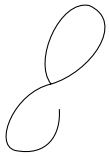


Diagram and typed description.
Self-intersecting contour.



Original source details: GELFAND-P000904

(they contain all the information about the interaction)

Trying to regularize the δ-function to give the equation meaning makes it meaningless (the contours open) — this must be done simultaneously with closing the contours.

$$\langle \Psi(c_1)\Psi(c_2)\dots\Psi(c_n)\rangle = \langle\Psi(c_1)\rangle\langle\Psi(c_2)\rangle\dots\langle\Psi(c_n)\rangle + O(N^{-2}) \tag{35}$$

Uncertain reading. The final argument on the right of (35) is overwritten. c_n is the preferred reading.

Original source details: GELFAND-P000904

$\Psi = \mathrm{Tr}\, U(c); \quad \Psi(0) = N.$

We must redefine: $\Psi = N\bar{\Phi}$.

For $\bar{\Phi}$ we obtain a certain eq., from which we get confinement

Uncertain reading. The crossed-out word below the last line is illegible. Two dots are visible to its right.

Original source details: GELFAND-P000904

Uncertain reading. The separate heavily crossed-out entry at the bottom is illegible.

Original source details: GELFAND-P000904

GELFAND-P000905

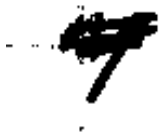
28.1.80. Gelfand Sem.

A. Zelevinsky. On the work of Kazhdan–Lusztig, reprs. of Coxeter groups, etc.

D. Kazhdan, G. Lusztig, Inv. Math, 53, $N = 2$, 1979

W — Coxeter group, S — simple reflections

W is generated by the generators S (a finite set).



Uncertain reading. A mark is crossed out between ‘finite’ and ‘set’. Its reading is unresolved.

Original source details: GELFAND-P000905

$$W = \langle S \mid s^2 = e,\ (ss')^{m(s,s')} = e \rangle \tag{1}$$

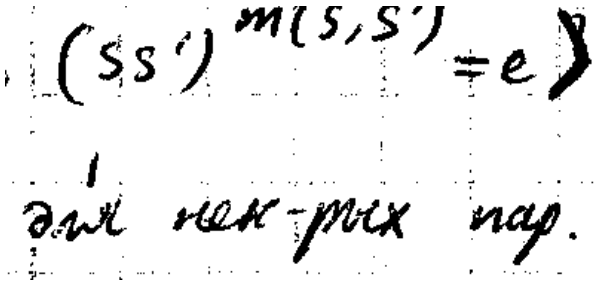
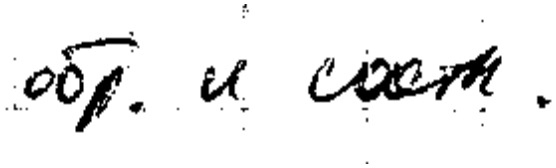


Diagram and typed description. Annotation beneath the second relation in (1).

/
для нек-рых пар.

Original source details: GELFAND-P000905

A Coxeter group is a group with such gens. and rels.



Uncertain reading. The last abbreviation reads «соотн.» (relations).

Original source details: GELFAND-P000905

Examples:

1) S_n: $\quad S = \{(i, i+1)\}$

$$((i,i+1)(i+1,i+2))^3 = e.$$

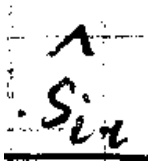
all the rest follow from these.

2) Weyl group W of a system of simple roots

$$w = s_1 \dots s_\ell \qquad \text{reduced, if the length is minimal} \qquad (2)$$
$$\ell = \ell(w)$$

If we discard any subset of the factors in a reduced expr. and multiply the rest, we obtain (def) the set of elements $\leq w$

$$w' = s_1 \dots \hat{s}_{i_1} \dots \hat{s}_{i_n} \dots s_\ell < w \qquad (3)$$



Uncertain reading. The last index on a deleted factor in (3) resembles i_n. The second letter is indistinct.

Original source details: GELFAND-P000905

This is independent of the reduced expression.

$$\mathcal{A} = \mathbb{Z}[q^{1/2}, q^{-1/2}]$$

Hecke algebra:

$\mathcal{H} \;\|\; \mathcal{H}_W$ — algebra over $\mathcal{A}$ with generators T_s, $s \in S$ and relations:

$$(T_s + 1)(T_s - q) = 0, \quad \forall s \qquad (4)_1$$

$$\underbrace{T_s T_{s'} T_s T_{s'} \dots}_{m(s,s')} = \underbrace{T_{s'} T_s T_{s'} T_s \dots}_{m(s,s')} \qquad (4)_2$$

Uncertain reading. A faint sloping stroke to the right of $(4)_2$ resembles part of a closing bracket.

Original source details: GELFAND-P000905

$q = 1 \Rightarrow$ we obtain the group algebra of the Coxeter group, since rels. (1) are written in the form

$$\underbrace{ss's's' \dots}_{m(s,s')} = \underbrace{s's \dots}_{m(s,s')} \qquad (5)$$

Uncertain reading. The crossed-out mark after the right-hand product in (5) is illegible.

Original source details: GELFAND-P000905

q — number of elements of a finite field; G — Chevalley grp. over q,

B — Borel subgroup

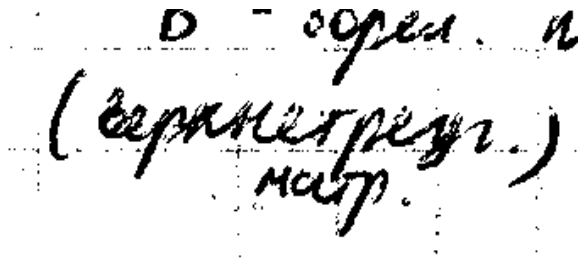


Diagram and typed description. Two-line annotation below ‘Borel subgroup’.

(upper triangular)
matrices

Original source details: GELFAND-P000905

$$\Rightarrow \quad \mathcal{H}(G,B) \cong \mathcal{H} \otimes C.$$

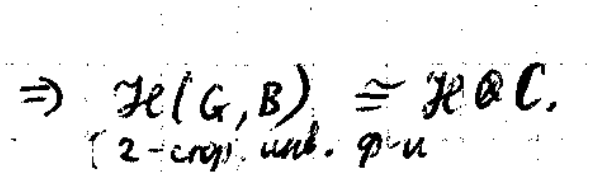


Diagram and typed description. Annotation under $\mathcal{H}(G,B)$.

(bi-inv. functions

Original source details: GELFAND-P000905

GELFAND-P000906

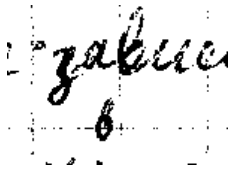
[source number: 8]

A basis is constructed in the alg. $\mathcal{H}$ (over $\mathcal{A}$)

Natural basis:

$$w = s_1 \dots s_\ell \rightsquigarrow \quad T_w = T_{s_1} \dots T_{s_\ell} \tag{6}$$

reduced (independent of the repr.)



Uncertain reading. A separate mark resembling Cyrillic «в» is visible under ‘independent’.

Original source details: GELFAND-P000906

$\{T_w\}$ — a basis in $\mathcal{H}$ (well known)

Uncertain reading. The crossed-out entry above ‘define’ may begin with «По».

Original source details: GELFAND-P000906

$$\text{Define an involution } \bar{\ } \text{ in } \mathcal{H}: \quad \overline{q^{1/2}} = q^{-1/2}, \quad \bar{T}_w = T_{w^{-1}}^{-1} \quad \left(\underset{\text{def}}{=} (T_{w^{-1}})^{-1}\right). \tag{7}$$

Invertibility follows from eq. $(4)_1$:

$$T_s^2 + T_s(1-q) = q, \quad \text{whence}$$

$$T_s^{-1} = q^{-1}(T_s + 1 - q) \tag{8}$$

Set

$$\varepsilon_w = (-1)^{\ell(w)}, \qquad q_w = q^{\ell(w)} \tag{9}$$

Theorem 1.1. $\forall w \in W,\ \exists!\ C'_w \in \mathcal{H}$:

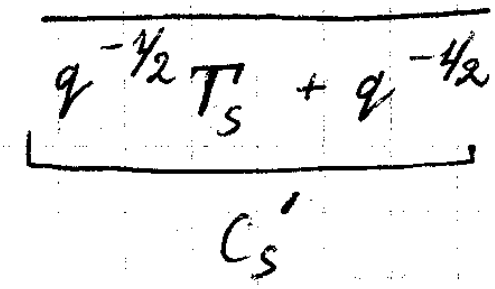
$$(1) \quad \overline{C'_w} = C'_w \tag{10}$$

$$(2) \quad C'_w = q_w^{-1/2} \sum_{y \le w} P_{y,w} \cdot T_y, \tag{11}$$

where

$$P_{w,w} = 1, \qquad \underset{(y<w)}{P_{y,w}} \text{ — polynomial in } q \text{ with coeffs. in } \mathbb{Z} \text{ of degree } \le \frac{1}{2}(\ell(w) - \ell(y) - 1) \tag{12}$$

$$\bar{T}_s = q^{-1}T_s + q^{-1} - 1$$
$$\overline{q^{-1/2}T_s} = q^{-1/2}T_s + q^{-1/2} - q^{1/2}$$
$$\underbrace{\overline{q^{-1/2}T_s + q^{-1/2}}}_{C'_s} = q^{-1/2}T_s + q^{-1/2}$$



Original source details: GELFAND-P000906

Uncertain reading. In the third line, a straight bracket beneath the expression on the left is labelled C'_s.

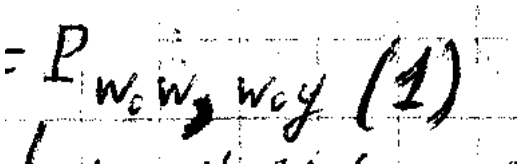
Or: $P_{s,s} = 1, \quad P_{e,s} = 1.$

Original source details: GELFAND-P000906

Uncertain reading. The first index on the second polynomial is heavily traced over and is read as e.

Conjecture. Let g be a complex semisimple Lie algebra, W the Weyl group;

$$M_w \text{ — Verma module;} \qquad [M_w : L_y] = P_{w_0 w, w_0 y}(1) \quad \text{— conjecture} \tag{13}$$



Uncertain reading. The indices read $w_0 w, w_0 y$. Both small zero glyphs have an open contour.

Original source details: GELFAND-P000906

L_w — the unique irred. quotient, $\quad L_y,\ y \le w$

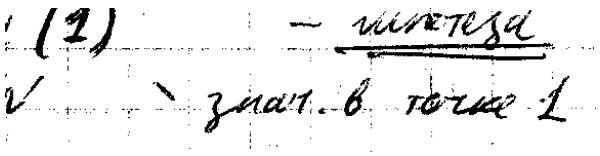


Diagram and typed description. Annotation beside (1) in (13).

↘ value at 1

Original source details: GELFAND-P000906

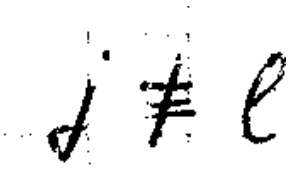
Checked for GL(4).

Vogan gave several equiv. reformulations.

$$P_{y,w} = \sum_{i \ge 0} q^i \dim \mathrm{Ext}^{\ell(w)-\ell(y)-2i}(M_y, L_w) \tag{14}$$

or

$$\mathrm{Ext}^j(M_y, L_w) = 0, \quad \text{if } j \not\equiv \ell(w) - \ell(y) \pmod 2 \tag{15}$$



Uncertain reading. In (15) a diagonal stroke crosses three horizontal bars.

Original source details: GELFAND-P000906

Set:

$$C_w = \varepsilon_w j(C'_w), \quad \text{where } j(q^{1/2}) = q^{-1/2}, \quad j(T_w) = \varepsilon_w q_w^{-1} T_w \tag{16}$$

(j — auxiliary involution)

$$y \prec w \overset{\text{def}}{\iff} y < w, \quad \varepsilon_y = -\varepsilon_w, \quad \deg P_{y,w} = \frac{1}{2}\big(\ell(w) - \ell(y) - 1\big). \tag{17}$$

Then denote by

$$\mu(y,w) \quad \text{the leading coefficient of } P_{y,w}.$$

Theorem 1.3. $s \in S,\ v \in W \Rightarrow$

$$T_s C_v = \begin{cases} -C_v, & sv < v, \\ qC_v + q^{1/2} C_{sv} + q^{1/2} \sum\limits_{\substack{z<v \\ sz<z}} \mu(z,v) C_z \end{cases} \tag{18}$$

if $sv > v$

(The action of the generators on the basis is written explicitly)

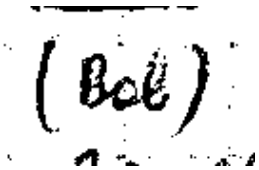
From this one can decompose the reg. repr. (block triangular form).

B. Mc Pherson (?) Макферсон.

Original source details: GELFAND-P000907

Uncertain reading. A sign, possibly an initial, is crossed out before «Макферсон».

(Bob)



Original source details: GELFAND-P000907

Uncertain reading. The annotation below the name is read tentatively as ‘Bob’.

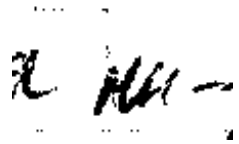
Geom. interp. of $P_{s,t}(q)$

— q the Poincaré polyn. of strange cohomology of the Schubert cells.

The groups $IH_i(X)$ — intersection-homologie groups. (for sets with singularities)

Original source details: GELFAND-P000907

Uncertain reading. There is a crossed-out entry before ‘homologie’. Its reading is unresolved.



Original source details: GELFAND-P000907

Uncertain reading. A short entry is crossed out between «с» (with) and «особ.» (singularities).

(defined by the speaker)

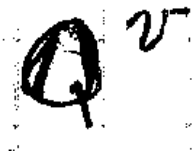
their properties:

$$1) \quad X \text{ nonsing.} \quad \Rightarrow \quad IH_i(X) = H_i(X) \tag{1}$$

Uncertain reading. The abbreviation after «не» reads «синг.» (sing.).

Original source details: GELFAND-P000907

$$2) \quad IH_i(X) \times IH_{n-i}(X) \longrightarrow \mathbb{Q}^v \quad \text{when } \dim X = n. \tag{2}$$



Uncertain reading. Above the heavily traced letter $\mathbb{Q}$ is a mark resembling v.

Original source details: GELFAND-P000907

Simplest topological def.

$$X = X_n \supset X_{n-2} \supset X_{n-4} \supset \cdots \supset X_0 \tag{3}$$

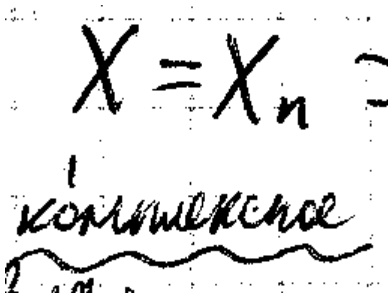


Diagram and typed description. Annotation below X in (3).

complex

Original source details: GELFAND-P000907

stratifications

Conditions:

a) $X_i - X_{i-2}$ — a nonsingular set of dim. i.

b) along X_{i-2} [inserted above: neighborhood] topologically triv. (locally) at points of $X_{i-2} - X_{i-4}$.

Diagram and typed description. Insertion above the line in condition b).

neighborhood

Original source details: GELFAND-P000907

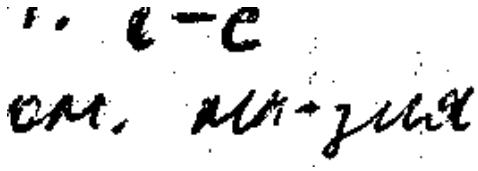
X — Schubert cell (closed), dim. $n \Rightarrow$ the smaller cells can be used for stratification.
Then by def.

Original source details: GELFAND-P000907

Uncertain reading. A sign resembling I is crossed out above $IH_i(X)$.

$$IH_i(X) \underset{\text{def}}{=} \left\{ \begin{array}{c} i\text{-cycles meeting each } X_{n-2c} \\ \text{in a set of dimension } i-c-1 \\ \hline \partial \text{ of } i+1 \text{ chains meeting each } X_{n-2c} \\ \text{in a set of dim. } i-c \end{array} \right. \tag{4}$$

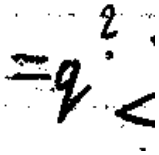
Theorem (using a special cohom. theory of sets of char. p)



Original source details: GELFAND-P000907

Uncertain reading. The abbreviation before ‘char.’ reads «мн-зий» (sets).

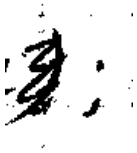
$$P_{s,t}(q) = q^{?} \sum_i \dim IH_i(X_t, X_t - y_s) q^i. \tag{5}$$



Original source details: GELFAND-P000907

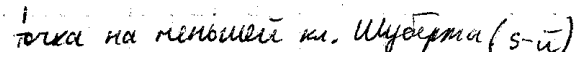
Uncertain reading. A question mark is written above the first q in (5).

s, t — elements of the Weyl group;



Original source details: GELFAND-P000907

Uncertain reading. A short mark is crossed out after ‘Weyl group’ and before the punctuation.



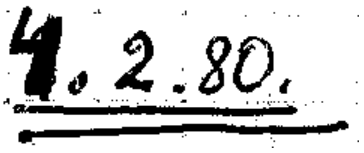
Original source details: GELFAND-P000907

Diagram and typed description. Annotation below y_s in (5).

|
point on the smaller Schubert cell (the sth)

GELFAND-P000908

4.2.80. Gelfand Sem.



Original source details: GELFAND-P000908

Uncertain reading. The first digit of the date is heavily traced over and reads as 4.

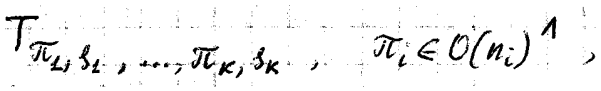
G. Olshanski.

$$G = \mathrm{GL}(\infty, \mathbb{R}) = \bigcup_{n \geq 1} \mathrm{GL}(n, \mathbb{R}) \tag{1}$$

$$K = \mathrm{SO}(\infty) = \bigcup_{n \geq 1} \mathrm{SO}(n) \tag{2}$$

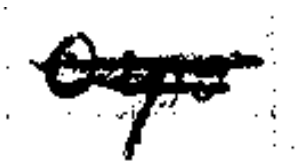
$\exists$ a family of reprs. of G for which K plays the role of a max. compact subgrp.

$$T_{\pi_1; s_1, \ldots, \pi_k, s_k}, \qquad \pi_i \in O(n_i)^\wedge, \quad s_i \in \mathbb{R}. \tag{3}$$



Original source details: GELFAND-P000908

Uncertain reading. Between π_1 and s_1 a comma and a separate dot above it are visible. The mark is rendered as a semicolon.



Original source details: GELFAND-P000908

Uncertain reading. The crossed-out abbreviation resembles «Огр» (restr.).

the restr. to K has discrete spectrum with finite multiplicities.

Analogous variant for $U(\infty)$.

$$\mathrm{SO}(p, \infty) = \overline{\bigcup_{n \geq 1} \mathrm{SO}(p, n)} \tag{4}$$

$$U(p, \infty) = \overline{\bigcup_{n \geq 1} U(p, n)} \tag{5}$$

$$\mathrm{Sp}(p, \infty) = \overline{\bigcup_{n \geq 1} \mathrm{Sp}(p, n)} \tag{6}$$

(completions)

(4)–(6) — type I groups, and a description of all irred. representations is given.

Original source details: GELFAND-P000908

Uncertain reading. Before ‘description’ the abbreviation reads «дан.» (given).

Mc Pherson.

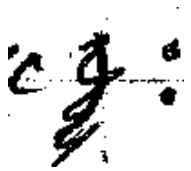
on the work of Gabrielov, Gelfand, and Losik on combin. Pontryagin classes,

which he understands better than the authors.

(on the first class)

X — a simplicial 4-manifold

(X — a simplicial complex g:



Original source details: GELFAND-P000908

Uncertain reading. After ‘complex’ there is a sign with a long descender, read tentatively as g.

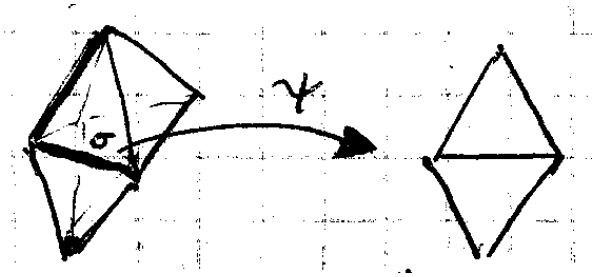
$$\forall \text{ simplex } \sigma \quad \exists \text{ map} \quad \psi : \underset{\text{star of } \sigma}{\operatorname{St}\sigma} \longrightarrow \mathbb{R}^4, \tag{1}$$

Original source details: GELFAND-P000908

Uncertain reading. A separate sign is crossed out before ψ.

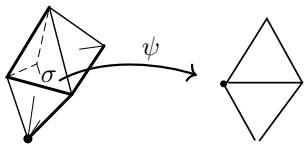
homeomorphic near the interior of σ, linear on each simplex.

The ideas are illustr. on a 2-manifold.



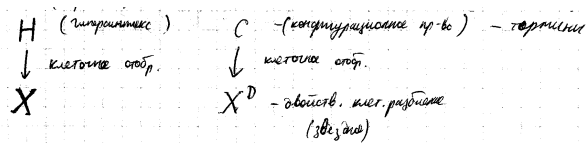
Original source details: GELFAND-P000908

Diagram and typed description. At left, a mesh with the front part σ marked and fine internal strokes. At right, two triangles with a common horizontal edge. The arrow is labelled ψ.



(def. of St)

H (hypersimplex) C — (configuration space) — terms.



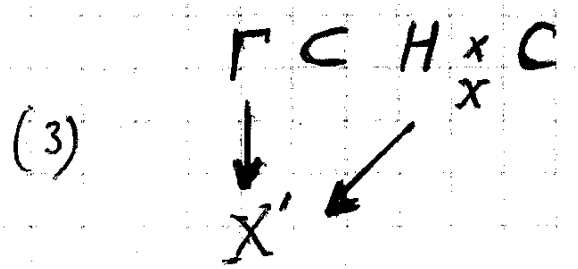
Original source details: GELFAND-P000908

Diagram and typed description. Cellular maps.

$$\begin{array}{ccc} & H & C \\ (2) & \downarrow \text{cellular map} & \downarrow \text{cellular map} \\ & X & X^D \end{array}$$

X^D — dual cell decomposition (star decomposition).

Cellularity: skeleton $\longrightarrow$ skeleton.



Original source details: GELFAND-P000908

Diagram and typed description. Inclusion into the fiber product and two maps to X'.

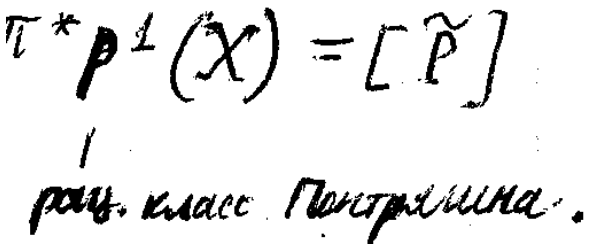
$$\begin{array}{cccc} (3) & \Gamma & \subset & H \underset{X}{\times} C \\ & \downarrow & \swarrow & \\ & X' & & \end{array}$$

GELFAND-P000909

[source number: 13]

$\widetilde{P}$ will be constr. — a cocycle in $C^4(\Gamma)$.

$$\pi^* p^1(X) = [\widetilde{P}]. \tag{4}$$



Original source details: GELFAND-P000909

Diagram and typed description. Annotation to $p^1(X)$ in (4).

rat. Pontryagin class.

Construction.

H

$\forall$ simplex σ in X, construct the polytope $H\sigma$ from maps

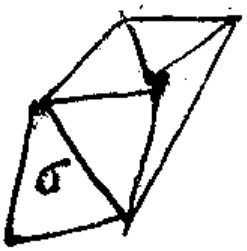
$$\{\text{vertices of the star of } \sigma\} \xrightarrow{h} [0,1]. \tag{5}$$

$$1) \quad h(v) = 1 \quad \text{when } v \in \sigma \tag{6}_1$$

$$2) \quad \sum_v h(v) = 5 = 4 + 1. \qquad (= \dim + 1) \tag{6}_2$$

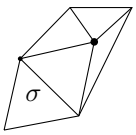
Examples.

a) simplex of highest dim. $\quad \operatorname{St}\sigma = \sigma$



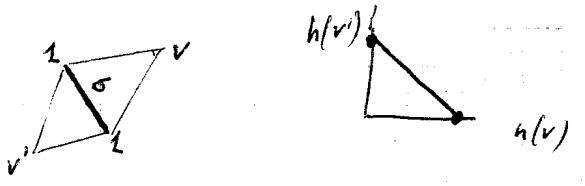
Original source details: GELFAND-P000909

Diagram and typed description. Star of a top-dimensional simplex, labelled σ.



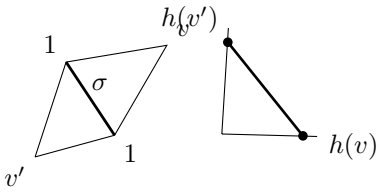
$H\sigma = 1$ point (the functions must be 1 everywhere).

b) simplex of codim. 1.



Original source details: GELFAND-P000909

Diagram and typed description. Two triangles with the common side σ, and a segment in the $h(v), h(v')$ plane.

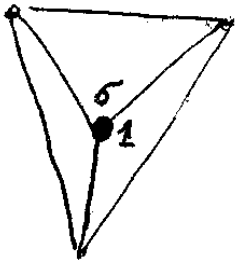


$H\sigma$ — a segment.

This also happens in the general case.

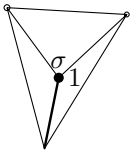
c) Codimension 2.

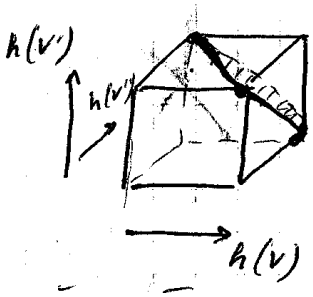
c′) (simple case)



Original source details: GELFAND-P000909

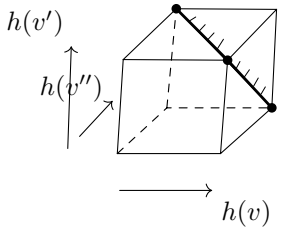
Diagram and typed description. Triangular mesh with the interior vertex marked σ and 1.





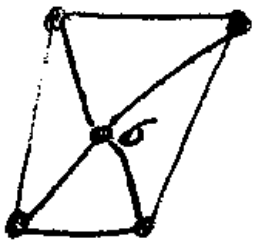
Original source details: GELFAND-P000909

Diagram and typed description. Cube section with three marked vertices and hatching. The arrows are labelled $h(v)$, $h(v')$, $h(v'')$.



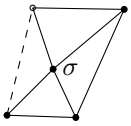
$H\sigma$ — a simplex. (2-dimensional simplex)

c″)



Original source details: GELFAND-P000909

Diagram and typed description. Quadrilateral mesh with σ at the interior vertex.



Cut the 4-dimensional cube by a plane equidist. from (some) vertices

Original source details: GELFAND-P000909

Uncertain reading. Below ‘cube’ a separate mark resembles 2.

Original source details: GELFAND-P000909

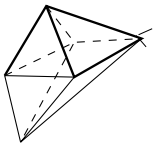
Uncertain reading. The crossed-out word before ‘equidist.’ is illegible. abc is written below it, beside a vertex of the drawing.

Original source details: GELFAND-P000909

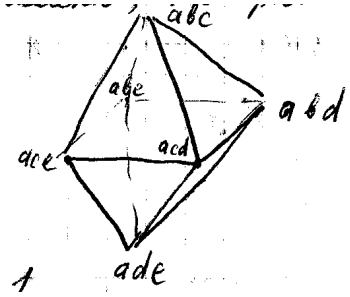
Uncertain reading. The word after the cancellation is abbreviated ‘equidist.’.

Original source details: GELFAND-P000909

Diagram and typed description. Octahedron with emphasized upper edges and broken interior strokes.

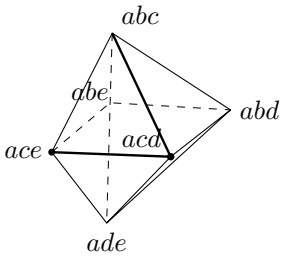


— an octahedron.



Original source details: GELFAND-P000909

Diagram and typed description. Octahedron with six labels, including abe by the broken rear connection.



Labels at the vertices of the polytopes.
The vertices are functions taking only the values 0 and 1.
The indices are those vertices where the value is 1.

$$H\sigma \times \sigma \qquad \sigma < \sigma' \quad \Rightarrow \quad H(\sigma') < H(\sigma) \quad \text{(the stars are nested this way)}$$

hence there is a common part

$$H\sigma \times \sigma \quad \text{and} \quad H\sigma' \times \sigma', \quad \text{namely}$$

$$\underline{H\sigma' \times \sigma}$$

Glue along this common part.
This gives the space H.
[Crossed out: Let us consider.]

Original source details: GELFAND-P000909

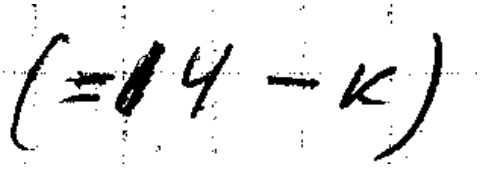
Uncertain reading. The crossed-out word is read tentatively as ‘Let us consider’.

GELFAND-P000910

Construction of C.
First we construct $C\sigma$.
By def. there is a map (at least one)

$$\operatorname{st}\sigma \xrightarrow{\psi} \mathbb{R}^4, \tag{7}$$

satisfying the conditions listed above.
$C\sigma$ — the set of all such ψ
Let σ have codimension k. $(= 4 - k)$



Original source details: GELFAND-P000910

Uncertain reading. In the parentheses after the equals sign and before $4 - k$, a short stroke resembling 1 is crossed out or traced over.

ψ is in general pos. if all sets

$$\left\{ \underbrace{v_1 \dots v_i}_{\text{all vertices of } \sigma} \qquad \underbrace{v'_1 \dots v'_k}_{k \text{ vertices from st}\,\sigma \setminus \sigma} \right\} \tag{8}$$

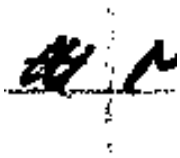
(5 points in all)

are in general position in $\mathbb{R}^4$.

ψ has speciality c if $\exists\, c$ sets for which this condition fails.

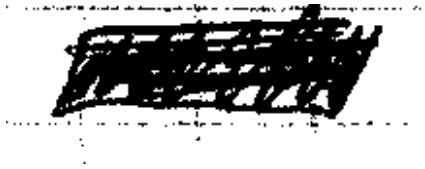
Original source details: GELFAND-P000910

Uncertain reading. The annotation beneath the second c is read tentatively as «набор» (collection). Its end is heavily traced over.



Original source details: GELFAND-P000910

Uncertain reading. An entry, possibly «и» (and), is crossed out after the second c.

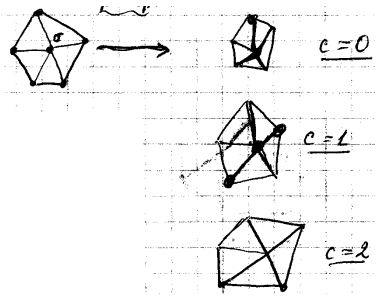


Original source details: GELFAND-P000910

Uncertain reading. The crossed-out word before ‘condition’ is illegible.

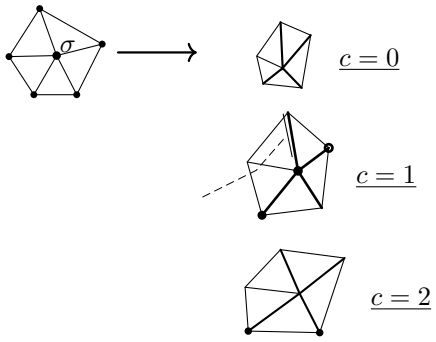
It is unknown: a stratification or a cell decomposition.

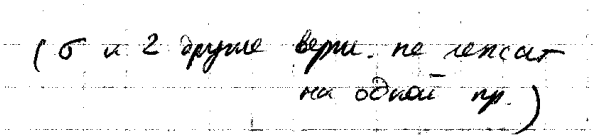
Example:



Original source details: GELFAND-P000910

Diagram and typed description. Five vertices around σ and three configurations. The faint sloping stroke at left in case $c = 1$ is drawn dashed.





Original source details: GELFAND-P000910

Diagram and typed description. Annotation beside $c = 0$.
(σ and 2 other vertices do not lie on one line)

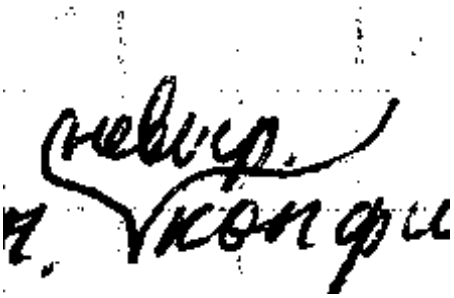
$C\sigma$ for points $(k = 4)$

$C\sigma$ — 1 point for each connected comp. of configurations in general position (discrete space of points)

$C\sigma$ for edges $(k = 3)$

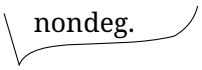
$C\sigma$ will be a graph

$C\sigma \ni$ a point for $\forall$ conn. comp. of [inserted above: nondeg.] configurations.



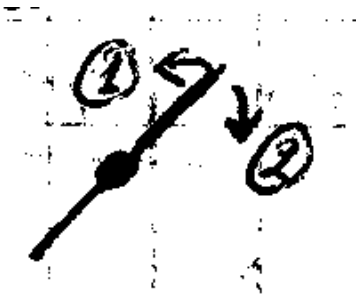
Original source details: GELFAND-P000910

Diagram and typed description. Insertion before ‘configurations’.



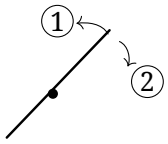
1 edge for each conn. component of the space of configurations with speciality 1

$c = 1 \Rightarrow \exists$ exactly 2 ways of bringing them into general pos.



Original source details: GELFAND-P000910

Diagram and typed description. Two arrows from the marked configuration.



There are no loops

Original source details: GELFAND-P000910

Diagram and typed description. A loop in parentheses.

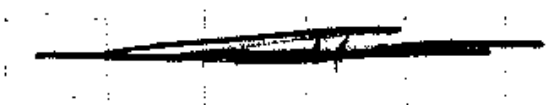
GELFAND-P000911

[source number: 14]

The graph is connected (a difficult theorem!)

$k = 2 \Rightarrow$ 2-dimensional (non-simpl.) complex.

points —



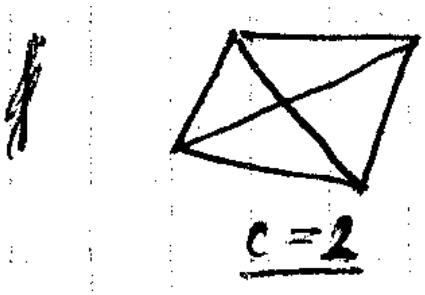
Original source details: GELFAND-P000911

Uncertain reading. Traces of writing appear under the long stroke. It may be a cancellation.

for connected comp. of config. in general pos.

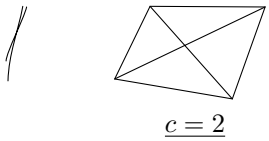
edges for config. of speciality 1

square for connected comp. —〃— 2



Original source details: GELFAND-P000911

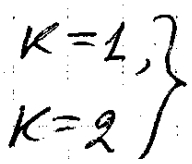
Diagram and typed description. Quadrilateral with both diagonals and the underlined label $c = 2$. The purpose of the double slanted stroke at left is uncertain.



break each one in 2 ways

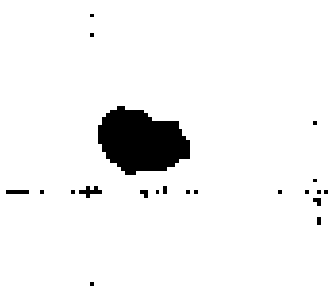
— these are the sides of the square.

$$\left.\begin{matrix} k=1, \\ k=2 \end{matrix}\right\} C\sigma \underset{\text{def}}{=} 1 \text{ point.}$$



Original source details: GELFAND-P000911

Uncertain reading. The second line reads $k = 2$.

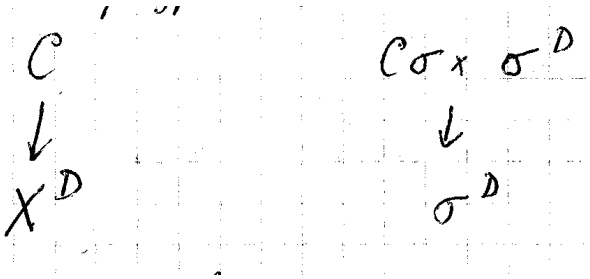


Original source details: GELFAND-P000911

Diagram and typed description. Isolated dot to the right of the definition.

•

(all config. in general pos. and form a connected set)

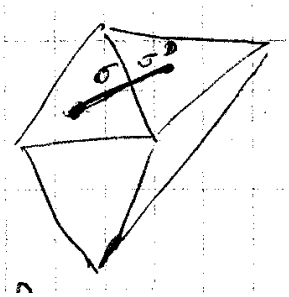


Original source details: GELFAND-P000911

Diagram and typed description. Two maps.

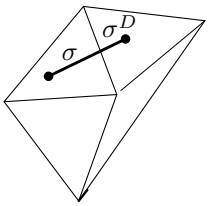
$$\begin{array}{ccc} C & & C\sigma \times \sigma^D \\ \downarrow & & \downarrow \\ X^D & & \sigma^D \end{array}$$

[At right:] do they glue together naturally?



Original source details: GELFAND-P000911

Diagram and typed description. Thickened segment with two marked ends and labels σ and σ^D. The faint interior edge on the right has a slight gap at the interior vertex.

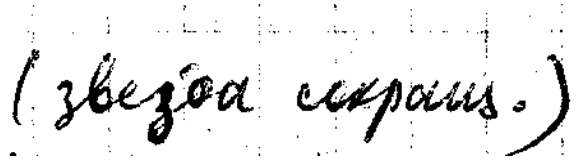


$$\sigma^D \subset \sigma'^D$$
$$\sigma \supset \sigma'$$

Then there is a map

$$\text{config. for } \sigma \longleftarrow \text{config. for } \sigma'$$

[At right:] (star shrinks)



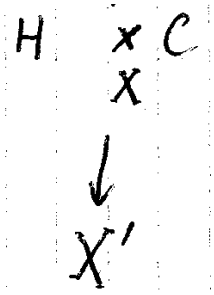
Uncertain reading. The abbreviated Russian word is tentatively read «сокращ.» (shrinks).

Original source details: GELFAND-P000911

(restriction of the maps ψ)

$$C\sigma \longleftarrow C\sigma'$$

We have a map:



Original source details: GELFAND-P000911

Diagram and typed description. Map to X'.

$$\begin{array}{c} H \underset{X}{\times} C \\ \downarrow \\ X' \end{array}$$

[At right:] $X' = X$ is decomposed into intersections of simplices and of the dual decomp.

$$\Gamma \subset H \underset{X}{\times} C$$

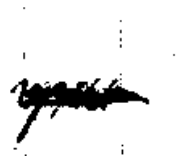
Γ — cells from H (a face of the hypersimplex)

for a vertex $\rightsquigarrow$ a point of the config. in general pos. $\Rightarrow$ then we include the direct products in Γ

σ — a simplex from X

$H\sigma$ $\quad$ $C\sigma$

face of $H\sigma$ ≈ [crossed out: illegible] set of five points $\xrightarrow{\psi}$ a set of five. If in general pos., include it in Γ

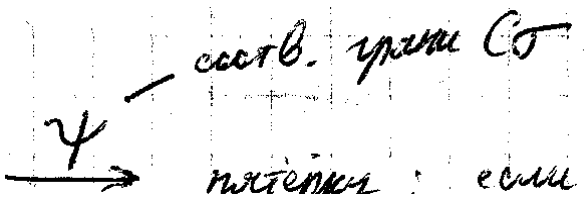


Original source details: GELFAND-P000911

Undeciphered passage. Short crossed-out word before ‘set of five’. It may be ‘face’.

Original source details: GELFAND-P000911

Uncertain reading. Two sloping strokes and a separate dot at the left of the upper stroke. The meaning of the sign is uncertain.

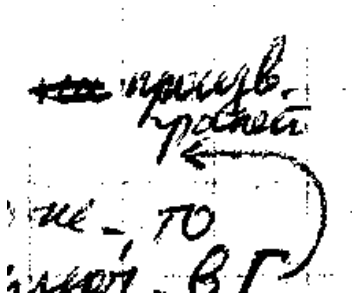


Original source details: GELFAND-P000911

Diagram and typed description. Sloping annotation line from ψ.

corresp. face of $C\sigma$

$\xrightarrow{\psi}$



Original source details: GELFAND-P000911

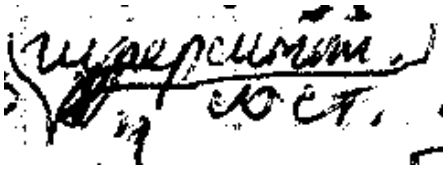
Diagram and typed description. Right-margin annotation. The short crossed-out mark before ‘prod.’ is illegible. It may be «c». The curved arrow comes from the lower line about Γ.

[crossed out] prod.

direct

$$\pm\frac{1}{48}\widetilde{P} \qquad \text{over the points of } X$$

Consider the 4-dimensional face [inserted above: hypersimpl.] consisting of the following maps

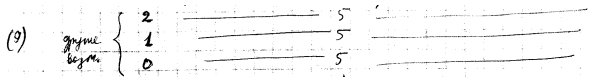


Original source details: GELFAND-P000912

Uncertain reading. There is a heavy stroke below the start of the insertion. It is unclear whether it cancels a separate entry.

$$h : \{\text{vert. of the star of } \sigma\} \longrightarrow [0,1]$$
$$h = 1 \text{ on } \sigma$$

$h = 1$ on 3 vertices (other ones), variable on 5, $= 0$ on the rest



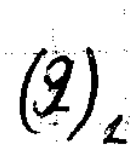
Original source details: GELFAND-P000912

Diagram and typed description. Three rows under the brace in (9).

$$(9)\ \text{other poss.}\ \begin{cases} 2 ____ 5 ____ \\ 1 ____ 5 ____ \\ 0 ____ 5 ____ \end{cases}$$

(there are no other 4-dimensional faces)

$$\widetilde{P} = 0 \quad \text{. on } (9)_1 \text{ and } (9)_4$$
$$\widetilde{P} = \pm\frac{1}{48} \quad \text{on } (9)_2 \text{ and } (9)_3.$$

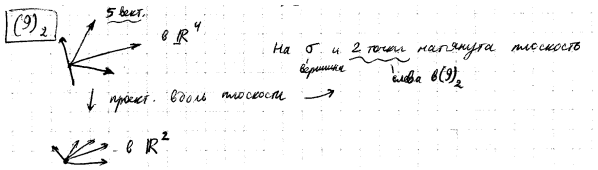


Original source details: GELFAND-P000912

Uncertain reading. The first label is tentatively $(9)_1$. There is an extra horizontal stroke below the 9 inside the parentheses.

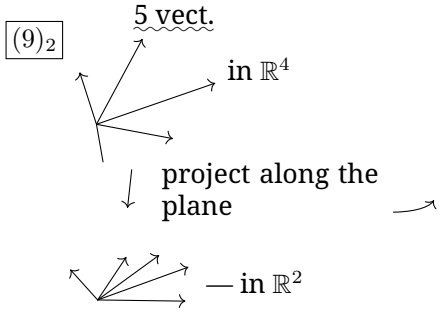
The number refers to an oriented cell.

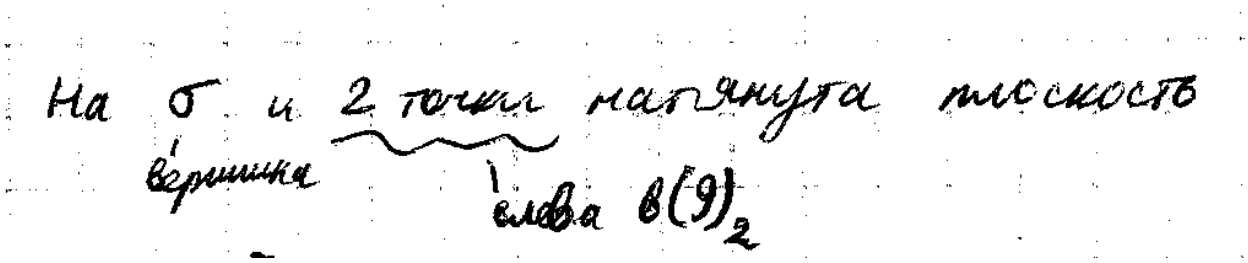
To the orientation of face $(9)_2$ corresponds the parity of the ordering of the 5 points.



Original source details: GELFAND-P000912

Diagram and typed description. Five vectors and their projection. The lowest stroke in the top group has no visible arrowhead.

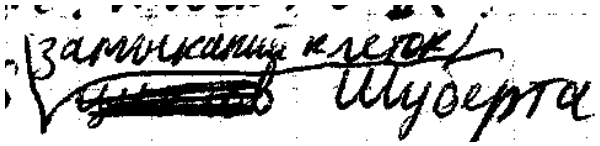




Original source details: GELFAND-P000912

Diagram and typed description. Annotation at the right of the figure.
The plane is spanned by σ [label below: vertex] and 2 points [label below: at left in $(9)_2$].

10 orientations of the plane (each pair)
We must take $(-1)^i \frac{1}{48}$, where i is the number of positive orientations
(it does not matter which are > 0, since there are 10 in all)
$\boxed{(9)_3}$ the same thing
we get 5 vectors in $\mathbb{R}^3$,
also 10 orient. in $\mathbb{R}^3$, and so on.
$G_{2,2}(\mathbb{R})$ — 2-dimensional planes in $\mathbb{R}^4$
there are 6 [crossed out: cycles?] [inserted above: closures of cells] of Schubert



Uncertain reading. A word beneath the insertion is crossed out, tentatively «циклов» (cycles).

Original source details: GELFAND-P000912

In $\mathbb{R}^4$: [mark crossed out] 6 coordinate 2-dimensional planes

Uncertain reading. The mark before 6 is heavily crossed out. The reading is unresolved.

Original source details: GELFAND-P000912

Planes [entry crossed out; inserted: not] transverse to the given coordinate plane

Uncertain reading. The crossed-out entry before the insertion ‘not’ is illegible.

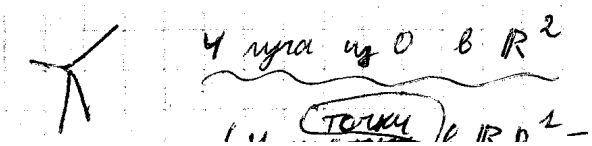
Original source details: GELFAND-P000912

“enhanced projective configur. of 4 points in $\mathbb{R}P^2$”

Uncertain reading. The exponent in the English phrase reads 2.

Original source details: GELFAND-P000912

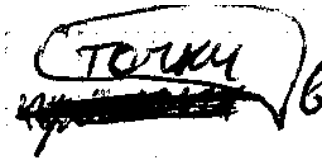
“improved — improved [Russian gloss: «улучшенные»]



Original source details: GELFAND-P000912

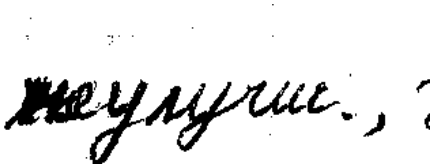
Diagram and typed description. 4 rays from 0 in $\mathbb{R}^2$

(4 [crossed out: illegible] [inserted above: points] in $\mathbb{R}P^1$ — unimproved, here rays)



Original source details: GELFAND-P000912

Undeciphered passage. The crossed-out word before the insertion ‘points’ is illegible.



Original source details: GELFAND-P000912

Uncertain reading. The beginning of «неулучш.» (unimproved) is heavily traced over.

GELFAND-P000913

[source number: 15]

Original source details: GELFAND-P000913

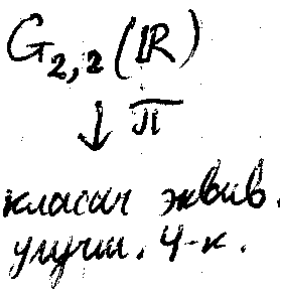
Diagram and typed description. Manuscript page number at upper right.

$$L \subset \mathbb{R}^4, \qquad \dim L = 2$$

in $\mathbb{R}^4/L$ there are 4 rays!!

We must take their cross ratio

2 such collections are equiv. if the action of $GL(2,\mathbb{R})$ carries one to the other



Original source details: GELFAND-P000913

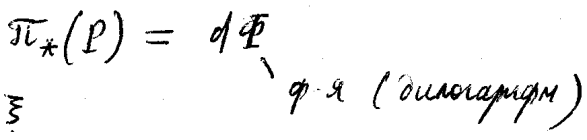
Diagram and typed description. Arrow labelled π.

$G_{2,2}(\mathbb{R})$

$\downarrow \pi$

classes of equiv.
enhanced quadruples

P — 4-form representing the Pontryagin class



Original source details: GELFAND-P000913

Diagram and typed description. Sloping annotation line from Φ.

$\pi_*(P) = d\Phi$

fn. (dilogarithm)

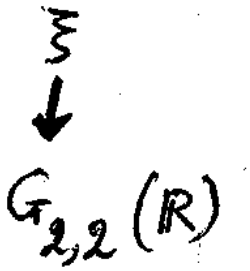


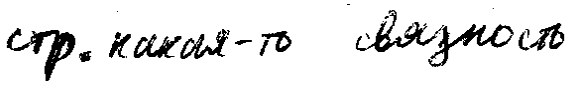
Diagram and typed description. Arrow from ξ to the Grassmannian.

$$\xi \downarrow G_{2,2}(\mathbb{R})$$

Original source details: GELFAND-P000913

$P = p(\xi)$, ξ — 2-dimensional (std.) bundle over $G_{2,2}(\mathbb{R})$

construct some connection.



Uncertain reading. The initial Russian verb is abbreviated «стр.», read as ‘construct’.

Original source details: GELFAND-P000913

GELFAND-P000914

11.2.80 Gelfand Sem.

R. Mc Pherson.

The Grassmannian $G_{n,k}$ of n-planes in $\mathbb{R}^{n+k}$,

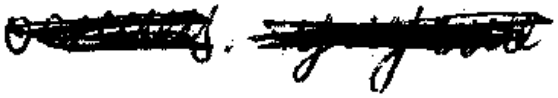
(unoriented)

Idea:

To an element of $G_{n,k}$ corresp. the pair

hypersimplex + configuration

$\widetilde{C}_{n,k}$ = [crossed out: enhanced?, [illegible]] enhanced project. configurations



Undeciphered passage. The first of the two crossed-out words may be «оснащ.» (enhanced). Two descending loops survive in the second word, which cannot be read in full.

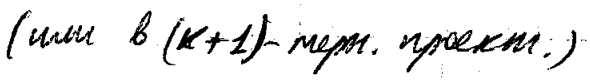
Original source details: GELFAND-P000914

Diagram and typed description. The English word ‘enhanced’ is written beneath the Russian abbreviation «оснащ.».

оснащ.
enhanced

Original source details: GELFAND-P000914

of $n + k$ points in a k-dimensional space (or in a $(k + 1)$-dimensional project. one)



Uncertain reading. The parentheses clearly contain $k + 1$.

Original source details: GELFAND-P000914

Original source details: GELFAND-P000914

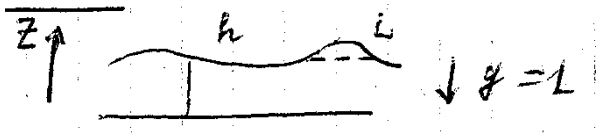
Undeciphered passage. Only isolated upper strokes remain below the last full line. No continuous text can be read there.

GELFAND-P000915

18.2.80 Gelfand Sem.

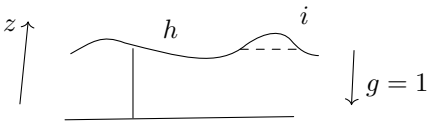
Zakharov V. E.

Benney eqs.



Original source details: GELFAND-P000915

Diagram and typed description. Two rises in the surface above the bed. A mark with a separate upper dot above the right crest is tentatively read i.



h — the shape of the surface, $h = h(\vec{r}, t)$.

$\vec{r}$ — a vector in the plane of the bed

Original source details: GELFAND-P000915

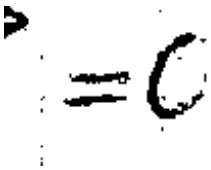
Uncertain reading. The horizontal coordinate reads $\vec{r}$.

$\vec{u} = \vec{u}(\vec{r}, z, t)$ — velocity in the plane of the bed

w — vert. velocity

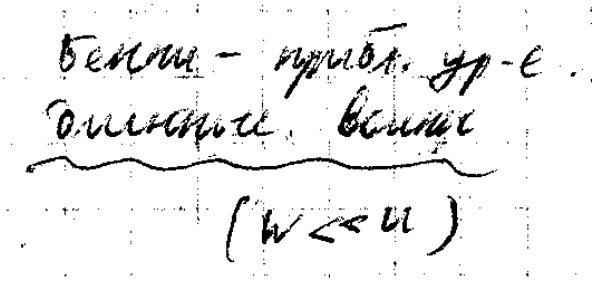
ideal incompress. fluid

$$\frac{\partial w}{\partial z} + \operatorname{div} \vec{u} = 0$$



Original source details: GELFAND-P000915

Uncertain reading. The glyph after the equals sign is open at the right. The context supports reading it as 0.



Original source details: GELFAND-P000915

Diagram and typed description. Annotation at the right margin.

Benney — approx. eq.

long waves

$(w \ll u)$

$$\frac{\partial w}{\partial t} + \underbrace{(u\nabla)w + w\frac{\partial w}{\partial z}}_{\text{neglect compared with the following terms}} - \frac{\partial p}{\partial z} + 1 = 0$$

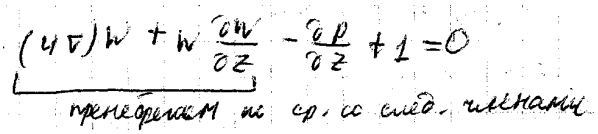
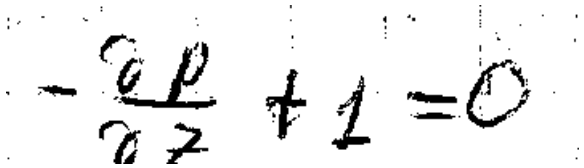


Diagram and typed description. A square bracket below the two transport terms.

Original source details: GELFAND-P000915



Uncertain reading. A minus sign is written before the pressure derivative.

Original source details: GELFAND-P000915

GELFAND-P000916

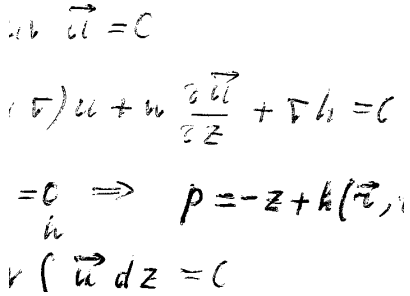
$$(1)\quad \begin{cases} \dfrac{\partial w}{\partial z} + \operatorname{div}\vec{u} = 0 \\ \vec{u}_t + (u\nabla)u + w\dfrac{\partial \vec{u}}{\partial z} + \nabla h = 0 \end{cases}$$

$$\frac{\partial p}{\partial z} + 1 = 0 \quad \Longrightarrow \quad p = -z + h(\vec{r}, t)$$

Uncertain reading. The coordinate in the argument of h reads $\vec{r}$.

Original source details: GELFAND-P000916

$$(1)_3 \quad \left\{ h_t + \operatorname{div}\int_0^h \vec{u}\,dz = 0 \right.$$



Uncertain reading. The four right-hand signs read as 0. Several have contours that are open at right.

Original source details: GELFAND-P000916

(1) — system of Benney eqs.

Manin [written below: and others] — the 1-dimensional case.

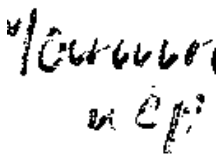


Diagram and typed description. ‘And others’ written below Manin’s name.

Original source details: GELFAND-P000916

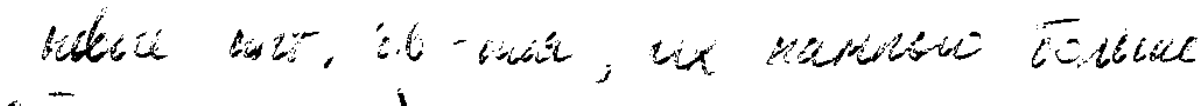
Zakharov a) in the 1-dimensional case derived a countable number of conserv. laws, there are many more of them

Uncertain reading. The verb reads ‘derived’.

Original source details: GELFAND-P000916

Uncertain reading. The abbreviation after ‘countable’ is read «з.с-ний» (conservation laws).

Original source details: GELFAND-P000916



Original source details: GELFAND-P000916

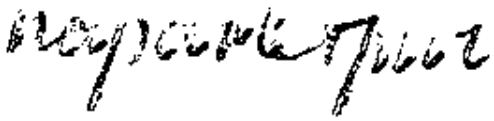
Diagram and typed description. End of item a). The final words read ‘there are many more of them’.

b) a different Ham. structure (simpler)

c) something for 2-dimensional dep. (with Hamiltonian character)

Idea: reinterpret. of the problem.

Uncertain reading. The abbreviation before ‘of the problem’ reads ‘reinterpret.’.



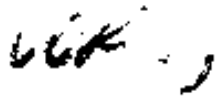
Original source details: GELFAND-P000916

I.M.G. Collective diff. eqs.:

when the particles do not interact dir., but [crossed out: illegible] [inserted above: are charged and] together create a field which [between the lines: (electr. field)] acts on each one separately.

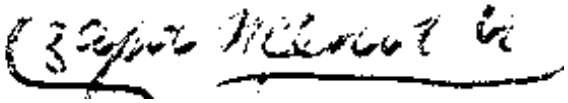
Undeciphered passage. A short entry after ‘but’ is crossed out. The reading is unresolved.

Original source details: GELFAND-P000916



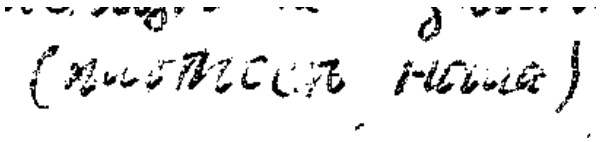
Original source details: GELFAND-P000916

Uncertain reading. The abbreviation after ‘interact’ is tentatively read «неп.» (dir.).



Original source details: GELFAND-P000916

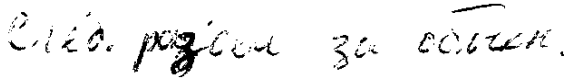
Diagram and typed description. The insertion ‘are charged and’ has a long leader.



Original source details: GELFAND-P000916

Uncertain reading. The first word in parentheses is tentatively read ‘electr.’.

Next section after ordinary diff. eqs.



Original source details: GELFAND-P000916

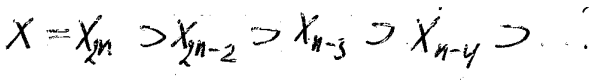
Uncertain reading. The reading ‘Next section after ordinary’ is tentative. There is a separate stroke above the last abbreviation.

GELFAND-P000917

22.9.80 Gelfand Sem.

Fuchs. There is a variety with sing. (algebr., complex) and there is a stratification.

$$(1) \quad X = X_{2n} \supset X_{2n-2} \supset X_{n-3} \supset X_{n-4} \supset \dots$$



Original source details: GELFAND-P000917

Uncertain reading. There is no 2 before n in the last two indices written out.

$X_i \setminus X_{i-1}$ — an i-dimensional variety.

Near a given stratum everything is arranged in the same way (the whole neighborhood).

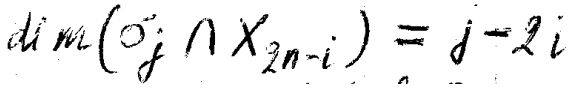
Consider sing. simplices of the space X with certain additional properties.

$\sigma_j \qquad \dim \sigma_j = j$

Original source details: GELFAND-P000917

Uncertain reading. An illegible mark between dim and σ_j is crossed out.

$$(2) \quad \dim(\sigma_j \cap X_{2n-i}) = j - 2i$$

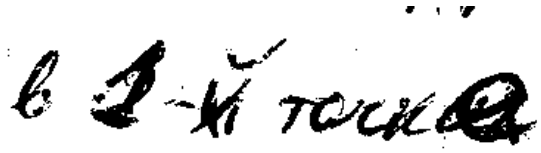


Original source details: GELFAND-P000917

Uncertain reading. In (2) the index on the left reads $2n - i$, and the expression on the right reads $j - 2i$.

There is Poincaré duality in these cohom. These are different cohomologies.

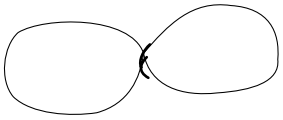
Example: Two 2-dimensional spheres glued at one point



Uncertain reading. The digit reads 1. The end of the Russian word for ‘point’ is traced over.

Original source details: GELFAND-P000917

Diagram and typed description. Two outlines meeting at the center. The strokes are retraced in the narrow contact region.



Original source details: GELFAND-P000917

These cohomologies are not homotopy invariant.

Ordinary homology:

$$\begin{cases} 0 & \mathbb{Z} \\ 1 & 0 \\ 2 & \mathbb{Z} \dot{+} \mathbb{Z} \end{cases}$$

Uncertain reading. A short entry at right of the first row is crossed out. Its beginning resembles $\mathbb{Z}$, but the entire entry is illegible.

Original source details: GELFAND-P000917

They do not satisfy Poincaré duality (immediately apparent).

MacPherson hom.

$$\begin{cases} 0 & \mathbb{Z} \dot{+} \mathbb{Z} \\ 1 & 0 \\ 2 & \mathbb{Z} \dot{+} \mathbb{Z}. \end{cases}$$

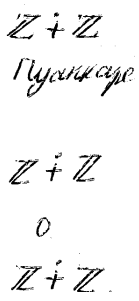


Diagram and typed description. In all three sums a dot is written above the plus sign.

Original source details: GELFAND-P000917

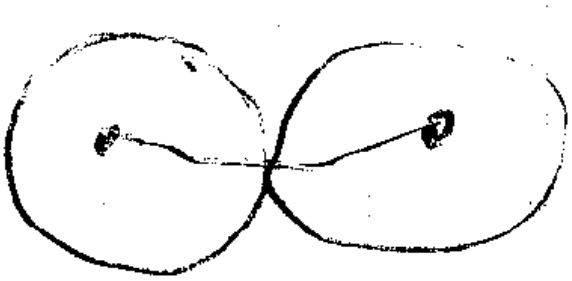
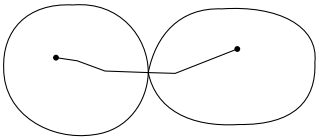


Diagram and typed description. A bent line joins two marked points and passes through the contact of the outlines.



Original source details: GELFAND-P000917

Such a simplex is forbidden.

Conjecture: they agree with L_2-cohomology.

(this interested Borel)

Suppose there is exactly one isol. singularity. Then MF-cohom.

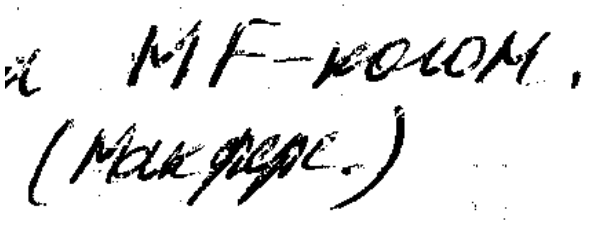


Diagram and typed description. '(MacPhers.)' is written under 'MF-cohom.'.

Original source details: GELFAND-P000917

$$\left.\begin{array}{ll} \text{in dim. } > \dfrac{\dim X}{2} & \text{agree with ordinary ones,} \\ \text{in dim. } < \dfrac{\dim X}{2} & \text{from Poincaré duality.} \end{array}\right] \quad \text{this is Deligne's def.}$$

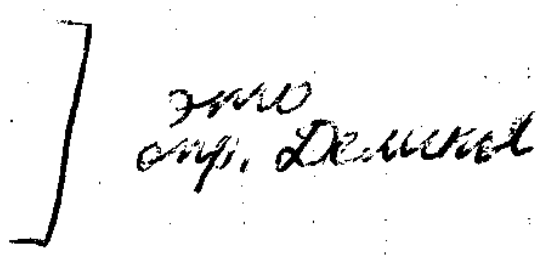


Diagram and typed description. 'This is Deligne's def.' is written to the right of the bracket.

Original source details: GELFAND-P000917

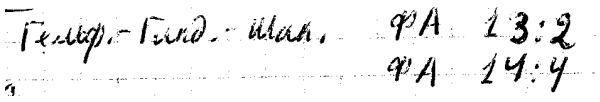
GELFAND-P000918

Deligne's def. in a more general situation (inductive)

M. I. Graev.

From the works of Gelf.-Gind.-Shap.

FA 13 : 2
FA 14 : 4

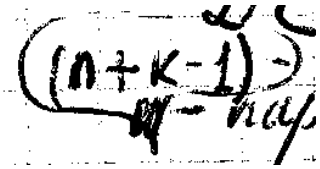


Uncertain reading. The last surname is abbreviated 'Shap.'. The right-hand entries read 'FA 13:2' and 'FA 14:4'.

Original source details: GELFAND-P000918

$\mathcal{D} \subset \mathbb{R}^n$,

$(n+k-1)$-param. family of curves



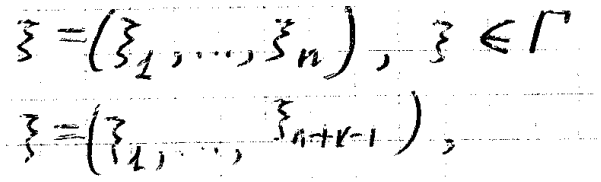
Original source details: GELFAND-P000918

Uncertain reading. $(n+k-1)$ is inserted above and linked to the crossed-out entry. The earlier mark is illegible.

$$x_i = \varphi_i(x_1, \xi), \qquad \xi = (\xi_1, \dots, \xi_n), \qquad \xi \in \Gamma$$

Integr. transform

$$\xi = (\xi_1, \dots, \xi_{n+k-1}),$$



Original source details: GELFAND-P000918

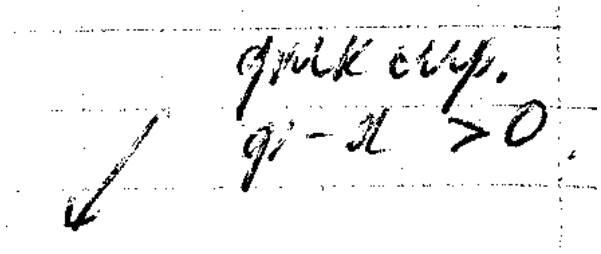
Uncertain reading. The first list ends at ξ_n, the second at ξ_{n+k-1}.

$f(x)$, supp $f \subset \mathcal{D}$

$$F(\xi) = I_\chi f = \int f(x_1, \varphi_1(x_1, \xi), \dots, \varphi_n(x_1, \xi))\chi(x_1, \xi)\, dx_1$$

Original source details: GELFAND-P000918

Uncertain reading. The first index on φ under the integral reads 1.



Original source details: GELFAND-P000918

Diagram and typed description. An arrow from the annotation to the factor χ.

fixed everywh.
fn. > 0.

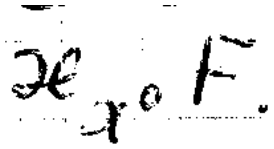
χ

It is of interest to find such sets of curves for which

$\exists$ a loc. inversion formula exists (recovery from integrals over curves passing through this point or an infinitesimally close one)

Such sets of curves are called admissible.

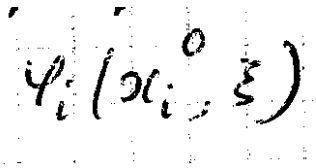
$\forall x^0 \in \mathcal{D}$ define an operator $F(\xi) \longrightarrow \mathcal{H}_{x^0} F$.



Original source details: GELFAND-P000918

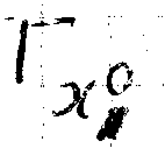
Uncertain reading. The operator's script letter is rendered $\mathcal{H}$.

$$\Gamma_{x^0} \qquad x_i^0 = \varphi_i(x_i^0, \xi) \qquad \dim \Gamma_{x^0} = k$$



Original source details: GELFAND-P000918

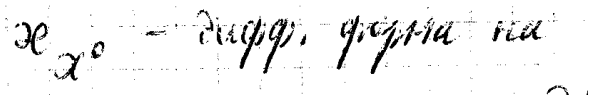
Uncertain reading. The dot on the subscript i is visible in the first argument of φ_i.



Original source details: GELFAND-P000918

Uncertain reading. An illegible mark below x in the subscript of the left-hand Γ is crossed out.

$\mathcal{H}_{x^0}$ — a diff. form on



Original source details: GELFAND-P000918

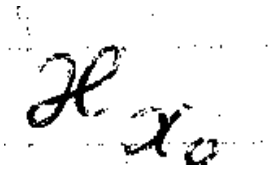
Undeciphered passage. The line ends with ‘on’. No object is written after it.

$$\mathcal{H}_{x^0} F = \sum a_{ij}(x_1^0, \xi) \frac{\partial F}{\partial \xi_j}\, d\xi_i + F \sum b_i(x_1^0, \xi)\, d\xi_i$$

The operator $\mathcal{H}_{x^0}$ is called admissible, if $\forall$ fn. f the diff. form
$\mathcal{H}_{x^0} I_\chi f$ is closed.
$k > 1 \Rightarrow$ this is the complete def.
If $k = 1$, it is automatically closed and it is req. that the functional

$$f \mapsto \int \mathcal{H}_{x_0} I_\chi f$$

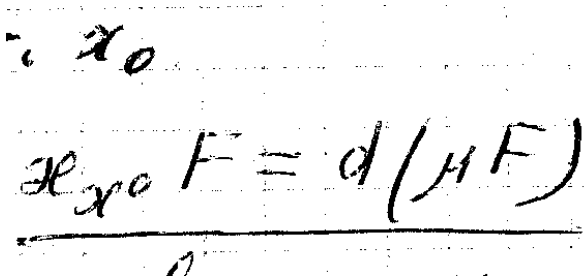
be concentrated at the point x_0.



Original source details: GELFAND-P000918

Uncertain reading. In the integral and the next line 0 is a subscript. In the preceding formulas the 0 on x is a superscript.

Trivial functional: $\underline{\mathcal{H}_{x^0} F = d(\mu F)}$



Original source details: GELFAND-P000918

Diagram and typed description. The formula for the trivial functional is underlined.

The quotient by the triv. functionals is important.

Chapter 18

GELFAND-P000919

13.10.80 Gelfand Sem.

I. M. Gelfand. Consider the equation

$$f(z_1, \dots, z_n; w_1, \dots, w_n) = 0 \tag{1}$$

Example: consider a 2nd-order diff. eq.

$$y'' = f(x, y, y') \tag{2}$$

Original source details: GELFAND-P000919

Uncertain reading. A superscript mark above the second argument y is crossed out. Its reading is unresolved.

Then

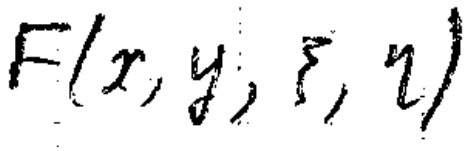
$$y = \varphi(x, \xi, \eta) \tag{3}$$

Original source details: GELFAND-P000919

Uncertain reading. The heavily traced letter reads φ.

Conversely, if we have family (3), a 2nd-order diff. eq. is constructed from it.
more generally:

$$F(x, y, \xi, \eta) = 0 \tag{4}$$



Original source details: GELFAND-P000919

Uncertain reading. In (4) there is a comma after y.

One can regard ξ, η as a point and construct the dual diff. eq.

(1) defines an n-param. family of hypersurfaces.

(5) Problem: find the geometry assoc. with this family.

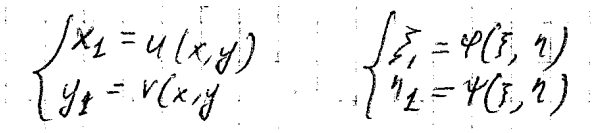
We can make changes of variables. E.g., in the eq. of straight lines

$$\xi x + \eta y = 1 \tag{6}$$

(7) Problem: how can we tell whether family (4) reduces to the family of straight lines (6).

Substitutions:

$$\begin{cases} x_1 = u(x,y) \\ y_1 = v(x,y \end{cases} \qquad \begin{cases} \xi_1 = \varphi(\xi,\eta) \\ \eta_1 = \psi(\xi,\eta) \end{cases}$$



Uncertain reading. The second line of the left-hand system has no closing parenthesis after y.

Original source details: GELFAND-P000919

(8) General definition: double fibration

(9) an element — a point and a curve passing through it

(curves from a 2-param. family)

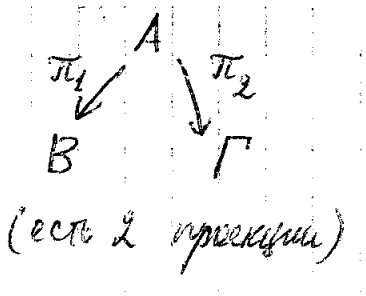
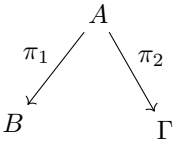


Diagram and typed description. Two projections labelled π_1 and π_2.



(there are 2 projections)

Original source details: GELFAND-P000919

(see the article by Gelfand, Gindikin, Shapiro in FA)

GELFAND-P000920

(10) Requirement: if

$$\begin{cases} & \pi_1 x = \pi_1 x' \\ \text{and} & \pi_2 x = \pi_2 x' \end{cases}, \quad \text{then } x = x'$$

a stronger requirement:

$$(11) \qquad A \subset B \times \Gamma \quad \text{as a submanifold}$$

$$x \mapsto (\pi_1 x, \pi_2 x)$$

Problem: classify up to diffeomorphism

$$f(z, w) = 0, \tag{12}$$

up to changes

$$\varphi(z) = z_1, \qquad \psi(w) = w_1$$

(everything analytic).

(13) Another example: $A = G$ — a group, K, H — 2 subgroups,

$$B = H\backslash G, \qquad \Gamma = K\backslash G.$$

(14) Important example:

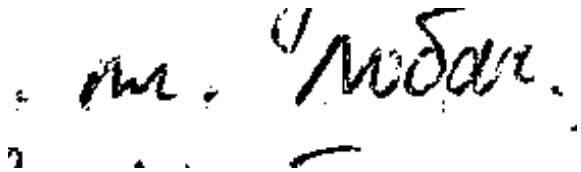
G — the group $GL(n, \mathbb{C})$,

K — subgr. of unitary matrices

H — —//— of triang. matrices

(15) Useful special case: $n = 2$.

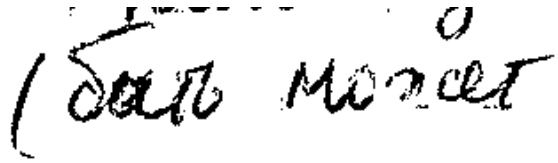
Then Lobach. geom. and horospheres arise.



Original source details: GELFAND-P000920

Uncertain reading. The abbreviation before ‘Lobach.’ reads «пл.», possibly ‘plane’.

(perhaps take 1 on the main diagonal in H)

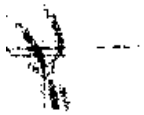


Original source details: GELFAND-P000920

Uncertain reading. The parentheses begin ‘perhaps’.

Work in Inventiones on Segre families.

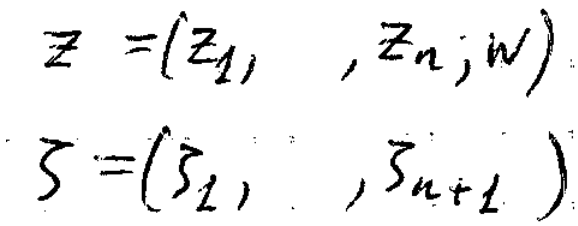
$$\tau(z,\zeta)=0 \tag{16}$$



Uncertain reading. Separate illegible strokes in the right margin level with (16).

Original source details: GELFAND-P000920

$$z=(z_1,\quad ,z_n;w)$$
$$\zeta=(\zeta_1,\quad ,\zeta_{n+1})$$



Uncertain reading. In both tuples, the spaces between the commas contain no visible ellipsis.

Original source details: GELFAND-P000920

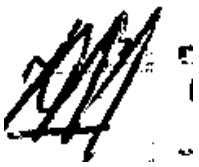
$$\theta=d_z\tau=\tau_\alpha\,dz^\alpha+\tau_w\,dw$$
$$\theta^\alpha=dz^\alpha$$
$$\theta_\alpha=d\zeta_\alpha$$

Under the map

$$(z,\zeta)\to(f(z),\bar f(\zeta)) \tag{17}$$

the forms transform as follows:

$$\theta^*=\theta,$$
$$\theta^{\alpha *}=\left(\frac{\partial f^\alpha}{\partial z^\beta}-\frac{\tau_\beta}{\tau_w}\frac{\partial f^\alpha}{\partial w}\right)\theta^\beta+\frac{1}{\tau_w}\frac{\partial f^\alpha}{\partial w}\theta, \tag{18}$$



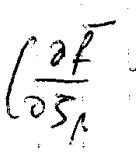
Undeciphered passage. An illegible expression is crossed out after the minus sign and before the fraction with τ_β.

Original source details: GELFAND-P000920

Uncertain reading. In the last term the glyph before the subscript w resembles τ with an angular top. The reading τ_w is tentative.

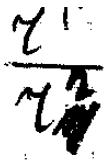
Original source details: GELFAND-P000920

$$\theta^*_\alpha = \left(\frac{\partial \bar{f}}{\partial \zeta_\beta} - \frac{\tau^\beta}{\tau^\eta}\frac{\partial f^\alpha}{\partial \eta}\right)\theta_\beta + \frac{1}{\tau^\beta}\frac{\partial \bar{f}_\alpha}{\partial \eta}.$$



Original source details: GELFAND-P000920

Uncertain reading. The first numerator of the lower line has a bar above f. No index is visible on f.

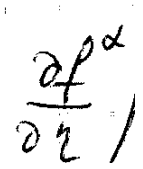


Original source details: GELFAND-P000920

Uncertain reading. The denominator after the minus sign tentatively has upper index η. The lower mark is heavily crossed out or overwritten.

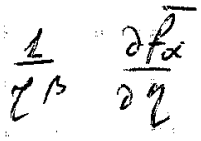
Original source details: GELFAND-P000920

Uncertain reading. A long looped cancellation lies between the fraction with τ^β and the next derivative.



Original source details: GELFAND-P000920

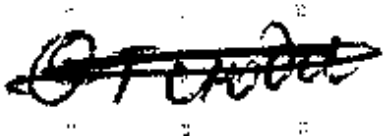
Uncertain reading. The middle numerator of the lower line has superscript α. No bar over f is visible.



Original source details: GELFAND-P000920

Uncertain reading. The last denominator reads τ^β, and the α on the last $\bar{f}$ is a subscript. No factor is visible after the derivative.

[Crossed out: From this.] Let



Original source details: GELFAND-P000920

Diagram and typed description. The crossed-out word reads ‘From this’.

$$\tau \mapsto s(z,\zeta)\tau(z,\zeta) \tag{19}$$

Then

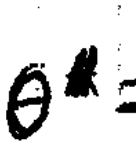
GELFAND-P000921

[source number: 3]

$$\begin{cases} \theta^* = s\theta \\ \theta^{\alpha *} = \theta^\alpha \\ \theta^*_\alpha = \theta_\alpha \end{cases} \tag{20}$$

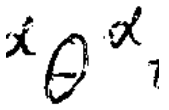
and

$$\begin{cases} \theta^* = u\theta^* \\ \theta^\alpha = u^\alpha \theta^\alpha + u^\alpha_\beta \theta^{\beta *} \\ \theta_\alpha = v_\alpha \theta^* + v^\beta_\alpha \theta^*_\beta \end{cases} \tag{21}$$



Uncertain reading. On the left of the first row a heavily traced superscript over θ is tentatively read as a star.

Original source details: GELFAND-P000921



Uncertain reading. In the second line the first term on the right reads $u^\alpha \theta^\alpha$, with no visible star.

Original source details: GELFAND-P000921

(22) A G-structure arises, where G is the group

$$\begin{pmatrix} u & 0 & 0 \\ u^\alpha & u^\alpha_\beta & 0 \\ v_\alpha & 0 & v^\beta_\alpha \end{pmatrix}$$

(23) Abstr. Segre family: [crossed out above: collection?] [inserted above: (a set with a G-struct. and)] if there is a collection of forms.

Uncertain reading. The word crossed out above the line is tentatively ‘collection’.

Original source details: GELFAND-P000921

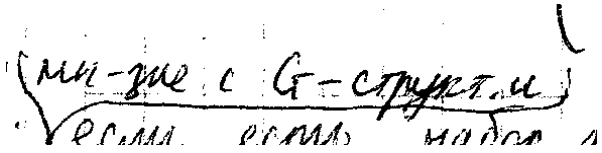


Diagram and typed description. A long insertion mark leads to ‘if’. The word after ‘struct.’ reads ‘and’.

a set with a G-struct. and

Original source details: GELFAND-P000921

and the conditions hold

$$\begin{cases} d\theta \equiv d\theta^{\alpha} \equiv 0 \pmod{\theta, \theta^1, \ldots, \theta^n} \\ d\theta \equiv d\theta_{\alpha} \equiv 0 \pmod{\theta, \theta_1, \ldots, \theta_n} \end{cases} \tag{24}$$

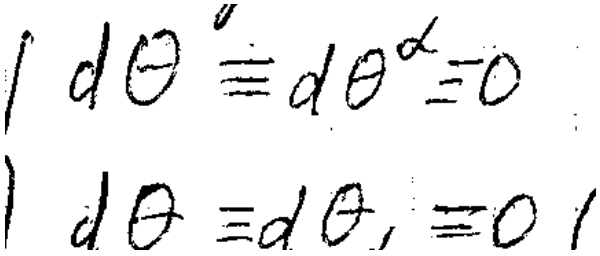


Diagram and typed description. Each of the four congruence signs in (24) has three strokes.

Original source details: GELFAND-P000921

Special case of the problem: what are the invariants of a family of curves in the plane, a family of surf. in 3-dimensional space, etc.

GELFAND-P000922

[source number: 6]

20.10.80 Gelfand Sem.

A. Goncharov. G-structures.

M — smooth manifold

Diagram and typed description. Bundle of tangent frames (principal $GL(n, R)$-bundle).

$$\begin{array}{c} E \\ \downarrow \\ M \end{array}$$

Original source details: GELFAND-P000922

$$G \subset GL(n, R)$$

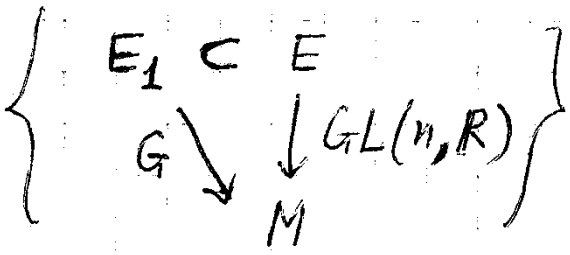


Diagram and typed description. G-structure.

$$\left\{ \begin{array}{ccc} E_1 & \subset & E \\ & \searrow & \downarrow GL(n,R) \\ & G & M \end{array} \right\}$$

Original source details: GELFAND-P000922

Problem: [crossed out] consider all geometries on S^2 associated with the different subgroups $G \subset GL(n, R)$.

Uncertain reading. The word before 'consider' is crossed out and illegible.

Original source details: GELFAND-P000922

D. B. Fuchs. On cohomologies according to Deligne, MacPherson, etc.

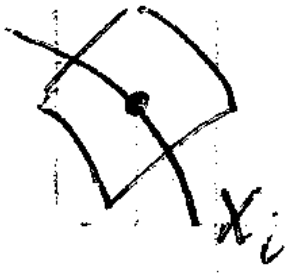
according to MacPherson.

X — set with sing., i.e. there is a stratification

$$X = X_n \supset X_{n-1} \supset \ldots \supset X_0. \tag{1}$$

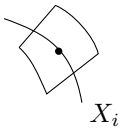
$X_i - X_{i-1}$ — smooth $2i$-dimensional set

and a certain natural condition of local triviality



Original source details: GELFAND-P000922

Diagram and typed description. Sketch accompanying the local triviality condition.



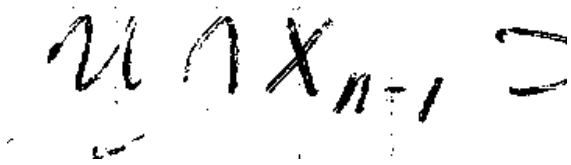
A point $x_0 \in X_i$ has a neighb. U such that

in the stratification

$$U \supset U \cap X_{n-1} \supset U \cap X_{n-1} \supset \ldots \supset U \cap X_i \tag{2}$$

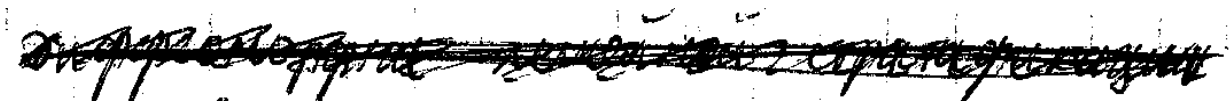
[crossed out: 'diffeomorphic' (?) [...] 'stratification' (?)]

has the same structure at every point of $U \cap X_i$



Original source details: GELFAND-P000922

Diagram and typed description. The term $U \cap X_{n-1}$ is written twice in succession in (2).



Original source details: GELFAND-P000922

Undeciphered passage. The middle of the crossed-out line is illegible. The beginning and ending are tentative readings.

Consider cycles of [circled and inserted above: sing.] simplices σ^q for which

$$\dim(\sigma^q \cap X_{n-i}) \leq q - i \qquad (\text{Codim } X_{n-i} = 2i\,!) \tag{3}$$

(like the condition that would hold in general pos. with respect to the distinguished part)

If the right side is q or $q - 2i$ (!), one gets ordinary cohomology.

Quotient by [crossed out above] the boundaries of chains subject to an analogous condition.

Original source details: GELFAND-P000922

Uncertain reading. The crossed-out sign before 'boundaries' is illegible.

Another defn., hypothetically equiv. to this one

Deligne wanted it to work over finite fields.

a more spec. reason: there is hope of proving a connection with L^2-cohomology

(L^2 [inserted below: -forms] on approaching the stratum)

A. Beilinson. The defn. above can be given a local form (pass to a complex of sheaves, its cohom. will be MacPherson cohom.).

GELFAND-P000923

Now the same using diff. operators

Embed

$$X \hookrightarrow \underset{\text{set [without singularities]}}{Y}$$

$$X_{\text{reg}} = X_n - X_{n-1} \hookrightarrow X \hookrightarrow Y. \tag{4}$$

On $X_n - X_{n-1}$ consider the triv. system of diff. eqs., i.e.

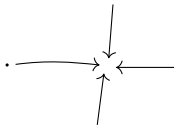
the sheaf $\mathcal{O}_{X_{\text{reg}}}$, the sheaf of fns., also a sheaf of modules over the ring of diff. op.

We want to extend this to all of Y.

Example:

X — a cross, Y — the plane

Diagram and typed description. Four arrows point inward. There is a separate dot to the left of the left arrow. Its role in the diagram is uncertain.



Original source details: GELFAND-P000923

extend the fns. by tensoring with δ in the transverse direction

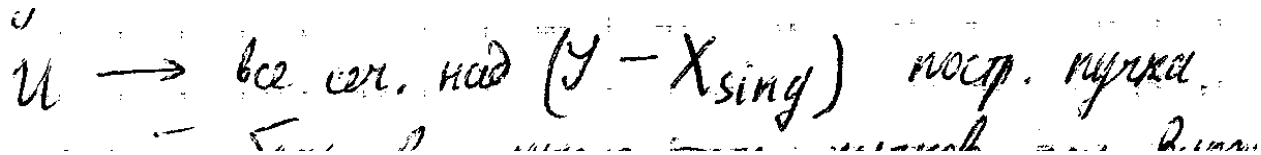
Diff. operators act on them.

(and the same in the general case)

Along X they must be reg. (and supported on X).

Consider the sheaf

$$U \longrightarrow \text{all sections over } (Y - X_{\text{sing}}) \text{ of the constructed sheaf.}$$



Original source details: GELFAND-P000923

Diagram and typed description. After 'over', the manuscript gives $(Y - X_{\text{sing}})$ without an intersection with U.

(5) (direct image in the sense of sheaf theory under the embedding $Y - X_{\text{sing}} \overset{i}{\hookrightarrow} Y$.)

(6) Thm. 1.

This sheaf of modules has a unique irred. submodule!

(Moral (I.M.G.) in the theory of sing. one should work with diff. operators.)

Thus a module over the ring of diff. op. over Y is constructed.

(sheaf of modules)

On X_{reg} consider the sheaf of fns. and extend them by δ-fns. in the transv. dir.

On $Y \setminus X_{\text{sing}}$ obtain a module over diff. op. (with holom. or algebr. coeffs.)

Original source details: GELFAND-P000923

Uncertain reading. The subscript on this line is overwritten. The final reading is ‘sing’.

Extend into the singularity X_{sing}.

The module is constr. in 2 steps

Step 1. Maximal — the direct image of the sheaf.

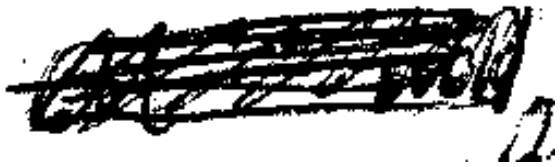
Step 2. Take the irred. submodule (it is unique!!)

Here: regular along X, but without compatibility!

Original source details: GELFAND-P000923

Diagram and typed description. (4 different limits!)

[crossed out above] Consider a resolution of this module M

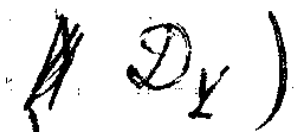


Original source details: GELFAND-P000923

Undeciphered passage. The word above ‘resolution’ is heavily crossed out and illegible.

$$\ldots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0 \tag{7}$$

(F_j — free module over diff. op. [crossed out] $\mathcal{D}_Y$)



Original source details: GELFAND-P000923

Uncertain reading. The cancelled sign before $\mathcal{D}_Y$ is illegible.

and map into fns. over Y (into $\mathcal{O}_Y$)

GELFAND-P000924

[source number: 7]

$$\bullet \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

A complex arises:

$$\mathrm{Hom}_{\mathcal{D}}(F_0, \mathcal{O}_Y) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(F_1, \mathcal{O}_Y) \longrightarrow \dots \tag{8}$$

Diagram and typed description. The abbreviation before ‘complex’ reads «Возн.», ‘arises’.

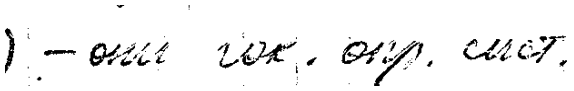
Original source details: GELFAND-P000924

up to $\mathrm{Codim}_Y X$ this is acyclic, then MacPherson cohomology

(9) The simplest overd. syst. of diff. eqs. — those for which there is locally a max. number of

(10) solutions (reduce to the system $\frac{\partial u}{\partial x_1} = 0, \dots, \frac{\partial u}{\partial x_n} = 0$) — they are wholly (?) determ. by the syst.

of their solns. — there will be a locally constant sheaf on the set.



Uncertain reading. The abbreviation after ‘they’ is tentatively read as «цел.» (‘wholly’).

Original source details: GELFAND-P000924

The solns. are locally const.

There may be no global solns.

Analogue in the general sit.

Consider any overd. syst.

(11) If the syst. has reg. singularities (there is a defn.), then the solutions are

— the MacPherson complex.

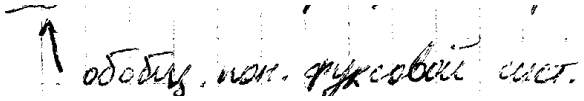


Diagram and typed description. Right-hand annotation. The arrow points to ‘reg. singularities’ in (11).

↑ generaliz. of the notion of a Fuchsian syst.

Original source details: GELFAND-P000924

This establ. a corresp. between these complexes and syst. with reg. sing.

Definition (Kashiwara) syst. with reg. sing.

Classical defn. (for syst. of ordinary eqs.) — reg. sing. (Fuchsian syst.):

$$\frac{dx}{dt} = \frac{1}{t}A(t)x, \quad \text{where } A \text{ is reg.} \tag{12}$$

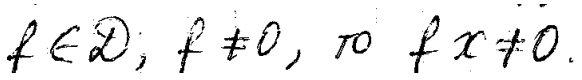
(first-order syst.)

Let [crossed out] a $\mathcal{D}$-module M be given (assoc. with an overd. syst.)

Original source details: GELFAND-P000924

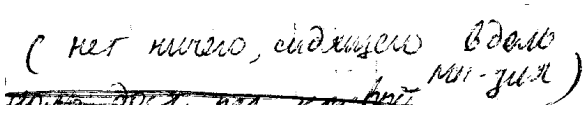
Uncertain reading. The writing crossed out before ‘given’ is illegible.

(13) a) Let M have no torsion, i.e. if $x \in M$, $x \neq 0$, $f \in \mathcal{D}$, $f \neq 0$, then $fx \neq 0$.



Original source details: GELFAND-P000924

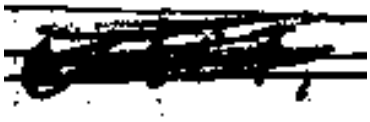
Diagram and typed description. The shape of the letter in $f \in \mathcal{D}$ matches $\mathcal{D}$ in the line about the module.



Original source details: GELFAND-P000924

Diagram and typed description. At right: (nothing is supported along sets)

[crossed out: Then [...] req. that the restr. along a suff. reg. curve ha[...] form (12)]



Original source details: GELFAND-P000924

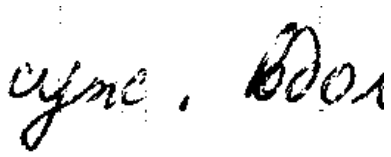
Undeciphered passage. The heavily crossed-out word after ‘Then’ is illegible.

Original source details: GELFAND-P000924

Undeciphered passage. Before ‘form (12)’ the beginning «име-» (‘ha-’) is visible.

(14) Then it reduces to syst. (10)

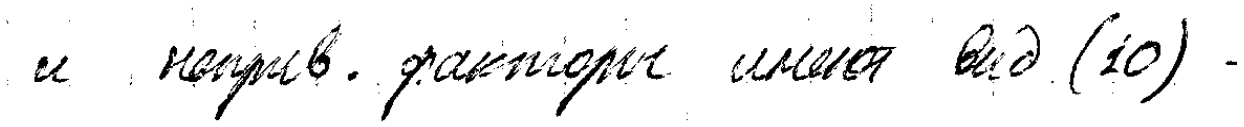
(15) Claim. M has reg. sing. $\Longleftrightarrow$ the restr. along a suff. reg. curve has form (12)



Original source details: GELFAND-P000924

Diagram and typed description. The abbreviation after the equivalence sign reads «суж.» (‘restr.’), as in the crossed-out line above.

General defn. In ∀ module there is a filtration
(finite length!) and the irred. factors has form (10) —
— integrable connection.



Original source details: GELFAND-P000924

Diagram and typed description. The line about the factors uses the singular verb ‘has’.

Defn. Each irred. compon. must have reg. sing.

GELFAND-P000925

27.10.80 Gelfand Sem.

Problems (for first-year students)

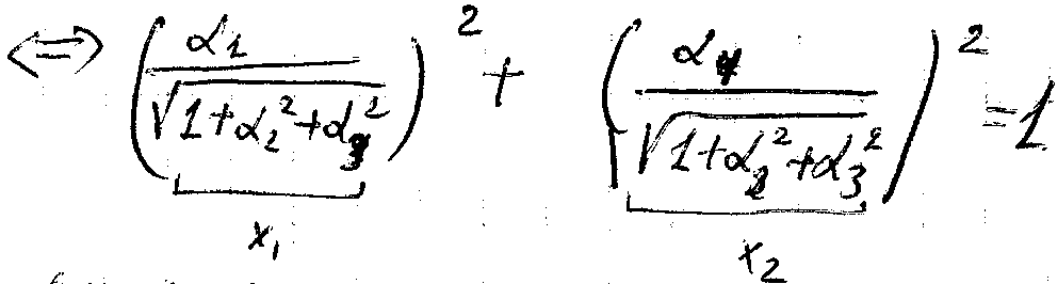
1. Find all subgroups and norm. subgroups of the symmetry group (?) of regular polyhedra and of the symmetric group.

Uncertain reading. The ending of the word before ‘symmetry’ is heavily overwritten. The construction suggests the genitive «группы» (‘of the group’). The ending is illegible.

Original source details: GELFAND-P000925

2. Is the group $SL(2,R)$ simply connected? Construct its univ. cover.

$$\alpha\delta-\beta\gamma=1 \iff \alpha_1^2-\alpha_2^2-\alpha_3^2+\alpha_4^2=1$$
$$\iff \alpha_1^2+\alpha_4^2=1+\alpha_2^2+\alpha_3^2 \iff$$
$$\iff \left(\underbrace{\frac{\alpha_1}{\sqrt{1+\alpha_2^2+\alpha_3^2}}}_{x_1}\right)^2+\left(\underbrace{\frac{\alpha_4}{\sqrt{1+\alpha_2^2+\alpha_3^2}}}_{x_2}\right)^2=1.$$



Original source details: GELFAND-P000925

Uncertain reading. The subscripts 3 in the first denominator, 4 in the second numerator and 2 in the second denominator are overwritten. The original signs are illegible.

$$(\alpha_1,\alpha_2,\alpha_3,\alpha_4) \underset{\text{homotopy equiv.}}{\longmapsto} (x_1,x_2)\in S^1$$

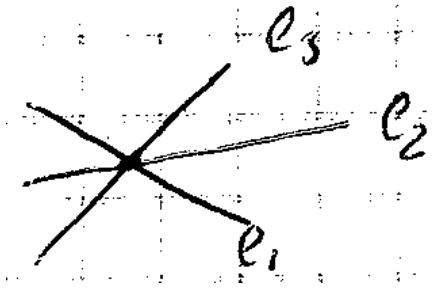
More precisely, one obtains

$$SL(2,R) \underset{\text{diffeom.}}{\cong} S^1\times R^2.$$

3. How is the univ. cover of $SL(2,R)$ structured as a group?

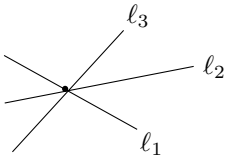
GELFAND-P000926

Suppose there are 3 lines

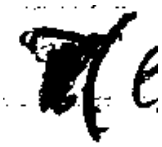


Original source details: GELFAND-P000926

Diagram and typed description. Three lines labelled ℓ_1, ℓ_2, ℓ_3.



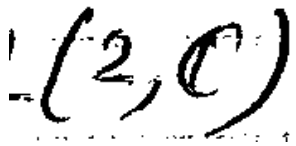
$$[?](\ell_1,\ell_2,\ell_3) = \begin{cases} +1 & \text{if } \ell_2 \text{ is inside } \widehat{\ell_1\ell_3} \text{ (angle } < \pi). \\ -1 & \text{otherwise} \end{cases}$$



Original source details: GELFAND-P000926

Uncertain reading. The sign before the arguments is heavily overwritten. Possible readings are τ and f. The right-hand note uses f for the three-argument function.

(g_j, n_j) — an element, $g_j \in SL(2,C)$

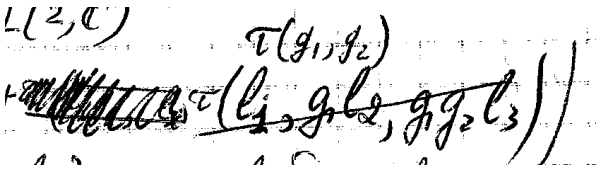


Original source details: GELFAND-P000926

Diagram and typed description. This line reads C. Question 3 on the preceding frame has R.

$$(g_1,n_1)(g_2,n_2) = (g_1g_2, n_1+n_2+\tau(g_1,g_2))$$

[crossed out $\tau(\ell_1, g_1\ell_2, g_1g_2\ell_3)$; inserted above the formula $\tau(g_1,g_2)$.]



Original source details: GELFAND-P000926

Undeciphered passage. A second, more heavily cancelled entry follows n_1+n_2+. Its reading is unresolved.

(ℓ_1, ℓ_2, ℓ_3) can be chosen arbitrarily, leading to a homolog. cocycle

$$g = \begin{pmatrix} \alpha & \beta \\ 0 & \alpha^{-1} \end{pmatrix} \qquad \varphi(g) = \begin{cases} 1, & \alpha > 0 \\ -1, & \alpha < 0 \end{cases}$$

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \varphi(g) = \begin{cases} i & c > 0 \\ -i & c < 0 \end{cases}$$

(if it does not reduce to triang.)

$$\varphi(g, n) = e^{i\frac{\pi}{2}n}\varphi(g)$$

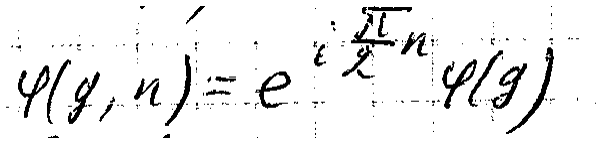


Diagram and typed description. The φ in the exponential formula has the same loop as in the two formulas above and in the definition of $\widetilde{G}$.

Original source details: GELFAND-P000926

Now take

$$\widetilde{G} = \{(g, n) : \varphi(g, n) = 1\}$$

$\widetilde{G}$ — this is a group and it is the univ. cover.

[at right] (Here $\ell_3 = \ell$ is the X-axis)

$$\tau(g_1, g_2) = f(\ell, g_1\ell\, g_1g_2\ell) = \pm 1$$

Uncertain reading. Between the second and third arguments a separate comma cannot be distinguished.

Original source details: GELFAND-P000926

ℓ — an oriented line

(invar. of a triple of [inserted below and circled: oriented] lines; index of a triple of lines;

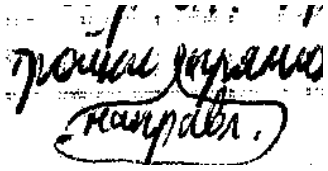


Diagram and typed description. The abbreviation for ‘oriented’ is circled and linked to the first occurrence of ‘lines’.

Original source details: GELFAND-P000926

$+1$ if $g_1\ell$ is between ℓ and $g_1g_2\ell$

This group is disconnected — there are 4 pieces; then take a conn. comp.

4. (Kirillov). Describe all locally projective structures on R

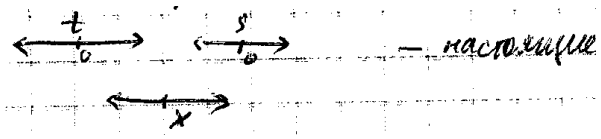
(covered by charts, transition — fractional-lin. transf.)

atlases are equiv. if the transition is a fract.-lin. transf.

Answer: there are countably many loc. project. structures

Example 1. R and param. t

Example 2:



Original source details: GELFAND-P000926

Diagram and typed description. Two upper axes t, s with zeros, and the lower axis x.

t (0) s (0) — genuine

x

Relations: $tx = -1, \quad xs = -1$

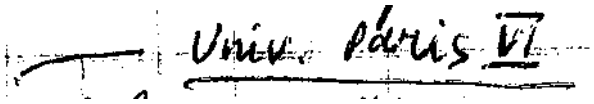
When one ends — go to the other, etc.

The second is not equiv. to the first.

A 2-sheeted cover of the project. line has a loc. project. structure.

A cover of the project. line — its project. transf. will be — transf. of the univ. cover.

Prof. Gaveau (from Paris) 3 possible topics.

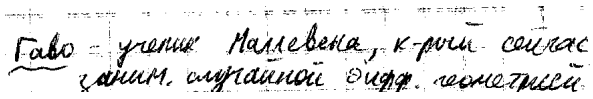


Original source details: GELFAND-P000926

Diagram and typed description. Annotation above the line with a leader to ‘from Paris’.

← Univ. Paris VI

1. Constr. of envelopes of holomorphy
2. Constr. of measures on certain ∞-dim. homog. sets
3. Scattering theory and functional integrals.



Original source details: GELFAND-P000926

Diagram and typed description. At right: Gaveau is a student of Malliavin, who is now working on stochastic diff. geometry.

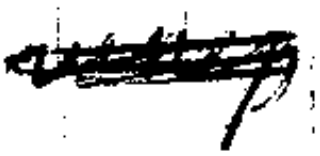
1. On constructions of envelopes of holomorphy.

Consider C^2 and try to constr. the envelope of holom. of a compact 2-dimensional surf. (in C^2)

Problem. Find such a [crossed out] Levi-flat hypersurf. M that

$$\partial M = S^2$$

(the Levi form on it must be $= 0$)

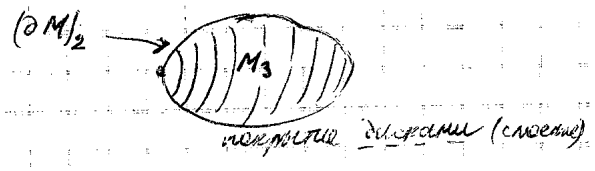


Original source details: GELFAND-P000926

Undeciphered passage. The word before ‘Levi-flat’ is heavily crossed out and illegible.

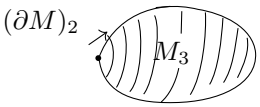
A Levi-flat region is foliated into compl. leaves

(On the torus this problem cannot be solved)



Original source details: GELFAND-P000926

Diagram and typed description. At the top of the oval is a short dark stroke whose direction is unclear.

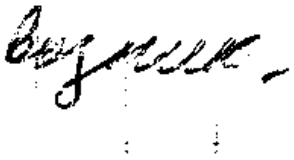


covering by discs (foliation)

GELFAND-P000927

[source number: 11]

There are sing. points of the foliation $\Rightarrow$ the index of the degen. (?) Riemann prob. $= 0 \Rightarrow$ it is solvable



Original source details: GELFAND-P000927

Uncertain reading. The abbreviation before ‘Riemann prob.’ is tentatively «вырожд.» or «вырож.» (‘degen.’).

(Bishop — there arises a sing. integro-diff. eq.)

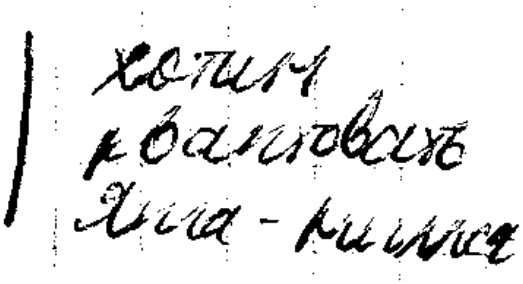
2-3.

M — a set, G — a comp. Lie group; g — a Lie alg.

$\Lambda^1(M, g)$ — 1-forms on M with vals. in g.

Original source details: GELFAND-P000927

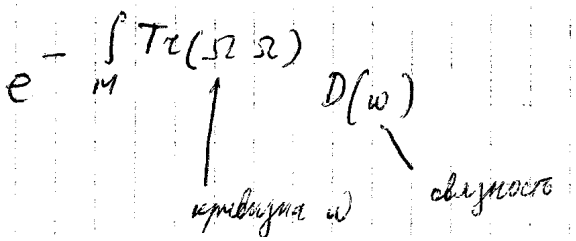
Uncertain reading. The subscript on Λ is crossed out. It may be the digit 2.



Original source details: GELFAND-P000927

Diagram and typed description. At right: ‘we want to quantize Yang–Mills’.

$$e^{-\int_M \mathrm{Tr}(\Omega\Omega)} D(\omega) \tag{1}$$



Original source details: GELFAND-P000927

Diagram and typed description. Curvature and connection are labelled below (1).

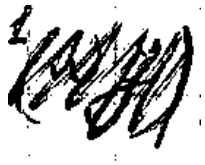
Ω ω

curvature of ωconnection

(Formal Feynman integral)

Group: $g \in C(M, G)$ (gauge)

$$g \cdot \omega = g^{-1} dg + g^{-1} \omega g \tag{2}$$



Original source details: GELFAND-P000927

Undeciphered passage. The expression crossed out immediately after the second g^{-1} is illegible.

Goal — construct a measure inv. under the gauge [crossed out] group.

Original source details: GELFAND-P000927

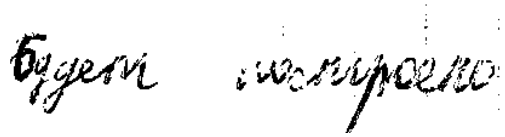
Uncertain reading. A short entry before ‘group’ is crossed out. It may read «гр.» (‘grp.’).

(easier than on the quotient space)

We must integrate over the set of paths in Λ^1.

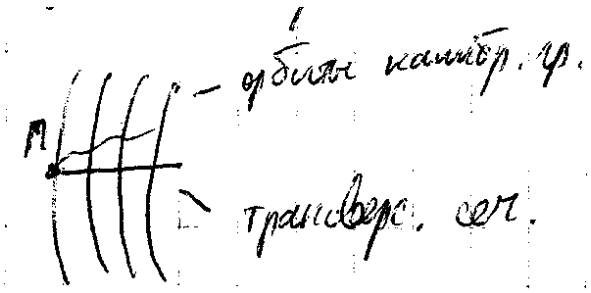
$$e^{-\int_0^t dt \int_M \mathrm{Tr}(\Omega_t \Omega_t)\, dx} D(\omega_t). \tag{3}$$

We will be constructed a family of measures, ‘approx.’ to the equil. measure.



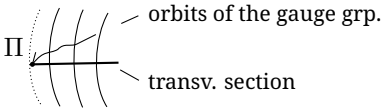
Original source details: GELFAND-P000927

Diagram and typed description. The manuscript reads «Будем построено», literally ‘we will be constructed’.



Original source details: GELFAND-P000927

Diagram and typed description. Orbits, the point Π and two lines issuing from it.

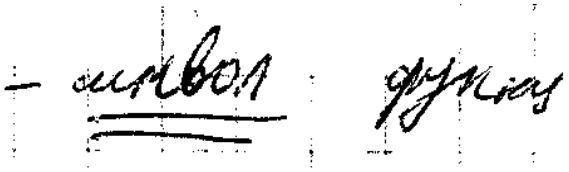


At first — a kind of Wiener measure on paths.

We need something like an ellipt. operator on the space of connections, inv. under the gauge group.

$$\Pi \to d_\Pi \text{ — covar. diff.}, \qquad \Box_\Pi = d_\Pi^* d_\Pi + d_\Pi d_\Pi^*$$

$$e^{-\tau \Box_\Pi}(x, y) = p_\tau^{(\Pi)}(x, y) \qquad \text{— symbol of the func.} \tag{4}$$



Original source details: GELFAND-P000927

Diagram and typed description. The word ‘symbol’ is double-underlined after (4).

$$L\varphi(\Pi) = \iint_{M\times M} e^{-\tau\Box_\Pi}(x,y)\frac{\delta^2\varphi}{\delta\Pi(x)\delta\Pi(y)}\,dx\,dy \tag{5}$$

[crossed out] τ fixed.

Original source details: GELFAND-P000927

Undeciphered passage. A sign or short word before τ is crossed out. The reading is unresolved.

One cannot work in Sob. spaces.

$$\Lambda^1_{k,\alpha}(M,g), \quad 0<\alpha<1, \tag{6}$$

— some space (Hölder)

The kth deriv. is Hölder with exponent α.

We need:

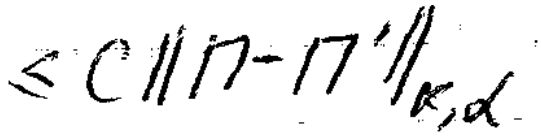
$$\left\|e^{-\tau\Box_\Pi} - e^{-\tau\Box_{\Pi'}}\right\|_{HS} \le C\|\Pi-\Pi'\| \tag{7}$$

[under HS: Hilb.–Schmidt.]

GELFAND-P000928

One obtains

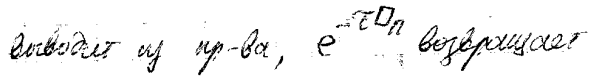
$$\left\|e^{-\tau\Box_\Pi} - e^{-\tau\Box_{\Pi'}}\right\|_{HS(L^2,L^2)} \le C\|\Pi-\Pi'\|_{k,\alpha} \tag{8}$$



Original source details: GELFAND-P000928

Diagram and typed description. In (8) there is a slanting stroke beneath $<$. The sign reads $\le$.

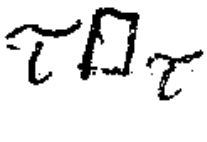
Techn. point: Brownian motion takes us out of the space, $e^{-\tau\Box_\Pi}$ brings us back.



Original source details: GELFAND-P000928

Diagram and typed description. Before the exponential the manuscript reads ‘takes us out of the space’.

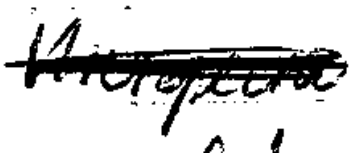
$$d\Pi_t = e^{-\tau\Box_\tau}\underbrace{d\beta_t}_{\text{white noise}}. \tag{9}$$



Original source details: GELFAND-P000928

Diagram and typed description. In (9) the subscript on $\Box$ has the same shape as the preceding τ.

[crossed out: ~~Interesting~~]



Original source details: GELFAND-P000928

Diagram and typed description. The word ‘Interesting’ is crossed out above (10).

$$\left(\tfrac{1}{2}\Delta - V\right)\Psi = \frac{\partial}{\partial t}\Psi \tag{10}$$

$$\Psi_t(x) = E_x\left(\exp\left(-\int_0^t V(b_\omega(s))\,ds\right)\Psi_0(b_\omega(t))\right) \tag{11}$$

In particular,

$$\pm\lambda_1 = \lim_{t\to\infty}\frac{\log E_x\left(\exp\left(-\int_0^t V(b_\omega(s))\,ds\right)\right.}{t} \tag{12}$$

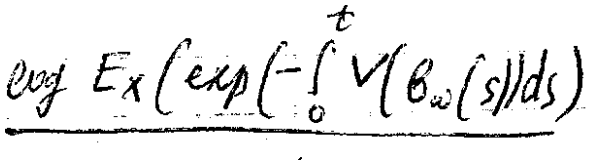


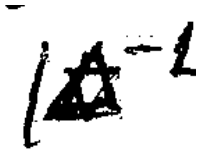
Diagram and typed description. The numerator of (12) has just one closing parenthesis after ds.

Original source details: GELFAND-P000928

(Interesting, e.g. when there are neg. levels)

Suppose

$$\left|\Delta^{-1}V\right| \le C < 1 \tag{13}$$

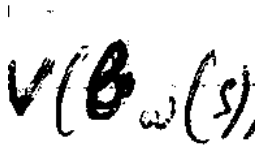


Uncertain reading. The sign before $^{-1}$ is overwritten. The outer contour reads as Δ. The earlier sign is unresolved.

Original source details: GELFAND-P000928

Then

$$E_x\left(\exp\left(-\int_0^\infty V(b_\omega(s))\,ds\right)\right) \le \frac{1}{1-C} \tag{14}$$



Uncertain reading. The letter in (14) is overwritten. The reading b_ω follows (11)–(12), with B_ω also possible.

Original source details: GELFAND-P000928

$\Rightarrow$ estimate the eigenvalue.

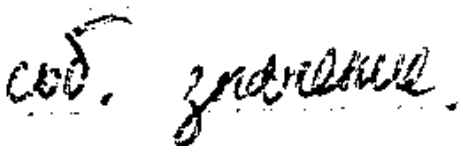
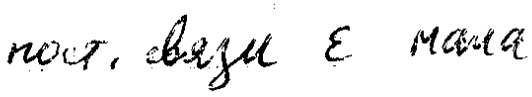


Diagram and typed description. The words ‘соб. значение’ read ‘eigenvalue’.

Original source details: GELFAND-P000928

$V \longmapsto \varepsilon V$ pot. (?) we took ε small $\Rightarrow$ something is estimated.



Uncertain reading. The abbreviation after εV is unclear. It may be ‘pot.’ (potential).

Original source details: GELFAND-P000928

GELFAND-P000929

3/XI-80 Gelfand Sem.

I.M.G. History of representation theory

1. Dedekind — characters of finite commut. groups (for L-fns.)
2. Frobenius — matrices and traces for noncommut. groups
3. Haar (died young)

(*)4. von Neumann — theory of C^*-algebras for repns. of noncommut. groups

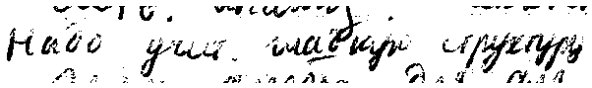
(**)5. Gelfand and Naimark — on repns. of the Lorentz grp.

Ideology (*): everything is set up in set-theoretic terms

(**). —//— smoothly and simply

Modern analysis: noncommut. diff. geometry and integral geometry.

One must take account of the smooth structure.



Original source details: GELFAND-P000929

Diagram and typed description. In the cramped line after ‘take account of’, the adjective reads ‘smooth’. A short stroke underlines its beginning.

Group algebra for a Lie alg. — enveloping algebra.

but there are still many of them (no Wedderburn thm.)

Original source details: GELFAND-P000929

Diagram and typed description. The manuscript spells the surname «Веддербарна».

Gelfand, Kirillov, skew field of fractions of the enveloping alg. ≅ skew field of fractions of the ring of diff. operators

This is unfinished. (Operators with polyn. coefficients are insufficient. This was partly understood in the work of [crossed out] Novikov, Krichever, Shabat, etc.)

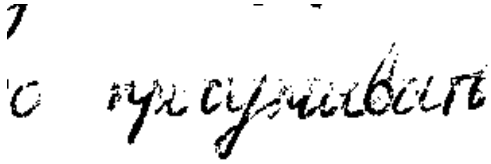
Original source details: GELFAND-P000929

Uncertain reading. The entry before ‘Novikov’ is heavily crossed out and illegible.

Bernstein, S. Gelfand, I.M. Gelfand in project. space.

One must work with sheaves.

Now one must listen to (?) what representation theory is being developed for, rather than to the theory itself.



Original source details: GELFAND-P000929

Uncertain reading. The opening strokes of the verb admit «прислушивать» and «прослушивать», both forms of ‘listen’.

GELFAND-P000930

[source number: 16]

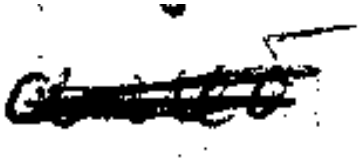
A. Beilinson

X — an algebr. set over $\mathbb{C}$

The simplest — affine.

They are uniquely determ. by their ring of fns.

Commut. ring = [crossed out: algeb. (?)] affine alg. set.



Original source details: GELFAND-P000930

Uncertain reading. The crossed-out entry after the equals sign is tentatively read «алгеб.» (‘algeb.’).

Modules over it = quasicoher. sheaves

$X = \operatorname{Spec} A$, M — an A-module (always the Zariski top.)

basis of open sets: $X_f = \{x : f(x) \neq 0\}$.

$O(X_f) = A_f$ — the localization of A at f

$$\mathcal{M}(X_f) = M_f = M \otimes_A A_f$$

We obtain a sheaf on Spec A. This gives the class of quasicoher. sheaves on X.

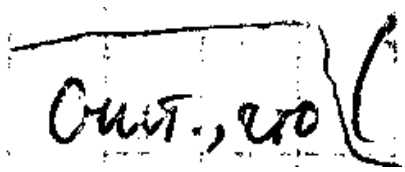
There will be a noncommut. analogue of localization. (passage from a mod. to a sheaf).

(1) Rings: univ. enveloping algs. of semisimple Lie algs. and their quotients.

X — an alg. set — is called $\mathcal{D}$-affine, if it has the property

($\mathcal{D}_X$ — the sheaf of rings of diff. operators on X)

X — smooth;



Original source details: GELFAND-P000930

Diagram and typed description. At left beside the smoothness line: ‘Ass., that’.

$\mathcal{M}$ — a sheaf of modules over $\mathcal{D}_X$, $O_X \subset \mathcal{D}_X$ (subsheaf, diff. operators of order 0)
$\mathcal{M}$ is called quasicoherent, if it is quasicoher. as a sheaf of O_X-modules.
(Example: $\mathcal{D}_X$ itself is quasicoherent).
X — an algebr. set, O_X — the sheaf of fns. on it; a sheaf $\mathcal{M}$ of O_X-modules on X
is called quasicoherent, if it has the following property:
$\forall$ affine set U and its aff. subset $U' \subset U$ there must be:

$$\mathcal{M}(U') = \mathcal{M}(U) \otimes_{O(U)} O(U') \tag{2}$$

(the right side of (2) clearly maps to the left, and we require this to be an isomorphism)
(3) Assertion. X is affine $\Longleftrightarrow \forall$ quasicoher. $\mathcal{M}$
(4) $H^i(X, \mathcal{M}) = 0$ and $\mathcal{M}$ is gen. over $O(X)$ by global sections, i.e.

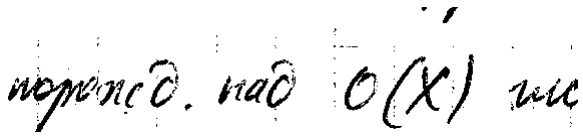


Diagram and typed description. In (4) the manuscript has $O(X)$. In the following line the tensor sign has no ring subscript.

Original source details: GELFAND-P000930

$$H^0(X, \mathcal{M}) \otimes O_X \underset{\text{epimorphism}}{\longrightarrow} \mathcal{M} \tag{5}$$

(A theorem that is not quite triv. — Serre, Grothendieck — see Serre's work on coher. sheaves)

GELFAND-P000931

(6). Definition. X is called $\mathcal{D}$-affine, if $\forall$ quasicoher. sheaf of $\mathcal{D}$-modules $\mathcal{M}$ the conditions (4), (5) hold, i.e.

Undeciphered passage. An earlier label is crossed out to the left of (7). It resembles (8), but the reading is unresolved.

Original source details: GELFAND-P000931

$$H^i(X, \mathcal{M}) = 0, \quad i > 0 \tag{7}$$

$$H^0(X, \mathcal{M}) \otimes \mathcal{D}_X \longrightarrow \mathcal{M} \longrightarrow 0 \tag{8}$$

(9) Examples.

1. $\forall$ aff. set — $\mathcal{D}$-affine.

2. $\forall$ project. space $\mathbb{P}^N$ — $\mathcal{D}$-affine.

(partic. case of the thm. to be discussed)

3. An elliptic curve — is not $\mathcal{D}$-affine.

(10) Consequence of $\mathcal{D}$-affinity

Let X be a set, $\mathcal{L}$ a sheaf of O-modules. Consider

(11) $\mathcal{D}_X \otimes_{O_X} \mathcal{L}$ — a quasicoher. sheaf of $\mathcal{D}_X$-modules

[crossed out above] (12) X — $\mathcal{D}$-aff. $\Rightarrow$ the cohomology of sheaf (11) is 0 and it is gen. by global sects. (i.e. (7) and (8) hold), even if $\mathcal{L}$ itself was bad

Undeciphered passage. A short word is crossed out above the start of (12). Its reading is unresolved.

Original source details: GELFAND-P000931

(13) (e.g., $O(-i)$ on $\mathbb{P}^N$, it has no sections and has higher cohomology)

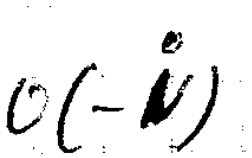


Diagram and typed description. A dot is visible above the heavily traced letter. It reads i.

Original source details: GELFAND-P000931

e.g.,

(14) $\mathcal{D}_X \otimes_{O_X} O(-i)$ on P^N is gen. by global sections

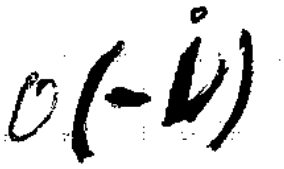


Diagram and typed description. In (14), the notation is again $O(-i)$, with i traced over.

Original source details: GELFAND-P000931

(the speaker does not know a direct proof)

Unfortunately there is no good notation for $\mathcal{D}_X$, since a diff. operator is not written globally as a sum of homogeneous ones.

(14) — a certain loc. free $\mathcal{D}_X$-module.

$\mathcal{D}_X$ in (11) is regarded as a right sheaf of O_X-modules

Uncertain reading. The last digit in the reference to (11) is heavily traced over.

Original source details: GELFAND-P000931

If we consider

$$O(-i) \otimes_{O_X} \mathcal{D}_X, \tag{15}$$

then this is no longer so. It is not gen. by global sects. and its higher cohom. is $\neq 0$ ($O(-i)$ is its direct summand)

[crossed out: ~~Elliptic curve~~] On the elliptic curve.

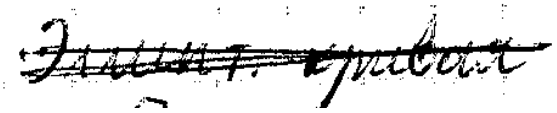


Diagram and typed description. The earlier heading is crossed out before the new one.

Original source details: GELFAND-P000931

(16) Suppose $\mathcal{D}_X$ has an involution, ident. on O_X, and $ab \overset{*}{\mapsto} ba$;

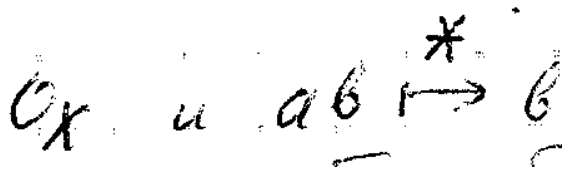


Diagram and typed description. An asterisk is written above the arrow from ab to ba.

Original source details: GELFAND-P000931

then (14) and (15) would be isomorphic and the set would not be $\mathcal{D}$-aff.

But on the ell. curve there is an involution — taking the formally adj. operator

$$f \cdot \frac{\partial}{\partial x} \mapsto \frac{\partial}{\partial x} f$$

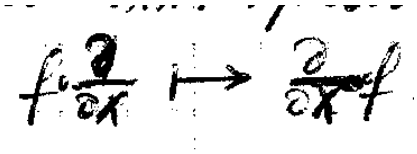


Diagram and typed description. There is no visible minus sign before the fraction on the right.

Original source details: GELFAND-P000931

(there are coords. — all operators are 'with const. coeffs.')

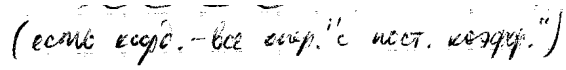


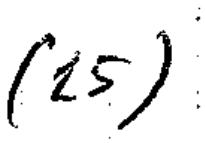
Diagram and typed description. The parenthesis opens with 'there are coords.'.

Original source details: GELFAND-P000931

GELFAND-P000932

[source number: 17]

The ring of fns. splits off from $\mathcal{D}_X \Rightarrow$ (15) is not acyclic



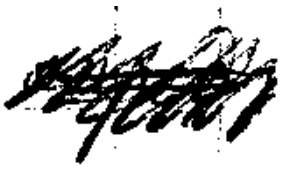
Uncertain reading. The first digit of the reference is overwritten. The link to the preceding frame supports reading (15).

Original source details: GELFAND-P000932

(17) Assertion. X — $\mathcal{D}$-affine, then [crossed-out writing unresolved] the section functor Γ

$$\mathcal{D}_X\text{-mod} \xrightarrow{\Gamma} \Gamma(\mathcal{D}_X)\text{-mod}, \tag{18}$$

is an equivalence of categories.



Original source details: GELFAND-P000932

Undeciphered passage. One crossed-out word between “then” and “the functor” is unresolved.

(The speaker does not know whether $\mathcal{D}$-affinity follows from this; it does in the commut. case)

Original source details: GELFAND-P000932

Uncertain reading. There is a small heavily inked mark after “The speaker”. Its function is unresolved.

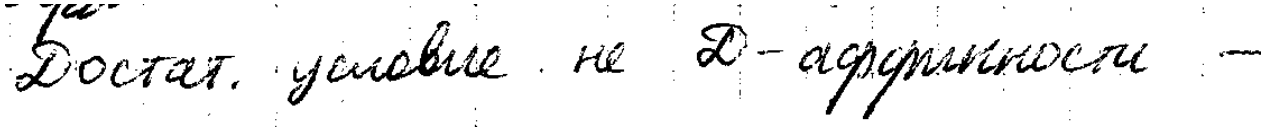
$$D_X \underset{\text{def}}{=} \Gamma(\mathcal{D}_X) \tag{19}$$

[crossed out: ~~Theorem~~ (?)]

Original source details: GELFAND-P000932

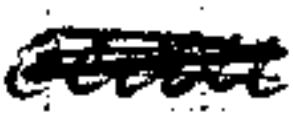
Uncertain reading. Crossed-out writing before (20) resembles “Theorem”.

(20) Suff. condition for failure of $\mathcal{D}$-affinity — [crossed-out writing unresolved] ch_1 (of the sheaf of forms of max. degree) $= 0$



Original source details: GELFAND-P000932

Diagram and typed description. No hyphen is visible between the Russian negation and $\mathcal{D}$ in the heading.



Original source details: GELFAND-P000932

Undeciphered passage. The crossed-out writing before ch_1 is unresolved.

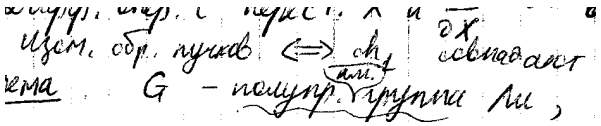
$$(\Leftrightarrow ch_1(\text{tangent bundle}) = 0) \tag{21}$$

$$(20) \Leftrightarrow (21) \Leftrightarrow \text{existence of an involution.} \tag{22}$$

(Diff. op. with x and $\frac{\partial}{\partial x}$ interchanged is a diff. op. on forms of max degree.)

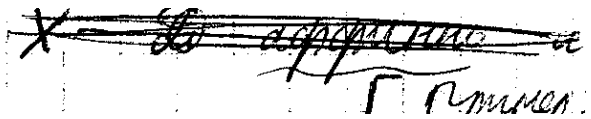
[interlinear insertion:] Isom. of inv. sheaves $\Leftrightarrow ch_1$ coincide

(23) Theorem. G — semisimple [inserted above and circled: alg. (?)] Lie group, B — Borel subgr., consid. $X = G/B$ — («flag space»). then [crossed out: ~~X — $\mathcal{D}$-affine and~~]



Original source details: GELFAND-P000932

Uncertain reading. The circled insertion resembles “alg.”. A leader descends toward “Lie group”. The exact reading is unresolved.



Original source details: GELFAND-P000932

Diagram and typed description. “X is $\mathcal{D}$-affine and” is crossed out after “Then”.

1. X — $\mathcal{D}$-affine
2. $D_X = U(\mathfrak{g})/U(\mathfrak{g})I_0$;

[in the right margin, with a square bracket at left]

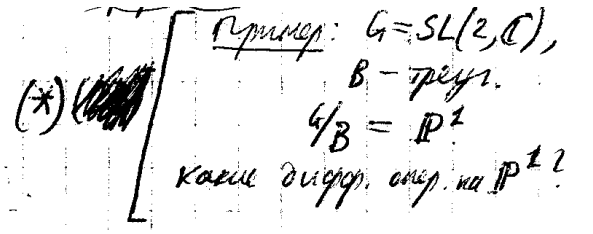
(*) ([crossed-out writing unresolved])

Example: $G = SL(2, \mathbb{C})$,

B — triang.

$G/B = \mathbb{P}^1$

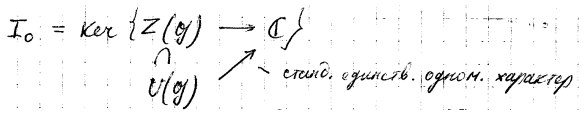
which diff. op. are there on $\mathbb{P}^1$?



Original source details: GELFAND-P000932

Undeciphered passage. The parenthesized writing before “Example” is crossed out. The exact reading is unresolved.

where



Original source details: GELFAND-P000932

Diagram and typed description. Diagram (24). The mark between $Z(\mathfrak{g})$ and $U(\mathfrak{g})$ resembles $\cap$.

$$I_0 = \ker\{\, Z(\mathfrak{g}) \longrightarrow \mathbb{C}\}$$
$$\cap \quad \nearrow$$
$$U(\mathfrak{g})$$

(24) — stand. unique one-dim. character

I_0 — Laplace operators without constant term

[crossed out: ~~Assertion~~ (?)] Proposition.

Original source details: GELFAND-P000932

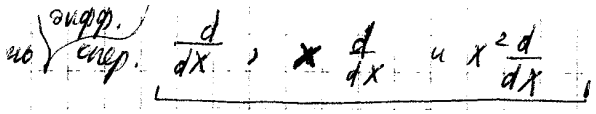
Uncertain reading. The crossed-out word before “Proposition” resembles “Assertion”.

(25) D_X-modules $\Leftrightarrow$ representations $\mathfrak{g}$, on which there acts trivially I_0.

($Z(\mathfrak{g})$ must act as on the triv. one-dimensional representation)

Example (*).

(26) On $\mathbb{P}^1$ there are [inserted above: diff.] op.



Original source details: GELFAND-P000932

Diagram and typed description. A bracket underneath groups the three operators in (26). A curved leader joins the inserted “diff.” to “op.”.

$$\frac{d}{dx}, \quad x\frac{d}{dx} \quad \text{and} \quad x^2\frac{d}{dx}$$

Take polynomials in them. This is the enveloping alg.

There is a unique op. commuting with all of them. (Shvartsman?)

Original source details: GELFAND-P000932

Uncertain reading. The surname is tentatively read as Shvartsman. The question mark is in the manuscript.

Moral: representation theory reduces to the theory of $\mathcal{D}_X$-modules.

Improvement over what was known: one can localize!!

GELFAND-P000933

10.11.80 Gelfand Sem.

Original source details: GELFAND-P000933

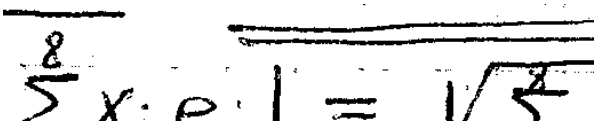
Uncertain reading. There are several isolated strokes above the date. No letters or digits can be distinguished.

I. M. G. Cayley numbers.

$$\left|\sum_{j=1}^{8} x_j e_j\right| = \sqrt{\sum_{1}^{8} x_j^2} \tag{1}$$

Original source details: GELFAND-P000933

Uncertain reading. A heavily overwritten or crossed-out stroke appears to the left of the digit in (1). The earlier mark is unresolved.



Original source details: GELFAND-P000933

Uncertain reading. The upper limit under the radical touches its bar. It is tentatively read as 8 by comparison with the left side.

In the Cayley numbers:

$$|ab| = |a||b| \tag{2}$$

How to prove there are no > 8 dim.?

From (2) one gets a problem in repr. theory and an explicit construction.

Question: perhaps it exists in an ∞-dimensional space? (with cond. (2))

Gindikin asserts that it does. We must repeat the proof that in the finite-dimensional case there are none besides $\mathbb{R}, \mathbb{C}, \mathbb{H}$ and Cayley.

Projective geometries (planes).

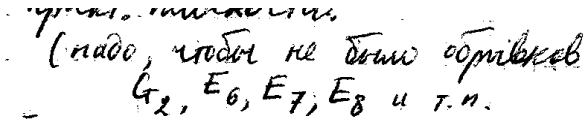
Introduce coord., associativity $\Leftrightarrow$ Desarguesian property

The Cayley proj. plane is the simplest non-Desarguesian one

It is interesting to study differ. and locally homogeneous proj. planes.

— The prob. of classifying Lie algebras is too particular (there should be no fragments

G_2, E_6, E_7, E_8 etc.



Original source details: GELFAND-P000933

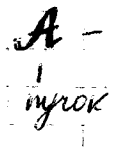
Diagram and typed description. The list of types is written below "there should be no fragments". No closing parenthesis is visible.

Beilinson. On localization of representations.

X — alg. variety (over $\mathbb{C}$),

$\mathcal{A}$ — quasi-coherent O_X-module

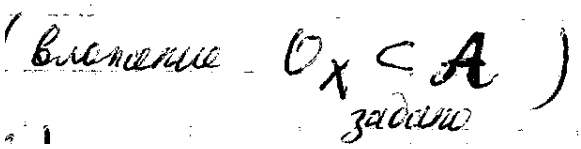
| a sheaf of rings containing O_X.



Original source details: GELFAND-P000933

Diagram and typed description. A short vertical stroke descends from $\mathcal{A}$ to the next line.

(the inclusion $O_X \subset \mathcal{A}$ [inserted below: is given])



Original source details: GELFAND-P000933

Diagram and typed description. "Is given" is inserted beneath the inclusion sign.

(E.g., $\mathcal{A}$ is a sheaf of diff. operators)

$\mathcal{A}$-mod — the category of sheaves of left $\mathcal{A}$-mod., that are quasi-coherent O_X-modules

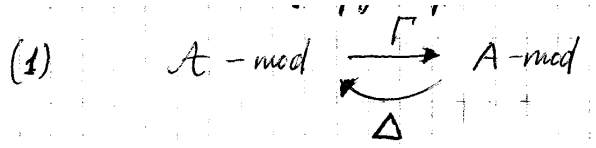
GELFAND-P000934

[source number: 20]

(Last time $\mathcal{A}$ was diff. op.)

$A = \Gamma(\mathcal{A})$, A-mod — categ. of A-modules (left).

There are 2 adj. functors.



Original source details: GELFAND-P000934

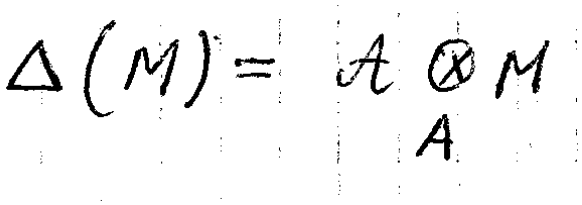
Diagram and typed description. In (1) the Γ arrow points right, with the curved Δ arrow pointing left below it.

$$\mathcal{A}\text{-mod} \underset{\Delta}{\overset{\Gamma}{\rightleftarrows}} A\text{-mod}$$

(1)

$$\Delta(M) = \mathcal{A} \underset{A}{\otimes} M, \tag{2}$$

M is consid. as a const. sheaf of $\mathcal{A}$-mod. over X (stalk $= M$ at each point) and the maps are trivial.



Original source details: GELFAND-P000934

Diagram and typed description. In (2) A is written directly beneath the tensor sign.

Original source details: GELFAND-P000934

Diagram and typed description. A calligraphic $\mathcal{A}$ precedes "mod. over X", as on the left in (1).

It is easy to see that they are adjoint.

(3) Defn.-lemma. [crossed out: ~~a)~~ (?)]

Original source details: GELFAND-P000934

Uncertain reading. The sign after the heading is crossed out. It may be "a)".

A variety X is called $\mathcal{A}$-affine, if one of the 2 equiv. conditions holds

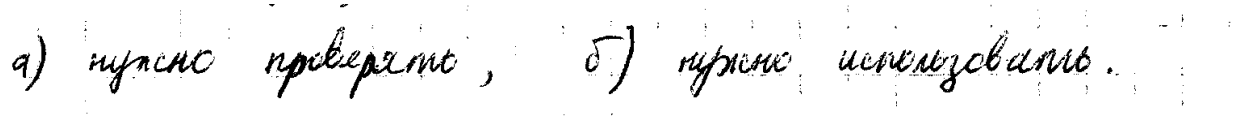
a) $\forall\, \mathcal{M} \in \mathcal{A}\text{-mod}, \quad H^i(X, \mathcal{M}) = 0, \quad i > 0$ and

$$H^0(X, \mathcal{M}) \neq 0$$
when $\mathcal{M} \neq 0$

b) The functors Γ and Δ are mutually inverse equivalences of categories.

(In fact a) is equiv. to what we had last time, we change the modules)

a) is what we need to check, b) is what we need to use.



Original source details: GELFAND-P000934

Diagram and typed description. The same word, “need”, is written before both “check” and “use”.

Proof of the equivalence of a) and b).

a) $\Rightarrow$ b).

$$\Gamma(\mathcal{M}) = \mathrm{Hom}_{\mathcal{A}\text{-mod}}(\mathcal{A}, \mathcal{M}) \tag{4}$$

(the image of 1 is a section of $\mathcal{M}$).

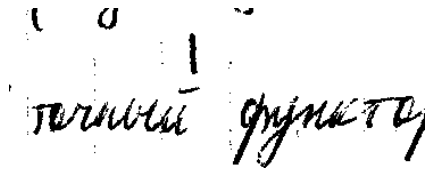
(5) a) $\Rightarrow \mathcal{A}$ — a projective generator in the categ. $\mathcal{A}$-mod.

(6) (an object such that i) $\mathrm{Hom}(\mathcal{A}, \mathcal{M})$ — exact functor

(7) ii) $\forall$ nonzero object $\mathcal{M}$; $\mathrm{Hom}(\mathcal{A}, \mathcal{M}) \neq 0$)

(i) follows from acyclicity, and ii) from the 2nd condition)

(exact functor — takes exact seq. to exact ones)



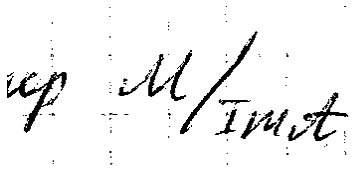
Original source details: GELFAND-P000934

Uncertain reading. A detached vertical stroke stands above the end of “exact”. It may be an exclamation mark.

i) — the property of projectivity

ii) — generator

(by the way, ii) $\Rightarrow$ we can map $\mathcal{A}$ to the quotient $\mathcal{M}/\operatorname{Im} t$ and so on)

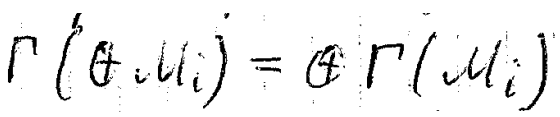


Original source details: GELFAND-P000934

Diagram and typed description. The symbol after Im in the denominator reads t.

(8) iii) Γ takes direct sums to direct sums.

$$\Gamma(\oplus_i \mathcal{M}_i) = \oplus \Gamma(\mathcal{M}_i)$$



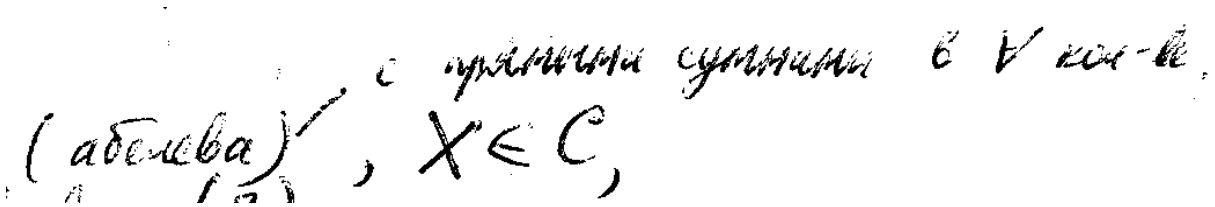
Original source details: GELFAND-P000934

Uncertain reading. A small stroke beside the left direct-sum sign is tentatively read as an index i. There is no visible counterpart at the right sign.

(a property of all quasi-coh. sheaves)

General categorical fact

(9) Lemma. Let there be given ∀ category C (abelian) [inserted above: with direct sums in ∀ quantity.], $X \in C$,



Original source details: GELFAND-P000935

Diagram and typed description. A leader from "(abelian)" points to the insertion about direct sums.

X — Proj. generator with property (8):

Uncertain reading. The X at the start of the line is heavily retraced. The writing underneath is unresolved.

Original source details: GELFAND-P000935

(10) $\mathrm{Hom}_C(X, \cdot)$ commutes with direct sums.

(11) [crossed out: ~~Then~~]

$$\mathrm{Hom}_C(X, \cdot) : C \longrightarrow \mathrm{End}_C X\text{-mod}$$

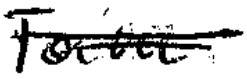


Diagram and typed description. "Then" is crossed out before formula (11).

Original source details: GELFAND-P000935

Then this functor is an equiv. of categories.

Everything follows from the lemma if we apply it to the case where

$$C = \mathcal{A}\text{-mod}, \quad X = \mathcal{A}, \quad \mathrm{End}_C X = A \tag{12}$$

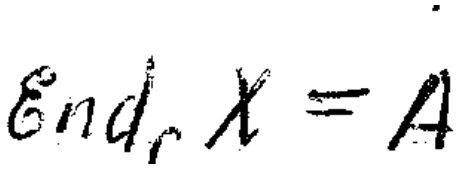


Diagram and typed description. The right side of (12) reads $\mathrm{End}_C X = A$.

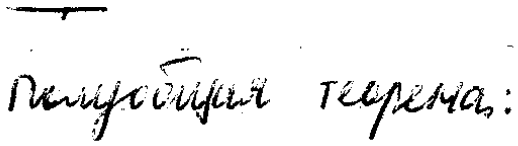
Original source details: GELFAND-P000935

The conv. implication b) ⇒ a) from the Čech resolution.

$\mathcal{D}$-affine variety, a variety that is $\mathcal{A}$-aff. for $\mathcal{A} = \mathcal{D}_X$.

(13) Thm. $\mathbb{P}^N$ — $\mathcal{D}$-affine.

The following theorem:



Original source details: GELFAND-P000935

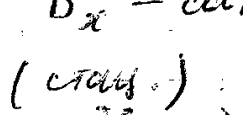
Diagram and typed description. The heading begins with “Следующая”, meaning “The following”.

G — a reductive group.

$X = G/B$ — flag space

(B — Borel subgroup)

(14) $x \in X \Rightarrow \underset{\text{(stab.)}}{B_x}$ — the corresp. Borel subgroup

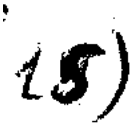


Original source details: GELFAND-P000935

Diagram and typed description. “(stab.)” is inserted beneath B_x.

G — acts on $X \Rightarrow \mathfrak{g}$ acts on X.

$$\mathfrak{g} \longrightarrow \text{Vect}\, X \tag{15}$$

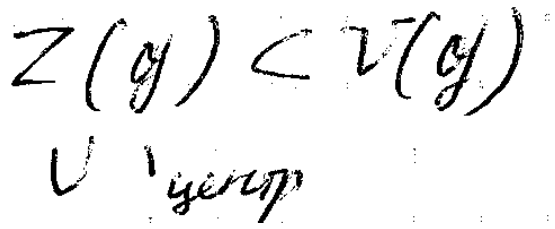


Original source details: GELFAND-P000935

Uncertain reading. The second digit in (15) is heavily retraced and reads 5.

hence.

$$U(\mathfrak{g}) \longrightarrow D_X = \Gamma(\mathcal{D}_X). \tag{16}$$



Original source details: GELFAND-P000935

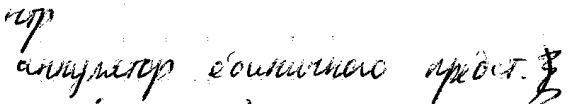
Diagram and typed description. A short hooked stroke under $Z(\mathfrak{g})$ stands next to the label “center”.

$Z(\mathfrak{g}) \subset U(\mathfrak{g})$

$\cup$ ' center

(17)

I — the annihilator of the unit repr. [?]



Original source details: GELFAND-P000935

Uncertain reading. The last sign after “repr.” is overwritten. The exact reading is unresolved.

(max ideal in the center)

$$J = I \cdot U(\mathfrak{g}). \tag{18}$$

(19) Theorem. a) X — $\mathcal{D}$-affine.

b) under the morphism (16) one obtains an isomorphism

$$U(\mathfrak{g})/J \xrightarrow{\sim} D_X \tag{20}$$

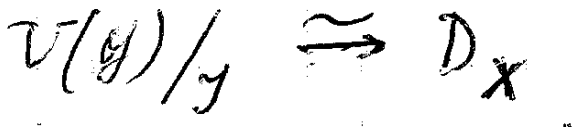


Diagram and typed description. In (20) a single arrow has a tilde above it.

Original source details: GELFAND-P000935

(21) hence, equiv. of the categories of 1) $\mathcal{D}$-sheaves on X and 2) representations of the Lie alg. $\mathfrak{g}$ with unit infinitesim. character.

GELFAND-P000936

[source number: 21]

[crossed out: ~~Definition~~ (?)] Digression.

Uncertain reading. The crossed-out heading resembles “Definition”. The exact reading is unresolved.

Original source details: GELFAND-P000936

Diagram and typed description. The new heading reads “Отступление”, or “Digression”.

Original source details: GELFAND-P000936

X, Y — both smooth alg. varieties.

Diagram and typed description. The word after X, Y reads “both”.

Original source details: GELFAND-P000936

$X \xrightarrow{f} Y$ a map.

Then there are the categories $\mathcal{D}_X$-mod and $\mathcal{D}_Y$-mod

There are 2 functors on sheaves f^* and f_*.

But in diff. op. there is no corresp. morphism.

But the (natural) functors do exist, they are def. at the level of the derived category

(actually there are 4 functors)

(22) f° — inverse image in the cat. of sheaves.

$$O_X \underset{f^\circ O_Y}{\otimes} f^\circ(\mathcal{D}_Y) \tag{23}$$

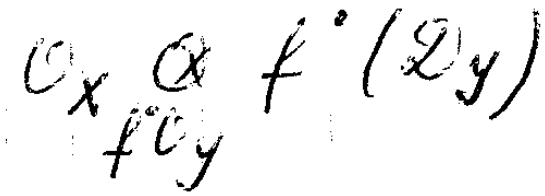


Diagram and typed description. In (23) the marks above f look like small hollow circles, as in (22).

Original source details: GELFAND-P000936

$$\mathcal{D}_{X\to Y} \underset{\text{def}}{=} O_X \underset{f^\bullet O_Y}{\otimes} f^\bullet \mathcal{D}_Y \tag{24}$$

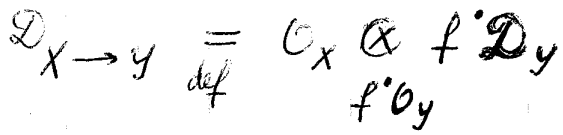
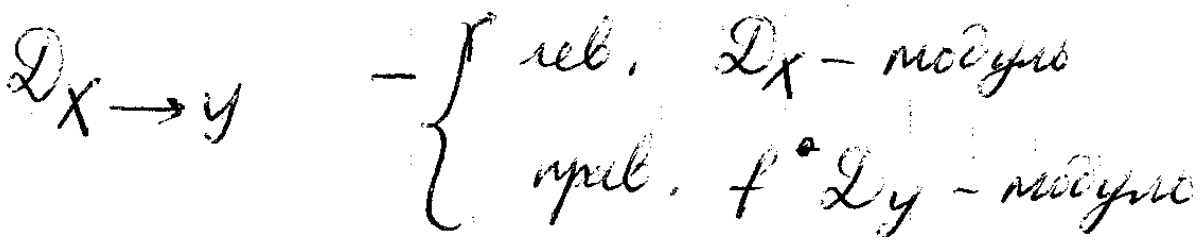


Diagram and typed description. In (24) the marks above both occurrences of f are filled dots.

Original source details: GELFAND-P000936

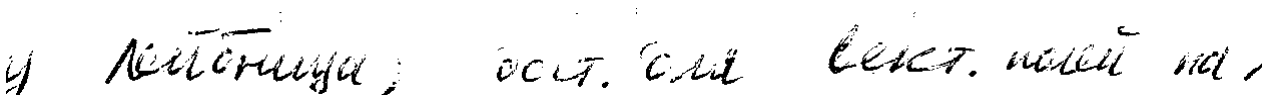
$$\mathcal{D}_{X\to Y} - \begin{cases} \text{left } \mathcal{D}_X\text{-module} \\ \text{right } f^\circ\mathcal{D}_Y\text{-module} \end{cases} \tag{25}$$



Original source details: GELFAND-P000936

Diagram and typed description. In the lower line of (25) the mark above f has an open center.

left $\mathcal{D}_X$ by the Leibniz rule; rem. for vec. fields on X:

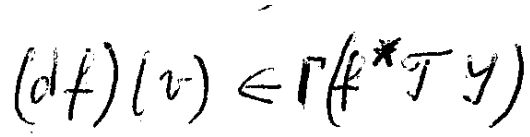


Original source details: GELFAND-P000936

Diagram and typed description. After the semicolon the manuscript has the abbreviation «ост.» (“rem.”).

$v \in$ Vect X;

$$(df)(v) \in \Gamma(f^* T\, Y)$$



Original source details: GELFAND-P000936

Diagram and typed description. After f^*, T and Y are written on the same line.

$(df)(v)$ acts on $\underline{\underline{f^\bullet \mathcal{D}_Y}}$.

Original source details: GELFAND-P000936

Uncertain reading. A heavy slanted stroke stands to the left of the last line. There is no visible separate dot below it.

Original source details: GELFAND-P000936

Diagram and typed description. A filled dot stands above f. The expression is double underlined.

GELFAND-P000937

17.11.80 Gelfand Sem.

I. M. G. problems.

1.

$$f(x) = \begin{cases} 1/q, & x = p/q \text{ — in lowest terms} \\ 0, & x \text{ — irrat.} \end{cases}$$

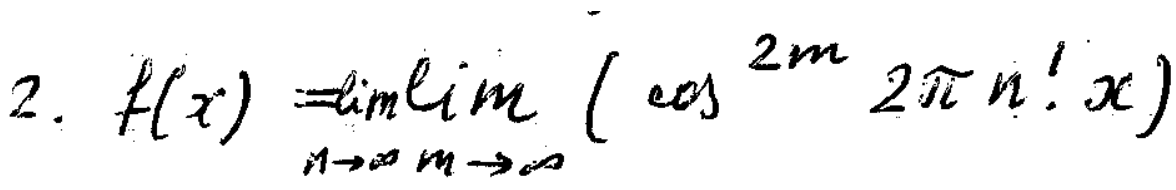
Where is it continuous?

where is it differentiable?

$$f(x) = \begin{cases} 1/q^\alpha, & x = p/q \text{ — in lowest terms} \\ 0, & x \text{ — irrat.} \end{cases}$$

What can be said about diff.?

2. $f(x) = \lim\limits_{n\to\infty} \lim\limits_{m\to\infty} (\cos^{2m} 2\pi n! x)$ — the Dirichlet fn.



Original source details: GELFAND-P000937

Diagram and typed description. The n in the cosine argument is followed by a factorial sign. The inner limit is over m.

3. Can the Dirichlet fn. be constructed with one limit sign?

(from cont. fns.)

[at upper right]

4. Calculate approximately

$$\int_{-\infty}^{\infty} e^{-x^2} \cos 1000x \, dx$$

The same with e^{-x^4} or e^{-x^6}

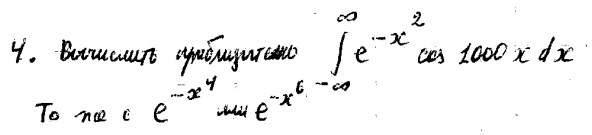


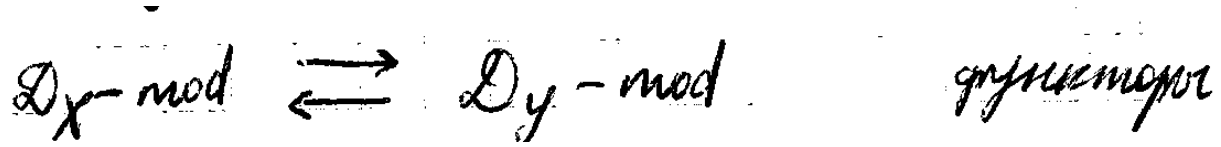
Diagram and typed description. Problem 4 is written in the upper right. The integral has $1000x$.

Original source details: GELFAND-P000937

A. Beilinson.

$X \xrightarrow{f} Y$, X, Y — varieties.

$\mathcal{D}_X$-mod, $\mathcal{D}_Y$-mod — categories of quasi-coher. sheaves of $\mathcal{D}_X$-mod and $\mathcal{D}_Y$-mod on X and Y



Original source details: GELFAND-P000937

Diagram and typed description. The upper arrow points right and the lower arrow points left. The arrows carry no labels.

$\mathcal{D}_X$-mod $\rightleftarrows$ $\mathcal{D}_Y$-mod

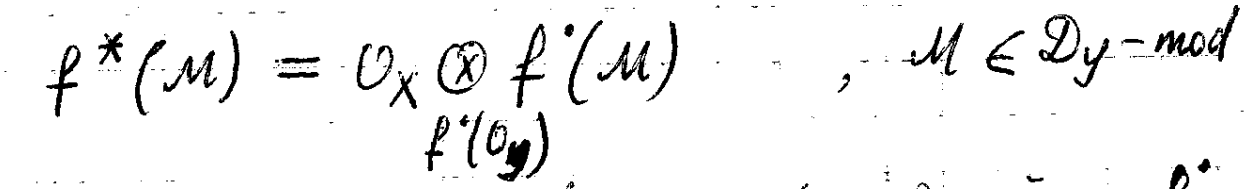
functors

Let us define:

$$f^* : \mathcal{D}_Y\text{-mod} \longrightarrow \mathcal{D}_X\text{-mod}$$

$$f^*(\mathcal{M}) = O_X \underset{f^\bullet(O_Y)}{\otimes} f^\bullet(\mathcal{M}), \qquad \mathcal{M} \in \mathcal{D}_Y\text{-mod} \tag{1}$$

(as a sheaf — inverse image in the categ. of O-modules, $f^\bullet$ — the abstract inverse image)



Original source details: GELFAND-P000937

Diagram and typed description. Both occurrences of f on the right of (1) have filled dots. The star is on the left.

Action of diff. operators: vec. fields suffice $v \in \mathrm{Vect}\, X$,

$$\ell \otimes m \in f^*(\mathcal{M}) \tag{2}$$

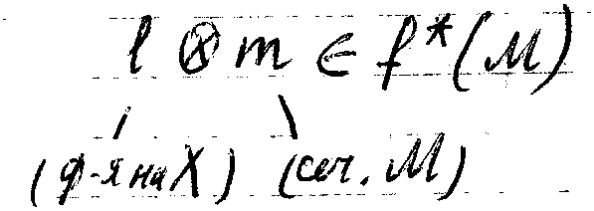


Diagram and typed description. Two leaders appear below the factors in (2).

$\ell \otimes m \in f^*(\mathcal{M})$

(fn. on X) (sect. of $\mathcal{M}$)

Original source details: GELFAND-P000937

$$df(v) = \sum_{\substack{\tau_i \\ | \\ \text{fn. on } X}} \otimes \frac{\partial}{\partial y_i} \in f^*(T_Y), \qquad df : T_X \longrightarrow f^* T_Y \tag{3}$$

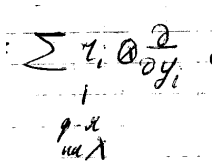


Diagram and typed description. The note under τ_i reads “fn. on X” in the singular.

Original source details: GELFAND-P000937

Instead of (1), one can write more simply

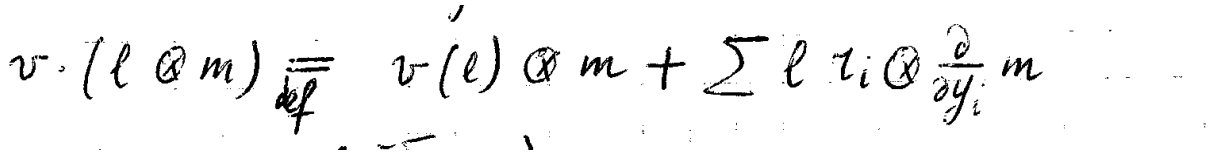
$$f^*(\mathcal{M}) = O_X \underset{O_Y}{\otimes} \mathcal{M} \tag{4}$$

In this same sense:

$$f^*(T_Y) = O_X \underset{O_Y}{\otimes} T_Y \tag{5}$$

$$v \cdot (\ell \otimes m) \underset{\text{def}}{=} v(\ell) \otimes m + \sum \ell \tau_i \otimes \frac{\partial}{\partial y_i} m \tag{6}$$

(by the Leibniz formula).



Original source details: GELFAND-P000937

Diagram and typed description. The equality in (6) has def below it. The coefficient in the sum reads $\ell\tau_i$.

GELFAND-P000938

[source number: 24]

If $\mathcal{M}$ comes from a vec. bundle with an integr. [inserted below: (flat)] connection, then this is the usual inverse image of the connection.

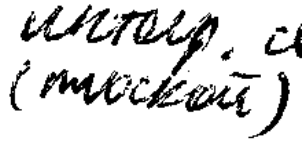


Diagram and typed description. “(Flat)” is inserted beneath “integr.”.

Original source details: GELFAND-P000938

(7) Examples.

1. $f^*O_Y = O_X$

2. From a sheaf [crossed out: unresolved sign] of fin. type there may result [inserted above: a sheaf] of infinite type

Undeciphered passage. The sign after "sheaf" is heavily crossed out. It may be $\mathcal{D}$.

Original source details: GELFAND-P000938

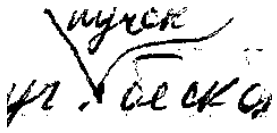


Diagram and typed description. "Sheaf" is inserted above the line with a leader.

Original source details: GELFAND-P000938

X — a point, Y — a var..

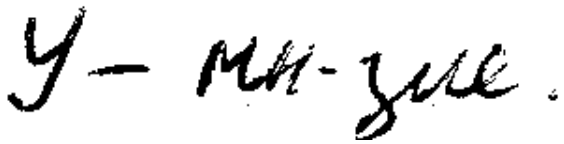


Diagram and typed description. The abbreviation after Y reads «мн-зие» ("var.").

Original source details: GELFAND-P000938

$f^*(\mathcal{D}_Y)$ is infinite-dimensional (as a vec. space)

$$\mathcal{D}_X = \mathbb{C}$$

$f^*(\mathcal{D}_Y)$ — diff. op. with const. coefficients

3. X, Y as above, $\mathcal{M}$ — δ-fns. at the point $X \in Y \underset{\text{def}}{=} \delta_x$

(generalized fns. concentr. at a point).

Then

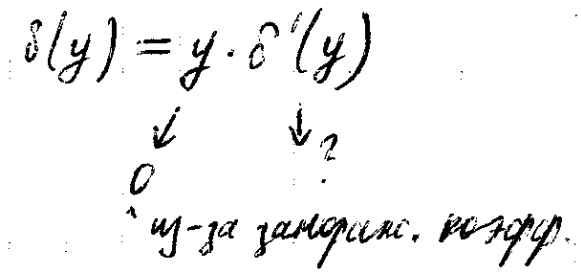
$$f^*(\delta_x) = 0,$$

since $\forall$ generalized fn. [inserted above: at a pt.] can be repres. in the form

$$\sum \alpha_i s_i, \qquad \alpha_i(X) = 0.$$

Diagram and typed description. "At a pt." is inserted above the line with a leader.

Original source details: GELFAND-P000938



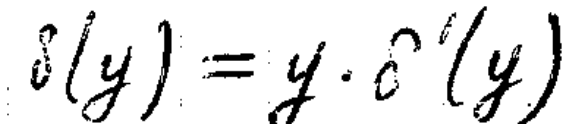
Original source details: GELFAND-P000938

Diagram and typed description. An arrow leads from y to 0, and one from $\delta'(y)$ to "?".

$\delta(y) = \; y \cdot \delta'(y)$

0 ?

due to freez... coeff.



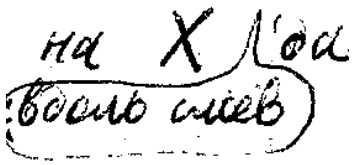
Original source details: GELFAND-P000938

Diagram and typed description. There is an equals sign with no visible minus before y.

Direct image is def. for right mod. (integr. forms only!, and they are a right mod.)

$$f_* \quad \text{mod-}\mathcal{D}_X \longrightarrow \text{mod-}\mathcal{D}_Y$$

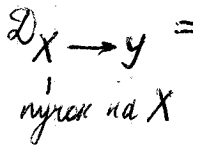
(integr. the volume forms on X [inserted below and circled: along the fibers] gives a volume form on Y)



Original source details: GELFAND-P000938

Diagram and typed description. The insertion "along the fibers" is circled below the line.

$$\underset{\text{a sheaf on } X}{\mathcal{D}_{X\to Y}} \quad = f^*(\mathcal{D}_Y) \in \mathcal{D}_X\text{-mod-}\mathcal{D}_Y \tag{8}$$



Original source details: GELFAND-P000938

Diagram and typed description. A vertical leader extends down from the sheaf symbol in (8).

($\mathcal{D}_X$-module and a right $\mathcal{D}_Y$-module)

$$f_*(\mathcal{M}) = f_\cdot\left(\mathcal{M} \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X\to Y}\right) \tag{9}$$

$f_\cdot$ — direct image on sheaves

[crossed out: earlier number unresolved] (10) Examples.

Original source details: GELFAND-P000938

Uncertain reading. An earlier number is crossed out before (10). Its digit is unresolved.

1. $X = x \in Y$. Consid. the stand. 1-dim. $\mathcal{D}_X$-module $(= \mathbb{C})$

Diagram and typed description. The abbreviation before "1-dim." reads "stand.".

Original source details: GELFAND-P000938

$f_*(\mathcal{M})$ — (generalized fns. concentr. at the point on Y) $= \delta_x$

$$f^*(\mathcal{D}_Y) = \mathcal{D}_Y / \sum y_i D_Y$$

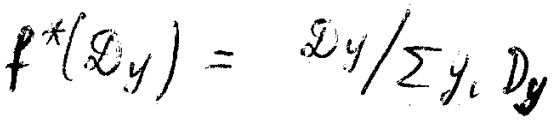
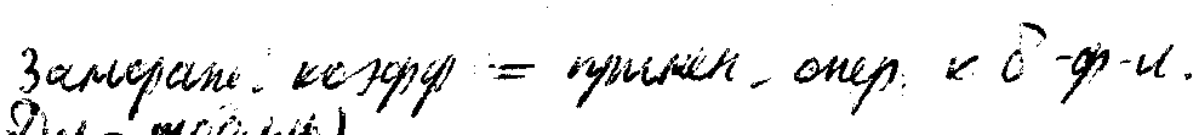


Diagram and typed description. The D in the denominator appears plain, without the calligraphic flourish of the numerator.

Original source details: GELFAND-P000938

[at right] Freez... (?) coeff. $=$ applying an op. to the δ-fn..



Original source details: GELFAND-P000938

Uncertain reading. The start of the word is tentatively read as "freez...". The exact ending is unresolved.

$(\Rightarrow \mathcal{D}_Y$-module)

GELFAND-P000939

2. Let us generalize:

$$X \xrightarrow{f} Y \qquad \text{embedding}$$

$f_*(\mathcal{M})$ by defn. — extension by means of δ-fns. in the transv. direction.

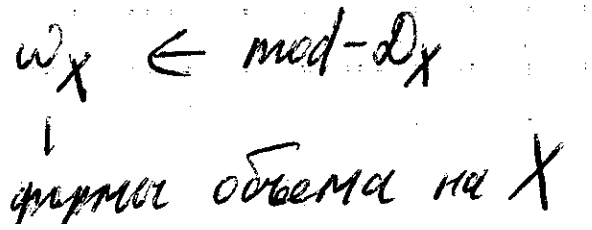


Diagram and typed description. A short stroke beneath ω_X leads to the caption.

$\omega_X \in$ mod-$\mathcal{D}_X$
|
volume forms on X

Original source details: GELFAND-P000939

$f_*(\omega_X)$ — generalized fns. concentr. on X and smooth in the dir. of X.

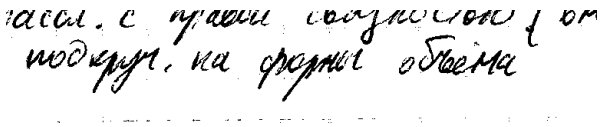
3. more generally:

[crossed out: earlier number unresolved] Vector bundle with a right connection (in place of volume forms)

Uncertain reading. The crossed-out digit resembles 4. The exact reading is unresolved.

Original source details: GELFAND-P000939

[added below: twist. by volume forms]



Original source details: GELFAND-P000939

Diagram and typed description. "Twist. by volume forms" is added beneath the line about the connection.

Relation to cohomology.

[crossed out: earlier number unresolved] Y — a point, X — a var.

Original source details: GELFAND-P000939

Uncertain reading. The earlier number before "Y is a point" is crossed out. It may be 4.

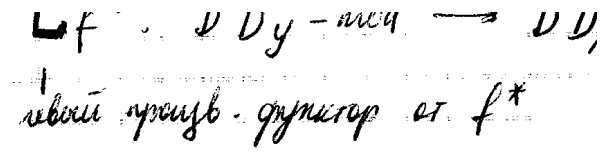
$f^*(\mathcal{M})$ — the sheaf of fns., rep. in the requisite number of copies.

f_* here computes the cohomology of the space X.

Higher direct images.

$$\begin{array}{c} Lf^* \\ | \end{array} : D\,D_Y\text{-mod} \longrightarrow D\,D_X\text{-mod} \tag{11}$$

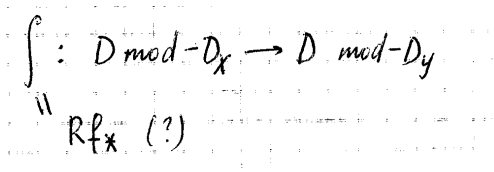
left derived functor of f^*.



Original source details: GELFAND-P000939

Diagram and typed description. A vertical stroke below L leads to the caption.

D — derived category

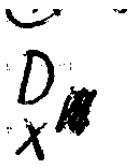


Original source details: GELFAND-P000939

Diagram and typed description. Two slanted strokes beneath the integral sign lead to "Rf_* (?)".

$$\begin{array}{l} \int : D\,\text{mod-}D_X \\ \quad \backslash\backslash \quad \longrightarrow D\,\text{mod-}D_Y \\ \quad Rf_*\,(?) \end{array} \tag{12}$$

$$\int \mathcal{M} = Rf_\bullet \left(\mathcal{M} \overset{L}{\underset{D_X}{\otimes}} \mathcal{D}_{X \to Y} \right) \quad \text{— a complex} \tag{13}$$



Original source details: GELFAND-P000939

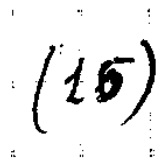
Uncertain reading. An X appears below D. The earlier sign crossed out beside it is unresolved.

$$\int^i \mathcal{M} = H^i\left(\int \mathcal{M}\right) \tag{14}$$

(Sato, Kashiwara)

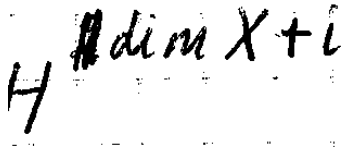
(15) Theorem. Y — a point $\Rightarrow$

$$\int^i \omega_X = H^{\dim X+i}(X) \tag{16}$$



Original source details: GELFAND-P000939

Diagram and typed description. The formula is numbered (16). The lower loop of the last digit is closed.



Original source details: GELFAND-P000939

Undeciphered passage. A sign is crossed out before dim $X + i$ in the exponent. The exact reading is unresolved.

(X — nonsingular over $\mathbb{C}$)

GELFAND-P000940

[source number: 25]

S. Gelfand

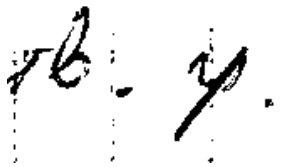
Derived category.

Original source details: GELFAND-P000940

Diagram and typed description. The heading has “category” in the singular.

Prob. 1. If there are 2 compl. with isom. cohomologies, one cannot always extend an isom. of cohom. to a map of compl. inducing an isomorphism.

Prob. 2. Consid. the sphere S^n and on it the cat. of sheaves of ab. groups C. Acting on it is $O(n)$. Consid. the cat. with the action of the grp. and the derived cat. $D(C)$. Consid. the dual S^{*n}. C_1 [crossed out: unresolved]



Original source details: GELFAND-P000940

Diagram and typed description. After “action” the abbreviation reads «гр.» (“grp.”).

Original source details: GELFAND-P000940

Undeciphered passage. A sign to the right of the first C_1 is heavily crossed out.

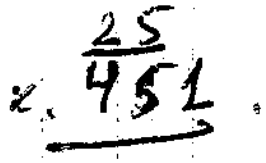
C_1 is not isom. to C (the action is different)

$$\underline{\underline{D(S) \sim D(S^*)}} \quad !!$$

24.11.80 Gelfand Sem.

Sectional sessions.

25 rm. $\underline{451}$ [crossed out: unresolved]



Original source details: GELFAND-P000940

Uncertain reading. The abbreviation before 451 is tentatively read as «к.» (“rm.”).

Original source details: GELFAND-P000940

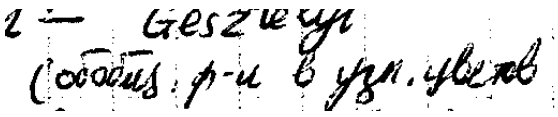
Undeciphered passage. The crossed-out writing before 15^{00} is unresolved.

$15^{\underline{00}}$ Sergeev
$15^{\underline{20}}$ Mebkhout (France)
Riem.-Hilb. prob. in higher dim.
$15^{\underline{40}}$ Henkin
$16^{\underline{00}}$ Kantor J.-M.

[boxed at right]

Conf. Plenary sessions.

24. $16^{\underline{20}}$ Sato
25 $10^{\underline{00}}$ Morimoto
$10^{\underline{40}}$ Maslov
$11^{\underline{40}}$ Gesztelyi
(generalized fns. in eqs. partial (?)
$12^{\underline{20}}$ Berezanskii et al.



Original source details: GELFAND-P000940

Uncertain reading. The ending after «в ур.» is tentatively read as «частн.» (“partial”). No closing parenthesis is visible.

26. [crossed out: unresolved] 10.00 Laugwitz
(from a nonstand. point of view)
$10^{\underline{40}}$ Ivanov
$11^{\underline{40}}$ Khromov, Danilov
$12^{\underline{20}}$ Berg multiplic. of distrib.

Original source details: GELFAND-P000940

Undeciphered passage. The earlier time before 10.00 is heavily crossed out.

28. $10^{\underline{00}}$ Lions
$10^{\underline{40}}$ Gelfand
$11^{\underline{40}}$ Komatsu — kernels in ultradistrib.
$12^{\underline{20}}$ Lassner
$13^{\underline{00}}$ Zharinov

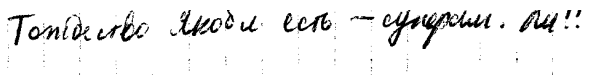
D. B. Fuchs. Whitehead mult.

$$\pi_n(X) \times \pi_m(X) \longrightarrow \pi_{n+m}(X)$$

$\sim$ commutator in a Lie alg.

$$\partial(I_n \times I_m) = (\partial I_n \times I_m) \cup (I_n \times \partial I_m)$$

There is a Jacobi identity — Lie supersym.!!



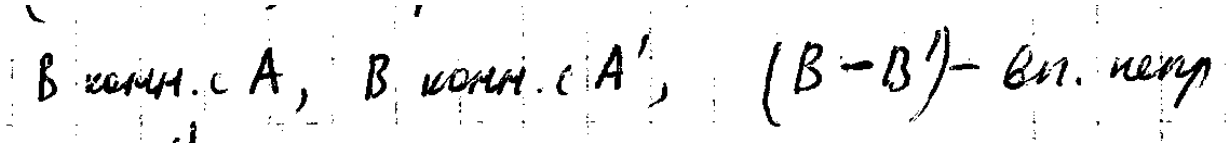
Original source details: GELFAND-P000940

Diagram and typed description. The abbreviation before “Lie” reads «суперсим.» (“supersym.”).

V. Lomonosov.

$(A - A')$ — nuclear

B comm. with A, B comm. with A', $(B - B')$ — compl. cont.



Original source details: GELFAND-P000940

Diagram and typed description. The B before A' has no visible prime.

$$\Rightarrow \quad B|_{\mathcal{H}_{ac}(B)} \underset{\text{unit.}}{\sim} B'|_{\mathcal{H}_{ac}(B')}$$

$\Rightarrow$ the theor. is consid. directly for commut. families of operators.

A. Beilinson.

$X \xrightarrow{f} Y$

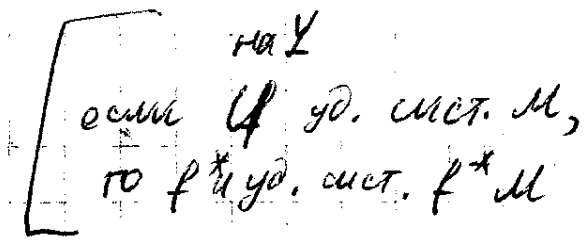
$$f^* : \mathcal{D}_Y\text{-mod} \longrightarrow \mathcal{D}_X\text{-mod} \tag{1}$$

Introduce

$$f_* : \text{mod-}\mathcal{D}_X \longrightarrow \text{mod-}\mathcal{D}_Y$$

[at right]

$$\left[\begin{array}{l} \text{if } \overset{\text{on } Y}{u} \text{ sat. the syst. } M, \\ \text{then } f^*u \text{ sat. the syst. } f^*M \end{array}\right.$$



Original source details: GELFAND-P000940

Diagram and typed description. “On Y” is inserted above u in the upper note on the right.

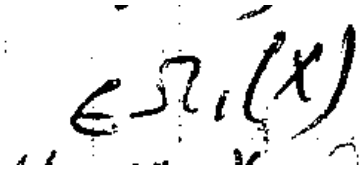
$$\mathcal{D}_{X\to Y} = f^*(\mathcal{D}_Y) \in \mathcal{D}_X\text{-mod-}\mathcal{D}_Y \qquad \text{(bimodule)} \tag{2}$$

$$f_*M = f_\bullet\left(M \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X\to Y}\right) \tag{3}$$

(f_* on volume forms of comp. support)
if

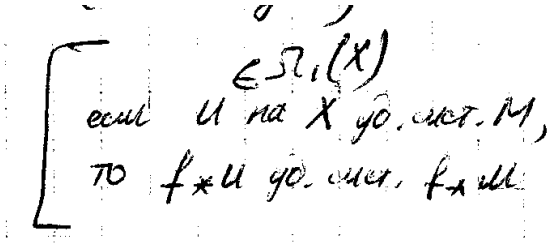
[at right]

$$\left[\begin{array}{l}\text{if } u \overset{\in\Omega_1(X)}{\text{ on }} X \text{ sat. the syst. } M,\\ \text{then } f_*u \text{ sat. the syst. } f_*M\end{array}\right.$$



Original source details: GELFAND-P000940

Uncertain reading. The small subscript on Ω resembles 1. The exact reading is unresolved.



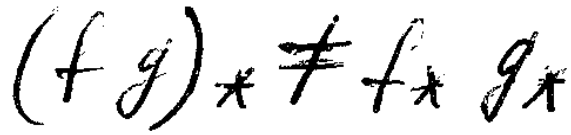
Original source details: GELFAND-P000940

Diagram and typed description. The lower note on the right begins with an opening square bracket.

GELFAND-P000941

(System of eqs. satisfied by improper integrals)

(4) f_* is not functorial: $(fg)_* \neq f_*g_*$



Original source details: GELFAND-P000941

Diagram and typed description. In (4) fg and f_*g_* are written in this order.

Let us construct.

$$Rf_* : D(\text{mod-}D_X) \longrightarrow D(\text{mod-}D_Y) \tag{5}$$

Example: X — a plane, Y — a line, f — projection,
$\mathcal{M}$ — a sheaf of forms on X (of top degree)

$Rf_*(\mathcal{M})$ — a complex, $\quad$ 0-comp. — 0
(-1)-comp. — forms on the line.

Original source details: GELFAND-P000941

Diagram and typed description. The example gives the 0-comp. and the (-1)-comp.

$$Rf_*(\mathcal{M}) = Rf_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) \tag{6}$$

$\overset{L}{\otimes}$ — the left derived functor of the tens. product.

$Rf_\bullet$ — the right derived functor of the functor (!) $f_\bullet$

(7) Prop.

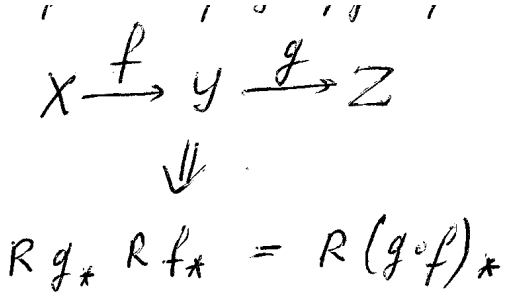


Diagram and typed description. A double arrow below the chain $X \to Y \to Z$ points down to the formula.

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$
$$\Downarrow$$
$$Rg_*Rf_* = R(g\circ f)_*$$

Original source details: GELFAND-P000941

[crossed out: ~~since~~] (f_* lacks this prop.)

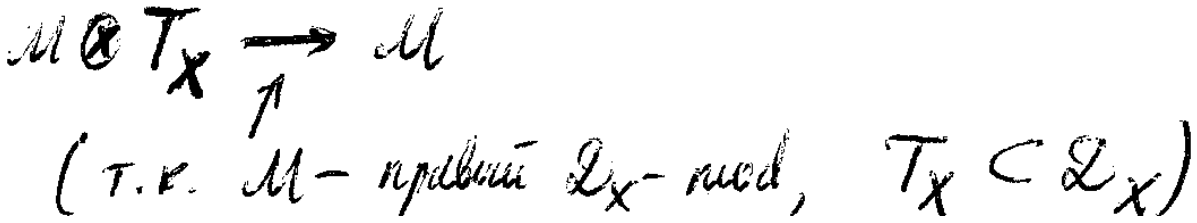
Uncertain reading. An opening parenthesis and the start of “since” are crossed out.

Original source details: GELFAND-P000941

Let

$$\mathcal{M} \in \text{mod-}\mathcal{D}_X$$
$$\ldots \longrightarrow \mathcal{M}\otimes\Lambda^2 T_X \longrightarrow \mathcal{M}\otimes T_X \underset{\nearrow}{\longrightarrow} \mathcal{M} \tag{8}$$

(T_X — tangent sheaf to X) (since $\mathcal{M}$ — a right $\mathcal{D}_X$-mod, $T_X \subset \mathcal{D}_X$)



Original source details: GELFAND-P000941

Diagram and typed description. An arrow from the right-hand note points toward the last differential in (8).

(9) Prop.

Consid. the cohomology of (8). It agrees with the cohom. of the compl. (6), i.e.

(de Rham!)

$$R\Gamma(\underbrace{DR}_{\text{De Rham}}\ \mathcal{M}) = Rf_*(\mathcal{M}) \tag{10}$$

(here $f : X \longrightarrow$ a point).

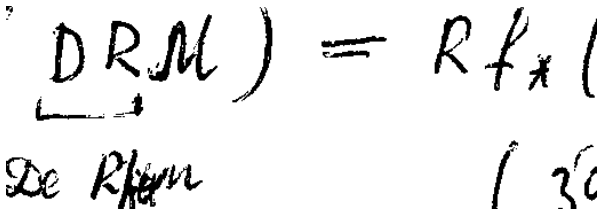


Diagram and typed description. Below DR a bracket leads to the label “De Rham”.

Original source details: GELFAND-P000941

$\mathcal{M}$ — the sheaf of volume forms $\Rightarrow$ (8) — the usual de Rham compl., shifted by $\mathcal{M}$ to the left

Uncertain reading. The sign after “by” resembles the $\mathcal{M}$ at the beginning of the sentence.

Original source details: GELFAND-P000941

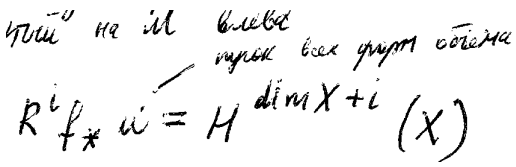


Diagram and typed description. A slanted stroke leads from the annotation toward ω.

Original source details: GELFAND-P000941

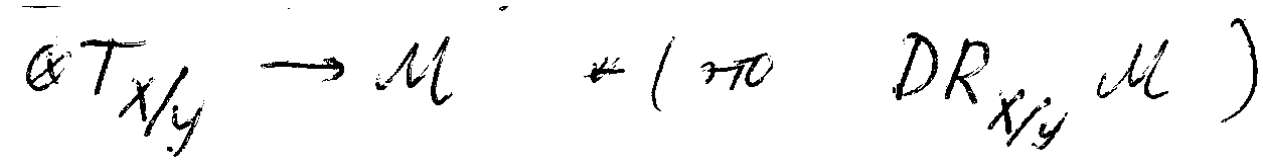
sheaf of all volume forms

$$R^i f_* \ \omega = H^{\dim X + i}(X)$$

(11)

More gen.: For a smooth morphism, sheaves of cohom. along the fibers!

$$\ldots \longrightarrow \mathcal{M} \otimes \Lambda^2 T_{X/Y} \longrightarrow \mathcal{M} \otimes T_{X/Y} \longrightarrow \mathcal{M} \quad \leftarrow \quad (\text{this is } DR_{X/Y}\mathcal{M}) \tag{12}$$



Original source details: GELFAND-P000941

Diagram and typed description. The short arrow points left from the parenthetical note toward the complex.

GELFAND-P000942

[source number: 26]

Then

$$Rf_\bullet(DR_{X/Y}\mathcal{M}) = Rf_*(\mathcal{M}) \tag{13}$$

Thus in the direct image of [inserted above: rel.] DeRhama one introduces the action of diff. operators (!)

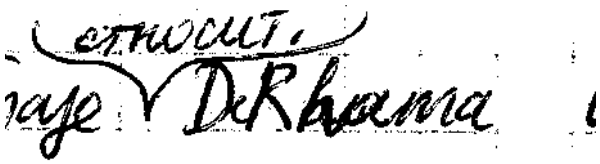


Diagram and typed description. The line reads “DeRhama”. The abbreviation for “relative” is inserted above its start.

Original source details: GELFAND-P000942

(this is not obvious, since in the orig. sheaf [crossed out: ~~it~~ (?)] [inserted above: unresolved] is absent). This action is called

Uncertain reading. The lower word before “absent” is heavily retraced and tentatively reads “it”. A short insertion above it is unresolved.

Original source details: GELFAND-P000942

the Gauss–Manin connection.

2nd case

$X \xrightarrow{f} Y$ embedding.

$$\begin{aligned} Rf_*(\mathcal{M}) &= Rf_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) \\ &= f_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) = \end{aligned} \tag{14}$$

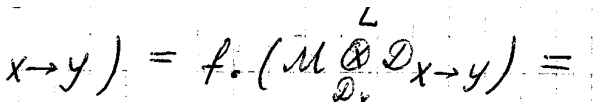


Diagram and typed description. At the end of the manuscript line (14) another equals sign follows the second expression.

Original source details: GELFAND-P000942

(The functor $f_\bullet$ is exact in this case —)

— no cohom. arises along the fibers

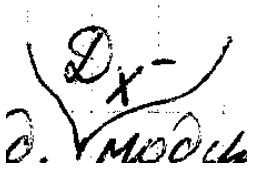
$$= f_\bullet \left(\mathcal{M} \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X \to Y} \right)$$

(since $\otimes$ is exact in this case)

$\omega \rightsquigarrow$ forms extended by δ-fns. in the transv. direction.

Holonomic modules.

X — a var., $\mathcal{M}$ — a finitely generated $\mathcal{D}_X$-module [crossed out: unresolved]



Original source details: GELFAND-P000942

Diagram and typed description. $\mathcal{D}_X$- is inserted above "module".

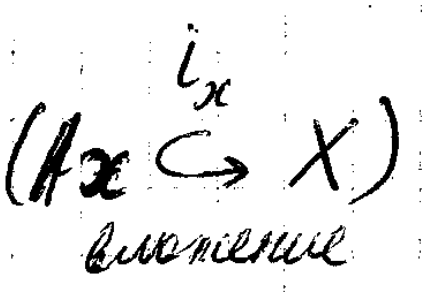
Original source details: GELFAND-P000942

Undeciphered passage. The writing after "module" is crossed out. It resembles $\mathcal{M}$, but the reading is unresolved.

It is called holonomic, if $\forall\, x \in X$,

$$\text{the cohomology of} \quad Li_x^* \mathcal{M} \quad \text{is finite-dimensional}$$

(stalk of the module over the point)



Original source details: GELFAND-P000942

Diagram and typed description. The arrow i_x goes from x to X. A heavy unresolved stroke precedes x.

$$([?]\, x \xrightarrow{i_x} X)$$

embedding

($i_x^* \mathcal{M}$ — "freezing coeff.", see above)

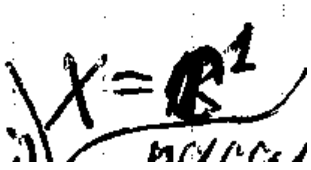
$\sim$ the complex $Li_x^* \mathcal{M}$ is finite-dimensional (since in the derived category).

A free module is not holonomic (see last time)

Holonomicity $\sim$ overdeterminacy.

Examples

a) [inserted above: $X = \mathbb{R}^1$ (?)] a bundle with an integr. connection



Original source details: GELFAND-P000942

Diagram and typed description. The field letter is retraced and has a slanting lower stroke. The tentative reading is $\mathbb{R}^1$.

The module is of this form except at a finite number of points where there may be sing.

b) $X = \mathbb{C}^2$ again on the complement of finitely many [crossed out: ~~points~~ (?)] [inserted above: curves] — a bundle with an integr. conn.

Original source details: GELFAND-P000942

Diagram and typed description. An earlier word is crossed out below the inserted “curves”. It may read “points”.

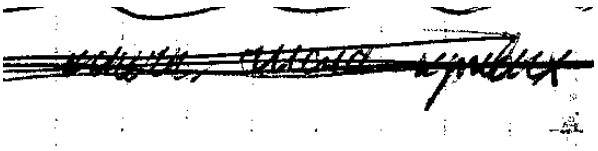
There are singularities on these curves.

(?) All are obtained by gluing from modules obtained from δ-fns. in the normal direction.

Original source details: GELFAND-P000942

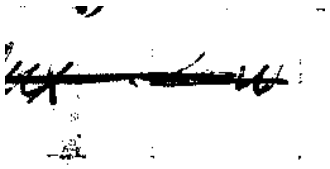
Diagram and typed description. A circled question mark stands to the left of the line about gluing.

[crossed out: ~~(except for points [...] curves~~ unresolved]



Original source details: GELFAND-P000942

Uncertain reading. “Points” is tentatively read in the middle of the crossed-out line. The next word is unresolved.



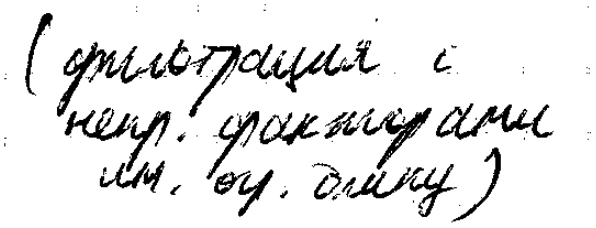
Original source details: GELFAND-P000942

Undeciphered passage. There is further crossed-out writing after “curves”. The ending is unresolved.

GELFAND-P000943

(15) Thm. 1) $\forall$ holon. module has finite length

(filtrations with irred. factors have the same length)



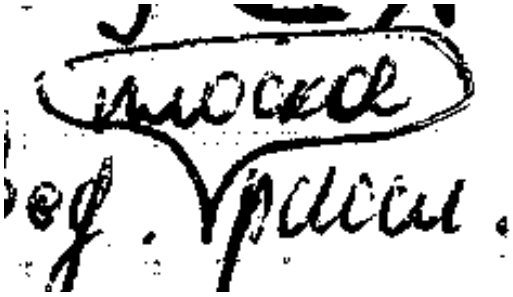
Original source details: GELFAND-P000943

Uncertain reading. The ending of the first word in parentheses is indistinct. The last line reads “have the same length”.

2) there is a descr. of all irreducibles

$$Y \subset X, \qquad Y \text{ irred.}$$

Irred. [inserted above and circled: flat] bundle $\mathcal{L}_Y$ with a connection on Y
(a vector bundle with an integr. conn.)



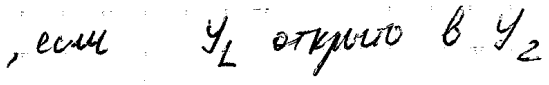
Original source details: GELFAND-P000943

Diagram and typed description. A leader points from the circled “flat” to “bundle”.

flat
Irred. bundle

2 pairs Y_1, Y_2 are called equiv. if Y_1 is open in Y_2
and

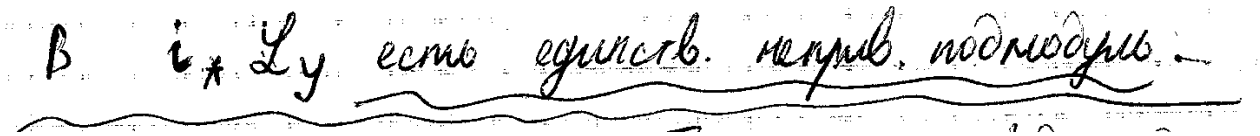
$$\mathcal{L}_{Y_1} = (\mathcal{L}_{Y_2})|_{Y_1}$$



Original source details: GELFAND-P000943

Diagram and typed description. The condition is written in the order “Y_1 is open in Y_2”.

(one can discard closed subsets, spec. over the generic point)
(16) Assertion. $i : Y \hookrightarrow X$
In $i_*\mathcal{L}_Y$ there is a unique irred. submodule —



Original source details: GELFAND-P000943

Diagram and typed description. Two wavy underlines of different lengths appear below the line.

this is the object sought, an irred. module
Y — smooth $\Rightarrow$ already irreducible
At sing. there appear δ-fns..

Original source details: GELFAND-P000943

Diagram and typed description. Below the last line are two tapered strokes with a dot between them.

GELFAND-P000944

[source number: 27]

1/XII-80 Gelfand Sem.

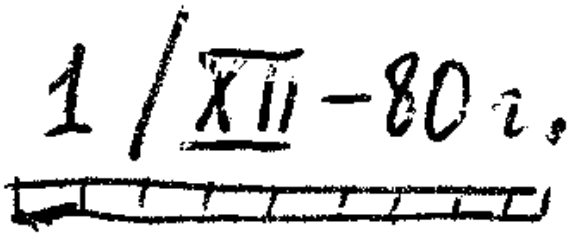


Diagram and typed description. A narrow bar with transverse strokes underlines the date. Separate lines appear above and below XII.

1/XII-80

Original source details: GELFAND-P000944

S. P. Novikov.

$$\hat{H} = \frac{1}{2}\sum_{\alpha=1}^{n}\left(i\frac{\partial}{\partial x^{\alpha}} - eA_{\alpha}\right)^{2} + cV(\vec{x}) \tag{1}$$

Undeciphered passage. Heavily crossed-out writing appears after the equals sign in (1), before the fraction. Its reading is unresolved.

Original source details: GELFAND-P000944

$$H_{\alpha\beta} = \partial_{\beta}A_{\alpha} - \partial_{\alpha}A_{\beta} \tag{2}$$

V — el. field, $H_{\alpha\beta}$ — magn. field

$$\hat{H}\psi = \varepsilon\psi \qquad \text{spectr. problem.} \tag{3}$$

The interesting case is when

(4) $H_{\alpha\beta}(x)$ and $V(x)$ are periodic with a lattice of periods.

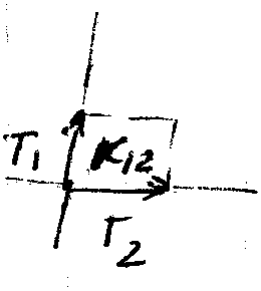
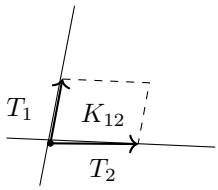
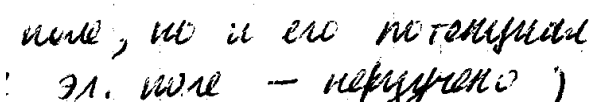


Diagram and typed description. Cell K_{12} lies between the arrows T_1, T_2. The top and right sides are broken lines.



Original source details: GELFAND-P000944

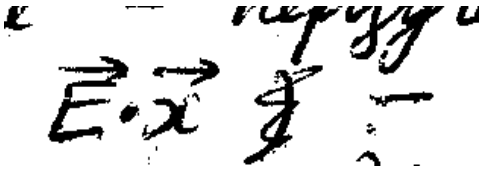
(not only the el. field but also its potential. A constant el. field — unreasonable)



Uncertain reading. The word after the dash is tentatively read as «неразумно» (“unreasonable”). Its ending is faint.

Original source details: GELFAND-P000944

— the potential then $\vec{E} \cdot \vec{x}$ [crossed out: unresolved] —



Original source details: GELFAND-P000944

Undeciphered passage. One sign is crossed out after $\vec{E} \cdot \vec{x}$. The exact reading is unresolved.

— even the 1-dim. problem is unstudied.)

A_α is nonperiodic (generally speaking).

$K_{\alpha\beta}$ — an elem. cell, spanned by the α and β coordinate

Original source details: GELFAND-P000944

Uncertain reading. The overwritten preposition after “spanned” is tentatively read as «на».

Original source details: GELFAND-P000944

Uncertain reading. The end of the word for “coordinate” is overwritten. The last letter reads «у».

$$\iint_{K_{\alpha\beta}} H_{\alpha\beta}\, dx^\alpha \wedge dx^\beta = \Phi_{\alpha\beta} \qquad \text{— magn. fluxes (basis)} \tag{5}$$

(no summation!)

$$i\nabla_\alpha = i\frac{\partial}{\partial x^\alpha} - eA_\alpha \qquad \text{has the meaning of a connection} \tag{6}$$

Rem. The spin here is 0, with spin $1/2$ one has

$$\hat{H} = (1) + e\sigma_j H^j, \qquad \text{where } H^1 = H_{23}, \quad H^2 = -H_{13}, \quad H^3 = H_{12}; \tag{7}$$

$\sigma_1, \sigma_2, \sigma_3$ — Pauli matr.

(8) Admissible case: $\frac{e}{2\pi}\Phi_{\alpha\beta} \in \mathbb{Z}$

The operator $\hat{H}$ is consid. in $L^2(\mathbb{R}^n)$.

(9) $(8) \Rightarrow \hat{H}$ — a univ. cover. op. on the torus (in a bundle), moreover

Original source details: GELFAND-P000944

Uncertain reading. The beginning of the word for “moreover” is retraced.

$$\left(c_1(\xi), \quad \underset{\substack{| \\ \text{basis cycle}}}{Z_{\alpha\beta}} \right) = \frac{e}{2\pi} \Phi_{\alpha\beta} \tag{10}$$

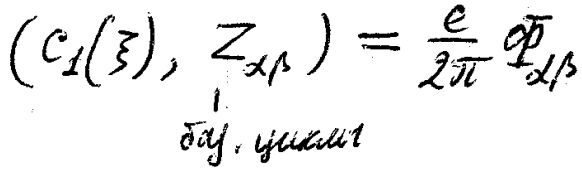
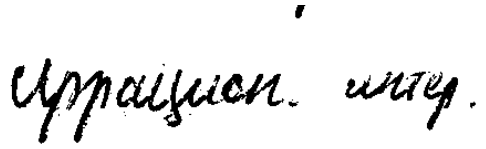


Diagram and typed description. A short stroke below $Z_{\alpha\beta}$ leads to “basis cycle”.

Original source details: GELFAND-P000944

But the op. always makes sense (phys.), yet there is no theory.

(11) Irrational is of int. Ration. reduces to it by coarsening the lattice



Uncertain reading. The abbreviation after “Irrational” is tentatively read as «интер.» (“of int.”).

Original source details: GELFAND-P000944

GELFAND-P000945

Magnetic translations.

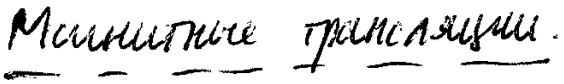
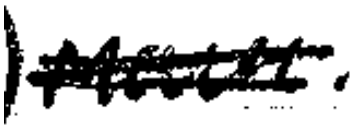


Diagram and typed description. A broken line underlines the heading.

Original source details: GELFAND-P000945

2 types of problems: a) [crossed out: ~~magn.~~ (?)] the lattice is imposed (solid state!), and the magn. field is external

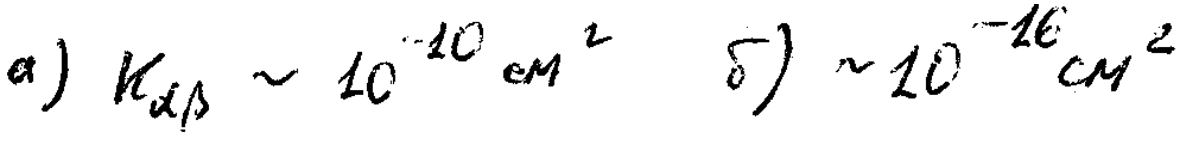


Uncertain reading. The crossed-out abbreviation before “lattice” is tentatively read as “magn.”.

Original source details: GELFAND-P000945

b) of superconduct. type: the magn. field arises on its own, they are automatically integral.

a) $K_{\alpha\beta} \sim 10^{-10}$ cm^2 b) $\sim 10^{-16}$ cm^2



Original source details: GELFAND-P000945

Diagram and typed description. The line reads 10^{-10} and 10^{-16} in square centimeters.

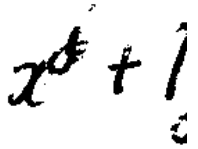
Magn. transl. We have:

$$A_\alpha(\dots, x^i + T_i, \dots) = A_\alpha(x) + \partial_\alpha f_i \tag{12}$$

(since dA is periodic)

$$\hat{T}_j^* \psi(x) = \psi(.. \; x^j + T_j, \;) \exp(-ie f_j) \tag{13}$$

(operator of magn. translation)

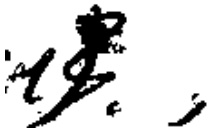


Uncertain reading. In (13) the superscript of x is retraced. The reading j is tentative and agrees with T_j on the same line.

Original source details: GELFAND-P000945

(14) Lemma 1 (known ~1960s [crossed out: unresolved], Zak and Brown)

$$\hat{T}_j^* \hat{T}_k^* = \hat{T}_k^* T_j^* \exp(-ie\Phi_{kj}), \qquad j, k = 1, \dots, n.$$



Uncertain reading. A sign after "1960s" is heavily retraced. Its exact reading is unresolved.

Original source details: GELFAND-P000945

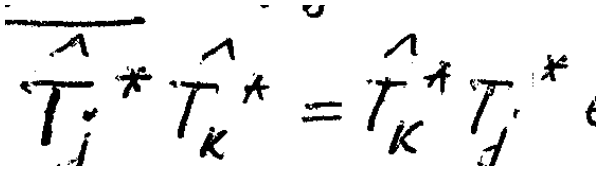


Diagram and typed description. The last T_j^* on the right side of the equality has no visible hat.

Original source details: GELFAND-P000945

This gives a projective repr. of an ab. group, with cocycles specified in advance.
Irrat. fluxes $\Rightarrow$ one gets groups not of type 1. (subgr. of the Heisenberg grp.).

$$\frac{e\Phi_{kj}}{2\pi} \in \mathbb{Z} \Rightarrow \text{commut. group} \tag{15}$$

$n = 2$
($n = 3$ is also interesting)
superconduct. is a 2-dim. problem
Problem: describe the spectrum of $\hat{H}$.
(16) $[\hat{H}, \hat{T}_j^*] = 0 \Rightarrow$ the eigenfns. must be sought as simultaneous eigenfns. for $\hat{T}_j^*$
[crossed out below the line: ~~magn.~~ (?)]

Uncertain reading. Below line (16) the crossed-out abbreviation resembles "magn.".

Original source details: GELFAND-P000945

Magnetic Bloch:

$$\hat{T}_j^* \psi_p = \exp(i p_j T_j) \psi_p \tag{17}$$

Uncertain reading. The left ψ subscript is heavily retraced. p is tentatively supplied from the neighboring line.

Original source details: GELFAND-P000945

Uncertain reading. The right ψ subscript is also heavily retraced. p is tentatively supplied.

Original source details: GELFAND-P000945

Op. in $L^2 =$ direct integral of op. on the torus with such bound. conds.

p_j — quasimomenta $\quad (p_1, p_2) \in \mathbb{T}^2 \quad$ (torus of the recip. lattice)

Fix $p_1, p_2 \Rightarrow$ an op. in a bundle on the torus with eigenvals.

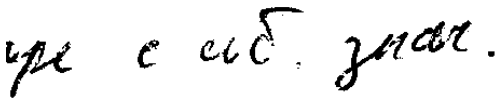


Diagram and typed description. The last two abbreviations read “eigenvals.”.

Original source details: GELFAND-P000945

$$\varepsilon_j(p_1, p_2) \qquad \text{— dispersion law} \tag{18}$$

$$j = 1, 2, \dots$$

GELFAND-P000946

[source number: 28]

(19) Theorem. Let $V = 0$, spin $s = \frac{1}{2}$,

$$n = 2, \qquad N \underset{\text{def}}{=} \frac{e\Phi_{12}}{2\pi}.$$

[In the right margin, beyond the double line]

Dubrovin, Novikov

ZhETF, Sept. 1980.

and there is also a paper in DAN

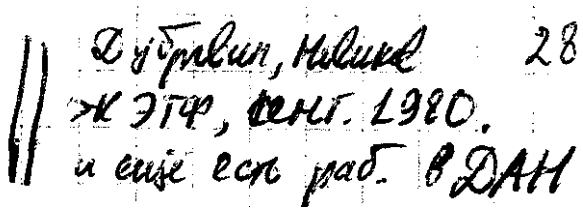
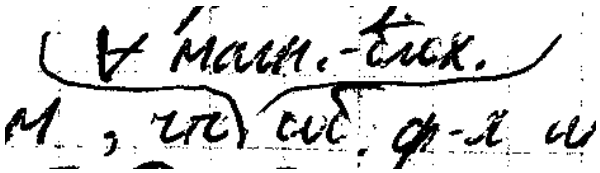


Diagram and typed description. References in the right margin.

Original source details: GELFAND-P000946

(magn. field directed along the Z axis — 2-dim. case — and periodic)

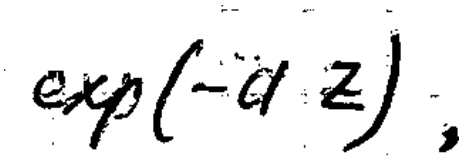
Then given $A = (a_1, \dots, a_N) \in C^N$, we get that the eigenfn. has the form [inserted above "eigen" ($\forall$ magn.-Bloch)]



Original source details: GELFAND-P000946

Diagram and typed description. A curved stroke links the insertion above the line to the main text.

$$\Psi_A = \exp(-e\varphi) \prod_{j=1}^{N} \sigma(z - a_j) \exp(-az). \tag{20}$$

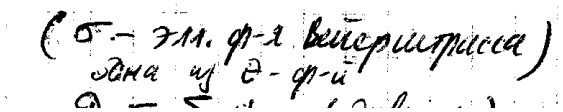


Original source details: GELFAND-P000946

Diagram and typed description. The sign before az in (20) is minus.

(σ — Weierstrass elliptic fn.)

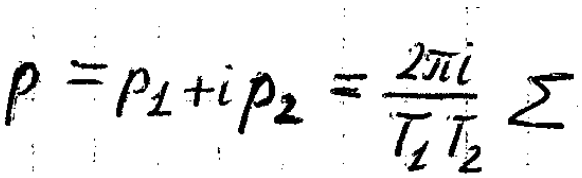
one of the θ-fns.



Original source details: GELFAND-P000946

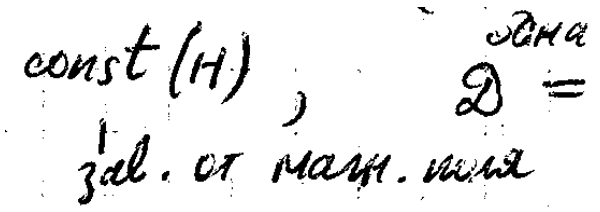
Diagram and typed description. Note at the right of (20).

$$p = p_1 + ip_2 = \frac{2\pi i}{T_1 T_2} \sum a_j + \underbrace{\text{const}(H)}_{\text{depends on the magn. field}}, \tag{21}$$



Original source details: GELFAND-P000946

Diagram and typed description. The summation term in (21) has a plus sign.



Original source details: GELFAND-P000946

Diagram and typed description. A line connects the annotation below const(H) to that term.

$$\mathcal{D} = \sum a_j. \qquad \text{(divisor)}$$

(22) [crossed out: ~~$\varphi(x,y)$~~ (?)]

$$\varphi(x,y) = \frac{1}{2\pi} \int_{K_{12}} \ln|\sigma(z-z')| H(z')\, d^2 z'.$$

Uncertain reading. The crossed-out entry before (22) is tentatively read as $\varphi(x,y)$.

Original source details: GELFAND-P000946

$z =$ [crossed out: entry unresolved] $x + iy$.

Undeciphered passage. Several signs are crossed out after the equals sign. Their exact reading is unresolved.

Original source details: GELFAND-P000946

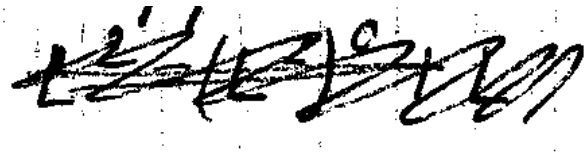
Fns. (20) — a complete basis of solutions to the eq. $\hat{H}\psi = 0$

(exact integr. at the lower zero energy level)

Diagram and typed description. The abbreviation after “exact” reads “integr.”

Original source details: GELFAND-P000946

[crossed out: $L^2 = (L^2)^0 + \ldots$ (?)]



Undeciphered passage. The beginning of the crossed-out formula is a tentative reading. The end after the plus sign is unresolved.

Original source details: GELFAND-P000946

$$L_2 = L_2 + L_2^{\perp}.$$

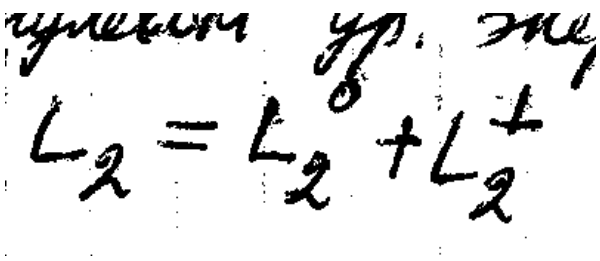


Diagram and typed description. The loop above the first L_2 on the right continues into a descender from the line above.

Original source details: GELFAND-P000946

$$a = a(a_1, \dots, a_N) \quad \text{— is calculated.} \tag{23}$$

Selecting the dependence (23) — is selecting real quasimomenta

(the fn. Ψ_A is always an eigenfn. and Bloch).

$\lambda\Psi_A$ is also an eigenfn.

$(\lambda, a_1, \dots, a_N)$ — the full set of parameters.

Rem.

$$(\lambda, a_1, \dots, a_N) \xrightarrow{P} \mathbb{T}^2. \tag{24}$$

(25) We get [crossed out: entry unresolved] a holom. vect. bundle (N-dimensional) over $\mathbb{T}^2$ with $c_1 = N$.

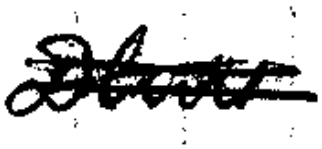
Original source details: GELFAND-P000946

Undeciphered passage. The heavily crossed-out entry before “holom.” is unresolved.

(= the number of quanta of magn. flux)

(26) Fibre — at fixed p — We get $\mathcal{L}(\mathcal{D})$, $\dim \mathcal{L}(\mathcal{D}) = N$

[crossed out: ~~Next~~ (?)] (mult. by a doubly periodic fn.)



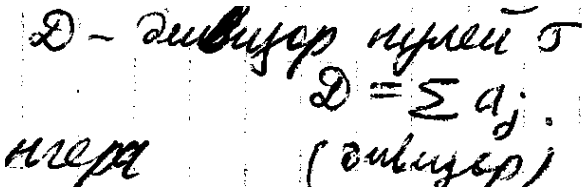
Original source details: GELFAND-P000946

Uncertain reading. The crossed-out entry is tentatively read as “Next”.

$\mathcal{D}$ — the divisor of zeros of σ

$\mathcal{D} = \sum a_j$

(divisor)



Original source details: GELFAND-P000946

Diagram and typed description. The note on the divisor in the right margin.

In fact (26) — from the Atiyah–Singer thm.

$$(\text{Coker} = 0)$$

infinite-dimensional machinery.

Magn. field here:

$$A_1 = -\partial_2\varphi, \quad A_2 = \partial_1\varphi, \quad \Delta\varphi = H. \tag{27}$$

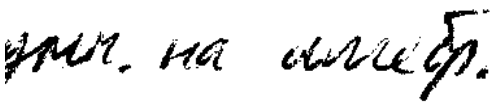
$a_1, \dots, a_N$ are determ. mod the lattice and symmetrized, i.e.

$$a \in S^N\Gamma.$$

GELFAND-P000947

The fibre (25) is a lin. space. (not immediately obvious)

All fns. Ψ_A in a fixed fibre are obt. from one by mult. by ellipt. (?) 2-periodic fns..



Uncertain reading. The abbreviation is tentatively read as the Russian “эллипт.”; “эллип.” is also possible.

Original source details: GELFAND-P000947

Corollary: Bloch fns. form, form a linear bundle.

Eigenfns. at fixed p form an N-dimensional space (!)

(the dimension is fixed)

In a uniform magn. field the assert. holds for all energy levels.

[crossed out: Question 2. ~~Where does the degeneracy come from?~~]

Original source details: GELFAND-P000947

Diagram and typed description. The whole question is crossed out.

Now let us perturb:

$$\hat{H} = \hat{H}_0 + e\,v(x,y) \qquad \text{(with the same grp. of periods).} \tag{28}$$

(29) Theorem 2. Perturb. op. — quadr. form of general type.

(2-param. family of quadr. forms in general pos. among all complex Hermitian forms).

I.e. if one excludes the set of potentials on whose Fourier coeffs. 3 eqs. are imposed, then the spectrum (Bloch eigenfns.) has the following properties

$$\hat{V}_{(p_1,p_2)}(\Psi_A) = \iint_{K_{12}} \bar{\Psi}_A\, v \Psi_A\, dx\, dy. \tag{30}$$

(complex Hermitian form, Ψ_A — eigenfns. of H_0)

at fixed quasimomenta.

(30) Is in general position (as an operator)

Quadr. form in general pos. — no pair of eigen [crossed out: ~~fns.~~ (?)] values coalesces

Uncertain reading. The crossed-out entry before “values” is tentatively read as “fns.”

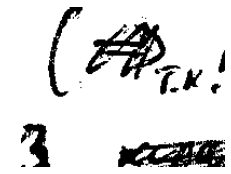
Original source details: GELFAND-P000947

The map of the torus into the set of quadr. forms is in general position.

(In the 1-dim. case — it is never in general position)

because the op. is of 2nd order.

([crossed out: ⇒ (?); inserted below: since] codim. 3, [crossed out: ~~then~~ (?)] then it does not intersect with mult. eigenvals.,



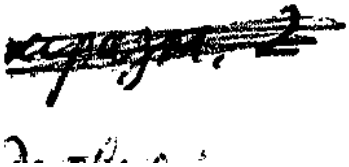
Uncertain reading. The crossed-out sign is tentatively read as ⇒.

Original source details: GELFAND-P000947

Uncertain reading. The heavily crossed-out entry before “then” is tentatively read as “then”.

Original source details: GELFAND-P000947

$n = 3$ [crossed out: ~~codim. 2~~ (?)] ⇒ [crossed out: initial Cyrillic “в” (?); the rest is unresolved] it inters. at isolated points.)

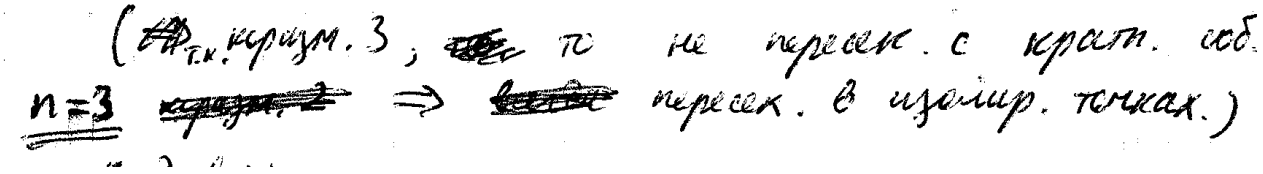


Uncertain reading. The crossed-out text is tentatively read as “codim. 2”.

Original source details: GELFAND-P000947

Undeciphered passage. After the arrow, the initial stroke resembles Cyrillic “в”. The cancellation obscures the rest.

Original source details: GELFAND-P000947



Original source details: GELFAND-P000947

Diagram and typed description. The two codimension lines, with “since” inserted below the first cancellation.

Corollary:

$n = 2$ ⇒ the bundle splits into 1-dimensional ones

I.M.G.:

The irrat. case is interesting!

(there everything is in fact noncommutative)

Idea: go into the complex domain (inside the circle!) — in a non-self-adjoint situation

i.e. carry out a regularization (parabolic)

GELFAND-P000948

[source number: 29]

Irrat. case:

$$H_{12} = \frac{\partial A_1}{\partial x_2} - \frac{\partial A_2}{\partial x_1}, \qquad A_1\,dx_1 + A_2\,dx_2.$$

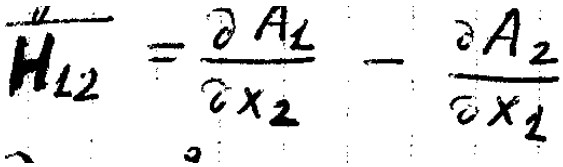


Diagram and typed description. The order of derivatives in H_{12} follows the manuscript.

Original source details: GELFAND-P000948

$$\left[\left(i\frac{\partial}{\partial x_1} - A_1\right)^2 + \left(i\frac{\partial}{\partial x_2} - A_2\right)^2\right] = \hat{H}.$$

$$H_{12}(x_1 + m_1T_1, x_2 + m_2T_2) = H_{12}(x_1, x_2)$$

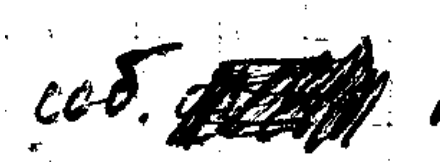
$$\Phi_{12} = \iint H_{12}\,dx_1 \wedge dx_2.$$

$$\Phi_{12} \neq 2\pi\frac{N}{M}$$

Group:

$$\hat{T}_1^*\hat{T}_2^* = \hat{T}_2^*\hat{T}_1^* \exp(-i\Phi_{12})$$

acts on the set of all fns. (fixed eigen [crossed-out entry unresolved] subspace)



Undeciphered passage. Dense hatching obscures the entry between “eigen” and “subspace”.

Original source details: GELFAND-P000948

$[H, \hat{T}_j^*] = 0$ still (choice of f_j in (12)–(13)).

At the basis of the theory [inserted above the line: of factors] of von Neumann — irreducibility ergodicity.

Diagram and typed description. A curved mark connects “of factors” above the line with the position after “theory”.

Original source details: GELFAND-P000948

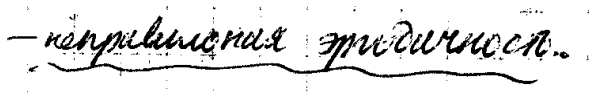


Diagram and typed description. One wavy line underlines both words.

Original source details: GELFAND-P000948

I.M.G. To know by next semester: G-structures, $\mathcal{D}$-structures and the Spencer resolution.

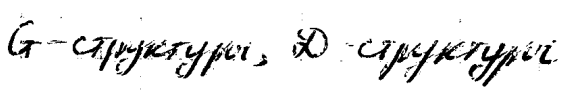


Diagram and typed description. The two structures are denoted by G and a calligraphic $\mathcal{D}$.

Original source details: GELFAND-P000948

GELFAND-P000949

[source number: 51]

9/II-81. Gelfand seminar.

MacPherson.

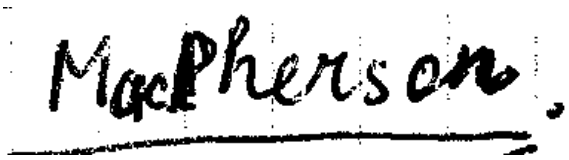


Diagram and typed description. The manuscript gives the name in Latin script.

Original source details: GELFAND-P000949

(On joint work with I. M. Gelfand.)

Problem — find interesting polyhedra in Grassmannians.

In proj. spaces polyhedra have flat faces (face $\subset$ subspace of smaller dim.)

$$P_1 \subset P_2 \subset P_3 \subset \dots \tag{0}$$

(projective)

G_n^m — n-planes in $\mathbb{R}^{m+n}$ (subspaces)

Embeddings:

$$\begin{array}{ccccccc} G_1^1 & \subset & G_1^2 & \subset & G_1^3 & \subset & \dots \\ \cap & & \cap & & \cap & & \\ G_2^1 & \subset & G_2^2 & \subset & G_1^3 & \subset & \dots \end{array} \tag{1}$$

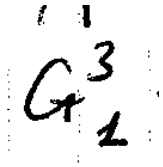


Diagram and typed description. The third entry in the lower row reads G_1^3.

Original source details: GELFAND-P000949

and so on

We want: faces — polyh. in smaller ones

Picture (0) corresp. to the upper row of (1).

Naïve method:

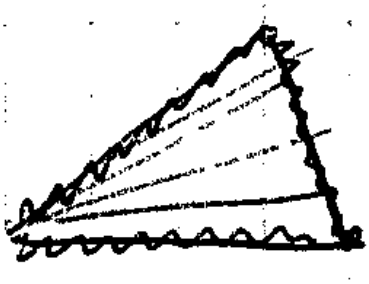
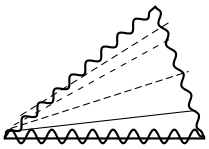


Diagram and typed description. Triangle with wavy sides and a fan of strokes. The broken interior strokes are drawn approximately.



Original source details: GELFAND-P000949

there is a triangle of 3 geod.

How to fill it in?

also with geod. (segments)

If one does this in the Grassmannian, it does not work.

Here one gets

$$\text{Rank} \geqslant 2$$

Diagram and typed description. The sign between “Rank” and 2 reads $\geqslant$.

Original source details: GELFAND-P000949

(2) Defin.

Hypersimplex Δ_n^m: set of points

$$(x_1, \ldots, x_{n+m}) \in \mathbb{R}^{n+m},$$

satisfying the rels.

$$0 \leq x_i \leq 1,$$
$$\sum_i x_i = n$$

[crossed out after n: ~~$+1$~~ (?)]

Uncertain reading. Hatching obscures the entry after n. The reading $+1$ is tentative; the letter m is also possible.

Original source details: GELFAND-P000949

(3) Def. Grassm. complex in P_{n+m-1}: choose $n+m$ points $e_1, \ldots, e_{n+m}$, not lying in a hypersubspace of smaller dim. (a hyperplane).

H^0 = conn. comp. of 1 of the subgroup of $PGL(n+m)$, which leaves these points fixed.

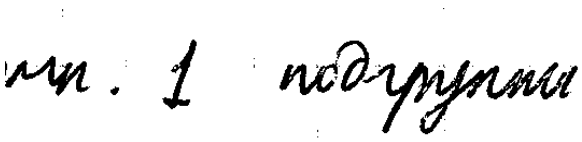


Diagram and typed description. The digit 1 follows the abbreviation for component.

Original source details: GELFAND-P000949

Grasm. complex — closure of an orbit in gen. pos. of the group H^0, acting on G_n^m. [written beneath “gen. pos.” *generic*.]

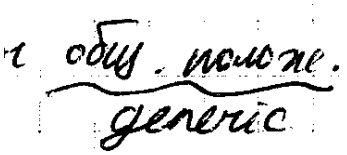


Diagram and typed description. The English gloss appears below the wavy underline.

Original source details: GELFAND-P000949

(4) *generic* — ?

this means $H^0\xi$

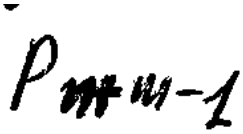
ξ — subspace transverse to all coord. subspaces:
if $e_1, \dots, e_{n+m}$ are taken as a basis (some points on them)
this is not the same as dim = max (!)
GELFAND-P000950

(5) Theorem. These 2 objects are diffeomorphic.
def. (3) can be given for $\forall$ semisimple group.

Original source details: GELFAND-P000950

Diagram and typed description. The abbreviated verb before “for” reads “прив.”.

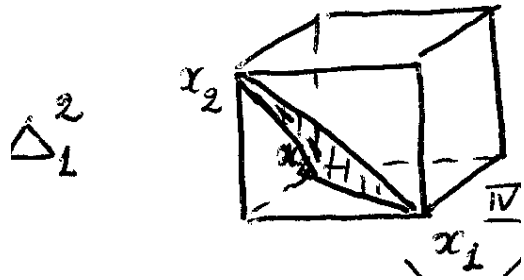
G_n^m — $(n-1)$-subspaces in P_{n+m-1} (?).



Original source details: GELFAND-P000950

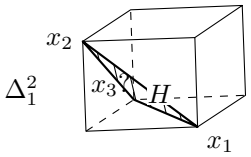
Uncertain reading. The first character of the index is heavily retraced. The reading P_{n+m-1} is tentative; P_{m+n-1} is also possible.

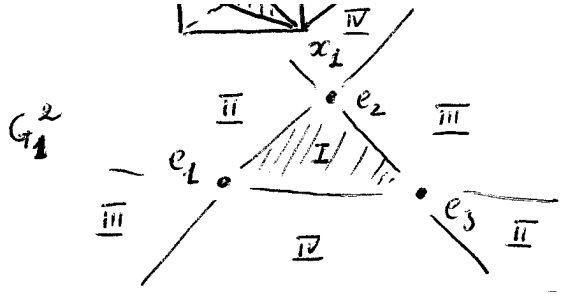
(6) Example. Δ_1^2 G_1^2



Original source details: GELFAND-P000950

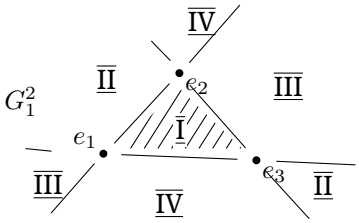
Diagram and typed description. Cube with hatched section H. The interior label, tentatively x_3, is overlapped by lines.





Original source details: GELFAND-P000950

Diagram and typed description. Three lines through the marked points e_1, e_2, e_3. The Roman numerals and hatching follow the manuscript.



transf. preserving lines as well

Original source details: GELFAND-P000950

Diagram and typed description. The character after “preserving” reads Russian “и” (“also”).

Interior of the Δ-gon — an orbit (and a conn. component of the corresp. subgrp.)
(since PGL is transit. on 4-tuples of points in general position)
Altogether — 4 triangles

Diagram and typed description. Small hatched triangle marked I.

Original source details: GELFAND-P000950

— example of a Grassm. simplex
There is a homeom., it even preserves the marks (numbers)

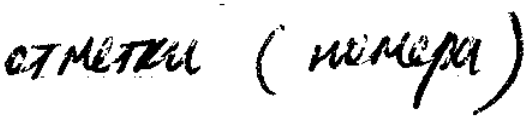


Diagram and typed description. The word before “(numbers)” reads “marks”.

Original source details: GELFAND-P000950

$\forall$ edge — a simplex in G_1^1
(7) Example. (another one !!).
G_2^1

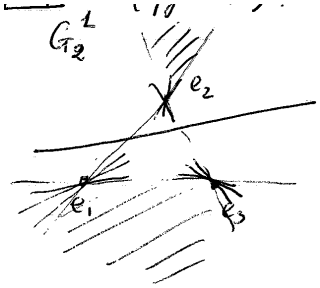
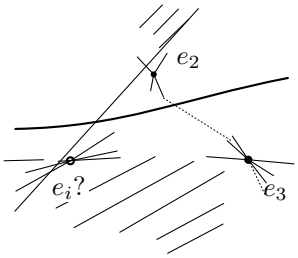


Diagram and typed description. Pencils through three points, a sloping transversal, and hatching. The faint lines are drawn approximately.



Original source details: GELFAND-P000950

Uncertain reading. The label at the lower left is tentatively e_i; the index may be 1.

Original source details: GELFAND-P000950

orbit of a line — $\forall$ lines which do not inters. the hatched region

This is a Δ-gon — with sides —

— lines passing through the vertices

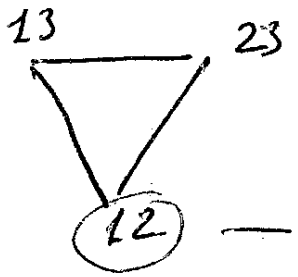
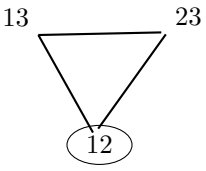


Diagram and typed description. Inverted triangle marked 13, 23, and circled 12.



Original source details: GELFAND-P000950

— corresp. to the side passing through e_1 and e_2.

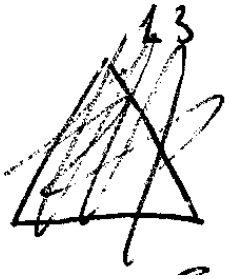
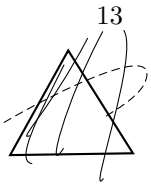


Diagram and typed description. Triangle marked 13 with curves crossing its sides. The faint curves are drawn approximately.



Original source details: GELFAND-P000950

Sides $\in G_1^1 \subset G_2^1$

Δ_2^1 — inters. of the cube with the plane $\sum x_i = 2$, with the vertices also denoted by pairs.

GELFAND-P000951

[source number: 52]

(8) Interesting example G_2^2 or Δ_2^2

[crossed out in the left margin: ~~(9)~~ (?)]

Uncertain reading. The upper crossed-out mark resembles “(9)”; “(8)” is also possible.

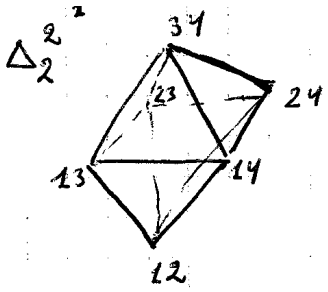
Original source details: GELFAND-P000951

$$\sum_1^4 x_i = 2 \quad - \quad (\perp \text{ main diag. section of a 4-dim. cube})$$

[crossed out in the left margin: sign unresolved]

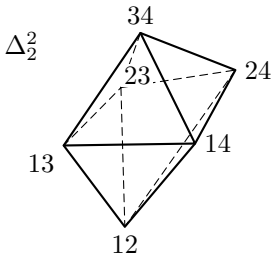
Uncertain reading. Several strokes obscure the sign to the left of the figure.

Original source details: GELFAND-P000951



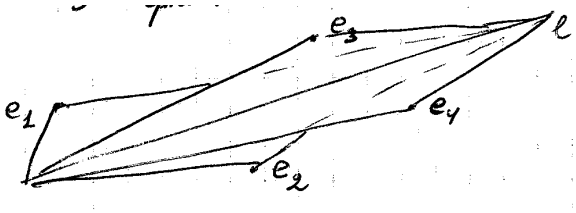
Original source details: GELFAND-P000951

Diagram and typed description. Labelled sketch of Δ_2^2. The faint interior connections are drawn approximately.



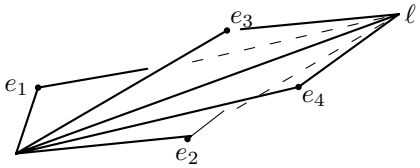
G_2^2 – 4-dimensional

H^0 – 3-dimensional.



Original source details: GELFAND-P000951

Diagram and typed description. Sketch with points e_1, e_2, e_3, e_4 and label ℓ. The faint broken strokes are drawn approximately.



$K(\ell) =$ cross-ratio of 4 planes

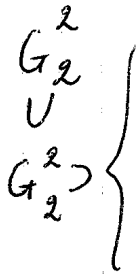
(anharm. ratio) passing through ℓ and the sel. points

(take the quotient of everything by $\ell \Rightarrow$ get it in the plane)

$$H^0\text{-orbit} \subset K(\ell) = \text{const}$$

$$\begin{array}{c|c} G_2^2 & \left\{\begin{array}{ccc|c} 12 & 13 & 14 & 1 \\ 21 & 23 & 24 & 2 \\ 31 & 32 & 34 & 3 \\ 41 & 42 & 43 & 4 \end{array}\right. \\ \cup & \\ G_2^2 \supset & \end{array}$$

$$\begin{array}{c|c} G_2^1 \supset & \left\{\begin{array}{ccc|c} 14 & 42 & 21 & \\ 13 & 32 & 21 & \\ 14 & 43 & 31 & \\ 24 & 43 & 32 & \end{array}\right. \\ \cap & \\ G_2^2 & \end{array}$$



Original source details: GELFAND-P000951

Diagram and typed description. Both upper labels read G_2^2, with a U-shaped sign between them.

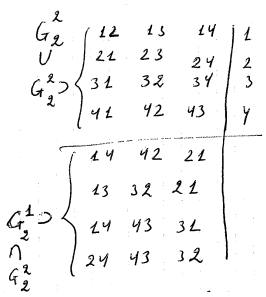
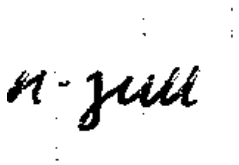


Diagram and typed description. The two blocks of the manuscript table, with the order of the numbers preserved.

Original source details: GELFAND-P000951

Each one — in its own [abbreviation unresolved] $K(\ell) = \text{const}$



Uncertain reading. The abbreviation after "its own" is tentatively read as "мн-зии". Its exact expansion is unresolved.

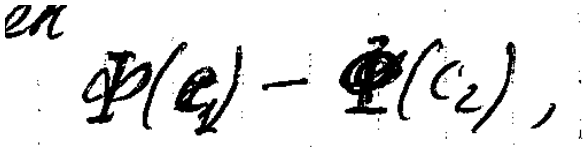
Original source details: GELFAND-P000951

(9) Theorem. The volume of the set, encl., between 2 Grassm. simplices

$$K(\ell) = c_1, \quad K(\ell) = c_2$$

equals

$$\Phi(c_1) - \Phi(c_2),$$



Uncertain reading. The second Φ is retraced. Its earlier form is unresolved.

Original source details: GELFAND-P000951

where

$$\Phi(x) = L_1(x) - L_1(1-x), \quad L_1(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}, \quad 0 < c_1 < c_2 < 1$$

(integr. logarithm)

GELFAND-P000952

(10) Theorem. Generally speaking, [inserted above the line: combinatorial] homeomorphism

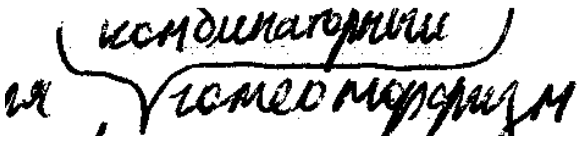


Diagram and typed description. "Combinatorial" is above the line, linked by a curved mark to the start of "homeomorphism".

Original source details: GELFAND-P000952

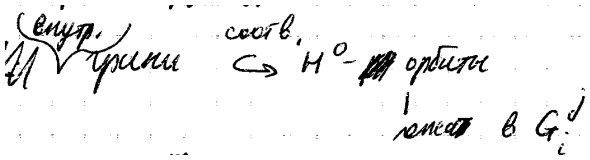
$\Delta_n^m \hookrightarrow$ Grassm. complex G_n^m

preserves a number of further structures.

(11) [crossed out: ~~Theorem~~]

Original source details: GELFAND-P000952

Diagram and typed description. The underlined word “Theorem” remains legible beneath the hatching.



Original source details: GELFAND-P000952

Diagram and typed description. Correspondence of faces and H^0-orbits. The stroke under “orbits” has no visible arrowhead.

inter. corresp.

Faces $\hookrightarrow H^0$-[?] orbits

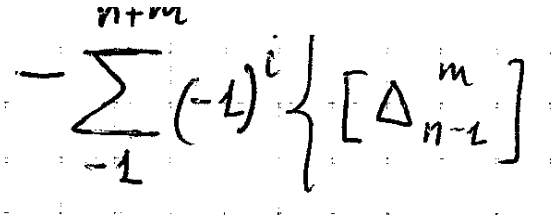
lie in G_i^j

Original source details: GELFAND-P000952

Uncertain reading. The crossed-out entry after H^0 and the hyphen is unresolved.

(12)

$$\partial[\Delta_n^m] = \sum_{i=1}^{n+m} (-1)^i \{[\Delta_n^{m-1}] \mid i\text{-th coord.} = 0\} - \sum_{-1}^{n+m} (-1)^i \{[\Delta_{n-1}^m]$$

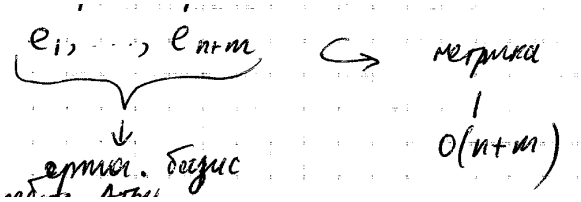


Original source details: GELFAND-P000952

Undeciphered passage. The lower limit in the second sum reads -1. The entry breaks off after $[\Delta_{n-1}^m]$.

(13) Theorem. $\forall$ harm. form in G_n^m vanishes on any Grassm. simplex.

The metric is ass. to be compatible in some way with the coordinates.



Original source details: GELFAND-P000952

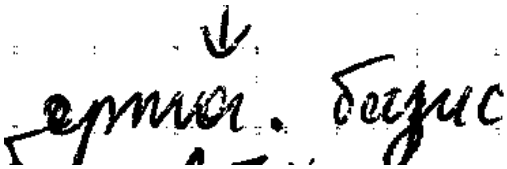
Diagram and typed description. A brace under the list leads to the basis. The stroke below “metric” has no visible arrowhead.

$\underbrace{e_i, \ldots, e_{n+m}}_{\downarrow}$ $\hookrightarrow$ metric

orthog. basis (?) $O(n+m)$

Diagram and typed description. The first e has a dotted subscript i.

Original source details: GELFAND-P000952



Uncertain reading. The abbreviation is tentatively “orthog.”; “orthon.” is also possible.

Original source details: GELFAND-P000952

On Atiyah’s work.

Diagram and typed description. The manuscript spells the name “Атьи”.

Original source details: GELFAND-P000952

Hence the funct. eq. for the dilogarithm.

Space of 5-tuples of points on P^1 up to proj. transf.

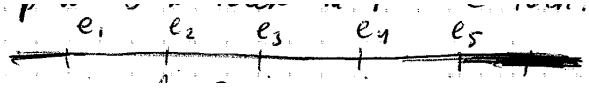
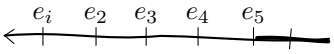


Diagram and typed description. Line with five labels and an unlabelled tick on the right. The darkened end is drawn approximately.



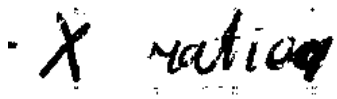
Original source details: GELFAND-P000952

Diagram and typed description. The first e is labelled with a dotted i.

Original source details: GELFAND-P000952

This space is 2-dimensional.

$$k_1 = \text{X ratio } (e_2\, e_3\, e_4\, e_5)$$
$$k_2 = \text{—”—} (e_1\, e_3\, e_4\, e_5)$$



Uncertain reading. The sign before “ratio” is tentatively read as X. The shapes of X and $\times$ are close.

Original source details: GELFAND-P000952

...

$$\varphi(x) = \mathrm{Li}(x) - \mathrm{Li}(1-x), \qquad \mathrm{Li}(x) = \sum_1^\infty \frac{x^n}{n^2}.$$

$$0 = d\varphi(k_1) - d\varphi(k_2) + d\varphi(k_3) - d\varphi(k_4) + d\varphi(k_5) \tag{14}$$

(Funct. eq. for the dilogarithm)

This follows from the previously formul. theorems.

G_3^2 — 3-dim. planes in $\mathbb{R}^5$.

Consider $P_1(\xi)$ (1st Pontryagin class of the tautol. 2- or 3-dim. bundle on the Grassm. G_3^2).

There is exactly 1 harm. 4-form.

GELFAND-P000953

[source number: 53]

$$\underset{\|}{d}\pi_* P_1(\xi) = \pi_* \underset{\|}{d} P_1(\xi) + \sum \pi_{i*}^{\partial} P_1(\xi)$$
$$0 \qquad\qquad 0$$

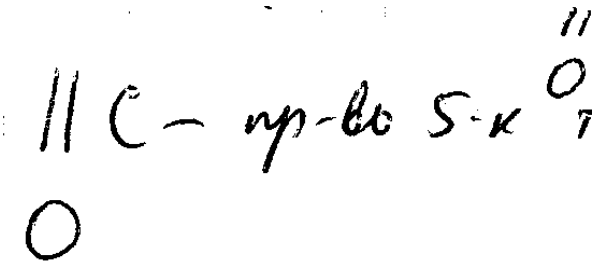


Diagram and typed description. Two strokes and a zero appear below each of the two d symbols.

Original source details: GELFAND-P000953

C — space of 5-tuples of points

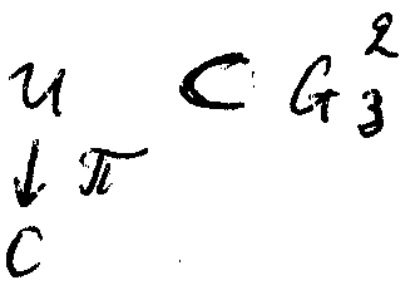


Diagram and typed description. The sign to the left of $\subset$ resembles lowercase u; uppercase U is also possible.

$$\begin{array}{ll} u? & \subset G_3^2 \\ \downarrow \pi & \\ C & \end{array}$$

Original source details: GELFAND-P000953

The last term is calc. with the aid of combinatorics

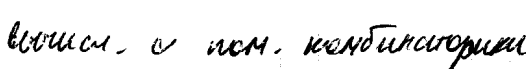


Diagram and typed description. The words before “combinatorics” read “with the aid of”.

Original source details: GELFAND-P000953

Generalization.

Let $(\mathbb{C}^*)^n$ act algebraically on a homogeneous algebr. [crossed out: entry unresolved] space with a fixed point

Undeciphered passage. The entry between “algebr.” and “space” is heavily crossed out.

Original source details: GELFAND-P000953

Then

1) $\forall$ orbit of $\left((\mathbb{R}^+)^*\right)^n$ is homeom. to a polyhedron. (theorem on toric embeddings)

(Diffeom. on $\forall$ orbit, but at the vertices it may have ∞ curvature; on the whole space — only a homeom.)

2) $\forall$ harm. form [crossed out: entry unresolved] invar. under $T^n \subset (\mathbb{C}^*)^n$

Undeciphered passage. Several strokes obscure the entry after “form”.

Original source details: GELFAND-P000953

[inserted below “harm. form”: (with respect to $\forall$ Kähler metric)]

Diagram and typed description. The condition on the metric is written on the line below.

Original source details: GELFAND-P000953

is equal to 0 on $\forall$ $\left((\mathbb{R}^+)^*\right)^n$-orbit.

(15) Theorem. (Atiyah).

M — symplect. manif., $f_1, \ldots, f_n \colon M \longrightarrow \mathbb{R}$ functions.

Diagram and typed description. The Russian abbreviation after “symplect.” reads “мн-зие” (manifold).

Original source details: GELFAND-P000953

$$\{f_i, f_j\} = 0$$

df_i all vanish to 0 at some point.

We get a map

$$M \underset{f_1,\ldots,f_n}{\longrightarrow} \mathbb{R}^n$$

Here

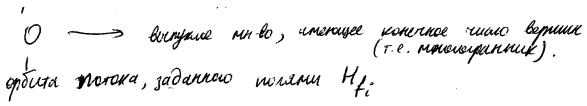


Diagram and typed description. An arrow leads from O to the convex set. A stroke below O leads to the orbit label.

Original source details: GELFAND-P000953

O ⟶ convex set having finitely many vertices (i.e. a polyhedron).

|

orbit of the flow given by the fields H_{f_i}

Diagram and typed description. Two strokes with a dot between them.

Original source details: GELFAND-P000953

Problem: relation to integr. by quadratures.

Chapter 19

GELFAND-P000954

16/II-81. Gelfand Sem.

Original source details: GELFAND-P000954

Diagram and typed description. A narrow hatched bar underlines the date. II has a stroke above and below it.

16/II-81.

Khovanskii. On Newton polyhedra.

[At the right: (reverse side of the prev. talk by Mc Pherson.)]

Original source details: GELFAND-P000954

Diagram and typed description. The note identifies this as the reverse side of the preceding talk.

$n = 2$ for simplicity.

refs.: 1) Mumford et al. Lect. Notes — toric embeddings.

2) D. Bernstein, FA.

1) Kempf, Knudsen, Mumford, Saint-Donat, Toroidal Embeddings,

Lect. Notes Math., 1972, $N = 339$.

Original source details: GELFAND-P000954

Diagram and typed description. The manuscript citation gives the year 1972 and number 339.

GELFAND-P000955

and also a survey by Danilov in UMN.

Articles by Khovanskii: FA, 1977, 11:4

1978, 12:1

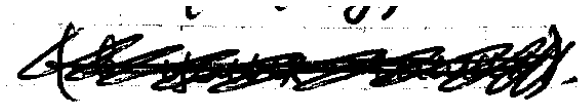
$$\{(x, y) : P(x, y) = 0\} \tag{1}$$

what does this curve look like? (handles, etc.)

How many solutions does the system have

$$\begin{cases} P(x, y) = 0 \\ Q(x, y) = 0 \end{cases} \tag{2}$$

[crossed out: (line unresolved).]



Original source details: GELFAND-P000955

Undeciphered passage. The parenthetical line is heavily crossed out.

$$y^2 = P_n(x) \tag{3}$$

has $\sim n/2$ handles (more precisely, $\left[\frac{n-1}{2}\right]$ handles).

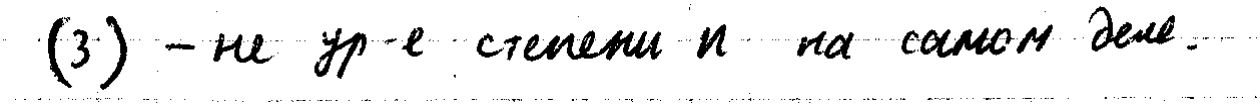
$$\begin{aligned} n = 3, 4 \quad & 1 \text{ handle}, \\ n = 5, 6 \quad & 2 \text{ handles}. \end{aligned}$$

Condition: P_n has no multiple roots.

For a general eq. (1) there is also a recipe for computing the number of handles:

$$\frac{(n-1)(n-2)}{2} \qquad \text{for an eq. in general position.} \tag{4}$$

(3) — is not an eq. of degree n in fact.

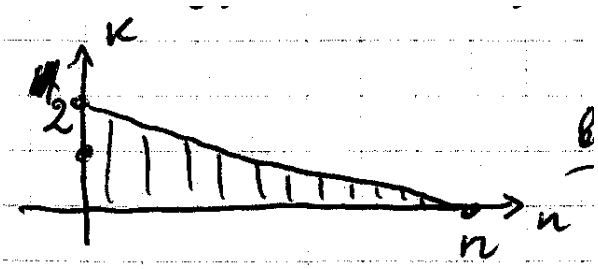


Original source details: GELFAND-P000955

Diagram and typed description. The sentence about the degree in (3) follows the manuscript wording.

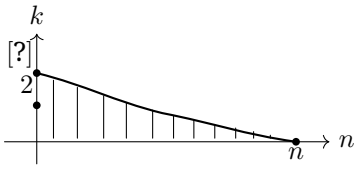
It all comes down to the Newton polyh..

$$P(x, y) = \sum a_{n,k} x^n y^k. \tag{5}$$



Original source details: GELFAND-P000955

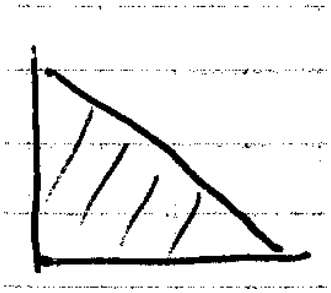
Diagram and typed description. Hatched triangle, axes k, n, and marked points. The label above 2 is crossed out.



Original source details: GELFAND-P000955

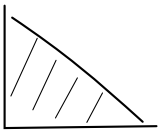
Undeciphered passage. The heavily crossed-out label above 2 is unresolved.

Convex hull of those (n, k), for which $a_{nk} \neq 0$.



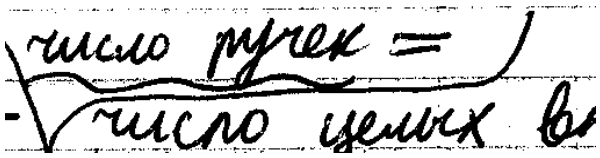
Original source details: GELFAND-P000955

Diagram and typed description. The second triangle, with four interior strokes.



— gen. case — simplex.

Answer for (1) — [inserted above the line: number of handles =] number of interior lattice points in the Newton polyh.



Original source details: GELFAND-P000955

Diagram and typed description. "number of handles" and the equals sign are inserted above the answer.

(for coeffs. $a_{n,k}$ in general position)

This is a theor. of Baker (that very one), rediscovered by Khovanskii in 1976.

(Baker taught Hodge, Hodge taught Atiyah)

In $\mathbb{C}^2$ some points of the curve (1) are hidden. We want to bring them out by certain substitutions.

Consid. the curve $P(x, y) = 0$ away from the coordinate cross.

x, y — meromorphic fns. on a Riem. surface.

Sometimes they become 0 or ∞. This gives a divisor:

at $\forall$ of its points there is a vector (n, m)

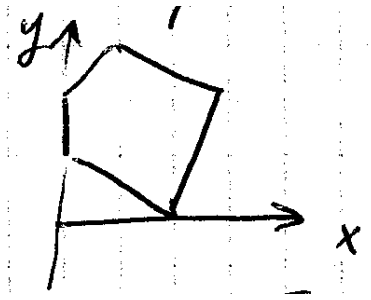
(order of the zero of $x = n$, order of the zero of $y = m$ at the given point)

GELFAND-P000956

[source number: 6]

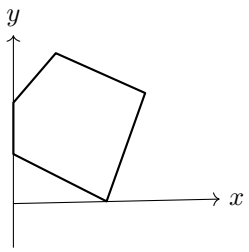
Points of order $(1, 0)$ or $(k, 0)$ are visible in the picture: they are the points of inters. with the axis $x = 0$.

We can transl. the polygon by multiplication by x^k and y^k.



Original source details: GELFAND-P000956

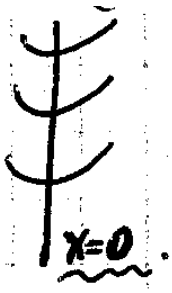
Diagram and typed description. The pentagon meets the y-axis along an edge and the x-axis at a vertex.



Translate so that it touches the axis $x = 0$.

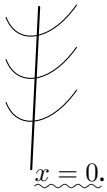
$$P(0, y) = 0 \tag{6}$$

Number of sols. $=$ length of the segment of intersection of NP with the axis $\{x = 0\}$.
(if there are no multiple roots



Original source details: GELFAND-P000956

Diagram and typed description. Three branches cross the vertical line labelled $x = 0$.



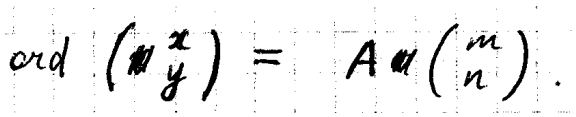
nonmultiple $\Rightarrow$ the implicit fn. thm. applies
$\Rightarrow$ there are no points of the form $(k, 0)$, $k \geqslant 2$.
Make a transf.

$$\begin{cases} x = u^{a_{11}} v^{a_{12}}, \\ y = u^{a_{21}} v^{a_{22}}, \end{cases} \qquad a_{ij} \in \mathbb{Z} \tag{7}$$

$$\det \begin{Vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{Vmatrix} = 1, \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{8}$$

the inverse is integer-valued $\Rightarrow$ there is an inverse transf. (also rational)
If $u \sim t^m$, $v \sim t^n$, then after the transformation

$$\text{ord} \begin{pmatrix} [\text{crossed out}]\, x \\ y \end{pmatrix} = A\, [\text{crossed out}] \begin{pmatrix} m \\ n \end{pmatrix}. \tag{9}$$



Original source details: GELFAND-P000956

Uncertain reading. The signs left of the column x, y and after A are heavily crossed out. Both resemble u, but the exact reading is unresolved.

We want a transf. such that at the pt. $\binom{n}{m}$ there lands $\binom{1}{0}$.
Let the vector (n, m) be primitive ($\sim$ on the segment $[0, (m, n)]$ there are no lattice points).

Original source details: GELFAND-P000956

Uncertain reading. The first character of the pair is heavily retraced and tentatively read as n. The next entry has the order (m, n).

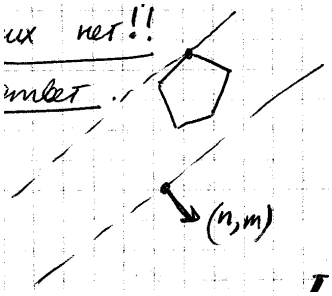
Moral: ∃ a matrix A.

If [crossed out: ~~it~~ (?); inserted above the line: (n, m)] is reducible, then this same transf. shows that there are no such divisors!!

Original source details: GELFAND-P000956

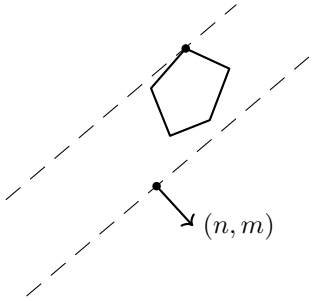
Uncertain reading. Above the crossed-out word, tentatively "it", is the insertion (n, m).

answer.



Original source details: GELFAND-P000956

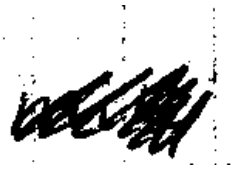
Diagram and typed description. One of the parallel lines meets a vertex of the pentagon. An arrow labelled (n, m) starts at the marked point on the other line.



move it until it touches

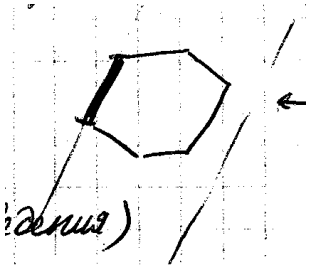
[crossed out: ~~if~~ (?)] We hit a point —

— then there are none of this kind.



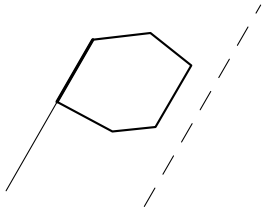
Original source details: GELFAND-P000956

Uncertain reading. The crossed-out word is tentatively read as "if".

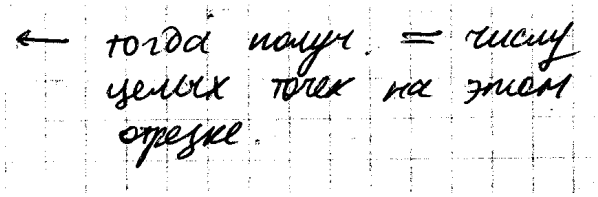


Original source details: GELFAND-P000956

Diagram and typed description. The left edge of the hexagon is emphasized on a sloping line. A parallel line is to the right.



← then we get = the number of lattice points on this segment.



Original source details: GELFAND-P000956

Diagram and typed description. The last line gives "the number of lattice points on this segment".

(take the min of the scalar product)

Total number of punctures = the number of lattice points on the boundary.

Consider now the system

$$\begin{cases} P(x,y)=0 \\ Q(x,y)=0 \end{cases} \tag{10}$$

and seek the number of sols. in $(\mathbb{C}\setminus 0)^2$

([crossed out: unresolved] $x\neq 0,\infty$; $y\neq 0,\infty$).

Original source details: GELFAND-P000957

Uncertain reading. A short entry before the condition on x is obscured by crossing-out.

Consid. the fn. Q on the curve $P(x,y)=0$.

Its divisor $=0$,

Original source details: GELFAND-P000957

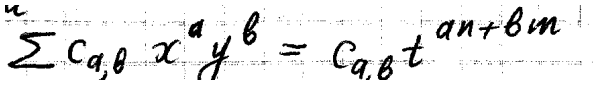
Diagram and typed description. The manuscript reads “Its divisor $=0$”.

Look at the order at the omitted pts.!!

(n,m) $\quad x\sim t^n,\ y\sim t^m$.

Then

$$\sum C_{a,b}x^a y^b = C_{a,b}t^{an+bm} \tag{11}$$



Original source details: GELFAND-P000957

Diagram and typed description. In (11) the summation sign stands to the left of the equality.

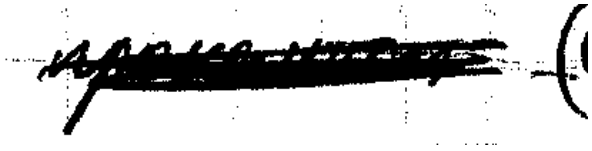
answer:

$$\underline{\min_{a,b}\big((n,m)\cdot(a,b)\big)} \quad \text{— a purely geom. answer.} \tag{12}$$

Punctured points — on the cross or at ∞

$\Rightarrow$ poles only at the punctured points.

$$\left[\sum_{m,n}(12)\right] = \text{the number of sols. of (10) in [crossed out: ~~the space~~ (?)] } (\mathbb{C}\setminus 0)^2. \tag{13}$$



Original source details: GELFAND-P000957

Uncertain reading. The crossed-out word before $(\mathbb{C}\setminus 0)^2$ is tentatively read as “the space”.

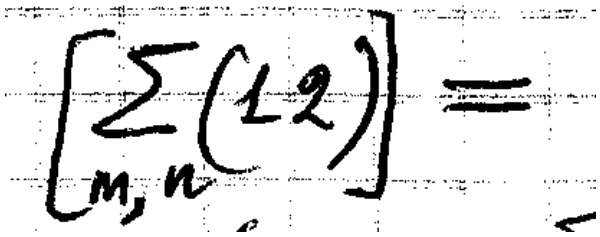


Diagram and typed description. There is no minus sign before the summation in (13).

Original source details: GELFAND-P000957

(for gen. pos. — otherwise cancellation may occur).

There are 2 polygons!! What do we get?

Answer — mixed volume!!

Add all the vects. (n, m) — we get a vector of length $=$ the length of the side.

Diagram and typed description. Three successive arrows point down and to the right.

Original source details: GELFAND-P000957

A vector $\perp$ to the side and equal to it in length.

Suppose the 2 polygons coincide (i.e. the same nonzero coefficients! of P and Q).

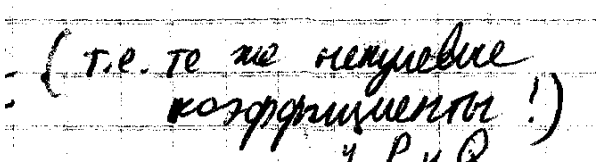


Diagram and typed description. The parenthesis reads “the same nonzero coefficients!”

Original source details: GELFAND-P000957

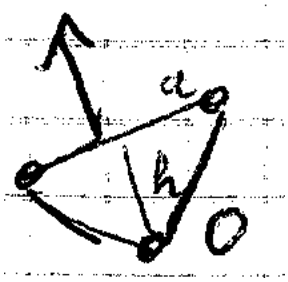
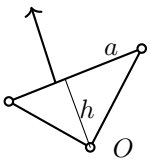


Diagram and typed description. Triangle with side a, height h, vertex O, and an outward arrow.



Original source details: GELFAND-P000957

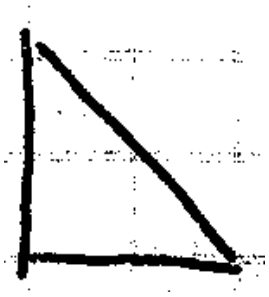
max for the triangle among all scalar prods. $=$

Diagram and typed description. The first word reads Latin max.

Original source details: GELFAND-P000957

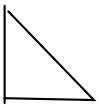
$= ah = 2S.$

(fix O)



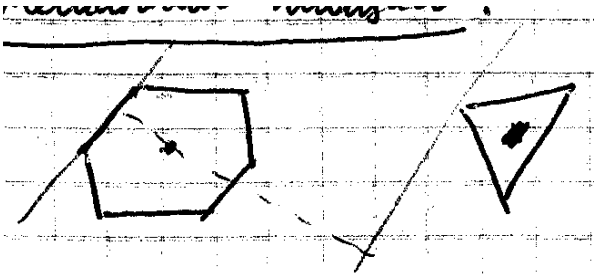
Original source details: GELFAND-P000957

Diagram and typed description. Small unlabelled right triangle.



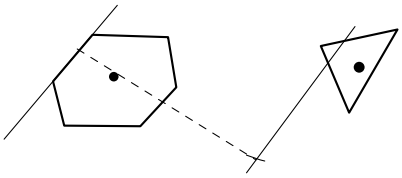
$\triangle$-gon $\Rightarrow$ we get n^2 (Bézout's theor.).

Mixed area.

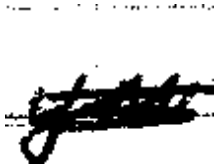


Original source details: GELFAND-P000957

Diagram and typed description. Hexagon and triangle with marked points. The dashed line and short strokes at the right-hand line are drawn approximately.



sides [crossed out: ~~mult.~~ (?)] of one mult. by the heights of the other.



Original source details: GELFAND-P000957

Uncertain reading. The crossed-out word after "sides" is tentatively read as "mult."

A common point is marked O.

$\forall$ side is moved until it supports the 2nd — this is the height.

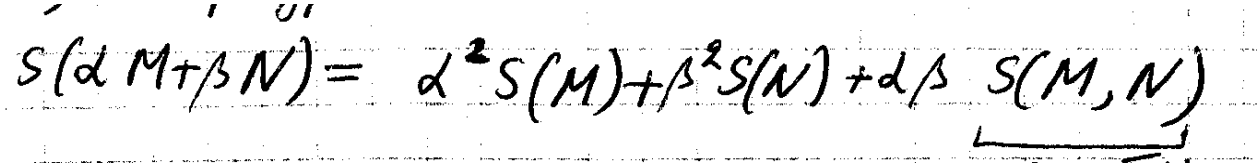
GELFAND-P000958

[source number: 7]

General defn.

M, N — figures.

$$S(\alpha M + \beta N) = \alpha^2 S(M) + \beta^2 S(N) + \alpha\beta \underbrace{S(M,N)}_{\text{mixed volume.}}$$



Original source details: GELFAND-P000958

Diagram and typed description. The first equality has $S(M,N)$ as the coefficient of $\alpha\beta$. The manuscript then uses V.

In n-dimensional space — mixed volume n figures.
$V(\Delta_1, \ldots, \Delta_n)$ is def. by the properties:

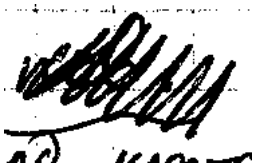
1) $V(\Delta, \ldots, \Delta) = V(\Delta)$ — volume.
2) $V(\Delta_1 + \Delta_1', \Delta_2, \ldots) = V(\Delta_1, \Delta_2, \ldots) + V(\Delta_1', \Delta_2, \ldots)$

Such a fn. [crossed out: entry unresolved] exists and is unique.

Original source details: GELFAND-P000958

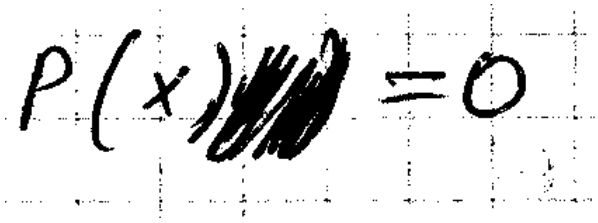
Uncertain reading. A short entry after "fn." is crossed out.

(14) Thm. (D. Bernstein)
Number of sols. (in $(\mathbb{C} \setminus 0)^2$) of a system in gen. pos. of form (2) $=$
$=$ mixed volume of their Newton polyh..
(15) Thm.
Euler char. [crossed out: ~~of the curve~~ (?)] of the hypersurf., $P(x)$ [crossed out: $, y)$ (?)] $= 0$



Original source details: GELFAND-P000958

Uncertain reading. The crossed-out word before "hypersurf." is tentatively read as "of the curve".

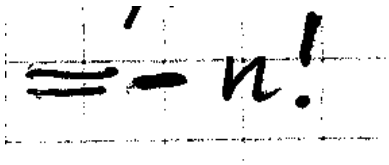


Original source details: GELFAND-P000958

Uncertain reading. A heavily crossed-out entry follows $P(x)$, possibly $, y)$.

(with the coord. cross removed)

$$= -n!(\text{volume of the Newton polyh.})$$



Original source details: GELFAND-P000958

Diagram and typed description. A minus sign is written before $n!$.

Corollary (geom.)

$$2 - 2B - \Gamma = -2S.$$

B — number of interior lattice pts..

Γ — —"— boundary —"—

$$B + \tfrac{1}{2}\Gamma = S + 1. \tag{16}$$

Relation of mixed volume to the isoperimetric inequality.

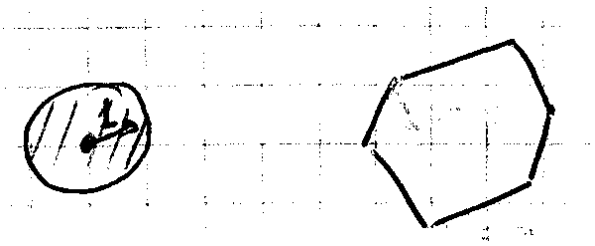
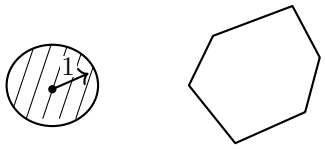


Diagram and typed description. Hatched circle with radius 1 and a hexagon.



Original source details: GELFAND-P000958

$$S(\Delta_1, \Delta_2) = S(\Delta_1 + \Delta_2) - S(\Delta_1) - S(\Delta_2). \tag{17}$$

O — centre of the circle.

Mixed area of the polygon and the circle = the perimeter of the polygon.

Inequality:

$$V^2(\Delta_1, \Delta_2) \geqslant V(\Delta_1)V(\Delta_2). \tag{18}$$

Attained on the circle.

Theorem for any alg. surface.

Γ_1, Γ_2 — two curves on an algebr. surface.

$$(\Gamma_1, \Gamma_2)^2 \geqslant (\Gamma_1, \Gamma_1) \cdot (\Gamma_2, \Gamma_2). \tag{19}$$

[label by (Γ_1, Γ_2): inters. index]

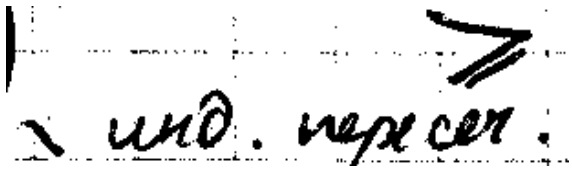
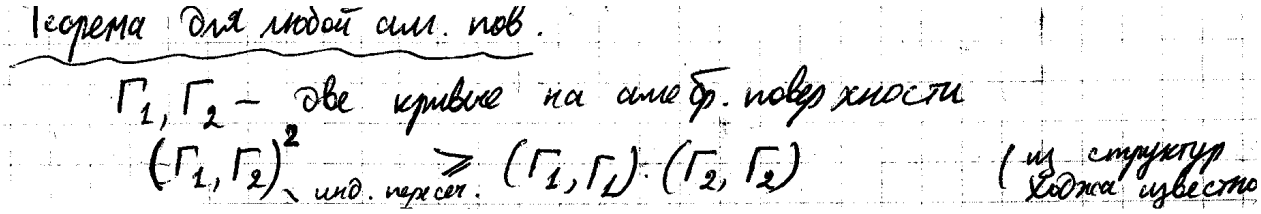


Diagram and typed description. A short stroke leads from the term on the left to "inters. index".

Original source details: GELFAND-P000958

(known from Hodge structures)



Original source details: GELFAND-P000958

Diagram and typed description. On this frame, (19) is accompanied by the two curves on an algebraic surface and the note on Hodge structures.

GELFAND-P000959

We construct a special compactif. of $(\mathbb{C} \setminus 0)^2$.

There is a set $P(x, y) = 0$.

We want to describe its charts so that the transitions are only monomial and yet it would be possible to consider everything,

what compact complex manifolds there are which are covered by a system of charts, the transition being given by unimod. matrices.

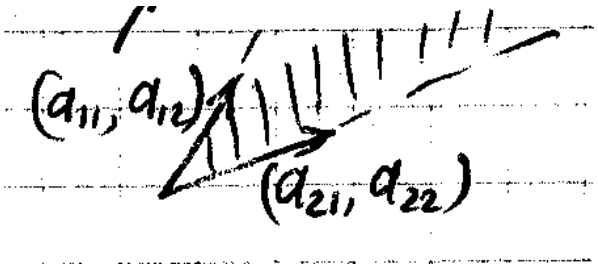
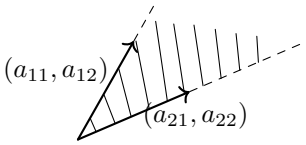


Diagram and typed description. Hatched cone with two labelled rays. The rays continue in broken strokes beyond the arrowheads.



Original source details: GELFAND-P000959

$$\binom{x}{y} \in \underline{\underline{\text{the torus}}}$$
$$(x \neq 0, y \neq 0)$$
$$(\mathbb{C} \setminus 0)^2$$

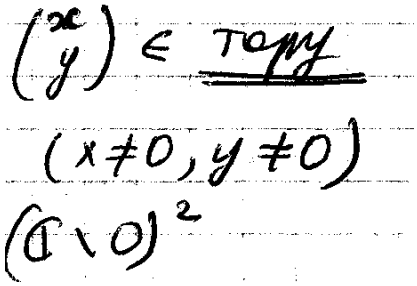


Diagram and typed description. The word “тору” (“the torus”) is underlined twice.

Original source details: GELFAND-P000959

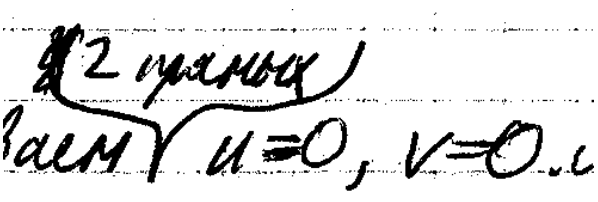
[under the figure on the left: crossed out ~~it~~ (?)]

Uncertain reading. The crossed-out entry resembles “он” (“it”); the reading is unresolved.

Original source details: GELFAND-P000959

For now it is noncompact.

With each transf. (7) we glue on $u = 0$, $v = 0$ [above the entry in a brace: “2 lines”] and draw a cone.



Uncertain reading. At the start of the brace, left of “2 lines”, there is a dark stroke. It is unclear whether a separate entry was crossed out here.

Original source details: GELFAND-P000959

If they intersect, the construction is impossible.

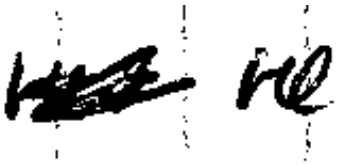
It is possible if the cones cover all of $\mathbb{R}^2$.

GELFAND-P000960

[source number: 12]

23/II–81. Gelfand seminar.

I.M.G. Dixmier found an example of 2 oper. A_1 and A_2 of orders 4 and 6, [crossed out: ~~not~~ (?)] that are not functions of one oper. A_0 of 2nd order; A_1, A_2 — with polynomial coeff. (and leading coeff. 1), such that $[A_1, A_2] = 0$.



Original source details: GELFAND-P000960

Uncertain reading. The crossed-out entry before “are not” tentatively reads “не” (“not”).

$$A_2 = \sqrt{\underbrace{\left(\frac{d^4}{dx^4} + f_0(x)\frac{d^3}{dx^3} + \dots\right)^3}_{A_1^3} + C}.$$

Grinevich found many such examples via Tyurin parameters.

Cherednik I.

On an event that brings soliton theory to a close.

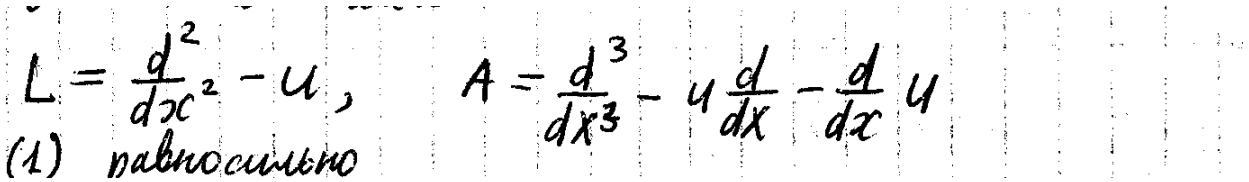
Stage I. Inverse problem method.

$$u_t = 6uu_x + u_{xxx}. \tag{1}$$

$$L = \frac{d^2}{dx^2} - u, \qquad A = \frac{d^3}{dx^3} - u\frac{d}{dx} - \frac{d}{dx}u. \tag{2}$$

Then (1) is equivalent to

$$[L, A] = L_t. \tag{3}$$



Original source details: GELFAND-P000960

Diagram and typed description. In (2), a minus sign precedes each of the two terms with u. No numerical coefficient precedes $\frac{d^3}{dx^3}$.

$$L \rightsquigarrow \text{scattering matr. } S(\lambda) \qquad (2\times 2 \text{ matrix}). \tag{4}$$

$S(\lambda)$ changes in a simple way as one varies t.

Stage II. (Zakharov et al.)

$$\left[\underbrace{\frac{d}{dx}-U}_{L}, \frac{d}{dt}-V\right]=0 \qquad \text{(zero-curvature representation)}. \tag{5}$$

U, V — rat. functions of λ.

For example,

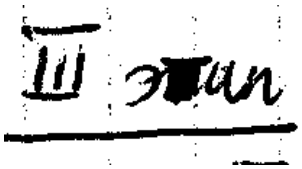
$$\begin{cases} U = U_0 + U_1\lambda + U_2\lambda^2, \\ V = V_0 + \lambda V_1 \end{cases} \qquad — \tag{6}$$

and one obtains a system of eqs.

$L \rightsquigarrow S$ and so on.

The Hamiltonian character was observed.

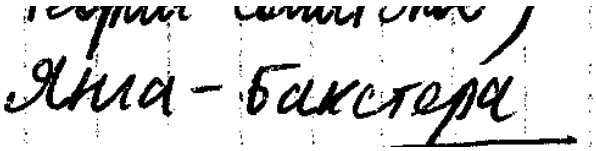
Stage III (the twilight of soliton theory)



Original source details: GELFAND-P000960

Uncertain reading. One stroke in the word “этап” (“stage”) is heavily retraced. The underlying entry is unresolved.

Yang–Baxter identities.

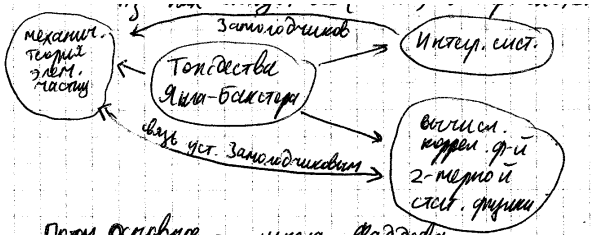


Original source details: GELFAND-P000960

Diagram and typed description. The manuscript spells the name “Яна-Бакстера”, without the Russian letter “г”.

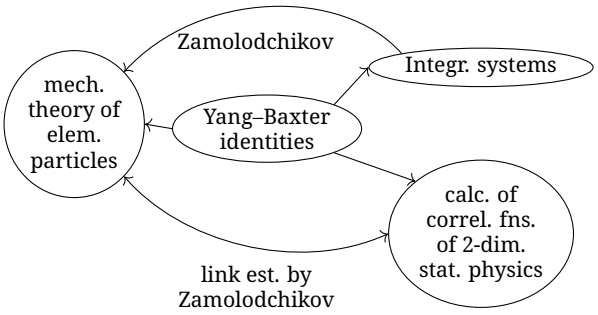
Algebr. object $\rightsquigarrow$ everything, with minor nuances.

From them one obtains all (almost) integr. systems.

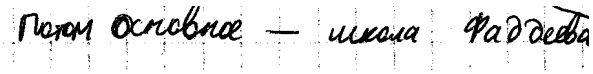


Original source details: GELFAND-P000960

Diagram and typed description. Three arrows leave the middle oval. The upper arc points left, and the lower arc has arrowheads at both ends.



Then (?) the main thing is the Faddeev school.



Uncertain reading. The first word of the last line looks closer to “Потом” (“Then”); the strokes remain indistinct.

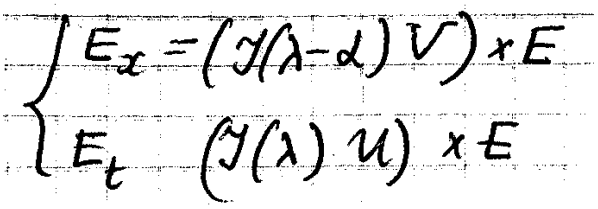
Original source details: GELFAND-P000960

GELFAND-P000961

Landau–Lifshitz eqs. and ... chir. fields

$$\left[\frac{d}{dt} - U, \frac{d}{dx} - V\right] = 0, \qquad \lambda \in \text{ell. curve!!} \tag{7}$$

$$\begin{cases} E_x = (J(\lambda - \alpha)V) \times E, \\ E_t \quad (J(\lambda)U) \times E \end{cases} \tag{8}$$



Uncertain reading. In the second line of (8) no sign is visible between E_t and the parenthesis.

Original source details: GELFAND-P000961

E, V, U — 3-vectors.

shift on the ell. curve

$$J = \text{diag}(\cdot, \cdot, \cdot).$$

hence

$$\begin{cases} U_x = (JV) \times U, \\ U_t = (JU) \times V \end{cases} \qquad \text{(asymmetric principal chiral field).} \tag{9}$$

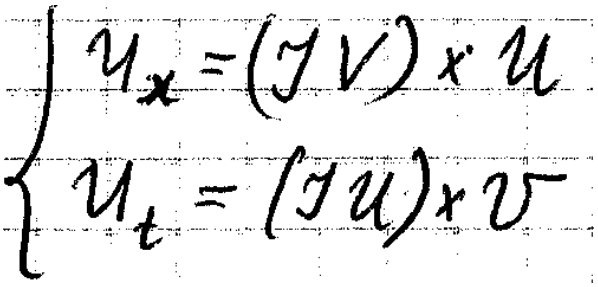


Diagram and typed description. The second line of (9) begins with U_t.

Original source details: GELFAND-P000961

J — const. diagon. 3-matrix.

$$J = J(\alpha).$$

The matrix J — arbitrary!!

(Attainable by choice of α and the ell. curve).

It is integrated using the ell. curve.

Landau–Lifshitz eq.:

$$U_t = U \times U_{xx} + JU \times U. \tag{10}$$

The Y.–B. identity at once gives the entire Hamiltonian formalism together with the brackets of the scattering data.

Khovanskii A.

We want:

$$\overset{\swarrow \text{manifold}}{M^n} \supset (\mathbb{C} \setminus 0)^n$$ we will construct a special compactif. of the torus.

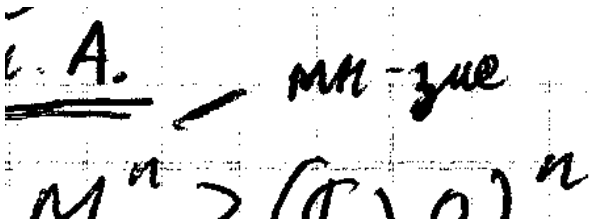


Diagram and typed description. The note “мн-зие” (“manifold”) above M^n is linked to it by a slanting stroke.

Original source details: GELFAND-P000961

and the complement is an analyt. set [crossed out: mark unresolved] of smaller dimension.

Uncertain reading. A short crossed-out mark is visible before “smaller”; its reading is unresolved.

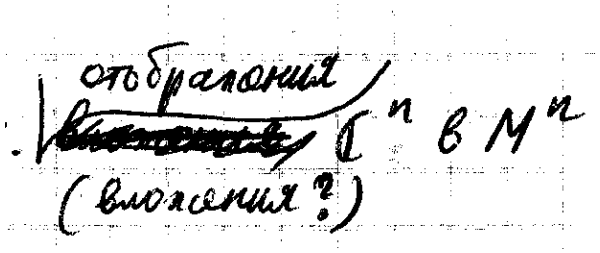
Original source details: GELFAND-P000961

Modular charts:

$$\begin{cases} x_1 = u_1^{a_{11}} \dots u_n^{a_{1n}} \\ \dots\dots\dots\dots \\ x_n = u_1^{a_{n1}} \dots u_n^{a_{nn}} \end{cases} \tag{1}$$

and extend to analyt. [crossed out: ~~embeddings~~ (?); inserted above: maps] $\mathbb{C}^n$ into M^n.

[beneath the revised word:] (embeddings?)



Uncertain reading. “Maps” is written above the cancellation. Below it one reads “(embeddings?)”; the cancelled word may be the same.

Original source details: GELFAND-P000961

One-param. subgroup (of the torus):

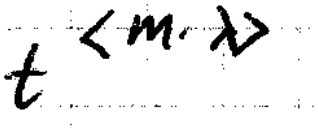
$$\begin{cases} x_1 = t^{\lambda_1} \\ \quad \vdots \\ x_n = t^{\lambda_n} \end{cases} \tag{2}$$

GELFAND-P000962

[source number: 13]

$$\underline{\text{Monomial}} \quad x_1^{m_1} \dots x_n^{m_n} \tag{3}$$

behaves as $t^{\langle m,\lambda\rangle}$ on subgroup (2).



Uncertain reading. The short stroke between m and λ is rendered as a comma. It is close to a dot.

Original source details: GELFAND-P000962

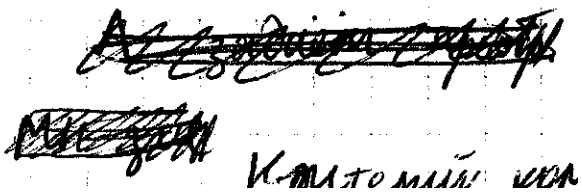
For $\forall$ chart we draw a cone in the space of one-parameter subgroups.
They are arrang. in a certain way.

$$\begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{nn} \end{pmatrix}$$ generate a conv. cone.

Intersect in their interiors $\Rightarrow$ they coincide. (condition for existence of the manifold)
Example:

$$\begin{cases} x_1 = u_1 u_2^{-1} \\ x_2 = u_2 \end{cases} \qquad \begin{cases} x_1 x_2 = u_1 \\ x_2 = u_2 \end{cases}$$

This is a σ-process.



Undeciphered passage. Two entries are crossed out, one before “If the faces” and one at the start of the next line. The words under the dense strokes are unresolved.

Original source details: GELFAND-P000962

If the faces intersect in their interiors, they likewise coincide.

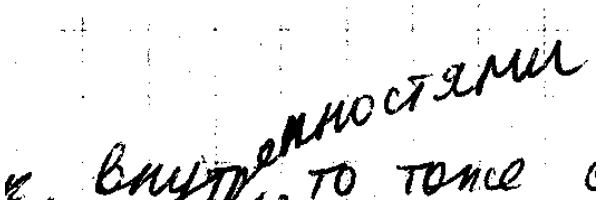
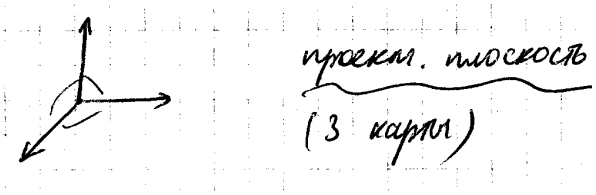


Diagram and typed description. The Russian ending “енностями”, completing “interiors”, is added diagonally after the abbreviation.

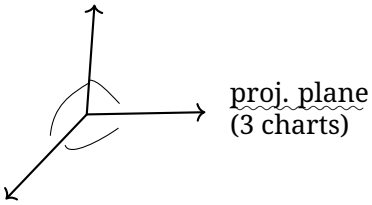
Original source details: GELFAND-P000962

Compactness criterion: nec. and suff. that the cones cover the entire space.



Original source details: GELFAND-P000962

Diagram and typed description. Three directed rays and three arcs without arrowheads.



The compactif. is specified by a subdivision of the integer space up to an integer automorphism.

Kazarnovskii.

Problem: Generalize the theorem on zeros of polynomials to functions of a more general form.

$$f = \sum_{\lambda \in \Lambda} C_\lambda \exp\langle z, \lambda \rangle \tag{1}$$

regarded as functions in $\mathbb{C}^n$; $\lambda \in \Lambda \subset \mathbb{C}^n$, $z \in \mathbb{C}^n$

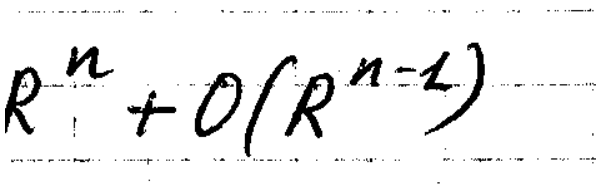
$$\Gamma = \mathrm{conv}(\Lambda) \qquad \text{— Newton polyh.}$$

Consider the system

$$F : \{f_1 = \ldots = f_n = 0\}.$$

The number of zeros in the ball of rad. R is denoted $N_F(R)$

$$N_F(R) = cV(\Gamma_1, \ldots, \Gamma_n)R^n + O(R^{n-1}). \tag{2}$$



Original source details: GELFAND-P000962

Diagram and typed description. The exponent in the remainder term of (2) is written $n - 1$.

$V(\Gamma_1, \ldots, \Gamma_n)$ — mixed volume if all the polyh. lie in the real parts of $\mathbb{C}^n$

1 variable $\Rightarrow$ the perimeter of a polygon appears

c — const, consisting of powers of 2 and π.

GELFAND-P000963

What is V in the general case?

$$\Gamma \subset \mathbb{C}^n,$$

$$V(\Gamma) = \sum_{\dim \Delta = n} \underset{\substack{\downarrow \\ \text{volume of face}}}{V(\Delta)} \quad \Omega(\Delta) S(\Delta).$$

Faces of middle dim. (n [mark unresolved])

Original source details: GELFAND-P000963

Uncertain reading. A heavy stroke follows n inside the parentheses. An exclamation mark is possible, but no separate dot is visible.

$\Omega(\Delta)$ — [crossed out: mark unresolved] ext. angle (volume of the spher. image)
[under "volume of the spher. image":] (n-dimensional).

Uncertain reading. A short entry before "ext." is crossed out. The strokes obscure the reading.

Original source details: GELFAND-P000963

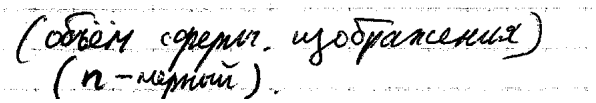


Diagram and typed description. "(n-dimensional)" is written below the parenthetical phrase about the volume of the spherical image.

Original source details: GELFAND-P000963

$S(\Delta)$ is def. as follows

Consider the real Grassmannian $G_{2n,n}$; more precisely, real n-dim. subsp. in $\mathbb{C}^n$.

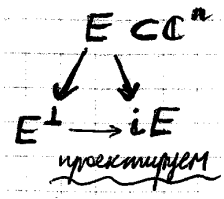


Diagram and typed description. Two arrows leave $E \subset \mathbb{C}^n$. The horizontal arrow goes from $E^\perp$ to iE.

Original source details: GELFAND-P000963

$E \subset \mathbb{C}^n$

$E^\perp \longrightarrow iE$

we project

area multiplied by the number denoted $S(\Delta)$.

$V(\Gamma)$ extends from polyhedra to arbitrary bodies $M \subset \mathbb{C}^n$.

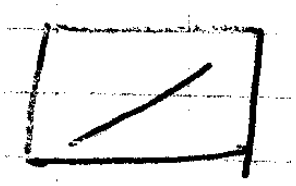
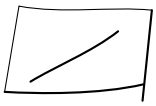


Diagram and typed description. Unlabelled quadrilateral with a slanting stroke inside. Its right side extends below the bottom edge.



Original source details: GELFAND-P000963

T_cM — complex tang. plane to ∂M.

B — 2nd quadr. form, def. on T_cM

The quadr. form $\dfrac{B(ix)+B(x)}{2}$ def. a Hermitian form

(Levi form)

$\kappa_1, \ldots, \kappa_{n-1}$ — eigenvalues.

$$S(\Delta) = \int \kappa_1 \ldots \kappa_{n-1}\, dS$$

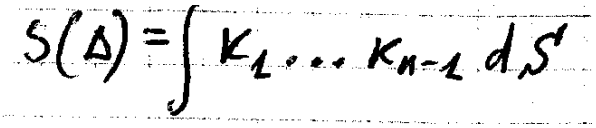


Diagram and typed description. The left side of the last formula is written $S(\Delta)$.

Original source details: GELFAND-P000963

Another version: via the support fn.

GELFAND-P000964

2.3.81. Gelfand seminar.

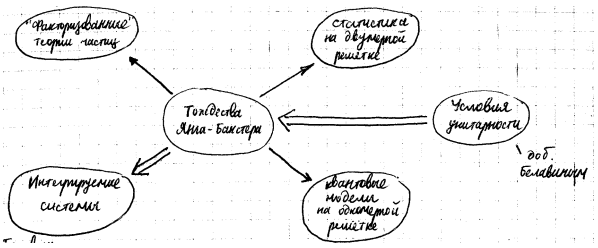
I.M.G. Problem. $\int_x^\infty e^{-t^2}\,dt$ expand in an asympt. series which diverges but is Borel summable.

Problem. Relate Kazarnovskii's result to the growth indicatrix.

GELFAND-P000965

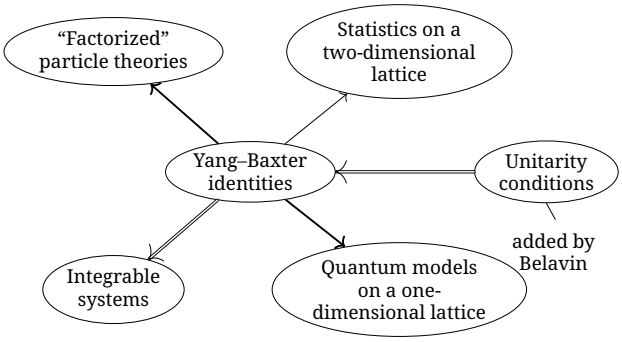
[source number: 19]

Cherednik I.

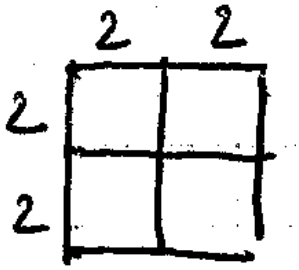


Original source details: GELFAND-P000965

Diagram and typed description. Six ovals. The double arrow on the right points toward the identities, and the one at the lower left points away from them. The middle oval spells the name "Янa-Бакстера".

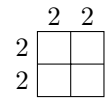


Belavin.



Original source details: GELFAND-P000965

Diagram and typed description. Four blocks, each with two rows and two columns.

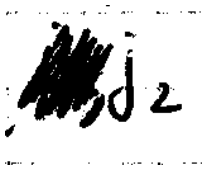
(1)  $= R^{j_1 j_2}_{i_1 i_2} \longrightarrow R^{1\,2}$

(abbrev. notation)

Gen. case: $n^2 \times n^2$.

i_1, j_1 index the blocks

i_2 [crossed out: entry unresolved], j_2 — entries within a block.



Original source details: GELFAND-P000965

Uncertain reading. The entry between i_2 and j_2 is obscured by strokes.

Interpr. as a tensor product.

$$R^{12} \xrightarrow{\Omega} (R^{12})^{-1} \qquad \underline{\text{unitarity transf.}} \tag{2}$$

$$R^{12} \xrightarrow{C} R^{\bar{1}2} \qquad \underline{\text{crossing}} \tag{3}$$

(Block transp. is crossing)
within the blocks it is unchanged

$$\Omega^2 = C^2 = 1 \tag{4}$$

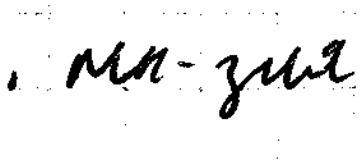
$$(\Omega C)^n \neq 1 \tag{5}$$

(6) [crossed out: entry unresolved] $R^{12} \longrightarrow (\Omega C) R^{12}$

Original source details: GELFAND-P000965

Uncertain reading. The first entry in (6) is crossed out. Its symbols cannot be made out reliably beneath the strokes.

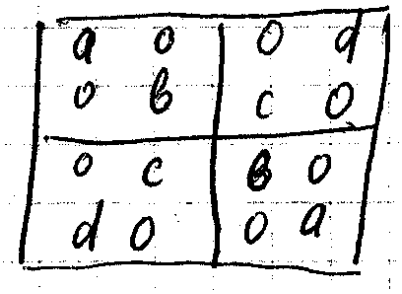
Invar. varieties — ? (for (6)).



Original source details: GELFAND-P000965

Diagram and typed description. The abbreviation after “Invar.” reads “мн-зия” (varieties).

Consider matr. of Baxter type



Original source details: GELFAND-P000965

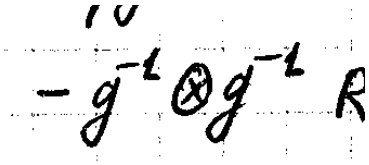
Diagram and typed description. Four matrix blocks. In row three, b is repeated on the diagonal.

(7)

a	0	0	d
0	b	c	0
0	c	b	0
d	0	0	a

(invar. with resp. to 2 groups Z_2 ⇓)

$$R^{12} \longrightarrow -g^{-1} \otimes g^{-1}\ R^{12}\ g \otimes g \tag{8}$$

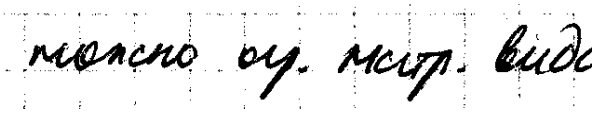


Original source details: GELFAND-P000965

Diagram and typed description. In (8) a separate minus sign precedes the first g^{-1}.

$$g_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad g_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{9}$$

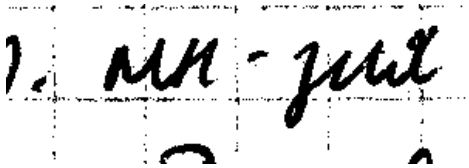
Transf. (8) together with $\Omega C \Rightarrow$ one can restrict [abbrev.] to matr. of form (7).



Original source details: GELFAND-P000965

Uncertain reading. The abbreviation before “matr.” resembles “or.”, tentatively expanded as “restrict to”.

Invar. varieties are as follows:



Original source details: GELFAND-P000965

Diagram and typed description. The abbreviation “мн-зия” (varieties) recurs before “are as follows”.

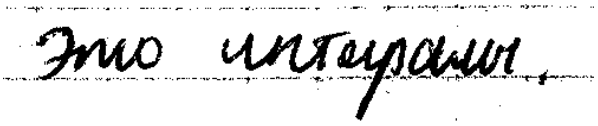
(with resp. to Ω and C simultaneously)

GELFAND-P000966

$$\frac{a^2 + b^2 - c^2 - d^2}{ab} = I_1 \tag{10a}$$

$$\frac{cd}{ab} = I_2 \tag{10б}$$

These are integrals.



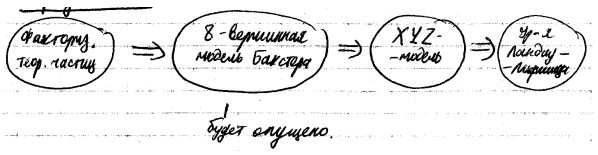
Original source details: GELFAND-P000966

Diagram and typed description. Under (10б) the manuscript reads “These are integrals”.

This is an ellipt. curve.

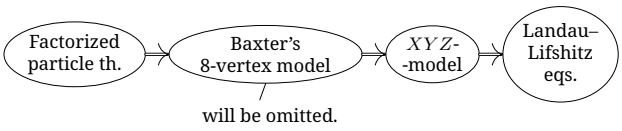
— · —

Cherednik I.



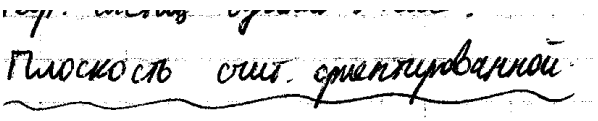
Original source details: GELFAND-P000966

Diagram and typed description. Four ovals with double arrows pointing right. Under the Baxter oval the note reads “will be omitted”.



A th. of particles of equal masses is constructed in 2-dim. space/time.

d types of particles The plane is taken as oriented



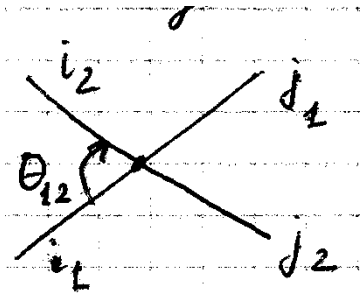
Original source details: GELFAND-P000966

Diagram and typed description. The final word of the wavy-underlined line reads “ориентированной” (“oriented”).

Momenta are conserved.

The only event is the intersect. of lines and their recoloring

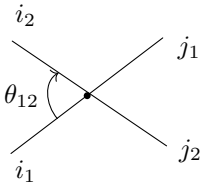
(elementary event)



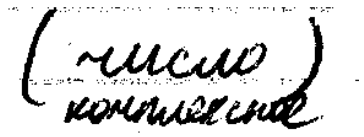
Original source details: GELFAND-P000966

Diagram and typed description. Two intersecting lines. The arc of angle θ_{12} on the left points upward.

(1)



(2) Assign an amplitude $S^{j_1 j_2}_{i_1 i_2}$ (number [inserted below: complex])



Original source details: GELFAND-P000966

Diagram and typed description. "Complex" is inserted below "number".

(3) $|S|^2$ — the probability of what took place

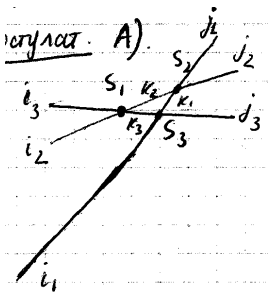
Supp. ($\Leftrightarrow$ relativism)

$$S^{j_1 j_2}_{i_1 i_2}(\theta_{12}), \tag{4}$$

θ_{12} — the usual Euclidean oriented angle

Nonrelativ. $\Leftrightarrow$ difference of velocities.

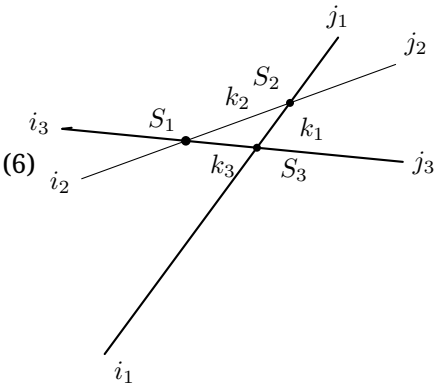
(5) Postulate. A).



Original source details: GELFAND-P000966

Diagram and typed description. Three pairwise intersections. The thick stroke at the left end of $i_3 j_3$ may be the start of the line or an incomplete arrowhead.

(6)



More complex event

It is assigned an amplitude S

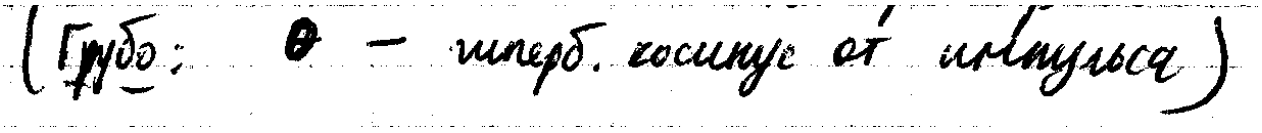
S_j — transition amplit. at the points indic. in the diagram

$$S = S_1 S_2 S_3 \tag{7}$$

B) The amplitude is independent of the location of events at a point in space,

i.e. it is unchanged by parallel translation of the lines.

(Roughly: θ — hyperb. cosine of the momentum) (we keep the colors of the outer ends)

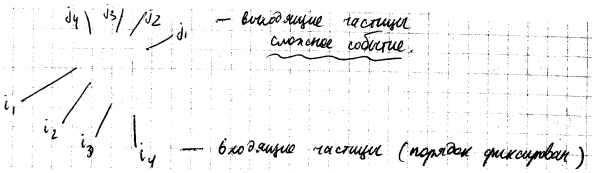


Original source details: GELFAND-P000966

Diagram and typed description. The lower entry reads “hyperb. cosine of the momentum”.

GELFAND-P000967

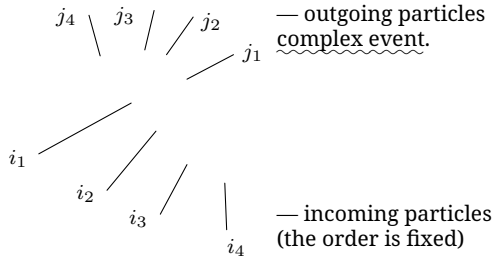
[source number: 20]



Original source details: GELFAND-P000967

Diagram and typed description. In (8) eight separate strokes are drawn, with a gap in the center.

(8)



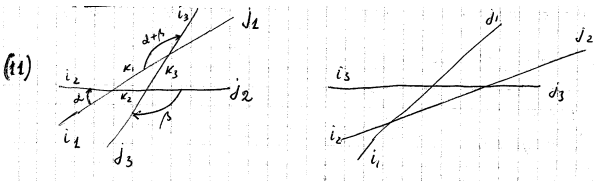
$$S^{j_1 \dots j_4}_{i_1 \dots i_4} = \sum_{\substack{\text{over all possible colorings of internal edges}}} \prod_{\substack{\text{over all vertices}}} S^{\cdot\cdot}_{\cdot\cdot} \tag{9}$$

(10) Lemma.

Postulate B) will always hold if it holds for the triangle.

Condition for the $\triangle$.

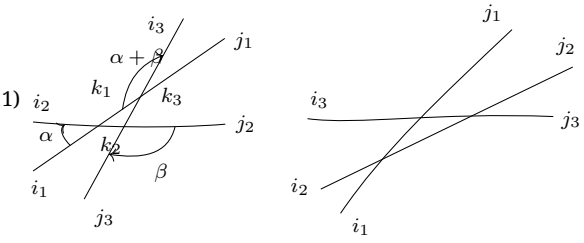
(11)



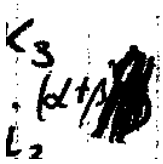
Original source details: GELFAND-P000967

Diagram and typed description. Two arrangements of three lines. The left one labels k_1, k_2, k_3 and the directed arcs $\alpha, \beta, \alpha+\beta$. Only the ends are labelled on the right.

(11)



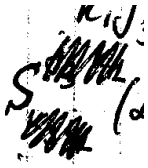
$$S^{k_1k_2}_{i_1i_2}(\alpha)S^{j_1k_3}_{k_1i_3}(\alpha+\beta\,[?])S^{j_2j_3}_{k_2k_3}(\beta) = S^{k_2k_3}_{i_2i_3}(\beta)S^{k_1j_3}_{j_1k_3}(\alpha+\beta)S^{j_1j_2}_{k_1k_2}(\alpha). \tag{12}$$



Original source details: GELFAND-P000967

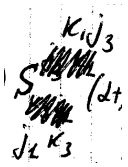
Uncertain reading. After $\alpha+$ part of a sign, probably β, is visible. Heavy cancellation obscures the right side of the argument.

[at the second factor on the right the earlier indices are crossed out. inserted above k_1j_3, below j_1k_3.]



Original source details: GELFAND-P000967

Uncertain reading. The earlier upper and lower indices are crossed out. Individual strokes remain visible, but the full reading is unresolved.



Original source details: GELFAND-P000967

Diagram and typed description. The replacement indices read k_1j_3 above and j_1k_3 below.

Problem: find S, for which these relations hold

The problem is strongly overdet.: eqs. d^6, unknowns d^4.

A simple invar. form of notation:

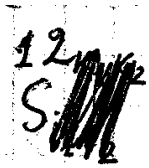
$$A = (a^j_i), \quad B = (b^j_i) \tag{13}$$

i — row, j — column

$$(A \otimes B)^{j\ell}_{ik} = a^j_i b^\ell_k \tag{14}$$

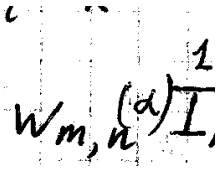
$$\overset{12}{S} = \sum_{m,n=1}^{d^2} w^{(\alpha)}_{m,n} \overset{1}{I}_m \otimes \overset{2}{I}_n \tag{15}$$

[at S the earlier indices are crossed out. 12 is written above S.]



Original source details: GELFAND-P000967

Uncertain reading. The former index pairs are heavily crossed out. Two digits 2 remain visible at the right. The readings j_1j_2 and i_1i_2 are tentative.



Original source details: GELFAND-P000967

Uncertain reading. The parenthesized α sits above m,n. It is set as an upper annotation on $w_{m,n}$.

(16) $\{I_n\}$ — a basis in the space of $d \times d$-matrices M_d.

Indices 1 and 2 indicate the space in which it is taken.

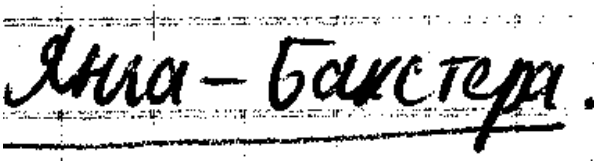
$$\overset{12}{S}(\alpha)\overset{13}{S}(\alpha+\beta)\overset{23}{S}(\beta) = \overset{23}{S}(\beta)\overset{13}{S}(\alpha+\beta)\overset{12}{S}(\alpha) \tag{17}$$

factorization or triangle rel.

(in the tens. prod. of 3 spaces)

(special case of the Yang–Baxter identity)

Yang–Baxter identity: the factorization rel. (i.e. (17)), spoiled in the following ways:



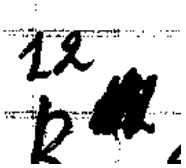
Original source details: GELFAND-P000968

Diagram and typed description. The underlined heading spells the name “Яна-Бакстера”.

$$\overset{121323'}{RSS} = \overset{23'1312}{S\ SR} \tag{18}$$

$$\overset{12}{R} \in \overset{1}{M}_d \otimes \overset{2}{M}_d(\mathbb{C}), \qquad R \in M_d \otimes M_d(\mathbb{C}), \tag{19}$$

[to the upper right of the first R crossed out: entry unresolved; 12 is written above R.]

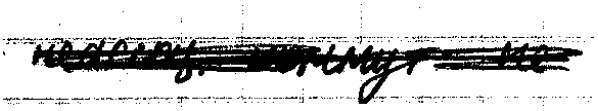


Original source details: GELFAND-P000968

Uncertain reading. The entry to the right of R is crossed out. 12 is clearly written above R.

$$S, S' \in M_d \otimes M_d(D), \tag{20}$$

(D — [crossed out: entry unresolved] an assoc., not necess. commut. alg. over $\mathbb{C}$)



Original source details: GELFAND-P000968

Uncertain reading. Several words before “assoc.” are crossed out. Some letters remain visible, but the phrase is unresolved.

(21) Theorem. From (18) it follows that

(22) [crossed out: entry unresolved]

$$[\mathrm{Sp}_3 \overset{13}{S},\ \mathrm{Sp}_3 \overset{13'}{S}\,] = 0$$

Original source details: GELFAND-P000968

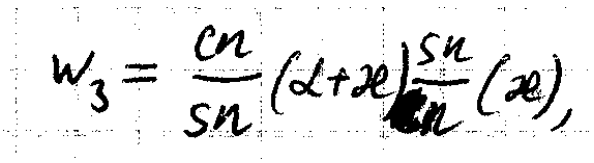
Uncertain reading. The entry before the square bracket is crossed out by diagonal strokes.

(commutator in the space of matrices over the 1st)

Nontriv. example of the factorization rel.:

$$\overset{12}{S}(\alpha) = \sum_{i=0}^{3} w_i(\alpha) \overset{1}{\sigma}{}^i \otimes \overset{2}{\sigma}{}^i \tag{23}$$

$$\begin{aligned} w_1 &= \frac{\operatorname{sn}(x)}{\operatorname{sn}(\alpha+x)}, \qquad w_2 = \frac{\operatorname{dn}}{\operatorname{sn}}(\alpha+x)\frac{\operatorname{sn}}{\operatorname{dn}}(x), \\ w_3 &= \frac{\operatorname{cn}}{\operatorname{sn}}(\alpha+x)\frac{\operatorname{sn}}{\operatorname{cn}\,[?]}(x), \qquad w_0 = 1 \end{aligned} \tag{24}$$



Uncertain reading. The first letter of the denominator in the last fraction for w_3 is heavily overwritten. Tentatively cn, with a possible trace of an earlier dn.

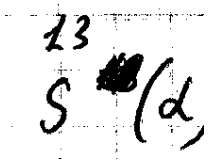
Original source details: GELFAND-P000968

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \text{— Pauli matr.}$$

$\alpha, x \in \mathbb{C}$.

Instead of (18):

$$\begin{aligned} &\overset{12}{R}(\alpha,\alpha')\overset{13}{S}(\alpha)\overset{23}{S}(\alpha') \\ &\qquad = \overset{23}{S}(\alpha')\overset{13}{S}(\alpha)\overset{12}{R}(\alpha,\alpha') \end{aligned} \tag{25}$$



Uncertain reading. A short entry next to the second S on the right, after the upper 13, is crossed out.

Original source details: GELFAND-P000968

GELFAND-P000969

23/III–81. Gelfand seminar.

I.M. Gelfand on Moser's art.

(coll. in honor of Chern's anniversary)

Ellipsoid

$$\langle A^{-1}x, x\rangle = 1, \qquad \text{semiaxes } \sqrt{\lambda_n} \tag{1}$$

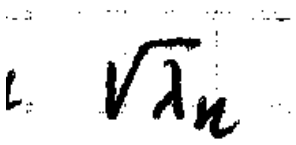


Diagram and typed description. The radical is written with λ_n.

Original source details: GELFAND-P000969

Geodesics:

$$\frac{d^2x}{dt^2} = -\nu A^{-1}x, \qquad x \in \mathbb{R}^n, \tag{2}$$

but for the ellipsoid the point remains on it.

where

$$\nu = \frac{\langle A^{-1}y, y\rangle}{|A^{-1}x|^2}, \qquad y = \frac{dx}{dt}. \tag{3}$$

Isospectral matrix

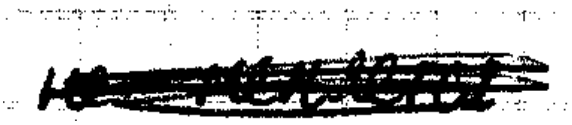
$$L(x,y) = P_y(A - x \otimes x) \cdot P_y \tag{4}$$

P_y — matrix of the orthog. proj. onto the orthog. complement to y:

$$P_y\xi = \xi - \frac{\langle \xi, y\rangle}{|y|^2} y \tag{5}$$

Thm. The matrix

(6) $\underbrace{L\left(x, \frac{dx}{dt}\right)}$ [crossed out: entry unresolved] does not change eigenvalues under motion along a geodesic.

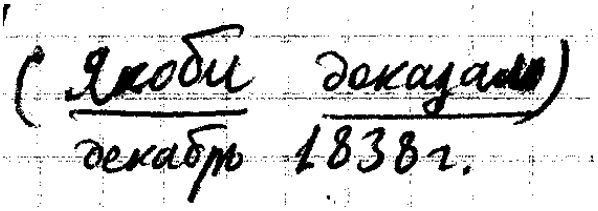


Original source details: GELFAND-P000969

Uncertain reading. The crossed-out entry appears to begin “не” (“not”). The following verb may be “changes” or “depends”, but the strokes obscure the reading.

(Jacobi proved it)

December 1838

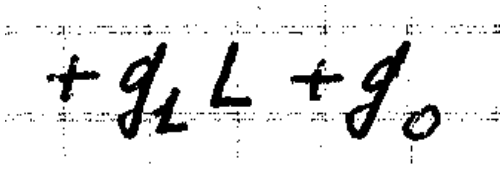


Original source details: GELFAND-P000969

Diagram and typed description. The end of “proved” is retraced, and the date is written separately below.

P. Grinevich.

$$L_6^2 = L_4^3 + g_1 L + g_0 \tag{1}$$



Original source details: GELFAND-P000969

Diagram and typed description. The L after g_1 has no subscript.

$$L_4 = \left(\frac{d^2}{dx^2} - u\right)^2 + 2c'_x\big(\wp(\gamma_2) - \wp(\gamma_1)\big)\frac{d}{dx} + \big[c'_x(\wp(\gamma_2) - \wp(\gamma_1)\big]'_x - - \wp(\gamma_1) - \wp(\gamma_2) \tag{2}$$

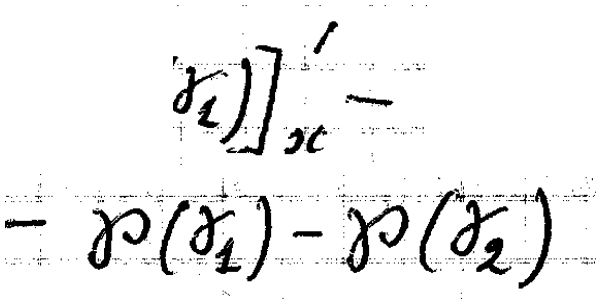
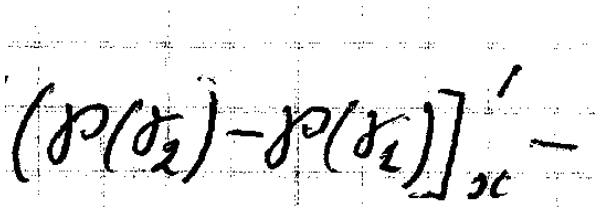


Diagram and typed description. A minus stands at the end of the line and is repeated at the beginning of the continuation.

Original source details: GELFAND-P000969



Uncertain reading. One round parenthesis is visible before the closing square bracket. A separate close for the parenthesis after c'_x is not distinguishable.

Original source details: GELFAND-P000969

$$\begin{cases}\gamma'_1 = \dfrac{1}{\alpha_1 - \alpha_2},\\ \gamma'_2 = \dfrac{1}{\alpha_2 - \alpha_1}\end{cases} \qquad \begin{cases}\alpha'_1 = \alpha_1^2 - u - \varphi(\gamma_1, \gamma_2, 0),\\ \alpha'_2 = \alpha_2^2 - u + \varphi(\gamma_1, \gamma_2, 0)\end{cases} \tag{3}$$

$$\begin{cases}\gamma_1(x) = \gamma_0 + c(x)\\ \gamma_2(x) = \gamma_0 - c(x)\end{cases}$$

$$u(x) = \frac{1}{4{c'_x}^2} - \frac{1}{4}\frac{{c''_{xx}}^2}{{c'_x}^2} - 2\varphi c''_{xx} - {c'_x}^2(\varphi'_c - \varphi^2) + \frac{1}{2}\frac{c'''_{xxx}}{c'_x}$$

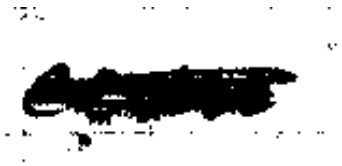
GELFAND-P000970

[source number: 43]

$g_1, g_0, c(x)$ are specified arbitrarily

$$c(x) = \int_{q(x)}^{\infty} \frac{dt}{\sqrt{t^3 + g_1 t + g_0}} \tag{4}$$

$q(x)$ — rat. fn. [crossed out: entry unresolved] $\Rightarrow L_4, L_6$ — oper. with rational coeff.



Uncertain reading. A short entry after “rat. fn.” is crossed out and remains unresolved.

Original source details: GELFAND-P000970

$\gamma_0 = 0 \Rightarrow q(x) = x$. (case of Dixmier)

$$\varphi(\gamma_1, \gamma_2, 0) = \zeta(\gamma_2 - \gamma_1) + \zeta(\gamma_1) - \zeta(\gamma_2)$$

$$\zeta'(z) = -\wp(z), \qquad \zeta(-z) = -\zeta(z).$$

Dixmier's example according to Kirillov

$$L_4 = \{A^2\}, \qquad L_6 = \{A^3\}, \qquad A = L + xL^{-1}, \qquad L = \frac{d^2}{dx^2} + x^3$$

$\{\cdot\}$ — differential part.

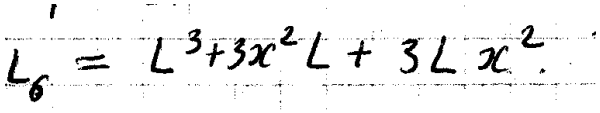
[crossed out: ~~Hence~~ (?)] Or:

Original source details: GELFAND-P000970

Uncertain reading. The cancelled entry before "Or" may read "Hence".

$$L_4 = L^2 + 2x$$

$$L_6 = L^3 + 3x^2L + 3Lx^2.$$



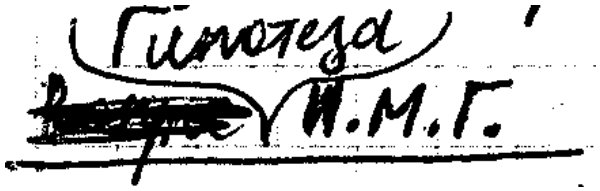
Original source details: GELFAND-P000970

Diagram and typed description. Both terms after L^3 have x^2.

For a curve of genus 2 $P(x, y) = 0$ it is unclear ∃(?) oper. L_1, L_2 with rat. coeff. for which $P(L_1, L_2) = 0$.

There are no coprime ords..

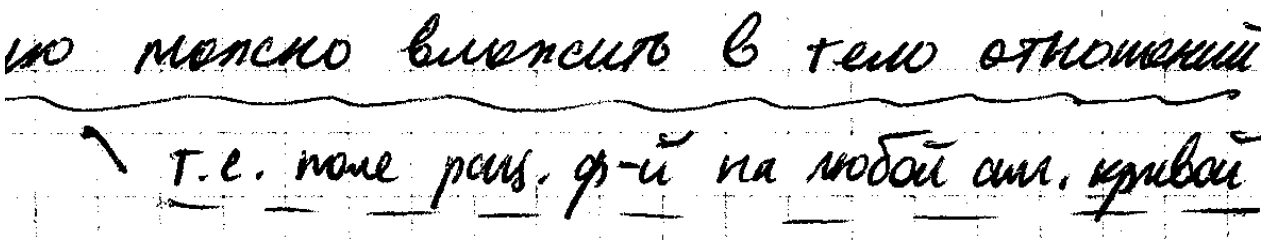
[inserted above the cancelled word: Conjecture] [crossed out: ~~Thm.~~ (?)] I.M.G.: any curve can be embedded in the division ring of fractions of the Weyl alg.



Original source details: GELFAND-P000970

Uncertain reading. "Conjecture" is inserted above the line with a brace. The crossed-out entry before "I.M.G." may read "Thm.".

[marginal addition with arrow] i.e. the field of rat. fns. on any alg. curve



Original source details: GELFAND-P000970

Diagram and typed description. The arrow from the gloss points to the phrase about embedding the curve.

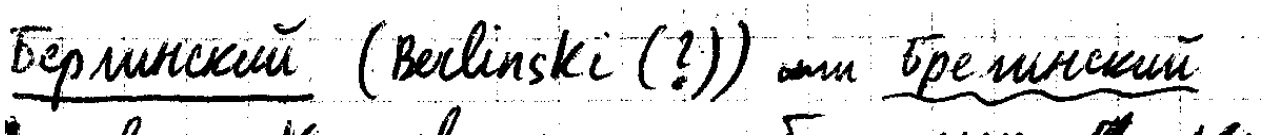
$$q, p, \qquad qp - pq = 1.$$

the generated [crossed out: entry unresolved] algebra embeds in a division ring (exercise)

Original source details: GELFAND-P000970

Uncertain reading. A short entry before "algebra" is crossed out.

Berlinskii (Berlinski (?)) or Brelinskii



Original source details: GELFAND-P000970

Diagram and typed description. The manuscript gives two Cyrillic forms of the surname and "Berlinski (?)" in Latin letters.

(Kashiwara's coauthor in the work on the conj. [crossed out: entry unresolved] of Kazhdan–Lusztig)

Original source details: GELFAND-P000970

Uncertain reading. A mark before "Kazhdan" is crossed out. Its reading is unresolved.

$\mathfrak{g}$ — semisimple Lie alg. over $\mathbb{C}$

$\cup$

$\mathfrak{b}$ — Borel alg., $\lambda\colon \mathfrak{b} \to \mathbb{C}$ Lie alg. character

Verma module M_λ (see Dixmier's book on envelop. algebras).

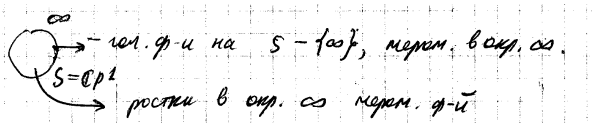
[the title underline is connected to the gloss] geom. objects on the space of flags

[crossed out: M_λ, $\mathfrak{g} = \mathfrak{b} \oplus \ldots$ (?)]

Original source details: GELFAND-P000970

Uncertain reading. Beneath the strokes one can see M_λ above and tentatively $\mathfrak{g} = \mathfrak{b} \oplus \ldots$ below. The rest is unresolved.

$G = SL(2)$



Original source details: GELFAND-P000970

Diagram and typed description. Circle labelled ∞ with two arrows toward the labels at right.

∞

→ — hol. fns. on $S - \{\infty\}$, merom. near ∞.

$S = \mathbb{CP}^1$

→ germs near ∞ of merom. fns.

GELFAND-P000971

Eckart Viehweg.

(after Hélène Esnault — Exp. Sem. Bourbaki)

(after work by Fujita, Kawamata, Iitaka, Ueno, Viehweg)

Uncertain reading. The initial in Iitaka could be read as I or T. The reading I is preferred.

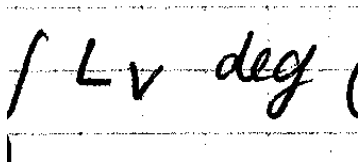
Original source details: GELFAND-P000971

On the classif. of 3-dimensional algebraic varieties.

V — proj. alg. variety over $\mathbb{C}$, $\omega_V = \bigwedge^{\max} \Omega^1_V$

$$K(V) = \begin{cases} L_v \text{ deg} \left(\bigoplus_{n \geq 0} H^0(V, \omega_V^n) \right) - 1 & \text{when } R(V) \neq \mathbb{C} \\ -\infty \end{cases}$$

(Kodaira number



Uncertain reading. The two marks before deg resemble L and a lowered v. The formula suggests the possible reading tr deg.

Original source details: GELFAND-P000971

$$q(V) = \dim H^0(\Omega^1).$$

$$1)\quad \left.\begin{array}{c} V \\ K(V) = 0 \\ q(V) = \dim V \end{array}\right\} \quad (\Leftarrow) \quad V \overset{\text{birat.}}{\sim} \underset{\text{ab. variety}}{A}$$

$$2)\quad \left.\begin{array}{c} \dim V = 3, \\ K(V) = 0 \\ q(V) = \begin{cases} 2 \\ 1 \end{cases} \end{array}\right\} \quad \Rightarrow \quad \alpha : V \longrightarrow \underset{\text{(Albanese variety)}}{\searrow} \quad A(V)$$

étale fibration

$$K(F) = 0.$$

$$3)\quad K(V) = 0, \quad q(V) = 0,$$

$$?$$

4) There is something about the case $K(V) = -\infty$.

Christian Kassel. (Strassburg)

Something about studying the homotopy type of the group Diff(M)

(Waldhausen's idea)

One can extend K-theory to objects A that are rings

$$A \times A \underset{\times}{\overset{+}{\longrightarrow}} A$$

up to homotopy (e.g., loop space)

(distributivity is genuine)

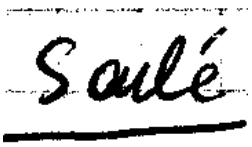
$$\mathbb{Z}[\Omega X]$$

K-functor is constr. using the construction of Quillen.

Reference: AMS Proc. Symp. Pure Math, v. 32 (1976)

Waldhausen, Hatcher.

Christophe Soulé (paris VII)



Uncertain reading. The first vowel in the manuscript surname can be read as o or a. Soulé is the preferred reading.

Original source details: GELFAND-P000971

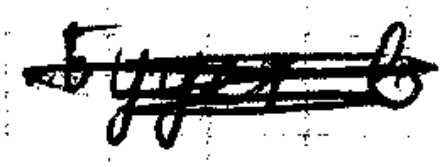
$K_*(\mathbb{Z}) \leftrightarrow \zeta(1-n)$ connection with topoi

GELFAND-P000972

[source number: 44]

$$\#K_{2i-2}(\mathbb{Z}) \geqslant \text{numerator}\,\zeta(1-i)$$
$$\#K_{2i-1}(\mathbb{Z}) \geqslant \text{denominator}\,\zeta(1-i)$$
$$i \equiv 0 \pmod 2$$

[crossed out: ~~Bourb...~~ (?)] Inventiones, 1980, February.



Uncertain reading. The crossed-out writing may begin with “Bourb...”. The rest is unclear.

Original source details: GELFAND-P000972

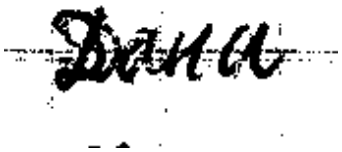
GELFAND-P000973

30/III-81. Gelfand sem.

I. M. Gelfand

Vlasov eq.

Given a mech. system — particles in a potential field.



Uncertain reading. The first letter has been overwritten. The strokes and agreement with “system” favor the reading “Дана” (“given”).

Original source details: GELFAND-P000973

Let there be many particles, and let them interact strongly.

The particles are sparse, so they do not collide, but they create a field — they are all charged. The potential is a function of the particle density.

Arnold. On work by A. Weinstein.

Hydrodynamics:

Consider a region filled with fluid, in a Riem. manifold G.

Motion of the fluid is a curve in the group of diffeom., preserving volume.

$$G = \text{SDiff}(G)$$

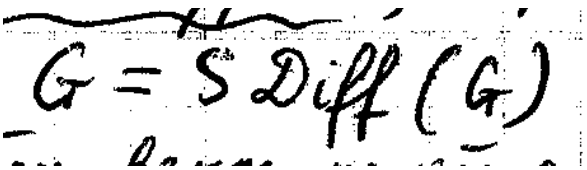


Diagram and typed description. Both letters have the form G. There is a short stroke beneath the letter in parentheses.

Original source details: GELFAND-P000973

$\mathfrak{g}$ — Lie alg. of G, i.e. the algebra of vector fields with divergence $= 0$.

Energy: $E = \frac{1}{2}\int |v(x)|^2\,dx$, $v(x)$ — is a vector field.

This is a quadr. form on $\mathfrak{g}$.

Extend it by left translations to all of G. We get a Riem. metric

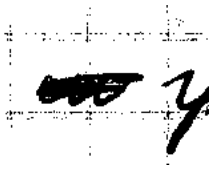
Def. Geodesics are flows of an ideal fluid

(Arnold, 1966)

Ann. Inst. Fourier and a survey in the book on mechanics.

There is curvature, a Hamiltonian formalism, etc.

Weinstein — extension to other [crossed out: writing unclear] groups (∞-dimensional)



Uncertain reading. The writing beneath the heavy cancellation before “groups” remains unclear.

Original source details: GELFAND-P000973

Symplectic group: gives the Vlasov eq.

(cold plasma)

Extend, add local degrees of freedom — get various eqs.

(including Yang–Mills).

if we take a gauge group.

Hamiltonian:

1) On T^*G there is the standard sympl. structure;

E defines the Hamiltonian.

$\Rightarrow$ Euler eq.

GELFAND-P000974

[source number: 59]

2) (2nd version)

Consider the map

$$T^*G \longrightarrow \mathfrak{g}^*$$

(one set of translations, left or right)

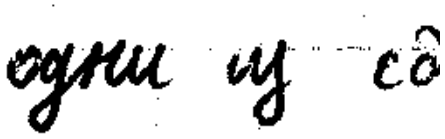


Diagram and typed description. The first word reads “одни” (“one set”).

Original source details: GELFAND-P000974

The curves from 1) project to the Euler eq. in $\mathfrak{g}^*$.

Symplectic struct. on the orbits of the coadjoint representation

(Kirillov struct.).

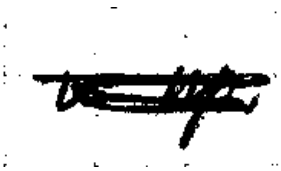
The Hamiltonian transfers by invariance.

The Kirillov form descends.

For a compressible fluid: Ebin and Marsden (later)

rot. is given $\Rightarrow$ only then do we get a Hamiltonian formalism

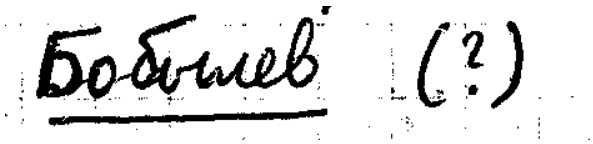
over G one must take bundles [crossed out: writing unclear] (where the density is)



Original source details: GELFAND-P000974

Undeciphered passage. The writing after "bundles" is heavily crossed out.

Someone named Bobylev (?)

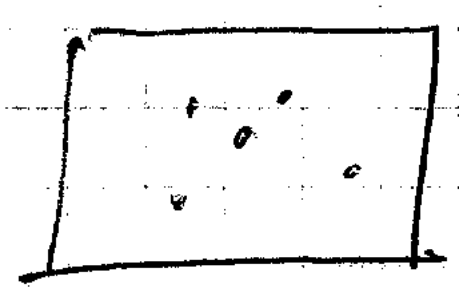


Original source details: GELFAND-P000974

Diagram and typed description. The parenthesized question mark is written beside the surname.

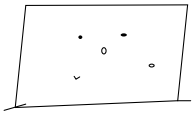
On the Boltzmann eq.

Gas in a vessel.



Original source details: GELFAND-P000974

Diagram and typed description. Vessel with five particle marks.



The spheres (hard) are the interaction potential $V(r)$.

Original source details: GELFAND-P000974

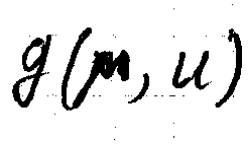
Uncertain reading. A short vertical stroke stands above the end of "interaction". Its meaning is unclear.

$f(\vec{v},t)$ — the number of particles with velocity $\vec{v}$ per unit volume at the given instant.

$$\frac{\partial f}{\partial t} = \int d\vec{w}\, d\vec{n}\, g\left(\frac{\vec{u}\vec{n}}{u}, u\right)\{f(\vec{v}')f(\vec{w}') - f(\vec{v})f(\vec{w})\} \tag{1}$$

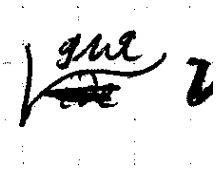
$$\begin{gathered} |\vec{n}| = 1, \qquad \vec{u} = \vec{v} - \vec{w}, \\ \vec{v}' = \tfrac{1}{2}(\vec{v} + \vec{w} + u\vec{n}), \qquad \vec{w}' = \tfrac{1}{2}(\vec{v} + \vec{w} - u\vec{n}) \end{gathered} \tag{2}$$

$$g(\mu, u) = \widetilde{g}(\mu)u^{1-\frac{4}{s}}, \quad \text{for} \quad V(r) = \frac{\alpha}{r^s} \tag{3}$$



Uncertain reading. The first argument of g is heavily retraced. The adjacent $\widetilde{g}(\mu)$ favors μ.

Original source details: GELFAND-P000974



Uncertain reading. "For" was inserted above a crossed-out word that may read "where".

Original source details: GELFAND-P000974

(this is for the spatially homog. case)

In fact

$$g = u\frac{d\sigma}{d\Omega}, \qquad \sigma \text{ — cross section} \tag{4}$$

$(\vec{v}', \vec{w}')$ — velocities after the collision

$(\vec{v}, \vec{w})$ — —‖— before —‖—

$\vec{u}$ — relative velocity, it turns and gets a new dir. $u\vec{n}$ (magnitude is conserved by conservation of energy)

Quadr. form — because the probability of collision is proportional

[crossed out: writing unclear] In recent years there has been much progress: if g depends only on the angle

Undeciphered passage. The short entry before "In recent years" is heavily crossed out.

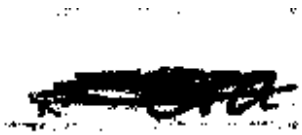
Original source details: GELFAND-P000974

(i.e. $V(r) = \frac{1}{r^4}$) $\Rightarrow$ greatly simplified using the Fourier transform. Then

we get exact solutions, an inf. number of conservation laws, etc.
Convolution with a Gaussian density does not change the eq.
Eigenfns. of the linearized eq. were obtained (new ones).
Consider

$$f_M = A \exp[-\alpha(\vec{v} - \vec{v}_0)^2] \tag{5}$$

[crossed out: writing unclear] — stationary



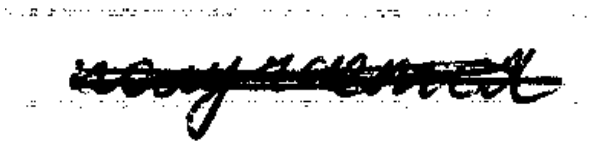
Undeciphered passage. The short entry between (5) and the dash is heavily crossed out.

Original source details: GELFAND-P000975

— a 3-param. family of fns., for the parameters $A, \alpha, \vec{v}_0$ [crossed out: writing unclear]

Diagram and typed description. The manuscript reads “3-param.” before “family”.

Original source details: GELFAND-P000975



Undeciphered passage. The entry after the list of parameters is heavily crossed out. The letters beneath the stroke are unclear.

Original source details: GELFAND-P000975

$$f = f_M[1 + F], \qquad F = F(\vec{v}, t), \qquad \|F\| \ll 1. \tag{6}$$

Hence for F we get the linearized eq.

$$\frac{\partial F}{\partial t} = \widehat{L}F, \tag{7}$$

where

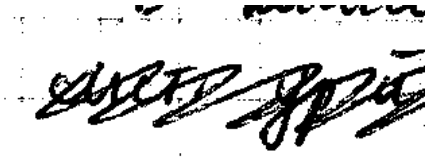
$$\widehat{L}F = \int d\vec{w}\, e^{-w^2/2}\, d\vec{n}\, g\left(\frac{\vec{u}\vec{n}}{u}, u\right)[F(\vec{v}') + F(\vec{w}') - F(\vec{v}) - F(\vec{w})] \tag{8}$$

The general solution is constructed in the class of rapidly decaying fns. [crossed out: ~~An infinite~~ (?) writing unclear]

Undeciphered passage. “An infinite” is a possible reading in the crossed-out entry after “the general solution”. The whole entry remains unclear.

Original source details: GELFAND-P000975

[crossed out: ~~system of eqs.~~ (?) writing unclear]



Original source details: GELFAND-P000975

Uncertain reading. The start of the crossed-out line tentatively reads “system of eqs.” The ending of the second abbreviation is unclear.

Original source details: GELFAND-P000975

Undeciphered passage. The rest of the crossed-out line remains unclear.

(9) $s = 4$; [crossed out: writing unclear] $f(v^2)$ — isotropic case $\Rightarrow$ reduces to an ∞ system of ordinary eqs. A transform takes it to a linear system.

Original source details: GELFAND-P000975

Uncertain reading. The separate entry between $s = 4$ and $f(v^2)$ is heavily crossed out.

Unknown in the other cases.

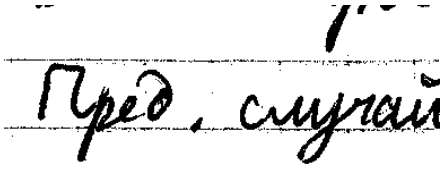
This is analogous to eqs. with const. coeffs. — here g does not depend on u.

$$\widehat{L} = CI + T, \qquad T \text{ — a compact operator.} \tag{10}$$

In this case $C = -1$.

The spatially inhomog. case is important.

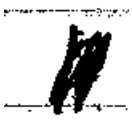
Lim. case (in mean free path) — Navier–Stokes eq.



Original source details: GELFAND-P000975

Uncertain reading. The manuscript has the abbreviation “Пред.”. It is taken here to mean “limiting”.

B. Lidskii. [crossed out: writing unclear]



Original source details: GELFAND-P000975

Uncertain reading. A short mark to the right of the heading is crossed out. Its reading remains unclear.

Consider [crossed out: $\mathbb{R}^{3n}$ (?)] $(x_1, \dots, x_n, y_1, \dots, y_n, z_1, \dots, z_n)$,

Original source details: GELFAND-P000975

Uncertain reading. The notation crossed out before the coordinates may be $\mathbb{R}^{3n}$.

$$\sum_{i=1}^{n}(x_i + y_i - z_i) = 0 \tag{1}$$

Consider there a subset singled out by inequalities

$$x_1 \geqslant \ldots \geqslant x_n, \quad y_1 \geqslant \ldots \geqslant y_n, \quad z_1 \geqslant \ldots \geqslant z_n \tag{2}$$

and $\exists$ 3 Hermitian matrices X, Y, Z, for which

$$X + Y - Z = 0 \tag{3}$$

and their spectra are subsets of (2).

GELFAND-P000976

[source number: 60]

(4) Definition: Let a set be given
(by induction)

$$I_n = \{1, 2, \ldots, n\} \qquad I_\ell = \{1, 2, \ldots, \ell\}$$

and 3 increasing seqs. of its elems.

$$\begin{gathered} a_1 < a_2 < \ldots < a_\ell \\ b_1 < b_2 < \ldots < b_\ell \\ c_1 < c_2 < \ldots < c_\ell \\ (a_i, b_i, c_i \in I_n) \end{gathered}$$

They are consistent if

$$1) \qquad \sum_{i=1}^{\ell}(a_i + b_i - c_i) = \frac{\ell(\ell+1)}{2}$$

2) Suppose 3 consistent seqs. are given

$$\begin{gathered} f_1 < f_2 < \ldots < f_k \\ g_1 < g_2 < \ldots < g_k \\ h_1 < h_2 < \ldots < h_k \end{gathered} \qquad \begin{gathered} (\text{of elements of } I_\ell). \\ \ell < n \end{gathered}$$

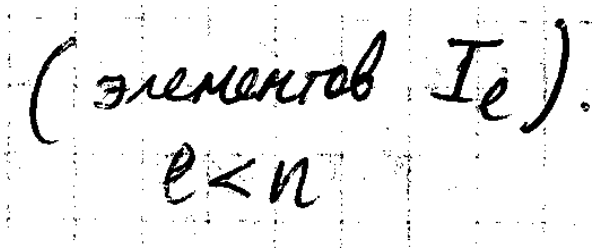


Diagram and typed description. Below "of elements of I_ℓ" the manuscript has $\ell < n$.

Original source details: GELFAND-P000976

and they are consistent. Then we must have

$$\sum_{j=1}^{k} a_{f_j} + \sum_{j=1}^{k} b_{g_j} \leqslant \sum_{j=1}^{k} c_{h_j} + \frac{k(k+1)}{2}$$

— · —

Final answer:
The inequalities that must hold are (2) and

$$\sum z_{c_j} \leqslant \sum x_{a_j} + \sum y_{b_j} \tag{5}$$

— · —

$a_i \mapsto (a_i - i) \Rightarrow$ then everything is written more simply.
Probability is also written as a convolution.

— · —

There is a dual theorem ⇒ an algorithm to determine which reps. occur in the tensor prod.

Uncertain reading. The abbreviation before “in the tensor” could be “встрет.” or “встреч.”, both indicating occurrence.

Original source details: GELFAND-P000976

GELFAND-P000977

6/IV-81. Gelfand sem.
S. G. Krein. On the eqs. of a compressible fluid.
(someone’s work) — Ebin, Marsden

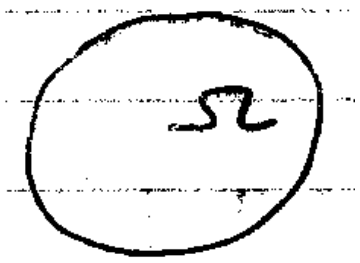
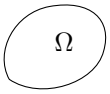


Diagram and typed description. Closed outline labelled Ω.



Original source details: GELFAND-P000977

diffeomorphisms of the region onto itself arise G.

Incompress. — preserve volume — group G_0.

G_0 — Riem. manifold, motion along geodesics.

Barotropic fluid: $p = p(\rho)$, e.g. $p = A\rho^\gamma$.

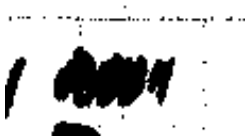
T is taken $=$ const.

Diagram and typed description. The line about taking T constant is written separately.

Original source details: GELFAND-P000977

Eqs.

$$\begin{cases} \dfrac{\partial \vec{u}}{\partial t} + (\vec{u}, \nabla)\vec{u} = -\dfrac{1}{\rho}\operatorname{grad} p \\ \dfrac{\partial \rho}{\partial t} + \operatorname{div}(\rho\vec{u}) = 0 \\ p = p(\rho) \end{cases} \tag{1}$$



Undeciphered passage. There are heavily crossed-out marks above p in the first line. Their reading remains unclear.

Original source details: GELFAND-P000977

I.M.G. thinks that barotropy is a poor restriction.

(heating, of a pump while pumping)

Next: Shock waves necessarily arise and must be taken into account.

$$u_t + cu_x = 0 \quad \Rightarrow \quad u = f(x - ct)$$

Diagram and typed description. A pulse with an almost flat top. A rightward arrow is labelled c.

Original source details: GELFAND-P000977

$$u_t + uu_x = 0 \quad \Rightarrow \quad u = f(x - ut)$$

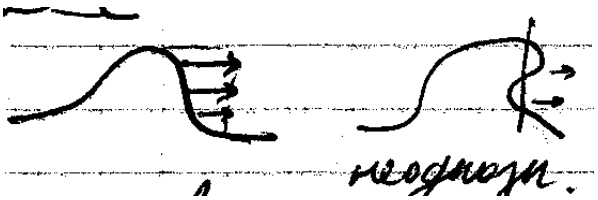
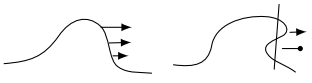


Diagram and typed description. At left, three rightward arrows lie on the falling side. At right, the curve folds around a vertical stroke. The upper mark to its right has a rightward arrowhead. The lower mark thickens at the right and its arrowhead is unclear.



multival.

Original source details: GELFAND-P000977

Riemann. At this point the soln. becomes discontinuous. We must write

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial (u^2)}{\partial x} = 0$$

and understand what a discontinuous solution is

GELFAND-P000978

[source number: 68]

Zakharov: eq.

$$\underbrace{\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} = 0}$$

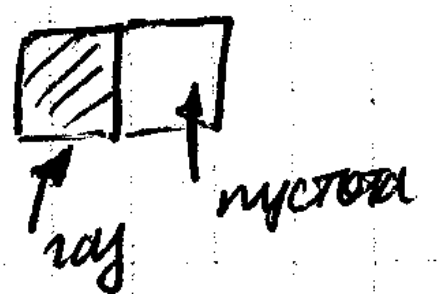
where U — a matrix,

can also be integrated completely.

[crossed out at left: ~~Ex~~ (?)] Discontinuous solutions are always needed in nonlinear hyperb. eqs.

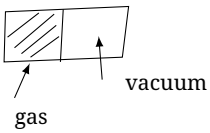
Original source details: GELFAND-P000978

Uncertain reading. The crossed-out mark at left may read “Пр” (“Ex”).



Original source details: GELFAND-P000978

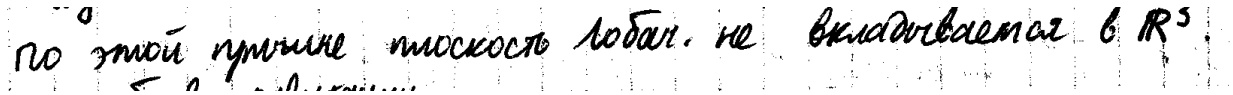
Diagram and typed description. A partition divides the vessel. The hatched left side is marked “gas” and the right side “vacuum”.



Remove the partition. How does it even out?

In fact, a system of shock waves going back and forth.

For this reason the Lobachevsky plane does not embed in $\mathbb{R}^3$.



Original source details: GELFAND-P000978

Diagram and typed description. The manuscript reads “does not embed”.

And esp. in gravitation.

S.G.K. weakly compress. fluid (there is a parameter.)

In the limit the solutions reduce to solns. of the eqs. of an incompress. fluid.

Hamiltonian system. Density is close to 1.

$g(x)$ — Jacobian of the map

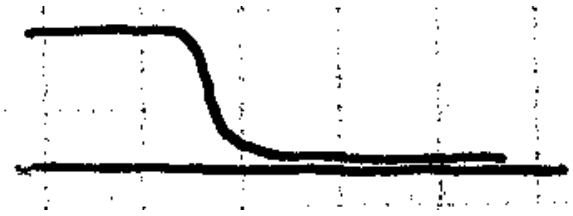
$$\int_\Omega \int_1^{g^{-1}(x)} \frac{p(\rho)}{\rho^2}\, d\rho\, dx \qquad \text{— potential energy}$$

The same exists for a viscous fluid (following Arnold).

Problem (I.M.G.).

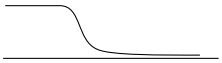
$$u_t + uu_x + p\left(\frac{\partial}{\partial x}\right) u = 0$$

$u|_{t=0} =$ [graph]



Original source details: GELFAND-P000978

Diagram and typed description. The curve drops from an upper horizontal stretch to a lower one, above a separate horizontal line.



What will happen?

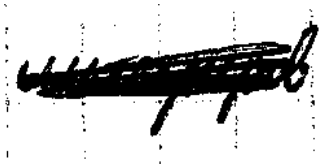
KdV — solitons keep splitting off.

G. Olshanskii. Lie semigroups.

[crossed out: ~~Everywhere~~ (?)] This is useful for [crossed out: writing unclear] the theory of reps. of ∞-dimensional groups.

Original source details: GELFAND-P000978

Undeciphered passage. The crossed-out word before “This” may read “Everywhere”.



Original source details: GELFAND-P000978

Undeciphered passage. The entry between “for” and “theory” is heavily crossed out.

$\mathfrak{g}$ — semisimple Lie algebra.

Example of a Lie semigroup.

$\mathbb{C}^n$, $n = p + q$, consider an indefinite scalar product

$$\langle x, x\rangle = -|x_1|^2 - \cdots - |x_p|^2 + |x_{p+1}|^2 + \cdots + |x_n|^2$$

$A\colon \mathbb{C}^n \to \mathbb{C}^n$ is called a J-contraction, if

$$\langle Ax, Ax\rangle \leqslant \langle x, x\rangle, \quad x \in \mathbb{C}^n.$$

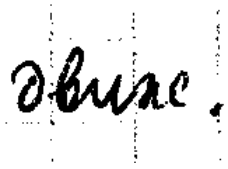
This is a semigroup (working example).

Real example.

Original source details: GELFAND-P000978

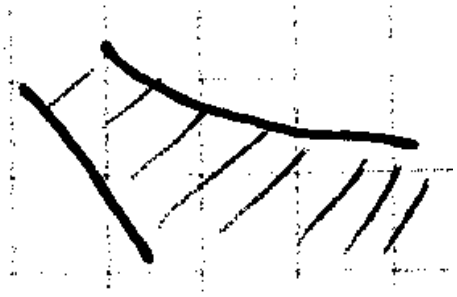
Diagram and typed description. A slanting boundary with seven strokes on the right.

consider motions of the Lobachevsky plane that send into itself some half-plane



Original source details: GELFAND-P000978

Diagram and typed description. The manuscript abbreviates the word for “motions”.



Original source details: GELFAND-P000978

Diagram and typed description. A slanting line at left and a curve above. Slanted hatching lies between them and below the curve.

Theorems

1) description of Lie semigroups.

2) description of all reps. of complex $\mathfrak{g}$.

3) —||— real forms.

4) D — bounded symm. domain, e.g., the ball $|z| < 1$.

$D \subset \widehat{D}$. (dual variety)

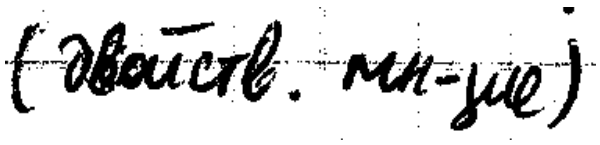


Diagram and typed description. The parenthesis reads “dual variety” in abbreviated form.

Original source details: GELFAND-P000979

G — semigr. of transformations of $\widehat{D}$, taking D into the domain

$z \in \mathrm{Mat}_{p,q} \Rightarrow$ a semigr. arises of J-contractions

— · —

Γ — semigr. of matrices A with $\|A\| \leqslant 1$.

$x \mapsto x^{\#}$ involution

$$(ab)^{\#} = b^{\#}a^{\#}, \qquad a^{\#\#} = a.$$

(here it is $*$).

A semigroup rep. is a homomorphism

$$T\colon x \mapsto T(x) \quad \text{into the set of contractions of a complex Hilbert space,}$$

$$T(x^{\#}) = T(x)^* \quad \text{— analogue of unitarity}$$

$$\|T(x)\| \leqslant 1.$$

$$T(xy) = T(x)T(y)$$

Semigr. of contractions $\mathbb{C}^n = V$.

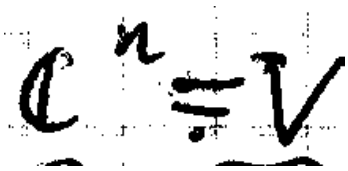


Diagram and typed description. A distinct dot lies below the equality sign.

Original source details: GELFAND-P000979

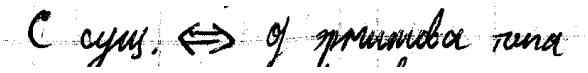
Reps. in $V^{\otimes m}$ are indexed by sequences — these are analytic.
There are also antianalytic ones.
More interesting are J-contractions.
Here analytic reps. arise.

— · —

$\mathfrak{g}$ — semisimple Lie algebra;

$$\underset{\substack{\text{cone in } \mathfrak{g}\\ \text{(convex)}\\ \text{with nonempty interior}}}{C} \subset \mathfrak{g}$$

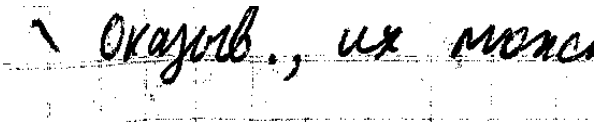
Invar. — invariant under the adjoint representation.
$\exists$ Do such cones exist? indep. Vinberg and Kostant.
Let $\mathfrak{g}$ be simple. C exists $\Leftrightarrow$ $\mathfrak{g}$ of Hermitian type



Original source details: GELFAND-P000979

Diagram and typed description. The two conditions are joined by an equivalence sign.

(the corresp. Riem. symm. space has a complex structure)
Vinberg found max and min.
[stroke leading to the remark] It turns out, they can be described!!

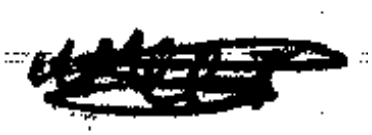


Original source details: GELFAND-P000979

Diagram and typed description. The remark starts with an abbreviation for “It turns out”.

$$\mathfrak{g} = \underset{\text{compact}}{k} + p$$

[crossed out at right: writing unclear]



Original source details: GELFAND-P000979

Undeciphered passage. The entry to the right of “compact” is heavily crossed out.

Cond. k has a 1-dimensional center.
their list:
1) $\mathrm{su}(p,q)$
2) $\mathrm{sp}(2n,\mathbb{R})$
3) $\mathrm{so}^*(2n)$
4) $\mathrm{so}(2,n)$
and 2 exceptional ones.

GELFAND-P000980

[source number: 69]

Consider the complexification $G_{\mathbb{C}}$ and the subset

$$G \cdot \exp iC \subset G_{\mathbb{C}}.$$

Uncertain reading. A subscript of the first G is crossed out. Its reading remains unclear.

Original source details: GELFAND-P000980

— · —

1) Kostant–Vinberg theorem.

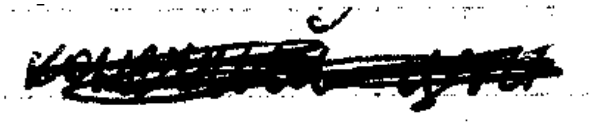
[crossed out at left: writing unclear] a cone C (conv. invar.) in a simple real Lie alg. $\mathfrak{g}$ with nonempty interior

exists $\Leftrightarrow$

Uncertain reading. A mark to the left of the line about the cone is crossed out.

Original source details: GELFAND-P000980

$\mathfrak{g}$ is of Hermitian type (i.e. the max. compact subalgebra has a [crossed out: writing unclear] 1-dimensional center)



Undeciphered passage. Several strokes are visible under the cancellation before "1-dimensional center". Their reading remains unclear.

Original source details: GELFAND-P000980

[crossed out at left: writing unclear] I.M.G. It is wrong to restrict to simple and semisimple ones (in ∞-dimensional theory)

Uncertain reading. A mark to the left of "I.M.G." is crossed out. Its reading remains unclear.

Original source details: GELFAND-P000980

2) Among all cones there are $C_{\max}$ and $C_{\min}$, $C_{\max} \supset C_{\min}$

(unique)

Uncertain reading. The digit 2 is retraced over other strokes.

Original source details: GELFAND-P000980

(G. Olshanskii and simult. S. M. Paneitz — a student of I. Sigal.)

There is a description ↗ [arrow to the preceding line] in terms related to the Cartan subalgebra.

3) Consider the set

$$G \exp iC \subset G_{\mathbb{C}}.$$

[crossed out at left: writing unclear] Thm. $\forall$ cones C the set $G \exp iC$ — a semigroup!

Original source details: GELFAND-P000980

Uncertain reading. A short entry to the left of "Thm." is heavily crossed out.

A weaker version holds for $\forall$ Lie algebras. We get a local Lie semigr.

$$\begin{array}{l} G \exp iC \\ \| \\ \Gamma(C) \end{array} \quad \text{— the class of complex semigroups considered}$$

$$\Gamma(C) \supset \underset{\text{(interior)}}{\Gamma^{\circ}(C)}$$

A holom. rep. if the matrix elems. are holom. in the interior.

Irred. = tensor prod. of holom. and antiholom..

[written below the abbreviation: irred. rep.]

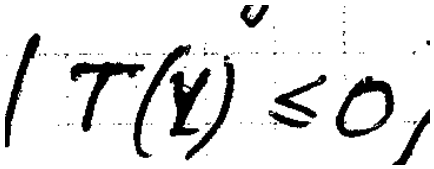
Holom.: distinguish (?) a highest-weight rep..

Original source details: GELFAND-P000980

Uncertain reading. The last letter after "выде" is heavily retraced. "выдел." and "выдер." are possible readings.

G — group; T — unit. representation, $t(x)$, $x \in \mathfrak{g}$.

$$C(T) = \{Y \mid T(Y) \leqslant 0\} \qquad t(y), \quad y \in i\mathfrak{g}.$$



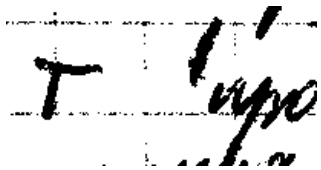
Original source details: GELFAND-P000980

Diagram and typed description. The cone formula shows an uppercase T. The nearby letters with shorter crossbars are rendered as t.

a highest-weight rep. $\Leftrightarrow C(T)$ is nontrivial.

Thm. T — a highest-weight rep., $C = C(T)$ — corresp. cone.

Then T 'extends to a holom. rep. of the corresp. semigroup $\Gamma(C)$

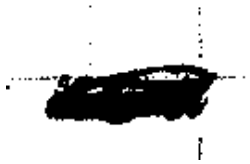


Original source details: GELFAND-P000980

Uncertain reading. There is a raised stroke before “extends”. Its meaning is unclear.

This is the exact domain of extendibility.

category of holom. reps. [crossed out: writing unclear] of the semigroup $\overset{\text{isom.}}{\Leftrightarrow}$ category of highest-weight reps. of the group.



Original source details: GELFAND-P000980

Undeciphered passage. The writing before “of the semigroup” is covered by crossing out.

— · —

Chapter 20

GELFAND-P000981

14/IX-81. Gelfand Sem.

I.M.G. On contragredient Lie algebras

21/IX-81. Gelfand Sem.

S. Gelfand.

Reading on Lie algebras:

1. Dixmier or Serre, part 3 (do not read 1 and 2 under any circumstances)

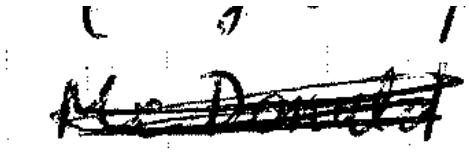
[Below the name Dixmier, linked by a short vertical line:] Enveloping algebras, ch. 1 — textbook literature.

2. Humphrey, Lect. Notes, 831

highest-weight modules for semisimple Lie algebras.

(Verma modules)

3. [Crossed out: Mc Donald[?]] Macdonald Sem. Bourbaki[?], June 1981.



Uncertain reading. Mc Donald

Original source details: GELFAND-P000981

Uncertain reading. Bourbaki

Original source details: GELFAND-P000981

4. Works of Kac and Gabber

5. I.M. Gelfand et al. Funct. Anal., 5:1; 10:2

(Bernstein, [crossed out: И[?]], I.M.G., S.I.G.)

Uncertain reading. И

Original source details: GELFAND-P000981

6. Jantzen, Lect. Notes, 750

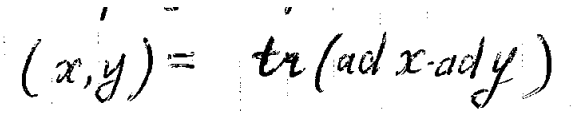
7. Kac, Kazhdan Advances Math., 1979, 34, $N = 1$.

I.M.G. h — vector space with basis $h_1, \ldots, h_n$

g — semisimple Lie algebra

$$\begin{aligned} \operatorname{ad} x \colon g &\longrightarrow g \\ y &\longmapsto [x, y] \end{aligned}$$

Cartan scalar product: $(x, y) = \operatorname{tr}(\operatorname{ad} x - \operatorname{ad} y)$?



Uncertain reading. $(x, y) = \operatorname{tr}(\operatorname{ad} x - \operatorname{ad} y)$

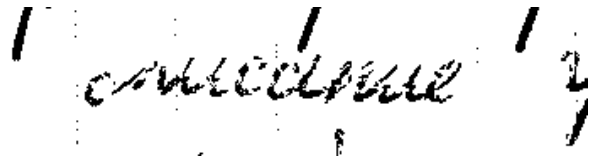
Original source details: GELFAND-P000981

(Check on the matrix algebra)

[Added below the line:] and the orthog. alg. of order 3.

I.M.G. Infinite-dimensional representations lie close to the standard representation of the Lorentz group.

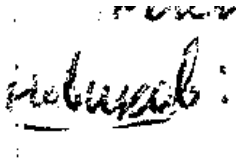
Idea: into physics, since the shift? of the Lorentz group is approximate (at our scales this is negligible)



Uncertain reading. shift

Original source details: GELFAND-P000981

Novikov?: a small shift should give the neutrino mass

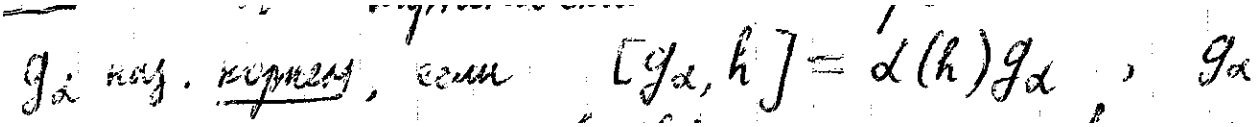


Uncertain reading. Novikov

Original source details: GELFAND-P000981

I.M.G. h — Cartan subalgebra

g_α is called a root if $[g_\alpha, h] = \alpha(h) g_\alpha$, g_α?



Original source details: GELFAND-P000981

Uncertain reading. g_α is called a root if $[g_\alpha, h] = \alpha(h) g_\alpha$, g_α

($\alpha(\cdot)$ — linear function on h, called a root)

GELFAND-P000982

[source number: 12]

On h a scalar product appears (dual to the Cartan one).

Then we obtain $(\alpha, \alpha) > 0$.

There exist roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that if α is another root, then

$$\alpha = \sum n_i \alpha_i, \qquad n_i \in \mathbb{Z}.$$

Here all the n_i have the same sign!

If all $n_i \geq 0$, the root is pos.

— " — $n_i \leq 0$ — " — neg.

Necessarily [crossed out: that?] all the n_i have the same sign!!

Uncertain reading. that

Original source details: GELFAND-P000982

They can be ordered

$\exists$ a greatest root α_0

Coroot:

$$\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}$$

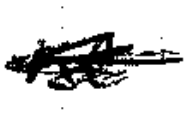
The set of coroots is again a root system (another algebra — Langlands dual).

We construct the reflection $w_\alpha \colon h \longrightarrow h$, $w_\alpha(\lambda) =$ [crossed out: unresolved sign] $\lambda - (\alpha^\vee, \lambda)\alpha$

Uncertain reading. unresolved sign

Original source details: GELFAND-P000982

We identify h and h^* [crossed out: unresolved word] by means of the metric



Uncertain reading. unresolved word

Original source details: GELFAND-P000982

[Above the following line, crossed out: α?]

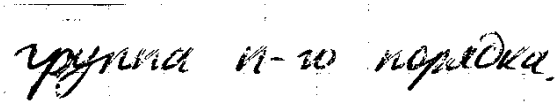
Uncertain reading. α

Original source details: GELFAND-P000982

w_α — reflection in the plane perpendicular to α.

$$w_\alpha(\alpha) = -\alpha.$$

The group generated by these reflections is finite. It is called the Weyl group W.
For $sl(n)$ this is the [crossed out: gr?] symmetric [crossed out: of permutations?] group of order n.

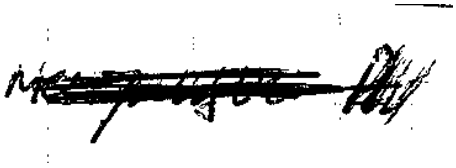


Original source details: GELFAND-P000982

Diagram and typed description. After "group" the manuscript reads "of order n".

Original source details: GELFAND-P000982

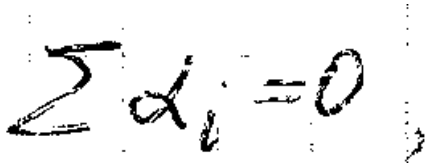
Uncertain reading. gr



Original source details: GELFAND-P000982

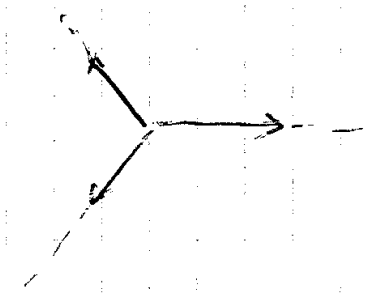
Uncertain reading. of permutations

$sl(3) \Rightarrow h$ — a 2-dimensional space
$\sum \alpha_i = 0$, we draw them all in space, but they lie in the plane $\sum_{i=1}^{3} x_i = 0$



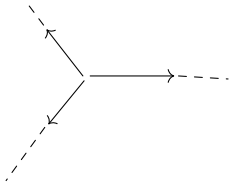
Original source details: GELFAND-P000982

Diagram and typed description. The first sum has no written limits.



Original source details: GELFAND-P000982

Diagram and typed description. ← we obtain 3 vectors at an angle of 120°.



Reduce [crossed out: unresolved sign] to reflections: permutations — to transpositions.

Original source details: GELFAND-P000982

Uncertain reading. unresolved sign

reflections of adjacent ones — simple roots.

All roots are obtained from the simple ones by transformations of the group W.

$a_{ij} = (\alpha_i^\vee, \alpha_j)$ — the Cartan matrix.

Claim. $a_{ij} \in \mathbb{Z}$; $a_{ij} \leq 0$; $a_{ij} = 0 \Rightarrow a_{ji} = 0$

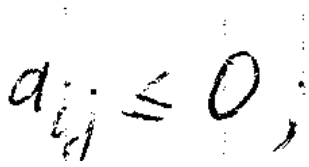


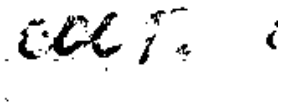
Diagram and typed description. The inequality $a_{ij} \leq 0$ has no written restriction on the indices.

Original source details: GELFAND-P000982

(a_{ij}) is positive definite.

Conversely: to every such matrix corresp.[?] a Lie alg.

Uncertain reading. corresp.



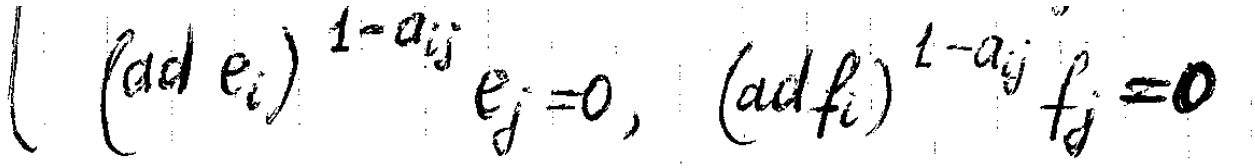
Original source details: GELFAND-P000982

Put $\alpha_i^\vee = h_i$.

Then we introduce the Lie alg. defined by the following relations:

GELFAND-P000983

$$\begin{cases} [e_i, f_j] = \delta_{ij} h_i \\ [h_i, h_j] = 0 \\ [h_i, e_j] = a_{ij} e_j, \quad [h_i, f_j] = -a_{ij} f_j \\ (\operatorname{ad} e_i)^{1-a_{ij}} e_j = 0, \quad (\operatorname{ad} f_i)^{1-a_{ij}} f_j = 0 \end{cases} \tag{*}$$



Original source details: GELFAND-P000983

Diagram and typed description. The last line of the system has no written condition on the indices.

V — a g-module

$\lambda \in h^*$ is called a weight if it is an eigenvalue, i.e. $\exists v \in V \setminus \{0\}$

$$hv = \lambda(h)v.$$

V_λ — the subspace of eigenvectors with weight λ.

Example of a g-module: g with ad.

$$\dim V < \infty \quad \Rightarrow \quad V = \bigoplus V_\lambda$$

and all λ lie in the lattice generated by the roots (coroots)

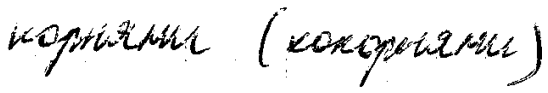


Diagram and typed description. The manuscript reads “roots (coroots)”.

Original source details: GELFAND-P000983

Ordering λ: lexicographically by the coordinates generated by the simple roots.

$\exists!$ finite-dimensional rep. generated by a highest-weight vector.

[Crossed out: $\forall^?$] α — a root, λ — a weight, v_λ has weight $\lambda \Rightarrow$

$e_\alpha v_\alpha$ has weight $\alpha + \lambda$.

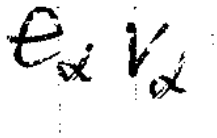


Diagram and typed description. The manuscript writes v_α after e_α.

Original source details: GELFAND-P000983

Uncertain reading. $\forall$

Original source details: GELFAND-P000983

Proof.

$$\begin{aligned} h(e_\alpha v_\alpha) &= e_\alpha(hv_\alpha) + [h, e_\alpha]v_\alpha \\ &= \lambda e_\alpha v_\alpha + \alpha e_\alpha v_\alpha = \\ &= (\lambda + \alpha) e_\alpha v_\alpha \end{aligned}$$

Cartan’s thm.

[Crossed out: $\exists!^?$; added above the line inside an outline: $\forall \lambda \geq 0$ $\exists!$] finite-dimensional rep.
[crossed out: if [remainder unresolved]] such that

$e_\alpha V_\lambda = 0$, $\forall \alpha \in \Delta_+$ (pos. roots)

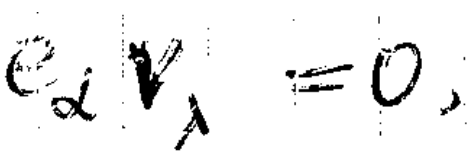
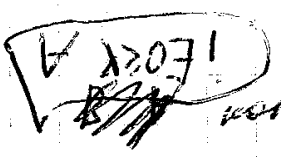


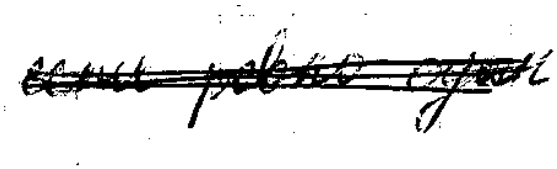
Diagram and typed description. The letter V in V_λ is retraced.

Original source details: GELFAND-P000983



Uncertain reading. ∃!

Original source details: GELFAND-P000983



Uncertain reading. if [remainder unresolved]

Original source details: GELFAND-P000983

(or $\sim \forall$ simple).

Introduce the group e^{λ} as follows:

$$e^{(\lambda+\mu)} = e^{\lambda} e^{\mu}$$

e — a letter.

$$e^{\sum n_i \alpha_i} = (e^{\alpha_1})^{n_1} \cdots (e^{\alpha_n})^{n_n}$$

Character of the representation:

$$\operatorname{ch} V \underset{\text{def}}{=} \sum_{\mu} (\dim V_{\mu}) e^{\mu}$$

Introduce [crossed out: root [fragment]$^{?}$] the following weight ρ:

$$\rho(h_i) = 1, \qquad i = 1, \ldots, n$$

Uncertain reading. root [fragment]

Original source details: GELFAND-P000983

$$h_i = \alpha_i^{\vee}$$

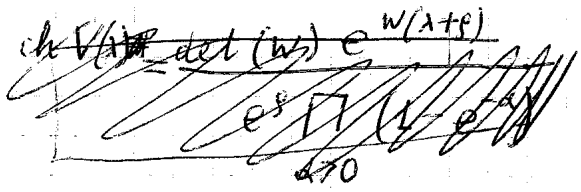
Weyl formula: for the rep. with highest weight λ:

GELFAND-P000984

[source number: 13]

[Crossed-out formula:]

$$\operatorname{ch} V(\lambda) = \frac{\det(w) e^{w(\lambda+\rho)}}{e^{\rho} \prod_{\alpha>0} (1 - e^{-\alpha})}.$$



Uncertain reading. In the cancelled formula no sum is visible before $\det(w)$. The strokes after $V(\lambda)$ overlap.

Original source details: GELFAND-P000984

$$\operatorname{ch} V(\lambda) = \frac{\sum_{w \in W} \det(w) e^{w(\lambda+\rho)}}{e^{\rho} \prod_{\alpha>0} (1 - e^{-\alpha})}.$$

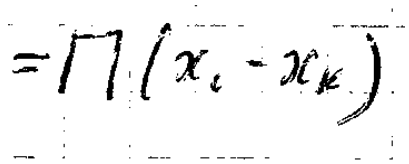
$w(\lambda+\rho)$ — result of the action of w on the weight $\lambda+\rho$.

Hence, in particular,

$$\sum \det(w) e^{w\rho} = e^{\rho} \prod_{\alpha>0} (1 - e^{-\alpha}).$$

For matrices this is the formula

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix} = \prod (x_i - x_k).$$



Original source details: GELFAND-P000984

Diagram and typed description. The product has no written index limits.

Rudakov. Kac–Moody algebras

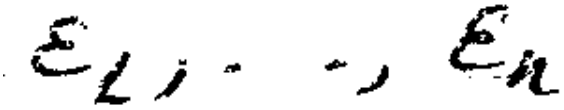
A — an $n \times n$ matr.

and a set of generators

$$e_1, \ldots, e_n, \quad f_1, \ldots, f_n, \quad h_1, \ldots, h_n, \quad \text{[crossed out] } \varepsilon_1, \ldots, \varepsilon_n$$

Original source details: GELFAND-P000984

Uncertain reading. The short entry before ε_1 is crossed out. Its reading remains unresolved.



Original source details: GELFAND-P000984

Diagram and typed description. The subscript on the first visible ε reads 1.

and rels.

$$\begin{cases} [h_i, h_j] = 0 \\ [h_i, e_j] = a_{ij} e_j \\ [h_i, f_j] = -a_{ij} f_j \\ [e_i, f_j] = \delta_{ij} h_i \end{cases}$$

$$\begin{cases} [\varepsilon_i, \varepsilon_j] = 0 \\ [\varepsilon_i, e_j] = \delta_{ij} e_j \\ [\varepsilon_i, f_j] = -\delta_{ij} f_j \\ [\varepsilon_i, h_j] = 0 \end{cases}$$

$F(A)$ — Lie algebra generated by these generators and relations.

$$F(A) \supset \widetilde{H} = \langle h_i, \varepsilon_i \rangle.$$

I — max ideal in $F(A)$ such that the image of $\{e_i, f_i, h_i, \varepsilon_i\}$ in $F(A)/I$ is a linearly indep. system of vectors.

It exists and is uniquely determined (max "non-injuring").

$$\widetilde{G}(A) \stackrel{\text{def}}{=} F(A)/I.$$

GELFAND-P000985

(2) — matrix of order 2 with trace 0.

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

[Crossed out: short entry unresolved.]

Uncertain reading. short entry unresolved

Original source details: GELFAND-P000985

Add. max non-injuring system

Criterion for finite dimension

matr. is symmetrizable (made symmetric by mult. by a diagonal one), and positive definite, 2 on the main diagonal

$$\widetilde{G}(A) \supset \underset{\text{center}}{Z}$$

$$Z \subset \langle h_i, \varepsilon_i \rangle$$

We still have to quotient by a max subspace in Z that does not intersect the linear span of $\{h_i\}$.

$$J \subset Z$$

(a_{ij}) is nonsingular $\Rightarrow$ this is no longer needed

(a_{ij}) is singular $\Rightarrow$ the roots are linearly dependent. We add [crossed out: some[?] remainder unresolved], to make them linearly independent.

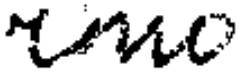
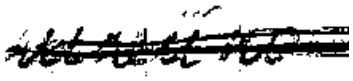


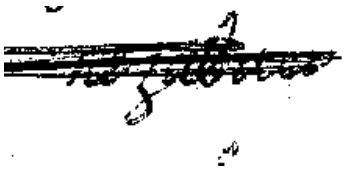
Diagram and typed description. After the cancellation the manuscript reads «что».

Original source details: GELFAND-P000985



Uncertain reading. some

Original source details: GELFAND-P000985



Original source details: GELFAND-P000985

Uncertain reading. remainder unresolved

For this we formally enlarge h with room to spare

$$J \subset Z, \qquad J \cap \langle h_i \rangle = 0.$$

$$G(A) \overset{\text{def.}}{=} \widetilde{G}(A)/J_{\max}.$$

$$H = \langle h_i, \varepsilon_i \rangle.$$

Define the set of vectors $x \in G(A)$ that are eigenvectors.

$$G_\alpha = \{x \in G(A),\ \forall h \text{ [crossed out]} \in H,\ [h, x] = \alpha(h)x\}.$$

Original source details: GELFAND-P000985

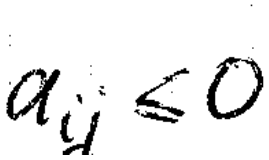
Uncertain reading. A symbol between h and $\in$ is crossed out. Its reading is unresolved.

$$\alpha \in H^*$$

Roots: those α for which G_α are nonzero.

A is a Cartan matrix if

1) $a_{ij} \in \mathbb{Z}, \qquad a_{ii} = 2, \qquad a_{ij} \leq 0$



Original source details: GELFAND-P000985

Diagram and typed description. There is no written index condition on $a_{ij} \leq 0$.

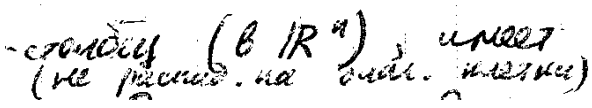
2) $a_{ij} = 0 \Rightarrow a_{ji} = 0$

[Crossed out before the addition: entry unresolved.]

Original source details: GELFAND-P000985

Uncertain reading. entry unresolved

[Added above the lemma: Let A be irreducible (does not split into diag. blocks[?]).]



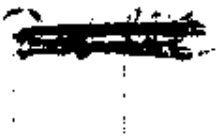
Original source details: GELFAND-P000985

Uncertain reading. does not split into diag. blocks

[Written above: u is a column vector (in $\mathbb{R}^n$), so $Au \in \mathbb{R}^n$ makes sense.]

Lemma.

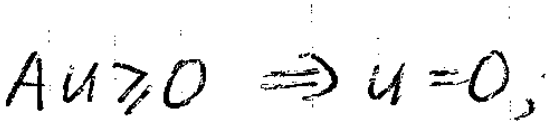
(I) If $Au \geq 0$, then $u > 0$ or $u = 0$; [crossed out: entry unresolved] $\det A \neq 0$.



Original source details: GELFAND-P000985

Uncertain reading. entry unresolved

(II) $Au \geq 0 \Rightarrow u = 0$; $\exists u > 0, Au = 0$; $\operatorname{rank} A = n - 1$



Original source details: GELFAND-P000985

Diagram and typed description. To the right of the arrow the manuscript reads $u = 0$.

(III) $Au \geq 0,\ u \geq 0 \Rightarrow u = 0$; $\exists u > 0 :\ Au < 0$

Assertion: exactly one of (I)–(III) holds.

GELFAND-P000986

[source number: 14]

(I) — finite-dimensional Lie algs.; they are classified

(II) — extended.

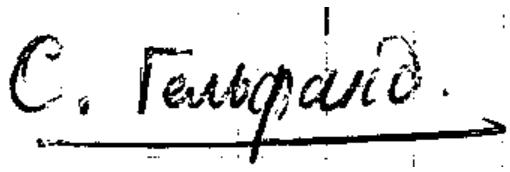
(III) — all the others.

I. M. G. Ramanujan's identities:

$$\frac{1}{\prod_{n=1}^{\infty}(1-q^{5n-2})(1-q^{5n-3})} = \sum_{m=0}^{\infty} \frac{q^{m(m+1)}}{(1-q)\cdots(1-q^m)}$$

$$\frac{1}{\prod_{n=1}^{\infty}(1-q^{5n-1})(1-q^{5n-4})} = \sum_{m=0}^{\infty} \frac{q^{m^2}}{(1-q)\cdots(1-q^m)}$$

S. Gelfand.



Original source details: GELFAND-P000986

Diagram and typed description. The line under the name ends in a right-pointing arrow.

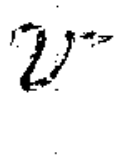
Verma module.

(see Dixmier, Jantzen)

In $G(A)$ there are generators $e_i, f_i, h_i \qquad H = \{h_i\}$

$\forall \lambda \in H^*$ we construct a module generated by one vector $v^{-?}$ with the conditions

$$(*) \quad \begin{cases} e_i v_0 = 0 \\ h v_0 = \lambda(h) v_0 \end{cases}$$



Uncertain reading. v^-

Original source details: GELFAND-P000986

$$N_+ = \{e_i\}, \qquad N_- = \{f_j\}$$
$$N_+ v_0 = 0$$

$f_{i_1} f_{i_2} \cdots f_{i_k} v_0$ — we write out such letters.

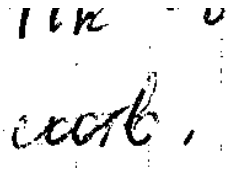
Impose rels. $(*)$ and also those dictated by f_j.

One readily obtains

$$h f_{i_1} \cdots f_{i_k} v_0 = [\lambda(h) - \alpha_{i_1}(h) - \cdots - \alpha_{i_k}(h)] f_{i_1} \cdots f_{i_k} v_0,$$

i.e. $f_{i_1} \cdots f_{i_k} v_0$ is also a weight vector

The module corresp.$^?$ to the char. λ is called $M(\lambda)$.



Uncertain reading. corresp.

Original source details: GELFAND-P000986

$$\mathrm{Hom}(M(\lambda), M(\mu)) = \begin{cases} 0 \text{ or} \\ \mathbb{C} \end{cases}$$

— and then every nontriv. homom. is an embedding

Theorem. If $\mathrm{Hom}(M(\lambda), M(\mu)) \neq 0$, then

$$1) \quad \lambda = \mu - \sum n_i \alpha_i \quad \text{for some integers } n_i \geq 0$$

$$2) \quad \lambda - \rho = w(\mu - \rho) \quad \text{for some } w \in W$$

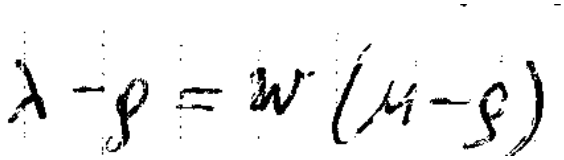


Diagram and typed description. Both occurrences of ρ are preceded by minus signs.

Original source details: GELFAND-P000986

(ρ is the half-sum of all pos. roots — see above).

$$3) \quad \lambda - \rho = s_{\delta_1} \cdots s_{\delta_k}(\mu - \rho)$$

(s_δ — reflections in any roots, not necess. simple)

and

$$- s_{\delta_i} \cdots s_{\delta_k}(\mu - \rho) + s_{\delta_{i-1}} s_{\delta_i} \cdots s_{\delta_k}(\mu - \rho) = m_i \delta_{i-1},$$
$$m_i \leq 0, \quad m_i \in \mathbb{Z}.$$

1)–3) are already sufficient for $\mathrm{Hom}(M(\lambda), M(\mu))$ to be nontrivial.

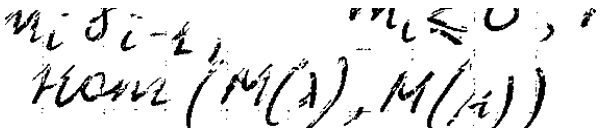


Diagram and typed description. In the final Hom the second weight reads μ.

Original source details: GELFAND-P000986

GELFAND-P000987

Problem (I.M.G.) (nontrivial)

Diagram and typed description. The first word of the heading reads “Problem”.

Original source details: GELFAND-P000987

(a_{ij}) — Cartan matr.; satisfies cond. 1) and 2)

What is the dimension of the root space?

For every root, lin. indep. words are constructed, etc.

Diagram and typed description. The abbreviation before “indep.” reads «лин.».

Original source details: GELFAND-P000987

28.9.81. Gelfand Sem.

Seifert (?) (Paris)

Diagram and typed description. The manuscript has “(?)” after “Seifert”.

Original source details: GELFAND-P000987

Diagram and typed description. A yellow comment icon appears to the right of the heading in the digital frame.

Original source details: GELFAND-P000987

(X, G) — an arbitrary transformation group

$\mathcal{P}(X)$ — the set of subsets of X

$\mathcal{F} \subset \mathcal{P}(X)$ is defined as follows:

[Crossed out above the start of the formula: entry unresolved.]

Uncertain reading. entry unresolved

Original source details: GELFAND-P000987

[Crossed out before S: entry unresolved.]

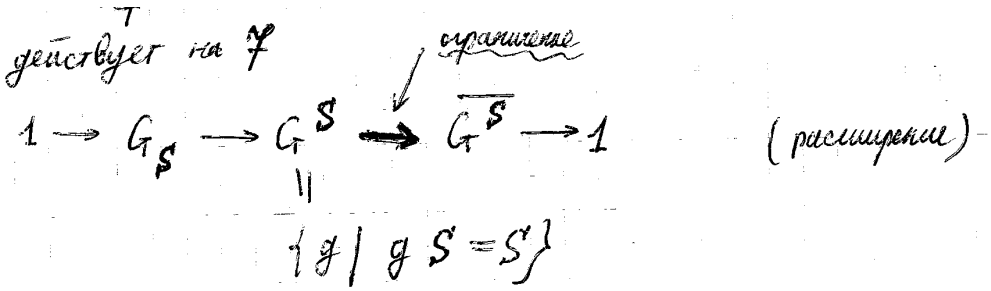
Uncertain reading. entry unresolved

Original source details: GELFAND-P000987

$$S \in \mathcal{F} \underset{\text{def}}{\Longleftrightarrow} \forall x \notin S,\ \exists g \in G,\ gs = s,\ \forall s \in S,\ gx \neq x$$

G acts on $\mathcal{F}$

$$1 \longrightarrow G_S \longrightarrow \underset{\underset{\{g \mid gS=S\}}{\|}}{G^S} \xrightarrow{\text{restriction}} \overline{G^S} \longrightarrow 1 \qquad \text{(extension)}$$



Original source details: GELFAND-P000987

Diagram and typed description. The pointer from “restriction” marks the arrow from G^S to $\overline{G^S}$.

Here

$$G_S = \bigcap_s G_s, \qquad G_s = \mathrm{St}_G(s)$$

$\mathcal{F}$ has the following formal property:
it is closed under arbitrary intersections

$$S_i \in \mathcal{F},\ \forall i \ \Rightarrow \bigcap_{i \in I} S_i \in \mathcal{F}$$

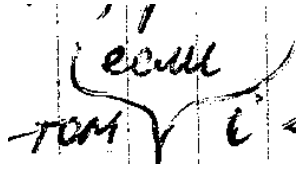
$(I, <)$ — a partially ordered set. Consider fns. $f : I \to \mathcal{F}$ with the property:

$$\{f : I \to \mathcal{F} \mid f(i) \subseteq f(j) \iff i \leq j\}$$

Then G acts on this set.
Fixing $(I, <)$, we can in particular consider all such fns.

$$(\overline{\mathcal{F}}_I, G) = \{f : I \to \mathcal{F} : \text{ as above and moreover}$$

["If" is inserted above the line with an insertion mark.]



Original source details: GELFAND-P000987

Diagram and typed description. The word "if" is above the V-shaped mark.

$i < i'$ [crossed out: entry unresolved] a minimal chain (2 adjacent elem.) in $(I, <)$
$\Rightarrow f(i) \underset{\neq}{\subset} f(i')$ is also a min. chain $\}$

Original source details: GELFAND-P000987

Uncertain reading. entry unresolved

Original source details: GELFAND-P000987

Diagram and typed description. The inclusion sign has $\neq$ beneath it.

Then G also acts on the set of such fns.

From the formal property of stability under intersections

$$S \subseteq X \rightsquigarrow \underset{\text{(closure)}}{S^{\mathrm{cl}}} = \bigcap_{\substack{T \in \mathcal{F} \\ T \supseteq S}} T = (0).$$

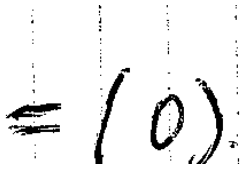


Diagram and typed description. The manuscript writes (0) to the right of the final equals sign.

Original source details: GELFAND-P000987

GELFAND-P000988

[source number: 15]

$$x \in X \Rightarrow \text{define } \{x\}^{\mathrm{cl}} \underset{\text{def}}{=} \text{line passing through this point}$$

G [inserted: acts] transitively $\Rightarrow \{x\}^{\mathrm{cl}}$ is the set of points with the same stabilizer as x

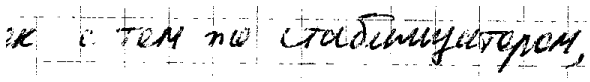


Diagram and typed description. The manuscript reads “with the same stabilizer”.

Original source details: GELFAND-P000988

G_x notation for the stabilizer.

Idea: single transitivity $\Rightarrow$ multiple transitivity

$$\exists x : \ \{x\}^{\mathrm{cl}} \underset{\neq}{\supseteq} x \Rightarrow \text{the group cannot act doubly transitively}$$

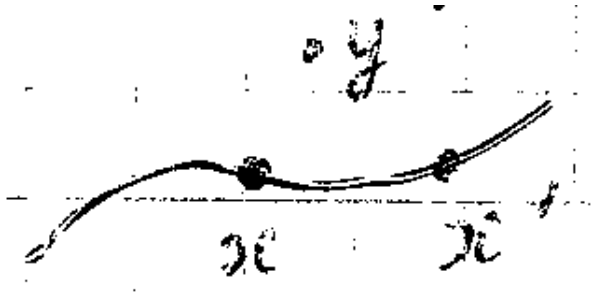
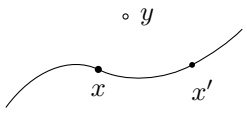


Diagram and typed description. Two marked points x, x' on the curve and a point y above it.



Original source details: GELFAND-P000988

$(x, y) \mapsto (x, x')$ is impossible, since $\{x\}^{\mathrm{cl}}$ is pres.[?]

Uncertain reading. pres.

Original source details: GELFAND-P000988

On $\overline{\mathcal{F}}_I$ it can already act doubly transitively

Example

doubly trans. on $F \iff$ trans. on $(F \times F) \setminus \Delta$

$$\forall x \in F, \quad G_x \text{ must act transitively on } F \setminus \{x\}$$

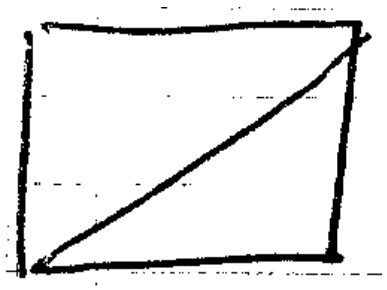
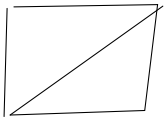


Diagram and typed description. A quadrilateral with a rising diagonal and a small gap in its upper left corner.



Original source details: GELFAND-P000988

The end will be as follows: for

B, N, T, W, P

an axiomatic generalization is given, for which the Bruhat decomp. makes sense.

Under certain conditions on the action:

(i)

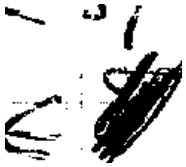
$$\mathcal{F}_{\max} = \{\max \text{ totally ordered chain} \quad S_1 < S_2 < \cdots < S_i < \cdots$$

[Crossed out at the end of the line: entry unresolved.]

Uncertain reading. entry unresolved

Original source details: GELFAND-P000988

$S_i \in$ [crossed out: entry unresolved] $\mathcal{F}\}$



Uncertain reading. entry unresolved

Original source details: GELFAND-P000988

[Below $\mathcal{F}_{\max}$:] (Flags)

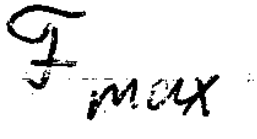


Diagram and typed description. There is no distinct bar above $\mathcal{F}_{\max}$.

Original source details: GELFAND-P000988

trans. with respect to G

(ii)

$$\forall S: \ \dim(S) = \text{length of a max flag}, \quad < +\infty$$

[Below S, $\mathcal{F}$ is added with an upward arrow.]

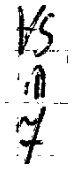


Diagram and typed description. The arrow points up from $\mathcal{F}$ to S.

Original source details: GELFAND-P000988

[Crossed out below $\dim(S)$: entry unresolved.]

Uncertain reading. entry unresolved

Original source details: GELFAND-P000988

[Below "length of a max flag": unresolved entry crossed out, then:] ending in S.

Uncertain reading. unresolved entry

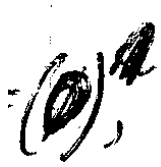
Original source details: GELFAND-P000988

(iii)

$$S, S' \subseteq T, \quad \dim S + \dim S' > \dim(T) \Rightarrow S \cap S' \neq \bigcap_{\mathcal{F} \ni T'} T'$$

(iv) $\forall S \in \mathcal{F}$, if $\{x\}^{\mathrm{cl}} \cap S = (\varnothing)^{\mathrm{cl}?}$, then

$$G^S \cap G^{\{x\}^{\mathrm{cl}}} \underset{\text{maps onto}}{\longrightarrow} \overline{G^S}$$



Uncertain reading. $(\varnothing)^{\mathrm{cl}}$

Original source details: GELFAND-P000988

Then

$$G = \coprod_W G^f w G^f \qquad \text{(disjoint union)}$$

(double coset)

where

$$G^f = \bigcap_I G^{f(i)}, \qquad f : I \to \mathcal{F}$$

[A leader from G^f to the caption:] stabilizer of a max flag.

$$W = N/T$$

[“Here” is written above f.]

$$\{ f : I \to \mathcal{F}; \quad S_1 < S_2 < S_3 < S_4 < \cdots$$

— any max chain $\}$

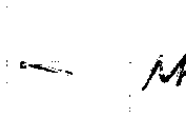


Diagram and typed description. A short stroke precedes “any”. No separate arrowhead is discernible.

Original source details: GELFAND-P000988

GELFAND-P000989

Problems. (I.M.G.)

Describe the 2-sided cosets

1) $U(n)\backslash GL(n)/U(n)$
2) $B(n)\backslash GL(n)/B(n)$

   triang.
3) $U(n)\backslash GL(n)/B(n)$

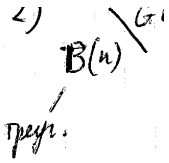


Diagram and typed description. An oblique leader connects “triang.” to the left $B(n)$ in item 2.

Original source details: GELFAND-P000989

Every max flag is obtained as follows:

$$f(i) = \mathrm{cl}(\{e_j\}_{j \le i}), \quad \text{where } \{e_j\} \text{ are points in } X.$$

$$C_i = \mathrm{cl}(\{e_i\})$$

$$N = \{ g \mid g \text{ permutes } \{C_i\} \}$$

and we have:

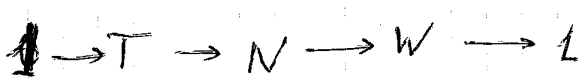


Diagram and typed description. The initial symbol is overwritten. Tentatively 1 over 0.

$[1?] \longrightarrow T \longrightarrow N \longrightarrow W \longrightarrow 1$

Original source details: GELFAND-P000989

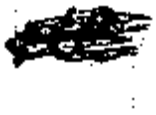
T consists of those that leave each C_i in place;

$W = N/T$ by def.

The speaker knows of no examples other than the classical ones where all the axioms hold.

But: there are examples where the decomp. [crossed out: not (?)] exists, although the axioms do not hold.

Uncertain reading. not (?)



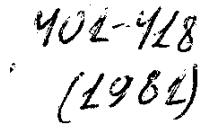
Original source details: GELFAND-P000989

Zelevinsky.

On the paper by S. Abeasis, A. Del Fra, H. Kraft.

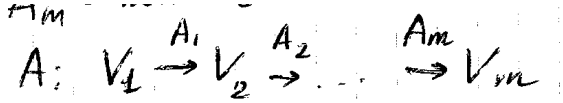
The geometry of repres. of A_m, Math. Ann., 256, N.3, 401–418, (1981).

Diagram and typed description. The opening page reads 401.



Original source details: GELFAND-P000989

A_m — a quiver



Original source details: GELFAND-P000989

Diagram and typed description. The last arrow has a right-pointing head and additional strokes at the left end. Its label is A_m.

$$A: \quad V_1 \xrightarrow{A_1} V_2 \xrightarrow{A_2} \cdots \xrightarrow{A_m} V_m$$

— this is a rep. of a quiver of series A_m.

Here the group acts

$$GL(V_1) \times \cdots \times GL(V_m).$$

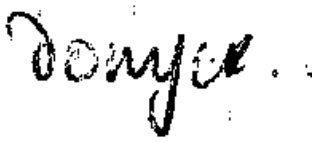
Diagram and typed description. The first subscript reads 1.

Original source details: GELFAND-P000989

Orbits are reps. up to equiv.

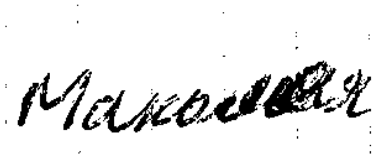
There are finitely many orbits. They admit[?] an explicit descr.

Uncertain reading. admit



Original source details: GELFAND-P000989

Their closures are good varieties (normal, Cohen—Macaulay[?], etc.).



Uncertain reading. Macaulay

Original source details: GELFAND-P000989

GELFAND-P000990

[source number: 16]

[Crossed out: Their (?)] They are analogous to Schubert cells, and the polynomials describe reps. of p-adic groups, as the geom. of Schubert cells is connected with reps. of classical groups.

Uncertain reading. Their (?)

Original source details: GELFAND-P000990

For

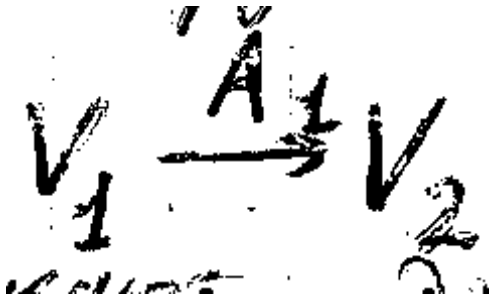


Diagram and typed description. The arrow from V_1 to V_2 is labelled A_1.

$$V_1 \xrightarrow{A_1} V_2$$

Original source details: GELFAND-P000990

determinantal varieties arise.

Description of orbits:

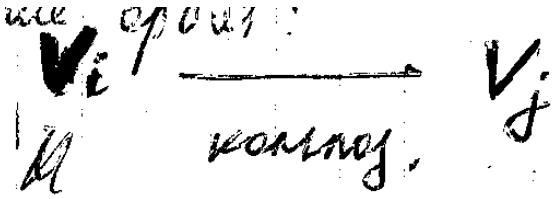


Diagram and typed description. A separate, overtraced mark lies below V_i. Its reading is unresolved.

$$V_i \xrightarrow[\text{compos.}]{} V_j$$

[?]

Original source details: GELFAND-P000990

Specify the ranks of all possible compositions r_{ij}.

A system of inequalities must hold.

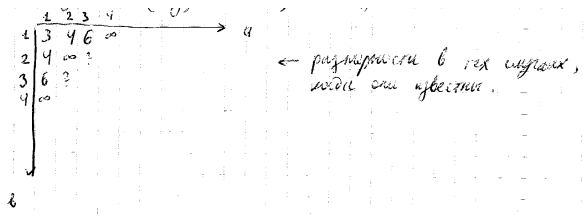
There is another descr. via multiplicities of occurrences.

19/X-81. Gelfand Sem.

Kirillov.

How is the Lie algebra with generators x, y and rels. structured?

$$(\operatorname{ad} x)^{a+1} y = 0, \qquad (\operatorname{ad} y)^{b+1} x = 0, \qquad a, b \in \mathbb{N}.$$



Original source details: GELFAND-P000990

Diagram and typed description. The a axis points right and b down. The entries $(a, b) = (3, 2), (2, 3)$ are question marks. Blank entries remain blank.

b \ a	1	2	3	4
1	3	4	6	∞
2	4	∞	?	$\leftarrow$
3	6	?		
4	∞			

dimensions in the cases where they are known.

Without relations, the number of commutators of bidegree k, ℓ is:

$$d(k, \ell) = \frac{1}{k + \ell} \sum_{d \mid (k, \ell)} \mu(d) \, \frac{\left(\frac{k}{d} + \frac{\ell}{d}\right)!}{\frac{k}{d}! \, \frac{\ell}{d}!}.$$

$k = n$, $\ell = n + 1 \Rightarrow$ the nth Catalan number

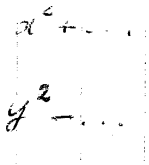
(= the number of ways of placing parentheses in the nonassoc. product $x \cdot x \cdot \ldots \cdot x$.)

[At left above the next line: $(\sim)$.]

Another question: the envelop. algebra.

Rels.

$$0 = x^{a+1} y - \binom{a+1}{1} x^a y x + \binom{a+1}{2} x^{a-1} y x^2 + \ldots,$$
$$0 = y^{b+1} x - \binom{b+1}{1} y^b x y + \binom{b+1}{2} y^{b-1} x y^2 - \ldots.$$



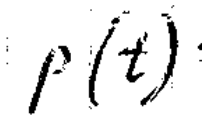
Original source details: GELFAND-P000990

Diagram and typed description. Before the ellipsis the first line has $+$ and the second $-$.

V — a vector space, $S(\bar{V}) = e^{\bar{V}}$ — the symm. algebra

[Below "V — a vector space": "(graded)".]

$p(t) = \sum t^k \dim V_k$ — the Hilbert polynomial.



Uncertain reading. $p(t)$

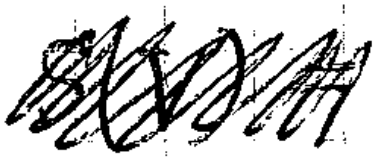
Original source details: GELFAND-P000990

GELFAND-P000991

k — a weight $\Rightarrow$ a weight arises on $S(V) \Rightarrow$ grading on $S(V) \Rightarrow$ the Hilbert polynomial is calculated $P(t)$ for $S(V)$ (or a formal series).

Relation:

[Crossed out: formula unresolved.]



Uncertain reading. formula unresolved

Original source details: GELFAND-P000991

$$\ln P(t) = \sum \frac{C_k}{k} t^k;$$

[Crossed out: short entry unresolved.]

Uncertain reading. short entry unresolved

Original source details: GELFAND-P000991

$$p(t) = \sum \frac{c_k}{k} t^k, \qquad C_k = \sum_{d|k} c_d.$$

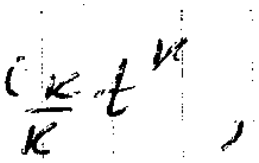


Diagram and typed description. In the expression for $p(t)$ the denominator is k and the numerator is c_k.

Original source details: GELFAND-P000991

$$\underline{x^{k_1}y^{\ell_1}x^{k_2}y^{\ell_2}\dots x^{k_n}y^{\ell_n}}$$

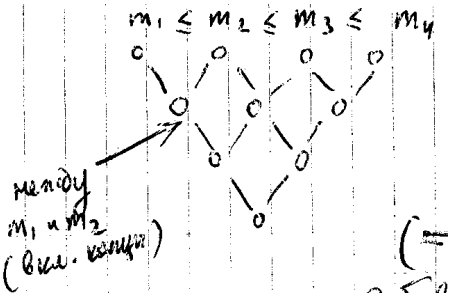
$$k_1+k_2+\dots+k_n=k,$$
$$\ell_1+\ell_2+\dots+\ell_n=\ell.$$

Lexicographic ordering.

y is higher than x. $x<y$.

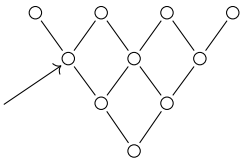
I.M.G. combin. problem

$m_1\le m_2\le m_3\le m_4$ — fixed



Original source details: GELFAND-P000991

Diagram and typed description. Four rows of circles with $4, 3, 2, 1$ entries. The pointer from the note at left marks the first circle of the second row.



between m_1 and m_2 (incl. endpoints)

how many tables?

(= how many reps. of $SL(4)$ with a given highest weight?)

see the appendix to the Gelfand–Minlos–Shapiro book.

GELFAND-P000992

16.11.81. Gelfand Sem.

A.S. Schwarz. Geometry of supergravity.

$\mathbb{R}^{p,q}$ — superspace $(x^1,\dots,x^p,\theta^1,\dots,\theta^q)$.

Λ — Grassmann alg. [Above the abbreviation: r or 4?.]

Uncertain reading. r or 4

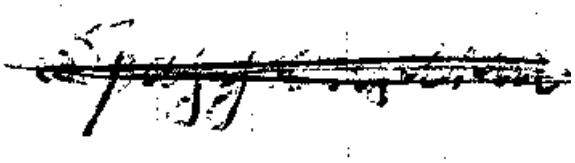
Original source details: GELFAND-P000992

[Crossed out: entry unresolved.]

Uncertain reading. entry unresolved

Original source details: GELFAND-P000992

[Crossed out: with grading?.]



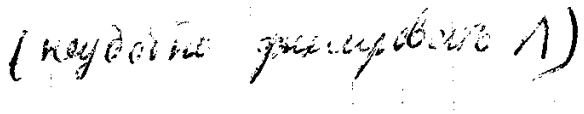
Uncertain reading. with grading

Original source details: GELFAND-P000992

(over $\mathbb{R}$).

$\mathbb{R}^{p,q}_\Lambda$ — the space of rows $(x^1,\dots,x^p,\theta^1,\dots,\theta^q)$, where x^j are even elems. of Λ, θ^j are odd elems. of Λ.

We need to consider this as a functor of Λ (it is inconvenient[?] to fix Λ).



Uncertain reading. inconvenient

Original source details: GELFAND-P000992

This gives $\mathbb{R}^{p,q}$.

$\mathbb{C}^{p,q}$ is def. analogously and an involution is def. in it.

Consider $\mathbb{C}^{4,2}$ $\quad (x^m,\theta^\alpha)=z^A$.

$$\dim_{\mathbb{C}}\mathbb{C}^{4,2}=(4,2),\qquad \dim_{\mathbb{R}}(\mathbb{C}^{4,2})=(8,4).$$

In it consider a real subvariety

$$\begin{array}{ll}\Omega & \subset \mathbb{C}^{4,2}\\ (4,4) & \leftarrow \underline{\mathbb{R}}\text{-dim}\end{array}$$

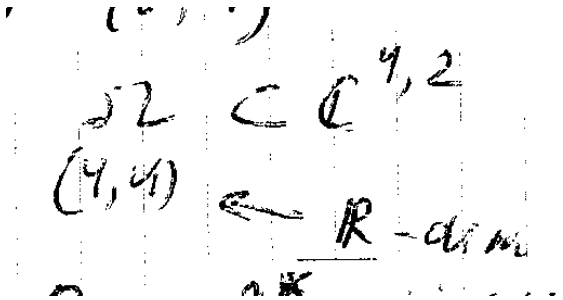


Diagram and typed description. Below Ω the manuscript has $(4,4)$, with an arrow pointing to it from $\underline{\mathbb{R}}$-dim.

Original source details: GELFAND-P000992

Ω is cut out by the eqs.

$$f^k(z,\bar z)=0,$$

f^k are even and real, $\quad k=1,2,3,4.$

Consider analytic transformations in $\mathbb{C}^{4,2}$:

$$\begin{cases}\widetilde{x}^a=\lambda^a(x,\theta),\\ \widetilde{\theta}^\alpha=\lambda^\alpha(x,\theta)\end{cases}\qquad \text{(indep. of } \bar x,\bar\theta).$$

GELFAND-P000993

[source number: 40]

They form a supergroup L.

(The group depends on Λ and all together — a functor from Grassmann algs. to ordinary groups — supergroups.)

Consider those that preserve the supervolume.

$$\widetilde{z}^A = \lambda^A(z).$$

Preservation of volume:

$$\operatorname{Ber} \frac{\partial \widetilde{z}^A}{\partial z^B} = 1.$$

We shall consider the functional

$$S(\Omega) = \int A(\text{first, second derivs. of } f^k) \prod_k \delta\big(f^k(z,\bar{z})\big)\, dz\, d\bar{z}.$$

[In the argument of A, before f^k, crossed out: $f^{k?}$.]

Uncertain reading. f^k

Original source details: GELFAND-P000993

[A leader to δ is labelled "δ-fn.".]

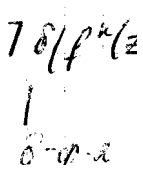


Diagram and typed description. The gloss "δ-fn." is below δ.

Original source details: GELFAND-P000993

Impose a condition on A so that $S(\Omega)$ depends only on Ω, rather than on its eqs.

In classical field theory, in place of the field φ one can consider the graph of the fn. — a surface.

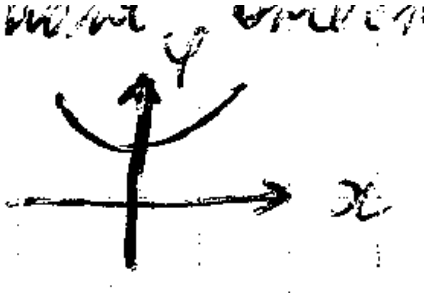
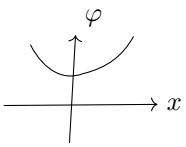


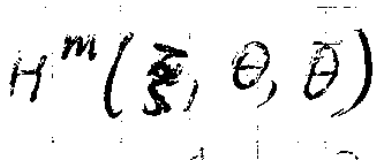
Diagram and typed description. A graph with its minimum above the horizontal x-axis and a vertical φ-axis.



Original source details: GELFAND-P000993

Consider Ω as the graph of the field

$$\mathbb{R}^{4,4} \longrightarrow \mathbb{R}^{4,0}, \qquad H^m(\xi, \theta, \bar{\theta}).$$



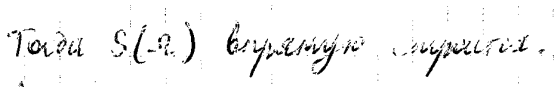
Uncertain reading. The first argument is heavily retraced. The tentative reading is ξ.

Original source details: GELFAND-P000993

The surf. Ω and the superfield are in one-to-one corresp.

Supergravity action: the unique (up to a factor) functional of this kind invariant with respect to the group L.

Then by varying $S(\Omega)$ we obtain[?].



Uncertain reading. obtain

Original source details: GELFAND-P000993

Namely:

$$A = |\det \Gamma|^{-1/2}, \qquad \Gamma \text{ is the Levi form of the surface } \Omega.$$

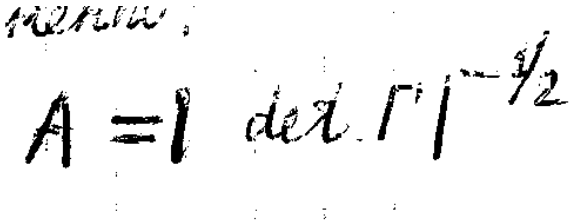


Diagram and typed description. A minus sign precedes the fraction $1/2$.

Original source details: GELFAND-P000993

["(super.)" is inserted beneath "form".]

The Levi form is the first struct. form of the CR-structure.

The form

$$\left.\frac{\partial^2 f^k}{\partial z^A \partial \bar{z}^B}\, dz^A\, d\bar{z}^B \right|_{\substack{\text{on the compl. tangent plane} \\ \frac{\partial f^k}{\partial z^A} dz^A = 0}}$$

(vector-valued)

[Crossed out: ∂f.]

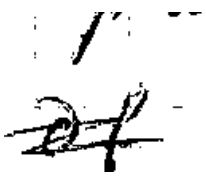


Diagram and typed description. Above "called the Levi form", ∂f is crossed out.

Original source details: GELFAND-P000993

is called the Levi form Γ.

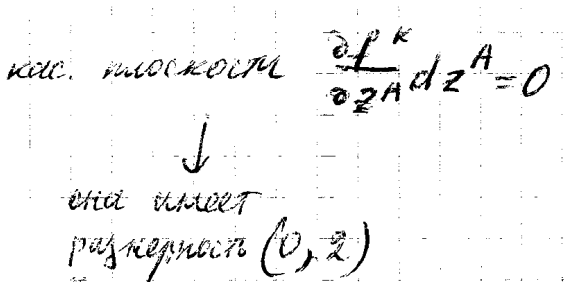


Diagram and typed description. A downward arrow from the words about the tangent plane leads to the dimensional note.

Original source details: GELFAND-P000993

it has dimension $(0, 2)$.

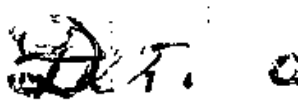
$$\frac{\partial f^k}{\partial x^a}\, dx^a + \frac{\partial f^k}{\partial \theta^\alpha}\, d\theta^\alpha = 0$$

$$\Rightarrow dx^a \text{ is expressed in terms of } d\theta^\alpha \Rightarrow \text{dim. } (0, 2).$$

GELFAND-P000994

The basis in the compl. tangent plane is chosen in different ways.

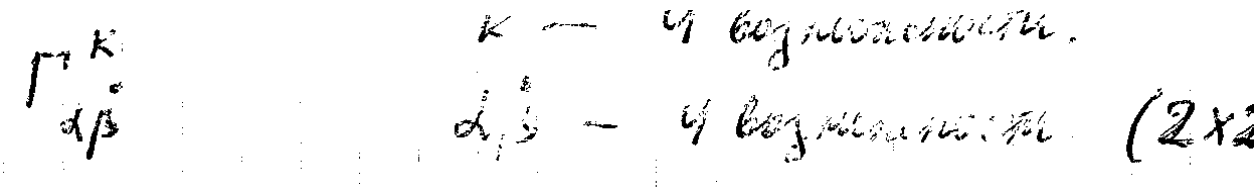
Det. (?) depends on this.



Uncertain reading. The initial strokes of “Det.” are overwritten.

Original source details: GELFAND-P000994

$$\Gamma^k_{\alpha\dot\beta} \qquad \begin{matrix} k \text{ — 4 possibilities,} \\ \alpha, \dot\beta \text{ — 4 possibilities} \quad (2 \times 2 = 4). \end{matrix}$$



Original source details: GELFAND-P000994

Diagram and typed description. In the subscript of Γ the dot is above β.

det — det such a 4×4 matrix.

It and det depend on the choice of basis.

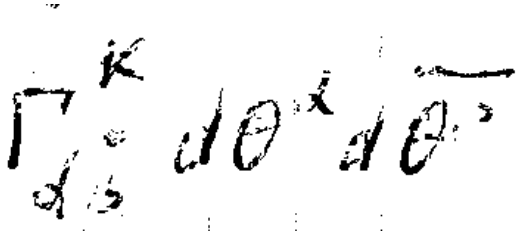
One must [crossed out: illegible] choose the basis correctly in the compl. tangent plane.

Uncertain reading. The entry after “One must” is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000994

For functions: natural choice of basis

$$\Gamma^k_{\alpha\dot\beta}\, d\theta^\alpha\, d\overline{\theta^\beta}.$$

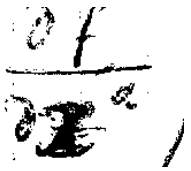


Original source details: GELFAND-P000994

Uncertain reading. The reach of the overbar and the dot at the superscript of the second differential are unclear: $d\overline{\theta^\beta}$ or $d\bar\theta^{\dot\beta}$.

and then one must write

$$A = \left(\det \frac{\partial f^k}{\partial z^a}\right)^{4/3} |\det \Gamma|^{-1/3}.$$



Original source details: GELFAND-P000994

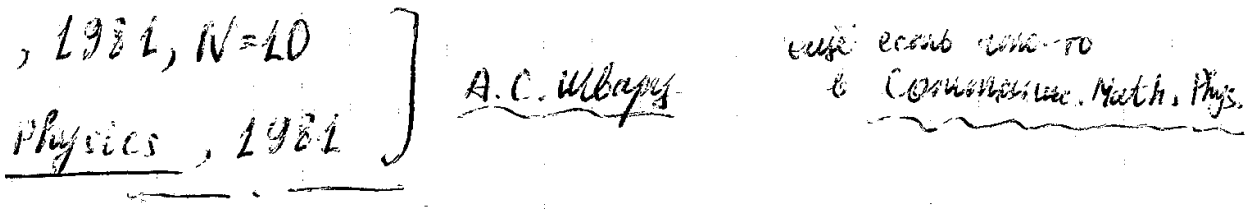
Uncertain reading. The denominator tentatively reads z^a, possibly x^a. The letter is overwritten.

see Yadernaya Fizika, 1981, $N = 10$.

prelim. large (?) work. Nuclear Physics, 1981.

[A right bracket groups the two references, with the label:] A.S. Schwarz.

[At right:] there is also something in Communic. Math. Phys.



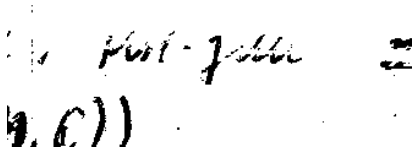
Original source details: GELFAND-P000994

Diagram and typed description. The name and a separate note on a journal stand to the right of the bracket.

Original source details: GELFAND-P000994

Uncertain reading. Before “Nuclear Physics”, tentatively “prelim. large work”.

G-struct. on a compl. manifold (abbrev. uncertain) $\Rightarrow CR$ structure on the surface.



Original source details: GELFAND-P000994

Uncertain reading. After “on a compl.” the manuscript uses a hyphenated abbreviation, tentatively “мн-зии”.

($\sim$ compl. str. $GL(n,\mathbb{C})$)

Here $SL(4,2\mid\mathbb{C})$ is the structure.

L is the autom. group of this struct.

An SCR structure arises on the surf., differing from CR by the det condition.

In $\mathbb{C}^{4,2}$ the coordinates and the field values are mixed up, and they cannot be considered separately.

— · —

GELFAND-P000995

23.11.81 Gelfand Sem.

Zelevinsky.

$\mathfrak{g}$ — compl. semis. Lie alg.

$$\mathfrak{g} \supset \underset{\text{Borel}}{\mathfrak{b}} \supset \underset{\text{Cartan}}{\mathfrak{h}} .$$

Example. $\mathfrak{sl}_n$ [below $\mathfrak{b}$: "upper triang.", below $\mathfrak{h}$: "diag."].

$$\underbrace{\overbrace{e_1,\dots,e_n}^{\mathfrak{n}_+},\overbrace{h_1,\dots,h_n}^{\mathfrak{h}}}_{\mathfrak{b}_+},\quad f_1,\dots,f_n.$$

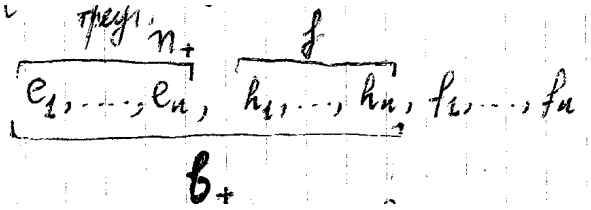
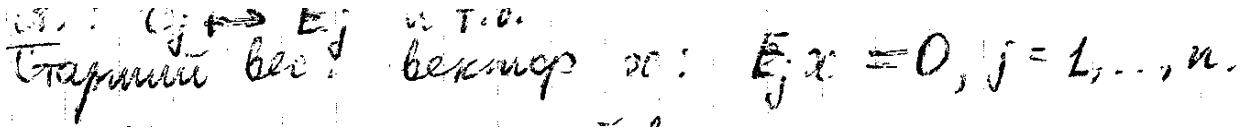


Diagram and typed description. The two upper brackets are marked $\mathfrak{n}_+$ and $\mathfrak{h}$ and the common lower bracket is marked $\mathfrak{b}_+$.

Original source details: GELFAND-P000995

Rep.: $e_j \mapsto E_j$ etc.

Highest weight! vector x: $E_j x = 0$, $j = 1,\dots,n$.



Original source details: GELFAND-P000995

Diagram and typed description. The mark after "weight" reads as an exclamation mark.

$\Rightarrow H_j x$ — also a highest weight.

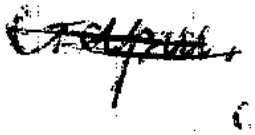
Always $\exists$ in a finite-dimensional rep.

Verma module

$\chi \in \mathfrak{h}^*$.

[Crossed out: "Highest" (abbrev., tentative)] Vector of highest weight χ:

$$\begin{cases} ev = 0, & \forall e \in \mathfrak{n}_+, \\ hv = \chi(h)v, & \forall h \in \mathfrak{h}. \end{cases}$$



Original source details: GELFAND-P000995

Uncertain reading. The crossed-out entry tentatively reads "Highest" in abbreviation.

Verma module: the largest module $M(\chi)$ gen. by highest weight χ.

(Universal module with these props.)

$M(\chi)$ is not irreduc. (generally speaking).

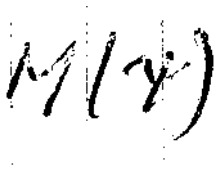
$L(\chi)$ — the unique irreduc. quotient module of $M(\chi)$.

Thm. $\exists$ a finite compos. series

$$M(\chi) = M_1 \supset M_2 \supset \ldots \supset M_n = 0, \qquad M_i/M_{i+1} = L(\chi_i).$$

$\Rightarrow$ every subquotient has this form.

Question: given ψ, what is the multiplicity of occurrence of $M(\psi)$ in the compos. series of a Verma mod.



Original source details: GELFAND-P000995

Diagram and typed description. The question has $M(\psi)$.

(This is equiv. to describing the Verma mod. [inserted above: "itself"].) [Below the parentheses: "— can be calculated".]

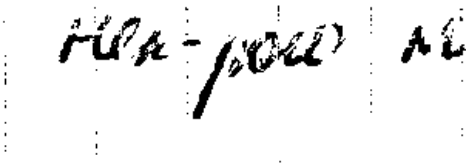
Original source details: GELFAND-P000995

Diagram and typed description. The word "itself" is above the line with an insertion mark.

Denote the multiplicities

$$m(\chi, \psi) = [M(\chi) : L(\psi)].$$

Multiplicities = value of a certain polynomial at pt. 1.



Uncertain reading. The word “certain” before “polynomial” is abbreviated.

Original source details: GELFAND-P000995

Condition:

$$\chi + \rho \in W(\psi + \rho), \quad \text{i.e.} \quad \chi + \rho = w(\psi + \rho).$$

[Label at W: “Weyl gr.”]

ρ is def. by the formula $\rho(h_i) = 1$.

Only then can the multiplicity $m(\chi, \psi)$ be $\neq 0$.

χ integral dominant if $\lambda(h_i) \in \{0, 1, 2, \ldots\}$ for all h_i.

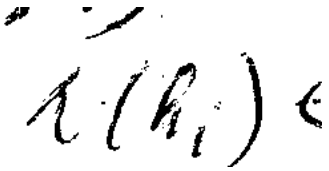


Diagram and typed description. After “if” the condition uses $\lambda(h_i)$, while the line begins with χ.

Original source details: GELFAND-P000995

GELFAND-P000996

[source number: 46]

Our case:

χ_0 — integral dominant, and

$$\chi + \rho = w_1(\chi_0 + \rho)$$
$$\psi + \rho = w_2(\chi_0 + \rho)$$

Then

$$m(\chi, \psi) = m(w_1, w_2)$$

(does not depend on the choice of dominant weight χ_0).

Kazhdan–Lusztig — hypothet. expression.

Now proved: Beilinson, Bernstein. Brylinski, Kashiwara.

They wrote down:

$$P_{w_1,w_2}(q)$$

— a polynomial (with nonneg. coefficients).

Conjecture

$$m(w_1, w_2) = P_{w_1,w_2}(1).$$

$$P_{w_1,w_2}(q) = \sum a_i q^i.$$

Meaning of a_i — ?

Explanation: $\exists$ a Jantzen filtration

$$M(\chi) \supset M_1 \supset M_2 \supset \cdots$$

$a_i =$ multiplicity of $L(\psi)$ in M_i/M_{i+1}.

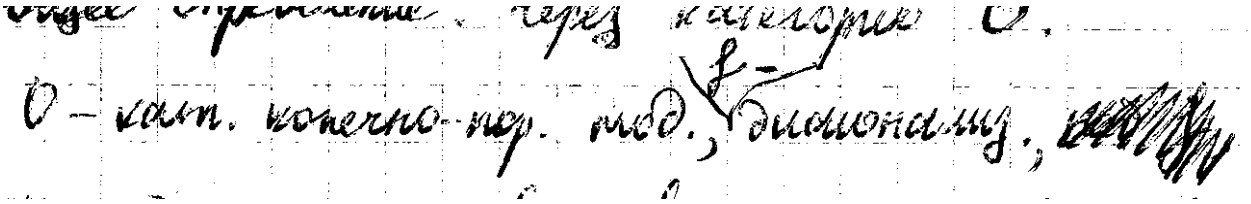
Beilinson.

Jantzen filtration.

Dual module ${}^*M(\chi)$.

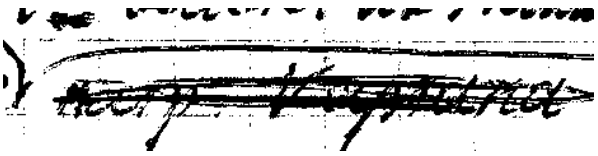
General definition: through category O.

O is the cat. of finitely gen. mod., [added above the line: $\mathfrak{h}$-]diagonaliz., [crossed out: illegible], where [crossed out: illegible] [replacement above the line: $\mathfrak{n}_-$ (?) acts locally nilpotently].



Original source details: GELFAND-P000996

Uncertain reading. Dense strokes cross out the entry after “diagonaliz.”.



Original source details: GELFAND-P000996

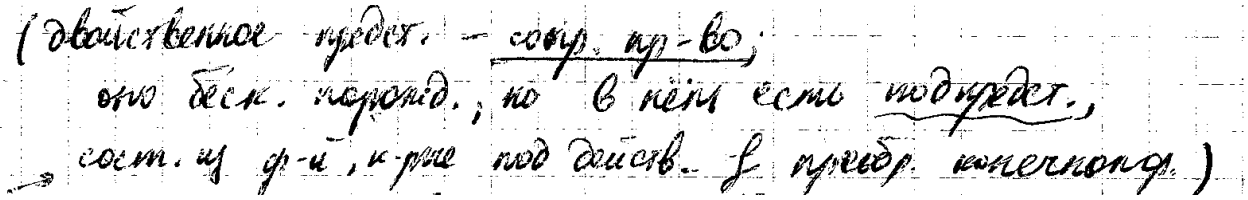
Uncertain reading. The entry after “where” is crossed out. Its exact reading is unresolved.

Original source details: GELFAND-P000996

Uncertain reading. The sign after $\mathfrak{n}$ is overwritten. The reading “$_-$” is tentative.

It has a natural involution $* : O \longrightarrow O$.

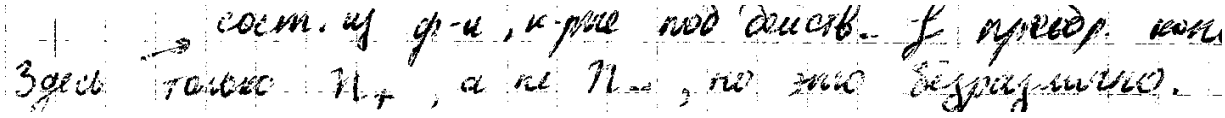
(The dual rep. is the dual space. It is infinitely gen., but contains a subrep. consisting of functions that transform finite-dimensionally under $\mathfrak{h}$.)



Original source details: GELFAND-P000996

Diagram and typed description. The end of the line reads “transform finite-dimensionally”, abbreviated in the manuscript.

[Added with an arrow to the preceding line:] Here only $\mathfrak{n}_+$, and not $\mathfrak{n}_-$, but this makes no difference.



Original source details: GELFAND-P000996

Diagram and typed description. The addition says $\mathfrak{n}_+$ followed by “and not $\mathfrak{n}_-$”.

S. Gelfand.

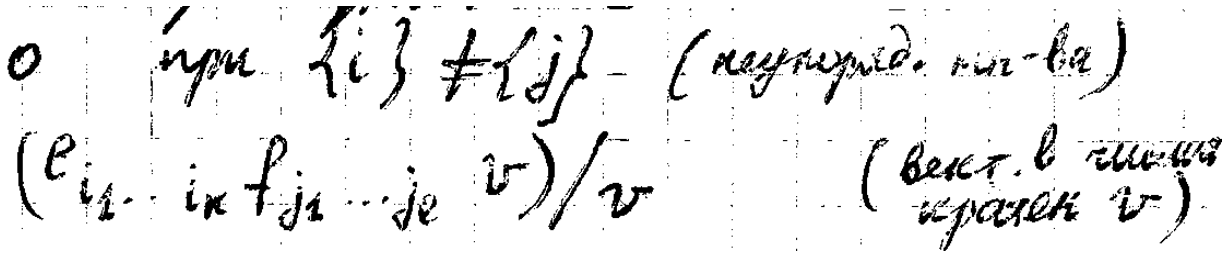
On each $M(\chi)$ introduce a bilinear form with numerical values.

$$M(\chi) \ni f_{i_1} \dots f_{i_k} v, \qquad v \text{ — highest-weight vector.}$$

Original source details: GELFAND-P000996

Diagram and typed description. The weight in the notation before $\ni$ is in parentheses.

$$(f_{i_1} \dots f_{i_k} v,\ f_{j_1} \dots f_{j_l} v) = \begin{cases} 0 \quad \text{when } \{i\} \neq \{j\} & \text{— (unordered sets),} \\ (e_{i_1 \dots i_k} f_{j_1 \dots j_l} v)/v & \text{(vector in numerator is a multiple of } v\text{).} \end{cases}$$



Original source details: GELFAND-P000996

Diagram and typed description. In the numerator each of the single letters e and f carries a sequence of indices.

We obtain a form invar. with respect to the Lie alg.

[Crossed out: illegible] We will identify all $M(\chi)$ (for different χ) as follows

$$f_{i_1} \dots f_{i_l} v_\chi \longrightarrow f_{i_1} \dots f_{i_l} v_{\chi'}.$$

Uncertain reading. A short entry before “We will” is crossed out.

Original source details: GELFAND-P000997

Consider $M(\chi)$ with [crossed out: illegible] fixed χ and then consider $M(\chi + t\rho)$.

Uncertain reading. A short entry is crossed out between “with” and “fixed”.

Original source details: GELFAND-P000997

We obtain many bilinear forms on one space:

$$(\ ,\ )_t.$$

Regard them all as forms on $M(\chi)$.

For small $t \neq 0$ the form is nondegenerate.

Def.

$$M_i = \{x \in M_\chi \mid (x, y)_t = O(t^i)\}.$$

[Above the condition, with an insertion mark:] $\forall y \in M(\chi)$:

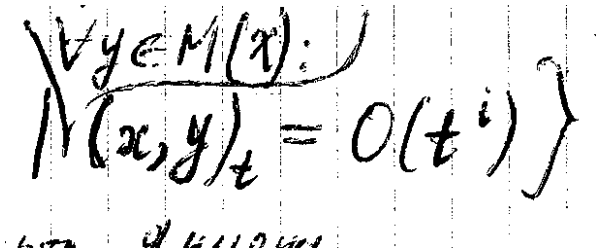


Diagram and typed description. The condition has the insertion $\forall y \in M(\chi)$ followed by a colon.

Original source details: GELFAND-P000997

This is the Jantzen filtr.

It is a filtr. by $\mathfrak{g}$-submodules.

What if we change ρ? Apparently the mult. will remain the same.

Polynomial

$$P_{w_1,w_2}(q) = \sum q^i \big([L_\psi : M_i/M_{i+1}]\big),$$

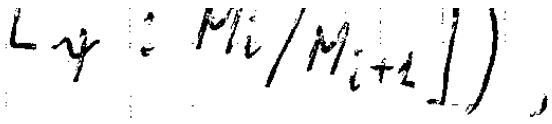


Diagram and typed description. The manuscript puts L_ψ before the colon and M_i/M_{i+1} after it.

Original source details: GELFAND-P000997

where

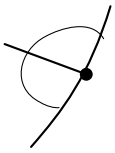
$$\psi = w_2(\chi_0 + \rho) - \rho,$$
$$M_i \subset M_\chi, \qquad \chi = w_1(\chi_0 + \rho) - \rho.$$

P_{w_1,w_2} — are determined recurr..

$$a_i = \dim H^i\big(\text{of one Schubert cell near another, according to MacPherson}\big).$$

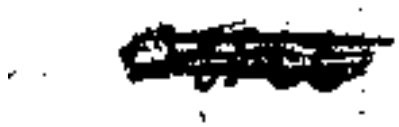
(local cohomol. of a normal section)

Diagram and typed description. A slanted line with a marked point, a curve and a short segment leading to the point. There are no labels.



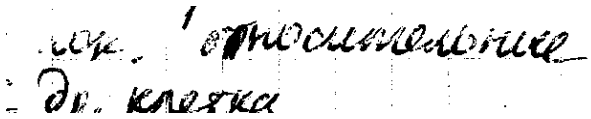
Original source details: GELFAND-P000997

or [crossed out: illegible] local rel. (?)
cell $\subset$ other cell.



Uncertain reading. The word after “or” is crossed out.

Original source details: GELFAND-P000997



Uncertain reading. Tentative reading “local relative”, with the first word abbreviated.

Original source details: GELFAND-P000997

Zelevinsky. Other problems, the same polynoms.

Original source details: GELFAND-P000997

Diagram and typed description. In the heading “the same” is followed by the abbreviation for “polynomials”.

Consider a finite-dimensional irreduc. rep.

$$L(\chi) = \bigoplus_\lambda L(\chi)_\lambda.$$

[Below $L(\chi)_\lambda$, joined by a stroke:] weight space with weight χ.

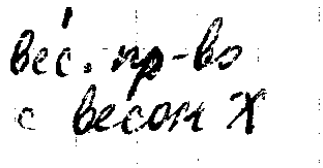
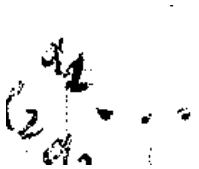


Diagram and typed description. The label ends with the letter χ.

Original source details: GELFAND-P000997

I. M. G. A good fn. for dim $L(\chi)_\lambda$ is known

$$\begin{vmatrix} x_1^{a_1} & x_2^{a_1} & \dots & x_n^{a_1} \\ x_1^{a_2} & x_2^{a_2} & \dots & x_n^{a_2} \\ x_1^{a_n} & x_2^{a_n} & \dots & x_n^{a_n} \end{vmatrix} : \begin{vmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \\ x_1^{n-1} & \dots & x_n^{n-1} \end{vmatrix} = \sum m(\chi,\lambda) x^\lambda.$$



Uncertain reading. The first exponent of x_2 reads as a_1 through the overlapping strokes.

Original source details: GELFAND-P000997

GELFAND-P000998

[source number: 47]

Here

$$\chi = (a_1, \dots, a_n), \quad \lambda = (\lambda_1, \dots, \lambda_n), \qquad x^\lambda = x_1^{\lambda_1} \dots x_n^{\lambda_n}.$$

$$m(\chi,\lambda) = \dim L(\chi)_\lambda.$$

Problem (for first-year students).

Calculate

$$\det \left\| \frac{1}{\lambda_i - \mu_j} \right\|_{i,j=1}^{n}, \qquad \lambda_i \neq \mu_j.$$

[A star to the right of the matrix.]

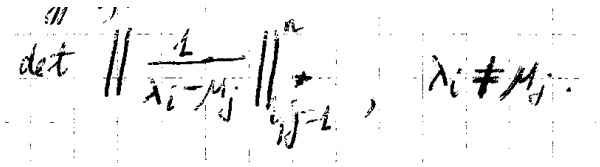


Diagram and typed description. The denominator is $\lambda_i - \mu_j$. A star is written to the right of the double bar.

Original source details: GELFAND-P000998

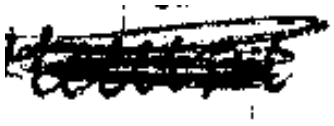
— · —

Zelevinsky.

[Crossed out before the formula: illegible.]

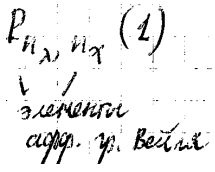
$$m(\lambda,\chi) = P_{n_\lambda, n_\chi}(1).$$

[Strokes from both indices lead to the label:] elements of the aff. Weyl gr.



Uncertain reading. The word before the formula is crossed out with dense strokes.

Original source details: GELFAND-P000998



Uncertain reading. The indices of P tentatively read n_λ, n_χ.

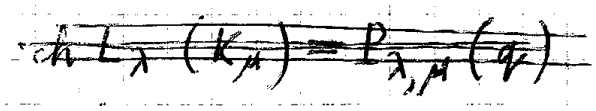
Original source details: GELFAND-P000998

Conjecture.

The same ∀ Kac–Moody algebras.

[The entire formula is crossed out.]

$$\operatorname{ch} L_\lambda(K_\mu) = P_{\lambda,\mu}(q).$$



Uncertain reading. The argument reads K_μ through the cancellation.

Original source details: GELFAND-P000998

Elashvili.

G — a lin. alg. group, i.e. $G \subset GL(n, \mathbb{C})$.

$L(G)$ — its Lie algebra.

$U(L)$ — the universal enveloping alg.

$U(L)$ is called primitive if the zero ideal is primitive (= the kernel of an irreduc. rep.), i.e. there is a faithful [added above the line: (irreduc.)] rep. of $U(L)$.

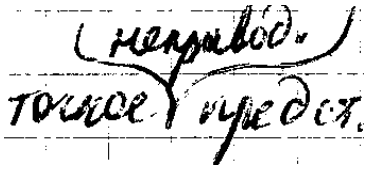


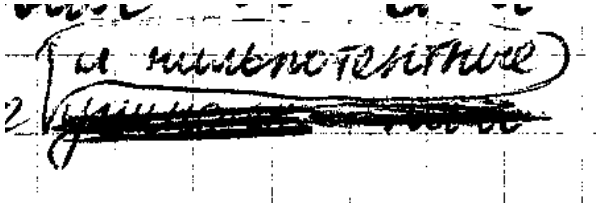
Diagram and typed description. "(irreduc.)" is inserted above "faithful rep." with a bracket.

Original source details: GELFAND-P000998

What conditions must G and $L(G)$ satisfy? (question of Jacobson)

[Below the name:] last year.

Semisimple ones all [crossed out: unipotent (?)] [added above the line: (and nilpotent ones)] are unsuitable.



Uncertain reading. The crossed-out word tentatively reads "unipotent". Above it, a bracket contains "and nilpotent ones".

Original source details: GELFAND-P000998

Example: 1) Lie alg. of the group of affine transf. ($L(G)$ consists of matr. with last column zero).

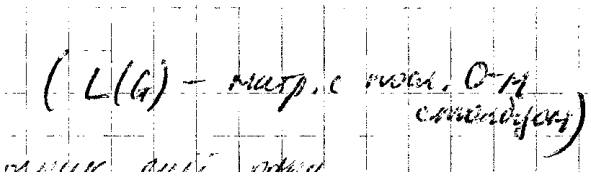


Diagram and typed description. The manuscript specifies the last column as zero.

Original source details: GELFAND-P000998

2) $L = gl(2, \mathbb{C}) + V_4$.

[Below V_4, joined by a stroke:] commut. ideal.

$gl(2, \mathbb{C})$ acts on V_4 by the unique 4-dim. irreduc. rep.

[At right after two vertical strokes:] Another interesting Γ-fn. arises here.

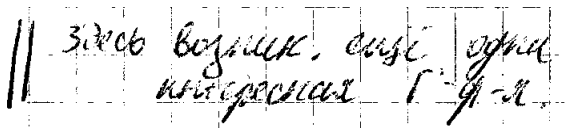
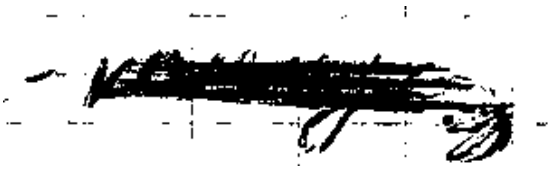


Diagram and typed description. The right-hand note reads “Γ-fn.”.

Original source details: GELFAND-P000998

Thm. There are no others if $L = R + V$, R is [crossed out: illegible] simple, V is commut., and the action of R on V is irreduc.



Uncertain reading. The word after R and before “simple” is crossed out with dense strokes.

Original source details: GELFAND-P000998

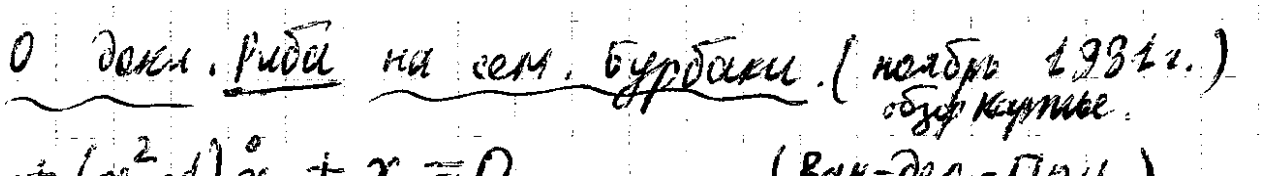
[Below the sum:] semidirect sum.

There is a method of construction beyond this.

GELFAND-P000999

30/XI-81 Gelfand Sem.

I. M. G. On Reeb’s talk at the Bourbaki sem. (November 1981) [Below the date:] Cartier’s survey.



Original source details: GELFAND-P000999

Diagram and typed description. The talk line names Reeb. Below the date is “Cartier’s survey”.

$$\varepsilon\ddot{x} + (x^2 - 1)\dot{x} + x = 0. \tag{1}$$

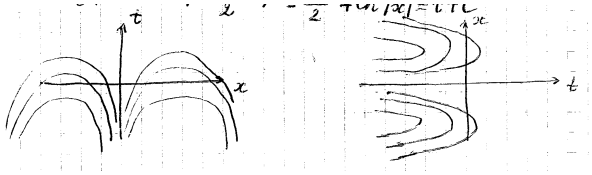
(Van der Pol)

What happens as $\varepsilon \to 0$?

$\varepsilon = 0$:

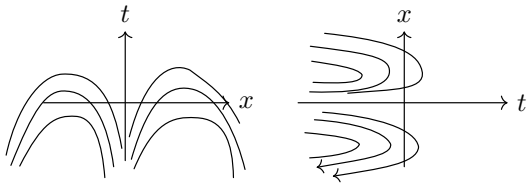
$$\frac{dx}{dt} = -\frac{x}{x^2-1}, \qquad \frac{dx(1-x^2)}{x} = dt, \qquad \frac{dx}{x} - x\,dx = dt.$$

$$C + t = \ln|x| - \frac{x^2}{2}, \qquad -\frac{x^2}{2} + \ln|x| = t + C.$$



Original source details: GELFAND-P000999

Diagram and typed description. The left plot has axes t and x. The right plot has axes x and t. Each plot has three branches on either side of the axis. Two lower branches at right have strokes resembling leftward arrows.

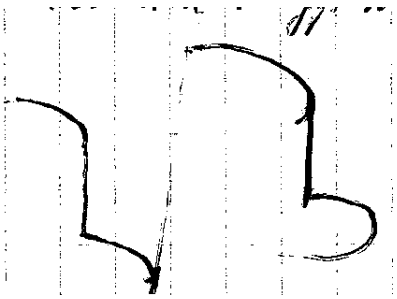


Another remarkable eq.:

$$\varepsilon\ddot{u} + \left(\frac{\dot{u}^3}{3} - \dot{u}\right) + u = 0. \tag{2}$$

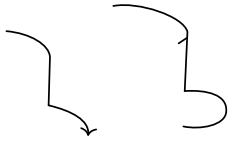
(Rayleigh)

It reduces to the Van der Pol eq.

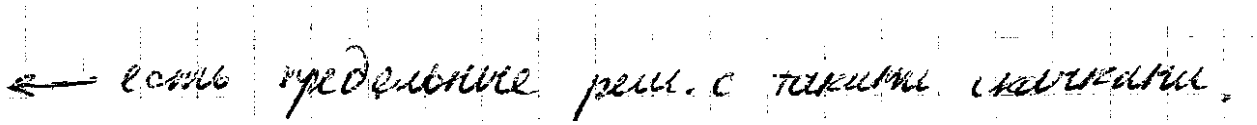


Original source details: GELFAND-P000999

Diagram and typed description. Two curves with vertical segments. At the lower left endpoint is a downward arrow. The short stroke near the upper bend at right is unclear.



[At right, with an arrow pointing left:] there are limiting solns. with such jumps.



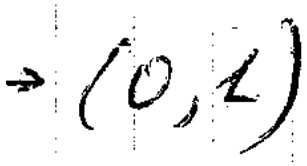
Original source details: GELFAND-P000999

Diagram and typed description. The arrow from the label points toward the two curves.

Eq. (1) is studied using nonstandard analysis!

Ω — a set,

$$F = 2^{\Omega}, \qquad p : F \to (0,1).$$



Original source details: GELFAND-P000999

Diagram and typed description. The interval is written with round brackets: $(0,1)$.

$$p(A \cup B) = p(A) + p(B) \quad \text{when } A \cap B = \varnothing.$$

X — a set,

$$x' : \Omega \to X, \qquad x'' : \Omega \to X.$$

A random element of X is a class of equivalent maps $\Omega \to X$.
[In small writing below the line:]

$$x' \sim x'' \Leftrightarrow P(x' = x'') = 1.$$

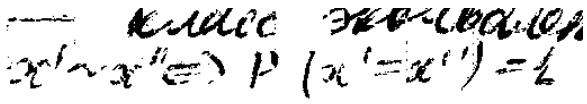


Diagram and typed description. The equivalence condition uses a capital P.

Original source details: GELFAND-P000999

[At right:] x is the equiv. class of x'. $\Rightarrow x'$ is a version of x.

GELFAND-P001000

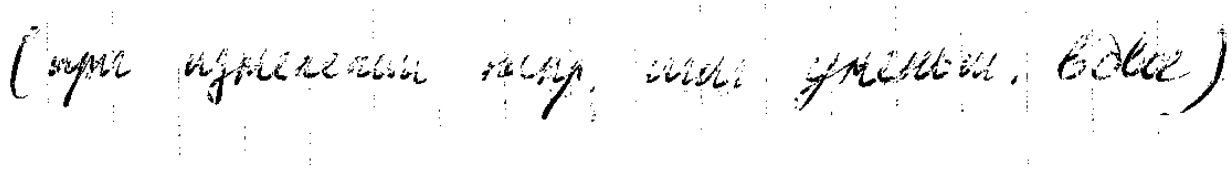
Kruskal's numbers: the simplest example of a field extending $\mathbb{R}$.

Diagram and typed description. The heading reads "Kruskal's".

Original source details: GELFAND-P001000

$$\begin{aligned} & \varnothing = 0. \\ \text{up} \quad & \uparrow = 1, \\ \text{down} \quad & \downarrow = -1, \\ & \overbrace{\uparrow\uparrow\uparrow \dots \uparrow}^{n} = n, \\ & \uparrow\downarrow = \frac{1}{2}, \\ & \uparrow\downarrow\uparrow = \frac{3}{4}. \end{aligned}$$

(when the direction changes the step is halved)



Original source details: GELFAND-P001000

Diagram and typed description. The manuscript says the step is halved when the direction changes.

$\hat{\uparrow}$ — infinite repetition of $\uparrow$.

$$\hat{\uparrow}\downarrow = \infty - 1; \qquad \hat{\uparrow} = \infty.$$

[A sign before $\infty - 1$ is crossed out.]

Uncertain reading. Strokes to the left of ∞ obscure the crossed-out sign.

Original source details: GELFAND-P001000

(there is no predecessor)

$$\hat{\uparrow}\hat{\downarrow} = \frac{1}{2}\infty.$$

$$\hat{\uparrow} : \hat{\uparrow}\hat{\downarrow} = 2 \qquad \text{— exact thm.}$$

$$\hat{\hat{\uparrow}} = \infty^2,$$

$$\uparrow\hat{\downarrow} = \varepsilon.$$

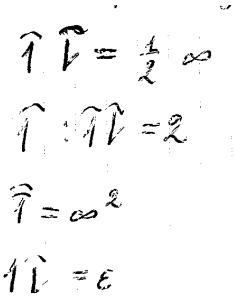
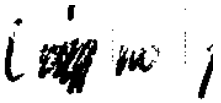


Diagram and typed description. In the last pair of formulas the first arrow has two arcs and the downward arrow has one.

Original source details: GELFAND-P001000

Krichever.

([Crossed out: illegible] from (?) work by Brozovskii, Dzyaloshinskii and Krichever)

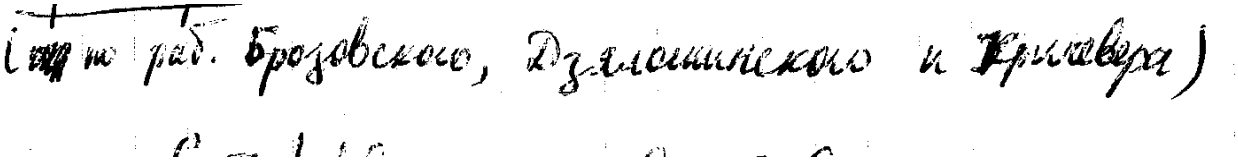


Uncertain reading. A short entry is crossed out at the start of the parenthesis.

Original source details: GELFAND-P001000

Uncertain reading. The preposition between the cancellation and “work” tentatively reads «по».

Original source details: GELFAND-P001000



Original source details: GELFAND-P001000

Diagram and typed description. The surname is written «Брозовского», with «о» in the first syllable.

$$G = \bigcup_n G_n, \qquad G_n \subset G_{n+1}.$$

Uncertain reading. A short entry follows G_{n+1}. Its reading is unresolved.

Original source details: GELFAND-P001000

$$u(z), \qquad \Gamma \ni z.$$

[Below $u(z)$, joined by a stroke:] matrix fn.

[Below Γ, joined by a stroke:] torus ($\mathbb{C}$/lattice).

$$u^{ij}(z) = \frac{\sigma(z - a_1^{ij}) \dots \sigma(z - a_n^{ij})}{\sigma^n(z)}\, e^{\zeta(z)(x_i - x_j)},$$
$$a_1^{ij} + \dots + a_n^{ij} = x_i - x_j.$$

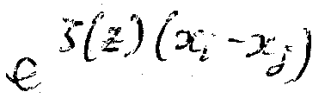


Diagram and typed description. The exponent reads $\zeta(z)(x_i - x_j)$.

Original source details: GELFAND-P001000

Chapter 21

GELFAND-P001001

– · –

8/II-82 Gelfand Sem..
Müller Werner.
Lattice gravity. (work with Cheeger and Schräder)

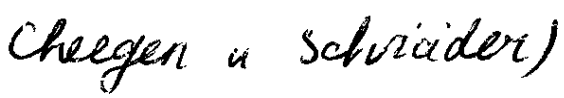


Diagram and typed description. The manuscript spells the name “Schräder”.

Original source details: GELFAND-P001001

1) Consider a one-dim. mech. system with Hamiltonian:

$$H(p,q) = +\frac{p^2}{2m} + V(q) \tag{1}$$

$q \to Q, \quad p \to P$ operators on $L^2(\mathbb{R})$

$$Qf(x) = xf(x), \qquad Pf(x) = i\bar{h}\frac{d}{dx}f(x)$$

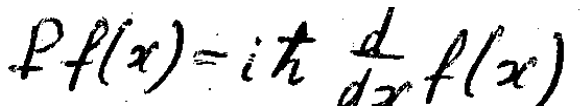


Diagram and typed description. There is no minus sign before $i\bar{h}$ in the expression for $Pf(x)$.

Original source details: GELFAND-P001001

$$H(p,q) \to \widehat{H}$$

Consider Schrödinger’s eq.

$$i\frac{\partial}{\partial t} = \bar{h}^{-1}\widehat{H}f \tag{2}$$

giving the evolution:

$$f(x,t) = e^{-i\frac{t}{\bar{h}}\widehat{H}}f(x,0) \tag{3}$$

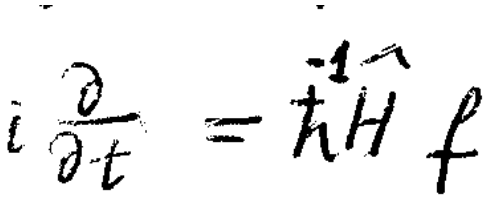


Diagram and typed description. On the left of (2) the manuscript has only $i\,\partial/\partial t$. The letter f is on the right.

Original source details: GELFAND-P001001

Prob. of transition from x to x' in time [added above the line: from t_0 to] t, according to Feynman

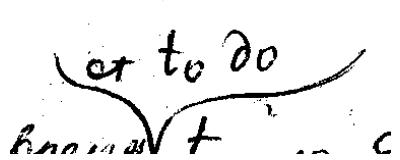


Diagram and typed description. The words “from t_0 to” are inserted before t.

Original source details: GELFAND-P001001

$$\langle x' \mid e^{-i\frac{t}{\hbar}\widehat{H}} \mid x\rangle = \int_{\substack{\text{over [inserted: all] paths } x \\ x(0)=x,\ x(t)=x'}} e^{-iS[x]} \prod_{\tau} \frac{dx(\tau)}{2\pi} \tag{4}$$

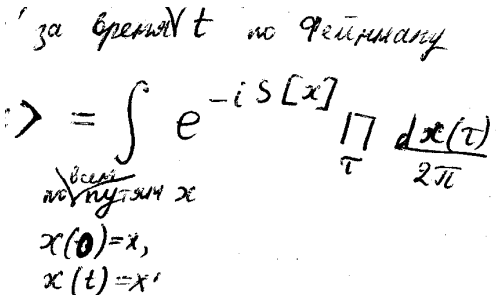
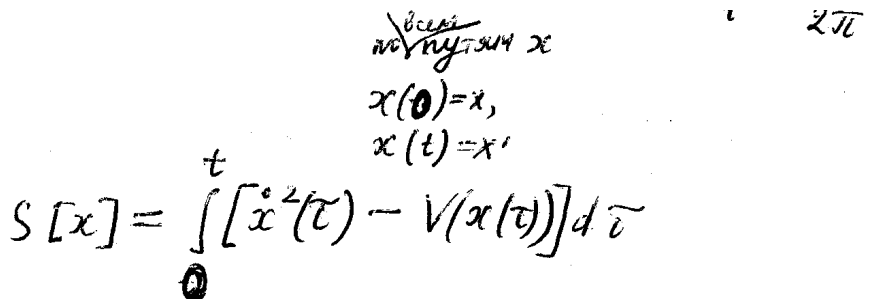


Diagram and typed description. The word “all” is inserted above “over paths”.

Original source details: GELFAND-P001001

$$S[x] = \int_0^t [\dot{x}^2(\tau) - V(x(\tau))]d\tau \tag{5}$$



Original source details: GELFAND-P001001

Diagram and typed description. The exponent in (4) is $-iS[x]$. In (5) there is no written coefficient before $\dot{x}^2(\tau)$.

(see last chapter of the book by Konopleva and Popov)

Uncertain reading. The Russian surname reads «Коноплевой», with no distinct dots above the «е».

Original source details: GELFAND-P001001

GELFAND-P001002

[source number: 4]

or: insert $t - t_0$ in place of t

in place of 0 — ditto — t_0

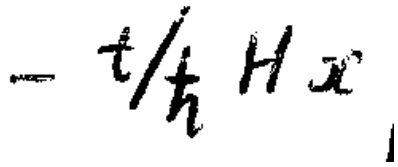
M. Kac gave some meaning to int. (4) with the substitution $i \to 1$

$$\overline{h}\frac{\partial f}{\partial t} = \widehat{H} f, \tag{6}$$

then

$$\langle x' \mid e^{-\frac{t}{\overline{h}} Hx} \mid x \rangle = \int_{\substack{\text{over all paths} \\ x:\, x(0)=x \\ x(t)=x'}} e^{-\int_0^t x(\tau)d\tau}\, dW(x) \tag{7}$$

[Arrow to $dW(x)$:] Wiener measure.



Uncertain reading. The letter x follows H. Its status as a subscript is unclear.

Original source details: GELFAND-P001002

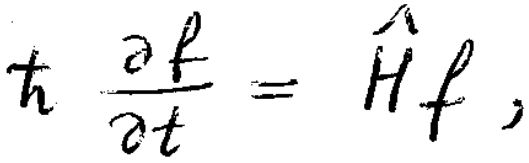


Diagram and typed description. In (6) there is no minus sign before $\widehat{H} f$ on the right.

Original source details: GELFAND-P001002

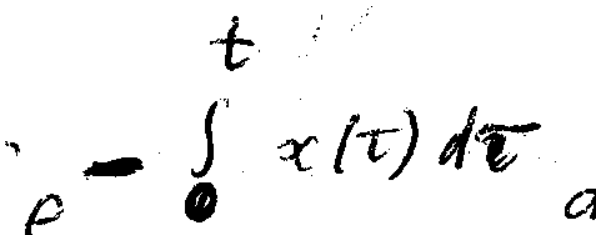


Diagram and typed description. The integrand in the exponent reads $x(\tau)$.

Original source details: GELFAND-P001002

2) Gravity.

$$i \mapsto 1 \qquad \rightsquigarrow$$

[Crossed out: “for” (?)] Lorentz. metric $\longmapsto$ Riemannian metric.

Uncertain reading. The crossed-out word before “Lorentz.” tentatively reads “for”.

Original source details: GELFAND-P001002

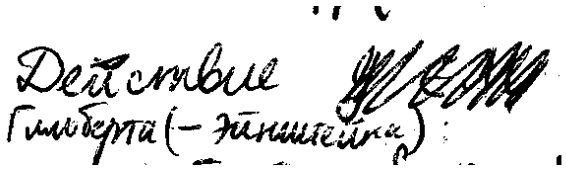
M — a compact C^∞ manifold

$$\mathcal{M} = \{\text{space of all Riem. metrics on } M\}$$

$$G = \mathrm{Diff}(M) \quad \text{— gauge group.}$$

Action [crossed out: illegible] [added below: Hilbert (– Einstein)]:

$$S[g] = \lambda \int_M R\, d\,\mathrm{vol}, \qquad R \text{ — scalar curvature} \tag{8}$$



Uncertain reading. The earlier entry after "Action" is crossed out. Below it is "Hilbert (– Einstein)".

Original source details: GELFAND-P001002

(nontrivial: Einstein thought there was no action)

— R contains 2nd derivs. ⇒ a priori a 4th-order eq.

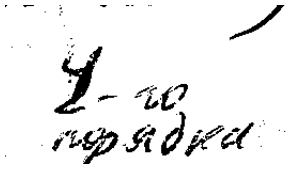
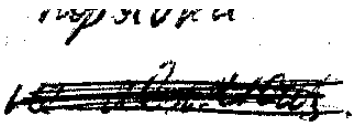


Diagram and typed description. The end of the line reads "4th order".

Original source details: GELFAND-P001002

But locally there is a noncovar. addition of divergence form, [crossed out: illegible] leaving S unchanged.



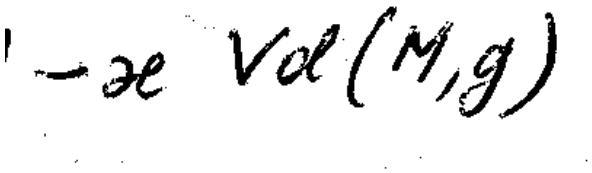
Uncertain reading. The reading under the cancellation after "form" is unresolved.

Original source details: GELFAND-P001002

Problem (I.M.G.) Find all comb. of curvatures yielding a lower-order eq.

"Answer": Einstein and Pontryagin classes. Action:

$$\int_{\mathcal{M}/G} e^{-S[g] - æ\,\mathrm{Vol}(M,g)} d\mu(g) \tag{9}$$



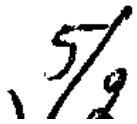
Uncertain reading. The coefficient before $\mathrm{Vol}(M,g)$ resembles "æ", possibly a handwritten $\varkappa$.

Original source details: GELFAND-P001002

Formally:

$$d\mu(g) = \prod_{x \in M} \det g(x)^{5/2} \prod_{\mu \le \nu} dg_{\mu\nu}(x) \tag{10}$$

Diagram and typed description. The exponent reads $5/2$, with no minus sign.



Original source details: GELFAND-P001002

(Faddeev – Popov)

Lattice approximation.

[A number is crossed out at left.]

$$P^n \subset \mathbb{R}^N \qquad P^n \underset{\varphi}{\longrightarrow} M \quad \text{triangulation}$$

[Stroke from P^n to the label:] piecewise linear subset.

Uncertain reading. Tentative reading of the crossed-out entry “(11)”.

Original source details: GELFAND-P001002

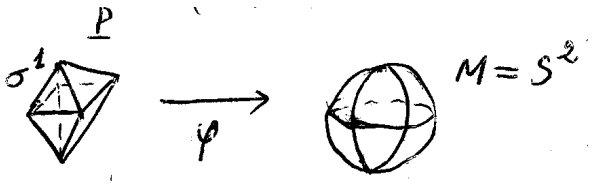
GELFAND-P001003

The metric on M induces a piecewise-lin. metr. on P.

$$(P, \ell), \qquad \ell : P^{(1)} \to \mathbb{R}_+$$

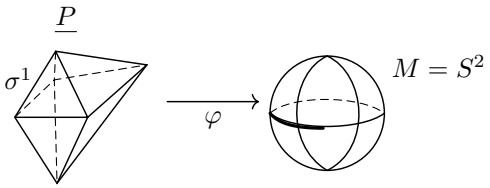
(simpl. $\rightsquigarrow$ its length)

(11)



Original source details: GELFAND-P001003

Diagram and typed description. Labels P, σ^1, φ and $M = S^2$. The dashed strokes follow the visible portions.



Discr. version of $\int_M R\,d\,\mathrm{vol}$.

$$P^n \longrightarrow M \qquad \text{triangulation} \tag{12}$$

$$\sigma^{n-2} \subset P^n$$

$\delta(\sigma^{n-2})$ — angle defect of σ^{n-2}

$$\sigma^n \supset \sigma^{n-2}$$

$[\sigma^{n-2}, \sigma^n]$ — dihedral angle

$$\delta(\sigma^{n-2}) = 2\pi - \sum_{\sigma^n \supset \sigma^{n-2}} [\sigma^{n-2}, \sigma^n] \tag{13}$$

$$R(P) = \sum_{\sigma^{n-2} \subset P} \delta(\sigma^{n-2})\,\mathrm{vol}(\sigma^{n-2}) \tag{14}$$

(Regge)

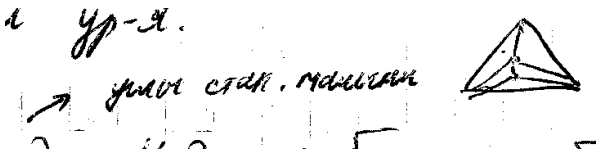
P_n — a seq. of triangulations, $P_n \to M$

$$\mathrm{mesh}(P_n) \underset{n\to\infty}{\longrightarrow} 0 \tag{15}$$

$$\text{Th.} \quad \text{Then} \quad R(P_n) \underset{n\to\infty}{\longrightarrow} \int_M R\,d\,\mathrm{vol} \tag{16}$$

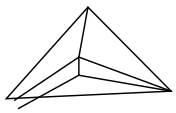
It is still unclear whether the eqs. converge.

Barycentric subdivision is unnecessary. [Upward arrow to a note:] angles become small.

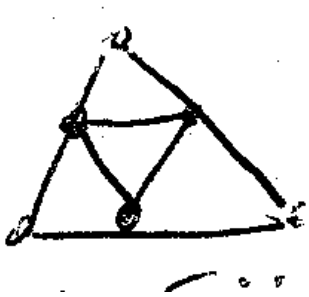


Original source details: GELFAND-P001003

Diagram and typed description. The interior strokes follow the visible joins. Some of the lower lines overlap.

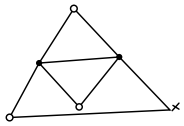


The angles must be bounded away from 0:



Original source details: GELFAND-P001003

Diagram and typed description. The marks at the vertices are reproduced as strokes. It is unclear whether they are letters.



a smooth triangulation is taken. (exists in Müller's work on Ray–Singer torsion)

– · –

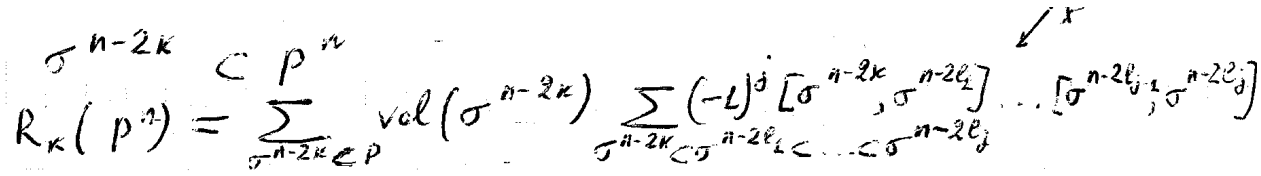
$$\int \operatorname{tr}(\Omega^k), \qquad \Omega \text{ — curvature tensor} \tag{17}$$

[Below Ω, with a stroke to it:] 2-form.

$$\sigma^{n-2k} \subset P^n$$

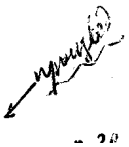
$$R_k(P^n) = \sum_{\sigma^{n-2k} \in P} \operatorname{vol}(\sigma^{n-2k}) \sum_{\sigma^{n-2k} \subset \sigma^{n-2\ell_1} \subset \cdots \subset \sigma^{n-2\ell_j}} (-1)^j \\ [\sigma^{n-2k}, \sigma^{n-2\ell_1}] \dots [\sigma^{n-2\ell_{j-1}}, \sigma^{n-2\ell_j}] \tag{18}$$

[Slanted above (18), with an arrow down and left:] norm. (?)



Original source details: GELFAND-P001003

Diagram and typed description. The last pair of indices in the product reads ℓ_{j-1}, ℓ_j.



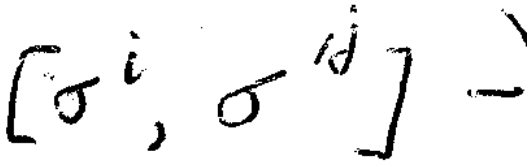
Uncertain reading. The note tentatively reads "norm.", with indistinct initial strokes.

Original source details: GELFAND-P001003

GELFAND-P001004

[source number: 5]

$[\sigma^i, \sigma^{ij}]$ — [inserted above: normalized] dihedral angle (solid) [Arrow to "solid":] normalized to 1.

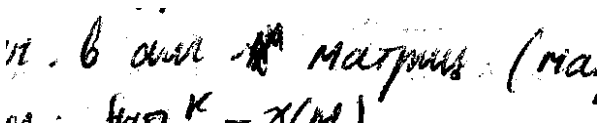


Uncertain reading. The second superscript reads ij, with a faint initial stroke.

Original source details: GELFAND-P001004

$$\underline{\text{Th.}} \quad R_k(P_n) \longrightarrow \int_M \operatorname{tr}(\Omega^k) \tag{19}$$

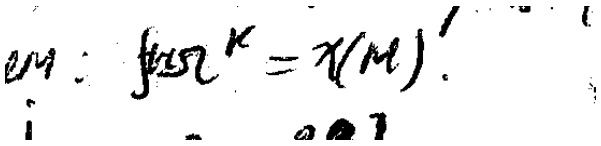
Ω — 2-form with values in the alg. [crossed-out sign] of matrices (a matrix of 2-forms).



Uncertain reading. Crossed-out sign, possibly A.

Original source details: GELFAND-P001004

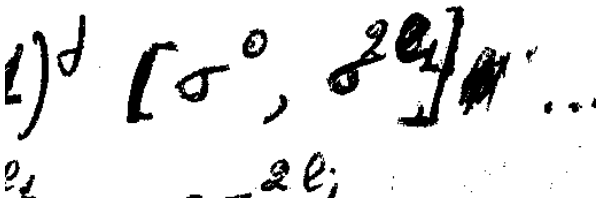
$$\underline{k = n/2} \quad \Rightarrow \quad \text{we obtain:} \quad \int \operatorname{tr} \Omega^k = \chi(M).$$



Uncertain reading. The manuscript reads $\int \operatorname{tr} \Omega^k = \chi(M)$.

Original source details: GELFAND-P001004

$$\chi(M) = \sum_{\substack{\sigma^0 \subset \sigma^{2\ell_1} \subset \cdots \subset \sigma^{2\ell_j} \\ \ldots [\sigma^{2\ell_{j-1}}, \sigma^{2\ell_j}]}} (-1)^j [\sigma^0, \sigma^{2\ell_1}] \tag{20}$$

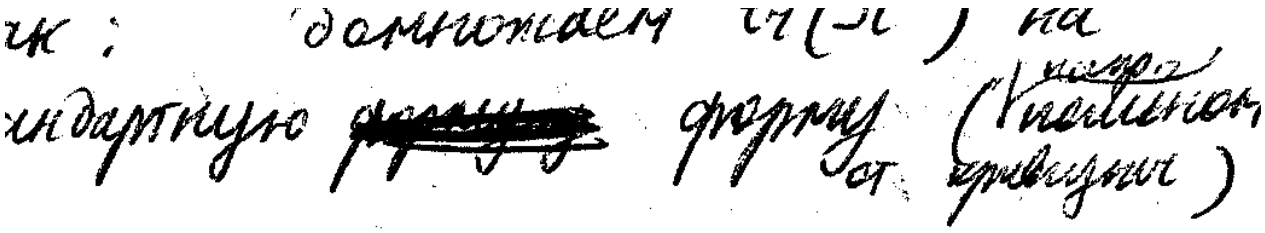


Uncertain reading. Illegible crossed-out sign after the first closing bracket.

Original source details: GELFAND-P001004

There is something similar in a work by Brin, presented at the seminar ca. 1946. It is unknown what it gives in the limit.

$\int_M \operatorname{tr}(\Omega^k)$ is def. as follows: multiply $\operatorname{tr}(\Omega^k)$ by the standard [crossed out: form?] form ([inserted above: inv.] polynomial in the curvature).



Original source details: GELFAND-P001004

Uncertain reading. The crossed-out word tentatively reads “form.”

$$\mathcal{M}(P^n) = \{\text{space of all piecewise-lin. metrics on } P^n\} \tag{21}$$

$\sigma^n \subset \mathbb{R}^N \longrightarrow n$-dim. simplex

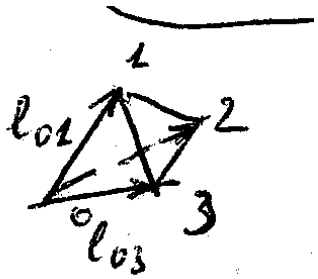
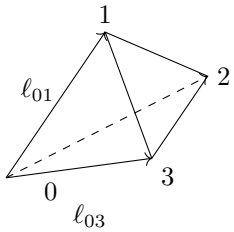


Diagram and typed description. Vertices $0, 1, 2, 3$, with edge labels ℓ_{01} and ℓ_{03}.



Original source details: GELFAND-P001004

$$\ell \rightsquigarrow g = (g_{ij}), \qquad g_{ij} = \frac{1}{2}(\ell_{0i}^2 + \ell_{0j}^2 - \ell_{ij}^2)$$

symm. and positive definite

$$\mathcal{P} = \{\text{symm. pos. def. matrices } n \times n\}$$

" $GL(n)/O(n)$

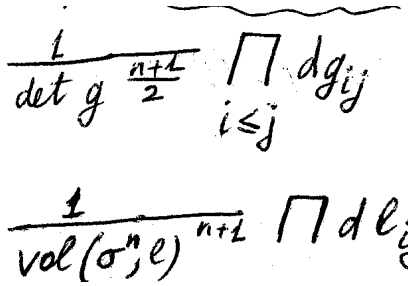
$$\mathcal{M}(\sigma^n) \equiv GL(n)/O(n)$$

Inv. measure on $\mathcal{M}(\sigma^n)$:

$$\frac{1}{\det g^{\frac{n+1}{2}}} \prod_{i \le j} dg_{ij} \tag{22}$$

$$\frac{1}{\mathrm{vol}(\sigma^n, \ell)^{n+1}} \prod d\ell_{ij} \tag{23}$$

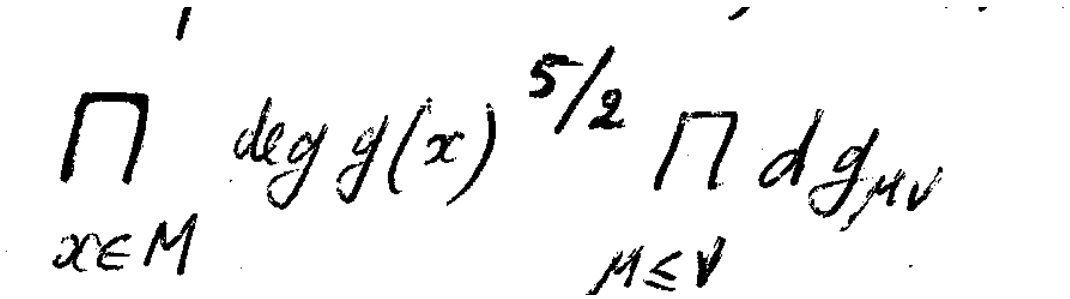
— measure on $\mathcal{M}(\sigma^n)$ (another way of writing it).



Uncertain reading. In (23) the manuscript gives $\prod d\ell_{ij}$.

Original source details: GELFAND-P001004

$$\prod_{x \in M} \deg g(x)^{5/2} \prod_{\mu \le \nu} dg_{\mu\nu} \tag{24}$$



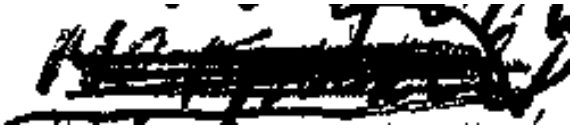
Original source details: GELFAND-P001004

Uncertain reading. The final stroke of "deg" descends below the line, unlike "det" in (22). The exponent in (24) is written as $5/2$.

$$\mathcal{M}(P^n) \hookrightarrow \prod_{\sigma^n \subset P^n} \mathcal{M}(\sigma^n) \tag{25}$$

(sub-set)

An induced [crossed out on the line: metric?] [circled and crossed out above: curvature?] [circled below: metric] arises on $\mathcal{M}(P^n)$ [crossed out on the right: not satisfy...].

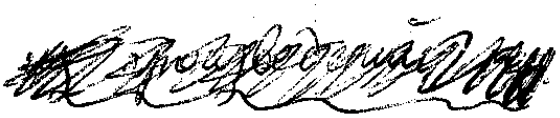


Original source details: GELFAND-P001005

Uncertain reading. On the line, tentatively "metric," with most of the word crossed out.

Original source details: GELFAND-P001005

Uncertain reading. The circled and crossed-out word above the line may read "curvature" or "metric."



Original source details: GELFAND-P001005

Uncertain reading. Under the cancellation, "not satisfy..." is legible. The ending and following strokes remain unclear.

($\rightsquigarrow$ gluing cond.)

A measure is introduced on the space of all metrizations. $\Rightarrow$ volume induced by the Riem. metric.

$$P_n \underset{n\to\infty}{\longrightarrow} M$$

(On $\mathcal{M}(\sigma^n)$ the inv. metric is the one on a symmetric space.)

Idea: $M^n \subset E^N$ isometric.

Chern's kinematic formula:

$$k = N - n + 2,$$

$\mathcal{G}_{N,k}$ = affine Grassmannian of $E^k \subset E^N$

dE^k — measure on $\mathcal{G}_{N,k}$

$$\int_{E^k \in \mathcal{G}_{N,k}} \chi(M^n \cap E^k)\, dE^k = c_{N,k} \int_M R\, d\,\mathrm{vol} \tag{26}$$

$$\int_{\mathcal{G}_{N,k}} \chi(P^n \cap E^k)\, dE^k = c_{N,k} R(P^n) \tag{27}$$

[Under P^n, with a vertical stroke:] simplicial sub-set

[Below the right-hand side:] (combinat. scalar curvature)

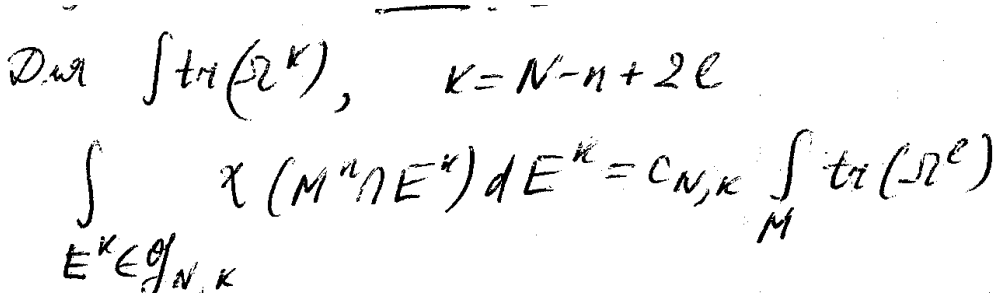
$$P_n \longrightarrow M$$

the integrands in (26) and (27) will agree a.e.

— . —

For $\int \operatorname{tr}(\Omega^k)$, $k = N - n + 2\ell$

$$\int_{E^k \in \mathcal{G}_{N,k}} \chi(M^n \cap E^k)\, dE^k = c_{N,k} \int_M \operatorname{tr}(\Omega^\ell) \tag{28}$$



Original source details: GELFAND-P001005

Uncertain reading. The line before (28) reads Ω^k, and the right-hand side of (28) reads Ω^ℓ.

$$\underset{\text{left-hand side of (27)}}{\text{— " —}} = c_{N,k} R_\ell(P^n). \tag{29}$$

GELFAND-P001006

[source number: 6]

15/II-82 Gelfand Sem.

Ginzburg.

M — sympl. manifold

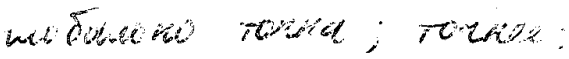
(ω — corresp. 2-form)

A — group acting on M in a Hamiltonian way.

1) A preserves ω

$\xi \in \mathfrak{A}$ — corresp. Lie algebra

$\xi \longrightarrow \omega(\xi, \cdot)$ globally (?) exact; exact:



Original source details: GELFAND-P001006

Uncertain reading. The word before "exact" tentatively reads "globally," with a tall medial loop. "Locally" is also possible.

2) $\mathfrak{A} \ni a \longmapsto H_a$ (Hamiltonian)
homomorphism of Lie algebras.
moment map: $\mu\colon M \longrightarrow \mathfrak{A}^*$

$$m \longmapsto \mu(m), \qquad \mu(m)(\underset{\uparrow}{a}) = H_a(m)$$
$$\mathfrak{A}$$

Original source details: GELFAND-P001006

Diagram and typed description. The arrow from $\mathfrak{A}$ below points to the argument a.

Commutes with the action of A and is compat. with the Poisson bracket.

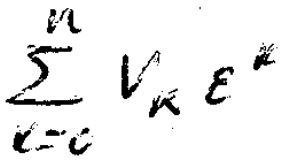
[Two short slanted strokes.]

GELFAND-P001007

22/II-82 Gelfand Sem.
Arnold.

$$V(\varepsilon) = \sum_{i=0}^{n} V_k \varepsilon^k, \qquad \sigma_i(k) = \sum_{C_n^i} k_1 \dots k_i$$

[Under $V(\varepsilon)$:] volume (ε-neighborhood of a conv. body). (Minkowski's theorem — this is a polynomial.)



Original source details: GELFAND-P001007

Uncertain reading. The summation starts at $i = 0$, while the summand reads $V_k \varepsilon^k$.

$$k_s \text{ — eigenvalues of } \frac{2 \text{ quadr. f.}}{1 \text{ quadr. f.}}$$

$$V_i = C_i \int_{\partial D} \sigma_i(k)\, dS$$

[Short vertical stroke below C_i.]

$$V_k = \widetilde{C}_k \underbrace{\int_{\xi \in G_{n,k}} \text{mes}(\pi_\xi D)\, d\xi}_{\text{“}}, \qquad \pi_\xi : \mathbb{R}^n \longrightarrow \xi$$

$$\int_{H_{n,k}} \chi(h \cap D)\, dh$$

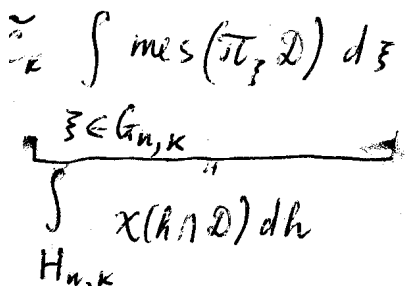


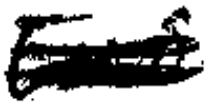
Diagram and typed description. A bracket with turned-up ends and two short slanting strokes appear below the upper integral.

Original source details: GELFAND-P001007

$H_{n,k}$ — all planes, not necessarily passing through 0

χ — Euler char..

This is the method of [crossed out: Buff...] Buffon (later generalized by Santaló) [Below “method of Buffon”:] for computing the number π.



Uncertain reading. The first strokes, “Buff...”, are visible under the cancellation.

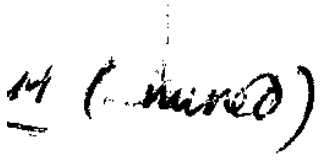
Original source details: GELFAND-P001007

Computation of the constants C_k, $\widetilde{C}_k$ — using balls.

This carries over to the lattice case.

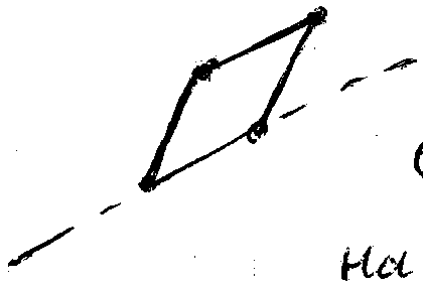
In place of volume, the number of integer points.

integral parallelogram (...piped?)



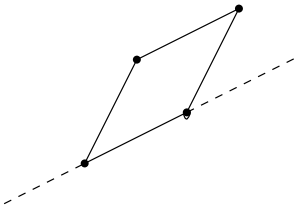
Uncertain reading. The letters after the dots tentatively read “piped.”

Original source details: GELFAND-P001007



Original source details: GELFAND-P001007

Diagram and typed description. The lower edge continues as a dashed line. A loop at the lower right vertex may be a separate 0.



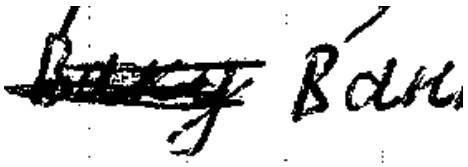
The volume of a parallelepiped is the number showing how many elem. par.-pipeds there are.

A lattice of intersections arises on the line.

Take the quotient space and this gives an analogue of height.

(there is an analogue of a lattice there too)

Mac Mullen a talk in [crossed out: Vancou...] Vancouver — there are refs. there.



Original source details: GELFAND-P001007

Uncertain reading. The first spelling is crossed out. The beginning reads “Vancou...”.

I. M. Gelfand.

On Mc Donald’s book “Symmetric Functions and Hall Polynomials”

Problem. Consider a partition (partition)

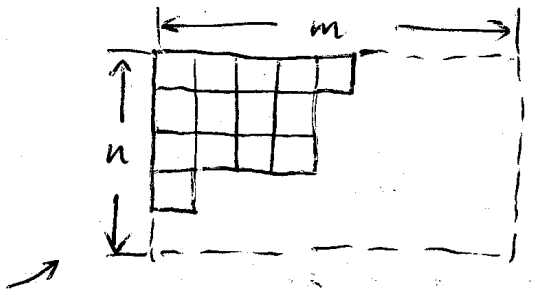
$$(L) \qquad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_k \qquad \text{(integers)}$$

Adding zeros is allowed.

Ways of writing it: (L) or

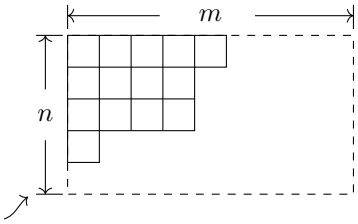
$$\begin{array}{ll} 5 & \bullet\ \bullet\ \bullet\ \bullet\ \bullet \\ 4 & \bullet\ \bullet\ \bullet\ \bullet \\ 4 & \bullet\ \bullet\ \bullet\ \bullet \\ 1 & \bullet \end{array}$$

Squares are better than dots (Young diagrams).



Original source details: GELFAND-P001007

Diagram and typed description. Rows of $5, 4, 4, 1$ cells in a dashed frame marked m and n.



Place it in a rectangle of dimensions $m \times n$ and consider the numbers

GELFAND-P001008

[source number: 11]

$$(*) \quad \begin{cases} \lambda_i + n - i, & 1 \le i \le n, \\ n - 1 + j - \lambda'_j, & 1 \le j \le m \end{cases}$$

$\lambda' = 4, 3, 3, 3, 1$ — dual diagram

Problem:

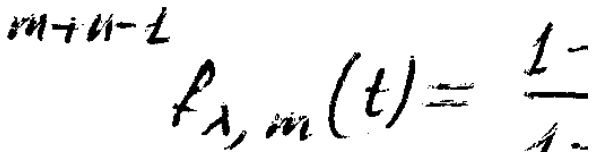
The list of $m + n$ numbers $(*)$ — is a permutation of the numbers $0, 1, 2, \ldots, m + n - 1$.

Corollary.

$$f_{\lambda,n}(t) \stackrel{\text{def}}{=} \sum_{i=1}^{n} t^{\lambda_i + n - i}$$

Then

$$f_{\lambda,n}(t) + t^{m+n-1} f_{\lambda,m}(t) = \frac{1 - t^{m+n}}{1 - t}$$



Uncertain reading. The second term reads $f_{\lambda,m}(t)$, with no visible prime on λ and no inverse argument.

Original source details: GELFAND-P001008

I. M. Gelfand from the work of Guillemin

Uncertain reading. The manuscript surname tentatively reads “Гийемина” (Guillemin).

Original source details: GELFAND-P001008

New cond. for Weyl’s asympt. formula

Sympl. cone: $\rho_a^* \omega = a\omega, \quad a > 0.$

Example 1. $\mathbb{R}^{2n} = \{(x, \xi)\}$, $x \in \mathbb{R}^n$, $\xi \in (\mathbb{R}^n)^*$, $\rho_a(x, \xi) = (a^{1/2}x, a^{1/2}\xi)$.

Example 2. $T^*X \qquad a(x, \xi) = (x, a\xi)$.

Uncertain reading. The base after T^* is heavily overwritten and tentatively reads X.

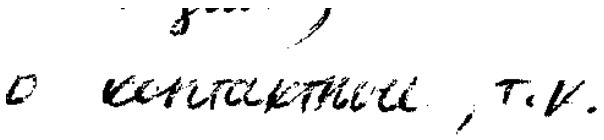
Original source details: GELFAND-P001008

Arnold.

Locally all cones are the same: the quotient is a contact manifold.

(Sympl. cones $\Longleftrightarrow$ contact manifolds)

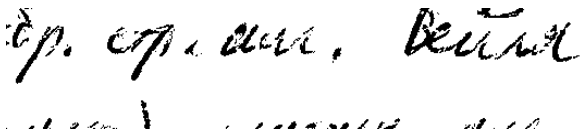
The quantization must be contact quantization, since we need the action — a 1-form!!



Uncertain reading. The ending of the word following "must be" tentatively gives the singular "contact."

Original source details: GELFAND-P001008

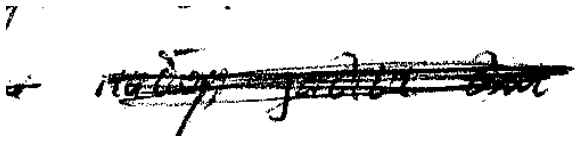
Claim. Can always be quantized and the Weyl alg. is constr. uniquely, microlocally unique, and topol. (microlocally) classical Weyl alg.



Uncertain reading. The abbreviation before "Weyl alg." reads "стр." (constr.).

Original source details: GELFAND-P001008

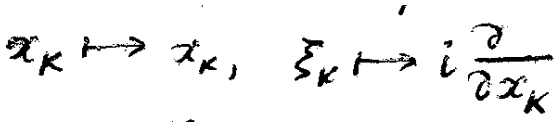
Quantization — [crossed out: a set...]



Uncertain reading. The beginning of the crossed-out line reads "a set...".

Original source details: GELFAND-P001008

$$x_k \longmapsto x_k, \qquad \xi_k \longmapsto i\frac{\partial}{\partial x_k}.$$



Uncertain reading. The image of ξ_k contains i with no minus sign.

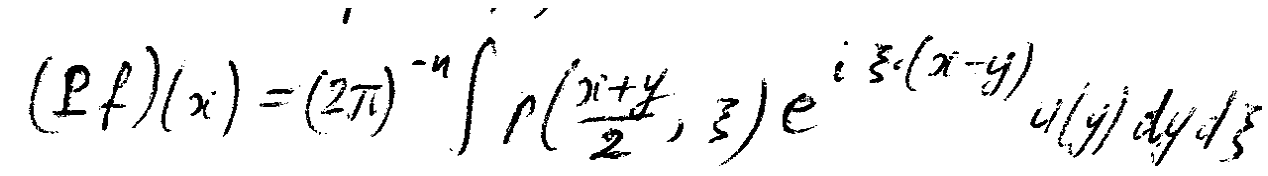
Original source details: GELFAND-P001008

$$p(x,\xi) = \sum_{i=0}^{\infty} p_i(x,\xi), \qquad p_i \text{ — homog. of degree } 2\mu - 2i$$

$$\rho_a(x,\xi) = (a^{1/2}x, a^{1/2}\xi); \qquad \mu \in \mathbb{C}.$$

(This is a symbol of order μ).

$$(Pf)(x) = (2\pi)^{-n} \int p\left(\frac{x+y}{2}, \xi\right) e^{i\xi\cdot(x-y)} u(y)\, dy\, d\xi$$



Original source details: GELFAND-P001008

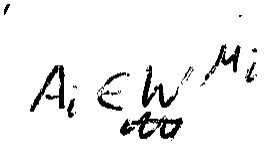
Uncertain reading. The left-hand side reads $(Pf)(x)$ and the integrand uses $u(y)$.

W^μ — space of such operators, $\underline{W^{\mu-1} \subset W^\mu}$.

Symbol: element of $W^\mu / W^{\mu-1}$.

H — Hilbert space, S — dense subspace on which all such op. act.

$\underline{W^\mu}$ — corresp. space of op. (of the form $A_1 + \cdots + A_k$, $A_i \in W^{\mu_i}$ in the example).



Uncertain reading. Below W are crossed-out looped strokes. The subscript reading is unresolved.

Original source details: GELFAND-P001008

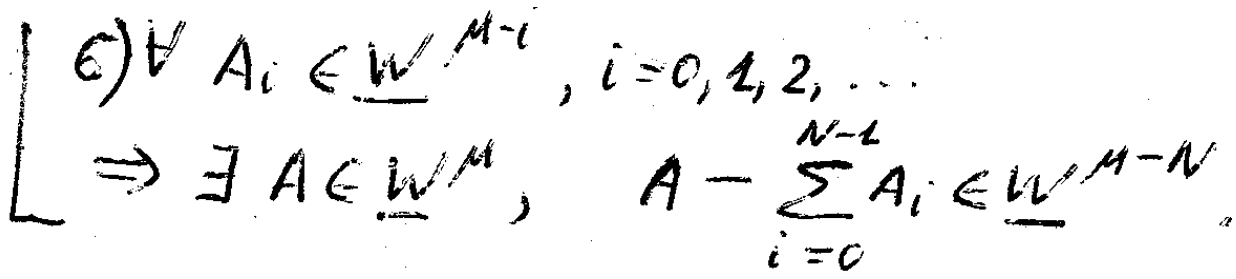
Axioms:

$$\underset{A}{\text{group}} \quad \begin{cases} 1) & \underline{W^\mu} \supset \underline{W^{\mu-1}} \\ 2) & A \in \underline{W^\mu}, \quad B \in \underline{W^\nu} \implies AB \in \underline{W^{\mu+\nu}} \\ 3) & A \in \underline{W^\mu}, \quad B \in \underline{W^\nu} \implies [A,B] \in \underline{W^{\mu+\nu-1}} \\ 4) & A \in \underline{W^\mu} \implies A^* \in \underline{W^{\bar\mu}} \end{cases}$$

$$5) \quad I \in \underline{W^0}$$

GELFAND-P001009

$$\left[\begin{array}{ll} 6) & \forall A_i \in \underline{W^{\mu-i}}, \quad i = 0,1,2,\ldots \\ & \implies \exists A \in \underline{W^\mu}, \qquad A - \sum_{i=0}^{N-1} A_i \in \underline{W^{\mu-N}}. \end{array} \right.$$



Original source details: GELFAND-P001009

Diagram and typed description. To the left of item 6) is the lower end of a square bracket.

Group $\underline{B}$.

1°. If $\operatorname{Re}\mu \le 0$, then $\forall A \in \underline{W^\mu}$ is bounded.

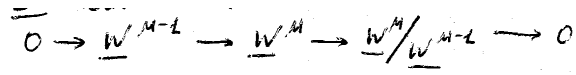
2°. If $\operatorname{Re}\mu < 0$, then $\forall A \in \underline{W^\mu}$ is compact.

3°. $\exists s_0 > 0$, if $A \in \underline{W^\mu}$, $\operatorname{Re}\mu < -s_0$; then A is nuclear.

Group $\underline{C}$.

$$A \in \underline{W^0}, \quad A \text{ is compact} \implies A \in \underline{W^{-1}}$$

$\underline{D}$. Exact seq.



Original source details: GELFAND-P001009

Diagram and typed description. Underlines on W and arrows of the sequence.

$$0 \longrightarrow \underline{W^{\mu-1}} \longrightarrow \underline{W^\mu} \longrightarrow \underline{W^\mu}/\underline{W^{\mu-1}} \longrightarrow 0$$

"Nicely" decomp. into a direct sum.

$E.$ $\exists A(\mu) \in \underline{W}^{\mu}$, depending analyt. on $\mu \in \mathbb{C}$, such that $A(0) - I \in \underline{W}^{-1}$ and

$$A(\mu)A(\nu) - A(\nu+\mu) \in \underline{W}^{\mu+\nu-1}.$$

GELFAND-P001010

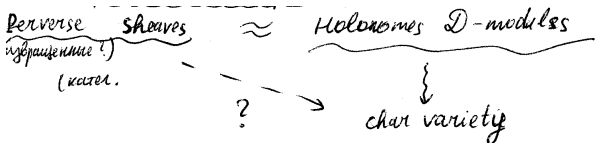
[source number: 15]

1/III-82. Gelfand Sem.

R. MacPherson. The title of a paper [crossed out: which?] he would speak about:

Original source details: GELFAND-P001010

Uncertain reading. The crossed-out English word tentatively reads “which.”



Original source details: GELFAND-P001010

Diagram and typed description. “Holonomes” and “char variety” retain the manuscript spellings. The wavy and dashed arrows point toward the lower label.

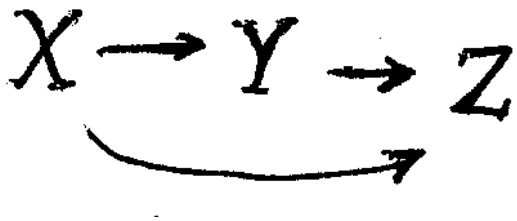
Perverse Sheaves (perverted?) (categ. $\approx$ Holonomes $\mathcal{D}$-modules

? char variety

Another title:

$H(X \longrightarrow Y)$ homol. theory of maps of spaces.

$$H(X \longrightarrow pt) = H_*(X)$$
$$H(X \xrightarrow{\text{id}} X) = H^*(X)$$



Original source details: GELFAND-P001010

Diagram and typed description. The lower arc points from X to Z.

$X \longrightarrow Y \longrightarrow Z$

there is a product, covering comp. of maps.

D. McDuff. Cohom. of the group of volume-preserving diffeom.

$$\mathcal{G} = \mathrm{Diff}^{c}_{\omega 0} M,$$

ω — volume form on M, M — a manifold;

c — with comp. support; 0 — connected component of Id.

$\mathcal{G}$ with the usual topology (C^∞).

G — the same group with the discr. top.

$$G \underset{\text{ident.}}{\longrightarrow} \mathcal{G}$$

$$\bar{B}\mathcal{G} \longrightarrow \overset{K(G,1)}{\overset{\shortparallel}{BG}} \longrightarrow B\mathcal{G} \qquad \text{(classifying spaces)}$$

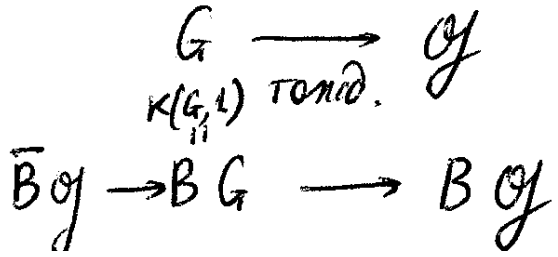


Diagram and typed description. Above BG is $K(G,1)$ with two short strokes. A stroke from $\bar{B}\mathcal{G}$ leads to the fiber annotation.

Original source details: GELFAND-P001010

homot. fiber (turn the map into a fibration, then $\bar{B}\mathcal{G}$ — fiber)

$\bar{B}\mathcal{G}$ depends only on a neighb. of 1 in $\mathcal{G}$.

This is a topological version of the Lie alg. in some sense.

Fuks:

$$G/H \longrightarrow BH \longrightarrow BG$$

(usually); if H is [inserted above: very] dense in G, then the fund. domain is an "infinitesim." neighb. of the identity.

$\bar{B}\mathcal{G} \times M$ has a foliation $\mathcal{F}$, transverse to all fibers $x \times M$ and defined by [inserted above: such] a closed n-form Ω, that $\Omega|_{x \times M} = \omega$.

Uncertain reading. The end of "closed" has heavily overwritten strokes.

Original source details: GELFAND-P001010

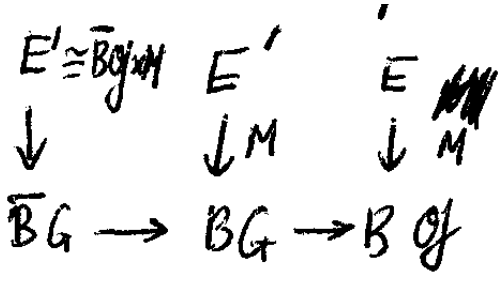


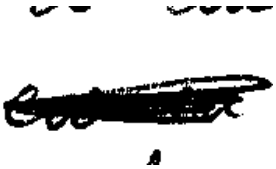
Diagram and typed description. The lower left label reads $\bar{B}G$ and the upper left uses $\bar{B}\mathcal{G}$. The product sign at top runs into the strokes of M. An entry above the right-hand M is heavily crossed out.

$$\begin{array}{ccccc} E' \cong \bar{B}\mathcal{G} \times M & & E' & & E \\ \downarrow & & \downarrow M & & \downarrow M \\ \bar{B}G & \longrightarrow & BG & \longrightarrow & B\mathcal{G} \end{array}$$

Original source details: GELFAND-P001010

E — std. bundle with fiber M.

E' — its lift, [crossed out: connection?] it has zero curvature, since the fiber of the corresp. principal bundle is discrete.



Original source details: GELFAND-P001010

Uncertain reading. The crossed-out word has an initial “c” and tentatively reads “connection.”

On the left, the restriction to the fiber becomes triv. by the property of the classifying space.

The zero-curvature connection is transferred and gives a foliation on $\bar{B}\mathcal{G} \times M$

GELFAND-P001011

$\Omega - \pi^*\omega = 0$ near ∞ on M.

Classes.

$$c_k(M) \in H^k\left(\bar{B}\,\mathrm{Diff}^c_{\omega_0} M; \underbrace{H_c^{n-k}(M;\mathbb{R})}_{\text{coefficients}}\right)$$

$n = \dim M$.

Let K — k-cycle in $\bar{B}\,\mathrm{Diff}^k_{\omega_0} M$.

Original source details: GELFAND-P001011

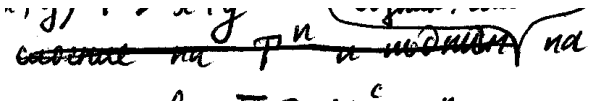
Uncertain reading. In the definition of the cycle the superscript on Diff is k. The neighboring formulas use c.

$$c_k(M)(K) = \text{class of the closed } (n-k)\text{-form } x \longmapsto \int_{K\times x} \Omega$$

Example. $M = T^n$, $K = T^n$.

$$T^n \times T^n \longrightarrow T^n$$
$$(x, y) \longmapsto x + y$$

Take [crossed out: a foliation on T^n and lift it?] [inserted above in a callout: a foliation arises] on $T^n \times T^n$. Obtain a foliation repr. a cycle in $\bar{B}\,\mathrm{Diff}^c_{\omega_0} T^n$.



Original source details: GELFAND-P001011

Uncertain reading. The crossed-out verb means “lift.” Its tense is uncertain.

$$c_n(T^n)(T^n) = \int_{T^n \times x} \Omega \qquad \text{indep. of the choice of } x.$$

Prop. 1

$$\begin{aligned} c_n(S^n) &= 0 \quad \text{for } n \text{ even} \\ &\neq 0 \quad \text{for } n \text{ odd} \end{aligned}$$

1st claim: $K^n \times S^n$ (K^n — any manifold)

Ω — form defining a foliation transv. to the spheres.

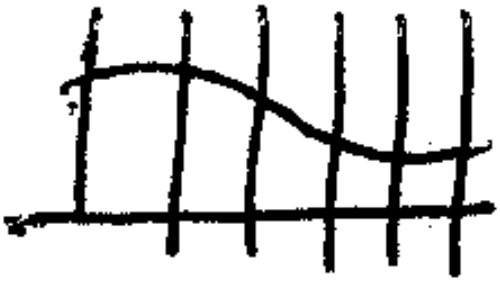
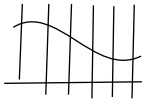


Diagram and typed description. Six nearly vertical strokes, a transverse straight line and a curve.



Original source details: GELFAND-P001011

$$\text{Then } \int_{\substack{\text{horiz.} \\ \text{leaf}}} \Omega = 0.$$

Constr. another class s

Prop. 2

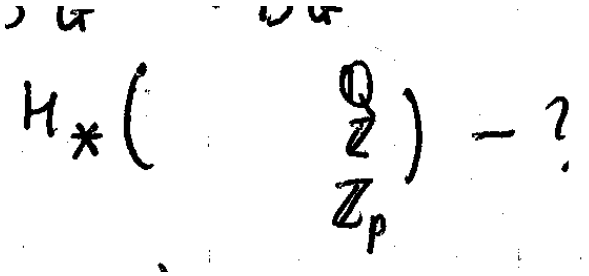
$$\begin{aligned} s(\mathbb{R}^n) &\neq 0 \quad n \text{ even} \\ &= 0 \quad n \text{ odd} \end{aligned}$$

Milnor.

G — Lie group, G^δ — the same with the discr. topology.

$$G^\delta \longrightarrow G$$
$$BG^\delta \longrightarrow BG$$

$$H_*\begin{pmatrix} \mathbb{Q} \\ \mathbb{Z} \\ \mathbb{Z}_p \end{pmatrix} - ?$$



Uncertain reading. The space to the left of the column $\mathbb{Q}, \mathbb{Z}, \mathbb{Z}_p$ in the parentheses is blank.

Original source details: GELFAND-P001011

$H_*(BG)$ are well known and well behaved

$H_*(BG^\delta)$ — ?

are unknown and badly behaved. It is unclear whether they are useful.

In algebraic K-theory, in foliation theory, in Hilbert's 3rd problem

— all this comes up. (equidecomposability)

GELFAND-P001012

[source number: 16]

Isomorphism conjecture:

$$(1) \qquad H_*(BG^\delta, \mathbb{Z}/p) \xrightarrow{\simeq} H_*(BG, \mathbb{Z}/p); \qquad \forall \text{ prime } p.$$

Applied to Hilb.'s 3rd problem by H. Sah, Dupont.

When are they equidecomposable? A partial answer comes from the cohomology of the classifying space.

Probably (1) is a very difficult assertion.

Chern–Weyl: G compact $\Rightarrow$

$$\widetilde{H}_*(BG^\delta, \mathbb{Q}) \xrightarrow{0} \widetilde{H}_*(BG, \mathbb{Q}).$$

(the map is trivial)

Sah–Wagoner:

Uncertain reading. The second surname reads "Wagoner."

Original source details: GELFAND-P001012

$$\underset{\text{connected}}{G} \neq 1 \quad \Rightarrow \quad \exists \text{ an epimorphism}$$

$$H_2(BG^\delta, \mathbb{Z}) \longrightarrow \underset{\substack{\text{uncount.}\\ \text{sum}}}{\bigsqcup} \mathbb{Q} \longrightarrow 0$$

At the same time:

$H_2(BG, \mathbb{Z})$ is finitely generated.

Let us discuss evid. for (1).

① (1) holds in the spec. case when the Lie alg. $\mathfrak{g}$ is solvable.

The proof is elementary.

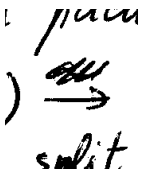
$$\bar{B}G \longrightarrow BG^\delta \longrightarrow BG$$

[Below $\bar{B}G$ with a vertical stroke:] prove by ind. that the cohom. of $\bar{B}G$ is trivial.

② Let G have finitely many components.

Then (1) is an epim. and splits.

$$H_*(BG^\delta, \mathbb{Z}/p) \underset{\text{split}}{\longrightarrow} H_*(BG, \mathbb{Z}/p) \longrightarrow 0$$



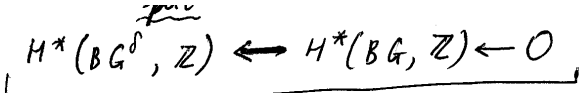
Uncertain reading. Short strokes above the arrow are crossed out, possibly "ep."

Original source details: GELFAND-P001012

[Crossed-out sign at left.] (2) Corollary.

Original source details: GELFAND-P001012

Uncertain reading. The crossed-out looped sign to the left of (2) is unresolved.



Original source details: GELFAND-P001012

Diagram and typed description. Both heads are visible on the middle arrow. A bracket appears below the expression.

$$H^*(BG^\delta, \mathbb{Z}) \longleftrightarrow H^*(BG, \mathbb{Z}) \longleftarrow 0$$

Gen. — thm. [crossed out: Du?] Bott–Heitsch.

Original source details: GELFAND-P001012

Uncertain reading. The cancellation before Bott–Heitsch may read “Du.”

③ (1) always holds [crossed out: when] for H_1 and in interesting cases for H_2:

$$\mathrm{SL}_n \mathbb{R}, \quad \mathrm{SL}_n \mathbb{C}.$$

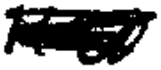
Original source details: GELFAND-P001012

Uncertain reading. A sign below the line between the index n and $\mathbb{R}$ is heavily crossed out, possibly $\mathbb{Z}$.

Haefliger.

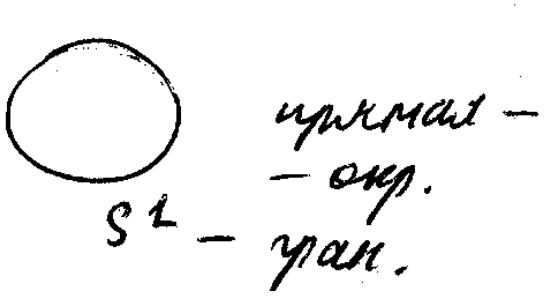
$\mathrm{PSL}_2(\mathbb{Z})$ on $\mathrm{Im}\, z > 0$.

(fractional-lin. transf. [crossed out: co?] with integral coeff.)



Original source details: GELFAND-P001012

Uncertain reading. The crossed-out fragment may begin “co.”



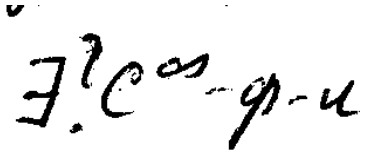
Original source details: GELFAND-P001012

Diagram and typed description. Circle and boundary label.

γ_1, γ_2 — generators.

$$\gamma_1 : z \longmapsto z + 1; \qquad \gamma_2 : z \longmapsto -\frac{1}{z}.$$

$\forall f \in C^\infty(S^1), \quad \exists?\, C^\infty$-fns. $f_1, f_2 \in C^\infty(S^1)$



Uncertain reading. A question mark with a separate dot follows $\exists$.

Original source details: GELFAND-P001012

GELFAND-P001013

$$(1) \qquad f(x) = f_1(x) - f_1(\gamma_1(x)) + f_2(x) - f_2(\gamma_2(x)).$$

This is related to foliations on a Riem. manifold whose leaves are min. surfaces.

Unknown!

Approximately true: approximation is possible (in the C^∞ top.) — this is easy.

GELFAND-P001014

15/III-82. Gelfand Sem.

Topics (for term papers, diploma theses, etc.)

Kirillov:

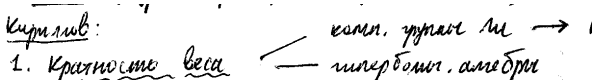
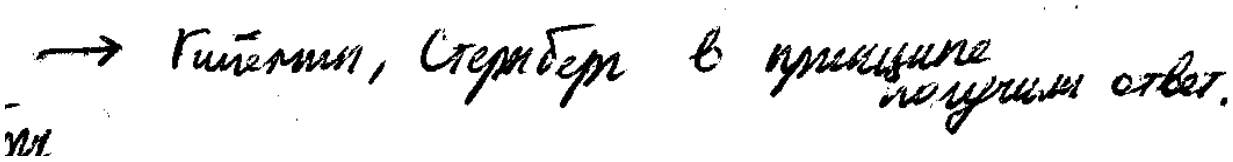


Diagram and typed description. The upper and lower strokes lead to two separate items.

1. Weight multiplicity
   - comp. Lie groups → Guillemin and Sternberg obtained an answer in principle.
   - hyperbolic algs.



Original source details: GELFAND-P001014

Diagram and typed description. The arrow after “comp. Lie groups” points to Guillemin and Sternberg.

Refs.: Macdonald (article)

Example problem in 2: a contragredient Lie algebra with a hyperbolic Cartan matrix (having a neg. eigenvalue) is considered.

Consider the action of $SU(3)$ on its Lie algebra:

$$X = \begin{pmatrix} i\varphi_1 & a & c \\ -\bar{a} & i\varphi_2 & b \\ -\bar{c} & -\bar{b} & i\varphi_3 \end{pmatrix}, \qquad \varphi_1 + \varphi_2 + \varphi_3 = 0$$

$$g \in SU(3) \Rightarrow g : X \to gXg^*.$$

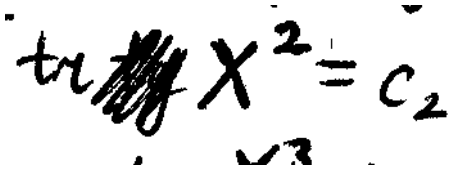
GELFAND-P001015

Consider the orbit Ω. [separate stroke] It has the form:

Uncertain reading. A separate slanted stroke follows the period. It could be 1 or part of an exclamation mark.

Original source details: GELFAND-P001015

$$\begin{cases} \text{tr [crossed-out sign] } X^2 = c_2, \\ \text{tr } X^3 = c_3 \end{cases} \qquad (\text{and, incidentally, } \operatorname{tr} X = 0)$$



Uncertain reading. Between tr and X^2 is a heavily crossed-out sign.

Original source details: GELFAND-P001015

(these traces recover the roots, and $\forall$ matr. can be brought into diag. form)

Denote it by Ω_{c_2,c_3}.

The orbit is of dimension 6 (gen. position).

Project onto the plane (2-dim.) $(\varphi_1, \varphi_2, \varphi_3)$, $\varphi_1 + \varphi_2 + \varphi_3 = 0$.

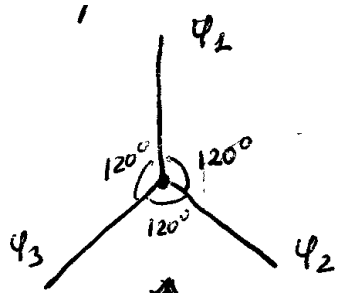
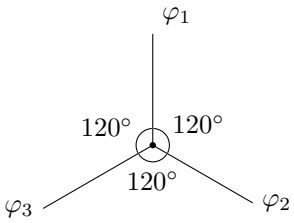


Diagram and typed description. Three axes labelled $\varphi_1, \varphi_2, \varphi_3$ with three $120°$ angles.



Original source details: GELFAND-P001015

Diagram and typed description. An upward arrow is visible in the crossed-out sketch. The underlying geometry and labels are unresolved.

Original source details: GELFAND-P001015

— project onto the plane $\varphi_1 + \varphi_2 + \varphi_3 = 0$ [crossed out: illegible] the coordinate axes.

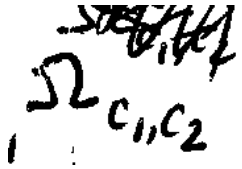
Uncertain reading. A short entry before "coordinate axes" is crossed out.

Original source details: GELFAND-P001015

— this is how it should be drawn. (p — projection)

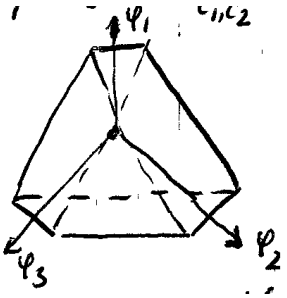
Facts:

a) The projection of Ω_{c_1,c_2} fills a hexagon.



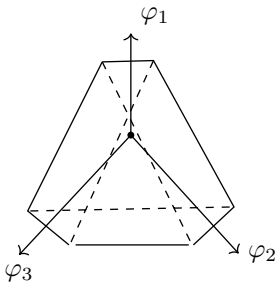
Original source details: GELFAND-P001015

Uncertain reading. The indices on this line are c_1, c_2. The designation above has c_2, c_3.



Original source details: GELFAND-P001015

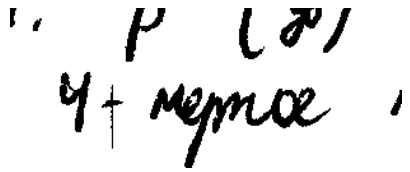
Diagram and typed description. Hexagon, three outgoing arrows and three dashed interior lines.



(this is the set of diag. entries of matrices with fixed eigenvalues $\lambda_1, \lambda_2, \lambda_3$)

b) Consider $p^{-1}(y_0)$, y_0 — a point of the hexagon.

It is a 4-dimensional manifold. Regularity fails on the dashed lines.



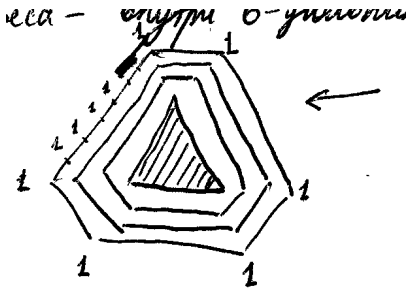
Original source details: GELFAND-P001015

Uncertain reading. A long thin vertical line crosses the horizontal stroke after 4. "4-dimensional" is the probable reading.

This is all in the paper of Berezin and Gelfand in Tr. MMO, 1956. ("Some remarks on the theory of spherical fns.")

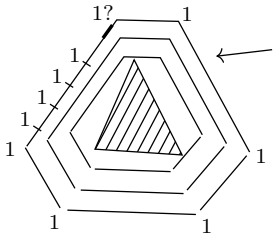
A rep. of $SU(3)$ is given by a pair of integers and the weights follow from them.

The weights are inside the hexagon. Multiplicities:



Original source details: GELFAND-P001015

Diagram and typed description. Three nested hexagonal contours surround a hatched triangle. Overlapping strokes obscure some marks at the upper-left corner.



$\leftarrow$ lines of changing multiplicity

$SU(3)$ — in Gelfand and Berezin.

The answer of Guillemin, Sternberg — for large c_2, c_3.

[Above the names, with an arrow to them:] preprint $\approx$ 1981

X — the preimage of *? a point, $X = p^{-1}(y_0)$.

Uncertain reading. A star-like sign between "preimage" and "point" may be a crossed-out entry.

Original source details: GELFAND-P001015

$\langle e^B \tau, X \rangle$ — multiplicity!!

[Below e^B the strokes of an equals sign are turned vertically.]

$$e^B = 1 + B + \frac{B \wedge B}{2} + \dots$$

B — sympl. form on the orbit.

τ — Todd class of the manifold.

I. M. G. Main question: duality between groups and their representations.

$$x^{\textcircled{n}} = x(x-1)\dots(x-n+1)$$

$$\textcircled{\frac{d}{dx}} f(x) = f(x+1) - f(x)$$

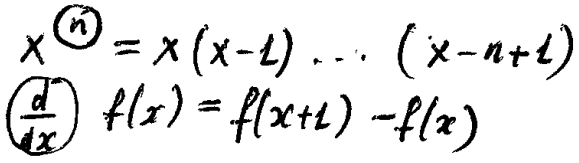


Diagram and typed description. The circles around n and $\frac{d}{dx}$ are part of the author's notation.

Original source details: GELFAND-P001015

GELFAND-P001016

[source number: 25]

Then

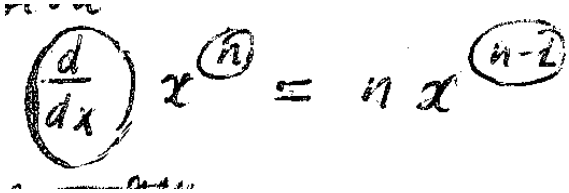


Diagram and typed description.

$\textcircled{\frac{d}{dx}} x^{\textcircled{n}} = n\, x^{\textcircled{n-1}}$

Original source details: GELFAND-P001016

Uncertain reading. A short entry is enclosed and crossed out. Its reading remains unresolved.

Original source details: GELFAND-P001016

Dual: for a representation, look at the weight multiplicities.

There is a (unique) invariant volume on the orbit.

Normalization: volume of the orbit = dimension of the linear representation associated with it.

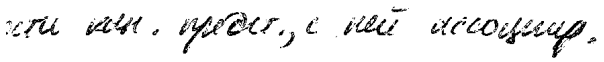
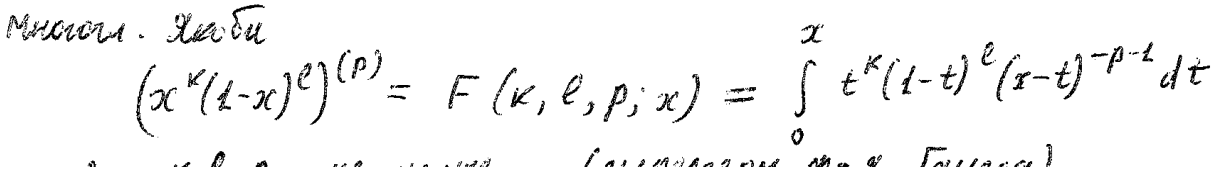


Diagram and typed description. of the linear representation associated with it.

Original source details: GELFAND-P001016

Jacobi polynomials

$$\left(x^k(1-x)^\ell\right)^{(p)} = F(k,\ell,p;x) = \int_0^x t^k(1-t)^\ell(x-t)^{-p-1}\,dt$$



Original source details: GELFAND-P001016

Uncertain reading. There is no prefactor between the equals sign and the integral.

All k, ℓ, p — are nonintegers (Gauss hypergeometric function)

Fractional derivative:

$$(x^n)^{(k)} = x^{n-k}\frac{n!}{(n-k)!} \qquad \text{or} \qquad \left(\frac{x^n}{n!}\right)^{(k)} = \frac{x^{n-k}}{(n-k)!}$$

and replace ! by Γ.

There is no satisfactory theory of hypergeometric functions of several variables (important for representations)

Atiyah: generalization of the statement that the points $\in$ a convex polygon, to the case where only the Cartan group acts.

Idea: recast all of representation theory in terms of the action of the Cartan group.

References: 1) Berezin, Gelfand 2) Atiyah 3) Guillemin, Sternberg 4) Khovanskii

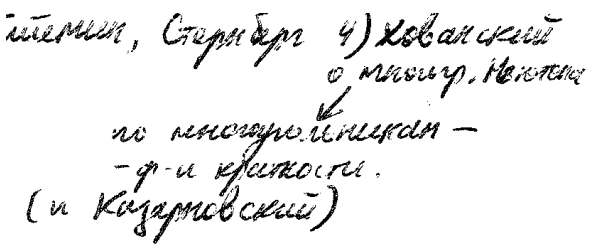


Diagram and typed description. on Newton polyhedra
↙
from polygons —
— multiplicity functions.
(and Kazarnovskii)

Original source details: GELFAND-P001016

GELFAND-P001017

[source number: 46]

29/III/82 Gelfand seminar.

Givental.

Thm. 1. M — a smooth manifold, N — a submanifold.

ω_0, ω_1 — two symplectic structures on M

$$\omega_0|_{TN} = \omega_1|_{TN}$$

Then $\forall x \in N$, $\exists$ a neighbourhood U and a diffeomorphism $g : U \to U$, such that

$$g|_N = \mathrm{id}_N, \qquad g^*\omega_1 = \omega_0.$$

— · —

A. Weinstein proved this if $\omega_0|_{TM|_N} = \omega_1|_{TM|_N}$. (he wrote $\omega_0|_N = \omega_1|_N$)

— · —

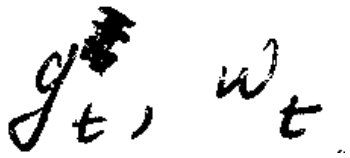
Moral: there is no extrinsic symplectic geometry for a submanifold.

— · —

There are also global formulations.

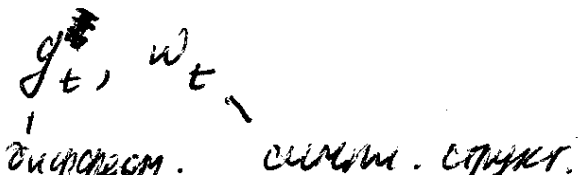
— · —

Idea of the proof: take a family g_t, ω_t



Original source details: GELFAND-P001017

Uncertain reading. Above g_t a mark is crossed out, possibly an asterisk.



Original source details: GELFAND-P001017

Diagram and typed description.

g_t — diffeomorphism

ω_t — symplectic structure

$$g_t^* \omega_t = \omega_0.$$

Differentiate and then use the existence and uniqueness theorem for differential equations. (reduces to a linear problem)

$N = \{*\} \Rightarrow$ a new proof of Darboux's theorem.

[Question: under what conditions is a 2-form obtained by restricting a symplectic form.

Uncertain reading. A short word is crossed out.

Original source details: GELFAND-P001017

necessary condition: $d\omega = 0$].

I. M. G. There is no need to add a closedness condition yet: a symplectic structure in $\forall$ tangent space.

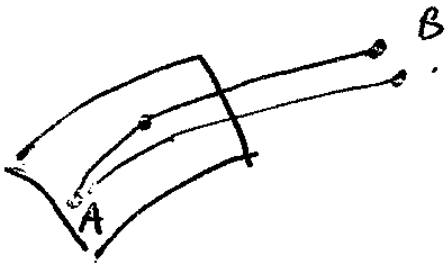
A symplectic form has no properties.

— · —

Symplectic structure in the space of polynomials.

($\Rightarrow$ Lagrangian manifolds with singularities)

Diagram and typed description.



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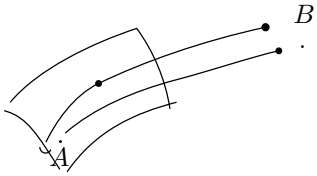


The small strokes at A leave the connection of the two curves unresolved.

A — a point on the surface.

B — an arbitrary point off the surface.

look for extremals (the geodesics break off)

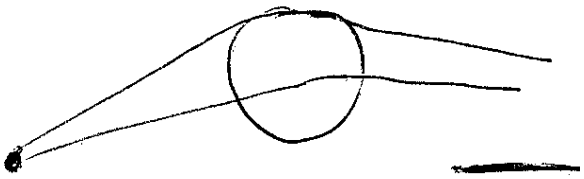
A. The space of lines in $\mathbb{R}^3$ has a natural symplectic structure!

B. Generalization: the space of extremals of any variational problem has a symplectic structure.

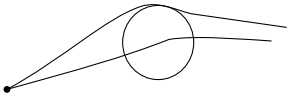
A. it is $T^*\mathbb{R}^2$ (Arnold's explanation)

Example:

Diagram and typed description.



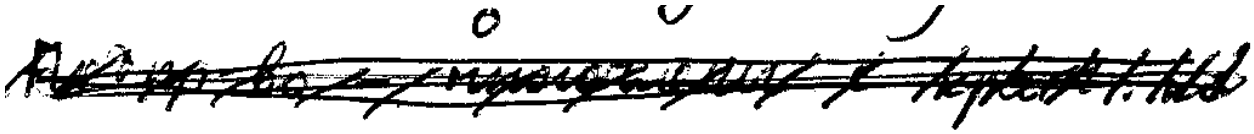
Original source details: GELFAND-P001017



a point and a sphere: a two-dimensional manifold of extremals arises.

— · —

Consider the space of binary forms $\{\sum_0^{2n-1} a_i x^i y^{2n-1-i}\}$ and on it the representation of the group $SL_2(\mathbb{R})$.



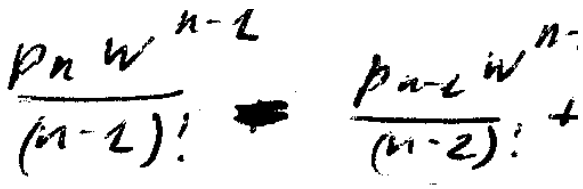
Original source details: GELFAND-P001017

Undeciphered passage. The sentence is crossed out. It may begin “This space” or “On the space”. Its full reading remains unresolved.

At $y = 1$ — simply a polynomial of degree $2n - 1$.

$w = \dfrac{x}{y}$ — an inhomogeneous coordinate. Write the polynomial in this variable using Darboux coordinates:

$$\frac{q_1 w^{2n-1}}{(2n-1)!} + \ldots + \frac{q_n w^n}{n!} + \frac{p_n w^{n-1}}{(n-1)!} \boxed{?} \frac{p_{n-1} w^{n-2}}{(n-2)!} + \ldots + (-1)^{n-1} \frac{p_1 w^0}{0!}$$



Uncertain reading. The sign before the term with p_{n-1} is overwritten. Two horizontal strokes and a lower vertical stroke are visible. The final sign remains unresolved.

Original source details: GELFAND-P001017

GELFAND-P001018

$$\omega = \sum_{i=1}^{n} dp_i \wedge dq_i$$

The action of $SL_2(\mathbb{R})$ preserves the form.

$w \mapsto w + t$ is a Hamiltonian flow; since it comes from $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$

Hamiltonian:

$$h = q_n^2/2 - q_{n-1}p_n - \ldots - q_1 p_2$$

q_1 — a first integral.

Consider the reduction $q_1 = 1$ and the space of characteristics on it.

(we also take $p_1 = 1$ — this is a convention)

Question of I. Goldsheid.

Consider a symplectic space with canonical basis $e_1, \ldots, e_n, f_1, \ldots, f_n$

Let L be a Lagrangian plane in it. Is there a (Lagrangian) plane L_1, spanned by vectors $\varphi_1, \ldots, \varphi_n$, where $\varphi_j = \alpha_j e_j + \beta_j f_j$, such that

$$\rho(L, L_1) \leq 1/\sqrt{2}$$

(it suffices to have $< 1 - \varepsilon$, where $\varepsilon > 0$ is independent of the dimension)

Here $\rho(L, L_1) = \|P_L - P_{L_1}\|$, where the (operator) norm is taken in the standard Euclidean metric in the basis $e_1, \dots, e_n, f_1, \dots, f_n$.

It is asserted that $\exists \varphi_j$ such that $\sin(\widehat{\varphi_j, L}) \le 1/\sqrt{2}$.

GELFAND-P001019

5/IV/82 Gelfand seminar.

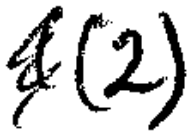
Wodzicki

M — a manifold (possibly with boundary ∂M)

D — a system of differential equations on M:

$$(1) \qquad D : C^\infty(M, E) \longrightarrow C^\infty(M, E)$$

elliptically determined (σ_D is invertible).



Uncertain reading. A mark before the equation number (2) is crossed out. Its reading remains unresolved.

Original source details: GELFAND-P001019

$$(2) \qquad \operatorname{index} D = \dim \ker D - \dim \operatorname{Coker} D = \int_M \alpha_0(x) \qquad \text{(Atiyah–Singer)}$$

[Below $\alpha_0(x)$: density.]

(2) — an example of a local formula.

The coefficient α_0 is expressed in terms of the symbol. Here there naturally appears a ζ-function

$$(3) \qquad \zeta_A(s) = \sum_{\lambda \in \operatorname{Spec} A} \lambda^{-s}$$

$$A = D^*D, \quad DD^*.$$

$$z_j = \frac{n-j}{m} \quad \text{— poles,} \qquad n = \dim M, \quad m = \text{order of } A.$$

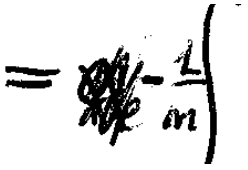
$$z_j = 0, -1, -2, \dots \quad \text{— there are no poles.}$$

$$\int_M \underbrace{\zeta_A(0; x)}_{\text{density}} = \zeta(0); \qquad \zeta_A(0; x) \text{ can be computed explicitly}$$

$$(4) \qquad \operatorname{index} D = \zeta_{1+D^*D}(0) - \zeta_{1+DD^*}(0)$$

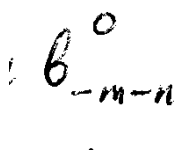
GELFAND-P001020

$$(5) \qquad \zeta_A(0; x) = -\frac{1}{m} \int_{|\xi|=1} d\xi' \int_0^\infty \operatorname{tr} b^0_{-m-n}(x, \xi, \lambda)\, d\lambda$$



Uncertain reading. A sign or factor is crossed out between the equals sign and $-1/m$.

Original source details: GELFAND-P001020



Uncertain reading. A small oval is written above b. It could be 0, o or $\circ$.

Original source details: GELFAND-P001020

[A line from b^0_{-m-n} to the explanation on the right:] component of the symbol of the parametrix for $(A-\lambda)$ in some chart.

$\zeta_A(0;x)\,dx$ — a density (!) — this is not obvious.

Suppose $\partial M \neq \varnothing$.

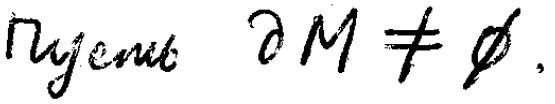


Diagram and typed description. The manuscript has $\partial M \neq \varnothing$.

Original source details: GELFAND-P001020

D — the signature operator (Dirac)

For this operator:

$$\text{(6)} \qquad \operatorname{index} D = \operatorname{sign} M$$

$$\text{(7)} \qquad \int_M L_k(p) = \operatorname{index} D = \int_M \alpha_0(x)$$

$$D : \Omega_+ \longrightarrow \Omega_-$$

$\Omega_\pm$ — forms even (odd) under the involution $\tau = i^{np+n+1}*$ (on p-forms).

$$\text{(8)} \qquad d + d^* = D|_{\Omega_+} : \Omega_+ \longrightarrow \Omega_-.$$

Example (I. M. G.)

$$\text{(9)} \qquad \begin{pmatrix} 0 & \operatorname{div} \\ \operatorname{grad} & \operatorname{rot} \end{pmatrix}^2 = \Delta_{\substack{I \\ \text{identity matrix}}} \qquad (\text{in } R^4)$$

$$\text{(10)} \qquad \text{Theorem.} \quad L_k(p) = \alpha_0(x) \qquad \text{(Atiyah–Bott–Patodi)}$$

$$n = 4 \quad \Rightarrow \quad \alpha_0(x) = \frac{1}{3} p_1$$

The case

Uncertain reading. Marks before D are crossed out.

Original source details: GELFAND-P001020

$D = \bar{\partial}$ and $\overset{\text{metric}}{M}$ is Kähler $\Rightarrow$ local agreement of $\alpha_0(\dots)$ and the component of the Todd genus $T(p)$ (in general this is false).

Original source details: GELFAND-P001020

Uncertain reading. The abbreviation before "Todd genus" most likely reads «соств.».

That is, there are two local formulas that do not coincide.

$\partial M \neq \varnothing \Rightarrow$ there is also a signature (degenerate form on middle cohomology, with zeros disregarded)

$$(11) \qquad \underset{\substack{\text{topological}\\ \text{invariant}}}{\operatorname{sign} M} = \underbrace{\int_M L_k(p)}_{\text{local formula}} - \underbrace{\eta(\partial M)}_{\substack{\text{depends}\\ \text{on the metric}\\ \text{at the boundary}}}$$

(the metric near the boundary — is a product).

I. M. G. Problem: is sign — the only additive invariant for manifolds

Original source details: GELFAND-P001020

Uncertain reading. A short word before "with" is crossed out.

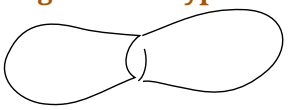
with boundary?

— · —

Pass to the double:

Original source details: GELFAND-P001020

Diagram and typed description.



There is a gap at the top of the right-hand stroke of the seam.

By Hirzebruch's theorem (equation (7)), we see that $\eta(\partial M)$ is independent of the filling.

(12) $\eta(\partial M)$ — a nonlocal invariant!!

Locality $\Rightarrow$ multiplicativity (with respect to [inserted above the line] passing to coverings)

There are examples (lens spaces) where this fails.

GELFAND-P001021

[source number: 58]

(13) Theorem. (Atiyah–Patodi–Singer)
$N = \partial M$, D — a first-order system

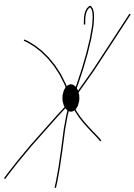
$$D = \frac{\partial}{\partial u} + A, \qquad A \text{ — an operator on } \partial M \tag{14}$$

(near the boundary)
Let A be normal with spectrum disjoint from a conical neighbourhood of $i\mathbb{R}$.

Uncertain reading. Between “not” and “intersecting” is a crossed-out abbreviation, possibly «сод.» or «соб.».

Original source details: GELFAND-P001021

Diagram and typed description.



Original source details: GELFAND-P001021

Then $\exists$ elliptic boundary conditions for D.
P_A — the projector onto the eigenvalues of A with Re $A \geq 0$.

Diagram and typed description. The condition reads Re $A \geq 0$.

Original source details: GELFAND-P001021

$$\left\{\begin{matrix} Df \\ P_A(f|_A) = 0 \end{matrix}\right\} \quad \text{— a nonlocal boundary-value problem} \tag{15}$$

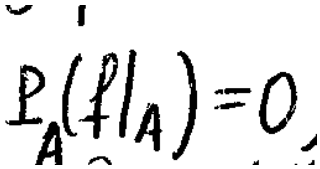


Diagram and typed description. The restriction subscript in (15) is written A.

Original source details: GELFAND-P001021

a local problem does not always exist.

But a nonlocal problem always exists!!

Original source details: GELFAND-P001021

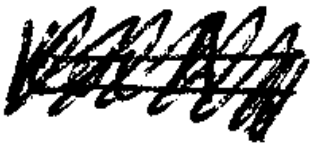
Uncertain reading. A mark after “nonlocal” is crossed out.

Then:

$$\operatorname{index} D = \int \alpha_0(x) - \frac{h_0 + \eta(A)}{2}, \tag{16}$$

where

$h_0 =$ dim (space of purely imaginary eigenvalues).



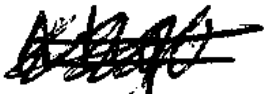
Original source details: GELFAND-P001021

Uncertain reading. The entry after dim is crossed out. It may read Ker A or Ker A_0.

$$\eta(A) \equiv \eta(0, A), \qquad \eta(s, A) = \sum_{\lambda \in \operatorname{Spec} A} \operatorname{sign} \operatorname{Re} \lambda \, |\operatorname{Re} \lambda|^{-s} \tag{17}$$

Point: generalization of the L-function.

In A.-P.-S. A — a self-adjoint operator

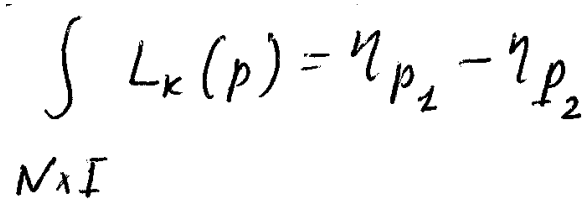


Original source details: GELFAND-P001021

Uncertain reading. The entry before A.-P.-S. is crossed out. It may read «Водз.» or «Водз».

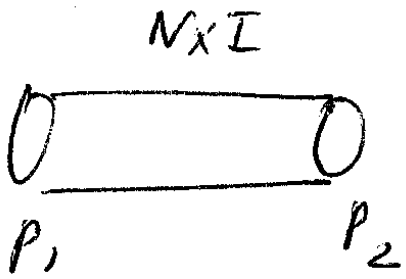
Theorem. A — an elliptic operator.

$$\int_{N \times I} L_k(p) = \eta_{p_1} - \eta_{p_2}$$



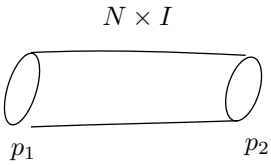
Original source details: GELFAND-P001021

Uncertain reading. The handwritten letters in $L_k(p)$ and the subscripts of η resemble p. The subscripts may be ρ.



Original source details: GELFAND-P001021

Diagram and typed description.



The end labels read p_1, p_2 or ρ_1, ρ_2.

GELFAND-P001022

12/IV/82 Gelfand seminar.

Wodzicki. (Mariusz)

X — a compact manifold with boundary.

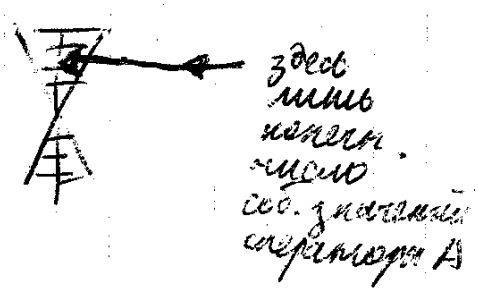
Original source details: GELFAND-P001022

Diagram and typed description. The manuscript abbreviates “manifold” as «мн-зие».

(1) $D\colon C^\infty(X,E) \to C^\infty(X,F)$ is a first-order elliptic system which (differential) near ∂X takes the form

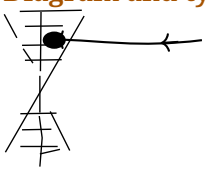
$$(2) \qquad D = \frac{\partial}{\partial u} + A, \qquad A \text{ — an operator on the boundary}$$

$$(3) \qquad A\colon C^\infty(Y, E|_Y) \longrightarrow C^\infty(Y, F|_Y)$$



Original source details: GELFAND-P001022

Diagram and typed description.



[Label with arrow:] only finitely many eigenvalues of the operator A.
Dense strokes at the tip obscure the exact joins.

Boundary conditions:

$$(4) \qquad P_+ f|_Y = 0,$$

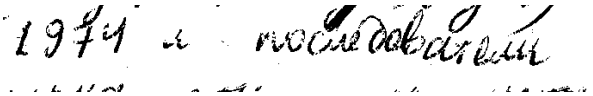
P_+ — the projector onto the positive eigenfunctions of the operator A.

Original source details: GELFAND-P001022

Uncertain reading. A mark is crossed out between the words abbreviated as “eigenfunctions”.

(5) Theorem (Atiyah–Patody–Singer)

1974 and followers



Original source details: GELFAND-P001022

Uncertain reading. The small word after "1974" most likely reads "and".

suppose

1) A is normal with respect to the Riemannian metric on the boundary $\quad (A^*A = AA^*)$

$$(6) \qquad \operatorname{index} D = \int_X \alpha_0(x)\,dx - \frac{h(A)+\eta(A)}{2}$$

$$(7) \qquad \eta(A,s) = \sum \operatorname{sign} \operatorname{Re}\lambda \cdot |\operatorname{Re}\lambda|^{-s}, \qquad \eta(A) = \eta(A,0)$$

Original source details: GELFAND-P001022

Uncertain reading. The number beside the formula for $\eta(A,s)$ is overwritten. The top stroke and the numbering sequence support (7).

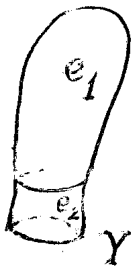
$$(8) \qquad h(A) = \dim \left\{ \begin{array}{c} \text{imaginary} \\ \text{eigenvalues of } A \end{array} \right\}$$

GELFAND-P001023

[source number: 64]

Proof. Use the formula:

$$(9) \qquad \operatorname{index} D = \underset{\sum_{\lambda \in \operatorname{Spec} D^*D} e^{-t\lambda}}{\operatorname{Tr} e^{-tD^*D}} - \underset{\sum_{\mu \in \operatorname{Spec} DD^*} e^{-t\mu}}{\operatorname{Tr} e^{-tDD^*}}$$



Original source details: GELFAND-P001023

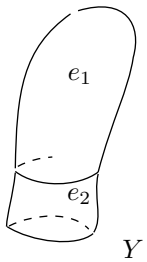


Diagram and typed description. The lower-right label reads Y. The rear arcs are broken.

— · —

A — a self-adjoint elliptic operator.

Uncertain reading. A word is crossed out before “elliptic”.

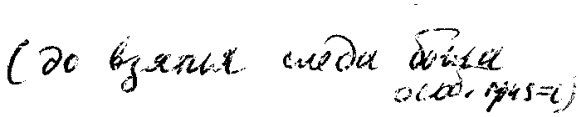
Original source details: GELFAND-P001023

$$(10) \qquad \sum_{\lambda>0} \lambda^{-s} = \zeta_+(s), \qquad \zeta_+(A) \text{ — the value at } s = 0.$$

(11) $\zeta_+(A) < \infty$, but locally there is a residue!! (nonlocal functional)

$$(12) \qquad \zeta_+(s) = \mathrm{Tr}(P_+ A^{-s}) = \int_X \mathrm{Tr}(P_+ A^{-s})(x,x)$$

[At right: (there were singularities before taking the trace at $s = i$).]



Uncertain reading. The sign after $s =$ resembles i, with a faint separate stroke above it. The reading remains uncertain.

Original source details: GELFAND-P001023

The local term has a residue, and $\int_X$ of this residue $= 0$.

Uncertain reading. The abbreviation after “local” most likely means “term”. The last strokes are faint.

Original source details: GELFAND-P001023

$$(13) \qquad \begin{cases} \zeta(A) = \zeta_+(A) + \zeta_-(A) \\ \eta(A) = \zeta_+(A) - \zeta_-(A) \end{cases}$$

I. M. G.: in the work of Gabrielov, Gelfand and Fuks there are algebraic reasons for nonlocality.

The most remarkable problem is unsolved: this is done to compute the values at the boundary of the forms of Pontryagin, but for general classes there is nothing.

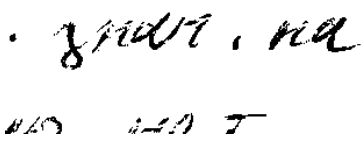


Diagram and typed description. The manuscript abbreviates “values” as «знач.» before “at the boundary”.

Original source details: GELFAND-P001023

Stredder Epstein (2 papers)	J. Diff. Geometry 10 (1975), 645	on natural differential operators.

— · —

Question of I. M. G.: why require self-adjointness of A?
Let a domain and an operator on it be given, the operator A.

Diagram and typed description.

Original source details: GELFAND-P001023

One constructs a theory of cobordisms.
The invariant [below it: $\eta(A)$] A is constructed with a membrane stretched over the manifold.

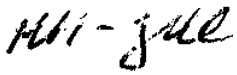


Diagram and typed description. The manuscript abbreviates “manifold” as «мн-зие».

Original source details: GELFAND-P001023

— · —

But in the absence of a filling this has yet to be proved!!

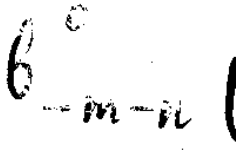
— · —

$$\alpha_0(x) = \zeta(D^*D + \varepsilon, x) - \zeta(DD^* + \varepsilon, x) \tag{14}$$

GELFAND-P001024

Formula for $\zeta(A, x)$

$$\frac{1}{m} \int_{|\xi|=1} d\xi' \int_0^\infty b^\circ_{-m-n}(x, \xi, \lambda)\, d\lambda \tag{15}$$



Uncertain reading. In (15), a small broken oval above b is rendered as a circle.

Original source details: GELFAND-P001024

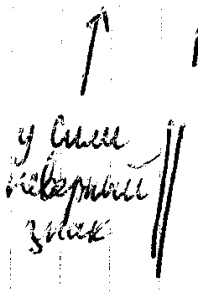


Diagram and typed description.
↑
Seeley has the wrong sign‖

Original source details: GELFAND-P001024

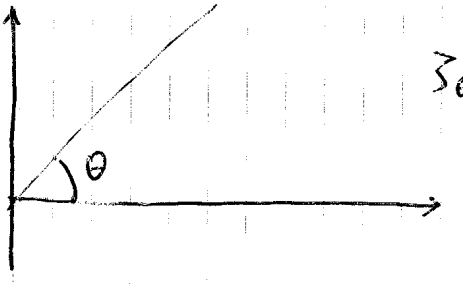
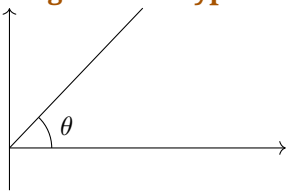


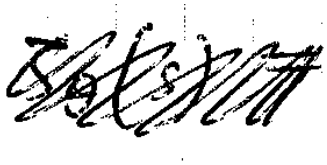
Diagram and typed description.



Original source details: GELFAND-P001024

$$\zeta_\theta(s) = \sum \lambda^{-s}$$

(with a cut along the ray at angle θ)

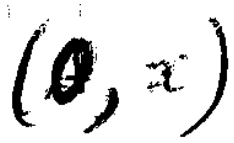


Uncertain reading. The crossed-out entry before (16) appears to read $\zeta_\theta(s) =$.

Original source details: GELFAND-P001024

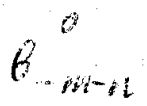
$$\zeta_\theta^A(0,x) = -\frac{e^{i\theta}}{m}\int_{|\xi|=1} d\xi' \int_0^\infty b^\circ_{-m-n}(x,\xi,e^{i\theta}\lambda)\,d\lambda \tag{16}$$

(this is the correct formula)



Uncertain reading. The first argument in (16) is overwritten. Both 0 and θ are possible.

Original source details: GELFAND-P001024



Uncertain reading. In (16), the superscript oval on b is rendered as a circle.

Original source details: GELFAND-P001024

I. M. G. A remarkable operator in $\mathbb{R}^3$ (elliptic):

$$\mathcal{A}\varphi = \begin{pmatrix} 0 & \operatorname{div} \\ \operatorname{grad} & \operatorname{rot} \end{pmatrix} \begin{pmatrix} \varphi_0 \\ \vec{\varphi} \end{pmatrix} = (\operatorname{div}\vec{\varphi}, \operatorname{grad}\varphi_0 + \operatorname{rot}\vec{\varphi}) \tag{17}$$

$$\mathcal{A}^2 = \begin{pmatrix} \Delta & 0 \\ 0 & \Delta \end{pmatrix}.$$

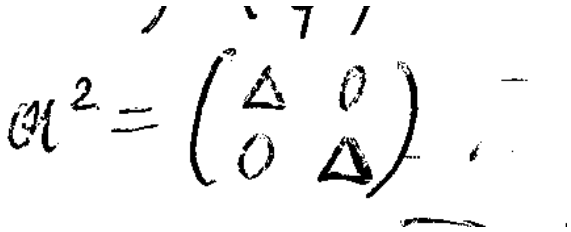
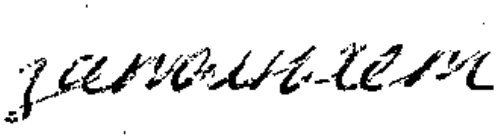


Diagram and typed description. Both diagonal entries read Δ.

Original source details: GELFAND-P001024

— · —

A — an elliptic operator on a closed manifold, the spectrum of A does not fill a sector around the ray (ray of min growth)



Uncertain reading. The word most likely means “fill”. “Meet” is also possible.

Original source details: GELFAND-P001024

$\underbrace{\zeta_{\theta,A}(s) = \sum \lambda_\theta^{-s}}$ define using the ray.

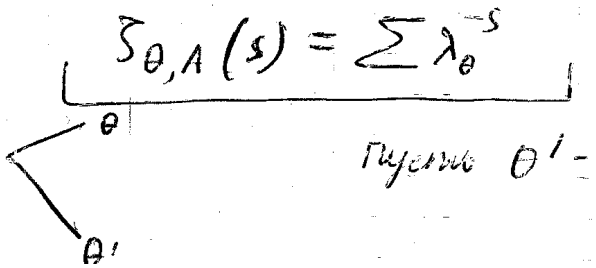
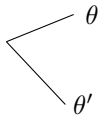


Diagram and typed description.

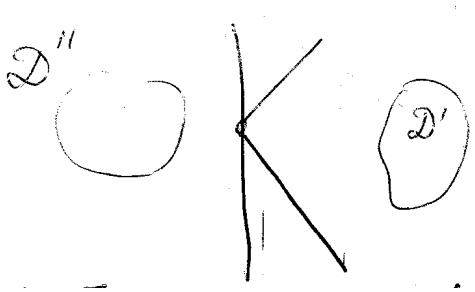


Original source details: GELFAND-P001024

Let θ' — be another ray of min growth.

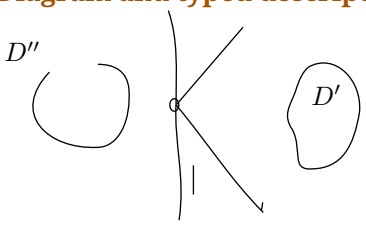
$$\zeta_{\theta',A}(s) = \sum \lambda_{\theta'}^{-s}$$

$\Rightarrow$ the spectrum lies in two sectors



Original source details: GELFAND-P001024

Diagram and typed description.



The left contour has a gap. The tick at the bottom of the slanted stroke may be part of an arrow.

(18) Theorem.

$$\rho(s) \stackrel{\text{def}}{=} \zeta_\theta(A,s) - \zeta_{\theta'}(A,s)$$

Then

a) $\rho(s)$ is meromorphic with residues at $z_j = \dfrac{-j+n}{m}$, $j = 0,1,2,\ldots$

b) $z_j \in \mathbb{Z} \Rightarrow$ the residue vanishes microlocally!!

(elementary!)

GELFAND-P001025

[source number: 65]

c)

$$\begin{array}{ccc} z_j & \in \mathbb{Z} \Rightarrow & \rho(z_j) \\ \| & & \| \\ \ell & & \rho(\ell) \end{array} \quad \text{is given by local formulas}$$

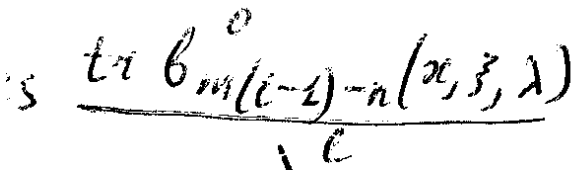
(a substantive assertion, — also valid for pseudodifferential operators!)

$$(19) \qquad \rho(i) \stackrel{\text{def}}{=} \int_X \rho_\ell(x)\,dx$$

$$= -\frac{2\pi i}{m} \int_X dx \int_{|\xi|=1} d\xi' \sum_{\lambda_\nu \in D'} \operatorname{Res} \frac{\operatorname{tr} b^\circ_{m(i-1)-n}(x,\xi,\lambda)}{\lambda^\ell}$$

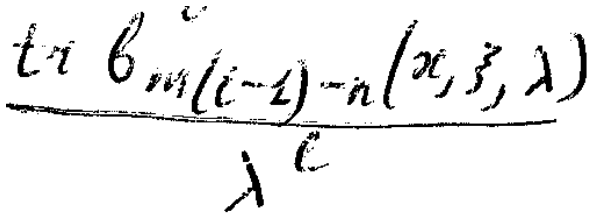
Original source details: GELFAND-P001025

Uncertain reading. The argument of $\rho(i)$ resembles i, and the subscript of $\rho_\ell(x)$ resembles ℓ. Part (c) uses ℓ.



Original source details: GELFAND-P001025

Uncertain reading. A broken oval above b is rendered as a circle.



Original source details: GELFAND-P001025

Uncertain reading. The index of b appears to contain i, and the exponent of λ appears to be ℓ.

d) $\ell \in \mathbb{Z}_- = \{0, -1, -2, -3, -4, \ldots\} \Rightarrow \rho(\ell) = 0$

(a stronger version of A.-P.-S.).

Hints (for (d)):

The proof should cover more than just the residues at 0, and should cover all integer points > 0

Original source details: GELFAND-P001025

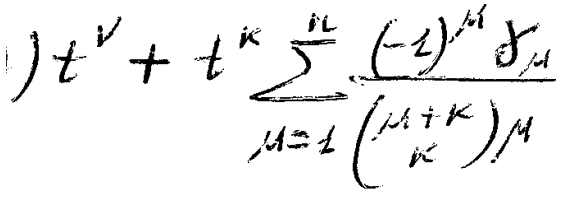
Uncertain reading. The word after "about" most likely reads "residues".

Original source details: GELFAND-P001025

Diagram and typed description. The inequality in the hint reads > 0.

At the point 0 it works out: use the Mahler transform (from the p-adic numbers):

$$\zeta_\theta(-k, A+t) = \underbrace{\sum_{\nu=0}^{k} \binom{k}{\nu} \zeta(\nu-k, A) t^\nu}_{\text{polynomial:}} + t^k \sum_{\mu=1}^{n} \frac{(-1)^\mu \delta_\mu}{\binom{\mu+k}{k} \mu} \tag{20}$$



Original source details: GELFAND-P001025

Uncertain reading. The second sum is preceded by t^k. No further power of t is visible in the summand.

δ_μ — the residue at the point μ for the ζ-function of the operator A
This is for first-order operators.

$$\text{Ker } A \neq \{0\} \Rightarrow \text{instead of } \zeta(0) \text{ write } \zeta(0) + h_0(A)$$

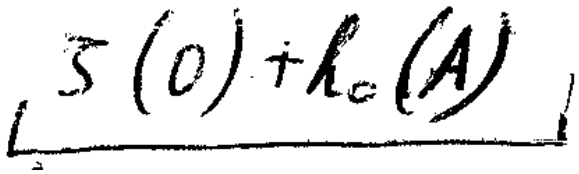
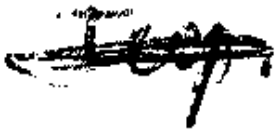


Diagram and typed description. $\zeta(0) + h_0(A)$

Original source details: GELFAND-P001025

($h_0(A) = \dim \text{Ker } A$).



Uncertain reading. The entry before “first sum” is crossed out and may read “Thm.”

Original source details: GELFAND-P001025

The first sum in (20) — the Mahler transform.
An automorphic function with respect to the Mahler transform extends to the p-adic domain.

Chapter 22

GELFAND-P001026

6/IX/82 Gelfand seminar.

(40th year of activity)

Kirillov on the work of Lepowsky and Wilson.

generaliz. (Lepowsky, Wilson
—"—, —"—, Kac, Kazdan)

Lie algebra with six generators.

$e_0 \quad h_0 \quad f_0$ | $\forall$ triple satisfies the standard relations
$e_1 \quad h_1 \quad f_1$ | of matrices of order two with trace 0.

GELFAND-P001027

I. M. G. Cartan algebras No need to learn the roots (everything follows from this)

$e_i, h_i, f_i, \quad i = 1, \dots, n$

$$\begin{cases} [e_i, f_k] = h_i \delta_{ik} \\ [h_i, f_k] = -a_{ik} f_k \\ [h_i, e_k] = a_{ik} e_k \\ [h_i, h_k] = 0 \end{cases} \quad \text{(without summing over } k\text{)} \tag{1}$$

(a_{ik}) — Cartan matrix

a_{ii} can be adjusted by normalizing f_i

Assume: $a_{ii} = 2$.

Important algebras: 1) $a_{ij} < 0$ for $i \neq j$

2) $a_{ij} \in \mathbb{Z}$

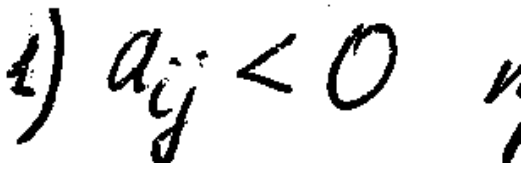


Diagram and typed description. The manuscript has a strict inequality $<$.

Original source details: GELFAND-P001027

As a rule there are no further relations (so that no linear combination of e_i, h_i, f_i is forced to be 0)

Sometimes one can add relations, and the theorem is that they are unique.

$$\begin{cases} e_i^{-a_{ik}+1} e_k = 0 \\ f_i^{-a_{ik}+1} f_k = 0 \end{cases} \tag{2}$$

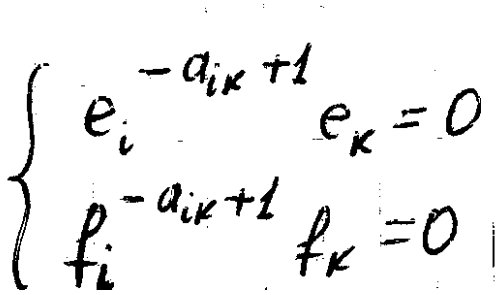


Diagram and typed description. The powers of e_i and f_i in (2) are written without the notation ad.

Original source details: GELFAND-P001027

Sometimes the algebra is finite-dimensional,

$n = 1$: e_1, f_1, h_1

$$[e_1, f_1] = h_1$$
$$[h_1, e_1] = 2e_1$$
$$[h_1, f_1] = -2f_1$$

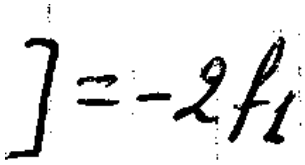


Diagram and typed description. The last subscript in $-2f_1$ has the foot of the numeral 1.

Original source details: GELFAND-P001027

this gives sl(2)

Uncertain reading. A short entry after sl(2) is heavily crossed out.

Original source details: GELFAND-P001027

Finite-dimensionality $\Leftrightarrow$ the matrix $(a_{ij}) > 0$ (strictly)

Affine $\Leftrightarrow$ $(a_{ij}) \geq 0$ and rank$(a_{ij}) = n - 1$.

$n = 2$

$$\underset{\text{sl}(3)}{\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}} \quad \underset{\text{so}(5)}{\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}} \quad \underset{G_2}{\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}}$$

This replaces Dynkin diagrams.

Relations (2) for sl(3):

$$[e_1[e_1e_2]] = 0$$
$$[e_2[e_2e_1]] = 0$$

GELFAND-P001028

[source number: 4]

Affine algebras:

$$\begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix} \qquad A_1^1$$

Diagram and typed description. The lower-right entry of the first matrix is written as -2.

Original source details: GELFAND-P001028

$$\begin{pmatrix} 2 & -4 \\ -1 & 2 \end{pmatrix}$$

Uncertain reading. An entry to the right of the second matrix is crossed out.

Original source details: GELFAND-P001028

Heisenberg algebra:

$$\underset{\text{symplectic}}{E} \oplus R, \qquad [x,y] = \langle x,y\rangle 1, \quad 1 \in R,$$

$\langle x,y\rangle$ — symplectic product.

— · —

Kirillov.

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \qquad \begin{matrix} f_1 & h_1 & e_1 \\ f_2 & h_2 & e_2 \end{matrix}$$

Nontrivial relation

$$[e_1[e_1[e_1,e_2]]] = 0$$

and analogous relations with the replacements $\begin{matrix} 1 \leftrightarrow 2 \\ e \leftrightarrow f \end{matrix}$.

I. M. G. Root vectors — monomials;

Uncertain reading. An entry before “monomials” is crossed out.

Original source details: GELFAND-P001028

weights are the numbers of occurrences of e_j

Uncertain reading. The first strokes of the crossed-out word may begin «вхо...», as in “occurrences”.

Original source details: GELFAND-P001028

K.

Grading: weights (pairs of numbers)

Uncertain reading. An entry after “Grading” is crossed out.

Original source details: GELFAND-P001028

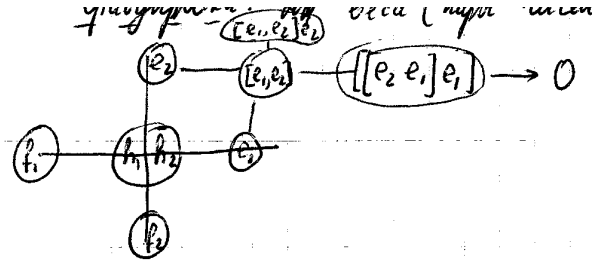
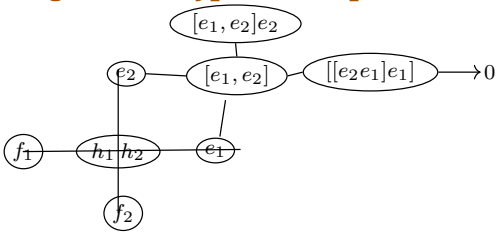


Diagram and typed description.

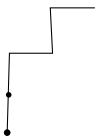


The upper oval reads $[e_1, e_2]e_2$. Any outer brackets remain uncertain.

Original source details: GELFAND-P001028

I. M. G.

Diagram and typed description.



Original source details: GELFAND-P001028

monomials: paths like $e_1e_1e_2e_1\ldots$

Question: how can the paths be made continuous?

K. Second realization. Functions on the circle with values in matrices of order two. $A(t)$ (polynomials in t and t^{-1}; defined for $|t| = 1$).

$$[A, B] = AB - BA + \frac{\mathbf{1}}{2\pi i}\int \mathrm{tr}(A\,dB), \qquad \mathbf{1} \text{ — a new generator} \tag{1}$$

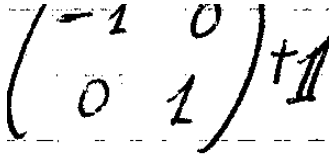
Uncertain reading. An entry above e_1 is crossed out.

Original source details: GELFAND-P001028

$$e_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$
$$e_2 = \begin{pmatrix} 0 & 0 \\ t & 0 \end{pmatrix}, \quad h_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \mathbf{1} \quad f_2 = \begin{pmatrix} 0 & t^{-1} \\ 0 & 0 \end{pmatrix} \tag{2}$$

Original source details: GELFAND-P001028

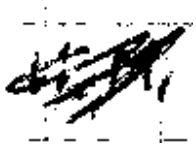
Uncertain reading. The entry between the matrix for h_1 and the comma is heavily crossed out.



Original source details: GELFAND-P001028

Diagram and typed description. The matrix for h_2 in (2) is followed by $+\mathbf{1}$.

They satisfy all the relations.



Original source details: GELFAND-P001028

Uncertain reading. An entry at the beginning of the line before “They” is crossed out.

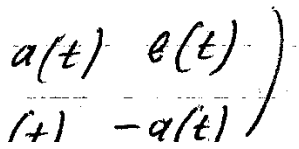
I. M. G. Pick out the Heisenberg subalgebra, the type of its representations, etc.

Original source details: GELFAND-P001028

Uncertain reading. The word before “its representations” most likely reads “type”.

K. This is convenient for writing a general element.

$$\begin{pmatrix} a(t) & e(t) \\ c(t) & -a(t) \end{pmatrix} + \alpha\mathbf{1}. \tag{3}$$



Original source details: GELFAND-P001028

Uncertain reading. The upper-right letter resembles e, with a short loop. A reading of b remains possible.

GELFAND-P001029

There are many representations; they have not all been described (and pointless) [tentative reading of the abbreviation]?

Uncertain reading. (and pointless) [tentative reading of the abbreviation]

Original source details: GELFAND-P001029

The highest-weight representations are described.

One of them is singled out and is remarkable:

cyclic vector — highest-weight vector v_0

$$\begin{cases} e_1 v_0 = 0, \\ e_2 v_0 = 0, \\ h_1 v_0 = 0, \\ h_2 v_0 = v_0 \end{cases} \tag{4}$$

The most economical highest-weight representation

The space is realized as the space of [Crossed out: illegible.]?

Uncertain reading. [Crossed out: illegible.]

Original source details: GELFAND-P001029

polynomials in $x_1, x_3, x_5, \ldots, x_{2n+1}, \ldots$

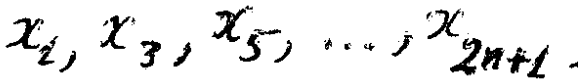


Diagram and typed description. The first indices in the list: $1, 3, 5$.

Original source details: GELFAND-P001029

A Heisenberg (infinite-dimensional) subalgebra is singled out:

$$B_{2k+1} = \begin{pmatrix} 0 & t^k \\ t^{k+1} & 0 \end{pmatrix}, \quad C_{2k+1} = \begin{pmatrix} 0 & -t^k \\ t^{k+1} & 0 \end{pmatrix}, \quad H_{2k} = \begin{pmatrix} t^k & 0 \\ 0 & -t^k \end{pmatrix}. \tag{5}$$

— · —

I. M. G.: This algebra arises naturally in the p-adic case.
Consider the case in which we quotient by $\{\mathbf{1}\}$
K. There are different (nonconjugate) commutative subalgebras.
1) diagonal
2) $\begin{pmatrix} 0 & f(t) \\ tf(t) & 0 \end{pmatrix}$
One cannot put it in diagonal form continuously (but one can for each fixed value).
I. M. G. Quadratic forms: a related problem; Mishchenko; Gelfand — the number is 2^n,
but the canonical forms are undescribed.
K. with **1** these algebras are now noncommutative

$$[B_{2k+1}, B_{2\ell+1}] = \delta(k+\ell)(2\ell+1)\mathbf{1}$$

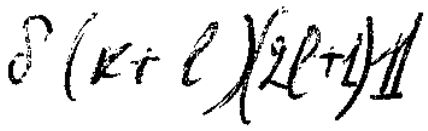


Diagram and typed description. The manuscript reads $\delta(k+\ell)(2\ell+1)\mathbf{1}$.

Original source details: GELFAND-P001029

(Heisenberg algebra)
Take its standard representation.

$B_1, B_3, \ldots$	— multiplication by $x_1, x_2, \ldots$	$B_k = x_k(2k+1)$
$B_{-1}, B_{-3}, \ldots$	— differentiation	$B_{-k} = \partial/\partial x_k$

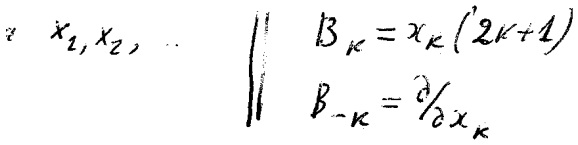


Diagram and typed description. The notation in the two manuscript columns.

Original source details: GELFAND-P001029

The restriction of the representation to the Heisenberg algebra remains irreducible.
Generating function for the action

$$e^{\sum x_k} e^{4\sum \partial/\partial x_k} \stackrel{df}{=} L = \sum L_m, \qquad L_m \text{ — homogeneous component of weight } m.$$

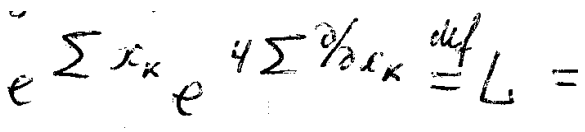


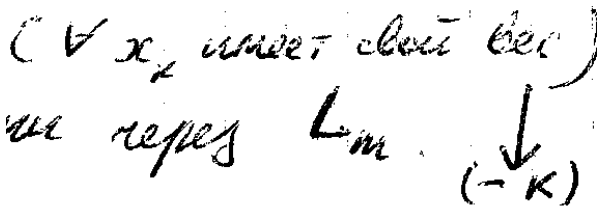
Diagram and typed description. The two exponential arguments in the manuscript.

Original source details: GELFAND-P001029

($\forall x_k$ has its own weight)
$\downarrow$
$(-k)$

Original source details: GELFAND-P001029

Uncertain reading. (not written out) [tentative reading]



Original source details: GELFAND-P001029

Diagram and typed description. Arrow labelled $(-k)$ below "weight".

C_{2k+1} and H_{2k} are expressed by simple formulas in terms of L_m.

GELFAND-P001030

[source number: 5]

Zamolodchikov (Sasha).

On the string.

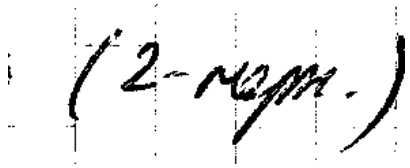
It arises from the theory of strong interactions (we neglect something).

There is a mechanical system with coordinates $x^\mu(\sigma)$, $\mu = 0, 1, \ldots, D-1$. σ — a continuous index; $\sigma \in (0, 2\pi)$ (a circle).

$$\dot{x}^\mu(\sigma) = \frac{\partial}{\partial\tau} x^\mu(\sigma) \qquad \text{(generalized velocities)},$$

$$\dot{x}^\mu(\sigma, \tau) = \frac{\partial}{\partial\tau} x^\mu(\sigma, \tau)$$

$x^\mu(\sigma, \tau)$ The trajectory is a surface (two-dimensional).



Original source details: GELFAND-P001030

Diagram and typed description. The manuscript abbreviation for "two-dimensional."

Action:

$$A[x^\mu] = \int d\tau \int d\sigma\, \mathcal{L}(x^\mu(\sigma,\tau), \dot{x}^\mu(\sigma,\tau)) \tag{1}$$

x^μ is regarded as a vector in a space with metric

$$\operatorname{diag}\underset{x^0}{(1}, \underset{x^1}{-1}, \ldots, \underset{x^{D-1}}{\underset{\ldots}{-1}})$$

Choose area as the action:

$$A[x^\mu] = \int d\sigma\, d\tau \sqrt{(\dot{x}_\mu)^2 (x'_\nu)^2 - (\dot{x}_\mu x'^\mu)^2} \tag{2}$$

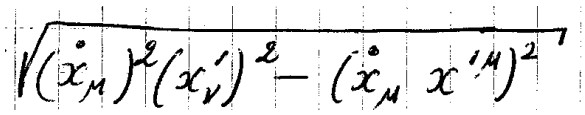


Diagram and typed description. The expression under the square root in (2).

Original source details: GELFAND-P001030

Here $x'_\mu = \dfrac{\partial}{\partial\sigma} x_\mu(\sigma,\tau)$.

$$\text{Invariance:} \quad \begin{matrix} \sigma \longrightarrow f(\sigma,\tau) \\ \tau \longrightarrow g(\sigma,\tau) \end{matrix} \quad // \quad \begin{matrix} \text{area} \\ \text{is unchanged} \end{matrix} \tag{3}$$

This is a relativistically invariant model.

Mass is proportional to length; a point interacts with its nearest neighbors. As it contracts, it reaches relativistic speeds.

Momenta:

$$p_\mu(s) = \frac{\partial \mathcal{L}}{\partial \dot{x}_\mu(\sigma)} \tag{4}$$

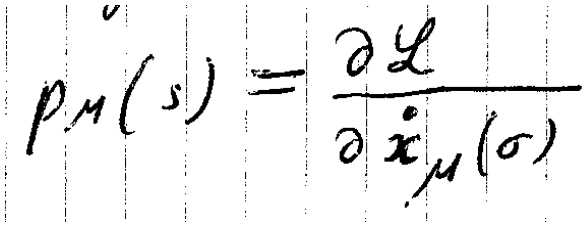
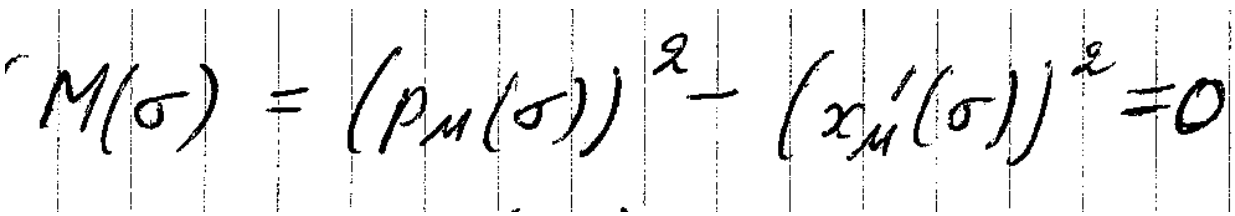


Diagram and typed description. The left side uses s and the right side uses σ.

Original source details: GELFAND-P001030

The Lagrangian is degenerate $\Rightarrow$ (4) cannot be solved for the velocities; there are relations between coordinates and momenta:

$$\begin{cases} M(\sigma) = (p_\mu(\sigma))^2 - (x'_\mu(\sigma))^2 = 0 \\ K(\sigma) = x'_\mu(\sigma) p^\mu(\sigma) = 0 \end{cases} \qquad \text{(constraints)} \tag{5}$$



Original source details: GELFAND-P001030

Diagram and typed description. The minus sign in the first constraint.

$$H = \sum p\dot{x} - L \qquad \Rightarrow \qquad H \equiv 0 \tag{6}$$

Worked on by: Barbashov, Chernikov (classical theory); Nambu — connections with dual models.

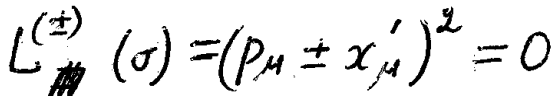
$(5), (6) \Rightarrow$ one must use Dirac's formalism (see Dirac's book).

GELFAND-P001031

$$L^{(\pm)}(\sigma) = (p_\mu \pm x'_\mu)^2 = 0 \tag{7}$$

Original source details: GELFAND-P001031

Uncertain reading. The cancelled subscript under $L^{(\pm)}$ is illegible.



Original source details: GELFAND-P001031

Diagram and typed description. The square in (7).

Standard commutation relations defining the Poisson bracket:

$$\{p_\mu(\sigma), x^\nu(\sigma')\} = \delta^\nu_\mu \delta(\sigma - \sigma') \tag{8}$$

Consider the Fourier expansion of x^ν and p_μ:

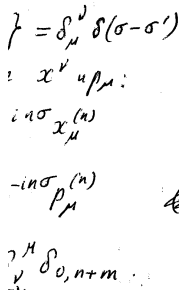
$$\begin{cases} x_\mu(\sigma) = \sum\limits_{n=-\infty}^{\infty} e^{in\sigma} x^{(n)}_\mu \\ p_\mu(\sigma) = \sum e^{-in\sigma} p^{(n)}_\mu \end{cases} \tag{9}$$

Original source details: GELFAND-P001031

Uncertain reading. Illegible cancelled entry to the right of the second line of (9).

Then

$$\{p^\mu_n, x_{m\nu}\} = \delta^\mu_\nu \delta_{0,n+m}. \tag{10}$$



Original source details: GELFAND-P001031

Diagram and typed description. The exponents $in\sigma, -in\sigma$ and the index $0, n+m$.

Expand $L^{(\pm)}(\sigma)$ from (7):

$$L^{(\pm)}(\sigma) = \sum L_n^{(\pm)} e^{in\sigma} \tag{11}$$

One can show:

$$\{L_n^{(+)}, L_m^{(-)}\} = 0, \qquad \{L_n^{(+)}, L_m^{(+)}\} = (n-m)L_{n+m}^{(+)}, \qquad n, m \in \mathbb{Z}, \tag{12}$$

(and the same upon replacing $+$ by $-$)
i.e. we obtain the algebra of vector fields on the circle.

$$\dot{p}_\mu = \{p_\mu, \widetilde{H}\}, \qquad \dot{x}_\mu = \{x_\mu, \widetilde{H}\} \tag{13}$$

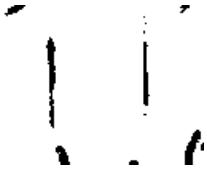


Diagram and typed description. | |

Original source details: GELFAND-P001031

$$\widetilde{H} = \alpha_+(\sigma)L^{(+)} + \alpha_-(\sigma)L^{(-)} \tag{14}$$

Changing $\alpha_\pm(\sigma)$ corresponds to gauge transformations. One can take

$$\widetilde{H} = \alpha_+(\sigma,\tau)L^{(+)} + \alpha_-(\sigma,\tau)L^{(-)} \tag{14'}$$

We must solve the equations (13) with initial data $p_\mu(\sigma,0)$, $x_\mu(\sigma,0)$, satisfying the condition:

$$L^{(\pm)}(p(\sigma,0), x(\sigma,0)) = 0 \tag{15}$$

($\Rightarrow$ they will remain satisfied)
Quantize. We obtain

$$[p_n^\mu, x_{m\nu}] = i\delta_\nu^\mu \delta_{0,n+m}, \qquad p_n^\mu, x_{m\nu} \text{ — Hermitian operators.} \tag{16}$$

A representation of these relations — ?
Introduce the operators

$$a_n^\mu = p_n^\mu + ix_{+n}^\mu, \qquad a_{-n} = a_n^+. \tag{17}$$

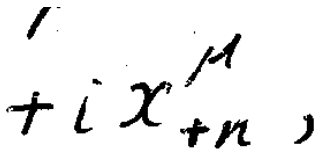


Diagram and typed description. The subscript $+n$ on x in (17).

Original source details: GELFAND-P001031

$$[a_n^\mu, a_n^{+\nu}] = \delta_{n,m}\delta_{\mu\nu}, \qquad n, m \geq 0. \tag{18}$$

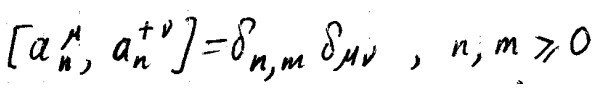


Diagram and typed description. Both operators on the left carry n. The right side has $\delta_{n,m}$.

Original source details: GELFAND-P001031

(the others commute).

GELFAND-P001032

[source number: 6]

Fock representation:

$$a_n|0\rangle = 0, \quad \forall n \geq 0. \tag{19}$$

$|0\rangle$ — the vector called the vacuum state.
All the other states should be obtained by acting with a_n^+ on $|0\rangle$.

$$[L_n^{(+)}, L_m^{(+)}] = (n-m)L_{n+m}^{(+)} + \frac{D}{12}(n^3 - n)\delta_{0,n+m}, \tag{20}$$

where

$$L_n^{(+)} = \sum_{k=-\infty}^{\infty} a_k a_{n-k} \tag{21}$$

(constraint operators)

Uncertain reading. Illegible cancelled entry before “constraint operators.”

Original source details: GELFAND-P001032

(22) Problem (quantum): find a state φ, such that $L_n^{(\pm)}|\varphi\rangle = 0$.
States of the system (stationary):

$$\begin{cases} H(p,q)|\varphi\rangle = E|\varphi\rangle \\ L^i(p,q)|\varphi\rangle = 0 \end{cases} \tag{23}$$

In (20) one must normal-order, with a_n^+ to the left of a_n.

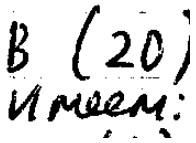


Diagram and typed description. “We have:” is written under the start of the line.

Original source details: GELFAND-P001032

We have: $L_n^{(+)}|0\rangle = 0$, $n \geq 0$, but for $n < 0$ this fails, and there is no state $|\varphi\rangle$, annihilated by all the $L_n^{(+)}$ (because of the last term in (20)).

Uncertain reading. The crossed-out word appears to read “Idea.”

Original source details: GELFAND-P001032

The result is that we satisfy $1/2$ of all the constraints, and that suffices since the matrix elements between all physical states will vanish.

(but only because there is no better option)

In the string problem there are ∞ operators commuting with all the constraints. These are the only ones that need quantizing! They have the form:

$$Q^{(n)}_{\mu_1 \ldots \mu_n} = \int_0^{2\pi} d\sigma_n \int d\sigma_1 \, d\sigma_2 \ldots d\sigma_{n-1} \, y_{\mu_1}(\sigma_1) \ldots y_{\mu_n}(\sigma_n) \qquad (24)$$
$$\sigma_n < \sigma_1 < \sigma_2 < \ldots < \sigma_n + 2\pi$$

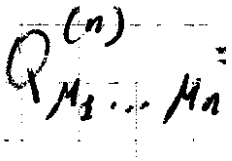


Diagram and typed description. The first subscript on Q in (24) is μ_1.

Original source details: GELFAND-P001032

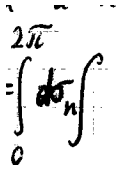


Diagram and typed description. The differential $d\sigma_n$ is written over a correction.

Original source details: GELFAND-P001032

(normal ordering is understood) in accordance with (26) below.

Here

$$y_\mu(\sigma) = p_\mu(\sigma) + i x'_\mu(\sigma)$$

Then

$$[Q, L_n] = 0 \qquad (25)$$

$$y_\mu = \sum_{n=0}^{\infty} a^n_\mu e^{in\sigma} + (a^n_\mu)^+ e^{-in\sigma} \qquad (26)$$

It is interesting to ask what commutation relations the $Q_{\mu_1 \ldots \mu_n}$ satisfy with one another.

Total momentum operator:

$$\int_0^{2\pi} p_\mu(\sigma) \, d\sigma = P_\mu \qquad (27)$$

GELFAND-P001033

$\{Q\}$ — a complete set of operators commuting with P

$$\{Q^{(n)}, Q^{(m)}\} = \sum Q^{(n+m-2)}$$

(classically)

13/IX–82 Gelfand Seminar.

A. S. Schwarz

Physicists: Witten, Wilson, Weinberg, t' Hooft

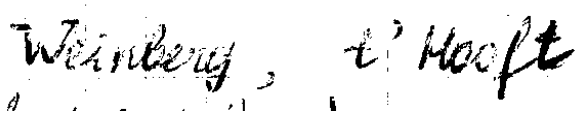


Diagram and typed description. Names written in Latin script: Weinberg, t' Hooft.

Original source details: GELFAND-P001033

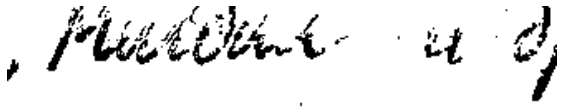
(+ Schwarz; Polyakov according to A. Migdal)

Witten — how topological invariants arise in supersymmetric models.

What is interesting to study and to hear about?

1) Gauge field theory.

String representation; Polyakov, Migdal [final stroke unclear]? and others.



Uncertain reading. Migdal [final stroke unclear]

Original source details: GELFAND-P001033

2) $SU(3), SU(2)$ and others — grand unification.

Schwarz worked on the theory of characteristic classes for supermanifolds.

3) Supersymmetry and supergravity.

— · —

Threads or strings in grand unification.

Gauge field: $A_\mu(x)$ with values in the Lie algebra of a compact Lie group

$$G_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \qquad \text{field strength} \tag{1}$$

Localization of internal symmetry.

$$L = L(\phi) = \langle \partial_\mu \phi, \partial^\mu \phi \rangle - U(\phi), \qquad \phi(x) \in \mathbb{R}^n, \quad S = \int L\, d^4x. \tag{2}$$

$G \subset SO(n)$, $\quad G$ acts on the field values $\Rightarrow$ internal symmetry.

(3) Required L — G-invariant

Sufficient: U — G-invariant.

Example: $U(\vec{\phi}) = \lambda(\vec{\phi}^2 - a^2)^2 \Rightarrow SO(3)$ model

n can be $126, 145$ (for $SO(10)$)

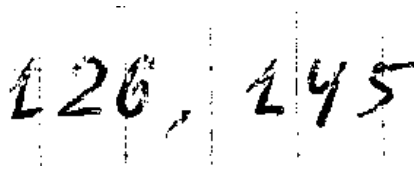


Diagram and typed description. The manuscript gives $126, 145$.

Original source details: GELFAND-P001033

Localize: add to ϕ a gauge field taking values in the Lie algebra of the group G of internal symmetry. We then treat the Lagrangian as follows:

$$\partial_\mu \to \partial_\mu + A_\mu$$

$$L \to \widetilde{L} \to \widetilde{L} + \text{const}\, \langle G_{\mu\nu}, G^{\mu\nu} \rangle \tag{4}$$

(barring cosmic miracles)

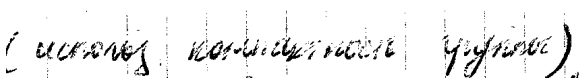
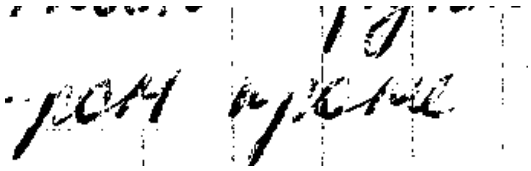


Diagram and typed description. The parenthetical remark below (4).

Original source details: GELFAND-P001033

We obtain a new Lagrangian in which the former [reading of the abbreviation]?



Uncertain reading. former [reading of the abbreviation]

Original source details: GELFAND-P001033

internal symmetries are localized symmetries.

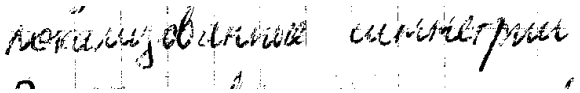
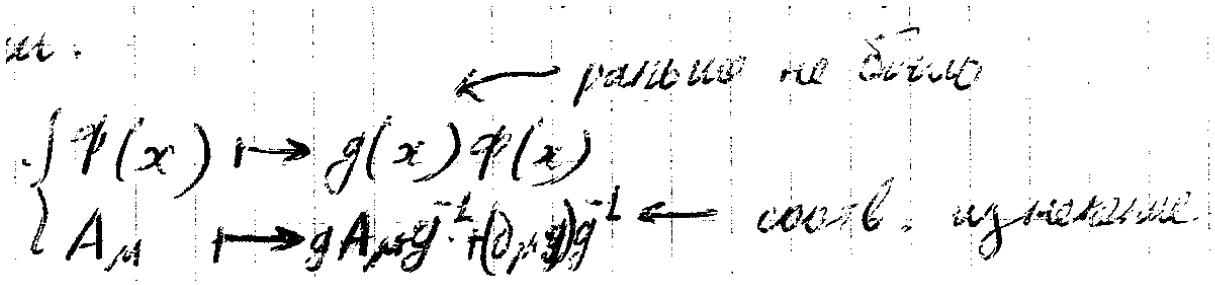


Diagram and typed description. The line ends with “localized symmetries.”

Original source details: GELFAND-P001033

We obtain invariance under

$$\begin{cases} \phi(x) \mapsto g(x)\phi(x) & \longleftarrow \text{absent before} \\ A_\mu \mapsto gA_\mu g^{-1} + (\partial_\mu g)g^{-1} & \longleftarrow \text{change accordingly} \end{cases} \tag{5}$$



Original source details: GELFAND-P001033

Diagram and typed description. In (5), a $+$ precedes the derivative of g.

GELFAND-P001034

[source number: 7]

t’ Hooft: if L is renormalizable, after localization we again obtain a renormalizable Lagrangian.

Classical analogue of the ground state: lowest energy. We obtain:

$$\phi(x) \equiv \phi, \qquad \text{where } \phi \text{ attains } \min U(\phi) \tag{6}$$

In example (3,5) one can take any $\vec{\phi}$ with $|\vec{\phi}| = a$; vacuum degeneracy

(vacuum — a two-dimensional sphere)

We obtain a set of vacua $\mathcal{R}$.

Uncertain reading. Illegible crossed-out entry before “nearby.”

Original source details: GELFAND-P001034

Nearby we expand in the deviations, retaining the quadratic part, and so on.

To agree with nature, one should take:

$$G = SU(3) \times SU(2) \times U(1) \tag{7}$$

+ add certain fermions and scalar fields

(this agrees brilliantly Illegible crossed-out entry.[?] with physics)

Uncertain reading. Illegible crossed-out entry.

Original source details: GELFAND-P001034

Furthermore, G is not simple $\Rightarrow$ extra parameters arise

(8) Idea: embed $G \subset \widetilde{G}$, so as to obtain the experimental results at low energies. — That is the idea of grand unification.

Electromagnetic subgroup $U(1) \subset SU(2) \times U(1)$

Glashow, Weinberg, Salam (before that, Gell-Mann).

Diagram and typed description. “Fritzsch” is written under “Gell-Mann” (manuscript spelling «Фритч»).

Original source details: GELFAND-P001034

Equation (7) is believed by 99% percent of physicists.

(8) is more doubtful.

Candidates for $\widetilde{G}$:

Uncertain reading. Illegible cancelled entry before $SU(5)$.

Original source details: GELFAND-P001034

$SU(5)$, $SO(10)$

still too small

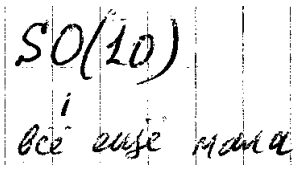
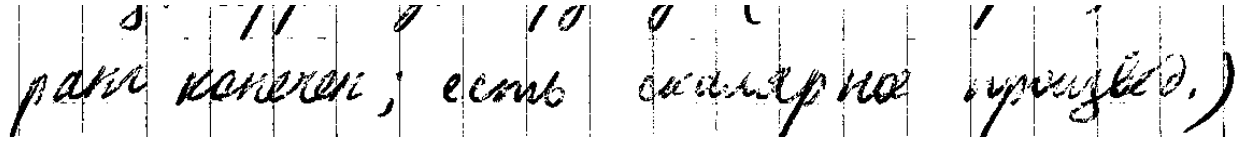


Diagram and typed description. Note under $SO(10)$.

Original source details: GELFAND-P001034

Tyupkin, Schwarz: E_8

I. M. G.'s idea: use an affine group (∞-dimensional, but in no way worse than finite-dimensional ones; the rank is finite; there is a scalar product).



Original source details: GELFAND-P001034

Diagram and typed description. At the end of the parenthesis: "there is a scalar product."

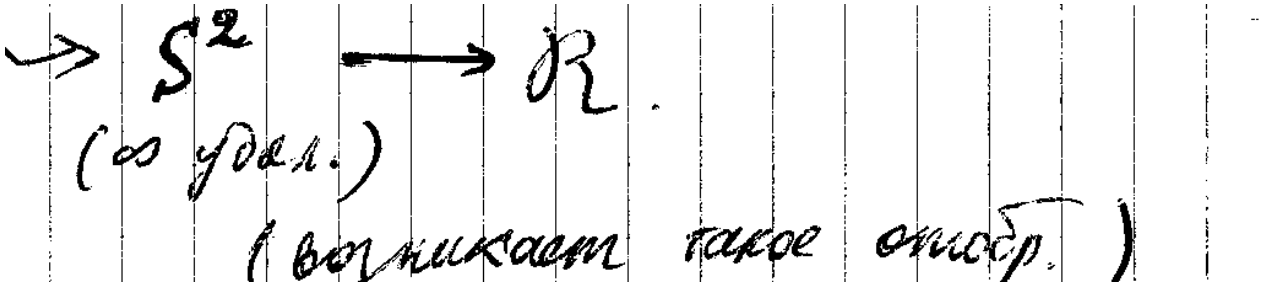
Sch. min possible rank = rank of $G = 4$.

I. M. G. Affine groups are defined using functions of one variable (cf. strings!)

— · —

Sch.: fields of finite energy must tend at ∞ to vacuum values

$$\phi(\vec{x}) \text{ on } \mathbb{R}^3 \quad \rightsquigarrow \quad \underbrace{S^2 \to \mathcal{R}}_{\substack{\text{(at infinity)} \\ \text{(such a map arises)}}} \tag{9}$$



Original source details: GELFAND-P001034

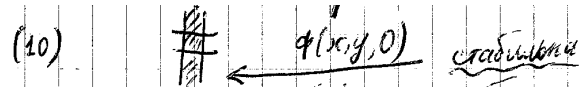
Diagram and typed description. Annotations below S^2 and the map into $\mathcal{R}$.

$\Rightarrow$ there are topological classes of fields; $\pi_2(\mathcal{R})$ — an element of this group arises.

$\Rightarrow$ magnetic monopoles.

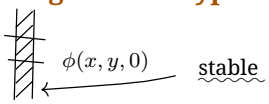
There are topologically nontrivial threads with finite energy density.

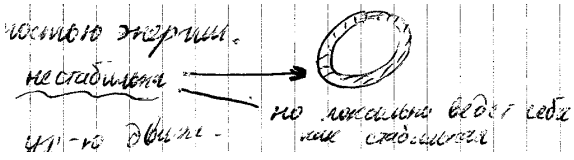
(10)



Original source details: GELFAND-P001034

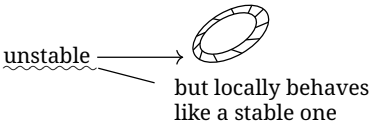
Diagram and typed description.





Original source details: GELFAND-P001034

Diagram and typed description.



One can require the thread to satisfy the equation of mo-

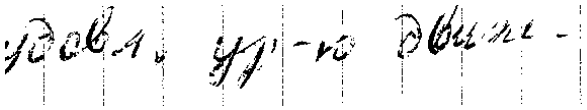


Diagram and typed description. The word breaks off at “mo-” at the end of the frame.

Original source details: GELFAND-P001034

GELFAND-P001035

Wonders with a thread: there is a notion of topological nontriviality.

(11) Thread: the field $\phi(x,y,t)$ (i.e. it depends on only two spatial variables)

Uncertain reading. Illegible cancelled entry before the abbreviated “functional.”

Original source details: GELFAND-P001035

The energy functional per unit of the coordinate z must be finite.

$$\int \mathcal{E}(x,y)\,dx\,dy < \infty$$

Diagram and typed description. The manuscript abbreviates “functional” as «ф-л».

Original source details: GELFAND-P001035

($\mathcal{E}$ — density, gives the energy)

Diagram and typed description. The gloss reads “density, gives the energy.”

Original source details: GELFAND-P001035

In (10) the region where the field is nontrivial is hatched.

(as $e^{-\alpha x^2}$ is concentrated in a region of size $1/\sqrt{\alpha}$).

(12)

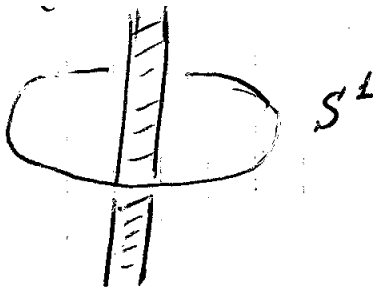
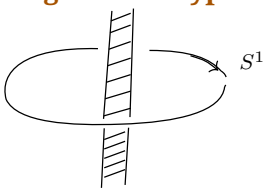


Diagram and typed description.



Original source details: GELFAND-P001035

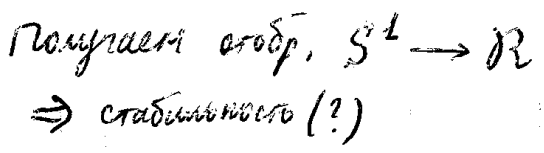


Diagram and typed description. We obtain a map $S^1 \to \mathcal{R}$.
$\Rightarrow$ stability (?)

Original source details: GELFAND-P001035

(13) On going around a thread, a particle may change type (become another particle)
Example: it may change its electric charge.

Diagram and typed description. | |

Original source details: GELFAND-P001035

$\forall$ vacuum $\phi \in \mathcal{R}$ $\exists$ a group of unbroken symmetries H_ϕ (preserving ϕ)
Assumption:

$$H_\phi = K \times U(1) \subset \widetilde{G} \tag{14}$$

(here $U(1)$ — the electromagnetic group).
Let $\widetilde{G}$ contain an element $\alpha \in \widetilde{G}$, which reverses $U(1)$:

$$\alpha u \alpha^{-1} = u^{-1} \qquad \text{for } u \in U(1) \tag{15}$$

(16) Claim. Then $\exists$ a topologically nontrivial thread (and even a solution) around which a particle changes its charge.

— · —

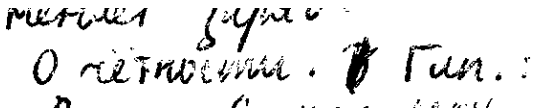
On parity.

Uncertain reading. The crossed-out mark is illegible.

Original source details: GELFAND-P001035

Hypothesis: mirror matter exists.

Diagram and typed description. “On parity,” then the abbreviation for “Hypothesis.”



Original source details: GELFAND-P001035

Inside the Sun there is a planet made of mirror matter. It interacts weakly but exerts a gravitational influence $\Rightarrow$ this explains certain experiments.
(17) Claim. 1) $\exists$ grand unifications that include mirror particles
2) Illegible cancelled entry after “2).”[?]

Uncertain reading. Illegible cancelled entry after “2).”

Original source details: GELFAND-P001035

∃ threads around which particles turn into mirror particles.

The interaction of our particles with mirror particles is weak (interaction through heavy bosons; if we neglect it — the Lagrangian splits).

Original source details: GELFAND-P001035

Diagram and typed description. The start of “neglect” is written over earlier strokes.

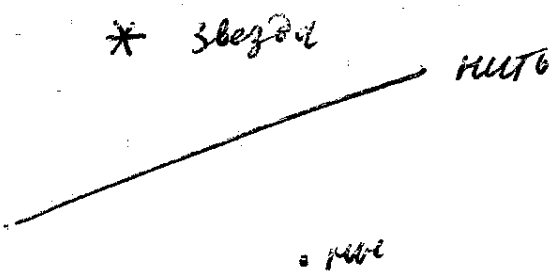
These are not antiparticles. There are mirror photons, mirror charge, and so on.

The coupling is very weak.

Thread of such unifications. Effect.

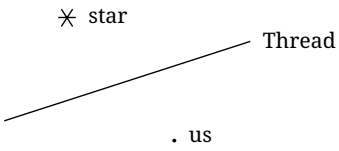
Original source details: GELFAND-P001035

Diagram and typed description. The manuscript underlines “Thread of such unifications.”



Original source details: GELFAND-P001035

Diagram and typed description.



We stop seeing the star because its photons

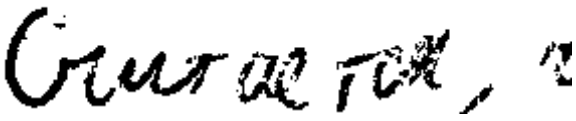
Original source details: GELFAND-P001035

Uncertain reading. Illegible cancelled entry before the abbreviated “become.”

become mirror photons.

The dynamics is semiclassical, which applies because we are far from the thread.

The threads are thought to have survived from the early universe.



Original source details: GELFAND-P001035

Diagram and typed description. The last sentence begins “It is thought.”

GELFAND-P001036

[source number: 8]

It is unclear whether there are quantum-mechanically stable threads (classically they are stable).

As the thread moves, the stars should be replaced (by mirror ones).

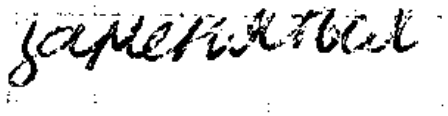


Diagram and typed description. The verb means “to be replaced.”

Original source details: GELFAND-P001036

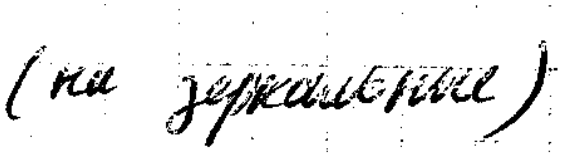
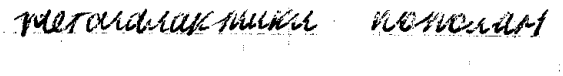


Diagram and typed description. In parentheses: “by mirror ones.”

Original source details: GELFAND-P001036

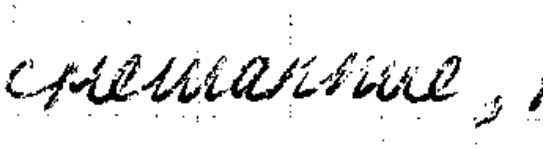
Khlopov: metagalaxies in half [tentative reading]?



Uncertain reading. metagalaxies in half [tentative reading]

Original source details: GELFAND-P001036

and nothing will change; the only chance — quasars; they are special [tentative reading]?,



Uncertain reading. special [tentative reading]

Original source details: GELFAND-P001036

but they have a characteristic rhythm of variation.

There is a preprint: one quasar over 100 years sharply changed its luminosity.

Diagram and typed description. The underlined word reads “preprint.”

Original source details: GELFAND-P001036

Khlopov’s remark: it is explained, if a thread passed.

GELFAND-P001037

20/IX-82. Gelfand Seminar.

Sinai.

Topics (in physics)

1. Renormalization group.

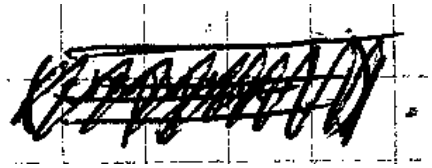
2. Debye screening as the principal mechanism of phase transitions in systems with continuous symmetry.

3. Gauge models.

4. Gravity, supersymmetry, supergravity.

A. A. Migdal.

5. Lattice models.



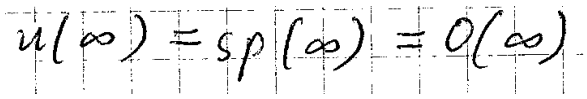
Original source details: GELFAND-P001037

Uncertain reading. Illegible cancelled expression in parentheses.

6. Strings.

7. Large N (N — rank of the gauge group):

$$u(\infty) = \mathrm{sp}(\infty) = O(\infty)$$



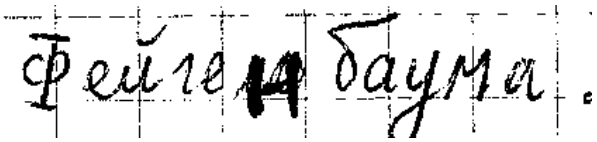
Original source details: GELFAND-P001037

Diagram and typed description. Manuscript notation: $u(\infty), \mathrm{sp}(\infty), O(\infty)$.

8. Quarks and leptons. Unified theories.

9. Anderson localization.

10. Feigenbaum universality.



Original source details: GELFAND-P001037

Diagram and typed description. The n in the handwritten surname “Feigenbaum” is written over earlier strokes.

Sinai on item 10.

In the one-dimensional case: Let

$$\begin{aligned} [-1,1] &\longrightarrow [-1,1] \\ x &\longmapsto f(x;\mu) \end{aligned}$$

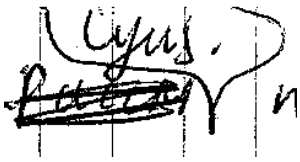
a) $f(-x;\mu) = f(x;\mu)$

b) $f(0;\mu) = 1.$

[At right:] Example: $f(x,\mu) = 1 - \mu x^2$, $1 < \mu < 2$.

GELFAND-P001038

[source number: 9]



Original source details: GELFAND-P001038

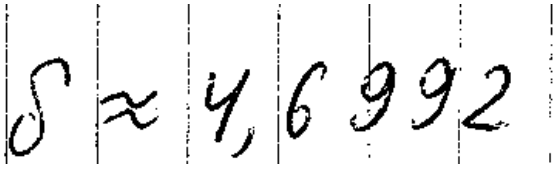
Uncertain reading. The cancelled word may be the abbreviation for “Consider.”

[inserted above the line] There is a sequence of values μ_n:

$$\forall \mu: \quad \mu_n < \mu < \mu_{n+1} \quad \Rightarrow \quad \exists \text{ a stable orbit of period } 2^n. \tag{1}$$

Consider

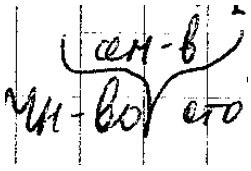
$$\frac{\mu_n - \mu_{n+1}}{\mu_{n-1} - \mu_n} \underset{n\to\infty}{\longrightarrow} \delta^{-1}, \qquad \delta \approx 4{,}6992 \tag{2}$$



Original source details: GELFAND-P001038

Diagram and typed description. The manuscript reads $\delta \approx 4{,}6992$.

The set of [inserted above the line] families of maps for which $\exists$ a sequence μ_n — an open set in the space of maps.

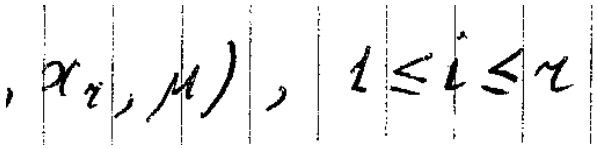


Original source details: GELFAND-P001038

Diagram and typed description. The abbreviation for “families,” with an insertion mark.

There is a generalization for differential equations.

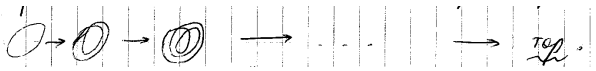
$$\frac{dx_i}{dt} = f(x_1, \ldots, x_r, \mu), \qquad 1 \le i \le r \tag{3}$$



Original source details: GELFAND-P001038

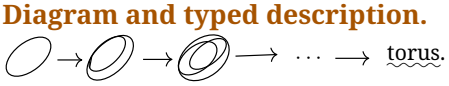
Uncertain reading. The subscript of the last x resembles the r in the upper bound. The strokes are small.

(4) μ_n, $\mu_n < \mu < \mu_{n+1} \Rightarrow$ there is a periodic orbit attracting everything, except a finite number of closed orbits.



Original source details: GELFAND-P001038

Diagram and typed description.



The exact joins in the two middle contours are unclear.

$$\mu_n - \mu_\infty \sim C(f)\delta^{-n}, \tag{5}$$

where δ — the same as above, and the set of such systems is open.

More parameters.

(6) $\mu_1, \mu_2 \Rightarrow$ period tripling occurs (and one also finds a const).

— · —

For the function f there is a complicated functional equation:

$$-\frac{1}{\alpha} f(f(\alpha x)) = f(x) \tag{7}$$

It is solved: with a polynomial (degree 10) and numerically on a computer.

The renormalization group arises from linearizing this equation.

(8) Question: what happens when $\mu > \mu_\infty$ for the vector field?

For maps, it is known that a map with an ergodic invariant measure arises (mixing and so forth $\Rightarrow$ onset of stochasticity).

A. A. Migdal.

Given a field $\varphi(x)$ with values in the algebra $u(N)$, i.e. (φ_{ij}) — an anti-Hermitian matrix,

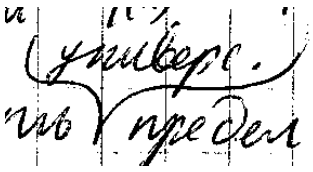
$$\varphi_{ij}(x) = -\varphi^*_{ji}(x) \qquad \text{or} \qquad \varphi(x) = -\varphi^+(x), \qquad \text{(for definiteness)} \tag{1}$$

As $N \to \infty$ there is a [inserted above the line] universal limit (non-Gaussian!) and for the limit there are certain equations and

Original source details: GELFAND-P001038

Uncertain reading. Illegible cancelled entry before “observations.”

observations.



Original source details: GELFAND-P001038

Diagram and typed description. The abbreviation for “universal” is inserted above the line.

$$L = \text{tr}\left[\frac{1}{2}(\partial_\mu \varphi)^2 + U(\varphi)\right] N/\lambda, \tag{2}$$

$U(\varphi)$ — a polynomial of degree 4 — for example, φ^4.

(3) • The classical equation $\dfrac{\delta S}{\delta \varphi} = 0$, where

$$S = \int L\, d^4x \tag{4}$$

(the space is Euclidean)

Consider the equation

$$\frac{\partial \varphi}{\partial \tau} + \frac{\delta S}{\delta \varphi} = J(x, \tau), \tag{5}$$

Original source details: GELFAND-P001039

Diagram and typed description. random current = Gaussian random noise

where

$$\langle J^{ik}(x,\tau) J^{i'k'}(x',\tau') \rangle = \delta^{ii'} \delta^{kk'} \delta(x-x') \delta(\tau - \tau') \tag{6}$$

(this describes them completely, since they are Gaussian)

(7) Claim. Let $\varphi|_{\tau=0} = 0$. Then the solution $\varphi(x,\tau)$ has the property that if

$$\overline{Q} = \frac{1}{T} \int_0^T Q(\varphi(x,\tau))\, d\tau$$

[At right:] (Q — a functional; e.g. $Q = \mathrm{tr}(\varphi^2)$). invariant under the group for the adjoint action

$$\varphi \longmapsto S^{+} \varphi S$$

then $\overline{Q}$ is given by the formula

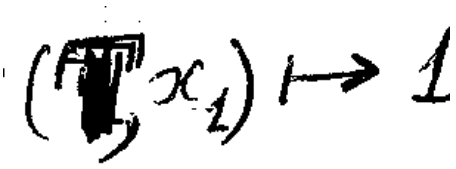
$$\overline{Q} = \int D\varphi \, \exp(-S)\, Q(\varphi(x)) \Big/ \int D\varphi \, e^{-S} \tag{8}$$

Instead of (8) we consider (5).

① Self-averaging. Consider

$$G(1, \ldots, n) = \frac{\mathrm{tr}}{N}(\varphi(1) \ldots \varphi(n)) \tag{9}$$

Here $(T, x_1) \mapsto 1, \ldots$ (for brevity); T is fixed.

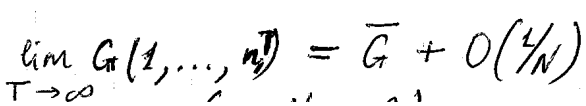


Original source details: GELFAND-P001039

Uncertain reading. The first character in the pair is overwritten and most likely reads T.

Then

$$\lim_{T \to \infty} G_T(1, \ldots, n, T) = \overline{G} + O(1/N), \qquad (\text{as } N \to \infty) \tag{10}$$



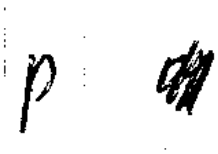
Original source details: GELFAND-P001039

Uncertain reading. A mark beside the base of G resembles T. After n there is an oblique stroke and a raised mark resembling T.

[At right:] here $\overline{G}(1,\dots,n)$ — a function of the points, independent of T and N.
(decrease in fluctuations).
The points coincide: $\frac{1}{N}\operatorname{tr}\varphi^n = \frac{1}{N}\sum_\lambda \lambda^n$ and the variance decreases.
In general there arises a $\sum$ of terms with identical probability distributions.
Thus: traces do not fluctuate.
② At $N=\infty$ a reduction is possible:

$$\varphi(x) \longmapsto e^{ipx}\varphi(0)e^{-ipx}, \qquad p = i\,\partial/\partial x \tag{11}$$

p An illegible mark after p is crossed out.?

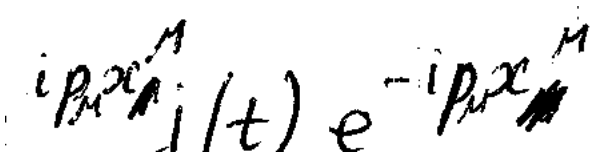


Uncertain reading. An illegible mark after p is crossed out.

Original source details: GELFAND-P001039

can be replaced by a matrix in the Cartan algebra (diagonal)

$$\widetilde{J}(x,t) = e^{ip_\mu x^\mu} j(t) e^{-ip_\mu x^\mu}, \tag{12}$$



Uncertain reading. In both exponents the x subscripts are crossed out and μ is written above. The cancelled subscripts are illegible.

Original source details: GELFAND-P001039

[At right:] p_μ — a diagonal matrix, $p = \operatorname{diag}(p_\mu^1,\dots,p_\mu^N)$
$j(t)$ — Gaussian noise

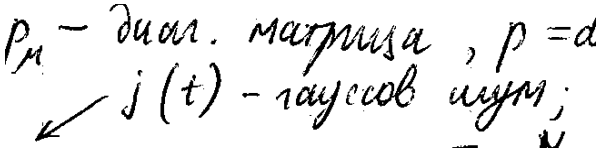


Diagram and typed description. $j(t)$ — Gaussian noise. ↙ to (13)

Original source details: GELFAND-P001039

Uncertain reading. A cancelled entry on the right begins with p.

Original source details: GELFAND-P001039

p^μ — a random quantity (independent of both t, and x)

Uncertain reading. The superscript resembles μ. The lower stroke may be a cancelled index.

Original source details: GELFAND-P001039

$$\langle j^{ik}(t), j^{i'k'}(t')\rangle = \delta(t-t')\delta^{ii'}\delta^{kk'} \tag{13}$$

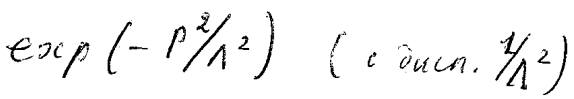
GELFAND-P001040

[source number: 10]

p^j — random variables, distributed [inserted above] (according to a Gaussian law) with density $\exp(-p^2/\Lambda^2)$ (with variance $1/\Lambda^2$)

Original source details: GELFAND-P001040

Uncertain reading. The superscript resembles j. It may be i.



Original source details: GELFAND-P001040

Diagram and typed description. The manuscript gives $\exp(-p^2/\Lambda^2)$ and "with variance $1/\Lambda^2$."

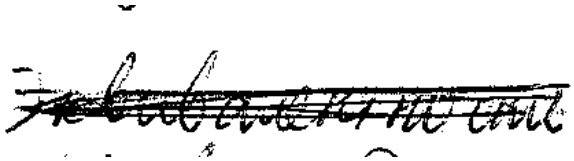
where $\underset{\text{cutoff radius}}{\Lambda} \to \infty$ (after $N \to \infty$).

In the limit as

$$N \to \infty$$
$$\Lambda \to \infty$$

we get the same white noise $J(x,t)$ (see above).

It reduces to an ordinary equation (a random matrix).



Original source details: GELFAND-P001040

Diagram and typed description. ~~Equivalence~~

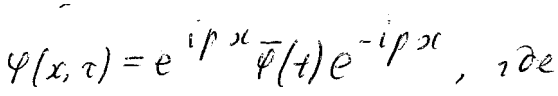
Assertion.

$$\underset{\text{with } J}{G} - \underset{\text{with } \widetilde{J}}{\widetilde{G}} \to 0$$

$$\frac{\partial \widetilde{\varphi}}{\partial \tau} + \frac{\delta S}{\delta \widetilde{\varphi}} = \widetilde{J}(x,\tau) \tag{14}$$

The solution is:

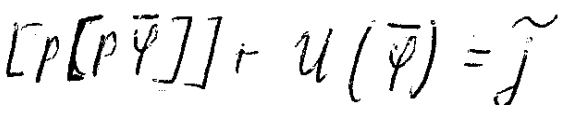
$$\varphi(x,\tau) = e^{ipx}\bar{\varphi}(t)e^{-ipx}, \qquad \text{where} \tag{15}$$



Original source details: GELFAND-P001040

Diagram and typed description. The left side uses τ. The argument of $\bar{\varphi}$ is t.

$$\frac{\partial \bar{\varphi}}{\partial \tau} + [p[p\bar{\varphi}]] + U(\bar{\varphi}) = \widetilde{j} \tag{16}$$



Original source details: GELFAND-P001040

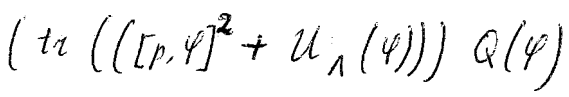
Diagram and typed description. There are no visible commas in the brackets or prime on U.

$$p = \mathrm{diag}(p_1, \ldots, p_N)$$

Thus the dependence on the coordinates has been eliminated.

The integrals pass into the integral:

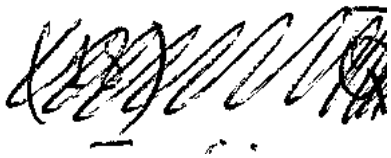
$$\bar{Q} = \int dp \int d\varphi \, \exp\big(\mathrm{tr}\big(([p,\varphi]^2 + U_\Lambda(\varphi))\big)Q(\varphi) \tag{17}$$



Original source details: GELFAND-P001040

Diagram and typed description. One opening parenthesis after exp, two after tr, and three closing ones after φ in $U_\Lambda(\varphi)$.

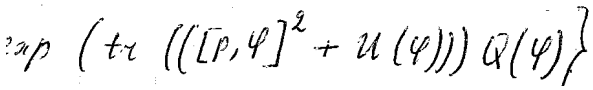
More precisely,



Original source details: GELFAND-P001040

Uncertain reading. The cancelled line may have started with (18) and $\bar{Q}$.

$$\begin{array}{l} \bar{Q} \\ \uparrow \\ \text{along trajectories (see above)} \end{array} = \lim_{\Lambda\to\infty}\lim_{N\to\infty} M_p \left\{\int_{U(N)} d\varphi \, \exp\big(\mathrm{tr}\big(([p,\varphi]^2 + U(\varphi))\big)Q(\varphi)\right\} \tag{18}$$

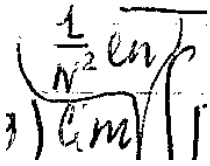


Original source details: GELFAND-P001040

Diagram and typed description. The parentheses follow the same pattern as in (17).

To be calculated:

$$\lim_{N\to\infty} \frac{1}{N^2} \ln \int_{U(N)} \prod_{\mu=1}^{4} dU_\mu \, \exp\Big(-\beta N \sum_{\mu,\nu} \mathrm{tr}(U_\mu U_\nu U_\mu^+ U_\nu^+)\Big) = F(\beta) \tag{19}$$



Original source details: GELFAND-P001040

Diagram and typed description. The manuscript inserts $\frac{1}{N^2}\ln$ before the integral.

(free energy)

dU_μ — Haar measure

(Partition function of lattice theory [inserted above] (field) as $N \to \infty$ in four-dimensional space)

This is $\sim$ field theory

Evaluated:

$$\int dU \ \exp\bigl(\beta N \operatorname{tr}(Uh + h^+U^+)\bigr) \tag{20}$$

for an arbitrary matrix h

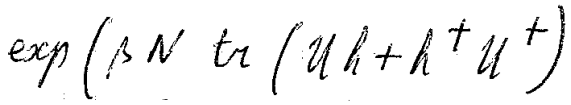


Diagram and typed description. Only one closing parenthesis is visible at the end of the formula.

Original source details: GELFAND-P001040

(a complicated expression)

First. Pirizen, Gross; then Fateev.

Uncertain reading. The first surname appears to read “Pirizen.” Its first vowel is unclear.

Original source details: GELFAND-P001040

GELFAND-P001041

27/IX-82 Gelfand Seminar.

Fuchs.

Representations of the diffeomorphism group of the circle.

(1) Consider the algebra $\{e_i,\ i \in \mathbb{Z}\}$

$$[e_i, e_j] = (j - i)e_{i+j}.$$

(of vector fields on the circle)

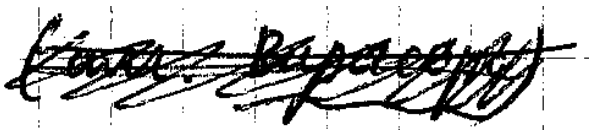


Diagram and typed description. [At right:] ~~(Virasoro alg.)~~

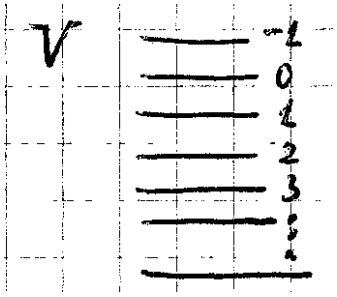
Original source details: GELFAND-P001041

There is a unique central extension (Gelfand–Fuchs):

$$\{e_i, z\}, \qquad [e_i, e_j] = \left[\begin{cases} (j-i)e_{i+j}, & i+j \neq 0, \\ \text{—"—} + \frac{1}{12}(j^3 - j)z, & i+j = 0, \end{cases} \right. \qquad \left. [e_i, z] = 0, \quad \forall i \right] \qquad \begin{matrix} \text{Virasoro} \\ \text{algebra} \end{matrix} \tag{2}$$

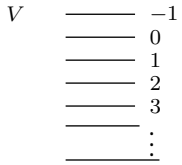
Representations of the Virasoro algebra.

(3) Verma modules,

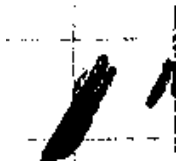


Original source details: GELFAND-P001041

Diagram and typed description.



graded The cancelled mark is illegible.[?]



Original source details: GELFAND-P001041

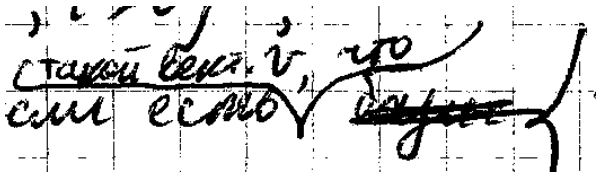
Uncertain reading. The cancelled mark is illegible.

modules bounded below

(graded and bounded on one side)

(4) Subalgebra $\{e_i,\ i > 0\}$;

Verma module: if there is [inserted above] a vector v, such that



Original source details: GELFAND-P001041

Diagram and typed description. ~~a basis~~

$$\{v,\ e_1v,\ e_1^2v,\ e_2v,\ e_1^3v,\ e_2e_1v,\ e_3v,\ \ldots\} \quad —$$

(5) — a basis, and $e_{-j}v = 0$ for $j > 0$. (it suffices that $e_{-1}v = e_{-2}v = 0$)

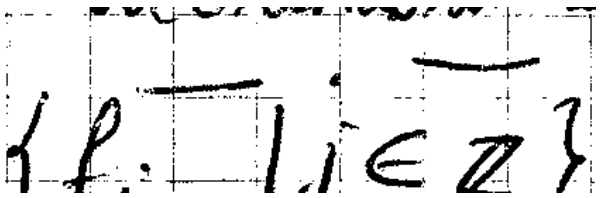
Fix complex numbers h and c.

$$e_0v = hv, \quad zv = cv \tag{6}$$

(additional condition). —

From h, c it is recovered uniquely. Notation $V_{h,c}$.

Construct a module $\{f_j \mid j \in \mathbb{Z}\}$



Uncertain reading. Two strokes appear above the entry, one to the right of f_j and one above $\mathbb{Z}$. Their relation to the entry is unclear.

Original source details: GELFAND-P001041

(auxiliary) by the formulas:

$$e_if_j = [j + \mu - (i+1)\lambda]f_{i+j} \tag{7}$$

(One can picture:

$$e_i = z^{i+1}\frac{d}{dz}, \quad f_j = z^{j+\mu}dz^{-\lambda}) \tag{8}$$

Denote it by $\overline{W}_{\lambda,\mu}$

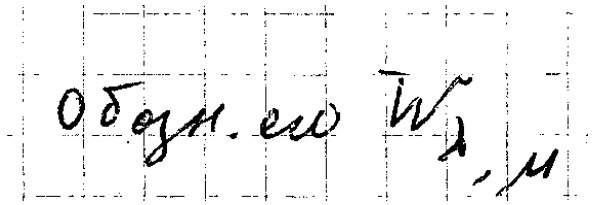


Diagram and typed description. A bar is visible over W.

Original source details: GELFAND-P001041

Consider the sequences.

$$\begin{gathered} \cdots \wedge f_{j_3} \wedge f_{j_2} \wedge f_{j_1} \\ \cdots < j_3 < j_2 < j_1 \\ \cdots \wedge f_{-4} \wedge \underset{-3}{f_2} \wedge \underset{-2}{f_5} \wedge \underset{-1}{f_9} \end{gathered} \tag{9}$$

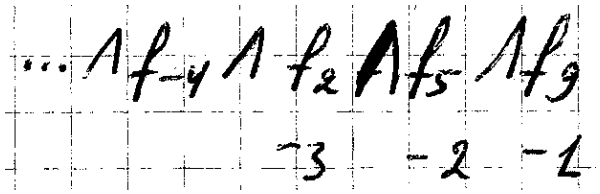


Diagram and typed description. The final factor is f_9, with -1 beneath it.

Original source details: GELFAND-P001041

(the numbers run consecutively up to a certain point; if the subsequent ones are replaced by

consecutive ones, then The short cancelled entry is illegible.?

Uncertain reading. The short cancelled entry is illegible.

Original source details: GELFAND-P001041

it ends at -1

e_i is applied successively from the right; one obtains a sum of infinitely many terms.

If the resulting f_{i+j} is already present, the product is 0.

GELFAND-P001042

[source number: 11]

In fact, one obtains a finite sum.

This defines e_i for $i \neq 0$, and e_0 cannot be defined this way.

$$e_0 = [e_{-1}, e_1] \quad \text{— define by this formula } e_0.$$

$$e_0 = [e_{-2}, e_2] \quad \text{— a different coefficient results! — contradiction.}$$

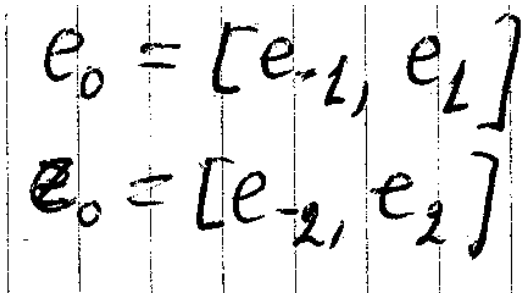


Diagram and typed description. The initial symbol of the second equality has been overwritten and reads as e_0.

Original source details: GELFAND-P001042

But one can determine e_0 and z simultaneously (and uniquely!).

z — multiplication by a number.

(see the work of Feigin and Fuchs; ΦA, 1982, $N = 2$)

e_0 — multiplication by $h + n$, $\quad h = \frac{1}{2}\mu(\mu - 2\lambda + 1)$. (?)

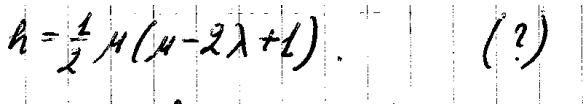


Diagram and typed description. The manuscript has $+1$ and "(?)" at right.

Original source details: GELFAND-P001042

(10) Highest-weight vector:

$$\cdots f_{-3} \wedge f_{-2} \wedge f_{-1}$$
(all consecutive)

If irreducible, then it is a Verma module.

Kac. (Lect. Notes Phys., 1979):

Which Verma modules $\overline{V}_{h,c}$ are irreducible?

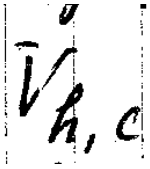


Diagram and typed description. A bar appears over V in the question.

Original source details: GELFAND-P001042

$$\overline{\Phi}_{k_1,k_2}(h,c) = \left(h - \frac{k_1^2-1}{24}(c-13) + \frac{k_1k_2-1}{2}\right)\left(h - \frac{k_2^2-1}{24}(c-13) + \frac{k_1k_2-1}{2}\right) + \\ + \frac{(k_1^2-k_2^2)^2}{16} \tag{11}$$

Diagram and typed description. A bar over Φ, and a minus sign before the first fraction.

Original source details: GELFAND-P001042

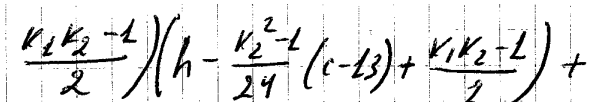
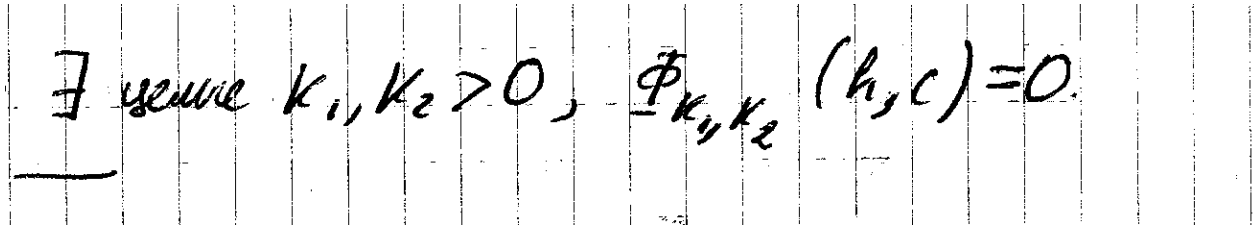


Diagram and typed description. The sign before the last 1 crosses a vertical grid line. It reads as a minus.

Original source details: GELFAND-P001042

(12) Thm. $V_{h,c}$ is reducible $(\Leftrightarrow)$ $\exists$ integers $k_1, k_2 > 0$, $\overline{\Phi}_{k_1,k_2}(h,c) = 0$.



Original source details: GELFAND-P001042

Uncertain reading. A line is visible below Φ. The short top stroke may be part of the letterform.

— . —

$$c = 0 \quad \Rightarrow \quad h = \frac{m^2 - 1}{24}$$

— . —

2nd construction.

(13) Consider polynomials in the variables

$$\begin{array}{rcccc} & x_1, & x_2, & x_3, & \dots \\ \text{of weights} & 1, & 2, & 3, & \dots \end{array}$$

There are two transformations: multiplication by x_i and $\partial/\partial x_i$.

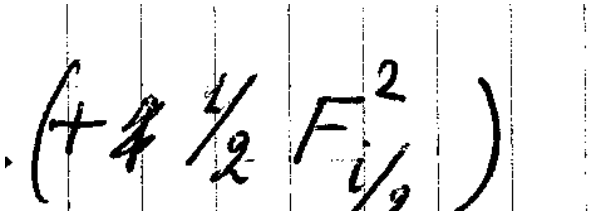
Set

$$F_k = \sum_i x_i \frac{\partial}{\partial x_{i+k}}. \tag{14}$$

Then

$$e_i = \alpha F_i + \sum_{\substack{p,q \\ p\neq 0,\ q\neq 0 \\ p+q=i \\ p>q}} F_p F_q \left(+\tfrac{1}{2} F_{i/2}^2\right), \quad \alpha \text{ — parameter.} \tag{15}$$

(if i is even)



Uncertain reading. Between $+$ and $1/2$ there is a cancelled or overwritten mark, possibly a 2.

Original source details: GELFAND-P001042

Here

$$\alpha = \mu - i(\lambda + \tfrac{1}{2}) \tag{16}$$

GELFAND-P001043

Zamolodchikov. (Sasha).

(1) Let there be a surface $\vec{x}(\xi^1, \xi^2)$, ξ^j — parameters.

We want to study sums over surfaces of different shapes.

Metric:

$$G_{ab} = \partial_a \vec{x} \cdot \partial_b \vec{x}, \qquad a, b = 1, 2. \tag{2}$$

The surface area has the form

$$A = \int \sqrt{\det G_{ab}}\, d\xi \tag{3}$$

The expression A is gauge invariant under $\xi^a \to f^a(\xi)$.

By a choice of gauge one can achieve:

$$G_{ab} = \rho \delta_{ab} \tag{4}$$

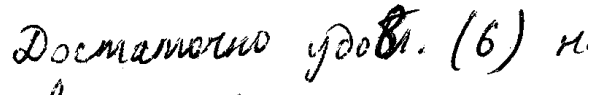
Equation of motion with Lagrangian A:

$$\partial_a \partial^a \vec{x} = 0, \qquad \text{(Laplace equation)} \tag{5}$$

$$\begin{cases} \partial_1 \vec{x} \cdot \partial_2 \vec{x} = 0, \\ (\partial_1 \vec{x})^2 = (\partial_2 \vec{x})^2 \end{cases} \tag{6}$$

(from (4)).

It suffices to satisfy (6) at the boundary $\Rightarrow$ the solution gives a minimal surface.



Original source details: GELFAND-P001043

Uncertain reading. The ending of the abbreviation for "satisfy" has been overwritten. The original strokes are indistinct.

We want to evaluate, in some sense, the sum

$$\sum_{\text{surfaces}} e^{-\mu_0^2 A(\text{surface})} \tag{7}$$

Measure — ?

Consider first

$$\sum_{\text{paths}} e^{-m L_{\text{path}}} \tag{8}$$

L_{path} — length of the path (in the usual sense; in d-dimensional Euclidean space)

For the analogy, introduce a parametrization $\vec{x}(t)$, $t \in (1, 2)$. Then

$$L = \int_1^2 \sqrt{(\dot{\vec{x}})^2}\, dt, \qquad \dot{\vec{x}} = \frac{d\vec{x}}{dt} \tag{9}$$

Minimizing (9), we get straight lines:

$$\ddot{\vec{x}} = 0 \qquad \text{(10) — analogue of (5)} \tag{10}$$

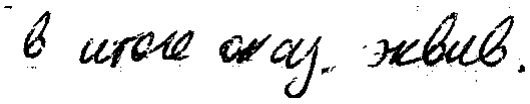
Freedom (gauge) under substitutions $t \mapsto f(t)$

$$\begin{cases} \dot{\vec{x}} = \vec{v} \\ \vec{v}^2 = 1 \end{cases} \qquad \text{(11) — analogue of (6)} \tag{11}$$

GELFAND-P001044

[source number: 12]

There are several ways to define the integral. They all turn out equivalent in the end.



Original source details: GELFAND-P001044

Uncertain reading. The final abbreviation reads as “оказ.” and may read “окаж.”. Both suggest “turn out.”

One:

(12) method replace R^d by a lattice.

$$x^\mu = 0, \pm 1, \pm 2, \dots$$

Length = number of steps.

The sum converges and can be evaluated by elementary means (explicitly!)

Another

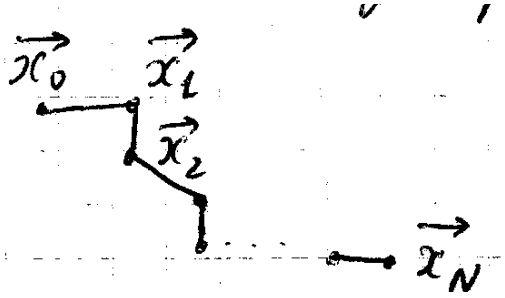
(13) method Replace the curve by a polygonal path of N links of length Δ.

Let Δ be fixed. The result is a finite-dimensional integral.

(a system of hinged rods).

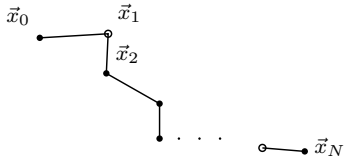
Then sum over N.

(14)



Original source details: GELFAND-P001044

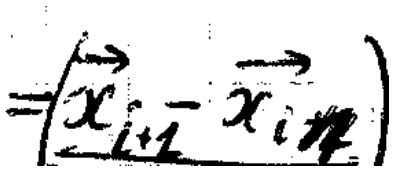
Diagram and typed description.



$$|\vec{x}_i - \vec{x}_{i+1}| = \Delta$$

Consider unit vectors

$$\vec{n}_i = \left(\frac{\vec{x}_{i+1} - \vec{x}_i}{\Delta} \right).$$



Original source details: GELFAND-P001044

Uncertain reading. After the index of the second $\vec{x}$, a symbol or group of symbols is crossed out, possibly $+1$.

$$\vec{x}_1 = \vec{x}_0 + n_1\Delta,$$
$$\vec{x}_2 = \vec{x}_0 + (\vec{n}_1 + \vec{n}_2)\Delta \qquad \sum \vec{n}_i = \frac{x_1 - x_2}{\Delta}$$
$$\cdots\cdots\cdots$$

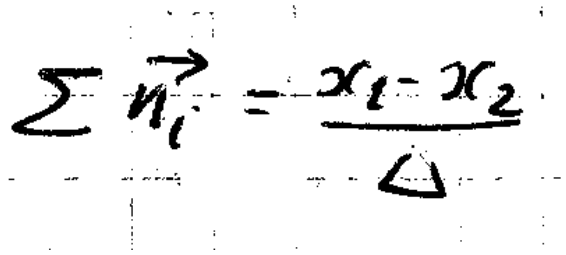


Diagram and typed description. The numerator reads $x_1 - x_2$ with no arrows.

Original source details: GELFAND-P001044

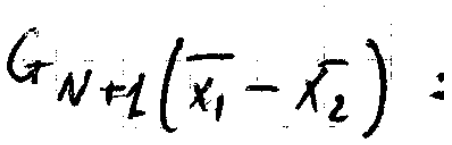
$\sum \vec{n}_i$ is given.

$\int$ over unit vectors. We are interested in the dependence on x_0 and x_N.

$$G_N(\vec{R}) = \int \prod_{i=1}^{N} d\vec{n}_i \, \delta\left(\sum \vec{n}_i - \frac{\vec{R}}{\Delta}\right), \qquad \vec{R} = \vec{x}_N - \vec{x}_0 \tag{15}$$

Recurrence relation:

$$G_{N+1}(\bar{x}_1 - \bar{x}_2) = \cdots \qquad \text{in terms of } G_N \text{ write out.}$$



Uncertain reading. There are short strokes over x_1, x_2 with no discernible arrowheads.

Original source details: GELFAND-P001044

$$G(x_1 - x_2) = \sum_{\text{paths}} e^{-N\Delta m}, \qquad G_N(x_1 - x_2) = \sum_{\substack{\text{paths} \\ \text{with } N \text{ links}}} \tag{16}$$

Consider the Fourier components

$$G(\vec{p}) = \int G(\vec{x}) e^{i\vec{p}\vec{x}} d\vec{x} \tag{17}$$

$$\begin{aligned} G_N(\vec{p}) &= \int G_N(\vec{R}) e^{i\vec{p}\vec{R}} d\vec{x} \\ &= \iint \prod d\vec{n}_i \, \delta\left(\sum \vec{n}_i - \frac{\vec{R}}{\Delta}\right) e^{i\vec{p}\vec{R}} d\vec{R} \\ &= \left(\int d\vec{n}\, e^{i\vec{p}\vec{n}\Delta}\right)^N = (C(\vec{p}, \Delta))^N \end{aligned} \tag{18}$$

Eliminate the δ-function. Independent integrations remain.

This is a Bessel function.

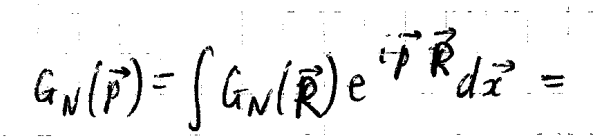


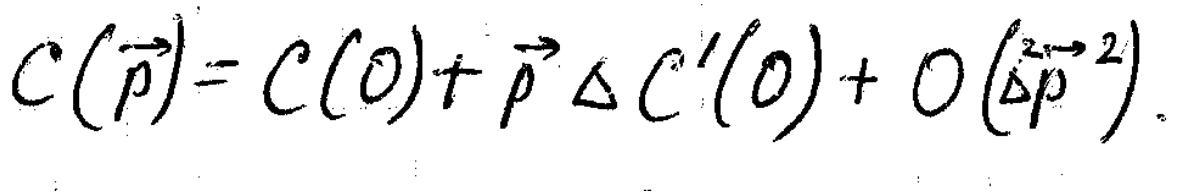
Diagram and typed description. The differential in the first integral in (18) is $d\vec{x}$.

Original source details: GELFAND-P001044

Total: geometric progression.

$$G(\vec{p}) = \sum G_N(\vec{p}) e^{-mN\Delta} = \sum [C(\vec{p}, \Delta) e^{-m\Delta}]^N \tag{19}$$

$$C(\vec{p}) = C(0) + \vec{p}\Delta C'(0) + O(\Delta^2 \vec{p}^2). \qquad (\text{take } \vec{p}\Delta \ll 1)$$



Original source details: GELFAND-P001044

Diagram and typed description. The manuscript has the term $\vec{p}\Delta C'(0)$.

GELFAND-P001045

Hence

$$G(\vec{p}) = \sum_N e^{(a+b\vec{p}-m)N\Delta} \tag{20}$$

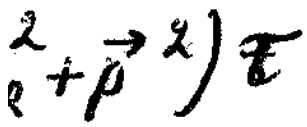
Set

$$m'^2 = m_R^2 = \frac{m-a}{b} \qquad (a, b \text{ — absolute constants})$$

(renormalized mass)

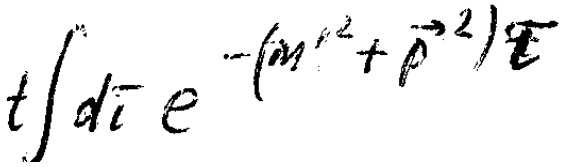
The sum ultimately passes to an integral:

$$\begin{aligned} G(\vec{p}) &= \text{const} \int d\tau\, e^{-(m_R^2 + \vec{p}^2)\tau} \\ &= \text{const} \int d\tau\, e^{-(m'^2 + \vec{p}^2)\tau} \end{aligned} \tag{21}$$



Uncertain reading. The mark after the parenthesis in the first exponent reads as τ over earlier strokes.

Original source details: GELFAND-P001045



Uncertain reading. The symbol in $d\tau$ has a detached top stroke. The τ in the exponent is overwritten, and its earlier strokes are indistinct.

Original source details: GELFAND-P001045

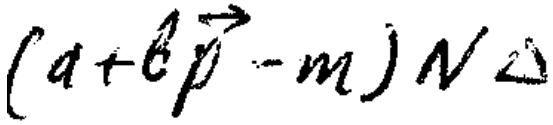
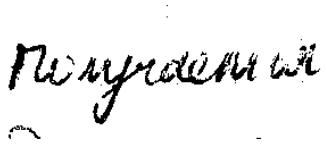


Diagram and typed description. Equation (20) has $b\vec{p}$. There is no square over $\vec{p}$.

Original source details: GELFAND-P001045

The result obtained is independent of the discretization, [crossed out: ~~one must?~~] one must carry out the appropriate mass renormalization.



Original source details: GELFAND-P001045

Uncertain reading. The first word reads “Получаемый” (“obtained”). The middle strokes are close together.

Original source details: GELFAND-P001045

Uncertain reading. The crossed-out word may read “надо” (“one must”).

3rd method. Treat the rods as springs. The length has a Gaussian distribution.

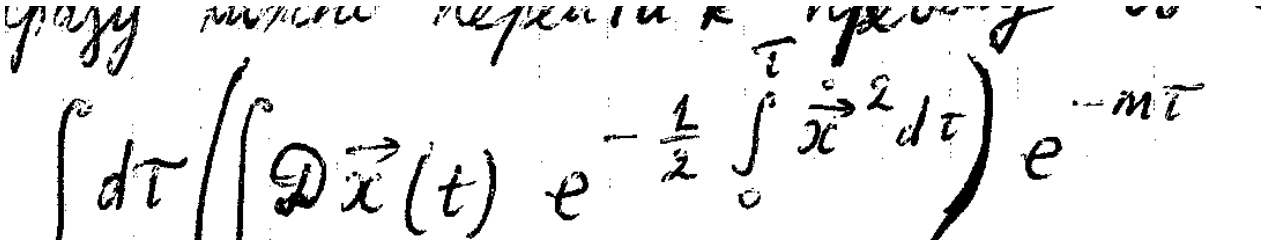
$$\sum_N e^{-Nm\Delta} \int \prod_i d\vec{x}_i \, e^{-\frac{1}{2\Delta}\sum_i^N (\vec{x}_i - \vec{x}_{i+1})^2} = G(\vec{R}) \tag{22}$$

(leads to the same answer).

Here we can pass to the limit straightaway, before evaluation.

$$\int d\tau \left(\int \mathcal{D}\vec{x}(t) e^{-\frac{1}{2}\int_0^\tau \dot{\vec{x}}^2 d\tau} \right) e^{-m\tau} \tag{23}$$

(integral with respect to Wiener measure).

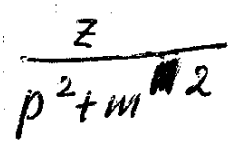


Original source details: GELFAND-P001045

Diagram and typed description. Equation (23) has $\vec{x}(t)$ and two differentials $d\tau$.

Answer:

$$\int G(\vec{R}) e^{i\vec{p}\vec{R}} d\vec{R} = \frac{z}{p^2 + m^{[?]\,2}} \qquad (z \text{ depends on normalization}) \tag{24}$$



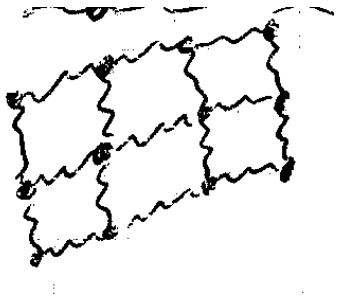
Original source details: GELFAND-P001045

Uncertain reading. Between m and the superscript 2 there is a heavily cancelled mark whose reading is uncertain.

$G(\vec{R})$ — inverse Fourier transform (Hankel function).

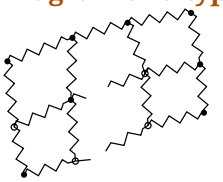
The case of surfaces.

(25)



Original source details: GELFAND-P001045

Diagram and typed description.



The gaps in the middle and lower rows are shown. The exact connections are unclear.

Connect the points with springs.

A regular lattice will not do:

Original source details: GELFAND-P001045

Diagram and typed description.

The drawing shows the visible gaps. The exact closure of the outline is unclear.

For the lattice:

$$\int \mathcal{D}\vec{x}(\xi^1,\xi^2)e^{-\frac{1}{2}\int \partial_a\vec{x}\cdot\partial^a\vec{x}\,d^2\xi} \tag{26}$$

(for a two-dimensional surface in 26-dimensional space, one obtains the correct answer)

Suppose there is an intrinsic geometry described by $g_{ab}(\xi)$. The integral will take the form:

GELFAND-P001046

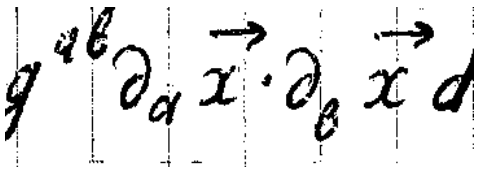
[source number: 13]

$$G=\int \mathcal{D}\vec{x}(\xi)\exp\left\{-\frac{1}{2}\int\sqrt{g}\,g^{ab}\partial_a\vec{x}\cdot\partial_b\vec{x}\,d^2\xi+\lambda\int\sqrt{g}\,d^2\xi\right\} \tag{27}$$

with the addition of λ, which we will choose.

Answer (correct):

$$\begin{aligned} G=&\int\mathcal{D}g_{ab}\int\mathcal{D}\vec{x}(\xi)\\ &\times\exp\left\{-\frac{1}{2}\int\sqrt{g}\,g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\,d^2\xi+\lambda\int\sqrt{g}\,d^2\xi\right\}\end{aligned} \tag{28}$$



Original source details: GELFAND-P001046

Diagram and typed description. A dot separates the derivatives in (27). In (28) they are juxtaposed.

By minimizing, we obtain the equations.

4/X̄-82 Gelfand Seminar.

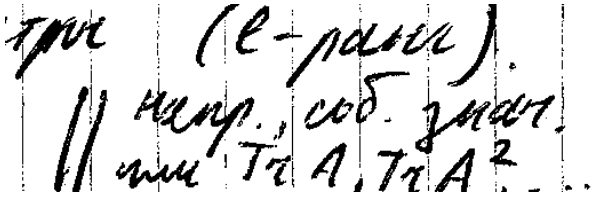
Kirillov on the work of Ginzburg.

G — a Lie group, $\mathfrak{g}$ — a Lie algebra, consider the orbits of the adjoint representation;

there are orbits in general position, and there are nilpotent ones (corresponding to nilpotent elements of the algebra $\mathfrak{g}$);

Ω_c — an orbit, $c = (c_1, \ldots, c_\ell)$ — its parameters (ℓ — the rank).

Example: $SL(2, \mathbb{R})$; $\mathfrak{g} \cong \mathbb{R}^3$; the orbits are hyperboloids.



Original source details: GELFAND-P001046

Diagram and typed description. [At right ||] e.g., eigenvalues or, Tr A, Tr $A^2, \ldots$

Desingularization according to Springer.

G — a semisimple group, $N \subset G$, N — the set of unipotent elements (with all eigenvalues 1).

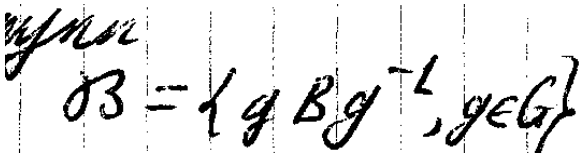
N — an algebraic set with singularities.

Original source details: GELFAND-P001046

Uncertain reading. A short entry before "singularities" is heavily crossed out. It may be "o."

B_0 — a Borel subgroup in G; $\mathcal{B}$ — the collection of all Borel subgroups (they are all conjugate), so that

$$\mathcal{B} = \{gBg^{-1},\ g \in G\}.$$



Original source details: GELFAND-P001046

Diagram and typed description. The formula has B without a subscript, while the preceding line has B_0.

A desingularization of the algebraic set X (with singularities) — a set Y (without singularities) of the same dimension, $Y \longrightarrow X$ is one-to-one almost everywhere (except in smaller dimension).

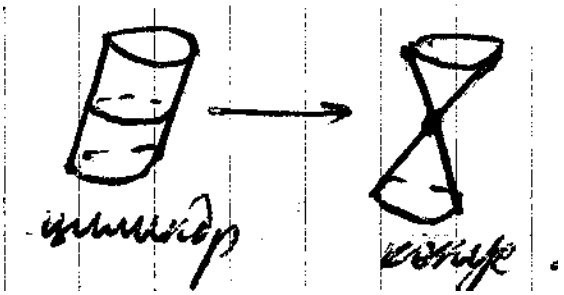
Original source details: GELFAND-P001046

Uncertain reading. The word before "Desingularization" is crossed out and may read "Всп.".

Original source details: GELFAND-P001046

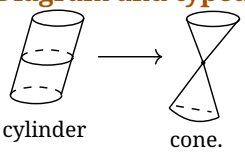
Uncertain reading. The word corresponding to "same" is written over a correction. The earlier strokes are indistinct.

Example:



Original source details: GELFAND-P001046

Diagram and typed description.

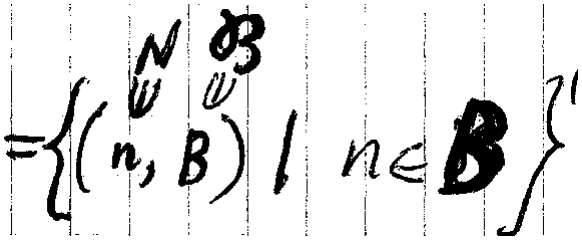


$$(x, y, z) \longmapsto (xz, yz, z).$$

Original source details: GELFAND-P001046

Uncertain reading. An indistinct symbol is crossed out after $\mathcal{M}$.

$$\mathcal{M} = \left\{ \left(\begin{matrix} N & \mathcal{B} \\ \cup & \cup \\ n, & B \end{matrix} \right) \middle| \, n \in B \right\} \qquad \text{(flags)}.$$



Original source details: GELFAND-P001046

Diagram and typed description. The membership signs above n and B are written vertically.

Then $\mathcal{M}$ — a desingularization of N.

Original source details: GELFAND-P001046

Uncertain reading. A symbol after "Then" is crossed out and illegible.

For $SL(2,\mathbb{R})$ this is precisely what occurs in the example.

On Ginzburg again.

$$f \in C^\infty(\mathfrak{g}) \longmapsto \psi(c) = \int_{\Omega_c} f$$

(the measure is chosen consistently on orbits in general position).

$$\int_{\Omega_0} f = P\psi(c)|_{c=0}\,, \qquad P \text{ — a differential operator}$$

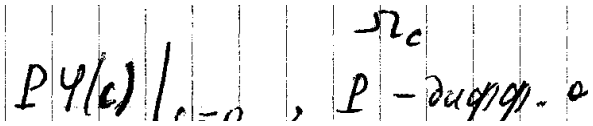


Diagram and typed description. The operator is written P in both places.

Original source details: GELFAND-P001046

GELFAND-P001047

I. M. G. Integrals over hyperbolas are given. Find the value of the function at 0.

(more generally — a hyperboloid).

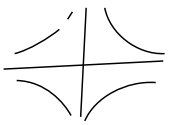


Diagram and typed description. The short bend at the right end of the horizontal line is unclear.

Original source details: GELFAND-P001047

$$xy = c; \qquad \text{distance from } 0 \text{ of order } \sqrt{c}.$$

For $|x| \geqslant 1$ distance $\sim c$.

Uncertain reading. The $>$ sign has a separate slanted stroke beneath it. The intended sign is likely $\geqslant$.

Original source details: GELFAND-P001047

$\Rightarrow$ we evaluate the integral over the lines (limit of the integral over the hyperbolas) and so on.

See the expansion of $\delta(xy - c)$ in powers of c in the first volume of Generalized Functions.

Zamolodchikov.

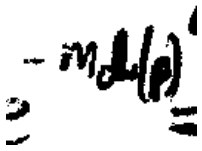
We want to find the sum

$$\sum_S e^{-\mu_0^2 A(S)} \tag{1}$$

S — surface

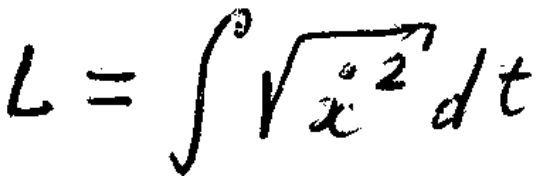
The integral over the set of all curves (see above) had the form

$$\sum_P e^{-m_0 L(P)} = \int_0^\infty d\tau\, e^{-m^2\tau} \int_{\substack{\vec{x}(0)=\vec{0}\\ \vec{x}(\tau)=\vec{R}}} \mathcal{D}\vec{x}(t)\, e^{-\frac{1}{2}\int_0^\tau \dot{\vec{x}}^2(t)\,dt},$$
$$L = \int \sqrt{\dot{x}^2}\, dt. \tag{2}$$



Original source details: GELFAND-P001047

Diagram and typed description. A subscript zero is visible between m and L, touching the stroke of L.



Original source details: GELFAND-P001047

Diagram and typed description. In the formula for L, $\dot{x}$ has no arrow.

($m \neq m_0$ — renormalization)

And it is equal to:

$$(\text{const}) \int_0^\infty d\tau \left(e^{-m^2\tau - \frac{\vec{R}^2}{2\tau}} \right) \frac{1}{\tau^{d/2}}, \qquad d \text{ — dimension of the space.} \tag{2'}$$

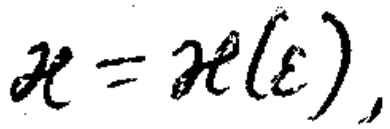
The relation between m and m_0 depends on the method of regularization [approximation — fine structure of the space].

$$\sum_P e^{-m_0 L} = \sum_{\substack{L=0 \\ \text{(integral)}}}^{\infty} f(L) e^{-m_0 L} \tag{3}$$

where

$$f(L) \sim L^\beta e^{\varkappa L} \qquad \text{when } L \gg \varepsilon \qquad (\varepsilon \text{ — lattice spacing})$$

$$\varkappa = \varkappa(\varepsilon), \qquad \beta \text{ is independent of } \varepsilon.$$



Uncertain reading. The looped symbol is rendered as $\varkappa$. A cursive x is also possible.

Original source details: GELFAND-P001047

Hence

$$(3) = \text{const} \sum (L)^\beta e^{-(m_0 - \varkappa)L} \tag{4}$$

$$m^2 = m_0 - \varkappa.$$

2 dangers:

a) concentration on straight lines

b) it diverges

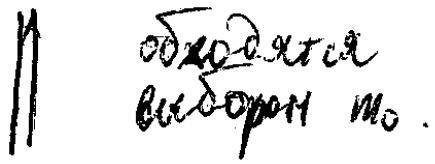


Diagram and typed description. [At right of both items ‖] are avoided by the choice of m_0.

Original source details: GELFAND-P001047

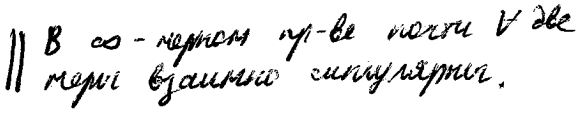


Diagram and typed description. [Farther right ‖] In an infinite-dimensional space almost any two measures are mutually singular.

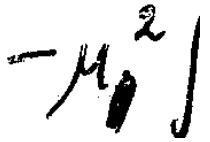
Original source details: GELFAND-P001047

GELFAND-P001048

[source number: 14]

Let us pass to surfaces.
$\vec{x}(\xi^1, \xi^2)$; $g_{ab}(\xi^1, \xi^2)$ — the metric on the surface.

$$\sum_S e^{-\mu_0^2 A(S)} = \int d\mu(\vec{x}, g) \exp\Big\{-\frac{1}{2}\int \sqrt{g}\, g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\, d^2\xi - \\ - \mu_0^2 \int \sqrt{g}\, d^2\xi\Big\}, \qquad g = \det g_{ab}. \tag{5}$$



Uncertain reading. The subscript of the second μ is heavily overwritten. It is probably 0.

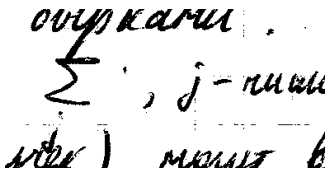
Original source details: GELFAND-P001048

(Polyakov formula) Here we sum over surfaces [inserted above: with] given topological properties.

Diagram and typed description. The preposition “с” (“with”) is inserted between “surfaces” and “given.”

Original source details: GELFAND-P001048

[In fact one has to take account of surfaces with handles and holes.
It is unclear how the contributions compare. One needs: $\sum$, j — the number of handles.
Foamy surfaces (with many handles) can make an overwhelming contribution.



Uncertain reading. A small mark is visible under $\sum$, without a discernible j there. The text to the right calls j the number of handles.

Original source details: GELFAND-P001048

Here we choose a parameter domain in $\mathbb{R}^2$. One introduces [inserted above: elements of] distance in the function space:

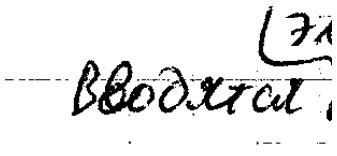


Diagram and typed description. The start of the phrase reads “Вводятся” (“are introduced”).

Original source details: GELFAND-P001048

Diagram and typed description. The word “elements” is written above the line.

Original source details: GELFAND-P001048

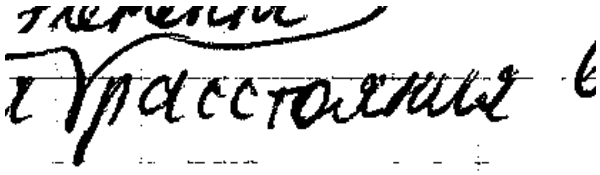


Diagram and typed description. The word “расстояния” (“distance” or “distances”) is written on the line.

Original source details: GELFAND-P001048

$$\|\delta\vec{x}(\xi^1,\xi^2)\|^2 = \int \sqrt{g}\,(\delta\vec{x}(\xi))^2\,d^2\xi$$
$$\|\delta g\|^2 = \int \sqrt{g}\,\left(g^{aa'}g^{bb'} + c\,g^{ab}g^{a'b'}\right)\delta g_{ab}\,\delta g_{a'b'}\,d^2\xi \tag{6}$$

Gauge invariance under the transformations:

$$\begin{cases}\xi^a \to \xi^a + \varepsilon^a(\xi),\\ g_{ab} \to g_{ab} + \nabla_a\varepsilon_b + \nabla_b\varepsilon_a,\end{cases} \qquad \nabla_a\varepsilon_b = \partial_a\varepsilon_b + \Gamma^c_{ab}\varepsilon_c. \tag{7}$$

nothing changes.

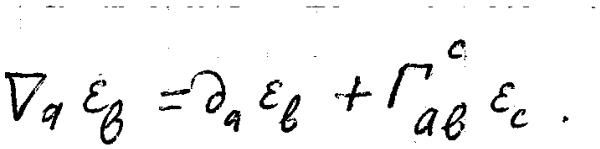


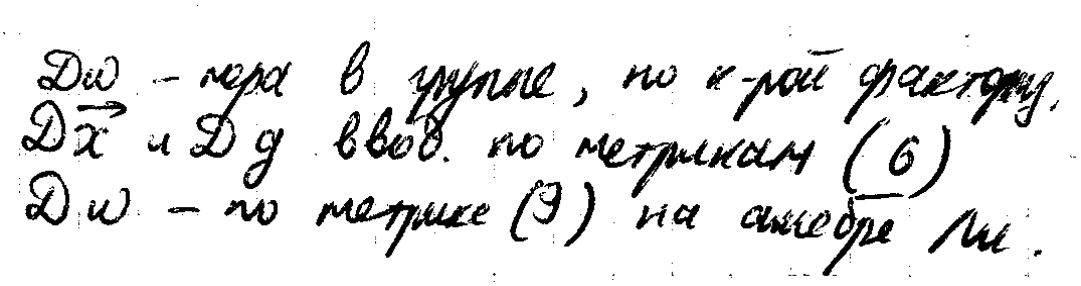
Diagram and typed description. A + appears before Γ^c_{ab} in (7).

Original source details: GELFAND-P001048

One must integrate over the space of orbits.

$$d\mu(x,g) = \frac{\mathcal{D}\vec{x}\,\mathcal{D}g}{\mathcal{D}\omega} \tag{8}$$

$$\|\varepsilon\|^2 = \int \sqrt{g}\,g^{ab}\varepsilon_a\varepsilon_b\,d^2\xi \tag{9}$$



Original source details: GELFAND-P001048

Diagram and typed description. [At right of (8) and (9)] $\mathcal{D}\omega$ — the measure on the group one quotients by,
$\mathcal{D}\vec{x}$ and $\mathcal{D}g$ are introduced using the metrics (6)
$\mathcal{D}\omega$ — using the metric (9) on the Lie algebra.

Now the integral [inserted above: (5)] is well defined.

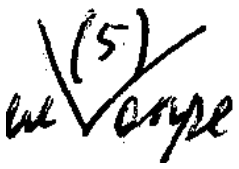


Diagram and typed description. The reference “(5)” is inserted above the line.

Original source details: GELFAND-P001048

$\int \mathcal{D}\vec{x}$ is Gaussian and reduces to the det of the operator.
Polyakov proposes choosing representatives as follows:

$$g_{ab} = \delta_{ab}\,\rho(\xi^1, \xi^2) \qquad \text{(conformally flat form).} \tag{10}$$

(choice of gauge). The conformal group remains (finite-dimensional).

Diagram and typed description. The opening strokes of “finite-dimensional” have been overwritten.

Original source details: GELFAND-P001048

det are the same. One obtains

$$\int d\mu(\rho)\,[\det(\Delta_g)]^{(26-d)/2}, \qquad \Delta_g \text{ — Laplacian of the metric} \tag{11}$$

(ρ — the coefficient in (10)).

GELFAND-P001049

One obtains:

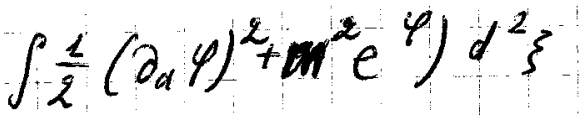
$$(11) = \int d\mu(\varphi)\, e^{\frac{d-26}{48\pi}\int \frac{1}{2}(\partial_a\varphi)^2 + m^2 e^{\varphi})\, d^2\xi} \tag{12}$$

Diagram and typed description. The reference “(11)” to the left of the equals sign is written over earlier strokes.

Original source details: GELFAND-P001049

Uncertain reading. The base of the power before e^{φ} is heavily overwritten and reads as m. The earlier strokes are indistinct.

Original source details: GELFAND-P001049



Uncertain reading. A closing parenthesis follows e^{φ}. An opening parenthesis for the whole sum is indistinct.

Original source details: GELFAND-P001049

$d\mu(\varphi)$ — the measure from the metric

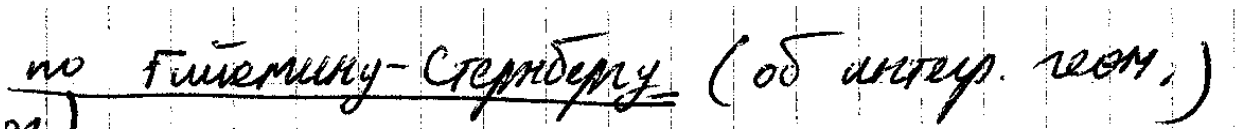
$$\|\delta\varphi\|^2 = \int e^{\varphi}(\delta\varphi)^2\, d^2\xi \tag{13}$$

GELFAND-P001050

11/X̄-82 Gelfand Seminar

V. Ginzburg.

after Guillemin–Sternberg (on integral geometry)

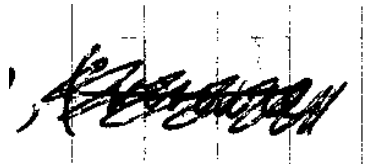


Original source details: GELFAND-P001050

Diagram and typed description. The right-hand heading reads “after Guillemin–Sternberg.”

X — a set (complex)

$\Sigma \subset T^*X$, Σ — a coisotropic subset,



Uncertain reading. The word after “subset” is crossed out and may read “which” or “conic.”

Original source details: GELFAND-P001050

has the following properties:

1. Σ — is conic;
2. $\Sigma \to X$ is a fibration.

On Σ there is a 0-foliation (kernel of the symplectic form), which is always integrable.

Locally arranged as follows: $q_1, \dots, q_n, p_1, \dots, p_k$

0-foliation

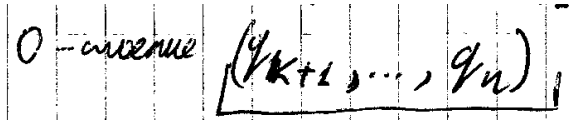


Diagram and typed description. $\underbrace{(q_{k+1}, \dots, q_n)}$

Original source details: GELFAND-P001050

Uncertain reading. There is a mark before “Suppose,” possibly “Мы” (“we”) or “Или” (“or”).

Original source details: GELFAND-P001050

Suppose it is a fibration (all leaves are closed, etc.)

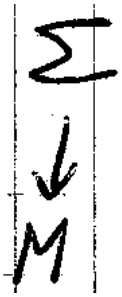


Diagram and typed description. $\begin{matrix}\Sigma \\ \downarrow \\ M\end{matrix}$

Original source details: GELFAND-P001050

and M — the quotient space.

From Σ construct a Lagrangian subset $\Lambda \subset T^*X \times T^*X$:

$$\Lambda = \{(\xi_1, \xi_2) : \quad \xi_1, \xi_2 \in \Sigma, \quad \xi_1 \sim \xi_2\}$$

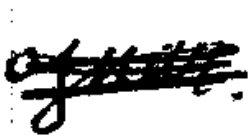
Properties:

1) $\Lambda^t = \Lambda$

2) $\Lambda \circ \Lambda = \Lambda$ (projector)

GELFAND-P001051

Apparently the converse: if Λ is Lagrangian and satisfies 1), 2), then it is obtained from Σ.



Uncertain reading. A word after "and" is heavily crossed out. A loop is visible beneath the cancellation.

Original source details: GELFAND-P001051

Fourier integral operators are constructed from Λ: denote their set by $\mathcal{E}_\Lambda$.

It is a ring by property 2).

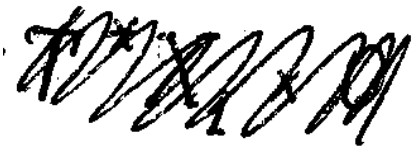
Uncertain reading. A short entry before "property" is crossed out.

Original source details: GELFAND-P001051

This is an analogue of the Radon transform.

$X = X_1 \times X_2$; Σ — directions normal to X_1 (along X_2) in the cotangent bundle.

$$\Lambda = \operatorname{diag}(T^*X_1 \times T^*X_1) \times 0$$
$$\subset T^*X_1 \times T^*X_1 \times T^*X_2 \times T^*X_2.$$



Uncertain reading. A formula before diag is crossed out. Individual signs resembling T^*, X_1, $\times$, and 0 are visible.

Original source details: GELFAND-P001051

Operators: in X_1 all pseudodifferential.

in X_2 smoothing.

Shmelyov.

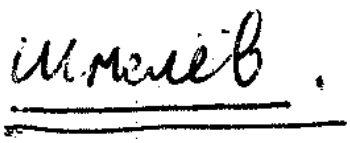
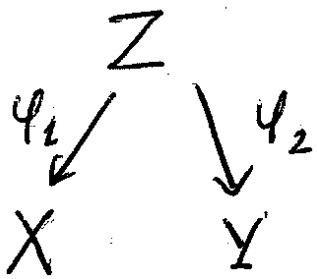


Diagram and typed description. The twice-underlined surname reads "Shmelyov."

Original source details: GELFAND-P001051



Original source details: GELFAND-P001051

Diagram and typed description.

$$\begin{array}{ccc} & Z & \\ {\scriptstyle\psi_1}\swarrow & & \searrow{\scriptstyle\psi_2} \\ X & & Y \end{array}$$ fibrations

Requirement: $\psi_2^* : N^*Z \longrightarrow T^*Y$ an embedding;

Fröhlich

1) Magnetic monopoles

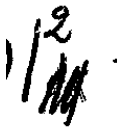
2) Gauge theories $G \overset{\text{e.g.}}{=} SU(2)$

Lagrangian: φ : isospin $= 1$ (3 components)

$$\begin{aligned} E &= \int \{(D_A\varphi)^2 + F^2\} \\ &= \int \left\{ \left|(\partial_\mu - eA_\mu\wedge)\varphi(x)\right|^2 + \operatorname{tr}(F_{\mu\nu}\cdot F^{\mu\nu})(x) \right\} dx, \end{aligned} \tag{1}$$

Original source details: GELFAND-P001051

Diagram and typed description. A sign resembling $\wedge$ follows A_μ.



Original source details: GELFAND-P001051

Uncertain reading. A mark beneath the squared norm is heavily crossed out.

Original source details: GELFAND-P001051

Uncertain reading. Below the dot between the two Fs are overwritten strokes and a wavy line. The earlier marks are indistinct.

$x \in \mathbb{R}^3$

Original source details: GELFAND-P001051

Uncertain reading. An equality to the right of $x \in \mathbb{R}^3$ is crossed out. It appears to begin $E = E$.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

Monopole: a solution of the Euler–Lagrange equations such that the degree of $\varphi|_{S^2} \longrightarrow S^2$ at ∞ equals 1. $\|$ (stationary solutions)

$\exists$ solutions attaining min or loc. min — explicitly, using twistor techniques.

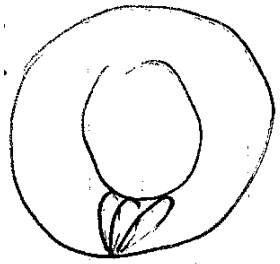
Uncertain reading. A short entry before "twistor" is crossed out.

Original source details: GELFAND-P001051

Question: other stationary points?

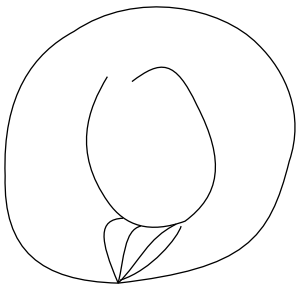
Answer: solutions have been found that are not minima

Idea:



Original source details: GELFAND-P001051

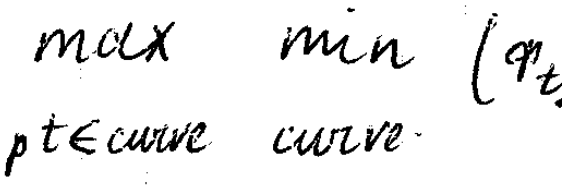
Diagram and typed description.



The inner loop has a gap. The exact connections of the thin lines at the bottom are indistinct.

one-parameter family:

$$\max_{[\rho?]\, t \in \text{curve}} \min_{\text{curve.}} (\varphi_t, F_t)$$



Uncertain reading. Before $t \in$ curve is a sign resembling ρ or p. Below min the manuscript has "curve.".

Original source details: GELFAND-P001051

C. Taubes

The new solution resembles a pair

monopole / antimonopole.

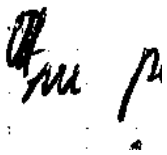
GELFAND-P001052

[source number: 16]

There is a manifold $\underset{\text{(compact)}}{M^4}$, $\pi_1(M^4) = \{\text{id}\}$; $H^2(M^4) = \{0\}$.

M^4 with a conformal structure.

$\exists\,[A_\mu?]$ a solution of the equation $F = \pm F^*$?

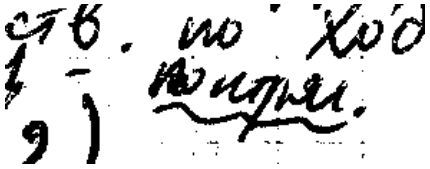


Uncertain reading. The symbol after $\exists$ is overwritten and may be A_μ.

Original source details: GELFAND-P001052

$*$ — Hodge duality;

with characteristic class 1 — [e.g.?]



Uncertain reading. The underlined word after "class 1" may read "напр." ("e.g."). Its opening has been overwritten.

Original source details: GELFAND-P001052

F is the curvature of the connection A; the group — $SU(2)$.

Answer: a solution exists!

Space of parameters labelling solutions $= \underset{\text{manifold } B}{5}$ (as on the sphere!)

(Taubes, Comm. Math. Phys, 1981).

Instanton parameters

B — a manifold with boundary, $\partial B \underset{\text{Donaldson}}{\simeq} M^4$ (the original manifold)

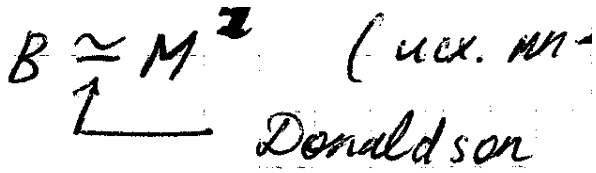
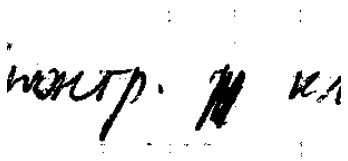


Diagram and typed description. The name "Donaldson" is connected to $\simeq$ by an angled arrow.

Original source details: GELFAND-P001052

M^4 — a boundary $\Longleftrightarrow$ (Pontr. [?] class $= 0$)

$\Longleftrightarrow$ (signature $= 0$ for the form in H^2).



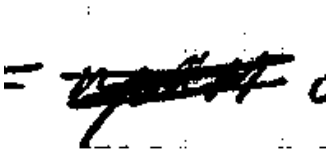
Uncertain reading. A sign between "Pontr." and "class" is heavily overwritten. It may be p.

Original source details: GELFAND-P001052

Donaldson: if the form on H^2 is positive definite, then the manifold = generalized projective plane $CP^2 \Rightarrow$ the form reduces to a sum of squares (over $\mathbb{Z}$).

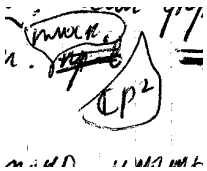
Original source details: GELFAND-P001052

Uncertain reading. A short entry between "positive" and "definite" is crossed out.



Original source details: GELFAND-P001052

Uncertain reading. The word immediately after the equals sign is crossed out.



Original source details: GELFAND-P001052

Uncertain reading. An entry after "projective" is crossed out. "Plane" is added above and CP^2 below. Both additions are outlined.

B must have singularities — cones over CP^2.

On the work of Witten.

M — n-dimensional manifold, B_p — the pth Betti number.

f — a function on M with isolated critical points.

M_p — the number of critical points of index p (the Hessian has p negative eigenvalues).

$$M_p \geqslant B_p \qquad \text{(Morse inequality)}$$

the simplest one

$d : \Omega_p \longrightarrow \Omega_{p+1}$ (exterior differential)

Original source details: GELFAND-P001052

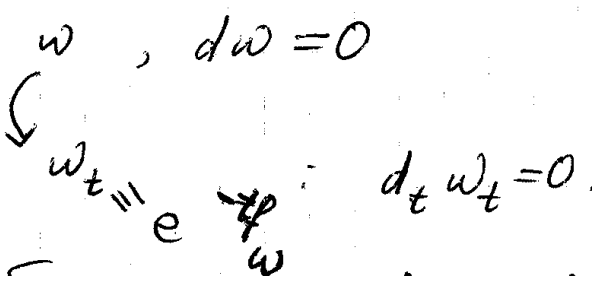
Uncertain reading. A mark before "exterior differential" is crossed out.

Introduce a metric on M; $*$ — the corresponding operator.

Choose in H^p a basis of harmonic p-forms (B_p in number)

Alternatively: set

$$d_t = e^{-tf} d\, e^{tf}$$



Original source details: GELFAND-P001052

Diagram and typed description.

$\omega, \quad d\omega = 0$

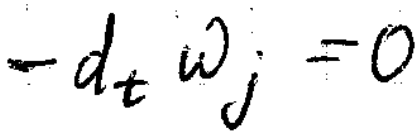
$\omega_t = e^{-tf}\omega; \quad d_t\omega_t = 0$

The exponent of e has been overwritten and reads as $-tf$.

Choose a basis in H^p, satisfying the conditions:

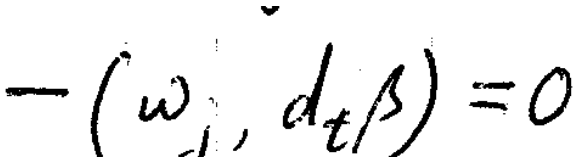
$$\left\{\begin{array}{c} -d_t\omega_j = 0 \\ -(\omega_j, d_t\beta) = 0, \quad \forall \beta \in \Omega_{p-1} \end{array}\right\} \Rightarrow (d_t^* d_t + d_t d_t^*)\omega_j = 0 \tag{1}$$

(closed in the new sense).



Original source details: GELFAND-P001052

Uncertain reading. The short stroke before $d_t\omega_j$ may be a minus or an item dash.



Original source details: GELFAND-P001052

Uncertain reading. A separate horizontal stroke also precedes the second line.

GELFAND-P001053

d_t-cohomology is the same as d-cohomology, since there is a homotopy.

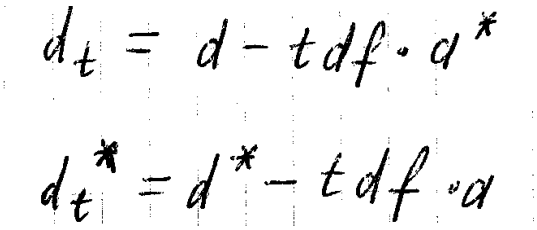
$$d_t^2 = 0 \quad \text{(obvious)}$$

Let us compute the operator $\Box_t = d_t^* d_t + d_t d_t^*$ and find its asymptotics as $t \to \infty$.

Original source details: GELFAND-P001053

Diagram and typed description. The manuscript reads “Let us compute the operator.” A diagonal stroke is visible inside the box sign.

$$\begin{cases} d_t = d - tdf \cdot a^*, \\ d_t^* = d^* - tdf \cdot a \end{cases} \qquad \begin{array}{c} df\, a^*\omega \equiv df \wedge \omega \\ \text{(physics notation)} \end{array} \tag{2}$$

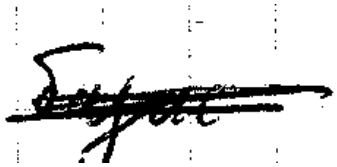


Original source details: GELFAND-P001053

Diagram and typed description. Both tdf terms in (2) carry minus signs.

$$\{a_j, a_i^*\} = \delta_{ij}. \tag{3}$$

a_i — ~~basis~~ correspond to the basis of 1-forms; (operators of exterior multiplication).



Original source details: GELFAND-P001053

Diagram and typed description. After a_i, "basis" is crossed out.

Choose so that one has (3).

Original source details: GELFAND-P001053

Uncertain reading. The entry before "one has" is heavily crossed out. It may contain a_i.

Kirillov:

$$a = (a^1, a^2, \ldots, a^n) \text{ — a vector}, \qquad df = (\partial_1 f, \partial_2 f, \ldots, \partial_n f)$$

a^i — interior multiplication (annihilation operator) — strike out dx^i, where it occurs, where it does not occur, annihilate.

Original source details: GELFAND-P001053

Uncertain reading. A word before "annihilation" is crossed out. It may be an abbreviation for "creation."

a^{i*} — exterior multiplication.

Indices can be lowered and raised using the metric.

a^j — interior multiplication by $\dfrac{\partial}{\partial x^j}$ (substitution in the first argument)

$df \cdot a$ — interior multiplication by the field obtained from df by raising the index

$$a_i^* = dx_i \wedge \qquad \text{(exterior multiplication)}$$

Fröhlich.

$$a_i = \partial/\partial x_i \llcorner \qquad \text{(interior multiplication)}$$

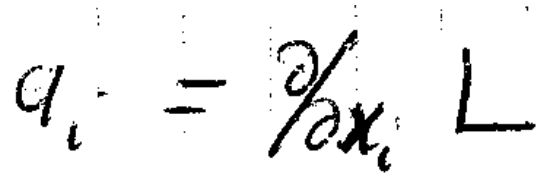


Diagram and typed description. An angle-shaped sign follows $\partial/\partial x_i$.

Original source details: GELFAND-P001053

Meaning:

One-particle space (fermions)

$$\{1, 2, \ldots, n\} \to \ell_2(\{1, 2, \ldots, n\}).$$

(ℓ_2 on a finite set).

$$m \text{ particles (fermions)} \Rightarrow f_{i_1 \ldots i_m}$$

(antisymmetric δ-functions — basis)

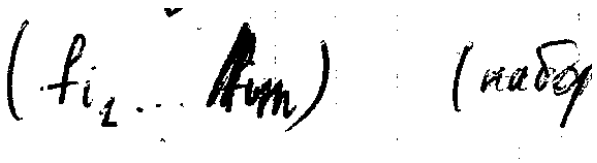
A way of writing elements of the Grassmann algebra.

(see Berezin: The Method of Second Quantization)

$$\sum_{i_1 \ldots i_m} a_{i_1}^* \ldots a_{i_m}^* f_{i_1 \ldots i_m} \Omega_0 \tag{4}$$

$$\updownarrow$$

$$(f_{i_1} \ldots f_{i_m}) \quad \text{(list of coefficients)}$$



Uncertain reading. The last base is overwritten and may read f_{i_m} or h_{i_m}.

Original source details: GELFAND-P001053

GELFAND-P001054

[source number: 17]

It is postulated:

$$\{a_i, a_j^*\} = \delta_{ij} \tag{5}$$

Exterior multiplication by df — a fermion creation operator.

$$a_i \Omega_0 = 0, \quad \forall i \tag{6}$$

$$\square_t = \square + t^2 (df)^2 + t \sum [a_i^*, a_j] \partial_i \partial_j f \tag{7}$$

In a neighborhood of the critical point

$$f(x) = \sum_{i=1}^{n-p} \lambda_i (x^i - x_0^i)^2 - \sum_{j=n-p+1}^{n} \lambda_j (x^j - x_0^j)^2 + O(|x - x_0|^3) \tag{8}$$

— . —

I. M. G. Idea: instead of changes in cycles — changes in forms!

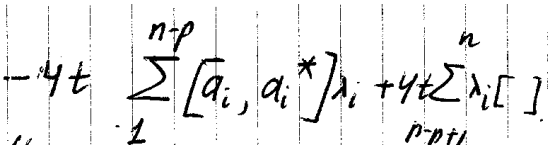
Connection between PDOs and Morse changes.

F. the principal term of $\square_t$ equals:

$$\square + 4t^2 \sum_{j=1}^{n} \lambda_j^2 (x^j - x_0^j)^2 + \overbrace{t \sum_{i<j} [a_i^*, a_j]}^{\text{const}} c_{ij} \tag{9}$$

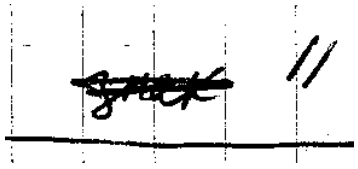
[Above the last term]

$$-4t \sum_{1}^{n-p} [a_i, a_i^*] \lambda_i + 4t \sum_{n-p+1}^{n} \lambda_i [\,].$$



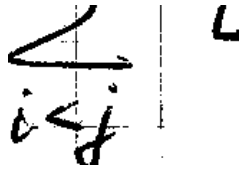
Original source details: GELFAND-P001054

Diagram and typed description. The square brackets in the second sum are empty.



Original source details: GELFAND-P001054

Uncertain reading. A short word above the bracket is crossed out, possibly “sign.” Two slanted strokes follow.



Original source details: GELFAND-P001054

Diagram and typed description. The index beneath the sum in (9) reads $i < j$.

$$\big|\approx \quad \textit{Annihilates}: \quad \underline{dx_{n-p+1}\wedge\cdots\wedge dx_n} \quad (p\text{-form}) \tag{10}$$

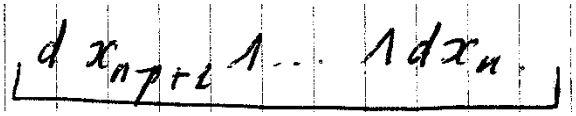


Diagram and typed description. A lower bracket surrounds the p-form.

Original source details: GELFAND-P001054

Let us look at eigenvalues other than the 0-th, the small ones.

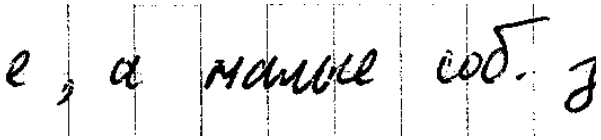
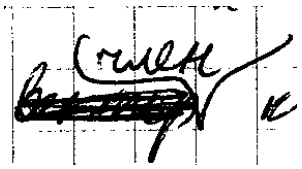


Diagram and typed description. The manuscript reads “small eigenvalues.”

Original source details: GELFAND-P001054

$\forall$ critical point gives an approximation to an eigenfunction.

The second [inserted above: term] cancels the 1-st exactly on the first vector (10).



Uncertain reading. After “second,” a word, possibly “vector,” is crossed out and “term” is inserted.

Original source details: GELFAND-P001054

The cancellation follows from supersymmetry considerations.

d_t plays the role of a supercharge; Yukawa model — the cancellation is known.

$$a = \frac{\delta}{\delta\varphi(x)} + ta^*(x)\varphi^n(x) \leftrightarrow d + tdf\cdot a^* \tag{11}$$

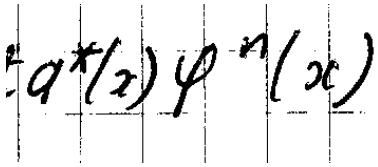


Diagram and typed description. In (11) the exponent of φ is n.

Original source details: GELFAND-P001054

$\underline{d=2}\Rightarrow$ fermions $=$ forms on the set of Bose-field configurations.

(since the Poincaré group is simple)

$\underline{d>2}\Rightarrow$ (?) gauge fields are needed for the interpretation. —

— there is no (such) interpretation yet.

V. Enss — asymptotic completeness of systems of particles with Coulomb interactions.

(three-particle) (worked on it for 15 years)

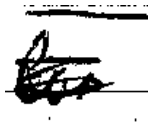
— a classical problem of mathematical physics.

GELFAND-P001055

18/X̄-82 Gelfand Seminar

Fröhlich. Statistical mechanics of surfaces.

Arises in the following settings:



Uncertain reading. A short entry after “в” (“in”) is crossed out and may read “во.”

Original source details: GELFAND-P001055

(a) Crystal growth

(b) Interfaces separating different phases

(c) (lattice) gauge theories

(d) dual resonance models (strings)

Simplified model for (a), (b):

$$\underset{\text{finite subset}}{\Lambda} \subset \mathbb{Z}^2; \qquad h : p \subset \mathbb{Z}^2 \to h_p \in \mathbb{Z}. \tag{1}$$

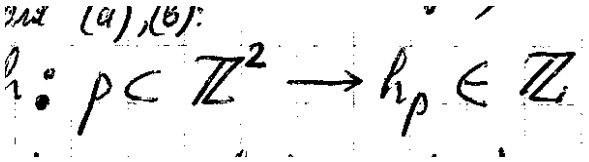


Diagram and typed description. The symbol following p is $\subset$.

Original source details: GELFAND-P001055

(solid-on-solid model)

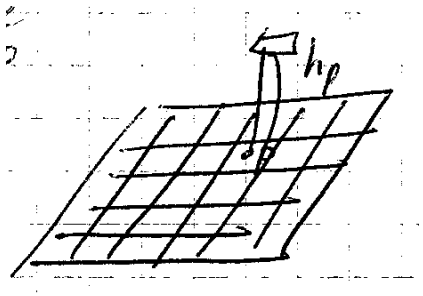
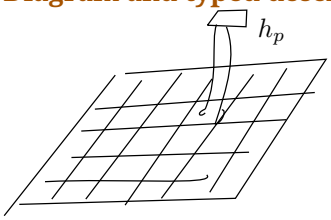


Diagram and typed description.



Loops appear at the lower ends of the vertical lines. It is unclear whether one is the letter p.

Original source details: GELFAND-P001055

p — a little square

h_p is interpreted as the height of the surface above the plane

$$w(S) \equiv e^{-\beta\,\mathrm{ares}(S)}, \quad \beta > 0 \tag{2}$$

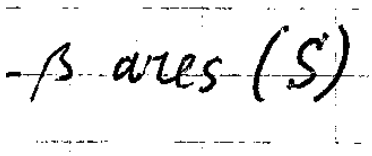
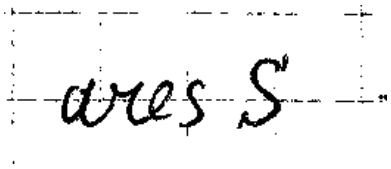


Diagram and typed description. The exponent has $\mathrm{ares}(S)$.

Original source details: GELFAND-P001055

ares S — the area of what is obtained, if supplemented with vertical walls



Original source details: GELFAND-P001055

Diagram and typed description. The explanation also has ares.

(build it out of solid cubes)

$$= e^{-\beta|\Lambda|} \prod_{\langle p,p'\rangle} e^{-\beta|h_p - h_{p'}|}, \tag{3}$$

$h_p = 0$ when $p \notin \Lambda$.

(boundary condition)

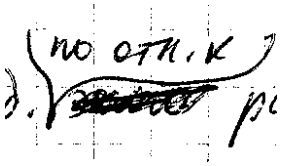
GELFAND-P001056

$$\mu(S) \equiv Z_\Lambda^{-1} w(S), \qquad Z_\Lambda = \sum_S{}' w(S). \tag{4}$$

— probability distribution.

Behavior of typical surfaces — ?

Results: let E — the expectation [inserted above the line: with respect to] the probability distribution μ as $\Lambda \nearrow Z^2$



Original source details: GELFAND-P001056

Uncertain reading. The word crossed out after “expectation” is difficult to read. It may be «этого» (“this”).

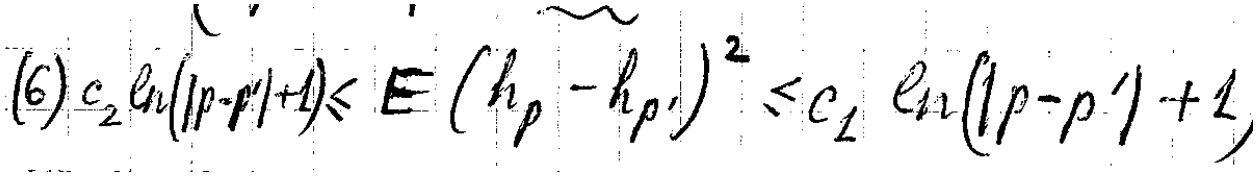
(a) for large β Eh_p^2 is bounded

$$\text{Prob}\{h : |h_p| = n\} \sim e^{-\text{const}\cdot n} \tag{5}$$

(a special case of Dobrushin’s general theorem)

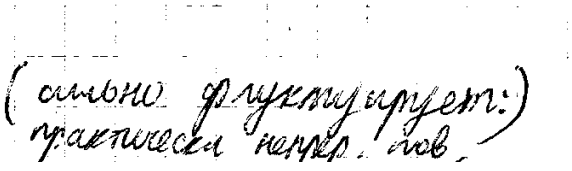
(b) β is small

$$c_2 \ln(|p-p'|+1) \leqslant E(h_p - h_{p'})^2 \leqslant c_1 \ln(|p-p'|+1) \tag{6}$$



Original source details: GELFAND-P001056

Diagram and typed description. The coefficient on the left is c_2, and the one on the right is c_1.

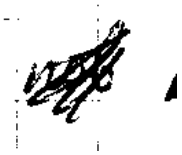


Original source details: GELFAND-P001056

Diagram and typed description. (fluctuates strongly.) Practically continuous behavior.

Enhancement of symm.

We shrink the little squares ⇒ a continuous symmetry group appears.



Original source details: GELFAND-P001056

Uncertain reading. A short entry is crossed out between “shrink” and “the little squares.” It may be «кв».

(too complicated to explain)

$$\mu(S) \sim \prod_{\langle p,p' \rangle} e^{-\beta|h_p - h_{p'}|} \tag{7}$$

$h_p \mapsto h_p + \text{const}$ symmetry (continuous)

Broken by the boundary conditions.

The same questions have yet to be addressed at small β

$$Z_{\beta,\gamma} \equiv \sum_{S:\partial S=\gamma} w_\beta(S), \qquad w_\beta(S) = e^{-\beta|S|} \tag{8}$$

$$\#\{S : \partial S = \gamma,\ |S| = \text{area } S = n\} \sim C_\gamma^n n^{\delta(\gamma)}, \quad n \text{ is large.} \tag{9}$$

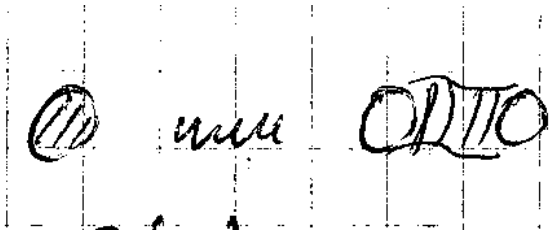
(from general combinatorial theorems)

Found in papers on the Peierls method.

$$\ln C_\gamma \equiv \lim_{n\to\infty} \frac{1}{n} \ln \#\{S \mid \partial S = \gamma,\ |S| = n\} \tag{10}$$

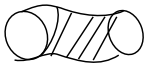
(11) Lemma C_γ does not depend on γ

Conjecture. Depends on the topological properties of γ:

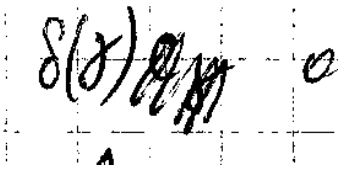


Original source details: GELFAND-P001056

Diagram and typed description.

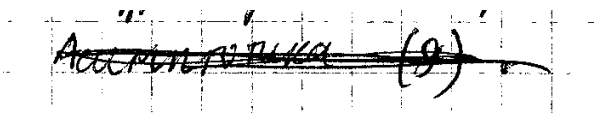
or 

$\delta(\gamma)$ determines the critical phenomena. $\delta(\gamma)$, apparently depends on the topology.



Original source details: GELFAND-P001056

Uncertain reading. A short entry after the first $\delta(\gamma)$ is crossed out. It may be «для» (“for”).



Original source details: GELFAND-P001056

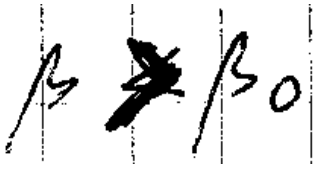
Diagram and typed description. ~~Asymptotics (9).~~

GELFAND-P001057

[source number: 27]

A difficult question: what if we take surfaces with $\forall$ topology.
How many handles does a typical surface have?

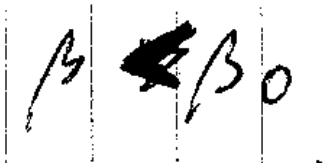
$$Z_{\beta,\gamma} < \infty \quad \text{when } \beta > \beta_0 \tag{12}$$



Uncertain reading. The relation sign has been overwritten. Possible readings are $\beta > \beta_0$ and $\beta \geqslant \beta_0$.

Original source details: GELFAND-P001057

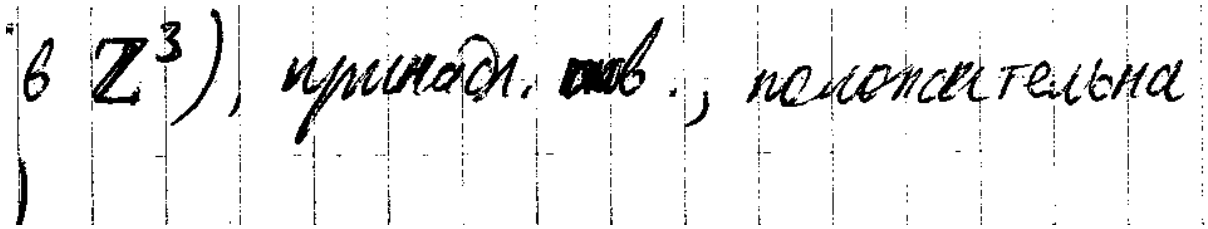
$$Z_{\beta,\gamma} = +\infty \quad \text{when } \beta < \beta_0 \qquad \text{(like a gas)} \tag{13}$$



Uncertain reading. The relation sign has been overwritten. Possible readings are $\beta < \beta_0$ and $\beta \leqslant \beta_0$.

Original source details: GELFAND-P001057

(13): The density of the squares (in Z^3), belonging to the surface, is positive
(like a Peano curve)



Original source details: GELFAND-P001057

Diagram and typed description. The abbreviation «пов.» (“surface”) is written over other strokes.

Conjecture, when $\beta_0 < \beta < \beta_R$ logarithmic fluctuations,
$\beta > \beta_R$ — roughening (coarsening)
— the shape becomes fixed, almost smooth (almost const)
Why bother?
1° NaCl is described by this model
2° This theory is equivalent to gauge theory $SU(N)$ in the limit $N \Rightarrow 0$ (!) (degenerate case)

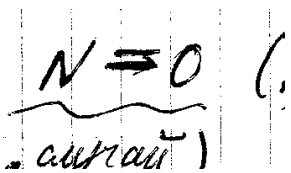


Diagram and typed description. A double arrow follows N.

Original source details: GELFAND-P001057

3° It is a fun problem.

Original source details: GELFAND-P001057

Uncertain reading. A short entry is crossed out after “fun.” It may begin with “pr.”

$$w_N(S) \underset{N\to 0}{\longrightarrow} w(S)$$

(Muscovites like to expand in $1/N$ when $N \to \infty$).

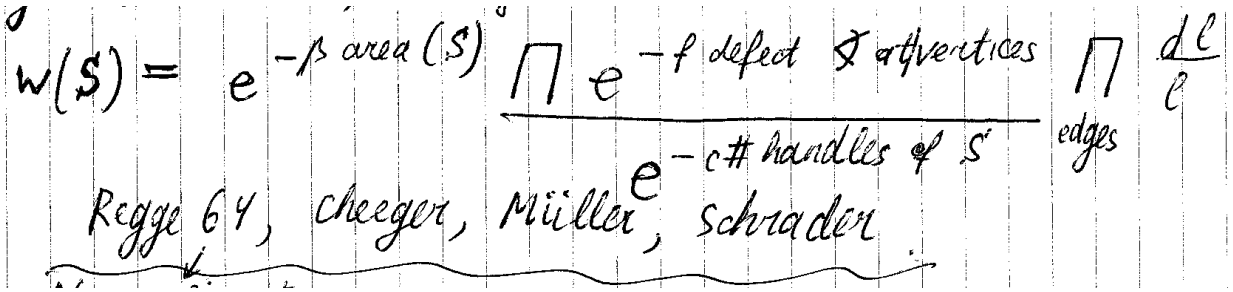
References. Phys. Lett. Amos Maritan, C. Omero

Generalizations→

Continuous case. Surfaces that do not lie on a lattice.

Piecewise linear surfaces made of Δs.

$$w(S) = e^{-\beta \,\text{area}(S)} \frac{\prod e^{-f \,\text{defect}\,\angle \,\text{at vertices}}}{e^{-c\,\#\,\text{handles of } S}} \prod_{\text{edges}} \frac{d\ell}{\ell} \tag{14}$$



Original source details: GELFAND-P001057

Uncertain reading. The line above the lower exponential is typeset as a fraction bar. Its placement is visible in the manuscript.

Original source details: GELFAND-P001057

Diagram and typed description.
Regge 64, Cheeger, Müller, Schrader

↓

Nuovo Cimento

Question: convergence when approaching a continuous surface.

$$R = \sum_f \text{area of } (d-2) \text{ simplex, } f \times \text{defect } \angle$$

$$\text{Theor.} \quad R \longrightarrow \int \sqrt{g(x)} R(x) d^d x. \tag{15}$$

$(14) \longrightarrow$ Polyakov's string theory

Exercise: Prove this!

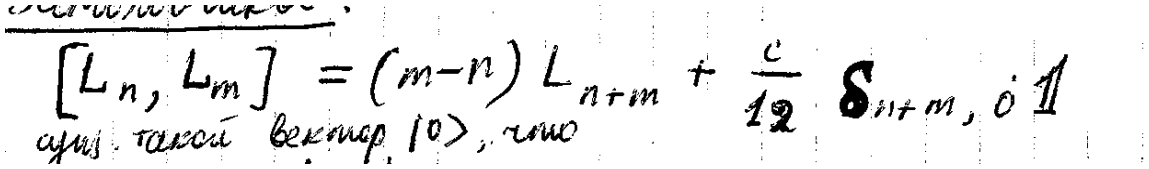
GELFAND-P001058

1/XI-82 Gelfand Seminar.

Zamolodchikov.

Suppose

$$[L_n, L_m] = (m-n)L_{n+m} + \frac{c}{12}\delta_{n+m,0}\mathbf{1} \qquad \text{(Virasoro algebra)}$$



Original source details: GELFAND-P001058

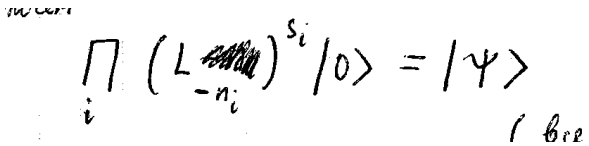
Uncertain reading. The central term visibly contains $\frac{c}{12}\delta_{n+m,0}\mathbf{1}$.

and there is a vector $|0\rangle$, such that

$$L_n|0\rangle = 0, \quad n > 0,$$
$$L_0|0\rangle = \Delta|0\rangle.$$

and

$$\prod_i (L_{-n_i})^{s_i}|0\rangle = |\psi\rangle$$



Uncertain reading. An index after L is crossed out. It may be n_i or m_i. The index $-n_i$ is inserted below.

Original source details: GELFAND-P001058

(all vectors are obtained in this way)

Suppose $\varphi(z)$ — an operator acting in this space, and

$$[L_n, \varphi(z)] = z^{n+1}\frac{d\varphi(z)}{dz} + (n+1)\Delta z^n \varphi(z)$$

It is uniquely determined by this formula.

First question: what can be said about it?

— how to calculate

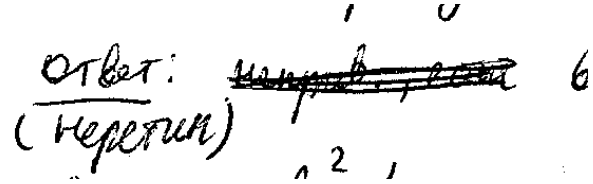
$$\langle 0|\varphi(z_1)\dots\varphi(z_n)|0\rangle \quad ?$$

$L_{-n}^{+} = L_n$, $\langle 0 \mid 0\rangle = 1 \Rightarrow$ recover the inner product.

Second question: what relations on the states can be written down?

Or: is this module irreducible?

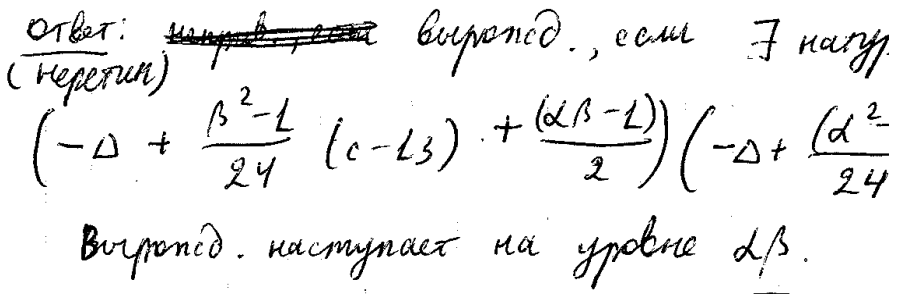
Answer: (Neretin) degenerate if $\exists$ natural numbers α, β, such that



Uncertain reading. The crossed-out words before “degenerate” appear to read “irreducible if.”

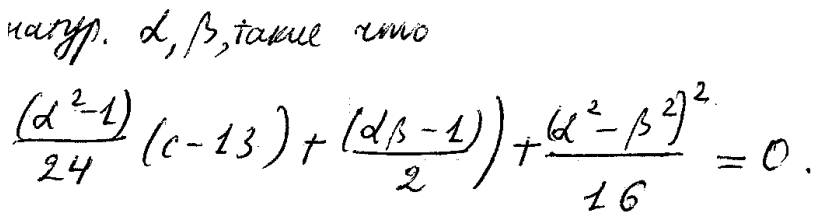
Original source details: GELFAND-P001058

$$\left(-\Delta+\frac{\beta^2-1}{24}(c-13)+\frac{(\alpha\beta-1)}{2}\right)\left(-\Delta+\frac{(\alpha^2-1)}{24}(c-13)+\frac{(\alpha\beta-1)}{2}\right)+\frac{(\alpha^2-\beta^2)^2}{16}=0.$$



Original source details: GELFAND-P001058

Diagram and typed description. The first factor contains β^2-1.



Original source details: GELFAND-P001058

Diagram and typed description. The second factor contains α^2-1.

Degeneracy occurs at level $\alpha\beta$.

$\vec{x}(\xi_1,\xi_2)$ — surface, $g_{ab}(\xi_1,\xi_2)$ — metric, $a,b=1,2$.

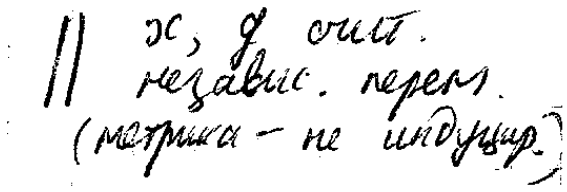


Diagram and typed description. // x, g are treated as independent variables (the metric is not induced)

Original source details: GELFAND-P001058

Suppose

$$Z \equiv \int \mathcal{D}\vec{x}\,\mathcal{D}g_{ab}\,\exp\left\{-\tfrac{1}{2}\int \sqrt{g}\,g^{ab}\partial_a\vec{x}\,\partial_b\vec{x}\,d^2\xi\right\} \exp\left\{-\mu_0^2\int\sqrt{g}\,d^2\xi\right\}. \quad (1)$$

Evaluation — ?

Invariant under the transformations

$$\begin{cases}\xi^a \mapsto \xi^a + \varepsilon^a(\xi),\\ g_{ab}(\xi) \mapsto g_{ab} + \nabla_a\varepsilon_b + \nabla_b\varepsilon_a.\end{cases}$$

We divide by this group (the volume element is understood to be factored) — one must integrate over the orbits.

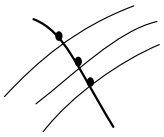


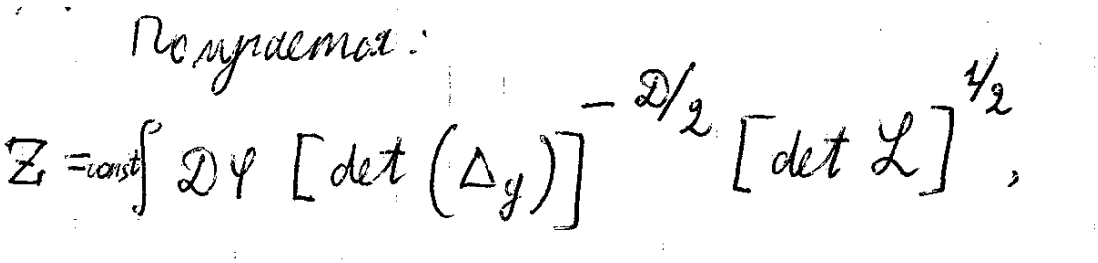
Diagram and typed description. The strokes thicken at the three intersections. It is unclear whether these are arrowheads.

Original source details: GELFAND-P001058

Representative: $\vec{x}(\xi)$ — arbitrary, $g_{ab}(\xi) = \delta_{ab}\rho(\xi)$

This gives:

$$Z = \text{const}\int \mathcal{D}\varphi\,[\det(\Delta_g)]^{-D/2}\,[\det\mathcal{L}]^{1/2}, \quad (2)$$



Original source details: GELFAND-P001058

Uncertain reading. The symbol after $\mathcal{D}$ has a loop on the right. It is provisionally transcribed as φ. Another possible reading is ψ.

GELFAND-P001059

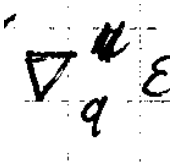
[source number: 38]

where

Uncertain reading. A short entry in the upper margin is crossed out. Its reading is uncertain.

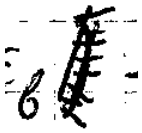
Original source details: GELFAND-P001059

$$(\mathcal{L}\varepsilon)_a = \nabla^b\big(\nabla^a_a\varepsilon_b + \nabla_b\varepsilon_a - g_{ab}\nabla^c\varepsilon_c\big). \tag{3}$$



Original source details: GELFAND-P001059

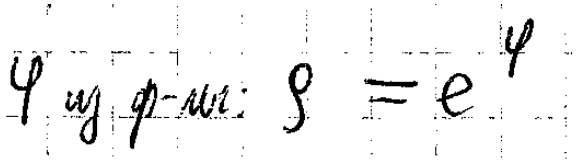
Uncertain reading. The first ∇ in parentheses has a subscript a and a thickened superscript resembling a. The superscript may be cancelled.



Original source details: GELFAND-P001059

Uncertain reading. A vertical entry after the first ε_b is crossed out.

On evaluating this, we obtain:



Original source details: GELFAND-P001059

Diagram and typed description. The same looped sign appears in (4), (6), and $\rho = e^{\varphi}$. It is transcribed as φ. Another possible reading is ψ.

D — dimension of the space, $D < 26$.

$$Z = \int \mathcal{D}\varphi \cdot \exp\left\{-\frac{26-D}{48\pi}\int\left[\frac{1}{2}\partial_a\varphi\,\partial_a\varphi + m_0^2 e^{\varphi}\right]d^2\xi\right\}. \tag{4}$$

φ from the formula: $\rho = e^{\varphi}$.

$$\Delta_g = \frac{1}{\sqrt{g}}\partial_a\big(\sqrt{g}\,g^{ab}\partial_b\big).$$

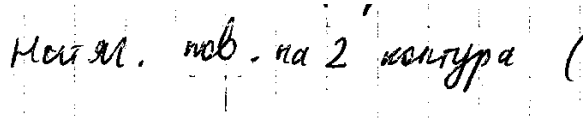
The string problem will be solved if we evaluate this integral.

$D = 26 \Rightarrow$ the answer is the same as for $g_{ab} = \delta_{ab}$.

This prescription is used in dual models. They are exact at $D = 26$.

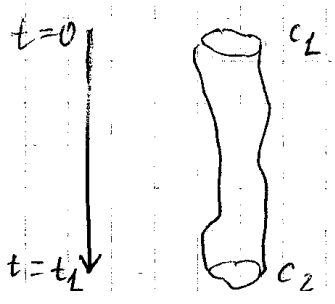
What surfaces should we integrate over?

Surfaces stretched across 2 contours (evolution of a contour).



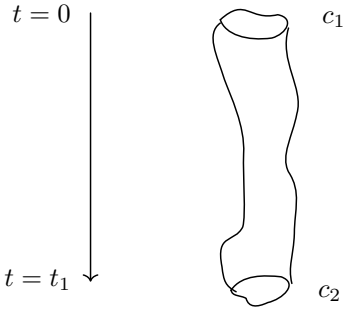
Original source details: GELFAND-P001059

Uncertain reading. The abbreviation at the start of the line appears to read «Натян.», for “stretched.”



Original source details: GELFAND-P001059

Diagram and typed description.



$$Z(c_1, c_2, t_1) \underset{t_1 \to \infty}{=} \sum_n G(c_1, n)\, G(c_2, n)\, e^{-E_n t_1}. \tag{5}$$

The spectrum of stationary states in Liouville theory is the QFT spectrum for the Lagrangian

Original source details: GELFAND-P001059

Uncertain reading. An entry above the words “Liouville theory” is crossed out.

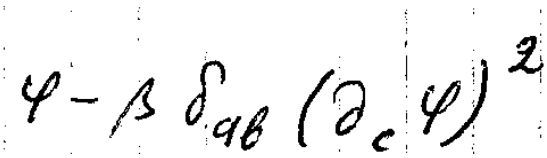
$$\int_M \left[\frac{1}{2}\partial_a\varphi\,\partial_a\varphi + m_0^2 e^{\varphi}\right] d^2\xi. \tag{6}$$

Conjecture: the spectrum is related to the system L_n, described on the previous page.

$$z = \xi_1 + i\xi_2,$$
$$\bar{z} = \xi_1 - i\xi_2.$$

Traceless two-dimensional conserved energy–momentum tensor:

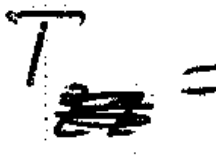
$$T_{ab} = \partial_a\varphi\,\partial_b\varphi - \tfrac{1}{2}\delta_{ab}(\partial_c\varphi)^2 + \alpha\partial_a\partial_b\varphi - \beta\delta_{ab}(\partial_c\varphi)^2.$$



Original source details: GELFAND-P001059

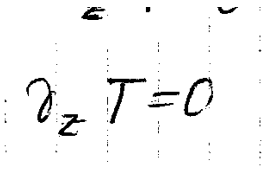
Diagram and typed description. The last term reads $-\beta\delta_{ab}(\partial_c\varphi)^2$.

$$T = \varphi_z^2 + 2\varphi_{zz}, \qquad \partial_{\bar{z}} T = 0,$$
$$\bar{T} = \varphi_{\bar{z}}^2 + 2\varphi_{\bar{z}\bar{z}}, \qquad \partial_z T = 0.$$



Original source details: GELFAND-P001059

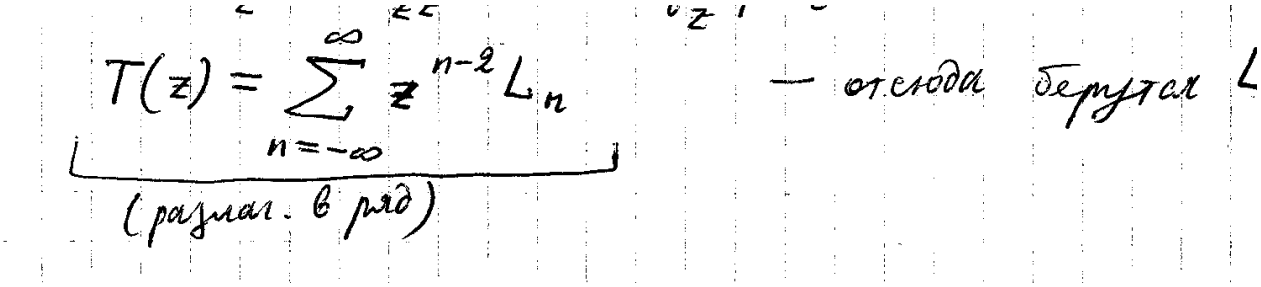
Uncertain reading. The subscript on the first T is crossed out. It may be zz.



Original source details: GELFAND-P001059

Diagram and typed description. The T in the second equality on the right has no additional bar.

$$\underbrace{T(z) = \sum_{n=-\infty}^{\infty} z^{n-2} L_n}_{\text{(expansion in a series)}} \quad \text{— this is where we obtain } L_n!$$



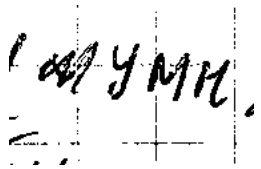
Original source details: GELFAND-P001059

Diagram and typed description. A lower bracket encloses the series and its left-hand side.

GELFAND-P001060

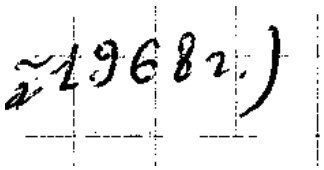
Beilinson A.

I.M.G. Gelfand, Ponomarev (UMN, $\underset{[2?]}{\sim}$ 1968) reduced the description of indecomposable representations of the Lorentz group to a problem in linear algebra.



Original source details: GELFAND-P001060

Uncertain reading. A short mark before “UMN” is crossed out.



Original source details: GELFAND-P001060

Uncertain reading. A mark resembling 2 appears below $\sim$.

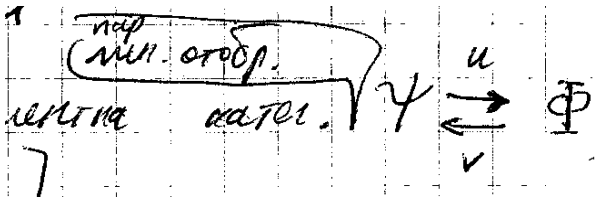
B. model example. — $\mathrm{SL}(2,\mathbb{R})$

On $\mathbb{P}^1$ we must classify D-modules with singularities at two fixed points.

Theorem 1. the category of D-modules on the disk with a fixed singularity is equivalent to the category

Original source details: GELFAND-P001060

Diagram and typed description.



Original source details: GELFAND-P001060

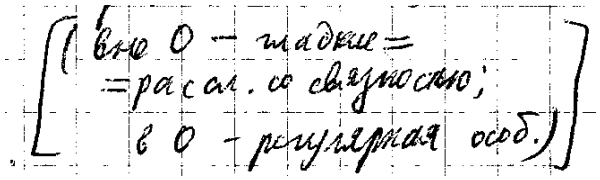
Diagram and typed description.

pairs of linear maps

$$\Psi \underset{v}{\overset{u}{\rightleftarrows}} \Phi$$

"Pairs of" is written above "linear maps."

such that $uv - \mathrm{Id}$ is invertible.

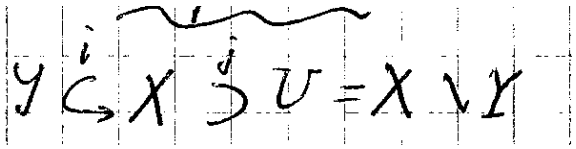


Original source details: GELFAND-P001060

Diagram and typed description.

$$\left[\begin{array}{l}\text{outside } 0 \text{ — smooth } = \\ = \text{bundles with connection;} \\ \text{at } 0 \text{ — regular singularity.}\end{array}\right]$$

Geometric problem.



Original source details: GELFAND-P001060

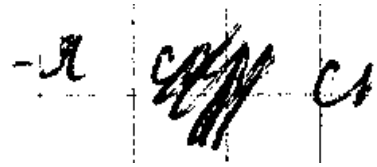
Diagram and typed description.

$$Y \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} U = X \setminus Y$$

A sheaf on X is given as follows:

$$\mathcal{F}_X = (\mathcal{F}_U, \mathcal{F}_Y, \quad \mathcal{F}_Y \longrightarrow i^* j_* \mathcal{F}_U)$$

(gluing function)



Original source details: GELFAND-P001060

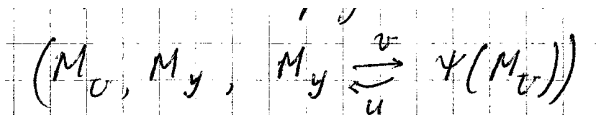
Uncertain reading. A short entry between "function" and "gluing" is crossed out. It may begin with «ск».

For representations we must solve the analogous problem on the space of flags.

Original source details: GELFAND-P001060

Uncertain reading. The crossed-out formula may read $Y = f^{-1}(0)$.

Theorem. the category of D-modules with regular singularities on X is equivalent to the category of tuples

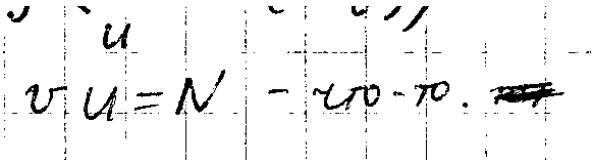


Original source details: GELFAND-P001060

Diagram and typed description.

$$(M_U,\ M_Y,\quad M_Y \underset{u}{\overset{v}{\rightleftarrows}} \Psi(M_U))$$

$vu = N$ — something.



Uncertain reading. A short entry after "something" is crossed out.

Original source details: GELFAND-P001060

GELFAND-P001061

[source number: 48]

15/XI-82 Gelfand Seminar.

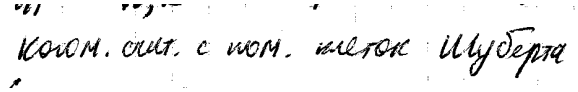
I.M.G. Problem. (imprecise)

$G_{n,k}$ — Grassmannian, k-dimensional planes in an n-dimensional space.

Uncertain reading. An entry at the start of the line is crossed out. It resembles G with additional marks.

Original source details: GELFAND-P001061

Cohomology is calculated using Schubert cells.



Uncertain reading. The first two abbreviations provisionally read «Когом. счит.», for "cohomology is calculated."

Original source details: GELFAND-P001061

(The cell is specified by the dimensions of intersections with the flag

$$0 = d_0 \leqslant d_1 \leqslant d_2 \leqslant \ldots \leqslant d_n = k,$$

$$\dim \text{ of the cell} = \sum d_j.$$

)

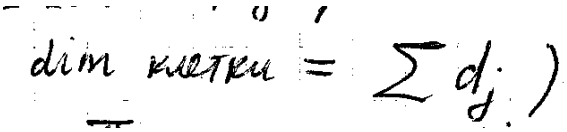


Diagram and typed description. The dimension formula reads $\sum d_j$.

Original source details: GELFAND-P001061

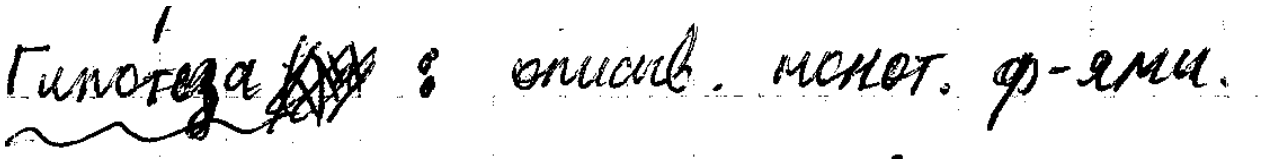
Consider a II_1 factor. In it, consider α-dimensional projections (a continuous analogue of Grassmannians). Is there an analogue of a Schubert cell?

Flag of continuous dimension $[0, 1]$.

Does this give a Grassmannian?

Continuous cohomology!?

Conjecture: described by monotone functions.



Original source details: GELFAND-P001061

Uncertain reading. A short entry after "Conjecture" is crossed out.

An even vaguer question: what topology? and what cohomology?

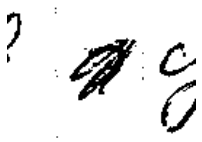
Uncertain reading. A mark before “An even” is crossed out.

Original source details: GELFAND-P001061

Yu. I. Manin.

1) Flag supermanifolds

2) The Radon–Penrose transform in a supermathematical setting.



Uncertain reading. A single mark is crossed out before “supermathematical.”

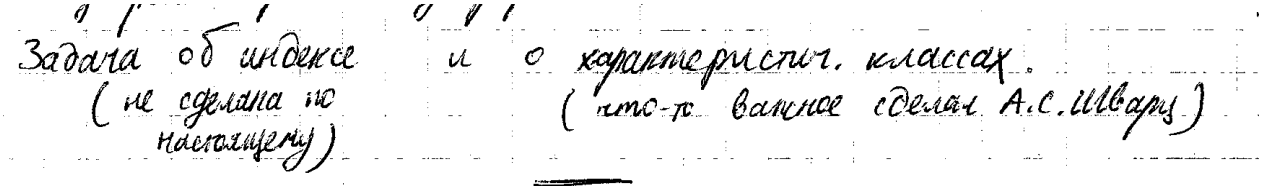
Original source details: GELFAND-P001061

3) Applications to solving nonlinear equations: new solutions of the non-self-dual Yang–Mills equations

(ordinary and supersymmetric)

4) Specific flag manifolds, their deformation, and so on in supersymmetry and supergravity.

5) The problem of the index and of characteristic classes.



Original source details: GELFAND-P001061

Diagram and typed description. [Below “of the index”] (has not really been done)
[Below “of characteristic classes”] (A. S. Schwarz did something important)

The talk will cover 1)–4)

1) Reminder about the self-dual Yang–Mills equations

$T = \mathbb{C}^4$ [antiholomorphic involution $T \oplus T^*$]

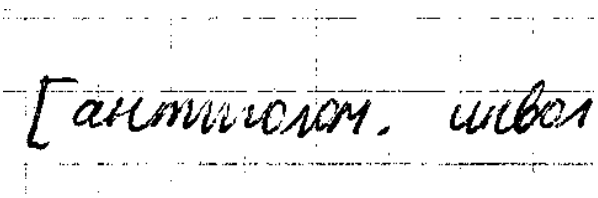


Diagram and typed description. The first abbreviation appears to read «антиголом.», for “antiholomorphic.”

Original source details: GELFAND-P001061

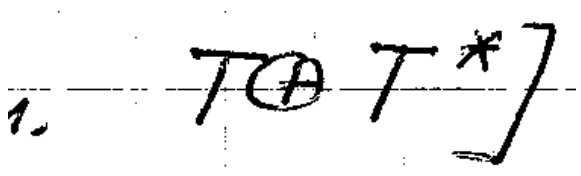
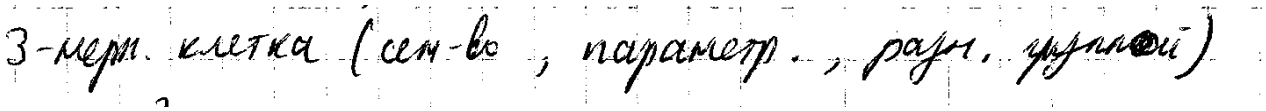


Diagram and typed description. The sign between T and T^* is $\oplus$.

Original source details: GELFAND-P001061

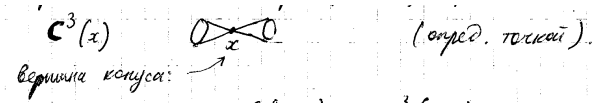
dim $G(2,T)=4$ = complex compact Minkowski space.

3-dimensional cell (family, parametrized by a group [one uncertain abbreviation])



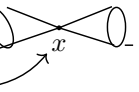
Original source details: GELFAND-P001061

Uncertain reading. The abbreviation before “group” may read «разн.» or «разл.».



Original source details: GELFAND-P001061

Diagram and typed description.

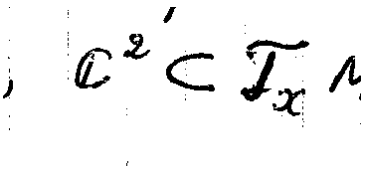


(determined by a point).

$$M^4 = \text{real part of } G(2,T)\setminus C^3(\infty) \leftarrow \infty\text{-cone}.$$

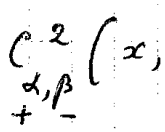
2-dimensional cells: 2 types

$$C^2_{\substack{\alpha,\beta\\+\ -}}(x, \mathbb{C}^2 \subset T_x M) \text{ — } \pm\text{-planes}, \quad (\alpha,\beta)\text{-planes}.$$



Original source details: GELFAND-P001061

Diagram and typed description. Inside the parentheses, the symbol before $\subset$ is a double-stroked $\mathbb{C}$ with superscript 2.

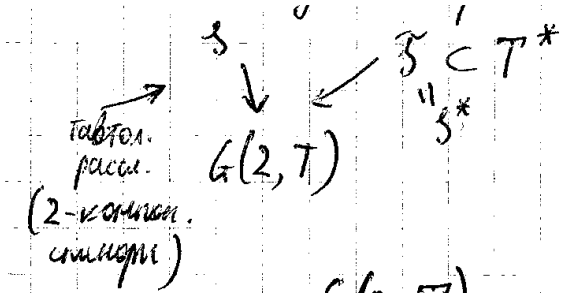


Original source details: GELFAND-P001061

Diagram and typed description. The $+$ is written below α, and the $-$ below β.

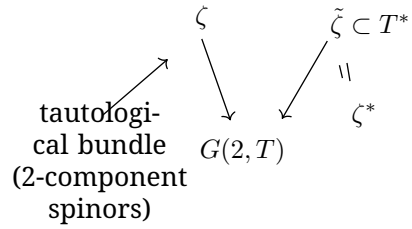
GELFAND-P001062

C^1 — null lines (light rays)



Original source details: GELFAND-P001062

Diagram and typed description.



The bundle symbol is provisionally transcribed as ζ. The meaning of the two strokes below the right-hand label is uncertain.

Problem. $G(2,T)$

Describe the superhomogeneous complex manifolds M, for which $M_{\text{red}} = G(2,T)$.

Condition: M — a flag manifold for simple Lie supergroups of classical type.

List (beginning).

a) $F(2|0,2|N,\mathbb{C}^{4|N}) = M$

Uncertain reading. A mark resembling 2 is crossed out after $2 \mid N$.

Original source details: GELFAND-P001062

(flags of dimensions $2|0, 2|N$ in $\mathbb{C}^{4|N}$).

$$N = 0, 1, 2, 3, \ldots \qquad \mathbb{C}^{4|N} = (\mathbb{C}^4)_0 \oplus (\mathbb{C}^N)_1.$$

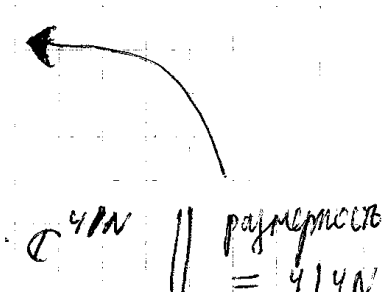


Diagram and typed description.

$\mathbb{C}^{4|N}$ — $\|$ dimension $= 4|4N$

The note "dimension $= 4 \mid 4N$" is to the right of the inclusions. The arrow points up and to the left.

Original source details: GELFAND-P001062

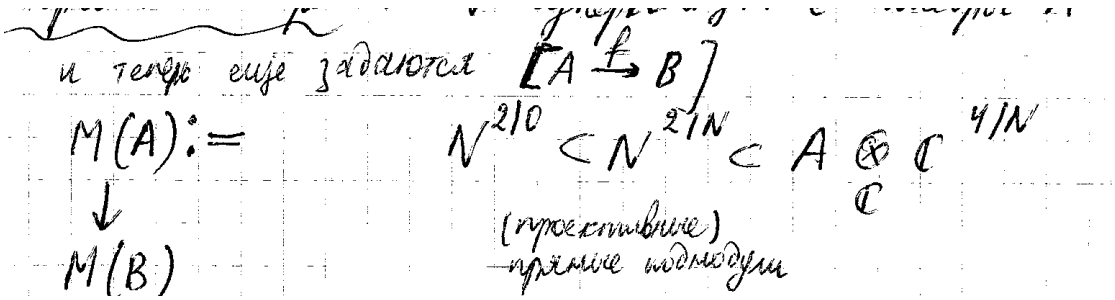
Naïve definition:

$$\text{points } x \in F(2|0,2|N,\mathbb{C}^{4|N}) = \underline{\mathbb{C}^{2|0} \subset \mathbb{C}^{2|N}} \subset \mathbb{C}^{4|N}.$$

But there are too few coordinates.

Correct definition: $\forall$ supercommutative $\mathbb{C}$-algebra A we specify $M(A)$ and now we also specify $[A \xrightarrow{f} B]$

$$\begin{array}{rl} M(A) := & N^{2|0} \subset N^{2|N} \subset A \otimes_{\mathbb{C}} \mathbb{C}^{4|N} \\ \downarrow & \text{(projective)} \\ M(B) & \text{direct submodules} \end{array}$$



Original source details: GELFAND-P001062

Diagram and typed description. The arrow goes from $M(A)$ to $M(B)$. The submodule description is written below the inclusions.

Coordinates: $G(1|0, \mathbb{C}^{m|n}) := P^{m-1|n} \supset P$.

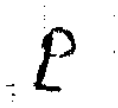


Diagram and typed description. The symbol to the right of $\supset$ appears to be P.

Original source details: GELFAND-P001062

Homogeneous coordinates: $(x_1, \dots, x_m; \xi_1, \dots, \xi_n)$.

$$\text{Cover } P^{m-1,n} = \bigcup_{i=1}^{m} U_i, \quad U_i : \left(\frac{x_1}{x_i}, \dots, \frac{x_m}{x_i}, \frac{\xi_1}{x_i}, \dots, \frac{\xi_m}{x_i} \right).$$

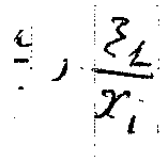


Diagram and typed description. The denominator of the first odd coordinate appears to be x_i.

Original source details: GELFAND-P001062

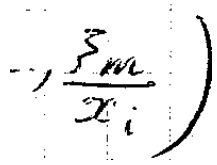


Diagram and typed description. The final fraction reads ξ_m/x_i. The list of homogeneous coordinates ends in ξ_n.

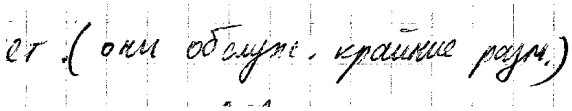
Original source details: GELFAND-P001062

Exercise: relate this to the preceding discussion (big cells).

We can have: $\xi_i = 0$ — a cell of codimension $0|1$.

It is unclear in which cohomology they lie?

Differential and integral forms are also insufficient. (they serve the extreme dimensions)



Uncertain reading. The abbreviation appears to be «обслуж.», for “serve.”

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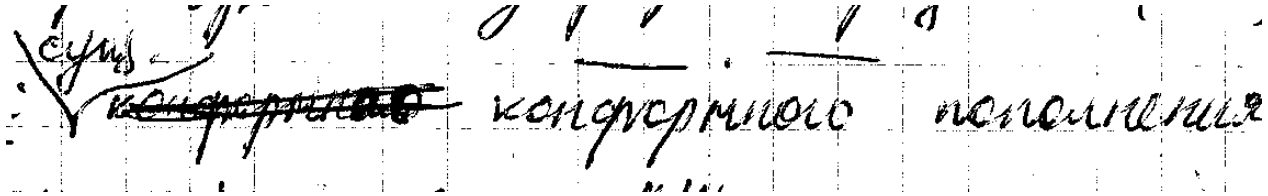
Schwarz’s objects are intermediate but have strange properties.

Real points of the big cell in $F(2|0, 2|N; \mathbb{C}^{4|N}) =$

$$= N\text{-extended physicists' superspace} : (x^m; \theta_{\alpha i}, \theta^{\beta i}).$$

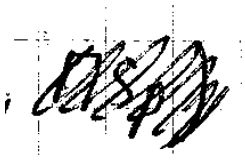
Requirement: [inserted existence of] ~~conformal~~ conformal completion.



Original source details: GELFAND-P001062

Diagram and typed description. “Existence” is written with an insertion mark. The earlier “conformal” is crossed out.

b) $FI(2|0, 2|2, \mathbb{C}^{4|4}, \Pi\mathrm{Sp})$ dimension $= 4|8$.



Uncertain reading. An entry before $\mathbb{C}^{4|4}$ is crossed out. It may read OSp.

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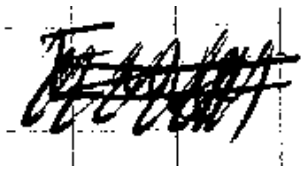
GELFAND-P001063

[source number: 49]

I — isotropic, ΠSp — the type of form with respect to which the flags are isotropic.
(odd symmetric form)
There are also series and isolated manifolds.



Uncertain reading. A word before “There are also” is crossed out. It may read «Бывают» (“there are”) or «Будут» (“there will be”).

Original source details: GELFAND-P001063

2) $M = G(2, T)$, $P(T) = G(1, T)$.

GELFAND-P001064

22/XI-82 Gelfand Seminar.
A. M. Polyakov.
(Alexander Markovich).
The string problem

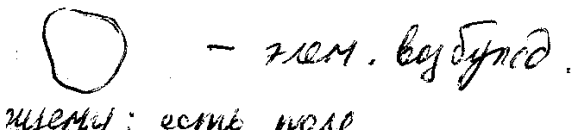
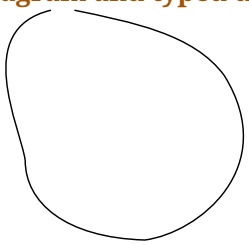


Diagram and typed description.



— elem. excitation.

Original source details: GELFAND-P001064

Reduces to the following: there is a field

$$\varphi = \varphi(\xi_1, \xi_2) \qquad (\sim \text{ surface in isothermal coordinates})$$

$$g_{ab}(\xi) = e^{\varphi(\xi)}\delta_{ab}. \tag{1}$$

$$\mathcal{L} = \frac{1}{\gamma}\left\{\frac{1}{2}(\partial_a\varphi)^2 + e^{\varphi}\right\}, \qquad S = \int \mathcal{L}\, d^2\xi, \tag{2}$$

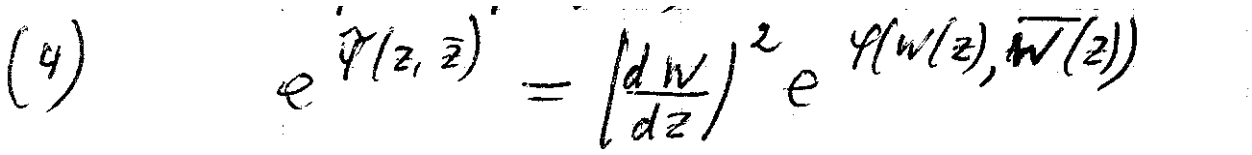
$$\frac{1}{\gamma} = \frac{26 - D}{48\pi} \tag{3}$$

(D — dimension of the space into which the two-dimensional surface under consideration is embedded)

Why does it have the form (2) (apart from γ)?

Fix the metric (1). The remaining freedom: conformal transformations.

$$e^{\widetilde{\varphi}(z,\bar{z})} = \left|\frac{dw}{dz}\right|^2 e^{\varphi(w(z),\overline{w}(z))}. \tag{4}$$



Original source details: GELFAND-P001064

Diagram and typed description. In the exponent on the right, the second w has a bar. The final z has none.

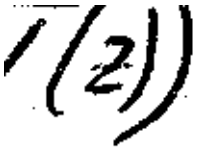
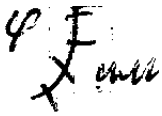


Diagram and typed description. At the right of the formula is "(2)" with an additional stroke.

Original source details: GELFAND-P001064

Lagrangian (2) — the only local Lagrangian (depending only on the value of the function and its derivatives at the point), invariant under transformation (4).

$\mathcal{L}$ is called the Liouville Lagrangian (the corresponding equation $\Delta\varphi = e^{\varphi}$ or $\partial^2\varphi = e^{\varphi}$).



Uncertain reading. A bracket-like mark after the first e^{φ} is crossed out.

Original source details: GELFAND-P001064

$$\partial^2\varphi \equiv \Delta\varphi = e^{\varphi}. \tag{5}$$

Can be solved explicitly. ($\sim$ curvature is constant)

Classical solution of equation (5):

$$\varphi = \log\left(\frac{|\partial_z w|^2}{(\operatorname{Im} w)^2}\right), \qquad w(z) \text{ — any holomorphic function.} \tag{6}$$

Consider a sphere with punctures that are at ∞ in the sense of the metric.

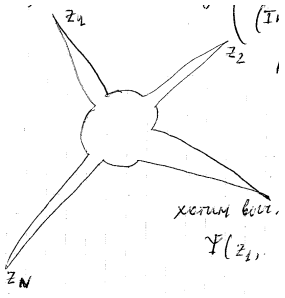


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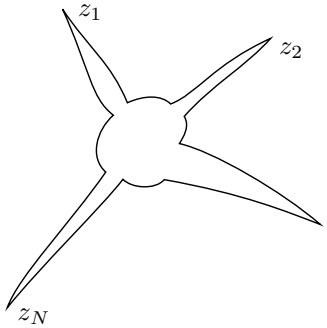
Diagram and typed description.



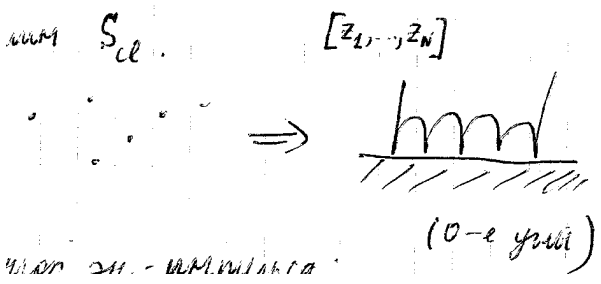
$$e^{\varphi} \underset{z\to z_i}{\longrightarrow} \frac{2}{|z-z_i|^2 \log^2|z-z_i|}.$$

we want to calculate the quantum quantity

$$\Psi(z_1,\dots,z_N) = \int \mathcal{D}\varphi(z) e^{-S(\varphi)} \underset{\gamma\to 0}{\simeq} e^{-S_{\text{cl}}(z_1,\dots,z_N)}. \tag{7}$$

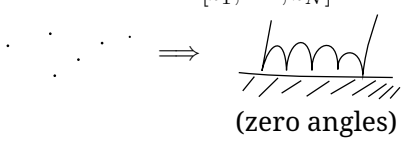
Let us calculate S_{cl}.

(8)



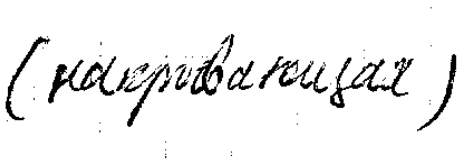
Original source details: GELFAND-P001065

Diagram and typed description.



(zero angles)

The caption below the drawing appears to read “zero angles.”



Original source details: GELFAND-P001065

Diagram and typed description. (covering surface)

Let us calculate the energy–momentum tensor:

$$T = \tfrac{1}{2}(\partial_z \varphi)^2 - \partial_z^2 \varphi, \qquad T = T_{11} - T_{22} + 2iT_{12}, \qquad T_{11} + T_{22} = 0. \tag{9}$$

(T_{ij} is recovered from T, since T_{ij} — real)

$$T(z|z_1, \ldots, z_N) = \underset{\text{(Schwarzian derivative)}}{\mathcal{D}}[w(z)] = \frac{w'''}{w'^2} - \frac{3}{2}\frac{(w'')^2}{(w')^4}. \tag{10}$$

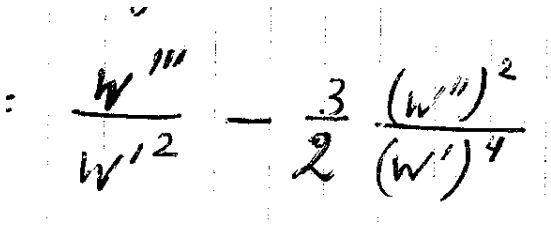


Diagram and typed description. The denominators read w'^2 and $(w')^4$.

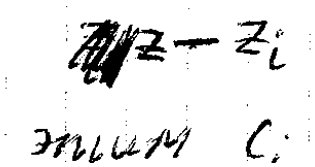
Original source details: GELFAND-P001065

(invariant under fractional linear transformations)

$$\widetilde{w}(z) = \frac{\alpha w + \beta}{\gamma w + \delta}.$$

Hence T — is single-valued and has only poles:

$$T(z|z_1, \ldots, z_N) = \frac{1}{4}\sum_1^N \frac{1}{(z - z_i)^2} + \sum_1^N \frac{c_i(z_1, \ldots, z_N)}{z - z_i}. \tag{11}$$



Uncertain reading. An entry before $z - z_i$ in the final denominator is heavily crossed out.

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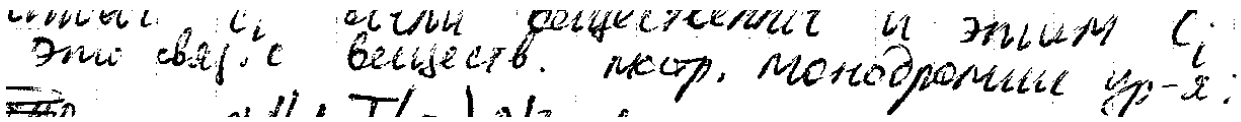
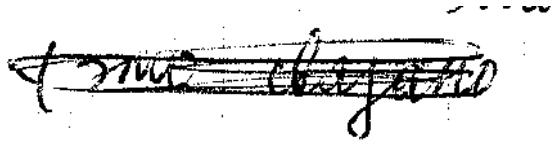
It is important that c_i be real, and this determines c_i .

(c_i — accessory coefficients)



Original source details: GELFAND-P001065

Diagram and typed description. In small handwriting above (12), “This is connected with the real monodromy coefficients of the equation.”



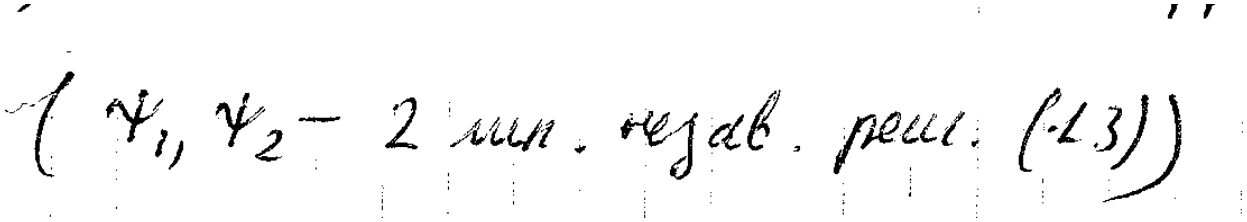
Uncertain reading. A phrase before (12) is crossed out. It may read “this is connected.”

Original source details: GELFAND-P001065

$$\psi'' + T(z)\psi = 0. \tag{12}$$

$$w(z) = \frac{\psi_1(z)}{\psi_2(z)}. \tag{13}$$

(ψ_1, ψ_2 — 2 linearly independent solutions of (13))



Original source details: GELFAND-P001065

Diagram and typed description. The reference after "solutions" is "(13)."

Condition on variation of c: the monodromy changes but remains real,

(this is the problem)

(in $\mathrm{SL}(2,\mathbb{R})$, as opposed to $\mathrm{SL}(2,\mathbb{C})$)

GELFAND-P001066

[source number: 55]

Relation:

$$c_i(z_1, \ldots, z_N) = \frac{\partial S_{\mathrm{cl}}}{\partial z_i}. \tag{14}$$

Diagram and typed description. ___ · ___

Original source details: GELFAND-P001066

The problem is solved via the quantum problem, then the semiclassical limit as $\gamma \to 0$!

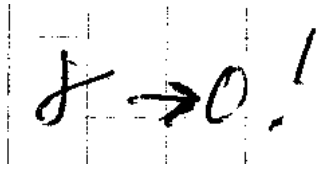
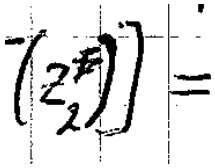


Diagram and typed description. The parameter in the semiclassical limit is transcribed as γ.

Original source details: GELFAND-P001066

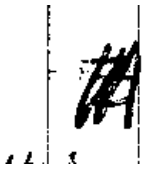
$$[T(z_1), T(z_2)] = \big(T(z_1) + T(z_2)\big)\delta'(z_1 - z_2) + \frac{c}{12}\delta'''(z_1 - z_2). \tag{15}$$



Uncertain reading. A small superscript over z_2 on the left of the first (15) is crossed out. It may be a digit.

Original source details: GELFAND-P001066

(the dependence on $z_1, \dots, z_N$ is left unwritten)



Uncertain reading. A mark before the parenthesis is heavily crossed out.

Original source details: GELFAND-P001066

or:

$$[T(z), T(w)] = \big(T(z) + T(w)\big)\delta'(z-w) + \frac{c}{12}\delta'''(z-w) \tag{15}$$

(in the classical case. Poisson brackets)

Let us calculate:

Lagrangian:

$$\mathcal{L} = \int \big(\partial_z\varphi\,\partial_{\bar z}\varphi - e^{\varphi}\big)$$

$$p = \frac{\partial\mathcal{L}}{\partial(\partial_{\bar z}\varphi)} = \partial_{\bar z}\varphi \tag{16}$$

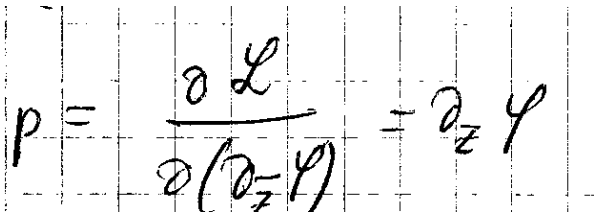


Diagram and typed description. On the right of (16), a separate bar is visible over the subscript z.

Original source details: GELFAND-P001066

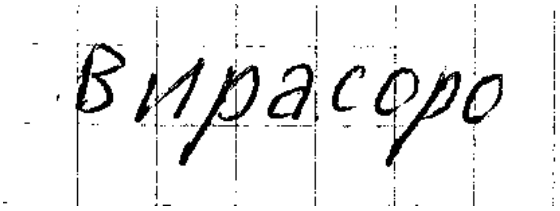
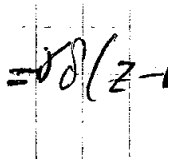


Diagram and typed description. [In the right margin] Virasoro

Original source details: GELFAND-P001066

whence

$$\{\varphi(z), \partial_w\varphi(w)\} = \gamma\delta(z-w). \tag{17}$$



Uncertain reading. A separate mark before δ appears to be γ.

Original source details: GELFAND-P001066

$\{T(z), T(w)\}$ and so on.

Expand in a Laurent series:

$$T(z) = \sum_n (L_n z^n + L_{-n} z^{-n}) \tag{18}$$

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12} n(n^2-1), \quad c = 26 - D. \tag{19}$$

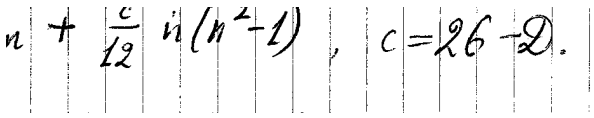


Diagram and typed description. A comma follows $n(n^2-1)$ in (19).

Original source details: GELFAND-P001066

(this could have been predicted from the Gelfand–Fuchs result, from the form of the central extension)

$$\widetilde{z} = z + \varepsilon(z) \qquad \text{(variation)}$$

$$\delta\varphi = \mathrm{Re}\big(\varepsilon(z)\partial_z\varphi + \partial_z\varepsilon\big)$$

A number is added to the action $\rightsquigarrow$ a central extension appears).

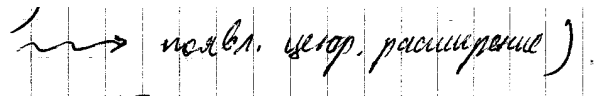


Diagram and typed description. A wavy arrow connects the two phrases.

Original source details: GELFAND-P001066

Let us try to find Ψ purely algebraically:

$$\text{Let} \qquad T_\varepsilon = \int \varepsilon(z) T(z)\, d^2 z \tag{20}$$

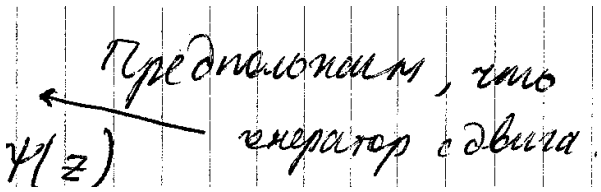


Diagram and typed description. $\longleftarrow$ Suppose it is the generator of the shift.

Original source details: GELFAND-P001066

$$\delta_\varepsilon \psi = [T_\varepsilon, \psi(z)] = \varepsilon(z)\frac{d\psi}{dz} + \Delta \cdot \varepsilon'(z)\psi(z) \tag{21}$$

(variation of the field ψ)

$$[L_n, \psi(z)] = z^{n+1}\frac{d\psi}{dz} + (n+1)\Delta\psi(z). \tag{22}$$

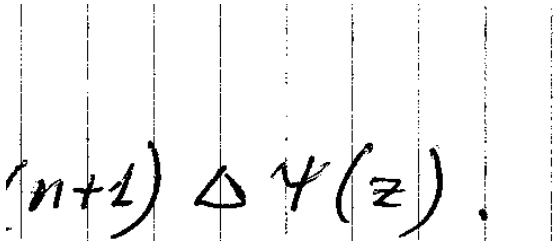


Diagram and typed description. The second term of (22) reads $(n+1)\Delta\psi(z)$.

Original source details: GELFAND-P001066

$\psi(z)$ — an operator in the same space on which the representation is defined.

$\psi(z)$ is the operator creating a horn on the sphere.

$$\Psi(z)|v\rangle.$$

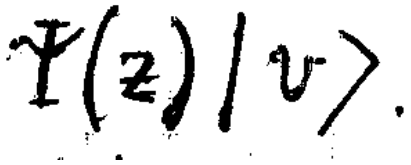


Diagram and typed description. Uppercase Ψ in the ket line.

Original source details: GELFAND-P001067

Consider the functions:

$$\langle v|\psi(z_1)\psi(z_2)\psi(z_3)\psi(z_4)|v\rangle \tag{23}$$

$$\Psi(x) = \langle v|\psi(\infty)\psi(1)\psi(x)\psi(0)|v\rangle$$

$$|0\rangle = \psi(0)|v\rangle, \qquad \langle v|\psi(\infty) = \langle 0|$$

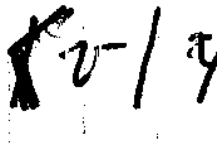


Diagram and typed description. The opening angle bracket of the bra is retraced.

Original source details: GELFAND-P001067

$$\Psi(x) = \langle 0|\psi(1)\psi(x)|0\rangle$$

$$\begin{cases} L_n|0\rangle = 0, \quad n > 0, \\ L_0|0\rangle = \Delta|0\rangle \end{cases} \tag{24}$$

Verma modules arise: such a representation without extraneous relations.

$\psi(x)$ (possibly $\Psi(x)$), overwritten symbol? is described above and maps the Verma module to itself.

Uncertain reading. $\psi(x)$ (possibly $\Psi(x)$), overwritten symbol

Original source details: GELFAND-P001067

$$\Psi(x) = e^{\Delta \cdot S_{cl}(x)} \quad \Rightarrow \quad \text{everything is determined by } \Psi(x) \tag{25}$$

crossed-out mark, possibly § or 4? Computation of Ψ.

Uncertain reading. crossed-out mark, possibly § or 4

Original source details: GELFAND-P001067

Add a relation (to the commutation relations):

$$\langle 0|\psi(x)|0\rangle = x^{-\Delta} \tag{26}$$

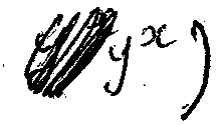
$$\langle 0|\psi(1)|0\rangle\langle 0|\psi(x)|0\rangle + \langle 0|\psi(1)|1\rangle\langle 1|\psi(x)|0\rangle + \\ + \langle 0|\psi(1)|2\rangle\langle 2|\psi(x)|0\rangle + \dots \tag{27}$$

$$I = \sum_n |n\rangle\langle n|$$

$$\langle 1|\psi(x)|0\rangle = \langle 0|L_1\psi(x)|0\rangle = \langle 0|[L_1, \psi(x)]|0\rangle = \\ = \left(x^2\frac{d}{dx} + 2\Delta x\right)\langle 0|\psi(x)|0\rangle = (\Delta \cdot x)x^{-\Delta}$$

[writing after 2Δ is crossed out, and x is inserted. The mark beneath the line may be an insertion mark or y?]

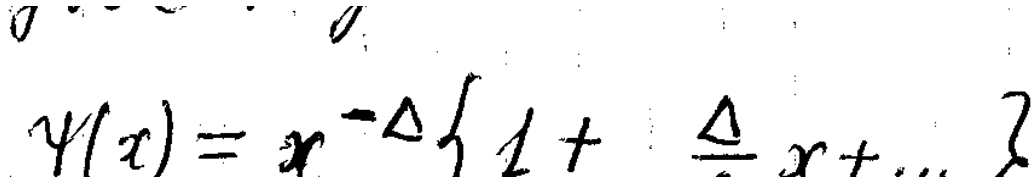
Uncertain reading. writing after 2Δ is crossed out, and x is inserted. The mark beneath the line may be an insertion mark or y



Original source details: GELFAND-P001067

and then one obtains:

$$\psi(x) = x^{-\Delta}\left\{1 + \frac{\Delta}{2}x + \dots\right\}$$



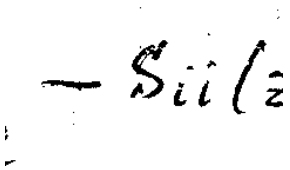
Original source details: GELFAND-P001067

Diagram and typed description. Lowercase ψ in the expansion.

$$\langle v|\psi(z_1)\dots\psi(z_N)|v\rangle = e^{-S_{ii}(z_1,\dots,z_N)}$$

[subscript of S, probably ii, possibly il?]

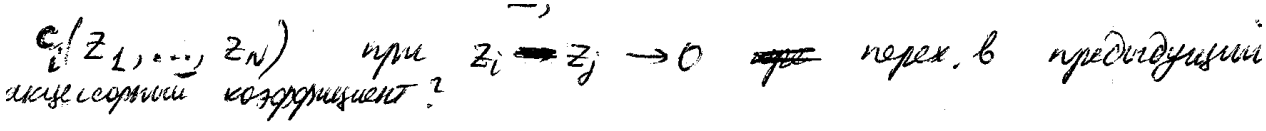
Uncertain reading. subscript of S, probably ii, possibly il



Original source details: GELFAND-P001067

$$\Delta, c \to \infty$$

Does $c_i(z_1, \dots, z_N)$ as $z_i - z_j \to 0$ [crossed-out «пр...». The mark between z_i and z_j is retraced and reads as minus?] go over to the preceding accessory coefficient?



Original source details: GELFAND-P001067

Uncertain reading. crossed-out «пр...». The mark between z_i and z_j is retraced and reads as minus

GELFAND-P001068

[source number: 56]

$S_{cl}(x)$

(3 other points $0, 1, \infty$)

~~The remaining ones~~ $\Rightarrow c_k = c_k(x)$

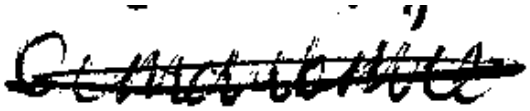


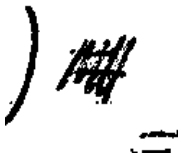
Diagram and typed description. Crossed-out word.

Original source details: GELFAND-P001068

$$c_k(x) = \frac{\partial S_{cl}(x)}{\partial x}$$

$$e^{S_{cl}(x)} = x^{-\Delta}\{1 + a_1 x + a_2 x^2 + \ldots\}.$$

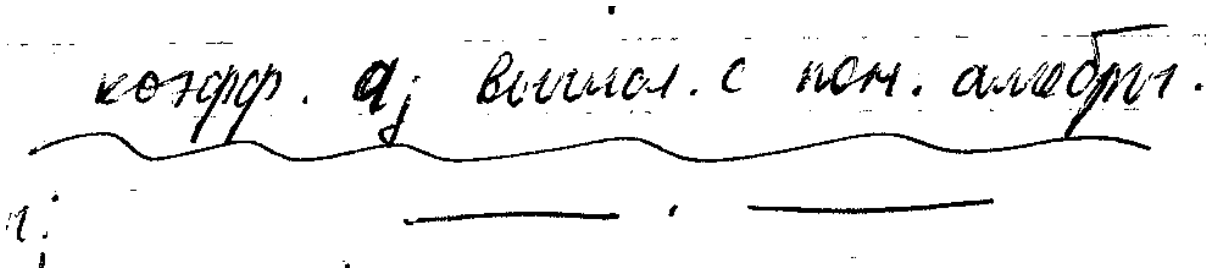
[crossed-out writing in the exponent after $S_{cl}(x)$?]



Uncertain reading. crossed-out writing in the exponent after $S_{cl}(x)$

Original source details: GELFAND-P001068

the coefficients a_j are computed by means of the algebra.



Original source details: GELFAND-P001068

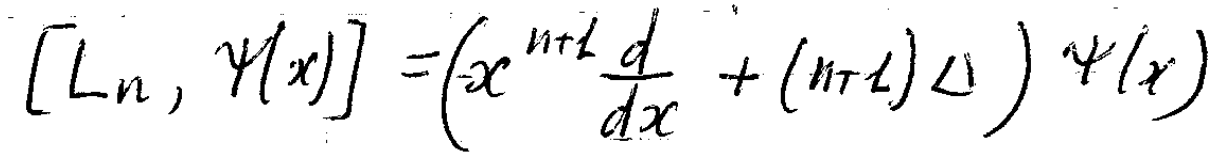
Diagram and typed description. Wavy underline and two shorter strokes underneath.

Axioms:

$$L_0|0\rangle = \Delta|0\rangle$$

$$L_n|0\rangle = 0, \quad n > 0$$

$$[L_n, \psi(x)] = \left(x^{n+1}\frac{d}{dx} + (n+1)\Delta\right)\psi(x)$$



Original source details: GELFAND-P001068

Diagram and typed description. Manuscript commutator. The factor x^n beside $(n+1)\Delta$ is absent in this line.

$$\langle 0|\psi(x)|0\rangle = x^{-\Delta}$$

$$\Psi(x) = \langle 0|\psi(1)\psi(x)|0\rangle$$

[illegible crossed-out mark after the ket?]

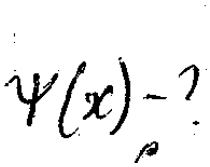
Uncertain reading. illegible crossed-out mark after the ket

Original source details: GELFAND-P001068

$$I = \sum_n |n\rangle\langle n|, \qquad |n\rangle = L_{-n}|0\rangle.$$

[Left margin: $\psi(x)$ — ?]

Diagram and typed description. Left-margin note.



Original source details: GELFAND-P001068

Difficulty. there is no orthogonality.

It is natural from the form of T:

$$L_n = (\alpha + i\beta n)a_n + \sum_{k=1}^{n-1} a_k a_{n-k} + \sum_{k=1}^{\infty} a_k^+ a_{n+k};$$

$$L_{-n} = L_n^+;$$

$$[a_n, a_m^+] = n\delta_{n,m};$$

$$\Delta = \frac{1}{2}(\alpha^2 + \beta^2), \quad c = 1 + 12\beta^2.$$

$$a_n|0\rangle = 0.$$

There is a correspondence between the bases:

$$a_{n_1}^+ a_{n_2}^+ \ldots |0\rangle$$

[illegible crossed-out mark to the right of the first basis line?]

Uncertain reading. illegible crossed-out mark to the right of the first basis line

Original source details: GELFAND-P001068

$$L_{-n_1} L_{-n_2} \ldots |0\rangle$$

expressible in terms of each other, but the first is orthogonal!!

The commutation relations of $\psi(x)$ in the first basis are more complicated

$$L(z) = T(z) = \frac{dp(z)}{dz} + p^2(z),$$

where

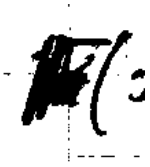
$$p(z) = \sum (a_n z^n + a_{-n} z^{-n})$$

GELFAND-P001069

Take $a_n = \dfrac{\partial}{\partial z_n}$, $a_n^+ = n z_n$.

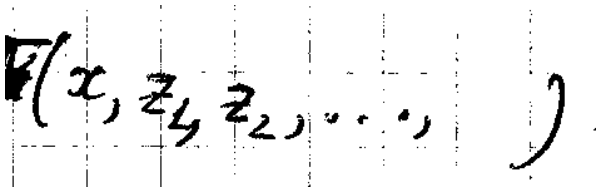
$$(\alpha + i\beta n)\frac{\partial}{\partial z_n} + \sum \frac{\partial^2}{\partial z_k \partial z_{n-k}} + \sum k z_k \frac{\partial}{\partial z_{n+k}}\Big) F(x, z_1, z_2, \dots,) =$$
$$= \left(x^{n+1}\frac{\partial}{\partial x} + (n+1)\Delta\right) F(x, z_1, , z_n).$$

[on the left F is written over another symbol. The underlying symbol is illegible?]



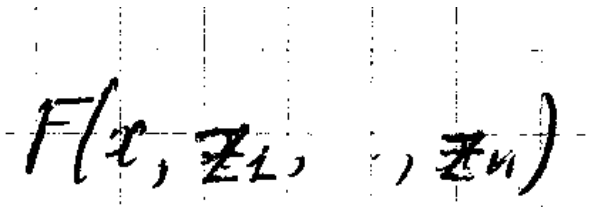
Original source details: GELFAND-P001069

Uncertain reading. on the left F is written over another symbol. The underlying symbol is illegible



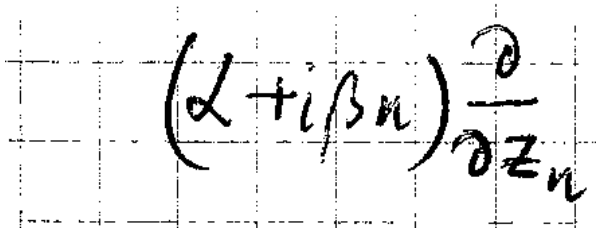
Original source details: GELFAND-P001069

Diagram and typed description. Arguments on the left. The comma touches z_1, and a gap follows the last comma.



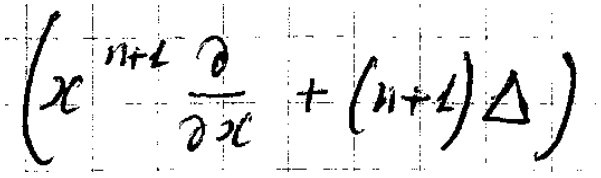
Original source details: GELFAND-P001069

Diagram and typed description. Arguments on the right, with a gap between commas.



Original source details: GELFAND-P001069

Diagram and typed description. Start of the left side. There is no separate outer opening parenthesis before the first parenthesis.



Original source details: GELFAND-P001069

Diagram and typed description. Right side. No factor x^n is written beside $(n+1)\Delta$.

GELFAND-P001070

29/XI–82 Gelfand Seminar.

Kobzarev (Igor Yuryevich)

On Strings.

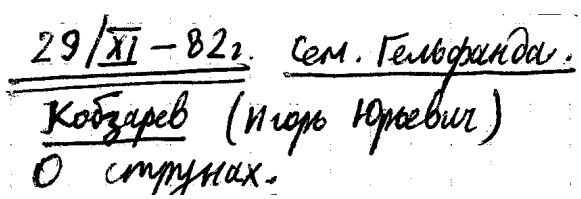


Diagram and typed description. Date, speaker's name and title.

Original source details: GELFAND-P001070

Field theory: there is a field φ, M^4 is Minkowski space, and $\mathcal{L}(\varphi)$ is the Lagrangian.

solutions and so on.

Another type of field theory

before 1970

$$\Box\psi + \mu^2\psi = 0 \tag{1}$$

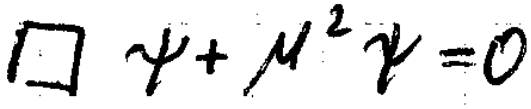


Diagram and typed description. The notation μ^2 in (1).

Original source details: GELFAND-P001070

[At right:] Interaction — perturbation theory. $\varphi_1\varphi_2\varphi_3, \ldots$

one must add $\lambda\varphi_1\varphi_2\varphi_3$ and so on.

after: of Higgs type

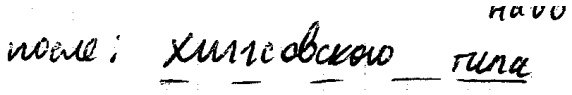


Diagram and typed description. Manuscript line "after: of Higgs type."

Original source details: GELFAND-P001070

$$\mathcal{L}(\varphi) = \frac{1}{2}(\partial_i\varphi)^2 - \lambda^2(\varphi^2 - \eta^2)^2 \tag{2}$$

Vacuum: for classical fields, the state with the least energy ||

Diagram and typed description. Double stroke to the right of the vacuum line.

Original source details: GELFAND-P001070

(corresponding Hamiltonian:

$$H = \frac{1}{2}(\partial_t\varphi)^2 + \frac{1}{2}(\bar{\nabla}\varphi)^2 + \lambda^2(\varphi^2 - \eta^2)^2 \tag{3}$$

Diagram and typed description. The digit 3 is retraced.

Original source details: GELFAND-P001070

Diagram and typed description. Bar over ∇ in (3).

Original source details: GELFAND-P001070

Degenerate vacuum: $\varphi = \eta$, $\varphi = -\eta$. (2 vacua)

$$\varphi \mapsto -\varphi \qquad \text{discrete symmetry} \tag{4}$$

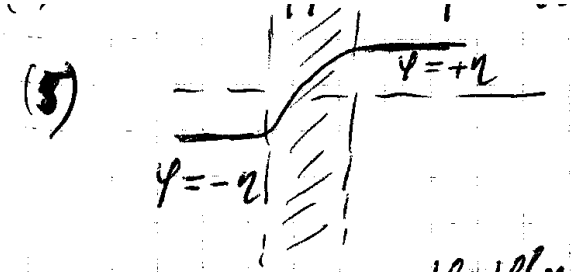
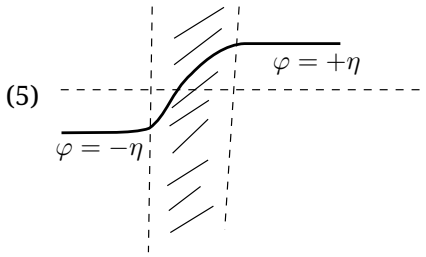


Diagram and typed description.



Original source details: GELFAND-P001070

[To the right of the figure:] Consider a wall — a solution localized on a plane.

$$\varphi = \varphi(x, y, z) \qquad \text{(independent of } t\text{)}$$

$$\Box\varphi + \varphi\lambda^2(\varphi^2 - \eta^2) = 0 \qquad \text{— Euler equation for Lagrangian (2)} \tag{6}$$

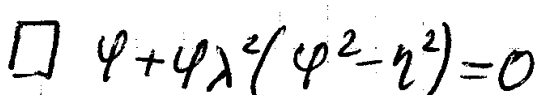


Diagram and typed description. The symbol before λ^2 is φ.

Original source details: GELFAND-P001070

(7) $\|$ $\varphi \underset{z\to\pm\infty}{=} \pm\eta$ independent of $x, y, z \Rightarrow$ it is easy to verify that such a solution exists.

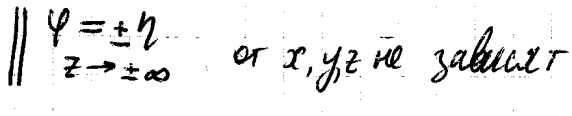


Diagram and typed description. Asymptotic condition and the manuscript's coordinate list.

Original source details: GELFAND-P001070

GELFAND-P001071

$\eta \sim m$

(8) Layer thickness [inserted below: (in (5))] $\sim \frac{1}{\eta}$.

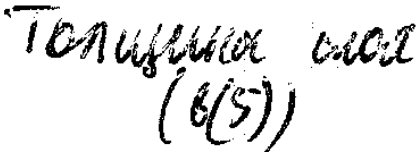


Diagram and typed description. Reference to (5) below the line on the layer thickness.

Original source details: GELFAND-P001071

[illegible crossed-out mark before “Solution”[?]] Solution (5) is found explicitly: th$(z\eta)$

Uncertain reading. illegible crossed-out mark before “Solution”

Original source details: GELFAND-P001071

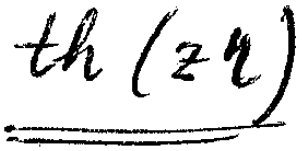


Diagram and typed description. The double-underlined th$(z\eta)$.

Original source details: GELFAND-P001071

Symmetry group: $\varphi \mapsto -\varphi$.

One now usually considers a continuous group G of symmetries of the Lagrangian.

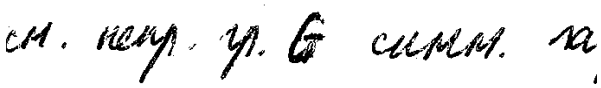


Diagram and typed description. Abbreviations in the line on the continuous group.

Original source details: GELFAND-P001071

$U(1)$-invariant theory:

$$\mathcal{L}(\varphi) = \frac{1}{2}\left|(\partial_i + iA_i)\varphi\right|^2 - (\varphi\varphi^* - \eta^2)^2 \tag{9}$$

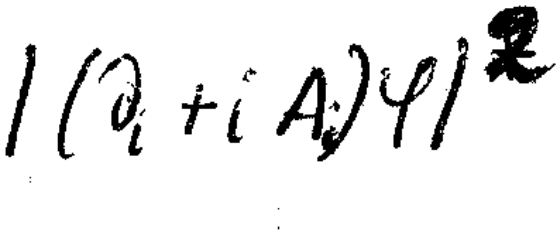


Diagram and typed description. The $+$ sign before iA_i in (9).

Original source details: GELFAND-P001071

$$\text{Vacuum: } (c_1, c_2): \quad c_1^2 + c_2^2 = \eta^2 \tag{10}$$

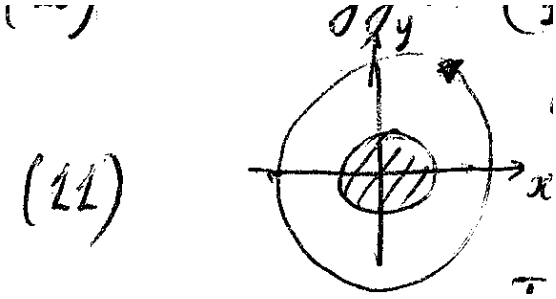


Diagram and typed description.

Original source details: GELFAND-P001071

[To the right of the figure:] A string is when the energy is concentrated along a line (one-dimensional).

On going around, the vector C turns through 2π

There is a topological invariant.

(Static solution: independent of t)

A is an additional vector field (electromagnetic).

$$\mathcal{L} = -\frac{1}{4}F_{ik}F_{ik} + \frac{1}{2}\left|(\partial_i + iA_i)\varphi\right|^2 - (\varphi\varphi^* - \eta^2)^2 \tag{12}$$

$$F_{ik} = \partial_i A_k - \partial_k A_i$$

Action of $U(1)$ on the field:

$$\begin{cases} \varphi \mapsto e^{i\alpha}\varphi \\ A_i \mapsto A_i + \nabla_i \alpha \end{cases} \tag{13}$$

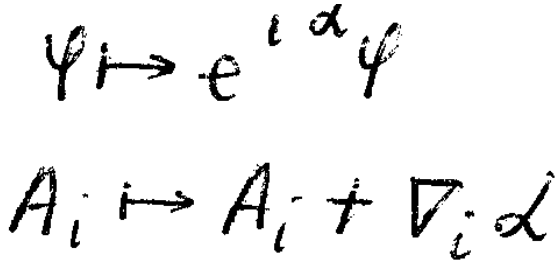
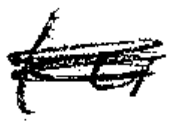


Diagram and typed description. Equation (13) has $e^{i\alpha}$ and a $+$ sign before $\nabla_i\alpha$.

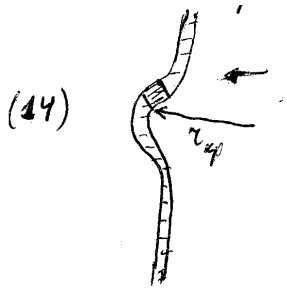
Original source details: GELFAND-P001071

Consider an approximate solution [crossed-out writing, possibly a number in parentheses[?]]



Uncertain reading. crossed-out writing, possibly a number in parentheses

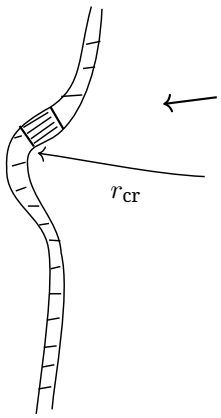
Original source details: GELFAND-P001071



Original source details: GELFAND-P001071

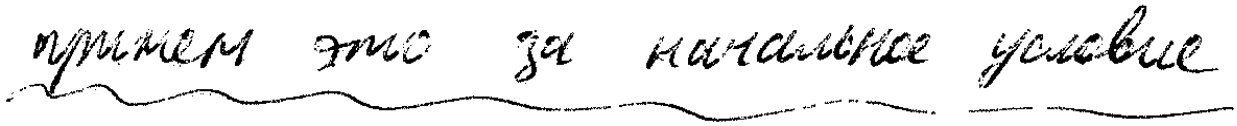
Diagram and typed description.

(14)



The arrow at the bend is labelled r_{cr}. The strokes inside the marked section may have arrowheads.

[To the right of the figure, left arrow:] take this as the initial condition



Original source details: GELFAND-P001071

Diagram and typed description. Wavy underline and singular “initial condition.”

Approximate form of solution (11):

$$\begin{cases} \varphi_1 = \dfrac{x}{r} f(r) \\ \varphi_2 = \dfrac{y}{r} f(r) \end{cases} \tag{15}$$

(for probably “rotation,” the first letters are only partly legible[?])

Uncertain reading. probably “rotation,” the first letters are only partly legible

Original source details: GELFAND-P001071

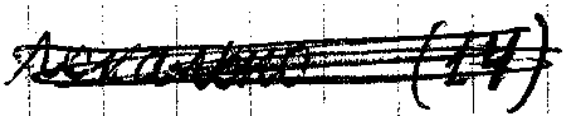
[At right:] Construct A in the same way.

$\varepsilon(r)$, the energy density in (11), can be calculated.

GELFAND-P001072

[source number: 67]

[Crossed out: possibly “Solution (14)”[?]]



Uncertain reading. possibly “Solution (14)”

Original source details: GELFAND-P001072

They must approach the vacuum: $\varphi_1^2 + \varphi_2^2 \to \eta^2$ as $r \to \infty$.

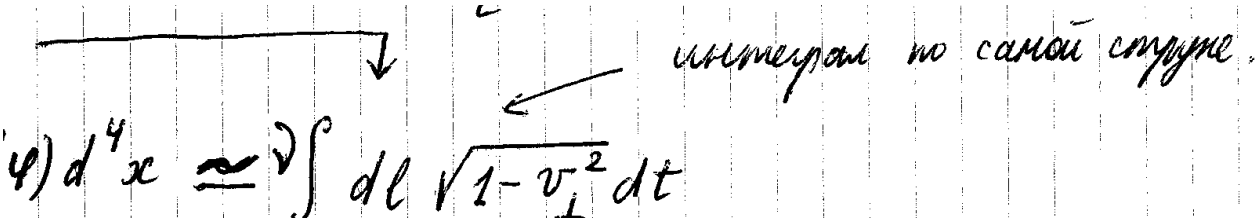
Locally (14) resembles (11).

(16) Let the radius of curvature satisfy $r_{\text{cr}} \gg \dfrac{1}{\eta}$

Effective action: [arrow rightward, then down to the right side of (17)]

$$S = \int \mathcal{L}(\varphi)\, d^4x \simeq \nu \int dl \sqrt{1 - v_\perp^2}\, dt \tag{17}$$

[Above right, arrow down-left toward the integral:] integral along the string itself.



Original source details: GELFAND-P001072

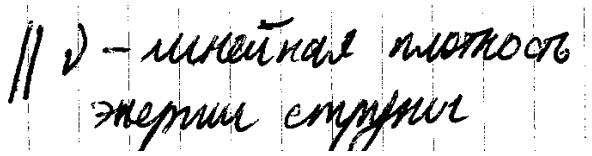
Diagram and typed description. Two arrows to the right side of (17).

Original source details: GELFAND-P001072

Diagram and typed description. The coefficient ν before the integral in (17).

$v_\perp$ is the transverse velocity of the string

|| ν is the linear energy density of the string.

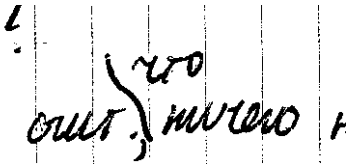


Original source details: GELFAND-P001072

Diagram and typed description. Double stroke and explanation of ν.

Evolution — ?

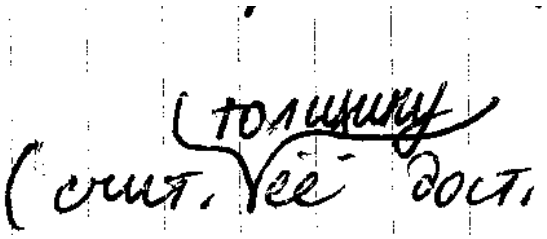
Inside the string we assume [inserted above: that] nothing happens



Original source details: GELFAND-P001072

Diagram and typed description. "That" is inserted above the line.

(we consider its [inserted above: thickness] sufficiently small — — compared with the radii of curvature)



Original source details: GELFAND-P001072

Diagram and typed description. Insertion of "thickness" before "its."

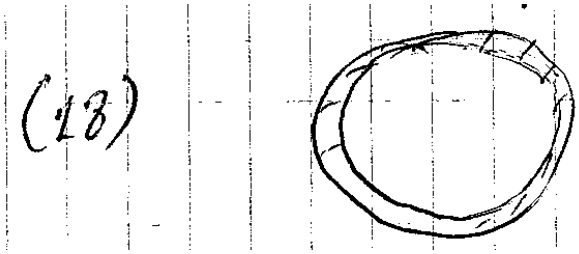
We evaluate the integral in each transverse section to obtain ν

$\sqrt{1-v_\perp^2}$ comes from d^4x (by a change of variables)

Or use relativistic invariance: ds (the interval), written in any coordinate system.

(17) is the formula of Nambu.

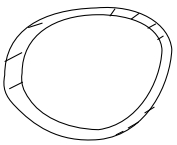
One can consider a ring-shaped string with Lagrangian given by the right side of (17)



Original source details: GELFAND-P001072

Diagram and typed description.

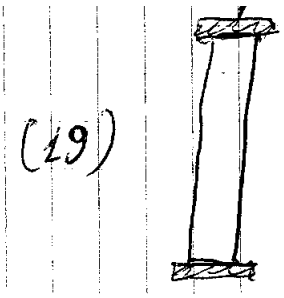
(18)



it will contract, quickly reach the speed of light, and shrink to a small lump.

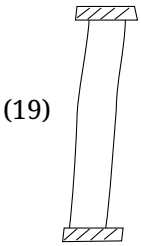
Variant.

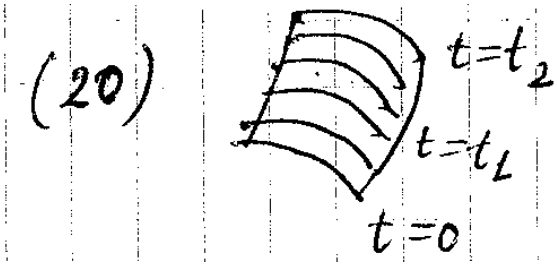
break the ends (put a monopole at each):



Original source details: GELFAND-P001072

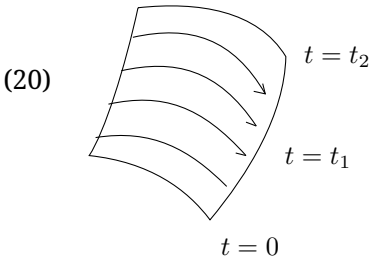
Diagram and typed description.





Original source details: GELFAND-P001072

Diagram and typed description.



Short strokes are visible at the ends of three arcs. It is uncertain whether they are arrowheads.

[To the right of the figure:] As it moves, the string sweeps out a sheet.

(21) The area of the swept surface is the Lagrangian (17_{right})

Original source details: GELFAND-P001072

Diagram and typed description. Wavy underline and reference 17_{right}.

GELFAND-P001073

$\dot{x} = \dfrac{\partial x}{\partial t}$, $x' = \dfrac{\partial x}{\partial s}$, $x(t, s)$ and calculate.

A min area arises.

Original source details: GELFAND-P001073

Diagram and typed description. The Russian word after min ends in «ь».

Strings and chromodynamics.

Quantum chromodynamics:

$$\mathcal{L}(\psi, A) = -\frac{1}{4}F^a_{ik}F^a_{ik} + \bar{\psi}(i\hat{\partial} - m)\psi \tag{22}$$

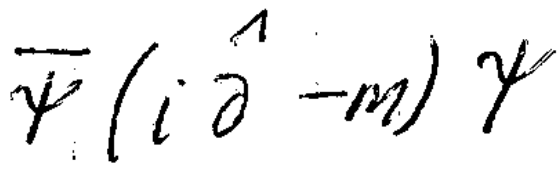


Diagram and typed description. Lowercase ψ on the right side of (22).

Original source details: GELFAND-P001073

$a = 1 \div 8$; $i, k = 0, 1, 2, 3$.

$$\hat{\partial} = \gamma^i(\partial_i + iI^a A^a_i) \tag{23}$$

[Beneath γ^i:] Dirac matrices

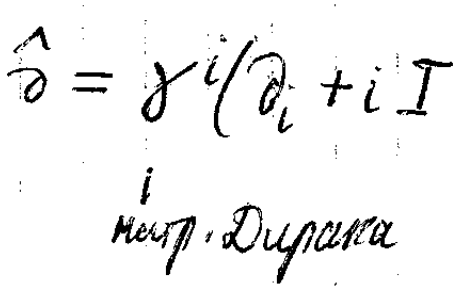


Diagram and typed description. A short dotted stroke appears above the note beneath γ^i.

Original source details: GELFAND-P001073

$$F^a_{ik} = \partial_i A^a_k - \partial_k A^a_i + g f_{abc} A^b_i A^c_k, \tag{24}$$

g is the coupling constant, and f_{abc} are the structure constants of the group.

Acts: the group $SU(3)$ (adjoint representation) — hence 8.

— acts on A

[Crossed out: illegible mark before I^a, possibly a?]

Uncertain reading. illegible mark before I^a, possibly a

Original source details: GELFAND-P001073

I^a is a generator of $SU(3)$, $I^a = I^a_{\alpha\beta}$.

in $\bar{\psi}$ and ψ there is also a color index (quarks are triplets) — — contraction is over that index.

Diagram and typed description. The note “quarks are triplets.”

Original source details: GELFAND-P001073

In fact several types of quarks are needed: at least 2 types

$$\mathcal{L}(\psi, A) = -\frac{1}{4}F^a_{ik}F^a_{ik} + \bar{u}(i\bar{\partial} - m)u + \bar{d}(i\hat{\partial} - m)d \tag{25}$$

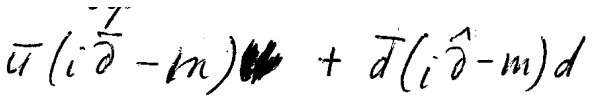


Diagram and typed description. There is a horizontal bar over ∂ in the first quark term, and a hat in the second.

Original source details: GELFAND-P001073

[heavily overwritten symbol after the first parenthesis, provisionally u?]

Uncertain reading. heavily overwritten symbol after the first parenthesis, provisionally u

Original source details: GELFAND-P001073

(u-quarks and d-quarks)

$$\hat{\partial}_i = \nabla_i + iA^a_i I^a, \qquad I^a = I^a_{\alpha\beta} \tag{26}$$

A is a Yang–Mills field, or a gluon field (it glues quarks together)

(gluon field)

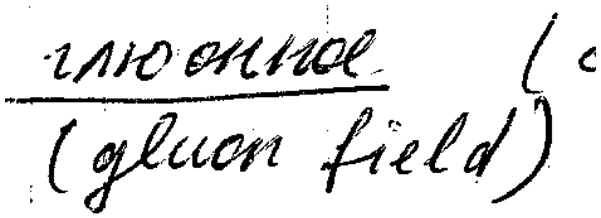


Diagram and typed description. English "gluon field" beneath the underlined word.

Original source details: GELFAND-P001073

Chapter 23

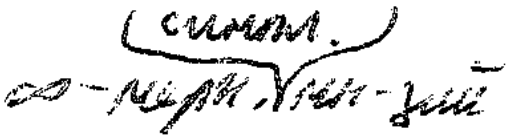
GELFAND-P001074

[source number: 1]

7/II–83

Gelfand Seminar

I. M. G. There is no familiar understanding of the ∞-dimensional symplectic group, ∞-dim. [inserted above: sympl.]



Uncertain reading. abbreviation apparently for “geometry,” ending indistinct

Original source details: GELFAND-P001074

and so on. “Fock space is a toy for mathematicians.”

There are popular lectures by Witten on supermathematics! (preprint; lectures)

↓

The Atiyah–Bott theorem follows from super considerations
(on fixed points)

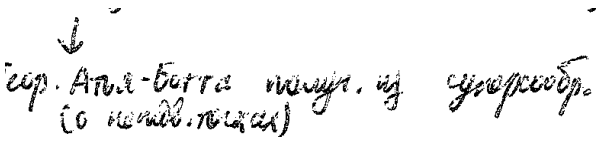


Diagram and typed description. Arrow from Witten’s lectures to the Atiyah–Bott theorem.

Original source details: GELFAND-P001074

Kirillov on Neretin’s work

$$T(g)f(z) = f\left(\frac{\alpha z+\gamma}{\beta z+\delta}\right)|\beta z+\delta|^{\lambda}$$

The representation of $SL(2,\mathbb{R})$ extends to a representation of Diff(S^1) (unitarily!).

$$f \longmapsto f(\varphi^{-1}(z))\left|\frac{d\varphi^{-1}(z)}{dz}\right|^{-\lambda/2}$$

[At left with a brace:] this is nonunitary, but

Diagram and typed description. Annotation and brace before “from this.”

Original source details: GELFAND-P001074

from this, unitary representations are constructed (by considering the action in Fock space —

Diagram and typed description. Separate closing parenthesis at the right of the line.

Original source details: GELFAND-P001074

— on coherent states.

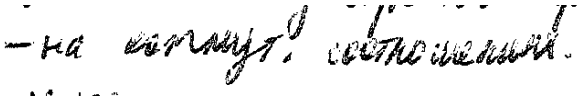


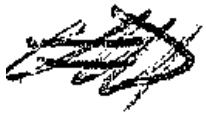
Diagram and typed description. Abbreviation for “coherent” before “states.”

Original source details: GELFAND-P001074

Idea:

$$\mathrm{Im} \int f(x_1)\overline{g(x_2)}\frac{|x_1 - x_2|^{-2+\lambda}}{\Gamma(-1+\lambda)}\, dx_1\, dx_2 \qquad \text{(symplectic form!)}$$

preserved under fractional linear transformations [crossed-out arrow →?]



Uncertain reading. crossed-out mark resembling a right arrow

Original source details: GELFAND-P001074

⇒ not preserved under all diffeomorphisms, but of similar form.

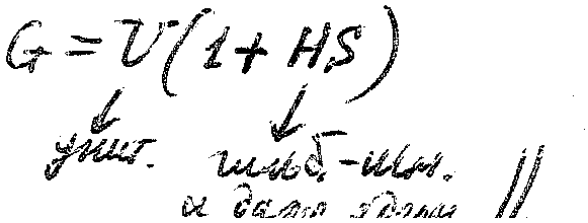


Diagram and typed description.

$G = U(1 + HS)$

unitary Hilbert–Schmidt

[and more general?]

The bottom annotation is provisional. Two oblique strokes appear at right.

Original source details: GELFAND-P001074

B. Feigin.

Strings and representations of Diff S^1.

V is the space of maps $S^1 \longrightarrow \mathbb{R}^d$. A metric is given on $\mathbb{R}^d$ (Riemannian or Lorentzian). Diff S^1 acts (changes of parameter).

$$f, g \longmapsto \int_{S^1} (f, dg) \tag{1}$$

On V/V_{const} one obtains a nondegenerate form. Put $V' = V/V_{\text{const}}$. [Below V_{const}:] constant maps.

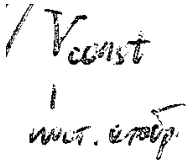


Diagram and typed description. Stroke from V_{const} toward its caption.

Original source details: GELFAND-P001074

On $V'/\operatorname{Diff} S^1$ there is a Hamiltonian structure (Poisson bracket).

Problem: quantize this situation. [marginal stroke]

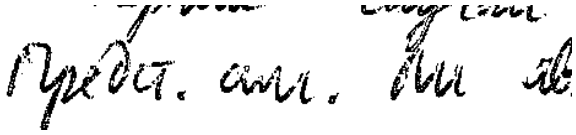
Uncertain reading. short oblique mark before a closing parenthesis in the right margin

Original source details: GELFAND-P001074

Kirillov. M is a symplectic manifold. G is a group acting on M and preserving its symplectic structure. $\Rightarrow$ One constructs the reduced manifold M_{red} (symplectic reduction).

idea: $\{\ \}$ on invariant functions is degenerate. One must take a quotient.

B. F. V' is a space with a nondegenerate symplectic form. The quantization method is to choose p, q and realize them in the usual way (Fock space). In finite dimension this is a satisfactory method. Where does it act? The space of invariants. The ∞-dimensional case is problematic. The representation of the symplectic manifold[?] is projective, and invariants do not exist. The space must be changed.



Uncertain reading. abbreviation provisionally read as «мн.» (“manifold”)

Original source details: GELFAND-P001074

(a Schwinger term appears)

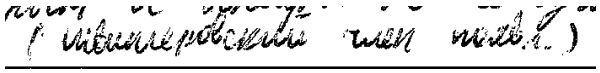


Diagram and typed description. Small note on the Schwinger term below the last line.

Original source details: GELFAND-P001074

GELFAND-P001075

$\operatorname{sp}(V')$ is the Lie algebra of symplectic transformations.

$\mathcal{L}$ is the algebra of vector fields on S^1.

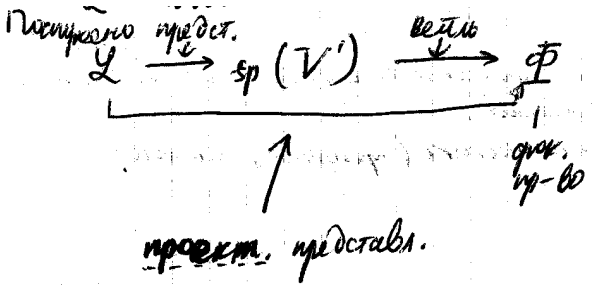
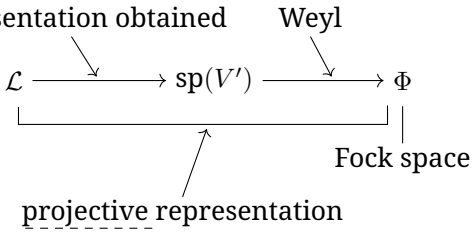


Diagram and typed description.



The label above the second arrow reads “Weyl.”

Original source details: GELFAND-P001075

Φ is constructed as follows:

V', [crossed out]

Uncertain reading. the crossed-out mark after V' is illegible

Original source details: GELFAND-P001075

$S^1 \longrightarrow \mathbb{R}^d$.

complexify and decompose

$$V' = \underset{\text{(annihilation operators)}}{H^+} \oplus \underset{\text{(creation operators)}}{H^-} \qquad \Phi = \bigoplus S^i H_+.$$

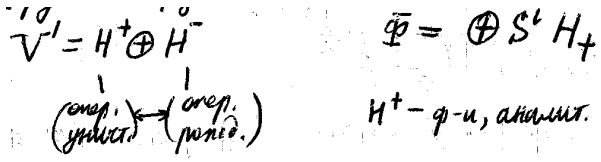
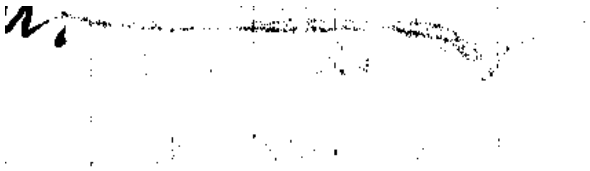


Diagram and typed description. Decomposition of V', operator labels and formula for Φ.

Original source details: GELFAND-P001075

H^+ consists of functions analytic inside. [faint mark in the margin]



Uncertain reading. a faint broken trace at the right of the line with no legible letters

Original source details: GELFAND-P001075

—— · ——

$$\mathcal{L} = \{L_i, c\}, \quad i \in \mathbb{Z}.$$

$$[L_i, L_j] = (j-i)L_{i+j} + \underbrace{\delta_{i,-j}\frac{i^3 - i}{12}\, c}_{\text{Schwinger term}}.$$

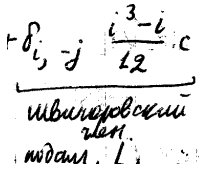


Diagram and typed description. Bracket and caption beneath the central term.

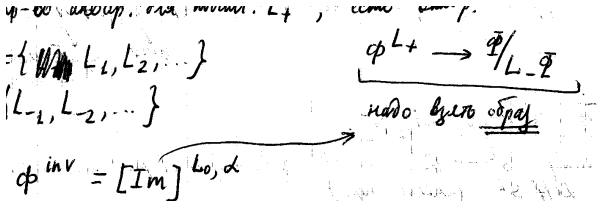
Original source details: GELFAND-P001075

ϕ^{L_+} is the space of invariants for the subalgebra L_+, and there is a map.

$$L_+ = \{\text{[crossed out] } L_1, L_2, \ldots\},$$
$$L_- = \{L_{-1}, L_{-2}, \ldots\}.$$

Uncertain reading. heavily crossed out before L_1, possibly L_0

Original source details: GELFAND-P001075



Original source details: GELFAND-P001075

Diagram and typed description.

$$\underbrace{\phi^{L_+} \longrightarrow \Phi/L_-\Phi}_{\text{take the image}}$$

$$\phi^{\text{inv}} = [\text{Im}]^{L_0,\alpha}$$

Nontrivial at least when $d = 26$.

Problem (I. M. G.). Construct a p-adic analogue of the representation of the diffeomorphism group.

GELFAND-P001076

14/II–83 Gelfand Seminar.

Yu. Neretin.

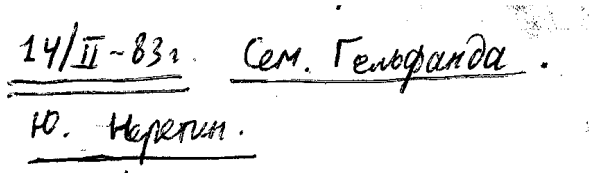


Diagram and typed description. Date and speaker's name.

Original source details: GELFAND-P001076

$\forall \rho$ almost preserves the inner product.

$$\frac{1}{\Gamma(\rho)} \int_{-\infty}^{\infty} |x_1 - x_2|^{-\rho} f(x_1) \overline{g(x_2)}\, dx_1\, dx_2. \tag{1}$$

([crossed out] the operator will be $O(1+T)$.)

Uncertain reading. illegible crossed-out writing before "operator"

Original source details: GELFAND-P001076

[Below T, a vertical stroke to the annotation:] H.–S.

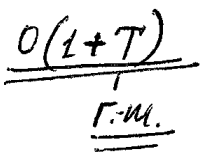


Diagram and typed description. Double underlines and annotation for T.

Original source details: GELFAND-P001076

Hilbert transform:

$$Hf(s) = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{ctg}\left(\frac{t-s}{2}\right) f(t)\,dt \qquad (f \text{ is a real function}) \tag{2}$$

$$H : \text{const} \longrightarrow 0$$

$\rho = 0, 2 \quad \Rightarrow$ in the space modulo const or with $\int = 0$.

H introduces a complex structure.

In Fourier modes:

$$\| \quad \begin{array}{l} \cos n\varphi \longrightarrow -\sin n\varphi \\ \sin n\varphi \longrightarrow \cos n\varphi \end{array}$$

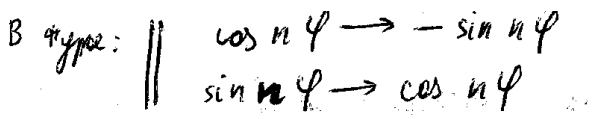


Diagram and typed description. Double stroke to the left of the two lines and overwritten n in the second.

Original source details: GELFAND-P001076

Inner product

$$\int \ln\left|\sin\frac{t-s}{2}\right| f_1(t) f_2(s)\,dt\,ds$$

$$\int \text{[crossed out]}\ \operatorname{ctg}\left(\frac{t-s}{2}\right) f_1(t) f_2'(s)\,dt\,ds$$

Uncertain reading. heavily crossed-out writing after the integral sign, possibly beginning with ln

Original source details: GELFAND-P001076

$\Rightarrow H$ defines a complex structure.

Uncertain reading. illegible crossed-out writing between the arrow and H

Original source details: GELFAND-P001076

[At right:] by decomposition into analytic and antianalytic functions

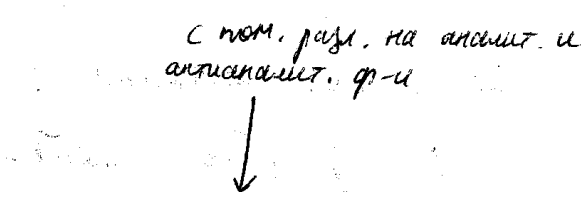


Diagram and typed description. Arrow down from the right-hand note to the remark on Hermitian extension.

Original source details: GELFAND-P001076

$\Rightarrow$ take the Hermitian extension using this H.

Diagram and typed description. Hatched double stroke at the left of the arrow.

Original source details: GELFAND-P001076

Im$\langle \cdot\,;\,\cdot \rangle$ is preserved under all diffeomorphisms.

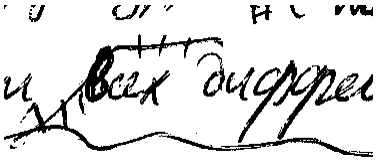


Diagram and typed description. Mark above "all" and wavy underlining.

Original source details: GELFAND-P001076

I. M. G. Problem: make this a general statement about pseudodifferential operators!! (invariant under [inserted above: all] diffeomorphisms!!)

Diagram and typed description. "All" is inserted above the line.

Original source details: GELFAND-P001076

GELFAND-P001077

Symplectic form — ?

Yu. N. Projections: $H \pm i$

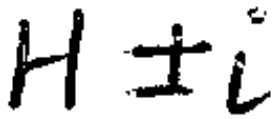


Diagram and typed description. The manuscript reads $H \pm i$.

Original source details: GELFAND-P001077

their images: Lagrangian manifolds.

Symplectic form:

$$\{f_1, f_2\} = \langle f_1, H f_2 \rangle$$

I. M. G. [crossed out]

Uncertain reading. illegible crossed-out writing after the initials

Original source details: GELFAND-P001077

[Inserted above: "We must also."] consider a generalization of $c[1 + (\text{comp.})]$ ($\Rightarrow$ write the whole series in symbols).

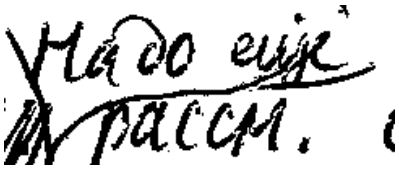


Diagram and typed description. "We must also" is inserted above the line.

Original source details: GELFAND-P001077

[Crossed-out formula: $\iint \ln \left|\sin \frac{t-s}{2}\right|$.]

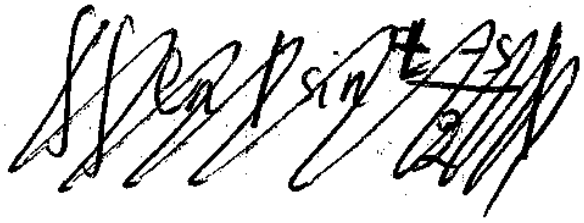
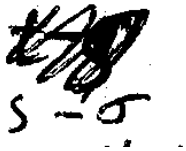


Diagram and typed description. Crossed-out formula containing sin.

Original source details: GELFAND-P001077

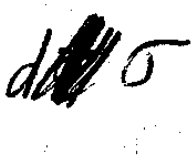
Claim

$$\iint \ln \left|\frac{t-s}{2}\right| f_1(t) f_2(\sigma) \frac{1}{s-\sigma}\, ds\, d\sigma\, dt$$



Uncertain reading. denominator corrected. $s-\sigma$ is written below. The cancelled beginning resembles $t-$

Original source details: GELFAND-P001077



Uncertain reading. a symbol after d is crossed out before σ, possibly t

Original source details: GELFAND-P001077

is preserved under diffeomorphisms

$\Rightarrow$ must be a differential operator (f').

Need: to calculate:

$$\int \ln \left|\frac{t-s}{2}\right| \frac{1}{s-\sigma}\, ds.$$

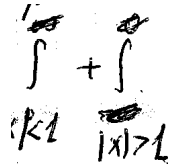
We should obtain δ'.

Diagram and typed description. The manuscript reads δ'.

Original source details: GELFAND-P001077

We must continue analytically!!

$$\int_{-\infty}^{\infty} |x|^{\lambda} f(x)\, dx = \int_{|x|<1} + \int_{|x|>1}$$



Uncertain reading. marks resembling ∞ are crossed out above both integrals on the right. The mark beneath the second may have been $-\infty$

Original source details: GELFAND-P001077

and we continue each analytically.

Idea:

$$\int |t-s|^{\lambda} \ \dots$$

and differentiate with respect to λ!!

— · —

~~V. Ginzburg~~

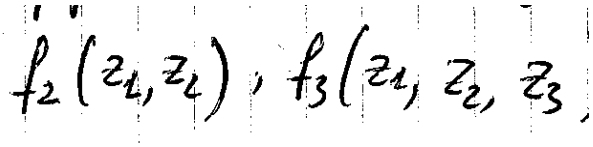
Diagram and typed description. Crossed-out name with a double underline.

Original source details: GELFAND-P001077

Another problem (following Feigin!):

How does a diffeomorphism of S^1 act on a sequence of analytic functions

$$\{f_1(z), f_2(z_1, z_1), f_3(z_1, z_2, z_3), \dots\}$$



Uncertain reading. the second subscript in f_2 looks more like 1. It may be 2

Original source details: GELFAND-P001077

$|z| < 1$.

What effect does smoothness have?

For φ fractional linear, this coincides with the usual one!

Uncertain reading. overwritten mark before "fractional linear," provisionally φ

Original source details: GELFAND-P001077

How can this be realized?

GELFAND-P001078

[source number: 5]

V. Ginzburg.

On the work of Duistermaat, Heckman (Invent. Math., 1982)

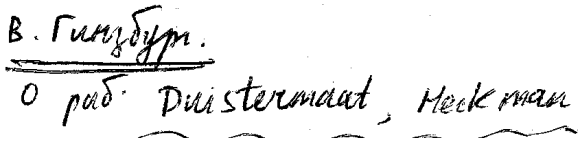


Diagram and typed description. Speaker's name and the authors of the work.

Original source details: GELFAND-P001078

$T = \mathbb{T}^k$ is a torus (k-dimensional). M is a compact symplectic manifold. T acts on M in a Hamiltonian way: it preserves the symplectic structure and every vector field

Uncertain reading. crossed out, possibly «не» or «на»

Original source details: GELFAND-P001078

from the Lie algebra of T determines a field given by a Hamiltonian, and these Hamiltonians commute (with respect to the Poisson bracket).

Diagram and typed description. The Russian text reads “a field given.”

Original source details: GELFAND-P001078

(Commuting fields do not imply commuting functions.)

(This is now called a Poisson Hamiltonian action.)

[At left, beyond the double stroke:] A map arises:

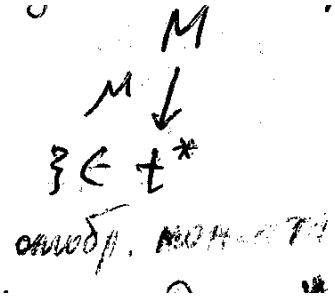


Diagram and typed description.

$$\begin{array}{c} M \\ \mu \big\downarrow \\ \xi \in t^* \end{array}$$

[moment map?]

Original source details: GELFAND-P001078

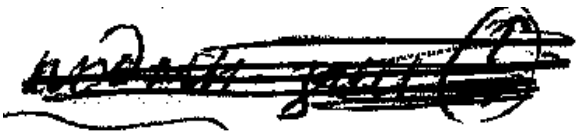
The caption beneath the diagram is faint. The end of the abbreviation is indistinct.

Or: give functions $f_1, \ldots, f_k$: $\{f_i, f_j\} = 0$, The flows of f_i are periodic.

Uncertain reading. mark over the end of the abbreviation “functions,” resembling an asterisk or a correction

Original source details: GELFAND-P001078

Local condition: the fields lie in one Lagrangian



Uncertain reading. crossed-out word beginning “sub…,” probably “subspace”

Original source details: GELFAND-P001078

subspace at every point.
[crossed-out heading]

Uncertain reading. crossed-out heading, possibly “Next” or “Another”

Original source details: GELFAND-P001078

$$\mu(m)(X) = H_X(m) \qquad (X \text{ is any field}) \tag{1}$$

Diagram and typed description. The X in the argument of $\mu(m)$ is retraced.

Original source details: GELFAND-P001078

Or: The cocycle $X_1, X_2 \longrightarrow \omega(X_1, X_2)(m)$ on the Lie algebra must be trivial.

Example: $\{y, -x\}$ give commuting fields $\dfrac{\partial}{\partial x}$, $\dfrac{\partial}{\partial y}$, but their bracket is $\neq 0$.

$$M_\xi = \mu^{-1}(\xi) \quad \text{; on it } \omega \text{ is degenerate;} \tag{2}$$

On M_ξ/T [inserted above: “it”] is already nondegenerate.

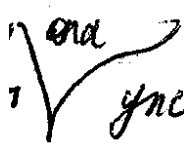


Diagram and typed description. “It” is inserted above the line.

Original source details: GELFAND-P001078

Denote it by ω_ξ. One can calculate volume with respect to it.

Statement of the lemma: ω_ξ is [inserted above: “piecewise”] linear in ξ (mod exact differentials).

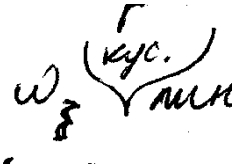


Diagram and typed description. Insertion of the abbreviation for “piecewise” before “linear.”

Original source details: GELFAND-P001078

Corollary. $\mathrm{vol}(M_\xi)$ is piecewise polynomial.

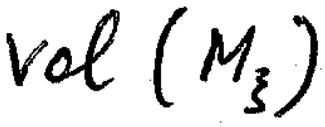
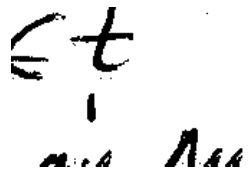


Diagram and typed description. The notation is $\mathrm{vol}(M_\xi)$.

Original source details: GELFAND-P001078

(3) Theorem. Consider a vector $X \in t$ and the corresponding Hamiltonian H_X



Original source details: GELFAND-P001078

Diagram and typed description. The abbreviation for "Lie algebra" appears beneath t.

(the lift of X as a function on t^* using μ). Then the integral

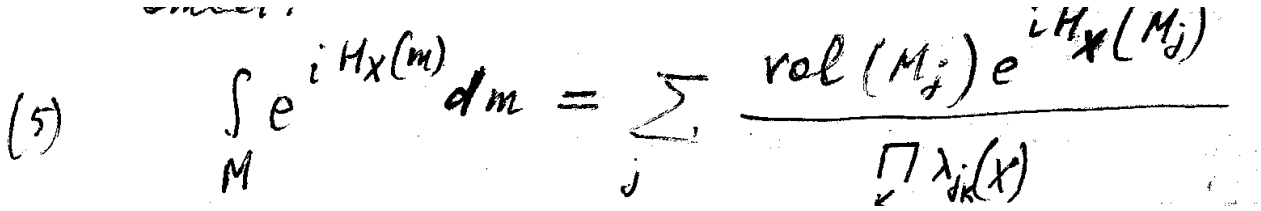
$$\int_M e^{iH_X(m)}\, dm \tag{4}$$

(a function on the Lie algebra)

can be calculated by stationary phase, and this gives an exact answer.

Answer:

$$\int_M e^{iH_X(m)}\, dm = \sum_j \frac{\mathrm{vol}(M_j) e^{iH_X(M_j)}}{\prod_k \lambda_{jk}(X)} \tag{5}$$



Original source details: GELFAND-P001078

Diagram and typed description. Equation (5), with $\mathrm{vol}(M_j)$ and a product in the denominator.

The fixed points of X (= stationary points) form a symplectic submanifold (possibly disconnected). M_j are its connected components. The $\lambda_{jk}(X)$ are linear forms that are computed (Jacobian of the action in transverse directions).

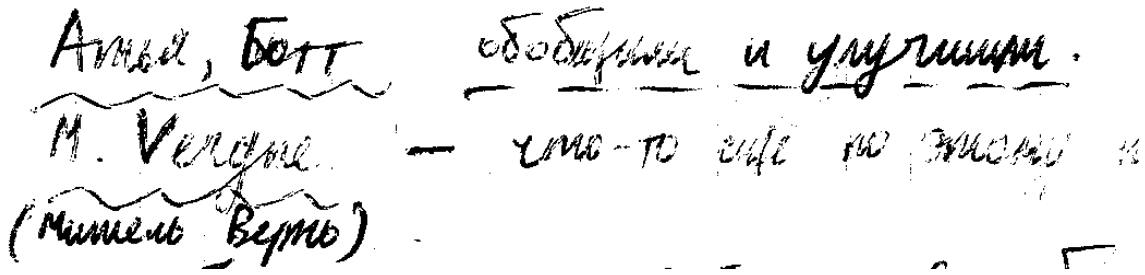
Original source details: GELFAND-P001078

Diagram and typed description. The bottom parenthesis reads "Jacobian of the action in transverse directions."

GELFAND-P001079

Atiyah, Bott generalized and improved it.

M. Vergne — something else on the subject (Michèle Vergne).



Original source details: GELFAND-P001079

Diagram and typed description. Atiyah, Bott, and Vergne in the manuscript.

An old result of Atiyah and Guillemin–Sternberg: the image of μ — is a convex region!!

(5) $\sim$ Weyl's character formula.

(~ 10 years ago there was a paper by Semenov-Tian-Shansky with similar content.)

(~10 лет назад была раб. Семенова – Тян-Шанского, близкая по содержанию,

Original source details: GELFAND-P001079

Diagram and typed description. Reference to the paper by Semenov-Tian-Shansky.

Idea of the proof: by stationary phase

$$\int_M e^{itH_X(m)}\,dm = \sum \frac{\mathrm{vol}(M_j)e^{itH_X(M_j)}}{\prod_k \lambda_{jk}(X)} + \text{lower-order terms}$$

$\int_M e^{itH_X(m)}\,dm = \sum \frac{\mathrm{vol}(M_j)\,e^{itH_X(M_j)}}{\prod_k \lambda_{jk}(X)}$ + мл. член

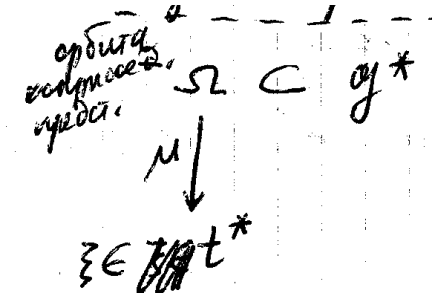
Original source details: GELFAND-P001079

Diagram and typed description. The parameter t in the exponents. The denominator reads $\prod_k \lambda_{jk}(X)$.

Locally quadratic $\Rightarrow$ the remainder is a rapidly decaying function.

Piecewise polynomial $\Rightarrow$ exponentials and polynomials $\Rightarrow$ remainder $= 0$.

Connection with representation theory:

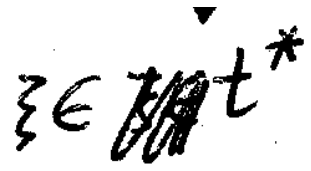


Original source details: GELFAND-P001079

Diagram and typed description.

orbit of the coadjoint representation

$$\begin{array}{c}\Omega \subset g^* \\ \mu \big\downarrow \\ \xi \in t^*\end{array}$$



Original source details: GELFAND-P001079

Uncertain reading. illegible crossed-out notation before t^*

G compact $\Rightarrow \Omega$ compact.

We obtain the Harish-Chandra integral.

Kirillov: this is — the character.

M. Vergne: proof for noncompact orbits of the discrete series.

A. A. Kirillov.

$SU(3)$ $H_0(3) \supset \Omega$

acts on Hermitian matrices (of trace 0), an 8-dimensional manifold

$$h \longrightarrow ghg^{-1}$$

Diagram and typed description. The conjugation formula.

Original source details: GELFAND-P001079

Orbit Ω: 2 invariants $\mathrm{Sp}(A^2)$, $\mathrm{Sp}(A^3)$. [Below «Orbit»: «6-dimensional».]

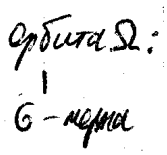


Diagram and typed description. The handwritten note on the orbit dimension.

Original source details: GELFAND-P001079

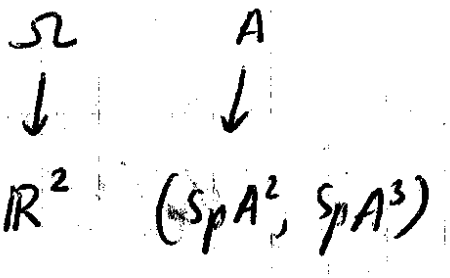


Diagram and typed description.

$$\begin{array}{cc} \Omega & A \\ \downarrow & \swarrow \\ \mathbb{R}^2 & (\mathrm{Sp}\,A^2, \mathrm{Sp}\,A^3) \end{array}$$

Original source details: GELFAND-P001079

The projection of Ω onto $\mathbb{R}^2$ — is an almost regular hexagon.

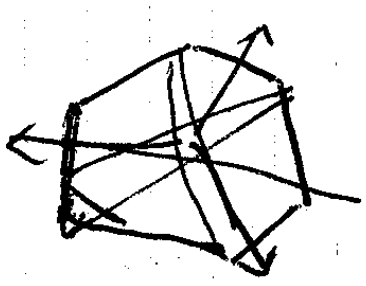
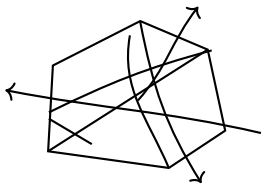


Diagram and typed description.

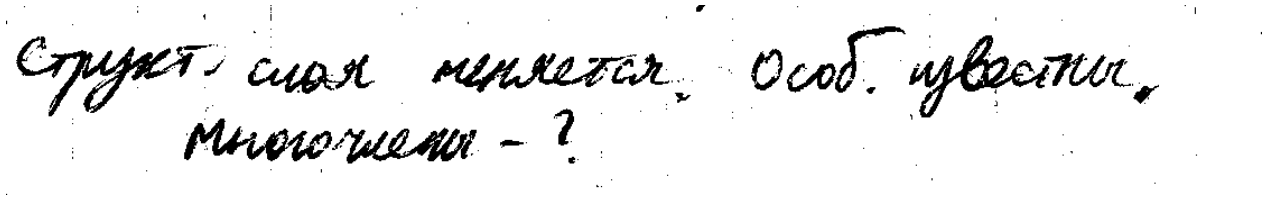


Some connections are uncertain where the strokes overlap.

Original source details: GELFAND-P001079

(All angles are $120°$; every other side has the same length.)

The structure of the fiber changes. The singularities are known,
Polynomials — ?



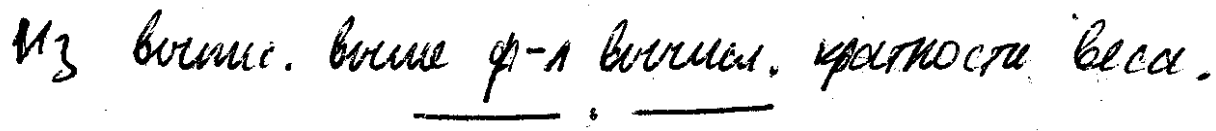
Original source details: GELFAND-P001079

Diagram and typed description. The concluding remark on the structure of the fiber.

GELFAND-P001080

[source number: 6]

From the formulas written out above, calculate the multiplicities of the weight.



Original source details: GELFAND-P001080

Diagram and typed description. The line on weight multiplicities with two underlined fragments.

GELFAND-P001081

21/II-83 Gelfand Seminar.
V. I. Arnold.

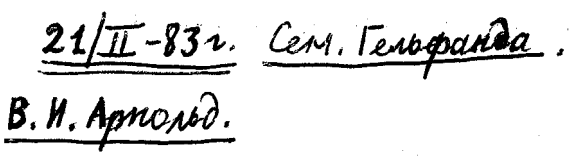
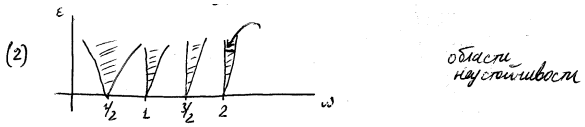


Diagram and typed description. The date, seminar name, and speaker.

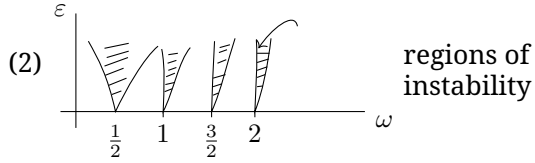
Original source details: GELFAND-P001081

$$(1) \qquad \ddot{x} + (\omega^2 + \varepsilon a(t))x = 0$$
$$a(t + 2\pi) = a(t)$$



Original source details: GELFAND-P001081

Diagram and typed description.



regions of instability

$a(t) = \cos t \quad \Rightarrow$ increasingly high order of contact (of the zone boundaries).

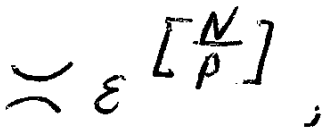
Diagram and typed description.

Sketch above the words “zone boundaries.”

Original source details: GELFAND-P001081

(3) Theorem. Width of the Nth zone $\asymp \varepsilon^{[\frac{N}{p}]}$; if a is a trigonometric polynomial

$$\deg a = p.$$



Uncertain reading. The exponent is written $[N/p]$. The brackets are left unexplained.

Original source details: GELFAND-P001081

Mathieu: $p = 1$.

Uncertain reading. The handwritten name appears to read «Матe».

Original source details: GELFAND-P001081

Hill’s equation (a in general position) $\Rightarrow$ first-order contact everywhere

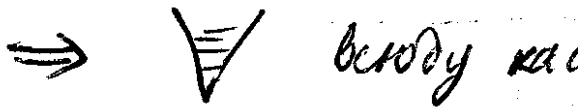


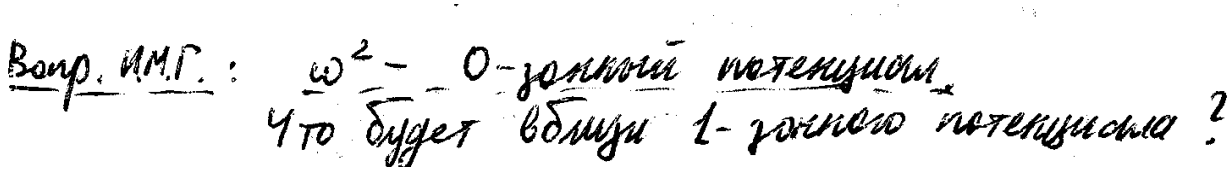
Diagram and typed description.

Sketch after the arrow.

Original source details: GELFAND-P001081

(but the width decreases exponentially)

Question by I. M. G.: ω^2 — a 0-zone potential. What happens near a 1-zone potential?

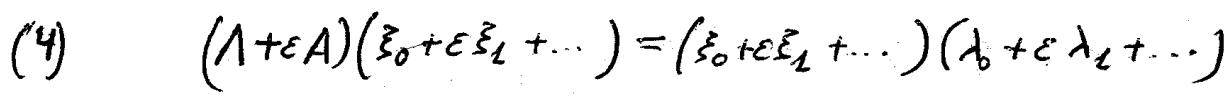


Original source details: GELFAND-P001081

Diagram and typed description. The question about a one-zone potential.

The proof depends on the structure of the Rayleigh–Schrödinger series. We seek the perturbation.

$$(4) \quad (\Lambda + \varepsilon A)(\xi_0 + \varepsilon\xi_1 + \ldots) = (\xi_0 + \varepsilon\xi_1 + \ldots)(\lambda_0 + \varepsilon\lambda_1 + \ldots)$$

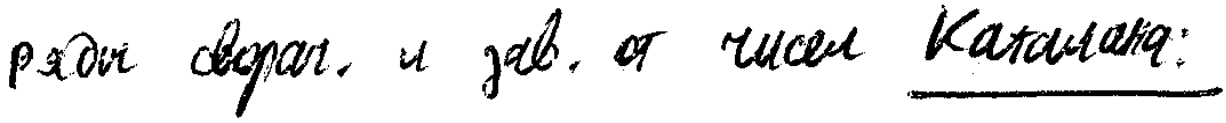


Original source details: GELFAND-P001081

Diagram and typed description. The perturbation series in equation (4).

GELFAND-P001082

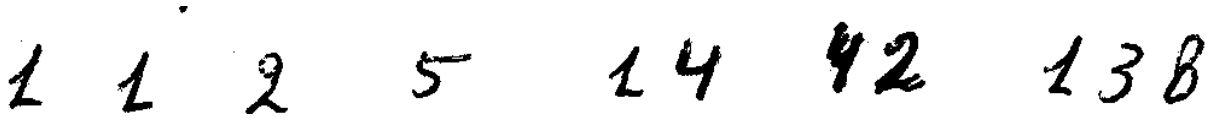
series are [combined?] and depend on the numbers named for Catalan:



Original source details: GELFAND-P001082

Uncertain reading. The abbreviation appears to read «сверт.».

$$1 \quad 1 \quad 2 \quad 5 \quad 14 \quad 42 \quad 138$$



Original source details: GELFAND-P001082

Diagram and typed description. The last number in the manuscript is 138.

Another occurrence, in Griffiths and Harris (they did not notice).

Formula:

$$(5) \qquad a_{n+1} = \sum_{j=1}^{n-1} a_j a_{n-j}$$

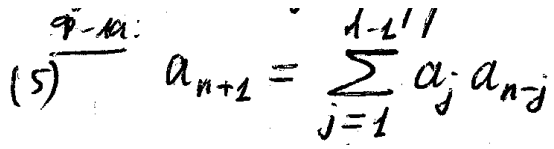


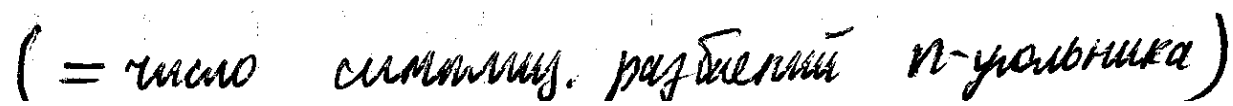
Diagram and typed description. The summation bounds and indices in the manuscript.

Original source details: GELFAND-P001082

Using a generating function we compute

$$a_n = \frac{(2n)!}{n!(n+1)!}$$

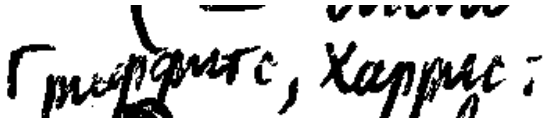
(= the number of simplicial subdivisions of an n-gon)



Original source details: GELFAND-P001082

Diagram and typed description. The polygon index in the manuscript.

[Inserted at the upper left, “Griffiths, Harris”:]
Riemann surfaces of genus g have $3g-3$ moduli.

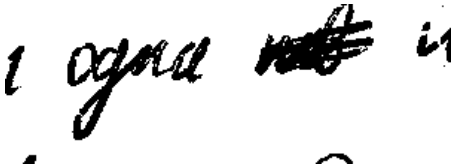


Original source details: GELFAND-P001082

Diagram and typed description. The names inserted above the line.

On $S^2 = \mathbb{C}P^1$ choose branch points, etc.

GH: the number of ways in which one [crossed out] and the same surface was obtained:



Original source details: GELFAND-P001082

Uncertain reading. The crossed-out word may be an abbreviation for “surface.”

$1, 1, 2, 5, \ldots$ — Catalan numbers (but it is not said why)

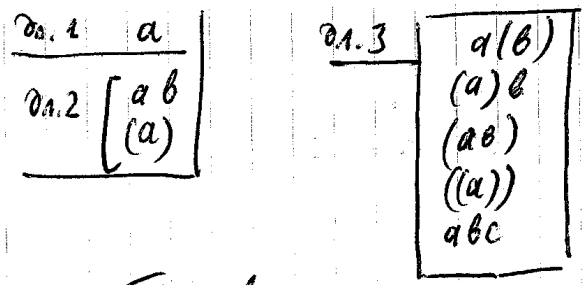
Bracket symbol: an expression properly formed from letters and brackets: example

$$((a))b\,(cd)e$$

$\sum$ of the counts of symbols and opening brackets $= n$ (length)

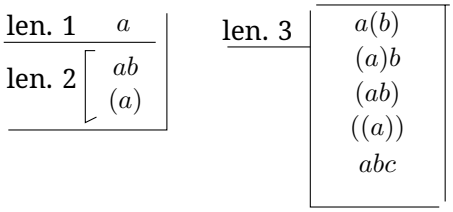
$\Downarrow$

count = the Catalan number



Original source details: GELFAND-P001082

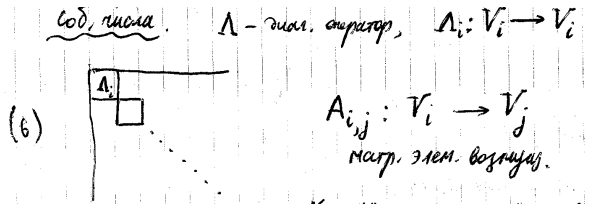
Diagram and typed description.



Bracket symbol $\rightsquigarrow$ term of the perturbation series.

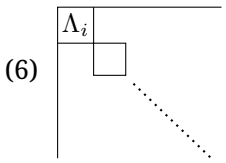
λ_i (eigenvalues) $\rightsquigarrow$ quiver

Eigenvalues. Λ — diagonal operator, $\Lambda_i : V_i \longrightarrow V_i$



Original source details: GELFAND-P001082

Diagram and typed description.



$A_{i,j} : V_i \longrightarrow V_j$
matrix element of the perturbation

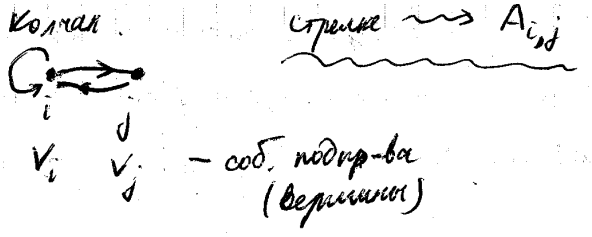
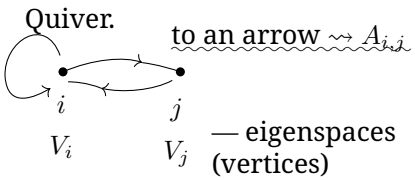


Diagram and typed description.



Original source details: GELFAND-P001082

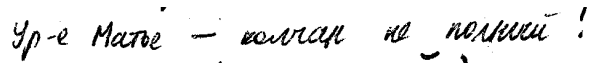
GELFAND-P001083

[source number: 9]

Suppose everything is finite or locally finite

(into $\forall$ vertex there lead finitely many)

Mathieu's equation — the quiver is incomplete!



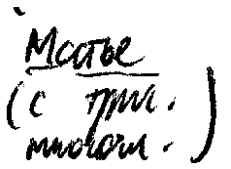
Uncertain reading. The handwritten name appears to read «Матe».

Original source details: GELFAND-P001083

(locally finite)

(the potential — cos and sin and the eigenfunctions too!)

Mathieu (with a trigonometric polynomial) $\Rightarrow$ a matrix of generalized Jacobi type.



Uncertain reading. The handwritten name appears to read «Матe».

Original source details: GELFAND-P001083

We seek the perturbation of λ_0.

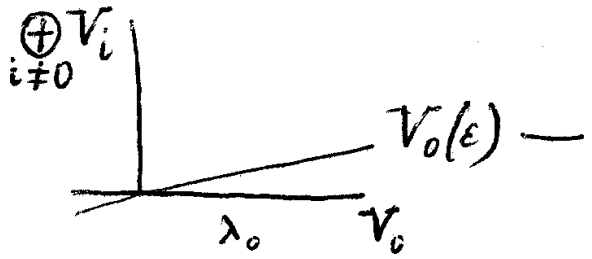
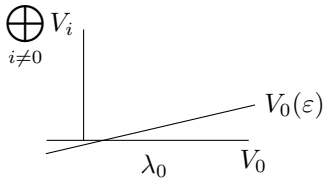
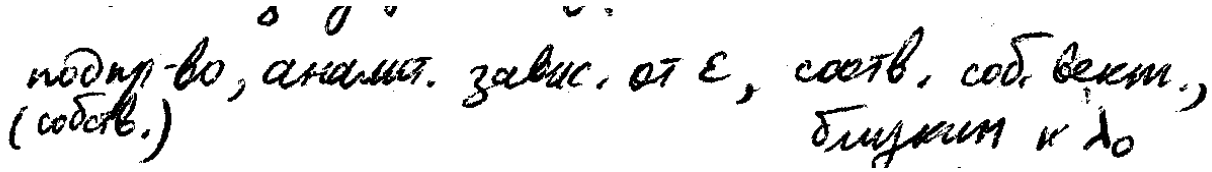


Diagram and typed description.



Original source details: GELFAND-P001083

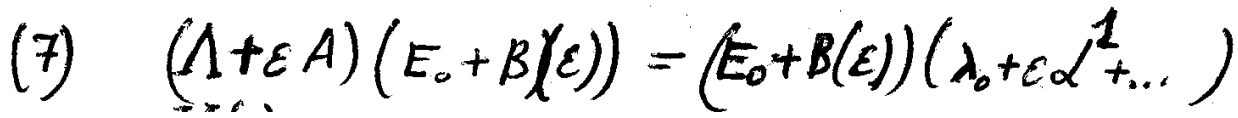
$V_0(\varepsilon)$ — a subspace, [below: "eigen"] depending analytically on ε, corresponding to eigenvectors close to λ_0.



Original source details: GELFAND-P001083

Diagram and typed description. The manuscript abbreviation for "eigenvectors."

$$(7) \quad (\Lambda + \varepsilon A)(E_0 + B(\varepsilon)) = (E_0 + B(\varepsilon))(\lambda_0 + \varepsilon\alpha^1 + \dots)$$



Original source details: GELFAND-P001083

Diagram and typed description. The coefficient α^1 in formula (7).

$V_0(\varepsilon)$ — the graph of the operator

$$B(\varepsilon) : V_0 \longrightarrow \bigoplus_{i \neq 0} V_i \qquad E_0 \text{ — the projector onto } V_0$$

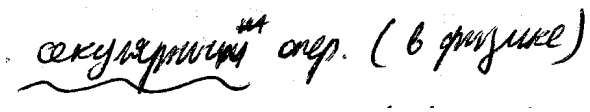
$$\alpha^1 : \text{ operator } V_0 \to V_0, \qquad \alpha^i : V_0 \to V_0$$

(7) — the condition of invariance of [crossed out] $V_0(\varepsilon)$

Uncertain reading. The crossed-out symbol is illegible.

Original source details: GELFAND-P001083

(8) operator $\lambda_0 + \varepsilon\alpha^1 + \dots$ is called the secular operator (in physics)



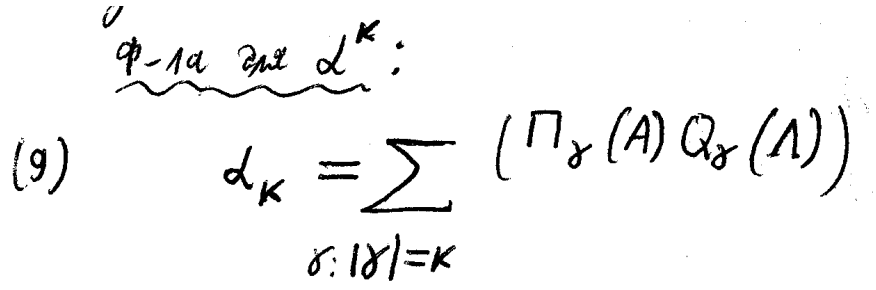
Uncertain reading. A correction is visible above the ending of the Russian word for "secular." The original ending is uncertain.

Original source details: GELFAND-P001083

The restriction of $\Lambda + \varepsilon A$ to $V_0(\varepsilon)$ is similar to the secular operator (8) $\Rightarrow$ they have the same eigenvalues.

Formula for α^k:

$$(9) \qquad \alpha_k = \sum_{\gamma : |\gamma| = k} \left(\Pi_\gamma(A) Q_\gamma(\Lambda)\right)$$

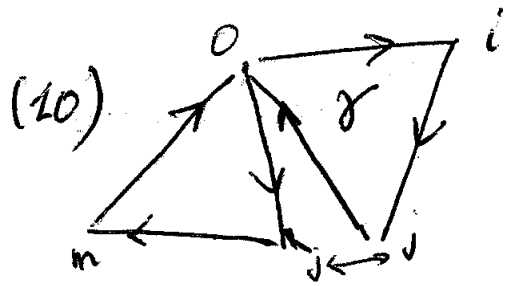


Original source details: GELFAND-P001083

Diagram and typed description. The heading uses a superscript and formula (9) a subscript.

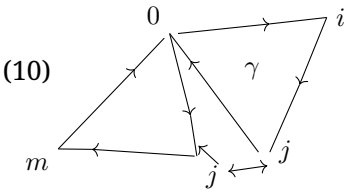
γ — a loop in the quiver from 0 to 0

k — the length of the loop



Original source details: GELFAND-P001083

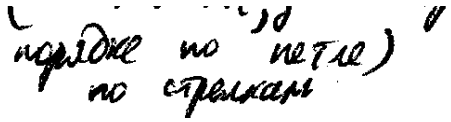
Diagram and typed description.



Both lower labels look like j. The connection of the short diagonal stroke to the adjacent arrows is uncertain.

$$\Pi_\gamma(A) = A_{m0} \dots A_{ij} A_{0i}$$

$\Pi_\gamma(A)$ — the chronological product around the loop ($= \prod A_{i,j}$ in the required order along the loop) [inserted below: «following the arrows».]



Original source details: GELFAND-P001083

Diagram and typed description. The insertion below the note about the order of factors.

$Q_\gamma(\Lambda)$ — a rational function of the eigenvalues

(computed using Catalan numbers!)

Consider a bracket symbol.

Filling it — insert vertices of the quiver in place of the letters.

GELFAND-P001084

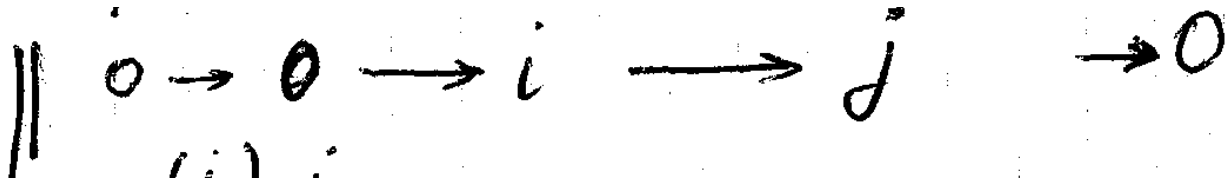
A loop is constructed from a filling.

Original source details: GELFAND-P001084

Uncertain reading. The crossed-out mark before “terms” is illegible.

The terms must be indexed by fillings

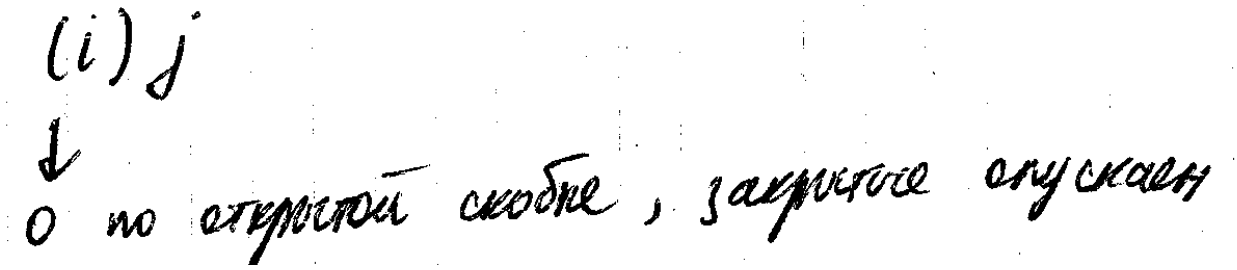
$$\| \quad 0 \longrightarrow 0 \longrightarrow i \longrightarrow j \longrightarrow 0$$



Original source details: GELFAND-P001084

Diagram and typed description. The chain of four arrows with two initial zeros.

$$\begin{matrix}(i)j\\ \downarrow\\ 0\end{matrix}$$ for an opening bracket, we omit the closing ones



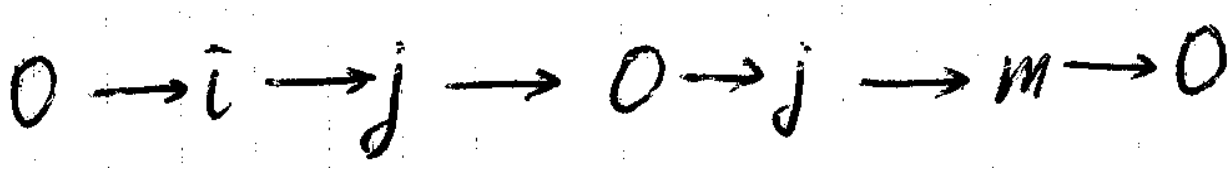
Original source details: GELFAND-P001084

Diagram and typed description. The arrow from the opening bracket to zero and its explanation.

There is one more arrow than the length of the filling.

One loop comes from different fillings.

$$0 \longrightarrow i \longrightarrow j \longrightarrow 0 \longrightarrow j \longrightarrow m \longrightarrow 0$$

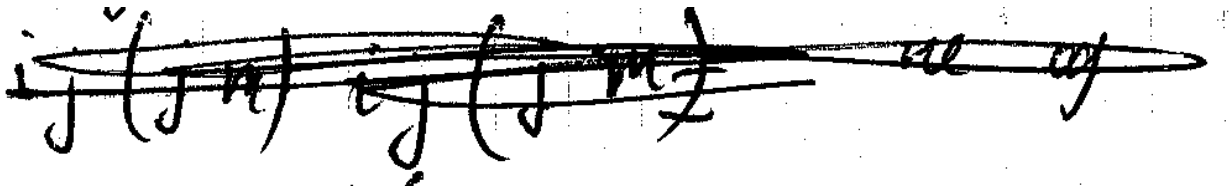


Original source details: GELFAND-P001084

Diagram and typed description. The loop with two occurrences of vertex j.

In fact, the sum runs over fillings rather than loops.

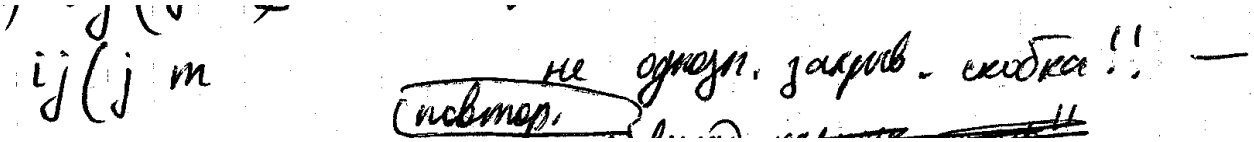
[Line crossed out: $ij(jm)$ $ij(jm)$ not unamb...]



Original source details: GELFAND-P001084

Uncertain reading. Two similar fillings are legible in the crossed-out line. At the right, possibly "not unamb...".

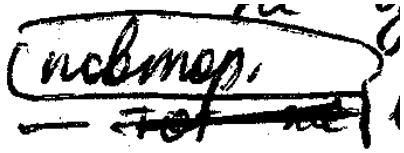
$ij(jm$ ambiguous closing bracket!! —



Original source details: GELFAND-P001084

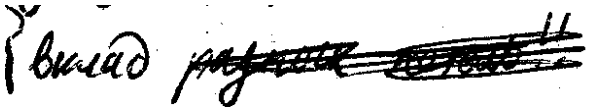
Diagram and typed description. The expression with the closing bracket omitted and its accompanying remark.

repeat [crossed out] contribution [crossed out]



Original source details: GELFAND-P001084

Uncertain reading. Below the circled word, the crossed-out text probably reads “— the same.”



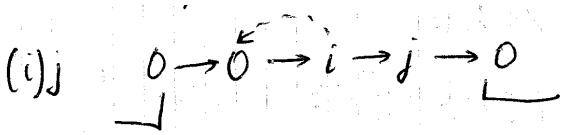
Original source details: GELFAND-P001084

Uncertain reading. After “contribution,” the crossed-out text probably reads “different brackets!!”.

Diagram of a filling: add dashed [~~brackets~~] [inserted below: arrows]

Original source details: GELFAND-P001084

Uncertain reading. The ending of the Russian word for “dashed” is overwritten. The original ending is uncertain.



Original source details: GELFAND-P001084

Diagram and typed description.

$$(i)j \qquad 0 \longrightarrow 0 \longrightarrow i \longrightarrow j \longrightarrow 0$$

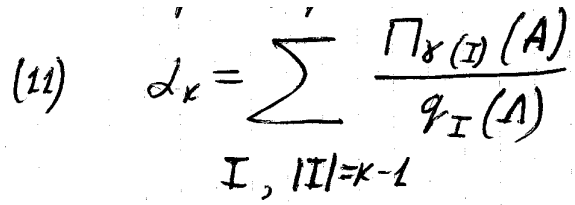
The outermost zeros do not signify anything

Arrow $\longrightarrow$ to 0, that corresponds to the opening bracket matching [crossed out] the closing bracket (from the letter before the closing bracket)

Original source details: GELFAND-P001084

Uncertain reading. The crossed-out word probably abbreviates “opening.”

$$\alpha_k = \sum_{I,\ |I|=k-1} \frac{\Pi_{\gamma(I)}(A)}{q_I(\Lambda)} \tag{11}$$



Original source details: GELFAND-P001084

Diagram and typed description. The left side of formula (11) reads α_k.

I
|
filling of a bracket symbol

$|I|$ — length of the bracket symbol

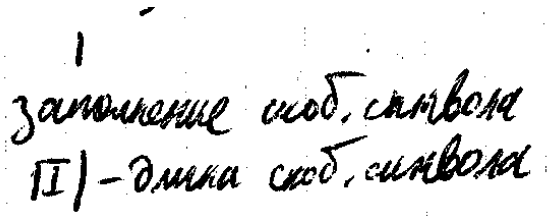
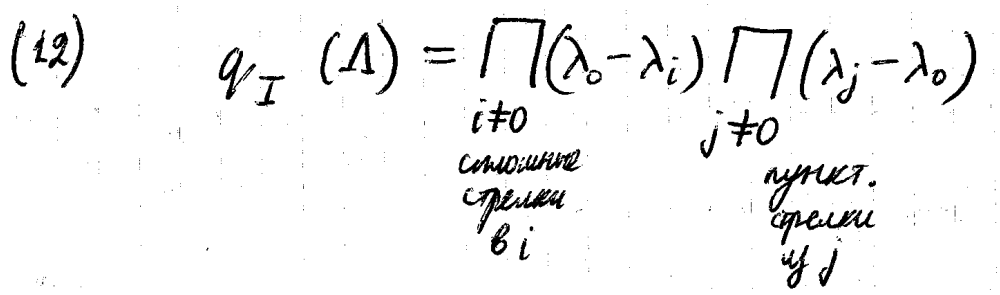


Diagram and typed description. The two annotations for the summation index.

Original source details: GELFAND-P001084

$\Pi_{\gamma(I)}(A)$ — the chronological product around the loop along the solid arrows.

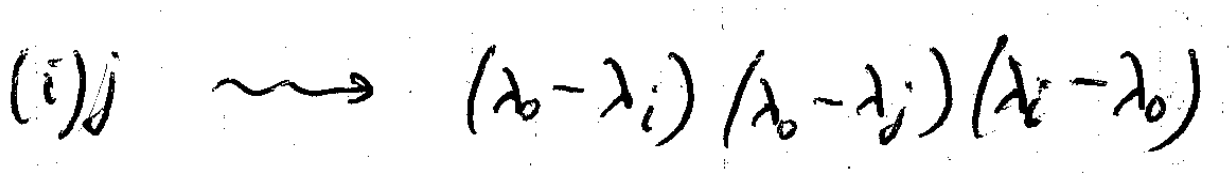
$$q_I(\Lambda) = \prod_{\substack{i\neq 0 \\ \text{solid arrows into } i}} (\lambda_0 - \lambda_i) \prod_{\substack{j\neq 0 \\ \text{dashed arrows from } j}} (\lambda_j - \lambda_0) \tag{12}$$



Original source details: GELFAND-P001084

Diagram and typed description. The factors and annotations for the two products in formula (12).

$$(i)j \rightsquigarrow (\lambda_0 - \lambda_i)(\lambda_0 - \lambda_j)(\lambda_i - \lambda_0)$$



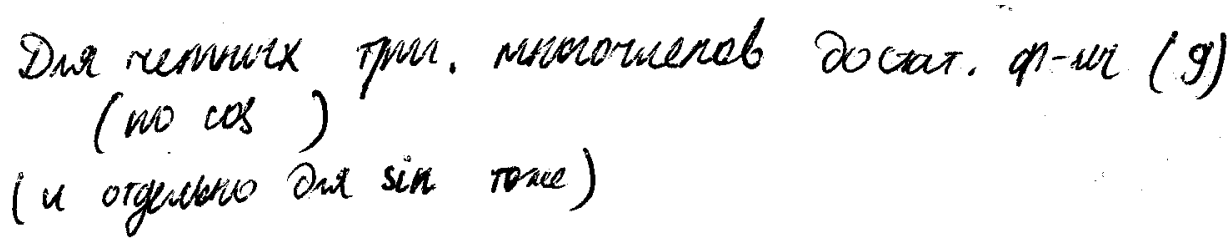
Original source details: GELFAND-P001084

Diagram and typed description. The product for the filling $(i)j$.

GELFAND-P001085

[source number: 10]

For even trigonometric polynomials, formula (9) suffices
[Below the word “even”: (for cos)]
(and separately for sin as well)



Original source details: GELFAND-P001085

Diagram and typed description. The cosine annotation below the word “even.”

$$V = \bigoplus V_i$$

$$\Lambda = \frac{d^2}{dt^2}, \qquad V_k = \{e_k, e_{-k}\}, \qquad e_k = e^{ikt} \tag{13}$$

Original source details: GELFAND-P001085

Diagram and typed description. The notation for the operator and basis in equation (13).

	V_0,	V_1,	V_2	V_3	$\cdots$	V_k
dim	1	2	2	2		2
						$\lambda = -k^2$

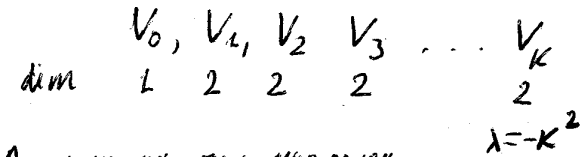


Diagram and typed description. The dimensions and the eigenvalue in the last column.

Original source details: GELFAND-P001085

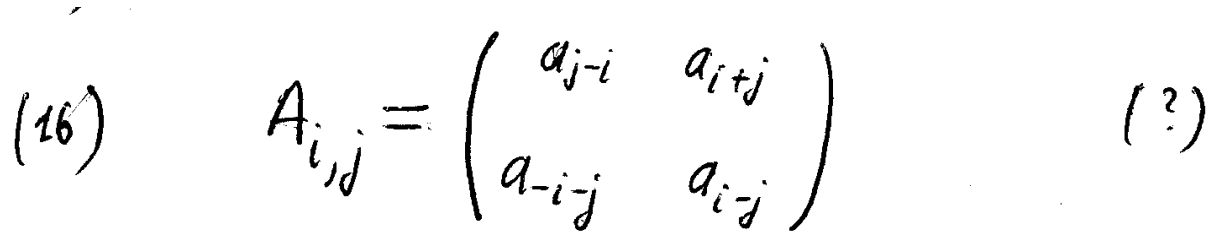
A — multiplication by a trigonometric polynomial

$$a(t) = \sum_{|s| \le p} a_s e_s \tag{14}$$

Let it be even, i.e.

$$a_{-s} = a_s \tag{15}$$

$$A_{i,j} = \begin{pmatrix} a_{j-i} & a_{i+j} \\ a_{-i-j} & a_{i-j} \end{pmatrix} \qquad (?) \tag{16}$$



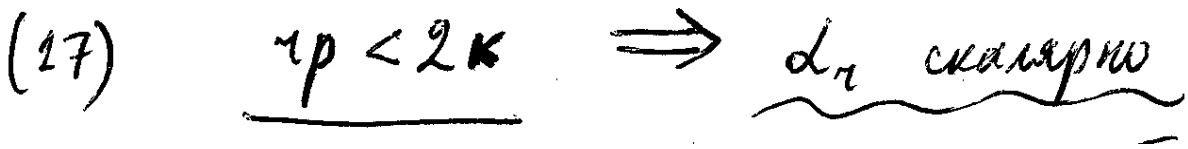
Original source details: GELFAND-P001085

Diagram and typed description. The signs in the subscripts and the question mark beside equation (16).

$(16) \Rightarrow$ if $i + j > p$, the matrices are diagonal and even scalar [written below: (by (15))] in this basis.

If all $A_{i,j}$ in the loop are scalar, then the product is also scalar.

$$\tau p < 2k \quad \Rightarrow \quad \alpha_\tau \text{ is scalar} \tag{17}$$

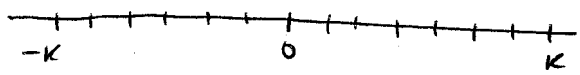


Original source details: GELFAND-P001085

Diagram and typed description. The coefficient α_τ and the underlining in equation (17).

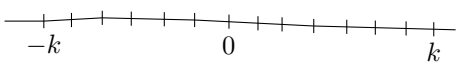
This is a topological lemma (from combinatorial topology).

Proof



Original source details: GELFAND-P001085

Diagram and typed description.



Consider a loop of τ arrows of length $\le p$.

Then for $\forall$ arrow $i + j > p$

(homotopic to 0!)

$i + j \le p \Rightarrow$ a jump across

Original source details: GELFAND-P001085

Uncertain reading. Provisional reading “a jump across.”

Original source details: GELFAND-P001085

Diagram and typed description.

The small mark at the right end looks like i. Arrowheads on the arches are indistinct.

$\Rightarrow$ length $\leq \tau p$ and so on.

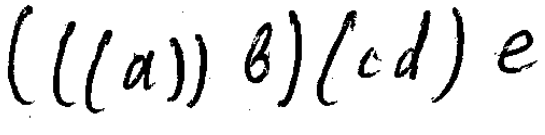
(17) $\Rightarrow$ in this approximation the eigenvalue has yet to split $\Rightarrow$

$\Rightarrow$ the required inequality!

GELFAND-P001086

An involution is also used:

$$(((a))b)(cd)e$$



Original source details: GELFAND-P001086

Diagram and typed description. The original bracket symbol.

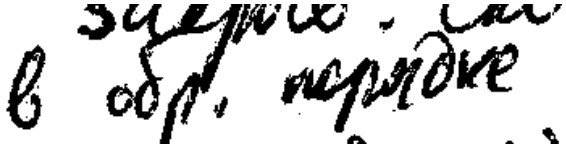
use the opening brackets to divide it into fragments

$$a))b) \qquad cd)e$$

Original source details: GELFAND-P001086

Diagram and typed description. The fragments with their opening brackets removed.

Attach the closing brackets to the letter. Rearrange the letters in the fragments [inserted below at the left: in reverse order]



Original source details: GELFAND-P001086

Diagram and typed description. The insertion specifying the reverse order of the letters.

$$b)a)) \qquad ed)c$$

Original source details: GELFAND-P001086

Diagram and typed description. The two fragments with their letters in reverse order.

[crossed out] Put the opening brackets back in their former places

Uncertain reading. The crossed-out abbreviation before “Put” is illegible.

Original source details: GELFAND-P001086

$$(((b)a))(ed)c$$

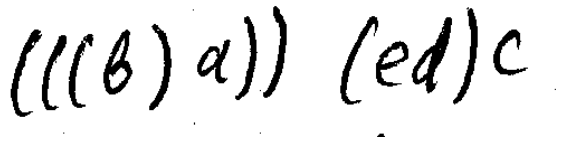
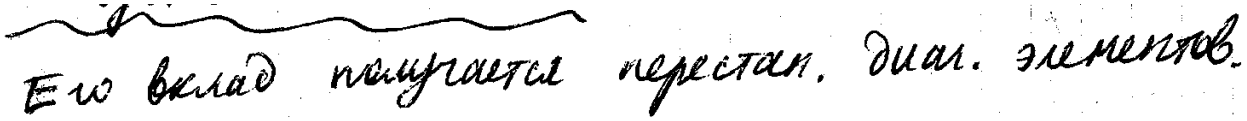


Diagram and typed description. The dual bracket symbol.

Original source details: GELFAND-P001086

We obtain the dual symbol

Its contribution is obtained by swapping the diagonal entries.



Original source details: GELFAND-P001086

Diagram and typed description. The Russian word for “diagonal” is abbreviated at the end of the line.

GELFAND-P001087

28/II-83 Gelfand Seminar.

I. Krichever.

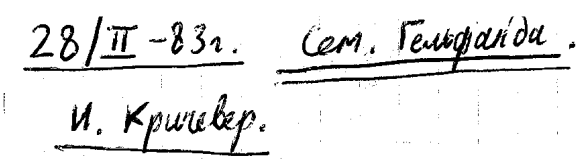


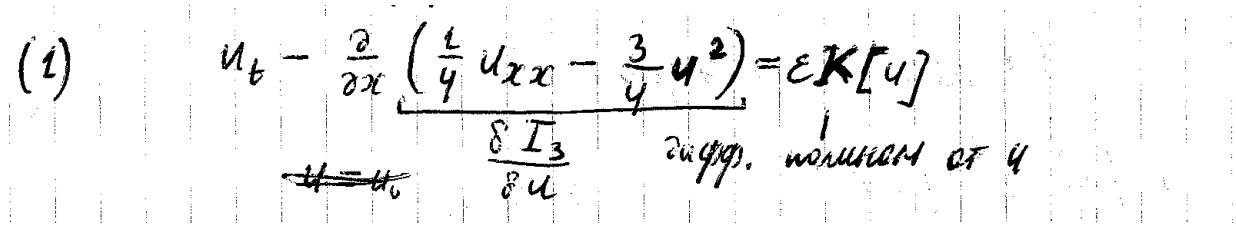
Diagram and typed description. The date, seminar name, and speaker.

Original source details: GELFAND-P001087

The aim is to construct a formal perturbation theory for the perturbation of the KdV equation.

$$(1) \qquad u_t - \frac{\partial}{\partial x}\underbrace{\left(\frac{1}{4}u_{xx} - \frac{3}{4}u^2\right)}_{\frac{\delta I_3}{\delta u}} = \varepsilon K[u]$$

[Below K: differential polynomial in u.]



Original source details: GELFAND-P001087

Diagram and typed description. Equation (1) with a square-ended bracket and the annotation below K.

[Crossed out: ~~$u = u_0$~~.]

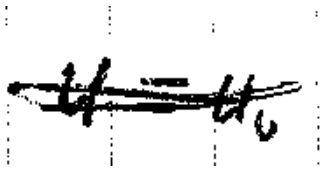


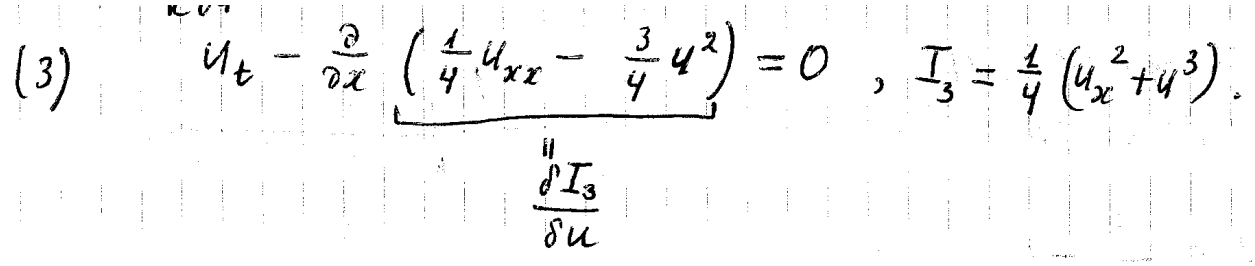
Diagram and typed description. The crossed-out equality before (2).

Original source details: GELFAND-P001087

$$(2) \qquad u = u_0(x,t) + \varepsilon(\ldots)$$

KdV

$$(3) \qquad u_t - \frac{\partial}{\partial x}\underbrace{\left(\frac{1}{4}u_{xx} - \frac{3}{4}u^2\right)}_{\parallel \frac{\delta I_3}{\delta u}} = 0, \qquad I_3 = \frac{1}{4}(u_x^2 + u^3).$$



Original source details: GELFAND-P001087

Uncertain reading. Two short vertical strokes stand between the lower bracket and its label. Their meaning is uncertain.

There is an infinite series of integrals in involution I_k: $I_1 = u$, $I_2 = u^2$, ...

Original source details: GELFAND-P001087

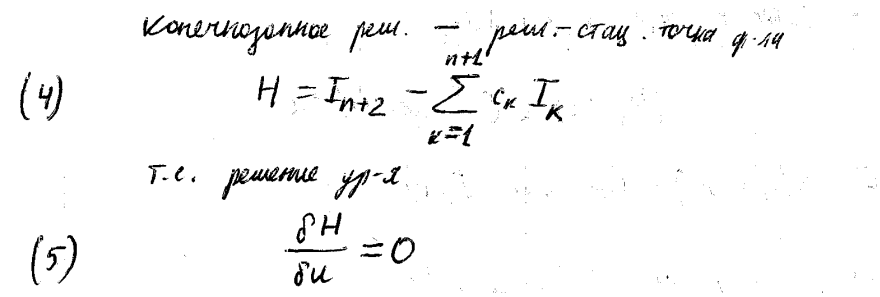
Diagram and typed description. The first integrals, including the overwritten u in I_1.

A finite-gap solution is a solution that is a stationary point of the functional

$$(4) \qquad H = I_{n+2} - \sum_{k=1}^{n+1} c_k I_k$$

i.e. a solution of the equation

$$(5) \qquad \frac{\delta H}{\delta u} = 0$$



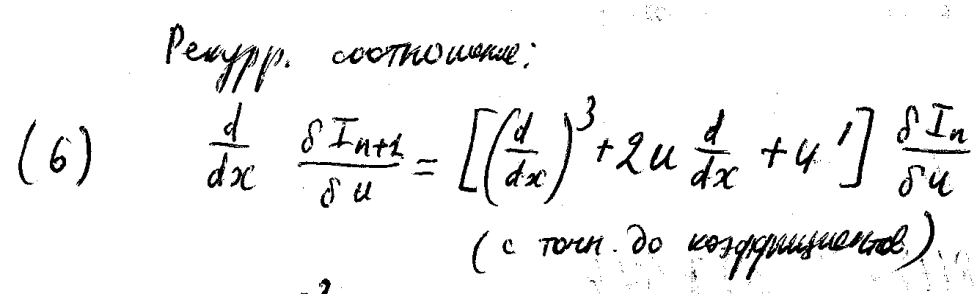
Original source details: GELFAND-P001087

Diagram and typed description. The functional and its stationarity condition in equations (4) and (5).

Recurrence relation:

$$(6) \qquad \frac{d}{dx}\frac{\delta I_{n+1}}{\delta u} = \left[\left(\frac{d}{dx}\right)^3 + 2u\frac{d}{dx} + u'\right]\frac{\delta I_n}{\delta u}$$

(up to coefficients).



Original source details: GELFAND-P001087

Diagram and typed description. The coefficients, prime, and indices in equation (6).

$$\delta^2 H - ?$$

at a specified stationary point.

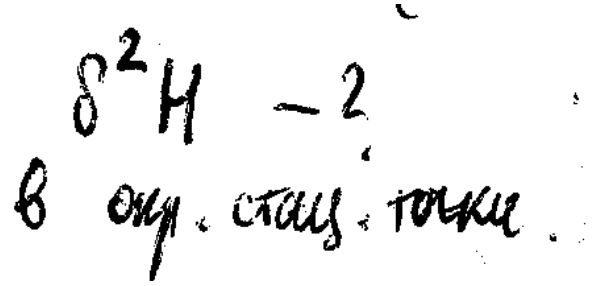


Diagram and typed description. The question about the second variation.

Original source details: GELFAND-P001087

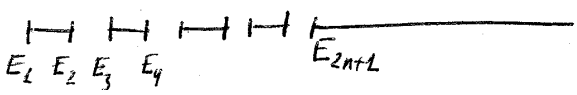
GELFAND-P001088

[source number: 15]

$$L = -\frac{d^2}{dx^2} + u_0(x) \qquad \text{spectrum:} \qquad (u_0(x+1) = u_0(x))$$

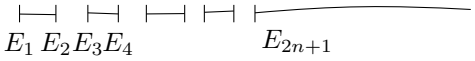
Original source details: GELFAND-P001088

Diagram and typed description. The operator and periodicity condition.



Original source details: GELFAND-P001088

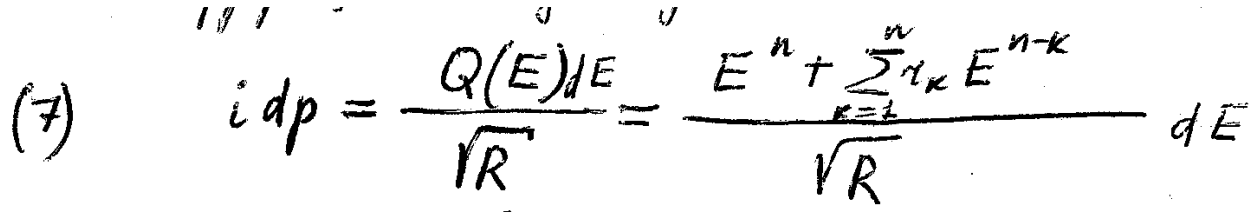
Diagram and typed description.



The remaining endpoints are simple eigenvalues of the periodic and antiperiodic problems.

Quasimomentum differential:

$$(7) \qquad i\,dp = \frac{Q(E)dE}{\sqrt{R}} = \frac{E^n + \sum_{k=1}^{n} r_k E^{n-k}}{\sqrt{R}}\,dE$$

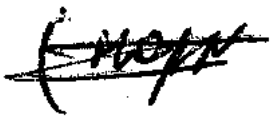


Original source details: GELFAND-P001088

Uncertain reading. The coefficient is provisionally read as r_k. The first letter is ambiguous.

$$R(E) = \prod_{i=1}^{2n+1} (E - E_i)$$

[Crossed out at the right: ~~(norm.~~]



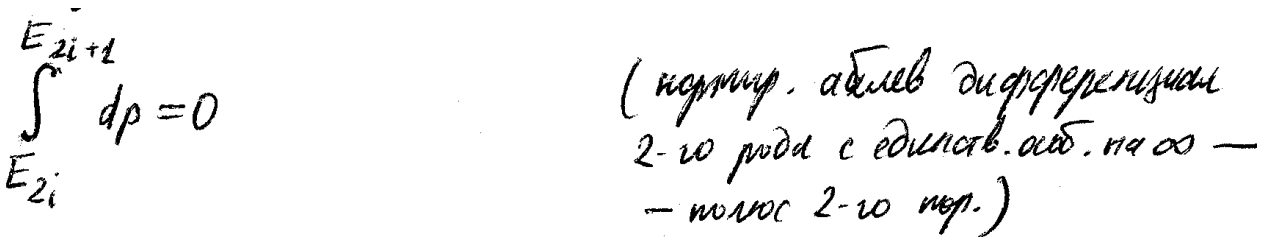
Uncertain reading. The beginning reads "(norm.". The rest of the abbreviation is uncertain.

Original source details: GELFAND-P001088

Conditions uniquely determining dp:

$$\int_{E_{2i}}^{E_{2i+1}} dp = 0$$

(normalized Abelian differential of the second kind with its sole singularity at ∞ — — a second-order pole.)



Original source details: GELFAND-P001088

Diagram and typed description. The vanishing integral and the note on the pole.

Now consider the perturbation $u = u_0 + \delta u$

E_i' — new band edges

$e_s^- < e_s^+$ — new band edges; $\delta e_s = e_s^+ - e_s^-$

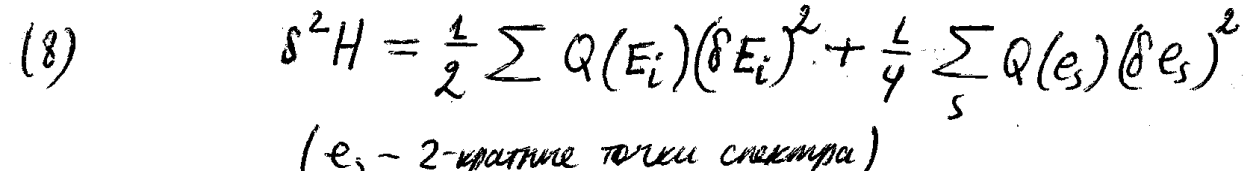
Original source details: GELFAND-P001088

Diagram and typed description. The two new edges and the definition of δe_s.

Then:

$$(8) \qquad \delta^2 H = \frac{1}{2}\sum Q(E_i)(\delta E_i)^2 + \frac{1}{4}\sum_s Q(e_s)(\delta e_s)^2$$

(e_s — double points of the spectrum)



Original source details: GELFAND-P001088

Diagram and typed description. The two sums of the second variation and the note on e_s.

Q changes sign once in $\forall$ of the forbidden zones $\Rightarrow$ one zero in each zone

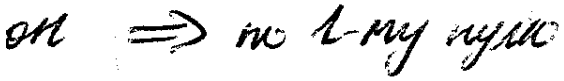
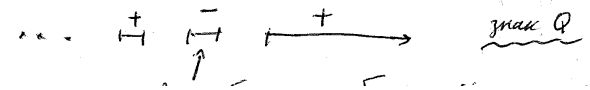


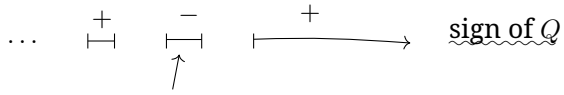
Diagram and typed description. The handwritten phrase “one zero each.”

Original source details: GELFAND-P001088

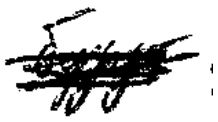


Original source details: GELFAND-P001088

Diagram and typed description.



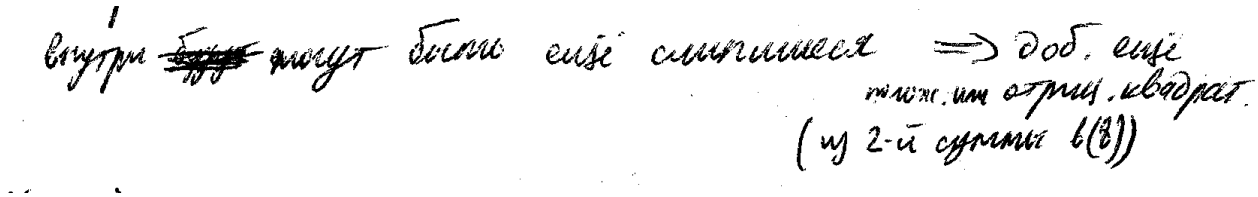
[Below the arrow:] inside [crossed out] there may be other coalesced ones $\Rightarrow$ add another positive or negative square



Uncertain reading. The crossed-out word may read “will be.”

Original source details: GELFAND-P001088

(from the second sum in (8))

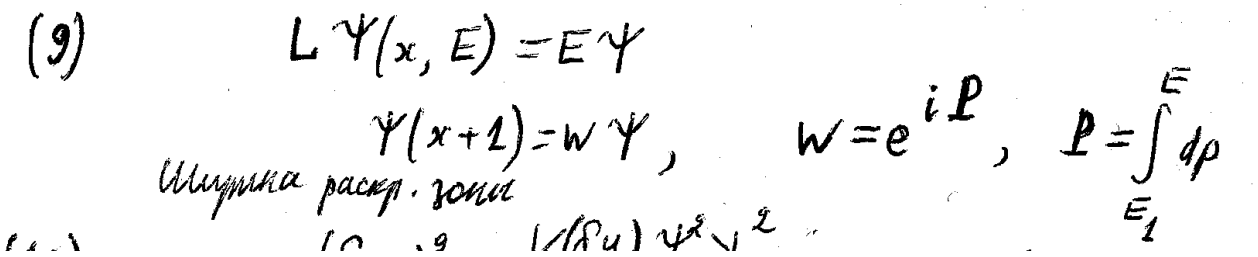


Original source details: GELFAND-P001088

Diagram and typed description. The note below the arrow in the sign diagram.

$$(9) \qquad L\Psi(x,E) = E\Psi$$

$$\Psi(x+1) = w\Psi, \qquad w = e^{iP}, \qquad P = \int_{E_1}^{E} dp$$



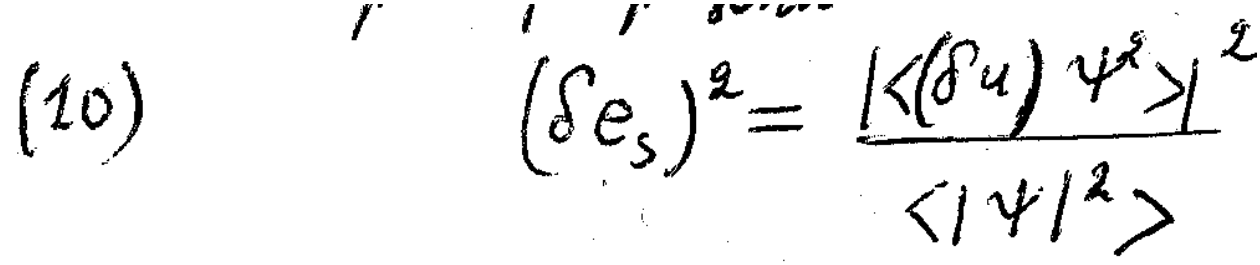
Original source details: GELFAND-P001088

Diagram and typed description. The Bloch equation and multiplier.

Width of the opened gap

$$(10) \qquad (\delta e_s)^2 = \frac{\left|\langle(\delta u)\Psi^2\rangle\right|^2}{\langle|\Psi|^2\rangle}$$

(Ψ — Bloch function, $\Psi(x, e_s)$)

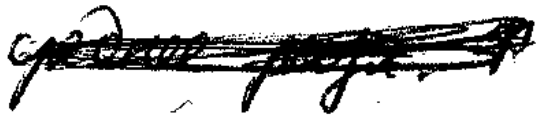


Original source details: GELFAND-P001088

Diagram and typed description. The written denominator is $\langle|\Psi|^2\rangle$ with no outer exponent.

$\langle \quad \rangle$ — averaging over a period

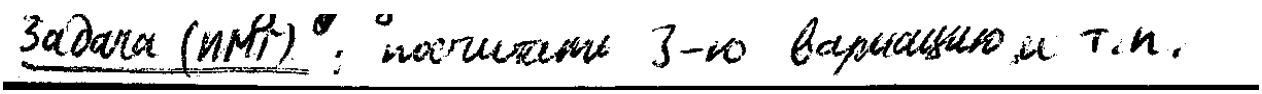
$u_0 = \text{const} \Rightarrow$ we obtain [crossed out: ~~averag.~~ ...] Fourier coefficients for the perturbation.



Undeciphered passage. After the start of "average," one or more words are obscured by the cancellation strokes.

Original source details: GELFAND-P001088

Problem (I. M. G.): compute the third variation, and so on.



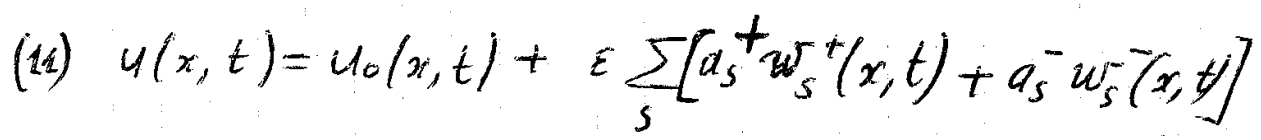
Original source details: GELFAND-P001088

Diagram and typed description. The problem concerning the third variation.

GELFAND-P001089

Return to (1) and let u be a solution. Then to first order in ε

$$u(x,t) = u_0(x,t) + \varepsilon \sum_s \left[a_s^+ w_s^+(x,t) + a_s^- w_s^-(x,t)\right] \tag{11}$$



Original source details: GELFAND-P001089

Diagram and typed description. Signs on the two terms in equation (11).

$w_s^\pm(x,t)$ — solutions of linearized KdV near an n-gap solution. They are found by opening a gap near e_s and taking the derivative with respect to the length of the new gap. [inserted above the line: «$(n+1)$-gap potential».]

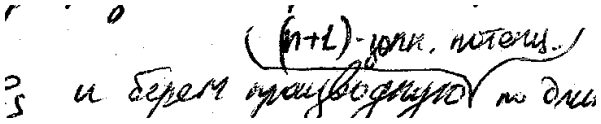
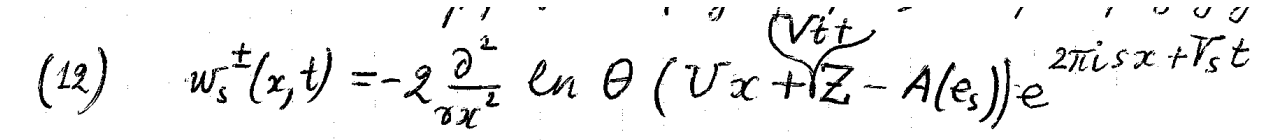


Diagram and typed description. The insertion about the $(n+1)$-gap potential.

Original source details: GELFAND-P001089

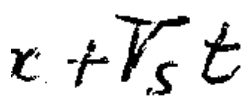
$$w_s^\pm(x,t) = -2\frac{\partial^2}{\partial x^2} \ln \theta\big(Ux + Vt + Z - A(e_s)\big) e^{2\pi i s x + V_s t} \tag{12}$$

[Inserted in the argument of θ: $Vt+$ before Z.]



Original source details: GELFAND-P001089

Diagram and typed description. Equation (12) with the insertion above the line.



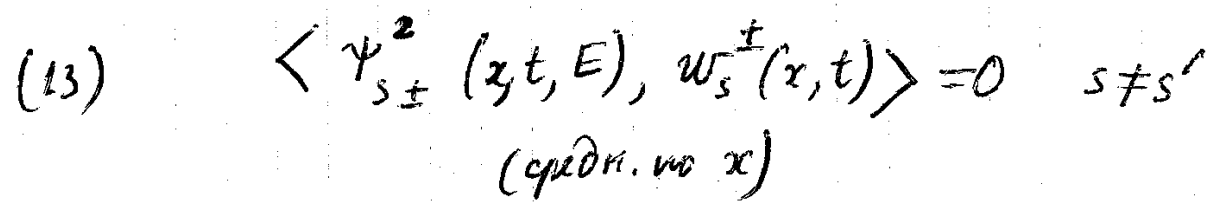
Uncertain reading. The exponent provisionally reads $V_s t$. The first letter is uncertain.

Original source details: GELFAND-P001089

$\Psi^2_{s\pm}(x,t,E)$ — a solution of the adjoint of the linearized KdV.

$$\left\langle \Psi^2_{s\pm}(x,t,E), w^{\pm}_s(x,t) \right\rangle = 0, \qquad s \neq s' \tag{13}$$

(averaged over x)

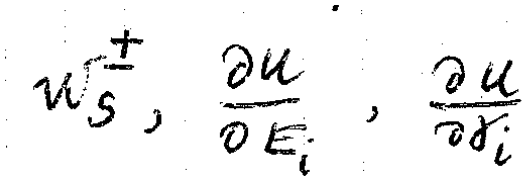


Original source details: GELFAND-P001089

Diagram and typed description. Both factors carry the index s. The condition on the right reads $s \neq s'$.

($\Rightarrow$ there is a basis expansion)

$$w^{\pm}_s, \quad \frac{\partial u}{\partial E_i}, \quad \frac{\partial u}{\partial \gamma_i}$$



Uncertain reading. The last denominator provisionally reads $\partial\gamma_i$. The first letter of the parameter is uncertain.

Original source details: GELFAND-P001089

Wodzicki (Mariusz).

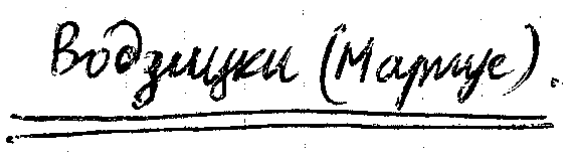


Diagram and typed description. The speaker's name in the manuscript.

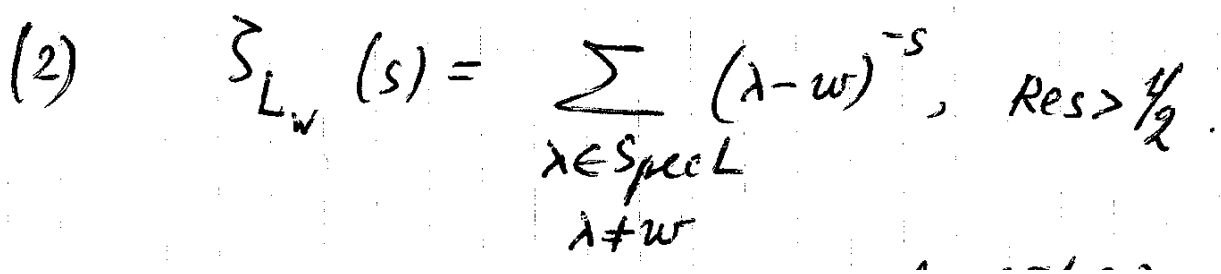
Original source details: GELFAND-P001089

$$L_w = -\frac{d^2}{dx^2} + u(x) - w, \qquad x \in [0,T] \tag{1}$$

(Hill operator; u — T-periodic)

ζ-function:

$$\zeta_{L_w}(s) = \sum_{\substack{\lambda \in \operatorname{Spec} L \\ \lambda \neq w}} (\lambda - w)^{-s}, \qquad \operatorname{Re} s > \tfrac{1}{2}. \tag{2}$$



Original source details: GELFAND-P001089

Diagram and typed description. The zeta function definition and convergence region.

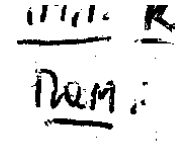
We are interested in the values at $s = 1, 2, 3, \ldots$

Suppose

$$\begin{gathered} u(x) \text{ — finite-gap,} \\ w \in \{\alpha_0, \ldots, \alpha_{2n}\} \text{ — band edges} \end{gathered} \tag{3}$$

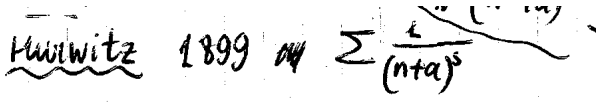
I. M. G. Kostya:

Доп: Hurwitz 1899 [overwritten] $\sum \frac{1}{(n+a)^s}$



Uncertain reading. The underlined abbreviation provisionally reads «Доп:».

Original source details: GELFAND-P001089



Uncertain reading. A short mark after “Hurwitz 1899” is heavily overwritten.

Original source details: GELFAND-P001089

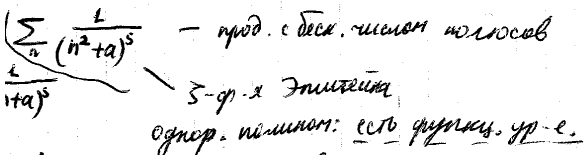


Diagram and typed description.

$\sum_n \frac{1}{(n^2+a)^s}$ — continuation with infinitely many poles

ζ-function of Epstein

Original source details: GELFAND-P001089

homogeneous polynomial: there is a functional equation.

$$\begin{aligned} \frac{1}{(n^2+a)^s} &= \frac{1}{\left(n^2(1+\frac{a}{n^2})\right)^s} = \left(\frac{1}{n^2} - \frac{a}{n^4} + \frac{a^2}{n^6} + \cdots\right)^s \\ &= \frac{1}{n^{2s}}\left(-\frac{as}{n^{2s+2}} + \cdots\right) \end{aligned}$$

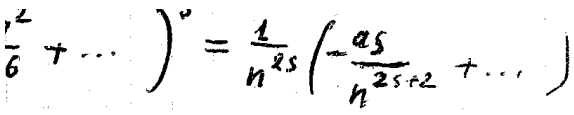
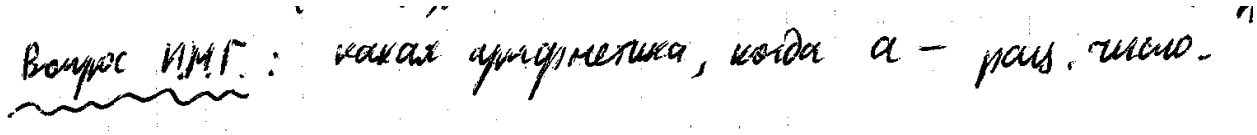


Diagram and typed description. The final equality in the manuscript. No separate 1 is visible before the minus.

Original source details: GELFAND-P001089

I. M. G.’s question: what arithmetic is involved when a — a rational number.



Original source details: GELFAND-P001089

Diagram and typed description. The question about arithmetic for a rational parameter.

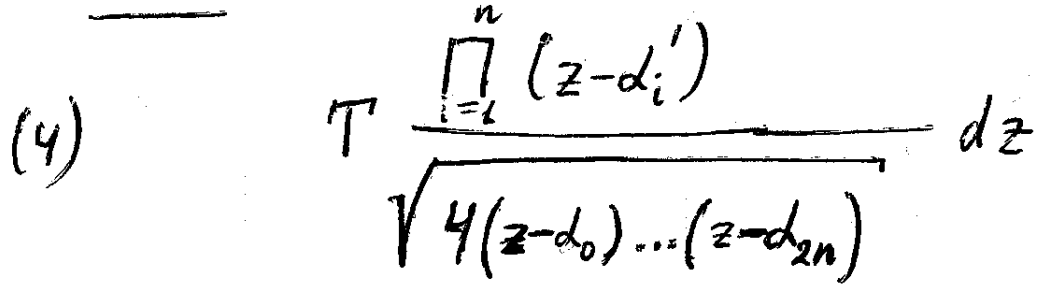
GELFAND-P001090

[source number: 16]

M. W.

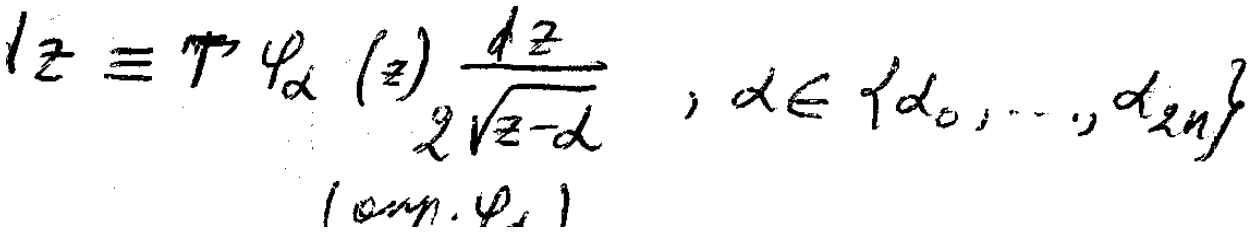
$$T\frac{\prod_{i=1}^{n}(z-\alpha_i')}{\sqrt{4(z-\alpha_0)\cdots(z-\alpha_{2n})}}\,dz \equiv T\varphi_\alpha(z)\frac{dz}{2\sqrt{z-\alpha}}, \qquad \alpha\in\{\alpha_0,\dots,\alpha_{2n}\} \tag{4}$$

(definition of φ_α)



Original source details: GELFAND-P001090

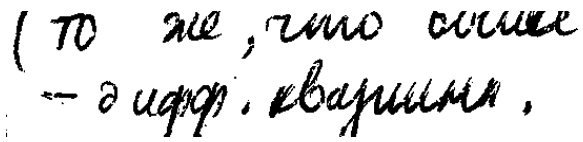
Diagram and typed description. The left-hand side of (4) with its product in the numerator.



Original source details: GELFAND-P001090

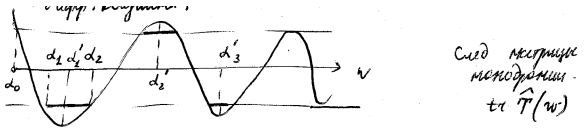
Diagram and typed description. The right-hand side of (4) and the choice of α.

(the same as above in Krichever)
— quasimomentum differential,



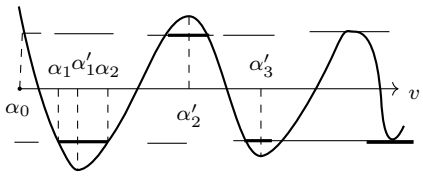
Original source details: GELFAND-P001090

Diagram and typed description. The abbreviated name of the differential.



Original source details: GELFAND-P001090

Diagram and typed description.



Trace of the monodromy matrix. $\operatorname{tr}\widehat{T}(w)$

The axis label beside the arrow resembles v or ν.

α'_j — min and max points of $\operatorname{tr} \widehat{T}(w)$.

Original source details: GELFAND-P001090

Diagram and typed description. The note on the minimum and maximum points.

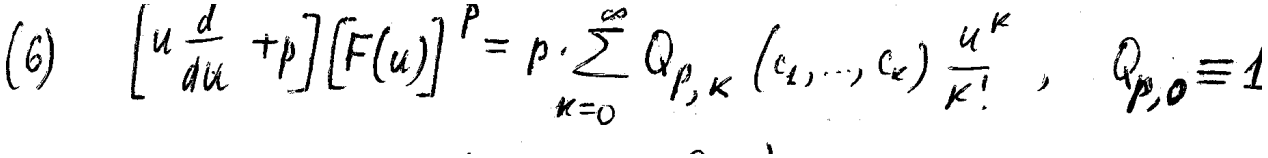
Introduce a universal family of polynomials:

$$F(u) = 1 + \sum_{i=1}^{\infty} c_i \frac{u^i}{i!} \tag{5}$$

and then

$$\left[u\frac{d}{du} + p\right][F(u)]^p = p \cdot \sum_{k=0}^{\infty} Q_{p,k}(c_1, \ldots, c_k)\frac{u^k}{k!}, \qquad Q_{p,0} \equiv 1. \tag{6}$$

(definition of $Q_{p,k}$)

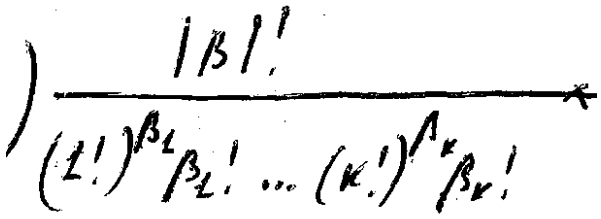


Original source details: GELFAND-P001090

Diagram and typed description. Defining relation (6).

$$\begin{aligned}\widehat{Q}_{p,k} &\equiv \frac{p+2k}{p+k} Q_{p,k}(c_1, \ldots, c_k) \\ &= \frac{p+2k}{p} k! \sum_{\substack{|\beta| \le \min(k,p) \\ \sum_{i=1}^{k} i\beta_i = k}} \binom{p}{|\beta|} \frac{|\beta|!}{(1!)^{\beta_1}\beta_1! \cdots (k!)^{\Lambda_k}\beta_k!} \\ &\quad \times c_1^{\beta_1} \cdots c_k^{\beta_k}\end{aligned} \tag{7}$$

Uncertain reading. The last exponent in the denominator resembles Λ_k.



Original source details: GELFAND-P001090

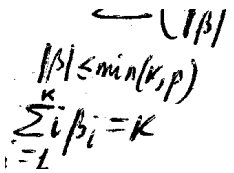


Diagram and typed description. The conditions below the sum.

Original source details: GELFAND-P001090

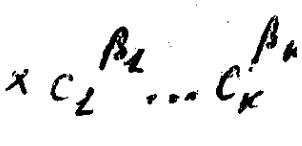


Diagram and typed description. The factors in c_1 and c_k.

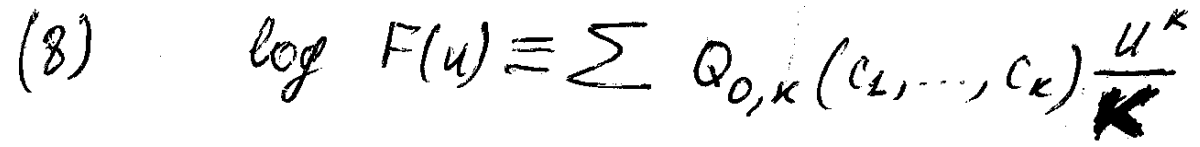
Original source details: GELFAND-P001090

$$\sum [F(u)]^p z^p = \frac{1}{1 - zF(u)} = \Phi(u, z)$$

and we need to take:

$$\left(u\frac{\partial}{\partial u} + z\frac{\partial}{\partial z} \right) \Phi(u, z).$$

$$\log F(u) \equiv \sum Q_{0,k}(c_1, \ldots, c_k) \frac{u^k}{k} \tag{8}$$



Original source details: GELFAND-P001090

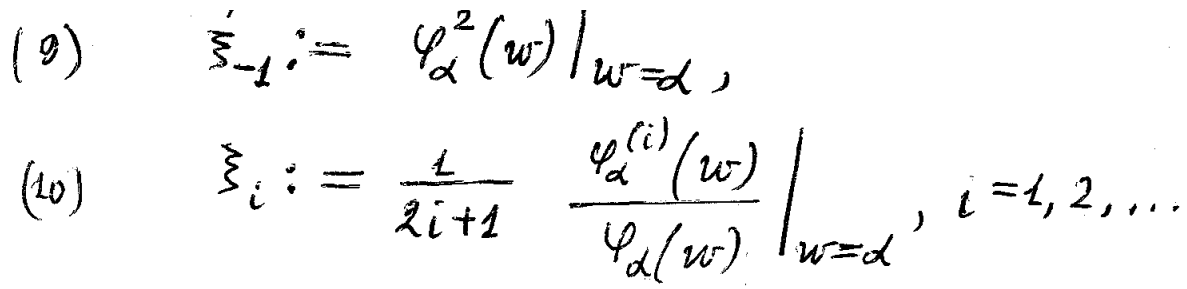
Uncertain reading. The denominator is overwritten and reads k. The earlier strokes are unclear.

(definition of $Q_{0,k}$)

Construct:

$$\xi_{-1} := \varphi_\alpha^2(w)\big|_{w=\alpha}, \tag{9}$$

$$\xi_i := \frac{1}{2i+1} \left. \frac{\varphi_\alpha^{(i)}(w)}{\varphi_\alpha(w)} \right|_{w=\alpha}, \qquad i = 1, 2, \ldots \tag{10}$$

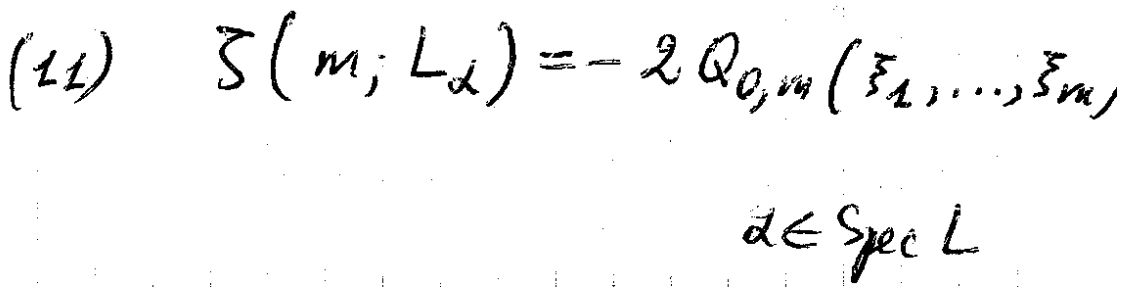


Original source details: GELFAND-P001090

Diagram and typed description. The evaluations and the factor $(2i+1)^{-1}$ in formulas (9) and (10).

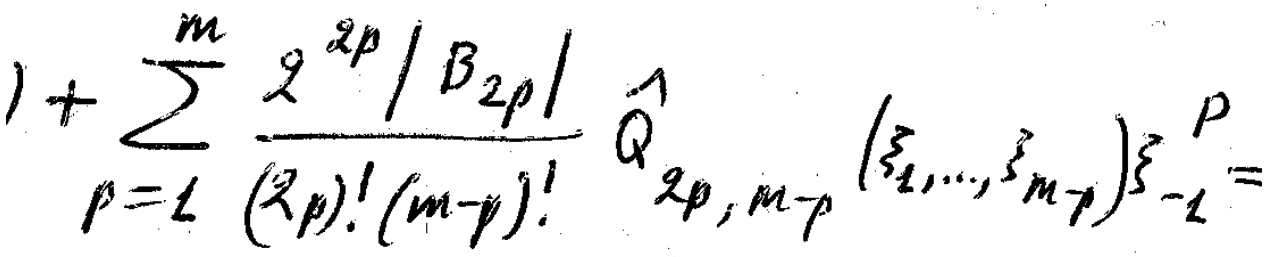
$$\begin{aligned}\zeta(m;L_\alpha) &= -2Q_{0,m}(\xi_1,\dots,\xi_m)\\ &\quad+\sum_{p=1}^{m}\frac{2^{2p}|B_{2p}|}{(2p)!(m-p)!}\widehat{Q}_{2p,m-p}(\xi_1,\dots,\xi_{m-p})\xi_{-1}^{p}\\ &= -2Q_{0,m}(\xi_1,\dots,\xi_m)\\ &\quad+\sum_{p=1}^{m}\frac{\zeta_0(p)}{(m-p)!}\widehat{Q}_{2p,m-p}(\xi_1,\dots,\xi_{m-p})\xi_{-1}^{p}\end{aligned}\tag{11}$$

[Below the first line: $\alpha \in \operatorname{Spec} L$.]



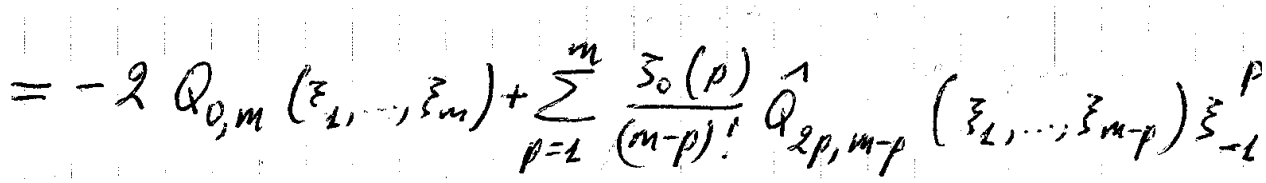
Original source details: GELFAND-P001091

Diagram and typed description. The start of equation (11) and the spectrum annotation.



Original source details: GELFAND-P001091

Diagram and typed description. The Bernoulli number, power of two, and factorials in the first line.



Original source details: GELFAND-P001091

Diagram and typed description. The second line of (11) with $\zeta_0(p)$.

ζ_0 — the ζ-function of $-\frac{d^2}{dx^2}$ on $[0,T]$.

$$\zeta_0(s) = \operatorname{Tr}\left(-\frac{d^2}{dx^2}, [0,T]\right)^{-s}$$

By ζ — the Riemann ζ-function

$$T = 2\pi, \qquad \zeta_0(s) = 2\zeta_{\text{Riemann}}(2s).$$

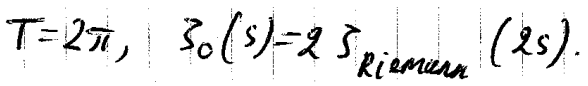


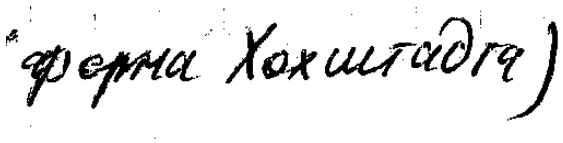
Diagram and typed description. The period is written as $T = 2\pi$.

Original source details: GELFAND-P001091

0-gap $\Rightarrow$ 1 term survives.

— · —

$u(x) \equiv u_0 = \text{const} \Rightarrow$ the formula becomes trivial
(Hochstadt form)



Uncertain reading. The first word is provisionally read as “form.”

Original source details: GELFAND-P001091

$$\frac{T\,dz}{\sqrt{4(z-u_0)}}$$

$$\zeta(m; L_{u_0}) = \frac{2^{2m}|B_{2m}|}{(2m)!}$$

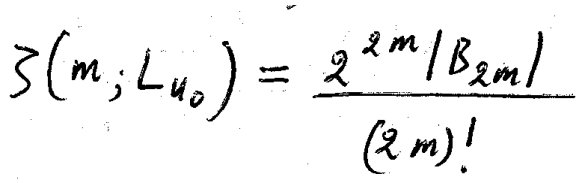


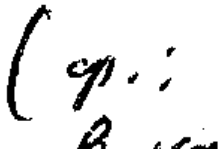
Diagram and typed description. The formula for a constant potential.

Original source details: GELFAND-P001091

1-gap

$$\omega\frac{z-\eta/\omega}{\sqrt{4z^3-g_2z+g_3}}$$

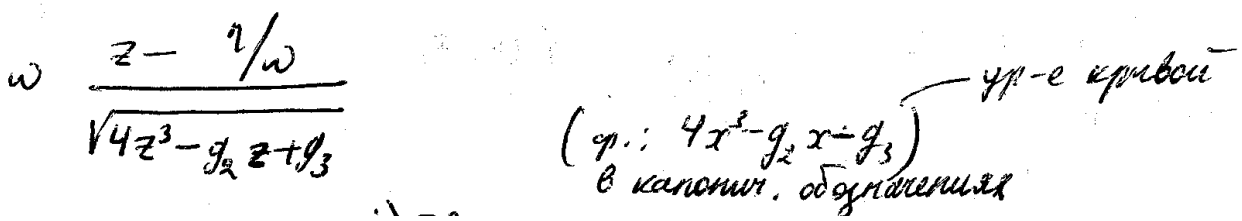
(f.: $4x^3 - g_2x - g_3$) [at the right: «equation of the curve».]



Uncertain reading. The mark before the colon appears to be the abbreviation “f.”.

Original source details: GELFAND-P001091

in canonical notation



Original source details: GELFAND-P001091

Diagram and typed description. The radicand has $+g_3$, and the formula beside it has $-g_3$.

$$\underline{u = 2\wp(x)}$$

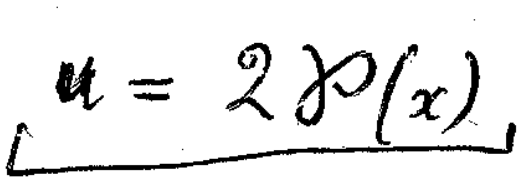


Diagram and typed description. The square-ended bracket below the potential.

Original source details: GELFAND-P001091

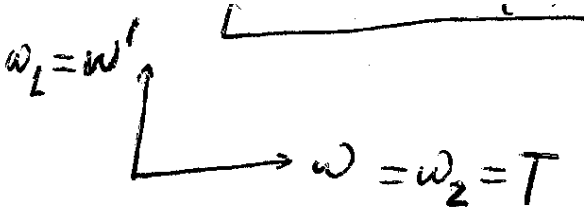


Diagram and typed description.

Original source details: GELFAND-P001091

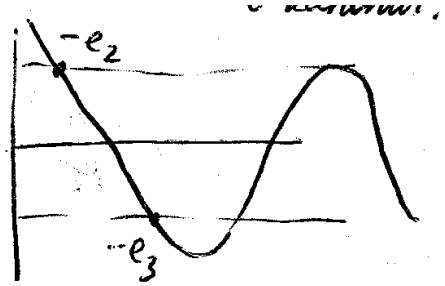
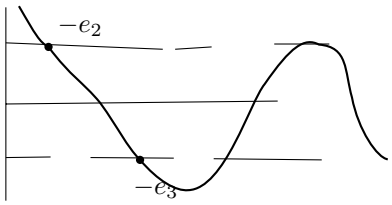


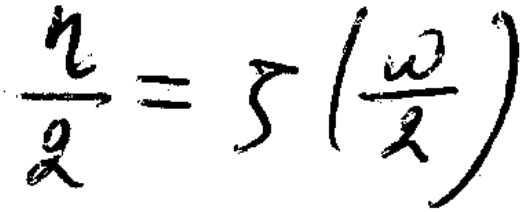
Diagram and typed description.



Original source details: GELFAND-P001091

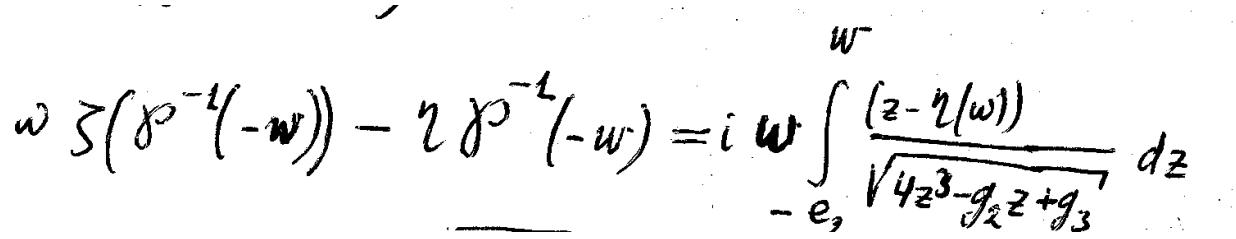
$$\frac{\eta}{2} = \zeta\left(\frac{\omega}{2}\right)$$

Diagram and typed description. The equality with half-period $\omega/2$.



Original source details: GELFAND-P001091

$$\omega\zeta\big(\wp^{-1}(-w)\big) - \eta\wp^{-1}(-w) = i\omega\int_{-e_2}^{w} \frac{\big(z-\eta(\omega)\big)}{\sqrt{4z^3-g_2z+g_3}}\,dz$$



Original source details: GELFAND-P001091

Uncertain reading. The numerator is provisionally read as $z-\eta(\omega)$. Two parentheses are visible after ω.

—·—

GELFAND-P001092

[source number: 17]

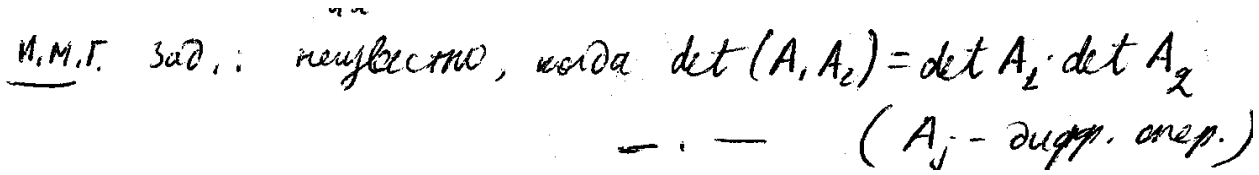
On the Riemann ζ function.

$$L_w \equiv -\frac{d^2}{dx^2} - w$$

I.M.G. problem: it is unknown when

$$\det(A_1A_2) = \det A_1 \cdot \det A_2$$

—·— $(A_j$ — differential op.)



Original source details: GELFAND-P001092

Uncertain reading. The first subscript on the right is read as 1. The handwriting also allows 2.

$$\zeta(s) = \sum \frac{1}{n^s}, \qquad \zeta(2k) = \frac{|B_{2k}|2^{2k}}{(2k)!}$$

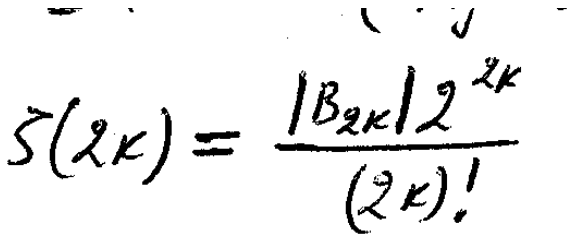


Diagram and typed description. The written value at the even argument $\zeta(2k)$.

Original source details: GELFAND-P001092

Apéry: $\sum \frac{1}{n^3}$ is irrational.

—·—

$$\sum n^2 = 0, \qquad \underbrace{\sum n^3 = -B_2}$$

a puzzle that it
[can be calculated?] and is $\neq 0$.

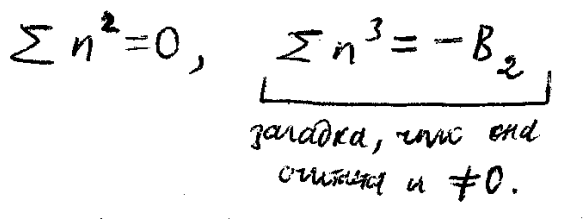


Diagram and typed description. The bracket and note under the sum of cubes.

Original source details: GELFAND-P001092

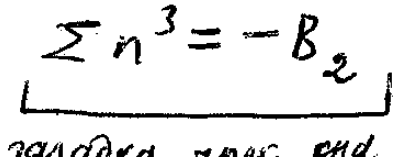
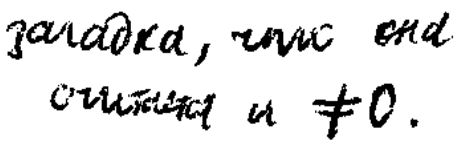


Diagram and typed description. Subscript 2 on B in the sum of cubes.

Original source details: GELFAND-P001092



Uncertain reading. The word before "and" is provisionally read as "can be calculated."

Original source details: GELFAND-P001092

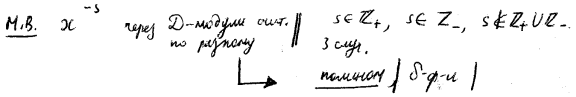


Diagram and typed description.

M.V. x^{-s}
calc. via D-modules
in different ways
$\Big\|$ $s \in \mathbb{Z}_+$, $s \in \mathbb{Z}_-$, $s \notin \mathbb{Z}_+ \cup \mathbb{Z}_-$.
3 cases

→ polynomial | δ-functions |

Original source details: GELFAND-P001092

ζ' corresponds to the logarithm.

$$\zeta'_{L_w}(-2) = -\log(-1)\cdot h_-(w)\,w^2 - \left(\frac{2}{T}\right)^4 \Big\{ T^2 w \underbrace{\mathrm{Cl}_3(T\sqrt{w})}_{\zeta(3)} - \\ - 3T\sqrt{w}\,[\text{cancelled}]\ \mathrm{Cl}_4(T\sqrt{w}) - 3\underbrace{\mathrm{Cl}_5(T\sqrt{w})}_{\zeta(-5)}\Big\}$$

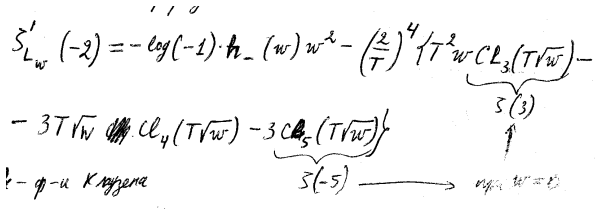
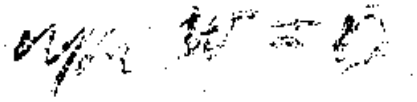


Diagram and typed description.

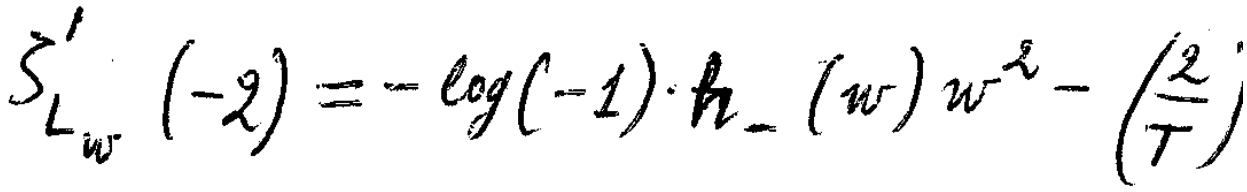
$$\begin{array}{c} \zeta(3) \\ \uparrow \\ \zeta(-5) \longrightarrow [\text{at } w = 0?] \end{array}$$

Original source details: GELFAND-P001092



Uncertain reading. The faint annotation provisionally reads “at $w = 0$.”

Original source details: GELFAND-P001092



Original source details: GELFAND-P001092

Diagram and typed description. The opening of the formula. The manuscript repeats the minus sign at the line break.

Uncertain reading. Cancelled marks before Cl_4. The start resembles Cl.

Original source details: GELFAND-P001092

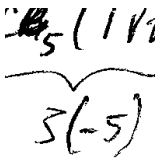


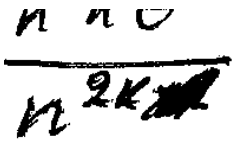
Diagram and typed description. The lower brace is labelled $\zeta(-5)$.

Original source details: GELFAND-P001092

Cl — Clausen functions

$$\mathrm{Cl}_{2k+1}(\theta) = \sum \frac{\cos n\theta}{n^{2k+1}}$$

$$\mathrm{Cl}_{2k}(\theta) = \sum \frac{\sin n\theta}{n^{2k}}$$



Uncertain reading. The cancelled marks after $2k$ in the exponent may read $+1$.

Original source details: GELFAND-P001092

GELFAND-P001093

21/III-83 Gelfand Sem.
S. P. Novikov

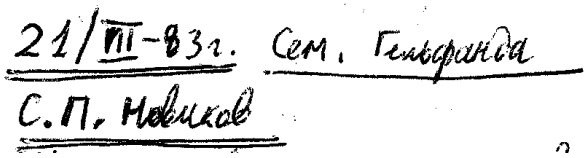


Diagram and typed description. Date, seminar name and speaker.

Original source details: GELFAND-P001093

What does the calculus of variations study?

multivalued functionals are important $\longrightarrow$ Dirac monopole (see Novikov's survey in Russian Mathematical Surveys)

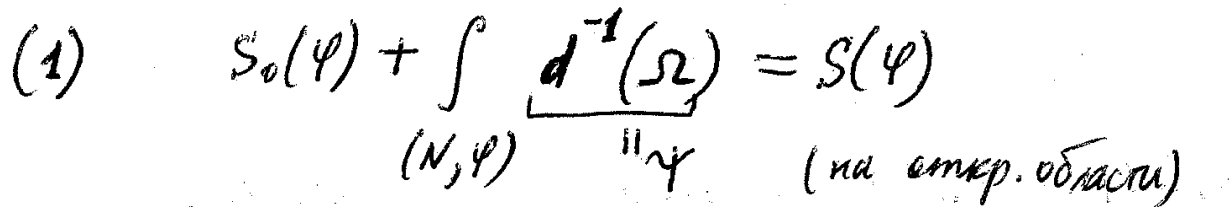
$$\begin{array}{ccc} x \in N^q & \varphi \in M^n & n > q \\ (x^1, \ldots, x^q) & (\varphi^1, \ldots, \varphi^n) & \\ & \text{coord.} & \end{array}$$

$$\varphi(x) : N \longrightarrow M$$

$S_0(\varphi)$ — regular functional

$$(1) \qquad S_0(\varphi) + \int_{(N,\varphi)} \underbrace{d^{-1}(\Omega)}_{\| \gamma} = S(\varphi)$$

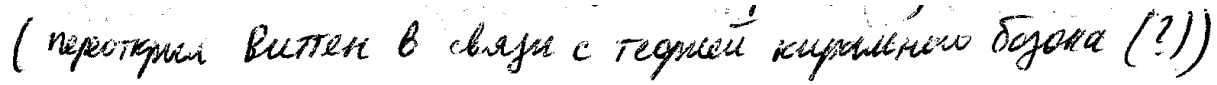
(on an open domain)



Original source details: GELFAND-P001093

Diagram and typed description. Formula (1), the square-ended bracket under $d^{-1}(\Omega)$ and the label γ.

(rediscovered by Witten in connection with the theory of the chiral boson (?))



Original source details: GELFAND-P001093

Diagram and typed description. The note on Witten with its parenthesized question mark.

Ω — $(q+1)$-form, $\quad d\Omega = 0$

Chiral fields. — this situation arises there.

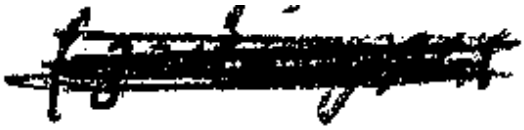
$\varphi \in \mathcal{F}$ — function space.

(2) Theorem. The following formula [1?] defines a form (well-defined)

Original source details: GELFAND-P001093

Uncertain reading. The mark after "formula" is overwritten. Possible readings are 1 and A.

δS — a 1-form on the function space (closed)



Original source details: GELFAND-P001093

Undeciphered passage. An illegible entry is cancelled below "function space."

[Volkov?]: $S^1 \to S^2$ (nontriv. case)

Original source details: GELFAND-P001093

Uncertain reading. The word before the colon is provisionally read as "Volkov."

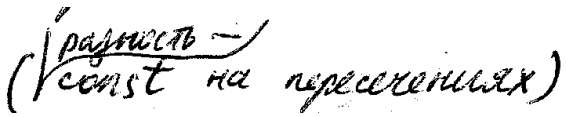
Ω — generalized external field (electromagnetic)

$$M^n = S^2, \qquad \int_{S^2} \Omega \neq 0 \qquad \Rightarrow \qquad \text{Dirac monopole}$$

Original source details: GELFAND-P001093

Uncertain reading. The cancelled notation above S^2 may read M^2.

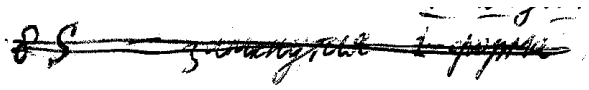
Multivalued function $\quad$ difference — (const on intersections)



Original source details: GELFAND-P001093

Diagram and typed description. The underlined insertion "difference —" above const.

~~δS closed 1-form~~



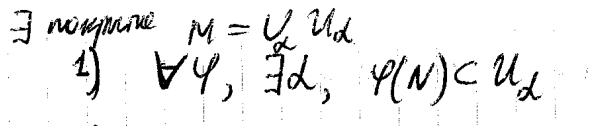
Original source details: GELFAND-P001093

Diagram and typed description. The cancelled line at the bottom of the frame.

GELFAND-P001094

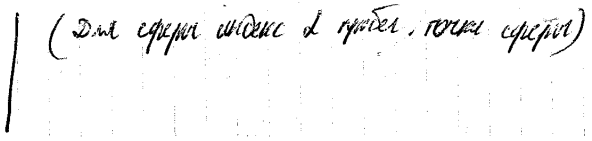
$\exists$ a cover $M = \bigcup_\alpha U_\alpha$.

$$\begin{aligned} &1) \quad \forall \varphi,\ \exists \alpha, \quad \varphi(N) \subset U_\alpha. \\ &2) \quad \Omega = d\psi_\alpha \quad \text{on } U_\alpha. \end{aligned}$$



Original source details: GELFAND-P001094

Diagram and typed description. The cover and the first condition on φ.



Original source details: GELFAND-P001094

Diagram and typed description.

> (For the sphere, the index α runs over its points)

$$W_\alpha \subset \mathcal{F}, \qquad \varphi(N) \subset U_\alpha.$$

$\{W_\alpha\}$ — a cover of $\mathcal{F}$.

$$S^\alpha = S_0 + \int_N \psi_\alpha.$$

Then $S^\alpha - S^\beta =$ const on $W_\alpha \cap W_\beta$, since

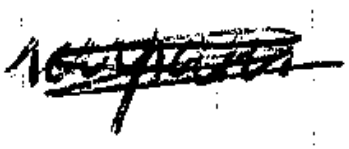
$$S^\alpha - S^\beta = \int_{(N,\varphi)} (\psi_\alpha - \psi_\beta)$$

depends only on the homotopy class, since $d(\psi_\alpha - \psi_\beta) = 0$.

Original source details: GELFAND-P001094

Uncertain reading. A short mark in the left margin may be the numeral 1.

Physical requirement:
$\exp(iS)$ — a single-valued function.
(Feynman amplitude)
Moral: [~~every?~~] the action need not be understood as single-valued.

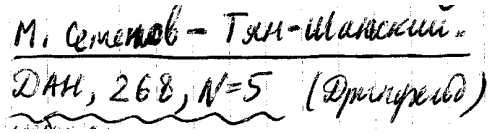


Original source details: GELFAND-P001094

Uncertain reading. The cancelled word may read “every.”

M. Semenov-Tian-Shansky.

DAN, 268, $N = 5$ (Drinfeld)



Original source details: GELFAND-P001094

Diagram and typed description. The speaker's name and the Drinfeld reference, with the issue number as written.

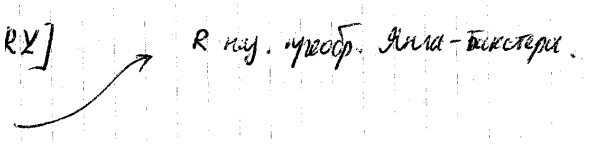
Lie bialgebra.

$\mathfrak{g}$ — Lie alg., $R \in \operatorname{End} \mathfrak{g}$.

$$[X, Y]_R = [RX, Y] + [X, RY]$$

also a Lie algebra

$\mathfrak{g}$, $\mathfrak{g}_R$.

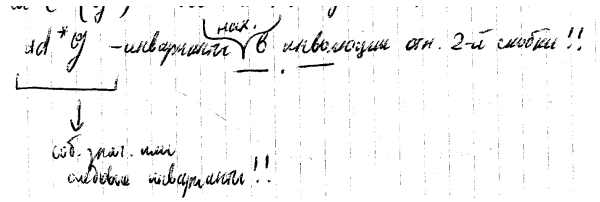


Original source details: GELFAND-P001094

Diagram and typed description.

R is called the
Yang–Baxter transformation.

On $C^\infty(\mathfrak{g}^*)$ — two Poisson brackets.



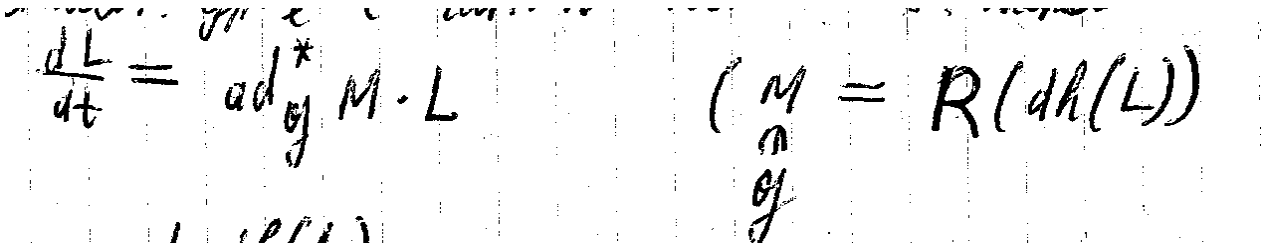
Original source details: GELFAND-P001094

Diagram and typed description.

$\underbrace{\text{ad}^* \mathfrak{g}}$-invariants $\overset{\text{are}}{\vee}$ in involution
w.r.t. the 2nd bracket!!
↓
eigenvalues or
trace invariants!!

Theorem. let h — $\text{ad}^* \mathfrak{g}$-invariant function on $\mathfrak{g}^*$.
then 1) the Hamiltonian eq. with Hamiltonian h has generalized Lax form

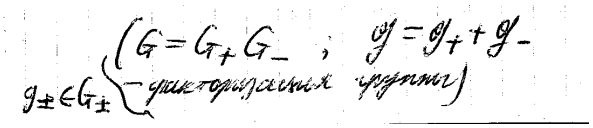
$$\frac{dL}{dt} = \text{ad}^*_{\mathfrak{g}} M \cdot L \qquad \left(\underset{\cup}{M}_{\mathfrak{g}} = R(dh(L))\right.$$



Original source details: GELFAND-P001094

Diagram and typed description. The vertical annotation $\in \mathfrak{g}$ below M.

$$2) \qquad \exp t\, dh(L) = \\ = g_+(t) \cdot g_-^{-1}(t)$$



Original source details: GELFAND-P001094

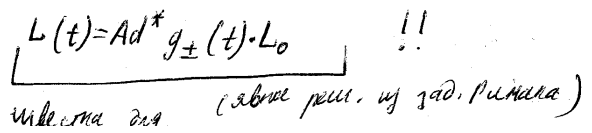
Diagram and typed description.

$g_\pm \in G_\pm$ { $(G = G_+G_-\ ;\ \mathfrak{g} = \mathfrak{g}_+ + \mathfrak{g}_-$ — group factorization)

GELFAND-P001095

[source number: 32]

Then



Original source details: GELFAND-P001095

Diagram and typed description.

$\underbrace{L(t) = \mathrm{Ad}^*\, g_\pm(t) \cdot L_0}$!!

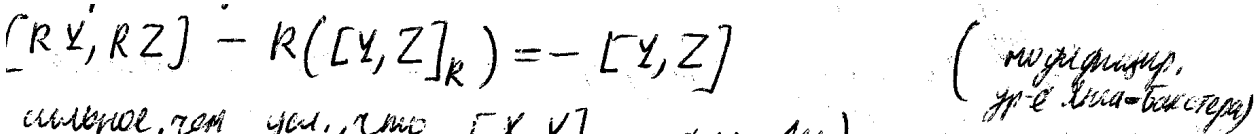
known for (explicit sol. from the Riemann problem)

$R = P_+ - P_-$.

Here: R-matrices satisfying the condition

$$[RY, RZ] - R([Y,Z]_R) = -[Y,Z]$$

(modified Yang–Baxter eq.)



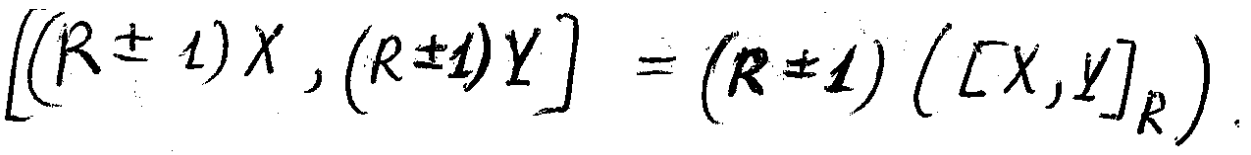
Original source details: GELFAND-P001095

Diagram and typed description. The equation and its right-hand note.

(stronger than the condition that $[X,Y]_R$ — is a Lie alg.).

— · —

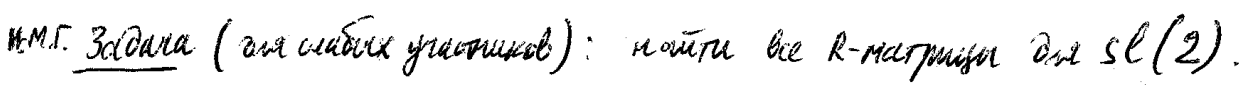
$$[(R \pm 1)X, (R \pm 1)Y] = (R \pm 1)([X,Y]_R).$$



Original source details: GELFAND-P001095

Diagram and typed description. The identity for $R \pm 1$.

I.M.G. Problem (for weaker participants): find all R-matrices for $sl(2)$.



Original source details: GELFAND-P001095

Diagram and typed description. The wording of the problem for $sl(2)$.

— · —

$$\mathfrak{g}_\pm = \operatorname{Im}(R \pm 1)$$

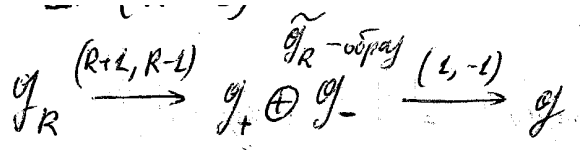


Diagram and typed description.

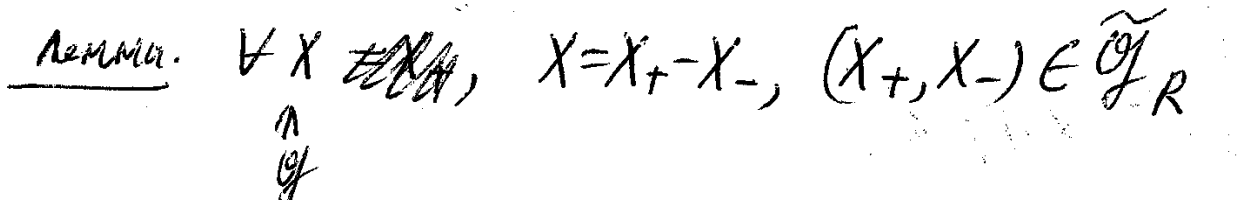
$\widetilde{\mathfrak{g}}_R$ — image

$$\mathfrak{g}_R \xrightarrow{(R+1,\,R-1)} \mathfrak{g}_+ \oplus \mathfrak{g}_- \xrightarrow{(1,\,-1)} \mathfrak{g}$$

Original source details: GELFAND-P001095

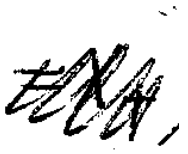
Lemma.

$$\forall\, \underset{\mathfrak{g}}{\cup} X \text{ [cancelled]}, \qquad X = X_+ - X_-, \qquad (X_+, X_-) \in \widetilde{\mathfrak{g}}_R.$$



Original source details: GELFAND-P001095

Diagram and typed description. The vertical annotation $\in \mathfrak{g}$ below X.



Uncertain reading. The cancelled notation after X may read $\in \mathfrak{g}$.

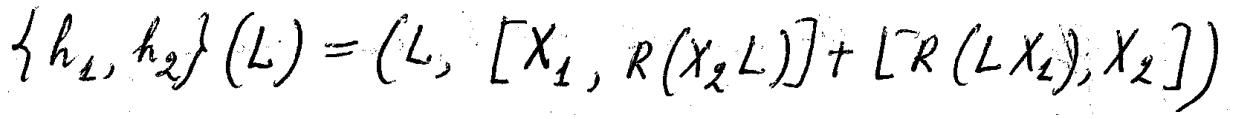
Original source details: GELFAND-P001095

— · —

2nd G.–D. bracket (Gelfand–Dickey)

$\mathfrak{g}$ — Lie alg. with assoc. multiplication and an (invar.) scalar product

$$\{h_1, h_2\}(L) = \big(L,\, [X_1, R(X_2 L)] + [R(L X_1), X_2]\big)$$



Original source details: GELFAND-P001095

Diagram and typed description. The order of the factors $X_2 L$ and $L X_1$ in the two terms.

$$X_i = \operatorname{grad} h_i$$

There are hierarchies of Hamiltonians in Lenard's sense.

— · —

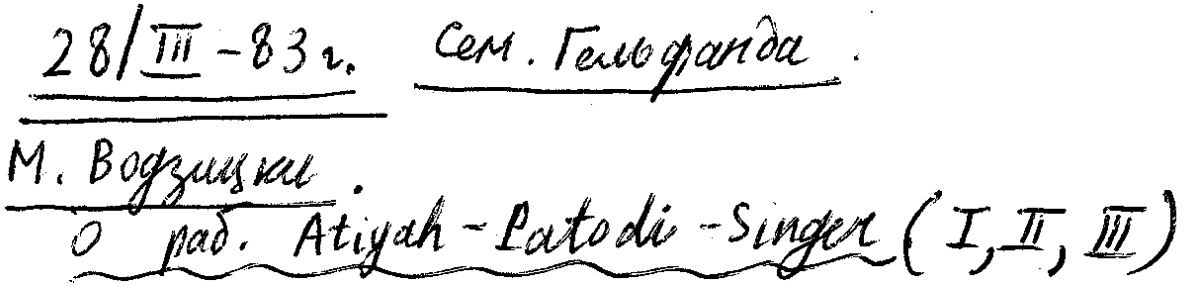
GELFAND-P001096

[source number: 39]

28/III-83 Gelfand Sem.

M. Wodzicki.

On the work of Atiyah–Patodi–Singer (I, II, III)



Original source details: GELFAND-P001096

Diagram and typed description. Date, speaker and reference to Atiyah, Patodi and Singer.

I. X [cancelled], $\partial X = Y$.

Undeciphered passage. The cancelled notation after the first X is illegible.

Original source details: GELFAND-P001096

D — 1st-order system on X (elliptic), near the boundary

$$D = \frac{\partial}{\partial u} + A, \qquad u \text{ — normal param.}$$

A — op. on Y.

A is self-adjoint.

Boundary cond. do not always exist (e.g., [inserted: for] the signature op. it is impossible). But nonlocal ones exist.

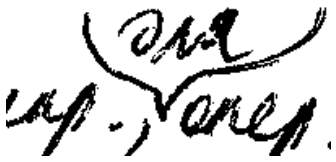


Diagram and typed description. The insertion “for” before “signature op.”

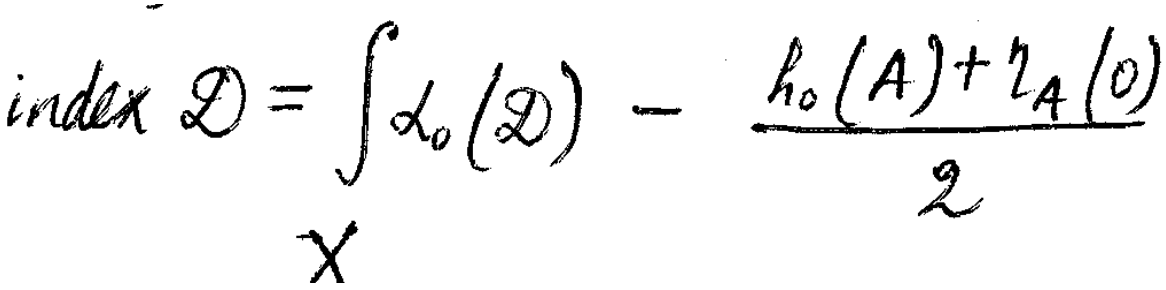
Original source details: GELFAND-P001096

$$(1) \qquad \begin{cases} Du = f \\ P(u|_Y) = 0 \end{cases}$$

(2) P — projection onto eigenvalues ≥ 0 of the operator A.

(3) Theorem. Problem (1) with P as in (2) has finite index and

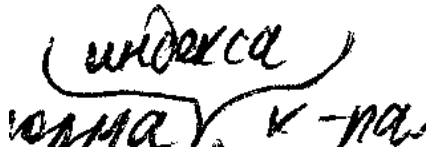
$$\operatorname{index} D = \int_X \alpha_0(D) - \frac{h_0(A) + \eta_A(0)}{2}$$



Original source details: GELFAND-P001096

Diagram and typed description. The index formula with its boundary term.

$\alpha_0(D)$ — differential form [inserted: of the index], written using the Riemannian metric, which has product form near the boundary.



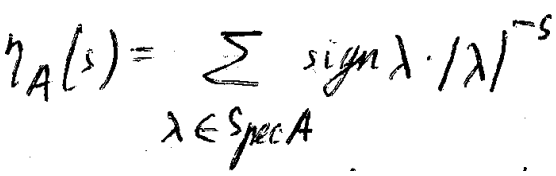
Original source details: GELFAND-P001096

Diagram and typed description. The insertion "of the index" above the line.

$$h_0(A) = \dim \ker A$$

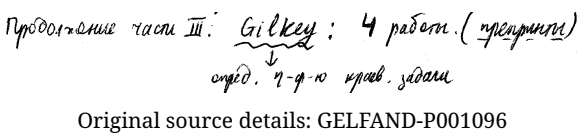
$\eta_A(0)$ — value at 0 of the analytic continuation of the function

$$\eta_A(s) = \sum_{\lambda \in \operatorname{Spec} A} \operatorname{sign} \lambda \cdot |\lambda|^{-s}$$



Original source details: GELFAND-P001096

Diagram and typed description. The summation index is written as $\lambda \in \operatorname{Spec} A$.



Original source details: GELFAND-P001096

Diagram and typed description.

Continuation of part III.

Gilkey: 4 papers. (preprints)

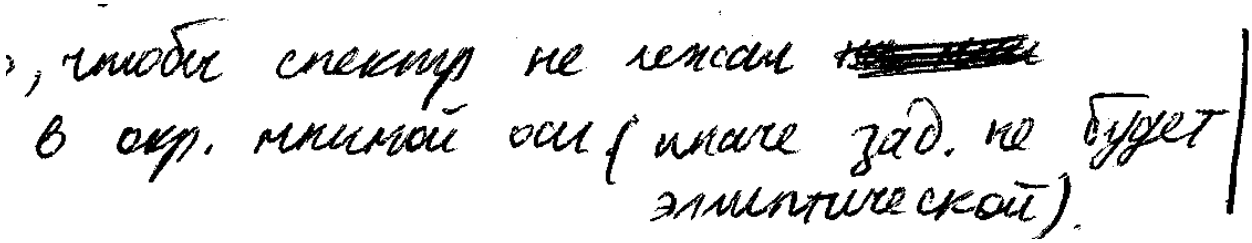
↓

defines the η-function
of the boundary problem

In place of self-adjointness one can require that the spectrum not lie [cancelled] near the imaginary axis (otherwise the problem will not be elliptic).

Undeciphered passage. The cancelled entry after “not lie” is illegible.

Original source details: GELFAND-P001096



Original source details: GELFAND-P001096

Diagram and typed description. The vertical rule to the right of the condition on the spectrum.

B. V. Fedosov.

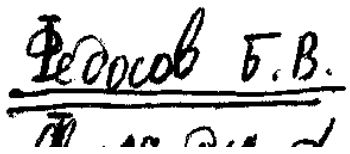


Diagram and typed description. Speaker named before the formula for α_0.

Original source details: GELFAND-P001096

Formula for α_0.

X — manifold (without boundary), SX — bundle of unit tangent vectors,

σ — symbol (function on SX). Using σ, construct a projector P defining the difference element (P — function on T^*X; $p(x,\xi)$)

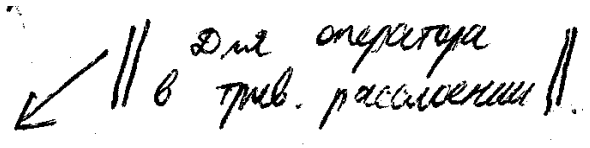


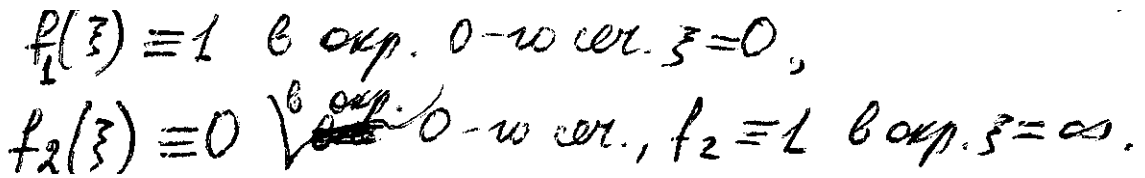
Diagram and typed description.

For the operator
in a trivial bundle

Original source details: GELFAND-P001096

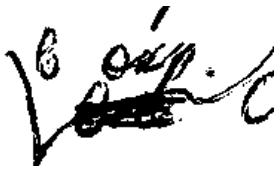
$f_1(\xi) \equiv 1$ near the zero section $\xi = 0$,

$f_2(\xi) \equiv 0$ [~~on?~~] [inserted: near] the zero section, $f_2 \equiv 1$ near $\xi = \infty$.



Original source details: GELFAND-P001096

Diagram and typed description. The conditions on f_1 and f_2 and the insertion “near.”



Uncertain reading. The cancelled word may read “on.”

Original source details: GELFAND-P001096

$f_1 + f_2 \equiv 1.$

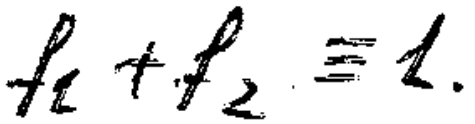


Diagram and typed description. The identity $f_1 + f_2 \equiv 1$ is written without exponents.

Original source details: GELFAND-P001096

$$p = \begin{pmatrix} f_1^2 & f_1 f_2 \sigma \\ f_1 f_2 \sigma^* & f_2^2 \end{pmatrix}$$

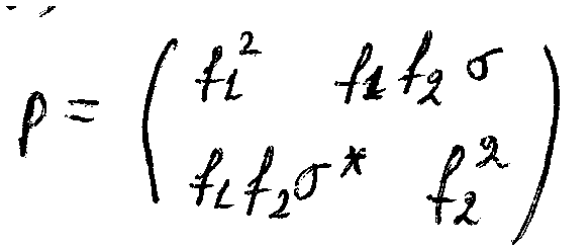


Diagram and typed description. The matrix p with squares on the diagonal.

Original source details: GELFAND-P001096

σ is extended homogeneously (degree 0)

GELFAND-P001097

Again: E_i — bundles over X;

$$A : \Gamma(E_1) \to \Gamma(E_2) \qquad \text{(differential operator)}$$

$p_1(x)$ $p_2(x)$ — projection functions defining the bundles (E_i — their images) (on X)

$\sigma = \sigma(x, \xi)$ — symbol of A (matrix function)

$\sigma p_1 \equiv p_2 \sigma$.

f_1, f_2 — cutoff functions (see above)

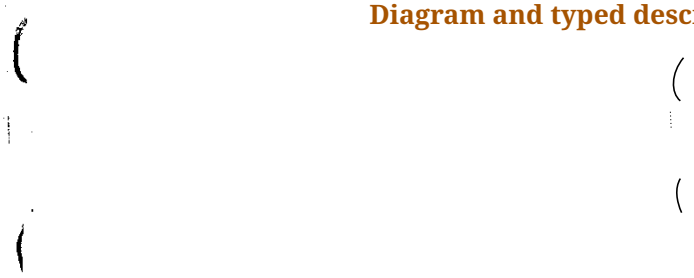


Diagram and typed description.

Original source details: GELFAND-P001097

$$P_1 = \begin{pmatrix} p_1 f_1^2 & \sigma f_1 f_2 \\ \sigma & \end{pmatrix}$$

— written explicitly and defined on all of T^*X.

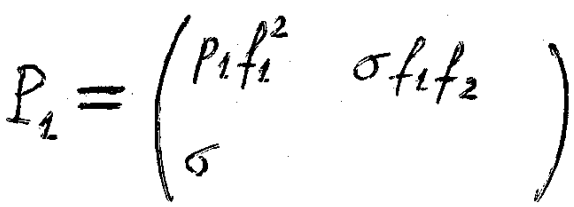


Diagram and typed description. The lower right cell is blank. Only σ is written in the lower left.

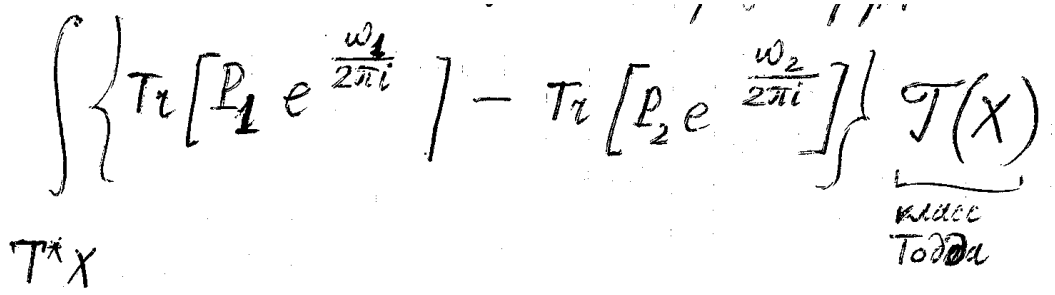
Original source details: GELFAND-P001097

$P_i = P_i(x, \xi)$

$P_1, P_2 \rightsquigarrow$ the curvature form is written from them

$$\omega_i = P_i\, dP_i\, dP_i \qquad \text{— matrix of 2-forms}$$

$$\int_{T^*X} \left\{ \mathrm{Tr}\left[P_1 e^{\frac{\omega_1}{2\pi i}}\right] - \mathrm{Tr}\left[P_2 e^{\frac{\omega_2}{2\pi i}}\right] \right\} \underbrace{\mathcal{T}(X)}_{\text{Todd class}}$$



Original source details: GELFAND-P001097

Diagram and typed description. The integral and the label below $\mathcal{T}(X)$.

$$a = f_1\sigma, \qquad \tau = f_2\sigma^{-1}$$

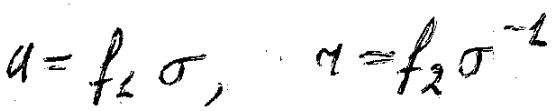
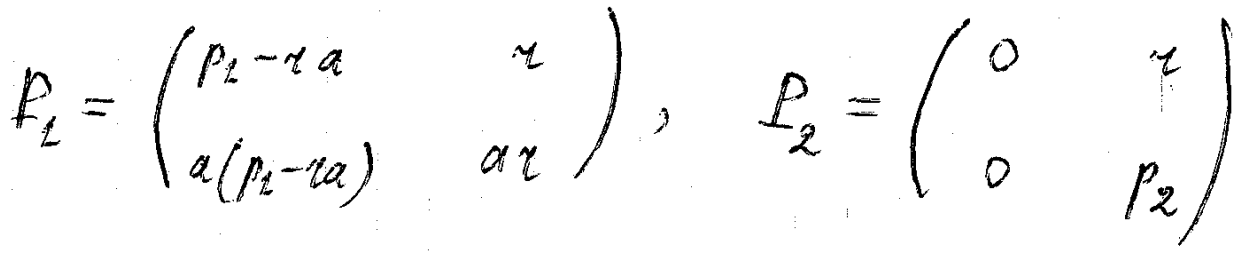


Diagram and typed description. The subscript on f in the definition of a reads 1.

Original source details: GELFAND-P001097

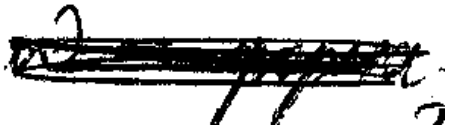
$$P_1 = \begin{pmatrix} p_1 - \tau a & \tau \\ a(p_1 - \tau a) & a\tau \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 0 & \tau \\ 0 & p_2 \end{pmatrix}$$



Original source details: GELFAND-P001097

Diagram and typed description. The full entries of matrices P_1 and P_2.

Todd class:

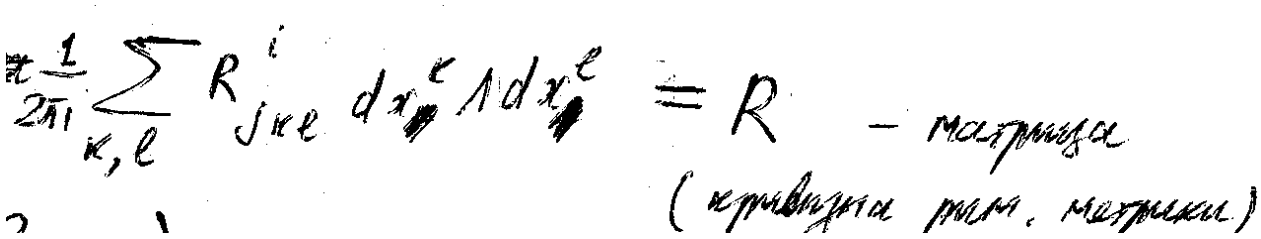


Undeciphered passage. A fragment before the coefficient is cancelled. It may read “2-form.”

Original source details: GELFAND-P001097

$$\frac{1}{2\pi i}\sum_{k,\ell} R^i_{jk\ell}\,dx^k\wedge dx^\ell = R$$

— matrix (curvature of the Riemannian metric)



Original source details: GELFAND-P001097

Diagram and typed description. The curvature matrix with the superscripts written above the differentials.

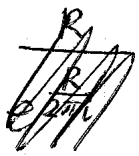
Original source details: GELFAND-P001097

Uncertain reading. An illegible subscript is cancelled beneath the first x. Its superscript is provisionally read as k.

Original source details: GELFAND-P001097

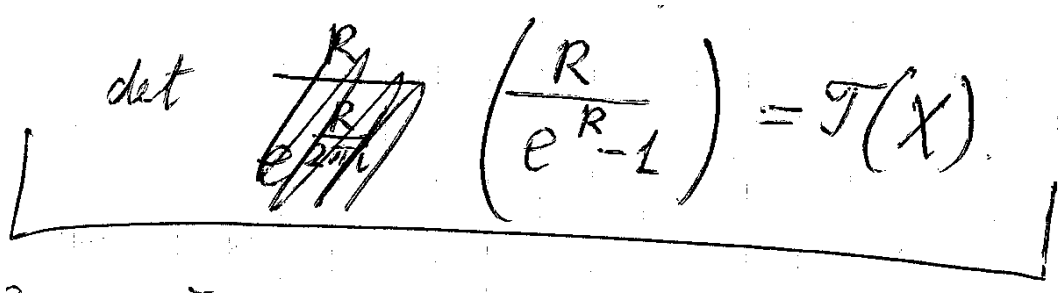
Uncertain reading. An illegible subscript is cancelled beneath the second x.

$$\det \frac{R/}{e^{\frac{R}{2\pi i}}}\left(\frac{R}{e^R-1}\right) = \mathcal{T}(X).$$



Original source details: GELFAND-P001097

Diagram and typed description. The cancelled fraction after det.



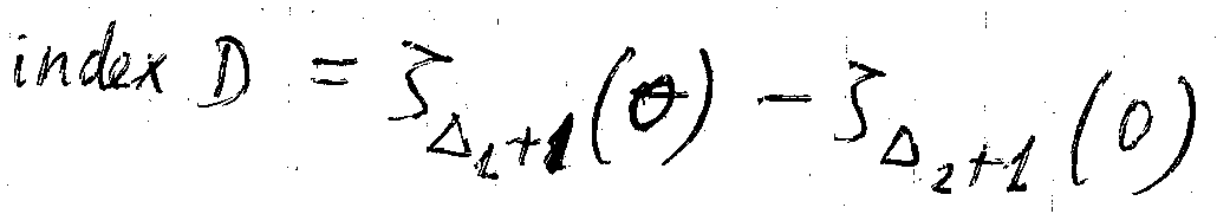
Original source details: GELFAND-P001097

Diagram and typed description. The square-ended bracket under the Todd class formula.

Wodzicki:

$$DD^* = \Delta_2, \qquad D^*D = \Delta_1.$$

$$\operatorname{index} D = \zeta_{\Delta_1+1}(\theta) - \zeta_{\Delta_2+1}(0)$$

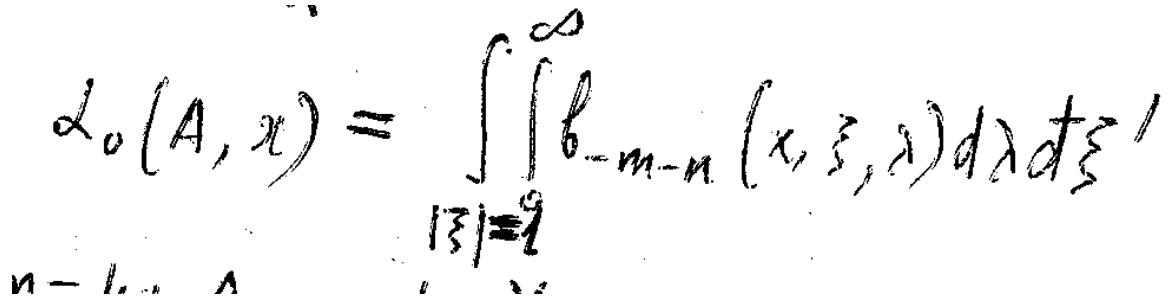


Original source details: GELFAND-P001097

Uncertain reading. The first argument resembles θ or a zero crossed by a horizontal stroke.

$$\zeta_A(0) = \int_X \alpha_0(A, x)$$

$$\alpha_0(A, x) = \int_{|\xi|=1} \int_0^\infty b_{-m-n}(x, \xi, \lambda)\, d\lambda\, d\xi'$$



Original source details: GELFAND-P001097

Diagram and typed description. The limits of integration and the subscript $-m-n$ on b.

$m = \deg A, \qquad n = \dim X$

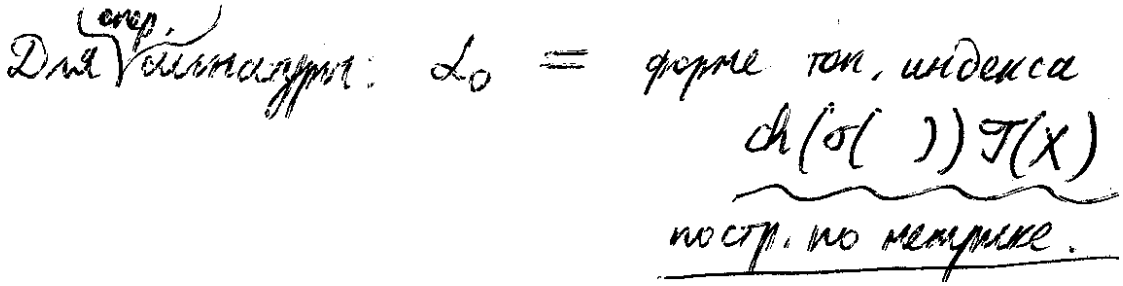
GELFAND-P001098

[source number: 40]

For the signature [inserted: op.]: $\alpha_0 =$ the form for the top. index

$$\underbrace{\operatorname{ch}(\sigma(\,))\mathcal{T}(X)}$$

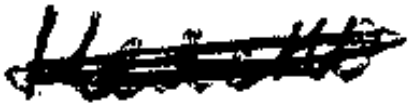
constructed from the metric.



Original source details: GELFAND-P001098

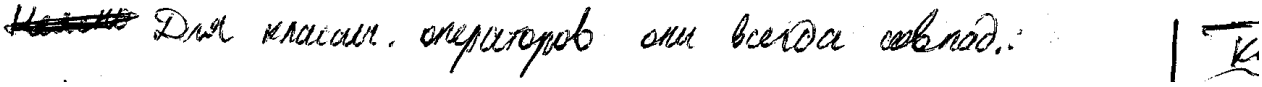
Diagram and typed description. The insertion “op.” and the empty parentheses after σ.

[cancelled] For classical operators they always coincide:



Undeciphered passage. A fragment is cancelled before “For.” Its beginning resembles the Russian «Не».

Original source details: GELFAND-P001098



Original source details: GELFAND-P001098

Diagram and typed description. The line on coincidence for classical operators.

a) signature op.

b) Dirac op.

There is a G-analogue, and there too the local formulas coincide.

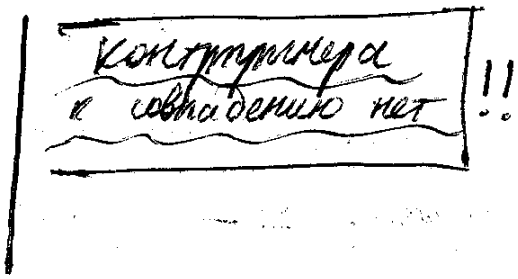


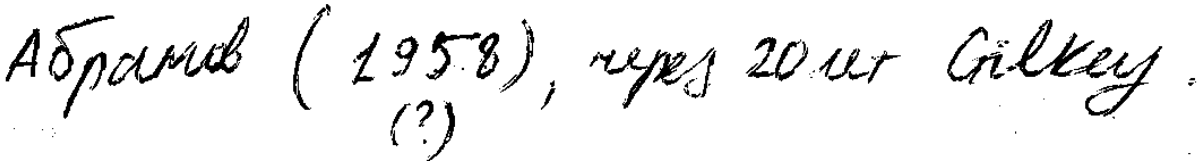
Diagram and typed description.

There is no counterexample to the formulas' agreement !!

Original source details: GELFAND-P001098

I.M.G. problems: construct formal cohomology for the metric and the symbol

Only for the metric: Abramov (1958), 20 years later Gilkey.
(?)



Original source details: GELFAND-P001098

Diagram and typed description. A question mark is written below the date 1958.

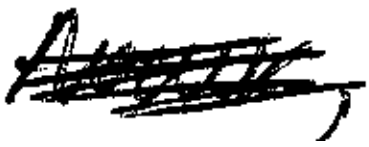
Are there other classes besides the index?

Perhaps this reduces to Gelfand–Fuks cohomology.

Or, the problem: construct local top. invariants.

(see Gabrielov, ~~Losik?~~, Gelfand, Losik

Gabrielov, Gelfand, Fuks — «Topology»).



Uncertain reading. The cancelled name is provisionally read as “Losik.”

Original source details: GELFAND-P001098

(см. Габриэлов, ~~Дынкин~~, Гельфанд, Лосик
Габриэлов, Гельфанд, Фукс — "Топология"

Original source details: GELFAND-P001098

Diagram and typed description. The two groups of names and the reference to the journal Topology.

In the jet space without a metric.

метрика → переход к лок. ф-лам;
↓ геод. сист. координат.

Original source details: GELFAND-P001098

Diagram and typed description.

metric ⟶ passage to local formulas.
↓
geod. coordinate system.

— · —

Wodzicki

A. P. S. I. Special case:

$$\operatorname{sign} X = \int_X L(x) - \eta(0), \qquad \eta \text{ — for } \pm(d^* - d)$$

sign X = ∫_X L(x) − η(0), η − для ±(d*−d)

Original source details: GELFAND-P001098

Diagram and typed description. The signature formula and the operator written at its right.

II. — preparatory part and discussion of the functions:

$$\eta(s, g) = \sum \operatorname{sign} \lambda \cdot \operatorname{Tr}(g|_{E_\lambda})|\lambda|^{-s}$$

η(s, g) = Σ sign λ · Tr(g|_{E_λ}) |λ|^{−s}
...ино

Original source details: GELFAND-P001098

Diagram and typed description. The trace of the restriction of g to E_λ.

III. [inserted: Prove that] $\eta(0)$ is finite for odd n. A problem like that of a higher index is posed.

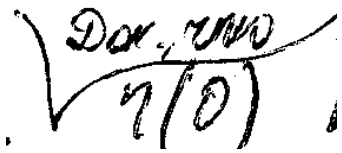


Diagram and typed description. The insertion "Prove that" before $\eta(0)$.

Original source details: GELFAND-P001098

The class is constructed

$$[\sigma_A] \in K^1(T^*X)$$

and from the repr. $\alpha : \pi_1 X \to U(n)$

one constructs

$$[\alpha] \in K^{-1}(T^*X, \mathbb{R}/\mathbb{Z})$$

$$K^0(T^*X, \mathbb{R}/\mathbb{Z}) \to \mathbb{R}/\mathbb{Z}$$

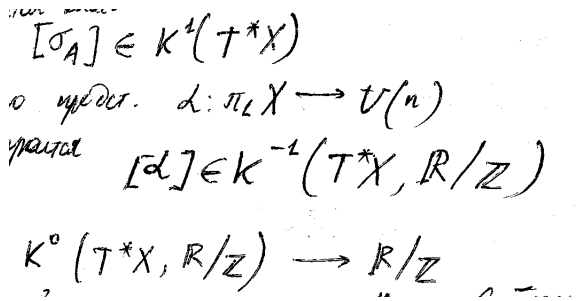


Diagram and typed description. The three K-theory degrees 1, -1 and 0.

Original source details: GELFAND-P001098

and so on.

Some Chern–Simons classes are obtained here.

GELFAND-P001099

Wilson. (integrable systems)

(joint work with G. Segal from Oxford)

[beneath G. Segal: (Graeme)]

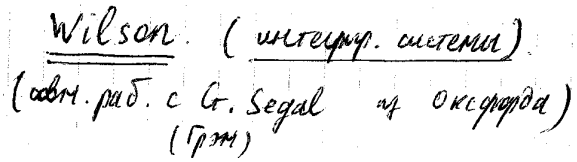


Diagram and typed description. Wilson's name and the note on joint work with G. Segal.

Original source details: GELFAND-P001099

An attempt to understand the work.

Miva; Jimbo, Kashiwara, Date [Sato.]

Original source details: GELFAND-P001099

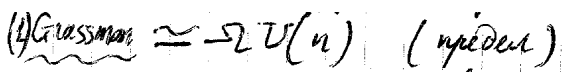
Diagram and typed description. The first name is written "Miva." Segal and Wilson's paper cites the author as Miwa.

Connection with an ∞-dimensional Grassmannian.

$$(1)\quad \underline{\text{Grassman}} \simeq \Omega U(n) \qquad \text{(limit)}$$

space of loops in the unitary group

$$\varinjlim_{n\to\infty} \Omega U(n)$$

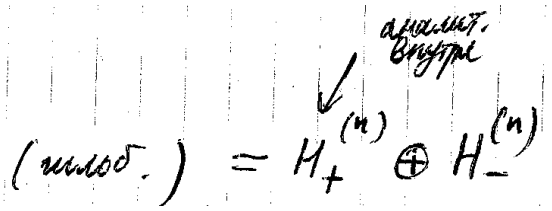


Original source details: GELFAND-P001099

Diagram and typed description. The spelling “Grassman” and the notation $\Omega U(n)$.

Explanation of the isomorphism.

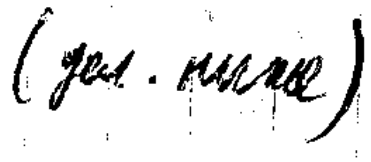
$$H^{(n)} = L^2(S^1;\mathbb{C}^n) \quad \text{(Hilbert)} = \overset{\text{analytic inside}}{\underset{\swarrow}{H_+^{(n)}}} \oplus H_-^{(n)}$$



Original source details: GELFAND-P001099

Diagram and typed description. The arrow from “analytic inside” to $H_+^{(n)}$.

$\gamma : S^1 \to GL(n)$ or $U(n)$ ([group of loops?])



Original source details: GELFAND-P001099

Uncertain reading. The annotation may read “group of loops.”

their group L (continuous)

Original source details: GELFAND-P001099

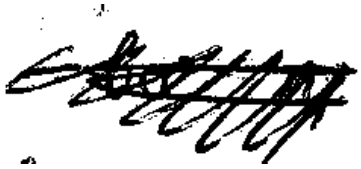
Diagram and typed description. The bracket with raised ends below “continuous.”

L acts on $H^{(n)}$

$$W_\gamma = \gamma(H_+^{(n)}) \qquad \text{— subspace}$$

[at the right] All such W_γ form the Grassmannian $Gr^{(n)}$

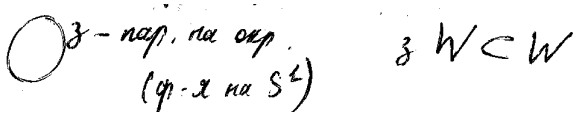
$$W \in Gr^{(n)} = \{W_\gamma \mid \gamma \in L\} \quad \text{[cancelled]}$$



Original source details: GELFAND-P001099

Undeciphered passage. The cancelled fragment after the set notation is illegible.

— homogeneous space with respect to the group of such loops L
Similarly for the unitary group.



Original source details: GELFAND-P001099

Diagram and typed description.

z — param. on the circle (function on S^1)

$zW \subset W$

Diagram and typed description.

Original source details: GELFAND-P001099

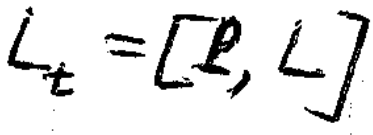
An isolated crossed mark at the right.

W has 2 properties

$$(i) \quad \text{pr} : W \to H_+^{(n)} \qquad \text{— Fredholm op.} \quad \text{(Toeplitz)}$$

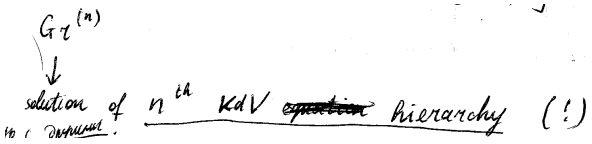
$$(ii) \quad \text{pr} : W \to H_-^{(n)} \qquad \text{compact operator.}$$

Open problem: characterize all loops satisfying $(i), (ii)$.
Connection with the KdV and G.–D. equations.
$Gr^{(n)}$ is related to $L = \partial^n + \ldots$; $\quad L_t = [\ell, L]$.



Original source details: GELFAND-P001099

Uncertain reading. The first entry of the commutator may be ℓ. The handwriting also allows P.



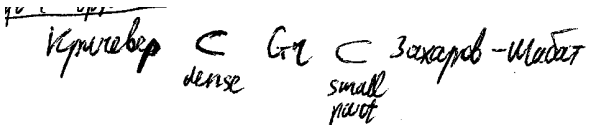
Original source details: GELFAND-P001099

Diagram and typed description.

$Gr^{(n)}$
↓
solution of n^{th} KdV ~~equation~~ hierarchy (!)

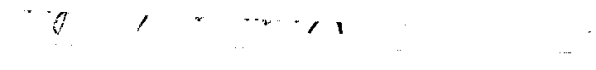
Connection with others.

Krichever $\underset{\text{dense}}{\subset}$ Gr $\underset{\text{small part}}{\subset}$ Zakharov–Shabat



Original source details: GELFAND-P001099

Diagram and typed description. The inclusions labelled “dense” and “small part.”



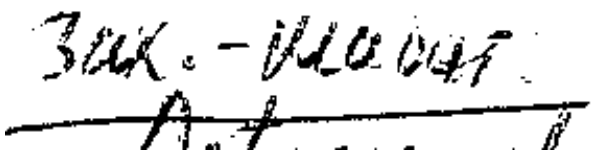
Original source details: GELFAND-P001099

Undeciphered passage. Only scattered strokes remain from the next line at the lower edge.

GELFAND-P001100

[source number: 41]

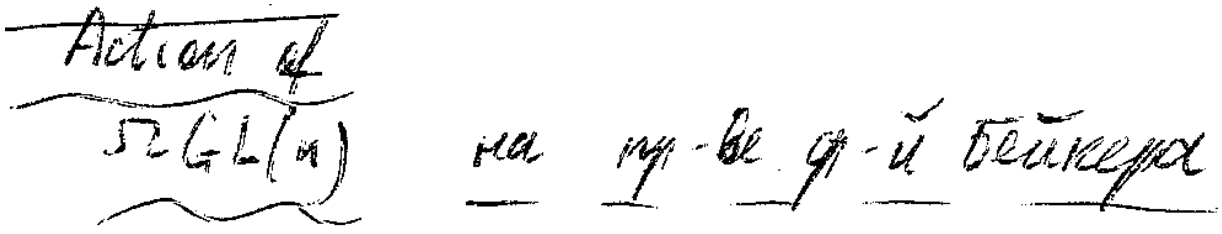
Zak.–Shabat.



Original source details: GELFAND-P001100

Uncertain reading. The first abbreviation may read “Zak.”

Action of
$\Omega GL(n)$ on the space of Baker functions



Original source details: GELFAND-P001100

Diagram and typed description. The heading “Action of” and the group notation.

Gr — orbit of the Z.–Sh. action (dressing of triv. solutions)

List of names:

(1) Gr [~~KdV?~~] and KdV

Original source details: GELFAND-P001100

Uncertain reading. The cancelled entry after Gr may read "KdV."

(2) $MKdV$ — modified KdV

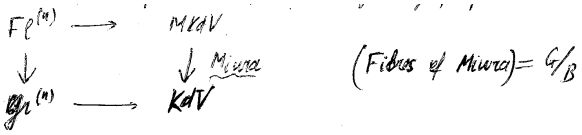
in Drinfeld and Sokolov: replacement of $GL(n)$ by a more general group.

Original source details: GELFAND-P001100

Diagram and typed description. The first divider.

$MKdV$ is associated with the full set of flags rather than Gr.

(quotient of all matrices by triangular matrices that extend analytically downwards)



Original source details: GELFAND-P001100

Diagram and typed description.

$$\begin{array}{ccc} Fl^{(n)} & \longrightarrow & MKdV \\ \downarrow & & \downarrow \text{Miura} \\ Gr^{(n)} & \longrightarrow & KdV \end{array}$$

The first letter at lower left is overwritten. The visible reading is $Gr^{(n)}$.

[at the right] (Fibres of Miura) $= G/B$

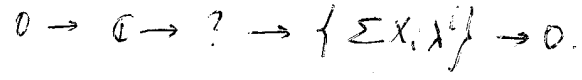
(3) τ-function (a Japanese invention)

Determinant function — on $Gr^{(n)}$.

In Kac–Moody alg., loops appear: $\sum_i X_i \lambda^i$

The extension is written explicitly:

$$0 \to \mathbb{C} \to ? \to \left\{ \sum X_i \lambda^i \right\} \to 0.$$



Original source details: GELFAND-P001100

Diagram and typed description. The exact sequence with a question mark in the middle term.

(4) [~~KdV?~~] Krichever $\subset Gr^{(n)}$

(rational curve)

Uncertain reading. The cancelled entry before "Krichever" may read "KdV."

Original source details: GELFAND-P001100

Diagram and typed description. The second divider.

Original source details: GELFAND-P001100

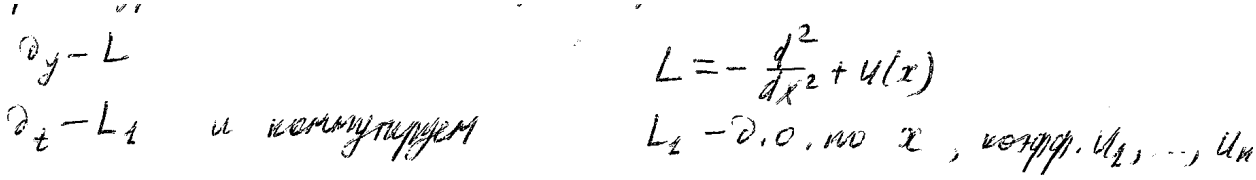
Krichever.

KP describes deformations of arbitrary rings (beyond the special form in KdV).

The hierarchy of KP equations can be understood from the τ-function.

$$\partial_y - L \qquad L = -\frac{d^2}{dx^2} + u(x)$$

$\partial_t - L_1$ and they commute $\qquad L_1$ — differential op. in x, coeffs. $u_1, \ldots, u_n$.



Original source details: GELFAND-P001100

Diagram and typed description. The two commuting operators and the notes at the right.

Property of the τ-function: the commut. system — a system of equations for one scalar function.

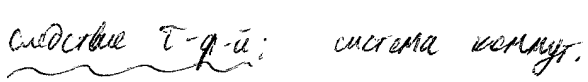


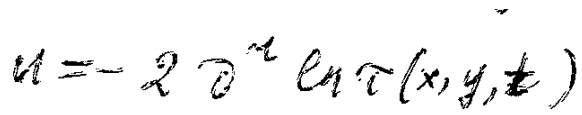
Diagram and typed description. The wavy underlining below the heading.

Original source details: GELFAND-P001100

Diagram and typed description. The abbreviation in "the commut. system."

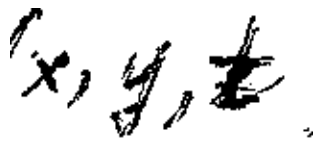
Original source details: GELFAND-P001100

$$u = -2\partial^2 \ln \tau(x, y, t)$$



Uncertain reading. The exponent on ∂ is read as 2. Additional strokes are visible beside the digit.

Original source details: GELFAND-P001100

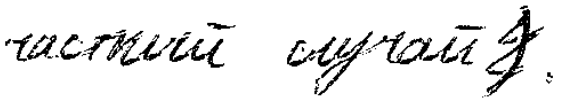


Uncertain reading. The third argument is overwritten as t. The earlier mark is illegible.

Original source details: GELFAND-P001100

τ-function is a function of infinitely many variables.

θ-function arises as a special case of ξ.



Uncertain reading. The final mark resembles ξ. The context allows τ.

Original source details: GELFAND-P001100

GELFAND-P001101

4/IV-83 Gelfand Sem.

Wilson.

(on his work with G. Segal)

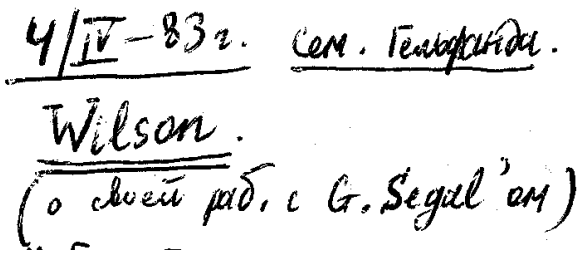


Diagram and typed description. The date, Wilson's name and the note on G. Segal.

Original source details: GELFAND-P001101

I.M.G. τ-function

$$\tau_g(W) \qquad \begin{array}{ll} W \subset H_+ \oplus H_- & \\ W \to H_+ & \text{Fred.} \\ W \to H_- & \text{— comp.} \end{array}$$

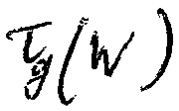


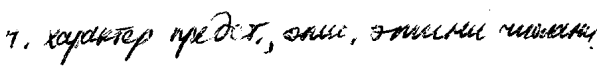
Diagram and typed description. The notation $\tau_g(W)$ before the definition.

Original source details: GELFAND-P001101

Sato: if it depends on $-n_1, -n_2, \ldots, -n_k$, then one obtains the character of the repr. [described?] by these numbers.

Uncertain reading. The abbreviation after "repr." may mean "described."

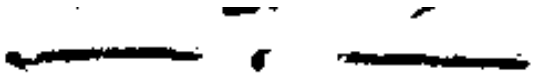
Original source details: GELFAND-P001101



Original source details: GELFAND-P001101

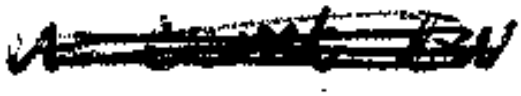
Diagram and typed description. The last part of the remark attributed to Sato.

—·—



Original source details: GELFAND-P001101

Diagram and typed description. Two horizontal strokes and a dot below the sequence of numbers.



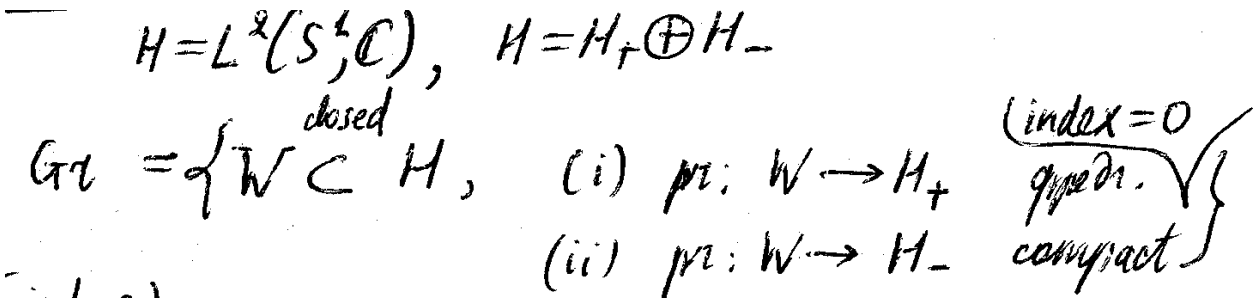
Original source details: GELFAND-P001101

Undeciphered passage. The cancelled entry before the second “Wilson.” is illegible.

Wilson.

Def. $H = L^2(S^1, \mathbb{C}), \quad H = H_+ \oplus H_-$

$$Gr = \left\{ \begin{matrix} \text{closed} \\ W \end{matrix} \subset H, \quad \begin{matrix} (i) & \text{pr} : W \to H_+ & \text{Fred.} \\ (ii) & \text{pr} : W \to H_- & \text{compact} \end{matrix} \right\}$$

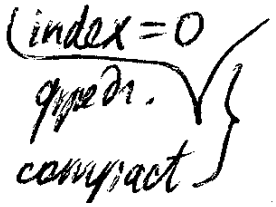


Original source details: GELFAND-P001101

Diagram and typed description. The definition of Gr and the note above condition (i).

Diagram and typed description.

$$\left\{ \begin{matrix} (\text{index} = 0 \\ \text{Fred.} \\ \text{compact} \end{matrix} \right.$$



Original source details: GELFAND-P001101

(without the requirement ind $= 0$)

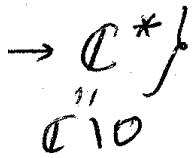
Gr is disconnected. Its connected components are indexed by the index of op. (i)

~~Another~~

Original source details: GELFAND-P001101

Diagram and typed description. The cancelled word “Another.”

$$\Gamma = \{\text{analytic functions } S^1 \to \underset{\mathbb{C}\backslash 0}{\mathbb{C}^*}\}$$



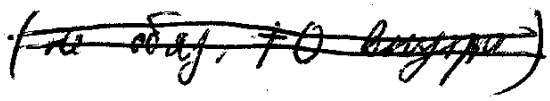
Original source details: GELFAND-P001101

Diagram and typed description. The annotation $\mathbb{C}\backslash 0$ beneath $\mathbb{C}^*$.

$\Gamma_+ \in \Gamma$ [~~KdV?~~] extends analytically inside ~~(not necessarily, $\neq 0$ inside)~~

Original source details: GELFAND-P001101

Uncertain reading. The cancelled entry after Γ may read “KdV.”



Original source details: GELFAND-P001101

Diagram and typed description. The cancelled parenthesis after “inside.”

Γ_+ acts on Gr.

$W \in Gr$

Def.

$$\tau_W : \Gamma_+ \longrightarrow \mathbb{C} \qquad \text{(analytic function!)}$$

$$\tau_W(g) = \det(\text{pr} : g^{-1}W \longrightarrow H_+)$$

Original source details: GELFAND-P001101

Diagram and typed description. The determinant formula with $g^{-1}W$.

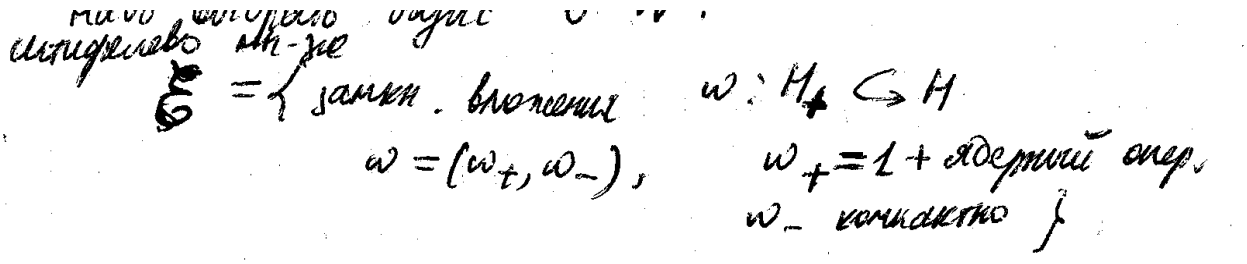
One must choose a basis in W.

$$\xi = \left\{ \begin{array}{ll} \text{closed embeddings } \omega : H_+ \hookrightarrow H & \\ \omega = (\omega_+, \omega_-), & \omega_+ = 1 + \text{nuclear op.} \\ & \omega_- \quad \text{compact} \end{array} \right\}$$

[above at the left: Stiefel set]

Uncertain reading. The mark before the definition of the Stiefel set may be ξ.

Original source details: GELFAND-P001101



Original source details: GELFAND-P001101

Diagram and typed description. The definition using an embedding and its two components.

GELFAND-P001102

[source number: 48]

One can prove that $\forall$ point of Gr [arises] in this way.

Uncertain reading. The abbreviation is provisionally read as the Russian word for "arises."

Original source details: GELFAND-P001102

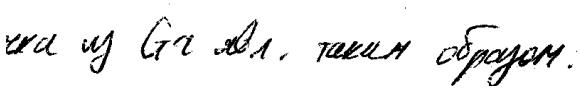


Diagram and typed description. The opening line and its abbreviations.

Original source details: GELFAND-P001102

One can speak of bases: $\omega_i = \omega(z^i)$.

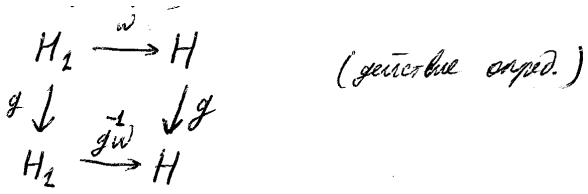


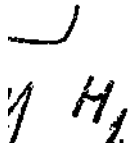
Diagram and typed description.

$$\begin{array}{ccc} H_1 & \xrightarrow{\omega} & H \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ H_1 & \xrightarrow{g^{-1}\omega} & H \end{array}$$

(action defined)

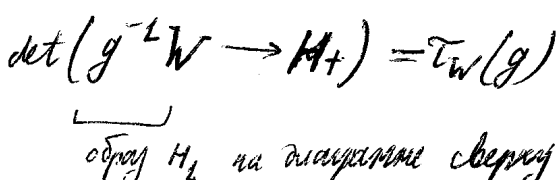
Original source details: GELFAND-P001102

$$\det(\underbrace{g^{-1}W}_{\substack{\text{image of } H_1 \\ \text{in the diagram above}}} \longrightarrow H_+) = \tau_W(g).$$



Original source details: GELFAND-P001102

Uncertain reading. The subscript in the note reads as 1, as in the left column of the diagram.



Original source details: GELFAND-P001102

Diagram and typed description. The determinant and the note below the square-ended bracket.

$g^{-1} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ — notation relative to the decomposition $H_+ \oplus H_-$.

Original source details: GELFAND-P001102

Diagram and typed description. [at the upper right of H_-:]

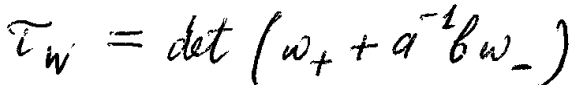
The τ-function is defined [crossed out: ~~for~~] only on such elements; in fact on the automorphism group of Gr.

Original source details: GELFAND-P001102

Uncertain reading. The word under the cancellation appears to mean “for.”

Then:

$$\tau_W = \det(\omega_+ + a^{-1}b\omega_-).$$



Original source details: GELFAND-P001102

Diagram and typed description. The formula for τ_W after the block decomposition.

Def. W is transverse $(tr\ H_-)$ if pr : $W \to H_+$ is an isomorphism or

$$W \cap H_- = \{0\}.$$

Such a W is the graph of $\omega_- : H_+ \to H_-$ $\Rightarrow$ canonical choice $\Rightarrow$

$$\tau_W(1) = 1 \qquad \text{(otherwise } 0\text{)}$$

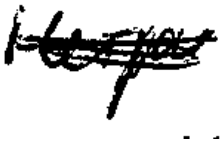
Remark. $\tau_W(g) = 0 \Leftrightarrow g^{-1}W$ is not transverse.

Lemma. $\forall W;\quad \tau_W \not\equiv 0.$

Diagram and typed description. The three crossed horizontal bars in the lemma.

Original source details: GELFAND-P001102

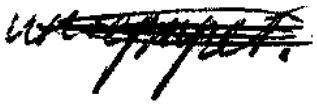
[crossed out: ~~Nontr.~~]



Uncertain reading. The crossed-out opening is provisionally read as an abbreviation for "nontransverse."

Original source details: GELFAND-P001102

Nontransverse subspaces W can be [crossed out:] — an analogue of the θ-divisor on the Jacobian variety of the curve.



Undeciphered passage. The cancelled word after "can be" is difficult to read. The beginning resembles Cyrillic "и" and two strokes descend below the line.

Original source details: GELFAND-P001102

More precisely: $\tau_W(e^{xz}) \not\equiv 0.$

Diagram and typed description. The formula for e^{xz} with the same crossed sign.

Original source details: GELFAND-P001102

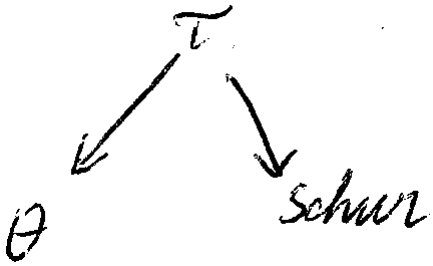
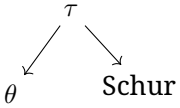


Diagram and typed description.



Original source details: GELFAND-P001102

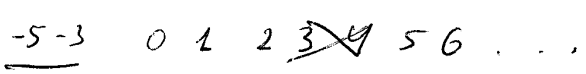
Schur. Consider special points

[crossed out: H]

Original source details: GELFAND-P001102

Uncertain reading. The cancelled sign before S resembles H.

$$S = \{s_0 < s_1 < s_2 < \ldots\}, \qquad s_i = i; \quad i \gg 0.$$



Original source details: GELFAND-P001102

Diagram and typed description.

$-5 \quad -3 \quad 0 \quad 1 \quad 2 \quad \cancel{3}\,\cancel{4} \quad 5 \quad 6 \quad \ldots$

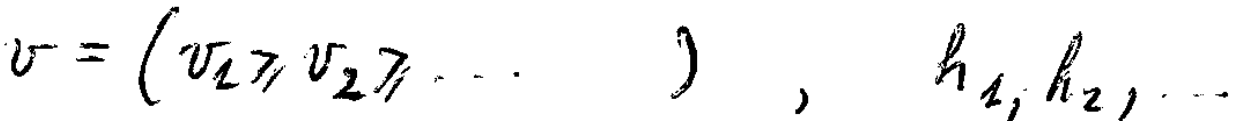
$$H_S = \{z^{s_i}\}_{s_i \in S}.$$

τ-function is a Schur function.

GELFAND-P001103

Suppose there is a partition

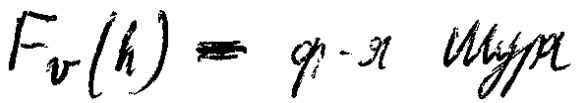
$$\nu = (\nu_1 \geq \nu_2 \geq \ldots), \qquad h_1, h_2, \ldots$$



Original source details: GELFAND-P001103

Diagram and typed description. The partition and the adjacent coefficients.

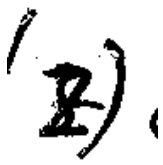
$F_\nu(h)$ [=?] Schur function.



Original source details: GELFAND-P001103

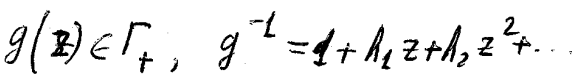
Uncertain reading. The sign is heavily overwritten. It could be an equals sign or a dash.

$$g(z) \in \Gamma_+, \qquad g^{-1} = 1 + h_1 z + h_2 z^2 + \ldots$$



Original source details: GELFAND-P001103

Uncertain reading. The overwritten variable in parentheses is provisionally read as z.



Original source details: GELFAND-P001103

Diagram and typed description. The expansion of g^{-1} with coefficients $h_1, h_2, \ldots$.

$$S \leftrightarrow \nu$$

$$i - s_i \leftrightarrow \nu_i$$

$$\underbrace{\tau_W(g) = F_S(h)}$$

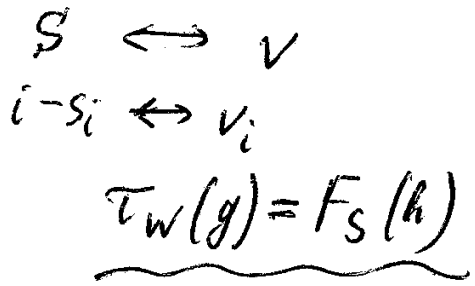


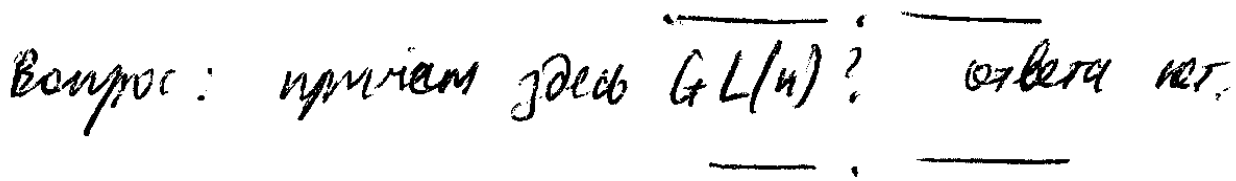
Diagram and typed description. The correspondence $S \leftrightarrow \nu$ and the wavy line beneath the formula for τ_W.

Original source details: GELFAND-P001103

— · —

Question: what does this have to do with $GL(n)$? there is no answer.

— · —

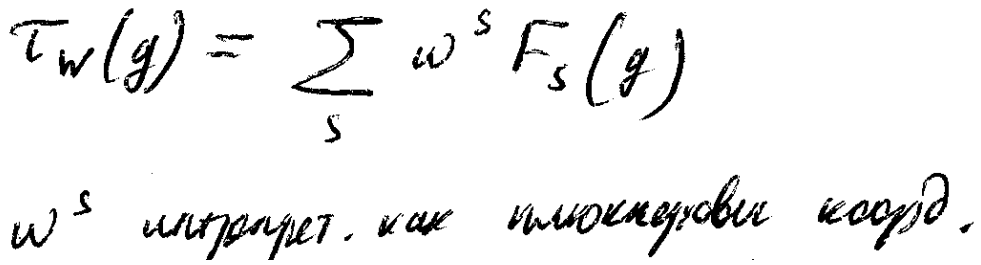


Original source details: GELFAND-P001103

Diagram and typed description. The question and response between two rules.

$$\tau_W(g) = \sum_S w^S F_S(g)$$

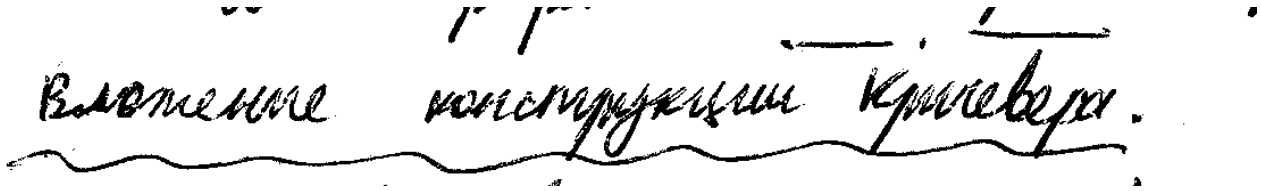
w^S is interpreted as a Plücker coordinate.



Original source details: GELFAND-P001103

Uncertain reading. Both occurrences are read as w^S. The handwritten letter could be ω, and the case of the superscript is uncertain.

Embedding of the Krichever construction.

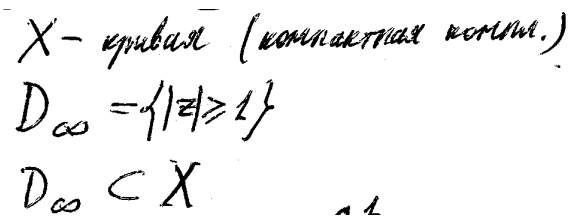


Original source details: GELFAND-P001103

Diagram and typed description. The heading on the Krichever construction.

X — curve (compact complex)

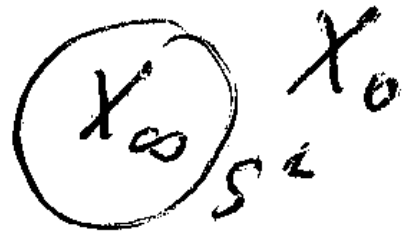
$$D_\infty = \{|z| \geq 1\}, \qquad D_\infty \subset X$$



Original source details: GELFAND-P001103

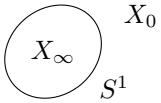
Diagram and typed description. The curve X and the weak inequality in the definition of D_∞.

choose an embedding S^1 — boundary



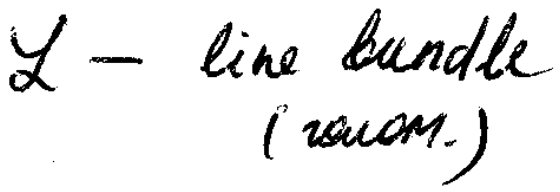
Original source details: GELFAND-P001103

Diagram and typed description.



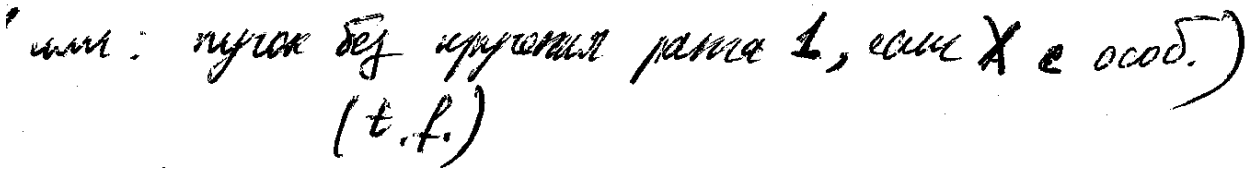
$$X_0 = X \setminus \operatorname{Int} X_\infty.$$

$\mathcal{L}$ — line bundle [below the line: (holom.)] (or: a torsion-free sheaf of rank 1, if X is singular). [Under "torsion-free": (t.f.)]



Original source details: GELFAND-P001103

Diagram and typed description. The original English wording "line bundle" and the abbreviation below it.



Original source details: GELFAND-P001103

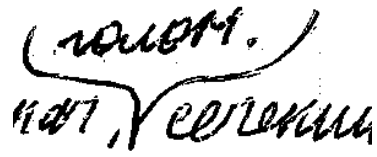
Diagram and typed description. The condition for a singular curve and the annotation "t.f."

$\varphi =$ trivialization of $\mathcal{L}$ over X_∞.

sections over $S^1 \leftrightarrow$ functions.

Construct a point on Gr:

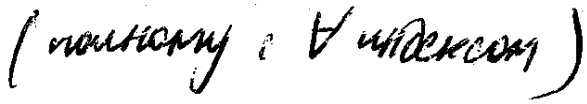
$$W = \{f \in H \mid f \text{ — boundary values of [(holom.)] sections of } \mathcal{L} \text{ over } X_0\}.$$



Original source details: GELFAND-P001103

Diagram and typed description. The insertion "holom." before "sections."

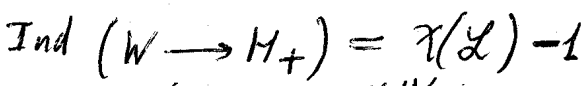
Assertion. $W \in Gr$. (the full one with $\forall$ index)



Original source details: GELFAND-P001103

Uncertain reading. The first word in parentheses is provisionally read as "the full one."

$$\mathrm{Ind}(W \longrightarrow H_+) = \chi(\mathcal{L}) - 1.$$

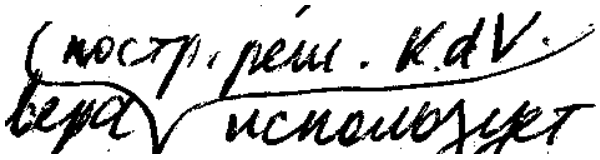


Original source details: GELFAND-P001103

Diagram and typed description. The index of the projection and the term -1.

Remark. Krichever's method [inserted above the line: (constructing KdV solutions)] uses only W, and does not use the curve!

$\Rightarrow$ solutions can be constructed for $\forall W$!!

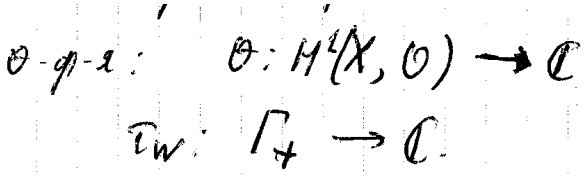


Original source details: GELFAND-P001103

Diagram and typed description. The insertion about constructing KdV solutions.

θ-function:

$$\theta : H^1(X, O) \longrightarrow \mathbb{C}$$
$$\tau_W : \Gamma_+ \longrightarrow \mathbb{C}.$$



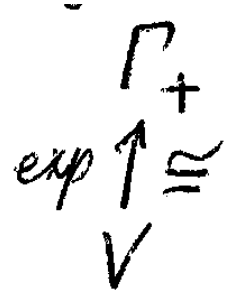
Original source details: GELFAND-P001103

Diagram and typed description. The domains of θ and τ_W.

GELFAND-P001104

[source number: 49]

Connection — ?

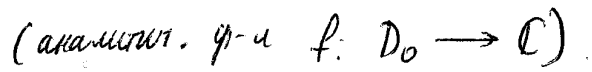


Original source details: GELFAND-P001104

Diagram and typed description.

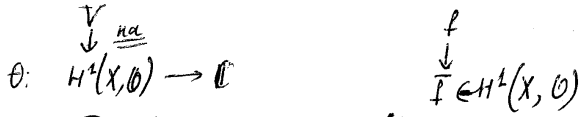
$$\begin{array}{c} \Gamma_+ \\ \exp \uparrow \cong \\ V \end{array}$$

(analytic functions $f\colon D_0 \longrightarrow \mathbb{C}$).



Original source details: GELFAND-P001104

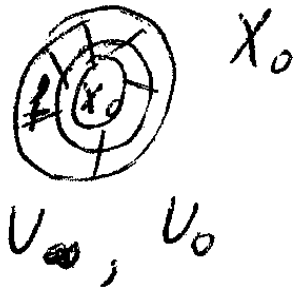
Diagram and typed description. The note with domain D_0.



Original source details: GELFAND-P001104

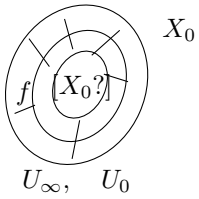
Diagram and typed description.

$$\begin{array}{ccc} V & & f \\ \downarrow \text{onto} & & \downarrow \\ \theta\colon\ H^1(X,O) & \longrightarrow \mathbb{C} & \bar{f} \in H^1(X,O) \end{array}$$



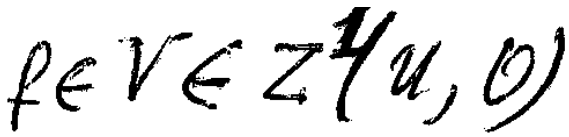
Original source details: GELFAND-P001104

Diagram and typed description.



The inner label is provisionally read as X_0.

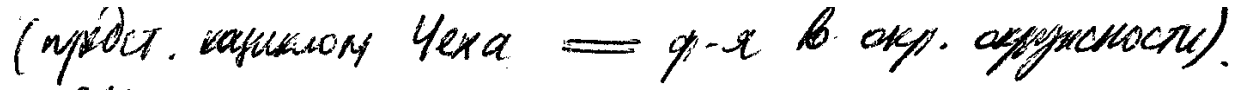
$$f \in V \in Z^1(U,O)$$



Original source details: GELFAND-P001104

Uncertain reading. The manuscript has a sign resembling $\in$ between V and Z^1.

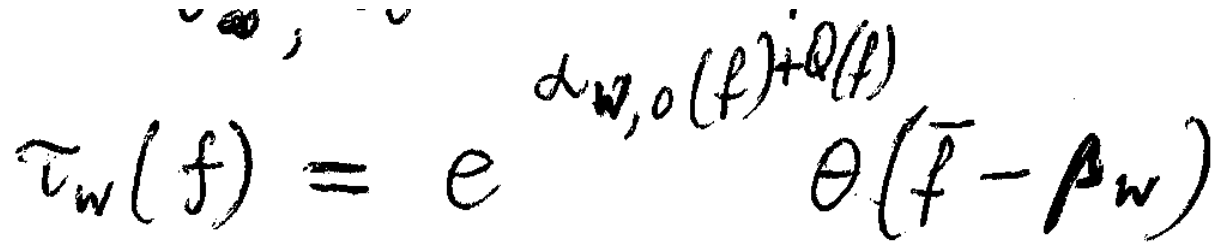
(represented by a Čech cocycle = function in a neighborhood of the circle).



Original source details: GELFAND-P001104

Diagram and typed description. The handwritten note on the Čech cocycle.

$$\widetilde{\tau}_W(f) = e^{\alpha_{W,0}(f)+Q(f)}\theta(\bar{f} - \beta_W)$$



Original source details: GELFAND-P001104

Diagram and typed description. The formula with exponent $\alpha_{W,0}(f) + Q(f)$.

Q depends only on (X, D_∞).

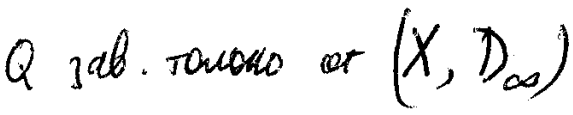
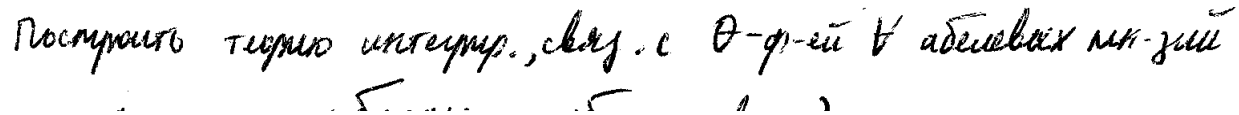


Diagram and typed description. The dependence of Q on (X, D_∞).

Original source details: GELFAND-P001104

Question from I. M. G. Construct a theory of integration connected with the θ-function of $\forall$ abelian varieties



Original source details: GELFAND-P001104

Diagram and typed description. The question about abelian varieties.

(here there arise [crossed out:] only Jacobians of algebraic curves)

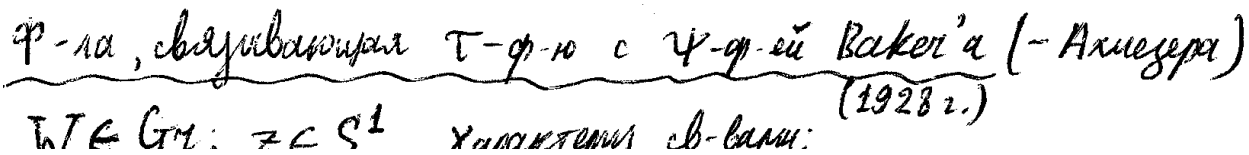
Undeciphered passage. The cancelled mark before “only” is illegible.

Original source details: GELFAND-P001104

—·—

Formula relating the τ-function to Baker’s Ψ-function (— Akhiezer)

[below Baker’s name: (1928)]



Original source details: GELFAND-P001104

Diagram and typed description. The wavy underline below the heading and the addition “— Akhiezer.”

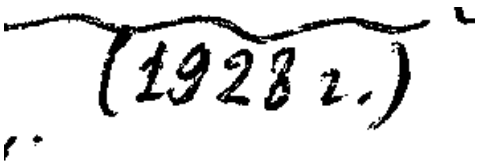


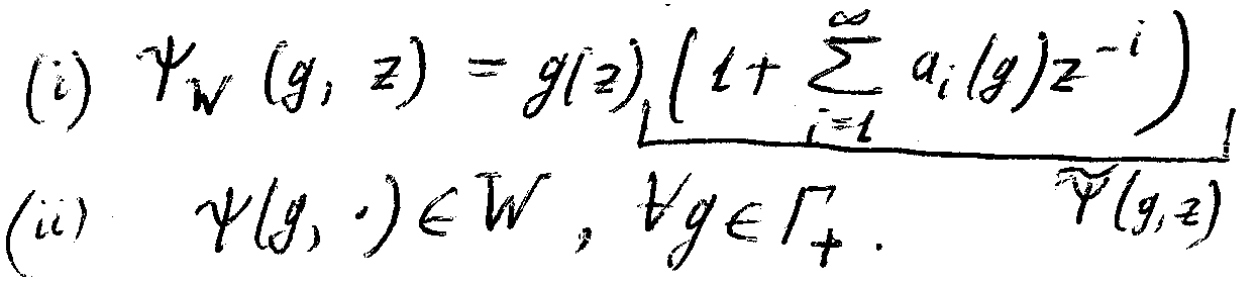
Diagram and typed description. The date below Baker’s name.

Original source details: GELFAND-P001104

$W \in Gr$; $z \in S^1$. Characterized by the properties:

$$\text{(i)} \quad \Psi_W(g,z) = g(z) \underbrace{\left(1 + \sum_{i=1}^{\infty} a_i(g) z^{-i}\right)}_{\widetilde{\Psi}(g,z)}$$

$$\text{(ii)} \quad \Psi(g,\cdot) \in W, \quad \forall g \in \Gamma_+.$$



Original source details: GELFAND-P001104

Diagram and typed description. The two properties and the square-ended bracket above $\widetilde{\Psi}(g,z)$.

Prop. $\forall W$, $\exists!$ such a Ψ_W [crossed out: ~~for any~~].

Uncertain reading. The cancelled ending provisionally reads “for any.”

Original source details: GELFAND-P001104

If $g^{-1}W \cap H_- = \{0\}$, then the value is constructed: [crossed out:] preimage of 1 under the isomorphism

Undeciphered passage. The mark before “preimage” is cancelled and illegible.

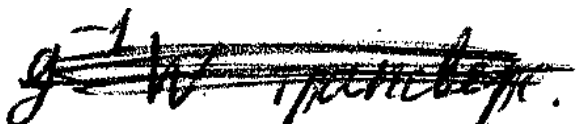
Original source details: GELFAND-P001104

$$g^{-1}W \xrightarrow{\cong} H_+$$

Diagram and typed description. The isomorphism to H_+.

Original source details: GELFAND-P001104

[crossed out: $g^{-1}W$ ~~transv.~~]



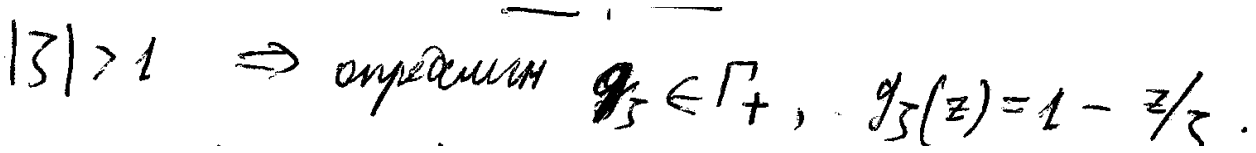
Uncertain reading. The cancelled line provisionally reads “$g^{-1}W$ transv.”

Original source details: GELFAND-P001104

Prop. $\exists!\ \Psi_W(g,z)$ — meromorphic on Γ_+.
Proof as in the case of curves
(there one uses the θ-function, and here the τ-function)

— · —

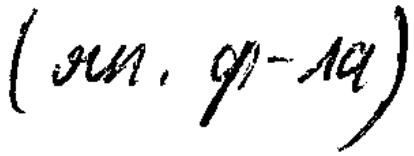
$|\zeta|>1 \quad \Rightarrow$ define $q_\zeta \in \Gamma_+, \quad q_\zeta(z) = 1 - z/\zeta$.



Original source details: GELFAND-P001104

Uncertain reading. The first letter is overwritten. In the formula at right it reads as q.

Prop. (explicit formula). For $|\zeta|>1$



Uncertain reading. The abbreviation is provisionally read as “explicit formula.”

Original source details: GELFAND-P001104

$$\widetilde{\Psi}_W(g,\zeta) = \frac{\tau_W(gq_\zeta)}{\tau_W(g)}$$

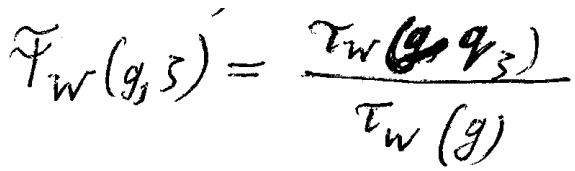


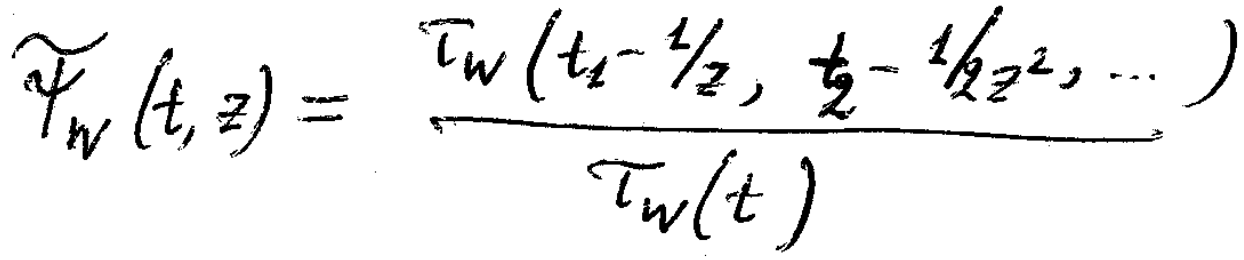
Diagram and typed description. The quotient formula, with an overwritten g in the numerator.

Original source details: GELFAND-P001104

GELFAND-P001105

Or: write it in the form exp and regard everything as functions of t (infinitely many)

$$\widetilde{\Psi}_W(t,z) = \frac{\tau_W(t_1 - 1/z,\ t_2 - 1/2z^2, \ldots)}{\tau_W(t)}$$



Original source details: GELFAND-P001105

Uncertain reading. The second shift is written with a slash as $1/2z^2$, without parentheses.

$$\left(g = e^{\sum_1^\infty t_i z^i}\right).$$

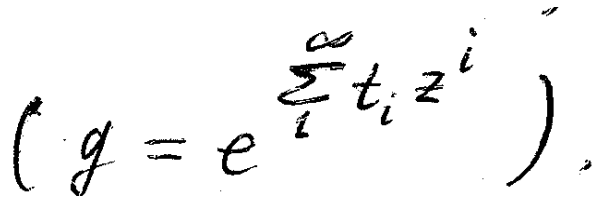


Diagram and typed description. The exponential form of g with lower summation limit 1.

Original source details: GELFAND-P001105

Or:

$$\Psi = \frac{(\text{vertex operator})\tau}{\tau}$$

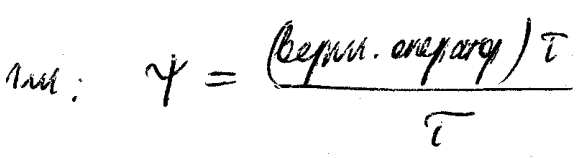
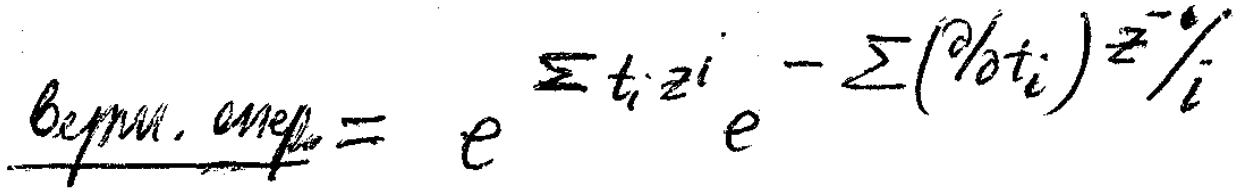


Diagram and typed description. The formula for Ψ in terms of the vertex operator.

Original source details: GELFAND-P001105

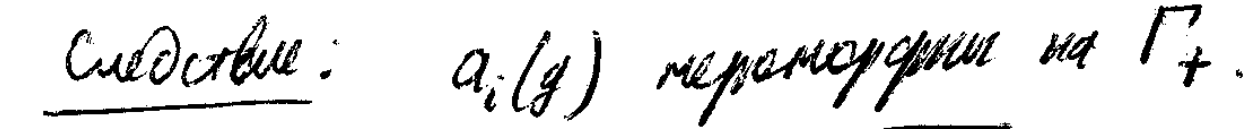
$$\underline{\text{vertex op.}} = e^{\sum t_i z^i} e^{-\sum(\partial/\partial t_i) z^{-i}/i}$$



Original source details: GELFAND-P001105

Diagram and typed description. The two exponentials and the exponent of the second one.

Corollary. $a_i(g)$ are meromorphic on Γ_+.



Original source details: GELFAND-P001105

Diagram and typed description. The corollary on meromorphicity of the coefficients.

—·—

The connection with nonlinear equations is as in Krichever!

—·—

[crossed out: ~~В...~~]

Undeciphered passage. The initial Cyrillic “В” is visible under the cancellation. The remainder is indistinct.

Original source details: GELFAND-P001105

$\forall W$ there is an infinite hierarchy of equations.

— · —

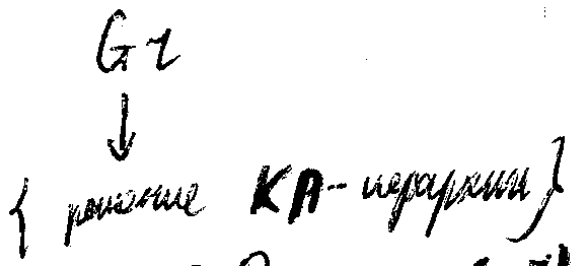


Diagram and typed description.

$$\begin{array}{c} Gr \\ \downarrow \\ \{\text{solution of the KP hierarchy}\} \end{array}$$

Original source details: GELFAND-P001105

$$D = \partial/\partial x; \qquad Q = D + \sum_1^\infty q_i D^{-i}$$

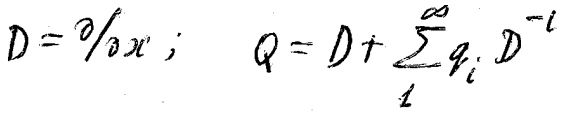


Diagram and typed description. The notation for D and Q.

Original source details: GELFAND-P001105

$$\frac{\partial Q}{\partial t_r} = [(Q^r)_+, Q]$$

[Note below the line from $(Q^r)_+$: differential part of Q^r.]

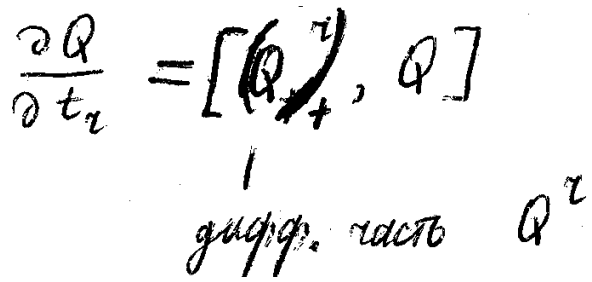


Diagram and typed description. The commutator and the annotation for the positive part.

Original source details: GELFAND-P001105

If $Q^n = L$ is differential, we obtain the Gelfand–Dickey system:

$$\frac{\partial L}{\partial t_r} = [L_+^{r/n}, L]$$

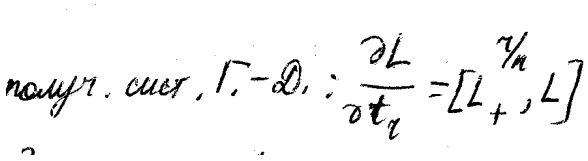
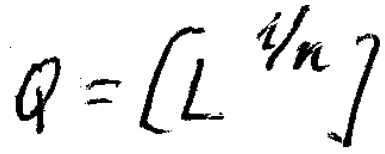


Diagram and typed description. The notation $L_+^{r/n}$ in the commutator.

Original source details: GELFAND-P001105

and conversely, if L is diff. $Q = [L^{1/n}]$, we obtain [crossed out:] the equation written above:

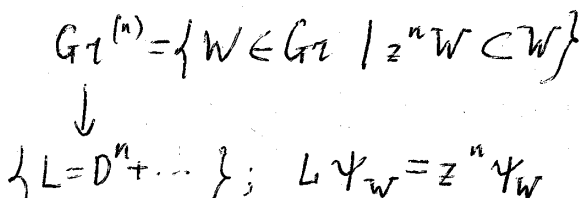


Original source details: GELFAND-P001105

Diagram and typed description. Square brackets in the notation $Q = [L^{1/n}]$.

Original source details: GELFAND-P001105

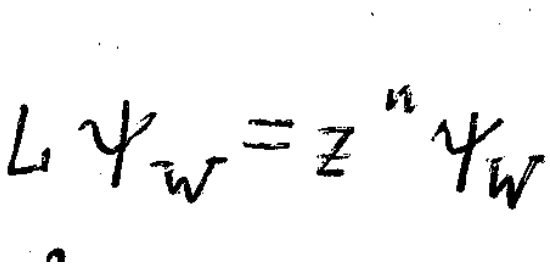
Undeciphered passage. The cancelled mark before “equation” is illegible.



Original source details: GELFAND-P001105

Diagram and typed description.

$$Gr^{(n)} = \{W \in Gr \mid z^n W \subset W\}$$
$$\downarrow$$
$$\{L = D^n + \ldots\}\,; \quad L\Psi_W = z^n \Psi_W$$



Original source details: GELFAND-P001105

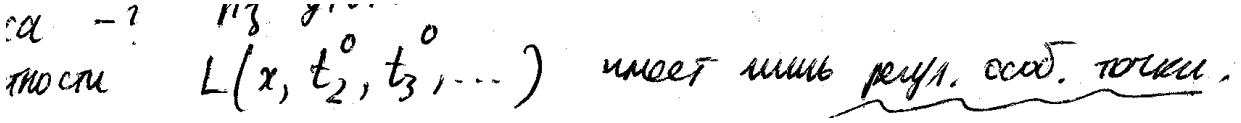
Diagram and typed description. The equation for Ψ_W.

Here:

$$g = e^{xz + t_2 z^2 + \ldots}$$

Poles — ? From the assertion about poles:

in particular $L(x, t_2^0, t_3^0, \ldots)$ has only regular singular points.

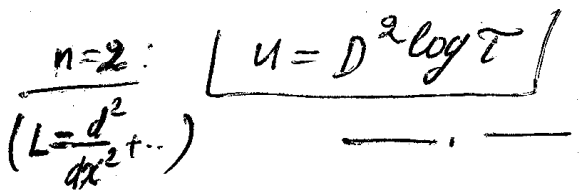


Original source details: GELFAND-P001105

Diagram and typed description. The fixed times t_2^0, t_3^0 and the underlined conclusion.

$$\underline{n = 2}: \quad \boxed{u = D^2 \log \tau}$$

$$\left(L = \frac{d^2}{dx^2} + \ldots\right)$$



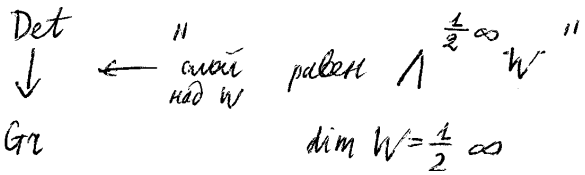
Original source details: GELFAND-P001105

Diagram and typed description. The open-topped bracket below the formula for u and the line about L.

GELFAND-P001106

[source number: 50]

Bundle:

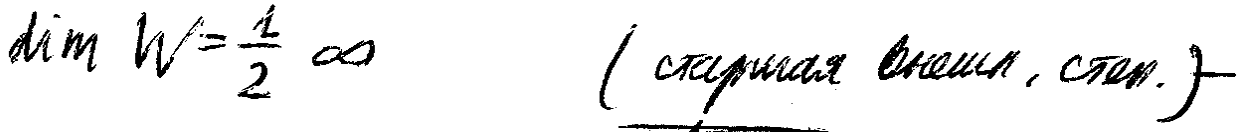


Original source details: GELFAND-P001106

Diagram and typed description.

$$\begin{array}{c} Det \\ \downarrow \\ Gr \end{array} \quad \longleftarrow \text{“the fibre over } W \text{ equals } \textstyle\bigwedge^{\frac{1}{2}\infty} W\text{”}$$

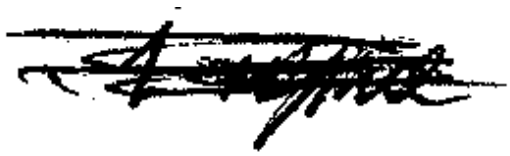
$$\dim W = \frac{1}{2}\infty \qquad \text{(top exterior power)—}$$



Original source details: GELFAND-P001106

Diagram and typed description. The dimension and the note on the top exterior power. A short stroke follows the parenthesis.

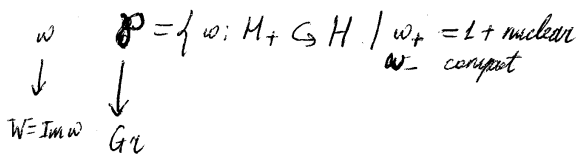
[crossed out below the note at right: ~~one-dimensional~~]



Original source details: GELFAND-P001106

Uncertain reading. The cancelled word is provisionally read as “one-dimensional.”

One-dimensional linear bundle.

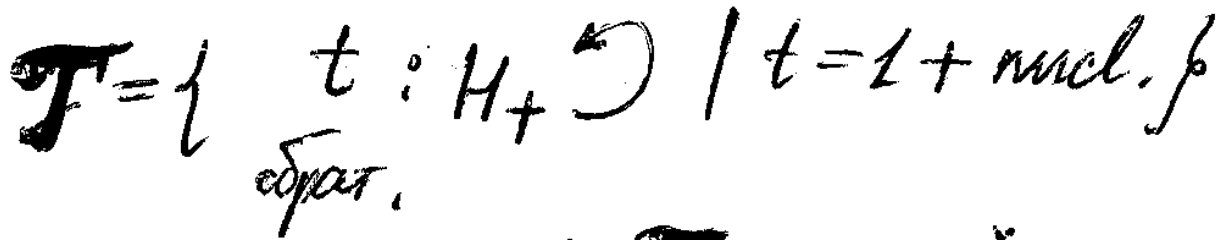


Original source details: GELFAND-P001106

Diagram and typed description.

$$\begin{array}{c} \omega \\ \downarrow \\ W = \operatorname{Im}\omega \end{array} \qquad \begin{array}{c} \mathcal{P} = \{\omega \colon H_+ \hookrightarrow H \mid \\ \quad \omega_+ = 1 + \text{nuclear} \\ \quad \omega_- \quad \text{compact} \\ \downarrow \\ Gr \end{array}$$

$$\mathcal{T} = \left\{ \underset{\text{inv.}}{t} : H_+ \circlearrowleft \;\middle|\; t = 1 + \text{nucl.} \right\}$$



Original source details: GELFAND-P001106

Diagram and typed description. The loop beside H_+ and the abbreviation below t.

$$\det\colon \mathcal{T} \longrightarrow \mathbb{C}^*$$

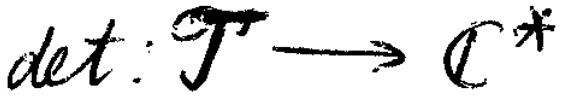


Diagram and typed description. The arrow from $\mathcal{T}$ to $\mathbb{C}^*$.

Original source details: GELFAND-P001106

$$Det = \mathcal{P} \underset{\mathcal{T}}{\times} \mathbb{C}.$$

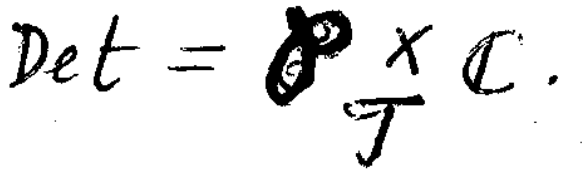
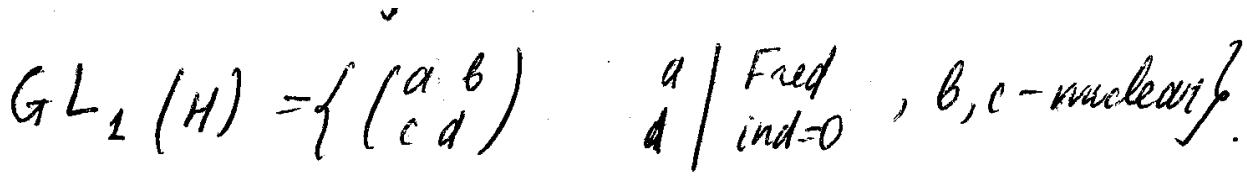


Diagram and typed description. The letter $\mathcal{T}$ is written below the product sign.

Original source details: GELFAND-P001106

$$GL_1(H) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{matrix} a \\ d \end{matrix} \;\middle|\; \begin{matrix} \text{Fred} \\ \text{ind} = 0 \end{matrix}, \quad b, c \text{ — nuclear} \right\}.$$



Original source details: GELFAND-P001106

Diagram and typed description. The blocks a, d and the Fred and ind $= 0$ conditions.

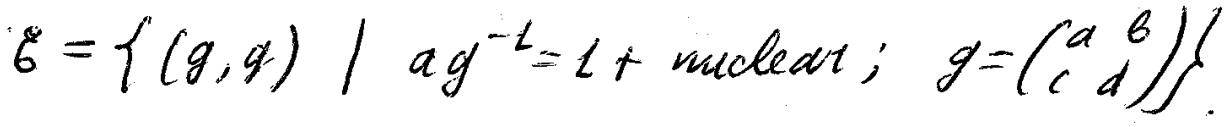
Its action on Gr will be lifted to an action on Det.

Uncertain reading. The middle of the word "action" has been overwritten.

Original source details: GELFAND-P001106

$$\mathcal{E} \subset GL_1(H) \times GL(H_+)$$

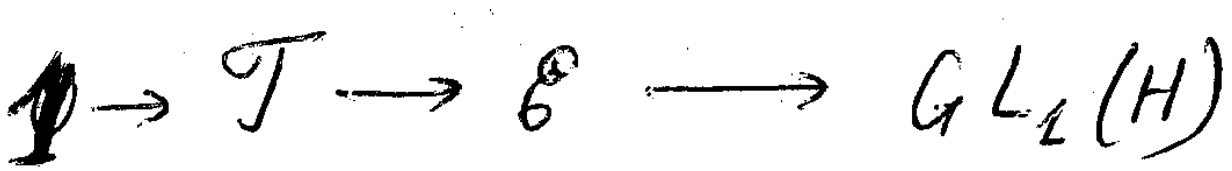
$$\mathcal{E} = \left\{ (g, q) \;\middle|\; aq^{-1} = 1 + \text{nuclear}; \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right\}.$$



Original source details: GELFAND-P001106

Diagram and typed description. The condition $aq^{-1} = 1 +$ nuclear for the pair (g, q).

$$1 \longrightarrow \mathcal{T} \longrightarrow \mathcal{E} \longrightarrow GL_1(H)$$



Original source details: GELFAND-P001106

Uncertain reading. The initial digit is overwritten and provisionally read as 1.

Assertion. the extension is nontrivial.

In the finite-dimensional case we obtain a trivial extension.

Action of $\mathcal{E}$ on Det:

$$(g,q)(\omega,\lambda) = (g\omega q^{-1},\lambda)$$

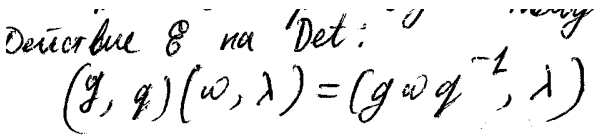


Diagram and typed description. The action on the pair (ω,λ).

Original source details: GELFAND-P001106

from the presentation $Det = \mathcal{P} \underset{\mathcal{T}}{\times} \mathbb{C}$.

Diagram and typed description. The note at the right giving the presentation of Det.

Original source details: GELFAND-P001106

$$\mathcal{T}_0 = \ker \det \mathcal{T} \longrightarrow 1$$

Uncertain reading. The symbols after ker det are $\mathcal{T}$, an arrow and the numeral 1, with no parentheses.

Original source details: GELFAND-P001106

[crossed out:] The elements of $\mathcal{T}_0$ (their image in $\mathcal{E}$) act trivially.

Undeciphered passage. The mark before "The elements" is heavily cancelled and illegible.

Original source details: GELFAND-P001106

$\Rightarrow$ there acts $\mathcal{E}/\mathcal{T}_0$.

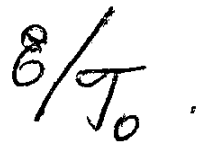
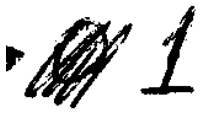


Diagram and typed description. The quotient $\mathcal{E}/\mathcal{T}_0$ on its own line.

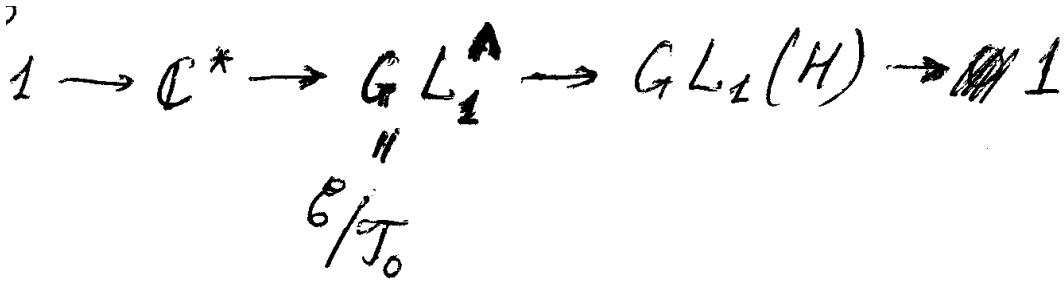
Original source details: GELFAND-P001106

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \widehat{GL_1} \longrightarrow GL_1(H) \longrightarrow 1$$
$$\widehat{GL_1} \;\|\; \mathcal{E}/\mathcal{T}_0$$



Uncertain reading. A sign resembling 0 is cancelled before the final 1.

Original source details: GELFAND-P001106



Original source details: GELFAND-P001106

Diagram and typed description. The exact sequence and the label beneath $\widehat{GL_1}$.

GELFAND-P001107

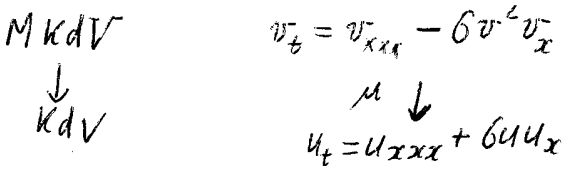


Diagram and typed description.

$$\begin{array}{ccc} \text{MKdV} & & v_t = v_{xxx} - 6v^2 v_x \\ \downarrow & & \mu \downarrow \\ \text{KdV} & & u_t = u_{xxx} + 6uu_x \end{array}$$

Original source details: GELFAND-P001107

$$\underline{u = v_x - v^2}$$

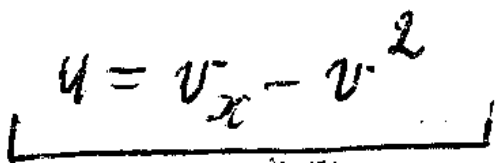
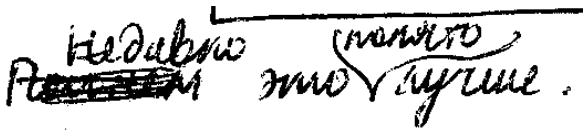


Diagram and typed description. The Miura formula and its square-ended bracket.

Original source details: GELFAND-P001107

[crossed out: ~~Let us consider~~; above it: Recently] this was [inserted: understood] better.



Uncertain reading. The cancelled word is provisionally read as “Let us consider.” “Recently” is written above it.

Original source details: GELFAND-P001107

KdV is connected with

$$\underbrace{D^2 + u = (D - v)(D + v)}_{(u = v_x - v^2)}$$

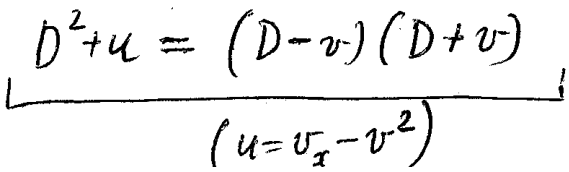
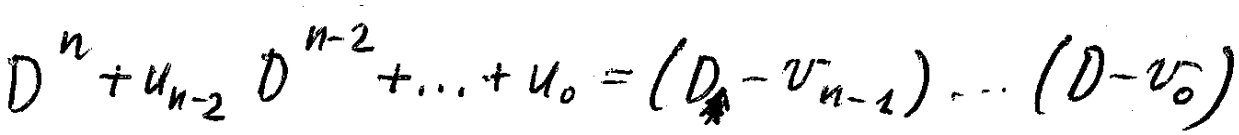


Diagram and typed description. The factorization and the note under the square-ended bracket.

Original source details: GELFAND-P001107

generalization of the right-hand side:

$$D^n + u_{n-2}D^{n-2} + \ldots + u_0 = (D - v_{n-1}) \ldots (D - v_0)$$



Original source details: GELFAND-P001107

Diagram and typed description. The general factorization in the order written.

Undeciphered passage. The cancelled subscript at the first D on the right is illegible.

Original source details: GELFAND-P001107

$$v \longmapsto u$$

generalized Miura transformation.

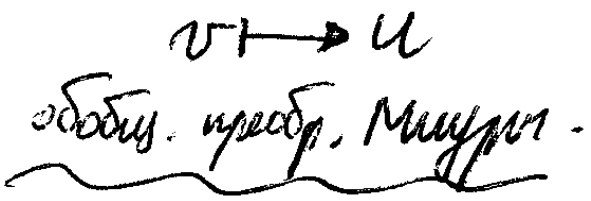


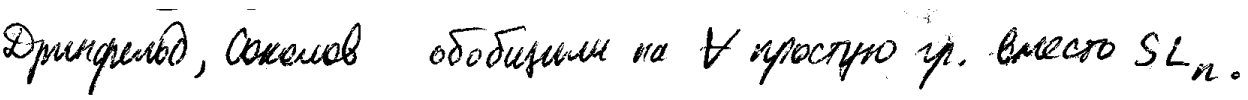
Diagram and typed description. The arrow from v to u and its underlined caption.

Original source details: GELFAND-P001107

[crossed out:] Drinfeld and Sokolov generalized this to $\forall$ simple group in place of SL_n.

Undeciphered passage. The heavily cancelled mark before Drinfeld's name is illegible.

Original source details: GELFAND-P001107



Original source details: GELFAND-P001107

Diagram and typed description. The statement about Drinfeld, Sokolov and a simple group.

The correspondence is not one-to-one. (diff. eq.)

[below the parentheses: one already has to solve it for $n = 2$.]

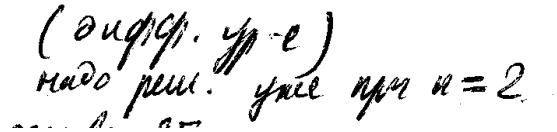


Diagram and typed description. The note below the line about the differential equation.

Original source details: GELFAND-P001107

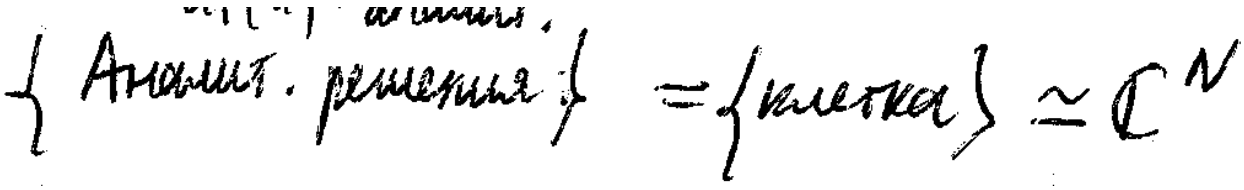
$$n = 2 \quad \Rightarrow \quad \forall u \text{ there is a one-parameter family of } v.$$

Original source details: GELFAND-P001107

Diagram and typed description. The line on the family for $n = 2$.

$u_i(x)$ — analytic.

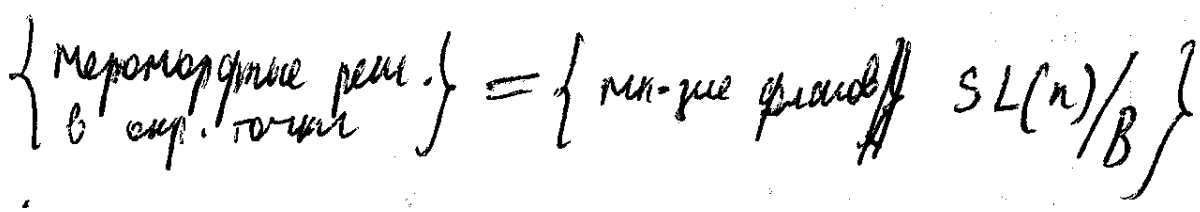
$$\{\text{Analytic solutions}\} = \{\text{cell}\} \simeq \mathbb{C}^N$$



Original source details: GELFAND-P001107

Diagram and typed description. The comparison of analytic solutions with a cell. The sign before $\mathbb{C}^N$ has one wavy and one horizontal stroke.

$$\left\{\begin{matrix}\text{Meromorphic sol.} \\ \text{near a point}\end{matrix}\right\} = \{\text{set of flags } SL(n)/B\}$$



Original source details: GELFAND-P001107

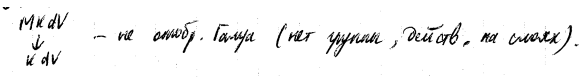
Diagram and typed description. The equality with the set of flags $SL(n)/B$.

Diagram and typed description. The closing brace after “flags” is cancelled. The retained brace follows $SL(n)/B$.

Original source details: GELFAND-P001107

$\forall n$ there is a system of nonlinear differential equations. Its solutions have the structure of a set of flags.

Main difficulty:

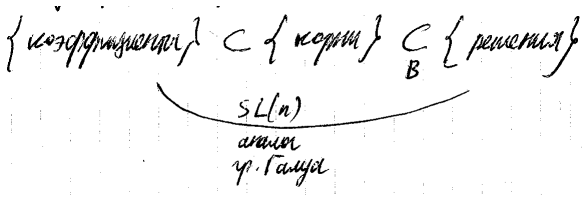


Original source details: GELFAND-P001107

Diagram and typed description.

MKdV
↓
KdV
— not a Galois map
(there is no group acting on the fibres).

Consider the differential field generated by $v_0, \dots, v_{n-1}$.



Original source details: GELFAND-P001107

Diagram and typed description.

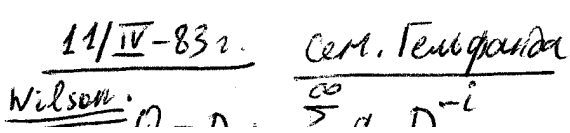
$$\underbrace{\{\text{coefficients}\} \subset \{\text{roots}\} \underset{B}{\subset} \{\text{solutions}\}}_{SL(n)}$$

analogue of the Galois group

GELFAND-P001108

11/IV-83 Gelfand Sem.

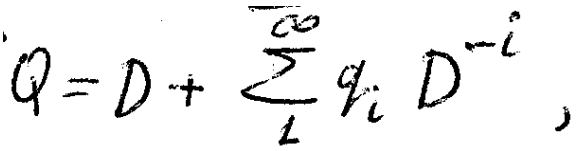
Wilson.



Original source details: GELFAND-P001108

Diagram and typed description. The date 11/IV-83 and the two underlines below Wilson's name.

$$Q = D + \sum_{1}^{\infty} q_i D^{-i}, \qquad L \text{ — diff.} \quad \Rightarrow$$



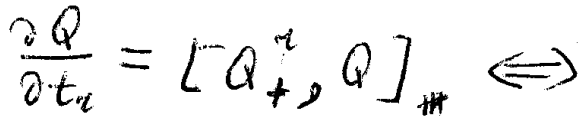
Original source details: GELFAND-P001108

Diagram and typed description. The expansion of Q with lower summation limit 1.

Original source details: GELFAND-P001108

Diagram and typed description. The condition on L and the arrow at the end of the line.

$$\frac{\partial Q}{\partial t_r} = [Q_+^r, Q] \text{ [cancelled]} \quad \Longleftrightarrow \quad L_{t_r} = [L_+^{r/n}, L] \qquad \text{(G.–D.)}$$



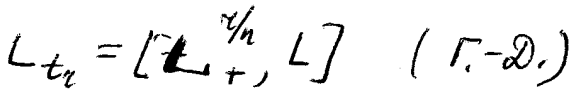
Original source details: GELFAND-P001108

Diagram and typed description. The first commutator and the equivalence sign.

Undeciphered passage. The small cancelled sign after the first square bracket is illegible.

Original source details: GELFAND-P001108

Diagram and typed description. The commutator for L and the abbreviation at the right.



Original source details: GELFAND-P001108

Uncertain reading. The L in the first entry of the second commutator is written over thinner strokes.

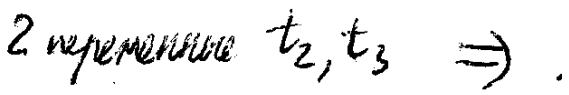
Original source details: GELFAND-P001108

2 variables t_2, t_3 $\Rightarrow$

$2q_1(x, t_2, t_3)$ — solution of K.–P. $\Rightarrow$ universal KP

y t

Diagram and typed description. The two variables and the rightward arrow.



Original source details: GELFAND-P001108

Original source details: GELFAND-P001108

Diagram and typed description. The factor 2 before q_1 and the labels y, t beneath the arguments.

GELFAND-P001109

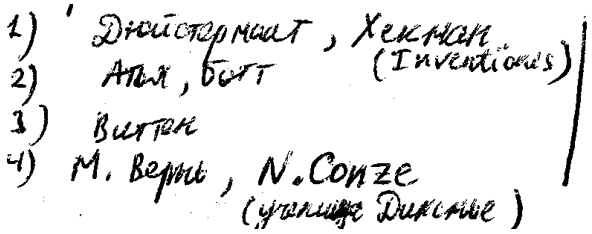
V. Ginzburg.

Diagram and typed description. The speaker's name and its underline.

Original source details: GELFAND-P001109

on the works:

1) Duistermaat, Heckman (Inventiones) 2) Atiyah, Bott 3) Witten 4) M. Vergne, N. Conze (Dixmier's pupils)	Berezin, Gelfand (vol. 5, Tr. Moscow Math. Soc.) 1956

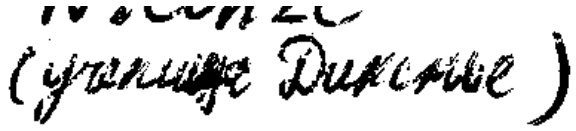


Original source details: GELFAND-P001109

Diagram and typed description. The four entries and the vertical line at the right.

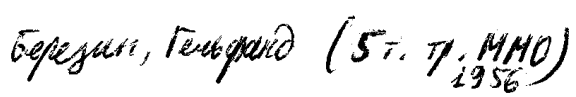
Original source details: GELFAND-P001109

Uncertain reading. The Russian spelling in the first entry is provisionally read as Duistermaat.



Original source details: GELFAND-P001109

Uncertain reading. The parenthesis is provisionally read as "Dixmier's pupils."

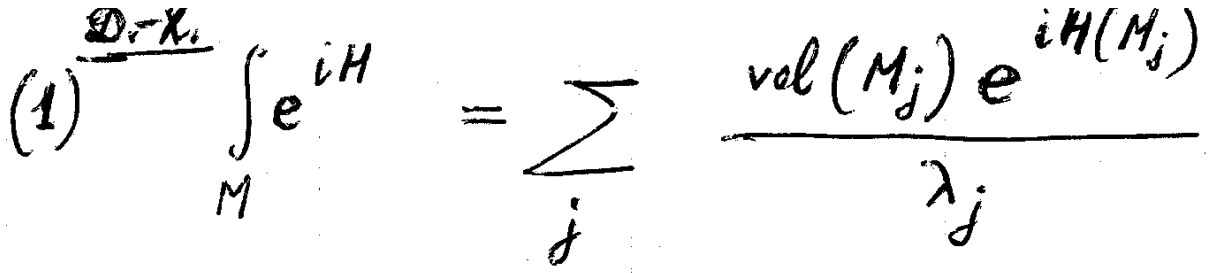


Original source details: GELFAND-P001109

Diagram and typed description. The reference to the right of the list, with the year below the parenthesis.

D.–H.

$$(1) \qquad \int_M e^{iH} = \sum_j \frac{\mathrm{vol}(M_j) e^{iH(M_j)}}{\lambda_j}$$



Original source details: GELFAND-P001109

Diagram and typed description. Formula (1), with exponent iH and denominator λ_j.

M — compact manifold, symplectic.

H — function on M; the condition on it: int. curves of the [inserted: Ham.] field [crossed out:] ξ_H are closed.

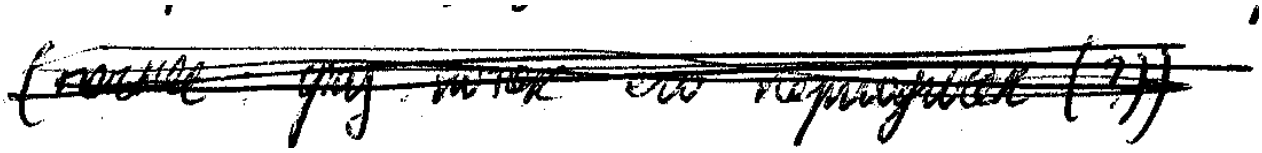
Original source details: GELFAND-P001109

Diagram and typed description. “Ham.” is inserted before “field.”

Original source details: GELFAND-P001109

Undeciphered passage. An illegible notation is cancelled before ξ_H.

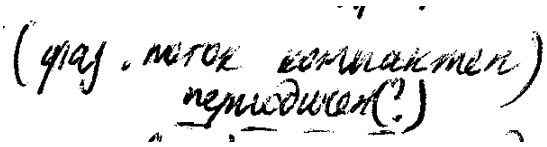
[crossed out: ~~(more precisely, its phase flow is periodic (?))~~.]



Original source details: GELFAND-P001109

Uncertain reading. The cancelled line is provisionally read as “(more precisely, its phase flow is periodic (?)).”

(the phase flow is compact) [added below: periodic (?)]

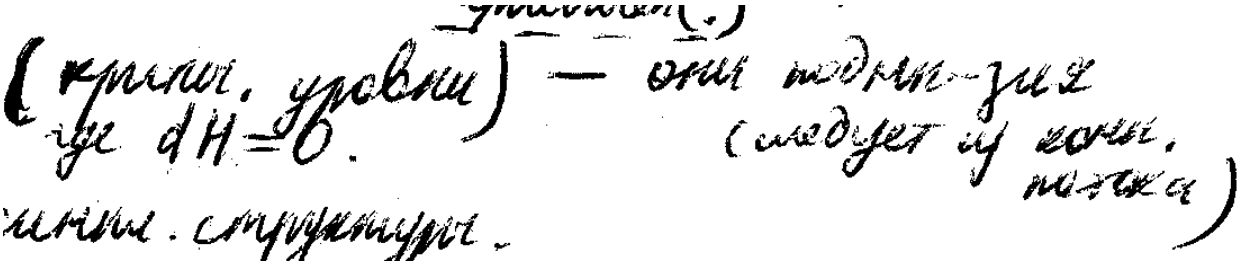


Original source details: GELFAND-P001109

Diagram and typed description. The addition “periodic (?)” below the line about compactness.

M_j — critical sets of the function H (critical levels) — they are submanifolds.

[below “critical levels”: where $dH = 0$. at right: (follows from the compact flow).]



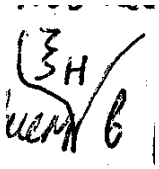
Original source details: GELFAND-P001109

Diagram and typed description. The two notes on the critical levels.

vol M_j — volume with respect to the induced sympl. structure.

H is constant on $M_j \Rightarrow H(M_j)$ — its value.

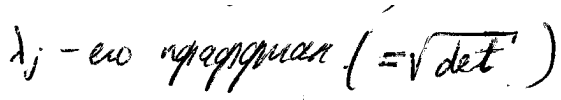
λ_j is determined by the action of [inserted: ξ_H] on (T_xM/T_xM_j) (T_xM_j preserved)



Original source details: GELFAND-P001109

Diagram and typed description. The insertion ξ_H in the line on the action on the quotient space.

The action of ξ_H is skew-symmetric: λ_j — its Pfaffian $(= \sqrt{\det})$.

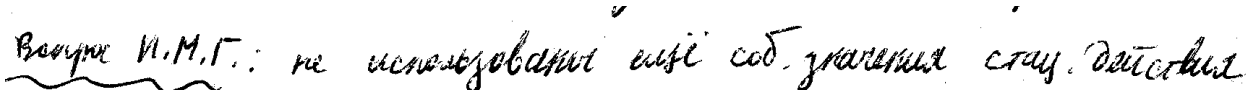


Original source details: GELFAND-P001109

Diagram and typed description. The Pfaffian and square root.

The action is equiv. to a linear one, and for a linear one everything works.

Question from I. M. G. the eigenvalues of the stationary action have yet to be used.



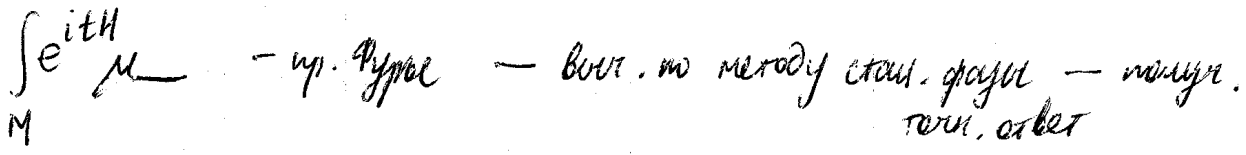
Original source details: GELFAND-P001109

Diagram and typed description. The question about the eigenvalues.

Idea of proof:

$$\int_M e^{itH}\mu$$

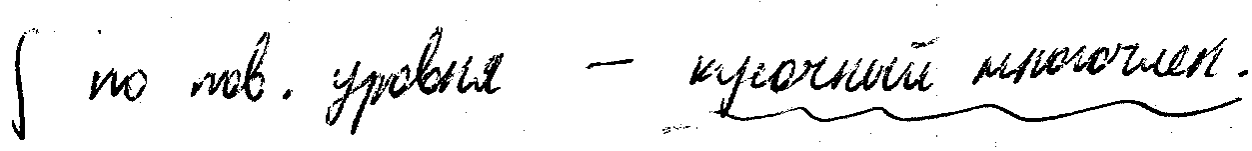
— Fourier transform — calculate by stationary phase — obtain [added below: exact answer].



Original source details: GELFAND-P001109

Diagram and typed description. The integral and the note about an exact answer.

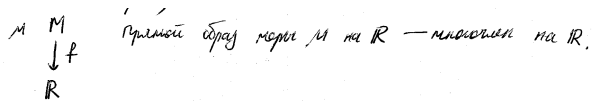
$\int$ over the level surface — piecewise polynomial.



Original source details: GELFAND-P001109

Diagram and typed description. The wavy underline below “piecewise polynomial.”

μ from the symplectic form $\Rightarrow$ everything is written in suitable local coordinates.
It suffices to prove that the sympl. form — is a polynomial (in fact a linear function)



Original source details: GELFAND-P001109

Diagram and typed description.

$$\begin{array}{c} \mu \quad M \\ \quad\quad \downarrow f \\ \quad\quad \mathbb{R} \end{array}$$

The pushforward of the measure μ onto $\mathbb{R}$ — is a polynomial on $\mathbb{R}$.

GELFAND-P001110

[source number: 58]

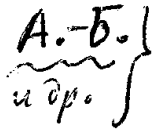


Diagram and typed description. A.–B. and others

Original source details: GELFAND-P001110

M — a compact manifold, G — a compact Lie group, acting on M.

$$H^*_G(M) \text{ — ?}$$

(cohomology)

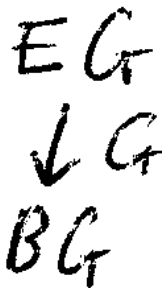
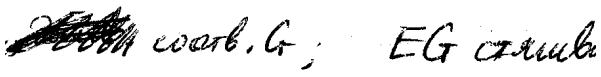


Diagram and typed description.

$$\begin{array}{c} EG \\ \quad\quad \downarrow G \\ \quad\quad BG \end{array}$$

Original source details: GELFAND-P001110

— universal bundle [crossed out] corresponding to G; EG is contractible.



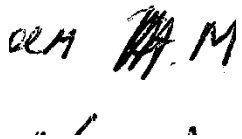
Uncertain reading. word after “bundle” illegible under the strokes

Original source details: GELFAND-P001110

Consider $EG \underset{G}{\times} M$

$$H^*_G(M) \stackrel{\text{def}}{=} H^*(EG \underset{G}{\times} M) \qquad \text{(Borel)}$$

$EG \underset{G}{\times} M$ — bundle over BG with fibre [crossed out] M.



Uncertain reading. the mark before M is heavily crossed out

Original source details: GELFAND-P001110

$H^*(EG \underset{G}{\times} M)$ — a module over $H^*(BG)$ (polynomials!)

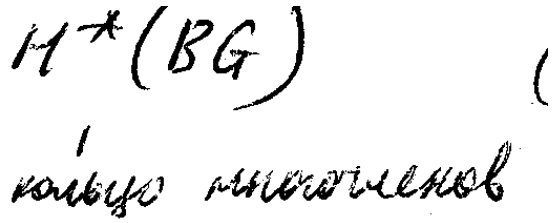


Diagram and typed description.

$H^*(BG)$

|

polynomial ring

Original source details: GELFAND-P001110

let $G = S^1$ and then $H^*(BG)$ — polynomial ring in 1 variable.

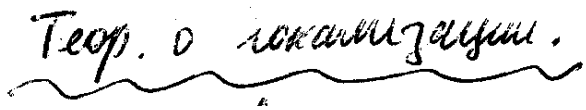


Diagram and typed description. Localization thm.

Original source details: GELFAND-P001110

M^G — set of fixed points of the action of G.

$$(1) \qquad H^*_G(M) \longrightarrow H^*_G(M^G)$$

On M^G the action is trivial $\Rightarrow$

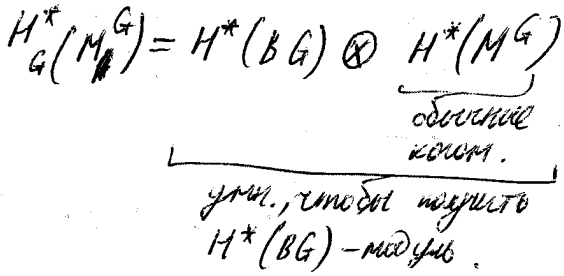


Diagram and typed description.

$H^*_G(M^G) = \underbrace{H^*(BG) \otimes \underbrace{H^*(M^G)}_{\text{ordinary cohom.}}}_{\text{mult. to obtain an } H^*(BG)\text{-module.}}$

Original source details: GELFAND-P001110

Uncertain reading. illegible crossed-out subscript at the first M

Original source details: GELFAND-P001110

Thm.

If we localize with respect to [crossed out ~~the whole~~ (?), with "the ring" inserted above[?]] of coefficients, we get an isomorphism

Uncertain reading. crossed out ~~the whole~~ (?), with "the ring" inserted above

Original source details: GELFAND-P001110

[crossed out]

Uncertain reading. illegible crossed-out mark after “isomorphism”

Original source details: GELFAND-P001110

$$(2) \qquad \mathrm{loc.}\big(H^*_G(M)\big) \xrightarrow{\sim} \mathrm{loc.}\big(H^*_G(M^G)\big)$$

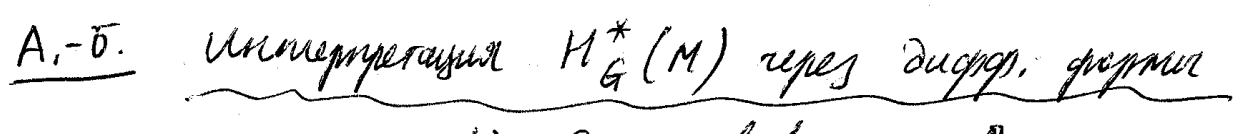
(i.e. replace [inserted $H^*(BG)$] by the field [crossed out] of fractions).

Uncertain reading. crossed-out writing between “field” and “of fractions”

Original source details: GELFAND-P001110

In K-theory — Atiyah, Segal.

In cohom. — Quillen.



Original source details: GELFAND-P001110

Diagram and typed description. A.–B. Interpretation of $H^*_G(M)$ via diff. forms

Original source details: GELFAND-P001110

Diagram and typed description. Localization thm. (2) is then equiv. to Witten’s thm. on a Killing field.

$G = S^1$

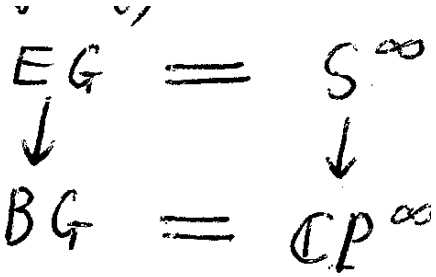


Diagram and typed description.

$$\begin{array}{ccc} EG & = & S^\infty \\ \downarrow & & \downarrow \\ BG & = & \mathbb{C}P^\infty \end{array}$$

Original source details: GELFAND-P001110

(inf.-dim. sphere)

$$H^*(BG) = \mathbb{Q}[u], \quad \deg u = 2.$$

ξ — the vector field determining the action of S^1.

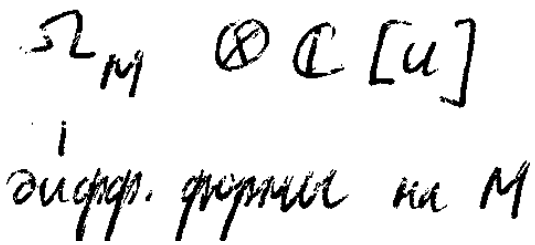
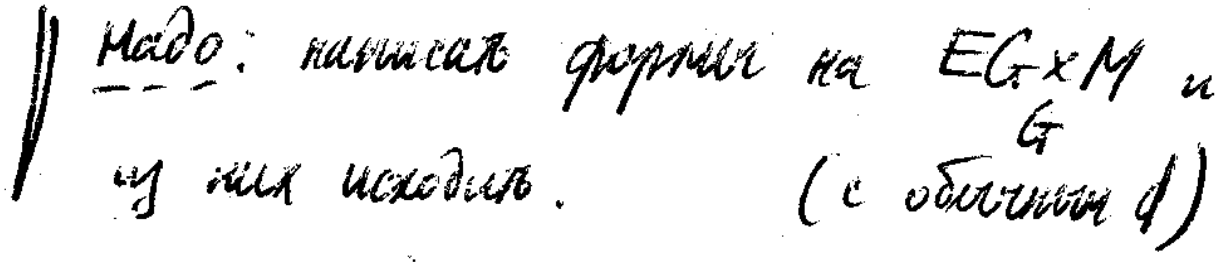


Diagram and typed description.
$\Omega_M \otimes \mathbb{C}[u]$
|
diff. forms on M

Original source details: GELFAND-P001111

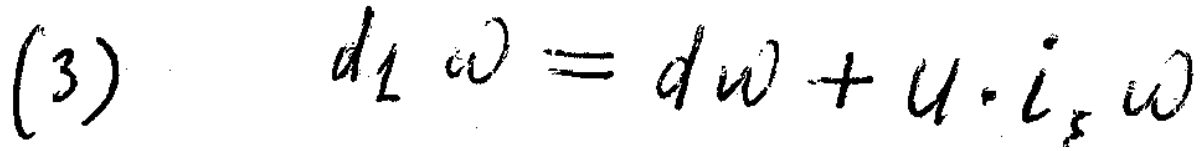
(to obtain a $\mathbb{C}[u]$-module)

Differential in $\Omega_M \otimes \mathbb{C}[u]$:



Original source details: GELFAND-P001111

Diagram and typed description. [at right] || Need to: write forms on $EG \underset{G}{\times} M$ and start from them. (with ordinary d)



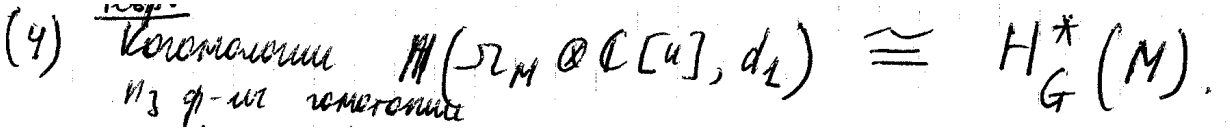
Original source details: GELFAND-P001111

Diagram and typed description. (3) $d_1\omega = d\omega + u \cdot i_\xi \omega$

(twist as in Witten)

The degrees agree!

Thm.



Original source details: GELFAND-P001111

Diagram and typed description. (4) Cohomology $H(\Omega_M \otimes \mathbb{C}[u], d_1) \cong H^*_G(M)$.

From the homotopy formula

$$L_\xi = d \circ i_\xi + i_\xi \circ d$$

it follows that

Diagram and typed description. [crossed out ~~from~~]

Original source details: GELFAND-P001111

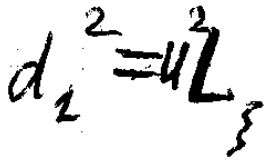


Diagram and typed description. $d_1^2 = u^2 L_\xi$

Original source details: GELFAND-P001111

Hence if $d_1\omega = 0$, then ω is invariant, i.e. $L_\xi\omega = 0$.

Twisted forms: [crossed out] ξ-invariant!! (from the very beginning).

Uncertain reading. heavily crossed-out mark before ξ

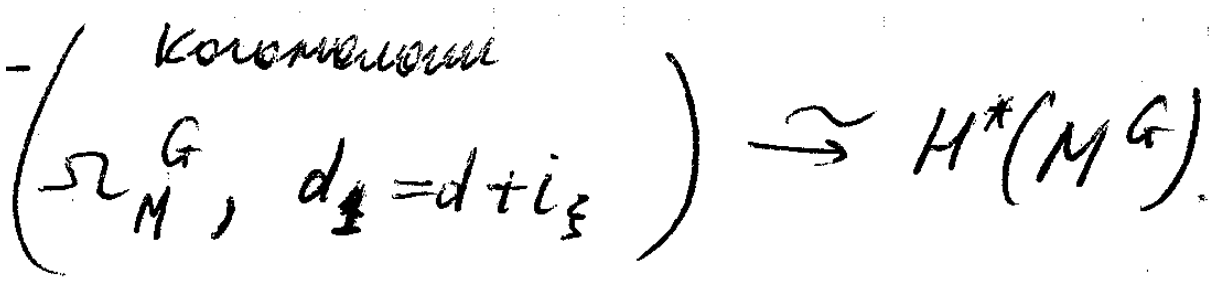
Original source details: GELFAND-P001111

$$\Omega^G = \Omega_M^G \otimes \mathbb{C}[u], \qquad \Omega_M^G \text{ — inv. forms on } M,$$

A. Goncharov: this is $\sim$ the Chern–Weil constr. of char. classes from connections.

Consequence of localization: u is erased.

(5) Thm.



Original source details: GELFAND-P001111

Diagram and typed description. $\left(\begin{matrix}\text{Cohomology} \\ \Omega_M^G, \quad d_1 = d + i_\xi\end{matrix}\right) \xrightarrow{\sim} H^*(M^G).$

I. M. G.'s problem. Formulate the same thing without G in the language of the double bundle.

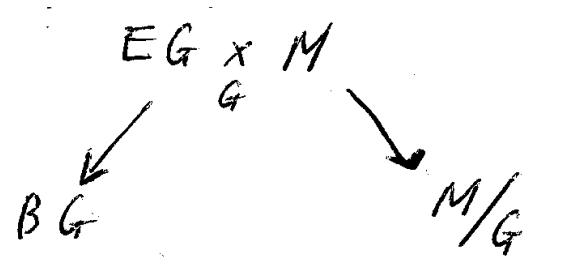


Diagram and typed description.

$$\begin{array}{ccc} & EG \underset{G}{\times} M & \\ \swarrow & & \searrow \\ BG & & M/G \end{array}$$

Original source details: GELFAND-P001111

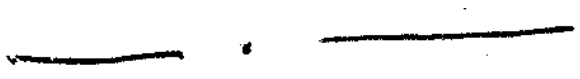


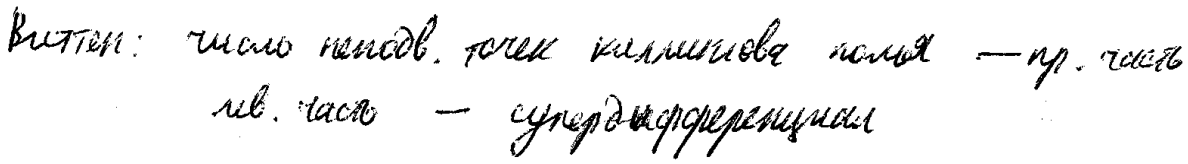
Diagram and typed description. — · —

Original source details: GELFAND-P001111

On the left in (5) there is no division into degrees — only parities.

We restrict to M^G: $i_\xi = 0$ there and we get the ordinary d!

This is precisely the map in (5)



Original source details: GELFAND-P001111

Diagram and typed description. Witten: the number of fixed points of a Killing field — right-hand side.
left-hand side — superdifferentials

GELFAND-P001112

[source number: 59]

Now the work of M. Vergne (its correct reading):

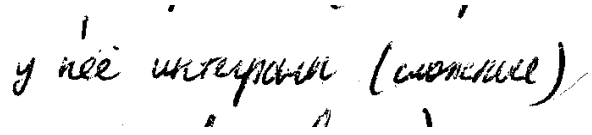


Diagram and typed description. [under the name] her integrals (complicated)

Original source details: GELFAND-P001112

Lemma (main). If there are no fixed points, the cohomology on the left is $= 0$.

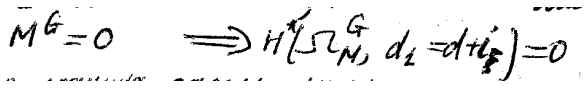


Diagram and typed description.
$M^G = 0 \implies H^*(\Omega_M^G, d_1 = d + i_\xi) = 0.$

Original source details: GELFAND-P001112

M. Vergne: wrote down explicitly what the cycle under consideration is a boundary of:

General argument: remove the fixed points and do the same thing.

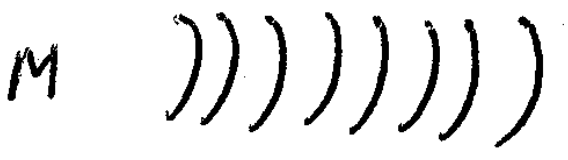
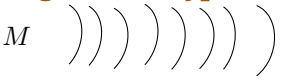


Diagram and typed description.



Original source details: GELFAND-P001112

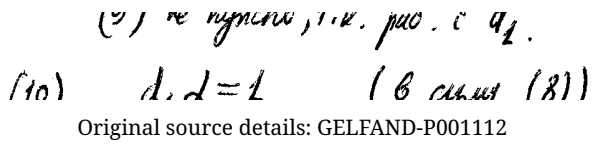
— circle S^1 acts without fixed points.

$$(6) \qquad \alpha = d\varphi \qquad (1\text{-form})$$

Properties (desired)

$$(7) \quad L_\xi \alpha = 0$$
$$(8) \quad i_\xi \alpha = 1$$
$$(9) \quad d\alpha = 0.$$

(*) Cond. (7), (8) can be satisfied $\alpha(\cdot) = \dfrac{\langle \xi, \cdot \rangle}{\langle \xi, \xi \rangle}$.



Original source details: GELFAND-P001112

Diagram and typed description. (9) is unnecessary, since we work with d_1. Let us rewrite (9):

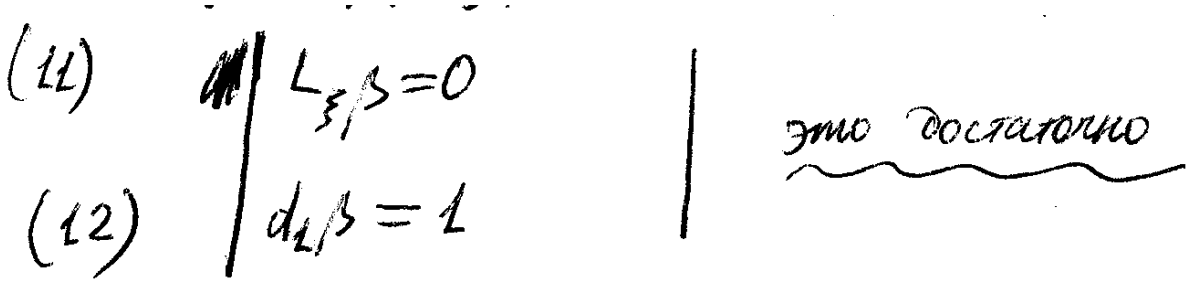
$$(10) \qquad d_1 \alpha = 1 \qquad \text{(by (8))}$$

$d\alpha = 1$ is impossible.

Make a form β:

Uncertain reading. heavily crossed-out mark before the left rule at (11)

Original source details: GELFAND-P001112



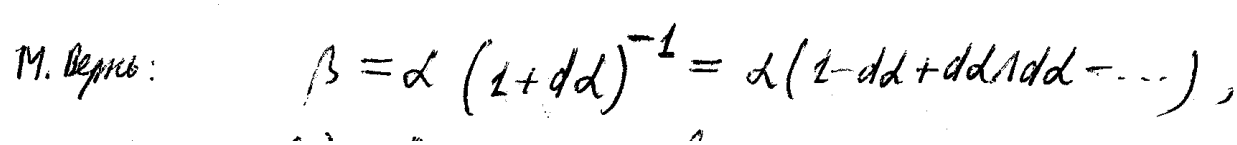
Original source details: GELFAND-P001112

Diagram and typed description.

$$\begin{vmatrix} (11) & L_\xi \beta = 0 \\ (12) & d_1 \beta = 1 \end{vmatrix}$$

this is sufficient

(13) M. Vergne:



Original source details: GELFAND-P001112

Diagram and typed description.

$$\beta = \alpha(1 + d\alpha)^{-1}$$
$$= \alpha(1 - d\alpha + d\alpha \wedge d\alpha - \ldots),$$

where α — is formula (*). Everything is easily checked.

Devise it: start with α and use perturbation theory.

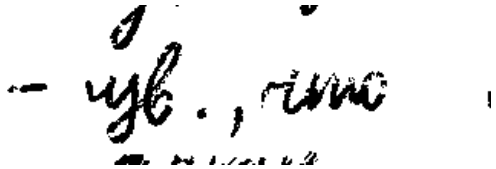
We have the identity:

$$\omega = d_1(\beta \wedge \omega) + \beta \wedge d_1\omega, \tag{14}$$

hence if $d_1\omega = 0$, then $\omega = d_1(\beta \wedge \omega)$.

Calculation of the integral over M (of a d_1-closed form): $\forall$ such integral reduces to integration over the set of fixed points. By Stokes' theorem! —

— "ув." (abbreviation unresolved), that $\omega = d_1(\beta \wedge \omega)$;



Uncertain reading. "ув." (abbreviation unresolved)

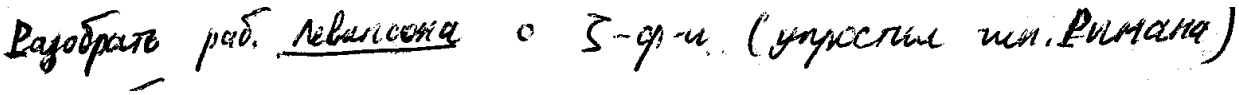
Original source details: GELFAND-P001112

the asymptotics must be calculated as we approach the fixed points.

GELFAND-P001113

Diagram and typed description. 18/IV-83 Gelfand Sem..

Original source details: GELFAND-P001113



Original source details: GELFAND-P001113

Diagram and typed description. Work through the paper by Levinson on the ζ-function. (simplified the Riemann hyp.)

V. Ginzburg.

M — a symplectic compact manifold.

Let the torus $\mathbb{T}^n$ act on it.

Moment map $M \xrightarrow{\mu} t^*$

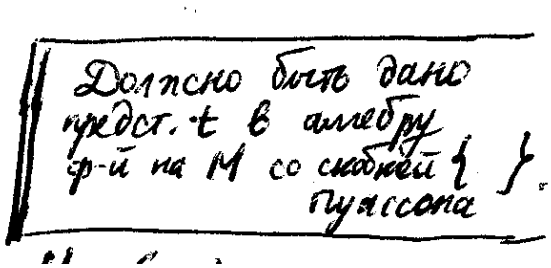


Diagram and typed description.

> A repr. must be given of t in the algebra of functions on M with the Poisson bracket $\{\}$.

Original source details: GELFAND-P001113

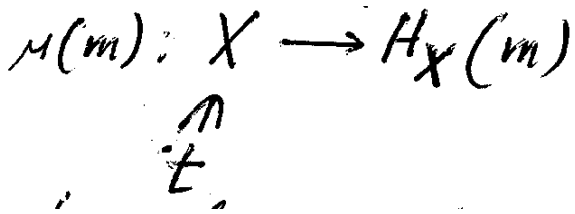


Diagram and typed description.

$$\mu(m): \quad X \longrightarrow H_X(m)$$
$$\uparrow$$
$$t$$

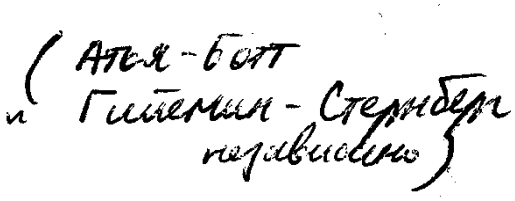
Original source details: GELFAND-P001113

H_X — the Hamiltonian fn. corresponding to the elem. X (as a vector field on M)
(we are given H_X for each X).
[in the left margin !!]

Original source details: GELFAND-P001113

Uncertain reading. abbreviation below "!!": "к." (?)

(1) Claim. $\mu(M)$ — a convex polytope.

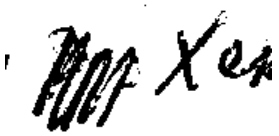


Original source details: GELFAND-P001113

Diagram and typed description. [at right] (Atiyah–Bott and Guillemin–Sternberg independently)

There is also the complex case.

(Duistermaat and [crossed out] Heckman — the fn. defined in the last talk — a piecewise polynomial on $\mu(M)$).



Original source details: GELFAND-P001113

Uncertain reading. illegible crossed-out writing before "Heckman"

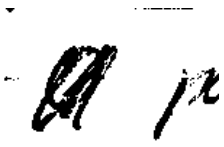
Faces (transition) — where $d\mu$ degenerates.

They are planes: apply to the degeneracy manifolds where the rank drops by several units.

A. Beilinson.

Values at integers of ζ-functions of varieties over number fields.

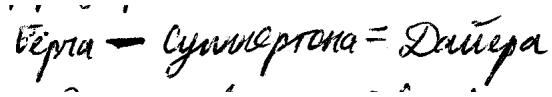
1. Statements up to multiplication by a [crossed out] rat. number $\neq 0$ (an element of $\mathbb{Q}^*$)



Original source details: GELFAND-P001113

Uncertain reading. crossed-out Q (?) before "rat."

2. Precise statements.



Original source details: GELFAND-P001113

Diagram and typed description. Conjecture of Birch — Swinnerton = Dyer

$X/\mathbb{Q}$ — a smooth curve; $\zeta(x,s)$ — its ζ-function.

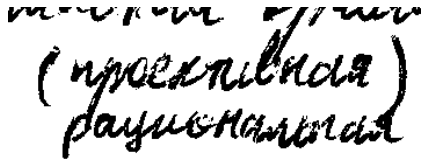
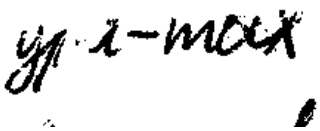


Diagram and typed description. [below "smooth curve"] (projective) rational

Original source details: GELFAND-P001113

$X/\mathbb{Z}$ "уд. л-щая" (?), satisfying (?) nondeg. reduction for any p.



Uncertain reading. "уд. л-щая" (?), satisfying (?)

Original source details: GELFAND-P001113

(the eqs. are chosen sufficiently well).

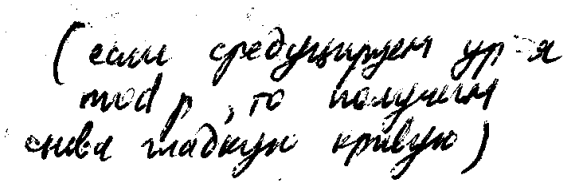


Diagram and typed description. [at right] (if we reduce the eqs. mod p, then we again obtain a smooth curve)

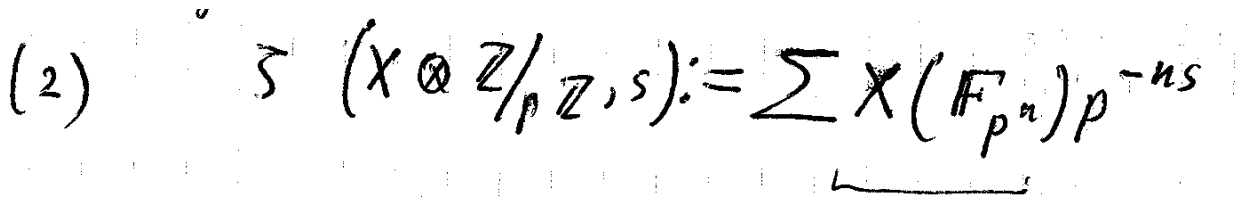
Original source details: GELFAND-P001113

GELFAND-P001114

$$(1)\quad \zeta(X,s) := \prod_p \zeta(X \otimes \mathbb{Z}/p\mathbb{Z}, s).$$

Here

$$(2)\quad \zeta(X \otimes \mathbb{Z}/p\mathbb{Z}, s) := \sum X(\mathbb{F}_{p^n}) p^{-ns}.$$



Original source details: GELFAND-P001114

Diagram and typed description. Formula (2) as written.

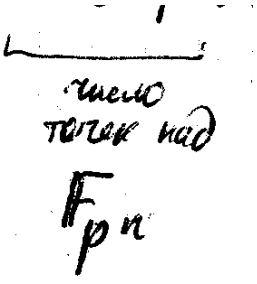


Diagram and typed description. $\underbrace{X(\mathbb{F}_{p^n})}$ number of points over $\mathbb{F}_{p^n}$

Original source details: GELFAND-P001114

Ordinary ζ-function

$$\prod \frac{1}{1-p^{-s}} = \prod (1 + p^{-s} + p^{-2s} + \ldots)$$

Here X — a point (i.e. $\mathbb{Q}$).

Diagram and typed description. — · —

Original source details: GELFAND-P001114

I. M. G.'s question.

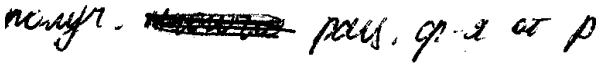
$$\prod_p \sum a_n(p) p^{-ns}$$

what are reasonable $a_n(p)$ (they are $\equiv 1$ for the ordinary ζ-function).

Weil conjecture (proved for a curve): one obtains [crossed out] a rat. function of p^{-s}

Uncertain reading. heavily crossed-out word between "obtains" and "rat."

Original source details: GELFAND-P001114

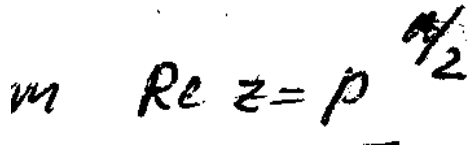


Undeciphered passage. crossed-out word illegible

Original source details: GELFAND-P001114

$$\frac{\prod (1 - \alpha_i p^{-s})}{\prod (1 - \beta_j p^{-s})}$$

Zeros at Re $z = p^{?/2}$.



Uncertain reading. overwritten numerator of the exponent: 1 or n (?)

Original source details: GELFAND-P001114

For curves even the fact of analytic continuation has not been proved for $\zeta(X, s)$.

(Over finite fields — full information [inserted below: is available]).

Diagram and typed description. inserted below: is available.

Original source details: GELFAND-P001114

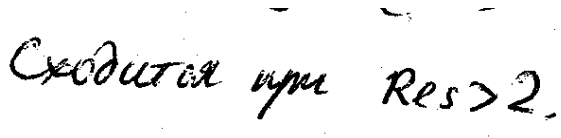
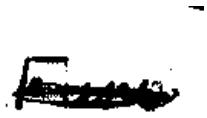


Diagram and typed description. [at right] Converges for Re $s > 2$.

Original source details: GELFAND-P001114

— question of compatibility for different p.

Known: ζ-functions of Langlands can be continued.



Uncertain reading. crossed-out word before “Conjecture”

Original source details: GELFAND-P001114

Undeciphered passage. initial stroke resembles “Б”, with the rest illegible

Original source details: GELFAND-P001114

Conjecture (strong): these 2 classes coincide, of ζ-functions.

Diagram and typed description. — · —

Original source details: GELFAND-P001114

Uncertain reading. crossed-out mark at left between the entries

Original source details: GELFAND-P001114

Wodzicki M. Which [crossed out ~~series~~ (?)] ζ-functions can be calculated?

Uncertain reading. crossed-out “series” (?)

Original source details: GELFAND-P001114

Undeciphered passage. the last letters cannot be read with confidence

Original source details: GELFAND-P001114

Shintani: Dedekind ζ-function [crossed out] reduces to ζ-functions of pseudodifferential operators.

Uncertain reading. crossed-out mark after “Dedekind”

Original source details: GELFAND-P001114

$$\zeta_{K/\mathbb{Q}}(s) = \sum N\mathfrak{a}^{-s} = \sum a_n n^{-s}.$$

Uncertain reading. heavily retraced sign after N: $\mathfrak{a}$ or a (?)

Original source details: GELFAND-P001114

Expr. via pseudodifferential operators on a torus (with constant coefficients).

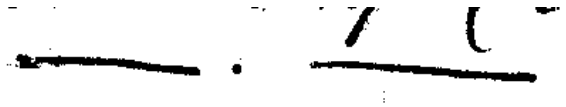


Diagram and typed description. — · —

Original source details: GELFAND-P001114

B. Birch — Swinn. = Dyer conjecture

Diagram and typed description. [in the left margin] X — a curve.

Original source details: GELFAND-P001114

1) Can be continued analytically.

2) functional eq.

$$(2\pi)^{-s}\Gamma(s)\zeta(X,s)$$

symmetric under the substitution $s \to 2 - s$

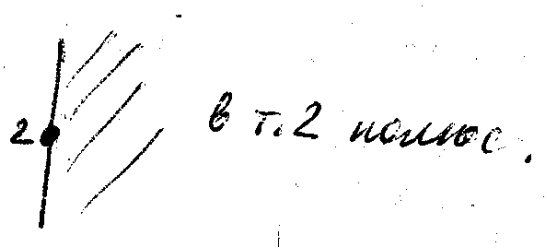
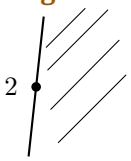


Diagram and typed description.



pole at pt. 2.

Original source details: GELFAND-P001114

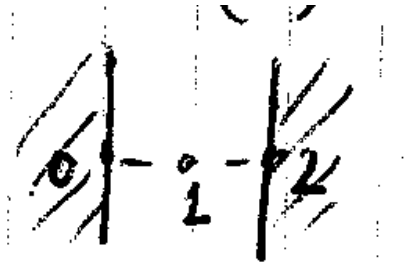
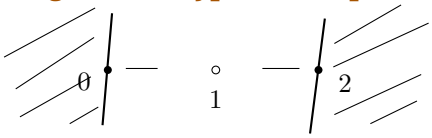


Diagram and typed description.



Original source details: GELFAND-P001114

$0 \leq \mathrm{Re}\, s \leq 2$ — critical strip.

GELFAND-P001115

Diagram and typed description. 5/IX-83 Sem. Gelfand.

Original source details: GELFAND-P001115

Diagram and typed description. Deligne.

Original source details: GELFAND-P001115

Finiteness theorems for the cohomology of families of algebraic varieties.

Work of FALTINGS and its variations. All over $\mathbb{C}$.

C — an alg. curve (compact Riem. surface)

S — a finite subset of C.

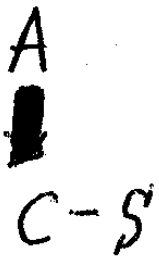


Diagram and typed description.

lower end of the connecting mark is blacked out.

Original source details: GELFAND-P001115

(family of alg. varieties parametrized by points of $C - S$)

Diagram and typed description.

Original source details: GELFAND-P001115

Néron model:

можно сделать так (после конечного накрытия C) / она: над $C \setminus S$ абелево мн-ие, а

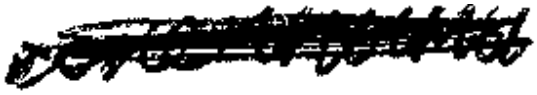
Original source details: GELFAND-P001115

Diagram and typed description. [inserted above] this can be done (after a finite cover of C).

$\mathcal{A}$ (?) over $C \setminus S$ — an abelian variety, and at S — an extension of an ab. variety by [crossed out] mult. groups $\mathbb{C}^*$.

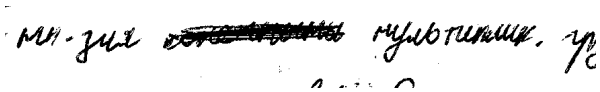
Original source details: GELFAND-P001115

Uncertain reading. inked-over letter before "over": $\mathcal{A}$ (?)



Original source details: GELFAND-P001115

Uncertain reading. long crossed-out word before "mult."



Original source details: GELFAND-P001115

Undeciphered passage. letters of the crossed-out word cannot be read with confidence

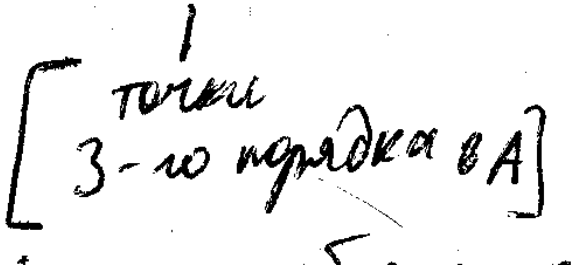
Problem — extend the bundle from $C - S$ to all of C.

$$A = \text{Lie } A/\Gamma, \qquad \Gamma \text{ — a lattice, varying from fibre to fibre.}$$

$$A_3 = \tfrac{1}{3}\Gamma/\Gamma \simeq \underline{\text{locale...}} \,.\, (\mathbb{Z}/3)^{2d}, \qquad d = \dim_{\mathbb{C}}.$$

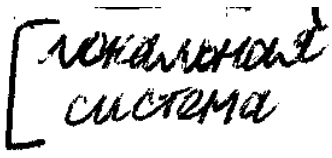
Original source details: GELFAND-P001115

Uncertain reading. "locale" is visible, with its ending heavily overwritten



Original source details: GELFAND-P001115

Diagram and typed description. [below A_3] [points of order 3 in A]



Original source details: GELFAND-P001115

Diagram and typed description. [below the Latin-script word] [local system

(any number is possible ≥ 3)

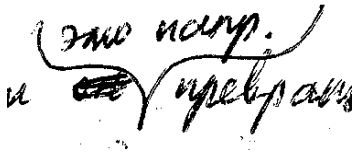
If the system is trivial, we can extend A to S.

$$(\mathbb{C}^*)^e \longrightarrow \mathcal{A} \longrightarrow \underset{b}{B}; \qquad b + e = d.$$

One period of the torus increases and $\to \infty$ (and [crossed out] [inserted: this dir.] becomes $\mathbb{C}^*$) (Néron).

Original source details: GELFAND-P001115

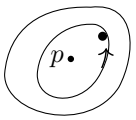
Uncertain reading. crossed-out word after "and"



Original source details: GELFAND-P001115

Diagram and typed description. inserted above this dir..

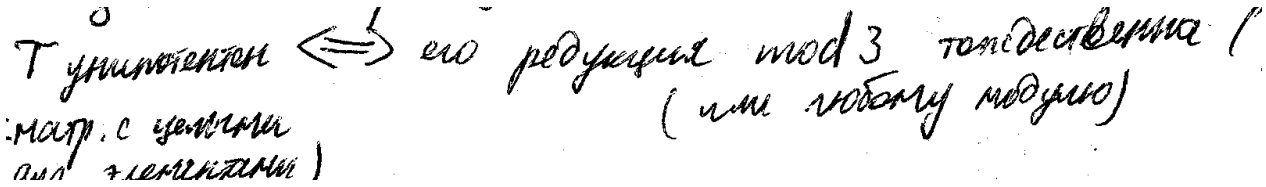
Original source details: GELFAND-P001115

Diagram and typed description. 

$T : \Gamma \to \Gamma$ (monodromy) around the point $p \in S$.

Eigenvalues of T — roots of 1.

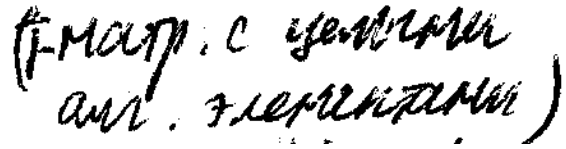
T is unipotent $\iff$ its reduction mod3 is the identity (unit) (or any modulus).



Original source details: GELFAND-P001115

Diagram and typed description. The two-way arrow and addition in the manuscript.

(Γ (?)-matrix with alg. integer entries)



Original source details: GELFAND-P001115

Uncertain reading. the initial looks like Γ; T is possible

$$\left(\bigwedge^d \underline{\text{Lie}}\right)^\vee = \omega = \text{line bundle}/C.$$

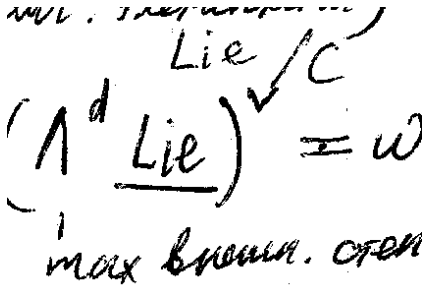
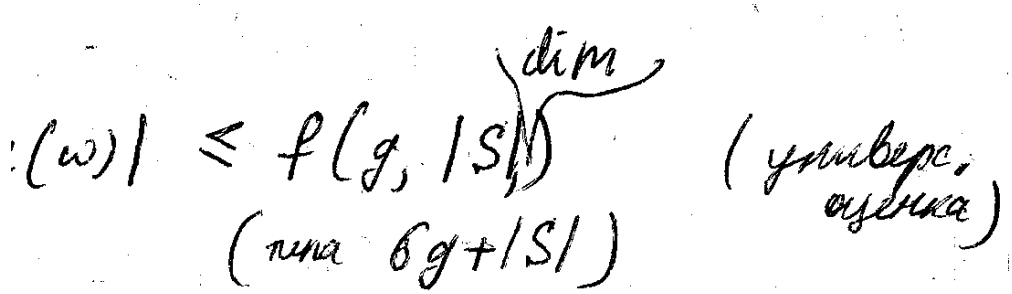


Diagram and typed description. [above] Lie $/C$.
[below $\bigwedge^d$] max exterior power.

Original source details: GELFAND-P001115

Thm. (Faltings)

$$|\deg(\omega)| \leq f(g, |S|, \overset{\text{ins.}}{\dim})$$



Original source details: GELFAND-P001115

Diagram and typed description. (universal bound)
(of the type $6g + |S|$)

Example.

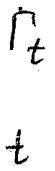


Diagram and typed description. [at left] Γ_t / t

Original source details: GELFAND-P001115

$$\pi_1(C - S, t) \longrightarrow GL(\Gamma_t)$$

This is one of the few ways to construct interesting discrete groups.

GELFAND-P001116

[source number: 76]

A way to construct a family of ab. varieties.
$C \subset$ [crossed out] $(\mathbb{P}^1)^5$

Uncertain reading. heavily crossed-out mark before $(\mathbb{P}^1)^5$

Original source details: GELFAND-P001116

$$y^N = \prod (x - x_i)^{m_i}$$

(curve depending on 5 [inserted above: distinct] points x_i)

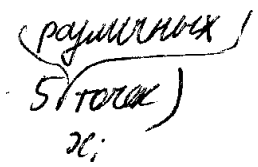


Diagram and typed description. inserted above: distinct.

Original source details: GELFAND-P001116

Take the Jacobian $\longrightarrow$ we obtain a family of abelian varieties.

(C — a curve in the set of quintuples of points)

S — the intersection of C with the set where some x_i coincide.

This is how, in $SU(2,1)$ and $SU(3,1)$ nonarithmetic subgroups were constructed...

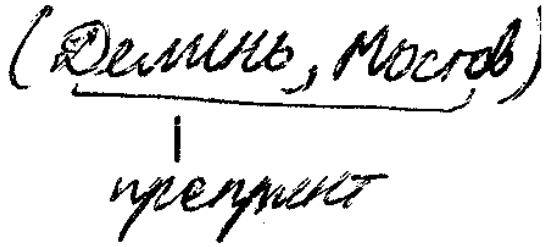


Diagram and typed description.

(Deligne, Mostow)

|

preprint

Original source details: GELFAND-P001116

This is related to hyperg. (?) functions

Uncertain reading. the abbreviation before “functions” reads “гип.” (?)

Original source details: GELFAND-P001116

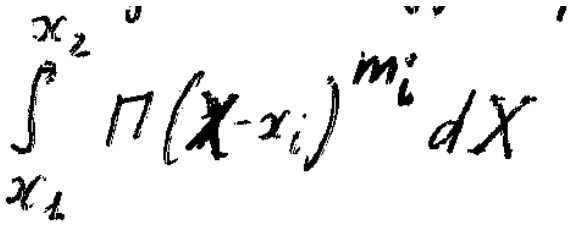


Diagram and typed description.

$$\int_{x_1}^{x_2} \prod (X - x_i)^{m_i}\, dX$$

Original source details: GELFAND-P001116

Terada (recently)

Lauricella (last century)

2 consequences of Faltings' thm.

a) Fix C, S and consider families with a principal polarization (e.g. Jacobians) $\longrightarrow$ only finitely many representations arise in this way (up to isom.)

b) vary the curve but fix the genus and the number of points. We get a repr. of another group. Still a "finite number": the curves and S are connected by a homotopy...

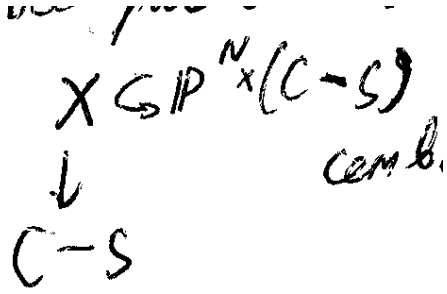


Diagram and typed description.

$$\begin{array}{c} X \subset \mathbb{P}^N \times (C-S) \\ \downarrow \\ C-S \end{array}$$

Original source details: GELFAND-P001116

a family of proj. nonsing. varieties (now nonabelian, generally speaking).

$$\pi_1(C-S,t) \longrightarrow GL\big(H_i(X_t,\mathbb{Z})/\text{Torsion}\big).$$

Diagram and typed description. inserted above: GL.

Original source details: GELFAND-P001116

c) dim H_i is bounded $\Longrightarrow$ we obtain a finite number of reprs..

Diagram and typed description. $\longrightarrow$ Hodge structure on homology.

Original source details: GELFAND-P001116

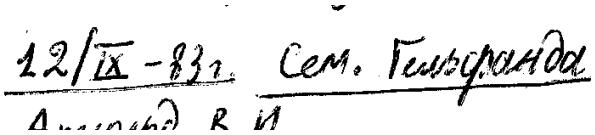


Diagram and typed description. 12/IX-83 Gelfand Sem.

Original source details: GELFAND-P001116

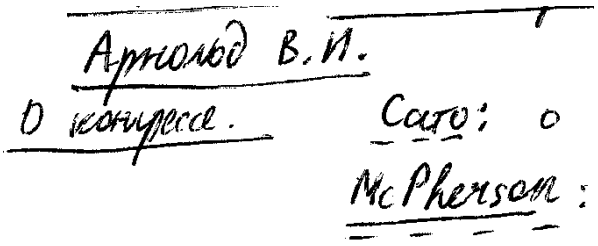


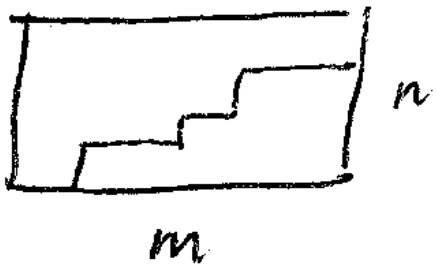
Diagram and typed description. Arnold V. I. On the congress. Sato: on correlation functions and the like
McPherson: global questions in the theory of singular spaces

Original source details: GELFAND-P001116

Stanley's talk. Combinatorial consequences of the hard Lefschetz thm.

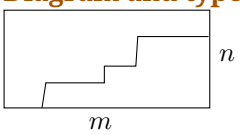
(on the hyperplane section)

$m, n \in \mathbb{Z}, \quad m, n > 0.$



Original source details: GELFAND-P001116

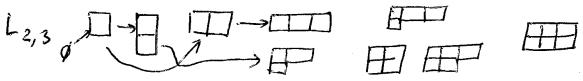
Diagram and typed description.



Consider all Young diagrams that fit inside.

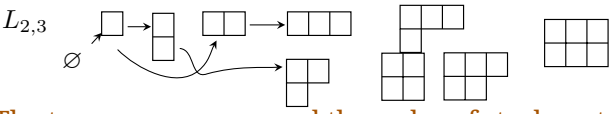
Order all the diagrams (partially, by inclusion).

$L_{m,n}$ — the resulting partially ordered set.



Original source details: GELFAND-P001116

Diagram and typed description.



The two curves cross, and the order of strokes at the crossing is unclear.

GELFAND-P001117

Width of the lattice = max. number of pairwise incomparable diagrams.

Thm. Width of the lattice $L_{m,n}$ = the number of diagrams of middle size (i.e. of size $[\frac{1}{2}mn]$).

See above; the width of $L_{2,3}$ equals 2.

(There is no simple combinatorial proof)

Proof: consider the Grassmannian $G_{m,m+n}$. Its homology is computed by means of Schubert cells (constructed by means of a frame)

(Flag; rows are dimensions of intersections with different spaces of the flag)

Cells = homology (since all the dimensions are even)

Area of the diagram = dimension of the cell.

Inclusion of diagr. = inclusion of cells.

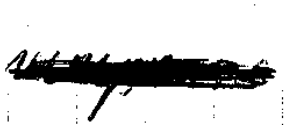
The variety is Kähler (and even projective — the Plücker map, a special case of the Veronese map for toric varieties)

$$H^i \xrightarrow{\wedge\omega^{N-i}} H^{2N-i}$$ This is an isomorphism (Lefschetz thm.)

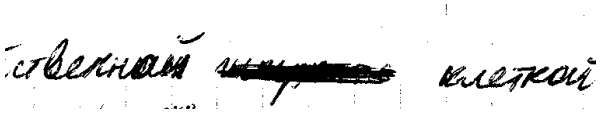
(exterior mult.)

In homology: intersection with the dual [crossed out: ~~Schubert~~ (?)] cell.

Uncertain reading. crossed-out "Schubert" (?)



Original source details: GELFAND-P001117



Original source details: GELFAND-P001117

Undeciphered passage. middle of the crossed-out word is illegible

I. M. G. Horrocks (comparatively recently) — computed [inserted above: exactly] the relation of the Chern classes (basis) to the basis obtained from Schubert cells.

Original source details: GELFAND-P001117

Diagram and typed description. inserted above: exactly.

A. 2nd part.

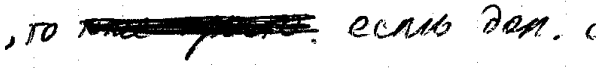
$$\alpha_0 - \alpha_1 + \alpha_2 = 2 \qquad \text{(Euler thm. for polyhedra)}$$

$$\alpha_0 - \alpha_1 + \ldots + (-1)^{d-1}\alpha_{d-1} = 1 + (-1)^{d-1}.$$

If all faces are simplicial, then [crossed out:] there are further relations.

Original source details: GELFAND-P001117

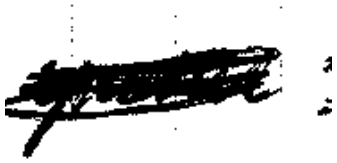
Uncertain reading. crossed-out words after “then”



Original source details: GELFAND-P001117

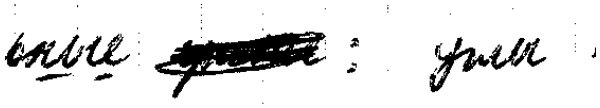
Undeciphered passage. words under the strokes cannot be read with confidence

Cosimplicial [crossed out:]: all angles are simplicial (dual to simplicial).



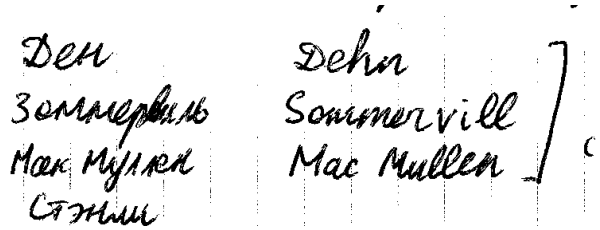
Original source details: GELFAND-P001117

Uncertain reading. crossed-out word before the colon



Original source details: GELFAND-P001117

Undeciphered passage. word beneath the strokes is illegible



Original source details: GELFAND-P001117

Diagram and typed description.

Dehn	Dehn
Sommerville	Sommervill
McMullen	Mac Mullen

Stanley

[to the right of the bracket] see Arnold's 1978 article from FA.

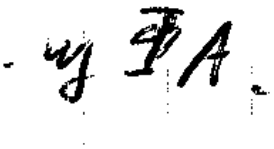


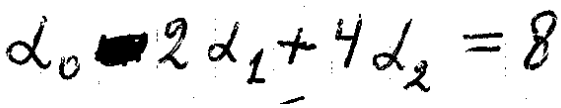
Diagram and typed description. Before "ФА" the word reads "из" (from).

Original source details: GELFAND-P001117

see references in Stanley's talk at the 1983 congress.

Mac Mullen — there is a book «Convex polyhedra»

$$\alpha_0 \,?\, 2\alpha_1 + 4\alpha_2 = 8$$



Uncertain reading. the sign before $2\alpha_1$ is covered by a solid stroke

Original source details: GELFAND-P001117

(see the cube, for example)

Def.

$$p(t) = \alpha_0 + (t-1)\alpha_1 + (t-1)^2\alpha_2 + \ldots + (t-1)^{d-1}\alpha_{d-1} + (t-1)^d$$

(Poincaré polynomial of the polyhedron)

Diagram and typed description. [crossed out: ~~Thm.~~]

Original source details: GELFAND-P001117

Dehn's thm.: 1) the polynomial $p(t)$ is reciprocal ($\Rightarrow$ identity).

2) coeffs.

Diagram and typed description.

Original source details: GELFAND-P001117

(unimodal — a hump) (Stanley)

Problem: determine whether a set of numbers can serve as the set $\alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_d$.

Thm. (Billera, Lee; another proof — Stanley) Some nec. and suff. condition.

GELFAND-P001118

[source number: 77]

h_j — coefficients.

$$h_0 = 1 = h_d, \qquad h_{d-i} = h_i \qquad \text{(reciprocity)}$$

Nec. and suff. that there exist a graded algebra (assoc., commut.)

$$R = R_0 \oplus R_1 \oplus \ldots \oplus , \qquad \text{generated by } R_1 \text{ over } R_0 = \mathbb{C},$$

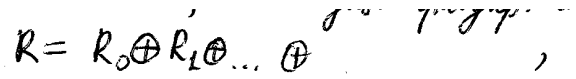


Diagram and typed description. There is a blank after the final direct sum sign.

Original source details: GELFAND-P001118

$$R_i R_j \subseteq R_{i+j}, \qquad R_0 = \mathbb{C}$$

and also

$$\dim_{\mathbb{C}} R_i = h_i - h_{i-1}, \qquad 0 \le i \le [d/2]. \quad \text{(from Lefschetz's thm.)}$$

($\Rightarrow$ unimodality, since $h_i \ge 0$)

Diagram and typed description. The manuscript has $h_i \ge 0$.

Original source details: GELFAND-P001118

Proof: from Lefschetz's thm..

From the polyhedron we construct a variety (algebraic).

Original source details: GELFAND-P001118

Diagram and typed description. [inserted between the lines] toric $\longrightarrow$ see Danilov in Uspekhi Mat. Nauk.

The dual polyhedron — simplicial.

Make it integral — approximation + dilation.

$\Rightarrow$ the variety can be constructed.

Algebra:

$$R = H^*(X)/(\omega), \qquad h_i = \dim H^{2i}(X) \qquad \text{(see Danilov)}$$

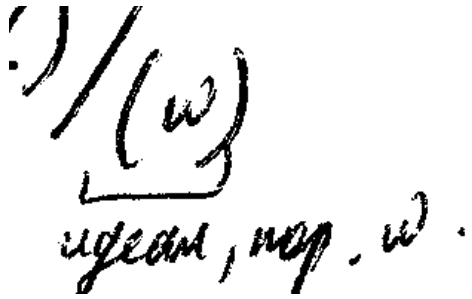


Diagram and typed description.

$\underbrace{(\omega)}$

ideal gen. by ω

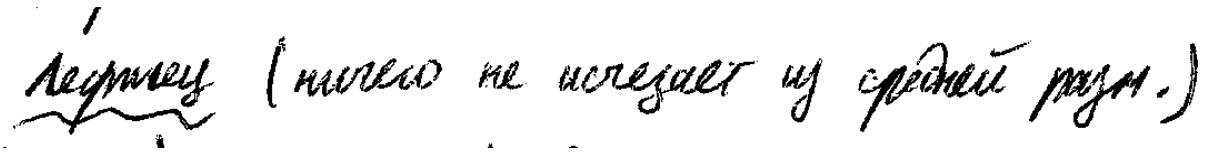
Original source details: GELFAND-P001118

It is important that simplicial ⇒ we get a Kähler variety.

Uncertain reading. mark at left resembling “7” or “4”

Original source details: GELFAND-P001118

$$\dim R_i = \dim H^{2i} - \dim(\omega H^{2i-2}) = \dim H^{2i} - \dim H^{2i-2} =$$



Original source details: GELFAND-P001118

Diagram and typed description. [below] Lefschetz (nothing disappears from the middle dim.).

There was also:

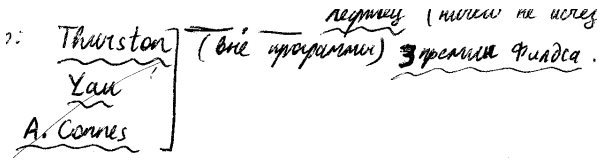


Diagram and typed description. Thurston
Yau
A. Connes
(outside the programme) 3 Fields Medals.

Original source details: GELFAND-P001118

Uncertain reading. at left, two indistinct marks before a closing parenthesis

Original source details: GELFAND-P001118

Yau — there is a survey and a list of unsolved problems (Princeton, geometry conf.).

Diagram and typed description. [below] 1–2 years ago.

Original source details: GELFAND-P001118

From the talk by Vinberg (after Mostow) it is clear these relations are the basis of the theorem of Vinberg: in Lobachevsky space of dim. ≥ 30 there are no groups gen. by reflections.

Thurston. $z \mapsto z^2 + i$

Iterations (rat. functions in general). There are works by Milnor, Douady, Sullivan. (dyn. system and so on)

Look at the image of the point 0 (crit.).

$$0 \mapsto i \mapsto -1+i \mapsto -i \mapsto -1+i \mapsto \ .$$

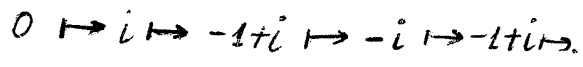


Diagram and typed description. The line in the manuscript ends with an arrow.

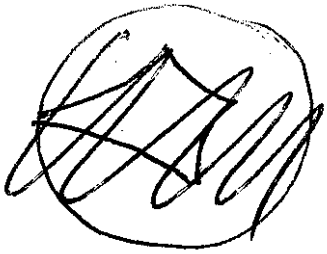
Original source details: GELFAND-P001118

The crit. point lands in a 2-cycle and stays there.

Conjecture: maps with this property are dense: the image of the crit. point becomes periodic from a certain point on.

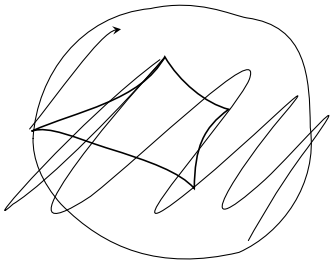
Main results — with the help of machines.

Thm. Consider a fund. domain in the Lobachevsky plane.



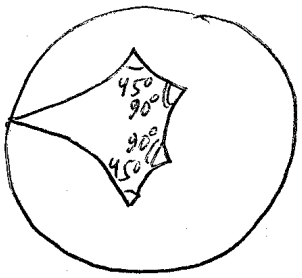
Original source details: GELFAND-P001118

Diagram and typed description.



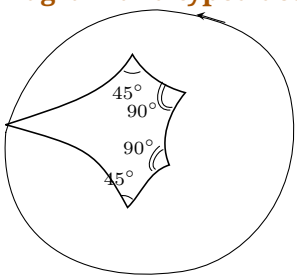
The outlines and the visible arrowhead at upper left are redrawn. The function of the overlapping diagonal strokes is unclear.

GELFAND-P001119



Original source details: GELFAND-P001119

Diagram and typed description.



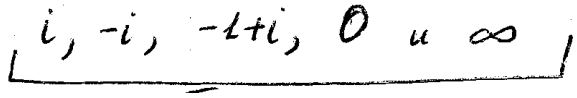
Map: there is a model with negative curvature,

Introduce a Riem. (?) metric of neg. curvature on the plane

Original source details: GELFAND-P001119

Uncertain reading. abbreviation before “metric” reads “рим.” or “рив.” (?)

without the points



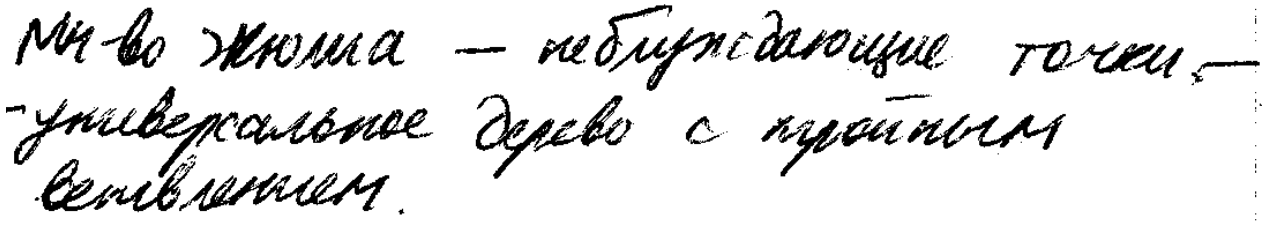
Original source details: GELFAND-P001119

Diagram and typed description.

$i,\ -i,\ -1+i,\ 0$ and ∞

(the crit. point and its images)

Take the universal cover and there a metric of const. neg. curvature.
Inverse map — expansion in this metric (uniform).
Useful — 2-sheeted cover.
On the covering surface the singular points disappear, and so on.
A tower of Riem. surfaces, and so on.
Final result:



Original source details: GELFAND-P001119

Diagram and typed description. [at right] Julia set — nonwandering points, — a universal tree with triple branching.

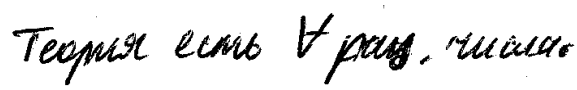


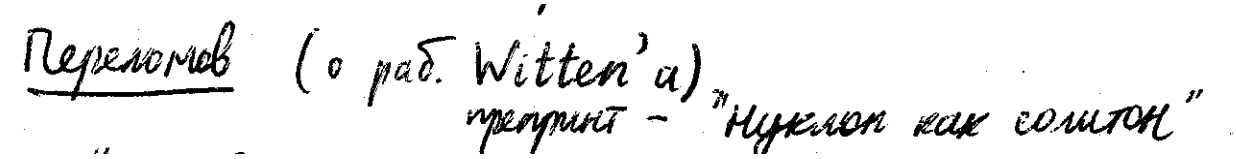
Diagram and typed description. There is a theory for ∀ rat. numbers.

Original source details: GELFAND-P001119

GELFAND-P001120

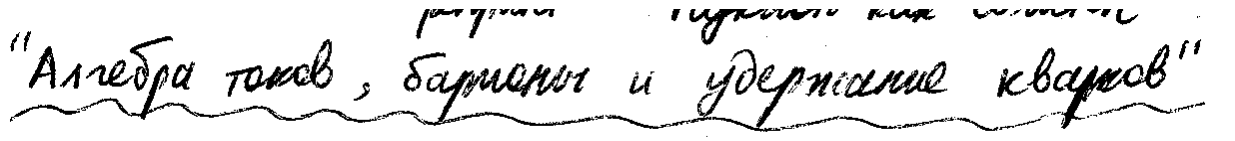
Diagram and typed description. 26/IX-83 Gelfand Sem.

Original source details: GELFAND-P001120



Original source details: GELFAND-P001120

Diagram and typed description. Perelomov (on Witten's work)
preprint — «Nucleon as a soliton»



Original source details: GELFAND-P001120

Diagram and typed description. «Current algebra, baryons, and quark confinement»

«A baryon is a particle with nonzero baryon charge»
Conservation of charges (from analysis of transformations):

Q — el. charge
B — bar. charge
ζ — strangeness
L — lepton charge

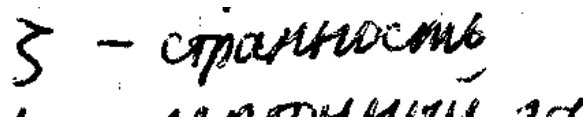
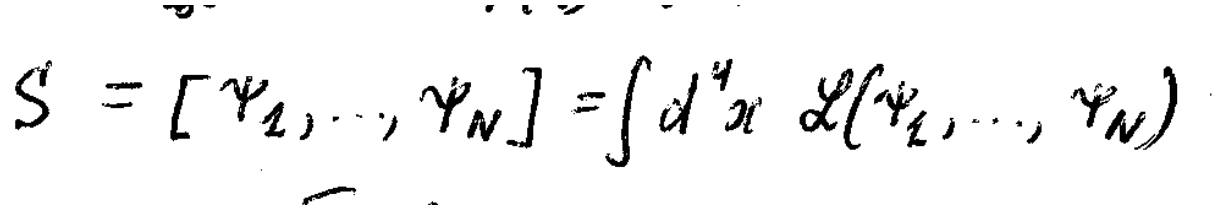


Diagram and typed description. ζ — strangeness.

Original source details: GELFAND-P001120

Before 1974 For a particle — a field $\Psi(x)$ and an action

$$S = [\Psi_1, \dots, \Psi_N] = \int d^4x\, \mathcal{L}(\Psi_1, \dots, \Psi_N).$$



Original source details: GELFAND-P001120

Diagram and typed description. The action line has an equals sign after S, before the brackets.

It is written so that all charges are conserved.
Particles enter in pairs $\Psi(x), \overline{\Psi}(x)$ (particle and antiparticle).

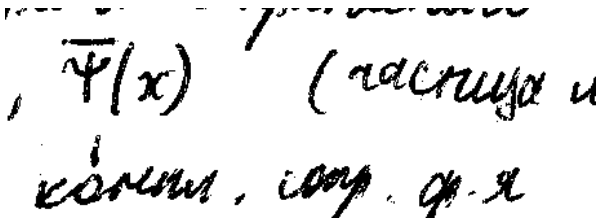


Diagram and typed description.

$\overline{\Psi}(x)$
|
complex conj. fn.

Original source details: GELFAND-P001120

$B = 1 \Rightarrow$ cannot write $\int \Psi^2(x) d^4x$; can write $\int \overline{\Psi}\Psi\, d^4x$.
(there would be a matrix elem. corresp. to a transition with creation of a baryon)
$\forall$ elem. particle $\rightsquigarrow$ field. The Lagrangian is written from general principles:

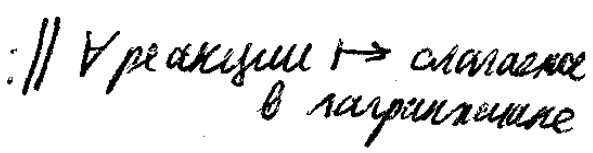
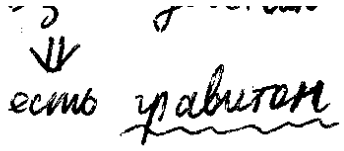


Diagram and typed description. $\|\, \forall$ reaction $\mapsto$ a summand in the Lagrangian

Original source details: GELFAND-P001120

∀ interaction also corresponds to a particle.

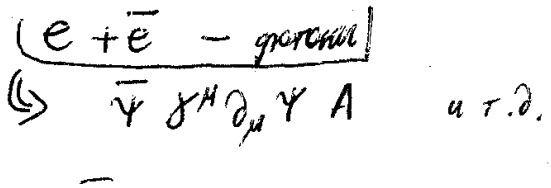


Original source details: GELFAND-P001120

Diagram and typed description.

⇓

there is a graviton



Original source details: GELFAND-P001120

Diagram and typed description.

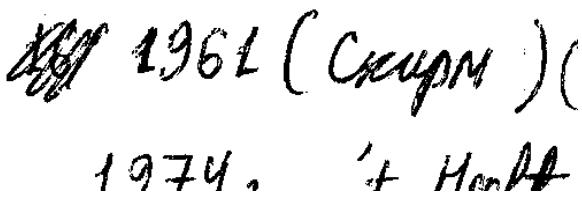
$e + \bar{e}$ — photon

$\overline{\Psi}\gamma^{\mu}\partial_{\mu}\Psi A$ and so on

[crossed out] 1961 (Skyrme) (no attention was paid to it) — there was a geom. [crossed out] structure.

Original source details: GELFAND-P001120

Uncertain reading. crossed-out digits before 1961

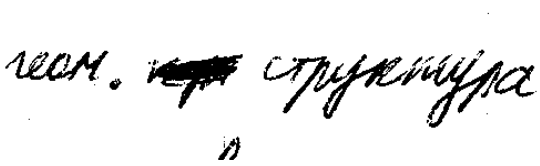


Original source details: GELFAND-P001120

Undeciphered passage. the full crossed-out date is illegible

Original source details: GELFAND-P001120

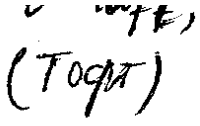
Uncertain reading. crossed-out writing before “structure”



Original source details: GELFAND-P001120

Undeciphered passage. a descending stroke is visible, but the crossed-out writing is illegible

1974 ’t Hooft, Polyakov ⇒ the geom. direction began to develop after them.



Original source details: GELFAND-P001120

Diagram and typed description. [below ’t Hooft] (Тофт).

Witten also belongs to the geom. direction.

GELFAND-P001121

[source number: 80]

The models are called chiral models or σ-models.

M^4 — Minkowski space,

$$\varphi\colon M^4 = \mathbb{R}^3 \times \mathbb{R}^1 \longrightarrow (\Phi, g).$$

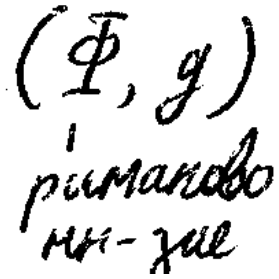


Diagram and typed description.

(Φ, g)

Riemannian manifold

Original source details: GELFAND-P001121

$\varphi(x)$ — field with values in a Riem. manifold ||

Action:

$$(1)\quad S = \int (\partial_\mu \varphi, \partial_\mu \varphi) d^4x$$
$$= \int (h_{\alpha\beta} \partial^\alpha \varphi, \partial^\beta \varphi) d^4x; \qquad \partial_\mu \varphi = \varphi_* \frac{\partial}{\partial x_\mu}.$$

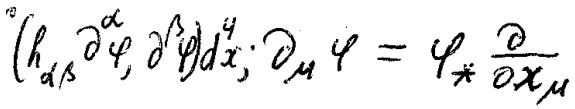


Diagram and typed description. The second integral has $h_{\alpha\beta}$, with the upper indices on ∂.

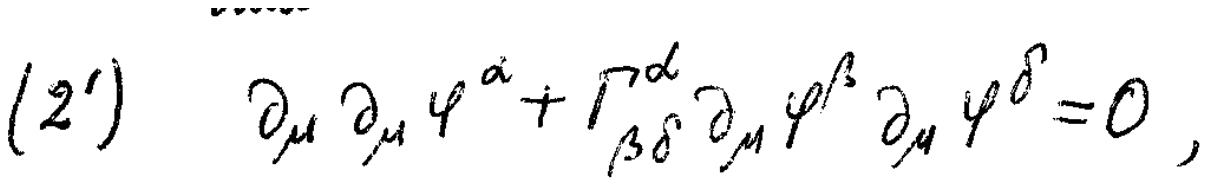
Original source details: GELFAND-P001121

Equation:

$$(2)\qquad \delta S = 0$$

or

$$(2')\quad \partial_\mu \partial_\mu \varphi^\alpha + \Gamma^\alpha_{\beta\delta} \partial_\mu \varphi^\beta \partial_\mu \varphi^\delta = 0,$$



Original source details: GELFAND-P001121

Diagram and typed description. hence this generalizes the geodesic eq..

Requirement:

$$(3) \quad \pi_3(\Phi) \neq 0, \qquad \varphi(x) = \varphi(\bar{x}, t)$$

For fixed t there is a map $\mathbb{R}^3 \to \Phi$.

$$(4) \quad \text{We require:} \quad \exists \lim_{|\bar{x}| \to \infty} \varphi(\bar{x}, t) = \varphi_0(t)$$

$$\mathbb{R}^3 \cup \{\infty\} = S^3.$$

The fields split into homot. classes (according to the map $S^3 \to \Phi$)

Let us require:

$$(5) \qquad S \geq \varepsilon \quad \text{on a homot. class}$$

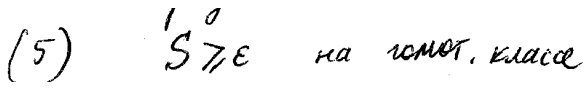


Diagram and typed description. The lower bound in (5).

Original source details: GELFAND-P001121

$\Rightarrow$ in $\forall$ homot. class there is a solution.

Examples.

(6) Φ — a simple [inserted above: compact] Lie group $\Phi = G$ (principal chiral field)

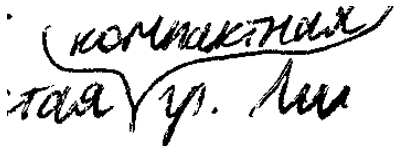


Diagram and typed description. inserted above: compact.

Original source details: GELFAND-P001121

or

(7) Φ — orbit of the adj. repr. of a compact Lie group $\mathfrak{g}$ (?)

Uncertain reading. mark after “Lie” resembles $\mathfrak{g}$ and may be crossed out

Original source details: GELFAND-P001121

(8) $\Phi = S^2$ — $\bar{n}$-field

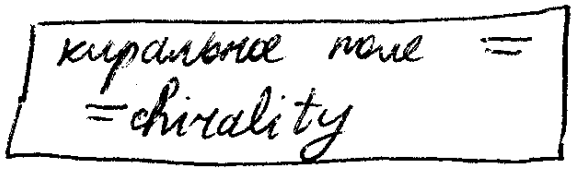
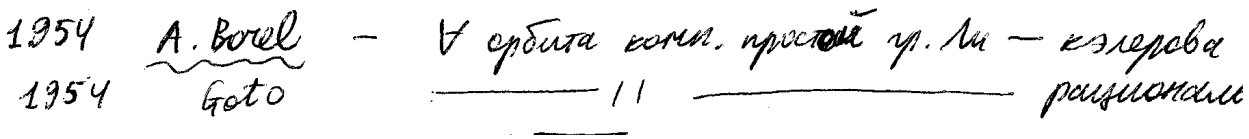


Diagram and typed description.

chiral field =
= chirality

Original source details: GELFAND-P001121



Original source details: GELFAND-P001121

Diagram and typed description. 1954 A. Borel — $\forall$ orbit of a comp. simple Lie group — is Kähler.

1954 Goto —— || —— is rational

Diagram and typed description. — · —

Original source details: GELFAND-P001121

2-dimensional case:

$$S^2 \to \Phi, \quad \Phi = G/H \qquad (G \text{ — simple compact})$$

$$\pi_2(\Phi) = \pi_1(H) \neq 0$$

Pass to a Kähler metric. The eqs. become the Cauchy–Riemann eqs.:

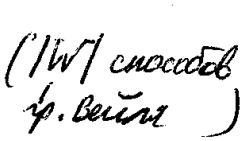
GELFAND-P001122

$$(9) \qquad \partial_{\bar{z}}\varphi = 0 \qquad (\Leftrightarrow \ \varphi \text{ is holomorphic})$$

(i.e. the solns. are holom. maps)

(Complex structure: $G/H \sim$ the flag space; in several ways ($|W|$ ways))

Diagram and typed description. [below W] Weyl group.



Original source details: GELFAND-P001122

Matr. = (triang.) × (unitary)

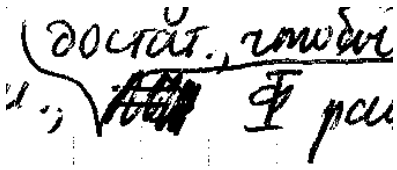
Φ is compact [crossed out]; it is unknown whether there are nonconstant solutions.

Uncertain reading. the crossed-out mark after "compact" is illegible

Original source details: GELFAND-P001122

To have a large number of solns.; [crossed out] [inserted above: (it suffices that)] Φ be rational (i.e. the field of fns. on Φ is isomorphic to the field of rat. fns. of some number of variables).

Uncertain reading. the crossed-out writing below the insertion is illegible



Original source details: GELFAND-P001122

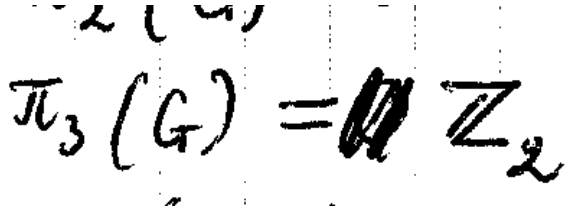
$$\mathbb{C}P^1 \longrightarrow \Phi \quad \text{(a Kähler manifold)}$$

$\Rightarrow$ there are many instanton solutions

$\Phi = S^2$ — Belavin, Polyakov. (from stereogr. projection)

$\Phi = G$ — a simple compact Lie group.

$$\pi_1(G) = 0$$
$$\pi_2(G) = 0$$
$$\pi_3(G) = \text{[crossed out] } \mathbb{Z}_2$$



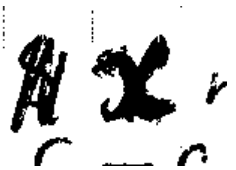
Uncertain reading. a mark resembling 0 is crossed out before $\mathbb{Z}_2$

Original source details: GELFAND-P001122

$$\pi_4(SU(3)) = 0, \quad \pi_4(SU(2)) = \mathbb{Z}_2, \quad \pi_5(SU(2)) = \mathbb{Z}_2.$$

There is no lower bound:

on replacing [crossed out] **x** by λ**x**: S by $\lambda^2 S \Rightarrow S$ can be made arbitrarily small.



Uncertain reading. the mark before **x** is crossed out, possibly φ

Original source details: GELFAND-P001122

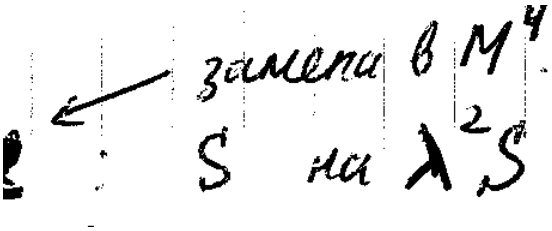


Diagram and typed description.
substitution in M^4

$\swarrow$ λ**x**

Original source details: GELFAND-P001122

(10) In fact $S = S_0 + S_1 + S_2$

$$(11) \quad S_0 = \int \operatorname{tr}(g^{-1}\partial_\mu g \cdot g^{-1}\partial_\mu g)\, d^4x$$

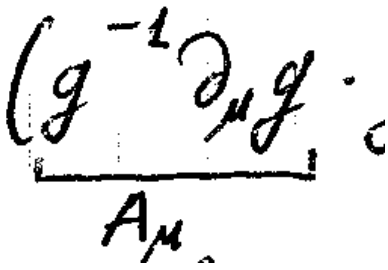


Diagram and typed description. $\underbrace{g^{-1}\partial_\mu g}_{A_\mu}$

Original source details: GELFAND-P001122

for unit. matrices ($SU(3)$ or $SU(2)$)

Diagram and typed description. [at right] A_μ — elem. of the Lie algebra.

Original source details: GELFAND-P001122

$$(12) \quad S_2 = \int d^4x \ \mathrm{tr}\big([A_\mu, A_\nu] : [A_\mu, A_\nu]\big)$$

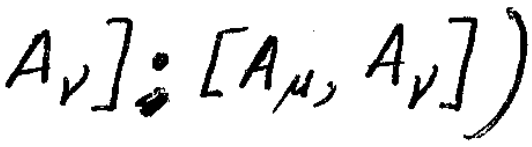
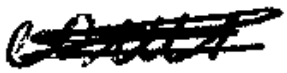


Diagram and typed description. Two dots are visible between the commutators in (12).

Original source details: GELFAND-P001122

Derivs. in t enter quadratically $\to$ already invariant under homothety in M^4

$S_0 + S_2$ already has [crossed out ~~soli...~~ (?)] instanton solutions.



Uncertain reading. the crossed-out beginning may read "soli..."

Original source details: GELFAND-P001122

Undeciphered passage. the remainder of the crossed-out word is illegible

Original source details: GELFAND-P001122

S_1 is added so as to be able to distinguish bosons from fermions

Wess, Zumino (1970) added S_1, but the formula is complicated

Witten greatly simplified this formula for S_1

Compactify time:

$$M^4 = S^3 \times S^1 \longrightarrow G = SU(3)$$
$$\|$$
$$\partial Q, \quad Q = S^3 \times D^2$$

(13) Claim. The map extends to a continuous map of Q

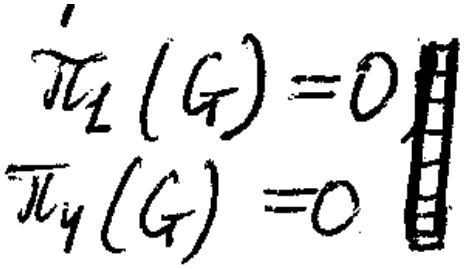


Diagram and typed description. (from the fact that $\pi_1(G) = 0$ $\pi_4(G) = 0$)

Original source details: GELFAND-P001122

[at right] Rewrite as a map

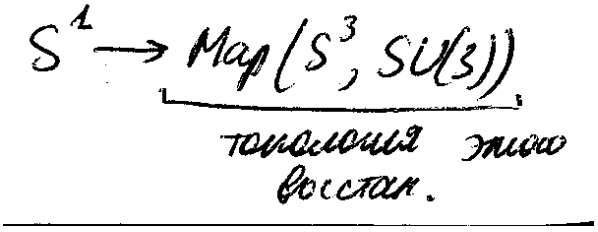


Diagram and typed description.
$S^1 \longrightarrow \underbrace{\mathrm{Map}(S^3, SU(3))}_{\text{the topology of this reconstr.}}$

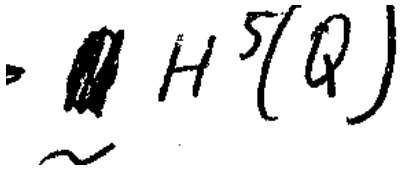
Original source details: GELFAND-P001122

GELFAND-P001123

[source number: 81]

We obtain:

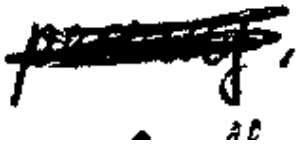
$$\varphi\colon Q \longrightarrow SU(3)$$
$$\varphi^*\colon H^5(SU(3)) \longrightarrow \text{[crossed out] } H^5(Q)$$
$$\omega \longmapsto \widetilde{\omega}$$



Uncertain reading. a mark resembling 0 is crossed out before $H^5(Q)$

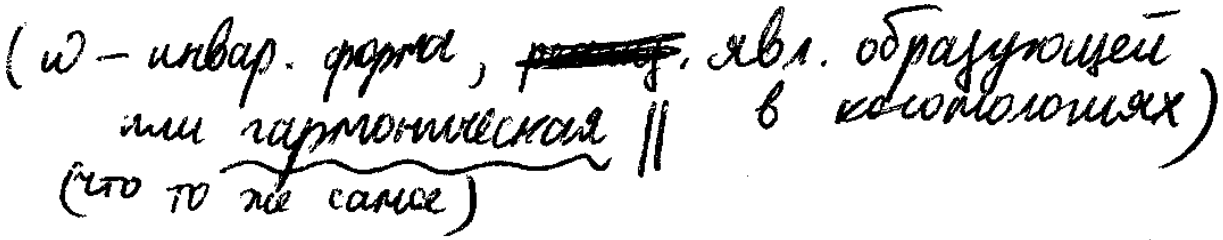
Original source details: GELFAND-P001123

[at right] (ω — an inv. form, [crossed out], generating the cohomology)



Uncertain reading. crossed-out writing after "inv. form"

Original source details: GELFAND-P001123



Original source details: GELFAND-P001123

Undeciphered passage. lower loops are visible, but the crossed-out word is illegible

or harmonic ‖ (which is the same thing)

$$(14) \qquad S_1 = N \int_Q \widetilde{\omega}$$

N is chosen to get an integer.

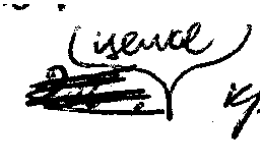
The choice of extension [inserted: φ] may change [crossed out] $\int$ by an [crossed out 2π (?); inserted: (integer)] multiple of 2π.

Diagram and typed description. inserted: φ.

Original source details: GELFAND-P001123

Uncertain reading. the crossed-out mark before the integral is illegible

Original source details: GELFAND-P001123



Uncertain reading. the crossed-out mark resembles 2π, with “integer” inserted above

Original source details: GELFAND-P001123

Cohomology of $SU(3)$: $(1+t^3)(1+t^5)$

Under deformations of [inserted: (the extn. of φ to)] Q the integral is unchanged

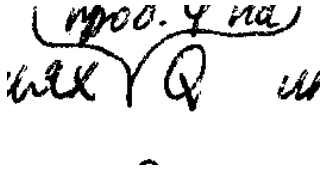


Diagram and typed description. inserted: (the extn. of φ to Q).

Original source details: GELFAND-P001123

$$(15) \qquad \int_Q \widetilde{\omega} = \int_{\varphi(Q)} \omega$$

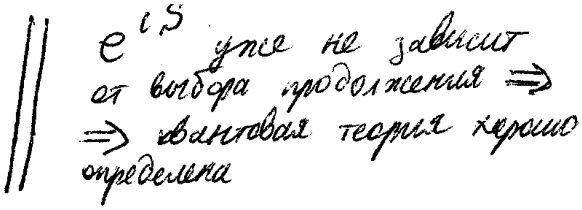


Diagram and typed description. [at right] ‖ e^{iS} is already independent of the choice of extension ⇒
⇒ the quantum theory is well defined

Original source details: GELFAND-P001123

Zumino — d-dimensional case already (preprint) and secondary char. classes of Chern–Simons appear there.

1) Baryon charge.

$$\pi_3(\Phi) = \mathbb{Z} \quad \text{and by Hurewicz's thm.} \quad H^3(\Phi) = \mathbb{Z}.$$

Expr. for baryon charge:

$$(16) \quad B = \int d^3x\, B_0 = \frac{1}{24\pi^2} \int d^3x\, \varepsilon^{ijk}\, \mathrm{tr}(A_i A_j A_k)$$

(Witten proposes calling this the baryon charge)

(invariant 3-form, the unique generator in $H^3(\Phi)$)

2) Fermion or boson?

$$\int e^{iS[\varphi]} d\varphi \qquad \text{— transition amplitude}$$

Suppose it is stationary $\Rightarrow$ the integral equals e^{-iMT}, M — mass, T — time

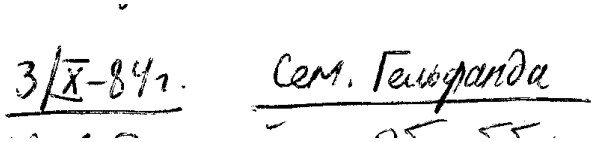


Diagram and typed description. 3/X-84 Gelfand Sem.

Original source details: GELFAND-P001123

Yu. L. Daletskii On a generalization of the Vlasov eq.

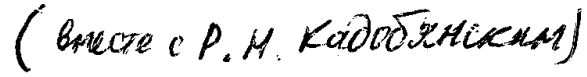
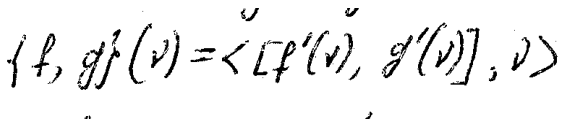


Diagram and typed description. (with R. N. Kadobianskii)

Original source details: GELFAND-P001123

$\mathfrak{g}$ — Lie alg., $\mathfrak{g}'$ — dual space, $\langle\,\rangle$ — duality.

$$\{f, g\}(\nu) = \langle [f'(\nu), g'(\nu)], \nu \rangle \qquad (f, g \text{ — fns. on } \mathfrak{g}')$$



Uncertain reading. the argument letter resembles ν or v

Original source details: GELFAND-P001123

$H(\nu)$ — a fn. on $\mathfrak{g}'$

$$\frac{\partial u}{\partial t} = \{u(\nu), H(\nu)\}$$

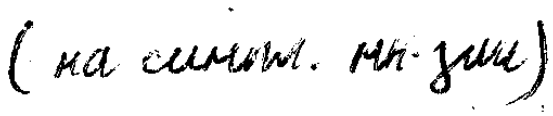
$$(*) \quad \frac{d\nu}{dt} \pm [\mathrm{ad}^*\, H'(\nu)]\nu = 0 \qquad (\text{adjoint eq.})$$

Uncertain reading. the sign is retraced, with a separate lower stroke beneath the crossbar; it may be $\pm$ or a corrected $+$

Original source details: GELFAND-P001123

GELFAND-P001124

$\mathfrak{g}$ is the Lie alg. of functions $\varphi(x)$, $x \in X$ (on a symplectic manifold).



Uncertain reading. manifold

Original source details: GELFAND-P001124

The dual space consists of measures $\nu(dx)$.

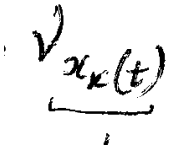
$h_1(x)$ is the particle energy.

$h_2(x_1, x_2)$ — // — of interaction.

$$H(\nu) = \int_X h_1(x)\nu(dx) + \frac{1}{2}\int_X \int_X h_2(x, x')\nu(dx)\nu(dx').$$

$$X^n \qquad y = (x_1, \ldots, x_n)$$

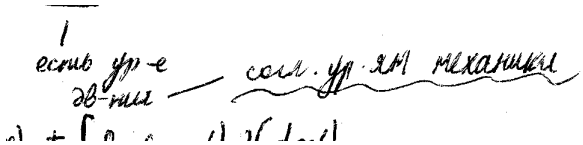
Claim. $\nu = \sum_{k=1}^{n} \nu_{x_k(t)}$ is a solution of the eq. (Klimontovich).



Uncertain reading. $\nu_{x_k(t)}$

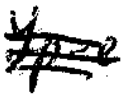
Original source details: GELFAND-P001124

[Under $x_k(t)$ a brace and an annotation:]



Uncertain reading. is the eq. of motion — according to the eqs. of mechanics.

Original source details: GELFAND-P001124



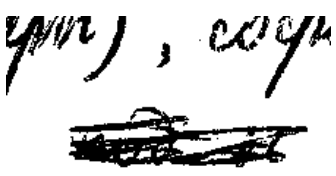
Uncertain reading. [Crossed out on the left: ~~Eq.?~~.]

Original source details: GELFAND-P001124

$$H'(\nu) = h_1(x) + \int h_2(x, x')\nu(dx').$$

One gets a flow on the space of generalized functions.

Generalized functions (measures) supported on a finite number of points. It turns out that [crossed out:] functions supported on n points [crossed out:] are invariant under the flow. This is equivalent to the Vlasov eq.



Uncertain reading. crossed out: illegible

Original source details: GELFAND-P001124

Uncertain reading. crossed out: illegible

Original source details: GELFAND-P001124

$$\nu(dx) = C(q,p)\,dp\,dq$$

$$\frac{\partial C}{\partial t} + \frac{p}{m}\frac{\partial C}{\partial q} = \frac{\partial C}{\partial p}\left(V'(q) + \int \frac{\partial \Phi(q,q')}{\partial q} C(q',p')\,dq'\,dp'\right).$$

(Vlasov eq. — collisionless plasma).

$$h_1(x) = \frac{p^2}{2m} + V(q); \qquad h_2(x,x') = \Phi(q,q').$$

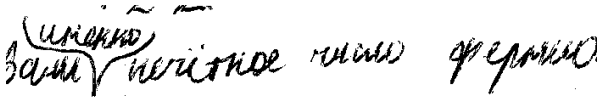
All this holds for Lie superalgebras (see the Berezin–Marinov eqs.).

A. Beilinson. On a paper by Witten.

There is no gauge $SU(2)$ theory with an odd number of fermions

Reason: $\pi_4(SU(2)) = \mathbb{Z}/2\mathbb{Z}$.

(Previously physicists considered [inserted above: specifically] an odd number of fermions)

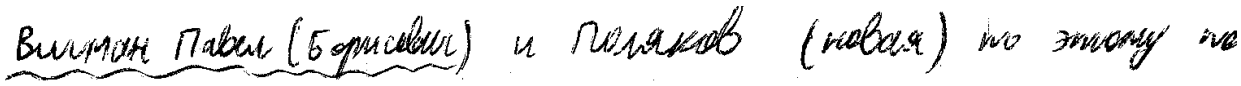


Uncertain reading. [inserted above: specifically] an odd number of fermions

Original source details: GELFAND-P001124

$$L = \int d^4x \left(\mathrm{Tr}\, F_{\mu\nu}^2 + \overline{\Psi} i D \Psi\right)$$

There is a paper by Wiegmann, Pavel (Borisovich), and Polyakov (a new one) on this subject.



Original source details: GELFAND-P001124

Uncertain reading. Wiegmann, Pavel (Borisovich),

$$\int d\Psi\, d\overline{\Psi}\, dA\, \exp(-L)$$

Uncertain reading. [Crossed out between exp and $(-L)$: illegible mark.]

Original source details: GELFAND-P001124

$$\pi_4(SU(2)) = \pi_0\left(SU(2)^{S^4}\right).$$

GELFAND-P001125

[source number: 82]

Fuchs:

Suspension: $\pi_3(S^2) \longrightarrow \pi_4(S^3)$ — an epimorphism with kernel $2\mathbb{Z}$.

I.M.G. Donaldson: on $\mathbb{R}^4$ there are smooth structures that depend on a parameter and are different for different values of the parameter!!

Fuchs:

$$0 \longrightarrow \mathbb{Z} \longrightarrow \pi_{4n-1}(S^{2n}) \xrightarrow{\text{epi}} \pi_{4n}(S^{2n+1}) \longrightarrow 0$$

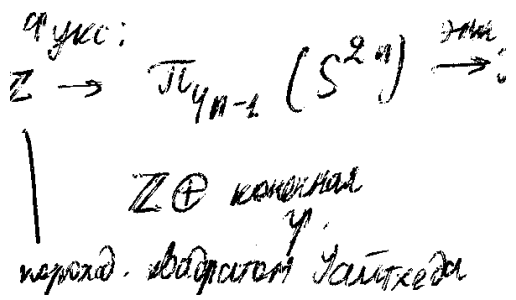


Diagram and typed description. [Under $\pi_{4n-1}(S^{2n})$] $\mathbb{Z} \oplus$ a finite group.
[Leader from $\mathbb{Z}$] generated by the Whitehead square.

Original source details: GELFAND-P001125

B. Theorem (Witten).

$\int d\Psi$ is interpreted as $(\det D)^{-1/2} \exp(\dots)$.

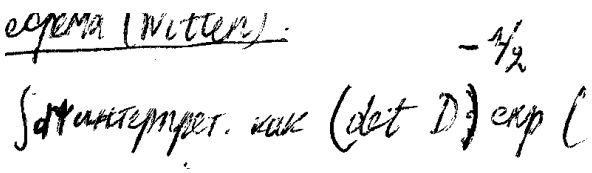


Diagram and typed description. Exponent in the manuscript $-1/2$.

Original source details: GELFAND-P001125

(A fixed)

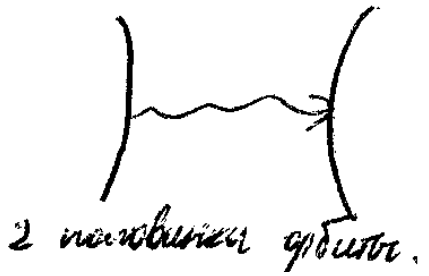
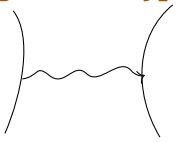


Diagram and typed description.

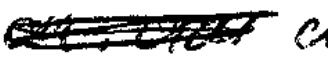


2 halves of the orbit.

Original source details: GELFAND-P001125

$\sqrt{\det D}$ will change sign, since some [crossed out] eigenvalues will change sign on passing through $0 \rightsquigarrow$

Uncertain reading. The crossed-out word is illegible.



Original source details: GELFAND-P001125

$\rightsquigarrow$ we have to count the zero modes of the Dirac operator on $S^4 \times S^1$.

The eigenvalues split into pairs [under the word "pairs": $(\lambda, -\lambda)$]; $\sqrt{\det}$ will change sign on passing through a zero mode.

Then one uses the Atiyah–Singer theorem. It gives the number of zero modes $\bmod 2$.

I.M.G. A general problem for mathematicians:

Use operators to do what is done with Feynman integrals.

Example. σ-model, simple (see Perelomov's earlier talk) $\longrightarrow$ write down an operator.

Diagram and typed description. simple.

Original source details: GELFAND-P001125

GELFAND-P001126

Diagram and typed description. 17 October 1983. Gelfand Seminar.

Original source details: GELFAND-P001126

A. Beilinson. On Witten's work

$$(1) \qquad S = \int (F, F) + i(\Psi, \mathcal{D}\Psi), \qquad F \text{ — curvature of the connection } A$$

Uncertain reading. [Before S an illegible mark is crossed out.]
[From S a leader points down to "action".]

Original source details: GELFAND-P001126

$$\mathcal{D} : \Gamma(E \otimes S) \longrightarrow \Gamma(E \otimes \bar{S}) \quad \text{— Dirac operator}$$

S — canonical spinor bundle on a 4-dimensional manifold.

Diagram and typed description. (2-dimensional)

Original source details: GELFAND-P001126

Ψ — 2-component spinor.

e^{iS} — a function on the space of pairs (A, Ψ), gauge invariant

$$(2) \qquad \int e^{iS}\, d(A, \Psi) \text{ — ?}$$

Claim. $(2) = 0$, if the number of components of Ψ is odd.

Consider.

$$(3) \qquad \int \left(\int e^{-S}\, d\Psi \right) dA \qquad \text{(integral over the set of orbits)}$$

Each orbit splits into 2 pieces. [Crossed out.] With any natural definition the function takes values of opposite sign on these components! $\Rightarrow$ hence $(2) = 0$.

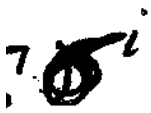
Uncertain reading. The crossed-out writing is illegible.

Original source details: GELFAND-P001126

$$(4) \qquad \int e^{-S}\, d\Psi = \sqrt{\det \mathcal{D}} \quad \text{— a function of } A_i$$

$$(5) \qquad \mathcal{D} = \sum \sigma^i \left(\frac{\partial}{\partial x_i} + A_i \right),$$

σ^i — matrices of order 2 (Pauli)



Uncertain reading. Overwritten symbol read as σ^i.

Original source details: GELFAND-P001126

$$[\sigma_1, \sigma_2] = i\sigma_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

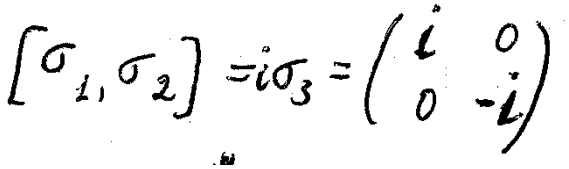


Diagram and typed description. $[\sigma_1, \sigma_2] = i\sigma_3$.

Original source details: GELFAND-P001126

$$(6) \qquad \pi_0 \left(SU(2)^{S^4} \right) = \pi_4(SU(2)) = \pi_4(S^3) = \mathbb{Z}_2$$

GELFAND-P001127

[source number: 84]

Consider $S^4 \times S^1$ and on it a bundle E with a connection.

We deform the connection [written underneath: A] into another [written underneath: A'] so that A and A' are related by a transformation from the 2nd component of the gauge group.

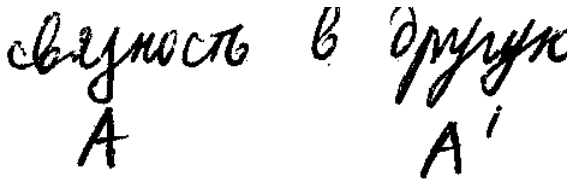


Diagram and typed description. [written underneath: A, A'.]

Original source details: GELFAND-P001127

(Connections form an affine space)

Then $\sqrt{\det \mathcal{D}}$ changes sign!

$\sqrt{\det \mathcal{D}}$ takes into account half the eigenvalues. [Unclear mark at right.] Eigenvalues occur in pairs $(\lambda, -\lambda)$.

Uncertain reading. The mark before "Eigenvalues" looks like a cancelled letter or an asterisk.

Original source details: GELFAND-P001127

An odd number of eigenvalues go from positive to negative and vice versa.

— · —

Dirac eq.

$$\sum \gamma^i \frac{\partial}{\partial x_i} + A_i(x_0, \ldots, x_3, t) \tag{7}$$

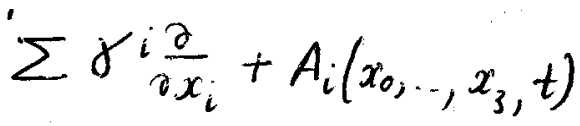


Diagram and typed description. Equation (7) as written, without additional parentheses.

Original source details: GELFAND-P001127

(Along S^4 it is the same as before)

t — the parameter for deforming the connection; let the start be the same as the end (gauge equiv.)

The bundle on $S^4 \times S^1$ is trivial on $S^4 \times [0, 1)$ and then glue using the chosen element of the gauge group $\Rightarrow$ it becomes nontrivial.

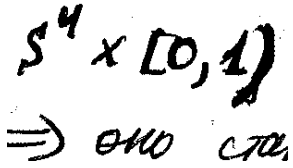


Diagram and typed description. $S^4 \times [0, 1)$.

Original source details: GELFAND-P001127

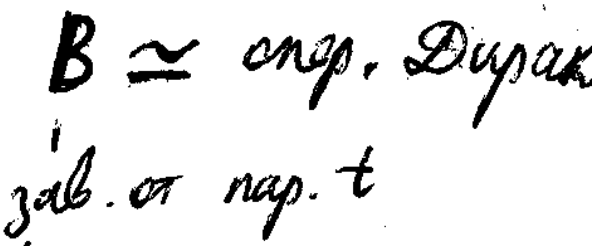
$$(8) \qquad \left(\sum \gamma^i \frac{\partial}{\partial x_i} + A_i(x_0, \ldots, x_3, t)\right) \Psi = 0$$

regard this as an evolution equation

$$(9) \qquad \frac{\partial}{\partial t}\Psi = B\Psi, \quad \text{where } B \simeq \text{ Dirac operator on a 4-dimensional manifold.}$$

Original source details: GELFAND-P001127

Uncertain reading. At the beginning of eq. (9) illegible marks are crossed out.



Original source details: GELFAND-P001127

Diagram and typed description. [Under B: depends on the parameter t.]

Let B be independent of t, and replace the interval by the line.

Choose as Ψ an eigenfunction with eigenvalue $\lambda > 0$.

$$\begin{aligned} t &\longrightarrow -\infty \quad \Rightarrow \quad \text{function} \longrightarrow 0. \\ t &\longrightarrow +\infty \quad \Rightarrow \quad \text{the function grows} \end{aligned}$$

It would be good if it decayed as $t \longrightarrow \pm\infty$.

We need: the eigenvalue to change sign.

A solution with these boundary conditions $\approx$ a function on the circle.

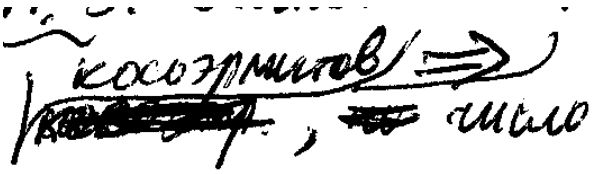
$\Rightarrow$ from an eigenvalue that changes sign one constructs a solution of the Dirac eq..

Thus the number of solutions to (8) on $S^4 \times S^1$ is related to changes in sign of the eigenvalues of operator (7).

Witten says «we will vary it adiabatically» (so that we have an eigenfunction throughout)

The A–S theorem counts mod2 the solutions of eq. (8) on $S^4 \times S^1$.

Operator [crossed out; inserted above: skew-Hermitian $\Rightarrow$], [crossed out] the number of solutions mod2 — a topological invariant.



Original source details: GELFAND-P001127

Uncertain reading. [crossed out; inserted above: skew-Hermitian $\Rightarrow$.]

Uncertain reading. The crossed-out writing before “the number” is illegible.

Original source details: GELFAND-P001127

I.M.G. Problem. Devise a general mathematical framework in which this is a special case.

GELFAND-P001128

A. Zelevinsky.

$\Lambda_n = \mathbb{Z}[x_1, \ldots, x_n]^{S_n}$ — symmetric polynomials

Λ_n^k — homogeneous [inserted above: symm.] polynomials of degree k.

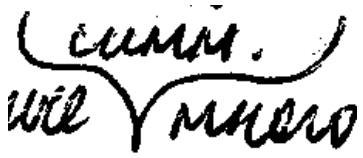


Diagram and typed description. [inserted above: symm.]

Original source details: GELFAND-P001128

$n \longrightarrow \infty \quad \Rightarrow \quad \Lambda_n^k$ ceases to depend on n.

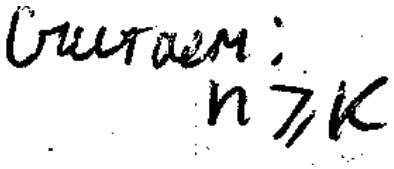


Diagram and typed description. [On the left:] Assume $n \geq k$.

Original source details: GELFAND-P001128

Λ — symm. polynomials in an “inf.” number of variables (formalized by a projective limit)

$$\begin{array}{rl} \text{1st degree} & x_1 + x_2 + \ldots = m_{(1)} \\ \text{2nd degree} & \begin{cases} x_1^2 + x_2^2 + \ldots = m_{(2)} \\ x_1x_2 + x_1x_3 + x_2x_3 + \ldots = m_{(1^2)} \end{cases} \end{array}$$

$$(1) \qquad \lambda = (\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq \ldots \geq 0)$$

(an inf. sequence that breaks off somewhere)

$(2) \qquad m_\lambda = x_1^{\lambda_1} x_2^{\lambda_2} \ldots +$ monomials obtained by permuting the letters

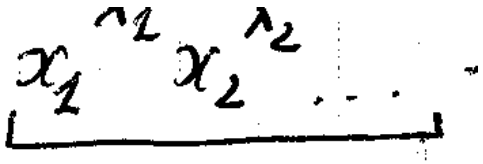


Diagram and typed description. Straight bracket below $x_1^{\lambda_1} x_2^{\lambda_2} \ldots$.

Original source details: GELFAND-P001128

Each is an integral linear combination, so, m_λ — a basis.

(monomial basis)

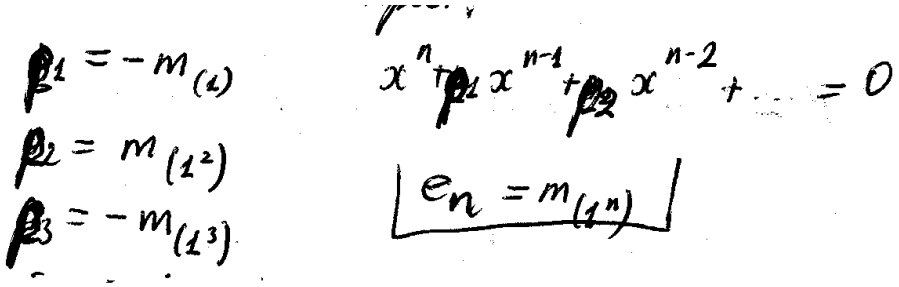
$$m_{(3\,1^2)} = x_1^3 x_2 x_3 + \dots$$

Diagram and typed description. Straight bracket below ... and label: perm.

Original source details: GELFAND-P001128

$$\beta_1 = -m_{(1)}$$
$$\beta_2 = m_{(1^2)}$$
$$\beta_3 = -m_{(1^3)}$$
$$\dots$$

$$x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \dots = 0$$



Original source details: GELFAND-P001128

Uncertain reading. The overwritten coefficient letters are tentatively read as β_i.

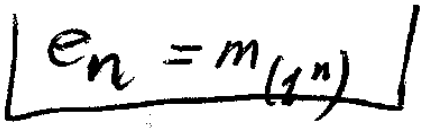


Diagram and typed description. [Boxed on three sides] $e_n = m_{(1^n)}$.

Original source details: GELFAND-P001128

$$(3) \qquad E(t) = \sum_{n\geq 0} e_n t^n = \prod_i (1 + t x_i) \qquad \text{(generating function)}, \quad e_0 = 1,$$

$$(4) \qquad h_n = \sum_{|\lambda| = n} m_\lambda,$$

$$h_2 = m_{(2)} + m_{(1^2)} = (x_1^2 + x_2^2 + \dots) + (x_1 x_2 + x_1 x_3 + \dots)$$

(sum of all monomials)

$$(5) \qquad H(t) = \sum_{n\geq 0} h_n t^n = \prod_i (1 - t x_i)^{-1} \qquad \text{(Exercise!)}, \quad h_0 = 1$$

$$(6) \qquad p_n = m_{(n)} \quad \text{— power sum}$$

$$(7) \qquad P(t) = \sum_{n\geq 1} p_n t^n = \sum_i \frac{t x_i}{1 - t x_i} = t H'(t)/H(t)$$

Bases: from $\lambda = (\lambda_1, \lambda_2, \dots)$ construct:

GELFAND-P001129

[source number: 85]

$$(8)\qquad \begin{cases} e_\lambda = \prod e_{\lambda_i} & (e_\lambda) \text{ — } \mathbb{Z}\text{-basis in } \Lambda \\ h_\lambda = \prod h_{\lambda_i} & (h_\lambda) \text{ — } \mathbb{Z}\text{-basis in } \Lambda \\ p_\lambda = \prod p_{\lambda_i} & (p_\lambda) \text{ — } \mathbb{Q}\text{-basis in } \Lambda \end{cases}$$

Diagram and typed description. [In the left margin by the second line] and.

Original source details: GELFAND-P001129

$$\Lambda = \mathbb{Z}[e_1, e_2, \ldots] = \mathbb{Z}[h_1, h_2, \ldots]$$

(fundamental theorem on symm. polynomials)

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[p_1, p_2, \ldots]$$

$$p_{(2,1^2)} = (x_1^2 + x_2^2 + \ldots)(x_1 + x_2 + \ldots)^2$$

5th basis. S-functions of Schur (appeared in Jacobi)

$$\alpha = (\alpha_1, \ldots, \alpha_n)$$

$$a_\alpha(x_1, \ldots, x_n) = \det(x_i^{\alpha_j}) = \begin{vmatrix} x_1^{\alpha_1} & x_1^{\alpha_2} & \ldots & x_1^{\alpha_n} \\ x_2^{\alpha_1} & x_2^{\alpha_2} & \ldots & x_2^{\alpha_n} \\ \ldots & \ldots & \ldots & \ldots \end{vmatrix}$$

[Below the determinant:] skew-symmetric function of $x_1, \ldots, x_n$.

$$a_\delta = \prod_{i<j}(x_i - x_j)$$

$$(\delta = (n-1, n-2, \ldots, 1, 0))$$

$\frac{a_\alpha}{a_\delta}$ — symm. polynomial.

Assume $\alpha_1 > \alpha_2 > \ldots > \alpha_n$ (one can always achieve this up to sign)

Diagram and typed description. Put differently:

Original source details: GELFAND-P001129

$$\alpha = \lambda + \delta, \qquad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$$

$$(9) \qquad s_\lambda \overset{\text{def}}{=} a_{\lambda+\delta}/a_\delta \quad \text{— another basis.}$$

[crossed out] (over $\mathbb{Z}$)

Uncertain reading. The start of the cancelled writing reads “в” (in). The rest is illegible.

Original source details: GELFAND-P001129

For $n \geq |\lambda|$ the ratio does not depend on n.

Theory: find relations between them and the structure constants (for multiplication)

Example — Theorem.

Uncertain reading. The crossed-out signs before the formula are illegible.

Original source details: GELFAND-P001129

$$s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$$

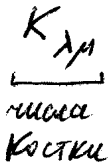
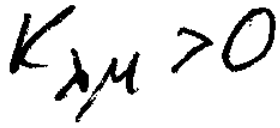


Diagram and typed description. Under the bracket below $K_{\lambda\mu}$: Kostka numbers.

Original source details: GELFAND-P001129

$$K_{\lambda\mu} \in \mathbb{Z}, \qquad K_{\lambda\mu} > 0$$

Diagram and typed description. $K_{\lambda\mu} > 0$.



Original source details: GELFAND-P001129

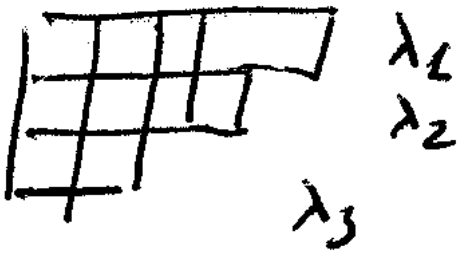
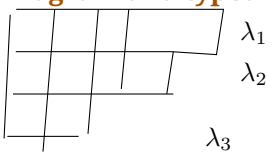


Diagram and typed description.

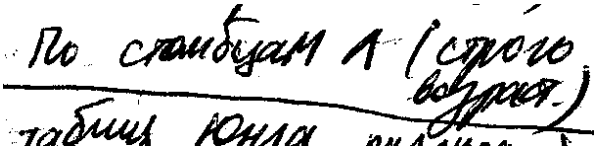


Original source details: GELFAND-P001129

λ defines a Young diagram.

μ — μ_1 ones, μ_2 twos, . . . ||

Fill in this diagram. Along the rows $\leq$ (nondecreasing). Along the columns $\nearrow$ (strictly increasing).



Original source details: GELFAND-P001129

Uncertain reading. The slanted mark is tentatively read as $\nearrow$.

Call this a tableau of Young.

Original source details: GELFAND-P001129

Diagram and typed description.

1	1	2
3	4	
5		

The number of Young tableaux of size λ and weight μ equals $K_{\lambda\mu}$.

GELFAND-P001130

Easy exercise: find the change-of-basis matrices between the others.

6th basis f_λ (from the word forgotten)

Original source details: GELFAND-P001130

Diagram and typed description. 24 October 1983. Gelfand Seminar.

I.M. Gelfand with work by Flashka.

Original source details: GELFAND-P001130

Diagram and typed description. Flashka.

$$L_n = \sum_{s=0}^{n} u_s \frac{\partial^s}{\partial x_1^s}, \qquad u_n = 1. \tag{1}$$

$$L_n \widetilde{\Psi} = k^n \widetilde{\Psi}. \tag{2}$$

Consider an asymptotic solution as $k \to \infty$

$$\widetilde{\Psi}^n = e^{kx} \sum_{0}^{\infty} c_j k^{-j}, \qquad c_0 = 1. \tag{3}$$

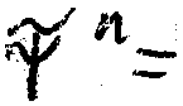


Diagram and typed description. $\widetilde{\Psi}^n$.

Original source details: GELFAND-P001130

Ex.: write recurrence formulas to determine c_j.

$$\chi = \widetilde{\Psi}_{x_i} \widetilde{\Psi}^{-1} = k + \sum_{1}^{\infty} \chi_j k^{-j}. \tag{4}$$

Diagram and typed description. $\widetilde{\Psi}_{x_i}$.

Original source details: GELFAND-P001130

Here χ_j does not depend on the choice of $\widetilde{\Psi}$

$\forall$ integer m, $\exists$ an operator L_m of order m ([crossed out] with leading coeff. 1)

Uncertain reading. The crossed-out writing before “with leading” is illegible.

Original source details: GELFAND-P001130

$$(L_m\widetilde{\Psi})\widetilde{\Psi}^{-1} = k^m + \varepsilon_m k^{-1} + \dots \tag{5}$$

Here:

$$L_m = [(L_n)^{m/n}] \quad \text{(Gelfand–Dikii)} \quad = [(L_n)^{m/n}]_+$$

(5) determines the operator L_m uniquely

Introduce variables $x_2, x_3, \dots$ such that

$$\frac{\partial L_n}{\partial x_m} = [L_n, L_m] \tag{6}$$

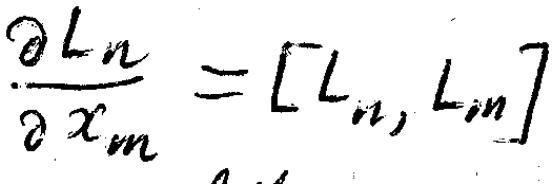


Diagram and typed description. $[L_n, L_m]$.

Original source details: GELFAND-P001130

or equivalently

Uncertain reading. The overwritten word is tentatively read as equivalently.

Original source details: GELFAND-P001130

$$\frac{\partial \Psi}{\partial x_m} = L_m \Psi \tag{7}$$

where

$$\Psi = e^{\sum_2^\infty x_j k^j}\widetilde{\Psi}. \tag{8}$$

Assume that u_j — functions of the variables $x_2, x_3, \dots$

$$\text{Put} \quad \varepsilon(k) \equiv -1 + \sum_{m=1}^{\infty} \varepsilon_m k^{-m-1} \tag{9}$$

$$\Lambda(k) = \sum_{0}^{\infty} k^{-m-1} L_m \tag{10}$$

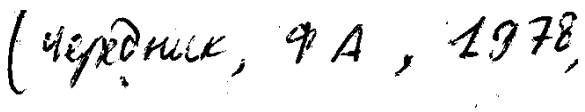


Diagram and typed description. [At right:] (Cherednik, FA, 1978)

Original source details: GELFAND-P001130

GELFAND-P001131

[source number: 86]

Cherednik's Thm. 3.1 states that

$$\left(\frac{\partial}{\partial x_1} - \chi(k)\right)\Lambda(k) = \varepsilon(k) \tag{11}$$

Cherednik:

$$\sum_{m=0}^{\infty} k^{-m-1}\left(\frac{\partial}{\partial x_1}(L_m\Psi(k))\Psi^{-1}(k)\right) = \frac{\partial\chi}{\partial k} + \mathcal{E}(k). \tag{12}$$

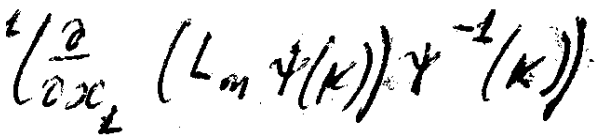
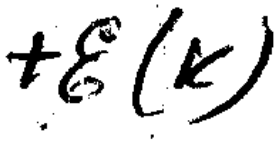


Diagram and typed description. Parentheses in eq. (12).

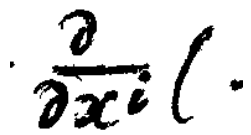
Original source details: GELFAND-P001131



Uncertain reading. The last letter is tentatively read as $\mathcal{E}(k)$.

Original source details: GELFAND-P001131

$$-\ell\chi_\ell + \varepsilon_\ell = \frac{\partial}{\partial x^i}(\dots) \tag{13}$$



Uncertain reading. The denominator is tentatively read with a superscript x^i.

Original source details: GELFAND-P001131

$$\begin{aligned}\frac{\partial}{\partial x_1}(L_m\Psi(k)\cdot\Psi^{-1}(k)) &= \frac{\partial}{\partial x_1}\left(\frac{\partial\Psi}{\partial x_m}\Psi^{-1}\right)\\ &= \frac{\partial}{\partial x_m}\left(\frac{\partial\Psi}{\partial x_1}\Psi^{-1}\right) = \frac{\partial\chi}{\partial x_m} \quad !!!\end{aligned} \tag{14}$$

Uncertain reading. Crossed-out number in the left margin (15?).

Original source details: GELFAND-P001131

$$-\ell\chi_\ell + \varepsilon_\ell = \frac{\partial\chi_{\ell-1}}{\partial x_1} + \ldots + \frac{\partial\chi_1}{\partial x_{\ell-1}}; \tag{15}$$

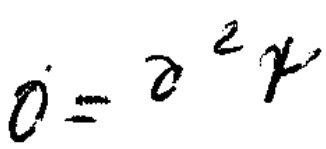


Diagram and typed description. [At right] $0 = \partial^2\Psi$.

Original source details: GELFAND-P001131

$$\varepsilon_j = \frac{\partial\tau}{\partial x_j} \qquad — \qquad \tau\text{-function} \tag{16}$$

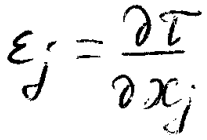


Diagram and typed description. $\varepsilon_j = \frac{\partial\tau}{\partial x_j}$.

Original source details: GELFAND-P001131

$$\Psi(x,k) = \frac{e^{\sum_{j=1}^{\infty} x_j k^j} e^{-\sum \frac{1}{\ell}\frac{\partial}{\partial x_\ell}} \tau(x)}{\tau(x)} \tag{17}$$

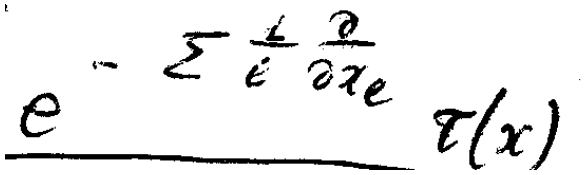


Diagram and typed description. Exponent of the second exponential $-\sum \frac{1}{\ell}\frac{\partial}{\partial x_\ell}$.

Original source details: GELFAND-P001131

(⇒ the structure of the recurrence formulas)
G. Olshanski.

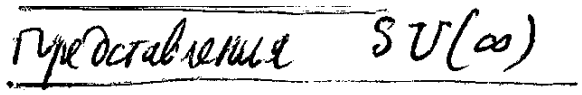


Diagram and typed description. Representations of $SU(\infty)$

Original source details: GELFAND-P001131

W — the Weil repr. of $Sp(\infty,\mathbb{R})^\sim$ on $\mathcal{H}(z_1, z_2, \ldots)$.

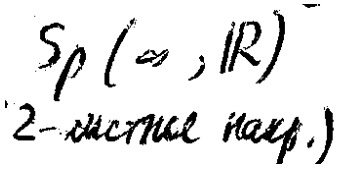


Diagram and typed description. [Under $Sp(\infty,\mathbb{R})^\sim$] (2-sheeted cover)

Original source details: GELFAND-P001131

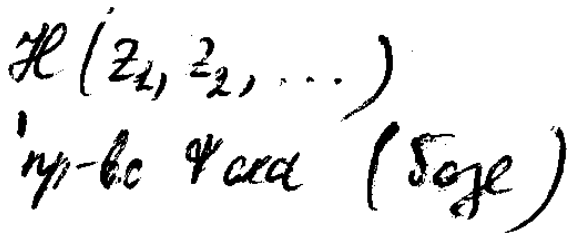


Diagram and typed description. [Under $\mathcal{H}(z_1, z_2, \ldots)$] Fock space (Bose)

Original source details: GELFAND-P001131

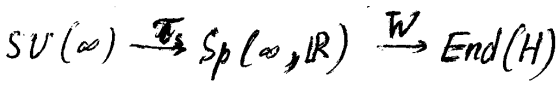


Diagram and typed description.

$$SU(\infty) \xrightarrow{\tau_s} Sp(\infty, \mathbb{R}) \xrightarrow{W} \mathrm{End}(H)$$

Original source details: GELFAND-P001131

[At right, past ||]

$$Sp(\infty, \mathbb{R}) = U(\infty, \infty) \cap Sp(\infty, \mathbb{C})$$

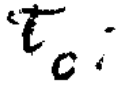
(complex realization)

$$T_s^+ = W \circ \tau_s \qquad \text{(fund. repr.)}$$

[At right] $s \neq 0 \Rightarrow$ irreducible

Embedding τ_0:

Uncertain reading. Tentative reading τ_0 (the subscript is open in shape).



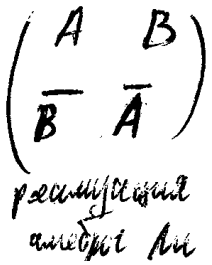
Original source details: GELFAND-P001131

$$\begin{pmatrix} \bar{g} & 0 \\ 0 & g \end{pmatrix} \subset Sp(\infty, \mathbb{R})^{\sim}; \qquad g \in SU(\infty).$$

$$\tau_s(g) = \begin{pmatrix} \mathrm{ch}\, s & \mathrm{sh}\, s \\ \mathrm{sh}\, s & \mathrm{ch}\, s \end{pmatrix} \begin{pmatrix} \bar{g} & 0 \\ 0 & g \end{pmatrix} \begin{pmatrix} \mathrm{ch}\, s & -\mathrm{sh}\, s \\ -\mathrm{sh}\, s & \mathrm{ch}\, s \end{pmatrix}$$

$$\begin{pmatrix} A & B \\ \bar{B} & \bar{A} \end{pmatrix} \qquad \begin{cases} A = -A^* \\ B = B' \end{cases}$$

Diagram and typed description. [Below the matrix] realization ↓ of the Lie algebra



Original source details: GELFAND-P001131

$$T_t^- = S \circ \tau_t^-, \qquad S \text{ — spinor repr. of the group } \mathrm{Spin}(\infty).$$

GELFAND-P001132

Weil repr.

$$W : \begin{cases} \begin{pmatrix} -E_{ji} & 0 \\ 0 & E_{ij} \end{pmatrix} \longmapsto z_i \partial_j + \dfrac{\delta_{ij}}{2} \\ \begin{pmatrix} 0 & E_{ij} + E_{ji} \\ 0 & 0 \end{pmatrix} \longmapsto \partial_i \partial_j \\ \begin{pmatrix} 0 & 0 \\ E_{ij} + E_{ji} & 0 \end{pmatrix} \longmapsto -z_i z_j \end{cases}$$

Uncertain reading. [At lower left] 26/...83... [tentative reading of the visible parts of the marks].

Original source details: GELFAND-P001132

Undeciphered passage. The lower parts of the date are cut off at the horizontal edge of the written area. The month and continuation cannot be read.

Original source details: GELFAND-P001132

Chapter 24

GELFAND-P001133

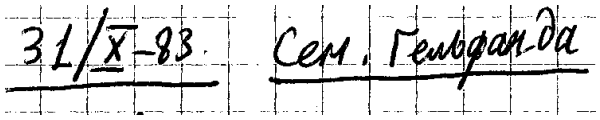


Diagram and typed description. 31 October 1983. Gelfand Seminar

Original source details: GELFAND-P001133

Zelevinsky.

Λ_n^k — homogeneous symm. polynomials in $x_1, \dots, x_n$ of degree k (over $\mathbb{Z}$).

For $n \geq k$, Λ_n^k does not depend on n.

Consider $\Lambda^k = \varprojlim_{\leftarrow n} \Lambda_n^k$ and the ring $\Lambda = \bigoplus_{k \geq 0} \Lambda^k$ of all symm. functions.

Elementary theory of symm. functions: single out 5 bases and establish the relations between them.

Partition:

$$\lambda = (\lambda_1 \geq \lambda_2 \geq \dots) \qquad \text{(all zero from some point on)}$$

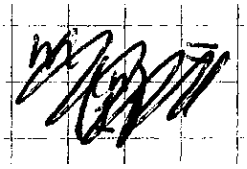
(all λ_j — nonnegative integers).

$\lambda = (4, 2, 1)$ example. λ_i — parts of the partition.

1°. Monomial functions:

$$m_\lambda(x) = x_1^{\lambda_1} x_2^{\lambda_2} \dots x_N^{\lambda_N} + \text{all possible permutations}$$

The partition can be written as $\lambda = (1^{m_1}, 2^{m_2}, \dots)$ (m_1 — the number of ones, m_2 — the number of twos, etc.).



Uncertain reading. [crossed out: $m_{\dots}$; the subscript is illegible.]

Original source details: GELFAND-P001133

$$m_{(n)} = x_1^n + x_2^n + \dots$$

$$m_{(1^n)} = x_1 x_2 \dots x_n + x_1 x_2 \dots x_{n-1} x_{n+1} + \dots$$

$m_\lambda(x)$ — a basis.

2°. $e_\lambda = e_{\lambda_1} e_{\lambda_2} \dots$

3°. $h_\lambda = h_{\lambda_1} h_{\lambda_2} \dots$

4°. $p_\lambda = p_{\lambda_1} p_{\lambda_2} \dots$

$e_n = m_{(1^n)}$ — elementary symm. functions

$h_n = \sum_{|\lambda|=n} m_\lambda$ — complete symm. function

$p_n = m_{(n)}$ — power sums

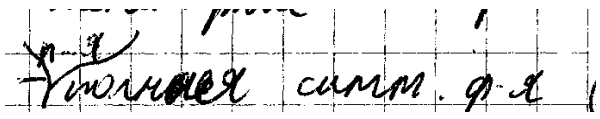
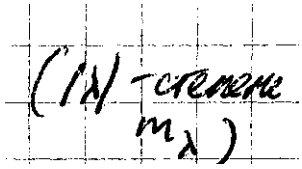


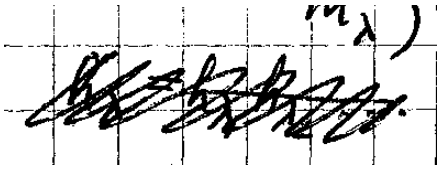
Diagram and typed description. [inserted above] nth.
The overwritten word reads «complete».

Original source details: GELFAND-P001133



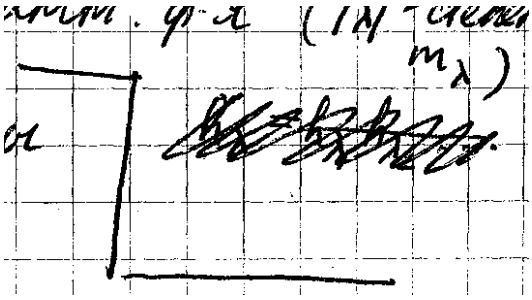
Original source details: GELFAND-P001133

Diagram and typed description. [At right] ($|\lambda|$ — degree of m_λ).



Original source details: GELFAND-P001133

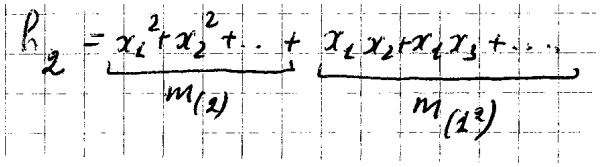
Uncertain reading. [At right, by the bracket] the crossed-out passage is illegible.



Original source details: GELFAND-P001133

Undeciphered passage. The writing under the strokes cannot be recovered. The right bracket is visible.

$$h_2 = \underbrace{x_1^2 + x_2^2 + \ldots +}_{m_{(2)}} \underbrace{x_1x_2 + x_1x_3 + \ldots}_{m_{(1^2)}}.$$



Original source details: GELFAND-P001133

Diagram and typed description. Brackets beneath the terms in the formula for h_2.

Original source details: GELFAND-P001133

Uncertain reading. [crossed out: h_2?]

$$h_{(1^2)} = h_1^2 = \left(\sum x_i\right)^2.$$

GELFAND-P001134

Claim.

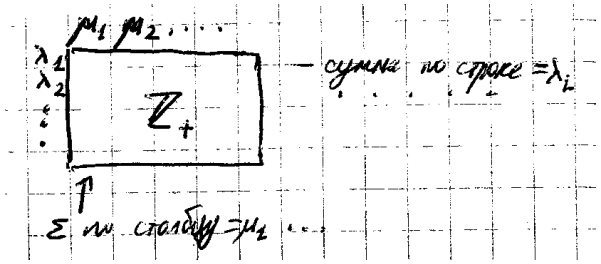
$$\Lambda = \bigoplus_\lambda \mathbb{Z} e_\lambda = \bigoplus_\lambda \mathbb{Z} h_\lambda,$$
$$\Lambda_{\mathbb{Q}} = \bigoplus_\lambda \mathbb{Q} p_\lambda.$$

[At right, beyond ||] Or:

$$\Lambda = \mathbb{Z}[e_1, e_2, \ldots] = \mathbb{Z}[h_1, h_2, \ldots],$$
$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[p_1, p_2, \ldots].$$

Relation between these 4 bases.

$$h_\lambda = \sum_\mu M_{\lambda\mu} m_\mu, \qquad M_{\lambda\mu} = \text{number of matrices.}$$



Original source details: GELFAND-P001134

Diagram and typed description.

$\mu_1, \mu_2, \ldots$

λ_1
λ_2
$\vdots$

$\mathbb{Z}_+$ — row sum $= \lambda_i$

$\uparrow$

$\sum$ down the column $= \mu_1 \ldots$

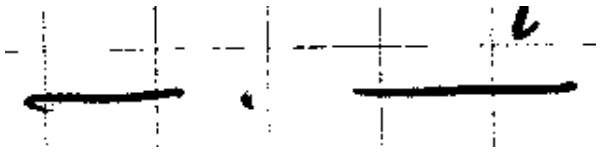
$$e_\lambda = \sum_\mu M^0_{\lambda\mu} m_\mu, \qquad M^0_{\lambda\mu} = \text{the same matrices, but with entries 0 and 1.}$$

Exercise: express p_λ in terms of m_μ (this is done similarly).

Relate h_λ, e_λ and p_λ. This is done using generating functions and Newton's formulas.

$$E(t) = \sum_{n \geq 0} e_n t^n, \qquad H(t) = \sum_{n \geq 0} h_n t^n, \qquad P(t) = \sum_{n \geq 1} p_n t^n.$$

$$E(t) = \prod_i (1 + x_i t), \qquad H(t) = \prod_i (1 - x_i t)^{-1}, \qquad P(t) = \frac{tH'(t)}{H(t)} = \sum_i \frac{t x_i}{1 - t x_i}.$$



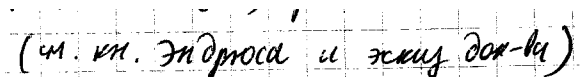
Original source details: GELFAND-P001134

Diagram and typed description. — · —

I.M.G. In Ramanujan

$p(n)$ — the number of partitions of n into a sum $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq 0$ ($|\lambda| = n$).

There is an explicit (!!!) formula — the result from the asymptotics, but with more terms. (see Andrews's book and [unclear abbreviation] proofs).



Original source details: GELFAND-P001134

Uncertain reading. The abbreviation between "and" and "proofs" is unclear.

$$\sum_{n\geq 0} p(n)t^n = \prod_{i\geq 1}(1-t^i)^{-1} = \frac{1}{(1-t)(1-t^2)(1-t^3)\dots}.$$

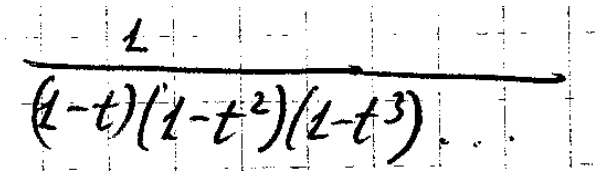


Diagram and typed description. Ellipsis under the fraction bar after $(1-t^3)$.

Original source details: GELFAND-P001134

Newton's formulas:

$$H(t)E(-t) = 1.$$

Hence

$$\sum_{r=0}^{n}(-1)^r e_r h_{n-r} = 0 \qquad \text{for } n \geq 1.$$

$$P(t)H(t) = tH'(t)$$

$$nh_n = \sum_{r=1}^{n} p_r h_{n-r}, \qquad n \geq 1.$$

$$a_\alpha = \det \begin{vmatrix} x_1^{\alpha_1} & x_1^{\alpha_2} & \dots & x_1^{\alpha_n} \\ x_2^{\alpha_1} & x_2^{\alpha_2} & \dots & x_2^{\alpha_n} \\ \dots & \dots & \dots & \\ x_n^{\alpha_1} & x_n^{\alpha_2} & \dots & x_n^{\alpha_n} \end{vmatrix}$$

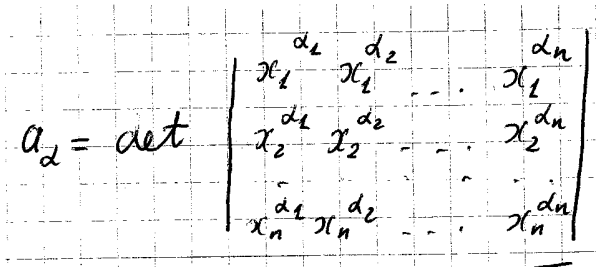


Diagram and typed description. The manuscript has "det" before the vertical bars and a line below the determinant.

Original source details: GELFAND-P001134

$$a_\delta = \prod_{i<j}(x_i - x_j)$$

[At right:] Schur function:

$$s_\lambda = a_{\lambda+\delta}/a_\delta.$$

Uncertain reading. [Before δ an illegible sign is crossed out.]

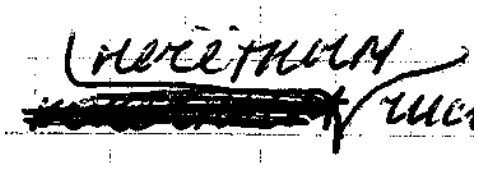
Original source details: GELFAND-P001134

$$\delta = (n-1, n-2, \dots).$$

GELFAND-P001135

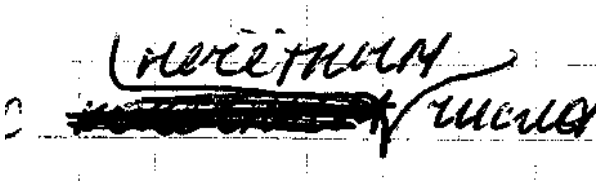
[source number: 4]

On Witten's work (nonexistence of gauge theories with [crossed out; inserted above: an odd] number of fermions).



Uncertain reading. Above the crossed-out word is «an odd».

Original source details: GELFAND-P001135



Undeciphered passage. The word beneath the crossing-out is illegible.

Original source details: GELFAND-P001135

$$\mathcal{L}(\Psi, A)$$

$$\int d\Psi\, d\bar{\Psi}\, dA\ \exp(-\mathcal{L}),$$

Uncertain reading. [After the comma crossed-out writing.]

Original source details: GELFAND-P001135

Undeciphered passage. The crossed-out signs after the comma cannot be recovered.

Original source details: GELFAND-P001135

$$\mathcal{L} = \int (\bar{\Psi}\, iD\Psi + \text{[crossed out]}\ \operatorname{Tr} F_{\mu\nu}^2)\,.$$

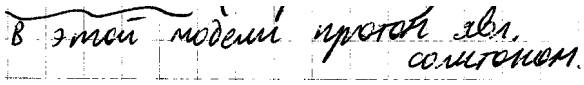
Uncertain reading. [Between + and Tr crossed-out writing.]

Original source details: GELFAND-P001135

There arises $(\det D)^{+1/2}$, where

$$D = \sum \gamma_i \left(A_i + \frac{\partial}{\partial x_i} \right).$$

Perelomov. On Witten's work on the σ-model (of the proton).

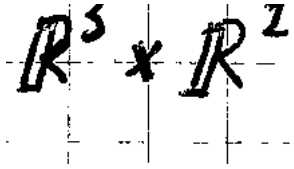


Original source details: GELFAND-P001135

Diagram and typed description. [At right below the heading] In this model the proton is a soliton.

$$S = S_0 + S_1 + S_2$$

$M_4 = \mathbb{R}^3 \times \mathbb{R}^1$ — space/time.



Original source details: GELFAND-P001135

Diagram and typed description. $\mathbb{R}^3 \times \mathbb{R}^1$.

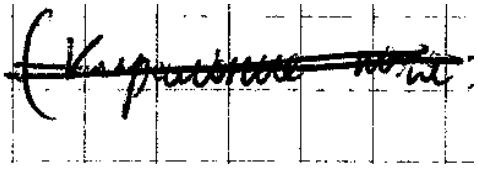
$G = SU(2)$ or $SU(3)$

$$\varphi\colon M^4 \longrightarrow G, \qquad g(x), \qquad \varphi_\mu = \partial_\mu g \cdot g^{-1}.$$

Original source details: GELFAND-P001135

Uncertain reading. [After φ before the colon crossed-out writing.]

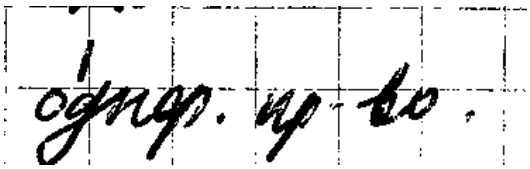
$$S_0 = \int d^4x \, \langle \varphi_\mu, \varphi_\mu \rangle.$$



Original source details: GELFAND-P001135

Diagram and typed description. [crossed out] ~~(Chiral field.~~

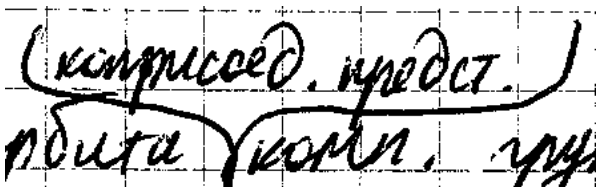
More general case: $\varphi\colon M^4 \longrightarrow X$.



Original source details: GELFAND-P001135

Diagram and typed description. [Under X] homogeneous space.

2-dimensional theory, X — orbit of a compact group [inserted above: (coadjoint repr.)] $\Longrightarrow$ all instantons have been found (Perelomov).



Original source details: GELFAND-P001135

Diagram and typed description. [inserted above:] (coadjoint repr.)

$X = S^2$ — Belavin, Polyakov (1975).

For chiral fields there is an analogue of d'Alembert's representation (local solutions found — Krichever). Soliton solutions have been written down.

$$S_2 = \int d^4x \, \langle [\varphi_\mu, \varphi_\nu], [\varphi_\mu, \varphi_\nu] \rangle.$$

$$M^4 \longrightarrow S^3 \times S^1 \qquad \text{(compactification)}$$

Original source details: GELFAND-P001135

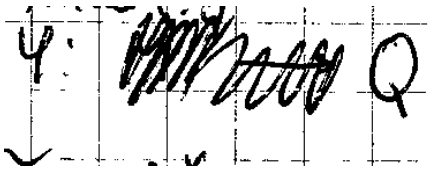
Uncertain reading. The crossed-out mark resembles φ?

$$M^4 = \partial Q, \qquad Q = S^3 \times D^2.$$

$$M^4 \longrightarrow SU(3)$$

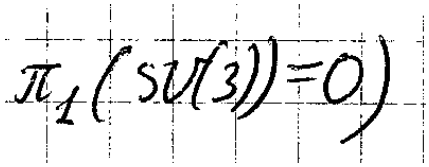
Extend:

$$\varphi \colon Q \longrightarrow SU(3) \qquad \text{(for } \pi_1(SU(3)) = 0\text{)}.$$



Original source details: GELFAND-P001135

Uncertain reading. [Between $\varphi\colon$ and Q crossed-out writing.]



Original source details: GELFAND-P001135

Diagram and typed description. $\pi_1(SU(3)) = 0$.

$$\widetilde{\omega} = \varphi^* \omega$$

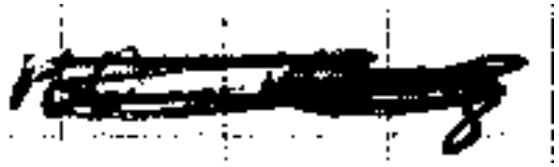
ω — generator of homology in the corresponding dimension.

$(1 + t^3)(1 + t^5)$ — Poincaré polynomial.

$$S_1 = N \int_Q \varphi^* \omega.$$

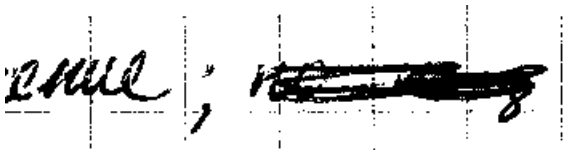
GELFAND-P001136

Wess and Zumino wrote down a complicated expression; [crossed out]



Original source details: GELFAND-P001136

Uncertain reading. A passage is crossed out after the semicolon. A lower loop is visible at the end.



Original source details: GELFAND-P001136

Undeciphered passage. The letters under the strokes are illegible.

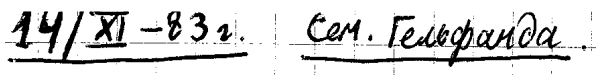
in particular, integrality was previously unknown.

S_1 — a special case of S. P. Novikov's multivalued functionals.

Original source details: GELFAND-P001136

Diagram and typed description. — · —

GELFAND-P001137

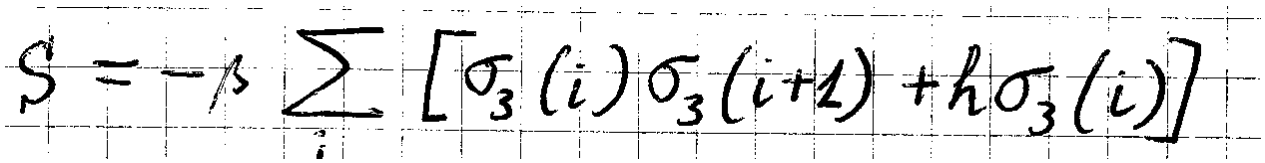


Original source details: GELFAND-P001137

Diagram and typed description. 14 November 1983. Gelfand Seminar.

I.M.G. Ising model Phys. Rev. D17, 1978

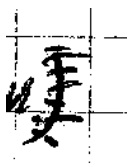
$$S = -\beta \sum_i [\sigma_3(i)\sigma_3(i+1) + h\sigma_3(i)]$$



Original source details: GELFAND-P001137

Diagram and typed description. First expression for S.

(with periodic conditions [crossed out] — on $\mathbb{Z}/N\mathbb{Z}$).

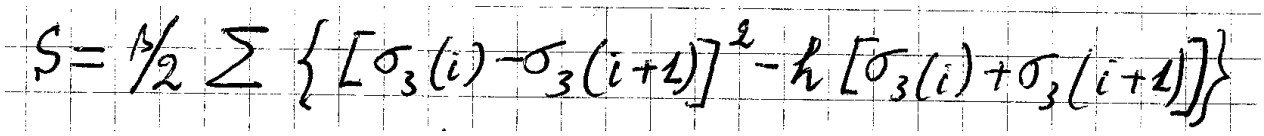


Original source details: GELFAND-P001137

Uncertain reading. Crossed-out mark, possibly a bracket.

$$\sigma_3(i) = \pm 1$$

$$S = \beta/2 \sum \{[\sigma_3(i) - \sigma_3(i+1)]^2 - h[\sigma_3(i) + \sigma_3(i+1)]\}$$



Original source details: GELFAND-P001137

Diagram and typed description. Squared-difference expression for S.

$$\mathcal{L}(i, i+1) = \frac{\beta}{2} \{[\sigma_3(i) - \sigma_3(i+1)]^2 - h[\sigma_3(i) + \sigma_3(i+1)]\}$$

$$S = \sum_i \mathcal{L}(i, i+1)$$

$$Z = \sum_\sigma e^{-S[\sigma]} = \sum_\sigma \prod_i e^{-\mathcal{L}(i,i+1)}.$$

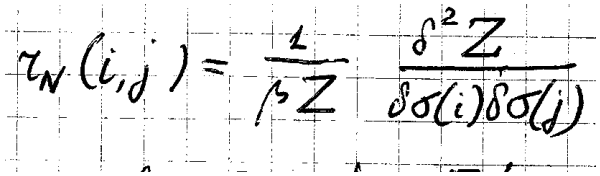
Original source details: GELFAND-P001137

Diagram and typed description. [In the left margin] Chulaevsky.

Correlation fn.:

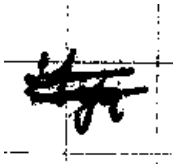
$$\tau_N(i,j) \stackrel{\text{def}}{=} E[\sigma(i)\sigma(j)]$$

$$\tau_N(i,j) = \frac{1}{\beta Z} \frac{\delta^2 Z}{\delta\sigma(i)\delta\sigma(j)}.$$



Original source details: GELFAND-P001137

Diagram and typed description. $\frac{1}{\beta Z} \frac{\delta^2 Z}{\delta\sigma(i)\delta\sigma(j)}$.



Original source details: GELFAND-P001137

Uncertain reading. [In the left margin] The crossed-out writing is illegible.

Probability space $\Omega_N = \{\sigma : \mathbb{Z}/N\mathbb{Z} \longrightarrow \pm 1\}$.

$$P_N\{\sigma\} = \frac{1}{Z_N} e^{-S[\sigma]}.$$

GELFAND-P001138

Zelevinsky.

$p(n) =$ the number of partitions of n into a sum of positive integers.

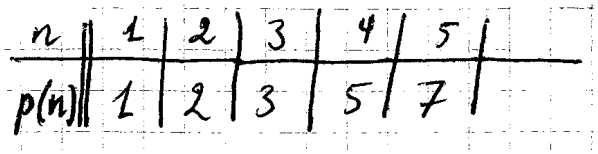


Diagram and typed description.

n	1	2	3	4	5	
$p(n)$	1	2	3	5	7	

Original source details: GELFAND-P001138

$$\sum p(n)t^n = \frac{1}{(1-t)(1-t^2)\dots}$$

$$\begin{aligned} p(n) &= (\text{number of conjugacy classes of elements in } S_n) \\ &= (\text{number of irreducible representations of } S_n) \end{aligned}$$

$$p(5n+4) \equiv 0 \pmod 5$$

$$p(7n+5) \equiv 0 \pmod 7$$

Dyson's problem (unsolved), connected with the first congruence.

$$\text{Invariant} \stackrel{\text{def}}{=} (\text{largest part of the partition}) - (\text{number of parts})$$

$$8 = 6+1+1 \rightsquigarrow 3\ (= 6-3)$$

Diagram and typed description. Retraced digit 3.

Original source details: GELFAND-P001138

$$8 = \underbrace{1+\dots+1} \rightsquigarrow -7$$

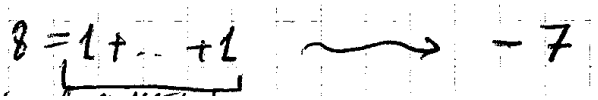


Diagram and typed description. Straight bracket below $1+\dots+1$.

Original source details: GELFAND-P001138

The remainder [inserted above (of the invariant)] on division by 5 gives a split into equal parts.

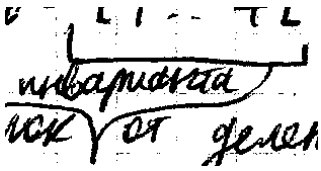
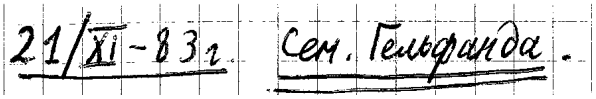


Diagram and typed description. [inserted above] (of the invariant).

Original source details: GELFAND-P001138

Proved, but by a noncombinatorial argument.



Original source details: GELFAND-P001139

Diagram and typed description. 21/XI-83 yr. Gelfand Sem..

A. M. Vershik. (on work with Kerov).

Original source details: GELFAND-P001139

Uncertain reading. [An illegible entry is struck out on the left.]

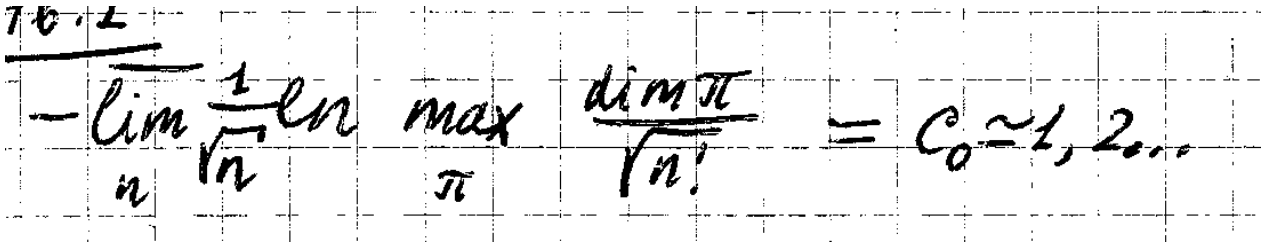
S_n is the symm. group.

Consider its representations (irred.) of max. dimension.

Question: asymptotics

$$\max_{\pi\in\widehat{S}_n} \dim \pi \qquad \text{as } n \longrightarrow \infty. \tag{1}$$

Prop. 1



Original source details: GELFAND-P001139

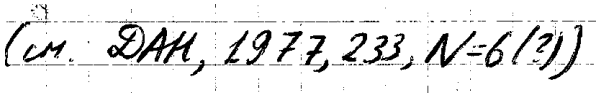
Diagram and typed description.

$$-\overline{\lim_n} \frac{1}{\sqrt{n}} \ln \max_\pi \frac{\dim \pi}{\sqrt{n!}} = c_0 \simeq 1.2\ldots \tag{2}$$

or

$$\max \dim \pi \simeq \sqrt{n!}\, e^{-1.2\sqrt{n}(1+o(1))} \tag{2'}$$

Sol. obtained by analysing Plancherel measure

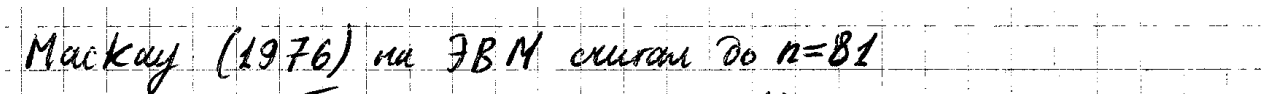


Original source details: GELFAND-P001139

Diagram and typed description. (see DAN, 1977, 233, $N = 6(?)$).

There, the asympt. of a typical Young diagram.

Idea: there is a limit of the Young diagrams (and it can be found!).

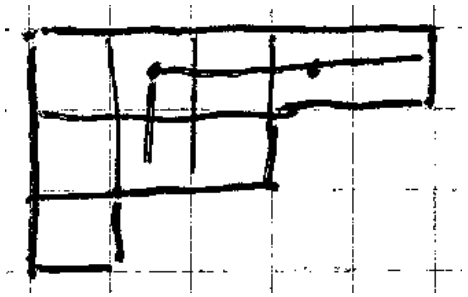


Original source details: GELFAND-P001139

Diagram and typed description. MacKay (1976) computed up to $n = 81$.

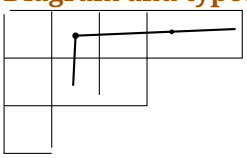
His conjecture was that there is a representation with $\dfrac{\dim \pi}{\sqrt{n!}} > \dfrac{1}{n}$.

It is false.



Original source details: GELFAND-P001139

Diagram and typed description.



Two heavier marks on the horizontal stroke. It is unclear whether they indicate points.

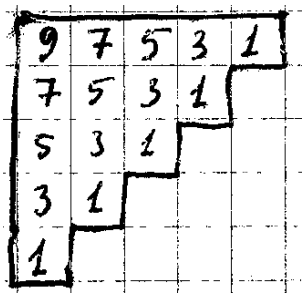
Dimension of the representation from the Young diagram.

Construct a hook at $\forall$ points.

$$\dim = \frac{n!}{\prod(\text{hook lengths})}.$$

$$\sum \left(\frac{n!}{\prod \dots}\right)^2 = n! \qquad \text{(consequence of a theorem of representation theory)}$$

Hook lengths:

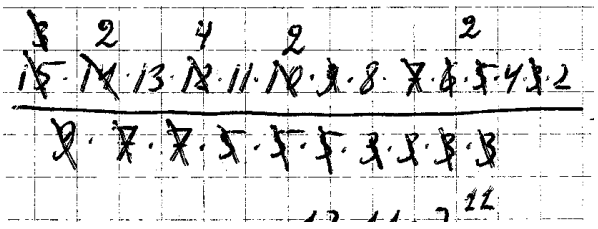


Original source details: GELFAND-P001139

Diagram and typed description.

9	7	5	3	1
7	5	3	1	
5	3	1		
3	1			
1				

$$\begin{aligned}\dim &= \frac{15!}{9 \cdot 7^2 \cdot 5^3 \cdot 3^4} \\ &= \frac{\overset{3}{\cancel{15}} \cdot \overset{2}{\cancel{14}} \cdot 13 \cdot \overset{4}{\cancel{12}} \cdot 11 \cdot \overset{2}{\cancel{10}} \cdot \cancel{9} \cdot 8 \cdot \cancel{7} \cdot \overset{2}{\cancel{6}} \cdot \cancel{5} \cdot 4 \cdot \cancel{3} \cdot 2}{\cancel{9} \cdot \cancel{7} \cdot \cancel{7} \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}} \\ &= \underline{13 \cdot 11 \cdot 2^{11}}.\end{aligned}$$

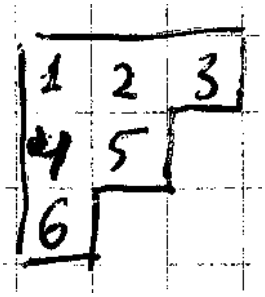


Original source details: GELFAND-P001139

Uncertain reading. The crossed-out digit above 15 reads as 3. The last superscript 2 sits to the right of 6.

or dim = the number of Young tableaux of the given shape; the tableau increases along rows and columns.

The numbers $1, 2, \ldots, k$, where k is the number of cells in the tableau.



Original source details: GELFAND-P001139

Diagram and typed description.

1	2	3
4	5	
6		

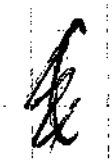
16 ways to fill it.

GELFAND-P001140

Fact: diagrams of representations of [~~max~~?] dimension

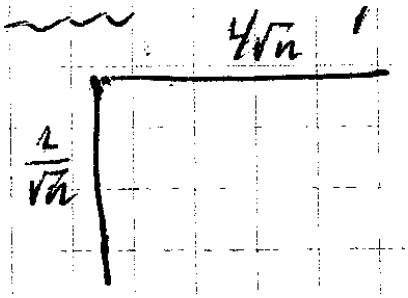
Original source details: GELFAND-P001140

Uncertain reading. The crossed-out entry tentatively reads “max”.



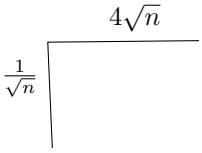
Original source details: GELFAND-P001140

Uncertain reading. [Left margin] The crossed-out entry resembles $\sqrt{n}$.

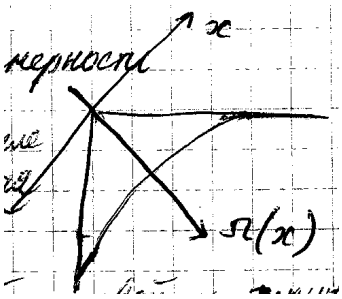


Original source details: GELFAND-P001140

Diagram and typed description.

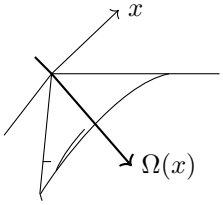


normalize: in the limit, a curve.



Original source details: GELFAND-P001140

Diagram and typed description.



A thin parallel stroke appears near the bottom of the arc. Its purpose is unclear.

The max representation tends to this curve too, as do typical ones.

Original source details: GELFAND-P001140

Diagram and typed description. The beginning of “tends” is retraced. The strokes underneath do not yield a separate reading.

$$\Omega(x) = \begin{cases} \frac{2}{\pi}\left(x \arcsin x + \sqrt{1-x^2}\right), & |x| \le 1, \\ |x|, & |x| > 1. \end{cases}$$

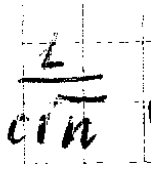
Histogram: according to Plancherel measure: $\frac{(\dim \Lambda)^2}{n!}$ — probability of a representation.

Almost all representations in this measure tend to this curve. Typical representations have a large fluctuation.

Prop. 2. Max. representations are so close to it $\Rightarrow$ computer calculation.

There are $p(n)$ conjugacy classes of elements. Their asymptotics

$$p(n) \simeq \frac{1}{c\sqrt{n}}\, e^{\frac{2\pi}{\sqrt{6}}\sqrt{n}}. \tag{3}$$



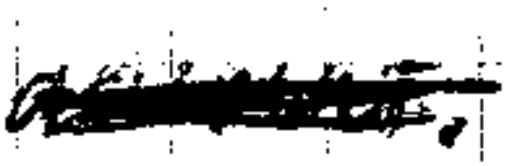
Original source details: GELFAND-P001140

Diagram and typed description. The manuscript gives the prefactor $\frac{1}{c\sqrt{n}}$.

This follows from

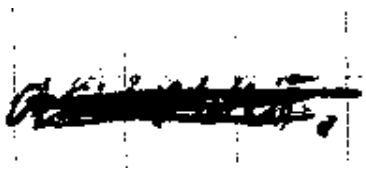
$$\sum_n p(n)x^n = \frac{1}{\prod_k (1-x^k)}.$$

([~~asympt.~~?], we estimate the integral)



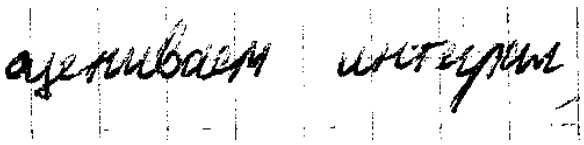
Original source details: GELFAND-P001140

Uncertain reading. Crossed-out writing, tentatively “asympt.”



Original source details: GELFAND-P001140

Undeciphered passage. Strokes obscure the writing after the initial “a”.

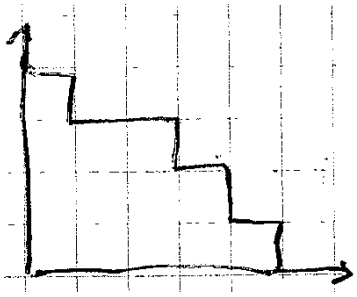


Original source details: GELFAND-P001140

Diagram and typed description. we estimate the integral

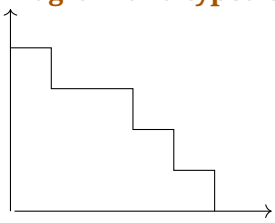
Rademacher (book on analyt. number theory) — asympt. of $p(n)$ (full) — $\simeq 40$ pp..

$p(n)$ — the number of curves with a given area.



Original source details: GELFAND-P001140

Diagram and typed description.



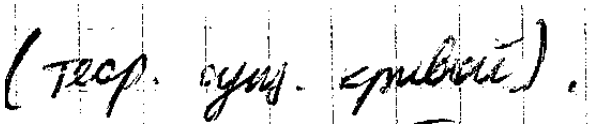
Consider tableaux with n cells and let each have probability $\frac{1}{p(n)}$ (all equally likely).

Typical diagram? (again shrink by a factor of $\sqrt{n}$)

Prop. 3. The resulting curve (in the same axes).

$$e^{-\frac{\pi}{\sqrt{6}}x} + e^{-\frac{\pi}{\sqrt{6}}y} = 1. \tag{4}$$

Idea: law of large numbers (theory of a random curve).



Original source details: GELFAND-P001140

Diagram and typed description. theory of a random curve

The sections of curve (4) are independent (in a weak sense).

GELFAND-P001141

[source number: 21]

Haar asympt.:

$$n = n_1 + \cdots + n_k.$$

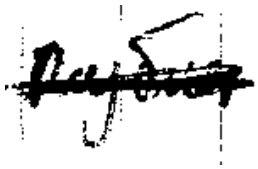
Probability proportional to the number of elements in the class. (a different statistic).

There is no limit here!

Bell statistics.

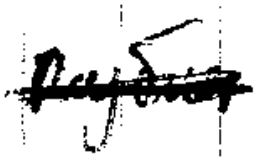
n individual objects split into subgroups

[~~Partit~~...] Number of partitions.



Original source details: GELFAND-P001141

Uncertain reading. The crossed-out entry begins “Partit...” [«Разби...»].



Original source details: GELFAND-P001141

Undeciphered passage. The stroke obscures the final letters.

Characters.

S_∞ — finitely supported permutations of the natural numbers.

$U(\infty)$

E. Thoma (FRG) — described the representations of S_∞. 1964; Vershik et al. 1981, FA, 15:4.

Diagram and typed description. FA, 15:4.

Original source details: GELFAND-P001141

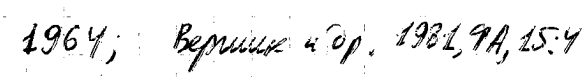


Diagram and typed description. 1964; Vershik et al. 1981, FA, 15:4.

Original source details: GELFAND-P001141

$U(\infty)$ Voiculescu 1976 and Vershik and Co. 1982 (DAN).

Earlier, in the 50s, in other terms (M. G. Krein, Gantmacher, Schoenberg, Karlin — totally positive functions)

Analytical problem — description of totally positive kernels

(Edrei [written «Эдрейн»] — by methods of the theory of entire functions)

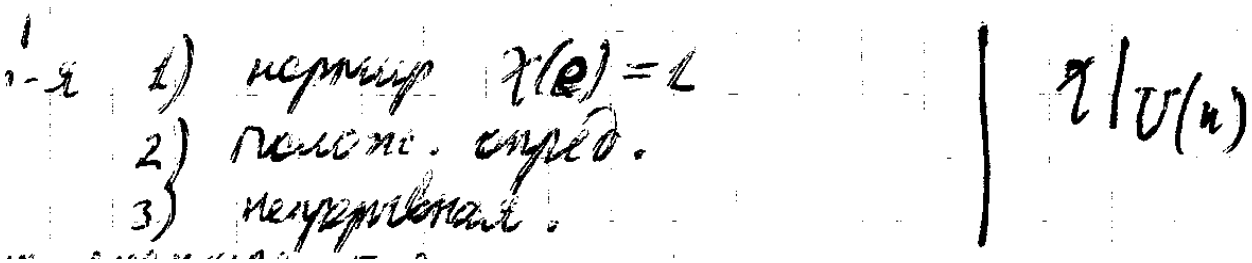
Diagram and typed description. Edrei [written «Эдрейн»]

Original source details: GELFAND-P001141

$$\begin{matrix} U(\infty) \\ \downarrow \\ g \end{matrix} \qquad \chi(g) \text{ — indecomposable character.}$$

Function:

1. normalized $\chi(e) = 1$
2. positive definite
3. continuous.



Original source details: GELFAND-P001141

Diagram and typed description. [At right, beyond the vertical stroke:] $\left| \chi|_{U(n)} \right.$.

$\Rightarrow \chi$ — a function of the spectrum, i.e.

$$\chi(g) = \prod_{\lambda_i \in \operatorname{Spec} g} \varphi(\lambda_i), \qquad \varphi(1) = 1.$$

Condition on φ: total positivity in the sense of Krein et al..

$$\varphi(\lambda) = \sum_{n \in \mathbb{Z}} c_n \lambda^n.$$

Answer: parameters $\gamma^{\pm}, \alpha_n^{\pm}, \beta_n^{\pm}$.

$$\varphi(z) = e^{\gamma^+(z-1)+\gamma^-(\bar{z}-1)} \prod_k \frac{(1-\beta_k^+ + \beta_k^+ z)}{(1+\alpha_k^+) - \alpha_k^+ z} \cdot \prod_k \frac{(1-\beta_k^- + \beta_k^- \bar{z})}{(1+\alpha_k^-) - \alpha_k^- \bar{z}}.$$

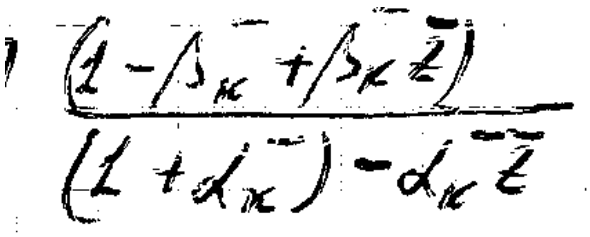


Diagram and typed description. The index in both terms of the denominator of the negative product reads k.

Original source details: GELFAND-P001141

$$\alpha_1^{\pm} \geq \cdots \geq \alpha_n^{\pm} \geq \cdots \geq 0, \qquad \gamma^{\pm} \geq 0,$$
$$\beta_1^{\pm} \geq \cdots \geq \beta_n^{\pm} \geq \cdots \geq 0,$$
$$\sum \alpha_k^+ + \alpha_k^- + \beta_k^+ + \beta_k^- < \infty,$$
$$\beta_1^+ + \beta_1^- \leq 1.$$

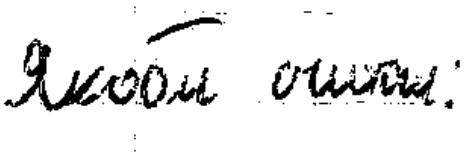
GELFAND-P001142

S_∞ — the difference is small, but the functions are analytic (without negative powers).

Uncertain reading. There is a retraced stroke beneath “without”. No distinct entry can be read beneath it.

Original source details: GELFAND-P001142

α_j, β_j — limiting parameters of Young diagrams.
Jacobi [interpretation?]:



Uncertain reading. The word after “Jacobi” tentatively reads “interpretation” [«смысл»]. The final strokes also permit «симв.» (“symbol”).

Original source details: GELFAND-P001142

$$\frac{\begin{vmatrix} x_1^{\lambda_1} & x_2^{\lambda_1} & \cdots & x_n^{\lambda_n} \\ x_1^{\lambda_2+1} & x_2^{\lambda_2+1} & \cdots & x_n^{\lambda_2+1} \\ \cdot & \cdot & \cdot & \cdot \end{vmatrix}}{\begin{vmatrix} 1 & \cdots & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \cdot & \cdot & & \\ x_1^{n-1} & \cdots & \cdots & x_n^{n-1} \end{vmatrix}}, \qquad \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n. \tag{5}$$

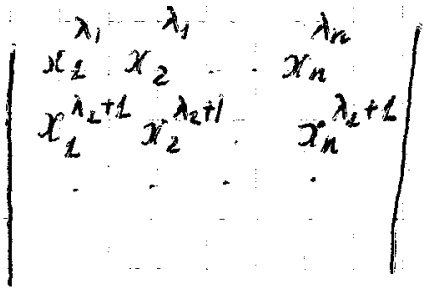


Diagram and typed description. The upper-right entry of the numerator is written $x_n^{\lambda_n}$.

Original source details: GELFAND-P001142

Suppose there exists $\lim_{n\to\infty} \frac{\lambda_k^{(n)}}{n} = \alpha_k \Rightarrow$ there exists a limit of det (5) and it is the generating function of the characters.

Prop. Let $|x_i| = 1$ and from some point on they are $= 1$.

Miwa, Jimbo: for each diagram, a differential operator.

K-functor:

Proc. Sem. LOMI, [crossed out] vol. 123

Uncertain reading. A mark before "vol. 123" is crossed out. It may be a number sign.

Original source details: GELFAND-P001142

— · —

$K^0(S_\infty)$ is constructed from projective modules over $\mathbb{C}(S_\infty)$
(finitely generated modules are direct summands).

$K^0(S_n)$ — additive group of characters.

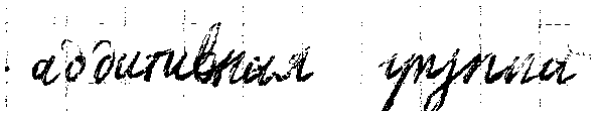
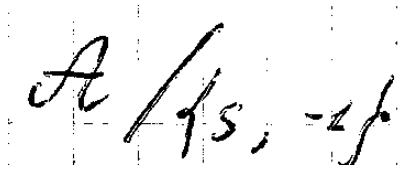


Diagram and typed description. additive group.

Original source details: GELFAND-P001142

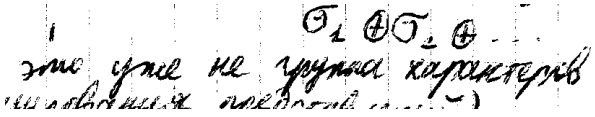
Thm.:

$$\varinjlim_{n\to\infty} K^0(S_n) = K^0(S_\infty) = A/\{s_1 - 1\}.$$



Uncertain reading. The subscript on s tentatively reads 1. The shape also allows i.

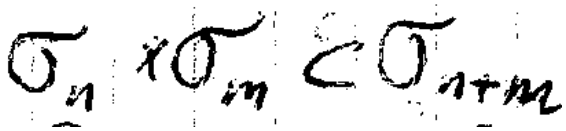
Original source details: GELFAND-P001142



Original source details: GELFAND-P001142

Diagram and typed description. [Beneath A:] $\sigma_1 \oplus \sigma_2 \oplus \cdots$.
[Beneath $K^0(S_\infty)$, marked by a vertical stroke:] this is no longer a group of characters.

(embedding by means of induced representations).



Original source details: GELFAND-P001142

Diagram and typed description. [At right:]

$$\sigma_n \times \sigma_m \subset \sigma_{n+m}.$$

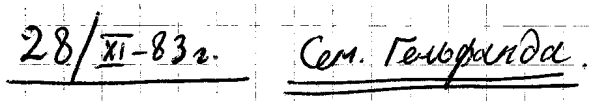
A description of the positive elements? It is given in terms of differential operators.
Positivity — positive combinations of characters.

Original source details: GELFAND-P001142

Uncertain reading. The beginning of the abbreviation “positive” is retraced. No distinct entry can be read under the strokes.

The cone of positive elements has a nontrivial structure.

GELFAND-P001143

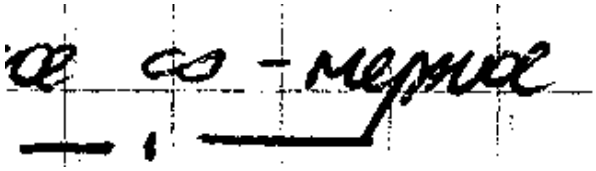


Original source details: GELFAND-P001143

Diagram and typed description. 28/XI-83 yr. Gelfand Sem..

I. M. G. It is unclear what an ∞-dimensional space is.

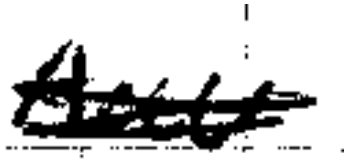
Repeats Vershik



Original source details: GELFAND-P001143

Diagram and typed description. Strokes under “∞-dimensional”. The right stroke turns up toward the word.

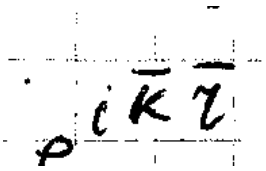
A. Migdal. [~~If~~?] The case of a deviation from the CLT (Gaussian limit theorem) is of interest



Original source details: GELFAND-P001143

Uncertain reading. The crossed-out word may read “If” [«Если»].

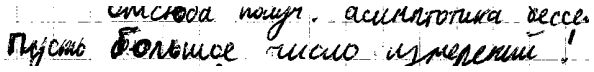
$$\int d\vec{\tau}\,\delta(1-\vec{\tau}^2)\,e^{i\bar{k}\bar{\tau}} \sim J_n(k) \tag{1}$$



Original source details: GELFAND-P001143

Diagram and typed description. The exponent has separate bars over k and τ: $i\bar{k}\bar{\tau}$.

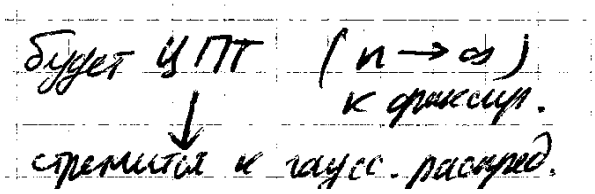
From this we get the asymptotics of the Bessel functions



Original source details: GELFAND-P001143

Diagram and typed description. Let there be a greater number of dimensions!

If k is held fixed, there will be



Original source details: GELFAND-P001143

Diagram and typed description.

CLT $\quad (n \to \infty)$

$\downarrow \quad k$ fixed

tends to a Gaussian distribution

Proof: Replace δ by a Fourier $\int$; integrate over τ.

$$\int d\alpha \int d\vec{\tau}\, e^{\alpha(1-\tau^2)+ik\tau}$$

Original source details: GELFAND-P001143

Uncertain reading. An illegible formula fragment after the first $\int$ and before $d\alpha$ is crossed out.

$\alpha \in i\mathbb{R}$ $\quad \| \quad$ evaluate the integral over $\tau \Rightarrow$ Gaussian distribution.

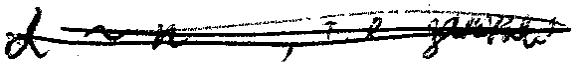
GELFAND-P001144

We obtain

$$\int \frac{d\alpha}{2\pi i}\,\alpha^{-n/2} e^{\alpha - \frac{k^2}{4\alpha}}.$$

Take $k \sim n$!! replace α by $n\alpha$ in the integral.

[~~$\alpha \sim n$~~, illeg.]



Original source details: GELFAND-P001144

Uncertain reading. The line is crossed out. The opening reads $\alpha \sim n$; the following signs and word remain unclear.

$$\int \frac{d\alpha}{2\pi i}\,\alpha^{-\frac{n}{2}} e^{n\alpha-\frac{k^2}{4\alpha n}}.$$

α fluctuates weakly $\Rightarrow$ replace it by a const for small k

n is large, $k \sim n \Rightarrow$ obtain some kind of arcsin.

An analogous problem arises for the
Fourier integral over the unitary group

$$\int du\ \exp\bigl(\beta\,\mathrm{tr}(\bar{n}u + u^{+}n)\bigr), \qquad u^{+}u = I \tag{2}$$

($n, \bar{n}$ — constant matrices)

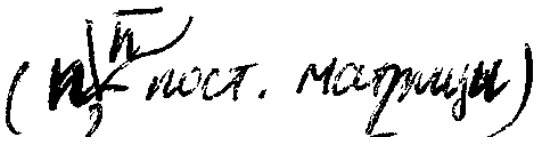


Diagram and typed description. [$\bar{n}$ is inserted above the line with an arrow to the position after n.]

Original source details: GELFAND-P001144

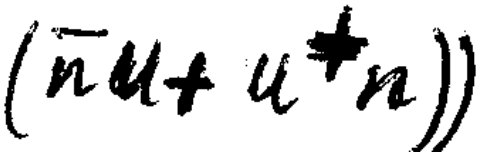


Diagram and typed description. The letter for the matrix reads n.

Original source details: GELFAND-P001144

Nontrivial asymptotics of large groups (there are no asymptotics).

Again.

$$\int d\vec{\tau}\,\delta(1-\vec{\tau}^{\,2})e^{i\vec{k}\vec{\tau}} \qquad 2\pi i\,\delta(x) = \int_{-i\infty}^{i\infty} e^{\alpha x}\,d\alpha$$

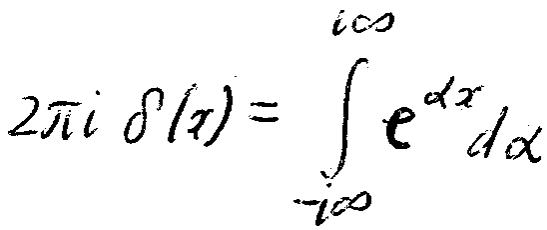


Diagram and typed description. $2\pi i\,\delta(x) = \int_{-i\infty}^{i\infty} e^{\alpha x}\,d\alpha$.

Original source details: GELFAND-P001144

||

Uncertain reading. The crossed-out entry before $\frac{1}{2\pi i}$ may read $\int d\alpha$.

Original source details: GELFAND-P001144

$$\frac{1}{2\pi i}\, n \int d\alpha \int d\vec{\tau}\, e^{n\alpha(1-\vec{\tau}^{\,2})+i\vec{k}\vec{\tau}} =$$
$$= \int d\alpha \, \exp\left(-\frac{n}{2}\ln\alpha + n\alpha - \frac{k^2}{4\pi\alpha}\right).$$

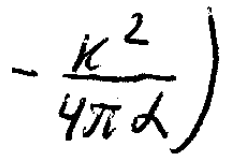


Diagram and typed description. The last denominator is written $4\pi\alpha$.

Original source details: GELFAND-P001144

$k \sim n \Rightarrow$ apply the saddle-point method

Entropy: n-fold integral with an exponential containing n, equals $e^{n(\ldots)}$

$\Rightarrow$ ln of the number of states $=$ ln of the integral $\sim n$.

Try this for (2) — replace it with $u \in U(N)$

$$\int \underbrace{du\, du^+\, \delta\overbrace{\left(I - u^+u\right)}^{\text{matrix}}}_{\text{inv. measure} \Rightarrow \substack{\text{measure}\\ \text{of Haar}}} \exp\left(N\beta\,\mathrm{tr}(\bar{n}u + u^+n)\right).$$

Uncertain reading. An illegible mark is crossed out after the first $\int$ and before du.

Original source details: GELFAND-P001144

Uncertain reading. A crossed-out entry between du^+ and δ may read $d\alpha$.

Original source details: GELFAND-P001144

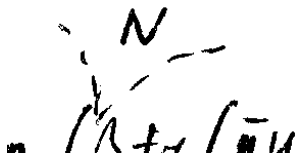


Diagram and typed description. N is inserted above β. Two dashed arrows point to the position before β.

Original source details: GELFAND-P001144

$$\int du\, du^+\, d\alpha\, \exp\left(N\,\mathrm{tr}(\alpha - u^+u\alpha) + N\beta\,\mathrm{tr}(\bar{n}u + u^+n)\right).$$

GELFAND-P001145

[source number: 28]

$\int d\alpha$ cannot be evaluated by the saddle-point method.
Entropy associated with α, is of the same order:
N^2 terms and N^2 integrals $\Rightarrow$ each integral ~ 1 and there are no asymptotics
(for the 1-dimensional integral)
(Rule of stat. phys.:
Multiple integral $\sim e^k$, where k — the number of dimensions)

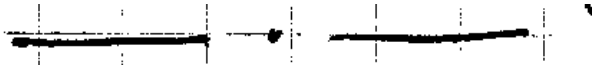


Diagram and typed description.

Original source details: GELFAND-P001145

Ambula.
Consider the problem

$$\int dA\ \exp\bigl(+N\beta\,\mathrm{Sp}(V(A))\bigr)\ \mathrm{Sp}(e^{At}) \tag{3}$$

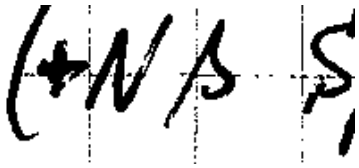


Diagram and typed description. Sign in (3): $+N\beta$.

Original source details: GELFAND-P001145

A — [inserted above: skew-]Hermitian matrices (alg. of the group $SU(N)$) $\quad A^+ = -A$.

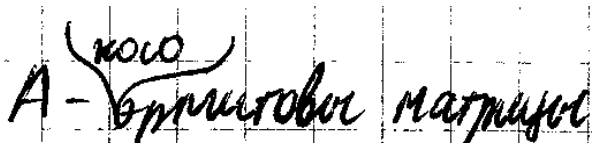
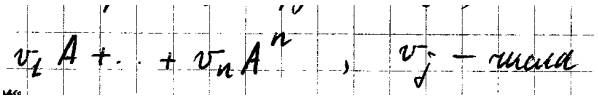


Diagram and typed description. “skew-” above the line, with an insertion mark before “Hermitian”.

Original source details: GELFAND-P001145

$$\underset{\text{— polynomial}}{V(A)} = v_1 A + \ldots + v_n A^n, \qquad v_j \text{ — numbers}$$



Uncertain reading. The coefficient symbol is read as v; ν is also possible.

Original source details: GELFAND-P001145

$$\Psi(t) = \frac{(3)}{\int dA\ \exp\bigl(-N\beta\,\mathrm{Sp}(V(A))\bigr)} \tag{4}$$

(generating function for moments)

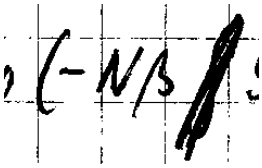


Diagram and typed description. Sign in the denominator of (4): $-N\beta$.

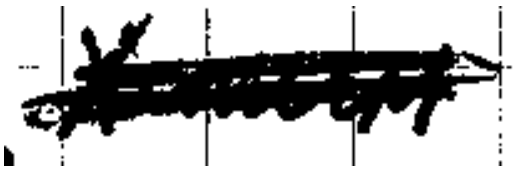
Original source details: GELFAND-P001145

Original source details: GELFAND-P001145

Uncertain reading. An illegible mark is crossed out between $N\beta$ and Sp.

A more complicated object: the Wilson average

$$\operatorname{Sp} T \exp\left(\int_C A_\alpha \, dx_\alpha\right). \qquad (T \text{ — matrix})$$



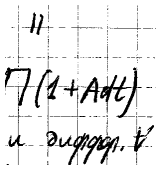
Original source details: GELFAND-P001145

Uncertain reading. An illegible word is crossed out. A branching initial stroke is visible.

Consider.

$$0 = \int dA \, \frac{\partial}{\partial A_{ij}} \left[e^{At} \exp\bigl(N\beta \operatorname{Sp} V(A)\bigr) \right] \tag{5}$$

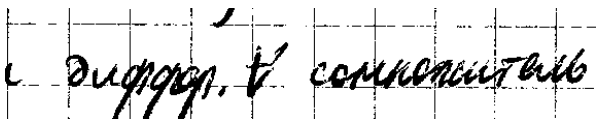
$$\sum_{i,j} \frac{\partial}{\partial A_{ij}} \underset{\parallel}{(e^{At})_{ij}} = \int_0^t d\tau \, \operatorname{Sp}(e^{A\tau}) \operatorname{Sp}(e^{A(t-\tau)})$$
$$\prod(1+A\,dt)$$



Original source details: GELFAND-P001145

Diagram and typed description. Beneath $(e^{At})_{ij} \parallel \prod(1 + A\,dt)$.

and differentiate the t' factor
(what comes before collects into exp;
what comes after also collects into exp).



Original source details: GELFAND-P001145

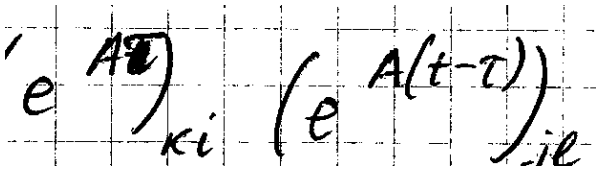
Uncertain reading. There is a slanted stroke above t, tentatively read as t'.

More generally:

Original source details: GELFAND-P001145

Uncertain reading. A summation sign before the derivative in (6) is crossed out, possibly $\sum_{i,j}$.

$$\frac{\partial}{\partial A_{ij}}(e^{At})_{k\ell} = \int_0^t d\tau\,(e^{A\tau})_{ki}\,(e^{A(t-\tau)})_{j\ell} \tag{6}$$



Uncertain reading. The last mark in the first exponent on the right, $A\tau$, is overwritten. The earlier mark is illegible.

Original source details: GELFAND-P001145

$$\frac{\partial}{\partial A_{ij}} \operatorname{Sp} V(A) = (V'(A))_{ji} \tag{7}$$

(V — polynomial)

Using (6), (7), differentiate in (5).

GELFAND-P001146

One calculates

$$\frac{\int dA\ \exp(N\beta \operatorname{Sp} V(A))\operatorname{Sp}(e^{At})/N}{\int dA\ \exp(N\beta \operatorname{Sp} V(A))} \overset{\text{def}}{=} \left\langle \frac{\operatorname{Sp} e^{At}}{N} \right\rangle \tag{8}$$

Diagram and typed description. Above the equals sign is written $\overset{\text{def}}{=}$.

Original source details: GELFAND-P001146

From (5)–(7) we find

$$0 = \left\langle \int_0^t d\tau\ \operatorname{Sp} e^{A\tau} \operatorname{Sp} e^{A(t-\tau)} + N\beta \operatorname{Sp}(V'(A)e^{At}) \right\rangle \tag{9}$$

$N\beta$ [crossed out ~~Sp~~?] $V'\left(\frac{\partial}{\partial t}\right)\operatorname{Sp}(e^{At})$

Uncertain reading. A crossed-out sign after $N\beta$ tentatively reads Sp.

Original source details: GELFAND-P001146

$$\operatorname{Sp} \mapsto \frac{\operatorname{Sp}}{N} \qquad \text{(mean of the eigenvalues)}$$

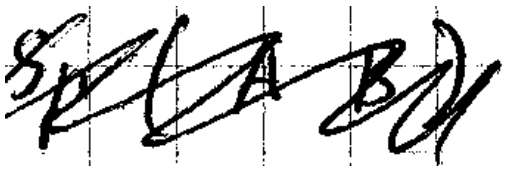


Diagram and typed description. [crossed out] ~~Sp(AB)~~.

Original source details: GELFAND-P001146

$$\langle \mathrm{Sp}\, A \cdot \mathrm{Sp}\, B \rangle \sim \langle \mathrm{Sp}(A \cdot B) \rangle$$

(statist. fact.)

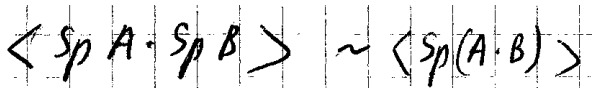


Diagram and typed description. The right-hand side is written $\langle \mathrm{Sp}(A \cdot B) \rangle$.

Original source details: GELFAND-P001146

We obtain

$$0 = \beta V' \left(\frac{\partial}{\partial t} \right) \Psi(t) + \int_0^t d\tau\, \Psi(\tau) \Psi(t - \tau) \tag{10}$$

Solved by the Fourier (Laplace) transform

Uncertain reading. The abbreviation before “Fourier” tentatively reads «пробр.» (“transform”).

Original source details: GELFAND-P001146

$$\beta V'(\omega) \Psi(\omega) + \Psi^2(\omega) = K(\omega) \tag{11}$$

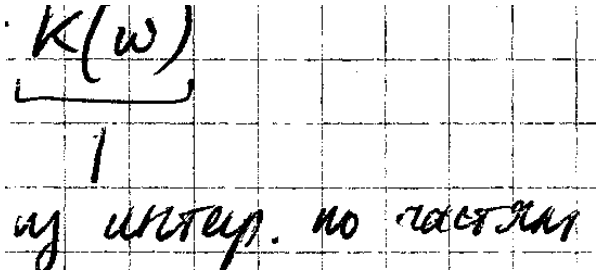


Diagram and typed description. [Under the bracket from $K(\omega)$:] from integration by parts.

Original source details: GELFAND-P001146

Hence

$$\Psi = -\frac{\beta}{2} V' + \sqrt{\frac{\beta^2}{4} V'^2 - K(\omega)} \tag{12}$$

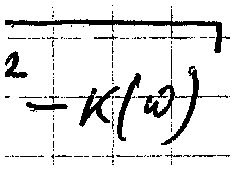


Diagram and typed description. Under the radical is written $-K(\omega)$.

Original source details: GELFAND-P001146

$K(\omega)$ is chosen so that the radical can almost be evaluated.

For the $\int$ over the group, analogously. The equation looks like this:

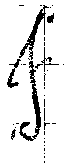
$$\int du\,[F(u) - F(u(1+\delta))] = 0 \qquad \text{(right-invariance condition for Haar measure)} \tag{13}$$

$$\int du \left(\frac{\partial F}{\partial u_{ij}} u_{jk}\right) = 0$$

and so on.

This can be done for arbitrary N and so on. $\Rightarrow$ hence an algorithm to find the series in powers of $1/N$.

Field theory — analogous equations, but functional.



Uncertain reading. [At the bottom] An isolated sign resembling $\int$.

Original source details: GELFAND-P001146

GELFAND-P001147

[source number: 29]

$$\frac{\int e^{N\beta \operatorname{Sp} V(A)} \frac{1}{N} \operatorname{Sp} e^{At}\, dA}{\int e^{N\beta \operatorname{Sp} V(A)}\, dA} = \Psi_N(t) = \Psi(t) + \frac{1}{N^2}\Psi_1(t) + \cdots \tag{14}$$

(asympt. expansion in $1/N$)

Consider also

$$\left\langle \frac{\operatorname{Sp} e^{At}}{N} \right\rangle = \frac{\int dA\, \exp\left(N\beta \operatorname{Sp} V(A) + \int J(t)\, dt\right) \operatorname{Sp} e^{At}/N}{\int dA\, \exp\left(N\beta \operatorname{Sp} V(A) + \int dt\, J(t) \operatorname{Sp}(e^{At})\right)} \tag{15}$$

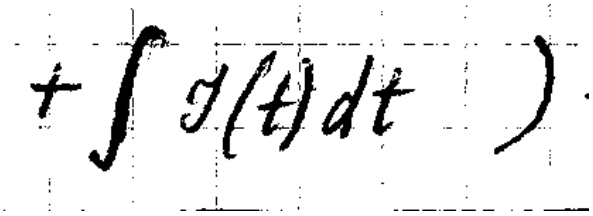
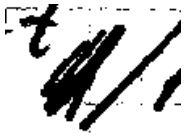


Diagram and typed description. The numerator exponent contains $\int J(t)\, dt$.

Original source details: GELFAND-P001147



Uncertain reading. An illegible mark is crossed out after $\operatorname{Sp} e^{At}$ and before $/N$.

Original source details: GELFAND-P001147

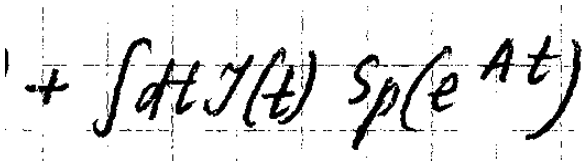


Diagram and typed description. The denominator exponent contains $\int dt\, J(t) \operatorname{Sp}(e^{At})$.

Original source details: GELFAND-P001147

The resulting equation: (something like the Fourier equation)

$$\beta V'\left(\frac{\partial}{\partial t}\right)\Psi(t,J) = \int_0^t d\tau\,\Psi(\tau)\Psi(t-\tau) + \\ + \frac{1}{N}\left\{\int_0^t d\tau\,\frac{\delta\Psi(t-\tau)}{\delta J(\tau)} + \int_0^\infty d\tau\,J(\tau)\Psi(t+\tau)\right\} \tag{16}$$

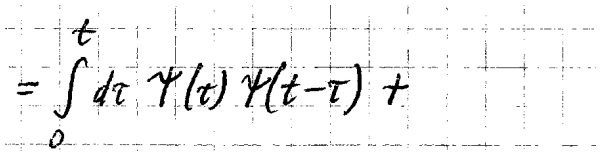


Diagram and typed description. In (16) the integral immediately follows the equals sign. The + at the end of the first line is repeated at the beginning of the next.

Original source details: GELFAND-P001147

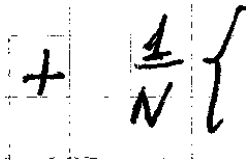


Diagram and typed description. Before the brace is written $\frac{1}{N}$.

Original source details: GELFAND-P001147

Field theory.

$$V(A) = F_{\mu\nu}^2, \qquad V'(A) \equiv \nabla_\mu F_{\mu\nu}$$

$$V'\left(\frac{\partial}{\partial t}\right) \mapsto \int_0^1 ds_1 \int_{s_1-0}^{s_1+0} ds_2\,\frac{\delta^2}{\delta x_\mu(s_1)\delta x_\nu(s_2)}.$$

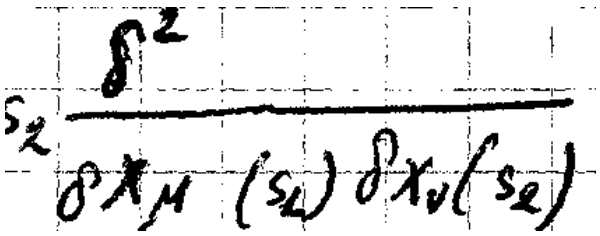


Diagram and typed description. The denominator reads $x_\mu(s_1)$, $x_\nu(s_2)$.

Original source details: GELFAND-P001147

Wiegmann P.

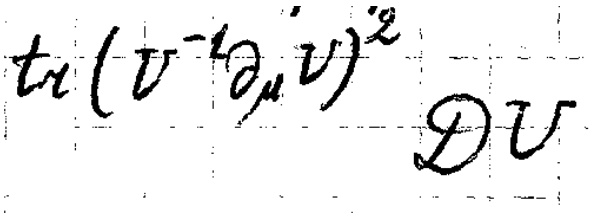
Exact sol. of the principal chiral field in 2-dimensional space

(jointly with Polyakov, following the programme he proposed 10 years ago)

$$\int e^{-\frac{1}{T}\int d^2x\ \mathrm{tr}(U^{-1}\partial_\mu U)^2}\,\mathcal{D}U \tag{1}$$

Diagram and typed description. The coefficient of the integral reads minus $\frac{1}{T}$.

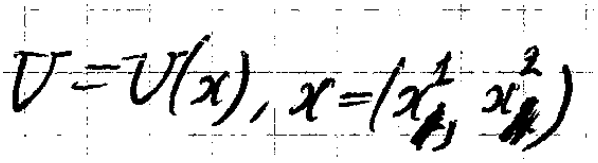
Original source details: GELFAND-P001147



Uncertain reading. The same letter in the action, the measure and the field definition is read as U. Its angular shape also allows V.

Original source details: GELFAND-P001147

$U = U(x), \quad x = (x^1, x^2)$



Uncertain reading. Illegible subscripts are crossed out below x^1, x^2.

Original source details: GELFAND-P001147

(from $n = 2$ to $n = 4$ there are ideas by A. B. Zamolodchikov, so far without success)

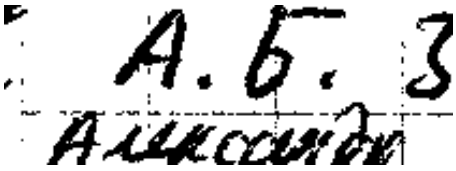
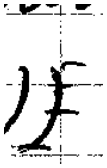


Diagram and typed description. [Under «A. B.»:] Alexander.

Original source details: GELFAND-P001147

What does it describe? π and K-mesons (if in $\mathbb{R}^4$)



Uncertain reading. After the parenthesis is a mark resembling a crossed-out $\int$.

Original source details: GELFAND-P001147

$(1+1)$ QFT or equivalent 2-dimensional stat. problems

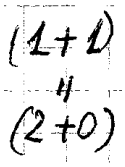


Diagram and typed description. [Under $(1+1)$:] $\|$ and below: $(2+0)$.

Original source details: GELFAND-P001147

Many 2-dimensional quantum models have been solved recently. This is one of them.
Glossary of the field theory terms needed

1. Renormalization group
2. Asymptotic freedom
3. Quantum anomaly of the axial current.
4. Bethe ansatz (Bethe) $\cong$ quantum inverse scattering method
5. $\mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu^{(x)}\gamma^\mu) = W\{A\}, \qquad A_\mu = \sum_a A_\mu^a X^a$

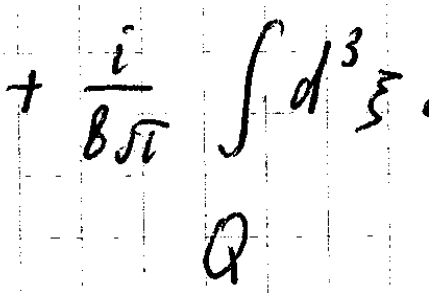
GELFAND-P001148

$$A_+ = a^{-1}\partial_+ a, \qquad A_- = b^{-1}\partial_- b.$$

$$g = ab^{-1}$$

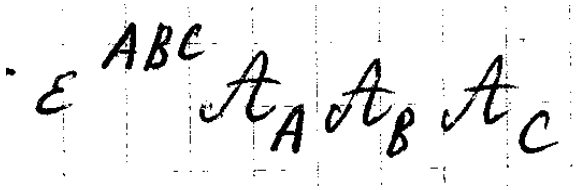
$$\mathcal{A}_\mu = g^{-1}\partial_\mu g$$

$$W\{A\} = \frac{1}{2}\int \operatorname{tr}\mathcal{A}_\mu^2\, d^2x + \frac{i}{8\pi}\int_Q d^3\xi\, \varepsilon^{ABC}\mathcal{A}_A\mathcal{A}_B\mathcal{A}_C$$



Original source details: GELFAND-P001148

Diagram and typed description. Coefficient of the cubic term: $\frac{i}{8\pi}$.

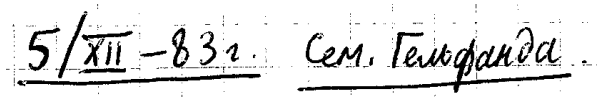


Original source details: GELFAND-P001148

Diagram and typed description. $\varepsilon^{ABC}\mathcal{A}_A\mathcal{A}_B\mathcal{A}_C$.

$$\partial Q = S^2,$$

GELFAND-P001149



Original source details: GELFAND-P001149

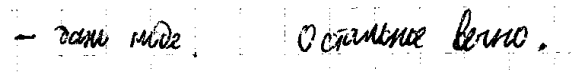
Diagram and typed description. 5/XII-83 yr. Gelfand Sem..

Original source details: GELFAND-P001149

Diagram and typed description. P. Wiegmann.

What does one need to know?

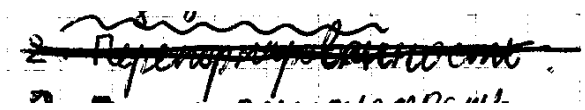
1. Bethe ansatz. — basic idea. The rest [вно.?].



Original source details: GELFAND-P001149

Uncertain reading. After «осн.» the manuscript reads «иде.» (“idea”). The abbreviation after “The rest” is uncertain and may read «вно.».

~~2. Renormalizability.~~



Original source details: GELFAND-P001149

Diagram and typed description. ~~2. Renormalizability.~~

2. Renormalizability.

3. Critical phenomena.

4. Asymptotic freedom.

5. Gauge theories: axial anomaly,

[crossed out] Instantons and topology, θ-vacuum.

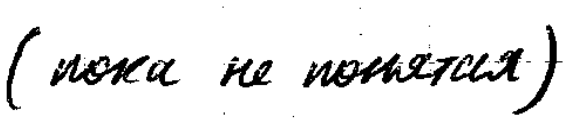
Original source details: GELFAND-P001149

Uncertain reading. An illegible mark to the left of “Instantons” is crossed out, possibly an item number.

6. String theory, which does not yet exist, the Wilson loop, minimal surfaces, statistics of random surfaces.

7. Virasoro algebra.

8. Solid-state physics: there is Anderson localization (as yet not understood).



Original source details: GELFAND-P001149

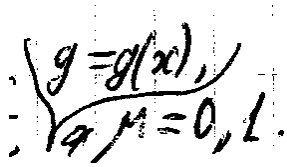
Diagram and typed description. (as yet not understood).

Original source details: GELFAND-P001149

Diagram and typed description. [Under «localization»:] (phenomenon).

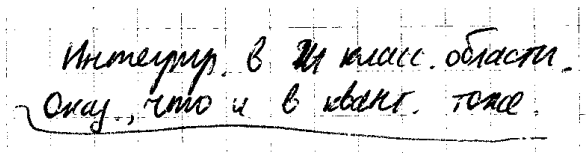
The principal chiral field in two dimensions.

$$S = \frac{1}{2\lambda}\int d^2x\ \mathrm{tr}(g^{-1}\partial_\mu g)^2, \qquad g \in G, \quad \begin{matrix} g = g(x), \\ \mu = 0, 1. \end{matrix} \tag{1}$$



Original source details: GELFAND-P001149

Diagram and typed description. [An insertion to the right of $g \in G$] $g = g(x); \quad \mu = 0, 1.$



Original source details: GELFAND-P001149

Uncertain reading. [At right] Integr. in the [crossed out] classical domain. It turns out this also holds in the quantum domain.

Quantum integrability: the possibility of calculating the integral

$$Z = \int e^{-S}\,\mathcal{D}g. \tag{2}$$

or diagonalizing the corresponding Hamiltonian

There will be standard calculations for this $\longrightarrow$ Bethe ansatz.

One needs to know an eigenstate (possibly other than the ground state). Such a state is unknown for the bosonic field. One must take the next step:

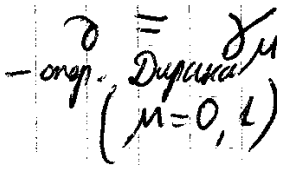
(3) Prop. The theory is equivalent to a fermionic theory with Lagrangian

$$\mathcal{L} = \bar{\psi} i \hat{\partial} \psi + \lambda j^a_\mu j^\mu_a, \qquad S = \int \mathcal{L}\, d^2x \tag{4}$$

$$\psi = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix} \quad \text{(Dirac spinor)},$$

transforms according to $O(2)$ (Lorentz group in the 2-dimensional case)

$$\hat{\partial} = \gamma_\mu \partial^\mu, \qquad \{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$$



Original source details: GELFAND-P001149

Diagram and typed description. [Under $\hat{\partial}$:] — Dirac operator $(\mu = 0, 1)$.

γ_0, γ_1

GELFAND-P001150

[source number: 36]

Isotopic indices

$$\alpha = 1, \dots, N \quad \text{— colour.}$$
$$(g \in SU(N)) \qquad f \quad \text{— scent = aroma (flavour)}$$
$$f = 1, \dots, N_f$$

$$j^a_\mu = \sum_f \bar{\psi}^\alpha_f \gamma_\mu X^a_{\alpha\beta} \psi^\beta_f, \qquad X^a \text{ — generators of the group } SU(N) \tag{5}$$

The interaction is constructed from the currents.

(6) Prop. As $N_f \longrightarrow \infty$ these theories are equivalent.

(theory (4) goes over into theory (1))

The colour groups must coincide.

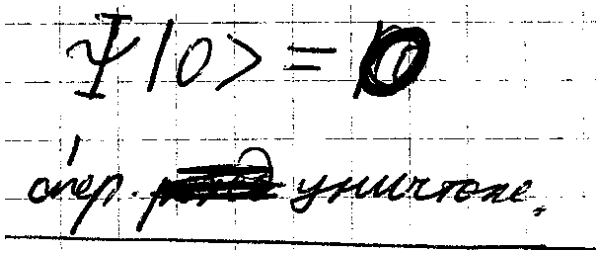
There is left and right invariance.

$$g \longrightarrow gG \quad \text{and so on; group } G \times G$$

For theory (4): $\underbrace{G \times F}$

Right invariance is absent and is restored at ∞ flavours.

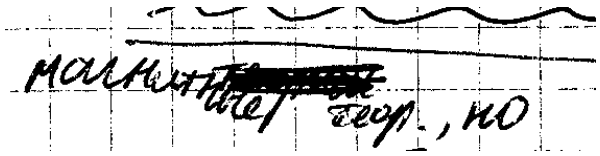
For (4) there is an eigenvector — the vacuum $|0\rangle$: $\Psi|0\rangle = 0$



Uncertain reading. [At the stroke under Ψ:] operator [~~creation~~?] annihil..

Original source details: GELFAND-P001150

(1) describes 2-dimensional magnetic [the ending is inserted below] [inserted below] theor., but such problems resemble 4-dimensional gauge theories (there are arguments, but only verbal ones so far).



Uncertain reading. The end of the original word is crossed out. The ending «ные» and then «теор.» (“theor.”) are written below.

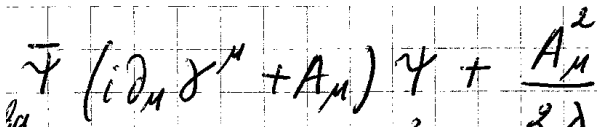
Original source details: GELFAND-P001150

Proof of equivalence.

Uncertain reading. An illegible sign is crossed out at left under the heading.

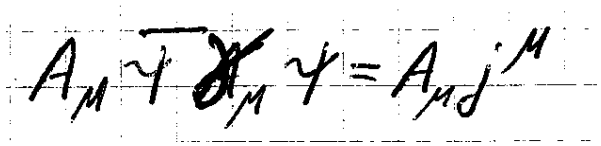
Original source details: GELFAND-P001150

$$(4) \Longrightarrow \bar{\psi}(i\partial_\mu \gamma^\mu + A_\mu)\psi + \frac{A_\mu^2}{2\lambda} \tag{7}$$



Original source details: GELFAND-P001150

Diagram and typed description. In (7) no γ^μ is written beside A_μ.

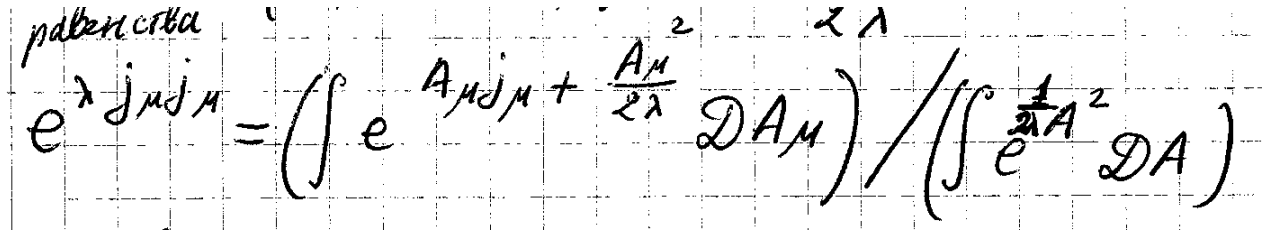


Original source details: GELFAND-P001150

Diagram and typed description. [At right] $A_\mu \bar{\psi}\gamma_\mu\psi = A_\mu j^\mu$. The γ is retraced.

from the equality

$$e^{\lambda j_\mu j_\mu} = \left(\int e^{A_\mu j_\mu + \frac{A_\mu^2}{2\lambda}} \mathcal{D}A_\mu\right) \Big/ \left(\int e^{\frac{1}{2\lambda}A^2} \mathcal{D}A\right) \tag{8}$$

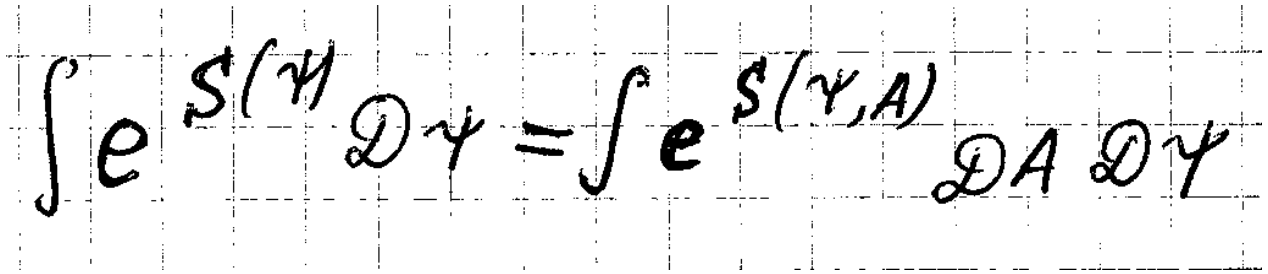


Original source details: GELFAND-P001150

Diagram and typed description. Both quadratic terms in (8) are written with positive signs.

(Gaussian integral with an intermediate field)

$$\int e^{S(\psi)}\,\mathcal{D}\psi = \int e^{S(\psi,A)}\,\mathcal{D}A\,\mathcal{D}\psi \tag{9}$$



Original source details: GELFAND-P001150

Diagram and typed description. The exponents in (9) are written $S(\psi)$, $S(\psi, A)$.

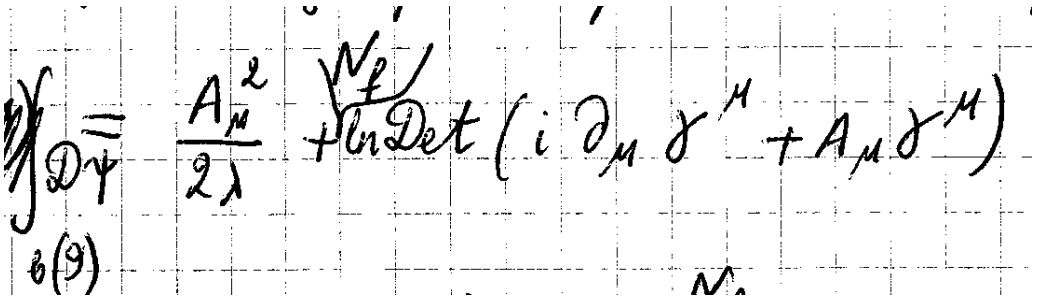
The index f is omitted for simplicity.

(Sometimes called the Stratonovich–Hubbard formula)

Original source details: GELFAND-P001150

Uncertain reading. An illegible sign is crossed out before the integral.

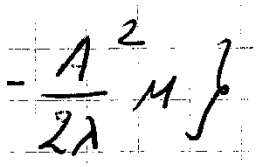
$$\int_{\mathcal{D}\psi} = \frac{A_\mu^2}{2\lambda} + \overset{N_f}{\ln}\,\mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu\gamma^\mu)$$



Original source details: GELFAND-P001150

Uncertain reading. Beneath the integral is written “in (9)”. N_f is inserted above ln Det. The left-hand side is in shorthand.

$$S(\psi, A) = \int d^2x \left\{ \sum_{f=1}^{N_f} \bar{\psi}_f (i\partial_\mu\gamma^\mu + \gamma^\mu A_\mu)\psi_f + \frac{A^2}{2\lambda}\,\mu \right\} \tag{10}$$



Uncertain reading. The μ is placed after the fraction and before the brace. It may be a displaced index on A^2.

Original source details: GELFAND-P001150

GELFAND-P001151

This gives

$$S\{A\} = \int \left[\frac{A_\mu^2}{2\lambda} + N_f \ln \mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu\gamma^\mu) \right] d^2x \tag{11}$$

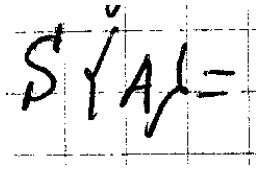


Diagram and typed description. $S\{A\}$.

Original source details: GELFAND-P001151

One needs to calculate

Uncertain reading. After “calculate” is a looped mark, possibly a crossed-out $\int$.

Original source details: GELFAND-P001151

$$\int e^{-S(A)}\,\mathcal{D}A_\mu \tag{12}$$

Diagram and typed description. [At right:] A_μ is already a bosonic field.

Original source details: GELFAND-P001151

and then $N_f \longrightarrow \infty$.

$\Longrightarrow$ in the limit, the trajectories that contribute are those where min ln Det

This gives a constraint on the fields A_μ: namely, min on the fields

$$A_\mu = g^{-1}\partial_\mu g \tag{13}$$

(pure gauge field); these are exactly the ones on which $F_{\mu\nu} = 0$

$$\lim_{N_f\to\infty} \int e^{-S(A)}\,\mathcal{D}A_\mu = \int e^{-S(A_\mu)}\delta(F_{\mu\nu})\,\mathcal{D}A_\mu \tag{14}$$

$$F_{\mu\nu} = \partial_\nu A_\mu - \partial_\mu A_\nu + [A_\mu, A_\nu]$$

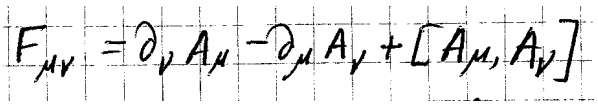
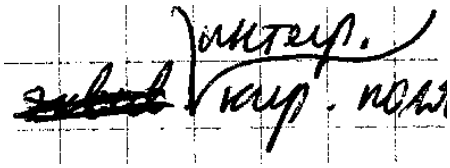


Diagram and typed description. The manuscript reads $\partial_\nu A_\mu - \partial_\mu A_\nu + [A_\mu, A_\nu]$.

Original source details: GELFAND-P001151

The problem with a finite number of flavours is integrable too. It is the one that is integrated, followed by passage to the limit.

Problem of classical integr. $\sim$ [~~equiv~~...] [inserted above: integr.] of the chiral field with the addition of a Wess–Zumino term.



Uncertain reading. The crossed-out word appears to begin “equiv...”. “Integr.” is inserted above the line.

Original source details: GELFAND-P001151

Gauge transform.:

$$A_\mu \longrightarrow U^{-1} A_\mu U + U^{-1} \partial_\mu U \tag{15}$$

$(13) \sim 0$ under this, or $F_{\mu\nu} = 0$ on (13).

det — gauge invariant; all eigenvalues of the Dirac operator are gauge inv.

$$(i\partial_\mu \gamma^\mu + A_\mu \gamma^\mu)\Psi = \lambda \Psi$$

The gauge transform. must be performed on A_μ and simultaneously on Ψ

(16) min — on fields gauge equiv. to 0.

Take the Abelian case for simplicity:

$$A_\mu = \partial_\mu \varphi + \epsilon_{\mu\nu} \partial_\nu \Phi \tag{17}$$

Transform. $A_\mu \longmapsto \partial_\mu \varphi + A_\mu$

In this case:

$$\ln \mathrm{Det}(i\partial_\mu \gamma^\mu + A_\mu \gamma^\mu) = \int d^2x \, (\partial_\mu \Phi)^2 \qquad \text{(Schwinger, 1961)} \tag{18}$$

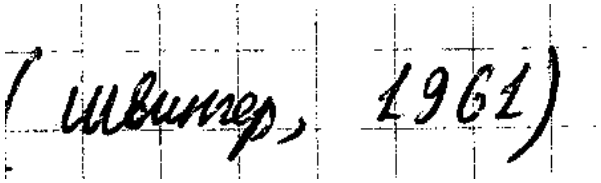


Diagram and typed description. (Schwinger, 1961).

Original source details: GELFAND-P001151

Hence the variation of det > 0, if there is Φ.

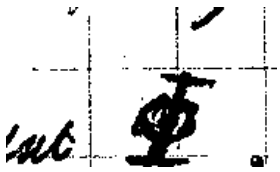


Diagram and typed description. The Φ is retraced. No separate reading can be established beneath the strokes.

Original source details: GELFAND-P001151

Moral: $A_\mu = \partial_\mu \varphi$ and so on.

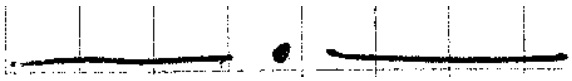


Diagram and typed description.

Original source details: GELFAND-P001151

GELFAND-P001152

[source number: 37]

Bethe ansatz method.

Simplify by dropping the flavours (leave 1), with colours $SU(2)$.

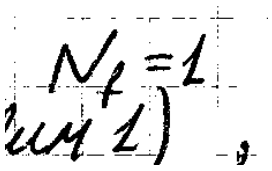
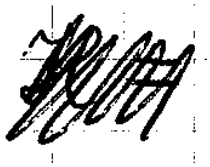


Diagram and typed description. [Above "leave 1"] $N_f = 1$.

Original source details: GELFAND-P001152

It will be a nonrelativistic problem. (for simplicity)

On diagonalizing the following fermionic Hamiltonian.

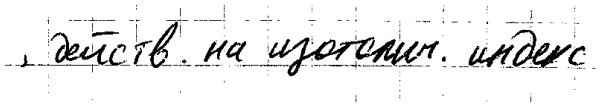


Original source details: GELFAND-P001152

Uncertain reading. A sign before (19) is crossed out, possibly $\mathcal{H}$.

$$\mathcal{L} = \bar{\psi}\, i\hat{\partial}\psi + \lambda\bar{\psi}\gamma_\mu\bar{\sigma}\psi\bar{\psi}\gamma_\mu\bar{\sigma}\psi \tag{19}$$

$\bar{\sigma}$ — the Pauli matrices (group $SU(2)$), acting on the isotopic index.

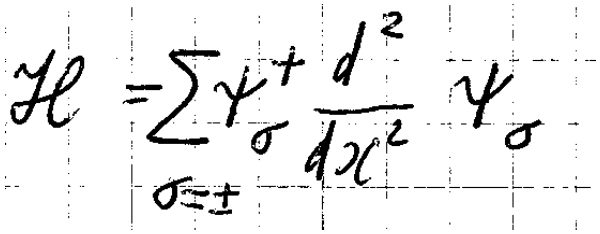


Original source details: GELFAND-P001152

Diagram and typed description. acting on the isotopic index.

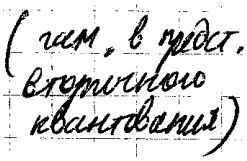
Hamiltonian density: (nonrelativ.):

$$\mathcal{H} = \sum_{\sigma=\pm} \psi_\sigma^+ \frac{d^2}{dx^2}\psi_\sigma + \lambda(\psi_\sigma^+\psi_\sigma)(\psi_{\sigma'}^+\psi_{\sigma'}) \tag{20}$$



Original source details: GELFAND-P001152

Diagram and typed description. The kinetic term in (20) is written with $\frac{d^2}{dx^2}$.



Original source details: GELFAND-P001152

Diagram and typed description. [At right of (20):] (Ham. in the second-quantized representation)

σ — spin.

The spectrum of $\mathcal{H}$ is already bounded below (unlike (19)) — this is in effect a regularization of (19) and the problems are equivalent up to this regularization.

$\psi_\sigma = \psi_\sigma(x)$.

Hamiltonian:

$$H = \int \mathcal{H}\, dx \qquad \psi_\sigma = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}. \tag{21}$$

ψ — a fermionic field, i.e.

$$\{\psi_\sigma^+(x), \psi_{\sigma'}(y)\} = \delta(x-y)\delta_{\sigma\sigma'} \tag{22}$$

Problem: find all eigenvalues and eigenfunctions of the Hamiltonian H.

$$H\Phi = \lambda\Phi. \tag{23}$$

$$N = \int \psi^+\psi\, dx \tag{24}$$

— number of particles. (commutes with H)

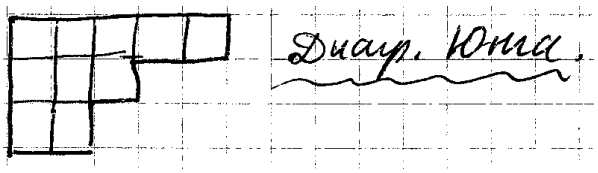
System of wave functions

$$\Phi(x_1, \dots, x_N, \sigma_1, \dots, \sigma_N)$$

$$S^z = \sum_\sigma \int \sigma\psi_\sigma^+\psi_\sigma\, dx \tag{25}$$

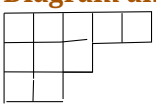
Consider a representation of the permutation group (of the pairs (x_j, σ_j)))

Fermions are antisymm..



Original source details: GELFAND-P001152

Diagram and typed description.

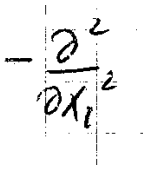


Young diagram.

$\Rightarrow$ on the n-particle subspace

$$\left[\sum_{i=1}^n -\frac{\partial^2}{\partial x_i^2} + \lambda \sum_{i<j} \delta(x_i - x_j)\right] \Phi = E\Phi \tag{26}$$

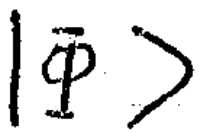
(in the first-quantized representation)



Original source details: GELFAND-P001152

Diagram and typed description. The kinetic term in (26) is written with $-\frac{\partial^2}{\partial x_i^2}$.

$$|\Phi\rangle = \int \Phi(x_1, \ldots, x_N) \prod_{i=1}^{N} \psi^+(x_i)|0\rangle \tag{27}$$



Uncertain reading. There is a stroke over the Φ in the ket, possibly part of the letterform.

Original source details: GELFAND-P001153

$|0\rangle$ — the vacuum, $\psi|0\rangle = 0$.

Φ — the same wave fn. as in (26).

Φ satisfies (26) $\Rightarrow |\Phi\rangle$ — a state vector.

Uncertain reading. Before “satisfies”, strokes overlap the Φ, one resembling a closing bracket.

Original source details: GELFAND-P001153

δ because the potential is of this form. In the general case:

$$\lambda\psi_\sigma^+(x)\psi_\sigma(x)V(x-y)\psi^+(y)\psi_\sigma(y)$$

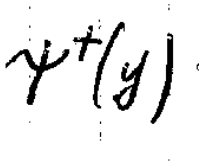
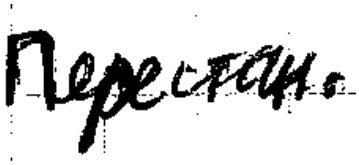


Diagram and typed description. The third field factor $\psi^+(y)$ has no subscript.

Original source details: GELFAND-P001153



Uncertain reading. Commutes

Original source details: GELFAND-P001153

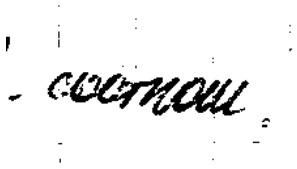
with the particle number ⇒ always a Schrödinger eq..

Integrability; Bethe ansatz — form of the wave fn..

1931 — Bethe wrote down the form of the wave fn. for our problem.

$$\Phi(x_1,\dots,x_N)=\sum_{\sigma_1,\dots,\sigma_N} A_{\sigma_{q_1}\dots\sigma_{q_N}}\sum_P (Q\cdot P)(-1)^P e^{i\sum_{j=1}^N x_j k_{P_j}} \tag{28}$$

k_{P_j} — arbitrary (characterize the state vector), $\sum k_j^2 = E$ — the only relation?



Uncertain reading. relation

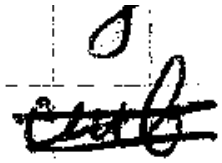
Original source details: GELFAND-P001153

Q, P — permutations of the numbers $\{1,\dots,N\}$, $(-1)^P$ — parity of the permutation, reflecting its fermionic nature.

(28) — in the region $X_Q = \{x_{q_1} < x_{q_2} < \dots < x_{q_N}\}$.

On passing [inserted above the line, circled and linked to this place: from one region X_Q to another] a collision occurs, the particles exchange places. A subtle point:

order the particles, there it is just a wave; we pass to another region, [crossed out: join?]



Uncertain reading. join

Original source details: GELFAND-P001153

joining the solns., will the momenta there be the same??

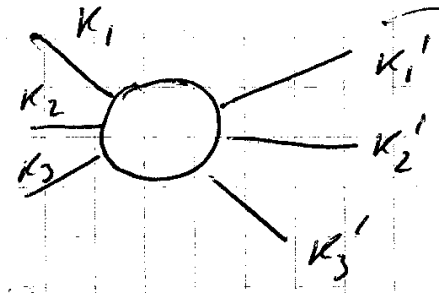
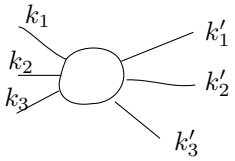


Diagram and typed description.



Original source details: GELFAND-P001153

[At the right of the diagram:] The laws of conservation of momentum and energy determine everything only for two.

Integrability: there are several more conservation laws.

Matching conditions — multiply by a matrix!

There is an analogous Bethe ansatz for any integrable system.

Nonintegrable case: $O(N)$ even with a δ-like potential.

GELFAND-P001154

12/XII-83 Gelfand Sem.

V. Ginzburg.

On a paper by Witten containing an attempt to unify gravitation and the electromagn. field.

I. $M = \mathbb{R}^4 \times S^1$, and S^1 — a small circle.

Uncertain reading. A crossed-out mark before the second S^1, possibly “and”.

Original source details: GELFAND-P001154

$$R_{ij} - \frac{1}{2} g_{ij} R = 0 \qquad \text{(Einstein's eq.)} \tag{1}$$

(write it on M)

g_{ij} — a 5×5 matrix (block matrix).

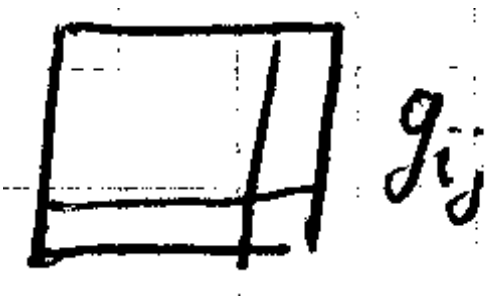
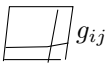


Diagram and typed description. 

Original source details: GELFAND-P001154

Expand g on S^1 in a Fourier series in $\varphi \in S^1$.

$$g = \sum g_k(x) e^{\frac{ik\varphi}{r}} \tag{2}$$

(r — radius of the small circle)

Grav. field is small $\rightsquigarrow$ Laplace's eq..

Write Laplace's eq.:

$$0 = \left(\Box_4 + \frac{\partial^2}{\partial \varphi^2} \right) g \tag{3}$$

from this and from (2):

$$\Box_4 g_k = \frac{k^2}{r^2} g_k \tag{4}$$

$\Rightarrow$ the kth term — a particle with mass $\frac{k^2}{r^2}$.

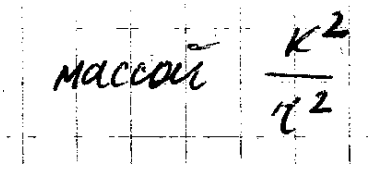


Diagram and typed description. The manuscript reads “mass” followed by $\frac{k^2}{r^2}$.

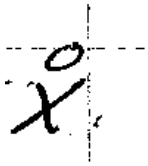
Original source details: GELFAND-P001154

$\underset{\sim}{k \neq 0} \Rightarrow$ with $r \sim$ radius of the electron one gets $m \sim 10^{37}$ [crossed out: illegible[?]]

Original source details: GELFAND-P001154

Uncertain reading. illegible

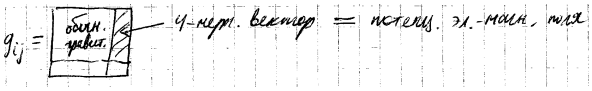
eV, which does not occur



Original source details: GELFAND-P001154

Diagram and typed description. After “does not occur” in the right margin, a slanted oval above two crossing strokes. The lower mark resembles x or χ.

Therefore $g = g(x)$ (but this does not save it either).



Original source details: GELFAND-P001154

Diagram and typed description.

(5) $g_{ij} =$ ordinary grav. — 4-dimensional vector = potential of the electromagn. field

(Kaluza–Klein theory)

The charge also comes out large (too large).

II part[?]

Original source details: GELFAND-P001154

Uncertain reading. part

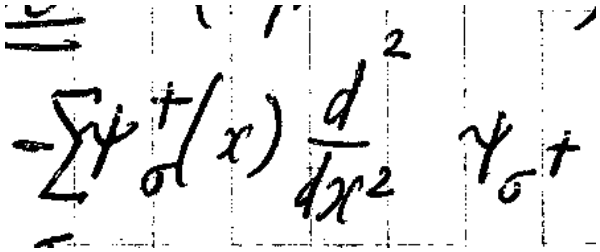
Positivity of the energy of the gravitational field.

P. Wiegmann (continued).

Quantum integrable systems by example, or
On a one-dimensional Fermi gas with δ-like attraction.

$$\mathcal{H} = -\sum_{\sigma} \psi_{\sigma}^{+}(x) \frac{d^2}{dx^2} \psi_{\sigma} + c \left(\sum_{\sigma} \psi_{\sigma}^{+} \psi_{\sigma} \right) \left(\sum_{\sigma'} \psi_{\sigma'}^{+} \psi_{\sigma'} \right) \tag{1}$$

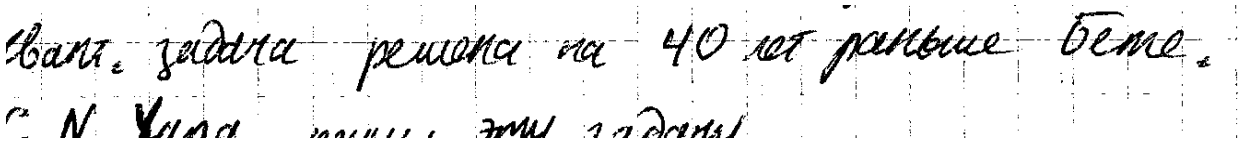
$\sigma = \pm 1$.



Original source details: GELFAND-P001154

Diagram and typed description. Here the kinetic term has a minus before the sum.

Classical analogue — NLS (nonlinear Schrödinger eq.), completely integr. (Zakh.–Shabat).
The quantum problem was solved 40 years before Bethe.



Original source details: GELFAND-P001154

Diagram and typed description. The quantum problem was solved 40 years before Bethe.

C. N. Yang solved this problem.

$$\psi_\sigma = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}$$

Complex field — there is also ψ^+.

GELFAND-P001155

[source number: 42]

$$\begin{aligned} \{\psi^+(x), \psi(y)\} &= \delta(x-y) \qquad \text{(fermions)} \\ \{\psi^+_\sigma(x), \psi_{\sigma'}(y)\} &= \delta(x-y)\delta_{\sigma\sigma'} \end{aligned} \tag{2}$$

Original source details: GELFAND-P001155

Uncertain reading. In the spin version of (2), the argument of the first operator, possibly (x), is unclear under overlapping strokes.

ψ^+ — creation op., ψ — annihilation op.

contents.

1. Bethe substitution and conservation laws
2. Integrability conditions (triangle eqs.)
3. Diagonalization of the transfer matrix (cumbersome computations)
   — the simplest method — Faddeev et al.

We have to solve the Schrödinger eq.:

$$\mathcal{H}|\phi\rangle = E|\phi\rangle \tag{3}$$

$$N = \int dx \sum_\sigma \psi^+_\sigma(x)\psi_\sigma(x) \qquad \text{(particle-number operator)} \tag{4}$$

$$[N, \mathcal{H}] = 0, \quad [\mathcal{H}, S] = 0, \quad \text{where} \tag{5}$$

Original source details: GELFAND-P001155

Uncertain reading. The $\mathcal{H}$ in the second commutator is overwritten.

$$S \text{ — spin op.:} \qquad S = \int dx \sum_{\sigma} \sigma \psi_{\sigma}^{+}(x) \psi_{\sigma}(x) \tag{6}$$

[crossed out: illegible mark?.]

Uncertain reading. illegible mark

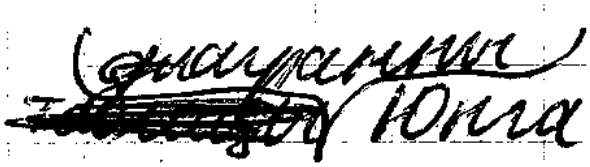
Original source details: GELFAND-P001155

(5_2) — spin conservation ($SU(2)$ symmetry)

We shall look for $|\phi\rangle$ with fixed particle number and spin.

We shall classify the repr. of the permutation group with the help of [crossed out: illegible word?;

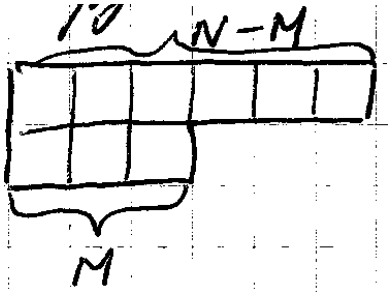
Uncertain reading. illegible word



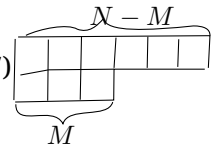
Original source details: GELFAND-P001155

inserted above the crossed-out word: “(diagram)”] of Young

Diagram and typed description.



(7)

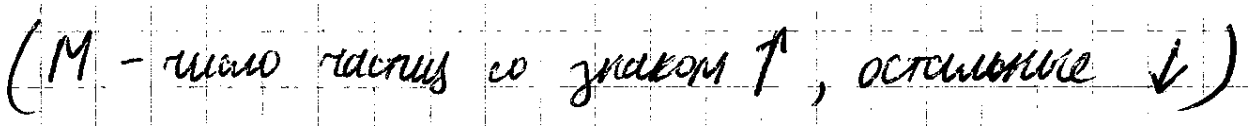


Original source details: GELFAND-P001155

Here the total spin

$$S = N - M - M = N - 2M \tag{8}$$

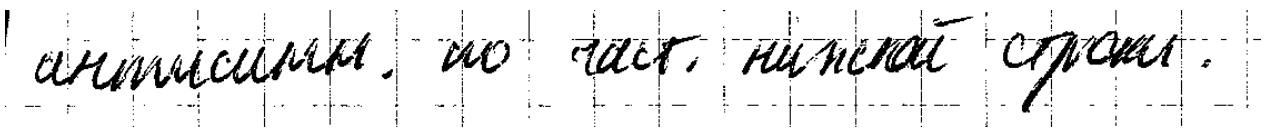
(M — number of particles with sign ↑, the others ↓)



Original source details: GELFAND-P001155

Diagram and typed description. The arrows in the line after (8) point up and down, respectively.

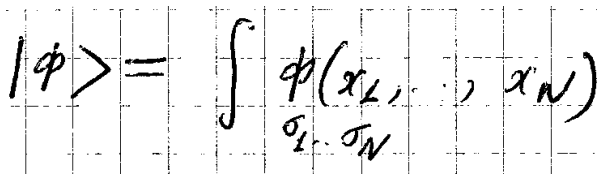
(9) ‖ Wave fn. symm. in the particles corresponding to the upper row and antisymm. in the particles of the lower row.



Original source details: GELFAND-P001155

Diagram and typed description. The manuscript reads “lower row”.

$$|\phi\rangle = \int \phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N)\psi^+_{\sigma_1}(x_1)\ldots\psi^+_{\sigma_N}(x_N) \, dx_1 \ldots dx_N \, |0\rangle \tag{10}$$



Original source details: GELFAND-P001155

Diagram and typed description. Equation (10) begins with an integral sign and no explicit spin-index sum.

$$|0\rangle \text{ — the vacuum;} \qquad \psi|0\rangle = 0 \tag{11}$$

(12) $\phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N)$ called the wave function.

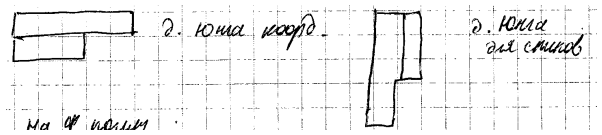
[In the right margin:] $\|$ antisymm. on swapping (x_i, σ_i) with (x_j, σ_j)

It must belong to repr. (7) of the symmetric group.

Expression (1) — second-quantized repr..

First quantization — an eq. for $\phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N)$

GELFAND-P001156



Original source details: GELFAND-P001156

Diagram and typed description.

(13)

Young d., coordinates

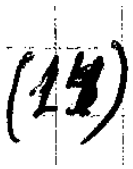
Young d. for spins

A short slanting stroke at the left of the first diagram resembles an arrowhead.

(width 2 because 2 spin values ± 1)

For ϕ we get:

$$\left\{ -\sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2} + c \sum_{i>j} \delta(x_i - x_j) \right\} \Phi = E\Phi \tag{14}$$



Original source details: GELFAND-P001156

Uncertain reading. The last digit of (14) is corrected and its previous form is illegible.

Explanation:

$\Phi_{\sigma_1 \dots \sigma_N}(x_1, \dots, x_N)$ — antisymm. under permutation of the pairs $(x_i, \sigma_i), (x_j, \sigma_j)$.

What happens if only the coordinates are permuted for a given set of σ_i?

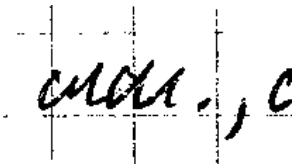
$$\phi = \sum f(x_1, \dots, x_N)\chi(\sigma_1, \dots, \sigma_N)$$

(sum of tensors; this can always be done; then we introduce only pairs f, χ with certain symmetries)

$$\Phi(x, \sigma) = \sum_{\tau_1, \tau_2} f_{\tau_1}(x) g_{\tau_2}(\sigma)$$

τ_1, τ_2 — repr. of the symmetric group.

Φ antisymm. $\Rightarrow$ only the terms[?],



Uncertain reading. terms

Original source details: GELFAND-P001156

corresp. to representations that complement one another are present.

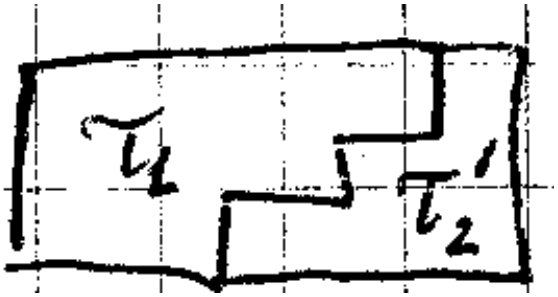
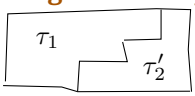


Diagram and typed description.



Original source details: GELFAND-P001156

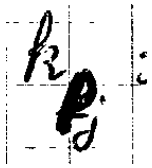
(see § in the chapter «Spin» book by Landau and Lifshitz, vol. 3)

Consider the region

$$X_Q = \{x_{q_1} < \dots < x_{q_N}\} \qquad (Q \text{ — permutation of } (1, 2, \dots, N)). \tag{15}$$

The answer in X_Q — Bethe substitution:

$$\Phi = \sum A_{\sigma_{q_1} \dots \sigma_{q_N}} (Q \cdot P)(-1)^P \exp i \sum_{j=1}^{N} k_{P_j} x_j \tag{16}$$



Uncertain reading. The subscript on k is overwritten, possibly P_j or p_j.

Original source details: GELFAND-P001156

$(k_1, \ldots, k_N)$ — momenta specifying the wave fn..

$$E = \sum_j k_j^2.$$

Nontriviality of formula (16):

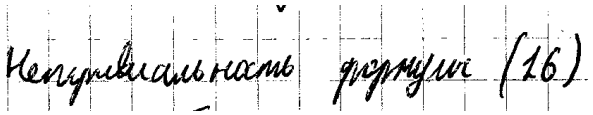


Diagram and typed description. Nontriviality of formula (16).

Original source details: GELFAND-P001156

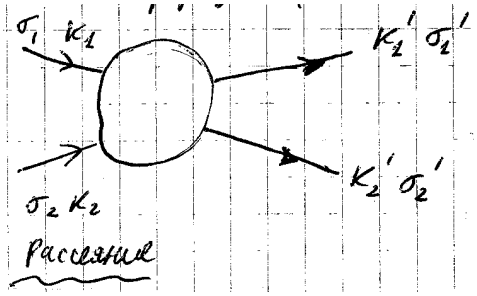
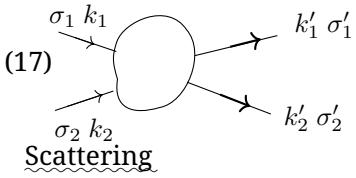


Diagram and typed description.



Original source details: GELFAND-P001156

GELFAND-P001157

[source number: 43]

$$\begin{aligned} k_1 + k_2 &= k_1' + k_2' \qquad \text{(cons. of momentum)} \\ k_1^2 + k_2^2 &= k_1'^2 + k_2'^2 \qquad \text{(cons. of energy)} \end{aligned} \tag{18}$$

(19) This is enough (in the 1-dimensional case) for fixing (k_1, k_2) (up to permutation). Only a change of spin can occur.

S-matrix (will be

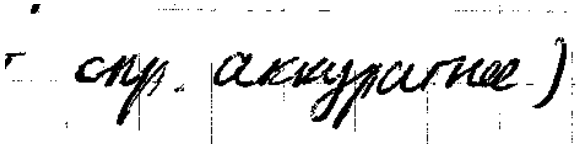
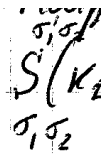


Diagram and typed description. defined

Original source details: GELFAND-P001157

more carefully later)

$$S^{\sigma_1'\sigma_2'}_{\sigma_1\sigma_2}(k_1, k_2) = bI + cP \qquad (P \text{ — rearrangement op.}) \tag{20}$$

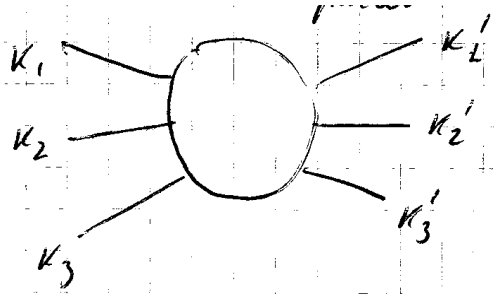


Uncertain reading. The first upper index in (20) is tightly written next to σ, possibly σ_1'.

Original source details: GELFAND-P001157

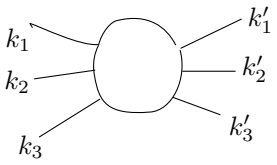
(σ_1, σ_2) — initial spins, (σ'_1, σ'_2) — final

Consider 3-particle scattering:



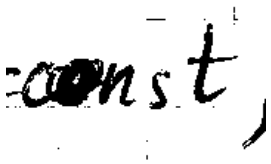
Original source details: GELFAND-P001157

Diagram and typed description.



A short angled stroke at the outer end of the upper left line has unclear significance.

$$\sum k_j = \text{const}, \qquad \sum k_j^2 = \text{const} \qquad \text{— insufficient} \tag{21}$$



Original source details: GELFAND-P001157

Uncertain reading. A letter in the first const in (21) is overwritten.

But because of integrability, here:

$$\sum k_j^n = \text{const}, \qquad n = 1, 2, 3, \ldots \tag{22}$$

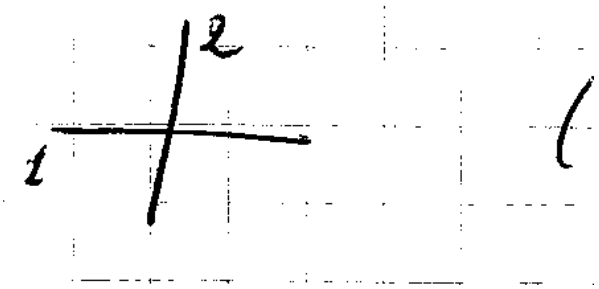
(absence of multiple production; the only event — a reshuffling of the particle spins)

$$\begin{matrix} \sigma_1 & k_1 \\ \vdots & \vdots \\ \sigma_n & k_n \end{matrix} \longmapsto \begin{matrix} \sigma_{q_1} & k_1 \\ \vdots & \vdots \\ \sigma_{q_n} & k_n \end{matrix} \tag{23}$$

Def. of the scattering matrix:

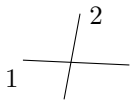
$$A(Q) = \widehat{S} A(Q') \tag{24}$$

Suppose 2 change places:



Original source details: GELFAND-P001157

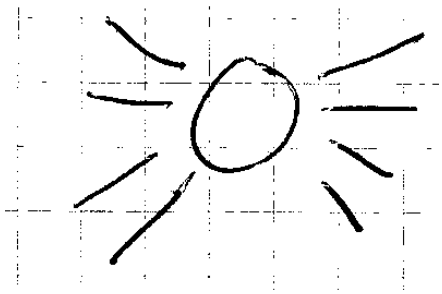
Diagram and typed description.

$S^{\sigma'_1\sigma'_2}_{\sigma_1\sigma_2}(k_1, k_2) =$ 

(25)

(notation)

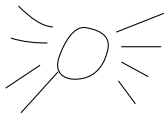
n-particle:



Original source details: GELFAND-P001157

Diagram and typed description.

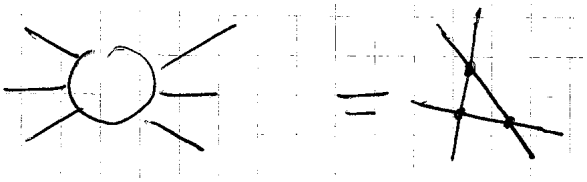
$$S^{\sigma'_1 \dots \sigma'_n}_{\sigma_1 \dots \sigma_n}(k_1, \dots, k_n) = \quad (26)$$



(27) Claim: (26) = a product of 2-particle scattering matrices.

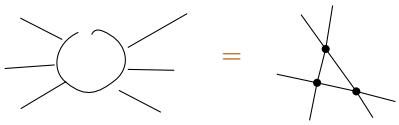
Original source details: GELFAND-P001157

Uncertain reading. Letters in “2-particle” have been corrected, obscuring their previous form.



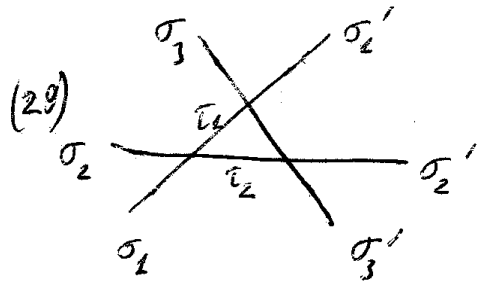
Original source details: GELFAND-P001157

Diagram and typed description.



(28)

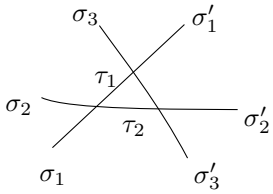
GELFAND-P001158



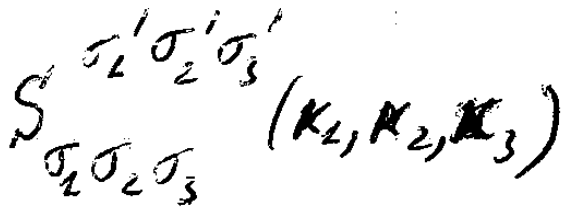
Original source details: GELFAND-P001158

Diagram and typed description.

(29)



$$S^{\sigma'_1 \sigma'_2 \sigma'_3}_{\sigma_1 \sigma_2 \sigma_3}(k_1, k_2, k_3) = \sum_{\tau} S^{\sigma'_2 \sigma'_3}_{\tau_2 \tau_3}(k_2, k_3) \quad (30)$$
$$S^{\sigma'_1 \tau_3}_{\tau_2 \sigma_3}(k_1, k_3) S^{\tau_1 \tau_2}_{\sigma_1 \sigma_2}(k_1, k_2)$$



Original source details: GELFAND-P001158

Uncertain reading. The momentum letters in the arguments on the left of (30) are heavily retraced or corrected.

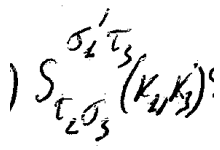


Diagram and typed description. The lower indices on the middle factor read $\tau_2\sigma_3$.

Original source details: GELFAND-P001158

Not every object [inserted below, circled and linked to “object”: (of type S)] is represented this way (by 2-particle scattering events).

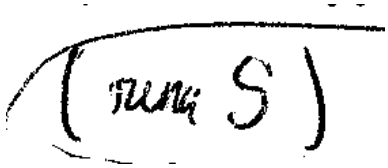


Diagram and typed description. (of type S)

Original source details: GELFAND-P001158

But this is so in integr. systems!!

Everything is as in a sys. of colliding hard balls, etc.

Condition for existence of repr. (30) — the $\triangle$ equation (nec. and suff.)

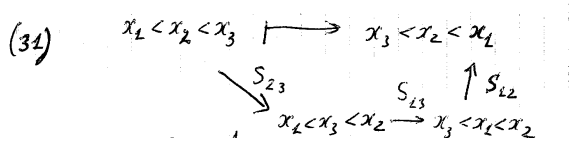


Diagram and typed description.

$$(31)\qquad \begin{array}{ccc} x_1<x_2<x_3 & \longmapsto & x_3<x_2<x_1 \\ \downarrow S_{23} & & \uparrow S_{12} \\ x_1<x_3<x_2 & \xrightarrow{S_{13}} & x_3<x_1<x_2 \end{array}$$

Original source details: GELFAND-P001158

(via pairwise steps)

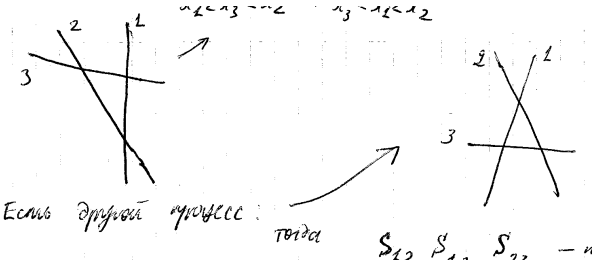
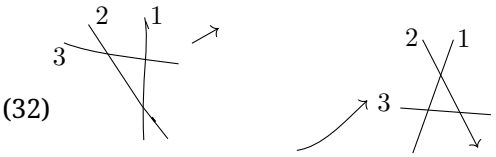


Diagram and typed description.



Original source details: GELFAND-P001158

In the left drawing, the short stroke at line 1 and the thickening on line 2 may be traces of arrowheads.

S_{12} S_{13} S_{23} — order of collisions

(33) Condition: the result of scattering must be the same in all ways: symbolically this is written as follows

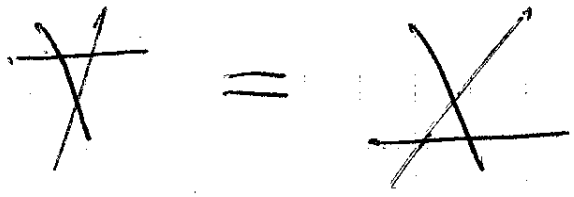
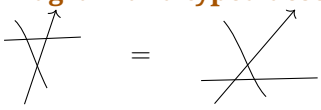


Diagram and typed description.



Original source details: GELFAND-P001158

Without isotopic indices: $S = e^{i\varphi}$ and everything holds automatically.

Uncertain reading. After i in the exponent is a looped sign, probably φ.

Original source details: GELFAND-P001158

With indices — a nontriv. relation:

$$\begin{aligned} &S_{23}(k_2, k_3) S_{13}(k_1, k_3) S_{12}(k_1, k_2) \\ &= S_{12}(k_1, k_2) S_{13}(k_1, k_3) S_{23}(k_2, k_3) \end{aligned} \tag{34}$$

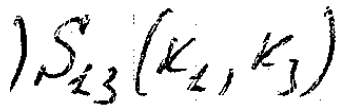


Diagram and typed description. On the right, the middle factor reads $S_{13}(k_1, k_3)$.

Original source details: GELFAND-P001158

This is a nec. and suff. condition for integrability of quantum systems. All quantum integr. sys. are described this way.

It is interesting that they exist!! (illegible abbreviation beginning with не?)

Uncertain reading. illegible abbreviation beginning with не

Original source details: GELFAND-P001158

They are still not fully classified.

GELFAND-P001159

[source number: 44]

Yang found the first soln..

2×2 Baxter found all solns.

(35) Claim. $\exists$ a parametrization $k(\theta)$, such that $S(k, k') = S(\theta - \theta')$

$$\left(\begin{array}{ll} \text{corresp.} & k = m \operatorname{sh} \theta \\ & \varepsilon = m \operatorname{ch} \theta \end{array} \right)$$

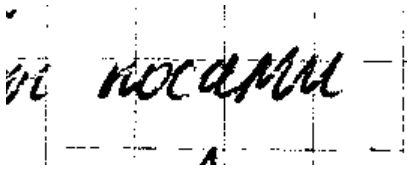
[At right:] $\| \ \theta - \theta'$ corresp. to the Galilei group and the Lorentz group

(36) Belavin, Drinfeld studied the quasi-class. $\triangle$ eqs.: $S = 1 + \hbar r$

(solved the corresp. quadr. eq.)

Tetrahedron: in multidimensional problems one must study strings, surfaces, etc.

Tetrahedra are sent? toward one another.



Uncertain reading. are sent

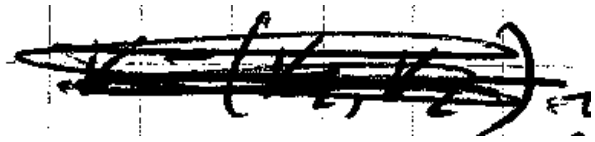
Original source details: GELFAND-P001159

No nontriv. solutions have been found so far (and it is suspected there are no nontriv. solutions; if there were, they would give integr. systems in high dimension)

The solns. found by Zamolodchikov describe a free particle.

Their existence would lead to multiple scattering (nontriv.)

Solution: [crossed out: $S(k_1, k_2)$ (?)]

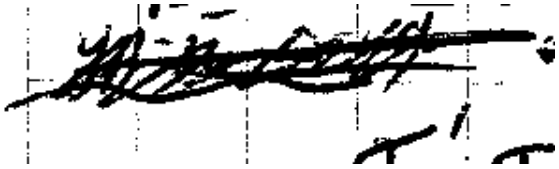


Uncertain reading. $S(k_1, k_2)$ (?)

Original source details: GELFAND-P001159

of the eqs. (34) for matrices of size 2×2 with the cond. of $SU(2)$ invariance for the upper and lower index pairs

[Under "Solution", crossed out: eqs. (34) (?)]



Uncertain reading. eqs. (34) (?)

Original source details: GELFAND-P001159

$$S^{\sigma_1'\sigma_2'}_{\sigma_1\sigma_2}(k) = I \cdot k + \text{[crossed-out coefficient]}\, \hat{P}, \qquad \hat{P} \text{ — permutation op.} \tag{37}$$

Uncertain reading. Between the plus and $\hat{P}$ is a heavily crossed-out, illegible coefficient.

Original source details: GELFAND-P001159

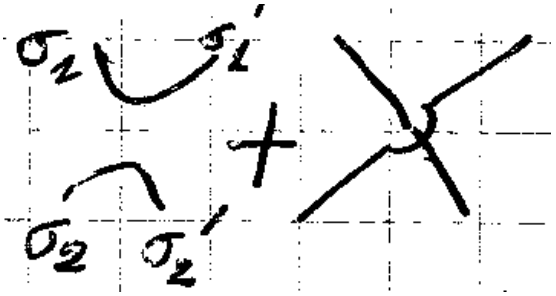
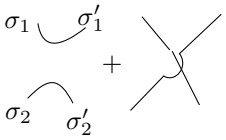


Diagram and typed description.

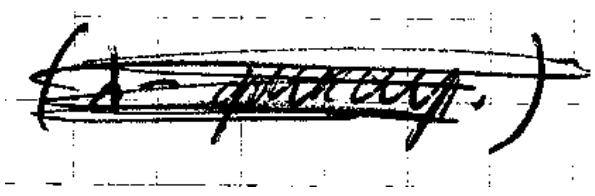


Original source details: GELFAND-P001159

A small bend is visible in the rising line at the crossing.

(here $k = k_1 - k_2$)

[crossed out: (λ — parameter.) (?)]



Uncertain reading. (λ — parameter.) (?)

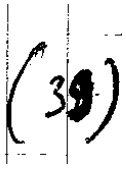
Original source details: GELFAND-P001159

$$I = \delta_{\sigma_1\sigma_1'}\delta_{\sigma_2\sigma_2'}, \qquad \hat{P} = \delta_{\sigma_1\sigma_2'}\delta_{\sigma_1'\sigma_2} \tag{38}$$

This is the $SU(2)$-invariant solution of the $\triangle$ eqs..

Compare with the soln. of the 2-particle problem.

$$\left[-\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} + V_{\sigma_1\sigma_2}^{\sigma_1'\sigma_2'}\delta(x_1 - x_2)\right]\Phi = E\Phi \tag{39}$$



Original source details: GELFAND-P001159

Uncertain reading. The last digit of (39) is overwritten.

V — a matrix taking $\sigma_1\sigma_2$ to $\sigma_1'\sigma_2'$

$$S = \frac{k + i\hat{V}\cdot\hat{P}}{k + i\hat{V}} \tag{40}$$

(this is from the soln. of eq. (39))

(Cayley transform of the potential $\hat{V}$).

Question: for which $\hat{V}$ does integrability hold?

(41) Solutions for $SU(2)$: $\hat{V} = I$ or $\hat{V} = \hat{P}$

(but they coincide after relabelling)

For other groups, other solutions and other potentials.

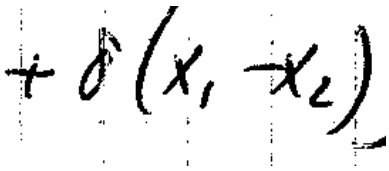
Therefore $\hat{V} = I$ and the system

GELFAND-P001160

$$\left[-\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} + \delta(x_1 - x_2)\right] \tag{42}$$

is integrable. Its 2-particle S-matrix has the form

$$S(k) = \frac{k + ic\hat{P}}{k + ic} \tag{43}$$



Original source details: GELFAND-P001160

Diagram and typed description. In (42) there is no visible coefficient before $\delta(x_1 - x_2)$, while (43) has c.

(44) For $\hat{V} \neq I, \hat{P}$ the system is nonintegrable!!

[Question: what then is the complete system of eigenfns.?].

(43) — S-matrix of bare particles.

We are interested in real particles [crossed out: illegible mark?].

Uncertain reading. illegible mark

Original source details: GELFAND-P001160

Spectrum — ?. Consider periodic boundary conditions (period $2L$)

$$\Phi(x_1, \ldots, x_i, \ldots, x_N)|_{x_i=-L} = \phi(x_1, \ldots, x_i, \ldots, x_N)|_{x_i=L}\,, \quad i = 1, \ldots, N \tag{45}$$

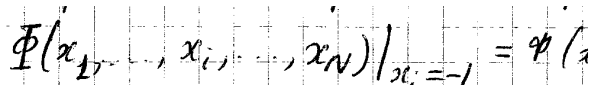


Diagram and typed description. In (45) the left has uppercase Φ and the right has lowercase ϕ.

Original source details: GELFAND-P001160

The conditions must be formulated in terms of the S-matrix { return to the initial phase on a segment of length $2L$):

$$\prod_i \hat{S}(k - k_i) \tag{46}$$

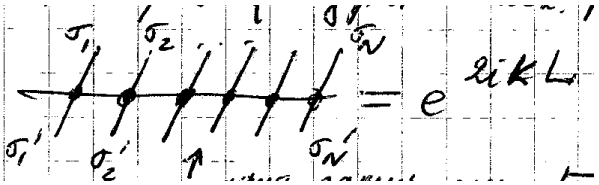
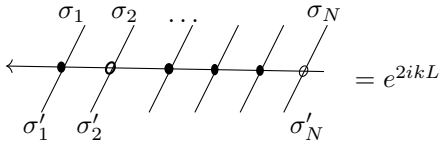


Diagram and typed description.



Original source details: GELFAND-P001160

Small openings remain inside the second and last crossings.

[Below the diagram, arrow to the diagram ↑:] graphical notation for the op. $\prod_{j \neq i} \hat{S}_{ji}(k_j - k_i)$

$$e^{ik_jL}A = \prod_{j \neq i} \hat{S}_{ji}(k_j - k_i)A \tag{47}$$

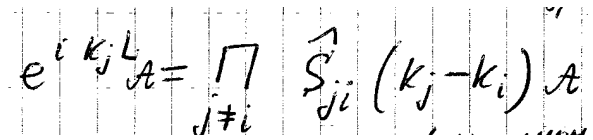


Diagram and typed description. Equation (47) writes e^{ik_jL} and has $j \neq i$ below the product sign.

Original source details: GELFAND-P001160

(actually a $2^N \times 2^N$ matrix, the initial amplitudes go into the final ones)
[An arrow from the parenthetical note points to the next line ↙]
A — coeff. (vector) in the Bethe wave fn., standing before the exponential.
Spectrum: quantization of momenta in a box.
Now solve the eigenvalue problem (47)

$$\prod_{j \neq i} \hat{S}_{ji}(k_j - k_i) \stackrel{\text{df}}{=} \text{transfer matrix} \tag{48}$$

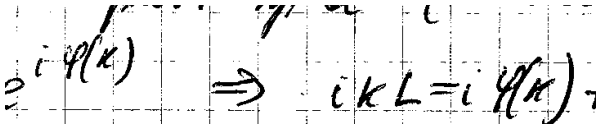
It transports [crossed out: Φ (?)]

Uncertain reading. Φ (?)

Original source details: GELFAND-P001160

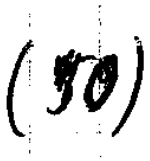
from one wall of the box to the other.
We have to diagonalize simultaneously N operators.
(49) A — a vector in a 2^N-dimensional space (eigenvector); $A = A_{\sigma_1 \ldots \sigma_N}$
Ordinary case:

$$e^{ikL} = e^{i\varphi(k)} \quad \Rightarrow \quad ikL = i\varphi(k) + i2\pi N \tag{50}$$



Uncertain reading. Both phase signs resemble φ, with a long right-hand loop.

Original source details: GELFAND-P001160



Uncertain reading. The first digit of (50) is heavily retraced or corrected.

Original source details: GELFAND-P001160

Here the operator has to be diagonalized.
[crossed out:]

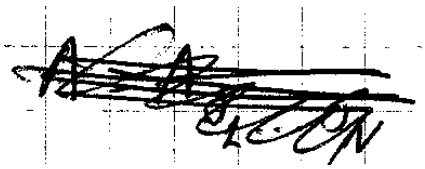
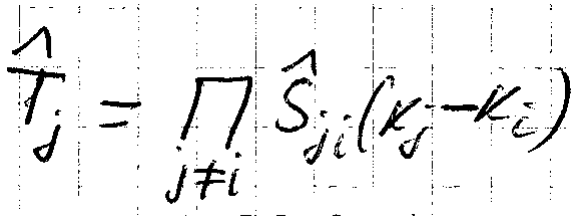


Diagram and typed description. $A = A_{\sigma_1 \ldots \sigma_N}$.

Original source details: GELFAND-P001160

$$\begin{cases} e^{ik_jL} = T_j(k_j, k_1, \dots, k_N), \\ \quad j = 1, \dots, N. \\ \hat{T}_j = \prod\limits_{j \neq i} \hat{S}_{ji}(k_j - k_i) \end{cases} \tag{51}$$



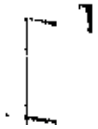
Original source details: GELFAND-P001160

Diagram and typed description. On the right of (51), j appears both on $\hat{T}$ and below the product sign.

Original source details: GELFAND-P001160

Uncertain reading. A mark above T_j in the first line is crossed out, possibly a hat.

[Above the formula, a diagonal mark pointing to T_j:] eigenvalue of the op. $\hat{T}_j$

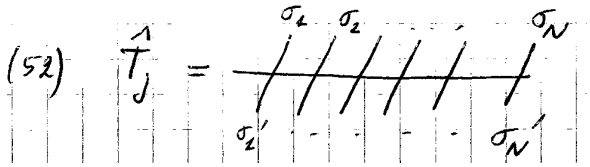


Original source details: GELFAND-P001160

Uncertain reading. In the upper right, a short angled mark by the edge of the sheet.

GELFAND-P001161

[source number: 45]



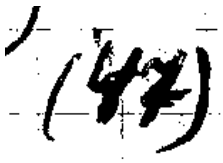
Original source details: GELFAND-P001161

Diagram and typed description.

$$\hat{T}_j = \begin{array}{ccccccc} \sigma_1 & \sigma_2 & & \dots & & \sigma_N \\ & & & & & \\ \sigma'_1 & & & \dots & & \sigma'_N \end{array} \tag{52}$$

$$[\hat{T}_j, \hat{T}_{j'}] = 0 \tag{53}$$

(consequence of the eqs. for $\triangle$s) $\Rightarrow$ consistency of (47)



Original source details: GELFAND-P001161

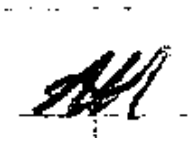
Uncertain reading. The reference number (47) is overwritten.

Diagonalization of $\hat{T}$ (the Leningrad group devised the most suitable method)

Yang was the first to solve it in 1967. Faddeev et al. started working on it in 1978. Their method is simpler than Yang's, but still cumbersome.

$\hat{T}$ corresp. to the Lax op. L ([crossed out: illegible?.]

Uncertain reading. illegible



Original source details: GELFAND-P001161

this is a quantum generalization of the Lax op. L)

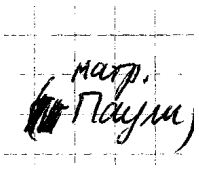
$$\begin{aligned} S^{\sigma_1'\sigma_2'}_{\sigma_1\sigma_2}(k) &= b(k)I + c(k)\hat{P} \\ &= \begin{pmatrix} b+\frac{1}{2}c+\frac{1}{2}c\sigma^z & c\sigma^- \\ c\sigma^+ & b+\frac{1}{2}c-\frac{1}{2}c\sigma^z \end{pmatrix}, \end{aligned} \tag{54}$$

where

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

([crossed out: m.?;

Uncertain reading. m.?

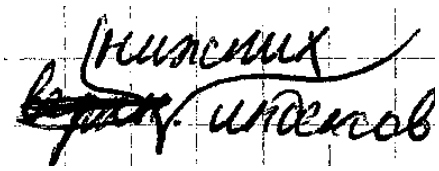


Original source details: GELFAND-P001161

inserted above the line: matrices] Pauli)

The table is in the space of [crossed out: upper?;

Uncertain reading. upper?

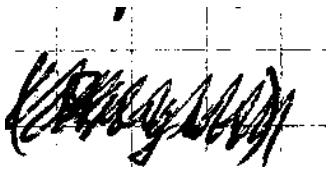


Original source details: GELFAND-P001161

inserted above the line: lower] indices

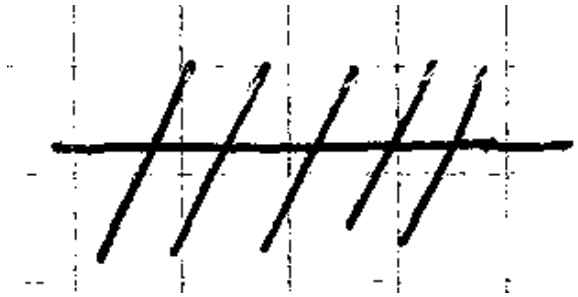
The matrices $\sigma^{\pm}, \sigma_z$ [crossed out: (Pauli) (?)]

Uncertain reading. (Pauli) (?)



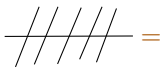
Original source details: GELFAND-P001161

act in the space of the upper indices.



Original source details: GELFAND-P001161

Diagram and typed description.

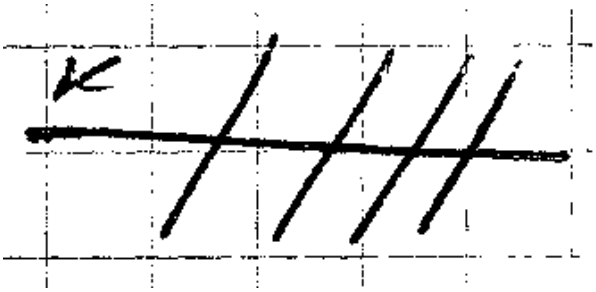


$$= \prod_i S_{ji}(k_j - k_i) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{55}$$

(notation accounting for the open ends)

$k_j = k$ is singled out. (analogue of the spectral param. in the Lax repr.)

$$\begin{pmatrix} A(k) & B(k) \\ C(k) & D(k) \end{pmatrix} = \prod_i S_{*i}(k - k_i) \tag{56}$$

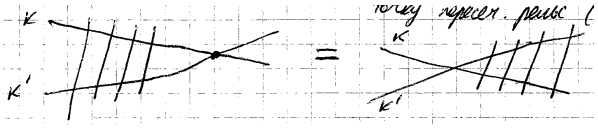


Original source details: GELFAND-P001161

Diagram and typed description.

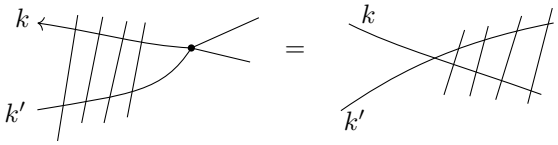
$T(k) \otimes T(k')$

All the sleepers can be moved through the point where the rails cross (eq. for $\triangle$s)



Original source details: GELFAND-P001161

Diagram and typed description.



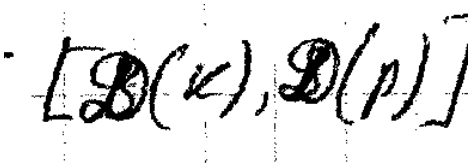
$$T(k) \otimes T(k') \cdot S(k - k') = S(k - k')T(k') \otimes T(k) \tag{57}$$

We get commut. relations between the entries A, B, C, D. They have the form:

GELFAND-P001162

Commut. relations in the entries A, B, C, D:

$$[A(k), A(p)] = [B(k), D(p)] = 0 \tag{58}$$



Original source details: GELFAND-P001162

Uncertain reading. The letters in the second commutator have different forms, with $B(k), D(p)$ a possible reading.

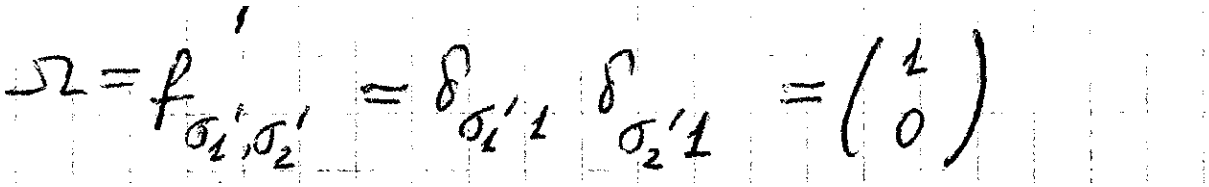
$$b(k-p)A(k)B(p) = B(p)A(k) - c(k-p)B(k)A(p) \tag{59}$$

and one more by analogy.

Here $b(k), c(k)$ are the same as in (54).

Consider the vector

$$\Omega = f_{\sigma_1',\sigma_2'} = \delta_{\sigma_1'1}\delta_{\sigma_2'1} = \begin{pmatrix}1\\0\end{pmatrix} \tag{60}$$



Original source details: GELFAND-P001162

Diagram and typed description. On the right of (60) is a single two-entry column $\begin{pmatrix}1\\0\end{pmatrix}$.

S-matrix [inserted above the line: (54).] on the vector Ω becomes triangular

Original source details: GELFAND-P001162

Diagram and typed description. inserted above the line: (54).

Hence the product of S-matrices has the eigenvector

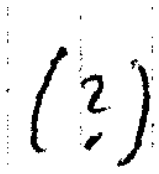
$$\Omega = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} \otimes \dots \tag{61}$$

Since

$$b = \frac{k}{k+ic}, \qquad c = \frac{ic}{k+ic} \tag{62}$$

(from unitarity and so on),

we find the eigenvalue: $b(k-k_1)b(k-k_2)\dots$ (?) (63)

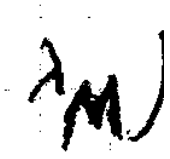


Original source details: GELFAND-P001162

Diagram and typed description. At the right of the product the manuscript has (?)

Other eigenvectors: we seek them in the form:

$$\Omega(\lambda_1, \dots, \lambda_M) = B(\lambda_1)\dots B(\lambda_N)\Omega \tag{64}$$



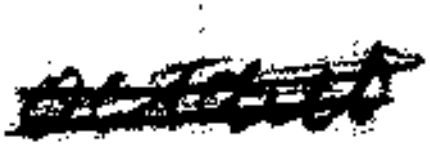
Original source details: GELFAND-P001162

Diagram and typed description. The last index on the left is overwritten and reads as M.

[Below the last $\Omega\uparrow$:] vacuum.
Now we choose $\lambda_1, \dots, \lambda_N$.

$$\Omega \text{ — eigenv. for } A \text{ with eigenval. } 1 \tag{65}$$

We apply A and use the commut. relations $\Rightarrow$ for the 1st term, always eigen.
For [crossed out: illegible word?.]



Uncertain reading. illegible word

Original source details: GELFAND-P001162

the 2nd [inserted above the line: in (59).] it is no longer so ("unwanted terms")
Eigenvalues:

$$T = \prod_{\alpha} \frac{1}{b(k_j - \lambda_\alpha)} \tag{66}$$

Result:

$$e^{2ik_jL} = \prod_{\alpha=1}^{M} \frac{k_j - \lambda_\alpha + ic}{k_j - \lambda_\alpha - ic}, \qquad j = 1, \dots, N. \tag{67}$$

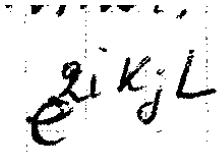


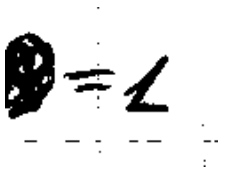
Diagram and typed description. The exponent on the left in (67) is $2ik_jL$.

Original source details: GELFAND-P001162

GELFAND-P001163

[source number: 46]

$$\prod_{j=1}^{N} \left(\frac{\lambda_\alpha - k_j + ic}{\lambda_\alpha - k_j - ic} \right) = \prod_{\beta=1}^{M} \left(\frac{\lambda_\alpha - \lambda_\beta + i2c}{\lambda_\alpha - \lambda_\beta - i2c} \right), \qquad \alpha = 1, \dots, M, \tag{68}$$



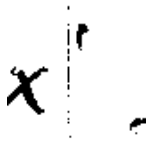
Uncertain reading. The letter in the lower limit of the product on the right is retraced and reads as β.

Original source details: GELFAND-P001163

We get a decomposition of the op. into a Clebsch–Gordan series

Diagram and typed description. $\square \times \square \times \dots \times \square = \Sigma$

Original source details: GELFAND-P001163



Original source details: GELFAND-P001163

Uncertain reading. After the second $\times$ there is a short raised mark of unclear significance.

These are algebr. eqs. for $k_1, \dots, k_N$, $E = \sum k_j^2$.

λ_j — auxiliary numbers to be determined.

We have to find λ and k from (67)–(68). (and we only need k)

We want to study the limit as $\begin{matrix} N \to \infty, \\ M \to \infty. \end{matrix}$

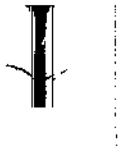
In the limit they can be solved!!

Yang proved the solvability of these eqs., the number of solutions — 2^N and so on.

In the limit — integral eqs., boundary-value problems and so on.

GELFAND-P001164

[source number: 56]



Original source details: GELFAND-P001164

Uncertain reading. At upper left is a mark resembling $\downarrow$. The middle of the mark overlaps a thick vertical band.

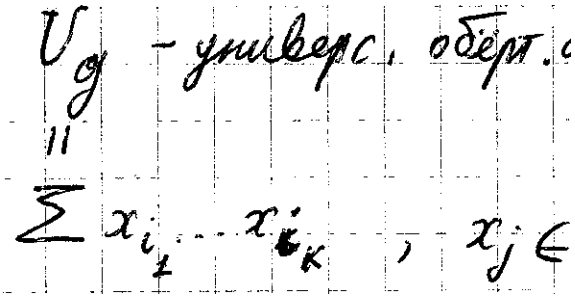
20/II-84 Cherednik. Nonlinear representation theory

$\mathfrak{g}$ — Lie alg. over $\mathbb{C}$.

$U_\mathfrak{g}$ — univ. enveloping alg.

$\| $

$\sum x_{i_1} \cdots x_{i_k}, \quad x_j \in \mathfrak{g}.$



Original source details: GELFAND-P001164

Uncertain reading. Two vertical strokes join $U_\mathfrak{g}$ to the sum. The last index is retraced and reads as i_k.

$$U_n = \left\{ \sum_{x_j \in \mathfrak{g}} x_1 \cdots x_n \right\}.$$

$$\mathrm{Gr} = \bigoplus U_n / U_{n-1} = S^\infty(\mathfrak{g})$$

(commut. alg., generated by $\mathfrak{g}$ as a vect. space)

$$[U_n, U_m] \subset U_{n+m-1} \quad \Rightarrow \quad \mathrm{Gr} \text{ is commutative.}$$

[ ] generates the Poisson bracket on $S^\infty(\mathfrak{g})$. (“quasiclassical limit”).

New def.

{ U — graded [inserted below: and filtered] with struct. in the param. $\overline{h}$ }

1. Zamolodchikov alg.
2. Sklyanin alg.
3. On some [inserted: finite] reprs.

$$\mathfrak{g} = \mathfrak{gl}_n, \qquad V = \bigoplus_{\substack{\alpha\in\mathbb{Z}_n^2\\ \lambda\in\mathbb{C}}} A_\alpha(\lambda).$$

$A_\alpha(\lambda)$ — a symbol (generator). [Inserted above "generator": basis of a vect. space.]

$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$.

(very infinite-dimensional)

$T(V)$ — the tensor alg. of the space V.

In $\mathfrak{gl}_n$ choose a basis $\{I_\alpha, \alpha \in \mathbb{Z}_n^2\}$ (any for now).

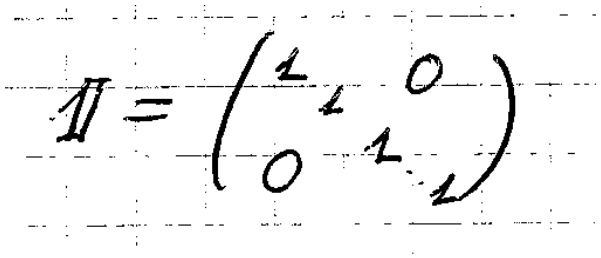
$$L(\lambda) = \sum_{\alpha\in\mathbb{Z}_n^2} A_\alpha(\lambda) I_\alpha \in \mathfrak{gl}_n(T(V)).$$

(matrix [struck-out mark] with entries from $T(V)$)

Original source details: GELFAND-P001164

Uncertain reading. After "matrix" there is a heavily struck-out mark. One possible reading is n.

$$L^1 = L \otimes \mathbf{1}, \qquad L^2 = \mathbf{1} \otimes L.$$



Original source details: GELFAND-P001164

Diagram and typed description.

$$\mathbf{1} = \begin{pmatrix} 1 & & & & 0 \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ 0 & & & & 1 \end{pmatrix}$$

Introduce a matrix [above the line: $n^2 \times n^2$]:

$$R(\lambda, \mu) \in \mathfrak{gl}_n \otimes \mathfrak{gl}_n.$$

Relations on $A_\alpha(\lambda)$:

$$R(\lambda,\mu)L^1(\lambda)L^2(\mu) = L^2(\mu)L^1(\lambda)R(\lambda,\mu). \tag{1}$$

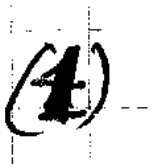
Original source details: GELFAND-P001164

Uncertain reading. The numeral in the label is overwritten. The reading (1) follows the order of the formulas on the leaf.

$$L^1(\lambda)L^2(\mu) \neq L^2(\mu)L^1(\lambda)$$

(in general) because of the noncommutativity of $T(V)$.

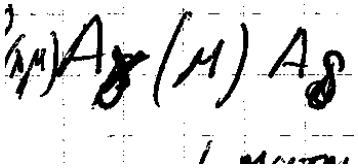
It follows from (1) that



Original source details: GELFAND-P001164

Uncertain reading. The numeral in the reference is overwritten in the same way as the label on the preceding formula.

$$A_\alpha(\lambda)A_\beta(\mu) = \sum r^{\gamma\delta}_{\alpha\beta}(\lambda,\mu)A_\gamma(\mu)A_\delta(\lambda). \tag{2}$$



Original source details: GELFAND-P001164

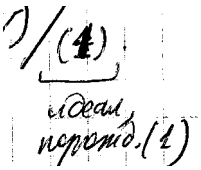
Uncertain reading. The subscript of the first A on the right is overwritten. The reading γ agrees with the coefficient's index pair $\gamma\delta$.

($r^{\gamma\delta}_{\alpha\beta}$ is not written out) (the matrix $R(\lambda,\mu)$ is assumed invertible)

GELFAND-P001165

(3) Thm. Let $A = T(V)/\underbrace{(1)}_{\substack{\text{ideal}\\ \text{generated by } (1)}}$. Then the conditions are equivalent

a) All the monomials $A_\alpha(\lambda_1)A_\beta(\lambda_2)\cdots A_\delta(\lambda_k)$ are linearly indep. in A (for fixed $\lambda_1,\lambda_2,\dots,\lambda_k$ and for arbitrary fixed k).



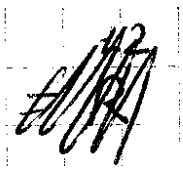
Original source details: GELFAND-P001165

Uncertain reading. The numeral in the denominator is overwritten. The note underneath reads "ideal generated by (1)."

b) R is factorizable, i.e. the Yang–Baxter or $\triangle$ relations hold.

$$\overset{12}{R}(\lambda,\mu)\overset{13}{R}(\lambda,\nu)\overset{23}{R}(\mu,\nu) = \overset{23}{R}(\mu,\nu)\overset{13}{R}(\lambda,\nu)\overset{12}{R}(\lambda,\mu). \tag{4}$$

$$R = \sum \rho_{\alpha\beta} I_\alpha \otimes I_\beta \quad \Rightarrow \quad \overset{12}{R} = R \otimes \mathbf{1}$$



Original source details: GELFAND-P001165

Uncertain reading. Between I_β and the arrow, $\overset{12}{R}$ is struck out, possibly preceded by $=$.

$$\overset{13}{R} = \sum \rho_{\alpha\beta} I_\alpha \otimes \mathbf{1} \otimes I_\beta \qquad \text{and simil. for } \overset{23}{R}.$$

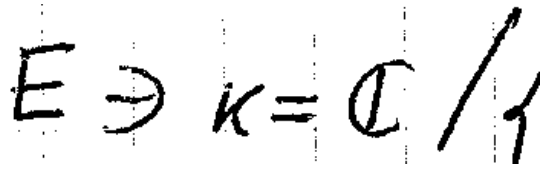
Idea: move across using rel. (1) and so on.

$$(4) \;\Rightarrow\; \underset{\lambda,\mu}{\overset{12}{R}}\; \underset{\mu,\lambda}{\overset{21}{R}} \sim \mathbf{1} \otimes \mathbf{1}. \tag{5}$$

Example connected with the ell. curve E.

$$E \ni \kappa = \mathbb{C}/\underbrace{\{\mathbb{Z} + \tau\mathbb{Z}\}}_{\Lambda}, \quad \underset{\Lambda}{\text{Im}\,\tau > 0}$$

$$E_n = \tfrac{1}{n}\mathbb{Z} + \tfrac{1}{n}\tau\mathbb{Z} \mod \Lambda$$
$$\| \\ \mathbb{Z}_n^2.$$



Uncertain reading. The letter after $\ni$ reads as κ or k.

Original source details: GELFAND-P001165

$w_\alpha(\lambda)$ — a function [inserted above: doubly per.] which

1) has poles of ord. ≤ 1 at the points $-\eta + E_n$

$$2) \quad w_\alpha(\lambda + \underset{\uparrow}{\underset{\mathbb{Z}_n^2 = E_n}{\beta}}) = \omega^{\alpha_1\beta_2 - \beta_1\alpha_2} w_\alpha(\lambda), \qquad \text{where } \omega^n = 1 \quad \text{(primitive root)}$$

$$3) \quad w_\alpha(0) = 1.$$

$w_\alpha(\lambda)$ is unique.

Let

$$I_\alpha I_\beta = \omega^{-\alpha_2\beta_1} I_{\alpha+\beta} \qquad \text{(we choose this basis)}. \tag{6}$$

($n = 2 \;\Rightarrow$ Pauli matrices;

$n > 2 \;\Rightarrow$ diag. + Coxeter element

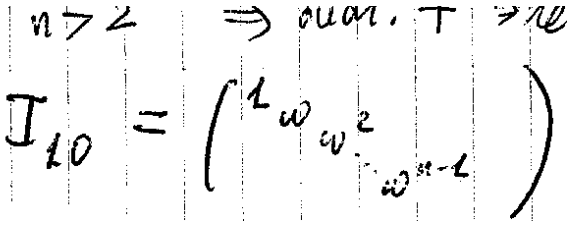


Diagram and typed description.

$$I_{10} = \begin{pmatrix} 1 & & & & \\ & \omega & & & \\ & & \omega^2 & & \\ & & & \ddots & \\ & & & & \omega^{n-1} \end{pmatrix}$$

Original source details: GELFAND-P001165

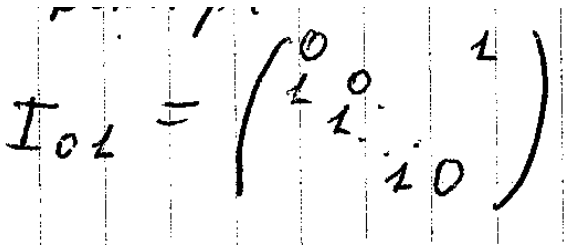


Diagram and typed description.

$$I_{01} = \begin{pmatrix} 0 & & & & 1 \\ 1 & 0 & & & \\ & 1 & \ddots & & \\ & & \ddots & \ddots & \\ & & & 1 & 0 \end{pmatrix}$$

Original source details: GELFAND-P001165

and the rest by property (6)

GELFAND-P001166

[source number: 57]

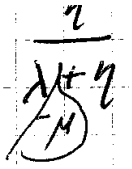
Prop.

$$R(\lambda,\mu)=\sum_{\alpha\in\mathbb{Z}_n^2} w_\alpha(\lambda-\mu)I_\alpha\otimes I_\alpha^{-1}. \tag{7}$$

Satisfies the Yang–Baxter rels. (Belavin's thm.).

Another example

$$\begin{aligned} R&=\mathbf{1}\otimes\mathbf{1}\frac{\lambda-\mu}{\eta+\lambda-\mu}+\frac{\eta}{\lambda-\mu+\eta}P,\\ P&=\frac{1}{n}\sum_{\alpha\in\mathbb{Z}_n^2} I_\alpha\otimes I_\alpha^{-1}. \end{aligned} \tag{8}$$



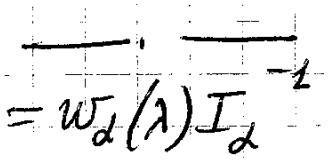
Uncertain reading. In the second denominator $-\mu$ is inserted from below.

Original source details: GELFAND-P001166

(9) Prop.

$$A_\alpha(\lambda)=\overline{w_\alpha}(\lambda)\overline{I_\alpha^{-1}} \quad\Rightarrow\quad \text{gives a finite-dimensional representation}$$

$$\mathcal{A}_R\longrightarrow \mathfrak{gl}_n.$$



Uncertain reading. Two separate horizontal strokes and a short mark between them appear above the right-hand side. Their purpose is unclear.

Original source details: GELFAND-P001166

Def. Sklyanin alg.

$$\begin{aligned} L(\lambda)&=\sum_{\alpha\in\mathbb{Z}_n^2} S_\alpha w_\alpha(\lambda)I_\alpha,\\ A_\alpha(\lambda)&=S_\alpha w_\alpha(\lambda). \end{aligned} \tag{10}$$

Reasonable rels.

$$\sum_{\alpha,\beta}\underset{\mathbb{C}}{\ell_{\alpha\beta}}S_\alpha S_\beta=0.$$

$$W=\bigoplus_{\alpha\in\mathbb{Z}_n^2}\mathbb{C}S_\alpha.$$

Important: there is a series of fin.-dimensional reprs. (Sklyanin: FA 16:4, 17:4).

$n=2$ $\quad S_0,S_1,S_2,S_3$

$$[S_i,S_j]=\{S_0,S_k\},\qquad [S_i,S_0]=J_i\{S_j,S_k\}.$$

GELFAND-P001167

27/II-84 Gelfand Sem.

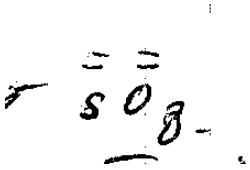
A. Zelevinsky (joint work with I. M. Gelfand)

Models of representations of classical groups and their hidden symmetries.

1. All reprs. of SO_3 are realized in functions on S^2 (in $L^2(S^2)$).

2. L. C. Biedenharn, D. Flath:

In the model of reprs. of SU_3 there is a mysterious action of $\overline{\overline{s}}\,\overline{o}_8$ — (the algebra only!) and the corresponding repr. of so_8 is irreducible.



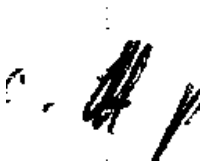
Original source details: GELFAND-P001167

Uncertain reading. There are two short strokes over each of s and o. The later so_8 is written without them.

3. First step in Penrose's twistor construction

$$\mathbb{R}^4 \subset \mathbb{C}^4 \subset G_2^4$$

Content [overwritten mark] of the work: all 3 constructions extend to all classical groups $\Rightarrow$



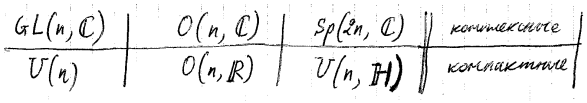
Original source details: GELFAND-P001167

Uncertain reading. Between "Content" and "of the work" is a heavily overwritten mark, possibly 4.

$\Rightarrow$ there arise by themselves supergroups, superalgebras and so on!!

1. Complex algebraic groups will take the place of compact groups

Model: direct sum of all finite-dimensional irreduc. reprs., each once



Original source details: GELFAND-P001167

Diagram and typed description.

$GL(n,\mathbb{C})$	$O(n,\mathbb{C})$	$Sp(2n,\mathbb{C})$	complex
$U(n)$	$O(n,\mathbb{R})$	$U(n,\mathbb{H})$	compact

G — one of the groups $GL(n,\mathbb{C})$, $O(n,\mathbb{C})$, $Sp(2n,\mathbb{C})$.

1 realization: induced representation (in functions or in sections of a bundle on the homogeneous space G/M).

GELFAND-P001168

[source number: 60]

$$\mathrm{Ind}_M^G \tau$$

Model (Bernstein + 2 Gelfands):

(1)

G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M	$O(n)$	$O(n+1)\times O(n)$	$O(n)\times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+1)\otimes 1$	$\lambda(n)\otimes 1$	$\lambda(n)$

Original source details: GELFAND-P001168

Diagram and typed description. (1)

G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M	$O(n)$	$O(n+1)\times O(n)$	$O(n)\times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+1)\otimes 1$	$\lambda(n)\otimes 1$	$\lambda(n)$

(all complex)

$\lambda(n)$ — exterior algebra of the standard representation

(actually because of supergroups)

Dual models (superduality)

(2)

G	$GL(2n)$	$O(2n)$	$Sp(4n+2)$	$Sp(4n)$
M	$Sp(2n)$	$GL(n)$	$Sp(2n+2)\times Sp(2n)$	$Sp(2n)\times Sp(2n)$
τ	$\sigma(2n)$	$\sigma(n)$	$\sigma(2n+2)\otimes 1$	$\sigma(2n)\otimes 1$

Original source details: GELFAND-P001168

Diagram and typed description.

(2)

G	$GL(2n)$	$O(2n)$	$Sp(4n+2)$	$Sp(4n)$
M	$Sp(2n)$	$GL(n)$	$Sp(2n+2)\times Sp(2n)$	$Sp(2n)\times Sp(2n)$
τ	$\sigma(2n)$	$\sigma(n)$	$\sigma(2n+2)\otimes 1$	$\sigma(2n)\otimes 1$

Here $\sigma(n)$ — the symmetric algebra of the standard representation.

(3) Transition:

$$G(n) \longrightarrow GL(2n)$$
$$O(n) \longrightarrow Sp(2n)$$
$$Sp(2n) \longrightarrow O(4n)$$
$$\lambda(n) \longrightarrow \sigma(2n)$$

$G(n) \longrightarrow GL(2n)$

Original source details: GELFAND-P001168

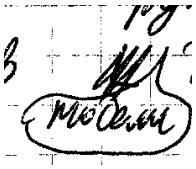
Uncertain reading. The left side of the first line reads $G(n)$. The L may have been omitted.

(4) Theorem 1. Table (2) consists of models.

Rem. It does not contain all the classical groups (unlike (1)): it lacks $GL(2n+1)$ and $O(2n+1)$, this comes after the comments.

$G \subset L$ — overgroup. L — is also a group.

Model in the sense of an overgroup: in [struck-out mark] [inserted below and circled: the model] a repr. of the Lie alg. ℓ of the group L acts.



Original source details: GELFAND-P001168

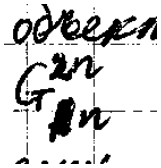
Uncertain reading. The writing before "acts" is struck out. The mark under the strokes may be M. Below it "the model" is inserted.

The models are constructed by one geom. scheme.

$\Gamma(L)$ — the basic Grassmannian of the group L.

I. M. G. Remarkable object G_2^4 — a book should be written about it (basic building block)

In general G_n^{2n} — is the main thing.



Original source details: GELFAND-P001168

Uncertain reading. Before n in the subscript is a heavily overwritten mark. It may be a cancelled 2.

What polyhedra are there?

What takes the place of their volumes?

Where does the dilogarithm come from?

Char. classes and why they do not localize, what takes the place of localization?

See the medieval book on the icosahedron (and Klein's book on it).

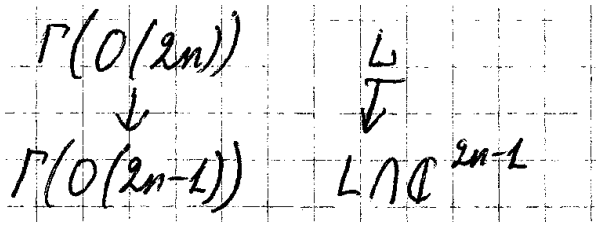
GELFAND-P001169

A. Z. List of Grassmannians (they are hom. spaces of the corresp. group):

$$\left[\begin{array}{rl} \Gamma(GL(2n)) = \Gamma_n^{2n} & (n\text{-dimensional planes in } 2n\text{-dimensional space}). \\ \Gamma(GL(2n+1)) = \Gamma_{n+1}^{2n+1} & \\ \Gamma(O(2n)) = & \text{Grassmannian of } n\text{-dimensional isotropic subspaces} \\ & \text{in } \mathbb{C}^{2n} \\ & \text{(isotr. — all vectors have length } 0) \\ & \text{It is disconnected.} \\ \Gamma(O(2n+1)) = & \text{Grassmannian of } n\text{-dimensional isotropic subspaces} \\ & \text{in } \mathbb{C}^{2n+1} \end{array}\right. \tag{5}$$

In $\mathbb{C}^2 \quad z_1^2 + z_2^2 = 0$, $z_1 = \pm i z_2 \Rightarrow \Gamma(O(2))$ consists of 2 points.

(6) Exercise. The map



Original source details: GELFAND-P001169

Diagram and typed description.

$$\begin{array}{ccc} \Gamma(O(2n)) & & L \\ \downarrow & & \downarrow \\ \Gamma(O(2n-1)) & & L \cap \mathbb{C}^{2n-1} \end{array}$$

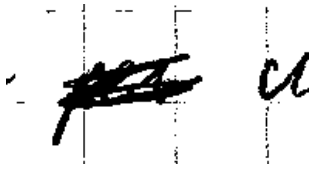
is an isomorphism on each component.

(5) cont. $\Gamma(Sp(2n))$ — Grassmannian of n-dimensional Lagrangian subspaces in $\mathbb{C}^{2n}$.

Next: $G \subset L$.

$\Gamma(L) \supset \Omega$. [Below Ω: open subset (so that there are holom. functions).]

It is in functions on Ω [struck-out writing] that the representation is constructed.

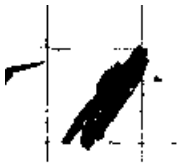


Original source details: GELFAND-P001169

Uncertain reading. After the first Ω the beginning of a word is struck out, possibly Russian “p...”.

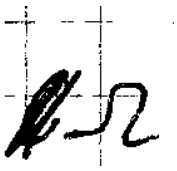
Ω — the unique open orbit.

L does not act, but ℓ acts [struck-out writing] on functions on [struck-out writing] Ω (infinites. shifts!).



Original source details: GELFAND-P001169

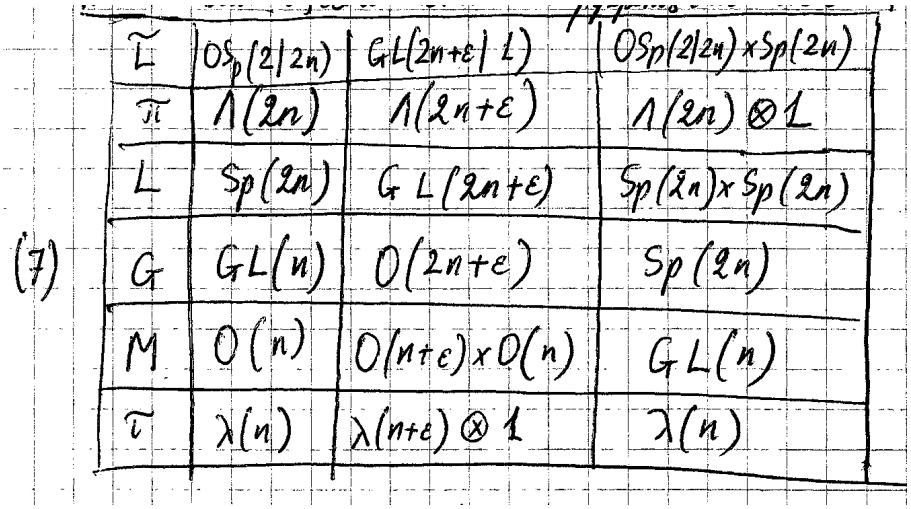
Uncertain reading. After “acts” there is an illegible struck-out mark.



Original source details: GELFAND-P001169

Uncertain reading. Immediately before Ω a mark is struck out, possibly ℓ.

I. M. G. Apparently there is an even differential in place of d (with $d^2 = 0$).



Original source details: GELFAND-P001169

Diagram and typed description. (7)

$\widetilde{L}$	$OSp(2\|2n)$	$GL(2n+\varepsilon\|1)$	$OSp(2\|2n) \times Sp(2n)$
$\widetilde{\pi}$	$\Lambda(2n)$	$\Lambda(2n+\varepsilon)$	$\Lambda(2n) \otimes 1$
L	$Sp(2n)$	$GL(2n+\varepsilon)$	$Sp(2n) \times Sp(2n)$
G	$GL(n)$	$O(2n+\varepsilon)$	$Sp(2n)$
M	$O(n)$	$O(n+\varepsilon) \times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+\varepsilon) \otimes 1$	$\lambda(n)$

$\varepsilon = 0, 1$

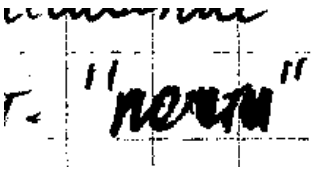
[To the right of the table:] $\Omega = G/M$.

τ — repr. in the fibre.

$$\mathrm{Ind}_M^G \tau$$

(8) Thm. Under the action of G on $\Gamma(L)$ there is a unique open orbit and a vect. bundle π on which there is an action of ℓ. (hidden symmetry)

The repr. is “almost” irreducible.

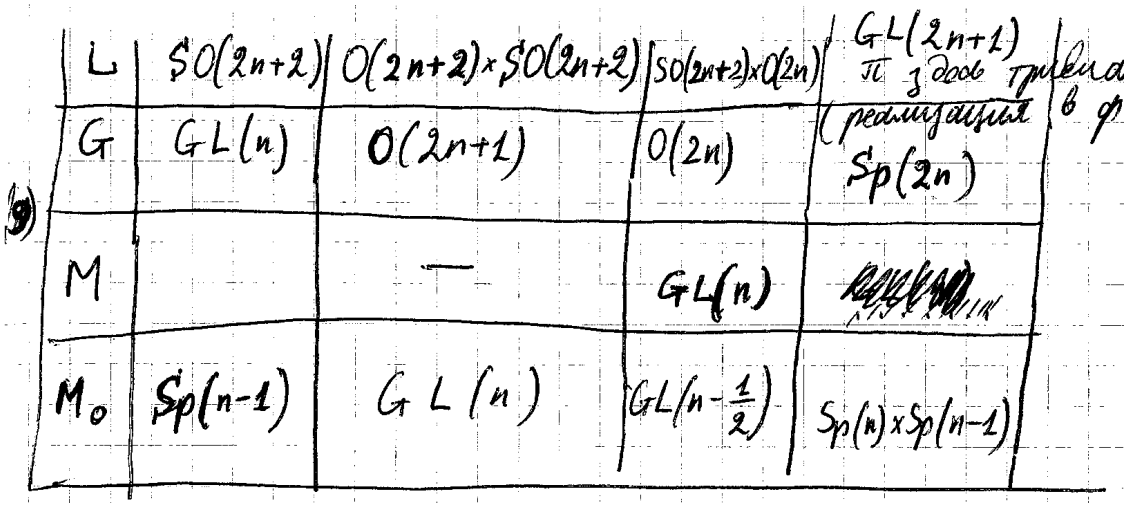


Original source details: GELFAND-P001169

Uncertain reading. The word in quotation marks is heavily retraced. It reads “almost”.

GELFAND-P001170

[source number: 61]



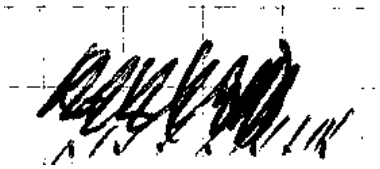
Original source details: GELFAND-P001170

Diagram and typed description.

(9)

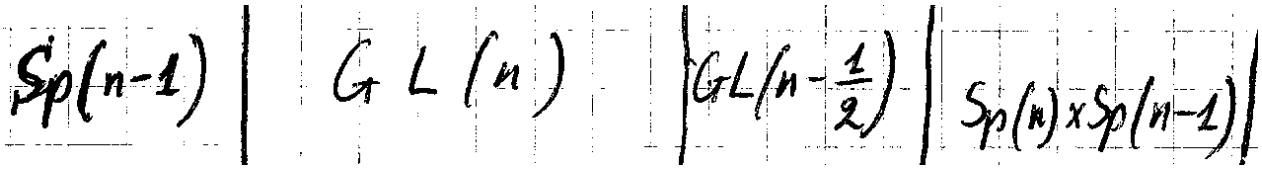
L	$SO(2n+2)$	$O(2n+2)\times SO(2n+2)$	$SO(2n+2)\times O(2n)$	$GL(2n+1)$
G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M		—	$GL(n)$	[struck out]
M_0	$Sp(n-1)$	$GL(n)$	$GL(n-\frac{1}{2})$	$Sp(n)\times Sp(n-1)$

[By the last column:] π is trivial here (realization in functions on the Grassmannian).



Uncertain reading. The last cell of row M is heavily struck out. It may read $GL(n)$.

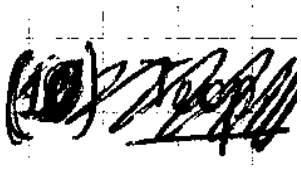
Original source details: GELFAND-P001170



Original source details: GELFAND-P001170

Uncertain reading. In row M_0 the manuscript shows $Sp(n-1)$, $GL(n-\frac{1}{2})$ and $Sp(n)\times Sp(n-1)$.

[struck out] ~~(10) Thm.~~

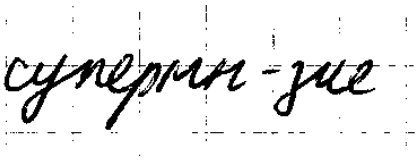


Uncertain reading. The struck-out heading appears to read “(10) Thm.”

Original source details: GELFAND-P001170

$$\tau = \mathrm{Ind}_{M_0}^{M} 1 \quad \Rightarrow \quad \text{table (7) reads } \mathrm{Ind}_{M_0}^{G} 1.$$

$\Rightarrow$ actually on L there arises a sheaf $\widetilde{L} \Rightarrow$ supermanif. and a super Grassmannian.



Uncertain reading. The abbreviation appears to mean “supermanifold”. The letters after the hyphen are uncertain.

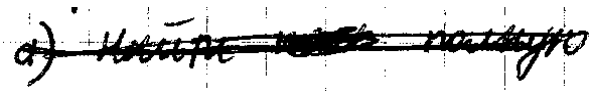
Original source details: GELFAND-P001170

[struck out] There are many unusual symmetries. For example, the top row in (9) resembles the bottom row in (7).

Uncertain reading. Before “There are” there is a short illegible struck-out entry.

Original source details: GELFAND-P001170

I. M. G. Open question. On G/M a large symmetry group acts.



Undeciphered passage. The crossed-out line begins “a)”. It may read “find ... such a”. The middle is heavily crossed out.

Original source details: GELFAND-P001170

What other symmetries are there?

Nikiforov. (joint with Uvarov and Suslov)

$$\sigma(x)y'' + \tau(x)y' + \lambda y = 0, \tag{1}$$

σ — polyn. of degree 2, τ — polyn. of degree ≤ 1.

For

$$\lambda = \lambda_n = -n\tau' - \frac{n(n-1)}{2}\sigma'' \tag{2}$$

— orthog. polynomials.

Many spec. functions are contained here:

$$P_n^{(\alpha,\beta)}(x), \quad L_n^{\alpha}(x), \quad H_n(x)$$

(orthog. polynomials), depending on the degrees.

Gen. formula:

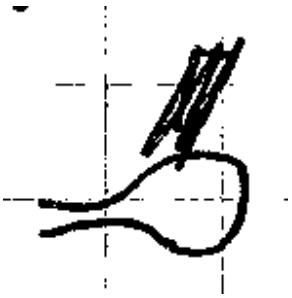
$$y_n(x) = \frac{A_n}{\rho(x)}\frac{d^n}{dx^n}[\sigma^n(x)\rho(x)]. \tag{3}$$

ρ — from the eq.: $(\sigma\rho)' = \tau\rho$.

[At the right:] The whole theory of spec. functions is in here — hypergeom., confluent hypergeom., and so on. We must take n nonintegral.

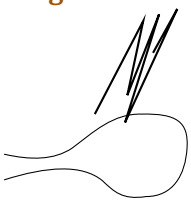
$$y(x) = \frac{A}{\rho(x)}\frac{1}{\Gamma(\alpha)}\int \frac{\sigma^{\alpha}(s)\rho(s)}{(s-x)^{\alpha+1}}. \tag{4}$$

α from a relation of type (2).



Original source details: GELFAND-P001170

Diagram and typed description.



There are heavy strokes at the upper branch. Their meaning and the direction of traversal are unclear.

(5) [crossed out with two intersecting strokes]

$$\frac{\sigma^{\alpha+1}(s)\rho(s)}{(s-x)^{\alpha+2}}.$$

Property of (1): the eq. for y' has the same form!! (with other params.)

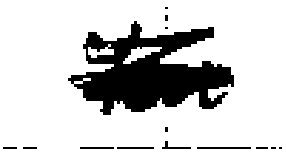
Generalization: pass to a difference eq. keeping property (5).

[In the left margin:]

Original source details: GELFAND-P001170

Uncertain reading. Struck out. The first letter may be “B”.

I. M. G. What is a multivariate hypergeom. function? I. M. G., Gindikin and others tried. [struck out] Cond.: that matrix entries be written in terms of it (via monodromy).



Uncertain reading. A short word before "Cond." is struck out.

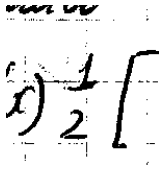
Original source details: GELFAND-P001170

Uncertain reading. The abbreviation reads "Cond.:".

Original source details: GELFAND-P001170

N.

$$\tau(x)\frac{1}{2}\left[\frac{y(x+h)-y(x)}{h}+\frac{y(x)-y(x-h)}{h}\right]+\lambda y(x)$$
$$+\sigma(x)\frac{1}{2}\left[\frac{y(x+h)-y(x)}{h}-\frac{y(x)-y(x-h)}{h}\right]=0. \tag{6}$$



Uncertain reading. The factor before the second square bracket is written $\frac{1}{2}$.

Original source details: GELFAND-P001170

GELFAND-P001171

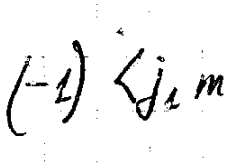
$\Rightarrow$ we obtain polynomials of a discr. variable.

In place of Jacobi polyns. we get Hahn polyns. $h_n(x)$ or $h_n^{(\alpha,\beta)}(x,N)$ $\left(\frac{2}{N}=h\right)$.

They are orthog. with a weight on a discr. set of points.

$\Rightarrow$ Clebsch–Gordan coeffs.

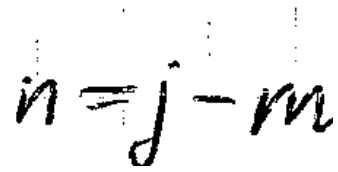
$$(-1)\quad \langle j_1 m_1\ j_2 m_2 \mid jm\rangle = \frac{\sqrt{\rho(x)}}{d_n} h_n^{(\alpha,\beta)}(x,N).$$



Uncertain reading. At the left is (-1). A small detached mark sits above the gap by the angle bracket. No exponent can be read.

Original source details: GELFAND-P001171

$$n = j - m, \quad x = j_2 - m_2, \quad N = j_1 + j_2 - m + 1,$$
$$\alpha = m - m', \quad \beta = m + m', \quad m' = j_1 - j_2.$$

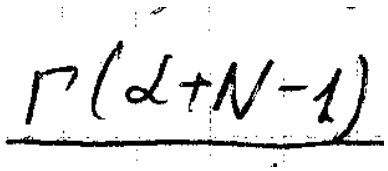


Uncertain reading. There is a dot above the left stroke of the initial n. Its purpose is unclear.

Original source details: GELFAND-P001171

Weight

$$\rho(x) = \frac{\Gamma(\alpha + N - 1)\Gamma(\beta + x + 1)}{\Gamma(x + 1)\Gamma(N - x)}.$$



Uncertain reading. The first factor in the numerator is written $\Gamma(\alpha + N - 1)$.

Original source details: GELFAND-P001171

In the book by Gelf.–Minl.–Shapiro — an expr. in terms of Jacobi polynomials. But the formula there is a polynomial depending on n (?)

(incorrect $\rho(x)$)

Uncertain reading. The word before $\rho(x)$ reads “incorrect”, possibly “incorrectly”.

Original source details: GELFAND-P001171

One can pass to a variable grid and write the diff. operator so that polynomials give polynomials of the corresp. degrees.

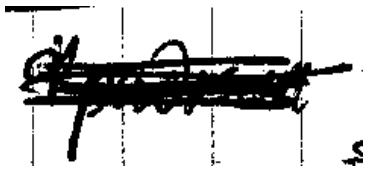
A quadr. grid gives Racah coeffs. in place of Clebsch–Gordan.

GELFAND-P001172

[source number: 66]

5/III–84 Gelfand Sem.

[struck out] ~~Fradkin~~ Berezansky Yu. M.



Uncertain reading. An entry before Berezansky’s name is struck out. It appears to read “Fradkin”.

Original source details: GELFAND-P001172

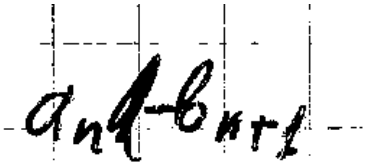
Applications of the ordinary inverse spectral problem in place of the inverse problem of scattering theory.

$$\begin{cases} \dot{a}_n = \frac{1}{2} a_n (-b_{n+1} \\ \dot{b}_n = a_n^2 - a_{n-1}^2 \end{cases} \tag{1}$$

(Toda latt.)

$n = 0, 1, \ldots$

$a_{-1} = 0.$



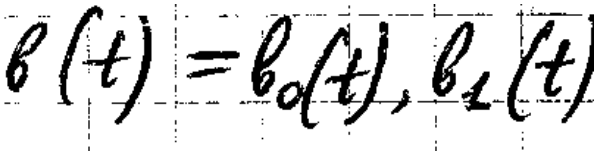
Uncertain reading. Before b_{n+1} are a retraced bracket-like stroke and a dash. The reading $(-$ is provisional.

Original source details: GELFAND-P001172

[At the right:]

$$a(t) = (a_0(t), a_1(t), \ldots)$$
$$b(t) = b_0(t), b_1(t), \ldots)$$

$a(0), b(0)$ are known.



Uncertain reading. There is no visible opening parenthesis before $b_0(t)$. The closing parenthesis is present.

Original source details: GELFAND-P001172

In the class of seqs.: $a(t) \in \ell^\infty$, $b(t) \in \ell^\infty$, $\forall t$ and $a_j(t) > 0$.

Prop. The soln. exists, is unique and is obtained by the following procedure:

Consider the Jacobi matr.

$$\begin{pmatrix} b_0(0) & a_0(0) & & 0 \\ a_0(0) & b_1(0) & a_1(0) & \\ \text{[struck out]} & a_1(0) & & \\ 0 & & & \end{pmatrix} = L(0). \tag{2}$$

Uncertain reading. The first entry in row three is struck out. The ending (0) is visible. Possible readings are $a_1(0)$ and $a_0(0)$.

Original source details: GELFAND-P001172

$$\mu(t) = \left[\int_{-\infty}^{\infty} e^{\lambda t} \, d\rho(\lambda, 0) \right]^{-1}; \tag{3}$$

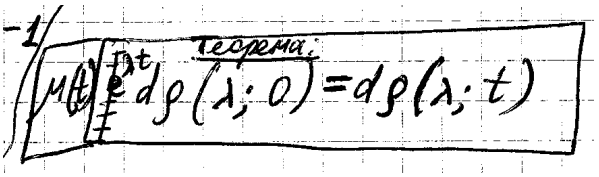


Diagram and typed description.

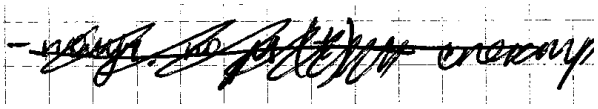
$$\boxed{\mu(t) e^{\lambda t} \, d\rho(\lambda; 0) = d\rho(\lambda; t)}$$

Theorem: under the evolution of $L(t)$ according to (1).

Original source details: GELFAND-P001172

$\rho(\lambda, 0)$ — spectr. function of the Jacobi matr. $L(0)$.

$\rho(\lambda, t)$ — [struck-out phrase] spectr. function of $L(t)$, its entries $a_n(t), b_n(t)$.



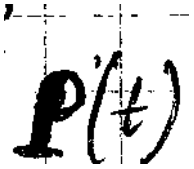
Original source details: GELFAND-P001172

Undeciphered passage. Several words before "spectr. function" are struck out. Writing resembling $\rho(t)$ is visible beneath the strokes.

I. M. G. (explanation)

$$\int_0^\infty t^n \, d\rho(t) = a_n \qquad \text{(suppose given)}$$

$$\int_0^\infty P'(t) \, d\rho(t) \qquad \text{is known too}$$



Original source details: GELFAND-P001172

Uncertain reading. A short stroke appears between the retraced P and the parenthesis. It may be a prime.

$\Rightarrow (P, Q)_{L^2_\rho}$ is known.

Condition: nonnegativity of this inner prod..

We obtain a Hilbert space and multiplication in it by $t \Rightarrow$ orthogonalizing, we construct an orthog. basis $P_n(t)$.

$$tP_n(t) = a_{n-1}P_{n-1}(t) + b_nP_n(t) + a_nP_{n+1}(t).$$

(there is nothing below P_{n-1} because the operator is self-adjoint).

$\{a_n\} \rightsquigarrow$ Jacobi matr..

Jacobi matr. $\rightsquigarrow$ moment problem.

There is a spectr. decomposition

$$\underline{(E(\lambda)1, 1) = \rho(\lambda)}$$

— this function solves the moment problem.

Uniqueness and nonuniqueness are deep matters.

Essential self-adj. does not always hold!

ρ nonunique $\Leftrightarrow$ the op. L has deficiency indices $\Leftrightarrow$

$\Leftrightarrow a_n$ does not grow too fast.

Connection with quasianalyt. classes and so on.

~~Spectr. decomp. of a self-adj. operator~~

Yu. M. B. a, b are bounded $\Rightarrow$ self-adjointness.

$$a_n(t) = \int_{-\infty}^{\infty} \lambda P_n(\lambda; t) P_{n+1}(\lambda, t)\, d_\lambda \rho(\lambda, t)$$

$$b_n(t) = \int_{-\infty}^{\infty} \lambda P_n^2(\lambda, t)\, d_\lambda \rho(\lambda, t)$$

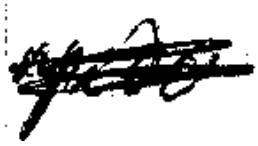
Difference: no asymptotics at infinity!!

Conditions: e^{tA} makes sense for the original Jacobi matr..

Fradkin E. S. How can one build a unified theory of all interactions?

Current point of view:

$\exists$ a broad group and all elem. [struck out] particles are embedded in one representation of it. (at high energies)



Original source details: GELFAND-P001173

Uncertain reading. A word between "elem." and "particles" is heavily struck out.

Basic particle. The others are its projections.

Breaking of the group [inserted above: (symmetry)] must be spontaneous.

It breaks down at low energies.

Gravitons will come in too.

Space/time remains

there will be a Riem. metric too.

The group will be a supergroup

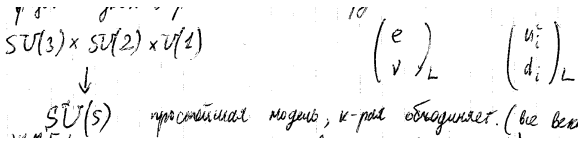
(superext. of the Poincaré group or of the conformal group).

1. Unification of the electromagn. and weak interactions (Weinberg, Salam, Glashow)

see the review: Phys. Reports, 1981, Lancaster.

The group $SU(2) \times U(1)$ unifies the weak and electromagn. interactions.

All low-energy physics is described by the group



Original source details: GELFAND-P001173

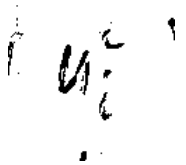
Diagram and typed description.

$SU(3) \times SU(2) \times U(1)$ $\begin{pmatrix} e \\ \nu \end{pmatrix}_L$ $\begin{pmatrix} u_i \\ d_i \end{pmatrix}_L$ — quarks

$\downarrow$

$SU(5)$

[To the right of $SU(5)$:] the simplest model that unifies. (all vector particles)



Uncertain reading. There is a separate hook above the index i of u_i. Its purpose is unclear.

Original source details: GELFAND-P001173

I. M. G. (see also Coleman's review in "gauge fields")

Supergroup — ext. of the Poincaré group Golfand, Likhtman

review Ferrari, Fayet Phys. Report, 1978.

Uncertain reading. The first surname in the manuscript reads "Ferrari".

Original source details: GELFAND-P001173

$$[M, M] = M \qquad \text{(Lorentz)}$$
$$[P, M] = P \qquad \text{(Poincaré)}$$
$$[Q^\alpha, \bar{Q}^\beta] = i(\gamma^\mu \bar{c})^\alpha_\beta P_\mu.$$

GELFAND-P001174

[source number: 67]

Gravitation is a gauge th. of the Poincaré grp..

Supergravity: review: Nieuwenhauzen

Phys. Report, vol. 68, $N = 4$ (1981)

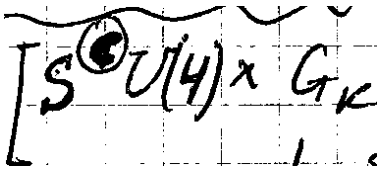
There are no other reviews.

Large unification (without grav.) — 24-dim. group in which all particles are unified.

(Grand unification (with gravitation))

Compare large unif. with the group:

$$\left[S^{(?)}U(4) \times \underbrace{G_k}_{\text{gauge}} \right.$$



Uncertain reading. A dark mark between S and $U(4)$ is circled. The reading is unresolved.

Original source details: GELFAND-P001174

rem. [struck out] the group that solves grand unif.

Uncertain reading. An illegible mark is crossed out after "rem."

Original source details: GELFAND-P001174

$G_k = E_8$ [inserted above: Fundam. repr.] (coincides with the adj.)

[At the right:] $E_8 \supset SU(5)$ as a subgroup.

Lagrangian: Letters to JETP vol. 32, 575 (Fradkin, Konshtein)

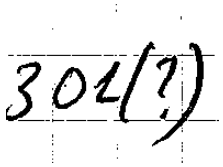
($SU(5)$ — "Quarks-80" (Fradkin))

E_8-model — is finite — has no divergences at all!!! —

— at all loops

Vainshtein, Shifman,

Fradkin, Tseytlin — Phys. Letters, 135 (1984), p. 301(?)



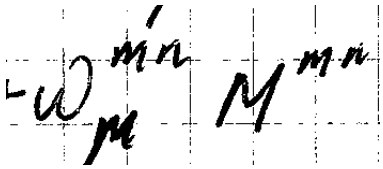
Original source details: GELFAND-P001174

Uncertain reading. The last digit resembles 1, and the number is followed by (?).

Gauging is an art (there is no technique yet)

From the group G we construct a connection V_μ (or vect. field)

$$V_\mu = e^a_\mu P_a + \omega^{mn}_\mu M^{mn} + \bar{\psi}^\alpha_\mu \Theta_\alpha.$$



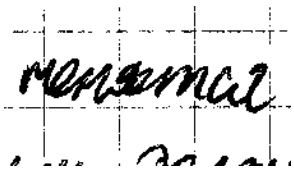
Original source details: GELFAND-P001174

Uncertain reading. In the first formula mn is written above M. In the next formula the same indices are below it.

$$\Lambda = \epsilon^a P_a + \lambda^{mn} M_{mn} + \bar{\epsilon}^\alpha \Theta_\alpha \qquad \text{— infinitesimal}$$

$$\left.\begin{matrix} \omega_{mn} & M_{mn} \text{ — gener.,} & \omega_{mn} \text{ — coeffs.} \\ e^a_\mu & P_\mu \end{matrix}\right]$$

To gauge: odd group and so on.



Original source details: GELFAND-P001174

Uncertain reading. The word before "group" is provisionally read as "odd". Another possible reading is "a certain".

BUT: translations must be replaced by something local.

Grav.: theory of particles with spins $\begin{pmatrix} 2 \\ 3/2 \end{pmatrix}$

gravitation is spin 2.

I. M. G. believes that space is neither 3- nor 4-dimensional, but ∞-dimensional.

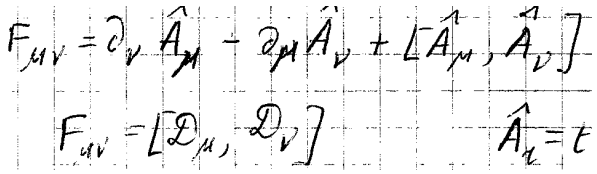
GELFAND-P001175

12/III–84 Gelfand Sem.

Fradkin E. S.

$$F_{\mu\nu} = \partial_\nu \hat{A}_\mu - \partial_\mu \hat{A}_\nu + [\hat{A}_\mu, \hat{A}_\nu]$$

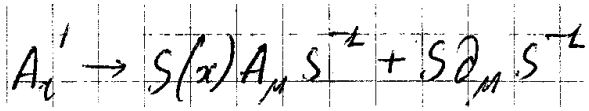
$$F_{\mu\nu} = [D_\mu, D_\nu] \qquad \hat{A}_\tau = t^a A^a_\tau$$



Uncertain reading. The index order of both formulas for $F_{\mu\nu}$ follows the manuscript.

Original source details: GELFAND-P001175

$$A'_\tau \longrightarrow S(x) A_\mu S^{-1} + S \partial_\mu S^{-1}$$



Uncertain reading. The index on the left resembles τ. On the right μ is written twice.

Original source details: GELFAND-P001175

How to introduce mass?

Adding $m^2 A^2_\mu$ does not work (no gauge invariance)

[At the right, with an arrow to the parenthesis:] and it is also nonrenormalizable.

It must be by [struck out] spontaneous symmetry breaking. //

Uncertain reading. An illegible mark is struck out in the phrase meaning “by”.

Original source details: GELFAND-P001175

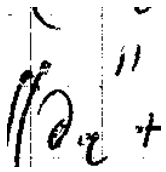
I. M. G.: we must state precisely what renormalizability is.

F. Higgs introduced a scalar (add.) field.

$$(D_\tau \varphi)^2 + \dots$$

$$\left((\partial''_\tau + t^\alpha \hat{A}_\mu)\varphi\right)^2$$

(covar. deriv.)



Uncertain reading. Two short strokes are written to the upper right of ∂_τ. Their purpose is unclear.

Original source details: GELFAND-P001175

If $\langle\varphi\rangle \neq 0$ (average over the vacuum), then

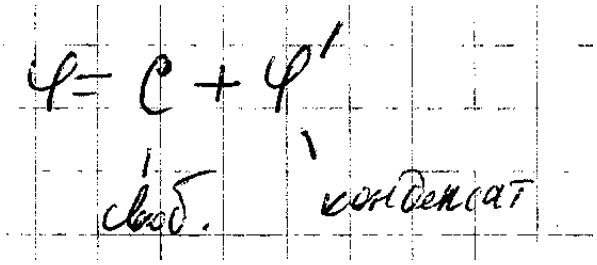


Diagram and typed description.

$\varphi = C + \varphi'$

free — condensate

Original source details: GELFAND-P001175

We obtain $(t^\alpha A_\mu \varphi(0))^2$

One can write:

$$(D_\tau \varphi)^2 + m_1^2 \varphi^2 + \lambda \varphi^4$$

We vary:

$$D_\tau^2 \varphi(0) + m^2 \varphi(0) + 4\lambda \varphi(0)^3 = 0$$

Simple soln. 0

There can also be a 2nd soln.:

$$m^2 \varphi(0) + 4\lambda \varphi(0)^3 = 0,$$

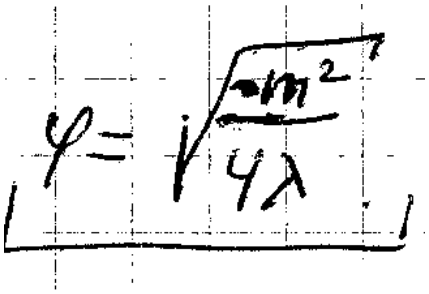


Diagram and typed description. $\varphi = \sqrt{\frac{-m^2}{4\lambda}}$

Original source details: GELFAND-P001175

$\Rightarrow$ a mass term for all the particles will arise.

GELFAND-P001176

[source number: 71]

Lagr.

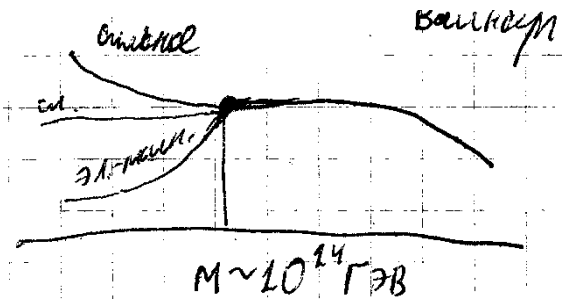
$$\mathcal{L} = F_{\mu\nu}^2 + (D_\mu \varphi)^2 + m_1^2 \varphi^2 + \lambda \varphi^4$$

We want: the remaining [abbr.]$^?$ massless grp. to be

Uncertain reading. remaining [abbr.]

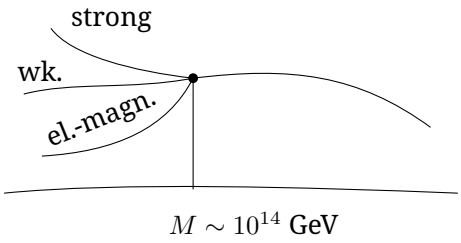
Original source details: GELFAND-P001176

$$SU(3) \times \underbrace{SU(2) \times U(1)}_{\text{Weinberg–Salam}}$$



Original source details: GELFAND-P001176

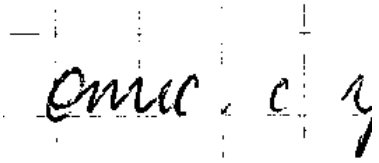
Diagram and typed description.



Sketch of the convergence of the interaction couplings

— · —

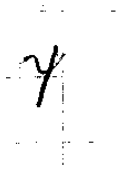
Abbreviated heading: list / description[?] with gravity:



Original source details: GELFAND-P001176

Uncertain reading. Abbreviated heading: list / description

				$\underline{N=8}$
$g_{\mu\nu}$	— 1 particle			1
ψ	N particles	(from $SO(N)$)	(spin $3/2$)	8
A_μ	$SO(N)$	(vector part.)		28
χ	·		(spin $1/2$)	56
φ	— scal. particles			70



Original source details: GELFAND-P001176

Uncertain reading. The symbol on the spin-$1/2$ row is tentatively read as χ.

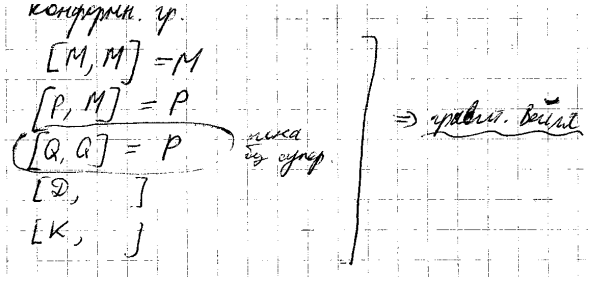
max spin determ. $N/4$ (taking chirality into account) $\Rightarrow N = 8$.

— · —

In 11-dimensional sp. there arises a multiplet with $N = 1$ and there are no conformal anomalies.

— · —

conform. grp.



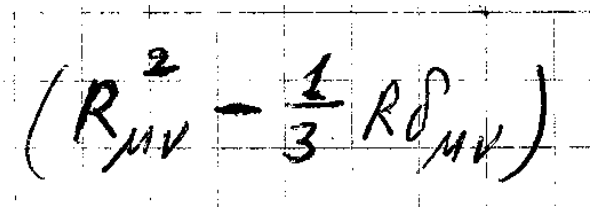
Original source details: GELFAND-P001176

Diagram and typed description.

$[M, M] = M$
$[P, M] = P$
$[Q, Q] = P$ for now without super $\Rightarrow$ Weyl gravit.
$[D, \ ]$
$[K, \ ]$

Conformal group brackets and the note beside the circled line

$$\left(R^2_{\mu\nu} - \frac{1}{3} R\delta_{\mu\nu}\right)$$



Original source details: GELFAND-P001176

Uncertain reading. The second term reads as $-\frac{1}{3} R\delta_{\mu\nu}$.

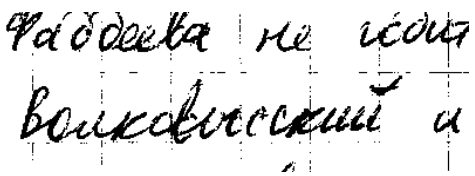
The simplest problem: compute the integral

$$\int e^{i \int d^4x\, R}\, dg_{\mu\nu}$$

highly degenerate. quantization?

Faddeev's method does not work.

Fradkin, Vilkovisskii [spelling uncertain]? and others —



Original source details: GELFAND-P001176

Uncertain reading. Vilkovisskii [spelling uncertain]

[Inserted above the line: to] gravity and others.

An answer arises, not expr. through det —

— the Lie bracket is not closed (closed only on the extremals).

— · —

GELFAND-P001177

19/III–84 yr. Gelfand Sem.

Leites.

Lie superalgebras.

$gl(n|m)$

Supersp. $V = V_{\bar{0}} \oplus V_{\bar{1}}$, grad. $\{\bar{0}, \bar{1}\} = \mathbb{Z}_2$, $\dim V = (n|m)$, $\dim V_{\bar{0}} = n$, $\dim V_{\bar{1}} = m$.

$\mathrm{Hom}(V, W)$ — supersp.;

(W — another supersp. $W = W_{\bar{0}} \oplus W_{\bar{1}}$)

$\mathrm{Hom}(V, W)_{\bar{0}}$ — pres. parity;

$$\backslash\backslash$$
$$\left\{ \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} \right\}$$

$\mathrm{Hom}(V, W)_{\bar{1}}$ — changing parity.

$$//$$
$$\begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}$$

Lie superalg.:

$$[x, y] = -(-1)^{\bar{x}\bar{y}}[y, x]$$
$$[x, [y, z]] = [[x, y], z] + (-1)^{\bar{x}\bar{y}}[y, [x, z]]$$

[In the bracket on the left before x, crossed out: illegible mark?]

Uncertain reading. illegible mark

Original source details: GELFAND-P001177

(Quillen's rule:)

if letters of parity p and q, are permuted, the sign arises $(-1)^{pq}$.

Superalg.: $A = A_{\bar{0}} \oplus A_{\bar{1}}$

$$m\colon A \otimes A \to A \qquad // \quad \text{mult. must be even}$$
$$p(m) = \bar{0}$$

Assoc. superalg. $\rightsquigarrow$ Lie superalg.

$$[a, b] = ab - (-1)^{\bar{a}\bar{b}}ba$$

[Crossed out: illegible writing?]

Uncertain reading. illegible writing

Original source details: GELFAND-P001177

$gl(n|m)$ — superalg. obt. from $\mathrm{Hom}(\mathbb{R}^{n/m}, \mathbb{R}^{n/m})$.

$$sl(n|m) = \{a \mid a \in gl(n|m),\ \mathrm{str}\, a = 0\},$$

$$\mathrm{str} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \mathrm{tr}\, A - \mathrm{tr}\, D$$

(Euler char. — supertrace)

Classif. in the ordinary case:

[Inserted below the first word: of algebras]

$$gl,\ sl,\ o,\ sp, \quad w,\ S,\ H,\ K.$$

$$osp(n|2m) = \{X \in gl(n|2m) \mid X^{st} B_{n,2m} + B_{n,2m} X = 0\},$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{st} = \begin{pmatrix} A^t & -C^t \\ B^t & D^t \end{pmatrix}$$

(one can also use $\begin{pmatrix} A^t & C^t \\ -B^t & D^t \end{pmatrix}$, this is equivalent)

GELFAND-P001178

$$B_{n,2m} = \left(\begin{array}{c|cc} 1_n & 0 & \\ \hline 0 & 0 & 1_m \\ & -1_m & 0 \end{array}\right) \qquad \text{(even form pres.)}$$

matrix — even.

(matrix of an even symm. or odd skew-symm. form)

I.M.G. isomorphic notation:

$$osp(2n|2m) = \left(\begin{array}{cc|cc} 0 & 1_n & & \\ 1_n & 0 & 0 & \\ \hline & & 0 & 1_m \\ 0 & & -1_m & 0 \end{array}\right)$$

$osp(2n+1|2m)$

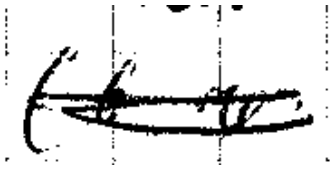
— · —

L.

$$\Pi(n) = \{X \in gl(n|n) \mid X^{st} J_{2n} + (-1)^{\bar{X}} J_{2n} X = 0\}$$

$$= \left\{ \begin{pmatrix} A & B \\ C & -A^t \end{pmatrix} \,\middle|\, \begin{array}{l} B = B^t \\ C = -C^t \\ A \in gl(n) \end{array} \right\}.$$

[Crossed out below the formula: illegible?]



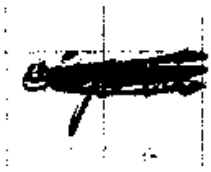
Uncertain reading. illegible

Original source details: GELFAND-P001178

(Π — in honor of Palamodov)

$$J_{2n} = \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix}$$

(odd [Crossed out: illegible word?] form pres.)



Uncertain reading. illegible word

Original source details: GELFAND-P001178

matrix odd.

$$q(n) = \{X \in gl(n|n) \mid [X, J_{2n}] = 0\} = \left\{ \begin{pmatrix} A & B \\ B & A \end{pmatrix} \right\}$$

$$\downarrow \quad q \cdot \mathrm{tr}$$
$$\mathrm{tr}\, B = \mathrm{str}(X \cdot J_{2n}) \qquad (q \text{ from queer, strange})$$

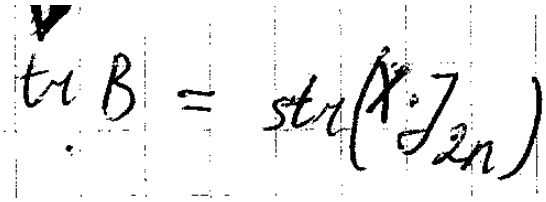


Diagram and typed description. The manuscript equation for the q-trace

Original source details: GELFAND-P001178

I.M.G. asks: what does the theor. on Jordan form look like?

[Crossed out: sq?]

Uncertain reading. sq?

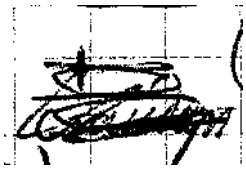
Original source details: GELFAND-P001178

$sq(n)$ also exists.

Projectivizations exist.

$$AB_3 = \underbrace{sl(2) \oplus O(7)}_{(AB_3)_{\bar{0}}} \oplus (AB_3)_{\bar{1}}$$

[Two short lines crossed out under the first term: illegible fragments?]



Uncertain reading. illegible fragments

Original source details: GELFAND-P001178

For $sl(2)$ there is a stand. [Inserted above: 2-dimensional] rep. V

$O(7)$ has the rep. spin_7 (stand.)

$$V \otimes \mathrm{spin}_7 \text{ — rep. } sl(2) \oplus O(7)$$

$$\mathcal{D}(\alpha) = osp(4|2;\alpha)$$

$$\mathcal{D}(\alpha)_{\bar{0}} = O(4) \oplus sp(2) = sl(2)_1 \oplus sl(2)_2 \oplus sl(2)_3$$

$$\mathcal{D}(\alpha)_{\bar{1}} = V_1 \otimes V_2 \otimes V_3$$

V_i — stand. rep. $sl(2)_i$

$$[v_1 \otimes v_2 \otimes v_3,\ u_1 \otimes u_2 \otimes u_3] = \sum_{\substack{(i,j,k)= \\ =\sigma(1,2,3)}} \lambda_k \underbrace{(v_i, u_i)_i}_{\text{skew-symm. prod.}} (v_j, u_j)_j \underbrace{v_k \cdot u_k}_{\text{symm. prod.}} .$$

Jacobi ident. satisf. when: $\lambda_1 + \lambda_2 + \lambda_3 = 0$.

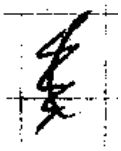
GELFAND-P001179

[source number: 79]

Penkov I.

$$G = SL(T^{n/m}) \qquad n \neq m$$

[Below T, a downward stroke labelled: compl.] [Mark crossed out at right: illegible?]



Uncertain reading. illegible

Original source details: GELFAND-P001179

$B =$ max. solv. closed subsupergroup, all of whose irred. finite-dimensional rep. have dimension $1|0$ or $0|1$.

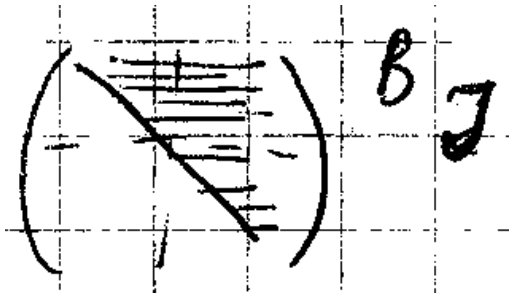
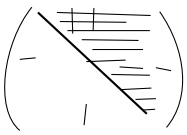


Diagram and typed description.



B_J

Matrix sketch B_J. The meaning of some short strokes is unclear.

Original source details: GELFAND-P001179

$$J \in \mathcal{J} = \{\text{permutations of } n+ \text{ and } m-\}$$

$$G/B_J = \underset{\text{(flags)}}{Fl}\,(J, T^{n/m})$$

$$\dim G/B_J = \frac{n(n-1)}{2} + \frac{m(m-1)}{2} \mid mn$$

$$\begin{array}{c} \operatorname{Pic} G/B \\ \| \\ \operatorname{Mor}(H, G_m) \end{array}$$ — Picard grp.

[Under G_m, a downward stroke: mult. group] [Crossed out: illegible?] $\mathbb{C}^*$

Uncertain reading. illegible

Original source details: GELFAND-P001179

$$\mathrm{Mor}(H, G_m) \xrightarrow{\sim} \mathrm{Pic}^G G/B_J$$
$$\stackrel{df}{=} \left\{ \begin{array}{c} \text{grp. of sheaves of rank } 1|0 \\ \text{with } G\text{-linearization} \end{array} \right\}.$$

[Under H, a downward stroke: Cartan subgrp. of the even part]

Theorem

$$\chi \in \mathrm{Mor}(H, G_m), \quad (\chi, \varepsilon_i - \varepsilon_j) \neq 0,$$
$$(\chi, \delta_i - \delta_j) \neq 0, \quad (\chi, \varepsilon_i - \delta_j) \neq 0.$$

Then $\forall J\ H^i(\mathcal{O}_J(\chi - \rho_J)) = 0$ for $i \neq l(w)$.

$$H^{l(w)}(\mathcal{O}_{J_1}(\chi - \rho_{J_1})) = H^{l(w)}(\mathcal{O}_{J_2}(\chi - \rho_{J_2}))$$
$$= V_{J_1}(\mu - \rho_{J_1}) = V_{J_2}(\mu - \rho_{J_2})$$

$\Rightarrow$ highest-weight rep. are obtained in cohomology.

$(\chi, \varepsilon_i - \delta_j) \neq 0$, but $(\chi, \alpha_0) = 0 \Rightarrow \forall J$ the sheaf $\mathcal{O}_J(\chi - \rho_J)$ is acyclic.

GELFAND-P001180

26/III–84. Gelfand Sem.

Penkov.

$$G = SL(n|m), \qquad G/B_J = Fl(J, T)$$

T — stand. rep.,

J — flag index

$$\mathrm{Pic}^G_J = \mathrm{Pic}^G(G/B_J)$$

Claim 1 $\mathrm{Pic}^G_J =$ characters of H

Thm. Let χ — a character, χ-regular; [Crossed out: illegible marks?]

Uncertain reading. illegible marks

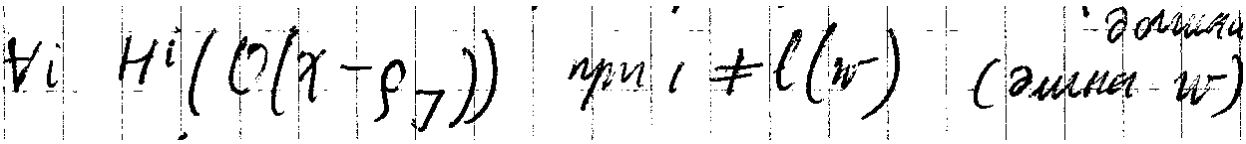
Original source details: GELFAND-P001180

$\exists \mu = w(\chi)$

[Under μ, a stroke labelled: dominant]

Then

$$\forall i \quad H^i(\mathcal{O}(\chi - \rho_J)) \quad \text{for } i \neq l(w) \qquad (\text{length of } w)$$



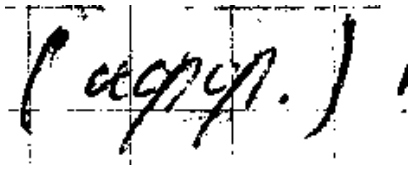
Original source details: GELFAND-P001180

Uncertain reading. There is a gap between $H^i(\mathcal{O}(\chi - \rho_J))$ and "for".

$$H^{l(w)}(\mathcal{O}(\chi-\rho_J))=V_J(\mu-\rho_J).$$

And there is a D-module version of this theorem.

Affine flags: also with area elements (aff.?[?]) on $\forall$ plane of the flag



Original source details: GELFAND-P001180

Uncertain reading. aff.?

T — lin. supersp.,

$$T=T^{n/m}=\underset{\text{even}}{T^{n/0}}\oplus\underset{\text{odd}}{T^{0/m}}$$

[Small mark crossed out above T: illegible[?]]

Original source details: GELFAND-P001180

Uncertain reading. illegible

Type.

$$T^{1/0}\subset T^{2/0}\subset\dots$$
$$\subset T^{1/1}\subset\dots$$

Original source details: GELFAND-P001180

Uncertain reading. The last superscript looks like $1/1$.

GELFAND-P001181

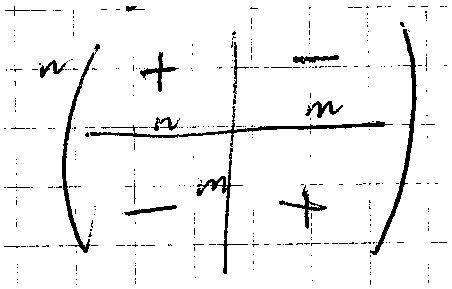
[source number: 87]

Underlying var.: prod. of two flag var.
(even part) (indep. of type)

$$Fl(J,T)_{\text{even}} = Fl(1,2,\dots,T^{n/0}) \times Fl(1,2,\dots,T^{0/m})$$

$$\dim Fl(J,T^{n/m}) = \frac{n(n-1)+m(m-1)}{2} \mid mn$$

(From the equality: $G/B_J = Fl(J,T)$).



Original source details: GELFAND-P001181

Diagram and typed description.

$$n\left(\begin{array}{c|c} + & - \\ n & m \\ \hline - & m \\ & + \end{array}\right)$$

GELFAND-P001182

2/IV–84 yr. Gelfand Sem.
Fuks.

Original source details: GELFAND-P001182

Uncertain reading. The name reads as “Fuks”, with overlapping middle strokes.

$\widehat{sl(2)}$ — algebra, gen. by 6 generators f_j, e_j, h_j $(j = 1, 2)$
K.-M.

$$\begin{matrix} \| \\ N_- \oplus H \oplus N_+ \end{matrix}$$

$$[e_i, f_j] = \delta_{ij} h_i$$
$$[e_1[e_1[e_1, e_2]]] = 0$$
$$[e_2[e_2[e_2, e_1]]] = 0$$

and something else.

$$[h_1, e_2] = -2e_2, \quad [h_2, e_1] = -2e_1,$$
$$[h_1, e_1] = 2e_1, \quad [h_2, e_2] = 2e_2$$

and the same fmls. with e replaced by f
and with the sign chang.,

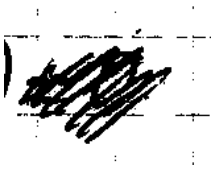
$$[h_1, h_2] = 0$$

I.M.G. This alg. almost coinc. with the algebra

$$\begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix}, \qquad a + d = 0, \qquad t \in S^1.$$

— · —

F. Claim. Verma modules over it depend on 2 compl. parameters. $M(\lambda_1, \lambda_2)$ [Crossed out: illegible fragment?]



Uncertain reading. illegible fragment

Original source details: GELFAND-P001182

GELFAND-P001183

N_- — words, comp. of f_j.

N_+ — words, comp. of e_j.

$\dim H < \infty$.

Verma module — rep. in which $\exists$ a gen. vector for which

$$\begin{cases} N_- v = 0 \\ h_j v = \lambda_j v \end{cases}$$

$e_1 v, \quad [e_1, e_2]v, \ldots$

May turn out to be reducible for certain λ_1, λ_2.

Thm. It is reducible $\Longleftrightarrow \exists w \in M(\lambda_1, \lambda_2)$; $f_i w = 0$.

It is also an eigenvector (sing. vector).

[In the left margin: then $\exists w \nparallel v$.]

(Kac — The spelling looks like "Katsdan", probably referring to Kazhdan.[?];

Uncertain reading. The spelling looks like "Katsdan", probably referring to Kazhdan.

Original source details: GELFAND-P001183

∞-dim. version of the theor. of Bernsh. — Gelf. — Gelf.)

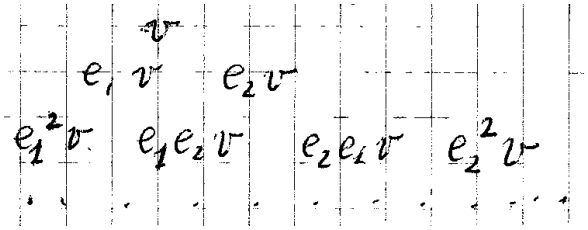


Diagram and typed description.

$$\begin{array}{ccccccccccc} & & & & v & & & & & & \\ & & & e_1 v & & & e_2 v & & & & \\ & & e_1^2 v & & & e_1 e_2 v & & & e_2 e_1 v & & & & e_2^2 v \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{array}$$

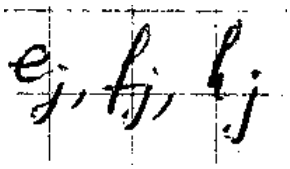
Original source details: GELFAND-P001183

Thm. $M(\lambda_1, \lambda_2)$ is reducible $\Longleftrightarrow \lambda_1, \lambda_2$ satisfy one of the eqs.

$$\begin{aligned} &(1) \quad (k+1)(\lambda_1 + 1) + k(\lambda_2 + 1) = n, \quad k \geq 0,\ n \geq 0. \\ &(2) \quad \lambda_1 \longleftrightarrow \lambda_2 \\ &(3) \quad \lambda_1 + \lambda_2 + 2 = 0 \end{aligned}$$

Problem. (I.M.G.) Find the Casimir op. of this algebra.

(polynomials (noncommut.) in the letters e_j, f_j, l_j



Uncertain reading. The last sign resembles l_j, possibly h_j.

Original source details: GELFAND-P001183

Idea of the constr.:

$$w = e_1^{n+\ell t} \dots e_2^{n+t} e_1^n e_2^{n-t} e_1^{n-2t} \dots e_1^{n-\ell t} v$$
$$t \in \mathbb{C} \ (!!)$$

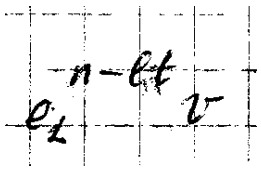
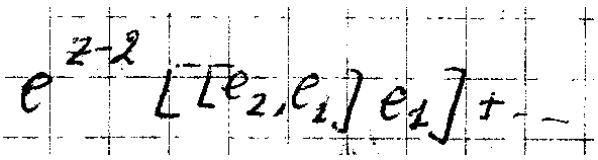


Diagram and typed description. The last factor has a subscript that reads 1.

Original source details: GELFAND-P001183

But: it turns out that this element lies in the enveloping algebra.

$$[e_2, e_1^z] = z e_1^{z-1} [e_2, e_1] + \frac{z(z-1)}{2} e^{z-2} [[e_2, e_1] e_1] + - -$$



Uncertain reading. There is no visible subscript on the base of e^{z-2}. The outer bracket is written without a comma, and two strokes follow the plus sign.

Original source details: GELFAND-P001183

Beilinson.

Hodge structures.

$n \in \mathbb{Z}$.

pure Hodge struct. of weight n — $H_{\mathbb{R}}$, [Under $H_{\mathbb{R}}$: real vect. sp.]

$$H_{\mathbb{C}} := H_{\mathbb{R}} \otimes \mathbb{C} = \bigoplus_{p+q=n} H^{p,q}, \qquad \bar{H}^{pq} = H^{qp}.$$

[Under $H_{\mathbb{C}}$, a stroke leading to “its c”]

Hodge filtration:

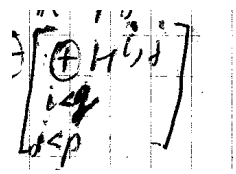
$$F^i H = \bigoplus_{p \geq i} H^{p,q}.$$

The structure is recovered from it:

$$H^{p,q} = F^p H \cap \overline{F^q H}.$$

Mixed Hodge struct. $H_{\mathbb{R}}$ and decomp. $H_{\mathbb{C}} = \bigoplus H^{p,q} \dots$, satisf. the conditions

$$(1) \qquad \bar{H}^{p,q} \subset H^{q,p} \oplus \left[\bigoplus_{\substack{i<q \\ j<p}} H^{i,j} \right].$$



Uncertain reading. The q in the bound $i < q$ is heavily retraced.

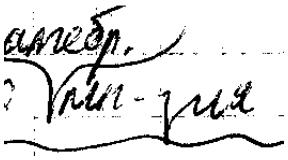
Original source details: GELFAND-P001183

Hereafter Hodge struct. means a mixed Hodge struct..

GELFAND-P001184

[source number: 91]

On the cohom. of any [Inserted above: algebr.]



Uncertain reading. var. [source abbr. uncertain]

Original source details: GELFAND-P001184

over $\mathbb{C}$ (not nec. smooth or compact) there is a mixed Hodge struct. (construction)

Split Hodge struct.: if $\bar{H}^{pq} = H^{qp}$.

This is a direct sum of pure ones (easy exercise). (the weight is not fixed at n).

Again — filtrations (Hodge filtr.).

$$F^i := \bigoplus_{p \geq i} H^{p,q} \qquad (\text{decr.})$$

Mixed Hodge struct. form an abelian category.

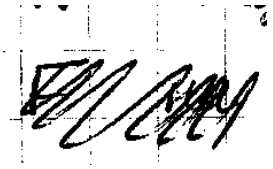
$$W_n := \bigoplus_{p+q \leq n} H^{p,q} \qquad \begin{matrix}(\text{weight filtr.}) \\ \text{incr.}\end{matrix}$$

Ex.* 1) This pair of filtrations uniquely determines the Hodge struct..

2) For a pair of filtr. to determ. a mixed Hodge struct., it is nec. and suff. to [Inserted above: fulf.] the foll. conditions:

A. W is real (=complexif. of a real filtration).

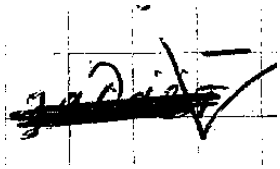
B. [Crossed out: $F \ldots$; the rest is illegible beneath the cancellation?]



Uncertain reading. $F \ldots$; the rest is illegible beneath the cancellation

Original source details: GELFAND-P001184

$(\mathrm{Gr}_n^W H, \mathrm{Gr}_n^W F)$ [Crossed out: defines? A short dash is visible alongside.?]



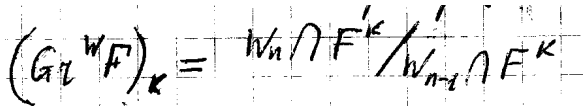
Uncertain reading. defines? A short dash is visible alongside.

Original source details: GELFAND-P001184

— pure Hodge struct. of weight n.

[Bracket under $\mathrm{Gr}_n^W F$:] this is a filtration on $\mathrm{Gr}_n^W H = W_n/W_{n-1}$.

$$(\mathrm{Gr}^W F)_k = \frac{W_n \cap F^k}{W_{n-1} \cap F^k}.$$



Original source details: GELFAND-P001184

Diagram and typed description. The left side is written $(\mathrm{Gr}^W F)_k$.

— · —

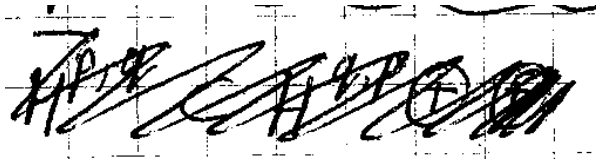
[Crossed out: Problem.; inserted above: Ex.] nontrivial.

(see Deligne, Hodge Theory II. Coll. of trans. “Matem.”, 1973)

— · —

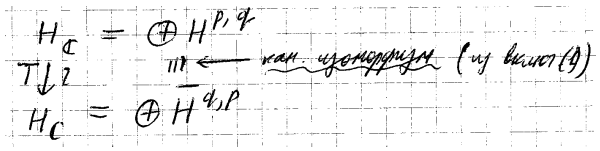
Arithmetic monodromy.

[Crossed out: $\bar{H}^{pq} \subset H^{qp} \oplus$ End of the crossed-out formula: $\bigoplus ?^{?}$]



Original source details: GELFAND-P001184

Uncertain reading. End of the crossed-out formula: $\bigoplus ?$



Original source details: GELFAND-P001184

Diagram and typed description.

$$\begin{array}{ccc} H_{\mathbb{C}} & = \bigoplus H^{p,q} & \\ T \downarrow & \downarrow 2 & \text{II} \longleftarrow \text{can. isomorphism (from incl. (1))} \\ H_{\mathbb{C}} & = \bigoplus \bar{H}^{q,p} & \end{array}$$

The meaning of the short strokes between the rows is uncertain.

$\bar{H}^{pq} \simeq H^{qp}$ (there is a can. isomorphism.), [Below: from (1)] T — the direct sum of these isomorphisms.

T is called the arithmet. monodromy operator.

$$T(H^{p,q}) \subset H^{pq} \oplus \bigoplus_{\substack{i<p \\ j<q}} H^{i,j}.$$

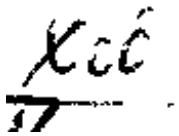
T is strictly unipotent (and even the identity on W_n/W_{n-2}) [Below: from the def.]

Claim.

The Hodge struct. is split $\iff T = \mathrm{Id}$.

GELFAND-P001185

Structures



Original source details: GELFAND-P001185

Uncertain reading. Hod. [source abbr. uncertain]

on [Crossed out: illegible fragment?]

Original source details: GELFAND-P001185

Uncertain reading. illegible fragment

(more gen.) over A (A-structure).

$A = \mathbb{Z}, \mathbb{Q}$.

$$(H_A,\ W_\bullet H_A \otimes \mathbb{C},\ H_{\mathbb{R}} = H_A \otimes \mathbb{R})$$

[Under $W_\bullet$ stroke leading to: incr. filtr.]

Defin. M — Hodge struct.. Abs. cohomology with coeff. in M.

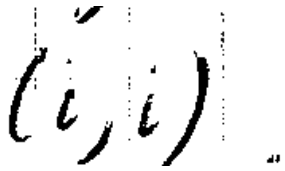
$$H^i_{\mathcal{H}}(M) = \mathrm{Ext}^i(\mathbb{R}, M) \qquad \text{(in the categ. of Hodge structures)}$$

Here $\mathbb{R}$ — the triv. Hodge struct. $\mathbb{R} = H^{0,0}$.

Tate structure

$$i \in \mathbb{Z} \ \Rightarrow \mathbb{R}(i) = (\sqrt{-1})^i \mathbb{R} \subset \mathbb{C}$$

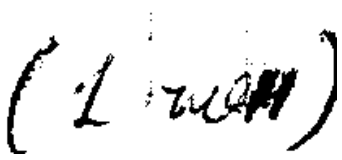
bigrading (i, i)



Original source details: GELFAND-P001185

Diagram and typed description. The manuscript writes (i, i).

(1 term [ending retraced]?)



Original source details: GELFAND-P001185

Uncertain reading. 1 term [ending retraced]

A-Hodge struct.:

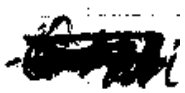
$$(2\pi\sqrt{-1})^i \cdot A = A(i).$$

Def.

$$H^i_{\mathcal{H}_A}(M) := \mathrm{Ext}^i_{\mathcal{H}_A}(A, M).$$

Lemma [Crossed out: illegible word or symbol?]

Uncertain reading. illegible word or symbol



Original source details: GELFAND-P001185

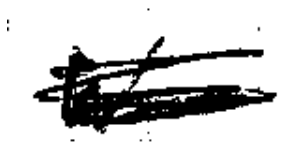
$H^i_{\mathcal{H}} = 0$ for all i, except 0 and 1.

Let $A = \mathbb{Q}, \mathbb{R}$; then

$$H^1_{\mathcal{H}}(M) = {}^{W_{\mathbb{C}\,0}}\!/_{W_{A\,0}+F^0\cap W_{\mathbb{C}\,0}}$$

[Under $W_{A\,0}$ stroke leading to: lattice in $W_{\mathbb{C}\,0}$.] [Crossed out at right: illegible mark?]

Uncertain reading. illegible mark

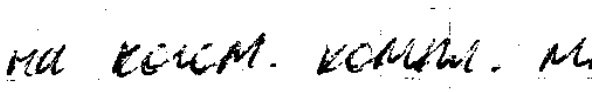


Original source details: GELFAND-P001185

Ex. Prove this fmla. (it is simple).

Aim: to put on the cohom. of

Uncertain reading. comp. compl. [tentative reading of the abbreviations]



Original source details: GELFAND-P001185

var. the same struct. as [Inserted above: there is] on étale cohom..

Chapter 25

GELFAND-P001186

16/IV–84 yr. Gelfand Sem.

Sinai. Ising model.

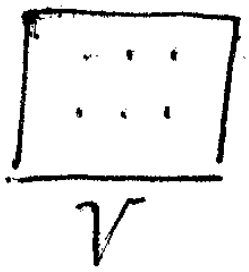
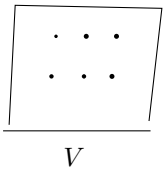


Diagram and typed description.



Original source details: GELFAND-P001186

$$\varphi(x) = \pm 1, \qquad H = -\sum_{\langle x,y\rangle} \varphi(x)\varphi(y),$$

$\langle x, y\rangle \leftarrow$ near. neighbors.

$$\frac{e^{-\beta H}}{\Xi(\beta \mid V)}, \qquad \Xi(\beta \mid V) = \sum_{\varphi(V)} e^{-\beta H}.$$

a) $\beta < \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\cdot\varphi(y)} \le Ce^{-\lambda\|x-y\|}, \quad \lambda = \lambda(\beta),$

b) $\beta > \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\varphi(y)} \longrightarrow m^2(\beta) > 0$ (long-range order),

c) $\beta = \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\varphi(y)} \sim \dfrac{C}{|x-y|^{\gamma}}, \quad \gamma = \dfrac{1}{4} \quad (d = 2).$

Uncertain reading. In line b), an additional stroke is visible inside the $>$ sign.

Original source details: GELFAND-P001186

In QFT a situation arises as with $\beta = \beta_{\text{cr}}$.

Explicitly computed for $d = 2$ are the free energy and the exponent γ.

— · —

Conjecture.

Let

$$\varphi_L(x) = \frac{\sum\limits_{y \in \Delta_L(x)} \varphi(y)}{L^\alpha}.$$

φ — rand. variables (see the Gibbs distr. above) with $\beta = \beta_{\text{cr}}$.

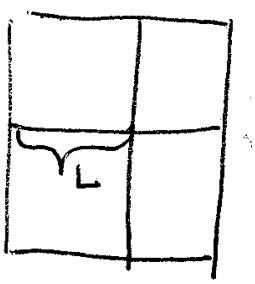
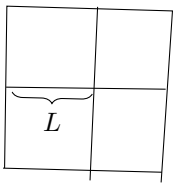


Diagram and typed description.



Original source details: GELFAND-P001186

Then the joint distribution of [inserted above: rand.] variables $\varphi_L(x)$ (at different x) has a limit and the lim. distr. does not depend on the model we started with.

$$x = (x_1, x_2),$$

$$\Delta_L(x) = \{\, y = (y_1, y_2) \mid Lx_1 \le y_1 < L(x_1 + 1),\ Lx_2 \le y_2 < L(x_2 + 1) \,\}$$

contains L^2 points. For different x these squares do not intersect.

Renormalization group method — [crossed out: illegible?]

Uncertain reading. illegible

Original source details: GELFAND-P001186

eq. and fixed points for distr..

Scale transf. induce transf. [crossed out: on??]

Uncertain reading. on?

Original source details: GELFAND-P001186

in the set of probability distr..

For $d \geq 4$ [crossed out: illegible?]

Uncertain reading. illegible

Original source details: GELFAND-P001186

the method is trivial — there is a Gaussian fixed point.

Next d is treated as a cont. parameter [crossed out: illegible?].

Uncertain reading. illegible

Original source details: GELFAND-P001186

$\underset{4-\varepsilon}{4} \longrightarrow 3$ cont. $\Rightarrow$ some things are calc. for $d = 3$.

McPhirson Perverse[d?] shea[fs?]s etc.

Uncertain reading. English inscription: after Perverse, a retraced mark resembling d; after shea, several overlapping strokes precede the final s.

Original source details: GELFAND-P001186

(perverse sheaves)

GELFAND-P001187

23/IV Gelfand Sem. Serre. On the work of [crossed out: $G \ldots ?$?]

Uncertain reading. $G \ldots ?$

Original source details: GELFAND-P001187

Delorme and Guichardet.

(1) Lemma. $A \subset$ aff. Hilbert H.

A bd. $\Rightarrow \exists!$ ball of min radius containing a.

Uncertain reading. The letter after "containing" resembles a lowercase a. The set is called A above.

Original source details: GELFAND-P001187

(min seq. ⇒ the centres converge)

(2) Theorem. [Mark crossed out: 1?[?]]

Uncertain reading. 1?

Original source details: GELFAND-P001187

A group of motions [inserted: in] aff. Hilbert sp. with [crossed out: illegible[?]]

Uncertain reading. illegible

Original source details: GELFAND-P001187

bd. orbits has a fixed point.

Proof: the centre of the min ball surrounding the orbit is fixed. ▶

Trees.

Prop. (FA): if G acts on a tree, then it has a fixed point.

[Note under (FA):]

Uncertain reading. arbre; a separate stroke sits above the beginning of the word.

Original source details: GELFAND-P001187

Examples 1) fin. [crossed out: illegible[?]]

Uncertain reading. illegible

Original source details: GELFAND-P001187

grp. 2) $SL_3(\mathbb{Z})$.

— · —

G. Margulis.

$$G = G_1 \times \ldots \times G_n \supset \Gamma \qquad \begin{matrix} G_i \text{ — simple} \\ \Gamma \text{ discrete} \end{matrix}$$

G/Γ is compact

or vol $G/\Gamma < \infty$.

Suppose there exist no such G', G'': $G = G' \times G''$

$$(G' \cap \Gamma) \text{ mult } (G'' \cap \Gamma) \quad \text{of fin. index.}$$

$$\text{rank } G \geq 2 \qquad (\text{rank } SL(n, \mathbb{R}) = n - 1)$$

$\Rightarrow \Gamma$ has prop. (FA).

— · —

GELFAND-P001188

3/IX–84 yr. Gelfand Sem.

I.M.G.

2 manif. M_1, M_2

Uncertain reading. A crossed-out or overwritten stroke is visible after M_2.

Original source details: GELFAND-P001188

are called Lipschitz equivalent, if $\exists$ a Lipsch. homeomorphism (rel. to 2 smooth metrics) (the inverse is also Lipsch.).

Example. surf. of a cube and a sphere.

Smoothness does not follow.

Sullivan's thm. On $\forall$ smooth manif. there is only one Lipsch. structure!!

(2 homeom. smooth manif. are Lipsch. equivalent)

Problem. Char. classes — integrals of curvatures.

g_{ik} must be C^2 fns.

(are 1st sufficient?)

How to express the classes in terms of g_{ik} without derivatives?

More precisely, it suffices to have the numbers, since only the res. of integration matters

(cf.: generalized fns.!)

Novikov's thm. on the agreement of char. numbers of homeom. manif. follows from Sullivan's thm..

— · —

$$\begin{pmatrix} \frac{\partial}{\partial x} & i\frac{\partial}{\partial y} \\ -i\frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{pmatrix}^2 = \begin{pmatrix} \Delta & 0 \\ 0 & \Delta \end{pmatrix}$$

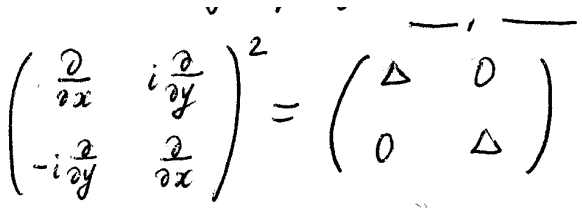
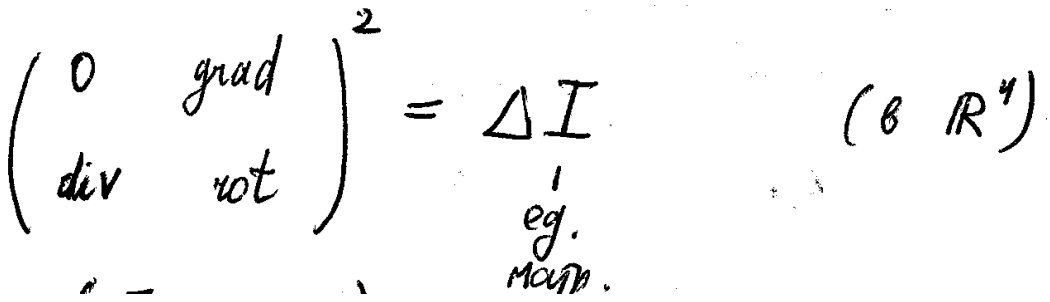


Diagram and typed description. Both diagonal entries of the matrix on the left are written without a minus sign.

Original source details: GELFAND-P001188

$$\begin{pmatrix} 0 & \text{grad} \\ \text{div} & \text{rot} \end{pmatrix}^2 = \Delta I \qquad (\text{in } \mathbb{R}^4)$$

[Under I: id. matr.]



Original source details: GELFAND-P001188

Diagram and typed description. The manuscript writes grad, div, rot and $\mathbb{R}^4$.

(Hamilton)

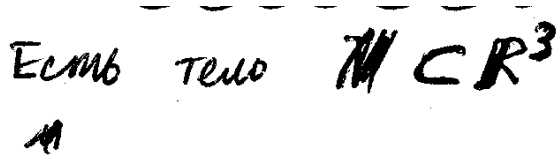
$$(d+\delta)^2 = \Delta \qquad (\delta = d^*)$$

(on forms)

GELFAND-P001189

The "almost everywhere" technique (Lebesgue).

There is a body [Overwritten: M or N?]?



Uncertain reading. [Overwritten: M or N?]

Original source details: GELFAND-P001189

$\subset R^3$. [Added below at left: A.] Volume — ? (via the areas of sections)

$S(x)$ — area of sections.

$$V = \int_a^b S(x)\,dx$$

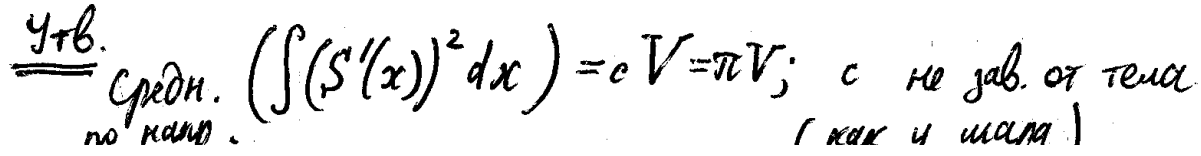
$\int (S'(x))^2\,dx$ also has the required dimension.

Average over all directions of the x-axis.

Assertion.

$$\underset{\text{over directions}}{\text{Mean}} \left(\int (S'(x))^2\,dx \right) = cV = \pi V;$$

c is independent of the body (as for a ball).



Original source details: GELFAND-P001189

Uncertain reading. The manuscript coefficient is π.

(see vol. 4 or 5 of Generalized Functions).

Problem. [Below the heading: (unsolved).] Take an arbitrary measurable set. Then $S(x)$ is measurable $(\forall x)$. Apparently, for almost all directions, $\exists S'(x)$. Question: in what sense is it differentiable, and what does the formula mean?

This is Sullivan's announcement.

On Grothendieck's preprint "Sketch of a Program".

Moved: Institute for Advanced Studies $\longrightarrow$ provincial university. Part has been written up as a 1,300-page manuscript.

Contents:

1. Message (dispatch)

2. The Lego–Teichmüller game and the Galois group Gal $\overline{Q}/Q$

[Below the name: Lego–Teichmüller.]

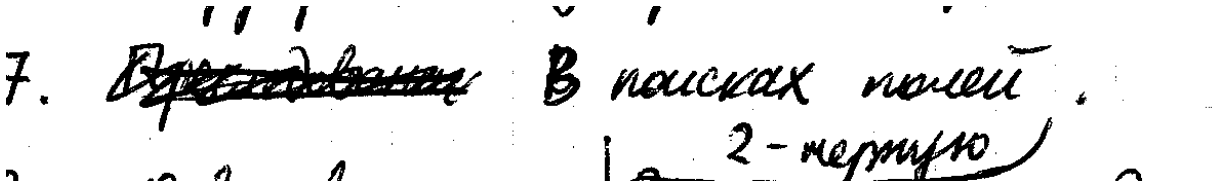
3. Number fields associated with a child's drawing.

4. ["Regular" crossed out and replaced above the line with a Russian synonym] polyhedra and finite fields.

5. Haro against general topology, and heuristic reflections on tame topology.

6. Differentiable and tame theory a là Nash.

7. [Crossed out: Pursuit?[?]]

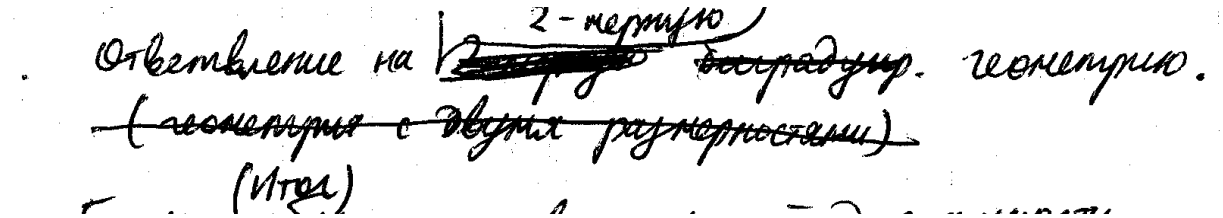


Original source details: GELFAND-P001189

Uncertain reading. Pursuit?

In search of fields.

8. Branching off into [Crossed out: two-dimensional? bigraded[?];



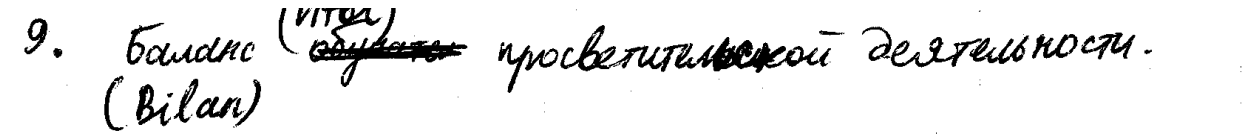
Original source details: GELFAND-P001189

Uncertain reading. two-dimensional? bigraded

inserted above: 2-dimensional] geometry.

[Line crossed out: (geometry with two dimensions).]

9. Balance sheet [crossed out: educational?[?];



Original source details: GELFAND-P001189

Uncertain reading. educational?

above the line: (Outcome)] of educational outreach.

[Below "Balance sheet": (Bilan).]

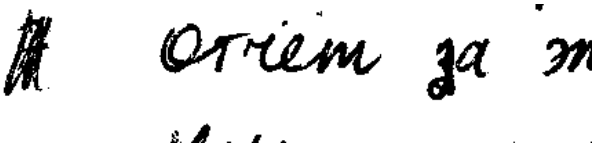
10. Epilogue.

GELFAND-P001190

[source number: 18]

In 1970 he stopped publishing.

[A sign crossed out in the left margin: И?[?]]



Uncertain reading. И?

Original source details: GELFAND-P001190

An account of the intervening period. Intuition of forms (geom.) precedes language.

(Topology and combinatorics — before differential)

Arnold.

Catalan numbers:

$$1, 1, 2, 5, 14, 42, \dots \qquad \text{(there are 34 definitions in all)}$$

$$\frac{(2n)!}{n!(n+1)!}$$

Harris (+ Mumford)

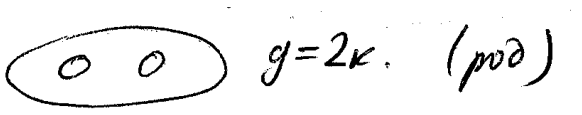
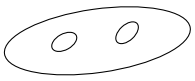


Diagram and typed description.

 $g = 2k$. (genus)

Original source details: GELFAND-P001190

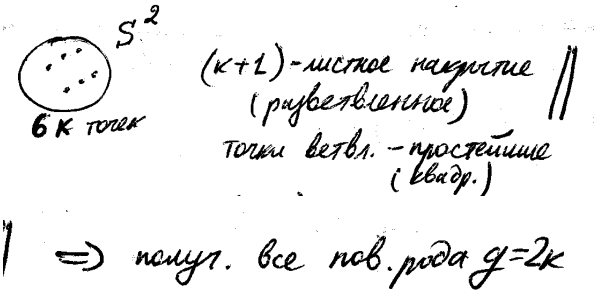


Diagram and typed description.

S^2 $(k+1)$-sheeted covering (branched) //

$6k$ points

$\Rightarrow$ obtain all surfaces of genus $g = 2k$.

branch points — simple (quadratic).

Original source details: GELFAND-P001190

$3g$ parameters (in $\mathbb{C}$).

$SL(2)$ action $\Rightarrow 3g - 3$ moduli.

Number of times one obtains a general surface (without symmetries) — — Catalan numbers!!

— · —

$\Gamma \subset SU(2)$

[Below Γ, linked by a stroke: finite group.] [Vertical dividing stroke.] they come in types A_k, D_k, E_6, E_7, E_8.

McKay noticed (+ another connection of Arnold's)

(+ order of the Monster is related to the modular group)

Consider the symmetry group of a regular polyhedron, i.e. a finite subgroup $\Gamma \subset SU(2)$.

It has finite-dimensional representations.

We $\otimes$ with others and decompose. The resulting square matrix — — Cartan matrix (with an inessential correction) of the affine system.

Arnold:

cases; finite groups Γ act on $\mathbb{C}^2$; there is an algebra of invariants, and it has been described. There are 3 dependent invariants and one relation. [Below the line: (since the space is 3-dimensional).]

$$P(x, y, z) = 0$$

In $\mathbb{C}^3$ consider the fiber $\{P = \varepsilon\} = M_\varepsilon$, $\dim_{\mathbb{R}} M_\varepsilon = 4$. Consider the intersection matrix in 2-dimensional homology. This is the matrix of Coxeter. (FA, 1972, no. 4)

GELFAND-P001191

Groupoid: a category in which every morphism is an isomorphism.

Fundamental groupoid of a topological space:

points (objects), homotopy classes of paths (morphisms)

Diagram and typed description.

Original source details: GELFAND-P001191

Relative to a collection of points: when some initial points are disallowed.

10/IX–84 Gelfand seminar.

M. Semenov–Tian–Shansky.

On the conference (in Japan)! (near Kyoto)

H. / M.? Morimoto

Uncertain reading. H. / M.

Original source details: GELFAND-P001191

(8 young Japanese + 8 Western Europeans)

Yosida (young) — integrability

K. Taubs (California)

V. Kac

B. Kupershmidt

Wilson

[To the right of the list, beyond a vertical stroke:] P. [crossed out: Van?[?]]

Uncertain reading. Van?

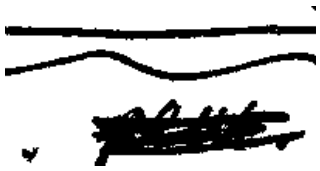
Original source details: GELFAND-P001191

van Moerbeke. — criterion for integrability and nonintegrability based on Kovalevskaya's ideas.

Main outcome: open problems (to be published).

On P. van Moerbeke's talk (with ["someone" crossed out, "Adler" inserted above])

On integrability of [crossed out: solutions?[?]] algebraic differential equations.



Uncertain reading. solutions?

Original source details: GELFAND-P001191

In time — meromorphic functions.

They worked out the left-invariant metrics on $SO(4)$

Mishchenko, Fomenko + 2 more tops.

$$\mathbb{R}^n \subset \mathbb{C}^n \qquad \dot{z} = J_{(z)} \frac{\partial H}{\partial z} =: \{z, H\}$$

$J(z)$ — a polynomial defining a Poisson bracket satisfying the Jacobi identity.

The bracket may be degenerate.

Algebraically [inserted above the line: completely] integrable:

1) $\exists\, k$ polynomial invariants $H_1, \ldots, H_k$ (Casimirs)

[At right, arrow toward "Casimirs":] in the center of the Poisson bracket (they commute with everything) (polynomials in z)

[Lower right, another arrow toward "Casimirs":] center of the Lie algebra generated by the Poisson bracket.

2) [Overwritten sign: $\exists$?[?]] $m = \frac{n-k}{2}$ polynomial invariants with $\{H_i, H_j\} = 0$

Uncertain reading. $\exists$?

Original source details: GELFAND-P001191

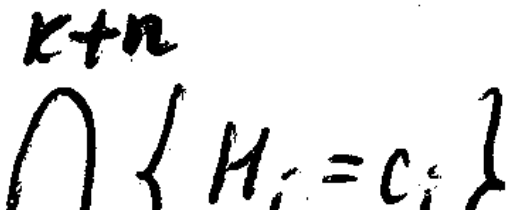
Then $\forall c_i$ (almost all) [crossed out: $E_c \ldots ?^?$]

Uncertain reading. $E_c \ldots ?$

Original source details: GELFAND-P001191

$$\underbrace{\bigcap_{i=1}^{k+n} \{H_i = c_i\} \cap \mathbb{R}^n}_{E_c} \quad \text{— Liouville tori.}$$

Uncertain reading. The upper limit is written $k+n$.



Original source details: GELFAND-P001191

(an affine torus — either T^ℓ or $T^{\ell_1} \times \mathbb{R}^{\ell_2}$)

(Liouville–Arnold theorem)

Algebraically completely integrable.

Main theorem.

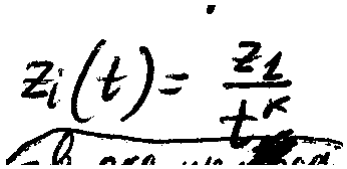
Algebraically completely integrable $\Rightarrow$ 1) solutions $z_i(t)$ are meromorphic in t; [crossed out: illegible$^?$]

Uncertain reading. illegible

Original source details: GELFAND-P001191

$$z_i(t) = \frac{z_1}{t^k}$$

Uncertain reading. The numerator reads z_1, possibly z_i.



Original source details: GELFAND-P001191

[Circled insertion above “expansions”: Laurent.] [Circled below the formula, arrow left toward “expansions of solutions”: near the pole.]

expansions of solutions have [circled insertion: exactly] $(n-1)$ free parameters.

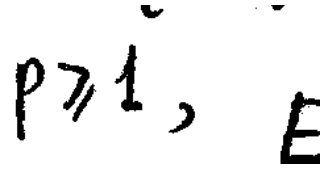
[At right, grouped with a brace:] $\exists$ $(n-1)$-dimensional family of Laurent solutions — Laurent series with a pole at zero (or another point), satisfying the system.

2)

$$D_c = \left\{ z_i = \frac{z_i^0}{t^p} + \dots, \quad H_i(z(t)) = c_i + \dots \right\}$$

[Below $H_i(z(t))$: (have no pole at zero).]

$p \geq 1$,



Uncertain reading. Pole order letter: p or ρ.

Original source details: GELFAND-P001191

$E_c \cup D_c$ — an abelian set (canonical compactification).

D_i — an algebraic set in the space of $(n-1)$-parameter solutions, moreover $\longrightarrow$

Uncertain reading. The subscript on D in the last line looks like i; above it is c.

Original source details: GELFAND-P001191

GELFAND-P001192

[source number: 19]

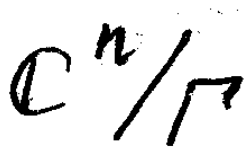
Difficulty: [overwritten sign: illegible?]

Uncertain reading. illegible

Original source details: GELFAND-P001192

constructing the formal series. Convergence is easier.

$$\longrightarrow \bigcap_i \{H_i = c_i\} \cup D_c = \mathbb{C}^n/\Gamma \qquad \text{(complex torus)}$$



Uncertain reading. The exponent is written n.

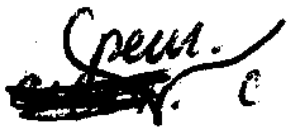
Original source details: GELFAND-P001192

Question: how is $\mathbb{C}^n/\Gamma$ related to the Jacobian (on which the equation is linearized).

They do not coincide!! But they differ by quotienting by a finite group.

The solution is written in terms of θ-functions.

In the practical construction the logic is reversed: we start with Laurent solutions depending on parameters. If ∃ [crossed out: illegible?;



Uncertain reading. illegible

Original source details: GELFAND-P001192

above the line: Family] with $(n-1)$ free parameters, then ([crossed out: hypothesis?] principle)

Uncertain reading. hypothesis?

Original source details: GELFAND-P001192

there is integrability. Moerbeke believes this is a general theorem.

— · —

[Crossed out: Example?]

Uncertain reading. Example?

Original source details: GELFAND-P001192

I. M. G.

$$y'' = f(t, y, y')$$

Uncertain reading. The first argument is overwritten and reads as t.

Original source details: GELFAND-P001192

[An isolated sign to the right of the formula: a?]

Uncertain reading. a?

Original source details: GELFAND-P001192

The singularities move. Painlevé equations: if the singularities are fixed.

Fixed singularities: for integrable systems. (Painlevé considered only essential singular points)

It is natural to regard as integrable those for which the essential singular points are fixed. Painlevé — order 2.

[Above the next line: Garnier — order 3 (?)]

Problem: give a definition of Painlevé-type integrability. There should be a classification.

S. T. S. Example.

In $\underbrace{\mathbb{R}^3}_{z_1,z_2,z_3} \oplus \underbrace{\mathbb{R}^3}_{z_4,z_5,z_6}$ there are 2 [crossed out: illegible?]

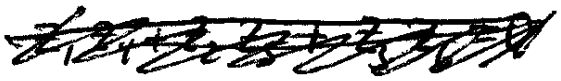
Uncertain reading. illegible

Original source details: GELFAND-P001192

sets of natural coordinates (Lie algebra $so(4)$)

[A semicolon separates the two coordinate annotations.] and z_{ij} (matrices)

[Line crossed out: $z_1z_4 + z_2z_5 + z_3z_6 = 0$?]



Uncertain reading. $z_1z_4 + z_2z_5 + z_3z_6 = 0$?

Original source details: GELFAND-P001192

$$H = \frac{1}{2}\sum \lambda_i z_i^2 + \sum_{i=1}^{3} \lambda_{i,i+3}\, z_i z_{i+3}$$

defines geodesics on $\underbrace{SO(4)}_{\text{6-dimensional}}$ (in a [inserted: left-] invariant metric); these are the equations for precession [inserted above: of momentum]; then one has to lift to $T^*SO(4)$.

[Two vertical strokes to the left of "6-dimensional."]

Only some are integrable. Some — Manakov; Mishchenko and Fomenko.

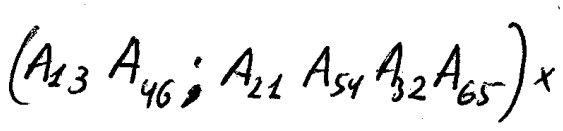
In each $\mathbb{R}^3$ the standard bracket (Lie–Poisson–Berezin–Kirillov)

$$\{z_i, z_j\} = \varepsilon_{ijk} z_k, \qquad i,k \in 1,2,3 \quad \text{or } 4,5,6$$

$$\{z_i, z_{j+3}\} = 0$$

2) $A_{ij} = \lambda_i - \lambda_j$

$$(\lambda_{14}^2, \lambda_{25}^2, \lambda_{36}^2)(A_{46}A_{32} - A_{65}A_{13}) = (A_{13}A_{46};\ A_{21}A_{54}A_{32}A_{65})\times$$



Uncertain reading. No separator is visible between A_{54} and A_{32}.

Original source details: GELFAND-P001192

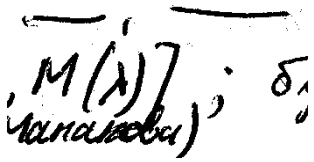
$$\times \left(\frac{(A_{25}-A_{36})^2}{A_{32}A_{65}}, \quad \frac{(A_{36}-A_{14})^2}{A_{13}A_{46}}, \quad \frac{(A_{14}-A_{25})^2}{A_{21}A_{54}} \right)$$

Here D — consists of 4 hyperelliptic curves (isomorphic) with the intersection described above.

Idea:

$$\dot{L}(\lambda) = [L(\lambda), M(\lambda)];$$

(Manakov's equation); we shall regard λ as an internal parameter.



Original source details: GELFAND-P001193

Uncertain reading. Above $M(\lambda)$ are two horizontal strokes and a short vertical mark.

$$\widetilde{sl}(n) = sl(n) \otimes \mathbb{C}(\lambda, \lambda^{-1})$$

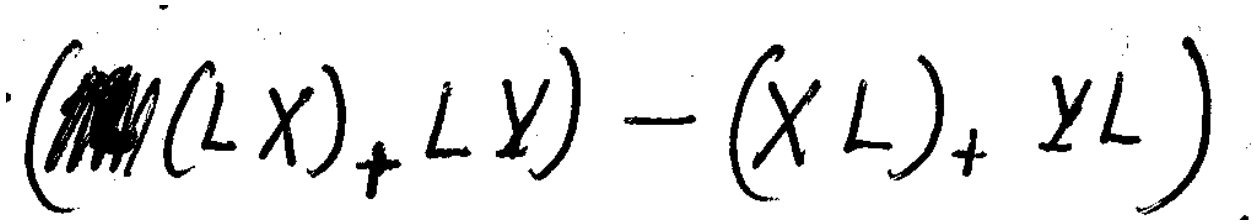
There are 2 brackets! The first is trivial (Berezin–Kirillov).

The second is constructed in R-matrix language

$$r = P_+ - P_-$$

$$\operatorname{Res}(\operatorname{tr} X \cdot Y)|_{\lambda=0}$$

$$\{f, g\}(L) = \big([\text{crossed out}]\,(LX)_+ LY\big) - (XL)_+ YL\big)$$

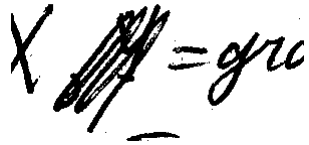


Original source details: GELFAND-P001193

Uncertain reading. The crossed-out fragment before (LX) is illegible. After the minus sign there is no visible opening parenthesis before $(XL)_+$.

(generalization of the Gelfand–Dickey bracket)

X [crossed out: Y?[?]]



Original source details: GELFAND-P001193

Uncertain reading. Y?

$=$ grad f, $Y =$ grad g.

For the series SO: the 2nd bracket cannot be obtained.

— · —

Drinfeld's theorem: description of the symplectic leaves in terms of double cosets in a larger Lie group.

$$G \times G \longrightarrow G \qquad \text{(Hamilton–Lie group)}$$

$d = g \oplus g^*$ — Lie algebra; g, g^* — subalgebras.

$$\{\text{orbits in } G\} = G^* \backslash D / G^*$$

Formula measuring the failure to preserve the Poisson bracket

$$X\{f,g\} - \{Xf, g\} - \{f, Xg\} = \langle X, [d_e \hat{f}, d_e \hat{g}]_* \rangle$$

This suffices to construct a variant of Hamiltonian reduction.

There is no general moment map.

On Grothendieck (Sosinsky). 3. Number fields associated with a child's drawing.

GELFAND-P001194

8/X–84. Gelfand seminar.

Beilinson on a paper by Atiyah and Singer

[At right, a curved line groups the title and journal line:]

"Dirac operators coupled to vector potentials."

Proc. Nat. Acad. Sci. USA,

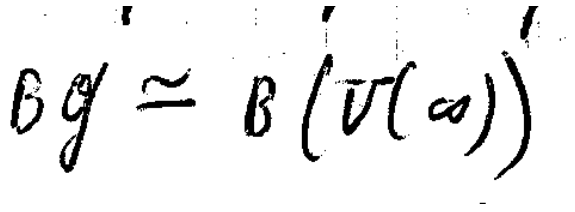
v. 81, pp. 2597–2600 (April 1984)

Construction of the classifying space of the gauge group.

Universal Dirac operator $\longrightarrow$ its index $\partial\!\!\!/ \in K_0(B\mathcal{G})$

$$B\mathcal{G} \simeq B(U(\infty))$$

$\Rightarrow$ on $B\mathcal{G}$ there is a virtual universal bundle — and this gives index $\partial\!\!\!/$.



Uncertain reading. The letter in $B(U(\infty))$ reads as U, with a pointed lower bend.

Original source details: GELFAND-P001194

$d \log(\det(\partial\!\!\!/^* \partial\!\!\!/))$ gives the class $t_1 \in H^1(\mathcal{G})$.

GELFAND-P001195

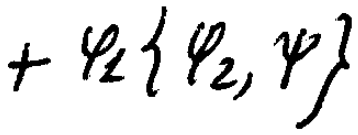
29/X–84 Gelfand seminar.

Semenov–Tian–Shansky M.

Drinfeld's theorem: a class of Poisson brackets.

M — set, suppose that on $C^\infty(M)$ there is a Poisson bracket $\{\varphi,\psi\}$ — a Lie algebra structure with the condition:

$$(1) \qquad \{\varphi_1\varphi_2,\psi\} = \varphi_2\{\varphi_1,\psi\} + \varphi_2\{\varphi_2,\psi\}$$



Uncertain reading. In the second term, the factor before the bracket is written φ_2.

Original source details: GELFAND-P001195

Then M is called Poisson.

This is a Category (morphisms — smooth maps preserving the brackets)

A Poisson Lie group G. — is a Lie group such that

1) G — a Poisson set

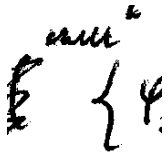
2)

$$\begin{aligned} G\times G &\longrightarrow G \qquad \text{— a Poisson map}\\ a,b &\longmapsto ab \end{aligned}$$

On $G\times G$ the direct product of brackets (there is a product in the category of Poisson sets)

$$\begin{aligned}\{\varphi(x,y),\psi(x,y)\} = &\{\varphi(\cdot,y),\psi(\cdot,y)\}(x)\\ &+\{\varphi(x,\cdot),\psi(x,\cdot)\}(y).\end{aligned}$$

[Above the line: or.] [At left, a crossed-out sign: {?$^?$]



Uncertain reading. {?

Original source details: GELFAND-P001195

$$\{\varphi_1,\varphi_2\}_{M_1\times M_2} = \{\varphi_1,\varphi_2\}_{M_1} + \{\varphi_1,\varphi_2\}_{M_2}.$$

Examples. 1°. G with the zero bracket

2°. g — a Lie algebra, g^* — its dual space; consider g^* as an additive Lie group. This is the Lie–Poisson group (with the canonical bracket!)

[Below "canonical bracket": Lie–Berezin–Kirillov.]

3°. G — a Lie group, where

(i) on the Lie algebra g there is an invariant inner product [Inserted above "invariant": nondegenerate.]

(ii) two subalgebras are given $g_+, g_- \subset g$, such that

$$g = g_+ \dot{+} g_-, \quad \text{and } g_+, g_- \text{ are isotropic}$$

(relative to the inner product).

GELFAND-P001196

Exercise. (I.M.G.) Suppose that on [inked over:

Undeciphered passage. The writing before E is obscured by heavy strokes.

Original source details: GELFAND-P001196

] E [below E, linked by a stroke: linear space] a Poisson bracket is introduced on linear functions. Then on E^* has a Lie algebra structure and the bracket on E is the one from the example.

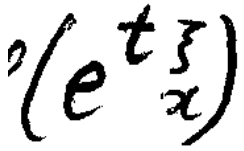
$$R \in \operatorname{End} \mathfrak{g}: \quad R = P_+ - P_-, \quad P_\pm \text{ — projections onto } \mathfrak{g}_\pm \text{ parallel to } \mathfrak{g}_\mp.$$

$$(2) \quad \{\varphi, \psi\} = (R(\nabla_\varphi), \nabla_\psi) - (R(\nabla'_\varphi), \nabla'_\psi)$$

[Callout under ∇_φ: left gradient $\in \mathfrak{g}$; under ∇'_φ: right gradient.]

Definition of the gradients.

$$(\nabla_\varphi(x), \xi) = \left(\frac{d}{dt}\right)_{t=0} \varphi(e_x^{t\xi})$$



Uncertain reading. x is written below $t\xi$, at the lower right of e. Possibly $e^{t\xi}x$.

Original source details: GELFAND-P001196

$$(\nabla'_\varphi(x), \xi) = \left(\frac{d}{dt}\right)_{t=0} \varphi(xe^{t\xi})$$

Theorem 1. (the proof is long). This bracket gives G the structure of a Poisson Lie group.

4°. $G = GL(n)$

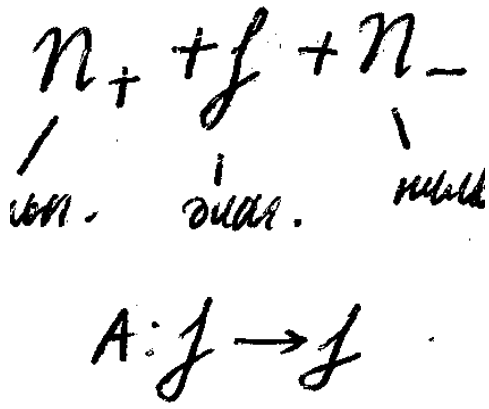
$$\mathfrak{g} = \underset{\text{nilp.}}{\mathfrak{n}_+} + \underset{\text{diag.}}{\mathfrak{f}} + \underset{\text{nilp.}}{\mathfrak{n}_-}$$

$$R = P_+ + AP_0 \text{ [inked over] } P_-,$$

Uncertain reading. The sign before P_- is heavily inked over. It could be $-$ or $+$.

Original source details: GELFAND-P001196

where $A\colon \mathfrak{f} \to \mathfrak{f}$ — a skew-symmetric operator.



Uncertain reading. The middle summand and both letters in the definition of A have the same shape. Rendered $\mathfrak{f}$, possibly $\mathfrak{h}$.

Original source details: GELFAND-P001196

$$((A, B) = \text{tr}(AB) \text{ on } gl(n))$$

Theorem 2 (in the setting of example 3°)
The symplectic leaves in G have the form

$$(G_+ x G_+) \cap (G_- y G_-) = M_{x,y};$$

(intersections of double cosets)
where $G_\pm$ — subgroups corresponding to the subalgebras $\mathfrak{g}_\pm$.
A Poisson set locally breaks up into leaves which are symplectic sets.
Symplectic leaves: locally a Poisson set looks like this

Diagram and typed description.
////

Original source details: GELFAND-P001196

(foliation; the leaves are symplectic, the bracket is recovered from the [inked over:

Undeciphered passage. The word before "symplectic" is heavily crossed out and remains illegible.

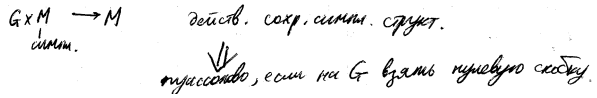
Original source details: GELFAND-P001196

] symplectic structure)

In fact there is a stratification! (the leaves can have different dim)

Def. Let G — a Poisson Lie group, M — a Poisson set; on $G \times M$ has a Poisson structure. We say that G acts on M in a Poisson way, if the action map $G \times M \to M$ is Poisson.

Such an action generally fails to preserve the Poisson bracket on M.



Original source details: GELFAND-P001196

Diagram and typed description.

$$\begin{array}{ccc} G \times M & \longrightarrow & M \\ \downarrow & & \\ \text{sympl.} & & \Downarrow \end{array}$$

the action preserves the symplectic structure

Poisson if we take the zero bracket on G

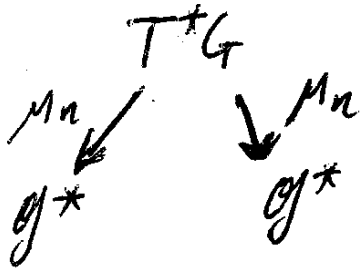
GELFAND-P001197

[source number: 34]

The condition that the action is Poisson is the equality

$$(3) \quad \{\varphi, \psi\}(g \cdot x) = \{\varphi(g\cdot), \psi(g\cdot)\}_M\big|_{\cdot=x} + \\ + \{\varphi(\cdot x), \psi(\cdot x)\}_G\big|_{\cdot=g}$$

The 2nd term measures the failure to preserve the Poisson bracket.



Original source details: GELFAND-P001197

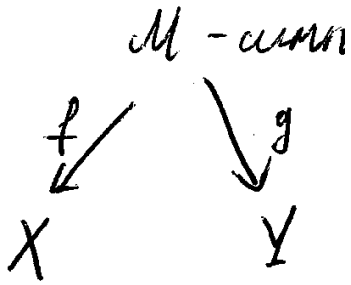
Diagram and typed description.

$$\begin{array}{ccc} & T^*G & \\ {\scriptstyle \mu_\Pi}\swarrow & & \searrow{\scriptstyle \mu_\Pi} \\ \mathfrak{g}^* & & \mathfrak{g}^* \end{array}$$

Both arrow labels look like μ_Π. The text at right names the left and right maps.

— the left and right moment maps; they form a Poisson pair.

M — symplectic



Original source details: GELFAND-P001197

Diagram and typed description.

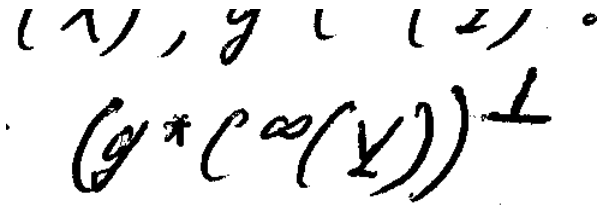
$$\begin{array}{ccc} & M & \\ {\scriptstyle f}\swarrow & & \searrow{\scriptstyle g} \\ X & & Y \end{array}$$

— Poisson maps. They form a Poisson pair, if [in the right margin, with an arrow to "Poisson pair": or dual pair]

$$\{f^*C^\infty(X), g^*C^\infty(Y)\} = 0$$

Call the pair complete, if one of the algebras $f^*C^\infty(X)$, $g^*C^\infty(Y)$ is the centralizer of the other:

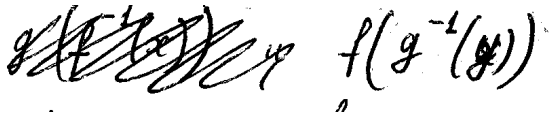
$$f^*C^\infty(X) = (g^*C^\infty(Y))^\perp$$



Uncertain reading. The symbol after the right parenthesis reads $\perp$.

Original source details: GELFAND-P001197

Theorem 3 (in the local form of the theorem). In this situation [crossed out: $g(f^{-1}(x))$] and $f(g^{-1}(y))$ — a symplectic leaf!



Uncertain reading. The letter inside g^{-1} is overwritten. A lower loop is visible and reads y, with a possible x beneath.

Original source details: GELFAND-P001197

(and locally all symplectic leaves are obtained this way)

We want to apply this to the proof of Drinfeld's theorem.

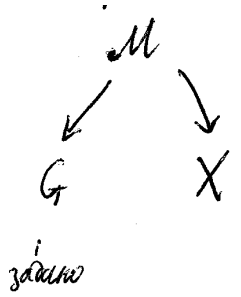
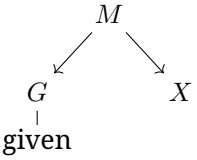


Diagram and typed description.



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For an arbitrary Poisson Lie group construct the double D (a group of twice the dim)

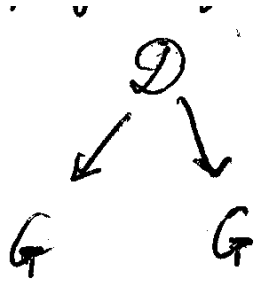
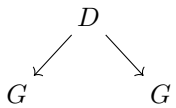


Diagram and typed description.



Original source details: GELFAND-P001197

$G = \mathfrak{g}^* \Rightarrow$ we obtain the Lie–Kirillov–Kostant construction

$$\mathfrak{d} = \mathfrak{g} \oplus \mathfrak{g}^*, \quad (\mathfrak{g}^* \text{ is also a Lie algebra, since there is a Poisson structure on } G)$$

D — the Lie group of the algebra $\mathfrak{d}$

On $\mathfrak{d}$ there is a natural inner product:

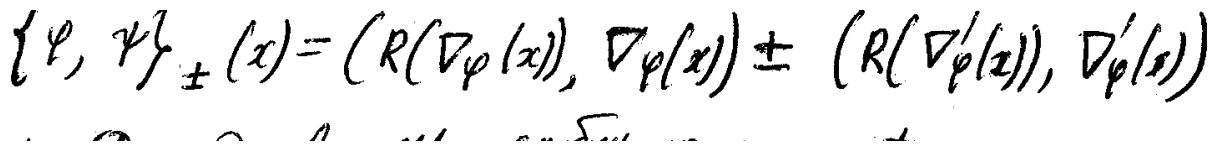
$$\langle (X, F), (X', F') \rangle = F(X') + F'(X)$$

This reduces to example 3°: the double acquires a Lie–Poisson group structure.

GELFAND-P001198

On G there are further 2 Poisson brackets:

$$\{\varphi,\psi\}_{\pm}(x) = (R(\nabla_\varphi(x)),\nabla_\varphi(x)) \pm (R(\nabla'_\varphi(x)),\nabla'_\varphi(x))$$



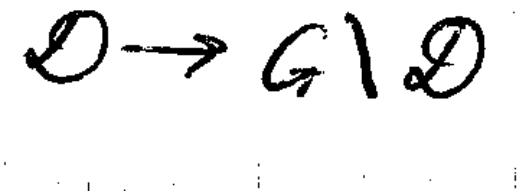
Original source details: GELFAND-P001198

Uncertain reading. All four subscripts on the right read φ. On the left, $\{\varphi,\psi\}$.

On D we should take the bracket with sign $+$.

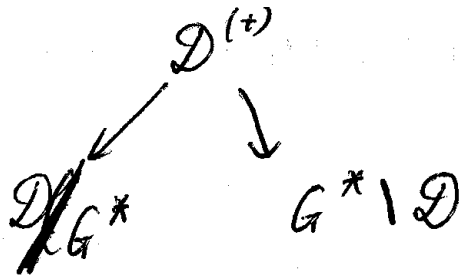
$\pm$ is related to homogeneity properties of the Sklyanin bracket.

Theorem 4. 1) Let G — be a Lie–Poisson group, D — its double. Then the canonical projections $D \to D/G^*$ and $D \to G\backslash D$ form a complete dual Poisson pair.



Original source details: GELFAND-P001198

Uncertain reading. The text has $G\backslash D$ with no asterisk. The diagram has $G^*\backslash D$.



Original source details: GELFAND-P001198

Diagram and typed description.

$$\begin{array}{ccc} & D^{(+)} & \\ \swarrow & & \searrow \\ D/G^* & & G^*\backslash D \end{array}$$

The slash at left is heavily overwritten and reads as /. The underlying strokes are indistinct.

2) $G \subset D/G^*$ and the structure is induced. [Under G, linked by a stroke: open cell.]

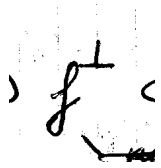
— · —

$H \subset G$ [Callout from H: subgroup.]

$$C^\infty(M)^H \subset C^\infty(M)$$

Thm. 5.

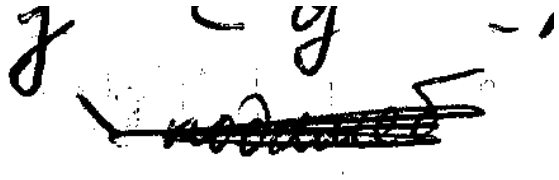
$$C^\infty(M)^H \text{ — Lie subalgebra} \iff \mathfrak{f}^\perp \subset \mathfrak{g}^* \text{ — Lie subalgebra}$$



Original source details: GELFAND-P001198

Uncertain reading. The base of $\perp$ has a lower loop and a cross stroke. Rendered $\mathfrak{f}^\perp$, possibly $\mathfrak{h}^\perp$.

[From the symbol with superscript $\perp$ a callout leads down to the crossed-out note]



Undeciphered passage. The note under the formula is heavily crossed out.

Original source details: GELFAND-P001198

Proof.

$$X\{\varphi,\psi\} - \{X\varphi,\psi\} - \{\varphi,X\psi\} = \langle [d_e\varphi(g\cdot x), d_e\psi(g\cdot x)]_*, X\rangle$$

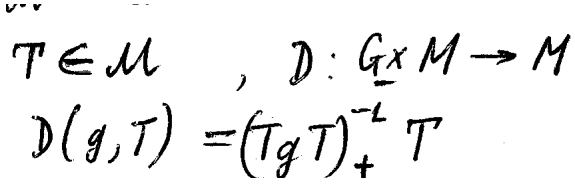
— · —

Let G — be the Poisson–Lie group from example 3°.

$M \subset G$ — a symplectic leaf

$$T \in M, \quad D\colon G_- \times M \to M$$

$$D(g,T) = (TgT)_+^{-1}T$$



Uncertain reading. The short stroke at the lower right of G reads as subscript $-$. Inside the parentheses is TgT. The exponent -1 appears only outside.

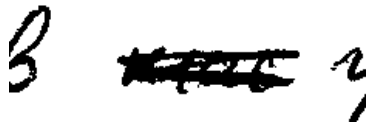
Original source details: GELFAND-P001198

(using the local factorization $x = x_+x_-$, $x_\pm \in G_\pm$)

Thm. 6. This action is Poisson

— · —

A quantum group is an object that in the classical limit goes over into [crossed out] group [abbr.].



Undeciphered passage. The word between "into" and "group" is obscured by cancellation.

Original source details: GELFAND-P001198

[Inserted below "goes over": algebra.]

(Drinfeld: group and cogroup. Both operations are series in h)

Sklyanin algebra: FA, 17:4

Chapter 26

GELFAND-P001199

9/IX–85 Gelfand seminar.

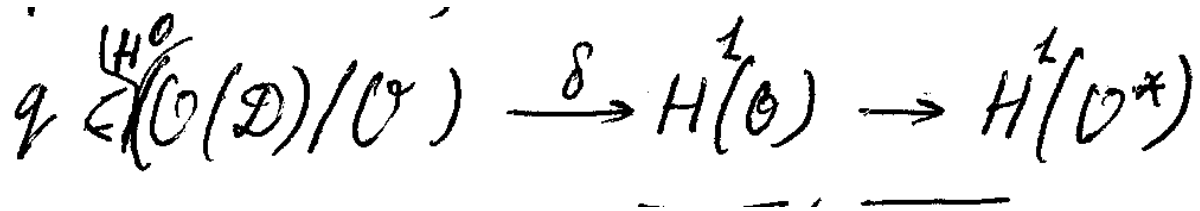
Kapranov, Misha.

Baker functions and their applications to the Yang–Mills equations.

Baker function of an arbitrary set — the simplest functions with essential singularities.

$$(1) \quad \varphi = e^{q(z)}\varphi_0, \qquad \varphi_0 \text{ — holomorphic}$$

$$(2) \quad q \in H^0(\mathcal{O}(D)/\mathcal{O}) \xrightarrow{\delta} H^1(\mathcal{O}) \longrightarrow H^1(\mathcal{O}^*)$$



Original source details: GELFAND-P001199

Uncertain reading. H^0 is inserted above the line, before the first parenthesis.

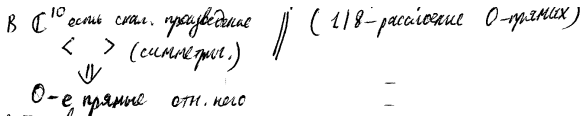
— · —

Now something inspired by Witten,

In $\mathbb{C}^{10|16}$ there is a family of lines of dimension $1/8$.

Original source details: GELFAND-P001199

Uncertain reading. The separator in the exponent $10 \mid 16$ is almost vertical. In the dimension $1/8$ it is slanted.



Original source details: GELFAND-P001199

Diagram and typed description.

In $\mathbb{C}^{10}$ there is an inner product $\langle\ \rangle$ (symmetric) $\Downarrow$ null lines with respect to it // (1|8-bundle of null lines)

[In the left margin:] A. Goncharov.

$SO(10)$ acts on $(\mathbb{C}^{10}, \langle\ ,\ \rangle)$

$V_0 \oplus V_1$, V_1 — spinor representation of the group $SO(10)$.

Uncertain reading. The line about the spinor representation reads $SO(10)$.

Original source details: GELFAND-P001199

dim $V_1 = 16$

Subspaces of dim. $(1, 8)$: $L_0 + L_1$

L_0 — a null line in $\mathbb{C}^{10} \Longrightarrow$ subgroup $SO(8) \subset SO(10)$

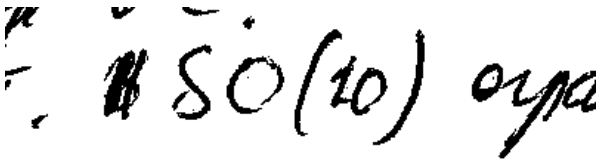
[Below the right-hand side:] (in the stabilizer of the line L_0)

(its semisimple part — $SO(8)$)

This is the group of rotations preserving L_0 and the orthogonal complement (which contains it)

$\Longrightarrow$ obtain rotations of the quotient space.

Now the spinor representation



Uncertain reading. The mark before $SO(10)$ is heavily overwritten. It may be the Russian preposition "в" (in).

Original source details: GELFAND-P001199

$SO(10)$ restrict to $SO(8)$; obtain 2 invariant subspaces (8-dimensional)

GELFAND-P001200

[source number: 59]

$\Longrightarrow$ 2 families of subspaces.

[Crossed out: illegible letters[?].]

Uncertain reading. illegible letters

Original source details: GELFAND-P001200

Consider the integrability condition for a connection along [inserted above: these] subspaces. This is a linear condition on the curvature, agreeing with the equations written earlier in a different context.

— · —

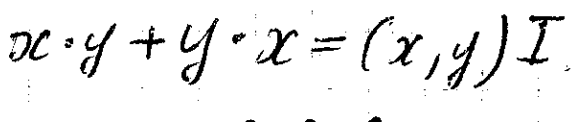
Spinor representation.

$$e_1, \ldots, e_{10}: \qquad e_i e_k + e_k e_i = 2\delta_{ik} I$$

and the ring they generate

[At right, beyond a double vertical stroke:]

$$x \cdot y + y \cdot x = (x, y) I$$



Uncertain reading. In the formula to the right of the stroke, $x \cdot y + y \cdot x = (x, y) I$ is written without a factor of 2.

Original source details: GELFAND-P001200

Dim. of the ring: $e_1, \dots, e_{10}$; $\quad e_i e_k, \quad e_i e_j e_k \qquad (e_i^2 = I)$

In all dim $= 2^{10}$; it is simple (— isomorphic to a matrix ring).

[Under "matrix":] $2^5 \times 2^5$

Obtain a representation of dimension 32.

$A \in SO(10)$ preserves the formulas $\Longrightarrow$ an automorphism of the matrix ring, and it is inner $\Longrightarrow$

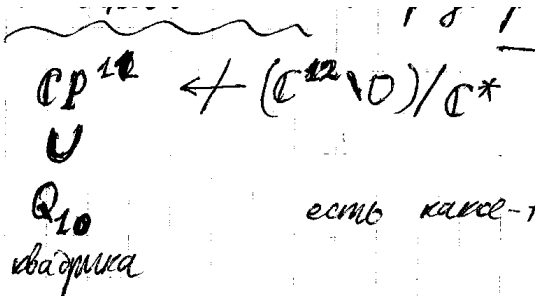
$\Longrightarrow$ a representation arises.

— · —

Curvature on the subspaces $= 0 \Longleftrightarrow$ generalized self-dual Yang–Mills equation.

Ordinary Y.-M.: dimensions 4 and 2.

— · —



Original source details: GELFAND-P001200

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{CP}^{12} & \longleftarrow & (\mathbb{C}^{12} \setminus 0)/\mathbb{C}^* \\ \cup & & \\ Q_{10} & & \end{array}$$

quadric

The final exponent digit on $\mathbb{CP}$ is overwritten. The reading 12 is tentative, possibly 11. A diagonal stroke crosses the arrow.

[To the right of Q_{10}:] there is some covering superspace [source abbr.][?].

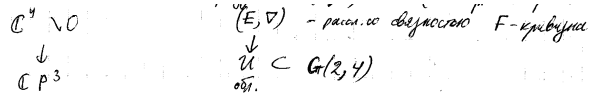
Uncertain reading. superspace [source abbr.]

Original source details: GELFAND-P001200

— · —

Radon–Penrose transform.

$\mathbb{CP}^3$; lines in $\mathbb{CP}^3$ form $G(2,4)$ (2-dimensional subspaces of $\mathbb{C}^4$)



Original source details: GELFAND-P001200

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^4 \setminus 0 & & (E, \nabla) \\ \downarrow & & \downarrow \\ \mathbb{CP}^3 & & U \subset G(2,4) \end{array}$$

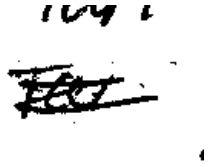
— bundle with connection

F — curvature

region

α-planes in T_xU: there is a quadratic cone defined by the inner product.

In it [inserted above the line: 2] families of 2-dimensional [crossed out: illegible letters?]



Uncertain reading. illegible letters

Original source details: GELFAND-P001200

subspaces. Choose one of the families.

$F|_{\Pi_\alpha} = 0 \Longleftrightarrow$ the bundle is called self-dual.

$G(2,4)$ has rank 2 (first after spheres). To match a pair of points one needs to specify 2 numbers (one on a sphere).

Curvature $= 0 \Longrightarrow$ self-dual Y.-M. equation.

$\rightsquigarrow$ another bundle over $\mathbb{CP}^3$; α-surface — integral subset; the bundle is trivialized $\Longrightarrow$ it suffices to take the fibre over 1 point.

[At right, beyond a double vertical stroke:] Now a bundle without connection on another set.

GELFAND-P001201

V.I. Arnold. On [inserted above: (some)] special functions arising in bifurcation theory.

H — a polynomial

$H(x,y) = c$ — a family of algebraic curves (real)

$\omega = P\,dx + Q\,dy$ — a form (P, Q — polynomials)

$$\oint_{H=c} \omega = I(c)$$

How many zeros are there for the integral $I(c)$?

In one special case this is solved completely.

Theorem (German Petrov-Tan'kin?)

Uncertain reading. German Petrov-Tan'kin

Original source details: GELFAND-P001201

5th-year student, expelled? from the university

Uncertain reading. expelled

Original source details: GELFAND-P001201

$$H = \frac{y^2}{2} + \underset{\text{a polynomial}}{P_3(x)}$$

c varies in the interval with ovals?

Uncertain reading. ovals

Original source details: GELFAND-P001201

$\deg P \le n, \quad \deg Q \le n.$

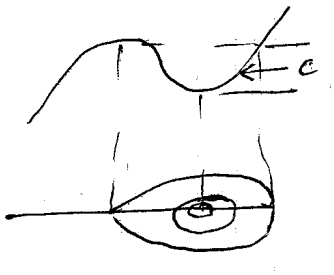
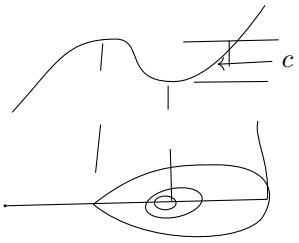


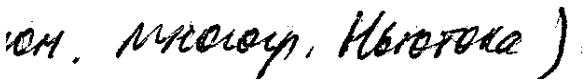
Diagram and typed description.



The thin vertical strokes between the curves are broken in places.

Original source details: GELFAND-P001201

$\dim\{I\} = n$ (count using the Newton polyh.?)



Uncertain reading. Newton polyh.

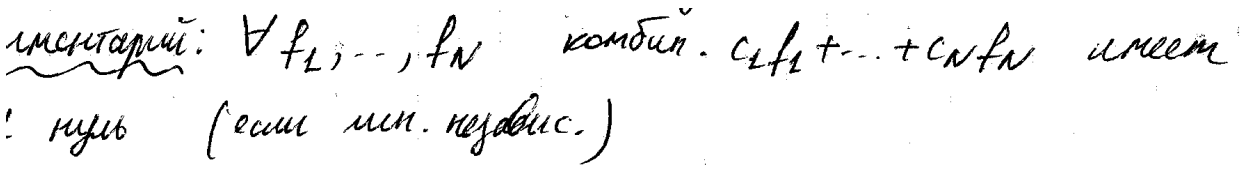
Original source details: GELFAND-P001201

(the integrals are functions of c. They form a vector space since one can add the forms ω)

(!) Assertion: number of zeros $\overset{\text{of any function considered}}{<} n$ (if it is $\neq 0$)

the functions satisfy a Picard–Fuchs equation (Gauss–Manin connection)

Comment: $\forall f_1, \dots, f_N$ the combination $c_1 f_1 + \dots + c_N f_N$ has $N-1$ zeros (if linearly independent)

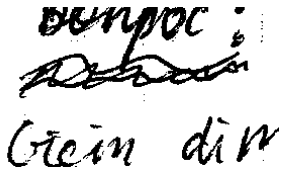


Original source details: GELFAND-P001201

Uncertain reading. The wording reads “has $N-1$ zeros”. No inequality sign is visible here.

If the number of zeros is $<$ dim, the system is called Chebyshev (nonoscillation)

Question: relation to oscillation theorems?



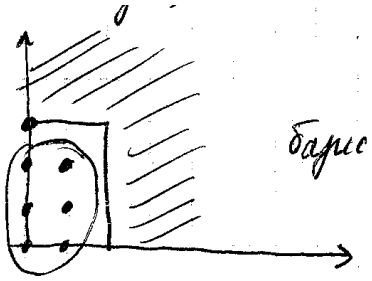
Uncertain reading. A short line beneath “Question” is hatched over. No letters can be read with certainty.

Original source details: GELFAND-P001201

— · —

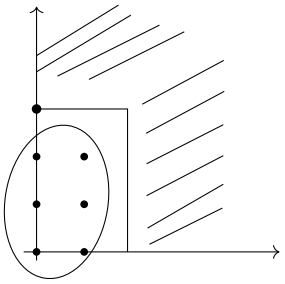
Counting dim: $\int = 0$ if a) an exact differential
b) multiply something by $H - c$.
Using the Newton polyhedron to compute dimensions

$$f = x^3 + y^4; \qquad \mathbb{C}[[x,y]]/(\tfrac{\partial f}{\partial x}, \tfrac{\partial f}{\partial y})$$

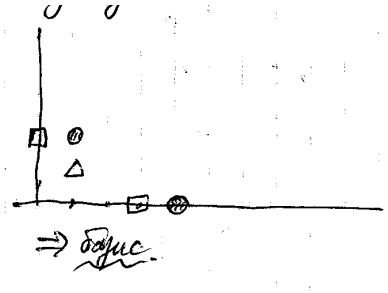


Original source details: GELFAND-P001201

Diagram and typed description.

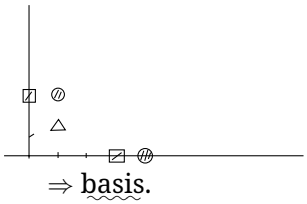


basis of the quotient space. dim $= 6$
$x^2y + y^4$



Original source details: GELFAND-P001201

Diagram and typed description.



The interior strokes in the squares and circles are schematic. Their reading remains uncertain.

GELFAND-P001202

[source number: 60]

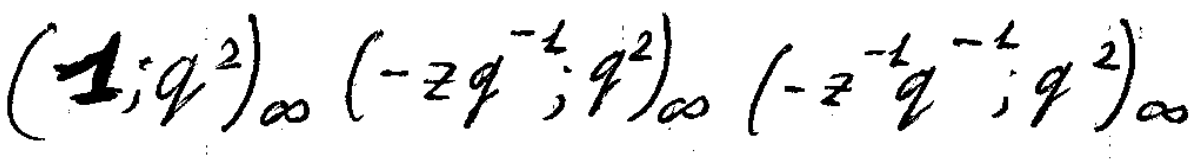
A. Zelevinsky. On “the most beautiful result in pure math.”
(Jacobi triple product)

$$(1) \qquad \sum_{r=-\infty}^{\infty} z^r q^{r^2} = \prod_{k=1}^{\infty} (1-q^{2k})(1+q^{2k-1}z)(1+q^{2k-1}z^{-1})$$

(Jacobi triple product)

$$\|$$

$$(1;q^2)_\infty (-zq^{-1};q^2)_\infty (-z^{-1}q^{-1};q^2)_\infty$$



Original source details: GELFAND-P001202

Uncertain reading. The first factor has 1 before the semicolon. The next two factors have q^{-1}.

Here

$$(a;q)_\infty := \prod_{n=0}^{\infty}(1-aq^n)$$

!! Follows from Ramanujan's identity.

Combinatorial proof

Move the first product to the left-hand side

$$(2)\qquad \left(\sum_{r=-\infty}^{\infty} z^r q^{r^2}\right)\prod_{k=1}^{\infty}(1-q^{2k})^{-1} = \prod_{k=1}^{\infty}(1+q^{2k-1}z)(1+q^{2k-1}z^{-1})$$

Uncertain reading. A mark above z^r on the left of (2) is heavily crossed out.

Original source details: GELFAND-P001202

Only the coefficient at the following type of monomial is nonzero $z^r q^{r^2+2n}$

(on the left and on the right)

$=$ number of partitions of n into summands $= p(n)$

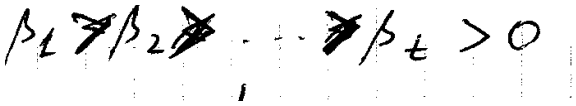
(on the left)

(3) On the right: exponents

$$\alpha_1 > \alpha_2 > \cdots > \alpha_s > 0$$

odd

$$\beta_1 > \beta_2 > \cdots > \beta_t > 0$$

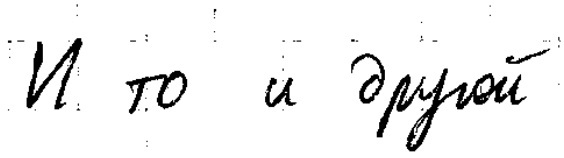


Uncertain reading. The three comparisons between the β terms are heavily retraced and read as $>$.

Original source details: GELFAND-P001202

$$\sum \alpha_j + \sum \beta_i = r^2 + 2n, \qquad s - t = r$$

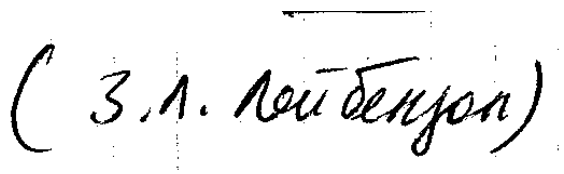
[Two diverging strokes from "odd" point toward the α and β lines. Vertical strokes flank the lines.]



Uncertain reading. Both the one and the other [irregular phrasing in the manuscript]

Original source details: GELFAND-P001202

(Z.L. Leibenzon[?]) — combinatorial description of the set:



Uncertain reading. Z.L. Leibenzon

Original source details: GELFAND-P001202

$00000\,1\,0000\,1\,00\,111111$

infinite words (linkages)

[Crossed out: on the left[?]]

Uncertain reading. on the left

Original source details: GELFAND-P001202

first infinitely many 0; then a jumble; then strokes

(infinitely many)

Height: $h(A)$ — number of inversions $\underbrace{1\,0}$ — arbitrarily far apart

$h(A) = n \quad \Rightarrow$ number of elements $p(n)$

— parametrized by tuples of the form (3).

1st method

$0000\,1\,0\,111$ — height 1

$p(n)$ — take the leftmost stroke and count zeros after it, and so on.

2nd method

The gaps have a [above "number": (natural)] number: (number of 1 on the left) $-$ (number of 0 on the right)

The number varies from $-\infty$ to $+\infty$!! (throughout!)

GELFAND-P001203

Arnold's remark:

Let us draw:

0 — a step to the right

1 — a step up

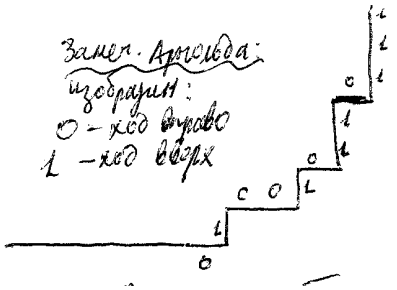
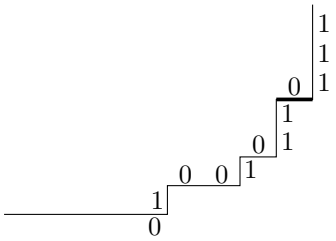


Diagram and typed description.



Original source details: GELFAND-P001203

Young diagram

the function $x - y$ numbers all

Consider the gap numbered r.

We characterize the linkage as follows:

the distance from the gap to the sticks, *[beneath the word, with a brace: lying to the left]* || odd numbers.

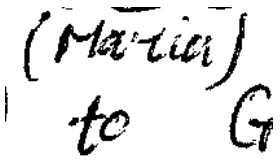
— " — the circles. *[beneath the word, with a brace: lying to the right]*

$$\sum \alpha_j + \sum \beta_j = r^2 + 2n$$

from the area under the curve.

16/IX–85 Gelfand Seminar.

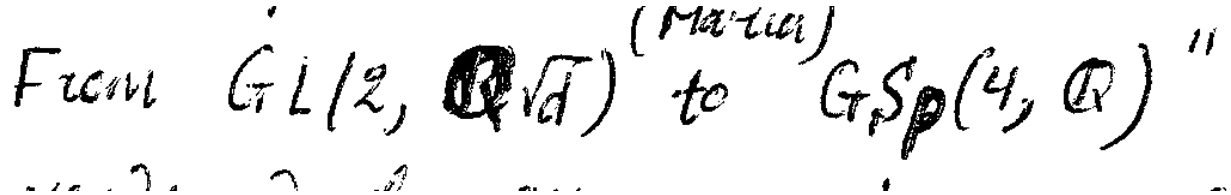
Zelevinsky (and Beilinson) as reported by M.-F. Vigneras (Marie-France Vigneras)



Original source details: GELFAND-P001203

Uncertain reading. Parenthetical note below the start of the name, provisionally read as "Ma-rie."

1) on the work "From $\mathrm{GL}(2, \mathbb{Q}\sqrt{d})$ to $\mathrm{GSp}(4, \mathbb{Q})$" (M.-F. Vigneras)



Original source details: GELFAND-P001203

Uncertain reading. The first letter in the field notation is overwritten and is read as $\mathbb{Q}\sqrt{d}$.

Construction of a correspondence between representations.

(lifting)

On passing to a p-adic field

this makes it possible to verify for $\mathrm{GSp}(4, F)$ some Langlands conjecture on functoriality.

[At right, with a pointer to GSp*:]* symplectic similitudes (preserve the symplectic form up to a scalar)

[Below F:] p-adic

2) Rallis, Piatetski-Shapiro and others

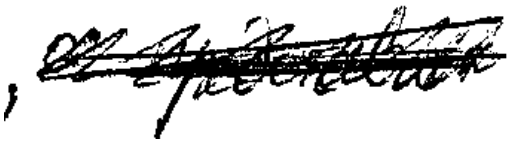
$$L(s, \pi_1 \times \pi_2 \times \pi_3), \quad \text{where } \pi_j \text{ — representations of } \mathrm{GL}_2(F)$$

3) According to Jacket "How to use the trace formula?"

Original source details: GELFAND-P001203

Uncertain reading. The surname in the manuscript reads "Jacket."

The latest important result is the Paley–Wiener theorem,



Original source details: GELFAND-P001203

Uncertain reading. Crossed out after the theorem name: “on ....”

proved by Kazhdan and Bernstein.

Original source details: GELFAND-P001203

Uncertain reading. The abbreviation at the start of the line is read as “latest.”

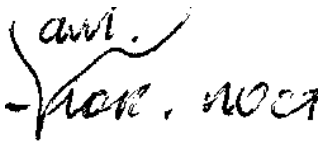
[In an oval linked to the name of the result:] on representations over F

G — a reductive p-adic group (e.g. GL_n)

$H^\bullet(G)$ — locally constant, compactly supported functions on G.

Original source details: GELFAND-P001203

Uncertain reading. The raised mark after H is read as a dot.



Original source details: GELFAND-P001203

Uncertain reading. “Alg.” is inserted above the dash, with a curved mark linking the insertion to the line.

[To the right of the vertical stroke:] locally constant functions take the place of C^∞ functions

From f we construct:

$$\pi \longrightarrow \operatorname{tr} \pi(f)$$

— a function on $\operatorname{Irr} G$

Here

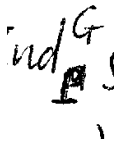
$$\pi(f) = \int_G f(g)\pi(g)\,dg$$

The theorem (Paley–Wiener, according to K.-B.) describes all such functions

Irreducible representations divide into sheets and families: from $\rho \in \operatorname{Irr} M$ one constructs $M \subset G$

$$\pi \in \mathrm{JH}\ \operatorname{Ind}_{\underline{P}}^{G} \rho$$

(Jordan–Hölder series)



Original source details: GELFAND-P001203

Uncertain reading. The overwritten lower index on induction is read as an underlined P.

‖ each family acquires a structure

M — the Levi subgroup of a parabolic subgroup of G

(semisimple part)

character of an irreducible representation $\Leftrightarrow$ algebraicity of a certain function

Idea in the real *[below "real," circled: and complex?]* case: decay on $\forall$ Cartan subalgebra

Uncertain reading. The abbreviation is read as "decay."

Original source details: GELFAND-P001203

GELFAND-P001204

[source number: 61]

M.-F. Vigneras. (on the same topic, but more precisely)

Uncertain reading. The last word of the note is read as the abbreviation for "precisely."

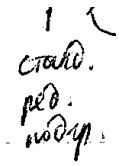
Original source details: GELFAND-P001204

G reductive p-adic group (e.g., GL_n)

$$\coprod_{(M,\rho)} \{\rho\chi\}^W = \operatorname{Irr} G \tag{1}$$

[Labels with pointers:] M — standard reductive subgroup, ρ — irreducible cuspidal representation, χ — unramified character, W — up to the action of the Weyl group.

Uncertain reading. The first line of the label for M is read as the abbreviation for "standard."



Original source details: GELFAND-P001204

$\{\rho\chi\}^W$ — finite-dimensional algebraic set (over $\mathbb{C}$).

from $f \in H(G)$ one constructs a function on [inserted: the set (1)]

$$\pi \longrightarrow \operatorname{trace} \operatorname{ind}_M^G \rho\chi(f)$$

Conditions

a) algebraic on $\forall$ such set

b) supported on finitely many of them

Beilinson.

Uncertain reading. The words below the heading are crossed out and cannot be read with certainty.

Original source details: GELFAND-P001204

Representations of $\mathrm{GSp}(4,\mathbb{Q})$ — ?

$$W_{\mathbb{Q}_p} \subset \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$$

$$\mathrm{Irr}\,\mathrm{GSp}(4,\mathbb{Q}) \longrightarrow \text{representations of the Weil–Deligne group}$$

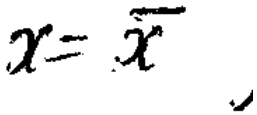
Uncertain reading. Unresolved crossed-out mark after the arrow, before "representations."

Original source details: GELFAND-P001204

[With a pointer to the right-hand side of the arrow:] the inverse image of $\forall$ point is finite (L packet)

$$\mathrm{GL}(2,\mathbb{Q}(\sqrt{d})) \longrightarrow \mathrm{GO}(4)$$

$$\left\{x \in M(2,\mathbb{Q}(\sqrt{d})) : x = \bar{x}\right\} \qquad x = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$$



Uncertain reading. The condition reads $x = \bar{x}$, with no visible transpose sign.

Original source details: GELFAND-P001204

det — canonical quadratic form (it is rational)

$$x \longmapsto gx\bar{g}^t \qquad \text{preserves this condition}$$

(analogue of the spinor representation)

Weil representation: representation of $\mathrm{GO}(4) \times \mathrm{GSp}(4)$

it is realized in θ functions

B. Feigin. On strings

Uncertain reading. The crossed-out word before "String theory" remains unresolved.

Original source details: GELFAND-P001204

String theory is the statistics of surfaces.

Quantum mechanics studies a space with a Lagrangian. The action depends on the trajectory.

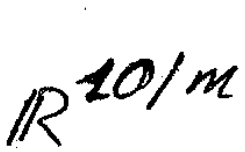
Strings: a surface takes the place of a trajectory.

$\mathcal{E}$ — a Riemann surface; consider the space $\mathbb{R}^d$

A trajectory is a map

$$\{\mathcal{E} \underset{x_1,\ldots,x_d}{\longrightarrow} \mathbb{R}^d\}.$$

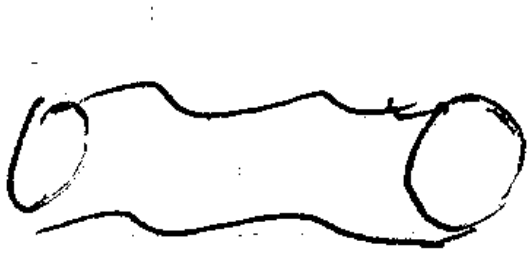
Idea: all elementary particles are described by maps from $\mathcal{E}$ to the superspace $\mathbb{R}^{10/m}$



Original source details: GELFAND-P001204

Uncertain reading. The exponent shows a slash and m: $10/m$.

The motion of a circle gives a surface



Original source details: GELFAND-P001204

Diagram and typed description.

The classical analogue is crystal growth.

GELFAND-P001205

Lagrangian:

$$\mathcal{L} = \int [\text{crossed out}]\,(\det g_{ij})^{1/2}\,d\sigma\,d\tau \qquad \text{(Nambu Lagrangian)} \tag{1}$$

Original source details: GELFAND-P001205

Uncertain reading. The passage crossed out immediately after the integral sign remains unresolved.

(area in the induced metric)

We need to consider the Feynman integral.

$$X : \mathcal{E} \hookrightarrow \mathbb{R}^d$$

Another way to write (1):

$$\mathcal{L}(X) = \int d\sigma\, d\tau \left[\left(\frac{\partial X}{\partial \sigma}\right)^2 \left(\frac{\partial X}{\partial \tau}\right)^2 - \left(\frac{\partial X}{\partial \sigma} \cdot \frac{\partial X}{\partial \tau}\right)^2 \right]^{1/2} \tag{1'}$$

Feynman integral

$$\int e^{\beta \mathcal{L}(X)}\, \mathcal{D}X \tag{2}$$

There are two approaches: via integrals and via ordinary operators (!). A correspondence has now been established.

Polyakov: another continuum integral

$\widetilde{X}$ — [inserted: a space] whose elements are pairs

1) $X : \mathcal{E} \longrightarrow \mathbb{R}^d$

2) g_{ij} on $\mathcal{E}$ (a Riemannian metric on $\mathcal{E}$)

Polyakov Lagrangian:

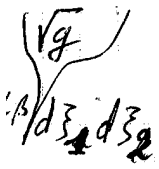
$$\text{[crossed out]}\ \mathcal{L} = \text{æ} \int (\det g_{ij})^{1/2} + \int \frac{\partial X_i}{\partial \xi_\alpha} \frac{\partial X_i}{\partial \xi_\beta} g^{\alpha\beta} \overset{\text{[inserted]}}{\sqrt{g}}\, d\xi_1\, d\xi_2 \tag{3}$$

Original source details: GELFAND-P001205

Uncertain reading. The crossed-out mark before $\mathcal{L}$ remains unresolved.

Original source details: GELFAND-P001205

Uncertain reading. The coefficient before the first integral resembles “æ.”



Original source details: GELFAND-P001205

Uncertain reading. $\sqrt{g}$ is inserted above the line. The overwritten subscripts in $d\xi_1\, d\xi_2$ are read as $1, 2$.

[Inserted: To it] the calculation technique applies

Procedure: fix g_{ij} and evaluate the integral over X — it is quadratic.

— · —

GELFAND-P001206

[source number: 62]

23.9.85 Gelfand Seminar.

I.M.G. Problem (for grade 9) What polyhedron results from cutting a 4-dimensional cube by a 3-dimensional hyperplane passing through the center and perpendicular to the main diagonal?

A. Beilinson. On a new series of papers by Piatetski-Shapiro and Rallis.

functional equations of L-functions associated with automorphic forms

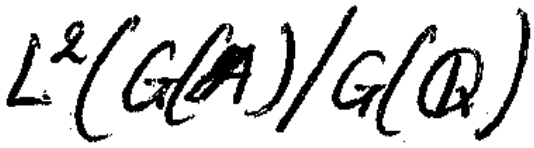
G — a semisimple group over $\mathbb{Q}$ (e.g., the group of matrices $n \times n$)

$G/\mathbb{Q}$.

Uncertain reading. The signs to the right of $G/\mathbb{Q}$ are heavily crossed out.

Original source details: GELFAND-P001206

Problem: describe the representations of the adelic group $G(A)$ in $L^2(G(A)/G(\mathbb{Q})$



Uncertain reading. The notation for the L^2 space has one closing parenthesis after $\mathbb{Q}$.

Original source details: GELFAND-P001206

The corresponding representations are called automorphic.

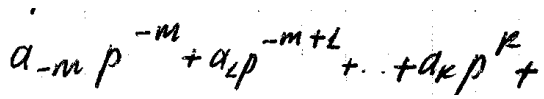
— · —

I.M.G. $\forall$ natural number can be written in the form

$$a_0 + a_1 p + a_2 p^2 + \ldots + a_n p^n$$

Any rational number

$$a_{-m} p^{-m} + a_1 p^{-m+1} + \ldots + a_k p^k +$$



Uncertain reading. The second term reads $a_1 p^{-m+1}$. A $+$ follows $a_k p^k$.

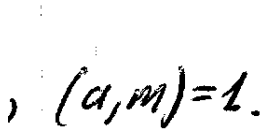
Original source details: GELFAND-P001206

For example,

$$\frac{1}{3} = 1 + 1 \cdot 2 + 1 \cdot 2^3 + 1 \cdot 2^5 + \dots$$

[At right:]

$$\|a\| = p^{-n}, \quad \text{if } a = p^n m, \quad (a, m) = 1.$$

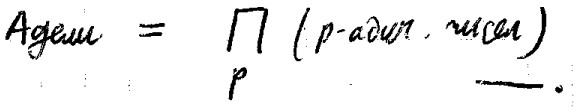


Uncertain reading. The coprimality condition reads $(a, m) = 1$.

Original source details: GELFAND-P001206

$$\underline{p = 2} \quad \Rightarrow \quad \|2\| = \frac{1}{2}; \quad \|3\| = 1, \quad \|4\| = \frac{1}{4}, \quad \left\|\frac{1}{2}\right\| = 2.$$

$$\text{Adeles} = \prod_p (p\text{-adic numbers})$$



Uncertain reading. The p appears below the product sign.

Original source details: GELFAND-P001206

— · —

$$\mathrm{SL}_2(\mathbb{R}) \supset \mathrm{SL}_2(\mathbb{Z}) \supset \Gamma_n$$

[Label pointing to Γ_n:] matrices of order 2, [crossed out: elem.?] that are congruent to 1 (matrix) mod$n$.

Uncertain reading. The abbreviation crossed out after "order" may read "elem."

Original source details: GELFAND-P001206

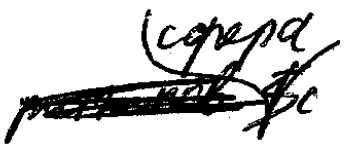
Uncertain reading. The ending of the word "congruent" is overwritten.

Original source details: GELFAND-P001206

$$\Gamma_{n_1} \supset \Gamma_{n_2} \quad \text{when } n_1 \mid n_2$$

$$G(A)/G(\mathbb{Q}) = \varprojlim_n G(\mathbb{R})/\Gamma_n.$$

Γ_n acts on the upper half-plane H; H/Γ_n — [crossed out: Riemann?] [inserted: a sphere] (with punctures)



Uncertain reading. The first crossed-out word may be "Riemann." "A sphere" is inserted above the line.

Original source details: GELFAND-P001206

$H/\mathrm{SL}_2(\mathbb{Z})$ — a sphere with two punctures

H/Γ_2 — — ″ — three punctures

Original source details: GELFAND-P001206

Uncertain reading. The first line says “a sphere with two punctures.”

$$\pi_{G(A)} = \pi_{G(\mathbb{R})} \otimes \pi_{G(\mathbb{Q}_2)} \otimes \dots$$

— · —

I.M.G. At Sem. Bourbaki there is a paper by A. Connes on the relation between the topology of knots and invariants of factors!!

Uncertain reading. The abbreviation before “knots” reads “top.”

Original source details: GELFAND-P001206

— · —

It is enough to know the representations of the p-adic group.

Uncertain reading. The crossed-out mark at the end of the line remains unresolved.

Original source details: GELFAND-P001206

Which occur in $L^2(G(A)/G(\mathbb{Q}))$? Far from all of them.

Representations of $G(\mathbb{Q}_p)$.

The simplest are spherical, when there is a vector invariant under a [inserted: max] compact subgroup.

$$G(\mathbb{Q}_p)/B(\mathbb{Q}_p)$$

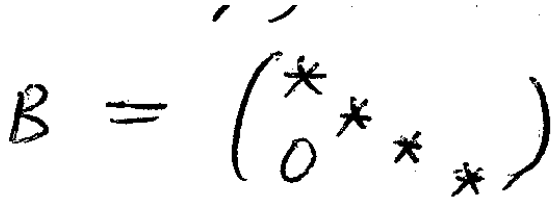
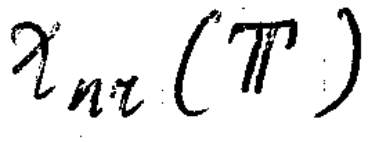


Diagram and typed description.

$$B = \begin{pmatrix} * & * & * \\ 0 & * & \\ & & * \end{pmatrix}$$

Original source details: GELFAND-P001206

are indexed by unramified characters of the torus $\chi_{nr}(T)$ = points of the dual torus $= T^0(\mathbb{C})$ (complex points)



Uncertain reading. The subscript on χ is read as nr.

Original source details: GELFAND-P001207

(trivial on the intersection of the torus with the maximal compact subgroup)

Spherical representations of $G(\mathbb{Q}_p) = T^0(\mathbb{C})/W$

unramified character $\longrightarrow$ conjugacy class in the matrix group

— · —

Let G — a split group (the maximal torus splits)

G^0 — dual group $\quad (G = \mathrm{GL} \Rightarrow G^0 = \mathrm{GL})$

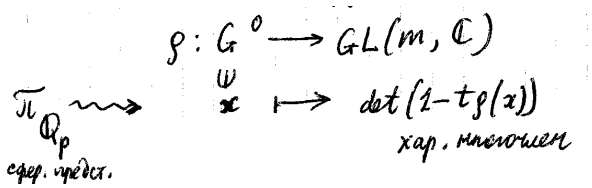


Diagram and typed description.

$$\begin{array}{ccccc} & & \rho: G^0 & \longrightarrow & \mathrm{GL}(m,\mathbb{C}) \\ & & \cup & & \\ \pi_{\mathbb{Q}_p} & \rightsquigarrow & x & \longmapsto & \det(1-t\rho(x)) \\ \text{spherical rep.} & & & & \text{characteristic polynomial} \end{array}$$

Original source details: GELFAND-P001207

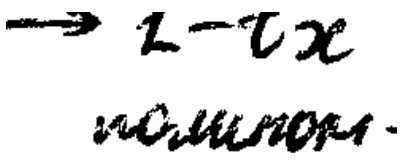
Suppose that for $\pi_{G(A)}$ almost all representations [Inserted above the line: (of the local groups $G(\mathbb{Q}_p)$)] are spherical (all but finitely many)

$$L(\pi,\rho,s) = \prod_p \frac{1}{\det\left(1 - p^{-s}\rho\left(\begin{array}{c} x_{\pi_{\mathbb{Q}_p}} \\ | \\ \text{semisimple element} \end{array}\right)\right)}$$

— · —

Example.

$$\rho : \mathrm{GL}_1 \underset{\text{identity}}{\longrightarrow} \mathrm{GL}_1 \quad \Rightarrow \quad x \longmapsto \underset{\text{polynomial}}{1-tx}$$



Uncertain reading. The abbreviation below $1-tx$ is read as “polynomial.”

Original source details: GELFAND-P001207

Uncertain reading. The crossed-out mark before $x = 1$ remains unresolved.

Original source details: GELFAND-P001207

$x = 1 \Rightarrow L(\pi, \rho, s) = \prod_p \frac{1}{1-p^{-s}}$ — the Riemann ζ-function.

Feigin B. On strings (continued).

$\mathcal{E}$ — a Riemann surface (a 2-dimensional set); $\mathbb{R}^d$ — a space with the Minkowski metric $\langle\,,\,\rangle$ (1 negative square) or the Euclidean metric.

Consider the space [Inserted above the line: X] of smooth maps $\mathcal{E} \xrightarrow{x} \mathbb{R}^d$

[Inserted above the line: The map x induces] on $\mathcal{E}$ [crossed out] a Riemannian metric.

Uncertain reading. The crossed-out word after $\mathcal{E}$ may start with the Russian letters “зад...”.

Original source details: GELFAND-P001207

Let ξ_1, ξ_2 — coordinates [crossed out] on $\mathcal{E}$.

Uncertain reading. The crossed-out writing between “coordinates” and “on” remains unresolved.

Original source details: GELFAND-P001207

$$g_{ij} = \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j}\right) \qquad \{g^{ij}\} = \{g_{ij}\}^{-1} \tag{1}$$

Nambu Lagrangian:

$$\mathcal{L}(x) = \int_{\mathcal{E}} \omega = \int_{\mathcal{E}} \sqrt{g}\, d\xi_1\, d\xi_2, \qquad g = \det\{g_{ij}\} \tag{2}$$

(area of the surface in the Riemannian metric)

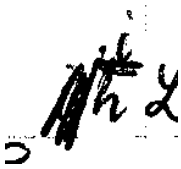
GELFAND-P001208

[source number: 63]

Path integral

$$\int e^{\frac{i}{h}\mathcal{L}(x)} \mathcal{D}x \tag{3}$$

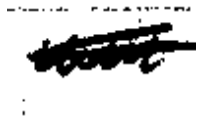
Uncertain reading. The overwritten coefficient in the exponent is read as i/h. The crossed-out mark on its left remains unresolved.



Original source details: GELFAND-P001208

Over all surfaces passing through given points [crossed out] etc. (to be specified below).

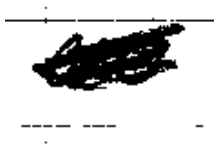
Uncertain reading. The word crossed out after "points" remains unresolved.



Original source details: GELFAND-P001208

We want to find the [crossed out] min of the Lagrangian $\mathcal{L}$.

Uncertain reading. The crossed-out mark before min remains unresolved.



Original source details: GELFAND-P001208

This is a minimal surface.

Properties of minimal surfaces.

a) $\mathcal{L}$ is independent of the choice of coordinates.

Variation of $\mathcal{L}$.

Let there be a 1-parameter family $x(t)$ of such surfaces.

$$\left.\frac{\partial \mathcal{L}}{\partial t}\right|_{t=0} = -\int (\dot{x}_t, H)\sqrt{g}\, d\xi_1\, d\xi_2. \tag{4}$$

Here

$$H = \Delta x = \frac{1}{\sqrt{g}} \sum_i \frac{\partial}{\partial \xi_i} \left(\sqrt{g} \sum_j g^{ij} \frac{\partial x}{\partial \xi_j} \right)$$

(the Laplace operator of the metric)

(5) Corollary:

if $x : \mathcal{E} \longrightarrow \mathbb{R}^d$ is given by harmonic functions, the surface is minimal.

— · —

I.M.G. Eells studies minimal surfaces.

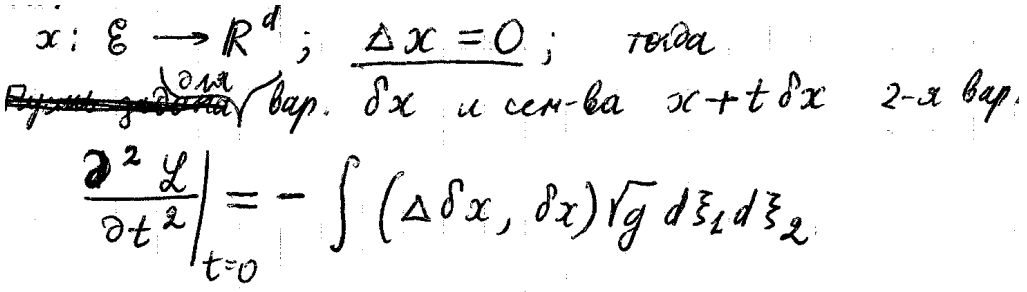
Lashof — " ——; which are constructed from groups.

Let;

$$x : \mathcal{E} \longrightarrow \mathbb{R}^d; \qquad \underline{\Delta x = 0;} \qquad \text{then} \tag{6}$$

[crossed out: Let there be] [Inserted above the line: For] a variation δx and the family $x + t\delta x$ the second variation takes the form

$$\left.\frac{\partial^2 \mathcal{L}}{\partial t^2}\right|_{t=0} = -\int (\Delta\delta x, \delta x)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{7}$$

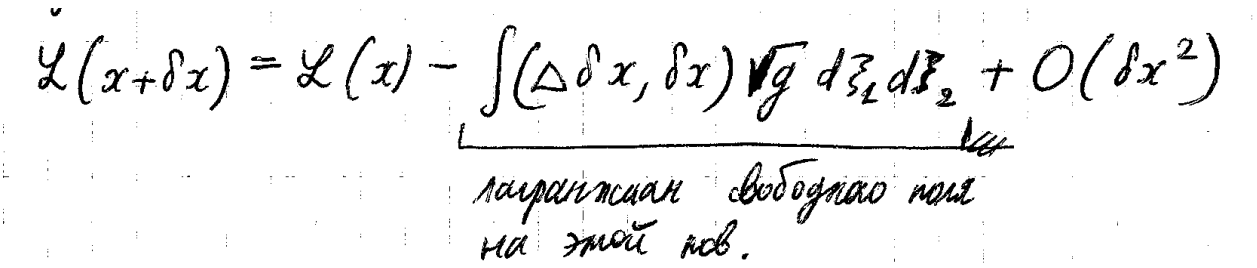


Original source details: GELFAND-P001208

Uncertain reading. The ∂ in the numerator is overwritten. The condition $\Delta x = 0$ appears above in (6).

We then obtain

$$\mathcal{L}(x + \delta x) = \mathcal{L}(x) - \underbrace{\int (\Delta\delta x, \delta x)\sqrt{g}\, d\xi_1\, d\xi_2}_{\substack{\text{Lagrangian of the free field} \\ \text{on this surface}}} + O(\delta x^2) \tag{8}$$



Original source details: GELFAND-P001208

Uncertain reading. The manuscript shows no factor $1/2$ before the integral and writes the remainder $O(\delta x^2)$.

Now the metric g_{ij} is fixed on $\mathcal{E}$. Replacing δx with x, we write the free-field Lagrangian in d form.

Uncertain reading. The mark before the Russian word for “form” resembles d.

Original source details: GELFAND-P001208

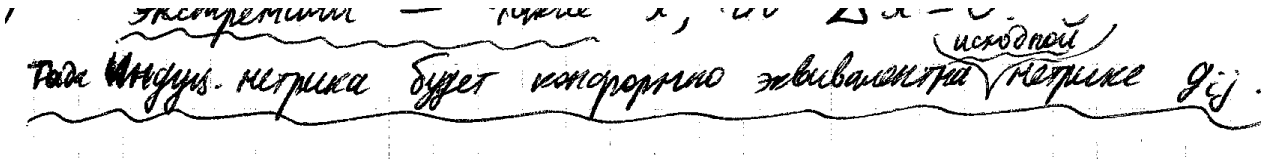
$$\mathcal{L} = \int \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{9}$$

(10) The extremals are those x, for which $\Delta x = 0$.

(11) then the induced metric will be conformally equivalent to the [Inserted above the line: original] metric g_{ij}.

Uncertain reading. The underlined line begins with “Then.”

Original source details: GELFAND-P001208



Original source details: GELFAND-P001208

Uncertain reading. “Original” is inserted above the line, before “metric.”

GELFAND-P001209

Let us rewrite this:

$$\mathcal{L}(x, g_{ij}) = \int \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{9'}$$

Consider the path integral

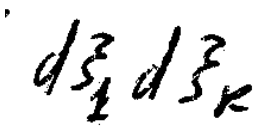
$$\int e^{ih^{-1}\mathcal{L}(x, g_{ij})} \mathcal{D}x\, \mathcal{D}g_{ij}. \tag{12}$$

Now x and g_{ij} are varied independently.

Let $\mathcal{D}$ — be the space of metrics. We give it a Riemannian metric

$$g_{ij} \longmapsto g_{ij} + \theta_{ij}$$

$$|\theta_{ij}|^2 = \int \theta_{ij} \theta_{k\ell} g^{ik} g^{j\ell} \sqrt{g}\, d\xi_1\, d\xi_k \tag{13}$$



Uncertain reading. The last differential in (13) reads $d\xi_k$.

Original source details: GELFAND-P001209

In (12) $\mathcal{D}g_{ij}$ — the (formal) volume form constructed from the metric (13).

The integral (12) [Inserted above the line: (undefined, since the integrand)] is invariant under the diffeomorphism group of the surface itself $\mathcal{E}$

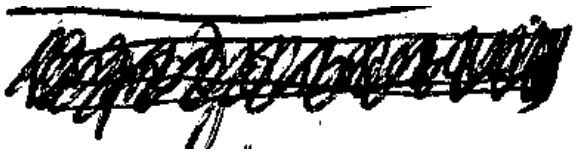
Uncertain reading. The surface symbol is read as $\mathcal{E}$, as on the preceding pages.

Original source details: GELFAND-P001209

$\Rightarrow$ constant on the orbits and undefined. One needs to use the Faddeev–Popov trick.

GELFAND-P001210

30/IX–85. B. Feigin (Gelfand Seminar)



Original source details: GELFAND-P001210

Undeciphered passage. The crossed-out line below the speaker's name remains undeciphered.

On Polyakov's work.

[I arrived $\frac{1}{2}$ an hour late]

Tools for calculating det.

$\mathcal{E}$ — a surface with a complex structure and a metric g_{ij}, compatible with the complex structure *[below the word "surface": (2-dimensional)]*.

T — the tangent bundle on $\mathcal{E}$, T^j — its jth tensor power

$$T^j \xrightarrow{\bar\partial} T^j \otimes \Omega^{0,1}, \qquad \Delta = \bar\partial^* \bar\partial$$

$$\zeta(s) = \sum \lambda_i^{-s}, \qquad \ln \operatorname{Det} \Delta = -\zeta'(0).$$

Det depends on g_{ij}

$$g_{ij} \rightsquigarrow g_{ij}(1+\varepsilon\rho) \qquad \text{(conformal change)}$$

$$\operatorname{Det}(\Delta + \varepsilon\delta\Delta) = \operatorname{Det}\Delta \cdot \operatorname{Det}(1 + \varepsilon\delta\Delta/\Delta)$$

$$\delta(\ln \operatorname{Det}\Delta) = \operatorname{Tr}(\delta\Delta/\Delta)$$

$\bar\partial$ does not depend on ρ; $\bar\partial^*$ depends on ρ

$$\begin{aligned}\Delta + \varepsilon\delta\Delta &= (1+\varepsilon\rho)^j \bar\partial^* (1+\varepsilon\rho)^{-j-1} \bar\partial \\ &= \bar\partial^*\bar\partial + \varepsilon \underbrace{\left(j\rho\bar\partial^*\bar\partial - (j+1)\bar\partial^*\rho\bar\partial\right)}_{\delta\Delta}\end{aligned}$$

Divide the second term by Δ.

$$\bar\partial^*\rho\bar\partial\bar\partial^{-1}(\bar\partial^*)^{-1} = \bar\partial^*\rho(\bar\partial^*)^{-1}$$

ρ is replaced by $\rho e^{-t\Delta}$ and we write $\operatorname{Tr}(\rho e^{-t\Delta})$ and then $t \to +0$; we have an expansion in powers of t

$$\operatorname{Tr}(\rho e^{-t\Delta}) = a_{-1}t^{-1} + a_0 + \dots \qquad \text{(2-dimensional surface)}$$

(see the standard textbooks: McKean and Singer)

Original source details: GELFAND-P001210

Uncertain reading. An undeciphered mark after the closing parenthesis is crossed out.

Original source details: GELFAND-P001210

Uncertain reading. Note below the names: "article in 'Matem.'"

a_{-1} — a linear functional of ρ; one can take the ρ, for which $a_{-1}(\rho) = 0 \Rightarrow$ for a subspace of codimension 1 it is well defined.

GELFAND-P001211

[source number: 66]

Otherwise

$$\operatorname{tr} \rho = \text{æ}(a_{-1}(\rho)) + a_0(\rho)$$

Uncertain reading. Here and below, the looped symbol most closely resembles "æ."

Original source details: GELFAND-P001211

$a_{-1}t^{-1}$ — the infinite term.

All reasonable methods lead to this

(this follows from linearity and agreement where $a_{-1}(\rho) = 0$)

æ may depend on the metric.

The definition of Det is related to the definition of the trace.

— · —

Let Δ act on functions. Then

$$\operatorname{Tr}(\rho e^{-t\Delta}) = \frac{1}{4\pi t}\left(\int \rho\sqrt{g}\, d\xi_1\, d\xi_2 + \frac{t}{3}\int \rho K\sqrt{g}\, d\xi_1\, d\xi_2 + \dots\right)$$

[Below K:] (scalar curvature).

For $\Delta_j : T^j \to T^j$ we have:

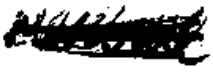
$$\operatorname{Tr}(\rho e^{-t\Delta_j}) = \frac{1}{4\pi t}\left(2\int \rho\sqrt{g}\, d\xi_1\, d\xi_2 + \left(j + \frac{2}{3}\right) t \int \rho K\sqrt{g}\, d\xi_1\, d\xi_2\right)$$

When $j = 0$ it is twice as [crossed out: small?] large, because we are in a complex bundle and take its realification.

Uncertain reading. The mark between j and 0 is overwritten and reads as equality.

Original source details: GELFAND-P001211

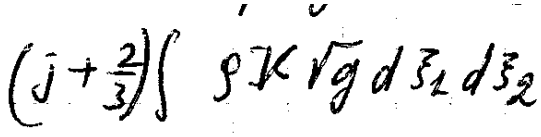
Uncertain reading. The crossed-out word before "large" may read "small."



Original source details: GELFAND-P001211

Hence the trace of the operator of multiplication by ρ equals

$$\left(j+\frac{2}{3}\right)\int \rho K\sqrt{g}\,d\xi_1\,d\xi_2$$



Uncertain reading. The factor before the integral on this line is written as $(j+2/3)$.

Original source details: GELFAND-P001211

On $T^j\otimes\Omega^{0,1}$ we need to use $\Delta=\bar\partial\bar\partial^*$. This is the same as computing with

$$T^{-j-1}\longrightarrow T^{-j-1}\otimes\Omega^{0,1}$$

$$\operatorname{tr}(\rho\cdot e^{-t\bar\partial_j\bar\partial_j^*})=\operatorname{tr}(\rho e^{-t\Delta_{-j-1}})$$

$$\operatorname{Tr}\left(\frac{\delta\Delta}{\Delta}\right)=\frac{1}{4\pi}\frac{6j^2+6j+1}{3}\int\rho K\sqrt{g}\,d\xi_1\,d\xi_2+\text{æ}\int\rho\sqrt{g}\,d\xi_1\,d\xi_2$$

If the metric is $\theta(z,\bar z)\,dz\,d\bar z$, then $K=\partial\bar\partial\ln\theta$

Let $\theta(z,\bar z)=e^{\varphi}$

This gives a linear [Inserted above the line: of the variation] dependence on $\varphi\Rightarrow$ we integrate it and obtain

Uncertain reading. Above the word "dependence," "of the variation" is inserted.

Original source details: GELFAND-P001211

$$\ln\operatorname{Det}\Delta(\theta)=\mathcal{K}+\int\left(\partial\varphi\wedge\bar\partial\varphi+\text{æ}\,e^{\varphi}\,dz\,d\bar z\right)$$

[Below $\mathcal{K}$:] some number (it depends on everything!).

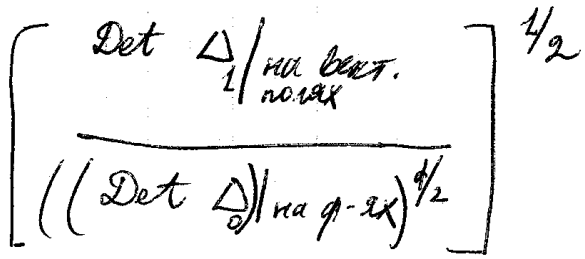
[At right:] (action of the Liouville equation).

We still need to be able to compute

$$\int e^{\mathcal{L}(\varphi)}\mathcal{D}\varphi,\qquad \mathcal{L}(\varphi)=\ln\operatorname{Det}\Delta$$

$\Leftrightarrow$ solve the Liouville equation.

$$\left[\frac{\operatorname{Det}\Delta_1|_{\text{on vector fields}}}{\left((\operatorname{Det}\Delta_0)|_{\text{on functions}}\right)^{d/2}}\right]^{1/2}$$



Uncertain reading. The denominator has exponent $d/2$, and the whole bracket has exponent $1/2$.

Original source details: GELFAND-P001212

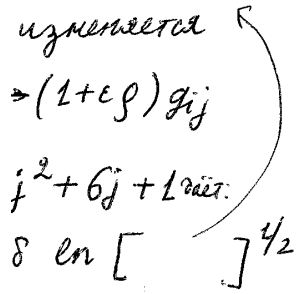
g_{ij} changes

$$g_{ij} \longrightarrow (1+\varepsilon\rho)g_{ij}$$

$6j^2+6j+1$ gives.

Hence [crossed out: ln?]

$$\delta \ln[\quad]^{1/2} = \left(13-\frac{d}{2}\right)\int \rho K\sqrt{g}\,d\xi_1\,d\xi_2$$



Uncertain reading. The brackets are empty. A curved arrow leads from them to the word “changes.”

Original source details: GELFAND-P001212

Uncertain reading. The crossed-out writing before δ may read ln.

Original source details: GELFAND-P001212

This suggests $d = 26$

(these were maps of the surface into $\mathbb{R}^d$).

Then it is independent of the metric!

(if the metric is compatible with the complex structure!)

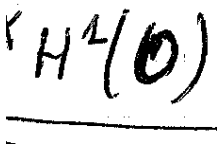
This is Polyakov's explanation of the role of 26 in string theory.

The constructions need to account for zero modes. We obtain

$$\mathcal{E} \rightsquigarrow \frac{\bigwedge^{\max} H^1(T)}{\bigwedge^{\max} H^1(O)}$$

[Below $\mathcal{E}$:] algebraic curve.

[Below the ratio, marked by a brace:] Here a canonical Hermitian metric arises.



Uncertain reading. The rounded sign in $H^1(O)$ resembles O or 0.

Original source details: GELFAND-P001212

GELFAND-P001213

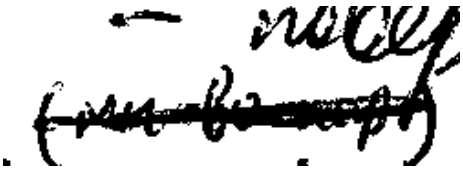
14/X–85. Gelfand Seminar.

Fuchs — recapitulation [crossed out] [inserted above the line: previous] talk by Feigin

Uncertain reading. The crossed-out mark between "recapitulation" and "talk" may be I or 1.

Original source details: GELFAND-P001213

$\mathcal{E}$ — a surface (a 2-dimensional [inserted: smooth] compact [inserted: orientable] manifold without boundary)



Uncertain reading. The crossed-out parenthesis below the surface label may read "(set of pairs)."

Original source details: GELFAND-P001213

$X = \{x : \mathcal{E} \to \mathbb{R}^d;\ g\}$; g — a Riemannian metric on $\mathcal{E}$

[Below X:] (X — the set of pairs).

$$\mathcal{L}(g, x) = \int_{\mathcal{E}} \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{1}$$

I.M.G. Strings $\leftrightarrow$ 2 constants (not only h) in quantum mechanics,

Statistics of thin tubes instead of paths.

The second constant is the thickness of the tube.

GELFAND-P001214

[source number: 79]

F. Goal: find

$$\int_X e^{\frac{i}{h}\mathcal{L}(g,x)}\mathcal{D}g\,\mathcal{D}x \tag{2}$$

Uncertain reading. The overwritten label below the integral sign reads X.

Original source details: GELFAND-P001214

g — fixed

$$\int_{\{\mathcal{E}\to\mathbb{R}^1\}} e^{-\mathcal{L}(x,g)}\mathcal{D}x = (\mathrm{Det}\,\Delta)^{-1/2}, \tag{3}$$

Uncertain reading. The overwritten exponent on $\mathbb{R}$ in (3) reads 1.

Original source details: GELFAND-P001214

Δ — the Laplace–Beltrami operator of the metric g.

[Is there an analogue of Wiener measure for surfaces?]

$$\int_{\{\mathcal{E}\to\mathbb{R}^d\}} e^{-\mathcal{L}(x,g)}\mathcal{D}x = (\mathrm{Det}\,\Delta)^{-d/2} \tag{4}$$

We need

$$\int_{\{g\}} (\mathrm{Det}\,\Delta)^{-d/2}\mathcal{D}g \tag{5}$$

$\mathcal{M} = \{g\}$ — the space of metrics. Acting on it is Diff($\mathcal{E}$) [crossed out] and the function $(\mathrm{Det}\,\Delta)^{-d/2}$ is invariant under the action of this group.

Uncertain reading. The crossed-out mark before “and” may have been another “and.”

Original source details: GELFAND-P001214

$\mathcal{M}^c$ — the space of metrics conformally equivalent to the standard metric

(constant negative curvature; genus $g \geq 2$)

$$\mathcal{M}^c \times \mathrm{Diff}\,\mathcal{E} \xrightarrow{\text{onto}} \mathcal{M} \tag{6}$$

[At right:] (Theorem: $\forall$ metric can be made conformally equivalent to the standard one by a change of variables).

Faddeev–Popov trick

f — Diff-invariant function on $\mathcal{M} \Rightarrow$

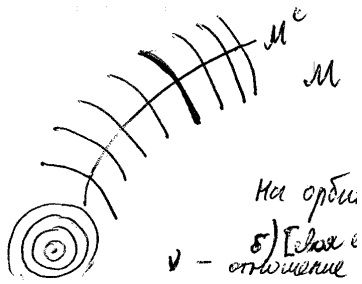
$$\int_{\mathcal{M}} f\mathcal{D}g = \left(\int_{\mathcal{M}^c} f \cdot \nu\right) \cdot (\text{volume of the group Diff}) \tag{7}$$

[In the right margin, beyond the double vertical line:] $\mathrm{vol}(\mathrm{Diff}) = \infty$, which we will later cross out.

Original source details: GELFAND-P001214

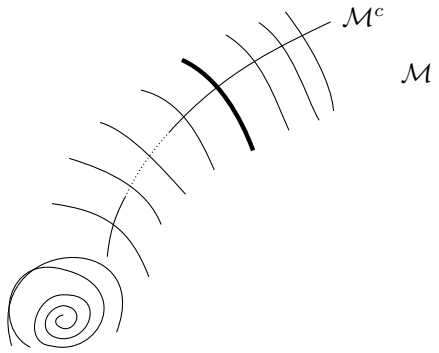
Uncertain reading. An undeciphered crossed-out mark is visible across the double line before “vol.”

Qualitative assertion: the main contribution comes from a small set (everything collapses to a point)



Original source details: GELFAND-P001214

Diagram and typed description.



The lower part of the sloping line is faint. The joins at the spiral are indistinct.

F.-P. trick

ν — the factor (at the integral over the orbits), which will be written out below.

There are two invariant metrics on the orbit; a) restriction of the metric of $\mathcal{M}$: dg

b) [its own natural metric (transferred from the group — the induced measure on G/H, H—the stabilizer subgroup)]

ν — the ratio of the two measures on each orbit.

GELFAND-P001215

δg — variation of the metric — tangent vector to the space of metrics.

$$|\delta g|^2 = \int_{\mathcal{E}} (\delta g, \delta g) \text{ vol. form} = \int_{\mathcal{E}} (\delta g, \delta g)\omega = \int_{\mathcal{E}} (\delta g, \delta g)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{8}$$

$\mathcal{D}g$ — the volume form [inserted: on $\mathcal{M}$], corresponding to $|\delta g|^2$ (9)

([overwritten mark] volume form constructed from the metric)

Uncertain reading. The overwritten mark before "volume form" may be an opening quotation mark or $=$.

Original source details: GELFAND-P001215

L — the Lie algebra of the diffeomorphism group, i.e. the Lie algebra of vector fields on $\mathcal{E}$.

$\varepsilon \in L$

$$|\varepsilon|^2 = \int_{\mathcal{E}} (\varepsilon, \varepsilon) \text{ vol. form} = \int_{\mathcal{E}} (\varepsilon, \varepsilon)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{10}$$

$$\delta_\varepsilon g \in S^2(T^*) \tag{11}$$

[At right:] (variation of the metric in the direction ε).

[Below the expression:] — Lie derivative.

Uncertain reading. The crossed-out words below $\delta_\varepsilon g$ may read "vector fields."

Original source details: GELFAND-P001215

define the operator:

$$L = \Gamma(T) \xrightarrow{D} \Gamma(S^2(T^*)), \qquad D(\varepsilon) = \delta_\varepsilon g \tag{12}$$

[Below $\Gamma(T)$*:]* (vector fields). (g is fixed).

Uncertain reading. The overwritten exponent on S on the right of (12) reads 2.

Original source details: GELFAND-P001215

$$\|\varepsilon\|^2 = \int_{\mathcal{E}} (D\varepsilon, D\varepsilon) \text{ vol. form} = \int_{\mathcal{E}} (D\varepsilon, D\varepsilon)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{13}$$

$$\frac{\text{vol. form corresponding to } \|\quad\|}{\text{vol. form corresponding to } |\quad|} = (\det \Delta)^{1/2}, \qquad \text{where } \Delta = D^*D \tag{14}$$

[In the right margin, beyond the double line:] Δ — Laplace operator on vector fields; $\Delta : \Gamma(T) \to \Gamma(T)$.

The ratio (14) is by definition the factor ν in (7)

$$\Gamma(S^2T) \subset \Gamma(\text{Hom}(T, T))$$

[Below Hom(T, T)*, brace:]* scalars $\oplus$ trace 0.

[At right:] (because of the metric); (identify S^2T, S^2T^* and embed in Hom(T, T)).

Corresponding decompositions of the sections:

$$\Gamma(S^2T) = \Gamma(S^2T)^c \oplus \Gamma(S^2T)^{\text{diff}} \tag{15}$$

[Below the first summand:] conformal changes of the metric.

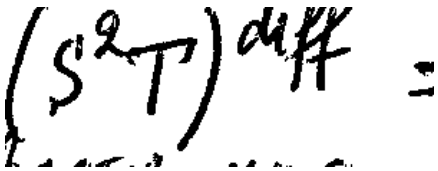
[Below the second summand:] changes under the action of Lie derivatives.

[crossed out]

$$(S^2T)^{\text{diff}} = T \otimes \Omega^{0,1} \tag{16}$$

Original source details: GELFAND-P001215

Uncertain reading. A mark before the left side of (16) is crossed out, possibly Γ or Φ.



Original source details: GELFAND-P001215

Uncertain reading. The superscript reads "diff."

[crossed out] It turns out that

$$D = \bar{\partial} : \Gamma(T) \longrightarrow \Gamma(T \otimes \Omega^{0,1}) \tag{17}$$

Original source details: GELFAND-P001215

Uncertain reading. An undeciphered mark before "It turns out that" is crossed out.

[The metric determines on $\mathcal{E}$ a complex structure, so it makes sense to write $\Omega^{0,1}$]

$$(S^2T)^{\text{diff}} \sim \text{-matrix } \begin{pmatrix} a & b \\ b & -a \end{pmatrix};$$

the meaning of (16) — the matrix is determined by its first row.

T — a holomorphic bundle, $\Omega^{0,1}$ — a 1-dimensional complex bundle.

on T acts the group of rotations.

S^2T — a 3-dimensional bundle, decompose it under the action of the rotation group.

GELFAND-P001216

[source number: 80]

Explanation of formula (14):

[Δ — the operator corresponding to quadratic form (13) relative to quadratic form (10);].

Incidentally:

$\delta_\varepsilon g$ — the Lie derivative of the metric g in the direction ε.

[A second way to define Δ on $\Gamma(T)$: identify T and T^* using the metric and take Δ on 1-forms. We get the same thing, since d (de Rham differential) and the Lie derivative are dual to one another].

Original source details: GELFAND-P001216

Uncertain reading. The last line in brackets reads “and the Lie derivative are dual to one another.”

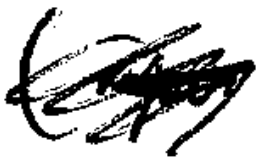
$S^2T = S^2T^*$ in the presence of the metric.

GELFAND-P001217

21/X–85. Gelfand Seminar.

Ryzhikova (A. Semenov’s student, in her 5th year).

An algorithm for checking any identity in the structure of subspaces [crossed out] of a finite-dimensional space.



Undeciphered passage. The crossed-out parenthesis before “finite-dimensional space” is undeciphered.

Original source details: GELFAND-P001217

We see which dimension suffices for checking [crossed out]

Uncertain reading. The crossed-out mark after “checking” may be 2 or a question mark.

Original source details: GELFAND-P001217

$$a(ab+c) = ab+ac \qquad \text{or} \qquad a(b+c) = ab+ac \quad \text{when } b \le a$$

[Below:] dimension 2 suffices.

General formula: it is enough to check in a space of dimension

$$n = \min[\max(d(p), d(q)), \max(k(p), k(q))] + 1$$

p, q — 2 given terms

$d(p)$ — the number of additions in p (disjunction)

$k(p)$ — ″ — multiplications in p (conjunction)

In a space of fixed dimension (over $\mathbb{R}$) there is Tarski's algorithm.

Fuchs (following Feigin). Continuation on strings.

We need to evaluate the integral:

$$\int_{\mathcal{M}^c} (\operatorname{Det}\Delta)^{-d/2}\nu \tag{1}$$

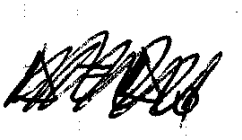
Feigin.

We need to compute the integral

$$\int_{\mathcal{M}_t} (\operatorname{Det}\Delta_0)^{-d/2}\mathcal{D}g = \int_{\mathcal{M}_t^c} \frac{\operatorname{Det}\Delta_1}{(\operatorname{Det}\Delta_0)^{d/2}}\mathcal{D}g, \tag{2}$$

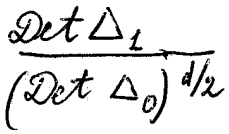
Original source details: GELFAND-P001217

Uncertain reading. In (2) the superscript on $\mathcal{M}_t$ under the first integral is crossed out, possibly c.



Original source details: GELFAND-P001217

Uncertain reading. The writing to the right of (2) is crossed out, possibly $\Delta = \Delta_0$ or $\Delta \neq \Delta_0$.



Original source details: GELFAND-P001217

Uncertain reading. The numerator reads $\operatorname{Det}\Delta_1$ with no visible exponent $1/2$.

$\mathcal{M}$ — the space of metrics, t — a point in the moduli space of curves of genus g

$\mathcal{M}_t$ — the space of metrics for the complex structure corresponding to modulus t.

GELFAND-P001218

[source number: 87]

$\mathcal{M}_t^c$ — the set of metrics [inserted above the line: (corresponding to the given modulus t)] of the form $e^{\varphi} g_0$, where g_0 — a metric of constant negative curvature $= -1$.

Original source details: GELFAND-P001218

Uncertain reading. The note below $\mathcal{M}_t^c$ appears to abbreviate "on the surface corresponding to t."

T — the tangent bundle on the curve (already equipped with a complex structure)

$$\bar{\partial} : \Gamma(T^j) \longrightarrow \Gamma(T^j \otimes \Omega^{0,1}), \qquad \Delta_j = \bar{\partial}^* \bar{\partial}$$

Original source details: GELFAND-P001218

Uncertain reading. A mark after T^j, before $\otimes$, is crossed out, probably a closing parenthesis.

In fact $\mathcal{M}_t^c$ [crossed out]; $\mathcal{M}_t^c \subset \mathcal{M}_t$ [crossed out]

Original source details: GELFAND-P001218

Uncertain reading. An undeciphered mark after the first $\mathcal{M}_t^c$ is crossed out.

Original source details: GELFAND-P001218

Uncertain reading. At right, the crossed-out expression may be $(\mathcal{M}^c \subset \mathcal{M}_t)$ or $(\mathcal{M}_t^c \subset \mathcal{M}_t)$.

(t is fixed).

Every metric in $\mathcal{M}_t$ can be (by the action of Diff)

Original source details: GELFAND-P001218

Uncertain reading. The sentence stops after "(by the action of Diff)."

The moduli space has dimension $3g - 3$.

The integral over it has yet to be evaluated (much work is devoted to it).

Original source details: GELFAND-P001218

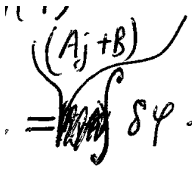
Undeciphered passage. The short crossed-out formula before the line about Det Δ_0 is undeciphered.

Det Δ_0 — is taken over the nonzero eigenvalues.

Det Δ_j = Det $\Delta_j(\varphi)$.

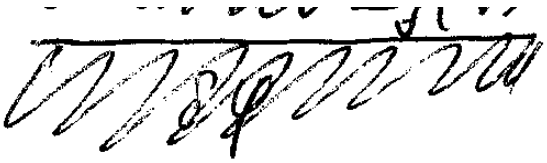
Next,

$$\delta \ln \operatorname{Det} \Delta_j(\varphi) = (Aj + B) \int_{\mathcal{E}} \delta\varphi \cdot \mathrm{Vol} + C(6j^2 + 6j + 1) \int_{\mathcal{E}} \delta\varphi \cdot K \cdot \mathrm{Vol} \tag{3}$$



Original source details: GELFAND-P001218

Uncertain reading. The original coefficient is crossed out. The insertion reads $(Aj + B)$, possibly $(A_j + B)$.



Original source details: GELFAND-P001218

Uncertain reading. The fraction bar and denominator $\delta\varphi$ under the left side are crossed out.

[Below the first Vol*, brace:]* volume form.

(j — "spin")

A, B, C — some constants; K — the Gaussian curvature of the surface.

$$j = 1 \Rightarrow 6j^2 + 6j + 1 = 13$$
$$j = 0 \Rightarrow 6j^2 + 6j + 1 = 1$$

Original source details: GELFAND-P001218

Uncertain reading. The result on the $j = 0$ line is overwritten and reads 1.

At $d = 26$ the anomaly cancels.

And then one has to integrate a constant (on the right-hand side of (2))

Corollary:

$$\ln \operatorname{Det} \Delta_j(\varphi) = A \int [\text{crossed out}]\, \varphi \cdot \partial\bar{\partial}\varphi + B \int e^{\varphi}\, \mathrm{Vol} + C \int \varphi\, \mathrm{Vol} + D \tag{4}$$

Original source details: GELFAND-P001218

Uncertain reading. The crossed-out expression immediately after the first integral sign in (4) may read $\partial\varphi \wedge \bar{\partial}\varphi$ or $\partial\varphi\, \bar{\partial}\varphi$.

(A, B, C, D — constants)

$\Rightarrow$ the calculation is equivalent to the Liouville equation (in (4) its Lagrangian)

Solving the Liouville equation is equivalent to computing

$$\int e^{\mathcal{L}_j(\varphi)} \mathcal{D}\varphi, \tag{5}$$

where $\mathcal{L}_j(\varphi)$ — left (= right) side of (4)
$\ln \operatorname{Det} \Delta_j$ is taken as the Lagrangian of the new theory. It is clear that

$$(2) = \int e^{\mathcal{L}_1(\varphi) - \frac{d}{2}\mathcal{L}_0(\varphi)} \mathcal{D}\varphi. \tag{6}$$

Chapter 27

GELFAND-P001219

28/$\overline{X}$—85 Gelfand Seminar

W. Müller.

Cheeger / Gromov

X — a complete Riemannian manifold, vol$(X) < \infty$, $|K| < 1$.

[Below K:] sectional curvature.

$$\int p(\Omega) - ?$$

Theorem. X — the interior of a compact manifold with boundary *[Below "interior": (diffeomorphic)]*

$$\Rightarrow 1) \int_X p(\Omega) = \chi(X).$$

Uncertain reading. An undeciphered mark above $p(\Omega)$ is crossed out.

Original source details: GELFAND-P001219

GELFAND-P001220

$$2) \int_X L_k(p_1, \ldots, p_k) \quad \text{— homotopy invariant}$$

Question related to the work of Cheeger

mfd [inserted above the line X] with cone like singularity

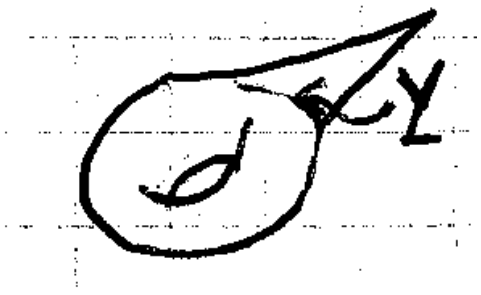
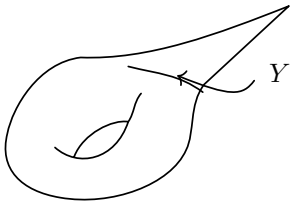


Diagram and typed description.



The pointer strokes overlap at the neck.

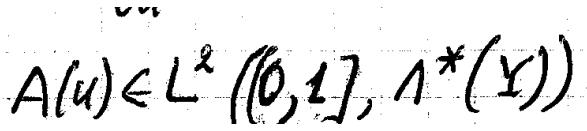
Original source details: GELFAND-P001220

Δ on Λ^*

Δ on cone

$$-\frac{\partial^2}{\partial u^2} + u^{-2}A(u)$$

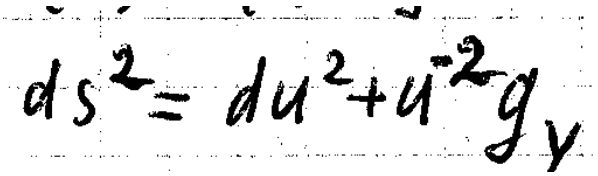
$$A(u) \in L^2((0,1], \Lambda^*(Y))$$



Uncertain reading. Reads $A(u) \in L^2((0,1], \Lambda^*(Y))$. Below, $A(u)$ is written as an operator.

Original source details: GELFAND-P001220

$$A(u) : \Lambda^*(Y) \longrightarrow \Lambda^*(Y),$$
$$C(Y) = (0,1] \times Y,$$
$$ds^2 = du^2 + u^{-2} g_Y.$$



Uncertain reading. The exponent on u before g_Y reads -2.

Original source details: GELFAND-P001220

$$\mathrm{Tr}(e^{-t\Delta}) \underset{t\to 0+}{\sim} \sum_{j\geq 0} a_j t^{-(n-j)/2} + c \log t + b$$

$$b = b_0 \zeta_Y(-1) + \sum_{j\geq 1} b_j \operatorname*{Res}_{s=2j-1} \zeta_Y(s)$$

$$\zeta_Y(s) = \sum_{\lambda_j} \lambda_j^{-s}$$

$D_s = d + \delta : \Lambda^*_+(X) \longrightarrow \Lambda^*_-(X)$ signature operator

[Below:] signature operator

$$(*) \qquad \mathrm{Ind}\, D_s = \int_X L_k(p_1, \ldots, p_k) + \eta_Y(B), \qquad \text{where } B \text{ is defined below.}$$

$\dim X = 4k$.

[Under the bracket at the integral] this would be the index without boundary.

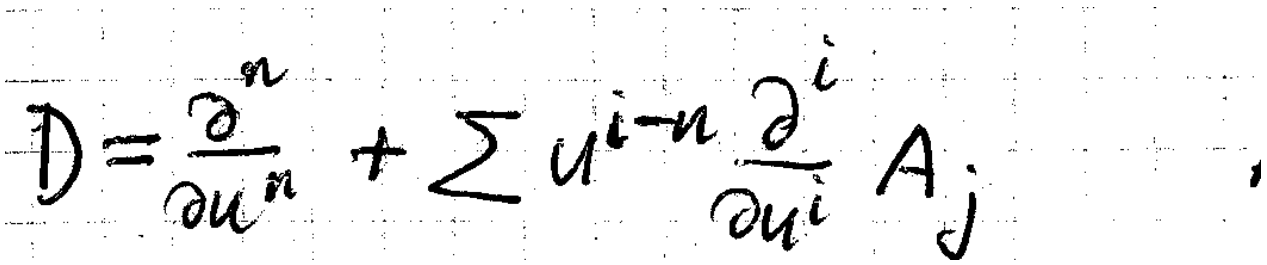
[Enclosed by a bracket] $L_k = 0$ on the cone.

On the cone:

$$d + \delta = D_s = -\frac{\partial}{\partial u} + u^{-1} B, \qquad B : \Lambda^*(Y) \longrightarrow \Lambda^*(Y).$$

[At right, beyond the double vertical line] On Λ^{ev}, B has the form $\pm(*d - d*)$.

$$D = \frac{\partial^n}{\partial u^n} + \sum u^{i-n} \frac{\partial^i}{\partial u^i} A_j$$



Original source details: GELFAND-P001220

Uncertain reading. The powers and derivative in the sum read i. The coefficient reads A_j.

A_j — diff. op. on Y.

index — ?

Cheeger's example gives η! — formula $(*)$

GELFAND-P001221

[source number: 6]

The index is defined on fns. satisfying [crossed out] the condition at the singularity:

Uncertain reading. The crossed-out word before “the condition” is illegible.

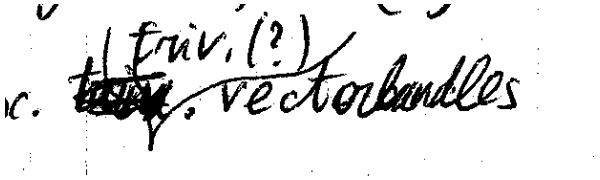
Original source details: GELFAND-P001221

$$\lim_{u\to 0} u^{-1/2}\varphi(u) = \lim_{u\to 0} u^{1/2}\varphi'(u)$$

Relation to representations.

$$X = K\backslash G/\Gamma \qquad \text{loc symm.}, \quad \text{vol}(X) < \infty$$

$\mathcal{E}, \mathcal{F} \longrightarrow X$ loc. [crossed out] [above the line: triv. (?)] vectorbundles



Uncertain reading. “triv. (?)” appears above the crossing-out. The crossed-out letters may also read “triv.”.

Original source details: GELFAND-P001221

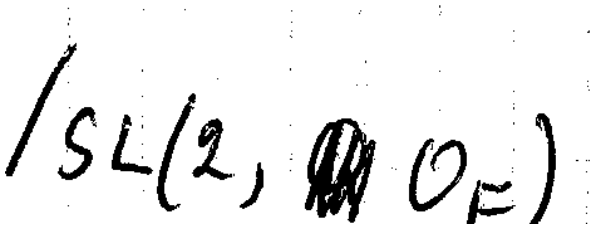
$$D : C_0^\infty(X, \mathcal{E}) \longrightarrow C_0^\infty(X, \mathcal{F}) \qquad \text{oper. loc inv.}$$

Th.

$$\ker \bar{D} < \infty,$$
$$\ker \bar{D}^* < \infty.$$
$$\text{Ind}_{L^2} D = \dim \ker \bar{D} - \dim \ker \bar{D}^*$$

— —

$$X = H \times H/SL(2, O_F), \qquad F = Q(\sqrt{D}), \quad D > 0.$$



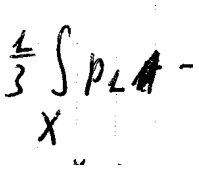
Uncertain reading. A mark before O_F in $SL(2, O_F)$ is heavily crossed out.

Original source details: GELFAND-P001221

only 1 cusp

$$D_s = d + \delta : \Lambda^*_+(X) \longrightarrow \Lambda^*_-(X)$$

$$\operatorname{ind} D_s = \frac{1}{3}\int_X p_1 - \frac{1}{2\pi} L(M, V, 1)\operatorname{vol}(\mathbb{R}^2/M)$$

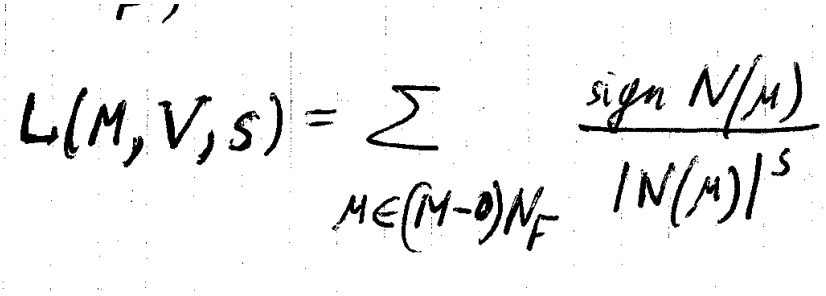


Uncertain reading. The heavily inked mark after p_1 may be an A.

Original source details: GELFAND-P001221

$$M = O_F, \qquad V = O_F^{*2}$$

$$L(M, V, s) = \sum_{\mu \in (M-0)N_F} \frac{\operatorname{sign} N(\mu)}{|N(\mu)|^s}$$



Original source details: GELFAND-P001221

Uncertain reading. The last marks beneath the sum appear to read N_F.

Feigin. On strings (work of Polyakov)

(Continued.)

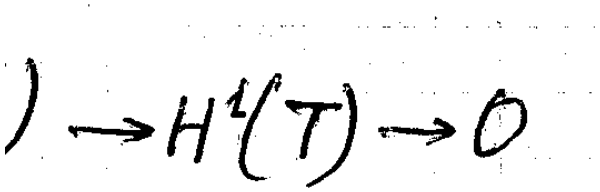
$\mathcal{E}$ — Riem. surface of genus g

T — tangent bundle

On $\mathcal{E}$ a complex structure induced by the Riem. metric is chosen, and the operator $\bar\partial : T^j \longrightarrow T^j \otimes \Omega^{0,1}$;

$$\Delta_j = \bar\partial^* \bar\partial$$

$$0 \longrightarrow H^0(T^j) \longrightarrow \Gamma(T^j) \xrightarrow{\bar\partial} \Gamma(T^j \otimes \Omega^{0,1}) \longrightarrow H^1(T) \longrightarrow 0. \tag{1}$$



Uncertain reading. The term before zero in (1) reads $H^1(T)$, with no exponent on T.

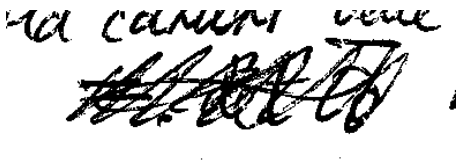
Original source details: GELFAND-P001221

The problem can be reformulated as computing the Ray–Singer torsion of this complex.

$$g_{ij} \longrightarrow g_{ij}(1+\varepsilon\rho) \tag{2}$$

$\bar\partial$ has ker and coker. [In the same interlinear note:] What was actually computed was the change

in Ray–Singer torsion [crossed out] of the complex (1) under deformation



Uncertain reading. The crossed-out notation after "Singer" may read Det $\bar\partial$.

Original source details: GELFAND-P001222

on $\Gamma(T^j)$ and on $\Gamma(T^j \otimes \Omega^{0,1})$ there are Hermitian metrics.

They induce Herm. metrics on $\underbrace{H^0(T^j)}_{\text{subspace}}$ and $H^1(T^j)$.

$\ker \bar\partial\bar\partial^*$ is identified with $H^1(T^j) \Rightarrow$ on $H^1(T^j)$ there is a Herm. metric.

$$\operatorname{Det} T^j := \Lambda^{\max} H^0(T^j) \otimes \left(\Lambda^{\max} H^1(T^j)\right)^* \tag{3}$$

(Quillen's terminol.) (1-dimensional space)

(4) On $\operatorname{Det} T^j$ there is also a Hermitian metric. Denote it by ε.

Ray–Singer torsion is another Herm. metric on $\operatorname{Det} T^j$. It equals the Herm. metric (4) divided by $\operatorname{Det} \Delta_j$:

$$\frac{\varepsilon}{\operatorname{Det} \Delta_j} \tag{5}$$

Consider the variation of the metric:

$$g_{ij} \longrightarrow g_{ij}(1+\varepsilon\rho) \tag{6}$$

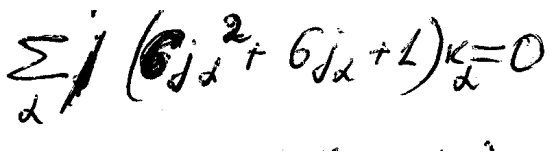
Given a seq. of numbers $j_1, j_2, \dots, j_n$, [inserted above the line: $k_1, \dots, k_n$] and 1-dimensional space

$$\operatorname{Det}(T^{j_1})^{k_1} \otimes \operatorname{Det}(T^{j_2})^{k_2} \otimes \cdots \otimes \operatorname{Det}(T^{j_n})^{k_n} \tag{7}$$

with Herm. metric = the product of the Ray–Singer torsions of each of these spaces.

(8) Claim. The metric on (7) is unchanged under variation (6) if

$$\sum_\alpha (6j_\alpha^2 + 6j_\alpha + 1)k_\alpha = 0. \tag{9}$$



Uncertain reading. There is a heavy diagonal stroke between $\sum_\alpha$ and the opening parenthesis. The first 6 is retraced.

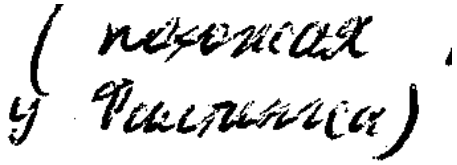
Original source details: GELFAND-P001222

Or: under condition (9) the metric on (7) depends only on the complex structure.

Let $\mathcal{M}$ — moduli manifold of complex curves of genus g.

The collection of spaces is a 1-dimensional bundle over $\mathcal{M}$.

A metric is constr. on this bundle (a similar constr. occurs in Faltings)



Uncertain reading. The surname at the end of the line is tentatively read as Faltings.

Original source details: GELFAND-P001222

GELFAND-P001223

$4/\overline{\text{XI}} - 85$ Gelfand Sem.

Ryzhikova On lattice representations. (work of Gelfand and Ponomarev)

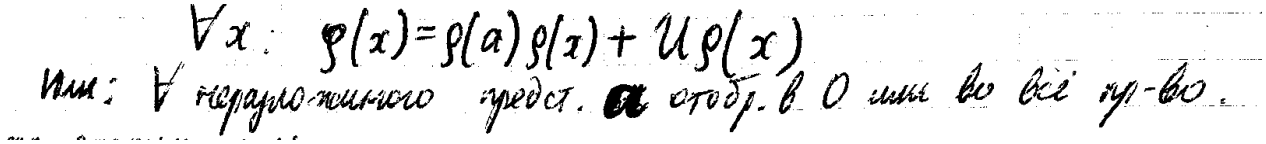
Def. An elem. $a \in L$ is called perfect, if

(L — a lattice)

$$\forall k, \quad \forall \rho : L \longrightarrow \mathcal{L}(V), \quad \exists U \subseteq V,$$
$$U + \rho(a) = V, \qquad U\rho(a) = 0,$$
$$\forall x : \quad \rho(x) = \rho(a)\rho(x) + U\rho(x).$$

[Above k — field. Above ρ — rep..]

Or: $\forall$ indecomposable rep., a maps to 0 or to the whole space.



Original source details: GELFAND-P001223

Uncertain reading. The overwritten sign before "maps" reads a.

Def. of a lattice.

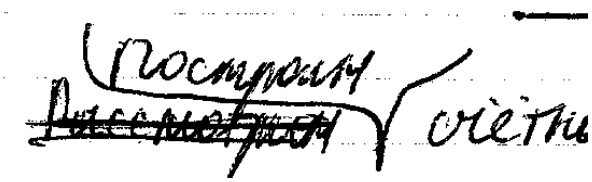
$a + b$, $a \cdot b$ — assoc. commut.

$$a \cdot a = a, \qquad a + a = a, \qquad a(a + b) = a, \qquad a + ab = a,$$
$$a \le b \Longleftrightarrow ab = a \Longleftrightarrow a + b = b,$$
$$a \le b \Longrightarrow b(a + c) = a + bc.$$

Rep. of a lattice $\rho : L \longrightarrow \mathcal{L}(V)$.

— —

[Crossed out: Consider.] [Above the line: Let us construct.] a countable number of cubes $B^+(\ell)$ and $B^-(\ell)$, $\ell \ge 1$,



Uncertain reading. "Let us construct" appears above the crossed-out "Consider".

Original source details: GELFAND-P001223

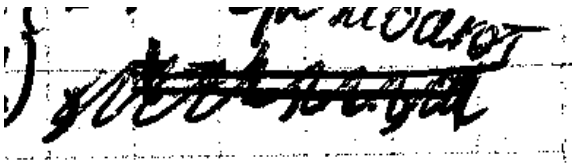
D^r — free modular lattice with r generators $e_1, \ldots, e_r$.

GELFAND-P001224

1. $\ell = 1$

Put $h_t(1) = \sum_{j \neq t} e_j$, $t = 1, \ldots, r$.

These $h_t(1)$ [inserted above the crossing-out: generate] the Boolean algebra $B^+(1)$ [crossed out]



Uncertain reading. The crossed-out text after $B^+(1)$ may begin with "already".

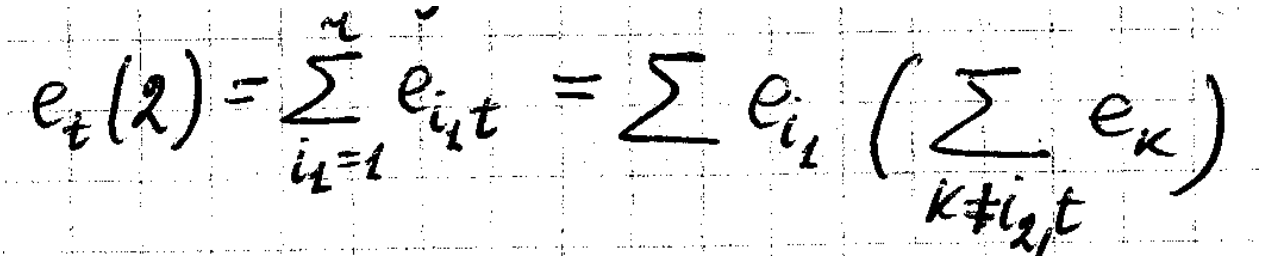
Original source details: GELFAND-P001224

$$0,\ h_1, \ldots, h_4, \quad h_1 \cdot h_2,\ h_2 \cdot h_3,$$
$$h_1 \cdot h_2 \cdot h_3, \ldots,$$
$$h_1 \cdot h_2 \cdot h_3 \cdot h_4.$$

The sum of products is already expr. as a product.

2. $\ell = 2$

$$\{(i_1, i_2) : i_1 \neq i_2\} = A(r, 2)$$
$$\{(i_1, t),\ i_1 \neq t\} = A_t(r, 2) \qquad (t \text{ — fixed})$$
$$e_t(2) = \sum_{i_1=1}^{r} e_{i_1 t} = \sum e_{i_1} \left(\sum_{k \neq i_2, t} e_k \right)$$



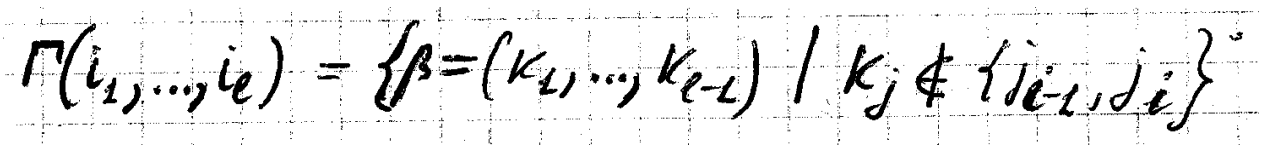
Original source details: GELFAND-P001224

Uncertain reading. The limits of the first sum read $i_1 = 1$ and r. The inner sum reads $k \neq i_2, t$.

$$h_t(2) = \sum_{j \neq t} e_j(2)$$

— —

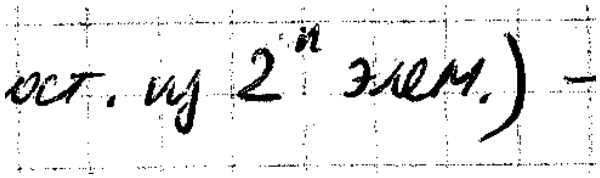
$$\{(i_1, \ldots, i_\ell),\ i_j \neq i_{j+1}\} = A(r, \ell)$$
$$\{(i_1, \ldots, i_{\ell-1}, t),\ i_{\ell-1} \neq t\} \cap A(r, \ell) = A_t(r, \ell)$$
$$\Gamma(i_1, \ldots, i_\ell) = \{\beta = (k_1, \ldots, k_{\ell-1}) \mid k_j \notin \{j_{i-1}, j_i\}$$



Original source details: GELFAND-P001224

Uncertain reading. The final subscripts read $i - 1$ and i. One closing brace is visible.

These are all cubes (consisting of 2^n elem.) — Boolean algebras; the lower ones are constructed dually.

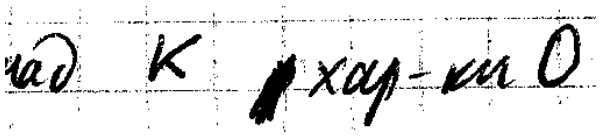


Uncertain reading. The exponent on 2 reads n.

Original source details: GELFAND-P001224

They are all ordered.

Def. V over K [inked over] of char. 0



Uncertain reading. There is a heavily inked mark between K and "of char.".

Original source details: GELFAND-P001224

$\rho : L \longrightarrow \mathcal{L}(V, K)$ is called completely irreducible if

$$\rho(L) \cong \mathcal{L}(\mathbb{Q}^n) \cong P(n-1, \mathbb{Q}).$$

GELFAND-P001225

[source number: 10]

B. L. Feigin On strings

How do the Virasoro alg. and hypergeometric fns. arise?

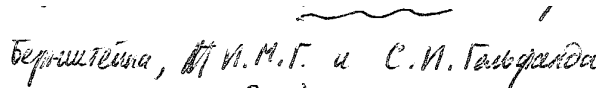
Virasoro algebra: Lie alg. $\mathcal{L}$ with basis $\{e_i\}$, $\underline{\underline{C}}$, $i \in Z$.

$$[e_i, e_j] = (j-i)e_{i+j} + \delta_{i,-j}\frac{i^3 - i}{12}\underline{\underline{C}}, \qquad (1)$$
$$[e_i, \underline{\underline{C}}] = 0.$$

(2) Quotient by the center = alg. of vector fields on the circle.

For Verma modules see the book of Dixmier on enveloping algebras.

and 2 papers by Bernstein, [crossed out] I. M. G. and S. I. Gelfand in FA



Uncertain reading. The crossed-out letter before "I. M. G." may be "И".

Original source details: GELFAND-P001225

(including on categories of Ore)

Uncertain reading. The word after "categories" reads "Ope".

Original source details: GELFAND-P001225

$$\mathfrak{h} = \{e_0, \underline{\underline{C}}\} \qquad \text{(Cartan)},$$
$$\mathcal{L}_+ = \{e_1, e_2, e_3, \ldots\} \qquad \text{(creat.)},$$
$$\mathcal{L}_- = \{e_{-1}, e_{-2}, \ldots\} \qquad \text{(annih.)}.$$

$\chi = (h, \bar{\bar{c}})$ — [crossed out] pair of numbers

Uncertain reading. The crossed-out abbreviation before “pair” may read “ord.”.

Original source details: GELFAND-P001225

Verma module: M_χ, there is a vac. vector v

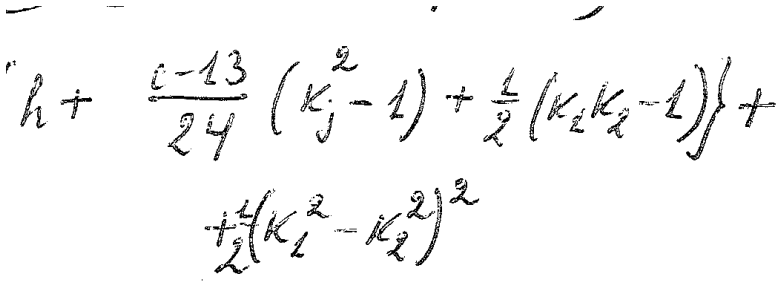
$$\begin{cases} e_0 v = hv, \\ Cv = \bar{\bar{c}}\, v \end{cases} \qquad \mathcal{L}_- v = 0. \tag{3}$$

Physicists write L_i in place of e_i.

(4) Given $M_\chi = M_{h,c}$, when is [inserted above the line: it] reducible?

Physicists call these mass formulas (mass spectrum)

$$0 = \Phi_{k_1 k_2}(h, c) = \prod_{j=1,2} \left\{ h + \frac{c-13}{24}(k_j^2 - 1) + \frac{1}{2}(k_1 k_2 - 1) \right\} + \\ + \frac{1}{2}(k_1^2 - k_2^2)^2. \tag{5}$$



Original source details: GELFAND-P001225

Uncertain reading. The coefficient of $(k_1^2 - k_2^2)^2$ in (5) reads $1/2$.

(Kac formula)

Here k_1, k_2 — natural numbers.

(6) Claim. [Below the heading: (Feigin, Fuchs).] If $\exists\, k_1, k_2$, satisfying (5), the rep. is reducible. If none exist, it is irreducible.

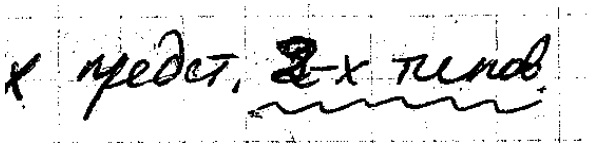
(7) Addendum: if M_χ is reducible, in it $\exists$ a vector different from the vac. vector and annihilated by all annih. operators (a “singular vector”).

$\mathcal{L}$ is naturally graded $\Rightarrow$ M_χ is too.

GELFAND-P001226

(8) Then the level of the sing. vector $= k_1 \cdot k_2$.

Proofs: constructions of various beautiful reps. of 2 types.



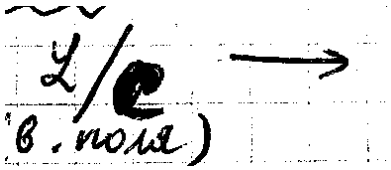
Uncertain reading. The overwritten numeral before “types” reads 2.

Original source details: GELFAND-P001226

1st type.

$$\mathcal{L}/C \longrightarrow gl_\infty,\ O_\infty,\ sp_\infty$$

[Below the quotient: (v. fields).]



Uncertain reading. The overwritten sign in the denominator reads C.

Original source details: GELFAND-P001226

2nd type

(9) $\mathcal{L} \hookrightarrow$ the enveloping alg. of an affine Kac–Moody algebra.

We construct the embedding

$\mathfrak{g}$ — finite-dim. simple Lie alg.

$\widehat{\mathfrak{g}}$ — 1-dimensional extension of the current algebra (usually called a Kac–Moody alg.)

[Between the lines:] ([crossed out] I.M.G. Current alg. — alg. of matrix fns. on the circle).

Uncertain reading. The crossed-out mark before “I.M.G.” is illegible.

Original source details: GELFAND-P001226

On $\mathfrak{g}$ fix the Killing form.

(10) Choose $u_\alpha \in \mathfrak{g}_\alpha$ (root space) so that $(u_\alpha, u_{-\alpha}) = 1$.

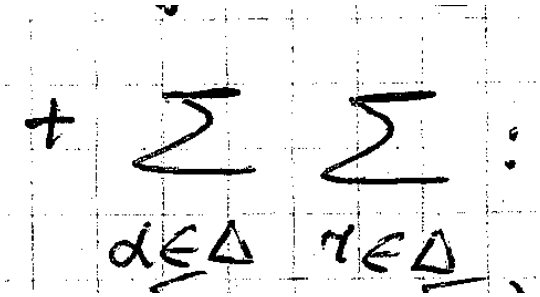
(11) Choose $u_1, \dots, u_\ell \in \mathfrak{h}$, $u^1, \dots, u^\ell$ — dual basis.

[Under $u_1, \dots, u_\ell$: (basis).]

Put

$$S(m) = \sum_{j=1}^{\ell} \sum_{r \in Z} : u^j(-r) u_j(r+m) : + \\ + \sum_{\alpha \in \Delta} \sum_{r \in \Delta} : u_{-\alpha}(-r) u_\alpha(r+m) : \tag{12}$$

(elements of the enveloping alg.)



Original source details: GELFAND-P001226

Uncertain reading. The inner sum on the second line of (12) reads $r \in \Delta$.

The order : : always in $u(r)u(s)$ is set so that $r < s$ or if $r = s$, we replace the product by the half-sum of the products.

$S(m)$ are called Graeme Segal operators.

$$[S(m), S(n)] = 2(n-m) \cdot (g+k) S(m+n) + \\ + \frac{\dim \mathfrak{g}}{12} \delta_{m,-n} (m^3 - m) 4(g+k)k. \tag{13}$$

(k — a number)

$S(m)$ act in a representation of the alg. $\widehat{\mathfrak{g}}$, in which the center acts by multiplication by k (for example, in a Verma module).

GELFAND-P001227

[source number: 11]

g — dual Coxeter number of the algebra $\mathfrak{g}$

(for $n \times n$ matrices this is n)

for $sl(2)$ $g = 2$.

The [crossed out] Virasoro algebra is given by the generators:

Original source details: GELFAND-P001227

Uncertain reading. The crossed-out text before “Virasoro” is illegible. It may start with “Ви...”.

$$e_m = \frac{S(m)}{2(g+k)}. \tag{14}$$

(15) Example. $\mathfrak{g} = sl_2$, $\widehat{\mathfrak{g}} = A_1^1$.

Verma modules are indexed by pairs of numbers (k_1, k_2): notation M_{k_1,k_2}.

The cent. charge in this case is $C = \frac{3k}{k+2}$, where $k = k_1 + k_2$,

$$h = \frac{k_1(k_1+1)}{k+2}.$$

[At right, beyond the vertical line:]

There was a construction: [inserted above "alg.": rep.] of the Virasoro alg. (embedding) is constructed from a rep. of the current alg.

Generalization of the construction of a representation

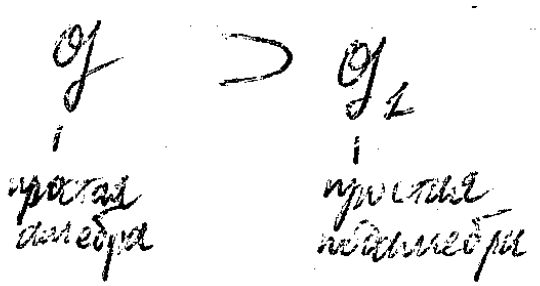


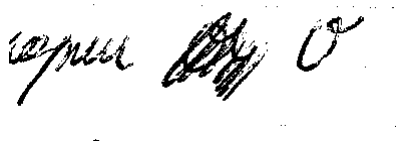
Diagram and typed description.

$\mathfrak{g}$	$\supset$	$\mathfrak{g}_1$
simple algebra		simple subalgebra

Original source details: GELFAND-P001227

$$\widehat{\mathfrak{g}} \supset \widehat{\mathfrak{g}}_1$$

M — a representation of $\widehat{\mathfrak{g}}$ from category [crossed out] O.



Uncertain reading. The mark before O is crossed out and illegible.

Original source details: GELFAND-P001227

$e_i^{\mathfrak{g}}$, $e_i^{\mathfrak{g}_1}$ — the corresponding e_i, constr. from $\mathfrak{g}$ and $\mathfrak{g}_1$.

Consider $e_i = e_i^{\mathfrak{g}} - e_i^{\mathfrak{g}_1}$.

(16) Theorem. 1) e_i [inserted above the line: also] generate a Virasoro alg.

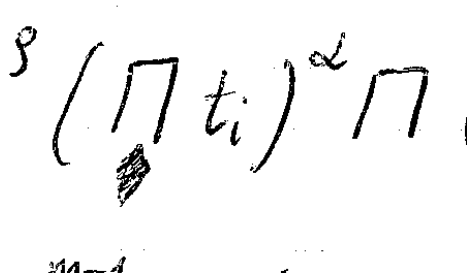
2) e_i commute with the algebra $\widehat{\mathfrak{g}}_i$.

Uncertain reading. The subscript resembles i, with a dot above its main stroke. The same letter takes subscript 1 above.

Original source details: GELFAND-P001227

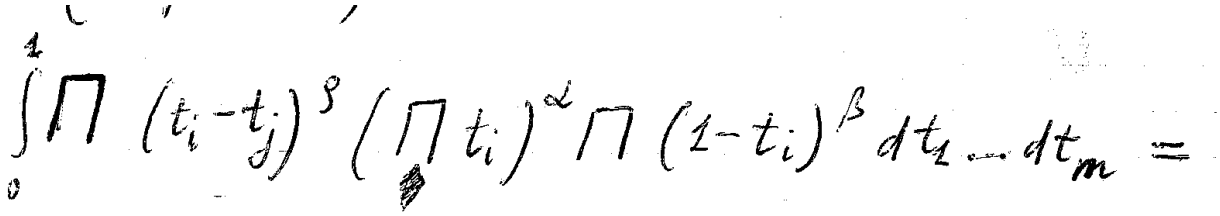
3) Cent. charge of the alg. $\{e_i\}$ = (cent. charge of $e_i^{\mathfrak{g}}$) − (cent. charge of $e_i^{\mathfrak{g}_1}$).
[Under “Cent. charge”, linked by two vertical strokes: (cent. element).]

$$\int_0^1 \prod (t_i - t_j)^\rho \left(\prod t_i\right)^\alpha \prod (1-t_i)^\beta \, dt_1 \dots dt_m = \\ = \prod_{i=1}^{m} \frac{\Gamma(i\rho)}{\Gamma(\rho)} \prod_{i=0}^{m-1} \frac{\Gamma(1+\alpha+i\rho)\,\Gamma(1+\beta+i\rho)}{\Gamma(2+\alpha+\beta+(m-1+i)\rho)}. \tag{17}$$



Uncertain reading. An illegible mark below the second product sign is crossed out.

Original source details: GELFAND-P001227



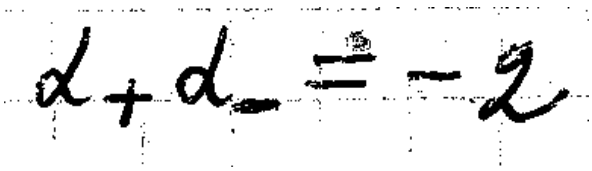
Original source details: GELFAND-P001227

Uncertain reading. The exponent on $(t_i - t_j)$ reads ρ. The first and third product signs have no indices.

Special case of a formula of V. Fateev and Dotsenko.

GELFAND-P001228

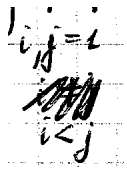
$\alpha_+, \alpha_-, \alpha_+\alpha_- = -2$.



Uncertain reading. The small mark above the equals sign is illegible.

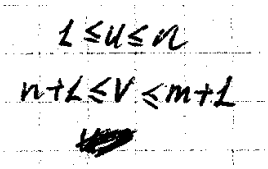
Original source details: GELFAND-P001228

$$\int_0^1 \prod_{\substack{i,j=1\\ i<j}}^{n} (t_i - t_j)^{\alpha_+^2} \prod_{\substack{n+1\le i,j\le n+m\\ i<j}} (t_i - t_j)^{\alpha_-^2} \left(\prod_{i=1}^{n+m} t_i\right)^{\beta} \cdot \prod_{\substack{1\le u\le n\\ n+1\le v\le m+1}} (t_u - t_v)^{-2} \cdot \prod_{1\le i\le n} (z - t_i)^{\gamma} \prod_{n+1\le i\le n+m} (z - t_i)^{\delta}\, dt_1 \dots dt_n. \tag{18}$$



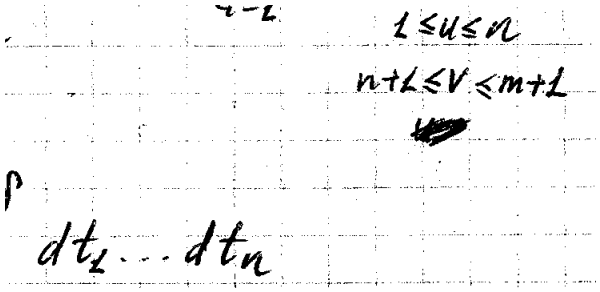
Original source details: GELFAND-P001228

Uncertain reading. Between $i, j = 1$ and $i < j$ beneath the first product, notation involving i and j is crossed out.



Original source details: GELFAND-P001228

Uncertain reading. An illegible mark is crossed out beneath the two constraints on the mixed product.

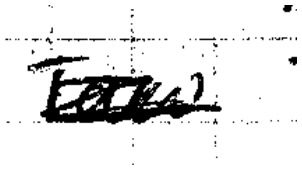


Original source details: GELFAND-P001228

Uncertain reading. The upper bound for v reads $m + 1$, and the final differential reads dt_n.

In this form the 2-point correlation fn. of the theory (conformal field theory) is represented.

[crossed out] This gives a countable number of correlation fns.



Original source details: GELFAND-P001228

Uncertain reading. The crossed-out word before “This gives” may read “If”.

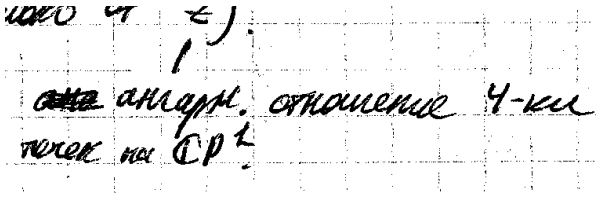
Original source details: GELFAND-P001228

Uncertain reading. The word before “number” is tentatively read as “countable”. The initial strokes are unclear.

The real fns. are their analytic continuation to complex n and m.

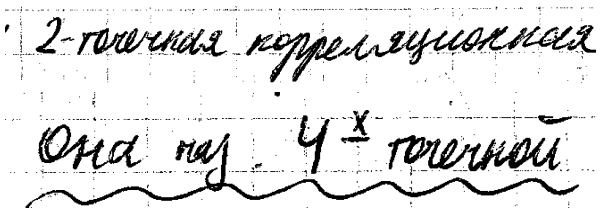
It is called 4-point (although it depends only on z).

[A short stroke leads down from z] [crossed out] cross-ratio of 4 points on CP^1.



Original source details: GELFAND-P001228

Uncertain reading. The crossed-out word before "cross-ratio" may read "it".



Original source details: GELFAND-P001228

Uncertain reading. The upper passage reads "2-point". The lower passage reads "4-point".

GELFAND-P001229

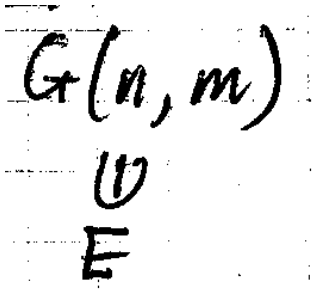
$11/\overline{\text{XI}} - 85$ Gelfand Sem.

Vershik A. M. Topology of configuration space.

Original source details: GELFAND-P001229

Uncertain reading. The last letter in the word for "space" is overwritten and reads "a".

1°.



Original source details: GELFAND-P001229

Diagram and typed description.

$G(n,m)$

$\cup$

E

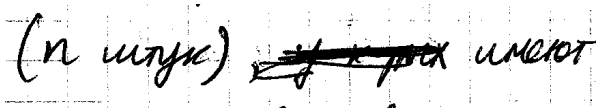
$e_1, \ldots, e_n$ — basis in $\mathbb{R}^n$

$$\tau_E = (Pe_1, \ldots, Pe_n)$$

What is the top. stratum in $G(n,m)$, where τ_E has a given combinat. type?

2°. Combinat. theory of convex polyhedra in $\mathbb{R}^n$.

2 polyhedra V_1, V_2 with ordered (numbered) vertices (n of them) [crossed out] have the same combinat. type if their faces correspond to each other.



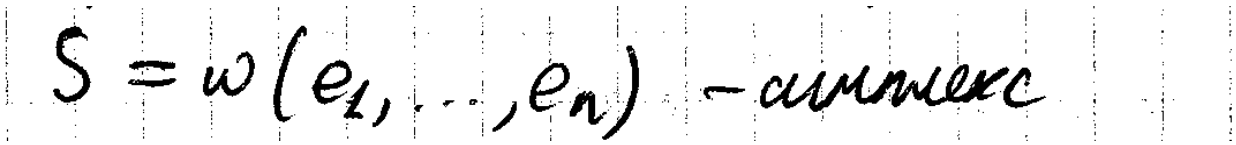
Original source details: GELFAND-P001229

Undeciphered passage. The crossed-out passage after "n of them" is undeciphered.

Grassmannization of the problem.

In $\mathbb{R}^n$ consider a subspace (m-dimensional) E

$$S = \omega(e_1, \ldots, e_n) \quad \text{— simplex}$$



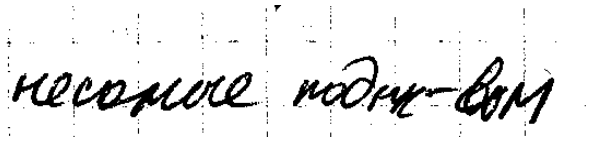
Original source details: GELFAND-P001229

Uncertain reading. The sign after the equality reads ω.

The projection of S onto E — polyhedron (convex)

Ex.: obtain combinatorially any polyhedron of dim. m with n vertices

On $G(n, m)$ consider the partition into subsets where the polyhedra carried by the subspace are combinatorially equivalent.

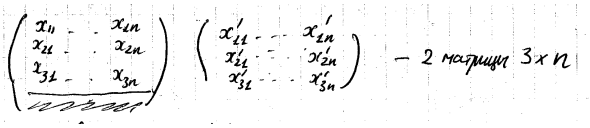


Original source details: GELFAND-P001229

Uncertain reading. The abbreviation after "carried" reads "подпр-вом", for "by the subspace".

What is the topology of these pieces?

3°.



Original source details: GELFAND-P001229

Diagram and typed description.

$$\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ x_{21} & \cdots & x_{2n} \\ x_{31} & \cdots & x_{3n} \end{pmatrix} \qquad \begin{pmatrix} x'_{11} & \cdots & x'_{1n} \\ x'_{21} & \cdots & x'_{2n} \\ x'_{31} & \cdots & x'_{3n} \end{pmatrix}$$

— 2 matrices $3 \times n$

A rule and looping hatching lie under the third row of the first matrix. The second matrix has neither.

Let all minors be $\neq 0$ and have the same signs for x and x'.

Does there exist a homotopy $x(t)$ of matrices with $x(0) = x$, $x(1) = x'$

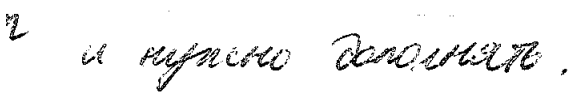
GELFAND-P001230

[source number: 15]

in the same class?

$$n \leq 7 \Rightarrow \text{possible}$$
$$n = 19 \Rightarrow \text{impossible!} \qquad \text{(N. E. Mnëv)}$$

Configurations also make sense in $\mathbb{R}P^n$ and one needs to supplement.



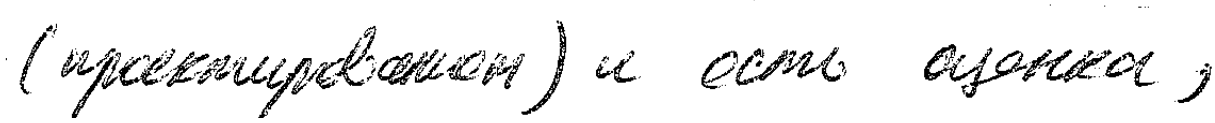
Original source details: GELFAND-P001230

Uncertain reading. The final word reads "дополнять" (to supplement).

One needs to probe the consequences (by projection) and there is a bound on how many one needs to take to have enough.

Uncertain reading. The verb is tentatively read as “пощупать” (to probe).

Original source details: GELFAND-P001230



Original source details: GELFAND-P001230

Uncertain reading. The word in parentheses reads “проектированием” (by projection). The next phrase has “оценка” (bound).

The topology can be that of any algebraic var.!

(This is over $\mathbb{R}$)

over C the analogue is computing π_1.

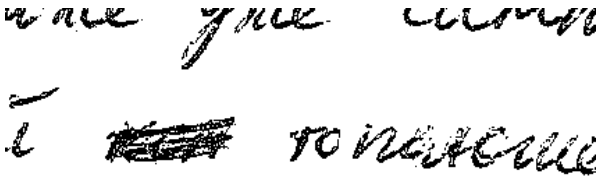
It is interesting what the convex analogue is over C.

Steinitz’s thm.: characterization of the 1-dim. skeleton of a convex polyhedron

(see any textbook of convex geom.) (in $\mathbb{R}^3$)

Corollary: the space of polyhedra of a given combinat. type in $\mathbb{R}^3$ is contractible.

But: in $\mathbb{R}^4$ there already exist even simplicial polyhedra with more or less any [crossed out] topology of combinat. type.



Undeciphered passage. The crossed-out word between “any” and “topology” is undeciphered.

Original source details: GELFAND-P001230

Problem: we want to construct a field of polyhedra on a manifold! (in each tangent sp.). There was [crossed out] a classific. problem for this one.

Undeciphered passage. The crossed-out passage after “There was” is undeciphered.

Original source details: GELFAND-P001230

Matroids m. Given a lattice. It is called a geometric lattice, if it is

1) Semimodular

2) Finite ($\forall$ elem. $=$ sup of finitely many atoms)

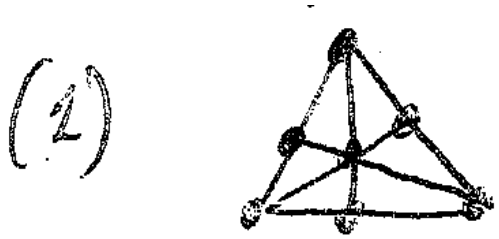
[Below this line:] atoms — those immed. after zero.

Uncertain reading. The overwritten mark after “Matroids” tentatively reads “м.”.

Original source details: GELFAND-P001230

Config. — geom. lattice.

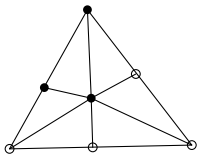
Conversely, let a geom. lattice be given. When is it given by a config.?



Original source details: GELFAND-P001230

Diagram and typed description.

(1)



Seven points and three interior lines are visible. No separate line through the three points on the sides is visible.

(Fano config.) over $\mathbb{R}, \mathbb{Z}$ is not realizable.

A faithful rep. of a lattice = a map from it

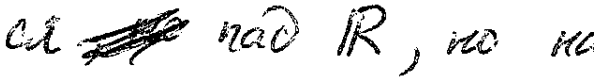
$$a : \Gamma \longrightarrow \mathcal{L}(F^n) \quad \text{— lattice of subspaces,}$$

such that

$$\gamma_1 < \gamma_2 \;\Rightarrow\; a(\gamma_1) < a(\gamma_2)$$

and sup is preserved

(1) is not realizable [crossed out] over $\mathbb{R}$, but over $\mathbb{Z}_2$ realiz..



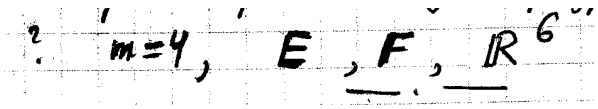
Uncertain reading. The crossed-out mark before “over” is illegible.

Original source details: GELFAND-P001230

GELFAND-P001231

A faithful rep. exists for config. (1):

$$? \qquad m = 4, \quad E, \quad \underline{F}, \quad \underline{\mathbb{R}^6}$$



Uncertain reading. The line with the question mark lists $m = 4$, E, F and $\mathbb{R}^6$ separately.

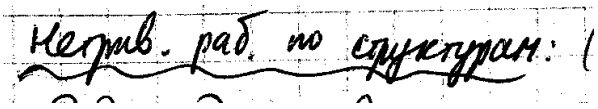
Original source details: GELFAND-P001231

Realizability problem $\Leftrightarrow$ problem of solving a system of algebraic inequalities.

Uncertain reading. The final abbreviation reads “нер-в” (inequalities).

Original source details: GELFAND-P001231

Nontriv. work on lattices: (according to Gelfand)



Uncertain reading. The first abbreviation in the heading appears to read “Нетрив.” (nontrivial).

Original source details: GELFAND-P001231

Dedekind: invented it

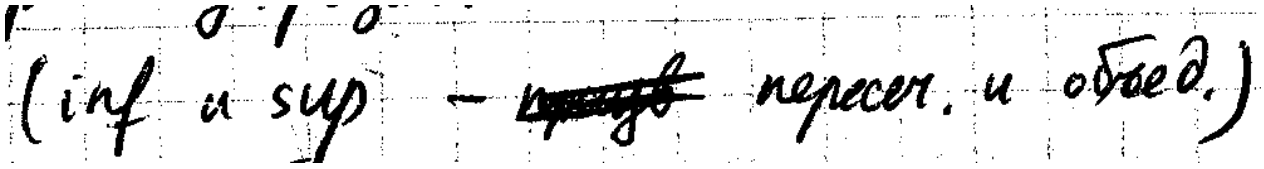
Kurosh, Ore.

von Neumann;

Gelfand, Ponomarev:

[In the left margin: Vershik's add.:] Pudlák and Tůma (Czechs): every finite lattice embeds into the partition lattice of a finite set (very difficult theorem).

(inf and sup — [crossed out] intersection and union)

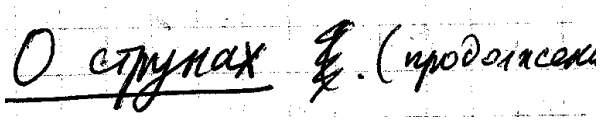


Original source details: GELFAND-P001231

Undeciphered passage. The word after "inf and sup" is crossed out and undeciphered.

0 — partition into points.

Feigin B. On strings [crossed out] (continued).

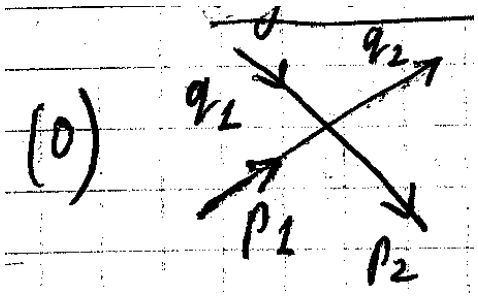


Original source details: GELFAND-P001231

Uncertain reading. The crossed-out vertical mark after "On strings" is undeciphered.

Operator theory of strings.

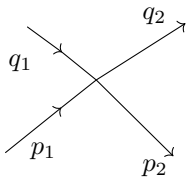
Dual amplitudes. Consider a collision of particles



Original source details: GELFAND-P001231

Diagram and typed description.

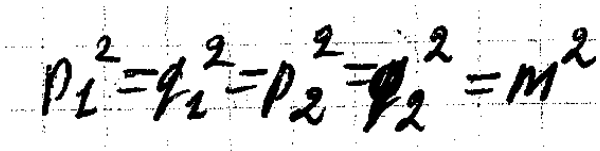
(0)



Let the masses all be the same and the total momentum $= 0$

All are vectors in space/time

$$\begin{cases} p_1 + q_1 = p_2 + q_2, \\ p_1^2 = q_1^2 = p_2^2 = q_2^2 = m^2 \end{cases} \qquad p_j, q_j \in \mathbb{R}^4. \tag{1}$$



Original source details: GELFAND-P001231

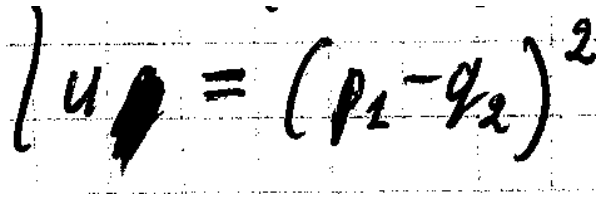
Uncertain reading. The overwritten letter in the penultimate square reads q_2.

Let an S-matrix characterizing this collision be given,

$$S(p_1, q_1, p_2, q_2) \tag{2}$$

Mandelstam variables:

$$\begin{cases} s = (p_1 + q_1)^2, \\ t = (p_1 - p_2)^2 \\ u \text{ [crossed out]} = (p_1 - q_2)^2 \end{cases} \tag{3}$$



Uncertain reading. An illegible mark between u and the equals sign is crossed out.

Original source details: GELFAND-P001231

[To the right of (3):] (Relativistically invariant pairwise scalar products are given) dependent:

$$s + t + u = 4m^2 \tag{4}$$

Define $A(s,t)$ (= expression for S in terms of s, t)

Requirement:

$$1) \quad A \text{ is analytic in } s, t \quad \text{[crossed out]} \tag{5}$$

Uncertain reading. A mark after t is crossed out. Its reading is unresolved.

Original source details: GELFAND-P001231

[source number: 16]

2) A is symmetric in s, t (indep. of the order)

3) For fixed t the fn. $A(s,t)$ has in s only simple poles, and [inserted between words: 4)] the locations of the poles do not depend on t. Usually one writes:

$$A(s,t) = \sum_{n=0}^{\infty} \frac{R_n(t)}{n - \alpha(s)} \tag{6}$$

5) $R_n(t)$ — polynomial

The graph of $\alpha(s)$ is called a Regge trajectory.

It turns out that from 1)—5) we uniquely determine the form of the fn. A:

$$A(s,t) = \frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s) - \alpha(t))} = B(-\alpha(s), -\alpha(t)) \tag{7}$$

It follows from analysis of the exper. data that

$$\alpha(s) \quad \text{— linear fn.} \tag{8}$$

A is called the Veneziano amplitude, or dual amplitude, and conds. 1)—5) are called duality conditions.

Operator method: we want to write $A(s,t)$ as an operator expectation

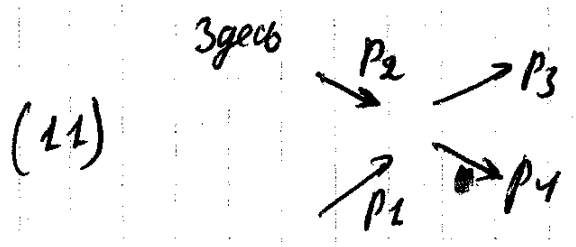
$$\langle 0|V(t)V(s)|0\rangle, \tag{9}$$

where $\langle 0|$ — vac. vector in some space, $V(t)$ — operators.

Another form of (7):

$$S(p_1,p_2,p_3,p_4) = \int_0^1 x^{-2\langle p_1,p_2\rangle}(1-x)^{-2\langle p_2,p_3\rangle}(x(1-x))^{-1-m^2}\,dx \tag{10}$$

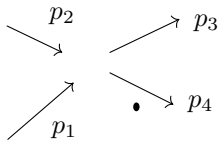
Here



Original source details: GELFAND-P001232

Diagram and typed description.

(11)



A separate ink dot is visible beside the lower-right arrow.

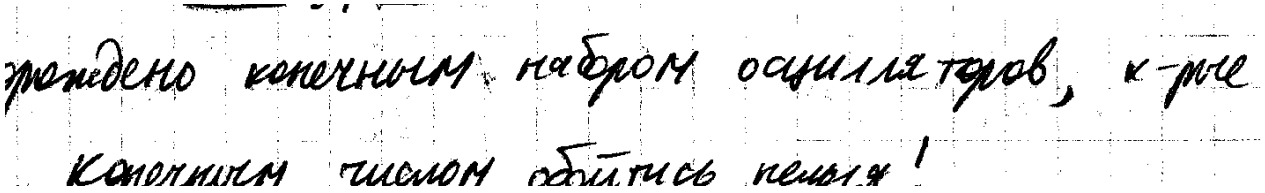
$$p_1 + p_2 + p_3 + p_4 = 0.$$

Similarly one can write amplitudes for a greater number of particles, for example:

$$A(p_1, p_2, p_3, p_4, p_5) \cong \int \int_0^1 x^{\cdots} y^{\cdots} (1-x)^{\cdots} (1-y)^{\cdots} (1-xy)^{\cdots} \, dx \, dy \tag{12}$$

This amplitude has not been calculated in general, only for some special values of the scalar products.

The space is generated by a finite set of oscillators connected as in a string. A finite number will not do!.

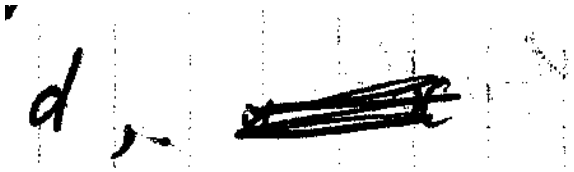


Original source details: GELFAND-P001232

Uncertain reading. The first sentence says “finite”. The second says “A finite number will not do!”

Consider the algebra with generators

$$a_m^\mu, x^\mu, \quad \mu = 0, 1, \ldots, d, \quad \text{[crossed out]} \tag{13}$$



Original source details: GELFAND-P001232

Undeciphered passage. The notation after d in (13) is crossed out and undeciphered.

where d — dim. of the space

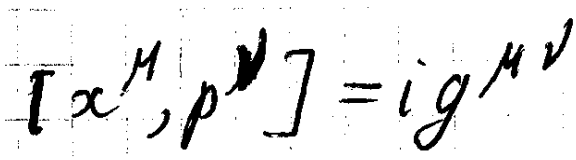
GELFAND-P001233

$$[a_m^\mu, a_n^\nu] = m g^{\mu\nu} \delta_{m,-n} \tag{14}$$

m, n — integers

$$a_0^\mu = p^\mu \tag{15}$$

$$[x^\mu, p^\nu] = i g^{\mu\nu} \tag{16}$$



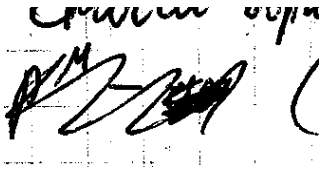
Original source details: GELFAND-P001233

Uncertain reading. The superscript on p in (16) is overwritten and reads ν.

$$[x^\mu, a^\nu_m] = 0, \quad m \neq 0 \tag{17}$$

Meaning of the generators x^μ, a^μ_m:

[Crossed out: p^μ — undeciphered]



Original source details: GELFAND-P001233

Undeciphered passage. The crossed-out continuation after p^μ is undeciphered.

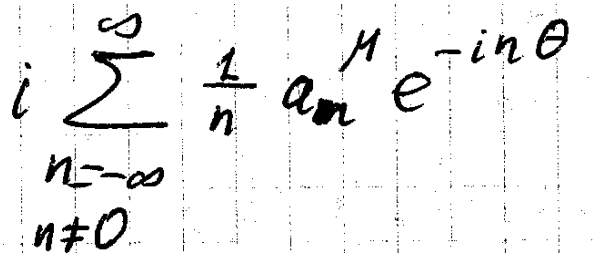
(x^μ, p^μ) — coord. and momentum of the center of gravity of the flying string, a^μ_m — Fourier series expansion of everything else.

Put:

$$P^\mu(\theta) = \sum_{-\infty}^{+\infty} a^\mu_m e^{-im\theta}, \tag{18}$$

(momentum of the point with param. θ)

$$Q^\mu(\theta) = x^\mu + a^\mu_0 \theta + i \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n} a^\mu_m e^{-in\theta}. \tag{19}$$



Original source details: GELFAND-P001233

Uncertain reading. The subscript on a in (19) is overwritten and tentatively reads m. The summation index is n.

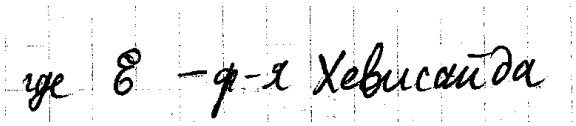
Then the commutation relations hold:

$$[Q_\mu(\theta), P_\nu(\theta')] = 2\pi i g_{\mu\nu} \delta(\theta - \theta') \tag{20}$$

$$[P_\mu(\theta), P_\nu(\theta')] = 2\pi i g_{\mu\nu} \delta'(\theta - \theta') \tag{21}$$

$$[Q_\mu(\theta), Q_\nu(\theta')] = -\pi i g_{\mu\nu} \varepsilon(\theta - \theta'), \tag{22}$$

where ε — Heaviside fn.



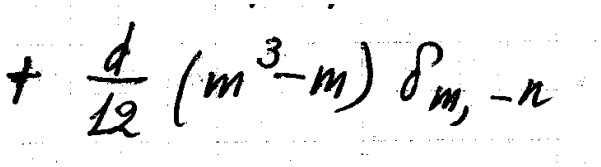
Original source details: GELFAND-P001233

Uncertain reading. The manuscript gives the name “Heaviside” without a definition of ε.

$$L_m \stackrel{\text{def}}{=} \frac{1}{4\pi}\int_0^{2\pi} e^{im\theta} : P^2(\theta) : \, d\theta \tag{23}$$

Then one obtains a representation of the Virasoro algebra

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{d}{12}(m^3 - m)\delta_{m,-n} \tag{24}$$



Uncertain reading. The numerator of the central coefficient reads d.

Original source details: GELFAND-P001233

Exercise: verify this.

Incidentally,

$$L_0 = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_{-n}a_n \tag{25}$$

GELFAND-P001234

[source number: 17]

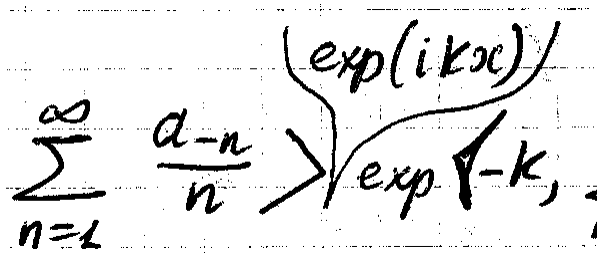
$$2L_m = : \sum_{r+s=m} a_r a_s : \tag{26}$$

Vertex operator: put

$$V(k,\theta) = \exp\langle ik, Q(\theta)\rangle, \tag{27}$$

where k — a vector, then

$$V(k) = V(k,0) = \exp\left\langle k, \sum_{n=1}^{\infty}\frac{a_{-n}}{n}\right\rangle \exp(ikx)\exp\left\langle -k, \sum_{n=1}^{\infty}\frac{a_n}{n}\right\rangle \tag{28}$$



Uncertain reading. The factor $\exp(ikx)$ is inserted above the line between the two exponentials.

Original source details: GELFAND-P001234

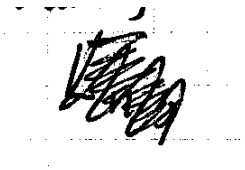
and

$$V(k,\theta) = \exp(i\theta L_0) V(k) \exp(-i\theta L_0). \tag{29}$$

Next,

[Crossed out before (30)]

$$[L_n - L_0, V(k)] = \frac{nk^2}{2} V(k). \tag{30}$$



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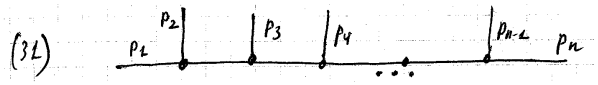
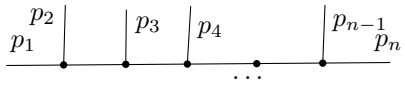
Undeciphered passage. The notation before the commutator in (30) is crossed out and undeciphered.

Consider a series of interactions:



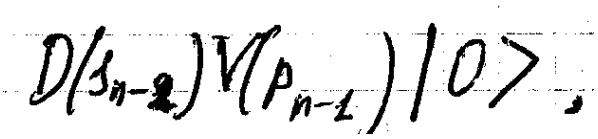
Original source details: GELFAND-P001234

Diagram and typed description.

(31)


and from such a picture construct the amplitude:

$$\langle 0|V(p_2)D(s_2)V(p_3)D(s_3)\dots D(s_{n-2})V(p_{n-1})|0\rangle, \tag{32}$$



Original source details: GELFAND-P001234

Uncertain reading. The overwritten numeral in the last D reads 2, giving the argument s_{n-2}.

where p — the same as k,

$$\begin{aligned} s_2 &= (p_2 + p_1)^2 \\ s_3 &= (p_3 + p_2 + p_1)^2, \\ &\dots \end{aligned} \tag{33}$$

$$D(s) = \int_0^1 x^{L_0 - \alpha(s) - 1} (1-x)^{\alpha(0)-1}\, dx, \tag{34}$$

where $\alpha(s)$ — linear fn. (Regge trajectory).

(32) — dual amplitude. It should agree with A, when there are 4 vertices.

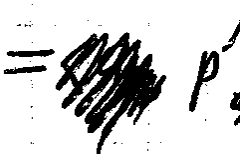
Expand $V(k,\theta)$ in a Fourier series:

$$V(k,\theta) = \sum V_n(k) e^{-in\theta}. \tag{35}$$

There are operators a_i^μ and a_0^μ and vacuum vectors $|0,p\rangle$, satisfying the condition

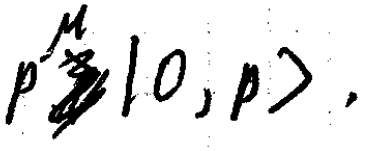
$$a_i^\mu |0,p\rangle = 0, \quad \text{if } i > 0, \tag{36}_i$$

$$a_0^\mu |0,p\rangle = \text{[crossed out] } p^\mu \text{ [crossed out] } |0,p\rangle. \tag{36}_0$$



Uncertain reading. crossed-out writing before p^μ

Original source details: GELFAND-P001235



Uncertain reading. crossed-out sign after p^μ

Original source details: GELFAND-P001235

(the vacua are indexed by momentum vectors)

We obtain the space

$$\bigoplus_p H(p) \tag{37}$$

The vertex operator acts as follows:

$$V_n(k) : H(p) \longrightarrow H(p+k). \tag{38}$$

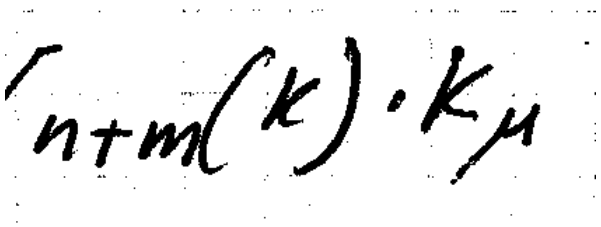
Commutation relations:

[Crossed out: illegible]

Undeciphered passage. The crossed-out writing under “Commutation relations” is illegible.

Original source details: GELFAND-P001235

$$[a_m^\mu, V_n(k)] = V_{n+m}(k) \cdot k_\mu \tag{39}$$



Uncertain reading. written as k_μ, with a lower index

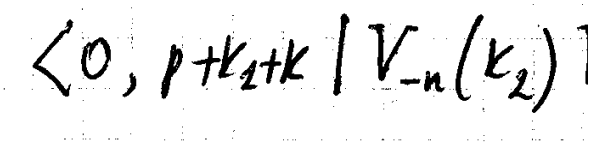
Original source details: GELFAND-P001235

[Crossed out: $V_n(k_1)|0,p\rangle$ / $V_1(k_1)|0,p\rangle$?] Set

Uncertain reading. $V_n(k_1)|0,p\rangle$ / $V_1(k_1)|0,p\rangle$

Original source details: GELFAND-P001235

$$\langle 0, p+k_1+k|V_{-n}(k_2)V_n(k_1)|0,p\rangle = A_n \qquad \text{(these are numbers)} \tag{40}$$



Uncertain reading. k without a subscript

Original source details: GELFAND-P001235

Then

$$\sum A_n z^n = (1-z)^{\langle k_1,k_2\rangle}. \tag{41}$$

This follows by a direct computation. The reader is asked to verify (41) as an exercise.

GELFAND-P001236

[source number: 26]

18 Nov. 1985 Gelfand Seminar

B. Feigin (continued).

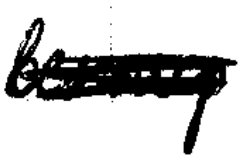
The L_i generate the Virasoro algebra. Let there be a large space on which this algebra acts.

Eigenvectors of L_0 are weight vectors.

By the commutation relations, L_i acts on the space of weight vectors.

(see (24) on p. 17, left)

Highest [crossed out: vector?] weight: a vector that cannot be raised any further



Uncertain reading. vector?

Original source details: GELFAND-P001236

$$\widetilde{\mathcal{H}} = \{v \mid L_0 v = v, \quad L_i v = 0 \text{ for } i > 0\}. \qquad \text{(weight 1)} \tag{1}$$

$$\overline{\mathcal{H}} = \{v \in \widetilde{\mathcal{H}}, \quad v = L_i v', \ i < 0\}, \tag{2}$$

[At right:] Here v' is an arbitrary weight vector from the large space.

(the space of "spurions," a term used by physicists some 15 years ago)

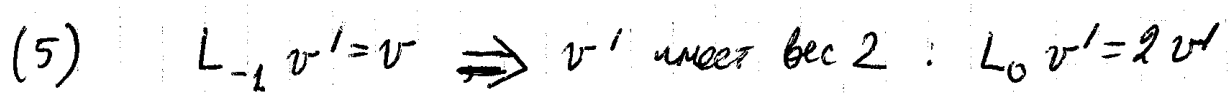
$$\mathcal{H} \stackrel{\text{def}}{=} \widetilde{\mathcal{H}}/\overline{\mathcal{H}} \tag{3}$$

$$\prod_{i=1}^{N} a_{m_i}^{\mu_i} |0, p\rangle \tag{4}$$

[At right:] (space of states)

a large module over the Virasoro algebra

$$L_{-1} v' = v \quad \Rightarrow \quad v' \text{ has weight } 2: \quad L_0 v' = 2v' \tag{5}$$



Original source details: GELFAND-P001236

Uncertain reading. $L_{-1} v' = v; \quad L_0 v' = 2v'$

The aim is to quotient by the diffeomorphism group throughout the space, but a central extension acts, and the centre acts nontrivially.

Another procedure is proposed for obtaining "invariant" vectors. They are invariant under one half, then we quotient by the action of the others and discard them.

The Virasoro algebra disappears, leaving the space of invariants.

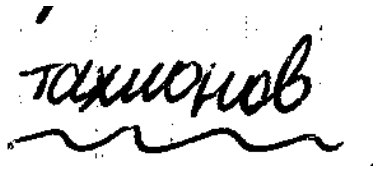
We want to

I. Construct a space generated by $d - 2$ oscillators.

II. Prove that all vectors in $\widetilde{\mathcal{H}}/\overline{\mathcal{H}}$ have norm > 0.

(this does not hold in the original d-dimensional space)

(absence of tachyons[?])



Uncertain reading. tachyons

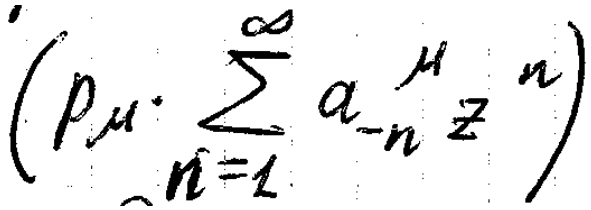
Original source details: GELFAND-P001236

All of this is achieved in dimension 26.

GELFAND-P001237

Vertex operators

$$X(p,z) = \exp\left(p_\mu \cdot \sum_{n=1}^{\infty} a^\mu_{-n} z^n\right) D_p \exp\left(-p_\mu \sum_{n=1}^{\infty} \frac{a^\mu_n}{n} z^{-n}\right), \tag{6}$$



Uncertain reading. $a^\mu_{-n} z^n$ without division by n

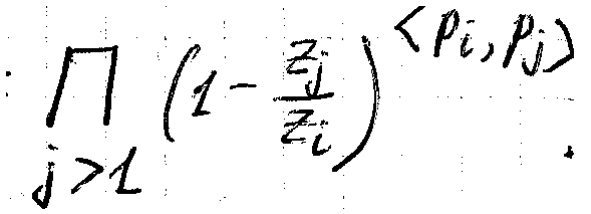
Original source details: GELFAND-P001237

where the operators D_p are defined by the formulas

$$D_p|0,k\rangle = |0,k+p\rangle, \quad [D_p, a^\mu_n] = 0,\ n \neq 0. \tag{7}$$

Correlation function

$$\begin{aligned}&\langle 0, k+\textstyle\sum p_i | X(z_1,p_1)\dots X(z_n,p_n)|0,k\rangle \\ &\quad = \prod_{j>1}\left(1-\frac{z_j}{z_i}\right)^{\langle p_i,p_j\rangle}.\end{aligned} \tag{8}$$



Uncertain reading. the index under the product reads $j > 1$

Original source details: GELFAND-P001237

Expanding $X(z,p)$ in a series gives

$$X(z,p) = \sum X_i(p) z^i \tag{9}$$

(this defines the operators $X_i(p)$)

We want to construct 24 oscillators.

Next we will construct the operators

$$\int X(p,z)\frac{dz}{z}, \tag{10}$$

which commute with the Virasoro algebra when $\langle p,p\rangle = 2$.

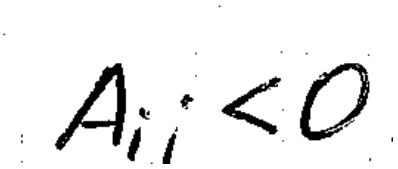
$$\begin{aligned} &\text{In } \mathbb{R}^{26} \text{ we want to have } p_1,\dots,p_n: \quad \langle p_i,p_i\rangle = 2 \\ &\langle p_i,p_j\rangle = A_{ij} \in \mathbb{Z}, \quad A_{ij} < 0. \end{aligned} \tag{11}$$

[In the left margin: crossed-out vertical sign?]

Uncertain reading. crossed-out vertical sign

Original source details: GELFAND-P001237

Uncertain reading. $A_{ij} < 0$



Original source details: GELFAND-P001237

Then the operators

$$e_i = \int X(p_i,z)\frac{dz}{z}, \quad f_i = \int X(-p_i,z)\frac{dz}{z} \tag{12}$$

generate an action of the Kac–Moody algebra with Cartan matrix A_{ij} on the space. We must also set

$$h_i|0,k\rangle = \langle k,p_i\rangle|0,k\rangle \tag{13}$$

Read the article by Macdonald (the beginning), Sem. Bourbaki.

Use it to study Lie algebras.

[In the left margin: I.M.G.] The largest finite simple group has $\sim 10^{54}$ elements. It is related to strings!

[Added at right:] (Monster “Moonshine”)

Moonshine

(there is work by Frenkel, Lepowsky and someone else)

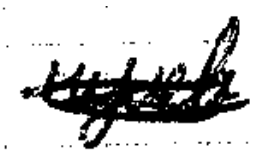
Somehow one obtains a fundamental representation of the Kac–Moody algebra.

GELFAND-P001238

[source number: 27]

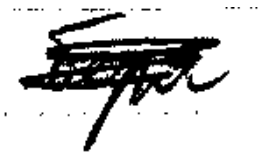
In $\mathbb{R}^{26}$ consider the metric $\langle -1, 1, 1, \ldots, 1\rangle$,

[Crossed out: null?[?]] vector of length 0, [crossed out: illegible]



Uncertain reading. null?

Original source details: GELFAND-P001238



Undeciphered passage. The crossed-out word after “vector of length 0” is illegible.

Original source details: GELFAND-P001238

[Before the next line: two vertical marks, the right one resembling a brace[?]] 24 oscillators

Uncertain reading. two vertical marks, the right one resembling a brace

Original source details: GELFAND-P001238

(Review by D. (?) Schwarz, 1973).

Uncertain reading. D. (?)

Original source details: GELFAND-P001238

— . —

Take $\mathbb{R}^d$ with an indefinite metric and a given vector p with $\langle p, p\rangle = 0$.

[Under “metric”:] (Minkowski, one minus sign)

Set

$$U_n(v) = \int v(z) X(np, z) \frac{dz}{z}, \tag{14}$$

where $v \in \mathbb{R}^d$, $v = \sum v_i e_i$, $v(z) = \sum v_\mu a_n^\mu z^n$

Let

$$v^1, v^2, \ldots, v^{d-2} \text{ be such that } \langle v^i, p\rangle = 0 \tag{15}$$

[“and” inserted over the crossed-out sign:] and [crossed out: crossed-out sign under the inserted “and”[?]]

Uncertain reading. crossed-out sign under the inserted “and”

Original source details: GELFAND-P001238

$v^1, v^2, \ldots, v^{d-2}, p$ are a basis of the space [crossed out: illegible] $\{x : \langle x, p\rangle = 0\}$.

Original source details: GELFAND-P001238

Undeciphered passage. The crossed-out word before the opening brace is illegible.

Set

$$b_n^\mu = U_n(v^\mu); \tag{16}$$

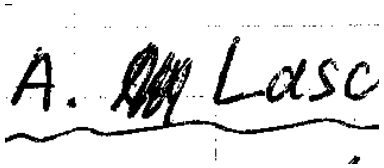
The b_n^μ form a Heisenberg algebra $\Rightarrow$ oscillators.

They generate the whole space.

There is also a review by Green and Schwarz (a volume of Physics Reports).

A. Zelevinsky Schubert cells [inserted above the line: (and polynomials)] and the cohomology of flag varieties.

On the work of A. [crossed out: crossed-out initial?] Lascoux, M.-P. Schützenberger



Original source details: GELFAND-P001238

Uncertain reading. crossed-out initial

(Schubert cells and the cohomology of Grassmannians and flag varieties)

Continuation of work by Bernstein, I.M.G. and S.I.G., Uspekhi Mat. Nauk, 1973, 28(3), 3–26.

X is the set of complete flags in V^n (flags of subspaces).

[Below X, above (1): $\|$, two vertical strokes?]

Original source details: GELFAND-P001238

Uncertain reading. $\|$, two vertical strokes

$$\{V^1 \subset V^2 \subset \ldots \subset V^n\} \tag{1}$$

The cohomology is described in two ways.

1°. Borel

$$H^*(X, \mathbb{Q}) = \mathbb{Q}[x_1, \ldots, x_n]/J$$

The ideal J is generated by symmetric polynomials with zero constant term.

2°. The Schubert cells (c_w) form a basis in homology $H_*(X, \mathbb{Q})$. w is a permutation of the numbers from 1 to n.

$$\begin{matrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{matrix}$$

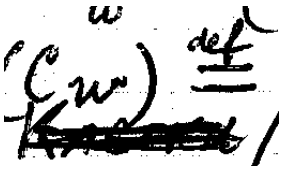
Consider a fixed complete flag

$$F = (\langle e_1\rangle,\ \langle e_1, e_2\rangle,\ \langle e_1, e_2, e_3\rangle, \dots,\ \langle e_1, e_2, \dots, e_n\rangle).$$

Act on it by the permutation w

$$F_w = (\langle e_{w(1)}\rangle,\ \langle e_{w(1)}, e_{w(2)}\rangle, \dots) = (\langle e_{i_1}\rangle,\ \langle e_{i_1}, e_{i_2}\rangle, \dots).$$

[Crossed out: Cell?[?]. Inserted above the crossing-out: $(C_w) \overset{\text{def}}{=}$.]



Uncertain reading. Cell?

Original source details: GELFAND-P001239

[Underlined: (Orbit of the flag F_w under the linear transformations preserving F).] [Crossed out: illegible notation[?].]

Uncertain reading. illegible notation

Original source details: GELFAND-P001239

$$\dim_{\mathbb{C}} c_w = \ell(w) \text{ is the number of inversions in } w.$$

The dual basis in cohomology

$$P_w \in H^{2\ell(w)}(X; \mathbb{Q}).$$

B.–G.–G. found explicit formulas for P_w in the Borel presentation, polynomials (!). [Under the first occurrence of "polynomials": $(\mathrm{mod}\ J)$.] [Crossed out: illegible[?].]

Uncertain reading. illegible

Original source details: GELFAND-P001239

They have now been lifted to actual polynomials. Their degrees are

$$\deg P_w = \ell(w).$$

$$P_{\mathrm{id}} = 1.$$

$$w_0 = (n, n-1, \dots, 1).$$

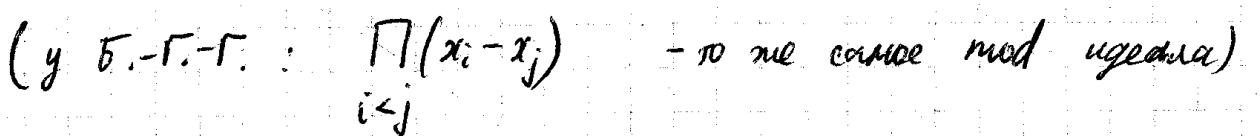
Then

$$\deg P_{w_0} = \frac{n(n-1)}{2}.$$

The choice is nonunique. Take

$$P_{w_0} = x_1^{n-1} x_2^{n-2} \dots x_{n-1}.$$

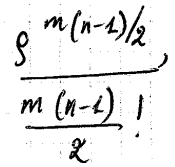
(In B.–G.–G. $\prod_{i<j}(x_i - x_j)$ is the same modulo the ideal.)



Original source details: GELFAND-P001239

Uncertain reading. The product has no normalization factor in the manuscript.

Or $\frac{\rho^{m(n-1)/2}}{\left(\frac{m(n-1)}{2}\right)!}$, where $\rho = (n-1)x_1 + (n-2)x_2 + \dots$



Uncertain reading. The letter before $(n-1)$ reads m both in the exponent and under the fraction bar.

Original source details: GELFAND-P001239

The rest are given by

$$P_w = A_{w_0 w^{-1}} P_{w_0},$$

GELFAND-P001240

[source number: 28]

A_w is the operator corresponding to the permutation

$$w = (i_1, i_1 + 1)(i_2, i_2 + 1) \dots (i_{\ell(w)}, i_{\ell(w)+1})$$

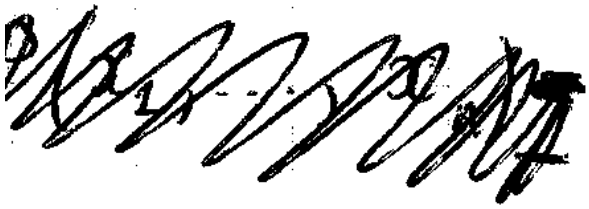
(a product of inversions).

Uncertain reading. In the last factor $+1$ is written at subscript height, $i_{\ell(w)+1}$.

Original source details: GELFAND-P001240

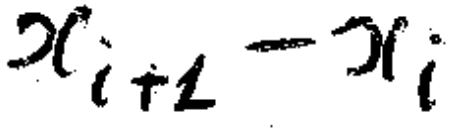
$$A_{(i,i+1)} = \frac{p(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n) - p(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n)}{x_{i+1} - x_i}.$$

[Crossed out after the equals sign, before the fraction: $p(x_1, \ldots$; the rest is illegible.]



Uncertain reading. $p(x_1, \ldots$; the rest is illegible

Original source details: GELFAND-P001240

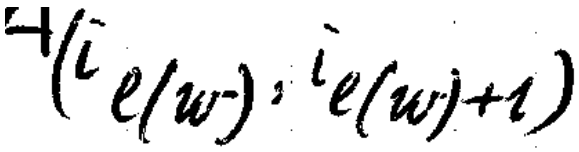


Uncertain reading. The manuscript denominator is $x_{i+1} - x_i$.

Original source details: GELFAND-P001240

The operator

$$A_w = A_{(i_1, i_1+1)} \ldots A_{(i_{\ell(w)}, i_{\ell(w)+1})}$$



Uncertain reading. In the last operator index $+1$ is at the height of $\ell(w)$.

Original source details: GELFAND-P001240

is well defined (independent of the decomposition into a product of inversions). This is the operator used in the formula.

Symmetry condition.

P_w is symmetric in x_i, x_{i+1} if $(i, i+1)$ is not an inversion of w.

GELFAND-P001241

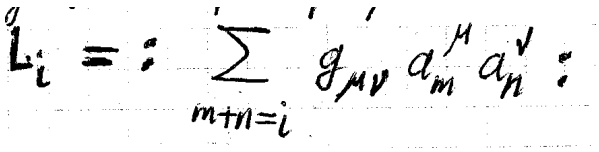
[source number: 35]

25 Nov. 1985 Gelfand Seminar.

Feigin (continued). (On strings).

Generators of the Virasoro algebra

$$L_i =: \sum_{m+n=i} g_{\mu\nu} a_m^\mu a_n^\nu : \tag{1}$$



Uncertain reading. there is no prefactor on the sum

Original source details: GELFAND-P001241

The colons : : make the operator well defined and only add a constant.

I.M.G. What is required for the operation : : to exist? Here it gives rise to the Virasoro algebra. What if we take a lattice or a torus? What happens on the sphere?

$$[L_i, L_j] = (i-j)L_{i+j} + \delta_{i,-j}\frac{i^3 - i}{12}\,c \tag{2}$$

$$p^\mu(t) = \sum a_m^\mu e^{-imt}, \qquad q^\mu(t) = \sum_{m\neq 0} \frac{a_m^\mu}{m} e^{-imt} + q_0^\mu \tag{3}$$

$$\begin{aligned} [x^\mu, a_m^\nu] &= 0, \quad m \neq 0, \\ [x^\mu, a_0^\nu] &= g^{\mu\nu}, \\ x^\mu(t) &= x^\mu + \sum_{n\neq 0} \frac{1}{n}\left(a_n^\mu e^{-int}\right). \end{aligned}$$

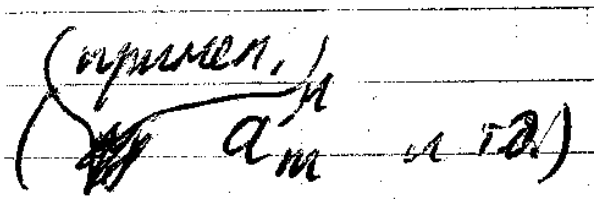
I.M.G. In the classical theory

$$L_n = e^{int}\frac{d}{dt}.$$

Quantum mechanics corresponds to a one-dimensional extension of the classical theory. Compare $[\frac{d}{dx}, x] = 1$.

(4) $\mathcal{H}_p$ is the module generated by the highest vector $|p\rangle$.

[At right: ([crossed out] a_m^μ etc.). "(apply)" is inserted above the crossed-out sign, with a line to the insertion point.]

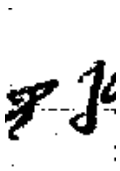


Uncertain reading. illegible crossed-out notation before a_m^μ

Original source details: GELFAND-P001241

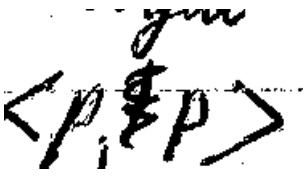
[Between the lines:] This is a module over the Virasoro algebra.

(5) Its structure [crossed out] depends only on $\langle p,$ [illegible] $p\rangle$; by definition



Uncertain reading. small crossed-out sign before "depends"

Original source details: GELFAND-P001241



Uncertain reading. an extra overwritten sign between the comma and the second p

Original source details: GELFAND-P001241

$$a_m^\nu|p\rangle = 0, \quad m > 0, \qquad a_0^\nu|p\rangle = p^\nu|p\rangle \tag{6}$$

(5) $\mathcal{H}_p \cong \mathcal{H}_{p'}$, if $\langle p, p\rangle = \langle p', p'\rangle$. || Denote it by $\mathcal{H}_{\langle p,p\rangle}$.

Consider

$$\bigoplus_{\langle p,p\rangle\in\mathbb{R}} \mathcal{H}_{\langle p,p\rangle} \tag{7}$$

GELFAND-P001242

Define the space of physical states from [crossed out] module (7), as was done [crossed out] above (p. (26)).

Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out sign before “module”

Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out sign before “above”

The construction depends [inserted above the line: only] on the [crossed out] action of the Virasoro algebra.

Original source details: GELFAND-P001242

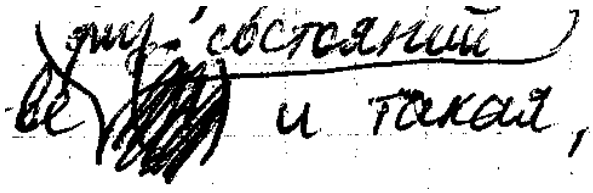
Uncertain reading. crossed-out word, possibly “definition”

(8) Theorem. (on ghosts).

1) There is a system of oscillators b^{μ}_{m}, $m \neq 0$, $\mu = 1, 2, \ldots, 24$ with standard commutation relations [undeciphered sign] on the space of [crossed out, inserted above the line: physical states], such that

Original source details: GELFAND-P001242

Uncertain reading. sign resembling Cyrillic д



Original source details: GELFAND-P001242

Uncertain reading. illegible crossed-out notation beneath the insertion “physical states”

$$[L_i, b_m^\mu] = 0.$$

The b_m^μ can be constructed from a.

2) $\exists$ a vacuum vector in the space of physical states [inserted above: $\widetilde{\mathcal{H}}$] $|0\rangle$, and $\widetilde{\mathcal{H}}$ is generated by the vectors $|0\rangle$ and the operators b_n^ν. This holds only for $\underline{\underline{d = 26}}$.

3) In $\widetilde{\mathcal{H}}$ there is a natural quadratic form inherited from space (7). It is positive definite on $\widetilde{\mathcal{H}}$.

$$X(p,z) = \exp\left(p_\mu \sum_{n=1}^{\infty} \frac{a_{-n}^\mu}{n} z^n\right) D_p \exp\left(-p_\mu \sum_{n=1}^{\infty} \frac{a_n^\mu}{n} z^{-n}\right), \tag{9}$$

Here D_p is defined by the formula

$$D_p|k\rangle = |k+p\rangle, \tag{10}$$

[Crossed out: and in this case.] Moreover

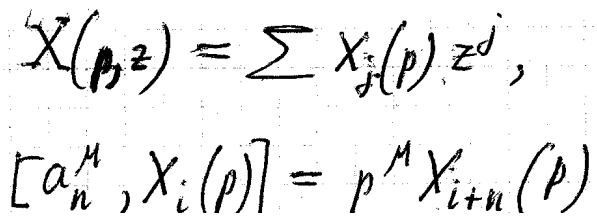
$$[D_p, a_n^\mu] = 0, \quad n \neq 0. \tag{11}$$

Commutation relations for $X(p,z)$ and a_n^μ. If

$$X(p,z) = \sum X_j(p) z^j,$$

then

$$[a_n^\mu, X_i(p)] = p^\mu X_{i+n}(p). \tag{12}$$



Original source details: GELFAND-P001242

Uncertain reading. The expansion reads z^j, and (12) reads $X_{i+n}(p)$.

Representation of the current algebra.

A space of dimension d. In $\mathbb{R}^d$ consider a lattice Q (a discrete subgroup). Assume that Q is integral and unimodular, so there is a basis

$$e_1, \dots, e_d, \quad \text{such that } \langle e_i, e_j \rangle \text{ is integral and } \det \langle e_i, e_j \rangle = 1. \tag{13}$$

Even lattice

$$\forall \gamma \in Q: \quad \langle \gamma, \gamma \rangle \text{ is even.} \tag{14}$$

[Narrow strip of the adjacent leaf at the right edge of the frame.]

Uncertain reading. From top to bottom, four clipped opening parentheses, a short vertical stroke and one more opening parenthesis are visible. Their contents are outside the frame.

Original source details: GELFAND-P001242

GELFAND-P001243

[source number: 36]

(15) Root of a lattice is a vector $\gamma \in Q$, such that $\langle \gamma, \gamma \rangle = 2$.

(16) [Crossed out] Such lattices occur from dimension [crossed out] 8 [crossed out], more precisely, [crossed out] only in dimensions divisible by 8.

Uncertain reading. illegible crossed-out mark after (16)

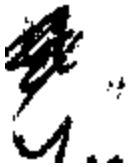
Original source details: GELFAND-P001243

Uncertain reading. illegible crossed-out sign before 8

Original source details: GELFAND-P001243

Uncertain reading. illegible crossed-out sign after 8

Original source details: GELFAND-P001243



Uncertain reading. illegible crossed-out sign before “only”

Original source details: GELFAND-P001243

Exercise Construct such a lattice in $\mathbb{R}^8$.

— . —

Q is a lattice (for now, it need not be even).

Incidentally [Crossed out] The root lattices of groups in series A, D, E have property (14).

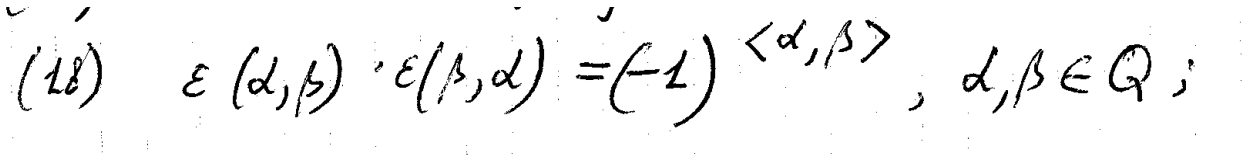
Uncertain reading. crossed-out sign before “root lattices”

Original source details: GELFAND-P001243

Consider the function

$$\varepsilon : Q \times Q \longrightarrow \{\pm 1\} \tag{17}$$

$$\varepsilon(\alpha,\beta)\cdot\varepsilon(\beta,\alpha) = (-1)^{\langle\alpha,\beta\rangle}, \qquad \alpha,\beta \in Q; \tag{18}$$



Original source details: GELFAND-P001243

Uncertain reading. In (18) the condition is written for all $\alpha, \beta \in Q$.

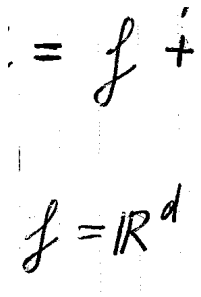
$$\varepsilon(\alpha,\alpha) = -1 \quad \text{if } \langle\alpha,\alpha\rangle = 2. \tag{19}$$

Δ is the set of roots of the lattice Q, i.e. $\Delta = \{\gamma \in Q \mid \langle\gamma,\gamma\rangle = 2\}$.

Let the quadratic form $\langle\ ,\ \rangle$ be positive definite. Introduce the Lie algebra

$$\mathfrak{g} = f + \sum_{\alpha\in\Delta} \mathbb{C}x_\alpha, \tag{20}$$

where $f = \mathbb{R}^d \otimes \mathbb{C} = \mathbb{C}^d$.



Uncertain reading. the two matching looped letters read as f

Original source details: GELFAND-P001243

Define the Lie algebra structure on $\mathfrak{g}$ by the formulas

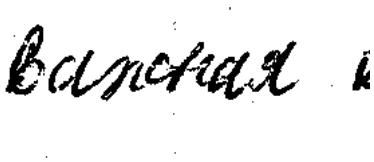
$$[h, x_\alpha] = \langle h,\alpha\rangle x_\alpha; \tag{21}_1$$

$$[x_\alpha, x_\beta] = 0 \quad \text{if } \alpha+\beta \notin \Delta\cup\{0\}; \tag{21}_2$$

$$[x_\alpha, x_\beta] = \varepsilon(\alpha,\beta)x_{\alpha+\beta} \quad \text{if } \alpha+\beta\in\Delta; \tag{21}_3$$

$$[x_\alpha, x_{-\alpha}] = -\alpha. \tag{21}_4$$

This is an important construction of finite-dimensional simple algebras.



Uncertain reading. reads “важная” (important)

Original source details: GELFAND-P001243

Now let the construction above have $p \in Q$, $|p\rangle$ be a vacuum vector, and let $\gamma \in \Delta$. Consider the vertex operator

$$X^\varepsilon(\gamma, z) = X(\gamma, z)\cdot\varepsilon_\gamma; \tag{22}_1$$

where

$$\varepsilon_\gamma|p\rangle = \varepsilon(\gamma,p)|p\rangle, \tag{22}_2$$

$$[\varepsilon_\gamma, a_m^\mu] = 0, \quad m \neq 0. \tag{22}_3$$

Now introduce

$$X^\varepsilon(\gamma, z) = \sum X_i^\varepsilon(\gamma) z^i. \tag{23}$$

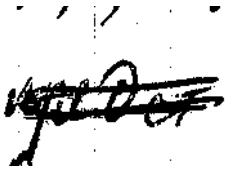
In $\mathbb{R}^{24}$ there is a special (exceptional) lattice that occurs here and in the theory of finite simple groups. There are 26 sporadic simple groups in all.

(24) $\{X_n^\varepsilon(\gamma), a_n^\mu\}$ generate a [inserted above the line: projective] representation of the current algebra on the circle.

$$[X_n^\varepsilon(\gamma_1), X_m^\varepsilon(\gamma_2)] = 0, \quad \text{if } \langle \gamma_1, \gamma_2 \rangle > 0 \tag{25}_1$$

$$[X_n^\varepsilon(\gamma_1), X_m^\varepsilon(\gamma_2)] = \varepsilon(\gamma_1, \gamma_2) X_{n+m}^\varepsilon(\gamma_1 + \gamma_2), \quad \text{if } \langle \gamma_1, \gamma_2 \rangle = -1 \tag{25}_2$$

and other [crossed out] relations can be written down.



Uncertain reading. possibly an abbreviation for “similar”

Original source details: GELFAND-P001244

In $\mathbb{R}^{26}$ choose a vector c, such that $\langle c, c \rangle = 0$, and consider the integrals

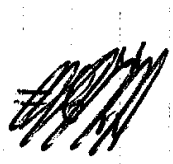
$$\int a(z) X(nc, z) \frac{dz}{z} \tag{26}$$

[A superscript on $a(z)$ is crossed out.]

Uncertain reading. possibly μ

Original source details: GELFAND-P001244

[Writing after the integral is crossed out.]



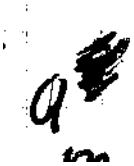
Uncertain reading. the start resembles $=$, and the rest is illegible

Original source details: GELFAND-P001244

Here

$$a(z) = \sum a_m z^m, \tag{27}$$

[A superscript on a_m is crossed out.]



Uncertain reading. possibly μ

Original source details: GELFAND-P001244

where

$$a_m = \sum \alpha_\mu a_m^\mu, \tag{27}'$$

(28) and the vector $(\alpha_1, \dots, \alpha_{26})$ is orthogonal to c.

Set

$$b_n^\mu = \int a^\mu(z) X(nc, z) \frac{dz}{z} \tag{29}$$

$$\langle \alpha, c \rangle = 0. \tag{30}$$

Choose $\alpha^1, \alpha^2, \dots, \alpha^{24}$, so that together with c they form a basis in the space of vectors orthogonal to c. Substituting them in $(27)'$ gives the family a^μ and then b_n^μ.

GELFAND-P001245

[source number: 56]

16 Dec. 1985 Gelfand Seminar

S. P. Novikov.

Soliton theory of the two-dimensional Schrödinger operator.

$n = 2$

$$L = -\partial\bar{\partial} + U = -\Delta + U \tag{1}$$

there are no Lax pairs

$$\frac{\partial L}{\partial t} = [A, L] \quad \Rightarrow \quad \text{dependence on } t \text{ is a one-parameter group of motions of } \mathbb{R}^2 \tag{2}$$

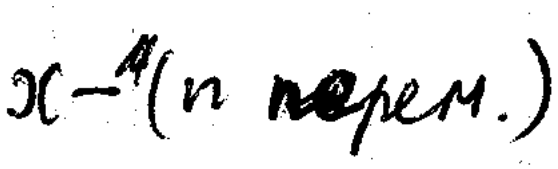
1976 1) Manakov, Uspekhi Mat. Nauk

2) Dubr., Krich., Nov., Dokl. Akad. Nauk.

Difference from the one-dimensional case.

$$U(x) \to 0 \quad \text{as } |x| \to \infty; \quad \Rightarrow \quad \text{there is a scattering amplitude } S(k_1, k_2), \quad |k_1| = |k_2|. \tag{3}$$

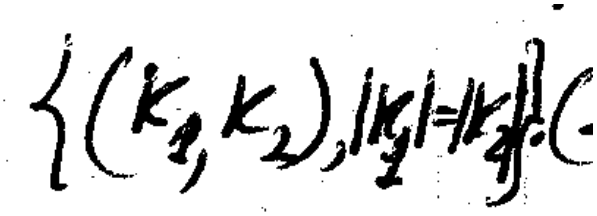
x — [?] (n variables).



Uncertain reading. horizontal stroke after x and an illegible raised sign

Original source details: GELFAND-P001245

[At right:] $\{(k_1, k_2), |k_1| = |k_2|\}$ $(2n-1)$ variables



Uncertain reading. the braces enclose a pair and equality of norms

Original source details: GELFAND-P001245

$$n = 2n - 1 \Leftrightarrow n = 1.$$

An overdetermined problem.

$\underline{n = 2}$ $\Rightarrow$ restriction to fixed $|k_1| = |k_2|$ — also two variables.

Require

$$\begin{cases} \left(\dfrac{\partial L}{\partial t} - [A, L] \right) \psi = 0, \\ L\psi = 0 \end{cases} \qquad (\text{or } L\psi = \varepsilon_0 \psi) \tag{4}$$

or

$$\frac{\partial L}{\partial t} = \underbrace{[A, L] + BL}_{\| \; C_1 L + L C_2}, \qquad A, B \text{ are differential operators.} \tag{5}$$

The inverse problem is solved with the same effectiveness.

Equation (5) is called the two-dimensional analogue of the Lax equation. For $n = 2$ it is now consistent.

$z = x + iy$, $\bar{z}$ and also t.

One should start with periodic problems (or quasiperiodic ones).

Γ is a Riemann surface of genus $g < \infty$, ∞_1, ∞_2 are two points on Γ,

$P_1 + \cdots + P_g$ is a divisor [abbreviation unclear]; w_1, w_2 are local coordinates near ∞_1, ∞_2,

(P_j are other points, $\neq \infty_k$ and distinct from one another), $k_j = w_j^{-1}$.

Uncertain reading. abbreviation before “divisor”, resembling Cyrillic «вхд.»

Original source details: GELFAND-P001245

(6) Seek an eigenfunction $\Psi(x, y, P)$ with the following properties

[Γ is written below P, with an arrow $\uparrow$ pointing to P.]

1) ψ is meromorphic in P away from ∞_1 and ∞_2. [At right: (x, y and possibly t are parameters).]

2) g poles P_j, independent of (x, y)

3)

$$\psi \sim c_1(x,y)e^{ik_1 z}(1+O(w_1)), \qquad \text{near } \infty_1$$
$$\psi \sim c_2(x,y)e^{ik_2 \bar{z}}(1+O(w_2)) \qquad \text{near } \infty_2$$

(7) Gauge transformations

$$\begin{matrix} c_1 \to cc_1 \\ c_2 \to cc_2 \end{matrix} \quad \Big| \quad c = c(x,y)$$

change nothing.

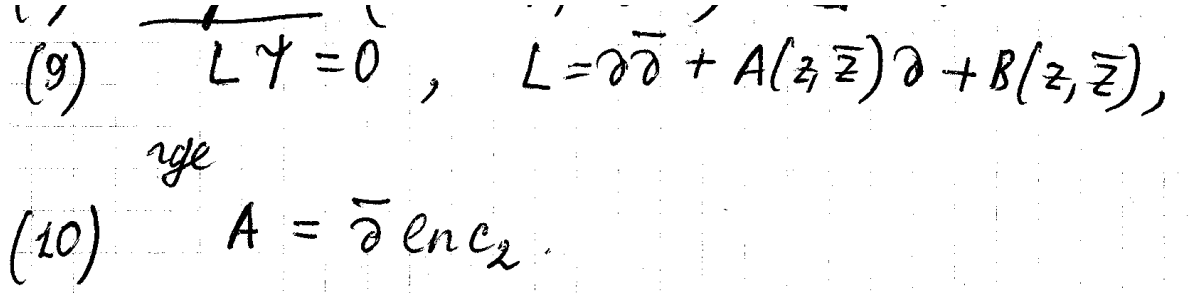
Take $c_1 \equiv 1 \Rightarrow$ it then becomes unique.

(8) Theorem (DKN, 1976) $\exists L$:

$$L\psi = 0, \qquad L = \partial\bar{\partial} + A(z,\bar{z})\partial + B(z,\bar{z}), \tag{9}$$

where

$$A = \bar{\partial} \ln c_2. \tag{10}$$



Original source details: GELFAND-P001246

Uncertain reading. In (9) the derivative after A is ∂ without a bar, and (10) reads $A = \bar{\partial} \ln c_2$.

(11) A, B are quasiperiodic for a Riemann surface in general position.

(12) The magnetic field must have [crossed out] zero flux through the cell (parasitic).

Original source details: GELFAND-P001246

Uncertain reading. crossed-out writing before "zero"

(uninteresting)

(13) Cherednik 1980, Dokl. Akad. Nauk

Veselov, Novikov, Dokl. Akad. Nauk, 1984 (end)

Sufficient conditions [inserted above, with a line to the point after “conditions”: for absence of the first-order term], later found to be necessary. [Below “found”: an arrow pointing left ⟵, a dot and a horizontal stroke.]

The Bloch functions form an n-dimensional complex set.

Consider them at one energy level This is the Fermi surface (complex).

(14) L is called finite-gap [below the line: (or algebro-geometric)] if this surface has finite genus.

P labels the eigenfunction. The P_j are a set of poles (in $\mathbb{C}$, off the spectrum).

∞_j, $j = 1, 2$, are for the operator with constant coefficients.

[Narrow strip of the adjacent leaf at right.]

Uncertain reading. From top to bottom, there is a short vertical stroke, an opening parenthesis, a small clipped mark, another opening parenthesis and a short vertical stroke.

Original source details: GELFAND-P001246

GELFAND-P001247

[source number: 57]

(15) Data for the inverse problem

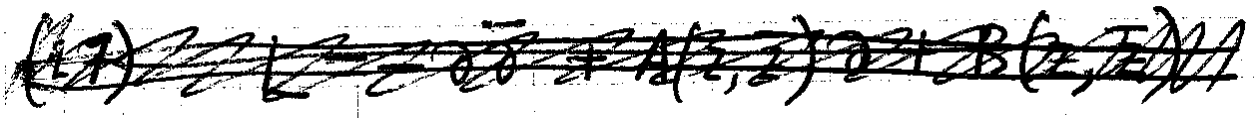
$$\Gamma, \infty_1, \infty_2, w_1, w_2; \qquad D = D_1 + \cdots + D_g$$

construct from them ψ.

(16) Difficulty. There is a first-order term (non-potential).

We want to eliminate the magnetic field. A sufficient condition for a pure potential and reality

[The line numbered (17) is crossed out.]



Original source details: GELFAND-P001247

Uncertain reading. $L = \partial\bar{\partial} + A(z, \bar{z})\partial + B(z, \bar{z})$?

(17) Γ has two involutions

$$\sigma^2 = 1, \quad \tau^2 = 1, \quad \sigma\tau = \tau\sigma \qquad (\text{the group } \mathbb{Z}_2 \times \mathbb{Z}_2)$$

[Below σ: holomorphic. Below τ: antiholomorphic.]

Consider a Riemann surface satisfying (17), and also assume

$$\begin{aligned} \sigma(\infty_i) &= \infty_i, \qquad \sigma(w_i) = -w_i \\ \tau(\infty_1) &= \infty_2, \qquad \tau(w_1) = \pm\bar{w}_2 \end{aligned} \tag{18}$$

$$\tau D = D, \qquad \sigma D + D \cong K + \infty_1 + \infty_2 \tag{19}$$

[A line from $\cong$ points to "linearly equiv."] [A line from K points to "canonical divisor of degree $2g-2$ (divisor of holomorphic differential forms)."]

Hyperelliptic curve.

σ is known (holomorphic). $\psi^*(-k) = \psi(k)$ [A barred sign is written above the equality.]

Original source details: GELFAND-P001247

Uncertain reading. resembles $\bar{\tau}$

(two adjoints, formal and Hermitian)

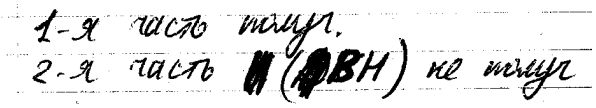
(20) Theorem. Conditions (17)–(19) are necessary and sufficient for reality and for a pure potential.

(21) ψ is expressed using Prym Θ-functions (there are more of them than Jacobi Θ-functions).

Multidimensional case.

Finite-gap property. The family of all Bloch functions (at any energy) contains a nontrivial algebraic curve.

[At right:] First part they write (?). Second part [crossed out] ([illegible]VN) they do not write (?).



Original source details: GELFAND-P001247

Uncertain reading. the word in both lines resembles "пишут" (they write)

Original source details: GELFAND-P001247

Uncertain reading. crossed-out sign after "part"

Original source details: GELFAND-P001247

Uncertain reading. the first letter in parentheses is obscured, the last two read Cyrillic ВН

(22) Example

$$L = \partial\bar{\partial} + V(z,\bar{z}), \qquad V \text{ is a real potential}$$

A direct check gives

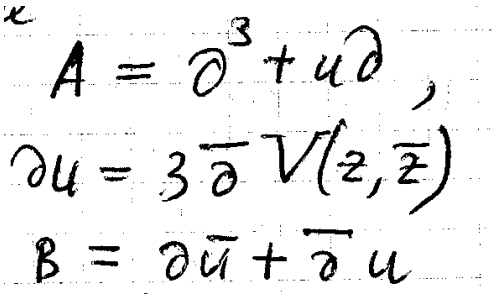
$$\frac{dL}{dt} = [A + \bar{A}, L] + BL,$$

where

$$A = \partial^3 + u\partial,$$

$$\partial u = 3\bar{\partial}V(z,\bar{z}) \qquad \text{(this determines } u\text{)},$$

$$B = \partial\bar{u} + \bar{\partial}u \quad \text{is a scalar function.}$$



Uncertain reading. In the line for u, the first ∂ has no bar and the second has a bar.

Original source details: GELFAND-P001247

[At lower left on the strip of the adjacent leaf.]

Uncertain reading. clipped Cyrillic ending “им)” or “ым)” with a closing parenthesis

Original source details: GELFAND-P001247

GELFAND-P001248

The corresponding system

$$V_t = (\partial^3 + \partial u)V + \text{complex conjugate} \tag{23}$$

there are exact solutions, rational [crossed out] solitons

Uncertain reading. short illegible crossed-out fragment

Original source details: GELFAND-P001248

The rapidly decaying problem is also essentially solved

(R. Novikov, P. Grinevich, Manakov)

D. Knizhnik.

(Dima)

Strings, [crossed out] moduli spaces, quadratic differentials.

Uncertain reading. crossed-out sign before "moduli spaces", possibly "and"

Original source details: GELFAND-P001248

Consider the partition function

$$\begin{aligned} Z_p &= \sum_{\text{surfaces of genus } p} e^{-A} \\ &= \int \mathcal{D}g_{ab}\,\mathcal{D}X_\mu\, e^{-\int \sqrt{g}\, g^{ab}\partial_b X^\mu \partial_a X_\mu \, d^2\xi} \end{aligned} \tag{1}$$

(A is the area)

$\xi^{1,2}$ are coordinates, $g_{ab}(\xi)$ is the metric,

$X^\mu(\xi)$, $\mu = 1, \dots, D$, parametrize the surfaces embedded in $\mathbb{R}^D$.

The action is generally covariant under changes of coordinates, with two arbitrary functions.

(2) Fix some of the degrees of freedom. What remains is $g_{ab} = g(\xi)\delta_{ab}$

Uncertain reading. the factor in (2) reads as $g(\xi)$ with an overwritten letter

Original source details: GELFAND-P001248

$$X^\mu(\xi), \quad \mu = 1, \dots, D = 26$$

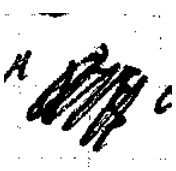
(3) then there is symmetry [inserted above: also] under conformal changes $g_{ab} \to \lambda(\xi) g_{ab}$.

Integrate over the orbits of the gauge transformations.

$D = 26$ $\Rightarrow$ this reduces to integration over the moduli space (Polyakov)

$$\int_{M_p} \frac{\prod dy^i \wedge \overline{dy^i}}{\det(f^i, f^j)} \frac{\int \mathcal{D}X_\mu\, e^{-\int \partial X_\mu \bar\partial X^\mu \, dz\wedge d\bar z}}{\int \mathcal{D}\varepsilon\, e^{-\int \rho\, \partial\bar\varepsilon\, \bar\partial\varepsilon\, dz\wedge d\bar z}} \tag{4}$$

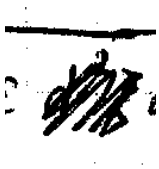
[Writing in the numerator before $dz \wedge d\bar z$ is crossed out.]



Uncertain reading. possibly $d^2\xi$

Original source details: GELFAND-P001248

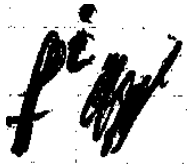
[In the denominator, $d^2\xi$ is crossed out before $dz \wedge d\bar{z}$.]



Uncertain reading. a raised 2 and a sign resembling ξ are visible

Original source details: GELFAND-P001248

f^i [crossed out] are holomorphic [inserted above the line: quadratic] differentials on the curve (a basis)



Uncertain reading. possibly dy^i

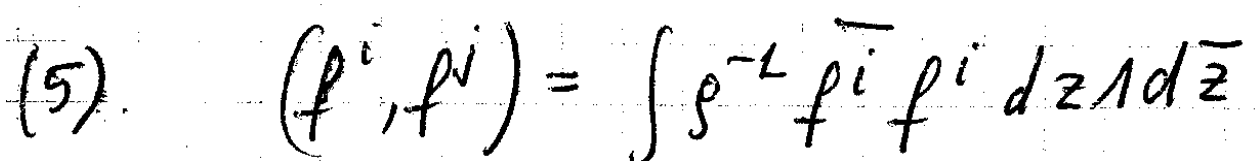
Original source details: GELFAND-P001248

[Writing below "basis" is crossed out.]

Uncertain reading. possibly an abbreviation for "quadratic"

Original source details: GELFAND-P001248

$$(f^i, f^j) = \int \rho^{-1} \overline{f^i} f^i \, dz \wedge d\bar{z} \tag{5}$$



Original source details: GELFAND-P001248

Uncertain reading. Both superscripts on the right of (5) are i, with i, j on the left.

$\rho \, dz \wedge d\bar{z}$ is a fixed metric on the curve (ρ is arbitrary)

$f(dz)^2$ is such a differential

$\varepsilon(dz)^{-1}$ is an invariant quantity.

[At lower right on the strip of the adjacent leaf.]

Uncertain reading. the fragments "(" and, below it, "(1" are visible. The rest is beyond the right edge of the image.

Original source details: GELFAND-P001248

GELFAND-P001249

[source number: 58]

For genus 2 there are three quadratic differentials

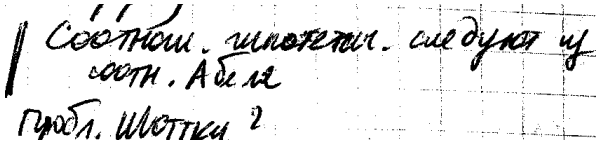
$$\omega_1^2, \quad \omega_2^2, \quad \omega_1\omega_2 \qquad (\omega_1, \omega_2 \text{ are holomorphic differentials})$$

They are always obtained this way, but for genus ≥ 4 there is a surplus.

τ-differentials form a symmetric form.

At most $3p - 3$ quadratic ones (by Riemann–Roch). Choose a dual basis.

[At right, beyond the vertical line:] Relations among products (?) follow from Abel's relations. The Schottky problem?

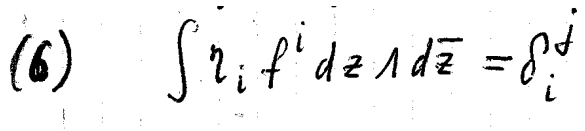


Uncertain reading. the abbreviation resembles Russian «множит.» ("products")

Original source details: GELFAND-P001249

S.P.N. Writing out the relations ⇔ the Schottky problem.

$$\int \eta_i f^i \, dz \wedge d\bar{z} = \delta_i^j \tag{6}$$



Uncertain reading. the index on f is written i, with δ_i^j on the right

Original source details: GELFAND-P001249

[Writing before ρ in (7) is crossed out.]

Undeciphered passage. illegible crossed-out writing before the metric

Original source details: GELFAND-P001249

$$\rho \, dz \, d\bar{z} \longrightarrow \widetilde{\rho} \, |dz + y^i \eta_i \, d\bar{z}|^2 \tag{7}$$

This is a way to parametrize the metric. [Writing after the full stop is crossed out.]

Undeciphered passage. crossed-out writing after "metric"

Original source details: GELFAND-P001249

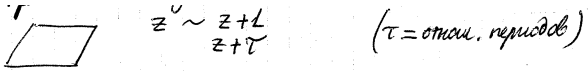
[At the left edge.]

Diagram and typed description. Near the binding is a shape resembling 2, with a vertical zigzag below. Their meaning is unclear.

Original source details: GELFAND-P001249

$p = 1$. The period matrix is a complex number (the ratio of periods along two cycles).

torus ≈ Jacobian



Original source details: GELFAND-P001249

Diagram and typed description.

$z \sim z + 1$
$z + \tau$

(τ = ratio of periods)

The moduli space is not the whole upper half-plane.

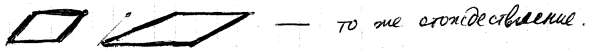


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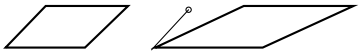
Diagram and typed description.



— the same identification

$$\text{Transformation} \quad \tau \longmapsto \frac{m_1\tau + n_1}{m_2\tau + n_2} \tag{8}$$

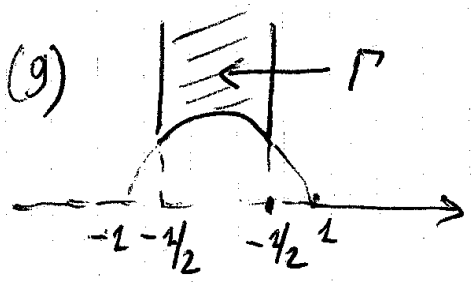
or

$$(\tau_1, \tau_2) \longmapsto (m_1\tau_1 + n_1\tau_2,\ m_2\tau_1 + n_2\tau_2)$$

matrices

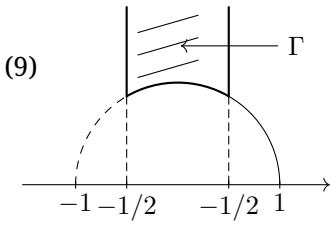
$$\begin{pmatrix} m_1 & n_1 \\ m_2 & n_2 \end{pmatrix}, \quad m_j, n_j \text{ are integers}, \quad m_1n_2 - m_2n_1 = 1$$

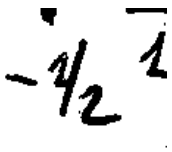
It suffices to use $\tau \longrightarrow \tau + 1$, $\tau \longrightarrow -1/\tau$.



Original source details: GELFAND-P001249

Diagram and typed description.





Original source details: GELFAND-P001249

Uncertain reading. the right-hand vertical is also labelled $-1/2$

$\triangle$ with angles $0°, 60°, 60°$ in the Lobachevsky plane.

Consider integrals over triangle (9).

Choose a basis of holomorphic quadratic differentials on the torus

$$f^1 = 1, \qquad \eta_1 = \frac{1}{\operatorname{Im}\tau} \tag{10}$$

$$du = dz + \delta y^1 \frac{d\bar{z}}{\operatorname{Im}\tau}$$

Uncertain reading. the letter u is heavily overwritten

Original source details: GELFAND-P001249

[Writing before the explanation of u is crossed out.]

Undeciphered passage. illegible crossed-out writing before the new complex symbol

Original source details: GELFAND-P001249

u is a new complex coordinate (complex structure)

y^1 is a new coordinate in moduli space.

The matrix has exactly one entry $\dfrac{dy^1 \wedge d\bar{y}^1}{\operatorname{Im}\tau}$ (11)

GELFAND-P001250

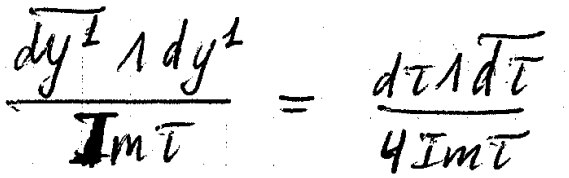
$$\tau \longrightarrow \frac{u(\tau)}{u(1)} = \frac{\tau + \dfrac{\delta y^1 \bar{\tau}}{\operatorname{Im}\tau}}{1 + \dfrac{\delta y^1}{\operatorname{Im}\tau}} \tag{12}$$

$$u(z) = z + \frac{\delta y^1}{\operatorname{Im}\tau}\bar{z} \tag{13}$$

y gives coordinates in the space of complex structures

y^1 is arbitrary, τ is calculated from y^1

$$\frac{\overline{dy^1} \wedge dy^1}{\operatorname{Im}\tau} = \frac{d\tau \wedge d\bar{\tau}}{4\operatorname{Im}\tau}$$



Uncertain reading. the barred differential comes first on the left and second on the right

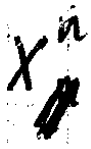
Original source details: GELFAND-P001250

Now it remains to compute

$$\int_\Gamma \frac{d\tau \wedge d\bar{\tau}}{\operatorname{Im}\tau} \frac{\int \mathcal{D}X_\mu \, e^{\int X_\mu \partial\bar{\partial} X^\mu \, dz\wedge d\bar{z}}}{\int \mathcal{D}\varepsilon \, e^{\int \bar{\varepsilon}\partial\bar{\partial}\varepsilon \, dz\wedge d\bar{z}}} \tag{14}$$

[Below the formula, an upward arrow to the measure.] ↑ (This is not the area! of the metric). This is unsurprising, since the starting point was not invariant.

$$X_\mu = a_\mu^n X^n_{\text{[crossed out]}}, \quad \Delta X^n_{\text{[crossed out]}} = \lambda^n X^n_{\text{[crossed out]}} \tag{15}$$



Undeciphered passage. illegible crossed-out subscript on the first X^n

Original source details: GELFAND-P001250

Undeciphered passage. illegible crossed-out subscript on the second X^n

Original source details: GELFAND-P001250

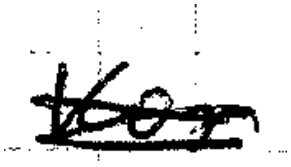
Undeciphered passage. illegible crossed-out subscript on the third X^n

Original source details: GELFAND-P001250

We obtain

$$\frac{\int dX^0_\mu (\det' \Delta)^{-13}}{\int d\varepsilon^0 (\det' \Delta)^{-1}} = \frac{V}{\operatorname{Im}\tau} (\det{}' \Delta)^{-12} \tag{16}$$

[Crossed out: Ko...]



Undeciphered passage. writing after “Ko” is crossed out and illegible

Original source details: GELFAND-P001250

The zero-mode integrals remain.

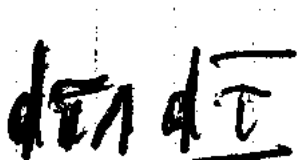
Above is the volume of the universe (26-dimensional).

Below is the area of the torus.

Temporarily set $V = 1$.

Result

$$-\int_\Gamma \frac{d\tau \wedge d\bar\tau}{(\operatorname{Im}\tau)^2}(\det{}' \Delta)^{-12} \tag{17}$$



Uncertain reading. the first $d\tau$ is overwritten

Original source details: GELFAND-P001250

[At bottom:] (from m.(?) parametrization) an extra $\operatorname{Im}\tau \Longrightarrow$ we obtain area.

Uncertain reading. the abbreviation resembles Cyrillic «м.»

Original source details: GELFAND-P001250

GELFAND-P001251

[source number: 59]

How to calculate

$$(\det{}' \Delta)^{-1/2} = \int \mathcal{D}\psi\, e^{-\int \partial\psi\bar\partial\psi\, dz\wedge d\bar z} \tag{18}$$

Calculate the variation of its ln with respect to the variation of complex structure.

Δ is the usual one, for the complex structure!

$$\delta \ln(\det{}' \Delta)^{-1/2} \tag{19}$$

The idea is to use a basis in which it is diagonal, and so on.

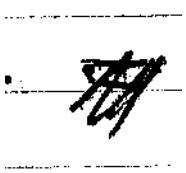
Answer

$$\begin{aligned}(\det{}' \Delta)^{-1/2} &= (\operatorname{Im}\tau)^{-1/2} e^{\frac{\pi}{6}\operatorname{Im}\tau} \prod_{n=1}^{\infty} |1 - e^{2\pi i n\tau}|^{-2} \\ &= (\operatorname{Im}\tau)^{-1/2} |\eta(q)|^{-2}\end{aligned} \tag{20}$$

Use formula (18) for the calculation

$$\delta(\ln \det{}^{-1/2} \Delta) = \frac{\delta \int \mathcal{D}\psi\, e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}}{\int \mathcal{D}\psi\, e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}} \tag{21}$$

[Writing after the fraction is crossed out.]



Undeciphered passage. illegible crossed-out writing after the fraction in (21)

Original source details: GELFAND-P001251

Here

[Writing before the variation is crossed out.]

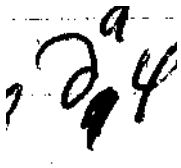
Undeciphered passage. illegible crossed-out writing before the metric variation

Original source details: GELFAND-P001251

$$\delta(\sqrt{g}\, g^{ab}\partial_a\psi\partial_b\psi) = \delta g^{ab} T_{ab},$$

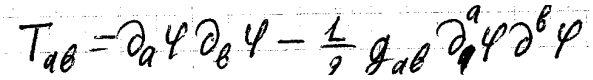
where

$$T_{ab} = \partial_a\psi\partial_b\psi - \frac{1}{2} g_{ab}\partial^a\psi\partial^b\psi$$



Undeciphered passage. illegible crossed-out subscript under the superscript a

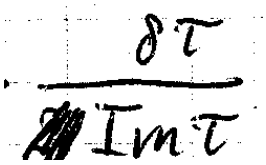
Original source details: GELFAND-P001251



Uncertain reading. the factors after g_{ab} carry superscripts a and b

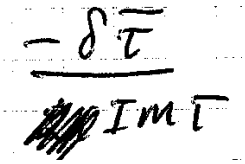
Original source details: GELFAND-P001251

$$\delta g^{++} = \frac{\delta\tau}{\operatorname{Im}\tau}, \qquad \delta g^{--} = \frac{-\delta\bar\tau}{\operatorname{Im}\tau} \qquad (?)$$



Undeciphered passage. crossed-out sign before Im in the denominator of δg^{++}

Original source details: GELFAND-P001251



Undeciphered passage. crossed-out sign before Im in the denominator of δg^{--}

Original source details: GELFAND-P001251

Thus

$$\delta(\ln \overset{-1/2}{\det}\,\Delta) = 2\operatorname{Re}\frac{i}{2}\delta\tau\langle T\rangle \tag{22}$$

where T is the energy–momentum tensor,

$$T(\xi) = \frac{\int \mathcal{D}\psi\, \partial\psi(\xi)\partial\psi(\xi) e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}}{\int \mathcal{D}\psi\, e^{-\int \partial\psi\bar\partial\psi\, d^2\xi}} \tag{23}$$

General formula (on a surface of any genus)

$$\int d^2\xi\, (\eta T + \bar\eta\bar T). \tag{24}$$

[At right, beyond the double vertical line.] $\partial\bar T = \bar\partial T_{+-}$.

(T is an almost holomorphic differential)
For a more precise form of (23), replace $\partial\psi(\xi)\partial\psi(\xi)$ by $:\partial\psi(z)\partial\psi(z):$ and take

$$\begin{aligned} T =: \partial\psi(z)\partial\psi(z) :\overset{\text{def}}{=} \oint \frac{dz'}{2\pi i}\frac{1}{z'-z}\partial\psi(z')\partial\psi(z) \quad \overset{\text{or}}{=} \\ = \langle\partial\psi(z)\partial\psi(z)\rangle \overset{\text{def}}{=} \left\langle \oint \frac{dz'}{2\pi i}\frac{1}{z'-z}\partial\psi(z')\partial\psi(z)\right. \end{aligned} \tag{25}$$

GELFAND-P001252

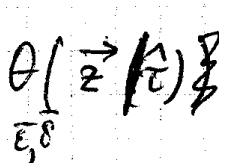
Formula for any genus.
There are p Abelian differentials ω_i and coordinates

$$\vec{z}(Q) = \int_{Q_0}^{Q}\vec{\omega} \tag{26}$$

$$\int_{b_i}\omega_j = \tau_{ij}, \qquad \int_{a_i}\omega_j = \delta_i^j \tag{27}$$

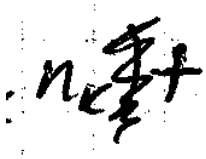
Also introduce θ-functions

$$\theta_{\vec{\varepsilon},\delta}(\vec{z}\mid\hat{\tau}) = \sum_{\vec{n}} e^{\pi i\left(n_i\hat{\tau}_{ik}n_k + 2\pi i\vec{z}\cdot(\vec{n}+\vec{\delta})\right.}$$



Original source details: GELFAND-P001252

Undeciphered passage. illegible crossed-out sign after the arguments of θ



Original source details: GELFAND-P001252

Undeciphered passage. illegible crossed-out sign after $n_i\hat{\tau}_{ik}n_k$ in the first exponent

$$\sum_{\vec{n}} e^{\pi i(\vec{n}+\vec{\delta},\hat{\tau},\vec{n}+\vec{\delta})+2\pi i(\vec{z}+\vec{\varepsilon})(\vec{n}+\vec{\delta})}$$

$$4\vec{\delta}\cdot\vec{\varepsilon} \pmod 2$$

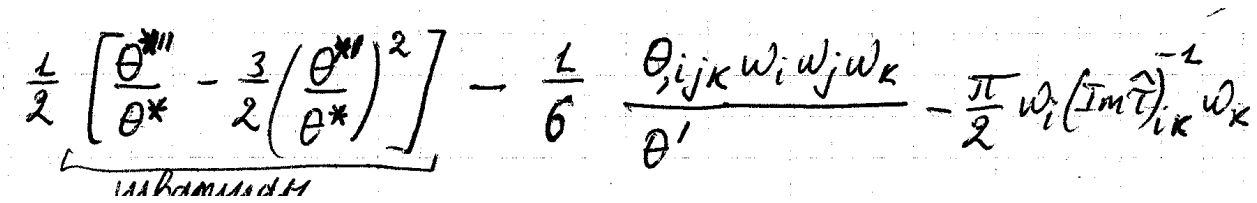
$$\theta^* \equiv \theta^{\bar{\varepsilon},\bar{\delta}}_{,k}\omega_k(z);$$

[Arrow to the coefficient $\theta^{\bar{\varepsilon},\bar{\delta}}_{,k}$.] ⟵ derivative of θ at zero.

[At right:] $\bar{\varepsilon},\bar{\delta}$ is an odd characteristic.

One can take the square root of this Abelian differential (all zeros are of order 2).

$$T = \underbrace{\frac{1}{2}\left[\frac{\theta^{*''}}{\theta^*} - \frac{3}{2}\left(\frac{\theta^{*'}}{\theta^*}\right)^2\right]}_{\text{Schwarzian}} - \frac{1}{6}\frac{\theta_{,ijk}\omega_i\omega_j\omega_k}{\theta'} - \frac{\pi}{2}\omega_i(\operatorname{Im}\hat{\tau})^{-1}_{ik}\omega_k \tag{28}$$



Original source details: GELFAND-P001252

Uncertain reading. the first numerator reads $\theta^{*''}$, and the second denominator reads θ'

All of this together is a quadratic differential.

ω does not include dz. The form is $\omega_i\,dz$.

Chapter 28

GELFAND-P001253

17 Feb. 1986 Gelfand Seminar.

A. Beilinson [Crossed out: On the theory (?)] The Quillen determinant, strings, and so on.

Uncertain reading. possibly “On the theory”

Original source details: GELFAND-P001253

We compute the partition function

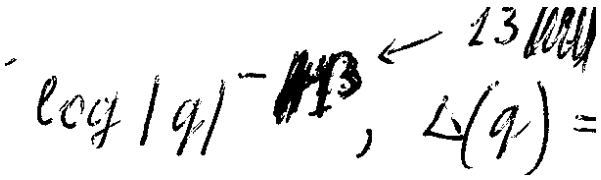
$$Z = \sum_{p \geq 0} Z_p. \tag{1}$$

p is the genus

$$Z_p = \int_{M_p} \mu_p, \qquad \begin{matrix} M_p \text{ is the moduli space of curves of genus } p. \\ \mu_p \text{ is the naturally arising measure} \end{matrix} \tag{2}$$

For $p = 1$ Kronecker (1890) and Shapiro (1972) calculated it independently (with the same result)

$$\mu_1 = \left|\frac{dq}{q}\right|^2 \Delta(q)^{-2} \log|q|^{-13}, \qquad \Delta(q) = q \prod (1-q^n)^{24}. \tag{3}$$



Uncertain reading. The exponent after $\log|q|$ is heavily overwritten and reads -13. Above are 13 and illegible writing in parentheses, with an arrow to the exponent.

Original source details: GELFAND-P001253

μ_1 is a measure on the moduli space of elliptic curves

— . —

X is a surface of genus p; g is a metric on X; L is a one-dimensional holomorphic bundle on X (holomorphic structure induced by the metric), h is a Hermitian metric on L

$$\bar\partial : \Gamma(L) \longrightarrow \Gamma(L \otimes \Omega^{0,1}). \tag{4}$$

Then

$$\underset{\substack{| \\ \text{holomorphic sections}}}{H^0(X,L)} = \operatorname{Ker} \bar\partial, \qquad H^1(X,L) = \operatorname{Coker} \bar\partial = \operatorname{Ker} \bar\partial^+. \tag{5}$$

(Holomorphic bundles are described by one discrete parameter, the Chern number, and p continuous parameters, a point on the Jacobian.)

H^0, H^1 are finite-dimensional

$$d(X,L) := \frac{\det H^0(X,L)}{\det H^1(X,L)}. \tag{6}$$

$\det H^0 =$ max exterior power of H^0 is a one-dimensional vector space

[The next line has long strokes and a repetition mark, with H^1 in place of H^0.]
Hence $d(X, L)$ is also a one-dimensional vector space.
(functions on $\det H^1$ with values in $\det H^0$)
It has a natural metric

$$\left.\begin{array}{l} e_1, \dots, e_k \text{ is a basis in } H^0 \\ g_1, \dots, g_d \text{ is a basis in } H^1 \end{array}\right| \; d(X, L) \ni \lambda \frac{e_1 \wedge \dots \wedge e_k}{g_1 \wedge \dots \wedge g_d} \quad \begin{array}{l} \uparrow \\ \mathbb{C} \end{array}$$

GELFAND-P001254

[source number: 4]

The metric will be defined in two stages.
First the stupid metric. Consider the [inserted above: natural] metric on

Uncertain reading. Crossed out. Possibly "пр."

Original source details: GELFAND-P001254

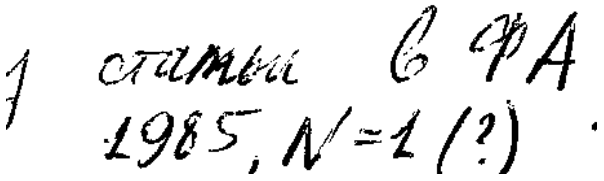
$\Gamma(L)$ and on $\Omega^{0,1}$, defining pre-Hilbert structures there.
On H^0 and $H^1 = \operatorname{Ker} \bar{\partial}^+$ metrics are also induced $\Rightarrow$
$\Rightarrow$ a metric arises in $d(X, L)$, called the stupid metric.
[Inserted above: Let us construct] the correct metric. [Downward arrow on the left.]
Consider $\Delta = \bar{\partial}^+ \bar{\partial}$ and its $\det \Delta$, calculated by Belavin and Knizhnik;
Δ' is the same Δ, with the zero modes removed.

$$\det \Delta' = \exp\bigl(-\zeta'_{\Delta'}(0)\bigr),$$

$$\zeta_{\Delta'}(s) = \sum_{\substack{\lambda \in \text{eigenvalues of } \Delta, \\ \lambda \neq 0}} \lambda^{-s}, \qquad \operatorname{Re} s > 1$$

$$\underline{\text{Quillen metric on } d(X, L)} \quad \left(\begin{array}{c}\text{from the paper in FA} \\ 1985,\ N{=}1\ (?)\end{array}\right) \tag{7}$$
$$= (\text{stupid metric}) \cdot \sqrt{\det \Delta'}.$$

Uncertain reading. The question mark after the issue number is in the manuscript.



Original source details: GELFAND-P001254

Explanation of definition (7). Let

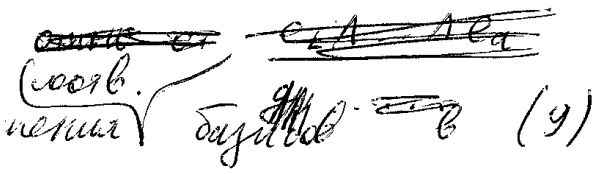
$$E_1 \xrightarrow{\bar\partial} E_2 \quad \text{be spaces with metrics (finite-dimensional)} \tag{8}$$

$$d = \frac{\det \operatorname{Ker} \bar\partial}{\det \operatorname{Coker} \bar\partial} \underset{\substack{\uparrow \\ \text{canonical} \\ \text{identification}}}{=} \frac{\det E_1}{\det E_2}. \tag{9}$$

Let bases be given. And suppose that

$$\begin{array}{l|l} e_1, \dots, e_k \text{ is a basis in } E_1 & \underbrace{e_1, \dots, e_a}_{\text{basis in } \operatorname{Ker} \bar\partial} \\ g_1, \dots, g_\ell \text{ — — " — } E_2 & g_{a+1}, \dots, g_k \\ & g_{a+1} = \bar\partial e_{a+1}, \dots, g_k = \bar\partial e_k. \end{array} \tag{10}$$

and



Uncertain reading. The crossed-out words are illegible. The product $e_1 \wedge \dots \wedge e_a$ is legible. There are more overwritten marks below.

Original source details: GELFAND-P001254

the ratios of the [inserted: corresponding] bases in (9) correspond to one another.

There are metrics on the right and on the left in (9). They do not coincide — [crossed out: difference; inserted: differ] by a factor $\sqrt{\det \Delta}$

On the left is the stupid metric, and on the right the one we need, but now we define

Uncertain reading. Crossed out. Possibly "it".

Original source details: GELFAND-P001254

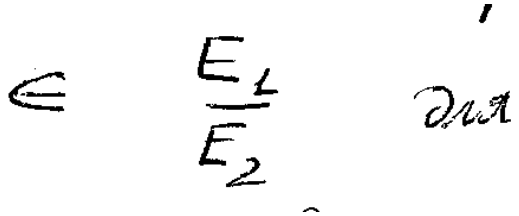
the right-hand side from the left-hand side in the infinite-dimensional situation.

GELFAND-P001255

Naturality of the "correct" metric: it does not change under variation of the operator $\bar\partial$.

In (9), if $\operatorname{Ker} \bar\partial = \operatorname{Coker} \bar\partial = 0$ on the right stands $\det \bar\partial$

$$\text{In general} \quad \det(\bar\partial : E_1 \to E_2) \in \frac{E_1}{E_2} \quad \text{for one-dimensional } E_1, E_2.$$



Uncertain reading. The quotient E_1/E_2 follows the manuscript.

Original source details: GELFAND-P001255

In the infinite-dimensional case there is only the modulus of the right-hand side of (9)

Unattainable ideal: define the right-hand side of (9) independently

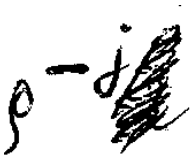
Uncertain reading. A short crossed-out passage is illegible.

Original source details: GELFAND-P001255

of $\bar{\partial}$ in a functorial way (isomorphisms of spaces $\Rightarrow$ isomorphisms of determinants)

Let L be the bundle of j-differentials, $L = \omega^j$, its sections are $f(z)(dz)^j$.

$$\begin{array}{c} \text{Riemannian metric} \quad \text{[crossed-out insertion: on } L\text{]} \\ \| \\ \rho\, dz\, d\bar{z} \\ \text{induces a metric on } \omega^j \quad \Rightarrow \quad |(dz)^j|^2 = \rho^{-j}. \end{array} \tag{11}$$



Uncertain reading. Illegible signs are crossed out after $-j$.

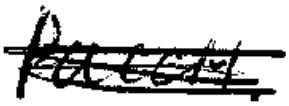
Original source details: GELFAND-P001255

Here $j \in \mathbb{Z}$.

The space $d(x, \omega^j)$ depends only on the complex structure on X.

The metric $|\ |_j(\underset{\substack{| \\ \text{metric on } X}}{\rho})$ is the Quillen metric on $d(X, \omega^j)$

(12) Theorem (Polyakov)



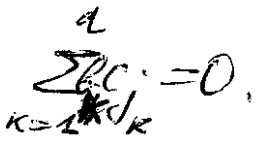
Uncertain reading. Crossed out: probably "Consider".

Original source details: GELFAND-P001255

$$\frac{\delta(\log|\ |_j(\rho))}{\delta\rho} = (6j^2 - 6j + 1) \qquad \underset{\substack{| \\ \text{curvature of the metric}}}{R(\rho)}$$

(13) Notation: $c_j = 6j^2 - 6j + 1$ (second Bernoulli polynomial)

(14) Corollary. Let there be a collection $j_1, \ldots, j_a \in \mathbb{Z}$, [inserted above: and integers $b_1, \ldots, b_a \in \mathbb{Z}$,] such that $\sum_{k=1}^{a} b_k c_{j_k} = 0$.



Uncertain reading. The subscript on b is overwritten. The summation and the exponent in the next line support reading it as k.

Original source details: GELFAND-P001255

Consider the Quillen metric on $\otimes d(X, \omega^{j_k})^{b_k}$.

Uncertain reading. A short crossed-out passage is illegible.

Original source details: GELFAND-P001255

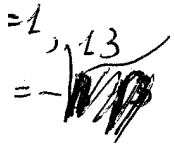
It is then independent of the [inserted: original] ρ, and depends only on the complex structure on X

(15) Definition. c_j is called the conformal anomaly.

(16) Most important example.

$$j_1 = 2, \quad b_1 = 1,$$
$$j_2 = 1, \quad b_2 = -13$$

— the corresponding number $\underline{\underline{13}}$



Uncertain reading. In b_2, the earlier writing after the minus sign is overwritten. The insertion above reads 13.

Original source details: GELFAND-P001255

(In 26-dimensional space there is no conformal anomaly)

— . —

GELFAND-P001256

[source number: 5]

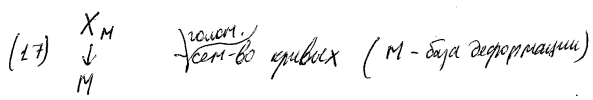


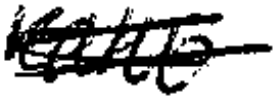
Diagram and typed description.

(17) $\begin{array}{c} X_M \\ \downarrow \\ M \end{array}$ [inserted: holomorphic] family of curves (M is the base of deformations)

Original source details: GELFAND-P001256

Then $\forall$ integer j there arises a one-dimensional bundle d_j over M with fiber $d(X_t, \omega^j)$ over $t \in M$.

This is



Uncertain reading. Crossed out: probably “is”.

Original source details: GELFAND-P001256

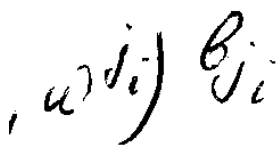
a one-dimensional holomorphic bundle over M.

(18)

Uncertain reading. Crossed out: possibly “On”.

Original source details: GELFAND-P001256

over $\otimes d(X_t, \omega^{j_i})^{b_{j_i}}$ there is a canonical Hermitian metric (b_j as in (14))



Uncertain reading. Here the exponent is written b_{j_i}.

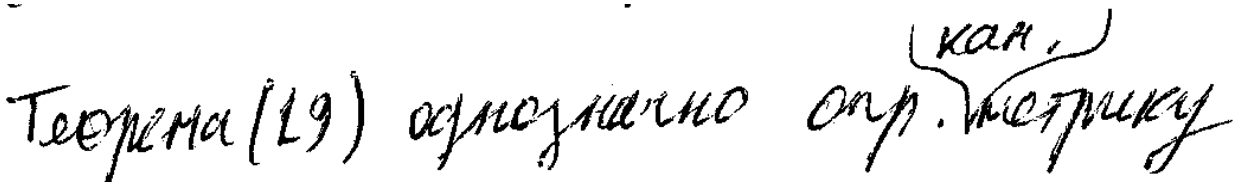
Original source details: GELFAND-P001256

(19) Theorem (Belavin, Knizhnik):

The curvature of the canonical metric on $\otimes d(X, \omega^{j_i})^{b_i}$ equals 0.

In words: Cancellation of the conformal anomaly implies cancellation of the gravitational one.

Theorem (19) uniquely determines the [inserted: canonical] metric (which will be determined by the genus alone)



Original source details: GELFAND-P001256

Uncertain reading. The statement of uniqueness follows the manuscript.

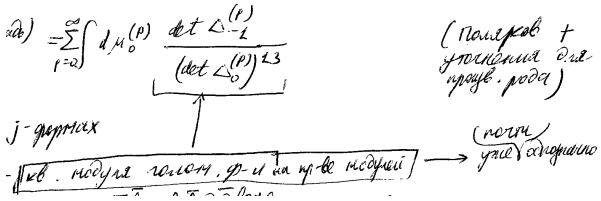
Belavin

$$\sum_{\text{surfaces}} e^{-a(\text{area})} = \sum_{p=2}^{\infty} \int d\mu_0^{(p)} \frac{\det \Delta_{-1}^{(p)}}{(\det \Delta_0^{(p)})^{13}} \qquad (20)$$

(Polyakov + refinements for arbitrary genus)

Δ_j is the Laplacian on j-forms

Important: on the right is



Original source details: GELFAND-P001256

Diagram and typed description.

$$\underbrace{\frac{\det \Delta_{-1}^{(p)}}{(\det \Delta_0^{(p)})^{13}}}$$

↑

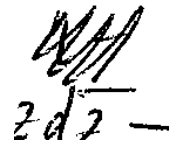
squared modulus of a holomorphic function on the moduli space

↓

already [inserted: almost] unique

$$\delta\bar{\delta} \log \det \Delta_j = c_j \int \frac{\partial f\, \bar{\partial} \bar{f} - f \bar{f}\, \partial\bar{\partial} \log \rho}{\rho^2}, \qquad f = \rho\eta, \qquad (21)$$

$$\delta_\rho \log \det \Delta_j = c_j R \qquad \text{Variation of the metric: } dz\, d\bar{z} \longrightarrow |dz + \eta\, d\bar{z}|^2 \qquad (21')$$



Uncertain reading. Illegible signs are crossed out above $d\bar{z}$ to the left of the arrow.

Original source details: GELFAND-P001256

Beilinson

The integral in (20) diverges and still has to be renormalized.

Belavin

$$Z_1 = \int \frac{d\tau\, d\bar{\tau}}{(\operatorname{Im}\tau)^2} \cdot \frac{1}{(\operatorname{Im}\tau)^{12}} \cdot \frac{1}{|\Delta(q)|^2}, \qquad q = e^{2\pi i \tau}$$

Uncertain reading. An illegible sign is crossed out after $(\operatorname{Im}\tau)^2$ in the first denominator.

Original source details: GELFAND-P001256

The integral diverges.

But there is a conjecture that everything converges in the superstring.

$$Z_2 = \int \frac{d^2\tau}{(\operatorname{Im}\tau)^3} \cdot \frac{1}{(\operatorname{Im}\tau)^{10}} \cdot \frac{1}{|F(\tau)|^2}, \qquad \tau = \begin{bmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{bmatrix}$$

τ is the period matrix

(Knizhnik, Morozov, Perelomov)

$$F(\tau) = \prod \theta_m^2(\tau)$$

Z_3 exists as well.

For general p there is no good description of the moduli space.

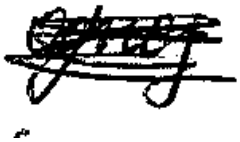
GELFAND-P001257

Beilinson

M is the moduli space.

$$p > 2 \Rightarrow \forall \underline{\text{holomorphic function}} = \text{const}$$

A metric [inserted above: of zero curvature] is determined up to the squared modulus of



Uncertain reading. Crossed out: possibly "function".

Original source details: GELFAND-P001257

a holomorphic function $\Rightarrow$ thus uniquely. It remains to calculate it. (This reduces to calculating the unique holomorphic section

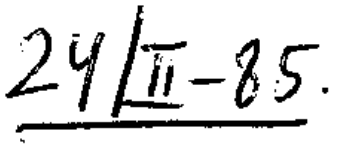
Uncertain reading. There is a short overwritten mark beneath the abbreviation for "holomorphic". Its reading is uncertain.

Original source details: GELFAND-P001257

of a certain bundle).

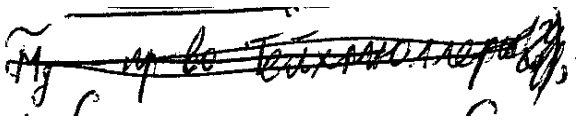
GELFAND-P001258

24 Feb. 85. Kontsevich (Maxim) (Gelfand Seminar)



Original source details: GELFAND-P001258

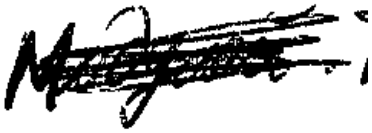
Uncertain reading. The manuscript reads "24/II-85". Catalogue record 667 dates the talk 24 February 1987.



Original source details: GELFAND-P001258

Uncertain reading. Crossed out: $\widetilde{M}_g$ is the space infinite-dimensional... [ending overwritten].

M_g is the space of



Original source details: GELFAND-P001258

Uncertain reading. Crossed out: probably "moduli".

Teichmüller

$$(\text{Curve } C,\ \text{point } p \in C,\ x \text{ is a coordinate at } p) = \widetilde{M}_g$$

$$\mathfrak{g} = \mathbb{C}((x))\frac{\partial}{\partial x}$$

$$\cup$$

$$\mathfrak{k} = \mathbb{C}[[x]]\frac{\partial}{\partial x}$$

γ is the Laurent series of vector fields on C with a pole at p.

$\gamma \subset \mathfrak{g}$

Original source details: GELFAND-P001258

Uncertain reading. The first γ is retraced and overwritten. It is repeated clearly below.

Original source details: GELFAND-P001258

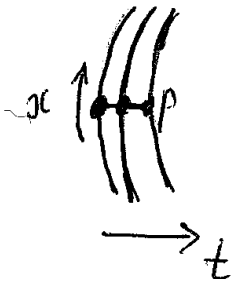
Uncertain reading. The ending of the word translated as "with a pole" is retraced and overwritten. A similar expression below supports the reading.

$\widetilde{M}_g$ in a neighborhood of a point is the homogeneous space $\mathfrak{g}/\gamma$

(1) Thm. 1. The algebra $\mathfrak{g}$ acts on $\widetilde{M}_g$, the stabilizer of the point [inserted above: t] is γ

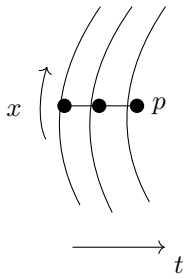
Description of the action of $\mathfrak{g}$ on $\widetilde{M}_g$

Let a one-parameter family of curves be given. On this nonlinear bundle there is a connection with a pole only at the point p.



Original source details: GELFAND-P001258

Diagram and typed description.



$$\nabla_t = \frac{\partial}{\partial t} + F(x,t)\frac{\partial}{\partial x}$$

The tangent space to $\widetilde{M}_g$ is identified with $\mathfrak{g}/\mathfrak{k}$.

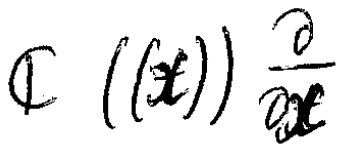
Original source details: GELFAND-P001258

Uncertain reading. The denominator in this line is $\mathfrak{k}$.

$$T(\widetilde{M}_g)_{(C,x,p)} \underset{\substack{\uparrow \\ \text{isomorphism} \\ \text{of Kodaira–Spencer}}}{=} \mathbb{C}((x))\frac{\partial}{\partial x}/$$

vector fields extending to $C \setminus \{p\}$

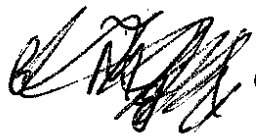
[In the margin: (all of C!!)]



Original source details: GELFAND-P001258

Uncertain reading. The x in the Laurent series and in the derivative denominator is retraced. The earlier strokes are illegible.

$\widetilde{\mathfrak{g}}$ is the Virasoro algebra. It acts



Original source details: GELFAND-P001258

Uncertain reading. Crossed out: in $\widetilde{M}_g$ [part of the subscript is overwritten].

in the bundle d_k.

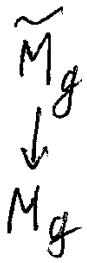


Diagram and typed description.

$$\begin{array}{c}\widetilde{M}_g\\ \downarrow\\ M_g\end{array}$$

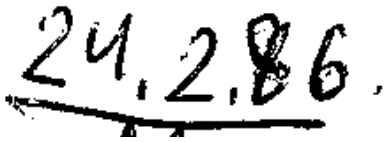
Original source details: GELFAND-P001258

d_k is the determinant bundle over M_g, lifted to $\widetilde{M}_g$

GELFAND-P001259

24 Feb. 86. Gelfand Seminar

Morozov Consequences of the Belavin–Knizhnik theorem.



Uncertain reading. The manuscript reads “24.2.86”. Catalogue record 668 dates the talk 24 February 1987.

Original source details: GELFAND-P001259

Explicit formulas for bosonic strings, etc.

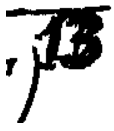
The bosonic string in the critical dimension $d = 26$.

Summary of Beilinson’s talk:

Uncertain reading. The first word overlaps strokes from the line above. The reading is “Summary”.

Original source details: GELFAND-P001259

$$\sum_p \int Dx\, Dg\ \exp \int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu\, d^2\xi \prod_\alpha V_\alpha = \\ = \sum_p \int_{M_p} \frac{|d\mu_p|^2}{(\det \operatorname{Im} T)^{13}}\, K\{V\} \tag{1}$$



Uncertain reading. The exponent 13 is heavily retraced.

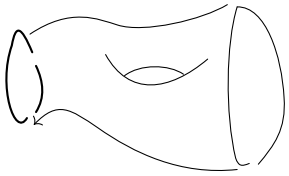
Original source details: GELFAND-P001259

V_α are vertex operators (this will be explained later)

Original source details: GELFAND-P001259

Diagram and typed description.

(2)



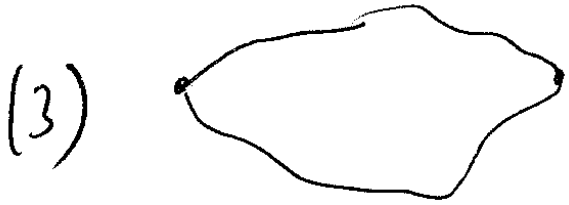
worldsheet of a propagating little ring

Original source details: GELFAND-P001259

Uncertain reading. The abbreviation is read as “распр.”, expanded in the translation as “propagating”.

Mutual transformations are accounted for by considering different topologies.

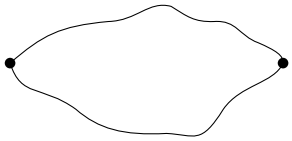
Technically it is convenient to take points at the beginning and end.



Original source details: GELFAND-P001259

Diagram and typed description.

(3)



Then V_α are δ-functions. Taking the Fourier transform, we obtain (α is the number of the vertex operator)

$$V_\alpha(p) = \int e^{ip_\alpha x(\xi)} d^2\xi \tag{4}$$

Original source details: GELFAND-P001259

Uncertain reading. An illegible sign, possibly μ, is crossed out after p_α in the exponent.

It is also convenient to take derivatives of δ-functions:

$$V_\alpha(p) = \int \partial x^\mu(\xi) e^{ip_\alpha x(\xi)} d^2\xi \tag{4'}$$

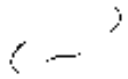
To obtain the right-hand side of (1):

a) Integrate over x to obtain

$$\frac{1}{(\det \Delta_0)^{13}} \prod_\alpha \int d^2\xi_\alpha \prod_{\alpha,\beta} \exp p_\alpha p_\beta \, G_{(reg)}(\xi_\alpha, \xi_\beta) \tag{5}$$

Δ_0 is the Laplacian on scalars

$G_{(reg)}$ is the regularized Green's function (smoothed when the points coincide)



Uncertain reading. Faint curved strokes appear in the right margin. Their reading is uncertain.

Original source details: GELFAND-P001259

GELFAND-P001260

[source number: 11]

b) Integrate over metrics.

$$g^{ab} = e^\varphi \delta^{ab} + (\text{traceless contributions}) \tag{6}$$

Uncertain reading. The ending of the first word in "traceless contributions" is overwritten.

Original source details: GELFAND-P001260

Eliminate the latter by the transformations

$$\xi \longrightarrow \xi + \varepsilon$$
$$g \longrightarrow g + \partial G$$

There arises $\underline{\det \Delta_{-1}}$. We arrive at

$$\int d\mu_p \frac{\det \Delta_{-1}}{(\det \Delta_0)^{13}} \tag{7}$$

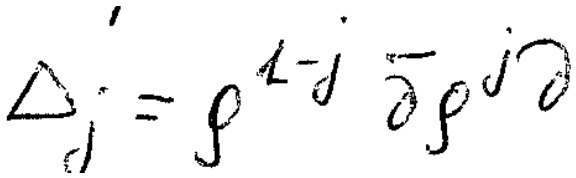
In the critical dimension, dependence on φ disappears.
Accounting for zero modes.

$$N^{(j)}_{\alpha\beta} = \langle \omega^{(j)}_\alpha \omega^{(j)}_\beta \rangle = \int \rho^{1-j} \bar{\omega}_\alpha \omega_\beta \, d^2\xi \tag{8}$$

There are j-differentials: $f(\xi) d\xi^j$,
the corresponding Laplace operator

$$\Delta_j = \rho^{1-j} \bar{\partial} \rho^j \partial \tag{9}$$

The zero modes are holomorphic differentials.

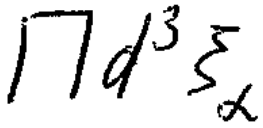


Original source details: GELFAND-P001260

Uncertain reading. In (9) the first ∂ bears a bar and the second is written without one.

Hence $\det N^{(j)}$ is defined.
We obtain

$$\int |d\nu_p|^2 \left(\frac{\det N^{(0)}}{\det' \Delta_0} \right)^{13} \det{}' \Delta_{-1} \prod d^3\xi_\alpha \prod_{\alpha,\beta} \exp p_\alpha p_\beta \, G_{(Reg)}(\xi_\alpha, \xi_\beta) \tag{10}$$

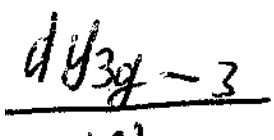


Original source details: GELFAND-P001260

Uncertain reading. Equation (10) reads $d^3\xi_\alpha$.

Here

$$d\nu_p = \frac{dy_1 \ldots dy_{3g-3}}{\det N^{(2)}} \tag{11}$$



Original source details: GELFAND-P001260

Uncertain reading. The last subscript in the numerator reads $3g-3$.

in the proper holomorphic coordinates on the moduli space.

$$dz\,d\bar z + f(z)dz^2 + f(\bar z)d\bar z^2 \qquad (12)$$
$$\| \quad \text{On the given surface } g\,dz\,d\bar z;$$

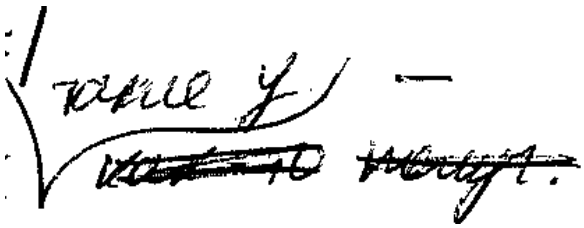
Uncertain reading. The coefficient of $dz\,d\bar z$ is retraced. It reads g.

Original source details: GELFAND-P001260

on neighboring surfaces

$$\rho\,|dz + y\eta\,d\bar z|^2 \qquad (13)$$

(14) Ahlfors–Bers theorem:

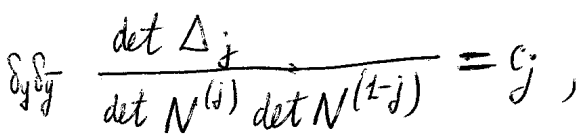


Uncertain reading. Crossed out: “some coord.” (the first word is partly illegible).

Original source details: GELFAND-P001260

[Inserted above: such y are] holomorphic coordinates on the moduli space
y parametrizes the deviation from the surface.
(15) Assertion. (Belavin–Knizhnik theorem)

$$\delta_y\delta_{\bar y}\frac{\det\Delta_j}{\det N^{(j)}\det N^{(1-j)}} = c_j, \qquad c_j = 6j^2 - 6j + 1. \qquad (16)$$



Uncertain reading. In (16) the operators $\delta_y\delta_{\bar y}$ stand directly before the quotient. No log is written.

Original source details: GELFAND-P001260

Uncertain reading. Several illegible mathematical signs are crossed out after $c_j =$.

Original source details: GELFAND-P001260

GELFAND-P001261

Gelfand’s question: there is no proper theory of det for elliptic operators

Uncertain reading. The abbreviation before “theory” is “прав.”, translated here as “proper”.

Original source details: GELFAND-P001261

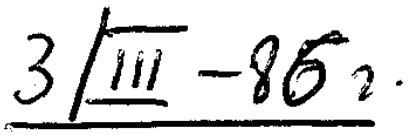
$$\det(AB) \stackrel{?}{=} \det A \cdot \det B$$

(A, B are elliptic operators of arbitrary order, invertible)

GELFAND-P001262

3 Mar. 86. Gelfand Seminar

Beilinson Mixed Hodge structures



Uncertain reading. The manuscript reads "3/III—86". Catalogue record 669 dates the talk 3 March 1987.

Original source details: GELFAND-P001262

(1) Def. A mixed Hodge structure is

1. A finitely generated abelian group L

and two filtrations

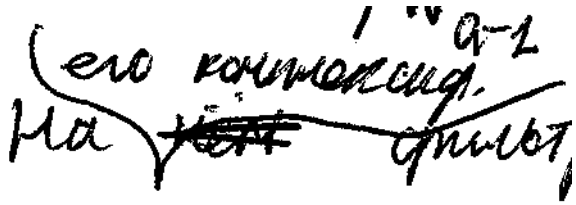
2. $L_{\mathbb{Q}}$: $\quad \ldots \subset W_i \subset W_{i+1} \subset \ldots$

3. $L_{\mathbb{C}}$: $\quad \ldots \supset F^j \supset F^{j+1} \supset \ldots$

Conditions:

W_a/W_{a-1} is a vector space over $\mathbb{Q}$.

On



Uncertain reading. Crossed out: probably "it".

Original source details: GELFAND-P001262

[inserted above: its complexification] filtration

Uncertain reading. Crossed out: possibly (3).

Original source details: GELFAND-P001262

$(1)_3$ induces a filtration:

$$F^j_{(a)} = \frac{F^j \cap (W_a)_{\mathbb{C}}}{(W_{a-1})_{\mathbb{C}} \cap F^j} \qquad \text{is the filtration on } (W_a/W_{a-1})_{\mathbb{C}}. \tag{4}$$

Required: $\forall a, j$

$$F^j_{(a)} \oplus \bar{F}^{a+1-j}_{(a)} \xrightarrow{\sim} (W_a/W_{a-1})_{\mathbb{C}} \tag{5}$$

or equivalently

$$\underline{\text{subspaces}}\ (W_a/W_{a-1})^{p,a-p}_{\mathbb{C}} = F^p_{(a)} \cap \bar{F}^{a-p}_{(a)} \tag{6}$$

Uncertain reading. The intersection sign is heavily retraced.

Original source details: GELFAND-P001263

as one varies p they define a grading on $(W_a/W_{a-1})_{\mathbb{C}}$

Another definition:

instead of $(1)_2$ and $(1)_3$ take the pair

$(7)_2$ A bigrading on $L_{\mathbb{C}}$: $\quad L_{\mathbb{C}} = \bigoplus L^{p,q}$

Uncertain reading. The sign is overwritten. It reads $=$.

Original source details: GELFAND-P001263

$(7)_3$ $\quad N \in \text{End}\, L_{\mathbb{C}}$

and the following properties must hold

(8) a) $\bar{L}^{p,q} = L^{q,p}$

b) $\bar{N} = -N$

c) $N(L^{p,q}) \subset \bigoplus_{\substack{a<p \\ b<q}} L^{a,b}$

d) the filtration

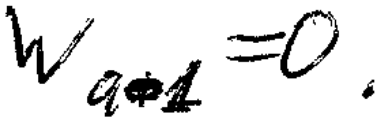
Undeciphered passage. Illegible crossed-out writing.

Original source details: GELFAND-P001263

$$W_i := \bigoplus_{p+q\le i} L^{p,q}$$

is rational (it is real by condition a)), that is, defined over $\mathbb{Q}$.

(9) Example 1. C is a smooth complex manifold. On $H^a(C,\mathbb{Z})$ there is a pure Hodge structure of weight a, that is, $W_a = H^a(C,\mathbb{Z})$, $W_{a+1} = 0$.

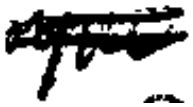


Uncertain reading. The sign before 1 is overwritten and resembles a plus. The next line clearly has $W_{a-1} = 0$.

Original source details: GELFAND-P001263

(10) (def. L is a pure Hodge structure of weight a, if $W_{a-1} = 0;\ W_a = L_{\mathbb{Q}}$).

In

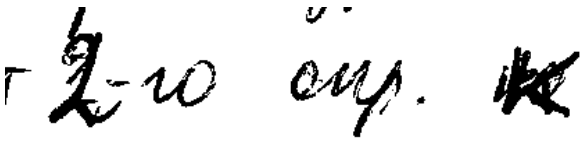


Undeciphered passage. Illegible crossed-out writing.

Original source details: GELFAND-P001263

Example 1 we have $N = 0$.

Passage from the second definition to the first:



Uncertain reading. The numeral 2 is retraced. The preposition “to” is overwritten.

Original source details: GELFAND-P001263

$$F^i = \exp(N)\Big(\bigoplus_{\substack{p\ge i \\ q \text{ arbitrary}}} L^{p,q}\Big) \tag{11}$$

$(10')$ def. 10 in the second version: L is a pure structure of weight a, if $L = \bigoplus_{p+q=a} L^{p,q}$

(12) In this case $N = 0$ (this follows from condition (8) c)).

GELFAND-P001264

[source number: 14]

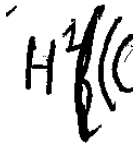
(13) Example 2. C is a compact Riemann surface with four given points x_1, x_2, y_1, y_2.

Uncertain reading. The ending of the word for “points” is retraced.

Original source details: GELFAND-P001264

On

$$H^1_{\mathbb{Q}}(C \setminus \{x_1, x_2\}, \{y_1, y_2\}) \tag{14}$$



Uncertain reading. The retraced subscript on H^1 reads $\mathbb{Q}$.

Original source details: GELFAND-P001264

there is a mixed Hodge structure:

$$W_0 \subset W_1 \subset W_2 \tag{15}$$

$$W_0 = \mathbb{Z}_{\mathbb{Q}}, \quad W_1/W_0 = H^1(C)_{\mathbb{Q}}, \quad W_2/W_1 = \mathbb{Z}_{\mathbb{Q}}, \tag{16}$$

[At right:]

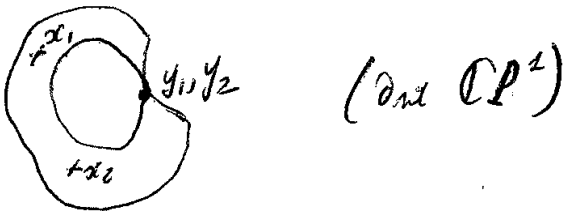
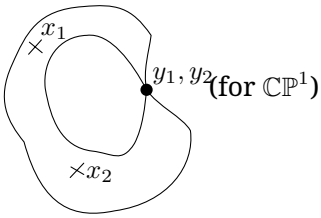


Diagram and typed description.



Original source details: GELFAND-P001264

namely

$$W_0 = \mathrm{Ker}\Big(\underset{\substack{| \\ \text{the group } (14)}}{H^1} \longrightarrow H^1(C \setminus \{x_1, x_2\}) \Big) \tag{17}$$

In terms of forms:

$$H^1 = \frac{\text{closed 1-forms on } C \setminus \{x_1, x_2\}}{\{d(f) : f(y_1) = f(y_2)\}} \tag{18}$$

Uncertain reading. The letter C in the numerator of (18) is retraced.

Original source details: GELFAND-P001264

$$W_1 = \mathrm{Ker}(H^1 \xrightarrow{\mathrm{Res}_{x_1}} \mathbb{Z}) \tag{19}$$

$$W_2 = H^1 \tag{20}$$

Passage to the second definition: On W_0 there is only type $(0,0)$

On $W_1/W_0 = H^1(C)$ there are types $(1,0)$ and $(0,1)$

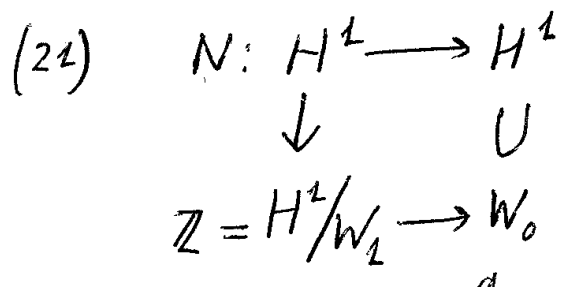


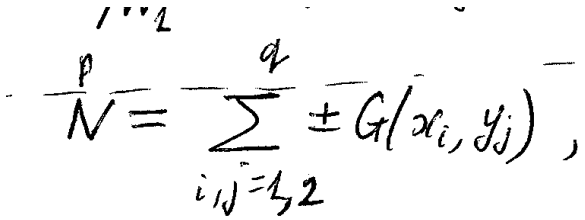
Diagram and typed description.

(21)
$$\begin{array}{ccc} N: H^1 & \longrightarrow & H^1 \\ \downarrow & & \cup \\ \mathbb{Z} = H^1/W_1 & \longrightarrow & W_0 \end{array}$$

Original source details: GELFAND-P001264

— here there is also a natural basis

$$\overset{p}{N} = \sum_{i,j=1,2}^{q} \pm G(x_i, y_j), \tag{22}$$



Uncertain reading. The letters p and q stand above (22), close to the broken horizontal line.

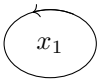
Original source details: GELFAND-P001264

where G is the Green's function of the surface C (the Green's function of the Laplacian on functions) for some metric

Standard bases:

$$p: \quad \int p = 1 \qquad \text{is a basis in } H^1/W_1$$

Diagram and typed description.



Original source details: GELFAND-P001264

$$q: \quad \int_{y_1}^{y_2} q = 1 \qquad \text{is a basis in } W_0$$

N is written in (22) in the bases p, q.

GELFAND-P001265

Varchenko applied this to calculate the asymptotics of oscillatory integrals (see [inserted above: his] paper in Izv. Akad. Nauk, about four years ago)

Morozov. Strings (continued)

Undeciphered passage. An illegible word is crossed out below the heading.

Original source details: GELFAND-P001265

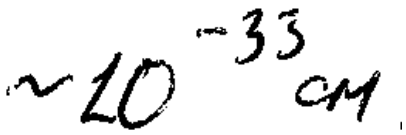
Size of the hypothetical string $\sim 10^{-33}$ cm

(Cf. the size of a proton

Undeciphered passage. The crossed-out passage is illegible.

Original source details: GELFAND-P001265

is the same $\sim 10^{-33}$ cm.



Original source details: GELFAND-P001265

Uncertain reading. Comparison with the two other occurrences on the leaf supports reading the exponent as -33.

—"— of an electron $\sim 10^{-18}$ cm)

All existing theories, except string theory, are internally inconsistent because of ultraviolet divergences

Measurable quantities should be independent of renormalizations. But everywhere at distances of $\sim 10^{-33}$ cm divergences occur (the inverse quantity is the Planck mass M_P)

$$V(k) = \int d^2\xi \, e^{ikx(\xi)} \tag{1}$$

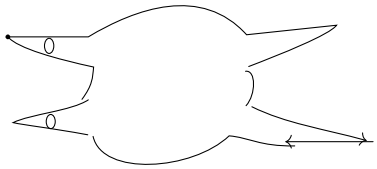
$$\langle V(k_1) \dots V(k_n) \rangle = \int Dx \, Dg \, e^{-\sqrt{g}\, g^{ab} \partial x \partial x} V(k_1) \dots V(k_n) \tag{2}$$

interpret as a scattering amplitude

Original source details: GELFAND-P001265

Diagram and typed description.

(3)

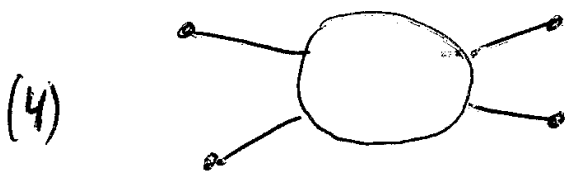


— physical picture

In the critical dimension it can be contracted

Original source details: GELFAND-P001265

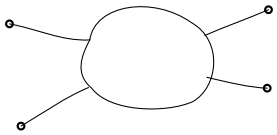
Uncertain reading. The last word is read as "contracted".



Original source details: GELFAND-P001265

Diagram and typed description.

(4)



It behaves like a small spring tending to contract to a point.

The main delight: the correlators will be gauge invariant.

$$V_\mu(k) = \int d^2\xi\, \partial x(\mu) e^{ikx(\xi)} \tag{5}$$

$$\langle V_{\mu_1}(k_1) \dots V_{\mu_n}(k_n) \rangle \tag{6}$$

have the property of gauge invariance!! and describe the interaction in Yang–Mills theory (7) + consistent with gravity!

Uncertain reading. The word after the plus sign reads “согласована”, translated as “consistent”.

Original source details: GELFAND-P001265

$\Rightarrow$ it is believed that this is a unified field theory (i.e. space is 10-dimensional, there is super., etc.)

GELFAND-P001266

[source number: 15]

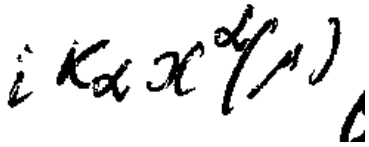
(8) Into field theory: a string is an infinite number of particles (harmonics). This means that ordinary field theory embeds here.

Diagram and typed description.

Original source details: GELFAND-P001266

$$\int_p Dx\, Dg\, e^{-\int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu d^2\xi} \prod_\alpha \int e^{ik_\alpha x^\mu(\xi_\alpha)} d^2\xi_\alpha \sim$$

$$\sim \int\limits_{\substack{M_p \\ y_1,\dots,y_{3p-3}}} \frac{|d\mu_p|^2}{|\det \operatorname{Im} T|^{13}} \underbrace{K(k_\alpha, y, \bar{y})}_{\substack{\| \\ \prod_\alpha \int d^2\xi_\alpha \prod_{\alpha,\beta} e^{k_\alpha k_\beta G_R(\xi_\alpha, \xi_\beta | y, \bar{y})}}} \tag{9}$$



Uncertain reading. Argument in the exponent on the first line: ξ_α?

Original source details: GELFAND-P001266

$\sum k_\alpha = 0$,

G_R — rel. Green’s function of the surface.

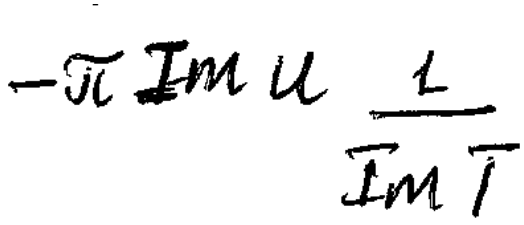
Uncertain reading. Abbreviation before ‘Green’s function’: «рел.» or «рег.»

Original source details: GELFAND-P001266

$$\exp G(\xi_\alpha, \xi_\beta) \sim e^{-\pi \operatorname{Im} u \frac{1}{\operatorname{Im} T}} |\Theta(u, T)|^2, \tag{10}$$

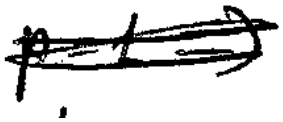
$u_i = \int_{\xi_\alpha}^{\xi_\beta} \omega_i$ | T — period matrix of the given curve;
$d\mu_p$ — a certain holomorphic measure on the moduli space

(see last time)



Uncertain reading. One factor of $\operatorname{Im} u$ is visible in the exponent.

Original source details: GELFAND-P001266



Uncertain reading. Crossed out: $p = 1 \Rightarrow$?

Original source details: GELFAND-P001266

$d\mu_p$ has been computed for $p \leq 4$, unknown beyond that (only Manin's less explicit formulas)

NB: a way of writing Green's function (of the Bergman metric) on a curve! — by means of a map into the Jacobian

Undeciphered passage. Crossed-out word after 'Jacobian'

Original source details: GELFAND-P001266

(Jacobi) and then use Θ-functions.

On the other hand

$$d\mu_p = \prod_i dy_i \frac{\det' \Delta_{-1}}{(\det' \Delta_0)^{13}} \tag{11}$$

$$\delta_{\eta_\alpha} \ln \det' \Delta_0 = \operatorname{Tr}(\partial \eta_\alpha)(\partial G) \stackrel{?}{=} \partial_\xi \eta_\alpha \partial_\xi G(\xi, \xi')\big|_{\xi'=\xi} \tag{12}$$

A full investigation has not been carried out.

$$\partial^2 G(\xi, \xi')\big|_{\xi'=\xi} = \frac{\Theta_{,i}\omega_i'' + \Theta_{,ijk}\omega_i\omega_j\omega_k}{\Theta_{,i}\omega_i} - \frac{3}{2}\left(\frac{\Theta_{,i}\omega_i'}{\Theta_{,i}\omega_i}\right)^2, \tag{13}$$

ω' — the derivative of ω with respect to ξ.

GELFAND-P001267

The realistic theory is the superstring. The impression is that the standard notion of superspace is not quite adequate.

Stat. sum after integration over the odd moduli?

Uncertain reading. The initial abbreviation reads «Стат.» (‘stat.’)

Original source details: GELFAND-P001267

It should be

$$\int_{M_p} \frac{|d\mu_p^{SS}|^2}{(\det \operatorname{Im} T)^5} \tag{14}$$

(Under modular transformations $T \to (AT+B)/(CT+D)$

$$\frac{d\mu_p^{SS}}{d\mu_p^{bos}} = f_8 \qquad \text{— a form of weight 8)} \tag{15}$$

It would be good if there were no cusp forms of weight 8

Genus $1, 2, 3$ — there are none.

Genus 4 — there are some

GELFAND-P001268

10/III–86 Gelfand Sem.

A. Morozov Low-energy limit of string theory

Gauge-theory Lagrangian

$$\operatorname{Tr} F_{\mu\nu}^2 \tag{1}$$

$$F_\mu^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + i \underbrace{f^{abc}}_{\text{structure const.}} A_\mu^b A_\nu^c \tag{2}$$

or

$$F = dA + [A, A] \tag{2'}$$

Gauge transformations:

$$A_\mu \to A_\mu + [A_\mu, \varepsilon] + \partial_\mu \varepsilon \tag{3}$$

How can we embed this in string theory? We flatten the strings into pointlike objects

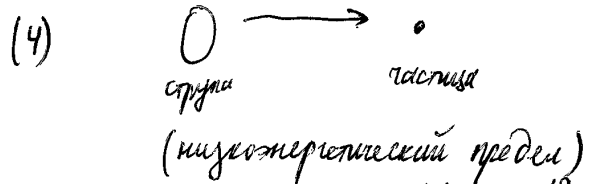


Diagram and typed description.

(4) string → particle

(low-energy limit)

Original source details: GELFAND-P001268

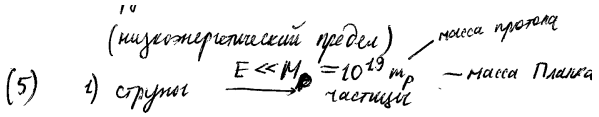


Diagram and typed description.

(5) 1) strings $\xrightarrow{E \ll M_P = 10^{19} m_p}$ particles (proton mass)

Original source details: GELFAND-P001268

— Planck mass

Undeciphered passage. Crossed-out passage

Original source details: GELFAND-P001268

($M_P^2 = \hbar c/\gamma$, γ — grav. const.)

2) particles different fields
different masses

3) massless vectors (particles of mass 0) are described precisely by Lagrangian (1) (this follows from string dynamics)

Where does the gauge group in string theory come from? We have to make the string live

Undeciphered passage. Crossed-out passage

Original source details: GELFAND-P001268

in non-flat space/time.

GELFAND-P001269

[source number: 18]

I. If $x^\mu(\xi) : C \to \mathbb{R}^{26}$ (flat), then vectors without the index i (in $F^i_{\mu\nu}$) $\Rightarrow$ abelian group $U(1)$

$$\int d^2\xi \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \tag{A}$$

II. The method of Chan–Paton gives $\forall$ group.

III.

Uncertain reading. Crossed-out mark: ‘A’?

Original source details: GELFAND-P001269

Consider $C \to T^k \otimes \mathbb{R}^{26-k}$

In the interesting case $k = 16 \rightsquigarrow$ a superstring from a bosonic string.

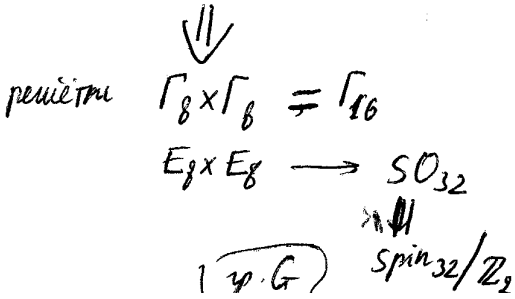


Diagram and typed description.

lattices $\Gamma_8 \times \Gamma_8 \neq \Gamma_{16}$

$E_8 \times E_8 \longrightarrow SO_{32}$

$\downarrow$

$Spin_{32}/\mathbb{Z}_2$

Original source details: GELFAND-P001269

Original source details: GELFAND-P001269

Uncertain reading. The sign between the lattices reads $\neq$?

Original source details: GELFAND-P001269

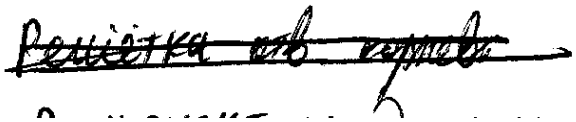
Uncertain reading. Two marks to the left of the arrow toward $Spin_{32}/\mathbb{Z}_2$.

Original source details: GELFAND-P001269

Undeciphered passage. Marks at the upper right of the lattice diagram

gp. G

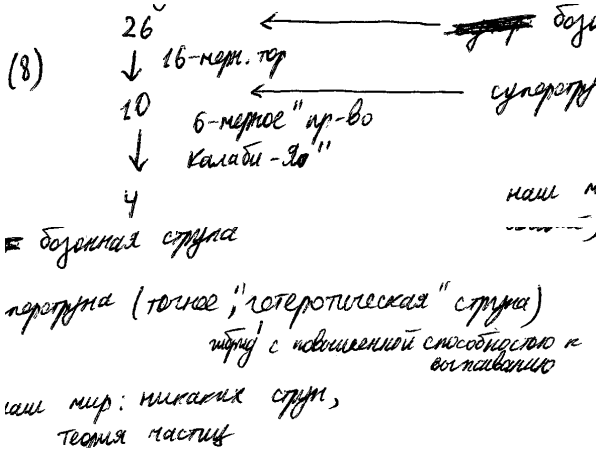
(6) T^k from the lattice of root vectors of the Lie alg. $\Rightarrow$ one gets the corresponding Yang–Mills theory with gauge group G



Original source details: GELFAND-P001269

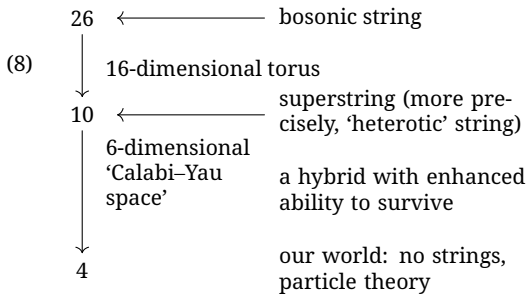
Uncertain reading. Crossed out: ‘Lattice [of?] roots’?

(7) At the birth of the world it was ∞-dimensional, then compactified into a thin film around 4-dimensional space (thickness $\sim 10^{-33}$ cm; the proton’s size is 10^{19} times larger).



Original source details: GELFAND-P001269

Diagram and typed description.



Original source details: GELFAND-P001269

Undeciphered passage. Crossed-out marks before ‘bosonic string’

(9) Question: according to what eqs. does compactification occur?

In diagram (8) the odd variables are left out (the dimensions are even)

Read: Kac, Frenkel — Invent. or Adv. in Math. (on the duality of bosonic and fermionic models)

Physics papers: Freund P.

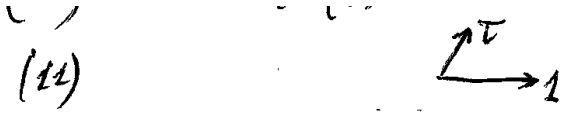
Gross, Harvey, Martinez, Rohm Nucl. Phys. B, 1985.

Original source details: GELFAND-P001269

Uncertain reading. The manuscript reads 'Martinez'.

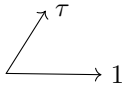
Consider the map

$$x^\mu(\xi) : T^2 \to T^{16} \times \mathbb{R}^{10} \tag{10}$$



Original source details: GELFAND-P001269

Diagram and typed description.

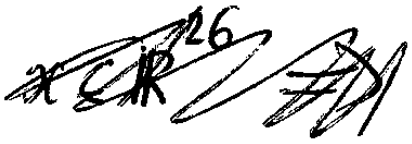
(11) 

— the lattice defining T^2 (one-loop string diagrams)

GELFAND-P001270

We shall compute

$$Z = \int Dx\, Dg_{ab}\ \exp \int d^2\xi \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \tag{12}$$



Original source details: GELFAND-P001270

Uncertain reading. Crossed out: $x \in \mathbb{R}^{26} \Rightarrow$

$x \in T^{16} \times \mathbb{R}^{10} \Rightarrow$

$$\left.\begin{cases} x^\mu(\xi + 1) = x^\mu(\xi) + \Lambda_2^\mu \\ x^\mu(\xi + \tau) = x^\mu(\xi) + \Lambda_1^\mu \end{cases}\right| \tag{13}$$

Original source details: GELFAND-P001270

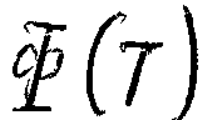
Uncertain reading. In the first line the sign before Λ_2^μ is heavily overwritten. $+$?

$$\Lambda_1^\mu = 0, \quad \mu = 1, \dots, 10; \quad \Lambda_1^\mu \in \text{the lattice for } \mu = 11, \dots, 26. \tag{14}$$

The answers obtained:

$$\underset{\Gamma_{16}}{\text{for}} \qquad \int_{M_p} Z_{[C \to T^{16} \times \mathbb{R}^{10}]} = \int_{M_p} Z_{[C \to \mathbb{R}^{26}]} \cdot \Phi(T) \tag{15}$$

Uncertain reading. One argument T is written in (15).



Original source details: GELFAND-P001270

$$\Phi(T, \bar{T}) = \sum_{\substack{\text{even} \\ \text{charact. } m}} |\Theta_m^{16}(T)|^2 \tag{16}$$

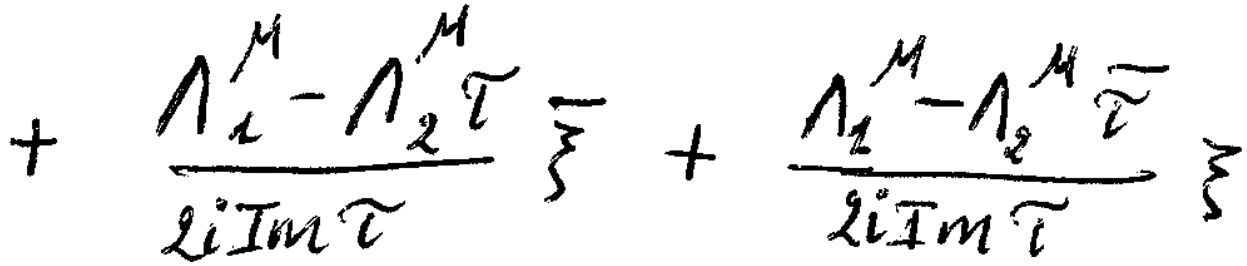
$$F(T) = \sum_{\substack{\text{even} \\ \text{charact.}}} \Theta_m^{16}(T) \tag{17}$$

Let

$$x^\mu(\xi) = \tilde{x}^\mu(\xi) + \frac{\Lambda_1^\mu - \Lambda_2^\mu \tau}{2i \operatorname{Im} \tau} \bar{\xi} + \frac{\Lambda_1^\mu - \Lambda_2^\mu \bar{\tau}}{2i \operatorname{Im} \tau} \xi \tag{18}$$

Uncertain reading. A $+$ is written before the second fraction term.

Original source details: GELFAND-P001270



Original source details: GELFAND-P001270

Uncertain reading. Both signs between the summands are $+$.

Then

(19) $\tilde{x}^\mu$ satisfies (13) without Λ_j^μ (as for a map into $\mathbb{R}^{26}$).

Integral (12) after integrating over x and g becomes

$$\int_M Z[T^2 \to T^{16} \times \mathbb{R}^{10}] \tag{20}$$

Uncertain reading. The sign $\times$ is crossed out in (20)

Original source details: GELFAND-P001270

(M — moduli space of Riemann surfaces)

$$Z[T^2 \to T^{16} \times \mathbb{R}^{10}] = \sum_{\Lambda_1, \Lambda_2} Z[T^2 \to \mathbb{R}^{26}] \exp\left(-\frac{|\Lambda_1 - \Lambda_2\tau|^2}{\operatorname{Im}\tau}\right) \tag{21}$$

Original source details: GELFAND-P001270

Uncertain reading. The sign $\times$ is crossed out in (21)

Original source details: GELFAND-P001270

Uncertain reading. The zero in the exponent 10 is circled or overwritten.

Integration over x (det), over $g \to e^{\varphi}\tilde{g}$

in dim. 26 the dependence on φ disappears etc.

GELFAND-P001271

[source number: 19]

Naively there is no dependence on φ since the dimension $= 2$. (The contributions of $\sqrt{g}$ and g^{ab} cancel)

So we need to know:

$$\Phi(\tau, \bar{\tau}) = \sum_{\Lambda_1, \Lambda_2 \in \Lambda} \exp -\frac{|\Lambda_1 - \Lambda_2\tau|^2}{\operatorname{Im}\tau} \tag{22}$$

Λ — the lattice corresponding to the 16-dimensional torus

Original source details: GELFAND-P001271

Uncertain reading. Crossed out: ‘чтоб’ (‘so that’)?

(does not possess any properties, but sometimes)

Generally speaking, it reduces to a finite sum of squared absolute values of anal. functions.

$$\varphi(\tau,\bar{\tau}) = \sum_{i=1}^{n} |f_i(\tau)|^2, \qquad f_i \text{ — analytic functions.} \tag{23}$$

There are 2 examples

This happens when

Original source details: GELFAND-P001271

Uncertain reading. The abbreviation before ‘when’ reads ‘быв.’.

(24)

Original source details: GELFAND-P001271

Uncertain reading. Crossed-out mark before ‘lattice’: ‘2)’?

the lattice is even and self-dual

For Γ_{16}, the formula apparently holds

$$\varphi(\tau,\bar{\tau}) = |\Theta_{01}^{16}(\tau)|^2 + |\Theta_{10}^{16}(\tau)|^2 + |\Theta_{00}^{16}(\tau)|^2 \tag{25}$$

(Apparently evenness is unnecessary)

Also put

$$F(\tau) \equiv \sum_{i=1}^{n} f_i(\tau) \tag{26}$$

Let us require that F behave well. Then the superstring and heterotic string are defined. It is interesting whether anything remains for lattices that are not self-dual. It turns out that

$$F(\tau) = \sum_{\lambda\in\Lambda} e^{i\pi\lambda^2\tau} \equiv \Theta \tag{27}$$

Next, we obtain the measures

$$|d\mu^{SS}(y)|^2 = |d\mu^{bos}(y) f^{ferm}(y)|^2 \tag{28}$$

(superstr.)

Original source details: GELFAND-P001271

Uncertain reading. After $d\mu^{bos}(y)$, $|^2$ is crossed out.

$$d\mu^{het}(y,\bar{y}) = |d\mu^{bos}(y)|^2 f^{ferm}(y)\bar{F}(\bar{y}) \tag{29}$$

Here

$$|d\mu^{bos}(y)|^2 = Z[T^2 \to \mathbb{R}^{26}] \tag{30}$$

(31) For even self-dual lattices the sum (26) transforms as a modular form

Uncertain reading. The end of the word ‘self-dual’ is overwritten.

Original source details: GELFAND-P001271

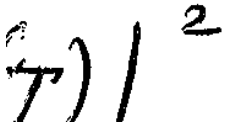
(32) It is unclear whether that holds only for even self-dual ones

GELFAND-P001272

(33) In $\mathbb{R}^{16}$ there are only 2 even self-dual lattices. (see Serre, ‘A Course in Arithmetic’)
In terms of $F(\tau)$ these lattices are indistinguishable

$$F_{\Gamma_{16}}(T) = \sum_{\substack{\text{even} \\ m}} \Theta_m^{16}(T), \tag{34}$$

$$F_{\Gamma_8 \times \Gamma_8}(T) = \left| \sum_{\substack{\text{even} \\ m}} \Theta_m^{8}(T) \right|^2. \tag{35}$$



Uncertain reading. In (35) the vertical bars and square are written.

Original source details: GELFAND-P001272

We want:

$$\text{Strings} \to \text{particles} \to \text{massless particles}. \tag{36}$$

1 More general integrals are needed:

Undeciphered passage. Crossed-out writing before ‘Vertex’

Original source details: GELFAND-P001272

Vertex opers. — under the cont. integral

Uncertain reading. The abbreviation before ‘integral’ is ‘конт.’ (‘cont.’).

Original source details: GELFAND-P001272

$$V(k_\alpha) = \int d^2\xi \, e^{ik^\mu x^\mu(\xi)} \tag{37}$$

$$V^\mu(k_\alpha) = \int d^2\xi \, \partial x^\mu(\xi) e^{ik^\mu x^\mu(\xi)} \tag{38}$$

(39)

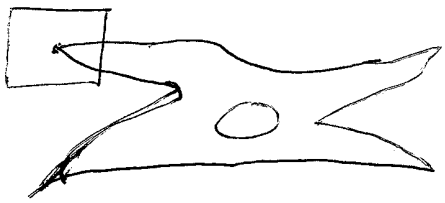
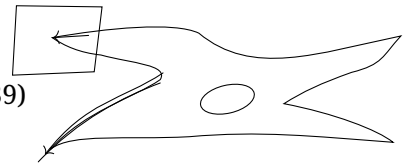


Diagram and typed description.

(39)



Original source details: GELFAND-P001272

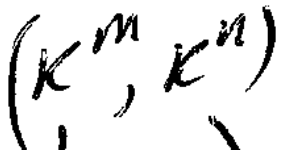
Weights can be added in the frames

Consider $T^{16} \times \mathbb{R}^{10}$ instead of $\mathbb{R}^{26}$.

Take the vertices corresponding to $\mathbb{R}^{10}$.

$$V^m(k) = \int d^2\xi \, \partial x^m e^{ik^m x^m} \cdot \underbrace{e^{ik^M x^M}}_{\substack{\text{This distinguishes it} \\ \text{from the 26-dimensional case}}} \tag{40}$$

[Under k:] $\left(\underset{\text{10-dimensional}}{k^m}, \underset{\text{16-dimensional}}{k^M}\right)$



Uncertain reading. The superscript of the second component reads M?

Original source details: GELFAND-P001272

We need to know [inserted: new] correlators. This leads to commut. relations,

$$[e^{ik^M x^M}, e^{ik'^M x^M}] \sim e^{i(k^M + k'^M)x^M}, \tag{41}$$

if $k^2 = k'^2 = 2$, $(k + k')^2 = 2$, and 0 in the remaining cases.

(Kac, Frenkel)

GELFAND-P001273

[source number: 20]

the commutators are understood as follows

$$x = \sum a_n z^n + \ldots \qquad \frac{\partial}{\partial z} \tag{42}$$

This leads to the appearance of groups in the string:

$$E_8 \times E_8 \quad \text{or} \quad SO_{32}$$

GELFAND-P001274

24.III.86. Gelfand Sem.

Uncertain reading. The manuscript reads ‘24.III.86.’. Catalogue record 672 dates the talk 24 March 1987.

Original source details: GELFAND-P001274

Fuchs. Cohomology

$\mathbb{R}^n \setminus \bigcup L_i$, L_i — (affine) planes

$\mathbb{R}^n \supset \bigcup L_i$, L_i — affine subspaces of arbitrary dimensions.

$$X = \mathbb{R}^n \setminus \bigcup L_i.$$

How to calculate $H_k(X)$

Suppose L_i — hyperplanes

Then $H_k = 0$ for $k \neq 0$.

Answer (Zaslawsky, 1975)

Construct

Uncertain reading. Crossed out: possibly ‘such a’.

Original source details: GELFAND-P001274

partially ordered set

Undeciphered passage. Crossed-out writing before A

Original source details: GELFAND-P001274

A with elements $\mathbb{R}^n$, L_i and all their nonempty intersections

Uncertain reading. The endings in the source read ‘непустым пересечения’.

Original source details: GELFAND-P001274

Möbius function

$$\mu(\mathbb{R}^n) = 1, \qquad \mu(L_i) = -1$$

$$\mu(v) = -\sum_{w>v} \mu(w)$$

$$h_0 = \sum_{v \in A} |\mu(v)| \qquad \leftarrow \text{Zaslavsky's formula.}$$

Uncertain reading. The letter A under the sum is overwritten.

Original source details: GELFAND-P001274

Example.

In $\mathbb{R}^3$ coordinate planes

$\mathbb{R}^3$,	L_x,	L_y,	L_z,	L_{xy},	L_{xz},	L_{yz},	0
1	-1	-1	-1	1	1	1	-1

Uncertain reading. The last letter in L_{yz} is overwritten.

Original source details: GELFAND-P001274

$$\sum_{1}^{8} |\pm 1| = 8, \quad \text{as expected (8 octants)}$$

Main case (Goresky).

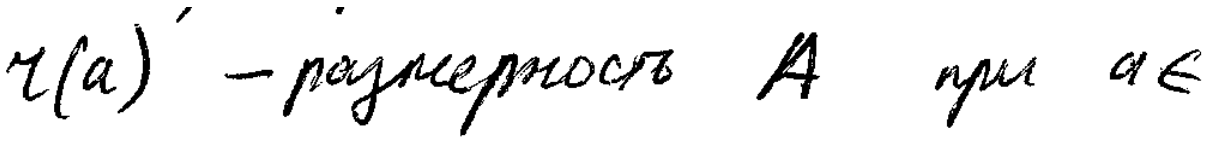
A

Undeciphered passage. Crossed-out writing after A

Original source details: GELFAND-P001274

is defined in the same way; A — partially ordered set

$\tau(a)$ — dimension of A for $a \in$



Original source details: GELFAND-P001274

Uncertain reading. The first symbol reads τ or r. The writing stops after $a \in$.

Theorem.

$$H_i(X) = \bigoplus_{a \in A} H_i(a),$$

(Goresky–MacPherson)

where

$$H_i(a) = H^{n-\tau(a)-i-1}\big(X_{>a}, X_{(a,\mathbb{R}^n)}\big),$$

Original source details: GELFAND-P001274

Uncertain reading. The letter in the exponent reads τ or r.

Original source details: GELFAND-P001274

Uncertain reading. The second X is overwritten. Its earlier form is unclear.

GELFAND-P001275

[source number: 29]

where

$$\left.\begin{matrix} > a \\ (a, \mathbb{R}^n) \end{matrix}\right\} \text{ subsets of the set } A \quad \begin{matrix} \leftarrow \{b \mid b > a\} \\ \leftarrow \{b \mid b > a\} - \{\mathbb{R}^n\} \end{matrix}$$

X_M, where M — a partially ordered set (in particular, $M \subset A$) is by definition a simplicial complex with vertices the elements of M, and p-dimensional simplices the subsets

$$\{a_0 < \cdots < a_p\} \subset M$$

(Comment: by definition we take

$$H^i(\varnothing, \varnothing) = \begin{cases} \mathbb{Z}, & i = -1, \\ 0, & i \neq -1. \end{cases}$$

The pair $(X_{>a}, X_{(a,\mathbb{R}^n)}) = (\varnothing, \varnothing)$ only if $a = \mathbb{R}^n$.

Indeed, $X_{>a}$ is always contractible, so $H^k(X_{>a}, X_{(a,\mathbb{R}^n)})$ reduces to the cohomology of $X_{(a,\mathbb{R}^n)}$, namely

$$H^i\begin{pmatrix}\text{contractible} \\ \text{space}\end{pmatrix}, B\Big) = \begin{cases} H^{i-1}(B), & i > 1 \\ H^0(B)/\mathbb{Z}, & i = 1. \end{cases}$$

In the Goresky

Original source details: GELFAND-P001275

Uncertain reading. Crossed out, possibly ‘and’.

— MacPherson theorem the coefficients are arbitrary.

Special case: Orlik–Solomon thm. (

Original source details: GELFAND-P001275

Uncertain reading. An overwritten sign before ‘when’, possibly ‘в’.

when everything is complex).

Example.

$\mathbb{R}^3 \setminus$ coordinate planes and one more line through 0, but lying in none of the hyperplanes.

$\underline{8}$ — 0-dimensional and $\underline{2}$ 1-dimensional.

Question: given an arbitrary partially ordered subset, what is the meaning of what is on
(I.M.G.)
the left (something like a toric variety

Original source details: GELFAND-P001275

Uncertain reading. Abbreviated ‘мн-зия’ or ‘м-зия’ in the source.

)

On the proof.

Consider a point [inserted above:] x_0 in general position in $\mathbb{R}^n$ and a ball centred at x_0; we shall expand it, tracking changes in the homology of what is inside (need relative homology).

Original source details: GELFAND-P001275

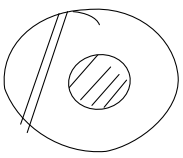
Uncertain reading. The insertion reads x_0 or a_0.

Original source details: GELFAND-P001275

Uncertain reading. After ‘need’ the visible letters read ‘на’, perhaps intended as ‘нам’ (‘we need’).

Original source details: GELFAND-P001275

Diagram and typed description.



ball with a line removed

mod the excised smaller ball.
GELFAND-P001276

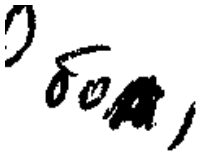
In fact the problem does not reduce to computing relative homology.

Uncertain reading. Crossed out: D_{small} and D_{large} with a sign that could be $\subset$ or $\supset$.

Original source details: GELFAND-P001276

$$D_{\text{small}} \subset D_{\text{large}}$$

$$H_*(D_{\text{large}}) = H_*(D_{\text{small}}) \oplus H_*(D_{\text{large}}, D_{\text{small}})$$



Uncertain reading. The end of the subscript 'large' in the relative group is overwritten.

Original source details: GELFAND-P001276

if the exact sequence splits, but here this is checked by purely geometric means.
(1)

Uncertain reading. Crossed out: 'When'.

Original source details: GELFAND-P001276

Example, where it is carried through to the end.

$$\operatorname{codim} L_i = k \qquad \text{(independent of } i\text{)}$$
$$\operatorname{codim}(L_{i_1} \cap \cdots \cap L_{i_q}) \quad \text{is divisible by } k$$

($k = 1$ is satisfied, and in the complex case it is satisfied)
But even

Uncertain reading. Crossed out, possibly 'with'.

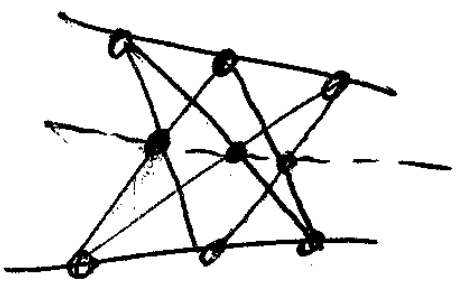
Original source details: GELFAND-P001276

in $\mathbb{R}^4$ with $k = 2$ this need not be a complex situation!! (There is an example of Goresky's)

Question (ИМГ): under what conditions does there exist a complex structure in which they all become complex?

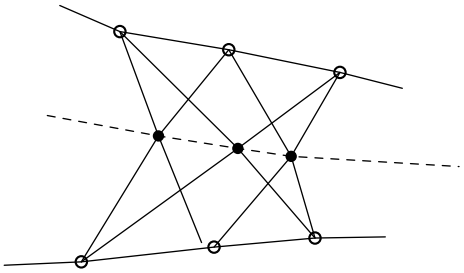
Goresky's example.

Pappus's thm..



Original source details: GELFAND-P001276

Diagram and typed description.

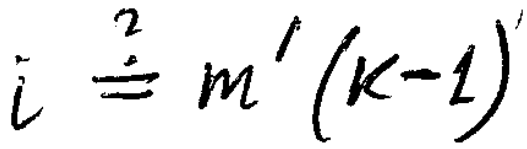


Under complex perturbation the situation persists, but under real perturbation it does not!!

Suppose the conditions of (1). hold. Then

$$\forall a,\ H_i(a) \neq 0 \quad \text{only for one } i:$$

$$i \stackrel{?}{=} m'(k-1)$$



Uncertain reading. A question mark stands above the equality. There is a faint stroke after the parenthesis, possibly another prime.

Original source details: GELFAND-P001276

$$H_i(a) = \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{|\mu(a)|}$$

Question of I.M.G.. How does one find the right basis for cohomology?

GELFAND-P001277

[source number: 30]

Undeciphered passage. Crossed-out writing before the name

Original source details: GELFAND-P001277

Varchenko. Mixed Hodge structures

These are 2 filtrations in a vector space, increasing and decreasing

(in the cohomology of a complex variety) Hodge and weight

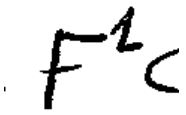
[Below 'weight', by the stroke:] topological (unchanged under deformation of the complex structure)

They are suitably compatible and vary suitably when the variety changes. There will be an example of this.

Cohomology of a nonsingular projective variety (compact)

In cohomology there is a pure Hodge structure

$$0 \subset F^k \subset \cdots \subset F^1 \subset F^0 = H^k \tag{1}$$



Uncertain reading. The superscript reads 1. The upper horizontal stroke joins F.

Original source details: GELFAND-P001277

Here

$$F^s = \{[\omega], \quad \omega \text{ — closed complex-valued differential form,} \\ \omega = \sum a_i \underbrace{dz_{i_1} \wedge \cdots \wedge dz_{i_s}}_{\text{in all } k} \wedge \underbrace{\cdots}_{}\} \tag{2}$$

$\exists$ a representative: ($\geq s$ holomorphic differentials in the local expression)

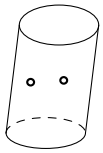
Change of complex structure $\Rightarrow$ the subspaces vary continuously.

Example of a weight filtration.

In $\mathbb{C} \setminus \{0\}$ identify 2 points $*, *$

Topologically:

Diagram and typed description.



Original source details: GELFAND-P001277

$$\dim H^1(X) = 2$$

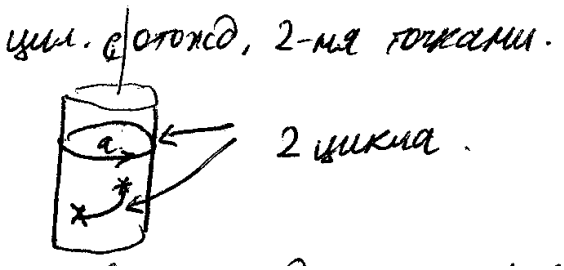
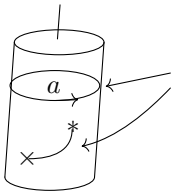


Diagram and typed description.

cyl. with 2 points identified.



2 cycles.

Original source details: GELFAND-P001277

In H_1 there is a 1-dimensional subspace generated by the cycle a

Uncertain reading. The superscript is crossed out, possibly 1.

Original source details: GELFAND-P001277

$W \subset H^1(X)$

Undeciphered passage. Crossed-out writing after W

Original source details: GELFAND-P001277

[Below the cancellation, by the oblique stroke:] the annihilator of this 1-dimensional subspace.

[By the vertical stroke under W:] weight space.

Another way mixed Hodge structures arise connected with asymptotics.

Consider a real-analytic function $f : \mathbb{R}^n \underset{\text{analytic}}{\longrightarrow} \mathbb{R}$.

$\omega = h\, dx_1 \wedge \cdots \wedge dx_n$ with compact supp

$$\omega \mapsto \int_{\mathbb{R}^n} f^\lambda \omega \tag{3}$$

extends

Uncertain reading. Crossed out before ‘на’, possibly ‘на’ or ‘не’.

Original source details: GELFAND-P001277

to all of $\mathbb{C}$ with poles at the points

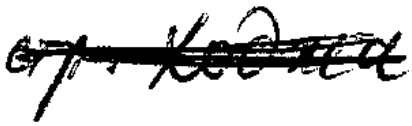
$$\int_{\mathbb{R}^n} f^\lambda \omega = \sum a_{k,\alpha}(\omega) \frac{1}{(\lambda+\alpha)^k} \tag{4}$$

Uncertain reading. A crossed-out mark to the right of (4), possibly α or a.

Original source details: GELFAND-P001277

GELFAND-P001278

k — a manifestation of the weight filtration



Original source details: GELFAND-P001278

Uncertain reading. Crossed out between ‘weight’ and ‘filtration’: ‘Hodge str.’.

α —”— Hodge —”—

2 parameters, the integer k and the rational α are always such a manifestation of a mixed Hodge str..

Original source details: GELFAND-P001278

Uncertain reading. The manuscript reads ‘таким’ (‘such a’).

Complex case.

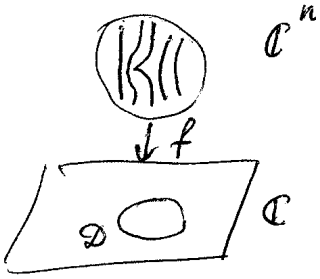
$f : (\mathbb{C}^n, 0) \longrightarrow (\mathbb{C}, 0), \quad f$ holom..

0 — an isolated critical point

Consider the fibre $f^{-1}(0)$ — singular, with isolated singular point 0.

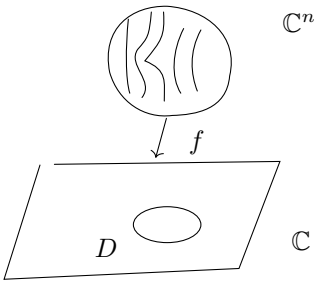
Consider in $\mathbb{C}^n$ a small ball and in $\mathbb{C}$ a small disc (much smaller) $\Rightarrow$

$\Rightarrow$ one gets holomorphic fibres (not algebraic at all)



Original source details: GELFAND-P001278

Diagram and typed description.



$X_t = f^{-1}(t)$ — one fibre.

Original source details: GELFAND-P001278

Uncertain reading. The X in X_t is overwritten.

Original source details: GELFAND-P001278

Uncertain reading. Crossed out after $f^{-1}(t)$, possibly ‘in...’.

$(n-1)$-dimensional complex variety.

Milnor's thm.:

$$X_t \underset{\text{homotopy equiv.}}{\sim} \underbrace{S^{n-1} \vee \cdots \vee S^{n-1}}_{\substack{\mu \text{ — the Milnor number} \\ \text{of the critical point}}}$$

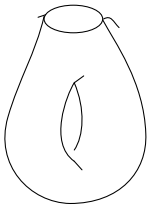
$n = 2 \Rightarrow X_t$ — a noncompact complex curve, connected

At the boundary something is cut off

Uncertain reading. Crossed out after 'boundary', possibly 'of the fibre'.

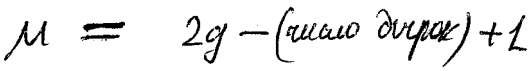
Original source details: GELFAND-P001278

Diagram and typed description.



Original source details: GELFAND-P001278

$$\mu = 2g - (\text{number of holes}) + 1 \tag{5}$$



Uncertain reading. In (5) a minus is visible before the parentheses and $+1$ after them.

Original source details: GELFAND-P001278

Diagram and typed description.

Original source details: GELFAND-P001278

One gets a bouquet of circles.

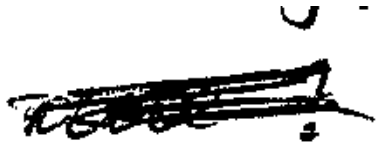
Going around produces the monodromy transformation $M : H^{n-1}(X_t) \longrightarrow H^{n-1}(X_t)$

Uncertain reading. Crossed-out heading, possibly 'Integrals'.

Original source details: GELFAND-P001278

Gelfand's question: what about $\int f_1^{\lambda_1} f_2^{\lambda_2} \cdots f_k^{\lambda_k}$ — ?

For $\mathbb{C}^n \to \mathbb{C}^k$ will something like a mixed Hodge str. arise?



Undeciphered passage. Crossed-out writing between 'will' and 'arise'

Original source details: GELFAND-P001278

I.M.G.'s question: what arises for a mapping given by a pair of quadratic forms?

Uncertain reading. Crossed out after 'what', possibly 'бы'.

Original source details: GELFAND-P001278

Consider the form

$\omega = h\, dz_1 \wedge \cdots \wedge dz_n, \qquad h$ — holom..

On a nonsingular fibre one can consider

ω/df — holom. $(n-1)$-form

Take a distinguished cycle in the fibre and extend it continuously to nearby fibres. We obtain a multivalued family of cycles $\delta(t)$

Uncertain reading. A short rising stroke after $\delta(t)$, possibly a comma.

Original source details: GELFAND-P001278

GELFAND-P001279

[source number: 31]

$$I(t) = \int_{\delta(t)} \frac{\omega}{df} \tag{6}$$

Passage through a critical value — Morse's thm.

going around 0 — Picard–Lefschetz thm.

These 2 aspects give 2 filtrations.

Weight filtration — a filtration of the linear operator M by its Jordan blocks.

Uncertain reading. A short overwritten mark over the end of 'Весовая' ('Weight'), possibly 'й'.

Original source details: GELFAND-P001279

Undeciphered passage. Crossed-out writing before ‘linear operator’

Original source details: GELFAND-P001279

For M we need to construct a nilpotent matrix and its filtration

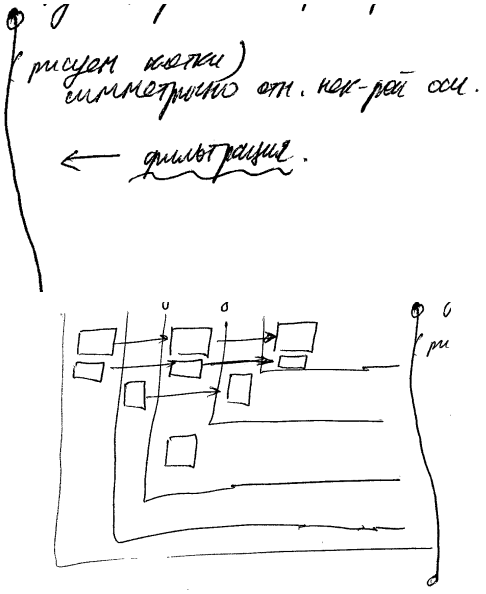
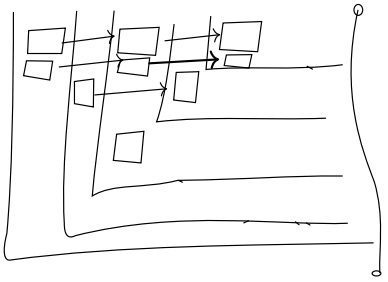


Diagram and typed description.



(draw the blocks) symmetrically about some axis.
⟵ filtration.

Original source details: GELFAND-P001279

1) Eigenvectors in the largest Jordan blocks

2) Add eigenvectors in smaller blocks

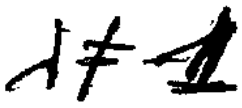
Undeciphered passage. Crossed out after ‘blocks’

Original source details: GELFAND-P001279

and so on

Different eigenvalues are grouped depending on whether

$$\underset{\text{middle weight } n}{\lambda = 1} \quad \text{or} \quad \underset{\text{weight } n-1 \text{ for all}}{\lambda \neq 1}$$



Uncertain reading. The 1 in $\lambda \neq 1$ is overwritten.

Original source details: GELFAND-P001279

$$I(t) = \int_{\delta(t)} \frac{\omega}{df} = \sum a_{k,\alpha}\, t^{\alpha} (\ln t)^k \tag{7}$$

Uncertain reading. Crossed out before $a_{k,\alpha}$, possibly ‘where’.

Original source details: GELFAND-P001279

$a_{k,\alpha}$ depend linearly on ω and on δ

$e^{2\pi i\alpha}$ = eigenvalues of M

k = [inserted above: sizes] of Jordan blocks of the operator M

Uncertain reading. The insertion reads ‘sizes’ or ‘size’.

Original source details: GELFAND-P001279

References:

1) Varchenko, DAN, 1980

2) the construction in VINITI

Undeciphered passage. Crossed-out mark before ‘the construction’

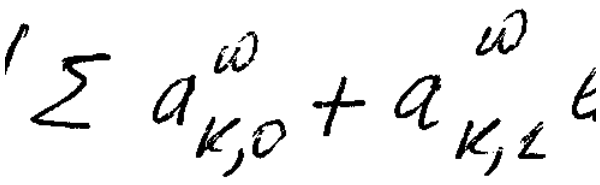
Original source details: GELFAND-P001279

3) Book on singularities

GELFAND-P001280

Leading term of the asymptotics.

$$I(t) = t^{\alpha(\omega)} \left(\sum a^{\omega}_{k,0} + a^{\omega}_{k,1} \ln t + \dots \right) + o\left(t^{\alpha(\omega)} \right) \tag{8}$$



Uncertain reading. The sum sign in (8) has no index. The coefficients show subscripts $k, 0$ and $k, 1$.

Original source details: GELFAND-P001280

The mixed Hodge str. is obtained from the first term.
The Bernstein polynomial comes from all the terms.

$V_\alpha = \{$classes of the leading parts of forms $\omega,\ \alpha(\omega) = \alpha\}$ — cohomology classes (α — rational)

$$V_\alpha \subset V_{\alpha+1}$$

Varchenko was interested in the semicontinuity of the leading parts of the asymptotics.

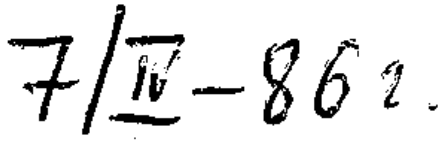
полунепр.

Uncertain reading. The abbreviation is ‘полунепр.’ (‘semicontinuity’).

Original source details: GELFAND-P001280

GELFAND-P001281

7/IV—86 Gelfand sem.


7/IV—86 г.

Uncertain reading. The handwritten year is ‘86’.

Original source details: GELFAND-P001281

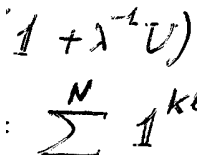
Cherednik. Yangian.
Definition of the Yangian.

$$\underset{\text{matr. } N\times N}{M_N} \supset \{\mathbf{1}^{k\ell}\}, \qquad \mathbf{1} \text{ — matrix identity}$$

[Slanting line from $\{\mathbf{1}^{k\ell}\}$ to ‘std. basis’.] (1 in position k, ℓ, the others are 0)

$$(\mathbf{1}^{k\ell})_{ij} = \delta_{ik}\delta_{j\ell}.$$

$$\det(\mathbf{1} + \lambda^{-1}U) \text{ — ?}$$


1 + λ⁻¹U)
: ∑ 1^k

Uncertain reading. The letter in the determinant reads as U.

Original source details: GELFAND-P001281

$$U = \sum_{k,\ell=1}^{N} \mathbf{1}^{k\ell} e_{\ell k} \in M_N\big(U(g\ell_N)\big)$$

$$\{e_{\ell k}\} \subset U(g\ell_N)$$

Uncertain reading. The initial letter of $U(g\ell_N)$ has the same form as the one in the determinant.

Original source details: GELFAND-P001281

(1) [Crossed out]

Uncertain reading. The strokes appear to cover $L_n(\lambda)$.

Original source details: GELFAND-P001281

$$L_n(\lambda) = E\mathbf{1} + \sum_{k,\ell} \sum_{q=1}^{n} \frac{1}{\lambda^q} \mathbf{1}^{k\ell} E_{\ell k}^{q-1} =$$

coeff. are the letters $E_{\ell k}^{q}$ (with 3 indices)

$$= \begin{pmatrix} E & & 0 \\ & E & \\ & \ddots & \\ 0 & & E \end{pmatrix} + \cdots$$

[At right:] (the index n is sometimes omitted below)

Set

$$R(\lambda) := \mathbf{1} \otimes \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{k\ell} \otimes \mathbf{1}^{\ell k} \in M_N^{\otimes 2} = M_{N^2} \tag{2}$$

(ordinary matrix function).

Consider the equation

$$R(\lambda_1 - \lambda_2) \cdot {}^1L_n(\lambda_1)\, {}^2L_n(\lambda_2) = {}^2L_n(\lambda_2)\, {}^1L_n(\lambda_1) R(\lambda_1 - \lambda_2), \tag{3}$$

Uncertain reading. There is a heavy ink mark between R and ${}^1L_n$. It is represented here by a multiplication dot.

Original source details: GELFAND-P001281

$$^1L_n = L_n \otimes \mathbf{1}, \qquad ^2L_n = \mathbf{1} \otimes L_n$$

The coefficients are quadratic in the letters $E^q_{\ell k}$

We require (3) to hold identically in λ_1, λ_2.

Eq. (3) is called the YBF equations (Yang–Baxter–Faddeev).

Uncertain reading. In ‘Фаддеева’ (Faddeev), one letter has been overwritten.

Original source details: GELFAND-P001281

It is equivalent to the algebraic system (4)–(5)

$$[E, E^q_{k\ell}] = 0 \tag{4}$$

$$[E^p_{ij}, E^q_{k\ell}] = \delta_{jk} E E^{p+q}_{i\ell} - \delta_{\ell i} E E^{p+q}_{kj} + \sum_{a+b=p+q-1}^{a<p\le b} \left(E^a_{kj} E^b_{i\ell} - E^b_{kj} E^a_{i\ell}\right) \tag{5}$$

(6) By definition $E^{n+i}_{k\ell} = 0$ for $i \ge 0$

GELFAND-P001282

[source number: 43]

(7) Claim. For $E = 1$ and on dropping the last term of (5), we obtain

$$U\big(g\ell_N[t]/(t^n)\big).$$

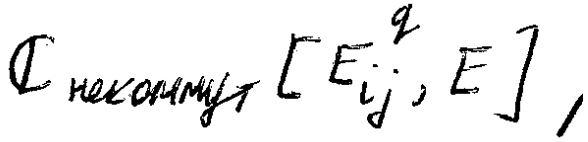
(envelop. algebra)

Uncertain reading. Before $g\ell_N$ there is a crossed-out sign, possibly g.

Original source details: GELFAND-P001282

Definition of the Yangian.

$$Y^n = \mathbb{C}_{\text{noncommut.}}[E^q_{ij}, E]\,/(4)\text{–}(5) \tag{8}$$

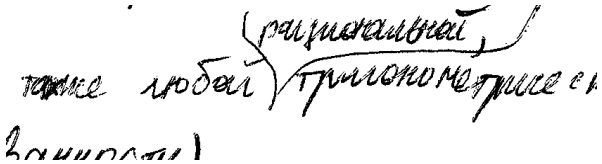


Uncertain reading. Below $\mathbb{C}$ the abbreviation reads ‘некоммут’ (‘noncommutative’). A comma follows E^q_{ij}.

Original source details: GELFAND-P001282

[!] IMG: Idea: commutation relations unlike those in a Lie alg., with $x_1x_2 - x_2x_1$ — a polynomial of deg. 2! Describing quadr. algebras is a proper math. question. (It is interesting when the Poincaré–Birkhoff–Witt theorem holds).

Ch. R-matrix (2) can also be taken to be any [Inserted above the line: rational,] trigonometric and elliptic one (with a compatibility rel.)



Uncertain reading. 'Rational,' is inserted above the line before 'trigonometric'.

Original source details: GELFAND-P001282

[!] Problem (IMG) Classify so that the nth degree has the requisite number of pieces.

Associativity is also optional

(Jackiw: associat. is a cocycle of deg. 3)

Ch.

(9) Claim. For $E = 1$ we have $Y^1 = U(g\ell_N)$

(univ. envelop. algebra)

(10) Zamolodchikov's theorem For rational R the P.B.W. theorem holds if and only if the relation holds $R\,R\,R = R\,R\,R$

Kogan Yan Strings and gauge fields.

Gauge symmetry $\overset{?}{\longleftarrow}$ string

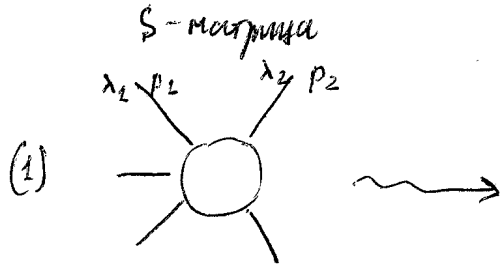
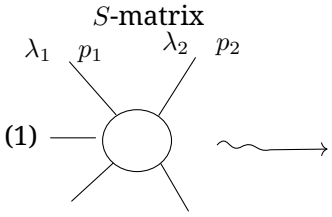


Diagram and typed description.



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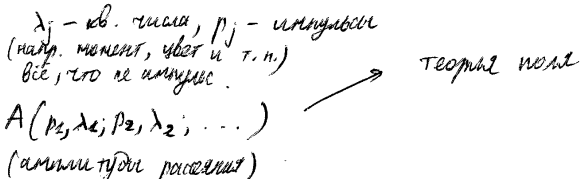

Diagram and typed description.

λ_j — quantum nos., p_j — momenta
(e.g. angular momentum, color, etc.)
everything other than momentum.

$A(p_1, \lambda_1; p_2, \lambda_2; \dots)$ → field theory
(scattering amplitudes)

Original source details: GELFAND-P001282

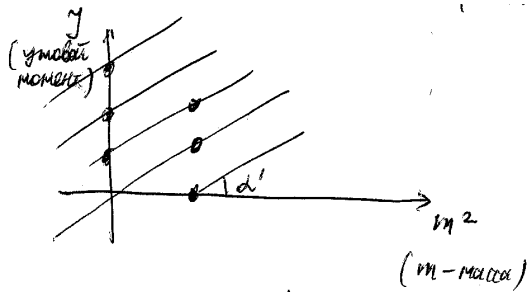

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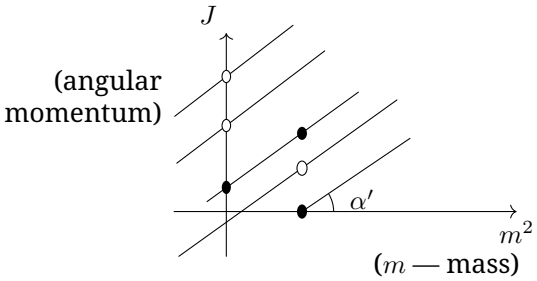
Diagram and typed description.



The six handwritten marks are shown as circles. The amount of fill is uncertain in some marks.

Ang. momentum is always quantized.
(α' — angle —
— string tens.)
[Crossed out]

Original source details: GELFAND-P001282

Uncertain reading. The crossed-out word appears to begin with ‘Угл.’ (‘ang.’).

Limit $\alpha' \to 0 \Rightarrow$ Yang–Mills or Einstein

GELFAND-P001283

Action for the string

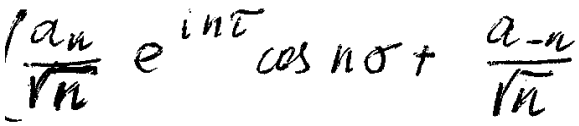
$$S = \frac{1}{2\pi\alpha'} \int g^{ab} \sqrt{g}\, \partial_a x^\mu \partial_b x^\mu \, d^2\xi \tag{2}$$

When passing to

$$S = \frac{1}{2\pi\alpha'} \int \partial_a x^\mu \partial_a x^\mu \, d^2\xi \tag{3}$$

the ghost problem arises from the wrong sign for x^0.
[Inserted between the lines: expand in oscillators]

$$x^\mu(\sigma,\tau) = x^\mu + p^\mu \tau + i \sum_{n=1}^{\infty} \left(\frac{a_n}{\sqrt{n}} e^{in\tau} \cos n\sigma + \frac{a_{-n}}{\sqrt{n}} e^{-in\tau} \cos n\sigma \right) \tag{4}$$

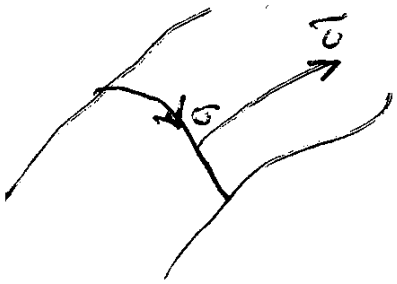


Original source details: GELFAND-P001283

Uncertain reading. Both denominators in (4) read $\sqrt{n}$.

(τ — time)

σ — parameter along the string



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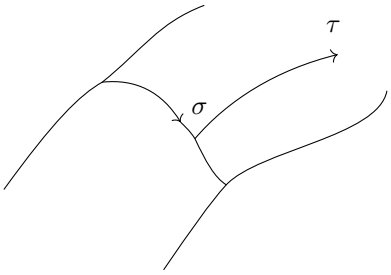
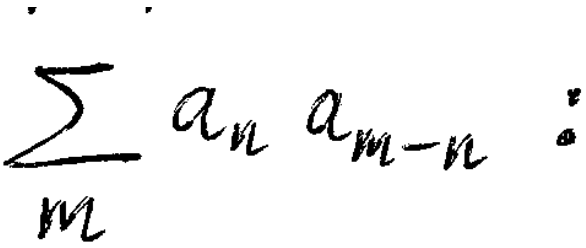


Virasoro operator

$$L_n =: \sum_m a_n a_{m-n} : \tag{5}$$



Original source details: GELFAND-P001283

Uncertain reading. In (5) the handwritten factors read $a_n a_{m-n}$, with summation over m.

a_k for $k > 0$ — annihilation op.

a_k for $k < 0$ —"— creation

(6) States

$$\underset{\text{(vacuum)}}{|0\rangle} \mapsto \text{tachyon}$$

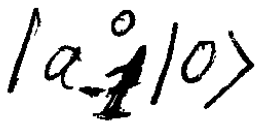
$$a^\mu_{-1}|0\rangle \quad \text{— massless vector particle}$$

$$a^\mu_{-1}a^\nu_{-1}|0\rangle, \qquad a^\mu_{-2}|0\rangle \qquad \text{etc.}$$

In general, it is easy to check that

$$M^2 = -1 + \sum_{n=1}^{\infty} : a^\mu_{-n} a^\mu_n : \tag{7}$$

Many states in (6) are extraneous. For example, $|a^0_{-1}|0\rangle$ has the wrong norm

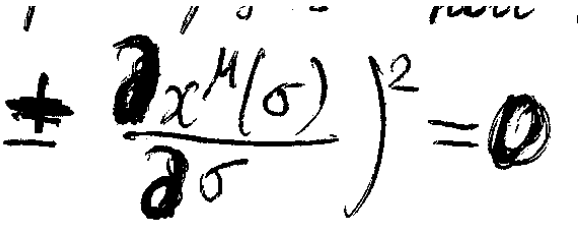


Original source details: GELFAND-P001283

Uncertain reading. The subscript -1 on a is retraced. A separate vertical stroke is visible before a.

[Inserted between the lines: classically there is]
Reparametrization invariance cond.

$$\left(p^{\mu}(\sigma) \pm \frac{\partial x^{\mu}(\sigma)}{\partial \sigma}\right)^2 = 0. \tag{8}$$



Uncertain reading. The $\pm$, the derivative and the zero in (8) are retraced.

Original source details: GELFAND-P001283

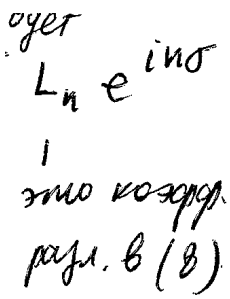
GELFAND-P001284

[source number: 44]

It follows from (8) that

$$\sum_n L_n e^{in\sigma} = 0 \quad \Rightarrow L_n = 0 \tag{9}$$

[Under the first L_n, attached by a short stroke: these are the coeffs. of the expansion in (8).]

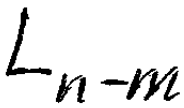


Uncertain reading. A short stroke connects the first L_n to the note below the formula.

Original source details: GELFAND-P001284

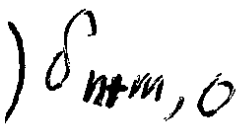
But in quantum theory this is impossible, since there is a central charge.

$$[L_n, L_m] = (n-m)L_{n-m} + \frac{D}{12}(n^3 - n)\delta_{n+m,0} \tag{10}$$



Uncertain reading. The index after $(n-m)$ is written $n-m$.

Original source details: GELFAND-P001284



Uncertain reading. The plus sign in the subscript of δ is retraced.

Original source details: GELFAND-P001284

[Inserted between the lines: note that]

$$[a_n^\mu, a_m^\nu] = n\delta_{n+m,0}\eta^{\mu\nu} \tag{11}$$

The condition on the state

$$L_n|\Psi\rangle = 0 \qquad \text{for all } n. \tag{12}$$

cannot be imposed.

Impose the conditions

$$L_n|\Psi\rangle = 0, \quad n \geq 1, \quad (L_0 - 1)|\Psi\rangle = 0 \tag{13}$$

The class. conditions (9) are to be interpreted through expectation values.

Uncertain reading. The first letter of ‘Классич.’ (‘class.’) is retraced.

Original source details: GELFAND-P001284

$$\langle\Psi|L_n|\Psi\rangle = 0, \tag{14}$$

which follows from (13) for any $n \neq 0$.

(15) String theory — quantum mech. of a large Hilbert space in which conds. (13) are imposed.

Uncertain reading. The abbreviation ‘к-ром’ (‘which’) has been overwritten.

Original source details: GELFAND-P001284

This is the old string theory.

Now string field theory.

(First in ’71–’72)

2 ways

Consider $|0\rangle \rightsquigarrow \varphi(x)$

$a_{-1}^\mu|0\rangle \rightsquigarrow A^\mu(x)$

Uncertain reading. A subscript on $A^\mu(x)$, possibly μ, has been crossed out.

Original source details: GELFAND-P001284

$$a_{-1}^\mu a_{-1}^\nu|0\rangle; \quad a_{-2}^\mu|0\rangle \rightsquigarrow h^{\mu\nu}(x), B^\mu(x)$$

In more accessible terms, consider

$$\left(\varphi(x) + A_\mu(x)a^\mu_{-1} + h_{\mu\nu}(x)a^\mu_{-1}a^\nu_{-1} + B_\mu(x)a^\mu_{-2} + \ldots\right)|0\rangle \tag{16}$$

In the old notation using $x^\mu(\sigma) \mapsto x^\mu, a^\mu_n$
this was a functional of the contour.
(16) — a state (a set of states),
to a point $x(\sigma) \rightsquigarrow$ the state $\Phi(x(\sigma))$. (field on the string)
An analogous expansion can be obtained

$$\Phi[x(\sigma)] = \Phi[x, a^\mu_n]$$

(17) Rule for the transition
Expand in a Taylor series in a^μ_n,
the coeffs. of a^μ_n with $n > 0$ are inessential (nonphysical),
discard them and retain only the operators of creation.
[Crossed out before (18)]

Original source details: GELFAND-P001285

Uncertain reading. The crossed-out fragment may read ‘Из’ (‘From’).

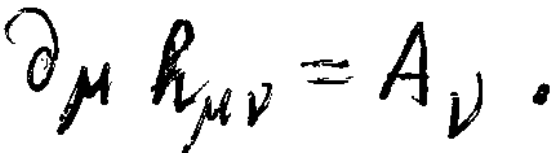
(18) There is reparametriz. invariance under conditions (13).
$\Rightarrow$ the fields are interdependent.
[Inserted between the lines: calculating by formula (5), we obtain]

$$L_1 = a^\mu_1 \frac{\partial}{\partial x^\mu} + \ldots \tag{19}$$

Applying this to (16), we obtain

$$\partial_\mu A_\mu(x) = 0, \tag{20}$$

$$\partial_\mu h_{\mu\nu} = A_\nu. \tag{21}$$



Original source details: GELFAND-P001285

Uncertain reading. The right side of (21) is written A_ν.

(up to coefficients)
This gives the Lorenz gauge cond. (20).
Thus, instead of imposing constraints, consider

$$\delta|\Psi\rangle = \sum_{n=1}^{\infty} b_n L_{-n}|\Phi\rangle, \tag{22}$$

[Below δ, with a short connecting stroke: variation of the state]
b_n — arbitrary parameters.
We will proceed so as to maintain this invariance.

$$\delta A_\mu = \partial_\mu \varepsilon + [\varepsilon, A] \tag{23}$$

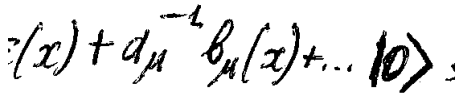
Uncertain reading. In the commutator, A is written without an index, after ε.

Original source details: GELFAND-P001285

GELFAND-P001286

[source number: 45]

$$\begin{aligned}(\delta\varphi(x) + \delta A_\mu(x) a_\mu^{-1} \ldots)|0\rangle = \\ b_1 \frac{\partial}{\partial x^\mu} a_\mu^{-1}(\varepsilon(x) + a_\mu^{-1} b_\mu(x) + \ldots|0\rangle = \\ = (\ldots\ b_1 \partial_\mu \varepsilon(x) a_\mu^{-1} + b_1 \partial_\mu b_\nu a_{-1}^\mu a_{-1}^\nu\ \ldots|0\rangle\end{aligned} \tag{24}$$



Uncertain reading. No closing round parenthesis is visible before $|0\rangle$ in either right-hand expression.

Original source details: GELFAND-P001286

and in the end we obtain

$$\delta A_\mu = \partial_\mu \varepsilon(x) \tag{25}$$

which gives a genuine gauge field.
Thus, [crossed out]

Undeciphered passage. The crossed-out mark after ‘Thus,’ is undeciphered.

Original source details: GELFAND-P001286

2-dimensional reparam. invariance turns into gauge symmetry.

Next, we want to write

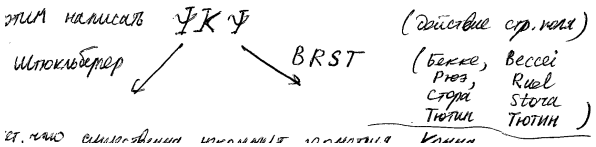


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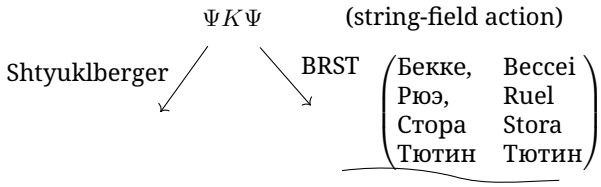
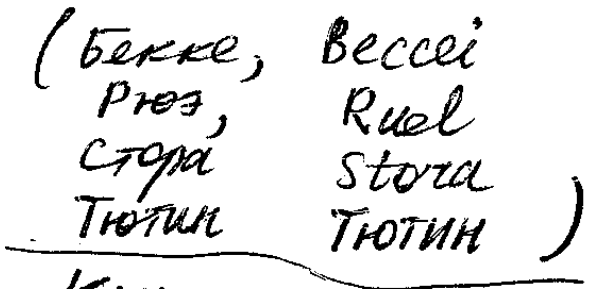
Diagram and typed description.





Original source details: GELFAND-P001286

Uncertain reading. The right column reads ‘Beccei’, ‘Ruel’, ‘Stora’, ‘Тютин’.

Original source details: GELFAND-P001286

Uncertain reading. The label by the left arrow reads ‘Штюкльбергер’ (‘Shtyuklberger’).

Witten writes that Connes’s noncommut. geometry is essential.

(nonabelianization, etc.)

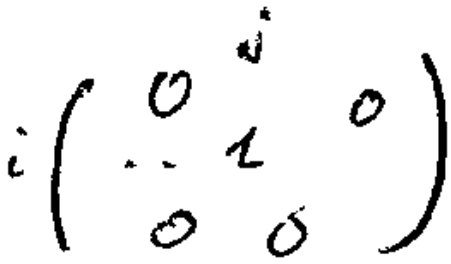
GELFAND-P001287

[source number: 61]

28.4.86 Cherednik. Yangian.

Original source details: GELFAND-P001287

Uncertain reading. The manuscript date is ‘28.4.86’. Catalogue record 676 gives 28 April 1987.



Original source details: GELFAND-P001287

Diagram and typed description.

$M_N \ni \mathbf{1}$, $\mathbf{1}^{ij}$ — std. matrix

$$i \begin{pmatrix} 0 & & 0 \\ \cdot\cdot & 1 & \\ 0 & 0 & \end{pmatrix}^{j} \quad N \times N$$

$$L = \mathbf{1} + \sum_{\substack{q=1 \\ 1\le i,j\le N}}^{n} \lambda^{-q} \mathbf{1}^{ij} E_{ji}^{q-1}, \tag{1}$$

E_{ji}^{q-1} [crossed out] — formal variables

Undeciphered passage. The word before ‘formal variables’ is obscured by crossing-out strokes.

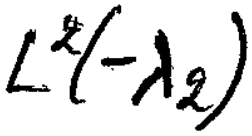
Original source details: GELFAND-P001287

n, N are fixed

$$R = \mathbf{1} \otimes \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{ij} \otimes \mathbf{1}^{ji} \in M_N^{\otimes 2} \tag{2}$$

$$\begin{aligned} \text{(YBF)} \quad & R(\lambda_1 - \lambda_2) L^1(\lambda_1) L^2(-\lambda_2) = L^2(\lambda_2) L^1(\lambda_1) R(\lambda_1 - \lambda_2), \\ & L^1 = L \otimes \mathbf{1}, \qquad L^2 = \mathbf{1} \otimes L \end{aligned} \tag{3}$$

(Yang–Baxter–Faddeev)



Uncertain reading. On the left there is a minus in $L^2(-\lambda_2)$.

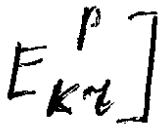
Original source details: GELFAND-P001287

Uncertain reading. On the right the factor reads $L^2(\lambda_2)$.

Original source details: GELFAND-P001287

$$\Updownarrow$$

$$\begin{aligned} [E_{ij}^q, E_{k\tau}^p] = {} & \delta_{jk} E_{i\tau}^{p+q} - \delta_{\tau i} E_{kj}^{p+q} \\ & + \sum_{a<p\le b}^{a+b=p+q-1} (E_{kj}^a \cdot E_{i\tau}^b - E_{kj}^b \cdot E_{i\tau}^a) \end{aligned} \tag{*}$$



Uncertain reading. The fourth matrix index is read as τ.

Original source details: GELFAND-P001287

This is a quantum algebra generalizing Lie algebra. If we insert the parameter $\mathbf{1} \rightsquigarrow i\mathbf{1}$, then in the limit we obtain the matrix Lie alg..

Uncertain reading. The sign before **1**, read as i, has a dot above it.

Original source details: GELFAND-P001287

Definition 1. Yangian of level n

$$Y_N^n = \mathbb{C}'[E_{ij}^q]/(*) \tag{4}$$

[Under $\mathbb{C}'$, with a short connecting stroke: ring of noncommut. polynomials]

$$Y_N^1 = U(g\ell_N), \qquad Y_N^1 \ni E_{ij}^0 \longrightarrow e_{ij} \in g\ell_N \tag{5}$$

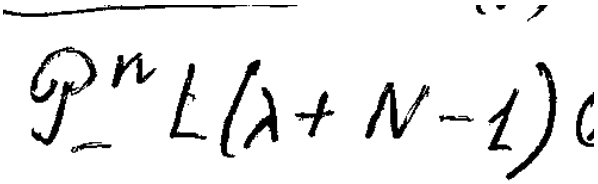
If the indices are outside the range of def., then $E_{\cdot\cdot}^{\cdot} = 0$, i.e.

$$E_{ij}^a = 0 \quad \text{for} \quad \begin{matrix} a < 0, \\ a > n-1 \end{matrix}$$

Definition 2. $H(\lambda)$ is defined from the relations

$$\mathcal{P}_-^n L(\lambda + N - 1) \otimes \ldots \otimes L(\lambda)\mathcal{P}_-^n \overset{\text{def}}{=} H(\lambda)\mathcal{P}_-^n \tag{6}$$

[Under $H(\lambda)$, bracketed: this is the determinant]



Uncertain reading. The antisymmetrizer bears n, and the first shift is $N-1$.

Original source details: GELFAND-P001287

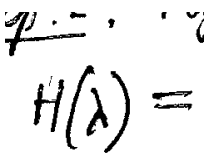
$\mathcal{P}_-^n$ — the antisymmetrizer $\in (M_N)^{\otimes n}$, acting in $(\mathbb{C}^N)^{\otimes n}$

$$\mathcal{P}_-^2 = \frac{1}{2}(\mathbf{1} \otimes \mathbf{1} - P)$$

$H(\lambda) \in Y_N^n$; on the right of (6) we obtain a scalar matr.

Thm. 1. Let

$$H(\lambda) = \sum_{q=1}^{n} (\lambda + i)^{-q} H_i^q + O(1) \quad \text{(in a nbhd of } \lambda = -i), \quad i = 0, \ldots, n-1 \tag{7}$$



Uncertain reading. The equals sign before the sum in (7) has two strokes.

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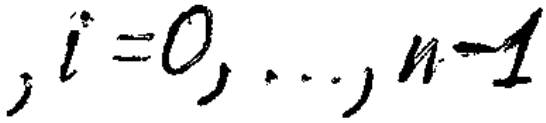


Uncertain reading. The upper end of the range of i is written $n-1$.

Original source details: GELFAND-P001287

Then the center of Y_N^n is given by the formula

$$c(Y_N^n) = \mathbb{C}[H_i^q] \tag{8}$$

GELFAND-P001288

In the absence of shifts [crossed out] (6) would be a pure def. of det.

Undeciphered passage. A word before '(6)' is crossed out. Its exact reading is uncertain.

Original source details: GELFAND-P001288

$$H(\lambda) \approx \det L.$$

Uncertain reading. The two wavy strokes between $H(\lambda)$ and $\det L$ are read as $\approx$.

Original source details: GELFAND-P001288

Rem. If [crossed out] U — a scalar matrix (if the letters in U commute), then

Undeciphered passage. A short crossed-out sign follows 'Если' ('If').

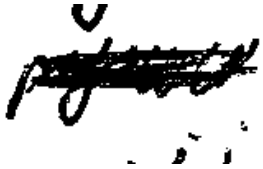
Original source details: GELFAND-P001288

$$\mathcal{P}_-^N \left(\mathbf{1} + \frac{1}{\lambda} U\right)^{\otimes N} \mathcal{P}_-^N = \det\left(\mathbf{1} + \frac{1}{\lambda} U\right) \mathcal{P}_-^N \tag{9}$$

Calculation of $H(\lambda)$ on the univ. envelop. algebra. Now

$$L = \mathbf{1} + \frac{1}{\lambda} \sum \mathbf{1}^{ij} e_{ji}, \qquad e_{ij} \in U(g\ell_N) \tag{10}$$

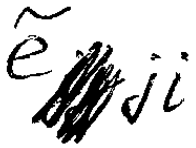
Representation $e_{ij} \longrightarrow \widetilde{e}_{ij} \in M_d$, satisf. [crossed out] the usual relations



Undeciphered passage. The word after 'удовл.' ('satisf.') is crossed out. Its reading is uncertain.

Original source details: GELFAND-P001288

$$V : g\ell_N \longleftrightarrow L_V = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij}\widetilde{e}_{ji} \in M_N(M_d) \tag{11}$$



Uncertain reading. The first subscript on $\widetilde{e}$ is crossed out, possibly ij. To its right is ji.

Original source details: GELFAND-P001288

V — a finite-dimensional rep. of $g\ell_N$ ($g\ell_N$-module).
From the $g\ell_N$-module V we construct the matrix func. $L_V(\lambda)$

$$\widetilde{e}_{ij} \in \operatorname{End} V = M_d$$

Example.

$$V = \mathbb{C}^N \longrightarrow L_{id} = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij} \otimes \mathbf{1}^{ji} \in M_N(M_N)$$

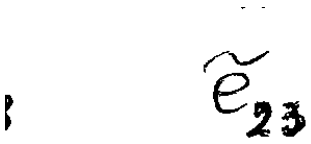
and there is an identification of tensor prod. and matrices
[crossed out]

Uncertain reading. The crossed-out word appears to be ‘Вопросы’ (‘Questions’).

Original source details: GELFAND-P001288

$$\begin{pmatrix} I + \frac{1}{\lambda}\widetilde{e}_{11} & \frac{1}{\lambda}\widetilde{e}_{21} & \dots \\ \frac{1}{\lambda}\widetilde{e}_{12} & I + \frac{1}{\lambda}\widetilde{e}_{22} & \dots \\ \frac{1}{\lambda}\widetilde{e}_{13} & \widetilde{e}_{23} & \dots \\ \cdot\ \cdot & \cdot & \cdot\ \cdot \end{pmatrix}$$

— the matrix on the right in (11)



Uncertain reading. No factor $1/\lambda$ is written before $\widetilde{e}_{23}$.

Original source details: GELFAND-P001288

(12) Problem. Compute L_V in terms of L_{id} for any V.
Let an irred. rep. assoc. with a Young diagram be given

$$\mu = (m_1 \geq m_2 \geq \ldots \geq m_\tau > 0), \quad \tau \leq N, \quad \sum m_i = n \tag{13}$$

$$V_\mu = \mathcal{P}_\mu(\mathbb{C}^N)^{\otimes n} \tag{14}$$

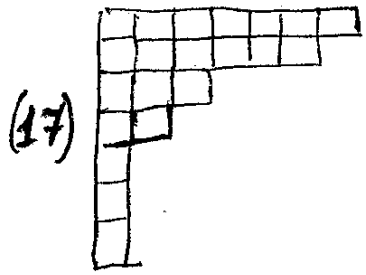
$\mathcal{P}_\mu$ — the Young projector (it is written out explicitly)

$$L_\mu = \mathbf{1} + \frac{1}{\lambda}\sum \mathbf{1}^{ij}\mathcal{P}_\mu e_{ji}\mathcal{P}_\mu \tag{15}$$

$$L_{V_\mu} = \widetilde{\mathcal{P}}_\mu\{L_{id}(\lambda - u_1)\cdot L_{id}(\lambda - u_2)\cdot \ldots \cdot L_{id}(\lambda - u_n)\}\widetilde{\mathcal{P}}_\mu \tag{16}$$

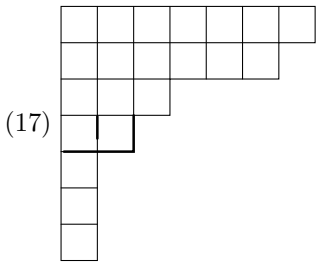
GELFAND-P001289

[source number: 62]



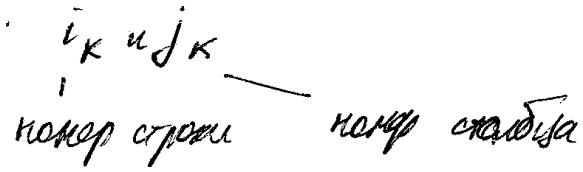
Original source details: GELFAND-P001289

Diagram and typed description.



The numbers from 1 to n are written into the diagram — we obtain — a std. tableau.

Each k acquires



Original source details: GELFAND-P001289

Diagram and typed description.

i_k and j_k

row number column number

$$u_k = i_k - j_k, \qquad u_1 = 0$$

Def. of the bold dot in (16).

The matrices have different coefficient rings. In multiplying them we take, in the matrix entries,

$$A \bullet B = A \otimes B$$

$\mathcal{P}_\mu$ acts only on the matrix coefficients ($\mathcal{P}_\mu$ — a scalar matrix)

$$\mathcal{P}_\mu = \mathcal{P}_\mu \mathbf{1}$$

Original source details: GELFAND-P001289

Uncertain reading. A superscript mark on the first $\mathcal{P}_\mu$ is crossed out.

def.

$$\mathcal{P}_\mu \in M_N^{\otimes n}$$

$\mathcal{P}_\mu$ — symmetrizer along the rows and antisymm. along the columns

On $M_N^{\otimes n}$ acts the symm. group S_n ord. n

Original source details: GELFAND-P001289

Uncertain reading. After S_n the manuscript reads ‘пор. n’ (‘ord. n’).

GELFAND-P001290

5/V–86 G. Olshanskii. Infinite-dimensional reps. and the Yangian.
(Gelfand sem.)

Original source details: GELFAND-P001290

Uncertain reading. The manuscript date is ‘5/V–86’. Catalogue record 677 gives 5 May 1987.

$o(n)$ — complex skew-symm. matrices $n \times n$

$$o(\infty) = \bigcup_{n=1}^{\infty} o(n),$$

ρ — tautological rep. of $o(\infty)$, acting on $\ell_2 = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix} = H$

$$\rho^{\otimes k} = \underbrace{\rho \otimes \cdots \otimes \rho}_{k \text{ times}},$$

Original source details: GELFAND-P001290

Undeciphered passage. A mark to the right of the tensor power is crossed out.

$$A = \Big\{ X = (X_n),\ X_n \in U(o(n))$$

[Below U] univ. envelop. alg.

Original source details: GELFAND-P001290

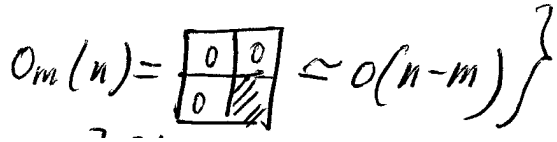
Undeciphered passage. A crossed-out mark stands by the explanation of U.

props.

1) convergence $\exists\, s\text{-}\lim\limits_{n\to\infty} \rho^{\otimes k}(X_n)$ in $H^{\otimes k}$, $\forall k$ (the limit is denoted $\rho^{\otimes k}(X)$)

2) const. $\exists M = M(X):\ \deg X_n \le M$

3) finite index $\exists m = m(X):\ X_n \in U(o(n))^{O_m(n)}$



Original source details: GELFAND-P001290

Diagram and typed description.

$O_m(n) = \begin{array}{|c|c|}\hline 0 & 0 \\ \hline 0 & \\ \hline \end{array} \simeq o(n-m) \Big\}$

Undeciphered passage. A mark in the left margin before ‘Theorem’ is crossed out.

Original source details: GELFAND-P001290

Theorem.

$$A = \mathbb{C}[x_1, x_2, \ldots] \otimes Y,$$

where Y — an alg. with generators $t_{ij}^{(M)}$, $M, i, j = 1, 2, \ldots$

$$(1) \quad [t_{ij}^{(M+1)}, t_{k\ell}^{(N)}] - [t_{ij}^{(M)}, t_{k\ell}^{(N+1)}] = t_{kj}^{(M)} t_{i\ell}^{(N)} - t_{kj}^{(N)} t_{i\ell}^{(M)}$$

Remark Y — the Yangian [inserted above with a line to this word] (for $g\ell(m)$).

Uncertain reading. The argument of $g\ell$ in the insertion is m.

Original source details: GELFAND-P001290

In fact this is an operator realization of the Yangian.

Uncertain reading. The abbreviation before ‘реализ.’ reads ‘операт.’ (‘operator’).

Original source details: GELFAND-P001290

4th prop. Sequences with identical limits in all representations are identified.

$$X \sim Y, \quad \text{if} \quad \rho^{\otimes k}(X) = \rho^{\otimes k}(Y)$$

$m = 0 \quad \Rightarrow$ 3) picks out the center

$m = 1 \quad \Rightarrow$ 3) [crossed out] matr. commuting with

Undeciphered passage. A short mark following ‘3)’ is crossed out.

Original source details: GELFAND-P001290

Diagram and typed description.

$\begin{vmatrix} 0 & 0 \\ 0 & \\ & \end{vmatrix}$ — a problem in lin. algebra.

Original source details: GELFAND-P001290

$Y = \bigcup Y_m$ We may fix m.

m is fixed $\Rightarrow$ we obtain

$$A_m = \mathbb{C}[x_1, x_2, \ldots] \otimes Y_m$$

$$t_{ij}^{(M)} \qquad (i, j \le m;\ M = 1, 2, \ldots)$$

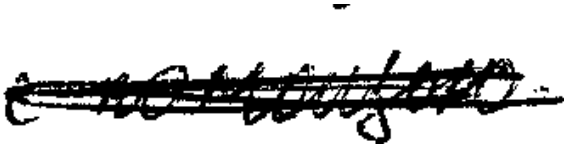
GELFAND-P001291

[source number: 63]

For $sp(\infty)$ we obtain the same A

For $u(\infty)$ —"— $A \otimes A$.

Thus the same A arises throughout, and there is an explanation [crossed out] from representation theory.

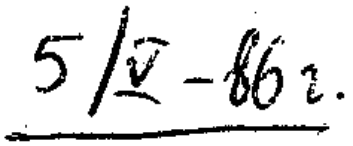


Uncertain reading. The crossed-out writing may read ‘с помощью’ (‘by means of’).

Original source details: GELFAND-P001291

Proposal I.M.G. to consider first the cases $m = 0$, $m = 1$

5/V–86 Cherednik I.



Uncertain reading. The manuscript date is ‘5/V–86’. Catalogue record 678 gives 5 May 1987.

Original source details: GELFAND-P001291

$$\mathbf{1},\ \mathbf{1}^{ij} \in M_N, \qquad e_{ij} \in U(g\ell_N); \qquad M_N \text{ — matrices } N \times N;$$

$$L(\lambda) = \mathbf{1} + \frac{1}{\lambda} \sum_{i,j} e_{ij} \mathbf{1}^{ji} \in M_N(U(g\ell_N));$$

$$H(\lambda) = \det_Q L(\lambda);$$

(see def. below) (Q from the word quantum)

$$\rho : g\ell_N \longrightarrow \operatorname{End} V \simeq M_d \qquad \text{(rep.)};$$

$$L_V = \mathbf{1} + \frac{1}{\lambda} \sum_{i,j} \rho(e_{ij}) \mathbf{1}^{ji} \in M_N(\operatorname{End} V).$$

Let V — be a rep. with [crossed out] highest weight $\mu = (m_1, \ldots, m_\tau, 0, \ldots, 0)$

Undeciphered passage. A short mark before ‘старшим’ (‘highest’) is crossed out.

Original source details: GELFAND-P001291

$$m_1 \geq m_2 \geq \cdots \geq 0, \qquad \tau \leq N, \qquad m_j \in \mathbb{Z}_+$$

$$V = V_\mu$$

For the group ∃! a vector, invariant under

Uncertain reading. The symbols ∃! stand before ‘vector’.

Original source details: GELFAND-P001291

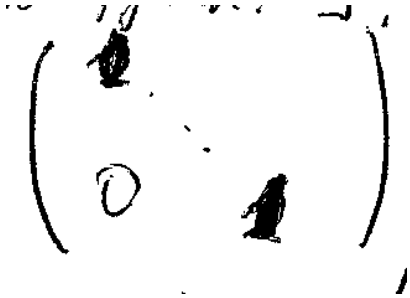


Diagram and typed description.

$$\begin{pmatrix} 1 & & \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Original source details: GELFAND-P001291

Then for

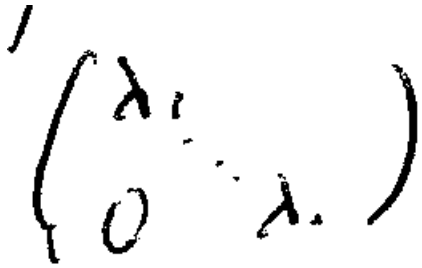


Diagram and typed description.

$$\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ 0 & & \lambda_\cdot \end{pmatrix}$$

A small dot follows the lower λ. No index letter is distinct.

Original source details: GELFAND-P001291

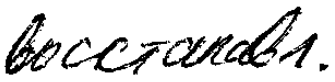
it is an eigenvector with [crossed out] eigenvalue $\lambda_1^{m_1}\lambda_2^{m_2}\cdots\lambda_\tau^{m_\tau}$

Undeciphered passage. The mark before ‘соб. знач.’ (‘eigenvalue’) is crossed out.

Original source details: GELFAND-P001291

and necessarily $m_1 \geq m_2 \geq \cdots \geq m_\tau$.

From it, uniquely reconstr. How?



Uncertain reading. The abbreviation reads ‘восстанавл.’ (‘reconstr.’).

Original source details: GELFAND-P001291

Construction by means of

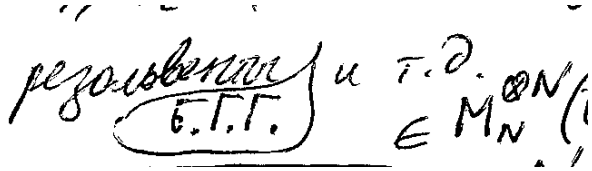


Diagram and typed description.

resolvents etc.

B.G.G.

Original source details: GELFAND-P001291

Def. of $H(\lambda)$

$$(1)\quad H(\lambda)\mathcal{P}_-^N = \mathcal{P}_-^N\big(L(\lambda+N-1)\otimes\cdots\otimes L(\lambda)\big)\mathcal{P}_-^N,$$
$$H(\lambda)\in U(g\ell_N)$$

[Above the tensor product, a bracket labelled $\in M_N^{\otimes N}(U(g\ell_N))$.]

(2) Thm.

$$H(\lambda)\big|_{V_\mu} \overset{\text{def}}{=} \det_Q L_{V_\mu} = \prod_{i=1}^{\tau}\frac{\lambda+N+m_i-i}{\lambda+N-i}$$

(in particular, it is asserted to be a scalar func.)

$\mathcal{P}_-^N$ — the antisymmetrizer

$$\mathcal{P}_-^N(x_1\otimes\cdots\otimes x_N) = \frac{1}{N!}\sum_{\sigma\in S_N}(-1)^{\epsilon(\sigma)}(x_{\sigma(1)}\otimes\cdots\otimes x_{\sigma(N)}),$$
$$x_j\in\mathbb{C}^N$$

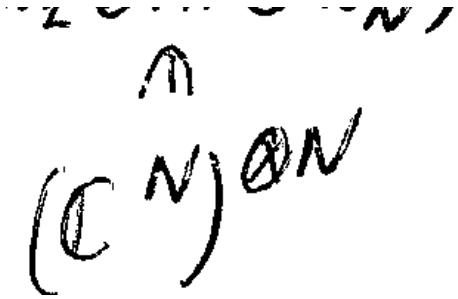


Diagram and typed description.

$x_1\otimes\cdots\otimes x_N$

$(\mathbb{C}^N)^{\overset{\wedge}{\otimes} N}$

A curved mark with three strokes stands below the argument. It may be a rotated $\in$.

Original source details: GELFAND-P001291

GELFAND-P001292

$$\mathcal{P}_-^N\in M_N^{\otimes N} = \mathrm{End}\big((\mathbb{C}^N)^{\otimes N}\big)$$

The image of $\mathcal{P}_-^N$ is a one-dimensional subspace.

On the right in (1) is the restriction of $L(\lambda+N-1)\otimes\cdots\otimes L(\lambda)$ to the 1-dimensional subspace.

Uncertain reading. The earlier lettering beneath the overwritten L in the last factor is illegible.

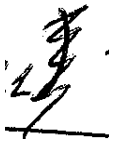
Original source details: GELFAND-P001292

$$H\mathcal{P}_-^2 = \mathcal{P}_-^2(A \otimes B)\mathcal{P}_-^2$$

$$\underbrace{\frac{1}{4}\left(a_{11}b_{22} + a_{22}b_{11} - a_{12}b_{21} - a_{21}b_{12}\right)}_{H \text{ in the case of arbitrary matrices}}$$

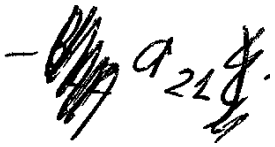
Original source details: GELFAND-P001292

Uncertain reading. Before $\frac{1}{4}$ is a cancelled sign, possibly $=$. Between the fraction and the parenthesis is an obscured stroke.



Original source details: GELFAND-P001292

Uncertain reading. After $a_{22}b_{11}$ a mark is cancelled, possibly $\frac{1}{2}$.



Original source details: GELFAND-P001292

Uncertain reading. Before a_{21} is a cancelled entry, possibly b_{12}. Between a_{21} and b_{12} a mark is cancelled, possibly $\frac{1}{2}$.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}.$$

$$a_{11} = 1 + \frac{1}{\lambda+1}e_{11}, \quad a_{12} = \frac{1}{\lambda+1}e_{21},$$
$$a_{22} = 1 + \frac{1}{\lambda+1}e_{22}, \quad a_{21} = \frac{1}{\lambda+1}e_{12}.$$

Original source details: GELFAND-P001292

Uncertain reading. The earlier letter under the correction in a_{21} is illegible.

$$b_{11} = 1 + \frac{1}{\lambda} e_{11}, \quad b_{12} = \frac{1}{\lambda} e_{21},$$
$$b_{22} = 1 + \frac{1}{\lambda} e_{22}, \quad b_{21} = \frac{1}{\lambda} e_{12}.$$

$$\begin{aligned} H(\lambda) = & \left(1 + \frac{1}{\lambda+1} e_{11}\right)\left(1 + \frac{1}{\lambda} e_{22}\right) \\ & + \left(1 + \frac{1}{\lambda+1} e_{22}\right)\left(1 + \frac{1}{\lambda} e_{11}\right) - \\ & - \left(\frac{1}{\lambda+1} e_{12}\right)\left(\frac{1}{\lambda} e_{21}\right) - \left(\frac{1}{\lambda+1} e_{21}\right)\left(\frac{1}{\lambda} e_{12}\right). \end{aligned}$$

Uncertain reading. The writing in $H(\lambda)$ before the equals sign is overwritten and the earlier strokes are illegible.

Original source details: GELFAND-P001292

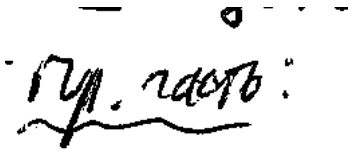
Uncertain reading. Before the minus at the line end, an indexed e is cancelled, possibly e_{11} or e_{22}.

Original source details: GELFAND-P001292

Claim. $H(\lambda)$ is a multiple of 1 on every irreducible rep. ($\in$ the center).

Uncertain reading. An illegible entry is cancelled before the heading.

Original source details: GELFAND-P001292

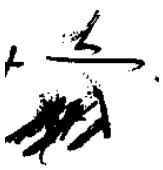


Uncertain reading. Pr. part (the abbreviation is uncertain).

Original source details: GELFAND-P001292

(G. Olshanskii)

$$\begin{aligned} H(\lambda) = & \sum \operatorname{sgn} \begin{pmatrix} i_1 & \cdots & i_N \\ j_1 & \cdots & j_N \end{pmatrix} \left(\delta_{i_1 j_1} + \frac{1}{\lambda + N - 1} e_{i_1 j_1}\right) \\ & \left(\delta_{i_2 j_2} + \frac{1}{\lambda + N - 2} e_{i_2 j_2}\right) \cdots \left(\delta_{i_N j_N} + \frac{1}{\lambda} e_{i_N j_N}\right) \end{aligned}$$



Uncertain reading. Illegible marks are cancelled before λ in the last denominator.

Original source details: GELFAND-P001292

If the letters e_{ij} commute (introduce h) $\Rightarrow$ we obtain the charact. polynomial.

If noncommut. this is the correct analogue of the charact. polynomial.

The following commute: $\sum e_{ii}$, $\sum e_{ij}e_{ji}$, etc.

GELFAND-P001293

1/IX - 86 Gelfand Sem.

At Princeton Witten gave a course on strings, and he had 2 listeners, Deligne and Kazhdan. ['strings' is underlined with a wavy line.]

I.M.G. and Serganova $\rightarrow$ matroids

Uncertain reading. Beneath the right-pointing arrow is an unclear sign, possibly a colon or a cancelled stroke.

Original source details: GELFAND-P001293

Beilinson and Ginzburg: quadr. algebras

Cherednik: Gelfand–Tsetlin basis

[The names in the three lines above are underlined with a wavy line. At left is a broken vertical line.]

I.M. Gelfand (and Serganova).

General definition of a matroid. [Underlined with a wavy line.]

Homolog. algebra is a branch of combinatorics

In Aigner's book 'Combinat. Theory' a third is devoted to matroids. ['Aigner' is underlined with a wavy line.]

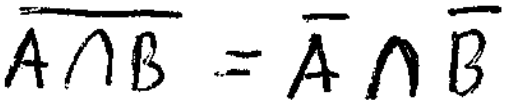
Def. 1. Matroid.

A finite set and a notion of 'closure' on it:

$$A \rightsquigarrow \bar{A} \supset A$$

such that

a) $\overline{A \cap B} = \bar{A} \cap \bar{B}$ (but this is not required for $A \cup B$)



Uncertain reading. The manuscript has an equals sign between $\overline{A \cap B}$ and $\bar{A} \cap \bar{B}$.

Original source details: GELFAND-P001293

b) [crossed out] $\bar{\bar{A}} = \bar{A}$.

Uncertain reading. The crossed-out formula may read $a \in A$ or $a \in \bar{A}$.

Original source details: GELFAND-P001293

[Inserted at right, above the crossed-out line.]

$$\text{c)} \quad a \notin \bar{A}, \quad a \in \overline{A + \{b\}} \quad \Rightarrow \quad b \in \overline{A + \{a\}}.$$

[[crossed out] c) $a \in \bar{A},\ b \notin \bar{A}\ \{a \in A,\ b \in \bar{A},\ b \neq a\} \Rightarrow a \in \overline{A \setminus \{b\}}$.]

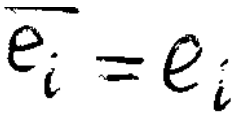
Uncertain reading. The start of the crossed-out line reads $a \in \bar{A},\ b \notin \bar{A}$, with crossing-out strokes over the membership signs.

Original source details: GELFAND-P001293

Example. In $\mathbb{R}^n$ consider k vectors $e_1, \ldots, e_k$.

$$A \subset \{1, 2, \ldots, k\};$$
$$\bar{A} \ni i \ \Leftrightarrow\ e_i \in \text{lin. span of } \{e_j,\ j \in A\}$$

$\bar{e}_i = e_i$; but $\overline{e_i \cup e_j} \neq \bar{e}_i \cup \bar{e}_j$ (in general).



Uncertain reading. The manuscript reads $\bar{e}_i = e_i$.

Original source details: GELFAND-P001293

Nondeg. matroid (or combinat. geometry) if $\bar{\varnothing} = \varnothing$. [Underlined with a wavy line.]

Where does a matroid arise?

Thm. (Gelf., Serg., Goresky, MacPherson). The Grassmann set is divided into strata (under the action of the Cartan subgroup), and there are also Schubert cells.

Uncertain reading. The surname appears to be written ‘Горецки’ (Goresky).

Original source details: GELFAND-P001293

In $\mathbb{R}^n$ diag. matrices act $\Rightarrow$ action on the Grassm.

The orbits are grouped into similar ones. These are the strata.

GELFAND-P001294

[source number: 71]

Each stratum in $G_{n,k}$ is uniquely specified by a matroid.

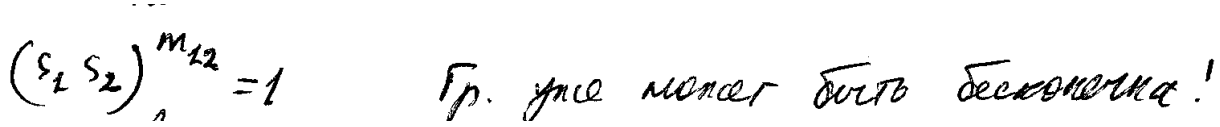
General def. of a matroid

Let a Coxeter grp. be given, gen. by $s_1, \dots, s_n$ (i.e., there are rel. $s_i^2 = 1$, $(s_i s_j)^{m_{ij}} = 1$, m_{ij} a given integer matrix)

Examples.

A. $1, s_1$ — 2 elements

B. $1, s_1, s_2$ — $(s_1 s_2)^{m_{12}} = 1$ The grp. can already be infinite!



Original source details: GELFAND-P001294

Uncertain reading. Both the relation with exponent m_{12} and the words 'The grp. can already be infinite!' are written.

Question. When? (for first-year students)

In general. When is it finite? Already difficult.

Original source details: GELFAND-P001294

Uncertain reading. Above 'In general' is a wavy stroke whose function is unclear.

Example. [wavy underlining] the permutation grp. S_n on n elements;

Original source details: GELFAND-P001294

Uncertain reading. 'Подстановок' (permutations) has thickened strokes in the middle.

s_j — [crossed out] transpos. of 2 elements (adjacent). (j and $j+1$)

Original source details: GELFAND-P001294

Uncertain reading. The crossed-out word may read 'это' ('is').

$$(s_i s_{i+1})^3 = 1, \qquad (s_i s_k)^2 = 1 \quad \text{for nonadjacent } i \text{ and } k.$$

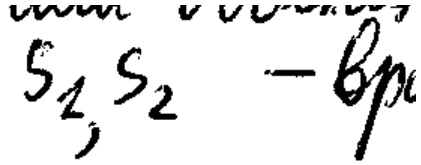
H^3: s_1, s_2, s_3:

$$(s_1 s_2)^5 = 1, \qquad (s_2 s_3)^3 = 1, \qquad (s_1 s_3)^2 = 1 \qquad \text{(120 elements)}$$

(symmetry grp. of the icosahedron!)

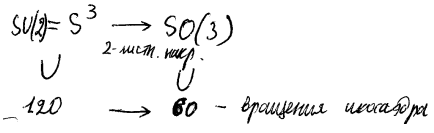
[Between the lines] or of the dodecahedron.

s_1, s_2 — rot. about a vertex



Original source details: GELFAND-P001294

Uncertain reading. There is a comma between s_1 and s_2.



Original source details: GELFAND-P001294

Diagram and typed description.

$$\begin{array}{ccc} SU(2) = S^3 & \xrightarrow[\text{2-sheeted cover}]{} & SO(3) \\ \cup & & \cup \\ 120 & \longrightarrow & 60 \end{array}$$

— rotations of the icosahedron

(See ch. 1 of the book by Gelf. — Minlos — Shapiro)

Original source details: GELFAND-P001294

Diagram and typed description.

The upper stroke starts below 120 and curves around the reference line at left. The lower stroke ends in an arrow before '120 points'. A join is indistinct.

120 points in $S^3 = SU(2)$

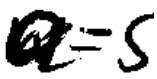
Convex hull — a regular polyh. with 600 faces, tetrahedra

Its symmetry grp. H^4. Its order is 120^2.

Flag matroid.

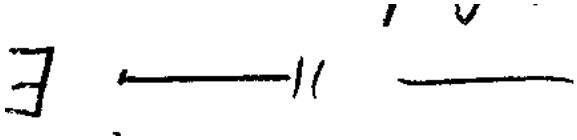
Let there be a Coxeter grp. and let $a = s_{i_1} \dots s_{i_\ell}$ be written minimally (as short as possible).

Uncertain reading. The thickened letter before $= s_{i_1} \dots s_{i_\ell}$ reads a.



Original source details: GELFAND-P001294

$a < b$ if $\exists$ a min. decomp. $a = w_{i_1} \dots w_{i_\ell}$.



Original source details: GELFAND-P001294

Diagram and typed description.

$\exists$ ——// —— $b = \dots.(w_{i_1} \dots w_{i_\ell}) \dots.$

(Bruhat order)

$a \overset{w}{<} b$ if $w^{-1}a < w^{-1}b$.

Uncertain reading. The thickened letter to the right of the $<$ marked w reads b.

Original source details: GELFAND-P001294

W is a Coxeter grp., $L \subset W$, Card $L < \infty$. L is called a flag matroid [wavy underlining] if for every w-ordering there is a min element in L.

One can enumerate all finite Coxeter grps. and the corresponding matroids.

GELFAND-P001295

A matroid of vectors is completely spec. by the list of max sets of lin. indep. vectors. [Both lines have wavy underlining.]

[Boxed and crossed out. Def. [crossed out] A parab. subgroup P of a Coxeter grp. is a subgroup gen. by a set $S \ldots$]

Uncertain reading. An illegible entry is cancelled after 'Def.'

Original source details: GELFAND-P001295

Is every matroid representable as a vector matroid?

Not every one!!

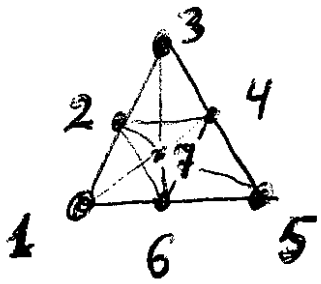
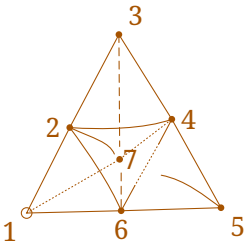


Diagram and typed description.



Some interior strokes break near point 7. Their exact joins are indistinct in places.

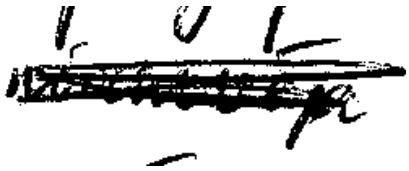
Original source details: GELFAND-P001295

$$\begin{array}{lll} 123 & 275 & 246 \\ 165 & 174 & \\ 345 & 376 & \end{array}$$

On the work of Beilinson and Ginzburg. 'Quadr. algebras'.

Consider a graded algebra with integer grading ($0, 1, 2, \ldots$ — degrees).

A_0 — [crossed out] a field;



Uncertain reading. Before 'field' a word is cancelled, possibly 'algebra'.

Original source details: GELFAND-P001295

Require that the algebra be generated by $A_1 = V$ ($\dim_{A_0} V < \infty$).

[crossed out] Relations?

Uncertain reading. Before ‘Relations?’ is a cancelled entry, possibly ‘Зад.’.

Original source details: GELFAND-P001295

[A broken vertical line at left.]

Example. $x_i x_k - x_k x_i = 0 \Rightarrow$ polynomial alg.

Requirement. All rel. are generated by elements of $V \otimes V$.

$$\underset{\text{subspace.}}{W} \subset V \otimes V$$

$$A_0 \oplus V \oplus (V \otimes V) \oplus \ldots \oplus V^{\otimes n} \oplus \ldots = T(V)$$

Uncertain reading. A superscript, possibly n, is cancelled after V, before $\otimes n$.

Original source details: GELFAND-P001295

Required. $\exists$ a lin. space $W \subset V \otimes V$ gen. all rel. in A.

$$(A = T(V)/J, \quad \text{where } J \text{ is gen. by } W$$
$$J = J(W))$$

Dual.

$$A_0 \oplus V^* \oplus V^* \otimes V^* \oplus \ldots \oplus (V^*)^{\otimes n} \oplus \ldots$$

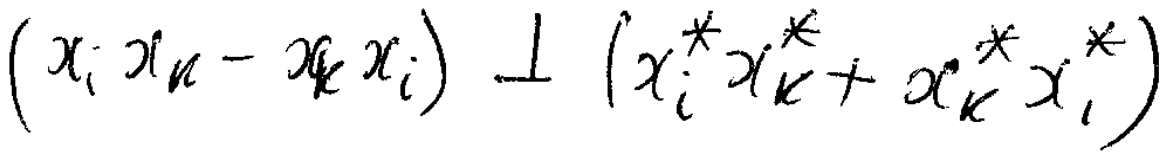
Consider in it $W^\perp \subset V^* \otimes V^*$ and the corresponding algebra

$$A^! = T(V^*)/J(W^\perp).$$

It is called the dual algebra. [Wavy underlining.]

Example 1. Let A be the commut. algebra of polynomials in $x_1, \ldots, x_n$. [‘polynomials’ has wavy underlining.]

$$(x_i x_n - x_k x_i) \perp (x_i^* x_k^* + x_k^* x_i^*)$$



Original source details: GELFAND-P001295

Uncertain reading. In the first product the second index reads n. In the next product the first index reads k.

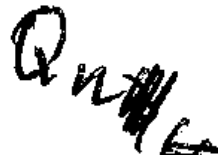
Rel. $\sum a_{ij}\, x_i \otimes x_j$ with $a_{ij} = -a_{ji}$.

$A^!$ is a Grassmann algebra.

GELFAND-P001296

[source number: 72]

Example 2. [Broken underlining.] Consider n quadr. forms $Q_1, \ldots, Q_n$ and the quotient of all commut. polynomials by $J\langle Q\rangle$, i.e.

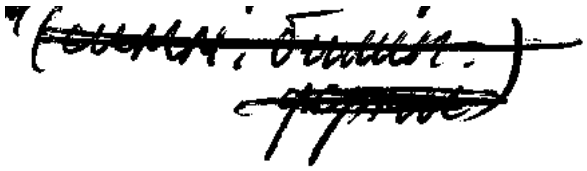


Original source details: GELFAND-P001296

Uncertain reading. An illegible mark is cancelled after Q_n.

Original source details: GELFAND-P001296

Uncertain reading. The delimiters around Q are overwritten. $J\langle Q\rangle$ follows the visible angle strokes. The formula below reads $J(Q)$.



Original source details: GELFAND-P001296

Uncertain reading. At right '(symm. bilin. forms)' is crossed out. The final word is heavily obscured.

Original source details: GELFAND-P001296

[Above the formula at right $T(V)$.]

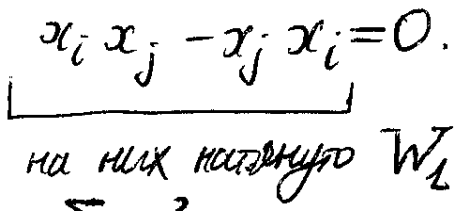
Original source details: GELFAND-P001296

Uncertain reading. The entry $T(V)$ stands by itself.

$$A = S(V)/J(Q)$$

For example. [Wavy underlining.]

$$\sum x_i^2 = 0,$$



Original source details: GELFAND-P001296

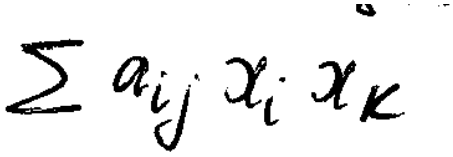
Diagram and typed description.

$x_i x_j - x_j x_i = 0.$

they span W_1

W is spanned by W_1 and $\sum x_i^2$.
Their lin. combinations?

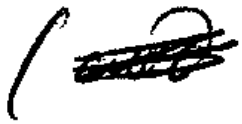
$$\sum a_{ij}x_ix_k, \qquad \begin{aligned} a_{ij} &= -a_{ji}, \quad i \neq j, \\ a_{ii} &= a_{jj}. \end{aligned}$$



Original source details: GELFAND-P001296

Uncertain reading. In the sum $\sum a_{ij}x_ix_k$ the second factor has index k.

$$W^{\perp} = \{b_{ij}x_i^*x_j^*; \quad b_{ij} = b_{ji}; \quad \sum b_{ii} = 0\}$$



Original source details: GELFAND-P001296

Uncertain reading. To the right of the formula is an opening parenthesis followed by an illegible cancelled entry.

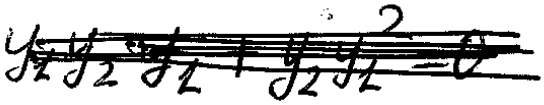
symm. matrices with trace 0.
Independent relations. [Wavy underlining.]

$$y_iy_j + y_jy_i = 0 \text{ for } i \neq j; \qquad y_i^2 = y_j^2 \quad (\forall i, j)$$

Put $y_i^2 = h$. Then

$$y_iy_j + y_jy_i = 2\delta_{ij}h.$$

[Crossed out $y_1y_2y_1 + y_2y_1^2 = 0$.]



Original source details: GELFAND-P001296

Uncertain reading. The crossed-out line reads $y_1y_2y_1 + y_2y_1^2 = 0$.

$$y_1^2y_i = y_iy_1^2 \qquad (\text{since } y_1^2 = y_i^2)$$

$\Rightarrow h \in$ the center! [Wavy underlining.]

Chapter 29

GELFAND-P001297

29/IX-86 Gelfand Sem.

V. Ginzburg [crossed out] Procesi, De Concini, Springer

Uncertain reading. Before ‘Procesi’ is an illegible cancelled entry.

Original source details: GELFAND-P001297

[crossed out] Progress in Math., $N = 60$ (1985), birkhäuser

Uncertain reading. Before ‘Progress in Math.’ the word ‘Springer’ is crossed out.

Original source details: GELFAND-P001297

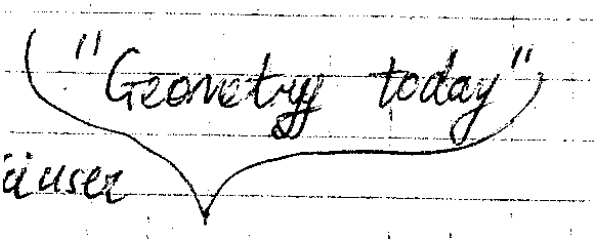


Diagram and typed description.

“Geometry today”

Original source details: GELFAND-P001297

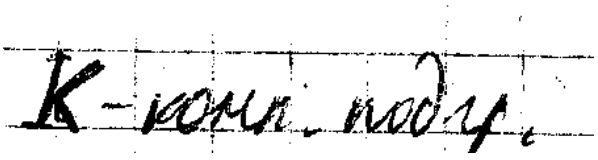
Compactification of quadr. forms

quadr. form [wavy underlining] (Space $\to$ dual space)

— a set of lin. subspaces

$$\underset{\text{compactification (min)}}{\bar{M}} \supset M = G/H$$

G a grp., K a comp. subgrp.



Uncertain reading. K is described using the abbreviation ‘комп.’ (‘comp.’).

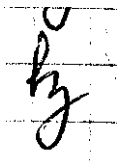
Original source details: GELFAND-P001297

H the normalizer of K

$$H/K = \oplus Z/2Z$$

$\mathfrak{g}$ the Lie alg.

h_g the subalg. corresponding to H



Uncertain reading. The subalgebra symbol has a loop below the baseline. The reading h_g is tentative.

Original source details: GELFAND-P001297

Properties of $\bar{M}$?

1) $\bar{M}$ is smooth

2) $\bar{M}$ consists of finitely many G-orbits

3) $\bar{M} \setminus M$ is a divisor with normal crossings

(a common sol. of k equations, where $k = \text{Codim}(\bar{M} \setminus M)$)

(like a system of coord. planes)

4) every component $D =$ closure of one orbit.

5) There is exactly 1 closed orbit, isom. to G/P, where P is some parab. subgrp. constructed from K.

6) Line bundles on $\bar{M} \xrightarrow{\sim}$ line bundles on G/P.

— · — · —

$$G = GL_n$$

Hecke algebra assoc. with the symm. group [wavy underlining]

H $\quad$ $T_1, \ldots, T_{n-1}$ — gen.

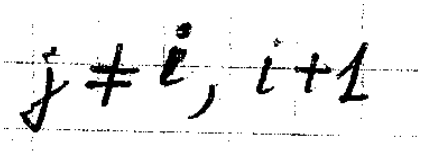
[At left] q-analogue.

$$1) \quad (T_i + 1)(T_i - q) = 0 \Rightarrow T_i^2 = 1 \quad \text{(at } q = 1\text{)} \qquad (q\text{-analogue})$$

$$2) \quad T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$$

(T_i the transpos. of the i-th and $(i+1)$-st elements)

$$3) \quad T_i T_j = T_j T_i, \qquad j \neq i, i+1$$



Uncertain reading. Condition (3) reads $j \neq i, i+1$.

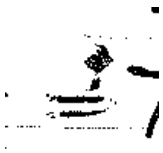
Original source details: GELFAND-P001297

Cherednik devised the embedding $H \subset \widetilde{H}$.

[Under $\widetilde{H}$] gen. $T_1, \ldots, T_{n-1};\ x_1, \ldots, x_n$

and rel.

$$x_i T_i = T_i x_{i+1} - (q-1)x_{i+1} \qquad \Big| \qquad x_i x_j = x_j x_i$$
$$x_{i+1} T_i \overset{?}{=} T_i x_i + (q-1)x_{i+1}$$



Uncertain reading. The mark above the second equals sign resembles a question mark.

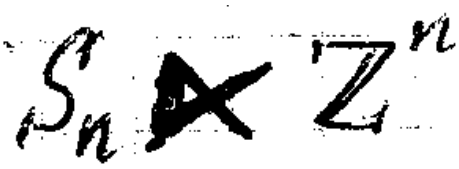
Original source details: GELFAND-P001297

GELFAND-P001298

[source number: 8]

This is related to the crossed prod.

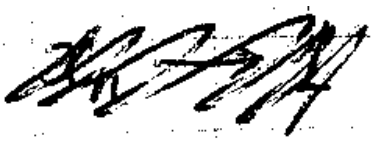
$$S_n \ltimes Z^n$$



Uncertain reading. The crossed-product sign has a vertical stroke at left.

Original source details: GELFAND-P001298

[Crossed out]



Uncertain reading. Below the formula is a crossed-out entry, possibly $x_n \to 1$.

Original source details: GELFAND-P001298

(x_j a basis in Z^n).

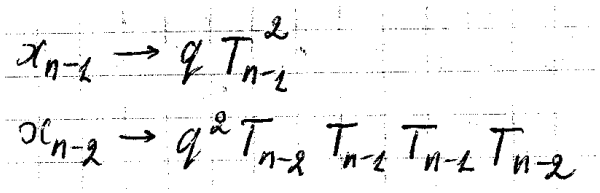
Uncertain reading. The overwritten index after x reads j.

Original source details: GELFAND-P001298

Map of the grp. into the algebra

$$(g, z) \mapsto g$$

$$x_n \to 1$$
$$x_{n-1} \to qT_{n-1}^2$$
$$x_{n-2} \to q^2 T_{n-2}T_{n-1}T_{n-1}T_{n-2}$$
$$\cdots\cdots\cdots$$



Uncertain reading. The first line reads qT_{n-1}^2. The next line has q^2 before the product of four T's.

Original source details: GELFAND-P001298

This is a homomorphism!

At $q = 1$ we obtain 1.

Problem (Cherednik)

(unsolved)

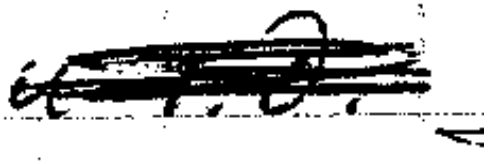
Find a homomorphism starting with $x_n^2 \to 1$

and what is the target of this homomorphism?

Or an arb. polynomial in x_n.

29.9.86 V. Ovsienko

On the connection of the KdV eq. with the Virasoro alg. [Crossed out] and of super KdV with the Schwarz and Ramond algebras.



Uncertain reading. After 'Virasoro' is a crossed-out entry, possibly 'KdV'.

Original source details: GELFAND-P001298

(geom. meaning)

KdV = the Euler eq. corresponding to the Virasoro alg., more precisely, to the Virasoro group.

Super KdV = — " — Schwarz and Ramond

Vect S^1 — alg. of vect. fields on S^1 and its central extension.

[Crossed out] KdV is an eq. on the dual space. It is tangent to the orbits and is Hamiltonian.

Uncertain reading. Before 'KdV' is a crossed-out entry whose reading is unresolved.

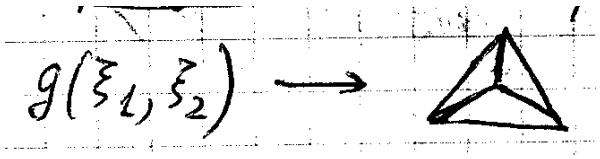
Original source details: GELFAND-P001298

— — — — —

29.9.86 Kazakov

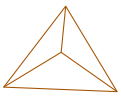
Discretization and exact solution of 2-dimensional gravity

(continued)



Original source details: GELFAND-P001298

Diagram and typed description.

$g(\xi_1, \xi_2) \longrightarrow$ 

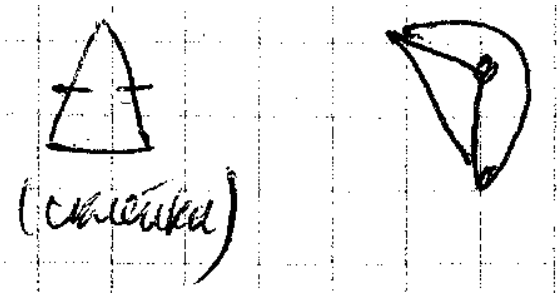
Segments join the interior point to all three vertices. The thickened ends do not show definite arrowheads.

$$Z = \underbrace{\int Dg(\xi) \exp\left\{-\int d^2\xi\sqrt{g}\right\}}_{\text{(surfaces with the topology of a sphere)}} \underset{\text{discretization}}{\Longrightarrow} Z = \sum_n^{\infty} \sum_{\{G^{(n)}\}} e^{-\beta n} \tag{1}$$

$G^{(n)}$ is a graph with n vertices, with the sum over all graphs.

GELFAND-P001299

$\{G^{(n)}\}$ is the set of combinatorial triangulations of the sphere (simplic. complexes with sphere topology up to permutations).



Original source details: GELFAND-P001299

Diagram and typed description.

(gluings)

The first sketch has two short transverse strokes. In the second, the joins and possible arrowheads at the loops are indistinct. The first caption appears to read ‘склейки’ (‘gluings’).

[To the right of the sketches] this can also be forbidden

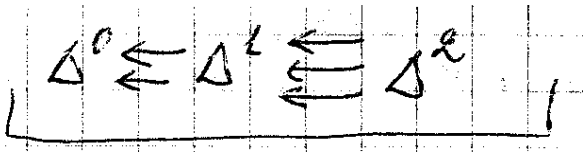
(we will leave it for now)

this will not affect the answer

Original source details: GELFAND-P001299

Uncertain reading. The abbreviation reads ‘сказ.’ (‘affect’).

$\underset{\text{sets}}{\Delta^0, \Delta^1, \Delta^2}$ and maps



Original source details: GELFAND-P001299

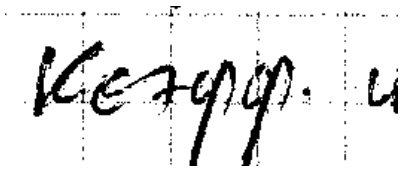
Diagram and typed description.

$\Delta^0 \leftarrow \Delta^1 \leftrightarrows \Delta^2$

A permutation group acts throughout. We quotient by it.
The singularity of the discr. sum and of the cont. integral is the same (near divergence).

$$Z(\beta) \underset{\beta\to\beta_c}{\approx} (\beta-\beta_c)^{5/2} \tag{2}$$

(β is called the cosmological constant)
Coeffs. and lower terms depend on the type of triangulations.



Original source details: GELFAND-P001299

Uncertain reading. The abbreviation is read as ‘Коэфф.’ (‘coeffs.’).

$$I(g) = \int d^{N^2}u \exp\left[-\operatorname{tr}\left(u^2 + \frac{g}{\sqrt{N}}u^3\right)\right] \tag{3}$$

(u an $N\times N$ [‘Hermitian’ inserted above ‘matrix’] matrix)
Claim. [wavy underlining]

$$Z \underset{N\to\infty}{=} \frac{1}{N^2}\ln I(e^{-\beta/2}) \tag{4}$$

Proof [wavy underlining] by expanding in g. The resulting [crossed out] Feynman graphs are triangulations of the sphere.

Original source details: GELFAND-P001299

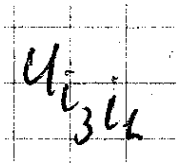
Uncertain reading. Before ‘Feynman graphs’ is an illegible crossed-out entry.

Integral (3) at $N\to\infty$ is computed exactly. We have

$$I(g) = \sum_n \frac{(-g)^n}{n!N^{n/2}} \int d^{N^2}u\, e^{-\operatorname{tr}u^2} (\operatorname{tr}u^3)^n \tag{5}$$

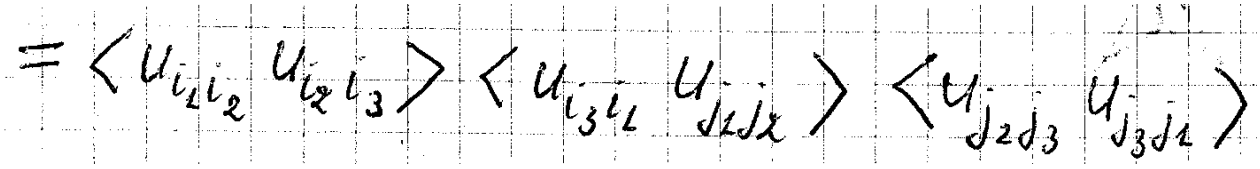
$\langle\ \ \rangle$ denotes av. with Gaussian weight.
Wick’s thm.

$$\begin{aligned}\langle \operatorname{tr}(uuu)\operatorname{tr}(uuu)\rangle &= \langle u_{i_1i_2}u_{i_2i_3}u_{i_3i_1}u_{j_1j_2}u_{j_2j_3}u_{j_3j_1}\rangle = \\ &= \langle u_{i_1i_2}u_{i_2i_3}\rangle\langle u_{i_3i_1}u_{j_1j_2}\rangle\langle u_{j_2j_3}u_{j_3j_1}\rangle\end{aligned} \tag{6}$$



Original source details: GELFAND-P001299

Uncertain reading. The third factor in the first average reads $u_{i_3i_1}$.

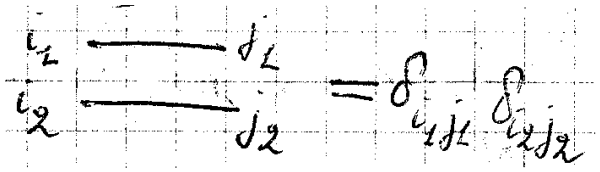


Original source details: GELFAND-P001299

Uncertain reading. The second line of (6) shows three consecutive pair averages. No summation sign is written on this line.

(A Gauss. integral is always expressed in terms of pairs!)

We shall denote the average $\langle u_{i_1 i_2} u_{j_1 j_2} \rangle$ by



Original source details: GELFAND-P001299

Diagram and typed description.

i_1 ——— j_1 $= \delta_{i_1 j_1} \delta_{i_2 j_2}$

i_2 ——— j_2

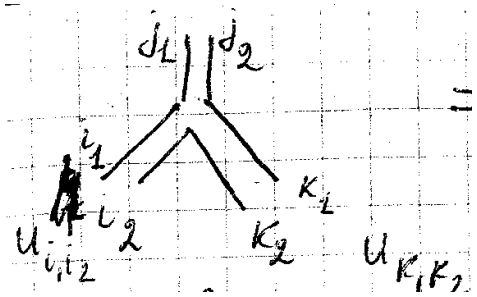
The upper pair is labelled i_1, j_1 and the lower pair i_2, j_2.

[Beneath the average] (normalization is understood)

[Below, with an arrow] in the coupling const.

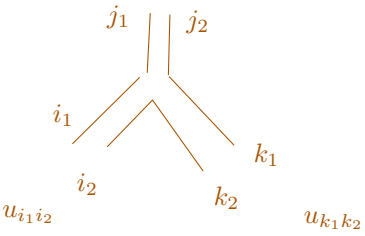
GELFAND-P001300

[source number: 9]



Original source details: GELFAND-P001300

Diagram and typed description.

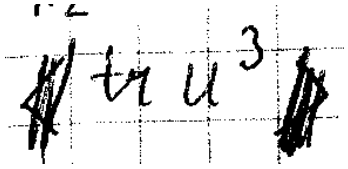


$$= \frac{g}{\sqrt{N}} \delta_{i_1 j_1} \delta_{k_1 j_2} \delta_{i_2 k_2}$$

Original source details: GELFAND-P001300

Uncertain reading. An illegible mark is crossed out to the left of i_1, i_2.

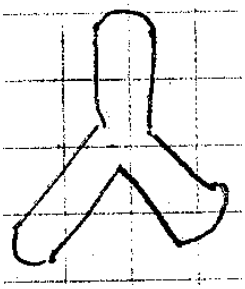
$\operatorname{tr} u^3$



Original source details: GELFAND-P001300

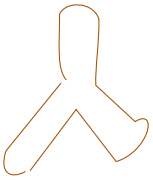
Uncertain reading. Illegible marks are crossed out on either side of $\operatorname{tr} u^3$.

Trace — closing the lines

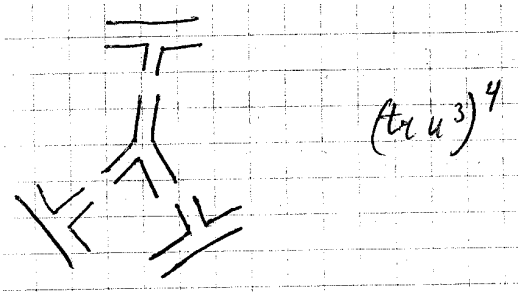


Original source details: GELFAND-P001300

Diagram and typed description.

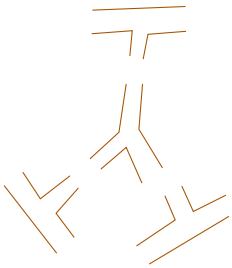


Small gaps are visible at the left of the neck and at the lower left end.



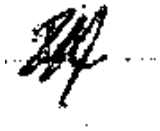
Original source details: GELFAND-P001300

Diagram and typed description.



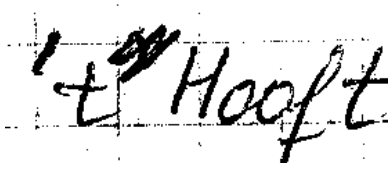
$(\mathrm{tr}\, u^3)^4$

The selection of planar graphs [crossed out] etc. was devised by ’t Hooft.



Original source details: GELFAND-P001300

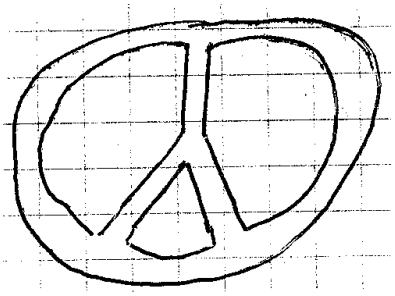
Uncertain reading. The mark after ‘graphs’ is crossed out and undeciphered.



Original source details: GELFAND-P001300

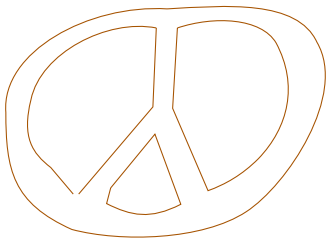
Uncertain reading. The crossed-out superscript mark between ‘’t’ and ‘Hooft’ is illegible.

We join them and sum over all possible ways of joining.



Original source details: GELFAND-P001300

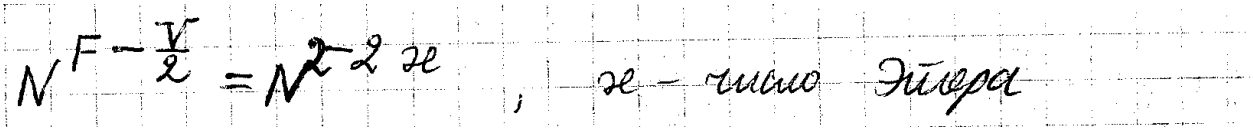
Diagram and typed description.



— tetrahedron. [wavy underlining]

$\forall$ index loop $=$ a factor N. Their number F is the number of faces.

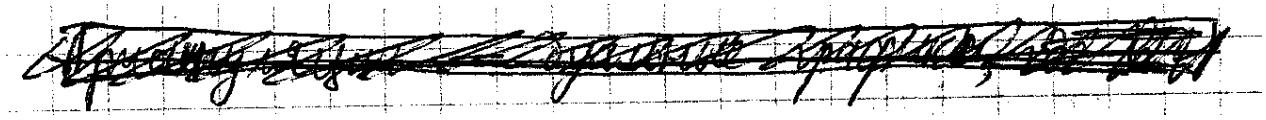
$$N^{F-\frac{V}{2}} = N^{2-2\chi}, \qquad \chi \text{ the Euler number}$$



Original source details: GELFAND-P001300

Uncertain reading. The same glyph appears in the exponent after $2-2$ and before 'Euler number'. The reading χ is tentative.

(to a vertex — $\frac{1}{\sqrt{N}}$)



Original source details: GELFAND-P001300

Undeciphered passage. The line above (7) is heavily crossed out with horizontal and oblique strokes. The original wording is undeciphered.

$$I(g) = \int d^{N^2}u \, \exp\left\{-\operatorname{tr}\left(u^2 + \frac{g}{\sqrt{N}}u^3\right)\right\} \tag{7}$$

(u a Hermitian matrix)

Computed in the work of Parisi, Itzykson, etc.

(in the lim. $N \to \infty$)

Pol. coord. (generalized)

$$u_{ij} = \sum_k \Omega^+_{ik}\lambda_k\Omega_{kj}, \tag{8}$$

Ω a unitary matrix

(separate out the eigenvalues)

$$I(g) = \int \exp\left\{-\sum_{i=1}^{N}\left(\lambda_i^2 + \frac{g}{\sqrt{N}}\lambda_i^3\right)\right\} d(\Omega)\prod d\lambda_j \prod_{i>j}(\lambda_i - \lambda_j)^2 \tag{9}$$

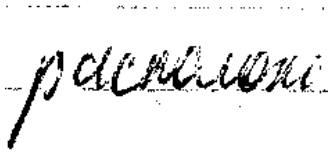
Uncertain reading. An illegible mark is cancelled between the integral sign and exp.

Original source details: GELFAND-P001300

$$\left(du = d(\Omega)\prod_{i>j}(\lambda_i - \lambda_j)^2 \prod_j d\lambda_j\right) \quad \text{— Dyson measure}$$

GELFAND-P001301

Dyson: [wavy underline] the most probable arrangement of the eigenvalues of a unitary matrix of high order is uniform on the circle.



Uncertain reading. The last strokes in the Russian abbreviation for “arrangement” run together.

Original source details: GELFAND-P001301

Rewrite

$$I(g) = \int d(\Omega) \prod d\lambda_j$$

$$\underbrace{\exp\left\{ -\sum_{i=1}^{N} \left(\lambda_i^2 + \frac{g}{\sqrt{N}} \lambda_i^3 \right) + 2 \sum_{i>j} \ln |\lambda_i - \lambda_j| \right\}}_{\sim N^2 \text{ terms}} \tag{10}$$

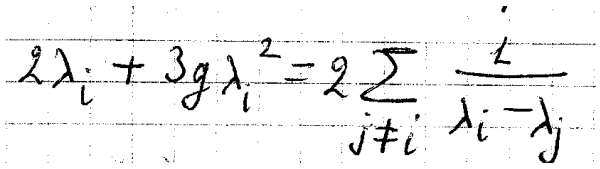
redefine $\lambda_i \rightsquigarrow$

$$I(g) = \int d(\Omega) \prod d\lambda_j$$

$$\exp\left\{ -N \sum_{i=1}^{N} (\lambda_i^2 + g\lambda_i^3) + 2 \sum_{i>j} \ln |\lambda_i - \lambda_j| \right\}$$

and it is calculated by the rules of the saddle-point method. The resulting eq. is

$$2\lambda_i + 3g\lambda_i^2 = 2 \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \tag{11}$$



Uncertain reading. The manuscript writes (11) without an additional factor of N or $1/N$.

Original source details: GELFAND-P001301

In the limit $N \to \infty$ the eq. reduces to an integral equation!

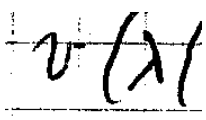
Introduce

$$x = \frac{i}{N}, \qquad \nu(\lambda(x)) = 2 \frac{dx}{d\lambda(x)} \quad \text{— eigenvalue density}$$

(in $\forall$ interval)

$$2\lambda + 3g\lambda^2 = \int\!\!\!\!\!\!- \frac{d\mu\, \nu(\mu)}{\lambda - \mu} \quad \text{(principal value)}$$

$$\int d\lambda\, \nu(\lambda) = 2 \quad \text{— normalization.}$$

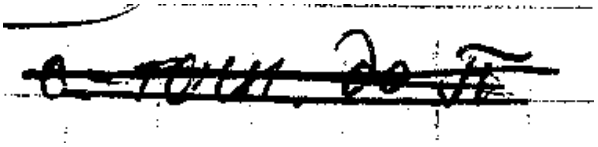


Uncertain reading. The density symbol in the numerator of the integral is overwritten. The reading ν follows the neighboring formulas.

Original source details: GELFAND-P001301

(Dispersion relation; representation of $\Re f(z)$ in terms of $\Im f(z)$)

$$F(\lambda) = 2\lambda + 3g\lambda^2 - \underbrace{[\alpha + \beta\lambda]\sqrt{(\lambda - 2a)(2b - \lambda)}}_{\pi\nu(\lambda)}$$

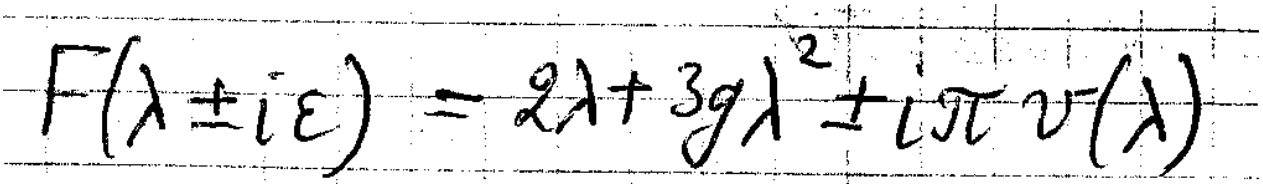


Original source details: GELFAND-P001301

Undeciphered passage. The note to the right of $\pi\nu(\lambda)$ is crossed out. “To” can be read; the remaining signs are uncertain.

a, b, α, β are functions of g; they are chosen.

$$F(\lambda \pm i\varepsilon) = 2\lambda + 3g\lambda^2 \pm i\pi\nu(\lambda)$$

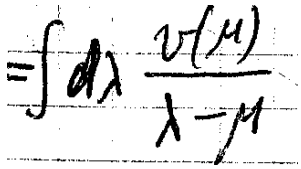


Original source details: GELFAND-P001301

Uncertain reading. The manuscript writes both signs as $\pm$.

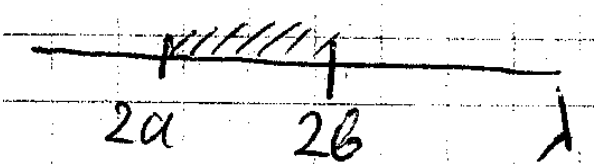
(Continuous spectrum! — analytic in a domain with a cut)

$$F(\lambda) = \int d\lambda \, \frac{\nu(\mu)}{\lambda - \mu}$$



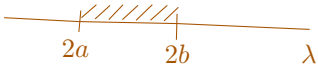
Original source details: GELFAND-P001301

Uncertain reading. The manuscript gives $d\lambda$ as the differential and μ in the fraction.



Original source details: GELFAND-P001301

Diagram and typed description.



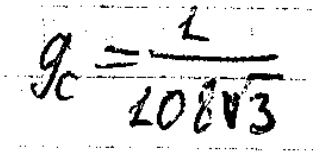
$$F(\lambda) \underset{\lambda\to\infty}{\longrightarrow} \frac{2}{\lambda}$$

α, β, a, b are found so as to cancel the second-degree polynomial.

Answer

$$F(g) = Z = -\frac{1}{2}\sum_{k=1}^{\infty} \frac{(72g^2)^k \Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)(k+2)!} \tag{12}$$

Critical constant $g_c = \frac{1}{108\sqrt{3}}$, singularity $F(g) \approx (g - g_c)^{5/2}$.



Uncertain reading. The last digit in 108 is read from the lower loop and upper stroke. The manuscript gives g_c without a square.

Original source details: GELFAND-P001301

Undeciphered passage. Beyond the sheet's right edge, three opening parentheses are visible with cropped strokes. Their continuations lie outside the frame.

Original source details: GELFAND-P001301

GELFAND-P001302

[source number: 10]

The number F_n of triangulations with n vertices is then found asymptotically

$$F_n \sim n^{-7/2} g_c^{-2n}$$

Possible generalization: [wavy underline] spin variables at $\forall$ vertex of the triangulation

GELFAND-P001303

Uncertain reading. A short isolated oblique stroke appears above the date and heading. Its meaning is uncertain.

Original source details: GELFAND-P001303

13.10.86 Gelfand Sem.

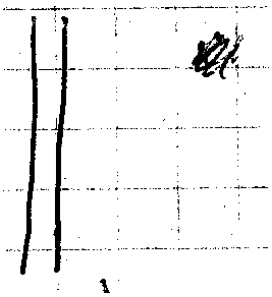
D. Gurevich. Generalized Lie algebras and groups.

Constant solutions of the Yang–Baxter equations

$\Downarrow$

generalization of Lie theory

analogous to the inclusion (supermathematics) $\supset$ (mathematics)



Original source details: GELFAND-P001303

Diagram and typed description.

Two parallel strokes to the right of the text. The small crossed-out mark at upper right is illegible.

$$L, \quad \dim L = n$$

$$L^{\otimes 2} \xrightarrow{S} L^{\otimes 2} \quad \text{(linear maps)}$$

$$L^{\otimes 3} \xrightarrow[\mathrm{id}\otimes S = S^{(2)}]{S\otimes \mathrm{id} = S^{(1)}} L^{\otimes 3}$$

(1) Y.–B. equation:

$$1) \quad S^{(1)}S^{(2)}S^{(1)} = S^{(2)}S^{(1)}S^{(2)}$$

$$2) \quad S^2 = \mathrm{id} \quad \overset{\text{def}}{\Longleftrightarrow} \quad S \text{ is a symmetry}$$

Generalization of the notion of a Lie algebra.

L is a finite-dimensional vector space

The structure of a generalized Lie algebra (or S-algebra) is a map [the end of the Russian word for “generalized” is below the line]

$$L^{\otimes 2} \xrightarrow{[\cdot,\cdot]} L \qquad \text{(linear) map}$$

Original source details: GELFAND-P001303

Uncertain reading. An illegible mark is crossed out before the formula.

Original source details: GELFAND-P001303

Uncertain reading. The superscript at the right of the final L is crossed out. The reading under the strokes is uncertain.

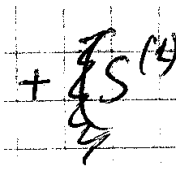
and the axioms hold

$$1) \quad [\cdot,\cdot] = -S[\cdot,\cdot]$$

(S is a symmetry operator satisfying axioms $(1)_{1,2}$ and is fixed)

(operators are written on the right)

$$2) \quad \big(\mathrm{id} + S^{(1)}S^{(2)} + S^{(2)}S^{(1)}\big)\big([\cdot,\cdot]\otimes \mathrm{id}\big)[\cdot,\cdot] = 0$$



Uncertain reading. A mark after the first $+$ and before $S^{(1)}$ is crossed out. Its reading is uncertain.

Original source details: GELFAND-P001303

Here $\mathrm{id} = \mathrm{id}_{L^{\otimes 3}}$

$$3) \quad S^{(1)}S^{(2)}\big([\cdot,\cdot]\otimes \mathrm{id}\big) = \big(\mathrm{id}\otimes[\cdot,\cdot]\big)S$$

Better to write $S^{(1)} = S^{(12)}, \quad S^{(2)} = S^{(23)}$.

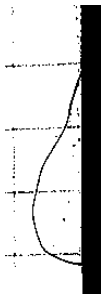


Diagram and typed description.

A thin line at the right edge. Its continuation beyond the visible edge is uncertain.

Original source details: GELFAND-P001303

GELFAND-P001304

S is a permutation of the factors $\Rightarrow$ what do the axioms mean?

Axioms (1) mean that S gives a rep. of the permutation group on 3 elements.

2) becomes the Jacobi identity [wavy underline] for S a permutation and [,] the commutator.

3) a tautology in the ordinary situation.

Supermathematics.

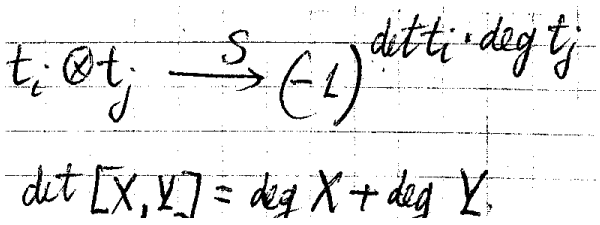
$t_1, \ldots, t_n$ is a homogeneous basis

Uncertain reading. A struck-out mark appears before t_1. Its reading is uncertain.

Original source details: GELFAND-P001304

$$t_i \otimes t_j \xrightarrow{S} (-1)^{\mathrm{dt}\, t_i \cdot \deg t_j}\, t_j \otimes t_i$$

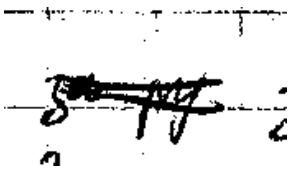
$$\mathrm{dt}[X, Y] = \deg X + \deg Y.$$



Uncertain reading. The two forms before t_i and $[X, Y]$ are rendered dt. The middle stroke is unclear. The other operators read deg.

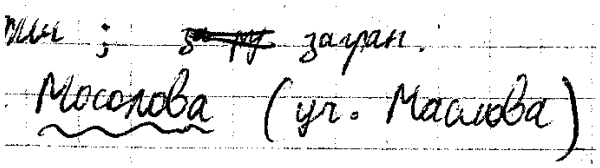
Original source details: GELFAND-P001304

Another example: [wavy underline] color symmetries; specified by Mosolov [wavy underline] (Maslov's student)



Uncertain reading. A word before "specified" is crossed out. It may begin with Russian "з".

Original source details: GELFAND-P001304



Uncertain reading. The surname is provisionally read "Мосолова" (Mosolov) and the abbreviation in parentheses "уч." (student).

Original source details: GELFAND-P001304

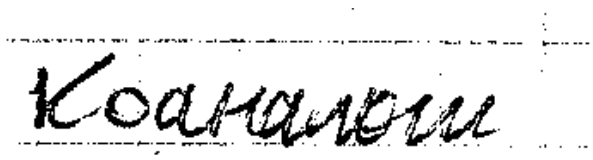
Color mathematics [broken underline]

$$S: \quad t_i \otimes t_j \longmapsto a(i,j) t_j \otimes t_i$$

(Latest work by Trostel, 1984, J. Math. Phys., 25, $N = 11$)

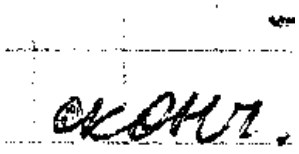
The $a(i, j)$ must satisfy certain conditions (written down from the axioms)

The braids in their final form were devised by Drinfeld.



Uncertain reading. The first word is provisionally read "Косички" (braids). Several letters in the middle run together.

Original source details: GELFAND-P001304



Uncertain reading. The abbreviation is read "оконч." (final). Its last strokes run together.

Original source details: GELFAND-P001304

The associative algebras will be the same.

An S-algebra is constructed from an assoc. algebra as follows

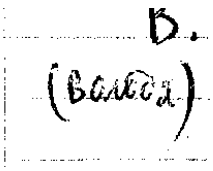
$$[A, B] = AB - (AB)S$$

More precisely, there is a multiplication operator $A^{\otimes 2} \overset{\circ}{\to} A$

$$[f, g] = (f \otimes g) \circ -((f \otimes g)S) \circ$$

It will satisfy all the axioms.

V. Kazakov. Another model of discrete 2-dimensional gravity. (Volodya)



Original source details: GELFAND-P001304

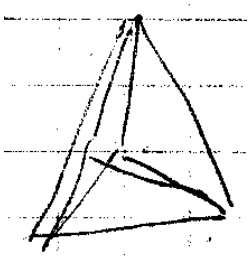
Uncertain reading. The parenthetical name under the initial on the left is provisionally read “Володя” (Volodya).

Consider a triangulation of S^2, assign additional variables $\sigma_i = \pm 1$ at the vertices and use Ising statistics

Original source details: GELFAND-P001304

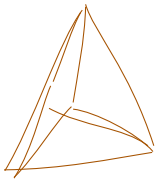
Uncertain reading. A sign before S^2 is crossed out. Its reading is uncertain.

$$Z_n = \sum_{\substack{\text{triang.} \\ \text{of the sphere}}} \sum_{\{\sigma_i\}} \exp[\beta G_{ij} \sigma_i \sigma_j]. \tag{1}$$



Original source details: GELFAND-P001304

Diagram and typed description.



Some slanted lines are retraced. The joins near the middle and at the lower-right corner are unclear.

G_{ij} is the adjacency matrix of the triangulation,

$$G_{ij} = \begin{cases} 1, & \text{if } i, j \text{ are neighbors} \\ 0 & \text{otherwise} \end{cases}$$

GELFAND-P001305

[source number: 14]

Uncertain reading. A short group of letters is crossed out before the first line. Its reading is uncertain.

Original source details: GELFAND-P001305

Why is this a model of QG (quantum grav.)? And of which QG?

Consider the continuous theory

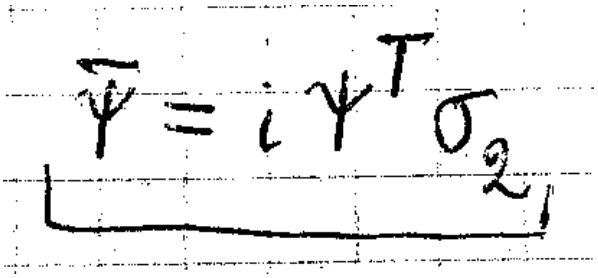
$$Z = \int \mathcal{D}g \int \mathcal{D}\Psi \, \exp\left(-\int d^2\xi \left[\mu\sqrt{g} + \bar{\Psi}\sigma_\mu\partial_\mu\Psi + g^{1/4}m\bar{\Psi}\Psi\right]\right) \tag{2}$$

Ψ is a 2-dimensional Majorana fermion in curved space/time

$(2) \sim (1)$ The proof is difficult, σ_μ are Pauli matrices

Solution of (2): Migdal and Bershadsky [wavy underline]

$\bar{\Psi} = i\Psi^T\sigma_2$ — relation for a Majorana fermion



Uncertain reading. A squared bracket is drawn beneath $\bar{\Psi} = i\Psi^T\sigma_2$.

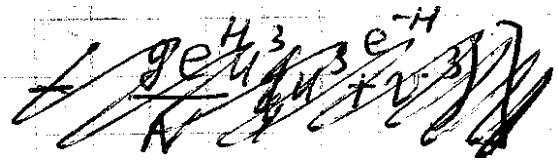
Original source details: GELFAND-P001305

($\Psi, \bar{\Psi}$ are not independent as they are for Dirac fermions)
Instead of (1), a more general one is considered below (with a magnetic field)

$$Z_n = \sum_{\{G^{(n)}\}} \sum_{\{\sigma_i\}} \exp\left[\beta \sum_{i,j=1}^{n} G_{ij}\sigma_i\sigma_j + H\sum_i \sigma_i\right] \tag{3}$$

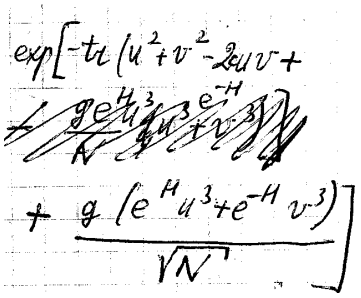
[At right, with a bracket around the note:] $G^{(n)}$ is a graph with n vertices
Introduce the model

$$\begin{gathered} Z(g) = \sum_n g^n Z_n(\beta) \overset{N\to\infty}{=} \frac{1}{N^2}\log \int d^{N^2}u \int d^{N^2}v \\ \exp\big[-\operatorname{tr}\big(u^2+v^2-2cuv+ \\ \text{[crossed-out term]} \\ + \frac{g(e^H u^3 + e^{-H}v^3)}{\sqrt{N}}\big] \end{gathered} \tag{4}$$



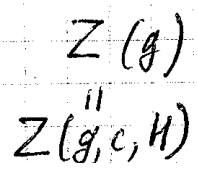
Original source details: GELFAND-P001305

Undeciphered passage. At left the crossed-out line may read $\frac{ge^H}{N}u^3$. Farther right, u^3, a raised e^{-H} and $+v^3$ can be distinguished. The grouping and the end of the entry are uncertain.



Original source details: GELFAND-P001305

Uncertain reading. The manuscript shows + at the end of the first exponent line, + at the start of the replacement line, and a square bracket after the fraction. A separate closing parenthesis for tr(is uncertain.



Original source details: GELFAND-P001305

Uncertain reading. Below the left-hand $Z(g)$ is $Z(g,c,H)$, with two short vertical strokes between the lines.

u, v are $N \times N$ Hermitian matrices

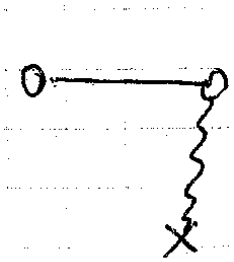
Here $c = e^{-2\beta}$

Solved in work by Zuber, Itzykson and . . .

The coupling constant g may be taken to be complex.

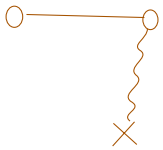
I.M.G. There is a book by Dyson (in Russian): all reasonable problems in symmetric spaces as $N \to \infty$ have an answer.

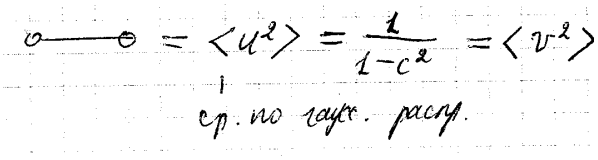
Proof of formula (4): expand exp $\Rightarrow$ diagrammatic technique. Vertices of two types: u-vertices and v-vertices.



Original source details: GELFAND-P001305

Diagram and typed description.



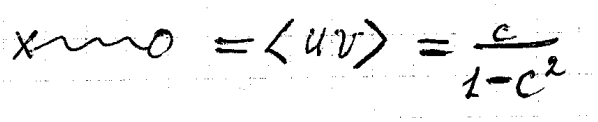


Original source details: GELFAND-P001305

Diagram and typed description.

$\circ\!\!-\!\!\circ = \langle u^2 \rangle = \dfrac{1}{1-c^2} = \langle v^2 \rangle$

[Below $\langle u^2 \rangle$, with a vertical stroke:] average over the Gaussian distribution.



Original source details: GELFAND-P001305

Diagram and typed description.

$\times\!\sim\!\circ = \langle uv \rangle = \dfrac{c}{1-c^2}$

GELFAND-P001306

What is there in model (1)?

Phase transition (spontaneous ordering of the spins) at the value

$$\beta_t^* = \tfrac{1}{2} \ln(2\sqrt{7} + 1).$$

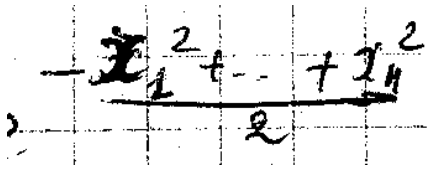
Original source details: GELFAND-P001306

Uncertain reading. The first strokes of the Russian abbreviation for “phase” are overwritten.

Varchenko

Wick's theorem [broken underline]

$$\frac{1}{(2\pi)^{n/2}} \int e^{-\frac{x_1^2+\cdots+x_n^2}{2}} \xi_1 \cdots \xi_N \, dx \stackrel{\text{def}}{=} \langle \xi_1 \cdots \xi_N \rangle$$



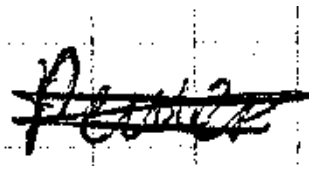
Original source details: GELFAND-P001306

Uncertain reading. The first sign in the exponent numerator is overwritten. The visible entry is rendered x_1^2. The earlier reading is uncertain.

Then

$$\langle \xi_1 \cdots \xi_N \rangle = \sum_{\substack{\text{arrangement} \\ \text{of brackets}}} \langle \xi_{i_1} \xi_{i_2} \rangle \cdots \langle \xi_{i_{N-1}} \xi_{i_N} \rangle$$

ξ_j are linear functions of the x_k

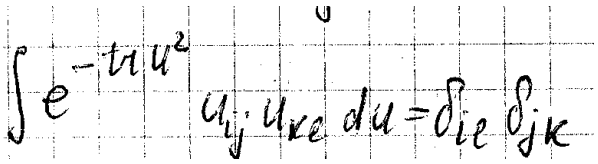


Original source details: GELFAND-P001306

Uncertain reading. The crossed-out word before "Proof" is provisionally read "Решение" (Solution).

Proof. [broken underline] first $\xi_i = x_{k_i} \Rightarrow$ integral = a product, and it remains to use multilinearity

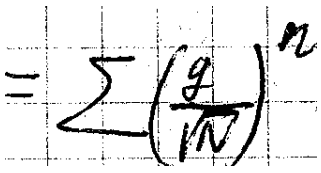
$$\int e^{-\operatorname{tr} u^2} u_{ij} u_{k\ell} \, du = \delta_{i\ell} \delta_{jk}$$



Original source details: GELFAND-P001306

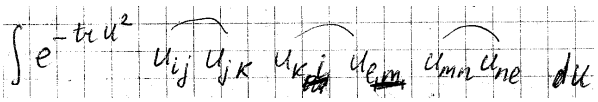
Uncertain reading. The measure is written du; there are no explicit normalization factors in the line.

$$\int du \, e^{-\operatorname{tr}(u^2 + \frac{g}{\sqrt{N}} u^3)} = \sum \left(\frac{g}{\sqrt{N}} \right)^n \int e^{-\operatorname{tr} u^2} (\operatorname{tr} u^3)^n \, du$$



Original source details: GELFAND-P001306

Uncertain reading. The sum has no written limits, and its coefficient has no written minus sign or factorial.



Original source details: GELFAND-P001306

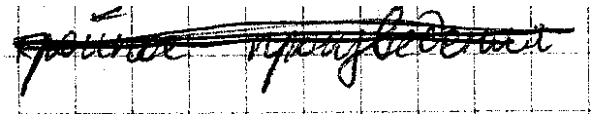
Diagram and typed description.

$$\int e^{-\operatorname{tr} u^2} \; \overbrace{u_{ij} u_{jk}} \; \overbrace{u_{ki} u_{\ell m}} \overbrace{u_{mn} u_{n\ell}} \; du$$

Original source details: GELFAND-P001306

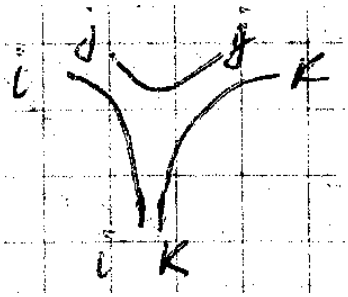
Uncertain reading. Under the third and fourth factors, some indices are crossed out. The uncrossed indices are read ki and ℓm.

Arrangement of brackets ⇒ [crossed out: triple product] graph.



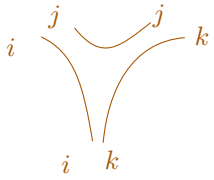
Original source details: GELFAND-P001306

Uncertain reading. The crossed-out words are provisionally read “тройное произведение” (triple product).



Original source details: GELFAND-P001306

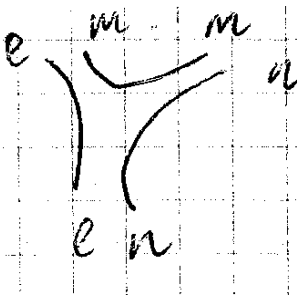
Diagram and typed description.



$$\delta_{ik}\delta_{kj}\delta_{km}\delta_{i\ell}\delta_{m\ell}\delta_{nn}$$

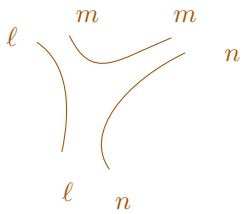
Original source details: GELFAND-P001306

Uncertain reading. The first index of the second factor is retraced. The visible strokes are rendered δ_{kj}.



Original source details: GELFAND-P001306

Diagram and typed description.



Original source details: GELFAND-P001306

Diagram and typed description.

A short, thin wavy stroke in the lower-left margin. It has no label.

GELFAND-P001307

20.10.86 Gelfand Sem.

V. Ginzburg.

On work by A. Givental. (and then on work by P. Deligne — L. Illusie)

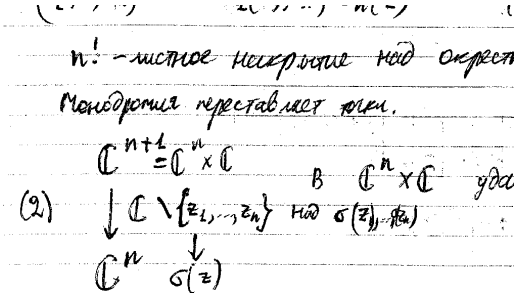
An explicit example of a representation of the Hecke algebra.

(arising from singularity theory)

$$\begin{aligned} &\mathbb{C}^n \longrightarrow \mathbb{C}^n \\ &(z_1, \dots, z_n) \longrightarrow \sigma_1(z), \dots, \sigma_n(z) \quad \text{(elem. symm. polynomials)} \end{aligned} \tag{1}$$

$n!$-sheeted covering over a neighborhood of a general point.

Monodromy permutes the points.



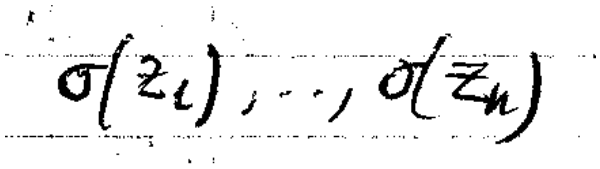
Original source details: GELFAND-P001307

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^{n+1} = \mathbb{C}^n \times \mathbb{C} & & \\ \Big\downarrow & \mathbb{C} \setminus \{z_1, \dots, z_n\} & \\ (2) & \text{over } \sigma(z_1, \dots, z_n) & \\ \mathbb{C}^n & \sigma(z) & \end{array}$$

The last argument in the label of the fiber is overwritten. The reading z_n follows the adjacent notation.

In $\mathbb{C}^n \times \mathbb{C}$, remove $\{z_1, \dots, z_n\}$ over $\sigma(z_1), \dots, \sigma(z_n)$.



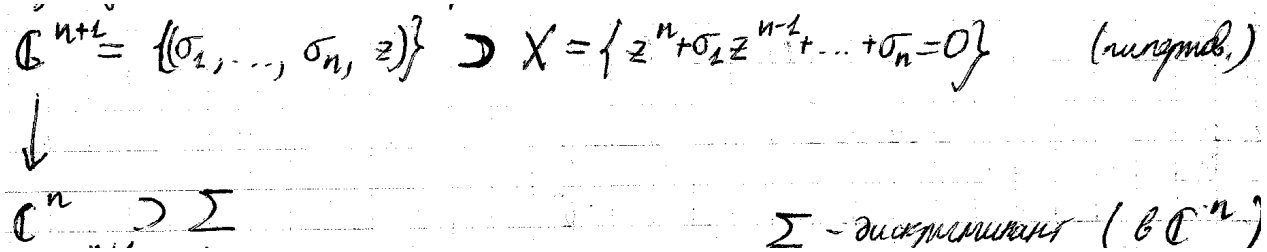
Original source details: GELFAND-P001307

Uncertain reading. The last line writes $\sigma(z_1), \dots, \sigma(z_n)$ separately. In the diagram a single σ takes the list of arguments.

GELFAND-P001308

[source number: 20]

Remove from the base the σ that correspond to roots that have coalesced.

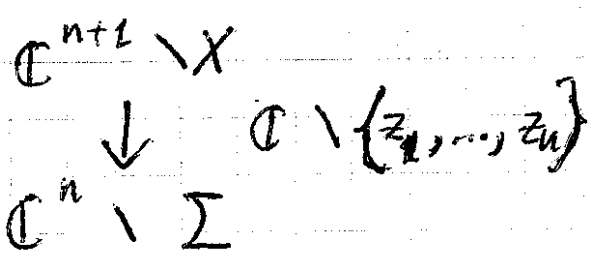


Original source details: GELFAND-P001308

Diagram and typed description.

$$\begin{array}{l} \mathbb{C}^{n+1} = \{(\sigma_1, \dots, \sigma_n, z)\} \supset \\ \qquad X = \{z^n + \sigma_1 z^{n-1} + \dots + \sigma_n = 0\} \\ \downarrow \text{ (hypersurface)} \\ \mathbb{C}^n \supset \Sigma \end{array}$$

Σ is the discriminant (in $\mathbb{C}^n$).



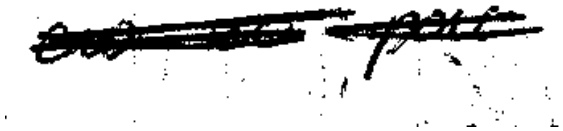
Original source details: GELFAND-P001308

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^{n+1} \setminus X & & \\ \downarrow & & \mathbb{C} \setminus \{z_1, \ldots, z_n\} \\ \mathbb{C}^n \setminus \Sigma & & \end{array}$$

Pass to 1-dimensional homology with coefficients

f^α — α is a fixed complex number; [crossed out] multivalued function



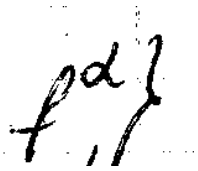
Original source details: GELFAND-P001308

Undeciphered passage. Two groups of letters are crossed out after “number”. Their reading is uncertain.

$$f = z^n + \sigma_1 z^{n-1} + \cdots + \sigma_n$$

f^α defines a covering; consider cohomology with coefficients in it

$$H^1\big(\mathbb{C} \setminus \{*, *, \ldots, *\}, f^\alpha\big)$$

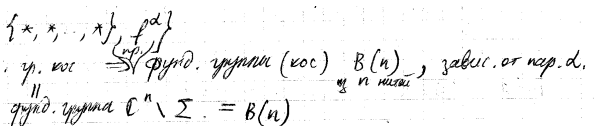


Original source details: GELFAND-P001308

Uncertain reading. The left delimiter of H^1 is round. The mark after f^α resembles a closing brace.

There is a rep. of the braid group $\Rightarrow$ of the fund. group (of braids) $B(n)$, depending on the parameter α. [Below $B(n)$: of n strands.]

fund. group of $\mathbb{C}^n \setminus \Sigma = B(n)$.



Original source details: GELFAND-P001308

Diagram and typed description.

$$\begin{array}{ccc} & \overset{f^\alpha}{\underset{\text{repr.}}{\swarrow}} & \\ \text{braid group} & \Rightarrow & \text{fund. group} \\ \| & & \| \\ \mathbb{C}^n \setminus \Sigma & = & B(n) \end{array}$$

The curved strokes below f^α meet in a point beside the arrow. The reading “repr.” and the attachment of the strokes to the words are provisional. Two vertical strokes appear below “braid group”.

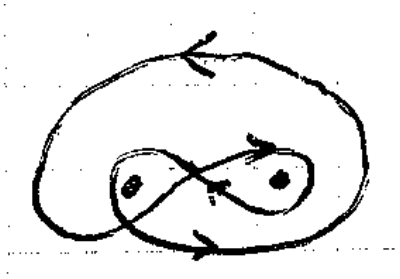
The braid group has generators $T_1, \ldots, T_{n-1}$ with rel.

$$T_1 T_2 T_1 = T_2 T_1 T_2.$$

(T_j corresponds to an elementary circuit)
This rep. has an additional relation.

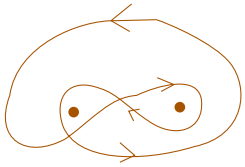
$$(T_i + 1)(T_i - \alpha) = 0 \quad \Rightarrow$$

$\Rightarrow$ the rep. factors through a rep. of the Hecke alg.



Original source details: GELFAND-P001308

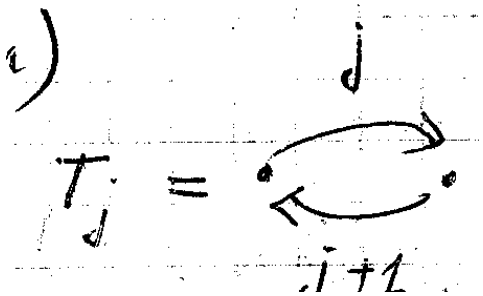
Diagram and typed description.



The points and arrow directions are visible. The exact connections at the two crossings are uncertain.

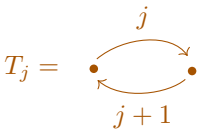
$$\pi_1(\mathbb{C}^n \setminus \Sigma) = B(n)$$

generators



Original source details: GELFAND-P001308

Diagram and typed description.



(lin. transformation corresponding to a transposition of 2 roots)

$\downarrow$

connect to Id.

1) $T_i T_j = T_j T_i$ when $|i - j| \geq 2$
2) $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$
3) $(T_i + 1)(T_i - \alpha) = 0$

The aim was to prove faithfulness of the rep. of the fund. group in a group of matrices whose entries are Laurent polynomials.

For [crossed out] $n = 3$ this is known.

Original source details: GELFAND-P001308

Uncertain reading. An entry before $n = 3$ is crossed out. The visible strokes could be n, but the reading is uncertain.

Uncertain reading. A faint curved mark appears to the right of “known”. Its meaning is uncertain.

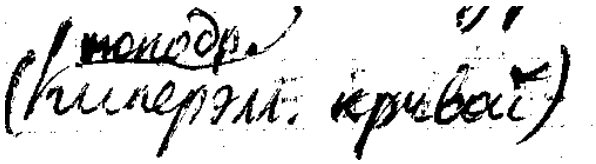
Original source details: GELFAND-P001308

Specialists in braid group theory call this the Burau representation.

Uncertain reading. The manuscript spells the name “Бурао” (Burau).

Original source details: GELFAND-P001308

$\alpha = -1 \Rightarrow$ monodromy group of elliptic integrals ([inserted above: monodr.] of a hyperelliptic curve)



Uncertain reading. The abbreviation “монодр.” (monodr.) is written above “hyperell.” and underlined with a curved connecting stroke.

Original source details: GELFAND-P001308

GELFAND-P001309

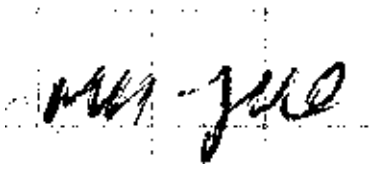
Beilinson

on work by P. Deligne and L. Illusie. Preprint Orsay

Algebraic proofs of facts formerly proved only by transcendental methods

(Laplace operator, etc.)

X is a smooth compact alg. variety over $\mathbb{C}$



Uncertain reading. The initial strokes in the Russian abbreviation “мн-ие” for “variety” run together. The reading follows the context.

Original source details: GELFAND-P001309

De Rham complex: [wavy underline]

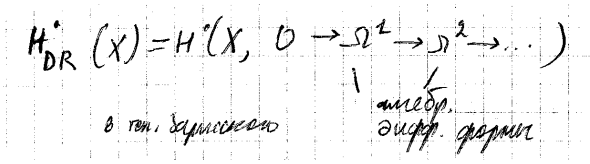
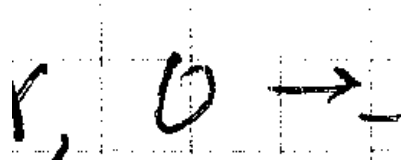


Diagram and typed description.

$H^{\bullet}_{\mathrm{DR}}(X) = H^{\bullet}(X, O \to \Omega^1 \to \Omega^2 \to \dots)$

in the Zariski top. — algebraic diff. forms

Original source details: GELFAND-P001309



Uncertain reading. The initial term of the complex is written with a sign resembling O.

Original source details: GELFAND-P001309

Thm.

$$E_1^{p,q} = H^p(X, \Omega^q)$$

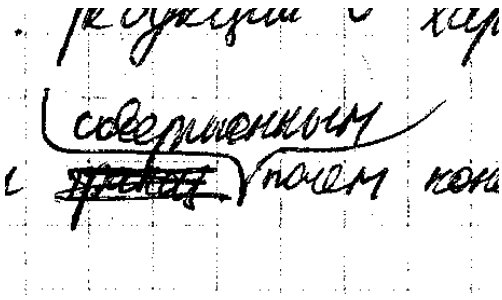
(the spectral sequence degenerates already at the first term)

1st [inserted above: alg.] proof — Faltings — nonelem.

Deligne and Illusie [wavy underline] — elementary algebraic

Idea of the proof: in place of Δ use reduction to char. p!!

X/k over a finite field k or a [crossed out] [inserted above: perfect] field of finite char.



Original source details: GELFAND-P001309

Undeciphered passage. The word before "field" is crossed out and its reading is uncertain. The insertion "perfect" is legible.

There is cohomology.

Kazakov V. (continued).

Formula on which the evaluation of the two-matrix integral is based

$$\int d(\Omega)_{U(N)} e^{\mathrm{tr}(\lambda\Omega^+\mu\Omega)} = \left(\prod_{k=1}^{N-1} k!\right) \frac{\det_{i,j} e^{\lambda_i\mu_j}}{\Delta(\lambda)\Delta(\mu)} \tag{1}$$

λ, μ are diagonal matrices

$$\Delta(\lambda) = \prod_{i>j}(\lambda_i - \lambda_j)$$

$$\lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_N \end{pmatrix}, \qquad \mu = \begin{pmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_N \end{pmatrix}$$

First in work by Itzykson, Zuber. Proof using the heat eq. on the matrix group.

I.M.G. There is a similar formula in the book by Gelfand, Naimark [wavy underline], "Unit. rep. of class. groups" (1951)

$$\Omega^{-1}\lambda\Omega = H \qquad \Delta_1^{\mu_1}\Delta_2^{\mu_2}\dots\Delta_n^{\mu_n}$$

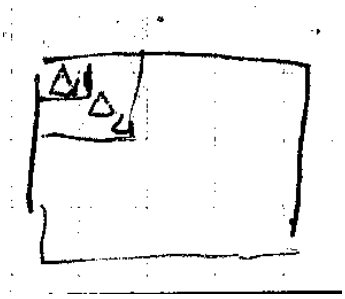
Original source details: GELFAND-P001309

Uncertain reading. The first glyph in $\Omega^{-1}\lambda\Omega$ is retraced. The visible outline reads Ω.

Ω is unitary

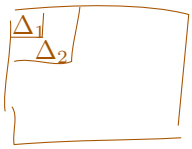
λ is diag. ($\lambda_i > 0$)

(principal minors)



Original source details: GELFAND-P001309

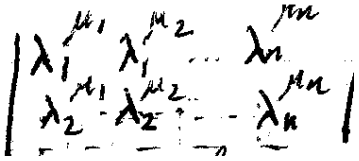
Diagram and typed description.



GELFAND-P001310

[source number: 21]

$$\int \Delta_1^{\mu_1} \Delta_2^{\mu_2} \dots \Delta_n^{\mu_n} \, d\Omega$$
$$= \begin{vmatrix} \lambda_1^{\mu_1} & \lambda_1^{\mu_2} & \dots & \lambda_n^{\mu_n} \\ \lambda_2^{\mu_1} & \lambda_2^{\mu_2} & \dots & \lambda_n^{\mu_n} \\ - & - & - & - \end{vmatrix} \left[\prod (\lambda_i - \lambda_k) \prod (\mu_i - \mu_k) \right]^{-1} \tag{2}$$



Original source details: GELFAND-P001310

Uncertain reading. In the right-hand column of both rows the visible indices read $\lambda_n^{\mu_n}$. Short horizontal strokes appear below the two rows.

Original source details: GELFAND-P001310

Uncertain reading. The lower-left part of Ω in $d\Omega$ is retraced. The earlier sign is indistinct.

(written in the paper of G. and N.)

From this we obtain Lidskii's [wavy underline] thm. on the eigenvalues of a product of (Hermitian) matrices [below the parentheses: $H_j > 0$].

What can they be?

The eigenvalues of $H_1 H_2$ are always > 0. The region is described.

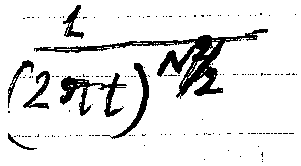
Proof using elliptic coordinates (generalized)

K. Proof of formula (1).

Write the heat equation on Hermitian matrices

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\Delta_u\right) f(t,u,v) = 0, \quad t > 0; \qquad f|_{t=0} = \delta(u-v) \tag{3}$$

$$f(t,u,v) = \frac{1}{(2\pi t)^{N^2/2}} e^{-\frac{1}{2t}\operatorname{tr}(u-v)^2} \tag{4}$$



Uncertain reading. The exponent of $(2\pi t)$ is retraced. A thick mark appears beside N above the slash. The dimension supports $N^2/2$, while $N/2$ remains possible from the strokes.

Original source details: GELFAND-P001310

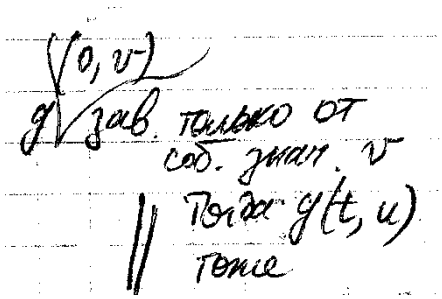
$\Delta_u = \sum$ of 2nd derivatives with respect to Re and Im of the entries (incl. diagonal)

$$g(t,u) = \int d^{N^2} v \, f(t,u,v) g(0,v) \tag{5}$$

Pass to Dyson's variables (variables in terms of eigenvalues), assuming g[inserted above: $(0,v)$] depends only on the eigenvalues of v

$$g(t,\mu) = \int d(\Omega) \int \prod d\lambda \, \Delta^2(\lambda) f(t,\mu,\Omega^+ \lambda \Omega) g(0,\lambda) \tag{6}$$

[In the right margin, after a double vertical stroke:] then $g(t,u)$ does too.



Uncertain reading. The insertion $(0,v)$ is linked to g by a curved stroke. The separate remark on the right is marked by two vertical strokes.

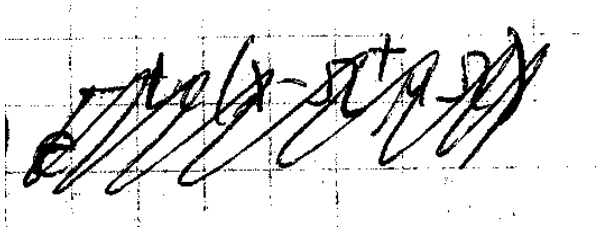
Original source details: GELFAND-P001310

$$u = \Omega_1^+ \mu \Omega_1, \qquad v = \Omega_2^+ \lambda \Omega_2; \quad \text{then} \quad \Omega = \Omega_2 \Omega_1^+$$

$$\underbrace{\Delta(\mu) g(t,\mu)}_{\xi(t,\mu)} = \int \prod d\lambda \, K(t;\mu,\lambda) \Delta(\lambda) g(0,\lambda) \tag{7}$$

Here

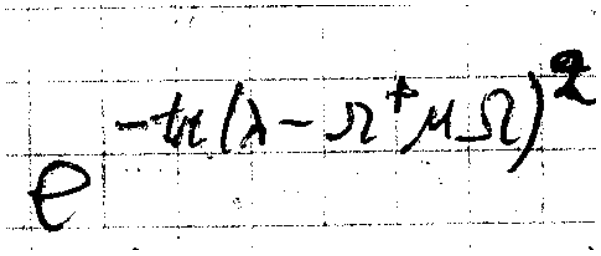
$$K(t,\mu,\lambda) = \Delta(\lambda)\Delta(\mu) \int d(\Omega)\, f(t,\mu,\Omega^+ \lambda \Omega) \tag{8}$$



Original source details: GELFAND-P001310

Uncertain reading. The expression crossed out after (8) provisionally reads $e^{-\operatorname{tr}(\lambda - \Omega^+ \mu \Omega)}$. Its final strokes are unclear.

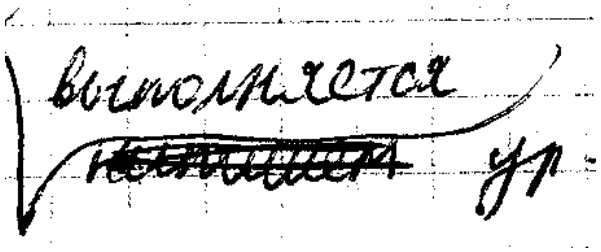
$$e^{-\operatorname{tr}(\lambda - \Omega^+ \mu \Omega)^2}$$



Original source details: GELFAND-P001310

Uncertain reading. Below (8) is a separate exponential with a square after the closing parenthesis. No factor of $1/(2t)$ is written in this line.

For $\xi(t,\mu)$ [crossed out] the eq. [inserted above: holds] (!)



Original source details: GELFAND-P001310

Uncertain reading. The crossed-out word may read “we will write”. The word “holds” is inserted above it with a pointer.

$$\left(\frac{\partial}{\partial t} - \frac{1}{2} \sum_{i=1}^{N} \frac{\partial^2}{\partial \lambda_i^2} \right) \xi(t,\lambda) = 0, \quad t > 0. \tag{9}$$

Original source details: GELFAND-P001310

Uncertain reading. A mark resembling f is crossed out before ξ. Its reading is uncertain.

Although

$$\left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta_\lambda \right) g(t,\lambda) = 0,$$
$$\text{where} \quad \Delta_\lambda = \frac{1}{\Delta(\lambda)^2} \sum_i \frac{\partial}{\partial \lambda_i} \left(\Delta^2(\lambda) \frac{\partial}{\partial \lambda_i} \right) \tag{10}$$

I.M.G. says this is in F. A. Berezin's candidate dissertation, Proc. Moscow Math. Soc. (≈ 1956)

GELFAND-P001311

Substituting $\dfrac{\xi(t,\lambda)}{\Delta(\lambda)} = g(t,\lambda)$ into (10), we arrive at (9).

Uncertain reading. The letter before (9) is heavily retraced. The visible strokes are read "к" (at).

Original source details: GELFAND-P001311

I.M.G. Berezin did it, and Harish-Chandra used it repeatedly.

K. Now write $\xi(t,\lambda)$ and so on.

Question: how to evaluate the integral

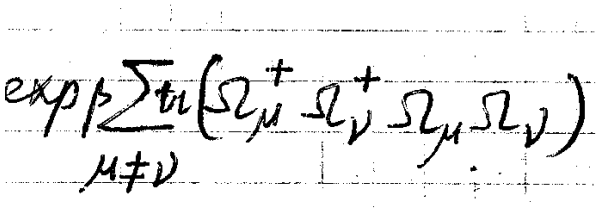
$$\int d\Omega\, e^{\mathrm{tr}(\lambda_1\Omega\mu_1\Omega^+)+\mathrm{tr}(\lambda_2\Omega\mu_2\Omega^+)} \tag{11}$$

— solution of the chiral field on $U(\infty)$

Evaluated in principle (Wiegmann and Polyakov solved the chiral field —
— Eguchi–Kawai reduction)

The most interesting case is connected with chromodynamics

$$\int \prod_{\mu=1}^{d} d\Omega_\mu \,\exp\beta \sum_{\mu\neq\nu} \mathrm{tr}(\Omega_\mu^+\Omega_\nu^+\Omega_\mu\Omega_\nu) \tag{12}$$



Uncertain reading. The sign after exp and before the sum reads β. The manuscript has no outer delimiters around the exponent.

Original source details: GELFAND-P001311

not evaluated, and gauge theory is unsolved [wavy underline]

All the d-dimensional physics is contained here ($d \leq 4$ suffices).

(12) is the Y.–M. partition function on the lattice.

GELFAND-P001312

27/X̄-86 Gelfand Sem.

Zakharevich

V is the space of $N \times N$ Hermitian matrices

Consider a function $g(t,x)$, $t \in \mathbb{R}$, $x \in V$

and the heat eq.

$$g_t = \Delta g \tag{1}$$

Claim.

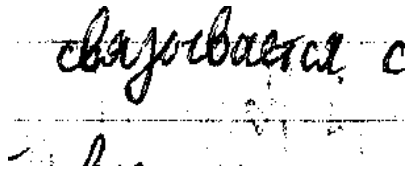
$$\int_{U(N)} d\Omega\, e^{\mathrm{tr}(\lambda\Omega^+\mu\Omega)} = \left(\prod_{k=1}^{N-1} k!\right) \frac{\det \|e^{\lambda_i\mu_j}\|}{\Delta(\lambda)\Delta(\mu)}, \tag{2}$$

$$\lambda = \mathrm{diag}(\lambda_i), \qquad \mu = \mathrm{diag}(\mu_i),$$

$$\Delta(\lambda) = \prod_{i>j}(\lambda_i - \lambda_j)$$

I.M.G. I.M. Gelfand, Naimark, Gindikin and Karpelevich, and Fateev also evaluated such integrals,

Important: this is connected with the combinatorial problem of triangulations of the sphere.



Original source details: GELFAND-P001312

Uncertain reading. The verb reads “связывается” (is connected).

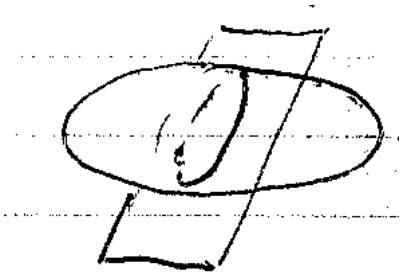
Problem from which representation theory arose

$$H_1 H_2 \qquad H_j^* = H_j > 0.$$

Given the eigenvalues of H_j, how are the eigenvalues of $H_1 H_2$ distributed?

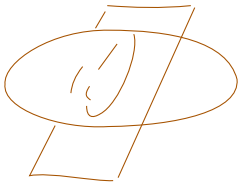
Elliptic coordinates on the set of unitary matrices

Consider an ellipsoid with axes a, b, c



Original source details: GELFAND-P001312

Diagram and typed description.



The inner line is broken at left. Two thickened ends of the quadrilateral are indistinct.

a_1, b_1 are the axes of the ellipse obtained in the section

$$a \le a_1 \le b \le b_1 \le c$$

$$\begin{array}{ccccccc} a & & b & & c & & d \\ & a_1 & & b_1 & & c_1 & \\ & & a_2 & & b_2 & & \\ & & & a_3 & & & \end{array}$$

We need to evaluate $d\Omega$ for matrices of order 2.

$$\int a_{11}^{\lambda_1} \left| \begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right|^{\lambda_2} \dots \, d\Omega,$$

λ_j are arbitrary parameters.

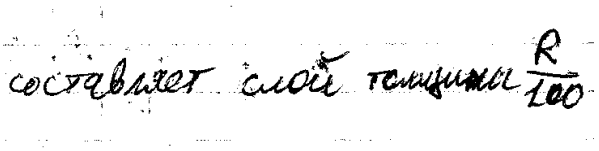
μ is a fixed diag. matrix.

$$A = \Omega^+ \mu \Omega$$

μ is given

Spectral analysis [broken underline]: analysis in $\mathbb{R}^N$ (or $\mathbb{C}^N$) as $N \to \infty$.

Question for schoolchildren [wavy underline]: what fraction of a 1000-dimensional watermelon is the layer of thickness $\frac{R}{100}$ around the equatorial plane? Answer [wavy underline]: almost all of it.



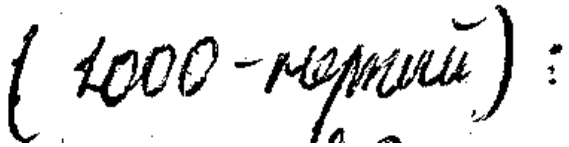
Uncertain reading. The manuscript gives the thickness as $R/100$.

Original source details: GELFAND-P001312

GELFAND-P001313

[source number: 24]

Given a simplex [broken underlining] (1000-dimensional), a particle reflects off the faces. [Crossed out] what fraction of the time will it spend in a $\frac{1}{100}$-neighborhood of the faces?

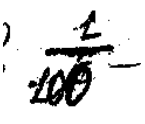


Uncertain reading. The digits read 1000.

Original source details: GELFAND-P001313

Undeciphered passage. A short word is crossed out before “what”. The opening loop allows the reading “она” (it).

Original source details: GELFAND-P001313



Uncertain reading. The fraction reads $\frac{1}{100}$. Its bar is thickened and strokes in the denominator meet in places.

Original source details: GELFAND-P001313

Answer [broken underlining]. Almost all the time.

B. Lidskii. Moment map [wavy underlining].

Original source details: GELFAND-P001313

Uncertain reading. The initial is provisionally read “Б.” (B.) from the lower stroke and catalogue record 690. The manuscript form can also be read “Е.” (E.).

Consider Hermitian matrices with fixed eigenvalues and the map on them

$$H \longrightarrow \text{diagonal of } H$$

The Fourier transform of the moment map is written down.

Original source details: GELFAND-P001313

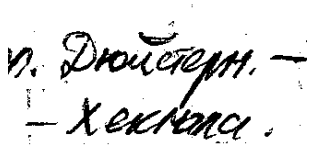
Uncertain reading. The Russian abbreviation reads “преобр.” (transform). The second stroke is thickened.

I.M.G. What is the superanalogue of the symmetric group?

Original source details: GELFAND-P001313

Uncertain reading. In the opening of “супераналог” (superanalogue), several strokes meet.

V. Ginzburg. Idea: find the critical points of the map. Take the first term of the asymptotics. It will be an exact formula by the Duisterm.-Heckman theorem.

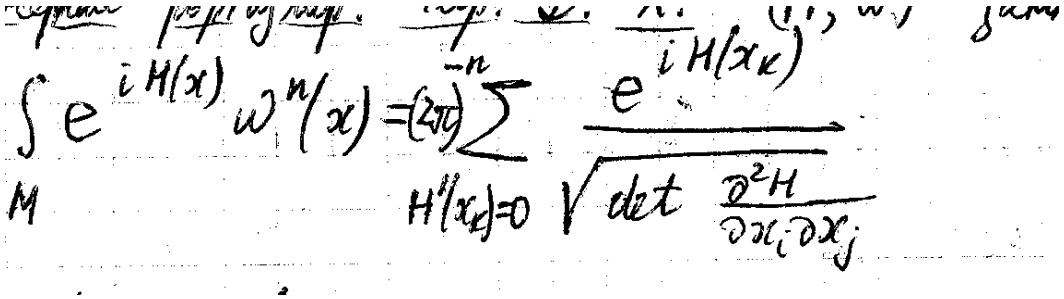


Original source details: GELFAND-P001313

Uncertain reading. The first surname is abbreviated, provisionally read “Дюйстерм.” (Duisterm.). “Хекмана” (Heckman) appears on the line below at right.

Approximate statement of the D.–H. theorem. (M, ω) is a closed symplectic manifold.

$$\int_M e^{iH(x)}\omega^n(x) = (2\pi)^{-n} \sum_{H'(x_k)=0} \frac{e^{iH(x_k)}}{\sqrt{\det \frac{\partial^2 H}{\partial x_i \partial x_j}}}$$

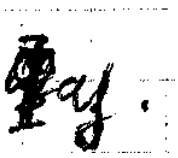


Original source details: GELFAND-P001313

Uncertain reading. The summation condition reads $H'(x_k) = 0$. The tall stroke to the right of the prime forms the opening parenthesis. The prefactor is written $(2\pi)^{-n}$.

$$dH(\cdot) = \omega(\xi, \cdot)$$

Condition. The phase flow ξ must be periodic.



Original source details: GELFAND-P001313

Uncertain reading. The first character of the Russian abbreviation “фаз.” (phase) is retraced several times.

Semenov-Tian-Shansky [broken underlining] showed that this agrees with Kirillov’s [inserted above the line: formula] in the case of an orbit.

Idea: $i\Omega^+\mu\Omega$ is an orbit with a symplectic structure

$$M = \text{quotient of unitary matrices by the diagonal}$$

$$X = \Omega^+\mu\Omega;$$

$$\omega(X)(\xi, \eta) = \operatorname{tr}(X \cdot [\xi, \eta]), \qquad \xi, \eta \in T_X$$

Diagram and typed description.

(!)

Original source details: GELFAND-P001313

I.M.G.'s question [wavy underlining]. When is the second term of the asymptotics also needed? Analyze the topological reasons why the second term is 0, etc. Connection with Morse inequalities.

For which integrals does the first term of the asymptotics give the exact answer?

Method [wavy underlining]. Look for cases where the second term of the asymptotics is $= 0$ at the current level.

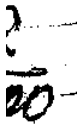
Subproblem [wavy underlining]. So that characteristic numbers arise.

Hypothesis (mine). In place of [crossed out] an S^1-action, specify an action of a nilpotent Lie grp. (e.g., 2-dimensional $\Rightarrow$ two terms will arise).

Original source details: GELFAND-P001313

Undeciphered passage. The cancelled mark before S^1 is obscured by slanted strokes.

[At the cropped left edge, near the bottom.]



Original source details: GELFAND-P001313

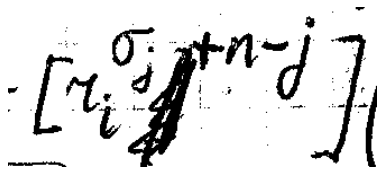
Undeciphered passage. The end of the fraction bar and two loops, provisionally "00", are visible in the denominator. The upper sign is clipped.

GELFAND-P001314

3/XI-86 Gelfand Sem.

G. Olshanskii.

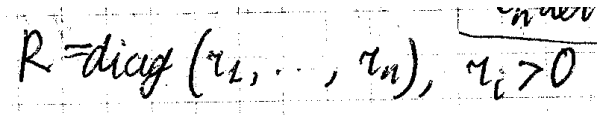
$$\int_{U(n)} \Delta_1(uRu^{-1})^{\sigma_1-\sigma_2}\cdots\Delta_{n-1}(uRu^{-1})^{\sigma_{n-1}-\sigma_n}\Delta_n(uRu^{-1})^{\sigma_n}\,du = \\ = c_n \det\left[r_i^{\sigma_j+n-j}\right]\left(\prod_{i<j}(r_i-r_j)\prod_{i<j}(\sigma_i-\sigma_j+j-i)\right)^{-1}, \tag{1}$$



Original source details: GELFAND-P001314

Uncertain reading. The exponent reads $\sigma_j + n - j$. Between the subscript j and the $+$, overlapping strokes are crossed out.

$$R = \operatorname{diag}(r_1,\dots,r_n), \quad r_i > 0$$

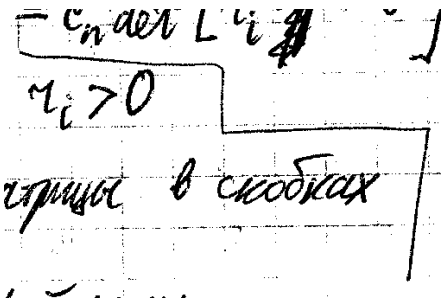


Original source details: GELFAND-P001314

Uncertain reading. The lowercase letters in the list of diagonal entries and before > 0 read r.

Δ_m — principal minors of the matrices in parentheses.

$$c_n = 1!2!\dots(n-1)!$$



Original source details: GELFAND-P001314

Diagram and typed description.

The callout starts beneath the second equals sign. Its final stroke passes to the right of the explanation of Δ_m.

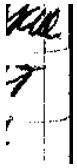
— an integral from the book by Gelfand and Naimark.

Original source details: GELFAND-P001314

Uncertain reading. The preposition before “книги” reads “из” (from).

Formula (2) on p. 24 is derived from (1).

[At the cropped left edge.]



Original source details: GELFAND-P001314

Uncertain reading. The cropped ending resembles “ми” or “ши”. Below is an arrow ↗, then a clipped hook and a separate mark, possibly part of a question mark.

GELFAND-P001315

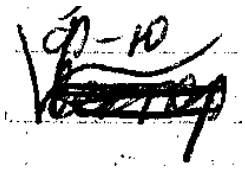
Proof of (1). [Wavy underlining.] (1) = H. Weyl’s character formula.

$e_1, \dots, e_n$ is the canonical basis.

$$ue_i = f_i$$

Consider GL_n and its unitary representation of class I (i.e., ∃ a vector invariant under the compact subgroup). Let ξ_0 be a vector with $T_u\xi_0 = \xi_0$.

Consider [crossed out: the vector] [above the line: the fn.] $(T_g\xi_0, \xi_0)$.



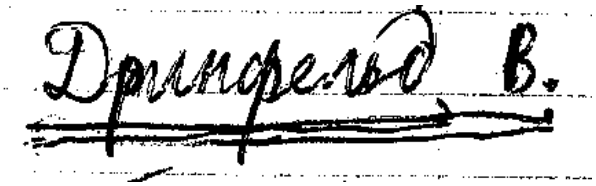
Original source details: GELFAND-P001315

Uncertain reading. The crossed-out entry reads “вектор” (vector). Above it is “ф-ю” (fn.). A curved line passes between the two entries.

If the representation is realized on the space of flags, this function is written as an integral of the form (1). The integral is a spherical fn. of GL_n relative to the unitary subgroup, etc.

I.M.G. General question. Which integrals can be evaluated in this way?

Drinfeld V. Quantization of KdV according to Belavin



Original source details: GELFAND-P001315

Diagram and typed description.
Drinfeld V.

Original source details: GELFAND-P001315

Uncertain reading. A short slanting stroke appears to the right of the heading. Its purpose is unclear.

On an idea of Belavin's.

KdV had previously been quantized incorrectly.

KdV is Hamiltonian with respect to two Gelfand–Dikii brackets.

Which one is needed? One must quantize with respect to both brackets.

Any linear combination of them is also a bracket.

One must introduce 2 Planck constants!

$$u_t = u_{xxx} + uu_x \tag{1}$$

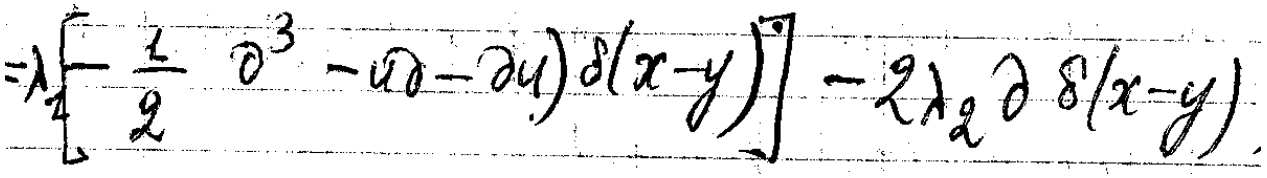
Brackets.

$$\text{II}: \ -\tfrac{1}{2}\partial^3 - u\partial - \partial u; \qquad \text{I}: \ -2\partial \tag{2}$$

— Hamiltonian operators for these brackets.

This means that

$$\{u(x), u(y)\} = \lambda_1\big[-\tfrac{1}{2}\partial^3 - u\partial - \partial u\big)\delta(x-y)\big] - 2\lambda_2\partial\delta(x-y), \tag{3}$$



Original source details: GELFAND-P001315

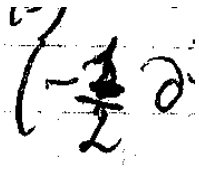
Uncertain reading. There is a square bracket after λ_1 and a closing parenthesis after $-\partial u$. The square bracket closes after $\delta(x-y)$.

where $\partial = \frac{\partial}{\partial x}$

The idea of the Hamiltonian operator goes back to Berezin (he proposed the matrix inverse to the symplectic one).

After quantizing (3) we obtain

$$[u(x), u(y)] = h_1 \left(-\tfrac{1}{2}\partial^3 - u\partial - \partial u\right) \delta(x-y) - 2h_2 \partial \delta(x-y). \tag{4}$$



Original source details: GELFAND-P001315

Uncertain reading. The numerator is heavily retraced. The fraction reads $\frac{1}{2}$, as in (2) and (3).

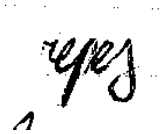
I.M.G. [Wavy underlining.] There is no nonlinear equation that has been properly quantized.

Original source details: GELFAND-P001315

Uncertain reading. The joined strokes before “следует” read “как”, forming the expression “properly”.

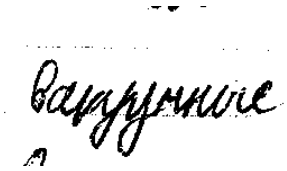
The first attempts are quantum groups.

Proposal. Do it by perturbing the measure. $P(\varphi)_2$ is done in this way,



Original source details: GELFAND-P001315

Uncertain reading. The full form “через” (through) is visible.



Original source details: GELFAND-P001315

Uncertain reading. The letters run together. The word reads “возмущение” (perturbation).

Dr. No one has compared quantization according to Faddeev and according to Albeverio–Høegh-Krohn.

Original source details: GELFAND-P001315

Uncertain reading. After “Kro” the scan appears to have “n” and a Russian case ending. The English line uses the published spelling Høegh-Krohn.

GELFAND-P001316

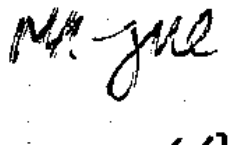
[source number: 28]

Dr. We will deal only with the 2nd bracket.

$$[u(x), u(y)] = h\left(-\frac{1}{2}\partial^3 - u\partial - \partial u\right)\delta(x-y) \tag{5}$$

This gives the Virasoro algebra (h is the central extension parameter).

The Poisson manifold is an orbit of the coadjoint rep. of the Virasoro alg. Quantization is according to Kirillov.



Uncertain reading. The Russian abbreviation after “Пуасс.” (Poisson) is provisionally read “мн-ие” (manifold). A long descender follows the dash.

Original source details: GELFAND-P001316

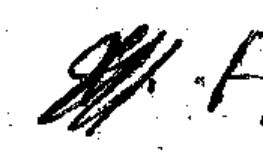
$\exists$ a countable number of integral functionals in involution.

$$F_n(u) = \int \varphi_n(u, u', \ldots, u^{(n)})\, dx \tag{6}$$

A quantum analogue?

u and u' do not commute.

Zamolodchikov. There is a quantum analogue [crossed out] of F_n.



Undeciphered passage. In the crossed-out entry before F_n, a loop, a long stroke and a separate mark at right are visible. Possible readings include φ and φ_n.

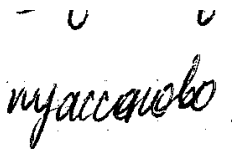
Original source details: GELFAND-P001316

We will leave the Hamiltonian aside.

Miura transform.

$$u = v' - v^2 \tag{7}$$

It is Poisson if on $\{u\}$ we take the 2nd G.–D. bracket,

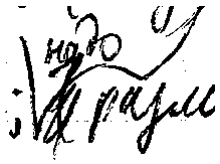


Uncertain reading. The Russian spelling is provisionally read “пуассаново” (Poisson). The vowel after “сс” merges with the next stroke.

Original source details: GELFAND-P001316

$$\text{on } \{v\} \quad — \quad \{v(x), v(y)\} = \delta'(x-y)$$
$$[v(x), v(y)] = h\delta'(x-y) \qquad (x \in S^1) \tag{8}$$

— the Heisenberg alg. [Crossed out] [above the line: one must] expand in a Fourier series [crossed out], and we obtain



Original source details: GELFAND-P001316

Uncertain reading. Before “разложить” (expand), a mark resembling “И” (and) is crossed out. Above it are “надо” (one must) and a curved callout stroke.

Original source details: GELFAND-P001316

Undeciphered passage. After “Fourier” a sign is crossed out. The visible loop permits v or ξ.

$$[v_m, v_n] = hm\delta_{m,-n}I \tag{9}$$

[To the right of (9)] (1-dimensional extension of a commutative algebra).

2 infinite-dimensional Lie algs. (enveloping algs.).

a) Heisenberg alg. b) Virasoro alg. [Both entries have wavy underlining.]

Claim. There is a simple homomorphism from the enveloping alg. in b) to the enveloping alg. in a).

Virasoro generators $\to$ quadratic expressions in Heisenberg generators.

Virasoro alg.

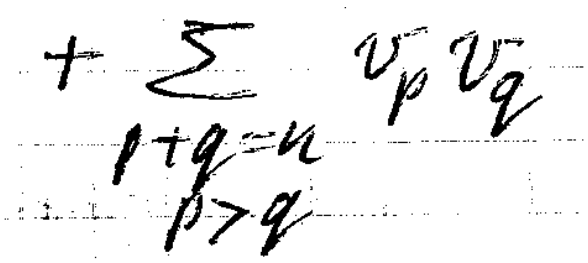
$$L_n = e^{in\varphi}\frac{d}{d\varphi} \qquad (\text{on } S^1) \tag{10}$$

and a 1-dimensional extension by means of the Gelfand–Fuchs cycle.

Fuchs. The corresponding transformation from (9) to (10).

$$v_i,\ i \neq 0, \qquad L_n = (\alpha n + \beta)v_n + \sum_{\substack{p+q=n \\ p>q}} v_p v_q \tag{11}$$

Heisenberg alg. $\quad n \neq 0$

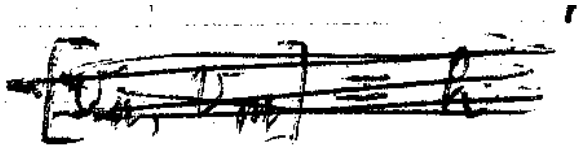


Original source details: GELFAND-P001316

Uncertain reading. The lower summation condition reads $p > q$. There is no second stroke below $>$.

This is the Miura transform.

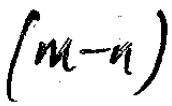
[An equation is crossed out.]



Undeciphered passage. Square brackets, two v signs, an equals sign and h are visible. The indices provisionally read $[v_n, v_m] = h$. The reverse order is also possible.

Original source details: GELFAND-P001316

$$[L_n, L_m] = (m-n)L_{n+m} + \delta_{n,-m}\frac{n^3-n}{12}c, \qquad c \text{ commutes with all } L_n$$



Uncertain reading. The factor before L_{n+m} reads $(m-n)$.

Original source details: GELFAND-P001316

Quantization = embedding in the enveloping alg. of the Heisenberg algebra [the ending of the Russian word “algebra” is below the line].

Uncertain reading. The ending “ры” is written below the line before “Гейзенберга” (Heisenberg).

Original source details: GELFAND-P001316

Problem (interesting). Write down all the Hamiltonians.

GELFAND-P001317

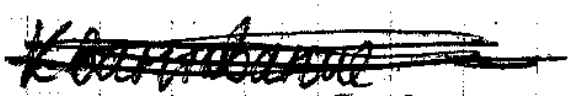
KdV is connected with the Sturm–Liouville operator.

Uncertain reading. The first surname is abbreviated “Шт.” (St.).

Original source details: GELFAND-P001317

$$L = \partial^2 + u \tag{12}$$

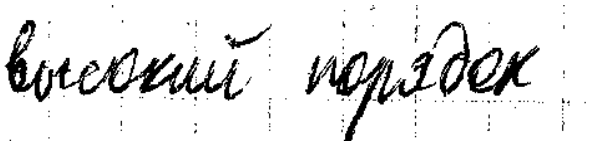
[Crossed out: Quantization.]



Uncertain reading. The lower curves and projecting strokes of the cancellation allow the reading “Квантование” (Quantization).

Original source details: GELFAND-P001317

how to generalize to high order



Uncertain reading. The word before “порядок” (order) reads “высокий” (high).

Original source details: GELFAND-P001317

$$L = \partial^n + u_{n-2}\partial^{n-2} + \ldots + u_0 \tag{13}$$

and there is an analogue of the Miura transf. ([above the line: Wilson,] Drinfeld, Sokolov).

Uncertain reading. “Wilson,” is inserted above “Drinfeld” with a curved callout stroke.

Original source details: GELFAND-P001317

Operator (13) is connected with $sl(n)$.

Instead of

$$u = v' - v^2$$

Undeciphered passage. In the left margin is a looped entry with overlapping diagonal strokes. Its reading is unresolved.

Original source details: GELFAND-P001317

it is more correct to write

$$L = \partial^2 + u = (\partial - v)(\partial + v) \tag{14}$$

Factor.

$$L = \partial^n + u_{n-2}\partial^{n-2} + \ldots + u_0 = (\partial + v_1)\ldots(\partial + v_n), \quad v_1 + \ldots + v_n = 0 \tag{15}$$

This gives the Miura transf.

This is connected with Picard–Vessiot.

$$\{u(x), u(y)\} = \left(-\tfrac{1}{2}\partial^3 - u\partial - \partial u\right)\delta(x - y) \tag{16}$$

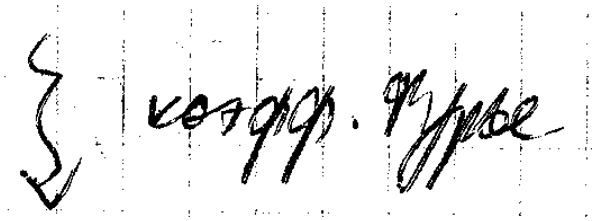
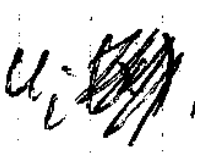


Diagram and typed description.

Fourier coeffs.

Original source details: GELFAND-P001317

$$\{u_i, u_j\} = \text{quadr. rels.} \tag{17}$$



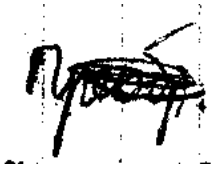
Undeciphered passage. After u_i a group of signs is crossed out. A curve is visible on the left of the group. Its interior signs overlap.

Original source details: GELFAND-P001317

Original source details: GELFAND-P001317

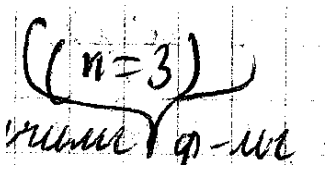
Undeciphered passage. After u_j, before the right brace, a second group is crossed out. A curved stroke is visible at its left edge.

[A crossed-out entry above “Fateev”.] Fateev and Zamolodchikov obtained [inserted above: $(n = 3)$] formulas that solve the problem.



Original source details: GELFAND-P001317

Undeciphered passage. An opening sign resembling “П” and long descenders are visible. The word is unresolved.

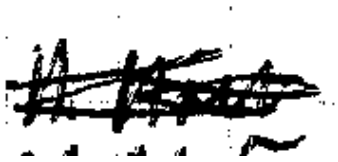


Original source details: GELFAND-P001317

Uncertain reading. The entry $(n = 3)$ is above the start of the abbreviation for “formulas”, with a curved callout stroke.

B. Feigin [wavy underlining] did something (1986, April [below the date: conf. at VDNKh]). Generalization to $sl(3)$.

Sklyanin [wavy underlining]. Quantization of KdV on a finite interval, with boundary conditions.

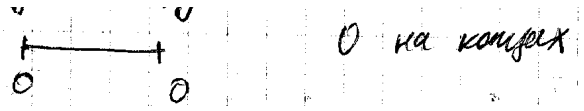


Original source details: GELFAND-P001317

Undeciphered passage. A short entry above “I.M.G.” is crossed out. The first sign resembles “И” (I). The other strokes overlap.

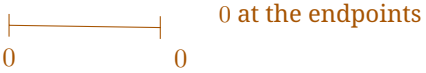
I.M.G. [Wavy underlining.] The sweep method (Gelf., Lokutsievskii) is analogous to the Miura transf.

$$y'' - a(x)y = 0$$



Original source details: GELFAND-P001317

Diagram and typed description.

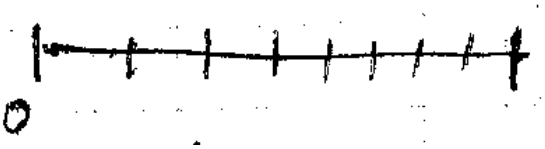


One cannot take an exp growing solution (accuracy is lost).

($\Rightarrow$ one cannot take 2 lin. indep. solutions)

GELFAND-P001318

[source number: 29]



Original source details: GELFAND-P001318

Diagram and typed description.

Move the boundary condition to the neighboring point.

$$y'(0) + \alpha y(0) = 0.$$

At the neighboring point α changes.

$$\alpha' - \alpha^2 = 0 \quad (!) \quad \text{— Miura.}$$

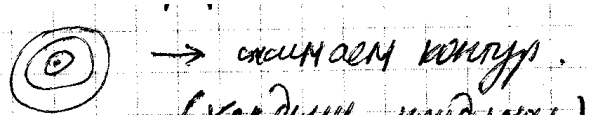
Original source details: GELFAND-P001318

Uncertain reading. The manuscript gives $\alpha' - \alpha^2 = 0$ followed by "(!)".

There is a sweep method for n-th order too!!

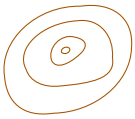
It fixes the correct factorization.

$$\Delta u = 0$$



Original source details: GELFAND-P001318

Diagram and typed description.



⟶ shrink the contour.

Inside the three contours is a small closed or nearly closed stroke.

(Keldysh devised this.)

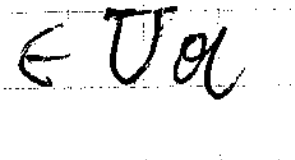
B. Feigin. [Wavy underlining.]

Finite-dimensional Lie alg. a and an inner product. [Wavy underlining.]

Original source details: GELFAND-P001318

Uncertain reading. The looped letter before "and" reads a.

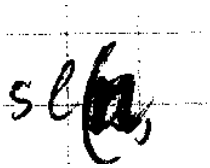
$\sum e_i e^i$ — Casimir elem. $\in Ua$ (from the invariant inner prod.)



Original source details: GELFAND-P001318

Uncertain reading. A looped a follows U, matching the heading.

$$sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$$



Original source details: GELFAND-P001318

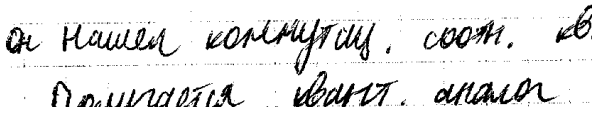
Uncertain reading. The first argument is retraced. The following line with $sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$ supports the reading n.

There are many inner products (invariant).

$$u, v \in sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$$

$$\text{coeff.}_k \operatorname{Tr}(uv)$$

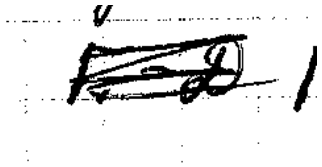
He found the commutation rels. of the quadr. and cubic Casimirs.



Original source details: GELFAND-P001318

Uncertain reading. The line begins “он нашёл коммутац. соотн.” (he found the commutation rels.). The first sign and the middle of the Russian abbreviation for commutation are joined.

This gives a quantum analogue of the formulas of [crossed out: G.–D.] Gelfand–Dikii.



Original source details: GELFAND-P001318

Uncertain reading. The crossed-out abbreviation has an angular first sign and a loop at the end. It is provisionally read “Г.-Д.” (G.–D.).

What problems are there?

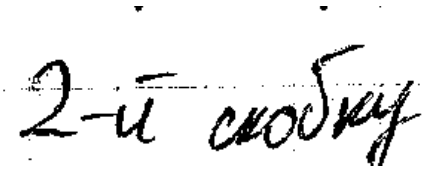
1° a) Quantize the KdV integrals.

[Crossed out: 2°] b) Compute the spectrum of irreducible reps. of the Virasoro alg.

Original source details: GELFAND-P001318

Uncertain reading. The mark before “b)” is crossed out and provisionally reads 2°.

2°. Quantize the 2nd G.–D. bracket in the general case.



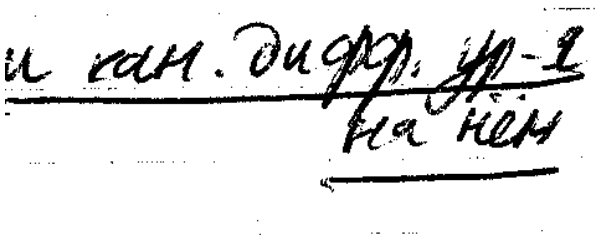
Original source details: GELFAND-P001318

Uncertain reading. Before “скобку” (bracket) the manuscript reads “2-й” (2nd), with a short stroke over its final letter.

GELFAND-P001319

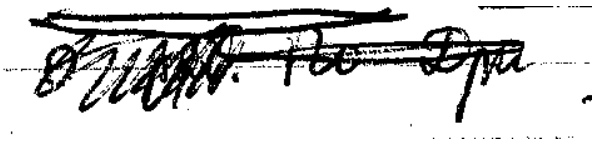
10.11.86 Gelfand Sem.

Beilinson. Action of the Virasoro alg. on the moduli space of algebraic curves and canonical differential eqs. on it.



Original source details: GELFAND-P001319

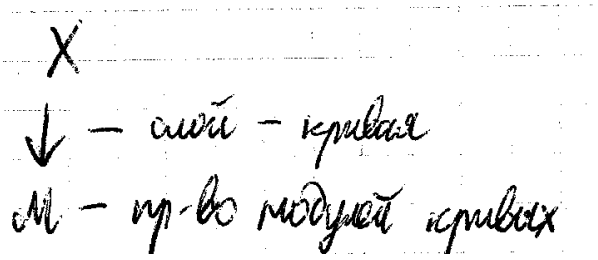
Uncertain reading. The abbreviation before “дифф.” (differential) reads “кан.” (canonical). The strokes under “and canonical differential eqs.” and “on it” have left-pointing tips.



Original source details: GELFAND-P001319

Undeciphered passage. An opening loop and a final group resembling “Дра” are visible through the cancellation. The full line remains unresolved.

On Belavin’s work according to Drinfeld.



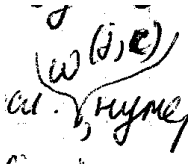
Original source details: GELFAND-P001319

Diagram and typed description.

X
$\downarrow$ — the fiber is a curve
M — moduli space of curves

$$\text{genus } g \quad \Rightarrow \quad \text{dimension } 6g-6$$

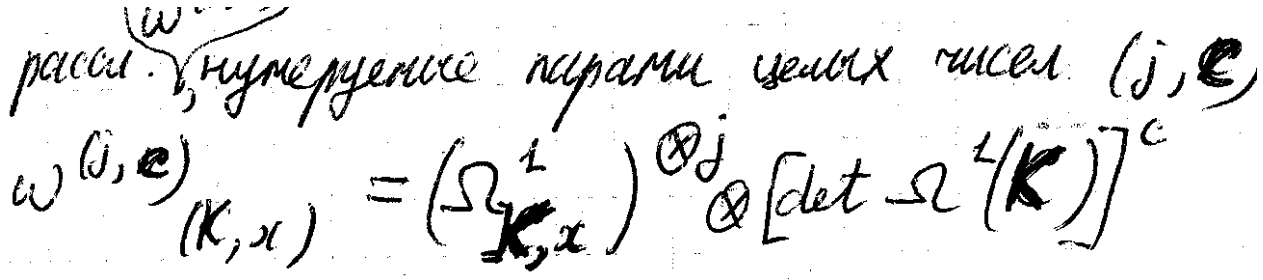
On X are line bundles [inserted above: $\omega^{(j,c)}$] indexed by pairs of integers (j,c).



Original source details: GELFAND-P001319

Uncertain reading. The notation $\omega^{(j,c)}$ is inserted above the line with a curved callout stroke. The c is thickened.

$$X \ni (K,x), \qquad \text{fiber } \omega^{(j,c)}_{(K,x)} = (\Omega^1_{K,x})^{\otimes j} \otimes [\det \Omega^1(K)]^c$$



Original source details: GELFAND-P001319

Uncertain reading. The c in (j,c) is retraced. In the equation its reading follows the exponent on $[\det \Omega^1(K)]^c$. The K signs on the right are also thickened.

[Below K: curve corresponding to a point of M, $x \in K$.]

Original source details: GELFAND-P001319

Uncertain reading. The first letter of (K,x) is heavily retraced. The remaining strokes and the note below support K.

[Below the right-hand side:] j-differentials of the curve K at x.

Differential eqs. are written for sections [wavy underlining] of this bundle.

Extend X.

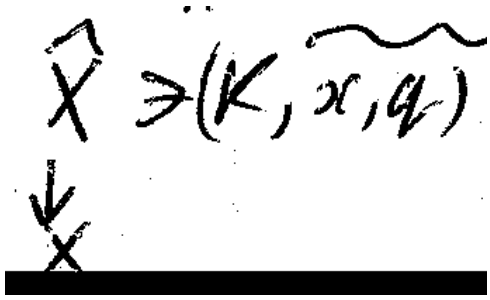


Diagram and typed description.

$\widehat{X} \quad \ni (K, x, q)$

$\downarrow$

X

Original source details: GELFAND-P001319

q is a formal parameter at x.

(choice of a formal coordinate system at x)

GELFAND-P001320

The Virasoro algebra acts on $\widehat{X}$.

(Kontsevich [crossed out] and Beilinson–Schechtman independently.)

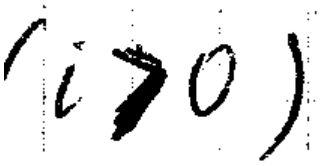
Undeciphered passage. A short entry between "Kontsevich" and "and" is crossed out. Oblique strokes and a projecting vertical line are visible.

Original source details: GELFAND-P001320

For now, the alg. of formal vector fields (without central charge).

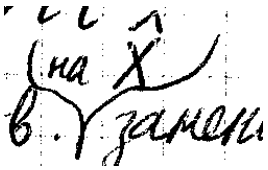
$$Vir = \left\{ \sum a_i t^i \frac{d}{dt} \right\} \supset Vir^+$$

Half of this alg. ($i > 0$) acts [inserted above: on $\widehat{X}$] by changing the parameter q.



Uncertain reading. The sign between i and 0 is heavily retraced. Its upper and lower strokes meet on the right, giving $>$.

Original source details: GELFAND-P001320

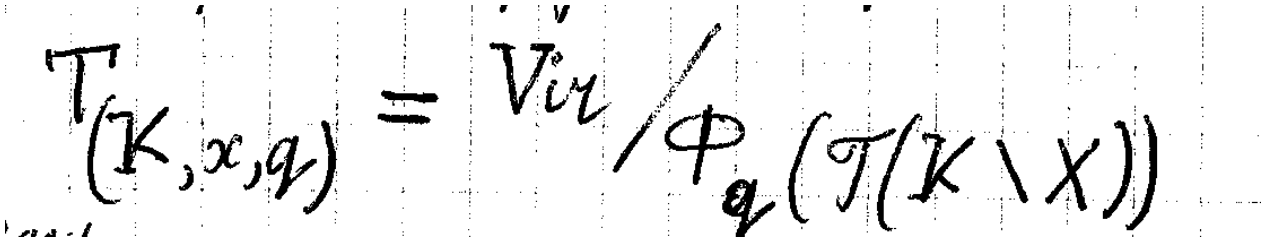


Uncertain reading. "on $\widehat{X}$" is inserted above the line with a curved callout stroke.

Original source details: GELFAND-P001320

(Taylor series in q.) This action extends to an action of the full alg. Vir on $\widehat{X}$.
The conformal class of the curve will then change.
According to Kodaira–Spencer [wavy underlining], the tangent space to $\widehat{X}$ has the form

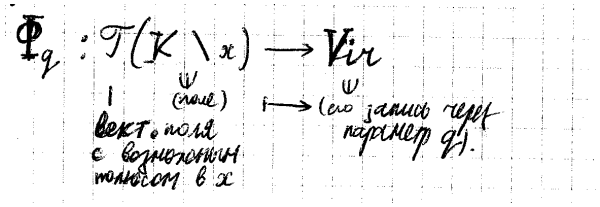
$$T_{(K,x,q)} = Vir/\Phi_q(\mathcal{T}(K\backslash X)) \tag{1}$$



Original source details: GELFAND-P001320

Uncertain reading. In (1), the letter after the backslash is capital X. The diagram below and (2) have lowercase x.

Here



Original source details: GELFAND-P001320

Diagram and typed description.

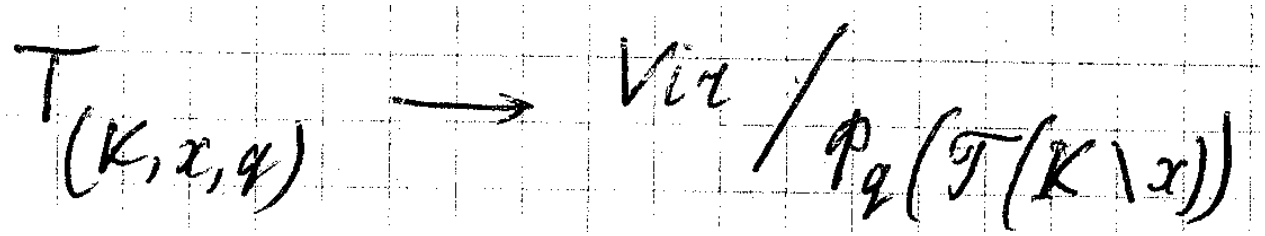
$\Phi_q : \mathcal{T}(K\backslash x) \longrightarrow Vir$

$\downarrow$ (field) $\longmapsto$ $\downarrow$ (its expression in the parameter q)

vect. fields with a possible pole at x

Explanation. [Wavy underlining.] Construct a map

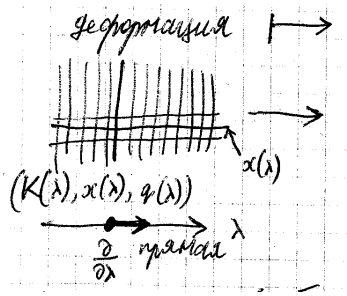
$$T_{(K,x,q)} \longrightarrow Vir/\varphi_q(\mathcal{T}(K\backslash x)) \tag{2}$$



Original source details: GELFAND-P001320

Uncertain reading. In (2) the map is written with lowercase φ_q. In (1) and in the diagram it has uppercase Φ_q.

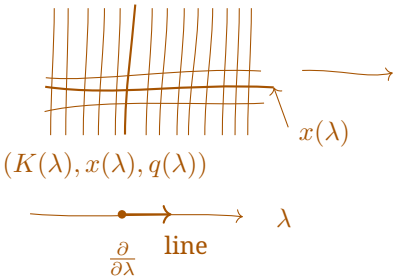
(which will be the isomorphism).



Original source details: GELFAND-P001320

Diagram and typed description.

deformation $\longmapsto$



$x(\lambda)$

$(K(\lambda), x(\lambda), q(\lambda))$

λ

$\frac{\partial}{\partial\lambda}$ line

The formal neighborhood (tubular) of the section is trivialized —
— by the coordinates λ and q.

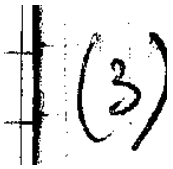
a) Lift $\dfrac{\partial}{\partial\lambda}$ horizontally to this tubular neighborhood.

b) $\dfrac{\partial}{\partial\lambda}$ can also be lifted to the complement of $x(\lambda)$ (another lift).

Original source details: GELFAND-P001320

Uncertain reading. Near the right edge of the main leaf, level with (2), there is a faint winding stroke. Its meaning is uncertain.

[On the strip of the facing leaf beyond the gutter: (3).]



Original source details: GELFAND-P001320

Uncertain reading. The number (3) on the facing leaf is visible in this frame.

GELFAND-P001321

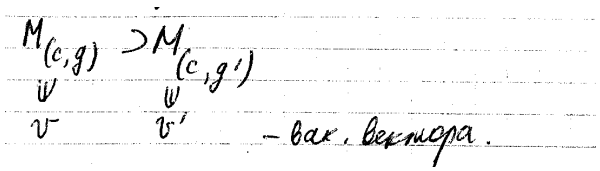
Consider a Verma module with highest weight (c, g).

$$M_{(c,g)}$$

Original source details: GELFAND-P001321

Uncertain reading. The second sign in the pair reads g, as in the module notation below.

Suppose it is reducible.



Original source details: GELFAND-P001321

Diagram and typed description.

$M_{(c,g)} \supset M_{(c,g')}$
$\downarrow \qquad \downarrow$
$v \qquad v'$ — vacuum vectors.

$$v' = L(v), \qquad L \in U(\mathit{Vir}).$$

The eqs. are canonical overdetermined differential eqs. on M.

Original source details: GELFAND-P001321

Uncertain reading. The first sign in the Russian abbreviation “Ур-я” (eqs.) is heavily retraced.

(Associated with singular vectors in Verma modules over *Vir*.)

$$\psi \in \Gamma(\omega_{X/M}^{\otimes j} \otimes \lambda^c)$$

Lift ψ to $\widehat{X} \Rightarrow$ obtain a singular vector $\Rightarrow$ apply L to it and obtain a new singular vector, i.e., a new section of $\omega_{X/M}^{\otimes j'} \otimes \lambda^c$.

Solutions [wavy underlining] form a local system of finite rank.

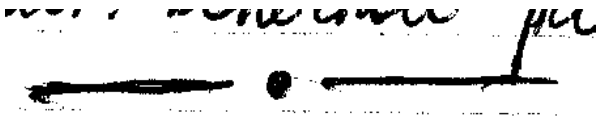


Diagram and typed description.

A dot separates the two lines. At the left end of the first line are two short slanted strokes.

Original source details: GELFAND-P001321

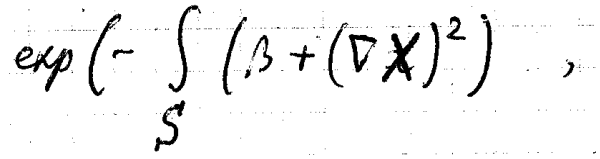
GELFAND-P001322

17.11.86 Gelfand Sem.

Kazakov

Zakharevich

$$Z = \int Dg \int DX \ \exp\Big(-\int_S (\beta + (\nabla X)^2\Big), \tag{1}$$

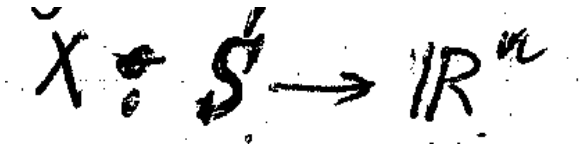


Uncertain reading. One closing parenthesis is visible after $(\nabla X)^2$, before the comma. The parenthesis opened after exp remains unclosed.

Original source details: GELFAND-P001322

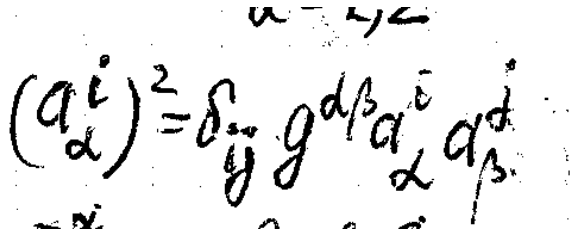
[To the right of the vertical line.] g is the metric on S.

$$X : S \longrightarrow \mathbb{R}^n$$
$$(\nabla X)^i_\alpha = \frac{\partial X^i}{\partial \xi^\alpha}$$
$$i = 1, \dots, n, \qquad \alpha = 1, 2$$
$$(a^i_\alpha)^2 = \delta_{ij} g^{\alpha\beta} a^i_\alpha a^j_\beta$$



Uncertain reading. Two dots appear between X and S. The upper dot is retraced and is read as part of the colon.

Original source details: GELFAND-P001322



Uncertain reading. The lower signs on δ read i, j. On the left the square is outside the parentheses enclosing a^i_α.

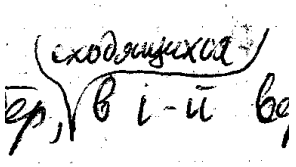
Original source details: GELFAND-P001322

ξ^α are coord. on S.

[Left column continued.]

$$Z = \sum_{\substack{T \\ \text{triang. of the sphere}}} \int \prod dx_i \exp\left(-\sum_i \beta\sigma_i - \sum_{\substack{i,j \\ \text{neighbors}}} (x_i - x_j)^2 \right) \tag{2}$$

$$\sigma_i = \frac{n_i}{3}, \qquad n_i = \#\text{ edges (meeting) at the } i\text{-th vertex} \tag{2'}$$

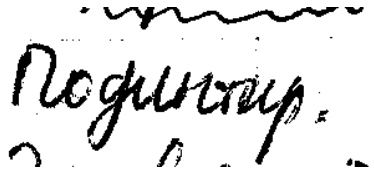


Original source details: GELFAND-P001322

Uncertain reading. “(meeting)” is inserted above the line after “edges”, with a curved callout stroke.

(this is called the defect [wavy underlining]).

(3) Claim. The integrand expressions agree for regular embeddings (when all $\triangle$s are regular).



Original source details: GELFAND-P001322

Uncertain reading. The opening abbreviation in (3) reads “Подынтгр.” (integrand). Some middle strokes are joined.

GELFAND-P001323

(1) is a bosonic string.

K. Wilson and Polyakov wrote the Yang–Mills Lagrangian in this way, but there the space is flat and one can take a cubic lattice.

I.M.G.’s question [wavy underlining]. How should one write the Y.–M. model? Characteristic classes arise there and one cannot restrict to a cubic lattice.

Z. Consider an arb. triangulated surface of the sphere. Then on average (for a random embedding X) all triangles of the embedding will be regular (the entire [crossed out] mass of a watermelon is in the rind!!).

Original source details: GELFAND-P001323

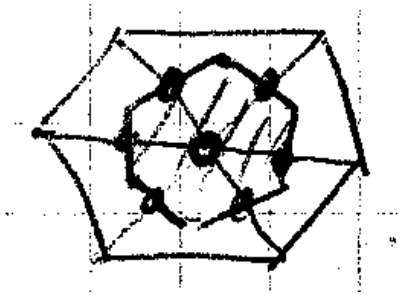
Undeciphered passage. An entry between “entire” and “mass” is crossed out. Projecting strokes are visible, including a long descender.

Thus it suffices to consider any metric g (fixed) and a linear map on each Δ.

K. [above at right: 4?] Why $\frac{n_i}{3}$?

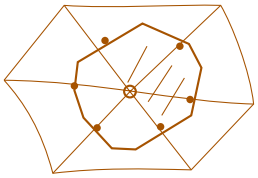
Original source details: GELFAND-P001323

Uncertain reading. The small sign to the right of “K.” resembles 4 or “a”. Its meaning is uncertain.



Original source details: GELFAND-P001323

Diagram and typed description.



Inside the hexagonal outline, six marks on the segments and a central mark are visible. Some slanted strokes inside are hard to distinguish from the notebook ruling.

Cut off [crossed out] $\frac{1}{2}$ of the edge and obtain area $= \frac{1}{3}$ (?).

Original source details: GELFAND-P001323

Undeciphered passage. A short sign is crossed out before $1/2$. A loop is visible at its top.

(all Δs are regular).

Z. In place of (2), now consider

$$Z_2 = \sum_m e^{-\beta m} \sum_{|T|=m} \left(\int \prod d\tilde{x}_i \cdot \exp - \sum_{i,j} (\tilde{x}_i - \tilde{x}_j)^2 \right)^n \tag{4}$$

K. It can only be evaluated at $n = 0$ and [crossed out] $n = -2$ (!!).

Original source details: GELFAND-P001323

Undeciphered passage. An entry between “and” and $n = -2$ is crossed out by diagonal strokes.

Z. Here the $\tilde{x}_i$ are already one-dimensional!

This follows because different components are integrated independently.

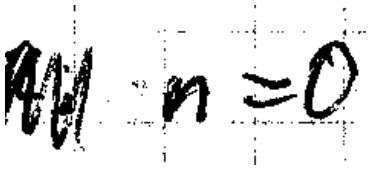
Original source details: GELFAND-P001323

Uncertain reading. A faint open loop appears in the right margin. Its purpose is uncertain.

The parameter n enters analytically!

At $n = 0$ and $n = -2$ the sum can be evaluated.

At $n = 0 \quad \Rightarrow$



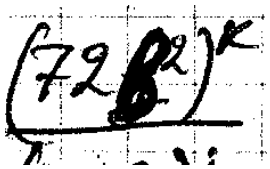
Original source details: GELFAND-P001323

Uncertain reading. The word before $n = 0$ is heavily retraced and provisionally reads “При” (at).

$$Z_2 = \sum_m \sum_{|T|=m} e^{-\beta m} = \sum_m e^{-\beta m} \#\{T : |T| = m\} \tag{5}$$

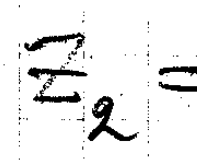
Answer.

$$Z_2 = -\frac{1}{2} \sum_{k=1}^{\infty} \frac{(72b^2)^k}{(k+2)!} \frac{\Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k+2}{2}\right)}, \quad b = b(\beta) \tag{6}$$



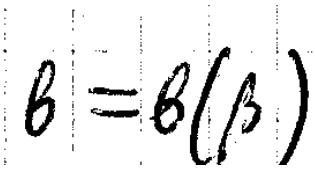
Original source details: GELFAND-P001323

Uncertain reading. The sign before the superscript 2 is retraced. The loops and the entry $b = b(\beta)$ at right support the reading b^2.



Original source details: GELFAND-P001323

Uncertain reading. The Z on the left of (6) has an additional horizontal stroke through its middle.



Original source details: GELFAND-P001323

Uncertain reading. To the right of the series is $b = b(\beta)$. The b and β have distinct forms.

$b(\beta)$ has not been evaluated, but physical quantities do not depend on this.

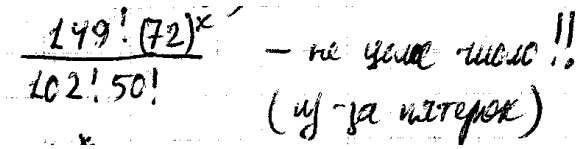
GELFAND-P001324

[source number: 37]

K.

$$\#\{T, |T| = 2k\} = \frac{(72)^k}{(k+2)!} \frac{\Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k+2}{2}\right)} \tag{7}$$

$k = 100 \quad \Rightarrow \quad \frac{149!\,(72)^k}{102!\,50!}$ is not an integer!! (because of the fives)



Original source details: GELFAND-P001324

Uncertain reading. The numerator reads 149!, and the denominator reads 102! 50!.

K. Radius of conv. of the series is b^* [below the line: (in (6))].

Original source details: GELFAND-P001324

Uncertain reading. Above the line about the radius is a thin winding stroke with a loop. Its purpose is uncertain.

$$Z_2 \underset{b\to b^*}{\cong} a + c(b-b^*)^{5/2} \tag{8}$$

Then

$$Z(\beta) = A + B(\beta-\beta^*)^{5/2} \tag{9}$$

The critical exponent matters.

This is a typical situation in field theory. For example, the mass spectrum

$$m_n = m_0(g) f(n)$$

Z. Combinatorial problem [wavy underlining]. Find

$$\frac{1}{n!}\#\{\tilde{T}, |\tilde{T}| = n\} = \text{[crossed out] } W(n)$$

Undeciphered passage. A group of signs between the equals sign and $W(n)$ is crossed out. Its interior strokes overlap.

Original source details: GELFAND-P001324

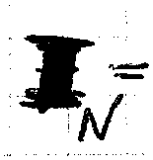
$\tilde{T}$ is a labeled triangulation of the sphere (with numbered Δs).

$\Rightarrow$ the answer need not be an integer.

K. Asymptotically $W(m) \sim m^\gamma e^{\beta^* m}, \quad \gamma = -7/2$. How does one compute the asymptotics?

Consider the integral

$$I_N = \int d^{N^2}u\, e^{-N\,\mathrm{tr}(u^2+gu^3)}, \quad u \text{ is Hermitian, } u^+ = u \tag{10}$$

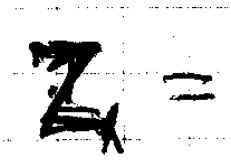


Uncertain reading. The main letter in (10) is retraced. Its shape and (11) support the reading I_N.

Original source details: GELFAND-P001324

Expanding in g and evaluating the Gaussian integrals, we obtain

$$Z_{[?]} = \sum_{n=0}^{\infty} \Pi_n g^n$$



Uncertain reading. Crossing strokes are visible at the lower right of Z. The subscript remains unresolved.

Original source details: GELFAND-P001324

$$Z = \lim_{N\to\infty} \log I_N \tag{11}$$

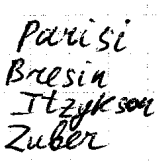
— the generating fn. for the number of triangulations of the sphere.

Idea [wavy underlining]. Interpret the integrals obtained by series expansion graphically. This interpretation gives triangulations of the sphere.

One can replace $u^2 + gu^3$ by $u^2 + gu^3 + \lambda u^4$.

This was evaluated by Feynman, t' Hooft, Parisi

Bresin
Itzykson
Zuber



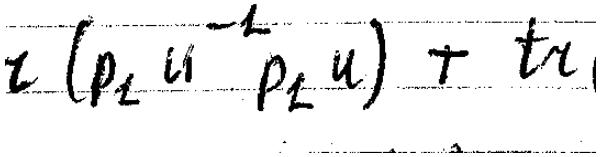
Original source details: GELFAND-P001324

Uncertain reading. The first surname in the lower-right list is written "Bresin".

GELFAND-P001325

Other integrals

$$\int du\, e^{\mathrm{tr}(p_1 u^{-1} p_1 u) + \mathrm{tr}(p_2 u^{-1} p_2 u)}, \qquad u^+ = u^{-1} \tag{12}$$



Original source details: GELFAND-P001325

Uncertain reading. The letter in (12) and the diagonal matrices reads as p, though its shape is close to ρ.

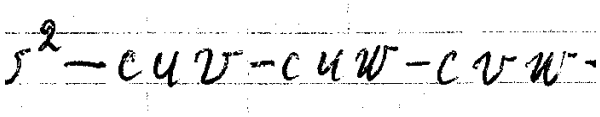
equiv. to the principal chiral field (2-dimensional) field.

$$S = \int d^2x\ \mathrm{tr}(\partial_\mu u^{-1}\ \partial_\mu u)$$

$$u^+ u = I, \qquad N \longrightarrow \infty$$

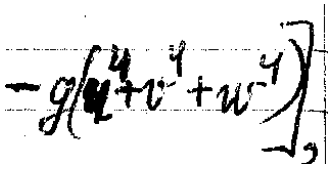
In principle solved by Polyakov and Wiegmann ⇒ such an integral ought to be calculable

$$\int d^{N^2}u\, d^{N^2}v\, d^{N^2}w\ \exp\big[-\mathrm{tr}(u^2 + v^2 + w^2 - cuv - cuw - cvw - g(u^4 + v^4 + w^4)\big], \tag{13}$$



Original source details: GELFAND-P001325

Uncertain reading. The final mixed term reads as $-cvw$. The final stroke of w matches the preceding term.



Original source details: GELFAND-P001325

Uncertain reading. The first letter after $g($ is heavily overwritten. It reads as u^4.

Original source details: GELFAND-P001325

Uncertain reading. After w^4 there is one closing parenthesis before the square bracket.

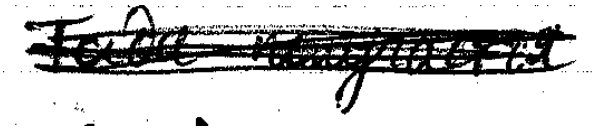
u, v, w are Hermitian

(problem of summing over triangulations of the sphere, but with a Z_3 spin at each vertex instead of Z_2)

In (12)

$$p_1 = \mathrm{diag}\big((p_1)_1, \ldots\big)$$
$$p_2 = \mathrm{diag}\big((p_2)_1, \ldots\big).$$

[Crossed out: Then the integral.]

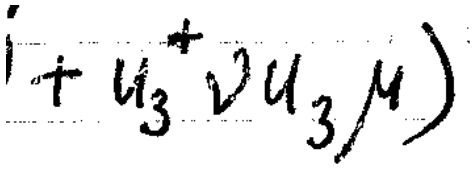


Original source details: GELFAND-P001325

Uncertain reading. The cancellation partly covers the final strokes of “integral”.

(13) reduces to an integral over the group (pass to eigenvalues)

$$\int d(u_1)d(u_2)d(u_3)\, e^{\mathrm{tr}(u_1^+ \lambda u_1 \mu + u_2^+ \mu u_2 \nu + u_3^+ \nu u_3 \mu)} \tag{14}$$



Original source details: GELFAND-P001325

Uncertain reading. The final factor in (14) is written as μ.

For Yang–Mills

$$Z = \int \prod_{\mu=1}^{d} du_\mu \, \exp\Big[\beta N \sum_{\mu>\nu} \mathrm{tr}(u_\mu u_\nu u_\mu^+ u_\nu^+)\Big] \tag{15}$$

$d = 2$ solved

Need $d = 3, 4$

GELFAND-P001326

[source number: 38]

[Strip of the facing leaf at the left.]

Uncertain reading. Only the tips of the brackets and the comma are visible,)], or],.

Original source details: GELFAND-P001326

I. M. Gelfand,

(+ V. B. Lidskii and Ponomarev)

In work that took 10 years.

There are 2 methods of working, lattices and categories.

Lattices are more convenient (but have been discredited by specialists in [crossed out: ...] [above the line: general] algebra)

Undeciphered passage. The cancellation covers the word under the insertion “general”.

Original source details: GELFAND-P001326

Nontrivial work on lattices [wavy underline]

1) Dedekind [wavy underline]

2) Weyl [wavy underline] on the axiomatics of projective geometry

dim. = max. length of an ordered chain.

$n \geq 3$ $\Rightarrow$ Desargues's axiom holds, etc.

3) von Neumann [wavy underline] removed the finiteness requirement and connected this with factors

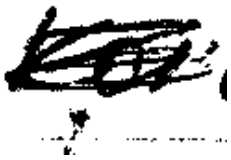
(recall in connection with the work of [crossed out: ...] Dison [?], which [crossed out: ...] Connes promotes)

Undeciphered passage. Several slanted strokes cover the short entry before the surname.

Original source details: GELFAND-P001326

Uncertain reading. The surname after the cancellation reads tentatively as “Dison”.

Original source details: GELFAND-P001326



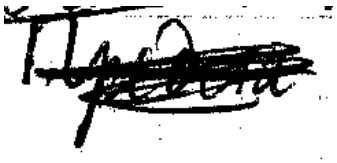
Undeciphered passage. The short entry before “Connes” is crossed out.

Original source details: GELFAND-P001326

4) Gelfand and Ponomarev [wavy underline]

Problem. There are k linear subspaces. How are they to be classified?

[Crossed out: Pred...]



Undeciphered passage. The initial fragment “Pred...” is visible. The cancellation covers the rest.

Original source details: GELFAND-P001326

$k = 2$ trivial

$k = 3$ a (good) problem in linear algebra

Decomp. into a direct sum of subspaces on a line and three

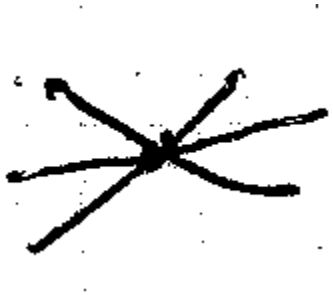
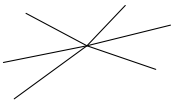


Diagram and typed description. 

Original source details: GELFAND-P001326

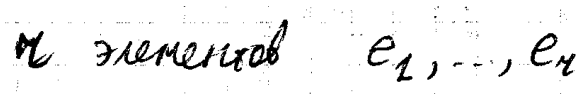
The corresponding lattice contains 28 elements

$k = 4$ took a long time (Gelf. and Ponomarev). See Manin’s book for the formulation [below the line: on logic]

Consider r elements $e_1, \ldots, e_r$ and the free modular lattice generated by [crossed out: ...] these letters.

Undeciphered passage. The cancellation covers the short entry before “these letters”.

Original source details: GELFAND-P001326



Uncertain reading. The number of generators and the last subscript read as r.

Original source details: GELFAND-P001326

(see Birkhoff, Lattice Theory)

operations, sum $+$ and intersection $\cdot$

equal if they can be transformed using the axioms

$r \geq 5$, the word identity problem is undecidable

Rep. $x \rightsquigarrow \underset{\text{subspace}}{\varrho(x)} \subset V$

$$\varrho(x_1 x_2) = \varrho(x_1) \cap \varrho(x_2), \qquad \varrho(x_1 + x_2) = \varrho(x_1) + \varrho(x_2)$$

Ryzhikova [wavy underline] (Semenov's diploma student). Can linear equivalence be formulated with finitely many axioms?

Uncertain reading. The surname reads as "Ryzhikova". Compare Yu. S. Ryzhikova, "Decidability of the identity problem in lattices of linear subspaces" (1987).

Original source details: GELFAND-P001326

Let there be 2 reps. ϱ_j in V_j, $j = 1, 2$.

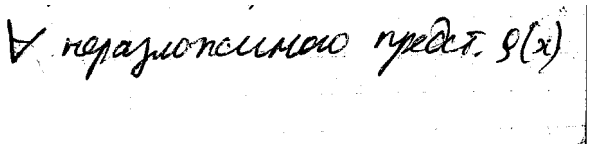
A morphism φ is a linear map $\varphi : V_1 \to V_2$ if $\varphi(\varrho_1(x)) \subset \varrho_2(x)$.

Indecomposable rep. if V_ϱ does not decompose into a direct sum ...

Uncertain reading. The heavily overwritten subscript on V reads tentatively as ϱ.

Original source details: GELFAND-P001326

A rep. ϱ is called preprojective [wavy underline] if $\forall$ indecomposable rep. $\varrho(x)$



Undeciphered passage. The writing at the foot of the leaf breaks off after $\varrho(x)$.

Original source details: GELFAND-P001326

GELFAND-P001327

$\exists$ only finitely many morphisms $\tau : \varrho \longrightarrow \varrho_0$

Uncertain reading. The word in the first line reads as "morphisms".

Original source details: GELFAND-P001327

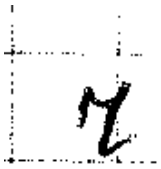
According to Arnold, these are reps. without parameters.

Lattice elts. are called preprojectively equiv. [wavy underline] if no preprojective rep. distinguishes them.

Identify them and we get the reduction.

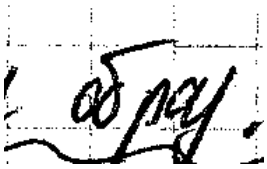
Problem (will be done) [below the line: actually already done]. Find the red. of the free lattice.

Thm. 1. All preprojective reps. of the [above the line: free] lattice on r gens. are described. [Wavy underline under the statement.]



Original source details: GELFAND-P001327

Uncertain reading. The letter reads as r, matching the notation for the number of generators on the preceding leaf.



Original source details: GELFAND-P001327

Uncertain reading. The abbreviation after r reads “образ.” (generators).

Thm. 2. The preprojective reps. of the lattice and of its reduction coincide. [Wavy underline under the statement.]

Def. A lattice elt. a is called perfect [wavy underline] if for every rep. [above the line: (in the space V)] either

$$\varrho(a) = V \quad \text{or} \quad \varrho(a) = 0$$

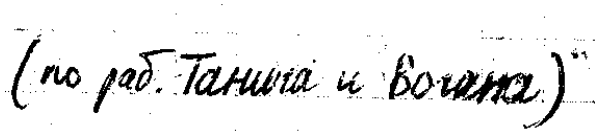
(this is an analogue of the centre of the group ring)

Perfect elements can be described.

GELFAND-P001328

24/XI-86 Gelfand Sem.

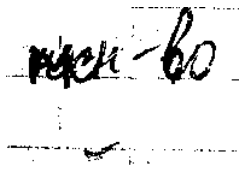
Zelevinsky A. Irreducible representations of $GL(n, \mathbb{C})$ (following the work of Tadić and Vogan)



Original source details: GELFAND-P001328

Uncertain reading. The first surname reads as “Tadić”. Two short marks follow the closing parenthesis.

Consider reps. of the group $GL(n, \mathbb{C})$. Denote the set of irreduc. unit. reps. by $GL(n, \mathbb{C})\hat{}$.



Original source details: GELFAND-P001328

Uncertain reading. The first strokes of the abbreviation for “set” are overwritten.

Uncertain reading. A mark before the superscript $\wedge$ is crossed out. Its reading as u is tentative.

Original source details: GELFAND-P001328

[Inserted above the line at left: Now] All unitary representations of this group are known. (it is proved that the ones found

$$D^u = (\text{multipl. chars. of the field } \mathbb{C}), \quad \text{i.e. } D^u \ni \delta :$$

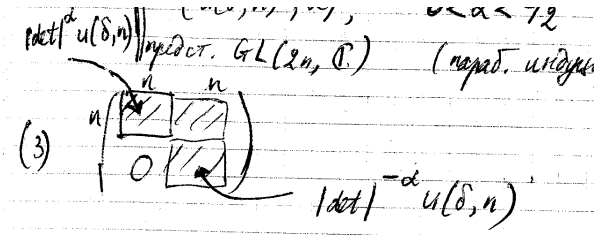
$$\delta(z) = \left(\frac{z}{|z|}\right)^k |z|^{i\rho}, \qquad k \in \mathbb{Z},\ \rho \in \mathbb{R} \tag{1}$$

$$u(\delta, n)(g) = \delta(\det g) \quad \text{is a unitary rep. of } GL(n, \mathbb{C}). \tag{2}$$

Additional family of representations [Double vertical stroke to the left of the following two lines.]

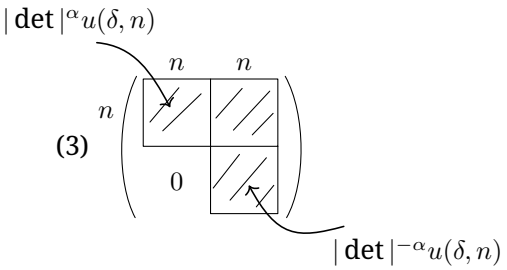
$$\pi(u(\delta, n), \alpha), \qquad 0 < \alpha < \frac{1}{2}$$

a rep. of $GL(2n, \mathbb{C})$ (parab. induction)



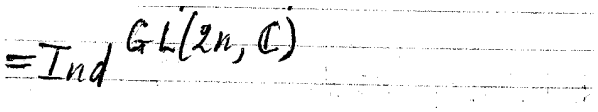
Original source details: GELFAND-P001328

Diagram and typed description.



and π is induced by this 1-dimensional representation.

$$\pi(u(\delta, n), \alpha) = \text{Ind}^{GL(2n,\mathbb{C})} \tag{4}$$



Undeciphered passage. The line ends at $\text{Ind}^{GL(2n,\mathbb{C})}$.

Original source details: GELFAND-P001328

I.M.G. Def. of an induced rep. [wavy underline] $G \supset H$, $\pi(h)$ is given Consider fns. $f(g)$ on the group G with values in the space of the rep. π such that $f(hg) = \pi(h)f(g)$

GELFAND-P001329

[source number: 42]

The space of such fns. carries an action of G and this rep. is called induced.

The restriction functor is dual to the induction functor.

3. Any representation [inserted below the line: (irreduc. unit.)] is obtained from these as follows

Consider a tuple $(n_1, \dots, n_k)$, $n_1 + n_2 + \dots + n_k = n$, and a collection $\sigma_i \in GL(n_i, \mathbb{C})^\vee$ of one of the 2 types described

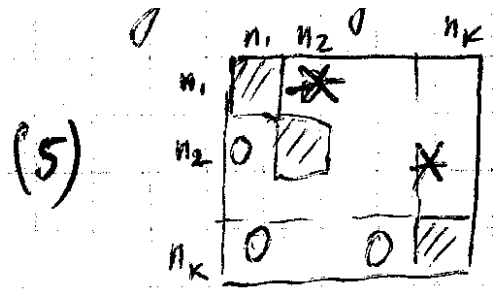
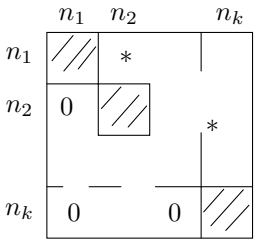


Diagram and typed description.

(5)



Original source details: GELFAND-P001329

Consider the rep. $\bigotimes \sigma_i$ for such a group and induce.

Theorem

a) The result is irreduc. and unitary.

b) Every irreduc. unit. rep. has this form.

c) equiv. $\iff$ the σ_i differ by a permutation

— . —

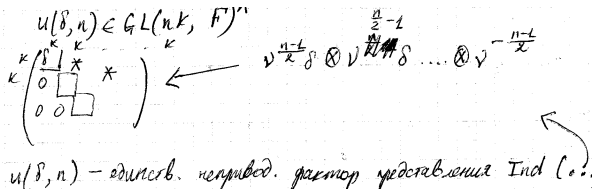
$D^u =$ square-integrable representations of the groups $GL(k, F)$ for some k

(with square-integrable matrix coefficients).

Then the statement of thm. (5) remains valid for every field.

(Previously $k = 1$.)

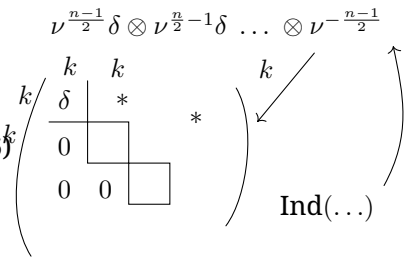
$$u(\delta, n) \in GL(nk, F)^\wedge$$

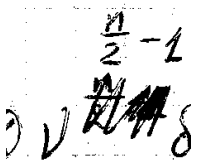


Original source details: GELFAND-P001329

Diagram and typed description.

(6)





Undeciphered passage. The exponent $\frac{n}{2} - 1$ is written above the cancellation at the second ν. Traces of a fraction remain beneath it. There is no δ after the final power of ν on the line.

Original source details: GELFAND-P001329

$u(\delta, n)$ is the unique irreduc. quotient of the representation Ind$(\dots)$

I.M.G.'s question [wavy underline]. What about other groups?

Kirillov [wavy underline] [crossed out nearby: Kir. (?)]

Uncertain reading. Tentatively "Kir." under the cancellation.

Original source details: GELFAND-P001329

In addition to $GL(n, \mathbb{C})$, all unit. irreduc. reps. are also known for the groups [wavy underline]

$$B_2 = C_2\,; \qquad G_2\,; \qquad SU(2,2)\,; \qquad SU(2,n)$$

$$SO(5) \sim Sp(4) \qquad \text{Knapp}$$

(Duflo)

Vogan did the quaternionic ones

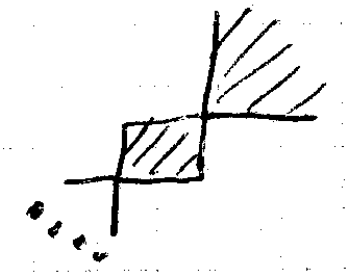
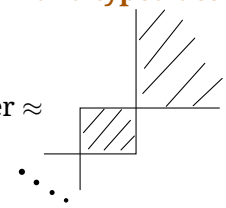


Diagram and typed description.

answer ≈



Original source details: GELFAND-P001329

GELFAND-P001330

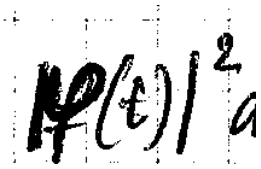
I.M.G. Introduce the inner prod.

$$\int |x_1 - x_2|^{-\varrho} f(x_1)\overline{f(x_2)}\, dx_1 dx_2$$

For $\varrho < 2$ the integral converges

$\varrho > 2 \Rightarrow$ need to continue analytically and the form will fail to be positive def.

$$\int t^{\tilde{\varrho}} |\varphi(t)|^2 \, dt$$



Uncertain reading. The function is written φ. The sign under the overwriting may be f.

Original source details: GELFAND-P001330

Positive when $\varphi(0) = 0$, and otherwise (regularization) we must also impose $\varphi'(0) = 0$, etc.
Degree of diverg. $\to$ number of minus signs
Kazakov.
Integrals that need to be evaluated

$$\int d(\Omega)_{U(N)} \exp[\mathrm{tr}\, \Omega^{+} p^{\mu} \Omega p^{\mu}], \qquad \Omega^{+} = \Omega^{-1} \tag{1}$$

Original source details: GELFAND-P001330

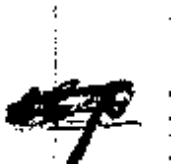
Uncertain reading. A superscript above the second Ω is crossed out, possibly $+$.

$$p^{\mu} = \mathrm{diag}(p_1^{\mu}, p_2^{\mu}, \ldots, p_N^{\mu}), \qquad \mu = 1, 2$$

The generalization is also of interest

$$\int d(\Omega)_{U(N)} \exp[\mathrm{tr}(\Omega^{-1} p^{\mu} \Omega q^{\mu})]. \tag{2}$$

Without μ, a 1-dimensional [crossed out: “кр.” (?)] chiral field (triv. problem)



Original source details: GELFAND-P001330

Uncertain reading. The short cancellation before the abbreviation for “chiral” tentatively reads “кр.”.

$\mu = 1, 2$, a 2-dimensional chiral field (not evaluated)

$$\downarrow$$

done by another method.

Perturbation theory shows that all [above the line: planar] diagrams of the 2-dimensional chiral field are reproduced. Namely, write

$$\Omega = e^{iA} \tag{3}$$

and expand in powers of A.

Approximation with $\Omega = 1$, then account for fluctuations.

$$(2) = Z(p, q) \tag{4}$$

Write the Haar measure and substitute $\Omega = e^{iA}$ in $d\Omega$ as well.

At leading order, a Gaussian term.

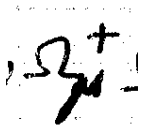
Uncertain reading. Beyond the right edge of the leaf, three clipped marks from the facing leaf are visible, including “(6”.

Original source details: GELFAND-P001330

GELFAND-P001331

[source number: 43]

$$Z = \int \prod_{\mu=1}^{d} d(\Omega_\mu) \exp\left[\beta \sum_{\mu,\nu} \operatorname{tr} \Omega_\mu \Omega_\nu \Omega_\mu^+ \Omega_\nu^+\right] \tag{5}$$



Uncertain reading. The index on the first adjointed factor in (5) is heavily retraced. It reads as μ.

Original source details: GELFAND-P001331

This is equiv. to ordinary gauge theory (as fns. of β) as $N \to \infty$.

— · —

Evaluation of integrals

Original source details: GELFAND-P001331

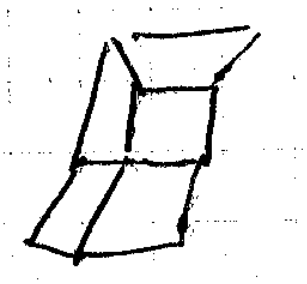
Uncertain reading. The ending of the word "integrals" is heavily retraced.

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int d^{N^2}u\, d^{N^2}v \exp \operatorname{tr}\left(-u^2 - v^2 + 2cuv - \frac{g}{N}(u^4+v^4)\right), \quad u^+ = u, \quad v^+ = v \tag{6}$$

$F(c,g)$ is the generating fn. of graphs, coverings of the sphere by quadrilaterals

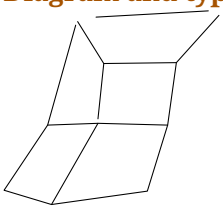
Original source details: GELFAND-P001331

Uncertain reading. Two horizontal strokes are visible above the word "graphs".



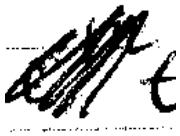
Original source details: GELFAND-P001331

Diagram and typed description.



All the quadrilaterals are coloured in 2 colours (white and black)

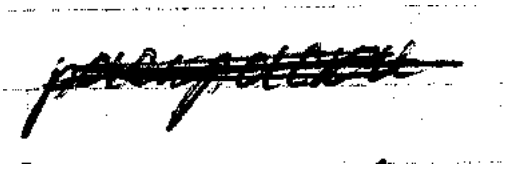
$$F(c,g) = \sum T_n^m g^n \quad \text{[crossed out: exp]} \quad e^{2cm}$$



Original source details: GELFAND-P001331

Uncertain reading. The cancelled fragment after g^n reads tentatively as exp.

T_n^m is the number of graphs for which the boundary length of the [crossed out: coloured (?)] coloured patches is m. n is the number of little squares

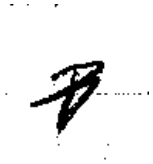


Original source details: GELFAND-P001331

Uncertain reading. The crossed-out word before “coloured patches” tentatively repeats “coloured”.

(edge length $\stackrel{\text{def}}{=} 1$)

This is also equiv. to the Ising model. ↗



Original source details: GELFAND-P001331

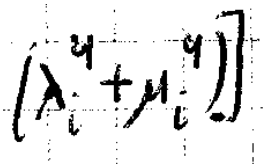
Uncertain reading. The mark after “Ising” resembles a retraced slanting arrow.

Let

$$u = \Omega_1^+ \lambda \Omega_1, \quad v = \Omega_2^+ \mu \Omega_2, \qquad \lambda, \mu \text{ are diagonal matrices}$$

Then

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \iint \left(\prod_{i=1}^{N} d\lambda_i \, d\mu_i \right) \Delta^2(\lambda)\Delta^2(\mu) \exp\Big[-\sum_{i=1}^{N} \Big[\lambda_i^2 + \mu_i^2 + \frac{g}{N}(\lambda_i^4 + \mu_i^4) \Big] \Big] \int d\Omega \, \exp 2c \, \mathrm{tr}(\Omega^+ \lambda \Omega \mu), \tag{7}$$



Original source details: GELFAND-P001331

Uncertain reading. After $\lambda_i^4 + \mu_i^4$, one closing parenthesis and one closing square bracket are visible.

$$\underbrace{\Omega = \Omega_1 \Omega_2^+}_{\substack{\text{relative matrix} \\ \text{angle}}}$$

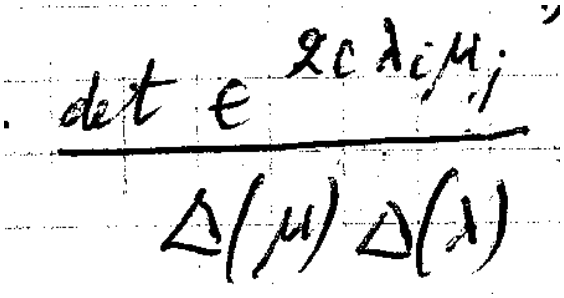
Original source details: GELFAND-P001331

Uncertain reading. The mark to the left of the formula resembles a crossed-out 4 or a slanting arrow.

Here $\Delta(\lambda) = \prod_{i<j}(\lambda_i - \lambda_j)$

But

$$\int d\Omega \, \exp 2c \operatorname{tr}(\Omega^{+}\lambda\Omega\mu) = \frac{\det e^{2c\lambda_i\mu_j}}{\Delta(\mu)\Delta(\lambda)} \tag{8}$$

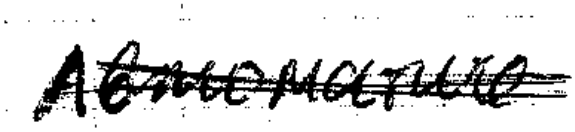


Uncertain reading. The right-hand side is written as a fraction with the determinant above two Δ factors.

Original source details: GELFAND-P001331

There are now $2N$ variables and the saddle-point method applies!

[Crossed out: Automatic...] We obtain



Uncertain reading. The beginning "Automatic..." is visible under the cancellation.

Original source details: GELFAND-P001331

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int \prod_{1}^{N} d\lambda_i \, d\mu_i \, \Delta(\lambda)\Delta(\mu) \times \underbrace{\times \exp\left[-\sum_{1}^{N}\left[\lambda_i^2 + \mu_i^2 + \frac{g}{N}(\lambda_i^4 + \mu_i^4) - 2c\lambda_i\mu_i\right]\right]}_{w(\lambda_i,\mu_i)} \tag{9}$$

Uncertain reading. The mark under the second line of (9) is labelled $w(\lambda_i, \mu_i)$.

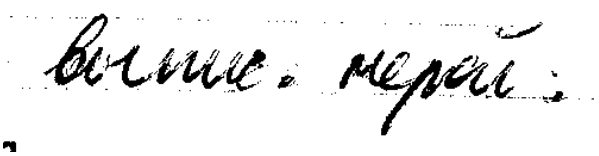
Original source details: GELFAND-P001331

$\Delta(\lambda)$ antisymm., $\Rightarrow$ antisymmetrization takes place automatically

GELFAND-P001332

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int\int \prod_{i=1}^{N} d\lambda_i \, d\mu_i \, w(\lambda_i,\mu_i)\Delta(\lambda)\Delta(\mu). \tag{10}$$

There are polynomials, orthog. with weight W and with a highest measure.



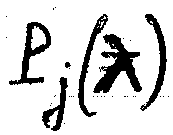
Uncertain reading. The word after "высше." (highest) reads tentatively as "мерой" (measure).

Original source details: GELFAND-P001332

$$\Delta(\lambda) = \det(\lambda_i^{j-1})_{i,j=1}^N = \det[P_{j-1}(\lambda_i)], \tag{11}$$

where

$$P_j(\lambda) = \lambda^j + \sum_{k=0}^{j-1} a_k \lambda^k$$



Uncertain reading. The argument of P_j reads as λ. There may be a z under the overwriting.

Original source details: GELFAND-P001332

Construct a system of orthog. polynomials P_j such that

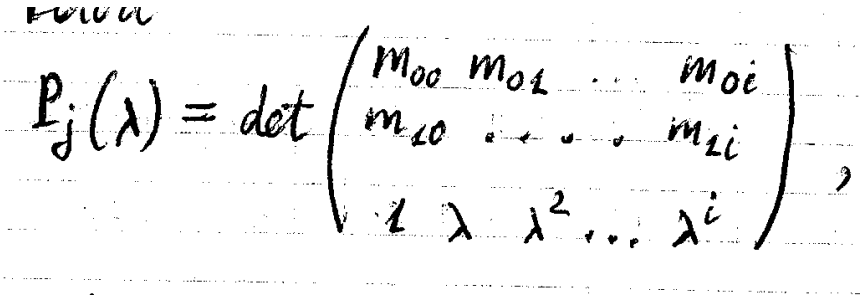
Uncertain reading. A separate mark before "Construct" is heavily overwritten, possibly Russian «И» ("And").

Original source details: GELFAND-P001332

$$\int_{-\infty}^{\infty} d\lambda\, d\mu\, W(\lambda,\mu) P_i(\lambda) P_j(\mu) = h_i \delta_{ij} \tag{12}$$

Then

$$P_j(\lambda) = \det \begin{pmatrix} m_{00} & m_{01} & \cdot & \cdot & m_{0i} \\ m_{10} & \cdot & \cdot & \cdot\cdot & m_{1i} \\ 1 & \lambda & \lambda^2 & \cdots & \lambda^i \end{pmatrix}, \tag{13}$$



Original source details: GELFAND-P001332

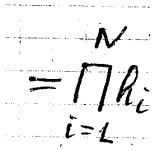
Uncertain reading. The left-hand side is $P_j(\lambda)$. The last column reads m_{0i}, m_{1i}, λ^i.

where

$$m_{ij} = \int d\lambda\, d\mu\, w(\lambda,\mu) \lambda^i \mu^j. \tag{14}$$

For h_j we get a system of recurrence eqs. In the limit of infinite N it becomes a single eq. Next,

$$\iint \prod d\lambda_i\, d\mu_i\, w(\lambda_i,\mu_i) \Delta(\lambda)\Delta(\mu) = \prod_{i=1}^N h_i \tag{15}$$



Uncertain reading. The right-hand side reads $\prod_{i=1}^N h_i$.

Original source details: GELFAND-P001332

Answer

$$\begin{cases} F(c,z) = \frac{1}{2}\ln\frac{z}{g(z)} + \frac{1}{2g^2(z)}\int_0^z \frac{dt}{t}g^2(t) - \frac{1}{g(z)}\int_0^z \frac{dt}{t}g(t) \\ g\ [\text{crossed out}] = \frac{z}{(1-3z)^2} - c^2z + 3c^2z^3 \end{cases} \tag{16}$$

Undeciphered passage. A short entry after g is crossed out, possibly (z).

Original source details: GELFAND-P001332

Eliminating z from this gives $F(c,g)$.

(this gives the answer parametrically)

This makes it possible to study the asymptotics of the number of graphs

([crossed out: ...] use $g'_z(z) = 0$ for the singularities)

Undeciphered passage. Slanted strokes cover the short fragment before “use”.

Original source details: GELFAND-P001332

GELFAND-P001333

9.2.87 Gelfand Sem.

Don Tsagir (Don Zagier)

I.M.G. [wavy underline] Faltings’s idea is to make p-adic strings.

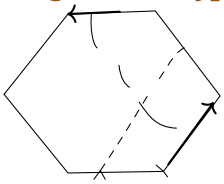
One piece of work (problem for the next seminar) Don Zagier.

Given an oriented [wavy underline] $2N$-gon, its sides are divided into pairs (each glued to some other side) [Inserted above the line: it cannot be glued to itself]

Identify in reverse (with a change of orientation)

We obtain a surface of genus g

Diagram and typed description.



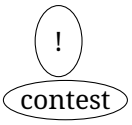
Some short strokes on the boundary and joins to the interior lines are uncertain.

Original source details: GELFAND-P001333

[In the left margin.]

Original source details: GELFAND-P001333

Diagram and typed description.

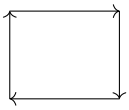


How many gluings yield a surface of genus g? [Wavy underline under the question.]

Write a formula. (2 gluings that differ by a rotation are counted as different) (a fn. of N and g).

Original source details: GELFAND-P001333

Diagram and typed description.



we can obtain a torus and a sphere.

This is the basis for computing the Euler characteristic of Teichmüller space!

D.Z. 2 things.

Modular forms and special values of zeta-functions

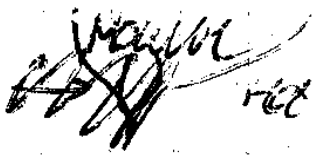
1. Classical Modular forms

2. Special values of zeta-functions

Modular forms. (parabolic) [wavy underline]

$$\Gamma = SL(2,\mathbb{Z})/\{\pm 1\}, \qquad H = \{z = x + iy;\ y > 0\}$$

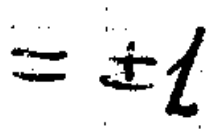
$f(z)$ holom. [above the line: on H], [crossed out: ...] [above the cancellation: zero] at ∞, and



Original source details: GELFAND-P001333

Undeciphered passage. The crossed-out fragment before "at ∞" is unclear. "Zero" is written above it.

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \tag{1}$$
$$a,b,c,d \in \mathbb{Z}, \qquad ad-bc = \pm 1$$



Uncertain reading. The right-hand side is written as ± 1.

Original source details: GELFAND-P001333

[At right:] Zeros at ∞ — cusp.

$$S_k = \{f\}, \qquad \dim S_k < \infty$$

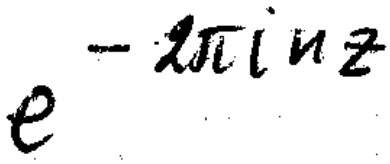
GELFAND-P001334

[source number: 69]

Hecke: $\exists$ many

$$f(z) = \sum_{n=1}^{\infty} a(n) e^{-2\pi i n z} \tag{2}$$

(since it follows from (1) that $f(z+1) = f(z)$)



Uncertain reading. In (2) there is a minus sign before $2\pi inz$.

Original source details: GELFAND-P001334

$\exists$ a basis in which $a(n) \in \mathbb{Z}$

(3) Example. [broken underline]

For $k = 2,4,6,8,10,14$, $S_k = \{0\}$ (no cusp forms)

Modular form [wavy underline] $O(1)$ at ∞ (as opposed to zero)

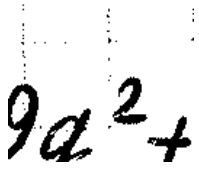
$$M_k \ni f(z) = \sum_{n=0}^{\infty} a(n) q^n, \qquad q = e^{2\pi i z} \tag{4}$$

Then $\exists$ an Eisenstein series [deletion between «ряд» and «Eisenstein'a»]

Undeciphered passage. Before «Eisenstein'a», $E\ldots$ is crossed out.

Original source details: GELFAND-P001334

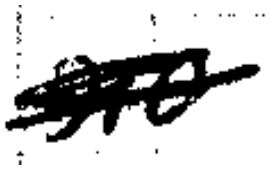
$$E_k(z) = \frac{1}{2} \sum_{(c,d)=1} \frac{1}{(cz+d)^2} \tag{5}$$



Original source details: GELFAND-P001334

Uncertain reading. In (5) the exponent in the denominator is written as 2.

Ex. Check [crossed out] automorphy.



Original source details: GELFAND-P001334

Undeciphered passage. The crossed-out word before “automorphy” may read «ero» (“its”).

$$\begin{aligned} E_4(z) &= 1 + 240(q + 9q^2 + 28q^3 + 73q^4 + \ldots) = \\ &= 1 + 240 \sum_{n=1}^{\infty} \underbrace{\sigma_3(n)}_{\| \; \sum_{d|n} d^3} q^n \end{aligned} \tag{6}$$

(7) k odd $\Rightarrow$ no cusp forms (obvious from the signs)

$$\begin{aligned} S_k \ni \Delta(z) &= q - 24q^2 + 252q^3 + \ldots - 6048q^6 + \ldots \\ &= \sum \tau(n) q^n \end{aligned} \tag{8}$$

(no constant term $\Rightarrow$ cusp form

$$24 \cdot 252 = 6048$$

$$\tau(nm) = \tau(n)\tau(m) \quad \text{when } (n,m) = 1. \tag{9}$$

(10) Claim. $\exists$ a basis of cusp forms whose coefficients have property (9)

$$\begin{aligned} \frac{E_4}{240} &= \frac{1}{240} + \sum_{n=1}^{\infty} \sigma_3(n) q^n, \\ \sigma_3(6) &= \sigma_3(2)\sigma_3(3) \end{aligned} \tag{11}$$

(12) Def. $f(z)$ is called a Hecke form [«Гекке» with “(Hecke)” underneath] if $f = \sum_0^\infty a(n)q^n$

$$a(nm) = a(n)a(m) \quad \text{when } (n,m) = 1$$

(13) Claim. The number of Hecke forms $=$ the dimension of the subspace, and they form a basis.

GELFAND-P001335

Proof: Hecke op. T_n:

Diagram and typed description. S_k

Original source details: GELFAND-P001335

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad f = f(z)$$

(14) notation:

$$(f \mid \gamma)(z) = \left(\frac{cz+d}{\sqrt{ad-bc}}\right)^{-k} f\left(\frac{az+b}{cz+d}\right)$$

(action of $SL_2(\mathbb{R})$) [added below] and even $GL_2^+(\mathbb{R})$

[The denominator $\sqrt{ad-bc}$ is inserted and the parenthesis extended.]

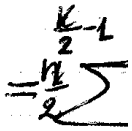
Uncertain reading. A stroke crosses the last letter of the numerator, which reads as b.

Original source details: GELFAND-P001335

$$M_k \ni f \iff f \mid \gamma = f \qquad (\gamma \in \Gamma).$$

$$T_1 = 1.$$

$$T_n\big(\overset{S_k}{\downarrow} f\big)(z) = \frac{n^{\frac{k}{2}-1}}{2} \sum_{\gamma \in \Gamma \backslash M_n} f \mid \gamma. \tag{15}$$



Uncertain reading. In (15), a denominator 2 is visible under $n^{k/2-1}$.

Original source details: GELFAND-P001335

$$M_n \subset M(2,\mathbb{Z}) \quad \text{— matrices with } \det = n.$$
$$2\times 2$$

$n = 2 \Rightarrow$ there are 3 terms

$$T_2(f) = 2^{\frac{k}{2}-1}\left[f \mid \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + f \mid \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} + f \mid \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\right] = \\ = 2^{k-1} f(2z) + \frac{1}{2} f\left(\frac{z}{2}\right) + \frac{1}{2} f\left(\frac{z+1}{2}\right) \tag{16}$$

Undeciphered passage. Strokes by the first f on the second line of (16) are crossed out. They may be a denominator 2.

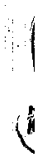
Original source details: GELFAND-P001335

Example: $k = 12$. Then for Δ, given by its Fourier series,

$$T_2(\Delta) = 2^{11} \sum_{n=1}^{\infty} \tau(n) q^{2n} + \sum_{n=1}^{\infty} \tau(2n) q^n \tag{17}$$

From the condition $T_2(\Delta) = \lambda_2 \Delta$ we find $\lambda_2 = -24$, etc.

(18) The eigenvalue λ_n is always a Fourier coeff. (follows directly)



Undeciphered passage. Fragmentary strokes are visible in the left margin by (18), without a secure reading.

Original source details: GELFAND-P001335

(19) Def. 2. f is a Hecke form $\iff T_n(f) = \lambda_n f, \forall n$.

The existence of a basis follows from 2 facts.

$$\text{a)} \quad T_n T_m = T_m T_n \tag{20}$$

b) S_k has an inner product (Petersson) in which the T_n are self-adjoint!

Then $\exists$ a basis of common eigenvectors.

Always:

$$a(1) = 1 \\ a(n) = \lambda_n \tag{21}$$

$$T_n T_m = T_{nm} \quad \underset{\text{when } (n,m)=1}{\Longrightarrow} \quad a(n)a(m) = a(nm) \tag{22}$$

(Mordell [wavy underline] introduced the Hecke operators in 1917)

Hecke proved implication (22).

GELFAND-P001336

[source number: 70]

Question [wavy underline]: compute $\mathrm{Tr}(T_n)$

Diagram and typed description. $T_n : S_k$

Original source details: GELFAND-P001336

$\underline{k = 12}$ $\quad \Delta = \sum_1^\infty \tau(n) q^n.$

$$T_n(\Delta) = \tau(n)\Delta, \quad T_n = (\tau(n)), \quad \mathrm{Tr}\, T_n = \tau_n. \tag{23}$$

The trace formula will give, in particular, a formula for $\tau(n)$

Inner product:

$$(f, g) = \int_{\Gamma\backslash H} f(z)\overline{g(z)} y^k \underbrace{dV}_{\frac{dx\,dy}{y^2}}, \qquad f, g \in S_k. \tag{24}$$

(inv. measure)

([Inserted above a deletion] $f(z)\overline{g(z)}y^k$ is Γ-invariant ($\Rightarrow$ the integral makes sense))

Undeciphered passage. The word below the insertion, possibly «оно» (“it”), is crossed out.

Original source details: GELFAND-P001336

(25) Lemma. Let $F(z)$ be holomorphic on H and sufficiently small at ∞, $k \geq 2$, $z_0 \in H$. Then

$$\int_H \frac{F(z)}{(\bar{z} - z_0)^k} y^k \, dV = C_k F(z_0), \tag{26}$$

where

$$C_k = \frac{(-1)^{k/2}\pi}{2^{k-2}(k-1)} \qquad \text{for } k \text{ even}.$$

Proof [broken underline]:

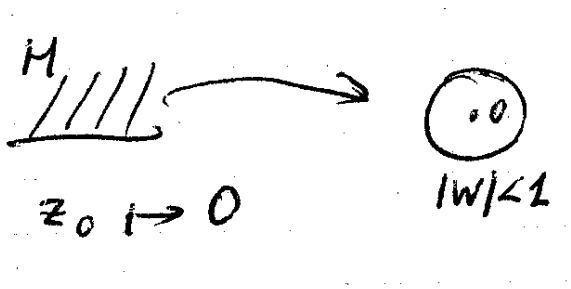
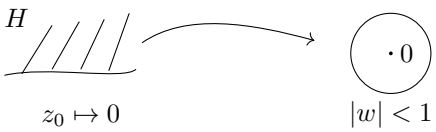


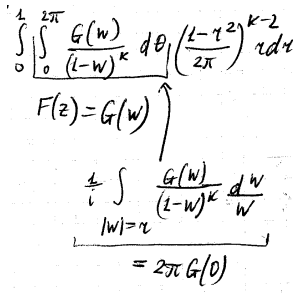
Diagram and typed description.



Original source details: GELFAND-P001336

$$w = \frac{z - z_0}{z - \bar{z}_0} = re^{i\theta}, \quad 0 \le r < 1,$$
$$z \in \mathbb{R} \Rightarrow |w| = 1$$

We obtain



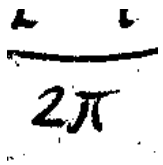
Original source details: GELFAND-P001336

Diagram and typed description.

$$\int_0^1 \left[\int_0^{2\pi} \frac{G(w)}{(1-w)^k}\, d\theta \right] \left(\frac{1-r^2}{2\pi} \right)^{k-2} r\, dr$$

$$F(z) = G(w)$$

$$\underbrace{\frac{1}{i} \int_{|w|=r} \frac{G(w)}{(1-w)^k} \frac{dw}{w}}_{= 2\pi G(0)}$$



Original source details: GELFAND-P001336

Uncertain reading. The denominator of the radial factor reads 2π.

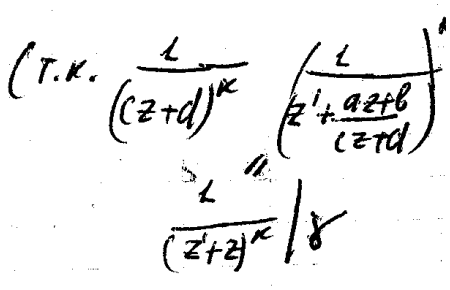
▷

$$K_n(\underset{\uparrow H}{z}, \underset{\uparrow H}{z'}) = \frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_n} \frac{1}{[(cz+d)z' + (az+b)]^k} \tag{27}$$
$$\in S_k \otimes S'_k.$$

(Bergman kernel) converges for $k \ge 4$.

[At lower right:]

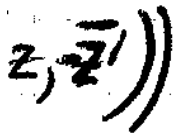
$$(\text{since } \frac{1}{(cz+d)^k} \left(\frac{1}{z' + \frac{az+b}{cz+d}} \right)^k$$
$$= \frac{1}{(z'+z)^k} \mid \gamma$$



Original source details: GELFAND-P001336

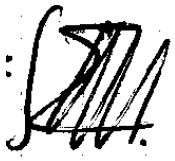
Uncertain reading. The two slanting strokes between the expressions have been read as an equals sign.

$$\big(f(z), K_n(z, -\bar{z}')\big) = \int_{\Gamma\backslash H} \sum_{\gamma\in M_n} \frac{f(z)}{(\bar{z}-z')^k} \mid \gamma\Big] y^k \, dV. \tag{28}$$



Uncertain reading. The short stroke before $\bar{z}'$ has been read as a minus sign.

Original source details: GELFAND-P001337

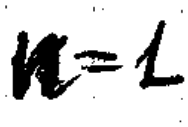


Undeciphered passage. An entry, possibly a $\sum$ sign, is crossed out after the integral sign in (28).

Original source details: GELFAND-P001337

$$n = 1 \quad \Rightarrow \quad \int_H \frac{f(z)}{(\bar{z}-z')^k} \, dz$$

and one has to use the lemma.



Uncertain reading. The first letter of the condition is overwritten. Tentative reading n.

Original source details: GELFAND-P001337

Any $n \Rightarrow$ we obtain $n^{1-k} C_k T_n(f)(z)$, i.e. K_n is the kernel of the Hecke op.

$f_1, \ldots, f_d$ is a basis of Hecke forms in S_k

$$f(z) = \sum_{n=1}^{\infty} a_j(n) q^n, \qquad a_j(1) = 1, \qquad T_n(f_j) = a_j(n) f_j$$

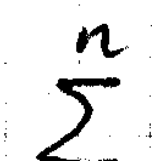
$$K_n(z, z') = \frac{C_k}{n^{k-1}} \sum_{j=1}^{d} \frac{a_j(n)}{(f_j, f_j)} f_j(z) f_j(z') \tag{29}$$

(obvious because the matrix is diagonal)

Undeciphered passage. A symbol, possibly $\sum$, is crossed out between the equals sign and the fraction in (29).

Original source details: GELFAND-P001337

$$\int_{\Gamma\backslash H} K_n(z,-\bar{z})y^k\,dV = \frac{C_k}{n^{k-1}} \underbrace{\sum_{j=1}^{n} \lambda_j(n)}_{\| \operatorname{Tr} T_n} \tag{30}$$

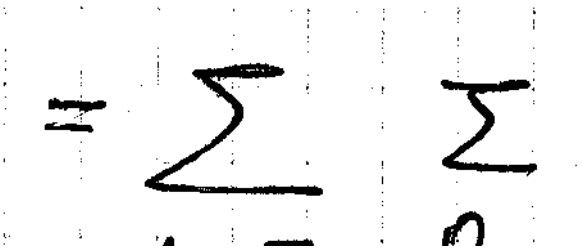


Original source details: GELFAND-P001337

Uncertain reading. The upper limit of the sum in (30) is written as n.

But:

$$\begin{aligned} K_n(z,-\bar{z}) &= \frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_n} \frac{1}{[c|z|^2 + d\bar{z} - az - b]^k} = \\ &= \sum_{\substack{t\in\mathbb{Z} \\ | \\ \text{trace}}} \sum_{\substack{Q \\ | \\ \operatorname{disc} Q = t^2-4n}} \frac{1}{(A|z|^2 + Bx + C - ity)^k} \end{aligned} \tag{31}$$



Original source details: GELFAND-P001337

Uncertain reading. In the second line of (31), the summation sign immediately follows the equals sign.

Original source details: GELFAND-P001337

Undeciphered passage. An isolated curved stroke is visible in the left margin next to (31).

[To the left of the second line of (31)] Put

$$\begin{aligned} d + a &= t \\ c &= A \\ d - a &= B \\ -b &= C \end{aligned}$$

$$\begin{aligned} Q = [A,B,C] &= Au^2 + Buv + Cv^2 \\ \operatorname{disc} Q = t^2 - 4n &= B^2 - 4AC = \\ &= (d-a)^2 + 4bc = (d+a)^2 - 4(ad-bc) \end{aligned}$$

Original source details: GELFAND-P001337

Undeciphered passage. A short isolated vertical stroke is visible in the lower left margin.

$K_n(z,-\bar{z})y^k$ must be Γ-invariant!

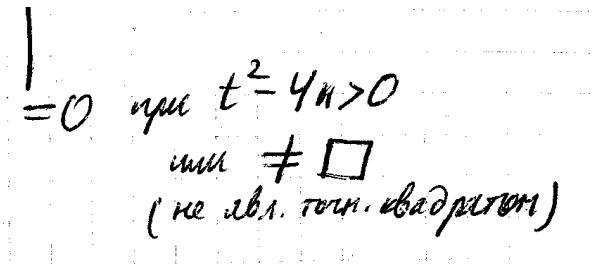
GELFAND-P001338

[source number: 71]

We obtain

$$K_n(z,-\bar{z})y^k = \sum_{t\in\mathbb{Z}} H(t^2-4n)(\cdot\ \cdot\ \cdot), \tag{32}$$

[Arrow to H] number of quadratic forms with the given discr. (class number)
[Arrow to the parentheses with dots] $=0$ when $t^2-4n>0$ or $\neq \square$ (is not a perfect square)



Original source details: GELFAND-P001338

Uncertain reading. The condition reads «или» (“or”).

and this is already a finite sum!

$$Q=[A,B,C], \qquad A,B,C\in\mathbb{Z}$$

$$Au^2+Buv+Cv^2$$

Γ acts [added below] on the set of such Q:

$$(Q\circ\gamma)(u,v) = Q\big((u,v)\gamma\big)$$

For $t\in\mathbb{R}$ put

$$R_{Q,t}(z) = \frac{y^k}{(A|z|^2+Bx+C-ity)^k} \tag{33}$$

$$K_n(z,-\bar{z})y^k = \sum_{t\in\mathbb{Z}} \sum_{\substack{Q \\ \operatorname{disc} Q=t^2-4n}} R_{Q,t}(z) \tag{34}$$

Original source details: GELFAND-P001338

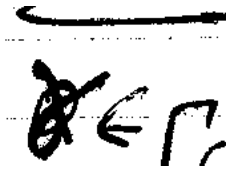
Undeciphered passage. A symbol at the end of (34) is crossed out. Two horizontal strokes are visible under the crossing-out.

Since

$$R_{Q,t}(\gamma z) = R_{Q\circ\gamma,t}(z), \tag{35}$$

the sum in (34) is automatically invariant

$$(34) = \sum_{t\in\mathbb{Z}} \sum_{\substack{[Q] \bmod \Gamma \\ \operatorname{disc} Q = t^2-4n}} \underbrace{\sum_{\gamma\in\Gamma_Q\backslash\Gamma} R_{Q\circ\gamma,t}(z)}_{\substack{\text{the integral of this is computed} \\ \text{— this is } (\cdots) \text{ in (32)}}} \tag{36}$$

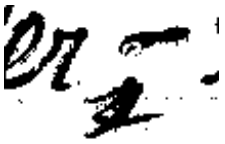


Uncertain reading. The index of the last sum is overwritten. It has been read as γ, as in the subscript of R in the same line.

Original source details: GELFAND-P001338

The important point is that the factor $H(t^2-4n)$ appears

(37) Theorem (Eichler-Selberg trace formula, 1956)



Undeciphered passage. A small sign resembling 2 or z is crossed out below the hyphen.

Original source details: GELFAND-P001338

Let $k \geq 4$ be even

$$\begin{aligned} \mathbb{Z} \ni \operatorname{Tr}(T_n \colon S_k \circlearrowleft) &= -\frac{1}{2} \sum_{\substack{t\in\mathbb{Z} \\ |t|\leq 2\sqrt{n}}} p_k(t,n) H(t^2-4n) \\ &\quad - \frac{1}{2} \sum_{d|n} \min\left(d, \frac{n}{d}\right)^{k-1} \end{aligned} \tag{38}$$

$$\begin{pmatrix} \tau(n) & k=12 \\ 0 & k=8 \end{pmatrix} \quad \text{[below the trace]}$$

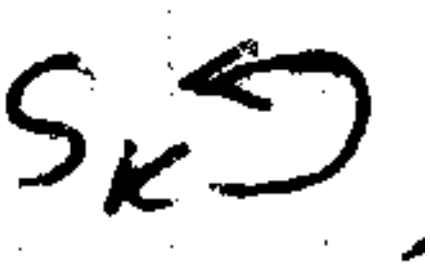


Diagram and typed description. $S_k \circlearrowleft$

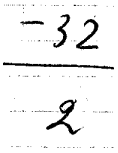
Original source details: GELFAND-P001338

GELFAND-P001339

Δ	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
$H(\Delta)$	$-\frac{1}{12}$	0	0	$\frac{1}{3}$	$\frac{1}{2}$	0	0	1	1	0	0	1	$\frac{4}{3}$	0

-14	-15	-16	-17	-18	-19	-20	-21	-22	-23	-24	-25
0	2	3/2	0	0	1	2	0	0	3	2	0

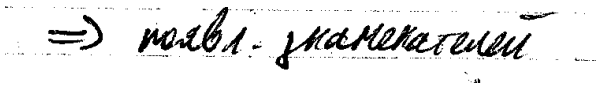
-26	-27	-28	-29	-30	-31	-32	-33
0	4/3	2	0	0	3	2	0



Uncertain reading. The table entry under -32 reads 2.

Original source details: GELFAND-P001339

Number of classes accounting for automorphisms ⇒ denominators appear



Uncertain reading. The words after the arrow read «появл. знаменателей» (“denominators appear”).

Original source details: GELFAND-P001339

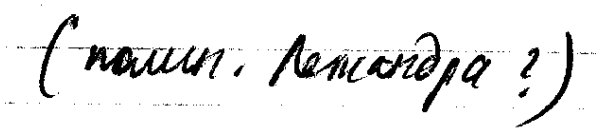
[At right, with a bracket] only the following occur

$$\Delta = B^2 - 4AC \equiv 0, 1 \pmod 4$$
otherwise $H(\Delta) = 0$.

$$\begin{aligned} p_2(t,n) &= 1 \\ p_3 &= t \\ p_4 &= t^2 - n \\ p_5 &= t^3 - 2tn \\ p_6 &= t^4 - 3t^2 n + n^2 \end{aligned}$$

$$p_k(t,n) = \frac{\alpha^{k-1} - \bar{\alpha}^{k-1}}{\alpha - \bar{\alpha}}$$

(Legendre polyn.?)



Uncertain reading. The manuscript has a question mark after “Legendre”.

Original source details: GELFAND-P001339

where α is such that

$$\alpha + \bar{\alpha} = t, \qquad \alpha\bar{\alpha} = n$$

(39) Corollary. $k = 4$, n prime. Then

$$-2 = \sum_{t^2 \le 4n} (t^2 - n) H(t^2 - 4n)$$

Uncertain reading. At the beginning of $(t^2 - n)$, the letter t is overwritten.

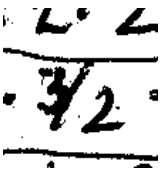
Original source details: GELFAND-P001339

Undeciphered passage. An isolated stroke under H has no secure reading as a symbol.

Original source details: GELFAND-P001339

$n = 5$ check

t	
0	$-5 \cdot 2$
± 1	$-4 \cdot 1 \cdot 2$
± 2	$-1 \cdot 3/2 \cdot 2$
± 3	$+4 \cdot 1 \cdot 2$
± 4	$+11 \cdot 4/2 \cdot 2$



Uncertain reading. In the row $t = \pm 2$ the fraction reads $3/2$. A stroke intersects the numerator.

Original source details: GELFAND-P001339

Uncertain reading. In the row $t = \pm 4$ the fraction reads $4/2$. The numerator may be 1 joined to the slash.

Original source details: GELFAND-P001339

In the general Selberg formula the sum is infinite.

GELFAND-P001340

[source number: 72]

Eisenstein series (— Maass) are nonholomorphic.

$$E(z, \underset{\uparrow C}{s}) = \frac{1}{2} \sum_{(c,d)=1} \frac{y^s}{|cz+d|^{2s}} \qquad \operatorname{Re} s > 1 \tag{40}$$

Uncertain reading. An upward arrow from the letter C appears under s.

Original source details: GELFAND-P001340

Undeciphered passage. A short fragment at the right of (40) is crossed out. The first stroke may belong to a c.

Original source details: GELFAND-P001340

$$E(\gamma z, s) = E(z, s), \quad \gamma \in \Gamma. \qquad \text{(easy)} \tag{41}$$

$$\Gamma_\infty = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \; b \in \mathbb{Z} \right\}$$

Then

$$E(z, s) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} \operatorname{Im}(\gamma z)^s \tag{42}$$

Let Φ be a Γ-invariant function on H (small at ∞). Consider the integral

$$\int_{\Gamma \backslash H} E(z, s) \varphi(z) \, \underset{\frac{dx\,dy}{y^2}}{dV} = \tag{43}$$

[At right, after a double vertical line:]

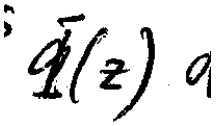
$$\varphi(z+1) = \varphi(z)$$

$$\Downarrow$$

$$\varphi(z) = \sum_{n \in \mathbb{Z}} c_n(y) e^{2\pi i n x} \qquad z = x + iy$$

$$= \int_{\Gamma_\infty \backslash H} \operatorname{Im}(z)^s \bar{\varphi}(z) \, dV =$$

$$= \int_0^\infty \int_0^1 \Phi(x+iy) y^{s-2} \, dx \, dy =$$



Uncertain reading. In the first integral continuing (43), a bar appears over φ.

Original source details: GELFAND-P001340

Original source details: GELFAND-P001340

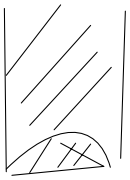


Diagram and typed description.

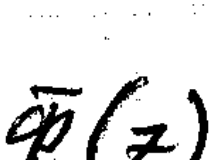
$$= \int_0^\infty c_0(y) y^{s-2}\, dy$$

(Rankin method)

Prop. [Arrow from above to $\varphi(z)$] Γ-invariant with decay

$$\int_{\Gamma\backslash H} \varphi(z) E(z,s)\, dV = \int_0^\infty c_0(y) y^{s-2}\, dy \tag{44}$$

[Arrow from below to $c_0(y)$] const. term in the Fourier exp. of $\bar{\varphi}(z)$.



Original source details: GELFAND-P001340

Uncertain reading. In the annotation to $c_0(y)$, φ also reads as barred.

Applications. [wavy underline]

(45) $E(z,s)$ properties

① merom. continuation in the s-plane (z fixed)

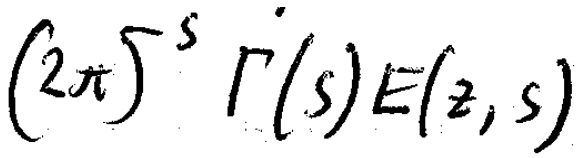
poles at $s = 1$ and $s = \frac{1}{2}\rho$, ρ a root of $\zeta(\rho) = 0$

(there are no other poles)

GELFAND-P001341

② Functional eq.

$$(2\pi)^s \Gamma(s) E(z,s) \quad \text{invariant under } s \mapsto 1-s$$

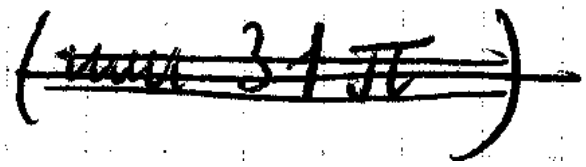


Original source details: GELFAND-P001341

Uncertain reading. The factor before $\Gamma(s)$ reads $(2\pi)^s$.

③

$$E(z,s) = \frac{3/\pi}{s-1} + O(1) \qquad \text{(the residue is independent of } z\text{)}$$

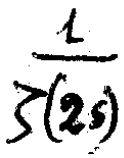


Original source details: GELFAND-P001341

Undeciphered passage. A parenthesized note below the formula is crossed out, possibly "(residue $3/\pi$)".

Reason for meromorphy. Consider

$$\frac{1}{\zeta(2s)} \sum_{\substack{(m,n)\in\mathbb{Z}^2 \\ \neq(0,0)}} \frac{y^s}{|mz+n|^{2s}} \tag{46}$$

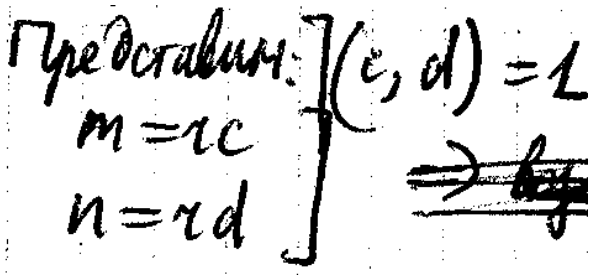


Uncertain reading. The factor in (46) is written $1/\zeta(2s)$.

Original source details: GELFAND-P001341

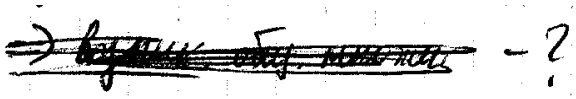
[At right] Write

$$\left.\begin{aligned} m &= rc \\ n &= rd \end{aligned}\right] \quad (c,d)=1 \quad \Rightarrow$$



Uncertain reading. The multiplier in both lines has been read as r.

Original source details: GELFAND-P001341



Undeciphered passage. A line after the arrow is crossed out. "— ?" remains at right.

Original source details: GELFAND-P001341

study using the Poisson summation formula.

$c_0(y)$ has surprising properties! (For every Γ-invariant function $\Phi(z)$!)

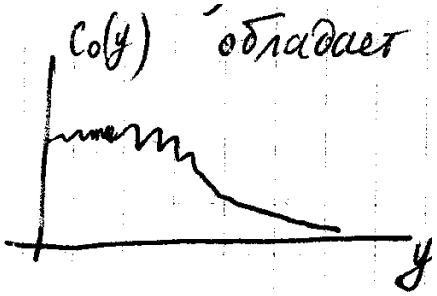
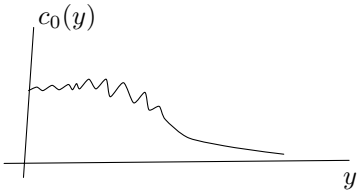


Diagram and typed description.



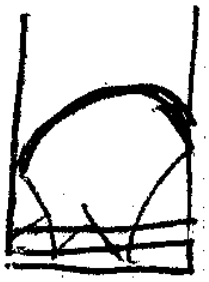
Original source details: GELFAND-P001341

the frequency of oscillations ["as $y \to 0$" inserted above] is given by ρ, a zero, $\rho \mapsto \zeta(2\rho)$

Uncertain reading. The sign in $\zeta(2\rho)$ and before the arrow reads ρ.

Original source details: GELFAND-P001341

(see the behavior of the Mellin transform)



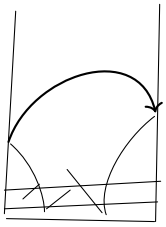


Diagram and typed description.

Original source details: GELFAND-P001341

$c_0(y)$ is obtained by integrating along const. horocycles!

Uncertain reading. The abbreviation before "horocycles" tentatively reads «пост.» ("const.").

Original source details: GELFAND-P001341

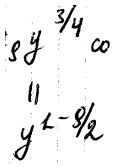
If

$$c_0(y) \sim \sum a_\lambda y^\lambda, \quad \text{then} \quad \int_0^1 y^{\lambda+s-2}\,dy = \frac{1}{\lambda-1+s}$$

$\Rightarrow$ a pole arises at $s = 1-\lambda$.

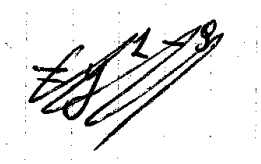
Take Φ to be the characteristic function of a region and then replicate it. We get [Circled insertion pointing to "region"] of volume $= \frac{1}{3}$ of the total

$$a_1 + \sum a_\rho \underbrace{y^{3/4}}_{\| \; y^{1-\rho/2}} \cos\left(\frac{\tau}{2}\log y - \varphi\right) \tag{47}$$



Uncertain reading. Below $y^{3/4}$ the manuscript gives $y^{1-\rho/2}$, with two vertical strokes.

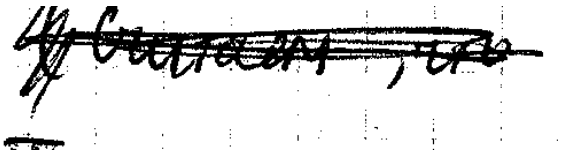
Original source details: GELFAND-P001341



Undeciphered passage. An expression at right is crossed out, possibly $+y^{1-\rho}$.

Original source details: GELFAND-P001341

The asymptotics of intersections are determined by the zeros of the ζ-function.



Undeciphered passage. The line is crossed out. The final word may read «что» ("that").

Original source details: GELFAND-P001341

(48) Theorem Let $\Phi(z)$ be Γ-invariant and $c_0(y)$ the constant term for Φ

$$\Rightarrow \quad \int_0^\infty c_0(y) y^{s-2}\,dy$$

poles at $s = 1, \rho/2$

functional eq. $s \mapsto 1 - s$

Residues [wavy underline]: at $s = 1$ this is a_1 in (47)

GELFAND-P001342

[source number: 73]

$$\frac{3}{\pi} \int_{\Gamma \backslash H} \Phi(z)\, dV = \lim_{y \to 0} c_0(y) = c_0(0) \tag{49}$$

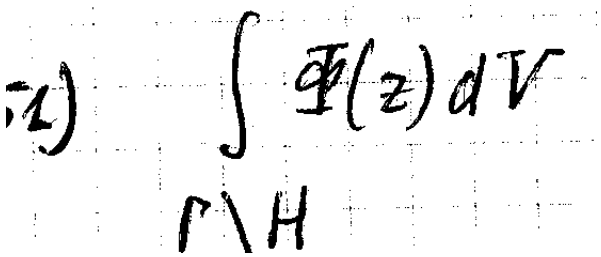
Use: instead of integrating, one can calculate $c_0(0)$, which is easy!

Application. [wavy underline] Let

$$\varphi(z) = y^k K_n(z, -\bar{z}), \quad k \geq 4 \text{ even}, \quad n \geq 1 \tag{50}$$

$$\int_{\Gamma \backslash H} \Phi(z)\, dV = c_0(0) \qquad \text{can be computed!} \tag{51}$$

(but this was known)



Uncertain reading. In (51), no coefficient is written before the integral.

Original source details: GELFAND-P001342

More interesting.

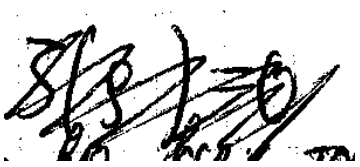
$$\begin{aligned} \int_{\Gamma \backslash H} \Phi(z) E(z,s)\, dV &= \sum_{t \in \mathbb{Z}} \int_{\Gamma \backslash H} \left(\sum_{\substack{Q \\ \operatorname{disc}(Q) = t^2 - 4n}} R_{Q,t}(z) \right) E(z,s)\, dV = \\ &= \sum_{t \in \mathbb{Z}} \underset{\substack{\uparrow \\ \text{some expression}}}{(\cdots)} \quad \zeta_{\mathbb{Q}(\sqrt{t^2-4n})}(s) \end{aligned} \tag{52}$$

$$\forall \text{ field ext. } \mathbb{Q} \subset K \qquad \zeta_K(s) = \sum \frac{1}{N(\,)^s} = \zeta(s) L(s)$$

$$\zeta(\rho) = 0 \quad \Rightarrow \quad \zeta_K(\rho) = 0$$

It follows that

$$\int_{\Gamma \backslash H} \Phi(z) E(z, \rho)\, dV = 0 \quad \text{provided that } \zeta(\rho) = 0! \tag{53}$$



Uncertain reading. Below (53), $\zeta(\rho) = 0$ is crossed out.

Original source details: GELFAND-P001342

Introduce the space of all such functions

$$V = \{\varphi(z) \text{ on } \Gamma\backslash H\}$$

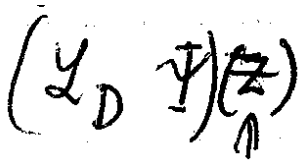
$$\int_{\Gamma\backslash H} \psi(z)\varphi(z) = 0$$

Original source details: GELFAND-P001342

Undeciphered passage. A fragment under the last integral sign and above $\Gamma\backslash H$ is crossed out.

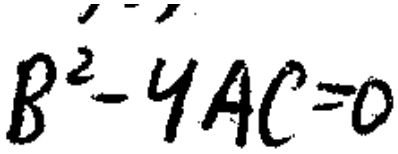
[At right after a double rule] This is a characterization of the roots ρ!!
(Spectrum of the representation in V!) (Laplace op.)
[Below and to the left] Spectrum of the Laplace op.
[Downward arrow] if it is unitary, we obtain the Riemann hypothesis.
[At left, with an arrow to (54)] The Φ form a large set
(The absence of other zeros is unproved)

$$(\mathcal{L}_D\Psi)(\underset{\Gamma\backslash H}{z}) = \sum_{\substack{A,B,C\in\mathbb{Z} \\ B^2-4AC=0}} \Psi\left(\frac{A|z|^2 + Bx + C}{y}\right) \tag{54}$$



Original source details: GELFAND-P001342

Uncertain reading. The looped letter with subscript D has been read as $\mathcal{L}_D$.



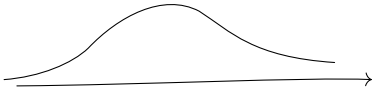
Original source details: GELFAND-P001342

Uncertain reading. The condition under the sum is written $B^2 - 4AC = 0$.

Automatically Γ-invariant

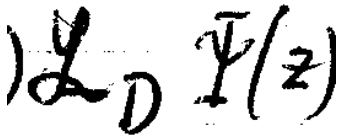
Original source details: GELFAND-P001342

Diagram and typed description.



Ψ is smooth and decaying
this describes W $\quad$ $V = W^\perp$

$$\int_{\Gamma\backslash H} E(z,s)\mathcal{L}_D\Psi(z)\,dV = (*)\zeta_{\mathbb{Q}(\sqrt{D})}(s). \tag{55}$$



Uncertain reading. The looped letter with subscript D has been read as $\mathcal{L}_D$.

Original source details: GELFAND-P001343

varying D, we obtain the Riemann ζ-function
one should take the standard operator.

$$\Delta = y^2\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right), \qquad \Delta(W)\subset W$$

Undeciphered passage. A clipped stroke is visible in the frame beyond the right edge of the page.

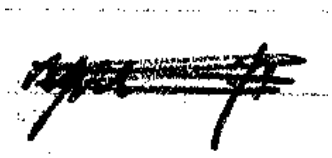
Original source details: GELFAND-P001343

16/II-87. Gelfand Sem.

Problem. (Kazakov). How many triangulations of the sphere [crossed out] with Δ's colored in 2 colors (white and black), with fixed numbers of black and white ones and fixed boundary length (between white and black).

Uncertain reading. The surname by the problem reads "Kazakov".

Original source details: GELFAND-P001343



Undeciphered passage. An entry is crossed out after "sphere".

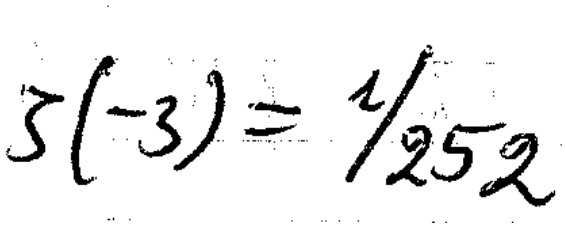
Original source details: GELFAND-P001343

Don Zagier.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \text{then } \zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90} \tag{1}$$

$$\zeta(2n) = \frac{(-1)^{n-1} 2^{2n-1} B_{2n}}{(2n)!} \pi^{2n}$$

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \zeta(-2) = 0, \quad \zeta(-3) = \frac{1}{252} \tag{2}$$



Uncertain reading. In (2) the value written for $\zeta(-3)$ is $1/252$.

Original source details: GELFAND-P001343

$$\zeta(1-2g) = \chi(\mathcal{M}_g) \qquad (?) \tag{3}$$

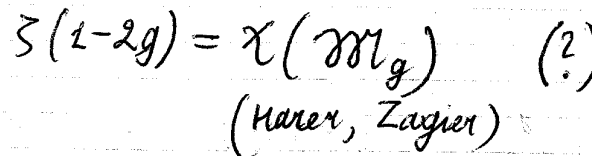
(Harer, Zagier)

$\mathcal{M}_g$ is the moduli space of [crossed out] surfaces of genus g

Undeciphered passage. A short fragment is crossed out before the abbreviation for "surfaces".

Original source details: GELFAND-P001343

$\chi(\mathcal{M}_g)$ is its Euler char.



Uncertain reading. A question mark in parentheses appears to the right of (3).

Original source details: GELFAND-P001343

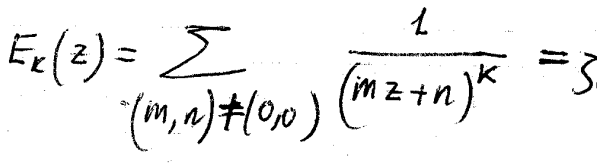
$k = 4, 6, 8, 10, \ldots$

$$E_k(z) = \frac{1}{2} \sum_{(c,d)=1} \frac{1}{(cz+d)^k} \tag{4}$$

$$\begin{aligned} \zeta(k) E_k(z) &= \sum_{(m,n)\neq(0,0)} \frac{1}{(mz+n)^k} = \\ &= \zeta(+k) + \frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n, \end{aligned} \tag{5}$$

Undeciphered passage. A symbol is crossed out before k in the first $\zeta(k)$ in (5).

Original source details: GELFAND-P001343



Uncertain reading. In (5) the sum over (m, n) directly follows the equals sign.

Original source details: GELFAND-P001343

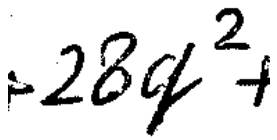
GELFAND-P001344

[source number: 74]

where

$$\sigma_{k-1}(n) = \sum_{d|n} d^{k-1}, \qquad q = e^{2\pi i z}$$

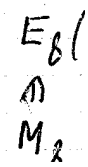
$$M_4 \ni E_4\zeta(4) = \zeta(4) + \frac{8\pi^4}{3}(q + 28q^2 + \ldots) \qquad \text{(weight 4)}$$



Uncertain reading. The formula of weight 4 reads $28q^2$.

Original source details: GELFAND-P001344

$$E_8(z)\zeta(8) = \zeta(8) + \underline{\ldots\pi^8}(q + 129q^2 + \ldots) \qquad \text{(weight 8)}$$
$$\Uparrow$$
$$M_8$$



Uncertain reading. Below E_8, M_8 is written with an upward arrow.

Original source details: GELFAND-P001344

Uncertain reading. The exponent of π has been read as 8. A similar glyph appears as the exponent of π in the formula for $\zeta(8)$ below.

Original source details: GELFAND-P001344

Undeciphered passage. A line is drawn below $\ldots\pi^8$. The space under it is blank.

Original source details: GELFAND-P001344

$$\dim M_8 = 1, \qquad \dim M_4 = 1$$

$$\Rightarrow E_4^2 = cE_8 \qquad \text{(the weights agree)}$$

Undeciphered passage. An entry to the left of the arrow to $E_4^2 = cE_8$ is crossed out.

Original source details: GELFAND-P001344

$$\zeta(4) = \pi^4/90 \quad \Rightarrow \quad \zeta(8) = \pi^8/9450$$

Thus:

From $\underset{\pi^4/90}{\zeta(4)}$ and $\underset{\pi^6/945}{\zeta(6)}$ one obtains all $\zeta(2n)$

$$E_4 E_6 = cE_{10} \quad \text{etc.}$$

Hecke L-series

Let

$$f(z) \in M_k(SL_2(\mathbb{Z}))$$

$$f(z) = \sum_{n=0}^{\infty} a_n q^n \qquad \text{mod. form of weight } k$$

$$a_0 = 0 \Rightarrow f \in S_k \qquad \begin{matrix}\text{(parabolic)} \\ \text{cusp form}\end{matrix}$$

$$L(f,s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

$$\Gamma(s) = \lambda^s \int_0^{\infty} y^{s-1} e^{-\lambda y}\, dy$$

$$\Gamma(s) \sum_{n=1}^{\infty} \frac{a_n}{(2\pi n)^s} = \int_0^{\infty} y^{s-1} \left(\sum_{n=1}^{\infty} a_n e^{-2\pi n y} \right) dy$$

Undeciphered passage. A short entry after a_n in the numerator is crossed out.

Original source details: GELFAND-P001344

$$\Gamma(s)(2\pi)^{-s} L(f,s) = \int_0^{\infty} y^{s-1} f(iy)\, dy \tag{6}$$

[At right] (if f is parabolic), $f \in S_k$

The integral on the right always converges,

$$f\left(-\frac{1}{z}\right) = z^k f(z), \qquad f\left(\frac{i}{y}\right) = (-1)^{k/2} y^k f(iy)$$

Hence

$$\underbrace{\Gamma(s)(2\pi)^{-s}L(f,s)}_{\|\ L^*(f,s)} = (-1)^{k/2}\,(s \to k-s) \tag{7}$$

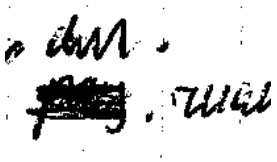
(functional eq.)

$$L^*(f,s) = (-1)^{k/2}L^*(f,k-s) \tag{8}$$

(9) Thm. (Eichler + Shimura + Manin)

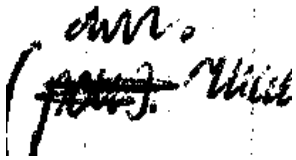
If $f(z) \in S_k(SL_2\mathbb{Z})$ is a Hecke form, then $\exists\,\omega_+,\omega_- \in \mathbb{R}$ (depending only on f) such that

$$L^*(f,n) = (\text{alg. number})\,\omega_-, \qquad n = 2,4,\ldots,k-2,$$
$$L^*(f,n) = (\text{alg. number})\,\omega_+, \qquad n = 1,3,\ldots,k-1.$$



Original source details: GELFAND-P001345

Undeciphered passage. In the first line, possibly «рац.» (“rat.”) is crossed out before “number”. “Alg.” is written above.



Original source details: GELFAND-P001345

Undeciphered passage. In the second line, possibly «рац.» (“rat.”) is crossed out before “number”. “Alg.” is written above.

ω_-, ω_+ are probably transcendental. This is proved for other groups (with complex multiplication) rather than $SL_2(\mathbb{Z})$.

$$\underset{\text{period}}{r_n(f)} = \int_0^\infty f(iy) y^n \, dy, \qquad n = 0, 1, \ldots, k-2. \tag{10}$$

Example.

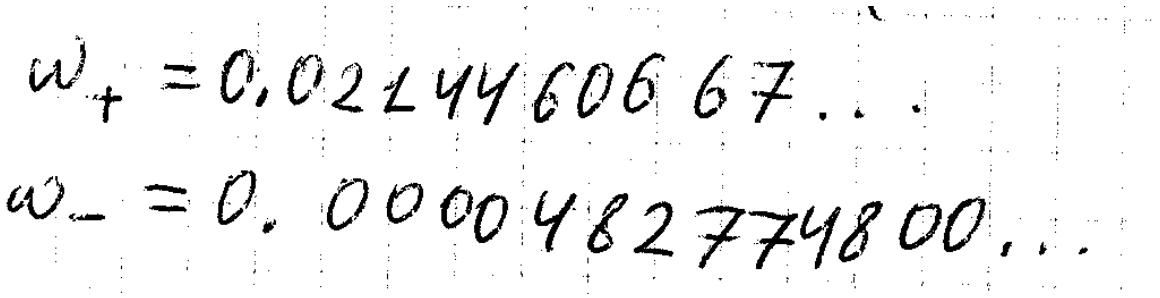
$$\Delta(z) \in S_{12}(SL_2(\mathbb{Z})) = q - 24q^2 + 252q^3 + \ldots =$$
$$= q \prod_1^\infty (1-q^n)^{24}$$

	10	9	8	7	6	
n	0	1	2	3	4	5
$r_n(\Delta)$		$\frac{384}{5}\omega_-$		$40\omega_-$		$\underbrace{32\omega_-}_{\text{def. of } \omega_-}$
	$\frac{192}{691}\omega_+$		$\frac{16}{135}\omega_+$		$\frac{8}{105}\omega_+$	

$$r_n = r_{10-n}$$

(only ratios matter)

$$\omega_+ = 0.0214460667\ldots$$
$$\omega_- = 0.0000482774800\ldots$$



Original source details: GELFAND-P001345

Uncertain reading. The decimal entries read $0.0214460667\ldots$ and $0.0000482774800\ldots$.

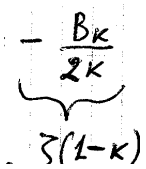
$$G_k(z) = \underbrace{-\frac{B_k}{2k}}_{\zeta(1-k)} E_k(z) \qquad \text{(Eisenstein series)} \tag{10'}$$

Original source details: GELFAND-P001345

Uncertain reading. The initial of $G_k(z)$ is overwritten. There may be an E underneath.

Original source details: GELFAND-P001345

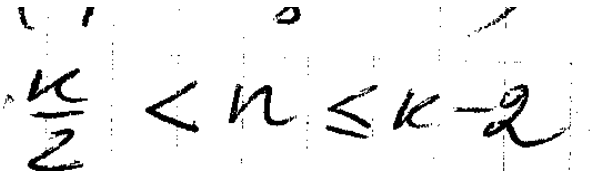
Undeciphered passage. An entry between the fraction and $E_k(z)$ is crossed out.



Uncertain reading. $\zeta(1-k)$ is written under $-B_k/(2k)$.

Original source details: GELFAND-P001345

$$\frac{k}{2} < n \leq k-2$$



Uncertain reading. The range at the right of (11) is $k/2 < n \leq k-2$.

Original source details: GELFAND-P001345

$$-\frac{4n}{B_{2n}} \left(\underset{\uparrow S_k}{f}\,,\ \underset{\uparrow M_{2n}}{G_{2n}} \underset{\uparrow M_{k-2n}}{G_{k-2n}} \right) \tag{11}$$

a. (one of them is parabolic ⇒ the inner product converges) [arrow down to the integrand] inner product:

$$\left((f,g) = \int_{\Gamma\backslash H} f(z)\overline{g(z)} y^k \frac{dx\,dy}{y^2} \right)$$

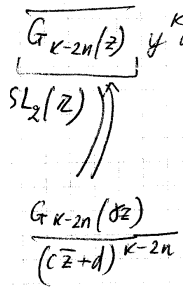
GELFAND-P001346

[source number: 75]

$$(12) \qquad (11) = \int_{\Gamma\backslash H} f(z) \sum_{\left(\begin{smallmatrix}\circ & \circ\\ c & d\end{smallmatrix}\right) \in \left(\begin{smallmatrix}1 & *\\ 0 & 1\end{smallmatrix}\right)\backslash SL_2(\mathbb{Z})} \frac{\overline{G_{k-2n}(z)}}{(c\bar{z}+d)^{2n}}\, y^k\, dV$$

A double arrow leads from the fraction to the brace.

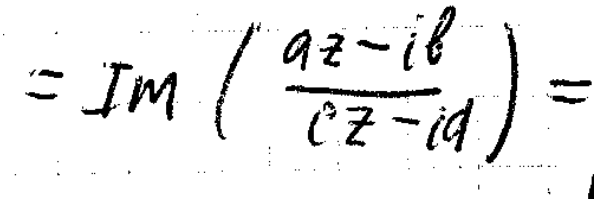
$$\frac{G_{k-2n}(\gamma z)}{(c\bar{z}+d)^{k-2n}} \qquad \left(\text{since } G(\gamma z) = (cz+d)^{k-2n} G(z)\right)$$



Uncertain reading. No overbar is visible over the numerator $G_{k-2n}(\gamma z)$.

Original source details: GELFAND-P001346

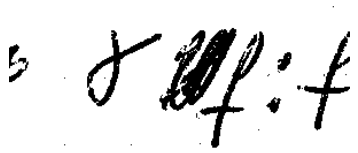
$$\operatorname{Im}(\gamma z) = \operatorname{Im}\left(\frac{az - ib}{cz - id}\right) = \frac{y}{|cz + d|^2}$$



Uncertain reading. The separate strokes in the fraction are read as $-i$.

Original source details: GELFAND-P001346

And one can insert γ into f, giving $f(\gamma z)$, by replacing the factor.



Uncertain reading. The preposition before f is overwritten, provisionally read as «в» (into).

Original source details: GELFAND-P001346

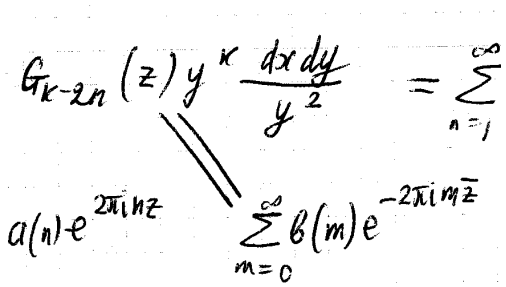
We obtain:

$$\sum_\gamma \int_{\mathcal{F}} f(\gamma z)\overline{G_{k-2n}(\gamma z)}\operatorname{Im}(\gamma z)^k\, dV =$$

$$= \sum_\gamma \int_{\gamma\mathcal{F}} f(z)\overline{G_{k-2n}(z)}(\operatorname{Im} z)^k\, dV =$$

$$= \int_{\underbrace{z\to z+1\setminus H}_{\int_0^\infty \int_0^1}} \underbrace{f(z)}_{\sum_{n=1}^\infty a(n)e^{2\pi i n z}} \underbrace{G_{k-2n}(z)}_{\sum_{m=0}^\infty b(m)e^{-2\pi i m\bar{z}}} y^k \frac{dx\,dy}{y^2}$$

$$= \sum_{n=1}^\infty a(n)\overline{b(n)} \int_0^\infty e^{-4\pi n y} y^{k-2}\, dy$$



Uncertain reading. The third integral has no overbar on $G_{k-2n}(z)$. In the series below it, $b(m)$ also has no overbar.

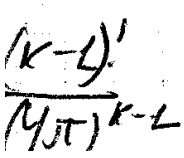
Original source details: GELFAND-P001346

Result:

$$f(z) = \sum_{n=1}^{\infty} a(n) q^n \in S_k,$$

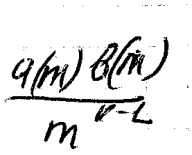
$$G(z) = \sum_{m=0}^{\infty} b(m) q^m \in M_{k-2n}, \qquad n \geq 2$$

$$-\frac{4n}{B_{2n}} \left(f, \underset{k-2n}{G} \cdot \underset{2n}{G_{2n}} \right) = \frac{(k-1)!}{(4\pi)^{k-1}} \underbrace{\sum_{m=1}^{\infty} \frac{a(m)b(m)}{m^{k-1}}}_{\text{(convolution)}} \tag{13}$$



Original source details: GELFAND-P001346

Uncertain reading. The numerator reads $(k-1)!$.



Original source details: GELFAND-P001346

Uncertain reading. No overbar is visible on $b(m)$ in (13).

$$G = G_{k-2n}(z) = (*) + \sum_{m=1}^{\infty} \sigma_{k-1-2n}(m) q^m$$

$$f = \sum_{m=1}^{\infty} a(m) q^m = \text{Hecke form}$$

[Crossed out: $= \prod_p [1 + \frac{a(p)}{p^s} +$]

$$\begin{pmatrix} a(mm') = a(m)a(m') & \text{when } (m, m') = 1 \\ a(2^a 3^b 5^c \ldots) = a(2^a)a(3^b)\ldots \end{pmatrix}$$

Then

$$(14) \qquad \sum \frac{a(m)}{m^s} = \prod_p \left[1 + \frac{a(p)}{p^s} + \frac{a(p^2)}{p^{2s}} + \ldots\right]$$

$$a(m)a(n) = \sum_{d|m,n} d^{k-1} a\left(\frac{mn}{d^2}\right)$$

$$b(p) = 1 + p^{k-1-2n}$$

$\Rightarrow$ the series is evaluated

$$\underbrace{\frac{(k-2)!}{(4\pi)^{k-1}} \sum_{m=1}^{\infty} \frac{a(m)\sigma_{k-1-2n}(m)}{m^{k-1}}}_{L(f,k-1)L(f,2n)/\zeta(2n)} = \underbrace{-\frac{4n}{B_{2n}} \left(f, \overbrace{G_{k-2n} \cdot G_{2n}}^{\alpha f + \beta f_1 + \ldots} \right)}_{\alpha \cdot (f,f)}$$

[Annotation to the expansion] and the f_j are a basis.

Original source details: GELFAND-P001347

Uncertain reading. The sign below $\alpha \cdot (f, f)$ is crossed out and its original shape cannot be read.

$$\alpha \in \overline{\mathbb{Q}}^* \quad \text{(algebraic)}$$

All coeffs. of the Eisenstein series are rational numbers.

Thus

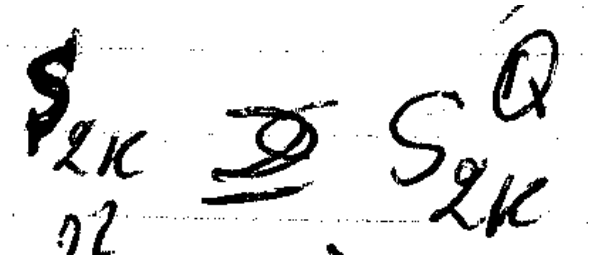
$$L(f, 2n)/\zeta(2n) \quad \text{indep. of } n \quad \Rightarrow \quad \text{the theorem (?) is proved } \triangle$$

Original source details: GELFAND-P001347

Uncertain reading. The sign in parentheses reads as “?” or “o”. The following word reads “доказана” (is proved). The final mark resembles a triangle.

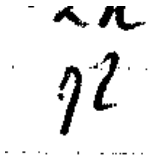
(Rankin, 1953)

$$S_{2k} \qquad S_{2k}^{\mathbb{Q}} = \left\{ f = \sum a(n) q^n,\ a(n) \in \mathbb{Q} \right\}$$



Uncertain reading. Between S_{2k} and $S_{2k}^{\mathbb{Q}}$ is a sign with overlapping strokes and a lower line.

Original source details: GELFAND-P001347



Uncertain reading. Below S_{2k} a mark resembling η precedes a raised 2.

Original source details: GELFAND-P001347

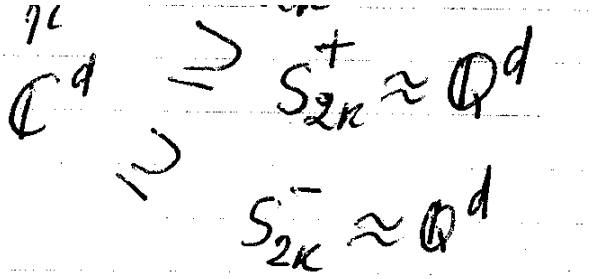


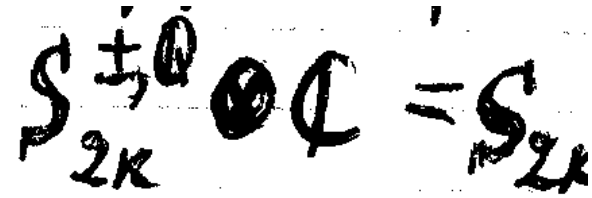
Diagram and typed description.

$$\begin{array}{ccc} & \supseteq & S_{2k}^{+} \approx \mathbb{Q}^d \\ \mathbb{C}^d & & \\ & \subseteq & S_{2k}^{-} \approx \mathbb{Q}^d \end{array}$$

Original source details: GELFAND-P001347

(e.g., 3 different rational structures in S_{2k})

$$S_{2k}^{\pm,\mathbb{Q}} \otimes \mathbb{C} = S_{2k}$$



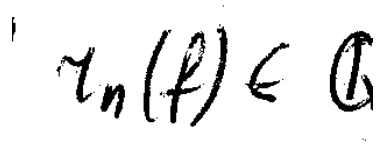
Uncertain reading. The sign between $S_{2k}^{\pm,\mathbb{Q}}$ and $\mathbb{C}$ is heavily overwritten, provisionally read as $\otimes$.

Original source details: GELFAND-P001347

GELFAND-P001348

[source number: 76]

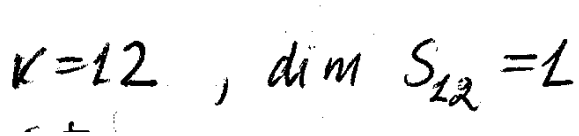
$$S^+_{2k} := \{f \in S_{2k} \mid r_n(f) \in \mathbb{Q},\ n = 0, 2, 4, \ldots, k-2\}$$
$$S^-_{2k} := \{\text{—//—} \qquad n = 1, 3, 5, \ldots\}$$



Uncertain reading. The period symbol reads r_n, as in the definition on frame GELFAND-P001345.

Original source details: GELFAND-P001348

$$k = 12, \qquad \dim S_{12} = 1$$



Uncertain reading. The example shows $k = 12$ and $\dim S_{12} = 1$ side by side.

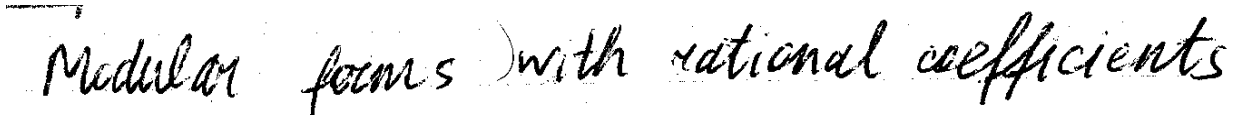
Original source details: GELFAND-P001348

$$S^+_{12} = \{f \mid r_0(f), r_2(f), r_4, r_6, r_8, r_{10} \in \mathbb{Q}\}$$

It is not obvious that the dimension is pres.!

Refs.: article by Kohnen + Zagier

Modular forms with rational coefficients



Original source details: GELFAND-P001348

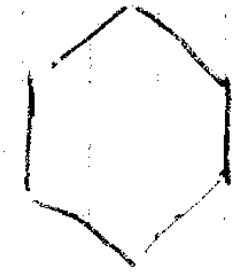
Uncertain reading. The manuscript reads “coefficients”.

(conf. Ed. Rankin — book)

GELFAND-P001349

2/III–87 V. Kazakov.

The problem of gluing surfaces of genus g from a $2n$-gon.



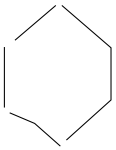
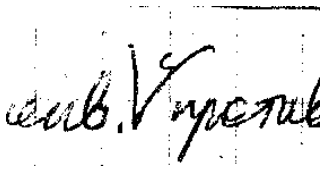


Diagram and typed description.

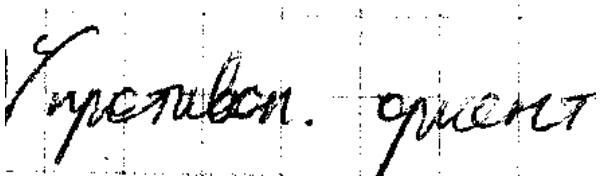
Original source details: GELFAND-P001349

$2n$-gon, gluing oppositely oriented sides (in pairs),



Original source details: GELFAND-P001349

Uncertain reading. Between “gluing” and “oppositely” is a separate mark resembling a check mark or γ.

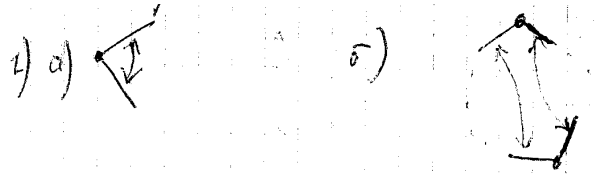


Original source details: GELFAND-P001349

Uncertain reading. The abbreviation before “oriented” is read as «противоп.» (oppositely).

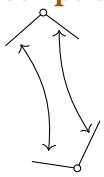
g is the genus of the resulting manifold. In how many different ways does a surface of genus g result? f_n^g is the number of such ways of gluing

(Some gluings may be forbidden. Prohibitions:



Original source details: GELFAND-P001349

Diagram and typed description.

1) a) b) 

2) [Crossed out: OR] Only b) is forbidden. 3) Only a) is forbidden.

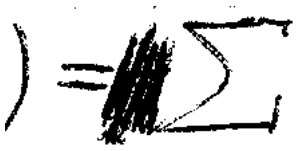
Everything reduces to the case without prohibitions.)

GELFAND-P001350

[source number: 85]

Generating fn.:

$$(1) \qquad f_n(N) = \sum_{g=0}^{\infty} N^{-2g} f_n^g$$



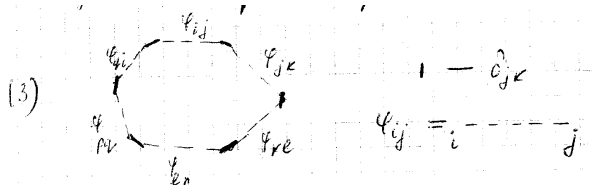
Original source details: GELFAND-P001350

Uncertain reading. A single sign before the summation sign is heavily crossed out.

Then

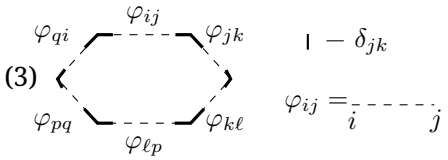
$$(2) \qquad f_n(N) = (2\pi)^{-N^2/2} \int d^{N^2}\varphi \, \frac{\mathrm{tr}}{N}(\varphi^{2n}) e^{-N \,\mathrm{tr}\, \varphi^2/2}$$
$$= \left\langle \frac{\mathrm{tr}}{N} \varphi^{2n} \right\rangle$$

where $\varphi^+ = \varphi$ is an $N \times N$ Hermitian matrix.



Original source details: GELFAND-P001350

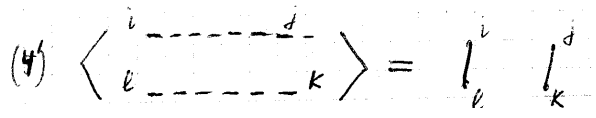
Diagram and typed description.



It is easy to check that

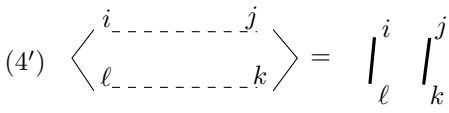
$$(4) \qquad \langle \varphi_{ij} \varphi_{k\ell} \rangle = \frac{1}{N} \delta_{i\ell} \delta_{jk}$$

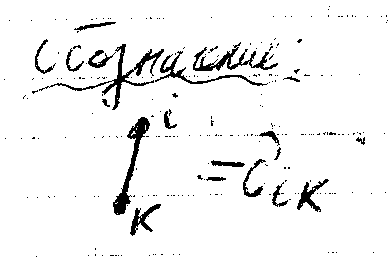
or



Original source details: GELFAND-P001350

Diagram and typed description.





Original source details: GELFAND-P001350

Notation:

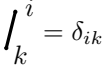
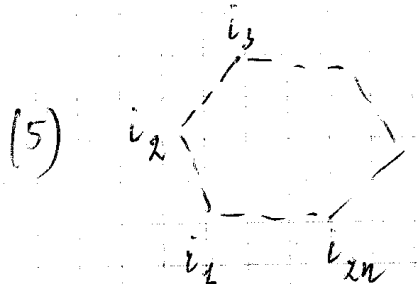
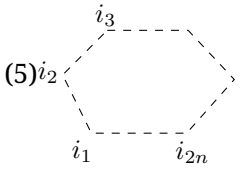


Diagram and typed description.

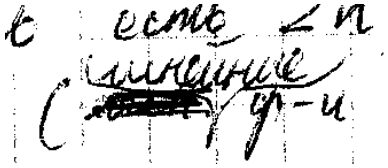


Original source details: GELFAND-P001350

Diagram and typed description.



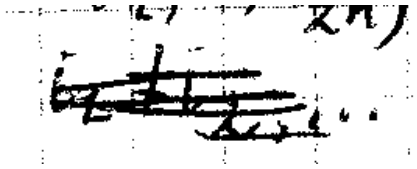
(6) Wick's theorem: Let there be $2n$ Gaussian variables $\xi_1, \xi_2, \ldots, \xi_{2n}$. Then ([Crossed-out writing] [Inserted above: linear] functions of φ_{ik})



Original source details: GELFAND-P001350

Undeciphered passage. The writing below "linear" is crossed out. Its reading is unresolved.

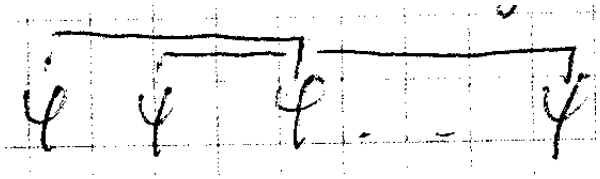
$$\langle \xi_1 \xi_2 \cdots \xi_{2n} \rangle = \sum_{P(i_1,\ldots,i_{2n})} \langle \xi_{i_1} \xi_{i_2} \rangle \langle \xi_{i_3} \xi_{i_4} \rangle \cdots$$



Original source details: GELFAND-P001350

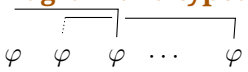
Undeciphered passage. A row of indices below the sum is crossed out. The initial index resembles i_2.

[Line from $P(i_1, \ldots, i_{2n})$] permutation [crossed out: $i_1, i_2, \ldots, i_{2n}$] $1, 2, \ldots, 2n$ The sum is over all partitions into pairs.



Original source details: GELFAND-P001350

Diagram and typed description.



(Exercise: prove?)

Original source details: GELFAND-P001350

Uncertain reading. The part «оказать» is clear. The initial stroke is faint. The likely reading is «доказать» (prove).

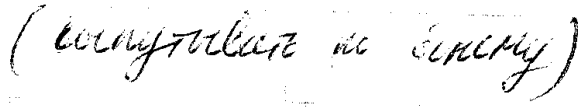
A more general theorem: let ξ be a Gaussian variable, then

$$\langle \xi f(\xi) \rangle = \langle \xi \xi \rangle \left\langle \frac{\partial f}{\partial \xi} \right\rangle \tag{7}$$

$$\langle \xi_i f(\xi) \rangle = \sum_j \langle \xi_i \xi_j \rangle \left\langle \frac{\partial f}{\partial \xi_j} \right\rangle \tag{8}$$

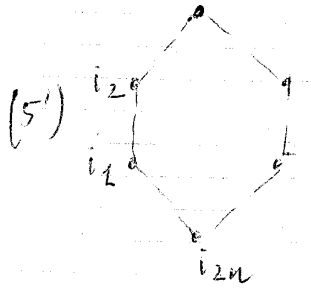
GELFAND-P001351

Using (7), (8), it is easy to prove Wick's theorem (by induction on the number of ξ's).



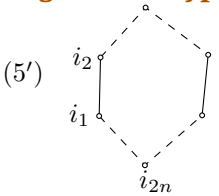
Original source details: GELFAND-P001351

Uncertain reading. The opening of the parenthesis is faint. "By induction on the number of ξ's" is provisional.



Original source details: GELFAND-P001351

Diagram and typed description.



We compute integral (2) using Wick's theorem. It is clear that

$$(9) \qquad \langle \operatorname{tr} \varphi^{2n} \rangle = \langle \varphi_{i_1 i_2} \varphi_{i_2 i_3} \cdots \varphi_{i_{2n} i_1} \rangle$$

Uncertain reading. The sign $\sum$ is crossed out between the equals sign and the bracket.

Original source details: GELFAND-P001351

Now let us apply Wick's theorem. It tells us to pair the sides! (of polygon $(5')$)
Gluing.

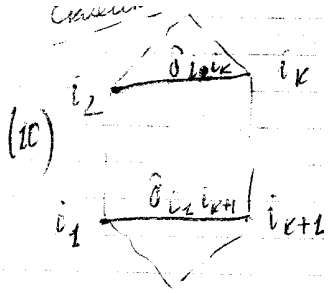
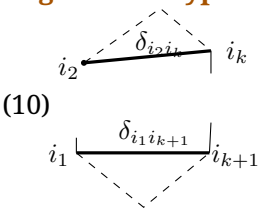


Diagram and typed description.



Original source details: GELFAND-P001351

Uncertain reading. To the right of (10), φ and another notation are crossed out. Lower strokes are visible, but the reading is unresolved.

Original source details: GELFAND-P001351

Formula (4) ensures the correct orientation in the gluing (because of Hermiticity!!)
(Choosing a matrix differently can give a nonorientable surface.)

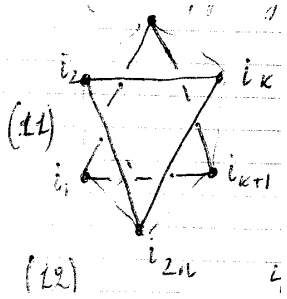
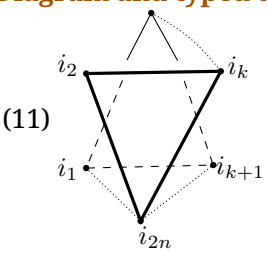


Diagram and typed description.



Original source details: GELFAND-P001351

Closed lines $\longrightarrow$ property of the Kronecker symbol, i.e. always N! For example,

$$N = \delta_{i_2 i_k} \, \delta_{i_k i_{2n}} \, \delta_{i_{2n} i_2}$$

$\forall$ closed cycles, the contribution is N
(12) Number of closed cycles = number of vertices in the glued closed figure.
(13) Contribution of each Wick pairing = (diagram)

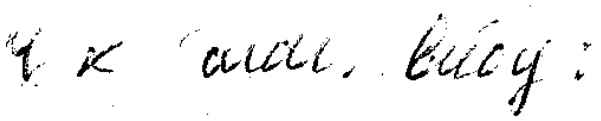
$$= N^{-(\#\text{ edges})+(\#\text{ vertices})-1} = N^{-2g}$$

(because of $\frac{N}{2} \operatorname{tr} \varphi^2$ in the exp)

GELFAND-P001352

[source number: 86]

Let us compute integral (2). We reduce it to diag. form:



Uncertain reading. The abbreviation before "form" is faint, provisionally read as "diag.".

Original source details: GELFAND-P001352

$$(14) \qquad \varphi_{ij} = \sum_k \Omega^+_{ik} \lambda_k \Omega_{kj}, \qquad \Omega \text{ is a unit. matrix}$$

$$(15) \qquad d^{N^2}\varphi = \left(\prod d\lambda_i \, [\Delta(\lambda)]^2\right) d\Omega,$$

where

$$\Delta(\lambda) = \prod_{i>j} (\lambda_i - \lambda_j)$$

Uncertain reading. A parenthesized number at left is crossed out, possibly (14) or (16).

Original source details: GELFAND-P001352

[In a box at right] (See H. Weyl: Classical groups, their invariants and representations.) (Dyson measure).

([Above the start of the list: Brezin(?)], Itzykson, Zuber, Parisi,)

Uncertain reading. The name above the start of the list is read as "Brezin". The question mark is in the manuscript.

Original source details: GELFAND-P001352

[Line down from Parisi] method of stoch. quantization quantum quantity $\longmapsto$ Markov process. The mean is computed by averaging over trajectories. [At left, arrow to (16)] works here.

$$(16) \qquad f_n(N) \sim \int \prod \left(d\lambda_i \, e^{-N\lambda_i^2/2}\right) \Delta^2(\lambda) \left(\frac{1}{N} \sum_{i=1}^N \lambda_i^{2n}\right) =$$

$$\underset{\text{symm. in } \lambda_j}{=} \int \prod \left(d\lambda_i \, e^{-N\lambda_i^2/2}\right) \Delta^2(\lambda) \lambda_1^{2n}$$

$$\sim \int \prod \left(d\lambda_i \, e^{-\lambda_i^2/2}\right) \Delta^2(\lambda) \lambda_1^{2n}$$

Uncertain reading. A separate mark is crossed out before the second integral in (16).

Original source details: GELFAND-P001352

(var. λ_i, we take out $\sqrt{N}$ in λ_i)

Uncertain reading. A separate mark is crossed out before $\sqrt{N}$.

Original source details: GELFAND-P001352

$$(17) \qquad \Delta(\lambda) = \det_{k,\ell} \|(\lambda_\ell)^k\| = \det \|H_k(\lambda_\ell)\|$$

H_k is any polynomial of degree k with leading coeff. 1. Let us choose H_k to be Hermite polynomials!! (with a normalizing factor so the leading coeff. $= 1$) (good, since H_k and H_ℓ are orthogonal)

Hence:

$$(18) \qquad f_n(N) \sim \frac{1}{C} \int \prod_{i=1}^{N} \left(d\lambda_i \, e^{-\lambda_i^2/2} \right) \lambda_1^{2n} \times \left[\sum_{\substack{p \\ \text{perm.}}} \prod_{k=1}^{N} H_k(\lambda_{p_k}) (-1)^{\operatorname{sgn} p} \right]^2 =$$

Uncertain reading. The number at left is faint, read as (18).

Original source details: GELFAND-P001352

(Parisi and Bessis devised the trick with Hermite polynomials) On squaring, every var. λ_{p_k} occurs exactly 2 times.

GELFAND-P001353

$$= \sum_{k=0}^{N-1} \frac{\int d\lambda \, e^{-\lambda^2} [H_k(\lambda)]^2 \lambda^{2k}}{\int d\lambda \, e^{-\lambda^2} (H_k(\lambda))^2}$$

Uncertain reading. The exponent on λ in the numerator is provisionally read as $2k$.

Original source details: GELFAND-P001353

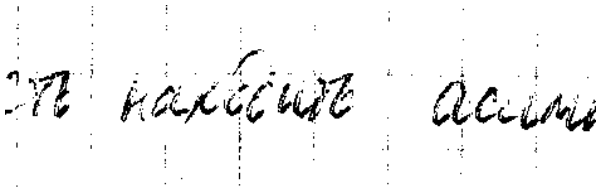
From a handbook (e.g., Abramowitz):

$$(19) \qquad \int d\lambda\, e^{i\lambda x} e^{-\lambda^2} [H_k(\lambda)]^2 = \underbrace{\sqrt{\pi}\, 2^n n!}_{\substack{\text{cancels} \\ \text{with the denominator}}} \quad e^{-x^2/4} \quad \underbrace{L_n(x^2/2)}_{\text{Laguerre poly.}}$$

In the end we get [arrow to (20)]:

$$(20) \qquad \sum_{n=0}^{\infty} \frac{(-x^2/2)^n}{(2n)!!} f_n(N) = e^{-x^2/4} \sum_{i=0}^{N-1} L_i(x^2/2)$$
$$= e^{-x^2/4} \quad \underbrace{L^1_{N-1}(x^2/2)}_{\text{assoc. Laguerre polynomials}}$$

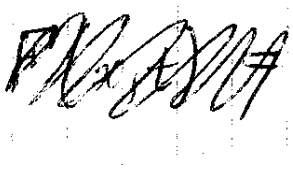
Here we are most [interested in] the asympt. as $N \to \infty$. Let us form the generating fn.:



Uncertain reading. The word before "asympt." is provisionally read as «наиболее» (most).

Original source details: GELFAND-P001353

[Crossed out before (21)] $F(x,t)$



Uncertain reading. Two horizontal strokes are visible under the crossing out after $F(x,t)$, possibly an equals sign.

Original source details: GELFAND-P001353

$$(21) \qquad f_n(t) = \sum_N t^N f_n(N)$$

The answer for it is:

$$(22) \qquad f_n(t) = \frac{(2n)!}{n!} \frac{(t+1)^n}{(1-t)^{n+2}}$$

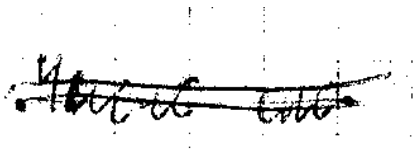
We are interested in the series expansion in t. Zakharevich: answer (after D. Zagier)

$$(23) \qquad \sum_{g=0}^{\infty} \frac{(n+1)!(n-2g)!}{(2n)!} f_g^n x^g = \left(\frac{\sqrt{x/2}}{\operatorname{th} \sqrt{x/2}} \right)^{n+1}$$

[An isolated left parenthesis in the margin] (when $n - 2g < 0$, multiply by $(2n)!$ and move it to the right-hand side) $f_g^n (n-2g)!$ is thereby defined also for $2g > n$(!).

Variant: 2 other combinatorial problems (worked on by Kazakov and Migdal).

① How to tile a $2n$-gon with k triangles so that the resulting surface has genus g.



Uncertain reading. "Number of gl." is visible beneath the crossing out.

Original source details: GELFAND-P001353

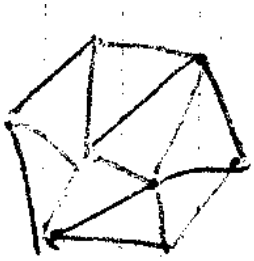
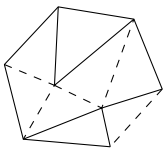
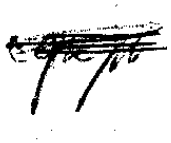


Diagram and typed description.



Original source details: GELFAND-P001353

(attaching handles is allowed!) (Simplicial complex of a given genus) $f_k^{2n}(g)$ (Bessis). (Number of Triangulations [crossed out] of a surface of genus g with k triangles, poss. $2n$-gons)



Undeciphered passage. The word between "Triangulations" and "surface" is crossed out and illegible.

Original source details: GELFAND-P001353

GELFAND-P001354

[source number: 87]

(1971–79, Itzykson et al.)

②

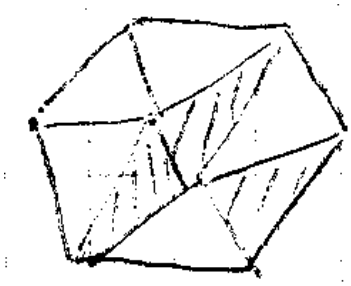
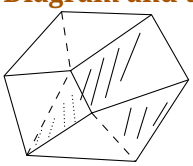
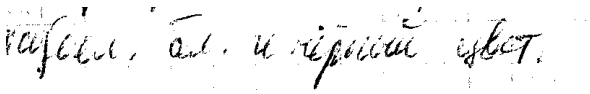


Diagram and typed description.



Original source details: GELFAND-P001354

The same, but we colour it and fix the boundary gluing, using white and black.



Uncertain reading. The faint abbreviation before "white and black" may read «раскр.» or «раздел.».

Original source details: GELFAND-P001354

GELFAND-P001355

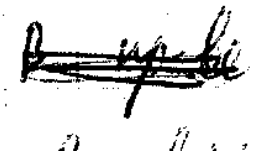
9/III–87. Gelfand Sem. I. M. G. There is a paper which says the following. Consider

$$z \longmapsto \frac{\alpha z + \beta}{\gamma z + \delta}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{Z}, \qquad \text{in the half-plane, } \operatorname{Im} z > 0$$

Uncertain reading. A short undeciphered entry is crossed out after $\mathbb{Z}$.

Original source details: GELFAND-P001355

$$\alpha\delta - \gamma\beta > 0$$

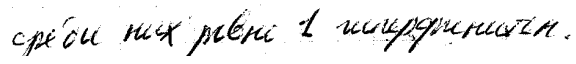


Original source details: GELFAND-P001355

Uncertain reading. A phrase is crossed out before “There exist”, possibly «в кр-ве» (abbreviation unresolved).

There exist sets of nonzero intermediate Hausd. measure, which arise in connection with the principal series.

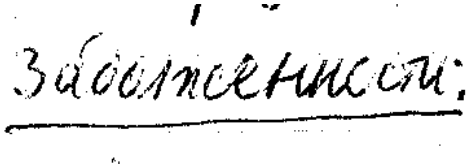
Conjecture: factors arise in reprs. of ∞-dimensional groups. In physics, factors of type $\overline{\text{III}}_\infty$ arise. Exactly 1 among them is hyperfinite.



Original source details: GELFAND-P001355

Uncertain reading. The last word in the line is read as «гиперфинитн.» (hyperfinite).

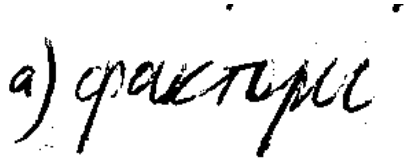
Talkativeness?:



Original source details: GELFAND-P001355

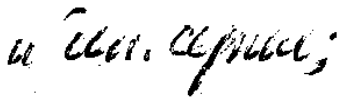
Uncertain reading. The heading is provisionally read as «Заболтенности» (talkativeness) or «Заболезненности» (ailments).

a) factors, b) attractors, c) chaos, Hausdorff measures and infin. series,



Original source details: GELFAND-P001355

Uncertain reading. The first word in the list reads “factors”.



Original source details: GELFAND-P001355

Uncertain reading. The abbreviation before “series” is provisionally read as «беск.» (infin.).

d) characteristic functions.

A. Beilinson. Basic facts about the Riemann ζ-function and its generalizations.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

1. Euler product:

$$\zeta(s) = \prod_{p} (1 - p^{-s})^{-1}$$

2. Analytic continuation: a simple pole at $s = 1$, holomorphic at the other points.
3. Functional equation. Introduce a more complete ζ-function:

$$\zeta^*(s) = \underbrace{\pi^{-s/2}\Gamma(s/2)}_{\text{Euler factor numbered } \infty} \zeta(s),$$

then $\zeta^*(s)$ is invariant under $s \leftrightarrow 1 - s$ (unlike the p-metric)

[In the left margin] Remark 1. Kirillov's. The π [crossed out] factor is the volume of an s-dimensional ball.

Original source details: GELFAND-P001355

Uncertain reading. Before the crossed-out sign the letter resembles π or Π.

Original source details: GELFAND-P001355

Undeciphered passage. A short sign is crossed out between the first letter and "factor".

Original source details: GELFAND-P001355

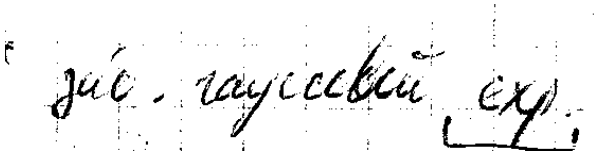
Uncertain reading. The bottom line after the dash reads "volume of an s-dimensional ball".

$$\mathbb{R}^* \qquad t^{-s} \qquad e^{-t^2}$$

Γ arises as a Mellin transform.

$$\mathbb{Q}_p^* \qquad |t|^s \qquad \text{char. fn. of} \quad \underbrace{(\mathbb{Z}_p \setminus 0)}_{\text{open subgroup}} \subset \mathbb{Q}_p^*$$

its analogue in $\mathbb{R}^*$ [abbr.] of a Gaussian exp.



Uncertain reading. After $\mathbb{R}^*$ is a faint abbreviation, possibly «дис.».

Original source details: GELFAND-P001356

4. Trivial zeros: the functional eq. makes it clear that $\zeta(s)$ has zeros (of multiplicity 1) at $s = -2k$, $k \in \mathbb{Z}$, $k > 0$. (Because of the Γ factor.)

Uncertain reading. A short entry under item 4 is heavily crossed out.

Original source details: GELFAND-P001356

5. Rational values:

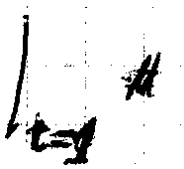
$$\zeta(1-n) = -\frac{B_n}{n}$$

(at negative points)

$$\zeta(-n) = \sum k^n = \left(t\frac{d}{dt}\right)^n \frac{1}{1-t}\bigg|_{t=1} \qquad \#$$

Uncertain reading. A short entry before $(t\,d/dt)^n$ is heavily crossed out.

Original source details: GELFAND-P001356



Uncertain reading. To the right of the evaluation $t = 1$ is a heavy mark of intersecting strokes, read as #.

Original source details: GELFAND-P001356

(more precisely, take the value of the first regular term)

6. Regulator formulas: $\forall n$, $\exists$ a natural subgroup in $\mathbb{R}$, $\Gamma_n \subset \mathbb{R}$, such that [Crossed out] $\zeta(1-2n) = (\text{vol}\,\mathbb{R}/\Gamma_n) \cdot (\text{rational number})$.

$$\zeta'(-2n) = (\text{vol}\,\mathbb{R}/\Gamma_n) \cdot (\text{rational number})$$

$\Gamma_n =$ image of $K_{4n+1}(\mathbb{Z})$ in $\mathbb{R}$.

7. Arithmetic properties of Bernoulli numbers: p-adic properties of $\zeta(1-n)$

$$\zeta(s)(1-p^{-s})$$

$$\zeta(1-n)(1-p^{-1+n}) = \int_{\mathbb{Z}_p^*} x^{1-n}\, d\mu$$

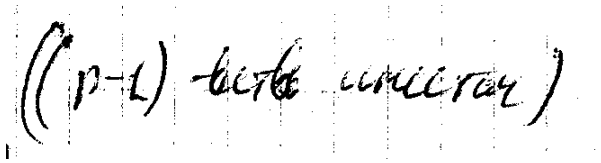
where $d\mu$ is a natural measure on $\mathbb{Z}_p^*$ (with values in $\mathbb{Z}_p$)

$$\mathbb{Z}_p^* = (1+p\mathbb{Z}_p) \times \sqrt[p-1]{1}$$

Uncertain reading. A mark resembling E is crossed out at left.

Original source details: GELFAND-P001356

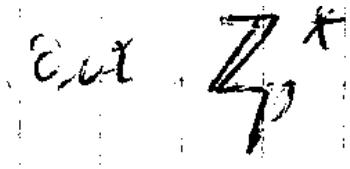
($(p-1)$th roots?)



Uncertain reading. The word after "$(p-1)$th" is faint, provisionally read as "roots".

Original source details: GELFAND-P001356

The set of characters [undeciphered] $\mathbb{Z}_p^*$ consists of $(p-1)$ components.



Uncertain reading. Before $\mathbb{Z}_p^*$ is a short word resembling «сил».

Original source details: GELFAND-P001356

8. Riemann hypothesis.

9. Connection with the distribution of prime numbers.

GELFAND-P001357

[source number: 91]

Generalizations of the Riemann ζ-function.

Artin L-functions $\quad L(\rho, s)$

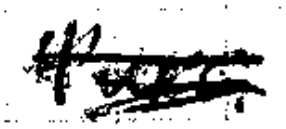
$$\rho : \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q} \underset{\text{continuous}}{\longrightarrow} GL(n, \mathbb{C})$$

Uncertain reading. A short entry is crossed out between Gal and $\overline{\mathbb{Q}}/\mathbb{Q}$.

Original source details: GELFAND-P001357

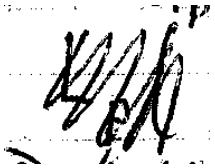
$$L(\rho, s) = \prod_p L_p(\rho, s), \qquad p \text{ is a prime number}$$

$\overline{\mathbb{Q}}_p$ is an alg. closure of the field $\mathbb{Q}_p$. Fix an embedding $\overline{\mathbb{Q}} \subset \overline{\mathbb{Q}}_p \quad \Rightarrow$



Uncertain reading. A word before $\overline{\mathbb{Q}}_p$ is crossed out.

Original source details: GELFAND-P001357



Uncertain reading. Below the start of the line a crossed-out entry resembles $\mathbb{Q}_p/\mathbb{Q}$ or $\overline{\mathbb{Q}}_p/\mathbb{Q}$.

Original source details: GELFAND-P001357

$\Rightarrow$ an embedding

$$\mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q}$$

Only the conjugacy class of these maps is invariantly determined.

Inertia group:

$$I \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p$$

[Line from I] acts trivially on $\overline{\mathbb{F}}_p = \overline{\mathbb{Z}/p\mathbb{Z}}$

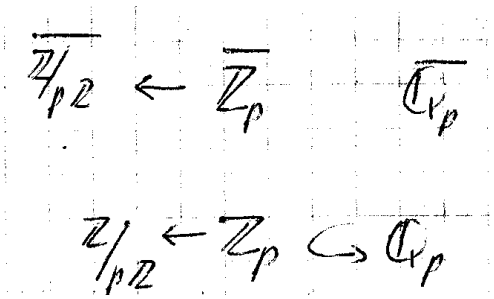
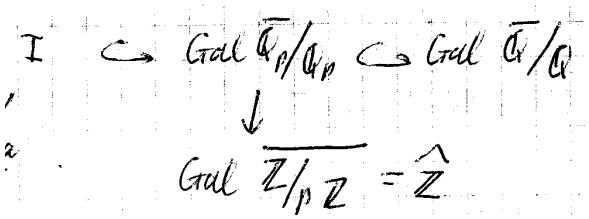


Diagram and typed description.

$\overline{\mathbb{Z}/p\mathbb{Z}} \longleftarrow \overline{\mathbb{Z}}_p \qquad \overline{\mathbb{Q}}_p$

$\mathbb{Z}/p\mathbb{Z} \longleftarrow \mathbb{Z}_p \hookrightarrow \mathbb{Q}_p$

Original source details: GELFAND-P001357



Original source details: GELFAND-P001357

Diagram and typed description.

$$I \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q}$$
$$\downarrow$$
$$\mathrm{Gal}\,\overline{\mathbb{Z}/p\mathbb{Z}} = \widehat{\mathbb{Z}}$$

[In the left margin, line to I] fixes $\sqrt[N]{1}$, $(N,p)=1$

Original source details: GELFAND-P001357

Diagram and typed description. Below $\widehat{\mathbb{Z}}$ before "generator" is a short vertical stroke over a slanting line. It may be an underlined 1.

generator Fr_p (Frobenius elem. $x \mapsto x^p$)

ρ continuous $\Rightarrow$ for a.e. p except a finite number, $\rho|_I$ is trivial. Unramified p: $\rho(I_p) = 1$

Refs.: Cassels, Fröhlich, "Alg. Number Theory"

$$L_p(\rho, s) = \det\left(1 - \rho(\mathrm{Fr}_p)p^{-s}\right)^{-1}$$

Example 1. Dirichlet L-functions

ρ is a 1-dimensional repr. given by a Dirichlet character

$$\chi : (\mathbb{Z}/N\mathbb{Z})^* \longrightarrow \mathbb{C}^*$$

From it one constructs a char. of the Galois group

$$\mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q} \longrightarrow (\mathbb{Z}/N\mathbb{Z})^* \xrightarrow{\chi} \mathbb{C}^*$$

$$L(\chi, s) = \sum_{(n,N)=1} \frac{\chi(n)}{n^s} = \prod_{(p,N)=1} \left(1 - \chi(p)p^{-s}\right)^{-1}$$

GELFAND-P001358

Example 2. $K/\mathbb{Q} \subset \overline{\mathbb{Q}}$

[In the upper right margin:]

Original source details: GELFAND-P001358

Uncertain reading. an isolated stroke; no legible page number

$$\operatorname{Gal}\overline{\mathbb{Q}}/K \subset \operatorname{Gal}\overline{\mathbb{Q}}/\mathbb{Q}$$

Take the induced representation, and then the corresponding L-function is $\zeta_K(s)$

$$L(\operatorname{Ind}_{K/\mathbb{Q}}, s) = \zeta_K(s) = \prod_{\substack{p \text{ prime} \\ \text{ideals in } K}} (1 - N(p)^{-s})^{-1}$$
$$= \sum_{\substack{\text{integral} \\ \text{ideals } I \text{ in } K}} N(I)^{-s}$$

Properties

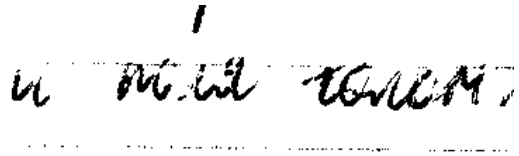
0. $L(\rho, s)$ uniquely determines ρ

! K is not recovered from $\zeta_K(s)$

$$L(\rho_1 \oplus \rho_2, s) = L(\rho_1, s)L(\rho_2, s)$$

1. Analytic continuation

L-functions of a field? almost all?[?] admit holomorphic continuation.



Uncertain reading. of a field? almost all?

Original source details: GELFAND-P001358

All admit meromorphic continuation

Conjecture. $\forall$ irreducible nontrivial ρ: $L(\rho, s)$ is holomorphic in s.

[Insertion beneath the line:] (if separate curved stroke; ρ?[?] passed through (abbreviated reading uncertain)[?] $GL(n, \mathbb{C})$ — an Artin function)

Uncertain reading. separate curved stroke; ρ?

Original source details: GELFAND-P001358

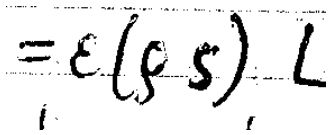
Uncertain reading. passed through (abbreviated reading uncertain)

Original source details: GELFAND-P001358

For l-adic representations, even meromorphicity is unknown

2. Functional equation. Known for Artin L-functions

$$L^*(\rho, s) = \underbrace{\varepsilon(\rho s)}_{\substack{\text{an important const is contained here} \\ \text{(Gauss sums)}}} L^*(\rho^*, 1-s)$$



Original source details: GELFAND-P001358

Uncertain reading. no comma can be made out in $\varepsilon(\rho s)$

Trivial zeros and so on.

Chapter 30

GELFAND-P001359

23.3.87 Gelfand seminar.

V. E. Zakharov. $\bar{\partial}$-problems and integrable systems.

Construction of integrable equations by the inverse-problem method. (actually neither integrable nor by the inverse-problem method)

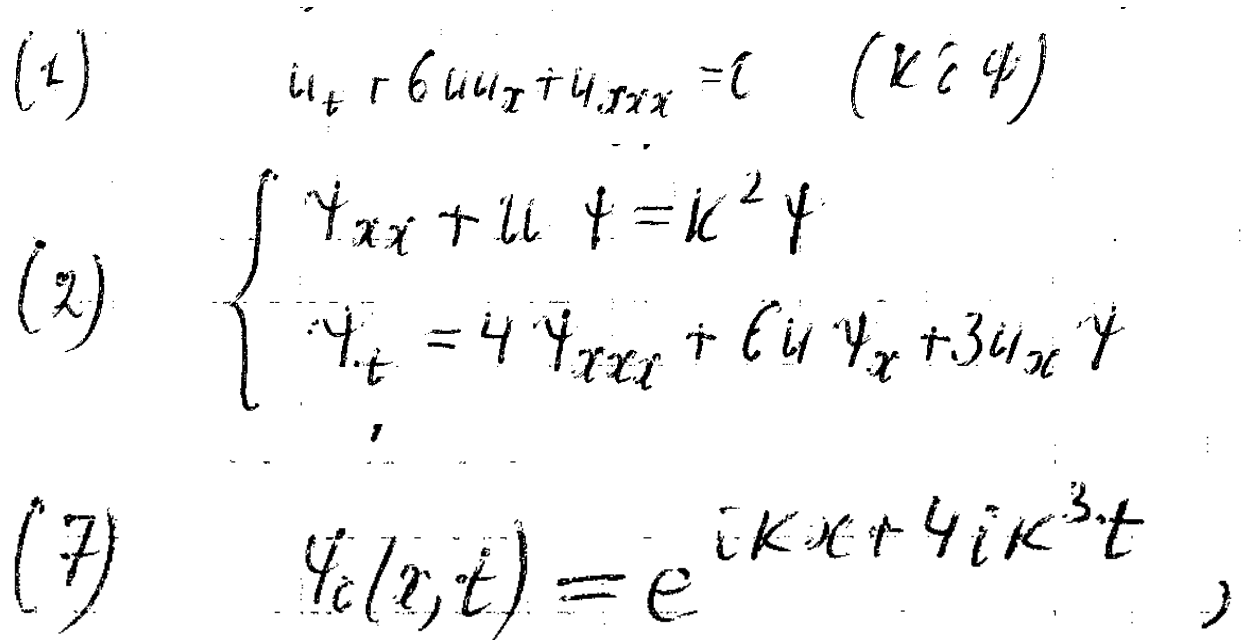
1973 A. B. Shabat formulated the dressing method.

Nonlinear equations are considered as compatibility conditions for overdetermined linear systems

$$u_t + 6uu_x + u_{xxx} = 0 \qquad \text{(KdV)} \tag{1}$$

— the compatibility condition for the equations

$$\begin{cases} \psi_{xx} + u\psi = k^2\psi, \\ \psi_t = 4\psi_{xxx} + 6u\psi_x + 3u_x\psi. \end{cases} \tag{2}$$

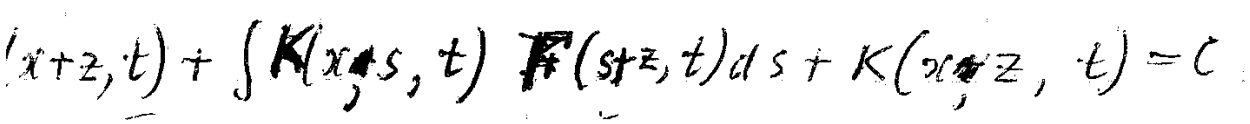


Original source details: GELFAND-P001359

Uncertain reading. The sign before $6uu_x$ is incomplete and is read as $+$. Compare the signs in (2) and (7).

The method consists in considering the equation

$$F(x+z,t) + \int K(x,s,t)F(s+z,t)\,ds + K(x,z,t) = 0. \tag{3}$$



Original source details: GELFAND-P001359

Uncertain reading. In the arguments of K, plus signs are crossed out and commas placed below. The letters underneath the retraced K and F are illegible.

for any function F for which

$$\frac{\partial}{\partial t}F(\xi,t)=8\frac{\partial^3 F}{\partial \xi^3}. \tag{4}$$

In (3), the integral is understood in one of three ways:

$$\int=\int_x^\infty \quad \text{or} \quad \int_{-\infty}^x \quad \text{or} \quad \frac{1}{2}\left(\int_x^\infty \ldots-\int_{-\infty}^x \ldots\right). \tag{5}$$

Then

$$u=2\frac{d}{dx}K(x,x,t) \quad \text{— a solution of (1).} \tag{6}$$

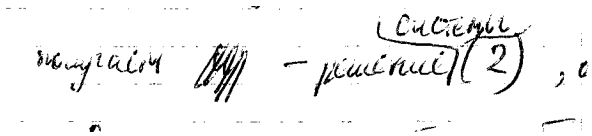
Taking

$$\psi_0(x,t)=e^{ikx+4ik^3\cdot t}, \tag{7}$$

and then

$$\psi=\psi_0(x)+\int K(x,s)\psi_0(s)\,ds, \tag{8}$$

we obtain [crossed out: illegible symbol, possibly $\psi^?$] — a solution of [inserted above the line: the system] (2), from which u is found.



Uncertain reading. illegible symbol, possibly ψ

Original source details: GELFAND-P001359

Consider also more general functions F satisfying the equations

$$\begin{cases}\dfrac{\partial^2 F}{\partial x^2}-\dfrac{\partial^2 F}{\partial z^2}=0,\\ \dfrac{\partial F}{\partial t}=4\left(\dfrac{\partial^3 F}{\partial x^3}+\dfrac{\partial^3 F}{\partial z^3}\right).\end{cases} \tag{9}$$

GELFAND-P001360

[source number: 24]

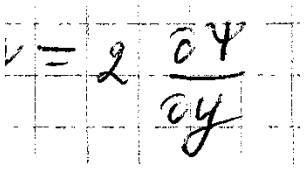
Then everything will hold as before.

Generalization. In place of $(9)_1$ we write:

$$\frac{\partial^2 F}{\partial x^2}-\frac{\partial^2 F}{\partial z^2}=\alpha\frac{\partial F}{\partial y}. \tag{10}$$

Then

$$\begin{cases}\psi_{xx}+u\psi=2\dfrac{\partial\psi}{\partial y},\\ \psi_t=4\psi_{xxx}+6u\psi_x+3u_x\psi\end{cases} \tag{11}$$



Uncertain reading. The coefficient in (11) reads 2. The α in (10) and (12) has a different shape.

Original source details: GELFAND-P001360

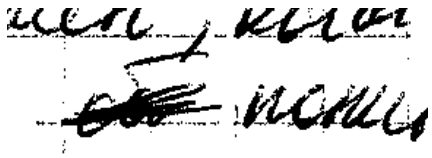
and for u we obtain:

$$\frac{\partial}{\partial x}\left[u_t + 6uu_x + u_{xxx}\right] = 3\alpha^2 u_{yy}. \tag{12}$$

y becomes a new parameter in (3).

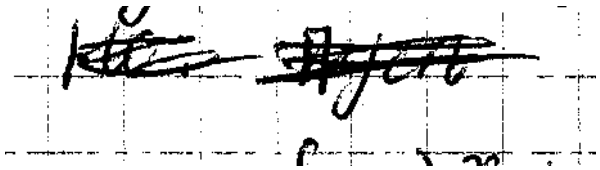
This was in 1974 in the work of Zakharov and Shabat.

The third integral in (5) is defined when the function decays in both directions. However, there is another way to [crossed out: illegible beginning of a word?] interpret the integral:



Uncertain reading. illegible beginning of a word

Original source details: GELFAND-P001360



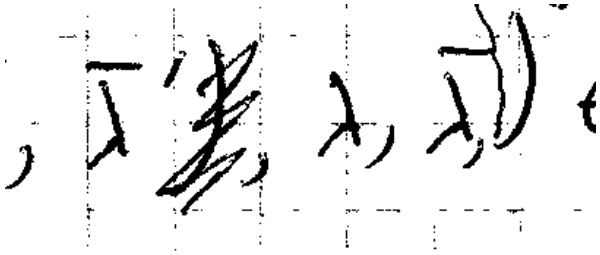
Uncertain reading. Crossed out on the line below: possibly “as Fourier”.

Original source details: GELFAND-P001360

$$\int e^{\lambda x}\,dx = \frac{1}{\lambda}e^{\lambda x} \qquad \text{and so on} \tag{13}$$

We will interpret the integral in the third way in (5). Introduce the function T by the formula

$$F(x,z) = \frac{1}{2\pi}\iint T(\lambda',\bar\lambda',\lambda,\bar\lambda,t,y)\, e^{i(\lambda' x - \lambda z)}\,d\lambda'\,d\bar\lambda'\,d\lambda\,d\bar\lambda. \tag{14}$$



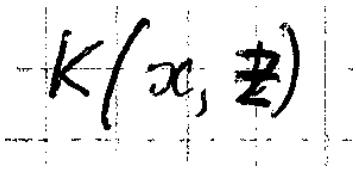
Uncertain reading. An illegible symbol after $\bar\lambda'$ is crossed out.

Original source details: GELFAND-P001360

[Inserted above the line before the closing parenthesis: t, y.]

and define the function K as

$$K(x,z) = -\frac{1}{2\pi}\int K(\lambda,\bar\lambda,x,t,y)e^{i\lambda(x-z)}\,d\lambda\,d\bar\lambda. \tag{15}$$



Uncertain reading. The second argument on the left is z, written over an earlier illegible symbol.

Original source details: GELFAND-P001360

[Inserted above the line before the closing parenthesis: t, y.]

This is a way to find growing solutions of the G.–L.–M. equation.

If K and F satisfy the integral equation, all is well (we use convention (13)).

[crossed out: $\chi(x, y)$.] Set:

$$\chi(\lambda, \bar{\lambda}, x, y, t) = 1 + \frac{1}{2\pi i} \int \frac{K(\eta, \bar{\eta}, x, y, t)}{\lambda - \eta}\, d\eta\, d\bar{\eta}. \tag{16}$$

Here:

$$\frac{1}{\lambda - \eta} = \lim_{\varepsilon \to 0} \frac{\bar{\lambda} - \bar{\eta}}{|\lambda - \eta|^2 + \varepsilon^2}. \tag{17}$$

Let us write the equation:

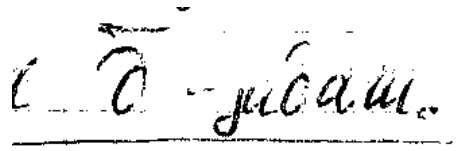
GELFAND-P001361

$$\frac{\partial \chi}{\partial \bar{\lambda}} = \int \chi(\eta, \bar{\eta}, x_i) T(\eta, \bar{\eta}, \lambda, \bar{\lambda}, x_i)\, d\eta\, d\bar{\eta}. \tag{18}$$

$$x_i = (x, y, t)$$

This is equivalent to the original approach.

The dressing method in the language of $\bar{\partial}$-problems.



Uncertain reading. The heading may end with "problems" or "problem".

Original source details: GELFAND-P001361

Let T satisfy the system.

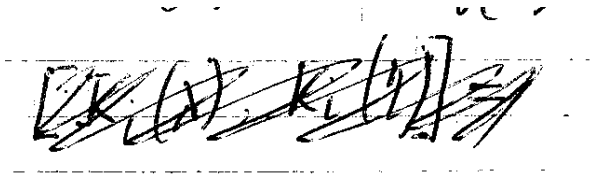
$$\begin{cases} \dfrac{\partial T}{\partial x} = -i(\lambda - \eta)T, \\ \dfrac{\partial T}{\partial y} = -\alpha(\lambda^2 - \eta^2)T, \\ \dfrac{\partial T}{\partial t} = -4(\lambda^3 - \eta^3)T, \end{cases} \tag{19}$$

from which

$$T = T_0(\eta, \bar{\eta}, \lambda, \bar{\lambda}) e^{-i(\lambda - \eta)x - \alpha(\lambda^2 - \eta^2)y - 4(\lambda^3 - \eta^3)t}. \tag{20}$$

In place of (19) one can introduce $x = x_i = (x_1, x_2, x_3)$ and the functions

$$K_i(\lambda) \quad \text{and} \quad I_i(\lambda):$$



Uncertain reading. In (21) a line resembling $[K_i(\lambda), K_i(\eta)] =$ is crossed out. The subscripts and the end are uncertain.

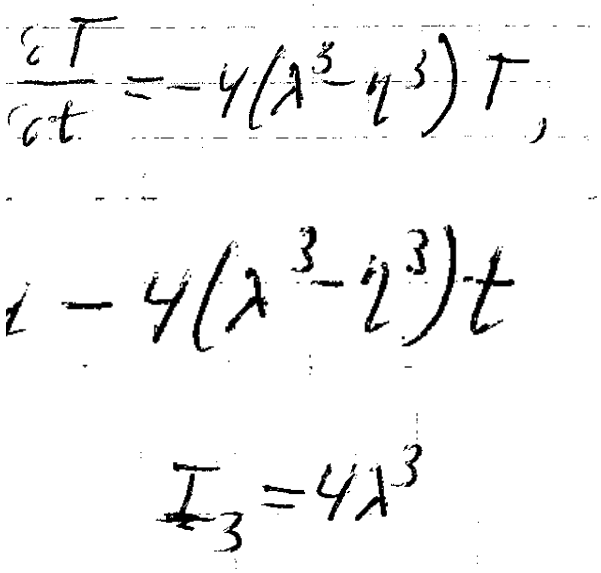
Original source details: GELFAND-P001361

$$[I_i(\lambda), I_j(\eta)] = 0. \tag{21}$$

$$\frac{\partial T}{\partial x_i} = I_i(\eta)T - TI_i(\lambda). \tag{22}$$

In the example

$$I_1 = i\lambda, \quad I_2 = \alpha\lambda^2, \quad I_3 = 4\lambda^3. \tag{23}$$



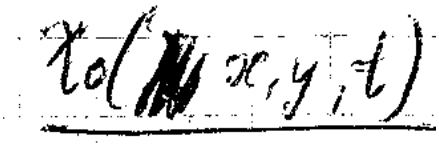
Uncertain reading. The cubic terms in (19), (20), and (23) have 4 without i.

Original source details: GELFAND-P001361

Now solve the nonlocal $\bar{\partial}$-problem (18) with the normalization condition $\chi(\infty) = 1$.

Let

$$\chi = 1 + \frac{\chi_0(x, y, t)}{\lambda} + \dots \tag{24}$$



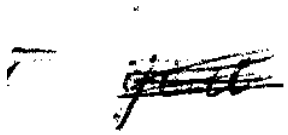
Uncertain reading. An argument before x, y, t in the numerator is crossed out, possibly λ.

Original source details: GELFAND-P001361

Then

$$u = 2i\frac{\partial \chi_0}{\partial x} \tag{25}$$

satisfies [crossed out: possibly "the formula"[?]] the KP equation, (12).



Uncertain reading. possibly "the formula"

Original source details: GELFAND-P001361

GELFAND-P001362

[source number: 25]

(26) Main assertion:

For K_j one can take any three commuting rational matrix functions.

Original source details: GELFAND-P001362

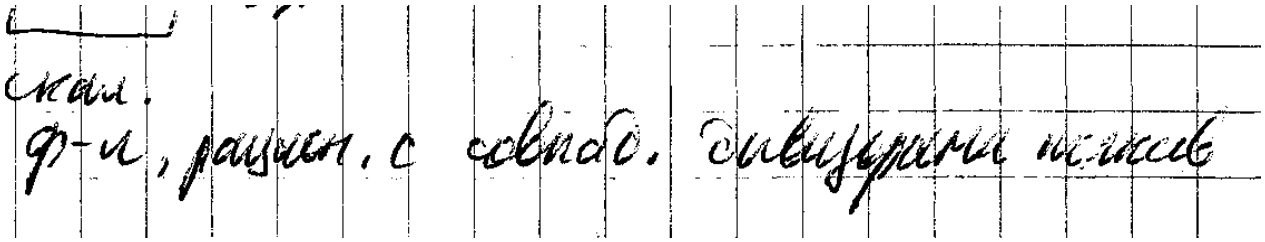
Uncertain reading. The K is written over an earlier illegible sign.

One can also take functions on a Riemann surface.

(27) New proposal.

take $K_i(\lambda)$ to be commuting differential operators with coefficients rational in λ

$$L_i = \frac{\partial}{\partial x_i} + K_i(\lambda) + \underbrace{f_i(\lambda)}_{\substack{\text{scalar functions, rational,} \\ \text{with freely moving poles}}} \frac{\partial}{\partial \lambda}. \tag{28}$$



Original source details: GELFAND-P001362

Uncertain reading. The final word of the caption is more likely “полюсов” than “полюсами” (both mean “poles” here).

(29) $[L_i, L_j] = 0 \Rightarrow$ conditions on K_i in which the number of equations $=$
$=$ the number of unknowns.

Many of the resulting equations are new (if everything has variable coefficients).

The same can be done on an elliptic curve.

In the work of Belinsky/Zakharov (1978), a special case is gravity.

New this year:

$$L_i = \frac{\partial}{\partial x_i} + M_i\left(\lambda, \frac{\partial}{\partial \lambda}\right). \tag{30}$$

$$[L_i, L_j] = 0 \quad \Longleftrightarrow \quad [M_i, M_j] = 0. \tag{31}$$

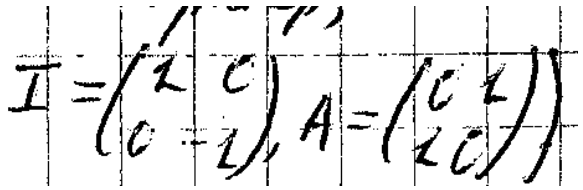
(32) Example:

$$M_1 = I\frac{\partial}{\partial \lambda} + \frac{A}{\lambda}, \qquad I^2 = 1, \quad A^2 = -1,$$
$$IA + AI = 0;$$
$$M_2 = M_1\frac{\partial}{\partial \lambda} + \frac{C}{\lambda^2}, \qquad C = -\frac{1}{2}(I + A).$$

[To the right, beyond a vertical rule:]

(I, A — matrices, for example,

$$I = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix})$$



Uncertain reading. The off-diagonal signs in A have the same shape as the ones in I.

Original source details: GELFAND-P001362

$$L_1 = \frac{\partial}{\partial x_1} + M_1, \qquad L_2 = \frac{\partial}{\partial x_2} + M_1^2, \qquad L_3 = \frac{\partial}{\partial x_3} + M_2.$$

A great deal of computation is needed.

Finding solutions of the equations:

$$L_i y = 0. \tag{33}$$

GELFAND-P001363

Begin with the G.–L.–M. integral equations in λ, take n-soliton solutions, and degenerate them to rational solutions.

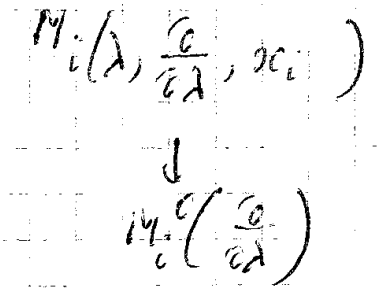


Diagram and typed description.

$$M_i(\lambda, \tfrac{\partial}{\partial \lambda}, x_i)$$
$$\downarrow$$
$$M_i^0(\tfrac{\partial}{\partial \lambda})$$

Original source details: GELFAND-P001363

construct $K(\lambda, \lambda', x_i)$, $F(\lambda, \lambda', x_i)$

and then it will remain (possibly “it remained”)[?] to construct rational solutions.

Uncertain reading. it will remain (possibly “it remained”)

Original source details: GELFAND-P001363

This includes dissipative systems too.

The next generalization:

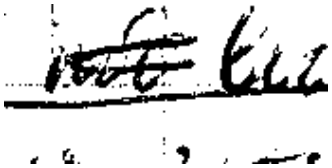
$$L_i = R_i\left(\lambda, \frac{\partial}{\partial\lambda}\right)\frac{\partial}{\partial x_i} - M_i\left(\lambda, \frac{\partial}{\partial\lambda}\right). \tag{34}$$

Uncertain reading. The number reads (34) beneath overlapping strokes.

Original source details: GELFAND-P001363

A. A. Migdal.

Another meaning of [crossed out: illegible start, possibly “вы...”[?]]



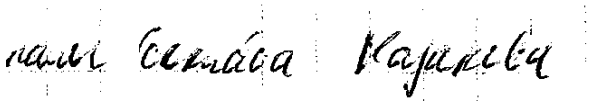
Uncertain reading. illegible start, possibly “вы...”

Original source details: GELFAND-P001363

calculating matrix integrals and the passage
from matrix integrals to continuum integrals

[“Matrix” is inserted above the line before “integrals”.]

A noncommutative analogue of Karasev’s symbol (likely reading)[?].



Uncertain reading. of Karasev’s symbol (likely reading)

Original source details: GELFAND-P001363

(Operators from the outset in place of $N \times N$ matrices)

$$\int dA\, \exp\left(-\frac{N}{\lambda}\operatorname{tr} V(A)\right), \qquad A^+ = -A, \quad N \to \infty. \tag{1}$$

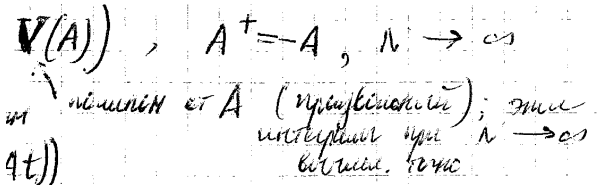


Diagram and typed description.

$V(A)$
\
a polynomial in A (arbitrary); these integrals as $N \to \infty$ are computed exactly

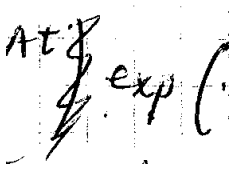
Original source details: GELFAND-P001363

A way to compute them. consider

$$\psi(t) = \frac{1}{N}\,\mathrm{tr}\big(\exp(At)\big) \tag{2}$$

and derive the equation of motion (Schwinger–Dyson). It is clear that

$$0 = \int dA\,\frac{1}{N^2}\,\mathrm{tr}\left(\frac{\partial}{\partial A}\left[e^{At}\exp\left(-\frac{N}{\lambda}\,\mathrm{tr}\,V(A)\right)\right]\right). \tag{3}$$



Uncertain reading. A bracket or several brackets are crossed out after e^{At}.

Original source details: GELFAND-P001363

(the integral of a total derivative $= 0$). This is equivalent to

$$0 = \int_0^t d\tau\,\psi(\tau)\psi(t-\tau) - \frac{1}{\lambda}V'\left(\frac{\partial}{\partial t}\right)\psi(t), \tag{4}$$

$$R(\omega) = \psi^2(\omega) - \frac{1}{\lambda}V'(\omega)\psi(\omega). \tag{5}$$

What is of interest is the average defined by the integral, $\langle f(A)\rangle$

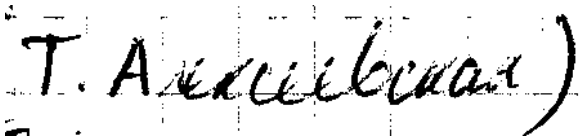
$$\langle f(A)\rangle = \frac{\int dA\,\exp(\ldots)f(A)}{\int dA\,\exp(\ldots)} \tag{6}$$

[On the right:]
($f(A)$ — some invariant function —
— a function of the eigenvalues.

GELFAND-P001364

[source number: 26]

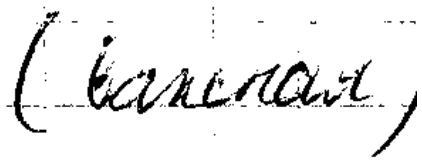
I. M. G. Problem. (T. Alekseevskaya / Aleksievskaya?)



Uncertain reading. Alekseevskaya / Aleksievskaya

Original source details: GELFAND-P001364

(uncertain reading, possibly “solved it”?)



Uncertain reading. uncertain reading, possibly “solved it”

Original source details: GELFAND-P001364

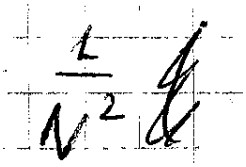
Given a cube (3-dimensional). Draw a plane through every three of its vertices. Into how many pieces do they cut the cube?

(Answer: 96)

The limit as $N \to \infty$ is the same for all the classical groups:

$$O(\infty) = SU(\infty) = SP(\infty). \tag{7}$$

$$0 = \int dA \, \frac{1}{N^2} \frac{\partial}{\partial A_{ij}} \left\{ (e^{At})^{ji} \exp\left(-\frac{N}{\lambda} \operatorname{tr} V(A) \right) \right\}. \tag{3'}$$



Uncertain reading. A symbol after $1/N^2$ is crossed out, possibly tr.

Original source details: GELFAND-P001364

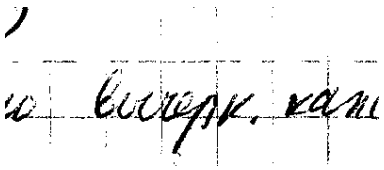
The formula used was:

$$\frac{\partial}{\partial A_{ij}} (e^{At})_{ji} = \int_0^t d\tau \, (e^{A\tau})_{jj} (e^{A(t-\tau)})_{ii}. \tag{8}$$

Derivation: differentiate

$$\prod (1 + A\Delta t) = T \exp\left(\int A \, dt \right)$$

and get the integral (in turn pulling out (likely reading)? each matrix element in its own place)



Uncertain reading. pulling out (likely reading)

Original source details: GELFAND-P001364

Another formula from matrix theory that was used:

$$\frac{\partial}{\partial A_{ij}} \operatorname{Tr} V(A) = (V'(A))_{ji}. \tag{9}$$

From (8) and (9), (4) follows:

$$(e^{At})^{ji} \frac{\partial}{\partial A_{ij}} \operatorname{Tr} V(A) = \\ = (V'(A))_{ji} (e^{At})_{ij} = V'\left(\frac{\partial}{\partial t} \right) \operatorname{tr}(e^{At}).$$

In (4) we should write the average ($\int dA$ is omitted).

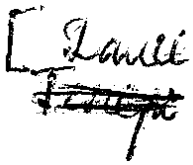
It was said that

$$0 = \int_0^t d\tau \, \langle \psi(\tau) \psi(t-\tau) \rangle - \frac{1}{\lambda} V'\left(\frac{\partial}{\partial t} \right) \langle \psi(t) \rangle. \tag{10}$$

[Next we must consider

$$\int \mathcal{D}A_\mu \exp\left(+\int d^4x \,\frac{N}{\lambda}\,\mathrm{tr}\,F_{\mu\nu}^2\right) \tag{11}$$

]



Uncertain reading. "Next" is inserted above a crossed-out word, probably "Now".

Original source details: GELFAND-P001365

We use the formula

$$\left\langle \frac{\mathrm{tr}\,A}{N}\frac{\mathrm{tr}\,B}{N}\right\rangle = \left\langle \frac{\mathrm{tr}\,A}{N}\right\rangle\left\langle \frac{\mathrm{tr}\,B}{N}\right\rangle + O\left(\frac{1}{N^2}\right). \tag{12}$$

Applying it, we obtain

$$0 = \int_0^t d\tau\,\langle\psi(\tau)\rangle\langle\psi(t-\tau)\rangle - \frac{1}{\lambda}V'\left(\frac{\partial}{\partial t}\right)\langle\psi(t)\rangle. \tag{13}$$

Now introduce the Laplace transform:

$$f(\omega) = \int_0^\infty dt\, e^{-\omega t}\langle\psi(t)\rangle; \tag{14}$$

then from (13) (convolution!)
[crossed out: and] we obtain the equation

$$\underbrace{R(\omega)}_{\text{polynomial in }\omega} = f^2(\omega) - \frac{1}{\lambda}V'(\omega)f(\omega) \tag{15}$$

or (taking the jumps):

$$0 = 2\,\mathrm{Im}\,f\left(\mathrm{Re}\,f - \frac{1}{2\lambda}V'(\omega)\right). \tag{16}$$

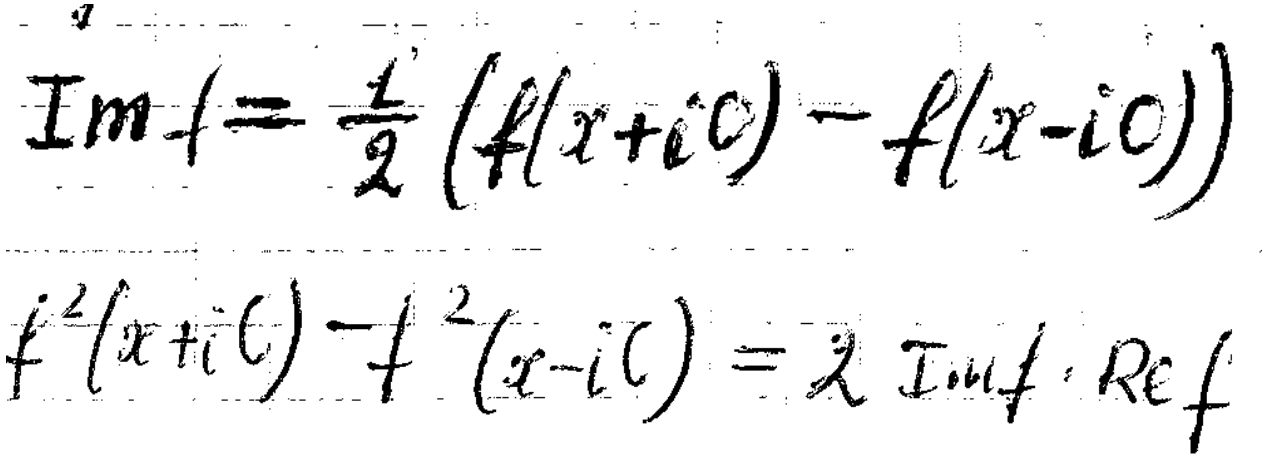
Here

$$\operatorname{Im} f = \frac{1}{2}\big(f(x+i0) - f(x-i0)\big). \tag{17}$$

But it is easy to see that

$$f(\omega) = \frac{1}{N}\left\langle \operatorname{tr} \frac{1}{\omega - A} \right\rangle. \tag{18}$$

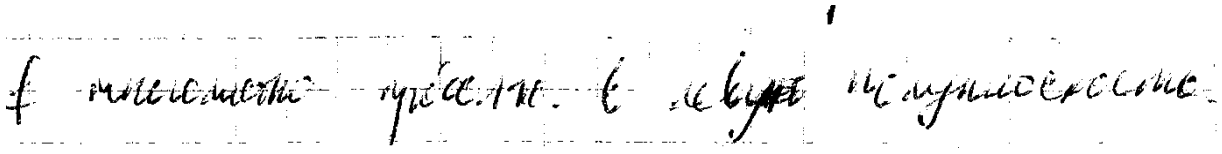
$$f^2(x+i0) - f^2(x-i0) = 2 \operatorname{Im} f \cdot \operatorname{Re} f. \tag{19}$$



Original source details: GELFAND-P001365

Uncertain reading. Equation (17) shows $1/2$, and (19) has a factor of 2. There is a faint separate stroke above the numerator in (17).

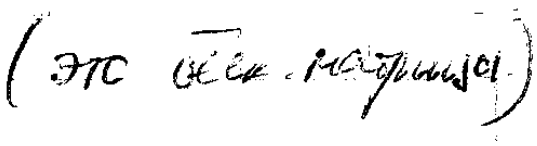
(20) f is fully represented in the left half-plane.



Original source details: GELFAND-P001365

Uncertain reading. The reading “is fully represented” is uncertain. The phrase may refer to analytic continuation.

$f(\omega)$ has jumps at the points in the spectrum of the operator A (this is an infinite matrix (likely reading)[?])



Uncertain reading. this is an infinite matrix (likely reading)

Original source details: GELFAND-P001365

GELFAND-P001366

[source number: 27]

Now we obtain the old (French) equation

$$f(\omega) = \int \frac{\varrho(\lambda)\, d\lambda}{\omega - \lambda}. \tag{21}$$

$$f = \frac{1}{2\lambda} V'(\omega) + M(\omega)\sqrt{(\omega - a)(\omega - b)}. \tag{22}$$

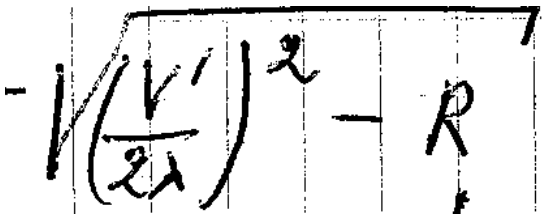
[Zakharevich.

$$A(x)(B(x) - \alpha(x)) = 0, \qquad iA(x) + B(x) \quad \text{continues analytically to } \operatorname{Im} x > 0$$

and so on.]

Instead one can solve the quadratic equation (15):

$$f = \frac{1}{2\lambda} V'(\omega) - \sqrt{\left(\frac{V'}{2\lambda}\right)^2 - R}. \tag{23}$$



Uncertain reading. In (23), the term under the square root is written $-R$.

Original source details: GELFAND-P001366

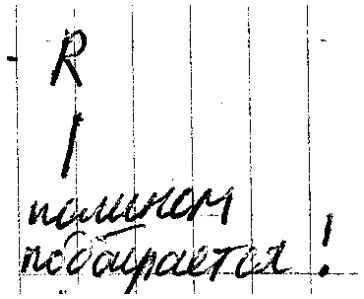
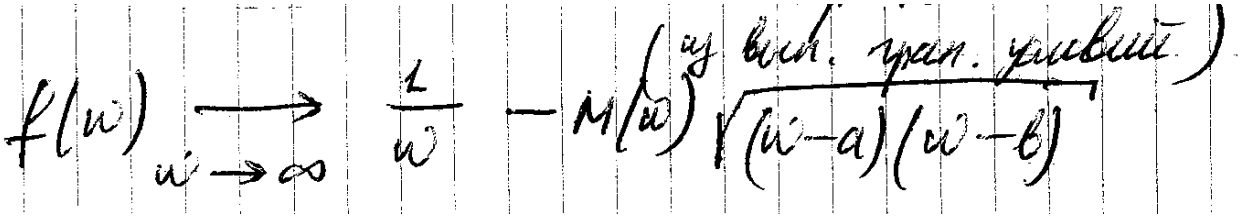


Diagram and typed description.

R
|
the polynomial
is chosen!

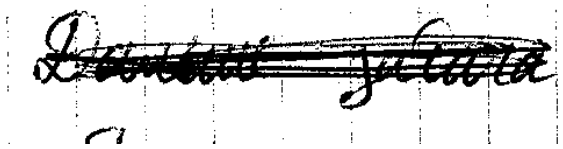
Original source details: GELFAND-P001366

$$f(\omega) \underset{\omega \to \infty}{\longrightarrow} \frac{1}{\omega} \quad \overset{\text{(from satisfying the boundary conditions)}}{-M(\omega)\sqrt{(\omega - a)(\omega - b)}} \tag{24}$$



Original source details: GELFAND-P001366

Uncertain reading. After $1/\omega$ in (24), the manuscript has $-M(\omega)\sqrt{(\omega - a)(\omega - b)}$.



Original source details: GELFAND-P001366

Uncertain reading. A crossed-out line, possibly “Next, the problem” or “The given equation”.

For field theory, Karasev’s method does not go through:

[crossed out: unclear start, possibly “ур” or “при”?] the components do not commute and cannot be put simultaneously in diagonal form.

Original source details: GELFAND-P001366

Uncertain reading. unclear start, possibly “ур” or “при”

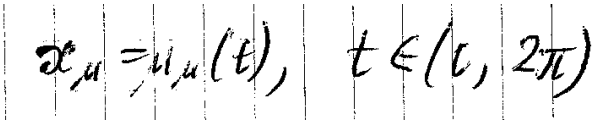
Consider the integral (11).

$$A_\mu(x) : \longrightarrow\ W(c) = \frac{1}{N} \operatorname{tr} P \exp\left(\oint_c A_\mu\, dx_\mu\right) \tag{25}$$

and repeat the same reasoning

$F_{\mu\nu}$ is a tensor in μ, ν and a matrix in the Lie algebra of $SU(N)$.

$$x_\mu = u_\mu(t), \qquad t \in (0, 2\pi)$$



Original source details: GELFAND-P001366

Uncertain reading. The right side more likely reads $u_\mu(t)$. Compare x_μ on the left.

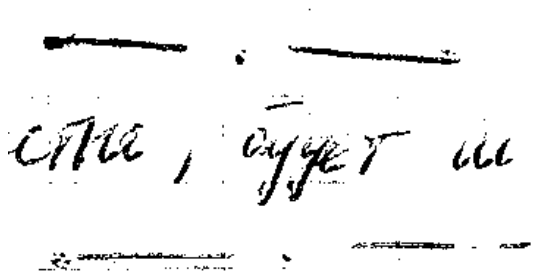
$$\int A_\mu(x(t))\dot{x}_\mu\, dt$$

$\sim \exp At$ earlier

A chain of identities: . . .

GELFAND-P001367

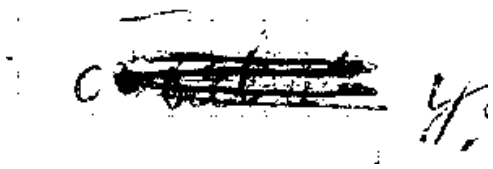
$$\int \mathcal{D}A\, \hat{P}\, \frac{\partial}{\partial A} \exp\left(\oint_c A\, dy\right) \tag{26}$$



Original source details: GELFAND-P001367

Diagram and typed description. Two strokes with a dot between them. Similar faint traces appear below the announcement.

30 March it is uncertain whether the seminar will take place (if cancelled, then [crossed out: initial “c” visible, remainder illegible[?]] ask at the lab 139-53-57)



Uncertain reading. initial “c” visible, remainder illegible

Original source details: GELFAND-P001367

GELFAND-P001368

13/IV-87 Gelfand seminar

Sinai

On spin glasses (following Parisi)

$$H = \sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j; \qquad J_{ij} \text{ are random.} \tag{1}$$

(Ising model of a spin glass)

$\langle i, j\rangle$ — a pair of neighbors

Spin glass: when the interaction forces are long-range and oscillate strongly

[crossed out: short entry, possibly “Par...” or “Theor...”[?]]

Uncertain reading. short entry, possibly “Par...” or “Theor...”

Original source details: GELFAND-P001368

Parisi: everything interacts, namely:

$$H = \frac{1}{\sqrt{N}} \sum J_{ij}\sigma_i\sigma_j \tag{2}$$

[On the right:] J_{ij} — a Gaussian random variable (all with the same variance)

There are many ground states (differing in energy by $\sim 1/N$);

$$\sigma_i = \pm 1.$$

One must study the case

$$N \to \infty \qquad \text{and} \qquad \lim_{N\to\infty} \min H$$
$$\text{is computed}$$

Entropy of the set of [inserted above the line: (ground)] configurations > 0 (in the limit $N \to \infty$)

$$\frac{\ln \#[O(N)]}{N} \underset{N\to\infty}{\longrightarrow} h > 0. \tag{3}$$

The property of ultrametricity for ground states (Parisi, Virasoro): if $\sigma', \sigma'', \sigma'''$ — are three ground states, then among the numbers

$$\frac{1}{N}(\sigma', \sigma''), \quad \frac{1}{N}(\sigma', \sigma'''), \quad \frac{1}{N}(\sigma'', \sigma''') \tag{4}$$

two are necessarily equal, and the third is greater.

GELFAND-P001369

Phase transition to a spin glass.

Parisi invented stochastic quantization.

A. S. Schwarz.

Superconformal transformations and a superconformal manifold.

There is a very complete and beautiful generalization of complex analysis to $(1,1)$-dimensional superconformal manifolds.

Definition

Let there be a superdomain (x,θ). Introduce the covariant derivative:

$$D = \frac{\partial}{\partial\theta} + \theta\frac{\partial}{\partial x}. \tag{1}$$

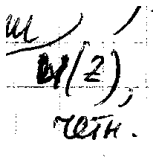
In dimension $(1,1)$.

A map $(x,\theta) \to (\widetilde{x},\widetilde{\theta})$ is called superconformal, if

$$\widetilde{D} = \underset{\text{function}}{F} D$$

(D is preserved up to a factor); here $\widetilde{D} = \frac{\partial}{\partial\widetilde{\theta}} + \widetilde{\theta}\frac{\partial}{\partial\widetilde{x}}$.

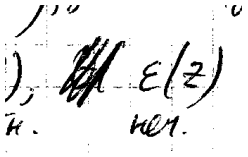
They are parametrized [inserted: by pairs]



Uncertain reading. $w(z)$, possibly $u(z)$

Original source details: GELFAND-P001369

[caption: even], [crossed out: illegible sign?]



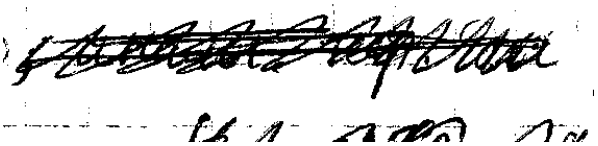
Uncertain reading. illegible sign

Original source details: GELFAND-P001369

$\varepsilon(z)$ [caption: odd] — analytic functions.

Superconformal manifold: glued from such superdomains.

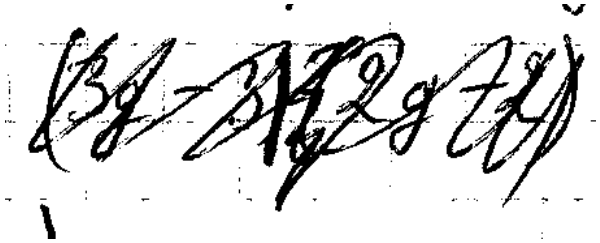
For them one can construct a theory [crossed out: illegible text?]



Uncertain reading. illegible text

Original source details: GELFAND-P001369

of superconformal curves, the moduli space has dimension [crossed out: $(3g-3 \mid 2g+2)$?[?]]



Uncertain reading. $(3g-3 \mid 2g+2)$?

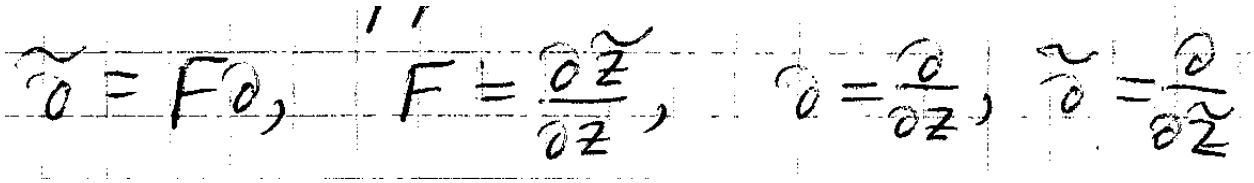
Original source details: GELFAND-P001369

$(3g-3 \mid 2g-2)$ (g — genus of the underlying surface).

There is an analogue of Abelian differentials and higher differentials.

For ordinary conformal maps:

$$\widetilde{\partial} = F\partial, \qquad F = \frac{\partial \widetilde{z}}{\partial z}, \qquad \partial = \frac{\partial}{\partial z}, \qquad \widetilde{\partial} = \frac{\partial}{\partial \widetilde{z}}.$$



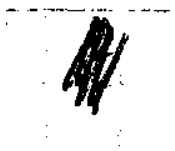
Original source details: GELFAND-P001369

Uncertain reading. The manuscript gives $\widetilde{\partial} = F\partial$ together with $F = \partial\widetilde{z}/\partial z$.

Question by I. M. G.: an analogue of the Riemann mapping theorem?

Answer (A. S. S.): if the underlying space is simply connected, it reduces to the standard one.

Question by I. M. G.: description of the ring of functions [crossed out: illegible sign[?]]

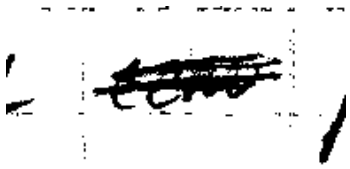


Uncertain reading. illegible sign

Original source details: GELFAND-P001369

on a compact superconformal manifold.

Answer (A. S. S.): there is a Riemann–Roch theorem if [crossed out: illegible word[?]]



Uncertain reading. illegible word

Original source details: GELFAND-P001369

the dimensions of all the objects make sense.

(2) Def. Superspace: a free Λ-module, where Λ is a Grassmann algebra.

$(x_1, \dots, x_n, \theta_1, \dots, \theta_n)$ — even and odd elements $\Lambda = \Lambda(e_1, \dots, e_k)$.

GELFAND-P001370

[source number: 49]

Specify a linear equation in this module. It singles out a submodule.

$$\sum a_k V_k = 0$$

Uncertain reading. Below “equation”: $\sum a_k V_k = 0$. The letter reads V or v.

Original source details: GELFAND-P001370

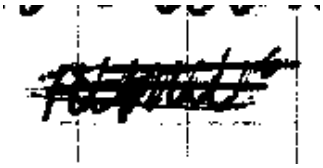
It will not be free and will not have [crossed out: illegible sign[?]] a dimension.

Uncertain reading. illegible sign

Original source details: GELFAND-P001370

[between the lines: functions with specified singularities may have no dimension.]

But in a [crossed out: possibly “general”[?]] typical situation it is a free module, and there is a standard form of the Riemann–Roch theorem.



Uncertain reading. possibly “general”

Original source details: GELFAND-P001370

Abelian differentials.

Uncertain reading. In the narrow left margin, a mark resembling e, 2, or ε.

Original source details: GELFAND-P001370

If Φ is a scalar function, then $D\Phi$ transforms accordingly in different local coordinates.

Fields that transform in this way, sections of Ω, are analogues of Abelian differentials. [above the line: sections of the bundle Ω.]

Suppose there is an Abelian differential V.

For a simply connected underlying surface, $V = D\Phi$.

Period: take the universal cover, where there is $\overline{\Phi}$, shifts, and so on.

There is a Jacobian, a Picard manifold, and so on.

The string partition function is expressed through an analogue of the Selberg ζ-function (joint work with Manin).

———— · ————

Fermionic string $g_{\alpha,\beta};\ X^a$.

Superanalogue of the Riemannian metric: an odd vector field $\widehat{e}$, for which

$$\{\widehat{e}, \overline{\widehat{e}}\} = \tau\widehat{e} + \overline{\tau}\,\overline{\widehat{e}}. \tag{3}$$

(4) Example: $\widehat{e} = \lambda D$.

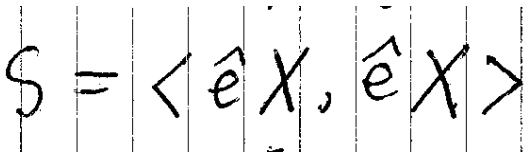
(5) Assertion. There always $\exists$ exists a coordinate system in which the super-Riemannian metric has the form (4).

Explanation: an ordinary Riemannian metric is given by the field $\lambda\frac{\partial}{\partial z}$, λ is determined up to the phase $e^{i\alpha}$.

(This is for any $(2 \mid 2)$-dimensional manifold.)

Fermionic string

$$S = \langle \widehat{e}X, \widehat{e}X \rangle. \tag{6}$$



Original source details: GELFAND-P001370

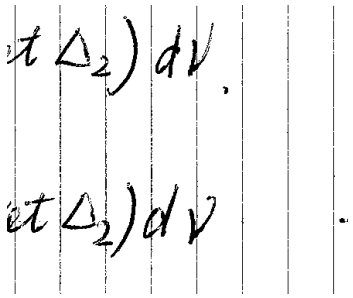
Uncertain reading. In (6) both e's carry a hat. The bar used in (3) is absent.

Original source details: GELFAND-P001370

Uncertain reading. An illegible sign below (6) is crossed out.

$$\int (\det \Delta_0)^{-13} (\det \Delta_2)\, dV, \qquad \text{for the bosonic string.} \tag{7}$$

$$\int (\det \Delta_0)^{-5} (\det \Delta_2)\, dV \qquad \text{— for the fermionic string.} \tag{8}$$



Uncertain reading. The differential in (7) and (8) reads dV or $d\nu$.

Original source details: GELFAND-P001370

The measure (7) has asymptotic properties (Belavin–Knizhnik).

GELFAND-P001371

Baranov, Schwarz: extension to the fermionic case, using the work of F.[?] Voronov

Uncertain reading. The initial resembles Cyrillic «Ф.»

Original source details: GELFAND-P001371

(reads as “supermanif.”[?]; computation of measures).

Uncertain reading. reads as “supermanif.”

Original source details: GELFAND-P001371

Another topic:

A noncommutative analogue of Ray–Singer torsion.

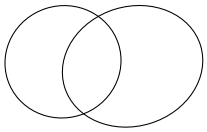
(no promise yet.)

Alekseevskii, D.

Quaternionic manifolds

Let

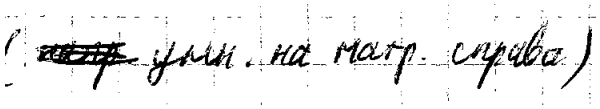
Diagram and typed description.



Original source details: GELFAND-P001371

$$U \subset \mathbb{H}^n, \qquad V \subset \mathbb{H}^n$$
$$f : U \to V$$
$$\forall x \in U \qquad df_x : \mathbb{H}^n \to \mathbb{H}^n \tag{1}$$

a morphism of left modules ([crossed out: possibly "matr."?]



Uncertain reading. possibly "matr."

Original source details: GELFAND-P001371

multiplication by a matrix on the right).

If it is simply connected, only a linear space results. [Below the line: "(".]

All automorphisms are linear.

One can write down the Lie algebra of infinitesimal transformations preserving it.

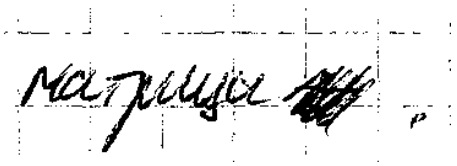
(2) Assertion: then the transition functions must be linear

The first approximation "is trivial," with the Russian ending partly illegible[?].

Uncertain reading. "is trivial," with the Russian ending partly illegible

Original source details: GELFAND-P001371

A vector field whose differential is a quaternionic matrix [crossed out: illegible mark[?]].



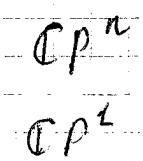
Uncertain reading. illegible mark

Original source details: GELFAND-P001371

Definition (1) is incorrect, since the projective space would then have no quaternionic structure.

$\mathbb{C}P^n$ — the space of one-dimensional quaternionic subspaces of $\mathbb{H}^{n+1}$.

$\mathbb{C}P^1$ — pairs (λ_1, λ_2) up to left multiplication $(\lambda\lambda_1, \lambda\lambda_2)$.



Uncertain reading. Both projective-space symbols visibly use the letter $\mathbb{C}$.

Original source details: GELFAND-P001371

2 charts

$$(\lambda_1, 1)$$
$$(1, \lambda_2)$$
$$(\lambda_1, 1) \sim (1, \lambda_1^{-1})$$

$\lambda \to \lambda^{-1}$ — the transition function

GELFAND-P001372

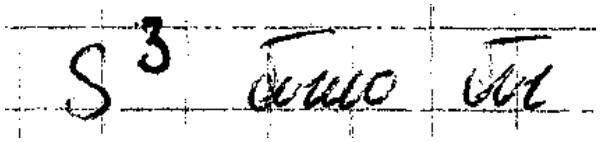
[source number: 50]

λ^{-1}

$$(\lambda_0 + i\lambda_1 + j\lambda_2 + k\lambda_3)^{-1} = \frac{\lambda_0 - i\lambda_1 - j\lambda_2 - k\lambda_3}{\lambda_0^2 + \lambda_1^2 + \lambda_2^2 + \lambda_3^2}.$$

$$d(\lambda^{-1}) = -\lambda^{-1} \cdot d\lambda \cdot \lambda^{-1}$$

Topological explanation: if it existed, then S^3 would have 3 complex structures.



Uncertain reading. The superscript on S reads 3.

Original source details: GELFAND-P001372

$$V = \mathbb{R}^{4n}$$

(3) Quaternionic structure — a subalgebra $Q \subset$ [crossed out: illegible mark?]

Uncertain reading. illegible mark

Original source details: GELFAND-P001372

End(V), isomorphic to $\mathbb{H}$.

(Or a subalgebra spanned by 3 anticommuting complex structures

$$Q = \langle I, J, K, E \rangle \qquad I^2 = J^2 = K^2 = -1, \quad IJ = -JI = K$$

and so on.

$\Rightarrow$ one can multiply on the left by quaternions

Nuance: The isomorphism is unspecified. There is no need to single out I, J and K.

($\mathbb{H}$ has nontrivial automorphisms)

Quaternionic structure on a manifold: a field of quaternionic structures (in each tangent space) (Or: an almost quaternionic structure)

Chern (1945) considered such structures

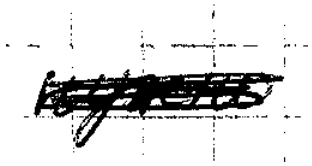
($\forall$ 4-dimensional Riemannian manifold has two, since

$$\mathrm{SO}(4) = \text{product}.$$

$$M^{4n}, \quad Q\colon x \mapsto Q_x$$

Integrability: Cauchy–Riemann equations in place of an atlas

(Up to second-order small terms [crossed out: illegible word?]



Uncertain reading. illegible word

Original source details: GELFAND-P001372

like a quaternionic transformation)

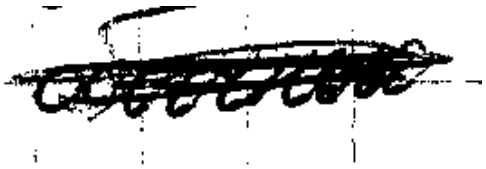
Automorphisms preserving the quaternionic structure

$$G = \mathrm{GL}(1, \mathbb{H}) \cdot \mathrm{GL}(n, \mathbb{H})$$

— group of automorphisms of the quaternionic structure

$$G = \mathrm{Sp}(1) \cdot \mathrm{Sp}(n)$$

— for [crossed out: possibly “ordinary”?]



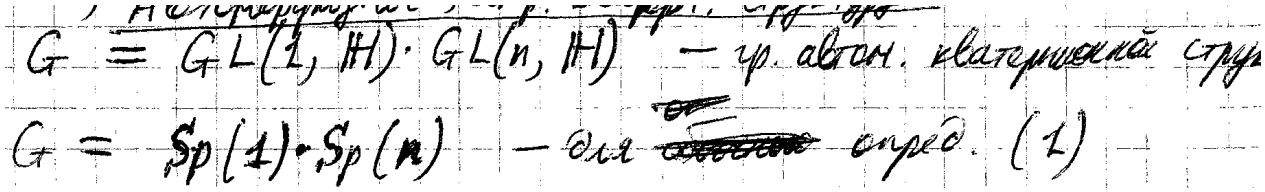
Original source details: GELFAND-P001372

Uncertain reading. possibly “ordinary”

definition (1). [Also crossed out above: illegible mark?.]

Original source details: GELFAND-P001372

Uncertain reading. illegible mark



Original source details: GELFAND-P001372

Uncertain reading. Both formulas begin with $G =$ in the manuscript. The second line refers to (1).

$$y_i = \sum x_k a_{ki}$$

[Crossed out between $\sum$ and x_k: illegible symbol?.]

Original source details: GELFAND-P001372

Uncertain reading. illegible symbol

$\|$ preserves the quaternionic structure (right).

Original source details: GELFAND-P001372

Uncertain reading. The word in parentheses reads “right.”

$$(x_1, \ldots, x_n) \to (\lambda x_1, \ldots, \lambda x_n)$$

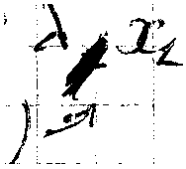
also preserves

[Crossed out before the first λ in the right-hand tuple: x_1?$^?$.

Original source details: GELFAND-P001372

Uncertain reading. x_1?

Crossed out between this λ and x_1: illegible mark$^?$.]



Original source details: GELFAND-P001372

Uncertain reading. illegible mark

[Below the formula, with an arrow pointing between the first λ and x_1:] (On the other side!)

GELFAND-P001373

Condition for 1-integrability:

$\exists$ a connection ∇ without torsion, preserving the quaternionic structure.

(Linear connection, not necessarily metric.)

[Crossed out in parentheses: An arrow $\to$ is visible. The words on either side may begin "integr..." and "preserv...".$^?$]

Original source details: GELFAND-P001373

Uncertain reading. An arrow $\to$ is visible. The words on either side may begin "integr..." and "preserv...".

Or: $\nabla Q = 0$.

Quaternionic subspace $\longrightarrow$ quaternionic subspace under parallel transport.

Goncharov, A.

Suppose there is a real-analytic manifold with an [inserted above: almost] quaternionic structure.

$$M^{4n}_{\mathbb{R}} \subset M^{4n}_{\mathbb{C}}$$

$$T_x M^{4n}_{\mathbb{C}} \supset K_x = \bigcup \{v + \sqrt{-1}\, Jv\}, \qquad v \in T M^{4n}_{\mathbb{R}}$$

[Above K_x: cone]

$$J \in Q_x, \qquad J^2 = -1$$

And there must be integrability.

D. A.

$$\gamma \text{ — a curve} \quad \Rightarrow \quad \tau_\gamma Q_{\gamma(0)} = Q_{\gamma(1)}$$

$$\gamma : [0,1] \to M$$

(τ_γ — parallel transport along γ)

Q_x — a subalgebra in Aut $T_x M$, it too can be transported.

Uncertain reading. The manuscript reads "a subalgebra in Aut $T_x M$."

Original source details: GELFAND-P001373

GELFAND-P001374

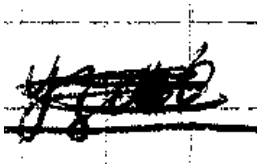
20/IV-87 Gelfand Seminar.

I.M.G. Question: there are 2 commuting isometries in an n-dimensional space. [Added above the line, with a pointer: not defined everywhere.] How can one turn them into 2 commuting unitary operators? What are the obstructions?

Kasparov has partly done this.

Kirillov on Jones's work.

Distinguishing [crossed out: possibly "knots"?]



Uncertain reading. possibly "knots"

Original source details: GELFAND-P001374

knots using factors

Consider the algebra generated by the generators T_i (infinitely many of them) with the relations

$$\begin{cases} (T_i+1)(T_i-q)=0, \\ T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \\ T_iT_j = T_jT_i, \quad |i-j|>1 \end{cases} \tag{1}$$

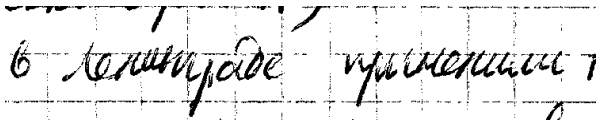
(Hecke algebra of series A)

$\Rightarrow$ [crossed out: illegible mark?]

Uncertain reading. illegible mark

Original source details: GELFAND-P001374

in Leningrad they likewise applied series B.



Uncertain reading. The verb reads «применили» (“applied”).

Original source details: GELFAND-P001374

1. Knot $\longrightarrow$ polynomial, equivalent knots $\longrightarrow$ equal polynomials.

2. Factor $\longrightarrow$ subfactor

GELFAND-P001375

Consider $\bigsqcup_n B_n$ ([crossed out: illegible abbreviation?]

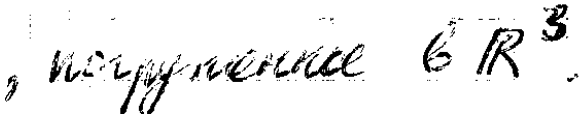
Uncertain reading. illegible abbreviation

Original source details: GELFAND-P001375

disjoint union). [Below B_n: braid group.]

Link: an oriented compact 1-dimensional manifold immersed in $\mathbb{R}^3$.

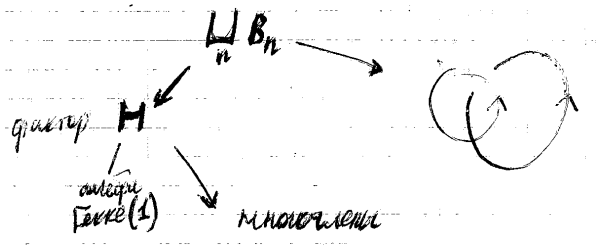
Uncertain reading. The definition uses “immersed.”



Original source details: GELFAND-P001375

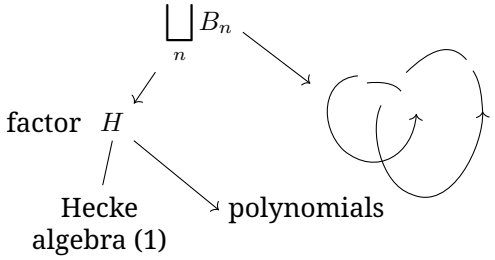
Uncertain reading. A short isolated stroke to the left of the diagram.

Original source details: GELFAND-P001375



Original source details: GELFAND-P001375

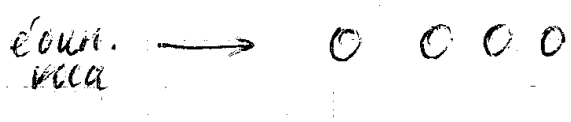
Diagram and typed description.



Original source details: GELFAND-P001375

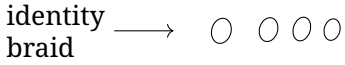
Uncertain reading. Both visible arrowheads point upward along the right branches. The upper arcs are broken or faint. Their joins and the upper crossing remain uncertain.

Braid $\longrightarrow$ link (join the beginning to the end).

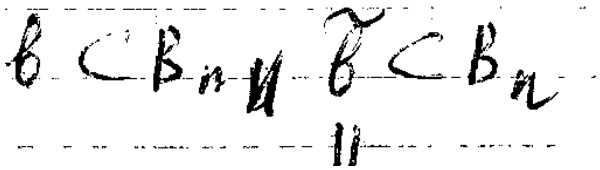


Original source details: GELFAND-P001375

Diagram and typed description.



(2) $b \subset B_n$ and $\widetilde{b} \subset B_n$ give equivalent links. [Below $\widetilde{b}$:] $\widetilde{b} = cbc^{-1}$. [To the right of the vertical line:] $B_n \subset B_{n+1}$.



Original source details: GELFAND-P001375

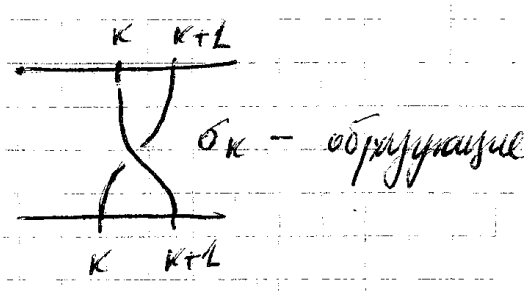
Uncertain reading. In (2) the signs preceding B_n look like $\subset$.

(3) $b \in B_n$, $\widetilde{b} \in B_{n+1}$, $\widetilde{b} = b \cdot \sigma_n^{\pm 1}$ (appending one generator from B_{n+1}).
Then b and $\widetilde{b}$ produce [crossed out: possibly "equiv."?]

Original source details: GELFAND-P001375

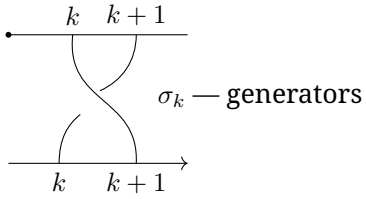
Uncertain reading. possibly "equiv."

equivalent links.



Original source details: GELFAND-P001375

Diagram and typed description.



(4) Markov's theorem (difficult). The operations described suffice to pass to any equivalent link.

Idea: construct a function invariant under these operations.

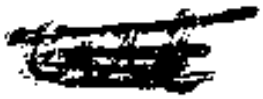
The map $\bigsqcup_n B_n$ into H is constructed as follows: $\sigma_i \to T_i$, $\sigma_i^{-1} \to T_i^{-1}$. [Below T_i^{-1}: found from the quadratic equation $(1)_1$.]

$(1)_{2,3}$ — a complete set of relations in the braid group.

$$\sigma_i^{-1} \longrightarrow \frac{T_i + (1-q)}{q} = T_i^{-1}$$

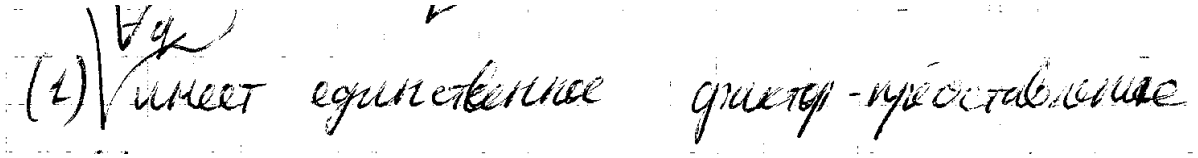
The set of generators (1) [inserted above: $\forall q$] has a unique factor representation with a trace, and [crossed out: illegible fragment?]

Uncertain reading. illegible fragment



Original source details: GELFAND-P001375

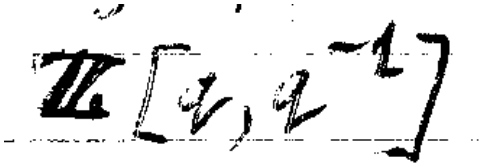
the trace is a polynomial in q, q^{-1}.



Original source details: GELFAND-P001375

Uncertain reading. The manuscript inserts $\forall q$ above the assertion.

$$\mathbb{Z}[z, z^{-1}]$$



Uncertain reading. The variable in the bottom formula resembles z or v. The q in the preceding line has a different shape.

Original source details: GELFAND-P001375

GELFAND-P001376

[source number: 57]

Exercise. Take $q = \pm 1$ and see what invariant of the braid group results (the ratio of the values of the Jones polynomial at these values).

Jones specifically excludes from consideration [crossed out: illegible fragment?]

Uncertain reading. illegible fragment

Original source details: GELFAND-P001376

roots of 1.

I.M.G.'s idea: consider precisely the roots of 1. It may turn out that there are no factors at all and that roots of 1 are what is needed.

A.M. Vershik has already done something on this.

A. Beilinson.

On work by Beilinson, MacPherson and V. Schechtman.

Notes on motivic cohomology

Let X — an analytic manifold, E — a vector bundle on X. [Under the description of X:] (or "complex," abbreviated[?])

Uncertain reading. "complex," abbreviated

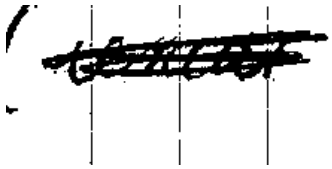
Original source details: GELFAND-P001376

(compact).

$$c_i(E) \in H^{2i}(X, \mathbb{C}) = H^{2i}(X, \mathcal{O} \to \Omega^1 \to \ldots)$$

[Above the final argument, with a pointer: cohomology with coefficients in a complex of sheaves.]

They are integral and satisfy the Hodge condition ([crossed out: illegible word[?]]



Uncertain reading. illegible word

Original source details: GELFAND-P001376

$c_i \in i$th term of the Hodge filtration).

(Since they can be computed via log derivatives in Čech cocycles.)

Uncertain reading. The word following log is abbreviated «произв.», read here as "derivatives."

Original source details: GELFAND-P001376

We will compute the manifold's cohomology from the de Rham complex.

$$0 \to \mathbb{C} \to \mathcal{O} \to \Omega^1 \to \Omega^2 \to \ldots \tag{1}$$

[In (1), between $\mathbb{C} \to$ and $\mathcal{O}$ crossed out: illegible mark[?].]

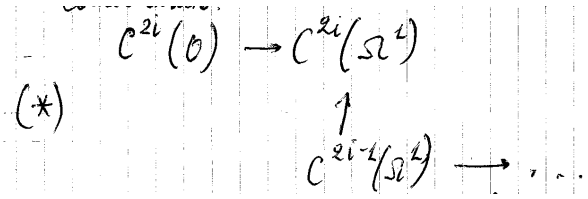
Uncertain reading. illegible mark

Original source details: GELFAND-P001376

[At right:] Ω^i — holomorphic i-forms (a sheaf).

$\{U_i\}$ — a cover of X by small open sets.

Compatibility:



Original source details: GELFAND-P001376

Diagram and typed description.

$$\begin{array}{cc} & C^{2i}(\mathcal{O}) \longrightarrow C^{2i}(\Omega^1) \\ (*) & \uparrow \\ & C^{2i-1}(\Omega^1) \longrightarrow \cdots \end{array}$$

Hodge filtration on $H^k(X, \mathbb{C})$

$$F^j H^k(X, \mathbb{C}) = H^k(X, 0 \to \ldots \to 0 \to \Omega^j \to \Omega^{j+1} \to \ldots)$$

Atiyah's construction of the characteristic class, from which it follows that $c_j \in F^j$

g_{ij} — transition functions

$$g_{ij} \colon U_i \cap U_j \longrightarrow \mathrm{GL}_n(\mathbb{C})$$

$$A = g_{ij}^{-1} dg_{ij} \in \Omega^1(U_i \cap U_j, \mathrm{Mat}_n(\mathbb{C}))$$

$$\in C^1(X, \Omega^1 \otimes \mathrm{Mat}_n)$$

[Below $g_{ij}^{-1}dg_{ij}$: a 1-form with values in a Lie algebra.]

Original source details: GELFAND-P001376

Uncertain reading. The formula for A has two successive $\in$ signs.

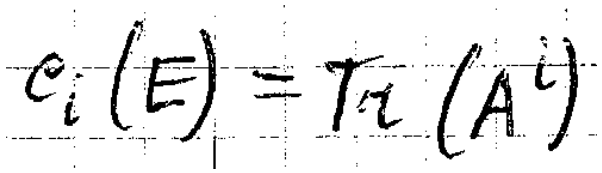
Take $(g_{ij}^{-1}dg_{ij})^k$ — a cochain with values in k-forms. [Below the power: (multiplication of cochains] ‖

Take the trace Tr $\longrightarrow$ the resulting form has one component.

GELFAND-P001377

$$c_i(E) = \mathrm{Tr}(A^i)$$

[At right:] A^i — a form of degree i on the intersection of $(i+1)$ sets.



Original source details: GELFAND-P001377

Uncertain reading. The manuscript writes $c_i(E)$ before the trace of the power.

[Crossed out: possibly «Вид» ("form")?]

Original source details: GELFAND-P001377

Uncertain reading. possibly «Вид» ("form")

$\Rightarrow$ an element of $2i$-dimensional cohomology (see diagram $(*)$).

Problem: explicitly construct a homotopy to an integral characteristic class.

For

$$c_1 = d\log(\det g_{ij})$$

We must drop d:

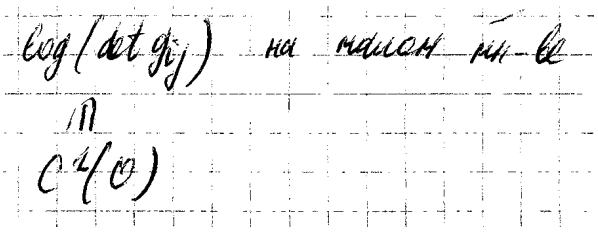


Diagram and typed description.

$\log(\det g_{ij})$ on a small set

$\cap$

$C^1(\mathcal{O})$

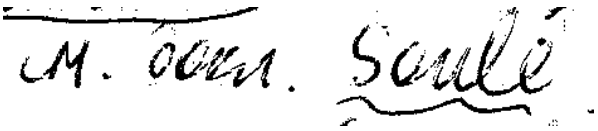
Original source details: GELFAND-P001377

GELFAND-P001378

27.4.87 Gelfand Seminar

A. Beilinson

Deligne cohomology [inserted above the end of the title — Beilinson]: see possibly an abbreviation for "survey"?



Uncertain reading. possibly an abbreviation for "survey"

Original source details: GELFAND-P001378

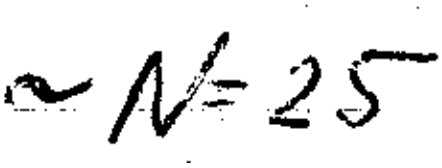
Soulé, Bourbaki Seminar, Feb. 85. [Below the surname:] (Regulateurs).

H. Esnault, E. Viehweg (preprint). [Below "preprint": Manin has it.]

Deligne–Beilinson cohomology

A. Beilinson: Current Problems in Mathematics, vol. 24.

Contempor. math. $\sim$ No. 25



Uncertain reading. After "Contempor. math." there is a $\sim$ sign and what appears to be "No. 25."

Original source details: GELFAND-P001378

Complex of sheaves on any manifold (complex analytic scheme).

$$\mathbb{Z}(i)_{\mathcal{D}} := \mathbb{Z} \overset{(2\pi\sqrt{-1})^i}{\longrightarrow} \mathcal{O} \longrightarrow \Omega^1 \longrightarrow \ldots \longrightarrow \Omega^{i-1} \tag{1}$$

(i — an integer.)
In particular

$$\mathbb{Z}(0)_{\mathcal{D}} = \mathbb{Z}$$

$$\mathbb{Z}(1)_{\mathcal{D}} = \left(\mathbb{Z} \overset{2\pi\sqrt{-1}}{\longrightarrow} \mathcal{O}\right) \overset{\exp}{\longrightarrow} \underbrace{\mathcal{O}^*}_{\text{degree } 1} [-1]$$

[At right:]

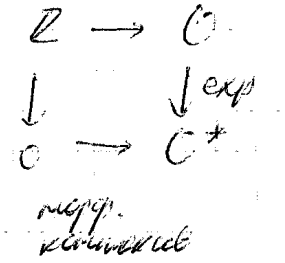


Diagram and typed description.

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & \mathcal{O} \\ \downarrow & & \downarrow \exp \\ 0 & \longrightarrow & \mathcal{O}^* \end{array}$$

morphism of complexes

Original source details: GELFAND-P001378

The cohomology sheaf is

$$H^1(\mathbb{Z}(1)) = \mathcal{O}^*$$

$$\mathbb{Z}(2)_{\mathcal{D}} = \left(\mathbb{Z} \overset{-4\pi^2}{\longrightarrow} \mathcal{O} \longrightarrow \Omega^1\right)$$

$$H^1(\mathbb{Z}(2)_{\mathcal{D}}) = \mathbb{C}/(2\pi\sqrt{-1})^2\mathbb{Z}$$

[Below the right-hand side:]

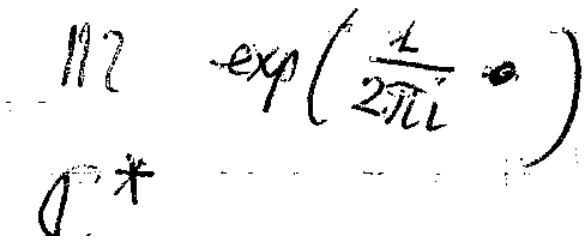
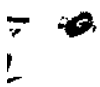


Diagram and typed description.

$$\begin{array}{ccc} \mathbb{R} & \exp(\frac{1}{2\pi i} \circ) \\ \| & \\ \mathbb{C}^* & \end{array}$$

Original source details: GELFAND-P001378



Uncertain reading. The last mark in the exponential argument resembles a small circle or dot.

Original source details: GELFAND-P001378

$$H^2(\mathbb{Z}(2)_{\mathcal{D}}) = \Omega^1/d\mathcal{O}$$

GELFAND-P001379

[source number: 67]

There is a map

$$\mathbb{Z}(i)_{\mathcal{D}} \longrightarrow \mathbb{Z} \qquad \text{(component 0)} \tag{2}$$

$$\mathbb{Z}(i)_{\mathcal{D}} \longrightarrow \Omega^i[-i] \tag{3}$$

(0 on all forms, the last one going to its differential)

$$\mathbb{Z}(i)_{\mathcal{D}} \otimes \mathbb{Z}(j)_{\mathcal{D}} \longrightarrow \mathbb{Z}(i+j)_{\mathcal{D}} \tag{4}$$

$$z_i \star \omega_j = z_i \omega_j$$

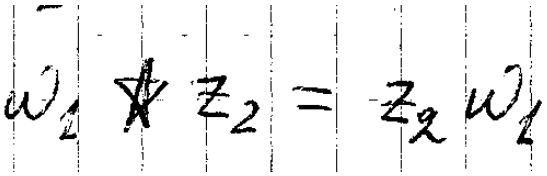
[To the right of (4):] arrows (2) and (3) send this multiplication to ordinary multiplication of numbers and forms.

Let z_j be from component 0.

$$z_1 \star z_2 = z_1 z_2 \qquad \text{(ordinary product.)} \tag{5'}$$

$$z_1 \star \omega_2 = z_1 \omega_2 \tag{5''}$$

$$\omega_1 \star z_2 = z_2 \omega_1 \tag{5'''}$$



Uncertain reading. The right-hand side reads $z_2\omega_1$. A short faint stroke appears before z_2.

Original source details: GELFAND-P001379

$$\omega_1 \star \omega_2 = \begin{cases} 0, & \text{if } \omega_2 \notin \Omega^{j-1}, \\ \omega_1 \wedge d\omega_2, & \text{if } \omega_2 \in \Omega^{j-1} \end{cases} \tag{6}$$

(7) Up to homotopy this multiplication is commutative. [Below «homotopy»: (of chain complexes).]

(8) Both maps (2) and (3) commute with multiplication.

(9) Let E — a holomorphic vector bundle [inserted above: on X]. Then its Chern classes are defined

$$c_i(E) \in H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$$

[At right:] (cohomology group with coefficients in a complex of sheaves). [Added above the beginning of the note: “abelian”?.]

Uncertain reading. “abelian”

Original source details: GELFAND-P001379

(10) Under map (2) they go to the ordinary topological Chern classes. This property and functoriality uniquely determine these classes in (9).

Description of the group $H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$.

Let X be compact.

Čech definition of the cohomology in (9)

C^{ij} — i-dimensional cochains with coefficients in the jth sheaf

[Crossed out to the left of the diagram: illegible mark?.]

Uncertain reading. illegible mark

Original source details: GELFAND-P001379

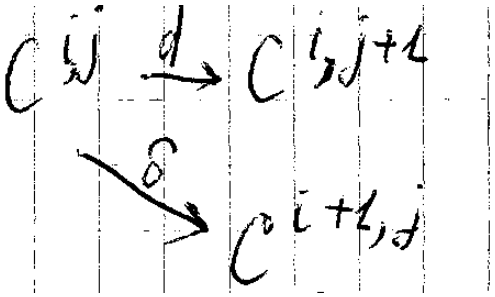


Diagram and typed description.

$$\begin{array}{ccc} C^{ij} & \xrightarrow{d} & C^{i,j+1} \\ & \searrow^{\delta} & \\ & & C^{i+1,j} \end{array}$$

Original source details: GELFAND-P001379

[At right:] d — the differential in the complex (de Rham).

Now for $d + \delta$ take ordinary cohomology.

(These are obtained with C^* [crossed out: illegible mark?]

Uncertain reading. illegible mark

Original source details: GELFAND-P001379

X (integral)).

GELFAND-P001380

For compact X

$$H^n(X,\mathbb{C}) = \bigoplus_{a+b=n} H^{a,b}(X) \qquad \text{(Hodge decomposition)} \tag{11}$$

$$F^i H^n(X,\mathbb{C}) = \bigoplus_{a\geq i} H^{a,b} \qquad \text{(Hodge filtration)} \tag{12}$$

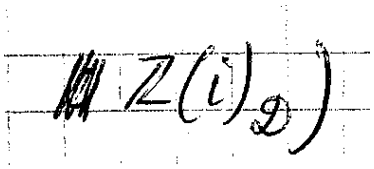
It is known that

$$H^{a,b}(X) = H^b(X,\Omega^a) \tag{13}$$

Then

$$\begin{aligned} 0 \longrightarrow \frac{H^{n-1}(X,\mathbb{C})}{H^{n-1}(X,\mathbb{Z}) + F^i H^{n-1}(X,\mathbb{C})} &\longrightarrow H^n\big(X,\mathbb{Z}(i)_{\mathcal{D}}\big) \\ &\longrightarrow H^n(X,\mathbb{Z}) \cap F^i H^n(X,\mathbb{C}) \longrightarrow 0 \end{aligned} \tag{14}$$

[Crossed out in the middle term of (14), before $\mathbb{Z}(i)_{\mathcal{D}}$: illegible mark?.]



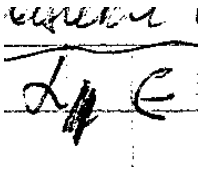
Uncertain reading. illegible mark

Original source details: GELFAND-P001380

Deligne cohomology class $\sim$ (difference of an integral cocycle and a de Rham cocycle) $+$ (a homotopy joining them).

Cochain with coefficients in the Deligne complex

A cocycle α [subscript crossed out or overwritten: illegible mark?]



Uncertain reading. illegible mark

Original source details: GELFAND-P001380

$\in Z^n(X, \mathbb{Z}(i)_{\mathcal{D}})$ consists of a collection

$$\begin{cases} \alpha_0 \in Z^n(X, \mathbb{Z}), \\ \alpha_1 \in C^{n-1}(X, \mathcal{O}), \\ \alpha_2 \in C^{n-2}(X, \Omega^1), \\ \vdots \\ \alpha_i \in C^{n-i}(X, \Omega^{i-1}), \end{cases} \tag{15}$$

compatible in the sense that

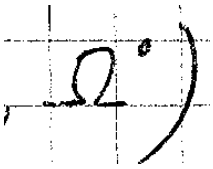
$$\begin{cases} \alpha_0 = \delta\alpha_1, \\ d\alpha_1 = \delta\alpha_2, \\ \vdots \\ \delta\alpha_i = d\alpha_{i-1} \end{cases} \tag{16}$$

Hence

$$d\alpha_i \in Z^{n-i}(X, \Omega^{i,\mathrm{cl}}) \qquad \text{(with values in closed forms)} \tag{17}$$

We can therefore consider

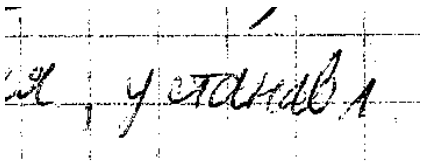
$$\alpha_0 - d\alpha_i \in Z^n(X, \Omega^\circ) \tag{18}$$



Uncertain reading. A small circle is visible above Ω.

Original source details: GELFAND-P001380

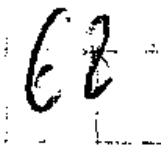
(The rest is a homotopy,



Uncertain reading. «устанавл», short for “establishing,” at the end of the line

Original source details: GELFAND-P001380

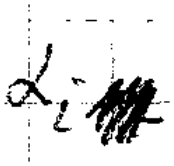
GELFAND-P001381



Uncertain reading. The number at the upper right reads 68 or 62.

Original source details: GELFAND-P001381

$\alpha_1 + \alpha_2 + \ldots + \alpha_i$ [crossed out: illegible mark[?]]

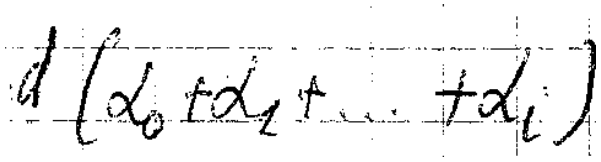


Uncertain reading. illegible mark

Original source details: GELFAND-P001381

$\in C$.

$$d(\alpha_0 + \alpha_1 + \ldots + \alpha_i) = \alpha_0 - d\alpha_i \tag{19}$$



Uncertain reading. In the parentheses in (19) the first term is α_0. The sum in the line above begins with α_1.

Original source details: GELFAND-P001381

So far these are general facts for $\forall$ double complexes.

(see also work by Gabrielov, Gelfand and Losik)

Let us describe the group $H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$ starting from (14).

$$\begin{aligned} 0 \longrightarrow \underbrace{\frac{H^{2i-1}(X,\mathbb{C})}{H^{2i-1}(X,\mathbb{Z}) + F^i}}_{\| \; J^i} \longrightarrow H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}}) \\ \longrightarrow \underbrace{H^{2i}(X,\mathbb{Z}) \cap F^i H^{2i}(X,\mathbb{C})}_{\| \; \mathrm{Hg}^i(X,\mathbb{Z})} \longrightarrow 0 \end{aligned} \tag{20}$$

[Below J^i:] (a complex torus called the ith Griffiths Jacobian)

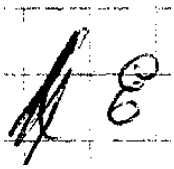
$$H^{2i-1} = F^i \oplus \overline{F}^i$$

$\Rightarrow$ hence the complex structure on J^i.

(21) Example.

$$c_1(\mathcal{E}) = c_1(\det \mathcal{E})$$

g_{ij} — transition functions of the bundle [crossed out: illegible mark?]



Uncertain reading. illegible mark

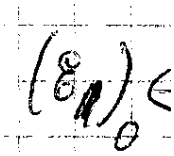
Original source details: GELFAND-P001381

$\mathcal{E}$

(with values in $\mathbb{C}^*$).

$$\begin{aligned} c_1(\mathcal{E}) &\in H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \\ c_1(\mathcal{E})_0 &\in Z^2(X, \mathbb{Z}) \end{aligned} \tag{22}$$

[In (22), the subscript-like mark before the parenthesis after $\mathcal{E}$: illegible mark?.]



Uncertain reading. illegible mark

Original source details: GELFAND-P001381

$$c_1(\mathcal{E})_1 \in C^1(X, \mathcal{O}), \tag{23}$$

and

$$\delta[c_1(\mathcal{E})_1] = c_1(\mathcal{E})_0 \tag{23'}$$

$$c_1(\mathcal{E})_1 = \log(\deg g_{ij}) \tag{24}$$

— any determination of log.

Original source details: GELFAND-P001381

Uncertain reading. The last letter of the operator in (24) resembles g. The earlier formula (21) uses det.

($c_1(\mathcal{E})_0$ — is found from $(23')$ [crossed out: the mark resembles an overwritten parenthesis?]

Original source details: GELFAND-P001381

Uncertain reading. the mark resembles an overwritten parenthesis

and gives the discrepancy).

This gives an isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \tag{25}$$

GELFAND-P001382

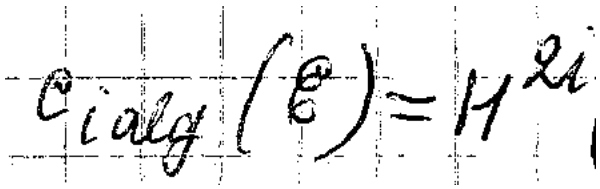
Construction of c_2. (this is already related to the dilogarithm)

c_1 was constructed in 2 steps

$$\mathcal{E} \overset{\det}{\longmapsto} \underset{\text{alg}}{c_1(\mathcal{E})} \in H^1(X, \mathcal{O}^*) \overset{\text{“log”}}{\longrightarrow} H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \tag{26}$$

In principle it should be

$$\begin{array}{ccc} c_i : \mathcal{E} \longmapsto c_{i\,\text{alg}}(\mathcal{E}) & = & H^{2i}(X, \mathbb{Z}(i)_M) \\ \| & & \\ \underset{\text{(motivic)}}{c_{iM}(\mathcal{E})} & & \\ \downarrow \mathcal{L}i_i & & \\ \underset{\text{polylogarithm}}{\uparrow} & & H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}}) \end{array} \tag{27}$$

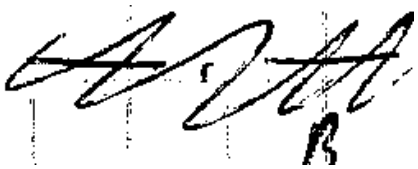


Original source details: GELFAND-P001382

Uncertain reading. In (27) an equals sign appears between $c_{i\,\text{alg}}(\mathcal{E})$ and H^{2i}.

This theory exists up to torsion.

[crossed out: illegible?.]



Original source details: GELFAND-P001382

Uncertain reading. illegible

in it $\mathbb{Z}(1)_M = \mathcal{O}^*[-1]$.

Some requirements

(28) a) $\mathbb{Z}(\text{negative number})_M = 0$

b) $\mathbb{Z}(0)_M = \mathbb{Z}$

c) $\mathbb{Z}(1)_M = \mathcal{O}^*[-1]$

d)

$$\mathbb{Z}(2)_M = \left(\mathbb{Z}(2)^1_M \longrightarrow \underset{\|}{\mathbb{Z}(2)^2_M} \right. \qquad \text{(up to 2-torsion)}$$
$$\wedge^2(\mathcal{O}^*)$$

Original source details: GELFAND-P001382

Uncertain reading. The outer parenthesis after the second term is left open.

$$i \geq 1: \quad H^j(\mathbb{Z}(i)_M) = 0 \qquad \text{when } j \notin [1, i] \tag{29}$$

$$H^i(\mathbb{Z}(i)_M) = \mathcal{K}_i^{\text{Milnor}} \qquad \text{(sheaf of Milnor } K\text{-functors)} \tag{30}$$

K-functor of Milnor.

For a smooth (algebraic) manifold these are sheaves.

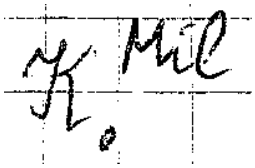
(31) a) $\mathcal{K}_0^{\text{Mil}} = \mathbb{Z}$,

b) $\mathcal{K}_1^{\text{Mil}} = \mathcal{O}^*$

c) [At left: (Steinberg relation).] let $t \in \mathcal{O}^*$ and $1 - t \in \mathcal{O}^*$, then

$$\underbrace{\{t, 1-t\}}_{\substack{\text{Steinberg symbol} \\ \swarrow \\ \text{product in the Milnor algebra}}} = 0$$

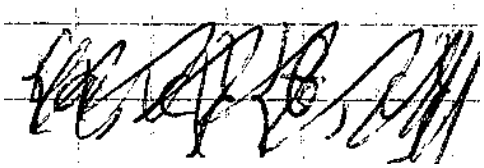
$\mathcal{K}_\circ^{\text{Mil}}$ = free graded algebra with generators [crossed out: $\mathcal{O}^*$] $\mathcal{O}^* = \mathcal{K}_1^{\text{Mil}}$ and the Steinberg relations.



Original source details: GELFAND-P001382

Uncertain reading. In the line on the graded algebra, $\mathcal{K}^{\text{Mil}}$ has a small circle below it that resembles 0.

$\{ab, c\} =$ [crossed out: There appear to be two groups, $\{a, c\}$ and $\{b, c\}$. The sign between them is uncertain.?]



Uncertain reading. There appear to be two groups, $\{a, c\}$ and $\{b, c\}$. The sign between them is uncertain.

Original source details: GELFAND-P001382

?.

[At right:] $\{\ \}$ — addition

GELFAND-P001383

[source number: 69]

(32) The operation $\{\ \}$ is supercommutative in $\mathcal{K}^{\text{Mil}}$.

In (28) d)

[Crossed-out number in the left margin: (33)?.]

Uncertain reading. (33)?

Original source details: GELFAND-P001383

[Entire line crossed out: $\mathbb{Z}(2)^1_M = \mathbb{Z}[\mathbb{P}^1(X)]$.]

$$\mathbb{Z}(2)^1_M = \mathbb{Z}\big[(\mathbb{P}^1 \setminus \{0, 1, \infty\})(X)\big] \,/\, (\text{relations}) \tag{33}$$

[From $\mathbb{P}^1$ a line runs down to the note:] sheaf of functions on X, avoiding the values $0, 1, \infty$.

Differential:

$$\begin{aligned} \mathbb{Z}(2)^1_M &\longrightarrow \mathbb{Z}(2)^2_M \\ t &\longmapsto t \wedge (1 - t) \end{aligned} \tag{34}$$

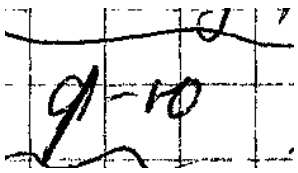
The relation comes from the observation that it lies in the kernel of the Steinberg symbol.

(5-term relation for the dilogarithm)

Let f, g — be 2 functions, nowhere equal to one another and avoiding the values $0, 1, \infty$.

$$(f, g, 0, 1, \infty) \tag{35}$$

(36) 5 distinct points on $\mathbb{P}^1$, omitting one at a time gives 5 cross-ratios, and one has to take their alternating function?.



Uncertain reading. function

Original source details: GELFAND-P001383

This is the 5-term relation.

Instead one can take 5 independent functions.

(37) Assertion. Relation (36) lies in the kernel of the differential.

First devised by Spencer, Bloch.

Original source details: GELFAND-P001383

Uncertain reading. The name before Bloch is written «Спенсер» (Spencer).

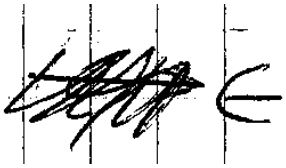
Suslin established the link with K-theory.

The complex is called the Bloch–Suslin complex.

One needs to construct a map

$$\mathcal{E} \longmapsto c_2(\mathcal{E})_M \in H^4(X, \mathbb{Z}(2)_M) \tag{38}$$

[In (38), between $c_2(\mathcal{E})_M$ and $\in$ crossed out: illegible mathematical fragment[?].]



Original source details: GELFAND-P001383

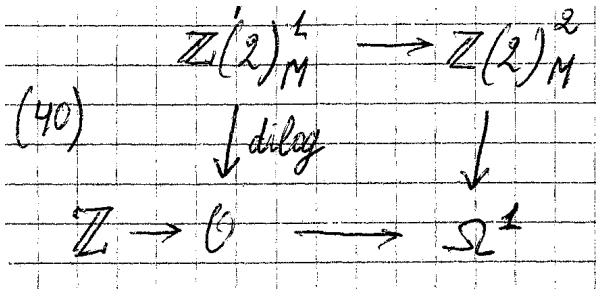
Uncertain reading. illegible mathematical fragment

and a function

$$\mathbb{Z}(2)_M \longrightarrow \mathbb{Z}(2)_{\mathcal{D}} \tag{39}$$

GELFAND-P001384

Let us construct map (39)

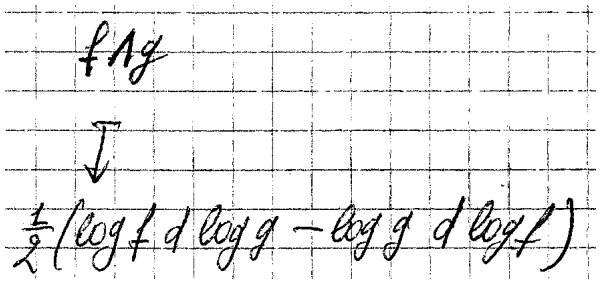


Original source details: GELFAND-P001384

Diagram and typed description.

$$\begin{array}{ccccc} & & \mathbb{Z}(2)^1_M & \longrightarrow & \mathbb{Z}(2)^2_M \\ (40) & & \downarrow{\scriptstyle \text{dilog}} & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathcal{O} & \longrightarrow & \Omega^1 \end{array}$$

[To the right of the diagram:]



Original source details: GELFAND-P001384

Diagram and typed description.

$$\begin{array}{c} f \wedge g \\ \downarrow \\ \frac{1}{2}(\log f \, d\log g - \log g \, d\log f) \end{array}$$

Chern class.

$$\mathcal{E} \longmapsto c_2(\mathcal{E})_M \in H^4(X, \mathbb{Z}(2)_M), \tag{41}$$

$\dim \mathcal{E} = 2$, $\{U_i\}$, s_i — a section of $\mathcal{E}$ over U_i.

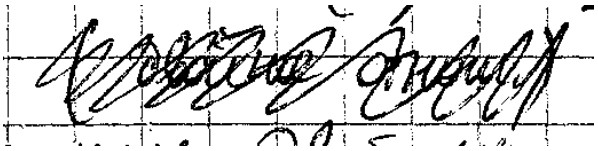
2 components.

$$c_2(\mathcal{E})^2_M \in C^2(\{U_i\}, \mathbb{Z}(2)^2_M)$$
$$c_2(\mathcal{E})^1_M \in C^3(\{U_i\}, \mathbb{Z}(2)^1_M)$$

$(s_1, \ldots, s_4)$ — 4 sections of $\mathcal{E}$.

Consider 4 points $\mathbb{P}(\mathcal{E}) \ni s_1, \ldots, s_4$

and so on. [Crossed out: “cross-ratio”?.]



Original source details: GELFAND-P001384

Uncertain reading. “cross-ratio”

It is constructed using the cross-ratio.

Chapter 31

GELFAND-P001385

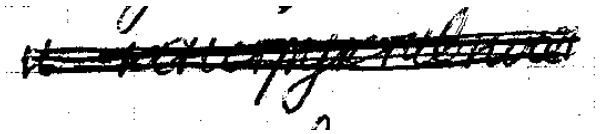
[source number: 1]

5/X-87 Gelfand Seminar.

A. Beilinson. Constructible sheaves (continued).

Syllabus on sheaves.

1. Sheaf, category of sheaves.
2. Operations on sheaves:
   a) Gluing over an open cover
   b) Gluing for an open and a closed set
   c) Direct and inverse images.
3. Locally constant sheaves.
4. Combinatorial [crossed out: possibly “constructible”[?]] sheaves = representations of ordered sets.



Uncertain reading. possibly “constructible”

Original source details: GELFAND-P001385

5. Constructible sheaves.

6. Cohomology with coefficients in sheaves.

7. Poincaré–Verdier duality

For items 1–5, see the preceding notebook.

Sheaves form an abelian category.

Let $\mathcal{F}, G$ — be sheaves, with a morphism $\mathcal{F} \xrightarrow{\varphi} G$.

All morphisms form a vector space (over $\mathbb{C}$)

Definition 1.

$\operatorname{Ker} \varphi \subset \mathcal{F}$ [Below the formula: subsheaf]

$$(\ker \varphi)(V) = \{a \in \mathcal{F}(V),\ \varphi(a) = 0\}$$

$\operatorname{Coker} \varphi$ — the sheaf associated to the presheaf C given by

$$C(V) = \operatorname{Coker}\big(\varphi : \mathcal{F}(V) \longrightarrow G(V)\big)$$

Constructing a sheaf from a presheaf.

Let C — be a presheaf.

Let V — be an open set, $V = \bigcup V_\alpha$ and suppose we are given $a_\alpha \in C(V_\alpha)$, compatible with one another:

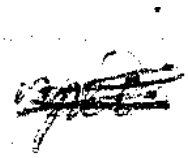
$$a_\alpha|_{V_\alpha \cap V_\beta} = a_\beta|_{V_\alpha \cap V_\beta}.$$

Such a family determines a section of the sheaf. There is an identification of sections "obtained"? from different families.

Uncertain reading. "obtained"

Original source details: GELFAND-P001385

C^+ — [crossed out: possibly «пред.» ("pre...")?] the sheaf associated to C.



Uncertain reading. possibly «пред.» ("pre...")

Original source details: GELFAND-P001385

GELFAND-P001386

There is a map $C \longrightarrow C^+$, universal in the following sense

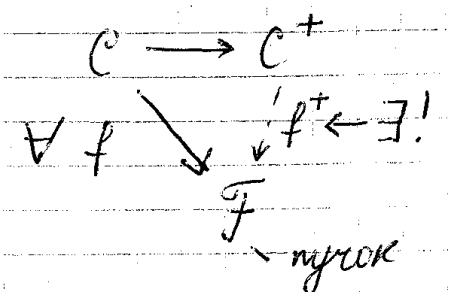
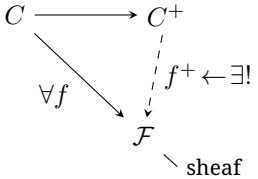


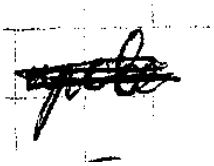
Diagram and typed description.



Original source details: GELFAND-P001386

$\forall$ map $f: C \longrightarrow \mathcal{F}$ [below $\mathcal{F}$: sheaf] $\exists!$ $f^+: C^+ \longrightarrow \mathcal{F}$, such that the diagram commutes

Exercise 1. [crossed out: "про..."?] construct C^+.



Uncertain reading. "про..."?

Original source details: GELFAND-P001386

There are maps

$$\operatorname{Ker}\varphi \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\varphi} G \xrightarrow{\beta} \operatorname{Coker}\varphi$$

Exercise 2. $\operatorname{Ker}\beta = \operatorname{Coker}\alpha$

An exact sequence of sheaves is defined in the natural way.

Exercise 3. The sequence $\mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3$ is exact $\Longleftrightarrow$ $\forall x \in X$ the sequence $\mathcal{F}_{1x} \longrightarrow \mathcal{F}_{2x} \longrightarrow \mathcal{F}_{3x}$ is exact.

(Consider the inclusion $x \xrightarrow{i_x} X$, then the stalk $\mathcal{F}_x$ at the point x has the form $\mathcal{F}_x = i_x^*\mathcal{F}$)

Definition 2 (Flabby sheaf.)

A sheaf $\mathcal{F}$ is called flabby if any section over an open set V extends to a section over all of X, i.e. $\forall$ open $V \subset X$

$$\mathcal{F}(X) \longrightarrow \mathcal{F}(V)$$

is an epimorphism.

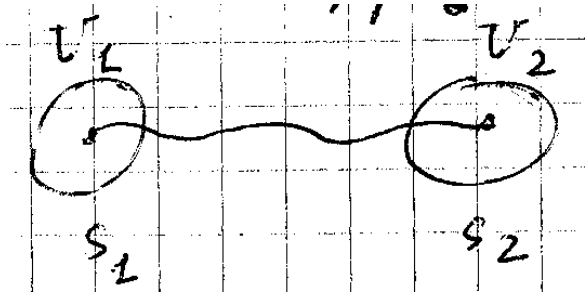
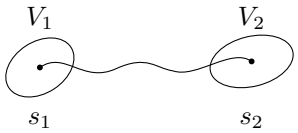


Diagram and typed description.



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In locally constant sheaves, sections can be transported [inserted above the line: (along a path)] and therefore such sheaves are not flabby.

Problem 4. Γ — a [inserted above the line: finite] ordered set; introduce the topology

$$\overline{x} := \{y : \ y \leq x\} \quad \text{— closure of the point } x$$

a) Sheaves on Γ = representations of Γ

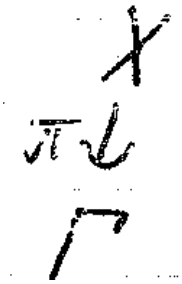
b) Describe the flabby sheaves on Γ

GELFAND-P001387

[source number: 2]

c) Connection with combinatorial sheaves

Suppose X has a stratification, consider the map



Original source details: GELFAND-P001387

Diagram and typed description.

$$\begin{array}{c} X \\ \pi \downarrow \\ \Gamma \end{array}$$

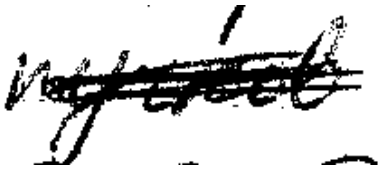
(to the point x — its stratum)

Γ — ordered set of strata.

[below Γ:] Give Γ the (Zariski) topology

Claim.

$\forall$ combinatorial sheaf on X is the inverse image of [crossed out: "пучков" (of sheaves)?] a sheaf on Γ (and conversely)



Original source details: GELFAND-P001387

Uncertain reading. "пучков" (of sheaves)?

$$\pi^*: \text{sheaves on } \Gamma \xrightarrow{\sim} \text{combinatorial sheaves on } X.$$

Exercise 5. a) Suppose

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is a short exact sequence of sheaves. Then the sequence of sections

$$0 \longrightarrow \mathcal{F}_1(X) \longrightarrow \mathcal{F}_2(X) \longrightarrow \mathcal{F}_3(X)$$

is exact.

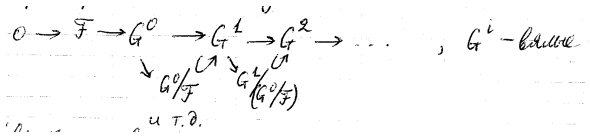
b) If $\mathcal{F}_1$ is flabby, the following sequence is exact

$$0 \longrightarrow \mathcal{F}_1(X) \longrightarrow \mathcal{F}_2(X) \longrightarrow \mathcal{F}_3(X) \longrightarrow 0$$

Flabby resolution of a sheaf.

Exercise 6 Let $\mathcal{F}$ be a sheaf. Then 1) It can be embedded in a flabby sheaf $\mathcal{F} \hookrightarrow G$.

2) Now construct from $\mathcal{F}$ an exact sequence



Original source details: GELFAND-P001387

Diagram and typed description.

$$0 \to \mathcal{F} \to G^0 \longrightarrow G^1 \longrightarrow G^2 \to \cdots, \quad G^i \text{ — flabby}$$

$$G^0/\mathcal{F} \qquad G^1/(G^0/\mathcal{F})$$

and so on

(flabby resolution)

3) $\forall$ two flabby resolutions can be embedded in a common third one.

Definition 3.

$$H^i(X, \mathcal{F}) = H^i\big(0 \longrightarrow G^0(X) \longrightarrow G^1(X) \longrightarrow \dots\big)$$

(here $0 \longrightarrow \mathcal{F} \longrightarrow G^0 \longrightarrow G^1 \longrightarrow \dots$ — flabby resolution $\mathcal{F}$)

On the right is the cohomology of a complex of vector spaces.

Definition 4. [crossed out: Dual sheaf.] Cohomology with compact support.

Suppose X is locally compact. Then we can consider sections with compact support: $\mathcal{F}_c(X)$ — sections with compact support on X.

GELFAND-P001388

$$H^i_c(X, \mathcal{F}) = H^i\big(0 \longrightarrow G^0_c(X) \longrightarrow G^1_c(X) \longrightarrow \dots\big)$$

Definition 5. Dual sheaf.

Presheaf: $\mathcal{F}^*(V) = \mathcal{F}_c(V)^*$ (analogue: “distribution” or possibly “distributions”[?])

Uncertain reading. “distribution” or possibly “distributions”

Original source details: GELFAND-P001388

(dual space in the sense of linear algebra)

$$V_1 \subset V_2 \;\Rightarrow\; \mathcal{F}_c(V_1) \longrightarrow \mathcal{F}_c(V_2) \quad \text{(embedding)}$$

and this gives $\mathcal{F}^*(V_2) \longrightarrow \mathcal{F}^*(V_1)$

Exercise 7.

a) $\mathcal{F}^*$ is a sheaf if $\mathcal{F}$ is flabby.

b) If $\mathcal{F}^*$ is a sheaf, then $\mathcal{F}^*$ is flabby, i.e. $\forall \underset{\text{open}}{V} \subset X$ the map $\mathcal{F}^*(X) \longrightarrow \mathcal{F}^*(V)$ is surjective

GELFAND-P001389

12 October 1987. Gelfand Seminar.

Vakhtangov told the actors in Princess Turandot to come on stage in fashionable clothes and change at once (put on a turban, etc.). The themes of mathematics are eternal, only the language changes.

I. M. G. On a combinatorial conjecture of Dyson, connected with polynomial coefficients

A paper from the thick green collection on combinatorics

Consider

$$F(x,a)=\prod_{i\neq j}\left(1-\frac{x_j}{x_i}\right)^{a_j},\qquad a_j\in\mathbb{N}. \tag{1}$$

Find the constant term

Dyson's conjecture (1962), the answer

$$\frac{(a_1+a_2\ldots+a_n)!}{a_1!a_2!\ldots a_n!} \tag{2}$$

In 1962 two proofs were given in J. Math. Phys. (long ones)

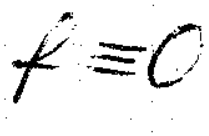
In 1970 in J. Math. Phys., a proof by

Uncertain reading. the name appears as "Вуда" (Vuda)

Original source details: GELFAND-P001389

in two lines.

Proof Applying Lagrange's interpolation formula to the function $f\equiv 0$



Uncertain reading. The manuscript reads $f\equiv 0$.

Original source details: GELFAND-P001389

and then setting $x=0$, we obtain

$$\sum_j\prod_{i\neq j}\left(1-\frac{x_j}{x_i}\right)^{-1}=1 \tag{3}$$

Multiplying by $F(x,a)$, we obtain

$$\sum_j F(x, a_1, \ldots, a_{j-1}, a_j - 1, a_{j+1}, \ldots, a_n) = F(x, a) \tag{4}$$

[crossed out: illegible opening mark?]

Uncertain reading. illegible opening mark

Original source details: GELFAND-P001390

Let $G(a)$ be the constant term. Then from (4)

$$G(a) = \sum G(a_1, \ldots, a_{j-1}, a_j - 1, a_{j+1}, \ldots, a_n), \tag{5}$$

which gives a recurrence formula.

When $a_j = 0$ we obtain

$$G(a_1, \ldots, a_{j-1}, 0, a_{j+1}, \ldots, a_n) = G(a_1, \ldots, a_{j-1}, a_{j+1}, \ldots, a_n) \tag{6}$$

Moreover,

$$G(0) = 1 \tag{7}$$

The relations (5)–(7) are sufficient as recurrence relations. They determine everything and yield (2).

A. Veselov

On geodesics on an ellipsoid and motion on a sphere

Following Jacobi and others, 1982. Knörrer's observation: equivalence of two problems.

[Two columns separated by a curved line. Left column.]

1) $(Aq, q) = 1$

geodesics on the ellipsoid

Weierstrass found them in θ-functions (1861)

Equation of the geodesics in the natural parameter:

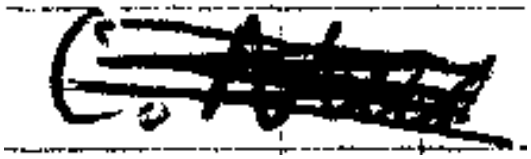
$$\ddot{q} = \lambda A q \tag{1}$$

(acceleration $\sim$ the normal)

[At right:] (λ is the Lagrange multiplier, found from the condition $(Aq, q) = 1$)

[Right column.]

2) Another problem [crossed out: $C.$ the name after the initial is struck through, possibly "Neu..."?]



Uncertain reading. $C.$ the name after the initial is struck through, possibly "Neu..."

Original source details: GELFAND-P001390

C. Neumann (1859)

Motion with potential energy

$$U(x) = \frac{1}{2}(Ax, x)$$

on the sphere $|x|^2 = 1$

Here the equation is

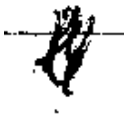
$$x'' = -Ax + ux \tag{1'}$$

or in coordinates

$$x_i'' = -a_i x_i + u x_i \tag{1''}$$

[a symbol is crossed out at right: q? / u?[?]]

Uncertain reading. q? / u?



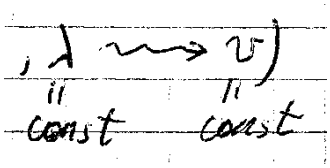
Original source details: GELFAND-P001390

[Below the columns.]

Switch to another parameter

$$q'' = -Aq + vq' \tag{2}$$

(the changes can be written out explicitly, $\underset{\text{const}}{\lambda} \rightsquigarrow \underset{\text{const}}{v}$)



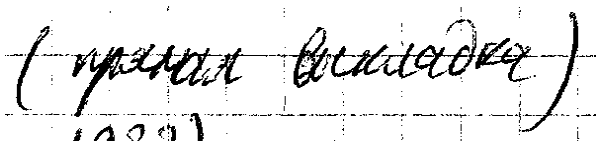
Uncertain reading. The manuscript has “const” under λ and v.

Original source details: GELFAND-P001390

and then the substitution

$$x = \frac{Aq}{|Aq|} \qquad \text{(Gauss substitution)} \tag{3}$$

takes solutions of problem 1) to solutions of problem 2) (direct computation)



Uncertain reading. The manuscript reads “прямая выкладка” (direct computation).

Original source details: GELFAND-P001390

(H. Knörrer, 1982)

Equation

[At right:] Introduce the operator:

$$L = -\frac{d^2}{dt^2} + u, \qquad Lx_i = a_i x_i \tag{4}$$

(x_i — eigenfunction!!)

GELFAND-P001391

[source number: 12]

At left:

$$\widetilde{L} = -\frac{d^2}{dt^2} + v\frac{d}{dt} \tag{5}$$

(conjugate to L by the Gauss substitution or the Miura substitution)

Knörrer arrived at it through the Moser–Trubowitz isomorphism

Consider problem 2)

There are two symplectic forms.

Problem (I. M. G.) There is a symmetric matrix A and a matrix K of rank one. How, from the eigenvalues of $A + K$ can one recover K? (A is fixed)

— this is exactly Jacobi's formula.

V. For $n = 2$ introduce new coordinates

$$M = [\dot{x}, x], \qquad \text{(angular momentum — cross product)}$$
$$p = x$$

1°. Lie–Berezin–Kirillov bracket

$$\{M_i, M_j\} = \varepsilon_{ijk} M_k$$
$$\{M_i, p_j\} = \varepsilon_{ijk} p_k$$
$$\{p_i, p_j\} = 0$$

Hamiltonian

$$2H = M^2 + (Ap, p)$$

Then

$$\begin{cases} \dot{M}_i = \{M_i, H\} \\ \dot{p}_i = \{p_i, H\} \end{cases} \qquad \text{— the Neumann equations}$$

Second bracket

$$\varepsilon_{ijk} a_k M_k = \{M_i, M_j\}$$
$$\varepsilon_{ijk} a_k p_k = \{M_i, p_j\}$$
$$\varepsilon_{ijk} M_k = \{p_i, p_j\}$$

Hamiltonian $F = \frac{1}{2}p^2$

I. M. G. This agrees with what Arnold discussed

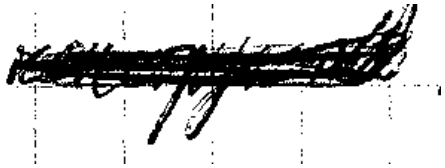
(see the previous notebook)

The system is Hamiltonian in both.

GELFAND-P001392

A. Beilinson. Constructible sheaves (continued)

Representations of a partially ordered set $\Longleftrightarrow$ [crossed out: possibly "constructible"?] combinatorial sheaves



Original source details: GELFAND-P001392

Uncertain reading. possibly "constructible"

Diagram and typed description.

$* \leq x$ $\qquad$ $\bullet \leq \bullet \geq \bullet$

Original source details: GELFAND-P001392

A combinatorial sheaf is a sheaf constant on each stratum.

Homological algebra.

$\mathcal{A}$ — abelian category (for example, of sheaves over a given space or of modules)

$C(\mathcal{A})$ — category of complexes over $\mathcal{A}$

one needs to know the derived category (Verdier, 1961)

Objects of [crossed out: possibly “из” (from) or a membership sign[?]] $C(\mathcal{A})$

Uncertain reading. possibly “из” (from) or a membership sign

Original source details: GELFAND-P001392

$$\ldots \longrightarrow X^i \xrightarrow{d} X^{i+1} \xrightarrow{d} X^{i+2} \longrightarrow \ldots$$
$$d^2 = 0$$

morphisms are morphisms of complexes (families of morphisms

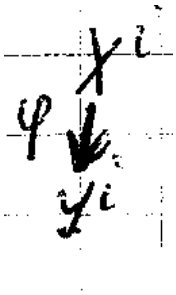


Diagram and typed description.

$$\begin{array}{c} X^i \\ \varphi \downarrow \\ Y^i \end{array}$$

Original source details: GELFAND-P001392

, commuting with d)

$C(\mathcal{A})$ — also an abelian category

Bounded complexes (only finitely many $\neq 0$) $C^b(\mathcal{A})$

bounded on the left or the right $C^{\pm}(\mathcal{A})$

$$C^+(\mathcal{A}) \text{ — bounded on the left}$$
$$C^-(\mathcal{A}) \text{ — bounded on the right}$$

Homotopy category $K(\mathcal{A})$

the objects of $K(\mathcal{A})$ are the same as in $C(\mathcal{A})$, and the morphisms are equivalence classes of morphisms of $C(\mathcal{A})$ with respect to homotopy.

$$\mathrm{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet)$$

A morphism $X^\bullet \xrightarrow{\varphi} Y^\bullet$ is a [inserted above the line: such] family $\varphi^i \colon X^i \longrightarrow Y^i$, for which the diagram commutes:

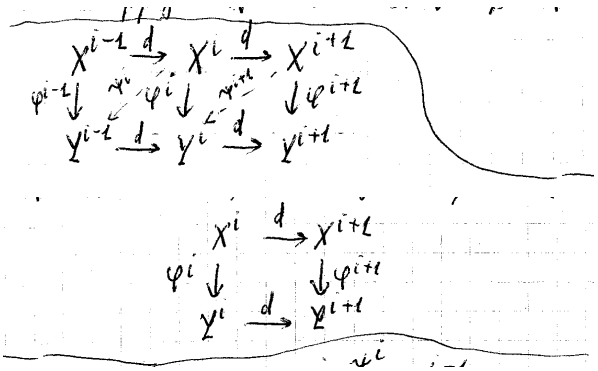


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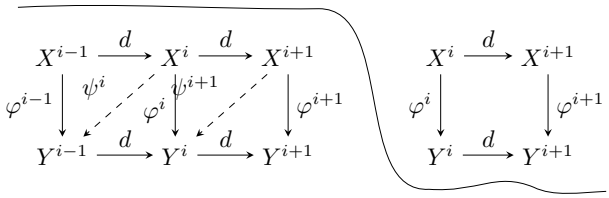
Diagram and typed description.



φ is called null-homotopic, if $\exists$ a family of maps $X^i \xrightarrow{\psi^i} Y^{i-1}$, such that

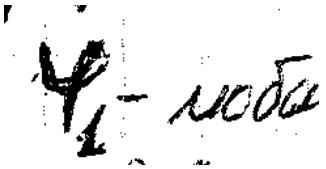
$$\varphi^i = d\psi^i + \psi^{i+1}d \tag{1}$$

GELFAND-P001393

[source number: 13]

Exercise 1. Check that for any system of maps ψ^i the family (1) gives a morphism of complexes, i.e. $d\varphi_i = \varphi_{i+1}d$.

Exercise 2. Morphisms homotopic to 0 form an ideal, i.e. if φ is homotopic to 0 and ψ_1 is any morphism,



Original source details: GELFAND-P001393

Uncertain reading. Here the symbol resembles ψ_1, and in the composition the symbol is φ_1.

then $\varphi \circ \varphi_1$ is homotopic to 0.
(and the same is true for the composition $\varphi_2 \circ \varphi$ with any morphism φ_2)

$$\mathrm{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet) = \mathrm{Hom}_{C(\mathcal{A})}(X^\bullet, Y^\bullet) \Big/ (\text{morphisms homotopic to } 0)$$

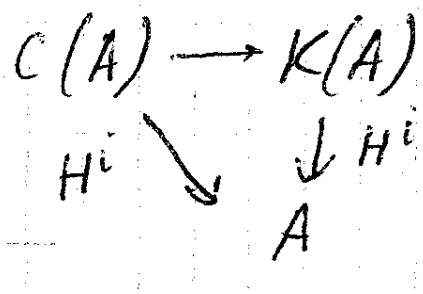
There are obvious functors

$$\mathcal{A} \longrightarrow C(\mathcal{A}) \longrightarrow K(\mathcal{A})$$

Cohomology functor

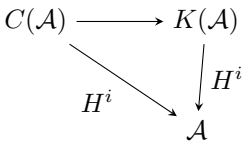
$$H^i \colon C(\mathcal{A}) \longrightarrow \mathcal{A} \qquad \left(H^i(X^\bullet) = \frac{\ker(d \colon X^i \longrightarrow X^{i+1})}{\mathrm{Im}(d \colon X^{i-1} \longrightarrow X^i)} \right)$$

Exercise 3. Prove that $\exists!$ a functor $H^i \colon K(\mathcal{A}) \longrightarrow \mathcal{A}$, such that



Original source details: GELFAND-P001393

Diagram and typed description.



On morphisms H^i is also defined in the obvious way.

Definition.

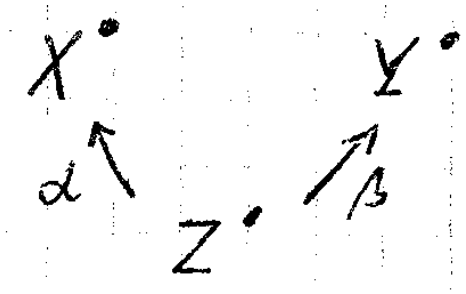
$$\varphi \colon X^\bullet \underset{\text{in the category } K(\mathcal{A})}{\longrightarrow} Y^\bullet$$

is called a quasi-isomorphism if it induces an isomorphism

$$H^i(X^\bullet) \longrightarrow H^i(Y^\bullet).$$

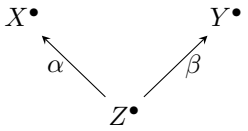
Derived category

Exercise 4. Suppose there are three complexes



Original source details: GELFAND-P001393

Diagram and typed description.



and morphisms, with α — a quasi-isomorphism.

Then there is a diagram (commutative)

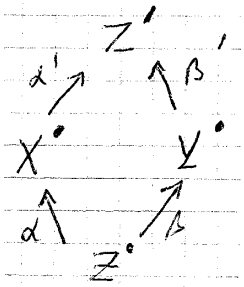


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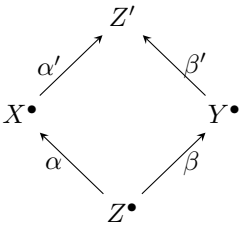
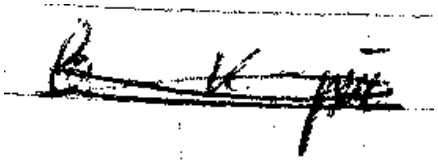
Diagram and typed description.



[crossed out to the left of the diagram: “in which”?[?].]

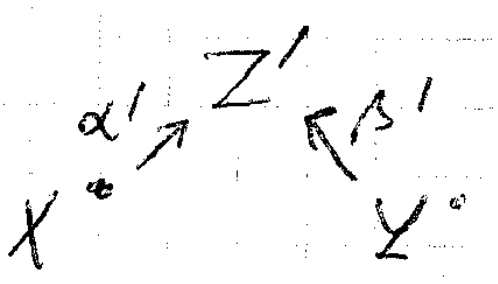


Original source details: GELFAND-P001393

Uncertain reading. “in which”?

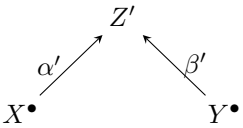
in which β' — a quasi-isomorphism.

The same is true if we start with the diagram



Original source details: GELFAND-P001393

Diagram and typed description.



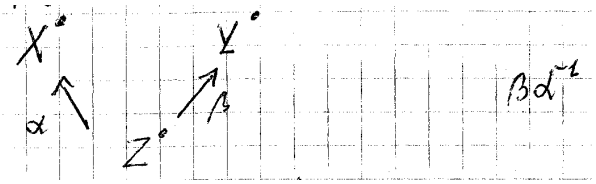
with quasi-isomorphism α'.

GELFAND-P001394

Derived category $D(\mathcal{A})$

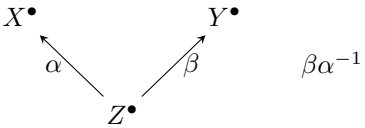
Objects = objects of $K(\mathcal{A})$ (i.e. complexes)

Morphisms [inserted above the line: $X^\bullet \longrightarrow Y^\bullet$] — equivalence classes of diagrams



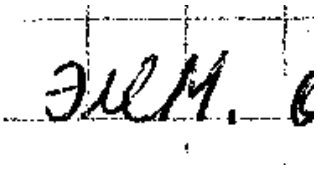
Original source details: GELFAND-P001394

Diagram and typed description.



(α — a quasi-isomorphism)

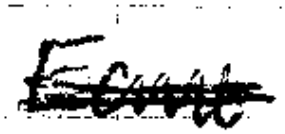
under the equivalence relation generated by the elementary relation



Uncertain reading. The abbreviation reads “элм.” (elementary).

Original source details: GELFAND-P001394

[crossed out: “There is”? / “If”?[?].]



Uncertain reading. “There is”? / “If”?

Original source details: GELFAND-P001394

(There is a functor $K(\mathcal{A}) \longrightarrow D(\mathcal{A})$)

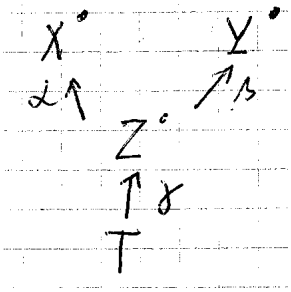
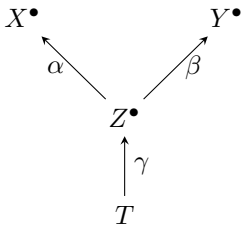
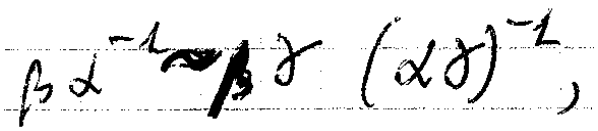


Diagram and typed description.



Original source details: GELFAND-P001394

$\beta\alpha^{-1} \sim \beta\gamma(\alpha\gamma)^{-1}$, where γ — a quasi-isomorphism.



Uncertain reading. The manuscript has reinforced strokes on $\sim$ and β.

Original source details: GELFAND-P001394

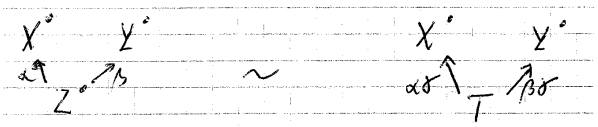
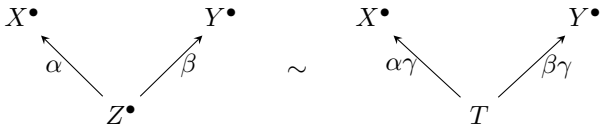


Diagram and typed description.



Original source details: GELFAND-P001394

We can now compose the morphisms.

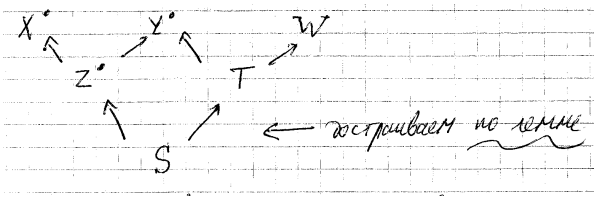
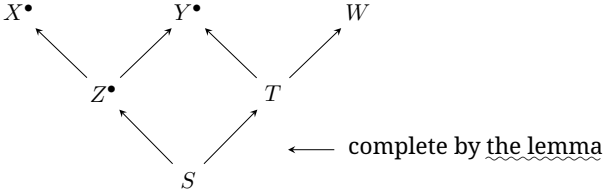


Diagram and typed description.



Original source details: GELFAND-P001394

the composition is the outside of the diagram.

GELFAND-P001395

19.10.87. Gelfand Seminar.

I. M. Gelfand Calculation of sums of kth powers and other sums

Uncertain reading. (following a recent idea?)

Original source details: GELFAND-P001395

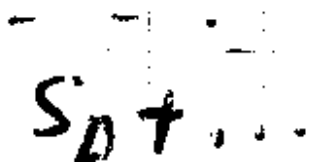
$$F'(x) = f(x) \qquad f(x) = x^k, \qquad F(x) = \frac{x^{k+1}}{k+1}$$

$$F(2) - F(1) = \frac{f(1)}{1!} + \frac{f'(1)}{2!} + \frac{f''(1)}{3!} + \dots$$
$$F(3) - F(2) = \frac{f(2)}{1!} + \frac{f'(2)}{2!} + \frac{f''(2)}{3!} + \dots$$
$$F(n+1) - F(n) = f(n) + \frac{f'(n)}{2!} + \frac{f''(n)}{3!} + \dots$$

$$F(n+1) - F(1) = S_0 + \frac{S_1}{2!} + \frac{S_2}{3!} + \dots$$

similarly,

$$f(n+1) - f(1) = S_1 + \frac{S_2}{2!} + \dots$$
$$f'(n+1) - f'(1) = S_2 + \frac{S_3}{2!} + \dots$$
$$\dots \qquad \dots$$
$$f^{(p)}(n+1) - f^{(p)}(1) = \dots \quad S_p + \dots$$



Uncertain reading. The index on the last line reads as S_p.

Original source details: GELFAND-P001395

[To the right of the system] Here

$$S_0 = f(1) + f(2) + \ldots + f(n)$$
$$S_1 = f'(1) + f'(2) + \ldots + f'(n)$$
$$\ldots$$
$$S_k = f^{(k)}(1) + f^{(k)}(2) + \ldots + f^{(k)}(n)$$

Hence [crossed out to the left of the determinant: “det”?[?]]

Uncertain reading. “det”?

Original source details: GELFAND-P001395

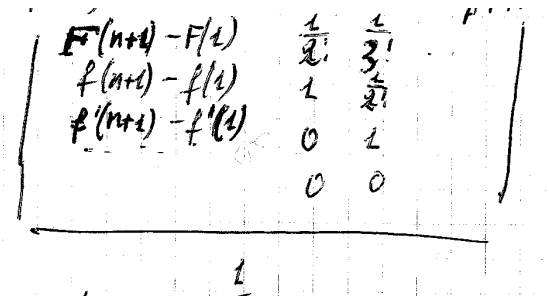


Diagram and typed description.

$$S_0 = \frac{\begin{vmatrix} F(n+1)-F(1) & \frac{1}{2!} & \frac{1}{3!} & \ldots \\ f(n+1)-f(1) & 1 & \frac{1}{2!} & \\ f'(n+1)-f'(1) & 0 & 1 & \\ \ldots & 0 & 0 & \end{vmatrix}}{1}$$

Original source details: GELFAND-P001395

Expand along the first column

$$S_0 = A_0\big(F(n+1) - F(1)\big) + A_1\big(f(n+1) - f(1)\big) + \ldots$$

coefficients are universal, it works for $\forall$ functions!!

Substitute $f(x) = e^{tx}$

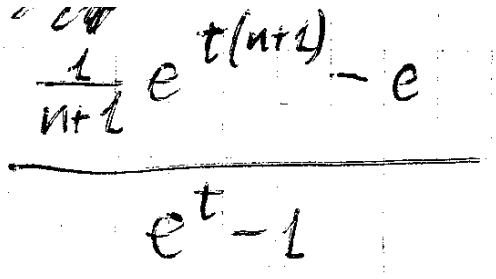
banded determinants appear!

[crossed out: possibly $S_0 = A_0$[?]]

Uncertain reading. possibly $S_0 = A_0$

Original source details: GELFAND-P001395

$$\frac{\frac{1}{n+1}e^{t(n+1)}-e}{e^t-1}$$



Original source details: GELFAND-P001395

Uncertain reading. In the manuscript the numerator reads $\frac{1}{n+1}e^{t(n+1)}-e$.

expand [crossed out: “series”?[?]]

Original source details: GELFAND-P001395

Uncertain reading. “series”?

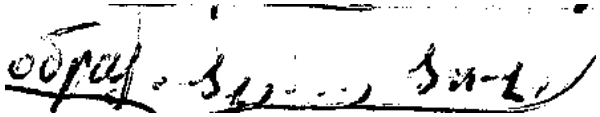
in $t \implies$ Bernoulli numbers

Problem. The symmetric group S_n is defined

Original source details: GELFAND-P001395

Uncertain reading. The verb appears to read “определяется” (is defined).

[inserted above the line: (generators $s_1, \ldots, s_{n-1}$)]



Original source details: GELFAND-P001395

Uncertain reading. The abbreviation reads “образ.” (generators).

by the relations

$$s_i^2 = 1, \qquad s_i s_j = s_j s_i \quad \text{when } |i-j| \geq 2, \qquad (s_i s_{i+1})^3 = 1$$

Consider the parabolic subgroup

$$G = \Sigma_n, \qquad H = \Sigma_k \times \Sigma_{n-k}, \qquad G \supset H$$

$$\mathcal{A} = \mathbb{C}[G] = \left\{ \sum_{g \in G} a_g g, \quad a_g \in \mathbb{C} \right\}$$

GELFAND-P001396

[source number: 23]

$$\mathcal{A} \supset \underset{\text{subalgebra}}{\mathcal{A}'} = \{x \in A,\ xh = hx = x \text{ for any } h \in H\}$$

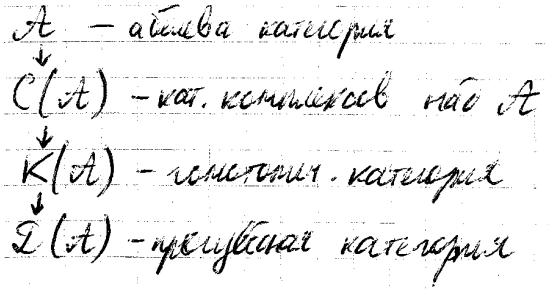
[crossed out: illegible mark?]

Uncertain reading. illegible mark

Original source details: GELFAND-P001396

Beilinson, A. (continued)

Derived categories.



Original source details: GELFAND-P001396

Diagram and typed description.

$\mathcal{A}$ — abelian category

$\downarrow$

$C(\mathcal{A})$ — category of complexes over $\mathcal{A}$

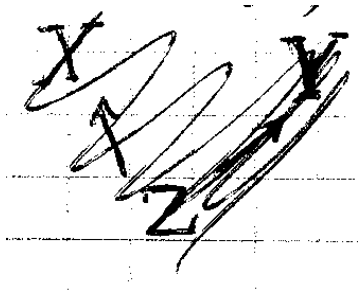
$\downarrow$

$K(\mathcal{A})$ — homotopy category

$\downarrow$

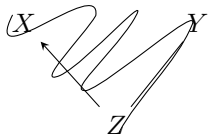
$D(\mathcal{A})$ — derived category

the objects of $D(\mathcal{A})$ are complexes, and the morphisms are equivalence classes of diagrams

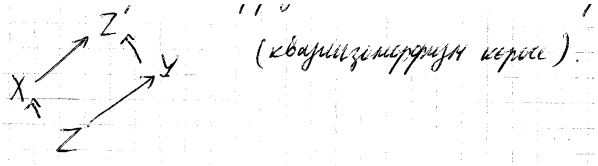


Original source details: GELFAND-P001396

Diagram and typed description.

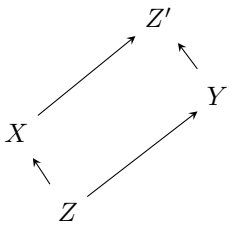


The diagram is crossed out. The right arrow is partly obscured.



Original source details: GELFAND-P001396

Diagram and typed description.



(quasi-isomorphisms are shorter).

Example 1. Let X be a complex with [crossed out: possibly "co" (with)?] $H^i(X) = 0$ for $i \neq 0$. Then X is canonically quasi-isomorphic to $H^0(X)$:

Original source details: GELFAND-P001396

Uncertain reading. possibly "co" (with)

$$X \cong H^0(X) \text{ in } D(\mathcal{A})$$

(In the homotopy category this is not so!)

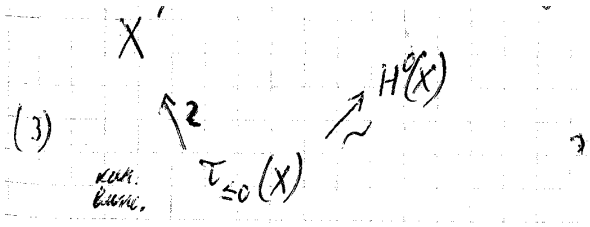
For a complex X introduce the subcomplex $\tau_{\leq 0}(X) \in X$:

$$\tau_{\leq 0}(X)^i = \begin{cases} X^i, & i < 0 \\ \ker(d\colon X^0 \to X^1), & i = 0 \\ 0, & i > 0 \end{cases} \tag{1}$$

It is easy to see that

$$H^j(\tau_{\leq 0}(X)) = \begin{cases} H^j(X), & j \leq 0 \\ 0, & j > 0 \end{cases} \tag{2}$$

The diagram



Original source details: GELFAND-P001396

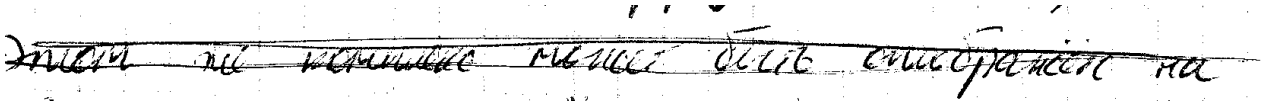
Diagram and typed description.

$$\begin{array}{ccc} & X & \\ (3) & \swarrow^{2?} & \searrow^{\sim} \\ \text{can. inclusion}_{\tau_{\leq 0}(X)} & & H^0(X) \end{array}$$

The mark by the left arrow resembles a 2 or a rotated tilde.

gives the required isomorphism in $D(\mathcal{A})$.

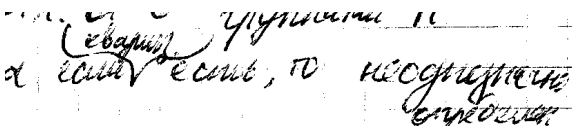
[crossed out: This same complex can be projected onto...?]



Original source details: GELFAND-P001396

Uncertain reading. This same complex can be projected onto...?

Remark. a complex X with several nonzero cohomology groups H^i is not quasi-isomorphic (in general) to the sum, and if there [inserted above the line: (quasi)] is one, the way is non-unique.



Original source details: GELFAND-P001396

Uncertain reading. The ending probably reads "in a non-unique way".

GELFAND-P001397

Structure of a triangulated category

$K(A)$ and $D(A)$ are not abelian categories.

Exercise 1. Check that

$$K(A) \text{ — abelian} \quad \Longleftrightarrow \quad \mathcal{A} \text{ is semisimple}$$

($\mathcal{A}$ — semisimple $\Longleftrightarrow$ [($\forall$ object) = (direct sum of irreducibles)]).

Cone of a morphism.

Let $f\colon X \longrightarrow Y$ be a morphism of complexes

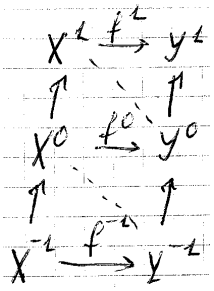
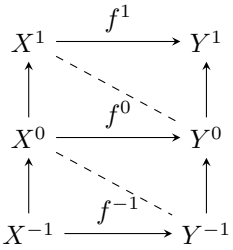


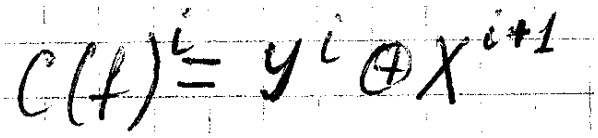
Diagram and typed description.



Original source details: GELFAND-P001397

[At right] consider all this as a bicomplex, i.e. write the complex

$$C(f)^i = Y^i \oplus X^{i+1}$$



Uncertain reading. The manuscript orders the summands as $Y^i \oplus X^{i+1}$.

Original source details: GELFAND-P001397

$$d(x,y) = (-d(x), d(y) + f(x))$$

There is a sequence

$$C(f[-1]) \longrightarrow X \xrightarrow{f} Y \underset{\text{incl.}}{\overset{g}{\longrightarrow}} C(f) \longrightarrow X[1] \longrightarrow \tag{4}$$

[Crossed out above the arrow $Y \to C(f)$: illegible entry?.]

Uncertain reading. illegible entry

Original source details: GELFAND-P001397

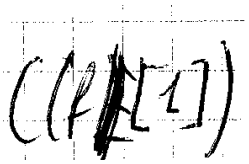
[Crossed out before $C(f)$: X? / Y??.]

Uncertain reading. X? / Y?

Original source details: GELFAND-P001397

[Below $X[1]$: the grading is shifted and the sign is changed on d.]

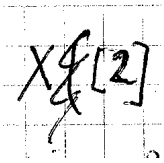
$$\longrightarrow Y(1) \longrightarrow C(f[1]) \longrightarrow X[2] \longrightarrow$$



Original source details: GELFAND-P001397

Uncertain reading. The closing parenthesis follows the shift. The expression reads $C(f[1])$.

[Between X and $[2]$ crossed out: a mark resembling a parenthesis?]



Original source details: GELFAND-P001397

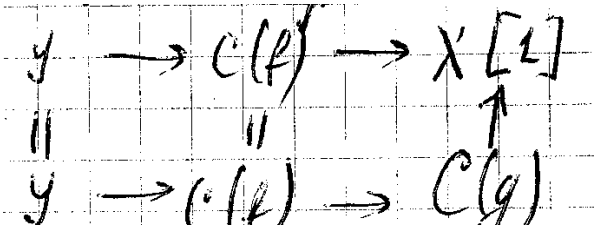
Uncertain reading. a mark resembling a parenthesis

it goes into itself under a shift of three steps. In the homotopy category it goes into itself under a shift of one step.

$$Y \xrightarrow{g} C(f) \longrightarrow C(g) \tag{5}$$

(6) Assertion. In $K(A)$ the complex $C(g) = X[1]$.

There is a map:



Original source details: GELFAND-P001397

Diagram and typed description.

$$\begin{array}{ccccc} Y & \longrightarrow & C(f) & \longrightarrow & X[1] \\ \| & & \| & & \uparrow \\ Y & \longrightarrow & C(f) & \longrightarrow & C(g) \end{array}$$

Namely,

$$C(g)^i = C(f)^i \oplus Y^{i+1} = X^{i+1} \oplus Y^i \oplus Y^{i+1}$$

projects to X^{i+1}.

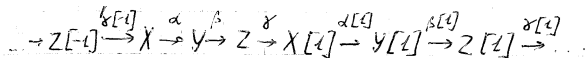
GELFAND-P001398

[source number: 24]

Exercise 2. Check that this is a morphism of complexes and an isomorphism in the homotopy category.

Exercise 3. Find in $K(A)$ the cones $C(\underset{\text{identity map}}{id})$ and $C(\underset{\text{zero map}}{0})$.

Definition 1. A triangle in $K(A)$ or $D(A)$ is the diagram



Original source details: GELFAND-P001398

Diagram and typed description.

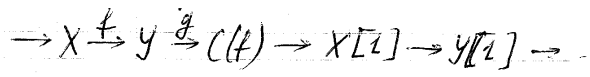
$$\cdots \longrightarrow Z[-1] \xrightarrow{\gamma[-1]} X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \xrightarrow{\gamma} X[1] \xrightarrow{\alpha[1]} Y[1] \xrightarrow{\beta[1]} Z[1] \xrightarrow{\gamma[1]} \cdots$$

Original source details: GELFAND-P001398

Diagram and typed description.

Crossed-out sketch. Two arrows are visible; any labels are illegible.

Def. 2. An exact triangle in $K(A)$ or $D(A)$ is a triangle isomorphic to the triangle



Original source details: GELFAND-P001398

Diagram and typed description.

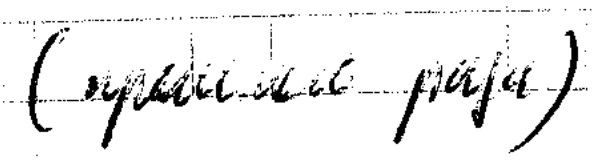
$\longrightarrow X \xrightarrow{f} Y \xrightarrow{g} C(f) \longrightarrow X[1] \longrightarrow Y[1] \longrightarrow \cdots$

GELFAND-P001399

9/XI-87 Gelfand seminar

[Crossed out: Lascoux.] Lascoux

I. M. G. after Lascoux (beautiful / regular[?] series)



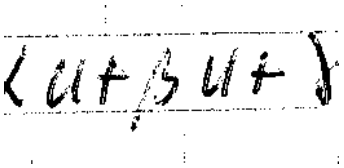
Original source details: GELFAND-P001399

Uncertain reading. beautiful / regular

Schubert polynomials.

Consider the polynomial

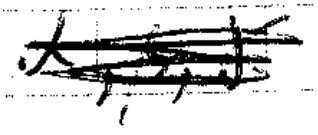
$$P = \alpha u + \beta u + \gamma w + \cdots \tag{1}$$



Original source details: GELFAND-P001399

Uncertain reading. Second monomial in (1): appears to read βu, possibly βv.

u, v, w [Crossed out: possibly α, β, γ[?]] are monomials.



Original source details: GELFAND-P001399

Uncertain reading. possibly α, β, γ

In our case we will have $u, v, w,$ [Crossed out: illegible mark] $\ldots = x_1, x_2, \ldots, x_n$.

Original source details: GELFAND-P001399

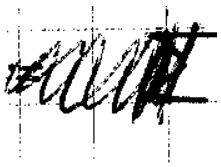
Uncertain reading. illegible mark

$\alpha, \beta, \gamma, \ldots$ are numbers.

The polynomials [Crossed out: fn. / fns.?] $S^m(P)$ [Crossed out: possibly Schur?] are defined by the formula

Original source details: GELFAND-P001399

Uncertain reading. fn. / fns.



Original source details: GELFAND-P001399

Uncertain reading. possibly Schur

$$\sum_{m\geq 0}^{\infty} z^m S^m(P) = \frac{1}{(1-zu)^\alpha (1-zv)^\beta \ldots} \tag{2}$$

Original source details: GELFAND-P001399

Uncertain reading. Lower limit in (2): reads as $m \geq 0$.

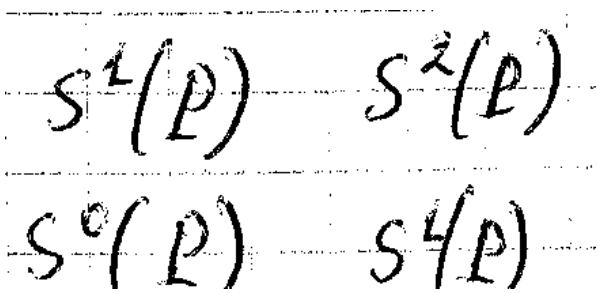
$$\alpha = \beta = \gamma = \cdots = 1 \quad \Rightarrow$$
$$S^m(x_1 + x_2 + \cdots + x_n) = \sum_{i_1 \leq i_2 \leq \cdots \leq i_m} x_{i_1} x_{i_2} \cdots x_{i_m} \tag{3}$$

For example,

$$S^2(x_1 + x_2) = x_1^2 + x_1 x_2 + x_2^2.$$

Consider a Hankel matrix (of infinite order)

$$\begin{pmatrix} S^0(P) & S^1(P) & S^2(P) & \ldots \\ S^{-1}(P) & S^0(P) & S^1(P) & \ldots \\ & & & \end{pmatrix}, \qquad S^{-i}(P) = 0 \text{ for } i > 0. \tag{4}$$



Original source details: GELFAND-P001399

Uncertain reading. Exponent in position $(2,3)$ of (4): more likely 1 than 4.

(see Akhiezer's book on the moment problem)

Multiplication of the series (2) corresponds to multiplication of the matrices (4).

GELFAND-P001400

[source number: 30]

A topic in operator theory that I. M. G. suggested to Birman / Burman[?]:

Uncertain reading. Birman / Burman

Original source details: GELFAND-P001400

Construct / relate[?] K-functor in operator theory

Uncertain reading. Construct / relate

Original source details: GELFAND-P001400

(Sato did this 15 years later)

and it came into fashion.

The only theorem from matrix theory

Consider the matrix A: $f_{\alpha\beta} = f(\alpha, \beta)$ and all its minors of order k: this is also a matrix (the associated matrix of order k) $\Lambda^k(A)$. Then

$$\Lambda^k(AB) = \Lambda^k(A)\Lambda^k(B) \tag{5}$$

It is convenient to denote the row indices by $i_1 + 1, i_2 + 2, \ldots, i_p + p$, and the column indices by $j_1 + 1, j_2 + 2, \ldots, j_p + p$

$$\begin{vmatrix} i_1 \leq i_2 \leq \cdots \leq i_p \\ j_1 \leq \cdots \leq j_p \end{vmatrix}$$

$$S_{J|I}(P) \tag{6}$$

is the corresponding minor, called the skew Schur function.

$S_{J|0}(P)$ is the ordinary Schur function.

[Crossed out: Then] One can consider the polynomial (Jacobi)

$$\frac{\begin{vmatrix} x_1^{j_1} & x_1^{j_2+1} & \ldots & x_1^{j_p+p-1} \\ x_2^{j_1} & x_2^{j_2+1} & \ldots & x_2^{j_p+p-1} \\ \ldots & \ldots & \ldots & \ldots \end{vmatrix}}{\begin{vmatrix} 1 & x_1 & \ldots & x_1^{p-1} \\ 1 & x_2 & \ldots & x_2^{p-1} \\ & & & \\ 1 & x_p & \ldots & x_p^{p-1} \end{vmatrix}} \tag{7}$$

Uncertain reading. In the first two rows of the numerator, probably $+1$ is crossed out after j_1.

Original source details: GELFAND-P001400

Uncertain reading. To the right of the numerator: 1 or a single oblique stroke.

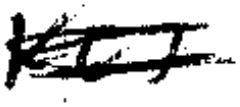
Original source details: GELFAND-P001400

(8 / $7^?$) ([Inserted above the line: Ex.] check that the denominator divides the numerator!)

Uncertain reading. 8 / 7

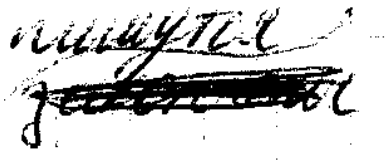
Original source details: GELFAND-P001400

[Crossed out: ко / но [but]?.] This is a symmetric function, but the coefficients are not [Crossed out: illegible] [Inserted above the line: written down explicitly] and are still being studied.



Uncertain reading. ко / но [but]

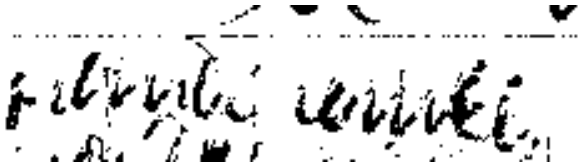
Original source details: GELFAND-P001400



Undeciphered passage. word under the inserted «are written down» is illegible

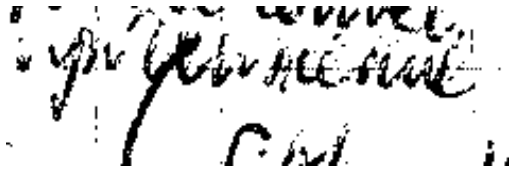
Original source details: GELFAND-P001400

[In the left margin: (first steps,? possibly «in solving it»?)]



Uncertain reading. first steps,

Original source details: GELFAND-P001400



Uncertain reading. possibly «in solving it»

Original source details: GELFAND-P001400

(see the paper by [crossed-out initials: I. G. / I. M.?; Inserted above the line: I. M.] Gelfand, Zelevinsky, 1986, FA; and the unpublished work by B. Lidskii / B. Litskii?)

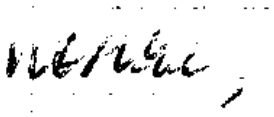
Uncertain reading. I. G. / I. M.

Original source details: GELFAND-P001400

Uncertain reading. B. Lidskii / B. Litskii

Original source details: GELFAND-P001400

It is clear so far / only?, that the coefficients [Crossed out: will be / all?] are integers.



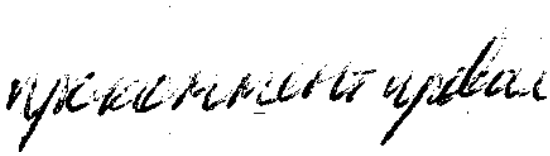
Uncertain reading. so far / only

Original source details: GELFAND-P001400

Original source details: GELFAND-P001400

Uncertain reading. will be / all

Schur realized that they are important: he commented on / having commented on[?] Jacobi, Frobenius and others.



Original source details: GELFAND-P001400

Uncertain reading. he commented on / having commented on

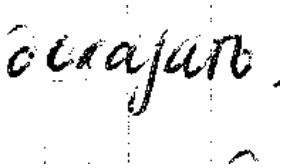
GELFAND-P001401

Frobenius found the characters of the symmetric group. This is a deep work for which H. Weyl's commentaries are no substitute.

(9) Theorem. (Jacobi–Trudi)

$$(7) = S_{J|0}(x_1 + \ldots + x_p) \tag{10}$$

(11) Problem: a) possibly «grasp it» (осознать)[?];



Original source details: GELFAND-P001401

Uncertain reading. possibly «grasp it» (осознать)

b) give its classical proof;

c) give an interpretation in representation theory.

(12) $S_{J|I}$ satisfies a system of identities;

these are Plücker coordinates (one of Sato's achievements!)

$$S^m(P+Q) = \sum_j S^{n-j}(P)S^j(Q); \tag{13}$$

Original source details: GELFAND-P001401

Uncertain reading. The exponent on the left of (13) is overwritten and looks like m or n.

$$S^n(PQ) = \sum_{|J|=n} S_J(\underline{P})S_J(Q), \qquad S'_J = S'_{J|0}; \tag{14}$$

$$S^n(S^m(P)) = \sum (n, m; J)S_J(P) \quad \|, \tag{15}$$

Uncertain reading. Two strokes resembling $\|$ follow $S_J(P)$ in (15).

Original source details: GELFAND-P001401

where $(n, m; J)$ are universal coefficients.

The cohomology of flag manifolds is computed in two ways

a) decomposition into Schubert cells (the dimensions are even!)

b) differential forms;

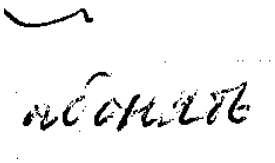
Which forms correspond to which cells?

Schubert polynomials arise there

(up to symmetric ones)

Lascoux, Schützenberger: one should forget topology and compute polynomials.

The polynomials should be housed[?] in the ring of all polynomials, and so on.



Uncertain reading. housed

Original source details: GELFAND-P001401

GELFAND-P001402

[source number: 31]

[The following passage is boxed and crossed out.]

(16) Theorem. Let us permute the rows

$$h_1 + 1,\ h_2 + 2, \ldots, h_p + p, \qquad h_1 \leq h_2 \leq \ldots \leq h_p;$$

Newton introduced the operations

$$\partial_i f(x_1, \ldots, x_n) = \frac{f(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n) - f(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n)}{x_{i+1} - x_i} \tag{16}$$

Then

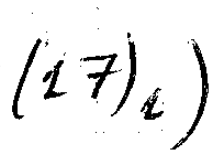
$$\begin{cases} \partial_i^2 = 0 \\ \partial_i \partial_j = \partial_j \partial_i, \qquad |i - j| \geq 2, \\ \partial_i \partial_{i+1} \partial_i = \partial_{i+1} \partial_i \partial_{i+1} \end{cases} \tag{17}$$

(18) Remark.

$(\partial_i + 1)(\partial_i - q) = 0 \quad \Rightarrow \quad \partial_i$ is a q-analogue of the generators of the symmetric group

$q = 1 \quad \Rightarrow \quad$ symmetric group

(instead of $(17)_1$)



Uncertain reading. The subscript on (17) reads as 1.

Original source details: GELFAND-P001402

Lascoux

Connection between Schubert polynomials and Schur functions.

(19) $\partial_i(fg) = g\partial_i(f)$, if [crossed out: appears to be f?] g is symmetric in [crossed out: two / all?] the variables x_i, x_{i+1}.

Uncertain reading. appears to be f

Original source details: GELFAND-P001402

Uncertain reading. two / all

Original source details: GELFAND-P001402

Reminder about Schubert polynomials

$$Y_{730} = x_1^7 x_2^3 x_3^0$$

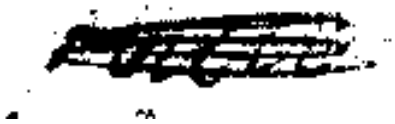
(20) Def. If $\underbrace{i_1 \geq i_2 \geq i_3 \geq \ldots}_{I}$, then $x_1^{i_1} x_2^{i_2} \ldots = Y_I$

(Schubert polynomial [crossed out: I / $=^?$] [crossed out: illegible] with index I is a monomial)

Uncertain reading. I / $=$

Original source details: GELFAND-P001402

Undeciphered passage. crossed-out writing is illegible



Original source details: GELFAND-P001402

All Schubert polynomials $=$ are the images of these under the action of an arbitrary operator ∂_i.

Uncertain reading. appears to be $=$

Original source details: GELFAND-P001402

[Inserted above «images»: «(all nonzero)».]

Rule for calculating$^?$ the indices

Uncertain reading. calculating

Original source details: GELFAND-P001402

$$Y_{\ldots i_p\, i_{p+1}\ldots} \xrightarrow{\partial_p} \begin{cases} Y_{\ldots i_{p+1}\, i_p - 1}, & i_p > i_{p+1} \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

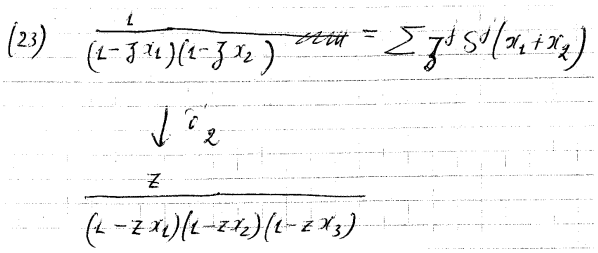
Ex. Prove that these definitions are well defined.

(22) Example

$$Y_{730} = x_1^7 x_2^3 x_3^0 = \begin{vmatrix} S^3(x_1+x_2) & S^8(x_1) \\ S^2(x_1+x_2) & S^7(x_1) \end{vmatrix}$$

GELFAND-P001403

(23)



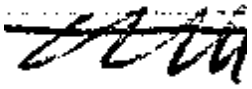
Original source details: GELFAND-P001403

Diagram and typed description.

$$\frac{1}{(1-zx_1)(1-zx_2)} = \sum z^j S^j(x_1+x_2)$$

$$\downarrow \partial_2$$

$$\frac{z}{(1-zx_1)(1-zx_2)(1-zx_3)}$$



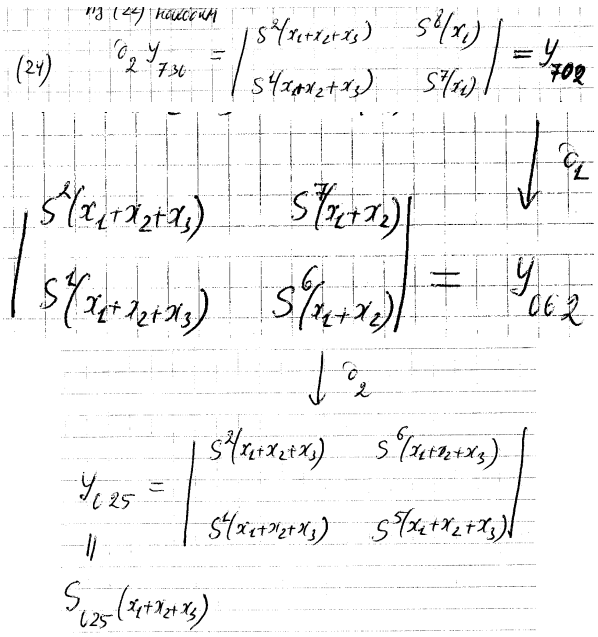
Original source details: GELFAND-P001403

Undeciphered passage. The crossed-out writing after the fraction, before the equals sign, is illegible.

$$S^n(x_1+x_2) \xrightarrow{\partial_2} S^{n-1}(x_1+x_2+x_3) \tag{23}$$

From (22) we obtain

(24)



Original source details: GELFAND-P001403

Diagram and typed description.

$$\partial_2 Y_{730} = \begin{vmatrix} S^2(x_1+x_2+x_3) & S^8(x_1) \\ S^1(x_1+x_2+x_3) & S^7(x_1) \end{vmatrix} = Y_{702}$$

$$\downarrow \partial_1$$

$$\begin{vmatrix} S^2(x_1+x_2+x_3) & S^7(x_1+x_2) \\ S^1(x_1+x_2+x_3) & S^6(x_1+x_2) \end{vmatrix} = Y_{062}$$

$$\downarrow \partial_2$$

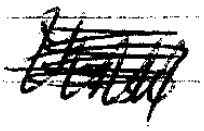
$$Y_{025} = \begin{vmatrix} S^2(x_1+x_2+x_3) & S^6(x_1+x_2+x_3) \\ S^1(x_1+x_2+x_3) & S^5(x_1+x_2+x_3) \end{vmatrix}$$

$$\| $$

$$S_{025}(x_1+x_2+x_3)$$

(Schur function, defined earlier; $S_{025|0}(x_1+x_2+x_3)$, sequences of zeros are omitted)

The expression of formula [crossed out: illegible] (22) as a determinant was used in elimination theory in the 19th century.



Original source details: GELFAND-P001403

Undeciphered passage. word illegible, possibly «Schur»

There are 5 (thick) volumes of the book

T. Muir: History of Determinants, Dover (1960)

GELFAND-P001404

[source number: 32]

There is another book (by the same author)

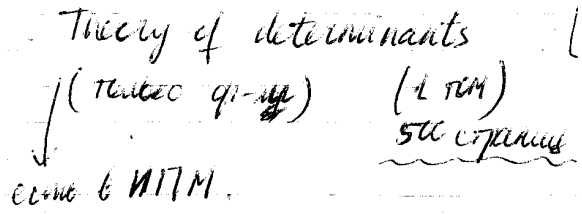
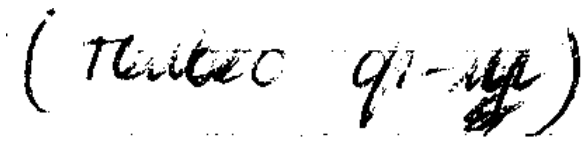


Diagram and typed description.
Theory of determinants
(formulas only) (1 volume)
↓ 500 pages
available at the IPM.

Original source details: GELFAND-P001404



Uncertain reading. formulas only / thousands of formulas

Original source details: GELFAND-P001404

[To the right of the title] everything up to 1900 | almost everything up to (1920)

Analogy: theory of integrable systems, construct / build[?] a theory to account for these formulas.

Uncertain reading. construct / build

Original source details: GELFAND-P001404

(25) Theorem.

$$S_{I|000}(x_1 + \cdots + x_n) = Y_{i_1 \ldots i_n 0000}$$

Uncertain reading. Index on the left: $I \mid 000$.

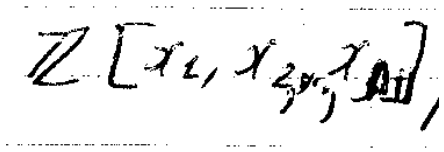
Original source details: GELFAND-P001404

$$0 \leq i_1 \leq \cdots \leq i_n$$

(26) Finite case: $\mathbb{Z}[x_1, x_2, \ldots, x_{n+1}]$ / ideal of symm. pol. in $(x_1, x_2,$ [crossed out: an x is visible, with an illegible subscript] $\ldots, x_{n+1})$

Uncertain reading. an x is visible, with an illegible subscript

Original source details: GELFAND-P001404



Original source details: GELFAND-P001404

Uncertain reading. The final subscript before the bracket reads as $n+1$.

(27) Theorem. The Schubert polynomials

$$Y_I, \quad 0 \le i_1 \le n, \quad 0 \le i_2 \le n-1, \quad \dots, \quad 0 \le i_{n+1} \le 0 \tag{28}$$

form a basis of the ring (26) as a free abelian group.

Example

$n = 2$ $\mathfrak{S}(3)$;

$$Y_{210} \quad Y_{110} \quad Y_{200} \quad Y_{010} \quad Y_{100} \quad Y_{000}$$

$\mathbb{Z}$-basis in $\mathbb{Z}[x_1, x_2, x_3]/(\quad)$

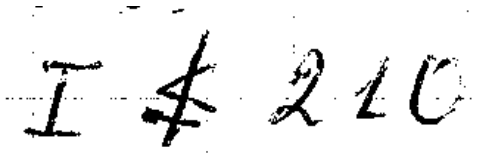
From (27) it is clear that the dimension is $n!$.

Original source details: GELFAND-P001404

Uncertain reading. The manuscript reads $n!$.

The others lie in the ideal:

$$Y_I \in \text{ideal} \quad \text{if} \quad I \not\le 210 \tag{29}$$



Original source details: GELFAND-P001404

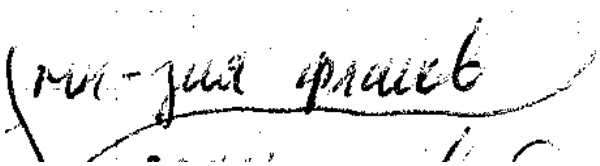
Uncertain reading. The sign in (29) is a struck-through $\le$.

(30) Question of I. M. G.: how is the representation of $\mathfrak{S}(n)$ [crossed out: possibly «в» (in)?] written when $\mathfrak{S}(n)$ acts on the variables $x_1, \dots, x_n$ in the Schubert polynomials

Original source details: GELFAND-P001404

Uncertain reading. possibly «в» (in)

In the cohomology ring [inserted above the line with an insertion mark: of the flag manifold?] there is a quadratic form, written as follows:



Original source details: GELFAND-P001404

Uncertain reading. of the flag manifold

$$PQ \xrightarrow{\partial_1 \partial_2 \partial_1} (P, Q) \tag{31}$$

(scalar product)

[To the right:] (a number in the quotient by the ideal)

(32) Chevalley theorem.

∂_1 is self-adjoint with respect to $(\ ,\ )$

∂_2 —"—

e.g.

$$(\partial_1(P), Q) = (P, \partial_1(Q)) \tag{33}$$

The proof is straightforward.

Lascoux.

$$Y_I, \qquad Y_{I'}, \qquad I + I' = n \quad n-1 \quad \dots \quad 1 \quad 0 \tag{34}$$

GELFAND-P001405

Then

$$(Y_I, Y_J) = 0 \quad \text{when } J \neq I'. \tag{35}$$

Uncertain reading. The symbol in (35) reads as 0.

Original source details: GELFAND-P001405

(all in the quotients)

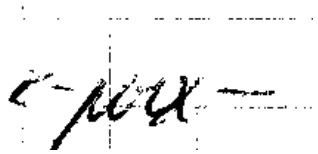
Theorem.

(36) $\mathbb{Z}[x_1, x_2, x_3]$ is a free module [inserted above the line: over] Sym(x_1, x_2, x_3) with the Schubert polynomials as a basis.

Not all Schubert polynomials are determinants, so divided differences are used in their place, ∂_i.

Problem of I. M. G. Generalize the theorem on characters: irreducible representations, characters of the

Uncertain reading. «с-ких» [uncertain abbreviation]



Original source details: GELFAND-P001405

— Schubert functions.

Next time, Takhtajan on quantum groups.

Kapranov (Misha)

2-categories.

This is an algebraic structure with objects of 4 types

a) objects •

b) horizontal morphisms • ⟶ •

c) vertical morphisms ↑

d) squares

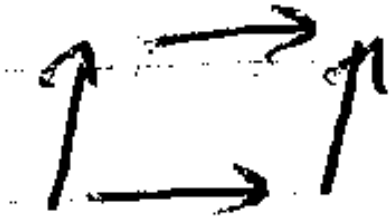


Diagram and typed description.

Original source details: GELFAND-P001405

[In the right margin:]

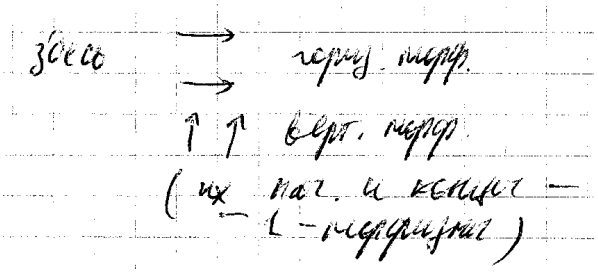


Diagram and typed description. Here ⟶ ⟶ horizontal morphisms,
↑↑ vertical morphisms
(their sources and targets are 1-morphisms)

Original source details: GELFAND-P001405

One can compose:

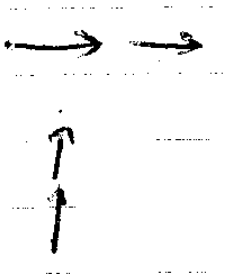


Diagram and typed description.
1) ⟶ ⟶ 2 horizontal morphisms
2) ↑ ↑ 2 vertical morphisms

Original source details: GELFAND-P001405

3) squares are multiplied in two ways

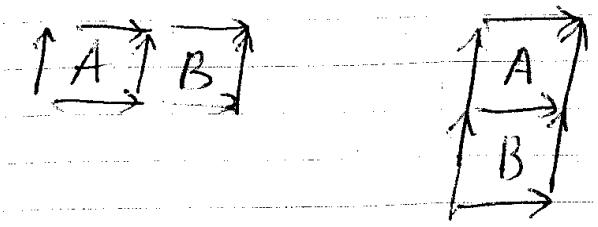
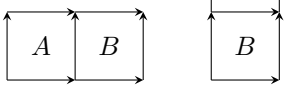
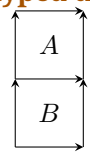


Diagram and typed description.





Original source details: GELFAND-P001405

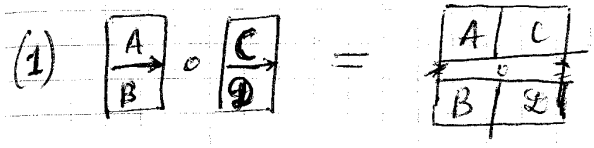
(see Gabriel's book)

GELFAND-P001406

[source number: 33]

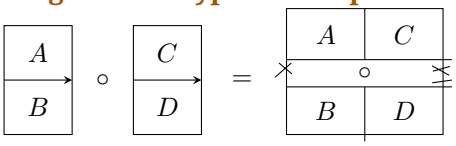
Main nontrivial axiom: 2-dimensional associativity

(1)



Original source details: GELFAND-P001406

Diagram and typed description.



The reading D is tentative. The strokes at the ends of the band on the right leave the arrows uncertain.

Example where [crossed out: illegible] ax. fails [crossed out: possibly an earlier «(1)»[?]] (1)

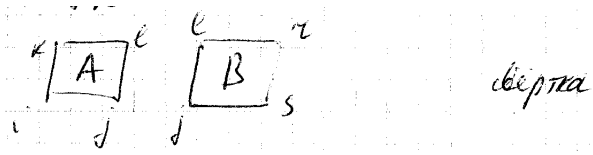
Original source details: GELFAND-P001406

Undeciphered passage. illegible

Original source details: GELFAND-P001406

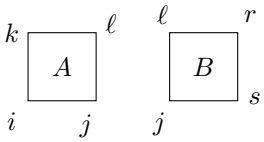
Uncertain reading. possibly an earlier «(1)»

Tensors:



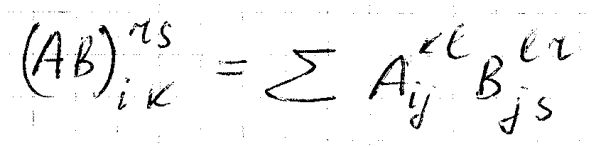
Original source details: GELFAND-P001406

Diagram and typed description.



contraction

$$(AB)^{rs}_{ik} = \sum A^{k\ell}_{ij} B^{\ell r}_{js}$$



Original source details: GELFAND-P001406

Uncertain reading. On the left, i, k are subscripts and r, s are superscripts.

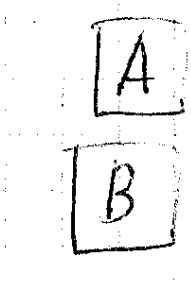


Diagram and typed description.
A
B

Original source details: GELFAND-P001406

Axiom (1) will fail.

Another def. (M. Kontsevich) (competing)

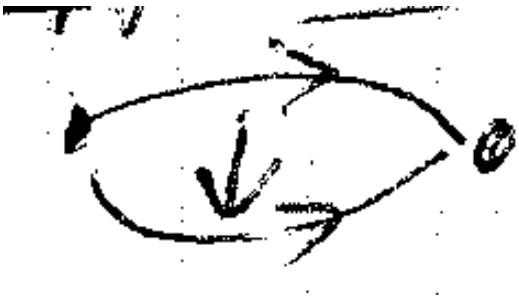
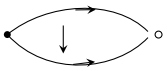


Diagram and typed description. 
morphisms between morphisms with the same endpoints.

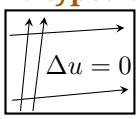
Original source details: GELFAND-P001406

Example: Cat (the category of categories. There is a morphism of functors)

This is a special case of the preceding one (contract the vertical arrows).

I. M. G.

Diagram and typed description.

св.? u and $\frac{\partial u}{\partial n}$ 

boundary conditions
The abbreviation above u is unclear.

Original source details: GELFAND-P001406

Request to Keldysh, Chentsov, and Gelfand

K. Higher fundamental groups. $X_0 \subset X_1 \subset X$ is a topological space

objects are points $x \in X_0$

morph. $\left(\underset{\text{interval}}{I}, \partial I\right) \longrightarrow (X_1, X_0)$ homotopic $(\text{mod } \partial I)$ are regarded as equivalent.

bimorphisms $(I^2, \partial I^2) \longrightarrow (X, X_1)$ (mod homotopy)

GELFAND-P001407

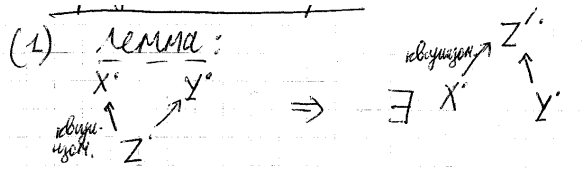
[source number: 44]

23/XI-87 Gelfand seminar.

Beilinson (continued)

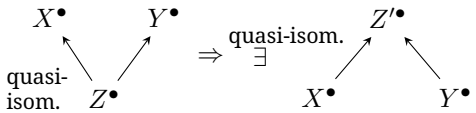
Derived categories.

(1) Lemma:



Original source details: GELFAND-P001407

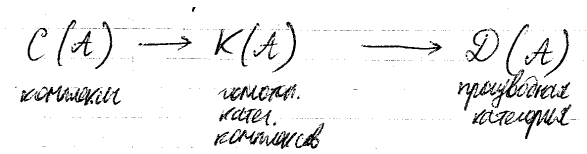
Diagram and typed description.



The lemma is proved using the mapping cone construction.

Reminder:

$\mathcal{A}$ is an ab. category



Original source details: GELFAND-P001407

Diagram and typed description.

$C(\mathcal{A})$ $\longrightarrow$ $K(\mathcal{A})$ $\longrightarrow$ $D(\mathcal{A})$

complexes

homotopy category of complexes

derived category

[At right:] (the objects are complexes throughout)

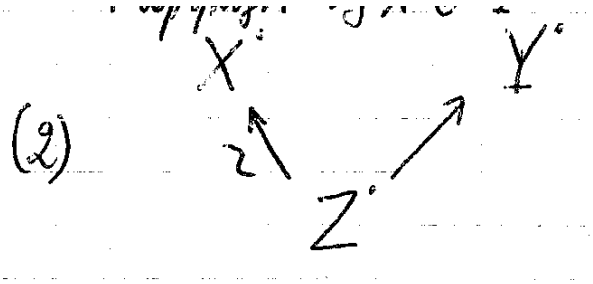
$f : X^\bullet \to Y^\bullet$ is homotopic to 0 if $\exists g^i : X^i \to Y^{i-1}$,

$$f^i = dg^i + g^{i+1}d.$$

$D(\mathcal{A})$ is $K(\mathcal{A})$ with the following inverted quasi-isomorphisms (morphisms inducing an isomorphism of cohomology)

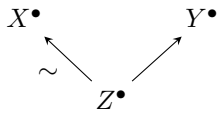
A morphism from $X^\bullet$ to $Y^\bullet$ is a diagram in $K(\mathcal{A})$

(2)



Original source details: GELFAND-P001407

Diagram and typed description.



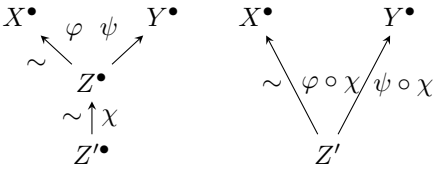
(the left arrow is a quasi-isomorphism)

more precisely, an equivalence class of such diagrams. Diagram (2) is equivalent to the diagram obtained using any quasi-isomorphism $\chi : Z'^{\bullet} \to Z^{\bullet}$

(3)

Original source details: GELFAND-P001407

Diagram and typed description.



(this relation extends to the smallest equivalence relation containing it)

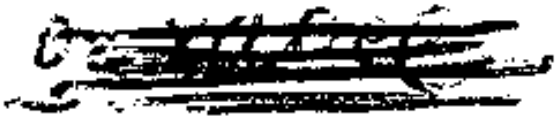
The cone of a morphism and exact triangles were defined [crossed out: illegible] already.

Uncertain reading. The short crossed-out passage before «already» is illegible.

Original source details: GELFAND-P001407

GELFAND-P001408

I. Zakharevich [crossed out: illegible] Improvisation on chaos.



Undeciphered passage. illegible

Original source details: GELFAND-P001408

Uncertain reading. The preposition in the title reads as «o» (on).

Original source details: GELFAND-P001408

Examples

1°. Apparently there is a morphism from one example to the other.

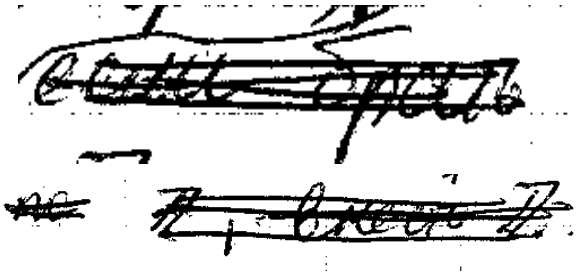
$$M = (\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}} \qquad \text{(sequences infinite in both directions)}$$

$T : M \to M$. M has the standard topology (a Cantor set).

$$a_i \to a_{i-1} \qquad \text{(shift transformation, Bernoulli transformation)}$$

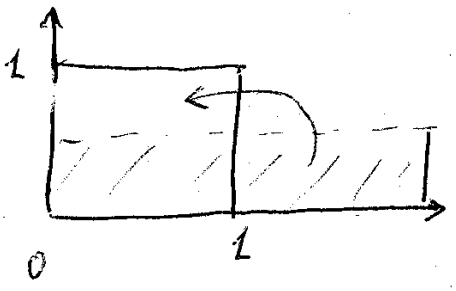
Consider the iterates T^N.

There is a measure on M. The baker's transformation coincides with this [inserted above the line: (up to a set of measure 0)] [crossed out: continuation on two lines].



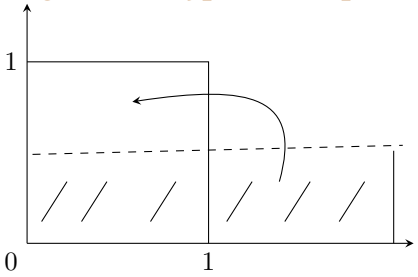
Original source details: GELFAND-P001408

Uncertain reading. The upper line appears to read «if one takes», the lower «not $\mathbb{Z}$, instead of $\mathbb{Z}$».



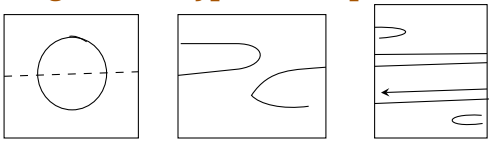
Original source details: GELFAND-P001408

Diagram and typed description.



Original source details: GELFAND-P001408

Diagram and typed description.



and so on.

The small strokes on the curves in the first and third drawings leave their direction uncertain.

Correspondence: write the coordinates of a point of the square in binary and take one digit to the left, the other to the right.

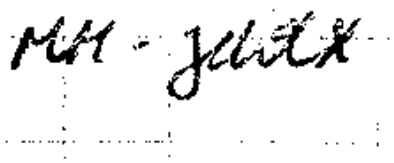
In what sense is this chaos

Consider $T^k x,\ k \leq 0$ — suppose this is known (to within 1 mm).

$$T^k x, \qquad k > 0$$

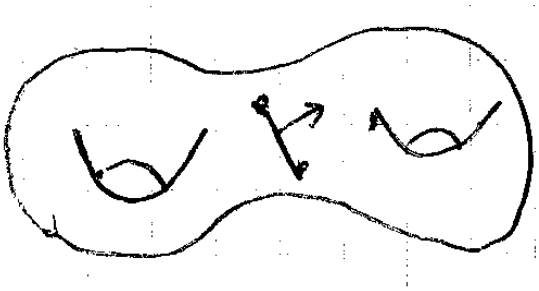
In a finite time all this information will be lost.

2°. Hyperbolic transformation: geodesic flows on manifolds of negative curvature.



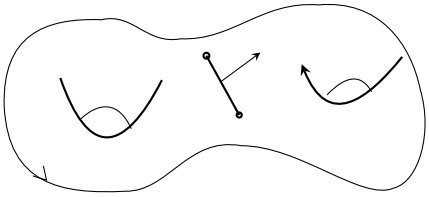
Original source details: GELFAND-P001408

Uncertain reading. The abbreviation reads as «мн-зиях» (manifolds).



Original source details: GELFAND-P001408

Diagram and typed description.



The middle branch points up and right. The stroke near the left edge of the outline and the small marks on the middle segment are uncertain.

GELFAND-P001409

[source number: 45]

G. Margulis. Iterations of maps of the complex plane.

Consider the transformation $z \to P(z)$, P is a polynomial

(more generally, a rational function)

$\deg P = 1$ [crossed out: illegible] uninteresting case

Original source details: GELFAND-P001409

Undeciphered passage. The short passage after $\deg P = 1$ is illegible.

$$z \mapsto az + b \tag{1}$$

Iterations:

$$z_{i+1} = P(z_i)$$

$|a| > 1 \Rightarrow$ one fixed point.

$|a| = 1 \Rightarrow$ everything splits into invariant circles on which the map is a rotation.

Original source details: GELFAND-P001409

Uncertain reading. The manuscript reads $|a| = 1$.

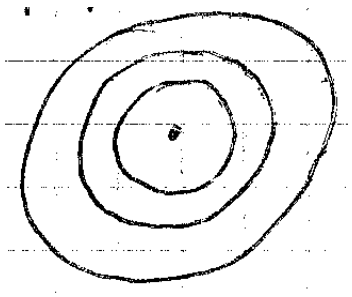
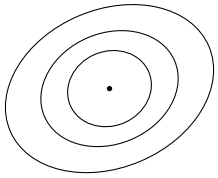


Diagram and typed description.



Original source details: GELFAND-P001409

I. M. G.: 1) the question of when a transformation of the circle is diffeomorphically conjugate to a rotation is nontrivial. Denjoy, Arnold, Herman.

2) What is a representation of an infinite discrete group?

There is no satisfactory answer.

Von Neumann's theorem is unaesthetic.

Chaos arises on the circle.

Conjecture: if one works with sets of intermediate Hausdorff dimension, some of the classical theory will be recovered!!

Question for school students: cover the nth rational number with an interval of length $\varepsilon/2^n$. Give an example of a number that will remain uncovered.

Problem: find the Hausdorff dimension of the perfect Cantor set.

(!) Problem. Construct a theory of representations of discrete groups on functions on sets of intermediate Hausdorff dimension.

To begin, look at physics, string theory and so on.

G. M.: $z \to z^2$ $\quad |z| = 1$ maps to itself

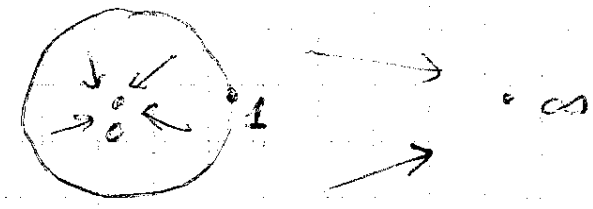
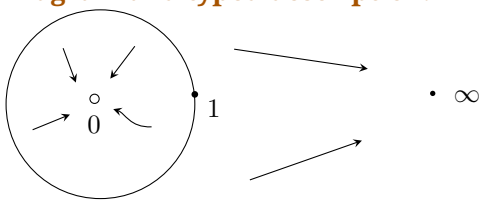


Diagram and typed description.



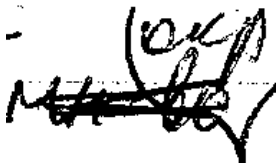
Original source details: GELFAND-P001409

on the circle, an analogue of the (one-sided) baker's transformation

$$f_n = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}$$

Def.

U is a [crossed out: set; inserted above the line: nbhd.] of normality if $f_n|_U$ is a compact set (normal family of analytic functions)



Original source details: GELFAND-P001410

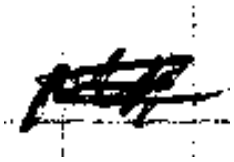
Uncertain reading. «set» is crossed out and «nbhd.» is inserted above it.

[Crossed out on a short line by itself:]

Original source details: GELFAND-P001410

Undeciphered passage. illegible

Set of normality [inserted above the line: X] consists of points x, for which $\exists$ nbhd. U — [crossed out: illegible] a nbhd. of normality



Original source details: GELFAND-P001410

Undeciphered passage. The crossed-out writing before «nbhd. of normality» is illegible.

[Under Set of normality: (it is open)]

The complement of the set of normality is the Julia set.

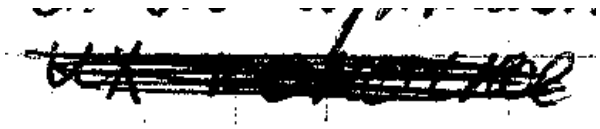
(it is closed)

The set of normality can be empty

$$X = X_1 \cup \cdots \cup X_n \cup \cdots$$

(connected components) of the set of normality

[crossed out below this caption: illegible]



Original source details: GELFAND-P001410

Undeciphered passage. Individual letters are visible below the caption, but the passage cannot be read as a whole.

$$f : (\text{connected comp.}) \to (\text{connected components})$$
$$X_i \mapsto X_{j(i)}$$

Sullivan proved that for the resulting [crossed out: illegible] map of a discrete set (countable or finite)

Undeciphered passage. The crossed-out writing before «map» is illegible.

Original source details: GELFAND-P001410

a) each orbit stabilizes,

b) the set of fixed points is finite.

Idea of the proof. Suppose there is an infinite orbit

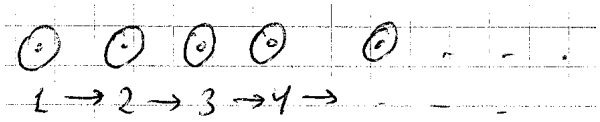
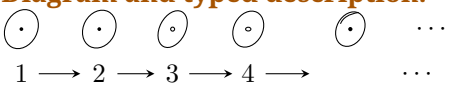
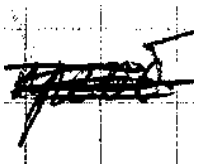


Diagram and typed description.



Original source details: GELFAND-P001410

Let us change the conformal [crossed out: possibly «transf.»] structure so that the transformation remains conformal.



Uncertain reading. The crossed-out word appears to read «transf.».

Original source details: GELFAND-P001410

Problem:

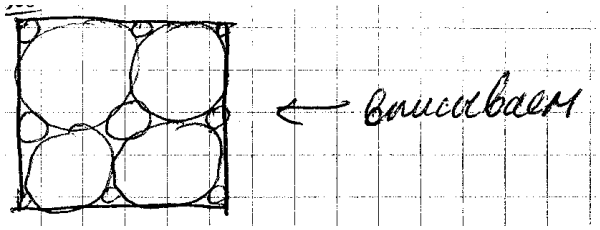
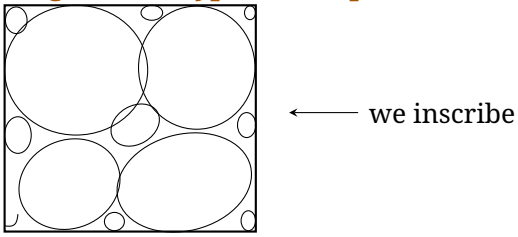


Diagram and typed description.



The contacts and some small arcs at the boundary are shown approximately.

Original source details: GELFAND-P001410

1) Prove that the remainder has measure 0

2) Prove that the remainder has Hausdorff dimension < 2.

GELFAND-P001411

[source number: 46]

I. M. Gelfand. Matroids.

In a K-dimensional space, consider n vectors. Number them and list which are linearly dependent and which are not.

Giving this list defines a matroid.

What conditions apply to this list?

Example: [Crossed out: illegible]

Undeciphered passage. The writing after «Example» is illegible.

Original source details: GELFAND-P001411

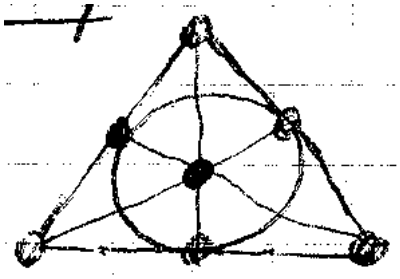
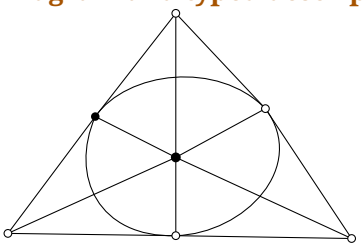


Diagram and typed description.



Original source details: GELFAND-P001411

If 3 points lie on one line, the vectors are linearly dependent.

The whole space is 3-dimensional (no 4 linearly independent vectors)

(1) Problem: 1) prove that over $\mathbb{R}$ and $\mathbb{C}$ there are no such vectors!

2) over $\mathbb{Z}/2\mathbb{Z}$ there are such vectors.

(all the nonzero vectors of $(\mathbb{Z}/2\mathbb{Z})^3$!)

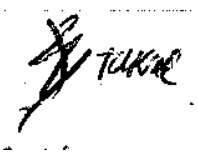
General (unsolved problem): which matroids are realizable a) over $\mathbb{R}$ b) over $\mathbb{C}$.?

Solved problem: which matroids are realizable over $\forall$ fields?

In Rota's book э...[At right: see book? by Aigner (?)] there are 15 different equivalent definitions of a matroid.

Uncertain reading. After «Rota» there is a mark resembling «э» followed by dots.

Original source details: GELFAND-P001411



Uncertain reading. The word after «see» has been overwritten and tentatively reads «книге» or «книжке» (book).

Original source details: GELFAND-P001411

(2) Axioms.

Certain subsets are singled out, called bases

B_1, B_2 are 2 bases

$$y \in B_2,\ y \notin B_1 \Rightarrow \exists x \in B_1; \qquad (B_1 \setminus x) \cup y, \quad (B_2 \setminus y) \cup x$$

are bases [Crossed out: writing between the two expressions]

Undeciphered passage. The writing after $(B_1 \setminus x) \cup y$ is crossed out and illegible.

Original source details: GELFAND-P001411

Exercise in linear algebra: verify this for a set of vectors in a linear space.

(3) Bimatroid: consider a matrix. It defines 2 matroids, one by columns and one by rows S and T

Another notion: list all the minors $\neq 0$

Bimatroid: a collection of subsets with the same number of elements.

(4) [Crossed out: illegible] Problem: give axioms for a bimatroid.

Undeciphered passage. The short passage before «Problem» is illegible.

Original source details: GELFAND-P001411

GELFAND-P001412

(5) Problem: how is the notion of a bimatroid from [Crossed out: illegible] Gelfand–Serganova (UMN) related to «Кунтом»

Undeciphered passage. The crossed-out writing after “from” is illegible.

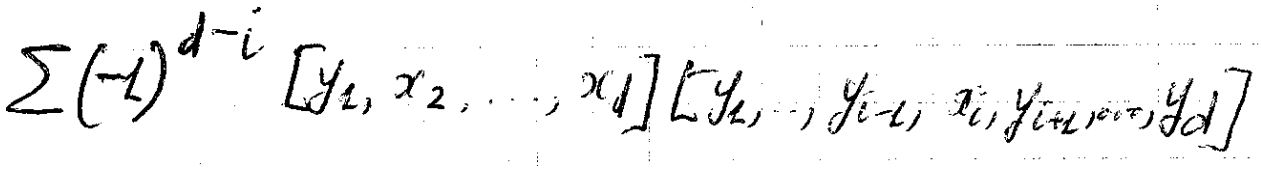
Original source details: GELFAND-P001412

Uncertain reading. The surname tentatively reads «Кунтом», possibly referring to Knuth.

Original source details: GELFAND-P001412

Problem: prove that

$$(6)\quad [x_1,\dots,x_d][y_1,\dots,y_d] = \sum (-1)^{d-i}[y_1,x_2,\dots,x_d]$$
$$[y_1,\dots,y_{i-1},x_i,y_{i+1},\dots,y_d]$$

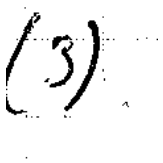


Original source details: GELFAND-P001412

Uncertain reading. The sign $(-1)^{d-i}$ and the indices y_1, x_i follow the manuscript.

$[x_1,\dots,x_d]$ is the determinant formed from the vectors.

(7) Remark: if $[x_1,\dots,x_d]$, $[y_1,\dots,y_d]$ are bases, then the right-hand side has a nonzero term, which gives the exchange axiom (3)



Uncertain reading. The reference (3) follows the manuscript.

Original source details: GELFAND-P001412

(8) Problem (tedious). Hilbert's chain of syzygies:

$$(\text{ring}) \longmapsto (\text{generators}) \underset{\text{— determinants}}{\longmapsto} (\text{relations among the generators})$$

Uncertain reading. The number has been overwritten and reads as 8.

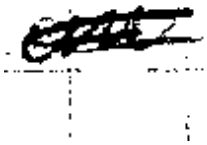
Original source details: GELFAND-P001412

(9) (Claim: (6) gives all the relations among the generators)

(10) a) Write down the whole chain!

b) Discretize the entire chain (write down the combinatorial relations).

(11) Problem. A bimatroid is a list of minors $\longrightarrow$ [Crossed out: illegible] 2 matroids (rows and columns)



Undeciphered passage. The crossed-out writing before "2 matroids" is illegible.

Original source details: GELFAND-P001412

A bimatroid is constructed from 2 matroids

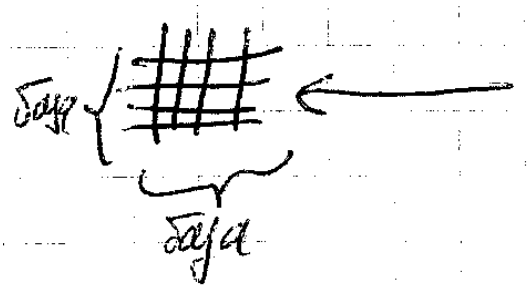
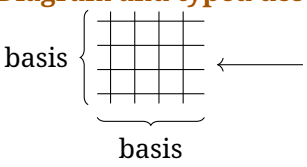


Diagram and typed description.



Original source details: GELFAND-P001412

by def. basis.

Uncertain reading. The word after “by def.” is underlined and reads “basis.”

Original source details: GELFAND-P001412

But this is not necessary.

There is more information in a bimatroid.

What additional information is there in a bimatroid?

This is an important cohomological concept.

Remark. Algebraic topology is a small island in the usual combinatorial technique.

GELFAND-P001413

30/XI-87 Gelfand Seminar.

I. M. Gelfand. Problem: find the characters of the symmetric group.

(following Frobenius).

On the paper by Kerov: «Realization of representations of the Hecke algebra etc.»

Hint: these are characters of the symmetric group.

The Hecke algebra is the algebra with generators $\{e_k\}$ [above the line: $k = 1, \dots, N-1$] and relations

$$(1) \quad \begin{cases} e_k e_{k+1} e_k - \tau e_k = e_{k+1} e_k e_{k+1} - \tau e_{k+1} \\ e_{k+j} e_k = e_k e_{k+j}, \quad j \geqslant 2, \\ e_k^2 = e_k \end{cases}$$

Other generators

$$(2) \quad \sigma_k = (q+1) e_k - 1, \qquad \tau = \frac{q}{(q+1)^2}$$

This algebra is denoted by $H_N(q)$.

Exercise: write down the relations for σ_k.

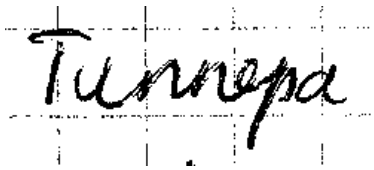
(3) For $q = 1$ (or $\tau = \frac{1}{4}$) we obtain the group algebra of the symmetric group.

(4) Problem: $\dim H_N(q) = N!$

[In the left margin: ⓘ] In the paper by Tipper and James

Proc. London Math. Soc. 1986, v. 52, p. 20–52

recommended!!



Original source details: GELFAND-P001413

Uncertain reading. The first name appears as «Типпера» (Tipper) in the manuscript. The cited publication gives R. Dipper.

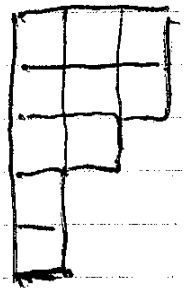
it is proved that the representation theory of $H_N(q)$ is analogous to the modular representation theory of the symmetric group.

This works even in the complex case (see the paper by Bernsh. — Gelf. — Gelf. on Verma modules).

Here it is assumed that $q \neq \sqrt[m]{1}$

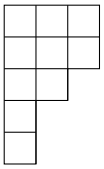
Construction of representations.

Young diagram $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$



Original source details: GELFAND-P001413

Diagram and typed description. (5)



$\sum \lambda_j = |\lambda|$

$\lambda_{i+1} \leqslant \lambda_i$

(λ_i is the number of boxes in the ith column)

This corresponds to a conjugacy class of elements.

GELFAND-P001414

[source number: 52]

Young tableau:

$$(6)\quad t = \{\lambda^{(k)}\}_{k=1}^{N} \quad \text{is a collection of diagrams in which } |\lambda^{(k)}| = k.$$

The usual definition (a different one): the tableau is filled with the numbers $1, 2, \ldots, |\lambda|$, with

$\rightarrow$
$\downarrow$ entries increasing in these directions

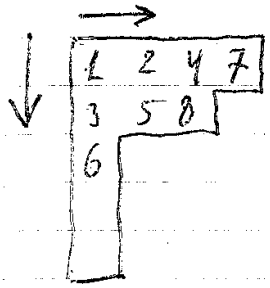
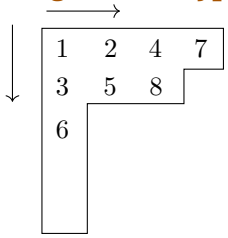


Diagram and typed description.



Original source details: GELFAND-P001414

In (6) they must increase

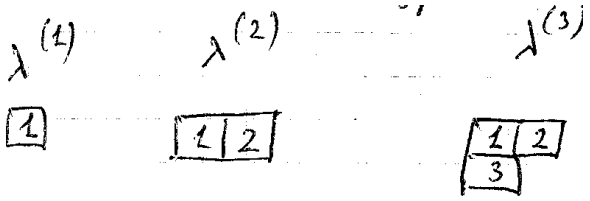
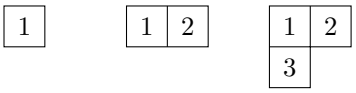


Diagram and typed description.

$\lambda^{(1)}$ $\lambda^{(2)}$ $\lambda^{(3)}$



Original source details: GELFAND-P001414

$$(7)\quad r_k(t) = (i_{k+1} - j_{k+1}) - (i_k - j_k)$$

point k has coordinates (i_k, j_k)

For the diagram:

$$(8)\quad h(\lambda) = r_\lambda + 1$$

$$(9)\quad r_\lambda = \max_{t \in T(\lambda)} \max_{1 \le k \le N} |r_k(t)|,$$

where $T(\lambda)$ is the set of tableaux constructed from the diagram λ, i.e., of shape λ.

Uncertain reading. The manuscript writes the upper bound as $k \le N$.

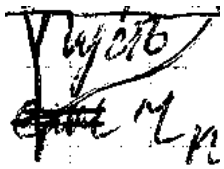
Original source details: GELFAND-P001414

Let $s = (\mu^{(k)})_{k=1}^{N}$, $\quad t = (\lambda^{(k)})_{k=1}^{N}$.

(10) Def. $\quad s \underset{n}{\approx} t$, if $\lambda^{(k)} = \mu^{(k)}$, $k \neq n$

Claim.

(11) [Crossed out: if?] [Inserted above the line: Let] $r_n(t) = \pm 1$. Then $s \underset{n}{\approx} t \Rightarrow s = t$.



Original source details: GELFAND-P001414

Uncertain reading. The crossed-out word tentatively reads "if." The insertion above the line reads "Let."

(12) Let $|r_n(t)| > 1$. Then $\exists!\ s:\ s \underset{n}{\approx} t,\ s \neq t$.

$$(13) \quad S_{\lambda}^{(k)} = \|S_{s,t}^{(k)}(\tau)\|_{s,t \in T(\lambda)} \quad \text{is a matrix.}$$

Original source details: GELFAND-P001414

Uncertain reading. An unclear mark follows the matrix index, possibly q or a corrected comma.

$$(14) \quad S_{s,t}^{(k)}(\tau) = \left(W_{r_k(s)}(\tau)\, W_{r_k(t)}(\tau)\right)^{1/2} \quad \text{if } s \underset{k}{\approx} t$$

Original source details: GELFAND-P001414

Undeciphered passage. There is a blotted mark after the first factor.

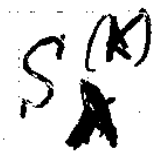
[Crossed out: if $s \underset{k}{\approx} t$?] and $S_{s,t}^{(k)}(\tau) = 0$ otherwise.

Original source details: GELFAND-P001414

Uncertain reading. The crossed-out condition tentatively reads "if $s \underset{k}{\approx} t$."

GELFAND-P001415

(15) Theorem. Let λ be a diagram with N elements and $q \neq \sqrt[m]{1}$. Then $S_{\lambda}^{(k)}$ give a representation of the Hecke algebra (in the generators e_k).



Original source details: GELFAND-P001415

Uncertain reading. The subscript of $S^{(k)}$ is heavily retraced and reads as λ.

Let us construct the functions W.

$W_{-r}(\tau)$ for $r = 1, 2, \ldots$ is determined from the relations

$$(16) \quad W_{-(r+1)}(\tau) = 1 - \frac{\tau}{W_{-r}(\tau)}, \qquad r \geqslant 0,$$

[Crossed out: and for $r \geqslant 0$?]

Uncertain reading. The crossed-out sign tentatively reads $\geqslant$.

Original source details: GELFAND-P001415

$$(17) \quad W_r(\tau) = 1 - W_{-r}(\tau), \qquad r \geqslant 0;$$

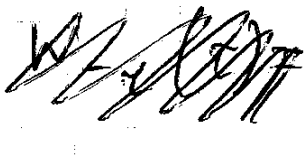
Or:

$W_{-r}(\tau)$ is determined [Crossed out: illegible] from its continued fraction expansion:

Undeciphered passage. The crossed-out word after "is determined" is illegible.

Original source details: GELFAND-P001415

$$(18) \quad \text{[Crossed out: } W_{-r}(\tau) = \text{]} \quad 1 - \cfrac{\tau}{1 - \cfrac{\tau}{1 - \ldots}}$$

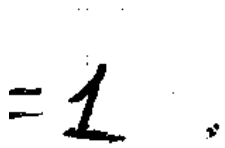


Uncertain reading. The crossed-out expression before the fraction tentatively reads $W_{-r}(\tau) =$.

Original source details: GELFAND-P001415

There is an equation:

$$(19) \quad W_r(\tau) W_{1-r}(\tau) = 1.$$



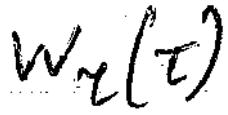
Uncertain reading. The right-hand side of (19) reads 1.

Original source details: GELFAND-P001415

General formula:

$$(20)\quad W_r(\tau) = \frac{q(q^{r-1}-1)}{(q+1)(q^r-1)}, \qquad r = 1, 2, \dots$$

$$(21)\quad W_{-r}(\tau) = \frac{q^{r+1}-1}{(q+1)(q^r-1)}.$$



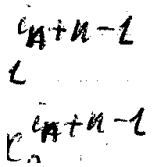
Uncertain reading. The subscript of W reads as r.

Original source details: GELFAND-P001415

(22) Problem: find the characters of the Hecke algebra.

Consider

$$(23)\quad \frac{\begin{vmatrix} x_1^{i_1} & x_1^{i_2+1} & \dots & x_1^{i_n+n-1} \\ x_2^{i_1} & x_2^{i_2+1} & \dots & x_2^{i_n+n-1} \\ \dots & \dots & \dots & \dots \\ x_n^{i_1} & & \dots & x_n^{i_n+n-1} \end{vmatrix}}{\begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & & \\ & & & & \\ 1 & x_n & x_n^2 & & x_n^{n-1} \end{vmatrix}} = \sum a(i,j) P_{j_1} P_{j_2} \dots P_{j_n}$$



Uncertain reading. The subscripts on i in the first two rows of the last column are overwritten. The exponent reads as $i_n + n - 1$.

Original source details: GELFAND-P001415

GELFAND-P001416

[source number: 53]

$j = (j_1, \dots, j_n)$, P_j is a power sum.

Then

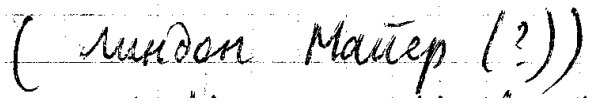
(24) $a(i,j)$ is the character value [inserted above the line: of the symmetric group] on an element with cycle lengths $j_1, \dots, j_n$

(up to a simple constant)

For the symmetric group, see Macdonald's book "Symmetric Functions and Hall Polynomials."

Location of the poles of Ruelle's zeta function.

(Lyndon Mayer (?))



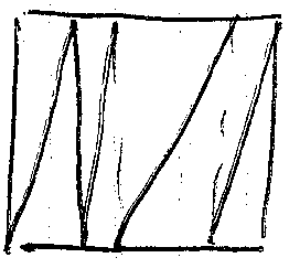
Uncertain reading. The manuscript appears to read "Lyndon Mayer (?)". The cited article is by Dieter H. Mayer.

Original source details: GELFAND-P001416

Letters in Math. Physics, $14:2$, 1987.

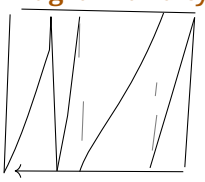
I is an interval, $T : I \to I$ is piecewise linear

$$I = \bigcup_j I_j, \qquad T(I_j) = I.$$



Original source details: GELFAND-P001416

Diagram and typed description.



The attachment of the faint strokes to the branches and the exact locations of the breaks are unclear.

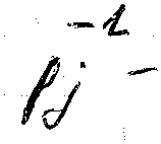
$$(1) \quad \zeta_T(z,\varphi) = \frac{\det(1 - zL_1)}{\det(1 - zL_2)},$$

where

$$L_1 f(z) = \varphi(z) \sum \frac{1}{\rho_j} f(\psi_j(z)),$$

$$L_2 f(z) = \varphi(z) \sum f(\psi_j(z)),$$

ψ_j are the local inverses of T on the interval I_j, and ρ_j^{-1} is the slope.

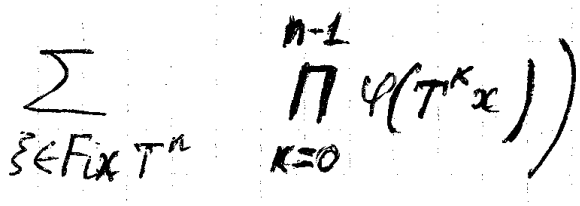


Original source details: GELFAND-P001416

Uncertain reading. The slope letter can be read as ρ_j or p_j.

In the general case:

$$(2) \quad \zeta_T(z,\varphi) = \exp\left(\sum_{n=1}^{\infty} \frac{z^n}{n} \sum_{\xi \in \operatorname{Fix} T^n} \prod_{k=0}^{n-1} \varphi(T^k x)\right)$$



Original source details: GELFAND-P001416

Uncertain reading. The second sum is indexed by $\xi \in \operatorname{Fix} T^n$. The product uses x.

GELFAND-P001417

7/XII-87 Gelfand Seminar. Some remarks on the paper by

D. Mayer Lett. in Math. Phys. 1987, 14:2

On Ruelle's ζ function

F is the set of spin values (finite).

We study spin configurations on a lattice

$$(\xi_i), \qquad -\infty < i < \infty, \qquad \xi_i \in F.$$

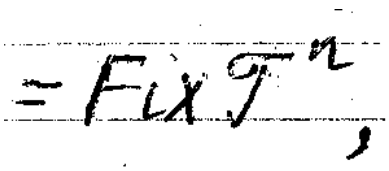
$M \subseteq F \times F$ is the admissible set. We require

$$(1) \quad (\xi_i, \xi_{i+1}) \in M$$

The simplest example of a Ruelle ζ function is the Artin–Mazur ζ function.

Consider the set of periodic configurations and their number:

$$(2) \quad N_n = \#\{(\xi_i) \mid \mathcal{T}^n(\xi_i) = (\xi_i)\} = \operatorname{Fix} \mathcal{T}^n,$$



Original source details: GELFAND-P001417

Uncertain reading. The last term of (2) is written as $\operatorname{Fix} \mathcal{T}^n$ without a cardinality sign.

where $\mathcal{T}$ is the transformation on the space of sequences

$$\mathcal{T}(\xi_i) = \xi_{i+1}$$

(shift)

Ruelle's ζ function:

$$(3) \quad \zeta_{\mathcal{T}}(s) = \exp\left(\sum_{n=1}^{\infty} \frac{N_n}{n} s^n\right)$$

the series converges for $|s| < \dfrac{1}{|F|}$.

Does an analytic continuation exist? Consider the matrix

Original source details: GELFAND-P001417

Uncertain reading. The end of the first word and the initial letter of "analytic" are retraced in the manuscript.

$$(4) \quad T = (T_{\alpha\beta}) = \begin{cases} 1, & (\alpha, \beta) \in M, \\ 0, & (\alpha, \beta) \notin M. \end{cases}$$

Then

$$(5) \quad \zeta_{\mathcal{T}}(s) = \frac{1}{\det(1 - sT)}$$

This shows that $\zeta_{\mathcal{T}}(s)$ is meromorphic.

The radius of convergence may exceed $1/|F|$.

For the proof, observe that

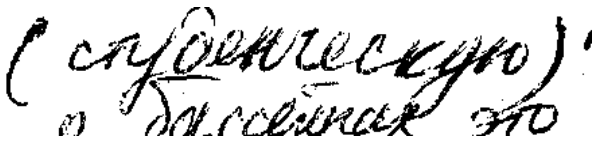
$$N_n = \operatorname{Tr} T^n. \tag{6}$$

GELFAND-P001418

[source number: 62]

Remark by I. M. G.: this resembles the work of Dynkin and Dmitriev:

consider a stochastic matrix (p_{ik}) [inserted above the line: (a student's?)] (in the problem about basins this is the [crossed-out sign: illegible] fraction of the water that flows from basin $i$to basin k).



Original source details: GELFAND-P001418

Uncertain reading. The insertion above the line tentatively reads «студенческую» (student's).

Original source details: GELFAND-P001418

Undeciphered passage. The sign before "fraction" is crossed out and illegible.

$$\underline{P} = (p_{ik}), \qquad \sum_k p_{ik} = 1.$$

This depends on the eigenvalues of P.

a) The eigenvalues lie in the disk $|\lambda| \leq 1$,

b) 1 is an eigenvalue.

What region do the eigenvalues fill?

It is the interior of some algebraic curve.

They went about it the wrong way.

There is a whole set of eigenvalues!!

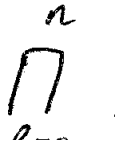
and one should plot the set of eigenvalues in $\mathbb{C}^n$ — it ought to be a beautiful region (Problem!!)

The projection onto a single axis can be complicated!

[Double line in the left margin.] In Ruelle's problem it is interesting to look at all the eigenvalues of the matrix T.

Speaker. Generalization of example (3)

$$(7) \quad \zeta_{\mathcal{T}}(s, \varphi) = \exp\left(\sum_{n=1}^{\infty} \frac{s^n}{n} \sum_{\xi \in \mathrm{Fix}(\mathcal{T}^n)} \prod_{\ell=0}^{n} \varphi(\mathcal{T}^\ell \xi)\right)$$



Original source details: GELFAND-P001418

Uncertain reading. The upper limit of the product in (7) is written as n.

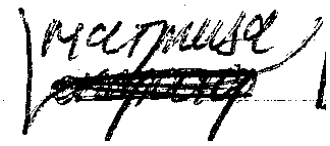
Here

$$\underset{\text{function}}{\varphi} = \varphi(\xi_1, \ldots, \xi_k), \qquad \xi_j \in F$$

(φ depends only on the first k values).

((3) is obtained for $\varphi = 1$)

(8) Claim. $\exists$ [Crossed out: operator?] [inserted above the line: matrix] L (finite-dimensional), such that:



Original source details: GELFAND-P001418

Uncertain reading. The crossed-out word tentatively reads "operator." The insertion above it reads "matrix."

$$\zeta_{\mathcal{T}}(s,\varphi) = \frac{1}{\det(1-sL)} \tag{9}$$

Specifically, consider

$$F' = F^k;$$

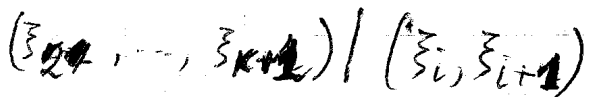
[Crossed out: illegible]

Original source details: GELFAND-P001418

Undeciphered passage. The writing before M' is crossed out and illegible.

$$M' \subset F' \times F'$$

$$M' = \{(\xi_1,\dots,\xi_k);(\xi_2,\dots,\xi_{k+1}) \mid (\xi_i,\xi_{i+1}) \in M\} \tag{10}$$



Original source details: GELFAND-P001418

Uncertain reading. The subscripts 2 and $k+1$ in the second tuple are overwritten. The $i+1$ in the condition is also retraced.

Then take

$$\ell_{\alpha\beta} = \begin{cases} \varphi(\alpha), & (\alpha,\beta) \in M', \\ 0, & (\alpha,\beta) \notin M', \end{cases} \qquad L = (\ell_{\alpha\beta}).$$

GELFAND-P001419

I. M. G.

NB: one needs to know the p-adic numbers

(be able to add, multiply, and divide them, [crossed out: illegible] and extract roots)

Original source details: GELFAND-P001419

Undeciphered passage. The mark after "divide" is crossed out and illegible.

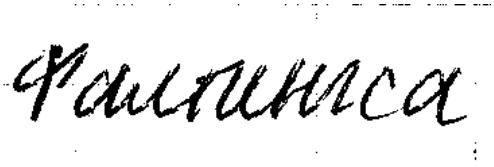
Beilinson A. p-adic numbers and physics.

On some of Manin's p-adic ideas in physics!

Original source details: GELFAND-P001419

Uncertain reading. The last mark resembles "!" and may read "?".

1°. Witten listened to the talk by Faltings with great interest and then bought all the books on number theory. The only thing he understood from the talk was the formula



Uncertain reading. The surname in the manuscript tentatively reads «Фалтингса» (Faltings).

Original source details: GELFAND-P001419

$$(1)\quad |a|_\infty \cdot |a|_2|a|_3 \ldots = 1, \qquad a \in \mathbb{Q}.$$

For $\frac{1}{2}$ we have here:

$$\frac{1}{2} \cdot 2 \cdot 1 \ldots = 1.$$

There are 3 papers:

1) Volovich (in Russian)

2) Yu. I. Manin (in English)

3) Witten and someone else

Uncertain reading. An isolated illegible mark stands to the right of the third entry.

Original source details: GELFAND-P001419

All 3 papers appeared in September.

Witten's text.

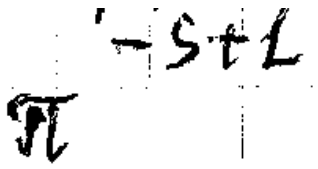
Consider the functional equation for the Riemann ζ function

$$(2)\quad \pi^{-s/2}\Gamma(s/2)\prod(1-p^{-s})^{-1} = \pi^{-(1-s)/2}\Gamma\left(\frac{1-s}{2}\right)\prod(1-p^{-(1-s)})^{-1}.$$

(analytic continuation is understood)

or (formally):

$$(3)\quad \frac{\pi^{-s+1}\Gamma(s/2)}{\Gamma\left(\frac{1-s}{2}\right)} = \prod_p \frac{1-p^{-s}}{1-p^{s-1}}.$$



Uncertain reading. The exponent of π is written as $-s+1$.

Original source details: GELFAND-P001419

On the left ≈ the Veneziano–Witten amplitude for an open string [inserted above the line: (with momentum s)]. On the right is a divergent expression, but what matters is that it is written as a product over the primes!

GELFAND-P001420

[source number: 63]

2°. On Manin's work

(1) Claim: The Tamagawa measure $= 1$.

In more detail:

$$\mathrm{SL}(n,\mathbb{R}) \supset \mathrm{SL}(n,\mathbb{Z}).$$

On $\mathrm{SL}(n,\mathbb{R})$ there is an invariant [inserted above "invariant": «left»] measure (the standard one).

Undeciphered passage. A crossed-out mark is visible below the end of "measure."

Original source details: GELFAND-P001420

Normalize [crossed out: illegible] using the integer matrices.

Undeciphered passage. The writing after "Normalize" is crossed out and illegible.

Original source details: GELFAND-P001420

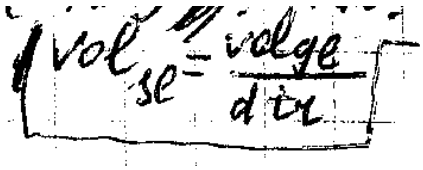
$$\mathrm{vol}\big(\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z})\big) - ?$$

[In the right margin:] (it is constructed from the standard basis in the Lie algebra $\mathfrak{sl}(n,\mathbb{R})$ or in $\mathfrak{gl}(n,\mathbb{R})$ [crossed out: illegible] standard basis.

Undeciphered passage. The mark after $\mathfrak{gl}(n,\mathbb{R})$ is crossed out and illegible.

Original source details: GELFAND-P001420

$$\boxed{\mathrm{vol}_{sl} = \frac{\mathrm{vol}_{gl}}{d\ \mathrm{tr}}}$$



Uncertain reading. The small letters next to "vol" read sl and gl.

Original source details: GELFAND-P001420

$\mathrm{SL}(2,\mathbb{R})$ consists of conformal transformations of the unit disk, acting transitively on S^*H^2. The invariant volume on S^*H^2 is [crossed out: illegible] $\mathrm{vol}_{H^2} \otimes d\varphi$.

Undeciphered passage. The mark before vol_{H^2} is crossed out and illegible.

Original source details: GELFAND-P001420

Uncertain reading. The letter after d reads as φ.

Original source details: GELFAND-P001420

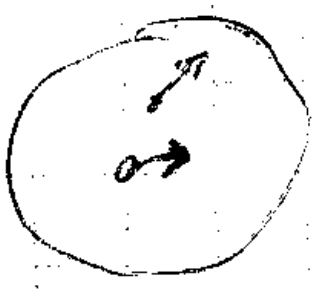
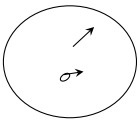


Diagram and typed description.



Original source details: GELFAND-P001420

Or take the quotient by the maximal compact subgroup

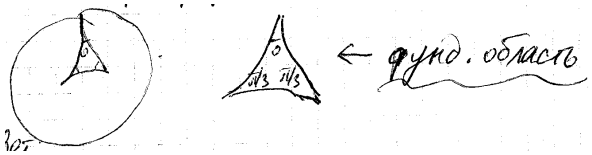
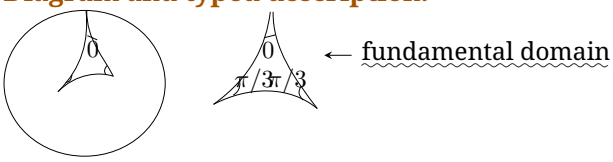


Diagram and typed description.



Original source details: GELFAND-P001420

Answer.

$$(2) \quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z}) = \zeta(2)\zeta(3)\ldots\zeta(n)$$

$\mathbb{A}$ denotes the adeles.

$$(3) \quad \mathrm{SL}(n,\mathbb{A}) \sim \mathrm{SL}(n,\mathbb{R}) \times \mathrm{SL}(n,\mathbb{Q}_2) \times \ldots$$
$$\cup$$
$$\mathrm{SL}(n,\mathbb{Q})$$

$$(4) \quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{A})/\mathrm{SL}(n,\mathbb{Q}) = 1.$$

$$\mathrm{vol}\,\mathrm{SL}(n,\mathbb{A})/\mathrm{SL}(n,\mathbb{Q}) \sim \mathrm{vol}\,\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z})$$
$$\prod \mathrm{vol}\,\mathrm{SL}(n,\mathbb{Z}_p).$$

$$(5) \quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{Z}_p) \sim \#\mathrm{SL}_n(\mathbb{F}_p)/p^{n^2-1}.$$

Exercise: compute $\#\mathrm{SL}_n(\mathbb{F}_p)$.

GELFAND-P001421

(6) Number of elements of $\mathrm{GL}(n, \mathbb{F}_p)$

$$(p^n - 1)(p^n - p)(p^n - p^2) \ldots = p^{n^2}(1 - p^{-n})(1 - p^{-n+1}) \ldots (1 - p^{-1}).$$

(7) For $\mathrm{SL}(n, \mathbb{F}_p)$ divide by $p - 1$:

$$p^{n^2-1}(1 - p^{-n})(1 - p^{-n+1}) \ldots (1 - p^{-2}).$$

This is a product of Euler factors.

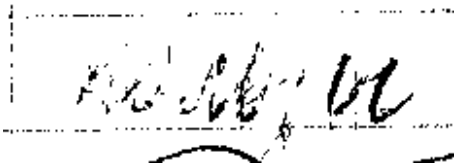
(8) Yu. I. Manin's conjecture: an analogue of formula (2) can be written for the moduli spaces of curves arising in string theory.

I. M. G.

Mathematical question: the volume of Teichmüller space has a p-adic analogue.

Or: what is adelic Teichmüller space

An example of the power of Witten's work:



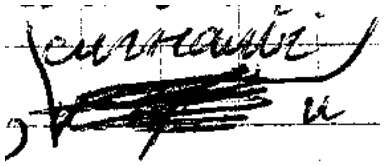
Original source details: GELFAND-P001421

Uncertain reading. The faint word between "example" and "work" may read «мощи» or «силы» (power).

Laumon did work in number theory based on Witten's ideas from Morse theory.

Idea: consider a function $u \in \mathcal{E}'(\mathbb{R})$, $\widetilde{u}(\xi) \in C^\infty$ and all the singularities will go into the asymptotics at infinity.

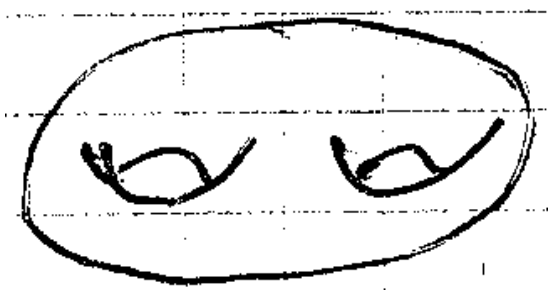
Advice from I. M. G.: read Kharkevich's book «Spectra, [crossed out: illegible] [inserted above the line: Signals] and Noise», to understand the Fourier transform.



Original source details: GELFAND-P001421

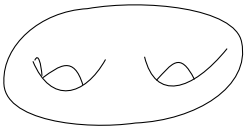
Undeciphered passage. The writing in the book title is crossed out. "Signals" is inserted above it.

On the work of Laumon.



Original source details: GELFAND-P001421

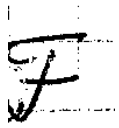
Diagram and typed description.



The short stroke at the left of the first loop follows the visible outline. Its purpose is unclear.

Consider a surface $\mathcal{F}$

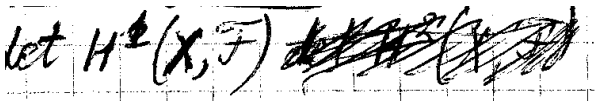
$$H^i(X, \mathcal{F}), \qquad i = 0, 1, 2.$$



Uncertain reading. After "surface" the manuscript writes $\mathcal{F}$, and then $H^i(X, \mathcal{F})$.

Original source details: GELFAND-P001421

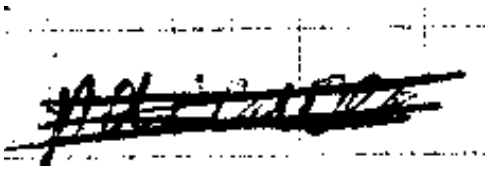
$$(1) \quad \det H^\bullet(X, \mathcal{F}) = \frac{\det H^0(X, \mathcal{F}) \det H^2(X, \mathcal{F})}{\det H^1(X, \mathcal{F}) \quad [\text{crossed out: } \det H^2(X, \mathcal{F})]}$$



Uncertain reading. The first superscript in the denominator is retraced and reads 1. The crossed-out factor tentatively reads $\det H^2(X, \mathcal{F})$.

Original source details: GELFAND-P001421

([crossed out: finite?] vector spaces. det is the top exterior power)



Uncertain reading. The crossed-out word before "vector spaces" tentatively reads "finite."

Original source details: GELFAND-P001421

Automorphisms?

Consider a triangulation. $\det H^\bullet$ is identified with the product of the det of the terms of the complex (an Euler-type formula): canonically

GELFAND-P001422

[source number: 64]

$$(2) \quad \det(C^\bullet) = \det(H^\bullet(C))$$

$2 - 2g$ is the Euler characteristic.

On the other hand

$$2g - 2 = \deg \underbrace{K}_{\text{canonical class}}$$

Let

D be a divisor.

$$(3) \quad \det \mathcal{F}_D = \bigotimes (\det \mathcal{F}_{x_i})^{\otimes a_i}, \qquad D = \sum a_i x_i$$

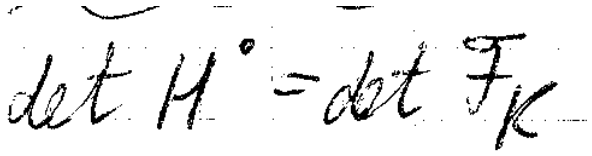
[crossed out: $D \longrightarrow D + K$?]

Uncertain reading. The crossed-out expression tentatively reads $D \to D + K$ or $D \mapsto D + K$.

Original source details: GELFAND-P001422

Deligne's formula:

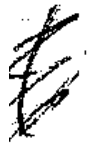
$$(4) \quad \det H^\bullet = \det \mathcal{F}_K$$



Original source details: GELFAND-P001422

Uncertain reading. The right-hand side of (4) reads $\det \mathcal{F}_K$ without the exponent -1.

[At right: crossed out: illegible]



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Undeciphered passage. The separate crossed-out mark to the right of the formula is illegible.

(fix the complex structure and the choice of a point in the canonical class)

For an arbitrary constructible sheaf one also computes det and there is an analogue of Deligne's formula!

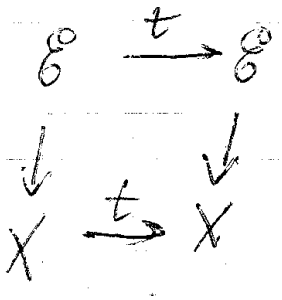
Here on $\mathbb{A}^1$ functions with singularities arise

Klyachko. Toric bundles.

A toric variety is a variety X on which an algebraic torus T, acts, with an orbit isomorphic to T.

$$X = \bigcup_\sigma O_\sigma, \quad \text{is the decomposition into orbits}$$

$\mathcal{E}$ is a bundle over X with a toric structure (and the action lifts! The diagram commutes)



Original source details: GELFAND-P001422

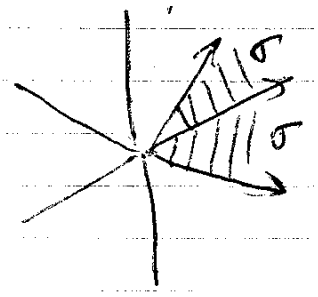
Diagram and typed description.

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{t} & \mathcal{E} \\ \downarrow & & \downarrow \\ X & \xrightarrow{t} & X \end{array}$$

One can describe the moduli of [inserted above the line: «such»] bundles in the language of linear algebra.

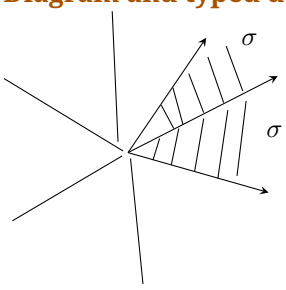
GELFAND-P001423

The structure of X is described by a fan Σ in the Lie algebra of the torus.



Original source details: GELFAND-P001423

Diagram and typed description.



(cones)

$|\Sigma|$ is the set of 1-dimensional cones in the fan Σ.

Theorem. (category of toric bundles) $\Longleftrightarrow$ (category of vector spaces [crossed-out sign: illegible] E with a collection of [inserted above the line: «decreasing $\mathbb{Z}$-indexed»] filtrations $E^\alpha(i)$, $\alpha \in |\Sigma|$, satisfying the condition

Undeciphered passage. The crossed-out mark before E is illegible.

Original source details: GELFAND-P001423

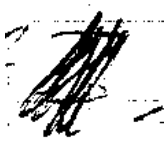
A) The filtrations E^α, $\alpha \in \sigma \in \Sigma$, are distributive.

Problems. 1) Describe the geometry of the bundle (cohomology, characteristic classes, and so on) in terms of the filtrations

2) classification — ?

Let Σ be a simplicial complex. Assign a filtration to a vertex

A representation of it is a collection of filtrations in a vector space [inserted above the line: «E:»] $E^\alpha(i)$, $\alpha \in$ [crossed-out sign: illegible] Σ,



Undeciphered passage. The mark between $\in$ and Σ is crossed out and illegible.

Original source details: GELFAND-P001423

$$E \ldots \supset E^\alpha(i) \supset E^\alpha(i+1) \supset \ldots 0$$

$$\alpha \in (\text{vertices of the complex}) = \Sigma.$$

The spaces $E^\alpha(i)$, $\alpha \in \underset{\text{simplex}}{\sigma}$, are distributive for fixed σ (for all i)

[Outlined insertion: (i.e., they generate a distributive lattice)]

For example, if

$\alpha_1, \alpha_2, \alpha_3$ are the vertices of a triangle, then

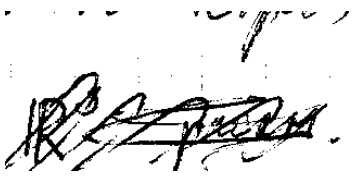
E^{α_1} E^{α_2} E^{α_3} are distributive, i.e.,

$$(E^{\alpha_1} + E^{\alpha_2}) \cap E^{\alpha_3} = E^{\alpha_1} \cap E^{\alpha_3} + E^{\alpha_2} \cap E^{\alpha_3}$$

Classify: a good problem

For a triangle, the solution is trivial

For the surface of a tetrahedron $\Sigma = \partial\Delta$, Δ is a simplex, and the answer is as follows. In [crossed out: an $\mathbb{R}^3$ space?] E, $\dim E = 3$, consider 4 2-dimensional subspaces in general position.



Uncertain reading. The crossed-out passage tentatively reads “an $\mathbb{R}^3$ space.”

Original source details: GELFAND-P001423

$$\mathcal{E} = (E;\ E^\alpha, E^\beta, E^\gamma, E^\delta)$$

Original source details: GELFAND-P001423

Uncertain reading. The last superscript is retraced and reads as δ.

GELFAND-P001424

[source number: 65]

we still need to take [above the line: their?] exterior powers

$$\Lambda^i\mathcal{E} = (\Lambda^i E, \Lambda^i E^\alpha, \Lambda^i E^\beta, \Lambda^i E^\gamma, \Lambda^i E^\delta)$$

Original source details: GELFAND-P001424

Uncertain reading. The insertion above the line tentatively reads «их» (“their”).

A general representation decomposes into a direct sum of 1-dimensional ones and ones of this kind.

I. M. G. The right problem: isolate in $\forall$ problem the solvable pieces. (Gelfand–Ponomarev philosophy)

V. [crossed-out initial: illegible] Kac describes all this in the language of graphs and Kac–Moody algebras: vertex dimensions = root coordinates, etc.

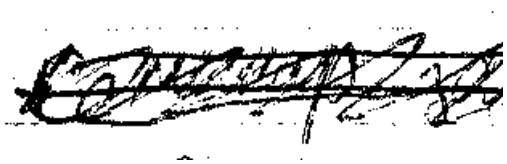
Original source details: GELFAND-P001424

Uncertain reading. The crossed-out initial between “V.” and “Kac” is illegible.

The number of parameters is determined.

Lect. Notes, 996; an earlier work — Inventiones. (conference talk)

[Crossed out: Seminar? ...]



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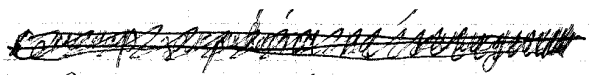
Uncertain reading. The first word of the crossed-out line tentatively reads “Seminar.”



Original source details: GELFAND-P001424

Undeciphered passage. The rest of the crossed-out line is illegible.

Next time: 8 February.

Chapter 32

GELFAND-P001425

[source number: 1]

8/II–88 Gelfand Seminar.

I. M. G. Problems. 1. Evaluate

$$\det \left\| \frac{1}{x_i - y_k} \right\|_{i,k=1}^{n}. \tag{1}$$

2. Prove that the Legendre polynomial

$$\left((x^2-1)^n\right)^{(n)}$$

has only real zeros.

3. Can a function that is the limit of continuous functions at every point of a closed interval

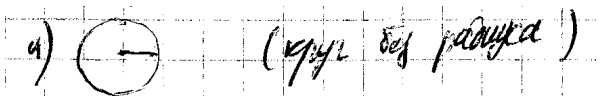
a) have discontinuities?

b) have an everywhere dense set of points of discontinuity?

c) be discontinuous everywhere (i.e., at every point)?

4. **Map conformally onto the disc**

a) (disc with a radius removed!)



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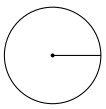

Diagram and typed description.

b)

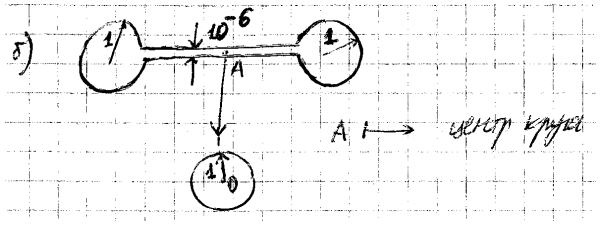

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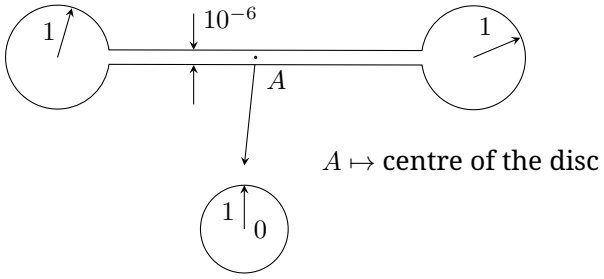
Diagram and typed description.



A. Beilinson. What does one need to know about string theory?

Conformal field theory. (2-dimensional)

Operator expansion.

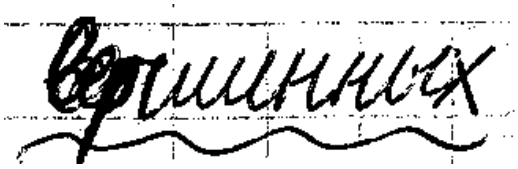
Connection with representations of natural infinite-dimensional Lie algebras.

W-algebras, parafermions.

[Crossed out: Be... / Bep...?] Algebra of vertex operators.

Original source details: GELFAND-P001425

Uncertain reading. The crossed-out beginning reads “Be...”, possibly “Bep...” (the start of “vertex”).



Original source details: GELFAND-P001425

Uncertain reading. The beginning of the word “vertex” is written over earlier strokes.

Various reductions of representations of $\widehat{SL}_2$ to the Virasoro algebra.

Connection with moduli spaces of algebro-geometric structures.

Explicit formulas for partition functions and [Crossed out: correlation functions; inserted above the line: correlators].

Branching property of characters.

GELFAND-P001426

D. Alekseevsky. G-structures

Scalar field quantity.

[crossed out: Field] — a function of the point $x \in \mathbb{R}^3$ with vector values

(e.g., with values in $\mathbb{R}$)

Field quantity: depends on the coordinate system

G-structure.

Def. 1 $G \subset GL(V), V = \mathbb{R}^n$.

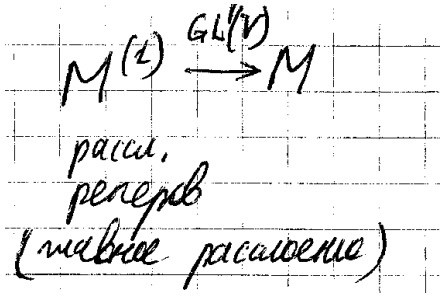
[Under G: linear group.]

G-structure [above the line: on M^n] — a principal subbundle of the manifold of all [crossed out: co?]frames.

[On the right, under «frames»: (coframes) — is better to say.]

Original source details: GELFAND-P001426

Uncertain reading. The crossed-out prefix before “frames” tentatively reads “co.”

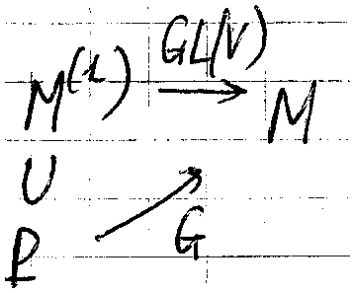


Original source details: GELFAND-P001426

Diagram and typed description.

$$M^{(1)} \xrightarrow{GL(V)} M$$

frame bundle
(principal bundle)

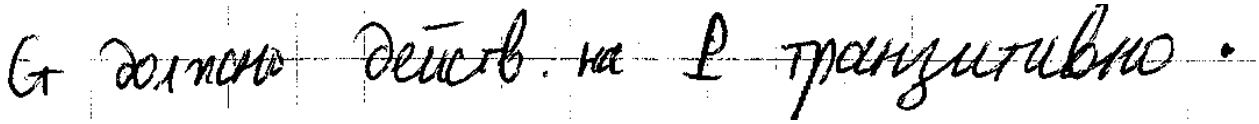


Original source details: GELFAND-P001426

Diagram and typed description.

$$\begin{array}{ccc} M^{(1)} & \xrightarrow{GL(V)} & M \\ \cup & \nearrow_{G} & \\ P & & \end{array}$$

G must act on P transitively.



Original source details: GELFAND-P001426

Uncertain reading. The manuscript says "on P transitively."

Then this is precisely a G-structure.

Examples: 1) a Riemannian metric is an $O(V)$-structure

2) Let S be a tensor field. When does it define a G-structure?

If they are all of the same canonical form

(at $\forall$ point $\exists$ a coframe [crossed out: θ / φ?], bringing S to a fixed form)

Uncertain reading. The crossed-out sign after "coframe" resembles θ or φ.

Original source details: GELFAND-P001426

Then if G — the group of automorphisms of the canonical form, then S defines a G-structure.

3) An affinor defines a G-structure, if at $\forall$ point the Jordan form is the same

4) Almost complex structure J: $G = GL\left(\frac{n}{2}, \mathbb{C}\right)$

5) Almost symplectic structure: 2-form σ of maximal rank,

(symplectic structure: $d\sigma = 0$)

Def. 2 G-structure — a principal G-bundle $\pi\colon P \xrightarrow{G} M$

$G \subset GL(V)$

equipped with a vector-valued 1-form $\theta\colon TP \to V$,

satisfying the following conditions:

GELFAND-P001427

[source number: 2]

a) G-equivariance

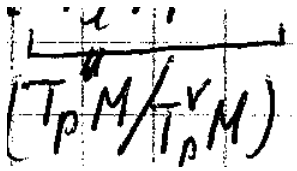
b) Ker $\theta = T^v P$ (vertical space), i.e., θ is strictly horizontal.

Equivalence of definitions 1 and 2.

$p \in P$, $x = \pi p$;

$$\theta_p \colon \underbrace{T_x M}_{\downarrow (T_p M / T_p^v M)} \xrightarrow{\simeq} V.$$

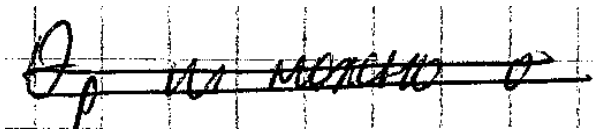
[On the right:] this is precisely the coframe with which we identify the point p.



Original source details: GELFAND-P001427

Uncertain reading. Below $T_x M$ is $T_p M / T_p^v M$, with M written in both terms.

[Crossed out: θ_p on the vertical [space]? 0.]



Original source details: GELFAND-P001427

Uncertain reading. The word between "on" and 0 tentatively begins "вертик..." ("vertical...").

Coframe $\simeq$ an isomorphism $\theta_p \colon T_x M \xrightarrow{\simeq} V$

θ is called the solder form

———— · ————

Suppose there is a G-structure

$$(\pi \colon P \xrightarrow{G} M; \theta)$$

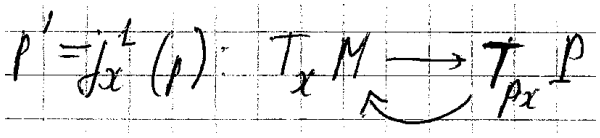
Consider the bundle of 1-jets of the bundle π

$$\pi' \colon P' \to P$$

$x \to p_x$ is a section

$$p' \colon T_x M \to T_{p_x} P$$

Def. A 1-jet is a map p' — inverse to the projection, $p' = j^1_x(p)$



Original source details: GELFAND-P001427

Diagram and typed description.

$p' = j^1_x(p) \colon \quad T_x M \longrightarrow T_{p_x} P$

i.e., $\pi_* \circ p' = \mathrm{id}$

Def. A horizontal subspace $H \subset T_pP$, [Crossed out: illegible] — is a subspace such that

Original source details: GELFAND-P001427

Undeciphered passage. The crossed-out mark after T_pP is illegible.

$$T_pP = T_p^v \oplus H$$

Second definition of a 1-jet

$$\underbrace{\text{Elements of } P'}_{(=\text{1-jets})} = \text{horizontal subspaces}$$

$$p' \in P' \quad \Longleftrightarrow \quad H(p') \text{ — a horizontal subspace}$$

Relation:

$$p' \colon T_xM \to H(p') \subset T_pP$$

Def. [Crossed out: P'?] π' is called the bundle of connections.

Original source details: GELFAND-P001427

Uncertain reading. The crossed-out mark before π' may read P'.

GELFAND-P001428

Space of connections: $\Gamma(\pi')^G$

(i.e., a connection is a G-equivariant section of the bundle $\pi' \colon P' \to P$)

Connection:

$$h \colon P \to P'$$
$$p \mapsto H(p)$$

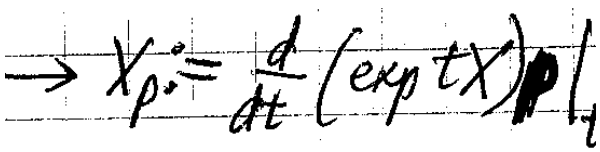
(G-invariant field of horizontal subspaces)

This gives parallel transport (restrict to a curve)

Torsion function (analogue of the structure constants of a Lie group)

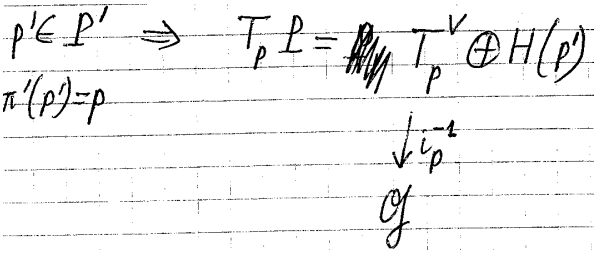
There is a vertical parallelism:

$$i_p \colon \mathfrak{g} \xrightarrow[\simeq]{} T_p^v P$$
$$X \mapsto X_p := \frac{d}{dt}(\exp tX)p\Big|_{t=0}$$



Original source details: GELFAND-P001428

Uncertain reading. The p following $(\exp tX)$ is retraced. It is written to the right of the exponential.

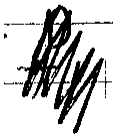


Original source details: GELFAND-P001428

Diagram and typed description.

$$p' \in P' \Longrightarrow T_pP = T_p^v \oplus H(p')$$
$$\pi'(p') = p \qquad \downarrow i_p^{-1}$$
$$\mathfrak{g}$$

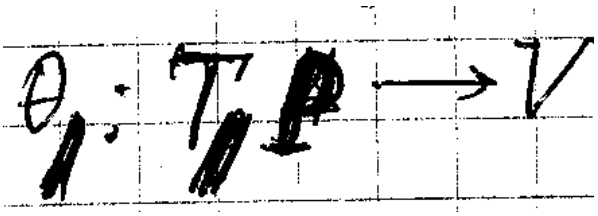
[After the $=$ sign, crossed out: illegible]



Original source details: GELFAND-P001428

Undeciphered passage. The mark between $=$ and T_p^v is crossed out and illegible.

Take the form $\theta_p \colon T_pP \to V$



Original source details: GELFAND-P001428

Uncertain reading. Both subscripts p and the letter P are retraced or overwritten.

and set

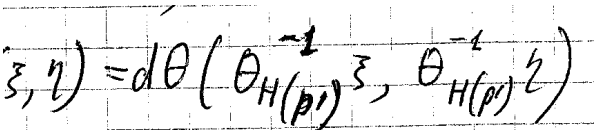
$$\theta_{H(p')} = \theta|_{H(p')} \colon H(p') \xrightarrow{\simeq} V$$

Def. The torsion function — is the map

$$t \colon P' \longrightarrow \mathrm{Hom}(V \wedge V, V)$$

(tensors of type $1, 2$)

$$p' \mapsto t_{p'}, \qquad t_{p'}(\xi, \eta) = d\theta\big(\theta^{-1}_{H(p')}\xi, \theta^{-1}_{H(p')}\eta\big)$$



Original source details: GELFAND-P001428

Uncertain reading. The last symbol has a differently shaped initial stroke and reads as η.

Properties of the torsion function.

1) G-equivariance and $G' = \mathrm{Hom}(V, \mathfrak{g})$-equivariance

GELFAND-P001429

[source number: 3]

15/II–88 Gelfand Seminar (from the notes of A. K. Zvonkin)

In I. M. G.'s lectures on quasilinear equations (UMN, 1959, vol. 14, no. 2) three pages present all of thermodynamics.

A concrete problem in geometry: what is the angle between two 2-dimensional subspaces in 4-dimensional space?

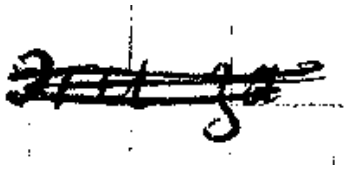
The angle is an invariant of relative motion (for lines).

There are 2 versions, for oriented lines and for unoriented lines.

For planes, how many such invariants are there and what are they?

Question: find all the invariants of two subspaces of a Hilbert space.

[Crossed out: Zam...?]



Original source details: GELFAND-P001429

Uncertain reading. The crossed-out writing above the name tentatively begins "Zam...".

Zamolodchikov, Sasha. Conformal (2-dimensional) field theory.

$$\langle A_1(x_1)A_2(x_2)\dots A_N(x_N)\rangle, \qquad x_i \in \mathbb{R}^2, \tag{1}$$

— «correlation functions»

$\mathcal{A} \ni A$ — fields (in general infinite-dimensional)

(1) — N-linear function on the space of fields $\mathcal{A}$ (also a function of $(x_1, \dots, x_N)$)

Let us specify the meaning of the average in (1):

$$(1) = Z^{-1} \int \mathcal{D}\varphi\, e^{-S[\varphi]} A_1(x_1)\dots A_N(x_N), \tag{2}$$

where $\varphi = \varphi(x)$,

$$S = \int \sigma\big(\varphi(x), \partial\varphi(x)\big)\, dx, \tag{3}$$

$$Z = \int \mathcal{D}\varphi\, e^{-S[\varphi]}. \tag{4}$$

Original source details: GELFAND-P001429

Uncertain reading. The number (4) is written over a digit resembling 3.

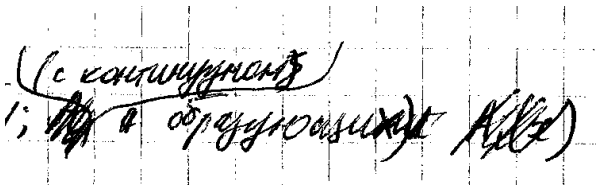
Properties of the correlation functions.

1) «Algebra of fields»

Let the fields form an associative algebra; [Crossed out: illegible] with [inserted above the line: (with a continuum)] generators $A_i(x)$.

Original source details: GELFAND-P001429

Undeciphered passage. The writing immediately after the semicolon is crossed out and illegible.



Original source details: GELFAND-P001429

Uncertain reading. The ending of the Russian word for “generators” is overwritten, possibly over a genitive plural. The strokes in $A_i(x)$ are also retraced.

Let $\{A_i\}$ — be a basis in $\mathcal{A}$. Then

$$A_i(x)A_j(y) = \sum_k \underbrace{c_{ij}^k(x,y)}_{\text{numerical functions of } x,y} A_k(y)$$

Set

$$\langle A_k(x)\rangle = a_k(x)$$

GELFAND-P001430

(Knowing the functions c_{ij}^k and a_k is sufficient to compute the correlation functions: we replace the pair A_1A_2 by $\sum c^k A_k$ and so on)

2) «Invariance»

Consider a field T of the form

$$T^{\mu\nu}, \quad \mu,\nu = 1,2; \quad T^{\mu\nu} \in \mathcal{A}.$$

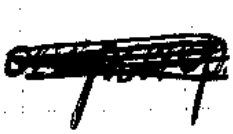
Incidentally

$$\frac{\partial}{\partial x^\mu} A_i(x)A_j(y) = \sum \frac{\partial}{\partial x^\mu} c_{ij}^k(x,y)A_k. \tag{5}$$

$$\frac{\partial T^{\mu\nu}}{\partial x^\mu} = 0 \tag{6}$$

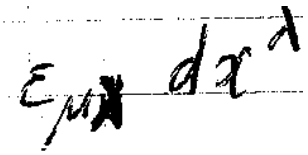
Consider [crossed out: operator?] the integral

$$\oint T^{\mu\nu}(x)A(y)\,\varepsilon_{\mu\lambda}\,dx^\lambda \tag{7}$$



Original source details: GELFAND-P001430

Uncertain reading. The crossed-out word tentatively reads “operator.”



Original source details: GELFAND-P001430

Uncertain reading. The second index of ε is overwritten. It reads λ, possibly over ν.

[To the right of the integral]

Original source details: GELFAND-P001430

Diagram and typed description.

[Callout to ε in (7):]

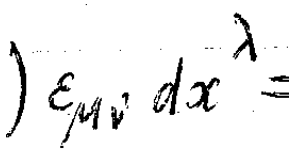
$$\varepsilon_{\mu\lambda} = -\varepsilon_{\lambda\mu}, \qquad \varepsilon_{12} = 1$$

It defines a linear operator in $\mathcal{A}$.

Required: $\exists$ a field T such that this operator equals $\dfrac{\partial}{\partial y}$.

$T^{\mu\nu}$:= «energy–momentum tensor»

$$\oint T^{\mu\nu}(x) A_i(y)\, \varepsilon_{\mu\nu}\, dx^\lambda = \frac{\partial}{\partial y^\nu} A_i(y)$$



Original source details: GELFAND-P001430

Uncertain reading. In the repeated formula the second index of ε reads ν, alongside dx^λ.

$1')$ $c_{ij}^k(x,y) = c_{ij}^k(x-y)$ — functions of one variable in $\mathbb{R}^2$.

3) «Positivity condition»

$A \mapsto A^+ \in \mathcal{A}$ — involution

$$A_i^+(x) A_j^+(y) = \sum \bar{c}_{ij}^k(\widetilde{x} - \widetilde{y}) A_k^+(y) \tag{8}$$

$$x = (x^1, x^2) \quad \Rightarrow \quad \widetilde{x} = (-x^1, x^2)$$

Requirement: the matrix $\langle A_i^+(\widetilde{x}) A_j(x) \rangle$ is positive definite for all x.

The problem is to construct substantive examples of all this.

GELFAND-P001431

[source number: 4]

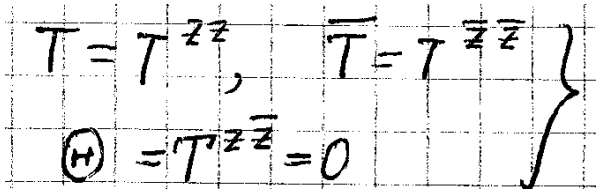
Let us introduce in $\mathbb{R}^2$ complex coordinates

$$z = x^1 + ix^2 \qquad ds^2 = dz\, d\bar{z}$$

$$T^{\nu\mu} = T^{\mu\nu} \tag{9}$$

$$\left.\begin{array}{l} T = T^{zz}, \quad \bar{T} = T^{\bar{z}\bar{z}} \\ \Theta = T^{z\bar{z}} = 0 \end{array}\right\} \quad \begin{array}{c} \text{under these additional conditions} \\ \text{the theory is called} \\ \text{«conformal field theory»} \end{array} \tag{10}$$

[Arrow from below to $\Theta = 0$:] Important special case: massless theory



Original source details: GELFAND-P001431

Uncertain reading. In (10) both upper indices in the definition of $\bar{T}$ have a bar: $\bar{T} = T^{\bar{z}\bar{z}}$.

Check question: write T^{zz}, $T^{z\bar{z}}$, $T^{\bar{z}\bar{z}}$ in terms of $T^{\mu\nu}$
(this is a linear algebra problem)

$$\frac{\partial T}{\partial \bar{z}} = 0, \quad \frac{\partial \bar{T}}{\partial z} = 0 \qquad \text{— Cauchy–Riemann equations} \tag{11}$$

$$\Updownarrow$$

T — an analytic function of z, $T = T(z)$, $\bar{T} = \bar{T}(\bar{z})$.

$$T(z)A_i(z_1, \bar{z}_1) = \sum f_{ij}(z - z_1)A_j(z_1, \bar{z}_1) \tag{12}$$

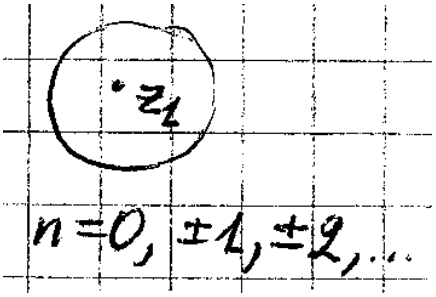
In electrodynamics.
ψ — electron
A — photon

$$\{A_i\} = \{\psi, \psi^2, \psi\partial\psi, \ldots, A, A^2, A\psi, \ldots\}$$

ψ^4 still exists, but $\psi^5 = \psi^6 = 0$ (since the spinor has only 4 components)
for A all powers exist
But, in general, A^2 does not exist.
Do not take the fields at the same point. Take them at different points.

$$\oint \frac{dz}{2\pi i} z^{n+1} T(z)A_i(z_1, \bar{z}_1) = L_n A_i(z_1, \bar{z}_1) \tag{13}$$

[To the right of (13):]



Original source details: GELFAND-P001431

Diagram and typed description.

$n = 0, \pm 1, \pm 2, \ldots$

$$L_{-1} = \frac{\partial}{\partial z_1}$$

$$\bar{T}(\bar{z}) \to \bar{L}_n; \quad \bar{L}_{-1} = \frac{\partial}{\partial \bar{z}_1} \tag{14}$$

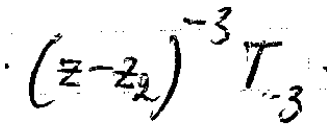
$T(z_1)T(z_2)$ — is a function of z_1, analytic everywhere except at z_2; let us expand it in a Laurent series.

$$T(z_1)T(z_2) = \sum_{m=-\infty}^{\infty} (z_1 - z_2)^m T_m(z_2), \qquad T_m \in \mathcal{A} \tag{15}$$

Let us make an assumption:

$$(15) = (z_1 - z_2)^{-4} T_{-4} + (z - z_2)^{-3} T_{-3} + \dots \tag{16}$$

(i.e., the series terminates at its lower end at term -4)



Uncertain reading. The second power in (16) is written $(z - z_2)^{-3}$, without a subscript on its first z.

Original source details: GELFAND-P001432

$$T_{-1} = \frac{dT}{dz_2}(z_2) \tag{17}$$

$$T_{-2} = 2T(z_2) \quad \leftarrow \quad \text{follows from the fact that } T \text{ is a rank-2 tensor} \tag{18}$$

The most important thing is to determine the most singular terms T_{-3} and T_{-4}.

It follows from the positivity condition that

$$T_{-4} \sim \text{identity operator}: \quad T_{-4} = \frac{c}{2}, \qquad T_{-3} = 0 \tag{19}$$

$$T(z)T(z') = \frac{c}{2(z - z')^4} + \frac{2T(z')}{(z - z')^2} + \frac{1}{z - z'} T'(z') + \dots \tag{20}$$

[Arrow to the ellipsis in (20):] the further terms are of no interest

From this expansion we obtain

$$\boxed{[L_n, L_m] = (n - m)L_{n+m} + \frac{c}{12}(n^3 - n)\delta_{n+m,0}} \tag{21}$$

These are the relations of the Virasoro algebra (central extension of the algebra of vector fields on the circle)

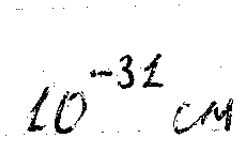
(In ordinary quantum mechanics, the Lie algebra of the group of translations on the line: $\frac{d}{dx}$ — small translation — momentum)

In general the energy–momentum tensor comes from coordinate transformations

In this sense Sophus Lie invented quantum mechanics. He was the first to replace coordinate transformations with infinitesimal transformations.

(The group for the Virasoro algebra is the group of diffeomorphisms of the circle.)

Lengths smaller than 10^{-31} cm have no meaning. This statement is already independent [crossed out: от / не?] of any theory — from the relations between universal constants.



Uncertain reading. The manuscript exponent reads as -31.

Original source details: GELFAND-P001432

Uncertain reading. The word crossed out after "independent" tentatively reads «от» ("of"), possibly «не» ("not").

Original source details: GELFAND-P001432

GELFAND-P001433

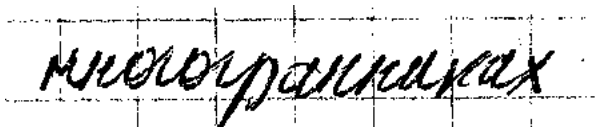
[source number: 5]

p-adic theory has computational advantages. Some determinants are best computed adelically

But there is another point. Physics can say nothing about distant points, or about nearby ones.

[Two vertical lines in the left margin] Everything is known about two-dimensional polyhedra, but as little is known about three- and four-dimensional ones as 2000 years ago.

This is new combinatorics.



Uncertain reading. The manuscript reads "polyhedra." The two rules at the left also cover the line "This is new combinatorics."

Original source details: GELFAND-P001433

How are we to think of space if small distances are ruled out? We lack imagination. (Thickness, discreteness, curvatures, ...)

Conformal invariance. The theory is unchanged under the substitution

$$\begin{cases} z \mapsto z + \varepsilon(z) \\ \bar{z} \mapsto \bar{z} + \bar{\varepsilon}(\bar{z}) \end{cases} \tag{22}$$

Taking $\varepsilon(z) \sim z^{n+1}$, we obtain the Virasoro algebra

Let us see how the average changes

$$\langle A_1(x_1) \dots A_N(x_N) \rangle = Z^{-1} \int A_1(x_1) \dots A_N(x_N) e^{-S[\varphi]} \, \mathcal{D}\varphi \tag{23}$$

under the substitutions

$$x^\mu \to x^\mu + \varepsilon^\mu(x) \tag{24}$$

Variation

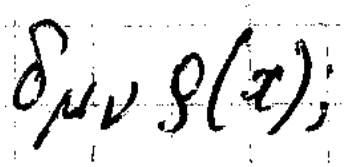
$$\begin{aligned} \delta \langle A_1(x_1) \dots A_N(x_N) \rangle = &\sum_{k=1}^{N} \langle A_1 \dots \delta_\varepsilon A_k \dots A_N \rangle \\ &+ \int \partial_\mu \varepsilon_\nu \langle T^{\mu\nu}(x) A_1 \dots A_N \rangle \, d^2x = 0 \end{aligned} \tag{25}$$

transfer (by parts) ∂_μ onto $T^{\mu\nu}$ and use $\partial_\mu T^{\mu\nu} = 0$;
integrating along small contours around $x_1, \dots, x_N$, we obtain

$$\delta_\varepsilon A_k(y) = \oint_{c_y} \varepsilon_\nu(x) T^{\lambda\nu}(x) \varepsilon_{\lambda\mu}\, dx^\mu A_k(y),$$

but

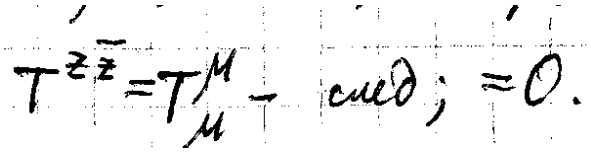
$$\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu = \delta_{\mu\nu} g(x);$$



Uncertain reading. The factor after $\delta_{\mu\nu}$ tentatively reads $g(x)$, possibly $\rho(x)$.

Original source details: GELFAND-P001433

$T^{z\bar{z}} = T^\mu_\mu$ — the trace; $= 0$.



Uncertain reading. The manuscript has an equals sign between $T^{z\bar{z}}$ and T^μ_μ.

Original source details: GELFAND-P001433

[At the lower right]

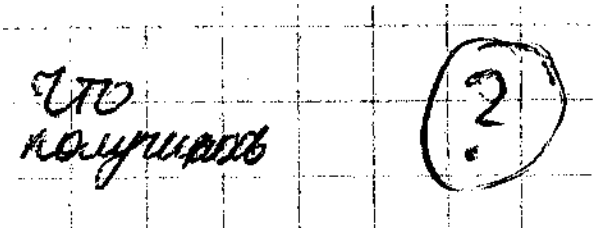


Diagram and typed description.
What did we get ?

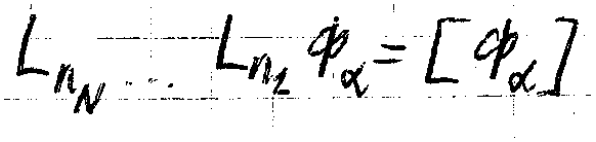
Original source details: GELFAND-P001433

GELFAND-P001434

Fields are classified by representations of the Virasoro algebra.

$$\begin{cases} L_n \phi_\alpha = 0, \quad n > 0; & \bar{L}_n \phi_\alpha = 0; \\ L_0 \phi_\alpha = \Delta_\alpha \phi_\alpha; & \bar{L}_0 \phi_\alpha = \bar{\Delta}_\alpha \phi_\alpha; \end{cases} \tag{26}$$

$$L_{n_N} \dots L_{n_1} \phi_\alpha = [\phi_\alpha]$$



Uncertain reading. The descendant line has $[\phi_\alpha]$ to the right of the equals sign, with a lowercase ϕ.

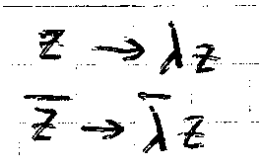
Original source details: GELFAND-P001434

$$n_i < 0; \quad \Delta_\alpha, \bar{\Delta}_\alpha \text{ — two numbers } \geq 0$$

$$\mathcal{A} = \bigoplus_\alpha [\Phi_\alpha]$$

$$\mathrm{I} \in \mathcal{A}; \qquad \mathrm{I} \cdot A_i(x) = A_i(x), \qquad \text{for it } \Delta, \bar{\Delta} = 0$$

$$\begin{matrix} z \to \lambda z \\ \bar{z} \to \bar{\lambda} z \end{matrix} \qquad \phi_\alpha \to \lambda^{\Delta_\alpha} \bar{\lambda}^{\bar{\Delta}_\alpha} \phi_\alpha,$$



Uncertain reading. The right-hand side of the second line is $\bar{\lambda} z$, without a bar over z.

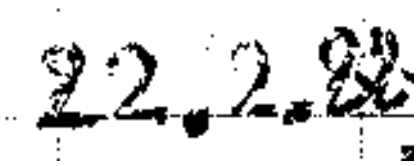
Original source details: GELFAND-P001434

$$\Delta_\alpha - \bar{\Delta}_\alpha = S_\alpha \text{ — the spin of } \phi_\alpha.$$

GELFAND-P001435

[source number: 10]

22.2.8? Angles between subspaces.



Uncertain reading. The final digit of the date is overwritten. Possible readings are 22.2.88 and 22.2.82. The MCCME catalogue dates Gelfand's talk to 22 February 1988.

Original source details: GELFAND-P001435

A series of problems on angles between subspaces

(I. M. Gelfand; from the notes of A. K. Zvonkin)

Some are accessible to an eighth-grader (problems in ($\mathbb{R}^2$ and $\mathbb{R}^3$), others are for a doctor of science (problems in Hilbert space).

What is the «angle» between two lines in the plane? It is an orthogonal invariant of this pair of lines, i.e., a numerical characteristic that is unchanged by every orthogonal transformation of the plane.

Moreover, the angle is a complete system of invariants, i.e., an orthogonal transformation takes one pair of lines to another if and only if the two pairs have the same angle.

Hereafter "angle" means a complete system of invariants of some object under the action of some group. Sometimes this is a number, sometimes a finite or countable set of numbers, and sometimes even a continuum of numbers.

(In cases where we are speaking, say, of four subspaces $\Rightarrow$ of a linear space, we will avoid the expression «angle between...», since it is unclear, between what?, and will say «angle for...».)

In every problem two things are of interest, what the angle is, and also what values it can take.

I. Problems for school pupils.

1. What is the angle between a line and a plane in three-dimensional space.

(Two versions, with the line oriented and unoriented).

2. What is the angle between two planes in three-dimensional space?

II. Problems for university students.

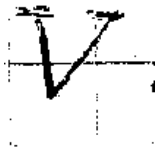
3. Given a 4-dimensional linear space with an inner product. (distinct from $\mathbb{R}^4$, since no axes are singled out). What is the angle between two of its two-dimensional subspaces?

4. Given $\{L, (\cdot,\cdot)\}$, $\dim L = n$, $v \in L$, $V \subset L$, $\dim V = k$.

Determine the angle between v and V.

GELFAND-P001436

5. Given $\{L, (\cdot,\cdot)\}$, $\dim L = n$, $V, W \subset L$, $\dim V = k$, $\dim W = m$. Determine the angle between V and W.



Original source details: GELFAND-P001436

Uncertain reading. The V at the end of problem 5 has two short strokes at its upper tips. A separate overbar cannot be distinguished.

6. Let L — just a linear space (without an inner product). Determine the angle for

(a) a subspace of L

(b) a pair of subspaces

(c) a triple of subspaces.

(Here the group is understood to be $GL(n)$)

7. Given the Minkowski plane, i.e., L, $\dim L = 2$ with the indefinite scalar product $-x_1^2 + x_2^2$. Determine the angle between two lines (i.e., an invariant under the group of transformations preserving this scalar product).

8. Further variations:

— for a Minkowski space of higher dimension and two of its subspaces

— for other spaces with an indefinite metric (i.e., when the scalar product has more than one minus sign)

— for spaces with a degenerate form ($\mathbb{R}^3$ with scalar product $x_1^2 + x_2^2$)

For all the listed problems consider the cases of the groups $O(n)$ and $SO(n)$.

III. Problems for graduate students.

9. Given a linear space L and four of its subspaces. Find the angle of this quadruple (this is the subject of the work of Gelfand and Ponomarev)

10. (A. A. Kirillov Jr.) Given $\{L, (\cdot,\cdot)\}$; then the tensor powers of L also carry a scalar product. Let $a, b \in L \otimes L$. Find the angle between the subspaces $a \otimes L$ and $L \otimes b$, embedded in $L \otimes L \otimes L$.

(This is needed for Hecke algebras and the Yang–Baxter equation)

GELFAND-P001437

[source number: 11]

IV. Problems for Candidates and Doctors of Science.

11. Determine the angle for two subspaces of a Hilbert space.

Give examples in which the value of the angle is a countable set of numbers and examples in which it is a continuum of numbers.

12. The same problem for a Hilbert space with an indefinite metric.

Comments.

1. The second part of these problems is particularly difficult. For example, given a subspace and an angle, construct another subspace that makes the given angle with the given subspace.

2. A linear operator can be specified by four subspaces.

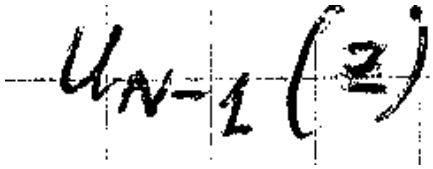
GELFAND-P001438

21/III-88 Gelfand Seminar

Radul, A.

Quantum Gelfand–Dickey brackets (following S. Lukyanov) Consider on S^1 the operator

$$L = \partial_z^N + \sum_{i=0}^{N-1} u_{N-1}(z)\partial_z^i. \tag{1}$$



Original source details: GELFAND-P001438

Uncertain reading. The coefficient subscript in (1) reads $N-1$.

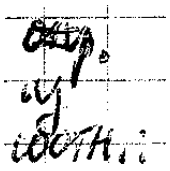
and the space of functionals of the form

$$f[u] = \oint P\big[z, u_i^{(k)}(z)\big]dz \tag{2}$$

and their finite products. P is a polynomial in the $u_i^{(k)}$.

Two Gelfand–Dickey brackets are defined on such functionals.

$$\frac{\delta f}{\delta u_i} \quad \text{defined by the relation:} \quad \frac{d}{d\varepsilon} f[u + \varepsilon\delta u]\Big|_{\varepsilon=0} = \oint dz\, \frac{\delta f}{\delta u_i}\delta u_i. \tag{3}$$



Original source details: GELFAND-P001438

Uncertain reading. The abbreviation «опр.» (“defined”) is written over earlier strokes.

I. M. Gelfand

A digression on functional analysis All the intuition of functional analysis proceeded from the idea that functionals are also functions, but on an infinite-dimensional space.

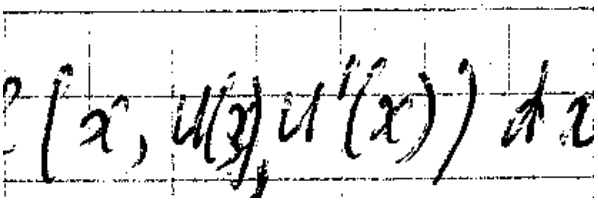
I. M. G. changed his point of view, and two consequences follow.

1) There are different types of functionals, and they must be classified.

2) Functionals are not functions ⇒ this affects geometry.

1) In the calculus of variations there are functionals

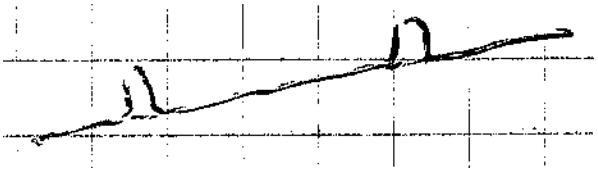
$$\int_0^L f\big(x, u(x), u'(x)\big)\, dx \qquad (*)$$



Original source details: GELFAND-P001438

Uncertain reading. The argument of the middle u tentatively reads x.

This is a functional of u, but it has the following property:



Original source details: GELFAND-P001438

Diagram and typed description.

A slanting line with two raised bumps. There are breaks in the strokes at the tops of the bumps.

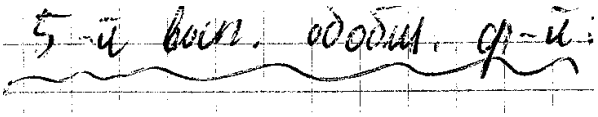
if there are increments [inserted above the line with an insertion mark: of u] at two places, the increment of the functional (*) will be the sum of [crossed out: illegible] the increments

Original source details: GELFAND-P001438

Uncertain reading. The writing after “sum” is crossed out. The strokes may conceal a 2 or the beginning of «двух» (“two”).

(locality!)

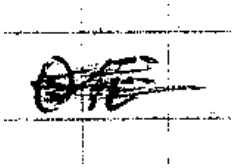
5th possible generalization of functions:



Original source details: GELFAND-P001438

Uncertain reading. The first sign tentatively reads 5. The manuscript abbreviates «обобщ.».

it follows from this (and from continuity) that the functional then [crossed out: «от» / «он»?] has this form



Original source details: GELFAND-P001438

Uncertain reading. The crossed-out writing after “then” tentatively reads «от» or «он».

[Crossed-out mark: illegible.] This is type 0.

Uncertain reading. The crossed-out mark before “This” resembles 1 or the Cyrillic letter И.

Original source details: GELFAND-P001438

GELFAND-P001439

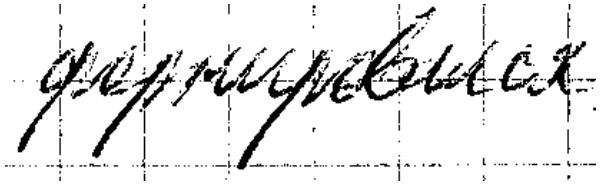
[source number: 27]

Type 1: if

$$\frac{\delta}{\delta u}F(u) \quad \text{is local.}$$

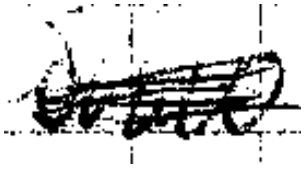
Such are the functionals that occur in Atiyah–Singer theory

The trouble with functional analysis is that it took shape when everyone [crossed out: «это» (“this”)?] was trying to rationalize.



Uncertain reading. The verb reads «формировался» (“took shape”).

Original source details: GELFAND-P001439



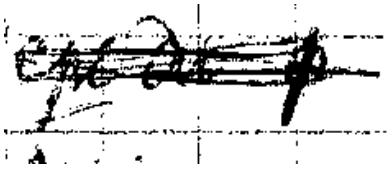
Uncertain reading. The crossed-out word after “everyone” may read «это» (“this”).

Original source details: GELFAND-P001439

Uncertain reading. The verb before “to rationalize” reads «пытались» (“were trying”).

Original source details: GELFAND-P001439

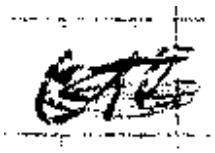
And [crossed out: spectrum?] in the theory of functions even continuity belongs to the domain of death.



Uncertain reading. The crossed-out passage after “And” tentatively reads «спектр» (“spectrum”).

Original source details: GELFAND-P001439

Functionals: a) length of a curve
b) area of a surface (local)
Transcendental functionals: (as in analysis — integrals that cannot be evaluated — elliptic functions)
passing from the coefficients to the eigenvalues!!
λ_1 — is such a functional
Functional of transcendence degree 1 — if the variational derivative is a local functional.
and so on.
Functional [crossed out: «ст.» / «от»?] of type 2: if $\frac{\delta F}{\delta u}$ is of type 1.



Uncertain reading. The crossed-out abbreviation before “type 2” resembles «ст.», possibly «от».

Original source details: GELFAND-P001439

Gelfand, Gabrielov and Loshik thought it through in their work, but it got stuck because it is difficult.

Uncertain reading. The handwritten surname resembles «Лошик» (Loshik). The paper on Atiyah–Patodi–Singer functionals lists M. V. Losik as the third author.

Original source details: GELFAND-P001439

Functionals of all types will appear in the theory of [inserted above the line: characteristic] classes.

Functionals must be considered on infinite-dimensional sets. But one must do more than simply replace an n-dimensional space with an infinite-dimensional one!

[Crossed out: «Род» / «Тут»?] The main thing is that locality plays an exceptional role!

Uncertain reading. The crossed-out writing before “The main thing” tentatively reads «Род» or «Тут».

Original source details: GELFAND-P001439

Then one must proceed differently.

Functions form an algebra.

Local functionals do not form an algebra!!

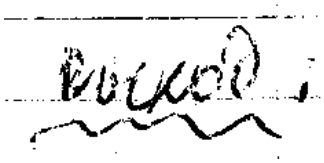
Problem: a calculus of variations is needed for sums and products of such functionals.

$$\int f\left(x, y; u_i(x), u_i'(x), v_i(y), v_i'(y)\right) dx\, dy$$

So one can no longer construct differential geometry.

Adding products formally is a bad idea!

A way out; structures introduced by G. and Dorfman and then change the connection, de Rham cohomology, Yang–Baxter brackets...



Uncertain reading. The first word tentatively reads «Выход» (“a way out”).

Original source details: GELFAND-P001440

Radul (continued)

Paper: IPM preprint. Translation in vol. 1 of the collected works.

Digression: symbols of pseudodifferential operators

$$A = \sum a_i(z)\partial_z^i, \qquad B = \sum b_j(z)\partial_z^j. \tag{4}$$

$$\partial_z \longrightarrow \xi$$

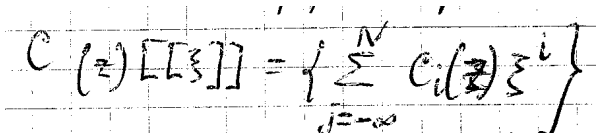
$$AB = A(z,\xi) \circ B(z,\xi)|_{\xi=\partial_z}, \tag{5}$$

where

$$A(z,\xi) \circ B(z,\xi) := \sum_{n\geq 0} \partial_\xi^n A(z,\xi)\, \partial_z^n B(z,\xi)/n!.$$

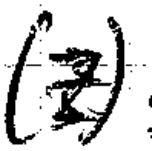
Let us switch to formal Laurent series:

$$C(z)[[\xi]] = \left\{ \sum_{j=-\infty}^{N} c_i(z)\xi^i \right\}. \tag{6}$$



Uncertain reading. The lower summation limit reads $j = -\infty$, while i is written twice in the summand $c_i(z)\xi^i$.

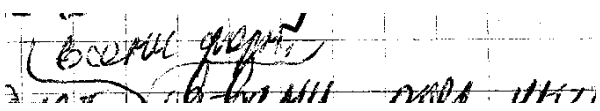
Original source details: GELFAND-P001440



Uncertain reading. The argument of c_i shows z written over earlier strokes.

Original source details: GELFAND-P001440

$\xi^{-1} = \partial_z^{-1}$ has [inserted above the line: all the formal] properties of the integration operator



Uncertain reading. «всеми форм.» (“all the formal”) is written above the line with an insertion mark.

Original source details: GELFAND-P001440

Operations on symbols

$$\left(\sum_{-\infty}^{N} a_i \xi^i\right)_+ = \sum_{0}^{N} a_i \xi^i \tag{7}$$

$$\operatorname{res}\left(\sum_{-\infty}^{N} a_i \xi^i\right) = a_{-1} \tag{8}$$

Gradient of f.

$$\nabla f := \sum_{j=0}^{N-1} \xi^{-j-1} \circ \frac{\delta f}{\delta u_j}. \tag{9}$$

GELFAND-P001441

[source number: 28]

Then

$$\delta f = \oint \operatorname{res} \delta L \circ \nabla f \tag{10}$$

here L is operator (1)

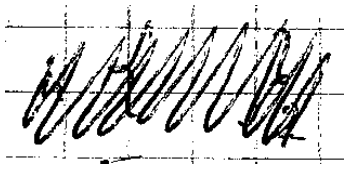
1st bracket (Hamiltonian structure)

$$1 \text{ HS:} \quad \{f, g\}_1 := \oint \operatorname{res}[L, \nabla f]\nabla g \tag{11}$$

2nd bracket:

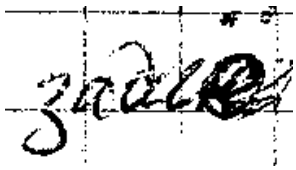
$$2 \text{ HS:} \quad \{f, g\}_2 = \oint \operatorname{res}\big(L(\nabla f L)_+ - (L\nabla f)_+ L\big)\nabla g \tag{12}$$

(13) Claim. $\{\,\}_i$ [crossed out: illegible] gives the space of functionals a Lie algebra structure.



Original source details: GELFAND-P001441

Undeciphered passage. The writing between $\{\,\}_i$ and “gives” is crossed out and illegible.



Original source details: GELFAND-P001441

Uncertain reading. The end of «задаёт» (“gives”) is written over.

(14) Example. $N = 2$, $u_1 = 0$, $\partial_z^2 + u_2$.

Let

$$u_2(n) = \frac{1}{2}\frac{1}{2\pi i}\oint u_2(z) z^{n+1}\, dz \tag{15}$$

Then

$$\{u_2(n), u_2(m)\}_2 = \frac{n^3 - n}{12}\delta_{n+m,0} + (n-m)u_2(n+m) \tag{16}$$

(this gives the commutation relations of the Virasoro algebra!)

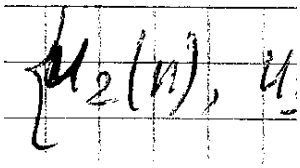
(17) Example. $N = 3$, $u_1 = 0$, $\partial_z^3 + u_2\partial_z + u_3$.

Uncertain reading. The second digit of (17) is overwritten and reads as 7.

Original source details: GELFAND-P001441

$$\{u_2(n), u_2(m)\}_2 = \frac{n^3 - n}{6}\delta_{n+m,0} + (n-m)u_2(n+m) \tag{18}_{\text{а}}$$

$$\{u_2(n), u_3(m)\}_2 = 3(n+1)u_3(m+n) - (m+n+3)u_3(m+n) \tag{18}_{\text{б}}$$



Uncertain reading. In (18)б the u of the first $u_2(n)$ is retraced, and the subscript reads 2.

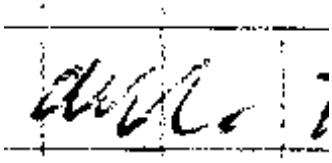
Original source details: GELFAND-P001441

$$\{u_3(n), u_3(m)\}_2 = \ldots + \sum_{k=-\infty}^{\infty} (m+n-k)u_2(m+n-k)u_2(k) \tag{18}_{\text{в}}$$

(square of the energy–momentum tensor)

Henceforth $\{\ \}$ always denotes $\{\ \}_2$.

Together this is the classical limit of some quantum current algebra.



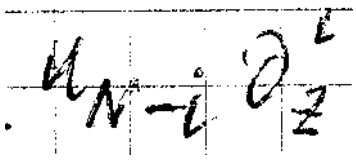
Uncertain reading. The abbreviation before «токов» (“currents”) tentatively reads «алг.» (“algebra”).

Original source details: GELFAND-P001441

Moral. one obtains an algebra with quadratic relations in place of a classical Lie algebra.
(Khovanova, T.)
Dynkin also introduced quadratic relations.
S. Lukyanov was working on the quantization of these brackets.
[On the left below the name: (Seryozha)] (S. Lukyanov is a graduate student of V. Fateev)

Digression The Miura transformation. Let

$$L(z) = \partial_z^N + \sum u_{N-i}\partial_z^i = (\partial_z + v_1)\dots(\partial_z + v_N) \tag{19}$$



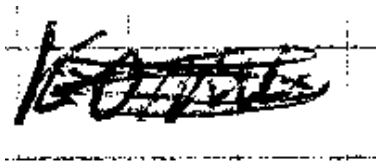
Original source details: GELFAND-P001442

Uncertain reading. In (19) the subscript on u reads $N - i$, with a dot over i.

— · —

remark
I. M. G. If $(\partial_z + v_1)(\partial_z + v_2)u = f$, then u is obtained by quadratures.

Algebraic equation in radicals — Galois theory
For differential equations the corresponding theory is Picard–Vessiot theory.
Kolchin [crossed out before the name: Kol...?] is working on this now.



Original source details: GELFAND-P001442

Uncertain reading. The crossed-out writing before «Колчин» (Kolchin) tentatively begins «Кол...».

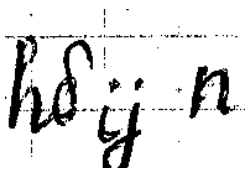
— · —

R.
Factorization (19): Miura
Kupershmidt — Wilson
Drinfeld — Sokolov

(20) Theorem. Let us give v_j the commutation relations

$$\{v_i(n), v_j(m)\} = h\delta_{ij}\, n\delta_{m+n,0}. \tag{21}$$

[On the right: (Lie algebra — [crossed mark] Heisenberg algebra)]



Original source details: GELFAND-P001442

Uncertain reading. The factor before δ_{ij} reads h.

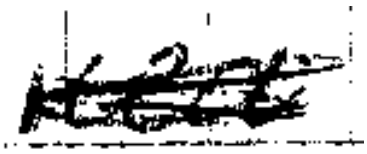
Uncertain reading. Before «Гейзенберга» (“Heisenberg”) is a crossed mark resembling H or #.

Original source details: GELFAND-P001442

Then u_i satisfy the algebraic G.–D. relations (Gelfand–Dickey)

(2nd G.–D. structure, here and below)

It turned out that the Lie algebra is isomorphic to a quadratic algebra [crossed out before “quadratic”: «квад...»?] which is not a Lie algebra.



Uncertain reading. The crossed-out writing before «квадр.» (“quadratic”) tentatively begins «ква...» or «квад...».

Original source details: GELFAND-P001442

Now let

$$L(z) = :(\partial_z + v_1)\dots(\partial_z + v_N): \tag{22}$$

What are the relations?

Introduce

GELFAND-P001443

[source number: 29]

$$M_0 = \mathrm{Hol}\big((\mathbb{C}\backslash 0)_z \times (\mathbb{C}\backslash 0)_w \longrightarrow \mathbb{C}\big)$$

[Below Hol: holomorphic maps.]

$$M = \mathrm{Hol}\big(((\mathbb{C}\backslash 0)_z \times (\mathbb{C}\backslash 0)_w \backslash \Delta) \longrightarrow U(v_1, \dots, v_N)\big)$$

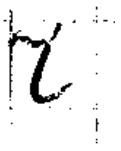
[A line connects U to the explanation: universal enveloping algebra of the Heisenberg algebra.]

An element of M is an operator-valued function.

Consider also the operator

$$\tau = \frac{\nu}{2}\ln(z-w)^2\,\sigma, \qquad \sigma f(z,w) = f(w,z).$$

on functions of two variables



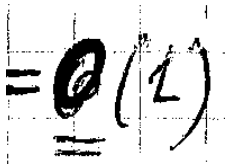
Uncertain reading. The sign with a top bar tentatively reads τ, possibly r. It recurs in (23).

Original source details: GELFAND-P001443

Then, if $\nu(\nu-1)=h$,

$$\exp(-\tau)L(z)L(w)\exp\tau[f]=\underline{\underline{O}}(1)\quad \text{as } z\longrightarrow w \tag{23}$$

$$\forall f\in M_0$$



Uncertain reading. The sign before (1) is retraced and double-underlined. It tentatively reads O, possibly 0.

Original source details: GELFAND-P001443

This is a set of relations on the u_i (on the coefficients)

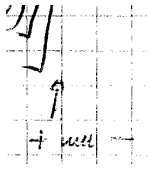
Result:

$$[u_i(z),u_j(w)]=\left[\sum_{k\geq 2}(z-w)^{-k}\sum_{p+q=i+j-k}c_{ij}^{pq}(k)[u_p(z),u_q(w)]\right] \tag{24}$$

[Below the outer right bracket is an arrow ↑ from the label "+ or −".]

Uncertain reading. The lower limit of the first sum tentatively reads $k\geq 2$. The sign between k and 2 is indistinct and may read $=$.

Original source details: GELFAND-P001443



Uncertain reading. The arrow from the label "+ or −" points to the outer right bracket of (24).

Original source details: GELFAND-P001443

$c_{ij}^{pq}(k)$ are expressed in terms of hypergeometric functions, but it is complicated

The form [below the word: «соотн.» ("relation")?] of (23) resembles a relation in the quantum inverse problem method.

Uncertain reading. The abbreviation below "The form" tentatively reads «соотн.» ("relation").

Original source details: GELFAND-P001443

$$L(z)L(w)=R^{-1}L(w)L(z)R. \tag{25}$$

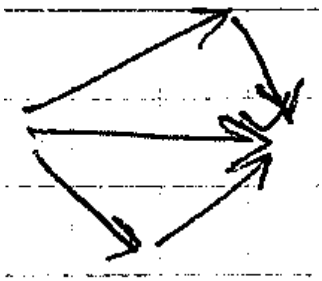
M. Kapranov. Quiver representations and the derived category.

(On Parthasarathy's work on quiver representations.)

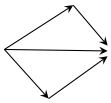
Original source details: GELFAND-P001444

Uncertain reading. The name tentatively reads «Партасарати» (Parthasarathy). See R. Parthasarathy's paper on t-structures in the derived category of quiver representations.

A quiver — a directed graph without cycles



Original source details: GELFAND-P001444

Diagram and typed description. 

Λ — graph

Ω — orientation

$M(\Lambda, \Omega)$ — [crossed out: «пр...»?] category of representations of the quiver

Original source details: GELFAND-P001444

Uncertain reading. The crossed-out writing before "category" tentatively begins «пр...», possibly «кат...».

$\mathcal{D}(\Lambda, \Omega)$ — derived category of this category.

(1) Lemma.

Then $\mathcal{D}(\Lambda, \Omega)$ does not depend on Ω

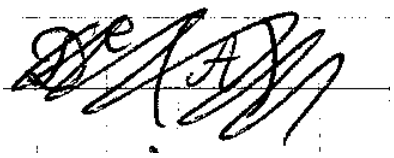
($\mathcal{D}(\Lambda, \Omega_1)$ and $\mathcal{D}(\Lambda, \Omega_2)$ are equivalent)

for different Ω_1 and Ω_2

Proof: application of symmetry.

Problem: find such [crossed out: $\mathcal{D}^{\ell}(\mathcal{A})$?] subcategories $\mathcal{A} \subset \mathcal{D}(\Lambda, \Omega)$, that

$$\mathcal{D}^{\ell}(\mathcal{A}) = \mathcal{D}(\Lambda, \Omega)$$

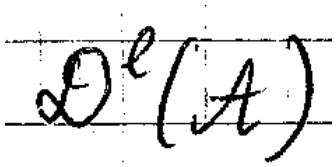


Original source details: GELFAND-P001444

Uncertain reading. The crossed-out formula tentatively reads $\mathcal{D}^{\ell}(\mathcal{A})$. The superscript also resembles e.

Original source details: GELFAND-P001444

Uncertain reading. The sign $\subset$ is retraced. An angular stroke is visible inside it.



Original source details: GELFAND-P001444

Uncertain reading. The sign above $\mathcal{D}$ in the manuscript resembles ℓ or e. Parthasarathy's paper uses $\mathcal{D}^b$ for the bounded derived category.

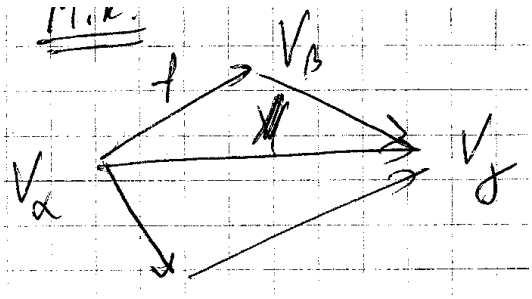
All of them have been found and are characterized by Dynkin graphs.

Reflections and gluings are used (and the philosophy of perverse sheaves)

Remark by I. M. G. The classification of vector bundles on CP^n can only be done in the language of derived categories.

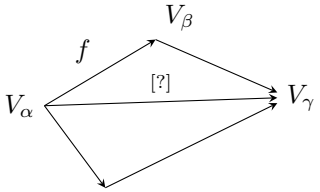
Grothendieck: through hard work he obtained the concepts leading to the derived category (like Michelangelo).

M. K.



Original source details: GELFAND-P001444

Diagram and typed description.



Above the middle arrow is a crossed-out mark, possibly g or φ. The lower vertex has no label.

Reflection a) $s_\alpha\Omega$ — reverse the arrows starting at α.

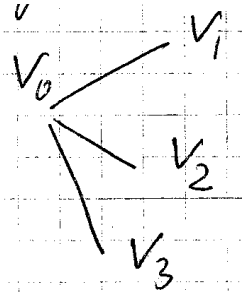
b)

$$(s_\alpha V)_{\beta\neq\alpha} = V_\beta$$

$$(s_\alpha V)_\alpha = \bigoplus_{\alpha\to\beta} V_\beta/(\text{the image of } V_\alpha)$$

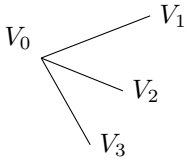
with the natural maps

There may be vectors in V_α that are mapped to 0 everywhere



Original source details: GELFAND-P001444

Diagram and typed description.

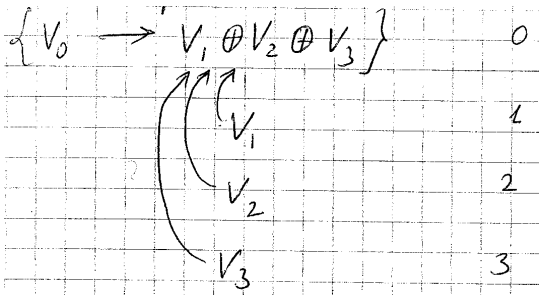


$$V_0 \longrightarrow V_1 \oplus V_2 \oplus V_3$$

GELFAND-P001445

[source number: 30]

Let us construct a complex from these data



Original source details: GELFAND-P001445

Diagram and typed description.

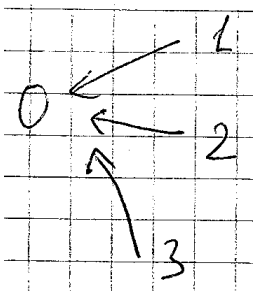
$\{V_0 \longrightarrow V_1 \oplus V_2 \oplus V_3\}$ 0

V_1 1

V_2 2

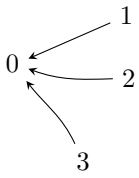
V_3 3

This is an object of the derived category (of quiver representations)



Original source details: GELFAND-P001445

Diagram and typed description.



To factorization in the ordinary category one must associate the map itself (the complex)

Parthasarathy poses and solves the question of when the derived category coincides with this one (for 3, but not for 4).

Original source details: GELFAND-P001445

Uncertain reading. The surname reads «Партасарати» (Parthasarathy), as on the preceding frame.

GELFAND-P001446

23. III. 88 Gelfand Seminar.

Zelevinsky, A. Matroids and Hall's theorem

I. M. G.

Matroid. [Above the next word, crossed out: illegible.] Example: linearly independent systems of vectors.

Original source details: GELFAND-P001446

Undeciphered passage. The writing above "Example" is crossed out and illegible.

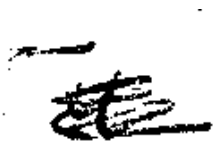
M_1, M_2 — sets, $\varphi \subset M_1 \times M_2$

(relation)

Suppose M_1 has the structure of a matroid.

How do we introduce a matroid structure on M_2?

$A \subset M_2$, $\pi_1(\pi_2^{-1}(A) \cap \varphi)$ — a set.



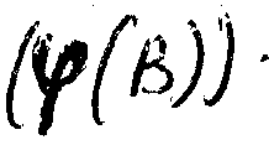
Undeciphered passage. The writing under the dash before "a set" is crossed out and illegible.

Original source details: GELFAND-P001446

The rank of the set A is determined

1°. Rado's theorem

$$r_{M_2}(A) = \min_{B \subset A}\big(r_{M_1}(\varphi(B)) + |A - B|\big), \qquad A \subset M_2.$$



Uncertain reading. The φ in $\varphi(B)$ is retraced in the rank formula.

Original source details: GELFAND-P001446

2°. Hall's theorem ≡ [Inserted above the surname: P.?]

Uncertain reading. The insertion before "Hall" tentatively reads «Ф.» (P.).

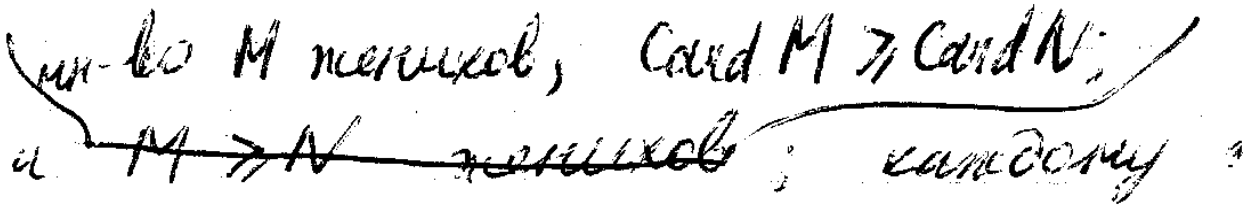
Original source details: GELFAND-P001446

[Crossed out at the left: illegible.] Problem of choosing brides.

Undeciphered passage. The writing to the left of "Problem" is crossed out and illegible.

Original source details: GELFAND-P001446

If there is [inserted above N: a set] N of brides and [crossed out: $M \geq N$ grooms] [inserted above: a set M of grooms, Card $M \geq$ Card N], each groom $m \in M$ likes the brides $A(m) \subset N$;



Original source details: GELFAND-P001446

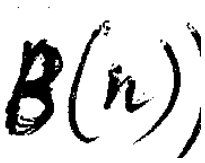
Uncertain reading. The insertion above the crossed-out "$M \geq N$ grooms" uses Card $M \geq$ Card N.

and [crossed out: $\forall A \subset N$] for every subset $K \subset N$,

$$\text{Card}\left(\bigcup_{n\in K} B(n)\right) \geq \text{Card}\, K.$$

Original source details: GELFAND-P001446

Undeciphered passage. The mark above Card is crossed out and illegible.



Original source details: GELFAND-P001446

Uncertain reading. The letter B is retraced in the last formula and reads $B(n)$.

GELFAND-P001447

[source number: 37]

(Here $B(n) \subset M$; [Below $B(n)$, joined by an equals sign:] the set of grooms who like n, i.e.

$$B(n) = \{m :\ n \in A(m)\}$$

[After $=$, crossed out: illegible.])

Original source details: GELFAND-P001447

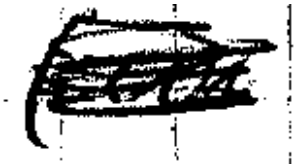
Undeciphered passage. The mark between $=$ and the opening brace is crossed out and illegible.

Then $\exists$ a one-to-one choice [crossed out: illegible]: a map

$$f : N \longrightarrow M,$$

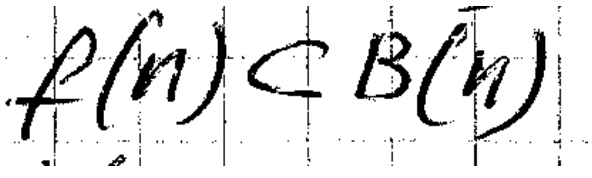
for which

$$f(n) \subset B(n) \qquad \forall n.$$



Original source details: GELFAND-P001447

Undeciphered passage. The writing after "choice" is crossed out and illegible.



Original source details: GELFAND-P001447

Uncertain reading. The manuscript uses $\subset$ between $f(n)$ and $B(n)$.

[Crossed-out line: Hall's theorem φ is a relation.]

[Crossed out: Theorem.] There exists a proof containing an algorithm.

Reduction to Rado's theorem: in N one must declare all sets independent.

Original source details: GELFAND-P001447

Uncertain reading. The letter after "in" is retraced heavily and reads N.

$2^\circ \Rightarrow 3^\circ$. Birkhoff's theorem.

It follows from Hall's theorem that if there is a subgroup $H \subset G$ [Below G: finite group], then $\exists$ a set of common representatives of left and right cosets.

I. M. G. There are several famous problems whose pointlessness is obvious to everyone.

Example. 1) Parallel axiom

(why this particular one?)

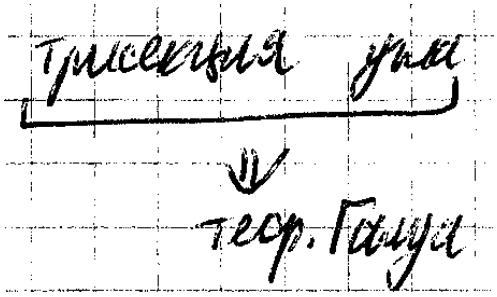
[Below the line:] they worked on it?

Original source details: GELFAND-P001447

Uncertain reading. Below the question the words tentatively read «занимались нею» ("they worked on it").

[To the right of the first two lines, beyond a vertical rule:] $\Rightarrow$ non-Euclidean geometry

2) Trisection of the angle and squaring the circle.



Original source details: GELFAND-P001447

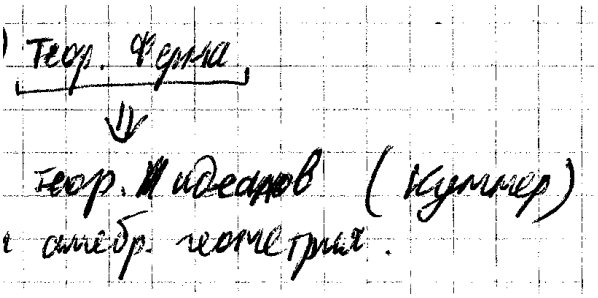
Diagram and typed description.

Trisection of the angle

$\Downarrow$

Galois theory

3) Fermat's theorem



Original source details: GELFAND-P001447

Diagram and typed description.

Fermat's theorem

$\Downarrow$

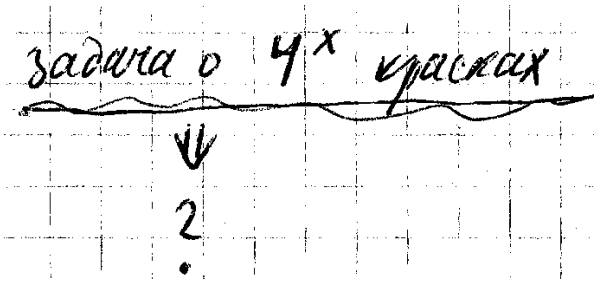
theory of [crossed out: illegible] ideals (Kummer) and algebraic geometry.

Original source details: GELFAND-P001447

Undeciphered passage. An illegible crossed-out mark stands before “ideals.”

[On the right, beyond a vertical rule:] work of Faltings.

4) The four-colour problem



Original source details: GELFAND-P001447

Diagram and typed description.
The four-colour problem
⇓
?

These works are worth going through.

5) Riemann hypothesis

GELFAND-P001448

A. Rudakov. Vector bundles, braid groups and Diophantine equations.

1°. On $\mathbb{P}^2$ $(= \mathbb{CP}^2)$

consider vector bundles (or coherent sheaves)

Original source details: GELFAND-P001448

Uncertain reading. The ending of the word for “sheaves” is heavily retraced.

$$E_1,\ E_2,\ E_3 \tag{1}$$

Original source details: GELFAND-P001448

Undeciphered passage. Illegible marks are crossed out after E_1.

$$\begin{aligned} &\operatorname{Ext}^k(E_j, E_i) = 0, \quad j > i, \quad k = 0, 1, 2, \\ &\operatorname{Ext}^k(E_i, E_i) = \begin{cases} 0, & k > 0, \\ \mathbb{C}, & k = 0. \end{cases} \end{aligned} \tag{2}$$

Original source details: GELFAND-P001448

Undeciphered passage. An illegible mark is crossed out before $k = 0, 1, 2$.

Such bundles exist, and if $\dim E_j = x_j$, then

$$x_1^2 + x_2^2 + x_3^2 = 3x_1x_2x_3 \qquad \text{(Markov equation)} \tag{3}$$

(and all [inserted above: integer] solutions are realized this way!)

The braid group acts on the collections [inserted above: (1) (there are countably many of them)], and all the collections (1) form 1 orbit!

2°.

$$Q = \mathbb{P}^1 \times \mathbb{P}^1.$$

Undeciphered passage. An illegible mark is crossed out after $Q =$.

Original source details: GELFAND-P001448

Then take 4 bundles E_1, E_2, E_3, E_4 with the same conditions (2). The braid group acts here too.

In place of (3) we get a system of Diophantine equations (but the correspondence goes one way. It has not been proved that all solutions are obtained). It is something like a root system, with a braid group in place of the Weyl group.

3°. On $\mathbb{P}^n$ one can consider bundles

$$E_1, E_2, \dots, E_{n+1} \qquad \text{with conditions (2)}$$

and again an action of the braid group arises.

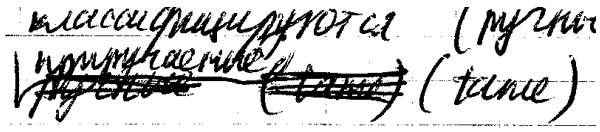
M. Kapranov: on the action of the braid group.

I. M. G.

Solvable and unsolvable classification problems.

No parameters $\Rightarrow$ can be classified (tame)

1 parameter $\Rightarrow$ [crossed out: «ручные» (“tame”) and illegible writing in parentheses$^?$; inserted above the line: semi-tame] (tame)

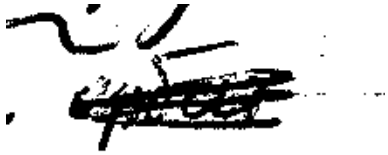


Uncertain reading. «ручные» (“tame”) and illegible writing in parentheses

Original source details: GELFAND-P001448

≥ 2 parameters $\Rightarrow$ unsolvable (e.g., pairs of matrices)

and classification up to



Undeciphered passage. Illegible writing is crossed out before the map.

Original source details: GELFAND-P001448

$$(A, B) \mapsto (C^{-1}AC, C^{-1}BC).$$

GELFAND-P001449

[source number: 38]

In unsolvable problems there is always a solvable part (an open orbit!)

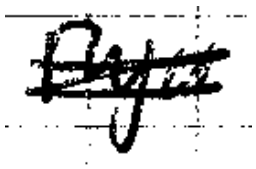
Original source details: GELFAND-P001449

Uncertain reading. The writing above the initials is crossed out. It resembles «Еф» or the start of Ext.

M. K.

$$\mathrm{Ext}^1(E,E)=0 \;\Rightarrow\; \text{rigidity}$$

(and such bundles have no parameters)



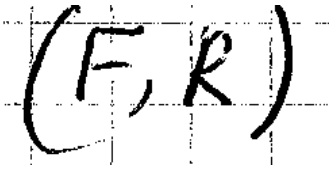
Original source details: GELFAND-P001449

Uncertain reading. The word crossed out above the beginning of the sequence tentatively reads «Пару» (“a pair”).

$$0 \longrightarrow R \longrightarrow E \otimes \mathrm{Hom}(E,F) \longrightarrow F$$

Call a pair (E,F) exceptional if $\mathrm{Ext}^j(F,E)=0$.

Then the pair (F,R) is also exceptional.



Original source details: GELFAND-P001449

Uncertain reading. The manuscript gives the order (F,R).

We obtain an action of $\mathbb{Z}$ on such pairs. The relations giving braids hold.

I. M. G. Work of Bernstein–Gelfand–Ponomarev:

in representations of graphs the Coxeter group acts.

What is needed from one's general education to work on, e.g., spectral theory.

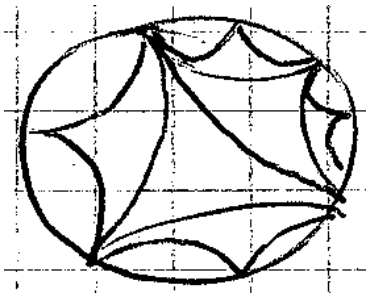
1) Representations of graphs (a quadruple of subspaces, etc.)

2) Derived category.

Beilinson, A. Perverse sheaves.

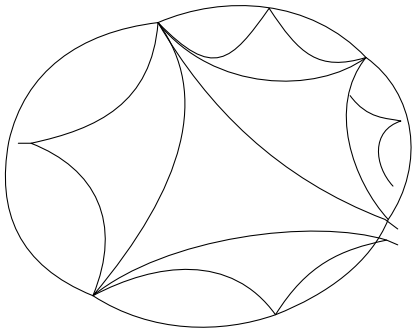
I. M. G.

(on MacPherson)



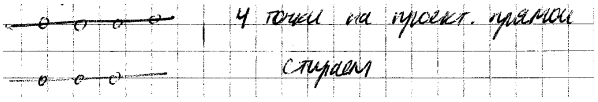
Original source details: GELFAND-P001449

Diagram and typed description.



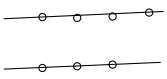
The strokes overlap at the lower-right junction. Their exact connections are unclear.

How are these regions to be parametrized?



Original source details: GELFAND-P001449

Diagram and typed description.



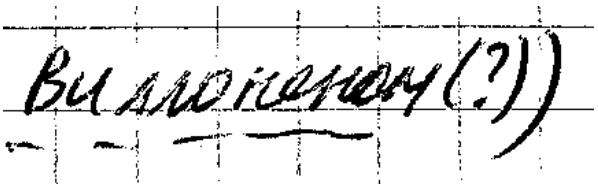
4 points on a projective line

erase

Siegel domains $\Rightarrow$ configurations of lines in $\mathbb{RP}^2$.

B.

On the work of MacPherson (on his work with Gelfand and Vilonen(?))



Original source details: GELFAND-P001449

Uncertain reading. The surname tentatively reads «Виллоненом» (Vilonen). The parenthesized question mark is in the manuscript.

Perverse sheaves?

A perverse sheaf corresponds to problems in linear algebra that describe it completely, but it is unclear how to single out the perverse sheaves.

GELFAND-P001450

X is a complex manifold (smooth)

Constructible sheaf.

A stratification is given $\{X_i\}$ [above X_i, with a leader: open], a sheaf $\mathcal{F}$ on X is called constructible [inserted below: with respect to it], if $\mathcal{F}|_{X_i}$ is a locally constant sheaf of finite-dimensional vector spaces.

(locally with respect to the usual topology)

Constructible complexes of sheaves

$$\ldots \longrightarrow \mathcal{F}^{-1} \longrightarrow \mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \mathcal{F}^2 \longrightarrow \ldots \tag{1}$$

a complex of sheaves is called constructible if all the sheaves $H^i(\mathcal{F}^\bullet)$ are constructible.

(there are no conditions on $\mathcal{F}^i$!)

Singular support of a sheaf or complex.

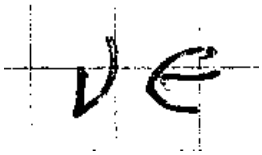
$\mathcal{F}^\bullet$ is a constructible complex.

$$SS(\mathcal{F}) \subset T^*X \tag{2}$$

[Below $SS(\mathcal{F})$: subset]

is defined as follows.

Let $\nu \in T^*_x X,\ \nu \neq 0$.



Uncertain reading. The covector letter reads as ν or Latin v.

Original source details: GELFAND-P001450

(3) Def.

$$\nu \notin SS(\mathcal{F}) \iff$$

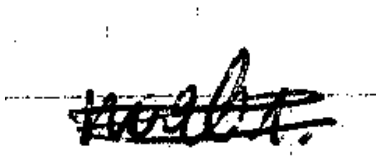
[Inserted above: if] one chooses a small neighbourhood $U \ni x$, a holomorphic function f on U, with $df_x = \nu$, $f(x) = 0$, then $\forall$ points $\varepsilon \in \mathbb{C}$ near 0

$$H^*\big(U, \underset{\cap U}{f^{-1}(\varepsilon)}; \mathcal{F}\big) = 0. \tag{4}$$

(relative cohomology groups mod the fibre)

(more precisely, one must pass to the inductive limit)

([crossed out: appearance] $\neq 0 \iff$ appearance of vanishing cycles)



Uncertain reading. The crossed-out writing tentatively reads «появл.» (“appearance”).

Original source details: GELFAND-P001450

(4) is independent of the choice of f

Zero section?

Add the zero section if $\mathcal{F}^\bullet$ is a nonzero complex.

$\mathcal{F}^\bullet$ is zero $\Rightarrow SS(\mathcal{F}^\bullet) \overset{\text{def}}{=} \varnothing$.

GELFAND-P001451

[source number: 39]

(5) Theorem. $SS(\mathcal{F})$ is a conic Lagrangian subset

(an analytic subset whose nonsingular points are Lagrangian in the usual sense [mark crossed out]; an equivalent condition is involutivity of the ideal)

Uncertain reading. The crossed-out mark before the semicolon resembles a closing parenthesis.

Original source details: GELFAND-P001451

Structure: a union of irreducible components (finitely many)

(6) Lemma. If $Y \subset T^*X$ is an irreducible Lagrangian conic subset, then

$$Y = N^*Z \qquad \text{for some irreducible } Z \subset X. \tag{7}$$

(conormal bundle)

(then $Z = \pi(Y)$, $\pi : T^*X \longrightarrow X$)

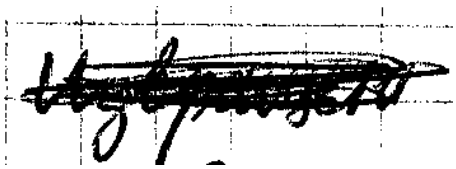
Here N^*Z is understood [crossed out: illegible] as the Zariski closure of N^*(smooth part of Z).

Undeciphered passage. The writing crossed out between “understood” and “as” is illegible.

Original source details: GELFAND-P001451

Undeciphered passage. An illegible mark is crossed out between the opening parenthesis and “smooth part.”

Original source details: GELFAND-P001451



Uncertain reading. The crossed-out writing above the heading tentatively begins «Извращ...» (“Perverse...”).

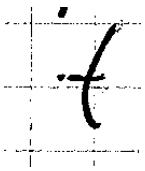
Original source details: GELFAND-P001451

(8) Definition. Let a conic Lagrangian subset be given $Y \subset T^*X$,

f is a holomorphic function on X.

f is called Morse with respect to Y, if it has the following property:

$df(X)$ (a Lagrangian subset (nonconic)) is transverse to Y,



Uncertain reading. The opening parenthesis after $df(X)$ has a short cross-stroke.

Original source details: GELFAND-P001451

that is, $df(X)$ intersects Y only at smooth points, and intersects transversely.

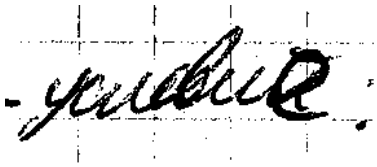
Examples.

1. $Y = X$ is the zero section.

then the Morse property of f with respect to $Y =$ the Morse property in the usual sense.

2. $Y =$ the conormal to Z, $Y = N^*Z$, Z is a smooth manifold.

Then the Morse property means satisfaction of [crossed out: two] the condition.



Original source details: GELFAND-P001451

Uncertain reading. The ending is overwritten. It tentatively reads «условие» (“condition”), possibly «условий» (“of conditions”).

[The label “a)” is crossed out.] $g = f|_Z$ is a Morse function (in the usual sense)

[The following two lines are boxed and crossed out with two diagonals.]

$$\text{b)} \quad dg(z) = 0 \;\Rightarrow\; df(z) \neq 0. \qquad (?)$$

(If Y contains the zero section, then this holds)

GELFAND-P001452

(9) Definition. Let $\mathcal{F}^\bullet$ be a constructible complex, and consider $SS(\mathcal{F}^\bullet)$, let f be Morse with respect to $SS(\mathcal{F}^\bullet)$ and let x be a critical point with respect to $SS(\mathcal{F}^\bullet)$ $\left(\overset{\text{def}}{\Longleftrightarrow} df(x) \in SS(\mathcal{F}^\bullet)\right)$.

Original source details: GELFAND-P001452

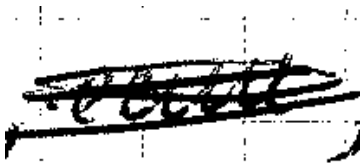
Uncertain reading. The equivalence arrow is heavily retraced, with “def” written above it.

Then

$$H^*\left(U, f^{-1}(\varepsilon); \mathcal{F}\right) \neq 0.$$

The complex $\mathcal{F}$ is called a perverse sheaf, if all such groups occur in only one degree (e.g., 0) [crossed out: if], that is, if

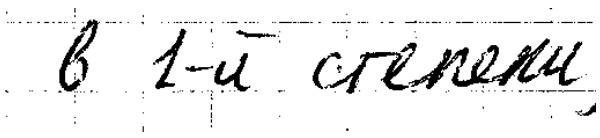
$$H^j\left(U, f^{-1}(\varepsilon), \mathcal{F}\right) = 0 \quad \text{for } j \neq 0. \qquad (10)$$



Original source details: GELFAND-P001452

Uncertain reading. The word crossed out after “e.g., 0” tentatively reads “if.”

(vanishing cycles occur only in the 1st degree)



Original source details: GELFAND-P001452

Uncertain reading. The note reads «в 1-й степени». The condition in (10) is written for $j \neq 0$.

Examples. a. $\mathbb{C}_{x_0}$ is a perverse sheaf. [Under the symbol: concentrated at a point.] (stalk $\mathbb{C}$ over x_0, 0 on the complement)

b. $\mathbb{C}_X[n]$ is the constant sheaf over X (shifted in grading) is a perverse sheaf.

$$n = \dim_{\mathbb{C}} X.$$

c. $Z \subset X$ [with a leader: smooth subset]

$$\mathbb{C}_Z[\dim Z]$$

[Under the symbol:] the constant sheaf on Z, extended by zero outside Z.

(or a shift of the Goresky–MacPherson complex)

Remark. [Under the heading: (approximate).] The strata with respect to which the sheaf $\mathcal{F}$ is constructible are the projections to the base of the irreducible components of its $SS(\mathcal{F})$.

Construction 1. Let $\mathcal{F}$ is a perverse sheaf. We construct a local system [inserted below: $\Phi(\mathcal{F})$] at the smooth points of its $SS(\mathcal{F})$.

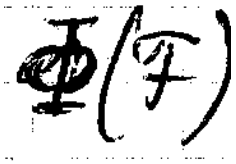
Let [crossed out: illegible] z be a smooth point of $SS(\mathcal{F})$.

Undeciphered passage. Illegible writing is crossed out before z.

Original source details: GELFAND-P001452

Then

$$\Phi(\mathcal{F}) = H^0\big(U, f^{-1}(\varepsilon), \mathcal{F}\big), \qquad df(\pi(z)) = z. \tag{11}$$



Uncertain reading. The Φ in (11) is heavily retraced. The insertion above shows it clearly.

Original source details: GELFAND-P001452

(the choice of f makes no difference)

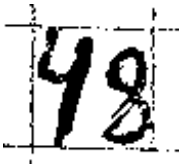
Next time in 2 weeks.

GELFAND-P001453

[source number: 48]

11/IV–88 Gelfand Seminar

D. B. Fuchs The four-colour problem. (Appel, Haken)



Uncertain reading. The number reads “48.” The lower loop of the second digit is thinner than the upper loop.

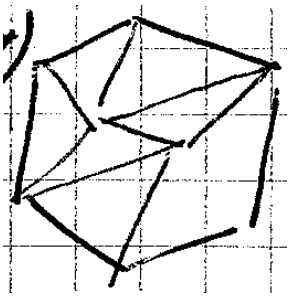
Original source details: GELFAND-P001453

Def. A configuration is a triangulation of a disc.

[Crossed out on the right: a subdivision of a sphere (disc) (triangulation).]

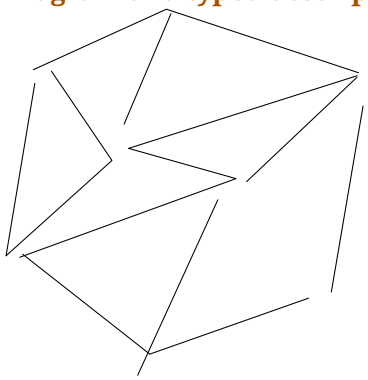
Original source details: GELFAND-P001453

Uncertain reading. The crossed-out word in the first parentheses tentatively reads “disc.”



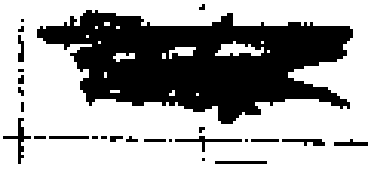
Original source details: GELFAND-P001453

Diagram and typed description.



The redraw preserves the visible gaps in the outline and interior strokes.

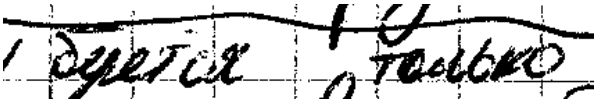
(1) We must prove that the vertices can be coloured with 4 colours [crossed out: illegible], so that adjacent vertices have different colours.



Original source details: GELFAND-P001453

Undeciphered passage. Illegible writing is crossed out after “4 colours.”

[Small annotation] (this is the dual problem to face colouring, so the other only



Original source details: GELFAND-P001453

Uncertain reading. The end of the note tentatively reads «другая только» (“the other only”). No closing parenthesis is visible.

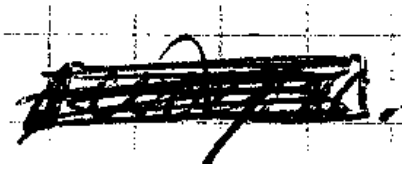
(2) A set of configurations is called unavoidable, if one of them occurs in every triangulation.

(3) A configuration is called reducible, if it cannot occur in a triangulation of the sphere that cannot be coloured with 4 colours and has the fewest vertices among such triangulations.

Reducible triangulation K: [crossed out: if; inserted above: suppose] there is a triangulation $T \supset K$ and it is known that $\forall$ configuration [crossed out: illegible] with fewer vertices (than T) can be coloured with 4 colours. Then T can also be coloured.

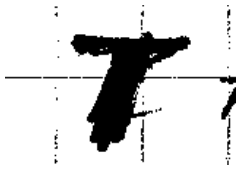
Original source details: GELFAND-P001453

Uncertain reading. The word crossed out before “there is” tentatively reads “if.”



Original source details: GELFAND-P001453

Undeciphered passage. The writing after “configuration” is crossed out and illegible.



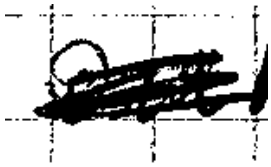
Original source details: GELFAND-P001453

Uncertain reading. The symbol before “also” is heavily retraced and reads T.

Induction on the number of vertices N.

Suppose it has already been proved for $N-1$ vertices.

configuration [crossed out: this is] K is called reducible, if [“from” inserted above] its occurrence in T with $|T| = N$ implies that T can be coloured.



Original source details: GELFAND-P001453

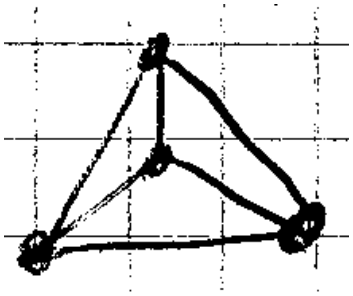
Uncertain reading. The word before K is crossed out, possibly «это» (“this is”).

[Crossed out: “Example.”]

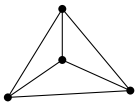
The main idea is to combine definitions (2) and (3).

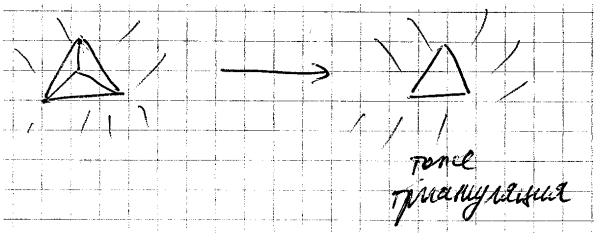
[Crossed out: “We need”; inserted above: “It suffices”] to construct an unavoidable set of reducible configurations.

(4) Example.



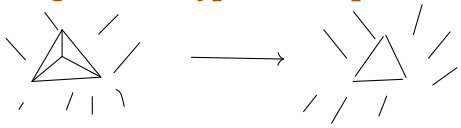
Original source details: GELFAND-P001453

Diagram and typed description. 
— a reducible configuration.



Original source details: GELFAND-P001453

Diagram and typed description.



GELFAND-P001454

I. M. G. One should invent everything using the simplest examples. (I. Bernstein learned this from I. M. G.)

I. M. G. devised the whole theory of spherical functions using the formula

$$\cos(\alpha+\beta)+\cos(\alpha-\beta)=2\cos\alpha\cdot\cos\beta.$$

On $\mathbb{R}$ one should consider the group generated by translations and reflections.

Uncertain reading. The writing above the heading is crossed out and may read «Утв.» (“Proposition”).

Original source details: GELFAND-P001454

(5) Theorem

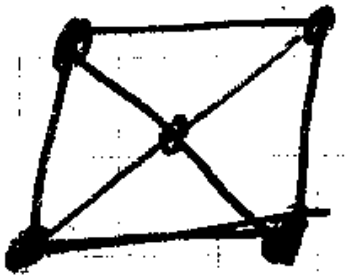
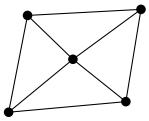


Diagram and typed description. 

reducible

Original source details: GELFAND-P001454

(Kempe, ~ 100 years ago)

Replace it with

(6)

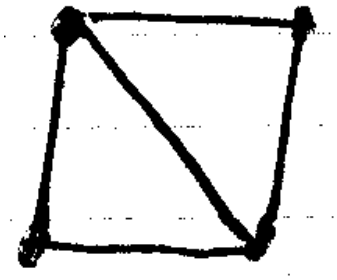
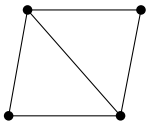
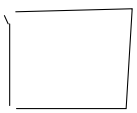


Diagram and typed description. 

Original source details: GELFAND-P001454

and colour the resulting triangulation.

(*) If any of the 4 have the same colour, colour the centre with the unused one.

Diagram and typed description. 

Original source details: GELFAND-P001454

it remains to consider the case

(7)

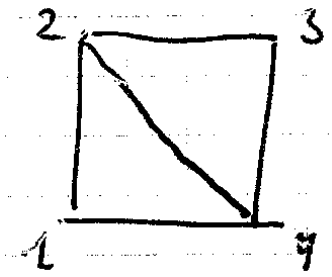
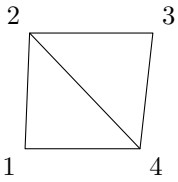


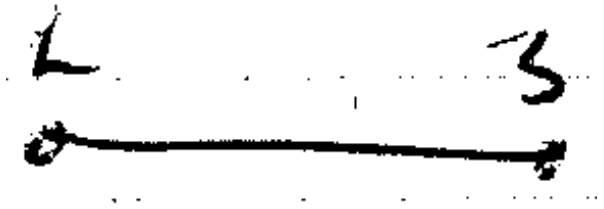
Diagram and typed description.



Original source details: GELFAND-P001454

Consider the subset K, consisting of vertices coloured 1 and 3 and edges

Diagram and typed description.



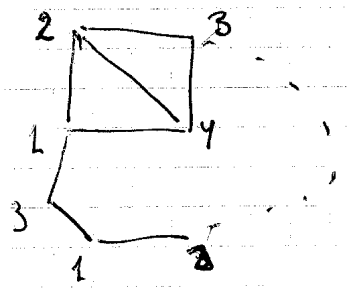
Original source details: GELFAND-P001454

(8) 1 may a) be connected to 3 in K

or b) be disconnected from it.

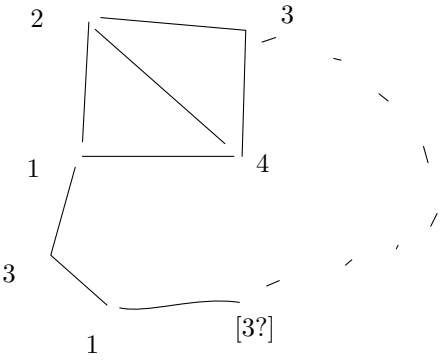
Case b). In the connected component of vertex 1 interchange 1 and 3 and reduce to the previous case (*).

(9)



Original source details: GELFAND-P001454

Diagram and typed description.



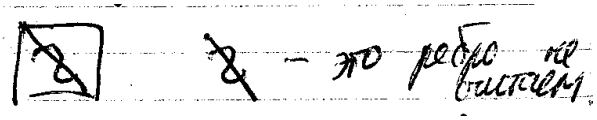
The lower-right digit is overwritten and may be 3. The continuation on the right is drawn as separate strokes.

[To the right of the figure] Suppose we are in case 8a).

Then there is no path from 2 to 4 with [crossed out: illegible] 2 and 4.

Undeciphered passage. The marks crossed out after “with” are illegible. The first resembles a 2.

Original source details: GELFAND-P001454



Original source details: GELFAND-P001454

Diagram and typed description.

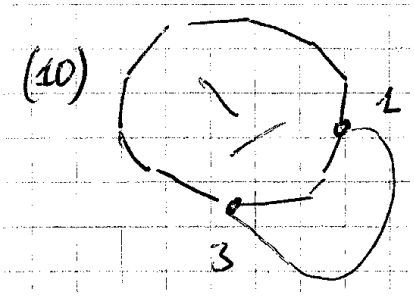
— do not count this edge.

(since it cannot cross the path of 1s and 3s)

and then reduce again to 8b).

GELFAND-P001455

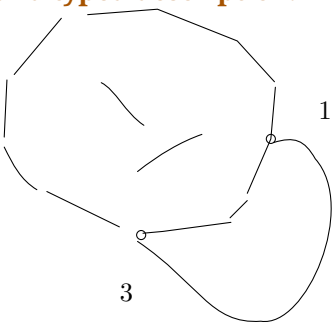
[source number: 49]



Original source details: GELFAND-P001455

Diagram and typed description.

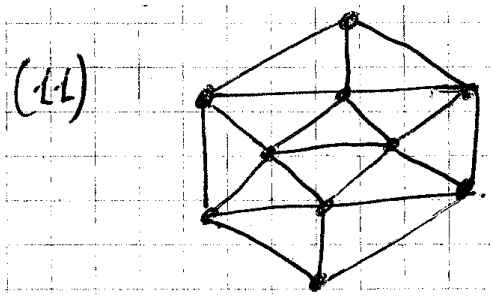
(10)



Consider a more complicated configuration and all possible colourings of its boundary. (With 4 colours)

Some extend to the interior and some do not.

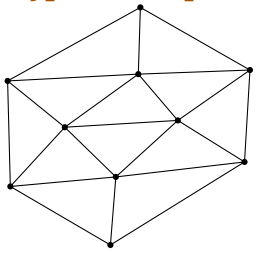
One can try recolouring as in (9) (using chains). The computer uses this method to look for reducible configurations.



Original source details: GELFAND-P001455

Diagram and typed description.

(11)



A reducible configuration (but Fuchs [crossed out: illegible mark] could not prove it)

Original source details: GELFAND-P001455

Uncertain reading. The crossed-out mark after “Fuchs” resembles «у» or «д».

I. M. G. Problem 6.

Original source details: GELFAND-P001455

Uncertain reading. The mark after “Problem” is heavily retraced. It may be the digit 6 or the letter «в».

Write a program for determining reducibility on a parallel computer.

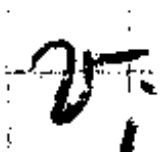
One of Appel and Haken's configurations (out of ~ 1700) was checked by hand (Allaire), since the computer could not handle it.

Unavoidable set.

Their construction is based on Euler's theorem

$$\mathrm{B} - \mathrm{P} + \Gamma = 2. \tag{12}$$

Kempe: let ν_i be the number of vertices joined to exactly i other vertices (with i edges issuing from each)

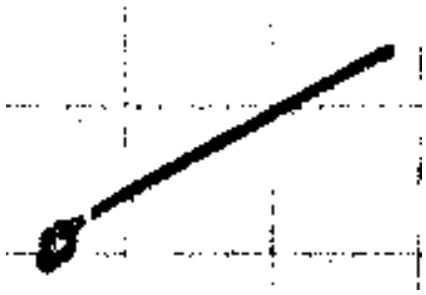


Original source details: GELFAND-P001455

Uncertain reading. The handwritten symbol can be read as ν_i or v_i.

$$\mathrm{B} = \nu_3 + \nu_4 + \nu_5 + \dots \tag{13}$$

$$\mathrm{P} = \frac{1}{2}(3\nu_3 + 4\nu_4 + 5\nu_5 + \dots) \tag{14}$$



Original source details: GELFAND-P001455

Diagram and typed description.

$$2\mathrm{P} = \text{the number of such pairs} = 3\nu_3 + 4\nu_4 + \dots$$

(a vertex and an edge issuing from it)

$$3\Gamma = 2\mathrm{P} \qquad \text{(since it is a triangulation)} \tag{15}$$

hence

$$\Gamma = \frac{3\nu_3 + 4\nu_4 + 5\nu_5 + \dots}{3} \tag{15'}$$

From (12) we find:

$$3\nu_3 + 2\nu_4 + \nu_5 - \nu_7 - 2\nu_8 - 3\nu_9 - \dots = 12, \tag{16}$$

that is,

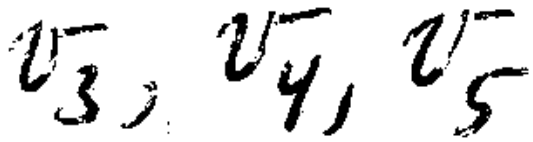
Original source details: GELFAND-P001455

Uncertain reading. A number resembling (17) or (16) is crossed out before (16').

$$\sum_{i=3}^{\infty}(6-i)\nu_i = 12. \tag{16'}$$

GELFAND-P001456

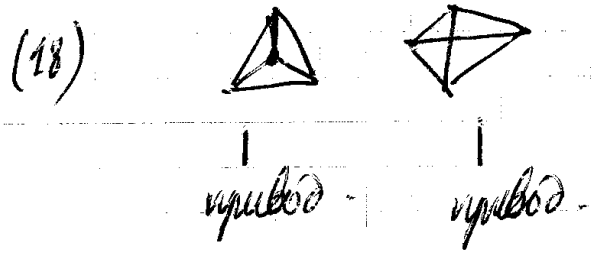
(17) Hence one of the numbers ν_3, ν_4, ν_5 is different from 0.



Original source details: GELFAND-P001456

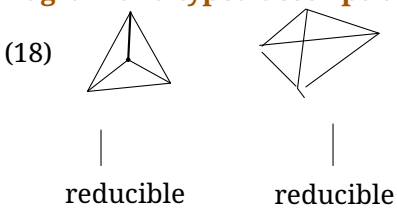
Uncertain reading. The letter in the three labels can be read as ν or v.

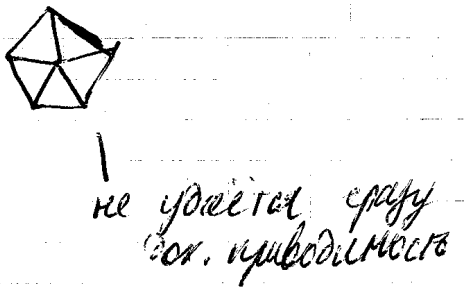
We obtain an unavoidable set



Original source details: GELFAND-P001456

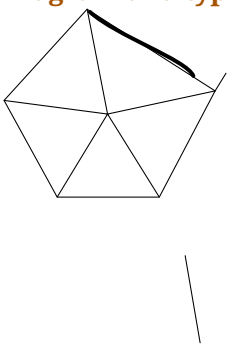
Diagram and typed description.





Original source details: GELFAND-P001456

Diagram and typed description.



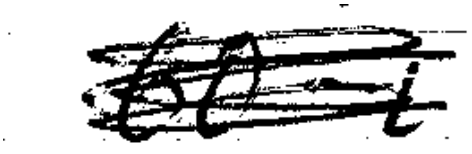
The upper-right side is heavily retraced. A short stroke of unclear purpose extends from its end.

Now one need only consider triangulations with

$$\nu_3 = \nu_4 = 0 \tag{19}$$

Discharging [Russian gloss: «разрядка»]

(20) Initial charge: to a vertex from which i edges issue, assign a charge [crossed out: $60 - i$] $60(6 - i)$.



Original source details: GELFAND-P001456

Uncertain reading. The crossed-out expression tentatively reads $60 - i$, possibly $6 - i$.

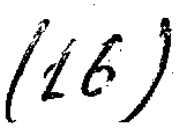
Redistribution procedure: construct another function on the vertices.

$$(16') \iff \text{sum of all charges} = 720.$$

(21) We want [inserted above: to make it so] that all charges are ≤ 0.
(which is impossible [crossed out: illegible] in view of equation (16))

Original source details: GELFAND-P001456

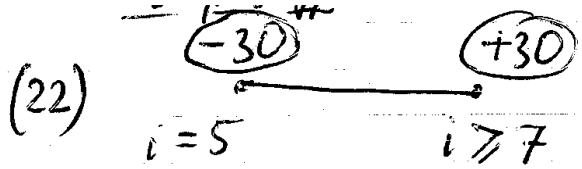
Undeciphered passage. Illegible writing is crossed out after “impossible.”



Original source details: GELFAND-P001456

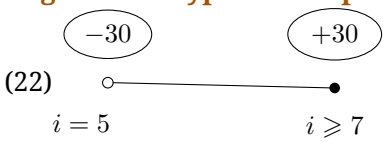
Uncertain reading. The equation number after “equation” is written without a prime.

Procedure 1.

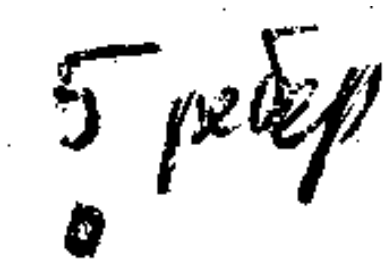


Original source details: GELFAND-P001456

Diagram and typed description.

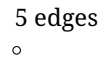


(decrease by 30 at the 1st vertex, increase by 30 at the 2nd)
Some situations in which a positive charge remains somewhere.
(23) 1°.



Original source details: GELFAND-P001456

Diagram and typed description.



is joined to 0 or 1 [crossed out: edg...] vertices with number of edges $\geqslant 7$.

Original source details: GELFAND-P001456

Uncertain reading. The crossed-out word resembles «рёбр» or «рёбер» (“edges”).

(⇒ a vertex with positive charge remains

Uncertain reading. A separate mark, possibly 2, is crossed out before (24).

Original source details: GELFAND-P001456

(24) 2°.

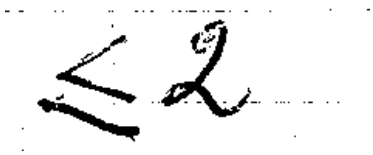
Diagram and typed description. deg. 7

Original source details: GELFAND-P001456

is joined [crossed out: edge with] to $\leqslant 2$ vertices of degree 5

Uncertain reading. The crossed-out writing tentatively reads «ребро с» (“edge with”), possibly «ребра с» (“edges with”).

Original source details: GELFAND-P001456



Uncertain reading. The manuscript has $\leqslant$ before 2.

Original source details: GELFAND-P001456

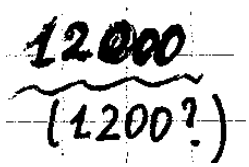
We end up with a finite set of configurations, which is unavoidable, since in the absence of all of them we can make every charge ≤ 0.

GELFAND-P001457

[source number: 50]

Now they must be checked for reducibility.

In 1977 the authors (Appel and Haken) thank the administration for 12 000 hours of computer time. [Written below the number: (1200?).]



Uncertain reading. The line gives “12000,” with “(1200?)” written below.

Original source details: GELFAND-P001457

Then the configurations are reworked into more complicated ones, and the transfer rules are reworked too.

A. Beilinson (continued)

(on work by K. Vilonen, S. Gelfand and R. D. MacPherson)

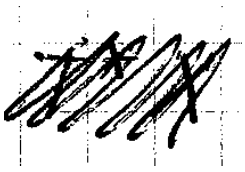
[Below "D.", with a leader: "Duncan."]

X is a smooth [inserted above: complex] algebraic variety.

$\mathcal{F}$ is a finite-dimensional constructible [crossed out: sheaf; inserted above: complex of sheaves] on X.

From these data one constructs

$SS(\mathcal{F})$ a Lagrangian conic set in [crossed out: T^*X] $\Omega X = T^*X$.



Original source details: GELFAND-P001457

Uncertain reading. The crossed-out notation before ΩX tentatively reads T^*X.

$$SS(\mathcal{F}) = \text{zero section} \iff H^i(\mathcal{F}) \text{ is smooth, i.e. locally constant.}$$

Def. $SS\mathcal{F}$

$$H^\bullet(\mathcal{F}) = 0 \quad \Rightarrow \quad SS(\mathcal{F}) = \varnothing \tag{1}$$

$$H^\bullet(\mathcal{F}) \neq 0 \quad \Rightarrow \quad SS(\mathcal{F}) \supset \text{zero section}, \quad \text{and}$$

$SS(\mathcal{F}) \setminus (\text{zero section})$ is described as follows

$$\nu \in \Omega^1_x, \qquad \text{choose } f\colon \quad df_x = \nu, \quad f(x) = 0$$

[Below ν, with a leader: "vector."]

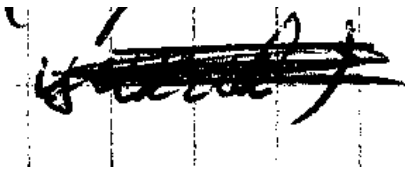
Let U be a small neighbourhood of the point x

$$H^\bullet\big(U, U \cap f^{-1}(\varepsilon); \mathcal{F}\big) = \Phi^\bullet_f(\mathcal{F})_x \tag{2}$$

Original source details: GELFAND-P001457

Uncertain reading. Before the semicolon in (2), a symbol resembling $\mathcal{F}$ is crossed out.

(space of vanishing [crossed out: cocycles...] cycles)

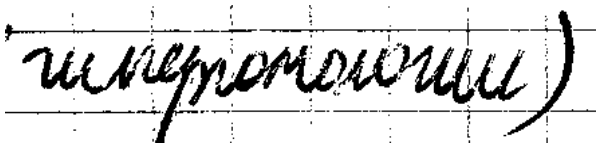


Original source details: GELFAND-P001457

Uncertain reading. The crossed-out word may read «коциклов» ("cocycles") or «когомол.» ("cohomology").

$\varepsilon \neq 0$ is small

(hyperhomology)



Uncertain reading. The manuscript tentatively reads “hyperhomology,” with “hypercohomology” another possible reading.

Original source details: GELFAND-P001457

(relative cohomology is defined via the cone) of the map, i.e. write down the bicomplex and totalize it

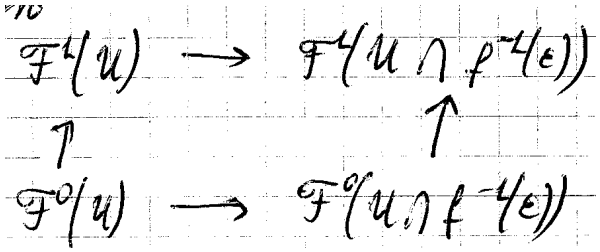


Diagram and typed description.

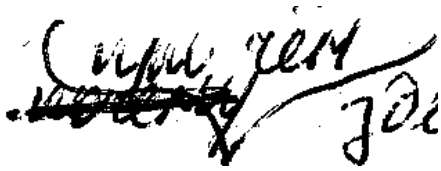
$$\begin{array}{ccc} \mathcal{F}^1(U) & \longrightarrow & \mathcal{F}^1(U \cap f^{-1}(\varepsilon)) \\ \uparrow & & \uparrow \\ \mathcal{F}^0(U) & \longrightarrow & \mathcal{F}^0(U \cap f^{-1}(\varepsilon)) \end{array}$$

Original source details: GELFAND-P001457

(if $\mathcal{F}$ is sufficiently acyclic)

GELFAND-P001458

an explanation of [crossed out: why; inserted above: what role] vanishing cycles play here.



Uncertain reading. The crossed-out word tentatively reads «почему» (“why”), possibly «зачем» (“what for”).

Original source details: GELFAND-P001458

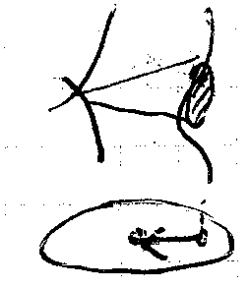
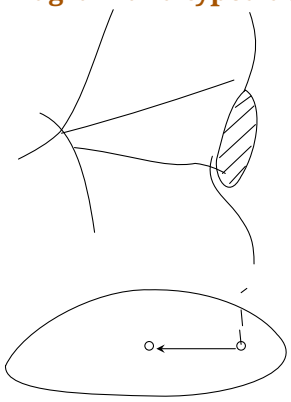


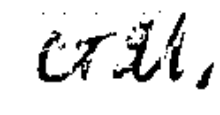
Diagram and typed description.



The lower arrow points left. Additional strokes inside the oval overlap it.

Original source details: GELFAND-P001458

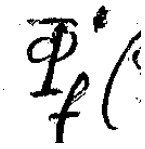
[On the right] the cycle contracts to a point



Uncertain reading. The abbreviation after “cycle” tentatively reads «стян.», possibly «стяг.».

Original source details: GELFAND-P001458

Df. $\nu \notin SS(\mathcal{F}) \iff$ vanishing cycles at ν are 0, i.e. $\Phi^{\bullet}_f(\mathcal{F})_x = 0$.



Uncertain reading. The superscript on Φ_f resembles a heavy dot. It could also be i or $*$.

Original source details: GELFAND-P001458

Consider the direct image of $\mathcal{F}|_U$ under the map $U \xrightarrow{f}$ disc.

Vanishing cycles measure how much the fibre at 0 differs from the neighbouring fibres.

f is called a Morse function on X, if [inserted above: f is holomorphic on X and] the graph $df(X) \subset$ [crossed out: illegible] Ω^1 intersects transversely $SS(\mathcal{F})$.

Undeciphered passage. An illegible mark is crossed out before Ω^1.

Original source details: GELFAND-P001458

A complex is called a perverse sheaf, if $\forall$ Morse with respect to $\mathcal{F}$ functions have vanishing cycles only in dimension 0.

(only $\Phi^0_f(\mathcal{F}) \neq 0$)

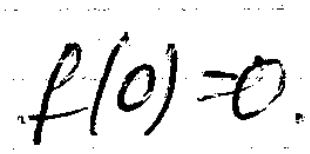
$\mathcal{F}$ is a perverse sheaf.

It gives a local system on the regular points of the singular support.

Its stalk over the point ν:

$$\Phi(\mathcal{F})_\nu = \Phi^0_f(\mathcal{F})_x$$

for a Morse function f with $df(x) = \nu$, $f(0) = 0$.



Uncertain reading. The manuscript gives $f(0) = 0$.

Original source details: GELFAND-P001458

What happens at singular points?

Problem. Describe perverse sheaves microlocally, i.e. in terms of vanishing cycles.

The work solves half of this problem.

Point. Where do perverse sheaves come from?

1°. $\mathcal{M}$ is an overdetermined system

[Under "system": (holonomic)]

[On the right] $\sum f_i \partial^i$

[Under f_i, with a vertical leader: holomorphic.]

$$\mathcal{F}^{0,0} \xrightarrow{\bar\partial} \mathcal{F}^{0,1} \longrightarrow \dots$$

Uncertain reading. Above the start of the line is a detached mark resembling 4 or an insertion sign.

Original source details: GELFAND-P001458

if $u \in \mathcal{F}^{0,0}$ is a solution of $\mathcal{M}$, then $\bar\partial u$ is too

The system is overdetermined, $\Rightarrow$ the solution complex is a perverse sheaf.

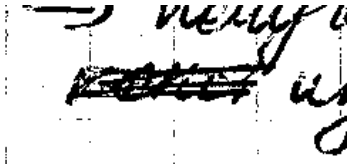
GELFAND-P001459

[source number: 51]

2°. Combinatorial method

(Goresky–MacPherson)

There is a [inserted above: general] method for extending sheaves to singular points $\Rightarrow$ this gives [word crossed out] perverse sheaves.

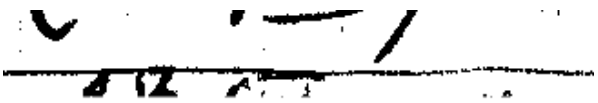


Uncertain reading. The cancellation obscures the word.

Original source details: GELFAND-P001459

The paper is devoted to the third method.

Next time 18/IV a talk by Reshetikhin on quantum groups.



Uncertain reading. Clipped strokes are visible below the line. The shape at left may be «M» and is too incomplete to read with confidence.

Original source details: GELFAND-P001459

GELFAND-P001460

18.4.1988 Gelfand Seminar.

Titles of Bourbaki Seminar talks (87–88)

Lerner. “Uncertainty principle and microlocalization” (after Fefferman and Phong)

“Geometry of polynomials and Newton’s method” (after Smale and Shub)

“Nonlinear mixed problems”

“Noncommutative stochastic calculus”

“Metaplectic representations and the Howe conjecture”

“Schottky problem and Novikov conjecture”

Douady. “Siegel disks and Herman ring”

“Deformations of plane curves”

“Large deviations”

“Classification of unitary representations of real groups”

“Classification theory”

“Stable sheaves and [crossed-out fragment]

Uncertain reading. The crossed-out fragment is illegible. No closing quotation mark can be made out.

Original source details: GELFAND-P001460

“Moonshine”

Verdier. “Quantum groups [crossed-out superscript mark] (after Drinfeld)”

Uncertain reading. The mark after “groups” may be crossed-out closing quotation marks or an asterisk.

Original source details: GELFAND-P001460

Malgrange. “Geometric Fourier transform”

“A new invariant of homology spheres of dimension 3”

“Elliptic cohomology”

“Recent advances in Fermat’s theorem”.

GELFAND-P001461

[source number: 63]

Beilinson on work by Atiyah and somebody else.

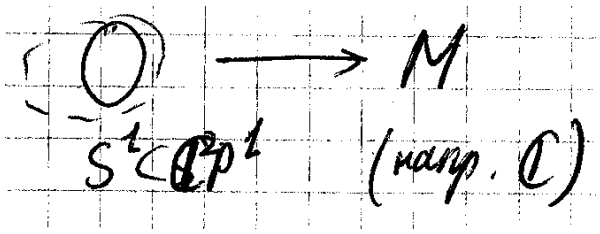
Semi-infinite Morse theory.

In the infinite-dimensional case there are 2 versions of Morse theory (for computing homology).

1) The geodesic problem with the energy functional resembles the finite-dimensional one. The negative index of inertia is finite and equals the number of conjugate points along the geodesic.

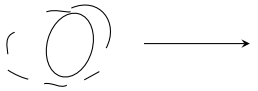
2) Infinite-dimensional manifold, a functional of energy type with isolated critical points but infinitely many $+$ and infinitely many $-$. But it turns out that at two distinct critical points the number of $-$ differs by a finite number. Homology is constructed by counting from $\infty/2$ (the topological version of Feigin and Tsygan)

Sensation: these invariants are related to Donaldson's invariants of 4-manifolds.



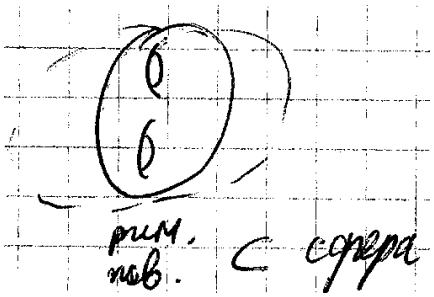
Original source details: GELFAND-P001461

Diagram and typed description.



$S^1 \subset \mathbb{C}P^1$ (e.g. $\mathbb{C}$)

For $S^1 \to \mathbb{C}$, distinguish 2 halves (extension inward and outward). They intersect in a finite-dimensional manifold.



Original source details: GELFAND-P001461

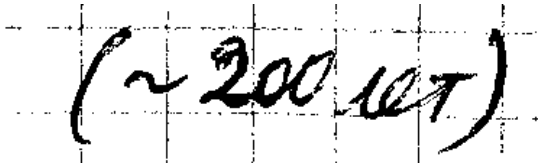
Diagram and typed description.

The exact junction of the upper inner stroke and the outline is unclear. The notes do not specify the meaning of the inner loops.

[Below the figure] Riemann surface $\subset$ sphere

[To the right of the figure] gauge fields; Y.–M. functional. critical points are flat connections.

I.M.G. The decomposition of $C^\infty(S^1)$ was used by many (~ 200 years)



Original source details: GELFAND-P001461

Uncertain reading. The apparent reading is "~ 200 years". Two rounded loops follow the 2.

(including τ-functions of Hirota–Sato, $\frac{1}{2}$-cohomology)

Remarkably, S^1 can be replaced by other manifolds. This will be for next time.

Reshetikhin Kolya.

~~Quantum groups.~~

$U_q(SU(2))$ and link invariants and representations of the braid group.

$$A = U_q(SU(2)); \qquad 1, H, X^{\pm} \quad \text{— generators}$$
assoc. alg.

$$[X^{\pm}, H] = \mp 2X^{\pm}; \tag{1}$$

$$[X^+, X^-] = \frac{\operatorname{sh}\left(\frac{h}{2}H\right)}{\operatorname{sh}(h/2)}; \qquad q = e^h$$

GELFAND-P001462

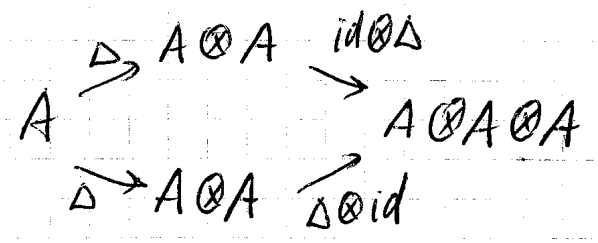
$$\left(q^{H/4}\right)^{\ell} H^n (X^+)^m (X^-)^k \quad \text{— basis as a vector space} \tag{2}$$

$$\Delta : A \to A \otimes A$$

$$\Delta(H) = H \otimes 1 + 1 \otimes H \tag{3}$$

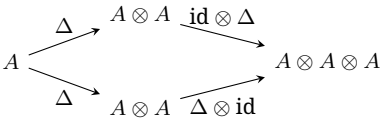
$$\Delta(X^{\pm}) = X^{\pm} \otimes q^{H/4} + q^{-H/4} \otimes X^{\pm}$$

1. Δ — an algebra homomorphism. $\Delta(ab) = \Delta(a)\Delta(b)$.
2. Δ — coassociative, i.e.



Original source details: GELFAND-P001462

Diagram and typed description.



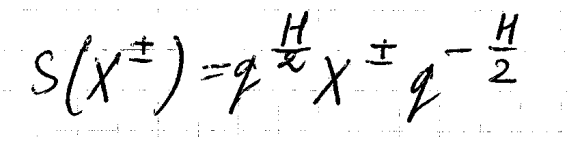
[At right] (Hopf alg.)

3. [crossed-out sign] $S : A \to A$ an antihomomorphism (analogous to passage to the inverse element in the algebra of functions on a group)

Original source details: GELFAND-P001462

Uncertain reading. The cancellation obscures the sign. Possibly Δ or «и».

$$S(X^{\pm}) = q^{H/2} X^{\pm} q^{-H/2} \tag{4}$$

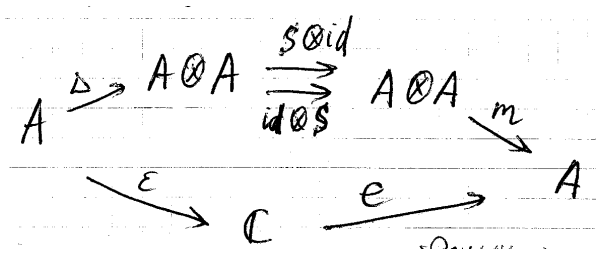


Original source details: GELFAND-P001462

Uncertain reading. In (4), $q^{H/2}$ immediately follows the equals sign. The formula follows the manuscript.

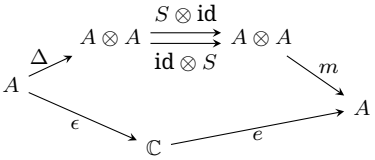
$$S(H) = -H$$

$$\epsilon(H) = \epsilon(X^{\pm}) = 0$$



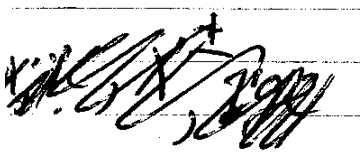
Original source details: GELFAND-P001462

Diagram and typed description.



[At right] (either arrow may be taken)

Turns into a quadratic algebra in [inserted above: other] generators [crossed-out fragment]



Original source details: GELFAND-P001462

Uncertain reading. Possibly K, then X^+ or $X^{\pm}$. The following strokes are illegible.

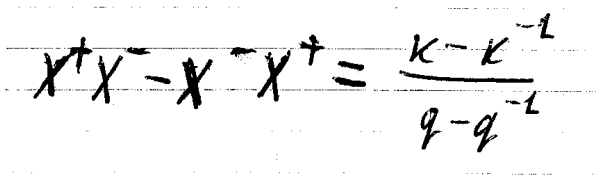
Consider.

$$K = q^{H/2}, \qquad K, X^{\pm}$$

[Crossed out at left: $K, X^{\pm}$.]

$$KX^{\pm} = q^{\pm 1} X^{\pm} K$$

$$X^+X^- - X^-X^+ = \frac{K - K^{-1}}{q - q^{-1}}$$



Original source details: GELFAND-P001462

Uncertain reading. The denominator $q - q^{-1}$ follows the manuscript.

GELFAND-P001463

[source number: 64]

(5) involution

$$(X^+)^* = X^-, \qquad H^* = H$$

finite-dimensional representations

$$\pi_j(X^\pm)e^j_m = \sqrt{\frac{\operatorname{sh}\left(\frac{h}{2}(j \mp m)\right)\operatorname{sh}\left(\frac{h}{2}(j \pm m + 1)\right)}{\operatorname{sh}^2\left(\frac{h}{2}\right)}}\, e_{m\pm 1}, \tag{6}$$

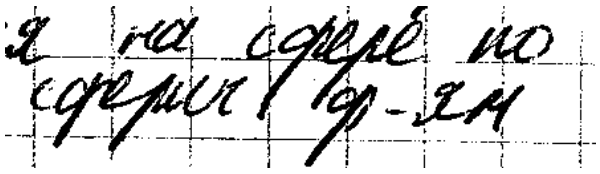
$$j \in \mathbb{Z}/2, \qquad -j \le m \le j.$$

$$\dim V^j = 2j + 1 \quad \text{(as for } \mathrm{SL}_2\text{)}$$

Sklyanin wrote in quadratic algebras.

Exercise (I. M. G.) $q = 1 \Rightarrow$ formulas from the book by Gelfand–Minlos–Shapiro

[At right, marked off by a bracket] specific problem: how to expand vector fields on a sphere in spherical harmonics



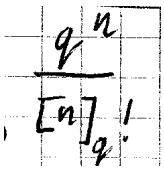
Original source details: GELFAND-P001463

Uncertain reading. The Russian abbreviation for “harmonics” appears to be «гар-ям», possibly «гар-м».

start with $\mathrm{SU}(2)$

(7) Thm. Set

$$R = \exp\left(\frac{h}{4}H \otimes H\right) \sum_{n\ge 0} \frac{q^n}{[n]_q!}\, e^n \otimes f^n \tag{8}$$



Original source details: GELFAND-P001463

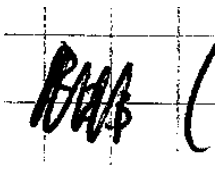
Uncertain reading. The numerator of the coefficient in (8) is q^n.

$$R \in U_q(\mathrm{SU}(2))^{\otimes 2},$$

where

$$e = q^{H/4}X^+, \qquad f = q^{-H/4}X^-, \tag{9}$$

$$1. \quad (\Delta \otimes \mathrm{id})R = R_{23}R_{13}; \qquad \left(\in U_q(\mathrm{SU}(2))^{\otimes 3}\right) \tag{10}$$



Original source details: GELFAND-P001463

Uncertain reading. An illegible fragment is crossed out before the parentheses in the first line of (10).

$$(\mathrm{id} \otimes \Delta)R = R_{12}R_{13};$$

$$2. \quad \sigma \circ \Delta(a) = R\,\Delta(a)R^{-1}; \qquad \text{(see (12))}$$

$$3.\quad R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

[At right] (Yang–Baxter equation with values in the tensor product of 3 copies of A)

In formula (8)

$$[n]_q! = [n]_q \dots [1]_q \qquad (q\text{-factorial}) \tag{11}$$

I. M. G. Quantum mechanics is the right way to introduce q.

Ex. what is Γ_q?

$$\sigma\colon A^{\otimes 2} \longrightarrow A^{\otimes 2}, \qquad \sigma(f \otimes g) = g \otimes f \tag{12}$$

In particular, it follows from $(10)_2$ that $\sigma \circ \Delta$ is also a comultiplication.

GELFAND-P001464

Define

$$R^{j_1 j_2}\colon V^{j_1} \otimes V^{j_2} \longrightarrow V^{j_2} \otimes V^{j_1}$$

Uncertain reading. The sign after $R^{j_1 j_2}$ reads as a colon over earlier strokes. The underlying sign is unclear.

Original source details: GELFAND-P001464

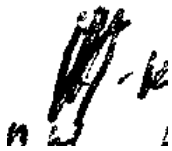
V^j — space of a finite-dimensional representation.

$$R^{j_1 j_2} = P^{j_1 j_2}\, \pi^{j_1} \otimes \pi^{j_2}(R) \tag{13}$$

[Beneath $P^{j_1 j_2}$, joined by a line] operator permuting the spaces

$$\begin{aligned}(1 \otimes R^{j_1 j_2})(R^{j_1 j_3} \otimes 1)(1 \otimes R^{j_2 j_3}) = \\ = (R^{j_2 j_3} \otimes 1)(1 \otimes R^{j_1 j_3})(R^{j_1 j_2} \otimes 1)\end{aligned} \tag{14}$$

q-analogues of the [crossed-out fragment] Clebsch–Gordan coefficients.

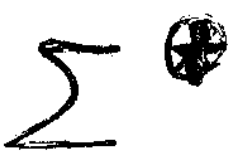


Uncertain reading. The crossed-out strokes before “coefficients” are illegible.

Original source details: GELFAND-P001464

all finite-dimensional representations are completely reducible.

$$V^{j_1} \otimes V^{j_2} = \sum_{|j_1 - j_2| \le j_3 \le j_1 + j_2}^{\oplus} V^{j_3} \tag{15}$$



Uncertain reading. The small sign above the sum reads $\oplus$, possibly over $\otimes$.

Original source details: GELFAND-P001464

V_j were found by Kulish and Reshetikhin (1981)
In the basis:

$$e^{j_1}_{m_1} \otimes e^{j_2}_{m_2} = \sum_{|j_1-j_2|\le j_3\le j_1+j_2} \begin{bmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{bmatrix} e^{j_3}_{m_3} \tag{16}$$

[A line joins the square bracket to the label] Clebsch–Gordan coefficients
where

$$m_3 = m_1 + m_2$$

Graphical notation:

$$A\colon V^{j_1} \otimes \ldots \otimes V^{j_N} \longrightarrow V^{j'_1} \otimes \ldots \otimes V^{j'_L}$$

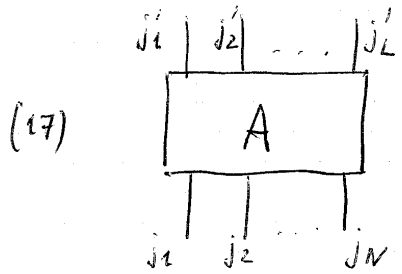
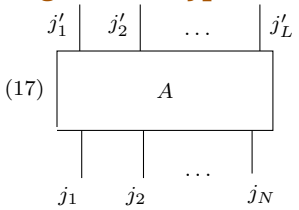
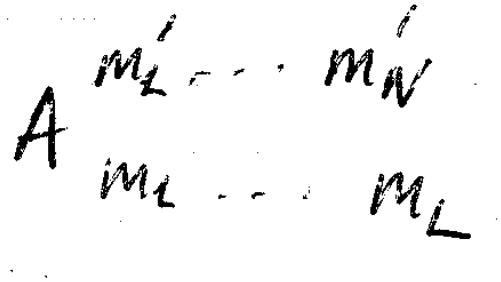


Diagram and typed description.



Original source details: GELFAND-P001464

$$A^{m'_1 \ldots m'_N}_{m_1 \ldots m_L}$$

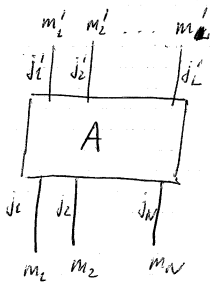


Uncertain reading. In the manuscript coefficient the upper indices end at m'_N, the lower at m_L.

Original source details: GELFAND-P001464

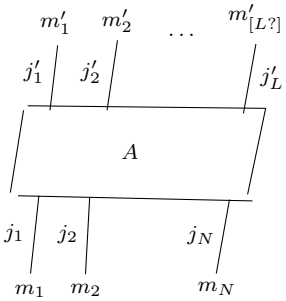
$$-j_i \le m_i \le j_i;$$

[At lower right]



Original source details: GELFAND-P001464

Diagram and typed description.



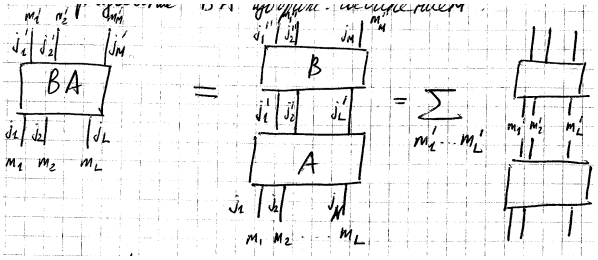
The upper-right m' subscript is overwritten. Its visible final stroke resembles L. The underlying form remains uncertain.

GELFAND-P001465

[source number: 65]

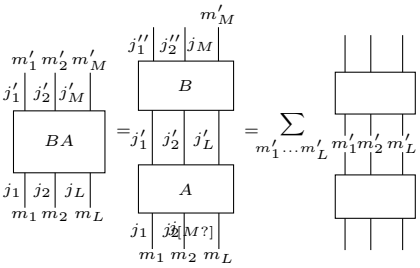
The product BA is shown by joining:

(18)



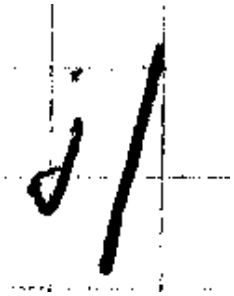
Original source details: GELFAND-P001465

Diagram and typed description.



In the middle drawing the first two upper j-labels at B have double primes. At the upper right j_M is legible and a prime is uncertain. The last lower j-index at A resembles M, though m_L stands below it.

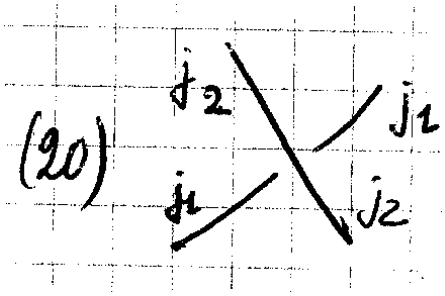
$V^j \xrightarrow{\text{id}} V^j$ is shown by a strand (19)



Original source details: GELFAND-P001465

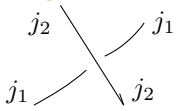
Diagram and typed description. j

(20)



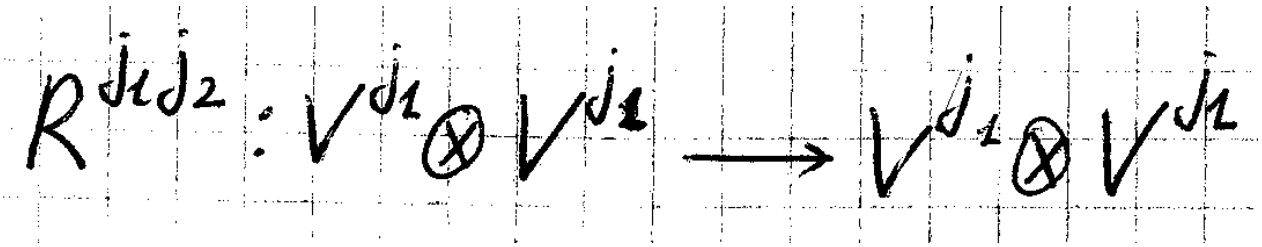
Original source details: GELFAND-P001465

Diagram and typed description.



A short side stroke touches the lower-right tip. A complete arrowhead cannot be made out.

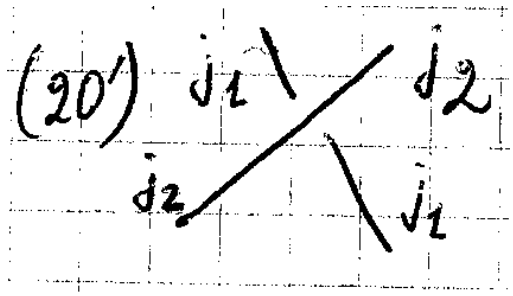
$$R^{j_1 j_2} : V^{j_1} \otimes V^{j_1} \longrightarrow V^{j_1} \otimes V^{j_1}$$



Original source details: GELFAND-P001465

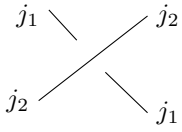
Uncertain reading. All four V factors read j_1, following the manuscript.

(20') $(R^{j_2 j_1})^{-1}$ in the same space.

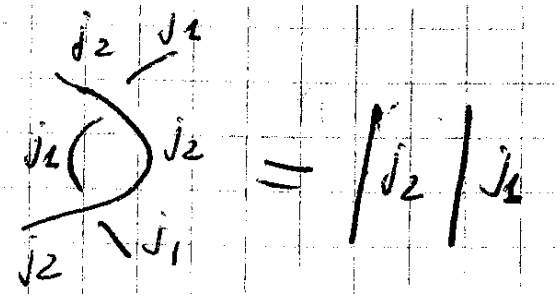


Original source details: GELFAND-P001465

Diagram and typed description.

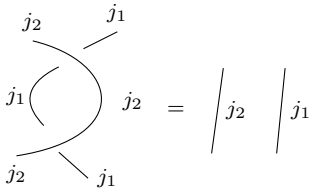


$$RR^{-1} = I$$



Original source details: GELFAND-P001465

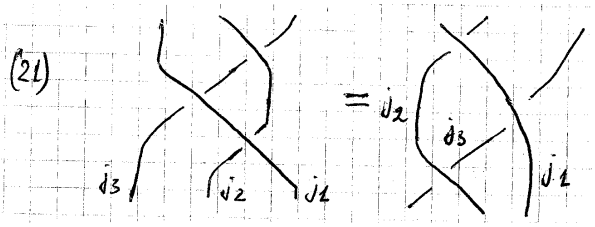
Diagram and typed description.



j_1, j_2 — colours of the strands

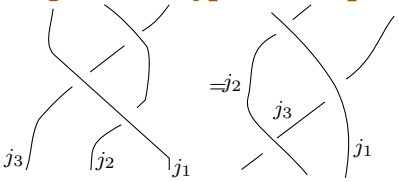
Yang–Baxter relation

(21)



Original source details: GELFAND-P001465

Diagram and typed description.



Additional vertices

(22)

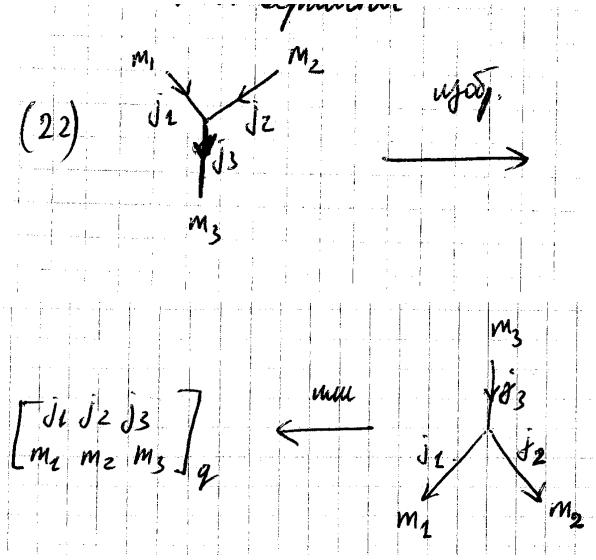
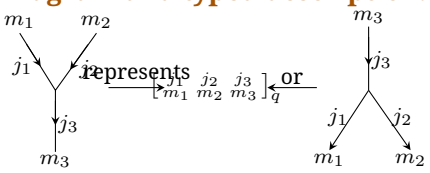


Diagram and typed description.



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GELFAND-P001466

(23)

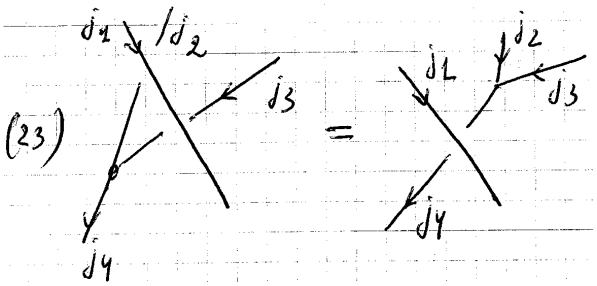
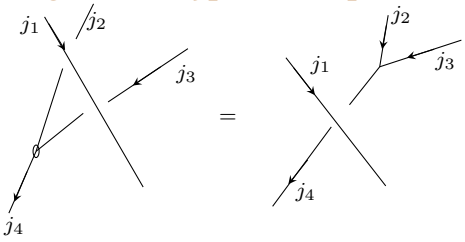


Diagram and typed description.



Original source details: GELFAND-P001466

(24)

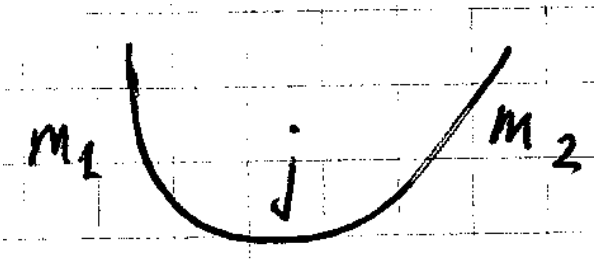


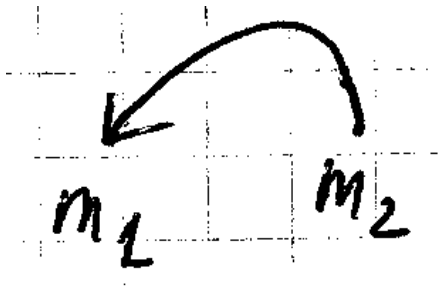
Diagram and typed description.

Original source details: GELFAND-P001466

$$\longrightarrow (-1)^{m_1} q^{m_1} \delta_{m_1,-m_2}$$

$-j \le m_i \le j, \quad i = 1, 2.$

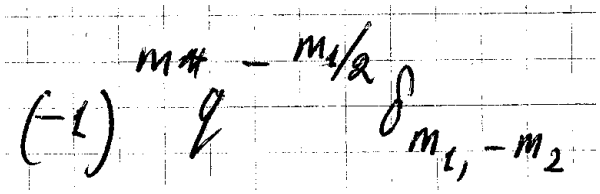
(25)



Original source details: GELFAND-P001466

Diagram and typed description.

$$\longrightarrow (-1)^{m[\text{crossed out}]}\, q^{-m_1/2} \delta_{m_1,-m_2}$$

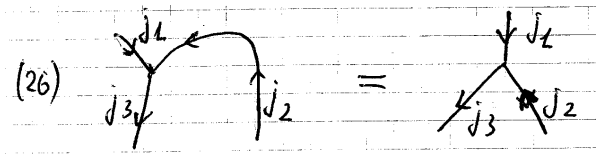


Original source details: GELFAND-P001466

Uncertain reading. A heavy cancellation follows m in the exponent of (-1). The exponent of q reads $-m_1/2$.

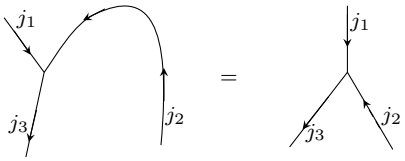
The following equalities hold:

(26)



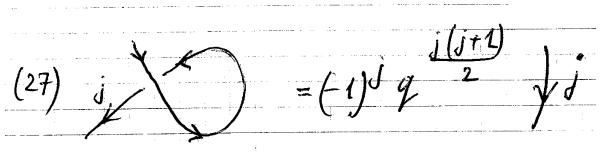
Original source details: GELFAND-P001466

Diagram and typed description.



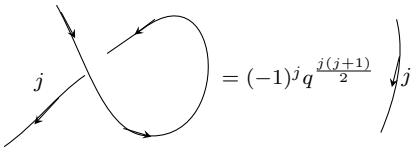
The arrow on the right j_2 branch is overwritten. The dotted arrow shows the possible direction towards the vertex.

(27)

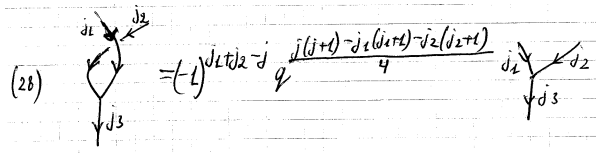


Original source details: GELFAND-P001466

Diagram and typed description.

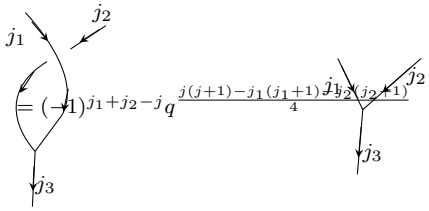


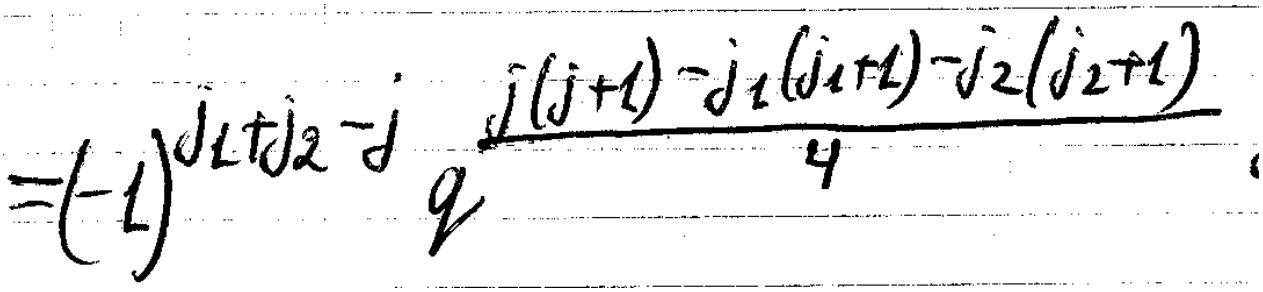
(28)



Original source details: GELFAND-P001466

Diagram and typed description.





Original source details: GELFAND-P001466

Uncertain reading. The coefficient uses j without a subscript. Both lower strands are labelled j_3.

This makes it possible to construct invariants of links, braids and knotted ribbons.

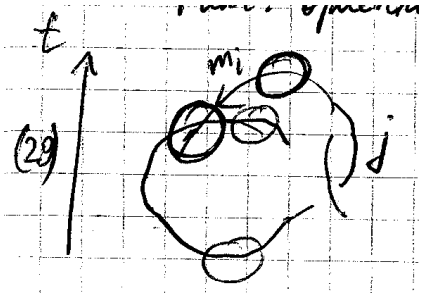
GELFAND-P001467

[source number: 66]

Link invariants.

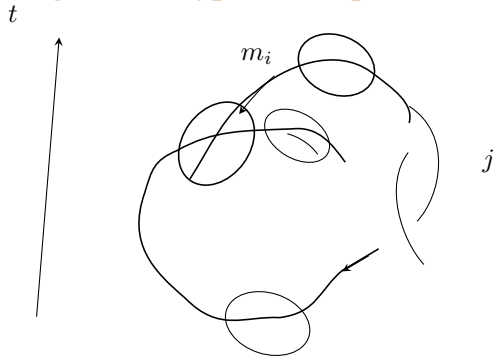
Fix an orientation and a direction t.

(29)



Original source details: GELFAND-P001467

Diagram and typed description.



The paths and individual arrow directions among the overlapping strokes are uncertain. The gaps at right are preserved.

Elementary pieces:

1) mark the points where the normal is $\|$ to the axis t (the chosen direction)

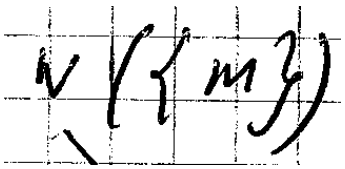
2) crossing points.

Place the numbers m_i at the joins between elementary pieces, so that

$$-j \le m_i \le j \qquad (j \text{ is chosen arbitrarily})$$

$$\sum_{\{m\}} \prod_{\substack{\text{over all} \\ \text{elementary} \\ \text{pieces}}} v(\{m\}) = \varphi_{j_1 \dots j_k}(D_L, q) \tag{30}$$

[From v a leader to the label] matrix entries computed above

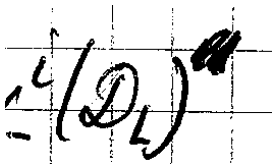


Uncertain reading. The letter before $(\{m\})$ may be v or ν.

Original source details: GELFAND-P001467

(31) Theorem

$$I_{j_1 \dots j_k}(L, q) = q^{\sum_i N_+^i(D_L) - N_-^i(D_L)} \varphi_{\{j\}}(D_L, q)$$



Uncertain reading. After $N_-^i(D_L)$ in the exponent, an illegible raised mark is crossed out.

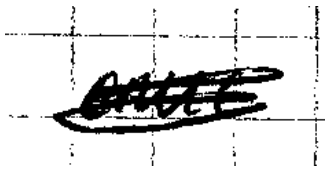
Original source details: GELFAND-P001467

— a link invariant (independent of the projection of the knot onto the plane).

$N_\pm^i$ — how many times the ith strand passes under and over

This contains the Jones invariant.

One can [crossed-out word] construct knot invariants from $\forall$ simple Lie algebras.



Uncertain reading. The word crossed out between “can” and “construct” is illegible.

Original source details: GELFAND-P001467

There is a conjecture that the SL_2-invariants suffice

There are also $6j$-symbols.

GELFAND-P001468

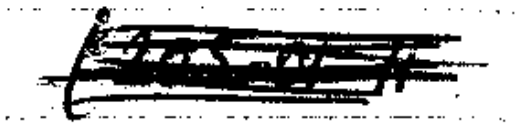
25.4.88. Gelfand Seminar

Ushveridze [crossed out] Alexander Grigoryevich. [crossed out]

The Riccati equation and quasi-exactly solvable problems.

Original source details: GELFAND-P001468

Uncertain reading. A short entry between the surname and the given name is heavily crossed out.



Original source details: GELFAND-P001468

Uncertain reading. Several horizontal strokes cross out the writing after the patronymic. The text is illegible.

$$y'(\lambda) + y^2(\lambda) + 2B(\lambda)y(\lambda) + C(\lambda) = 0, \tag{1}$$

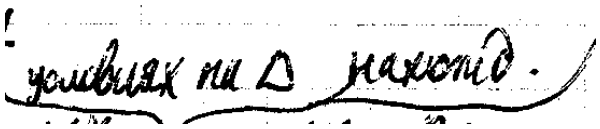
$$B(\lambda) = \sum_{\alpha=1}^{N} b_\alpha \Delta(\lambda - a_\alpha), \tag{2}$$

$$C(\lambda) = \sum_{\alpha=1}^{N} c_\alpha \Delta(\lambda - a_\alpha). \tag{3}$$

We seek a solution to (1) of the form

$$y(\lambda) = \sum_{i=1}^{M} \Delta(\lambda - \lambda_i) \tag{4}$$

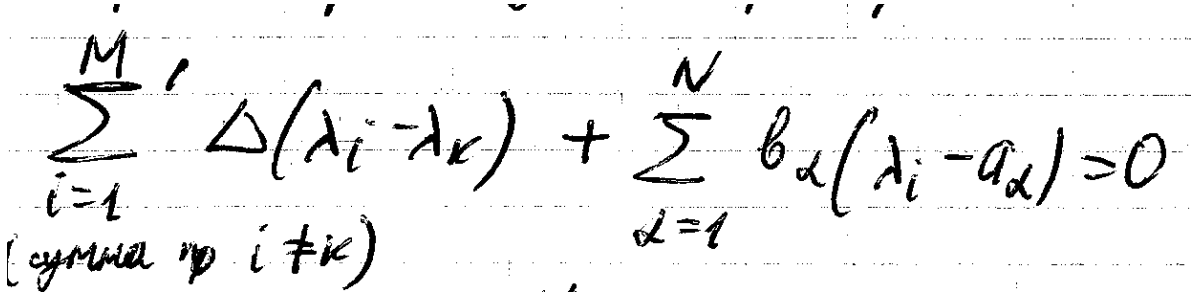
Then, under certain [inserted above: conditions on Δ, finding the] parameter values reduces to algebraic equations:



Original source details: GELFAND-P001468

Uncertain reading. The final Russian abbreviation in the insertion appears to read «нахожд.», for “finding”.

$$\begin{cases} \sum_{i=1}^{M}{}'\Delta(\lambda_i-\lambda_k)+\sum_{\alpha=1}^{N} b_\alpha(\lambda_i-a_\alpha)=0 \\ \text{(sum over } i\neq k\text{)} \\ c_\alpha = 2b_\alpha \sum_{i=1}^{M}\Delta(\lambda_i-a_\alpha) \end{cases} \tag{5}$$



Original source details: GELFAND-P001468

Uncertain reading. In the first line of (5), $(\lambda_i - a_\alpha)$ immediately follows b_α. The first argument is written $\lambda_i - \lambda_k$.

(6) Properties of Δ and Ω (generalization of the addition theorem for elliptic functions)

$$1) \quad \Delta(\lambda_1)\Delta(\lambda_2)+\Delta(\lambda_2)\Delta(\lambda_3)+\Delta(\lambda_3)\Delta(\lambda_1) + \Omega(\lambda_1)+\Omega(\lambda_2)+\Omega(\lambda_3)=0$$

when $\lambda_1+\lambda_2+\lambda_3=0$

$$2) \quad \Delta(\lambda)+\Delta(-\lambda)=0$$

3) [crossed out] $\Omega(\lambda)-\Omega(-\lambda)=0$

Original source details: GELFAND-P001468

Uncertain reading. In the crossed-out formula before $\Omega(\lambda)$, one sign resembles Δ. The remainder is illegible.

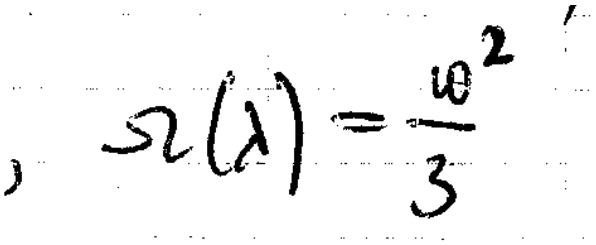
(7) Examples of solutions of system (6)

$$1) \quad \Delta(\lambda)=\frac{1}{\lambda}, \qquad \Omega(\lambda)=0;$$

Original source details: GELFAND-P001468

Uncertain reading. At right of the first pair of formulas, the crossed-out fragment may begin with Δ.

$$2) \quad \Delta(\lambda) = \omega \operatorname{ctg} \omega\lambda, \qquad \Omega(\lambda) = \frac{\omega^2}{3}$$

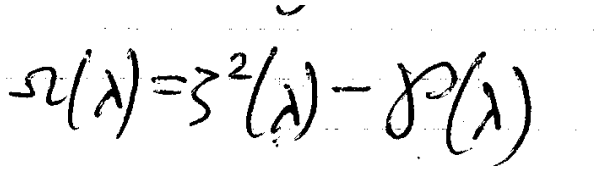


Uncertain reading. The manuscript gives $\omega^2/3$ with no minus sign.

Original source details: GELFAND-P001468

$$3) \quad \Delta(\lambda) = \zeta(\lambda), \qquad \Omega(\lambda) = \zeta^2(\lambda) - \wp(\lambda)$$

[Below $\zeta(\lambda)$, with a short leader] Weierstrass function



Uncertain reading. The manuscript gives $\zeta^2(\lambda) - \wp(\lambda)$ without a numerical factor.

Original source details: GELFAND-P001468

(quasiperio

Uncertain reading. The word and parenthesis are unfinished on this leaf.

Original source details: GELFAND-P001468

GELFAND-P001469

[source number: 67]

Applications

A. $y = \dfrac{f'}{f}$ $\Rightarrow$ Schrödinger equation for f;

The solution of (6) with a singularity at 0 satisfies the equation

$$\Delta'(\lambda) + \Delta^2(\lambda) = 2\Omega(\lambda) + \Omega(0) \tag{8}$$

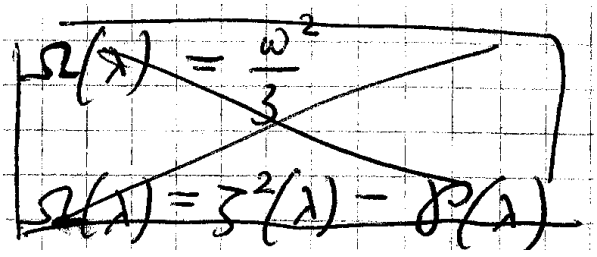


Diagram and typed description.

$\Omega(\lambda) = \dfrac{\omega^2}{3}$

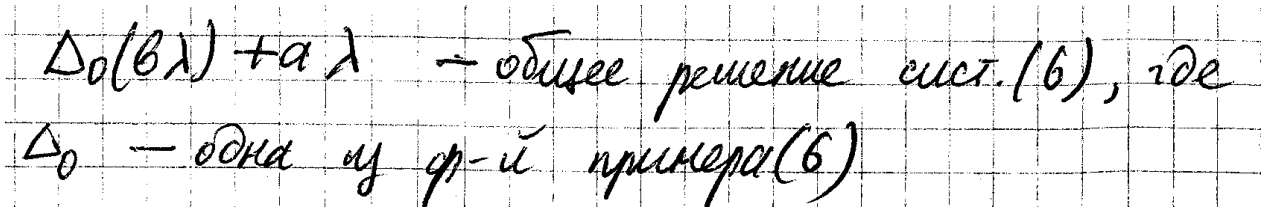
$\Omega(\lambda) = \zeta^2(\lambda) - \wp(\lambda)$

Both formulas are boxed and crossed out by two diagonals.

Original source details: GELFAND-P001469

(9) $\Delta_0(b\lambda) + a\lambda$ — the general solution of system (6), where

Δ_0 — one of the functions from example (6)

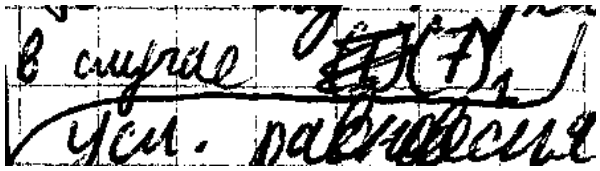


Original source details: GELFAND-P001469

Uncertain reading. There is no factor before $\Delta_0(b\lambda)$. The reference after "example" reads (6).

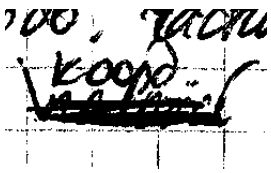
B. This also gives the Bethe ansatz solution for spin systems.

C. (5) $\Leftrightarrow$ [inserted above: in case [crossed out] $(7)_1$] equilibrium conditions for a classical system of charged particles of 2 kinds: fixed particles with coordinates a_α and charges b_α; mobile particles with [crossed out; inserted above: coordinates] λ_i and charge 1.



Original source details: GELFAND-P001469

Uncertain reading. An illegible entry is crossed out in the insertion "in case" before $(7)_1$.



Original source details: GELFAND-P001469

Uncertain reading. The abbreviation «коорд.» for "coordinates" is written above the crossed-out word.

[At right] Potential:

$$q_i q_k \ln|\lambda_i - \lambda_k|$$

In case $(7)_2$ the particles are on a circle with potential

$$q_i q_k \ln|\sin(\lambda_i - \lambda_k)|$$

In case $(7)_3$ the interaction potential is

$$q_i q_k \ln\theta(\lambda_i - \lambda_k)$$

(10) For the Schrödinger equation

λ_i — nodes of the wavefunction

In the Coulomb system

[crossed out] λ_i — coordinates of the charged particles

Original source details: GELFAND-P001469

Uncertain reading. A separate mark before λ_i is crossed out and illegible.

In the Bethe equation λ_i — rapidities

Stability of the quantum-mechanical system ⇔ stability of the Coulomb system

What to do with partial integrability? (with an incomplete system of integrals)

$$H = H_0 + L_1(H_1 - c_1) + L_2(H_2 - c_2), \qquad L_j \text{ — operators} \tag{11}$$

Let H_j commute for $j = 0, 1, 2$.

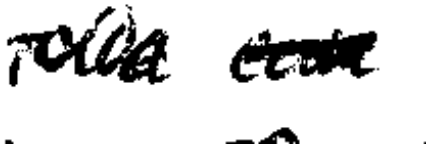
GELFAND-P001470

[Crossed out in the left margin]

Uncertain reading. The crossed-out mark is illegible.

Original source details: GELFAND-P001470

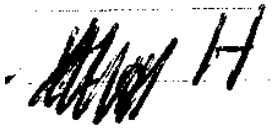
Consider a common eigenvector of H_0, H_1, H_2; then [crossed out]



Uncertain reading. The crossed-out word after “then” appears to be «если» (“if”).

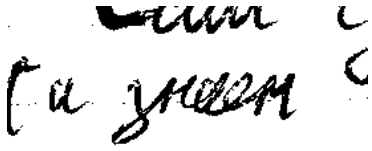
Original source details: GELFAND-P001470

If c_j — eigenvalues of H_j, they give an eigenvalue of [crossed out] H (and we know the eigenvectors — as many as the degeneracy)



Uncertain reading. The crossed-out strokes before H are illegible.

Original source details: GELFAND-P001470



Uncertain reading. The verb before “eigenvectors” tentatively reads «знаем» (“know”), possibly «умеем» (“can”).

Original source details: GELFAND-P001470

In this language

quantum nonlinear Schrödinger $\Longleftrightarrow$ harmonic oscillator

~~oper.~~ energy $\Longleftrightarrow$ particle-number operator

———— · ————

The spectrum of H_i is found from the Bethe ansatz equations

———— · ————

Reference:

Brief Reports on Physics, FIAN.

From the Riccati equation to the Schrödinger equation

This can be done in countless ways

$$y = \frac{f'}{f} \quad \Longrightarrow$$

$$f'' + 2Bf' + Cf = 0 \qquad \text{(the independent variable here is } \lambda\text{)} \tag{12}$$

Multiply [inserted above: (12)] by an arbitrary function A and then eliminate $f' \Longrightarrow$ [below the line (by changing the independent variable!)] $\Longrightarrow$ we obtain the Schrödinger equation

$$\left[-\frac{\partial^2}{\partial x^2} + V(x)\right] \Psi(x) = 0 \tag{13}$$

Each A gives its own potential V.

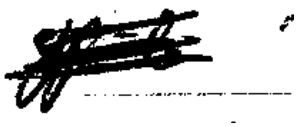
The function A can be chosen so that information about the numbers [crossed out] c_α, λ_i is concentrated in the x-independent part of the potential:

Uncertain reading. A small illegible entry is crossed out before c_α, λ_i.

Original source details: GELFAND-P001470

$$V(x) = U(x) - E(\lambda_i) \tag{14}$$

Then (13) [crossed out] $\sim$ the Schrödinger equation



Uncertain reading. An illegible entry is crossed out between (13) and the sign $\sim$.

Original source details: GELFAND-P001470

For this we need

$$A(\lambda) = \prod_{\alpha=1}^{N} \Xi(\lambda - a_\alpha), \qquad \frac{\Xi'(\lambda)}{\Xi(\lambda)} = \Delta(\lambda) \tag{15}$$

Change to x;

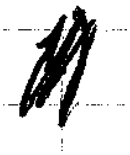
$$x = \int^{\lambda(x)} \frac{d\lambda}{\sqrt{A(\lambda)}} \tag{16}$$

GELFAND-P001471

[source number: 68]

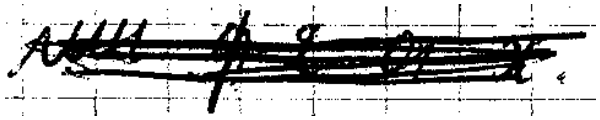
(17) [crossed out] In case $(7)_1$:

$$A(\lambda) = \prod_{\alpha=1}^{N} (\lambda - a_\alpha)$$



Original source details: GELFAND-P001471

Uncertain reading. The short crossed-out entry before “In case” is illegible.



Original source details: GELFAND-P001471

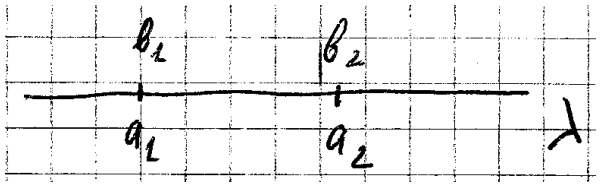
Uncertain reading. The crossed-out line reads “linear function of λ.”

($\boxed{\Xi}$ is a linear function of λ)

from here we get the Jacobi transformation (16)

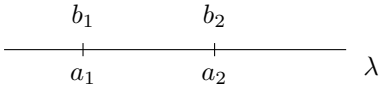
Use the Coulomb analogy to classify exactly and quasi-exactly solvable problems.

Let $N = 2$ in (2)–(3)



Original source details: GELFAND-P001471

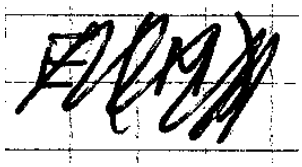
Diagram and typed description.



a_1, a_2 are the coordinates of the fixed particles (classical)

Mobile particles $\{\lambda_i\}$ — there are M of them
coord.

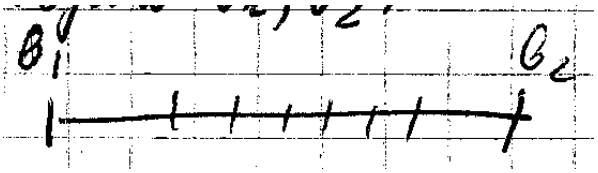
(18) [crossed out] The potential V does not contain M (!) $\Longrightarrow$ we have an exactly solvable model.



Original source details: GELFAND-P001471

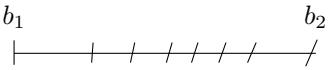
Uncertain reading. The crossed-out word before “The potential” tentatively reads «Если» (“If”).

(19) Let $b_1, b_2 > 0$



Original source details: GELFAND-P001471

Diagram and typed description.

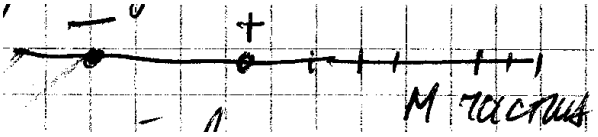


The particles inside cannot leave because of repulsion $\Longrightarrow$ $\exists$ min $\Longrightarrow$ $\exists$ a solution of the Schrödinger equation with M nodes $\Longrightarrow$ integrability!!!

(all the wave functions are found!!)

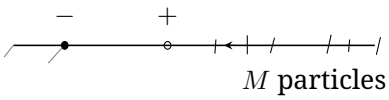
Another case

(20)

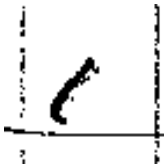


Original source details: GELFAND-P001471

Diagram and typed description.



The left end of the line is thickened. Faint strokes beneath it and the left point are shown in grey. An arrowhead at the left end cannot be made out with confidence.



Original source details: GELFAND-P001471

Uncertain reading. There is a short separate oblique stroke in the left margin below (20).

Stability condition for the system of M particles to the right of b_2:

$$b_1 + b_2 + M < 0 \tag{21}$$

(they do not approach b_2, and do not escape to ∞)

But this condition fails for large M! $\Longrightarrow$ only finitely many levels occur.

GELFAND-P001472

Choosing the intervals differently gives different problems.

Here we obtain exactly solvable models (magnets).

Beilinson A.

on the work of Atiyah

Original source details: GELFAND-P001472

Uncertain reading. The surname appears to read «Ати», rendered here as Atiyah. The final strokes also allow «Атии».

New invariants of 3- and 4-dimensional manifolds

(semi-infinite Morse theory)

$$\underset{\substack{\text{natural}\\ \infty\text{-dimensional manifolds}}}{X} \xrightarrow{f} \mathbb{R} \qquad \text{— Morse function.}$$

The critical points will contain ∞ many $+$ and $-$ squares, but different points differ by a finite number.

New homologies arise.

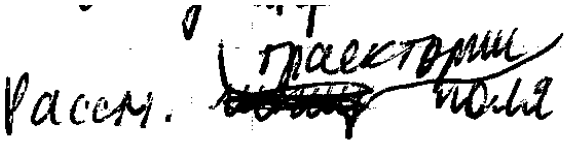
On the manifold X put a Riemannian metric

Suppose first that $\dim X < \infty$ and X is compact

g — metric,

$$\tau = \operatorname{grad} f$$

Consider [crossed out] [inserted above: trajectories] of the field τ.



Uncertain reading. The crossed-out word beneath "trajectories" is illegible.

Original source details: GELFAND-P001472

There are two types: points [below the line: (critical)] and trajectories connecting two critical points.

If the metric is in general position, consider the space $T(x_1, x_2)$ of all trajectories going from x_1 to x_2. Then

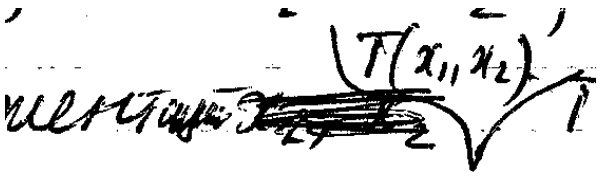
$$\dim T(x_1, x_2) = \operatorname{ind}_{x_1} f - \operatorname{ind}_{x_2} f - 1 \tag{1}$$

In particular, if the indices differ by 1, it is a discrete set.

Suppose

$$\operatorname{ind}_{x_1} f = \operatorname{ind}_{x_2} f + 1 \tag{2}$$

Then with each trajectory connecting x_1 and x_2 one associates a number ± 1, depending on the given orientations [crossed out] [inserted above: $T(x_1, x_2)$].



Uncertain reading. The crossed-out entry may contain x_1, x_2. The insertion above reads $T(x_1, x_2)$.

Original source details: GELFAND-P001472

Now construct the complex

GELFAND-P001473

[source number: 69]

$$\longrightarrow C_i \xrightarrow{d} C_{i-1} \longrightarrow \dots \tag{3}$$

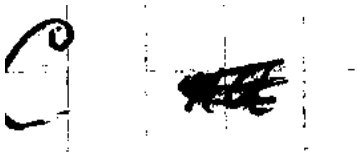
C_i — vector space with basis $\{x_\alpha\}$, ind $x_\alpha = i$
d is given by a matrix;
index of an entry = pair of points of adjacent index x_1, x_2

$$d = \|d_{x_1 x_2}\|,$$

$$d_{x_1 x_2} = \sum_{\substack{\text{trajectories} \\ \text{connecting } x_1, x_2}} \pm 1 \tag{4}$$

———— · ————

Let M be a compact 3-dimensional manifold.
Consider the space of all [inserted above: irreducible] connections C [crossed out]

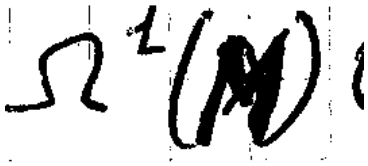


Original source details: GELFAND-P001473

Uncertain reading. The short entry to the right of C is crossed out and illegible.

[Beneath "connections": (Hermitian)]
on the trivial 2-dimensional bundle over M
Such a connection has the form $d + A$, where

$$A \in \Omega^1(M) \otimes \mathrm{su}_2$$



Original source details: GELFAND-P001473

Uncertain reading. The M is heavily retraced. The earlier strokes cannot be distinguished.

Consider them up to gauge equivalence

$$G \text{ — gauge group } SU_2^M, \qquad X = C/G \text{ — orbit space}$$

X — the main manifold.
On X define a Morse function, or more precisely its differential ν — a 1-form on C.
Tangent vector to C: matrix-valued 1-form
(C — affine space)

$$B \in \Omega^1(M) \otimes \mathrm{su}_2.$$

$$\langle \nu, B_\nabla \rangle = \int_M \mathrm{Tr}(F_\nabla \cdot B), \qquad B \in \Omega^1(M) \otimes \mathrm{su}_2 \tag{5}$$

[Beneath F_∇ with a leader: curvature of A]

[At right:] $F_\nabla \cdot B$ — exterior product, (3-form !!)

$$F_\nabla = dA + \tfrac{1}{2}[A, A] \tag{6}$$

GELFAND-P001474

(7) Properties of ν.

1. ν is pulled back from X

2. ν — closed 1-form

3. [inserted above: Primitive] ν — secondary characteristic class for c_2 (Chern class)

— Chern–Simons class for c_2:

$$\int_{M\times[0,1]} \mathrm{Tr}(F_{\widetilde{\nabla}})^2 \tag{8}$$

Uncertain reading. The first symbol with subscript 2 is heavily retraced and may be c_2 or e_2. The next line clearly has c_2.

Original source details: GELFAND-P001474

where

(9) [crossed out A_∇?] $\widetilde{\nabla}$ — connection on the trivial bundle over $M \times [0, 1]$

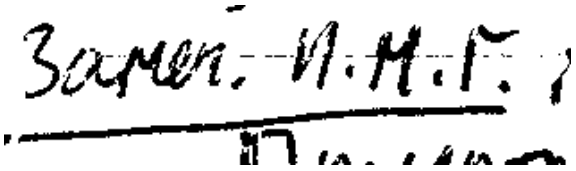
with potential at the point (x, t), equal to tA.

Uncertain reading. The crossed-out notation before $\widetilde{\nabla}$ resembles A_∇, possibly with a tilde in the subscript.

Original source details: GELFAND-P001474

This connection is on C, and not on X.

[small insertion between lines] Remark by I. M. G.: All of this is contained in the "simple" part of the work by Gabrielov–Gelfand–Losik

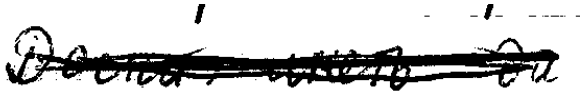


Uncertain reading. The middle initial resembles M. (M.), with H. (N.) also possible.

Original source details: GELFAND-P001474

Let us look at the critical points.

[crossed out]

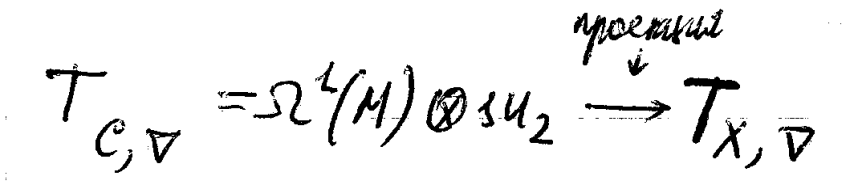


Uncertain reading. The short crossed-out line is illegible. Individual strokes remain above and below the cancellations.

Original source details: GELFAND-P001474

These are connections with curvature 0, i.e. integrable ones, i.e. representations of $\pi_1(M)$ Hessian.

$$\nabla \in C, \qquad T_{C,\nabla} = \Omega^1(M) \otimes \mathrm{su}_2 \xrightarrow{\text{projection}} T_{X,\nabla} \tag{10}$$

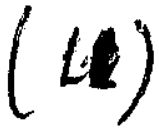


Original source details: GELFAND-P001474

Uncertain reading. Faint strokes above ∇ in the right subscript coincide with the ruling. A separate overbar cannot be established.

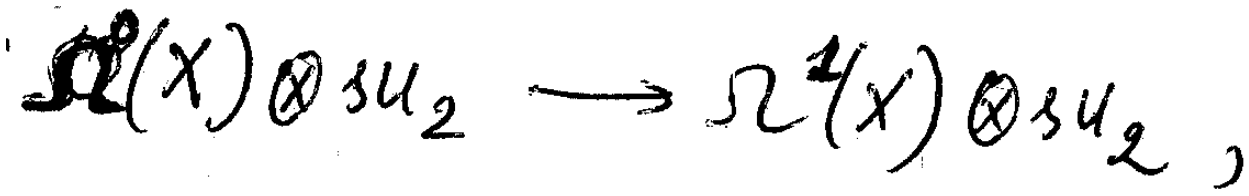
Introduce

$$\nabla_{\mathrm{ad}} \colon \Omega^0(X) \otimes \mathrm{su}_2 \longrightarrow \Omega^1(X) \otimes \mathrm{su}_2, \tag{11?}$$



Original source details: GELFAND-P001474

Uncertain reading. The second digit is heavily overwritten. The sequence of formulas suggests (11).



Original source details: GELFAND-P001474

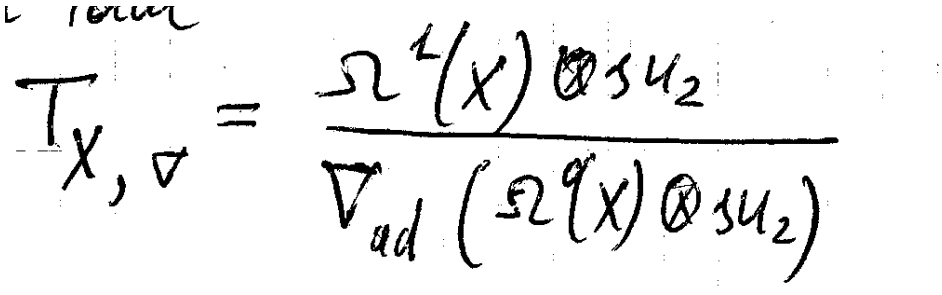
Uncertain reading. Both sides of the formula use X. The Ω^0 on the left is heavily retraced.

by the formula

$$\nabla_{\mathrm{ad}}(\varphi) = d\varphi + [A, \varphi] \tag{12}$$

and then

$$T_{X,\nabla} = \frac{\Omega^1(X) \otimes \mathrm{su}_2}{\nabla_{\mathrm{ad}}\left(\Omega^0(X) \otimes \mathrm{su}_2\right)} \tag{13}$$



Original source details: GELFAND-P001474

Uncertain reading. The numerator and denominator use X. Equation (10) uses M.

This holds for any connection.

GELFAND-P001475

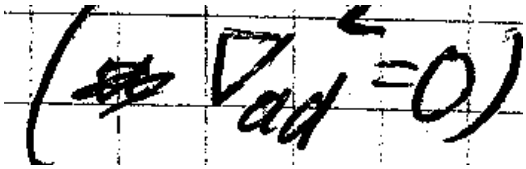
[source number: 70]

The Hessian is a quadratic form on the space (13), defined invariantly at a critical point.

$$H(\varphi) = \int_M \mathrm{Tr}(\nabla_{\mathrm{ad}}\varphi \cdot \varphi), \qquad \varphi \in T_{X,\nabla}. \tag{14}$$

$$\Omega^1(M) \otimes \mathrm{su}_2 \xrightarrow{\nabla_{\mathrm{ad}}} \Omega^2(M) \otimes \mathrm{su}_2. \tag{15}$$

If the connection was integrable, then ∇_{ad} is a complex ([crossed out] $\nabla^2_{\mathrm{ad}} = 0$)

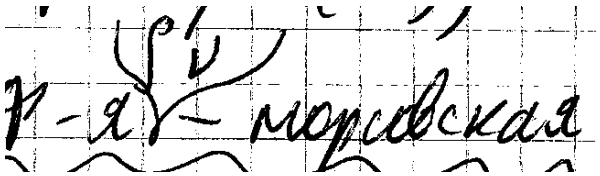


Original source details: GELFAND-P001475

Uncertain reading. The crossed-out sign before $\nabla^2_{\mathrm{ad}} = 0$ resembles φ or Φ.

well-defined $\nabla_{\mathrm{ad}}\varphi$ (independent of the representative in the quotient (13))
the function is Morse [inserted above: $\int \nu$?] $\iff$ ∇_{ad} has no degree-one cohomology, i.e.

$$H^1\left(M, \underset{\text{or } \forall \text{ local systems}}{\text{any representation of } \pi_1}\right) = 0 \tag{16}$$

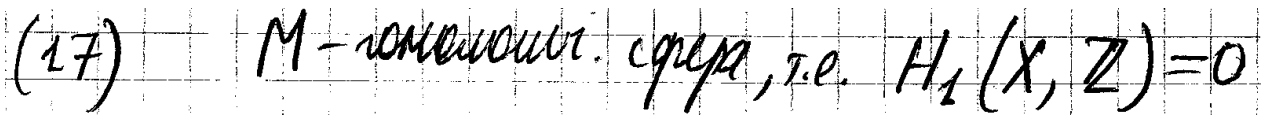


Original source details: GELFAND-P001475

Uncertain reading. The symbol after the integral sign may be ν or v.

This is equivalent to

$$M \text{ is a homology sphere, i.e. } H_1(X, \mathbb{Z}) = 0 \tag{17}$$



Original source details: GELFAND-P001475

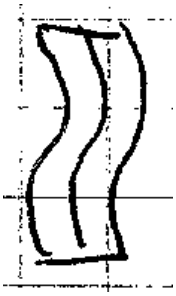
Uncertain reading. Equation (17) reads $H_1(X, \mathbb{Z}) = 0$, with M at the start of the line.

Following the standard scheme, consider gradient flows.

2 critical points $\longleftrightarrow$ 2 representations of π_1.

Join them by a trajectory of nonintegrable connections.

Choose a single coordinate system in a neighbourhood of the curve



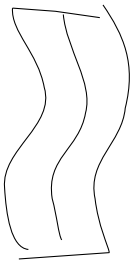


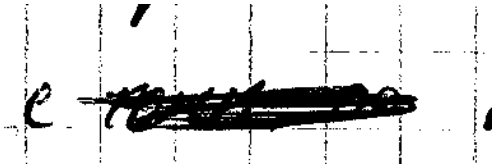
Diagram and typed description.

Original source details: GELFAND-P001475

and track the change in the matrix of second derivatives.

$$\begin{aligned}\text{Difference [inserted: of indices]} &= \text{number of points [inserted: on the path]} \\ &\quad \text{where a zero mode appears} \\ &= \text{spectral asymmetry.}\end{aligned} \tag{18}$$

This is done with [crossed out] the usual index theorem.



Uncertain reading. The crossed-out word after "with" is illegible.

Original source details: GELFAND-P001475

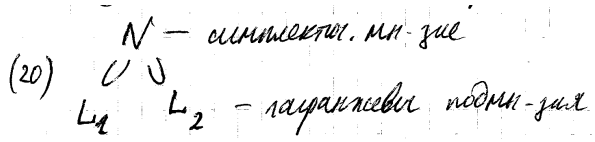
$$\text{The difference of indices } \underline{\mod 8} \text{ is well-defined!!} \tag{19}$$

GELFAND-P001476

Next example.

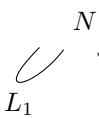
Computing the index from the finite-dimensional point of view. (Also for a homology sphere).

(20)



Original source details: GELFAND-P001476

Diagram and typed description.



N — symplectic manifold

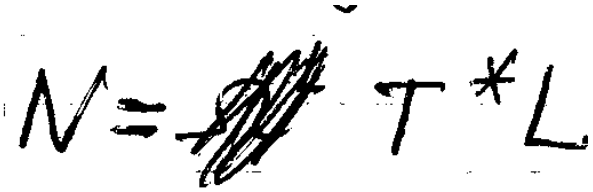
L_1 L_2 — Lagrangian submanifolds

Suppose

$$L_1 \cap L_2 \quad \text{is finite.} \tag{21}$$

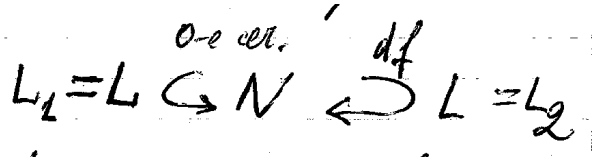
We want to compute the weighted number of intersection points.

(22) Example. $N =$ [crossed out] T^*L,



Original source details: GELFAND-P001476

Uncertain reading. The crossed-out sign between $N =$ and T^*L may be Ω.



Original source details: GELFAND-P001476

Diagram and typed description.

zero section $\qquad df$

$L_1 = L \hookrightarrow N \hookleftarrow L = L_2$

[Beneath $L_1 = L$ crossed out]

Original source details: GELFAND-P001476

Uncertain reading. The entry beneath $L_1 = L$ is crossed out. It may be H_2 or L_2.

f is Morse

$L_1 \cap L_2$ — critical points of f

they intersect transversely.

to an intersection point — an index

$$L_1 \cap L_2 \xrightarrow{\text{index}} \{0, 1, \dots, \dim L\}$$

$$\chi(N) = \sum_{x \in L_1 \cap L_2} (-1)^{\text{index}\, x} \tag{23}$$

General situation (20): define a relative index!

(24) X — paths joining L_1 and L_2

$$\begin{aligned} [0,1] &\longrightarrow N \\ 0 &\longrightarrow L_1 \\ 1 &\longrightarrow L_2 \end{aligned}$$

On such an X there is a multivalued Morse function

Critical points — points of intersection of the L_j

$\varphi \in X$, τ — tangent vector at φ, i.e. a vector field along φ

[From φ a line leads down to the label «path».]

GELFAND-P001477

[source number: 71]

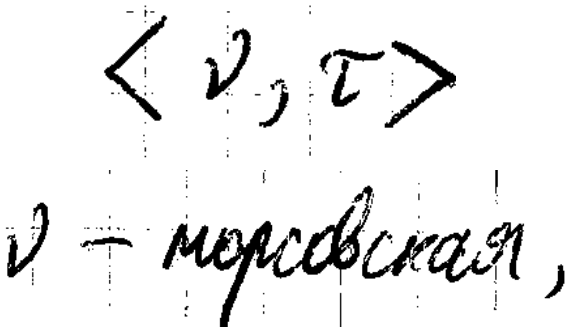
Then

$$\langle \nu, \tau \rangle := \int_0^1 \omega(\dot{\varphi}, \tau), \tag{25}$$

where

ω — symplectic form.

ν is Morse if L_1 and L_2 intersect transversely.



Uncertain reading. The two signs have the same shape and are rendered ν. The strokes also allow Latin v.

Original source details: GELFAND-P001477

The difference of indices is defined, but there is also an elementary construction (using the Maslov index)

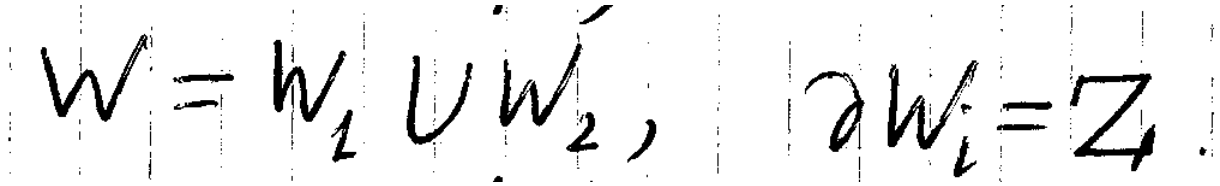
Consider a manifold M — a homology sphere. !

(26) Theorem

M can be cut along the Riemann surface Z,

(Heegaard splitting)

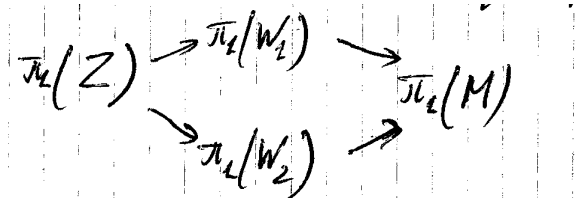
$$W = W_1 \cup W_2, \qquad \partial W_i = Z.$$



Original source details: GELFAND-P001477

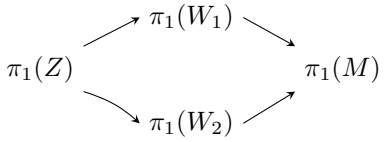
Uncertain reading. The formula has W. The preceding text has M.

(27)



Original source details: GELFAND-P001477

Diagram and typed description.



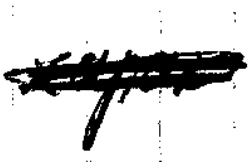
[at right] images determine $\pi_1(M)$

Original source details: GELFAND-P001477

Uncertain reading. The first word of the note at right tentatively reads «образы» (“images”).

$\pi_1(W_j)$ — free group on g generators

Representations of $\pi_1(M)$ [crossed out] — in it $\pi_1(W_j)$ are Lagrangian manifolds.

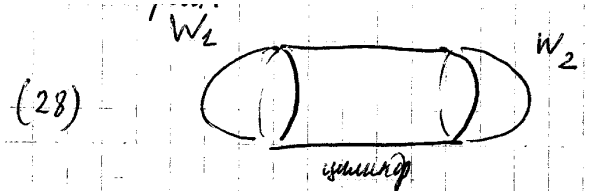


Original source details: GELFAND-P001477

Uncertain reading. The word after $\pi_1(M)$ is crossed out. Individual letters cannot be distinguished reliably.

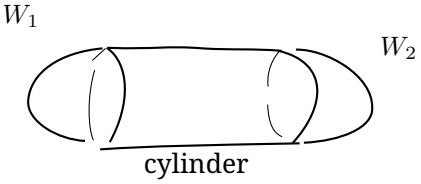
Let us construct

(28)



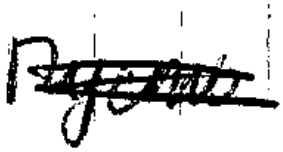
Original source details: GELFAND-P001477

Diagram and typed description.



Faint curves are shown as thin lines. The short slanting strokes at the upper joins may be part of the outline.

[crossed out] A path is a family of connections.



Original source details: GELFAND-P001477

Uncertain reading. The crossed-out word before “A path” starts with a letter resembling «P» (R). The remaining strokes cannot be read with confidence.

Second half of the work:

replace flat connections with instantons,

pass to M^4 and

Donaldson invariants

(M^4 is cut into 2 three-dimensional ones).

Original source details: GELFAND-P001477

Uncertain reading. The final words inside the parentheses are “2 three-dimensional ones,” followed by the closing parenthesis.

GELFAND-P001478

12.9.88 Gelfand Seminar.

1. «What is the smallest dimension of a space in which two two-dimensional planes can be skew?»

2. What is a determinant?

Two first-order equations can be replaced by [inserted above: one] second-order equation. How?

$$\begin{cases} \left(\frac{d}{dx} + q_{11}\right)\varphi_1 + q_{12}\varphi_2 = 0 \\ q_{21}\varphi_1 + \left(\frac{d}{dx} + q_{22}\right)\varphi_2 = 0 \end{cases} \tag{1}$$

a) First method: solve the first for φ_2 and substitute into the second

Second method: solve the second for φ_1 and substitute into the first

We get different equations!

but having solved one, we can solve the second.

This is known as the “Bäcklund transformation”.

We can take a linear combination of φ_1 and φ_2 and then a family of equations arises.

Uncertain reading. The first subscript is tentatively 1. It resembles 2, but its top stroke is closer to the 1 in (1).

Original source details: GELFAND-P001478

The answers are very beautiful

b) What about three first-order equations?

This is more complicated and the beauty disappears.

Gelfand and Retakh did it.

There is work by Sato and Kashiwara on noncommutative determinants.

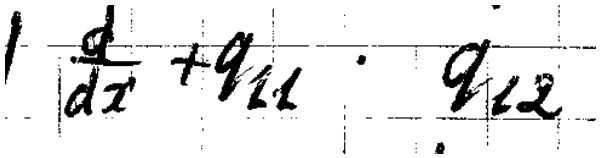
c) Let $q_{ij} = \text{const}$

[crossed out at left] Then there is a determinant

Uncertain reading. The crossed-out label before the closing parenthesis may be «r)» or «1)».

Original source details: GELFAND-P001478

$$\begin{vmatrix} \frac{d}{dx} + q_{11} & \cdot & q_{12} \\ q_{21} & & \frac{d}{dx} + q_{22} \end{vmatrix} = \left(\frac{d}{dx}\right)^2 + (q_{11} + q_{22})\frac{d}{dx} + (q_{11}q_{22} - q_{12}q_{21}) \tag{2}$$



Uncertain reading. A separate dot appears between the top entries of the determinant. Its purpose is unclear.

Original source details: GELFAND-P001478

$\frac{d}{dx} \longleftrightarrow \lambda \quad \Longrightarrow$ we get the characteristic determinant of the matrix

in the general case we get [inserted above: (noncommutative analogue of)] the characteristic determinant.

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{— ?}$$

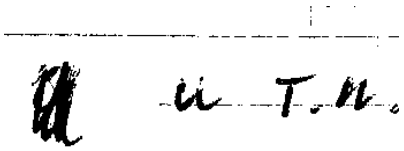
a_{ij} — a map $E_j \to E_i$,

so $a_{11}a_{22} - a_{12}a_{21}$ does not make sense.

There are 4 determinants in all:

$$a_{11} - a_{12}a_{22}^{-1}a_{21} \tag{3}$$

[crossed out] and so on.



Uncertain reading. Before “and so on” a heavily crossed-out sign resembles a or «и».

Original source details: GELFAND-P001478

GELFAND-P001479

[source number: 84]

Third order: 36 ways of writing it.

There arise noncommutative K-theory and noncommutative homologies.

c) 8 letters arranged in a cube $2 \times 2 \times 2$.

Uncertain reading. The overwritten label resembles «в)» (letter c), with «г)» (letter d) also possible.

Original source details: GELFAND-P001479

What is the determinant?

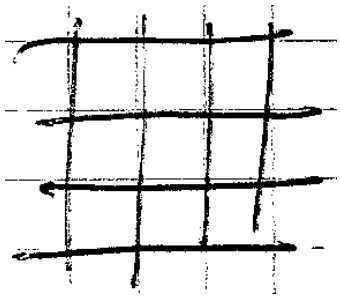
Cayley did this.

It is thought that after Hilbert the results of Gordan, Cayley, Clebsch and others lost their interest. They were declared old-fashioned nonsense. But that is untrue.

Moore. (GREGORY MOORE) (I.AS. Princeton)

Conformal field theory. (CFT)

Ising model



Original source details: GELFAND-P001479

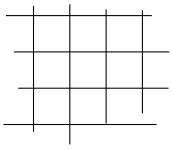
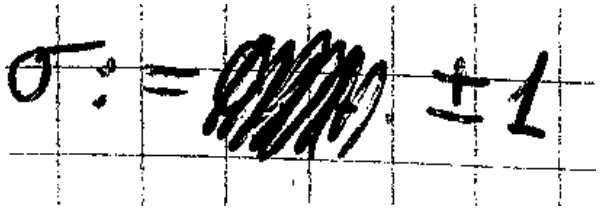


Diagram and typed description.

2-dimensional lattice [beneath “lattice”: (square)], at every point a variable $\sigma :=$ [crossed out] ± 1



Original source details: GELFAND-P001479

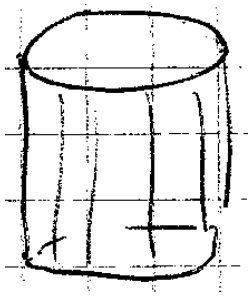
Uncertain reading. The entry between $\sigma :=$ and ± 1 is crossed out and illegible.

$$\text{Energy} \quad E = \sum_{\langle ij \rangle} \sigma_i \sigma_j \tag{1}$$

$\langle ij \rangle$ — neighbouring points

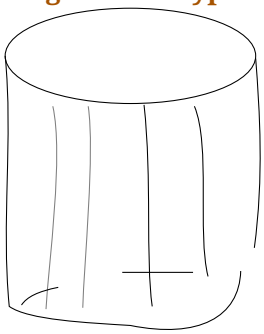
Partition function

$$Z = \sum e^{-\beta E} \tag{2}$$



Original source details: GELFAND-P001479

Diagram and typed description.

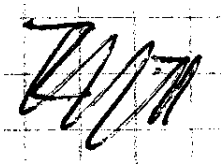


Two faint inner lines are shown in grey. The lower junctions and the purpose of the short strokes are unclear.

Consider it on a cylinder and refine the lattice

The asymptotics are expressed in terms of the functions

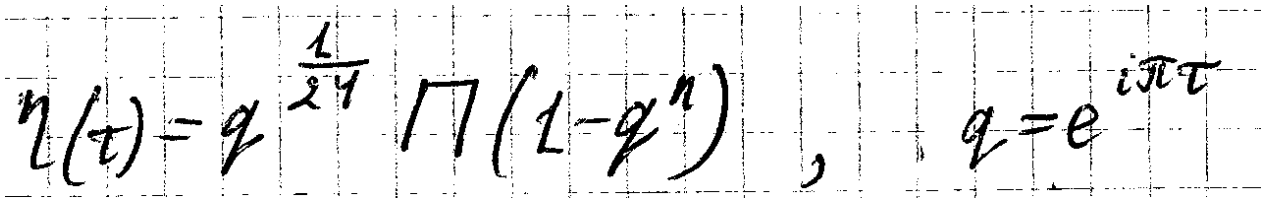
[crossed out at left]



Uncertain reading. The crossed-out entry begins with a sign resembling Z. The rest is illegible.

Original source details: GELFAND-P001479

$$\eta(\tau) = q^{1/24} \prod (1 - q^n), \qquad q = e^{i\pi\tau}$$



Original source details: GELFAND-P001479

Uncertain reading. The manuscript combines $q = e^{i\pi\tau}$ with $q^{1/24} \prod (1 - q^n)$. No product limits are written.

$$\theta_3(\tau) = \sum e^{i\pi\tau n^2}$$

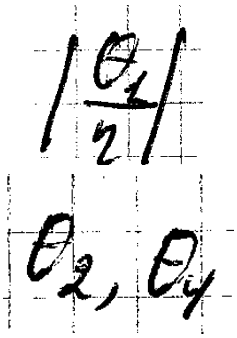
and similarly θ_2, θ_4

[above the line: $\tau = i\beta$]

Invariant under the substitution

$$\tau \longrightarrow -1/\tau$$

$$Z \longrightarrow \left|\frac{\theta_2}{\eta}\right| + \left|\frac{\theta_3}{\eta}\right| + \left|\frac{\theta_1}{\eta}\right| \qquad \text{(up to a factor)}$$



Uncertain reading. The final subscript resembles 1. In the upper excerpt it differs from the 4 of the earlier θ_4, shown below.

Original source details: GELFAND-P001479

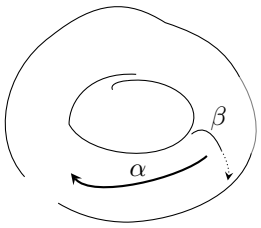
$$Z(\beta) = Z\left(\frac{1}{\beta}\right)$$

GELFAND-P001480

In CFT we associate real analytic functions to surfaces

Original source details: GELFAND-P001480

Diagram and typed description.



A short hooked stroke appears at the upper edge of the hole. An arrowhead there cannot be made out with confidence. The arc β has gaps.

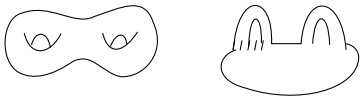
$$= \sum_{i=1}^{N} f_i(\tau)\bar{g}_i(\tau), \qquad f_i := \left\{ \sqrt{\frac{\theta_2}{\eta}}, \sqrt{\frac{\theta_3}{\eta}}, \sqrt{\frac{\theta_4}{\eta}} \right\}$$

$$\tau \to -\frac{1}{\tau}, \qquad \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} \to \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix}$$

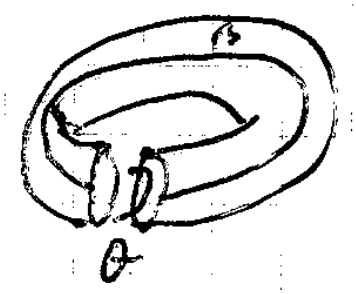
Other variants:

Original source details: GELFAND-P001480

Diagram and typed description.

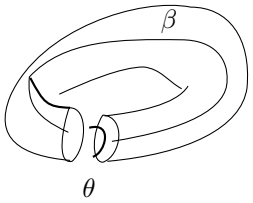


On every surface there is a corresponding model of CFT!!



Original source details: GELFAND-P001480

Diagram and typed description.



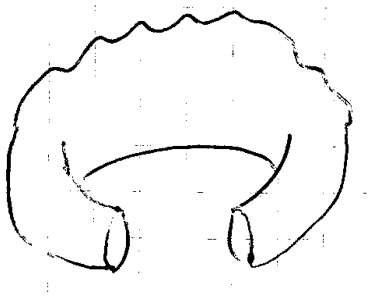
A heavy angular stroke appears at the left inner edge. Its purpose is unclear.

$$= f(\beta + i\theta, \beta - i\theta) \qquad \text{(real analytic functions)}$$

one can twist by θ

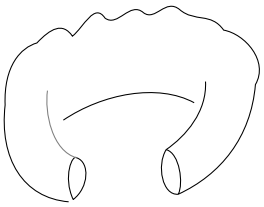
The torus parameters are 2 real numbers

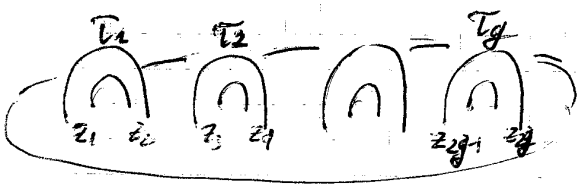
$$\tau = L + i\theta$$



Original source details: GELFAND-P001480

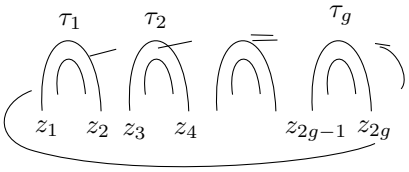
Diagram and typed description.





Original source details: GELFAND-P001480

Diagram and typed description.



Above the handles are τ_1, τ_2, τ_g. The rightmost subscript is heavily retraced and tentatively reads $2g$.

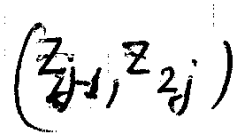
the lengths and twist rotations of the handles matter (handles)

$$\begin{array}{ccc} 0 & 1 & \infty \\ \uparrow & \uparrow & \uparrow \\ z_1 & z_2 & z_3 \end{array}$$

We obtain $3g - 3$ complex parameters. $\qquad (g \geq 2)$

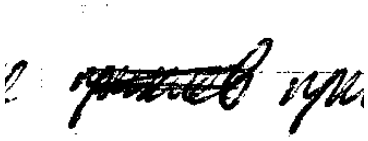
$$f(z_i, \tau_i; \bar{z}_i, \bar{\tau}_i)$$

(z_{2j-1}, z_{2j}) — points at which [crossed out] the handles are glued (thin).



Original source details: GELFAND-P001480

Uncertain reading. The first subscript tentatively reads $2j - 1$. The 2 overlaps the base z. The second subscript is $2j$.



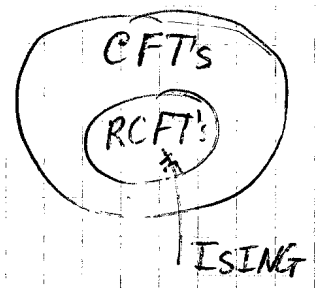
Original source details: GELFAND-P001480

Uncertain reading. The word before "glued" is crossed out. Loops and a long final stroke are visible.

τ_j — as in the torus.

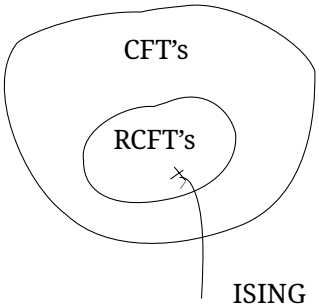
GELFAND-P001481

[source number: 85]



Original source details: GELFAND-P001481

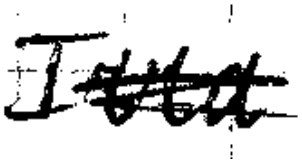
Diagram and typed description.



Inside the smaller oval, the line from ISING meets a mark resembling a cross or an arrowhead.

R — Rational

I [crossed out] RATIONAL. (the name is historical)



Original source details: GELFAND-P001481

Uncertain reading. The letters after the initial Latin I are crossed out and cannot be read with confidence.

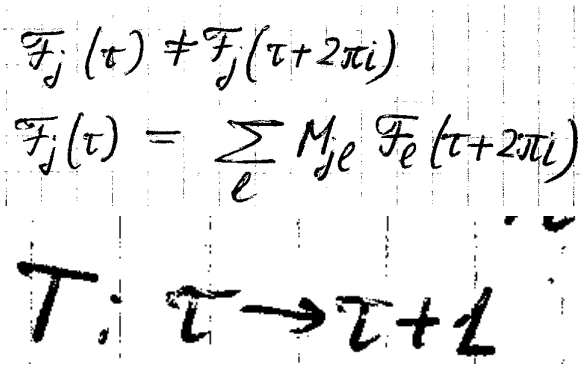
$$f(z_i,\tau_i,\bar{z}_i,\bar{\tau}_i)=\sum_{j=1}^{N<\infty}\mathcal{F}_j(z_i,\tau_i)\overline{\mathcal{F}}_j(\bar{z}_i,\bar{\tau}_i)$$

Torus:

$$\theta\to\theta+2\pi$$

$$\mathcal{F}_j(\tau)\neq\mathcal{F}_j(\tau+2\pi i)$$

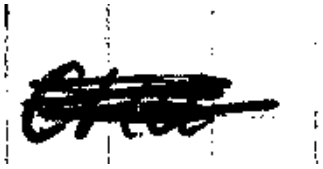
$$\mathcal{F}_j(\tau)=\sum_{\ell}M_{j\ell}\mathcal{F}_\ell(\tau+2\pi i)$$



Original source details: GELFAND-P001481

Uncertain reading. The two upper formulas have $\tau+2\pi i$. At the foot of the page, $T:\tau\to\tau+1$.

$M =$ monodromy matrix. [crossed out]



Original source details: GELFAND-P001481

Uncertain reading. The word after "monodromy matrix" is crossed out and illegible.

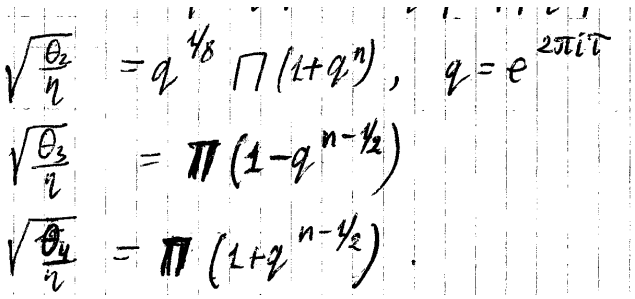
E.g. Ising

$$Z(\tau,\bar\tau) = \left|\sqrt{\frac{\theta_2}{\eta}}\right|^2 + \left|\sqrt{\frac{\theta_3}{\eta}}\right|^2 + \left|\sqrt{\frac{\theta_4}{\eta}}\right|^2$$

$$\sqrt{\frac{\theta_2}{\eta}} = q^{1/8}\prod(1+q^n), \qquad q = e^{2\pi i\tau}$$

$$\sqrt{\frac{\theta_3}{\eta}} = \prod(1-q^{n-1/2})$$

$$\sqrt{\frac{\theta_4}{\eta}} = \prod(1+q^{n-1/2}).$$



Original source details: GELFAND-P001481

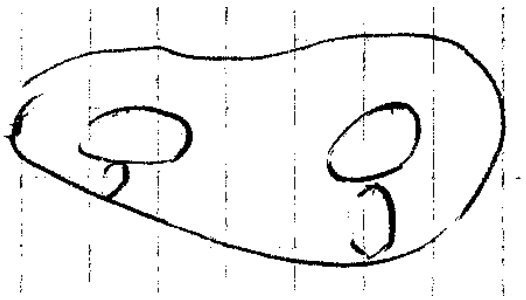
Uncertain reading. The manuscript has exponent $1/8$, minus in the θ_3 row and plus in the θ_4 row. The products have no written limits.

I. M. G. Elliptic functions are trigonometry with a parameter q.

$\mathcal{F}_j$ — are not invariant, but f is invariant

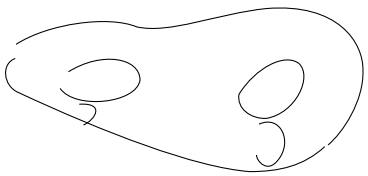
($\overline{\mathcal{F}}_j$ need not be complex conjugates of $\mathcal{F}_j$!)

The braid group appears.



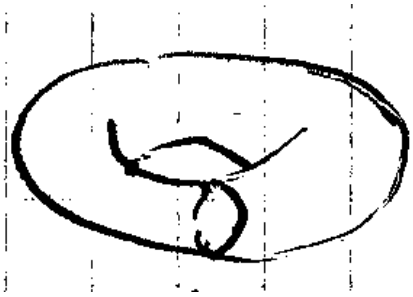
Original source details: GELFAND-P001481

Diagram and typed description.



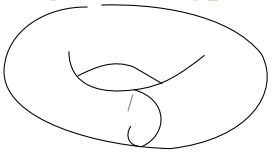
Generalize.

$g = 1$



Original source details: GELFAND-P001481

Diagram and typed description.



$$S : \tau \to -1/\tau \qquad T : \tau \to \tau + 1$$
$$S^2 = 1, \qquad (ST)^3 = I$$

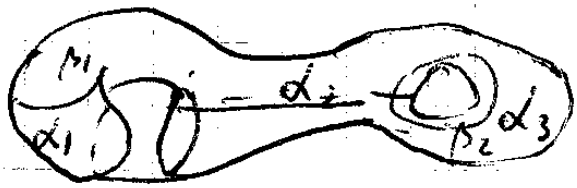
GELFAND-P001482

"DEHNTWIST"

Original source details: GELFAND-P001482

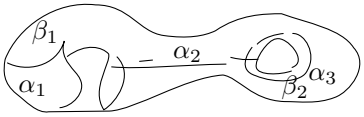
Uncertain reading. The H is overwritten. The earlier strokes cannot be read separately.

$g = 2$



Original source details: GELFAND-P001482

Diagram and typed description.



Where the strokes overlap near the holes, some of the curve junctions are indistinct.

$$z_i, \tau_i \to \tilde{z}_i, \tilde{\tau}_i$$

$$\alpha_1\beta_1\alpha_1 = \beta_1\alpha_1\beta_1$$
$$\alpha_2\beta_2\alpha_2 = \beta_2\alpha_2\beta_2$$
$$\alpha_2\beta_1\alpha_2 = \beta_1\alpha_2\beta_1 \qquad (\beta_1 \to \beta_2)$$
$$\alpha_1\alpha_2 = \alpha_2\alpha_1$$
$$\beta_1\beta_2 = \beta_2\beta_1 \qquad \text{etc.}$$
$$(\alpha_1\beta_1\alpha_2)^4 = \alpha_3^2$$
$$(\alpha_3\beta_2\alpha_2\beta_1\alpha_1^2\beta_1\alpha_2\beta_2\alpha_3)^2 = 1$$
$$\beta_1^{-1}\alpha_1\beta_1 = \alpha_1\beta_1\alpha_1^{-1}.$$

——— · ———

CFT.

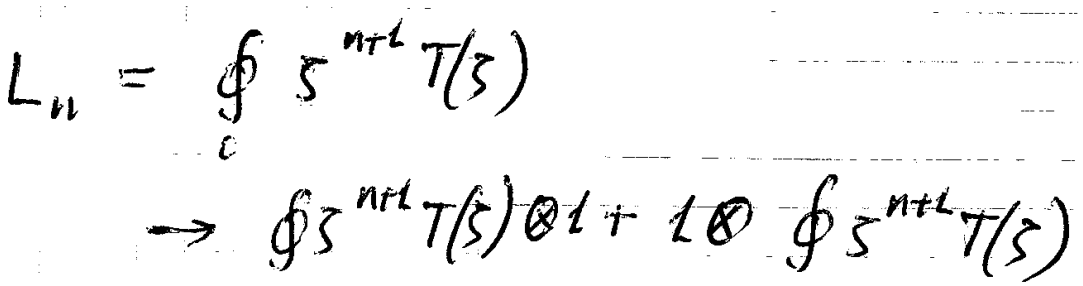
$$\mathcal{A} = \{(\sigma, \Delta) \text{ fields}\} = \text{operator algebra}$$

$$\mathcal{A} \qquad \{T(z), T'(z), \ldots\}$$

$$\mathcal{H} = \bigoplus_{i=1}^{N} \mathcal{H}_i \underset{\mathcal{A}}{\otimes} \mathcal{H}_{\bar{i}} \underset{\bar{\mathcal{A}}}{}$$

$$\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$$

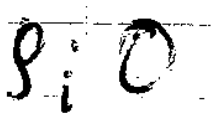
$$L_n = \oint_c \zeta^{n+1} T(\zeta)$$
$$\to \oint \zeta^{n+1} T(\zeta) \otimes 1 + 1 \otimes \oint \zeta^{n+1} T(\zeta)$$



Original source details: GELFAND-P001482

Uncertain reading. The variable resembles ζ or ξ, and the mark below the first integral resembles c or C. The differential is unwritten.

$\rho_i : \mathcal{A} \to \operatorname{End} \mathcal{H}_i, \qquad \rho_i\, C$



Original source details: GELFAND-P001482

Uncertain reading. The last mark resembles C or O. There is no visible continuation.

. . . . . .

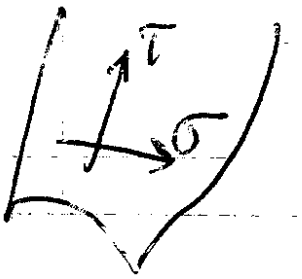
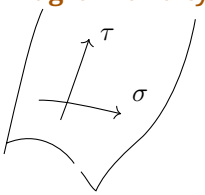
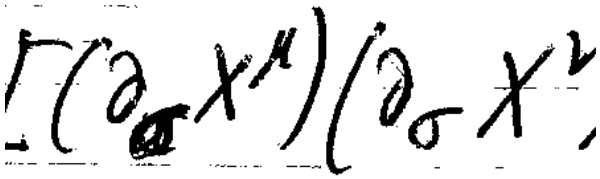


Diagram and typed description.



Original source details: GELFAND-P001482

$$S = \int g_{\mu\nu}(X)[(\partial_\sigma X^\mu)(\partial_\sigma X^\nu) - (\partial_\tau X^\mu)(\partial_\tau X^\nu)]$$



Uncertain reading. The first subscript σ is heavily retraced. The second subscript in the same product supports this reading.

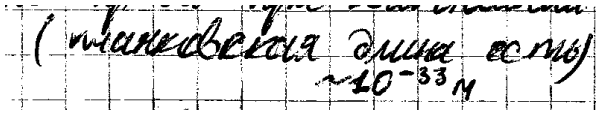
Original source details: GELFAND-P001482

Conformal Invariance $\rightarrow R_{\mu\nu} = 0$.

GELFAND-P001483

[source number: 86]

I. M. G. What should replace the notion of space in quantization?
(the Planck length is $\sim 10^{-33}$ m)



Uncertain reading. After 10^{-33}, the manuscript has a single sign read as “m”.

Original source details: GELFAND-P001483

1st variant: something discrete.

Riemann understood that this was important and gave thought to the notion of space.

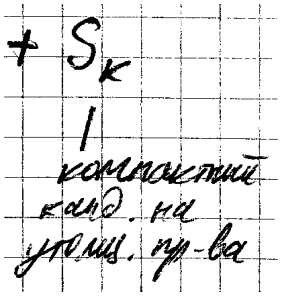
A thin film around a 4-dimensional surface.

A sensible exercise: naive geometry of 4-dimensional polyhedra.

—— · ——

$$S = \underbrace{\int_{R^4} g_{\mu\nu}(X)\left[(\partial_\sigma X^\mu)(\partial_\sigma X^\nu) - (\partial_\tau X^\mu)(\partial_\tau X^\nu)\right]}_{\substack{c_{R^4}\ +c_K\ =26 \\ \| \\ 4}} + S_K$$

[Note joined by a stroke to S_K:] compact candidate for a thickening of space

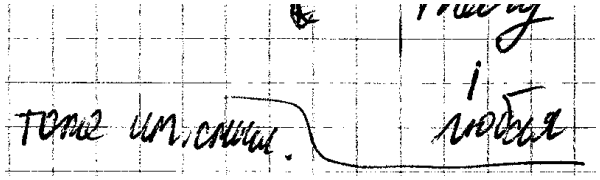


Original source details: GELFAND-P001483

Uncertain reading. Tentatively read as «канд. на утолщ. пр-ва». The first abbreviation may be «конд.», and the middle strokes of the abbreviation before «пр-ва» are indistinct.

can be replaced by ∀ other model with certain invariance properties

$$c\begin{pmatrix}\text{conformal Field}\\ \text{Theory}\end{pmatrix} — ?$$

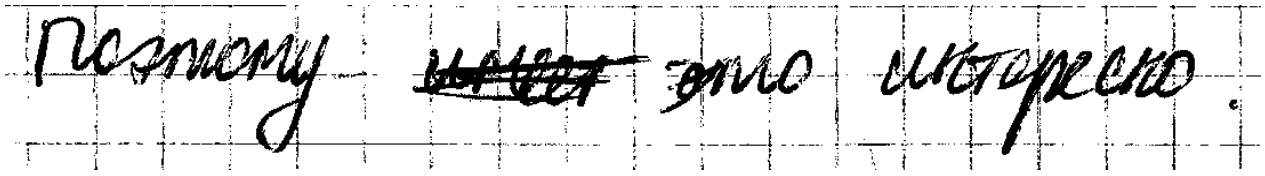


Original source details: GELFAND-P001483

Diagram and typed description.

Theory
/
also makes sense. any

Therefore [crossed out] this is interesting.



Original source details: GELFAND-P001483

Uncertain reading. The crossed-out word resembles «имеет» or «имеют», both forms of “has/have”.

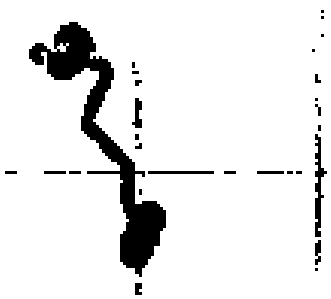
19.9.88 Gelfand Seminar.

Morozov A.

Conformal field theory (following Moore's talk).

String theory has been developing since 1984.

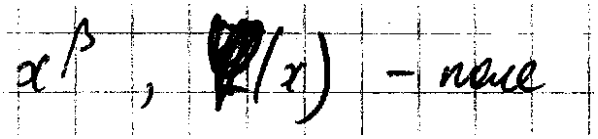
Starting picture



Original source details: GELFAND-P001483

Diagram and typed description.

x^β, $\varphi(x)$ — a field.



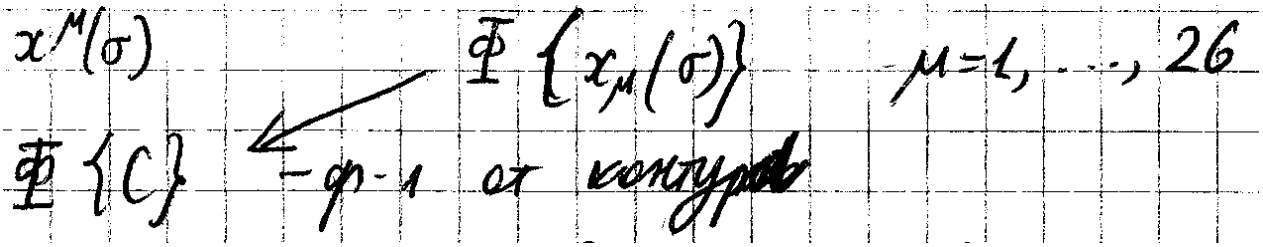
Original source details: GELFAND-P001483

Uncertain reading. The letter before (x) is overwritten and resembles φ or Φ.

$\Phi\{x_\mu(\sigma)\}$ $\mu = 1, \dots, 26$

$\Phi\{C\}$ ↙

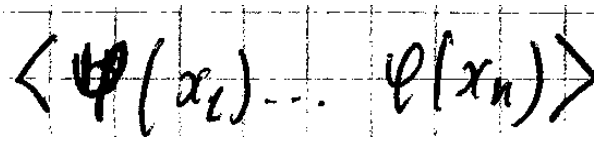
— functional of contours



Original source details: GELFAND-P001483

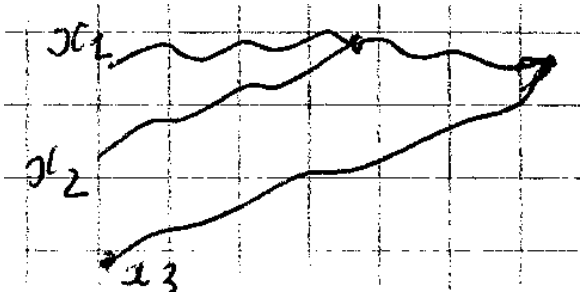
Uncertain reading. The ending is overwritten. "Of the contour" is also possible.

$$(1) \quad \langle \varphi(x_1) \dots \varphi(x_n) \rangle = \int D\varphi \, e^{\Sigma(\varphi)} \varphi(x_1) \dots \varphi(x_n)$$



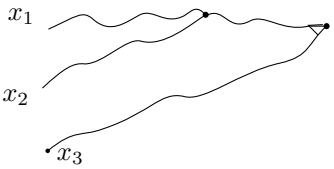
Original source details: GELFAND-P001483

Uncertain reading. The first field symbol in (1) is retraced. The second factor supports φ. Φ is also possible.



Original source details: GELFAND-P001483

Diagram and typed description.



The dot at the first junction is visible. At the right, short strokes form a dark tip or dot.

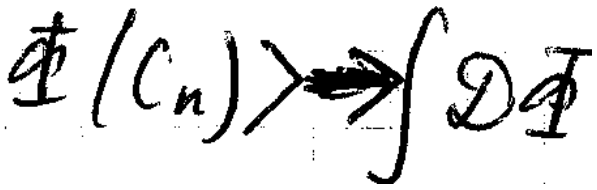
(There were 3 particles at the start, and nothing was left at the end)

$\Sigma(\varphi)$ is called the quantum action

GELFAND-P001484

We want the same in string theory

$$(2) \quad \langle \Phi(C_1) \dots \Phi(C_n) \rangle \longrightarrow \int D\Phi \, e^{\Sigma(\Phi)} \Phi(C_1) \dots \Phi(C_n);$$



Uncertain reading. The sign before the integral is overwritten. An arrowhead is visible at the right. The earlier strokes beneath it are indistinct.

Original source details: GELFAND-P001484

but there is no reasonable progress in this direction!

But the left-hand side of (2) can be described in other terms.

$$(3) \quad \Phi(C) = \sum \varphi_n(x) e^{in\sigma} \qquad \text{(closed string)}$$

φ_n are identified with the fields in (1)

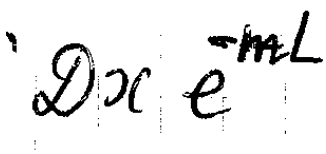
Consider

$$(4) \quad \langle \varphi_{N_1}(x_1) \dots \varphi_{N_n}(x_n) \rangle$$

and write another representation for it.

$$(5) \qquad \langle \varphi(x_1)\varphi(x_2) \rangle = \int Dx \, e^{-mL}, \qquad L = \int \sqrt{\dot{x}^2}\, dt$$

$$\| \qquad \int D\varphi \, e^{\Sigma_0(\varphi)} \varphi(x_1)\varphi(x_2)$$

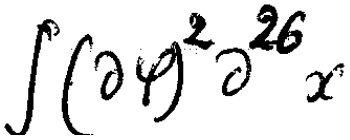


Uncertain reading. The letter after the minus is overwritten. The visible reading is $-mL$.

Original source details: GELFAND-P001484

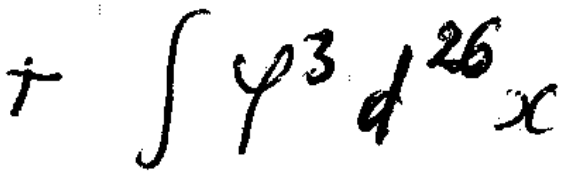
for the [inserted above: simplest] action Σ_0, describing a free particle.

$$(6) \quad \Sigma_0(\varphi) = \int (\partial\varphi)^2 \partial^{26}x + \int \varphi^3 d^{26}x$$



Original source details: GELFAND-P001484

Uncertain reading. In the first term the manuscript has ∂ before ^{26}x.



Original source details: GELFAND-P001484

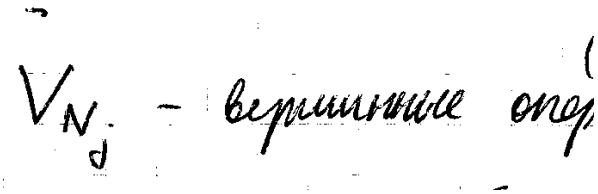
Uncertain reading. The second term is written $+\int \varphi^3 d^{26}x$.

Now

$$(7) \quad \langle \varphi_{N_1}(x_1) \dots \varphi_{N_n}(x_n) \rangle \sim \int Dx\, e^S\, V_{N_1}(x) \dots V_{N_n}(x),$$

[Below the integral] over worldsheets $x^\mu(\sigma, t)$ (nonsingular) and over all topologies

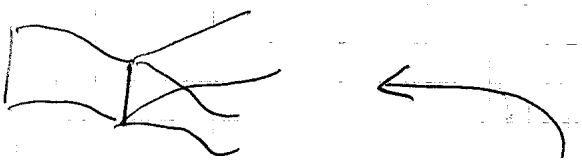
V_{N_j} — vertex operators



Original source details: GELFAND-P001484

Uncertain reading. The small index on N resembles j or d. There is a detached short stroke above V.

(8)



Original source details: GELFAND-P001484

Diagram and typed description.

The two inner curves cross. Their junctions with the middle stroke are partly indistinct. The arrow at the right points from the word “branching” toward the drawing.

String theory: [crossed out] we forbid branching

Original source details: GELFAND-P001484

Uncertain reading. The crossed-out fragment resembles «зап» or «запр», the start of “forbid”.

(9)

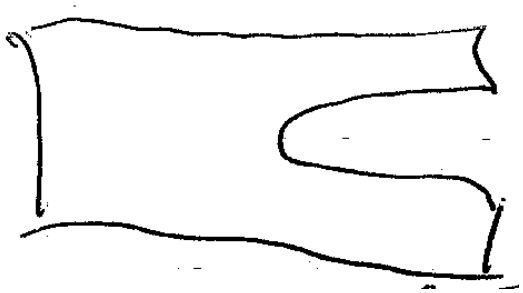
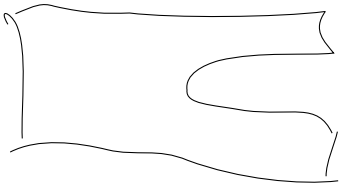


Diagram and typed description.



Original source details: GELFAND-P001484

(but such transformations remain!)
and we get a theory of free particles.

GELFAND-P001485

[source number: 87]

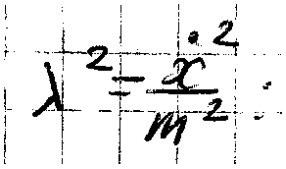
$$\int Dx\, e^{mL} \longrightarrow \int Dx\, D\lambda\, e^{\int \lambda^{-1}\dot{x}^2\, dt + \int m\lambda\, dt} \tag{10}$$

Uncertain reading. A separate sign before $m\lambda\, dt$ is crossed out. It may be m.

Original source details: GELFAND-P001485

L as in (5)
Integrating on the right by the saddle-point method (over λ), we obtain

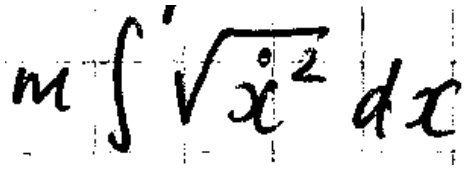
$$\lambda^2 = \frac{\dot{x}^2}{m^2}\ :$$



Uncertain reading. The denominator reads m^2. In (10), the writing after the cancellation reads $m\lambda$.

Original source details: GELFAND-P001485

$$\int Dx\, e^{m\int \sqrt{\dot{x}^2}\, dx} \tag{11}$$

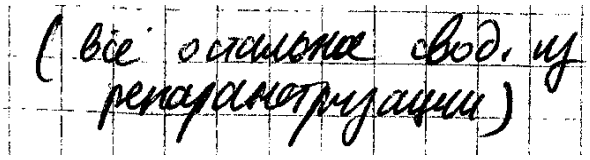


Uncertain reading. The last letter resembles x, as in Dx on the left. Equation (5) has dt.

Original source details: GELFAND-P001485

((11) ⇒ (10) gets rid of the square root). It turns out that the equality is exact!!

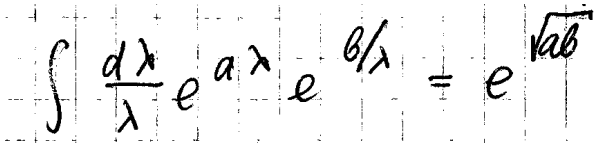
The number λ has the meaning of a one-dimensional metric (everything else reduces from reparametrization)



Original source details: GELFAND-P001485

Uncertain reading. The abbreviation after “else” resembles «свод.» (reduces). «своб.» (free/freedom) is also possible.

$$\int \frac{d\lambda}{\lambda} e^{a\lambda} e^{b/\lambda} = e^{\sqrt{ab}} \tag{12}$$



Original source details: GELFAND-P001485

Uncertain reading. The manuscript has $e^{\sqrt{ab}}$ on the right. No integration limits or additional factors are written.

Green and others did a similar thing in string theory, passing to

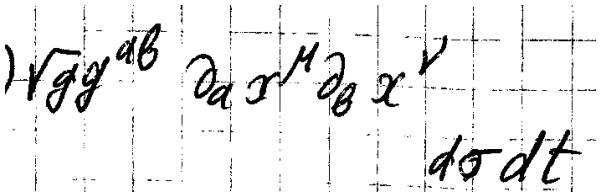
$$\sum_{\text{topol.}} \int Dx\, Dg\, e^{\int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \, d\sigma\, dt} \underbrace{e^{\int \sqrt{g}\, d\sigma\, dt}}_{\text{this can be removed}} \tag{13}$$

Generalization:

curved 26-dimensional space

$$\sum_{\text{topol.}} \int Dx\, Dg\, e^{\int G_{\mu\nu}(x) \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\nu} \tag{14}$$

$$d\sigma\, dt$$



Original source details: GELFAND-P001485

Uncertain reading. The manuscript puts $d\sigma\, dt$ below the right end of the exponent. It is unclear whether it belongs within the exponent.

We want Einstein’s equations to hold

$$R_{\mu\nu}(x) = 0 \tag{15}$$

$(15) \Leftrightarrow$ dilatation invariance!

$\Downarrow$

Yang–Mills equations

All invariant equations for external fields are consequences of dilatation invariance (in the one-loop approximation)

Original source details: GELFAND-P001485

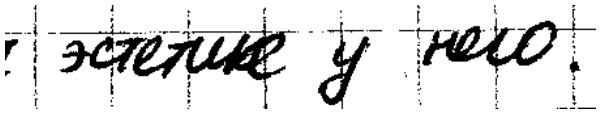
Uncertain reading. The word after “for” is tentatively read as «внешних» (“external”).

26/IX-88 Gelfand Seminar.

I. M. G. Read Feynman's book on QED and learn aesthetics from him.

Original source details: GELFAND-P001486

Uncertain reading. The verb tentatively reads «учите». The final strokes also allow «учить». The sense could be "to learn".



Original source details: GELFAND-P001486

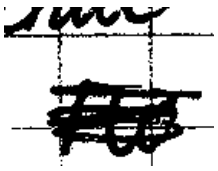
Uncertain reading. The last mark resembles «е», giving «эстетике». «эстетику» is also possible.

There QED is presented without formulas, with pictures alone.

Kazakov. Combinatorics and new work on strings

Developments have taken place (in combinatorics)

Polyakov and [crossed out] Zamolodchikov — Knizhnik — Polyakov



Original source details: GELFAND-P001486

Uncertain reading. The crossed-out strokes may begin with «по».

showed that the models given above really do describe two-dimensional quantum gravity.

(Knizhnik — Polyakov — Zamolodchikov)

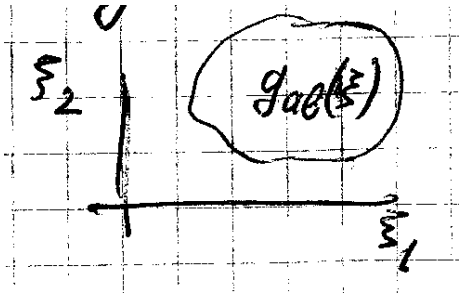
[Below the names] in English alphabetical order.

Z.–K.–P. considered a model of the bosonic string.

$$S = \int d^2\xi \, \sqrt{g} \, \partial_a x^\mu \partial_b x^\mu \, g^{ab} + \beta \int d^2\xi \, \sqrt{g} \tag{1}$$

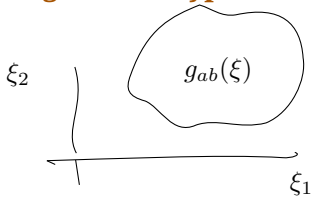
$$Z(\beta) = \int Dg_{ab} \int Dx^\mu \, e^{-S} \tag{2}$$

$\mu = 1, \ldots, D$.



Original source details: GELFAND-P001486

Diagram and typed description.

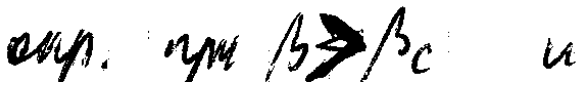


The right end of the horizontal line curls. Distinct arrowheads are not fully discernible at the ends of the axes.

GELFAND-P001487

Integral (2) is defined for $\beta > \beta_c$ and there behaves like

$$(\beta - \beta_c)^{2-\gamma_{\text{str}}(D)} \tag{3}$$



Uncertain reading. The inequality sign is retraced. The angle reads as $>$. A stroke of $\geq$ is also possible.

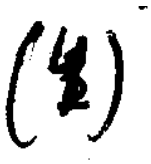
Original source details: GELFAND-P001487

(at β_c the volume of the surface becomes infinite)

str = string

K.–P.–Z. proved that

$$\gamma_{\text{str}} = \frac{1}{12}\left(D - 1 - \sqrt{(D-1)(D-25)}\right). \tag{4}$$



Uncertain reading. The digit is overwritten. Its upper strokes and the sequence of formulas support (4).

Original source details: GELFAND-P001487

This makes sense for $D \leq 1$ and $D \geq 26$.

Uncertain reading. The right part of the line reads $D \geq 26$. There is a separate slanting stroke below $>$.

Original source details: GELFAND-P001487

This was obtained using Kac–Moody algebras arising from the study of the Lagrangian.

Kazakov and others predicted several values of γ_{str}:

for $D = -2, 0, \frac{1}{2}, 1$

[Mark in left margin] It all agreed!!

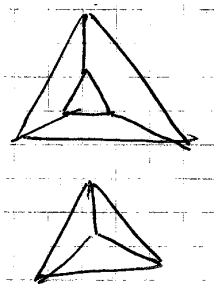
Uncertain reading. The intertwined strokes resemble the monogram «ИГ» or «ИМ».

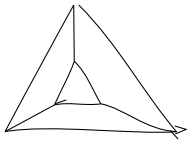
Original source details: GELFAND-P001487

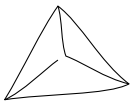
Define

$$Z(\beta) = \sum_{n=0}^{\infty} e^{-2n\beta} Z_{2n}, \tag{5}$$

where Z_n — the number of planar graphs [inserted above: with n vertices] with the topology of a sphere (3 edges must meet at a vertex)







Original source details: GELFAND-P001487

Diagram and typed description. Both drawings are in the right margin. Two thickened marks in the upper drawing resemble arrowheads. In the lower drawing the left segment stops short of the other two segments' junction.

Then

$$Z(\beta) \sim (\beta - \beta_c)^{2-\gamma_{\text{str}}(D)} \tag{6}$$

[Bracket above $\gamma_{\text{str}}(D)$: $= -\frac{1}{2}$.]

This corresponds to $D = 0$

[crossed out] Asymptotics

Original source details: GELFAND-P001487

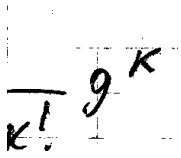
Uncertain reading. The crossed-out word before "Asymptotics" may read «число» ("number").

$$Z_{2n} \sim \left(\frac{256}{27}\right)^n n^{-3+\gamma_{\text{str}}} \tag{7}$$

γ_{str} — a universal number, independent of the way the planar graphs are defined (one can have 4 edges at 1 vertex, and so on)

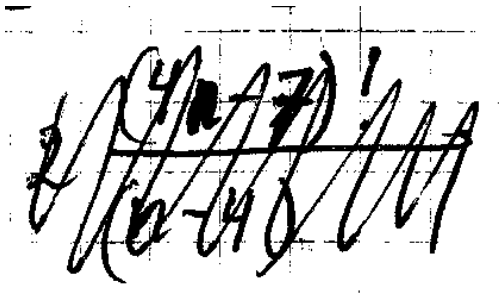
Formula for Z_n:

$$Z_{2k} = 2\frac{(4k-3)!}{(3k-1)!\,k!}\,9^k \tag{8}$$



Original source details: GELFAND-P001487

Uncertain reading. The last factor is tentatively read as 9^k. The base also resembles g.



Original source details: GELFAND-P001487

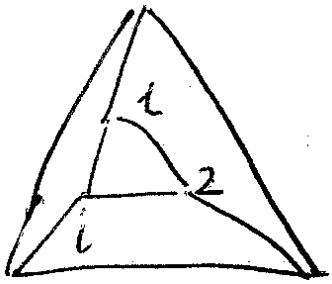
Uncertain reading. A fraction is crossed out after the equals sign. The numerator tentatively reads $(4n+7)!$, and $(n-4)$ is visible in the denominator. The indices and remaining strokes are indistinct.

Z_{2k} — the number of planar graphs inscribed in a triangle with the number of internal vertices $= 2k$.

GELFAND-P001488

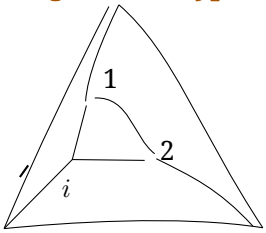
[source number: 91]

Tutte solved this problem by purely combinatorial means.



Original source details: GELFAND-P001488

Diagram and typed description.



A thickening on the left edge of the outer triangle resembles a downward arrowhead. It may be overlapping strokes.

Consider

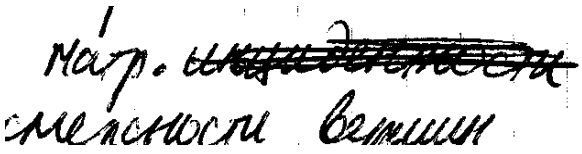
$$Z_n = \sum_{\substack{\text{graphs}\\ \text{with } n\\ \text{vertices}}} \int \prod_{i=1}^{n} d\vec{x}_i \, e^{-\sum_{\langle ij\rangle}(x_i^\mu - x_j^\mu)^2} \tag{9}$$

$\langle ij\rangle$ — nearest neighbours in the graph

internal vertices

(10) The integral in (9) equals $\det^{-D/2}(3I - G)$.

[Stroke leading from G:] Matrix of [crossed out] vertex adjacency

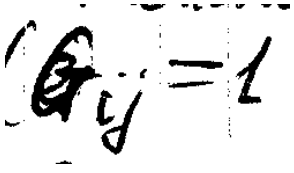


Original source details: GELFAND-P001488

Uncertain reading. The crossed-out word after “Matrix of” tentatively reads “incidence”.

($G_{ij} = 1 \Longleftrightarrow i, j$ are joined by an edge

$G_{ij} = 0$ [crossed out $\Longleftrightarrow$] otherwise).



Original source details: GELFAND-P001488

Uncertain reading. The first G is overwritten. Strokes beneath it resemble E.

Original source details: GELFAND-P001488

Uncertain reading. After $G_{ij} = 0$, a two-headed arrow is visible beneath dark strokes.

Kapranov Misha.

On the work by [crossed out B.–G.–G.] on $\mathcal{D}$-modules. [Inserted below the abbreviation: Bernst.–Gelf.–Gelf.]

V — a vector [crossed out] space, $P(V)$ — its projectivization.

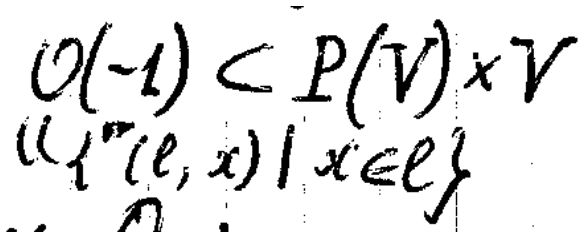
Original source details: GELFAND-P001488

Uncertain reading. A short fragment between "vector" and "space" is crossed out. It may be the start of «пространство» ("space").

$O(-1)$ — [inserted above: tautological] bundle on $P(V)$ (fundamental);

[At right]

$$O(-1) \subset P(V) \times V$$
$$\| \{(\ell, x) \mid x \in \ell\}$$



Original source details: GELFAND-P001488

Uncertain reading. Two curved strokes below $O(-1)$ may be a vertical equals sign. A separate short mark follows the opening brace.

The points of $P(V)$ — are the lines ℓ in V, passing through 0;

the line itself is the fibre over ℓ.

$$O(-k) = O(-1)^{\otimes k}, \qquad k > 0;$$

$O(+k)$ — the dual bundle of $O(-k)$

A way to construct new bundles:

$\Lambda(V)$ — the exterior algebra on V;

$\Lambda^k(V)$ — the homogeneous degree k part of $\Lambda(V)$.

We consider modules over $\Lambda(V)$ with a grading.

$$\{\dots M_{-1};\ M_0;\ M_1;\ \dots\}, \qquad \Lambda^k M_i \subset M_{i+k}$$

Exercise for next time: write the answers for [crossed out] $\dim V = 1$ and 2.

Original source details: GELFAND-P001488

Uncertain reading. The crossed-out sign before $\dim V$ resembles M or V.

GELFAND-P001489

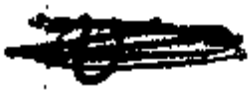
$V \ni v$, $v^2 = 0$ in $\Lambda(V)$.

Consider multiplication by v:

$$(1) \qquad \ldots \to M_{-1} \xrightarrow{v} M_0 \xrightarrow{v} M_1 \xrightarrow{v} M_2 \to \ldots$$

$v^2 = 0 \Rightarrow$ we obtain a complex.

$$\operatorname{Ker} v \supset \operatorname{Im} v, \qquad \text{[crossed out]}$$

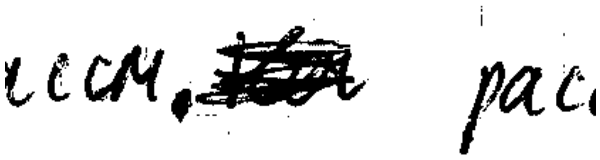


Undeciphered passage. The fragment to the right of Im v is crossed out with several horizontal lines and is illegible.

Original source details: GELFAND-P001489

Consider [crossed out] the bundle

$$v \longmapsto \frac{\operatorname{Ker}\{v : M_0 \to M_1\}}{\operatorname{Im}\{v : M_{-1} \to M_0\}}, \qquad v \neq 0.$$



Undeciphered passage. The crossed-out fragment between “Consider” and “the bundle” is illegible.

Original source details: GELFAND-P001489

This is a “bundle”, or more precisely a “sheaf”, over $\mathbb{P}(V)$.

Examples.
of modules

1). The exterior algebra itself $\Lambda(V)$

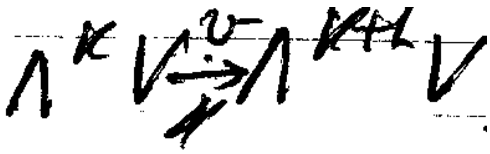
$$\underset{0}{\mathbb{C}} \xrightarrow{v} \underset{1}{V} \xrightarrow{v} \underset{2}{\Lambda^2 V} \xrightarrow{v} \ldots \xrightarrow{v} \underset{N}{\Lambda^N V}$$

The zero bundle corresponds to it; Ker $v =$ Im v

2) The truncated algebra Λ, i.e. $\Lambda^{\geq k}(V)$

(keep only the components of degrees $\geq k$).

$$\{0;\ \Lambda^k V \xrightarrow{v} \Lambda^{k+1} V;\ \ldots\}$$

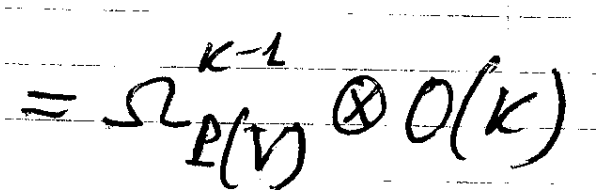


Uncertain reading. A short sign below the arrow is crossed out. It resembles v or k.

Original source details: GELFAND-P001489

Bundle:

$$\mathrm{Ker}\{v : \Lambda^k V \to \Lambda^{k+1} V\} = \Omega^{k-1}_{\mathbb{P}(V)} \otimes O(k)$$

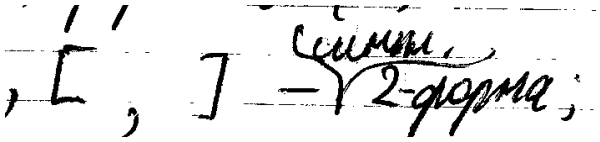


Uncertain reading. The end of the formula clearly reads $O(k)$.

Original source details: GELFAND-P001489

[Below $\Omega^{k-1}_{\mathbb{P}(V)}$:] $(k-1)$-forms on $\mathbb{P}(V)$

3) Let V — be a symplectic space, $[\ ,\ ]$ — [inserted above: symm.] 2-form;



Uncertain reading. The abbreviation «симм.» (“symm.”) is above “2-form” and joined to it by a bracket.

Original source details: GELFAND-P001489

consider the module

$$0 \to \underset{-1}{\mathbb{C}} \xrightarrow{v} \underset{0}{V} \xrightarrow{v} \underset{1}{\mathbb{C}} \to 0$$

$$v : w \longmapsto [v, w]$$

$$v \longmapsto v^{\perp}/\mathbb{C}v, \qquad \perp \text{ — orthogonality with respect to } [\cdot, \cdot]$$

Theorem. Every algebraic vector bundle over $\mathbb{P}(V)$ arises from some $\Lambda(V)$-module

$$E = \frac{\mathrm{Ker}\{v : M_0 \to M_1\}}{\mathrm{Im}\{v : M_{-1} \to M_0\}}, \quad \text{and the other factors } = 0.$$

GELFAND-P001490

[source number: 92]

Proof see the appendix by S. Gelfand to the book by Okonek and others, “Bundles over Projective Space”.

Statement: note in FA in 1978.

The proof uses derived categories. Can this be avoided? S. Gelfand thinks it cannot.

Kapranov’s result: a description of bundles over a quadric in $\mathbb{P}(V)$.

$$V, \quad \langle\,\rangle, \qquad \mathbb{P}(V) \supset Q = \{v : \langle v, v\rangle = 0\}$$

[Below $\langle\,\rangle$: symmetric form.]

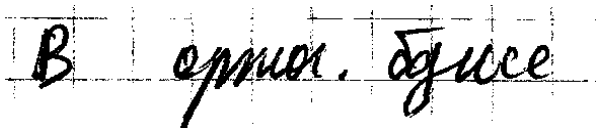
We construct a graded Clifford algebra [inserted above with a bracket: $A(V)$], generated by the generators

$$v, \quad \deg v = 1 \text{ and } h, \quad \deg h = 2$$

[Below h: fixed.]

$$vv' + v'v = \langle v, v'\rangle h$$

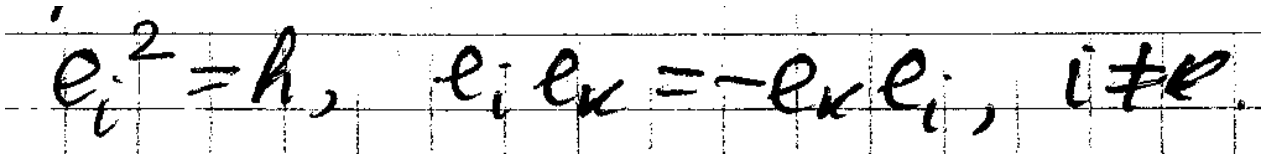
In an orthonormal basis $e_1, \dots, e_r$



Uncertain reading. The abbreviation tentatively reads «ортон.» (“orthonormal”). «ортог.» (“orthogonal”) is also possible.

Original source details: GELFAND-P001490

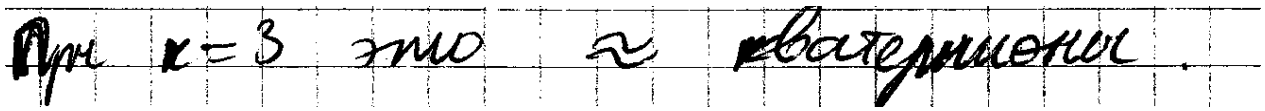
$$e_i^2 = h, \qquad e_i e_k = -e_k e_i, \quad i \neq k.$$



Original source details: GELFAND-P001490

Uncertain reading. The basis line reads $e_i^2 = h$. The preceding relation has $\langle v, v'\rangle h$ on the right.

For $r = 3$ this is $\approx$ the quaternions.



Original source details: GELFAND-P001490

Uncertain reading. The letter before $= 3$ is heavily retraced. The preceding notation e_r supports a tentative r.

$$\langle v, v\rangle = 0 \;\Rightarrow\; v^2 = 0$$

and we construct the same kind of bundle.

Then the same kind of theorem holds.

Chapter 33

GELFAND-P001491

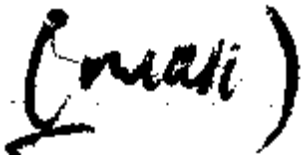
3/X̄-88. Gelfand Seminar.

Uncertain reading. In the digital image a yellow icon sits to the left of the digit 3. The area beneath it is obscured.

Original source details: GELFAND-P001491

What should one read in mathematics? (for first-year students)

Beilinson. (outline)



Uncertain reading. The short entry in parentheses tentatively reads «план» (“outline”).

Original source details: GELFAND-P001491

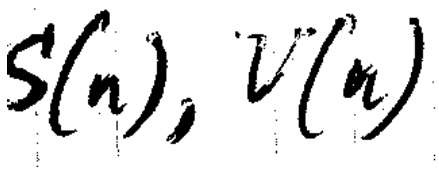
1. Linear algebra.
2. Functions of a complex variable.
3. Galois theory.
4. Elementary algebraic topology.
5. Representation theory.
6. Algebraic geometry (elliptic curves, etc.).
7. Arithmetic.

GELFAND-P001492

[source number: 4]

Kirillov (outline)

1. Linear algebra
2. Analysis — real, complex and p-adic
3. Operator theory + quantum mechanics.
4. Harmonic analysis + integral geometry
5. Representations of $S(n)$, $V(n)$ and combinatorics.



Uncertain reading. The second letter is shaped like V. U is also possible.

Original source details: GELFAND-P001492

6. Elements of algebraic geometry (commutative and noncommutative).
7. Classical works (a selection).

Vasiliev Vitya

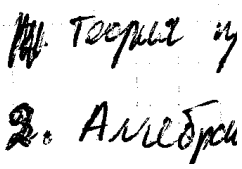
1. Linear [crossed out] algebra.

Original source details: GELFAND-P001492

Uncertain reading. The crossed-out fragment before “algebra” resembles «алг.» (“alg.”).

[number crossed out] Representation theory.

2. Algebraic and differential topology, homological algebra.



Original source details: GELFAND-P001492

Uncertain reading. The number before “Representation theory” is crossed out. The following number is overwritten and reads 2.

3. Dynamical systems, ergodic theory, mechanics.
4. Algebraic geometry (geometric chapters).
5. Functional analysis.

Goncharov Sasha

1. Linear algebra.
2. Theory of functions of a complex variable [ТФКП].
3. Analysis (generalized functions ...)
4. Algebraic topology, homological algebra, algebraic geometry (classical)
5. Algebraic geometry (modern)
6. Differential geometry.
7. Representation theory (finite-dimensional and ∞-dimensional)
8. Arithmetic
   a) classical
   b) class field theory
9. F. Klein, “Lectures on the Development of Mathematics in the XIX Century”

GELFAND-P001493

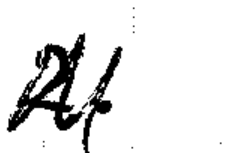
Kapranov Misha

[At upper right: (here linear algebra [crossed out] is omitted as uninformative)]

Original source details: GELFAND-P001493

Uncertain reading. The word after “algebra” is crossed out and illegible.

1. Elementary topology.
   (π_1, surfaces)



Original source details: GELFAND-P001493

Uncertain reading. The crossed-out number at left may be 2.

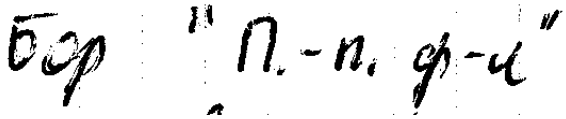
2. Galois theory.
3. Algebraic geometry.
4. Algebraic topology
5. Physics
6. Arithmetic.

Zelevinsky Andrei.

1. Algebra: Galois theory.
   etc.

I. M. Gelfand

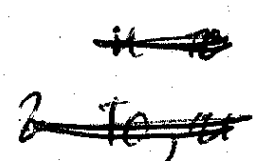
1. Feynman for schoolchildren
2. Bohr, “П.-п. ф-и” [abbreviated title]



Original source details: GELFAND-P001493

Uncertain reading. The surname reads «Бор» (Bohr), possibly «Бэр» (Ber).

3. Minkowski, “Geometrie der Zahlen”
4. Hurwitz, “Elliptic Functions”
5. Gelfand, “Linear Algebra” (chapters 1 and 2)
6. Mechanics, “Landau and Lifshitz”,
   calculus of variations, symplectic geometry,
   (especially Gromov)
7. Quantum mechanics



Original source details: GELFAND-P001493

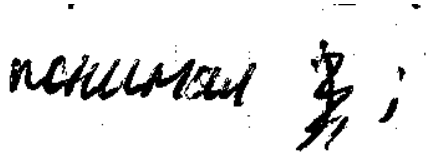
Uncertain reading. Two crossed-out lines. The first appears to start with “and …” (и); the second with «2. Те…».

8. Spectral theory of operators.

9. Group representation theory

(Serre, “Theory of Representations of Finite Groups” — only chapters 1 and 2!

[Below at right:] in the third he understood nothing [crossed out];

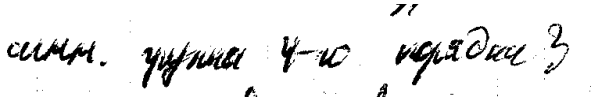


Uncertain reading. The crossed-out mark after “understood” is illegible.

Original source details: GELFAND-P001493

[Below at left:] Gelfand, Minlos, Shapiro)

Why is the group of motions of the cube the symmetric group of order 4?



Uncertain reading. The manuscript reads «4-го порядка» (of order 4).

Original source details: GELFAND-P001493

There are 2 equally valid geometries, Euclidean and symplectic,

GELFAND-P001494

[source number: 5]

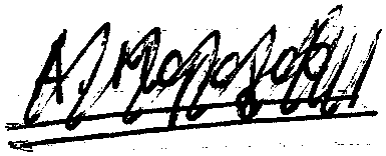
10. Statistical mechanics, chaos, ТФДП [source abbreviation], ergodic theory

11. p-adic analysis, strings, etc.

(especially, Morozov’s lectures)

12. Quantization in symplectic geometry, theory of D-modules

A. Yu. Morozov [oblique strokes]



Uncertain reading. The heading is double-underlined and crossed by oblique strokes.

Original source details: GELFAND-P001494

I. M. Gelfand

What is a particle’s spin?

[Inserted above: in the theory of relativity]

Consider the mass hyperboloid:

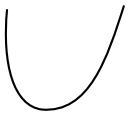
$$-p_1^2 - p_2^2 - p_3^2 + p_4^2 = c^2$$

[At right: (with [abbrev.] rest mass)]

Uncertain reading. The abbreviation before “rest mass” is unclear. Possibly «нул.» (“zero”).

Original source details: GELFAND-P001494

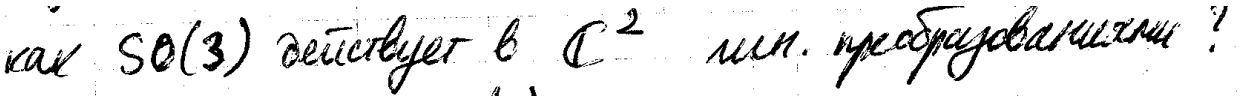
Diagram and typed description.



Original source details: GELFAND-P001494

At each point of the hyperboloid one must put a linear space characterizing its degree of freedom.

Exercise: how does $SO(3)$ act on $\mathbb{C}^2$ by linear transformations?



Original source details: GELFAND-P001494

Uncertain reading. The manuscript reads $SO(3)$ and $\mathbb{C}^2$.

(see Gelfand–Minlos–Shapiro, chapter 1)

Problem. Construct the mathematics corresponding to the case of arbitrary spin (spin 0 — ordinary; spin $1/2$ — super).

Fröhlich explains how the braid group is applied.

A. Morozov.

In 3-dimensional space there is linking of contours.

The Bohm–Aharonov effect

(Bohm-Aharonov)

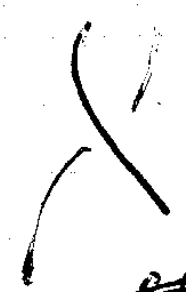


Diagram and typed description.

Original source details: GELFAND-P001494

[crossed out]

Undeciphered passage. The word above "Spin?" is illegible.

Original source details: GELFAND-P001494

Spin?

spin θ

(in the future)

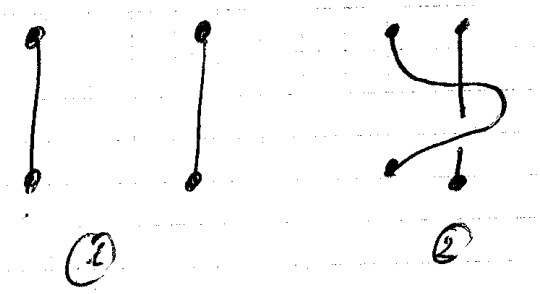
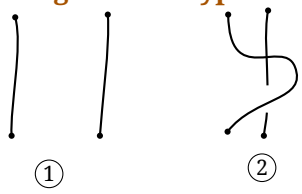


Diagram and typed description.



Original source details: GELFAND-P001494

worldlines

$\mathbb{R}^1_t \times \mathbb{R}^2$

[At right: (in $\mathbb{R}^3$ and only there is there a fundamental difference between ① and ②)]

	①	②
Amplitude	1	$e^{2\pi i\theta}$

[At right: (earlier θ — spin)]

This applies to the quantum Hall effect.

On [taking it?] downward, some number of circuits arises.

Original source details: GELFAND-P001494

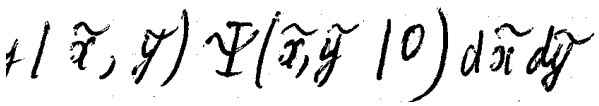
Uncertain reading. The word before “downward” is tentatively read «унесении» (taking away).

GELFAND-P001495

What happens to the wave function

$$\begin{aligned} d > 3 &\implies \theta = \frac{\text{integer}}{2} \\ d = 3 &\implies \theta \text{ — arbitrary!} \end{aligned}$$

$$\Psi(x, y; T) = \int A_T(x, y \mid \tilde{x}, y)\, \Psi(\tilde{x}, \tilde{y} \mid 0)\, d\tilde{x}\, d\tilde{y}$$



Uncertain reading. In A_T the second argument after the bar has no distinct tilde. The top stroke joins the y.

Original source details: GELFAND-P001495

[Crossed out beneath the integral sign]

Uncertain reading. Strokes resembling d and y are visible beneath the cancellation.

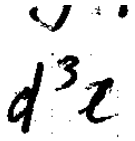
Original source details: GELFAND-P001495

$$A_T = \sum_{\substack{\text{over particle} \\ \text{trajectories}}} e^{L_1} e^{L_2} \underbrace{F(\text{trajectories})}_{\substack{\| \\ e^{2\pi i \theta \text{ (index)}}}} =$$

(how many times they became linked)

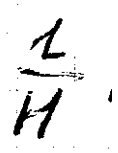
θ half-integer except [inserted: the case] $d = 3$

$$= \int \mathcal{D}\varphi \, \mathcal{D}A \quad \exp \int_{d^3 z} |D\varphi|^2 + \frac{1}{H} F_{\mu\nu}^2 + \epsilon_{\mu\nu\lambda} A_\mu F_{\nu\lambda}$$



Uncertain reading. The letter after d^3 looks like z. Possibly x.

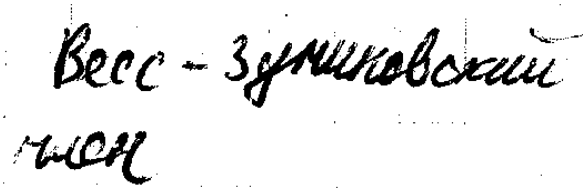
Original source details: GELFAND-P001495



Uncertain reading. The denominator looks like H.

Original source details: GELFAND-P001495

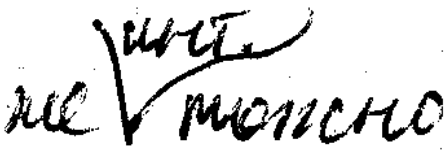
[Beneath the last summand:] Wess–Zumino term



Uncertain reading. The last summand is labelled «Весс-Зуминовский член» (Wess–Zumino term).

Original source details: GELFAND-P001495

The same [inserted: integrals [abbrev.]] can be used for high-temperature conductivity [source abbreviation: провод.].



Uncertain reading. The insertion appears to read «инт.».

Original source details: GELFAND-P001495

Witten recently used it to describe knot invariants

(Jones polynomials)

S. Shatashvili (LOMI)

P. Wiegmann

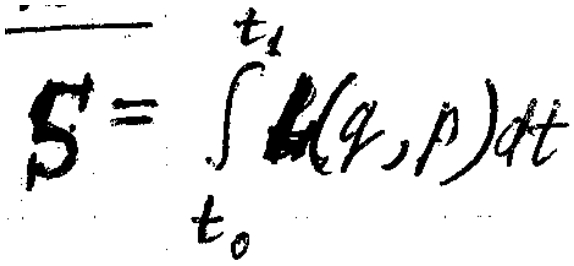
GELFAND-P001496

17/X-88. Gelfand seminar.

I. M. Gelfand. 1) Hierarchy of Functionals

Local functionals:

$$S = \int_{t_0}^{t_1} H(q,p)\,dt$$



Uncertain reading. The letter before (q,p) is overwritten. It is read tentatively as H, possibly L.

Original source details: GELFAND-P001496

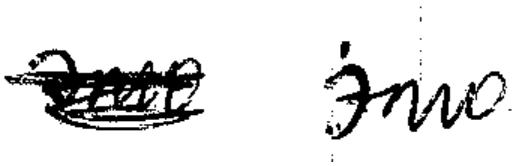
Functionals of the next level: if the variation δS is local.

(of the 1st kind)

[At right: ‖ Example: λ_1]

Functionals of the 2nd kind: if δS — of the 1st kind.

~~This~~ This is a new subject of the homological-algebra type.



Uncertain reading. The crossed-out word before “This is a new subject” appears to be a repeated “This”.

Original source details: GELFAND-P001496

2) Quantum mechanics: amplitude

$$e^{\frac{i}{\hbar}\int_{t_0}^{t_1} L(p,q)\,dt}$$

Read Einstein and Bohr’s discussion of the uncertainty principle.

One must write

$$\int e^{\frac{i}{\hbar}\int_{t_0}^{t_1} L(p,q)\,dt}\,Dp\,Dq$$

The stationary-phase method.

GELFAND-P001497

Problem: calculate approximately

$$\int_0^1 (\sin 10^5 x^3)\, dx$$

An engaging problem: a dual exposition of quantum mechanics, through which one can understand much mathematics

One should read: a) Landau, Lifshitz, “Quantum Mechanics”

Feigin B. Strings.

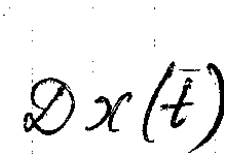
Quantum mechanics:

M is a manifold, to every map of the circle $S^1 \to M$ one assigns

$$e^{\frac{i}{\hbar}\int \frac{m\dot{x}^2}{2}\, dt} = e^{\frac{i}{\hbar}\int \mathcal{L}(x,\dot{x})\, dt} \tag{1}$$

and one must write

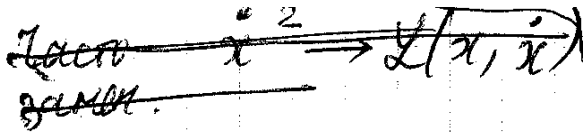
$$\int e^{\frac{i}{\hbar}\int \frac{m\dot{x}^2}{2}\, dt} \mathcal{D}x(\bar{t}) = \int e^{\frac{i}{\hbar}\int \mathcal{L}(x,\dot{x})\, dt} \mathcal{D}x(t) \tag{1}'$$



Uncertain reading. A separate bar is visible over t in the first $\mathcal{D}x(\bar{t})$. The second $\mathcal{D}x(t)$ has no such bar.

Original source details: GELFAND-P001497

[Crossed out: Част. / Чисто; замен. [source wording] $\dot{x}^2 \longrightarrow \mathcal{L}(x, \dot{x})$.]

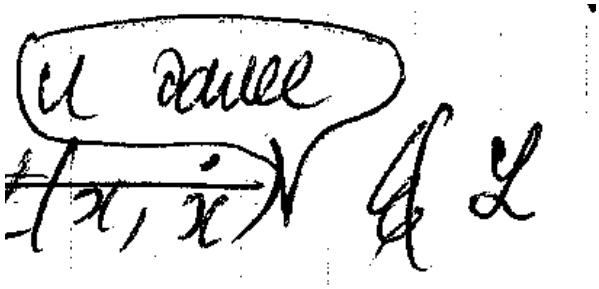


Uncertain reading. The first crossed-out word looks like «Част.» or «Чисто». The second looks like «замен.» (replace).

Original source details: GELFAND-P001497

[Inserted and circled: And even]

& $\mathcal{L}$ must be taken more general.



Uncertain reading. The sign before $\mathcal{L}$ resembles &. It may be a rewritten conjunction.

Original source details: GELFAND-P001497

Let M be a symplectic manifold

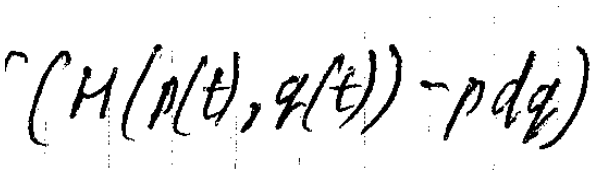
p, q are coordinates, H is a function on M

Write the action for a trajectory $p(t), q(t)$

$$\int [H(p(t), q(t))\, dt - p\, dq] \tag{2}$$

and

$$\int e^{\frac{i}{\hbar}\int (H(p(t),q(t)) - p\, dq)}\, Dp\, Dq \tag{2'}$$



Uncertain reading. Equation (2′) gives no dt after $H(p(t), q(t))$. Equation (2) has dt there.

Original source details: GELFAND-P001497

The approaches using (1′) and (2′) are related by the Legendre transformation

~~Now one must~~

A more invariant expression: $S^1 \to M$

$$\int e^{\frac{i}{\hbar}\left[\int_{S^1} H\, dt - \int_{\oslash} \omega\right]} \tag{3}$$

Diagram and typed description. ⊘

Original source details: GELFAND-P001497

[At right:] (ω is the symplectic form,

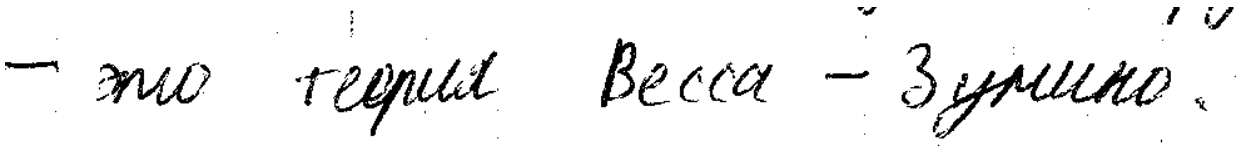
⊘ — [Crossed out: disk?; inserted: membrane], spanning S^1

Uncertain reading. The crossed-out word beneath “membrane” may be “disk”.

Original source details: GELFAND-P001497

The generalization of (3) to strings in a certain special case —

— is the Wess–Zumino theory.



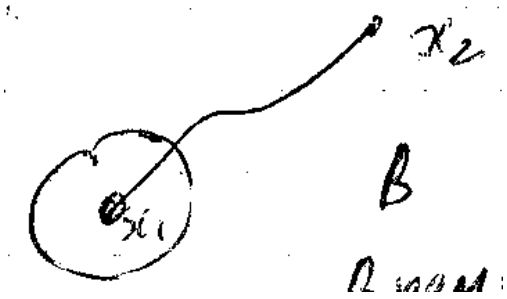
Original source details: GELFAND-P001497

Uncertain reading. The word between «это» and «Весса» reads «теория» (theory).

GELFAND-P001498

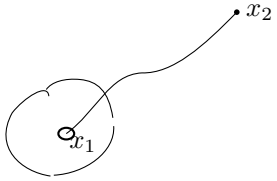
[source number: 41]

I. M. G. Legendre transformation: see Gelfand–Fomin's book on the calculus of variations
(appendix)
(via wavefronts)



Original source details: GELFAND-P001498

Diagram and typed description.



The outline has gaps and a short hooked stroke. The purpose of the stroke is unclear.

A "wind rose" is specified at every point
Time of motion

$$\int_{x_1}^{x_2} L(x, \dot{x})\, dt$$

(independent of parametrization)
Take wavefronts.
Trajectories are characteristics.

— · —

F. $\mathcal{E}$ is a Riemannian 2-dimensional manifold; g_{ij} is the metric tensor on $\mathcal{E}$.
Free Lagrangian (analogue of $\dot{x}^2$)

$$\mathcal{L}(X) = \int_{\mathcal{E}} \sqrt{g}\, g^{\alpha\beta} \frac{\partial X^a}{\partial x^\alpha} \frac{\partial X^b}{\partial x^\beta} G_{ab}\, dx^0\, dx^1 \tag{4}$$

Original source details: GELFAND-P001498

Uncertain reading. Equation (4) shows G_{ab} and a cancelled superscript. The following line uses G^{ab}.

Original source details: GELFAND-P001498

Uncertain reading. The first numeral in $dx^0\, dx^1$ is heavily retraced and looks like 0. The line below gives x^1, x^2.

$\{X^a(x^1, x^2)\}$ is an embedding of $\mathcal{E}$ into $\mathbb{R}^d$; G^{ab} is the metric in $\mathbb{R}^d$

Feynman integral

[Crossed out at left]

$$\int e^{\frac{i}{\hbar}\mathcal{L}(X)}\,\mathcal{D}X\,\mathcal{D}g \tag{4'}$$

Uncertain reading. The crossed-out label before (4′) resembles (4).

Original source details: GELFAND-P001498

I. M. G.'s question: what is the Legendre transformation associated with surfaces rather than curves?

This is a fundamental and substantial question

Wess–Zumino theory.

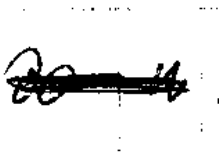
$\mathcal{E}$ with a complex structure. Write in local coordinates:

$z, \bar{z}$ are coordinates

$$\left.\begin{aligned}\frac{\partial}{\partial z} + A_0(z,\bar{z})\\ \frac{\partial}{\partial \bar{z}} + A_1(z,\bar{z})\end{aligned}\right\} \quad \text{connection} \tag{5}$$

Choose a 3-dimensional membrane whose boundary is $\mathcal{E}$

Complete [crossed out] the components (5) to flat connections

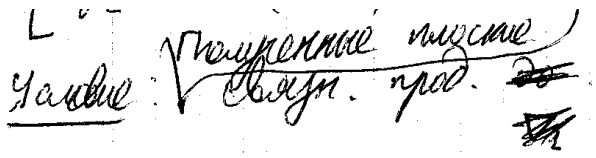


Uncertain reading. After "Complete," «до» ("to") and one further stroke are crossed out.

Original source details: GELFAND-P001498

$$\begin{aligned}\left[\frac{\partial}{\partial \bar{z}} + B_0(z,\bar{z}),\ \frac{\partial}{\partial z} + A_0(z,\bar{z})\right] = 0\\ \left[\frac{\partial}{\partial z} + B_1(z,\bar{z}),\ \frac{\partial}{\partial \bar{z}} + A_1(z,\bar{z})\right] = 0\end{aligned} \tag{6}$$

(7) Condition: [inserted above: The resulting flat] conn. extend [crossed out] onto the membrane as flat connections ∇_1, ∇_2



Uncertain reading. After «прод.» ("extend"), «до» ("to") is crossed out. The cancelled symbol below it resembles ∇_1.

Original source details: GELFAND-P001498

GELFAND-P001499

Lagrangian:

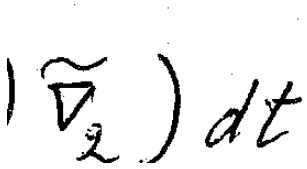
$$\mathcal{L}(A_0, A_1) = \int (A_0 - B_1)(B_0 - A_1)\, dz \wedge d\bar{z} + \underset{\text{(Wess-Zumino)}}{WZ}(\widetilde{\nabla}_1, \widetilde{\nabla}_2), \tag{8}$$

The notation used above is

$$\nabla_1 = \left\{ \frac{\partial}{\partial z} + A_0,\ \frac{\partial}{\partial \bar{z}} + B_0 \right\},$$
$$\nabla_2 = \left\{ \frac{\partial}{\partial z} + B_1,\ \frac{\partial}{\partial \bar{z}} + A_1 \right\};$$

$\widetilde{\nabla}_j$ are flat extensions of ∇_j to the membrane

$$WZ(\widetilde{\nabla}_1, \widetilde{\nabla}_2) = \int_0^1 \int_{\text{membrane}} \mathrm{tr}((\widetilde{\nabla}_1 - \widetilde{\nabla}_2) \cdot \Phi(t\widetilde{\nabla}_1 + (1-t)\widetilde{\nabla}_2)\, dt \tag{9}$$



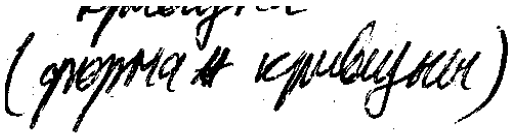
Original source details: GELFAND-P001499

Uncertain reading. Two opening parentheses follow tr. Before dt only the parenthesis closing the argument of Φ is visible.

[A stroke from Φ leads to the caption]

curvature

(form [crossed out] of curvature)



Original source details: GELFAND-P001499

Uncertain reading. A short mark between “form” and “of curvature” is crossed out.

I. M. Gelfand

Determinant of a cubic matrix.

(after [crossed out] I. M. G., Zelevinsky and Kapranov)

Original source details: GELFAND-P001499

Uncertain reading. The crossed-out surname may start «Зелевин…» (Zelevin…).

introduction to a future talk

$$a_{i,j,k}, \quad i,j,k = 1,2.$$

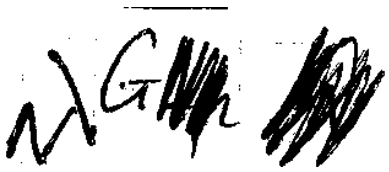
$$\det \|\lambda a_{ij1} + \mu a_{ij2}\|$$

a quadratic form in λ and μ.
Take its discriminant in λ, μ. It is called the determinant of the cubic matrix.
In det A there is the action XAY,
and in the quadratic form X^*AX.
The group action is important.
The whole of group theory arose from the concept of det.
From cubic [crossed out] matrices new objects should arise.

Original source details: GELFAND-P001499

Undeciphered passage. The fragment before "matrices" is illegible.

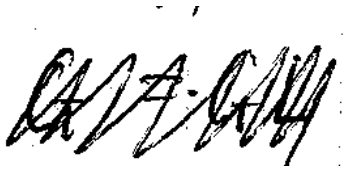
Representations of GL_n or SL_n.
N — a unipotent subgroup; ${}_N\backslash G$ [crossed out] — the space of affine flags



Original source details: GELFAND-P001499

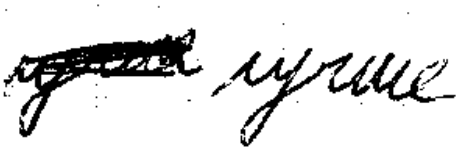
Undeciphered passage. Two groups of signs after ${}_N\backslash G$ are crossed out.

(look in detail at GL_2); [crossed out] $G = SL_2$ [crossed out] is better



Original source details: GELFAND-P001499

Uncertain reading. The crossed-out notation before $G = SL_2$ contains shapes resembling G and A.



Original source details: GELFAND-P001499

Undeciphered passage. The crossed-out word before "better" is illegible.

On ${}_N\backslash G$ consider functions (polynomials), homogeneous under left shifts.

GELFAND-P001500

[source number: 42]

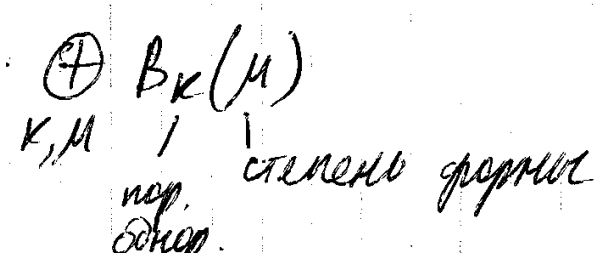
Example: the affine plane and polynomials of degree n

But there are also differential forms. Denote by B the space of forms ω,

such that $i_\lambda \omega = 0$ (λ is an element of the Lie algebra)

Now one must take $B + dB$

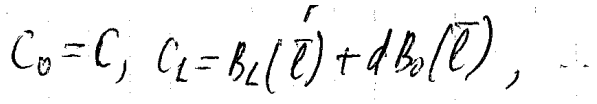
$$B = \bigoplus_{k,\mu} B_k(\mu)$$



Uncertain reading. A stroke from k points to "order of homogeneity" and one from μ to "degree of the form".

Original source details: GELFAND-P001500

$$C_0 = C, \quad C_1 = B_1(\bar{\ell}) + dB_0(\bar{\ell}), \ \ldots$$



Uncertain reading. The arguments in the formula for C_1 read as $\bar{\ell}$.

Original source details: GELFAND-P001500

Whitehead torsion arises: hyperdeterminant.

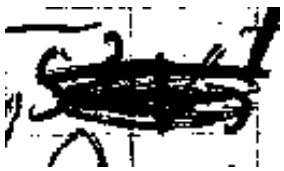
Apparently this makes it possible to replace Young diagrams.

GELFAND-P001501

24.10.88. Gelfand Seminar

I. M. G. Problems on representations.

There are representations, [crossed out] the trivial one; the tautological one in V; in S^2V (homogeneous polynomials of degree 2).



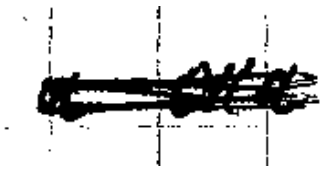
Undeciphered passage. The entry before "the trivial one" is crossed out and illegible.

Original source details: GELFAND-P001501

Prove irreducibility of $SL(2,\mathbb{R})$ in S^3V (polynomials of degree 3).

$\dim V = 2$, in V the tautological representation acts.

What about $SL(3,\mathbb{R})$? This is an important problem [crossed out] and it needs to be thought through.



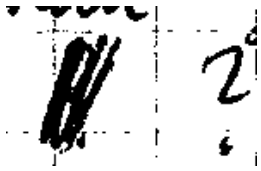
Undeciphered passage. The entry after "problem" is crossed out. Its strokes are illegible.

Original source details: GELFAND-P001501

Original source details: GELFAND-P001501

Uncertain reading. The last word of the line is read «продумать» (think through).

Hypothesis: homogeneous polynomials here too [crossed out]?



Original source details: GELFAND-P001501

Undeciphered passage. A short mark is heavily crossed out before the question mark.

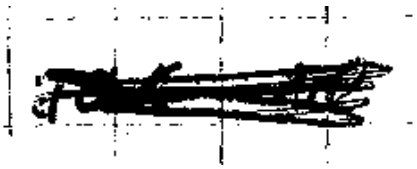
Gelfand, Naimark

3 important problems.

1) S_n (symmetric group)

2) $SL_n(\mathbb{R})$ — finite-dimensional representations

3) SO_n [crossed out] in the same detail as [crossed out] in the book by G.–M.–S., [below:] for SO_3.



Original source details: GELFAND-P001501

Undeciphered passage. An illegible fragment after SO_n is crossed out.

Original source details: GELFAND-P001501

Uncertain reading. The crossed-out entry after “as” may begin with “for”.

Original source details: GELFAND-P001501

Uncertain reading. A small detached mark appears below the line. Its status is unclear.

Representations of $SL(3,\mathbb{R})$.

The answer is obtained in Gelfand–Naimark’s book “Unitary Representations of the Classical Groups”.

N, B — subgroups in $G = SL(3,\mathbb{R})$.

GELFAND-P001502

[source number: 46]

$$\left\{ \begin{pmatrix} 1 & 0 & 0 \\ a_{21} & 1 & 0 \\ a_{31} & a_{32} & 1 \end{pmatrix} \right\} = N \tag{1}$$

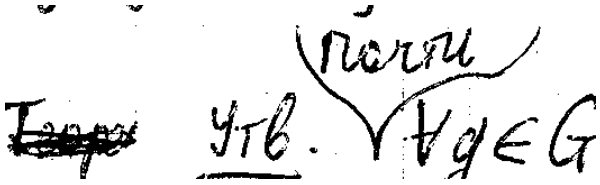
$$\left\{ \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \right\} = B \quad \begin{matrix} \text{(Borel)} \\ b_{11}b_{22}b_{33} = 1. \end{matrix} \tag{2}$$

Consider the cosets

$$\begin{matrix} N\backslash G \\ \| \\ \{Ng \mid g \in G\} \end{matrix} \quad \text{and} \quad B\backslash G \tag{3}$$

(4) ~~Thm.~~ Claim. [Inserted above ∀: Almost]

$$\forall g \in G,\ \exists \text{ a decomposition } g = nz, \quad \text{where } n \in N, \qquad z = \begin{pmatrix} z_{11} & z_{12} & z_{13} \\ 0 & z_{22} & z_{23} \\ 0 & 0 & z_{33} \end{pmatrix}.$$



Original source details: GELFAND-P001502

Uncertain reading. "Thm." before "Claim" is crossed out. "Almost" is inserted above ∀.

(denote by Z the group of such matrices)

(Gauss decomposition) (upper triangular)

The decomposition of any matrix generalizing this is called the Bruhat decomposition.

[crossed out] We will use matrices z as representatives in $N\backslash G$.

Original source details: GELFAND-P001502

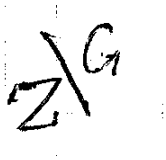
Undeciphered passage. The initial mark before "We will" is crossed out and illegible.

Define functions $f(z)$ [crossed out] on $N\backslash G$.

Original source details: GELFAND-P001502

Undeciphered passage. The mark after $f(z)$ is crossed out and illegible.

$Z\backslash G$ — the space of flags. (compactify further)

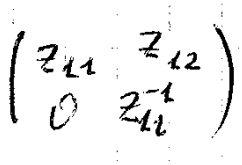


Uncertain reading. The manuscript here reads $Z\backslash G$.

Original source details: GELFAND-P001502

$$\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ \alpha a + c & \alpha b + d \end{pmatrix}$$

$$\begin{pmatrix} z_{11} & z_{12} \\ 0 & z_{11}^{-1} \end{pmatrix}$$



Uncertain reading. The lower-right subscript reads 11. The final digit differs from the 2 above it.

Original source details: GELFAND-P001502

Construct the polynomial $f(z_{11}, z_{12})$

For order 3 one must take polynomials in the first two rows, homogeneous in each. They should be considered on the appropriate compactification.

Uncertain reading. The word before "compactification" reads «нужной» (appropriate).

Original source details: GELFAND-P001502

$$\lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}, \tag{5}$$

multiplication [inserted: on the left] by λ takes a coset to a coset:

$$g_1 = ng_2 \;\Rightarrow\; \lambda g_1 = \lambda n g_2 = \underbrace{\lambda n \lambda^{-1}}_{n_1} \lambda g_2, \quad \text{as required.}$$

Uncertain reading. Two short fragments near the implication arrow are crossed out.

Original source details: GELFAND-P001502

GELFAND-P001503

General definition.

$Z = N\backslash G$ — the basic affine space.

Consider on Z all analytic functions. These are polynomials. Consider functions homogeneous under left translations:

$$f(\lambda z) = \lambda_1^{k_1}\lambda_2^{k_2}\lambda_3^{k_3} f(z) \tag{6}$$

$$(\lambda_1\lambda_2\lambda_3 = 1)$$

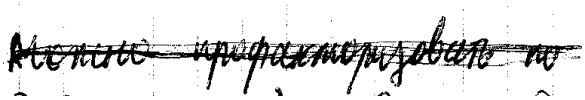
The space of homogeneous polynomials is finite-dimensional.

k_1, k_2, k_3 — the highest weight of the representation.

Formulae for the representation:

$$(T_g f)(z) = f(zg)$$

~~One can factor by~~.



Uncertain reading. "One can factor by" is crossed out.

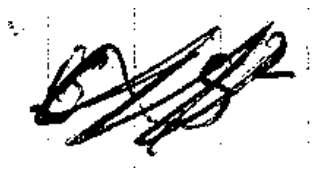
Original source details: GELFAND-P001503

The representation is specified by a triple of numbers (k_1, k_2, k_3).

Problem: 1) when is the representation trivial?

2) when is the representation tautological?

3) when is it [crossed out: in S^2] equivalent to the representation in S^2V?

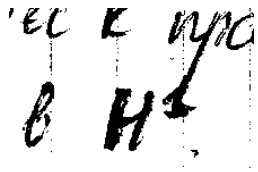


Uncertain reading. The crossed-out entry before "equivalent" appears to read "in S^2".

Original source details: GELFAND-P001503

Find the right basis in this space.

If the highest weight k is right, the representation is realized in sections of a bundle, i.e. in H^1.



Uncertain reading. The superscript of H resembles 1. A long stroke from the preceding line meets it.

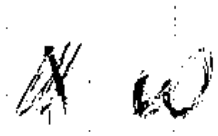
Original source details: GELFAND-P001503

Otherwise Bott discovered that the representation passes to higher cohomology of the flag space.

Gelf., Zelevinsky, Kapranov.

An important observation: on Z one must consider functions as well as differential forms!!!

Consider the [inserted: vector] field λ and all forms [crossed out] ω, for which $i_\lambda \omega = 0$.



Uncertain reading. The crossed-out sign before ω resembles λ.

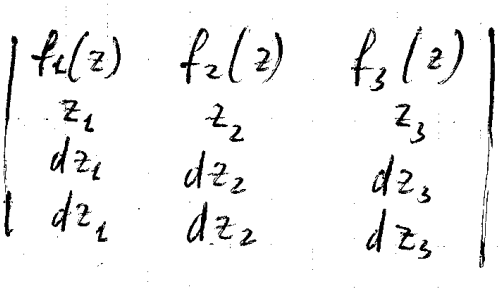
Original source details: GELFAND-P001503

Superalgebras of forms arise here.

Some important forms:

$$\Psi(z) \begin{vmatrix} z_1 & z_2 & z_3 \\ dz_1 & dz_2 & dz_3 \\ dz_1 & dz_2 & dz_3 \end{vmatrix}$$

$$\begin{vmatrix} f_1(z) & f_2(z) & f_3(z) \\ z_1 & z_2 & z_3 \\ dz_1 & dz_2 & dz_3 \\ dz_1 & dz_2 & dz_3 \end{vmatrix}$$



Uncertain reading. The final expression has four rows and three columns between vertical bars.

Original source details: GELFAND-P001503

GELFAND-P001504

[source number: 47]

All this is connected with the problem of what is meant by det.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\det A \text{ — ?}$$

The denominator that arises when solving linear systems?

What if a_{ij} do not commute?

Another approach.

Vectors can be transformed by the action of the group GL_n.

There are 2 [crossed out] orbits: 0 and $V \setminus \{0\}$.

Undeciphered passage. The short entry between 2 and "orbits" is crossed out and illegible.

Original source details: GELFAND-P001504

Let X, Y be nonsingular matrices. One can replace A with XAY. Then all nonsingular matrices are equivalent.

det — a function $f(A)$, vanishing on singular matrices.

Or:

$$\sum a_{ik}\xi_i\eta_k$$

Uncertain reading. A sign resembling η is crossed out between ξ_i and η_k.

Original source details: GELFAND-P001504

X acts on ξ and Y on η.

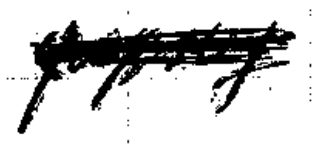
det — [inserted: homogeneous] function of the coefficients (a_{ij}) of least degree, equal to 0 on all singular surfaces.

(All this is on projective spaces).

Consider the cubic matrix a_{ijk} and [crossed out] the cone

$$\sum a_{ijk}\xi_i\eta_j\zeta_k = 0 \tag{7}$$

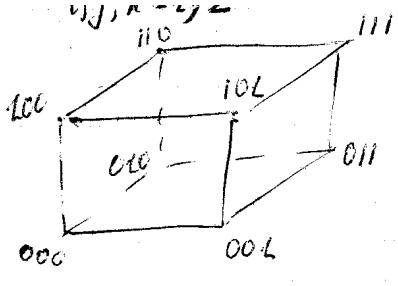
in projective space.



Uncertain reading. The word crossed out before "cone" may be "form".

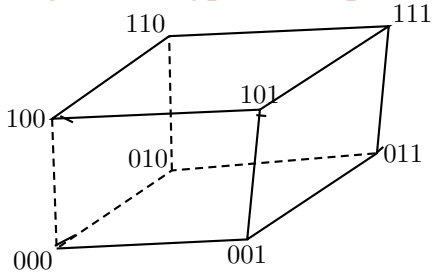
Original source details: GELFAND-P001504

Cubic det — a homogeneous function of least degree of a_{ijk}, equal to 0 on singular surfaces.
2-dimensional case: Cayley did it, but with an error
$i, j, k = 1, 2$



Original source details: GELFAND-P001504

Diagram and typed description.

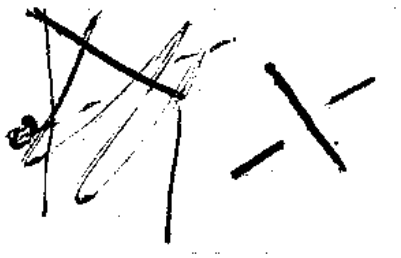


Several edges have gaps and short strokes. Some strokes resemble arrowheads.

deg $= 4$
Elements of 3 types: 1) squares (6 in all):

$$\sum \text{products over squares}$$

2) antisquares (inscribed tetrahedron)
(2 in all)



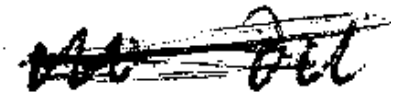
Original source details: GELFAND-P001504

Diagram and typed description.

The left sketch has several overlapping strokes. In the right sketch one diagonal has a gap at the crossing.

GELFAND-P001505

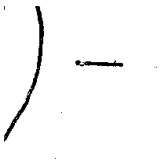
3) $\sum$ ~~over diag.~~ a^2b^2 (4 in all)
[beneath the sum: over the diagonals]



Original source details: GELFAND-P001505

Uncertain reading. “Over diag.” before a^2b^2 is crossed out.

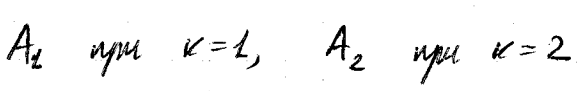
$$\mathrm{Det}(a_{ijk}) = \left(\begin{vmatrix} a_{111} & a_{212} \\ a_{121} & a_{222} \end{vmatrix} + \begin{vmatrix} a_{211} & a_{112} \\ a_{221} & a_{122} \end{vmatrix}\right)^2 - \\ - 4 \begin{vmatrix} a_{111} & a_{112} \\ a_{121} & a_{122} \end{vmatrix} \cdot \begin{vmatrix} a_{211} & a_{212} \\ a_{221} & a_{222} \end{vmatrix} \tag{8}$$



Uncertain reading. The minus sign is written twice, at the end of the first line and the beginning of the second.

Original source details: GELFAND-P001505

Consider 2 matrices A_1 at $k = 1$, A_2 at $k = 2$.



Uncertain reading. The line specifying the slices reads $k = 1$, $k = 2$.

Original source details: GELFAND-P001505

$\det |\xi_1 A_1 + \xi_2 A_2|$ — a quadratic form in ξ_1, ξ_2.

$$\mathrm{Discr.}\,[\det(\xi_1 A_1 + \xi_2 A_2)] = (8). \tag{9}$$

From (9) symmetry in all indices is not obvious.

(a_{ijk}) — product rep. $SL_n \times SL_n \times SL_n$

(on the three indices)

(tensor product of three tautological representations)

Hypothesis: det — Whitehead torsion of the Cayley–Koszul complex

(see p. 42)

Another approach

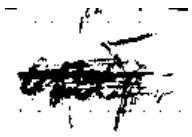
Consider the embedding

$$X = P^{m_1-1} \times P^{m_2-1} \times P^{m_3-1} \subset P^{m_1 m_2 m_3 - 1} = P$$

by Segre (given by the sheaf $O(1,1,1)$) — by multilinear forms.

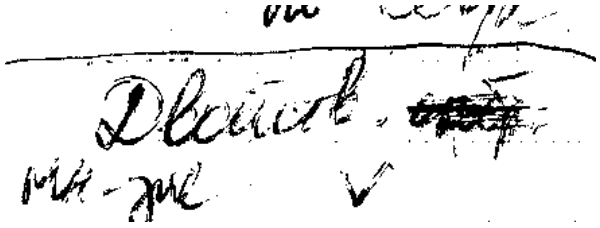
[At left, with a line to $\check{X}$: Dual [crossed out] set [abbrev.]]

Then $\check{X} \subset \check{P}$.



Undeciphered passage. An illegible word is crossed out after “Dual”.

Original source details: GELFAND-P001505



Uncertain reading. The abbreviation below “Dual” reads «мн-же» (set).

Original source details: GELFAND-P001505

$\check{X}$ — a hypersurface. deg — ? Equation — ?

$\check{P}$ — the space of hyperplanes in P.

$\check{X}$ — the set of hyperplanes tangent to X at some point.

[marked by a bracket:] tangency:

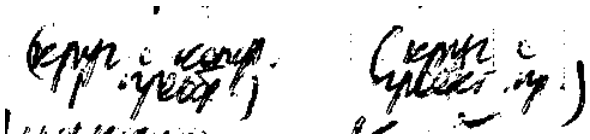
$$\pi \text{ is tangent to } X \ \Leftrightarrow \ T_x\pi \supset T_xX$$

One should read Klein

a) Hist. development of math. b) Higher geom. c) Elem. math. from the higher point of view.

Isomorphism of the Poincaré and Klein models for H^2?

[Above “Poincaré”: (disk with angles preserved). Above “Klein”: (disk with straight ...).]



Uncertain reading. The end of the note above “Klein” is unclear. It may read «yr.» (angles).

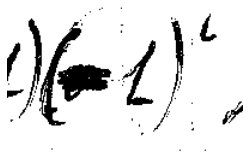
Original source details: GELFAND-P001505

GELFAND-P001506

[source number: 48]

$$N_d(2) = (-1)^d \sum_{i=0}^{d} (i+1)(-1)^i 2^{d-i} \binom{d}{i} i! \tag{10}$$

[In (10), the sign before 1 under the sum: minus?[?].]



Uncertain reading. minus?

Original source details: GELFAND-P001506

(degree of det in d-dimensional space)

[Inserted below:] of matrices $\underbrace{2 \times 2 \times \ldots \times 2}_{d \text{ times}}$.

Correct formula.

$$N_d(2) = (d+3)d!e_d(-2) + (d+1)(-2)^d \in \mathbb{Z}, \tag{11}$$

where

$$e_d(z) = \sum_{k=0}^{d-1} \frac{z^k}{k!}.$$

$$N_d(2) \sim \frac{(d+3)d!}{e^2} \quad \text{for large } d$$

d	$N_d(2)$	
2	2	
3	4	
4	24	(Schläfli)
5	128	

A. S. Schwarz.

Noncommutative analogue of Ray–Singer torsion.

Idea for constructing topological invariants in QFT

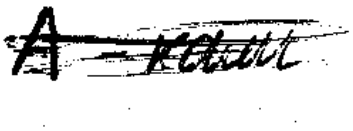
The action

$$S(A) = \int_{M^3} A \wedge dA \tag{1}$$

[Crossed out after 1 in the equation number: 2?[?]. Crossed out at right: A — gau...[?].]

Uncertain reading. 2?

Original source details: GELFAND-P001506



Uncertain reading. A — gau...

Original source details: GELFAND-P001506

gives Ray–Singer torsion.

There is a generalization when A is a form with values in the Lie algebra $\mathfrak{g}$

(gauge field or connection in a principal bundle (trivial))

$M^3 \times G$

G is a compact Lie group

$$S(A) = \operatorname{Tr} \int_{M^3} A \wedge dA + \frac{2}{3} \operatorname{Tr} \int_{M^3} A \wedge A \wedge A \tag{2}$$

[The lettering in the second term has been overwritten: Tr?[?].]

Uncertain reading. Tr?

Original source details: GELFAND-P001506

[Below the second term:] $\operatorname{Tr}[A, A] \wedge A$

one should consider the partition function

$$Z = \int e^{-S(A)} \tag{3}$$

$$\delta S = 0 \quad \text{(equation of motion)} \quad \Leftrightarrow \quad F = 0 \tag{4}$$

[Crossed out after $\delta S = 0$: yp?[?].]

Uncertain reading. yp?

Original source details: GELFAND-P001506

[Below F:] curvature of the connection.

GELFAND-P001507

One can write

$$S(A) = \int_{N^4} F \wedge F, \qquad \partial N = M \tag{5}$$

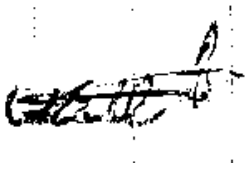
(extend the connection A to N^4 and take the curvature F)

Here unresolved word arise [wavy underline].

Uncertain reading. unresolved word

Original source details: GELFAND-P001507

Link $\rightsquigarrow$ neighbourhood of the knot, [crossed out: glu...?] glue it to the double — — obtain M^3.



Uncertain reading. glu...?

Original source details: GELFAND-P001507

Also consider [Crossed out before P: Tr?.]

Uncertain reading. Tr?

Original source details: GELFAND-P001507

$$P \exp \oint_\Gamma A\,dx \quad \text{(monodromy)}$$

and modify

$$S(A) = \int e^{-S(A)} \operatorname{Tr} P \exp \oint_\Gamma A\,dx \tag{6}$$

[An opening parenthesis is in the margin before (6). Left-hand side of (6): $S(A)$?.]

Uncertain reading. $S(A)$

Original source details: GELFAND-P001507

GELFAND-P001508

[source number: 53]

31/X-88. Gelfand Seminar.

V. E. Zakharov. Equations of hydrodynamics in Clebsch form.

Let S be a matrix from a semisimple Lie algebra. Consider the equation

$$\frac{\partial S}{\partial t} = \left[S, \frac{\partial H}{\partial S}\right] = S_{\alpha\gamma}\frac{\partial H}{\partial S_{\gamma\beta}} - \frac{\partial H}{\partial S_{\alpha\gamma}}S_{\gamma\beta} \tag{1}$$

[Crossed out after the second equals sign: H?.]

Uncertain reading. H?

Original source details: GELFAND-P001508

$$\operatorname{Sp} S^n = I_n \quad \text{— integral of motion} \tag{2}$$

$$S = [p, q], \qquad H = H(S) = H(p, q] \qquad (p, q \text{ are letters}) \tag{3}$$

[The closing delimiter in $H(p, q]$: square bracket or sharply curved parenthesis[?].]

Uncertain reading. square bracket or sharply curved parenthesis

Original source details: GELFAND-P001508

$$\begin{cases} \dot{p} = \left[p, \dfrac{\partial H}{\partial S}\right] = -\dfrac{\partial H}{\partial q}, \\ \dot{q} = \left[q, \dfrac{\partial H}{\partial S}\right] = \dfrac{\partial H}{\partial p}. \end{cases} \tag{4}$$

$$\operatorname{Sp}(p^{n_1} q^{n_2} p^{n_3} \ldots) = \text{const} \quad \text{— integrals} \tag{5}$$

Clebsch variables

$\varphi(x, y)$

$$A = -\varphi_y \frac{\partial}{\partial x} + \varphi_x \frac{\partial}{\partial y} \tag{6}$$

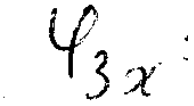
Suppose there are 2 such A_1, A_2.

Then

$$[A_1, A_2] = -\varphi_{3y} \frac{\partial}{\partial x} + \varphi_{3x} \frac{\partial}{\partial y} \tag{7}$$

$$\varphi_{3x} = \varphi_{1x}\varphi_{2y} - \varphi_{1y}\varphi_{2x} = \{\varphi_1, \varphi_2\}$$

[Left-hand side of the bracket definition: φ_{3x}[?].]



Uncertain reading. φ_{3x}

Original source details: GELFAND-P001508

Equations of hydrodynamics in Clebsch form

$$\frac{\partial \Omega}{\partial t} + \{\Psi, \Omega\} = 0 \tag{8}$$

[Crossed out at right: illegible[?].]

Uncertain reading. illegible

Original source details: GELFAND-P001508

$v_x, v_y, \quad \operatorname{div} v = 0$

$$v_x = -\Psi_y, \quad v_y = \Psi_x \qquad (\Psi \text{ is the stream function}) \tag{9}$$

$$\Omega = \Delta\Psi \tag{10?}$$

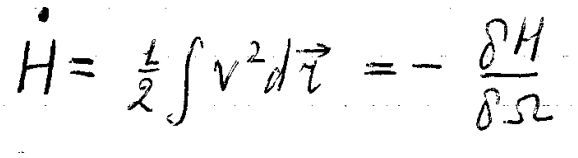
[Overwritten equation number: 10?[?].]

Uncertain reading. 10?

Original source details: GELFAND-P001508

$$\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta \Omega} \tag{11}$$

[Line with a dot above H and two equals signs: $\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta \Omega}$?.]

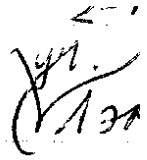


Uncertain reading. $\dot{H} = \frac{1}{2}\int v^2\, d\vec{r} = -\frac{\delta H}{\delta \Omega}$

Original source details: GELFAND-P001508

(Hamiltonian formulation of 2-dimensional hydrodynamics)

Known since 1844 [inserted below: (Clebsch)] ([written above the parenthesis: eqs.[?]] of Lamb et al.)



Uncertain reading. eqs.

Original source details: GELFAND-P001508

Let $\Omega = \{\lambda, \mu\}$. One gets, as in (4):

[Below $\{\lambda, \mu\}$:] (Poisson bracket)

One gets:

$$\begin{cases} \dfrac{\partial\lambda}{\partial t} + \{\Psi, \lambda\} = 0 \\ \dfrac{\partial\mu}{\partial t} + \{\Psi, \mu\} = 0 \end{cases} \quad \Bigg| \quad \begin{cases} \dfrac{\partial\lambda}{\partial t} = -\dfrac{\delta H}{\delta\mu} \\ \dfrac{\partial\mu}{\partial t} = \dfrac{\delta H}{\delta\lambda} \end{cases} \tag{12}$$

Integrals of motion?

We have

$$\Psi = -\frac{\delta H}{\delta\Omega} \tag{13}$$

and

$$\frac{\partial\Omega}{\partial t} + \{\Psi, \Omega\} = 0 \tag{14}$$

(15) Lemma 1 If

$$\frac{\partial\lambda_i}{\partial t} + \{\Psi, \lambda_i\} = 0, \qquad i = 1, 2, \tag{16}$$

then $\{\lambda_1, \lambda_2\}$ satisfies the same equation

$$\frac{\partial}{\partial t}\{\lambda_1, \lambda_2\} + \{\Psi, \{\lambda_1, \lambda_2\}\} = 0. \tag{17}$$

Lemma 2.

If λ_i, $i = 1, \ldots, N$ satisfy (16), then

$$\int F(\lambda_1, \ldots, \lambda_N)\, d\vec{r} = \text{const} \quad \text{— integral of motion} \tag{18}$$

Thus the following are conserved

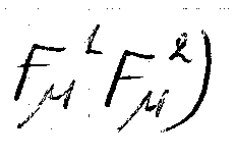
$$\int F(\lambda, \mu, \Omega, \{\lambda, \Omega\}, \{\mu, \Omega\}, \ldots)\, d\vec{r} \tag{19}$$

The Lie algebra of functionals (19) (with respect to the "Poisson bracket") contains the subalgebra (!)

$$J_0 = \left\{ \int F(\lambda, \mu)\, d\vec{r} \right\} \tag{20}$$

$$\begin{aligned} [J_0^1, J_0^2] &= \int (F_\lambda^1 F_\mu^2 - F_\mu^1 F_\mu^2)\, d\vec{r} \\ &= \int \langle F_1, F_2 \rangle\, d\vec{r} \end{aligned} \tag{21}$$

[Last factor in parentheses in (21): $F_\mu^{2?}$.]



Uncertain reading. F_μ^2

Original source details: GELFAND-P001509

[Below $\langle F_1, F_2 \rangle$:] ordinary Poisson bracket on phase space.

Consider the algebra

$$J_1 = \left\{ \int F(\lambda, \mu, \Omega)\, d\vec{r} \right\} \tag{22}$$

This will be a current algebra over the algebra (20).

GELFAND-P001510

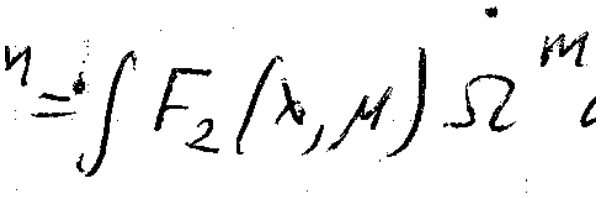
[source number: 54]

Consider

$$J_1^n = \int F_1(\lambda,\mu)\Omega^n\, d\vec{r}, \qquad J_2^m = \int F_2(\lambda,\mu)\Omega^m\, d\vec{r} \tag{23}$$

[In the right-hand formula in (23) there are dots near the equals sign and above Ω: their significance is unclear?.]

Uncertain reading. their significance is unclear

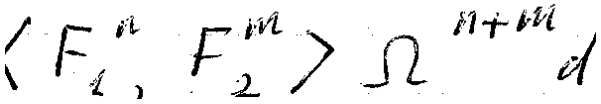


Original source details: GELFAND-P001510

$$[J_1^n, J_2^m] = \frac{(n-1)(m-1)}{n+m-1}\int \langle F_1^n, F_2^m\rangle \Omega^{n+m}\, d\vec{r} \tag{24}$$

[In (24): $\langle F_1^n, F_2^m\rangle\Omega^{n+m}$?.]

Uncertain reading. $\langle F_1^n, F_2^m\rangle\Omega^{n+m}$



Original source details: GELFAND-P001510

(one obtains the Virasoro algebra with zero central charge)

There is a 2-parameter family of integrals. There is no integrability in the general case (H is arbitrary!!) What kind of incompleteness?

Let us try applying Liouville's theorem.

We make changes of variables.

$$\begin{cases} x' = x'(x,y) \\ y' = y'(x,y) \end{cases} \qquad \frac{\partial(x',y')}{\partial(x,y)} \equiv 1. \tag{25}$$

preserve area. All the integrals (23) are unchanged by this.

Fix λ. Which changes preserve λ

One has to slide along the level curves of λ. $\varphi = f(\lambda)$.

Introduce

$$a_k = \frac{1}{\sqrt{2}}(\lambda_k + i\mu_k)$$

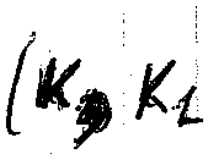
and consider the Hamiltonian

$$H = \int W_{kk_1,k_2k_3}\bar{a}_k\bar{a}_{k_1}a_{k_2}a_{k_3} \tag{26}$$

For its special form (the exact expression is left unwritten here)

$$W'(k,k_1,k_2,k_3) = \{k_1,k_2\}\{k_2,k_3\}f(k_1-k_3)+\dots \tag{27}$$

[After the first k in (27): overwritten comma or subscript followed by a comma?.]



Uncertain reading. overwritten comma or subscript followed by a comma

Original source details: GELFAND-P001510

$(28)_1$ [Crossed out: For] When $f = \frac{1}{\vec{k}^2}$, there are too few integrals.

$(28)_2$ When $f = \frac{\vec{\alpha}\,\vec{k}}{\vec{\beta}\,\vec{k}}$, one obtains an ∞-dimensional top and it is integrable.

Zakharov wants to conduct a numerical experiment on the onset of chaos.

[Inserted at left:] In the 3-dimensional case

$$\vec{\Omega} = [\vec{A}, \operatorname{rot}\vec{B}] - [\vec{B}, \operatorname{rot}\vec{A}]$$

Now a vector, and integrals do not appear in sufficient number.

At the beginning of the next seminar Zakharov will speak about turbulence.

GELFAND-P001511

I. M. Gelfand.

[Crossed out and hatched below the name: illegible?.]

Uncertain reading. illegible

Original source details: GELFAND-P001511

Dual curve

$$y = kx + b$$

Diagram and typed description.

Original source details: GELFAND-P001511

this is inconvenient because k and b have different dimensions.

$$\frac{x}{a}+\frac{y}{b}=1$$ also less than satisfactory.

Better:

$$\xi x+\eta y=1$$

Original source details: GELFAND-P001511

Diagram and typed description.

A straight line is a point in the ξ,η plane

Similarly:

$$\xi_1 x_1+\ldots+\xi_n x_n=1$$ — a hyperplane.

Then

the dual curve is formed by all the tangents in the (ξ,η) plane

Example 1. $y=x^3$

$$y-y_0=3x_0^2(x-x_0)$$
$$k=3x_0^2,\quad b=-2x_0^3$$

Original source details: GELFAND-P001511

Diagram and typed description.

dual curve

From this one can understand the Legendre transform.

Example 2.

$$|x|^p+|y|^p=1$$

It is best to take

$$\xi x+\eta y=1$$ and find when it is tangent.

One obtains

$$|\xi|^q+|\eta|^q=1,\qquad \text{where}\quad \frac{1}{p}+\frac{1}{q}=1$$

[The first exponent has been overwritten: q; the earlier mark is illegible?.]

Original source details: GELFAND-P001511

Uncertain reading. q; the earlier mark is illegible

(dual Banach space)

Problem (for enthusiasts of the geometry of Banach spaces)

A sufficiently high-dimensional cube has a section arbitrarily close to a circle.

Complex point of view.

Dual set — tangent planes

GELFAND-P001512

[source number: 55]

In $\mathbb{C}^3$:

$$\underline{\text{line}} \quad \longleftrightarrow \quad \underline{\text{line}}$$

The dimension of the dual set is an important problem.

Problem Consider the space

$$\mathbb{P}^{(m_1-1)} \times \mathbb{P}^{(m_2-1)} \times \ldots \times \mathbb{P}^{(m_k-1)} \subset \mathbb{P}^{m_1 m_2 \ldots m_k - 1}$$

[Crossed out: Problem.] The problem of hyperdeterminants is equivalent to the following problem:

when does the dual surface have codim $= 1$?

The left-hand side of the equation defining the dual surface is the hyperdeterminant.

Degree — ?

Gelfand and coauthors can compute this. The answer is as follows: for

$$\mathbb{P}^{\ell_1} \times \mathbb{P}^{\ell_2} \times \ldots \times \mathbb{P}^{\ell_d} \subset \mathbb{P}^{(\ell_1+1)(\ell_2+1)\ldots(\ell_d+1)-1}$$

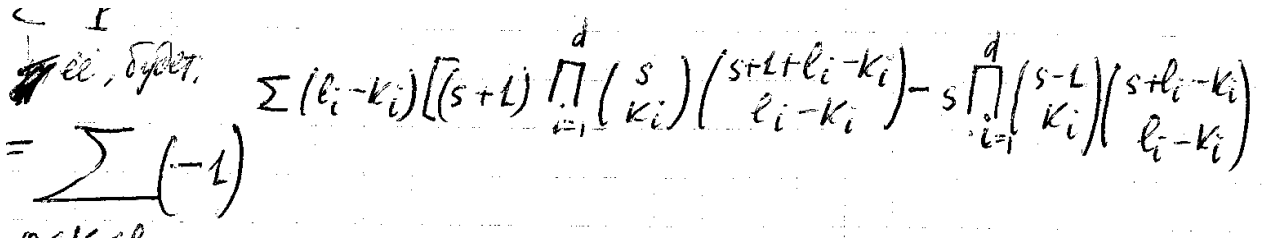
the degree of the polynomial defining [crossed out: ϕ?[?]] it will be

Uncertain reading. ϕ?

Original source details: GELFAND-P001512

$$N(\ell_1, \ell_2, \ldots, \ell_d) = \sum_{\substack{0 \le k_1 \le \ell_1 \\ \ldots \\ 0 \le k_d \le \ell_d}} (-1)^{\sum(\ell_i - k_i)} \Big[(s+1) \prod_{i=1}^{d} \binom{s}{k_i} \binom{s+1+\ell_i-k_i}{\ell_i - k_i} - s \prod_{i=1}^{d} \binom{s-1}{k_i} \binom{s+\ell_i-k_i}{\ell_i-k_i} \tag{1}$$

[In (1), the continuation after (-1) is raised to the level of the exponent and the closing square bracket is not visible: the expression after the sum in the exponent is typeset as a multiplier[?].]



Original source details: GELFAND-P001512

Uncertain reading. the expression after the sum in the exponent is typeset as a multiplier

$$s = k_1 + \ldots + k_d$$

D. Leshchiner.

Generating function for this number

$$\sum N(\ell_1, \ldots, \ell_d, \ldots)\, x_1^{\ell_1} x_2^{\ell_2} \ldots x_d^{\ell_d} x_{d+1}^0 \ldots = \left(\sum_{k=0}^{\infty} (1-k) e_k \right)^{-2} \tag{2}$$

[The first variable in (2) has been overwritten: x_1 or z_1?.]

Uncertain reading. x_1 or z_1

Original source details: GELFAND-P001512

[Crossed out after $x_2^{\ell_2}$: ℓ?.]

Uncertain reading. ℓ?

Original source details: GELFAND-P001512

(after ℓ_d we append zeros successively?)

Uncertain reading. successively?

Original source details: GELFAND-P001512

$$e_0 = 1$$

e_k are the elementary symmetric functions:

$$e_1 = \sum_i x_i,$$
$$e_2 = \sum_{i<j} x_i x_j$$
$$e_3 = \sum_{i<j<k} x_i x_j x_k$$
$$\ldots$$

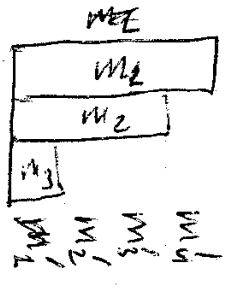
M. Kapranov. Another formula:

$$\lambda = (\ell_1 \geq \ell_2 \geq \ldots)$$

$$N(\lambda) = \sum_{\mu} \frac{(m_1+1)! \prod_{i \geq 1} (i-1)^{m_i - m_{i+1}}}{(m_1 - m_2)!\,(m_2 - m_3)! \ldots} \cdot M(\mu', \lambda),$$

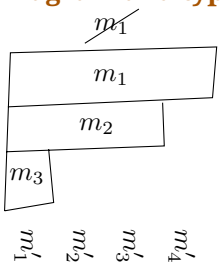
$$\mu = (m_1 \geq m_2 \geq \ldots)$$

where μ' is the conjugate partition



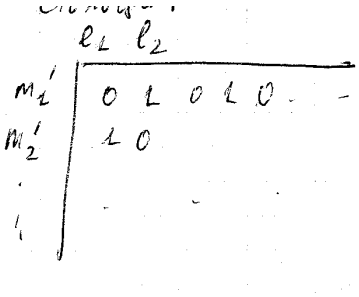
Original source details: GELFAND-P001513

Diagram and typed description.



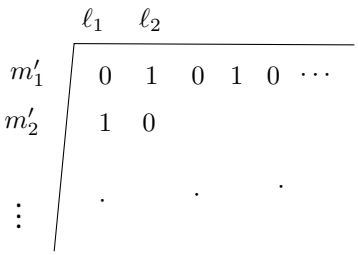
[Above the top strip a label resembling m_1 is crossed out. Below the diagram the first rotated label, m'_1, is overwritten.]

$M(\mu', \lambda)$ is the number of matrices of 0 and 1 whose row and column sums



Original source details: GELFAND-P001513

Diagram and typed description.



$$M(\mu', \lambda) > 0 \quad \Rightarrow \quad \lambda \geq \mu \quad \text{with respect to the natural order.}$$

and, moreover, then

$$\forall i: \quad \ell_i \leq \sum_{j \neq i} \ell_j$$

Zelevinsky

Boundary case:

$$\ell_1 = \ell_2 + \ldots + \ell_d$$

Then

$$N = \frac{(\ell_1 + 1)!}{\ell_2! \ell_3! \ldots \ell_d!}$$

Apparently this is where one should compute det

(1st nontrivial case: [crossed out: illegible mark?] $2 \times 3 \times 4$)

Uncertain reading. illegible mark

Original source details: GELFAND-P001513

GELFAND-P001514

13/II-88.

Zelevinsky. On multidimensional determinants.

$$A = (a_{i_1 \ldots i_d})_{0 \le i_k \le \ell_k} \quad \text{— a multidimensional matrix}$$

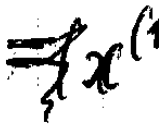
$$\dim V_k = \ell_k + 1, \qquad x_i^{(1)} \in V_1, \ldots, x_i^{(d)} \in V_d.$$

$$A \in (V_1 \otimes \ldots \otimes V_d)^*$$

$$x = x^{(1)} \otimes \ldots \otimes x^{(d)}$$

[A stroke crosses the two bars before $x^{(1)}$: $=$ or $\neq$?.]

Uncertain reading. $=$ or $\neq$

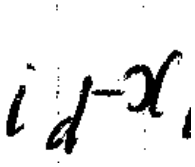


Original source details: GELFAND-P001514

$$A(x) = \sum_{i_1, \ldots, i_d} a_{i_1 \ldots i_d} - x_{i_1}^{(1)} \ldots x_{i_d}^{(d)}$$

[After $a_{i_1 \ldots i_d}$ there is a short stroke: minus or linking stroke?.]

Uncertain reading. minus or linking stroke



Original source details: GELFAND-P001514

$$\text{Det } A = 0 \iff A \text{ is degenerate}$$

$$A \text{ is degenerate} \overset{\text{def}}{\iff} \exists x^{(1)}, \ldots, x^{(d)} \neq 0$$

$$\ell_1 \ge \ell_2, \ell_3, \ldots, \ell_d$$

$$A(x^{(1)}, \ldots, x^{(i-1)}, \cdot, x^{(i+1)}, \ldots, x^{(d)}) \equiv 0 \quad \text{for all } i$$

$$\text{Det is defined} \overset{\text{def}}{\iff} \text{Codim}\{\text{degenerate matrices}\} = 1$$

$$\Updownarrow$$

$$\ell_1 \le \ell_2 + \ell_3 + \ldots + \ell_d$$

$$(\ell_1 + 1) \times (\ell_1 + 1) \quad \Rightarrow \quad \deg = \ell_1 + 1$$

(1) Claim. Suppose $\ell_1 = \ell_2 + \ldots + \ell_d$

Then Det has degree

$$\frac{(\ell_1 + 1)!}{\ell_2! \ell_3! \ldots \ell_d!}$$

Let the degree of Det for $(\ell_1 + 1) \times \ldots \times (\ell_d + 1)$ be $N(\ell_1, \ldots, \ell_d)$

and $N(\ell_1, \ldots, \ell_d) = 0$ if Det is undefined

GELFAND-P001515

[source number: 69]

(2) Claim.

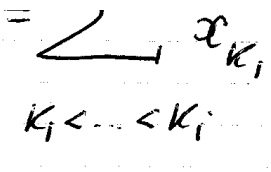
$$\sum_{\ell_1,\ldots,\ell_d \geq 0} N(\ell_1,\ldots,\ell_d) x_1^{\ell_1} \ldots x_d^{\ell_d} =$$

$$= \left[1 - \sum_{i \geq 2} (i-1) e_i(x_1,\ldots,x_d) \right]^{-2},$$

where e_i is the elementary symmetric polynomial

$$e_i(x_1,\ldots,x_d) = \sum_{k_i < \ldots < k_i} x_{k_i} \ldots x_{k_i}$$

[The first lower index in the summation condition and in the first factor: i or j?.]



Uncertain reading. i or j

Original source details: GELFAND-P001515

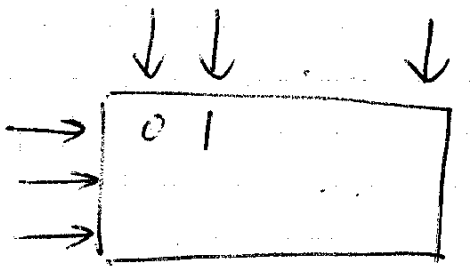
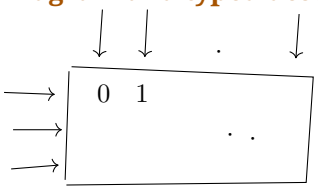


Diagram and typed description.



Original source details: GELFAND-P001515

Consider $0,1$-matrices with given row and column sums.

Everything is expressed in terms of such numbers.

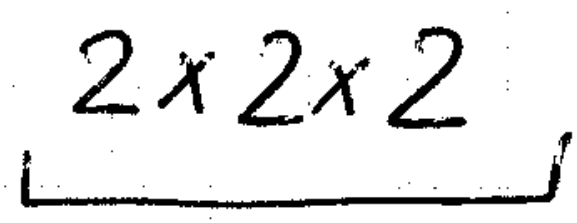
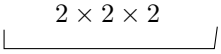


Diagram and typed description.



Original source details: GELFAND-P001515

A complex will be constructed

$$0 \longrightarrow C_0 \xrightarrow{d(A)} C_1 \xrightarrow{d(A)} \dots \xrightarrow{d(A)} C_n \longrightarrow 0 \tag{3}$$

[In the right margin:] ‖ $d(A)$ depends linearly on the matrix A

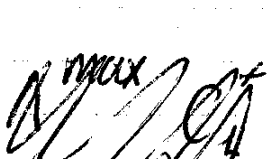
$$A \text{ is degenerate} \iff \text{failure of exactness}$$

Let $0 \to C_0 \to C_1 \to \dots \to C_n \to 0$

$$\operatorname{Det} C_\bullet = \frac{\Lambda^{\max} C_0 \otimes \Lambda^{\max} C_2 \otimes \dots}{\Lambda^{\max} C_1 \otimes \Lambda^{\max} C_3 \otimes \dots}$$

[Crossed out: $\Lambda^{\max} C_1$ or $\Lambda^{\max} C_i$?.]

Uncertain reading. $\Lambda^{\max} C_1$ or $\Lambda^{\max} C_i$



Original source details: GELFAND-P001515

$$\sum (-1)^i \dim C_i = \sum (-1)^i \dim H_i$$

A natural isomorphism follows from the proof.

$$\operatorname{Det} C_\bullet \cong \operatorname{Det} H_\bullet$$

Acyclicity of $C_\bullet \Rightarrow \operatorname{Det} H_\bullet = \mathbb{C}$.

GELFAND-P001516

Choose bases in all the $C_i \Rightarrow$ this gives an element [crossed out: in?]

Uncertain reading. in?

Original source details: GELFAND-P001516

$$\alpha \in \operatorname{Det} C_\bullet$$

and, consequently, a number.

α depends on A rationally and often polynomially.

For the 2×2 matrix determinant $ad - bc$, one obtains

$$0 \to \mathbb{C} \to \mathbb{C}^4 \to \mathbb{C}^7 \to \mathbb{C}^4 \to 0$$

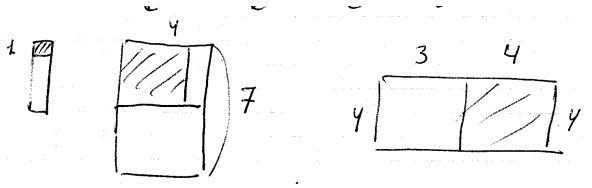
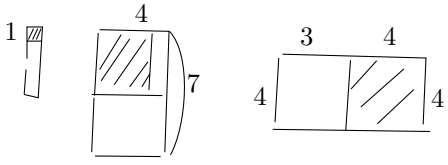
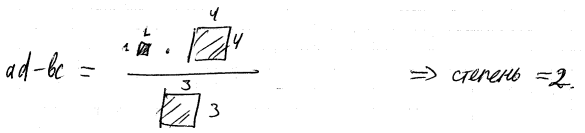


Diagram and typed description.



Original source details: GELFAND-P001516



Original source details: GELFAND-P001516

Diagram and typed description.

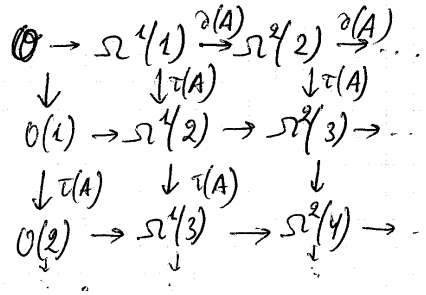
$$ad - bc = \frac{1\frac{1}{\square} \cdot \overset{4}{\square}\, 4}{\underset{3}{\square}\, 3} \Rightarrow \text{degree} = 2.$$

det = GCD of the maximal minors of the 4×7 matrix

— · —

Kapranov

$$X = P(V_1^*) \times \cdots \times P(V_d^*), \qquad O(1) = O(1, \ldots, 1)$$



Original source details: GELFAND-P001516

Diagram and typed description.

$$\begin{array}{ccccccc}
O & \longrightarrow & \Omega^1(1) & \xrightarrow{\partial(A)} & \Omega^2(2) & \xrightarrow{\partial(A)} & \cdots \\
\downarrow & & \downarrow{\scriptstyle \tau(A)} & & \downarrow{\scriptstyle \tau(A)} & & \\
O(1) & \longrightarrow & \Omega^1(2) & \longrightarrow & \Omega^2(3) & \longrightarrow & \cdots \\
\downarrow{\scriptstyle \tau(A)} & & \downarrow{\scriptstyle \tau(A)} & & \downarrow & & \\
O(2) & \longrightarrow & \Omega^1(3) & \longrightarrow & \Omega^2(4) & \longrightarrow & \cdots \\
\downarrow & & \downarrow & & \downarrow & &
\end{array}$$

[Upper-left node: O beneath the overdrawn strokes?[?].]

Uncertain reading. O beneath the overdrawn strokes?

Original source details: GELFAND-P001516

[At the upper end of the arrow $\Omega^1(1) \to \Omega^1(2)$: a slanted stroke resembling part of a second arrowhead[?].]

Uncertain reading. a slanted stroke resembling part of a second arrowhead

Original source details: GELFAND-P001516

1) $\partial(A)^2 = 0, \qquad \partial(A)\partial(B) = -\partial(B)\partial(A)$
2) $\tau(A)\partial(A) = \partial(A)\tau(A)$
3) $\tau(A)\tau(B) = \tau(B)\tau(A)$

GELFAND-P001517

[source number: 70]

Then fixing $m \geq 0$, we obtain

$$C_i^{(m)} \cong \Gamma(X, \Omega^i(i+m)) \oplus \Gamma(X, \Omega^{i-1}(i+m))$$

[Sign between the left- and right-hand sides: $\cong^?$.]

Uncertain reading. $\cong$

Original source details: GELFAND-P001517

[Below the formula, at right:] (totalization of a bicomplex consisting of 2 rows)

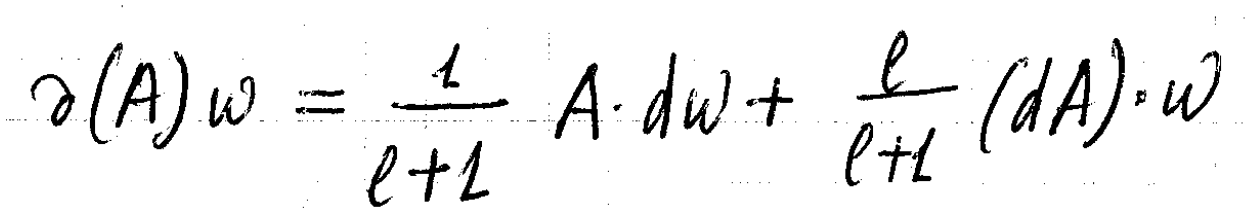
$$A \in \Gamma(X, O(1))$$

$\tau(A)$ is multiplication by $A \in \Gamma(X, O(1))$

$\partial(A)$ on $\Omega^k(\ell)$

$$\partial(A)\omega = \frac{1}{\ell+1} A \cdot d\omega + \frac{\ell}{\ell+1}(dA) \cdot \omega$$

[The manuscript reads: a $+$ between the terms and coefficients $\frac{1}{\ell+1}$ and $\frac{\ell}{\ell+1}$?.]



Original source details: GELFAND-P001517

Uncertain reading. a $+$ between the terms and coefficients $\frac{1}{\ell+1}$ and $\frac{\ell}{\ell+1}$

— · —

Zelevinsky

$$X \subset \mathbb{P}(V^*)$$

[Above the abbreviation for “linearly”: 1 or a vertical stroke?.]

Uncertain reading. 1 or a vertical stroke

Original source details: GELFAND-P001517

linearly normal, i.e. $H^0(X, O(1)) =$ [crossed out: illegible mark?] V.

Uncertain reading. illegible mark

Original source details: GELFAND-P001517

Examples

1) V is an irreducible representation of a semisimple group G

$X \subset \mathbb{P}(V^*)$ is the projectivization of the orbit of the highest-weight vector.

2) Let a complex[?] torus $(\mathbb{C}^*)^n = H$ act on V

Uncertain reading. complex

Original source details: GELFAND-P001517

Characters $\mathbb{Z}^n$, with a distinguished collection in it

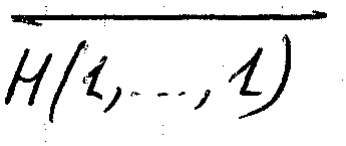
$$\mathbb{Z}^n = ((\mathbb{C}^*)^n)^\wedge \ni \chi_1, \dots, \chi_N, \qquad N = \dim V, \quad V = \mathbb{C}^N$$

All the χ_j lie on an affine hyperplane in $\mathbb{Z}^n$

$$\begin{aligned} X &= (\text{closure of a generic orbit of this complex torus}) \\ &= \overline{H(1, \dots, 1)} \end{aligned}$$

[Ends of the bar above $H(1, \dots, 1)$: strokes resembling arrowheads[?].]

Uncertain reading. strokes resembling arrowheads



Original source details: GELFAND-P001517

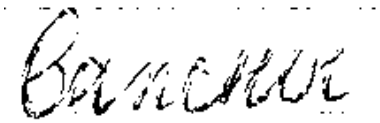
Intersection of examples 1 and 2 — hyperdeterminants.

GELFAND-P001518

[source number: 73]

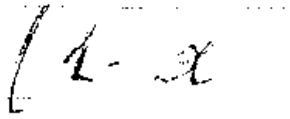
The behavior of the Betti number on the space of representations has been little studied! Jumps in the Betti number are important[?] (first?[?] triv. unresolved word!)

Uncertain reading. important



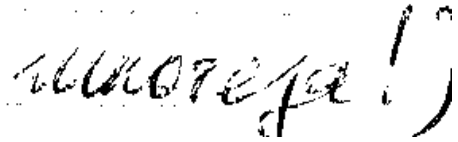
Original source details: GELFAND-P001518

Uncertain reading. first?



Original source details: GELFAND-P001518

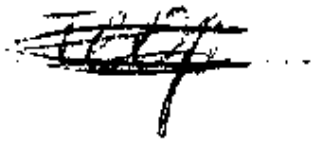
Uncertain reading. unresolved word



Original source details: GELFAND-P001518

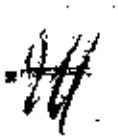
Conjecture: [Crossed out: illegible[?].] The Jones polynomial is merely [crossed out: illegible[?].] a scheme for computing the invariants that have to be computed first.

Uncertain reading. illegible



Original source details: GELFAND-P001518

Uncertain reading. illegible



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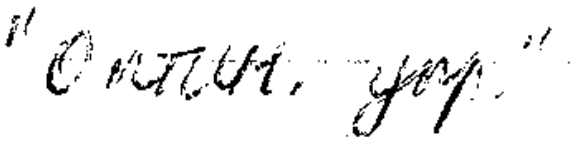
20/II-89 Plan for Vishik's seminar

[Left column:]

1. Fursikov "Optimal control[?]"
2. Vainberg
3. Kuksin[?].
4. Agranovich

Uncertain reading. Optimal control



Original source details: GELFAND-P001518

Uncertain reading. Kuksin

Original source details: GELFAND-P001518

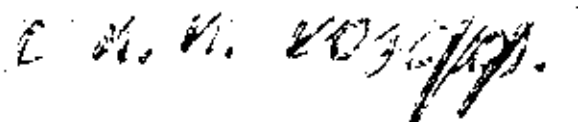
[Crossed out below item 4: illegible[?].]

Uncertain reading. illegible

Original source details: GELFAND-P001518

[Right column, beyond the vertical rule:]

5. Sinai “Schrödinger operator with q.p.[?] coefficients”
6. Birman or Solomyak (spectral asymptotics[?] for the Schrödinger operator)
7. Babin, Vishik: on attractors for unbounded domains.
8. Tuzhilin “Minimal surfaces”
9. Gelfand, Zakharevich.



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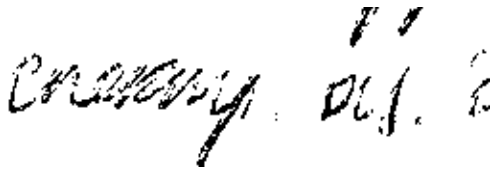
Uncertain reading. q.p.



Original source details: GELFAND-P001518

Uncertain reading. spectral asymptotics

20/II-89 Beilinson.
Lusztig.
The following objects are roughly equivalent

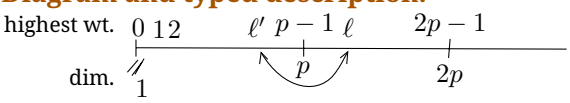
1. Representations of [crossed out: illegible[?]] semisimple algebraic groups in finite characteristic.
2. Representations of quantum groups when $q =$ a root of 1.
3. Representations of loop groups with integral level.
4. Geometry of affine Schubert cells

Original source details: GELFAND-P001518

Uncertain reading. illegible

Original source details: GELFAND-P001518

Diagram and typed description.



[An illegible mark is crossed out below 0. The arrowheads on the arc are faint.]

X_ℓ is the irreducible representation of highest weight ℓ

V_ℓ is the representation on homogeneous polynomials.

Let ℓ be an integer, then

$$0 \to X_{\ell'} \to V_\ell \to X_\ell \to 0$$

I. M. Gelfand.

Consider 6 lines in the projective plane

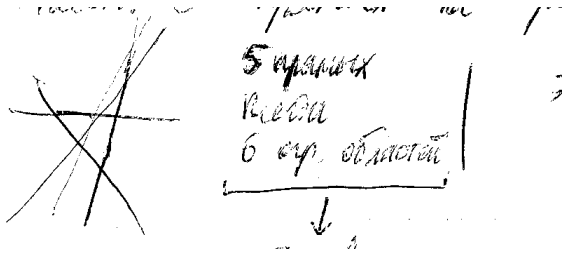


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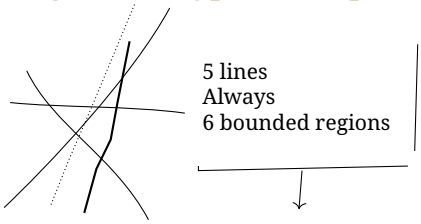
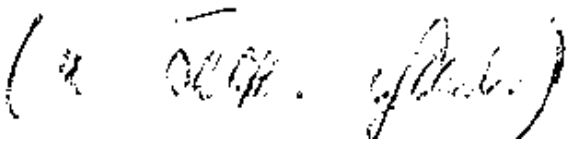
Diagram and typed description.



[The thin steep line is visible in places. Some intersections are indistinct.]

[Below the figure:] (and[?] the line at infinity)



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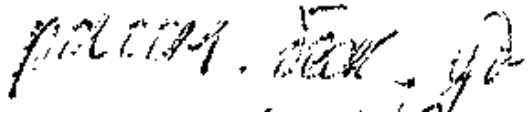
Uncertain reading. and

[Further right:] This occurs in the integral

$$\int \frac{dx}{\sqrt{(x-a)(x-b)(x-c)(x-d)}}$$

[Below the arrow:] Proof: consider the infinitely distant[?] line.

Uncertain reading. infinitely distant



Original source details: GELFAND-P001518

GELFAND-P001519

Kapranov

X is a $K3$ surface if

1) $\pi_1(X) = 0$

2) Ω^2_X is trivial (as a holomorphic bundle)

[Written below Ω^2_X:] (bundle of volume forms)
(i.e. $\exists$ a holomorphic volume form [crossed out: illegible?] which is nowhere zero)

Uncertain reading. illegible

Original source details: GELFAND-P001519

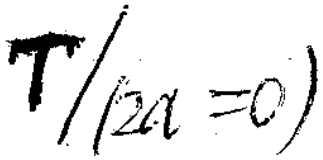
They are all homeomorphic and even diffeomorphic.

Example: 2-dimensional complex torus

$$\begin{array}{c} T = \mathbb{C}^2/\mathbb{Z}^4 \\ \downarrow \qquad\qquad (x \text{ and } -x \text{ identified}) \\ T/(2x = 0) \end{array}$$

[Notation for the quotient: $T/(2x = 0)$?.]

Uncertain reading. $T/(2x = 0)$



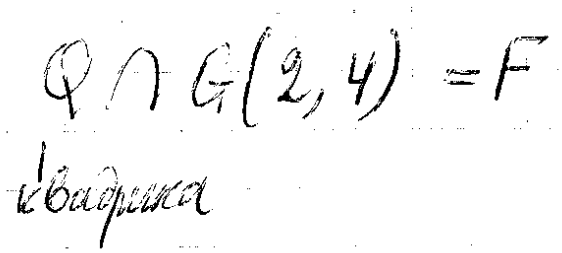
Original source details: GELFAND-P001519

16 singular points arise (Kummer surface)
It will be smooth after resolving the singularities.

— · —

$G(2,4) \subset \mathbb{P}^5$ (quadric)

Diagram and typed description.
$Q \cap G(2,4) = F$
|
quadric



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Chapter 34

GELFAND-P001520

15/$\overline{V}$-89. Gelfand Sem.

V. Drinfeld.

Quasi-Hopf algebras and the Knizhnik–Zamolodchikov equations.

The Kn.–Zam. eqs. arose in conformal field theory

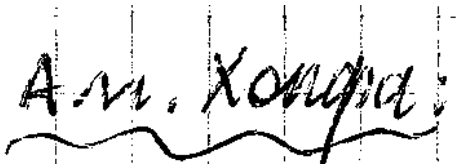
Theorem [above the line, pointing to “Theorem”: (Kohno and somebody else)] on the equiv. of two classes of reprs. of the Artin braid group.

1st class — matrices

2nd class — monodromy of the Kn.–Zam. eqs.

Hopf alg.: an assoc. algebra A with an algebra homomorphism

$$\Delta\colon A \longrightarrow A \otimes A$$



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Uncertain reading. alg.

there are axioms for the counit and antipode

Coassociativity: the diagram commutes

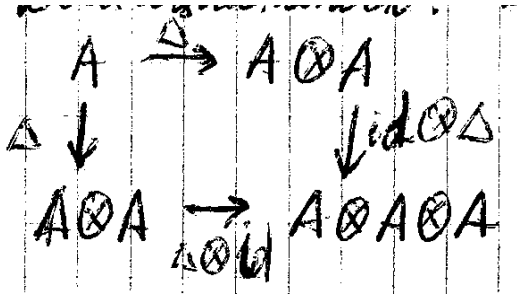

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Diagram and typed description.

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \otimes A \\ {\scriptstyle\Delta}\downarrow & & \downarrow{\scriptstyle id \otimes \Delta} \\ A \otimes A & \xrightarrow[\Delta \otimes id]{} & A \otimes A \otimes A \end{array}$$

or

$$(id \otimes \Delta)(\Delta(a)) = (\Delta \otimes id)(\Delta(a)) \tag{1}$$

Quasi-Hopf property: equality (1) holds up to conjugation

$$(id \otimes \Delta)(\Delta(a)) = \phi \cdot (\Delta \otimes id)(\Delta(a))\phi^{-1}, \tag{2}$$

where $\phi \in A \otimes A \otimes A$

5-gon axiom

$$\begin{aligned} &(id \otimes id \otimes \Delta)(\phi) \cdot (\Delta \otimes id \otimes id)(\phi) \\ &\quad = (1 \otimes \phi) \cdot (id \otimes \Delta \otimes id)(\phi)(\phi \otimes 1) \end{aligned} \tag{3}$$

GELFAND-P001521

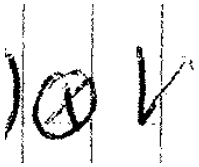
[source number: 36]

Meaning of axiom (3): the composition

$$A \xrightarrow{\Delta} A \otimes A \longrightarrow \mathrm{End}(V_1 \otimes V_2) \tag{4}$$

will be a representation (if the V_j are representations), and there is weak associativity: an isomorphism

$$(V_1 \otimes V_2) \otimes V \xrightarrow{\sim} V_1 \otimes (V_2 \otimes V_3) \tag{5}$$

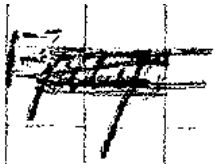


Original source details: GELFAND-P001521

Uncertain reading. V

[Below the arrow, pointing to it:] not the identity! (constructed using ϕ).

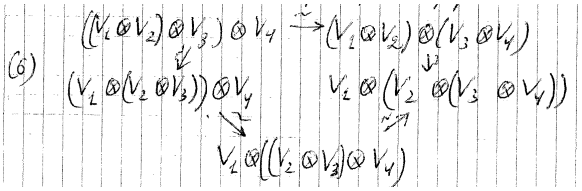
Suppose there are 4 factors. Then there are 5 ways to place the parentheses.



Original source details: GELFAND-P001521

Undeciphered passage. Crossed out before the diagram: a short word, the letters obscured by strokes.

We draw the diagram:



Original source details: GELFAND-P001521

Diagram and typed description.

$$\begin{array}{ccc} ((V_1 \otimes V_2) \otimes V_3) \otimes V_4 & \xrightarrow{\sim} & (V_1 \otimes V_2) \otimes (V_3 \otimes V_4) \\ \downarrow \sim & & \downarrow \sim \\ (6)\ (V_1 \otimes (V_2 \otimes V_3)) \otimes V_4 & & V_1 \otimes (V_2 \otimes (V_3 \otimes V_4)) \\ & \searrow^{\sim} \quad \swarrow_{\sim} & \\ & V_1 \otimes ((V_2 \otimes V_3) \otimes V_4) & \end{array}$$

(all the isomorphisms are constructed using ϕ).

Requirement: diagram (6) commutes.

Quasitriangular quasi-Hopf algebras.

A / Δ [sign heavily overwritten] is commutative: $R\Delta = \Delta R$, $R \in A \otimes A$

Original source details: GELFAND-P001521

Uncertain reading. A / Δ

Original source details: GELFAND-P001521

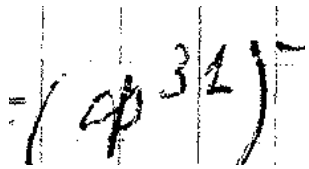
Uncertain reading. $R\Delta = \Delta R$

One requires $\Delta'(a) = R\Delta(a)R^{-1}$ — also a comultiplication

6-gon axioms

$$(\Delta \otimes id)(R) = \Phi^{312}R^{13}(\Phi^{132})^{-1}R^{23}\phi \tag{7}_1$$

$$(id \otimes \Delta)(R) = (\phi^{31})^{-1}R^{13}\phi^{213}R^{12}\Phi^{-1} \tag{7}_2$$



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Uncertain reading. 31

In fact quasi-Hopf algebras are simpler [inserted above the line: (than Hopf algebras)], since there is an operation of Twisting. [Below "Twisting": ($\sim$ gauge transformation)] Let there be given A, Δ, ϕ (a quasi-Hopf algebra)

Suppose there is an invertible element $F \in A \otimes A$

[I.M.G.'s question: a) what is the dual object? — dual space

Original source details: GELFAND-P001521

Uncertain reading. dual object

b) what is the quasi-Hopf property on the dual space?]

GELFAND-P001522

The new quasi-Hopf algebra:

$$\widetilde{\Delta}(a) = F \cdot \Delta(a) \cdot F^{-1}, \tag{8}$$

$$\widetilde{\phi} = F^{23}(id \otimes \Delta)(F) \cdot \phi(\Delta \otimes id)(F^{-1})(F^{12})^{-1}, \tag{9}$$

$$\widetilde{R} = F^{21}RF^{-1}. \tag{10}$$

It is natural to regard such objects as equivalent.

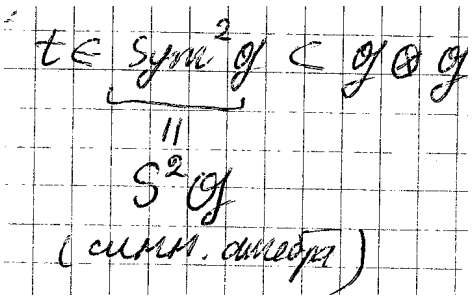
Original source details: GELFAND-P001522

Uncertain reading. Before the paragraph: a heavy vertical mark.

Next, to study Hopf algebras it is useful to consider quasi-Hopf algebras.

Construction of quasi-Hopf algebras.

Initial data: Lie alg. g,



Original source details: GELFAND-P001522

Diagram and typed description.

$$t \in \underbrace{\mathrm{Sym}^2 g}_{\| \; S^2 g} \subset g \otimes g$$

(symm. algebra)

$$A = (U_g)[[h]] \quad \text{— an algebra over } \mathbb{C}[[h]] \tag{11}$$

[Below U_g, with a connecting stroke: “univ. enveloping”.]

Comultiplication

$$x \in g \Rightarrow \Delta(x) = x \otimes 1 + 1 \otimes x \tag{12}$$

$$R = e^{ht/2} \tag{13}$$

and now to determine ϕ one writes down a tricky system of eqs. which has a solution, unique up to twists that leave the form of the comultiplication and t unchanged.

Consider the ordinary diff. eq.:

$$G'(x) = \bar{h}\left(\frac{t^{12}}{x} + \frac{t^{23}}{x-1}\right) G(x), \quad \bar{h} = \frac{h}{2\pi i}, \tag{14}$$

where

$$G(x) \in (U_g \otimes U_g \otimes U_g)[[h]].$$

This construction yields all deformations of quasi-Hopf algebras

Now ϕ is determined as follows: on $(0,1)$ consider 2 solutions

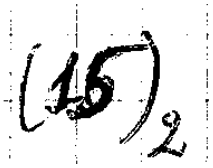
GELFAND-P001523

[source number: 37]

G_j, $j = 1, 2$, with standard asymptotics at 0 and 1:

$$G_1(x) \sim x^{\bar{h}t^{12}}, \quad x \to 0 \tag{15}_1$$

$$G_2(x) \sim (1-x)^{\bar{h}t^{23}}, \quad x \to 1 \tag{15}_2$$



Original source details: GELFAND-P001523

Uncertain reading. Number of the second line: the digit 5 is retraced or corrected.

Then

$$G_1 = G_2 \cdot \Phi, \quad \phi \in (U_g \otimes U_g \otimes U_g)[[h]]. \tag{16}$$

and this ϕ satisfies all the required conditions.

I.M.G. The Kn.–Zam. eqs. are a very simple and deep example of a hypergeom. function.

Dotsenko. Ising model.

Motivated by an old work of A.M. Polyakov on the 3-dimensional Ising model

The Ising model on a lattice of any dimension: the partition function has the form

$$Z(\beta) = \sum_{\{\sigma\}} \exp\left\{-\beta \sum_{x,\alpha} \sigma_x \sigma_{x+\alpha}\right\} = \sum_{\{\sigma\}} \prod_{x,\alpha} e^{-\beta \sigma_x \sigma_{x+\alpha}} \tag{1}$$

$$x \in \mathbb{Z}^d; \qquad \alpha \in \mathbb{Z}^d, \quad |\alpha| = 1, \qquad \beta > 0$$

$\{\sigma\}$ — spin configurations, $\sigma_x = \pm 1$.

(Partition sum = Partition function)

Hamiltonian:

[K. Khanin]

$$H = -\sum_{x,\alpha} \sigma_x \sigma_{x+\alpha} \tag{2}$$

One should take everything in a finite container V and the free energy

$$f(\beta) = \lim_{V \to \infty} \frac{\ln Z_V(\beta)}{|V|} \tag{3}$$

is independent of the boundary conditions and determines phase transitions

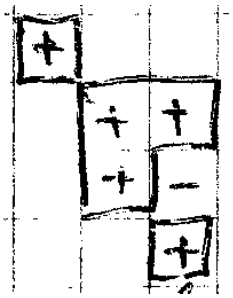
One can also take

$$H = -\sum_{x,\alpha} \sigma_x \sigma_{x+\alpha} + h \sum_x \sigma_x \tag{4}$$

and calculate correlation functions.

GELFAND-P001524

A configuration [inserted above the line: (for $d = 2$)] is uniquely characterized by a set of non-intersecting contours (what separates the pluses and minuses).



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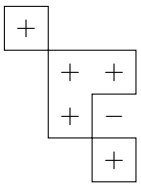


Diagram and typed description.

It is easy to check that in the 2-dimensional case

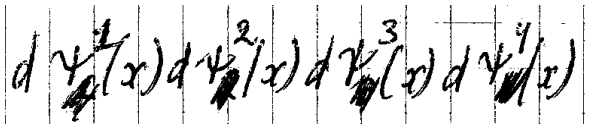
$$Z(\beta) = \sum_{\text{contours}} (\lambda)^L; \tag{5}$$

where

$$\lambda = e^{-2\beta}; \qquad L \text{ — contour length}$$

Next, one can write (from (5))

$$Z(\beta) = \prod_x \int d\psi^1(x)\, d\psi^2(x)\, d\psi^3(x)\, d\psi^4(x)\ \exp\{A[\psi]\}, \tag{6}$$



Original source details: GELFAND-P001524

Uncertain reading. Crossed-out subscripts under the four ψ symbols: reading undetermined.

where ψ^α — Grassmann variables (4 at each site);

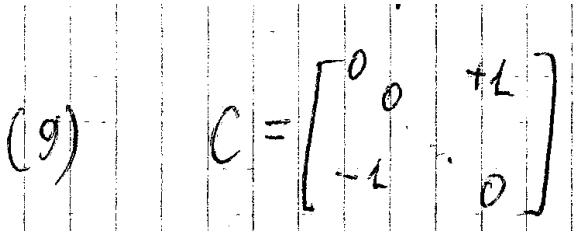
Original source details: GELFAND-P001524

Uncertain reading. Separate mark at left above (7): x? / z?.

$$A[\psi] = -\frac{1}{2}\sum_{x,\alpha}(\psi(x)\, C\, \psi(x)) + \lambda \sum_{x,\alpha} \psi(x+\alpha) P_\alpha \psi(x) \tag{7}$$

$$(\psi C \psi) = \psi^\alpha C^{\alpha\beta} \psi^\beta; \tag{8}$$

$C^{\alpha\beta}$ — a certain matrix (not uniquely determined)



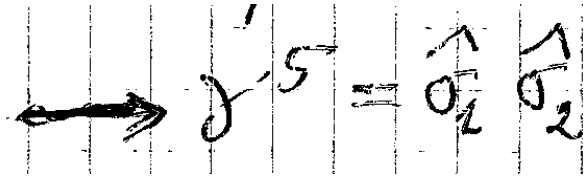
Original source details: GELFAND-P001524

Diagram and typed description.

(9) $$C = \begin{bmatrix} 0 & 0 & +1 \\ & 0 & \\ -1 & & 0 \end{bmatrix}$$

In the continuum limit

$$C \longleftrightarrow \gamma^5 = \hat{\sigma}_1 \hat{\sigma}_2 \tag{10}$$



Original source details: GELFAND-P001524

Uncertain reading. γ^5

σ_j — Pauli matrices

$$P_\alpha = \frac{1 + \vec{\alpha}\,\vec{\sigma}}{2} \quad \text{— projector} \tag{11}$$

The contours reduce to directed walks. In the walk a factor $\lambda e^{i\Delta\varphi/2}$ arises, which

[source number: 38]

enters the amplitude (complex probability); $\Delta\varphi$ — angle of turn.
$e^{i\Delta\varphi/2}$ — Kac–Ward factors
Explanation by a walk problem — Feynman.

$$\sum_{\mathcal{P}_{x_1,x_2}} \prod \lambda e^{i\Delta\varphi/2} = \langle \psi^\alpha(x_1)\psi^\beta(x_2)\rangle \tag{12}$$

Diagram and typed description. $= \sum_{\mathcal{P}_{x_1,x_2}}$

Original source details: GELFAND-P001525

$$= W^{\alpha\beta}(x_1, x_2)$$

$$\mathcal{P}_{x_1,x_2} = \{\text{paths from } x_1 \text{ to } x_2\}$$

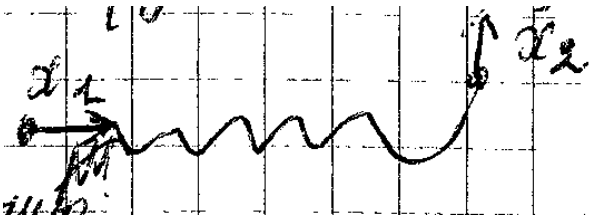


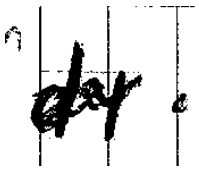
Diagram and typed description.

[Unclear mark below the first arrow.]

Original source details: GELFAND-P001525

By definition:

$$\int d\psi = 0, \qquad \int d\psi \cdot \psi = 1 \tag{13}$$



Uncertain reading. $d\psi$

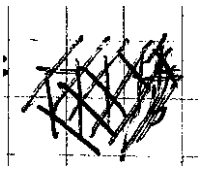
Original source details: GELFAND-P001525

Parafermionic variables are sometimes needed (see V. Fateev, A. Zamolodchikov, ZhETF [below the journal title: 1985]). [Below the line:] constructed explicitly in conformal field theory

Remark. Calculation of the specific heat in repr. (6)

Set

$$F(\beta) = \ln Z(\beta) \tag{14}$$



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Uncertain reading. Crossed out before $\ln Z(\beta)$: reading undetermined.

$$\frac{d^2F(\beta)}{d\beta^2} \sim \frac{Z''}{Z} - \frac{(Z')^2}{Z^2} = \underset{\text{specific heat}}{\underline{\underline{C(\beta)}}} \sim (\beta - \beta_c)^{-\alpha} \tag{15}$$

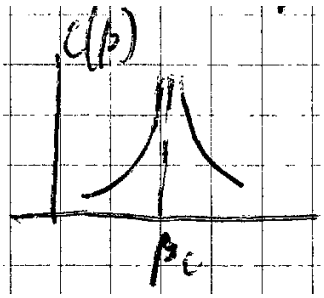

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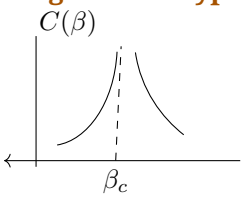
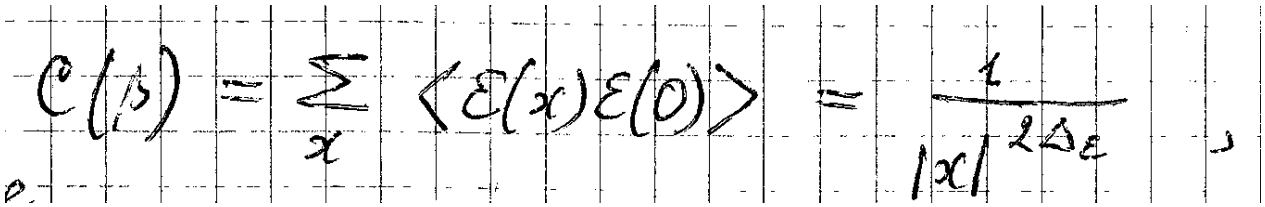
Diagram and typed description.



The exponent α is a critical index. We have:

$$C(\beta) = \sum_x \langle \varepsilon(x)\varepsilon(0)\rangle = \frac{1}{|x|^{2\Delta_\varepsilon}}, \tag{16}$$

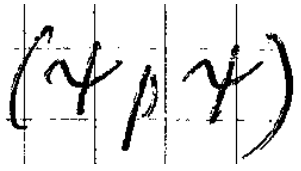


Original source details: GELFAND-P001525

Uncertain reading. $\frac{1}{|x|^{2\Delta_\varepsilon}}$

where

$$\varepsilon = (\psi\,\rho\,\psi)$$



Original source details: GELFAND-P001525

Uncertain reading. ρ

Here Δ_ε is uniquely related to α and is called the anomalous dimension:

$$\alpha = \frac{d - 2\Delta_\varepsilon}{d - \Delta_\varepsilon}, \qquad d \text{ — dimension of space} \tag{17}$$

(the usual scaling relation)

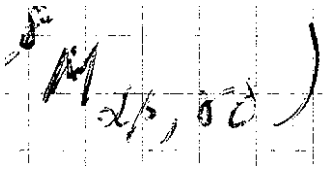
3-dimensional case.

$$Z(\beta) = \prod_{\text{edges}} \int d\psi^\alpha_{x,\mu} \, \exp\{A[\psi]\} \tag{18}$$

$$\begin{aligned} A[\psi] = {} & \frac{1}{2} \sum_{\text{edges}} \left(\psi^\alpha C^{\alpha\beta} \psi^\beta \right) \\ & + \lambda \sum_{\text{faces}} \left(\psi^\alpha \psi^\beta \psi^\gamma \psi^\delta M_{\alpha\beta,\gamma\delta \,/\, \delta\gamma} \right) \end{aligned} \tag{19}$$

Uncertain reading. β

Original source details: GELFAND-P001526



Uncertain reading. $\gamma\delta \,/\, \delta\gamma$

Original source details: GELFAND-P001526

[In the first term the index of the last ψ and the closing parenthesis are heavily overwritten.]

$$Z(\beta) = \sum_S \lambda^{A(S)}, \tag{20}$$

where S — closed surface in R^3,

(encl. different phases)

Uncertain reading. encl.

Original source details: GELFAND-P001526

$\psi^\alpha_{x,\mu}$ — Grassmann variables

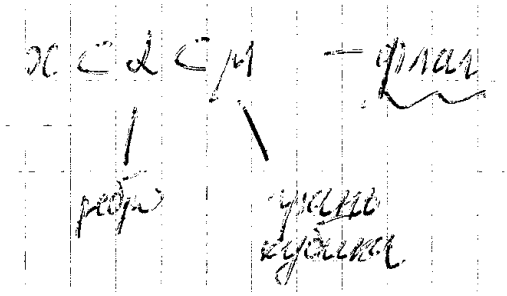


Diagram and typed description.

$x \subset \alpha \subset \mu$ — flag

edge

face of a cube

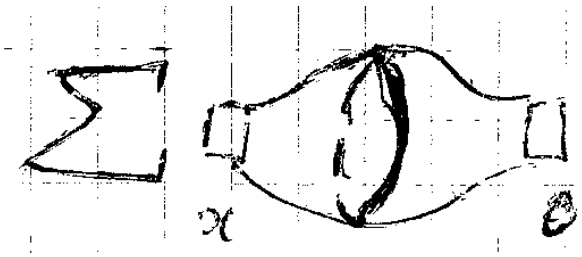
Original source details: GELFAND-P001526

Let us calculate

$$\ln Z = F, \tag{21}$$

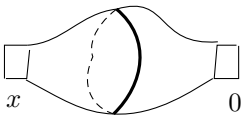
then

$$F'' = \sum \langle \varepsilon(x)\varepsilon(0)\rangle \tag{22}$$



Original source details: GELFAND-P001526

Diagram and typed description. $= \sum$



[The thick strokes at the junctions are indistinct.]

(birth of a closed fermionic string)

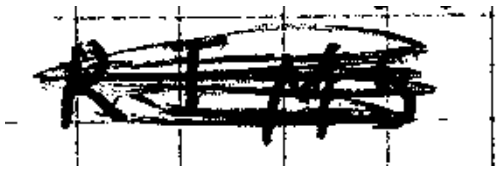
GELFAND-P001527

[source number: 39]

Next one has to solve the problem in the continuum limit.

References: Dotsenko, Polyakov (~ 60 pp.)

(in the book Advanced Studies in Pure Mathematics,



Original source details: GELFAND-P001527

Undeciphered passage. Crossed out below "in the book": a short fragment, the letters obscured by strokes.

v. 16, 1988) [The first digit of the year is heavily retraced.]

Original source details: GELFAND-P001527

Uncertain reading. Crossed-out sign between Pure and Mathematics: &